Dark Matter Detection in Solids

Claude
Edited by To Chin Yu

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Conventions and units

We use natural units, $\hbar = c = k_{\mathrm{B}} = 1$, unless stated otherwise, and quote physical values in conventional units with SI prefixes.

A widely used formula from the literature appears in a gray box, a Limit Check, that plots it against the book's more complete result, with the range of agreement shaded; Sect. 9.4 explains how a box is read, and Appendix F records the formal details of each.

Part I

Foundations

Chapter 1

Light dark matter

Most of the matter in the universe has been detected only through its gravity. It shapes galaxies and the cosmic microwave background and outweighs the atoms we know five to one, yet decades of searches for a heavy particle that recoils off a nucleus have found nothing. Those null results have shifted attention to lighter particles, below a GeV, which carry too little momentum to move a nucleus measurably but enough energy to lift an electron across a band gap or to set a crystal lattice vibrating. Detecting them turns a dark-matter search into a problem in condensed-matter physics: the target is no longer a collection of free nuclei but a solid, with all of its electronic and vibrational structure. What such a detector can see is set by a single quantity, the rate at which it registers dark-matter events, and that rate factorizes into a dark-matter part, the halo and the coupling, and the response of the solid.

1.1 The master equation of direct detection

A dark-matter detector counts events. Each one is a small burst of electron–hole pairs, phonons, quasiparticles or photons, released when a particle from the Galactic halo gives up an energy ω and a momentum $\boldsymbol{q}$ to the target. What theory has to predict is how many such events a kilogram of target collects per year, the *event rate* $\mathcal{R}$, and how they are distributed in ω. For dark matter χ of mass m_χ that scatters on the electrons of the target through an interaction that couples to their density and not to their spins, the rate has the form

$$\mathcal{R} = \frac{1}{\rho_T} \underbrace{\frac{\rho_\chi}{m_\chi} \int \mathrm{d}^3 v \, f_\chi(\boldsymbol{v})}_{\text{halo}} \int \frac{\mathrm{d}^3 q}{(2\pi)^3} \underbrace{|\tilde{V}(\boldsymbol{q})|^2}_{\text{coupling}} \int \mathrm{d}\omega \, \underbrace{S(\boldsymbol{q},\omega)}_{\text{target}} \underbrace{\delta\!\left(\omega - \boldsymbol{q}\cdot\boldsymbol{v} + \frac{q^2}{2m_\chi}\right)}_{\text{kinematics}} . \tag{1.1}$$

We will derive it in the chapters that follow: Chap. 2 obtains it from Fermi's golden rule (Eq. (2.67)), and once the directions of $\boldsymbol{q}$ are averaged it reduces to Eq. (1.47), the form used for limits [1].

Under the integrals sit the halo, the coupling and the response of the target, tied together by a delta function. The halo enters through ρ_χ/m_χ, the number density of dark-matter particles near the Earth, with ρ_χ their mass density, and through $f_\chi(\boldsymbol{v})$, the distribution of their velocities $\boldsymbol{v}$ in the laboratory, normalized to one. The coupling is $\tilde{V}(\boldsymbol{q})$, the Fourier transform of the potential between a dark-matter particle and an electron, and its square weights each momentum transfer by the strength of the dark-matter–electron interaction at that momentum. The response is the *dynamic structure factor* $S(\boldsymbol{q},\omega)$ of the electron density, which measures how readily the solid absorbs a momentum $\boldsymbol{q}$ together with an energy ω. At zero temperature it is a sum over the excited states of the solid that lie an energy ω above its ground state. Each term is the squared matrix element of the electron density at wavevector

$\boldsymbol{q}$ between the two states. A factor 2π comes from the golden rule, and a division by the volume of the target makes S independent of its size (Eq. (2.67); Sect. 4.2). The delta function is energy conservation. A particle whose momentum changes by $\boldsymbol{q}$ loses the kinetic energy $\boldsymbol{q}\cdot\boldsymbol{v} - q^2/2m_\chi$, and the target must take up exactly that amount. Finally, $1/\rho_T$, with ρ_T the mass density of the target, turns a rate per unit volume into a rate per unit target mass, the unit in which experiments quote it.

The value of Eq. (1.1) is that each factor belongs to a different body of physics. The halo is astrophysics, the coupling is particle physics, and the response is condensed-matter physics; only the kinematics of the delta function connects them, and it involves only $\boldsymbol{q}$, $\boldsymbol{v}$, ω and m_χ. As a result, a new dark-matter model changes $|\tilde{V}|^2$ and leaves S alone, while a new target changes S and leaves the coupling alone. A response computed once for silicon serves every model that couples to the electron density, and once S is tabulated, a limit can be recast for another model by weighting it with the new coupling, without computing the solid again. The response involves of order 10^{23} interacting electrons per cubic centimeter in a periodic lattice, and computing it is most of what this book does.

The halo needs no new physics. Its local density and velocity distribution are adopted from astronomy by convention, $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$ and a Maxwellian with speeds near 10^{-3} in units of the speed of light, seen from a laboratory that moves through the halo (Sect. 1.2). Since the delta function involves no property of the target, much of what a detector can reach follows before any target is specified (Sect. 1.3). Kinematics fixes the smallest speed that can deposit a given ω at a given q and the lightest mass that can get over a detector threshold. Once the velocity integral is done, all that remains of the halo is $\eta(v_{\min})$, the mean inverse speed above that minimum speed. The coupling is a choice of model. Its simplest form is the exchange of a *mediator*, a particle that carries the interaction between the dark matter and the electron. The coupling then condenses into $|\tilde{V}(q)|^2 = \pi\bar{\sigma}_e F_{\mathrm{DM}}(q)^2/\mu_{\chi e}^2$ (Eq. (1.34)), with a reference cross-section $\bar{\sigma}_e$ that sets the strength, a form factor $F_{\mathrm{DM}}(q)$ that carries the momentum dependence and the reduced mass $\mu_{\chi e}$ of the dark matter and the electron (Sect. 1.4). An experiment then quotes its result as a limit on $\bar{\sigma}_e$ at each mass. The response, however, is a many-body problem. It needs the electronic states of a crystal, the linear response of its electrons, the vibrations of its lattice and the coupling between the two (Chaps. 3 to 7), a framework that holds for any coupling (Chap. 8) and a separate treatment of each excitation channel (Chaps. 10 to 17). Even an exact S does not finish the calculation, because a detector does not record ω. It records the charges, phonons, quasiparticles or photons into which the deposit relaxes, and that conversion is the last link between the rate and a measured count (Sect. 1.5; Chaps. 18 to 21).

The product in Eq. (1.1) is the simplest case, one coupling to one density, and the book generalizes it in several directions without losing the separation. Dark matter can couple to more than the electron density, to nuclei, spins or currents as well, and the single S then becomes a matrix of response functions, one for each pair of couplings, whose off-diagonal entries make the contributions interfere (Chap. 8). The electrons of the target also rearrange to screen the dark-matter potential. For a mediator that mixes with the photon, the screened response is the *loss function*, the imaginary part of the inverse dielectric function, which takes the place of S (Sect. 10.2). A bosonic dark-matter particle can instead be absorbed. It deposits its whole mass m_ϕ at almost zero momentum, and its rate is then set by the optical conductivity of the target at $\omega = m_\phi$ instead of by a scattering response (Sects. 1.4 and 10.7). Other processes need more than one response at once. In the Migdal effect a recoiling nucleus shakes off an electron, which takes the joint response of electrons and lattice (Chap. 12). An electron excited across an indirect gap, where the highest filled and the lowest empty electron states differ in crystal momentum (Sect. 1.3), needs a phonon to supply the missing momentum, a second-order process (Chap. 13). Finally, no computed response is exact. Each

S in the book names the approximation behind it and comes with an estimate of its error, and Chap. 21 carries those errors into the exclusion limits.

1.1.1 Inputs from cosmology

All of the evidence for dark matter is gravitational. Stars and gas in spiral galaxies orbit at a circular speed v_c that stays flat far outside the luminous disk, where the visible mass alone would make it fall as $r^{-1/2}$ with the galactocentric radius r. A flat curve needs an enclosed mass that grows linearly with r where almost no light is emitted [2, 3]. In galaxy clusters, the velocities of the galaxies, the temperature of the hot gas and gravitational lensing all require several times more mass than is seen [3]. In the colliding clusters 1E 0657-558 the lensing mass is displaced from the X-ray-emitting gas that holds most of the baryons, which shows that the extra mass is not a modification of gravity tied to the visible matter [4]. The anisotropies of the cosmic microwave background (CMB) fix the total amount, because dark matter does not take part in the acoustic oscillations of the photon–baryon plasma before recombination, and its potential wells shape the relative heights of the acoustic peaks [5].

None of this identifies the particle. The observations measure a mass density, and for detection they supply two numbers. The local density ρ_χ comes from the motion of stars near the Sun and from the rotation curve of the Milky Way [6, Sect. 3.1]; Sect. 1.2 gives the measured range and the value we use. The cosmological abundance is quoted as $\Omega_c h^2$, where Ω_c is the mean density of cold dark matter as a fraction of the critical density, the density of a spatially flat universe, and h is the present Hubble rate in units of $100\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. The CMB anisotropies, with CMB lensing, give [5, Table 2 and Eq. (23)]

$$\Omega_c h^2 = 0.1200 \pm 0.0012,$$

usually quoted as 0.120 ± 0.001. The baryon density from the same fit, 0.02237 ± 0.00015 in the same units, is smaller by a factor of 5.4.

Gravity cannot fix the mass m_χ of the dark-matter particle: by the equivalence principle, a test body falls the same way whatever its mass, and every observation above measures the mass density $n_\chi m_\chi$ alone, with n_χ the number density. The mass enters only where the particle picture itself fails. At the light end, a boson with $m_\chi \approx 10^{-22}\,\mathrm{eV}$ has a de Broglie wavelength comparable to a dwarf galaxy, and lighter fields cannot cluster on those scales. Fermions need m_χ above a few hundred eV so that halo phase-space densities stay below the Pauli limit, the largest density that the exclusion principle allows; the argument is related to the Tremaine–Gunn bound [7, 8]. A particle once in equilibrium with the plasma must be heavier than roughly a keV, or it moves fast enough to erase observed small-scale structure [8, Sect. I A]. At the heavy end, fundamental particles reach the Planck mass and compact objects such as primordial black holes extend to tens of solar masses [8, Sect. I C]. The allowed range spans about 90 decades (Fig. 1.1).

Within this range, the window between a keV and a GeV has been tested least. Dark matter bound to the Galaxy moves at $v \approx 10^{-3}$, which puts its kinetic energy $\frac{1}{2} m_\chi v^2$ at 0.5 keV for 1 GeV, 0.5 eV for 1 MeV and 0.5 meV for 1 keV. A 1 GeV particle can deposit at most about 0.1 keV in a xenon nucleus (Sect. 1.3). That is below the thresholds of ton-scale xenon detectors, which read the scintillation and ionization of the recoiling nucleus: Ref. [1, Sect. I] places them at the few-hundred-eV scale, and analyses that require both signals sit nearer a keV. Solids, however, have excitations across the range of these kinetic energies: typical electronic band gaps of an eV and optical phonons of tens of meV [1, Table I], and acoustic phonons and superconducting gaps at the meV scale and below (Sects. 1.3 and 1.5). Particles in this window, between a keV and a GeV, are what the field calls *light dark matter*, or sub-GeV dark matter.

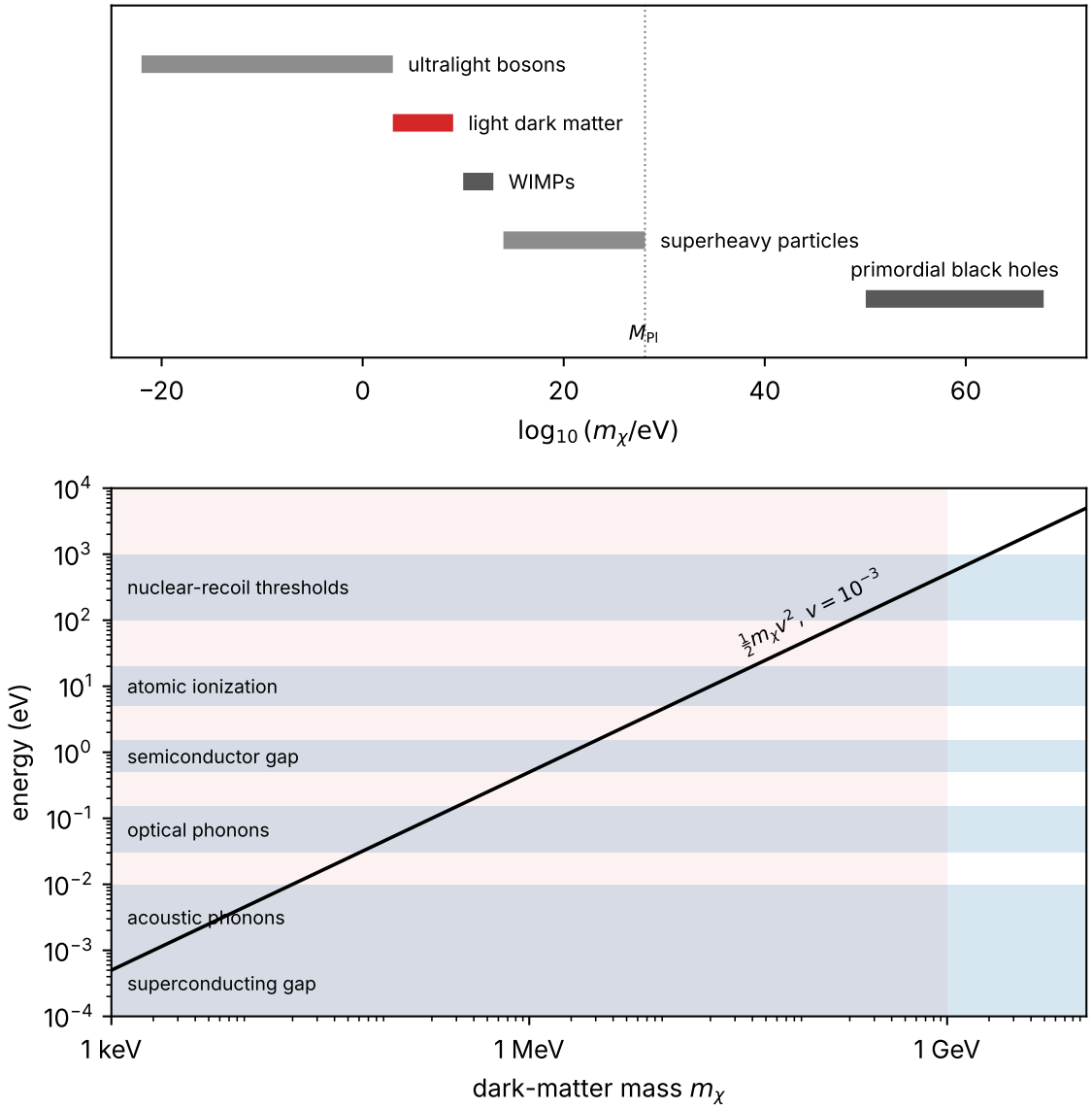


Figure 1.1: Top: masses of dark-matter candidates, from ultralight bosonic fields to primordial black holes; the dotted line marks the Planck mass. Ranges follow Ref. [8, Sect. I]. Bottom: the keV–GeV window (shaded red). The solid line is the kinetic energy $\frac{1}{2}m_\chi v^2$ of dark matter moving at $v = 10^{-3}$. The blue bands are the energy scales of excitations in solids and the thresholds of nuclear-recoil detectors [1].

Why a particle in this window should have the observed abundance is a separate question, and the simplest answer, thermal freeze-out, constrains how light dark matter interacts. Suppose χ was once in equilibrium with the hot plasma of the early universe. Once the temperature drops below m_χ, the equilibrium density falls exponentially, and annihilation stops when a particle no longer meets a partner within one expansion time. The number left over is then inversely proportional to the thermally averaged annihilation cross-section times relative speed, $\langle\sigma v\rangle$ [8, Eq. (32)]: the more efficiently dark matter annihilates, the less of it survives. The observed $\Omega_c h^2$ requires $\langle\sigma v\rangle \approx 2\times10^{-26}\,\mathrm{cm^3\,s^{-1}}$ to $3\times10^{-26}\,\mathrm{cm^3\,s^{-1}}$ for masses above about 10 GeV. The commonly quoted benchmark is $3\times10^{-26}\,\mathrm{cm^3\,s^{-1}}$, while a numerical solution of the Boltzmann equation for the number density gives $2.2\times10^{-26}\,\mathrm{cm^3\,s^{-1}}$ for a Majorana particle, one that is its own antiparticle, rising to $5.2\times10^{-26}\,\mathrm{cm^3\,s^{-1}}$ near 0.3 GeV [9, abstract and Sect. II C]. A particle with weak-scale mass and couplings lands near this value, which made the weakly interacting massive particle (WIMP) the historical benchmark.

The same requirement rules out light dark matter that annihilates only through heavy particles. If χ annihilates through the weak bosons, or through any mediator much heavier than itself, $\langle\sigma v\rangle$ scales as the square of the dark-matter mass divided by the fourth power of the mediator mass, as in Fermi's theory of the weak interaction [8, Eq. (44)]. Below a few GeV it then falls short of the thermal value, too many particles survive, and the universe is overclosed; for a heavy neutrino this is the Lee–Weinberg bound, $m_\chi \gtrsim 2\,\mathrm{GeV}$ [10]. Thermal light dark matter therefore needs a new, light mediator, with a mass at or below the GeV scale, to carry the interaction between χ and ordinary matter [11]. Alternatively, χ was never in equilibrium and was produced by *freeze-in*, slow leakage from the plasma through a feeble coupling [12], or by another non-thermal history. Both histories fix the coupling factor of Eq. (1.1), and Sect. 1.4 develops them. Whatever its history, a detector sees the dark matter that is in the Galaxy today, and the next step is the velocity distribution of the particles that reach it (Sect. 1.2).

1.2 The local halo

Every rate in this book is an average over the dark matter that passes through the detector. That average is fixed by the local mass density, which sets how many particles there are, and by the velocity distribution in the laboratory, which sets how fast they move and from which direction they come. We will fix both in the conventions of Appendix A.8 and then work out how the motion of the Earth makes the distribution change with the seasons and, for a crystal, with the time of day.

1.2.1 Density and velocity distribution

Measurements of the local dark-matter density, from the motion of stars near the Sun and from the rotation curve of the Galaxy, give values between 0.2 and $0.6\,\mathrm{GeV\,cm^{-3}}$ [13, Sect. 3.1.1]. Mass models with different halo shapes and data sets disagree by more than their statistical errors [6, Sect. 3.1]. For this reason direct-detection results are quoted for a fixed reference value; we use the recommended one [13, Table 1],

$$\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}} = 2.31\times10^{-6}\,\mathrm{eV^4}, \tag{1.2}$$

where the value in natural units follows from multiplying by $(\hbar c)^3$, with $\hbar c = 1.973\times10^{-5}\,\mathrm{eV\,cm}$ (Appendix A.1). Since the density enters every rate as an overall factor, a different value rescales a limit without changing its shape [13, Sect. 3.1.1]. The number density of particles

of mass m_χ is $n_\chi = \rho_\chi/m_\chi$: three particles per cubic centimeter at 100 MeV, three thousand at 100 keV. For comparison, $0.3\,\mathrm{GeV\,cm^{-3}}$ is the mass of one hydrogen atom in roughly $3\,\mathrm{cm^3}$, while a crystal holds about 10^{23} atoms in each cubic centimeter.

The velocity distribution is modeled first in the rest frame of the Galaxy, in which the halo as a whole does not rotate. We write $\boldsymbol{v}_{\mathrm{gal}}$ for a dark-matter velocity in that frame and use Cartesian galactic axes: the first points from the Sun to the Galactic center, the second along the direction of Galactic rotation, and the third to the north Galactic pole [6, Sect. 5]. The *standard halo model* (SHM) takes the halo to be a singular isothermal sphere, a self-gravitating gas with density $\rho_{\mathrm{halo}}(r) \propto r^{-2}$ at galactocentric radius r, which encloses a mass growing linearly with r and so gives the flat rotation curve of Sect. 1.1 [6, Sect. 2.1]. Its particles have an isotropic Maxwellian distribution,

$$f_{\mathrm{gal}}(\boldsymbol{v}_{\mathrm{gal}}) = \frac{1}{N_{\mathrm{esc}}}\, \mathrm{e}^{-v_{\mathrm{gal}}^2/v_0^2}\, \Theta(v_{\mathrm{esc}} - v_{\mathrm{gal}}), \tag{1.3}$$

normalized to $\int \mathrm{d}^3 v_{\mathrm{gal}}\, f_{\mathrm{gal}} = 1$, with $v_{\mathrm{gal}} = |\boldsymbol{v}_{\mathrm{gal}}|$ and $\Theta(x)$ the Heaviside step function [13, Eq. (10)]. It is specified by the speed parameter v_0 and the escape speed v_{esc}, which together fix the normalization N_{esc}.

The speed parameter v_0 is the local circular speed. Each Cartesian component of $\boldsymbol{v}_{\mathrm{gal}}$ in Eq. (1.3) has variance $v_0^2/2$. For collisionless particles this velocity variance plays the role of a pressure, $\rho_{\mathrm{halo}} v_0^2/2$: the Jeans equation, the first velocity moment of the collisionless Boltzmann equation, has the form of the equation of hydrostatic equilibrium. Jeans equilibrium of an isotropic, spherical system balances the pressure gradient against gravity, $\mathrm{d}(\rho_{\mathrm{halo}} v_0^2/2)/\mathrm{d}r = -\rho_{\mathrm{halo}} v_{\mathrm{c}}^2/r$ [6, Eq. (11)]. For $\rho_{\mathrm{halo}} \propto r^{-2}$ the left side is $-\rho_{\mathrm{halo}} v_0^2/r$, hence $v_0 = v_{\mathrm{c}}$. We take the speed of the local standard of rest, the circular orbit at the Sun's radius, $v_0 = 238\,\mathrm{km\,s^{-1}}$ (derived below Eq. (1.5)); much of the literature uses the older $220\,\mathrm{km\,s^{-1}}$.

The escape speed v_{esc} truncates the distribution. A particle faster than the local escape speed is not bound to the Galaxy and is assumed absent. Since the isothermal sphere itself has an infinite escape speed, the cut is imposed by hand, and the sharp step is an idealization [6, Sect. 2.1]. The value $v_{\mathrm{esc}} = 544\,\mathrm{km\,s^{-1}}$ comes from the fastest stars in the Radial Velocity Experiment (RAVE) survey; later surveys give medians from 485 to $580\,\mathrm{km\,s^{-1}}$ [13, Sect. 3.1.2 and Table 2].

The normalization N_{esc} follows in spherical coordinates [14, Eq. (15)]. Since the integrand depends only on v_{gal}, the angles contribute 4π, and

$$N_{\mathrm{esc}} = 4\pi \int_0^{v_{\mathrm{esc}}} \mathrm{d}y\, y^2 \mathrm{e}^{-y^2/v_0^2} = \pi^{3/2} v_0^3 \left[\mathrm{erf}\left(\frac{v_{\mathrm{esc}}}{v_0}\right) - \frac{2}{\sqrt{\pi}}\, \frac{v_{\mathrm{esc}}}{v_0}\, \mathrm{e}^{-v_{\mathrm{esc}}^2/v_0^2} \right], \tag{1.4}$$

with the error function $\mathrm{erf}(x) = (2/\sqrt{\pi}) \int_0^x \mathrm{e}^{-y^2} \mathrm{d}y$ (Appendix C.8). The second equality integrates $y^2 \mathrm{e}^{-y^2/v_0^2}$ once by parts; Appendix D.1 writes the step out. Without the cut, N_{esc} would be the Gaussian integral $\pi^{3/2} v_0^3$; the bracket is the fraction of that Gaussian below v_{esc}.

The Sun moves through this distribution. Its velocity in the galactic frame is the circular velocity of the local standard of rest plus the Sun's peculiar motion relative to it [13, Table 1 and Eq. (14)],

$$\boldsymbol{v}_\odot = (0, 238, 0) + (11.1, 12.2, 7.3)\ \mathrm{km\,s^{-1}} = (11.1, 250.2, 7.3)\ \mathrm{km\,s^{-1}}, \tag{1.5}$$

in the galactic axes above. The in-plane component is what is measured: the Sun's angular velocity about the Galactic center, $30.24\,\mathrm{km\,s^{-1}\,kpc^{-1}}$ from the proper motion of the central radio source, times the Sun's distance from the center, 8.275 kpc, is $250.2\,\mathrm{km\,s^{-1}}$, and subtracting the peculiar $12.2\,\mathrm{km\,s^{-1}}$ leaves $v_0 = 238\,\mathrm{km\,s^{-1}}$ [13, Sect. 3.1.5]. The magnitude is $|\boldsymbol{v}_\odot| = (11.1^2 + 250.2^2 + 7.3^2)^{1/2}\,\mathrm{km\,s^{-1}} = 250.55\,\mathrm{km\,s^{-1}}$, which we round to the reference value $250.5\,\mathrm{km\,s^{-1}}$. Table 1.1 collects the parameters.

Table 1.1: Standard-halo-model parameters adopted in this book, with their sources. Vectors are given in galactic axes (toward the Galactic center, along Galactic rotation, toward the north Galactic pole). The last two rows follow from the others. The reference Earth speed is the adopted magnitude of the solar velocity $(0, 238, 0) + (11.1, 12.2, 7.3)\,\mathrm{km\,s^{-1}}$. The largest laboratory speed adds it to the escape speed.

Symbol	Quantity	Value	Source
ρ_χ	local dark-matter density	$0.3\,\mathrm{GeV\,cm^{-3}}$	[13, Table 1]
v_0	circular speed at the Sun's radius	$238\,\mathrm{km\,s^{-1}}$	[13, Table 1]
v_{esc}	Galactic escape speed	$544\,\mathrm{km\,s^{-1}}$	[13, Table 1]
–	solar peculiar velocity	$(11.1, 12.2, 7.3)\,\mathrm{km\,s^{-1}}$	[13, Table 1]
–	mean orbital speed of the Earth	$29.8\,\mathrm{km\,s^{-1}}$	[13, Eq. (13)]
$\boldsymbol{v}_\oplus(t)$	Earth velocity relative to the Sun	Eq. (1.12)	[13, Eq. (11)]
$v_{\mathrm{E}}^{\mathrm{ref}}$	reference Earth speed (adopted $\lvert\boldsymbol{v}_\odot\rvert$)	$250.5\,\mathrm{km\,s^{-1}}$	Eq. (1.5)
v_{max}	largest laboratory speed $v_{\mathrm{esc}} + v_{\mathrm{E}}^{\mathrm{ref}}$	$794.5\,\mathrm{km\,s^{-1}}$	Eq. (1.9)

1.2.2 The laboratory frame

A detector shares the motion of the Earth. Its velocity in the galactic frame is

$$\boldsymbol{v}_{\mathrm{E}}(t) = \boldsymbol{v}_\odot + \boldsymbol{v}_\oplus(t), \tag{1.6}$$

where $\boldsymbol{v}_\oplus(t)$ is the Earth's orbital velocity relative to the Sun. Since galactic speeds are below 10^{-2} in natural units, the change of frame is Galilean, with relativistic corrections of relative size $v^2 \approx 10^{-6}$. A particle with laboratory velocity $\boldsymbol{v}$ has galactic velocity

$$\boldsymbol{v}_{\mathrm{gal}} = \boldsymbol{v} + \boldsymbol{v}_{\mathrm{E}}(t). \tag{1.7}$$

The shift has unit Jacobian, $\mathrm{d}^3 v_{\mathrm{gal}} = \mathrm{d}^3 v$, and a velocity cell holds the same number of particles in both frames. The laboratory distribution is the galactic one of Eq. (1.3) [13, Eq. (10)] evaluated at the boosted velocity, $f_\chi(\boldsymbol{v}) = f_{\mathrm{gal}}(\boldsymbol{v} + \boldsymbol{v}_{\mathrm{E}}(t))$:

$$f_\chi(\boldsymbol{v}, t) = \frac{1}{N_{\mathrm{esc}}}\, \mathrm{e}^{-|\boldsymbol{v}+\boldsymbol{v}_{\mathrm{E}}(t)|^2/v_0^2}\, \Theta\big(v_{\mathrm{esc}} - |\boldsymbol{v} + \boldsymbol{v}_{\mathrm{E}}(t)|\big). \tag{1.8}$$

We write $f_\chi(\boldsymbol{v})$ when the time is not needed, and $v = |\boldsymbol{v}|$ for the laboratory speed. The normalization is unchanged, $\int \mathrm{d}^3 v\, f_\chi = 1$, and we write angle brackets for an average over the distribution, $\langle \cdots \rangle = \int \mathrm{d}^3 v\, f_\chi(\boldsymbol{v})\, (\cdots)$. In the laboratory the distribution is no longer centered on zero: the most probable velocity is $-\boldsymbol{v}_{\mathrm{E}}$. Dark matter arrives as a wind from the direction in which the Earth moves. Because f_{gal} is even in $\boldsymbol{v}_{\mathrm{gal}}$, its mean velocity vanishes, and by Eq. (1.7) the mean laboratory velocity is exactly $\langle \boldsymbol{v} \rangle = -\boldsymbol{v}_{\mathrm{E}}(t)$, cut or no cut.

The spread about this mean is the galactic one: $\langle v_{\mathrm{gal}}^2 \rangle^{1/2} = (3/2)^{1/2} v_0 = 291\,\mathrm{km\,s^{-1}}$ for the Gaussian and $284\,\mathrm{km\,s^{-1}}$ with the cut. The spread is larger than the drift: the wind is broad. A numerical integration of Eq. (1.8) at $v_{\mathrm{E}} = v_{\mathrm{E}}^{\mathrm{ref}}$ shows that 94 % of the particles move with a positive component along $-\boldsymbol{v}_{\mathrm{E}}$, and 59 % within 45° of it; the mean cosine of the angle to $-\boldsymbol{v}_{\mathrm{E}}$ is 0.66.

The fastest particle in the laboratory is one that moves at v_{esc} in the galactic frame, opposite to $\boldsymbol{v}_{\mathrm{E}}$; its laboratory speed is $v_{\mathrm{esc}} + |\boldsymbol{v}_{\mathrm{E}}(t)|$. We quote thresholds and kinematic edges at a fixed reference speed, the rounded solar speed $v_{\mathrm{E}}^{\mathrm{ref}} = 250.5\,\mathrm{km\,s^{-1}}$ (Eq. (1.5); Appendix A.8). The *maximum speed* is then

$$v_{\mathrm{max}} = v_{\mathrm{esc}} + v_{\mathrm{E}}^{\mathrm{ref}} = 794.5\,\mathrm{km\,s^{-1}} = 2.650 \times 10^{-3}. \tag{1.9}$$

Over a year $v_{\rm esc} + |\boldsymbol{v}_{\rm E}(t)|$ ranges from 781 to 810 km s^{-1} (see the annual modulation below). With the reference value, thresholds stay time-independent, while integrals over f_χ use the full $\boldsymbol{v}_{\rm E}(t)$, averaged over the exposure. Note that the annual mean of $|\boldsymbol{v}_{\rm E}(t)|$ is 252.1 km s^{-1}, not 250.5 km s^{-1}, because the orbital velocity perpendicular to $\boldsymbol{v}_\odot$ adds to the speed in quadrature.

Many rates depend only on the speed. The *speed distribution* $F_\chi(v)$, normalized to $\int_0^\infty \mathrm{d}v\, F_\chi = 1$, is v^2 times the integral of f_χ over directions. Take the polar axis along $\boldsymbol{v}_{\rm E}$, write ϑ for the angle between $\boldsymbol{v}$ and $\boldsymbol{v}_{\rm E}$, and $v_{\rm E} = |\boldsymbol{v}_{\rm E}|$. Then $|\boldsymbol{v} + \boldsymbol{v}_{\rm E}|^2 = v^2 + v_{\rm E}^2 + 2vv_{\rm E}\cos\vartheta$, and the azimuth contributes 2π:

$$F_\chi(v) = \frac{2\pi v^2}{N_{\rm esc}}\, \mathrm{e}^{-(v^2+v_{\rm E}^2)/v_0^2} \int_{-1}^{\cos\vartheta_{\rm cut}} \mathrm{d}(\cos\vartheta)\, \mathrm{e}^{-2vv_{\rm E}\cos\vartheta/v_0^2}.$$

The upper limit $\cos\vartheta_{\rm cut}$ is where $|\boldsymbol{v} + \boldsymbol{v}_{\rm E}| = v_{\rm esc}$, or 1 if that never happens. For $v < v_{\rm esc} - v_{\rm E}$ no direction reaches the cut and $\cos\vartheta_{\rm cut} = 1$. For $v_{\rm esc} - v_{\rm E} < v < v_{\rm esc} + v_{\rm E}$ the cut removes the directions close to $\boldsymbol{v}_{\rm E}$, and at $\cos\vartheta_{\rm cut}$ the exponent equals $-v_{\rm esc}^2/v_0^2$. Doing the elementary integral, we find

$$F_\chi(v) = \frac{\pi v_0^2\, v}{v_{\rm E} N_{\rm esc}} \times \begin{cases} \mathrm{e}^{-(v-v_{\rm E})^2/v_0^2} - \mathrm{e}^{-(v+v_{\rm E})^2/v_0^2}, & v < v_{\rm esc} - v_{\rm E}, \\ \mathrm{e}^{-(v-v_{\rm E})^2/v_0^2} - \mathrm{e}^{-v_{\rm esc}^2/v_0^2}, & v_{\rm esc} - v_{\rm E} < v < v_{\rm esc} + v_{\rm E} \end{cases}, \tag{1.10}$$

and $F_\chi = 0$ above $v_{\rm esc} + v_{\rm E}$. The two branches join continuously at $v = v_{\rm esc} - v_{\rm E}$, and the second vanishes at $v_{\rm esc} + v_{\rm E}$. Eq. (1.10) assumes $v_{\rm E} < v_{\rm esc}$, which holds by a factor of two. Fig. 1.2 shows F_χ on the days of largest and smallest Earth speed. In the galactic frame the speed distribution $4\pi v_{\rm gal}^2 f_{\rm gal}$ peaks at v_0; in the laboratory the drift moves the peak to 335 km s^{-1} (at $v_{\rm E} = v_{\rm E}^{\rm ref}$) and stretches the tail to $v_{\rm esc} + v_{\rm E}$.

The tail of F_χ matters for light dark matter, which must often use the fastest particles to deposit enough energy (Sect. 1.3). We integrate Eq. (1.10) numerically at $v_{\rm E} = v_{\rm E}^{\rm ref}$. The fraction of particles faster than 400, 500, 600 and 700 km s^{-1} is 36 %, 15 %, 4.2 % and 0.58 %; above 750 km s^{-1} it is 0.10 %. A threshold that needs $v \gtrsim 700$ km s^{-1} sees a rate suppressed by more than two orders of magnitude, and one that depends on $v_{\rm esc}$ and on the exact shape of the cut, the most uncertain parts of the model.

Scattering rates contain the speed through $1/v$: the energy-conservation delta function of a collision, integrated over the direction of $\boldsymbol{v}$, leaves a factor $1/v$ (Sect. 1.3). In the SHM this average has a closed form.

Worked example 1.1: Normalization and mean inverse speed

With $v_{\rm esc}/v_0 = 544/238 = 2.286$, the error function is $\mathrm{erf}(2.286) = 0.99877$ and the second term in the bracket of Eq. (1.4) is $(2/\sqrt{\pi})(2.286)\,\mathrm{e}^{-5.224} = 2.579 \times 0.005383 = 0.01388$. The bracket is 0.98489: the cut removes 1.5 % of the Gaussian. With $\pi^{3/2} v_0^3 = 5.5683 \times (238)^3\,(\mathrm{km/s})^3 = 7.507 \times 10^7\,(\mathrm{km/s})^3$,

$$N_{\rm esc} = 0.98489 \times 7.507 \times 10^7\,(\mathrm{km/s})^3 = 7.393 \times 10^7\,(\mathrm{km/s})^3.$$

The mean inverse speed is $\langle 1/v \rangle = \int_0^\infty \mathrm{d}v\, F_\chi(v)/v$. Dividing Eq. (1.10) by v leaves Gaussians in $v \mp v_{\rm E}$ and a constant. Substituting $y = v - v_{\rm E}$ in the first exponential and $y = v + v_{\rm E}$ in the second, the two cancel on $v_{\rm E} < y < v_{\rm esc}$ and leave $\int_{-v_{\rm E}}^{v_{\rm E}} \mathrm{e}^{-y^2/v_0^2}\mathrm{d}y = \sqrt{\pi}\, v_0\, \mathrm{erf}(v_{\rm E}/v_0)$; the constant term, integrated over an interval of length $2v_{\rm E}$, contributes $-2v_{\rm E}\mathrm{e}^{-v_{\rm esc}^2/v_0^2}$ (Appendix D.1). Hence

$$\left\langle \frac{1}{v} \right\rangle = \frac{\pi v_0^2}{v_{\rm E} N_{\rm esc}} \left[\sqrt{\pi}\, v_0\, \mathrm{erf}\left(\frac{v_{\rm E}}{v_0} \right) - 2v_{\rm E}\, \mathrm{e}^{-v_{\rm esc}^2/v_0^2} \right]. \tag{1.11}$$

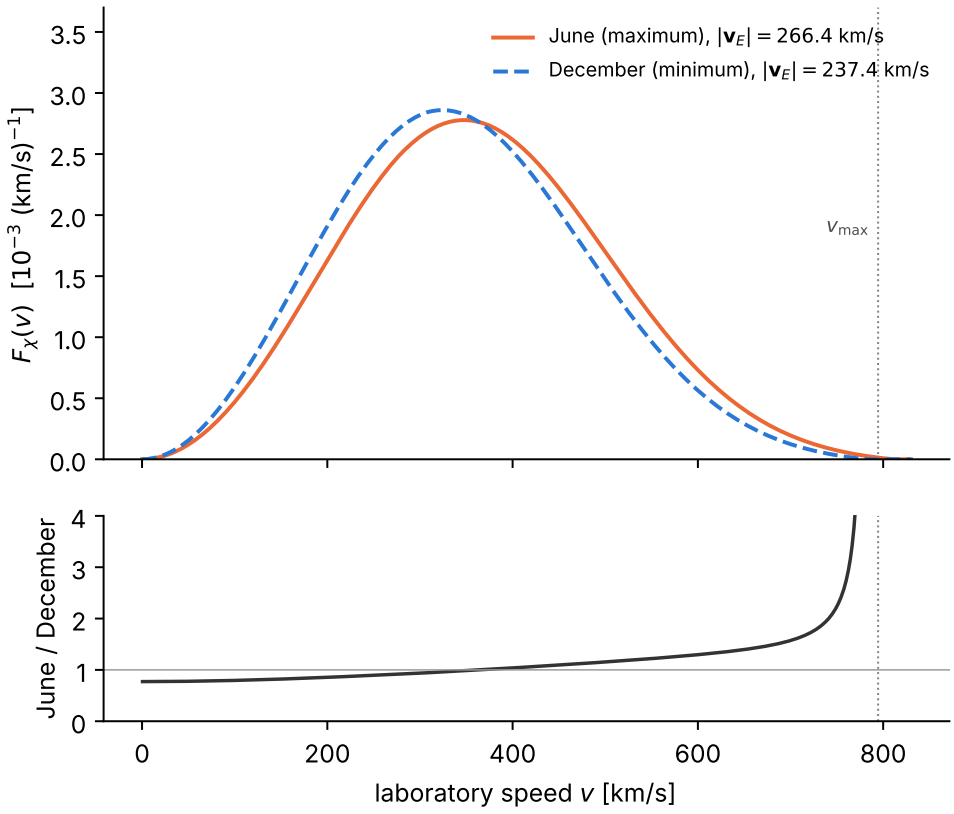


Figure 1.2: Top: laboratory-frame speed distribution $F_\chi(v)$ of the standard halo model with $v_0 = 238\,\mathrm{km\,s^{-1}}$ and $v_{\mathrm{esc}} = 544\,\mathrm{km\,s^{-1}}$, in June (solid, Earth speed at its annual maximum $266.4\,\mathrm{km\,s^{-1}}$) and December (dashed, $237.4\,\mathrm{km\,s^{-1}}$), the days of largest and smallest Earth speed for the Earth velocity of Eq. (1.12) [13, Eq. (11)]. Bottom: ratio of the June to the December distribution. The dotted line marks $v_{\mathrm{max}} = 794.5\,\mathrm{km\,s^{-1}}$; in June the distribution extends to $810\,\mathrm{km\,s^{-1}}$, in December to $781\,\mathrm{km\,s^{-1}}$.

At $v_{\mathrm{E}} = v_{\mathrm{E}}^{\mathrm{ref}} = 250.5\,\mathrm{km\,s^{-1}}$, $v_{\mathrm{E}}/v_0 = 1.0525$ and $\mathrm{erf}(1.0525) = 0.86338$. The bracket is $\sqrt{\pi} \times 238 \times 0.86338 - 2 \times 250.5 \times 0.005383 = 364.21 - 2.70 = 361.51\,\mathrm{km\,s^{-1}}$, and the prefactor is $\pi(238)^2/(250.5 \times 7.393 \times 10^7) = 9.608 \times 10^{-6}\,(\mathrm{km/s})^{-2}$. The result is

$$\left\langle \frac{1}{v} \right\rangle = 3.474 \times 10^{-3}\,(\mathrm{km/s})^{-1} = \frac{1}{287.9\,\mathrm{km\,s^{-1}}}, \quad \text{or 1041 in natural units.}$$

Without the cut, Eq. (1.11) reduces to $\mathrm{erf}(v_{\mathrm{E}}/v_0)/v_{\mathrm{E}} = 1/(290.1\,\mathrm{km\,s^{-1}})$; the cut removes fast particles and raises $\langle 1/v \rangle$ by 0.8 %. In June and December the same formula yields $1/(298.3\,\mathrm{km\,s^{-1}})$ and $1/(279.8\,\mathrm{km\,s^{-1}})$, a 6 % swing.

The mean inverse speed is the value at zero of the function $\eta(v_{\mathrm{min}}) = \langle \Theta(v - v_{\mathrm{min}})/v \rangle$ that Sect. 1.3 introduces, with v_{min} the minimum velocity of Eq. (1.23); Appendix D.1 gives η for every v_{min}.

1.2.3 Annual modulation

The Earth's orbital velocity, in the galactic axes of Eq. (1.5), is [13, Eq. (11)]

$$\boldsymbol{v}_{\oplus}(t) = 29.8\,\mathrm{km\,s^{-1}} \left[\begin{pmatrix} 0.9941 \\ 0.1088 \\ 0.0042 \end{pmatrix} \cos \omega_{\mathrm{yr}} t + \begin{pmatrix} -0.0504 \\ 0.4946 \\ -0.8677 \end{pmatrix} \sin \omega_{\mathrm{yr}} t \right], \tag{1.12}$$

with $\omega_{\mathrm{yr}} = 0.0172\,\mathrm{d^{-1}}$ the orbital angular frequency and t counted in days from 22 March. The two column vectors are orthogonal unit vectors, to 10^{-4}, that span the plane of the orbit, and $29.8\,\mathrm{km\,s^{-1}}$ is the mean orbital speed [13, Eq. (13)]. We treat the orbit as a circle; its eccentricity moves the date of the largest Earth speed by 1.2 days and the modulation amplitude by 0.4 % [15, Table I], both too small to matter here.

Since the orbital speed is small next to the solar speed, $29.8/250.5 = 0.12$, we can expand the Earth's speed to first order in it:

$$|\boldsymbol{v}_{\mathrm{E}}(t)| = \left(|\boldsymbol{v}_{\odot}|^2 + 2\boldsymbol{v}_{\odot} \cdot \boldsymbol{v}_{\oplus} + |\boldsymbol{v}_{\oplus}|^2\right)^{1/2} \approx v_{\mathrm{E}}^{\mathrm{ref}} + \hat{\boldsymbol{v}}_{\odot} \cdot \boldsymbol{v}_{\oplus}(t),$$

with $\hat{\boldsymbol{v}}_{\odot} = \boldsymbol{v}_{\odot}/|\boldsymbol{v}_{\odot}| = (0.0443, 0.9986, 0.0291)$ from Eq. (1.5). Projected onto $\hat{\boldsymbol{v}}_{\odot}$, the two vectors of Eq. (1.12) have components 0.1528 and 0.4664, and

$$|\boldsymbol{v}_{\mathrm{E}}(t)| \approx v_{\mathrm{E}}^{\mathrm{ref}} + 14.6\,\mathrm{km\,s^{-1}} \cos\left[\omega_{\mathrm{yr}}(t - t_{\mathrm{peak}})\right], \tag{1.13}$$

with amplitude $29.8 \times (0.1528^2 + 0.4664^2)^{1/2} = 29.8 \times 0.4908 = 14.6\,\mathrm{km\,s^{-1}}$ and phase $\omega_{\mathrm{yr}} t_{\mathrm{peak}} = \arctan(0.4664/0.1528) = 71.9^\circ = 1.254\,\mathrm{rad}$, which is $t_{\mathrm{peak}} = 72.9$ days after 22 March, that is, 2 June. Only half of the orbital speed modulates the Earth's speed because the Sun's velocity makes an angle of $\arccos 0.4908 = 60.6^\circ$ with the plane of the orbit. The second-order term, $(|\boldsymbol{v}_{\oplus}|^2 - (\hat{\boldsymbol{v}}_{\odot} \cdot \boldsymbol{v}_{\oplus})^2)/(2v_{\mathrm{E}}^{\mathrm{ref}})$, adds $1.3\,\mathrm{km\,s^{-1}}$ at both extremes and $1.6\,\mathrm{km\,s^{-1}}$ on average over the year. The exact speed from Eqs. (1.6) and (1.12) is largest on 2 June, $266.4\,\mathrm{km\,s^{-1}}$, smallest on 2 December, $237.4\,\mathrm{km\,s^{-1}}$, and averages $252.1\,\mathrm{km\,s^{-1}}$ over the year. Fig. 1.2 shows the two extreme distributions.

The rate follows the change of F_χ with the season, and its sign depends on which speeds the process can use. In June the distribution is shifted to higher speeds: it holds more particles above $364\,\mathrm{km\,s^{-1}}$, where the curves of Fig. 1.2 cross, and fewer below. A rate that needs fast particles, such as one near a kinematic threshold, peaks in June. A rate that weights all speeds by $1/v$ peaks in December, since $\langle 1/v \rangle$ is 6 % larger then (Worked example 1.1). For the

integral $\eta(v_{\min})$ of Sect. 1.3, built from Appendix D.1, the phase reverses at $v_{\min} = 211\,\mathrm{km\,s^{-1}}$; above, the modulation grows from a few percent to $+21\,\%$ and $-16\,\%$ about the reference value at $v_{\min} = 600\,\mathrm{km\,s^{-1}}$, and to $+42\,\%$ and $-28\,\%$ at $700\,\mathrm{km\,s^{-1}}$. Averaged over the year, $\langle 1/v\rangle$ lies 0.3 % below its value at $v_{\mathrm{E}}^{\mathrm{ref}}$ and agrees to 10^{-4} with its value at the annual-mean speed. In the tail the difference grows: at $v_{\min} = 600$ and $700\,\mathrm{km\,s^{-1}}$ the yearly average of η exceeds its value at $v_{\mathrm{E}}^{\mathrm{ref}}$ by 2.6 % and 5.6 %, because η there is a convex function of v_{E}.

The size of the modulation follows from the closed forms by differentiation. For a rate that depends on time only through v_{E}, Eq. (1.13) makes the relative modulation about the yearly mean

$$(14.6\,\mathrm{km\,s^{-1}})\,\frac{\partial \ln(\mathrm{rate})}{\partial v_{\mathrm{E}}}\,\cos\bigl[\omega_{\mathrm{yr}}(t - t_{\mathrm{peak}})\bigr],$$

with the derivative taken at $v_{\mathrm{E}}^{\mathrm{ref}}$. For a rate proportional to $\langle 1/v\rangle$, write Eq. (1.11) as $\pi v_0^2 \mathcal{I}(v_{\mathrm{E}})/(v_{\mathrm{E}} N_{\mathrm{esc}})$ with $\mathcal{I}(v_{\mathrm{E}})$ the bracket. Since $\mathrm{d}\,\mathrm{erf}(x)/\mathrm{d}x = (2/\sqrt{\pi})\mathrm{e}^{-x^2}$, $\mathrm{d}\mathcal{I}/\mathrm{d}v_{\mathrm{E}} = 2\mathrm{e}^{-v_{\mathrm{E}}^2/v_0^2} - 2\mathrm{e}^{-v_{\mathrm{esc}}^2/v_0^2} = 0.6498$, and

$$\frac{\partial \ln\langle 1/v\rangle}{\partial v_{\mathrm{E}}} = -\frac{1}{v_{\mathrm{E}}} + \frac{1}{\mathcal{I}}\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}v_{\mathrm{E}}} = -\frac{1}{250.5} + \frac{0.6498}{361.51} = -2.195 \times 10^{-3}\,(\mathrm{km/s})^{-1}.$$

The amplitude is $14.6 \times (-2.195 \times 10^{-3}) = -3.2\,\%$: the mean inverse speed is largest in December, as we saw above. The same reversal and its position, $v_{\min} \approx 0.89\,v_0$, are known for nuclear recoils [6, Sect. 5.2]. Since light dark matter often needs the fastest particles to cross a threshold (Sect. 1.3), its rate typically sits on the June side, with a large modulation.

Annual modulation is a signature because each of its features is fixed by the halo and the orbit, not by the detector. The period is one year, the phase is set by the June maximum and December minimum, the phase flips at a definite $v_{\min}$, and the amplitude grows toward the kinematic edge. A background would have to reproduce all four. The amplitude is also more sensitive to the shape of the speed distribution than the time-averaged rate is, and an observed modulation would test the halo model itself [6, Sect. 6.2].

1.2.4 Daily modulation

The Earth also rotates, and a crystal fixed in the laboratory turns with it relative to $\boldsymbol{v}_{\mathrm{E}}$. The rotation speed, $2\pi \times 6378\,\mathrm{km}/86\,164\,\mathrm{s} = 0.47\,\mathrm{km\,s^{-1}}$ at the equator and less by the cosine of the latitude elsewhere, is 3 % of the annual amplitude of Eq. (1.13) and leaves the speed distribution practically unchanged. What changes, however, is the direction from which the wind arrives in crystal axes.

The rotation is simplest in equatorial coordinates. Their Cartesian axes are fixed relative to the distant stars: the third points along the Earth's rotation axis to the north celestial pole, the first to the vernal equinox in the celestial equator, and the second completes a right-handed set. A direction on the sky is given by its *declination*, the angle above the celestial equator, and its *right ascension*, the angle measured eastward in the equatorial plane from the vernal equinox. Right ascension is often quoted in hours ($24\,\mathrm{h} = 360°$). In equatorial Cartesian axes a direction has the polar angle 90° minus its declination and the azimuth equal to its right ascension. A laboratory turns about the third axis once per sidereal day, the rotation period relative to the stars. Its *meridian* is the half-plane bounded by the rotation axis that contains the local vertical, its *local sidereal time* is the right ascension of that meridian, and the *hour angle* of a direction is the local sidereal time minus the direction's right ascension.

The rotation from equatorial to galactic Cartesian axes is fixed by three angles: the equatorial coordinates of the north Galactic pole, right ascension 192.85948° and declination 27.12825°, and the galactic longitude of the north celestial pole, 122.932° [15, App. B, Eq. (B3)].

Built from these angles, it takes a unit vector with equatorial components $(x_{\rm eq}, y_{\rm eq}, z_{\rm eq})$ to its galactic components,

$$\begin{pmatrix} x_{\rm gal} \\ y_{\rm gal} \\ z_{\rm gal} \end{pmatrix} = \begin{pmatrix} -0.0549 & -0.8734 & -0.4838 \\ 0.4941 & -0.4448 & 0.7470 \\ -0.8677 & -0.1981 & 0.4560 \end{pmatrix} \begin{pmatrix} x_{\rm eq} \\ y_{\rm eq} \\ z_{\rm eq} \end{pmatrix} . \tag{1.14}$$

The rows are the galactic axes in equatorial components; the first, the direction of the Galactic center, has right ascension 266.4° and declination −28.9°. The third column is the north celestial pole in galactic axes. Its scalar product with $\hat{\boldsymbol{v}}_\odot$ is 0.7378, which means the solar velocity makes the angle

$$\theta_{\rm E} = \arccos 0.7378 = 42.5^\circ, \tag{1.15}$$

with the rotation axis, in agreement with the 42° of Ref. [14, Sect. II D and Fig. 4]. The transpose of Eq. (1.14), which is its inverse, takes $\hat{\boldsymbol{v}}_\odot$ to the equatorial components $(0.4657, -0.4887, 0.7378)$, that is, right ascension $\arctan(-0.4887/0.4657) + 360^\circ = 313.6^\circ = 20.9\,{\rm h}$ and declination $47.5^\circ = 90^\circ - \theta_{\rm E}$, in the constellation Cygnus. The orbital motion moves this direction during the year, to declination 42.8° in June and 52.6° in December, so the angle to the rotation axis varies between 47.2° and 37.4°. We use $\hat{\boldsymbol{v}}_\odot$ below; a prediction at the level of a few degrees computes the direction of $\boldsymbol{v}_{\rm E}(t)$ from Eqs. (1.6) and (1.12) on each day, which combines the annual and daily modulations.

Let $\hat{\boldsymbol{n}}$ be an axis fixed in the laboratory, for example a crystal axis, at polar angle $\theta_{\hat{n}}$ from the north celestial pole. As the Earth turns, $\hat{\boldsymbol{n}}$ keeps its polar angle and its right ascension advances at the sidereal rate $\omega_{\rm sid} = 2\pi/(23\,{\rm h}\,56\,{\rm min}\,4\,{\rm s})$. In equatorial Cartesian axes the two unit vectors are $\hat{\boldsymbol{v}}_\odot = (\sin\theta_{\rm E}\cos\alpha_{\rm E}, \sin\theta_{\rm E}\sin\alpha_{\rm E}, \cos\theta_{\rm E})$, with $\alpha_{\rm E}$ its right ascension, and $\hat{\boldsymbol{n}}(t) = (\sin\theta_{\hat{n}}\cos\alpha_{\hat{n}}(t), \sin\theta_{\hat{n}}\sin\alpha_{\hat{n}}(t), \cos\theta_{\hat{n}})$, with $\alpha_{\hat{n}}(t) = \alpha_{\rm E} + \omega_{\rm sid}(t - t_{\rm E})$ and $t_{\rm E}$ the time at which both lie in one meridian plane. Their scalar product fixes the angle $\theta_{\rm lab}(t)$ between $\boldsymbol{v}_{\rm E}$ and the laboratory axis,

$$\cos\theta_{\rm lab}(t) = \cos\theta_{\rm E}\cos\theta_{\hat{n}} + \sin\theta_{\rm E}\sin\theta_{\hat{n}}\cos\big[\omega_{\rm sid}(t - t_{\rm E})\big]. \tag{1.16}$$

Over one sidereal day $\theta_{\rm lab}$ oscillates between $|\theta_{\rm E} - \theta_{\hat{n}}|$ and $\theta_{\rm E} + \theta_{\hat{n}}$ (for $\theta_{\rm E} + \theta_{\hat{n}} \le 180^\circ$). An axis along the rotation axis, $\theta_{\hat{n}} = 0$, keeps the fixed angle $\theta_{\rm E}$; an axis in the equatorial plane swings between 47.5° and 132.5°. For the local vertical at latitude $\ell_{\rm lat}$ north, $\theta_{\hat{n}} = 90^\circ - \ell_{\rm lat}$ and $\alpha_{\hat{n}}$ is the local sidereal time. Fig. 1.3 plots Eq. (1.16) for the vertical at four latitudes. At 45°N the angle runs from 2.5°, when Cygnus is nearly overhead and the wind comes from above, to 87.5° twelve sidereal hours later.

A crystal is usually mounted with its axes along the local vertical and the local north and east directions. The full direction of $\hat{\boldsymbol{v}}_\odot$ in these axes follows from the same construction. The local vertical at latitude $\ell_{\rm lat}$ is $(\cos\ell_{\rm lat}\cos\alpha_{\rm z}, \cos\ell_{\rm lat}\sin\alpha_{\rm z}, \sin\ell_{\rm lat})$ in equatorial axes, with $\alpha_{\rm z}$ the local sidereal time. The local north is the derivative of this vector with respect to $\ell_{\rm lat}$, $(-\sin\ell_{\rm lat}\cos\alpha_{\rm z}, -\sin\ell_{\rm lat}\sin\alpha_{\rm z}, \cos\ell_{\rm lat})$. The local east is the unit vector $(-\sin\alpha_{\rm z}, \cos\alpha_{\rm z}, 0)$ in the direction of increasing $\alpha_{\rm z}$. With $\delta_{\rm E} = 90^\circ - \theta_{\rm E}$ the declination of $\hat{\boldsymbol{v}}_\odot$ and $h_{\rm E} = \alpha_{\rm z} - \alpha_{\rm E}$ its hour angle, the scalar products of these three vectors with $\hat{\boldsymbol{v}}_\odot$ are its components

$$\begin{aligned} \hat{v}_{\rm east} &= -\cos\delta_{\rm E}\sin h_{\rm E}, \\ \hat{v}_{\rm north} &= \cos\ell_{\rm lat}\sin\delta_{\rm E} - \sin\ell_{\rm lat}\cos\delta_{\rm E}\cos h_{\rm E},, \\ \hat{v}_{\rm up} &= \sin\ell_{\rm lat}\sin\delta_{\rm E} + \cos\ell_{\rm lat}\cos\delta_{\rm E}\cos h_{\rm E} \end{aligned} \tag{1.17}$$

with $h_{\rm E} = \omega_{\rm sid}(t - t_{\rm E})$ advancing at the sidereal rate. The vertical component is Eq. (1.16) for $\theta_{\hat{n}} = 90^\circ - \ell_{\rm lat}$. At 45°N and $h_{\rm E} = 0$ the (east, north, up) components are $(0, 0.04, 1.00)$, a nearly vertical direction; six sidereal hours later, $h_{\rm E} = 90^\circ$, they are $(-0.67, 0.52, 0.52)$,

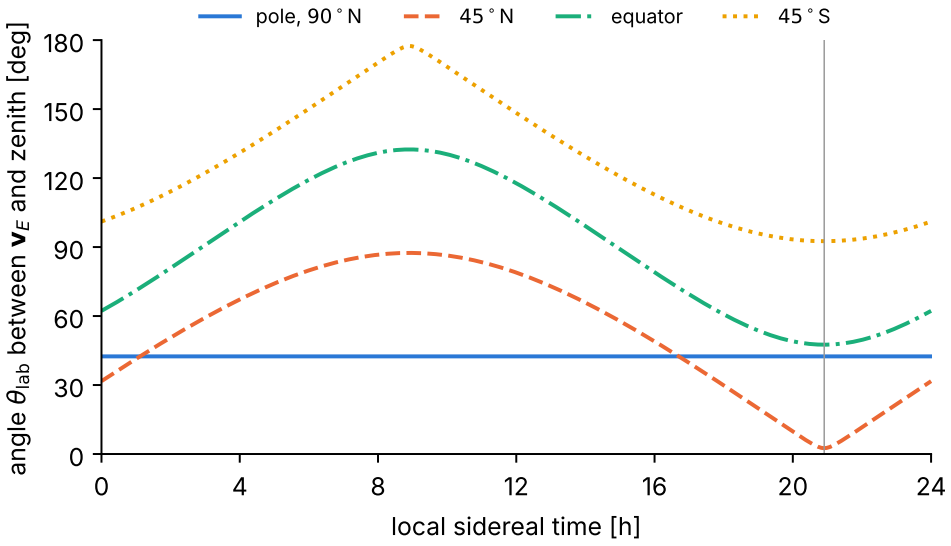


Figure 1.3: Angle θ_{lab} between the Earth's galactic-frame velocity and the local vertical over one sidereal day, as a function of local sidereal time, at latitudes 90°N, 45°N, 0° and 45°S. The velocity direction is that of the solar velocity, right ascension 20.9 h and declination 47.5° (vertical line: its transit), from Ref. [13, Table 1] and the coordinate rotation of Ref. [15, Eq. (B3)]. The dark-matter wind arrives from this direction.

toward the northwest at 31° above the horizon. The wind arrives from the direction $\hat{\boldsymbol{v}}_\odot$; the dark-matter velocities point along $-\hat{\boldsymbol{v}}_\odot$.

For an isotropic target, such as a liquid or a polycrystal, the rate depends on $\boldsymbol{v}_{\mathrm{E}}$ only through its magnitude, and the rotation changes nothing. A single crystal is anisotropic: its response to a momentum transfer depends on the direction of $\boldsymbol{q}$ relative to the crystal axes, and slow dark matter transfers momentum preferentially along the wind. As θ_{lab} sweeps through its daily range, the wind samples different crystal directions and the rate varies with the sidereal period.

The simplest anisotropy shows how Eq. (1.16) enters a rate. Take a uniaxial crystal, one in which a single axis $\hat{\boldsymbol{n}}$ (the optic axis) differs from the two directions perpendicular to it, and whose response is unchanged under $\hat{\boldsymbol{n}} \to -\hat{\boldsymbol{n}}$. Since the halo of Eq. (1.8) is symmetric under rotations about $\boldsymbol{v}_{\mathrm{E}}$, after the average over velocities the rate can depend on the two directions only through the scalar $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{v}}_{\mathrm{E}} = \cos\theta_{\mathrm{lab}}$, and evenness in $\hat{\boldsymbol{n}}$ makes it a function of $\cos^2\theta_{\mathrm{lab}}$. To lowest order in the anisotropy the rate is a constant plus a constant times $\cos^2\theta_{\mathrm{lab}}(t)$; the crystal response fixes both constants. With the optic axis in the equatorial plane, $\theta_{\hat{n}} = 90°$ and Eq. (1.16) reduces to $\cos\theta_{\mathrm{lab}} = \sin\theta_{\mathrm{E}}\cos[\omega_{\mathrm{sid}}(t - t_{\mathrm{E}})]$; $\cos^2\theta_{\mathrm{lab}}$ then swings between 0 and $\sin^2\theta_{\mathrm{E}} = 0.456$ twice each sidereal day. With the optic axis along the Earth's axis the rate is constant. The orientation of the crystal sets both the amplitude and the period of the signal, and only the coefficient of $\cos^2\theta_{\mathrm{lab}}$, the anisotropy of the target, is left to compute.

Polar crystals, whose ions carry opposite charges, show sizable daily modulation of phonon rates when the lattice is strongly anisotropic, as in sapphire, while the cubic crystal GaAs shows almost none. The sidereal period is a distinctive signature: the sidereal day is about four minutes shorter than the solar day, and a background that follows the solar day, or does not vary at all, falls out of phase with the signal over many periods [14, Sect. III C]. Chap. 10 treats the target side, the angular structure of the crystal response that the sweep of Eq. (1.16) probes.

1.2.5 Number flux

The number flux of dark matter is the number of particles that cross a unit area per unit time. Each particle is counted across an area perpendicular to its own velocity:

$$\Phi_\chi = n_\chi \langle v \rangle = \frac{\rho_\chi}{m_\chi} \langle v \rangle. \tag{1.18}$$

Multiplied by a velocity-independent cross-section, it is the rate per target particle. A target particle at rest presents a fixed cross-section to each incoming particle, and the particles with velocity in $\mathrm{d}^3 v$ sweep a volume v per unit time per unit of cross-section; they hit it at the rate $n_\chi f_\chi(\boldsymbol{v})\,\mathrm{d}^3 v\, v$ times the cross-section. When the cross-section does not depend on v, summing over velocities replaces v by $\langle v \rangle$, and the rate is Φ_χ times the cross-section. When it does depend on v, as in most of this book, the average must be taken with the cross-section inside, and Φ_χ serves only as a measure of how much dark matter passes through. A detector face of fixed orientation intercepts a different number, which depends on the angle between the face and the wind; Eq. (1.18) avoids that geometric factor. The mean speed follows from Eq. (1.10) as $\langle v \rangle = \int_0^\infty \mathrm{d}v\, v F_\chi(v)$. It is a sum of Gaussian moments, evaluated in Appendix D.1:

$$\begin{aligned} \langle v \rangle = \frac{\pi v_0^2}{v_\mathrm{E} N_\mathrm{esc}} \Bigg[& v_\mathrm{E} v_0^2\, \mathrm{e}^{-v_\mathrm{E}^2/v_0^2} + \sqrt{\pi}\, v_0 \left(\frac{v_0^2}{2} + v_\mathrm{E}^2 \right) \mathrm{erf}\left(\frac{v_\mathrm{E}}{v_0} \right) \\ & - 2 v_\mathrm{E}\, \mathrm{e}^{-v_\mathrm{esc}^2/v_0^2} \left(v_0^2 + v_\mathrm{esc}^2 + \frac{v_\mathrm{E}^2}{3} \right) \Bigg] \end{aligned} . \tag{1.19}$$

Without the cut ($v_\mathrm{esc} \to \infty$, $N_\mathrm{esc} \to \pi^{3/2} v_0^3$) the bracket loses its last term and Eq. (1.19) becomes the mean speed of a shifted Maxwellian, $v_0[\mathrm{e}^{-x^2}/\sqrt{\pi} + (x + 1/(2x))\,\mathrm{erf}\, x]$ with $x = v_\mathrm{E}/v_0$.

Worked example 1.2: Number flux of 100 MeV dark matter

We take $v_\mathrm{E} = v_\mathrm{E}^\mathrm{ref}$, so $v_\mathrm{E}/v_0 = 1.0525$, $\mathrm{e}^{-1.1078} = 0.33028$ and $\mathrm{erf}(1.0525) = 0.86338$ (Worked example 1.1). In units of $(\mathrm{km\,s^{-1}})^3$ the three terms in the bracket of Eq. (1.19), in the order written, are

$$\begin{aligned} \text{first:} \quad & 250.5 \times 238^2 \times 0.33028 = 4.687 \times 10^6, \\ \text{second:} \quad & 1.7725 \times 238 \times (28322 + 62750) \times 0.86338 = 3.3170 \times 10^7, \\ \text{third:} \quad & -501 \times 0.005383 \times (56644 + 295936 + 20917) = -1.0073 \times 10^6. \end{aligned}$$

Their sum, 3.6849×10^7, times the prefactor $9.608 \times 10^{-6}\,(\mathrm{km/s})^{-2}$ of Worked example 1.1 gives $\langle v \rangle = 354.1\,\mathrm{km\,s^{-1}}$; without the cut it would be $358.2\,\mathrm{km\,s^{-1}}$. At $m_\chi = 100\,\mathrm{MeV}$ the number density is $n_\chi = (0.3\,\mathrm{GeV\,cm^{-3}})/(0.1\,\mathrm{GeV}) = 3\,\mathrm{cm^{-3}}$, so

$$\Phi_\chi = 3\,\mathrm{cm^{-3}} \times 3.541 \times 10^7\,\mathrm{cm\,s^{-1}} = 1.06 \times 10^8\,\mathrm{cm^{-2}\,s^{-1}}.$$

About 10^8 particles cross each square centimeter every second, 3.4×10^{15} in a year. A kilogram of silicon, of density $2.33\,\mathrm{g\,cm^{-3}}$, is a cube of volume $429\,\mathrm{cm^3}$ and side 7.5 cm; it holds about 1300 such particles at any instant. The flux scales as $1/m_\chi$: at 1 MeV it is $10^{10}\,\mathrm{cm^{-2}\,s^{-1}}$. These large numbers are what makes feeble couplings testable; the cross-section, not the flux, keeps the rate small.

Eq. (1.8) leaves out the gravitational pull of the Sun and of the Earth. Gravity bends and accelerates the particles that approach the Sun and the Earth. By energy conservation, a

particle with speed v far away arrives with speed $(v^2 + u_{\mathrm{esc}}^2)^{1/2}$, where u_{esc} is the local escape speed of the attracting body: $42.1\,\mathrm{km\,s^{-1}}$ from the Sun at the Earth's orbit and $11.2\,\mathrm{km\,s^{-1}}$ from the Earth at its surface. For $v = 354\,\mathrm{km\,s^{-1}}$ the speed increases by $u_{\mathrm{esc}}^2/2v = 2.5\,\mathrm{km\,s^{-1}}$ and $0.18\,\mathrm{km\,s^{-1}}$, below 1 % and comparable to the eccentricity effect; the resulting modulation from solar focusing is small [6, Sect. 5]. We neglect these shifts, as we neglected the Earth's rotation.

The standard halo model is a reference, not a measured distribution. Its density is uncertain by roughly a factor of two, its speed parameters by several percent, and stellar surveys reveal kinematic substructure, debris of galaxies that merged with ours, that a single Maxwellian does not describe [13, Sect. 3.1]. Near a kinematic threshold the rate probes only the tail of F_χ, where these changes matter most [6, Sect. 6]. The closed form of η in Appendix D.1 makes the sensitivity explicit. Changing v_0 by $\pm 6\,\mathrm{km\,s^{-1}}$ changes $\langle 1/v \rangle$ by ∓ 1 %, but η at $v_{\mathrm{min}} = 600\,\mathrm{km\,s^{-1}}$ by factors 1.11 and 0.89. Changing v_{esc} by $\pm 25\,\mathrm{km\,s^{-1}}$ leaves $\langle 1/v \rangle$ within 0.4 % and changes $\eta(600\,\mathrm{km\,s^{-1}})$ by factors 1.06 and 0.92, but $\eta(760\,\mathrm{km\,s^{-1}})$ by factors 2.26 and 0.10. Close to v_{max}, the escape speed is the dominant uncertainty, larger than the annual modulation. Results in this book use the parameters of Table 1.1. The rate of Eq. (1.1) takes ρ_χ and f_χ from here, and Sect. 1.3 combines them and v_{max} with the kinematics of a single collision.

1.3 Kinematics of slow dark matter

A dark-matter particle that crosses a detector carries at most 3.5 eV of kinetic energy at 1 MeV and 3.5 meV at 1 keV, and about a seventh of that at a typical speed (Sect. 1.1). Whether it can excite a target, and which excitation it can excite, is decided by energy and momentum conservation before any matrix element is computed. Kinematics alone already tells us a lot: how much energy a slow particle can deposit at a given momentum transfer, the minimum velocity v_{min} it needs to do so, and from that the lightest mass that can get over a detector threshold. We then combine these with the velocity distribution of Sect. 1.2 into the speed integral $\eta(v_{\mathrm{min}})$ that every scattering rate in this book contains. Sign conventions follow Appendix A.4.

1.3.1 Energy transfer and minimum velocity

We start from the elastic collision with a free particle at rest. A dark-matter particle of mass m_χ and laboratory velocity $\boldsymbol{v}$ has momentum $m_\chi \boldsymbol{v}$ and kinetic energy $\frac{1}{2} m_\chi v^2$; at speeds $v \lesssim 3 \times 10^{-3}$ the nonrelativistic forms are exact to a part in 10^5. It scatters on a nucleus of mass M at rest. We write $\boldsymbol{q}$ for the momentum transferred *from* the dark matter *to* the target; the dark matter leaves with $m_\chi \boldsymbol{v} - \boldsymbol{q}$ and the nucleus with $\boldsymbol{q}$. In a crystal, $\boldsymbol{q}$ brings some lattice vocabulary with it; Sect. 3.1 develops it properly, and for now four working definitions are enough. A *reciprocal-lattice vector* $\boldsymbol{G}$ is a wavevector whose plane wave $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{x}}$ takes the same value at every lattice site, and a crystal conserves momentum only up to such a vector. As a result, a state of an electron or phonon in a crystal is labeled by a *crystal momentum*, defined only up to a reciprocal-lattice vector. The *first Brillouin zone* is the set of wavevectors closer to the origin than to any nonzero $\boldsymbol{G}$, and *folding* a momentum means subtracting the $\boldsymbol{G}$ that brings it into that zone. We never fold $\boldsymbol{q}$: it is the physical momentum lost by the dark matter (the reason is given below Eq. (1.27)). Energy conservation reads

$$\tfrac{1}{2} m_\chi v^2 = \frac{(m_\chi \boldsymbol{v} - \boldsymbol{q})^2}{2 m_\chi} + \frac{q^2}{2M} \quad \Longrightarrow \quad \boldsymbol{q} \cdot \boldsymbol{v} = \frac{q^2}{2 \mu_{\chi N}}, \tag{1.20}$$

with $q = |\boldsymbol{q}|$ and $\mu_{\chi N} = m_\chi M/(m_\chi + M)$ the dark-matter–nucleus reduced mass [8, Eqs. (75)–(77)]. We write $\mu_{\chi e}$ and $\mu_{\chi p}$ for the reduced masses with an electron and a proton, defined in the same way. Since $\boldsymbol{q}\cdot\boldsymbol{v} \le qv$, there is a maximum momentum transfer, $q \le 2\mu_{\chi N} v$, and the energy deposited in the nucleus, the *nuclear recoil energy* $E_R = q^2/2M$, obeys

$$E_R \le \frac{2\mu_{\chi N}^2 v^2}{M}. \tag{1.21}$$

This is the classic result of nuclear-recoil searches [8, Eq. (78)]. For $m_\chi \ll M$ it is $(4m_\chi/M) \times \frac{1}{2}m_\chi v^2$: a light particle bounces off a heavy nucleus like a ball off a wall and keeps almost all of its energy.

The free particle at rest is an idealization, and in a solid a poor one: electrons are bound, with typical momenta $\sim \alpha m_e$, and nuclei are tied to a lattice. The target can absorb a momentum $\boldsymbol{q}$ and an energy ω that are not related by a free dispersion $q^2/2M$, and it pays to separate the two parties. We write ω for the *energy deposited* in the target. In this book it is defined as the energy given to the target, so it can be negative at $T > 0$. All of the target's physics then sits in which pairs $(\boldsymbol{q}, \omega)$ it can absorb, and how often. The dark-matter side alone fixes ω once $\boldsymbol{q}$ and $\boldsymbol{v}$ are given.

To keep the models of Sect. 1.4 in view we allow the scattering to change the dark-matter state, $\chi_1 \to \chi_2$, with a mass splitting $\Delta_\chi = m_{\chi_2} - m_{\chi_1}$. The energy the dark matter loses is its initial kinetic energy minus its final kinetic and extra rest energy:

$$\omega = \tfrac{1}{2}m_\chi v^2 - \frac{(m_\chi \boldsymbol{v} - \boldsymbol{q})^2}{2m_\chi} - \Delta_\chi.$$

We have neglected Δ_χ against m_χ in the final kinetic energy, a relative error $|\Delta_\chi|/m_\chi$, negligible for splittings small compared with the mass. Expanding the square, the terms $\frac{1}{2}m_\chi v^2$ cancel, and energy conservation becomes $\omega = \omega_{\boldsymbol{q}}$, with

Energy transfer and minimum velocity

$$\omega_{\boldsymbol{q}} = \boldsymbol{q}\cdot\boldsymbol{v} - \frac{q^2}{2m_\chi} - \Delta_\chi, \tag{1.22}$$

$$v_{\min}(q,\omega) = \left|\frac{\omega + \Delta_\chi}{q} + \frac{q}{2m_\chi}\right|. \tag{1.23}$$

A pair $(\boldsymbol{q}, \omega)$ is open to a particle of speed v if and only if $v \ge v_{\min}(q,\omega)$.

Eq. (1.22) is the form used for electron and phonon targets [1, Eq. (9), 8, Eq. (116), 16, Eq. (3.2)]. The splitting is endothermic for $\Delta_\chi > 0$, when part of the kinetic energy is spent on the heavier final state, and exothermic for $\Delta_\chi < 0$, when the dark matter releases $|\Delta_\chi|$. For elastic scattering $\Delta_\chi = 0$. A free nucleus at rest absorbs only $\omega = q^2/2M$, and Eq. (1.22) with $\Delta_\chi = 0$ then returns Eq. (1.20). Some papers take $\boldsymbol{q}$ as the momentum gained by the dark matter instead, a difference that matters for a crystal without inversion symmetry or with magnetic order, where $\boldsymbol{q}$ and $-\boldsymbol{q}$ are physically different (Appendix A.4).

Bound electrons show why the separation is needed. Treat an outer electron as free but moving with a typical velocity $\sim \alpha$, about ten times faster than the dark matter. Its momentum is then $\sim \alpha m_e$, and a momentum transfer of that size is what changes its state. The typical momentum transfer is set by the electron, not by the dark matter: $q \approx \mu_{\chi e}\alpha$, which is αm_e for $m_\chi \gg m_e$ [8, Eqs. (117) and (118)]. By Eq. (1.22) the energy deposited is then $\sim qv \approx \alpha m_e v \approx 4\,\mathrm{eV}$ at $v = 10^{-3}$, independent of m_χ, and is not tied to q by any dispersion of the dark matter. For $m_\chi < m_e$ the estimate $q \approx m_\chi\alpha$ fails: with $\Delta_\chi = 0$,

Eq. (1.22) would allow at most $qv - q^2/2m_\chi = m_\chi\alpha(v - \alpha/2)$, which is negative because $\alpha/2 = 3.6 \times 10^{-3}$ exceeds every halo speed. The momentum transfer is then bounded by roughly $m_\chi v$, and the dark matter can give up at most its kinetic energy [8, Eq. (119)]. A real target has a spectrum of states rather than a single electron velocity, and Eq. (1.22) holds for all of them: it constrains only the dark-matter side.

The minimum velocity follows from the angle. Let θ be the angle between $\boldsymbol{q}$ and $\boldsymbol{v}$. Eq. (1.22) with $\omega = \omega_{\boldsymbol{q}}$ fixes

$$\cos\theta = \frac{1}{v}\left(\frac{\omega + \Delta_\chi}{q} + \frac{q}{2m_\chi}\right),$$

and a direction with this cosine exists only if the right side lies between -1 and 1. That is the condition $v \geq v_{\min}$ of Eq. (1.23). The absolute value matters only when $\omega + \Delta_\chi < 0$, which happens in two cases: for exothermic dark matter, and for a target at temperature $T > 0$ that gives energy to the dark matter, $\omega < 0$ (the thermal weights of such transitions are set in Sect. 2.5). Refs. [17, Eq. (2)] and [1, Eq. (10)], which treat $\omega \geq 0$ and $\Delta_\chi = 0$, write the same expression without the absolute value. For $\omega + \Delta_\chi < 0$, however, the signed expression can be negative, and reading a negative value as "every particle qualifies" would be wrong: at $\omega = 0$ and small q, Eq. (1.23) demands $v \geq |\Delta_\chi|/q$, which no halo particle meets once $q < |\Delta_\chi|/v_{\max}$. With the absolute value, $v_{\min} \geq 0$ always. At $\omega = -\Delta_\chi = |\Delta_\chi|$ it reduces to $q/2m_\chi$, which vanishes as $q \to 0$: an exothermic particle deposits its splitting at any speed, the case of the exothermic models of Sect. 1.4.

Without any knowledge of the target, Eq. (1.22) fixes the largest energy a particle can deposit and the minimum velocity needed to deposit a given energy. First, at fixed v, Eq. (1.22) is largest for $\boldsymbol{q}$ parallel to $\boldsymbol{v}$ and $q = m_\chi v$, where

$$\omega \leq \tfrac{1}{2}m_\chi v^2 - \Delta_\chi. \tag{1.24}$$

For elastic scattering the dark matter can give up its entire kinetic energy, provided the target absorbs the momentum $m_\chi v$ at that energy [8, Eq. (119)]. A free target of mass M cannot, because part of the energy goes into its recoil: the most energy that can go into an internal excitation of a free target is $\frac{1}{2}\mu_{\chi T}v^2$, with $\mu_{\chi T}$ the reduced mass of the pair [16, Eq. (3.4)]. A crystal absorbs momentum as a whole, its mass is macroscopic, and the full kinetic energy is available. Second, for elastic scattering at fixed $\omega > 0$, Eq. (1.23) as a function of q is smallest where its derivative $-\omega/q^2 + 1/2m_\chi$ vanishes, at $q = (2m_\chi\omega)^{1/2}$, with the value $v_{\min} = (2\omega/m_\chi)^{1/2}$. A particle of speed v can deposit ω at some q only if $\frac{1}{2}m_\chi v^2 \geq \omega$.

1.3.2 Threshold mass

A detector registers an event only if the deposited energy exceeds a *threshold energy* $E_{\rm th}$, set by the target (the lowest excitation energy) or by the readout. No particle in the laboratory is faster than $v_{\max} = 794.5\,\mathrm{km\,s^{-1}} = 2.650 \times 10^{-3}$ (Eq. (1.9)). Combining with Eq. (1.24), an elastic scatter above threshold requires $\frac{1}{2}m_\chi v_{\max}^2 \geq E_{\rm th}$, that is $m_\chi \geq m_{\min}$ with the *threshold mass*

$$m_{\min} = \frac{2E_{\rm th}}{v_{\max}^2}. \tag{1.25}$$

The bound is attained only if the target has an excitation of energy $E_{\rm th}$ at the momentum $q = m_{\min}v_{\max}$. If the lowest excitation energy of the target depends on q, the condition becomes the requirement that $v_{\min}(q,\omega) \leq v_{\max}$ for some q at which the target can absorb $\omega \geq E_{\rm th}$; Eq. (1.25) with the global minimum of the excitation energy is then a strict lower bound. Sect. 3.3 applies this to silicon, whose lowest electronic excitations sit at a nonzero

crystal momentum. For inelastic dark matter E_{th} is replaced by $E_{\mathrm{th}} + \Delta_\chi$ when this is positive; if $E_{\mathrm{th}} + \Delta_\chi \leq 0$, kinematics alone sets no lower mass. The two signs change the spectrum in opposite ways. An endothermic splitting removes Δ_χ from every deposit and raises m_{min}. An exothermic splitting adds $|\Delta_\chi|$: by Eq. (1.22) the deposits then lie between $|\Delta_\chi| - qv - q^2/2m_\chi$ and $|\Delta_\chi| + qv - q^2/2m_\chi$, a band of width $2qv$ about $|\Delta_\chi|$, and for small q the spectrum approaches a line at the splitting.

The scale of m_{min} follows from $v_{\mathrm{max}}^2 = 7.02 \times 10^{-6}$. A threshold of 1 eV needs $m_\chi \geq 0.28\,\mathrm{MeV}$, and a threshold of 1 meV needs $m_\chi \geq 0.28\,\mathrm{keV}$. Every factor of ten in threshold is a factor of ten in reach. The threshold mass depends on the halo through v_{max}^2 alone. A shift of the escape speed by $25\,\mathrm{km\,s^{-1}}$, the size of its uncertainty (Sect. 1.2), changes v_{max} by 3.1 % and m_{min} by 6.3 %. Over the year the largest speed moves between 781 and $810\,\mathrm{km\,s^{-1}}$ (Eq. (1.9)), and the threshold mass computed with it moves by −3.9 % in June and 3.4 % in December about its reference value.

Worked example 1.3: Threshold masses for an electronic and a phonon threshold

The band gap of silicon, the lowest energy that frees an electron from the occupied states, depends on temperature: it is 1.12 eV at 300 K [18, Table I] and 1.1692 eV near zero temperature [18, Eq. (11)], where detectors run. With the low-temperature gap, $E_{\mathrm{g}} = 1.17\,\mathrm{eV}$, as the threshold, Eq. (1.25) gives

$$m_{\mathrm{min}} = \frac{2 \times 1.17\,\mathrm{eV}}{(2.650 \times 10^{-3})^2} = \frac{2.34\,\mathrm{eV}}{7.023 \times 10^{-6}} = 3.33 \times 10^5\,\mathrm{eV} = 0.333\,\mathrm{MeV}.$$

The optimal momentum transfer is $q = m_{\mathrm{min}} v_{\mathrm{max}} = 3.33 \times 10^5\,\mathrm{eV} \times 2.650 \times 10^{-3} = 0.88\,\mathrm{keV}$. Many projections use $E_{\mathrm{g}} = 1.11\,\mathrm{eV}$ [16, Eq. (5.2)], close to the room-temperature gap, which puts m_{min} 5 % lower, at 0.32 MeV; Sect. 3.3 computes the gap. For a phonon threshold of 30 meV,

$$m_{\mathrm{min}} = \frac{2 \times 0.030\,\mathrm{eV}}{7.023 \times 10^{-6}} = 8.5 \times 10^3\,\mathrm{eV} = 8.5\,\mathrm{keV},$$

at $q = 8.5 \times 10^3\,\mathrm{eV} \times 2.650 \times 10^{-3} = 23\,\mathrm{eV}$. Lowering the threshold from the electronic to the phonon scale lowers the reach by a factor of $1.17/0.030 = 39$. Both values are limits set by kinematics alone: at $m_\chi = m_{\mathrm{min}}$ only particles moving at exactly v_{max} in exactly the right direction contribute, and the rate vanishes (Sect. 1.3.5).

1.3.3 Momentum and energy scales

Eq. (1.22) defines, for each mass, the region of the (q, ω) plane that the halo can reach. With $\Delta_\chi = 0$ and the fastest particles moving along $\boldsymbol{q}$,

$$\omega \leq q v_{\mathrm{max}} - \frac{q^2}{2m_\chi}. \tag{1.26}$$

The boundary is a parabola that leaves the origin with slope v_{max}, peaks at $\omega = \frac{1}{2} m_\chi v_{\mathrm{max}}^2$ at $q = m_\chi v_{\mathrm{max}}$, and returns to zero at $q = 2m_\chi v_{\mathrm{max}}$, the maximum momentum transfer. Whatever the mass, every allowed point lies below the line $\omega = q v_{\mathrm{max}}$, and depositing ω needs $q \geq \omega / v_{\mathrm{max}}$ [1, Eq. (122)]; no point lies beyond $q = 2m_\chi v_{\mathrm{max}}$. Fig. 1.4 draws the region for four masses, after Ref. [1, Fig. 1], and Table 1.2 lists the corresponding extremes. The energy

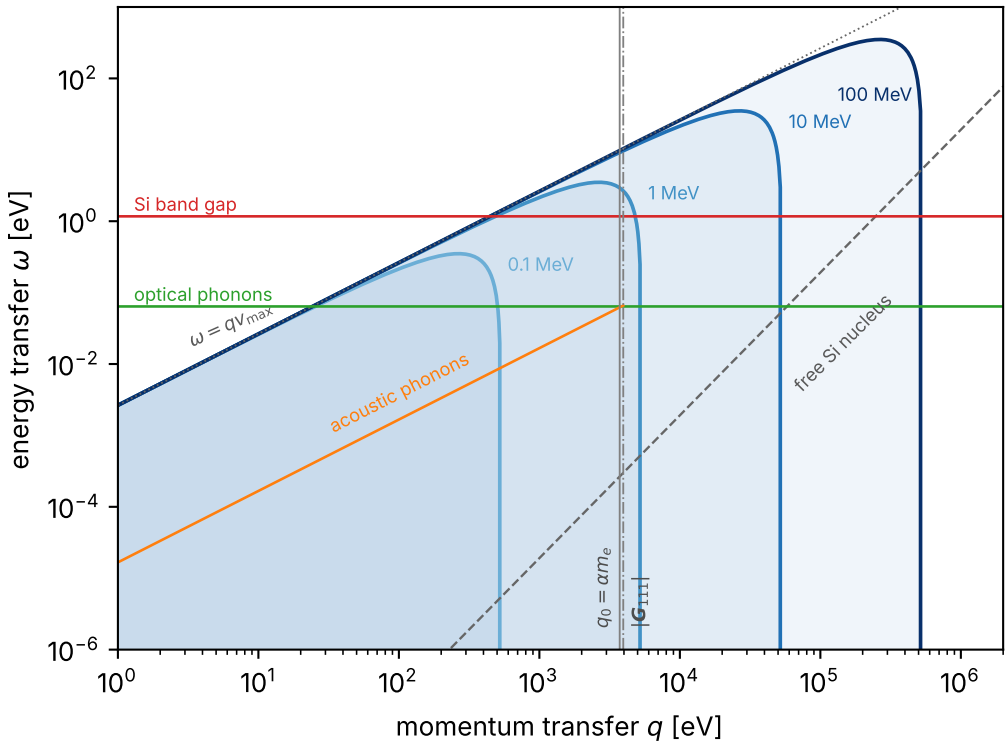


Figure 1.4: Momentum transfer q and energy transfer ω accessible to elastic dark matter of mass 0.1, 1, 10 and 100 MeV (blue curves, light to dark; shaded below), from $\omega \leq qv_{\max} - q^2/2m_\chi$ with $v_{\max} = 794.5\,\mathrm{km\,s^{-1}}$. The dotted line $\omega = qv_{\max}$ bounds all masses. Overlaid for silicon are the low-temperature band gap, 1.17 eV (red), and the optical-phonon energy at the zone center, 64 meV (green). The orange line is an acoustic branch $\omega =$ sound speed $\times\, q$ for a sound speed of $5\,\mathrm{km\,s^{-1}}$, drawn up to $q = |\boldsymbol{G}_{111}|$, and the dashed curve is the recoil $q^2/2M$ of a free silicon nucleus, with $M = 26.16\,\mathrm{GeV}$ the mean atomic mass of natural silicon. Vertical lines: the electron momentum scale $q_0 = \alpha m_e = 3.73\,\mathrm{keV}$ and the length $|\boldsymbol{G}_{111}| = \sqrt{3}\,2\pi/a \approx 4.0\,\mathrm{keV}$ of the shortest reciprocal-lattice vector of silicon, whose lattice constant is $a = 5.431$ Å. Scales from Refs. [1, Eqs. (50) and (121), Sect. IV A, 18, Eq. (11), 19, lattice parameter of silicon].

deposited is small compared with the momentum transfer, $\omega \leq qv_{\max} < 3 \times 10^{-3} q$. As a result, scattering of halo dark matter is always far from the light cone $\omega = q$, which is why the static potential of Sect. 1.4 suffices.

Fig. 1.4 answers at a glance which excitations each mass can reach. At 0.1 MeV the largest deposit, 0.35 eV, is below the silicon gap: only phonons, and other excitations below an eV, are open. At 1 MeV the gap is open in a narrow band of momentum transfers, from $q = 0.49\,\mathrm{keV}$ to 4.8 keV, the two roots of Eq. (1.26) at $\omega = 1.17\,\mathrm{eV}$. At 10 MeV the band widens to 0.45–53 keV, and deposits up to 35 eV are allowed. A free silicon nucleus, by contrast, receives 1.17 eV only for $m_\chi \gtrsim 47\,\mathrm{MeV}$, where $2\mu_{\chi N}^2 v_{\max}^2/M$ reaches that value.

Two momentum scales of the target, one for bound electrons and one for the lattice, decide what these momentum transfers probe. The first is the *electron momentum scale*

$$q_0 = \alpha m_e = 3.73\,\mathrm{keV}, \tag{1.27}$$

the inverse Bohr radius [1, Eq. (121)]. A bound outer electron has a wavefunction of size $1/(\alpha m_e)$, typical momenta $\sim q_0$ and a velocity $\sim \alpha$, about ten times the dark-matter speed [8,

Table 1.2: Kinematic extremes for elastic dark matter at $v_{\max} = 794.5\,\mathrm{km\,s^{-1}}$. Columns: the dark-matter mass; the largest momentum transfer to a target that absorbs any momentum; the largest energy transfer; the largest recoil energy of a free silicon nucleus ($M = 26.16\,\mathrm{GeV}$, the mean atomic mass of natural silicon). The $\frac{1}{2}m_\chi v_{\max}^2$ column is also the threshold energy for which $m_{\min} = m_\chi$.

m_χ	$2m_\chi v_{\max}$	$\frac{1}{2}m_\chi v_{\max}^2$	free Si nucleus
0.1 MeV	0.53 keV	0.35 eV	5.4 µeV
1 MeV	5.3 keV	3.5 eV	0.54 meV
10 MeV	53 keV	35 eV	54 meV
100 MeV	530 keV	351 eV	5.3 eV
1 GeV	5.3 MeV	3.5 keV	0.50 keV

Sect. V A]. A momentum transfer much smaller than q_0 barely changes the electron's state; a much larger one reaches only the high-momentum tail of its wavefunction. The second is the *interatomic scale*, the length of the shortest reciprocal-lattice vector. For a cubic crystal with lattice constant a the natural unit is $2\pi/a$. Silicon has $a = 5.431\,\text{Å}$ [19, lattice parameter of silicon], about ten Bohr radii [1, Sect. V], which makes $2\pi/a = 2.28\,\mathrm{keV} = 0.61\,q_0$. Its lattice is face-centered cubic, and the shortest reciprocal-lattice vectors, along the cube diagonals, are longer by $\sqrt{3}$: $|\boldsymbol{G}_{111}| = \sqrt{3}\,2\pi/a = 3.95\,\mathrm{keV} \approx 1.06\,q_0$. A momentum transfer well below $|\boldsymbol{G}_{111}|$ has a wavelength longer than the interatomic spacing and sees the crystal as a whole; a larger one resolves the atoms. Sect. 3.1 makes this precise with the reciprocal lattice.

The two scales are close to each other, and light dark matter straddles both. For an electronic excitation of a few eV, $q \geq \omega/v_{\max}$ requires $q \gtrsim 1\,\mathrm{keV}$, roughly $0.3\,q_0$; the silicon gap alone needs $q \geq 1.17\,\mathrm{eV}/2.650 \times 10^{-3} = 0.44\,\mathrm{keV} = 0.12\,q_0$ [1, Eq. (123)]. At the upper end, interpolating Table 1.2, $2m_\chi v_{\max} = q_0$ at $m_\chi = 0.70\,\mathrm{MeV}$. Between about 1 MeV and 1 GeV, then, the momentum transfers match the momenta at which electrons in atoms and solids have most of their weight. Lighter dark matter cannot reach q_0 and probes the long-wavelength response of the crystal. Heavier dark matter can reach far above q_0, but momentum transfers of that size find little electronic weight.

The comparison with $|\boldsymbol{G}_{111}|$ explains why we never fold $\boldsymbol{q}$. Above about 0.75 MeV, $2m_\chi v_{\max}$ exceeds $|\boldsymbol{G}_{111}|$. Because the crystal conserves momentum only up to a reciprocal-lattice vector, the same excitation can be reached by momentum transfers that differ by such a vector, and these have different magnitudes. The dark-matter side, Eqs. (1.22) and (1.23), depends on the magnitude of the physical momentum transfer $\boldsymbol{q}$, not on its image in the first Brillouin zone. Folding $\boldsymbol{q}$ would assign the wrong $v_{\min}$ and the wrong dark-matter form factor $F_{\mathrm{DM}}(q)$ of Sect. 1.4. Sect. 3.1 writes $\boldsymbol{q}$ as a first-zone part plus a reciprocal-lattice vector and keeps both.

On the energy side, electronic excitations and phonons set the scales. *Electronic excitations* move an electron from an occupied to an empty state and typically cost an eV or more. In an insulator or semiconductor the lowest such energy is the *band gap* E_g (Worked example 1.3; Sect. 3.3). *Phonons* are the quantized vibrations of the lattice, with energies from zero to roughly 100 meV [1, Sect. III A]; the optical phonon of silicon at the zone center has 64 meV, or 15.5 THz [20] (Sect. 5.3). Two kinds, optical and acoustic, matter here. Optical phonons, in which neighboring atoms move against each other, keep a finite energy as their momentum goes to zero. Acoustic phonons, long-wavelength sound waves, have energy equal to the sound speed times the momentum, with sound speeds of 3 to $10\,\mathrm{km\,s^{-1}}$ [1, Eq. (50)], that is 1 to 3×10^{-5}.

The kinematics treats optical and acoustic phonons differently. An optical phonon branch that is nearly flat, with energy ω, can be excited at any q between $\omega/v_{\max}$ and the upper root

of Eq. (1.26), close to $2m_\chi v_{\max}$. Such a branch can absorb nearly the full kinetic energy, and Eq. (1.25) applies; the 30 meV threshold of Worked example 1.3 needs a momentum transfer of just $q \approx 23\,\mathrm{eV}$, below a hundredth of $|\boldsymbol{G}_{111}|$. An acoustic phonon of branch ν has $\omega = c_\nu q$, with c_ν its sound speed (Sect. 5.2). By Eq. (1.23) it needs $v_{\min} = c_\nu + q/2m_\chi \le v$, which bounds $q \le 2m_\chi(v - c_\nu)$ and $\omega \le 2c_\nu m_\chi v$ to leading order in c_ν/v [1, Eq. (93)]. The acoustic branch takes at most a fraction $4c_\nu/v$, a few percent, of the kinetic energy. In Fig. 1.4 it is the orange line, which for the lightest mass leaves the parabola near its right end.

Both energy scales are material properties, and later chapters use computed values for them. The band gap is computed from first principles with density-functional theory, corrected for the systematic underestimate of gaps that the method makes (Appendices E.1 and E.3); phonon energies come from density-functional perturbation theory (Appendix E.4). Measured values serve as checks.

Nuclear recoils. Eq. (1.21) shows why scattering on free nuclei loses its reach for light dark matter. The free nucleus is the dashed parabola $\omega = q^2/2M$ of Fig. 1.4. It meets each dark-matter region only where $q^2/2M$ is small, because M is large; for $m_\chi \ll M$ the nucleus receives at most the fraction $4m_\chi/M$ of the kinetic energy, 1.5×10^{-2} at 100 MeV on silicon. The same parabola explains when a nucleus in a crystal is not free. A recoil energy comparable to the phonon energies, tens of meV, is reached at $m_\chi \approx 10\,\mathrm{MeV}$ (Table 1.2). Below that mass the nucleus cannot be treated as free: the dark matter excites one or a few phonons of the crystal instead, with the energies set by the lattice, and Chap. 5 develops that description.

For a free nucleus, Eq. (1.23) at $\omega = E_R = q^2/2M$ gives $v_{\min} = q/2\mu_{\chi N} = (ME_R/2\mu_{\chi N}^2)^{1/2}$. The crossover to phonons is already visible for a single nucleus held in a harmonic well of frequency ω_0. At zero temperature a transfer q excites n quanta with the Poisson probability $P_n = \mathrm{e}^{-E_R/\omega_0}(E_R/\omega_0)^n/n!$, where $E_R = q^2/2M$ is the free recoil energy [1, Eqs. (90) and (91)]. The energy deposit $n\omega_0$ therefore has mean exactly E_R and rms spread $(E_R\omega_0)^{1/2}$. The free formula is the limit $E_R \gg \omega_0$, in which the spread is small compared with E_R.

Limit Check 1.1. Elastic nuclear recoil

A free nucleus of mass M at rest that receives momentum q absorbs the energy $\omega = E_R = q^2/2M$. A dark-matter particle of speed v gives it at most $E_R \le 2\mu_{\chi N}^2 v^2/M$, with $\mu_{\chi N}$ the dark-matter–nucleus reduced mass, and a recoil E_R needs the speed $v \ge v_{\min} = (ME_R/2\mu_{\chi N}^2)^{1/2}$ [8, Eqs. (77) and (78)].

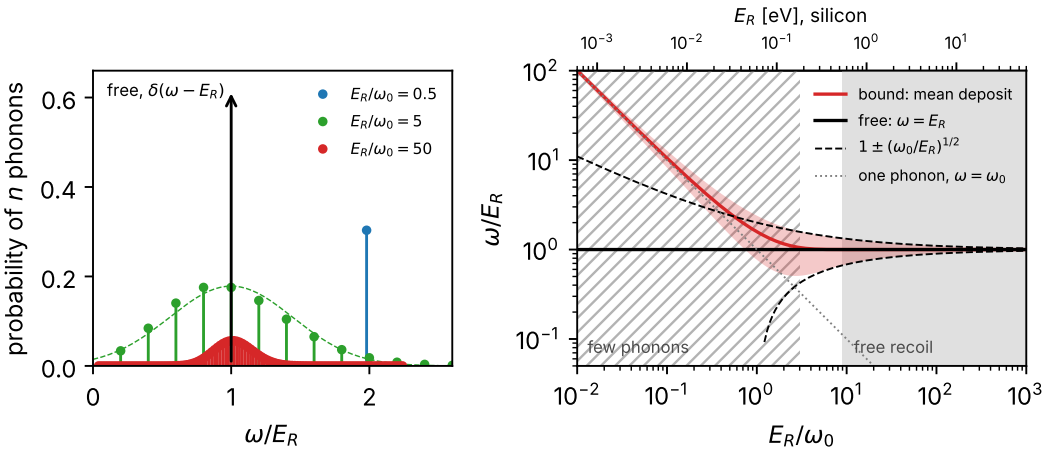


Figure 1.5: Free nuclear recoil against a nucleus bound in a harmonic (Einstein) potential

of frequency $\omega_0 = 60\,\mathrm{meV}$, a typical silicon phonon energy, at zero temperature [1, Sect. IV A, Eqs. (80), (90) and (91)]. Left: probability of exciting n phonons, a deposit $\omega = n\omega_0$, for $E_R/\omega_0 = 0.5$, 5 and 50 (dots; the term $n = 0$, which deposits nothing, is omitted), with Gaussians of rms width $(E_R\omega_0)^{1/2}$ (dashed) and the free response $\delta(\omega - E_R)$ (arrow). Right: mean deposit of the scatterings that excite the lattice (red line) and its rms spread (red band), against E_R/ω_0, with the free value $\omega = E_R$ (black), the estimate $1 \pm (\omega_0/E_R)^{1/2}$ (dashed) and one phonon, $\omega = \omega_0$ (dotted); the top axis gives E_R for silicon. Gray: $E_R \geq 9\omega_0 = 0.54\,\mathrm{eV}$, where the rms width is below a third of E_R and the free formula holds. Hatched: $E_R \leq 3\omega_0$, the single- and few-phonon regime.

Below a few ω_0 the deposit comes in whole phonons and spreads as widely as E_R itself; below ω_0 most scatterings leave the lattice unexcited, and the rest deposit at least one phonon, ω_0, well above E_R.

Worked example 1.4: Largest recoil energy of a silicon nucleus

For $m_\chi = 1\,\mathrm{GeV}$ on silicon, $M = 26.16\,\mathrm{GeV}$ (the mean atomic mass of natural silicon) and $\mu_{\chi N} = (1 \times 26.16/27.16)\,\mathrm{GeV} = 0.963\,\mathrm{GeV}$. The maximum momentum transfer at v_max is $q = 2\mu_{\chi N} v_\mathrm{max} = 2 \times 0.963\,\mathrm{GeV} \times 2.650 \times 10^{-3} = 5.10\,\mathrm{MeV}$, and Eq. (1.21) gives

$$E_R = \frac{q^2}{2M} = \frac{(5.10\,\mathrm{MeV})^2}{2 \times 26\,160\,\mathrm{MeV}} = 5.0 \times 10^{-4}\,\mathrm{MeV} = 0.50\,\mathrm{keV}.$$

The kinetic energy at v_max is $\frac{1}{2} m_\chi v_\mathrm{max}^2 = 3.5\,\mathrm{keV}$, so the nucleus receives at most 14 % of it; the exact fraction is $4\mu_{\chi N}^2/(m_\chi M) = 0.142$. A typical particle, at $v \approx 300\,\mathrm{km\,s^{-1}}$ and with an average over angles, deposits far less. This momentum transfer, 5.1 MeV, is below the inverse nuclear radius, tens of MeV, so the nucleus recoils coherently (Sect. 1.4). On xenon, $M = 122.3\,\mathrm{GeV}$, the same formula gives 0.11 keV, the value quoted in Sect. 1.1.

1.3.4 The speed integral

A rate sums the probability of each momentum transfer over the dark-matter velocities. Energy conservation enters it as a delta function $\delta(\omega - \omega_{\boldsymbol{q}})$, integrated against $f_\chi(\boldsymbol{v})$ (Sect. 2.2 derives the rate). The integral over $\boldsymbol{v}$ can be done once for all targets whose response depends on the magnitude q but not on the direction of $\boldsymbol{q}$, as for a liquid, an atom, or a crystal averaged over orientations. For such a target only the angle θ between $\boldsymbol{q}$ and $\boldsymbol{v}$ matters. Its average is

$$\frac{1}{2}\int_{-1}^{1} \mathrm{d}(\cos\theta)\, \delta\left(\omega + \Delta_\chi + \frac{q^2}{2m_\chi} - qv\cos\theta\right) = \frac{1}{2qv}\,\Theta(v - v_\mathrm{min}(q,\omega))\,.$$

The delta function fires once if its root $\cos\theta = (\omega + \Delta_\chi + q^2/2m_\chi)/qv$ lies in $[-1, 1]$, which is the condition $v \geq v_\mathrm{min}$. The factor $1/qv$ is the derivative of its argument with respect to $\cos\theta$. This is the origin of the factor $1/v$ promised in Sect. 1.2. Averaging over the halo, we find $\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\, \Theta(v - v_\mathrm{min})/(2qv) = \eta(v_\mathrm{min})/(2q)$, with

$$\eta(v_\mathrm{min}) \equiv \int \mathrm{d}^3v\, \frac{f_\chi(\boldsymbol{v})}{v}\, \Theta(v - v_\mathrm{min}) = \left\langle \frac{\Theta(v - v_\mathrm{min})}{v} \right\rangle. \tag{1.28}$$

The function $\eta(v_\mathrm{min})$ is the *mean inverse speed above* v_min. It carries all of the halo dependence of a rate for such a target [17, below Eq. (5), 1, Eqs. (11) and (12), 16, Eqs. (A.12)–(A.15)];

nuclear-recoil papers call the same function $g(v_{\min})$ [8, Eq. (103)]. Its value at zero is the mean inverse speed $\langle 1/v\rangle$ of Eq. (1.11), and it decreases monotonically to zero at the largest speed in the halo. Because $v_{\min} \geq 0$ by Eq. (1.23), η is never needed at negative arguments.

Eq. (1.28) fixes how the halo enters the integral over momentum transfers. For an integrand that depends only on q, $\mathrm{d}^3q/(2\pi)^3 = 4\pi q^2\mathrm{d}q/(2\pi)^3$, and with the factor $\eta/2q$ the measure becomes [1, Eq. (11), 16, Eq. (A.14)]

$$\int \frac{\mathrm{d}^3q}{(2\pi)^3}\,\frac{\eta(v_{\min})}{2q}\,(\cdots) = \int \frac{q\,\mathrm{d}q}{4\pi^2}\,\eta(v_{\min}(q,\omega))\,(\cdots).$$

Small momentum transfers carry the weight $q\,\mathrm{d}q$, and η cuts the integral off wherever $v_{\min}(q,\omega) > v_{\max}$, at both small and large q. Sect. 1.5 assembles the full rate from this measure, the cross-section of Sect. 1.4 and the response of the target. For an anisotropic crystal at a fixed orientation the angular average does not factor out; the rate then depends on the direction of $\boldsymbol{v}_{\mathrm{E}}$ relative to the crystal axes, which is the daily modulation of Sect. 1.2.

In the standard halo model the integral is elementary. With $F_\chi(v)$ the speed distribution of Eq. (1.10), $\eta(v_{\min}) = \int_{v_{\min}}^{\infty} \mathrm{d}v\, F_\chi(v)/v$, and the factor v in F_χ cancels the $1/v$. Appendix D.1 integrates the Gaussians and finds three branches in closed form: Eq. (D.3) for $0 \leq v_{\min} < v_{\mathrm{esc}} - v_{\mathrm{E}}$, Eq. (D.4) for $v_{\mathrm{esc}} - v_{\mathrm{E}} \leq v_{\min} < v_{\mathrm{esc}} + v_{\mathrm{E}}$, and $\eta = 0$ above. Both are differences of error functions (Appendix C.8) minus a constant from the cut, with $v_{\mathrm{E}} = |\boldsymbol{v}_{\mathrm{E}}(t)|$. The same branches appear, in other notation, in Ref. [16, Eqs. (B.4) and (B.5)].

Fig. 1.6 shows η at the reference Earth speed and on the days of largest and smallest Earth speed. Three features matter for light dark matter: a plateau, a Gaussian fall and a second branch. Up to $v_{\min} \approx 200\,\mathrm{km\,s^{-1}}$, η stays within a factor of 1.41 of $\langle 1/v\rangle = 1/(287.9\,\mathrm{km\,s^{-1}})$: almost every particle qualifies. Between 300 and $700\,\mathrm{km\,s^{-1}}$ it falls roughly as a Gaussian, $\mathrm{e}^{-(v_{\min}-v_{\mathrm{E}})^2/v_0^2}$, by a factor of 190, from $1.53\times10^{-3}\,(\mathrm{km/s})^{-1}$ to $7.95\times10^{-6}\,(\mathrm{km/s})^{-1}$. Above $v_{\mathrm{esc}} - v_{\mathrm{E}}^{\mathrm{ref}} = 293.5\,\mathrm{km\,s^{-1}}$ the second branch applies; every $v_{\min}$ near a kinematic threshold lies on it. In natural units, $\eta(0) = 1041$, $\eta(600\,\mathrm{km\,s^{-1}}) = 19.2$ and $\eta(700\,\mathrm{km\,s^{-1}}) = 2.38$.

Note that inside η the Earth speed is the actual $|\boldsymbol{v}_{\mathrm{E}}(t)|$, while reference scalars such as $m_{\min}$ use $v_{\mathrm{E}}^{\mathrm{ref}}$ (Sect. 1.2). Near the end point the moving cut dominates the seasonal change. Relative to the reference curve, η in June and December is 1.42 and 0.72 times as large at $v_{\min} = 700\,\mathrm{km\,s^{-1}}$, 2.2 and 0.37 times at $760\,\mathrm{km\,s^{-1}}$, and 4.5 and 0.009 times at $780\,\mathrm{km\,s^{-1}}$, where the December halo has almost no particles left.

1.3.5 The kinematic edge

The *kinematic edge* of a rate is the boundary beyond which no particle in the halo is fast enough. At fixed m_χ the spectrum ends at $\omega = \frac{1}{2}m_\chi v_{\max}^2 - \Delta_\chi$, by Eq. (1.24). At a fixed threshold the rate, as a function of mass, ends at $m_{\min}$. Both edges are reached as $v_{\min} \to v_{\max}$, and the shape of a rate near an edge is controlled by the shape of η there.

Close to $v_{\max}$, η vanishes as a power. Write $v_{\min} = v_{\mathrm{esc}} + v_{\mathrm{E}} - \delta$ with small $\delta > 0$. On the second branch, Eq. (D.4), the difference of error functions is an integral, $(\sqrt{\pi}v_0/2)[\mathrm{erf}(v_{\mathrm{esc}}/v_0) - \mathrm{erf}((v_{\mathrm{esc}}-\delta)/v_0)] = \int_{v_{\mathrm{esc}}-\delta}^{v_{\mathrm{esc}}} \mathrm{e}^{-y^2/v_0^2}\mathrm{d}y$, and the constant term is $\delta\,\mathrm{e}^{-v_{\mathrm{esc}}^2/v_0^2} = \int_{v_{\mathrm{esc}}-\delta}^{v_{\mathrm{esc}}} \mathrm{e}^{-v_{\mathrm{esc}}^2/v_0^2}\mathrm{d}y$. With $s = v_{\mathrm{esc}} - y$, the integrand of their difference is

$$\mathrm{e}^{-v_{\mathrm{esc}}^2/v_0^2}\left[\mathrm{e}^{(2v_{\mathrm{esc}}s - s^2)/v_0^2} - 1\right] = \mathrm{e}^{-v_{\mathrm{esc}}^2/v_0^2}\left[\frac{2v_{\mathrm{esc}}s}{v_0^2} + \left(\frac{2v_{\mathrm{esc}}^2}{v_0^4} - \frac{1}{v_0^2}\right)s^2 + O(s^3)\right].$$

Integrating over $0 < s < \delta$ and multiplying by the prefactor $\pi v_0^2/(v_{\mathrm{E}}N_{\mathrm{esc}})$,

$$\eta(v_{\min}) = \frac{\pi v_{\mathrm{esc}}\,\mathrm{e}^{-v_{\mathrm{esc}}^2/v_0^2}}{v_{\mathrm{E}}N_{\mathrm{esc}}}\,\delta^2\left[1 + \frac{(2v_{\mathrm{esc}}^2 - v_0^2)\,\delta}{3v_{\mathrm{esc}}v_0^2} + O(\delta^2)\right], \qquad \delta = v_{\mathrm{esc}} + v_{\mathrm{E}} - v_{\min}. \quad (1.29)$$

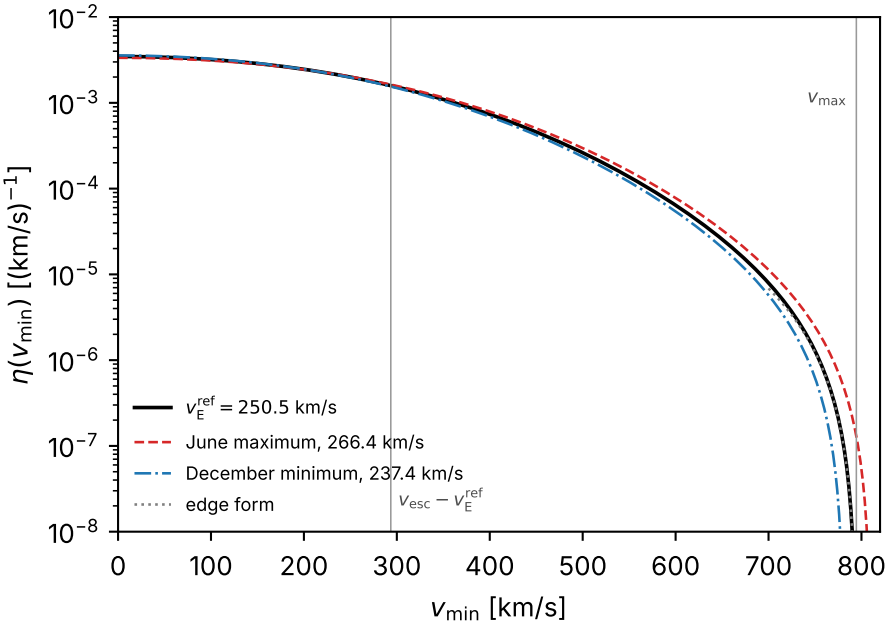


Figure 1.6: Mean inverse speed above $v_{\rm min}$, $\eta(v_{\rm min}) = \langle \Theta(v - v_{\rm min})/v \rangle$, in the standard halo model ($v_0 = 238\,{\rm km\,s^{-1}}$, $v_{\rm esc} = 544\,{\rm km\,s^{-1}}$). Solid: Earth speed $v_{\rm E}^{\rm ref} = 250.5\,{\rm km\,s^{-1}}$; dashed: June maximum ($266.4\,{\rm km\,s^{-1}}$); dash-dotted: December minimum ($237.4\,{\rm km\,s^{-1}}$). Vertical lines: $v_{\rm esc} - v_{\rm E}^{\rm ref} = 293.5\,{\rm km\,s^{-1}}$, where the closed form changes branch, and $v_{\rm max} = 794.5\,{\rm km\,s^{-1}}$, where η vanishes. The dotted curve above $700\,{\rm km\,s^{-1}}$ is the edge form of Eq. (1.29) with its first correction; it lies within 10 % of the exact curve above $720\,{\rm km\,s^{-1}}$, within 5 % above $745\,{\rm km\,s^{-1}}$, and merges with it toward $v_{\rm max}$. At $v_{\rm min} = 0$, $\eta = 1/(287.9\,{\rm km\,s^{-1}})$, or 1041 in natural units. Closed form from Appendix D.1; halo parameters from Ref. [13, Table 1].

The scale of the correction is $3v_{\rm esc}v_0^2/(2v_{\rm esc}^2 - v_0^2) = 173\,{\rm km\,s^{-1}}$. At the reference Earth speed and $v_{\rm min} = 760$ and $780\,{\rm km\,s^{-1}}$ ($\delta = 34.5$ and $14.5\,{\rm km\,s^{-1}}$), the quadratic term alone captures 0.81 and 0.92 of the exact value, and with the correction 0.98 and 1.00. The edge is quadratic because two things vanish together: the speed window $v_{\rm max} - v_{\rm min}$, and the density of particles inside it, which drops linearly to zero at the escape speed of the truncated Maxwellian. The prefactor $\mathrm{e}^{-v_{\rm esc}^2/v_0^2}/N_{\rm esc}$ is the height of the distribution at the cut. Both factors are fixed by the most uncertain parts of the halo model (Sect. 1.2); a smoother truncation would raise the power of δ.

The edge in mass follows. For a threshold $E_{\rm th}$ the smallest $v_{\rm min}$ available to a particle of mass m_χ is $(2E_{\rm th}/m_\chi)^{1/2}$ (below Eq. (1.24)). Its distance from $v_{\rm max}$ is

$$v_{\rm max} - \left(\frac{2E_{\rm th}}{m_\chi}\right)^{1/2} = v_{\rm max}\left[1 - \left(\frac{m_{\rm min}}{m_\chi}\right)^{1/2}\right] \simeq \frac{v_{\rm max}}{2}\,\frac{m_\chi - m_{\rm min}}{m_{\rm min}},$$

so a mass one percent above $m_{\rm min}$ uses only particles within $4\,{\rm km\,s^{-1}}$ of $v_{\rm max}$. Just above the edge the rate is the product of η, quadratic in this window, and of the phase space of pairs (q, ω) that fit inside it, which shrinks as a further power of the window. How fast that phase space shrinks depends on how the target's response begins at threshold. For silicon the electronic response begins at a crystal momentum away from the zone center, and the question becomes which excitations the dark matter can reach at the edge (Sect. 3.3).

The recovery above the edge is slow. Take the silicon gap of Worked example 1.3 as threshold, $m_{\rm min} = 0.333\,{\rm MeV}$, and evaluate η at the most favorable speed $(2E_{\rm g}/m_\chi)^{1/2}$, an upper bound on what the halo offers at each mass. Relative to $\eta(0)$, it is 6.0×10^{-5} at 0.35 MeV, 1.1×10^{-3} at 0.4 MeV, 0.76 % at 0.5 MeV, 12 % at 1 MeV and 84 % at 10 MeV. Between 0.35 MeV, 5 % above $m_{\rm min}$, and $2m_{\rm min}$ the halo supply grows by a factor of more than 500. The whole of that range is set by particles faster than about $560\,{\rm km\,s^{-1}}$, the most favorable speed at $2m_{\rm min}$, which are 7 % of the halo. A limit in this mass range is a statement about the tail of the velocity distribution as much as about the target.

The position of the edge, by contrast, depends on the target and the halo only through $E_{\rm th}/v_{\rm max}^2$. Their uncertainties enter as $\delta \ln m_{\rm min} = \delta \ln E_{\rm th} - 2\,\delta \ln v_{\rm max}$. A 5 % change in the gap moves $m_{\rm min}$ by 5 %; the escape-speed uncertainty moves it by about 6 %. A precise edge needs both a precise threshold from the target side and a stated halo. The edge recurs throughout the book: the band structure (Sect. 3.3), the error of computed gaps (Sect. 3.4) and the meV gaps of superconductors (Sect. 7.2) each move it.

1.4 Light dark-matter models

Kinematics (Sect. 1.3) decides which pairs $(\boldsymbol{q}, \omega)$ are open; the interaction with electrons and nuclei decides how often they occur. We will set up the mediators, among them the dark photon, and the absorbed bosons used in the rest of the book. Their strength is quoted through the reference cross-section $\bar\sigma_e$ and the dark-matter form factor $F_{\rm DM}(q)$ for scattering on electrons, the nucleon and nucleus cross-sections, and the couplings through which bosonic dark matter is absorbed. Model building itself is covered by reviews [8, Sect. III, 1]; we keep only what fixes a rate.

1.4.1 Mediators and the Yukawa potential

Thermal light dark matter needs a mediator with a mass at or below the GeV scale (Sect. 1.1). We take it to be either a massive vector field A'_μ or a real scalar field ϕ_V, with mass m_V and couplings y_χ to dark matter, y_e to electrons and y_N to nucleons. The vector couples to a

current of each species and the scalar to a density ($\bar\psi\gamma^\mu\psi$ and $\bar\psi\psi$ in field-theory notation). For slow particles both reduce, at leading order, to the number density [21, Eq. (7)], and we use only that limit. The interaction Lagrangian densities are then

$$\mathcal{L}_{\rm int} = -A'_\mu\left(y_\chi J^\mu_\chi + y_e J^\mu_e + y_N J^\mu_N\right) \quad \text{or} \quad \mathcal{L}_{\rm int} = -\phi_V\left(y_\chi n_\chi + y_e n_e + y_N n_N\right), \tag{1.30}$$

where J^μ_χ, J^μ_e and J^μ_N are number currents whose time components are the number densities n_χ, n_e and n_N. The spatial components carry a factor of the particle velocity; they lead to velocity-dependent interactions, mentioned at the end of this section, and do not enter the potential below.

The mediator can be treated as a classical field sourced by the particles. A dark-matter particle at rest at the origin is a point source $y_\chi\delta^3(\boldsymbol{x})$. For this static source the Euler–Lagrange equation of the vector field reduces to the static Klein–Gordon equation $(-\nabla^2 + m_V^2)A'^0 = y_\chi\delta^3(\boldsymbol{x})$ (Eq. (C.80)). Its solution is y_χ times the Green's function of Eq. (C.82), $A'^0(\boldsymbol{x}) = y_\chi\,\mathrm{e}^{-m_V r}/(4\pi r)$ with $r = |\boldsymbol{x}|$. By Eq. (1.30), an electron at $\boldsymbol{x}$ has the potential energy $y_e A'^0(\boldsymbol{x})$. The dark-matter–electron potential is then

$$V(\boldsymbol{x}) = y_\chi y_e\,\frac{\mathrm{e}^{-m_V|\boldsymbol{x}|}}{4\pi|\boldsymbol{x}|}, \tag{1.31}$$

the Yukawa potential. It has range $1/m_V$ and becomes Coulomb-like as $m_V \to 0$. A scalar mediator leads to the same function with the opposite sign: vector exchange repels sources whose couplings have equal signs, and scalar exchange attracts them (Eq. (C.84), where the signs are derived from the field energy). With the Fourier convention of Appendix A.3 and Eq. (C.81), the transform is

Yukawa potential in momentum space

$$\tilde V(\boldsymbol{q}) = \int \mathrm{d}^3x\,\mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}\,V(\boldsymbol{x}) = \frac{y_\chi y_e}{q^2 + m_V^2}. \tag{1.32}$$

For a scalar mediator the sign is reversed. Since every rate in this book depends on $|\tilde V(\boldsymbol{q})|^2$, the sign does not enter.

Here $\boldsymbol{q}$ is the momentum the dark matter transfers to the target (Sect. 1.3). Because $V(\boldsymbol{x})$ depends only on the separation, the amplitude for transferring $\boldsymbol{q}$ is proportional to $\tilde V(\boldsymbol{q})$ (Appendix C.9; derived in Sect. 2.2). Eq. (1.32) has two limits, which the field calls the heavy- and light-mediator limits. For a heavy mediator, with m_V above every relevant q, $\tilde V$ is a constant: a contact interaction. For a light mediator, with m_V below every relevant q, $\tilde V \propto 1/q^2$, as in Coulomb scattering, and the rate is weighted toward the smallest momentum transfers. The potential is static because the energy deposited is small: a moving source adds $-\omega^2$ to $q^2 + m_V^2$ in the denominator, and in scattering $\omega \le qv$ with $v \lesssim 3\times10^{-3}$, which keeps the correction below 10^{-5} (Appendix C.9).

Which couplings a mediator has depends on the model. A vector coupled to electric charge, the dark photon below, couples to electrons and protons and not to neutrons. A vector coupled to baryon minus lepton number couples to electrons, protons and neutrons with equal magnitude. A scalar that mixes with the Higgs boson couples to each fermion in proportion to its mass, which makes y_N/y_e large [8, Sect. III C]. Each mediator is constrained by accelerator, stellar and cosmological data, which bound the products $y_\chi y_e$ and $y_\chi y_N$ that enter rates and the couplings to ordinary matter separately [8, Sect. III C]. We will not need these bounds, since the rates below depend on the models only through $\tilde V(\boldsymbol{q})$ and the absorption couplings.

1.4.2 Reference cross-section and form factor

Experiments compare models through the reference cross-section $\bar{\sigma}_e$ and the dark-matter form factor $F_{\rm DM}(q)$, both fixed by $\tilde{V}(\boldsymbol{q})$. The reference momentum is $q_0 = \alpha m_e = 3.73\,\text{keV}$ (Sect. 1.3), the inverse Bohr radius, typical of electrons bound in atoms and solids. We define [17, Eqs. (3) and (4), 21, Eqs. (11) and (12)]

$$\bar{\sigma}_e \equiv \frac{\mu_{\chi e}^2}{\pi} |\tilde{V}(q_0)|^2, \qquad F_{\rm DM}(q) \equiv \frac{\tilde{V}(q)}{\tilde{V}(q_0)}, \tag{1.33}$$

with $\mu_{\chi e}$ the dark-matter–electron reduced mass (Sect. 1.3). Equivalently,

$$|\tilde{V}(q)|^2 = \frac{\pi \bar{\sigma}_e}{\mu_{\chi e}^2} F_{\rm DM}(q)^2, \tag{1.34}$$

the form in which the interaction enters every electron-scattering rate in this book. For the Yukawa potential of Eq. (1.32),

$$\bar{\sigma}_e = \frac{\mu_{\chi e}^2\, y_\chi^2 y_e^2}{\pi\,(m_V^2 + \alpha^2 m_e^2)^2}, \qquad F_{\rm DM}(q) = \frac{m_V^2 + \alpha^2 m_e^2}{m_V^2 + q^2}, \tag{1.35}$$

in agreement with Ref. [21, Eqs. (11) and (12)].

Which process does $\bar{\sigma}_e$ describe? For a potential whose transform is a constant $\tilde{V}$, the first Born approximation yields the free elastic cross-section $\mu_{\chi e}^2 |\tilde{V}|^2/\pi$ (Appendix C.9; derived in Sect. 2.2). It is that cross-section with $\tilde{V}$ frozen at q_0; for a heavy mediator it is the actual cross-section of dark matter on a free electron. For a light mediator, however, it is the cross-section of no process: the free cross-section of a $1/q^2$ potential diverges at small angles, as Rutherford's does, and $\bar{\sigma}_e$ merely normalizes $|\tilde{V}(q_0)|^2$. In a target the divergence is cut off by the minimum momentum transfer needed to excite it and by screening. Since $\bar{\sigma}_e$ is defined in vacuum, with screening and every other in-medium effect assigned to the target response (Sect. 4.4), a quoted $\bar{\sigma}_e$ means the same thing for every target. Other papers use other reference points and normalizations (Appendix A.9).

Two limits of Eq. (1.35) cover most of the literature. A heavy mediator, m_V above both q_0 and every momentum transfer that contributes, gives $F_{\rm DM} = 1$ and $\bar{\sigma}_e = \mu_{\chi e}^2 y_\chi^2 y_e^2/(\pi m_V^4)$. A light mediator, m_V below both, gives $F_{\rm DM} = (q_0/q)^2$ and $\bar{\sigma}_e = \mu_{\chi e}^2 y_\chi^2 y_e^2/(\pi q_0^4)$. To first order in the small ratio, $F_{\rm DM} \simeq 1 + (q_0^2 - q^2)/m_V^2$ for the heavy mediator and $F_{\rm DM} \simeq (q_0/q)^2\,(1 + m_V^2/q_0^2 - m_V^2/q^2)$ for the light one. The momentum transfers that contribute run from $\omega/v_{\rm max}$, $0.44\,\text{keV} = 0.12\,q_0$ for the silicon gap, up to $2 m_\chi v_{\rm max}$ (Sect. 1.3).

Limit Check 1.2. Heavy and light mediators

For a mediator of mass m_V and a momentum transfer q, the form factor of Eq. (1.33) is $F_{\rm DM}(q) = 1$ when $m_V \gg q, q_0$ (heavy mediator) and $F_{\rm DM}(q) = (q_0/q)^2$ when $m_V \ll q, q_0$ (light mediator), with $q_0 = \alpha m_e$ [17, Eq. (15) and the display above it].

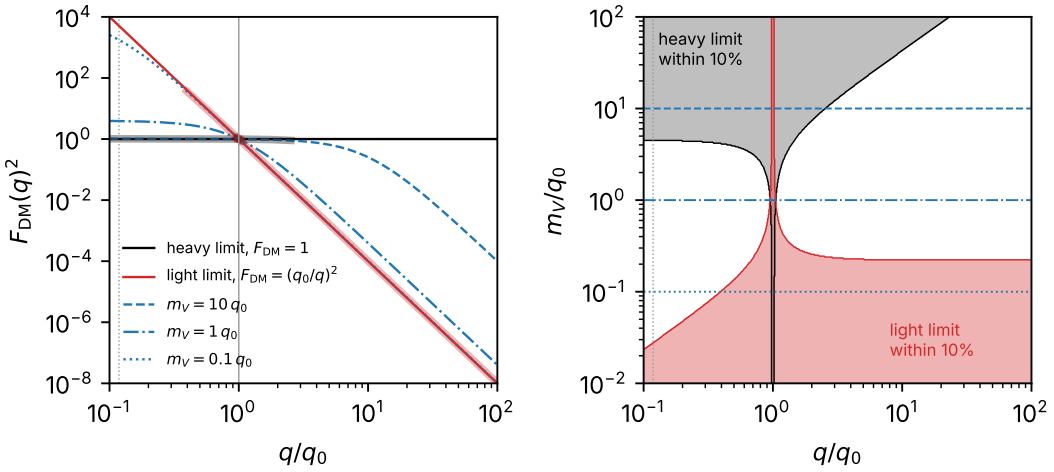


Figure 1.7: Squared dark-matter form factor $F_{\rm DM}(q)^2 = [(m_V^2 + q_0^2)/(m_V^2 + q^2)]^2$ [21, Eq. (12)], with $q_0 = \alpha m_e = 3.73\,\text{keV}$. Left: $F_{\rm DM}^2$ against q/q_0 for $m_V = 10\,q_0$ (dashed), q_0 (dash-dotted) and $0.1\,q_0$ (dotted), between the heavy limit $F_{\rm DM} = 1$ (black) and the light limit $(q_0/q)^4$ (red). A curve is overdrawn in gray or light red where it lies within 10 % of the heavy or light limit, $|F_{\rm DM}^2/\text{limit} - 1| \leq 0.1$. Right: the same two regions in the plane of q/q_0 and m_V/q_0; the horizontal lines are the three mediator masses of the left panel. Every curve equals both limits at $q = q_0$ by construction, which produces the narrow vertical strip. Dotted vertical line: $q = 0.12\,q_0$, the smallest momentum transfer that crosses the low-temperature silicon gap. Across $0.12\,q_0 \leq q \leq q_0$ the heavy limit holds for $m_V \geq 4.5\,q_0 = 17\,\text{keV}$; for all $q \geq 0.12\,q_0$ the light limit holds for $m_V \leq 0.028\,q_0 = 0.10\,\text{keV}$.

For m_V within a factor of a few of the momentum transfers that the target samples, neither limit holds, and rates must use the full form of Eq. (1.35).

A light mediator weights the rate toward small momentum transfers, by $(q_0/q)^4$, and favors excitations near the kinematic threshold at small q, which is also where the target screens most strongly (Sect. 4.4).

1.4.3 The dark photon

The most studied vector mediator is the dark photon: the gauge boson of a new U(1) symmetry under which the dark matter carries charge y_χ and ordinary matter carries none. It reaches ordinary matter through kinetic mixing, a term $(\zeta/2)F_{\mu\nu}F'^{\mu\nu}$ that couples the photon field strength $F_{\mu\nu}$ to its own, $F'_{\mu\nu}$, with a dimensionless strength ζ [22]. The literature writes ϵ, κ or χ for ζ (Appendix B.2). Gauge invariance permits the term, and loops of heavy particles charged under both symmetries generate it [22]. The mixing is removed by shifting the photon field, $A_\mu = \tilde{A}_\mu + \zeta A'_\mu$, which leaves the mass term of A'_μ unchanged (Eqs. (C.85) and (C.86)).

Dark-photon couplings

To first order in ζ, the massive dark photon A'_μ couples to the dark-matter current with strength y_χ and to the electromagnetic current $J^\mu_{\rm em}$ with strength ζe, through $\mathcal{L} \supset -\zeta e\, A'_\mu J^\mu_{\rm em} - y_\chi A'_\mu J^\mu_\chi$. Here $e > 0$ is the elementary charge, $e^2 = 4\pi\alpha$ (Appendix A.2), and $J^\mu_{\rm em}$ counts each particle with its charge in units of e. In the notation of Eq. (1.30)

the electron coupling is $y_e = -\zeta e$, since the electron charge is $-e$; protons have $+\zeta e$ and neutrons zero. The dark matter stays neutral under the photon.

Inserting $y_e^2 = 4\pi\alpha\zeta^2$ into Eq. (1.35), we find [8, Eq. (74)]

$$\bar{\sigma}_e = \frac{4\alpha\,\zeta^2 y_\chi^2\,\mu_{\chi e}^2}{(m_V^2 + \alpha^2 m_e^2)^2}. \tag{1.36}$$

Ref. [17, display above Eq. (15)] writes it with $y_\chi^2 = 4\pi\alpha_D$, where α_D is a dark fine-structure constant.

Worked example 1.5: Reference cross-section of a heavy dark photon

We take the thermal-relic benchmark of Ref. [8, Sect. III D]: $m_\chi = 10\,\text{MeV}$, $m_V = 30\,\text{MeV}$, $y_\chi = 0.3$ and $\zeta = 10^{-4}$. The reduced mass is $\mu_{\chi e} = m_e m_\chi/(m_e + m_\chi) = (0.511 \times 10/10.511)\,\text{MeV} = 0.486\,\text{MeV}$, so $\mu_{\chi e}^2 = 0.236\,\text{MeV}^2$. The maximum momentum transfer is $2m_\chi v_{\max} = 0.053\,\text{MeV}$, smaller than m_V by a factor of 570: the mediator is heavy, $F_{\text{DM}} = 1$, and $\alpha^2 m_e^2/m_V^2 = 1.5 \times 10^{-8}$ can be dropped in Eq. (1.36). With $4\alpha = 0.0292$, $\zeta^2 = 10^{-8}$ and $y_\chi^2 = 0.09$,

$$\bar{\sigma}_e = \frac{4\alpha\,\zeta^2 y_\chi^2\,\mu_{\chi e}^2}{m_V^4} = \frac{0.0292 \times 10^{-8} \times 0.09 \times 0.236\,\text{MeV}^2}{(30\,\text{MeV})^4} = \frac{6.21 \times 10^{-12}\,\text{MeV}^2}{8.10 \times 10^5\,\text{MeV}^4} = 7.67 \times 10^{-18}\,\text{MeV}^{-2}.$$

With $(\hbar c)^2 = 3.894 \times 10^{-22}\,\text{MeV}^2\,\text{cm}^2$ this is $\bar{\sigma}_e = 3.0 \times 10^{-39}\,\text{cm}^2$. Ref. [8, Sect. III D] quotes about $10^{-37}\,\text{cm}^2$ for both of its benchmarks, although its own Eq. (74) leads to the value found here for this one. At fixed couplings $\bar{\sigma}_e$ scales as m_V^{-4}, and for $m_\chi \gg m_e$ it hardly depends on m_χ, because $\mu_{\chi e} \to m_e$.

Freeze-out and freeze-in, the two production histories of Sect. 1.1, lead to the heavy- and light-mediator limits of the form factor. Thermal freeze-out through a dark photon needs couplings large enough to deplete the dark matter, which favors a heavy mediator as in Worked example 1.5. Freeze-in, by contrast, needs couplings so feeble that the dark matter never reaches equilibrium; with an ultralight dark photon the rate of scattering on electrons is still observable, because the light-mediator limit enhances small momentum transfers. The freeze-in benchmark of Ref. [8, Sect. III D], $m_\chi = 1\,\text{MeV}$, $m_V = 10^{-12}\,\text{eV}$, $y_\chi = 3 \times 10^{-6}$ and $\zeta = 10^{-6}$, has $F_{\text{DM}} = (q_0/q)^2$ and, from the light limit of Eq. (1.36) with $\mu_{\chi e} = 0.338\,\text{MeV}$, $\bar{\sigma}_e = 4\alpha\zeta^2 y_\chi^2 \mu_{\chi e}^2/q_0^4 = 6.1 \times 10^{-38}\,\text{cm}^2$.

The field redefinition holds in vacuum. In a medium the photon has a self-energy Π_{em} from the charges it polarizes, and the medium modifies the mixing. For absorption of a dark photon at rest, with $\omega = m_V$ and $q \to 0$, the effective mixing is $\zeta_{\text{eff}}^2 = \zeta^2 m_V^4/\{[m_V^2 - \text{Re}\,\Pi_{\text{em}}(m_V)]^2 + [\text{Im}\,\Pi_{\text{em}}(m_V)]^2\}$ [23, Eqs. (5) and (6), with κ for ζ]. It is suppressed where m_V lies below the scale of $|\Pi_{\text{em}}|$, which in metals and semiconductors is set by the plasma frequency (Sect. 4.3), and approaches ζ well above it. For scattering, the same physics is the screening of the dark photon's potential by the target. In the convention of this section, where $\tilde{V}(\boldsymbol{q})$ is the unscreened potential, the screening is a factor $1/|\varepsilon(\boldsymbol{q},\omega)|^2$ inside the target response [21, Eqs. (6) and (8)], and $\bar{\sigma}_e$ stays a vacuum quantity. The self-energy Π_{em} and the longitudinal dielectric function $\varepsilon(\boldsymbol{q},\omega)$ are material quantities: they are measured optically or computed from first principles, in the random-phase approximation with local fields or

in time-dependent density-functional theory (Appendix E.6). Sect. 4.4 treats screening, and Chap. 10 the in-medium dark photon.

1.4.4 Scattering on nuclei

A mediator that couples to nucleons with strength y_N, the same for protons and neutrons, leads to a dark-matter–nucleon potential of the form of Eq. (1.32) with $y_e \to y_N$. For a heavy mediator the dark-matter–proton cross-section is $\sigma_p = \mu_{\chi p}^2 y_\chi^2 y_N^2/(\pi m_V^4)$ [8, Eq. (99)], the analogue of $\bar\sigma_e$, with $\mu_{\chi p}$ the dark-matter–proton reduced mass. A nucleus of mass number A and atomic number Z holds A nucleons within a radius of a few fm. When q is small compared with the inverse nuclear radius, $1/R \approx 57\,\mathrm{MeV}$ for silicon, whose nuclear radius is $R \approx 3.47\,\mathrm{fm}$ (Sect. 2.2), the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}$ is the same for every nucleon, and the Fourier transform of the summed potential is A times the single-nucleon transform. The amplitudes add coherently.

Replacing y_N by Ay_N and $\mu_{\chi p}$ by the dark-matter–nucleus reduced mass $\mu_{\chi N}$ in σ_p gives

$$\sigma_N = \sigma_p \left(\frac{\mu_{\chi N}}{\mu_{\chi p}}\right)^2 A^2, \tag{1.37}$$

the standard convention for spin-independent scattering [24, 8, Eq. (104)]. If protons and neutrons couple differently, A becomes $Z + (A-Z)y_n/y_p$, with y_p and y_n the two couplings [8, Eq. (104)]. Since a dark photon couples only to charge, A^2 becomes Z^2 for it. For silicon, $A = 28$ and $Z = 14$, with the mean atomic mass of natural silicon, 26.16 GeV. At $m_\chi = 100\,\mathrm{MeV}$, $\mu_{\chi p} = 90.4\,\mathrm{MeV}$ and $\mu_{\chi N} = 99.6\,\mathrm{MeV}$, so $(\mu_{\chi N}/\mu_{\chi p})^2 = 1.215$ and $\sigma_N = 953\,\sigma_p$; at 1 GeV the factor grows to 3.96 and $\sigma_N = 3100\,\sigma_p$.

At larger q the nuclear form factor $F_{\mathrm{Helm}}(q)$ of Sect. 2.2 multiplies the amplitude. For the light dark matter of this book the momentum transfer stays below $2m_\chi v_{\max} \approx 5\,\mathrm{MeV}$ even at $m_\chi = 1\,\mathrm{GeV}$, and the nucleus is always coherent. At q below the inverse size of an atom, the electrons screen the nuclear charge seen by a dark photon; this enters through the target response (Sect. 4.4). In a crystal, however, the nuclei are bound, and at low q the dark matter excites phonons instead of free recoils (Chap. 5).

1.4.5 Absorption of bosonic dark matter

Dark matter lighter than about a keV cannot be a thermal relic (Sect. 1.1), but bosons in this range can be produced non-thermally, for example by misalignment, and behave as a classical field oscillating at their mass [8, Sect. I B]. Such a boson ϕ of mass m_ϕ can be absorbed: it disappears and deposits its whole energy in the target. Bound to the Galaxy, it moves at $v \approx 10^{-3}$ and carries energy $m_\phi(1 + v^2/2) \approx m_\phi$ and momentum $\boldsymbol{p}_\phi = m_\phi \boldsymbol{v}$. Absorption then deposits

$$\omega \approx m_\phi, \qquad \boldsymbol{q} = \boldsymbol{p}_\phi, \qquad |\boldsymbol{p}_\phi| \approx m_\phi v \approx 10^{-3}\,\omega. \tag{1.38}$$

The momentum is much smaller than the energy, the reverse of scattering, where $\omega \le qv$ (Sect. 1.3). Absorption probes the response of the target at $q \to 0$, the regime of optical absorption, at a single energy, $\omega = m_\phi$. Compared to scattering, absorption also reaches lower masses. A scattering particle can deposit at most its kinetic energy, and crossing a gap E_g needs $m_\chi \ge 2E_\mathrm{g}/v_{\max}^2$ (Sect. 1.3), about 0.3 MeV for $E_\mathrm{g} \approx 1\,\mathrm{eV}$; an absorbed boson deposits its rest energy and needs only $m_\phi \ge E_\mathrm{g}$, about 1 eV. Below the gap, absorption on phonons extends the reach to m_ϕ of tens of meV (Chap. 5).

The standard absorbed bosons are the dark photon, the axion-like particle and the scalar. Appendix C.9 derives the classical-field steps that the dark photon shares with scattering, Chap. 2 derives absorption rates from the golden rule, and Chap. 8 treats each channel. A

dark photon that is itself the dark matter, with $m_V = m_\phi$, couples to $J^\mu_{\rm em}$ exactly as a photon does with e replaced by ζe. Its absorption rate is the photon absorption rate times $\zeta^2_{\rm eff}$. Per unit target mass, the normalization of the event rate $\mathcal{R}$ of Sect. 1.5, it is [23, Eq. (7)]

$$\mathcal{R} = \frac{1}{\rho_T}\,\frac{\rho_\chi}{m_\phi}\,\zeta^2_{\rm eff}\,\sigma_1(m_\phi), \tag{1.39}$$

with ρ_χ the local dark-matter density (Sect. 1.2), ρ_T the target mass density and σ_1 the real part of the optical conductivity (Sect. 4.4), measured or computed at $\omega = m_\phi$. The flux of dark matter does not enter, only its number density ρ_χ/m_ϕ: an absorbed field deposits energy at a rate set by its amplitude, whatever its speed.

An axion-like particle (ALP), a light pseudoscalar, couples to electrons through $\mathcal{L} \supset g_{aee}\,\phi\,\bar\psi \mathrm{i}\gamma^5\psi$ [25, Eq. (3)] and to photons through $\mathcal{L} \supset -\frac{1}{4} g_{a\gamma\gamma}\,\phi\,F_{\mu\nu}\tilde F^{\mu\nu} = g_{a\gamma\gamma}\,\phi\,\boldsymbol{E}\cdot\boldsymbol{B}$, with $\tilde F^{\mu\nu}$ the dual field strength [26, Eqs. (2) and (3)]; the subscript a names the ALP. At low energy the electron field ψ reduces to a two-component field $\hat\psi$ (the field operator of Sect. 2.3), and the electron coupling becomes [25, Eq. (18) and Table I]

$$\mathcal{L}_{aee} = -\frac{g_{aee}}{2m_e}\,(\nabla\phi)\cdot(\hat\psi^\dagger\boldsymbol{\sigma}\hat\psi) + \frac{\mathrm{i}g_{aee}}{4m_e^2}\,(\partial_t\phi)\,\hat\psi^\dagger\boldsymbol{\sigma}\cdot\overleftrightarrow{\nabla}\hat\psi, \tag{1.40}$$

with $\boldsymbol{\sigma}$ the Pauli matrices and $\hat\psi^\dagger\overleftrightarrow{\nabla}\hat\psi \equiv \hat\psi^\dagger(\nabla\hat\psi) - (\nabla\hat\psi^\dagger)\hat\psi$. The first term couples $\nabla\phi$, of size $|\boldsymbol{p}_\phi|\phi$, to the electron spin density and is suppressed by $|\boldsymbol{p}_\phi|$. The second couples $\partial_t\phi$, of size $m_\phi\phi$, to the electron spin-current. Compared with the first, it trades the dark-matter velocity, typically 10^{-3}, for the electron velocity $|\nabla|/m_e$, typically $\sim\alpha$ in a solid; as a result it dominates absorption, although it is formally of higher order in $1/m_e$ [25, Sect. II]. When spin–orbit coupling is negligible its rate follows from the optical conductivity, like that of a dark photon [25, Table I]. The photon coupling contributes to absorption in a solid only in a static magnetic field $\boldsymbol{B}_0$: there ϕ drives an electric field along $\boldsymbol{B}_0$ that oscillates at m_ϕ, and the ALP is absorbed like a dark photon polarized along $\boldsymbol{B}_0$.

A scalar couples through $\mathcal{L} \supset g_{\phi ee}\,\phi\,\bar\psi\psi$ [25, Eq. (3)], which at low energy becomes [25, Eq. (18) and Table I]

$$\mathcal{L}_{\phi ee} = g_{\phi ee}\,\phi\,\hat\psi^\dagger\hat\psi + \frac{g_{\phi ee}}{8m_e^2}\,\phi\,\hat\psi^\dagger\overleftrightarrow{\nabla}^2\hat\psi, \tag{1.41}$$

where a third term of Ref. [25, Eq. (18)], involving the photon field, is omitted because it does not contribute to absorption in an isotropic medium [25, Table I], and $\hat\psi^\dagger\overleftrightarrow{\nabla}^2\hat\psi \equiv \hat\psi^\dagger(\nabla^2\hat\psi) + (\nabla^2\hat\psi^\dagger)\hat\psi - 2(\nabla\hat\psi^\dagger)\cdot(\nabla\hat\psi)$. The first term couples to the electron number density, whose matrix elements between different states vanish as $\boldsymbol{p}_\phi \to 0$ by charge conservation. The second, a kinetic-energy density, dominates, and its rate is not fixed by the optical conductivity [25, Table I]. Table 1.3 collects the interactions of this section.

1.4.6 Inelastic dark matter and further interactions

The dark matter may have two nearly degenerate states, χ_1 and χ_2, split by $\Delta_\chi = m_{\chi_2} - m_{\chi_1}$. This happens, for example, when a small mass term splits one dark-matter particle into two states of slightly different mass, and the mediator couples only across them, so every scattering changes the state [27]; in field-theory language, a Dirac fermion splits into two Majorana states. The potential and $\bar\sigma_e$ keep their form; only the kinematics changes, through the term $-\Delta_\chi$ in the energy transfer $\omega_{\boldsymbol{q}}$ of Eq. (1.22), with the endothermic and exothermic cases of Sect. 1.3.

The mediators of this section lead to the simplest non-relativistic interaction, a spin-independent potential. A general basis of non-relativistic interactions exists, built from $\boldsymbol{q}$, the

Table 1.3: Interactions of light dark matter used in this book. Each row gives the coupling, the target quantity that sets the rate and the excitation that dominates in a semiconductor. The density response and conductivity are defined in Chap. 4, the spin and kinetic-energy responses in Chap. 8, and phonons in Chap. 5.

Interaction	Coupling	Target quantity	Dominant excitation
Scattering, heavy mediator	$\bar{\sigma}_e$, $F_{\mathrm{DM}} = 1$	electron density response at all q	electron–hole pairs
Scattering, light mediator	$\bar{\sigma}_e$, $F_{\mathrm{DM}} = (q_0/q)^2$	electron density response at small q, screened	electron–hole pairs near threshold
Scattering on nuclei	σ_p, σ_N	nuclear density response	phonons; nuclear recoils above the binding scale
Dark-photon absorption	ζ	optical conductivity at $\omega = m_\phi$	electron–hole pairs above the gap; optical phonons in polar crystals below it
ALP absorption	g_{aee}; $g_{a\gamma\gamma}$	electron spin-current response; field $\boldsymbol{B}_0$	electron–hole pairs; photons in $\boldsymbol{B}_0$
Scalar absorption	$g_{\phi ee}$	electron kinetic-energy response	electron–hole pairs

relative velocity and the spins of the dark matter and the target [28]; the Yukawa potential is its first, spin-independent operator multiplied by $F_{\mathrm{DM}}(q)$. Chap. 8 treats this basis and the target responses it requires.

1.5 Detector signals

A detector records none of the dark-matter flux (Sect. 1.2), the energy and momentum it can transfer (Sect. 1.3) or the strength of its interaction (Sect. 1.4) directly. It counts quanta: electrons, phonons, quasiparticles or photons produced as the deposited energy relaxes. We will follow the energy from the deposit to the quanta a sensor counts, assemble the event rate, and turn a count of events into a limit on $\bar{\sigma}_e$. Fig. 1.8 shows the chain that the rest of the book follows.

1.5.1 Readout quanta

A deposit of a few eV or less does not stay in the excitation that the dark matter created. Hot electrons and holes lose energy to further electron–hole pairs and to phonons; quasiparticles in a superconductor emit phonons; phonons decay into lower-energy phonons. Detectors read the products at one of four stages: as charge, as heat carried by phonons, as quasiparticles of a superconductor, or as photons [1, Sect. VII]. Each channel counts a different quantum, and the energy per quantum sets how finely a deposit is resolved.

Charge. In a semiconductor an excitation across the band gap promotes an electron to the conduction band and leaves a vacancy, a *hole*, in the valence band. The pair is called an *electron–hole pair*. Both carriers are mobile, and an applied electric field drifts them to electrodes before they recombine [1, Sect. VII A]. The number of pairs produced by a deposit ω is the *ionization yield* N_{eh}. A primary carrier with energy well above the gap creates further pairs by impact ionization, knocking further electrons across the gap, and also emits phonons; on average, then, each pair costs more than E_{g}. For deposits well above 20 eV the mean yield is proportional to the deposit [18, abstract and Eq. (1)],

$$\langle N_{\mathrm{eh}} \rangle = \frac{\omega}{\varepsilon_{\mathrm{eh}}}, \tag{1.42}$$

with the mean energy per pair $\varepsilon_{\mathrm{eh}}$. Measurements in silicon give 3.62 eV to 3.66 eV at 300 K and 3.70 eV to 3.75 eV between 5 K and 123 K [18, Table I], roughly three times the gap. The excess goes into phonons. The fluctuations of N_{eh} are smaller than Poisson fluctuations: the variance is 0.1 to 0.15 times the mean [16, Eq. (5.4)]. A 100 eV deposit produces on average $100/3.7 = 27$ pairs with a spread of $(0.12 \times 27)^{1/2} = 1.8$, against $27^{1/2} = 5.2$ for a Poisson count. The pairs are not produced independently: the energy is fixed, and a cascade that makes one more pair has less energy left for the next.

Light dark matter, however, deposits only a few eV, where Eq. (1.42) fails. Below $2E_{\mathrm{g}}$ energy conservation allows one pair at most; between that and a few tens of eV the yield depends on how the primary carriers share the energy, and the silicon data cover 3 to 8 eV and then 50 eV upward, with nothing in between [18, Sect. III]. A widely used step model assigns one pair for the first E_{g} and one more for each further $\varepsilon_{\mathrm{eh}}$ [16, Eq. (5.1)],

$$N_{\mathrm{eh}}(\omega) = 1 + \left\lfloor \frac{\omega - E_{\mathrm{g}}}{\varepsilon_{\mathrm{eh}}} \right\rfloor \qquad (\omega \geq E_{\mathrm{g}}), \tag{1.43}$$

where $\lfloor x \rfloor$ is the largest integer not above x. With the low-temperature gap $E_{\mathrm{g}} = 1.17$ eV of Worked example 1.3 and $\varepsilon_{\mathrm{eh}} = 3.6$ eV [16, Eq. (5.2)], a 5 eV deposit makes $1 + \lfloor 1.06 \rfloor = 2$ pairs and a 10 eV deposit $1 + \lfloor 2.45 \rfloor = 3$. Chap. 18 derives the yield from the carrier cascade.

A detector that counts pairs states its threshold as a number of pairs, and Eq. (1.43) turns that number into an energy. The n-th pair appears at $\omega = E_{\mathrm{g}} + (n-1)\varepsilon_{\mathrm{eh}}$, and thresholds of one, two and three pairs then correspond to 1.17, 4.77 and 8.37 eV. Through

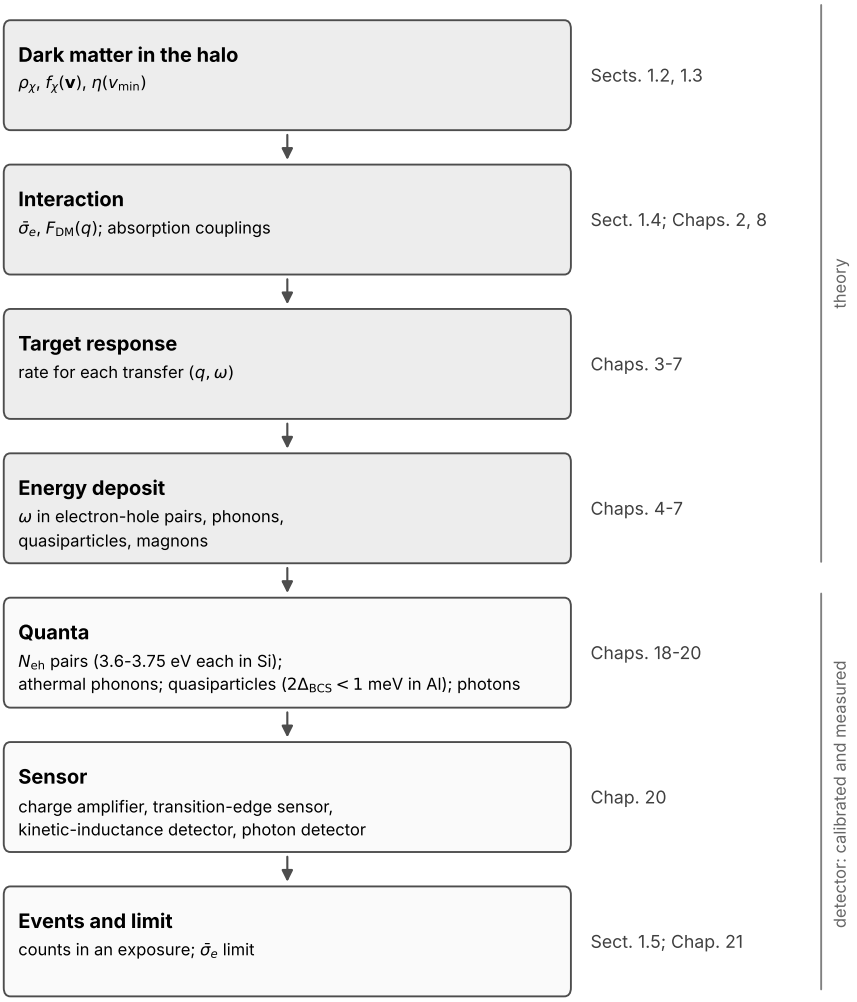


Figure 1.8: The signal chain from dark matter to sensor. The upper four steps are computed from theory: the halo (Sects. 1.2 and 1.3), the interaction (Sect. 1.4; Chaps. 2 and 8), the target response and the primary excitations (Chaps. 3 to 7). The lower three are fixed by calibration and measurement: the conversion of the deposit into countable quanta (Chaps. 18 and 20), the sensor (Chap. 20) and the statistical step from counts to a limit (Sect. 1.5; Chap. 21). Energies per quantum are from Refs. [18, Table I, 29, Sect. 2.2.1].

$m_{\min} = 2E_{\mathrm{th}}/v_{\max}^2$ (Eq. (1.25)) with $v_{\max} = 2.650 \times 10^{-3}$, these translate into threshold masses of 0.33, 1.36 and 2.38 MeV. A search that discards the single-pair bin, where the dark counts described below accumulate, gives up a factor of four in mass reach for a cleaner sample. Since the yield has not been measured in silicon between 8 and 50 eV, the energy assigned to a three-pair threshold, and with it $m_{\min}$, carries a model uncertainty (Sect. 1.3.5).

Skipper charge-coupled devices and calorimetric detectors count single charges. A skipper charge-coupled device measures the charge in each pixel many times without destroying it; the noise falls with the square root of the number of measurements, below 0.1 electron per pixel [1, Sect. VII A]. A calorimetric detector instead drifts the carriers across a bias voltage V_{bias}. Each carrier then emits phonons in proportion to the work done on it (the Neganov–Trofimov–Luke effect). The phonon energy is [1, Eq. (195)]

$$E_{\mathrm{ph}} = E_{\mathrm{prim}} + N_{\mathrm{eh}}\, eV_{\mathrm{bias}}, \tag{1.44}$$

where E_{prim} is the part of the deposit that went directly into phonons, at most ω; the second term is usually much larger. At $V_{\mathrm{bias}} = 150$ V each pair adds 150 eV, and the spectrum shows peaks at integer N_{eh} [1, Sect. VII A]. In both devices the lowest bins are polluted by a dark rate, charge that leaks across the gap without any deposit, and single-pair events carry the largest background.

Phonons. Every excitation of a solid ends as phonons, the quantized lattice vibrations of Sect. 1.3. A calorimeter measures this heat and with it the total deposit, whatever the primary excitation [1, Sect. VII B]. The phonons first created in silicon are optical phonons of about 64 meV; through several anharmonic decays they eventually become acoustic phonons of a few meV. At an operating temperature of 15 mK the thermal phonons carry about 1 μeV, and the signal phonons are far from equilibrium (*athermal*) and cross the crystal ballistically before they reach a sensor on the surface [1, Sect. VII B]. The quanta are too numerous and too soft to count one by one. The sensor integrates their energy, and its noise sets the threshold: a 10.6 g silicon crystal read by a tungsten transition-edge sensor reaches a resolution of 3.86 eV and a threshold of 16.3 eV [1, Sect. VII B]. A transition-edge sensor is a superconducting film held at the edge of its transition, where a small heat input produces a large change in resistance. Phonons are treated precisely in Sect. 5.3.

Quasiparticles. In a superconductor the electrons near the Fermi level, the energy that separates occupied from empty states, bind into Cooper pairs. Breaking a pair costs $2\Delta_{\mathrm{BCS}}$, where Δ_{BCS} is the superconducting gap, and creates two *quasiparticles*, the excitations of the paired state (made precise in Sect. 7.2). The Bardeen–Cooper–Schrieffer (BCS) theory relates the gap to the critical temperature, $\Delta_{\mathrm{BCS}} \approx 1.76\, k_{\mathrm{B}} T_{\mathrm{c}}$ [30, Sect. 3.3]. With $T_{\mathrm{c}} = 1.2$ K for bulk aluminum [31], $2\Delta_{\mathrm{BCS}} \approx 0.36$ meV. Thin films have higher critical temperatures [31], and dark-matter projections adopt the film value $2\Delta_{\mathrm{BCS}} = 0.6$ meV [29, Sect. 2.2.1, 25, Table II]. A deposit in a superconducting absorber cascades through quasiparticles and phonons until no phonon can break another pair; about 60 % of the energy ends as quasiparticles and the rest as phonons near $2\Delta_{\mathrm{BCS}}$ [29, Sect. 2.2.1]. Quasiparticles also convert phonons into a signal. In transition-edge sensors and in kinetic-inductance detectors, superconducting resonators whose frequency shifts when pairs are broken, phonons from the crystal break Cooper pairs in a superconducting film, and the change in the film's resistance or impedance is read out [30, Sect. 3.3.1]. The pair-breaking energy is the smallest quantum in Table 1.4. If a single broken pair in an aluminum absorber could be detected, Eq. (1.25) with the bulk value $E_{\mathrm{th}} = 0.36$ meV would place the threshold mass at 0.10 keV (0.17 keV for the film value), below the keV lower end of the window of Sect. 1.1. Present sensors are far from this: their thresholds are set by noise, not by the quantum.

Table 1.4: Readout channels, the quantum each counts and its energy in silicon or aluminum, with the sensors that read them. The phonon and quasiparticle energies set the scale of the quanta; calorimetric sensors integrate many of them.

Channel	Quantum	Energy per quantum	Sensor
Charge	electron–hole pair	$\varepsilon_{\mathrm{eh}} = 3.6$ to $3.75\,\mathrm{eV}$ in Si (at least E_{g} for the first pair)	skipper charge-coupled device (CCD); phonon calorimeter with bias
Heat	athermal phonon	$60\,\mathrm{meV}$ optical at creation, a few meV after decay, in Si	transition-edge sensor
Quasiparticles	broken Cooper pair	$2\Delta_{\mathrm{BCS}} \approx 0.36\,\mathrm{meV}$ in bulk Al	transition-edge sensor; kinetic-inductance detector
Light	photon	a few eV (optical, near-ultraviolet); infrared in GaAs	photomultiplier; transition-edge sensor; Si photodetector

Photons. A scintillator returns part of the deposit as photons. In NaI and CsI, with band gaps of about 6 eV, these are optical or near-ultraviolet photons of a few eV; GaAs, with its gap of 1.52 eV, emits in the infrared [1, Sect. VII C]. The photons are counted by a photomultiplier, by a transition-edge sensor or by a silicon device, in which each photon above the silicon gap, 1.17 eV at low temperature (Worked example 1.3), creates electron–hole pairs.

Table 1.4 collects the four channels: charge, phonons, quasiparticles and photons. In every case the detector sees a number of quanta, or their summed energy, after the cascade. The relation between that number and ω is an input to every limit, and for deposits of a few eV it is the most poorly known step of the chain.

1.5.2 Event rate

The event rate $\mathcal{R}$ of Eq. (1.1), the expected number of events per unit time per unit target mass, is quoted in $\mathrm{kg^{-1}\,yr^{-1}}$. Its *spectrum* $\mathrm{d}\mathcal{R}/\mathrm{d}\omega$ is the rate per unit deposited energy, and $\mathcal{R}$ is the integral of the spectrum above the threshold E_{th} (Sect. 1.3). A detector of mass M_{det} that runs for a live time t_{exp} collects the *exposure* $M_{\mathrm{det}} t_{\mathrm{exp}}$, quoted in kg yr. The expected number of signal events is

$$N_{\mathrm{sig}} = M_{\mathrm{det}}\, t_{\mathrm{exp}}\, \mathcal{R}, \tag{1.45}$$

where $\mathcal{R}$ includes the probability that an event passes the selection. The absorption rate of Eq. (1.39) uses the same normalization per unit target mass.

A charge detector does not measure ω but N_{eh}. The rate of events with n pairs collects the spectrum over the deposits that Eq. (1.43) maps to n,

$$\mathcal{R}_n = \int_{E_{\mathrm{g}}+(n-1)\varepsilon_{\mathrm{eh}}}^{E_{\mathrm{g}}+n\varepsilon_{\mathrm{eh}}} \mathrm{d}\omega\, \frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega}, \tag{1.46}$$

and a search compares the counts in each bin with the expected background in that bin. Since for scattering through a heavy mediator in silicon the typical deposit is a few eV, the spectrum peaks in the two-pair bin [16, Fig. 7].

The schematic rate of Eq. (1.1) becomes concrete for a target whose response depends on the magnitude q of the momentum transfer but not on its direction, as for a liquid, an atom or a crystal averaged over orientations. The velocity integral and the average over the direction of $\boldsymbol{q}$ then act on the delta function alone, and Sect. 1.3.4 turned them into the measure $q\,\mathrm{d}q\,\eta(v_{\mathrm{min}})/4\pi^2$. With the coupling of Eq. (1.34) and the halo of Sect. 1.2, the

factors of Sect. 1.1 combine into [1, Eq. (11)]

$$\mathcal{R} = \frac{1}{\rho_T} \underbrace{\frac{\rho_\chi}{m_\chi}}_{\text{halo}} \int \frac{q\,\mathrm{d}q}{(2\pi)^2} \int \mathrm{d}\omega \; \underbrace{\eta(v_{\min}(q,\omega))}_{\text{halo, kinematics}} \underbrace{\frac{\pi \bar{\sigma}_e F_{\mathrm{DM}}(q)^2}{\mu_{\chi e}^2}}_{\text{coupling}} \underbrace{S(q,\omega)}_{\text{target}}. \tag{1.47}$$

Sect. 2.5 derives it term by term from the thermal golden rule, and Sect. 4.2 develops the properties of $S(q,\omega)$. Here we read off how $\mathcal{R}$ depends on the dark-matter parameters.

For a crystal, $S(q,\omega)$ is computed from the electronic states of the material. Density-functional theory supplies the bands and wavefunctions as Kohn–Sham states, the eigenstates of an effective one-electron problem rather than of the interacting electrons. Their overlaps at momentum transfer q, the crystal form factors, fix the transition probabilities; Appendix E.7 describes how these are evaluated and converged, and Appendix E.6 adds screening. This step is where most of the theory uncertainty of a limit originates.

Define the response-weighted average of the squared form factor,

$$\langle F_{\mathrm{DM}}^2 \rangle = \frac{\int q\,\mathrm{d}q \int \mathrm{d}\omega \; \eta(v_{\min})\, S(q,\omega)\, F_{\mathrm{DM}}(q)^2}{\int q\,\mathrm{d}q \int \mathrm{d}\omega \; \eta(v_{\min})\, S(q,\omega)}, \tag{1.48}$$

which is 1 for a heavy mediator and $\langle (q_0/q)^4 \rangle$ for a light one. Then Eq. (1.47) reads

$$\mathcal{R} = \frac{\pi \bar{\sigma}_e\, \rho_\chi\, \langle F_{\mathrm{DM}}^2 \rangle}{\rho_T\, m_\chi\, \mu_{\chi e}^2} \int \frac{q\,\mathrm{d}q}{(2\pi)^2} \int \mathrm{d}\omega \; \eta(v_{\min})\, S(q,\omega). \tag{1.49}$$

We see that the rate is linear in $\bar{\sigma}_e$ and in ρ_χ, and $\bar{\sigma}_e$ appears nowhere else. The mass enters through the number density, the reduced mass and the region of the last integral. The number density brings $1/m_\chi$. The reduced mass tends to m_e once $m_\chi \gg m_e$, and the factor $\mu_{\chi e}^2$ stops changing above a few MeV. The last integral depends on m_χ only through the momentum transfers allowed by $v_{\min}(q,\omega) \le v_{\max}$ (Eq. (1.26)): it vanishes for $m_\chi \le m_{\min}$ and stops growing once $2 m_\chi v_{\max}$ exceeds the momenta at which $S(q,\omega)$ is appreciable. Well above threshold, then,

$$\mathcal{R} \propto \frac{\bar{\sigma}_e\, \rho_\chi\, \langle F_{\mathrm{DM}}^2 \rangle}{m_\chi}. \tag{1.50}$$

This is a statement about scaling only; the absolute rate needs $S(q,\omega)$ and the full integral. As a check of Eq. (1.47), in natural units $\rho_\chi/(\rho_T m_\chi)$ has dimension eV^{-1} (an inverse mass), $q\,\mathrm{d}q\,\mathrm{d}\omega$ has eV^3, $\bar{\sigma}_e/\mu_{\chi e}^2$ has eV^{-4}, $S(q,\omega)$ has eV^2 with the normalization of Sect. 4.2, and η, an inverse speed, is dimensionless. The product is dimensionless, as a rate (eV) per unit mass (eV) must be.

1.5.3 Counting and the upper limit

Since dark-matter events are rare and independent, their number in a fixed exposure follows the Poisson distribution. To see this, divide the exposure into K equal parts, each so small that it holds one event with probability $p = N_{\mathrm{sig}}/K$ and two events with negligible probability. The probability of N_{obs} events is binomial,

$$\binom{K}{N_{\mathrm{obs}}} p^{N_{\mathrm{obs}}} (1-p)^{K-N_{\mathrm{obs}}} = \frac{K(K-1)\cdots(K-N_{\mathrm{obs}}+1)}{K^{N_{\mathrm{obs}}}} \frac{N_{\mathrm{sig}}^{N_{\mathrm{obs}}}}{N_{\mathrm{obs}}!} \left(1 - \frac{N_{\mathrm{sig}}}{K}\right)^{K-N_{\mathrm{obs}}}.$$

As $K \to \infty$ at fixed N_{obs}, the first fraction tends to 1 and the last factor to $\mathrm{e}^{-N_{\mathrm{sig}}}$, and we are left with

$$P(N_{\mathrm{obs}} \,|\, N_{\mathrm{sig}}) = \frac{N_{\mathrm{sig}}^{N_{\mathrm{obs}}}}{N_{\mathrm{obs}}!}\, \mathrm{e}^{-N_{\mathrm{sig}}}. \tag{1.51}$$

Suppose a search sees no events and expects no background. A large $N_{\rm sig}$ would make this outcome improbable, since $P(0\,|\,N_{\rm sig}) = \mathrm{e}^{-N_{\rm sig}}$. The upper limit at 90 % confidence, N_{90}, is the expected count at which the observed outcome, or one with fewer events, would occur in only 10 % of repeated experiments. Every larger $N_{\rm sig}$ would yield at least one event more than 90 % of the time, and is excluded at that confidence. The confidence level describes the procedure, not the parameter: if the experiment were repeated many times and the limit computed the same way each time, the interval from 0 to N_{90} would contain the true $N_{\rm sig}$ in at least 90 % of the repetitions. It is not the probability that $\bar{\sigma}_e$ lies below the limit, which would require a prior distribution for $\bar{\sigma}_e$. Setting $\mathrm{e}^{-N_{90}} = 0.1$ gives

$$N_{90} = \ln 10 = 2.303. \tag{1.52}$$

At 95 % confidence the same condition leads to $\ln 20 = 3.00$, the three events assumed in many projections [25, Sect. V]. If instead $N_{\rm obs}$ events are seen and all are kept as possible signal, the 90 % limit solves $\sum_{n=0}^{N_{\rm obs}} P(n\,|\,N_{90}) = 0.1$. A numerical solution yields $N_{90} = 3.89$ for one event and 5.32 for two. Such a limit holds whatever the origin of the observed events.

Since $\mathcal{R}$ is proportional to $\bar{\sigma}_e$ (Eq. (1.49)), we write $\mathcal{R} = \bar{\sigma}_e\, r(m_\chi)$ with $r(m_\chi)$ the rate per unit cross-section. Setting $N_{\rm sig} = N_{90}$ in Eq. (1.45) gives the limit:

Zero-background limit

A search with exposure $M_{\rm det} t_{\rm exp}$ that observes no events and expects no background excludes at 90 % confidence

$$\bar{\sigma}_e > \bar{\sigma}_e^{\rm lim}(m_\chi) = \frac{N_{90}}{M_{\rm det}\, t_{\rm exp}\, r(m_\chi)}, \qquad N_{90} = 2.303. \tag{1.53}$$

The limit improves in proportion to the exposure, $\bar{\sigma}_e^{\rm lim} \propto 1/(M_{\rm det} t_{\rm exp})$.

Background changes the scaling. Suppose the background is known to give $N_{\rm bkg} = \mathcal{R}_{\rm bkg}\, M_{\rm det} t_{\rm exp}$ events, with $\mathcal{R}_{\rm bkg}$ the background rate per unit exposure, and cannot be removed event by event. The observed count then fluctuates by roughly $N_{\rm bkg}^{1/2}$, and a signal is excluded only when it exceeds a fixed multiple k of that fluctuation, $\bar{\sigma}_e\, r\, M_{\rm det} t_{\rm exp} \geq k\,(\mathcal{R}_{\rm bkg}\, M_{\rm det} t_{\rm exp})^{1/2}$. Solving for $\bar{\sigma}_e$, we find $\bar{\sigma}_e^{\rm lim} = k\,\mathcal{R}_{\rm bkg}^{1/2}/[r\,(M_{\rm det} t_{\rm exp})^{1/2}]$: the limit now improves only as the inverse square root of the exposure.

Counting alone cannot tell signal from background, but the annual modulation of Sect. 1.2 can. For momentum transfers that need $v_{\rm min} = 600\,\mathrm{km\,s^{-1}}$, η is 21 % above its reference value in June and 16 % below it in December (Sect. 1.2). Take the modulation as a cosine with amplitude $A_{\rm mod} = 0.185$, the mean of the two. Averaged over the half-year centered on June, a cosine of amplitude $A_{\rm mod}$ has mean $2A_{\rm mod}/\pi = 0.118$; with N events expected per half-year, the June half then exceeds the December half by $2 \times 0.118\, N = 0.236\, N$. The Poisson fluctuation of the difference of two counts of about N each is $(2N)^{1/2}$. A difference of two standard deviations requires $0.236\, N \geq 2(2N)^{1/2}$, that is $N \geq 144$ events per half-year. A modulation test needs a detected signal of a few hundred events, far beyond the few events that set a limit. Chap. 21 treats the statistics in full.

Worked example 1.6: Expected events and a limit for a toy rate

We take a silicon target, $m_\chi = 100\,\mathrm{MeV}$, a heavy mediator and $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$. For the toy rate we keep the prefactor of the silicon rate of Ref. [16, Eq. (3.13)]. That equation is written for the spectrum per unit logarithm of the deposit, $\mathrm{d}\mathcal{R}/\mathrm{d}\ln\omega$, and the reference describes its remaining integral as roughly unity for typical q and ω. We set that integral

equal to one between $E_g = 1.17\,\mathrm{eV}$ and $30\,\mathrm{eV}$, the range of the toy response of Fig. 1.9, and to zero elsewhere. Integrating over $\ln\omega$ then multiplies the prefactor by $\ln(30/1.17) = 3.24$, and the toy rate is

$$\mathcal{R}_{\mathrm{toy}} = \ln\!\left(\frac{30\,\mathrm{eV}}{E_g}\right) \frac{\rho_\chi}{m_\chi}\,\frac{\bar\sigma_e}{M_{\mathrm{cell}}}\,\alpha\,\frac{m_e^2}{\mu_{\chi e}^2}.$$

Here $M_{\mathrm{cell}} = 52.33\,\mathrm{GeV}$ is the mass of the two-atom silicon unit cell [16, below Eq. (3.13)], so $1/M_{\mathrm{cell}}$ is the number of cells per unit mass. The factor $\alpha = q_0/m_e$ is the typical speed of a bound electron, αc in conventional units. The factors are: $\rho_\chi/m_\chi = 3\,\mathrm{cm}^{-3}$; $1\,\mathrm{kg} = 5.610\times10^{26}\,\mathrm{GeV}$, so a kilogram holds 1.072×10^{25} cells; $\alpha c = 2.188\times10^{8}\,\mathrm{cm\,s^{-1}}$; $\mu_{\chi e} = 0.5084\,\mathrm{MeV}$, so $m_e^2/\mu_{\chi e}^2 = 1.010$. Without the logarithm their product is $7.106\times10^{33}\,\mathrm{cm^{-2}\,s^{-1}\,kg^{-1}}$; with it, $\mathcal{R}_{\mathrm{toy}}/\bar\sigma_e = 2.305\times10^{34}\,\mathrm{cm^{-2}\,s^{-1}\,kg^{-1}}$. With $1\,\mathrm{yr} = 3.156\times10^{7}\,\mathrm{s}$ this is $r = 7.28\times10^{41}\,\mathrm{cm^{-2}\,kg^{-1}\,yr^{-1}}$. For an exposure of $1\,\mathrm{kg\,yr}$ and $\bar\sigma_e = 10^{-40}\,\mathrm{cm}^2$, Eq. (1.45) predicts $N_{\mathrm{sig}} = 73$ events. With no events observed, Eq. (1.53) gives

$$\bar\sigma_e^{\mathrm{lim}} = \frac{2.303}{(1\,\mathrm{kg\,yr})(7.28\times10^{41}\,\mathrm{cm^{-2}\,kg^{-1}\,yr^{-1}})} = 3.2\times10^{-42}\,\mathrm{cm}^2,$$

and ten times the exposure brings it to $3.2\times10^{-43}\,\mathrm{cm}^2$. Since the toy has no material response and no yield model, these numbers show only the arithmetic and the scaling; a real silicon search must compute $S(q,\omega)$ and fold in Eq. (1.43) and the efficiency.

1.5.4 Exclusion plots

Results of light-dark-matter searches are shown as *exclusion limits*: the curve $\bar\sigma_e^{\mathrm{lim}}(m_\chi)$ of Eq. (1.53) in the plane of m_χ and $\bar\sigma_e$, drawn separately for a heavy and a light mediator. Every point above the curve would have produced more than N_{90} events and is excluded. By Eqs. (1.49) and (1.53),

$$\bar\sigma_e^{\mathrm{lim}}(m_\chi) \propto \frac{m_\chi\,\mu_{\chi e}^2}{\rho_\chi\,\langle F_{\mathrm{DM}}^2\rangle\,M_{\mathrm{det}}t_{\mathrm{exp}}}\left[\int\frac{q\,\mathrm{d}q}{(2\pi)^2}\int\mathrm{d}\omega\;\eta(v_{\mathrm{min}})\,S(q,\omega)\right]^{-1}, \tag{1.54}$$

and the shape of the curve follows from the mass dependence discussed after Eq. (1.49). Fig. 1.9 shows the shapes for a toy response; Chap. 21 treats real limits.

The low-mass edge. Below $m_{\mathrm{min}} = 2E_{\mathrm{th}}/v_{\mathrm{max}}^2$ (Eq. (1.25)) no dark-matter particle in the halo can deposit E_{th}, the integral in Eq. (1.54) vanishes and the limit is infinite. Just above m_{min} only the fastest particles contribute, and η grows from zero quadratically in the distance from the escape speed (Eq. (1.29)): the curve falls steeply from a vertical wall. The position of the wall depends only on the threshold and on v_{max}; a lower threshold moves it to lower mass in proportion (Sect. 1.3.5).

The heavy-mediator slope. With $F_{\mathrm{DM}} = 1$ the integrand of Eq. (1.54) grows with q through the factor $q\,\mathrm{d}q$, and the maximum momentum transfer, $2m_\chi v_{\mathrm{max}}$, grows with m_χ. It equals q_0 at $0.70\,\mathrm{MeV}$ (Sect. 1.3). The integral keeps growing until $2m_\chi v_{\mathrm{max}}$ passes the momenta of the valence electrons, a few q_0, which happens at masses around $10\,\mathrm{MeV}$; the curve falls over this whole range. Above it the integral is constant, $\mu_{\chi e} = m_e$, and only the factor m_χ remains: $\bar\sigma_e^{\mathrm{lim}} \propto m_\chi$, a line of slope one on logarithmic axes. The heavier dark matter is less numerous at fixed ρ_χ.

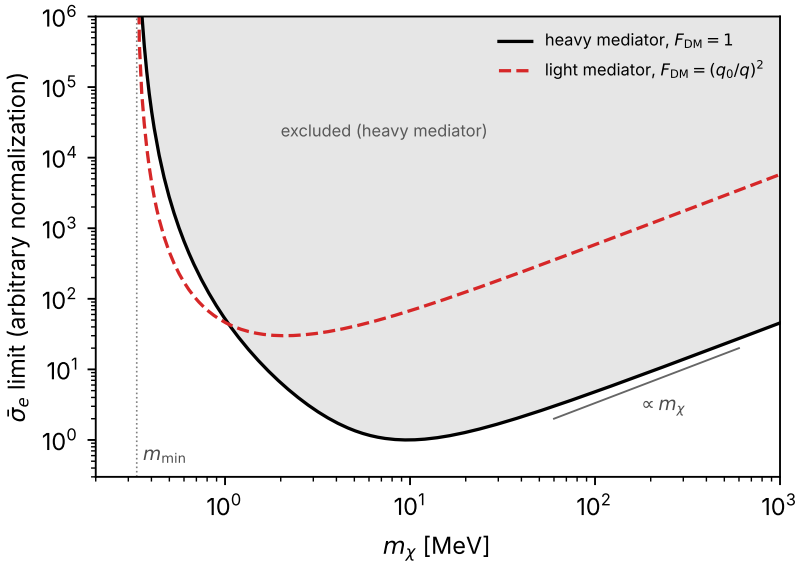


Figure 1.9: Schematic exclusion limits on $\bar{\sigma}_e$ for a silicon-like target. The threshold is the low-temperature gap $E_{\rm g} = 1.17\,\text{eV}$. The curves follow from Eq. (1.54) with the standard-halo $\eta(v_{\rm min})$ of Sect. 1.3 and a toy response: $S(q,\omega)$ flat between $E_{\rm g}$ and 30 eV, multiplied by the Gaussian cutoff $\exp[-q^2/2(2q_0)^2]$ with $q_0 = \alpha m_e = 3.73\,\text{keV}$. Each curve has an arbitrary normalization (the light-mediator curve is shifted up for legibility). Both curves rise without bound at the threshold mass $m_{\rm min} = 0.33\,\text{MeV}$ (dotted) and grow as m_χ at large mass. The heavy-mediator curve ($F_{\rm DM} = 1$) reaches its minimum at 9.7 MeV, where the allowed momenta cover the cutoff; the light-mediator curve ($F_{\rm DM} = (q_0/q)^2$), dominated by the smallest q, reaches its minimum at 2.1 MeV. The shaded region is excluded by the heavy-mediator search.

The light-mediator slope. With $F_{\mathrm{DM}} = (q_0/q)^2$ the weight $(q_0/q)^4\, q\, \mathrm{d}q$ favors the smallest momentum transfer that can deposit the threshold, $q \approx E_{\mathrm{th}}/v_{\max}$ (Sect. 1.3). That momentum transfer is reachable as soon as m_χ is a few times $m_{\min}$, and the light-mediator curve turns over close to the edge, growing as m_χ from a few MeV upward. Because the rate comes from small q, the light-mediator limit is also the one most affected by screening (Sect. 4.4) and by the low-energy end of the ionization yield.

As a result, the two curves are not related by a constant factor, and a limit quoted for one mediator cannot be rescaled to the other without the response. In the toy of Fig. 1.9 the heavy-mediator limit at 1 MeV stands a factor of 49 above its minimum at 9.7 MeV, while the light-mediator limit is already within a factor of 1.5 of its minimum at 2.1 MeV. A two-pair threshold would move both walls from 0.33 to 1.36 MeV. Real exclusion plots add curves from several experiments and the benchmark cross-sections of thermal and freeze-in models (Worked example 1.5).

1.5.5 Inputs of a limit

Every exclusion curve combines measured inputs from the detector with theory inputs, and Fig. 1.8 marks the boundary. The measured inputs belong to the detector: the exposure $M_{\mathrm{det}} t_{\mathrm{exp}}$, the threshold, the efficiency with which events pass the selection, the background and the observed count. Their errors are statistical or come from calibration, and experiments propagate them.

The theory inputs are the halo parameters, the particle-physics model and the response of the target. The halo parameters ρ_χ, v_0 and v_{esc} are fixed by convention to the values of Sect. 1.2 so that experiments can be compared (Appendix A.8); the limit scales as $1/\rho_\chi$, and v_{esc} moves the edge (Sect. 1.3.5). The particle physics enters only through F_{DM}, which is a definition of the model rather than an approximation (Sect. 1.4). The target enters through $S(q,\omega)$ and through the conversion from ω to quanta, $N_{\mathrm{eh}}(\omega)$ for charge detectors. Both are computed, not measured, in the eV range that matters for light dark matter. Their errors are not statistical: they come from approximations, such as the treatment of exchange and correlation in the band structure, the neglect of screening or of phonons, or the step model of Eq. (1.43). A longer exposure does not reduce them. Their effect on a limit follows from Eq. (1.54). A fractional error in $S(q,\omega)$ at the momentum transfers that dominate the integral moves the limit by the same fraction. An error in the gap or in the threshold energy moves the edge, $\delta \ln m_{\min} = \delta \ln E_{\mathrm{th}}$ (Sect. 1.3.5), and near the edge it changes the limit by a much larger factor, because η varies steeply there. Together these errors make up the *theory uncertainty* of a limit.

Two lines of argument start here: the signal chain and the theory uncertainty. The first follows the signal chain of Fig. 1.8 from the response of the target to the measured signal: Chaps. 2 to 7 build the response, and later chapters follow the deposit through the cascade to the sensor (Chaps. 18 and 20). The second follows the theory uncertainty: each approximation, when it is introduced, comes with an estimate of its error, and Chap. 21 turns these estimates into uncertainties on exclusion limits.

Summary

The event rate is an integral over a product of three factors, the halo, the coupling $|\tilde{V}(\boldsymbol{q})|^2$ and the dynamic structure factor $S(\boldsymbol{q},\omega)$ of the target, tied together by energy conservation (Sect. 1.1). The particle physics and the condensed-matter physics therefore separate, and most of the book computes S. Gravity measures the density of dark matter but not its mass: a local density of a few tenths of a GeV cm^{-3} (Sect. 1.2) and a cosmological density

$\Omega_c h^2 = 0.120$ from the cosmic microwave background. The mass is open over some 90 decades, and the keV–GeV window is the range where the kinetic energy of halo dark matter, $\frac{1}{2} m_\chi v^2$ with $v \approx 10^{-3}$, matches the meV–eV excitations of solids. A thermal relic needs $\langle \sigma v \rangle \approx 2 \times 10^{-26}\,\mathrm{cm^3\,s^{-1}}$ to $3 \times 10^{-26}\,\mathrm{cm^3\,s^{-1}}$, which below a few GeV requires a light mediator; freeze-in is the main alternative.

The local halo is described by the standard halo model (Sect. 1.2): a truncated Maxwell–Boltzmann distribution with $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$, $v_0 = 238\,\mathrm{km\,s^{-1}}$ and $v_\mathrm{esc} = 544\,\mathrm{km\,s^{-1}}$, seen in the laboratory through the Earth's motion. The largest laboratory speed is $v_\mathrm{max} = 794.5\,\mathrm{km\,s^{-1}}$. The Earth's orbit modulates the rate annually, with a phase that reverses at $v_\mathrm{min} \approx 211\,\mathrm{km\,s^{-1}}$, and its rotation turns a crystal through the dark-matter wind once per sidereal day.

Kinematics (Sect. 1.3) relates the momentum q and energy ω given to the target through $\omega_{\boldsymbol{q}} = \boldsymbol{q} \cdot \boldsymbol{v} - q^2/2m_\chi - \Delta_\chi$, and every scattering rate contains $\eta(v_\mathrm{min})$, the mean inverse speed above the minimum speed v_min. A threshold E_th is reachable only above $m_\mathrm{min} = 2E_\mathrm{th}/v_\mathrm{max}^2$, 0.33 MeV for the low-temperature silicon gap. Near m_min the rate vanishes quadratically with the tail of the halo. Electron transitions involve momentum transfers comparable to $q_0 = \alpha m_e = 3.73\,\mathrm{keV}$, close to the shortest reciprocal-lattice vector of silicon, and free nuclear recoils deposit little energy for sub-GeV dark matter.

A mediator of mass m_V produces the potential $\tilde{V}(q) = y_\chi y_e/(q^2 + m_V^2)$ (Sect. 1.4). Its strength is quoted as the vacuum reference cross-section $\bar{\sigma}_e$ and its momentum dependence as $F_\mathrm{DM}(q)$, equal to 1 for a heavy and $(q_0/q)^2$ for a light mediator. The dark photon is the standard example; bosonic dark matter can instead be absorbed, depositing $\omega \approx m_\phi$ at $q \approx 0$.

A detector counts electron–hole pairs (3.6 eV to 3.75 eV each in silicon), phonons, quasiparticles (about 0.36 meV per pair in bulk aluminum) or photons (Sect. 1.5). The rate is linear in $\bar{\sigma}_e$ and ρ_χ and falls as $1/m_\chi$ well above threshold. With no events and no background the 90 % upper limit is 2.3 expected events, and the cross-section limit improves in proportion to the exposure. Exclusion curves have a vertical edge at m_min and grow as m_χ at large mass. Their accuracy rests on the computed response of the target, which Chaps. 2 to 7 develop. The errors of that computation, the theory uncertainty, do not shrink with exposure, and the book estimates them with each approximation it makes.

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Chapter 2

Transition rates

About 10^8 dark-matter particles of 100 MeV cross each square centimeter of a detector every second (Worked example 1.2), each with at most 350 eV of kinetic energy (Sect. 1.3), and Eq. (1.47) turns them into an event rate that Chap. 1 quoted without derivation. Deriving it is a question for quantum mechanics: how often does a weak interaction move a kilogram of silicon, with some 10^{26} electrons, from its ground state into one of its excited states? Scattering theory for two particles in vacuum answers only part of it, because a solid is not a free electron. Its excited states are states of many identical electrons, some of them can be reached only through an intermediate state that a phonon completes, and at any finite temperature the crystal can also hand energy back to the dark matter. Bosonic dark matter raises a related question, why its absorption rate depends on the local density and not on the flux. The answers need Fermi's golden rule and four extensions of it: the Born approximation, second quantization, second-order perturbation theory and the thermal average. With them we derive the cross-sections and form factors behind the reference quantities of Sect. 1.4; Fig. 2.1 shows how each extends the golden rule.

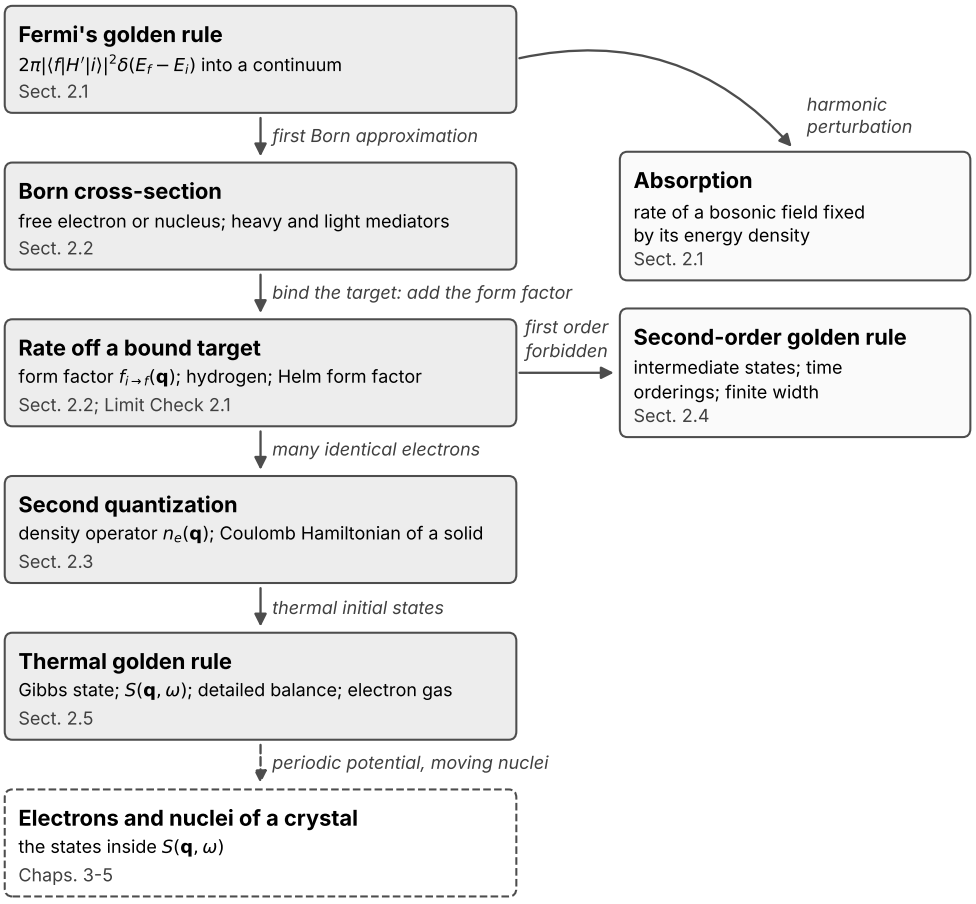


Figure 2.1: From Fermi's golden rule to the dynamic structure factor. Arrows name the step between levels: the first Born approximation gives the cross-section on a free electron or nucleus (Sects. 2.1 and 2.2); binding the target adds the transition form factor $f_{i\to f}(\boldsymbol{q})$, checked on the ionization of hydrogen (Sect. 2.2; Limit Check 2.1); identical electrons call for occupation numbers and the density operator $n_e(\boldsymbol{q})$ (Sect. 2.3); thermal initial states turn the rate into the dynamic structure factor $S(\boldsymbol{q}, \omega)$ with detailed balance (Sect. 2.5). Side branches: a harmonic perturbation gives the absorption rate of a bosonic field, set by its energy density (Sect. 2.1), and a transition forbidden at first order proceeds through an intermediate state (Sect. 2.4). The dashed box is the crystal whose states fill S (Chaps. 3 to 5).

2.1 Fermi's golden rule

A kilogram of silicon exposed for a year to 100 MeV dark matter with $\bar{\sigma}_e = 10^{-40}\,\mathrm{cm}^2$ expects 73 events in the toy rate of Worked example 1.6, and twice as many in two years: the count is a rate times an exposure (Eq. (1.45)). Quantum mechanics gives probabilities, not rates. A probability that grows linearly in time appears only when a weak perturbation couples an initial state to a continuum of final states, and only within a window of times. To see how a rate emerges, and when the idea stops making sense, we derive Fermi's golden rule; along the way we fix the normalization of states that every later rate uses. The derivation follows Refs. [1, Chap. 5] and [2, Chap. XIII].

2.1.1 First-order amplitude

The target, together with a dark-matter particle far from it, evolves under an unperturbed Hamiltonian H_0 with eigenstates $|n\rangle$ and energies E_n. At $t = 0$ a perturbation $H'(t)$ is switched on, and the system starts in the eigenstate $|i\rangle$. We expand the state in the unperturbed basis, with the free phases taken out:

$$|\psi(t)\rangle = \sum_n c_n(t)\, \mathrm{e}^{-\mathrm{i}E_n t}\, |n\rangle, \qquad c_n(0) = \delta_{ni}.$$

Inserting this into $\mathrm{i}\,\partial_t|\psi\rangle = (H_0 + H')|\psi\rangle$, the time derivative of each phase cancels the action of H_0. Projecting onto $\mathrm{e}^{-\mathrm{i}E_f t}|f\rangle$ leaves an exact set of equations for the amplitudes,

$$\mathrm{i}\,\dot{c}_f(t) = \sum_n \langle f|H'(t)|n\rangle\, \mathrm{e}^{\mathrm{i}(E_f - E_n)t}\, c_n(t). \tag{2.1}$$

To first order in H' we replace $c_n(t)$ on the right by its initial value δ_{ni} and integrate. For $f \neq i$,

$$c_f(t) = -\mathrm{i}\int_0^t \mathrm{d}t'\, \langle f|H'(t')|i\rangle\, \mathrm{e}^{\mathrm{i}(E_f - E_i)t'}, \tag{2.2}$$

and the transition probability is $P_{i\to f}(t) = |c_f(t)|^2$. The replacement is consistent only while the initial state is barely depleted, $\sum_{f\neq i} P_{i\to f}(t) \ll 1$. Eq. (2.2) is the first term of the Dyson series of the interaction picture; Sect. 2.4 takes the second.

2.1.2 Constant perturbation

Let H' be independent of time for $t > 0$. This is the case of dark-matter scattering: the interaction H_{int} does not depend on time, and energy passes between the dark matter and the target, whose kinetic and internal energies are both in E_i and E_f. With $\xi = E_f - E_i$ the integral in Eq. (2.2) is elementary, $c_f(t) = -\langle f|H'|i\rangle(\mathrm{e}^{\mathrm{i}\xi t} - 1)/\xi$. Since $|\mathrm{e}^{\mathrm{i}\xi t} - 1|^2 = 2 - 2\cos\xi t = 4\sin^2(\xi t/2)$,

$$P_{i\to f}(t) = 4\,|\langle f|H'|i\rangle|^2\, \frac{\sin^2(\xi t/2)}{\xi^2}. \tag{2.3}$$

All the time dependence sits in the function $\sin^2(\xi t/2)/\xi^2$, drawn in Fig. 2.2(a). At $\xi = 0$ it equals $t^2/4$. Its zeros are at $\xi = 2\pi k/t$ with k a nonzero integer, so the central peak has width $4\pi/t$. The substitution $u = \xi t/2$ gives its integral,

$$\int_{-\infty}^{\infty} \mathrm{d}\xi\, \frac{\sin^2(\xi t/2)}{\xi^2} = \frac{t}{2}\int_{-\infty}^{\infty} \mathrm{d}u\, \frac{\sin^2 u}{u^2} = \frac{\pi t}{2},$$

with $\int \mathrm{d}u\, \sin^2 u/u^2 = \pi$ (Appendix C.1). The height grows as t^2, the width shrinks as $1/t$, and the area grows as t; the central peak holds 90.3 % of it. Integrated against any function $w(\xi)$ that varies little over a range $2\pi/t$, the function returns $(\pi t/2)\, w(0)$, with a relative error $\sim 1/(t\,\Delta_\mathrm{w})$, where Δ_w is the range over which w varies. In the sense of distributions,

$$\frac{\sin^2(\xi t/2)}{\xi^2} \;\longrightarrow\; \frac{\pi t}{2}\,\delta(\xi) \qquad (t \to \infty), \tag{2.4}$$

which Appendix C.1 proves. Applied to Eq. (2.3), it gives a probability that grows linearly in time, $P_{i\to f}(t) \to 2\pi t\,|\langle f|H'|i\rangle|^2\,\delta(E_f - E_i)$. The same limit can be taken after differentiating. The time derivative of Eq. (2.3) is $2|\langle f|H'|i\rangle|^2 \sin(\xi t)/\xi$, a function of ξ whose area is π at every $t > 0$ and whose width shrinks as $1/t$; it tends to $\pi\delta(\xi)$. Since the area stays fixed while the width shrinks, the probability per unit time stops depending on t, and that is what we mean by a rate. The transition rate is

Fermi's golden rule

$$\Gamma_{i\to f} = 2\pi\,|\langle f|H'|i\rangle|^2\,\delta(E_f - E_i). \tag{2.5}$$

The rate is defined only after summation over a continuum of final states f, for times in the window of Eq. (2.7).

In natural units the rate is an energy: $|\langle f|H'|i\rangle|^2$ carries eV^2 and the delta function eV^{-1}. Dividing by $\hbar$ converts it to a frequency: a rate of 10 meV is $1.52 \times 10^{13}\,\mathrm{s}^{-1}$. The delta function is energy conservation, enforced to within about $2\pi/t$ after a time t. For a dark-matter particle it becomes $\delta(\omega - \omega_{\boldsymbol{q}})$ of Sect. 1.3 once the dark-matter part of $E_f - E_i$ is written out (Sect. 2.2).

The sudden switch-on at $t = 0$ adds a transient but does not change the rate. A perturbation turned on smoothly over a time τ_s replaces $\sin^2(\xi t/2)/\xi^2$ by a function whose tails fall off faster in ξ. Its area still grows as $\pi t/2$ once $t \gg \tau_\mathrm{s}$, because the area counts only the time during which the perturbation acts at full strength. The rate is the slope of the probability at long times, and the slope is the same for every switching.

2.1.3 Conditions for a rate

For a single discrete final state, however, Eq. (2.3) does not grow linearly. Off resonance it oscillates below $4|\langle f|H'|i\rangle|^2/\xi^2$; exactly on resonance it grows as t^2 until first order fails. A rate requires a continuum. Let $\mathcal{D}_f(E)\,\mathrm{d}E$ be the number of final states with energies in $[E, E + \mathrm{d}E]$, the *density of final states*, and let $\overline{|\langle f|H'|i\rangle|^2}(E)$ be the average of the squared matrix element over those states. The total transition probability is then

$$\sum_f P_{i\to f}(t) = \int \mathrm{d}E\, \mathcal{D}_f(E)\; \overline{|\langle f|H'|i\rangle|^2}(E)\; \frac{4\sin^2[(E - E_i)t/2]}{(E - E_i)^2}.$$

By Eq. (2.4) it becomes $\Gamma_i t$, with the total rate

$$\Gamma_i = 2\pi\, \overline{|\langle f|H'|i\rangle|^2}(E_i)\; \mathcal{D}_f(E_i). \tag{2.6}$$

The catch is that t has to sit in a window. It must be long compared with the inverse energy scale of the continuum, so that energy is resolved, and short compared with the inverse level spacing, so that the levels still look continuous. It must also be short enough that the initial state is barely depleted, so that first order holds. To make the first precise, let E_w be the

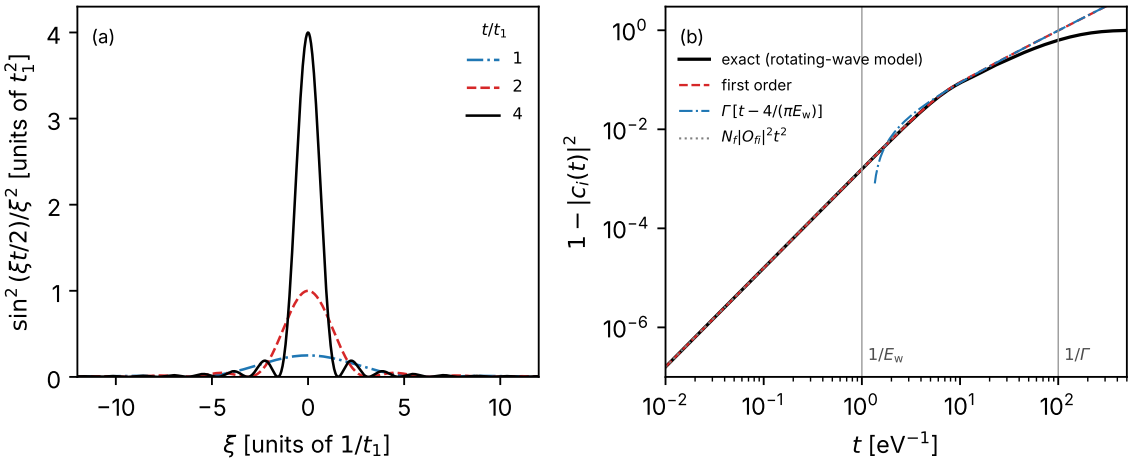


Figure 2.2: (a) The function $\sin^2(\xi t/2)/\xi^2$ of the first-order transition probability, Eq. (2.3), against the energy mismatch $\xi = E_f - E_i$, for times $t = t_1$, $2t_1$ and $4t_1$, where t_1 is an arbitrary unit of time ($\hbar = 1$). The peak height grows as t^2, the width as $1/t$, and the area, $\pi t/2$, as t. (b) Depletion $1 - |c_i(t)|^2$ of a discrete state coupled to a flat continuum by the absorption term of a harmonic perturbation, for the model of Worked example 2.1. It has 2000 final states spread uniformly over a width $E_w = 1\,\text{eV}$ centered on resonance, all with the same matrix element, chosen so that the golden-rule rate is $\Gamma = 10\,\text{meV}$. Solid: exact solution of this model with the counter-rotating (emission) term dropped, by diagonalization of the 2001-level Hamiltonian in the frame rotating with the perturbation. Dashed: first-order probability, Eq. (2.9). Dash-dotted: its linear long-time form. Dotted: its quadratic short-time form. Vertical lines mark $t = 1/E_w$ (0.66 fs) and $t = 1/\Gamma$ (66 fs).

energy scale over which the product $\mathcal{D}_f\,\overline{|\langle f|H'|i\rangle|^2}$ changes appreciably: the width of the continuum, or of a structure within it. The delta-function limit needs the peak to be narrow on that scale, $t \gg 1/E_{\mathrm{w}}$. The sum over discrete levels becomes an integral only when the peak covers many levels, whose spacing is $1/\mathcal{D}_f$; this requires $t \ll 2\pi\mathcal{D}_f$. First-order perturbation theory requires $\Gamma_i t \ll 1$. Together,

$$\frac{1}{E_{\mathrm{w}}} \ll t \ll \min\!\left(\frac{1}{\Gamma_i},\, 2\pi\mathcal{D}_f\right). \tag{2.7}$$

In dark-matter scattering, however, the continuum is always present, even when the target has discrete levels, such as an atom below its ionization threshold. The final state contains the scattered dark-matter particle, whose momentum ranges over a continuum, and $\mathcal{D}_f$ counts target levels and dark-matter momenta together. Energy conservation then selects, for each target level, a shell of outgoing momenta, and the delta function of Eq. (2.5) is integrated over that shell rather than over the target spectrum (Sect. 2.2).

The upper bound $2\pi\mathcal{D}_f$ grows with the volume of the system, because $\mathcal{D}_f$ does (Eq. (2.11)); it drops out when the infinite-volume limit is taken before the long-time limit, as we do throughout. The window then exists whenever $\Gamma_i \ll E_{\mathrm{w}}$: the transition must be slow compared with the inverse energy scale of the continuum.

At times beyond $1/\Gamma_i$ the initial state is depleted. The exact evolution then becomes exponential decay at the same rate Γ_i (Wigner–Weisskopf; Sect. 2.4), and Γ_i keeps its meaning beyond the first-order window. For a dark-matter particle crossing a detector the condition $\Gamma_i t \ll 1$ holds by a wide margin, because the probability that it scatters at all during its passage is the small number that makes the searches of Sect. 1.5 difficult.

The golden rule also fails when the first-order matrix element vanishes and when the perturbation shifts the energy of the initial state. If $\langle f|H'|i\rangle$ vanishes for every final state at energy E_i, by a selection rule or because no such state exists, Eq. (2.6) vanishes, and the leading rate comes from the second-order amplitude through intermediate states (Sect. 2.4). If the perturbation shifts the energy of $|i\rangle$ itself, by $\langle i|H'|i\rangle \neq 0$, the delta function must be evaluated at the shifted energy; the shift is of first order, while the rate is of second order in H'. For dark-matter scattering the diagonal element is the $\boldsymbol{q} = 0$ term of Eq. (2.12), $\tilde{V}(\mathbf{0})/\mathcal{V}$, which for a massive mediator vanishes as the volume grows, and the shift plays no role. For a massless mediator $\tilde{V}(\mathbf{0})$ itself diverges; in a neutral target the charges cancel this term, as Sect. 2.3 shows for the Coulomb interaction.

2.1.4 Harmonic perturbations

An absorbed bosonic dark-matter field oscillates in time at its mass (Sect. 1.4), and an applied field oscillates at its frequency. Such a perturbation is harmonic,

$$H'(t) = \hat{O}\,\mathrm{e}^{-\mathrm{i}\omega t} + \hat{O}^\dagger\,\mathrm{e}^{\mathrm{i}\omega t}, \qquad \omega > 0,$$

where $\hat{O}$ is an operator on the target and the second term makes H' Hermitian. Eq. (2.2) now has two terms of the static form, with ξ shifted by $\mp\omega$:

$$c_f(t) = -\langle f|\hat{O}|i\rangle\,\frac{\mathrm{e}^{\mathrm{i}(\xi-\omega)t}-1}{\xi-\omega} - \langle f|\hat{O}^\dagger|i\rangle\,\frac{\mathrm{e}^{\mathrm{i}(\xi+\omega)t}-1}{\xi+\omega}.$$

The first term peaks at $\xi = \omega$, the second at $\xi = -\omega$, each with width of order $2\pi/t$. For $\omega t \gg 1$ the peaks do not overlap. The cross term in $|c_f|^2$ is then bounded in time instead of growing, and each squared term turns into a delta function by Eq. (2.4):

$$\Gamma_{i\to f} = 2\pi\,|\langle f|\hat{O}|i\rangle|^2\,\delta(E_f - E_i - \omega) + 2\pi\,|\langle f|\hat{O}^\dagger|i\rangle|^2\,\delta(E_f - E_i + \omega). \tag{2.8}$$

The first term is absorption: the target gains the energy ω. The second is emission: the target gives ω to the field, so the energy given to the target is $-\omega$. With the temporal Fourier convention of Appendix A.3, $f(\omega) = \int \mathrm{d}t\, \mathrm{e}^{+\mathrm{i}\omega t} f(t)$, the factor $\mathrm{e}^{-\mathrm{i}\omega t}$ is the component of H' at positive frequency, and positive frequency means energy given to the target. Since $|\langle f|\hat{O}^\dagger|i\rangle| = |\langle i|\hat{O}|f\rangle|$, emission from i to f and absorption from f to i have the same matrix element; Sect. 2.5 combines this with thermal populations into detailed balance.

For a classical field the emission term is stimulated emission, driven by the field itself. When the field is a quantized mode, the dark-matter or phonon state is part of $|i\rangle$ and $|f\rangle$, the perturbation is static, and Eq. (2.5) applies to the combined system. Absorption removes a quantum and emission adds one, with matrix elements proportional to $\sqrt{n}$ and $\sqrt{n+1}$ for a mode holding n quanta (Sects. 2.3 and 2.5). The extra 1 is spontaneous emission, which a classical field misses. At $\omega = 0$ the two peaks coincide, the cross term no longer averages out, and Eq. (2.8) reduces to Eq. (2.5) with $H' = \hat{O} + \hat{O}^\dagger$.

The harmonic form explains why the absorption rates of Sect. 1.4 depend on the local density of dark matter and not on its flux. A bosonic field of mass m_ϕ that makes up the local dark matter oscillates as $\phi(t) = \phi_0 \cos(m_\phi t)$, with energy density $\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}m_\phi^2\phi^2 = \frac{1}{2}m_\phi^2\phi_0^2 = \rho_\chi$. Its spatial phase $\mathrm{e}^{\mathrm{i}\boldsymbol{p}_\phi\cdot\boldsymbol{x}}$ changes over a length $1/|\boldsymbol{p}_\phi| \approx 1/(m_\phi v)$, a thousand times the Compton wavelength $1/m_\phi$, and we neglect it here. An interaction linear in the field, $H' = \phi(t)$ times a target operator, is harmonic with $\omega = m_\phi$, and $\hat{O}$ is $\phi_0/2$ times that operator. The absorption term of Eq. (2.8) is therefore proportional to $\phi_0^2 = 2\rho_\chi/m_\phi^2$: the rate is fixed by the energy density of the field, and the speed of the dark matter does not enter. Chap. 8 evaluates the target operators for each boson.

Worked example 2.1: A harmonic perturbation coupled to a flat continuum

A discrete state $|i\rangle$ is coupled by the absorption term $\hat{O}\,\mathrm{e}^{-\mathrm{i}\omega t}$ to N_f final states spread uniformly over an energy width E_w centered on the resonance $E_f = E_i + \omega$. The density of final states is $\mathcal{D}_f = N_f/E_\mathrm{w}$, and every final state has the same $|\langle f|\hat{O}|i\rangle|^2 \equiv |O_{fi}|^2$. With $\xi_f = E_f - E_i - \omega$, Eq. (2.3) with $H' \to \hat{O}$ and $\xi \to \xi_f$, summed over f, gives the first-order depletion. For $t \ll 2\pi\mathcal{D}_f$ the sum is an integral, and the substitution $u = \xi t/2$ gives

$$P(t) = 4|O_{fi}|^2\mathcal{D}_f \int_{-E_\mathrm{w}/2}^{E_\mathrm{w}/2} \mathrm{d}\xi\, \frac{\sin^2(\xi t/2)}{\xi^2} = 4|O_{fi}|^2\mathcal{D}_f\, t \int_0^{E_\mathrm{w}t/4} \mathrm{d}u\, \frac{\sin^2 u}{u^2}.$$

Integration by parts, with $\mathrm{d}(-\sin^2 u/u)/\mathrm{d}u = \sin^2 u/u^2 - \sin 2u/u$, gives $\int_0^U \mathrm{d}u\, \sin^2 u/u^2 = \mathrm{Si}(2U) - \sin^2 U/U$, where $\mathrm{Si}(z) = \int_0^z \mathrm{d}s\, \sin s/s$ is the sine integral. Hence

$$P(t) = 4|O_{fi}|^2\mathcal{D}_f\, t \left[\mathrm{Si}\left(\frac{E_\mathrm{w}t}{2}\right) - \frac{\sin^2(E_\mathrm{w}t/4)}{E_\mathrm{w}t/4}\right]. \tag{2.9}$$

At short times, $E_\mathrm{w}t \ll 1$, the expansions $\mathrm{Si}(z) = z - z^3/18 + \ldots$ and $\sin^2 U/U = U - U^3/3 + \ldots$ make the bracket $E_\mathrm{w}t/4$, so $P(t) = N_f|O_{fi}|^2t^2$. Every final state is populated, as if each were resonant: the energy is not yet resolved. At long times, $E_\mathrm{w}t \gg 1$, the asymptotic form $\mathrm{Si}(z) = \pi/2 - \cos z/z + O(z^{-2})$ and the identity $2\cos 2U + 4\sin^2 U = 2$ make the bracket $\pi/2 - 2/(E_\mathrm{w}t)$, so

$$P(t) = \Gamma\left(t - \frac{4}{\pi E_\mathrm{w}}\right), \qquad \Gamma = 2\pi|O_{fi}|^2\mathcal{D}_f,$$

up to terms that decay as $1/t$. We see that the probability grows linearly at the golden-rule rate Eq. (2.6), delayed by $4/(\pi E_\mathrm{w})$, the time the system needs to resolve the energy.

For numbers we take $E_\mathrm{w} = 1\,\mathrm{eV}$, $N_f = 2000$ (level spacing $0.5\,\mathrm{meV}$, $\mathcal{D}_f = 2000\,\mathrm{eV}^{-1}$) and $|O_{fi}| = 0.892\,\mathrm{meV}$, for which $\Gamma = 10\,\mathrm{meV}$. With $\hbar = 6.582 \times 10^{-16}\,\mathrm{eV\,s}$, the time scales are $1/E_\mathrm{w} = 0.66\,\mathrm{fs}$, the delay $4/(\pi E_\mathrm{w}) = 0.84\,\mathrm{fs}$, $1/\Gamma = 66\,\mathrm{fs}$ and $2\pi\mathcal{D}_f = 8.3\,\mathrm{ps}$, so the window Eq. (2.7) spans two orders of magnitude, from 0.66 fs to 66 fs. Fig. 2.2(b) compares Eq. (2.9) with the exact evolution of the same model, obtained by solving Eq. (2.1) with the emission term dropped. The substitution $\tilde{c}_f = c_f\,\mathrm{e}^{\mathrm{i}\xi_f t}$ (and $\tilde{c}_i = c_i$) removes every time-dependent phase from Eq. (2.1), since the absorption term carries $\mathrm{e}^{-\mathrm{i}\omega t}\mathrm{e}^{\mathrm{i}(E_f - E_i)t} = \mathrm{e}^{\mathrm{i}\xi_f t}$; in this frame, which rotates with the perturbation, the coefficients are constant and the model is a time-independent Hamiltonian of 2001 levels, which we diagonalize numerically. First order exceeds the exact depletion by 10 % at $t = 21.9\,\mathrm{eV}^{-1}$, where $\Gamma t = 0.22$; at $t = 1/\Gamma$ it predicts 0.99 against the exact 0.63. First order therefore breaks down at $\Gamma t \approx 0.2$. Beyond it the exact depletion follows $1 - \mathrm{e}^{-\Gamma t}$, within 0.4 % from $t = 1/\Gamma$ on, which is the exponential decay of Sect. 2.4 with the golden-rule rate as its decay constant.

2.1.5 Box normalization and final-state sums

With plane waves normalized in infinite volume, $\langle \boldsymbol{p}|\boldsymbol{p}'\rangle = (2\pi)^3\delta^3(\boldsymbol{p}-\boldsymbol{p}')$, momentum conservation in a matrix element is a delta function, and Eq. (2.5) squares it. The square $[\delta^3(\boldsymbol{P}_f - \boldsymbol{P}_i)]^2$ has no meaning on its own. A finite box makes every step well defined: momenta are discrete, momentum conservation is a Kronecker delta, whose square is itself, and the golden rule needs countable states to sum over. We therefore place the system in a cube of volume $\mathcal{V} = L^3$ with periodic boundary conditions. The plane waves $\langle \boldsymbol{x}|\boldsymbol{p}\rangle = \mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{x}}/\sqrt{\mathcal{V}}$, with $\boldsymbol{p} = 2\pi\boldsymbol{n}/L$ and $\boldsymbol{n}$ a vector of integers, obey $\langle \boldsymbol{p}|\boldsymbol{p}'\rangle = \delta_{\boldsymbol{p}\boldsymbol{p}'}$: each state holds one particle in the volume $\mathcal{V}$. The script $\mathcal{V}$ always denotes this volume; an italic V is a potential and always carries its argument, $V(\boldsymbol{x})$, or a tilde, $\tilde{V}(\boldsymbol{q})$. Since each allowed momentum occupies a cell of volume $(2\pi)^3/\mathcal{V}$, for any function that varies little over $2\pi/L$ [3, Eqs. (A1) and (A2)]

$$\sum_{\boldsymbol{p}} w(\boldsymbol{p}) \;\to\; \mathcal{V}\int \frac{\mathrm{d}^3p}{(2\pi)^3}\, w(\boldsymbol{p}), \qquad \delta_{\boldsymbol{p}\boldsymbol{p}'} \;\to\; \frac{(2\pi)^3}{\mathcal{V}}\,\delta^3(\boldsymbol{p}-\boldsymbol{p}'). \tag{2.10}$$

Appendix A.3 collects these conventions and Appendix C.2 the continuum limit of such sums, including sums over the Brillouin zone. For one free particle of mass m with energy $E = p^2/2m$, the number of states in a shell is $\mathcal{V}\,4\pi p^2\,\mathrm{d}p/(2\pi)^3$, and with $\mathrm{d}E = p\,\mathrm{d}p/m$ we find

$$\mathcal{D}_f(E) = \frac{\mathcal{V}\,m\,p}{2\pi^2}, \qquad p = (2mE)^{1/2}. \tag{2.11}$$

The density of final states is proportional to $\mathcal{V}$. In a crystal the box is a block of N unit cells, and the same counting applies to crystal momenta (Sect. 3.1). Discrete quantum numbers, such as spin, are summed separately in the final state. When the initial state is not known exactly, the rate is averaged over it: over the two spin states of an unpolarized particle, and over a thermal ensemble of target states (Sect. 2.5).

The volume must drop out of every observable, and in a finite box we can follow it through every rate as a check. Consider a dark-matter particle and an electron, both box-normalized, interacting through a potential $V(\boldsymbol{x}_\chi - \boldsymbol{x}_e)$ of their separation (Sect. 1.4). Writing $\boldsymbol{r} = \boldsymbol{x}_\chi - \boldsymbol{x}_e$ and integrating over $\boldsymbol{x}_e$ at fixed $\boldsymbol{r}$,

$$\langle \boldsymbol{p}'_\chi, \boldsymbol{p}'_e|V(\boldsymbol{x}_\chi - \boldsymbol{x}_e)|\boldsymbol{p}_\chi, \boldsymbol{p}_e\rangle = \frac{1}{\mathcal{V}^2}\int \mathrm{d}^3x_e\,\mathrm{e}^{-\mathrm{i}(\boldsymbol{P}_f - \boldsymbol{P}_i)\cdot\boldsymbol{x}_e}\int \mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,V(\boldsymbol{r}) = \frac{\tilde{V}(\boldsymbol{q})}{\mathcal{V}}\,\delta_{\boldsymbol{P}_f\boldsymbol{P}_i}, \tag{2.12}$$

with $\boldsymbol{q} = \boldsymbol{p}_\chi - \boldsymbol{p}'_\chi$ the momentum transferred to the electron, $\boldsymbol{P}_{i,f}$ the initial and final total momenta, and $\tilde{V}(\boldsymbol{q})$ the transform of Appendix A.3. Each plane wave brings $\mathscr{V}^{-1/2}$ and the integral over the position of the pair brings $\mathscr{V}$ times momentum conservation. In Eq. (2.5) the squared matrix element brings $1/\mathscr{V}^2$, and the sum over one final momentum, the other being fixed by $\delta_{\boldsymbol{P}_f \boldsymbol{P}_i}$, brings $\mathscr{V}$. We see that the rate is proportional to $1/\mathscr{V}$: it is the rate at which two particles placed in the same box find each other.

A cross-section removes this factor. Since the incident particle has density $1/\mathscr{V}$ and speed v relative to the target, its flux is $v/\mathscr{V}$ (Eq. (1.18), with a single velocity), and the rate per target particle is that flux times the cross-section σ_T for scattering on one target particle. Hence

$$\sigma_T = \frac{\mathscr{V}}{v}\,\Gamma = \frac{\mathscr{V}}{v}\sum_f 2\pi\,|\langle f|H'|i\rangle|^2\,\delta(E_f - E_i), \tag{2.13}$$

which is independent of $\mathscr{V}$. The rate in a detector follows by reversing the argument. Dark matter with velocity in d^3v crosses a target particle with flux $n_\chi f_\chi(\boldsymbol{v})\,\mathrm{d}^3v\,v$ (Sect. 1.2). Integrating over velocities and multiplying by the number of target particles per unit mass, $1/m_T$ with m_T the mass of one, leads to an event rate per unit mass $(\rho_\chi/m_\chi m_T)\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\, v\,\sigma_T(v)$, the structure that Sect. 1.5 assembled for a target with a response. Sect. 2.2 evaluates the cross-section with Eq. (2.12), for a free target and for a bound one.

For readers comparing with particle-physics papers, we also translate to relativistic normalization, $\langle \boldsymbol{p}|\boldsymbol{p}'\rangle_{\mathrm{rel}} = 2E_{\boldsymbol{p}}(2\pi)^3\delta^3(\boldsymbol{p}-\boldsymbol{p}')$ [4, Chap. 2]; it is used again only to compare with the literature (Sect. 2.4 and Appendix A.9). In the box, $(2\pi)^3\delta^3(\boldsymbol{0}) \to \mathscr{V}$ by Eq. (2.10), and $\langle \boldsymbol{p}|\boldsymbol{p}\rangle_{\mathrm{rel}} = 2E_{\boldsymbol{p}}\mathscr{V}$: the state holds $2E_{\boldsymbol{p}}$ particles per unit volume, and $|\boldsymbol{p}\rangle_{\mathrm{rel}} = (2E_{\boldsymbol{p}}\mathscr{V})^{1/2}|\boldsymbol{p}\rangle$. At first order the invariant amplitude $\mathcal{M}$ of a process $1+2\to 3+4$ is defined by $\langle f|H'|i\rangle_{\mathrm{rel}} = -\mathcal{M}\,(2\pi)^3\delta^3(\boldsymbol{P}_f - \boldsymbol{P}_i)$, the first-order term of $\langle f|\mathrm{i}\hat{T}|i\rangle = \mathrm{i}(2\pi)^4\delta^4(P_f - P_i)\,\mathcal{M}$ once the time integral has produced $2\pi\delta(E_f - E_i)$ [4, Chap. 4]. Dividing by the four normalization factors and using $(2\pi)^3\delta^3 \to \mathscr{V}\,\delta_{\boldsymbol{P}_f\boldsymbol{P}_i}$ gives the box matrix element $\langle f|H'|i\rangle = -\mathcal{M}\,\delta_{\boldsymbol{P}_f\boldsymbol{P}_i}/[\mathscr{V}\prod_{j=1}^4(2E_j)^{1/2}]$. Inserted into Eq. (2.13), with the sum over $\boldsymbol{p}_3$ turned into an integral,

$$\sigma_T = \frac{1}{2E_1\,2E_2\,v}\int \frac{\mathrm{d}^3p_3}{(2\pi)^3\,2E_3}\,\frac{2\pi}{2E_4}\,|\mathcal{M}|^2\,\delta(E_f - E_i),$$

the relativistic formula with its flux factor and two-body phase space [4, Chap. 4]. The factors $2E$ that the relativistic normalization puts into the states reappear as $1/2E$ in the flux and the phase space. We use box normalization throughout, because a crystal is not Lorentz invariant and its states are naturally normalized to one per crystal volume. In the nonrelativistic limit $2E_j \to 2m_j$, and for elastic dark-matter–electron scattering comparison with Eq. (2.12) leads to $|\mathcal{M}|^2 = 16m_\chi^2 m_e^2|\tilde{V}(\boldsymbol{q})|^2$. The definition of $\bar{\sigma}_e$ in Eq. (1.33) then takes the form $\mu_{\chi e}^2|\mathcal{M}(q_0)|^2/(16\pi m_\chi^2 m_e^2)$ used in the particle-physics literature [5, Eq. (3)].

2.2 Born approximation and form factors

A dark-matter particle of 100 MeV that meets a free electron scatters with the cross-section $\mu_{\chi e}^2|\tilde{V}|^2/\pi$ when the mediator is heavy. Sect. 1.4 defined $\bar{\sigma}_e$ through this cross-section. The same golden-rule calculation yields the rate off an electron bound in an atom, where the electron's wavefunction enters through a single overlap, the transition form factor, and the rate off a nucleus, whose nucleons scatter in phase and multiply the cross-section by A^2. We derive all three from Eqs. (2.5) and (2.12), to first order in the potential: the first Born approximation.

2.2.1 The Born cross-section

Take a dark-matter particle and an electron in a box, interacting through a potential $V(\boldsymbol{x}_\chi - \boldsymbol{x}_e)$ of their separation. Since the total momentum is conserved, it is simplest to work in the center-of-mass frame, where the dark matter has momentum $\boldsymbol{p}$, the electron $-\boldsymbol{p}$, and the relative speed is $v = p/\mu_{\chi e}$. The kinetic energy of relative motion is $p^2/2\mu_{\chi e}$. In general the Hamiltonian splits as $P_{\rm tot}^2/2(m_\chi + m_e) + p_{\rm rel}^2/2\mu_{\chi e} + V(\boldsymbol{r})$, with $\boldsymbol{P}_{\rm tot}$ the total momentum, $\boldsymbol{p}_{\rm rel}$ the relative momentum and $\boldsymbol{r}$ the separation. The potential acts on the relative motion alone, which is a single particle of mass $\mu_{\chi e}$; the center of mass moves freely and drops out of the cross-section. A transition to $\boldsymbol{p}' = \boldsymbol{p} - \boldsymbol{q}$ has the box matrix element $\tilde{V}(\boldsymbol{q})/\mathscr{V}$ of Eq. (2.12), with $\boldsymbol{q}$ the momentum transferred to the electron. The potential does not act on spin, and the spins need no sum. Inserting this into the cross-section of Eq. (2.13) and summing over the one free final momentum,

$$\sigma_T = \frac{\mathscr{V}}{v} \sum_{\boldsymbol{p}'} 2\pi \, \frac{|\tilde{V}(\boldsymbol{q})|^2}{\mathscr{V}^2} \, \delta\!\left(\frac{p'^2}{2\mu_{\chi e}} - \frac{p^2}{2\mu_{\chi e}} \right) = \frac{1}{v} \int \frac{p'^2 \, \mathrm{d}p' \, \mathrm{d}\Omega_{\boldsymbol{q}}}{(2\pi)^3} \, 2\pi \, |\tilde{V}(\boldsymbol{q})|^2 \, \frac{\mu_{\chi e}}{p} \, \delta(p' - p),$$

where we used Eq. (2.10) for the sum and $\delta(p'^2/2\mu_{\chi e} - p^2/2\mu_{\chi e}) = (\mu_{\chi e}/p)\,\delta(p' - p)$, the rule $\delta(g(x)) = \delta(x - x_0)/|g'(x_0)|$ for a simple root x_0 (Appendix C.1). Here $\mathrm{d}\Omega_{\boldsymbol{q}}$ is the element of solid angle of the scattered momentum $\boldsymbol{p}'$ (the literature writes $\mathrm{d}\Omega$). The volume has dropped out, as it must. Doing the radial integral and using $p = \mu_{\chi e} v$,

Born cross-section

$$\frac{\mathrm{d}\sigma_T}{\mathrm{d}\Omega_{\boldsymbol{q}}} = \left(\frac{\mu_{\chi e}}{2\pi} \right)^2 |\tilde{V}(\boldsymbol{q})|^2, \qquad |\boldsymbol{q}| = 2p \sin(\theta_{\rm s}/2), \tag{2.14}$$

with $\theta_{\rm s}$ the scattering angle between $\boldsymbol{p}$ and $\boldsymbol{p}'$. For dark-matter–nucleus scattering $\mu_{\chi e}$ becomes $\mu_{\chi N}$.

This is the textbook first Born approximation [1, Chap. 6]: the scattering amplitude is $-(\mu_{\chi e}/2\pi)\tilde{V}(\boldsymbol{q})$, and its square is the differential cross-section. The minus sign makes the amplitude positive for an attractive potential, and it drops out of every rate. The momentum transfer $\boldsymbol{q} = \boldsymbol{p} - \boldsymbol{p}'$ is the same in the laboratory frame, because a Galilean boost shifts $\boldsymbol{p}$ and $\boldsymbol{p}'$ by the same amount (Sect. 1.2).

A quick check is Coulomb scattering. For $m_V \to 0$ and couplings $y_\chi y_e \to e^2 = 4\pi\alpha$, Eq. (1.32) becomes $\tilde{V} = 4\pi\alpha/q^2$, and Eq. (2.14) becomes $\mathrm{d}\sigma_T/\mathrm{d}\Omega_{\boldsymbol{q}} = 4\mu_{\chi e}^2\alpha^2/q^4 = \alpha^2/[4\mu_{\chi e}^2 v^4 \sin^4(\theta_{\rm s}/2)]$, Rutherford's formula. Its divergence at small angle, $\theta_{\rm s}^{-4}$, is the divergence of $\bar{\sigma}_e$ for a light mediator noted in Sect. 1.4.

The approximation keeps one power of $V(\boldsymbol{x})$ in the amplitude. The next term, of second order (Sect. 2.4), is small when the potential barely distorts the incoming wave over its range. For a Yukawa potential of range $1/m_V$ and strength $y_\chi y_e$ (Eq. (1.31)) this needs, up to $O(1)$ factors, $y_\chi y_e \, \mu_{\chi e}/(4\pi m_V) \ll 1$ at low energy and $y_\chi y_e/(4\pi v) \ll 1$ when the de Broglie wavelength is shorter than the range [1, Chap. 6]. The second condition is the familiar $\alpha/v \ll 1$ of Coulomb scattering. For the heavy dark photon of Worked example 1.5, $y_\chi y_e = y_\chi \zeta e = 9 \times 10^{-6}$, and both parameters are below 10^{-3} at $v = 10^{-3}$. Couplings of dark matter to ordinary matter are always this weak, because larger ones are excluded (Sect. 1.4), and first order is sufficient for our purposes. The exception is the dark sector itself, where self-interactions can be strong; it does not enter a detector. Screening in a target is not a higher Born term either. It comes from the target's electrons rearranging in response to the potential, at first order in $V(\boldsymbol{x})$ but to all orders in the Coulomb interaction among themselves, and it belongs to the target response (Sect. 4.4).

The rate needs the distribution in q rather than in angle. From $q^2 = 2p^2(1-\cos\theta_\mathrm{s})$, the solid angle is $\mathrm{d}\Omega_{\boldsymbol{q}} = 2\pi\,\mathrm{d}\cos\theta_\mathrm{s} = \pi\,\mathrm{d}q^2/p^2 = 2\pi q\,\mathrm{d}q/(\mu_{\chi e}^2 v^2)$ after the trivial azimuthal integral. Eq. (2.14) then gives

$$\frac{\mathrm{d}\sigma_T}{\mathrm{d}q} = \frac{q\,|\tilde{V}(\boldsymbol{q})|^2}{2\pi v^2} = \frac{q\,\bar{\sigma}_e}{2\mu_{\chi e}^2 v^2}\,F_\mathrm{DM}(q)^2, \qquad 0 \le q \le 2\mu_{\chi e} v, \tag{2.15}$$

where the form in $\bar{\sigma}_e$ uses $|\tilde{V}|^2 = \pi\bar{\sigma}_e F_\mathrm{DM}^2/\mu_{\chi e}^2$ (Eq. (1.34)). The upper limit is backscattering, $\theta_\mathrm{s} = \pi$. For a contact interaction, $F_\mathrm{DM} = 1$, the distribution rises linearly in q and integrates to

$$\sigma_T = \frac{|\tilde{V}|^2}{2\pi v^2}\,\frac{(2\mu_{\chi e} v)^2}{2} = \frac{\mu_{\chi e}^2 |\tilde{V}|^2}{\pi} = \bar{\sigma}_e, \tag{2.16}$$

independent of the speed. This is the statement behind the definition of $\bar{\sigma}_e$ in Eq. (1.33): for a heavy mediator, $\bar{\sigma}_e$ is the total cross-section of dark matter on a free electron (also Eq. (A.48)). For a nucleus, $\mu_{\chi e} \to \mu_{\chi N}$ and $\bar{\sigma}_e \to \sigma_N$; we return to it at the end of the section.

2.2.2 Scattering off a bound electron

An electron bound in an atom or a crystal does not conserve momentum by itself. The nucleus, or the lattice, absorbs whatever momentum the electron does not keep, and the electron jumps between states $|i\rangle$ and $|f\rangle$ of its own Hamiltonian instead of between plane waves. We treat the binding potential as fixed in space, which neglects the recoil of a nucleus thousands of times heavier than the electron. The dark matter is still a plane wave. The matrix element follows from the same step as Eq. (2.12): integrating over the dark-matter position at fixed electron position $\boldsymbol{r}$ (Eq. (A.9)) gives

$$\langle \boldsymbol{p}-\boldsymbol{q}, f|V(\boldsymbol{x}_\chi - \boldsymbol{r})|\boldsymbol{p}, i\rangle = \frac{\tilde{V}(\boldsymbol{q})}{\mathscr{V}}\,f_{i\to f}(\boldsymbol{q}), \qquad f_{i\to f}(\boldsymbol{q}) \equiv \langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle = \int \mathrm{d}^3 r\;\psi_f^*(\boldsymbol{r})\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,\psi_i(\boldsymbol{r}). \tag{2.17}$$

The *transition form factor* $f_{i\to f}(\boldsymbol{q})$ is the amplitude for the electron to absorb the momentum $\boldsymbol{q}$ and end up in $|f\rangle$. It is a property of the target alone; the dark matter enters only through $\tilde{V}(\boldsymbol{q})$. Its sign convention follows that of $\tilde{V}$ (Appendix A.3). For free electrons, ψ_i and ψ_f are plane waves and $f_{i\to f}$ is a Kronecker delta that conserves momentum, so Eq. (2.12) is the special case.

The energy that the dark matter loses, $p^2/2m_\chi - (\boldsymbol{p}-\boldsymbol{q})^2/2m_\chi = \boldsymbol{q}\cdot\boldsymbol{v} - q^2/2m_\chi$, is the energy deposit $\omega_{\boldsymbol{q}}$ of Eq. (1.22) (with $\Delta_\chi = 0$; an inelastic dark-matter particle adds $-\Delta_\chi$). The signs follow Appendix A.4: $\boldsymbol{q}$ points from the dark matter to the target, and ω is the energy the target gains, which can be negative for a target that starts in an excited state. The golden rule, summed over the final dark-matter momentum by Eq. (2.10) and over the final electron state, yields the rate Γ per dark-matter particle in the box. The box holds one dark-matter particle, a density $1/\mathscr{V}$. The volume-independent combination is $\mathscr{V}\Gamma$, which is $\sigma_T v$ by Eq. (2.13):

Rate off a bound target

$$\sigma_T v = \mathscr{V}\Gamma = \int \frac{\mathrm{d}^3 q}{(2\pi)^3}\,|\tilde{V}(\boldsymbol{q})|^2 \sum_f |f_{i\to f}(\boldsymbol{q})|^2\;2\pi\,\delta(E_f - E_i - \omega_{\boldsymbol{q}}). \tag{2.18}$$

The target enters only through $|f_{i\to f}(\boldsymbol{q})|^2$ and the excitation energies $E_f - E_i$; the dark matter only through $|\tilde{V}(\boldsymbol{q})|^2$ and $\omega_{\boldsymbol{q}}$.

As in Sect. 2.1, energy conservation fixes, for each final state, a shell of momentum transfers $\boldsymbol{q}$, and the integral over $\boldsymbol{q}$ supplies the continuum that the golden rule needs even when the levels E_f are discrete. With $|\tilde{V}|^2$ written through Eq. (1.34), Eq. (2.18) becomes $\sigma_T v = (\bar{\sigma}_e/\mu_{\chi e}^2) \int \mathrm{d}^3q\, (4\pi)^{-1} F_{\mathrm{DM}}^2 \sum_f |f_{i\to f}|^2\, \delta(E_f - E_i - \omega_{\boldsymbol{q}})$ [6, Eqs. (3.9) and (3.10)].

Depositing an energy ω needs $q \geq \omega/v$, since $\omega_{\boldsymbol{q}} \leq qv$, and this decides which momentum transfers Eq. (2.18) weights. A bound electron absorbs momentum transfers up to about its own momentum, $q_0 = \alpha m_e$ for an outer electron, and beyond that $|f_{i\to f}|^2$ collapses. The two requirements meet at $\omega \approx q_0 v \approx \alpha m_e v \approx 4\,\mathrm{eV}$ for $v \approx 10^{-3}$, independent of m_χ (Sect. 1.3). This is why eV-gap targets are a good kinematic match to sub-GeV dark matter: the silicon gap of 1.11 eV used in projections sits below this energy, and the momentum transfers that cross it, $q \gtrsim 0.42\,\mathrm{keV}$ at $v_{\max}$ (Sect. 1.3), are ones the valence electrons absorb readily. Ionizing hydrogen, $\omega \geq 13.6\,\mathrm{eV}$, needs $q \geq 5.1\,\mathrm{keV} = 1.4\,q_0$ even at the largest halo speed $v_{\max} = 2.65 \times 10^{-3}$, and a dark-matter mass of at least $2 \times 13.6\,\mathrm{eV}/v_{\max}^2 = 3.9\,\mathrm{MeV}$ (Eq. (1.25)). The form factor at such q is not yet small, as the hydrogen example shows, but the halo factor $\eta(v_{\min})$ is, because only the fastest dark matter can supply the energy.

Closure fixes the total size of the form factor. Summed over a complete set of final states, bound and free, and including $|i\rangle$ itself,

$$\sum_f |f_{i\to f}(\boldsymbol{q})|^2 = \langle i|\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle = 1. \tag{2.19}$$

The electron absorbs the momentum $\boldsymbol{q}$ with total probability one; the form factor distributes that probability over final states. The term $f = i$ is the *elastic form factor* $f_{i\to i}(\boldsymbol{q}) = \int \mathrm{d}^3r\, |\psi_i(\boldsymbol{r})|^2\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, the Fourier transform of the electron's charge cloud. The orbital radius fixes the range of momentum transfers that the form factor accepts. An electron bound in an orbital of radius r has momenta $\sim 1/r$. A momentum transfer much smaller than $1/r$ barely changes the state, so the elastic term takes almost all of the probability, $f_{i\to i} \to 1$. A momentum transfer much larger than $1/r$ must be absorbed by the high-momentum tail of the wavefunction, and every overlap with a bound final state falls off. This is the physical picture behind $q_0 = \alpha m_e$ in Sect. 1.3, and the hydrogen atom checks it below.

An atom with Z electrons shows the same split on a larger scale. The target operator is now $\sum_{j=1}^{Z} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$, and its elastic matrix element in the atomic ground state is the transform of the total electron density, which tends to Z as $q \to 0$. Elastic scattering at small q is coherent, with a cross-section proportional to Z^2. Inelastic transitions, in which one electron changes its orbital, add incoherently, with total weight at most Z by closure applied to each electron. The nucleus repeats this pattern with A nucleons, at the end of the section. In a crystal the coherent part is the Bragg-like elastic response at reciprocal-lattice vectors, which deposits no energy, and the incoherent part is the electronic excitation a detector sees (Chap. 4).

The rate in a detector averages Eq. (2.18) over the dark-matter velocities. For a target whose $\sum_f |f_{i\to f}(\boldsymbol{q})|^2$ at fixed energy depends only on $q = |\boldsymbol{q}|$, such as an atom with closed shells, only the angle between $\boldsymbol{q}$ and $\boldsymbol{v}$ enters the delta function. Its average over that angle is $\Theta(v - v_{\min})/(2qv)$, and the halo average turns $\Theta(v - v_{\min})/v$ into $\eta(v_{\min})$, the mean inverse speed of the dark matter above $v_{\min}$ (Eq. (1.28)). Inserting $1 = \int \mathrm{d}\omega\, \delta(\omega - E_f + E_i)$ and writing $\mathrm{d}^3q = 4\pi q^2\, \mathrm{d}q$ for the remaining integral,

$$\frac{\mathrm{d}\langle\sigma_T v\rangle}{\mathrm{d}\omega} = \frac{1}{2\pi} \int q\, \mathrm{d}q\, |\tilde{V}(q)|^2\, \eta\big(v_{\min}(q,\omega)\big) \sum_f |f_{i\to f}(q)|^2\, \delta(\omega - E_f + E_i), \tag{2.20}$$

with $\langle\cdot\rangle$ the average over the halo. The assumption that the response depends on $q = |\boldsymbol{q}|$ alone holds for a closed shell. A potential that does not act on spin leaves the spin of the electron unchanged, and each occupied orbital contributes once for each of its two spin states.

A full shell of angular momentum l therefore holds $2(2l+1)$ electrons. Its density is spherical, because $\sum_m |Y_{lm}|^2 = (2l+1)/4\pi$ for the spherical harmonics Y_{lm} (Appendix C.3), and so is its summed response to $\boldsymbol{q}$. A crystal is not spherical, and Eq. (2.18) must then be kept with its angular integral; Sect. 1.2 gave the resulting daily modulation.

For a gas of such atoms with number density n_T, the event rate per unit target mass is $(\rho_\chi/m_\chi)(n_T/\rho_T)\langle\sigma_T v\rangle$. Comparing with Eq. (1.47) identifies the target response of a gas of independent atoms: $S(q,\omega) = 2\pi\, n_T \sum_i \sum_f |f_{i\to f}(q)|^2\, \delta(\omega - E_f + E_i)$, summed over the occupied initial states i of each atom. For any target, $S(q,\omega)$ is the dynamic structure factor of the density operator $n_e(\boldsymbol{q}) = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$ of Eq. (A.8) (Eqs. (2.67) and (A.31)), whose single-electron matrix elements are the $f_{i\to f}$ of Eq. (2.17).

For several electrons the sum over i runs over occupied states and the final states must be empty; second quantization (Sect. 2.3) is the bookkeeping that handles this, and Sect. 4.2 recovers Eq. (2.20) as the independent-particle limit. A target at finite temperature starts in a mixture of states $|i\rangle$, and the rate is averaged over them with thermal weights (Sect. 2.5).

2.2.3 The hydrogen atom

The hydrogen ground state is the one bound electron whose form factor we can write in closed form. Its wavefunction is $\psi_{1s}(\boldsymbol{r}) = \mathrm{e}^{-r/a_0}/(\pi a_0^3)^{1/2}$, with the Bohr radius $a_0 = 1/(\alpha m_e) = 0.5292\,\text{Å}$ (Appendix A.1). The reference momentum of Sect. 1.3 is its inverse, $q_0 = \alpha m_e = 1/a_0$, so $q_0 a_0 = 1$ exactly.

Worked example 2.2: The hydrogen 1s elastic form factor

The elastic form factor is the Fourier transform of $|\psi_{1s}|^2 = \mathrm{e}^{-2r/a_0}/(\pi a_0^3)$. Since the density is spherical, the integral of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ over the directions of $\boldsymbol{r}$ gives $4\pi\sin(qr)/(qr)$ (Appendix C.3), and

$$F_{1s}(q) \equiv f_{1s\to 1s}(\boldsymbol{q}) = \frac{4\pi}{\pi a_0^3}\int_0^\infty \mathrm{d}r\, r^2\, \mathrm{e}^{-2r/a_0}\,\frac{\sin qr}{qr}$$
$$= \frac{4}{a_0^3 q}\int_0^\infty \mathrm{d}r\, r\, \mathrm{e}^{-\lambda_{1s} r}\sin qr, \qquad \lambda_{1s} = \frac{2}{a_0}.$$

The remaining integral is the imaginary part of $\int_0^\infty \mathrm{d}r\, r\, \mathrm{e}^{-(\lambda_{1s}-\mathrm{i}q)r} = 1/(\lambda_{1s}-\mathrm{i}q)^2 = (\lambda_{1s}+\mathrm{i}q)^2/(\lambda_{1s}^2+q^2)^2$, which is $2\lambda_{1s}q/(\lambda_{1s}^2+q^2)^2$. So $F_{1s}(q) = (4/a_0^3 q)\,(4q/a_0)/(4/a_0^2+q^2)^2$, or

$$F_{1s}(q) = \frac{16}{(4+q^2a_0^2)^2}. \tag{2.21}$$

At small q, $F_{1s} = (1+q^2a_0^2/4)^{-2} \simeq 1 - q^2a_0^2/2$. This is the general small-q form $1 - q^2\langle r^2\rangle/6$ of an elastic form factor, with $\langle r^2\rangle = 3a_0^2$ for the 1s state. At large q, $F_{1s} \simeq 16/(qa_0)^4$: the elastic amplitude dies as the fourth power of the momentum transfer. The value at the reference momentum is $F_{1s}(q_0) = 16/25 = 0.64$: the electron stays in its ground state with probability $F_{1s}^2 = 0.41$ and, by Eq. (2.19), is excited or ionized with probability 0.59. The form factor falls to one half at $qa_0 = 1.29$ and to 0.25 at $qa_0 = 2$; its square falls to one half at $qa_0 = 0.87$. The 1s electron is the most compact state of hydrogen, and q_0 is close to the momentum transfer at which it stops responding as a whole and starts to be knocked out.

Where does the inelastic probability go? The lowest excitation, $1s \to 2p$ at 10.2 eV, has a closed form too. Take $\boldsymbol{q}$ along z. Only the $m = 0$ state $\psi_{2p} = (32\pi a_0^5)^{-1/2}\, z\, \mathrm{e}^{-r/2a_0}$ couples, because the operator $\mathrm{e}^{\mathrm{i}qz}$ does not change the angular momentum about z. The angular integral is $2\pi\int_{-1}^{1}\mathrm{d}u\, u\, \mathrm{e}^{\mathrm{i}qru} = 4\pi\mathrm{i}\, j_1(qr)$, with $j_1(x) = (\sin x - x\cos x)/x^2$ the

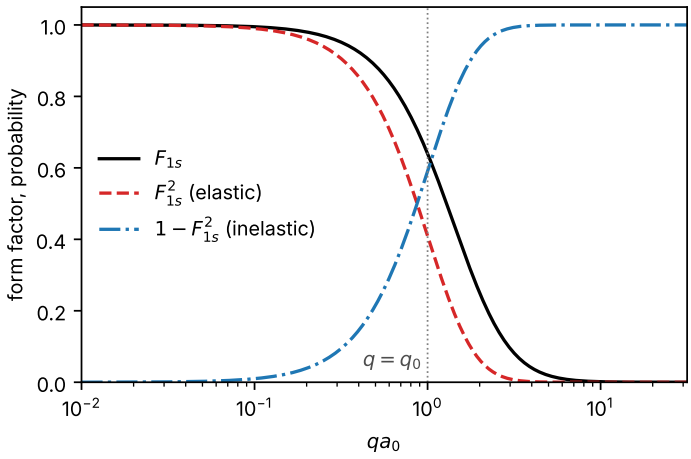


Figure 2.3: Hydrogen 1s elastic form factor $F_{1s}(q) = 16/(4 + q^2a_0^2)^2$ (solid), its square, the probability that the electron stays in the ground state after absorbing momentum q (dashed), and the probability $1 - F_{1s}^2$ that it is excited or ionized (dash-dotted), against qa_0, with a_0 the Bohr radius. The dotted line marks the reference momentum $q_0 = \alpha m_e = 3.73\,\text{keV}$, where $qa_0 = 1$ exactly and $F_{1s} = 0.64$. The inelastic probability follows from the closure relation $\sum_f |\langle f|e^{i\boldsymbol{q}\cdot\boldsymbol{r}}|1s\rangle|^2 = 1$.

spherical Bessel function of order one (Appendix C.3), and the radial integral, done as in the elastic case with $\lambda_{2p} = 3/2a_0$, is $\int_0^\infty dr\, r^3 e^{-\lambda_{2p}r} j_1(qr) = 8\lambda_{2p}q/(\lambda_{2p}^2 + q^2)^3$. Collecting the normalizations,

$$|f_{1s\to 2p}(q)|^2 = \frac{72\,(qa_0)^2}{(9/4 + q^2a_0^2)^6}.$$

At small q it grows as $(2^{15}/3^{10})\,q^2a_0^2 = 0.555\,q^2a_0^2$, which is $q^2|\langle 2p|z|1s\rangle|^2$: a momentum transfer much smaller than $1/a_0$ acts on the atom as a weak uniform field, through its dipole. The square peaks at $qa_0 = 0.67$ with the value 0.084 and is 0.061 at q_0. So at q_0 the $2p$ level takes about a tenth of the inelastic probability of 0.59, and the rest goes to higher levels and, mostly, to ionization.

Fig. 2.3 shows the crossover. Below $q \approx q_0/3$ the atom absorbs the momentum transfer elastically: the electron and the nucleus move together, and the energy deposit is the recoil of the whole atom. Above $q \approx 3q_0$ almost every scattering excites or ionizes the atom. The elastic form factor also explains how a neutral atom hides its charge. A dark photon couples to the electron and to the proton with opposite signs (Sect. 1.4), and the elastic amplitude off the atom is proportional to $1 - F_{1s}(q)$: the proton, a point, contributes 1, and the electron cloud $-F_{1s}$. At $q \ll q_0$ the two cancel and the dark matter sees a neutral atom; at q_0 the squared factor is $(1 - 0.64)^2 = 0.13$. This is screening in its simplest form, and Sect. 4.4 treats it in solids.

For ionization the final state is an unbound electron of momentum p_{ej} and kinetic energy $E_{\text{ej}} = p_{\text{ej}}^2/2m_e$, and the energy transfer is $\omega = E_B + E_{\text{ej}}$, with $E_B > 0$ the binding energy of the shell. Far from the atom the ejected electron is a spherical wave; near it, the wave is distorted by the ion. The sum over final states in Eq. (2.20) becomes an integral over p_{ej} and a sum over the angular quantum numbers of the outgoing wave, and the result is conventionally packaged as the *atomic ionization form factor* $|f_{\text{ion}}(p_{\text{ej}}, q)|^2$ of each shell [5, Eq. (6)].

The outgoing waves are labeled by p_{ej} and by the angular quantum numbers l', m', and

normalized to $\langle\tilde\psi_{p'l'm'}|\tilde\psi_{plm}\rangle = (2\pi)^3\delta_{l'l}\delta_{m'm}\,\delta(p'-p)/p^2$. The sum over final states is then $\sum_f = \sum_{l'm'}\int p_{\rm ej}^2\,\mathrm{d}p_{\rm ej}/(2\pi)^3$. With $\mathrm{d}E_{\rm ej} = p_{\rm ej}\,\mathrm{d}p_{\rm ej}/m_e$, $p_{\rm ej}^2\,\mathrm{d}p_{\rm ej} = (p_{\rm ej}^3/2)\,\mathrm{d}\ln E_{\rm ej}$. Define

$$|f_{\rm ion}(p_{\rm ej},q)|^2 = \frac{2p_{\rm ej}^3}{(2\pi)^3}\sum_{l'm'}\sum_i\left|\int \mathrm{d}^3r\;\tilde\psi^*_{p_{\rm ej}l'm'}(\boldsymbol{r})\,\psi_i(\boldsymbol{r})\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\right|^2, \tag{2.22}$$

with i running over the occupied states of the shell, both spins included. At fixed E_B, $\mathrm{d}\omega = \mathrm{d}E_{\rm ej} = E_{\rm ej}\,\mathrm{d}\ln E_{\rm ej}$. In Eq. (2.20) the delta function $\delta(\omega - E_B - E_{\rm ej})$ then turns $(p_{\rm ej}^3/2)\,\mathrm{d}\ln E_{\rm ej}$ into $p_{\rm ej}^3/(2E_{\rm ej})$, and passing from $\mathrm{d}/\mathrm{d}\omega$ to $\mathrm{d}/\mathrm{d}\ln E_{\rm ej}$ multiplies by $E_{\rm ej}$ again. The sum over final states is therefore $|f_{\rm ion}|^2/4$ per unit $\ln E_{\rm ej}$. Inserting this and $|\tilde V|^2 = \pi\bar\sigma_e F_{\rm DM}^2/\mu_{\chi e}^2$ into Eq. (2.20) leaves the prefactor $(1/2\pi)(\pi\bar\sigma_e/\mu_{\chi e}^2)(1/4) = \bar\sigma_e/(8\mu_{\chi e}^2)$:

$$\frac{\mathrm{d}\langle\sigma v\rangle}{\mathrm{d}\ln E_{\rm ej}} = \frac{\bar\sigma_e}{8\mu_{\chi e}^2}\int q\,\mathrm{d}q\,|f_{\rm ion}(p_{\rm ej},q)|^2\,F_{\rm DM}(q)^2\,\eta(v_{\rm min}), \qquad v_{\rm min} = \frac{E_B + E_{\rm ej}}{q} + \frac{q}{2m_\chi}, \tag{2.23}$$

the ionization rate of Ref. [5, Eqs. (5) and (6)]. It rests on the first Born approximation in the dark-matter potential and treats one electron at a time in a fixed atomic potential, with no correlation and no recoil of the ion.

The exact outgoing waves are tedious to compute, and the standard shortcut replaces them by plane waves. This also shows where the spin factor sits. Replace the outgoing wave by $\mathrm{e}^{\mathrm{i}\boldsymbol{p}_{\rm ej}\cdot\boldsymbol{r}}$. The overlap in Eq. (2.17) is then the momentum-space wavefunction $\tilde\psi_i(\boldsymbol{p}_{\rm ej} - \boldsymbol{q})$, with $\tilde\psi_i(\boldsymbol{k}) = \int \mathrm{d}^3r\,\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\psi_i(\boldsymbol{r})$. For a full shell, $\sum_m |\tilde\psi_{nlm}(\boldsymbol{k})|^2 = (2l+1)|\tilde\psi_{nl}(k)|^2/4\pi$, where $\tilde\psi_{nl}(k)$ is the radial momentum wavefunction of the shell with principal and orbital quantum numbers n, l (Ref. [5] writes χ_{nl}), normalized to $\int k^2\,\mathrm{d}k\,|\tilde\psi_{nl}|^2 = (2\pi)^3$, and the two spin states double this. The sum over final states is $\int \mathrm{d}^3p_{\rm ej}/(2\pi)^3$. At fixed $p_{\rm ej}$ the integral over its direction relative to $\boldsymbol{q}$ becomes an integral over $k = |\boldsymbol{p}_{\rm ej} - \boldsymbol{q}|$, since $k^2 = p_{\rm ej}^2 + q^2 - 2p_{\rm ej}q\cos\theta'$ gives $2\pi\,\mathrm{d}\cos\theta' = 2\pi k\,\mathrm{d}k/(p_{\rm ej}q)$, with θ' the angle between $\boldsymbol{p}_{\rm ej}$ and $\boldsymbol{q}$ and k running between $|p_{\rm ej} - q|$ and $p_{\rm ej} + q$. Collecting the factors as for Eq. (2.22),

$$|f_{\rm ion}(p_{\rm ej},q)|^2_{\rm PW} = \frac{(2l+1)\,p_{\rm ej}^2}{4\pi^3 q}\int_{|p_{\rm ej}-q|}^{p_{\rm ej}+q} k\,\mathrm{d}k\,|\tilde\psi_{nl}(k)|^2, \tag{2.24}$$

which is the expression given in Ref. [5, text after Eq. (6)]; it counts all $2(2l+1)$ electrons of a full shell, and for the single electron of hydrogen 1s the rate is half of it. The result has a direct reading: the ejected electron leaves with the momentum it had in the atom plus the momentum transfer, and the form factor counts how much of the bound momentum distribution lies on the shell $|\boldsymbol{k} + \boldsymbol{q}| = p_{\rm ej}$.

The plane wave ignores the ion. The ion attracts a slow electron and raises its wavefunction near the nucleus, where the overlap with ψ_i is taken, and the effect grows as $E_{\rm ej}$ falls. For an electron of momentum p leaving a point charge $Z_{\rm eff}e$, the ratio of the exact to the free probability density at the origin is the Fermi factor familiar from beta decay,

$$F_{\rm Fermi}(p) = \frac{2\pi\xi_{\rm C}}{1 - \mathrm{e}^{-2\pi\xi_{\rm C}}}, \qquad \xi_{\rm C} = \frac{Z_{\rm eff}\,\alpha m_e}{p}, \tag{2.25}$$

which grows as $1/p$ for a slow electron [5, Eq. (1)]. Multiplying Eq. (2.24) by $F_{\rm Fermi}(p_{\rm ej})$ with $Z_{\rm eff} = 1$ is the standard approximation. For the outer shells of xenon and argon, which dominate their rates, its authors report agreement with continuum states computed in the atomic potential to within $O(30\,\%)$ in the total rate [5, text after Eq. (6)]. Hydrogen 1s is the one case in which the exact continuum states are known in closed form, and Fig. 2.4 uses it to test the approximation on a single compact shell. There the same recipe reaches only about half of the exact rate, which does not contradict the xenon and argon result: the quality of the Fermi factor depends on the shell.

Limit Check 2.1. Atomic ionization rate

The ionization rate of a shell with binding energy E_B, per unit logarithm of the ejected electron's kinetic energy $E_{\rm ej} = p_{\rm ej}^2/2m_e$, is

$$\frac{\mathrm{d}\langle\sigma v\rangle}{\mathrm{d}\ln E_{\rm ej}} = \frac{\bar{\sigma}_e}{8\mu_{\chi e}^2}\int q\,\mathrm{d}q\,|f_{\rm ion}|^2\,F_{\rm DM}(q)^2\,\eta(v_{\rm min}),$$
$$|f_{\rm ion}|^2 \simeq \frac{2\pi\xi_{\rm C}}{1-\mathrm{e}^{-2\pi\xi_{\rm C}}}\,\frac{(2l+1)\,p_{\rm ej}^2}{4\pi^3 q}\int_{|p_{\rm ej}-q|}^{p_{\rm ej}+q} k\,\mathrm{d}k\,|\tilde{\psi}_{nl}(k)|^2,$$

with $\xi_{\rm C} = \alpha m_e/p_{\rm ej}$ (the Fermi factor of Eq. (2.25) for $Z_{\rm eff} = 1$), q the momentum transfer, $\tilde{\psi}_{nl}(k)$ the radial momentum wavefunction of the shell with principal and orbital quantum numbers n, l, and $v_{\rm min} = (E_B + E_{\rm ej})/q + q/2m_\chi$ [5, Eqs. (1), (5) and (6)].

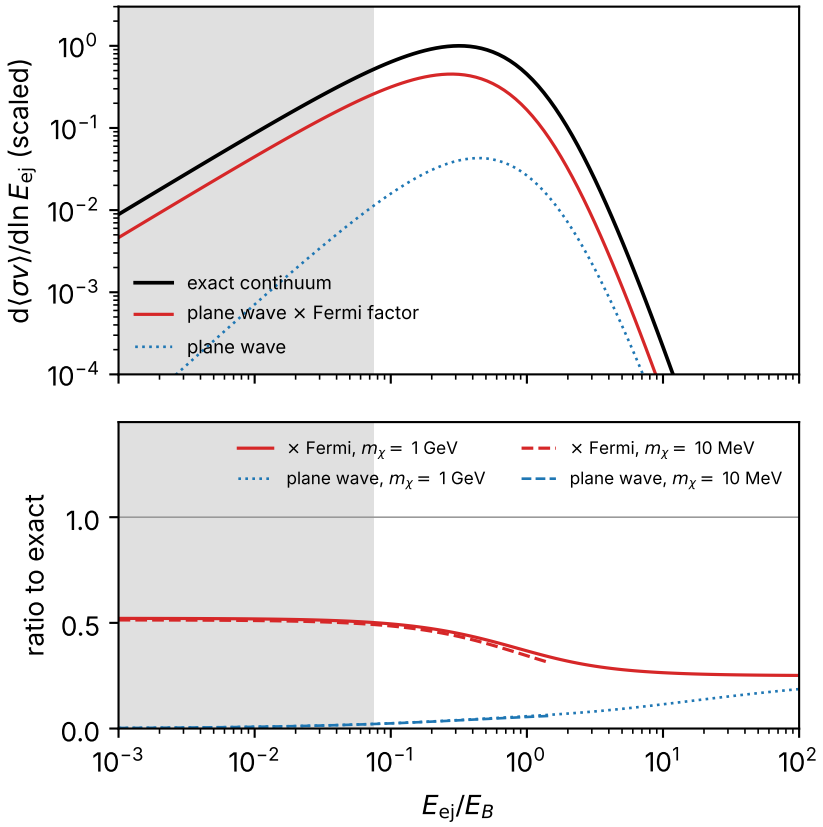


Figure 2.4: Ionization spectrum of hydrogen 1s (one electron, so half of the two-spin formula above; $E_B = 13.6\,\mathrm{eV}$) for a heavy mediator, $F_{\rm DM} = 1$, in the standard halo model with the parameters of Table 1.1, against $E_{\rm ej}/E_B$. Top, for $m_\chi = 1\,\mathrm{GeV}$, scaled to the peak of the exact result: exact Coulomb continuum states (black), plane wave times the Fermi factor with $Z_{\rm eff} = 1$ (red), plane wave alone (blue dotted). Bottom: the two approximations divided by the exact spectrum, for $m_\chi = 1\,\mathrm{GeV}$ and $10\,\mathrm{MeV}$. Gray: $E_{\rm ej} \leq 0.075\,E_B$, where the Fermi-corrected plane wave reaches about half of the exact spectrum. Integrated over $E_{\rm ej}$ at 1 GeV, it gives 0.46 of the exact rate and the plane wave alone 0.036. The exact form factor sums the overlaps with Coulomb partial waves of the hydrogen continuum.

Above the shaded region, where most of the rate lies, the halo forces momentum transfers $q \geq (E_B + E_{\rm ej})/v$ well above $p_{\rm ej}$, which probe the Coulomb wave at $r \sim 1/q$, where it is not a rescaled plane wave, and the approximation falls to between a quarter and a third of the exact spectrum for $E_{\rm ej}$ of several E_B.

Hydrogen is the only atom with closed-form orbitals. For every other atom the initial orbitals ψ_i are not known exactly: they are computed self-consistently, each electron moving in the average field of the others, as Hartree–Fock orbitals, and the ionization form factors are integrals over them. Ref. [5] used tabulated Roothaan–Hartree–Fock wavefunctions, Hartree–Fock orbitals expanded in a finite set of analytic radial functions; Appendix E.7 describes how such form factors are computed and converged, and Sect. 3.4 makes the self-consistent-field method precise. In a crystal the same overlap is taken between Bloch states of the valence and conduction bands, summed over their crystal momenta, and the atomic $|f_{\rm ion}|^2$ becomes the crystal form factor [6, Eqs. (3.13) and (3.17)] of Chap. 10.

2.2.4 Free electrons

Eq. (2.18) contains free scattering as a special case, and the check, now in the laboratory frame, shows how the mediator mass enters the free cross-section.

Worked example 2.3: Scattering off a free electron

Take an electron at rest, $|i\rangle = |\mathbf{0}\rangle$, with final states $|f\rangle = |\boldsymbol{k}\rangle$ in the box. By Eq. (2.17), $f_{i\to f}(\boldsymbol{q}) = \langle \boldsymbol{k}|e^{i\boldsymbol{q}\cdot\boldsymbol{r}}|\mathbf{0}\rangle = \delta_{\boldsymbol{k}\boldsymbol{q}}$: the electron takes exactly the momentum it is given. The sum over f in Eq. (2.18) keeps $\boldsymbol{k} = \boldsymbol{q}$, with $E_f - E_i = q^2/2m_e$, and with $\omega_{\boldsymbol{q}} = qv\cos\theta - q^2/2m_\chi$ the delta function is $\delta(q^2/2\mu_{\chi e} - qv\cos\theta)$, where θ is the angle between $\boldsymbol{q}$ and $\boldsymbol{v}$ and $1/\mu_{\chi e} = 1/m_e + 1/m_\chi$. Its root is $\cos\theta = q/(2\mu_{\chi e}v)$, which lies in $[-1,1]$ for $q \leq 2\mu_{\chi e}v$, and its weight is $1/(qv)$. With $\mathrm{d}^3q = 2\pi q^2\,\mathrm{d}q\,\mathrm{d}\cos\theta$,

$$\sigma_T v = \int_0^{2\mu_{\chi e}v} \frac{2\pi q^2\,\mathrm{d}q}{(2\pi)^3}\,|\tilde{V}(q)|^2\,\frac{2\pi}{qv} = \frac{1}{2\pi v}\int_0^{2\mu_{\chi e}v} q\,\mathrm{d}q\,|\tilde{V}(q)|^2.$$

Dividing by v gives $\mathrm{d}\sigma_T/\mathrm{d}q = q|\tilde{V}(q)|^2/(2\pi v^2)$, which is Eq. (2.15), obtained before in the center-of-mass frame.

For a mediator of mass m_V, Eq. (1.35) has $F_{\rm DM}^2 = (m_V^2 + q_0^2)^2/(m_V^2 + q^2)^2$, and the integral over q^2 from 0 to $q_{\rm max}^2 = (2\mu_{\chi e}v)^2$ is elementary, $\int_0^{q_{\rm max}^2} \mathrm{d}x\,(m_V^2 + x)^{-2} = q_{\rm max}^2/[m_V^2(m_V^2 + q_{\rm max}^2)]$. So

$$\sigma_T = \bar{\sigma}_e\,\frac{(m_V^2 + q_0^2)^2}{m_V^2\,(m_V^2 + q_{\rm max}^2)}.$$

For concreteness take $m_\chi = 100\,\mathrm{MeV}$, with $\mu_{\chi e} = 0.508\,\mathrm{MeV}$, and the largest halo speed, $v_{\rm max} = 2.65\times10^{-3}$ (Eq. (1.9)). Then $q_{\rm max} = 2.69\,\mathrm{keV} = 0.72\,q_0$. A free electron at rest cannot even absorb the reference momentum q_0 from such a particle: $\bar{\sigma}_e$ is defined at a momentum transfer that free kinematics does not reach. For a heavy mediator this does not matter. With $m_V = 10\,q_0$ we find $\sigma_T = 1.015\,\bar{\sigma}_e$, and at $m_V = 30\,\mathrm{MeV}$ the difference is 2.3×10^{-8}. At $m_V = q_0$, $\sigma_T = 2.63\,\bar{\sigma}_e$, and at $m_V = 0.1\,q_0$, $\sigma_T = 192\,\bar{\sigma}_e$. As $m_V \to 0$ it grows as $q_0^4/(m_V^2 q_{\rm max}^2)$: the free cross-section of a light mediator diverges at small q, like Rutherford's, and $\bar{\sigma}_e$ is then just a normalization (Sect. 1.4). In a target the divergence is cut off by $q \geq \omega/v_{\rm max}$ for an excitation of energy ω (Sect. 1.3) and by screening (Sect. 4.4).

The free electron at rest is a poor model of a bound one. A bound electron moves with a typical

speed $\sim \alpha$, about ten times the dark-matter speed (Sect. 1.3), so it can absorb momentum transfers well above $2\mu_{\chi e}v$, and it needs an energy $E_f - E_i$ that free kinematics does not supply. Both effects sit in $f_{i\to f}$. The free cross-section serves to fix the units: Eqs. (2.18) and (2.20) carry $\bar\sigma_e$ as an overall factor, and all target dependence is in $\sum_f |f_{i\to f}|^2$.

For a light mediator the bound target also removes the divergence of Worked example 2.3. In Eq. (2.20) the factor $F_{\rm DM}^2 = (q_0/q)^4$ multiplies $q\,\mathrm{d}q$, and the integrand grows as q^{-3} toward small q. It stops at $q = \omega/v_{\max}$, below which no dark-matter particle can deposit ω. In addition, the squared form factor of every inelastic transition vanishes at small q, as q^2 for a dipole-allowed one (Worked example 2.2). The rate is then dominated by the smallest q that kinematics allows, just above the threshold, which is why light-mediator rates are sensitive to the lowest excitations of the target and to screening.

2.2.5 Nuclei and the Helm form factor

A nucleus of mass number A is a bound target of A nucleons, and Eq. (2.17) applies with one change: every nucleon couples. With a potential $V_N(\boldsymbol{x}_\chi - \boldsymbol{r}_j)$ between the dark matter and nucleon j, the same for protons and neutrons, the target operator is $\sum_{j=1}^{A} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$. Write $\boldsymbol{r}_j = \boldsymbol{R}_{\rm cm} + \boldsymbol{r}'_j$, with $\boldsymbol{R}_{\rm cm}$ the center of mass of the nucleus and $\boldsymbol{r}'_j$ the internal coordinates. The center of mass is a free particle and its plane waves give momentum conservation, as in Eq. (2.12). The internal state is the nuclear ground state $|0\rangle$, before and after. The lowest excited levels lie at 1.78 MeV in ^{28}Si, 1.27 MeV in ^{29}Si and 2.24 MeV in ^{30}Si, and between 0.56 MeV and 1.04 MeV in the even germanium isotopes. The exception is ^{73}Ge, 7.8 % of natural germanium, whose first excited state is at 13.3 keV and its second at 66.7 keV [7]. Sub-GeV dark matter carries at most a few keV of kinetic energy (Sect. 1.3), below all of these. The internal matrix element is

$$\Big\langle 0\Big|\sum_{j=1}^{A} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}'_j}\Big|0\Big\rangle = \int \mathrm{d}^3r\; \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\, n_N(\boldsymbol{r}) \equiv A\, F_N(q), \tag{2.26}$$

with $n_N(\boldsymbol{r})$ the nucleon number density of the nucleus, normalized to $\int \mathrm{d}^3r\, n_N = A$, and $F_N(q)$ the *nuclear form factor*, the elastic form factor of the nucleus with $F_N(0) = 1$. This is the coherent sum: at $q \to 0$ the phases of all nucleons agree and the amplitude is A times that of one nucleon. The two-body result Eq. (2.15), with $\mu_{\chi e} \to \mu_{\chi N}$ and $\tilde V \to A\tilde V_N F_N$, gives

$$\frac{\mathrm{d}\sigma_N}{\mathrm{d}q} = \frac{q\,A^2|\tilde V_N(q)|^2}{2\pi v^2}\,F_N(q)^2 = \frac{q\,\sigma_N}{2\mu_{\chi N}^2 v^2}\,F_N(q)^2, \qquad \sigma_N = \frac{\mu_{\chi N}^2 A^2 |\tilde V_N|^2}{\pi} = \sigma_p \left(\frac{\mu_{\chi N}}{\mu_{\chi p}}\right)^2 A^2, \tag{2.27}$$

for a heavy mediator, $\tilde V_N = y_\chi y_N/m_V^2$. The expression through σ_p uses the proton cross-section $\sigma_p = \mu_{\chi p}^2|\tilde V_N|^2/\pi$, which is Eq. (2.16) with $\mu_{\chi e} \to \mu_{\chi p}$. This derives Eq. (1.37), and σ_N is the cross-section at $q \to 0$; the form factor then cuts off the distribution at large q. Unequal proton and neutron couplings replace A by $Z + (A-Z)y_n/y_p$ (Sect. 1.4).

Experiments measure the recoil energy $E_R = q^2/2M$ rather than q, with M the nuclear mass. With $\mathrm{d}E_R = q\,\mathrm{d}q/M$, $\mathrm{d}\sigma_N/\mathrm{d}E_R = M\sigma_N F_N^2/(2\mu_{\chi N}^2 v^2)$. We multiply by the number of nuclei per unit mass, $1/M$, and by the dark-matter flux $(\rho_\chi/m_\chi)\, v\, f_\chi(\boldsymbol{v})\,\mathrm{d}^3v$. Integrating over velocities above $v_{\min} = (ME_R/2\mu_{\chi N}^2)^{1/2}$ (Sect. 1.3) then turns the factor v/v^2 into $\eta(v_{\min})$:

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}E_R} = \frac{\rho_\chi\,\sigma_N}{2m_\chi\mu_{\chi N}^2}\,F_N(q)^2\,\eta(v_{\min}), \qquad q = (2ME_R)^{1/2},$$

the standard spin-independent recoil spectrum, Ref. [3, Eq. (104)] with σ_N written through Eq. (1.37) and equal couplings to protons and neutrons. The halo, the coupling and the nucleus each enter as one factor.

The nucleon density is close to uniform inside the nucleus and falls off over a skin about 1 fm thick. Helm's model [8] writes it as a uniform sphere of radius R_{nuc} smeared by a Gaussian of width s_{skin}. The transform of a convolution is the product of the transforms. For the uniform sphere, the angular integral is $4\pi \sin(qr)/(qr)$ again (Appendix C.3), and $\int_0^R \mathrm{d}r\, r \sin qr = (\sin qR - qR\cos qR)/q^2$, so its normalized transform is $3(\sin x - x\cos x)/x^3 = 3j_1(x)/x$ with $x = qR_{\mathrm{nuc}}$ and j_1 the spherical Bessel function. The normalized Gaussian $\mathrm{e}^{-r^2/2s_{\mathrm{skin}}^2}/(2\pi s_{\mathrm{skin}}^2)^{3/2}$ transforms to $\mathrm{e}^{-q^2 s_{\mathrm{skin}}^2/2}$. Multiplying the two gives [9, Eq. (17)]

$$F_{\mathrm{Helm}}(q) = \frac{3j_1(qR_{\mathrm{nuc}})}{qR_{\mathrm{nuc}}}\, \mathrm{e}^{-q^2 s_{\mathrm{skin}}^2/2}. \tag{2.28}$$

The standard parameters are those of Lewin and Smith [10]: $R_{\mathrm{nuc}}^2 = c_{\mathrm{nuc}}^2 + \frac{7}{3}\pi^2 a_{\mathrm{nuc}}^2 - 5s_{\mathrm{skin}}^2$, with $c_{\mathrm{nuc}} = (1.23A^{1/3} - 0.60)\,\mathrm{fm}$, $a_{\mathrm{nuc}} = 0.52\,\mathrm{fm}$ and $s_{\mathrm{skin}} = 0.9\,\mathrm{fm}$ [9, Eqs. (11) and (12), quoting Lewin and Smith]. Here c_{nuc} and a_{nuc} are the half-density radius and surface diffuseness of a Woods–Saxon density, and the formula for R_{nuc} matches the Helm form to it. The literature writes r_n or R_1 for R_{nuc}, s for s_{skin}, and c, a for c_{nuc}, a_{nuc}. At small q, $3j_1(x)/x \simeq 1 - x^2/10$ and $F_{\mathrm{Helm}} \simeq 1 - q^2(R_{\mathrm{nuc}}^2/10 + s_{\mathrm{skin}}^2/2)$, the form $1 - q^2\langle r^2\rangle/6$ with $\langle r^2\rangle = \frac{3}{5}R_{\mathrm{nuc}}^2 + 3s_{\mathrm{skin}}^2$. For silicon, $c_{\mathrm{nuc}} = 3.14\,\mathrm{fm}$ and $R_{\mathrm{nuc}} = 3.47\,\mathrm{fm}$; for germanium ($A = 73$), $c_{\mathrm{nuc}} = 4.54\,\mathrm{fm}$ and $R_{\mathrm{nuc}} = 4.77\,\mathrm{fm}$. The simpler fit $R_{\mathrm{nuc}} \approx 1.14A^{1/3}\,\mathrm{fm}$ of Ref. [3, Sect. IV C] yields 3.46 fm and 4.76 fm, the same to within a percent.

The Helm form is a model, not a measurement. Electron scattering measures the charge density of the nucleus, that is, the protons. The form factor of Eq. (2.26) needs the density of all nucleons. The standard assumption is that the two are proportional, in which case the nucleon form factor equals the charge form factor [9, Eqs. (7) and (8)]; for the dark photon, which couples to charge, the charge form factor is exact. The physical form factor is then the transform of the measured charge density, and Ref. [9, Sect. V] compares the two. For ^{28}Si the Lewin–Smith Helm form factor differs from the measured one negligibly at small q and by up to 5 % at the momentum transfer of a 100 keV recoil; for ^{70}Ge the difference is 2 % at 10 keV and 20 % at 100 keV. We use the Helm form as the convention of the field, and it is accurate wherever light dark matter is concerned.

Fig. 2.5 shows that light dark matter never resolves the nucleus. The silicon radius $R_{\mathrm{nuc}} = 3.47\,\mathrm{fm}$ corresponds to $1/R_{\mathrm{nuc}} = 57\,\mathrm{MeV}$, and germanium's 4.77 fm to 41 MeV. The form factor matters once qR_{nuc} approaches one, at tens of MeV, which needs dark matter of several GeV.

Worked example 2.4: The A^2 enhancement for silicon

For ^{28}Si, $A^2 = 784$. With the reduced masses of Sect. 1.4, $\mu_{\chi p} = 90.4\,\mathrm{MeV}$ and $\mu_{\chi N} = 99.6\,\mathrm{MeV}$ at $m_\chi = 100\,\mathrm{MeV}$, Eq. (2.27) gives $\sigma_N/\sigma_p = 1.215 \times 784 = 953$; at 1 GeV, $\mu_{\chi p} = 484\,\mathrm{MeV}$, $\mu_{\chi N} = 963\,\mathrm{MeV}$, and $\sigma_N/\sigma_p = 3.96 \times 784 = 3100$, the values quoted in Sect. 1.4. Coherence turns the A of 28 nucleons scattering independently into A^2, a gain of 28; the reduced-mass factor is a separate effect of the heavier recoiling target. Per kilogram of target the rate still grows as A, because A^2 in the cross-section beats the $1/A$ in the number of nuclei per unit mass. The form factor costs almost nothing. At $m_\chi = 1\,\mathrm{GeV}$ the maximum momentum transfer is $q_{\max} = 2\mu_{\chi N} v_{\max} = 5.10\,\mathrm{MeV}$ (Worked example 1.4), and $q_{\max}R_{\mathrm{nuc}} = 0.090$ and $F_{\mathrm{Helm}}^2 = 0.998$ at the maximum momentum transfer, where the suppression is greatest. The same calculation yields $F_{\mathrm{Helm}}^2(q_{\max}) = 0.89$ at 10 GeV and 0.36 at 100 GeV. For germanium at 100 GeV, $q_{\max} = 214\,\mathrm{MeV}$ lies beyond the first zero and $F_{\mathrm{Helm}}^2 = 0.002$ there. Form factors shape the recoil spectra of weak-scale dark matter but are a 0.2 % correction at 1 GeV and far less for the sub-GeV dark matter of this book.

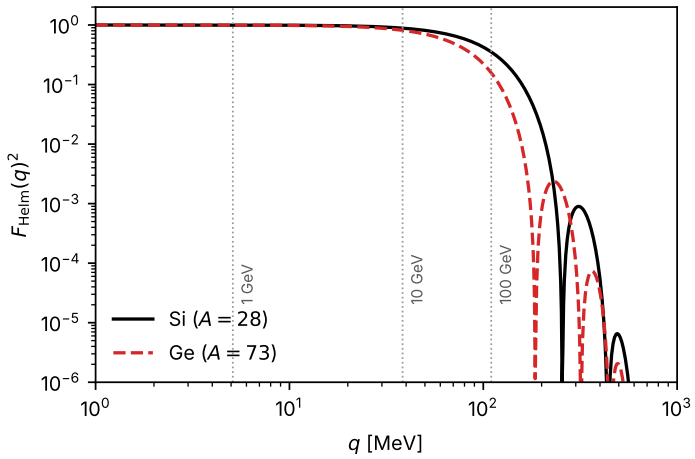


Figure 2.5: Squared Helm nuclear form factor $F_{\mathrm{Helm}}(q)^2 = [3j_1(qR_{\mathrm{nuc}})/(qR_{\mathrm{nuc}})]^2 \mathrm{e}^{-q^2 s_{\mathrm{skin}}^2}$ of silicon ($A = 28$, solid) and germanium ($A = 73$, the integer nearest the mean mass number 72.6 of natural germanium; dashed), against the momentum transfer q. Parameters of Lewin and Smith: $R_{\mathrm{nuc}}^2 = c_{\mathrm{nuc}}^2 + \frac{7}{3}\pi^2 a_{\mathrm{nuc}}^2 - 5s_{\mathrm{skin}}^2$, $c_{\mathrm{nuc}} = (1.23A^{1/3} - 0.60)\,\mathrm{fm}$, $a_{\mathrm{nuc}} = 0.52\,\mathrm{fm}$, $s_{\mathrm{skin}} = 0.9\,\mathrm{fm}$, which give $R_{\mathrm{nuc}} = 3.47\,\mathrm{fm}$ (Si) and 4.77 fm (Ge). Dotted lines: the maximum momentum transfer $2\mu_{\chi N} v_{\mathrm{max}}$ to a silicon nucleus for dark-matter masses of 1, 10 and 100 GeV, with $v_{\mathrm{max}} = 794.5\,\mathrm{km/s}$. The first zero is at $qR_{\mathrm{nuc}} = 4.49$, the first zero of j_1: $q = 256\,\mathrm{MeV}$ for Si and 186 MeV for Ge.

For sub-GeV dark matter the nucleus is a point of charge A, or Z for a dark photon. What limits the reach of nuclear scattering is kinematics, not nuclear structure: the small fraction $4m_\chi/M$ of the kinetic energy that reaches a heavy nucleus (Sect. 1.3). For dark-matter masses below about 10 MeV, a nucleus bound in a crystal no longer recoils freely, and the dark matter excites phonons instead (Chap. 5).

2.3 Second quantization

When dark matter lifts one electron of silicon across the gap, the other 10^{26} electrons stay where they were. Written as a wave function, the final state is still an antisymmetric function of all the electron coordinates, and so is the ground state. Written as a list of which single-particle orbitals are filled, the two states differ in two entries: one orbital emptied, one filled. Second quantization is the bookkeeping that works with such lists. Its operators create and destroy particles in orbitals, they take care of the exchange signs automatically, and every operator we need, from the density $n_e(\boldsymbol{q})$ that dark matter couples to, to the Coulomb Hamiltonian of electrons and nuclei, takes a compact form in them. Phonons and magnons later fit into the same language (Sects. 5.3 and 7.1). We follow the standard treatments of Refs. [11, Chap. 1], [12, Chap. 1] and [13, Chap. 1].

2.3.1 Identical particles

Two electrons cannot be told apart, and swapping their labels changes no measurement. Write $x_i = (\boldsymbol{r}_i, s_i)$ for the position and spin projection $s_i = \uparrow, \downarrow$ of particle i. The N-particle wave function $\Psi(x_1, \ldots, x_N)$ can then change under the exchange of two arguments only by a phase, and for the particles of this book that phase is $+1$ or -1. Electrons, and fermions generally, take -1: Ψ is antisymmetric under every exchange. Bosons, such as phonons and photons,

take +1.

For particles that do not interact, the eigenstates are built from single-particle orbitals $\psi_\gamma(x) = \langle x|\gamma\rangle$, where γ labels a complete orthonormal set of one-particle states, spin included: plane waves $\gamma = (\boldsymbol{k}, s)$ in a box, atomic orbitals, or the Bloch states of Sect. 3.2. The product $\psi_{\gamma_1}(x_1)\cdots\psi_{\gamma_N}(x_N)$ is not antisymmetric. Summing it over all permutations P of the arguments, with the sign $(-1)^P$ of each, fixes that and gives the Slater determinant

$$\Psi_{\gamma_1\ldots\gamma_N}(x_1,\ldots,x_N) = \frac{1}{\sqrt{N!}}\sum_P (-1)^P\,\psi_{\gamma_1}(x_{P1})\cdots\psi_{\gamma_N}(x_{PN}) = \frac{1}{\sqrt{N!}}\det\bigl[\psi_{\gamma_j}(x_i)\bigr]. \quad (2.29)$$

Two equal labels give two equal columns, and the determinant vanishes: this is the Pauli principle. Swapping two labels swaps two columns and flips the sign. Orthonormal orbitals make Ψ normalized, and Slater determinants with different sets of labels are orthogonal. The determinants built from all N-element subsets of the labels form a basis of the antisymmetric N-particle states, interacting or not. A correlated ground state is a superposition of many of them.

A determinant is fixed, up to its sign, by which orbitals it contains. We therefore fix an order of the labels once and for all, and write the state as a list of occupation numbers,

$$|n_{\gamma_1}, n_{\gamma_2}, n_{\gamma_3}, \ldots\rangle, \qquad n_\gamma \in \{0,1\},$$

with the determinant's columns in the fixed order. For bosons the same construction uses the permanent (all signs +), any $n_\gamma \geq 0$ is allowed, and a normalization factor $(\prod_\gamma n_\gamma!)^{-1/2}$ is needed because repeated labels make some permutations identical. The space spanned by the occupation-number states of every total number $N = 0, 1, 2, \ldots$ is Fock space. Its state with no particles is the vacuum $|0\rangle$. Fock space lets the particle number change, which the operators below need even when the physical number of electrons is fixed: removing an electron and putting it back elsewhere passes through the space with one electron fewer.

2.3.2 Fermion operators

The creation operator $c_\gamma^\dagger$ adds an electron in orbital γ and puts it at the front of the list:

$$c_\gamma^\dagger\,|\gamma_1\ldots\gamma_N\rangle = |\gamma\,\gamma_1\ldots\gamma_N\rangle, \quad (2.30)$$

where $|\gamma_1\ldots\gamma_N\rangle$ is the determinant of Eq. (2.29) as a ket. Every determinant is then a string of creation operators acting on the vacuum, $|\gamma_1\ldots\gamma_N\rangle = c_{\gamma_1}^\dagger c_{\gamma_2}^\dagger\cdots c_{\gamma_N}^\dagger|0\rangle$. Since swapping two labels flips the sign of the determinant, $c_\gamma^\dagger c_{\gamma'}^\dagger = -c_{\gamma'}^\dagger c_\gamma^\dagger$, and in particular $(c_\gamma^\dagger)^2 = 0$. The annihilation operator c_γ is the adjoint of $c_\gamma^\dagger$.

To see what c_γ does, act on an occupation-number state. Let $S_\gamma = \sum_{\gamma'<\gamma} n_{\gamma'}$ count the occupied orbitals that come before γ in the fixed order. To bring a new electron in γ from the front of the string to its own place, $c_\gamma^\dagger$ must pass S_γ creation operators. Since each pass costs a sign,

$$\begin{aligned} c_\gamma^\dagger\,|\ldots n_\gamma\ldots\rangle &= (-1)^{S_\gamma}\,(1-n_\gamma)\,|\ldots 1_\gamma\ldots\rangle,\\ c_\gamma\,|\ldots n_\gamma\ldots\rangle &= (-1)^{S_\gamma}\,n_\gamma\,|\ldots 0_\gamma\ldots\rangle. \end{aligned} \quad (2.31)$$

The second relation follows from the first: the only nonzero matrix element of $c_\gamma^\dagger$ takes 0_γ to 1_γ and equals $(-1)^{S_\gamma}$, and its complex conjugate is the matrix element of c_γ between the same states in the other direction. In words, c_γ empties a full orbital and kills an empty one, and neither operator can act twice on the same orbital.

The algebra follows in two lines. For $\gamma = \gamma'$, Eq. (2.31) leads to $c_\gamma c_\gamma^\dagger|\ldots n_\gamma\ldots\rangle = (1-n_\gamma)|\ldots n_\gamma\ldots\rangle$ and $c_\gamma^\dagger c_\gamma|\ldots n_\gamma\ldots\rangle = n_\gamma|\ldots n_\gamma\ldots\rangle$, and the two add to one. For $\gamma \neq \gamma'$,

the two orders of c_γ and $c^\dagger_{\gamma'}$ reach the same final state, but in one of them the second operator sees the occupation of the other orbital changed by one, and the sign factors differ by -1. With $\{A,B\} = AB + BA$,

$$\{c_\gamma, c^\dagger_{\gamma'}\} = \delta_{\gamma\gamma'}, \qquad \{c_\gamma, c_{\gamma'}\} = \{c^\dagger_\gamma, c^\dagger_{\gamma'}\} = 0. \tag{2.32}$$

These anticommutation relations contain the whole of Fermi statistics: the Pauli principle is $(c^\dagger_\gamma)^2 = 0$, and the exchange signs of Eq. (2.29) are the signs picked up by reordering operators. From here on we rarely need a determinant.

The product $c^\dagger_\gamma c_\gamma$ counts the electrons in γ: its eigenvalue on $|\ldots n_\gamma \ldots\rangle$ is n_γ. The total number operator is $\hat{N}_e = \sum_\gamma c^\dagger_\gamma c_\gamma$, with eigenvalue N_e on every state with N_e electrons. A Hamiltonian of independent electrons with orbital energies E_γ is $\sum_\gamma E_\gamma\, c^\dagger_\gamma c_\gamma$: each filled orbital adds its energy. The excitation of the opening paragraph, an electron moved from a filled orbital γ_v to an empty one γ_c, is $c^\dagger_{\gamma_c} c_{\gamma_v}$ acting on the ground state, and its energy is $E_{\gamma_c} - E_{\gamma_v}$ in this approximation.

The commutator of a bilinear with a single operator does most of the work later. By Eq. (2.32),

$$[c^\dagger_\gamma c_{\gamma'},\, c^\dagger_{\gamma''}] = \delta_{\gamma'\gamma''}\, c^\dagger_\gamma, \qquad [c^\dagger_\gamma c_{\gamma'},\, c_{\gamma''}] = -\delta_{\gamma\gamma''}\, c_{\gamma'}, \tag{2.33}$$

because $c^\dagger_\gamma c_{\gamma'} c^\dagger_{\gamma''} = \delta_{\gamma'\gamma''} c^\dagger_\gamma - c^\dagger_\gamma c^\dagger_{\gamma''} c_{\gamma'}$ and the last term equals $-c^\dagger_{\gamma''} c^\dagger_\gamma c_{\gamma'}$ after one swap of creation operators. The sign problems of fermions cancel in pairs here: a bilinear behaves like an ordinary operator that moves one particle. As a result, every bilinear commutes with $\hat{N}_e$, and operators built from bilinears conserve the number of electrons. And for independent electrons, $H_0 = \sum_\gamma E_\gamma c^\dagger_\gamma c_\gamma$, the Heisenberg equation $\mathrm{i}\,\mathrm{d}c^\dagger_\gamma/\mathrm{d}t = [c^\dagger_\gamma, H_0] = -E_\gamma c^\dagger_\gamma$ integrates to $\mathrm{e}^{\mathrm{i}H_0 t} c^\dagger_\gamma \mathrm{e}^{-\mathrm{i}H_0 t} = \mathrm{e}^{\mathrm{i}E_\gamma t} c^\dagger_\gamma$. An operator that moves an electron from γ' to γ then oscillates as $\mathrm{e}^{\mathrm{i}(E_\gamma - E_{\gamma'})t}$, at the energy of the transition it causes, which is how the energy denominators of Sect. 2.4 and the response functions of Sect. 4.1 arise.

2.3.3 Boson operators

Bosons work the same way with every sign turned positive. Generic boson operators d_γ, $d^\dagger_\gamma$ act on occupation numbers $n_\gamma = 0, 1, 2, \ldots$ as

$$d^\dagger_\gamma\,|\ldots n_\gamma \ldots\rangle = \sqrt{n_\gamma + 1}\,|\ldots n_\gamma + 1 \ldots\rangle, \qquad d_\gamma\,|\ldots n_\gamma \ldots\rangle = \sqrt{n_\gamma}\,|\ldots n_\gamma - 1 \ldots\rangle, \tag{2.34}$$

with no sign factor, because exchanging bosons costs nothing. The square roots come from the normalization factor of the symmetrized states: the state with $n_\gamma + 1$ quanta, built by $d^\dagger_\gamma$ from the state with n_γ, has norm $n_\gamma + 1$ before it is normalized. Then $d_\gamma d^\dagger_\gamma - d^\dagger_\gamma d_\gamma$ gives $(n_\gamma + 1) - n_\gamma = 1$ on every state, and operators of different modes commute:

$$[d_\gamma, d^\dagger_{\gamma'}] = \delta_{\gamma\gamma'}, \qquad [d_\gamma, d_{\gamma'}] = [d^\dagger_\gamma, d^\dagger_{\gamma'}] = 0. \tag{2.35}$$

The number of quanta in mode γ is $d^\dagger_\gamma d_\gamma$.

These are the ladder operators of the harmonic oscillator. An oscillator of frequency E_γ has $H = E_\gamma(d^\dagger_\gamma d_\gamma + \frac{1}{2})$, and its nth level reads as n quanta in a single mode. A set of uncoupled oscillators, such as the normal modes of a crystal, is a set of bosonic modes. This is why the quantized lattice vibrations of Sect. 5.3 are particles, phonons, with operators $b_{\boldsymbol{Q}\nu}$. The magnon operators $a_{\boldsymbol{k}\nu}$ of Sect. 7.1 are bosons in the same sense. Eq. (2.34) is the origin of the matrix elements $\sqrt{n}$ for absorption and $\sqrt{n+1}$ for emission quoted in Sect. 2.1, and Sect. 2.5 puts thermal occupations into n.

Unlike electrons, these bosons are created and destroyed in physical processes, and the Hamiltonians that describe this are not built from bilinears $d^\dagger d$ alone. An electron that emits or absorbs a phonon is described by a term of the form $c^\dagger_\gamma c_{\gamma'}(b_{\boldsymbol{Q}\nu} + b^\dagger_{-\boldsymbol{Q}\nu})$, with phonon wavevector $\boldsymbol{Q}$ and branch ν (Sect. 6.2): the electron bilinear conserves electron number and the boson operator changes the number of phonons by one. Absorption of bosonic dark matter fits the same pattern. If the dark-matter field is quantized, the oscillating amplitude ϕ_0 of Sect. 2.1 becomes an annihilation operator for the dark-matter mode. Its matrix element $\sqrt{n}$ between states with n and $n-1$ quanta leads to a rate proportional to the number of quanta in the mode, that is, to the density ρ_χ. The classical field of Sect. 2.1 is the limit of large n, where $\sqrt{n}$ and $\sqrt{n+1}$ no longer differ.

2.3.4 One-body operators and the density

The operator that dark matter couples to in electron scattering is $n_e(\boldsymbol{q}) = \sum_i \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$ (Eq. (A.8)). It is a sum over electrons of the same single-particle operator, and so are the kinetic energy $\sum_i \boldsymbol{p}_i^2/2m_e$ and the potential energy $\sum_i U_1(\boldsymbol{r}_i)$ in an external potential U_1. A one-body operator has this form, $\hat{O}_1 = \sum_i \hat{o}_i$, where $\hat{o}_i$ is a fixed one-particle operator $\hat{o}$ acting on particle i. In Fock space it reads

One-body operators

$$\hat{O}_1 = \sum_{i=1}^{N} \hat{o}_i \;=\; \sum_{\gamma\gamma'} \langle\gamma|\hat{o}|\gamma'\rangle\, c^\dagger_\gamma c_{\gamma'}. \tag{2.36}$$

The right-hand side is the same for every N, holds for bosons with $c \to d$, and has the same form in every orthonormal single-particle basis.

Acting on a determinant, $\sum_i \hat{o}_i$ replaces each occupied orbital in turn by $\hat{o}|\gamma_j\rangle = \sum_\gamma |\gamma\rangle\langle\gamma|\hat{o}|\gamma_j\rangle$ and leaves the rest alone. The bilinear $c^\dagger_\gamma c_{\gamma_j}$ does exactly this replacement: c_{γ_j} removes the orbital γ_j from its place in the string at a sign $(-1)^{j-1}$, $c^\dagger_\gamma$ puts γ at the front, and moving it back to place j costs the same sign again, which cancels the first. If γ is already occupied elsewhere, both sides vanish. Appendix D.2 writes out this argument, extends it to bosons, and proves the basis independence stated in the box.

Expressed through the field operator,

$$\hat{\psi}_s(\boldsymbol{r}) = \sum_\gamma \psi_\gamma(\boldsymbol{r}, s)\, c_\gamma, \qquad \{\hat{\psi}_s(\boldsymbol{r}), \hat{\psi}^\dagger_{s'}(\boldsymbol{r}')\} = \delta_{ss'}\, \delta^3(\boldsymbol{r} - \boldsymbol{r}'), \tag{2.37}$$

Eq. (2.36) takes a basis-free form. The anticommutator follows from Eq. (2.32) and the completeness of the orbitals, $\sum_\gamma \psi_\gamma(\boldsymbol{r}, s)\psi^*_\gamma(\boldsymbol{r}', s') = \delta_{ss'}\delta^3(\boldsymbol{r} - \boldsymbol{r}')$. We see that $\hat{\psi}^\dagger_s(\boldsymbol{r})$ creates an electron at the point $\boldsymbol{r}$ with spin s, and the density of electrons at $\boldsymbol{r}$, written $n_e(\boldsymbol{r}) = \sum_i \delta^3(\boldsymbol{r} - \boldsymbol{r}_i)$ in Appendix A.3, becomes $\sum_s \hat{\psi}^\dagger_s(\boldsymbol{r})\hat{\psi}_s(\boldsymbol{r})$. For an operator that is local in position and spin-independent, such as $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ or $U_1(\boldsymbol{r})$, Eq. (2.36) becomes $\sum_s \int \mathrm{d}^3r\, \hat{\psi}^\dagger_s(\boldsymbol{r})\, o(\boldsymbol{r})\, \hat{\psi}_s(\boldsymbol{r})$; the kinetic energy puts $-\nabla^2/2m_e$ between the two field operators.

In a box with plane waves, $\psi_{\boldsymbol{k}s}(\boldsymbol{r}, s') = \delta_{ss'}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}/\sqrt{\mathcal{V}}$ (Eq. (2.10)), and the field operator is $\hat{\psi}_s(\boldsymbol{r}) = \mathcal{V}^{-1/2}\sum_{\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\, c_{\boldsymbol{k}s}$. The matrix element of the density is a Kronecker delta,

$$\langle \boldsymbol{k}'s'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\boldsymbol{k}s\rangle = \frac{\delta_{ss'}}{\mathcal{V}} \int \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{k}')\cdot\boldsymbol{r}} = \delta_{\boldsymbol{k}',\boldsymbol{k}+\boldsymbol{q}}\, \delta_{ss'},$$

so

$$n_e(\boldsymbol{q}) = \sum_{\gamma\gamma'} \langle\gamma|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma'\rangle\, c^\dagger_\gamma c_{\gamma'} = \sum_{\boldsymbol{k}s} c^\dagger_{\boldsymbol{k}+\boldsymbol{q},s}\, c_{\boldsymbol{k}s}. \tag{2.38}$$

The density operator takes an electron from $\boldsymbol{k}$ and puts it at $\boldsymbol{k}+\boldsymbol{q}$, keeping its spin. That is the momentum bookkeeping of a scattering event: the dark matter goes from $\boldsymbol{p}$ to $\boldsymbol{p}-\boldsymbol{q}$, and Eq. (A.9) shows that its interaction with the electrons, between these two dark-matter states, is $\tilde{V}(\boldsymbol{q})\, n_e(\boldsymbol{q})/\mathscr{V}$. We can therefore write every electronic scattering rate of this book as a matrix element of $n_e(\boldsymbol{q})$ between many-electron states, squared and summed (Sect. 4.2). Note that $n_e(\boldsymbol{q})^\dagger = n_e(-\boldsymbol{q})$, and that $n_e(\boldsymbol{0}) = \hat{N}_e$. In a crystal the same formula in the Bloch basis gives the density vertex of Sect. 3.2.

The plane-wave form also shows the Pauli principle at work in a scattering event. Acting on a state in which the electrons fill all plane waves inside some region of $\boldsymbol{k}$ space, the term $c^\dagger_{\boldsymbol{k}+\boldsymbol{q},s} c_{\boldsymbol{k}s}$ survives only if $\boldsymbol{k}$ is filled and $\boldsymbol{k}+\boldsymbol{q}$ is empty. For a momentum transfer q small compared with the size of the filled region, only electrons within about q of its surface can take the momentum, and the rest are blocked. This is why the electrons of a metal respond to slow dark matter as a thin shell rather than as the whole electron density, a point Sects. 2.5 and 4.5 make quantitative. At q larger than the filled region no electron is blocked, and every term contributes. The same counting, with the filled region replaced by the valence bands and the empty one by the conduction bands, sets the electronic excitations of a semiconductor.

For independent electrons the many-body matrix element of $n_e(\boldsymbol{q})$ reduces to a single-particle one. Let $|\Phi_0\rangle$ be a determinant with orbital γ_v filled and γ_c empty, and $c^\dagger_{\gamma_c} c_{\gamma_v}|\Phi_0\rangle$ the excited determinant, whose bra is $\langle\Phi_0|c^\dagger_{\gamma_v} c_{\gamma_c}$. In $\langle\Phi_0|c^\dagger_{\gamma_v} c_{\gamma_c}\, c^\dagger_{\gamma} c_{\gamma'}|\Phi_0\rangle$ the bilinear $c^\dagger_{\gamma} c_{\gamma'}$ must undo the excitation, and only $\gamma=\gamma_c$, $\gamma'=\gamma_v$ survives, with weight one. With Eq. (2.38),

$$\langle\Phi_0|c^\dagger_{\gamma_v} c_{\gamma_c}\; n_e(\boldsymbol{q})\; |\Phi_0\rangle = \langle\gamma_c|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma_v\rangle, \tag{2.39}$$

which is the transition form factor $f_{i\to f}(\boldsymbol{q})$ of Sect. 2.2 for the electron that moves, whatever the other electrons do. The spectators enter only through which orbitals are blocked. Interactions spoil this simple picture, and Sects. 4.2 and 4.6 show what replaces it.

A one-body potential follows from the inverse transform. With Eqs. (A.6) and (A.10), $U_1(\boldsymbol{r}) = \mathscr{V}^{-1}\sum_{\boldsymbol{q}} \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} U_1(\boldsymbol{q})$, and $\sum_i U_1(\boldsymbol{r}_i) = \mathscr{V}^{-1}\sum_{\boldsymbol{q}} U_1(\boldsymbol{q})\, n_e(-\boldsymbol{q})$: a potential with a Fourier component at $\boldsymbol{q}$ moves electrons by $-\boldsymbol{q}$. For plane waves the kinetic energy is diagonal, $\sum_{\boldsymbol{k}s} (k^2/2m_e)\, c^\dagger_{\boldsymbol{k}s} c_{\boldsymbol{k}s}$. Table 2.1 collects these forms with the two-body operators derived below.

2.3.5 Change of basis

A calculation often starts in one basis and ends in another: plane waves for the Coulomb interaction, Bloch states for the band structure (Sect. 3.2), localized Wannier orbitals for interpolation (Sect. 3.6). A new orthonormal basis $|\tilde{\gamma}\rangle = \sum_\gamma |\gamma\rangle\langle\gamma|\tilde{\gamma}\rangle$ comes with new operators, defined by the same rule. A creation operator adds a particle in a given orbital, so it expands like the orbital:

$$c^\dagger_{\tilde{\gamma}} = \sum_\gamma \langle\gamma|\tilde{\gamma}\rangle\, c^\dagger_\gamma, \qquad c_{\tilde{\gamma}} = \sum_\gamma \langle\tilde{\gamma}|\gamma\rangle\, c_\gamma. \tag{2.40}$$

The coefficients form a unitary matrix, so the anticommutators survive,

$$\{c_{\tilde{\gamma}}, c^\dagger_{\tilde{\gamma}'}\} = \sum_\gamma \langle\tilde{\gamma}|\gamma\rangle\langle\gamma|\tilde{\gamma}'\rangle = \delta_{\tilde{\gamma}\tilde{\gamma}'}.$$

Inserting Eq. (2.40) into Eq. (2.36) and using completeness twice leads to $\sum_{\tilde{\gamma}\tilde{\gamma}'}\langle\tilde{\gamma}|\hat{o}|\tilde{\gamma}'\rangle c^\dagger_{\tilde{\gamma}} c_{\tilde{\gamma}'}$: one-body operators look the same in every basis, with the matrix elements of the new one. The field operator of Eq. (2.37) is the special case in which the new basis is the position eigenbasis, $\langle\gamma|\boldsymbol{r}s\rangle = \psi^*_\gamma(\boldsymbol{r},s)$. In the basis that diagonalizes $\hat{o}$ the bilinear collapses to $\sum_\gamma o_\gamma c^\dagger_\gamma c_\gamma$. A Slater

Table 2.1: Operators used in this book in first quantization (sums over electrons i and nuclei I) and in second quantization, in a general orthonormal single-particle basis $\{|\gamma\rangle\}$ (spin included in γ) and in box-normalized plane waves $|\boldsymbol{k}s\rangle$ of a box of volume $\mathcal{V}$. Fourier transforms carry $\mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$; $v_{\mathrm{C}}(k) = e^2/k^2$ is the Heaviside–Lorentz Coulomb kernel, $e^2 = 4\pi\alpha$; Z_κ is the charge number of nuclear species κ and $n_\kappa(\boldsymbol{k}) = \sum_{I\in\kappa} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{X}_I}$ its density, with $\boldsymbol{X}_I$ the position of nucleus I. Two-body elements are $\langle\gamma_1\gamma_2|U_2|\gamma_3\gamma_4\rangle = \int \mathrm{d}x\,\mathrm{d}x'\,\psi^*_{\gamma_1}(x)\psi^*_{\gamma_2}(x')U_2(\boldsymbol{r}-\boldsymbol{r}')\psi_{\gamma_3}(x)\psi_{\gamma_4}(x')$, with $x = (\boldsymbol{r}, s)$ and $\int \mathrm{d}x = \sum_s \int \mathrm{d}^3r$. The $\boldsymbol{k} = 0$ Coulomb terms are removed by charge neutrality (Worked example 2.5).

First quantization	General basis	Plane waves
$N_e = \sum_i 1$	$\hat{N}_e = \sum_\gamma c^\dagger_\gamma c_\gamma$	$\sum_{\boldsymbol{k}s} c^\dagger_{\boldsymbol{k}s} c_{\boldsymbol{k}s}$
$\sum_i \boldsymbol{p}_i^2/2m_e$	$\sum_{\gamma\gamma'}\langle\gamma\|\frac{\boldsymbol{p}^2}{2m_e}\|\gamma'\rangle c^\dagger_\gamma c_{\gamma'}$	$\sum_{\boldsymbol{k}s} \frac{k^2}{2m_e} c^\dagger_{\boldsymbol{k}s} c_{\boldsymbol{k}s}$
$\sum_i U_1(\boldsymbol{r}_i)$	$\sum_{\gamma\gamma'}\langle\gamma\|U_1\|\gamma'\rangle c^\dagger_\gamma c_{\gamma'}$	$\frac{1}{\mathcal{V}}\sum_{\boldsymbol{q}} U_1(\boldsymbol{q})\, n_e(-\boldsymbol{q})$
$n_e(\boldsymbol{q}) = \sum_i \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$	$\sum_{\gamma\gamma'}\langle\gamma\|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\|\gamma'\rangle c^\dagger_\gamma c_{\gamma'}$	$\sum_{\boldsymbol{k}s} c^\dagger_{\boldsymbol{k}+\boldsymbol{q},s} c_{\boldsymbol{k}s}$
dark matter $\boldsymbol{p} \to \boldsymbol{p} - \boldsymbol{q}$: $\frac{1}{\mathcal{V}}\tilde{V}(\boldsymbol{q})\sum_i \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$	$\frac{1}{\mathcal{V}}\tilde{V}(\boldsymbol{q})\, n_e(\boldsymbol{q})$	$\frac{1}{\mathcal{V}}\tilde{V}(\boldsymbol{q})\sum_{\boldsymbol{k}s} c^\dagger_{\boldsymbol{k}+\boldsymbol{q},s} c_{\boldsymbol{k}s}$
$\frac{1}{2}\sum_{i\neq j} U_2(\boldsymbol{r}_i - \boldsymbol{r}_j)$	$\frac{1}{2}\sum\langle\gamma_1\gamma_2\|U_2\|\gamma_3\gamma_4\rangle$ $\times c^\dagger_{\gamma_1} c^\dagger_{\gamma_2} c_{\gamma_4} c_{\gamma_3}$	$\frac{1}{2\mathcal{V}}\sum_{\boldsymbol{k}\boldsymbol{p}\boldsymbol{p}'ss'} U_2(\boldsymbol{k})$ $\times c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s} c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'} c_{\boldsymbol{p}'s'} c_{\boldsymbol{p}s}$
$\sum_{i<j} \dfrac{e^2}{4\pi\lvert\boldsymbol{r}_i - \boldsymbol{r}_j\rvert}$	$U_2 \to e^2/(4\pi\lvert\boldsymbol{r} - \boldsymbol{r}'\rvert)$	$\frac{1}{2\mathcal{V}}\sum_{\boldsymbol{k}\neq 0} v_{\mathrm{C}}(k)\big[n_e(\boldsymbol{k})^\dagger n_e(\boldsymbol{k}) - \hat{N}_e\big]$
$-\sum_{iI} \dfrac{Z_{\kappa(I)}e^2}{4\pi\lvert\boldsymbol{r}_i - \boldsymbol{X}_I\rvert}$	$\sum_{\gamma\gamma'}\langle\gamma\|U_1\|\gamma'\rangle c^\dagger_\gamma c_{\gamma'}$, $U_1(\boldsymbol{r}) = -\sum_I \dfrac{Z_{\kappa(I)}e^2}{4\pi\lvert\boldsymbol{r} - \boldsymbol{X}_I\rvert}$	$-\frac{1}{\mathcal{V}}\sum_{\boldsymbol{k}\neq 0} v_{\mathrm{C}}(k)\sum_\kappa Z_\kappa\, n_\kappa(\boldsymbol{k})\, n_e(\boldsymbol{k})^\dagger$

determinant in the old basis becomes a sum of determinants in the new one (Appendix D.2), which is why the occupation-number language, and not the wave function, is the economical one.

Two orbitals, one on each of two neighboring atoms, show what this means for silicon's covalent bond. Take two orthonormal orbitals $|A\rangle$ and $|B\rangle$ on the two atoms, and the bonding and antibonding combinations $|\pm\rangle = (|A\rangle \pm |B\rangle)/\sqrt{2}$, a caricature of the covalent bond of silicon. Two electrons of opposite spin in the bonding orbital form one determinant, $c^\dagger_{+\uparrow}c^\dagger_{+\downarrow}|0\rangle$. By Eq. (2.40) it is

$$c^\dagger_{+\uparrow}c^\dagger_{+\downarrow}|0\rangle = \tfrac{1}{2}(c^\dagger_{A\uparrow}c^\dagger_{A\downarrow} + c^\dagger_{B\uparrow}c^\dagger_{B\downarrow} + c^\dagger_{A\uparrow}c^\dagger_{B\downarrow} + c^\dagger_{B\uparrow}c^\dagger_{A\downarrow})|0\rangle,$$

four determinants in the atomic basis: two with both electrons on one atom, and two with one electron on each. The same state is simple in one basis and a superposition in the other, and the operators of Eqs. (2.36) and (2.42) handle both without any change of form.

2.3.6 Two-body operators and normal ordering

The Coulomb repulsion $\frac{1}{2}\sum_{i\neq j} U_2(\boldsymbol{r}_i - \boldsymbol{r}_j)$, with $U_2(\boldsymbol{r}) = e^2/(4\pi r)$, is a sum over pairs, and the condition $i \neq j$ matters: an electron does not repel itself. To see what it becomes, split the pair sum into all pairs minus the diagonal, $\sum_{i\neq j} = \sum_{i,j} - \sum_{i=j}$, and write each sum through the density $n_e(\boldsymbol{r}) = \sum_i \delta^3(\boldsymbol{r} - \boldsymbol{r}_i)$:

$$\frac{1}{2}\sum_{i\neq j} U_2(\boldsymbol{r}_i - \boldsymbol{r}_j) = \frac{1}{2}\int \mathrm{d}^3r\,\mathrm{d}^3r'\,U_2(\boldsymbol{r}-\boldsymbol{r}')\big[n_e(\boldsymbol{r})n_e(\boldsymbol{r}') - \delta^3(\boldsymbol{r}-\boldsymbol{r}')\,n_e(\boldsymbol{r})\big].$$

Each density is now the bilinear $\sum_s \hat\psi^\dagger_s\hat\psi_s$. Moving the inner annihilation operator in $\hat\psi^\dagger_s(\boldsymbol{r})\hat\psi_s(\boldsymbol{r})\hat\psi^\dagger_{s'}(\boldsymbol{r}')\hat\psi_{s'}(\boldsymbol{r}')$ to the right with Eq. (2.37) produces a term $\delta_{ss'}\delta^3(\boldsymbol{r}-\boldsymbol{r}')\hat\psi^\dagger_s(\boldsymbol{r})\hat\psi_s(\boldsymbol{r})$, which cancels the self-interaction term exactly, and leaves, for $\hat O_2 = \frac{1}{2}\sum_{i\neq j} U_2(\boldsymbol{r}_i - \boldsymbol{r}_j)$,

$$\hat O_2 = \frac{1}{2}\sum_{ss'}\int \mathrm{d}^3r\,\mathrm{d}^3r'\,U_2(\boldsymbol{r}-\boldsymbol{r}')\,\hat\psi^\dagger_s(\boldsymbol{r})\hat\psi^\dagger_{s'}(\boldsymbol{r}')\hat\psi_{s'}(\boldsymbol{r}')\hat\psi_s(\boldsymbol{r}). \tag{2.41}$$

All creation operators stand to the left of all annihilation operators. An operator product rearranged into this order, with the sign of the permutation for fermions, is called normal ordered and is written $:\cdots:$. Thus $\hat O_2 = \frac{1}{2}\int\!\int U_2 :n_e(\boldsymbol{r})n_e(\boldsymbol{r}'):$, and normal ordering is what removes the self-interaction. Since a normal-ordered product of two annihilation operators vanishes on any state with fewer than two particles, a single electron feels no Coulomb energy of its own, as it should, and the vacuum has none either. Expanding the field operators in a basis gives the general form of Table 2.1,

$$\hat O_2 = \frac{1}{2}\sum_{\gamma_1\gamma_2\gamma_3\gamma_4}\langle\gamma_1\gamma_2|U_2|\gamma_3\gamma_4\rangle\,c^\dagger_{\gamma_1}c^\dagger_{\gamma_2}c_{\gamma_4}c_{\gamma_3}. \tag{2.42}$$

The order of the last two indices matters: particle 2 is annihilated first and created last, so the two operators of each particle sit symmetrically around the middle. Reversing them flips the sign. Appendix D.2 proves Eq. (2.42) for any two-body operator, spin-dependent or not, by applying the one-body result twice, which is the operator form of the split $\sum_{i,j} - \sum_{i=j}$ used above.

2.3.7 Coulomb interaction in momentum space

The Coulomb potential $e^2/(4\pi r)$ has the Fourier transform (Eq. (A.3))

$$v_{\mathrm{C}}(k) = \frac{e^2}{k^2} = \frac{4\pi\alpha}{k^2} \qquad \text{(Heaviside–Lorentz)}, \tag{2.43}$$

the $m_V \to 0$ limit of the Yukawa transform of Eq. (C.81). Condensed-matter sources usually work in Gaussian units, where the electron charge e_{G} obeys $e_{\mathrm{G}}^2 = \alpha$ and the same kernel reads $4\pi e_{\mathrm{G}}^2/k^2$; SI sources write $e_{\mathrm{SI}}^2/(\varepsilon_0^{\mathrm{SI}} k^2)$, with $\varepsilon_0^{\mathrm{SI}}$ the vacuum permittivity and $\hbar = c = 1$. All three are the same number, by Eq. (A.5): a formula converts by replacing our e^2 with $4\pi e_{\mathrm{G}}^2$ or $e_{\mathrm{SI}}^2/\varepsilon_0^{\mathrm{SI}}$.

In the box, the inverse transform is a sum, $e^2/(4\pi|\boldsymbol{r}|) = \mathcal{V}^{-1}\sum_{\boldsymbol{k}} v_{\mathrm{C}}(k)\,\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$, and the pair sum factorizes into densities at once. Using the split $\sum_{i\neq j} = \sum_{i,j} - \sum_{i=j}$ again, with $\sum_{i,j}\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{r}_i-\boldsymbol{r}_j)} = n_e(-\boldsymbol{k})\,n_e(\boldsymbol{k})$ and $\sum_{i=j} 1 = N_e$,

$$H_{ee} = \frac{1}{2}\sum_{i\neq j}\frac{e^2}{4\pi|\boldsymbol{r}_i-\boldsymbol{r}_j|} = \frac{1}{2\mathcal{V}}\sum_{\boldsymbol{k}} v_{\mathrm{C}}(k)\big[n_e(\boldsymbol{k})^\dagger n_e(\boldsymbol{k}) - \hat{N}_e\big]. \tag{2.44}$$

The energy of the electrons is a sum over Fourier components of the density fluctuation, each weighted by $v_{\mathrm{C}}(k)$, with the self-energy of each point charge subtracted. Dimensions check: v_{C} is in eV^{-2}, $1/\mathcal{V}$ in eV^3, and the densities are numbers. To get the plane-wave form, insert Eq. (2.38) with $n_e(\boldsymbol{k})^\dagger = \sum_{\boldsymbol{p}'s'} c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'}c_{\boldsymbol{p}'s'}$ and normal order. The anticommutator $c_{\boldsymbol{p}'s'}c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s} = \delta_{\boldsymbol{p}',\boldsymbol{p}+\boldsymbol{k}}\delta_{ss'} - c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s}c_{\boldsymbol{p}'s'}$ gives a one-body piece equal to $\hat{N}_e$, which cancels the subtraction. In the two-body piece that remains, swapping the two creation operators removes the minus sign:

$$n_e(\boldsymbol{k})^\dagger n_e(\boldsymbol{k}) - \hat{N}_e = \sum_{\boldsymbol{p}\boldsymbol{p}'ss'} c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s}\,c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'}\,c_{\boldsymbol{p}'s'}\,c_{\boldsymbol{p}s}. \tag{2.45}$$

This is the matrix element of Eq. (2.42) for plane waves: two electrons with momenta $\boldsymbol{p}$ and $\boldsymbol{p}'$ exchange a momentum $\boldsymbol{k}$, each keeps its spin, and the amplitude is $v_{\mathrm{C}}(k)/\mathcal{V}$. Without normal ordering, the $\hat{N}_e$ term would carry $\mathcal{V}^{-1}\sum_{\boldsymbol{k}} v_{\mathrm{C}}(k) = \int \mathrm{d}^3k\,(2\pi)^{-3}e^2/k^2$, which diverges linearly at large k: the infinite Coulomb self-energy of a point charge, which has nothing to do with the solid.

The exchange sign shows up as soon as we take a matrix element. Let two electrons start in $c^\dagger_{\boldsymbol{p}s}c^\dagger_{\boldsymbol{p}'s'}|0\rangle$ and end in $c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s}c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'}|0\rangle$, with $\boldsymbol{k}\neq 0$. The two annihilation operators of Eq. (2.45) can remove the initial pair in two ways, with signs $+1$ and -1. Each of these pairs with two matchings of the creation operators to the final pair, one at momentum exchange $\boldsymbol{k}$ and the other at $\boldsymbol{p}'-\boldsymbol{p}-\boldsymbol{k}$. The four terms add to $2[v_{\mathrm{C}}(k) - \delta_{ss'}v_{\mathrm{C}}(|\boldsymbol{p}'-\boldsymbol{p}-\boldsymbol{k}|)]/\mathcal{V}$, and the factor $\frac{1}{2}$ of Eq. (2.42) cancels the 2, which counts the two equivalent labelings of the pair. The result is

$$\langle \boldsymbol{p}+\boldsymbol{k}\,s,\ \boldsymbol{p}'-\boldsymbol{k}\,s'|H_{ee}|\boldsymbol{p}s,\ \boldsymbol{p}'s'\rangle = \frac{1}{\mathcal{V}}\Big[v_{\mathrm{C}}(k) - \delta_{ss'}\,v_{\mathrm{C}}(|\boldsymbol{p}'-\boldsymbol{p}-\boldsymbol{k}|)\Big]. \tag{2.46}$$

The first term is the direct process: electron $\boldsymbol{p}$ takes the momentum $\boldsymbol{k}$. The second is the exchange process, in which the electrons end up in the same final orbitals but each in the other's: electron $\boldsymbol{p}$ goes to $\boldsymbol{p}'-\boldsymbol{k}$, so the momentum it takes is $\boldsymbol{p}'-\boldsymbol{p}-\boldsymbol{k}$. Because the two processes lead to the same final state, their amplitudes add, and Fermi statistics attaches a minus sign to the exchange. Electrons with opposite spins, however, can be told apart by their spins, and for them the exchange term is absent. The exchange term lowers the energy of parallel spins, and it is the origin of the exchange energy of Sect. 3.4 and of the exchange coupling of magnetic order (Sect. 7.1).

The same counting gives the Coulomb energy of a single determinant, the starting point of the Hartree–Fock method. In $\langle\Phi_0|c^\dagger_{\gamma_1}c^\dagger_{\gamma_2}c_{\gamma_4}c_{\gamma_3}|\Phi_0\rangle$ the annihilation operators must remove two occupied orbitals and the creation operators must put the same two back, either in the same places or swapped. For orthonormal orbitals Eq. (2.42) then reduces to

$$\langle\Phi_0|\hat{O}_2|\Phi_0\rangle = \frac{1}{2}\sum_{\gamma,\gamma'}\sum_{\text{occupied}}\Big[\langle\gamma\gamma'|U_2|\gamma\gamma'\rangle - \langle\gamma\gamma'|U_2|\gamma'\gamma\rangle\Big],$$

a direct (Hartree) term, the classical electrostatic energy of the charge distribution, minus an exchange (Fock) term. The diagonal $\gamma = \gamma'$ cancels between the two: no orbital interacts with itself. Sect. 3.4 minimizes this energy and then improves on it.

2.3.8 Coulomb Hamiltonian of a solid

Eq. (2.44) has one more infinity, at $\boldsymbol{k} = 0$, where $v_{\mathrm{C}}(0)$ diverges. It is the energy of a uniform charge spread over the whole box, which grows faster than the volume, and in a neutral system it is cancelled by the positive charge. The simplest model of a metal or of a valence electron gas, jellium, replaces the nuclei by a uniform positive background. Jellium (used again in Sects. 2.5 and 4.5) shows how the cancellation works, in Worked example 2.5 below.

How large is the Coulomb term? A rough answer comes from the mean spacing $d = \boldsymbol{n}_e^{-1/3}$ of the electrons. For the four valence electrons per atom of silicon, $\boldsymbol{n}_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$ (Appendix A.2), and $d = 1.71\,\text{Å} = 3.2\,a_0$, with $a_0 = 1/(\alpha m_e)$ the Bohr radius (Sect. 2.2). The Coulomb energy of two neighbors, α/d, is then about 8.4 eV. The kinetic energy must respect the Pauli principle, and for a free gas at this density its mean is $\frac{3}{5}E_{\mathrm{F}} = 7.5\,\mathrm{eV}$ (Sect. 2.5), about the same. The field measures the ratio by $r_s = (3/4\pi\boldsymbol{n}_e)^{1/3}/a_0$, the radius per electron in units of a_0: the Coulomb energy scales as $1/r_s$ and the kinetic energy as $1/r_s^2$, and their ratio grows as r_s. For silicon's valence density $r_s = 2.0$. The interaction is therefore not a small perturbation of free electrons. It is tamed instead by screening (Sect. 4.4) and by the self-consistent fields of Sect. 3.4, and all of them start from the Coulomb Hamiltonian of Worked example 2.5.

Worked example 2.5: The Coulomb Hamiltonian of jellium and of a solid

Take N_e electrons in the box with a uniform positive background of number density $N_e/\mathcal{V}$ and charge $+e$ per unit, so the system is neutral. To give the $\boldsymbol{k} = 0$ terms a meaning, screen the Coulomb potential at a large length $1/k_{\mathrm{reg}}$, so $e^2/(4\pi r) \to e^2\mathrm{e}^{-k_{\mathrm{reg}}r}/(4\pi r)$ and $v_{\mathrm{C}}(k) \to e^2/(k^2 + k_{\mathrm{reg}}^2)$ by Eq. (C.81). We take $\mathcal{V} \to \infty$ at fixed $N_e/\mathcal{V}$ first, with $\mathcal{V}^{1/3} \gg 1/k_{\mathrm{reg}}$, and $k_{\mathrm{reg}} \to 0$ at the end. The integral of the screened potential over all space is $v_{\mathrm{C}}(0) = e^2/k_{\mathrm{reg}}^2$. The uniform background has a Fourier component only at $\boldsymbol{k} = 0$, where three terms meet:

- the background with itself, $+\frac{1}{2}(N_e/\mathcal{V})^2\,\mathcal{V}\,v_{\mathrm{C}}(0) = +N_e^2\,v_{\mathrm{C}}(0)/2\mathcal{V}$;
- the electrons with the background, $-(N_e/\mathcal{V})\sum_i \int \mathrm{d}^3r\, e^2\mathrm{e}^{-k_{\mathrm{reg}}|\boldsymbol{r}_i-\boldsymbol{r}|}/(4\pi|\boldsymbol{r}_i-\boldsymbol{r}|) = -N_e^2\,v_{\mathrm{C}}(0)/\mathcal{V}$;
- the $\boldsymbol{k} = 0$ term of Eq. (2.44), $+v_{\mathrm{C}}(0)\,(N_e^2 - N_e)/2\mathcal{V}$, since $n_e(\boldsymbol{0}) = \hat{N}_e = N_e$.

The terms in N_e^2, each of order $N_e \cdot (N_e/\mathcal{V})/k_{\mathrm{reg}}^2$ and so proportional to the volume, cancel exactly. What is left is $-N_e\,v_{\mathrm{C}}(0)/2\mathcal{V} = -N_e e^2/(2Vk_{\mathrm{reg}}^2)$, and per electron it vanishes as $\mathcal{V} \to \infty$. Then $k_{\mathrm{reg}} \to 0$ restores $v_{\mathrm{C}}(k) = e^2/k^2$ at every $\boldsymbol{k} \neq 0$. With the kinetic energy

of Table 2.1 and Eq. (2.45), the jellium Hamiltonian is [11, Chap. 2]

$$H_{\text{jel}} = \sum_{\boldsymbol{k}s} \frac{k^2}{2m_e}\, c^\dagger_{\boldsymbol{k}s} c_{\boldsymbol{k}s} + \frac{1}{2\mathscr{V}} \sum_{\boldsymbol{k}\neq 0} \sum_{\boldsymbol{p}\boldsymbol{p}' ss'} v_{\text{C}}(k)\, c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s}\, c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'}\, c_{\boldsymbol{p}'s'}\, c_{\boldsymbol{p}s}. \tag{2.47}$$

The background has disappeared, apart from the instruction to drop $\boldsymbol{k} = 0$. The order of limits is part of the answer: with $k_{\text{reg}} \to 0$ first, the screening length would exceed the box, and the three terms would each be infinite.
A solid replaces the background by point nuclei. Nucleus I sits at $\boldsymbol{X}_I$, belongs to a species $\kappa(I)$ with charge number Z_κ and mass M_κ (for silicon one species, $Z_\kappa = 14$; species are treated systematically in Sect. 5.2), and the species density is $n_\kappa(\boldsymbol{k}) = \sum_{I\in\kappa} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{X}_I}$. Every pair of point charges interacts through the product of their charge numbers, -1 for an electron and Z_κ for a nucleus, times $e^2/(4\pi r)$ at separation r. Splitting the pair sum as in Eq. (2.44) leads to

$$H_{\text{Coul}} = \frac{1}{2\mathscr{V}} \sum_{\boldsymbol{k}} v_{\text{C}}(k) \Big[n_Z(\boldsymbol{k})^\dagger n_Z(\boldsymbol{k}) - N_e - \sum_\kappa Z_\kappa^2 N_\kappa \Big], \qquad n_Z(\boldsymbol{k}) = -n_e(\boldsymbol{k}) + \sum_\kappa Z_\kappa\, n_\kappa(\boldsymbol{k}),$$

where N_κ counts the nuclei of species κ. Neutrality, $\sum_\kappa Z_\kappa N_\kappa = N_e$, now holds as an operator identity, $n_Z(\boldsymbol{0}) = 0$. The $\boldsymbol{k} = 0$ term reduces to the self-energy subtraction, $-v_{\text{C}}(0)(N_e + \sum_\kappa Z_\kappa^2 N_\kappa)/2\mathscr{V}$, which vanishes per particle in the same limit as before. The nuclei are their own neutralizing background. Expanding $n_Z^\dagger n_Z$ leads to three pieces. The electron–electron piece is Eq. (2.45). The cross terms $-Z_\kappa[n_e(\boldsymbol{k})^\dagger n_\kappa(\boldsymbol{k}) + n_\kappa(\boldsymbol{k})^\dagger n_e(\boldsymbol{k})]$ are equal after $\boldsymbol{k} \to -\boldsymbol{k}$, since v_{C} is even and electron and nuclear coordinates commute, and add to twice the first. The nuclear piece keeps its own self-energy subtraction.

Collecting the pieces, with the nuclei left in first quantization because their quantization belongs to the lattice (Sect. 5.3):

Coulomb Hamiltonian of electrons and point nuclei

$$\begin{aligned} H_C = \sum_{\boldsymbol{k}s} \frac{k^2}{2m_e} c^\dagger_{\boldsymbol{k}s} c_{\boldsymbol{k}s} - \sum_I \frac{\nabla_I^2}{2M_{\kappa(I)}} + \frac{1}{2\mathscr{V}} \sum_{\boldsymbol{k}\neq 0} v_{\text{C}}(k) \Big[\sum_{\boldsymbol{p}\boldsymbol{p}' ss'} c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s} c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'} c_{\boldsymbol{p}'s'} c_{\boldsymbol{p}s} \\ - 2\sum_\kappa Z_\kappa\, n_\kappa(\boldsymbol{k}) \sum_{\boldsymbol{p}s} c^\dagger_{\boldsymbol{p}-\boldsymbol{k},s} c_{\boldsymbol{p}s} + \sum_{\kappa\kappa'} Z_\kappa Z_{\kappa'} \big(n_\kappa(\boldsymbol{k})^\dagger n_{\kappa'}(\boldsymbol{k}) - \delta_{\kappa\kappa'} N_\kappa \big) \Big]. \end{aligned} \tag{2.48}$$

The three terms in the bracket are the electron–electron, electron–nucleus and nucleus–nucleus interactions; ∇_I acts on $\boldsymbol{X}_I$, and $\boldsymbol{k} = 0$ is removed by neutrality.

The powers of $\mathscr{V}$ in Eq. (2.48) are a useful check, in the spirit of Sect. 2.1. The energy of a large system must be extensive, proportional to $\mathscr{V}$ at fixed density. In the electron–electron term, take the expectation value in a determinant of plane waves. Only terms that put both electrons back where they started contribute. The direct one has $\boldsymbol{k} = 0$ and is gone with the background; the exchange one, the diagonal case of Eq. (2.46) with $\boldsymbol{k} = \boldsymbol{p}' - \boldsymbol{p}$, leaves $-(1/2\mathscr{V}) \sum_{\boldsymbol{p}\neq\boldsymbol{p}'} v_{\text{C}}(|\boldsymbol{p} - \boldsymbol{p}'|)$ for each spin, summed over occupied momenta. By Eq. (2.10) each sum over momenta brings a factor $\mathscr{V}$, and we see that the total is $\mathscr{V}^2/\mathscr{V} = \mathscr{V}$ times an integral over the occupied region: extensive, as it must be. The same count applied to the subtracted self-energy, $(1/2\mathscr{V}) \sum_{\boldsymbol{k}} v_{\text{C}}(k) N_e$, yields N_e times a divergent integral, the reason it is removed rather than kept. A stray power of $\mathscr{V}$ in a Coulomb expression nearly always

means a misplaced $\boldsymbol{k} = 0$ term or a forgotten box normalization.

The electron–nucleus term is the one-body potential $\sum_i U_1(\boldsymbol{r}_i)$ of Table 2.1 with $U_1(\boldsymbol{k}) = -v_{\mathrm{C}}(k) \sum_\kappa Z_\kappa n_\kappa(\boldsymbol{k})$. As a check, its matrix element for an electron going from $\boldsymbol{p}$ to $\boldsymbol{p} - \boldsymbol{k}$,

$$\langle \boldsymbol{p} - \boldsymbol{k} | U_1 | \boldsymbol{p} \rangle = -\frac{1}{\mathcal{V}} \sum_I Z_{\kappa(I)}\, v_{\mathrm{C}}(k)\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{X}_I},$$

is what Eq. (A.9) gives for the potential $-\sum_I Z_{\kappa(I)} e^2/(4\pi|\boldsymbol{r} - \boldsymbol{X}_I|)$, with the nuclei in the role of the dark matter. When the nuclei are held at fixed positions, $n_\kappa(\boldsymbol{k})$ is a number, and for a perfect crystal it is nonzero only at reciprocal-lattice vectors, which leads to the periodic potential of the band structure (Sects. 3.1 and 3.2).

Momentum conservation reads off Eq. (2.48) directly. In every electron–electron term the electrons gain $\boldsymbol{k}$ and lose $\boldsymbol{k}$, and the total electron momentum is conserved by the Coulomb repulsion. The electron–nucleus term moves electron momentum by $-\boldsymbol{k}$ and multiplies by $n_\kappa(\boldsymbol{k})$, which, for mobile nuclei, shifts the nuclear momenta by $+\boldsymbol{k}$: the total momentum of electrons and nuclei is conserved. With the nuclei clamped in a crystal, the electrons can gain or lose only reciprocal-lattice vectors, and their momentum is conserved only modulo these. This is the origin of crystal momentum (Sect. 3.1).

Eq. (2.48) is the starting point of everything the book computes about the target. It is nonrelativistic and treats the nuclei as point charges; spin–orbit coupling and the relativistic corrections of core electrons are added where they matter (Sects. 3.3 and 3.5). Sect. 5.1 separates the motion of the nuclei from that of the electrons, and the dark-matter coupling enters only through $n_e(\boldsymbol{q})$ for electrons, or $n_\kappa(\boldsymbol{q})$ for nuclei. No one, however, solves Eq. (2.48) for 10^{26} electrons as it stands. Density-functional theory replaces the electron–electron term by an effective one-body potential (Sect. 3.4), and the codes that do this in practice are described in Appendix E.1.

2.4 Second-order processes

A photon of 1.2 eV carries a momentum of about $6 \times 10^{-4}\,\text{Å}^{-1}$, while the lowest conduction states of silicon sit at a crystal momentum (Sect. 1.3) a large fraction of $2\pi/a = 1.16\,\text{Å}^{-1}$ away from the top of the valence band (Sect. 3.3). The photon has the energy to cross the gap, 1.12 eV at 300 K, but not the momentum, and the golden rule of Sect. 2.1 gives zero. Silicon absorbs such photons all the same, because a lattice vibration can supply the missing momentum. The photon lifts an electron into a state that conserves crystal momentum but not energy, and a phonon emitted or absorbed a moment later takes it to a final state that conserves both. The state in between is never observed. It enters the rate through an energy denominator, the mismatch between its energy and that of the initial state. We will derive these amplitudes at second order, count the time orderings that contribute when one quantum is absorbed and another emitted, and see what happens when a denominator vanishes and the intermediate state becomes real. That last case needs a finite lifetime, a Lorentzian line and, in the rate, a distribution that has no principal value: the Hadamard finite part. The treatment follows Refs. [1, Chap. 5] and [14, Chaps. I and III].

2.4.1 Interaction picture and Dyson series

To go beyond first order we need the amplitude equations Eq. (2.1) in a form that iterates, and the interaction picture provides it. Define the state $|\psi_{\mathrm{I}}(t)\rangle = \mathrm{e}^{\mathrm{i}H_0 t}|\psi(t)\rangle$, whose components are exactly the amplitudes, $c_n(t) = \langle n|\psi_{\mathrm{I}}(t)\rangle$, and the perturbation $H'_{\mathrm{I}}(t) = \mathrm{e}^{\mathrm{i}H_0 t} H'(t)\, \mathrm{e}^{-\mathrm{i}H_0 t}$.

The Schrödinger equation becomes

$$\mathrm{i}\,\partial_t|\psi_\mathrm{I}(t)\rangle = H'_\mathrm{I}(t)\,|\psi_\mathrm{I}(t)\rangle, \tag{2.49}$$

whose matrix elements in the basis $|n\rangle$ are Eq. (2.1). The free evolution sits in the operator: $\langle m|H'_\mathrm{I}(t)|n\rangle = H'_{mn}(t)\,\mathrm{e}^{\mathrm{i}(E_m-E_n)t}$, with $H'_{mn}(t) \equiv \langle m|H'(t)|n\rangle$. For independent electrons this is the oscillation $\mathrm{e}^{\mathrm{i}(E_\gamma-E_{\gamma'})t}$ of $c^\dagger_\gamma c_{\gamma'}$ found in Sect. 2.3.

Integrating Eq. (2.49) from t_0 to t and inserting the result into its own right-hand side, again and again, leads to the evolution operator of the interaction picture as a power series in H', the Dyson series,

$$U_\mathrm{I}(t,t_0) = 1 - \mathrm{i}\int_{t_0}^{t}\mathrm{d}t_1\,H'_\mathrm{I}(t_1) + (-\mathrm{i})^2\int_{t_0}^{t}\mathrm{d}t_1\int_{t_0}^{t_1}\mathrm{d}t_2\,H'_\mathrm{I}(t_1)\,H'_\mathrm{I}(t_2) + \ldots, \tag{2.50}$$

with $|\psi_\mathrm{I}(t)\rangle = U_\mathrm{I}(t,t_0)|\psi_\mathrm{I}(t_0)\rangle$. In each term the operators stand in order of decreasing time from left to right, which is why the series is also called the time-ordered exponential of $-\mathrm{i}\int H'_\mathrm{I}$. The first-order term reproduces Eq. (2.2). A complete set of states $|m\rangle$ between the two factors of the second-order term leads to the amplitude for $f \neq i$,

$$c^{(2)}_f(t) = (-\mathrm{i})^2\sum_m\int_{t_0}^{t}\mathrm{d}t_1\int_{t_0}^{t_1}\mathrm{d}t_2\;H'_{fm}(t_1)\,\mathrm{e}^{\mathrm{i}(E_f-E_m)t_1}\;H'_{mi}(t_2)\,\mathrm{e}^{\mathrm{i}(E_m-E_i)t_2}. \tag{2.51}$$

It reads from right to left as a story: at t_2 the perturbation takes the system from i to m, it evolves freely in m until t_1, and at t_1 the perturbation takes it to f. Every state m contributes, $m = i$ and $m = f$ included, and the amplitudes of the different m add. Each further order adds one more factor of H' and one more time integral. For a constant perturbation acting for a time t, the terms of Eq. (2.50) grow as powers of $|H'|t$, the expansion parameter of Sect. 2.1. Once the time integrals are done in the long-time limit, the powers of t turn into powers of $|H'|$ divided by energy differences.

2.4.2 Second-order golden rule

For a constant perturbation the time integrals in Eq. (2.51) are elementary once we decide how H' is switched on. Sect. 2.1 showed that a rate does not depend on the switching. The most convenient choice is to turn H' on slowly in the remote past, $H'(t) = H'\,\mathrm{e}^{t/\tau_\mathrm{s}}$ with $t_0 \to -\infty$, and to let the switching time τ_s go to infinity at the end, at fixed t. The inner integral is then

$$\int_{-\infty}^{t_1}\mathrm{d}t_2\,\mathrm{e}^{[\mathrm{i}(E_m-E_i)+1/\tau_\mathrm{s}]t_2} = \frac{\mathrm{i}\,\mathrm{e}^{[\mathrm{i}(E_m-E_i)+1/\tau_\mathrm{s}]t_1}}{E_i - E_m + \mathrm{i}/\tau_\mathrm{s}}.$$

The free phase that the system accumulates in m has turned into a factor $1/(E_i - E_m)$, the *energy denominator*, and the switching has put a small positive imaginary part into it. The rest of the integrand of Eq. (2.51) has the form of the first-order one, with $\xi = E_f - E_i$ and the switching rate doubled. The amplitude to second order is

$$c_f(t) = -\mathrm{e}^{\mathrm{i}\xi t}\left[\frac{A_1\,\mathrm{e}^{t/\tau_\mathrm{s}}}{\xi - \mathrm{i}/\tau_\mathrm{s}} + \frac{A_2\,\mathrm{e}^{2t/\tau_\mathrm{s}}}{\xi - 2\mathrm{i}/\tau_\mathrm{s}}\right], \qquad A_1 = H'_{fi}, \quad A_2 = \sum_m \frac{H'_{fm}H'_{mi}}{E_i - E_m + \mathrm{i}0^+},$$

where $\mathrm{i}0^+$ stands for $\mathrm{i}/\tau_\mathrm{s}$ in the limit. The transition rate is the time derivative of $|c_f|^2$. The squared first-order term contributes $(2/\tau_\mathrm{s})\,\mathrm{e}^{2t/\tau_\mathrm{s}}|A_1|^2/(\xi^2 + 1/\tau_\mathrm{s}^2)$, a Lorentzian of area $2\pi|A_1|^2$ whose width vanishes with $1/\tau_\mathrm{s}$, and it tends to $2\pi|A_1|^2\delta(\xi)$; the squared second-order term

does the same with $2/\tau_\mathrm{s}$ in place of $1/\tau_\mathrm{s}$. The cross term needs more care, since the two terms carry different switching rates. Its derivative is

$$2\,\mathrm{Re}\left[A_1 A_2^*\,\mathrm{e}^{3t/\tau_\mathrm{s}}\,\frac{3/\tau_\mathrm{s}}{(\xi-\mathrm{i}/\tau_\mathrm{s})(\xi+2\mathrm{i}/\tau_\mathrm{s})}\right].$$

The fraction falls off as $1/\xi^2$, and its only pole in the upper half plane is at $\xi=\mathrm{i}/\tau_\mathrm{s}$. Closing the contour there, the integral over ξ is $2\pi\mathrm{i}\,(3/\tau_\mathrm{s})/(3\mathrm{i}/\tau_\mathrm{s})=2\pi$, independent of τ_s, while its width shrinks as $1/\tau_\mathrm{s}$. Its imaginary part is odd in ξ and drops out against any smooth function. The cross term therefore tends to $2\pi\cdot 2\,\mathrm{Re}(A_1A_2^*)\,\delta(\xi)$. The three terms combine into a single squared modulus:

Second-order golden rule

$$\Gamma_{i\to f}=2\pi\,|T_{fi}|^2\,\delta(E_f-E_i),\qquad T_{fi}=H'_{fi}+\sum_m\frac{H'_{fm}\,H'_{mi}}{E_i-E_m+\mathrm{i}0^+}+\dots. \tag{2.52}$$

The sum runs over all intermediate states m; the rate is defined after summation over a continuum of final states, as in Eq. (2.5).

The quantity T_{fi} is the element of the transition matrix, or T-matrix. Continued to all orders, the Dyson series becomes a geometric series in the energy denominators, which sums to the Lippmann–Schwinger equation [1, Chap. 6]

$$T_{fi}=H'_{fi}+\sum_m\frac{H'_{fm}\,T_{mi}}{E_i-E_m+\mathrm{i}0^+}. \tag{2.53}$$

Iterating it once reproduces Eq. (2.52). In the language of Sect. 2.1, the S-matrix element is $\delta_{fi}-2\pi\mathrm{i}\,\delta(E_f-E_i)\,T_{fi}$, and T_{fi} equals the invariant amplitude $-\mathcal{M}$ divided by the normalization factors of the box matrix element of Sect. 2.1, at every order.

Energy is conserved between the initial and the final state, through $\delta(E_f-E_i)$, but not in the intermediate state: E_m can be anything. The intermediate state is *virtual*. The phase $\mathrm{e}^{\mathrm{i}(E_m-E_i)t_2}$ in Eq. (2.51) oscillates with period $2\pi/|E_i-E_m|$, and contributions from times t_2 further than about $1/|E_i-E_m|$ before t_1 cancel. The system spends a time of that size in m. The further m is from the initial energy, the shorter the time and the smaller the amplitude. The physical content is the energy–time relation. A state m with $E_m=E_i$ is *on shell*, any other one *off shell*, and an on-shell intermediate state makes the denominator vanish.

The second-order term is small compared to the first by a factor $\sim|H'_{fm}|/|E_i-E_m|$, the coupling measured against the energy mismatch. For dark-matter scattering this factor is the Born parameter of Sect. 2.2. The second-order term of Eq. (2.52), with plane-wave intermediate states, is the second Born term, and it is suppressed by $y_\chi y_e\,\mu_{\chi e}/(4\pi m_V)$ or $y_\chi y_e/(4\pi v)$, numbers below 10^{-3} for the couplings that are still allowed. Second-order processes matter in this book for another reason: when first order is closed, an internal interaction of the target, such as the coupling of electrons to phonons, can open the channel.

Classical fields fit the same formula. If H' contains harmonic terms $\hat{O}_j\mathrm{e}^{-\mathrm{i}\omega_j t}$, as in Sect. 2.1, each factor H'_I in Eq. (2.51) carries its own frequency. The inner time integral then produces the denominator $E_i+\omega_j-E_m$ for the field j that acts first. The outer one leads to $\delta(E_f-E_i-\omega_j-\omega_k)$ in the rate, with ω_k the frequency of the field that acts second. An emission term enters with $-\omega$ in place of ω. The result is the same as if each field were a quantized mode whose photon energy is part of E_i, E_m and E_f, and we use whichever language is clearer.

The second-order golden rule also inherits the time window of Eq. (2.7), with one more condition. The energy denominator replaces the free phase in m only after that phase has

oscillated many times, and the derivation took $\tau_{\mathrm{s}} \to \infty$ to ensure this. As long as we watch for longer than $1/|E_i - E_m|$, about a femtosecond below a direct gap of a few eV, the denominator form holds, and every measurement satisfies this. Near a resonance the condition fails, and it is replaced by $t \gg 1/\Gamma_{\mathrm{w}}$, where Γ_{w} is the decay width of the real intermediate state (Eq. (2.57)).

2.4.3 Time orderings

The phonon-assisted absorption of the opening paragraph has the general structure of a second-order process in which a target absorbs one quantum and emits another. Let quantum 1, of energy ω_1, be absorbed, and quantum 2, of energy ω_2, be emitted, while the target goes from a state $|i_{\mathrm{T}}\rangle$ to a state $|f_{\mathrm{T}}\rangle$, with the subscript T for target. In the language of Sect. 2.3 the two quanta are boson modes with operators d_1 and d_2, and the perturbation is a sum of two terms that each change one occupation by one,

$$H' = H_1 + H_2, \qquad H_1 = \hat{O}_a\, d_1 + \hat{O}_a^\dagger\, d_1^\dagger, \qquad H_2 = \hat{O}_b\, d_2 + \hat{O}_b^\dagger\, d_2^\dagger,$$

with $\hat{O}_a$ and $\hat{O}_b$ operators on the target. The initial state $|i\rangle = |i_{\mathrm{T}};\, n_1 = 1, n_2 = 0\rangle$ has energy $E_i = E_{i_{\mathrm{T}}} + \omega_1$, and the final state $|f\rangle = |f_{\mathrm{T}};\, 0, 1\rangle$ has $E_f = E_{f_{\mathrm{T}}} + \omega_2$. Neither H_1 nor H_2 connects them, since each changes only one occupation, and $H'_{fi} = 0$. At second order there are two routes, drawn in Fig. 2.6. In the absorb-first ordering, H_1 acts first and removes quantum 1; the intermediate state $|m;\, 0, 0\rangle$ has energy E_m. In the emit-first ordering, H_2 acts first and creates quantum 2 while quantum 1 is still present; the intermediate state $|m;\, 1, 1\rangle$ has energy $E_m + \omega_1 + \omega_2$. Inserting both into Eq. (2.52), with $d_1|1\rangle = |0\rangle$ and $d_2^\dagger|0\rangle = |1\rangle$,

$$T_{fi} = \sum_m \left[\frac{\langle f_{\mathrm{T}}|\hat{O}_b^\dagger|m\rangle\langle m|\hat{O}_a|i_{\mathrm{T}}\rangle}{E_{i_{\mathrm{T}}} + \omega_1 - E_m + \mathrm{i}0^+} + \frac{\langle f_{\mathrm{T}}|\hat{O}_a|m\rangle\langle m|\hat{O}_b^\dagger|i_{\mathrm{T}}\rangle}{E_{i_{\mathrm{T}}} - \omega_2 - E_m + \mathrm{i}0^+} \right], \tag{2.54}$$

and energy conservation, $E_{f_{\mathrm{T}}} + \omega_2 = E_{i_{\mathrm{T}}} + \omega_1$, holds between i and f but not in between.

Both orderings connect the same states, and their amplitudes add before squaring and can interfere. If i_{T} is the ground state of the target, $E_m \geq E_{i_{\mathrm{T}}}$ for every m, and the emit-first denominator $E_{i_{\mathrm{T}}} - \omega_2 - E_m$ is at most $-\omega_2$: it never vanishes, and this ordering is always off shell. The absorb-first denominator vanishes when $\omega_1 = E_m - E_{i_{\mathrm{T}}}$, that is, when quantum 1 alone could excite the target to a real state m. This is the resonance, and near it the absorb-first ordering dominates. Far from every resonance the two orderings are of similar size, and dropping one changes the rate by a factor of a few or more, as Worked example 2.6 shows. In silicon, with quantum 1 a photon and quantum 2 a phonon, Eq. (2.54) is the bracket of the theory of phonon-assisted absorption of Hall, Bardeen and Blatt [15], which also includes phonon absorption in place of emission; Ref. [16, Eq. (200)] writes it in first-principles form. The photon-first term has the denominator of a direct, vertical transition at fixed crystal momentum, and the phonon-first term the denominator of an electron scattered by the phonon before the photon arrives. Sect. 6.5 evaluates both. When the modes hold n_1 and n_2 quanta rather than one and zero, the matrix elements of d_1 and $d_2^\dagger$ bring factors $\sqrt{n_1}$ and $\sqrt{n_2 + 1}$, in both orderings alike; Sect. 2.5 averages them over thermal populations.

In a target with many electrons, i_{T}, m and f_{T} are the many-electron states of Sect. 2.3, and the denominators are differences of total energies. For independent electrons they reduce to differences of orbital energies, because an operator that moves one electron from γ' to γ changes the total energy by $E_\gamma - E_{\gamma'}$, the frequency at which $c_\gamma^\dagger c_{\gamma'}$ oscillates in the interaction picture. If the photon moves an electron from an occupied valence orbital to an empty conduction orbital at the same crystal momentum, the absorb-first denominator is the photon energy minus that vertical transition energy, whatever the other electrons do. The Pauli principle changes the list of intermediate states. A state in which the moved electron would

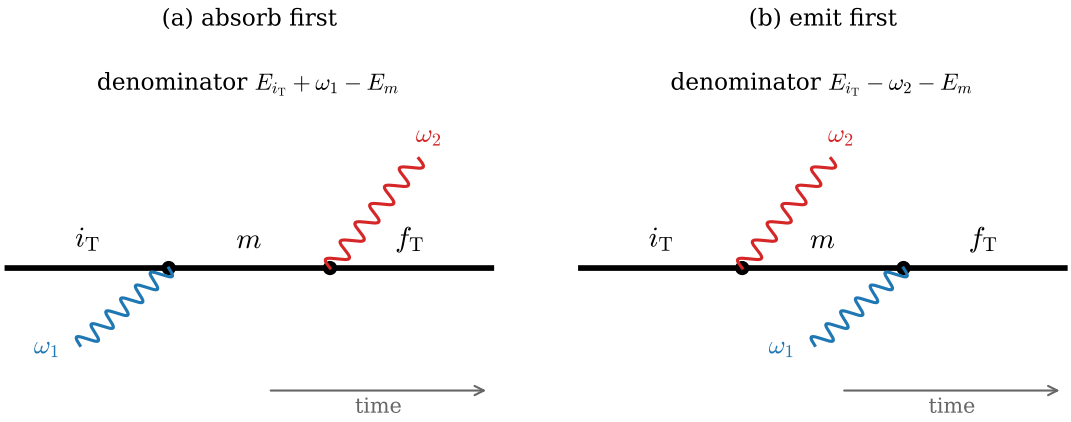


Figure 2.6: The two time orderings of a second-order process in which a target goes from state i_T to state f_T while absorbing a quantum of energy ω_1 and emitting one of energy ω_2. Time runs to the right; the thick line is the target and the wavy lines are the quanta (absorbed from below, emitted upward). (a) Absorb first: the intermediate state is the target in m with no quanta, of energy E_m. (b) Emit first: the target is in m while both quanta are present, of energy $E_m + \omega_1 + \omega_2$. Each panel gives the energy denominator, the initial energy $E_{i_T} + \omega_1$ minus the intermediate energy, with which its amplitude enters Eq. (2.54). Both orderings connect the same initial and final states, and their amplitudes add.

enter an occupied orbital does not exist. In its place there are orderings in which a different electron moves first and the electron moved by the photon then fills the hole it left, with the anticommutation sign of Sect. 2.3 attached to the amplitude. The single-particle picture of one electron passing through an intermediate band is a convenient way of organizing these terms, not a literal history, and Sect. 6.5 counts them for silicon.

2.4.4 Resonance and finite width

At resonance the absorb-first term of Eq. (2.54) is infinite, and the rate with it. Nothing in nature diverges here. The trouble is that Eq. (2.51) lets the system stay in m forever with unit probability, while a real excited state decays. Quantum 1 can take the target to m, and m then empties into f_T by emitting quantum 2, or into any other state it is coupled to. We need the time dependence of a state that decays into a continuum, and this is the Wigner–Weisskopf problem [17]. A state that empties at rate Γ_w survives for a time of about $1/\Gamma_w$, so its energy is defined only to within about Γ_w, and a denominator that would vanish should instead saturate at a size of order Γ_w. The calculation confirms this and fixes the factors.

Take a discrete state $|i\rangle$ coupled by a constant H' to a continuum of states $|f\rangle$, with no other couplings, as in Sect. 2.1. Appendix D.3 solves the amplitude equations Eq. (2.1) for this case without assuming that c_i stays close to one. Under the condition that made the golden rule possible, $\Gamma_i \ll E_w$, the memory of the continuum is short compared with the decay time. At times $t \gg 1/E_w$ the amplitude of the initial state is then

$$c_i(t) = \mathrm{e}^{-(\Gamma_i/2 + \mathrm{i}\Delta E_i)\,t}, \qquad \Delta E_i = \mathrm{PV} \sum_f \frac{|H'_{fi}|^2}{E_i - E_f}, \tag{2.55}$$

where Γ_i is the golden-rule rate Eq. (2.6) and PV denotes the principal value of the sum, made precise in the next subsection. The probability $|c_i|^2 = \mathrm{e}^{-\Gamma_i t}$ decays exponentially at the

golden-rule rate, which justifies the statement of Sect. 2.1 that Γ_i keeps its meaning beyond the first-order window (Worked example 2.1). The energy of the state is shifted by ΔE_i, the second-order level shift of stationary perturbation theory, now with the principal value that a continuum requires; in Worked example 2.1 the shift vanishes because the continuum is symmetric about resonance. The populations of the final states at late times trace out the line,

$$|c_f(\infty)|^2 = \frac{|H'_{fi}|^2}{(E_f - E_i - \Delta E_i)^2 + \Gamma_i^2/4}, \tag{2.56}$$

a Lorentzian of full width Γ_i at half maximum, centered at the shifted energy. Summed over the continuum with density $\mathcal{D}_f$, it integrates to $\overline{|H'_{fi}|^2}\mathcal{D}_f \cdot 2\pi/\Gamma_i = 1$: the decaying state has gone somewhere, and the golden-rule rate fixes both how fast and how sharply in energy. The same shape describes the cross-section near a resonance in scattering, where it is called a Breit–Wigner line. Appendix D.3 also states the limits of the exponential law. It fails at times shorter than $1/E_\mathrm{w}$, where the decay starts quadratically as in Worked example 2.1. It also fails after many lifetimes, where the edges of the continuum leave a slow power-law tail, by then far below any measurable level when $\Gamma_i \ll E_\mathrm{w}$.

Back to the intermediate state. Between t_2 and t_1 in Eq. (2.51) the system evolves in m. Eq. (2.55) applied to m replaces its free phase $\mathrm{e}^{-\mathrm{i}E_m(t_1-t_2)}$ by $\mathrm{e}^{-\mathrm{i}(\tilde{E}_m - \mathrm{i}\Gamma_\mathrm{w}/2)(t_1-t_2)}$, where $\tilde{E}_m = E_m + \Delta E_m$ is the shifted energy and Γ_w the total decay rate of m, summed over all of its decay channels. The inner time integral then converges without any switching and gives

$$\frac{1}{E_i - E_m + \mathrm{i}0^+} \;\longrightarrow\; \frac{1}{E_i - \tilde{E}_m + \mathrm{i}\Gamma_\mathrm{w}/2}. \tag{2.57}$$

The infinitesimal becomes a finite width, and at resonance the amplitude is large, $-2\mathrm{i}/\Gamma_\mathrm{w}$ in place of the denominator, but finite. In the language of the Dyson series, Eq. (2.57) keeps, from every order, the repeated emission and reabsorption that happen while the system sits in m. These are the terms that grow near resonance, and summing them is consistent under the same condition as before, Γ_w small compared with the energy over which the decay continuum of m changes. Off resonance, with $|E_i - \tilde{E}_m| \gg \Gamma_\mathrm{w}$, the width is a small correction and the shift is usually absorbed in the measured energy $\tilde{E}_m$. From now on we write E_m for the shifted energy. In a solid the width of an excited electron comes from the emission of phonons and from scattering with other electrons, and it depends on how far above the band edge the electron sits; Sect. 6.3 computes the phonon part. For the off-resonant denominators that dominate below a direct gap, an infinitesimal $\mathrm{i}0^+$ is enough.

2.4.5 Principal value and finite part

An energy denominator regularized by a width, $1/(x + \mathrm{i}\Gamma_\mathrm{w}/2)$ with $x = E_i - E_m$, has a real and an imaginary part,

$$\frac{1}{x + \mathrm{i}\Gamma_\mathrm{w}/2} = \frac{x}{x^2 + \Gamma_\mathrm{w}^2/4} - \mathrm{i}\,\frac{\Gamma_\mathrm{w}/2}{x^2 + \Gamma_\mathrm{w}^2/4},$$

drawn in Fig. 2.7(a). The real part equals $1/x$ for $|x| \gg \Gamma_\mathrm{w}$ and turns over smoothly through zero inside $|x| < \Gamma_\mathrm{w}/2$, with extrema $\pm 1/\Gamma_\mathrm{w}$. The imaginary part is a Lorentzian of area π and depth $2/\Gamma_\mathrm{w}$. As $\Gamma_\mathrm{w} \to 0$ the Lorentzian becomes $\pi\delta(x)$. The real part, integrated against a smooth function $w(x)$, becomes $\int \mathrm{d}x\, x[w(x) - w(0)]/(x^2 + \Gamma_\mathrm{w}^2/4)$, since $x/(x^2 + \Gamma_\mathrm{w}^2/4)$ is odd, and this tends to $\int \mathrm{d}x\,[w(x) - w(0)]/x$. That is the principal value of $\int \mathrm{d}x\, w(x)/x$: the integral with a symmetric interval $(-x_\mathrm{c}, x_\mathrm{c})$ cut out around the pole, in the limit $x_\mathrm{c} \to 0$, in

which the divergences on the two sides cancel. Together these limits are the Sokhotski–Plemelj formula,

$$\frac{1}{x+\mathrm{i}0^+} = \mathrm{PV}\,\frac{1}{x} - \mathrm{i}\pi\,\delta(x), \tag{2.58}$$

which Appendix C.4 proves for the general case. Applied to a second-order amplitude, it splits the sum over intermediate states into two parts with different physics. The principal-value part collects the off-shell intermediate states, and it is real: it shifts energies, as ΔE_i does in Eq. (2.55), and it produces the dispersive effects of a medium. The delta part picks out the on-shell intermediate states, those with $E_m = E_i$; it is imaginary and produces the decay. Eq. (2.55) is the example: the self-coupling of i through the continuum, $\sum_f |H'_{fi}|^2/(E_i - E_f + \mathrm{i}0^+)$, is $\Delta E_i - \mathrm{i}\Gamma_i/2$ by Eq. (2.58). The same statement can be read off the T-matrix. The forward element at second order, $T_{ii} = \sum_f |H'_{fi}|^2/(E_i - E_f + \mathrm{i}0^+)$, has imaginary part $-\pi \sum_f |H'_{fi}|^2 \delta(E_f - E_i) = -\Gamma_i/2$. The total decay rate of a state is $-2\,\mathrm{Im}\,T_{ii}$, the lowest-order form of the optical theorem, which follows in general from the unitarity of the evolution: the probability that leaves i is the probability that arrives in all the states f.

The principal-value part has a familiar instance in light scattering. Let quanta 1 and 2 be photons of the same frequency, so that the target absorbs one photon and emits another and returns to $f_\mathrm{T} = i_\mathrm{T}$. Eq. (2.54) is then the amplitude of elastic (Rayleigh) scattering, and below every resonance both orderings are off shell and real. Summed over intermediate states, the two orderings combine into $\sum_m 2(E_m - E_{i_\mathrm{T}})|\langle m|\hat{O}|i_\mathrm{T}\rangle|^2/[\omega^2 - (E_m - E_{i_\mathrm{T}})^2]$, for a Hermitian $\hat{O}$. For a dipole coupling, $\hat{O}$ is the field amplitude times the dipole operator along the field, and the sum is minus the dynamic polarizability of the target, whose denominators are $(E_m - E_{i_\mathrm{T}})^2 - \omega^2$, times the square of the field amplitude. Its imaginary part, which appears once a width or a continuum puts the absorb-first denominator on shell, is absorption. The dielectric function of Sect. 4.4 has the same structure, with a real part made of off-shell transitions and an imaginary part made of on-shell ones.

The rate needs the squared modulus, and near resonance this brings a double pole. With $|E_i - E_m + \mathrm{i}\Gamma_\mathrm{w}/2|^{-2} = 1/(x^2 + \Gamma_\mathrm{w}^2/4)$, the limit $\Gamma_\mathrm{w} \to 0$ cannot be taken in Eq. (2.58): the square of $\mathrm{i}\pi\delta(x)$ has no meaning, as the square of the momentum delta function had none in Sect. 2.1. The function $1/x^2$ has no principal value either. Its singularity is even, the divergences on the two sides add instead of cancelling, and $\int_{|x|>x_\mathrm{c}} \mathrm{d}x/x^2 = 2/x_\mathrm{c}$. The Hadamard finite part removes exactly this divergence:

$$\mathrm{fp}\int \mathrm{d}x\,\frac{w(x)}{x^2} = \lim_{x_\mathrm{c}\to 0}\left[\int_{|x|>x_\mathrm{c}} \mathrm{d}x\,\frac{w(x)}{x^2} - \frac{2w(0)}{x_\mathrm{c}}\right] = \int_0^\infty \mathrm{d}x\,\frac{w(x)+w(-x)-2w(0)}{x^2}. \tag{2.59}$$

The integrand on the right is finite at $x = 0$, where it tends to $w''(0)$. With it, the Lorentzian of the rate splits as follows. Write $\int \mathrm{d}x\, w(x)/(x^2 + \Gamma_\mathrm{w}^2/4)$ as $w(0)\int \mathrm{d}x/(x^2 + \Gamma_\mathrm{w}^2/4) = 2\pi w(0)/\Gamma_\mathrm{w}$ plus $\int \mathrm{d}x\,[w(x) - w(0)]/(x^2 + \Gamma_\mathrm{w}^2/4)$. The remainder tends to the finite part as $\Gamma_\mathrm{w} \to 0$, since its integrand is integrable at $x = 0$. Its difference from the limit is $-(\Gamma_\mathrm{w}^2/4)\int \mathrm{d}x\,[w(x) - w(0)]/[x^2(x^2 + \Gamma_\mathrm{w}^2/4)]$, which is dominated by $|x| \lesssim \Gamma_\mathrm{w}$. There the odd part of $w(x) - w(0)$ drops out and the even part is $w''(0)x^2/2$, and the difference is $-(\Gamma_\mathrm{w}^2/8)\,w''(0)\cdot 2\pi/\Gamma_\mathrm{w} = -\pi\Gamma_\mathrm{w} w''(0)/4$. In the sense of distributions,

$$\frac{1}{x^2 + \Gamma_\mathrm{w}^2/4} = \frac{2\pi}{\Gamma_\mathrm{w}}\,\delta(x) + \mathrm{fp}\,\frac{1}{x^2} - \frac{\pi\Gamma_\mathrm{w}}{4}\,\delta''(x) + O(\Gamma_\mathrm{w}^2). \tag{2.60}$$

For a function w that varies on an energy scale Δ_w, the delta term is of size w/Γ_w, the finite part of size w/Δ_w, and the correction a fraction $(\Gamma_\mathrm{w}/\Delta_\mathrm{w})^2$ of the delta term. Appendix C.4 collects these definitions.

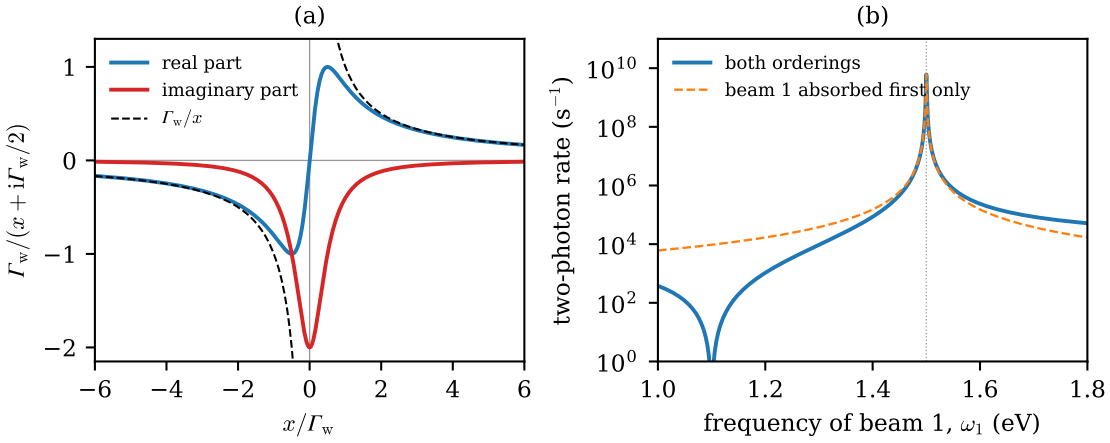


Figure 2.7: (a) Real and imaginary parts of an energy denominator regularized by a width, $1/(x+\mathrm{i}\Gamma_\mathrm{w}/2)$, multiplied by Γ_w and plotted against x/Γ_w, where x is the initial energy minus the intermediate energy. Dashed: Γ_w/x, to which the real part tends for $|x| \gg \Gamma_\mathrm{w}$. As $\Gamma_\mathrm{w} \to 0$ the real part becomes the principal value of $1/x$ and the imaginary part $-\pi\delta(x)$. (b) Rate of two-photon absorption in the three-level model of Worked example 2.6, against the frequency ω_1 of beam 1 at fixed $\omega_2 = 1.90\,\mathrm{eV}$, for the channel with final energy $\omega_1 + \omega_2$. The model has a ground level i_T, an intermediate level m at 1.50 eV with total width $\Gamma_\mathrm{w} = 1\,\mathrm{meV}$, and a flat band of final states above 3.20 eV. Either beam drives $i_\mathrm{T} \to m$ with matrix element 0.1 meV, and beam 2 empties a real level m into the band at the rate 0.1 meV. Solid: both time orderings. Dashed: the ordering in which beam 1 is absorbed first, alone. The peak at $\omega_1 = 1.50\,\mathrm{eV}$ is the resonance with m; the zero at 1.10 eV is destructive interference between the two orderings.

The two terms of Eq. (2.60) are the on-shell and the off-shell parts of a resonant second-order rate, and the on-shell part has a direct interpretation. Inserted in Eq. (2.52), the delta term of the absorb-first ordering is the first-order golden-rule rate $2\pi|\langle m|\hat{O}_a|i_\mathrm{T}\rangle|^2\delta(E_{i_\mathrm{T}}+\omega_1-E_m)$, at which quantum 1 creates a real state m, multiplied by $2\pi|\langle f_\mathrm{T}|\hat{O}_b^\dagger|m\rangle|^2\delta(E_f-E_i)/\Gamma_\mathrm{w}$. Summed over final states, this factor is the fraction of the decays of m that end in f_T. The on-shell part is a cascade of two first-order processes, the excitation of m and its decay into f_T, the decay weighted by its branching ratio. The finite part is what remains of the genuinely second-order process, the virtual passage through m. It is not positive: against a function w peaked at $x=0$ with width Δ_w, Eq. (2.59) is about $-w(0)/\Delta_\mathrm{w}$, because $w(x)-w(0)<0$ near the peak. Only the sum of the two parts is a rate. This is the reason for care in any calculation that counts the cascade separately, as two first-order events, and also includes a second-order rate: the delta part of the second-order rate is then counted twice. The resonant denominators of phonon-assisted absorption return in Sect. 6.5, where the electron in the intermediate state can be real.

2.4.6 Forbidden first-order transitions

Since each order costs a factor $|H'|/|E_i-E_m|$, a second-order process competes with a first-order one only when the first-order rate vanishes, by a selection rule or by kinematics. The book meets both.

If a symmetry of the target makes $\langle f|H'|i\rangle=0$ for every final state at the right energy, the leading amplitude is the second-order sum, which involves different matrix elements. The classic case is parity. The electric-dipole operator is odd, and a single photon cannot connect two states of the same parity. Two photons can, through intermediate states of the opposite parity. The metastable $2s$ level of hydrogen decays by emitting two photons, and two-photon absorption spectroscopy reaches levels that one photon cannot. Spin, angular momentum and the symmetries of a crystal impose rules of the same kind.

Kinematics closes the first-order channel when no final state conserves both energy and momentum. The opening example is of this type. Compared to the Brillouin zone of silicon, an absorbed photon has essentially zero momentum, $6\times10^{-4}\,\text{Å}^{-1}$ against $2\pi/a=1.16\,\text{Å}^{-1}$. A vertical transition at fixed crystal momentum therefore needs an energy well above the 1.12 eV gap at 300 K, because the conduction-band minimum and the valence-band maximum sit at different crystal momenta (Sect. 3.3). Below that energy the first-order rate is zero, and phonon-assisted absorption, with the amplitude Eq. (2.54), is the leading process. The phonon costs or supplies tens of meV of energy [18, Table I], but it can carry any crystal momentum in the zone. The same holds for absorbed dark matter. A boson of mass m_ϕ is absorbed with momentum $m_\phi v\approx10^{-3}m_\phi$ (Sect. 2.1), even smaller compared to its energy than a photon's, and in an indirect-gap material its absorption between the indirect and the direct gap is a second-order process (Chap. 8). Dark-matter scattering is different. The momentum transfer is typically $q\sim q_0=\alpha m_e=3.73\,\text{keV}$, which is $1.89\,\text{Å}^{-1}$, larger than $2\pi/a$, so scattering reaches the indirect transitions at first order. At small momentum transfer, however, scattering runs into the same obstacle as a photon, and there the simultaneous emission of a phonon, the dark-matter analogue of phonon-assisted absorption, takes over [18, Sect. VI]. The Migdal effect is a process of the same structure, with a nuclear recoil in place of the phonon (Sect. 12.2).

In each case the size of the second-order rate relative to a first-order one is set by $|H_2|^2/(E_i-E_m)^2$, with H_2 the coupling that the second-order route adds and E_i-E_m the off-shell distance of the intermediate state. Sect. 8.5 names this ratio for the electron–phonon coupling and uses it to organize the expansion.

When the first-order amplitude does not vanish, second order is a correction to it, and it

enters the rate through the cross term $2\,\mathrm{Re}(A_1 A_2^*)$, at relative order $|H'|/|E_i - E_m|$ rather than its square. Corrections of this kind are part of the target response when H_2 is internal to the target. They are already contained in an exact treatment of the target's states, such as the band structure with its phonon renormalization (Sect. 6.3), and are not added again as second-order rates. A second-order rate is a separate channel only when it leads to a final state that first order cannot reach, such as a state with one more phonon.

Worked example 2.6: Two-photon absorption in a three-level system

A target has a ground level a at zero energy, an intermediate level m at $E_m = 1.50\,\mathrm{eV}$, and a flat band of final states f_T above 3.20 eV, with density of states $\mathcal{D}_{f_\mathrm{T}}$. Two light beams of frequencies ω_1 and ω_2 couple to the target through the same operator $\hat{O}$ (equal polarization and intensity); Sect. 2.1 showed that a classical beam acts like a quantized mode holding many photons, and we use the language of Eq. (2.54). Here, however, both photons are absorbed, so the ordering that absorbs ω_2 first has the denominator $E_{i_\mathrm{T}} + \omega_2 - E_m$, with $+\omega_2$ in place of the $-\omega_2$ of emission in Eq. (2.54). We fix $\omega_2 = 1.90\,\mathrm{eV}$ and tune ω_1 near 1.50 eV. Neither photon alone reaches the band, and neither do two photons of beam 1 for $\omega_1 < 1.60\,\mathrm{eV}$: the first-order rate is kinematically zero. The pair $\omega_1 + \omega_2 \approx 3.40\,\mathrm{eV}$ reaches the band. Two photons of beam 2 reach it too, at 3.80 eV; that final state differs from ours and does not interfere, and we follow the channel with final energy $\omega_1 + \omega_2$. The model numbers are $|\langle m|\hat{O}|i_\mathrm{T}\rangle| = 0.1\,\mathrm{meV}$, a rate $\Gamma_{m\to f_\mathrm{T}} = 2\pi|\langle f_\mathrm{T}|\hat{O}|m\rangle|^2\,\mathcal{D}_{f_\mathrm{T}} = 0.1\,\mathrm{meV}$ at which beam 2 empties a real level m into the band (beam 1 cannot, since $E_m + \omega_1 < 3.20\,\mathrm{eV}$), and a total width $\Gamma_\mathrm{w} = 1\,\mathrm{meV}$ of m, which includes $\Gamma_{m\to f_\mathrm{T}}$ and all other decays.

Both orderings absorb one photon from each beam. Absorbing ω_1 first leaves the target in m with photon ω_2 still present, and the denominator is $\delta_1 = E_{i_\mathrm{T}} + \omega_1 - E_m = \omega_1 - 1.50\,\mathrm{eV}$; absorbing ω_2 first leads to $\delta_2 = \omega_2 - 1.50\,\mathrm{eV} = 0.40\,\mathrm{eV}$. With Eq. (2.57),

$$T_{fi} = \langle f_\mathrm{T}|\hat{O}|m\rangle\langle m|\hat{O}|i_\mathrm{T}\rangle\,\mathcal{A}(\omega_1), \qquad \mathcal{A}(\omega_1) = \frac{1}{\delta_1 + \mathrm{i}\Gamma_\mathrm{w}/2} + \frac{1}{\delta_2 + \mathrm{i}\Gamma_\mathrm{w}/2},$$

and summing Eq. (2.52) over the flat band gives $\Gamma_{i_\mathrm{T}\to f_\mathrm{T}} = \Gamma_{m\to f_\mathrm{T}}\,|\langle m|\hat{O}|i_\mathrm{T}\rangle|^2\,|\mathcal{A}|^2$. Fig. 2.7(b) plots the rate against ω_1.

Off resonance, at $\omega_1 = 1.20\,\mathrm{eV}$, $\delta_1 = -0.30\,\mathrm{eV}$ and $\mathcal{A} = -3.33 + 2.50 = -0.833\,\mathrm{eV}^{-1}$. The two orderings have denominators of opposite sign and interfere destructively. Beam 1 is below the resonance of m and beam 2 above it, and the virtual excitation of m follows the two beams with opposite phases, as a driven oscillator follows a force below and above its natural frequency. The rate is $6.9 \times 10^{-13}\,\mathrm{eV}$, or $1.1 \times 10^3\,\mathrm{s}^{-1}$; keeping the absorb-$\omega_1$-first ordering alone would overestimate it by a factor of 16. At $\omega_1 = 1.10\,\mathrm{eV}$ the two denominators are equal and opposite, and the rate vanishes up to terms of relative order $(\Gamma_\mathrm{w}/\delta_2)^2$.

On resonance, at $\omega_1 = 1.50\,\mathrm{eV}$, the first denominator is $-2\mathrm{i}/\Gamma_\mathrm{w}$ and $|\mathcal{A}|^2 = 4.0 \times 10^6\,\mathrm{eV}^{-2}$, so $\Gamma_{i_\mathrm{T}\to f_\mathrm{T}} = 4.0 \times 10^{-6}\,\mathrm{eV} = 6.1 \times 10^9\,\mathrm{s}^{-1}$, larger than the off-resonant rate by 5.8×10^6. The cascade picture reproduces this number. On resonance, beam 1 excites m at the first-order rate $2\pi|\langle m|\hat{O}|i_\mathrm{T}\rangle|^2$ times the peak $2/(\pi\Gamma_\mathrm{w})$ of the Lorentzian of m, that is $4|\langle m|\hat{O}|i_\mathrm{T}\rangle|^2/\Gamma_\mathrm{w} = 4.0 \times 10^{-5}\,\mathrm{eV}$; a fraction $\Gamma_{m\to f_\mathrm{T}}/\Gamma_\mathrm{w} = 0.10$ of the excited levels end in the band. The product agrees with the second-order rate to five parts in 10^6, the contribution of the off-resonant ordering. Perturbation theory holds, since the probability of finding the target in m on resonance is $|\langle m|\hat{O}|i_\mathrm{T}\rangle|^2/(\Gamma_\mathrm{w}/2)^2 = 0.04$.

A broadband beam 1 tests Eq. (2.60). Let its spectrum be a Gaussian of rms width 10 meV in δ_1, normalized to one; the measured rate is $|\mathcal{A}|^2$ averaged over it. Centered on

resonance, the exact average is $2.410 \times 10^5\,\mathrm{eV}^{-2}$. The delta term of Eq. (2.60) is $(2\pi/\Gamma_\mathrm{w})$ times the spectral density at resonance, $2.507 \times 10^5\,\mathrm{eV}^{-2}$, an overestimate by 4.0 %. The finite part, $-1/(10\,\mathrm{meV})^2 = -1.00 \times 10^4\,\mathrm{eV}^{-2}$ for a centered Gaussian, is negative, as it must be for a spectrum peaked at the pole. Delta and finite part together are within 0.13 % of the exact value, and the remainder matches the term $-\pi\Gamma_\mathrm{w} w''(0)/4$. Centered 30 meV away from resonance, the delta term is $2785\,\mathrm{eV}^{-2}$, the finite part $1795\,\mathrm{eV}^{-2}$, now positive, the principal-value cross term between the two orderings $197\,\mathrm{eV}^{-2}$, and the off-resonant ordering $6\,\mathrm{eV}^{-2}$. Their sum is within 0.6 % of the exact average, $4753\,\mathrm{eV}^{-2}$, while the delta term alone misses it by 41 %. The cascade accounts for nearly the whole rate only when the spectrum sits on the resonance and is broad compared with Γ_w. Even then the finite part corrects it by about Γ_w divided by the spectral width. Away from resonance, the finite part and the cross term between the orderings are as large as the cascade itself.

In silicon the intermediate electron of Eq. (2.54) becomes real once the photon energy exceeds the direct gap, and Sect. 6.5 handles that region with these tools.

2.5 Thermal states and detailed balance

At 300 K each optical phonon mode of silicon at the zone center, at 64 meV, holds on average 0.09 quanta; at the 4 K of a liquid-helium cryostat it holds 10^{-81}. A crystal at a finite temperature is not in an eigenstate of its Hamiltonian, as the rates of Sects. 2.1 and 2.2 assumed, but in a statistical mixture of them, and an excited component of the mixture can give its energy to the dark matter. This is the negative ω that Sect. 1.3 allowed. We will write the thermal state, average the golden rule over it, and derive detailed balance, the relation that ties energy gain by the dark matter to energy loss. At zero temperature the same occupation numbers produce the filled Fermi sea of a free electron gas, whose Fermi momentum and density of states Chap. 4 uses throughout. We follow Refs. [19] and [11, Chap. 1] for the ensembles and Ref. [20, Chap. 2] for the free electron gas.

2.5.1 Gibbs state

A mixture that holds the eigenstate $|i\rangle$ with probability p_i is described by the density matrix

$$\hat{\varrho} = \sum_i p_i\, |i\rangle\langle i|, \qquad \langle \hat{A} \rangle = \mathrm{Tr}(\hat{\varrho}\,\hat{A}) = \sum_i p_i\, \langle i|\hat{A}|i\rangle, \tag{2.61}$$

with $\sum_i p_i = 1$; the thermal average $\langle \hat{A} \rangle$ is the quantum expectation value averaged over the mixture. A target in equilibrium with a bath at temperature T has the Gibbs weights

$$p_i = \frac{\mathrm{e}^{-\beta E_i}}{\mathcal{Z}}, \qquad \mathcal{Z} = \sum_i \mathrm{e}^{-\beta E_i} = \mathrm{Tr}\ \mathrm{e}^{-\beta H}, \qquad \beta = \frac{1}{T}, \tag{2.62}$$

so that $\hat{\varrho} = \mathrm{e}^{-\beta H}/\mathcal{Z}$, with $\mathcal{Z}$ the partition function. In natural units temperature is an energy, $1\,\mathrm{K} = 86.17\,\mu\mathrm{eV}$, and a detector at 10 mK has $T = 0.86\,\mu\mathrm{eV}$.

Since the Gibbs state is a function of H, it commutes with H, and its time evolution $\hat{\varrho}(t) = \mathrm{e}^{-\mathrm{i}Ht}\,\hat{\varrho}\,\mathrm{e}^{\mathrm{i}Ht}$ leaves it unchanged: it is a stationary state, and every thermal average is independent of time. A correlation $\langle \hat{A}(t)\hat{B}(t') \rangle$ of Heisenberg operators $\hat{A}(t) = \mathrm{e}^{\mathrm{i}Ht}\,\hat{A}\,\mathrm{e}^{-\mathrm{i}Ht}$ depends only on $t - t'$, which is what makes its Fourier transform in time a function of one frequency (Sect. 4.1). Note that stationarity does not need the Gibbs form; any $\hat{\varrho}$ diagonal in the energy basis has it. Thermal equilibrium fixes the weights.

Particles can also be exchanged with a reservoir, or we may prefer to fix their mean number instead of the exact one. The grand-canonical state then replaces H by $H - \mu_{\text{ch}}\hat{N}$, with $\hat{N}$ the particle-number operator and μ_{ch} the chemical potential:

$$\hat{\varrho} = \frac{\mathrm{e}^{-\beta(H-\mu_{\text{ch}}\hat{N})}}{\mathcal{Z}}, \qquad p_i = \frac{\mathrm{e}^{-\beta(E_i-\mu_{\text{ch}}N_i)}}{\mathcal{Z}}. \tag{2.63}$$

The value of μ_{ch} is fixed by $\langle\hat{N}\rangle = N$. A detector holds a fixed number of electrons, and the canonical state is the physical one. However, the two ensembles differ in a macroscopic body by fluctuations of relative size $N^{-1/2}$, and the grand-canonical state is much easier to use, since for independent particles it factorizes into one factor per single-particle state (Eq. (2.69) below).

2.5.2 Thermal golden rule

The probability of finding the system in $|f\rangle$ at time t is linear in the initial density matrix. With $U(t)$ the evolution operator,

$$\langle f|U(t)\,\hat{\varrho}\,U(t)^\dagger|f\rangle = \sum_i p_i\,|\langle f|U(t)|i\rangle|^2 = \sum_i p_i\,P_{i\to f}(t).$$

Different initial states do not interfere, because $\hat{\varrho}$ has no off-diagonal elements between them. Each $P_{i\to f}$ grows at its golden-rule rate (Eq. (2.5)), and the rate out of the thermal state is the weighted sum

$$\Gamma = \sum_i p_i \sum_f \Gamma_{i\to f}. \tag{2.64}$$

This is the thermal average over initial states (Sect. 2.1); the final states are summed, as before.

For dark-matter scattering the initial state is the product of the thermal target and a dark-matter plane wave of momentum $\boldsymbol{p}$, and the final state has the target in $|f\rangle$ and the dark matter at $\boldsymbol{p}-\boldsymbol{q}$. The interaction between these two dark-matter states is $\tilde{V}(\boldsymbol{q})\,n_e(\boldsymbol{q})/\mathscr{V}$ (Eq. (2.38)), and energy conservation, with the dark-matter energies written out, is $\delta(\omega_{\boldsymbol{q}} - E_f + E_i)$, where $\omega_{\boldsymbol{q}}$ is the energy lost by the dark matter (Eq. (1.22)). Summing Eq. (2.64) over the final momenta,

$$\Gamma = \sum_{\boldsymbol{q}} \frac{2\pi}{\mathscr{V}^2}\,|\tilde{V}(\boldsymbol{q})|^2\,\mathcal{S}_{n_e(\boldsymbol{q})}(\omega_{\boldsymbol{q}}),$$

where all the target physics sits in the transition sum of an operator $\hat{O}$,

$$\mathcal{S}_O(\omega) = \sum_{i,f} p_i\,|\langle f|\hat{O}|i\rangle|^2\,\delta(\omega - E_f + E_i). \tag{2.65}$$

It counts the transitions that $\hat{O}$ drives from the thermal state, weighted by their probability, at each energy ω given to the target. The same object is the Fourier transform of a thermal correlation function. Inserting a complete set of states between the operators of $\langle\hat{O}^\dagger(t)\hat{O}(0)\rangle$, each term oscillates as $\mathrm{e}^{\mathrm{i}(E_i-E_f)t}$, and the time integral turns it into a delta function:

$$\int_{-\infty}^{\infty} \mathrm{d}t\;\mathrm{e}^{\mathrm{i}\omega t}\,\langle\hat{O}^\dagger(t)\,\hat{O}(0)\rangle = 2\pi\,\mathcal{S}_O(\omega). \tag{2.66}$$

The rate measures how the fluctuations of $\hat{O}$ in the equilibrium state are correlated in time, the viewpoint that Sect. 4.2 develops. With Eq. (2.10) for the sum over $\boldsymbol{q}$,

Thermal golden rule

$$\Gamma = \int \frac{\mathrm{d}^3 q}{(2\pi)^3} \underbrace{|\tilde{V}(\boldsymbol{q})|^2}_{\text{coupling}} \underbrace{S(\boldsymbol{q}, \omega_{\boldsymbol{q}})}_{\text{target}}, \qquad S(\boldsymbol{q}, \omega) = \frac{2\pi}{\mathscr{V}} \mathcal{S}_{n_e(\boldsymbol{q})}(\omega). \tag{2.67}$$

Γ is the scattering rate of one dark-matter particle of momentum $\boldsymbol{p}$ in a box of volume $\mathscr{V}$ filled by the target, which is in thermal equilibrium at temperature T.

The function $S(\boldsymbol{q}, \omega)$ is the dynamic structure factor of the electron density (Eq. (A.31)) [21, Eq. (4)]; that reference writes the density at $-\boldsymbol{k}$ where we write $n_e(\boldsymbol{q})$ (Appendix A.7); Sect. 4.2 develops it. The volume drops out as it should. The transition sum is extensive, since a target twice as large offers twice as many transitions, and S is intensive. The rate of one dark-matter particle, present with density $1/\mathscr{V}$ in a target of volume $\mathscr{V}$, is then independent of $\mathscr{V}$.

The rate assembled in Sect. 1.5 now follows. For a target whose response depends only on $q = |\boldsymbol{q}|$, we write $S(q, \omega_{\boldsymbol{q}}) = \int \mathrm{d}\omega\, S(q, \omega)\, \delta(\omega - \omega_{\boldsymbol{q}})$. The average of the delta function over the direction of $\boldsymbol{q}$ and over the halo is $\eta(v_{\min})/2q$ (Sect. 1.3; Eq. (2.20)), and the measure becomes $\int \mathrm{d}^3 q/(2\pi)^3\, (2q)^{-1} = \int q\, \mathrm{d}q/(2\pi)^2$. Multiplying by the number density ρ_χ/m_χ of dark matter, dividing by the target mass density ρ_T and inserting $|\tilde{V}|^2 = \pi \bar{\sigma}_e F_{\mathrm{DM}}^2/\mu_{\chi e}^2$ (Eq. (1.34)) reproduces Eq. (1.47) term by term. Unlike the version of Sect. 1.5, the derivation holds at any temperature, with the thermal weights inside S, and its integral over ω runs over negative values too.

2.5.3 Detailed balance

Negative ω means that the target gives energy to the dark matter. The weight of such transitions is tied to the weight of the reverse ones by relabeling the sum. Write the transition sum of $\hat{O}^\dagger$ at $-\omega$ and exchange the names of the summed states i and f:

$$\mathcal{S}_{O^\dagger}(-\omega) = \sum_{i,f} p_i\, |\langle f|\hat{O}^\dagger|i\rangle|^2\, \delta(-\omega - E_f + E_i) = \sum_{i,f} p_f\, |\langle i|\hat{O}^\dagger|f\rangle|^2\, \delta(\omega - E_f + E_i).$$

Since $|\langle i|\hat{O}^\dagger|f\rangle| = |\langle f|\hat{O}|i\rangle|$, the identity of Sect. 2.1, the matrix elements are those of $\mathcal{S}_O(\omega)$. The weights differ: the delta function sets $E_f = E_i + \omega$, where $p_f = p_i\, \mathrm{e}^{-\beta\omega}$. Hence

Detailed balance

$$\mathcal{S}_{O^\dagger}(-\omega) = \mathrm{e}^{-\beta\omega}\, \mathcal{S}_O(\omega). \tag{2.68}$$

For the electron density, $n_e(\boldsymbol{q})^\dagger = n_e(-\boldsymbol{q})$, and $S(-\boldsymbol{q}, -\omega) = \mathrm{e}^{-\beta\omega} S(\boldsymbol{q}, \omega)$. For a target with inversion or time-reversal symmetry, $S(\boldsymbol{q}, -\omega) = \mathrm{e}^{-\beta\omega} S(\boldsymbol{q}, \omega)$. The density form is the diagonal element of Eq. (A.34).

In the grand-canonical state the weights carry $-\mu_{\mathrm{ch}} N$ as well, and the step $p_f = p_i \mathrm{e}^{-\beta\omega}$ needs $N_f = N_i$. It holds for every operator that conserves particle number, such as $n_e(\boldsymbol{q})$; an operator that adds ΔN particles acquires the factor $\mathrm{e}^{-\beta(\omega - \mu_{\mathrm{ch}}\Delta N)}$ instead. The symmetric form uses $S(-\boldsymbol{q}, \omega) = S(\boldsymbol{q}, \omega)$, which inversion or time reversal guarantees (Appendix A.4); silicon has both.

Detailed balance holds for every target in equilibrium, whatever its interactions, because the argument used only the Gibbs weights. In a two-level system with splitting Δ_{tl}, driven by a Hermitian $\hat{O}$ with matrix element $O_{10} = \langle 1|\hat{O}|0\rangle$, the transition sum is

$$\mathcal{S}_O(\omega) = |O_{10}|^2 [p_0\, \delta(\omega - \Delta_{\mathrm{tl}}) + p_1\, \delta(\omega + \Delta_{\mathrm{tl}})], \qquad \frac{p_1}{p_0} = \mathrm{e}^{-\beta\Delta_{\mathrm{tl}}}.$$

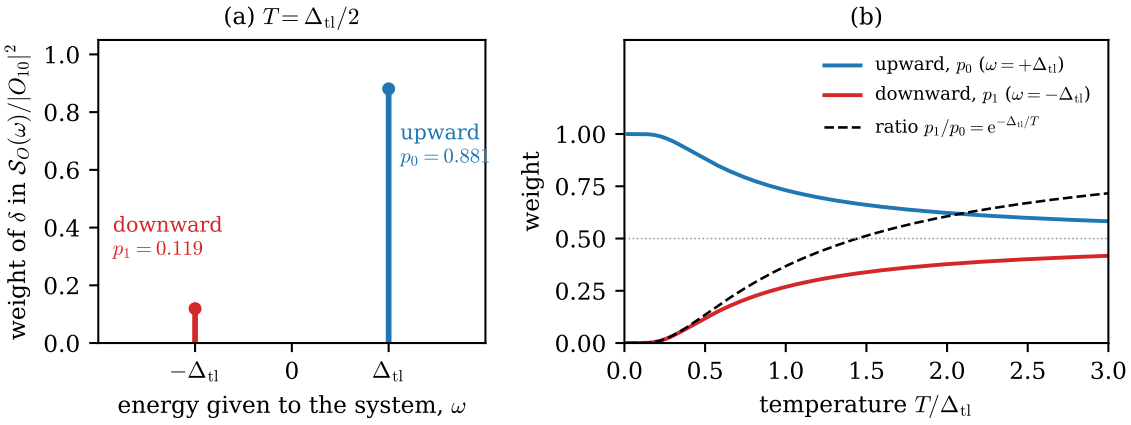


Figure 2.8: Upward and downward transitions of a two-level system with splitting Δ_{tl} in thermal equilibrium at temperature T, driven by a Hermitian operator $\hat{O}$ with matrix element O_{10}. (a) At $T = \Delta_{tl}/2$ the transition sum $\mathcal{S}_O(\omega)$ consists of two delta functions, drawn as stems with their weights in units of $|O_{10}|^2$. Upward, at energy $\omega = +\Delta_{tl}$ given to the system, with the ground-state population $p_0 = 0.881$; downward, at $\omega = -\Delta_{tl}$, with the excited-state population $p_1 = 0.119$. (b) The two weights against T/Δ_{tl}, and their ratio $p_1/p_0 = e^{-\Delta_{tl}/T}$ (detailed balance). The weights approach $\frac{1}{2}$ (dotted line) from either side as T grows and never cross.

The upward transition, at $\omega = +\Delta_{tl}$, starts from the ground state; the downward one, at $\omega = -\Delta_{tl}$, starts from the excited state and carries its population (Fig. 2.8). At $T = \Delta_{tl}/2$ the weights are 0.881 and 0.119. At $T = 0$ the downward weight vanishes and $\omega \geq 0$. As $T \to \infty$ both tend to $\frac{1}{2}$, but the upward weight never falls below the downward one: a thermal target always absorbs more often than it emits, by the factor $e^{\beta\omega}$.

The negative-ω part of the rate is suppressed by the Boltzmann factor of the excitation it removes, $e^{-\beta|\omega|}$ relative to the positive-ω part at the same $|\omega|$. For a 1 meV acoustic phonon at 1 K, $\beta\omega = 11.6$ and the factor is 9×10^{-6}; Appendix A.4 gives the case of an electronic gap. Note that such an event deposits no energy in the detector. The target loses energy to the dark matter, and a calorimeter or a charge readout registers nothing. Such transitions nevertheless matter in three places: in the total scattering rate, in the ω integrals of sum rules (Sect. 4.3), and in the factor $1 - e^{-\beta\omega}$ that relates S to the dissipative part of the response, the fluctuation–dissipation theorem (Eq. (A.33), Sect. 4.2).

2.5.4 Occupation of a mode

The thermal factors n and $n+1$ of a bosonic mode (Sect. 2.3) follow from the grand-canonical state of independent particles. For $H - \mu_{ch}\hat{N} = \sum_\gamma (E_\gamma - \mu_{ch})\, c_\gamma^\dagger c_\gamma$ the exponential in Eq. (2.63) is a product over single-particle states γ, and so is $\mathcal{Z}$. Each fermion factor has $n_\gamma = 0$ or 1, and each boson factor $n_\gamma = 0, 1, 2, \ldots$, a geometric series. The mean occupation of one state follows by weighting n_γ with its factor:

$$\langle c_\gamma^\dagger c_\gamma \rangle = \frac{e^{-\beta(E_\gamma - \mu_{ch})}}{1 + e^{-\beta(E_\gamma - \mu_{ch})}} = n_F(E_\gamma - \mu_{ch}), \qquad \langle d_\gamma^\dagger d_\gamma \rangle = \frac{\sum_n n\, e^{-\beta n E_\gamma}}{\sum_n e^{-\beta n E_\gamma}} = n_B(E_\gamma), \tag{2.69}$$

with the Fermi and Bose functions $n_F(x) = (e^{\beta x} + 1)^{-1}$ and $n_B(x) = (e^{\beta x} - 1)^{-1}$ [19]. A mode with $E_\gamma \ll T$ holds $n_B \approx T/E_\gamma$ quanta, the classical equipartition value, and a mode with $E_\gamma \gg T$ holds $n_B \approx e^{-\beta E_\gamma}$. The boson form has no chemical potential. Phonons

and magnons are created and destroyed freely, their number is not conserved, and the free energy is at its minimum when $\mu_{\mathrm{ch}} = 0$. The reversed products follow from Eqs. (2.32) and (2.35): $\langle c_\gamma c_\gamma^\dagger \rangle = 1 - n_{\mathrm{F}}$, the probability that the state is free to receive an electron, and $\langle d_\gamma d_\gamma^\dagger \rangle = n_{\mathrm{B}} + 1$. Absorbing a boson from a thermal mode has the weight n_{B} and emitting one has $n_{\mathrm{B}} + 1$, and $(n_{\mathrm{B}} + 1)/n_{\mathrm{B}} = \mathrm{e}^{\beta E_\gamma}$ is detailed balance for a single mode. Off-diagonal averages $\langle c_\gamma^\dagger c_{\gamma'} \rangle$ with $\gamma \neq \gamma'$ vanish, since $\hat{\varrho}$ conserves each occupation.

The fermion factors decide the transition sum of the density for independent electrons. By Eq. (2.36), $n_e(\boldsymbol{q}) = \sum_{\gamma\gamma'} \langle \gamma' | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \gamma \rangle \, c_{\gamma'}^\dagger c_\gamma$. Acting on an occupation-number state $|i\rangle$, each term with $\gamma' \neq \gamma$ moves one electron and leads to its own final state, and the sum over those final states of $|\langle f | c_{\gamma'}^\dagger c_\gamma | i \rangle|^2$ is $\langle i | c_\gamma^\dagger c_{\gamma'} c_{\gamma'}^\dagger c_\gamma | i \rangle = n_\gamma (1 - n_{\gamma'})$. The terms with $\gamma' = \gamma$ leave the state unchanged and contribute only at $\omega = 0$. The Gibbs state of independent electrons is a product over orbitals, the average of $n_\gamma(1 - n_{\gamma'})$ factorizes, and for $\omega \neq 0$

$$\mathcal{S}_{n_e(\boldsymbol{q})}(\omega) = \sum_{\gamma \neq \gamma'} |\langle \gamma' | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \gamma \rangle|^2 \, n_{\mathrm{F}}(E_\gamma - \mu_{\mathrm{ch}}) \big[1 - n_{\mathrm{F}}(E_{\gamma'} - \mu_{\mathrm{ch}})\big] \, \delta(\omega - E_{\gamma'} + E_\gamma). \tag{2.70}$$

An electron must be present in γ, with probability n_{F}, and γ' must be free to receive it, with probability $1 - n_{\mathrm{F}}$: this is Pauli blocking at finite temperature. The identity $n_{\mathrm{F}}(E')[1 - n_{\mathrm{F}}(E)] = \mathrm{e}^{-\beta(E'-E)} \, n_{\mathrm{F}}(E)[1 - n_{\mathrm{F}}(E')]$, with both energies measured from μ_{ch}, makes Eq. (2.68) visible term by term. In a semiconductor at $T \ll E_{\mathrm{g}}$ the factors are one for a filled valence state γ and an empty conduction state γ' and zero otherwise, and Eq. (2.70) becomes a sum over valence-to-conduction transitions of the squared single-particle form factor of Eq. (2.39). With the spin sum supplying the factor 2 of a spin-independent vertex, and written in Bloch states, that sum is the crystal form factor of Chap. 10.

Worked example 2.7: Scattering off a thermal harmonic oscillator

Let the target be a single mode, $H = \omega_0 (d^\dagger d + \frac{1}{2})$, in equilibrium at temperature T, and let the dark matter couple to its displacement, $\hat{O} = \lambda_{\mathrm{osc}} (d + d^\dagger)$, with a coupling λ_{osc} that depends on $\boldsymbol{q}$. A phonon mode coupled to a nuclear recoil has this form (Sect. 5.3). By Eq. (2.34), $\hat{O}$ connects $|n\rangle$ to $|n+1\rangle$ with the matrix element $\lambda_{\mathrm{osc}} \sqrt{n+1}$ and to $|n-1\rangle$ with $\lambda_{\mathrm{osc}} \sqrt{n}$. The transition sum Eq. (2.65) is therefore

$$\begin{aligned} \mathcal{S}_O(\omega) &= |\lambda_{\mathrm{osc}}|^2 \sum_n p_n \big[(n+1)\, \delta(\omega - \omega_0) + n \, \delta(\omega + \omega_0)\big] \\ &= |\lambda_{\mathrm{osc}}|^2 \big[(n_{\mathrm{B}} + 1)\, \delta(\omega - \omega_0) + n_{\mathrm{B}} \, \delta(\omega + \omega_0)\big], \end{aligned}$$

with $n_{\mathrm{B}} = n_{\mathrm{B}}(\omega_0) = \sum_n p_n n$ by Eq. (2.69). Emission of a quantum, which takes energy from the dark matter, has the weight $n_{\mathrm{B}} + 1$; absorption of a quantum, which gives energy to the dark matter, has the weight n_{B}. At the same $\boldsymbol{q}$ their ratio is

$$\frac{\mathcal{S}_O(+\omega_0)}{\mathcal{S}_O(-\omega_0)} = \frac{n_{\mathrm{B}} + 1}{n_{\mathrm{B}}} = \mathrm{e}^{\beta\omega_0},$$

as Eq. (2.68) requires, since $\hat{O}$ is Hermitian. The ratio of the two scattering rates also contains the dark-matter kinematics: the energy-loss rate at fixed q carries $\eta(v_{\min}(q, +\omega_0))$ and the energy-gain rate $\eta(v_{\min}(q, -\omega_0))$ (Eq. (1.23)). The minimum velocity for energy gain, $|q/2m_\chi - \omega_0/q|$, vanishes at $q = (2m_\chi \omega_0)^{1/2}$, and gain is kinematically allowed for every dark-matter mass.

For numbers, take the 64 meV optical phonon of silicon at the zone center. At 300 K, $\beta\omega_0 = 2.48$, $n_{\mathrm{B}} = 0.092$, and the loss-to-gain ratio of the target factors is $\mathrm{e}^{2.48} = 11.9$. At 4 K, $\beta\omega_0 = 186$ and $n_{\mathrm{B}} = 2.3 \times 10^{-81}$. For a 1 meV acoustic phonon, $n_{\mathrm{B}} = 9.1 \times 10^{-6}$ at

1 K and 10^{-252} at 20 mK, but 1.27 at 20 K, where the ratio is 1.79. A cryogenic detector is in its ground state for every mode above a few tens of μeV, and the "+1" of spontaneous emission is the whole signal: the dark matter creates a phonon in an empty mode.
The total weight $\int \mathrm{d}\omega\, \mathcal{S}_O$ and the first moment $\int \mathrm{d}\omega\, \omega\, \mathcal{S}_O$ respond differently to temperature. The total weight is $\int \mathrm{d}\omega\, \mathcal{S}_O = |\lambda_{\mathrm{osc}}|^2(2n_{\mathrm{B}}+1)$, which grows with T because the thermal fluctuations of the displacement grow; at 300 K it is 1.18 times the value at $T = 0$ for the optical phonon. The first moment is $\int \mathrm{d}\omega\, \omega\, \mathcal{S}_O = |\lambda_{\mathrm{osc}}|^2\omega_0[(n_{\mathrm{B}}+1)-n_{\mathrm{B}}] = |\lambda_{\mathrm{osc}}|^2\omega_0$, independent of T, because the thermal populations cancel between emission and absorption. This is the oscillator's version of the f-sum rule (Eq. (A.39), Sect. 4.3); it requires the negative-ω weight.

2.5.5 Free electron gas at zero temperature

Jellium with its Coulomb interaction dropped (Worked example 2.5) is a gas of free electrons, $H = \sum_{\boldsymbol{ks}} (k^2/2m_e)\, c^\dagger_{\boldsymbol{ks}} c_{\boldsymbol{ks}}$, and its occupations are $n_{\mathrm{F}}(k^2/2m_e - \mu_{\mathrm{ch}})$. As $T \to 0$ the Fermi function becomes a step, $n_{\mathrm{F}}(x) \to \Theta(-x)$: every state below μ_{ch} is filled and every state above it is empty. The ground state is the Fermi sea, $\prod_{k<k_{\mathrm{F}},\, s} c^\dagger_{\boldsymbol{ks}}|0\rangle$, the filled sphere of radius k_{F} in $\boldsymbol{k}$ space, and the chemical potential at $T = 0$ is the Fermi energy $E_{\mathrm{F}} = k_{\mathrm{F}}^2/2m_e$. Counting two spin states and the volume $(2\pi)^3/\mathcal{V}$ per wavevector (Eq. (2.10)), $N_e = 2\mathcal{V}(4\pi k_{\mathrm{F}}^3/3)/(2\pi)^3 = V k_{\mathrm{F}}^3/3\pi^2$. The density of states per spin and per unit volume at energy E is $\mathcal{D}_f(E)/\mathcal{V} = m_e k/2\pi^2$, from Eq. (2.11), and at the Fermi level it is $\mathcal{N}_{\mathrm{F}}$.

Free electron gas at $T = 0$

$$k_{\mathrm{F}} = (3\pi^2 \boldsymbol{n}_e)^{1/3}, \qquad E_{\mathrm{F}} = \frac{k_{\mathrm{F}}^2}{2m_e}, \qquad v_{\mathrm{F}} = \frac{k_{\mathrm{F}}}{m_e}, \qquad \mathcal{N}_{\mathrm{F}} = \frac{m_e k_{\mathrm{F}}}{2\pi^2}, \tag{2.71}$$

where $\boldsymbol{n}_e = N_e/\mathcal{V}$ is the mean electron density and $\mathcal{N}_{\mathrm{F}}$ counts one spin. As a check, $2\mathcal{N}_{\mathrm{F}} E_{\mathrm{F}} = \frac{3}{2}\boldsymbol{n}_e$.

These are the results of Ref. [20, Chap. 2]. For the four valence electrons per atom of silicon, $\boldsymbol{n}_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$ (Appendix A.2), and Eq. (2.71) yields $k_{\mathrm{F}} = 3.57\,\mathrm{keV} = 0.96\, q_0$, $E_{\mathrm{F}} = 12.5\,\mathrm{eV}$, $v_{\mathrm{F}} = 7.0 \times 10^{-3} = 2090\,\mathrm{km/s}$ and $\mathcal{N}_{\mathrm{F}} = 9.2 \times 10^7\,\mathrm{eV}^2$. Silicon is of course not a metal: its valence electrons fill bands separated from the empty ones by a gap (Sect. 3.3). The gas sets the scales, however. Its Fermi momentum is the typical electron momentum $q_0 = \alpha m_e$ of Sect. 1.3, and its Fermi velocity is 2.6 times the maximum dark-matter speed $v_{\max} = 2.65 \times 10^{-3}$. Temperature hardly matters here: T/E_{F} is 2×10^{-3} at 300 K and 7×10^{-8} at 10 mK, and the step is sharp. In a real crystal the same occupations apply to the computed bands; for metals, codes replace the step by a smooth function of width much larger than T to converge the sums over the Brillouin zone (Appendix E.1).

The Fermi sea quantifies the Pauli blocking of Sect. 2.3. The term $c^\dagger_{\boldsymbol{k}+\boldsymbol{q},s} c_{\boldsymbol{ks}}$ of $n_e(\boldsymbol{q})$ survives if $\boldsymbol{k}$ lies inside the sphere and $\boldsymbol{k}+\boldsymbol{q}$ outside it. The blocked electrons are those in the overlap of the sphere with its copy displaced by $\boldsymbol{q}$, a lens of volume $\pi(4k_{\mathrm{F}} + q)(2k_{\mathrm{F}} - q)^2/12$. The fraction of the sea that can take the momentum q is therefore

$$\frac{3q}{4k_{\mathrm{F}}} - \frac{q^3}{16k_{\mathrm{F}}^3} \quad (q < 2k_{\mathrm{F}}), \qquad 1 \quad (q \geq 2k_{\mathrm{F}}). \tag{2.72}$$

At small q it grows linearly: the active electrons form a shell of thickness $\sim q$ at the Fermi surface. At $q = 0.1\, q_0$, where a light mediator puts much of its weight, 7.8 % of the silicon-density sea is active. Energy conservation narrows this further, since the dark matter supplies

at most $qv_{\max}$ and $v_{\max} < v_{\mathrm{F}}$. Combining the two constraints with the occupations of Eq. (2.69) is the Lindhard calculation of Sect. 4.5.

Summary

The rate at which dark matter excites a target follows from one formula, Fermi's golden rule, $\Gamma_{i\to f} = 2\pi|\langle f|H'|i\rangle|^2\delta(E_f - E_i)$ (Eq. (2.5)). It is a rate only after a sum over a continuum of final states and within a window of times, which in dark-matter scattering is always open because the outgoing dark matter supplies the continuum. In a periodic box every factor of the volume can be tracked, and a rate turns into a cross-section by dividing by the flux (Eq. (2.13)). A harmonic perturbation separates absorption from emission, and the absorption rate of a bosonic field is fixed by its energy density, not its flux.

In the first Born approximation, which suffices because every allowed coupling of dark matter to ordinary matter keeps the Born parameters below 10^{-3}, the cross-section on a free electron is $\bar{\sigma}_e$ for a heavy mediator (Eq. (2.16)). On a bound target the potential factorizes from the transition form factor $f_{i\to f}(\boldsymbol{q}) = \langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle$ (Eq. (2.18)), whose squares sum to one over final states, and the halo enters only through $\eta(v_{\min})$. Nucleons add coherently at small q, which is the origin of the A^2 of nuclear scattering, and the Helm form factor describes how the coherence is lost at large q.

Second quantization writes the many-electron states as occupation numbers, and the operator that dark matter couples to as $n_e(\boldsymbol{q}) = \sum_{\boldsymbol{k}s} c^\dagger_{\boldsymbol{k}+\boldsymbol{q},s} c_{\boldsymbol{k}s}$ (Eq. (2.38)). The Coulomb Hamiltonian of electrons and nuclei, with its $\boldsymbol{k} = 0$ terms removed by neutrality (Eq. (2.48)), is the starting point of Chaps. 3 and 5.

Where first order vanishes, by a selection rule or by kinematics, the rate comes from the second-order T-matrix (Eq. (2.52)), with virtual intermediate states and energy denominators. When one quantum is absorbed and another emitted, two time orderings connect the same states, and their amplitudes interfere (Eq. (2.54)). A real intermediate state decays exponentially at its golden-rule rate (Wigner–Weisskopf, Eq. (2.55)), and its width turns a vanishing denominator into a finite one. In the rate the resonance then splits into a cascade of two first-order processes and a finite-part remainder (Eq. (2.60)), and counting the cascade separately as well would count it twice. Phonon-assisted absorption in an indirect-gap crystal is the case that returns in Sect. 6.5.

A target at temperature T is a Gibbs mixture, and its rate is the thermal average of the golden rule. For scattering, all the target physics then sits in the dynamic structure factor $S(\boldsymbol{q},\omega)$ (Eq. (2.67)), which reproduces the rate formula of Sect. 1.5. Its negative-ω weight, energy given to the dark matter, is fixed by detailed balance, $S(-\boldsymbol{q},-\omega) = \mathrm{e}^{-\beta\omega}S(\boldsymbol{q},\omega)$; it is suppressed by $\mathrm{e}^{-\beta|\omega|}$ and negligible in a cryogenic detector for every excitation above a few tens of µeV. At $T = 0$ the free electron gas fills a Fermi sphere with $k_{\mathrm{F}} \approx q_0$ at silicon's valence density, and only a shell of it can absorb a small momentum transfer. Chap. 4 builds on these results the response of an interacting target: $S(\boldsymbol{q},\omega)$, its sum rules, and screening.

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Chapter 3

Electrons in crystals

A plane wave with the typical momentum of a valence electron, $q_0 = 3.73\,\mathrm{keV}$ (Sect. 1.3), has a wavelength of 3.3 Å, not much longer than the 2.35 Å bond between two silicon atoms. Dark matter that scatters on electrons with such momentum transfers sees the arrangement of the atoms. Chap. 2 reduced every scattering rate to matrix elements of the density operator $n_e(\boldsymbol{q})$ between states of the target and collected them in the dynamic structure factor, but it left open what those states are. The free electron gas of Sect. 2.5 has the right density and the right momentum scale, $k_\mathrm{F} \approx q_0$, yet it can take up any energy, while the electrons of silicon cannot be excited by less than 1.17 eV. The difference comes from the periodic potential of the nuclei. It groups the electron states into bands separated by gaps, it decides which momentum transfers connect which states, and it shapes the wavefunctions on the scale that a few-keV momentum transfer resolves. The threshold of an electron-recoil detector, and how fast its rate grows above threshold, are therefore properties of the electronic structure of the crystal. Density-functional theory turns the Coulomb Hamiltonian of Eq. (2.48) into a problem of independent electrons in an effective potential, accurate enough to compute bands and wavefunctions of real materials, and we will see where it falls short. By the end of the chapter, along the path of Fig. 3.1, we will be able to compute the matrix elements of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ in silicon from first principles and judge how far to trust them.

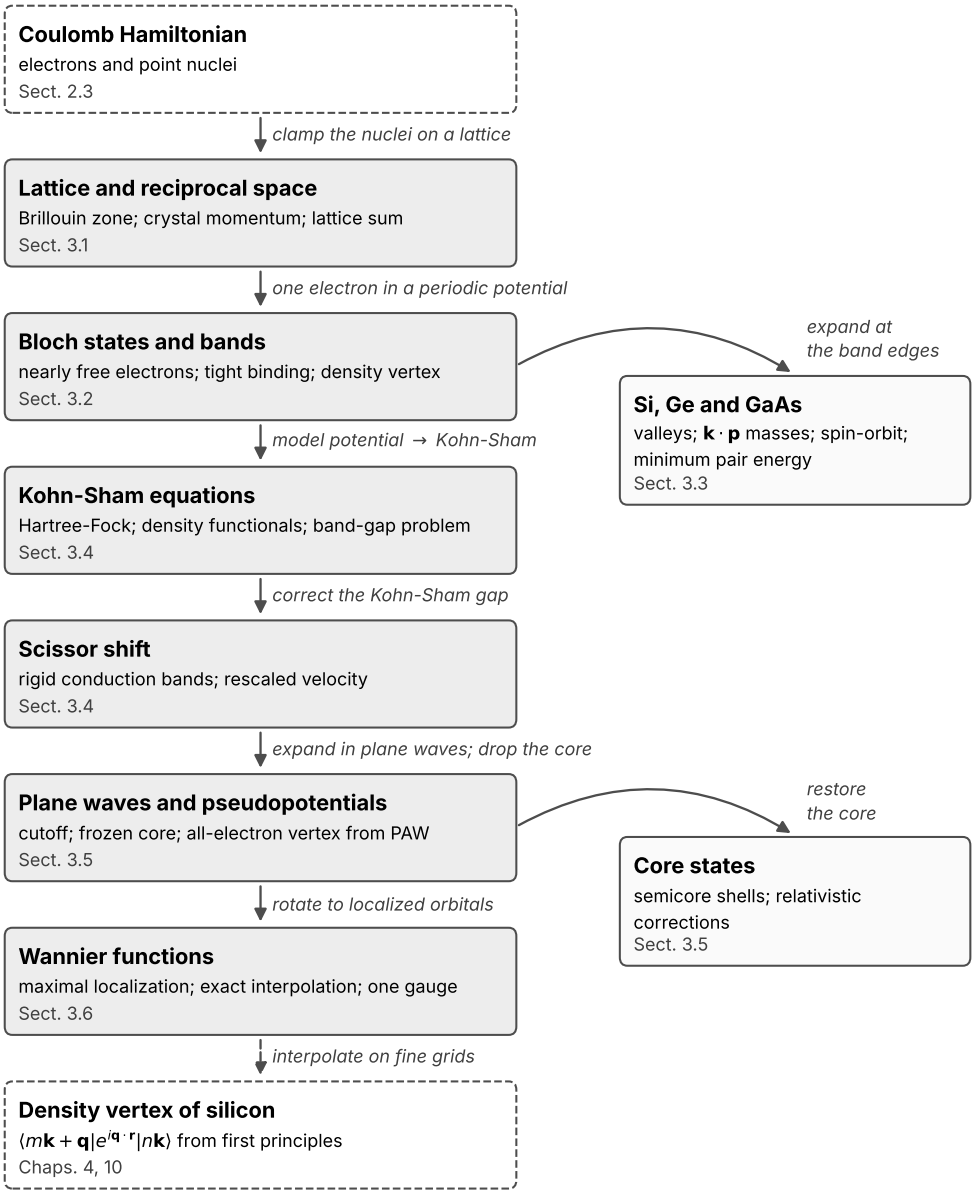


Figure 3.1: From the Coulomb Hamiltonian to the density vertex of silicon. Clamping the nuclei on a lattice gives reciprocal space and crystal momentum (Sect. 3.1); one electron in a periodic potential gives Bloch bands and the density vertex (Sect. 3.2), whose band edges fix the valleys, masses and minimum pair energy of Si, Ge and GaAs (Sect. 3.3). The Kohn–Sham potential replaces the model potential, and the scissor shift corrects its gap (Sect. 3.4); plane waves and pseudopotentials make the calculation practical, with the core restored where large momentum transfers need it (Sect. 3.5); Wannier functions interpolate the result (Sect. 3.6). Dashed boxes: the starting Hamiltonian (Sect. 2.3) and the use of the vertex (Chaps. 4 and 10).

3.1 Lattices and reciprocal space

A kilogram of silicon holds 2.1×10^{25} atoms, and every one of them sits at a point of a lattice that repeats the same pair of atoms in every direction. The repetition has two consequences that every later rate uses: a crystal conserves momentum only up to a reciprocal-lattice vector, the working definition of Sect. 1.3, and the momenta it can tell apart fill a single cell of reciprocal space, the Brillouin zone. We construct both for the diamond structure of silicon, count the states in the zone, and measure the momentum transfers of Sect. 1.3 against the size of the zone. The material is standard, and we follow Refs. [1, Chaps. 4–6 and 8] and [2, Chaps. 1–2].

Throughout Chaps. 3 and 4 the nuclei sit at fixed equilibrium positions: we work with a *static lattice*. The approximation rests on the mass ratio. A silicon nucleus is $M/m_e = 5.1 \times 10^4$ times heavier than an electron, and the electrons rearrange long before the nuclei have moved appreciably. Sect. 5.1 makes this precise with the Born–Oppenheimer expansion. What the static lattice leaves out are the vibrations themselves, phonons, and their coupling to the electrons, which Chaps. 5 and 6 add back.

3.1.1 Bravais lattice and basis

A *Bravais lattice* is the set of points

$$\boldsymbol{R} = n_1\boldsymbol{a}_1 + n_2\boldsymbol{a}_2 + n_3\boldsymbol{a}_3, \qquad n_i \in \mathbb{Z}, \tag{3.1}$$

generated by three linearly independent *primitive vectors* $\boldsymbol{a}_i$. Every lattice vector $\boldsymbol{R}$ maps the lattice onto itself. The parallelepiped spanned by the $\boldsymbol{a}_i$ is a primitive cell: its translations by all $\boldsymbol{R}$ fill space without overlap, and it contains one lattice point. Its volume,

$$\Omega = \boldsymbol{a}_1 \cdot (\boldsymbol{a}_2 \times \boldsymbol{a}_3), \tag{3.2}$$

is the *cell volume*, the only meaning of a bare Ω in this book (a solid angle always carries a subscript, as in $\mathrm{d}\Omega_q$). The choice of primitive vectors is not unique, but every primitive cell has the same volume, since each holds one lattice point.

A crystal places a group of atoms, the *basis*, at every lattice point. We label the atoms of the basis by $\kappa = 1, \ldots, n_{\mathrm{at}}$ and their positions within the cell by $\boldsymbol{\tau}_\kappa$, so that the nuclei sit at

$$\boldsymbol{X}_{\boldsymbol{R}\kappa} = \boldsymbol{R} + \boldsymbol{\tau}_\kappa. \tag{3.3}$$

This is the κ that labels Z_κ and $n_\kappa(\boldsymbol{k})$ in Sect. 2.3; two sites of one element still carry different κ.

Silicon has the *diamond structure* drawn in Fig. 3.2. Its Bravais lattice is face-centered cubic (fcc): the corners and face centers of a cube of side a, the *lattice constant*. A symmetric choice of primitive vectors joins a corner to the three nearest face centers,

$$\boldsymbol{a}_1 = \frac{a}{2}(0,1,1), \qquad \boldsymbol{a}_2 = \frac{a}{2}(1,0,1), \qquad \boldsymbol{a}_3 = \frac{a}{2}(1,1,0), \tag{3.4}$$

and Eq. (3.2) yields $\Omega = a^3/4$: the cube contains four lattice points. The basis consists of two silicon atoms,

$$\boldsymbol{\tau}_1 = 0, \qquad \boldsymbol{\tau}_2 = \frac{a}{4}(1,1,1), \tag{3.5}$$

displaced along the cube diagonal by a quarter of its length. Each atom then has four nearest neighbors at the corners of a regular tetrahedron, at the distance $\sqrt{3}\,a/4$ and with bond angles of 109.47°. These are the four covalent bonds of a silicon atom, one for each valence electron,

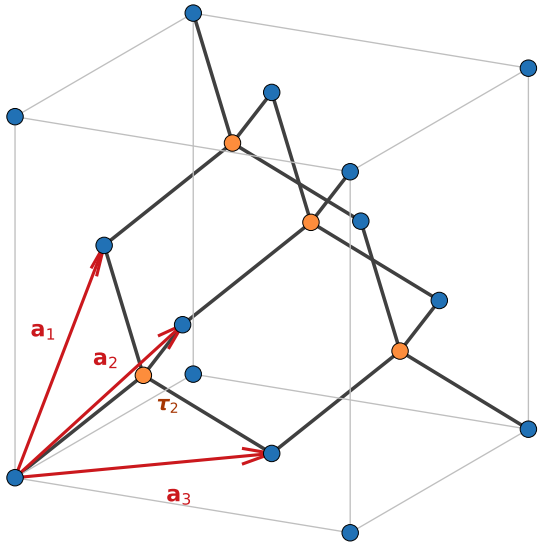


Figure 3.2: The diamond structure, drawn in a cube of side a, the lattice constant. Blue: the face-centered cubic lattice of sites $\boldsymbol{R}$ (cube corners and face centers). Orange: the second atom of the basis, at $\boldsymbol{R}+\boldsymbol{\tau}_2$ with $\boldsymbol{\tau}_2 = (a/4)(1,1,1)$; the four inside the cube are shown. Dark lines join nearest neighbors, at a distance $\sqrt{3}\,a/4$, 2.352 Å in silicon ($a = 5.431$ Å); each atom is bonded to four others at the corners of a regular tetrahedron. Red arrows: the primitive vectors $\boldsymbol{a}_1 = (a/2)(0,1,1)$, $\boldsymbol{a}_2 = (a/2)(1,0,1)$, $\boldsymbol{a}_3 = (a/2)(1,1,0)$, which span a primitive cell of volume $a^3/4$ containing two atoms. In the zincblende structure (GaAs) the blue and orange sites hold different elements.

and Sect. 3.2 returns to the sp^3 orbitals that form them. Note that the diamond structure is not a Bravais lattice: the environment of an atom at $\boldsymbol{\tau}_2$ is that of an atom at $\boldsymbol{\tau}_1$ inverted through the bond center, and no lattice vector maps one onto the other.

The lattice constant of silicon is $a = 5.431\,\text{Å}$ [3, lattice parameter of silicon], or $2.752 \times 10^{-3}\,\text{eV}^{-1}$ in natural units. The cell volume is $\Omega = 40.05\,\text{Å}^3$, the nearest-neighbor distance is $2.352\,\text{Å}$, and the cube of side a holds eight atoms. These numbers fix the densities used in the earlier chapters. The atomic density is $8/a^3 = 4.99 \times 10^{22}\,\text{cm}^{-3}$, which with the atomic weight 28.0855 yields the mass density $2.33\,\text{g}\,\text{cm}^{-3}$ of Sect. 1.2. With four valence electrons per atom, the valence density $32/a^3 = 2.00 \times 10^{23}\,\text{cm}^{-3}$ is the one used for the free electron gas of Sect. 2.5. Germanium has the same structure with a lattice constant a few percent larger. Gallium arsenide has the *zincblende structure*, the same lattice and the same $\boldsymbol{\tau}_\kappa$, but with gallium at $\boldsymbol{\tau}_1$ and arsenic at $\boldsymbol{\tau}_2$ [1, Chap. 4]. Since the two sites now differ, the inversion that exchanged them in silicon is no longer a symmetry. As a result GaAs is polar, a property that matters for its phonons (Sect. 5.5). Sect. 3.3 gives the band structures of all three materials.

A first-principles calculation, in density-functional theory (Sect. 3.4), either takes a from experiment or finds it by minimizing the computed total energy of the crystal with respect to a (Appendix E.1). For silicon in the local-density approximation the minimum lies at 5.401 Å, 0.5 % below the measured value [4, Sect. 10.2]. A rate is usually computed at the measured a. Since band energies depend on a, the choice shifts the computed gap, one of the reasons why computed and measured thresholds differ (Sect. 3.4).

3.1.2 The reciprocal lattice

A plane wave $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$ that takes the same value at every lattice point, $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{R}} = 1$ for all $\boldsymbol{R}$, is the building block of every function with the periodicity of the lattice. The wavevectors $\boldsymbol{G}$ with this property are the *reciprocal-lattice vectors*. By Eq. (3.1) the condition holds for all $\boldsymbol{R}$ if and only if $\boldsymbol{G}\cdot\boldsymbol{a}_i$ is a multiple of 2π for each i. We therefore look for vectors $\boldsymbol{b}_j$ with

$$\boldsymbol{a}_i \cdot \boldsymbol{b}_j = 2\pi\delta_{ij}. \tag{3.6}$$

They are

$$\boldsymbol{b}_1 = \frac{2\pi}{\Omega}\,\boldsymbol{a}_2 \times \boldsymbol{a}_3, \qquad \boldsymbol{b}_2 = \frac{2\pi}{\Omega}\,\boldsymbol{a}_3 \times \boldsymbol{a}_1, \qquad \boldsymbol{b}_3 = \frac{2\pi}{\Omega}\,\boldsymbol{a}_1 \times \boldsymbol{a}_2. \tag{3.7}$$

To check Eq. (3.6), note that $\boldsymbol{a}_1 \cdot \boldsymbol{b}_1 = 2\pi\,\boldsymbol{a}_1 \cdot (\boldsymbol{a}_2 \times \boldsymbol{a}_3)/\Omega = 2\pi$ by Eq. (3.2), while $\boldsymbol{a}_2 \cdot \boldsymbol{b}_1$ and $\boldsymbol{a}_3 \cdot \boldsymbol{b}_1$ vanish because a triple product with a repeated vector is zero; the other components follow by cyclic permutation. Since the $\boldsymbol{b}_j$ are linearly independent, any wavevector can be written as $\boldsymbol{G} = \sum_j m_j \boldsymbol{b}_j$ with real m_j, and then $\boldsymbol{G}\cdot\boldsymbol{a}_i = 2\pi m_i$. The condition $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{R}} = 1$ thus requires integer m_i, and the reciprocal-lattice vectors form a Bravais lattice of their own,

$$\boldsymbol{G} = m_1\boldsymbol{b}_1 + m_2\boldsymbol{b}_2 + m_3\boldsymbol{b}_3, \qquad m_i \in \mathbb{Z}, \tag{3.8}$$

with primitive vectors $\boldsymbol{b}_i$. In the book's convention (Appendix A.3), a function with the periodicity of the lattice is a sum $f(\boldsymbol{r}) = \sum_{\boldsymbol{G}} f_{\boldsymbol{G}}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$ over these vectors, with $f_{\boldsymbol{G}} = \Omega^{-1}\int_\Omega \mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}} f(\boldsymbol{r})$ (Eq. (A.12)). The plane wave $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$ is constant on planes perpendicular to $\boldsymbol{G}$, and for the shortest $\boldsymbol{G}$ in a given direction its wavelength $2\pi/|\boldsymbol{G}|$ is the spacing of the lattice planes perpendicular to it.

The volume of the cell spanned by the $\boldsymbol{b}_i$ follows from Eq. (3.7). With the identity $(\boldsymbol{u}\times\boldsymbol{v})\times(\boldsymbol{u}\times\boldsymbol{w}) = \boldsymbol{u}\,[\boldsymbol{u}\cdot(\boldsymbol{v}\times\boldsymbol{w})]$ for $\boldsymbol{u} = \boldsymbol{a}_1$,

$$\boldsymbol{b}_2 \times \boldsymbol{b}_3 = \frac{(2\pi)^2}{\Omega^2}\,(\boldsymbol{a}_3 \times \boldsymbol{a}_1) \times (\boldsymbol{a}_1 \times \boldsymbol{a}_2) = \frac{(2\pi)^2}{\Omega^2}\,\boldsymbol{a}_1\,[\boldsymbol{a}_1 \cdot (\boldsymbol{a}_2 \times \boldsymbol{a}_3)] = \frac{(2\pi)^2}{\Omega}\,\boldsymbol{a}_1,$$

where the sign of the first cross product was reversed to bring $\boldsymbol{a}_1$ to the front of both factors, and the sign of the second triple product with it. Then Eq. (3.6) gives

$$\boldsymbol{b}_1 \cdot (\boldsymbol{b}_2 \times \boldsymbol{b}_3) = \frac{(2\pi)^3}{\Omega}. \tag{3.9}$$

A small cell in real space is a large one in reciprocal space, as the Fourier transform requires.

For the fcc vectors of Eq. (3.4), Eq. (3.7) yields

$$\boldsymbol{b}_1 = \frac{2\pi}{a}(-1,1,1), \qquad \boldsymbol{b}_2 = \frac{2\pi}{a}(1,-1,1), \qquad \boldsymbol{b}_3 = \frac{2\pi}{a}(1,1,-1), \tag{3.10}$$

as we can confirm directly from Eq. (3.6): for instance, $\boldsymbol{a}_1 \cdot \boldsymbol{b}_1 = \pi(0+1+1) = 2\pi$ and $\boldsymbol{a}_2 \cdot \boldsymbol{b}_1 = \pi(-1+0+1) = 0$. These are the primitive vectors of a body-centered cubic lattice. Writing $\boldsymbol{G} = (2\pi/a)(h,k,l)$, Eq. (3.8) reads $h = -m_1 + m_2 + m_3$ and its cyclic permutations, so that $h + k = 2m_3$, $k + l = 2m_1$ and $l + h = 2m_2$. The reciprocal-lattice vectors of the fcc lattice are the vectors $(2\pi/a)(h,k,l)$ with integers h,k,l that are all even or all odd, and every such triple occurs. The integers are the Miller indices of the lattice planes perpendicular to $\boldsymbol{G}$, and we write $\boldsymbol{G}_{hkl}$. The shortest nonzero vectors are the eight $\boldsymbol{G}_{111}$ along the cube diagonals, with $|\boldsymbol{G}_{111}| = \sqrt{3}\,2\pi/a$, followed by the six $\boldsymbol{G}_{200}$ along the cube axes, with $|\boldsymbol{G}_{200}| = 2 \cdot 2\pi/a$. In silicon $2\pi/a = 2.28\,\mathrm{keV} = 1.16\,\text{Å}^{-1}$, $|\boldsymbol{G}_{111}| = 3.95\,\mathrm{keV}$ and $|\boldsymbol{G}_{200}| = 4.57\,\mathrm{keV}$, the scales already met in Sects. 1.3 and 2.4.

3.1.3 The Brillouin zone

Two wavevectors that differ by a reciprocal-lattice vector give the same phase at every lattice point, $\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{R}} = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}$, and a lattice cannot tell them apart. The wavevectors that a crystal can distinguish fill one primitive cell of the reciprocal lattice. The conventional choice of cell is the *first Brillouin zone*, the set of points closer to the origin than to any other reciprocal-lattice vector (Sect. 1.3). The same construction around a point of the direct lattice, the points closer to it than to any other lattice point, is the *Wigner–Seitz cell*, so the first Brillouin zone is the Wigner–Seitz cell of the reciprocal lattice. Its boundary is made of pieces of the planes $\boldsymbol{k} \cdot \boldsymbol{G} = |\boldsymbol{G}|^2/2$ that bisect the vectors $\boldsymbol{G}$, and a point on the boundary is equidistant from the origin and from at least one $\boldsymbol{G}$. The zone has the full point symmetry of the lattice, which is why it is preferred to the parallelepiped spanned by the $\boldsymbol{b}_i$. Both cells tile reciprocal space under translations by $\boldsymbol{G}$, and the zone has the volume of Eq. (3.9), $(2\pi)^3/\Omega$. For fcc it is $4(2\pi/a)^3$.

For the fcc lattice the zone is the truncated octahedron of Fig. 3.3. The planes that bisect the eight $\boldsymbol{G}_{111}$ cut out an octahedron, $|k_x| + |k_y| + |k_z| \le \frac{3}{2}(2\pi/a)$, and the planes that bisect the six $\boldsymbol{G}_{200}$, $|k_x|, |k_y|, |k_z| \le 2\pi/a$, cut off its six corners; no longer $\boldsymbol{G}$ contributes a face. Points of high symmetry carry standard labels. The zone center is Γ. The centers of the square faces are the six $\mathrm{X} = (2\pi/a)(1,0,0)$, at a distance $2\pi/a$, and the centers of the hexagonal faces are the eight $\mathrm{L} = (2\pi/a)(\frac{1}{2},\frac{1}{2},\frac{1}{2})$, at $\frac{\sqrt{3}}{2}\,2\pi/a$, the closest boundary points to Γ. The midpoints of the edges shared by two hexagons are the points $\mathrm{K} = (2\pi/a)(\frac{3}{4},\frac{3}{4},0)$, and the corners are the points $\mathrm{W} = (2\pi/a)(1,\frac{1}{2},0)$, at $\frac{\sqrt{5}}{2}\,2\pi/a$, the farthest. The zone points are set upright, Γ, X, L, K, W, to keep them apart from the italic rates Γ of Chap. 2 and the loss function L of Sect. 4.4. A band structure is plotted along straight lines between these points, such as L–Γ–X, and the band extrema of silicon, germanium and GaAs lie on them (Sect. 3.3).

For silicon the zone is about 2.3 keV in radius: every momentum shorter than $|\Gamma\mathrm{L}| = 1.98\,\mathrm{keV}$ lies inside it, and every momentum longer than $|\Gamma\mathrm{W}| = 2.55\,\mathrm{keV}$ lies outside. A sphere of the same volume has the radius $(3/\pi)^{1/3}\,2\pi/a = 0.985\,(2\pi/a) = 2.25\,\mathrm{keV}$.

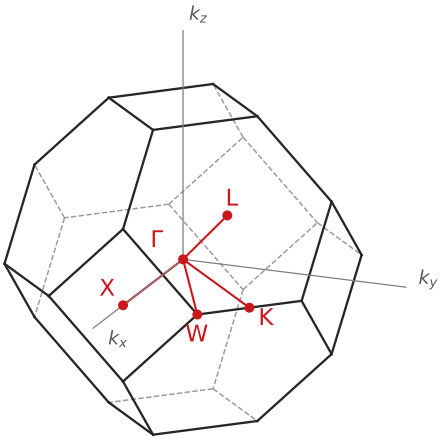


Figure 3.3: The first Brillouin zone of the face-centered cubic lattice, a truncated octahedron with eight hexagonal faces, which bisect the vectors $\boldsymbol{G}_{111} = (2\pi/a)(\pm1, \pm1, \pm1)$, and six square faces, which bisect $\boldsymbol{G}_{200} = (2\pi/a)(\pm2, 0, 0)$ and its permutations. Dashed edges are hidden. The high-symmetry points, in units of $2\pi/a$, are $\Gamma = (0,0,0)$, $\mathrm{X} = (1,0,0)$, $\mathrm{L} = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, $\mathrm{K} = (\frac{3}{4}, \frac{3}{4}, 0)$ and $\mathrm{W} = (1, \frac{1}{2}, 0)$; their distances from Γ are 1, 0.866, 1.061 and 1.118, or 2.28, 1.98, 2.42 and 2.55 keV for silicon ($a = 5.431$ Å). The zone volume is $4(2\pi/a)^3 = (2\pi)^3/\Omega$.

3.1.4 Crystal momentum and the lattice sum

With the nuclei clamped at the sites of Eq. (3.3), the Hamiltonian of the electrons, Eq. (2.48) without the nuclear kinetic energy, is unchanged when all electrons are moved by the same lattice vector $\boldsymbol{R}$, but not by an arbitrary vector. Let $T_{\boldsymbol{R}}$ be the operator that does this, $(T_{\boldsymbol{R}}\Psi)(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_{N_e}) = \Psi(\boldsymbol{r}_1 + \boldsymbol{R}, \ldots, \boldsymbol{r}_{N_e} + \boldsymbol{R})$ for a wavefunction of N_e electrons (spins are not moved). It is unitary, it commutes with the Hamiltonian, and translations compose, $T_{\boldsymbol{R}} T_{\boldsymbol{R}'} = T_{\boldsymbol{R}+\boldsymbol{R}'}$. As a result, the eigenstates of the Hamiltonian can be chosen as eigenstates of all $T_{\boldsymbol{R}}$ at once, with eigenvalues $c(\boldsymbol{R})$ that are phases and obey $c(\boldsymbol{R} + \boldsymbol{R}') = c(\boldsymbol{R})c(\boldsymbol{R}')$. Writing $c(\boldsymbol{a}_i) = \mathrm{e}^{2\pi \mathrm{i} x_i}$ with real x_i, the composition rule requires $c(\boldsymbol{R}) = \mathrm{e}^{2\pi \mathrm{i} \sum_i n_i x_i}$, and with Eq. (3.6) this is

$$T_{\boldsymbol{R}}\Psi = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\,\Psi, \qquad \boldsymbol{k} = \sum_i x_i \boldsymbol{b}_i. \tag{3.11}$$

The wavevector $\boldsymbol{k}$ is the *crystal momentum* of the state. It is defined only up to a reciprocal-lattice vector, since $\boldsymbol{k}$ and $\boldsymbol{k} + \boldsymbol{G}$ give the same eigenvalues, and we take it in the first Brillouin zone. This makes precise the working definition of Sect. 1.3. For a free particle every translation is a symmetry, and $\boldsymbol{k}$ is the momentum; in a crystal just the lattice translations survive, and momentum is conserved modulo $\boldsymbol{G}$, as Sect. 2.3 found from the Coulomb Hamiltonian. The literature also writes "quasi-momentum"; the two terms mean the same. For a single electron, Eq. (3.11) becomes Bloch's theorem, which Sect. 3.2 derives together with the form of the states.

A finite crystal needs boundary conditions, and the *Born–von Karman* conditions are the periodic ones of the box of Sect. 2.1, adapted to the lattice. The crystal is a block of $N = N_1 N_2 N_3$ cells, the vectors $\boldsymbol{R} = \sum_i n_i \boldsymbol{a}_i$ with $0 \le n_i < N_i$, and a translation by $N_i \boldsymbol{a}_i$

acts as the identity. Its volume is

$$\mathcal{V} = N\Omega, \tag{3.12}$$

the normalization volume of Sect. 2.1. The condition $c(N_i \boldsymbol{a}_i) = \mathrm{e}^{\mathrm{i}N_i \boldsymbol{k}\cdot\boldsymbol{a}_i} = 1$ restricts the crystal momenta to

$$\boldsymbol{k} = \sum_i \frac{m_i}{N_i}\,\boldsymbol{b}_i, \qquad m_i \in \mathbb{Z}. \tag{3.13}$$

Adding N_i to m_i adds $\boldsymbol{b}_i$, a reciprocal-lattice vector. The distinct crystal momenta are those with $0 \le m_i < N_i$, and there are N of them, one per cell of the crystal. They form a uniform grid that fills one primitive cell of the reciprocal lattice, of volume $(2\pi)^3/\Omega$, and each occupies $(2\pi)^3/(N\Omega) = (2\pi)^3/\mathcal{V}$: the density of allowed wavevectors is the $\mathcal{V}/(2\pi)^3$ of a plane wave in a box of the same volume. Shifting points by reciprocal-lattice vectors maps the grid onto the first Brillouin zone, which then holds the same N points.

The identity behind every crystal-momentum selection rule in this book is the lattice sum. For $\boldsymbol{k}$ on the grid of Eq. (3.13) the sum over the block factorizes, since $\boldsymbol{k}\cdot\boldsymbol{R} = 2\pi\sum_i m_i n_i/N_i$, into three geometric sums. Each of them is

$$\sum_{n=0}^{N_i-1} \mathrm{e}^{2\pi\mathrm{i}m_i n/N_i} = \begin{cases} N_i, & m_i \text{ a multiple of } N_i, \\ \dfrac{1-\mathrm{e}^{2\pi\mathrm{i}m_i}}{1-\mathrm{e}^{2\pi\mathrm{i}m_i/N_i}} = 0, & \text{otherwise,} \end{cases}$$

by the geometric series. The product is nonzero only when all three m_i are multiples of the N_i, that is, when $\boldsymbol{k}$ is a reciprocal-lattice vector. It then equals N:

Lattice sum

$$\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} = N \sum_{\boldsymbol{G}} \delta_{\boldsymbol{k},\boldsymbol{G}}. \tag{3.14}$$

The sum runs over the N cells of the Born–von Karman block and $\boldsymbol{k}$ over the grid of Eq. (3.13). The crystal momenta form N points in a zone of volume $(2\pi)^3/\Omega$, and $\sum_{\boldsymbol{k}\in\mathrm{BZ}} \to \mathcal{V}\int_{\mathrm{BZ}} \mathrm{d}^3k/(2\pi)^3$.

Physically, the N phases add up only when they agree at every lattice site, and cancel otherwise. Appendix C.2 derives the infinite-crystal form, $\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} = [(2\pi)^3/\Omega]\sum_{\boldsymbol{G}} \delta^3(\boldsymbol{k}-\boldsymbol{G})$, and the error made when a calculation replaces the integral over the zone by a sum over a coarse grid.

The count of states fixes the filling of the bands. Each band of a crystal (Sect. 3.2) has one state per crystal momentum and spin, $2N$ states in all, two per cell. A primitive cell of silicon holds two atoms and eight valence electrons, which fill exactly four bands. A kilogram of silicon has $N = 1.07 \times 10^{25}$ cells, and as many crystal momenta in each band. The same count connects to the free electron gas of Sect. 2.5. Eight electrons per cell fill four zone volumes, and a Fermi sphere of four zone volumes has the radius $4^{1/3} \times 2.25\,\mathrm{keV} = 3.57\,\mathrm{keV}$, the k_F of silicon's valence density. Silicon's valence electrons are not a free gas, as Sect. 2.5 noted, and a gap opens at the zone boundary (Sect. 3.2). However, the comparison shows that the occupied states reach well beyond the first zone in the free-electron picture, and that the momentum scale of the valence electrons, $q_0 = \alpha m_e = 3.73\,\mathrm{keV}$, is 1.7 times the radius of a sphere with the volume of the zone.

3.1.5 Momentum transfer in a crystal

A dark-matter particle that scatters on the crystal transfers a momentum $\boldsymbol{q}$ through the density operator $n_e(\boldsymbol{q}) = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$ (Sect. 2.3). Translating the electrons multiplies each term

by $e^{i\boldsymbol{q}\cdot\boldsymbol{R}}$, $T_{\boldsymbol{R}} n_e(\boldsymbol{q}) T_{\boldsymbol{R}}^{-1} = e^{i\boldsymbol{q}\cdot\boldsymbol{R}} n_e(\boldsymbol{q})$. If $|i\rangle$ and $|f\rangle$ have crystal momenta $\boldsymbol{k}_i$ and $\boldsymbol{k}_f$, the unitarity of $T_{\boldsymbol{R}}$ and Eq. (3.11) give

$$\langle f|n_e(\boldsymbol{q})|i\rangle = \langle T_{\boldsymbol{R}} f|\, T_{\boldsymbol{R}} n_e(\boldsymbol{q}) T_{\boldsymbol{R}}^{-1}\, |T_{\boldsymbol{R}} i\rangle = e^{i(\boldsymbol{k}_i+\boldsymbol{q}-\boldsymbol{k}_f)\cdot\boldsymbol{R}} \langle f|n_e(\boldsymbol{q})|i\rangle$$

for every $\boldsymbol{R}$. The matrix element vanishes unless the phase is one at every lattice site, that is, unless

$$\boldsymbol{k}_f = \boldsymbol{k}_i + \boldsymbol{q} - \boldsymbol{G} \tag{3.15}$$

for some reciprocal-lattice vector $\boldsymbol{G}$. The argument rests on symmetry alone, and it holds for the interacting many-electron states as well as for single electrons; Sect. 3.2 works out the single-electron matrix element.

Since only $\boldsymbol{q}$ modulo $\boldsymbol{G}$ enters Eq. (3.15), it is useful to split the momentum transfer into its part in the first zone and a reciprocal-lattice vector,

$$\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q, \qquad \boldsymbol{q}_r \in \mathrm{BZ}, \tag{3.16}$$

where $\boldsymbol{G}_q$ is the reciprocal-lattice vector closest to $\boldsymbol{q}$. The split is unique except on the zone boundary, a set of measure zero. Fig. 3.4 shows an example. The crystal momentum of the target changes by $\boldsymbol{q}_r$, up to a reciprocal-lattice vector, and the rest of the momentum transfer, $\boldsymbol{G}_q$, goes to the lattice as a whole. The static lattice takes it with no recoil energy, since its mass is infinite. A transition in which a nonzero $\boldsymbol{G}$ appears in Eq. (3.15) is an *Umklapp* process, from the German for "flipping over", and one with $\boldsymbol{G} = 0$ is a normal process. Whether a transition is an Umklapp process depends on $\boldsymbol{k}_i$ as well as on $\boldsymbol{q}$: even for $\boldsymbol{q}$ inside the zone, $\boldsymbol{k}_i + \boldsymbol{q}$ can leave it and be folded back. For $|\boldsymbol{q}|$ larger than the diameter of the zone, $2|\Gamma\mathrm{W}|$, $\boldsymbol{k}_i + \boldsymbol{q}$ always lies outside, and every transition is an Umklapp process.

The momentum transfer itself is never reduced to the first zone. The dark matter loses $\boldsymbol{q}$, not $\boldsymbol{q}_r$, and its kinematics, $v_{\min}$ and the dark-matter form factor $F_{\mathrm{DM}}(q)$ (Sects. 1.3 and 1.4) depend on $|\boldsymbol{q}|$. The target does not treat $\boldsymbol{q}$ and $\boldsymbol{q}_r$ alike either. Both change the crystal momentum by the same amount. However, the plane wave $e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$ varies on the scale $2\pi/|\boldsymbol{q}|$ and resolves the electron density inside the cell on that scale. At large $|\boldsymbol{G}_q|$ it probes the region near the nuclei, where the core electrons and the steep part of the valence wavefunctions sit (Sect. 3.5). The response of the crystal is therefore a function of $\boldsymbol{q}_r$ and $\boldsymbol{G}_q$ separately, not of $\boldsymbol{q}_r$ alone, and the book keeps both (Appendix A.3).

Worked example 3.1: Momentum transfers against the zone of silicon

With $\hbar c = 1973.27\,\mathrm{eV\,\AA}$ and $a = 5.431\,\mathrm{\AA}$,

$$\frac{2\pi}{a} = \frac{2\pi \times 1973.27\,\mathrm{eV\,\AA}}{5.431\,\mathrm{\AA}} = 2283\,\mathrm{eV} = 2.28\,\mathrm{keV}.$$

The shortest reciprocal-lattice vectors are $|\boldsymbol{G}_{111}| = 3.95\,\mathrm{keV}$ and $|\boldsymbol{G}_{200}| = 4.57\,\mathrm{keV}$. An electronic excitation across the low-temperature gap, $E_{\mathrm{g}} = 1.17\,\mathrm{eV}$ (Sect. 3.3), needs $q \geq E_{\mathrm{g}}/v_{\max} = 1.17\,\mathrm{eV}/(2.650 \times 10^{-3}) = 0.44\,\mathrm{keV}$, or $0.19\,(2\pi/a)$, deep inside the zone. The largest momentum transfer that can still deposit E_{g} follows from Eq. (1.26), $q = m_\chi v_{\max}[1 + (1 - 2E_{\mathrm{g}}/m_\chi v_{\max}^2)^{1/2}]$. It is $4.81\,\mathrm{keV} = 2.11\,(2\pi/a)$ for $m_\chi = 1\,\mathrm{MeV}$, $52.6\,\mathrm{keV} = 23.0\,(2\pi/a)$ for 10 MeV and $530\,\mathrm{keV} = 232\,(2\pi/a)$ for 100 MeV. The lower end of the range is 0.49 keV at 1 MeV and tends to $E_{\mathrm{g}}/v_{\max} = 0.44\,\mathrm{keV}$ for heavier particles, hardly depending on the mass; the upper end grows in proportion to m_χ.
Already at 1 MeV the range extends past the eight $\boldsymbol{G}_{111}$ and the six $\boldsymbol{G}_{200}$, and the typical momentum of a valence electron, $q_0 = 3.73\,\mathrm{keV} = 1.63\,(2\pi/a)$, lies outside the zone. A ball of radius q contains about $(4\pi/3)q^3/[4(2\pi/a)^3]$ reciprocal-lattice vectors, one per

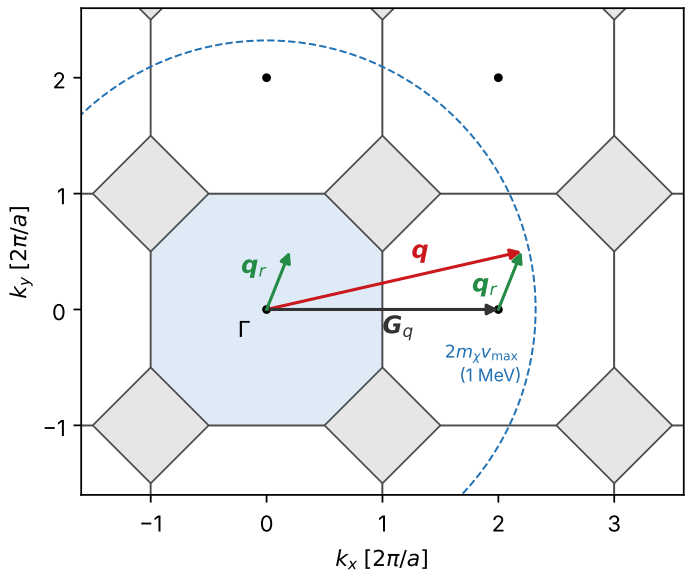


Figure 3.4: A momentum transfer reduced to the first Brillouin zone, in the plane $k_z = 0$ of the fcc reciprocal lattice, with momenta in units of $2\pi/a$ (2.28 keV for silicon). Dots: the reciprocal-lattice vectors in this plane, $(2\pi/a)(2m, 2n, 0)$. Octagons: the cuts of their Brillouin zones by the plane; the first zone is shaded blue. The grey squares belong to zones centered on reciprocal-lattice vectors $(2\pi/a)(h, k, \pm 1)$ with h, k odd, which lie outside the plane. The momentum transfer $\boldsymbol{q} = (2\pi/a)(2.2, 0.5, 0)$, of length 5.15 keV, is the sum of the reciprocal-lattice vector $\boldsymbol{G}_q = (2\pi/a)(2, 0, 0)$ nearest to it and of $\boldsymbol{q}_r = (2\pi/a)(0.2, 0.5, 0)$, of length 1.23 keV, drawn both at $\boldsymbol{G}_q$ and at the origin. Dashed circle: the maximum momentum transfer $2m_\chi v_{\max} = 5.30\,\text{keV}$ of dark matter of mass 1 MeV, with $v_{\max} = 794.5\,\text{km}\,\text{s}^{-1}$.

zone volume. For the maximum momentum transfer $2m_\chi v_{\max}$ this is 1.3×10^4 at 10 MeV, which a direct count confirms, and 1.3×10^7 at 100 MeV. Above about a MeV, most of the kinematically allowed momentum transfers have $\boldsymbol{G}_q \neq 0$, and a calculation of the rate must carry the response out to reciprocal-lattice vectors whose number grows as m_χ^3 (Sect. 3.5 and Appendix E.7).

3.1.6 The basis structure factor

The positions of the atoms within the cell enter every Fourier component of a lattice-periodic quantity. Consider a function built from one term per atom, $f(\boldsymbol{r}) = \sum_{\boldsymbol{R}}\sum_\kappa v_\kappa(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\kappa)$, such as the potential of the clamped nuclei or a superposition of atomic charge densities. Its components, Eq. (A.12), are

$$f_{\boldsymbol{G}} = \frac{1}{\Omega}\int_\Omega \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}} \sum_{\boldsymbol{R}}\sum_\kappa v_\kappa(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\kappa) = \frac{1}{\Omega}\sum_\kappa \int \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}\, v_\kappa(\boldsymbol{r}-\boldsymbol{\tau}_\kappa).$$

In the second step, the translate of the cell by $-\boldsymbol{R}$ was used for each term; since $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{R}}=1$, the phase is unchanged, and the cells together fill all of space. Shifting $\boldsymbol{r}$ by $\boldsymbol{\tau}_\kappa$ then gives

$$f_{\boldsymbol{G}} = \frac{1}{\Omega}\sum_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_\kappa}\,\tilde{v}_\kappa(\boldsymbol{G}), \qquad \tilde{v}_\kappa(\boldsymbol{G}) = \int \mathrm{d}^3x\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{x}}\, v_\kappa(\boldsymbol{x}). \tag{3.17}$$

Each atom contributes the transform of its own term at $\boldsymbol{G}$, with a phase set by its position in the cell. When all atoms of the basis are of one element, $\tilde{v}_\kappa = \tilde{v}$, the positions factor out into the *basis structure factor*

$$S^{\mathrm{bas}}(\boldsymbol{G}) = \sum_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_\kappa}, \tag{3.18}$$

and $f_{\boldsymbol{G}} = S^{\mathrm{bas}}(\boldsymbol{G})\,\tilde{v}(\boldsymbol{G})/\Omega$. A zero of S^{bas} is an *extinction*: the component vanishes at that reciprocal-lattice vector, whatever the atomic term, because the contributions of the atoms in the cell cancel.

The nuclei of Sect. 2.3 are the simplest case. Their density for site κ, $n_\kappa(\boldsymbol{k}) = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_\kappa)}$, becomes with Eq. (3.14)

$$\sum_\kappa Z_\kappa\, n_\kappa(\boldsymbol{k}) = N\sum_{\boldsymbol{G}} \delta_{\boldsymbol{k},\boldsymbol{G}} \sum_\kappa Z_\kappa\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_\kappa},$$

nonzero only at reciprocal-lattice vectors, as Sect. 2.3 anticipated. The electron–nucleus potential of Eq. (2.48) then has the components of Eq. (3.17) with $\tilde{v}_\kappa(\boldsymbol{G}) = -Z_\kappa v_{\mathrm{C}}(|\boldsymbol{G}|)$. The same expression describes dark matter that couples to the clamped nuclei: it can transfer only a reciprocal-lattice vector, elastically, with no energy deposited. This is the coherent, Bragg-like part of the response mentioned in Sect. 2.2, and a static lattice has no other nuclear response. Energy reaches the nuclei once they are allowed to move (Chap. 5).

Worked example 3.2: The diamond structure factor and the absent (200) reflection

For $\boldsymbol{G} = (2\pi/a)(h,k,l)$ and the basis of Eq. (3.5), $\boldsymbol{G}\cdot\boldsymbol{\tau}_2 = (\pi/2)(h+k+l)$, and Eq. (3.18) gives

$$S^{\mathrm{bas}}(\boldsymbol{G}_{hkl}) = 1 + \mathrm{e}^{\mathrm{i}\pi(h+k+l)/2}.$$

For h,k,l all odd, $h+k+l$ is odd and $S^{\mathrm{bas}} = 1\pm\mathrm{i}$, with $|S^{\mathrm{bas}}|^2 = 2$; for (111), $h+k+l=3$

and $S^{\mathrm{bas}} = 1 - \mathrm{i}$. For h, k, l all even, $h + k + l$ is even, and two cases arise. If $h + k + l$ is a multiple of four, as for (220) and (400), the two atoms add in phase, $S^{\mathrm{bas}} = 2$ and $|S^{\mathrm{bas}}|^2 = 4$. If $h + k + l = 4n + 2$, as for (200) and (222), $S^{\mathrm{bas}} = 1 + \mathrm{e}^{\mathrm{i}\pi} = 0$. The plane wave at $\boldsymbol{G}_{200}$ has its crests on the atoms at $\boldsymbol{\tau}_1$ and its troughs on those at $\boldsymbol{\tau}_2$, a quarter cube diagonal away, and the two contributions cancel.
The (200) and (222) reflections of silicon are therefore absent in X-ray diffraction from spherical atoms, although $\boldsymbol{G}_{200}$ and $\boldsymbol{G}_{222}$ are reciprocal-lattice vectors of its fcc lattice and $\boldsymbol{G}_{200}$ is the second-shortest. The same cancellation removes these components from any part of the crystal potential that is a sum of identical spherical atomic terms. The (222) component returns through the part of the potential that is not of this form, such as the charge that piles up in the bonds (Sect. 3.4), and is observed as a weak X-ray reflection. The (200) component does not. The diamond structure is mapped onto itself by *glides*, each a reflection in a plane parallel to a cube face followed by a translation by a quarter of a face diagonal within that plane. The glides force it to vanish for every function with the symmetry of the crystal (Appendix C.17). In zincblende GaAs the two sites hold different nuclei, $Z_\kappa = 31$ (gallium) and 33 (arsenic), and the nuclear factor $\sum_\kappa Z_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_\kappa}$ equals $31 - 33 = -2$ at (200), against $31 + 33 = 64$ at (220). The (200) component is weak, smaller in square by a factor of $4/4096 \approx 10^{-3}$, but not zero, and it is what distinguishes the zincblende structure from diamond in diffraction.

Extinctions also explain why the fcc reciprocal lattice lacks the vectors $(2\pi/a)(h, k, l)$ of mixed parity. The fcc lattice can also be described as a simple cubic lattice of side a with a basis of four points, at a corner and the three adjacent face centers. Its reciprocal lattice is then simple cubic, $(2\pi/a)(h, k, l)$ with any integers, and the structure factor of the four points is $1 + \mathrm{e}^{\mathrm{i}\pi(h+k)} + \mathrm{e}^{\mathrm{i}\pi(k+l)} + \mathrm{e}^{\mathrm{i}\pi(l+h)}$, which is 4 when h, k, l have one parity and 0 otherwise. A basis structure factor is the price of using a cell larger than the primitive one, and its zeros remove exactly the vectors that the smaller cell does not have.

The electrons move in the periodic potential whose components Eq. (3.17) describes. Sect. 3.2 shows that their states are labeled by a crystal momentum in the zone and a band index, and computes the matrix element of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ between them, with the Umklapp vector kept in the plane wave.

3.2 Bloch states and band structure

An electron in silicon moves in a potential that repeats itself after every lattice vector: the attraction of the clamped nuclei, the repulsion of the other electrons, and the exchange and correlation effects that come with them (Sect. 2.3). The full problem is a many-electron one. Most of what a dark-matter rate needs, however, can be understood with each electron moving independently in an effective potential $U(\boldsymbol{r})$ with the periodicity of the lattice, $U(\boldsymbol{r} + \boldsymbol{R}) = U(\boldsymbol{r})$. Density-functional theory supplies the best such potential and tells us what it leaves out (Sect. 3.4). Here U is any lattice-periodic function, and its periodicity alone fixes the form of the states and the momentum transfers that connect them. Bands and gaps are understood in two opposite limits, nearly free electrons and tight binding. We then compute the two matrix elements that every electronic rate in this book contains: that of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, to which dark matter couples, and that of the velocity, to which light couples. We follow Refs. [1, Chaps. 8–10], [5, Chap. 3] and [6, Chap. 2].

3.2.1 Bloch's theorem

The one-electron Hamiltonian is

$$H_{\mathrm{e}} = \frac{\hat{\boldsymbol{p}}^2}{2m_e} + U(\boldsymbol{r}), \qquad U(\boldsymbol{r}+\boldsymbol{R}) = U(\boldsymbol{r}), \tag{3.19}$$

with $\hat{\boldsymbol{p}} = -\mathrm{i}\nabla$. We keep the electron spinless and count the two spin states with an explicit factor of 2; spin–orbit coupling, which mixes them, enters in Sect. 3.3. The translation operator of Sect. 3.1, acting now on one electron, is $(T_{\boldsymbol{R}}\psi)(\boldsymbol{r}) = \psi(\boldsymbol{r}+\boldsymbol{R})$. The kinetic energy is unchanged by any translation, and U by a lattice translation, so every $T_{\boldsymbol{R}}$ commutes with H_{e}. The argument that led to Eq. (3.11) then applies to a single electron: the eigenstates of H_{e} can be chosen so that $\psi(\boldsymbol{r}+\boldsymbol{R}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\psi(\boldsymbol{r})$ for every $\boldsymbol{R}$, with $\boldsymbol{k}$ in the first Brillouin zone and on the Born–von Karman grid of Eq. (3.13). Define $u(\boldsymbol{r}) = \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\psi(\boldsymbol{r})$. It is periodic, since $u(\boldsymbol{r}+\boldsymbol{R}) = \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\psi(\boldsymbol{r}) = u(\boldsymbol{r})$. Each eigenstate is therefore a plane wave modulated by a function with the period of the lattice:

Bloch's theorem

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{N}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\,u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad u_{n\boldsymbol{k}}(\boldsymbol{r}+\boldsymbol{R}) = u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad \int_{\Omega}\mathrm{d}^3r\,|u_{n\boldsymbol{k}}|^2 = 1. \tag{3.20}$$

Equivalently, $\psi_{n\boldsymbol{k}}(\boldsymbol{r}+\boldsymbol{R}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\psi_{n\boldsymbol{k}}(\boldsymbol{r})$. The state is normalized to one over the crystal of volume $\mathcal{V} = N\Omega$, and its periodic part to one over a cell.

The two normalizations agree because $|\psi_{n\boldsymbol{k}}|^2 = |u_{n\boldsymbol{k}}|^2/N$ is periodic and the crystal holds N cells; these are the conventions of Appendix A.5. The label n, the *band index*, appears because there are many solutions at each $\boldsymbol{k}$. Inserting $\psi = N^{-1/2}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}u$ into $H_{\mathrm{e}}\psi = \varepsilon\psi$ and using $\hat{\boldsymbol{p}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}u = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}(\hat{\boldsymbol{p}}+\boldsymbol{k})u$ leaves an equation for u alone,

$$H_{\boldsymbol{k}}\,u_{n\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}}\,u_{n\boldsymbol{k}}, \qquad H_{\boldsymbol{k}} \equiv \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}H_{\mathrm{e}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} = \frac{(\hat{\boldsymbol{p}}+\boldsymbol{k})^2}{2m_e} + U(\boldsymbol{r}), \tag{3.21}$$

to be solved in one cell with periodic boundary conditions. Like a particle in a box, $H_{\boldsymbol{k}}$ has a discrete spectrum at each $\boldsymbol{k}$, which we order as $\varepsilon_{1\boldsymbol{k}} \le \varepsilon_{2\boldsymbol{k}} \le \dots$. Since $\boldsymbol{k}$ enters $H_{\boldsymbol{k}}$ as a parameter, each *band energy* $\varepsilon_{n\boldsymbol{k}}$ varies continuously with $\boldsymbol{k}$ and sweeps out an energy band; the set of all of them, plotted along lines through the zone, is the *band structure*. The bands are periodic in reciprocal space: $\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}u_{n\boldsymbol{k}}$ is periodic and satisfies $H_{\boldsymbol{k}+\boldsymbol{G}}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}u_{n\boldsymbol{k}} = \mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}H_{\boldsymbol{k}}u_{n\boldsymbol{k}}$, so $\varepsilon_{n,\boldsymbol{k}+\boldsymbol{G}} = \varepsilon_{n\boldsymbol{k}}$. We use this freedom to identify the states themselves, $\psi_{n,\boldsymbol{k}+\boldsymbol{G}} = \psi_{n\boldsymbol{k}}$, which is the *periodic gauge* of Eq. (A.16). A crystal momentum outside the first zone can then be used directly.

A Bloch state is not a state of definite momentum. Its periodic part is a Fourier series over reciprocal-lattice vectors, and we write it in the convention of plane-wave codes (Appendix A.3),

$$u_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{\Omega}}\sum_{\boldsymbol{G}} c_{n\boldsymbol{k}}(\boldsymbol{G})\,\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}, \qquad \psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{\mathcal{V}}}\sum_{\boldsymbol{G}} c_{n\boldsymbol{k}}(\boldsymbol{G})\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}, \qquad \sum_{\boldsymbol{G}}|c_{n\boldsymbol{k}}(\boldsymbol{G})|^2 = 1. \tag{3.22}$$

The coefficient $c_{n\boldsymbol{k}}(\boldsymbol{G})$ multiplies $\mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$, so it is the lattice Fourier component of Eq. (A.12) at $-\boldsymbol{G}$, times $\Omega^{1/2}$. The electron in state $n\boldsymbol{k}$ has the momenta $\boldsymbol{k}+\boldsymbol{G}$, with probabilities $|c_{n\boldsymbol{k}}(\boldsymbol{G})|^2$. A smooth valence state is dominated by a few small $\boldsymbol{G}$; the part of the wavefunction near the nuclei, where it oscillates rapidly, puts weight at large $|\boldsymbol{G}|$, and it is this tail that a momentum transfer of many keV probes (Sect. 3.5).

The same expansion turns Eq. (3.21) into an algebraic equation and provides a second proof of Bloch's theorem. Write the potential as $U(\boldsymbol{r}) = \sum_{\boldsymbol{G}''} U_{\boldsymbol{G}''}\, \mathrm{e}^{-\mathrm{i}\boldsymbol{G}''\cdot\boldsymbol{r}}$, with the components $U_{\boldsymbol{G}} = \Omega^{-1} \int_\Omega \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}} U(\boldsymbol{r})$ of Eq. (A.12). Acting on a plane wave, U produces $U(\boldsymbol{r})\, \mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G}')\cdot\boldsymbol{r}} = \sum_{\boldsymbol{G}''} U_{\boldsymbol{G}''}\, \mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G}'-\boldsymbol{G}'')\cdot\boldsymbol{r}}$. Collecting the coefficient of $\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}$ in $H_\mathrm{e}\psi_{n\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}}\psi_{n\boldsymbol{k}}$ picks the term $\boldsymbol{G}'' = \boldsymbol{G}' - \boldsymbol{G}$, and

$$\Big[\frac{(\boldsymbol{k}+\boldsymbol{G})^2}{2m_e} - \varepsilon_{n\boldsymbol{k}}\Big] c_{n\boldsymbol{k}}(\boldsymbol{G}) + \sum_{\boldsymbol{G}'} U_{\boldsymbol{G}'-\boldsymbol{G}}\, c_{n\boldsymbol{k}}(\boldsymbol{G}') = 0. \tag{3.23}$$

A periodic potential couples a plane wave only to those that differ from it by a reciprocal-lattice vector. A state built from one $\boldsymbol{k}$ and its translates $\boldsymbol{k}+\boldsymbol{G}$ stays among them, which is Bloch's theorem again. The matrix $(\boldsymbol{k}+\boldsymbol{G})^2\delta_{\boldsymbol{G}\boldsymbol{G}'}/2m_e + U_{\boldsymbol{G}'-\boldsymbol{G}}$ is Hermitian, since a real U has $U_{-\boldsymbol{G}} = U_{\boldsymbol{G}}^*$. Its eigenvalues are the $\varepsilon_{n\boldsymbol{k}}$, and its eigenvectors the $c_{n\boldsymbol{k}}(\boldsymbol{G})$. A plane-wave code diagonalizes this matrix, truncated at a cutoff in $|\boldsymbol{k}+\boldsymbol{G}|$ (Sect. 3.5).

Bloch states with different $\boldsymbol{k}$ are orthogonal, because they belong to different eigenvalues of the unitary $T_{\boldsymbol{R}}$, and states with the same $\boldsymbol{k}$ and different n are orthogonal as eigenvectors of the Hermitian $H_{\boldsymbol{k}}$. Together they form an orthonormal basis, and the change of basis of Eq. (2.40) turns the plane-wave operators of Sect. 2.3 into operators $c^\dagger_{n\boldsymbol{k}s}$ and $c_{n\boldsymbol{k}s}$ that create and destroy an electron of spin s in a Bloch state. For independent electrons the Hamiltonian is $\sum_{n\boldsymbol{k}s} \varepsilon_{n\boldsymbol{k}}\, c^\dagger_{n\boldsymbol{k}s} c_{n\boldsymbol{k}s}$, and in thermal equilibrium the *occupation* of a state is, by Eq. (2.69),

$$f_{n\boldsymbol{k}} = \langle c^\dagger_{n\boldsymbol{k}s} c_{n\boldsymbol{k}s}\rangle = n_\mathrm{F}(\varepsilon_{n\boldsymbol{k}} - \mu_\mathrm{ch}). \tag{3.24}$$

Each band holds $2N$ electrons, as Sect. 3.1 counted. In silicon the eight valence electrons per cell fill the four lowest bands, and a gap of about an electronvolt separates them from the next ones (Sect. 3.3). At any temperature at which a detector runs, $f_{n\boldsymbol{k}} = 1$ in these four bands and 0 above them, to an accuracy set by $\mathrm{e}^{-E_\mathrm{g}/2T}$, the factor that the intrinsic carrier density of Eq. (3.48) carries. Two questions follow: where such gaps come from, and how $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ connects the states on either side of them.

3.2.2 Nearly free electrons

Switch the potential off. The eigenstates of Eq. (3.23) are then single plane waves, $c_{n\boldsymbol{k}}(\boldsymbol{G}) = \delta_{\boldsymbol{G},\boldsymbol{G}_n}$ for some vector $\boldsymbol{G}_n$ that labels the band, with energies $\varepsilon^0_{\boldsymbol{k}+\boldsymbol{G}_n}$. Here $\varepsilon^0_{\boldsymbol{k}} = k^2/2m_e$ is the free-electron energy at wavevector $\boldsymbol{k}$, inside the zone or not. Nothing physical has changed: the free parabola has been cut into pieces and each piece shifted by a reciprocal-lattice vector into the first zone, so that at a given $\boldsymbol{k}$ the bands are the parabolas centered on all the $-\boldsymbol{G}$. This is the *empty-lattice* band structure, drawn dashed in Fig. 3.5(a). Its bands cross wherever two plane waves have equal energy, $\varepsilon^0_{\boldsymbol{k}} = \varepsilon^0_{\boldsymbol{k}-\boldsymbol{G}}$, that is, on the planes $\boldsymbol{k}\cdot\boldsymbol{G} = |\boldsymbol{G}|^2/2$ that bisect the reciprocal-lattice vectors. These are the Bragg planes whose pieces bound the Brillouin zone (Sect. 3.1).

Now let U be weak. Away from a crossing, the coupling $U_{\boldsymbol{G}'-\boldsymbol{G}}$ mixes plane waves whose energies differ by much more than $|U|$, and second-order perturbation theory shifts each level by $\sum_{\boldsymbol{G}\neq 0} |U_{\boldsymbol{G}}|^2/(\varepsilon^0_{\boldsymbol{k}} - \varepsilon^0_{\boldsymbol{k}-\boldsymbol{G}})$, a small correction. Near a crossing, however, two plane waves are degenerate and perturbation theory in this form fails. We keep just those two, $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ and $\mathrm{e}^{\mathrm{i}(\boldsymbol{k}-\boldsymbol{G})\cdot\boldsymbol{r}}$. Eq. (3.23) for the coefficients $c(0)$ and $c(-\boldsymbol{G})$ reads

$$\begin{pmatrix} \varepsilon^0_{\boldsymbol{k}} & U_{-\boldsymbol{G}} \\ U_{\boldsymbol{G}} & \varepsilon^0_{\boldsymbol{k}-\boldsymbol{G}} \end{pmatrix} \begin{pmatrix} c(0) \\ c(-\boldsymbol{G}) \end{pmatrix} = \varepsilon \begin{pmatrix} c(0) \\ c(-\boldsymbol{G}) \end{pmatrix}, \tag{3.25}$$

where the diagonal U_0, a constant shift, has been absorbed into the energy zero. The eigenvalues are

$$\varepsilon_\pm = \frac{\varepsilon^0_{\boldsymbol{k}} + \varepsilon^0_{\boldsymbol{k}-\boldsymbol{G}}}{2} \pm \sqrt{\Big(\frac{\varepsilon^0_{\boldsymbol{k}} - \varepsilon^0_{\boldsymbol{k}-\boldsymbol{G}}}{2}\Big)^2 + |U_{\boldsymbol{G}}|^2}. \tag{3.26}$$

Far from the plane the square root tends to $|\varepsilon^0_{\boldsymbol{k}} - \varepsilon^0_{\boldsymbol{k}-\boldsymbol{G}}|/2$, and the two free parabolas are recovered. On the plane the two free energies are equal, and the levels are pushed apart symmetrically:

Nearly-free-electron gap

On the Bragg plane $\boldsymbol{k}\cdot\boldsymbol{G} = |\boldsymbol{G}|^2/2$, a weak periodic potential splits the two degenerate plane waves into $\varepsilon_\pm = \varepsilon^0_{\boldsymbol{k}} \pm |U_{\boldsymbol{G}}|$. The gap,

$$\varepsilon_+ - \varepsilon_- = 2|U_{\boldsymbol{G}}|, \tag{3.27}$$

is set by the Fourier component of the potential at the reciprocal-lattice vector that the plane bisects. It holds to first order in U, as long as $|U_{\boldsymbol{G}}|$ is small compared with the energy separation to the next plane waves.

At the zone boundary the band structure therefore has a gap, and no state of the crystal has an energy inside it, whatever the direction of $\boldsymbol{k}$ along the plane. Whether the gaps on different faces of the zone overlap in energy and leave a gap for the whole crystal depends on the sizes of the $U_{\boldsymbol{G}}$ and on the shape of the zone. In a nearly-free metal such as aluminium they do not, and a Fermi surface survives [1, Chap. 9]. Note that a component that vanishes by extinction (Worked example 3.2) opens no gap at first order. In silicon, the square faces of the zone bisect $\boldsymbol{G}_{200}$, and $U_{\boldsymbol{G}_{200}} = 0$ exactly, by the glide symmetry of the diamond structure (Worked example 3.2, Appendix C.17).

The states at the gap are standing waves. For a potential with inversion symmetry and real $U_{\boldsymbol{G}} = U_{-\boldsymbol{G}}$, the eigenvectors of Eq. (3.25) on the plane are $c(-\boldsymbol{G}) = \pm c(0)$, so that $\psi \propto \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}/2} \pm \mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}/2}$ for $\boldsymbol{k} = \boldsymbol{G}/2$: a cosine and a sine of $\boldsymbol{G}\cdot\boldsymbol{r}/2$. The two have the same kinetic energy, but the cosine piles the electron density up at the points where $\boldsymbol{G}\cdot\boldsymbol{r}$ is a multiple of 2π, the lattice planes, and the sine piles it up halfway between them. If the potential is attractive at the planes, $U_{\boldsymbol{G}} < 0$, the cosine lies lower by $|U_{\boldsymbol{G}}|$ and the sine higher by the same amount. A running wave at the boundary would be Bragg-reflected into its partner at $\boldsymbol{k} - \boldsymbol{G}$; the two standing waves are what remains, and both carry no current. The group velocity, the slope of the band (Eq. (3.41) below), vanishes on the Bragg plane in the direction of $\boldsymbol{G}$, and the bands meet the zone boundary with zero slope, as Fig. 3.5(a) shows.

Worked example 3.3: The one-dimensional gap

Take the one-dimensional potential $U(x) = 2U_{G_1}\cos(G_1 x)$, whose only components are $U_{\pm G_1} = U_{G_1}$, with $U_{G_1} = -0.5\,\mathrm{eV}$ and the period $a = 5.431\,\text{Å}$, so that $G_1 = 2\pi/a = 2283\,\mathrm{eV}$. The potential is attractive at the lattice sites, $U(0) = -1\,\mathrm{eV}$. The free-electron energy at the zone boundary, $k = G_1/2$, is

$$\varepsilon^0_{G_1/2} = \frac{(G_1/2)^2}{2m_e} = \frac{(1141.5\,\mathrm{eV})^2}{2\times 511.0\,\mathrm{keV}} = 1.275\,\mathrm{eV},$$

and Eq. (3.26) puts the band edges at $1.275\,\mathrm{eV} \mp 0.5\,\mathrm{eV}$, that is, at 0.775 and 1.775 eV, with the gap $2|U_{G_1}| = 1.000\,\mathrm{eV}$. The lower edge is the cosine, concentrated on the sites, and the upper edge the sine, concentrated between them; the full calculation of Fig. 3.5

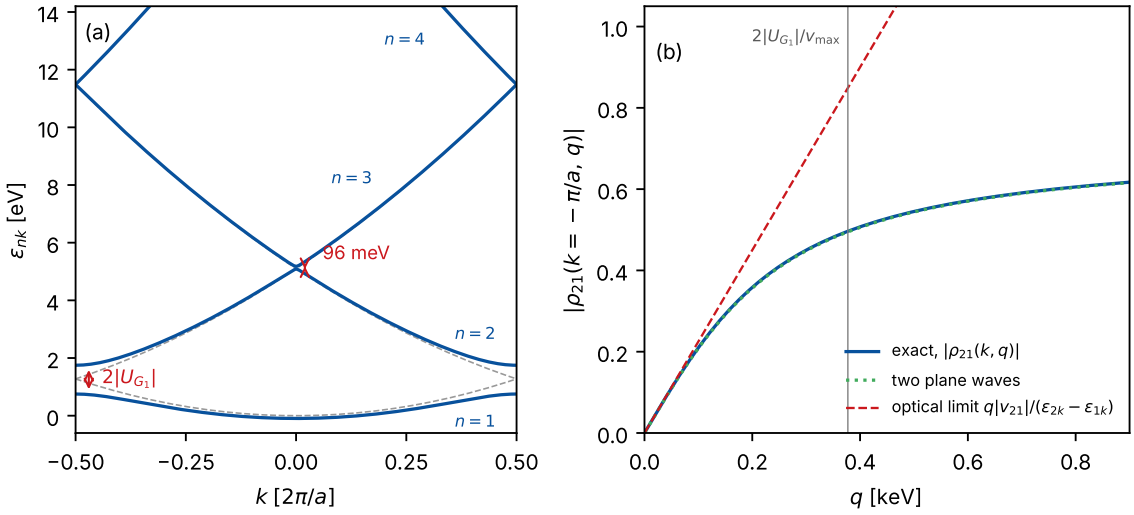


Figure 3.5: Nearly free electrons in one dimension, for the potential $U(x) = 2U_{G_1}\cos(G_1 x)$ with $G_1 = 2\pi/a$, $a = 5.431\,\text{Å}$ (the lattice constant of silicon, used only to set the scale) and $U_{G_1} = -0.5\,\text{eV}$. The bands are computed by diagonalizing Eq. (3.23) in 25 plane waves $e^{i(k+jG_1)x}$, $|j| \leq 12$. (a) The four lowest bands ε_{nk} (solid) and the free-electron parabolas $(k + jG_1)^2/2m_e$ folded into the first zone (dashed). The first gap, at the zone boundary $k = \pm\pi/a$, is 0.998 eV, close to $2|U_{G_1}| = 1\,\text{eV}$; the second, at $k = 0$, is 96 meV, second order in U_{G_1}, since the potential has no component at $2G_1$. (b) The density vertex $|\rho_{21}(k, q)|$ between the two lowest bands at $k = -\pi/a$, for momentum transfers $0 < q < 0.9\,\text{keV}$, which keep $k + q$ in the first zone. Solid: the full calculation. Dotted: the two-plane-wave model, indistinguishable at this scale. Dashed: the optical limit $q|v_{21}(k)|/(\varepsilon_{2k} - \varepsilon_{1k})$. The vertical line marks the smallest momentum transfer at which dark matter can deposit 1 eV, $2|U_{G_1}|/v_{\max} = 0.38\,\text{keV}$ for $v_{\max} = 2.650 \times 10^{-3}$.

confirms that $|\psi|^2$ of the lower state vanishes at $x = a/2$, and that of the upper state at $x = 0$.
Diagonalizing Eq. (3.23) in 25 plane waves gives the edges 0.752 and 1.749 eV, and the gap 0.998 eV. The lower edge has moved down by 0.023 eV and the upper by 0.026 eV. The shift comes from the next plane waves, $e^{i(G_1/2+G_1)x}$ and $e^{i(-G_1/2-G_1)x}$, which each of the two boundary states can reach with one factor of U_{G_1}. Their energy is $9 \times 1.275\,\text{eV} = 11.47\,\text{eV}$, and second-order perturbation theory yields $U_{G_1}^2/(1.275\,\text{eV} - 11.47\,\text{eV}) = -0.025\,\text{eV}$ for each edge, the average of the two computed shifts. The gap $2|U_{G_1}|$ is thus accurate to 0.2 %, although $|U_{G_1}|$ is 40 % of the kinetic energy at the boundary.
The second gap is a different matter. It opens at $k = 0$, where the plane waves $e^{\pm iG_1 x}$ are degenerate at $\varepsilon^0_{G_1} = 5.099\,\text{eV}$. They differ by $2G_1$, and $U_{2G_1} = 0$, so they are not coupled directly. They are coupled at second order through the plane wave e^{i0x}, with the effective matrix element $U_{G_1}^2/(\varepsilon^0_{G_1} - \varepsilon^0_0) = 0.049\,\text{eV}$, and the two-level splitting is twice this, 0.098 eV. The full calculation finds 0.096 eV. A gap at a Bragg plane is first order in the component of U at that $\boldsymbol{G}$, and higher order if that component vanishes. This is what happens on the square faces of the silicon zone, where the extinction of $\boldsymbol{G}_{200}$ removes the first-order term.

Silicon is not in the regime where these estimates are quantitative. A three-dimensional gap requires the gaps on all faces of the zone to overlap in energy, and in silicon the band edges lie at different crystal momenta (Sect. 3.3). Its potential is not weak either: the free-electron energy at L is $|\boldsymbol{G}_{111}|^2/8m_e = 3.82\,\text{eV}$, and the strong attraction of the ion cores mixes many plane waves into each state, a problem that the pseudopotentials of Sect. 3.5 remove. What survives is the mechanism. Gaps open where the free-electron energy surfaces cross the Bragg planes, and they are set by the Fourier components of the potential. The opposite limit, in which electrons hop between atomic orbitals, explains why silicon has a gap where it does.

3.2.3 Tight binding and the sp^3 bond

A silicon atom in the gas has two $3s$ and two $3p$ electrons in sharp atomic levels. Bring the atoms together into a crystal of N cells, and each atomic level of each atom in the cell broadens into a band of N states, because the electrons can now hop from atom to atom. When the hopping is weak compared with the spacing of the atomic levels, the natural starting point is the atomic orbitals, not plane waves. This is the *tight-binding* method, or the linear combination of atomic orbitals (LCAO).

Let $\varphi^{\text{at}}_j(\boldsymbol{r})$ be atomic orbitals, with j labeling both the orbital and the site $\boldsymbol{\tau}_\kappa$ of the basis on which it is centered, and $\varphi^{\text{at}}_{j\boldsymbol{R}}(\boldsymbol{r}) = \varphi^{\text{at}}_j(\boldsymbol{r} - \boldsymbol{R})$ its copy in cell $\boldsymbol{R}$. No single orbital has the form Eq. (3.20), but the *Bloch sum*

$$|j\boldsymbol{k}\rangle = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}}\, |\varphi^{\text{at}}_{j\boldsymbol{R}}\rangle, \tag{3.28}$$

does: translating it by a lattice vector $\boldsymbol{R}'$ relabels the sum and multiplies it by $e^{i\boldsymbol{k}\cdot\boldsymbol{R}'}$. Since H_e commutes with lattice translations, it does not connect Bloch sums of different $\boldsymbol{k}$, and we look for the states as $|\psi_{n\boldsymbol{k}}\rangle = \sum_j C_{nj}(\boldsymbol{k})\,|j\boldsymbol{k}\rangle$, with one coefficient per orbital in the cell. The matrix elements between Bloch sums reduce to sums over one lattice vector. Translation invariance gives $\langle\varphi^{\text{at}}_{j\boldsymbol{R}}|H_e|\varphi^{\text{at}}_{j'\boldsymbol{R}'}\rangle = \langle\varphi^{\text{at}}_{j\boldsymbol{0}}|H_e|\varphi^{\text{at}}_{j',\boldsymbol{R}'-\boldsymbol{R}}\rangle$. The double sum over $\boldsymbol{R}$ and $\boldsymbol{R}'$ in $\langle j\boldsymbol{k}|H_e|j'\boldsymbol{k}\rangle$ becomes N times a single sum over $\boldsymbol{R}'' = \boldsymbol{R}' - \boldsymbol{R}$:

$$H^{\text{TB}}_{jj'}(\boldsymbol{k}) = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}}\, \langle\varphi^{\text{at}}_{j\boldsymbol{0}}|H_e|\varphi^{\text{at}}_{j'\boldsymbol{R}}\rangle, \qquad S^{\text{ov}}_{jj'}(\boldsymbol{k}) = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}}\, \langle\varphi^{\text{at}}_{j\boldsymbol{0}}|\varphi^{\text{at}}_{j'\boldsymbol{R}}\rangle. \tag{3.29}$$

Orbitals on neighboring atoms overlap, so the Bloch sums are not orthonormal, and the *overlap matrix* $S^{\mathrm{ov}}(\boldsymbol{k})$ is not the identity. Requiring the energy $\langle\psi|H_{\mathrm{e}}|\psi\rangle/\langle\psi|\psi\rangle$ to be stationary with respect to the coefficients C^*_{nj} turns the Schrödinger equation into the generalized eigenvalue problem

$$\sum_{j'} H^{\mathrm{TB}}_{jj'}(\boldsymbol{k})\, C_{nj'}(\boldsymbol{k}) = \varepsilon_{n\boldsymbol{k}} \sum_{j'} S^{\mathrm{ov}}_{jj'}(\boldsymbol{k})\, C_{nj'}(\boldsymbol{k}). \tag{3.30}$$

There are as many bands as orbitals in the cell. Of course, the result is only as good as the basis, which in practice is a few orbitals per atom; the payoff is that a handful of numbers, the on-site energies and the hopping and overlap integrals between neighbors, fix the band structure.

The simplest case is one s orbital per site of a Bravais lattice, which yields one band, $n = 1$. Its on-site element $\langle\varphi^{\mathrm{at}}_{\boldsymbol{0}}|H_{\mathrm{e}}|\varphi^{\mathrm{at}}_{\boldsymbol{0}}\rangle = E_s$ is the atomic level, shifted slightly by the potential of the neighbors. For nearest neighbors we write $\langle\varphi^{\mathrm{at}}_{\boldsymbol{0}}|H_{\mathrm{e}}|\varphi^{\mathrm{at}}_{\boldsymbol{R}}\rangle = -t_{\mathrm{hop}}$, where the *hopping amplitude* t_{hop} is positive for s orbitals in an attractive potential. We neglect hopping beyond nearest neighbors and the overlap between different sites, which sets $S^{\mathrm{ov}} = 1$. Eq. (3.30) then reduces to

$$\varepsilon_{1\boldsymbol{k}} = E_s - t_{\mathrm{hop}} \sum_{\boldsymbol{R}\in\mathrm{nn}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}. \tag{3.31}$$

The face-centered cubic lattice has 12 nearest neighbors, at $(a/2)(\pm1, \pm1, 0)$ and its permutations. The four with a given pair of nonzero components sum to $4\cos(k_x a/2)\cos(k_y a/2)$, and

$$\varepsilon_{1\boldsymbol{k}} = E_s - 4t_{\mathrm{hop}}\Big[\cos\frac{k_x a}{2}\cos\frac{k_y a}{2} + \cos\frac{k_y a}{2}\cos\frac{k_z a}{2} + \cos\frac{k_z a}{2}\cos\frac{k_x a}{2}\Big]. \tag{3.32}$$

Fig. 3.6 plots this band along the lines between the high-symmetry points of Sect. 3.1. At Γ all phases are one and $\varepsilon = E_s - 12t_{\mathrm{hop}}$: the orbitals add in phase and the state is bonding between every pair of neighbors. At $\mathrm{L} = (2\pi/a)(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ every cosine is $\cos(\pi/2) = 0$, and $\varepsilon = E_s$. At $\mathrm{X} = (2\pi/a)(1, 0, 0)$ two of the three products are -1 and one is $+1$, and $\varepsilon = E_s + 4t_{\mathrm{hop}}$; the band stays at this maximum along the whole line from X to W. The bandwidth is $16t_{\mathrm{hop}}$, proportional to the number of neighbors and to the hopping between them. Near Γ, expanding each cosine to second order leads to $\varepsilon_{1\boldsymbol{k}} = E_s - 12t_{\mathrm{hop}} + t_{\mathrm{hop}}a^2k^2$, the dispersion of a free particle with the *effective mass* $1/(2t_{\mathrm{hop}}a^2)$. A narrow band is a heavy band: an electron that tunnels slowly between atoms responds to a force as if it were massive. Sect. 3.3 defines effective masses precisely and lists those of silicon.

Silicon needs more than one orbital per atom, and the reason is chemical. A silicon atom has its $3s$ level several electronvolts below its $3p$ level, and in the crystal it uses both: it forms four bonds to the four neighbors of Eq. (3.5), at the corners of a tetrahedron. The orbitals that point along these bonds are the four sp^3 *hybrids*

$$\begin{aligned} h_1 &= \tfrac{1}{2}(s + p_x + p_y + p_z), & h_2 &= \tfrac{1}{2}(s + p_x - p_y - p_z), \\ h_3 &= \tfrac{1}{2}(s - p_x + p_y - p_z), & h_4 &= \tfrac{1}{2}(s - p_x - p_y + p_z). \end{aligned} \tag{3.33}$$

These are orthonormal combinations of one s and three p orbitals on the same atom. Each is directed along one of the cube diagonals $(1,1,1)$, $(1,-1,-1)$, $(-1,1,-1)$ and $(-1,-1,1)$ that join an atom at $\boldsymbol{\tau}_1$ to its neighbors. Each hybrid has the on-site energy $(E_s + 3E_p)/4$, a quarter of the way from E_p toward E_s. Forming the hybrids costs promotion energy, since an electron is moved from s to p character, and the bonds pay it back [5, Chap. 3].

Two hybrids that point at each other across a bond, one on each atom, are coupled by a large matrix element, which we write $-V_{\mathrm{cov}}$ with $V_{\mathrm{cov}} > 0$ (Harrison's covalent energy). Treated alone, the pair is the two-orbital problem of Sect. 2.3: a bonding combination $(h + h')/\sqrt{2}$ at $(E_s + 3E_p)/4 - V_{\mathrm{cov}}$ and an antibonding combination $(h - h')/\sqrt{2}$ at $(E_s + 3E_p)/4 + V_{\mathrm{cov}}$. A

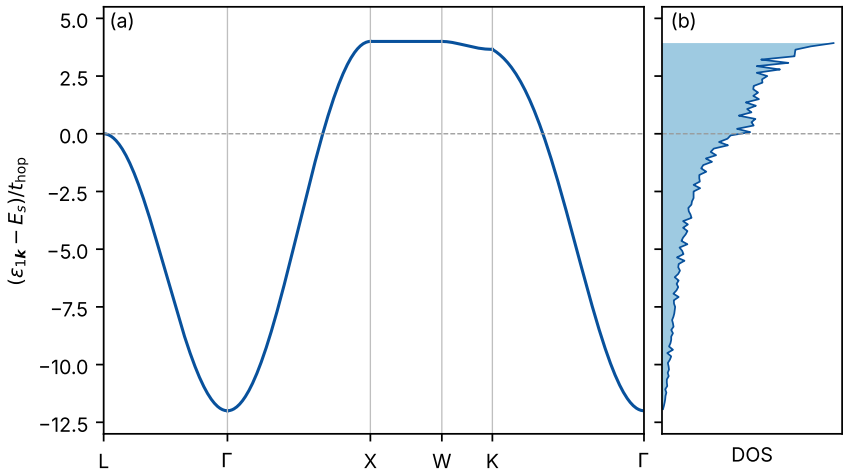


Figure 3.6: Tight-binding s band of the face-centered cubic lattice, Eq. (3.32): one orbital per site with on-site energy E_s, hopping $-t_{\mathrm{hop}}$ between the 12 nearest neighbors, no longer-range hopping, and the overlap between orbitals on different sites neglected. (a) $(\varepsilon_{1\boldsymbol{k}} - E_s)/t_{\mathrm{hop}}$ along the path L–Γ–X–W–K–Γ, with L $= (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, X $= (1, 0, 0)$, W $= (1, \frac{1}{2}, 0)$ and K $= (\frac{3}{4}, \frac{3}{4}, 0)$ in units of $2\pi/a$. The band runs from $-12t_{\mathrm{hop}}$ at Γ to $+4t_{\mathrm{hop}}$, which it reaches at X and keeps along the line X–W; its value at L is 0 and at K $3.66\,t_{\mathrm{hop}}$. (b) Density of states, in arbitrary units, from a uniform grid of 160^3 crystal momenta. Its peak at the top of the band comes from the flat line X–W.

primitive cell holds two atoms and eight hybrids, which make four bonds, hence four bonding and four antibonding orbitals. Its eight valence electrons exactly fill the four bonding orbitals, two per bond, with opposite spins. The smaller couplings, between hybrids on the same atom and between neighboring bonds, then broaden the four bonding levels into four valence bands and the four antibonding levels into four conduction bands. The gap of silicon is the remnant of the bonding–antibonding splitting $2V_{\mathrm{cov}}$, reduced by these broadenings, and it survives because the splitting is larger than the bands are wide [5, Chap. 3]. This picture accounts for the count of Sect. 3.1, four filled bands, without any reference to Bragg planes. It also explains where the valence charge sits: the bonding orbitals concentrate it between the atoms, in the bonds. This bond charge is not a sum of spherical atomic terms, and it gives the potential of silicon its (222) component; the (200) component stays zero by the glide symmetry (Worked example 3.2, Appendix C.17). Germanium is bonded the same way, more weakly. In GaAs the two hybrids of a bond have different energies, the bond is partly ionic, and the gap is wider than the covalent splitting alone would give (Sect. 3.3).

The nearly-free and the tight-binding pictures are two limits of the same problem, Eq. (3.21), and both are approximations to it. Real valence states of silicon are well described by neither: they have the bonding character of the tight-binding picture, and the plane-wave content near the zone boundary that the nearly-free picture predicts. A first-principles calculation solves Eq. (3.21) in a plane-wave basis (Sects. 3.4 and 3.5), and Wannier functions (Sect. 3.6) recover from it a set of localized orbitals in which the band structure has the tight-binding form of Eq. (3.30), with orthonormal orbitals and hopping integrals computed rather than guessed.

3.2.4 The density vertex

A dark-matter particle couples to the electrons through $n_e(\boldsymbol{q}) = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$, and for independent electrons each transition moves one electron, with the single-particle matrix element $f_{i\to f}(\boldsymbol{q}) =$

$\langle f|e^{i\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle$ of Eq. (2.17) (Eq. (2.39)). Between Bloch states this matrix element has a simple structure. Take $\psi_{n\boldsymbol{k}}$ as the initial state and $\psi_{m\boldsymbol{k}'}$ as the final one, with $\boldsymbol{k}$ and $\boldsymbol{k}'$ in the first zone. The momentum transfer lies on the same grid as the crystal momenta, since the dark-matter states are normalized in the same periodic box of volume $\mathcal{V}$ (Sect. 2.1). With Eq. (3.20),

$$\langle\psi_{m\boldsymbol{k}'}|e^{i\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_{n\boldsymbol{k}}\rangle = \frac{1}{N}\int_V \mathrm{d}^3r\; e^{i(\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{k}')\cdot\boldsymbol{r}}\, u^*_{m\boldsymbol{k}'}(\boldsymbol{r})\, u_{n\boldsymbol{k}}(\boldsymbol{r}).$$

Split the crystal into cells, $\boldsymbol{r} = \boldsymbol{R} + \boldsymbol{x}$ with $\boldsymbol{x}$ in the cell at the origin. The product u^*u is the same in every cell, and the phase factorizes into $e^{i(\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{k}')\cdot\boldsymbol{R}}$ times its value at $\boldsymbol{x}$. The sum over cells is the lattice sum of Eq. (3.14), which equals N if $\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{k}'$ is a reciprocal-lattice vector $\boldsymbol{G}_\mathrm{U}$ and vanishes otherwise:

$$\langle\psi_{m\boldsymbol{k}'}|e^{i\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_{n\boldsymbol{k}}\rangle = \delta_{\boldsymbol{k}',\,\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{G}_\mathrm{U}}\int_\Omega \mathrm{d}^3x\; e^{i\boldsymbol{G}_\mathrm{U}\cdot\boldsymbol{x}}\, u^*_{m\boldsymbol{k}'}(\boldsymbol{x})\, u_{n\boldsymbol{k}}(\boldsymbol{x}).$$

This is the selection rule Eq. (3.15), now for one electron. The vector $\boldsymbol{G}_\mathrm{U}$ is the one that folds $\boldsymbol{k}+\boldsymbol{q}$ back into the first zone. It depends on $\boldsymbol{k}$ as well as on $\boldsymbol{q}$, and it is not in general the vector $\boldsymbol{G}_q$ nearest to $\boldsymbol{q}$ of Eq. (3.16). With the periodic gauge, $u_{m,\boldsymbol{k}+\boldsymbol{q}} = e^{-i\boldsymbol{G}_\mathrm{U}\cdot\boldsymbol{r}}u_{m\boldsymbol{k}'}$ by Eq. (A.16), and the phase is absorbed into the final state:

Density vertex

For $\boldsymbol{k}$ in the first zone and any $\boldsymbol{q}$, $e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$ connects $\psi_{n\boldsymbol{k}}$ only to the states at crystal momentum $\boldsymbol{k}+\boldsymbol{q}$, folded back as $\boldsymbol{k}' = \boldsymbol{k}+\boldsymbol{q}-\boldsymbol{G}_\mathrm{U}$. The amplitude is the *density vertex*

$$\begin{aligned}\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) \equiv \langle\psi_{m,\boldsymbol{k}+\boldsymbol{q}}|e^{i\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_{n\boldsymbol{k}}\rangle &= \int_\Omega \mathrm{d}^3r\; u^*_{m,\boldsymbol{k}+\boldsymbol{q}}(\boldsymbol{r})\, u_{n\boldsymbol{k}}(\boldsymbol{r}) \\ &= \int_\Omega \mathrm{d}^3r\; u^*_{m\boldsymbol{k}'}(\boldsymbol{r})\, e^{i\boldsymbol{G}_\mathrm{U}\cdot\boldsymbol{r}}\, u_{n\boldsymbol{k}}(\boldsymbol{r}),\end{aligned} \tag{3.34}$$

with $\rho_{nn}(\boldsymbol{k},0) = 1$. It is of order one and carries no factor of N. The Umklapp vector $\boldsymbol{G}_\mathrm{U}$ stays in the plane wave.

The normalization $\rho_{nn}(\boldsymbol{k},0) = \int_\Omega |u_{n\boldsymbol{k}}|^2 = 1$ is that of Eq. (3.20). The plane wave $e^{i\boldsymbol{G}_\mathrm{U}\cdot\boldsymbol{r}}$ in Eq. (3.34) cannot be dropped. The overlap $\int_\Omega u^*_{m\boldsymbol{k}'}u_{n\boldsymbol{k}}$ without it is a different number, and it would make the vertex depend on $\boldsymbol{q}$ only through its reduced part. The crystal momentum changes by $\boldsymbol{q}$ modulo a reciprocal-lattice vector, but the wavefunction is probed by the whole $\boldsymbol{q}$, which is why Sect. 3.1 never folds it.

The plane-wave form of the vertex shows what happens to the momentum. We insert Eq. (3.22) for both states in the cell integral of Eq. (3.34) that carries $e^{i\boldsymbol{G}_\mathrm{U}\cdot\boldsymbol{r}}$. The cell integral of $e^{i(\boldsymbol{G}_\mathrm{U}+\boldsymbol{G}-\boldsymbol{G}')\cdot\boldsymbol{r}}$ is $\Omega\,\delta_{\boldsymbol{G}',\boldsymbol{G}+\boldsymbol{G}_\mathrm{U}}$, and

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) = \sum_{\boldsymbol{G}} c^*_{m\boldsymbol{k}'}(\boldsymbol{G}+\boldsymbol{G}_\mathrm{U})\, c_{n\boldsymbol{k}}(\boldsymbol{G}), \qquad \boldsymbol{k}' = \boldsymbol{k}+\boldsymbol{q}-\boldsymbol{G}_\mathrm{U}. \tag{3.35}$$

Each plane-wave component of the initial state, with momentum $\boldsymbol{k}+\boldsymbol{G}$, is shifted by exactly $\boldsymbol{q}$ to $\boldsymbol{k}'+\boldsymbol{G}+\boldsymbol{G}_\mathrm{U}$, and the vertex is the overlap of the shifted state with the final one. Momentum is conserved component by component; the crystal only decides how the result is labeled. In the empty lattice, where each state is one plane wave, Eq. (3.35) reduces to the momentum conservation of free electrons, and the distinction between normal and Umklapp processes is a matter of band labels. In a crystal the coefficients spread each state over many $\boldsymbol{G}$, and a large $|\boldsymbol{q}|$ picks out the overlap between the high-momentum tail of the initial state and the low-momentum part of the final one. In the notation of Ref. [7, Eqs. (3.14)–(3.16)], $c_{n\boldsymbol{k}}(\boldsymbol{G})$ is their $u_n(\boldsymbol{k}+\boldsymbol{G})$ and Eq. (3.35) is their $f_{[n\boldsymbol{k},m\boldsymbol{k}',\boldsymbol{G}_\mathrm{U}]}$.

The vertices out of one state exhaust the probability. The periodic functions $u_{m,\boldsymbol{k}+\boldsymbol{q}}$, $m = 1, 2, \ldots$, are the eigenfunctions of the Hermitian operator $H_{\boldsymbol{k}+\boldsymbol{q}}$ on one cell and form a complete orthonormal set there. Expanding $u_{n\boldsymbol{k}}$ in them and using its normalization,

$$\sum_m |\rho_{mn}(\boldsymbol{k},\boldsymbol{q})|^2 = \sum_m \Big| \int_\Omega u^*_{m,\boldsymbol{k}+\boldsymbol{q}}\, u_{n\boldsymbol{k}} \Big|^2 = \int_\Omega |u_{n\boldsymbol{k}}|^2 = 1 \tag{3.36}$$

for every $\boldsymbol{q}$. This is the closure relation of Eq. (2.19) for a Bloch electron: a momentum transfer redistributes the electron over the bands at $\boldsymbol{k}+\boldsymbol{q}$ without changing its total weight. At small q the weight stays in band n; at q beyond a few reciprocal-lattice vectors it spreads over many high bands, and the transitions that a detector sees, into the conduction bands just above the gap, receive the fraction that Worked example 3.4 illustrates.

In second quantization, Eq. (2.38) in the Bloch basis becomes

$$n_e(\boldsymbol{q}) = \sum_s \sum_{\boldsymbol{k}\in \mathrm{BZ}} \sum_{mn} \rho_{mn}(\boldsymbol{k},\boldsymbol{q})\, c^\dagger_{m,\boldsymbol{k}+\boldsymbol{q},s}\, c_{n\boldsymbol{k}s}, \tag{3.37}$$

with $c_{m,\boldsymbol{k}+\boldsymbol{q},s} = c_{m\boldsymbol{k}'s}$ in the periodic gauge. The particle–hole matrix element of Eq. (2.39) that takes an electron from a valence state $v\boldsymbol{k}$ to a conduction state $c\boldsymbol{k}'$ is $\rho_{cv}(\boldsymbol{k},\boldsymbol{q})$, and the independent-electron transition sum of Eq. (2.70) becomes a sum of $|\rho_{cv}(\boldsymbol{k},\boldsymbol{q})|^2$ over $\boldsymbol{k}$, the valence and conduction bands, and spin. That sum, weighted by energy conservation, is the crystal form factor of Ref. [7, Eq. (3.17)], which Sect. 10.1 derives in the book's normalization. The counting of Appendix A.3 explains why no factor of N appears in Eq. (3.34): one sum over $\boldsymbol{k}$ supplies a factor N, and the prefactor $2\pi/\mathscr{V}$ of the structure factor turns it into the intensive $2\pi/\Omega$.

3.2.5 Velocity matrix elements

Light of energy ω has the momentum $q = \omega$, a few electronvolts for visible and ultraviolet photons, and 10^{-3} of the zone size. Optical absorption probes the density vertex at small q, and there it is controlled by a second matrix element, that of the velocity. The momentum and velocity matrix elements between Bloch states at the same $\boldsymbol{k}$ are

$$\begin{aligned} \boldsymbol{p}_{mn}(\boldsymbol{k}) &= \langle \psi_{m\boldsymbol{k}} | \hat{\boldsymbol{p}} | \psi_{n\boldsymbol{k}} \rangle = \int_\Omega \mathrm{d}^3 r\; u^*_{m\boldsymbol{k}}\, (\hat{\boldsymbol{p}} + \boldsymbol{k})\, u_{n\boldsymbol{k}}, \\ \boldsymbol{v}_{mn}(\boldsymbol{k}) &= \langle \psi_{m\boldsymbol{k}} | \boldsymbol{v} | \psi_{n\boldsymbol{k}} \rangle, \qquad \boldsymbol{v} = \mathrm{i}[H_\mathrm{e}, \boldsymbol{r}]. \end{aligned} \tag{3.38}$$

The velocity operator is the time derivative of the position in the Heisenberg picture. For the local potential of Eq. (3.19), $U(\boldsymbol{r})$ commutes with $\boldsymbol{r}$ and $\boldsymbol{v} = \hat{\boldsymbol{p}}/m_e$, so that $\boldsymbol{v}_{mn} = \boldsymbol{p}_{mn}/m_e$. In plane waves, $\boldsymbol{p}_{mn}(\boldsymbol{k}) = \sum_{\boldsymbol{G}} c^*_{m\boldsymbol{k}}(\boldsymbol{G})(\boldsymbol{k}+\boldsymbol{G})c_{n\boldsymbol{k}}(\boldsymbol{G})$. The matrix elements of $\boldsymbol{r}$ itself between extended states are not defined as integrals over the crystal, but for $m \neq n$ the commutator gives them a meaning. Taking $\langle \psi_{m\boldsymbol{k}} | \cdot | \psi_{n\boldsymbol{k}} \rangle$ of $\boldsymbol{v} = \mathrm{i}[H_\mathrm{e}, \boldsymbol{r}]$,

$$\langle \psi_{m\boldsymbol{k}} | \boldsymbol{r} | \psi_{n\boldsymbol{k}} \rangle = \frac{-\mathrm{i}\, \boldsymbol{v}_{mn}(\boldsymbol{k})}{\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}}, \qquad m \neq n, \tag{3.39}$$

which is the form in which interband position matrix elements enter first-principles codes (Appendix E.4).

The dependence of these matrix elements on $\boldsymbol{k}$ is easiest to handle through the operator $H_{\boldsymbol{k}}$ of Eq. (3.21). Differentiating its definition, $\nabla_{\boldsymbol{k}}(\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} H_\mathrm{e} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}) = \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}(-\mathrm{i}\boldsymbol{r} H_\mathrm{e} + \mathrm{i} H_\mathrm{e}\boldsymbol{r})\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$, so

$$\nabla_{\boldsymbol{k}} H_{\boldsymbol{k}} = \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\, \boldsymbol{v}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}, \qquad \boldsymbol{v}_{mn}(\boldsymbol{k}) = \int_\Omega \mathrm{d}^3 r\; u^*_{m\boldsymbol{k}}\, (\nabla_{\boldsymbol{k}} H_{\boldsymbol{k}})\, u_{n\boldsymbol{k}}. \tag{3.40}$$

For the local potential, $\nabla_{\boldsymbol{k}} H_{\boldsymbol{k}} = (\hat{\boldsymbol{p}} + \boldsymbol{k})/m_e$; the identity holds for any Hamiltonian, nonlocal ones included. The group velocity and the $\boldsymbol{k}\cdot\boldsymbol{p}$ relation both follow by differentiating $H_{\boldsymbol{k}} u_{n\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}} u_{n\boldsymbol{k}}$ with respect to $\boldsymbol{k}$,

$$(\nabla_{\boldsymbol{k}} H_{\boldsymbol{k}})\, u_{n\boldsymbol{k}} + H_{\boldsymbol{k}} \nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}} = (\nabla_{\boldsymbol{k}} \varepsilon_{n\boldsymbol{k}})\, u_{n\boldsymbol{k}} + \varepsilon_{n\boldsymbol{k}} \nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}},$$

and projecting onto $u_{m\boldsymbol{k}}$. For $m = n$ the terms with $\nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}}$ cancel, because $\langle u_{n\boldsymbol{k}} | H_{\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}} \langle u_{n\boldsymbol{k}}|$, and normalization leaves

Group velocity

$$\boldsymbol{v}_n^{\mathrm{g}}(\boldsymbol{k}) \equiv \nabla_{\boldsymbol{k}} \varepsilon_{n\boldsymbol{k}} = \boldsymbol{v}_{nn}(\boldsymbol{k}). \tag{3.41}$$

The slope of a band is the expectation value of the velocity in the Bloch state.

This is the Hellmann–Feynman theorem for the parameter $\boldsymbol{k}$. A wave packet built from states of band n near $\boldsymbol{k}$ moves with $\boldsymbol{v}_n^{\mathrm{g}}$, the *group velocity* [1, Chap. 8 and App. E]. For free electrons it is $\boldsymbol{k}/m_e$. On a Bragg plane its component along $\boldsymbol{G}$ vanishes, as the standing waves of the nearly-free picture require, and in the tight-binding band Eq. (3.32) it vanishes at Γ, X and L and is at most of order $t_{\mathrm{hop}} a$. For $m \neq n$, and $\varepsilon_{m\boldsymbol{k}} \neq \varepsilon_{n\boldsymbol{k}}$, the projection gives instead

$$\langle u_{m\boldsymbol{k}} | \nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}} \rangle = \frac{\boldsymbol{v}_{mn}(\boldsymbol{k})}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}}}, \tag{3.42}$$

the first-order change of $u_{n\boldsymbol{k}}$ along band m when $\boldsymbol{k}$ moves; this is $\boldsymbol{k}\cdot\boldsymbol{p}$ perturbation theory [6, Chap. 2].

The optical limit of the vertex is now one line. Expand the final state, $u_{m,\boldsymbol{k}+\boldsymbol{q}} = u_{m\boldsymbol{k}} + \boldsymbol{q}\cdot\nabla_{\boldsymbol{k}} u_{m\boldsymbol{k}} + O(q^2)$, in the overlap $\int_\Omega u^*_{m,\boldsymbol{k}+\boldsymbol{q}} u_{n\boldsymbol{k}}$ of Eq. (3.34). The zeroth-order term $\langle u_{m\boldsymbol{k}} | u_{n\boldsymbol{k}} \rangle$ vanishes for $m \neq n$, and the first-order term is $\boldsymbol{q}\cdot\langle \nabla_{\boldsymbol{k}} u_{m\boldsymbol{k}} | u_{n\boldsymbol{k}} \rangle = \boldsymbol{q}\cdot\langle u_{n\boldsymbol{k}} | \nabla_{\boldsymbol{k}} u_{m\boldsymbol{k}} \rangle^*$. With Eq. (3.42), with n and m exchanged, and the hermiticity of the velocity, $\boldsymbol{v}^*_{nm} = \boldsymbol{v}_{mn}$:

Optical limit of the density vertex

$$\rho_{mn}(\boldsymbol{k}, \boldsymbol{q}) = \frac{\boldsymbol{q}\cdot\boldsymbol{v}_{mn}(\boldsymbol{k})}{\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}} + O(q^2), \qquad m \neq n. \tag{3.43}$$

An interband vertex vanishes linearly as $q \to 0$, with a slope set by the velocity matrix element and the transition energy. The intraband vertex tends to one, $\rho_{nn}(\boldsymbol{k}, \boldsymbol{q}) = 1 + O(q)$.

The same result follows from expanding the plane wave, $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \simeq 1 + \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}$, between states at the same $\boldsymbol{k}$, and using Eq. (3.39), which is the route of Eq. (A.18). The expansion assumes that the phases of the states are chosen smoothly in $\boldsymbol{k}$ between $\boldsymbol{k}$ and $\boldsymbol{k}+\boldsymbol{q}$. Any other choice multiplies both sides of Eq. (3.43) by phases (Eq. (A.19)), and leaves $|\rho_{mn}|^2$, which is what a first-order rate contains, unchanged. When band m is degenerate at $\boldsymbol{k}$, only the sum of $|\rho_{mn}|^2$ over the degenerate partners is defined (Appendix A.5).

The interband probability $|\rho_{mn}|^2$ therefore grows as q^2 at small q, like the dipole transition $1s \to 2p$ of hydrogen in Sect. 2.2. The intraband vertex does not vanish, but in a full band it cannot act: the state at $\boldsymbol{k}+\boldsymbol{q}$ in the same band is occupied, and the term is Pauli blocked. Intraband transitions matter in metals, at the Fermi surface (Sect. 4.5). For photons, Eq. (3.43) is all that is needed. The absorption spectrum of a semiconductor is a sum over the squared components of $\boldsymbol{v}_{cv}(\boldsymbol{k})$ along the polarization of the light (Sects. 4.4 and 4.6). Absorbed bosonic dark matter, whose momentum is also far below the zone scale (Sect. 3.3), is governed by the same matrix elements (Sect. 10.7).

The expansion of $u_{m,\boldsymbol{k}+\boldsymbol{q}}$ is useful while its first-order term is small, and by Eq. (3.42) this requires

$$q \ll \frac{|\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}|}{|\boldsymbol{v}_{mn}(\boldsymbol{k})|}. \tag{3.44}$$

For a photon, with $q = \omega \simeq |\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}|$, the condition reads $|\boldsymbol{v}_{mn}| \ll 1$, the speed of light, and it always holds. For dark matter that deposits the same energy, the momentum transfer is at least $\omega/v_{\max}$ (Sect. 1.3), and the condition becomes $|\boldsymbol{v}_{mn}| \ll v_{\max} = 2.650 \times 10^{-3}$. Valence electrons, however, move with typical velocities $\sim \alpha$, and the Fermi velocity of silicon's valence density is 7.0×10^{-3} (Sect. 2.5). Parametrically, then, the momentum transfers at which dark matter can excite an electron across the gap lie at or beyond the edge of the optical limit. Eq. (3.43) is a check on a calculation of ρ_{mn} at small q, and a scattering rate needs the full vertex of Eq. (3.35).

In practice, the band energies $\varepsilon_{n\boldsymbol{k}}$ and the periodic parts $u_{n\boldsymbol{k}}$ of a real material come from density-functional theory (Sect. 3.4). A code solves Eq. (3.23) for the Kohn–Sham potential in a basis of plane waves up to a cutoff (Sect. 3.5) and stores the coefficients $c_{n\boldsymbol{k}}(\boldsymbol{G})$ on a grid of crystal momenta. The vertex is then the sum Eq. (3.35), with the coefficients of the two states matched by the integer coordinates of their $\boldsymbol{G}$. For all $\boldsymbol{G}_\mathrm{U}$ at once it is a fast Fourier transform of the product $u^*_{m\boldsymbol{k}'}u_{n\boldsymbol{k}}$, and the momentum matrix element is the corresponding sum over $\boldsymbol{k} + \boldsymbol{G}$ (Appendix E.7). The pseudopotentials of these codes are nonlocal, and then $\boldsymbol{v} \neq \hat{\boldsymbol{p}}/m_e$: by Eq. (3.40), $\boldsymbol{v}_{mn}$ acquires the derivative of the nonlocal potential with respect to $\boldsymbol{k}$, a correction that Appendix E.7 evaluates and that changes the computed absorption of diamond by 15 % to 20 % [8, Fig. 1].

Worked example 3.4: The vertex between the two lowest nearly-free bands

Return to the one-dimensional crystal of Worked example 3.3, and take the initial state at the left zone boundary, $k = -\pi/a = -G_1/2$, in the lowest band, with the final state in the second band. Near the boundary the two bands are made of the plane waves $\mathrm{e}^{\mathrm{i}kx}$ and $\mathrm{e}^{\mathrm{i}(k+G_1)x}$, and Eq. (3.25), with $\boldsymbol{G} \to -G_1$, mixes them. Its lower eigenvector is $(c(0), c(G_1)) = (\cos\theta_k, \sin\theta_k)$ and its upper one $(-\sin\theta_k, \cos\theta_k)$, with $\tan 2\theta_k = 2|U_{G_1}|/(\varepsilon^0_{k+G_1} - \varepsilon^0_k)$; for $U_{G_1} < 0$ both components of the lower state have the same sign. At the boundary the two free energies are equal and $\theta = \pi/4$.

Small q. For $0 < q < G_1$, $k + q$ stays in the first zone, $\boldsymbol{G}_\mathrm{U} = 0$, and Eq. (3.35) reads $\rho_{21}(k, q) = -\sin\theta_{k+q}\cos\theta_k + \cos\theta_{k+q}\sin\theta_k = \sin(\theta_k - \theta_{k+q})$. With $\varepsilon^0_{k+G_1} - \varepsilon^0_k = G_1(k + G_1/2)/m_e$, the angle falls from $\pi/4$ with the slope $\mathrm{d}\theta/\mathrm{d}k = -G_1/(4m_e|U_{G_1}|)$ at the boundary, and

$$|\rho_{21}(k, q)| \simeq \frac{qG_1}{4m_e|U_{G_1}|} = \frac{q}{448\,\mathrm{eV}}.$$

The optical limit agrees. In the two-wave model $|v_{21}| = G_1/2m_e = 2.23 \times 10^{-3}$, the velocity of a free electron at the zone boundary, and Eq. (3.43) gives $q|v_{21}|/2|U_{G_1}|$, the same expression. The full calculation of Fig. 3.5(b) has $|v_{21}| = 2.24 \times 10^{-3}$ and $\varepsilon_{2k} - \varepsilon_{1k} = 0.998\,\mathrm{eV}$. At $q = 10\,\mathrm{eV}$ the vertex is 0.0225 in the full calculation and in the optical limit; at 100 eV the two are 0.210 and 0.225. At $q = 2|U_{G_1}|/v_{\max} = 377\,\mathrm{eV}$, the smallest momentum transfer with which dark matter can deposit the 1 eV gap, the full vertex is 0.496 and the optical limit 0.849, too large by a factor of 1.7, and $|\rho_{21}|^2$ by a factor of 2.9. This q is a reference scale, not an allowed event, since the vertex is evaluated at fixed k: at $q = 377\,\mathrm{eV}$ the final state from this k has moved up the second band, the transition costs 1.6 eV, and depositing 1.6 eV needs $q \geq 0.61\,\mathrm{keV}$. The vertex saturates instead of growing: as $k + q$ moves away from the boundary, the final state becomes a

single plane wave, and $|\rho_{21}|$ tends to $\sin(\pi/4) = 0.71$ in the two-wave model. The scale on which the linear law fails, $4m_e|U_{G_1}|/G_1 = 448\,\mathrm{eV}$, is the $|\varepsilon_{2k} - \varepsilon_{1k}|/|v_{21}|$ of Eq. (3.44), and $|v_{21}| = 0.85\,v_{\max}$ puts the momentum transfers of dark matter right at it.
Umklapp. Now take $q = G_1 = 2.28\,\mathrm{keV}$. Then $k + q = +\pi/a$, which is folded back to $k' = -\pi/a$ with $\boldsymbol{G}_\mathrm{U} = G_1$: the final state has the same reduced crystal momentum as the initial one, and the transition energy is the same 0.998 eV as at $q \to 0$. At the boundary the states are $(c(0), c(G_1)) = (1,1)/\sqrt{2}$ in band 1 and $(-1,1)/\sqrt{2}$ in band 2. In Eq. (3.35) only the term $\boldsymbol{G} = 0$ survives in the two-wave model, $c^*_{2k'}(G_1)\,c_{1k}(0) = \frac{1}{2}$, so $|\rho_{21}(k, G_1)| = \frac{1}{2}$; the full calculation finds 0.501. The same pair of states, with the same transition energy, has the vertex 0.02 at $q = 10\,\mathrm{eV}$ and 0.50 at $q = G_1$. The difference comes from $\boldsymbol{G}_\mathrm{U}$ in the plane wave: the momentum transfer G_1 moves the component $\mathrm{e}^{\mathrm{i}kx}$ of the initial state onto the component $\mathrm{e}^{\mathrm{i}(k'+G_1)x}$ of the final one, and these overlap at order one. In the two-wave model band 2 receives a quarter of the probability and band 1 keeps another quarter. The remaining half, from the component $\mathrm{e}^{\mathrm{i}(k+G_1)x}$ shifted to $\mathrm{e}^{\mathrm{i}(k+2G_1)x}$, goes into bands 3 and 4, whose energies at the boundary are near $(3G_1/2)^2/2m_e = 11.5\,\mathrm{eV}$. The full calculation finds $|\rho_{m1}|^2 = 0.298$, 0.251, 0.226 and 0.225 for $m = 1$ to 4, and the closure relation Eq. (3.36) holds to 10^{-10}.

A one-dimensional cosine potential has two parameters and silicon has many. The lessons carry over, however: interband vertices are suppressed as q at small momentum transfer, they saturate on the scale where the states change across the zone, and an Umklapp vector in the plane wave can make them of order one. Where the band edges of silicon, germanium and GaAs lie, and how large the relevant velocities and gaps are, is the subject of Sect. 3.3.

3.3 Silicon, germanium and gallium arsenide

The highest occupied electron state of silicon sits at the center of the Brillouin zone, and the lowest empty one sits 85 % of the way to the square face of the zone, at a crystal momentum of 1.94 keV. A photon with the 1.17 eV needed to cross the gap carries a momentum a thousand times smaller and cannot connect the two states by itself. A dark-matter particle that scatters with a momentum transfer of a few keV can. Which momentum transfers reach which excitation energies is the target's half of the kinematic question of Sect. 1.3, and the answer lies in the bands just below and just above the gap. We locate these band edges in silicon, germanium and GaAs, the book's running materials, derive their effective masses and spin–orbit splittings, and compare the smallest pair energy at each momentum transfer with what the halo delivers; Table 3.1 collects the numbers.

3.3.1 Band edges and valleys

At the temperatures at which detectors run, the four lowest bands of silicon are full and all higher bands are empty (Sect. 3.2). The full bands are the *valence bands* and the empty ones the *conduction bands*. We label them with the subscripts v and c, as in $\varepsilon_{v\boldsymbol{k}}$ and $\varepsilon_{c\boldsymbol{k}}$, and the two letters never appear on their own. The highest valence energy is the *valence-band maximum* (VBM), and the lowest conduction energy the *conduction-band minimum* (CBM). The band gap is the distance between the two,

$$E_\mathrm{g} = \min_{c,\boldsymbol{k}} \varepsilon_{c\boldsymbol{k}} - \max_{v,\boldsymbol{k}} \varepsilon_{v\boldsymbol{k}}, \tag{3.45}$$

wherever in the zone the two extremes occur. When they occur at different crystal momenta the gap is *indirect*. A transition at fixed crystal momentum is called *vertical*, after its appearance

in a band diagram, and the smallest vertical transition energy is the *minimum direct gap*,

$$E_{\text{dir}} = \min_{v,c,\boldsymbol{k}} (\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}) \geq E_{\text{g}}. \tag{3.46}$$

In a direct-gap crystal the two are equal.

Fig. 3.7(a) shows the bands of silicon along the lines of Sect. 3.1. The VBM lies at Γ and is threefold degenerate, sixfold with spin. These are the bonding states of the sp^3 picture of Sect. 3.2 with pure p character: the three states transform like x, y and z under the rotations of the crystal, and an s orbital, unchanged by every rotation, cannot enter them. The CBM lies on the line from Γ to X, at the crystal momentum

$$\boldsymbol{k}_{\text{c}} = 0.85 \, \frac{2\pi}{a} \, (1, 0, 0) \tag{3.47}$$

and at the five equivalent points along $\pm x$, $\pm y$ and $\pm z$; the position was measured by electron–nuclear double resonance on donor electrons, which occupy these states [9]. The six minima are the *valleys* of the conduction band. An electron excited across the gap relaxes into one of them, and near its bottom it behaves like a free particle with an anisotropic mass, which we derive below. The gap is $E_{\text{g}} = 1.17\,\text{eV}$ at low temperature, 1.1692 eV in the extrapolation to zero temperature [10, Eq. (11)], and 1.12 eV at 300 K [11, Table I]. It shrinks on warming because the lattice expands and the electron–phonon interaction renormalizes the band energies (Sect. 13.7). Detectors run at millikelvin to kelvin temperatures, and the book uses 1.17 eV; the value 1.11 eV of Worked example 1.3 is the one commonly used in projections. The vertical transitions of silicon cost much more. The smallest, $E_{\text{dir}} = 3.4\,\text{eV}$, joins the valence and conduction states at Γ [12, Table II], and vertical transitions near L start only slightly higher, at 3.45 eV [12, Table VI]. Silicon is thus indirect by a wide margin, $E_{\text{dir}} \approx 3E_{\text{g}}$.

Germanium has the same diamond structure, with $a = 5.658\,\text{Å}$ at room temperature [12, Sect. II A], and the same threefold VBM at Γ. Its CBM, however, lies at L, the center of a hexagonal face of the zone (Fig. 3.8(a)). The eight L points come in pairs that differ by a vector $\boldsymbol{G}_{111}$ and are the same crystal momentum, so germanium has four valleys, each made of two half-ellipsoids on opposite faces of the zone. The gap is 0.744 eV at low temperature [12, Table VI] and 0.66 eV at 300 K [11, Table I]. The conduction state at Γ is only 0.15 eV higher, $E_{\text{dir}} = 0.89\,\text{eV}$ [12, Table II]: germanium is indirect by a narrow margin, and a calculation can easily get the order of the two states wrong (Appendix E.1).

GaAs has the zincblende structure, with $a = 5.653\,\text{Å}$ at room temperature [16], and both band edges at Γ (Fig. 3.8(b)). Its gap is direct, 1.52 eV at low temperature and 1.42 eV at 300 K [11, Table I], and the conduction band has a single, isotropic valley. Valleys at L and X follow at 1.815 and 1.981 eV, 0.30 and 0.46 eV above the minimum [16]. Gallium and arsenic flank germanium in the periodic table, yet the gap of GaAs is twice that of germanium. The difference comes from the partly ionic bond of Sect. 3.2: the two hybrids of a bond have different energies, and this difference adds to their covalent splitting.

The bands in Fig. 3.7 and Fig. 3.8(b) come from density-functional theory, which yields the potential from the electron density (Sect. 3.4); Appendix E.1 sets out the calculation. An older route, which we use for the germanium bands of Fig. 3.8(a) and below as a model to test the velocity sum rules, is the empirical pseudopotential. It solves the central equation Eq. (3.23) with a potential that has nonzero Fourier components $U_{\boldsymbol{G}}$ on just the three shortest shells of reciprocal-lattice vectors, and fits these three numbers per element to measured optical transitions [17]. That the fit works at all shows how much of the band structure the lattice fixes and how little of the potential it needs; the price is that the potential is not computed but adjusted, and it says nothing about wavefunctions at large momentum. Its

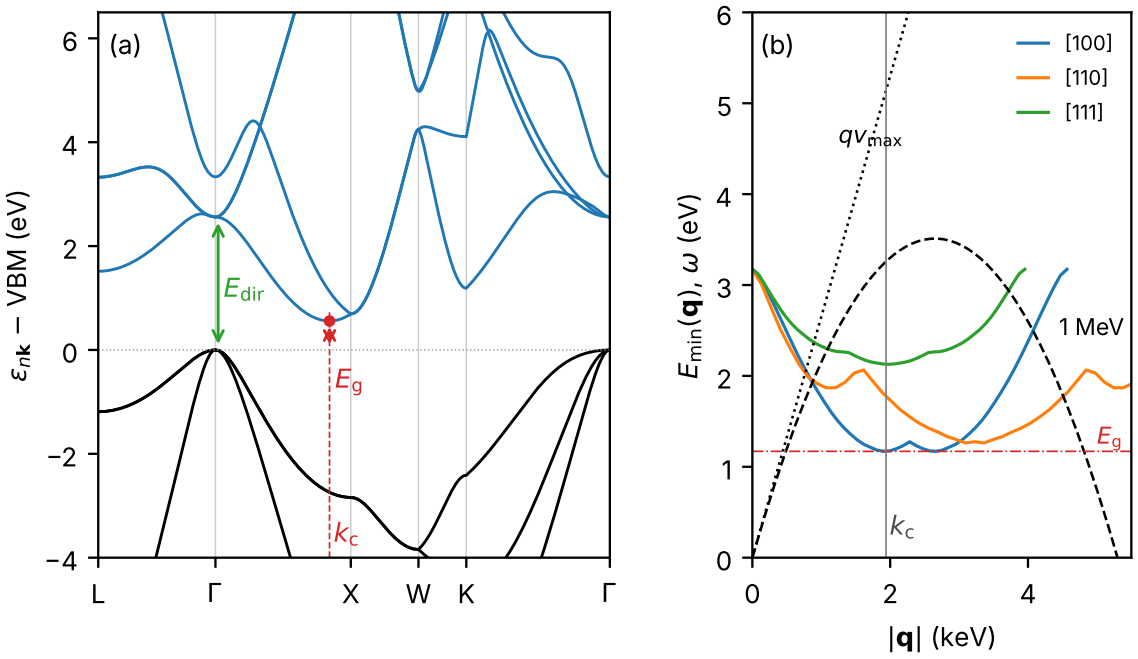


Figure 3.7: Silicon. (a) Band structure along L–Γ–X–W–K–Γ (points of Sect. 3.1), valence bands black and conduction bands blue, energies from the valence-band maximum. The bands are the Kohn–Sham bands of the plane-wave calculation of Appendix E.1 at the measured lattice constant: generalized-gradient functional of Perdew, Burke and Ernzerhof (PBE), SG15 norm-conserving pseudopotential, cutoff 60 Ry, 12^3 grid for the density, $a = 5.431$ Å, no spin–orbit coupling [13–15]. They put the conduction-band minimum at $k_{\mathrm{c}} = 0.843\,(2\pi/a)$ (measured 0.85) and the indirect gap E_{g} at 0.557 eV (measured 1.17 eV), the gap problem of Sect. 3.4. The vertical gap at Γ, marked E_{dir}, is 2.56 eV (measured 3.4 eV) and is the smallest vertical gap in the zone; the one at L is 2.70 eV (measured 3.45 eV). (b) Minimum pair energy $E_{\mathrm{min}}(\boldsymbol{q})$, Eq. (3.57), for $\boldsymbol{q}$ along [100], [110] and [111], from the same calculation on a grid of 48^3 crystal momenta, with the conduction bands shifted rigidly up by 0.61 eV (a scissor, Sect. 3.4) so that the indirect gap equals the measured 1.17 eV (dash-dotted line); at $\boldsymbol{q} = 0$ the shifted bands give $E_{\mathrm{min}} = 3.17$ eV. Dashed: the largest energy that dark matter of 1 MeV can deposit, $qv_{\mathrm{max}} - q^2/2m_\chi$; dotted: the bound qv_{max} for any mass, with $v_{\mathrm{max}} = 2.650 \times 10^{-3}$. The vertical line marks $|\boldsymbol{k}_{\mathrm{c}}| = 1.94$ keV. Along [100], E_{min} returns to its value at $\boldsymbol{q} = 0$ at $|\boldsymbol{G}_{200}| = 4.57$ keV.

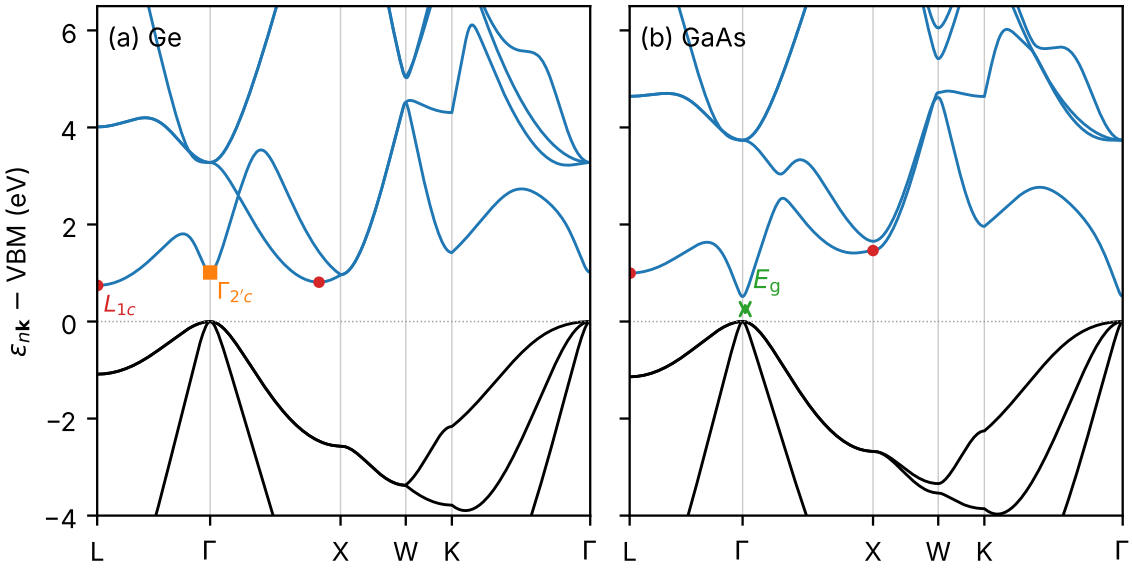


Figure 3.8: Band structures of (a) germanium and (b) GaAs along the path of Fig. 3.7(a). Energies from the valence-band maximum; red dots mark the lowest conduction states away from Γ. (a) Local empirical pseudopotential of Cohen and Bergstresser [17] at $a = 5.66\,\text{Å}$, the lattice constant of their fit, without spin–orbit coupling, with the conduction bands shifted down by 0.21 eV so that the L minimum sits at the measured 0.744 eV. The crystal is indirect, with the s-like state $\Gamma_{2'c}$ (square) at 1.01 eV (measured 0.89 eV) and the Δ minimum, at $0.83\,(2\pi/a)$ along Γ–X, at 0.81 eV. Plain PBE with spin–orbit coupling at the measured lattice constant closes the germanium gap, with Γ_{7c} 0.05 eV below the valence maximum (scalar-relativistic bands in Appendix E.1). (b) PBE plane-wave calculation of Appendix E.1 (SG15 pseudopotentials with the Ga $3d$ shell in the valence, 60 Ry, 12^3 grid, no spin–orbit coupling) at the room-temperature lattice constant $a = 5.653\,\text{Å}$ [13–15]. The gap is direct, as measured, but PBE underestimates it: 0.51 eV against the measured 1.52 eV at low temperature, with the L and X states at 1.00 and 1.46 eV (measured 1.815 and 1.981 eV).

Table 3.1: Band parameters of silicon, germanium and GaAs. Lattice constant a at room temperature: silicon [3, lattice parameter of silicon], germanium [12, Sect. II A], GaAs [16]. Gaps at low temperature: silicon [10, Eq. (11)], germanium [12, Table VI], GaAs [11, Table I]; at 300 K [11, Table I]. Minimum direct gaps E_{dir} and spin–orbit splittings Δ_{SO} of silicon and germanium [12, Table II]. Conduction valleys: position and number; masses along ($m_{\parallel}^{*}$) and across ($m_{\perp}^{*}$) the valley axis, silicon [19] as listed in Ref. [12, Table VII], germanium [12, Table VII]. Heavy- and light-hole masses m_{hh}, m_{lh} along [100] and [111] from Eq. (3.56) with the Luttinger parameters $(\gamma_1^{\mathrm{Lu}}, \gamma_2^{\mathrm{Lu}}, \gamma_3^{\mathrm{Lu}}) = (4.285, 0.339, 1.446)$ for silicon and $(13.25, 4.20, 5.56)$ for germanium [12, Table VII], and $(6.98, 2.06, 2.93)$ for GaAs. All GaAs entries not credited otherwise are from Ref. [16]. Masses in units of m_e.

	Si	Ge	GaAs
a (Å)	5.431	5.658	5.653
E_{g}, low T / 300 K (eV)	1.17 / 1.12	0.744 / 0.66	1.52 / 1.42
gap	indirect	indirect	direct
E_{dir} (eV)	3.4	0.89	1.52
valleys	0.85 X, 6	L, 4	Γ, 1
$m_{\parallel}^{*}$ / $m_{\perp}^{*}$	0.916 / 0.191	1.59 / 0.0815	0.067 (isotropic)
Δ_{SO} (eV)	0.044	0.296	0.341
m_{hh} [100] / [111]	0.277 / 0.718	0.206 / 0.469	0.350 / 0.893
m_{lh} [100] / [111]	0.201 / 0.139	0.046 / 0.041	0.090 / 0.078

potential is local, which is what makes it a convenient test bed for the sum rules. The band parameters of Table 3.1 are measured. A first-principles calculation is checked against exactly these numbers: the measured gaps, the valley positions, the masses from cyclotron resonance, and the electronic dielectric constant of silicon, $\varepsilon_\infty = 11.7$, the square of its long-wavelength refractive index, 3.42 [18] (Sect. 4.4). For the gaps the comparison needs one more step, a *quasiparticle correction* that turns the Kohn–Sham eigenvalues into the energies to add or remove an electron (Sect. 3.4 and Appendix E.3). The PBE functional of Fig. 3.7 is *semilocal*: its exchange–correlation energy at each point depends on the density and its gradient there and nowhere else (Sect. 3.4). It places the CBM of silicon at $k_{\mathrm{c}} = 0.843\,(2\pi/a)$ and its longitudinal mass, $0.95\,m_e$, within 4 % of experiment, but it puts the gap at 0.56 eV, half the measured one (Appendix E.1).

A gap of an electronvolt is enormous compared with the temperature of a detector, and the number of electrons excited across it by heat alone is a useful check of what "empty" means. Far from the chemical potential μ_{ch} the occupation Eq. (3.24) of a conduction state is the Boltzmann factor $\mathrm{e}^{-(\varepsilon_{c\boldsymbol{k}}-\mu_{\mathrm{ch}})/T}$. For a parabolic band of mass m its sum over crystal momenta, $\int \mathrm{d}^3k\,(2\pi)^{-3}\mathrm{e}^{-k^2/2mT} = (mT/2\pi)^{3/2}$, times 2 for spin, gives the electron density $2(m_{\mathrm{dc}}T/2\pi)^{3/2}$ times the Boltzmann factor of the CBM. The empty valence states, the holes, give the same expression with the valence mass and the factor $\mathrm{e}^{-(\mu_{\mathrm{ch}}-\varepsilon_{\mathrm{VBM}})/T}$. In a pure crystal the two densities are equal, the chemical potential drops out of their product, and the *intrinsic carrier density* is [1, Chap. 28]

$$n_{\mathrm{i}} = 2\left(\frac{T}{2\pi}\right)^{3/2}(m_{\mathrm{dc}}m_{\mathrm{dv}})^{3/4}\,\mathrm{e}^{-E_{\mathrm{g}}/2T}. \tag{3.48}$$

Here m_{dc} and m_{dv} are the density-of-states masses of the conduction and valence bands: for silicon $m_{\mathrm{dc}} = 6^{2/3}(m_{\parallel}^{*}m_{\perp}^{*2})^{1/3} = 1.06\,m_e$, which counts the six valleys, and $m_{\mathrm{dv}} = (m_{\mathrm{hh}}^{3/2} + m_{\mathrm{lh}}^{3/2})^{2/3} = 0.50\,m_e$, which adds the heavy and light holes with their spherically averaged masses $m_e/(\gamma_1^{\mathrm{Lu}} \mp 2\bar{\gamma}^{\mathrm{Lu}})$, $\bar{\gamma}^{\mathrm{Lu}} = (2\gamma_2^{\mathrm{Lu}} + 3\gamma_3^{\mathrm{Lu}})/5$, from the Luttinger parameters

of Eq. (3.56) below. At 300 K, with $E_g = 1.12\,\mathrm{eV}$, Eq. (3.48) yields $n_i \approx 6 \times 10^9\,\mathrm{cm}^{-3}$ for parabolic bands. With the low-temperature gap it falls to $10^{-20}\,\mathrm{cm}^{-3}$ at 77 K and to $10^{-720}\,\mathrm{cm}^{-3}$ at 4 K. A kilogram of silicon has a volume of $429\,\mathrm{cm}^3$, so a cold crystal holds no thermally excited carriers at all, and the dark counts of a charge detector (Sect. 1.5) come from other sources.

3.3.2 Effective masses from $k \cdot p$ theory

Near the bottom of a valley the band is a quadratic function of the distance $\delta\boldsymbol{k} = \boldsymbol{k} - \boldsymbol{k}_0$ from the minimum $\boldsymbol{k}_0$. Its curvature decides how an electron there moves, and how many states lie within an energy ω of the edge. The curvature follows from the states at $\boldsymbol{k}_0$ alone. With Eq. (3.21), the Bloch Hamiltonian a little away from $\boldsymbol{k}_0$ is

$$H_{\boldsymbol{k}_0+\delta\boldsymbol{k}} = \frac{(\hat{\boldsymbol{p}} + \boldsymbol{k}_0 + \delta\boldsymbol{k})^2}{2m_e} + U(\boldsymbol{r}) = H_{\boldsymbol{k}_0} + \delta\boldsymbol{k} \cdot \frac{\hat{\boldsymbol{p}} + \boldsymbol{k}_0}{m_e} + \frac{|\delta\boldsymbol{k}|^2}{2m_e}.$$

The operator multiplying $\delta\boldsymbol{k}$ is $\nabla_{\boldsymbol{k}} H_{\boldsymbol{k}}$ at $\boldsymbol{k}_0$, whose matrix elements between the periodic parts are the velocity matrix elements $\boldsymbol{v}_{mn}(\boldsymbol{k}_0)$ of Eq. (3.40). Treating the two $\delta\boldsymbol{k}$-dependent terms as a perturbation of $H_{\boldsymbol{k}_0}$, whose eigenstates are the $u_{m\boldsymbol{k}_0}$, and going to second order in $|\delta\boldsymbol{k}|$ for a nondegenerate band n,

$$\varepsilon_{n,\boldsymbol{k}_0+\delta\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}_0} + \delta\boldsymbol{k} \cdot \boldsymbol{v}_{nn} + \frac{|\delta\boldsymbol{k}|^2}{2m_e} + \sum_{m\neq n} \frac{|\delta\boldsymbol{k} \cdot \boldsymbol{v}_{mn}|^2}{\varepsilon_{n\boldsymbol{k}_0} - \varepsilon_{m\boldsymbol{k}_0}} + O(|\delta\boldsymbol{k}|^3), \tag{3.49}$$

with all velocities at $\boldsymbol{k}_0$. The linear term is the group velocity Eq. (3.41), which vanishes at an extremum. Writing $|\delta\boldsymbol{k} \cdot \boldsymbol{v}_{mn}|^2 = \sum_{\alpha\beta} \delta k_\alpha\, \delta k_\beta\, v^\alpha_{nm} v^\beta_{mn}$, with $v^\alpha_{nm} = (v^\alpha_{mn})^*$, and comparing with $\varepsilon_{n\boldsymbol{k}_0} + \frac{1}{2}\sum_{\alpha\beta} \delta k_\alpha (m^{*-1})_{\alpha\beta} \delta k_\beta$:

Effective mass from $k \cdot p$ theory

At a nondegenerate extremum $\boldsymbol{k}_0$ of band n, the inverse effective-mass tensor is

$$(m^{*-1})_{\alpha\beta} \equiv \frac{\partial^2 \varepsilon_{n\boldsymbol{k}}}{\partial k_\alpha\, \partial k_\beta} = \frac{\delta_{\alpha\beta}}{m_e} + 2 \sum_{m\neq n} \frac{\mathrm{Re}\big[v^\alpha_{nm}(\boldsymbol{k}_0)\, v^\beta_{mn}(\boldsymbol{k}_0)\big]}{\varepsilon_{n\boldsymbol{k}_0} - \varepsilon_{m\boldsymbol{k}_0}}, \tag{3.50}$$

and near the extremum $\varepsilon_{n,\boldsymbol{k}_0+\delta\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}_0} + \frac{1}{2}\,\delta\boldsymbol{k} \cdot m^{*-1} \delta\boldsymbol{k}$.

The same result follows from differentiating the group velocity Eq. (3.41) once more and using Eq. (3.42), which is how $\boldsymbol{k} \cdot \boldsymbol{p}$ theory is usually presented [6, Chap. 2]. For a nonlocal potential the term $\delta_{\alpha\beta}/m_e$ becomes the expectation value of $\partial^2 H_{\boldsymbol{k}}/\partial k_\alpha \partial k_\beta$, while the sum keeps its form with the velocities of Eq. (3.40) (Appendix E.7).

Eq. (3.50) reads like level repulsion. Every other band m pushes band n away from itself as $\boldsymbol{k}$ moves, and pushes harder the closer it lies and the larger the velocity matrix element between the two. Bands below a conduction minimum push it up on both sides and make it light; bands above a valence maximum push it down and make the hole light. A small gap therefore means a light mass, provided the velocity matrix element across it is sizable.

Silicon provides a check. The sum rule is exact only for a local potential, so we test it in the empirical pseudopotential of Cohen and Bergstresser described above. The model has form factors of -0.21, $+0.04$ and $+0.08$ Ry on the shells $|\boldsymbol{G}|^2 = 3$, 8 and $11\,(2\pi/a)^2$, atoms at $\pm(a/8)(1,1,1)$, $a = 5.43\,\text{Å}$, the 113 plane waves with $|\boldsymbol{G}|^2 \le 20\,(2\pi/a)^2$, and no spin–orbit coupling [17]. Its conduction minimum, at $k_c = 0.853\,(2\pi/a)$, has the curvature masses $0.872\,m_e$ along the valley axis and $0.184\,m_e$ across it, from finite differences of the

band energies. The sum Eq. (3.50) over all 113 bands of the model reproduces both to three digits, as it must in a complete basis. The PBE bands of Fig. 3.7 give $0.95\,m_e$ and $0.193\,m_e$, and the measured values are $0.916\,m_e$ and $0.191\,m_e$ [19]. In the model, the light transverse mass comes mostly from the doubly degenerate valence level 3.73 eV below the minimum. Its two states together contribute 4.53 of the total $m_e/m_\perp^* = 5.43$, with a summed squared velocity matrix element across the valley axis of $(4.1 \times 10^{-3})^2$.

By the symmetry of the valley, the mass tensor of silicon is diagonal in axes along and across the line Γ–X on which the minimum sits, with two equal transverse components. The same holds for the L valleys of germanium, with axes along ⟨111⟩. Measured from the VBM, each valley is an ellipsoid of revolution,

$$\varepsilon_{c,\boldsymbol{k}_c+\delta\boldsymbol{k}} = E_g + \frac{(\delta k_\parallel)^2}{2m_\parallel^*} + \frac{(\delta k_\perp)^2}{2m_\perp^*}, \tag{3.51}$$

where $\delta k_\parallel$ is the component of $\delta\boldsymbol{k}$ along the valley axis and $\delta k_\perp$ the one across it. The ellipsoids of silicon are elongated by $(m_\parallel^*/m_\perp^*)^{1/2} = 2.2$ along their axis, and those of germanium, with a mass ratio of 20, are needles of aspect ratio 4.4. The mass of GaAs at Γ is isotropic.

GaAs shows the level repulsion at work. Its conduction minimum at Γ is an s-like state, 1.52 eV above the three p-like valence states, while the next conduction states at Γ lie about 3 eV higher still. The sum in Eq. (3.50) is then dominated by the valence states. By symmetry, only the one among them that transforms like x, denoted $|X\rangle$, has a nonzero matrix element of v^x with the conduction state. Its square defines the *Kane energy*

$$E_P = 2m_e\,|v_{cX}^x|^2.$$

Keeping only the valence terms of Eq. (3.50), of which this one survives,

$$\frac{m_e}{m^*} \simeq 1 + \frac{E_P}{E_g}. \tag{3.52}$$

In the empirical pseudopotential of Cohen and Bergstresser, whose velocity is $\boldsymbol{p}/m_e$ exactly, $E_P = 18.32$ eV and the gap is 1.417 eV, so that Eq. (3.52) predicts $m^* = 0.0718\,m_e$; the curvature of the computed band is $0.0721\,m_e$. The measured mass is $0.067\,m_e$ [16]; with the measured gap and the spin–orbit splitting of the valence states, to which we turn next, the same argument reproduces it. Since E_P varies little, between 16 and 23 eV for the phosphides, arsenides and antimonides of boron, aluminium, gallium and indium [20, Table VIII], Eq. (3.52) makes the mass at a direct gap roughly proportional to the gap. As a result, a calculation that underestimates the gap also underestimates the mass: the PBE bands of Fig. 3.8(b), with a gap of 0.51 eV, have a Γ-valley mass of $0.030\,m_e$ (Appendices E.1 and E.3).

The Kane energy also fixes the interband velocity that enters the optical limit of Sect. 3.2. With $E_P = 22.9$ eV [20, Table VIII], $|v_{cX}^x| = (E_P/2m_e)^{1/2} = 4.7 \times 10^{-3}$, or $1.8\,v_{\max}$. By Eq. (3.44), the momentum transfers at which dark matter excites GaAs across its gap lie beyond the reach of the optical limit, as in the one-dimensional model of Worked example 3.4.

3.3.3 Spin–orbit coupling

An electron moving through the electric field of a nucleus sees, in its rest frame, a magnetic field proportional to its orbital angular momentum $\boldsymbol{L}$, and its spin magnetic moment couples to that field. This spin–orbit coupling, which in hydrogen produces the fine structure, is the largest effect of spin on the bands near the gap, which we have treated as spinless so

far (Sect. 3.2). Including the Thomas precession, the interaction for an electron of potential energy $U_{\rm at}(r)$ in a central field is [21, Chap. 5]

$$H_{\rm so} = \frac{1}{2m_e^2}\,\frac{1}{r}\,\frac{{\rm d}U_{\rm at}}{{\rm d}r}\,\boldsymbol{L}\cdot\boldsymbol{S}. \tag{3.53}$$

In a level of orbital angular momentum l, $\boldsymbol{L}\cdot\boldsymbol{S} = \frac{1}{2}[j(j+1) - l(l+1) - \frac{3}{4}]$ equals $l/2$ for $j = l + \frac{1}{2}$ and $-(l+1)/2$ for $j = l - \frac{1}{2}$. With the radial average $\xi_{\rm SO} = \langle (2m_e^2)^{-1} r^{-1} {\rm d}U_{\rm at}/{\rm d}r\rangle$, a p level ($l = 1$) splits into a fourfold $j = \frac{3}{2}$ level at $+\xi_{\rm SO}/2$ and a twofold $j = \frac{1}{2}$ level at $-\xi_{\rm SO}$, a splitting of $\frac{3}{2}\xi_{\rm SO}$; the weighted mean, $4 \times \frac{1}{2} - 2 \times 1 = 0$, stays where it was. For the $2p$ level of hydrogen, $U_{\rm at} = -\alpha/r$ and $\langle r^{-3}\rangle = 1/24a_0^3$ with $a_0 = 1/\alpha m_e$, so that $\xi_{\rm SO} = \alpha^4 m_e/48$ and the splitting is $\alpha^4 m_e/32 = 4.5 \times 10^{-5}\,{\rm eV}$, or $0.365\,{\rm cm}^{-1}$.

In a heavier atom the valence electron spends part of its time inside the core, where it sees the nearly unscreened nuclear charge and ${\rm d}U_{\rm at}/{\rm d}r$ is large. The splitting then grows steeply with the atomic number. The cleanest measurement is in an ion with a single p electron outside closed shells, whose ground term is a ${}^2P$ doublet split by exactly $\frac{3}{2}\xi_{\rm SO}$. The $3s^2 3p$ ion Si^+ has a splitting of $287.24\,{\rm cm}^{-1}$, or 35.6 meV, 800 times that of hydrogen. For the $4s^2 4p$ ion Ge^+ it is 219 meV, six times more [22].

In the crystal the valence-band maximum inherits the atomic splitting. The three VBM states at Γ transform like x, y and z and are built from p orbitals on the atoms of the cell. Within these three states, the orbital angular momentum of each atom acts on the p orbitals as it acts on an atomic p level, turning p_x into p_y and so on. The spin–orbit operator of the crystal, $\sum_\kappa \xi_{{\rm SO},\kappa} \boldsymbol{L}_\kappa \cdot \boldsymbol{S}$ summed over the atoms, reduces to $\xi_{\rm eff}\boldsymbol{L}\cdot\boldsymbol{S}$, with $\xi_{\rm eff} = \sum_\kappa w_\kappa \xi_{{\rm SO},\kappa}$ and w_κ the p weight of the VBM states on atom κ. Cross terms between orbitals on different atoms drop out, because ${\rm d}U_{\rm at}/{\rm d}r$ is large only near each nucleus, where the orbitals of that atom alone have weight. The six states, three orbital states times two spin states, then split like an atomic p level. A fourfold $j = \frac{3}{2}$ level forms the new VBM, and a twofold $j = \frac{1}{2}$ level lies below it by the *spin–orbit splitting*

$$\Delta_{\rm SO} = \tfrac{3}{2}\,\xi_{\rm eff} = \tfrac{3}{2}\sum_\kappa w_\kappa\,\xi_{{\rm SO},\kappa}. \tag{3.54}$$

In an elemental crystal with w_κ summing to nearly one, $\Delta_{\rm SO}$ is close to the atomic splitting. The measured values bear this out to about 30 %: 0.044 eV in silicon against the ionic 0.036 eV, and 0.296 eV in germanium against 0.219 eV [12, Table II], with a growth from silicon to germanium by a factor of 6.7 in the crystal and 6.2 in the ions. A calculation of the crystal that includes spin–orbit coupling closes the remaining gap; the quasiparticle calculation of Ref. [12, Table II] finds 0.050 and 0.312 eV. In GaAs the valence states sit mostly on the arsenic atom, the anion, whose hybrid is the lower one in the ionic bond of Sect. 3.2. Its splitting of 0.341 eV [16] lies close to that of the $4s^2 4p$ ion As^{2+}, 0.365 eV, and far from that of the gallium atom, 0.102 eV [22].

Spin–orbit coupling lowers the gap. A calculation without it places the VBM at the mean of the $j = \frac{3}{2}$ and $j = \frac{1}{2}$ levels, while the true VBM is the $j = \frac{3}{2}$ level, $\Delta_{\rm SO}/3$ higher. The correction is 15 meV in silicon, about 1 % of the gap, which is why silicon is often computed without spin–orbit coupling. In germanium it is 0.10 eV, 13 % of the gap, and cannot be omitted (Appendix E.3).

The splitting also refines the Kane estimate Eq. (3.52). The conduction state of GaAs is coupled by v^x to the orbital state $|X\rangle$ with either spin, and the spin–orbit levels share that state between them. For $l = 1$ the projector onto $j = \frac{3}{2}$ is $\frac{2}{3}(1 + \boldsymbol{L}\cdot\boldsymbol{S})$, since $\boldsymbol{L}\cdot\boldsymbol{S}$ is $\frac{1}{2}$ there and -1 on $j = \frac{1}{2}$. A real orbital such as p_x has $\langle \boldsymbol{L}\rangle = 0$, so $|X\uparrow\rangle$ has weight $\frac{2}{3}$ in the $j = \frac{3}{2}$ level and $\frac{1}{3}$ in the $j = \frac{1}{2}$ level. The valence terms of Eq. (3.50) then carry the energy

denominators E_g and $E_g + \Delta_{SO}$ with these weights,

$$\frac{m_e}{m^*} \simeq 1 + \frac{E_P}{3}\left(\frac{2}{E_g} + \frac{1}{E_g + \Delta_{SO}}\right), \tag{3.55}$$

the result of Kane [23]. With $E_g = 1.52\,\text{eV}$, $\Delta_{SO} = 0.341\,\text{eV}$ and $E_P = 22.9\,\text{eV}$, it yields $m^* = 0.066\,m_e$, against the measured $0.067\,m_e$. The same formula with the gap of a semilocal functional, about 0.5 eV, yields $0.025\,m_e$, and with the 1.77 eV of a calculation that overestimates the gap it yields $0.076\,m_e$, close to the $0.077\,m_e$ such a calculation finds (Appendix E.3).

At the VBM itself Eq. (3.50) does not apply, because the states are degenerate and the perturbation mixes them among themselves. Second-order perturbation theory within the fourfold $j = \frac{3}{2}$ level yields a 4×4 matrix in $\delta\boldsymbol{k}$, and cubic symmetry leaves three independent constants in it, the Luttinger parameters γ_1^{Lu}, γ_2^{Lu} and γ_3^{Lu}. Its eigenvalues, measured downward from the VBM, are [24][6, Chap. 2]

$$\begin{aligned}\varepsilon_{hh,lh}(\delta\boldsymbol{k}) = -\frac{1}{2m_e}\Big[\gamma_1^{Lu}|\delta\boldsymbol{k}|^2 \mp 2\Big\{(\gamma_2^{Lu})^2|\delta\boldsymbol{k}|^4 + 3\big((\gamma_3^{Lu})^2 - (\gamma_2^{Lu})^2\big) \\ \times\big((\delta k_x)^2(\delta k_y)^2 + (\delta k_y)^2(\delta k_z)^2 + (\delta k_z)^2(\delta k_x)^2\big)\Big\}^{1/2}\Big],\end{aligned} \tag{3.56}$$

two doubly degenerate branches that touch at Γ: the *heavy holes* (upper sign) and the *light holes*. Along [100] the cross terms vanish and the masses are $m_e/(\gamma_1^{Lu} \mp 2\gamma_2^{Lu})$; along [111] the square root is $\gamma_3^{Lu}|\delta\boldsymbol{k}|^2$ and they are $m_e/(\gamma_1^{Lu} \mp 2\gamma_3^{Lu})$. The $j = \frac{1}{2}$ level forms the *split-off* band, Δ_{SO} lower. The constant-energy surfaces of the heavy holes are strongly warped spheres: in silicon the heavy-hole mass is $0.28\,m_e$ along [100] and $0.72\,m_e$ along [111] (Table 3.1). An electron removed near the VBM leaves behind a hole with this direction-dependent mass, and the hole's share of the momentum of a pair matters in the next subsection.

3.3.4 The minimum pair energy

A momentum transfer $\boldsymbol{q}$ moves an electron from $v\boldsymbol{k}$ to $c, \boldsymbol{k}+\boldsymbol{q}$ (Eq. (3.34)), and the energy it deposits is $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}}$. The smallest such energy at fixed $\boldsymbol{q}$ is the *minimum pair energy*,

Minimum pair energy

$$E_{min}(\boldsymbol{q}) = \min_{v,c,\boldsymbol{k}} \left(\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}}\right). \tag{3.57}$$

No electron–hole pair with momentum transfer $\boldsymbol{q}$ costs less. E_{min} is periodic, $E_{min}(\boldsymbol{q} + \boldsymbol{G}) = E_{min}(\boldsymbol{q})$; it equals E_{dir} at $\boldsymbol{q} = 0$, and its smallest value over all $\boldsymbol{q}$ is E_g.

The periodicity follows from that of the bands, $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{G}} = \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}}$, and the value at $\boldsymbol{q} = 0$ is the definition Eq. (3.46). The minimum over $\boldsymbol{q}$ is reached when $\boldsymbol{k}$ sits at the VBM and $\boldsymbol{k} + \boldsymbol{q}$ at the CBM: in silicon at $\boldsymbol{q} = \boldsymbol{k}_c$, at its five images and at all vectors that differ from these by a $\boldsymbol{G}$. E_{min} depends only on the reduced momentum $\boldsymbol{q}_r$ of Eq. (3.16), although the strength of the transition, $|\rho_{cv}(\boldsymbol{k}, \boldsymbol{q})|^2$, does not (Sect. 3.2).

Near a valley, the shape of E_{min} follows from the masses. Take $\boldsymbol{q}$ along the axis of a silicon valley, at a distance $\Delta q = q - k_c$ from its minimum, and a hole of mass m_h in the same direction. The pair energy $E_g + (k+\Delta q)^2/2m_\parallel^* + k^2/2m_h$ is smallest at $k = -m_h\Delta q/(m_\parallel^* + m_h)$, where

$$E_{min}(\boldsymbol{q}) \simeq E_g + \frac{(q - k_c)^2}{2(m_\parallel^* + m_h)}. \tag{3.58}$$

The electron and the hole share the momentum that the valley does not absorb, in proportion to their masses, and the pair recoils like a single particle of mass $m_{\parallel}^{*} + m_{\mathrm{h}}$. In silicon, with $m_{\mathrm{h}} = 0.28\, m_e$ for heavy holes along [100], this mass is $1.2\, m_e$, and E_{min} rises by 0.1 eV within 0.35 keV of the minimum. Far from the valley the parabola fails, and Fig. 3.7(b) shows E_{min} computed from the PBE bands, with the gap set to the measured one, along three directions. Along [100] it falls from the vertical gap at $\boldsymbol{q} = 0$ to E_{g} at $|\boldsymbol{q}| = k_{\mathrm{c}}$, stays near E_{g} up to $|\boldsymbol{G}_{200}| - k_{\mathrm{c}} = 2.63$ keV, where an Umklapp process reaches the opposite valley at $-\boldsymbol{k}_{\mathrm{c}}$, and returns to the vertical gap at $|\boldsymbol{G}_{200}| = 4.57$ keV. Along [110] and [111] it stays higher, since those directions pass the valleys at a distance: its lowest values are 1.26 eV near 3.1 keV along [110] and 2.13 eV along [111].

Elastic dark matter of mass m_χ deposits at most $q v_{\mathrm{max}} - q^2/2m_\chi$ at momentum transfer q (Eq. (1.26)), and the comparison with the target is now direct. An electron–hole pair at $\boldsymbol{q}$ is therefore kinematically allowed only if

$$E_{\mathrm{min}}(\boldsymbol{q}) \leq q v_{\mathrm{max}} - \frac{q^2}{2m_\chi} \tag{3.59}$$

for some $\boldsymbol{q}$. This is the condition of Sect. 1.3 with the target's lowest excitation energy at each q in place of a single threshold. The threshold mass $m_{\mathrm{min}} = 2E_{\mathrm{g}}/v_{\mathrm{max}}^2$ of Eq. (1.25) assumes that the target can absorb E_{g} at the momentum transfer $m_\chi v_{\mathrm{max}}$ where the reach peaks. In silicon it cannot. At that momentum transfer E_{min} is well above E_{g}, and the true threshold of the indirect gap lies above m_{min}. Worked example 3.5 puts numbers on this.

A photon is the opposite extreme. Its momentum equals its energy, $q = \omega$, which is 6×10^{-4} of k_{c} for $\omega = E_{\mathrm{g}}$, and on the scale of the zone every photon absorption is vertical. A photon needs at least $E_{\mathrm{min}}(\boldsymbol{q} \approx 0) = E_{\mathrm{dir}}$ to create a pair on its own. Between E_{g} and E_{dir} silicon absorbs light only with the help of a phonon, which supplies the crystal momentum $\boldsymbol{k}_{\mathrm{c}}$ at a cost or gain of tens of meV (Sect. 2.4); Sect. 6.5 derives the rate of this *phonon-assisted* absorption. Absorbed bosonic dark matter, with momentum $m_\phi v \approx 10^{-3} m_\phi$, is in the same position. Scattering dark matter lies in between. Above a few hundred keV in mass it reaches k_{c} with energy to spare, and it excites the indirect gap of silicon directly, at first order. Just above m_{min}, however, it delivers enough energy only at momentum transfers well below k_{c}, and there a phonon can again supply the missing crystal momentum. Between the threshold mass of the phonon-assisted process and that of the zero-phonon one lies a window of masses in which silicon detects dark matter only through phonon-assisted scattering. Its kinematics and rates are the subject of Sects. 13.1 and 13.3, and the window of silicon is evaluated in Sect. 13.9.

Worked example 3.5: The momentum to reach the silicon conduction-band minimum

With $2\pi/a = 2283$ eV (Worked example 3.1), the CBM of silicon lies at $k_{\mathrm{c}} = 0.85 \times 2283\,\mathrm{eV} = 1.94$ keV, or $0.98\,\text{Å}^{-1}$, along each of the six directions $\langle 100 \rangle$. A transition from the VBM to the CBM needs a momentum transfer of this size, up to a reciprocal-lattice vector.

Dark matter of 1 MeV deposits $\omega = E_{\mathrm{g}} = 1.17$ eV for momentum transfers between the two roots of $q v_{\mathrm{max}} - q^2/2m_\chi = E_{\mathrm{g}}$,

$$q = m_\chi v_{\mathrm{max}} \pm \sqrt{(m_\chi v_{\mathrm{max}})^2 - 2 m_\chi E_{\mathrm{g}}} = 2650\,\mathrm{eV} \pm 2164\,\mathrm{eV},$$

that is, from 0.49 keV to 4.81 keV, as in Sect. 1.3. The momentum k_{c} lies well inside this

window. At $q = k_c$ a particle of 1 MeV can deposit up to

$$k_c v_{max} - \frac{k_c^2}{2m_\chi} = 1940\,\text{eV} \times 2.650 \times 10^{-3} - \frac{(1940\,\text{eV})^2}{2 \times 1\,\text{MeV}} = 5.14\,\text{eV} - 1.88\,\text{eV} = 3.26\,\text{eV},$$

almost three times the gap. Its largest momentum transfer, $2m_\chi v_{max} = 5.30\,\text{keV}$, also covers the images $\boldsymbol{k}_c + \boldsymbol{G}$ nearest to the origin. For such a particle, the indirect gap of silicon is open at first order.

Lighter particles have less room. The lightest mass that deposits E_g at exactly $q = k_c$ solves $k_c v_{max} - k_c^2/2m_\chi = E_g$:

$$m_\chi = \frac{k_c^2}{2(k_c v_{max} - E_g)} = \frac{(1940\,\text{eV})^2}{2\,(5.14\,\text{eV} - 1.17\,\text{eV})} = 0.474\,\text{MeV}.$$

The kinematic bound of Worked example 1.3 at the low-temperature gap is $m_{min} = 0.333\,\text{MeV}$, attained at $q = m_{min} v_{max} = 0.88\,\text{keV} = 0.39\,(2\pi/a)$. That momentum transfer reaches less than halfway to the valley, and there E_{min} is about 1.9 eV in Fig. 3.7(b), not 1.17 eV.

A particle of 0.333 MeV cannot create a pair at all without a phonon, whatever the details of the bands: it deposits 1.17 eV only at $q = 0.88\,\text{keV}$, and that momentum transfer does not connect the VBM to any valley, so that $E_{min} > E_g$ there. Slightly below 0.474 MeV, particles can still reach the valley at a momentum transfer a little smaller than k_c, where E_{min} is a little higher, so the threshold mass of the indirect gap lies between 0.333 and 0.474 MeV; Sect. 13.1 locates it.

Worked example 3.6: Threshold masses for the direct gaps of silicon and germanium

Worked example 1.3 evaluated $m_{min} = 2E_{th}/v_{max}^2$ for the indirect gap of silicon. With $v_{max}^2 = 7.023 \times 10^{-6}$, the other gaps of Table 3.1 lead to

$$\begin{aligned}
&\text{Si, direct:} && m_{min} = \frac{2 \times 3.4\,\text{eV}}{7.023 \times 10^{-6}} = 0.968\,\text{MeV}, && q = m_{min} v_{max} = 2.57\,\text{keV},\\
&\text{Ge, direct:} && m_{min} = \frac{2 \times 0.89\,\text{eV}}{7.023 \times 10^{-6}} = 0.253\,\text{MeV}, && q = 0.67\,\text{keV},\\
&\text{Ge, indirect:} && m_{min} = \frac{2 \times 0.744\,\text{eV}}{7.023 \times 10^{-6}} = 0.212\,\text{MeV}, && q = 0.56\,\text{keV}.
\end{aligned}$$

The direct gap of silicon would triple the threshold mass, but nothing confines dark-matter scattering to vertical transitions: at $q = 2.57\,\text{keV}$, a particle of 0.968 MeV reaches pairs well below 3.4 eV in every direction of Fig. 3.7(b). The direct gap matters for photons and for absorbed dark matter, not for the threshold of scattering. In germanium the two gaps differ by 0.15 eV, and the bound of the indirect gap, 0.212 MeV, is 36 % below that of silicon. As in silicon, however, the bound assumes that the target absorbs the gap at $q = 0.56\,\text{keV}$, while the L valleys sit at $|\Gamma\text{L}| = \frac{\sqrt{3}}{2}\,(2\pi/a) = 1.90\,\text{keV}$, almost exactly the k_c of silicon. Depositing 0.744 eV at $q = 1.90\,\text{keV}$ requires $m_\chi \geq (1898\,\text{eV})^2/[2\,(5.03\,\text{eV} - 0.744\,\text{eV})] = 0.420\,\text{MeV}$, the counterpart of the 0.474 MeV of silicon. As there, lighter particles reach the valleys at a smaller momentum transfer, where E_{min} is a little higher, so the threshold mass of the indirect gap lies between 0.212 and 0.420 MeV; Sect. 13.1 locates it by minimizing over the momentum transfer and the crystal momentum of the hole. The direct gap does

not lower it. The bound of 0.253 MeV needs a pair at Γ that absorbs $q = 0.67$ keV without kinetic energy. At that mass the particle can deposit only the 0.89 eV of the gap itself, while carrying that momentum costs the pair, a light electron with a heavy hole and about half an electron mass in total, a further 0.8 eV. The route through Γ therefore opens only far above the L valleys, at a few MeV for parabolic bands.

The band edges, masses and splittings of Table 3.1 are all a threshold calculation needs from the band structure, and every one of them is measured. A rate needs more: the full bands and the wavefunctions behind $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q})$ at every $\boldsymbol{k}$ and up to large $\boldsymbol{q}$, which no table provides. Computing them requires a potential $U(\boldsymbol{r})$ that is derived rather than fitted, and the next section obtains it from the many-electron problem.

3.4 Many electrons and density-functional theory

A density-functional calculation of silicon, the source of the bands behind computed dark-matter rates, puts the gap at about 0.5 eV, less than half the measured 1.17 eV. Density-functional theory computes the potential $U(\boldsymbol{r})$ of Sect. 3.2 from the electrons themselves instead of fitting it, and the gap is its best-known failure. Since the threshold mass of Eq. (1.25) is proportional to the gap, the error enters the reach directly: the computed gap would put the reach of a silicon detector at 0.15 MeV instead of 0.33 MeV. The many-electron problem of Eq. (2.48) can be reduced to one electron in an effective potential, first by the best single Slater determinant and then by the Kohn–Sham construction. The eigenvalues that come out are not the energies to add or remove an electron, and the standard repair, a rigid shift of the conduction bands, changes the energies and some of the matrix elements of a rate.

3.4.1 Hartree–Fock theory

The simplest many-electron state with the right antisymmetry is a single Slater determinant $|\Phi_0\rangle$, Eq. (2.29), built from N_e orthonormal orbitals $\psi_\gamma(x)$, with $x = (\boldsymbol{r}, s)$ the position and spin. Its Coulomb energy is a direct (Hartree) term minus an exchange (Fock) term (Sect. 2.3). By the variational principle the energy of any state is at least the exact ground-state energy E_0, so the best determinant is the one of lowest energy. The orbitals are varied under the constraint that they stay orthonormal. The minimum condition is the set of *Hartree–Fock equations* (Appendix D.4):

$$\Big[-\frac{\nabla^2}{2m_e} + U_1(\boldsymbol{r}) + v_{\mathrm{H}}(\boldsymbol{r})\Big]\psi_\gamma(x) + \int \mathrm{d}x'\, \Sigma_{\mathrm{x}}(x, x')\, \psi_\gamma(x') = E_\gamma\, \psi_\gamma(x). \tag{3.60}$$

Here U_1 is the attraction of the clamped nuclei, $v_{\mathrm{H}}(\boldsymbol{r}) = \int \mathrm{d}^3r'\, e^2 \boldsymbol{n}(\boldsymbol{r}')/4\pi|\boldsymbol{r} - \boldsymbol{r}'|$ is the electrostatic potential of the electron density $\boldsymbol{n}(\boldsymbol{r})$, and the exchange term

$$\Sigma_{\mathrm{x}}(x, x') = -\frac{e^2}{4\pi|\boldsymbol{r} - \boldsymbol{r}'|} \sum_{\gamma' \,\mathrm{occ}} \psi_{\gamma'}(x)\psi^*_{\gamma'}(x'),$$

couples each electron only to occupied orbitals of the same spin. The equations are nonlinear, since the potentials are built from the occupied solutions, and they are iterated until the orbitals that go in are the ones that come out. This is the *self-consistent field*. The orbitals of the atoms behind the ionization form factors of Sect. 2.2 are of this kind.

Each term has a plain meaning. The Hartree potential treats the other electrons as a smooth charge cloud. On its own it would also let each electron repel its own charge. The

exchange term removes that self-repulsion exactly. Around every electron it digs a hole in the density of the electrons of the same spin, the exchange hole, which integrates to one missing electron. The exchange term is, however, an integral operator, not a function of $\boldsymbol{r}$. It acts on ψ_γ through its values everywhere, and it is the first example in the book of a nonlocal one-electron operator, with the consequences for the velocity and the f-sum rule that Sect. 4.3 works out.

The eigenvalues E_γ have a precise meaning, *Koopmans' theorem* [25]. If the other orbitals are held fixed, $-E_\gamma$ is the energy needed to remove an electron from the occupied orbital γ, and the eigenvalue E_γ of an empty orbital is the change in energy when an electron is added to it (Eq. (D.16)). In a crystal the frozen-orbital hypothesis costs nothing. A Bloch state is spread over all N cells, and removing it changes the density in each cell by a relative amount of order $1/N$. Since the energy is stationary in the orbitals, letting them relax lowers it by a term of second order, which vanishes as $N \to \infty$ (Appendix D.4). Hartree–Fock bands are therefore the removal and addition energies of the Hartree–Fock state, and their gap is its fundamental gap.

The trouble is that a single determinant has no correlation, and the electron gas shows how much that matters. There the Hartree–Fock bands can be computed in closed form, Eq. (D.17). At the valence density of silicon the exchange energy is $-6.22\,\mathrm{eV}$ per electron, comparable to the kinetic energy of $7.5\,\mathrm{eV}$ (Eq. (D.18)). Exchange widens the occupied band from the free $12.5\,\mathrm{eV}$ to $20.8\,\mathrm{eV}$, and it makes the group velocity diverge logarithmically at the Fermi surface. Neither is seen in real metals. Both come from the long range of the bare exchange, $v_\mathrm{C}(k) \propto 1/k^2$, which in a real solid the other electrons screen. In a semiconductor the same unscreened exchange makes the Hartree–Fock gap far too large. An added electron is not allowed to push the others away, and a removed one does not let them close in, so both the addition and the removal energies come out too high in magnitude. We return to this picture when we compare gaps below.

3.4.2 Density-functional theory

The electron density of silicon is a function of three variables, while its ground-state wavefunction depends on the $3N_e$ coordinates of all the electrons. Hohenberg and Kohn showed that the density is enough [26]. Their first theorem states that the ground-state density $n(\boldsymbol{r})$ determines the external potential U_1 up to a constant. With it, n fixes the Hamiltonian and every property of the system, excited states included. Their second theorem turns this into a variational principle. In the constrained-search form of Levy [27], define the universal functional

$$\mathcal{F}[n] = \min_{\Psi \to n} \langle \Psi | H_\mathrm{kin} + H_{ee} | \Psi \rangle,$$

the lowest kinetic plus Coulomb energy of any state of N_e electrons with density n. It does not depend on the nuclei and is the same for silicon, germanium and a single atom. The ground-state energy in the potential U_1 is then the minimum over densities,

$$E_0 = \min_n \Big(\mathcal{F}[n] + \int \mathrm{d}^3 r \, U_1(\boldsymbol{r})\, n(\boldsymbol{r}) \Big), \tag{3.61}$$

reached at the ground-state density. Appendix D.5 proves both statements. The proof of the first is a two-line contradiction between the variational principles of two Hamiltonians that would share a density; the second follows from doing the minimization over states in two steps, first at fixed density and then over densities. Neither says how to find $\mathcal{F}$.

Kohn and Sham made Eq. (3.61) usable by computing the largest and hardest piece of $\mathcal{F}$ exactly [28]. That piece is the kinetic energy. For independent electrons with the density n it is $K_\mathrm{s}[n]$, the lowest kinetic energy of a single determinant with that density (written T_s in

the literature). Together with the Hartree energy $E_{\mathrm{H}}[n]$, the classical electrostatic energy of the density, it leaves a remainder,

$$\mathcal{F}[n] = K_{\mathrm{s}}[n] + E_{\mathrm{H}}[n] + E_{\mathrm{xc}}[n],$$

which defines the *exchange–correlation energy* E_{xc}. It contains the exchange energy, which as in Hartree–Fock removes the self-repulsion in E_{H}, the correlation energy that a determinant misses, and the part of the true kinetic energy that independent electrons do not have. It is the one piece that must be approximated. Minimizing over the orbitals of the determinant, as in Appendix D.4, leads to one-electron equations.

Kohn–Sham equations

The ground-state density is $n(\boldsymbol{r}) = \sum_s \sum_{\gamma\,\mathrm{occ}} |\psi_\gamma(\boldsymbol{r}, s)|^2$, summed over the N_e lowest solutions of

$$\begin{gathered} \Big[-\frac{\nabla^2}{2m_e} + V_{\mathrm{KS}}(\boldsymbol{r})\Big]\psi_\gamma(x) = E_\gamma\,\psi_\gamma(x), \\ V_{\mathrm{KS}}(\boldsymbol{r}) = U_1(\boldsymbol{r}) + v_{\mathrm{H}}(\boldsymbol{r}) + v_{\mathrm{xc}}(\boldsymbol{r}), \qquad v_{\mathrm{xc}}(\boldsymbol{r}) = \frac{\delta E_{\mathrm{xc}}}{\delta n(\boldsymbol{r})}. \end{gathered} \tag{3.62}$$

With the exact E_{xc}, the density and the ground-state energy are exact. In a crystal $\gamma \to n\boldsymbol{k}$ and $E_\gamma \to \varepsilon_{n\boldsymbol{k}}$.

Eq. (3.62) has the form of the Hartree–Fock equations Eq. (3.60), with one difference: the Kohn–Sham potential is a local function of $\boldsymbol{r}$, where the Fock term was an integral operator. It is the potential $U(\boldsymbol{r})$ of Eq. (3.19). It is lattice-periodic, because the density is, and every result of Sects. 3.2 and 3.3 applies to its Bloch states. Unlike the fitted potential of Sect. 3.3, it is computed. The density that enters it is that of the bonds, not a sum of spherical atoms. Through v_{H} and v_{xc}, the charge piled up between neighboring atoms gives V_{KS} a (222) component, which the basis structure factor removes from the nuclear part; the (200) component vanishes for any potential with the symmetry of the crystal (Worked example 3.2, Appendix C.17).

The ground-state energy, unlike the eigenvalues, has a physical meaning, and it is Eq. (D.22). Minimizing it with respect to the lattice constant, or the positions of the atoms, yields the equilibrium structure, and its second derivatives yield the force constants of the phonons (Sect. 5.2). In the local-density approximation, introduced next, the computed lattice constant of silicon is 5.401 Å, 0.5 % below the measured one (Sect. 3.1).

Exchange–correlation functionals. The *local-density approximation* (LDA) assumes that each small volume of the solid contributes the exchange–correlation energy of a uniform electron gas of the local density,

$$E_{\mathrm{xc}}^{\mathrm{LDA}}[n] = \int \mathrm{d}^3r\, n(\boldsymbol{r})\, \varepsilon_{\mathrm{xc}}(n(\boldsymbol{r})), \tag{3.63}$$

with $\varepsilon_{\mathrm{xc}}(n)$ the energy per electron of the gas [28]. Its exchange part is the Hartree–Fock result Eq. (D.18), $\varepsilon_{\mathrm{x}} = -3e^2k_{\mathrm{F}}/16\pi^2$ with the local $k_{\mathrm{F}} = (3\pi^2 n)^{1/3}$, and the exchange potential that follows is $v_{\mathrm{x}}^{\mathrm{LDA}} = \frac{4}{3}\varepsilon_{\mathrm{x}}$, the Hartree–Fock exchange of an electron at the Fermi surface of the gas: $-8.29\,\mathrm{eV}$ at the valence density of silicon (Appendix D.5). The correlation part comes from quantum Monte Carlo energies of the gas [29], fitted as a function of the density [30]. The approximation is exact for a uniform density and good when the density changes little over the local $1/k_{\mathrm{F}}$. Valence densities do not satisfy this, yet LDA works far better than the

condition suggests, because the exchange–correlation hole it implies obeys the same sum rule as the exact one. The *generalized-gradient approximations* (GGA) add a dependence on $|\nabla \boldsymbol{n}|$; PBE [31] is the one used most in solids. *Hybrid functionals* replace part of the local exchange by the Fock term of Eq. (3.60) built from the Kohn–Sham orbitals, which makes the potential nonlocal again; the hybrid functional of Heyd, Scuseria and Ernzerhof (HSE) [32] takes one quarter of the exchange from Fock, and only at short range. Each step from LDA to GGA to hybrids costs more, the hybrid step by one to three orders of magnitude (Appendix E.1). Semilocal functionals give lattice constants within 1 % to 2 %, LDA slightly too small and PBE slightly too large (Table E.1), and they give gaps that are much too small.

In practice, a Kohn–Sham calculation of silicon expands the periodic parts $u_{n\boldsymbol{k}}$ in plane waves, replaces the nuclei and core electrons by pseudopotentials (Sect. 3.5), solves Eq. (3.62) on a grid of crystal momenta, and iterates the density until it no longer changes (Appendix E.1). The calculation of Worked example E.1, with the PBE functional and a norm-conserving silicon pseudopotential of the SG15 library (Sect. 3.5) [15], takes seconds. At the measured lattice constant it places the conduction-band minimum at $k_\mathrm{c} = 0.843\,(2\pi/a)$ against the measured 0.85, with a longitudinal valley mass $0.95\,m_e$ against $0.92\,m_e$ and a gap of 0.557 eV, half the measured value (Fig. 3.7). Minimized in its total energy, the lattice constant comes out at 5.477 Å, 0.85 % above the measured one, where the gap is 0.602 eV. The electronic dielectric constant it computes is 13.04 against the measured 11.7 (Sect. 3.3). The positions and curvatures of the bands are trustworthy; the gap is not, and the screening inherits part of its error.

3.4.3 The band-gap problem

A dark-matter threshold is an energy difference between many-electron states. The gap that a detector sees is the energy to take an electron out of the crystal at the top of the valence band and put it back at the bottom of the conduction band, far enough away that the two do not interact. With E_{N_e} the ground-state energy of N_e electrons, it is the *fundamental gap*

$$E_\mathrm{g} = (E_{N_e+1} - E_{N_e}) - (E_{N_e} - E_{N_e-1}) = I - A, \tag{3.64}$$

the ionization energy I minus the electron affinity A. More generally, the energy to add an electron in an empty state $n\boldsymbol{k}$, or minus the energy to remove one from an occupied state, is the *quasiparticle energy* $E_\mathrm{qp}(n\boldsymbol{k})$, which Sect. 4.7 defines precisely as a pole of the one-electron Green's function. The bands that photoemission and inverse photoemission measure are quasiparticle bands, and the threshold of Sect. 3.3 is the difference of two of them.

The Kohn–Sham eigenvalues $\varepsilon_{n\boldsymbol{k}}$ are not quasiparticle energies. They are the Lagrange multipliers of a construction designed to reproduce the density, and only one of them has an exact meaning. By Janak's theorem [33], $\varepsilon_{n\boldsymbol{k}}$ is the derivative of the total energy with respect to the occupation of the state, and since the exact energy is a straight line between integer electron numbers [34], the highest occupied eigenvalue of the exact functional equals $-I$ (Appendix D.5). The lowest empty eigenvalue would equal $-A$ only if the potential were the same on both sides of N_e. It is not. Since the exact energy is a straight line between integers, its slope, the chemical potential, jumps at N_e from $-I$ to $-A$, by the fundamental gap. The slope of the Kohn–Sham kinetic energy K_s jumps only by the Kohn–Sham gap, because an added electron enters the lowest empty orbital. The Hartree energy is a smooth function of the density, so the rest of the jump must sit in E_xc. When the electron number crosses N_e, the exact exchange–correlation potential jumps by a constant Δ_xc, the *derivative discontinuity*, and even with the exact functional

$$E_\mathrm{g} = E_\mathrm{g}^\mathrm{KS} + \Delta_\mathrm{xc}, \tag{3.65}$$

with E_g^{KS} the gap between the Kohn–Sham eigenvalues [35, 36]. LDA and GGA are smooth functions of the density and have no discontinuity at all. For silicon, most of the difference between the LDA gap and the measured one is the missing Δ_{xc}, not an error of the LDA potential [37]. This is the *band-gap problem*: the Kohn–Sham gap is too small for a reason that a better semilocal functional cannot remove.

LDA puts the silicon gap at 0.52 eV, PBE at 0.56 to 0.60 eV, against the measured 1.17 eV (Table E.3 and Worked example E.1). The error is similar in other semiconductors. GaAs comes out at 0.43 eV in LDA against 1.52 eV, and germanium, with a measured gap of 0.744 eV, loses it: in LDA its gap closes (Table E.1), and in PBE with spin–orbit coupling its s-like conduction state at Γ falls below the valence-band maximum, which makes it a semimetal (Fig. 3.8). Fig. 3.9(a) shows LDA bands of silicon. Their minimum gap, 0.40 eV (lowered by about 0.1 eV by the PBE-generated pseudopotential, see the caption), sits at $0.848\,(2\pi/a)$ along Γ–X, at the measured position, while quasiparticle energies and measured ones lie higher across the zone. In the figure, the G_0W_0 squares, from a separate calculation, sit 0.66 to 0.70 eV above the solid LDA bands at the conduction states, and the dashed bands, shifted by one constant, pass close to all of them. The correction is not the same for every state, but within about 0.1 eV the LDA conduction bands of silicon are right in shape and wrong in position.

Why are quasiparticle gaps larger than Kohn–Sham gaps, and smaller than Hartree–Fock ones? A physical picture follows from what each theory lets the other electrons do, as anticipated above. In Hartree–Fock, an electron in an occupied state is bound by the exchange with the other electrons of its spin, while an electron in an empty state has no exchange with itself; this difference, of order the exchange energy of several eV, opens the gap. But the other electrons are frozen. A real added electron repels the electrons around it and polarizes the crystal, and the polarization lowers its energy; a real hole attracts them, with the same effect on the energy to create it. Both corrections close the gap. They come from screening, which at long range cuts the exchange interaction down by $1/\varepsilon_\infty$, a factor of 12 in silicon. The result is screened exchange: the exchange of Hartree–Fock with the bare Coulomb interaction replaced, at long range, by the interaction screened by ε_∞, together with the energy of the polarization cloud that the particle carries along, the Coulomb-hole term of Appendix E.3. A local Kohn–Sham potential is at the other extreme. It acts in the same way on occupied and empty states, and does not contain the discontinuity in the cost of adding an electron. The quasiparticle gap lies between the two limits, and in a strongly screening semiconductor it lies closer to the Kohn–Sham one. The *GW* approximation of Sect. 4.8 makes this picture quantitative: its self-energy is the exchange with the dynamically screened interaction W^{scr}. Started from LDA, it gives the silicon gap at 1.24 to 1.29 eV in the calculations of the 1980s and early 1990s [39, 40][38, Table I], and at 1.10 eV when converged with current methods [42, Sect. IV C]. Hybrid functionals use the same idea crudely, with a fixed fraction of the Fock exchange; HSE gives 1.22 eV (Fig. 3.9(b)).

The gap error reaches every quantity of a dark-matter rate that depends on energy differences across the gap. The threshold mass $m_{min} = 2E_g/v_{max}^2$ is proportional to the gap, and the minimum pair energy $E_{min}(\boldsymbol{q})$ of Eq. (3.57) moves with the conduction bands. Effective masses at a direct gap scale with it through Eq. (3.52), as the Kane estimate of Sect. 3.3 shows for GaAs. The silicon valleys are less sensitive, since their mass comes from distant bands, and the PBE longitudinal mass is only 4 % too heavy (Appendix E.1). The screening is affected as well. Every transition contributes to $\varepsilon_\infty - 1$ more strongly the smaller its energy, and a gap that is too small makes the screening too strong, 13.04 against 11.7 in the PBE calculation of Appendix E.1. The in-medium screening of Chap. 4 suppresses a rate by $|\varepsilon|^{-2}$, and it then suppresses it too much. A computed rate is therefore only as good as the gap behind it. Where the gap is measured, it can be imposed; where it is not, the spread

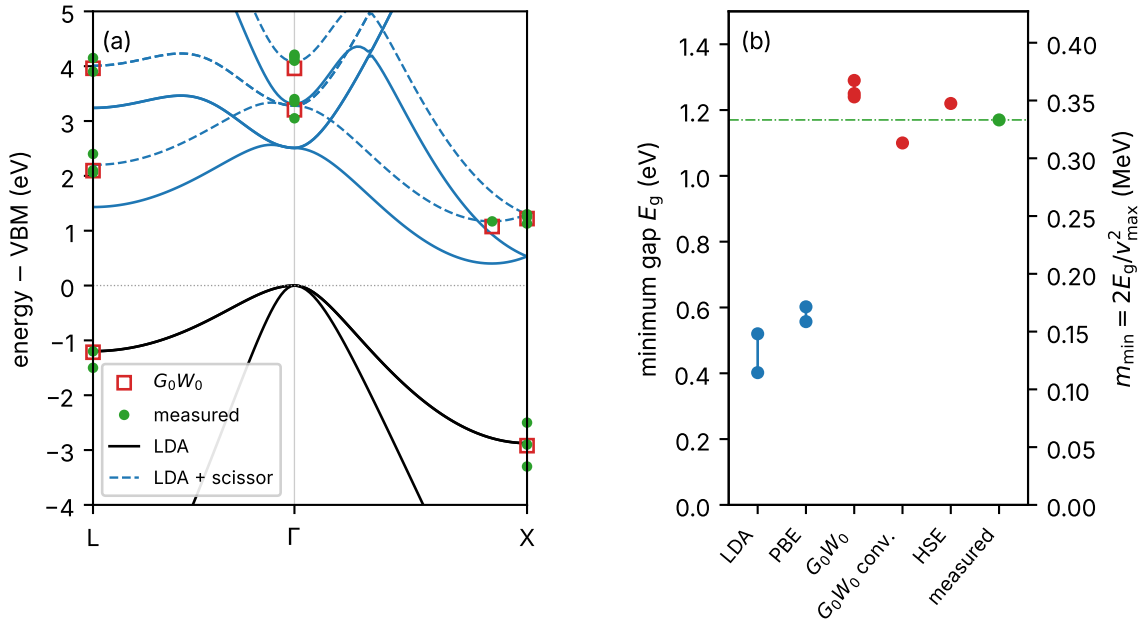


Figure 3.9: The gap of silicon in density-functional theory, in quasiparticle calculations and in experiment. (a) Solid: Kohn–Sham bands in the local-density approximation along L–Γ–X, valence black and conduction blue, from the VBM; Slater exchange and Perdew–Zunger correlation [30], in the plane-wave calculation of Appendix E.1 (SG15 norm-conserving pseudopotential, 60 Ry, 12^3 grid, $a = 5.431$ Å, no spin–orbit coupling) [13–15]. Their minimum gap is 0.40 eV at $0.848\,(2\pi/a)$; it changes by less than 1 meV at 80 Ry or on a 16^3 grid, and the conduction states at Γ, X and L lie at 2.51, 0.53 and 1.43 eV. The pseudopotential was generated with PBE; a consistent LDA potential would raise the gap to about 0.50 eV, close to the tabulated LDA value of 0.52 eV (Table E.3). Dashed: the same conduction bands shifted up by $\Delta_{scis} = 0.77$ eV, so that the minimum gap equals the measured 1.17 eV. Squares: one-shot G_0W_0 quasiparticle energies, and circles: measured energies (all values listed there), both as given in Ref. [12, Tables II and VI], at Γ, L, X and the conduction-band minimum. The G_0W_0 calculation is a different one from the LDA bands shown and includes spin–orbit coupling, whose splittings, below 0.05 eV, are not resolved. (b) Minimum gap from LDA [38, Table I] and from the LDA bands of panel (a), from PBE at the measured lattice constant (Fig. 3.7) and at the PBE lattice constant (Worked example E.1), from G_0W_0 calculations started from LDA [39–41] as collected in Ref. [38, Table I] and converged [42, Sect. IV C], from the hybrid functional HSE [11, Table I], and measured at low temperature [10, Eq. (11)]. Right axis: the threshold mass $m_{min} = 2E_g/v^2_{max}$ of Eq. (1.25), $v_{max} = 2.650 \times 10^{-3}$.

of the methods in Fig. 3.9(b), about 0.1 eV in silicon among calculations that treat the gap seriously, is a theory uncertainty of the threshold.

3.4.4 The scissor shift

Since the LDA or PBE conduction bands of silicon are nearly right in shape, the cheapest repair is to move them. The *scissor shift* raises every conduction band by the same amount Δ_{scis} and leaves the valence bands and all the states as they are:

$$H^{\text{s}}_{\boldsymbol{k}} = H_{\boldsymbol{k}} + \Delta_{\text{scis}}\, P_{\text{c}}(\boldsymbol{k}), \qquad P_{\text{c}}(\boldsymbol{k}) = \sum_{c} |u_{c\boldsymbol{k}}\rangle\langle u_{c\boldsymbol{k}}|, \tag{3.66}$$

with $H_{\boldsymbol{k}}$ the Bloch Hamiltonian Eq. (3.21) of the Kohn–Sham potential and the sum over all conduction bands. Its eigenstates are those of $H_{\boldsymbol{k}}$, with eigenvalues $\varepsilon_{v\boldsymbol{k}}$ and $\varepsilon_{c\boldsymbol{k}} + \Delta_{\text{scis}}$. The shift is chosen to bring the computed gap to a measured or a quasiparticle one: for the PBE calculation of Worked example E.1, $\Delta_{\text{scis}} = 1.170 - 0.602 = 0.568$ eV at the PBE lattice constant, and 0.613 eV at the measured one. Against full quasiparticle calculations, a rigid shift is good to about 0.1 eV across the bands in silicon, and to 0.2 and 0.4 eV in GaAs and diamond [40][38, Sect. V B].

What does the shift do to the ingredients of a rate? Energies and the density vertex are easy. Every interband energy grows by Δ_{scis}, and so does $E_{\text{min}}(\boldsymbol{q})$ at every $\boldsymbol{q}$. The density vertex $\rho_{cv}(\boldsymbol{k},\boldsymbol{q})$ of Eq. (3.34) is an overlap of states, and the states do not change. The velocity matrix elements are less obvious. P_{c} is a nonlocal operator, and the velocity, $\mathrm{i}[H,\boldsymbol{r}]$, picks up a commutator with it. By Eq. (3.40), the velocity matrix elements of H^{s} are those of $\nabla_{\boldsymbol{k}} H^{\text{s}}_{\boldsymbol{k}}$. This operator contains the derivative of the projector,

$$\nabla_{\boldsymbol{k}} H^{\text{s}}_{\boldsymbol{k}} = \nabla_{\boldsymbol{k}} H_{\boldsymbol{k}} + \Delta_{\text{scis}} \sum_{c'} \Big(|\nabla_{\boldsymbol{k}} u_{c'\boldsymbol{k}}\rangle\langle u_{c'\boldsymbol{k}}| + |u_{c'\boldsymbol{k}}\rangle\langle \nabla_{\boldsymbol{k}} u_{c'\boldsymbol{k}}| \Big).$$

Between a conduction state c and a valence state v at the same $\boldsymbol{k}$, the first term in the bracket contributes $\sum_{c'} \langle u_c|\nabla_{\boldsymbol{k}} u_{c'}\rangle\langle u_{c'}|u_v\rangle = 0$, since conduction and valence states are orthogonal. The second contributes $\langle \nabla_{\boldsymbol{k}} u_c|u_v\rangle$, the term $c' = c$ of $\sum_{c'} \langle u_c|u_{c'}\rangle\langle \nabla_{\boldsymbol{k}} u_{c'}|u_v\rangle$. Differentiating $\langle u_c|u_v\rangle = 0$ gives $\langle \nabla_{\boldsymbol{k}} u_c|u_v\rangle = -\langle u_c|\nabla_{\boldsymbol{k}} u_v\rangle$, and Eq. (3.42) gives $\langle u_c|\nabla_{\boldsymbol{k}} u_v\rangle = \boldsymbol{v}_{cv}/(\varepsilon_{v\boldsymbol{k}} - \varepsilon_{c\boldsymbol{k}})$. Altogether,

$$\boldsymbol{v}^{\text{s}}_{cv}(\boldsymbol{k}) = \boldsymbol{v}_{cv}(\boldsymbol{k}) - \Delta_{\text{scis}}\, \frac{\boldsymbol{v}_{cv}(\boldsymbol{k})}{\varepsilon_{v\boldsymbol{k}} - \varepsilon_{c\boldsymbol{k}}} = \boldsymbol{v}_{cv}(\boldsymbol{k}) \Big(1 + \frac{\Delta_{\text{scis}}}{\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}} \Big).$$

Between two conduction states the two terms of the bracket give

$$\Delta_{\text{scis}} \big(\langle u_c|\nabla_{\boldsymbol{k}} u_{c'}\rangle + \langle \nabla_{\boldsymbol{k}} u_c|u_{c'}\rangle \big) = \Delta_{\text{scis}} \nabla_{\boldsymbol{k}} \langle u_c|u_{c'}\rangle = 0,$$

and between two valence states the projector does not act, so neither changes.

Scissor rescaling of the velocity

Under the scissor shift Eq. (3.66), energies and states transform as

$$\begin{gathered} \varepsilon_{c\boldsymbol{k}} \to \varepsilon_{c\boldsymbol{k}} + \Delta_{\text{scis}}, \qquad \rho_{mn}(\boldsymbol{k},\boldsymbol{q}) \to \rho_{mn}(\boldsymbol{k},\boldsymbol{q}), \\ \boldsymbol{v}_{cv}(\boldsymbol{k}) \to \boldsymbol{v}_{cv}(\boldsymbol{k})\, \frac{\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}} + \Delta_{\text{scis}}}{\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}}, \end{gathered} \tag{3.67}$$

while the velocity matrix elements within the valence bands and within the conduction bands, the group velocities included, are unchanged.

The interband velocity grows in proportion to the transition energy, and the optical limit Eq. (3.43) shows why it must. The vertex $\rho_{cv} = \boldsymbol{q} \cdot \boldsymbol{v}_{cv}/(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}})$ is fixed because the states are fixed, so a larger denominator needs a larger numerator. The same follows from Eq. (3.39): the position matrix element between the unchanged states stays the same, and the velocity is i times the energy difference times it. Appendix E.3 extends the rescaling to quasiparticle corrections that differ from state to state. At the direct gap of silicon at Γ, 2.55 eV in the PBE calculation at its own lattice constant, the shift of 0.568 eV multiplies the velocity by $3.12/2.55 = 1.22$ and its square by 1.50. An absorption rate built from the shifted energies but the unshifted velocities comes out too small by that factor.

A scissor keeps the order of the Kohn–Sham bands, and where a functional gets that order wrong, as PBE does in germanium, whose conduction state at Γ falls below the L valleys, no rigid shift repairs it (Appendix E.1). It leaves the states unchanged, while the energies of all transitions rise, so that the f-sum rule of Sect. 4.3, a sum of transition energies times $|\rho_{cv}|^2$, exceeds its exact value by $\Delta_{\text{scis}} \sum_c |\rho_{cv}|^2$, the signature of a nonlocal operator in the Hamiltonian (Appendix E.3). And a scissor to a measured gap imports everything the measurement contains, including the renormalization of the gap by the zero-point motion of the lattice (Sect. 13.7), which must then not be added a second time.

Worked example 3.7: The silicon threshold mass from computed gaps

The threshold mass of Eq. (1.25) is $m_{\text{min}} = 2E_{\text{g}}/v_{\text{max}}^2$, with $v_{\text{max}}^2 = 7.023 \times 10^{-6}$. For the gaps of Fig. 3.9(b),

$$\begin{aligned}
&\text{LDA, } 0.52\,\text{eV}: && m_{\text{min}} = \frac{2 \times 0.52\,\text{eV}}{7.023 \times 10^{-6}} = 0.148\,\text{MeV},\\
&\text{PBE, } 0.602\,\text{eV}: && m_{\text{min}} = 0.171\,\text{MeV},\\
&G_0W_0\text{, } 1.10 \text{ to } 1.29\,\text{eV}: && m_{\text{min}} = 0.313 \text{ to } 0.367\,\text{MeV},\\
&\text{measured, } 1.17\,\text{eV}: && m_{\text{min}} = 0.333\,\text{MeV}.
\end{aligned}$$

Taken at face value, an LDA band structure lowers the threshold mass by a factor $1.17/0.52 = 2.25$, to less than half its true value. A projection computed with uncorrected bands would claim reach between 0.148 and 0.333 MeV, where a silicon detector has none. The bound of Worked example 3.5, which requires the particle to deposit the gap at the conduction-band minimum $k_{\text{c}} = 1.94\,\text{keV}$, responds much less. With the LDA gap,

$$m_\chi = \frac{k_{\text{c}}^2}{2(k_{\text{c}} v_{\text{max}} - E_{\text{g}})} = \frac{(1940\,\text{eV})^2}{2\,(5.14\,\text{eV} - 0.52\,\text{eV})} = 0.407\,\text{MeV},$$

against 0.474 MeV with the measured gap. Near k_{c} the momentum needed to reach the valley, not the energy, limits light particles, and the error of the gap is diluted by the kinetic term $k_{\text{c}} v_{\text{max}} = 5.14\,\text{eV}$. Where the true threshold lies between these bounds is a question for Sect. 13.1. Both bounds, however, move down when the gap is too small.
Once the gap is fixed by a scissor to the measured value, the remaining uncertainty is that of the measurement and of its temperature, a few meV. Without a measurement, the spread of 0.1 eV among quasiparticle methods translates through $\delta \ln m_{\text{min}} = \delta \ln E_{\text{g}}$ (Sect. 1.3.5) into $0.1/1.17 = 8.5\,\%$ in m_{min}, more than the 6 % that the uncertainty of the escape speed contributes. Just above the edge, where the halo supply grows by orders of magnitude over a factor of two in mass, such a shift changes the expected rate by large factors.

The Kohn–Sham bands, corrected for the gap, are the input of every electronic rate in

the rest of the book. They come from a calculation with pseudopotentials, whose states are smooth where the true ones oscillate near the nuclei, and the density vertex at the momentum transfers that dark matter reaches depends on exactly those oscillations. Sect. 3.5 restores them.

3.5 Plane waves, pseudopotentials and core states

A momentum transfer of 30 keV resolves the electron density on a length scale $1/q = 0.066$ Å. That is smaller than the $2p$ shell of silicon, whose radial density peaks 0.21 Å from the nucleus, and it is the scale on which the valence states of silicon change sign twice to stay orthogonal to the core. Dark matter of a few MeV reaches such momentum transfers, and a heavy mediator, unlike a light one, does not suppress them against the few-keV momentum transfers of Sect. 1.3. The density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})$ of Eq. (3.34) at large q is therefore a statement about the wavefunctions within a fraction of an angstrom of the nuclei. The calculations of Sect. 3.4, on the other hand, are built to describe the bonds between the atoms, and they do so by throwing that region away. What a plane-wave code keeps, what a pseudopotential removes and how the projector-augmented-wave method puts it back decide the vertex at large q, together with the core states of silicon and germanium and the relativistic corrections that their electrons carry.

3.5.1 Plane waves and the cutoff

A plane-wave code stores each periodic part as the Fourier series Eq. (3.22), truncated to the reciprocal-lattice vectors inside a sphere,

$$u_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{\Omega}} \sum_{|\boldsymbol{k}+\boldsymbol{G}| \le k_{\mathrm{cut}}} c_{n\boldsymbol{k}}(\boldsymbol{G})\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}, \qquad E_{\mathrm{cut}} = \frac{k_{\mathrm{cut}}^2}{2m_e}. \tag{3.68}$$

The kinetic-energy cutoff E_{cut} is the one parameter of the basis. Every plane wave with kinetic energy $|\boldsymbol{k}+\boldsymbol{G}|^2/2m_e$ below it is kept, so the basis treats all regions of the cell alike and improves systematically as E_{cut} is raised; the total energy, which is variational, converges from above. The number of plane waves per state grows as $E_{\mathrm{cut}}^{3/2}$, and in silicon it is 2121 at 60 Ry, or 816 eV with 1 Ry = 13.6 eV (Eq. (E.2)). In Eq. (3.23) the kinetic energy is diagonal in $\boldsymbol{G}$, while the potential is diagonal in $\boldsymbol{r}$, and a fast Fourier transform between a grid of $\boldsymbol{G}$ and a grid of points in the cell applies each where it is diagonal. This is what makes plane waves the standard basis for crystals.

The price is paid near the nuclei. A state confined to a region of size r_0 has momentum components up to a few times $1/r_0$, and the innermost orbitals of an atom are small. A hydrogen-like $1s$ orbital of nuclear charge Z, $(p_0^3/\pi)^{1/2}\mathrm{e}^{-p_0 r}$ with $p_0 = Z\alpha m_e$, has the radius $1/p_0$. Its Fourier transform follows from $\int \mathrm{d}^3r\, \mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}}\mathrm{e}^{-p_0 r} = 8\pi p_0/(p^2+p_0^2)^2$, and its momentum density, normalized to $\int \mathrm{d}^3p = 1$, is $(8/\pi^2)\, p_0^5/(p^2+p_0^2)^4$. For silicon, $p_0 = 14 \times 3.73\,\mathrm{keV} = 52\,\mathrm{keV}$. Worse, the density falls as slowly as p^{-8}, a power law that comes from the cusp of the orbital at the nucleus, and so do the squared coefficients $|c_{n\boldsymbol{k}}(\boldsymbol{G})|^2$ of any state that has weight there. The valence states inherit a version of this tail. A $3s$ electron of silicon must be orthogonal to the $1s$ and $2s$ orbitals, and it does this by changing sign at 0.08 and 0.38 Å from the nucleus; a $3p$ electron changes sign at 0.38 Å to stay orthogonal to $2p$. These oscillations, and the cusp, are what a momentum transfer of tens of keV sees. To resolve the silicon $1s$ orbital alone, a plane-wave basis would need a cutoff near 1800 Ry and 3×10^5 plane waves per state (Appendix E.2). Before we see how to avoid that cost, we ask which cutoff a given momentum transfer actually needs.

The vertex in plane waves is the sum Eq. (3.35). A term contributes only if both coefficients exist: the component $\boldsymbol{k}+\boldsymbol{G}$ of the initial state and the component $\boldsymbol{k}+\boldsymbol{G}+\boldsymbol{q}$ of the final state must both lie inside the cutoff sphere. Two points of a sphere of radius k_{cut} are at most $2k_{\mathrm{cut}}$ apart, and the computed vertex vanishes identically for $q > 2k_{\mathrm{cut}}$, which at 60 Ry is 58 keV. This bound would be reached if the two states had their weight at opposite ends of the sphere, and the kinematics of dark matter keeps them from it. By Eq. (1.23), a momentum transfer q deposits at most $\omega = qv_{\max}$, 80 eV at $q = 30$ keV, far below the free-electron energy $q^2/2m_e = 0.88$ keV at that momentum. The final state is a low-lying conduction state. Its plane-wave components are concentrated at momenta below a scale p_f set by its energy above the bottom of the valence band, $p_f \simeq [2m_e(\omega + E_{\mathrm{F}})]^{1/2}$ with $E_{\mathrm{F}} = 12.5$ eV the free-electron value at the valence density (Sect. 2.5). In Eq. (3.35) the final-state coefficient at $\boldsymbol{k}+\boldsymbol{G}+\boldsymbol{q}$ is then appreciable for $|\boldsymbol{k}+\boldsymbol{G}+\boldsymbol{q}| \lesssim p_f$, and the initial state must supply the momentum, through its components at $|\boldsymbol{k}+\boldsymbol{G}| \simeq q$. The same holds with the two states exchanged. The vertex at momentum transfers up to $q_{\max}$ is converged once the basis contains these components,

$$k_{\mathrm{cut}} \gtrsim q_{\max} + p_f, \qquad E_{\mathrm{cut}} \gtrsim \frac{(q_{\max}+p_f)^2}{2m_e}. \tag{3.69}$$

The large-q vertex is the overlap of the high-momentum tail of one state with the low-momentum body of the other, as Sect. 3.2 anticipated. A cutoff converged for total energies says nothing about whether that tail is in the basis, and Eq. (3.69) has to be checked separately for every rate calculation. Whether the tail is physical is another matter, which depends on the potential the states were computed in.

3.5.2 Pseudopotentials

The core electrons of silicon, $1s^2 2s^2 2p^6$, are bound by 99 eV and more and take no part in bonding; their orbitals hardly change between the atom and the crystal. The valence electrons, however, feel them through orthogonality. A $3s$ state must be orthogonal to $1s$ and $2s$, and its two nodes inside 0.4 Å carry a large kinetic energy there. That kinetic energy acts like a repulsion that nearly cancels the attraction of the nucleus inside the core, and outside the core the valence electron sees a nucleus screened by ten electrons. Phillips and Kleinman made this precise: subtracting the core states from a valence state leaves a smooth function that obeys a Schrödinger equation with a weak, repulsive correction added to the potential inside the core [43]. The cancellation is why the nearly-free-electron picture of Sect. 3.2 works for silicon at all, and why the empirical pseudopotential of Sect. 3.3 needs only three Fourier components.

A *pseudopotential* turns the cancellation into a construction. It replaces the nucleus and the core electrons by an effective potential whose valence states are smooth and nodeless inside a *core radius* r_c and identical to the true, all-electron states outside it. The core states are dropped from the calculation, and with them the oscillations of the valence states that kept them orthogonal. The construction starts from an all-electron calculation of the atom and is carried out for each angular momentum l separately. Hamann, Schlüter and Chiang formulated the conditions that make the result useful [44]. Two of them do the work. For a reference configuration of the atom, the pseudo and all-electron valence states must have the same eigenvalue and agree beyond r_c, and they must hold the same charge inside it. Their logarithmic derivatives at r_c then agree, and so do the first energy derivatives of these. The charge condition, *norm conservation*, is the one with content, and it implies the last.

To see this, write the radial Schrödinger equation for $u_l(r;\varepsilon) = rR_l(r)$ at energy ε in a

potential $V_l(r)$, which may differ from one l to the next,

$$-\frac{1}{2m_e}\,u_l'' + \Big[V_l(r) + \frac{l(l+1)}{2m_e r^2} - \varepsilon\Big]u_l = 0, \tag{3.70}$$

with primes for $\partial/\partial r$, and differentiate it with respect to ε. The derivative $\dot{u}_l = \partial u_l/\partial\varepsilon$ obeys the same equation with the source u_l on the right-hand side. Multiplying the equation for u_l by $\dot{u}_l$, the equation for $\dot{u}_l$ by u_l, and subtracting, the potential drops out and leaves $(\dot{u}_l u_l' - u_l \dot{u}_l')' = 2m_e u_l^2$. Both functions vanish at the origin, and integrating to r_c,

$$\frac{\partial}{\partial\varepsilon}\,\frac{u_l'(r_c)}{u_l(r_c)} = \frac{\dot{u}_l' u_l - u_l' \dot{u}_l}{u_l^2}\bigg|_{r_c} = -\frac{2m_e}{u_l(r_c)^2}\int_0^{r_c} \mathrm{d}r\, u_l^2. \tag{3.71}$$

The logarithmic derivative at r_c fixes how a wave arriving from outside is scattered by everything inside, and Eq. (3.71) ties its energy dependence to the charge inside the sphere. A pseudo-state that matches the all-electron state in value and slope at r_c at the reference energy, and has the same charge inside, therefore scatters like the true atom to first order in the energy away from the reference. A potential fitted to the free atom then works in the crystal, where the valence energies are shifted by the bonding. Two or more reference energies per channel extend the range of energies over which it works. Vanderbilt introduced them in his ultrasoft scheme, which relaxes the norm-conservation condition of Hamann, Schlüter and Chiang and restores the missing charge with augmentation functions [45]. Hamann's optimized potentials keep norm conservation in a generalized form: the overlaps $\int_0^{r_c} \mathrm{d}r\, u_i u_j$ of the pseudo and all-electron reference states agree for every pair i, j [14, Eq. (23)].

The potential this produces is *nonlocal*. Orthogonality acts only within one angular momentum: a $3s$ electron is pushed out by the $1s$ and $2s$ cores, a $3p$ electron by $2p$, and a $3d$ electron by nothing, since silicon has no d core. The effective potential is then different in each channel, $\sum_l V_l(r) P_l$ with P_l the projector on angular momentum l about the nucleus, and in practice it is cast into a local part plus a few separable projections, the form Eq. (E.8) of Kleinman and Bylander [46]. A nonlocal potential does not commute with $\boldsymbol{r}$, and we have already met two consequences: the velocity is no longer $\boldsymbol{p}/m_e$ (Sect. 3.2), and the f-sum rule acquires a correction (Sect. 4.3). The density vertex needs none, since $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ couples to the pseudo-states by multiplication, as in any Hamiltonian; its only error is the missing core region.

The core radius sets the trade-off. A large r_c makes the pseudo-states smooth and the cutoff low, but asks one potential to describe a large part of the atom, and it transfers less well to other environments. The cutoff is decided by the Fourier tail of the pseudo-states, and the optimized construction of Hamann chooses the shape inside r_c to minimize the kinetic energy above a wavevector q_{opt} (Eq. (E.9)). In practice, the potentials of the plane-wave calculations of this book (Appendices E.1 and E.2) come from the SG15 library [15]. The silicon potential keeps the four $3s$ and $3p$ electrons, with $r_c = 1.27$ Å for s and 1.89 Å for p, and $q_{\mathrm{opt}} = 19$ and 13 keV. With it, the total energy converges to 0.1 meV per atom at about 60 Ry (Table E.2). Appendix E.2 describes how such potentials are built and tested.

Fig. 3.10(a) shows what is lost, for a norm-conserving silicon potential built with the SG15 core radii. The pseudo-orbitals agree with the all-electron ones beyond r_c and are smooth and nodeless inside, where the true orbitals oscillate; the charge inside r_c is the same, 0.662 of the $3s$ electron and 0.767 of the $3p$ electron. In momentum space, panel (b), the two agree to 10 % below 2 keV for $3s$ and 5 keV for $3p$, where most of the weight is. Above, the pseudo-orbitals fall off much faster. Above 10 keV the all-electron $3s$ orbital keeps 7 % of its weight and the pseudo-orbital 0.14 %; at 30 keV the all-electron momentum densities exceed the pseudo ones by more than four orders of magnitude. A pseudopotential calculation converged in E_{cut} converges to the pseudo answer, and at large q the pseudo answer is wrong.

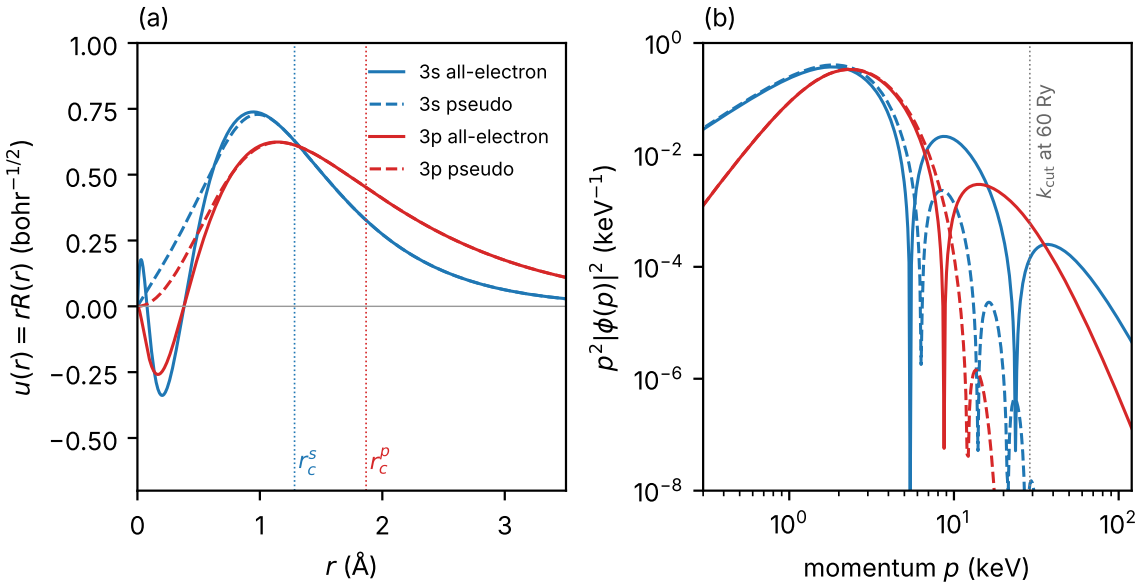


Figure 3.10: The valence orbitals of the silicon atom near the nucleus, all-electron (solid) and pseudo (dashed), for $3s$ (blue) and $3p$ (red). (a) Radial functions $u = rR$. The all-electron orbitals change sign at 0.08 and 0.38 Å ($3s$) and at 0.38 Å ($3p$) to stay orthogonal to the core; the pseudo-orbitals are nodeless inside the core radii r_c (dotted). Outside them they equal the all-electron ones, and they carry the same charge inside. (b) Momentum distributions $p^2|\phi(p)|^2$, normalized to $\int \mathrm{d}p\, p^2|\phi|^2 = 1$; dotted, the plane-wave cutoff $k_{\mathrm{cut}} = 28.9\,\mathrm{keV}$ at which the total energy of silicon converges (60 Ry). Free atom, all-electron calculation with the PBE functional, scalar-relativistic (mass–velocity and Darwin terms kept, spin–orbit coupling dropped), configuration [Ne] $3s^2 3p^2$. The pseudopotential is a single-projector norm-conserving potential of the Troullier–Martins form [47], generated for this figure at the core radii of the SG15 potential of the plane-wave calculations, 1.27 and 1.89 Å for s and p, to within 0.02 Å; it reproduces the all-electron eigenvalues. The SG15 potential has the same radii but a different shape inside them.

3.5.3 Projector augmented waves

The core region that the pseudopotential discards can be restored. Outside the core spheres the pseudo-state is the true state, and inside each sphere the true state is fixed, up to a few coefficients, by the atom from which the pseudopotential was built. Blöchl's *projector-augmented-wave* (PAW) method makes the map from one to the other explicit and linear [48]. For each atom it uses three sets of functions, labeled by an index j that collects the site, the angular momentum lm and a reference energy. The *all-electron partial waves* ϕ_j^{AE} are solutions of the radial equation of the atom at the reference energies. The *pseudo partial waves* ϕ_j^{PS} are smooth functions that coincide with them outside the sphere. The *projectors* $\tilde{p}_j$ are localized inside the sphere and dual to the pseudo partial waves, $\langle \tilde{p}_i | \phi_j^{\mathrm{PS}} \rangle = \delta_{ij}$. The pseudo partial waves are assumed complete for smooth functions inside the sphere, $\sum_j |\phi_j^{\mathrm{PS}}\rangle\langle \tilde{p}_j| = 1$ there. The all-electron state is then

$$|\psi_{n\boldsymbol{k}}\rangle = |\tilde{\psi}_{n\boldsymbol{k}}\rangle + \sum_j \big(|\phi_j^{\mathrm{AE}}\rangle - |\phi_j^{\mathrm{PS}}\rangle\big)\langle \tilde{p}_j | \tilde{\psi}_{n\boldsymbol{k}}\rangle, \tag{3.72}$$

where $\tilde{\psi}_{n\boldsymbol{k}}$ is the smooth pseudo-state of the calculation. Outside the spheres the bracket vanishes and $\psi = \tilde{\psi}$. Inside a sphere, completeness lets us write the pseudo-state as its one-center expansion $\tilde{\psi}^1 = \sum_j \phi_j^{\mathrm{PS}} \langle \tilde{p}_j | \tilde{\psi}\rangle$, so that Eq. (3.72) subtracts it and puts in its place $\psi^1 = \sum_j \phi_j^{\mathrm{AE}} \langle \tilde{p}_j | \tilde{\psi}\rangle$, the same combination of all-electron partial waves. The nodes and the cusp come back with the ϕ_j^{AE}.

We can compute matrix elements without ever building ψ on a fine grid. Write $\psi = \tilde{\psi} + \psi^1 - \tilde{\psi}^1$ for both states, and take an operator A that is local, so that it does not connect the inside of a sphere with the outside. In a cross term such as $\langle \tilde{\psi}_m | A | \psi_n^1 - \tilde{\psi}_n^1 \rangle$ the difference lives inside the sphere, where $\tilde{\psi}_m = \tilde{\psi}_m^1$, and the cross term equals $\langle \tilde{\psi}_m^1 | A | \psi_n^1 - \tilde{\psi}_n^1 \rangle$. Doing the same in the other cross term, the three terms that contain a difference combine into $\langle \psi_m^1 | A | \psi_n^1 \rangle - \langle \tilde{\psi}_m^1 | A | \tilde{\psi}_n^1 \rangle$, and

$$\langle \psi_m | A | \psi_n \rangle = \langle \tilde{\psi}_m | A | \tilde{\psi}_n \rangle + \sum_{ij} \langle \tilde{\psi}_m | \tilde{p}_i \rangle \big(\langle \phi_i^{\mathrm{AE}} | A | \phi_j^{\mathrm{AE}} \rangle - \langle \phi_i^{\mathrm{PS}} | A | \phi_j^{\mathrm{PS}} \rangle \big) \langle \tilde{p}_j | \tilde{\psi}_n \rangle, \tag{3.73}$$

with the partial-wave matrix elements taken inside the spheres. The smooth part is computed in plane waves at a modest cutoff. The rest is a sum of one-center terms, computed once for each species on a radial grid fine enough for the cusp.

All-electron density vertex

With $A = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ and $\boldsymbol{r} = \boldsymbol{\tau}_\kappa + \boldsymbol{x}$ in the sphere of atom κ of the cell, Eq. (3.73) gives the vertex of Eq. (3.34) between all-electron states as

$$\rho_{mn}(\boldsymbol{k}, \boldsymbol{q}) = \langle \tilde{\psi}_{m,\boldsymbol{k}+\boldsymbol{q}} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \tilde{\psi}_{n\boldsymbol{k}} \rangle + \sum_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa} \sum_{ij} \langle \tilde{\psi}_{m,\boldsymbol{k}+\boldsymbol{q}} | \tilde{p}_{\kappa i} \rangle \, \Delta_{\kappa,ij}(\boldsymbol{q}) \, \langle \tilde{p}_{\kappa j} | \tilde{\psi}_{n\boldsymbol{k}} \rangle, \tag{3.74}$$

with the one-center matrix $\Delta_{\kappa,ij}(\boldsymbol{q}) = \langle \phi_{\kappa i}^{\mathrm{AE}} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}} | \phi_{\kappa j}^{\mathrm{AE}} \rangle - \langle \phi_{\kappa i}^{\mathrm{PS}} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}} | \phi_{\kappa j}^{\mathrm{PS}} \rangle$.

Here we have written the atom index explicitly, $\tilde{p}_{\kappa i}$ for the projectors and $\phi_{\kappa i}$ for the partial waves of atom κ. The projections $\langle \tilde{p}_{\kappa j} | \tilde{\psi}_{n\boldsymbol{k}} \rangle$ are taken with the states normalized over one cell (Appendix A.5). Every one of the N cells contributes the same one-center terms by Bloch's theorem, each with weight $1/N$ for states normalized over the crystal, and the cell normalization adds them up; the phase is that of the atom in the reference cell. The plane wave about the nucleus expands into spherical Bessel functions $j_l(qx)$ times spherical

harmonics (Eq. (C.23)), so each element of $\Delta_{\kappa,ij}$ is a sum of radial integrals of $j_l(qx)$ against products of partial waves, times integrals of three spherical harmonics (Appendix E.2).

How large is the correction? Expand $\Delta_{\kappa,ij}(\boldsymbol{q})$ in $\boldsymbol{q}\cdot\boldsymbol{x}$. The zeroth-order term is the difference of overlaps inside the sphere. For partial waves that obey the generalized norm-conservation condition it vanishes; a PAW data set does not require that condition, and the term is then the charge that the smooth part lacks inside the sphere. The first-order term is a difference of dipole moments of partial-wave products, of relative size qr_c, weighted by the fraction of the state inside the sphere; the next is a quadrupole difference, of order $(qr_c)^2$. The pseudo vertex is accurate for $qr_c \ll 1$, below about 1 keV for the silicon potential above, and the correction grows from there. Well above q_{opt} it is the whole answer, since the pseudo vertex has decayed. Griffin and coworkers computed the all-electron vertices of silicon and germanium in this way, from PAW calculations, and the valence-to-conduction rates for dark matter of 1 GeV. The reconstruction raises the heavy-mediator rate well above the pseudo-state result even for thresholds as low as about 10 eV. Together with the core states it raises it by more than an order of magnitude at higher thresholds, especially in germanium, while the light-mediator rate, dominated by small q, changes less [49, Sects. 3.1 and 4]. A norm-conserving potential can be reconstructed with the same formula, using the partial waves of the atom from which it was built. The reconstruction is as good as its ingredients. It assumes a frozen core. It restores the core region only within the span of a few partial waves per channel, which is accurate near the reference energies and less so far above them. And it does not add the core states themselves, which are absent from the pseudopotential calculation. Appendix E.7 gives the practice and its convergence.

Worked example 3.8: The cutoff for momentum transfers up to 30 keV

A momentum transfer $q = 30\,\text{keV}$ needs $2m_\chi v_{\max} \geq q$, that is $m_\chi \geq q/2v_{\max} = 5.7\,\text{MeV}$ for $v_{\max} = 2.650 \times 10^{-3}$, and it deposits at most $\omega = qv_{\max} = 79.5\,\text{eV}$. The final state lies at most 79.5 eV above the valence-band maximum, or about 92 eV above the bottom of the valence band. Its plane-wave components are concentrated below

$$p_f = [2m_e(\omega + E_\text{F})]^{1/2} = (2 \times 0.511\,\text{MeV} \times 92\,\text{eV})^{1/2} = 9.7\,\text{keV}.$$

By Eq. (3.69), the basis must hold the initial-state components up to $q + p_f = 39.7\,\text{keV}$:

$$E_\text{cut} \gtrsim \frac{(39.7\,\text{keV})^2}{2 \times 511\,\text{keV}} = 1.54\,\text{keV} = 113\,\text{Ry}.$$

With $k_\text{cut} = 10.6/a_0$ and the silicon cell of $270.3\,a_0^3$, in units of the Bohr radius $a_0 = 0.529\,\text{Å}$, Eq. (E.2) gives about 5500 plane waves per state, 2.6 times the 2121 at which the total energy has converged to 0.1 meV per atom. The same estimate counts the bands below ω: the free-electron states inside a sphere of radius p_f number $\Omega p_f^3/6\pi^2 \approx 80$ per spin.
At 60 Ry, $k_\text{cut} = 28.9\,\text{keV}$. The vertex at 30 keV is not zero there, since it vanishes identically only beyond $2k_\text{cut} = 58\,\text{keV}$, but it lacks the initial-state components between 28.9 and 39.7 keV. Raising the cutoff to 113 Ry converges it, to the pseudo answer. In the atom of Fig. 3.10, the pseudo-orbitals carry less than 10^{-7} of their weight above 30 keV, the all-electron orbitals 0.7 ($3s$) and 0.4 % ($3p$): the converged pseudo vertex at 30 keV misses almost all of the true one. Only the reconstruction Eq. (3.74) restores the tail, and the cutoff estimate applies to the reconstructed states. The silicon and germanium calculation of Griffin and coworkers meets these requirements: pseudo-states at a cutoff of 1 keV, reconstructed all-electron states up to 2 keV, which hold momentum components up to $(2m_e \times 2\,\text{keV})^{1/2} = 45\,\text{keV}$ [49, Sect. 2.1].

3.5.4 Core and semicore states

Below the valence bands of silicon lie the atomic shells $2p$, $2s$ and $1s$. Their binding energies relative to the valence-band maximum, measured by photoemission, are 99.4 eV for $2p$ (the $2p_{3/2}$ level; $2p_{1/2}$ is at 99.8 eV), 150 eV for $2s$ and 1839 eV for $1s$ [50, Table 1-1]. Their radial densities peak at 0.21, 0.24 and 0.038 Å from the nucleus, against about 1 Å for the $3s$ and $3p$ valence orbitals. Half of the $2p$ electrons have momenta above 14 keV, and half of the $1s$ electrons above 37 keV; for the valence electrons the median is 2 to 3 keV. These are free-atom numbers, from the all-electron calculation of Fig. 3.11, but the core shells change little in the crystal. A pseudopotential calculation contains none of these states.

Whether they matter is decided by kinematics. An electron in a shell at depth E_{core} below the valence-band maximum needs at least $\omega = E_{\text{g}} + E_{\text{core}}$ to reach the conduction band. By Eq. (1.23), a particle of mass m_χ deposits ω at momentum transfer q only if $\omega/q + q/2m_\chi \leq v_{\text{max}}$. Since the second term is positive, the momentum transfer must exceed ω/v_{max}; minimizing the left-hand side over q, at $q = (2m_\chi\omega)^{1/2}$, gives the threshold mass $2\omega/v_{\text{max}}^2$ of Eq. (1.25). With $v_{\text{max}} = 2.650 \times 10^{-3}$, the silicon $2p$ shell needs $q \geq 38$ keV for any mass and opens at $m_\chi = 29$ MeV, where the only usable momentum transfer is $(2m_\chi\omega)^{1/2} = 2\omega/v_{\text{max}} = 76$ keV. The $2s$ shell needs $q \geq 57$ keV and opens at 43 MeV, with 114 keV there, and the $1s$ shell opens above 0.5 GeV. Below about 29 MeV, dark matter scatters only on the four valence electrons of each silicon atom. Above, the $2p$ shell is reached through the tail of its momentum distribution: 3 % of its weight lies above 38 keV.

Germanium is different. Its $3d$ shell lies only 29 eV below the valence-band maximum [50, Table 1-1], with a radial density peaked at 0.27 Å. It needs $q \geq 11$ keV for any mass, 22 keV at the threshold mass $m_\chi = 8.5$ MeV, and three quarters of its momentum distribution lies above 11 keV. Ten electrons per atom, kinematically accessible to light dark matter and probed near the peak of their momentum distribution, are what the field calls a *semicore* shell: too deep to bond, too shallow and too extended to be frozen without loss. A germanium pseudopotential can freeze it, with four valence electrons, or keep it in the valence, with fourteen and a higher cutoff (Appendix E.2). Only the second choice puts the $3d$ states into the calculated response. For a heavy mediator, and once their large-momentum components are restored, the $3d$ electrons dominate the total rate of germanium for $m_\chi \gtrsim 30$ MeV at a one-pair threshold [49, Sects. 1 and 3.3], a point to which Sect. 10.6 returns. The deeper shells, $3p$ and below in germanium and all core shells of silicon, are added as atomic orbitals arranged in Bloch sums (Eq. (E.48)).

Fig. 3.11 also explains why the valence states need the core region. Above about 5 keV the core electrons outnumber the valence electrons in both atoms, and at 30 keV the silicon $2p$ shell holds forty times more weight per keV than the valence shell. What weight the valence shell has there comes from the inner lobes that keep it orthogonal to the core. These are the lobes that a pseudopotential removes and that Eq. (3.74) restores.

3.5.5 Relativistic corrections

A $1s$ electron of germanium moves at about $Z\alpha = 0.23$ of the speed of light, and the core electrons that a large momentum transfer probes are the fastest electrons of the atom. How large are the relativistic corrections to their states? For a single electron in the potential energy $U_1(\boldsymbol{r})$ of the nuclei and the other electrons, the Dirac equation $(\boldsymbol{\alpha}_{\text{D}} \cdot \boldsymbol{p} + \beta_{\text{D}} m_e + U_1)\psi = E\psi$ (Appendix C.15) couples a large upper component ψ_{A} to a small lower one ψ_{B},

$$\boldsymbol{\sigma} \cdot \boldsymbol{p}\, \psi_{\text{B}} = (\epsilon - U_1)\, \psi_{\text{A}}, \qquad \boldsymbol{\sigma} \cdot \boldsymbol{p}\, \psi_{\text{A}} = (2m_e + \epsilon - U_1)\, \psi_{\text{B}},$$

with $\epsilon = E - m_e$ the energy without the rest mass. The second equation gives $\psi_{\text{B}} = (2m_e + \epsilon - U_1)^{-1} \boldsymbol{\sigma} \cdot \boldsymbol{p}\, \psi_{\text{A}}$. Wherever $|\epsilon - U_1| \ll 2m_e$, which fails within $Z\alpha/2m_e$ of a nucleus,

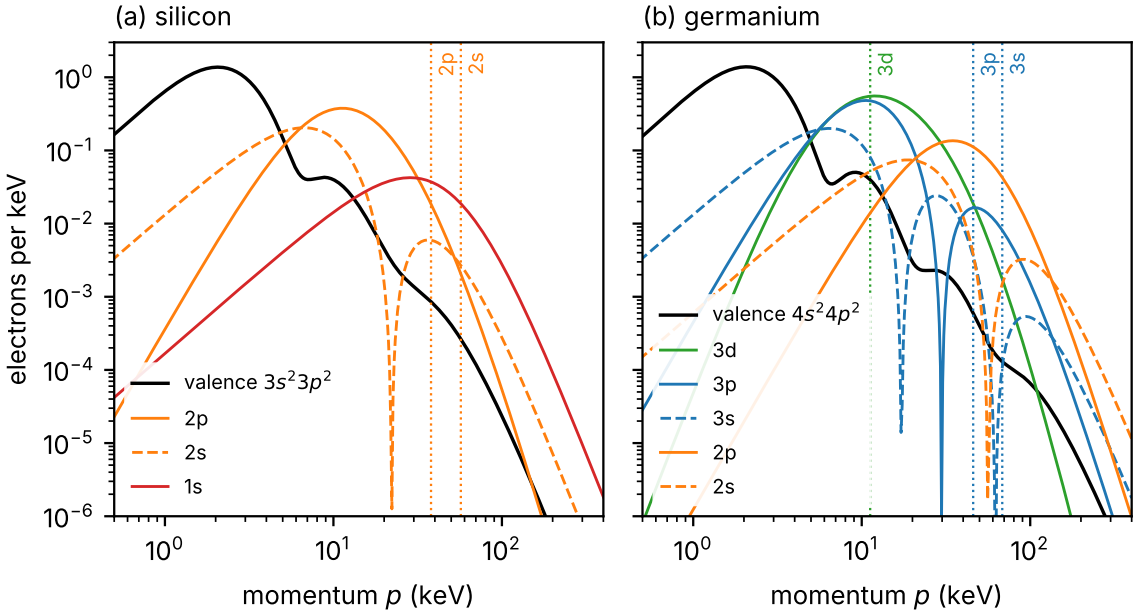


Figure 3.11: How the electrons of each atomic shell are distributed in momentum, for (a) silicon and (b) germanium: electrons per keV of momentum magnitude, $N_{nl}\, p^2 |\phi_{nl}(p)|^2$, with N_{nl} the occupation of the shell and ϕ_{nl} the radial momentum-space orbital normalized to $\int \mathrm{d}p\, p^2 |\phi_{nl}|^2 = 1$. The orbitals are those of the free atoms, from an all-electron, scalar-relativistic calculation with the PBE functional (configurations [Ne] $3s^2 3p^2$ and [Ar] $3d^{10} 4s^2 4p^2$; germanium $1s$, which opens above $q = 4\,\mathrm{MeV}$, is not shown). Black: the valence shell. Dotted vertical lines: the smallest momentum transfer at which dark matter of any mass can lift an electron of the shell of the same color into the conduction band, $(E_\mathrm{g} + E_\mathrm{core})/v_\mathrm{max}$ (twice that at the threshold mass), with $v_\mathrm{max} = 2.650 \times 10^{-3}$, gaps of 1.17 and 0.744 eV and measured binding energies relative to the valence-band maximum [50, Table 1-1]. Above about 5 keV the core electrons outnumber the valence electrons in both atoms. The silicon $2p$ shell opens at 38 keV, in the tail of its distribution; the germanium $3d$ shell opens at 11 keV, near its peak.

we expand to first order, $\psi_\mathrm{B} \simeq (2m_e)^{-1}[1-(\epsilon-U_1)/2m_e]\boldsymbol{\sigma}\cdot\boldsymbol{p}\,\psi_\mathrm{A}$, and insert it into the first equation. With $(\boldsymbol{\sigma}\cdot\boldsymbol{p})^2 = p^2$,

$$\Big[\frac{p^2}{2m_e} + U_1 - \frac{\boldsymbol{\sigma}\cdot\boldsymbol{p}\,(\epsilon-U_1)\,\boldsymbol{\sigma}\cdot\boldsymbol{p}}{4m_e^2}\Big]\psi_\mathrm{A} = \epsilon\psi_\mathrm{A}.$$

This is not yet a Schrödinger equation, since the energy appears on both sides and ψ_A is not normalized: the probability is $\int(|\psi_\mathrm{A}|^2+|\psi_\mathrm{B}|^2) = \int\psi_\mathrm{A}^\dagger(1+p^2/4m_e^2)\psi_\mathrm{A}$ to this order. The normalized two-component state is $\Phi = (1+p^2/8m_e^2)\psi_\mathrm{A}$. We substitute $\psi_\mathrm{A} = (1-p^2/8m_e^2)\Phi$ and multiply the equation from the left by $1-p^2/8m_e^2$, keeping terms to order $1/m_e^2$ relative to $p^2/2m_e$. The kinetic term becomes $p^2/2m_e - p^4/8m_e^3$; the potential becomes $U_1 - (p^2U_1 + U_1p^2)/8m_e^2$; the term $-\epsilon p^2/4m_e^2$ on the left cancels the same term that the right-hand side, $\epsilon(1-p^2/4m_e^2)\Phi$, acquires. What remains is

$$H\Phi = \Big[\frac{p^2}{2m_e} + U_1 - \frac{p^4}{8m_e^3} - \frac{p^2U_1+U_1p^2}{8m_e^2} + \frac{\boldsymbol{\sigma}\cdot\boldsymbol{p}\,U_1\,\boldsymbol{\sigma}\cdot\boldsymbol{p}}{4m_e^2}\Big]\Phi = \epsilon\Phi.$$

The last two terms simplify with $[\boldsymbol{p},U_1] = -\mathrm{i}\nabla U_1$. Moving $\boldsymbol{p}$ to the right, $p^2U_1 = U_1p^2 - 2\mathrm{i}\nabla U_1\cdot\boldsymbol{p} - \nabla^2U_1$, and with $(\boldsymbol{\sigma}\cdot\boldsymbol{a})(\boldsymbol{\sigma}\cdot\boldsymbol{b}) = \boldsymbol{a}\cdot\boldsymbol{b} + \mathrm{i}\boldsymbol{\sigma}\cdot(\boldsymbol{a}\times\boldsymbol{b})$, $\boldsymbol{\sigma}\cdot\boldsymbol{p}\,U_1\,\boldsymbol{\sigma}\cdot\boldsymbol{p} = U_1p^2 - \mathrm{i}\nabla U_1\cdot\boldsymbol{p} + \boldsymbol{\sigma}\cdot(\nabla U_1\times\boldsymbol{p})$. The terms in U_1p^2 and in $\nabla U_1\cdot\boldsymbol{p}$ cancel between the two, and

$$H = \frac{p^2}{2m_e} + U_1 \underbrace{-\frac{p^4}{8m_e^3}}_{H_\mathrm{mv}} \underbrace{+\frac{\nabla^2U_1}{8m_e^2}}_{H_\mathrm{Dar}} \underbrace{+\frac{\boldsymbol{\sigma}\cdot(\nabla U_1\times\boldsymbol{p})}{4m_e^2}}_{H_\mathrm{so}}. \tag{3.75}$$

The *mass–velocity* term H_mv is the next term of the expansion of $(p^2+m_e^2)^{1/2}$: a fast electron is heavier. The *Darwin* term H_Dar smears the potential over the Compton wavelength $1/m_e$. For the nuclear attraction $-Z\alpha/r$ it is the contact term $(\pi Z\alpha/2m_e^2)\,\delta^3(\boldsymbol{r})$, felt only by s electrons. In a central potential $\nabla U_1\times\boldsymbol{p} = r^{-1}(\mathrm{d}U_1/\mathrm{d}r)\boldsymbol{L}$, and with $\boldsymbol{\sigma} = 2\boldsymbol{S}$ the last term is the spin–orbit coupling Eq. (3.53), now with its coefficient, including the Thomas factor, derived rather than quoted. The Foldy–Wouthuysen transformation reaches the same three operators by a unitary transformation, systematically to any order, and also gives the corresponding correction to $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ (Appendix C.15).

The size of each term follows from the velocity. For a hydrogen-like orbital, the virial theorem makes the mean kinetic energy equal to the binding energy, $\langle p^2\rangle/2m_e = (Z\alpha)^2m_e/2n^2$, so that $\langle v^2\rangle = (Z\alpha/n)^2$. The mass–velocity and Darwin terms are of relative order $\langle v^2\rangle$, and so is the change of the wavefunction near the nucleus. For $1s$ both can be evaluated. Since $p^2/2m_e = \epsilon - U_1$ for the eigenstate, $\langle p^4\rangle = 4m_e^2\langle(\epsilon-U_1)^2\rangle$. With $\epsilon = -(Z\alpha)^2m_e/2$, $\langle U_1\rangle = 2\epsilon$ and $\langle U_1^2\rangle = (Z\alpha)^2\langle r^{-2}\rangle = 2(Z\alpha)^4m_e^2$, this is $5(Z\alpha)^4m_e^4$, and $\langle H_\mathrm{mv}\rangle = -\frac{5}{8}(Z\alpha)^4m_e$. The Darwin term is $(\pi Z\alpha/2m_e^2)|\psi_{1s}(0)|^2 = +\frac{1}{2}(Z\alpha)^4m_e$, with $|\psi_{1s}(0)|^2 = (Z\alpha m_e)^3/\pi$. Each is a fraction of order $(Z\alpha)^2$ of the binding energy $(Z\alpha)^2m_e/2$, and they nearly cancel: their sum deepens the level by a quarter of $(Z\alpha)^2$ of its binding energy, the $1s$ term of the Dirac fine-structure formula. Relativistic effects are therefore of order $(Z\alpha)^2$ in the state and of order $(Z\alpha)^2/4$ in the energy of a $1s$ shell.

In practice, pseudopotentials and PAW data sets are generated from *scalar-relativistic* atoms [51]: the mass–velocity and Darwin terms are kept in the radial equation, which stays independent of spin, and spin–orbit coupling is dropped. The core-region shape of the valence states, and the partial waves of Eq. (3.72), then contain the leading relativistic contraction, and a reconstructed vertex inherits it. Spin–orbit coupling is added, when needed, with fully relativistic potentials (Appendix E.1).

For the core shells, spin–orbit coupling is the visible effect. A shell with $l \geq 1$ splits into $j = l \pm \frac{1}{2}$, by $(l+\frac{1}{2})\xi_\mathrm{SO}$, the p splitting $\frac{3}{2}\xi_\mathrm{SO}$ of Sect. 3.3 generalized. Near the nucleus

$r^{-1}\mathrm{d}U_1/\mathrm{d}r \simeq Z\alpha/r^3$, and for an inner shell, which lives there, $\langle r^{-3}\rangle$ grows as Z^3, so that ξ_{SO} grows as Z^4. The hydrogen-like $2p$ splitting $(Z\alpha)^4 m_e/32$ of Sect. 3.3 is an upper bound for a real atom, in which the outer part of the orbital sees a screened charge: it would be 1.7 eV for silicon and 47 eV for germanium. A Dirac–Kohn–Sham calculation of the atoms yields 0.65 and 31.9 eV, and the measured germanium $2p$ doublet, at 1217.0 and 1248.1 eV, is split by 31.1 eV [50, Table 1-1]. A core edge is therefore a doublet. In the $2p$ edge of silicon the two components lie a few tenths of an eV apart, while those of the germanium $3d$ edge, the one light dark matter reaches, lie 0.6 eV apart, with the statistical weights 4 : 6 of $3d_{3/2}$ and $3d_{5/2}$.

Worked example 3.9: Relativistic corrections for silicon and germanium core states

With $\alpha = 1/137.036$, the squared velocity of a $1s$ electron is

$$(Z\alpha)^2 = \Big(\frac{14}{137.036}\Big)^2 = 0.0104 \quad \text{(silicon)},$$
$$(Z\alpha)^2 = \Big(\frac{32}{137.036}\Big)^2 = 0.0545 \quad \text{(germanium)}.$$

These are the relative sizes of the mass–velocity and Darwin terms, and of the relativistic change of the $1s$ orbitals: 1 % in silicon, 5 % in germanium. The binding energy shifts by $(Z\alpha)^2/4$ of itself, 0.26 and 1.36 %. All-electron calculations of the free atoms with the PBE functional, with and without the mass–velocity and Darwin terms, check this. The $1s$ level of silicon moves from −1781.2 to −1785.9 eV, by 0.26 %, and that of germanium from −10 803 to −10 961 eV, by 1.46 %. The screening of the nucleus by the other electrons changes the binding energies, but hardly the ratio.

The germanium $3d$ shell is not hydrogen-like, and we estimate an effective charge from its size. A hydrogen-like orbital has $\langle r\rangle = [3n^2 - l(l+1)]/2Z_{\mathrm{eff}}$ in bohr, $10.5/Z_{\mathrm{eff}}$ for $3d$. The atomic calculation gives $\langle r\rangle = 0.737$ bohr, so $Z_{\mathrm{eff}} = 14.2$, and

$$\langle v^2\rangle = \Big(\frac{Z_{\mathrm{eff}}\alpha}{n}\Big)^2 = \Big(\frac{14.2}{3\times 137.036}\Big)^2 = 1.2\times 10^{-3}.$$

The direct relativistic correction to the $3d$ states is about 0.1 %. The atomic calculation shows a larger, indirect effect of the opposite sign: the scalar-relativistic Kohn–Sham $3d$ level lies at −29.31 eV instead of −30.15 eV, 2.8 % less bound. The s and p shells inside it contract, screen the nucleus more strongly, and push the $3d$ level up. Spin–orbit coupling splits the level by 0.59 eV in the Dirac–Kohn–Sham atom, against the measured 0.6 eV, or 2 % of the binding energy [50, Table 1-1].

Dark matter lighter than 0.3 GeV reaches every shell except silicon $1s$ and germanium $2p$ and below. Each shell it reaches needs relativistic corrections of at most a few percent in energy, and less in its states. A scalar-relativistic pseudopotential or PAW data set includes the leading part, and the spin–orbit doublet of the $3d$ edge is the one correction that a spectrum resolves.

The density vertex of a real crystal can now be computed at every momentum transfer that dark matter reaches. A rate, however, needs it on a fine grid of crystal momenta, where the bands near the gap vary on scales much finer than any affordable self-consistent calculation resolves. The localized orbitals of Sect. 3.6 turn a calculation on a coarse grid into one on any grid.

3.6 Wannier functions

A conduction electron of silicon 10 meV above the band minimum has a crystal momentum within $(2m_\perp^* \times 10\,\mathrm{meV})^{1/2} = 44\,\mathrm{eV}$ of $\boldsymbol{k}_\mathrm{c}$ in the transverse direction (Table 3.1), 2 % of $2\pi/a$. A uniform grid that places a point in every such pocket needs about $|\boldsymbol{b}_i|/44\,\mathrm{eV} \approx 90$ points along each reciprocal primitive vector, close to 10^6 in the zone, and a phonon-assisted rate pairs each of them with as many phonon wavevectors (Sects. 6.5 and 13.3). Every point would be a plane-wave diagonalization, and every pair a perturbation calculation. Such grids are affordable only because the Bloch states of a group of bands can be recombined into orbitals localized within a cell or two, the Wannier functions. In their basis the Hamiltonian couples nearby cells and little else, and its Fourier series returns the bands at any $\boldsymbol{k}$ for the price of diagonalizing an 8×8 matrix. They are the computed tight-binding orbitals that Sect. 3.2 anticipated, orthonormal, with hopping integrals taken from a first-principles calculation instead of fitted. They are not unique, and asking for the most localized set fixes the freedom; the interpolation they allow is exact when the Hamiltonian has finite range in their basis, and every quantity interpolated with them must be built in one gauge. We follow Refs. [52–54].

3.6.1 From Bloch states to localized orbitals

Take first a single band n that is separated from all others by gaps throughout the zone. Its N Bloch states each spread over the whole crystal with the same weight in every cell. A superposition of them with phases that add up in one cell and cancel elsewhere is a wave packet localized near that cell:

$$w_n(\boldsymbol{r}-\boldsymbol{R}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\, \psi_{n\boldsymbol{k}}(\boldsymbol{r}), \tag{3.76}$$

with the sum over the N crystal momenta of the Born–von Karman grid, Eq. (3.13). The left side is written as a function of $\boldsymbol{r}-\boldsymbol{R}$ because it is one: Bloch's theorem in the form $\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\psi_{n\boldsymbol{k}}(\boldsymbol{r}-\boldsymbol{R})$ turns the right side into $N^{-1/2}\sum_{\boldsymbol{k}} \psi_{n\boldsymbol{k}}(\boldsymbol{r}-\boldsymbol{R})$, the function for $\boldsymbol{R}=0$ shifted by $\boldsymbol{R}$. All the Wannier functions of a band are copies of one function, one per cell. The transform inverts with the lattice sum. On the grid of Eq. (3.13) the same geometric sums that proved Eq. (3.14) give $\sum_{\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} = N\delta_{\boldsymbol{R},0}$ for lattice vectors of one Born–von Karman crystal. Multiplying Eq. (3.76) by $N^{-1/2}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}$, summing over $\boldsymbol{R}$ and using this identity to collapse the sum over crystal momenta,

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\, w_n(\boldsymbol{r}-\boldsymbol{R}). \tag{3.77}$$

This is the Bloch sum of Eq. (3.28), with the Wannier function in place of the atomic orbital. Unlike atomic orbitals, Wannier functions are orthonormal. Writing $w_{n\boldsymbol{R}}$ for $w_n(\boldsymbol{r}-\boldsymbol{R})$,

$$\langle w_{n\boldsymbol{R}} | w_{n'\boldsymbol{R}'} \rangle = \frac{1}{N} \sum_{\boldsymbol{k}\boldsymbol{k}'} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R} - \mathrm{i}\boldsymbol{k}'\cdot\boldsymbol{R}'} \langle \psi_{n\boldsymbol{k}} | \psi_{n'\boldsymbol{k}'} \rangle = \frac{1}{N} \sum_{\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{R}-\boldsymbol{R}')}\, \delta_{nn'} = \delta_{\boldsymbol{R}\boldsymbol{R}'}\delta_{nn'}.$$

The overlap matrix of the tight-binding problem, Eq. (3.29), is then the identity. The normalization of Eq. (3.20) gives w_n the dimension $\Omega^{-1/2}$, the same as $u_{n\boldsymbol{k}}$, and a function localized within a few cells has a norm that does not depend on the size of the crystal.

The construction has a loose end. A Bloch state is defined only up to a phase, and $\psi_{n\boldsymbol{k}} \to \mathrm{e}^{\mathrm{i}\theta_n(\boldsymbol{k})}\psi_{n\boldsymbol{k}}$ describes the same physical state at every $\boldsymbol{k}$. It changes the Wannier function, however, since Eq. (3.76) adds the states with new relative phases. In the periodic gauge of Eq. (A.16), with $\theta_n(\boldsymbol{k}+\boldsymbol{G}) = \theta_n(\boldsymbol{k})$, the summand $\psi_{n\boldsymbol{k}}(\boldsymbol{r})$ at fixed $\boldsymbol{r}$ is a periodic

function of $\boldsymbol{k}$. The value of w_n in cell $\boldsymbol{R}$ is then a Fourier coefficient of that function, and its decay with $|\boldsymbol{R}|$ is set by how smooth the function is. If θ_n jumps at random from one $\boldsymbol{k}$ to the next, as the phases of a numerical diagonalization do, the sum is a random walk and w_n spreads over the whole crystal. If the periodic parts $u_{n\boldsymbol{k}}$ are smooth, the Wannier function decays faster than any power of $|\boldsymbol{R}|$, and if they are analytic in a strip of half-width $1/\xi_W$ around the real $\boldsymbol{k}$ axis it decays as $\mathrm{e}^{-|\boldsymbol{R}|/\xi_W}$. In two and three dimensions such an analytic choice exists if and only if the Chern invariants of the bands vanish [55]. A Chern invariant is a topological integer of a group of bands, a count of how often the phases of its Bloch states must wind across a plane of the zone, and it vanishes in every crystal with time-reversal symmetry, silicon among them. A Wannier function is not a property of the band alone: it depends on a choice of gauge, and a good choice makes it exponentially localized.

A single isolated band is the exception. The four valence bands of silicon touch at Γ, where three of them are degenerate (Sect. 3.3), and cross elsewhere; no one of them is smooth across the zone. Their sum, the projector $P(\boldsymbol{k}) = \sum_{n=1}^{4} |u_{n\boldsymbol{k}}\rangle\langle u_{n\boldsymbol{k}}|$ on the four-dimensional space they span, is smooth, since the group is separated from the conduction bands by the gap at every $\boldsymbol{k}$. The way out is to give up Wannier functions per band and build N_W functions for the whole group of N_W bands, mixing the bands at each $\boldsymbol{k}$ with a unitary $N_W \times N_W$ matrix:

$$w_n(\boldsymbol{r}-\boldsymbol{R}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\, \psi^W_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad \psi^W_{n\boldsymbol{k}} = \sum_m U_{mn}(\boldsymbol{k})\, \psi_{m\boldsymbol{k}}, \tag{3.78}$$

which is the convention of Eq. (A.20). The gauge matrix $U(\boldsymbol{k})$ is periodic, $U(\boldsymbol{k}+\boldsymbol{G}) = U(\boldsymbol{k})$; the phases $\mathrm{e}^{\mathrm{i}\theta_n(\boldsymbol{k})}$ are its diagonal special case. The rotated states $\psi^W_{n\boldsymbol{k}}$ are still Bloch functions at $\boldsymbol{k}$, but they are no longer eigenstates of H_e, and n is no longer a band index: it labels a Wannier function. Translations, the inverse transform and orthonormality carry over from the single band, with ψ^W in place of ψ. Since a unitary mixing within the group leaves $P(\boldsymbol{k})$ unchanged, the N_W Wannier functions of a cell, together with their copies in all other cells, span exactly the states of the group [54, Eqs. (8)–(10)].

A first smooth gauge comes from projection. Choose N_W trial orbitals $\varphi^\mathrm{tr}_n(\boldsymbol{r})$ located where the Wannier functions are expected, such as sp^3 hybrids on the atoms of silicon or Gaussians at the bond centers, and compute the $N_W \times N_W$ matrix $A^\mathrm{pr}_{mn}(\boldsymbol{k}) = \langle \psi_{m\boldsymbol{k}} | \varphi^\mathrm{tr}_n \rangle$. Its Löwdin orthonormalization

$$U(\boldsymbol{k}) = A^\mathrm{pr}(\boldsymbol{k}) \left[A^{\mathrm{pr}\dagger}(\boldsymbol{k}) A^\mathrm{pr}(\boldsymbol{k}) \right]^{-1/2}, \tag{3.79}$$

is unitary, and it is smooth in $\boldsymbol{k}$ because the trial orbitals are localized [54, Eqs. (16) and (17)]. The arbitrary phases of the $\psi_{m\boldsymbol{k}}$ cancel in Eq. (3.79): a phase on $\psi_{m\boldsymbol{k}}$ multiplies row m of A^pr and is undone in ψ^W. The construction fails where $A^{\mathrm{pr}\dagger}A^\mathrm{pr}$ becomes nearly singular, that is, where the group has almost no weight on some trial orbital. For the four valence bands of silicon, four Gaussians at the bond centers are enough, and the resulting Wannier functions are centered on the four bonds of a cell, each a bonding combination of two sp^3 hybrids [54, Fig. 2].

3.6.2 Maximal localization

Projection leaves the result tied to the trial orbitals. Marzari and Vanderbilt removed this arbitrariness with a criterion. They chose the gauge that minimizes the total quadratic spread of the Wannier functions of one cell around their centers [52, Eq. (11)],

$$\Omega_W = \sum_{n=1}^{N_W} \left[\langle r^2 \rangle_n - \bar{\boldsymbol{r}}_n^{\,2} \right], \qquad \bar{\boldsymbol{r}}_n = \langle w_{n\boldsymbol{0}} | \boldsymbol{r} | w_{n\boldsymbol{0}} \rangle, \qquad \langle r^2 \rangle_n = \langle w_{n\boldsymbol{0}} | r^2 | w_{n\boldsymbol{0}} \rangle. \tag{3.80}$$

The functions that minimize Ω_W are the *maximally localized Wannier functions*, the solid-state version of the Foster–Boys orbitals of quantum chemistry.

The position operator needs care in a crystal. On the Born–von Karman torus $\boldsymbol{r}$ is not a periodic function. It does not map Bloch states into Bloch states, and $\langle\psi_{n\boldsymbol{k}}|\boldsymbol{r}|\psi_{n\boldsymbol{k}}\rangle$ depends on where the box is cut. For a localized function the matrix elements of $\boldsymbol{r}$ and r^2 are well defined, as long as the function vanishes near the edges of the box, and expressed through Bloch states the position operator becomes a derivative with respect to $\boldsymbol{k}$. To see how, let the crystal grow, $N \to \infty$, so that $N^{-1}\sum_{\boldsymbol{k}} \to \Omega(2\pi)^{-3}\int_{\mathrm{BZ}} \mathrm{d}^3k$, and write $u^W_{n\boldsymbol{k}} = \sum_m U_{mn}(\boldsymbol{k}) u_{m\boldsymbol{k}}$ for the periodic parts of the rotated states. Eq. (3.78) with Eq. (3.20) reads $w_n(\boldsymbol{r}) = \Omega(2\pi)^{-3}\int \mathrm{d}^3k\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} u^W_{n\boldsymbol{k}}(\boldsymbol{r})$. Writing $\boldsymbol{r}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} = -\mathrm{i}\nabla_{\boldsymbol{k}}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ and integrating by parts in $\boldsymbol{k}$,

$$\boldsymbol{r}\, w_n(\boldsymbol{r}) = \frac{\Omega}{(2\pi)^3}\int \mathrm{d}^3k\; \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\; \mathrm{i}\nabla_{\boldsymbol{k}} u^W_{n\boldsymbol{k}}(\boldsymbol{r}).$$

The boundary terms cancel because $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}u^W_{n\boldsymbol{k}}$ is periodic in $\boldsymbol{k}$ in the periodic gauge. Multiplying by $w^*_m(\boldsymbol{r}-\boldsymbol{R})$, written in the same way with a second momentum $\boldsymbol{k}'$, and integrating over all space leaves $\int \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}(\boldsymbol{k}-\boldsymbol{k}')\cdot\boldsymbol{r}} f(\boldsymbol{r})$ with $f = u^{W*}_{m\boldsymbol{k}'}\nabla_{\boldsymbol{k}} u^W_{n\boldsymbol{k}}$ periodic. Only the $\boldsymbol{G}=0$ component of f survives for $\boldsymbol{k}$ and $\boldsymbol{k}'$ in the zone, and the integral is $(2\pi)^3\Omega^{-1}\delta^3(\boldsymbol{k}-\boldsymbol{k}')\int_\Omega f$. Hence [52, Eqs. (4) and (8)]

$$\langle w_{m\boldsymbol{R}}|\boldsymbol{r}|w_{n\boldsymbol{0}}\rangle = \mathrm{i}\,\frac{\Omega}{(2\pi)^3}\int \mathrm{d}^3k\; \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\,\langle u^W_{m\boldsymbol{k}}|\nabla_{\boldsymbol{k}} u^W_{n\boldsymbol{k}}\rangle_\Omega, \qquad \langle r^2\rangle_n = \frac{\Omega}{(2\pi)^3}\int \mathrm{d}^3k\; \|\nabla_{\boldsymbol{k}} u^W_{n\boldsymbol{k}}\|^2_\Omega, \tag{3.81}$$

where the second form is the norm of $\boldsymbol{r}w_n$ obtained in the same way, and $\langle\cdot|\cdot\rangle_\Omega$ is an integral over one cell. The center $\bar{\boldsymbol{r}}_n$ is the zone average of the Berry connection $\mathrm{i}\langle u^W_{n\boldsymbol{k}}|\nabla_{\boldsymbol{k}} u^W_{n\boldsymbol{k}}\rangle$. A gauge change $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}'}$ on $u^W_{n\boldsymbol{k}}$, which keeps $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}u^W_{n\boldsymbol{k}}$ periodic in $\boldsymbol{k}$ only for a lattice vector $\boldsymbol{R}'$, moves the center by $\boldsymbol{R}'$. The center is defined only modulo a lattice vector. This is the ambiguity of the Berry-phase polarization (Eq. (E.26)): the sum of the centers of the valence group, times $-2e/\Omega$ for spin-degenerate bands, is the electronic polarization.

A calculation knows the states on a grid, and the derivatives in Eq. (3.81) become finite differences between neighboring points $\boldsymbol{k}$ and $\boldsymbol{k}+\Delta\boldsymbol{k}$. All the information is in the overlaps $M_{mn}(\boldsymbol{k},\Delta\boldsymbol{k}) = \langle u^W_{m\boldsymbol{k}}|u^W_{n,\boldsymbol{k}+\Delta\boldsymbol{k}}\rangle_\Omega$, and with weights $\lambda_{\Delta\boldsymbol{k}}$ fixed by the geometry of the grid, $\sum_{\Delta\boldsymbol{k}}\lambda_{\Delta\boldsymbol{k}}\Delta k_\alpha\Delta k_\beta = \delta_{\alpha\beta}$, the forms used in practice are [52, Eqs. (31) and (32)]

$$\begin{aligned}\bar{\boldsymbol{r}}_n &= -\frac{1}{n_k}\sum_{\boldsymbol{k},\Delta\boldsymbol{k}} \lambda_{\Delta\boldsymbol{k}}\,\Delta\boldsymbol{k}\,\operatorname{Im}\ln M_{nn}(\boldsymbol{k},\Delta\boldsymbol{k}),\\ \langle r^2\rangle_n &= \frac{1}{n_k}\sum_{\boldsymbol{k},\Delta\boldsymbol{k}} \lambda_{\Delta\boldsymbol{k}}\Big\{1-|M_{nn}|^2+\big[\operatorname{Im}\ln M_{nn}\big]^2\Big\},\end{aligned} \tag{3.82}$$

for a grid of n_k points, which is a crystal of n_k cells. They agree with Eq. (3.81) to first and second order in Δk, and they are chosen, among many such expressions, because the logarithm makes the center shift by exactly $\boldsymbol{R}'$ when the Wannier function is translated by $\boldsymbol{R}'$. The branch of the logarithm, defined modulo 2π, is the discrete form of the ambiguity of the center.

The spread splits into a part that no gauge can change and a part that the gauge controls. Inserting $1 = P + Q$ between the two factors of $r^2 = \sum_\alpha r_\alpha r_\alpha$, with $P = \sum_{m\boldsymbol{R}} |w_{m\boldsymbol{R}}\rangle\langle w_{m\boldsymbol{R}}|$ the projector on the whole group, and $Q = 1 - P$, gives $\langle r^2\rangle_n = \sum_\alpha \langle w_{n\boldsymbol{0}}|r_\alpha Q r_\alpha|w_{n\boldsymbol{0}}\rangle + \sum_{m\boldsymbol{R}}|\langle w_{m\boldsymbol{R}}|\boldsymbol{r}|w_{n\boldsymbol{0}}\rangle|^2$. The term with $(m,\boldsymbol{R}) = (n,\boldsymbol{0})$ is $\bar{\boldsymbol{r}}_n^2$, and [52, Eqs. (13)–(16)]

$$\Omega_W = \Omega_W^{\mathrm{I}} + \tilde{\Omega}_W, \qquad \Omega_W^{\mathrm{I}} = \sum_\alpha \operatorname{Tr}_{\mathrm{c}}[P r_\alpha Q r_\alpha], \qquad \tilde{\Omega}_W = \sum_n \sum_{(m,\boldsymbol{R})\neq(n,\boldsymbol{0})} \big|\langle w_{m\boldsymbol{R}}|\boldsymbol{r}|w_{n\boldsymbol{0}}\rangle\big|^2, \tag{3.83}$$

with $\mathrm{Tr_c}$ a trace per cell. Both parts are positive. The first depends on the projector P alone, which no unitary mixing within the group changes, and it is gauge invariant; it measures how fast the space spanned by the group turns as $\boldsymbol{k}$ moves through the zone. Minimizing Ω_W therefore means minimizing $\tilde{\Omega}_W$, the sum of the squared matrix elements of $\boldsymbol{r}$ between different Wannier functions. In one dimension $\tilde{\Omega}_W$ can be made to vanish, by choosing the Wannier functions as eigenstates of the projected position operator PxP. In three dimensions PxP, PyP and PzP do not commute, and the maximally localized functions are the best compromise between diagonalizing all three [52, Sect. III]. The condition for that compromise follows from an infinitesimal gauge change $U(\boldsymbol{k}) \to U(\boldsymbol{k})[1 + \mathrm{d}W(\boldsymbol{k})]$, with $\mathrm{d}W$ anti-Hermitian. Maximal localization requires $\mathrm{d}\Omega_W = 0$ for every $\mathrm{d}W(\boldsymbol{k})$, that is [52, Eq. (52)]

$$\sum_{\Delta\boldsymbol{k}} \lambda_{\Delta\boldsymbol{k}} \Big(\mathcal{A}\big[\mathsf{R}(\boldsymbol{k},\Delta\boldsymbol{k})\big] - \mathcal{S}\big[\mathsf{T}(\boldsymbol{k},\Delta\boldsymbol{k})\big] \Big) = 0 \quad \text{at every } \boldsymbol{k}, \tag{3.84}$$

with $\mathsf{R}_{mn} = M_{mn} M^*_{nn}$, $\mathsf{T}_{mn} = (M_{mn}/M_{nn})\,[\mathrm{Im}\ln M_{nn} + \Delta\boldsymbol{k}\cdot\bar{\boldsymbol{r}}_n]$, and the superoperators $\mathcal{A}[B] = (B - B^\dagger)/2$ and $\mathcal{S}[B] = (B + B^\dagger)/2\mathrm{i}$. The left side, multiplied by four, is the gradient of Ω_W with respect to $\mathrm{d}W(\boldsymbol{k})$; codes follow it downhill from the projected gauge of Eq. (3.79), which also steers them away from the false minima that a random starting gauge can fall into.

For the eight sp^3 functions of silicon a further step is needed. The four valence bands are an isolated group, but the four lowest conduction bands are not: above the conduction minimum they cross and mix with higher bands of other character, and no set of eight bands is separated from the rest everywhere in the zone. The eight-dimensional space to be Wannierized must first be chosen at each $\boldsymbol{k}$ from a larger set of bands, which makes $U(\boldsymbol{k})$ a rectangular matrix, with more rows than columns. This is *disentanglement*. Souza, Marzari and Vanderbilt chose the space by the same smoothness criterion. Among all N_W-dimensional subspaces of the bands in an outer energy window, they take the one that minimizes the gauge-invariant spread [53, Eqs. (8) and (9)],

$$\Omega_W^{\mathrm{I}} = \frac{1}{n_k} \sum_{\boldsymbol{k},\Delta\boldsymbol{k}} \lambda_{\Delta\boldsymbol{k}}\, \mathrm{Tr}\big[P(\boldsymbol{k})\, Q(\boldsymbol{k}+\Delta\boldsymbol{k})\big], \tag{3.85}$$

where $P(\boldsymbol{k})$ projects on the chosen subspace at $\boldsymbol{k}$ and $Q = 1 - P$. Each trace is the spillage of the subspace at one point out of the subspace at its neighbor, and the minimum is the subspace that turns least from point to point. States inside an inner, or frozen, window are kept in the subspace unchanged, so that the bands there are reproduced exactly; $\tilde{\Omega}_W$ is then minimized within the chosen subspace as before.

Fig. 3.12 shows one of the resulting functions for silicon. It is an sp^3-like hybrid on one atom, with its center 0.80 Å from the nucleus. Its main lobe points away from the bonds, opposite to one of them and toward the empty tetrahedral site of the diamond structure, and small lobes of opposite sign sit on the bonds. Its amplitude falls by two orders of magnitude within 6 Å. The hybrids that chemistry would draw, pointing along the bonds, are also a stationary point of Ω_W, but a less localized one. The more localized set is the one that points away from the bonds. On a 6^3 grid their total spread is 19.8 Å^2, against 16.2 Å^2 for the functions of Fig. 3.12, with the same $\Omega_W^{\mathrm{I}} = 14.0$ Å^2. The order does not depend on these settings: with outer windows up to 16 eV, a frozen window up to 1 eV, or an 8^3 grid, the totals move by up to 2.6 Å^2, but the set that points away from the bonds stays the lower one by 3.5 to 3.7 Å^2, and a minimization started from randomly placed trial orbitals ends in it. The silicon examples distributed with `wannier90` agree [56]. Started from symmetric sp^3 hybrids, the eight functions end pointing away from the bonds. Started from hybrids with the same orientation on both atoms, which point toward the bonds on one atom and away from them on the other, the minimization stops at a mixed stationary point. The four functions on

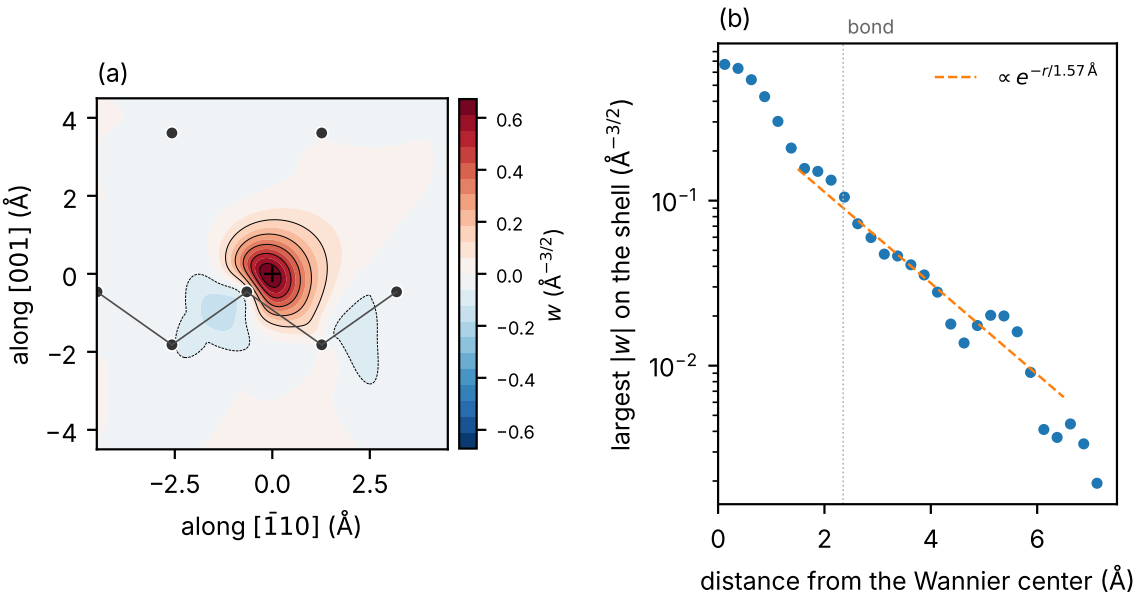


Figure 3.12: A maximally localized sp^3 Wannier function of silicon, one of eight disentangled from the 16 lowest bands. (a) The function on the (110) plane through the atom on which it sits (cross: Wannier center; dots: atoms in the plane; lines: bonds in the plane). The plane contains the [$\bar{1}$11] axis of the function and the bond opposite to it. The function is real and normalized to one; red and blue are opposite signs. (b) Largest $|w|$ on spherical shells 0.25 Å wide around the center, against distance, with an exponential fit between 2 and 5.5 Å (dashed), whose decay length is 1.57 Å. Calculation: PBE, norm-conserving SG15 pseudopotential, 40 Ry cutoff, $a = 5.431$ Å, 6^3 coarse grid, outer window up to 12 eV and frozen window up to 2.5 eV above the valence-band maximum, minimization started from sp^3 hybrids on both atoms pointing away from the bonds [13–15, 56]. The spread of each function is 2.02 Å^2; of the total $\Omega_W = 16.2$ Å^2, the gauge-invariant part Ω_W^{I} is 14.0 Å^2.

one atom point away from the bonds, with spreads of 2.02 Å^2 and centers 0.79 Å from the nucleus, and the four on the other point along them, with 2.37 Å^2. Both of our sets are real. Maximally localized functions in a crystal with time-reversal symmetry are found to be real, as Marzari and Vanderbilt observed, but no general proof exists [52]; reality, a common check that a minimization has converged, therefore does not tell the two sets apart. Which of the two a minimization finds depends on the trial orbitals it starts from, an example of the false minima mentioned above. Where the windows go, which trial orbitals start the minimization and how the result is checked are questions of practice, taken up in Appendix E.5.

3.6.3 Wannier interpolation

The Hamiltonian between Wannier functions follows from Eq. (3.78). Since H_{e} does not connect different $\boldsymbol{k}$, and $\langle \psi^W_{m\boldsymbol{k}} | H_{\mathrm{e}} | \psi^W_{n\boldsymbol{k}} \rangle = \sum_{m'} U^*_{m'm}(\boldsymbol{k}) \varepsilon_{m'\boldsymbol{k}} U_{m'n}(\boldsymbol{k})$, the double sum over $\boldsymbol{k}$ collapses to one:

$$H^W_{mn}(\boldsymbol{R}) \equiv \langle w_{m\boldsymbol{0}} | H_{\mathrm{e}} | w_{n\boldsymbol{R}} \rangle = \frac{1}{N} \sum_{\boldsymbol{k}} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} \left[U^\dagger(\boldsymbol{k})\, E(\boldsymbol{k})\, U(\boldsymbol{k}) \right]_{mn}, \qquad E(\boldsymbol{k}) = \mathrm{diag}(\varepsilon_{n\boldsymbol{k}}). \tag{3.86}$$

This real-space Hamiltonian is an $N_W \times N_W$ matrix for each lattice vector. Its diagonal elements at $\boldsymbol{R} = 0$ are the on-site energies of the Wannier functions, the others are hopping integrals, and since the Wannier functions are localized it decays with $|\boldsymbol{R}|$. Its Fourier series is, by the lattice sum,

$$H^W(\boldsymbol{k}) \equiv \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\, H^W(\boldsymbol{R}) = U^\dagger(\boldsymbol{k})\, E(\boldsymbol{k})\, U(\boldsymbol{k}), \tag{3.87}$$

a Hermitian $N_W \times N_W$ matrix whose eigenvalues are the band energies $\varepsilon_{n\boldsymbol{k}}$ and whose eigenvectors, the columns of $U^\dagger(\boldsymbol{k})$, convert the Wannier gauge back into band states [54, Eqs. (98) and (99)]. Eq. (3.87) is the tight-binding eigenvalue problem of Eq. (3.30), with orthonormal orbitals, so that $S^{\mathrm{ov}} = 1$, and with every hopping computed. In practice the Bloch states come from a plane-wave Kohn–Sham calculation (Sect. 3.4) on a uniform grid; the code supplies the overlaps M_{mn} and the projections A^{pr}_{mn}, a separate program minimizes the spread, and $H^W(\boldsymbol{R})$ follows from Eq. (3.86) on the same grid. How the windows and the grid are chosen, and how the result is tested, is in Appendix E.5.

The point of Eq. (3.87) is that its left side can be evaluated at any $\boldsymbol{k}$, not only at the points of the grid on which $H^W(\boldsymbol{R})$ was computed. On a coarse grid of $n_k = n_1 n_2 n_3$ points, the sum in Eq. (3.86) defines the coarse-grid matrix

$$\tilde{H}^W(\boldsymbol{R}) = \frac{1}{n_k} \sum_{j=1}^{n_k} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{R}}\, H^W(\boldsymbol{k}_j).$$

Inserting Eq. (3.87) for $H^W(\boldsymbol{k}_j)$ and using the lattice sum on the coarse grid, $n_k^{-1}\sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{R}} = \sum_{\boldsymbol{L}_{\mathrm{sc}}} \delta_{\boldsymbol{R},\boldsymbol{L}_{\mathrm{sc}}}$, where $\boldsymbol{L}_{\mathrm{sc}} = \sum_i l_i n_i \boldsymbol{a}_i$ (l_i integers) are the lattice vectors of the supercell that the coarse grid makes periodic, we find

$$\tilde{H}^W(\boldsymbol{R}) = \sum_{\boldsymbol{L}_{\mathrm{sc}}} H^W(\boldsymbol{R} + \boldsymbol{L}_{\mathrm{sc}}). \tag{3.88}$$

The coarse grid does not lose the hoppings; it folds them onto one supercell, so that a hopping to $\boldsymbol{R} + \boldsymbol{L}_{\mathrm{sc}}$ is added to the hopping to $\boldsymbol{R}$. This aliasing is the only error of the interpolation. Suppose $H^W(\boldsymbol{R})$ vanishes beyond some range $R_{\max}$, and the supercell is large enough that every $\boldsymbol{R}$ with $|\boldsymbol{R}| \le R_{\max}$ lies inside its Wigner–Seitz cell. Then at most one term of Eq. (3.88) is nonzero for each $\boldsymbol{R}$ of that cell, and $\tilde{H}^W(\boldsymbol{R}) = H^W(\boldsymbol{R})$ there. A lattice vector on the boundary of the Wigner–Seitz cell belongs to two or more cells; its term is shared among its images with weights that add up to one, which restores the same result.

Exactness of Wannier interpolation

If the real-space Hamiltonian $H^W(\boldsymbol{R})$ has finite range $R_{\max}$, and the supercell of the coarse grid contains the sphere of radius $R_{\max}$ in its Wigner–Seitz cell, then

$$\sum_{\boldsymbol{R}\in\mathrm{WS}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\, \tilde{H}^W(\boldsymbol{R}) = H^W(\boldsymbol{k}) \quad \text{for every } \boldsymbol{k}, \tag{3.89}$$

and its eigenvalues are the exact bands $\varepsilon_{n\boldsymbol{k}}$, on the coarse grid and between its points. The sum runs over the lattice vectors of the Wigner–Seitz supercell, with fractional weights on its boundary.

Real Wannier functions are not of finite range. They decay exponentially, and so does $H^W(\boldsymbol{R})$. The error of the interpolation is then the aliased tail of Eq. (3.88), of the size of H^W at a distance of half a supercell, and it falls exponentially as the coarse grid is refined; Worked

example 3.10 shows both behaviors in a model. The gauge enters at this point. The bands at the grid points do not depend on $U(\boldsymbol{k})$, but the range of $H^W(\boldsymbol{R})$ does, and the most localized Wannier functions make the shortest-ranged Hamiltonian and the fastest convergence. With disentanglement, the interpolated bands are exact only inside the frozen window; outside it they are the bands of the selected subspace.

Fig. 3.13 applies this to silicon, where the convergence turns out slower than the size of the Wannier functions suggests. The four bond-centered functions of the valence bands have a total spread of 7.58 Å^2 on a 6^3 grid, close to the 7.61 Å^2 of Marzari and Vanderbilt [52, Table I]. Interpolated from 4^3, 6^3 and 8^3 grids, their bands deviate from a direct calculation by at most 350, 120 and 51 meV, about a factor 2.5 for every two added points per direction. The reason is visible in $H^W(\boldsymbol{R})$. Its elements fall to 0.02 eV at 5.4 Å in most directions, but along the twelve $\langle 110 \rangle$ directions, in which the bonds of the diamond structure form zigzag chains, they are 0.12 eV two lattice vectors away (7.7 Å) and 0.03 eV three away (11.5 Å). These are the hoppings that the coarse grids fold back. For the eight sp^3 functions on the 8^3 grid, the valence bands are within 40 meV and the conduction bands below 1.5 eV within 84 meV. The largest error sits near the conduction minimum, which lies between the grid points, and calculations of rates use denser coarse grids for the conduction bands for this reason (Appendix E.5).

Derivatives come for free. Differentiating Eq. (3.87) analytically, $\nabla_{\boldsymbol{k}} H^W(\boldsymbol{k}) = \sum_{\boldsymbol{R}} \mathrm{i}\boldsymbol{R}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} H^W(\boldsymbol{R})$, and by the Hellmann–Feynman argument of Eq. (3.41) its diagonal elements in the band basis are the group velocities. The interband velocities need more. By Eq. (3.42), $\boldsymbol{v}_{mn} = (\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}}) \langle u_{m\boldsymbol{k}} | \nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}} \rangle$ for $m \neq n$, and the overlap on the right is interpolated from the matrix elements of $\boldsymbol{r}$ between Wannier functions, Eq. (3.81), rotated with the same $U(\boldsymbol{k})$ [54, Eqs. (106)–(109)]; the Hamiltonian alone would treat the Wannier functions as point-like orbitals.

3.6.4 One gauge for all interpolated quantities

Any operator that conserves crystal momentum interpolates the same way. If $O_{mn}(\boldsymbol{k}) = \langle \psi_{m\boldsymbol{k}} | O | \psi_{n\boldsymbol{k}} \rangle$ is known on the grid, its Wannier-gauge form $U^\dagger(\boldsymbol{k}) O(\boldsymbol{k}) U(\boldsymbol{k})$ is transformed to real space as in Eq. (3.86), summed back at the new $\boldsymbol{k}$ as in Eq. (3.87), and rotated into the band basis with the eigenvectors of $H^W(\boldsymbol{k})$ at that point. Vertices that change the crystal momentum, the density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})$ and the electron–phonon vertex, carry one gauge matrix at each end, $U^\dagger(\boldsymbol{k}+\boldsymbol{q})\, \rho(\boldsymbol{k}, \boldsymbol{q})\, U(\boldsymbol{k})$, and are interpolated in the same way (Eq. (E.36)). The step back into the band basis uses the eigenvectors of the interpolated Hamiltonian. As a result, the band index of an interpolated vertex refers to the interpolated bands only if the vertex and the Hamiltonian were rotated into the Wannier gauge with the same $U(\boldsymbol{k})$. That is the rule of Appendix A.5, and the reason is short. Suppose $O(\boldsymbol{k})$ was computed from states that differ from those behind $U(\boldsymbol{k})$ by phases, $\psi_{m\boldsymbol{k}} \to \mathrm{e}^{\mathrm{i}\theta_m(\boldsymbol{k})} \psi_{m\boldsymbol{k}}$, as two independent diagonalizations produce. With $D(\boldsymbol{k}) = \operatorname{diag} \mathrm{e}^{\mathrm{i}\theta_m(\boldsymbol{k})}$, the matrix that is transformed is $U^\dagger D^\dagger O D\, U$, the operator in the gauge DU. That gauge jumps at random from point to point, its real-space form does not decay, and Eq. (3.88) folds an unbounded tail onto the supercell. The interpolated matrix elements between the grid points are then wrong by amounts as large as the elements themselves. At the grid points the moduli $|O_{mn}|$ of nondegenerate bands survive, because there the error is only a phase. A check made on the grid does not reveal it. For degenerate bands, such as the valence-band maximum of silicon at Γ, D is a unitary mixing and even the moduli are lost. The interpolated energies carry no trace of the mismatch at all, since H^W was rotated with its own states.

A first-order rate contains only $|\rho_{mn}|^2$ between fixed bands, so a gauge mismatch spoils it between the grid points but not on them. A second-order amplitude is not protected even on

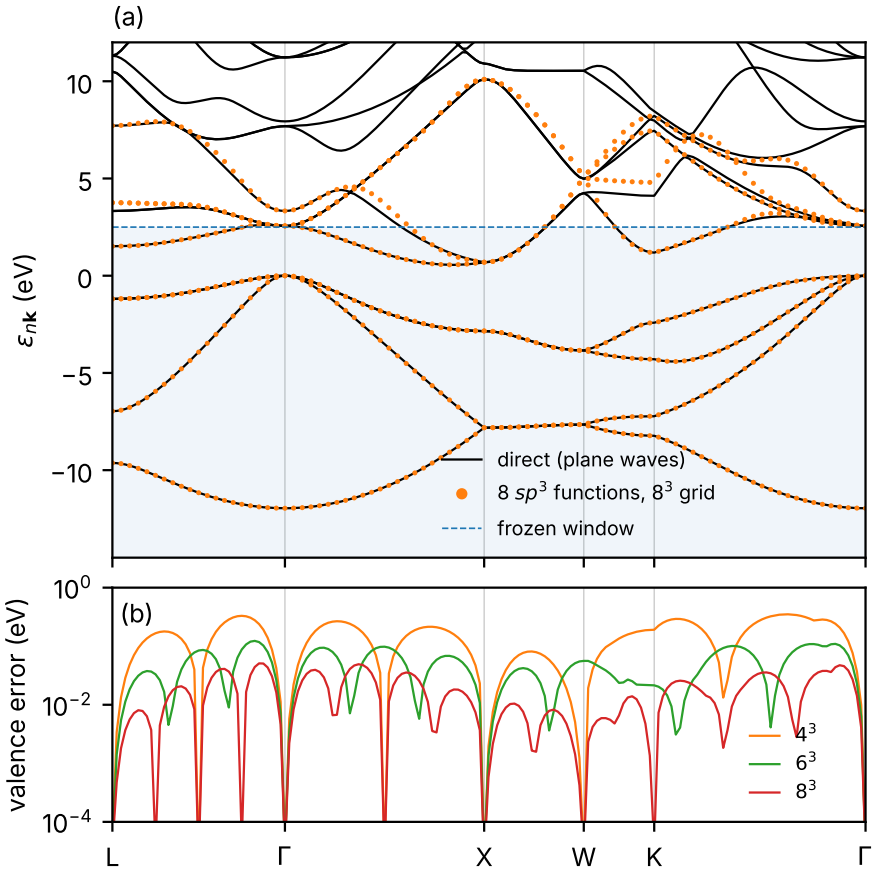


Figure 3.13: Wannier interpolation of the bands of silicon. (a) Bands computed directly in plane waves at each point of the path (lines) and interpolated from the eight sp^3 Wannier functions of Fig. 3.12, built on an 8^3 coarse grid (dots), relative to the valence-band maximum. Inside the frozen window (shaded, up to 2.5 eV) the Wannier bands follow the four valence and the lowest conduction bands; above it they follow the selected subspace, not the true bands, and the higher bands (lines without dots) are outside the Wannier manifold. (b) Largest error of the four valence bands interpolated from four bond-centered Wannier functions, an isolated group, for coarse grids of 4^3, 6^3 and 8^3 points. It vanishes at Γ, X and L, which lie on all three grids. PBE calculation as in Fig. 3.12; its gap, 0.557 eV at $0.843\,(2\pi/a)$ along ΓX, is the Kohn–Sham gap of Sect. 3.4, not the measured 1.17 eV.

the grid. It sums products such as $g_{cm\nu}\rho_{mv}$ over the intermediate band m (Sect. 6.5), and the phase of band m cancels between the two factors only if both were built from the same state. Every quantity that enters one amplitude is therefore computed from one set of wavefunctions and rotated with one set of gauge matrices; Appendix E.5 describes how this is organized in practice. One further restriction concerns dark matter. The density vertex depends on the momentum transfer $\boldsymbol{q}$, which is a parameter, not a crystal momentum of the grid, and at $q \sim q_0$ the factor $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ varies over a single Wannier function. Its real-space form must then be built separately for each $\boldsymbol{q}$ (Appendix E.5).

Worked example 3.10: A two-band chain interpolated from a coarse grid

A chain of lattice constant a has two orthonormal orbitals per cell, on sites a and b, with on-site energies $E_\mathrm{a} = -E_\mathrm{b} = 0.5\,\mathrm{eV}$, a hopping matrix element $-t_\mathrm{hop}$ between the two sites of a cell and $-t'_\mathrm{hop}$ between site a of one cell and site b of the cell to its left, with $t_\mathrm{hop} = 2.0\,\mathrm{eV}$ and $t'_\mathrm{hop} = 1.0\,\mathrm{eV}$. The numbers are illustrative, not fitted to a material. In the basis of Bloch sums of the two orbitals, Eq. (3.29) gives

$$H(k) = \begin{pmatrix} E_\mathrm{a} & -t_\mathrm{hop} - t'_\mathrm{hop}\mathrm{e}^{-\mathrm{i}ka} \\ -t_\mathrm{hop} - t'_\mathrm{hop}\mathrm{e}^{\mathrm{i}ka} & -E_\mathrm{a} \end{pmatrix},$$
$$\varepsilon_{\pm,k} = \pm\left(E_\mathrm{a}^2 + t_\mathrm{hop}^2 + t_\mathrm{hop}'^2 + 2t_\mathrm{hop}t'_\mathrm{hop}\cos ka\right)^{1/2}.$$

The two bands span $\pm 1.118\,\mathrm{eV}$ to $\pm 3.041\,\mathrm{eV}$, with a gap of $2.236\,\mathrm{eV}$ at $k = \pi/a$. We treat the model as a code would. On a grid of n_k points $k_j = 2\pi j/(n_k a)$ we diagonalize $H(k_j)$ and give each eigenvector a random phase. We then build the gauge by projection, Eq. (3.79), on the two orbitals, and compute $\tilde{H}^W(R)$ on the Wigner–Seitz supercell, $R = -\lfloor n_k/2\rfloor a, \ldots, \lfloor n_k/2\rfloor a$, with weight $\frac{1}{2}$ on the two end points when n_k is even. Finally we diagonalize the interpolated $H^W(k)$ at 2001 points between $-\pi/a$ and π/a and compare with $\varepsilon_{\pm,k}$.

Two Wannier functions. For the pair of bands the projection returns $U(k_j) = C^\dagger(k_j)$, with C the eigenvector matrix, and the random phases cancel. The Wannier functions are the two orbitals themselves, and $H^W(R)$ has range a: on-site energies and $-t_\mathrm{hop}$ at $R = 0$, $-t'_\mathrm{hop}$ at $R = \pm a$, nothing beyond. The exactness condition of Eq. (3.89) requires the supercell to hold $R = -a, 0, a$ inside its Wigner–Seitz cell, that is $n_k \geq 3$. With $n_k = 3$, 4 and 6 the largest band error is $1 \times 10^{-15}\,\mathrm{eV}$ to $3 \times 10^{-15}\,\mathrm{eV}$, rounding error. With $n_k = 2$ the hoppings to $R = +a$ and $R = -a$ fall on the same boundary point, they are not equal (one couples a to b, the other b to a), and the error is $0.24\,\mathrm{eV}$.

One Wannier function. The lower band alone is an isolated band, and its Wannier function is a combination of both orbitals in many cells. Its $H^W(R)$ is the set of Fourier coefficients of $\varepsilon_{-,k}$, which do not depend on the gauge. It decays as $|R|^{-3/2}\mathrm{e}^{-|R|/\xi_W}$, where $1/\xi_W$ is the distance of the nearest branch point of $\varepsilon_{-,k}$ from the real axis, at $\cos ka = -(E_\mathrm{a}^2 + t_\mathrm{hop}^2 + t_\mathrm{hop}'^2)/(2t_\mathrm{hop}t'_\mathrm{hop}) = -1.3125$, so that $a/\xi_W = \operatorname{arccosh} 1.3125 = 0.771$. In numbers, $|H^W(R)| = 2.19$, 0.468, 0.0525 and $0.0120\,\mathrm{eV}$ for $R/a = 0$ to 3, and the ratio of successive values approaches $(R/(R+1))^{3/2}\mathrm{e}^{-a/\xi_W}$, with $\mathrm{e}^{-a/\xi_W} = 0.462$. The interpolation error is now never zero. It is $51\,\mathrm{meV}$ for $n_k = 4$, $4.7\,\mathrm{meV}$ for 8, $0.64\,\mathrm{meV}$ for 12 and $2.4\,\mu\mathrm{eV}$ for 24, approaching a factor $\mathrm{e}^{-a/2\xi_W} = 0.68$ per added grid point, as the aliased tail at half a supercell predicts; at small n_k the power-law factor makes the fall somewhat faster. A tolerance of $1\,\mathrm{meV}$ needs 12 points.

One gauge. For the gauge test we take the matrix $\partial H/\partial k$ of the model, whose band-diagonal elements are the group velocities, and interpolate its interband element $|(\partial H/\partial k)_{+-}|$, which ranges between 0.27 and $1.00\,\mathrm{eV}\,a$ along the zone. With $n_k = 6$ we compute it in

the band basis of one set of random phases. Rotated into the Wannier gauge with the eigenvectors of that set, it interpolates to within $10^{-15}\,\mathrm{eV}\,a$ everywhere. Rotated instead with the eigenvectors of a second, independent set, which describe the same states with other phases, it is correct in modulus at the six grid points and off between them by amounts that depend on the random phases, up to 0.3 to 0.8 eV a in different draws, as large as the matrix element itself. The interpolated bands are exact in both cases.

The interpolated energies, velocities and vertices on fine grids are the raw material of the phonon-assisted rates of Sects. 6.5 and 13.3. Wannier functions make the electron–phonon vertex as short-ranged as the Hamiltonian, which is what lets those grids be filled (Sect. 6.2), except for the long-range dipole part of polar crystals, which is split off analytically (Appendix E.5).

Summary

With the nuclei clamped at their equilibrium sites, a crystal is a Bravais lattice with a basis (Sect. 3.1). Silicon is a face-centered cubic lattice with two atoms per cell, $a = 5.431\,\text{Å}$ and $\Omega = 40.05\,\text{Å}^3$. Its reciprocal lattice is body-centered cubic, and the shortest reciprocal-lattice vector, $|\boldsymbol{G}_{111}| = \sqrt{3}\,2\pi/a = 3.95\,\mathrm{keV}$, is close to the typical momentum q_0 of electron recoils. Translation symmetry assigns every state a crystal momentum, defined modulo $\boldsymbol{G}$. The lattice sum $\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} = N \sum_{\boldsymbol{G}} \delta_{\boldsymbol{k},\boldsymbol{G}}$ turns it into a selection rule: dark matter that transfers $\boldsymbol{q}$ changes the crystal momentum of the target by $\boldsymbol{q}$ up to a reciprocal-lattice vector, which the lattice absorbs without recoil energy. The momentum transfer itself is never folded into the zone, since the halo integral needs $|\boldsymbol{q}|$.

For independent electrons in a periodic potential the states are Bloch states, $\psi_{n\boldsymbol{k}} = N^{-1/2}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}u_{n\boldsymbol{k}}$, with band energies $\varepsilon_{n\boldsymbol{k}}$ (Sect. 3.2). Nearly free electrons and tight binding are the two limits; the second, with sp^3 bonds, accounts for the four filled bands of silicon and for its bond charge. Every electronic rate in the book contains the density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$, an overlap of periodic parts that keeps the Umklapp vector in its plane wave and obeys $\sum_m |\rho_{mn}|^2 = 1$. At small q it reduces to a velocity matrix element, but the momentum transfers of light dark matter lie at or beyond the edge of that optical limit, and the full vertex is needed.

Silicon has an indirect gap of 1.17 eV at low temperature, between the valence-band maximum at Γ and six conduction valleys at $0.85\,(2\pi/a)$ along the cube axes, with valley masses $0.916\,m_e$ and $0.191\,m_e$; its smallest vertical gap is 3.4 eV (Sect. 3.3). Germanium is indirect with its minima at L, GaAs direct. Effective masses follow from $\boldsymbol{k}\cdot\boldsymbol{p}$ theory, and spin–orbit coupling splits the valence-band maximum by an amount inherited from the atoms. The minimum pair energy $E_{\min}(\boldsymbol{q})$ combines bands and kinematics into the condition for a zero-phonon transition. In silicon it shows that dark matter just above the kinematic threshold mass of 0.333 MeV cannot reach the conduction minimum without a phonon, which opens a phonon-assisted window of masses.

The effective potential comes from the many-electron problem (Sect. 3.4). Hartree–Fock exchange is unscreened and opens the gap far too wide; the Kohn–Sham equations of density-functional theory reproduce the ground-state density and energy, and with local or semilocal functionals give good lattice constants, band shapes and masses. Their eigenvalues are not quasiparticle energies, however, and the Kohn–Sham gap of silicon is about half the measured one, the band-gap problem. A gap of 0.52 eV would put the threshold mass at 0.148 MeV instead of 0.333 MeV. The standard repair, a scissor shift of the conduction bands, leaves

the density vertex unchanged and rescales interband velocities by the ratio of corrected to Kohn–Sham transition energies.

The states themselves are computed in plane waves with pseudopotentials (Sect. 3.5). A vertex at momentum transfer q needs plane waves up to about q plus the momentum of the final state, far beyond the cutoff that converges a total energy. A pseudopotential replaces the core region by a smooth potential whose valence states agree with the true ones outside a core radius, and norm conservation fixes their charge there. Above about a keV, the density vertex depends on the oscillations of the valence states near the nuclei, which the all-electron reconstruction of the projector-augmented-wave method restores, and core and semicore shells are added as atomic orbitals where kinematics opens them. Relativistic corrections grow as $(Z\alpha)^2$ and stay at a few percent for every shell that dark matter lighter than about 0.3 GeV can reach.

Wannier functions, finally, recombine the Bloch states of a group of bands into orthonormal orbitals localized within a cell or two (Sect. 3.6). They depend on a gauge $U(\boldsymbol{k})$, which maximal localization of the spread Ω_W fixes, and disentanglement extends them to bands that cross others, as the conduction bands of silicon do. The Hamiltonian between them, $H^W(\boldsymbol{R})$, has short range, and its Fourier series interpolates the bands exactly when that range fits inside the supercell of the coarse grid, and with an exponentially small error otherwise. Every quantity interpolated together must be rotated with one gauge. The chapter has produced the independent-particle states and energies of silicon, corrected for the gap, and the means to sample them as finely as a rate requires. What it has not included is the response of the other electrons to the perturbation itself, the screening that Chap. 4 adds.

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Chapter 4

Linear response

A dark-matter particle that scatters in silicon meets 2.00×10^{23} valence electrons per cubic centimeter (Appendix A.2), and they do not respond independently. They repel one another, so when the particle's potential pushes on some of them the others move to screen it, and at long wavelengths all of them can oscillate together, near 16.4 eV (Sect. 4.4). The thermal golden rule of Sect. 2.5 has room for all of this, since it puts the whole target into one function, the dynamic structure factor $S(\boldsymbol{q}, \omega)$, built from exact many-body states. However, nobody can write down the exact states of the 10^{26} interacting electrons in a kilogram of silicon, and every computed S of a real crystal is an approximation. Two questions follow: which properties of S hold exactly, so that they can test an approximation, and how the approximations are organized. Linear response answers both. It describes how a target in equilibrium reacts to a weak external field, and the correlation functions that measure that reaction are the ones the golden rule needs. Causality then ties the real part of a response to its imaginary part, and commutators fix its frequency moments whatever the interactions. The long-range Coulomb force can be separated once and for all as screening by a dielectric function, the same function that modifies a dark photon inside the medium (Sect. 1.4). We start from the response to a weak field and end with the Green's functions, diagrams and excitons from which the computed response of silicon is built, a hierarchy of approximations, each a limit of the next (Fig. 4.1).

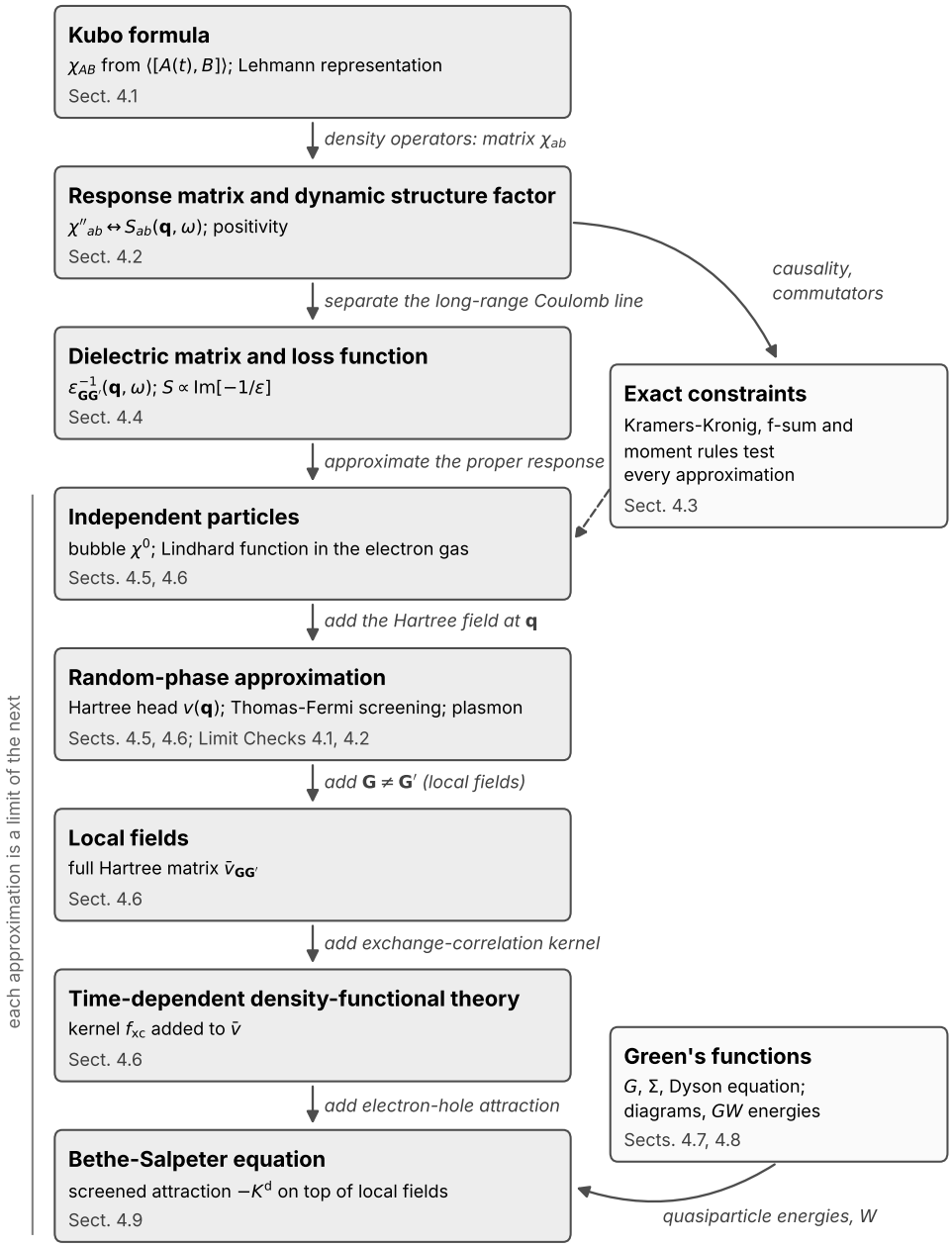


Figure 4.1: The exact backbone and the hierarchy of approximations of the density response. The Kubo formula (Sect. 4.1) leads to the response matrix and the dynamic structure factor (Sect. 4.2), whose exact constraints, Kramers–Kronig relations and sum rules (Sect. 4.3), hold in every approximation; separating the long-range Coulomb line gives the dielectric matrix and loss function (Sect. 4.4). The approximations to the proper response add, in turn, the Hartree field at $\boldsymbol{q}$ (random-phase approximation, Sect. 4.5, with Thomas–Fermi screening and the plasmon, Limit Checks 4.1 and 4.2), the local fields $\boldsymbol{G} \neq \boldsymbol{G}'$ and an exchange–correlation kernel (Sect. 4.6), and the screened electron–hole attraction (Sect. 4.9), with quasiparticle energies and the screened interaction W from the Green's functions and diagrams of Sects. 4.7 and 4.8.

4.1 The Kubo formula

Apply to a crystal a weak potential that varies in space as a plane wave of wavevector $\boldsymbol{q}$ and in time at frequency ω, and its electron density develops a ripple with the same wavevector and frequency. The amplitude of the ripple is proportional to the potential, and in general the ripple lags in phase. The ratio of ripple to potential is a response function. It is a property of the target alone, and it is fixed by correlations that the target has in equilibrium, before any potential is applied. The potential can be an applied electric field, a passing charge or the dark matter itself: between two of its plane-wave states, a dark-matter particle acts on the electrons through the density operator $n_e(\boldsymbol{q})$ (Eq. (2.38)), the same operator through which a plane-wave potential acts. We will derive the response of any target in thermal equilibrium to first order in the field, the Kubo formula [1], and write it in the eigenstates of the target. We will then see why the requirement that an effect follow its cause makes the response analytic in the upper half of the complex frequency plane. A driven harmonic oscillator and a two-level system then show what the general formulas contain. The treatment follows Refs. [2, Chap. 3] and [3, Chap. 6].

4.1.1 Response to a weak field

The target has the Hamiltonian H, with all of its interactions, eigenstates $|i\rangle$ and energies E_i. Before the field acts, the target is in the Gibbs state $\hat{\varrho} = \mathrm{e}^{-\beta H}/\mathcal{Z}$ of Eq. (2.62). The field couples to a target operator O_b,

$$H_{\mathrm{tot}}(t) = H + O_b\,\phi_{\mathrm{ext}}(t), \qquad \phi_{\mathrm{ext}}(t) \to 0 \ \text{ as } t \to -\infty, \tag{4.1}$$

where the external potential $\phi_{\mathrm{ext}}(t)$ is a number, not an operator: the source of the field is outside the target and is not changed by it. For a dimensionless O_b, such as a density, ϕ_{ext} is an energy.

A real potential energy $\phi_{\mathrm{ext}}(t)\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} + \text{c.c.}$, acting on every electron, has exactly this form. Summed over electrons with Eq. (A.8), it is

$$\sum_j \left[\phi_{\mathrm{ext}}(t)\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j} + \text{c.c.}\right] = n_e(\boldsymbol{q})\,\phi_{\mathrm{ext}}(t) + n_e(\boldsymbol{q})^\dagger\,\phi_{\mathrm{ext}}(t)^*,$$

two terms of the form Eq. (4.1), with $O_b = n_e(\boldsymbol{q})$ and $O_b = n_e(\boldsymbol{q})^\dagger$. To first order the responses to the two terms add, and we treat one at a time. The dark matter, for its part, supplies one of them. Between its plane-wave states $\boldsymbol{p}$ and $\boldsymbol{p}-\boldsymbol{q}$ the interaction is $\tilde{V}(\boldsymbol{q})\,n_e(\boldsymbol{q})/\mathcal{V}$ (Eq. (A.9)): the term with $O_b = n_e(\boldsymbol{q})$, which gives the target the momentum $\boldsymbol{q}$.

The thermal state is a mixture, and each of its components obeys the Schrödinger equation with $H_{\mathrm{tot}}(t)$. The density matrix therefore evolves as $\mathrm{i}\,\partial_t\hat{\varrho}(t) = [H_{\mathrm{tot}}(t), \hat{\varrho}(t)]$. As in Sect. 2.4, we take the known part of the evolution out by going to the interaction picture, with one difference: the part we take out is the evolution under the full Hamiltonian H of the target, interactions included, not under a free H_0. With $\hat{\varrho}_{\mathrm{I}}(t) = \mathrm{e}^{\mathrm{i}Ht}\hat{\varrho}(t)\,\mathrm{e}^{-\mathrm{i}Ht}$,

$$\mathrm{i}\,\partial_t\hat{\varrho}_{\mathrm{I}}(t) = \big[O_b(t)\,\phi_{\mathrm{ext}}(t),\ \hat{\varrho}_{\mathrm{I}}(t)\big], \qquad O_b(t) = \mathrm{e}^{\mathrm{i}Ht}O_b\,\mathrm{e}^{-\mathrm{i}Ht},$$

the counterpart of Eq. (2.49) for a density matrix. The Heisenberg operator $O_b(t)$ evolves under the target's own Hamiltonian, without the field. In the remote past $\hat{\varrho}_{\mathrm{I}}$ is the Gibbs state, which commutes with H and is unchanged by the transformation. Integrating once from $-\infty$ and keeping the term linear in ϕ_{ext}, the first-order term of the Dyson series Eq. (2.50), gives

$$\hat{\varrho}_{\mathrm{I}}(t) = \hat{\varrho} - \mathrm{i}\int_{-\infty}^{t} \mathrm{d}t'\,\big[O_b(t'),\,\hat{\varrho}\big]\,\phi_{\mathrm{ext}}(t') + O(\phi_{\mathrm{ext}}^2).$$

The average of an operator A at time t is $\mathrm{Tr}[\hat{\varrho}(t)A] = \mathrm{Tr}[\hat{\varrho}_\mathrm{I}(t)A(t)]$, with $A(t)$ the Heisenberg operator. The zeroth-order term is the equilibrium average. In the first-order term the cyclic property of the trace moves $\hat{\varrho}$ to the front, $\mathrm{Tr}([B,\hat{\varrho}]A) = \mathrm{Tr}(\hat{\varrho}[A,B])$, and the change of the average is

$$\delta\langle A\rangle(t) = -\mathrm{i}\int_{-\infty}^{t} \mathrm{d}t'\,\langle[A(t),\,O_b(t')]\rangle\,\phi_\mathrm{ext}(t'), \tag{4.2}$$

with the average taken in the Gibbs state. The field at time t' disturbs the target, the disturbance is carried to time t by the target's own dynamics, and the commutator measures how much the disturbance shows up in A. Times $t' > t$ do not contribute. Since the Gibbs state is stationary, the commutator depends on $t - t'$ alone (Sect. 2.5).

Which average responds to O_b? Suppose the target is invariant under translations, as the electron gas is. Then $O_b = n_e(\boldsymbol{q})$ adds the momentum $\boldsymbol{q}$ to every state it acts on, and the equilibrium average of $[A(t), O_b(t')]$ vanishes unless A removes the same momentum. The operator that does so is the conjugate of an operator at $\boldsymbol{q}$, such as $n_e(\boldsymbol{q})^\dagger = n_e(-\boldsymbol{q})$. For the same reason the term with $n_e(\boldsymbol{q})^\dagger$ in the real potential above cannot change $\langle n_e(\boldsymbol{q})^\dagger\rangle$: both operators remove $\boldsymbol{q}$, and the average of their commutator would need the target to lose $2\boldsymbol{q}$. In a crystal, momentum is conserved only up to a reciprocal-lattice vector $\boldsymbol{G}$ (Sect. 3.1). The field at $\boldsymbol{q}$ then also changes the averages at $\boldsymbol{q}+\boldsymbol{G}$, which are the local fields of Sect. 4.4, and the two terms of a real potential mix only when $2\boldsymbol{q}$ is a reciprocal-lattice vector. Taking $A = O_a(\boldsymbol{q})^\dagger$ in Eq. (4.2) for a second target operator $O_a(\boldsymbol{q})$, with the Heisenberg operator $O_a(\boldsymbol{q},t)$ defined as $O_b(t)$, we arrive at the result.

Kubo formula

$$\begin{aligned}\delta\langle O_a(\boldsymbol{q})^\dagger\rangle(t) &= \mathscr{V}\int_{-\infty}^{\infty}\mathrm{d}t'\,R_{ab}(\boldsymbol{q},t-t')\,\phi_\mathrm{ext}(t'),\\ R_{ab}(\boldsymbol{q},t) &= -\frac{\mathrm{i}}{\mathscr{V}}\,\Theta(t)\,\langle[O_a(\boldsymbol{q},t)^\dagger,\,O_b(\boldsymbol{q},0)]\rangle.\end{aligned} \tag{4.3}$$

The change of $\langle O_a(\boldsymbol{q})^\dagger\rangle$, to first order in a field coupled as $O_b(\boldsymbol{q})\,\phi_\mathrm{ext}(t)$, is fixed by the commutator of the two operators in the unperturbed equilibrium state of the target.

The step function $\Theta(t)$ carries the restriction $t' < t$, and R_{ab} is called the retarded response function. The equilibrium state enters through the average alone: the formula holds for any H, with any interactions, at any temperature. The factor $1/\mathscr{V}$ makes R_{ab} intensive. For density operators the commutator is a sum over all the electrons and grows with the size of the target, and so does the induced $\delta\langle n_e(\boldsymbol{q})^\dagger\rangle$; their ratio to $\mathscr{V}$ does not. In a box, the inverse of Eq. (A.6) writes the electron density as $\mathscr{V}^{-1}\sum_{\boldsymbol{q}'}\mathrm{e}^{-\mathrm{i}\boldsymbol{q}'\cdot\boldsymbol{r}}n_e(\boldsymbol{q}')$, and the induced change of the density in position space is the ripple

$$\frac{1}{\mathscr{V}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,\delta\langle n_e(\boldsymbol{q})^\dagger\rangle(t) + \text{c.c.} = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\int\mathrm{d}t'\,R_{ee}(\boldsymbol{q},t-t')\,\phi_\mathrm{ext}(t') + \text{c.c.},$$

where R_{ee} is the density response, with $O_a = O_b = n_e(\boldsymbol{q})$, in the labeling of Appendix A.7. The response function is the induced density per unit applied potential energy, which is why $R_{ee}(\boldsymbol{q},\omega)$ has the dimension $\mathrm{eV}^3/\mathrm{eV} = \mathrm{eV}^2$; the function of time, $R_{ee}(\boldsymbol{q},t)$, has one power of energy more, eV^3.

Kubo derived Eq. (4.2) in this generality [1], and it carries his name. Many texts write the response function as χ_{AB} and call it a susceptibility; others write C^R_{AB}. They usually omit the factor $1/\mathscr{V}$ and write the responding operator A in place of our $O_a^\dagger$. A text that couples the field with a minus sign, $-O_b\,\phi_\mathrm{ext}$, has a response function of the opposite sign. In this book χ is the dark-matter particle and response functions are always R; Appendix B.2 lists the correspondence.

4.1.2 Frequency dependence

A field that oscillates at one frequency produces a response at the same frequency, with an amplitude set by the transform of R_{ab}. With the time convention of Appendix A.3, the convolution in Eq. (4.3) becomes a product,

$$\delta\langle O_a(\boldsymbol{q})^\dagger\rangle(\omega) = \mathscr{V} R_{ab}(\boldsymbol{q},\omega)\,\phi_{\text{ext}}(\omega), \qquad R_{ab}(\boldsymbol{q},\omega) = -\frac{\mathrm{i}}{\mathscr{V}}\int_0^\infty \mathrm{d}t\; \mathrm{e}^{\mathrm{i}\omega t}\,\langle[O_a(\boldsymbol{q},t)^\dagger,\, O_b(\boldsymbol{q},0)]\rangle, \tag{4.4}$$

which is the definition Eq. (A.30). For a target with a discrete spectrum the commutator oscillates forever, and the integral needs a prescription to converge, which the switching supplies. Turn the field on slowly, $\phi_{\text{ext}}(t) = \phi_0\,\mathrm{e}^{-\mathrm{i}\omega t+\eta_\text{b} t}$ with a small rate $\eta_\text{b} > 0$. Substituting $s = t - t'$ in Eq. (4.3),

$$\delta\langle O_a(\boldsymbol{q})^\dagger\rangle(t) = \mathscr{V}\phi_{\text{ext}}(t)\int_0^\infty \mathrm{d}s\; \mathrm{e}^{\mathrm{i}(\omega+\mathrm{i}\eta_\text{b})s}\,R_{ab}(\boldsymbol{q},s).$$

The response follows the field, and its amplitude is the transform of R_{ab} at the complex frequency $\omega + \mathrm{i}\eta_\text{b}$. The physical response function is the limit $\eta_\text{b} \to 0^+$, and ω in Eq. (4.4) always stands for $\omega + \mathrm{i}0^+$. This is the adiabatic switching of Sect. 2.4, which put $+\mathrm{i}0^+$ into the energy denominators there.

For operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$, the response at $-\boldsymbol{q}$ is the complex conjugate of the response at $\boldsymbol{q}$. Since $\langle X\rangle^* = \langle X^\dagger\rangle$ and $[A,B]^\dagger = -[A^\dagger, B^\dagger]$, the complex conjugate of $R_{ab}(\boldsymbol{q},t)$ is $-(\mathrm{i}/\mathscr{V})\,\Theta(t)\,\langle[O_a(\boldsymbol{q},t), O_b(\boldsymbol{q})^\dagger]\rangle$. For densities and other operators with this property, it is the response at $-\boldsymbol{q}$:

$$R_{ab}(\boldsymbol{q},t)^* = R_{ab}(-\boldsymbol{q},t), \qquad R_{ab}(\boldsymbol{q},\omega)^* = R_{ab}(-\boldsymbol{q},-\omega) \quad (\omega \text{ real}). \tag{4.5}$$

It is what makes the ripple produced by a real potential a real function of $\boldsymbol{r}$ and t. In a target with inversion or time-reversal symmetry, where $R_{ee}(-\boldsymbol{q},\omega) = R_{ee}(\boldsymbol{q},\omega)$, the real part of the density response is even in ω and its imaginary part odd.

For the electron gas R_{ee} can be computed in closed form at the level of independent electrons (Sect. 4.5). For silicon it is a material quantity. It is computed from Kohn–Sham states (Sect. 3.4), as an independent-particle sum corrected for the Coulomb interaction in the random-phase approximation or in time-dependent density-functional theory (Sects. 4.4 and 4.6); Appendix E.6 describes the calculation and its convergence.

4.1.3 Lehmann representation

In the eigenstates of H every Heisenberg operator oscillates at the energies of the transitions it causes, $\langle i|O(t)|f\rangle = \mathrm{e}^{\mathrm{i}(E_i-E_f)t}\langle i|O|f\rangle$, and the commutator in Eq. (4.4) becomes a sum of such oscillations. Writing out its two products, with the Gibbs weights $p_i = \mathrm{e}^{-\beta E_i}/\mathscr{Z}$, and inserting a complete set of states $|f\rangle$ between the operators,

$$\langle[O_a(\boldsymbol{q},t)^\dagger, O_b(\boldsymbol{q})]\rangle = \sum_{i,f} p_i\Big[\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle\,\mathrm{e}^{-\mathrm{i}(E_f-E_i)t} - \langle i|O_b|f\rangle\langle f|O_a^\dagger|i\rangle\,\mathrm{e}^{+\mathrm{i}(E_f-E_i)t}\Big],$$

where the operators without a time argument are taken at $t = 0$ and we drop the argument $\boldsymbol{q}$. The time integral of each oscillation, with $\omega + \mathrm{i}0^+$, is elementary, $-\mathrm{i}\int_0^\infty \mathrm{d}t\,\mathrm{e}^{\mathrm{i}(\omega\mp(E_f-E_i)+\mathrm{i}0^+)t} = 1/[\omega \mp (E_f - E_i) + \mathrm{i}0^+]$.

Lehmann representation

$$R_{ab}(\boldsymbol{q},\omega) = \frac{1}{\mathcal{V}} \sum_{i,f} p_i \left[\frac{\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle}{\omega - (E_f - E_i) + \mathrm{i}0^+} - \frac{\langle i|O_b|f\rangle\langle f|O_a^\dagger|i\rangle}{\omega + (E_f - E_i) + \mathrm{i}0^+} \right]. \tag{4.6}$$

The response is a sum of simple poles at the transition energies of the target, each displaced by $-\mathrm{i}0^+$ below the real axis and weighted by the thermal population of its initial state and a product of matrix elements.

This is Eq. (A.32). The term with the pole at $\omega = E_f - E_i$ is what the optics literature calls the resonant term, and the one with the pole at $\omega = -(E_f - E_i)$ the antiresonant term. For a target in its ground state, $E_f - E_i \geq 0$, and the resonant term has its poles at positive ω. The interactions of the target, however strong, enter through the exact energies and matrix elements. No approximation has been made beyond first order in ϕ_{ext}.

The poles split the response into two parts with different meanings. By the Sokhotski–Plemelj formula Eq. (2.58) (Appendix C.4), each denominator is a principal value minus $\mathrm{i}\pi$ times a delta function. The principal value collects transitions that the field drives off resonance. They are never completed, they shift the target's energy, and they make the target polarizable, as the off-shell intermediate states of Sect. 2.4 did. The delta functions collect the transitions that conserve energy, and they make up the imaginary part. For a diagonal element, $a = b$, the weights are squared matrix elements, $|\langle f|O_a|i\rangle|^2$ in the resonant term and $|\langle f|O_a^\dagger|i\rangle|^2$ in the antiresonant one. The imaginary part is then a difference of two transition sums Eq. (2.65):

$$\mathrm{Im}\, R_{aa}(\boldsymbol{q},\omega) = -\frac{\pi}{\mathcal{V}} \Big[\mathcal{S}_{O_a}(\omega) - \mathcal{S}_{O_a^\dagger}(-\omega) \Big]. \tag{4.7}$$

The transition sum $\mathcal{S}_{O_a}(\omega)$ counts the transitions in which the target takes the energy ω; $\mathcal{S}_{O_a^\dagger}(-\omega)$ counts those in which it gives up ω. At $T = 0$ the target starts in its ground state, nothing can give up energy, and $\mathrm{Im}\, R_{aa} < 0$ for $\omega > 0$. At finite temperature both sums contribute, and detailed balance Eq. (2.68) makes $\mathcal{S}_{O_a}(\omega)$ larger by the factor $\mathrm{e}^{\beta\omega}$. Sect. 4.2 turns this into the fluctuation–dissipation theorem, which ties $\mathrm{Im}\, R_{aa}$ to the dynamic structure factor.

Absorbed power. The sign of $\mathrm{Im}\, R$ is the statement that a target in equilibrium absorbs energy from a field. Take a Hermitian O_b, so that $R_{bb}(t)$ is real (the average of the commutator of two Hermitian operators is imaginary), and a real field $\phi_{\mathrm{ext}}(t) = \phi_0 \cos\omega t$ with $\omega > 0$. The field does work on the target at the rate $\langle \partial H_{\mathrm{tot}}/\partial t\rangle = \langle O_b\rangle(t)\, \dot{\phi}_{\mathrm{ext}}(t)$. The field has components $\frac{1}{2}\phi_0$ at the frequencies $\pm\omega$, and by Eq. (4.4) the induced $\delta\langle O_b\rangle$ has components $\frac{1}{2}\phi_0 V R_{bb}(\pm\omega)$, with $R_{bb}(-\omega) = R_{bb}(\omega)^*$ since $R_{bb}(t)$ is real. In the time average of the product the cross terms between $+\omega$ and $-\omega$ alone survive, and the mean absorbed power is

$$P = -\frac{\omega\, \phi_0^2}{2}\, \mathcal{V}\ \mathrm{Im}\, R_{bb}(\omega) = \frac{\pi}{2}\, \omega\, \phi_0^2 \Big[\mathcal{S}_{O_b}(\omega) - \mathcal{S}_{O_b}(-\omega) \Big], \tag{4.8}$$

where the expression in transition sums follows from Eq. (4.7). This is ω times the difference of the absorption and emission rates of the golden rule for a harmonic perturbation, Eq. (2.8) with $\hat{O} = \frac{1}{2}\phi_0 O_b$, averaged over the thermal state as in Sect. 2.5. The imaginary part of a diagonal response is therefore its dissipative part. The same formula describes a bosonic dark-matter field that couples to O_b. With $\omega = m_\phi$, the amplitude ϕ_0 is then the field amplitude of Sect. 2.1, $\phi_0^2 = 2\rho_\chi/m_\phi^2$, times the coupling constant of the field to O_b, which we set to one here. Then P/m_ϕ is the rate at which the field is absorbed in the volume $\mathcal{V}$, which is why absorption rates are proportional to $-\,\mathrm{Im}\, R$ (Chap. 8).

Static limit. At $\omega \to 0$ the Lehmann form measures how much the target deforms in a static field. For a Hermitian O_b the resonant and antiresonant terms of a pair i, f have the same weight, so pairs with $E_f = E_i$ cancel, and the rest add:

$$\mathcal{V} R_{bb}(0) = -2 \sum_{i,f} p_i \, \frac{|\langle f|O_b|i\rangle|^2}{E_f - E_i} \qquad (E_f \neq E_i).$$

At $T = 0$ every term is negative. Compare second-order perturbation theory for a static $H_{\text{ext}} = O_b \phi_0$: the ground-state energy is shifted by $-\phi_0^2 \sum_f |\langle f|O_b|0\rangle|^2/(E_f - E_0)$, which is $\frac{1}{2} V R_{bb}(0)\, \phi_0^2$. A static response is negative because the target always lowers its energy by yielding to a static field; for the density, electrons leave the regions where the potential energy is high. At finite temperature the limit $\omega \to 0$ describes a target that is deformed by a slowly applied field while isolated from any bath, Kubo's isolated response. The isothermal response, in which the target stays in equilibrium with a bath, has an extra term proportional to β from matrix elements of O_b between states of equal energy; the two agree when there are none [1].

Worked example 4.1: The driven harmonic oscillator

Take one mode of frequency ω_0, $H = \omega_0 d^\dagger d$, and the operator $O_a = O_b = \lambda_{\text{osc}}(d + d^\dagger)$ of Worked example 2.7, with λ_{osc} real. A single oscillator has no volume of its own, and the combination VR is its response, independent of the box. The Heisenberg operators follow from the commutator $[d, H] = \omega_0 d$, which integrates, as for the fermion operators of Sect. 2.3, to $d(t) = d\, \mathrm{e}^{-\mathrm{i}\omega_0 t}$. The commutator in Eq. (4.3) is then a number,

$$[O(t), O] = \lambda_{\text{osc}}^2 ([d, d^\dagger]\, \mathrm{e}^{-\mathrm{i}\omega_0 t} + [d^\dagger, d]\, \mathrm{e}^{\mathrm{i}\omega_0 t}) = -2\mathrm{i}\lambda_{\text{osc}}^2 \sin \omega_0 t,$$

the same in every state, and $VR(t) = -2\lambda_{\text{osc}}^2\, \Theta(t) \sin \omega_0 t$. With $\int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}zt} \sin \omega_0 t = \omega_0/(\omega_0^2 - z^2)$ for $\operatorname{Im} z > 0$, its transform is

$$VR(\omega) = \frac{2\lambda_{\text{osc}}^2 \omega_0}{(\omega + \mathrm{i}0^+)^2 - \omega_0^2} = \lambda_{\text{osc}}^2 \left[\frac{1}{\omega - \omega_0 + \mathrm{i}0^+} - \frac{1}{\omega + \omega_0 + \mathrm{i}0^+} \right]. \tag{4.9}$$

The Lehmann form gives the same. From the level n, O reaches $n+1$ with $|\langle n+1|O|n\rangle|^2 = \lambda_{\text{osc}}^2 (n+1)$ and $n-1$ with $\lambda_{\text{osc}}^2 n$, and averaged with the thermal weights of Eq. (2.69) the resonant term of Eq. (4.6) is $\lambda_{\text{osc}}^2 [(n_{\text{B}}(\omega_0) + 1)/(\omega - \omega_0) + n_{\text{B}}(\omega_0)/(\omega + \omega_0)]$, with $n_{\text{B}}(\omega_0)$ the Bose occupation of the mode. The antiresonant term has the same weights with the denominators exchanged. Their difference is Eq. (4.9), and $n_{\text{B}}(\omega_0)$ cancels. The oscillator absorbs and emits more often at higher temperature, by the factors $n_{\text{B}}(\omega_0) + 1$ and $n_{\text{B}}(\omega_0)$ of Worked example 2.7, but the net response to a field does not depend on temperature at all.

The reason is that the oscillator is a linear system, and the quantum response is the classical one. For a particle of mass M_{osc} in a harmonic well, the displacement is $x = (2M_{\text{osc}}\omega_0)^{-1/2}(d + d^\dagger)$, so $O = x$ has $\lambda_{\text{osc}}^2 = 1/(2M_{\text{osc}}\omega_0)$. The field $H_{\text{ext}} = x\, \phi_{\text{ext}}$ is a force $-\phi_{\text{ext}}$, and Newton's equation $M_{\text{osc}}(\ddot{x} + \omega_0^2 x) = -\phi_{\text{ext}}$, with $x \propto \mathrm{e}^{-\mathrm{i}\omega t}$, gives $x(\omega) = \phi_{\text{ext}}(\omega)/[M_{\text{osc}}(\omega^2 - \omega_0^2)]$, which is Eq. (4.9) with this λ_{osc}. The static limit is the stretched spring, $x = -\phi_{\text{ext}}/(M_{\text{osc}}\omega_0^2)$, and $VR(0) = -2\lambda_{\text{osc}}^2/\omega_0$; the energy $\frac{1}{2} VR(0)\phi_0^2 = -\lambda_{\text{osc}}^2 \phi_0^2/\omega_0$ is the second-order shift of the ground state by $\lambda_{\text{osc}}\phi_0(d + d^\dagger)$, as the static limit above requires.

A real oscillator also loses energy, to other modes or to electrons. With a damping term $M_{\text{osc}}\Gamma_{\text{w}}\dot{x}$ added to Newton's equation, where Γ_{w} is the rate at which the energy of the

free oscillation decays, the response becomes

$$VR(\omega) = \frac{2\lambda_{\mathrm{osc}}^2\omega_0}{\omega^2 + \mathrm{i}\Gamma_{\mathrm{w}}\omega - \omega_0^2}, \qquad -\operatorname{Im} VR(\omega) = \frac{2\lambda_{\mathrm{osc}}^2\omega_0\,\Gamma_{\mathrm{w}}\omega}{(\omega^2-\omega_0^2)^2 + \Gamma_{\mathrm{w}}^2\omega^2}.$$

The infinitesimal 0^+ has become the finite width Γ_{w}, as in Eq. (2.57). The absorptive part $-\operatorname{Im} VR$ is a resonance of height $2\lambda_{\mathrm{osc}}^2/\Gamma_{\mathrm{w}}$ at $\omega = \omega_0$, ten times $\lambda_{\mathrm{osc}}^2/\omega_0$ for $\Gamma_{\mathrm{w}} = 0.2\,\omega_0$, and it is positive at every $\omega > 0$ (Fig. 4.2b). The real part is negative below the resonance, where the oscillator follows the force in phase, and positive above it, where it moves against the force. At large ω it falls as $2\lambda_{\mathrm{osc}}^2\omega_0/\omega^2$.

4.1.4 Causality and analyticity

A response cannot precede its cause. In Eq. (4.3) this is the step function $\Theta(t)$, and in frequency it becomes a statement about complex ω. Define the transform at a complex frequency z,

$$R_{ab}(\boldsymbol{q}, z) = \int_0^\infty \mathrm{d}t\; \mathrm{e}^{\mathrm{i}zt}\, R_{ab}(\boldsymbol{q}, t). \tag{4.10}$$

For $\operatorname{Im} z > 0$ the factor $\mathrm{e}^{\mathrm{i}zt} = \mathrm{e}^{\mathrm{i}t\operatorname{Re} z}\mathrm{e}^{-t\operatorname{Im} z}$ decays, while $R_{ab}(\boldsymbol{q}, t)$ stays bounded: in a stable target all energies are real, and the commutator average oscillates without growing. The integral then converges, it may be differentiated under the integral sign with respect to z, and $R_{ab}(\boldsymbol{q}, z)$ is analytic everywhere in the upper half plane. Two facts went into this: causality, which restricts the integral to $t > 0$, and stability, which keeps $R_{ab}(\boldsymbol{q}, t)$ from growing. The physical response is the boundary value on the real axis approached from above, $R_{ab}(\boldsymbol{q}, \omega + \mathrm{i}0^+)$, which is the prescription found from the switching. For $\operatorname{Im} z < 0$ the integral diverges, and the response there is defined, if at all, by analytic continuation. Appendix C.5 states these results in general.

The Lehmann form shows the same thing pole by pole. With $\omega + \mathrm{i}0^+$ replaced by z, Eq. (4.6) is a sum of terms $1/[z \mp (E_f - E_i)]$, whose poles lie on the real axis, and the physical frequency passes just above them (Fig. 4.2a). In a finite target the poles are isolated, and the imaginary part on the real axis is a comb of delta functions. In a macroscopic target the transition energies crowd together, with spacings that shrink as the volume grows. If the limit $\mathcal{V} \to \infty$ is taken first, at a fixed small η_{b}, the sums over f become integrals, and $R_{ab}(\boldsymbol{q}, \omega + \mathrm{i}\eta_{\mathrm{b}})$ tends to a smooth function as $\eta_{\mathrm{b}} \to 0^+$. The poles have merged into a branch cut along the parts of the real axis where the spectrum is continuous. The opposite order of limits returns the comb. This is the time window of the golden rule (Sect. 2.1) once more. The switching rate η_{b} must be small compared with the scale on which the response varies, and large compared with the level spacing, which a macroscopic target makes arbitrarily small.

Across the cut, the continuation of R_{ab} into the lower half plane can have poles at complex frequencies. They are resonances, long-lived collective or single-particle excitations. A pole at z_{p} contributes $\mathrm{e}^{-\mathrm{i}z_{\mathrm{p}}t}$ to $R_{ab}(\boldsymbol{q}, t)$, which decays only if $\operatorname{Im} z_{\mathrm{p}} < 0$. The damped oscillator of Worked example 4.1 shows them in closed form. Its response has poles where $\omega^2 + \mathrm{i}\Gamma_{\mathrm{w}}\omega - \omega_0^2 = 0$,

$$\omega = -\frac{\mathrm{i}\Gamma_{\mathrm{w}}}{2} \pm \Big(\omega_0^2 - \frac{\Gamma_{\mathrm{w}}^2}{4}\Big)^{1/2}.$$

For $\Gamma_{\mathrm{w}} < 2\omega_0$ they lie on the circle $|\omega| = \omega_0$ in the lower half plane, at a depth $\Gamma_{\mathrm{w}}/2$ that is the decay rate of the amplitude. At critical damping, $\Gamma_{\mathrm{w}} = 2\omega_0$, they meet at $-\mathrm{i}\omega_0$, and for larger Γ_{w} they move apart along the negative imaginary axis. For no positive Γ_{w} do they cross into the upper half plane, where a pole would describe a response that grows exponentially

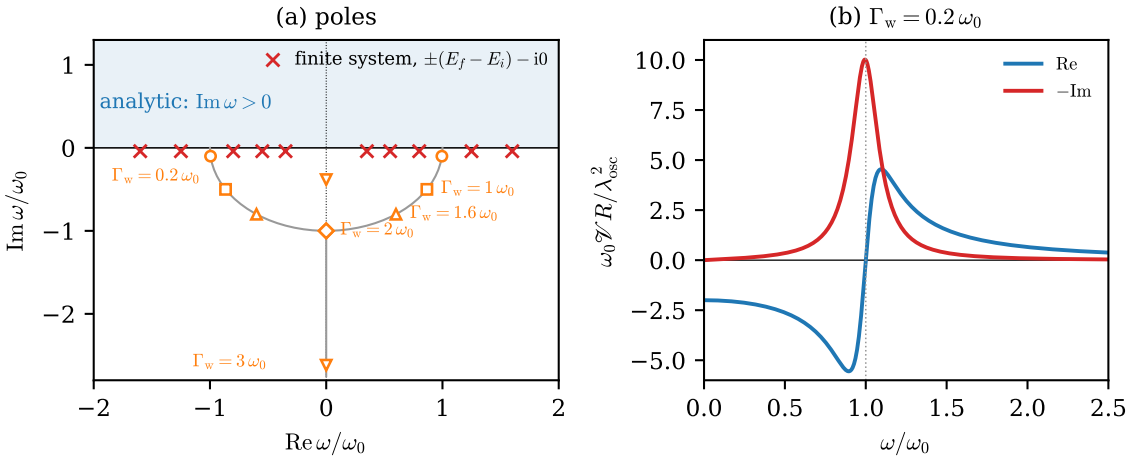


Figure 4.2: Poles of the retarded response function in the complex frequency plane ω, in units of an oscillator frequency ω_0. (a) The response is analytic in the upper half plane (shaded). A finite target has poles at its transition energies $\pm(E_f - E_i)$, displaced by an infinitesimal $-\mathrm{i}0$ below the real axis (crosses, at illustrative transition energies). A harmonic oscillator with energy damping rate Γ_w has poles at $-\mathrm{i}\Gamma_\mathrm{w}/2 \pm (\omega_0^2 - \Gamma_\mathrm{w}^2/4)^{1/2}$ (open symbols, for the values of Γ_w marked). They lie on the circle $|\omega| = \omega_0$ below critical damping, $\Gamma_\mathrm{w} = 2\omega_0$, and on the negative imaginary axis above it; for $\Gamma_\mathrm{w} = 3\omega_0$ the upper of the two is at $-0.38\,\mathrm{i}\omega_0$. (b) Real part and minus the imaginary part of VR for the damped oscillator with $\Gamma_\mathrm{w} = 0.2\,\omega_0$ and coupling $\lambda_\mathrm{osc}(d + d^\dagger)$, on the real axis. The absorptive part peaks at $2\lambda_\mathrm{osc}^2/\Gamma_\mathrm{w}$ at $\omega = \omega_0$; the real part is negative below the resonance and positive above it, and tends to $-2\lambda_\mathrm{osc}^2/\omega_0$ at $\omega = 0$.

after the field is gone. The plasmon (Sect. 4.5) and the quasiparticles of Sect. 4.7 are poles of this kind.

How fast R_{ab} falls at large $|z|$ follows from the Lehmann form too. Expanding $1/[z \mp (E_f - E_i)] = 1/z \pm (E_f - E_i)/z^2 + \dots$ in Eq. (4.6), the coefficient of $1/z$ is the equal-time commutator, and in the coefficient of $1/z^2$ the resonant and antiresonant terms combine into a double commutator:

$$R_{ab}(\boldsymbol{q}, z) = \frac{\langle [O_a^\dagger, O_b] \rangle}{\mathscr{V} z} + \frac{\langle [O_a^\dagger, [H, O_b]] \rangle}{\mathscr{V} z^2} + O(z^{-3}). \tag{4.11}$$

The coefficient of $1/z^2$ is checked by writing out $\langle i|O_a^\dagger H O_b - O_a^\dagger O_b H - H O_b O_a^\dagger + O_b H O_a^\dagger|i\rangle$ in eigenstates, which gives $\sum_f (E_f - E_i)[\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle + \langle i|O_b|f\rangle\langle f|O_a^\dagger|i\rangle]$. Density operators at different wavevectors commute, since both are functions of the electron positions alone, and the density response therefore falls as $1/z^2$. When the potential acting on the electrons is local, only the kinetic energy contributes to the coefficient, and the result, the f-sum rule of Sect. 4.3, depends only on the electron density. For the oscillator, $[O, [H, O]] = 2\lambda_\mathrm{osc}^2\omega_0$, the coefficient found in Worked example 4.1. Analyticity in the upper half plane together with this decay leads to the Kramers–Kronig relations, which reconstruct the real part of R_{ab} on the real axis from its imaginary part (Sect. 4.3).

Worked example 4.2: The two-level system

Take the two-level system of Sect. 2.5: states $|0\rangle$ and $|1\rangle$ with splitting Δ_tl, Gibbs weights p_0 and p_1, and a Hermitian operator $O_a = O_b = O$ with only the off-diagonal element

$O_{10} = \langle 1|O|0\rangle$. The Lehmann form Eq. (4.6) has four entries. In the resonant term, $i = 0$ contributes $p_0|O_{10}|^2/(\omega - \Delta_{\rm tl})$ and $i = 1$ contributes $p_1|O_{10}|^2/(\omega + \Delta_{\rm tl})$. In the antiresonant term the same weights appear with the poles exchanged and a minus sign. Altogether,

$$VR(\omega) = (p_0 - p_1)\,|O_{10}|^2 \left[\frac{1}{\omega - \Delta_{\rm tl} + \mathrm{i}0^+} - \frac{1}{\omega + \Delta_{\rm tl} + \mathrm{i}0^+}\right] = \frac{2\Delta_{\rm tl}\,|O_{10}|^2 \tanh(\beta\Delta_{\rm tl}/2)}{(\omega + \mathrm{i}0^+)^2 - \Delta_{\rm tl}^2}, \tag{4.12}$$

with $p_0 - p_1 = \tanh(\beta\Delta_{\rm tl}/2)$ from Eq. (2.62). It has the form of the oscillator, Eq. (4.9), with $\omega_0 \to \Delta_{\rm tl}$ and $\lambda_{\rm osc}^2 \to (p_0 - p_1)|O_{10}|^2$, but now the temperature matters. At $T = \Delta_{\rm tl}/2$ the weights are 0.881 and 0.119 (Fig. 2.8), the response is reduced by the factor 0.762, and $VR(0) = -1.523\,|O_{10}|^2/\Delta_{\rm tl}$. At $T \gg \Delta_{\rm tl}$, $\tanh(\beta\Delta_{\rm tl}/2) \to \Delta_{\rm tl}/2T$ and $VR(0) \to -|O_{10}|^2/T$. For a spin $\frac{1}{2}$ in a magnetic field, with $\Delta_{\rm tl}$ the Zeeman splitting and O a transverse component of its magnetic moment, this is the Curie law of a paramagnet: the static susceptibility falls as $1/T$.

The physics is in the weight $p_0 - p_1$. The field drives the upward transition from $|0\rangle$ and stimulated emission from $|1\rangle$ with the same matrix element, so the excited population cancels part of the absorption, and the absorbed power Eq. (4.8) is proportional to $p_0 - p_1$. Since $p_1 < p_0$ at every temperature (Sect. 2.5), a two-level system in equilibrium always absorbs, and becomes transparent only as $T \to \infty$. The oscillator escapes this saturation because its matrix elements grow up the ladder: the extra absorption from level n, weighted by $n + 1$, exactly compensates the emission, weighted by n, which is why $n_{\rm B}(\omega_0)$ dropped out of Eq. (4.9). The tail coefficient of Eq. (4.11) is $\langle[O,[H,O]]\rangle = 2\Delta_{\rm tl}(p_0 - p_1)|O_{10}|^2$, temperature dependent here and constant for the oscillator.

The two examples also mark where linear response stops. For the oscillator, the commutator $[O(t), O(t')]$ is a number, every nested commutator of higher order vanishes, and with it every higher-order term of the Dyson series, so Eq. (4.9) holds at any field strength. The two-level system saturates. On resonance the field moves population between $|0\rangle$ and $|1\rangle$, and Eq. (4.12) holds only while that transfer is small: the drive $|O_{10}\phi_0|$ must be small compared with the width of the transition, the weak-drive condition of Appendix D.3. A stronger field makes the populations oscillate at the Rabi frequency, a response that is not linear in the field.

Eq. (4.6) is built from the same thermal weights, matrix elements and transition energies as the thermal golden rule of Sect. 2.5, and Eq. (4.7) already writes the imaginary part of a diagonal response as a difference of transition sums. Sect. 4.2 completes the identification. Every exact statement about R, its analyticity, the coefficient of its tail and, later, its screening, then becomes a statement about the dynamic structure factor and hence about the dark-matter scattering rate.

4.2 Correlations and the dynamic structure factor

A dark-matter particle in silicon scatters at a rate fixed by the dynamic structure factor $S(\boldsymbol{q}, \omega)$ of Eq. (2.67), which counts the transitions the target makes. A weak potential applied from outside, acting through the same density operator (Sect. 4.1), makes the density ripple by an amount fixed by the response function R_{ee} of Eq. (4.3). Both are built from the same matrix elements and the same Gibbs weights, and we will see that at each $\boldsymbol{q}$ and ω either one fixes the other. This is the fluctuation–dissipation theorem, and it lets every method that computes a response also compute a rate. Before we get there, we look at S from two other

sides. In space and time it is the Fourier transform of a density–density correlation function, Van Hove's picture of a scattering experiment. Over several target operators it is a positive matrix, which keeps every rate positive and bounds the interference between two couplings. We end with independent particles, for which S has a closed form in the electron gas, and the hydrogen atom of Sect. 2.2 comes back as a special case. The treatment follows Refs. [2, Chaps. 3 and 4] and [4, Chap. 2].

4.2.1 The response matrix

A dark photon couples to the charge of the electrons and of the nuclei, with opposite signs, and other mediators couple to spin or to the nucleon number (Sect. 1.4). In general the interaction between the dark-matter plane waves $\boldsymbol{p}$ and $\boldsymbol{p}-\boldsymbol{q}$ is a sum $\mathscr{V}^{-1}\sum_b \tilde{V}_b(\boldsymbol{q})\, O_b(\boldsymbol{q})$ over target operators $O_b(\boldsymbol{q})$, each with its own potential $\tilde{V}_b(\boldsymbol{q})$; the density coupling of Eq. (A.9) is the case of a single term, $O_e = n_e(\boldsymbol{q})$ with $\tilde{V}_e = \tilde{V}$. The golden-rule amplitude to reach $|f\rangle$ from $|i\rangle$ is then $\mathscr{V}^{-1}\sum_b \tilde{V}_b \langle f|O_b|i\rangle$. Its square has cross terms $\tilde{V}_a^* \tilde{V}_b\, \langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle$, and the target enters through the matrix of their thermal sums.

Response matrix

$$\begin{aligned} S_{ab}(\boldsymbol{q},\omega) &= \frac{2\pi}{\mathscr{V}} \sum_{i,f} p_i\, \langle i|O_a(\boldsymbol{q})^\dagger|f\rangle\langle f|O_b(\boldsymbol{q})|i\rangle\, \delta(\omega - E_f + E_i) \\ &= \frac{1}{\mathscr{V}} \int_{-\infty}^{\infty} \mathrm{d}t\; \mathrm{e}^{\mathrm{i}\omega t}\, \big\langle O_a(\boldsymbol{q},t)^\dagger\, O_b(\boldsymbol{q},0)\big\rangle. \end{aligned} \tag{4.13}$$

Its diagonal elements are the transition sums of Eq. (2.65), $S_{aa} = (2\pi/\mathscr{V})\, \mathcal{S}_{O_a}$, and the dynamic structure factor is the density element, $S \equiv S_{ee}$.

The correlation-function form follows as Eq. (2.66) did. Inserting a complete set of states between the two operators, each term oscillates as $\mathrm{e}^{-\mathrm{i}(E_f - E_i)t}$, and the time integral turns the oscillation into $2\pi\,\delta(\omega - E_f + E_i)$. This is the definition Eq. (A.31), and its density element is $S(\boldsymbol{q},\omega)$ of Eq. (2.67), Ref. [5, Eq. (4)], with the same normalization. Exchanging a and b complex-conjugates each term, so $S_{ab}^* = S_{ba}$ and the matrix is Hermitian. Like R_{ab}, it is intensive, and for density operators it has the dimension eV^2. Detailed balance carries over with the indices exchanged. For operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$, relabeling $i \leftrightarrow f$ as in Eq. (2.68) gives $S_{ab}(-\boldsymbol{q},-\omega) = \mathrm{e}^{-\beta\omega} S_{ba}(\boldsymbol{q},\omega)$, which is Eq. (A.34).

The sum over final dark-matter momenta is the one of Sect. 2.5, term by term, and the rate per dark-matter particle is

$$\Gamma = \int \frac{\mathrm{d}^3 q}{(2\pi)^3} \sum_{a,b} \tilde{V}_a(\boldsymbol{q})^*\, \tilde{V}_b(\boldsymbol{q})\, S_{ab}(\boldsymbol{q}, \omega_{\boldsymbol{q}}), \tag{4.14}$$

with $\omega_{\boldsymbol{q}}$ the energy deposited at $\boldsymbol{q}$ (Sect. 2.5). For a single operator it reduces to $\int \mathrm{d}^3 q\, (2\pi)^{-3} |\tilde{V}_a|^2 S_{aa}$, and for the density to Eq. (2.67). The off-diagonal elements carry the interference between two couplings. For a dark photon, for instance, the electron and nuclear densities enter with potentials of opposite sign, and S_{ab} between them measures how the electrons follow a recoiling nucleus; Chap. 8 builds the general rate on Eq. (4.14).

4.2.2 Correlations in space and time

At a typical momentum transfer $q_0 = \alpha m_e = 3.73\,\mathrm{keV}$ the dark matter probes the electron density on a length $2\pi/q_0 = 3.32\,\text{Å}$, and at an energy of $10\,\mathrm{eV}$ over a time $1/\omega = 0.066\,\mathrm{fs}$. The structure factor can be read as a statement about the density on these scales. Van Hove

introduced this reading for neutron scattering [6]: the rate measures how the density at one place and time is correlated with the density at another. To see it, write the density in position space, $\hat{n}_e(\boldsymbol{r}) = \sum_j \delta^3(\boldsymbol{r} - \boldsymbol{r}_j)$, the operator whose transform Eq. (A.6) is $n_e(\boldsymbol{q})$, and define the Van Hove correlation function

$$\mathcal{G}_{\text{VH}}(\boldsymbol{r}, t) = \frac{1}{N_e} \int \mathrm{d}^3 r' \, \langle \hat{n}_e(\boldsymbol{r}', 0) \, \hat{n}_e(\boldsymbol{r}' + \boldsymbol{r}, t) \rangle, \tag{4.15}$$

with N_e the number of electrons in the volume $\mathcal{V}$ and the Heisenberg operators of Sect. 4.1. In the correlation form Eq. (4.13), stationarity moves the time from one operator to the other, $\langle n_e(\boldsymbol{q}, t)^\dagger n_e(\boldsymbol{q}, 0) \rangle = \langle n_e(\boldsymbol{q}, 0)^\dagger n_e(\boldsymbol{q}, -t) \rangle$. Writing both operators as integrals over position, $n_e(\boldsymbol{q})^\dagger = \int \mathrm{d}^3 r' \, \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}'} \hat{n}_e(\boldsymbol{r}')$, and writing the position in $n_e(\boldsymbol{q}, -t)$ as $\boldsymbol{r}' + \boldsymbol{r}$,

$$\begin{aligned} \langle n_e(\boldsymbol{q}, 0)^\dagger n_e(\boldsymbol{q}, -t) \rangle &= \int \mathrm{d}^3 r \, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \int \mathrm{d}^3 r' \, \langle \hat{n}_e(\boldsymbol{r}', 0) \, \hat{n}_e(\boldsymbol{r}' + \boldsymbol{r}, -t) \rangle \\ &= N_e \int \mathrm{d}^3 r \, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \, \mathcal{G}_{\text{VH}}(\boldsymbol{r}, -t). \end{aligned}$$

Dividing by $\mathcal{V}$, integrating over t with $\mathrm{e}^{\mathrm{i}\omega t}$ and reversing the sign of t,

$$S(\boldsymbol{q}, \omega) = n_e \int_{-\infty}^{\infty} \mathrm{d}t \int \mathrm{d}^3 r \, \mathrm{e}^{\mathrm{i}(\boldsymbol{q}\cdot\boldsymbol{r} - \omega t)} \, \mathcal{G}_{\text{VH}}(\boldsymbol{r}, t), \tag{4.16}$$

with $n_e = N_e/\mathcal{V}$. The dynamic structure factor is the space–time Fourier transform of the Van Hove function, per unit volume. Inserted into Eq. (2.67), this is the Born rate of any target written through its density correlations, the form Van Hove derived for neutrons scattering on nuclei.

The classical reading of $\mathcal{G}_{\text{VH}}$ is simple. For commuting densities, Eq. (4.15) becomes $N_e^{-1} \sum_{j,l} \langle \delta^3(\boldsymbol{r} + \boldsymbol{r}_j(0) - \boldsymbol{r}_l(t)) \rangle$: given an electron at the origin at $t = 0$, it is the density of electrons at $\boldsymbol{r}$ at time t. The terms with $j = l$ follow the electron itself. They integrate to one and spread out as the electron moves away. The terms with $j \neq l$ describe all the others, and they tend to n_e at large r. At $t = 0$ the function is $\delta^3(\boldsymbol{r}) + n_e \, g_{\text{pair}}(\boldsymbol{r})$, with $g_{\text{pair}}(\boldsymbol{r})$ the pair distribution, the probability of finding a second electron at $\boldsymbol{r}$ relative to the first, normalized to one at large separation. Quantum mechanically, however, the densities at different times do not commute, and $\mathcal{G}_{\text{VH}}$ is complex for $t \neq 0$. Its imaginary part is what makes $S(\boldsymbol{q}, \omega)$ and $S(-\boldsymbol{q}, -\omega)$ differ by the factor $\mathrm{e}^{-\beta\omega}$ of detailed balance, and it drops out when $|\omega| \ll T$.

A single free electron shows how the time dependence encodes the recoil. Take it at rest, in the plane wave $\boldsymbol{p} = 0$. Then $n_e(\boldsymbol{q}, t) = \mathrm{e}^{\mathrm{i}Ht} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \mathrm{e}^{-\mathrm{i}Ht}$ acting on $|\boldsymbol{0}\rangle$ gives $\mathrm{e}^{\mathrm{i}q^2 t/2m_e} |\boldsymbol{q}\rangle$, and the correlation $\langle n_e(\boldsymbol{q}, t)^\dagger n_e(\boldsymbol{q}, 0) \rangle = \mathrm{e}^{-\mathrm{i}q^2 t/2m_e}$ oscillates at the recoil energy forever. Its time transform is $2\pi \, \delta(\omega - q^2/2m_e)$: the electron takes the momentum $\boldsymbol{q}$ and exactly the free recoil energy, as in Worked example 2.3. In position space this is the self term of $\mathcal{G}_{\text{VH}}$, spreading as the struck electron moves off. A bound or interacting electron cannot keep oscillating at one frequency, its correlation decays, and the delta function broadens into the spectrum of the target.

Integrating Eq. (4.16) over ω sets $t = 0$, since $\int \mathrm{d}\omega \, \mathrm{e}^{-\mathrm{i}\omega t} = 2\pi \, \delta(t)$. The result is the static structure factor,

$$\int_{-\infty}^{\infty} \mathrm{d}\omega \; S(\boldsymbol{q}, \omega) = 2\pi \, n_e \, S^{\text{st}}(\boldsymbol{q}), \qquad S^{\text{st}}(\boldsymbol{q}) = \frac{1}{N_e} \langle n_e(\boldsymbol{q})^\dagger n_e(\boldsymbol{q}) \rangle = \int \mathrm{d}^3 r \, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \, \mathcal{G}_{\text{VH}}(\boldsymbol{r}, 0), \tag{4.17}$$

where the middle form follows from summing Eq. (4.13) over f at fixed i. S^{st} is the total weight, per electron, of the transitions in which the target takes up the momentum $\boldsymbol{q}$ at any energy. It is not a probability and can exceed one, as it does at a Bragg vector. It is the

many-electron version of the closure relation Eq. (2.19), and it tells us how the electrons share a momentum transfer. With $\mathcal{G}_{\mathrm{VH}}(\boldsymbol{r},0)=\delta^3(\boldsymbol{r})+n_e g_{\mathrm{pair}}(\boldsymbol{r})$, the constant part of g_{pair} transforms into a term $(2\pi)^3 n_e\,\delta^3(\boldsymbol{q})$ alone, and at $\boldsymbol{q}\neq 0$ $S^{\mathrm{st}}=1+n_e\int \mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}[g_{\mathrm{pair}}(\boldsymbol{r})-1]$. At large q the phase averages the pair term away and $S^{\mathrm{st}}\to 1$: the dark matter scatters incoherently, on one electron at a time. At small q, however, the pair term matters. In a uniform gas it cancels most of the self term, because each electron digs a hole in the density of the others, and long-wavelength density fluctuations are suppressed (Worked example 4.3). In a crystal $\langle n_e(\boldsymbol{q})\rangle$ itself is nonzero at $\boldsymbol{q}=\boldsymbol{G}$, the elastic Bragg term of Sect. 2.2, which in the ground state contributes $(2\pi/\mathcal{V})|\langle n_e(\boldsymbol{q})\rangle|^2\,\delta(\omega)$ to S at a fixed lattice and deposits no energy.

4.2.3 The fluctuation–dissipation theorem

The response function measures dissipation: by Eq. (4.8), $-\operatorname{Im} R_{bb}$ is the power a field coupled to O_b loses to the target. The structure factor measures fluctuations, the correlations of O_b in the undisturbed equilibrium state. The two were already close in Sect. 4.1. The imaginary part of a diagonal response is a difference of two transition sums, $\operatorname{Im} R_{aa}=-(\pi/\mathcal{V})[\mathcal{S}_{O_a}(\omega)-\mathcal{S}_{O_a^\dagger}(-\omega)]$ (Eq. (4.7)), and detailed balance Eq. (2.68) turns the emission sum $\mathcal{S}_{O_a^\dagger}(-\omega)$ into $\mathrm{e}^{-\beta\omega}\mathcal{S}_{O_a}(\omega)$. Hence $\operatorname{Im} R_{aa}=-(\pi/\mathcal{V})(1-\mathrm{e}^{-\beta\omega})\,\mathcal{S}_{O_a}(\omega)$, and with $S_{aa}=(2\pi/\mathcal{V})\mathcal{S}_{O_a}$ the diagonal relation is $(1-\mathrm{e}^{-\beta\omega})S_{aa}=-2\operatorname{Im} R_{aa}$.

The off-diagonal elements follow from the Lehmann form. Write $c_{if}=\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle$ and $\omega_{fi}=E_f-E_i$, so that Eq. (4.6) reads $R_{ab}=\mathcal{V}^{-1}\sum_{i,f}p_i\,[c_{if}/(\omega-\omega_{fi}+\mathrm{i}0^+)-\langle i|O_b|f\rangle\langle f|O_a^\dagger|i\rangle/(\omega+\omega_{fi}+\mathrm{i}0^+)]$. For real ω the complex conjugate of R_{ba} has the same numerators, since $(\langle i|O_b^\dagger|f\rangle\langle f|O_a|i\rangle)^*=c_{if}$, and the same denominators with $-\mathrm{i}0^+$. In the difference $R_{ab}-R_{ba}^*$ the principal values cancel, and each pair of denominators leaves $1/(x+\mathrm{i}0^+)-1/(x-\mathrm{i}0^+)=-2\pi\mathrm{i}\,\delta(x)$ (Eq. (2.58)). The resonant terms give $-\mathrm{i}S_{ab}$ directly. In the antiresonant terms, exchanging the names of i and f turns the weight p_i into $p_f=p_i\,\mathrm{e}^{-\beta\omega}$ on the support of the delta function, as in the proof of detailed balance, and they give $+\mathrm{i}\,\mathrm{e}^{-\beta\omega}S_{ab}$.

Fluctuation–dissipation theorem

$$\begin{aligned}(1-\mathrm{e}^{-\beta\omega})\,S_{ab}(\boldsymbol{q},\omega)&=\mathrm{i}\big[R_{ab}(\boldsymbol{q},\omega)-R_{ba}(\boldsymbol{q},\omega)^*\big],\\ (1-\mathrm{e}^{-\beta\omega})\,S_{aa}(\boldsymbol{q},\omega)&=-2\,\operatorname{Im} R_{aa}(\boldsymbol{q},\omega).\end{aligned}\tag{4.18}$$

The spectrum of equilibrium fluctuations of the target fixes its dissipative response, and conversely, for any H and any temperature.

This is Eq. (A.33), and the diagonal density element is Ref. [5, Eq. (3)], whose χ is our R_{ee} (Appendix A.7). The relation also runs the other way, and fixes all of R, not only its imaginary part. Writing the antiresonant terms of the Lehmann form with i and f exchanged, $R_{ab}=\mathcal{V}^{-1}\sum_{i,f}(p_i-p_f)\,c_{if}/(\omega-\omega_{fi}+\mathrm{i}0^+)$, and since $(p_i-p_f)=p_i(1-\mathrm{e}^{-\beta\omega_{fi}})$,

$$R_{ab}(\boldsymbol{q},\omega)=\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega'}{2\pi}\,\frac{(1-\mathrm{e}^{-\beta\omega'})\,S_{ab}(\boldsymbol{q},\omega')}{\omega-\omega'+\mathrm{i}0^+}.\tag{4.19}$$

Given S_{ab} at all frequencies, this reconstructs the response, real part included; Sect. 4.3 turns it into the Kramers–Kronig relations.

The factor $1-\mathrm{e}^{-\beta\omega}$ is one at $T=0$ and $\beta\omega$ for $|\omega|\ll T$. At $T=0$ the theorem reads $S_{aa}=-2\operatorname{Im} R_{aa}$ for $\omega>0$, and both sides vanish for $\omega<0$. For an electronic excitation in a detector the zero-temperature form is all we need: across the silicon gap of 1.12 eV at 300 K, $\mathrm{e}^{-\beta\omega}=2\times10^{-19}$. For $|\omega|\ll T$, $S_{aa}=-2T\operatorname{Im} R_{aa}/\omega$ grows with temperature. This is the

classical form of the theorem, the one behind the thermal noise of a resistor, and it applies to the low-lying phonon and magnon excitations of Chaps. 5 and 7.

The thermal oscillator shows the two sides in closed form. Its response, Eq. (4.9), does not depend on temperature, and its imaginary part is $\mathcal{V}\,\mathrm{Im}\,R = -\pi\lambda_{\mathrm{osc}}^2[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$. Its structure factor, from Worked example 2.7, is $VS = 2\pi\lambda_{\mathrm{osc}}^2[(n_{\mathrm{B}}(\omega_0)+1)\,\delta(\omega-\omega_0) + n_{\mathrm{B}}(\omega_0)\,\delta(\omega+\omega_0)]$ and does. The theorem connects them through $1/(1-\mathrm{e}^{-\beta\omega_0}) = n_{\mathrm{B}}(\omega_0)+1$ at $\omega=\omega_0$ and $1/(1-\mathrm{e}^{\beta\omega_0}) = -n_{\mathrm{B}}(\omega_0)$ at $\omega=-\omega_0$. The thermal populations sit in S alone. A temperature-independent response is compatible with a rate that grows with temperature, because the target's fluctuations grow while its net absorption does not.

For dark matter, Eq. (4.18) is the bridge between a rate and a response. The rate needs S, while most of the theory of solids, and every optical and electron-energy-loss measurement, delivers a response. Sect. 4.4 shows that the density response of the electrons is the inverse dielectric function, and the theorem then writes S as the energy-loss function times $2q^2/e^2$ [5, Eq. (6)], which carries screening into the rate. For silicon, S is computed in two ways. At the level of independent electrons, it is a sum over transitions between Kohn–Sham bands weighted by the density vertex $\rho_{cv}(\boldsymbol{k},\boldsymbol{q})$ (Eq. (A.38); Appendix E.7). With screening and local fields, a first-principles calculation computes R_{ee} in the random-phase approximation or with a kernel of time-dependent density-functional theory, and the theorem converts it (Sect. 4.6; Appendix E.6).

4.2.4 Positivity

A rate cannot be negative, and for one operator the structure factor is manifestly a sum of positive terms. With several operators the off-diagonal elements are complex and of either sign, and positivity becomes a statement about the matrix. Take any coefficients x_b and the combined operator $O_x = \sum_b x_b O_b$. Its transition sum is

$$\sum_{a,b} x_a^*\, S_{ab}(\boldsymbol{q},\omega)\, x_b = \frac{2\pi}{\mathcal{V}} \sum_{i,f} p_i \left|\sum_b x_b \langle f|O_b|i\rangle\right|^2 \delta(\omega - E_f + E_i) \geq 0,$$

since $\sum_a x_a^* \langle i|O_a^\dagger|f\rangle$ is the complex conjugate of $\sum_b x_b\langle f|O_b|i\rangle$.

Positivity of the response matrix

At every $\boldsymbol{q}$ and ω, $S_{ab}(\boldsymbol{q},\omega)$ is a Hermitian positive semidefinite matrix: $\sum_{ab} x_a^* S_{ab} x_b \geq 0$ for all x_b. In particular $S_{aa} \geq 0$ and

$$|S_{ab}(\boldsymbol{q},\omega)|^2 \leq S_{aa}(\boldsymbol{q},\omega)\, S_{bb}(\boldsymbol{q},\omega). \tag{4.20}$$

The inequality is the Cauchy–Schwarz inequality for a positive semidefinite matrix, the nonnegative determinant of its 2×2 principal submatrices (Eq. (C.55); Appendix C.7 gives the equivalent conditions). Positivity constrains rates, interference and absorption. The rate Eq. (4.14) is nonnegative at every $\boldsymbol{q}$ for any set of couplings, since its integrand has the form $x^\dagger S x$ with $x_b = \tilde{V}_b$. The interference term between two couplings is bounded in magnitude by twice the geometric mean of their separate rates, which Chap. 8 uses to bound mixed electron–nuclear signals. Through Eq. (4.18), the matrix $\mathrm{i}(R_{ab} - R_{ba}^*)$ is positive semidefinite for $\omega > 0$, at any temperature, since there $1-\mathrm{e}^{-\beta\omega} > 0$. A target in equilibrium therefore absorbs power from any weak field at positive frequency. Eq. (4.8) showed this for a single Hermitian operator; positivity extends it to operators that are not Hermitian, such as $n_e(\boldsymbol{q})$, and to any combination of them. On the diagonal it reads $\mathrm{Im}\,R_{aa} \leq 0$ for $\omega > 0$, at any temperature. The real part of R obeys no such constraint, and neither does the static response,

which can have either sign off the diagonal. In practice positivity is a check: a computed S_{ab} with a negative eigenvalue at some $(\boldsymbol{q},\omega)$ contains an error.

4.2.5 Independent particles

When the electrons do not interact, the many-body states are occupation-number states of single-particle orbitals γ, and the structure factor is a sum over moves of one electron. The transition sum was worked out in Eq. (2.70), and multiplying by $2\pi/\mathcal{V}$,

$$S^0(\boldsymbol{q},\omega) = \frac{2\pi}{\mathcal{V}} \sum_{\gamma\neq\gamma'} \left|\langle\gamma'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma\rangle\right|^2 n_{\mathrm{F}}(E_\gamma-\mu_{\mathrm{ch}})\left[1-n_{\mathrm{F}}(E_{\gamma'}-\mu_{\mathrm{ch}})\right]\delta(\omega-E_{\gamma'}+E_\gamma), \quad (4.21)$$

with the spin included in γ. To it adds the elastic term, the many-body diagonal $f=i$ of Eq. (4.13), which at $T=0$ is $(2\pi/\mathcal{V})|\langle n_e(\boldsymbol{q})\rangle|^2\delta(\omega)$; it contains cross terms between different occupied orbitals, and it is not part of the single-particle sum, whose terms with $\gamma'=\gamma$ carry $n(1-n)=0$ at $T=0$. We write S^0 for the independent-particle structure factor; through Eq. (4.18) it is the imaginary part of the independent-particle response R^0 of Sect. 4.5. Each term is a single-particle transition form factor $f_{\gamma\to\gamma'}(\boldsymbol{q})$ of Eq. (2.17), squared, with the Pauli factors of the many-electron state. For a real target S^0 is the starting point, not the answer. Built from Kohn–Sham orbitals it is the independent-particle structure factor of Eq. (A.38), and it misses the screening and the collective oscillation that the Coulomb interaction produces (Sects. 4.4 and 4.5). The free electron gas and a gas of atoms show what it contains.

Worked example 4.3: The independent-particle structure factor of the electron gas

For free electrons at $T=0$ the orbitals are plane waves, $|\langle\boldsymbol{k}'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\boldsymbol{k}\rangle| = \delta_{\boldsymbol{k}',\boldsymbol{k}+\boldsymbol{q}}$, and the occupations are those of the Fermi sea of Eq. (2.71). A transition moves an electron from $\boldsymbol{k}$ inside the sphere to $\boldsymbol{k}+\boldsymbol{q}$ outside it, leaving a hole behind, and deposits $(\boldsymbol{k}+\boldsymbol{q})^2/2m_e - k^2/2m_e = \boldsymbol{k}\cdot\boldsymbol{q}/m_e + q^2/2m_e$. With two spins and $\sum_{\boldsymbol{k}} \to \mathcal{V}\int \mathrm{d}^3k/(2\pi)^3$,

$$S^0(q,\omega) = 4\pi \int \frac{\mathrm{d}^3k}{(2\pi)^3}\,\Theta(k_{\mathrm{F}}-k)\,\Theta(|\boldsymbol{k}+\boldsymbol{q}|-k_{\mathrm{F}})\,\delta\!\left(\omega-\frac{kqu}{m_e}-\frac{q^2}{2m_e}\right),$$

with u the cosine of the angle between $\boldsymbol{k}$ and $\boldsymbol{q}$. The delta function fixes $u=u_0=(m_e\omega/q-q/2)/k$ with the weight $m_e/(kq)$. The angle must exist, $|u_0|\le 1$, which is $k\ge|m_e\omega/q-q/2|$. The final state must be empty, $k^2+2kqu_0+q^2 = k^2+2m_e\omega > k_{\mathrm{F}}^2$. With $\mathrm{d}^3k = 2\pi k^2\,\mathrm{d}k\,\mathrm{d}u$, the integral over k runs from the larger of the two lower limits, $k_{\min}$, to k_{F}:

$$S^0(q,\omega) = \frac{m_e}{\pi q}\int_{k_{\min}}^{k_{\mathrm{F}}} k\,\mathrm{d}k = \frac{m_e}{2\pi q}\left(k_{\mathrm{F}}^2-k_{\min}^2\right), \qquad k_{\min} = \max\Big(\Big|\frac{m_e\omega}{q}-\frac{q}{2}\Big|,\,\left(k_{\mathrm{F}}^2-2m_e\omega\right)^{1/2}\Big), \quad (4.22)$$

for $k_{\min}<k_{\mathrm{F}}$ and zero otherwise, with the square root set to zero when $2m_e\omega > k_{\mathrm{F}}^2$. Through Eq. (4.18) it fixes the imaginary part of the independent-particle response R^0 of the gas, the Lindhard function of Sect. 4.5 [7].

The weight is confined to the particle–hole continuum (Fig. 4.3a). Its upper edge, $\omega_+ = qv_{\mathrm{F}} + q^2/2m_e$, is an electron at the top of the sea kicked along $\boldsymbol{q}$. Its lower edge is zero for $q<2k_{\mathrm{F}}$, since an electron at the Fermi surface can be moved to an empty state just outside at almost no cost. For $q>2k_{\mathrm{F}}$ even an electron moving against $\boldsymbol{q}$ at v_{F} ends up outside the sea, and the edge rises to $\omega_- = q^2/2m_e - qv_{\mathrm{F}}$. Inside, Eq. (4.22) has a linear and a parabolic form. Below the curve $\omega = qv_{\mathrm{F}} - q^2/2m_e$, which exists for $q<2k_{\mathrm{F}}$,

the Pauli condition decides, $k_{\min}^2 = k_F^2 - 2m_e\omega$, and

$$S^0(q,\omega) = \frac{m_e^2\,\omega}{\pi q}.$$

The weight grows linearly with ω, because only electrons within a shell of thickness $\sim m_e\omega/k_F$ below the Fermi surface can be promoted. Above the curve the angle decides, and $S^0 = (m_e/2\pi q)[k_F^2 - (m_e\omega/q - q/2)^2]$, a parabola in ω. The two forms meet at $S^0 = m_e(2k_F - q)/2\pi$; at $q = k_F/2$ this is $0.239\,m_e k_F$, at $\omega = 0.75\,E_F$. For $q \gg k_F$ only the parabola is left. It is centered on the free recoil $\omega = q^2/2m_e$ of Worked example 2.3, with the half-width qv_F of the Doppler shift from the electron's motion in the sea: this is the impulse approximation, in which the struck electron recoils as if free (Fig. 4.3b).
The total weight and the first moment check the result. Integrating the sum form over ω removes the delta function, and what is left counts the electrons that can take the momentum q, $\int \mathrm{d}\omega\, S^0 = 2\pi n_e \times (3q/4k_F - q^3/16k_F^3)$ for $q < 2k_F$ and $2\pi n_e$ above. By Eq. (4.17) the static structure factor of the free gas is the unblocked fraction Eq. (2.72): Pauli exclusion is the hole around each electron that suppresses the long-wavelength fluctuations, $S^{\mathrm{st}} \to 3q/4k_F$. The first moment weights each transition with its energy $\Delta_{\boldsymbol{k}} = (\boldsymbol{k}+\boldsymbol{q})^2/2m_e - k^2/2m_e$. With the occupations $n_{\boldsymbol{k}} = \Theta(k_F - k)$, the sum $\sum_{\boldsymbol{k}s} n_{\boldsymbol{k}}(1-n_{\boldsymbol{k}+\boldsymbol{q}})\Delta_{\boldsymbol{k}}$ splits in two. The part with $n_{\boldsymbol{k}}n_{\boldsymbol{k}+\boldsymbol{q}}$ changes sign under $\boldsymbol{k} \to -\boldsymbol{k} - \boldsymbol{q}$, which exchanges the two occupations and reverses $\Delta_{\boldsymbol{k}}$, and it vanishes. In the rest, $\Delta_{\boldsymbol{k}} = \boldsymbol{k}\cdot\boldsymbol{q}/m_e + q^2/2m_e$, the term in $\boldsymbol{k}\cdot\boldsymbol{q}$ averages to zero over the sphere, and $N_e q^2/2m_e$ is left. Multiplying by $2\pi/\mathcal{V}$, $\int \mathrm{d}\omega\,\omega\, S^0 = \pi n_e q^2/m_e$ at every q, the f-sum rule Eq. (A.39) that Sect. 4.3 derives for any target.
For dark matter at the silicon valence density, $k_F = 3.57\,\mathrm{keV}$, $E_F = 12.5\,\mathrm{eV}$ and $v_F = 7.0\times10^{-3}$ (Sect. 2.5), and the energy deposit is at most $qv_{\max}$ with $v_{\max} = 2.65\times10^{-3} = 0.379\,v_F$. The dark matter samples only the corner of the continuum below this line. For $q < 2k_F(1 - v_{\max}/v_F) = 1.24\,k_F = 4.43\,\mathrm{keV}$ the line lies in the linear region, where on the line itself $S^0 = m_e^2 v_{\max}/\pi = 2.2\times10^8\,\mathrm{eV}^2$, or $0.121\,m_e k_F$, independent of q. A slow probe excites only electrons near the Fermi surface. A real semiconductor differs in the corner that matters most: its gap removes all weight below 1.17 eV, and the Coulomb interaction moves much of the weight at small q into the plasmon (Sect. 4.5).

A gas of atoms is the other independent-particle target, and the one of Sect. 2.2. There the rate off a bound electron was written with single-electron form factors, and the structure factor of a gas of atoms was identified by comparison with the rate formula of Sect. 1.5. The many-body definition now produces it directly.

Worked example 4.4: Hydrogen from the response matrix

Take N_{at} hydrogen atoms at fixed positions $\boldsymbol{R}_\ell$, far enough apart that their wavefunctions do not overlap, in a box of volume $\mathcal{V}$, with number density $n_T = N_{\mathrm{at}}/\mathcal{V}$. At any temperature far below the 10.2 eV of the lowest excitation, every atom is in its 1s state. The density operator is a sum over atoms, $n_e(\boldsymbol{q}) = \sum_\ell \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_\ell}$, with $\boldsymbol{r}_\ell$ the position of the electron of atom ℓ. Acting on the ground state it changes one atom at a time. A final state in which atom ℓ is in the level f and all others are in 1s is reached by the term of atom ℓ alone, with the matrix element $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_\ell} f_{1s\to f}(\boldsymbol{q})$, since $\boldsymbol{r}_\ell = \boldsymbol{R}_\ell + \boldsymbol{r}$ with $\boldsymbol{r}$ measured from the nucleus. The phase drops out of the square, every atom contributes the same,

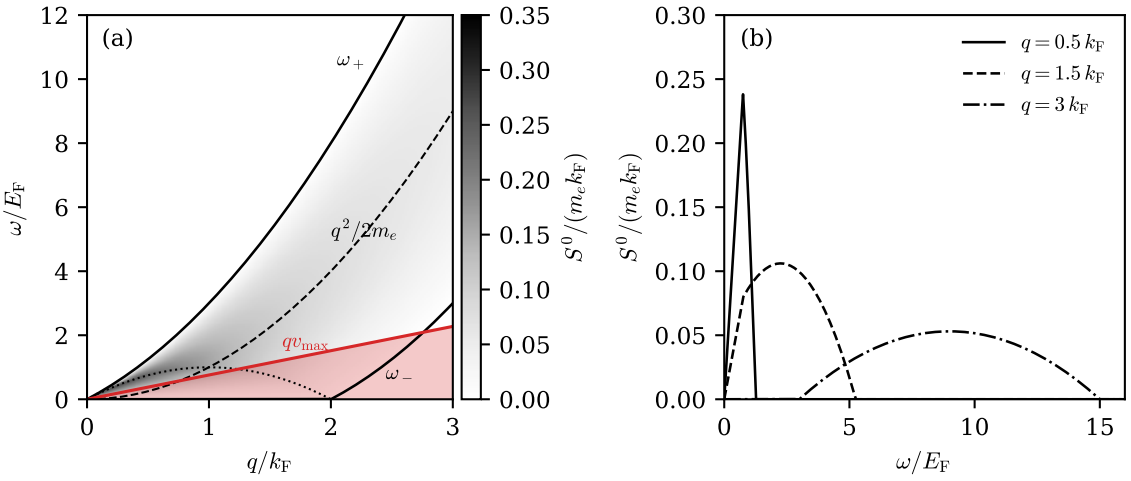


Figure 4.3: Dynamic structure factor $S^0(q,\omega)$ of the non-interacting electron gas at zero temperature, in units of $m_e k_F$, against the momentum transfer q in units of the Fermi wavevector k_F and the energy ω in units of the Fermi energy $E_F = k_F^2/2m_e$. (a) Gray scale: S^0 inside the particle–hole continuum, bounded above by $\omega_+ = qv_F + q^2/2m_e$ and below by $\omega_- = q^2/2m_e - qv_F$ for $q > 2k_F$ (solid). Below the dotted curve $\omega = qv_F - q^2/2m_e$, $S^0 = m_e^2\omega/\pi q$ grows linearly in ω; above it S^0 is a parabola in ω centered on the free recoil $q^2/2m_e$ (dashed). Red: the energies $\omega \le qv_{\max}$ that dark matter with the maximum halo speed $v_{\max} = 2.65 \times 10^{-3}$ can deposit, for the valence density of silicon, $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$ ($k_F = 3.57\,\mathrm{keV}$, $E_F = 12.5\,\mathrm{eV}$, $v_{\max} = 0.379\,v_F$). (b) S^0 against ω at $q = 0.5$, 1.5 and $3\,k_F$. At $q = 0.5\,k_F$ the triangle peaks at $0.239\,m_e k_F$ at $\omega = 0.75\,E_F$; at $q = 3\,k_F$ the parabola spans $\omega_- = 3\,E_F$ to $\omega_+ = 15\,E_F$. Every curve integrates to $2\pi n_e$ times the fraction of the Fermi sea that can take the momentum q, and its first moment is $\pi n_e q^2/m_e$.

and Eq. (4.13) becomes

$$S(\boldsymbol{q},\omega) = 2\pi\, n_T \sum_{f\neq 1s} |f_{1s\to f}(\boldsymbol{q})|^2\, \delta(\omega - E_f + E_{1s}) + \frac{2\pi}{\mathcal{V}}\, F_{1s}(q)^2 \left|\sum_\ell \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_\ell}\right|^2 \delta(\omega), \quad (4.23)$$

with F_{1s} the elastic form factor Eq. (2.21). The inelastic term is the structure factor that Sect. 2.2 identified by comparing rates. Inserted into Eq. (2.67), it gives

$$\Gamma = n_T \int \frac{\mathrm{d}^3 q}{(2\pi)^3}\, |\tilde{V}(\boldsymbol{q})|^2 \sum_f |f_{1s\to f}(\boldsymbol{q})|^2\, 2\pi\, \delta(\omega_{\boldsymbol{q}} - E_f + E_{1s}),$$

which is n_T times the $\sigma_T v$ of one atom, Eq. (2.18). Each atom scatters on its own, and the rate counts atoms; the angle and halo averages of Sect. 2.2 then produce the spectrum Eq. (2.20). For an atom with several electrons, Eq. (4.21) supplies what Sect. 2.2 added by hand: the initial orbitals must be occupied and the final ones empty.

The elastic term is where the atoms act together. At a fixed $\boldsymbol{q}$ its weight depends on the arrangement of the nuclei. For atoms at random positions the cross terms of $|\sum_\ell \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_\ell}|^2$ average to zero at $q \neq 0$ and the sum is N_{at}. In a crystal the phases add at the reciprocal-lattice vectors instead. With fixed nuclei the elastic term deposits nothing; free atoms would take up the recoil of the whole atom.

The numbers of Worked example 2.2 distribute the weight. At $q = q_0$ the integral of the inelastic term over ω is $2\pi n_T(1 - F_{1s}^2) = 2\pi n_T \times 0.59$, of which the $2p$ level takes 0.061, about a tenth. The elastic term of the random gas carries $2\pi n_T \times 0.41$, and by Eq. (4.17) the two add to $S^{\mathrm{st}}(q_0) = 1$. A gas of independent atoms scatters incoherently at every q, and the form factor decides only how much of each scattering excites the atom.

The free gas and the isolated atom are both exact for independent particles, and both miss what makes a solid's response many-body. The f-sum check above is one of a set of exact constraints on S that hold whatever the interactions; Sect. 4.3 derives them from causality and commutators, and they are the tests that any computed structure factor of silicon has to pass.

4.3 Causality and sum rules

No calculation has the exact excited states of silicon, yet one number about its dynamic structure factor is known exactly at every momentum transfer: the energy-weighted integral $\int \mathrm{d}\omega\, \omega\, S(\boldsymbol{q},\omega)$ equals $\pi n_e q^2/m_e$, fixed by the electron density alone. Another exact statement ties the real part of every response function to its imaginary part. The energy-weighted integral follows from a commutator, the link between real and imaginary parts from causality, and neither depends on the interactions, the temperature or the crystal structure. We will derive both, and then use the f-sum rule to test a computed or tabulated spectrum: hydrogen, the free electron gas of Worked example 4.3, and silicon, whose 14 electrons per atom share the sum over energies from a few eV to tens of keV. The treatment follows Refs. [2, Sects. 3.2–3.4] and [4, Chaps. 2 and 3].

4.3.1 Kramers–Kronig relations

The spectral form Eq. (4.19) already reconstructs all of R from S, and for a diagonal element it becomes a relation between the real and imaginary parts of R itself. By the fluctuation–dissipation theorem Eq. (4.18), the numerator $(1 - \mathrm{e}^{-\beta\omega'})S_{aa}(\boldsymbol{q},\omega')$ is $-2\,\mathrm{Im}\, R_{aa}(\boldsymbol{q},\omega')$, a real

function of ω', and

$$R_{aa}(\boldsymbol{q},\omega) = -\frac{1}{\pi}\int_{-\infty}^{\infty} \mathrm{d}\omega'\, \frac{\operatorname{Im} R_{aa}(\boldsymbol{q},\omega')}{\omega - \omega' + \mathrm{i}0^+}.$$

The Sokhotski–Plemelj formula Eq. (2.58) splits the denominator into a principal value and $-\mathrm{i}\pi\,\delta(\omega-\omega')$. The delta function returns $\operatorname{Im} R_{aa}(\omega)$ on the right, as it must, and the principal value is the real part.

Kramers–Kronig relations

For a response function analytic in the upper half plane and falling off at least as $1/|\omega|$,

$$\begin{aligned} \operatorname{Re} R_{aa}(\boldsymbol{q},\omega) &= \frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \mathrm{d}\omega'\, \frac{\operatorname{Im} R_{aa}(\boldsymbol{q},\omega')}{\omega' - \omega}, \\ \operatorname{Im} R_{aa}(\boldsymbol{q},\omega) &= -\frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \mathrm{d}\omega'\, \frac{\operatorname{Re} R_{aa}(\boldsymbol{q},\omega')}{\omega' - \omega}. \end{aligned} \tag{4.24}$$

The dissipative part of a response, at all frequencies, fixes its reactive part at each frequency, and conversely.

These are Eq. (C.38), found by Kronig and by Kramers for the dispersion of light [8, 9]. The route through Eq. (4.19) yields the relation for the real part from the Lehmann form. The pair of relations needs less, just the two properties of Sect. 4.1: R is analytic in the upper half plane, and it falls off at large $|z|$, as $1/z^2$ for a density by Eq. (4.11). Integrate $R(\omega')/(\omega'-z)$, with z in the upper half plane, along the real axis and back over a large semicircle above it (Fig. 4.4a). The only singularity inside is the pole at $\omega' = z$; the singularities of R, its poles at the transition energies and the cut they merge into, lie below the axis. On the semicircle the integrand falls off faster than $1/|\omega'|$, and its contribution vanishes as the radius grows. Cauchy's formula then reads $R(z) = (2\pi\mathrm{i})^{-1}\int \mathrm{d}\omega'\, R(\omega')/(\omega'-z)$. Letting $z \to \omega + \mathrm{i}0^+$ and applying Eq. (2.58) once more, $R(\omega) = (\mathrm{i}\pi)^{-1}\,\mathrm{PV}\int \mathrm{d}\omega'\, R(\omega')/(\omega'-\omega)$, whose real and imaginary parts are Eq. (4.24). Appendix C.5 writes out the steps in full, and Toll showed that the relations are in fact equivalent to causality [10]. Since nothing in this argument refers to eigenstates, the relations hold for a measured response, for a model such as the damped oscillator of Worked example 4.1, and for any approximation that respects causality. Fig. 4.4b rebuilds the real part of the oscillator response from its imaginary part alone. For an off-diagonal element the same argument applies to the analytic function R_{ab}, with the combination $\mathrm{i}(R_{ab} - R_{ba}^*)$ of Eq. (4.18) in the role of the absorptive part.

Most responses have a symmetry that folds the integral onto positive frequencies. For the density response of a target with inversion or time-reversal symmetry, the real part is even in ω and the imaginary part odd (Sect. 4.1). Splitting the integral at zero and combining $1/(\omega'-\omega) + 1/(\omega'+\omega) = 2\omega'/(\omega'^2 - \omega^2)$,

$$\operatorname{Re} R_{ee}(\boldsymbol{q},\omega) = \frac{2}{\pi}\,\mathrm{PV}\int_0^{\infty} \mathrm{d}\omega'\, \frac{\omega'\,\operatorname{Im} R_{ee}(\boldsymbol{q},\omega')}{\omega'^2 - \omega^2}, \tag{4.25}$$

which is Eq. (C.39). Two limits of it carry physics. At $\omega = 0$, with $-2\operatorname{Im} R_{ee} = (1 - \mathrm{e}^{-\beta\omega})S$,

$$R_{ee}(\boldsymbol{q},0) = -\frac{1}{\pi}\int_0^{\infty} \mathrm{d}\omega\, \frac{(1-\mathrm{e}^{-\beta\omega})\,S(\boldsymbol{q},\omega)}{\omega}. \tag{4.26}$$

The static response is an inverse moment of the structure factor, and it is negative because $S \geq 0$. At $T = 0$, inserting Eq. (4.13) returns the static limit of Sect. 4.1, $VR_{ee}(0) = -2\sum_f |\langle f|n_e(\boldsymbol{q})|0\rangle|^2/(E_f - E_0)$. Transitions at low energy dominate, since they are weighted

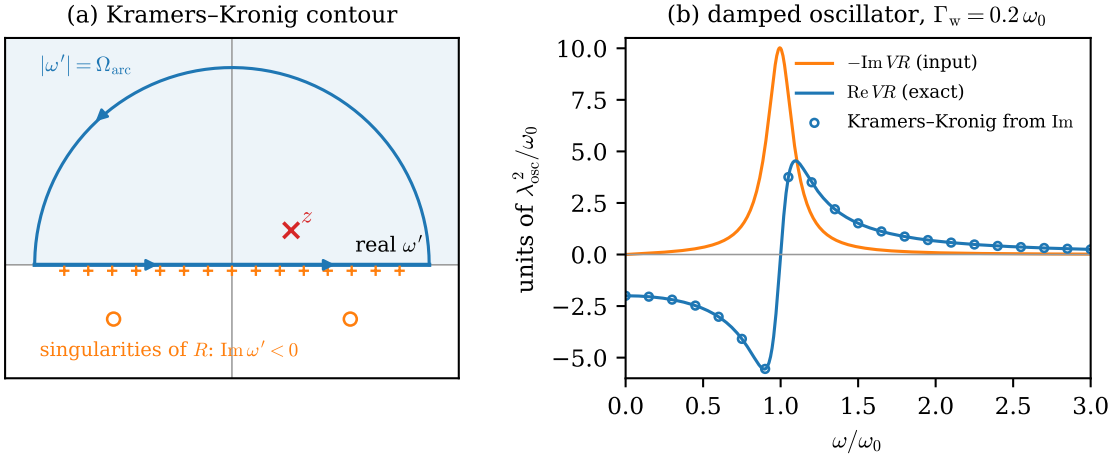


Figure 4.4: The Kramers–Kronig relations. (a) The contour in the complex frequency plane ω': the real axis, closed by a semicircle of radius $\Omega_{\rm arc} \to \infty$ in the upper half plane (shaded), where the retarded response R is analytic. The only singularity enclosed is the pole of $1/(\omega' - z)$ at a point z of the upper half plane (cross). The singularities of R, the poles at transition energies that merge into a branch cut and the resonance poles of its continuation (schematic), all lie below the axis. Letting z approach the real axis yields the relations. (b) Test on the damped oscillator with energy damping rate $\Gamma_{\rm w} = 0.2\,\omega_0$ and coupling $\lambda_{\rm osc}(d + d^\dagger)$. The real part of VR computed from $-\,\mathrm{Im}\,VR$ alone, with the folded relation for an even real part and an odd imaginary part (circles), agrees with the exact real part (line) to better than $10^{-11}\,\lambda^2_{\rm osc}/\omega_0$. At $\omega = 0$ both give $-2\lambda^2_{\rm osc}/\omega_0$.

by $1/\omega$: a target with a large gap is hard to deform. At large ω, on the other hand, the denominator of Eq. (4.25) becomes $-\omega^2$, and $\mathrm{Re}\,R_{ee} \to -(2/\pi\omega^2)\int_0^\infty \mathrm{d}\omega'\,\omega'\,\mathrm{Im}\,R_{ee}$. The tail Eq. (4.11) fixes the same coefficient by a double commutator. Equating the two, the energy-weighted integral of the absorptive part is fixed by an equal-time average. This is the first of the sum rules.

4.3.2 Moments from commutators

The k-th frequency moment of S_{ab} weights each transition $i \to f$ with ω_{fi}^k, where $\omega_{fi} = E_f - E_i$. The same power appears in a nested commutator. For any operator X, $\langle f|[H,X]|i\rangle = \omega_{fi}\langle f|X|i\rangle$, and k commutators with H give ω_{fi}^k. Take operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$, as for every density, and write $[H, O_b]_k$ for O_b commuted k times with H. In the average of $[O_a^\dagger, [H, O_b]_k]$ over any state diagonal in the energy basis, with weights p_i, we insert a complete set $|f\rangle$ between the two operators. The term $O_a^\dagger [H, O_b]_k$ contributes $\sum_{i,f} p_i\,\omega_{fi}^k \langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle$, the k-th moment of $V S_{ab}(\boldsymbol{q},\omega)/2\pi$ by Eq. (4.13). The term $[H, O_b]_k O_a^\dagger$ contributes $\sum_{i,f} p_i\,\omega_{if}^k \langle i|O_b|f\rangle\langle f|O_a^\dagger|i\rangle$. Since $O_b = O_b(-\boldsymbol{q})^\dagger$ and $O_a^\dagger = O_a(-\boldsymbol{q})$, this is $(-1)^k$ times the k-th moment of $V S_{ba}(-\boldsymbol{q},\omega)/2\pi$. Altogether,

$$\frac{1}{\mathcal{V}}\langle [O_a(\boldsymbol{q})^\dagger, [H, O_b(\boldsymbol{q})]_k]\rangle = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi}\,\omega^k\,\big[S_{ab}(\boldsymbol{q},\omega) - (-1)^k S_{ba}(-\boldsymbol{q},\omega)\big]. \tag{4.27}$$

Since the derivation used only the stationarity of the state, Eq. (4.27) holds in the Gibbs state at any temperature, in any eigenstate, and with any interactions. The integral runs over negative ω too, where a thermal target gives up energy. For $k = 0$ the left side is an

equal-time commutator, and for densities, which commute with one another, it vanishes. The case $k = 1$ is the one with content.

First-moment identity

For operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$ and any stationary state,

$$\mathscr{M}^1_{ab}(\boldsymbol{q}) \equiv \frac{1}{\mathscr{V}}\langle [O_a(\boldsymbol{q})^\dagger, [H, O_b(\boldsymbol{q})]]\rangle = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi}\, \omega\, [S_{ab}(\boldsymbol{q},\omega) + S_{ba}(-\boldsymbol{q},\omega)]. \qquad (4.28)$$

The energy-weighted spectrum, summed over $\pm\boldsymbol{q}$, is an equal-time double commutator.

The first-moment matrix $\mathscr{M}^1_{ab}$ is the coefficient of $1/z^2$ in the tail Eq. (4.11), now read as a statement about S. On the diagonal, for a target with $S_{aa}(-\boldsymbol{q},\omega) = S_{aa}(\boldsymbol{q},\omega)$, it reads $\int \mathrm{d}\omega\, \omega\, S_{aa} = \pi \mathscr{M}^1_{aa}$. A double commutator is often far simpler than the states it sums over, and this is why the identity is useful. The oscillator of Worked example 4.1 is the smallest example: $[O,[H,O]] = 2\lambda^2_{\mathrm{osc}}\omega_0$ is a number, and the first moment of its structure factor, $2\pi\lambda^2_{\mathrm{osc}}\omega_0$, does not depend on temperature, the result found from the populations in Worked example 2.7. Chap. 8 builds the higher moments into matrices for several operators.

4.3.3 The f-sum rule

For the electron density the double commutator is a number. The Hamiltonian of electrons and nuclei of Sect. 2.3 is the kinetic energy of the particles plus a potential energy that depends on their positions alone, and so do any external static potentials. The density $n_e(\boldsymbol{q}) = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$ commutes with every position and with every nuclear momentum. Only the electron kinetic energy is left in $[H, n_e(\boldsymbol{q})]$. For one electron, $[\boldsymbol{p}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}] = \boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, and $[\boldsymbol{p}^2, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}] = \boldsymbol{p}\cdot\boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} + \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,\boldsymbol{q}\cdot\boldsymbol{p}$. Summed over electrons,

$$[H, n_e(\boldsymbol{q})] = \boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q}), \qquad \boldsymbol{j}_e(\boldsymbol{q}) = \sum_j \frac{1}{2}\Big\{\frac{\boldsymbol{p}_j}{m_e},\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\Big\}, \qquad (4.29)$$

exactly, with $\{A,B\} = AB + BA$. This is the continuity equation in operator form. In the Heisenberg picture $\partial_t n_e(\boldsymbol{q},t) = \mathrm{i}[H, n_e(\boldsymbol{q},t)] = \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q},t)$, the transform with $\mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ of $\partial_t \hat{n}_e(\boldsymbol{r}) + \nabla\cdot\boldsymbol{j}_e(\boldsymbol{r}) = 0$ for the density operator of Sect. 4.2 and the current density in position space: density at $\boldsymbol{q}$ changes only because electrons carry it away. The symmetrized $\boldsymbol{j}_e(\boldsymbol{q})$ is what the literature calls the paramagnetic current, the current of the electrons when no vector potential acts.

The outer commutator, with $n_e(\boldsymbol{q})^\dagger$, is a number. Conjugating by the plane wave shifts the momentum, $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\boldsymbol{p}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = \boldsymbol{p} + \boldsymbol{q}$, so $[\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}, \boldsymbol{p}] = \boldsymbol{q}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, and each electron contributes q^2/m_e to $[n_e(\boldsymbol{q})^\dagger, \boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q})]$ (Appendix D.6). The commutator is $N_e q^2/m_e$ in every state, and $\mathscr{M}^1_{ee} = n_e q^2/m_e$.

f-sum rule

For electrons that move in local potentials, at any temperature and with any interactions,

$$\int_{-\infty}^{\infty} \mathrm{d}\omega\, \omega\, S(\boldsymbol{q},\omega) = \frac{\pi\, n_e\, q^2}{m_e} \qquad \big(S(-\boldsymbol{q},\omega) = S(\boldsymbol{q},\omega)\big), \qquad (4.30)$$

and in general the same for the average of S over $\boldsymbol{q}$ and $-\boldsymbol{q}$. The energy-weighted spectrum of density fluctuations counts the electrons.

This is Eq. (A.39) [2, Sect. 3.4]. The number $\boldsymbol{n}_e$ counts every electron, core electrons included, and the nuclei may move. The normalization is π over the whole real line. Many-body texts write the rule per particle and over positive frequencies, and the conversion is one line. At $T = 0$, S vanishes for $\omega < 0$, and dividing by $2\pi\boldsymbol{n}_e$ leads to $\int_0^\infty \mathrm{d}\omega\, \omega\, S/(2\pi\boldsymbol{n}_e) = q^2/2m_e$: per electron, and without the factor 2π of Eq. (4.13), the first moment is the free recoil energy [4, Chap. 2]. At $T > 0$, detailed balance $S(\boldsymbol{q}, -\omega) = \mathrm{e}^{-\beta\omega} S(\boldsymbol{q}, \omega)$ folds the negative frequencies onto the positive ones, $\int_0^\infty \mathrm{d}\omega\, \omega\, (1 - \mathrm{e}^{-\beta\omega})\, S = \pi\boldsymbol{n}_e q^2/m_e$. By the fluctuation–dissipation theorem the integrand is $-2\omega\, \mathrm{Im}\, R_{ee}$, which closes the loop with the large-ω limit of Eq. (4.25).

The rule has a direct reading at large q. Divided by the total weight $2\pi\boldsymbol{n}_e S^{\mathrm{st}}(\boldsymbol{q})$ of Eq. (4.17), it gives the mean energy deposited at $T = 0$, $q^2/[2m_e S^{\mathrm{st}}(\boldsymbol{q})]$. Where $S^{\mathrm{st}} \to 1$ and the dark matter scatters on one electron at a time, the mean energy deposited tends to the free recoil energy, however strongly the electron is bound. Binding moves the weight around, broadening it as in the impulse approximation of Worked example 4.3, but it cannot shift its center. At small q, where S^{st} is suppressed, the mean energy lies far above $q^2/2m_e$, and in the interacting electron gas the plasmon carries all of the sum (Sect. 4.5).

The right side of Eq. (4.30) defines the frequency scale of the electrons. With $e^2 = 4\pi\alpha$,

$$\frac{\pi\boldsymbol{n}_e q^2}{m_e} = \frac{\pi q^2}{e^2}\,\omega_p^2, \qquad \omega_p^2 = \frac{e^2\boldsymbol{n}_e}{m_e}, \tag{4.31}$$

where ω_p is the plasma frequency, the frequency at which the electrons, treated as free, would oscillate together against a uniform positive background (Sect. 4.5). For the valence density of silicon, $\boldsymbol{n}_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$, $\omega_p = 16.6\,\mathrm{eV}$; for all 14 electrons per atom, $7.00 \times 10^{23}\,\mathrm{cm}^{-3}$, it is 31.1 eV. Sect. 4.4 converts Eq. (4.30) into the form $\int_0^\infty \mathrm{d}\omega\, \omega\, L = (\pi/2)\,\omega_p^2$ for the loss function, the form that a computed or measured spectrum is usually tested against.

4.3.4 Nonlocal potentials and the f-sum rule

The derivation of Eq. (4.30) needed one hypothesis: every potential energy in H is a function of positions. The Hamiltonian of electrons and nuclei satisfies it, but the effective Hamiltonians with which solids are computed often do not. A pseudopotential (Sect. 3.5) replaces the nucleus and the core electrons by a local potential and a nonlocal part V_{NL}, the sum of projections Eq. (E.8), which acts on a valence wavefunction through its values over a whole core region rather than at one point. The Fock exchange operator of Sect. 3.4 is nonlocal in the same sense. Such an operator does not commute with $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, the commutator $[H, n_e(\boldsymbol{q})]$ acquires $[V_{\mathrm{NL}}, n_e(\boldsymbol{q})]$ besides $\boldsymbol{q}\cdot\boldsymbol{j}_e$, and

$$\mathscr{M}^1_{ee}(\boldsymbol{q}) = \frac{\boldsymbol{n}_e q^2}{m_e} + \frac{1}{\mathscr{V}}\langle [n_e(\boldsymbol{q})^\dagger, [V_{\mathrm{NL}}, n_e(\boldsymbol{q})]]\rangle, \tag{4.32}$$

with $\boldsymbol{n}_e$ now the density of the valence electrons that the calculation keeps. The exact rule is not violated, since it belongs to the full Hamiltonian. The pseudo-Hamiltonian is a device that reproduces the valence states outside the cores, not their every commutator. A pseudo-wavefunction is smooth where the true valence state oscillates to stay orthogonal to the core states, and part of what that oscillation does is carried by V_{NL}. The extra term is a sum of projections and costs almost nothing to evaluate. Appendix E.2 writes it for one electron, Eq. (E.16), and finds that in a representative silicon calculation this term and the truncated band sum together put the computed sum 12 to 19 percent below the local value at small q. A sum-rule test of a pseudopotential calculation therefore compares with Eq. (4.32), not with Eq. (4.30). At $q \to 0$ the same algebra turns the momentum into the velocity $\mathrm{i}[H, \boldsymbol{r}] = \boldsymbol{p}/m_e + \mathrm{i}[V_{\mathrm{NL}}, \boldsymbol{r}]$, the form that Ismail-Beigi, Chang and Louie derived for the coupling of a nonlocal potential to a long-wavelength field [11].

Relativistic electrons break the hypothesis in a different way. To order $1/m_e^2$, an electron in a one-body potential $U_1(\boldsymbol{r})$ has three corrections, derived in Sect. 3.5: the mass–velocity term $-\boldsymbol{p}^4/8m_e^3$, the Darwin term $\nabla^2 U_1/8m_e^2$ and the spin–orbit term $\boldsymbol{\sigma}\cdot(\nabla U_1\times\boldsymbol{p})/4m_e^2$. The Darwin term is a function of position and commutes with $n_e(\boldsymbol{q})$. The spin–orbit term is linear in $\boldsymbol{p}$, so its commutator with $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ is a function of position and spin, which commutes with $n_e(\boldsymbol{q})^\dagger$; it drops out of the double commutator. The mass–velocity term does not. Shifting $\boldsymbol{p}$ by $\pm\boldsymbol{q}$ as before, it contributes $-[(\boldsymbol{p}+\boldsymbol{q})^4+(\boldsymbol{p}-\boldsymbol{q})^4-2\boldsymbol{p}^4]/8m_e^3 = -[4p^2q^2+8(\boldsymbol{p}\cdot\boldsymbol{q})^2+2q^4]/8m_e^3$ per electron. In an atom, or in a crystal of cubic symmetry, the average of $(\boldsymbol{p}\cdot\hat{\boldsymbol{q}})^2$ is a third of that of p^2, and (Appendix D.6)

$$\mathscr{M}^1_{ee}(\boldsymbol{q}) = \frac{n_e q^2}{m_e}\Big[1-\frac{5}{3}\,\frac{\bar{K}}{m_e}-\frac{q^2}{4m_e^2}\Big], \tag{4.33}$$

where $\bar{K}$ is the mean kinetic energy per electron. A moving electron is heavier, and the f-sum, which counts electrons weighted by $1/m_e$, is reduced by the relative order $\bar{K}/m_e$. For silicon the virial theorem makes the total kinetic energy of the atom minus its total energy, 7.86 keV from the Hartree–Fock value of −288.854 hartree [12], so $\bar{K} = 561$ eV and $(5/3)\bar{K}/m_e = 1.8\times10^{-3}$, or 0.026 electrons per atom. The two 1s electrons dominate: each has a kinetic energy near $Z^2\alpha^2 m_e/2 = 2.67$ keV, which reduces its own share of the sum by 8.7×10^{-3}. The term in q^2 is 2.7×10^{-3} at the 53 keV that a 10 MeV particle can transfer at most (Appendix E.2). The two-body Breit terms, which depend on the momenta of pairs of electrons, contribute at the same relative order, and we do not estimate them. All of these corrections lie below the accuracy of any calculation of S for dark matter. They do mean, however, that an all-electron relativistic calculation with positive-energy states alone should not reproduce Eq. (4.30) to better than a few parts in 10^3. The expansion in $1/m_e$ is this positive-energy statement. The full Dirac Hamiltonian is linear in the momentum, its double commutator with the density vanishes, and the f-sum is then carried by the creation of electron–positron pairs, at energies near $2m_e$.

The nuclei obey the same algebra as the electrons. With the density $n_\kappa(\boldsymbol{q}) = \sum_{I\in\kappa}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$ of the nuclei of species κ, their mass M_κ and number density n_κ (Sect. 2.3), the kinetic energy of the nuclei gives $[H, n_\kappa(\boldsymbol{q})] = \boldsymbol{q}\cdot\boldsymbol{j}_\kappa(\boldsymbol{q})$, with a current $\boldsymbol{j}_\kappa$ built like $\boldsymbol{j}_e$ from nuclear momenta and M_κ. A current of one species contains only that species' momenta, and it commutes with the density of every other species. The mixed double commutators therefore vanish.

Species-resolved f-sum rule

For electrons and nuclei in local potentials, with $s, s' \in \{e, \kappa\}$ and m_s equal to m_e or M_κ,

$$\mathscr{M}^1_{ss'}(\boldsymbol{q}) = \delta_{ss'}\,\frac{q^2\,n_s}{m_s}. \tag{4.34}$$

Each species has its own f-sum, fixed by its density and mass, and the energy-weighted electron–nucleus interference vanishes.

The nuclear rule, $\int \mathrm{d}\omega\,\omega\,S_{\kappa\kappa} = \pi n_\kappa q^2/M_\kappa$, makes the mean energy deposited on a nucleus equal to its free recoil energy $q^2/2M_\kappa$ wherever S^{st} of the nuclei is one, the incoherent limit of the lattice response of Chaps. 5 and 6. The mixed rule has a less obvious consequence. For a target with inversion symmetry, $S_{\kappa e}(-\boldsymbol{q},\omega) = S_{\kappa e}(\boldsymbol{q},\omega)$, which Hermiticity makes $S_{e\kappa}(\boldsymbol{q},\omega)^*$, and Eq. (4.28) with Eq. (4.34) reads $\int \mathrm{d}\omega\,\omega\,\mathrm{Re}\,S_{e\kappa}(\boldsymbol{q},\omega) = 0$. At $T = 0$ the integral runs over $\omega > 0$ alone, and the real part of the electron–nucleus interference must change sign: the electrons follow a recoiling nucleus in phase at some energies and against it at others. A computed interference spectrum of one sign, for the Migdal effect for instance (Chaps. 8

and 12), is wrong. Finally, for the total charge density $n_Z = -n_e + \sum_\kappa Z_\kappa n_\kappa$ of Sect. 2.3, the diagonal terms add: $\mathscr{M}^1_{ZZ} = q^2[n_e/m_e + \sum_\kappa Z_\kappa^2 n_\kappa/M_\kappa]$. For silicon, with $Z_\kappa = 14$ and all 14 electrons per atom, the nuclear share is $14\,m_e/M_\kappa = 2.73 \times 10^{-4}$. A probe of the total charge, such as a dark photon, sees an f-sum set by the electrons to that accuracy (Appendix D.6).

4.3.5 Oscillator strengths

For a single atom the f-sum rule becomes a statement about its spectral lines. Take an atom with Z electrons in its ground state $|0\rangle$. Its double commutator is Zq^2/m_e, and inserting the excited states $|n\rangle$ as in Eq. (4.27) turns it into $\sum_n (E_n - E_0)\big[|\langle n|n_e(\boldsymbol{q})|0\rangle|^2 + |\langle n|n_e(-\boldsymbol{q})|0\rangle|^2\big]$. For a ground state with inversion symmetry the two terms are equal, and

$$\sum_n (E_n - E_0)\,\big|\langle n|n_e(\boldsymbol{q})|0\rangle\big|^2 = \frac{Zq^2}{2m_e}, \tag{4.35}$$

at every $\boldsymbol{q}$. For the gas of atoms of Worked example 4.4, whose structure factor is $2\pi n_T$ times the sum of these transition weights, it is Eq. (4.30) with $n_e = Z n_T$. As $q \to 0$, the matrix element to an excited state starts at first order in q, $\langle n|n_e(\boldsymbol{q})|0\rangle = \mathrm{i}\boldsymbol{q}\cdot\langle n|\sum_j \boldsymbol{r}_j|0\rangle$, since the zeroth order vanishes by orthogonality. The dipole matrix elements that remain define the oscillator strengths,

$$f^{\mathrm{osc}}_{n0} = 2m_e\,(E_n - E_0)\,\Big|\langle n|\sum_j \hat{\boldsymbol{q}}\cdot\boldsymbol{r}_j|0\rangle\Big|^2, \tag{4.36}$$

and Eq. (4.35) divided by $q^2/2m_e$ becomes the Thomas–Reiche–Kuhn rule,

$$\sum_n f^{\mathrm{osc}}_{n0} = Z, \tag{4.37}$$

where the sum includes the continuum, as an integral of the oscillator-strength density $\mathrm{d}f^{\mathrm{osc}}/\mathrm{d}\omega$ over the energy $\omega = E_n - E_0$ of the absorbed quantum. It was found independently in 1925 by Thomas, by Reiche and Thomas, and by Kuhn, before the Schrödinger equation, from the theory of dispersion [13–15]. In the classical picture an atom was a set of electrons bound by springs, and f^{osc}_{n0} counted how many of them oscillated at the frequency of line n. The rule says that the lines of an atom share exactly Z such classical electrons. Its use is as a check. A table of computed or measured oscillator strengths, or of photoabsorption cross-sections, must add up to Z, and a set of ionization form factors (Eq. (2.22)) must satisfy Eq. (4.35) at each q, which tests the normalization of the continuum.

Worked example 4.5: The Thomas–Reiche–Kuhn sum of hydrogen

For hydrogen, $Z = 1$, and the oscillator strengths from 1s are known in closed form [16, Sect. 61]. The dipole operator reaches only p states. For the bound levels,

$$f^{\mathrm{osc}}_{np,1s} = \frac{2^8\,n^5\,(n-1)^{2n-4}}{3\,(n+1)^{2n+4}},$$

which is 0.4162 for 2p, and 0.0791, 0.0290 and 0.0139 for 3p to 5p. For large n, $[(n-1)/(n+1)]^{2n} \to \mathrm{e}^{-4}$ and $f^{\mathrm{osc}}_{np,1s} \to (2^8/3)\,\mathrm{e}^{-4}/n^3$. The levels crowd toward the binding energy $E_B = \alpha^2 m_e/2 = 13.6\,\mathrm{eV}$ of Sect. 2.2 with spacing $2E_B/n^3$, and their strength per unit energy tends to $(2^7/3)\,\mathrm{e}^{-4}/E_B = 0.782/E_B$. The continuum takes over at the same value. With the Coulomb parameter of Sect. 2.2 for the ejected electron,

$\xi_C = \alpha m_e/p = [E_B/(\omega - E_B)]^{1/2}$ for $\omega > E_B$, its density is [16, Sect. 61]

$$\frac{\mathrm{d}f^{\mathrm{osc}}}{\mathrm{d}\omega} = \frac{2^7}{3E_B}\Big(\frac{E_B}{\omega}\Big)^4 \frac{\exp(-4\xi_C \operatorname{arccot}\xi_C)}{1 - \mathrm{e}^{-2\pi\xi_C}},$$

which tends to $0.782/E_B$ at threshold, where $\xi_C \to \infty$ and $\xi_C \operatorname{arccot}\xi_C \to 1$.
Adding up (Fig. 4.5a), the bound levels carry 0.565, of which 2p alone carries 0.416 and the first three 0.524. The continuum carries 0.435: 0.316 within E_B of the threshold, and only 0.0042 above $10E_B$, since at large ω, where $\xi_C \to 0$, the density falls as $\omega^{-7/2}$. The total is 1.000, as Eq. (4.37) requires. A calculation that keeps the bound states alone fails the check by 43 percent: the continuum, which is the ionization channel of Sect. 2.2, is not a correction.
The finite-q rule Eq. (4.35) tests the same states at a dark-matter momentum transfer. At $q_0 = \alpha m_e$ its right side is exactly $q_0^2/2m_e = \alpha^2 m_e/2 = E_B$. The 2p level has $|f_{1s\to 2p}(q_0)|^2 = 0.061$ (Worked example 2.2) at the energy $\frac{3}{4}E_B$, and contributes $0.046\,E_B$, under 5 percent of the sum, although it takes a tenth of the inelastic weight at that q (Worked example 4.4). The rest of the energy-weighted sum lies in the continuum, at energies $\sim E_B$ and above.

Worked example 4.6: The f-sum of the free electron gas

Worked example 4.3 obtained the first moment of S^0 from the sum over occupied states. The closed form Eq. (4.22) must reproduce it, and integrating it shows where the weight sits. For $q > 2k_F$, S^0 is the parabola between ω_- and ω_+. We change variable to $x = m_e\omega/q - q/2$, which runs from $-k_F$ to k_F. Then $\omega = (q/m_e)(x + q/2)$, $\mathrm{d}\omega = (q/m_e)\,\mathrm{d}x$ and $S^0 = (m_e/2\pi q)(k_F^2 - x^2)$, so that

$$\int \mathrm{d}\omega\, \omega\, S^0 = \frac{q}{2\pi m_e}\int_{-k_F}^{k_F} \mathrm{d}x \Big(x + \frac{q}{2}\Big)(k_F^2 - x^2) = \frac{q}{2\pi m_e}\,\frac{q}{2}\,\frac{4k_F^3}{3} = \frac{q^2 k_F^3}{3\pi m_e}.$$

The term odd in x, the Doppler shift from the motion of the struck electron, integrates to zero, and with $n_e = k_F^3/3\pi^2$ the result is $\pi n_e q^2/m_e$. For $q < 2k_F$ the linear region, $0 < \omega < \omega_1 = (q/m_e)(k_F - q/2)$, contributes $\int \mathrm{d}\omega\, \omega\, m_e^2\omega/\pi q = m_e^2\omega_1^3/3\pi q = q^2(k_F - q/2)^3/3\pi m_e$. That is the fraction $(1 - q/2k_F)^3$ of the sum, $27/64 = 0.42$ at $q = k_F/2$, and the parabola carries the rest. As $q \to 0$ the linear region carries the whole sum, and the mean energy $(q^2/2m_e)/S^{\mathrm{st}}$, with $S^{\mathrm{st}} \to 3q/4k_F$, is $\frac{2}{3}qv_F$, which vanishes. A gapless target can place its f-sum at vanishing energies, but only because its long-wavelength fluctuations are suppressed by the Pauli hole of Sect. 4.2.
A gap changes the balance. Take the valence electrons of an insulator at $T = 0$, with the nuclei clamped and the potentials local, and their share of the f-sum, collected below the core edges (Worked example 4.7). Their inelastic S vanishes below the gap E_g, so $E_g \int \mathrm{d}\omega\, S \le \int \mathrm{d}\omega\, \omega\, S$. By Eqs. (4.17) and (4.30), the inelastic part of the static structure factor then obeys, to the accuracy with which the valence electrons carry their own f-sum (about 2 percent in silicon, Worked example 4.7),

$$S^{\mathrm{st}}(\boldsymbol{q}) \le \frac{q^2}{2m_e E_g}.$$

At the valence density of silicon, $k_F = 3.57\,\mathrm{keV}$, and with $E_g = 1.17\,\mathrm{eV}$ this bound lies below the free-gas value $3q/4k_F$ for $q < 3m_e E_g/2k_F = 0.25\,\mathrm{keV}$; at $q = 0.1\,\mathrm{keV}$ it is 0.40

of it. The long-wavelength density fluctuations of an insulator vanish as q^2, not as q, and the f-sum rule is what turns the gap into this statement.

Worked example 4.7: The f-sum of silicon

A silicon atom has 14 electrons, four in the valence shell and ten in the core, two 1s and eight 2s and 2p. With the full, local Hamiltonian the f-sum counts all of them, and at $q \to 0$ the Thomas–Reiche–Kuhn rule Eq. (4.37), applied per atom of the crystal, gives 14. The four valence electrons absorb from the gap upward; the core electrons cannot absorb below their binding energies, which put the L edge near 100 eV and the K edge at 1.84 keV. We want to know over which energies each part of the sum is collected.

Henke, Gullikson and Davis tabulated the photoabsorption of silicon from 50 eV to 30 keV. Their current distribution extends down to 30 eV, and we integrate it from there. They express it through the imaginary part $f_2(\omega)$ of the atomic scattering factor, which in the convention of the tables is $f_2 = (\pi/2)\,\omega\,\mathrm{d}f^{\mathrm{osc}}/\mathrm{d}\omega$ per atom [17]. The two edges appear in the tables as steps of f_2, at 99.8 eV and 1.839 keV. Integrating $2f_2/\pi\omega$ over the regions they bound gives 0.35 between 30 eV and the L edge, 8.46 between the L and K edges, and 1.61 above the K edge. Of the last, 0.007 comes from extending the last decade of the tables beyond 30 keV as a power law. Above 30 eV the tables therefore hold 10.42 of the 14. The rule then requires 3.58 below 30 eV (3.55 with the mass–velocity reduction of Eq. (4.33)), and 3.93 below the L edge, against the four valence electrons (Fig. 4.5b).

The saturation is slow and has the same pattern in each region. The valence part is complete to 2 percent below the L edge, at six times ω_p. The region between the edges collects 97 percent of its strength by 1 keV, ten times its edge energy, and the region above the K edge collects 84 percent by 5 keV and 96 percent by 10 keV. Each shell gathers its strength over energies from its edge to several times it, with the slow $\omega^{-5/2}$ approach of the remaining strength that hydrogen showed.

The regions are not the shells. The region above the K edge holds 1.61, not 2. The 1s electrons cannot be promoted into the full 2p shell, a transition that carries 0.42 of the sum in hydrogen. In a picture of independent electrons, the rule holds for each electron only when transitions into occupied orbitals are counted too, upward with positive and downward with negative strength. In the atom both members of such a pair are blocked by the Pauli principle, and since their strengths cancel, the total stays 14 while strength moves from the inner shells to the outer ones. The excess of the region between the edges over 8 is the receiving side of this transfer, together with the overlap of tails: the L absorption continues above the K edge, and the valence absorption continues above the L edge.

For dark matter the split matters at finite q. At $q \to 0$ the core part of the sum lies above 100 eV, far from the valence part. At larger q the valence weight moves toward the free recoil energy $q^2/2m_e$, which is 98 eV at $q = 10$ keV, where the dark matter can deposit at most $qv_{\max} = 26.5$ eV. A valence-only calculation tested against its f-sum at such momentum transfers must follow the valence spectrum to energies where the core shells it omits absorb as well. The sum rule constrains the total over all energies, not the low-energy corner that the dark matter samples.

Sect. 4.4 carries them over to the dielectric and loss functions, where the Coulomb interaction between the electrons decides how the fixed weight is distributed in energy.

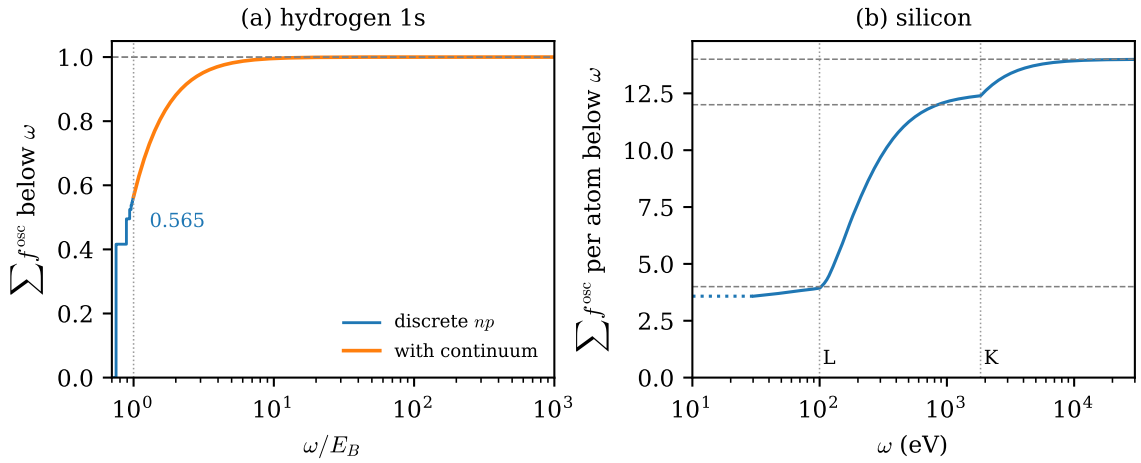


Figure 4.5: The Thomas–Reiche–Kuhn sum accumulated with energy. (a) Hydrogen in its ground state: oscillator strength of all transitions below the absorbed energy ω, in units of the binding energy $E_B = 13.6\,\text{eV}$, from the closed forms of Bethe and Salpeter. The bound 1s–np lines (steps, starting with 2p at $\frac{3}{4}E_B$) carry 0.565 and the continuum 0.435; the total is one electron (dashed). (b) Silicon, per atom: oscillator strength below ω from the photoabsorption tables of Henke and coworkers above 30 eV, integrated as $\mathrm{d}f^{\rm osc}/\mathrm{d}\omega = 2f_2/\pi\omega$, and offset so that the total, with the region above 30 keV extrapolated as a power law, is 14. The offset, 3.58, is the strength below 30 eV that the rule requires (dotted). Dashed lines mark 4 valence electrons, 4 + 8 and 14; dotted vertical lines the L and K edges at 99.8 eV and 1.839 keV.

4.4 Screening and the dielectric function

A dark-matter particle that transfers the typical momentum $q_0 = \alpha m_e = 3.73\,\text{keV}$ to silicon does not meet the bare valence electrons. Its potential pushes the electrons it reaches, the displaced charge produces a potential of its own, and the electrons respond to the sum. A crude estimate (Worked example 4.8) reduces the potential at q_0 by a factor of about two, and the rate, which goes as its square, by about five. Full calculations for silicon and germanium find that screening changes the rate by a factor of a few [5]. The response function R_{ee} of Sect. 4.1 contains all of this, but it mixes two kinds of physics: the long-range Coulomb field of the displaced charge, which is classical and universal, and everything else, which depends on the material. Separating the two defines the dielectric function. We will see that the structure factor that sets the rate is, up to a factor q^2, the imaginary part of the inverse dielectric function, the loss function, which is measured in optical and electron-energy-loss experiments. Along the way we carry the sum rules of Sect. 4.3 over to the loss function and to optical absorption. We also derive two classic applications: the screening length of an electron gas and the energy lost by a charged particle crossing a solid. The treatment follows Refs. [2, Chaps. 3 and 5] and [18].

4.4.1 Proper response and the dielectric function

Apply to the electrons an external potential energy that varies as $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, with amplitude $\phi_{\rm ext}(\boldsymbol{q},\omega)$ at frequency ω, and keep the nuclei clamped. By the Kubo formula Eq. (4.3), the electron density develops a ripple with amplitude $\delta\boldsymbol{n}_e = R_{ee}(\boldsymbol{q},\omega)\,\phi_{\rm ext}$, where $\delta\boldsymbol{n}_e$ is $\delta\langle n_e(\boldsymbol{q})^\dagger\rangle/\mathscr{V}$, the coefficient of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ in the induced density (Sect. 4.1). The ripple is a charge, and by Coulomb's law it acts on every electron with the potential energy $v_{\rm C}(q)\,\delta\boldsymbol{n}_e$, with the

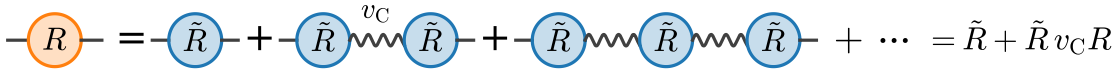


Figure 4.6: Schematic of the relation between the density response R and its Coulomb-proper part $\tilde{R}$. Each shaded blob is the proper response of the electrons, which includes every interaction effect except the long-range Coulomb field at the momentum transfer $\boldsymbol{q}$. Each wavy line is the Coulomb kernel $v_C(q) = e^2/q^2$, which carries the potential of the charge induced in one blob to the next. Summed, the chain is the geometric series $R = \tilde{R} + \tilde{R}v_C\tilde{R} + \cdots$, whose closed form is $R = \tilde{R} + \tilde{R}v_C R$. The drawing is a schematic, not a set of Feynman rules.

kernel Eq. (2.43). Each electron then feels the total potential energy

$$\phi_{\text{tot}}(\boldsymbol{q},\omega) = \phi_{\text{ext}}(\boldsymbol{q},\omega) + v_C(q)\,\delta n_e(\boldsymbol{q},\omega), \tag{4.38}$$

the external part plus the Hartree potential of the induced density. The induced density is a response to ϕ_{tot} as much as to ϕ_{ext}, and the response to the total potential is the one that excludes the long-range field. We define it as the Coulomb-proper (or irreducible) response $\tilde{R}_{ee}$,

$$\delta n_e = \tilde{R}_{ee}(\boldsymbol{q},\omega)\,\phi_{\text{tot}}(\boldsymbol{q},\omega). \tag{4.39}$$

It contains every effect of the interactions except the classical Hartree field at the wavevector $\boldsymbol{q}$: exchange, correlation, the band structure, the excitations of the target. In contrast, R_{ee} is the improper response.

Inserting Eq. (4.38) into Eq. (4.39) and comparing with $\delta n_e = R_{ee}\phi_{\text{ext}}$, for any ϕ_{ext},

$$R_{ee} = \tilde{R}_{ee} + \tilde{R}_{ee}\,v_C\,R_{ee}, \qquad R_{ee} = \frac{\tilde{R}_{ee}}{1 - v_C\tilde{R}_{ee}}. \tag{4.40}$$

This is a Dyson equation of the form Eq. (C.59), with $\tilde{R}_{ee}$ as the bare object and v_C as the kernel. Iterating it gives the geometric series

$$R_{ee} = \tilde{R}_{ee} + \tilde{R}_{ee}v_C\tilde{R}_{ee} + \tilde{R}_{ee}v_C\tilde{R}_{ee}v_C\tilde{R}_{ee} + \cdots,$$

which has a direct reading. The external potential polarizes the electrons, the polarization creates a Coulomb potential, that potential polarizes them again, and so on. Fig. 4.6 draws the series with the proper parts as blobs joined by Coulomb lines. The name "proper" says that $\tilde{R}_{ee}$ is the part of the response that cannot be cut into two pieces joined by a single Coulomb line at the momentum $\boldsymbol{q}$; Sect. 4.8 reads the figure as a sum of Feynman diagrams. The closed form holds beyond the radius of convergence of the series, and its poles, at $1 - v_C\tilde{R}_{ee} = 0$, are oscillations that the electrons sustain without an external field.

The ratio of the external to the total potential is the dielectric function. From Eqs. (4.38) and (4.39), $\phi_{\text{ext}} = (1 - v_C\tilde{R}_{ee})\,\phi_{\text{tot}}$ and $\phi_{\text{tot}} = (1 + v_C R_{ee})\,\phi_{\text{ext}}$. Hence

$$\varepsilon(\boldsymbol{q},\omega) \equiv \frac{\phi_{\text{ext}}}{\phi_{\text{tot}}} = 1 - v_C(q)\,\tilde{R}_{ee}(\boldsymbol{q},\omega), \qquad \varepsilon^{-1}(\boldsymbol{q},\omega) = 1 + v_C(q)\,R_{ee}(\boldsymbol{q},\omega), \tag{4.41}$$

and the two forms are each other's inverse by Eq. (4.40). These are the relations Eqs. (A.35) and (A.36) stated in Appendix A.7, and the second is Ref. [5, Eq. (5)] in our units. We write $\varepsilon = \varepsilon_1 + \mathrm{i}\varepsilon_2$. The classical reading is the familiar one. In Heaviside–Lorentz units the external charge is the source of $\boldsymbol{D}$ and the total charge the source of $\boldsymbol{E}$. For a longitudinal field the

potential energies are proportional to the potentials, so ε is the ratio D/E of the Fourier components, the dielectric function of classical electromagnetism, now for every $\boldsymbol{q}$ and ω. It is the longitudinal dielectric function, because a potential produces a field along $\boldsymbol{q}$.

The inverse function $\varepsilon^{-1} - 1 = v_{\mathrm{C}} R_{ee}$ is a retarded response, the induced potential per unit external one, and it has every property Sects. 4.1 and 4.3 proved for R_{ee}: analyticity in the upper half plane, the Kramers–Kronig relations, and $\operatorname{Im} \varepsilon^{-1} \leq 0$ for $\omega > 0$. The function $\varepsilon - 1$ obeys the Kramers–Kronig relations only because a stable target has no zeros of ε in the upper half plane (Eq. (C.41)). The static limit of $\varepsilon^{-1} v_{\mathrm{C}}$ is the statically screened Coulomb interaction: an external point charge, or a second electron held at rest, acts through $v_{\mathrm{C}}(q)/\varepsilon(\boldsymbol{q}, 0)$ rather than $v_{\mathrm{C}}(q)$ (Sect. 4.8 uses it in the electron self-energy). At small q, v_{C} grows as $1/q^2$, and this is where screening matters most.

In an insulator the static dielectric function at long wavelength is finite. With the lattice clamped, its value $\varepsilon_\infty = \varepsilon(\boldsymbol{q} \to 0, \omega \to 0)$ is the electronic, or high-frequency, dielectric constant, so called because it is measured at frequencies above those of the lattice vibrations and below the gap; the static constant ε_0 with the lattice included is the subject of Sect. 5.5. For silicon $\varepsilon_\infty = 11.7$ [19], the square of the long-wavelength refractive index 3.414 of Li's tabulation at 293 K [20], as reproduced in Ref. [21]: a static charge in the crystal is screened to a twelfth of its bare strength at distances of many cells. Where ε_1 crosses zero and ε_2 is small, the external potential produces a large total one, and the electrons oscillate collectively. This is the plasmon, a pole of Eq. (4.40) near the real axis, which appears as a peak of $\operatorname{Im}(-\varepsilon^{-1})$; Sect. 4.5 makes it precise.

The dielectric function of silicon is computed from Kohn–Sham states. The proper response is replaced by the independent-particle response built from the Kohn–Sham orbitals, which is the random-phase approximation, or corrected by an exchange–correlation kernel (Sect. 4.6), and the inversion of Eq. (4.41) is done as a matrix in the reciprocal-lattice vectors introduced next (Appendix E.6). Semilocal functionals overestimate ε_∞ (Appendices E.1 and E.4). At small q the result is tested against measurements: optical absorption at $q \to 0$ and electron-energy-loss spectra at finite q.

4.4.2 Local fields and the dielectric matrix

In a crystal a potential at $\boldsymbol{q}$ also induces density at $\boldsymbol{q} + \boldsymbol{G}$ for every reciprocal-lattice vector $\boldsymbol{G}$ (Sect. 4.1): the electrons pile up near the atoms and in the bonds, on the scale of the cell, not uniformly. The density and the potentials are then vectors indexed by $\boldsymbol{G}$, at a wavevector $\boldsymbol{q}$ in the first zone, and the response is a matrix,

$$R_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}, \omega) = R_{ab}(\boldsymbol{q}, \omega)\big|_{O_a = n_e(\boldsymbol{q}+\boldsymbol{G}),\; O_b = n_e(\boldsymbol{q}+\boldsymbol{G}')}, \tag{4.42}$$

the density response at $\boldsymbol{q}+\boldsymbol{G}$ to a potential at $\boldsymbol{q}+\boldsymbol{G}'$, from the Kubo formula of Sect. 4.1. Each Fourier component of the induced density produces its own Hartree potential, $v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\, \delta n_{e,\boldsymbol{G}}$, and Eqs. (4.38) and (4.39) become matrix equations. The same steps lead to $R = \tilde{R} + \tilde{R}\, v_{\mathrm{C}}\, R$, with v_{C} the diagonal matrix of $v_{\mathrm{C}}(|\boldsymbol{q} + \boldsymbol{G}|)$, and to the dielectric matrix

$$\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}, \omega) = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\, \tilde{R}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}, \omega), \qquad \left(\varepsilon^{-1}\right)_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} + v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\, R_{\boldsymbol{G}\boldsymbol{G}'}, \tag{4.43}$$

for any proper response $\tilde{R}$, exact or approximate. That the two matrices are inverses of each other takes one line of matrix algebra (Appendix D.7). This is Eq. (A.35), the unsymmetrized counterpart of the symmetrized longitudinal dielectric matrix of Ref. [5, Eqs. (A5)–(A7)], which is built with $v_{\mathrm{C}}^{1/2} \tilde{R} v_{\mathrm{C}}^{1/2}$ and has the same diagonal elements of the inverse (Appendix D.7).

An experiment with a long-wavelength field sees the average of the total field over a cell. The external field has only the component $\boldsymbol{G}' = 0$, and the cell average of the total field is

its component $\boldsymbol{G} = 0$, which is $(\varepsilon^{-1})_{\boldsymbol{00}}$ times the external one. The macroscopic dielectric function is therefore

$$\varepsilon_{\text{mac}}(\boldsymbol{q},\omega) = \frac{1}{(\varepsilon^{-1})_{\boldsymbol{00}}(\boldsymbol{q},\omega)}, \tag{4.44}$$

the inverse of the head of the inverse matrix, not the head $\varepsilon_{\boldsymbol{00}}$ itself [22, 23]. The two differ because the macroscopic field drives density at all $\boldsymbol{G}$, and the Hartree potentials of these microscopic components feed back on the $\boldsymbol{G} = 0$ component. Inverting the matrix in blocks (Eq. (C.57)) gives the correction explicitly,

$$\varepsilon_{\text{mac}} = \varepsilon_{\boldsymbol{00}} - \sum_{\boldsymbol{G},\boldsymbol{G}'\neq 0} \varepsilon_{\boldsymbol{0G}} \left[(\varepsilon_{\neq 0})^{-1}\right]_{\boldsymbol{GG}'} \varepsilon_{\boldsymbol{G}'\boldsymbol{0}},$$

with $\varepsilon_{\neq 0}$ the block with $\boldsymbol{G}, \boldsymbol{G}' \neq 0$. The sum is the local-field effect: the difference between the response to the average field and the response to the field that the electrons actually feel inside the cell. Appendix D.7 derives it, shows that it stays finite as $\boldsymbol{q} \to 0$ although single elements of the matrix diverge there, and writes ε_{mac} in the equivalent form $1 - v_{\text{C}}(q)\bar{R}_{\boldsymbol{00}}$, with $\bar{R}$ the response computed with the Coulomb kernel at $\boldsymbol{G} = 0$ removed [18]. In a homogeneous system the off-diagonal elements vanish and $\varepsilon_{\text{mac}} = \varepsilon_{\boldsymbol{00}} = \varepsilon$; how large the local-field effects of silicon are in the random-phase approximation is the subject of Sect. 4.6. The macroscopic function is the one measured optically, and ε_∞ is its static limit at $\boldsymbol{q} \to 0$. For a momentum transfer beyond the first zone, $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$ as in Eq. (3.16), the same reasoning applies at the row and column of $\boldsymbol{G}_q$, which is the unreduced indexing of Eq. (A.13): $(\varepsilon^{-1})_{00}(\boldsymbol{q},\omega)$ means $(\varepsilon^{-1})_{\boldsymbol{G}_q\boldsymbol{G}_q}(\boldsymbol{q}_r,\omega)$.

4.4.3 Structure factor and loss function

The rate needs S, and the fluctuation–dissipation theorem Eq. (4.18) ties S to $\operatorname{Im} R_{ee}$, which Eq. (4.41) ties in turn to the inverse dielectric function. Taking the imaginary part of $\varepsilon^{-1} = 1 + v_{\text{C}} R_{ee}$, with v_{C} real, leads to $\operatorname{Im}\varepsilon^{-1} = v_{\text{C}} \operatorname{Im} R_{ee}$, and with $S = -2 \operatorname{Im} R_{ee}/(1 - \mathrm{e}^{-\beta\omega})$,

Structure factor and loss function

$$S(\boldsymbol{q},\omega) = \frac{2q^2}{e^2}\,\frac{L(\boldsymbol{q},\omega)}{1-\mathrm{e}^{-\beta\omega}} = \frac{q^2}{2\pi\alpha}\,\frac{L(\boldsymbol{q},\omega)}{1-\mathrm{e}^{-\beta\omega}}, \qquad L(\boldsymbol{q},\omega) = \operatorname{Im}\left[-(\varepsilon^{-1})_{00}(\boldsymbol{q},\omega)\right], \tag{4.45}$$

for the density response of the electrons with the nuclei clamped. The loss function L is nonnegative for $\omega > 0$, and the rate of any density coupling is fixed by it.

This is Eq. (A.37), Ref. [5, Eq. (6)], and at $T = 0$ Ref. [24, Eqs. (S.7) and (S.24)]. In a crystal the derivation goes through element by element. By Eq. (4.42) the diagonal element of $R_{\boldsymbol{GG}'}(\boldsymbol{q}_r,\omega)$ at $\boldsymbol{G} = \boldsymbol{G}' = \boldsymbol{G}_q$ is the density response $R_{ee}(\boldsymbol{q},\omega)$ at the unreduced $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$, and Eq. (4.43) relates it to the diagonal element of ε^{-1} at the same place. The loss function is that element, local fields included, and the relation holds at each vector $\boldsymbol{q}$ with no assumption of isotropy. The name comes from the energy loss of fast electrons in thin films, whose spectrum measures L directly; Eq. (4.54) below makes the connection.

The relation needs weak single scattering, a coupling to the density, and a probe that sees the electrons alone, and each of these fails somewhere in this book. The probe acts once and weakly, so that the golden rule and linear response apply; this is the Born approximation of Sect. 2.2. It couples to the electrons through their density alone, through a potential that depends on their positions, and not through spin or current [24]; spin couplings need the spin response of Chap. 7, and velocity-dependent couplings the current response of Chap. 8. Finally, it couples to the electrons only, or the nuclei are clamped.

The last condition separates two kinds of probe. A test charge, a photon or a dark photon, whose coupling to matter is the photon's scaled by the mixing ζ (Sect. 1.4), acts on the total charge, $n_Z = -n_e + \sum_\kappa Z_\kappa n_\kappa$ of Sect. 2.3. The dielectric function measured with such probes is $\varepsilon^{-1} = 1 + v_C R_{ZZ}$, with the response of the total charge, and its loss function L fixes the total-charge structure factor exactly, $S_{ZZ} = (2q^2/e^2)\, L/(1 - \mathrm{e}^{-\beta\omega})$. A mediator that couples to electrons but not to nucleons sees $S = S_{ee}$ instead, and the relation Eq. (4.45) holds exactly with the electron-channel function $\varepsilon_e^{-1} = 1 + v_C R_{ee}$ and its loss L_e (Appendix A.7). With the nuclei clamped, n_κ contributes only static Bragg terms, $R_{ZZ} = R_{ee}$, and $L_e = L$. With moving nuclei the two differ at phonon energies, where the nuclear charge oscillates and the electrons partly follow it (Chap. 5). Optical and electron-energy-loss experiments measure the total-charge L. They yield the dark-photon rate exactly at every energy [24], and the rate of an electron-only coupling at energies well above the phonon band, which is where electronic excitations live. The interference between electron and nuclear densities between those two cases is the business of Chaps. 8 and 12.

For a scalar dielectric function, with the local fields neglected or in a homogeneous system, $L = \varepsilon_2/|\varepsilon|^2$. Since $\varepsilon_2 = -v_C \operatorname{Im} \tilde{R}_{ee}$, the absorptive part of the proper response, Eq. (4.45) becomes $S = S^{\mathrm{prop}}/|\varepsilon|^2$, where $S^{\mathrm{prop}} = (2q^2/e^2)\, \varepsilon_2/(1 - \mathrm{e}^{-\beta\omega})$ is the structure factor built from $\tilde{R}_{ee}$ (Eq. (A.37)). The rate is the rate of the proper response, screened by $1/|\varepsilon|^2$. In the random-phase approximation the proper response is the independent-particle one, $S^{\mathrm{prop}} = S^0$ of Eq. (4.21), and the factor $1/|\varepsilon|^2$ is what earlier calculations of the dark-matter–electron rate left out [5].

Fig. 4.7 shows what the factor does in silicon at $q \to 0$. Absorption, ε_2, is strong between the gap and about 10 eV, with peaks at 3.4 and 4.2 eV from interband transitions. There, however, $|\varepsilon|$ is large, and the loss function is small: at 3 eV, $\varepsilon_2 = 2.4$ but $L = 3.1 \times 10^{-3}$. The real part ε_1 is negative from 4.2 eV and crosses zero again at 16.3 eV, where ε_2 has fallen below one. The loss function peaks there, at 16.4 eV, with a height of 4.5 and a full width of 3.7 eV. This is the plasmon of the valence electrons; how its position relates to the free-electron ω_p is the subject of Sect. 4.5. For a dark-matter particle, which can deposit at most $qv_{\max}$, the consequence is that the energies it reaches at typical momentum transfers, a few eV, are those where the target absorbs strongly and screens strongly at once.

4.4.4 Sum rules for the loss function

The sum rules of Sect. 4.3 carry over. For the f-sum we start from its form at any temperature, $\int_0^\infty \mathrm{d}\omega\, \omega(1 - \mathrm{e}^{-\beta\omega})S = \pi \boldsymbol{n}_e q^2/m_e$, which folds the negative frequencies onto the positive ones with detailed balance. By Eq. (4.45) the factor $(1 - \mathrm{e}^{-\beta\omega})S$ is $(2q^2/e^2)L$, the temperature drops out, and with $\omega_p^2 = e^2 \boldsymbol{n}_e/m_e$ from Eq. (4.31),

$$\int_0^\infty \mathrm{d}\omega\, \omega\, L(\boldsymbol{q}, \omega) = \frac{\pi}{2}\, \omega_p^2, \tag{4.46}$$

at every $\boldsymbol{q}$ and temperature, with L for the electron channel, or the total one at clamped nuclei. This is Eq. (A.40) and Ref. [24, Eq. (S.10)]. With moving nuclei the total-charge L collects in addition the nuclear share of the charge f-sum of Sect. 4.3, 2.73×10^{-4} of the total for silicon.

The absorptive part obeys the same rule. The macroscopic ε_2 is $-v_C(q) \operatorname{Im} \bar{R}_{\mathbf{00}}$ (Appendix D.7), and $\bar{R}$ is the density response of the electrons with the Coulomb kernel at the single wavevector $\boldsymbol{q}$ removed from the Hamiltonian. Removing one Fourier component out of a continuum of them changes the equilibrium state by nothing in the thermodynamic limit (Sect. 4.8.3), while the double commutator of Sect. 4.3 does not involve the interaction at all.

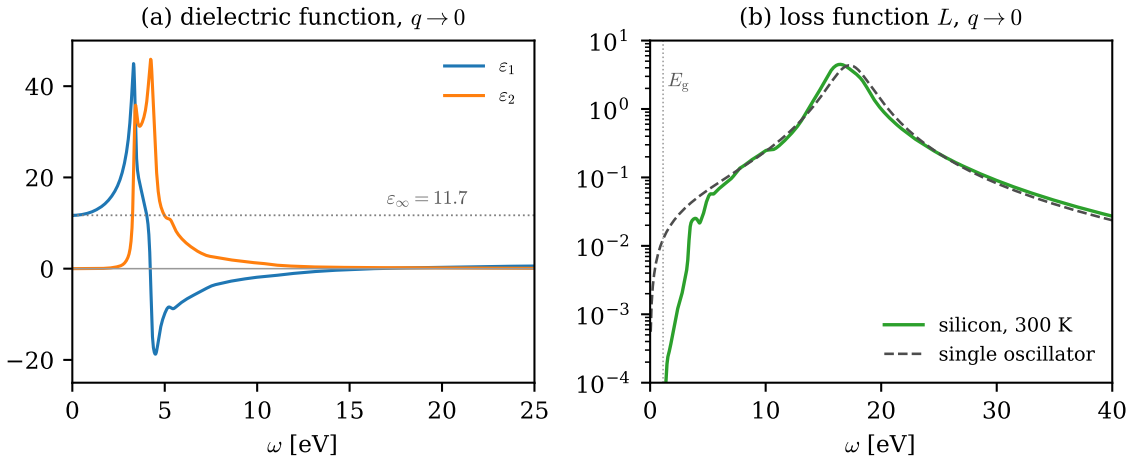


Figure 4.7: Dielectric function and loss function of crystalline silicon at $q \to 0$ and 300 K, from the optical constants of Franta and coworkers, a Kramers–Kronig-consistent dispersion model fitted to ellipsometric and spectrophotometric measurements and tabulated from 4 meV to 40 eV [19]. The data are optical, that is transverse; at $q \to 0$ the transverse and longitudinal functions coincide (Eq. (4.52)), and $L = \varepsilon_2/|\varepsilon|^2$. (a) ε_1 and ε_2; at low energy $\varepsilon_1 = \varepsilon_\infty = 11.7$ (dotted). The interband peaks of ε_2 lie at 3.4 and 4.2 eV, and ε_1 crosses zero upward at 16.3 eV. (b) The loss function on a logarithmic scale (solid), with its peak of 4.5 at 16.4 eV. Dashed: the single damped oscillator of Worked example 4.9 at $q = 0$, which has the same f-sum and the same ε_∞, and a damping equal to the measured full width, 3.66 eV. Dotted: the room-temperature gap, 1.12 eV, to match the data. Below 40 eV the tabulated spectrum collects 96 percent of the f-sum of the four valence electrons in $\int \omega\varepsilon_2 \, \mathrm{d}\omega$ and 92 percent in $\int \omega L \, \mathrm{d}\omega$.

The response $\bar{R}$ therefore has the same f-sum as R, and

$$\int_0^\infty \mathrm{d}\omega\, \omega\, \varepsilon_2(\boldsymbol{q},\omega) = \frac{\pi}{2}\,\omega_p^2, \tag{4.47}$$

which Appendix C.5 also obtains from the Kramers–Kronig relation for $\varepsilon - 1$ at large frequency. The two functions ε_2 and L distribute the same weight differently: ε_2 puts it at the transition energies, and $L = \varepsilon_2/|\varepsilon|^2$ moves it up to where $|\varepsilon|$ is small, toward the plasmon. Tested on the tabulated silicon data of Fig. 4.7, the spectrum below 40 eV holds 96 percent of $(\pi/2)\omega_p^2 = 433\,\mathrm{eV}^2$ for the four valence electrons in $\int \omega\varepsilon_2$, and 92 percent in $\int \omega L$, which has more of its weight at high energy. The rest lies above the table, where the valence absorption overlaps that of the core (Worked example 4.7).

The inverse moments follow from the Kramers–Kronig relations at $\omega = 0$ (Eqs. (C.40) and (C.41)). For an insulator at $\boldsymbol{q} \to 0$,

$$\frac{2}{\pi}\int_0^\infty \mathrm{d}\omega\, \frac{L(\omega)}{\omega} = 1 - \frac{1}{\varepsilon_\infty}, \qquad \frac{2}{\pi}\int_0^\infty \mathrm{d}\omega\, \frac{\varepsilon_2(\omega)}{\omega} = \varepsilon_\infty - 1. \tag{4.48}$$

In a metal static screening is perfect at long wavelength, $\varepsilon^{-1} \to 0$, and the first integral becomes $\pi/2$ (Appendix C.5). Weighted by $1/\omega$, these integrals converge much faster than the f-sum: the tabulated data of Fig. 4.7 return 99.8 percent of $\varepsilon_\infty - 1$ and 97.8 percent of $1 - 1/\varepsilon_\infty$ below 40 eV.

Together, the f-sum and the inverse moment already say where the weight of L sits at small q. Their ratio defines a mean excitation energy, $\bar{E}^2 = \int \omega L / \int (L/\omega)$, and

$$\bar{E}^2 = \frac{\omega_p^2}{1 - 1/\varepsilon_\infty}. \tag{4.49}$$

With $\omega_p = 16.6\,\mathrm{eV}$ for the valence electrons and $\varepsilon_\infty = 11.7$, $\bar{E} = 16.6/0.956 = 17.4\,\mathrm{eV}$. If the loss spectrum were a single line, it would sit there, more than fifteen times above the gap of 1.17 eV. By Eq. (4.45), the same holds for the density fluctuations S at $T = 0$: at long wavelengths the valence electrons respond collectively, at energies $\sim \omega_p$, and not at the gap, a point Sect. 4.5 makes precise.

4.4.5 Optical conductivity and transverse response

Light, and a dark photon absorbed at rest, act through a transverse field, not a potential, and what the experiment measures is a current. The induced current density $\boldsymbol{J}$ is linear in the total field $\boldsymbol{E}$, $\boldsymbol{J} = \sigma_{\mathrm{cond}}\boldsymbol{E}$, which defines the optical conductivity $\sigma_{\mathrm{cond}}(\omega) = \sigma_1 + \mathrm{i}\sigma_2$. With fields varying as $\mathrm{e}^{-\mathrm{i}\omega t}$ (Appendix A.3), the Ampère–Maxwell law in Heaviside–Lorentz units, $\nabla \times \boldsymbol{B} = \boldsymbol{J} + \partial_t \boldsymbol{E}$, written as $\nabla \times \boldsymbol{B} = \partial_t \boldsymbol{D}$ with $\boldsymbol{D} = \varepsilon\boldsymbol{E}$, requires $-\mathrm{i}\omega\varepsilon = \sigma_{\mathrm{cond}} - \mathrm{i}\omega$, so

$$\varepsilon = 1 + \frac{\mathrm{i}\,\sigma_{\mathrm{cond}}}{\omega}, \qquad \sigma_1 = \omega\,\varepsilon_2, \qquad \sigma_2 = \omega\,(1 - \varepsilon_1). \tag{4.50}$$

Texts in Gaussian units write $\varepsilon = 1 + 4\pi\mathrm{i}\sigma_{\mathrm{cond}}/\omega$ (Appendix A.2). The real part σ_1 is the dissipative one, and by Eq. (4.47) it obeys $\int_0^\infty \mathrm{d}\omega\, \sigma_1 = (\pi/2)\omega_p^2$, the f-sum rule in its optical form. Written with the loss function, $\sigma_1 = \omega|\varepsilon|^2 L$ [24]. For the absorption of a dark photon, the in-medium photon self-energy of Sect. 1.4 is $\Pi_{\mathrm{em}} = \omega^2(1 - \varepsilon^{\mathrm{T}}) = -\mathrm{i}\omega\sigma_{\mathrm{cond}}$: the wave equation $\omega^2\varepsilon^{\mathrm{T}}\boldsymbol{A} = k^2\boldsymbol{A}$ for a transverse vector potential in the medium is $(\omega^2 - k^2 - \Pi_{\mathrm{em}})\boldsymbol{A} = 0$. Its imaginary part, $-\omega\sigma_1$, sets the absorption rate of Eq. (1.38) (Chap. 10).

The function that enters here is the transverse dielectric function ε^{T}, the response to a field perpendicular to $\boldsymbol{q}$, while Eq. (4.41) defines the longitudinal one. Both come from one tensor.

A vector potential $\boldsymbol{A}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ in the gauge with no scalar potential, $\boldsymbol{E} = -\partial_t \boldsymbol{A} = \mathrm{i}\omega\boldsymbol{A}$, enters the kinetic energy of each electron as $(\boldsymbol{p}+e\boldsymbol{A})^2/2m_e$. To first order it couples to the paramagnetic current $\boldsymbol{j}_e(\boldsymbol{q})$ of Eq. (4.29) through $e\boldsymbol{A}\cdot\boldsymbol{j}_e(\boldsymbol{q})$, and it changes the velocity of every electron from $\boldsymbol{p}/m_e$ to $(\boldsymbol{p}+e\boldsymbol{A})/m_e$. The induced particle current per volume therefore has two parts. The diamagnetic part is the density times the added velocity, $\boldsymbol{n}_e\, e\boldsymbol{A}/m_e$. The paramagnetic part is the Kubo response Eq. (4.3) of $\boldsymbol{j}_e$ to the perturbation $O_b\phi_{\mathrm{ext}} = j_e^\beta(\boldsymbol{q})\, eA^\beta$, which is $\tilde{R}_{jj}^{\alpha\beta}\, eA^\beta$, with $\tilde{R}_{jj}^{\alpha\beta}$ the proper current–current response, proper because $\boldsymbol{E}$ is the total field. Together,

$$\frac{\delta\langle j_e^\alpha(\boldsymbol{q})^\dagger\rangle}{\mathscr{V}} + \boldsymbol{n}_e\frac{eA^\alpha}{m_e} = e\Big[\frac{\boldsymbol{n}_e}{m_e}\delta^{\alpha\beta} + \tilde{R}_{jj}^{\alpha\beta}(\boldsymbol{q},\omega)\Big]A^\beta.$$

Multiplying by the electron charge $-e$ and inserting $\boldsymbol{A} = -\mathrm{i}\boldsymbol{E}/\omega$,

$$\varepsilon^{\alpha\beta}(\boldsymbol{q},\omega) = \delta^{\alpha\beta} - \frac{e^2}{\omega^2}\Big[\frac{\boldsymbol{n}_e}{m_e}\delta^{\alpha\beta} + \tilde{R}_{jj}^{\alpha\beta}(\boldsymbol{q},\omega)\Big].$$

For free electrons $\tilde{R}_{jj}$ vanishes at $\boldsymbol{q} = 0$, since the total momentum is conserved, and $\varepsilon = 1 - \omega_p^2/\omega^2$. The transverse function is $\varepsilon^{\mathrm{T}} = \hat{\boldsymbol{e}}\cdot\varepsilon\cdot\hat{\boldsymbol{e}}$ for a unit vector $\hat{\boldsymbol{e}} \perp \boldsymbol{q}$, and the longitudinal one is $\hat{\boldsymbol{q}}\cdot\varepsilon\cdot\hat{\boldsymbol{q}}$.

That the longitudinal projection is the ε of Eq. (4.41) follows from the continuity equation Eq. (4.29), $\partial_t n_e(\boldsymbol{q},t) = \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q},t)$. In the Kubo integral for $R_{ee}(\boldsymbol{q},\omega)$ we integrate by parts in t twice, using $\omega + \mathrm{i}0^+$ so that the boundary terms at large t vanish. The first boundary term contains the equal-time commutator of two densities, which is zero. The first integration replaces $n_e(\boldsymbol{q},t)^\dagger$ by $-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q},t)^\dagger$; the second, after moving the time to the other operator by stationarity, replaces $n_e(\boldsymbol{q},0)$ by $\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q},0)$ and leaves the boundary term $\langle[\boldsymbol{q}\cdot\boldsymbol{j}_e^\dagger, n_e]\rangle/\mathscr{V} = \boldsymbol{n}_e q^2/m_e$, the double commutator of the f-sum rule. The result is a Ward identity, the statement of charge conservation for the response functions,

$$\omega^2 R_{ee}(\boldsymbol{q},\omega) = \frac{\boldsymbol{n}_e q^2}{m_e} + q_\alpha q_\beta\, R_{jj}^{\alpha\beta}(\boldsymbol{q},\omega), \tag{4.51}$$

exactly, and the same identity holds for the proper functions, by the argument that gave Eq. (4.47) [2, Chap. 3]. Inserted into the tensor, $\hat{\boldsymbol{q}}\cdot\varepsilon\cdot\hat{\boldsymbol{q}} = 1 - (e^2/q^2)\tilde{R}_{ee} = \varepsilon$. As $\boldsymbol{q}\to 0$, the proper tensor $\tilde{R}_{jj}^{\alpha\beta}$ tends to a limit that does not depend on the direction of $\boldsymbol{q}$, since the proper response has no long-range Coulomb kernel and so no singularity at $\boldsymbol{q} = 0$. Both projections are then taken of the same tensor $\varepsilon^{\alpha\beta}(\omega)$. In a cubic crystal that tensor is a multiple of the unit matrix (Appendix C.3), and

$$\varepsilon^{\mathrm{T}}(\boldsymbol{q},\omega) \to \varepsilon(\boldsymbol{q},\omega) \qquad (\boldsymbol{q}\to 0). \tag{4.52}$$

In a crystal of lower symmetry the longitudinal function depends on the direction $\hat{\boldsymbol{q}}$ even at $\boldsymbol{q}\to 0$, but it is still the projection of the optical tensor. This is why optical data at $q\to 0$ give the longitudinal loss function, as in Fig. 4.7, and why the absorption of a dark photon and the scattering of dark matter at small q are governed by the same ε_2. At finite q the two functions separate; for dark-matter scattering, which needs momentum transfers of keV, the longitudinal function is the one that matters.

4.4.6 Thomas–Fermi screening

The simplest proper response is that of a gas of electrons in a static potential that varies slowly in space. Each region of the gas then sits in local equilibrium, with the potential energy $\phi_{\mathrm{tot}}(\boldsymbol{r})$ shifting all its levels. The electrochemical potential must be the same everywhere, so the local chemical potential is $\mu_{\mathrm{ch}} - \phi_{\mathrm{tot}}(\boldsymbol{r})$, and the density changes by $\delta\boldsymbol{n}_e = -(\partial\boldsymbol{n}_e/\partial\mu_{\mathrm{ch}})\,\phi_{\mathrm{tot}}$.

For the free gas, $\partial n_e/\partial\mu_{\rm ch} = 2\mathcal{N}_{\rm F}$ at $T = 0$, with $\mathcal{N}_{\rm F}$ per spin from Eq. (2.71). The proper response is $\tilde{R}_{ee}(q,0) = -2\mathcal{N}_{\rm F}$, and Eq. (4.41) gives

$$\varepsilon_{\rm TF}(q) = 1 + \frac{k_{\rm TF}^2}{q^2}, \qquad k_{\rm TF}^2 = 2e^2\mathcal{N}_{\rm F} = \frac{4\alpha\, m_e k_{\rm F}}{\pi} = \frac{3\,\omega_p^2}{v_{\rm F}^2}, \tag{4.53}$$

where the last two forms use Eqs. (2.71) and (4.31). The statically screened interaction is

$$\frac{v_{\rm C}(q)}{\varepsilon_{\rm TF}(q)} = \frac{e^2}{q^2 + k_{\rm TF}^2},$$

the transform of a Yukawa potential $\alpha\, {\rm e}^{-k_{\rm TF}r}/r$, the same form as a massive mediator (Eq. (1.31)) with mass $k_{\rm TF}$. An impurity charge in a metal is screened within a distance $1/k_{\rm TF}$. This is the Thomas–Fermi theory of screening, after the semiclassical model of the atom by Thomas and Fermi. Its assumptions are a static potential and a slow variation, $q \ll k_{\rm F}$, so that the local-equilibrium picture holds; Sect. 4.5 recovers Eq. (4.53) as the static long-wavelength limit of the Lindhard function and gives the corrections.

Worked example 4.8: The Thomas–Fermi length of the silicon valence electrons

Treat the four valence electrons per atom of silicon as a free gas, $n_e = 2.00 \times 10^{23}\,{\rm cm}^{-3}$, with $k_{\rm F} = 3.57\,{\rm keV}$ and $\mathcal{N}_{\rm F} = 9.2 \times 10^7\,{\rm eV}^2$ (Sect. 2.5). By Eq. (4.53), $k_{\rm TF}^2 = 4\alpha m_e k_{\rm F}/\pi = 1.69 \times 10^7\,{\rm eV}^2$, or $k_{\rm TF} = 4.12\,{\rm keV}$, and a screening length $1/k_{\rm TF} = 0.48\,{\rm Å}$, a fifth of the bond length. At the typical momentum transfer $q_0 = \alpha m_e = 3.73\,{\rm keV}$, $\varepsilon_{\rm TF} = 1 + (4.12/3.73)^2 = 2.22$, and the squared potential, which sets the rate, is reduced by $|\varepsilon_{\rm TF}|^2 = 4.9$. This is the estimate quoted at the start of the section.

The estimate is crude because $k_{\rm TF}$ is not small against $k_{\rm F}$ and because silicon is an insulator. With $k_{\rm TF} = 1.15\,k_{\rm F}$, the screening happens at the wavevectors where the condition $q \ll k_{\rm F}$ fails, and the Lindhard function of Sect. 4.5 screens less than Eq. (4.53) at $q \sim k_{\rm F}$. In an insulator, moreover, the electrons cannot follow a potential that varies over many cells, and its static screening at long wavelength saturates at $\varepsilon_\infty = 11.7$, while the gas formula diverges. The two descriptions meet where $1 + k_{\rm TF}^2/q^2 = \varepsilon_\infty$, at $q = k_{\rm TF}/(\varepsilon_\infty - 1)^{1/2} = 1.26\,{\rm keV}$, a length $1/q$ of 1.6 Å. As a rough picture, then, silicon screens like a metal at the momentum transfers of a few keV that dark matter typically deposits, and like an insulator with a dielectric constant of 11.7 below about 1 keV. Both statements concern the static response; the rate needs ε at the energy deposited, which is why the computed loss function, not Eq. (4.53), enters the calculations of Chap. 10.

4.4.7 Dielectric stopping

A charged particle crossing a solid slows down by exciting it, and the loss function fixes how fast. Take a particle of charge $Z_1 e$, heavy enough that its recoil can be neglected, moving with velocity $\boldsymbol{v}$. Its potential energy with an electron is $-Z_1 e^2/(4\pi r)$, so $\tilde{V}(\boldsymbol{q}) = -Z_1 v_{\rm C}(q)$, and a momentum transfer $\boldsymbol{q}$ deposits $\omega = \boldsymbol{q}\cdot\boldsymbol{v}$, the energy change $\boldsymbol{q}\cdot\boldsymbol{v} - q^2/2M_1$ of a particle of mass $M_1 \to \infty$. Weighting each momentum transfer in the rate Eq. (4.14) by the energy it deposits gives the energy lost per unit time, and dividing by v the energy lost per unit path. At $T = 0$, with Eq. (4.45) and $|\tilde{V}|^2 = Z_1^2 e^4/q^4$,

$$-\frac{{\rm d}E}{{\rm d}x} = \frac{1}{v}\int \frac{{\rm d}^3q}{(2\pi)^3}\,\frac{Z_1^2 e^4}{q^4}\,\frac{2q^2}{e^2}\,\omega\, L(\boldsymbol{q},\omega)\Big|_{\omega=\boldsymbol{q}\cdot\boldsymbol{v}}.$$

For an isotropic L we use polar coordinates about $\boldsymbol{v}$, $\mathrm{d}^3q = 2\pi q^2 \mathrm{d}q\,\mathrm{d}u$ with u the cosine of the angle, and change variables from u to $\omega = quv$, $\mathrm{d}u = \mathrm{d}\omega/(qv)$. Only $\omega > 0$ contributes at $T = 0$, and

$$-\frac{\mathrm{d}E}{\mathrm{d}x} = \frac{Z_1^2 e^2}{2\pi^2 v^2}\int_0^\infty \frac{\mathrm{d}q}{q}\int_0^{qv} \mathrm{d}\omega\;\omega\,L(q,\omega) = \frac{2Z_1^2\alpha}{\pi v^2}\int_0^\infty \frac{\mathrm{d}q}{q}\int_0^{qv} \mathrm{d}\omega\;\omega\,L(q,\omega). \tag{4.54}$$

This is Lindhard's dielectric formula for the stopping power [7], in Heaviside–Lorentz units. The fast and the slow limits show what it contains. For a fast particle, qv reaches beyond the whole loss spectrum at all but the smallest q, and the inner integral is the f-sum $(\pi/2)\omega_p^2$. The stopping power is $Z_1^2\alpha\omega_p^2/v^2$ times a logarithm of the range of q that contributes: the Bethe formula, set by the electron density alone. For a slow particle, qv stays below the main peak, and the inner integral samples only the low-energy side of L, which the sum rules do not fix. The same is true of a dark-matter particle, which is a slow probe in exactly this sense.

Worked example 4.9: A single damped oscillator

The simplest loss function consistent with the f-sum and with ε_∞ puts all the weight in one damped oscillator. Write

$$\varepsilon^{-1}(q,\omega) = 1 + \frac{\omega_p^2}{\omega^2 - \omega_{\mathrm{res}}^2 + \mathrm{i}\Gamma_{\mathrm{w}}\omega}, \qquad \omega_{\mathrm{res}}(q)^2 = \bar{E}^2 + \Big(\frac{q^2}{2m_e}\Big)^2,$$

with a damping rate Γ_{w}. The resonance frequency moves from $\bar{E}$ at small q to the free recoil energy $q^2/2m_e$ at large q; this dispersion is our modeling choice, made so that the large-q limit has the free-recoil center that Sect. 4.3 requires. The function $\varepsilon^{-1} - 1$ has the form of the damped-oscillator response of Worked example 4.1, with ω_p^2 for $2\lambda_{\mathrm{osc}}^2\omega_0$ and ω_{res} for ω_0. It is causal, and its imaginary part is

$$L(q,\omega) = \frac{\omega_p^2\,\Gamma_{\mathrm{w}}\,\omega}{(\omega^2 - \omega_{\mathrm{res}}^2)^2 + \Gamma_{\mathrm{w}}^2\omega^2}.$$

Integrating, $\int_0^\infty \omega L\,\mathrm{d}\omega = (\pi/2)\omega_p^2$ for any Γ_{w} and q, the f-sum Eq. (4.46), and $\int_0^\infty (L/\omega)\,\mathrm{d}\omega = (\pi/2)\omega_p^2/\omega_{\mathrm{res}}^2$. At $q \to 0$ the inverse moment Eq. (4.48) then requires $1 - 1/\varepsilon_\infty = \omega_p^2/\bar{E}^2$, which is Eq. (4.49): the single oscillator that satisfies both sum rules has its loss peak at the mean excitation energy, $\bar{E} = 17.4\,\mathrm{eV}$ for silicon. Inverting, ε itself has its resonance lower, at $(\bar{E}^2 - \omega_p^2)^{1/2} = \bar{E}/\varepsilon_\infty^{1/2} = 5.08\,\mathrm{eV}$, and $\varepsilon(0,0) = 11.7$. We take $\Gamma_{\mathrm{w}} = 3.66\,\mathrm{eV}$, the measured full width of the silicon loss peak, which sets the peak height $\omega_p^2/(\Gamma_{\mathrm{w}}\bar{E}) = 4.34$ against the measured 4.5 (Fig. 4.7b).

By Eq. (4.45), the structure factor at $T = 0$ is

$$S(q,\omega) = \frac{q^2}{2\pi\alpha}\,\frac{\omega_p^2\,\Gamma_{\mathrm{w}}\,\omega}{(\omega^2 - \omega_{\mathrm{res}}^2)^2 + \Gamma_{\mathrm{w}}^2\omega^2},$$

and its first moment is $(q^2/2\pi\alpha)(\pi/2)\omega_p^2 = \pi n_e q^2/m_e$, the f-sum Eq. (4.30). As $\Gamma_{\mathrm{w}} \to 0$, the weight collapses onto $\omega = \omega_{\mathrm{res}}$ with $\int \mathrm{d}\omega\, S = \pi n_e q^2/(m_e\omega_{\mathrm{res}})$, and by Eq. (4.17) $S^{\mathrm{st}} = q^2/(2m_e\omega_{\mathrm{res}})$: the density fluctuations vanish as q^2 at small q, below the bound $q^2/(2m_e E_{\mathrm{g}})$ of Sect. 4.3 since $\omega_{\mathrm{res}} > E_{\mathrm{g}}$.

For the stopping power of a slow particle, take $qv \ll \omega_{\mathrm{res}}$ at every q. Since ω_{res}/q is smallest, $(\bar{E}/m_e)^{1/2} = 5.8\times10^{-3}$, at $q = (2m_e\bar{E})^{1/2} = 4.2\,\mathrm{keV}$, this means $v \ll 5.8\times10^{-3}$. Then $L \simeq \omega_p^2\Gamma_{\mathrm{w}}\omega/\omega_{\mathrm{res}}^4$ on the whole range of the inner integral of Eq. (4.54), which becomes $\omega_p^2\Gamma_{\mathrm{w}}q^3v^3/(3\omega_{\mathrm{res}}^4)$. With $q = (2m_e\bar{E})^{1/2}x$, the remaining integral is $\int_0^\infty q^2\mathrm{d}q/\omega_{\mathrm{res}}^4 =$

$(2m_e\bar{E})^{3/2}\bar{E}^{-4}\int_0^\infty x^2\mathrm{d}x/(1+x^4)^2$, and the last integral is $\pi/(8\sqrt{2})$. Altogether

$$-\frac{\mathrm{d}E}{\mathrm{d}x} \simeq \frac{Z_1^2\,\alpha\,\Gamma_\mathrm{w}\,\omega_p^2\,m_e^{3/2}}{6\,\bar{E}^{5/2}}\,v.$$

The slow particle feels a friction proportional to its speed, with a coefficient set by the damping. For silicon's numbers and $Z_1 = 1$ it is 0.18 eV/Å at $v = 10^{-3}$; the full integral Eq. (4.54) gives 0.186 eV/Å, 3 percent more, and 22 percent more at $v_\mathrm{max} = 2.65 \times 10^{-3}$, where the condition $v \ll 5.8 \times 10^{-3}$ is only marginal.
The coefficient tests only the tail of the model. The energies that matter are up to $qv \approx 4\,\mathrm{eV}$ at the dominant $q \approx 4\,\mathrm{keV}$, and there the oscillator and silicon disagree badly: at 3 eV the model has $L = 0.035$, eleven times the measured value, and it has weight below the gap, where silicon has none (Fig. 4.7b). Both loss functions satisfy the f-sum and give the same ε_∞. The sum rules fix the main peak, while slow particles, and dark matter, sample the low-energy side of L, which only a calculation or a measurement of the actual material can supply.

Where the loss function comes from in the simplest interacting system, the electron gas, and how its peak and its low-energy side follow from the Lindhard function, is the subject of Sect. 4.5.

4.5 The homogeneous electron gas

Spread the four valence electrons of each silicon atom uniformly over the crystal, replace the ions by a uniform positive background, and ask where the loss function of this electron gas peaks at long wavelength. The answer is the plasma frequency, $\omega_p = 16.6\,\mathrm{eV}$, and the measured loss peak of silicon in Fig. 4.7 sits at 16.4 eV. A model with no lattice, no gap and no bonds places the main feature of the spectrum within 1.2 percent of the measured peak. The reason is that at long wavelength and high frequency the valence electrons respond collectively, and the details of the crystal matter less than the Coulomb force between them. The electron gas is also the one interacting system in which the independent-particle response and its screened version can be written in closed form, and this is why it serves as the reference model for valence electrons, plasmons and screening. We will compute where the plasmon of the gas lies, at which momentum it dissolves into electron–hole pairs, and where on the (q,ω) plane a dark-matter particle can reach it. The same functions are used in the dark-matter literature as a first model of the loss function of semiconductors and metals [5, 24, 25], and we will see which corners of the (q,ω) plane they describe and which they do not.

The model is jellium (Worked example 2.5) at the valence density of silicon, $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$. Its density parameter, the radius per electron in units of the Bohr radius, is $r_s = 2.0$ (Sect. 2.3), and the free-gas scales of Eq. (2.71) are $k_\mathrm{F} = 3.57\,\mathrm{keV}$, $E_\mathrm{F} = 12.5\,\mathrm{eV}$ and $v_\mathrm{F} = 7.0 \times 10^{-3}$. In these units the plasma frequency is $\omega_p = 1.33\,E_\mathrm{F}$ and the Thomas–Fermi wavevector of Eq. (4.53) is $k_\mathrm{TF} = 1.15\,k_\mathrm{F}$. Both ratios grow with r_s, since $k_\mathrm{TF}^2/k_\mathrm{F}^2 = 4\alpha m_e/(\pi k_\mathrm{F}) = 0.663\,r_s$ with $k_\mathrm{F} = (9\pi/4)^{1/3}/(r_s a_0)$: the denser the gas, the weaker the Coulomb force compared with the kinetic energy.

4.5.1 The independent-particle bubble

Without the Coulomb interaction the electrons of jellium are free, and the Lehmann representation Eq. (4.6) of their density response is a sum over the moves of one electron. Take

an occupation-number state $|i\rangle$ of plane waves, with the spin included in the label $\boldsymbol{k}$. The operator $n_e(\boldsymbol{q}) = \sum_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}+\boldsymbol{q}} c_{\boldsymbol{k}}$ of Eq. (2.38) moves one electron from $\boldsymbol{k}$ to $\boldsymbol{k}+\boldsymbol{q}$, and the move is possible only if $\boldsymbol{k}$ is occupied and $\boldsymbol{k}+\boldsymbol{q}$ empty. In the resonant term of Eq. (4.6) the final state $|f\rangle$ is this move, with weight $n_{\boldsymbol{k}}(1-n_{\boldsymbol{k}+\boldsymbol{q}})$ and energy $E_f - E_i = \xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}}$. In the antiresonant term $n_e(\boldsymbol{q})^\dagger$ makes the reverse move, from $\boldsymbol{k}+\boldsymbol{q}$ to $\boldsymbol{k}$, with weight $n_{\boldsymbol{k}+\boldsymbol{q}}(1-n_{\boldsymbol{k}})$ and energy $\xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}}$. The two terms then share the denominator $\omega + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}} + \mathrm{i}0^+$, and their numerators combine to $n_{\boldsymbol{k}}(1-n_{\boldsymbol{k}+\boldsymbol{q}}) - n_{\boldsymbol{k}+\boldsymbol{q}}(1-n_{\boldsymbol{k}}) = n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}}$. The Pauli factors have cancelled: a move into an occupied state appears in both terms with opposite signs. The thermal average of $n_{\boldsymbol{k}}$ over the Gibbs state is $n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$, and

$$R^0(\boldsymbol{q},\omega) = \frac{1}{\mathscr{V}} \sum_{\boldsymbol{k}} \frac{n_{\mathrm{F}}(\xi_{\boldsymbol{k}}) - n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}})}{\omega + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}} + \mathrm{i}0^+}, \tag{4.55}$$

the independent-particle bubble. This is the Matsubara bubble Eq. (4.93) continued to real frequencies, obtained here without imaginary time. Its imaginary part counts the transitions of Eq. (2.70), and by the fluctuation–dissipation theorem it is $-\frac{1}{2}S^0$ at $T=0$, with the structure factor of Worked example 4.3.

At $T=0$ the sum is a set of elementary integrals over the Fermi sphere, which Appendix D.8 carries out for any complex frequency z in the upper half plane. The result is the Lindhard function [7].

Lindhard function

$$R^0(q,\omega) = -2\mathcal{N}_{\mathrm{F}} \left\{ \frac{1}{2} + \frac{k_{\mathrm{F}}}{4q} \Big[(1-\nu_+^2)\,\ell(\nu_+) - (1-\nu_-^2)\,\ell(\nu_-) \Big] \right\}, \tag{4.56}$$

with $\nu_\pm = (\omega + \mathrm{i}0^+)/qv_{\mathrm{F}} \pm q/2k_{\mathrm{F}}$, $\ell(w) = \ln[(w+1)/(w-1)]$ on the principal branch and $\mathcal{N}_{\mathrm{F}} = m_e k_{\mathrm{F}}/2\pi^2$ per spin. It is Eq. (D.38) at $z = \omega + \mathrm{i}0^+$, and it holds with $\omega + \mathrm{i}0^+$ replaced by any z with $\operatorname{Im} z > 0$.

Apart from the prefactor $2\mathcal{N}_{\mathrm{F}}$, the response depends on two numbers only, the momentum in units of the diameter of the Fermi sphere, $y = q/2k_{\mathrm{F}}$, and the frequency in units of the largest Doppler shift, ω/qv_{F}. On the real axis the logarithms take the branch Eq. (C.75), $\ell(\nu + \mathrm{i}0^+) = \ln|(\nu+1)/(\nu-1)| - \mathrm{i}\pi\,\Theta(1-|\nu|)$. The real part, Eq. (D.39), keeps the form of Eq. (4.56) with the logarithms of the moduli. The imaginary part is a sum of steps, and it is nonzero only between the edges

$$\omega_+ = qv_{\mathrm{F}} + \frac{q^2}{2m_e}, \qquad \omega_- = \max\Big(0,\ \frac{q^2}{2m_e} - qv_{\mathrm{F}}\Big),$$

the particle–hole continuum of Fig. 4.3. Inside it, Eq. (D.40) has two forms. Below the curve $\omega = qv_{\mathrm{F}} - q^2/2m_e$, which exists for $q < 2k_{\mathrm{F}}$, $\operatorname{Im} R^0 = -m_e^2\omega/(2\pi q)$, linear in ω because only electrons in a shell of thickness $\sim m_e\omega/k_{\mathrm{F}}$ below the Fermi surface can be promoted. Above it, $\operatorname{Im} R^0 = -(m_e/4\pi q)[k_{\mathrm{F}}^2 - (m_e\omega/q - q/2)^2]$, a parabola. Both are $-\frac{1}{2}$ times the forms of S^0 found by counting transitions in Worked example 4.3, as the fluctuation–dissipation theorem requires. The real part, which counts the same transitions off resonance through their principal values, is the new information.

The static limit and the high-frequency tail of the real part are needed below. At $\omega = 0$ (Eq. (D.41)),

$$R^0(q,0) = -2\mathcal{N}_{\mathrm{F}}\, F_{\mathrm{L}}\Big(\frac{q}{2k_{\mathrm{F}}}\Big), \qquad F_{\mathrm{L}}(y) = \frac{1}{2} + \frac{1-y^2}{4y} \ln\Big|\frac{1+y}{1-y}\Big|, \tag{4.57}$$

with the function F_{L} of Eq. (C.74), which falls from one at $q = 0$ to $\frac{1}{2}$ at $q = 2k_{\mathrm{F}}$ and decays as $4k_{\mathrm{F}}^2/(3q^2)$ at large q. At high frequency it is simplest to start again from the sum. Let $\Delta_{\boldsymbol{k}} = q^2/2m_e + kqu/m_e$ be the energy of the move, with u the cosine of the angle between $\boldsymbol{k}$ and $\boldsymbol{q}$, and let $\langle\cdot\rangle$ be the average over the Fermi sphere. Relabeling $\boldsymbol{k} \to -\boldsymbol{k} - \boldsymbol{q}$ in the term with $n_{\boldsymbol{k}+\boldsymbol{q}}$, as Appendix D.8 does, puts Eq. (4.55) at $T = 0$ in the form $R^0 = n_e\langle 2\Delta_{\boldsymbol{k}}/(z^2 - \Delta_{\boldsymbol{k}}^2)\rangle$. For $|z| > \omega_+$, the largest $|\Delta_{\boldsymbol{k}}|$, the geometric series converges,

$$R^0(q,z) = \frac{2n_e}{z^2}\sum_{j=0}^{\infty}\frac{\langle\Delta_{\boldsymbol{k}}^{2j+1}\rangle}{z^{2j}} = \frac{n_e q^2}{m_e z^2}\left[1 + \frac{\frac{3}{5}q^2v_{\mathrm{F}}^2 + (q^2/2m_e)^2}{z^2} + \frac{\frac{3}{7}q^4v_{\mathrm{F}}^4 + O(q^6)}{z^4} + O(z^{-6})\right]. \tag{4.58}$$

The averages are elementary. Odd powers of u vanish, and $\langle k^2u^2\rangle = \frac{1}{5}k_{\mathrm{F}}^2$ and $\langle k^4u^4\rangle = \frac{3}{35}k_{\mathrm{F}}^4$ from $\langle k^{2n}\rangle = 3k_{\mathrm{F}}^{2n}/(2n+3)$ and $\langle u^{2n}\rangle = 1/(2n+1)$. Hence $\langle\Delta\rangle = q^2/2m_e$, $\langle\Delta^3\rangle = (q^2/2m_e)[(q^2/2m_e)^2 + \frac{3}{5}q^2v_{\mathrm{F}}^2]$ and $\langle\Delta^5\rangle = (q^2/2m_e)[\frac{3}{7}q^4v_{\mathrm{F}}^4 + O(q^6)]$. The leading term is the free-particle tail whose coefficient is the f-sum, and the term in $1/z^4$ is the correction of Eq. (D.43), found there from the closed form. The coefficients are the odd moments of the energies of the moves, the frequency moments of S^0 of Sect. 4.3.

For silicon itself the same Lehmann sum runs over Kohn–Sham bands instead of plane waves, with the crystal form factors $\langle n'\boldsymbol{k}+\boldsymbol{q}|\mathrm{e}^{\mathrm{i}(\boldsymbol{q}+\boldsymbol{G})\cdot\boldsymbol{r}}|n\boldsymbol{k}\rangle$ in the numerators, and it is a matrix in the reciprocal-lattice vectors (Sect. 4.4). It is computed by summing over a grid of $\boldsymbol{k}$ points and over occupied and empty bands, with a small broadening in place of $\mathrm{i}0^+$ (Appendix E.6). The closed form Eq. (4.56) has no such parameters, and the gas is the standard test of the codes that compute it.

4.5.2 The random-phase approximation

The Coulomb interaction enters through the proper response of Sect. 4.4. The random-phase approximation replaces it by the independent-particle bubble,

$$\tilde{R}_{ee} \simeq R^0, \qquad \varepsilon_{\mathrm{Lin}}(q,\omega) = 1 - v_{\mathrm{C}}(q)\,R^0(q,\omega), \qquad R_{ee} = \frac{R^0}{\varepsilon_{\mathrm{Lin}}}, \tag{4.59}$$

by Eqs. (4.40) and (4.41), and $\varepsilon_{\mathrm{Lin}}$ with R^0 of Eq. (4.56) is the Lindhard dielectric function. The physical content is that each electron responds as a free particle to the total potential, external plus Hartree, and to nothing else. What is dropped is the hole that exchange and correlation dig around each electron, which makes the potential an electron feels differ from the average Hartree potential; Sect. 4.6 puts it back as an exchange–correlation kernel. The name goes back to Bohm and Pines, who derived the collective oscillation from the equation of motion of $n_e(\boldsymbol{q})$ by dropping the terms that couple it to density fluctuations at other wavevectors, on the argument that their phases are random and average out [26]. The approximation is exact to leading order at high density, $r_s \ll 1$, where the kinetic energy dominates and the correlation hole matters least. At $r_s = 2$ it is not exact, but it gets the long-wavelength physics right, since at small q the Hartree term $v_{\mathrm{C}}R^0 \propto 1/q^2$ dominates every correction.

In the static limit the random-phase approximation screens a slowly varying potential exactly as Thomas–Fermi theory does, and it corrects that theory at shorter wavelengths. By Eq. (4.57) and $k_{\mathrm{TF}}^2 = 2e^2\mathcal{N}_{\mathrm{F}}$,

$$\varepsilon_{\mathrm{Lin}}(q,0) = 1 + \frac{k_{\mathrm{TF}}^2}{q^2}\,F_{\mathrm{L}}\Big(\frac{q}{2k_{\mathrm{F}}}\Big),$$

and $F_{\mathrm{L}} = 1 - q^2/12k_{\mathrm{F}}^2 + O(q^4)$ recovers Eq. (4.53) at $q \ll k_{\mathrm{F}}$. As the wavelength shrinks to the spacing of the electrons, the gas screens less than the local-equilibrium picture of Sect. 4.4

assumes, because a potential that varies on the scale $1/k_{\mathrm{F}}$ cannot be followed by a shift of the local Fermi level. At the silicon valence density and the typical momentum transfer $q_0 = 3.73\,\mathrm{keV}$, $\varepsilon_{\mathrm{Lin}}(q_0, 0) = 2.10$ against $\varepsilon_{\mathrm{TF}} = 2.22$, and the screening factor $1/|\varepsilon|^2$ of the static potential is 1/4.41 rather than 1/4.9 (Worked example 4.8). At $q = 2k_{\mathrm{F}}$, the momentum that spans the Fermi sphere, F_{L} has a logarithmic kink (Appendix C.8). In position space it turns the screening cloud around an impurity into a damped oscillation at the wavevector $2k_{\mathrm{F}}$, the Friedel oscillation; in the lattice dynamics of a metal the same kink is the Kohn anomaly.

Limit Check 4.1. Thomas–Fermi screening

A static potential that varies slowly in an electron gas is screened by the dielectric function $\varepsilon_{\mathrm{TF}} = 1 + k_{\mathrm{TF}}^2/q^2$, with $k_{\mathrm{TF}}^2 = 3\omega_p^2/v_{\mathrm{F}}^2$ [24, Eq. (S.14)].

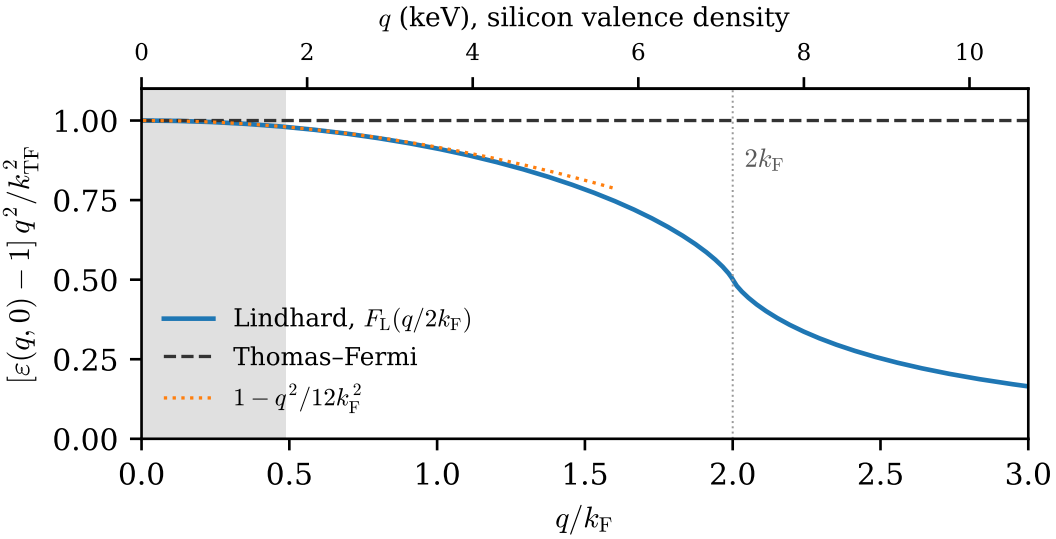


Figure 4.8: Static screening of the electron gas in the random-phase approximation against Thomas–Fermi theory. The induced part of the dielectric function, $[\varepsilon(q,0)-1]\, q^2/k_{\mathrm{TF}}^2$, is the Lindhard function $F_{\mathrm{L}}(q/2k_{\mathrm{F}})$ (solid) and one for Thomas–Fermi (dashed); dotted, the leading correction $1 - q^2/12k_{\mathrm{F}}^2$. The curve depends on q/k_{F} alone; the top axis gives q for the silicon valence density, $k_{\mathrm{F}} = 3.57\,\mathrm{keV}$. Gray: $q < 0.49\,k_{\mathrm{F}}$, where the two agree within 2 percent; they differ by 10 percent at $q = 1.06\,k_{\mathrm{F}}$. The kink at $2k_{\mathrm{F}}$ is the edge of the Fermi sphere.

At $q \gtrsim k_{\mathrm{F}}$ the local-equilibrium picture fails and the gas screens less; beyond $2k_{\mathrm{F}}$ the induced term falls as $1/q^4$ instead of $1/q^2$, and screening dies out.

Static screening at long wavelength is perfect. As $q \to 0$, $\varepsilon_{\mathrm{Lin}}(q, 0) \to \infty$, and an external charge is completely neutralized by the charge it induces: the induced density at $q \to 0$ is $R_{ee}\phi_{\mathrm{ext}} = -(1-1/\varepsilon)\,\phi_{\mathrm{ext}}/v_{\mathrm{C}}$, the opposite of the external charge. Through the inverse-moment relation Eq. (C.40) this becomes a sum rule for the loss function, $(2/\pi)\int_0^\infty \mathrm{d}\omega\, L(q,\omega)/\omega = 1 - 1/\varepsilon(q,0) \to 1$, the perfect-screening sum rule of a metal. At the silicon valence density and $q = 0.1\,k_{\mathrm{F}}$ the right side is 0.9925. In an insulator the same integral is $1 - 1/\varepsilon_\infty$ (Eq. (4.48)), 0.915 for silicon; the gas, with no gap, screens like a metal.

Fig. 4.10a shows the loss function $L = \mathrm{Im}(-1/\varepsilon_{\mathrm{Lin}})$ of the gas at $r_s = 2$ over the (q, ω) plane. Inside the particle–hole continuum, $\varepsilon_2 = -v_{\mathrm{C}}\,\mathrm{Im}\,R^0$ and $L = \varepsilon_2/|\varepsilon_{\mathrm{Lin}}|^2$. At $q \gg k_{\mathrm{F}}$, where $v_{\mathrm{C}}R^0$ is small, $\varepsilon_{\mathrm{Lin}} \to 1$, and the loss function is the free-recoil parabola of Worked

example 4.3, centered on $q^2/2m_e$: at short wavelength the electrons scatter one at a time, and screening is irrelevant. At $q \lesssim k_{\mathrm{F}}$ the factor $1/|\varepsilon_{\mathrm{Lin}}|^2$ suppresses the continuum, and the weight moves out of it, into a line above it. Since the f-sum rule fixes the total weight at every q, the weight that screening removes from the pairs has to go somewhere, and the line is where it goes.

4.5.3 The plasmon

The line is the plasma oscillation, and its frequency follows from a classical argument. Displace all the electrons of a slab of jellium by a small distance s along x, against the fixed positive background. The two faces of the slab acquire surface charges $\pm e\boldsymbol{n}_e s$ per unit area, and between them the field, in Heaviside–Lorentz units, is $E = e\boldsymbol{n}_e s$. Each electron feels the restoring force $-eE$, and $m_e\ddot{s} = -e^2\boldsymbol{n}_e s$: the electrons oscillate together at $\omega_p = (e^2\boldsymbol{n}_e/m_e)^{1/2}$, the plasma frequency of Eq. (4.31), whatever the size of the slab. No electron moves relative to its neighbors, and no kinetic energy of the Fermi sea enters. At finite wavelength it does, and this is what the random-phase approximation adds.

A pole of the response with nothing to drive it is a zero of the dielectric function, by Eq. (4.40). Above the continuum, $\omega > \omega_+$, $\operatorname{Im} R^0 = 0$ and $\varepsilon_{\mathrm{Lin}}$ is real, so a zero there is a pole of $1/\varepsilon_{\mathrm{Lin}}$ on the real axis, a mode that does not decay. Inserting Eq. (4.58) into Eq. (4.59), with $v_{\mathrm{C}}\boldsymbol{n}_e q^2/m_e = \omega_p^2$,

$$\varepsilon_{\mathrm{Lin}}(q,\omega) = 1 - \frac{\omega_p^2}{\omega^2}\left[1 + \frac{A}{\omega^2} + \frac{B}{\omega^4} + \dots\right], \qquad A = \tfrac{3}{5}q^2v_{\mathrm{F}}^2 + \Big(\frac{q^2}{2m_e}\Big)^2, \qquad B = \tfrac{3}{7}q^4v_{\mathrm{F}}^4 + O(q^6).$$

Setting it to zero gives $\omega^2 = \omega_p^2 + A\omega_p^2/\omega^2 + B\omega_p^2/\omega^4 + \dots$, which we solve by iteration. At zeroth order $\omega = \omega_p$. Substituting $\omega^2 = \omega_p^2 + A$ on the right, $A\omega_p^2/\omega^2 = A - A^2/\omega_p^2 + O(q^6)$ and $B\omega_p^2/\omega^4 = B/\omega_p^2 + O(q^6)$, so

Plasmon dispersion in the random-phase approximation

$$\omega_{\mathrm{pl}}(q)^2 = \omega_p^2 + \frac{3}{5}v_{\mathrm{F}}^2q^2 + \Big(\frac{1}{4m_e^2} + \frac{12}{175}\,\frac{v_{\mathrm{F}}^4}{\omega_p^2}\Big)q^4 + O(q^6), \tag{4.60}$$

where $\frac{12}{175} = \frac{3}{7} - \frac{9}{25}$ collects B and $-A^2$. The plasmon of the electron gas starts at ω_p and rises quadratically with q.

The q^2 term is the pressure of the Fermi sea. A classical plasma at temperature T has $3T/m_e$ in place of $\frac{3}{5}v_{\mathrm{F}}^2$. In the degenerate gas the electrons already move at speeds up to v_{F} at $T = 0$, and $\frac{3}{5}v_{\mathrm{F}}^2$ is three times their mean square velocity along $\boldsymbol{q}$, $\langle k^2u^2\rangle/m_e^2 = \frac{1}{5}v_{\mathrm{F}}^2$. Written as $\omega_{\mathrm{pl}} \simeq \omega_p + \alpha_{\mathrm{pl}}q^2/m_e$, the dispersion coefficient is $\alpha_{\mathrm{pl}} = \frac{3}{5}E_{\mathrm{F}}/\omega_p$, 0.45 at the silicon valence density. The expansion converges only outside the continuum, $\omega_{\mathrm{pl}} > \omega_+$, and Fig. 4.9 compares it with the exact zero of $\varepsilon_{\mathrm{Lin}}$.

Limit Check 4.2. Plasmon dispersion

In the random-phase approximation the plasmon of an electron gas disperses as $\omega_{\mathrm{pl}}^2 \simeq \omega_p^2 + \frac{3}{5}q^2v_{\mathrm{F}}^2$ and exists up to a cutoff $q_{\mathrm{c}} \simeq \omega_p/v_{\mathrm{F}}$ [25, Eq. (171) and the text following it].

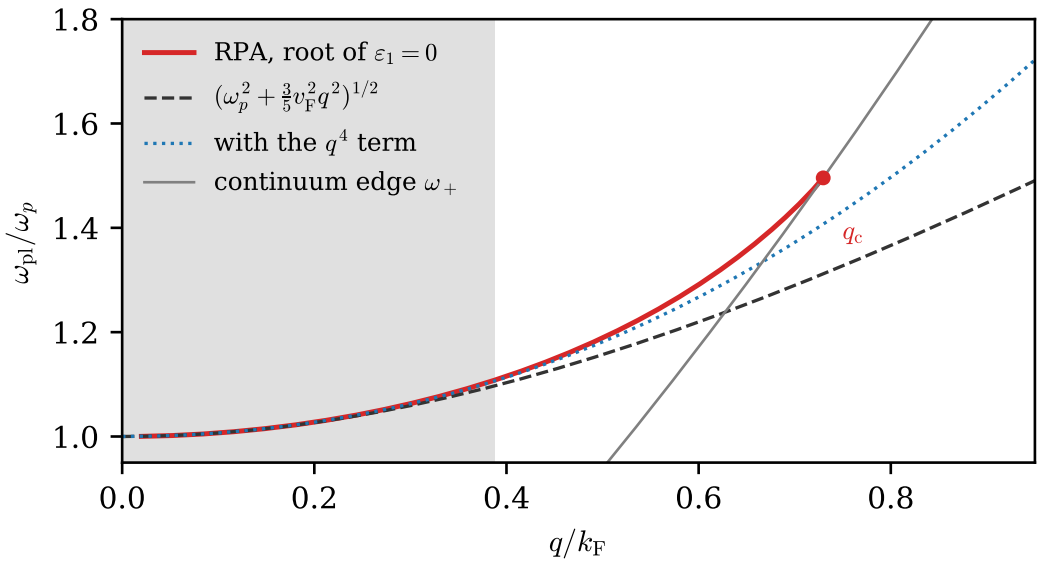


Figure 4.9: Plasmon dispersion of the electron gas at $r_s = 2$, the silicon valence density, in the random-phase approximation. Solid: the zero of $\mathrm{Re}\,\varepsilon_{\mathrm{Lin}}$ above the continuum, computed from Eq. (4.56), ending at $q_c = 0.730\,k_F$ where it meets the continuum edge $\omega_+ = qv_F + q^2/2m_e$ (thin line). Dashed: the quadratic formula; dotted: with the q^4 term of Eq. (4.60). Gray: $q < 0.39\,k_F$, where the quadratic formula is within one percent; there the q^4 term reduces the error to below 1.6×10^{-3}, and to below 3×10^{-4} for $q < 0.30\,k_F$. The estimate $q_c \simeq \omega_p/v_F = 0.666\,k_F$ is 9 percent low.

Near q_c the series in q^2 fails, since the plasmon approaches the pair energies that set its radius of convergence; beyond q_c there is no undamped plasmon at all.

The weight of the plasmon follows from the slope of $\varepsilon_{\mathrm{Lin}}$ at its zero. Near ω_{pl}, $\varepsilon_{\mathrm{Lin}} \simeq (\partial\varepsilon_1/\partial\omega)(\omega - \omega_{\mathrm{pl}} + \mathrm{i}0^+)$, with the $\mathrm{i}0^+$ carried from ω, and the Sokhotski–Plemelj formula gives

$$L(q,\omega) = \frac{\pi}{\partial\varepsilon_1/\partial\omega|_{\omega_{\mathrm{pl}}}}\,\delta(\omega - \omega_{\mathrm{pl}}(q)) \;\to\; \frac{\pi\omega_p}{2}\,\delta(\omega - \omega_p) \qquad (q \to 0), \tag{4.61}$$

since $\varepsilon_1 \to 1 - \omega_p^2/\omega^2$ with slope $2/\omega_p$ at ω_p. At long wavelength the line alone satisfies both sum rules. Its first moment is $(\pi/2)\omega_p^2$, the whole f-sum Eq. (4.46), and its inverse moment is $\pi/2$, the whole perfect-screening rule. The continuum, whose weight vanishes as $q \to 0$, has nothing left. By Eq. (4.45), the structure factor at $T = 0$ is then $S = (\pi q^2\omega_p/e^2)\,\delta(\omega - \omega_p)$. Its integral over ω is $2\pi n_e S^{\mathrm{st}}$ by Eq. (4.17), and with $e^2 n_e = m_e\omega_p^2$

$$S^{\mathrm{st}}(q) \to \frac{q^2}{2m_e\omega_p} \qquad (q \to 0).$$

The mean energy $q^2/(2m_e S^{\mathrm{st}})$ of Sect. 4.3 is ω_p at small q, and the density fluctuations of the interacting gas vanish as q^2, much faster than the $3q/4k_F$ of the free gas. At $q = 0.1\,k_F$ the two are 7.5×10^{-3} and 7.5×10^{-2}. The Coulomb force suppresses long-wavelength fluctuations of the charge, since any such fluctuation costs the energy of a plasma oscillation. At finite q the plasmon shares the f-sum with the continuum. Integrating the gridded loss function of Fig. 4.10 and adding the weight of Eq. (4.61) returns $(\pi/2)\omega_p^2$ to 10^{-5} at $q = 0.1\,k_F$ and $0.5\,k_F$; the plasmon carries $1 - 4 \times 10^{-5}$ of it at $0.1\,k_F$ and 0.957 at $0.5\,k_F$.

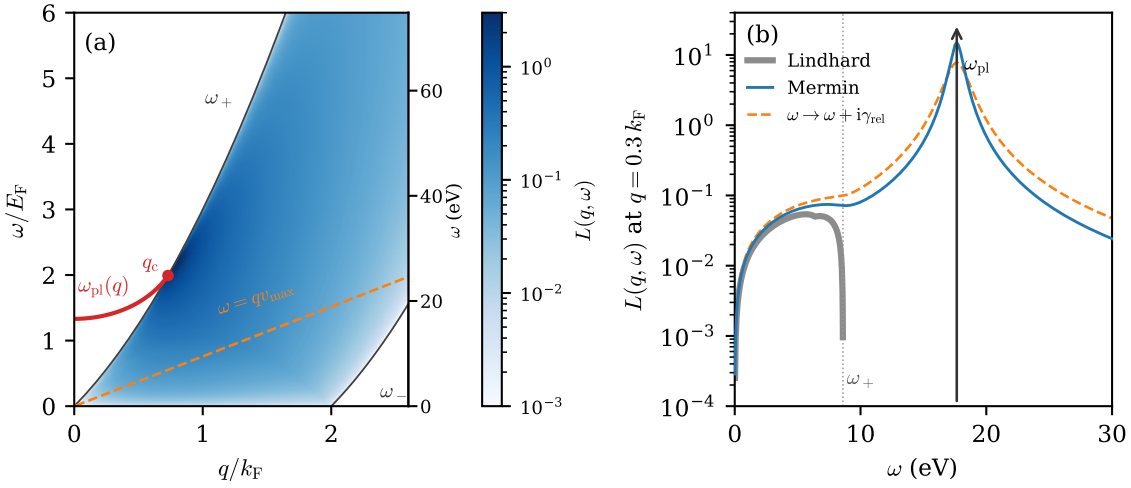


Figure 4.10: Loss function of the electron gas at $r_s = 2$, the valence density of silicon ($k_F = 3.57\,\text{keV}$, $E_F = 12.5\,\text{eV}$, $\omega_p = 16.6\,\text{eV}$), in the random-phase approximation at $T = 0$. (a) $L(q,\omega) = \text{Im}[-1/\varepsilon_{\text{Lin}}]$, logarithmic color scale, inside the particle–hole continuum bounded by $\omega_\pm$ (thin lines); the right axis gives ω in eV. Red: the undamped plasmon $\omega_{\text{pl}}(q)$, a delta function in L, drawn as a line up to $q_c = 0.730\,k_F$ (dot), where it enters the continuum and is Landau damped. Dashed: the largest energy a dark-matter particle can deposit at momentum q, $\omega = qv_{\text{max}}$ with $v_{\text{max}} = 0.379\,v_F$; it stays below the plasmon at every q. (b) L at $q = 0.3\,k_F$ on a logarithmic scale. Thick gray: Lindhard, with the continuum ending at ω_+ and the plasmon as a delta function (arrow, at 17.65 eV). Solid: the Mermin function with $\gamma_{\text{rel}} = 1\,\text{eV}$, an illustrative relaxation rate. Dashed: the naive substitution $\omega \to \omega + i\gamma_{\text{rel}}$ in R^0 with the same rate. Its plasmon peak is twice as wide, and its tail between the continuum and the plasmon is up to 1.7 times higher.

The plasmon cannot decay as long as no electron–hole pair has its energy and momentum, that is, as long as $\omega_{\text{pl}}(q) > \omega_+(q)$. The dispersion rises as q^2 from ω_p, while the continuum edge rises linearly from zero, and the two meet at the momentum q_c defined by $\varepsilon_{\text{Lin}}(q_c, \omega_+(q_c)) = 0$. A rough estimate sets $\omega_+ \simeq qv_F$ equal to ω_p, $q_c \simeq \omega_p/v_F$. At $r_s = 2$ the exact root is $q_c = 0.730\,k_F = 2.60\,\text{keV}$, at $\omega_{\text{pl}} = 24.8\,\text{eV}$; the estimate is $0.666\,k_F$. Beyond q_c, $\varepsilon_2 \neq 0$ where ε_1 vanishes, the zero of ε_{Lin} continued below the real axis moves into the lower half plane, and the plasmon becomes a resonance of the kind described in Sect. 4.1, with a width that grows quickly with q. Its energy goes into single electron–hole pairs. This is Landau damping, named after the analogous damping of waves in a classical plasma by particles that move with the wave.

In Fig. 4.10a the plasmon line enters the continuum at q_c, where the loss function of the continuum is largest. From there a broad Landau-damped ridge continues toward the free-recoil parabola. The figure also shows the line $\omega = qv_{\text{max}}$ of the largest energy a halo particle can deposit, which stays below the plasmon at every q (Worked example 4.10). Dark matter samples the low-energy corner of the continuum, where the electrons near the Fermi surface absorb the energy and the factor $1/|\varepsilon_{\text{Lin}}|^2$ screens it. At small q and $\omega \ll qv_F$ the corner has a simple form. There $\varepsilon_2 = (e^2/q^2)\, m_e^2\omega/(2\pi q)$ from the linear form of $\text{Im}\, R^0$, and $\varepsilon_1 \simeq k_{\text{TF}}^2/q^2 \gg \varepsilon_2$, so

$$L(q,\omega) \simeq \frac{\varepsilon_2}{\varepsilon_1^2} = \frac{e^2 m_e^2}{2\pi k_{\text{TF}}^4}\, q\,\omega \qquad (q \ll k_F,\ \omega \ll qv_F),$$

linear in both q and ω, and suppressed by the fourth power of the screening wavevector. At the typical momentum transfer q_0 and the largest energy $q_0 v_{\max} = 9.88\,\mathrm{eV}$, the full Lindhard function yields $|\varepsilon_{\mathrm{Lin}}|^2 = 3.96$ and $L = 0.183$: dynamic screening at the energy deposited is slightly weaker than the static estimate of Worked example 4.8.

4.5.4 Collisions and the Mermin function

In a real metal, or a doped semiconductor, the electrons scatter off impurities and phonons at some rate γ_{rel}, and the sharp edges of the Lindhard function and its undamped plasmon are smeared on that scale. The obvious way to include the collisions is to give the frequency an imaginary part, $\omega \to \omega + \mathrm{i}\gamma_{\mathrm{rel}}$, in R^0. This substitution is equivalent to letting the electrons relax toward the uniform equilibrium state at the rate γ_{rel}. It has an unphysical consequence: the excess density at a crest of the disturbance decays at that rate even when no current flows to carry it away. Electrons are destroyed at the crests and created in the troughs. Collisions with impurities and phonons conserve the number of electrons, and what they relax is the shape of the distribution at each point, not the number of electrons there. Mermin [27] therefore let the electrons relax toward a local equilibrium that has, at each instant, the same density as the disturbed gas. Appendix D.9 solves the equation of motion of the one-electron density matrix with this relaxation term and finds the response $\delta n_e/\phi = R^{\mathrm{Mer}}$, which in the random-phase approximation gives the Mermin dielectric function.

Mermin dielectric function

$$\varepsilon_{\mathrm{Mer}}(q,\omega) = 1 + \frac{(1 + \mathrm{i}\gamma_{\mathrm{rel}}/\omega)\,\big[\varepsilon_{\mathrm{Lin}}(q,\omega+\mathrm{i}\gamma_{\mathrm{rel}}) - 1\big]}{1 + (\mathrm{i}\gamma_{\mathrm{rel}}/\omega)\,\big[\varepsilon_{\mathrm{Lin}}(q,\omega+\mathrm{i}\gamma_{\mathrm{rel}}) - 1\big]/\big[\varepsilon_{\mathrm{Lin}}(q,0) - 1\big]}, \tag{4.62}$$

with $\varepsilon_{\mathrm{Lin}}$ at the complex frequency $\omega + \mathrm{i}\gamma_{\mathrm{rel}}$ from Eq. (4.56) and its static value from Eq. (4.57). It conserves the number of electrons at every q and ω, and it reduces to $\varepsilon_{\mathrm{Lin}}$ as $\gamma_{\mathrm{rel}} \to 0$.

This is Eq. (D.46), Mermin's form [27], and Refs. [5, Eq. (19)] and [25, Eq. (168)], where the rate is written Γ and Γ_p. Analyticity of $R^0(q, z + \mathrm{i}\gamma_{\mathrm{rel}})$ in the upper half plane does not by itself exclude a zero of the denominator there. Appendix D.9 shows that relaxation toward local equilibrium has no growing free mode, so the Mermin response has no pole in the upper half plane and is causal.

Conservation is a statement about the continuity equation, and it is where the two prescriptions part. Summing the equation of motion of the coherences $\langle c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}+\boldsymbol{q}}\rangle$ over $\boldsymbol{k}$ gives the Fourier transform of $\partial_t n_e + \nabla\cdot\boldsymbol{j} = 0$, with the paramagnetic current of Eq. (4.29), plus whatever the collision term contributes to the sum (Eq. (D.47)). With the naive substitution that contribution is $-\mathrm{i}\gamma_{\mathrm{rel}}$ times the density itself, a source and sink of electrons. With Mermin's reference state it vanishes identically, because the local equilibrium is chosen to have the actual density. Collisions then change only the current, and the density changes only as fast as the current carries electrons in or out.

A static potential and the long-wavelength limit tell the two prescriptions apart, as Appendix D.9 works out; the f-sum rule does not. For a static potential, $\omega \to 0$ at fixed γ_{rel}, $\varepsilon_{\mathrm{Mer}} \to \varepsilon_{\mathrm{Lin}}(q,0)$: a static potential leaves the electrons in equilibrium in that potential, with or without collisions, so the static screening, perfect screening included, is that of the collisionless gas. The naive substitution yields $\varepsilon_{\mathrm{Lin}}(q,\mathrm{i}\gamma_{\mathrm{rel}})$ instead; at $q = 0.3\,k_{\mathrm{F}}$ and $\gamma_{\mathrm{rel}} = 1\,\mathrm{eV}$ in the silicon-density gas it is 12.8 against the correct 15.7. At long wavelength,

$q \to 0$ at fixed ω, the Mermin function becomes the Drude dielectric function,

$$\varepsilon_{\mathrm{Mer}} \to 1 - \frac{\omega_p^2}{\omega(\omega + \mathrm{i}\gamma_{\mathrm{rel}})} = 1 + \frac{\mathrm{i}\sigma_{\mathrm{cond}}}{\omega}, \qquad \sigma_{\mathrm{cond}}(\omega) = \frac{\omega_p^2}{\gamma_{\mathrm{rel}} - \mathrm{i}\omega},$$

by Eq. (4.50), with the dc conductivity $\omega_p^2/\gamma_{\mathrm{rel}}$ of a metal whose electrons are scattered at the rate γ_{rel}. The naive substitution leads to $1 - \omega_p^2/(\omega + \mathrm{i}\gamma_{\mathrm{rel}})^2$, which tends to $1 + \omega_p^2/\gamma_{\mathrm{rel}}^2$ as $\omega \to 0$: no dc current and a finite static dielectric constant, the response of an insulator. Finally, the f-sum rule does not distinguish the two. Both functions are analytic in the upper half plane and both fall off as ω_p^2/ω^2, so both obey Eq. (4.46) by the argument of Appendix C.5. The f-sum constrains the response at high frequency, while particle conservation acts at low frequency and long wavelength, where the collisions matter.

At $q \to 0$ the Drude form puts the whole loss function in one Lorentzian line [25, Eq. (172)],

$$L(0, \omega) = \frac{\omega_p^2 \, \gamma_{\mathrm{rel}} \, \omega}{(\omega^2 - \omega_p^2)^2 + \gamma_{\mathrm{rel}}^2 \omega^2}.$$

It is the damped oscillator of Worked example 4.9 with $\omega_{\mathrm{res}} = \omega_p$, no gap, and γ_{rel} in the role of Γ_{w}. The plasmon now has the full width γ_{rel}, and below it the loss function has a tail, $L \simeq \gamma_{\mathrm{rel}}\omega/\omega_p^2$ for $\omega \ll \omega_p$, linear in ω down to zero frequency. The naive substitution puts the pole at $\pm\omega_p - \mathrm{i}\gamma_{\mathrm{rel}}$, with twice the width and twice the tail, $2\gamma_{\mathrm{rel}}\omega/\omega_p^2$. Fig. 4.10b compares the three functions at $q = 0.3\, k_{\mathrm{F}}$. For a dark-matter particle, which deposits energies far below the plasmon, the tail is the part that matters. It is weight that the Lindhard function does not have at small q, and whether a real material has it is a question about the material, not about the gas [24, 25].

In practice the Mermin function is used as a fitting form. The plasmon of silicon has a width of a few eV, but it does not come from collisions: the plasmon decays into interband electron–hole pairs, as Worked example 4.10 shows. A rate γ_{rel} fitted to that width is a parameter of the fit, not a collision rate. The Mermin oscillator method of Ref. [5, Eqs. (20) and (21)] writes the measured optical loss function as a sum of Mermin terms,

$$L(q, \omega) = \sum_i A_i(q) \, \mathrm{Im}\big[-1/\varepsilon_{\mathrm{Mer}}(q, \omega; \omega_{p,i}, \gamma_i)\big] \, \Theta(\omega - \omega_{\mathrm{edge},i}).$$

Each term has its own plasma frequency $\omega_{p,i}$, rate γ_i and onset $\omega_{\mathrm{edge},i}$. The parameters are fitted at $q = 0$, and the Mermin function carries each term to finite q. The amplitudes $A_i(q)$ are rescaled at each q so that every term keeps its share of the f-sum. The step functions remove the weight below the gap that a single gas would put there. The method is a way to extrapolate optical data to the momentum transfers of dark-matter scattering while keeping the sum rules; the physics of the extrapolation is that of the gas.

Worked example 4.10: The silicon valence electrons as an electron gas

Treat the four valence electrons per atom of silicon as jellium at $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$, $r_s = 2.0$. The plasma frequency is $\omega_p = 16.6\,\mathrm{eV}$ (Eq. (4.31)), and by Eq. (4.60) the plasmon disperses as $\omega_{\mathrm{pl}} \simeq \omega_p + 0.45\, q^2/m_e$, up to $q_{\mathrm{c}} = 2.60\,\mathrm{keV}$ and $\omega_{\mathrm{pl}} = 24.8\,\mathrm{eV}$. The measured loss function at $q \to 0$ (Fig. 4.7) peaks at 16.4 eV, and ε_1 crosses zero at 16.3 eV; the gas places the zero 1.9 percent higher. Three questions follow: why the agreement is so good for a crystal with a gap and covalent bonds, why the measured peak has a width of 3.7 eV when the gas has none, and what the comparison means for a dark-matter particle.

Position. Near 16 eV the valence electrons of silicon respond almost as if they were

free. In the tabulated data, 87 percent of their f-sum weight $\int \omega\varepsilon_2 \,\mathrm{d}\omega$ lies below 16.3 eV, at a root-mean-square energy of 6.5 eV, in the interband transitions of Fig. 4.7a. A transition at an energy ω_i well below ω contributes to ε_1 as an oscillator driven above its resonance, $-f_i/(\omega^2-\omega_i^2) \simeq -(f_i/\omega^2)(1+\omega_i^2/\omega^2)$, which is the free-electron term with a correction of order $(\omega_i/\omega)^2 \approx 0.16$. The slope of the data confirms the picture: at the zero, $\partial\varepsilon_1/\partial\omega = 0.128\,\mathrm{eV}^{-1}$, against $2\omega_p^2/\omega^3 = 0.127\,\mathrm{eV}^{-1}$ for the gas. The correction from binding has the sign that pushes the zero up; if all the weight sat in one oscillator at 5.08 eV, the resonance of the single-oscillator model, the zero would move to $\bar{E} = 17.4\,\mathrm{eV}$ (Worked example 4.9). The remaining weight, 9.4 percent between 16.3 eV and 40 eV and the rest above, lies above the plasmon and contributes positively to ε_1, which pulls the zero down. The two shifts nearly cancel.

Width. In the gas at $q \to 0$ the plasmon cannot decay, since the continuum shrinks to $\omega < qv_\mathrm{F} \to 0$. In silicon, vertical interband transitions exist at all energies up to well above 16 eV, and at the zero of ε_1 the data have $\varepsilon_2 = 0.224$. Near the zero $L = \varepsilon_2/(\varepsilon_1^2+\varepsilon_2^2)$ with $\varepsilon_1 \simeq (\partial\varepsilon_1/\partial\omega)(\omega-\omega_\mathrm{pl})$, a Lorentzian of full width $2\varepsilon_2/(\partial\varepsilon_1/\partial\omega) = 3.5\,\mathrm{eV}$, against the measured 3.66 eV. The plasmon of silicon decays into interband electron–hole pairs: the crystal potential supplies the momentum that, in the gas, only $q > q_\mathrm{c}$ could provide. A Mermin function with $\gamma_\mathrm{rel} = 3.66\,\mathrm{eV}$ reproduces the line: at $q \to 0$ its peak height is $\omega_p/\gamma_\mathrm{rel} = 4.54$, against the measured 4.5. Its tail, however, is $L = 0.042$ at 3 eV, 14 times the measured 3.1×10^{-3}, and it continues below the gap, where silicon absorbs nothing, just as the single oscillator of Worked example 4.9 did.

Dark matter. The plasmon is out of reach. A halo particle deposits at most qv_max, and with $v_\mathrm{max} = 0.379\,v_\mathrm{F}$ the line $\omega = qv_\mathrm{max}$ reaches $\omega_p = 16.6\,\mathrm{eV}$ only at $q = \omega_p/v_\mathrm{max} = 6.26\,\mathrm{keV} = 1.75\,k_\mathrm{F}$, far beyond q_c, where the gas has no plasmon left (Fig. 4.10a). The plasmon, which carries most of the weight at small q, is kinematically inaccessible [25]. What the particle sees is the low-energy side of the loss function, at a few eV, and there the gas and the crystal differ most. The gas has a continuum down to $\omega = 0$, screened by $|\varepsilon_\mathrm{Lin}|^2 \approx 4$ at q_0, while silicon has a gap, interband peaks, and an optical loss function of 3.1×10^{-3} at 3 eV. The electron gas is a good model of the peak that holds most of the f-sum and a poor one of the corner that sets the rate. Its use in the dark-matter literature is accordingly limited to energies well above the gap, between about 5 and 15 eV, where it describes the computed spectrum of silicon and germanium fairly well [25].

The electron gas has no gap, no lattice, and no attraction between an excited electron and the hole it leaves behind. Each of these changes the low-energy side of the loss function, which is the part that dark matter samples. The lattice adds local fields, the response at $\boldsymbol{q}+\boldsymbol{G}$ to a potential at $\boldsymbol{q}$, which the random-phase approximation in a crystal keeps and the gas cannot have. The attraction between electron and hole returns in the excitons of Sect. 4.9. Sect. 4.6 computes the random-phase approximation for silicon, local fields included, and adds the exchange–correlation kernel.

4.6 Local fields and exchange–correlation kernels

The absorption spectrum of silicon has two peaks, at 3.42 and 4.24 eV (Fig. 4.7), and its static screening saturates at $\varepsilon_\infty = 11.7$. A calculation that lets each electron respond on its own, with band energies fitted to experiment, puts the upper peak in nearly the right place, turns the lower one into a shoulder, and finds a dielectric constant within a few percent of 11.7. Adding the microscopic fields that the induced charge produces inside each cell lowers both peaks and lowers ε_∞ by about a tenth. Adding the simplest exchange–correlation kernel

undoes about half of that and changes the shape hardly at all. Only the attraction between the excited electron and its hole, the subject of Sect. 4.9, restores the lower peak. Each of these steps is a different approximation to the proper response $\tilde{R}$ of Sect. 4.4. For dark matter the differences matter where the rate is largest for light masses: in the first few eV above the gap, where the loss function is small and the screening strong. We will build each step from the crystal version of the bubble of Sect. 4.5, and we will see that the spread between them is what a computed loss function can quote as its many-body uncertainty.

4.6.1 The crystal bubble

Take the Kohn–Sham states of Sect. 3.4, with band energies $\varepsilon_{n\boldsymbol{k}}$ and occupations $f_{n\boldsymbol{k}}$ (Eq. (3.24)), as a system of independent electrons, and compute its density response matrix $R_{\boldsymbol{G}\boldsymbol{G}'}$ of Eq. (4.42) from the Lehmann representation Eq. (4.6). The steps are those of the electron gas in Sect. 4.5, with Bloch states in place of plane waves. In the Bloch basis the density operator at $\boldsymbol{q}+\boldsymbol{G}'$ is Eq. (3.37) with $\boldsymbol{q}\to\boldsymbol{q}+\boldsymbol{G}'$,

$$n_e(\boldsymbol{q}+\boldsymbol{G}') = \sum_s \sum_{\boldsymbol{k}} \sum_{mn} \rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G}')\, c^\dagger_{m,\boldsymbol{k}+\boldsymbol{q},s}\, c_{n\boldsymbol{k}s},$$

with $\boldsymbol{q}$ in the first zone. The reciprocal vector does not change the crystal momentum of the final state, $\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{G}' \equiv \boldsymbol{k}+\boldsymbol{q}$; it only changes the plane wave that probes the two states, through the density vertex ρ_{mn} of Eq. (3.34). Acting on an occupation-number state $|i\rangle$ of Bloch states, the operator moves one electron from $n\boldsymbol{k}$ to $m,\boldsymbol{k}+\boldsymbol{q}$. In the resonant term of Eq. (4.6) the final state is this move, with weight $f_{n\boldsymbol{k}}(1-f_{m,\boldsymbol{k}+\boldsymbol{q}})$, energy $\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}}-\varepsilon_{n\boldsymbol{k}}$ and matrix elements $\langle f|n_e(\boldsymbol{q}+\boldsymbol{G}')|i\rangle = \rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G}')$ and $\langle i|n_e(\boldsymbol{q}+\boldsymbol{G})^\dagger|f\rangle = \rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})^*$. In the antiresonant term the final state has the reverse move, from $m,\boldsymbol{k}+\boldsymbol{q}$ to $n\boldsymbol{k}$, with weight $f_{m,\boldsymbol{k}+\boldsymbol{q}}(1-f_{n\boldsymbol{k}})$; its two matrix elements are the same pair, and its denominator $\omega+(\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}})+\mathrm{i}0^+$ is the resonant one. As in the gas, the Pauli factors cancel in the sum of the two numerators, $f_{n\boldsymbol{k}}(1-f_{m,\boldsymbol{k}+\boldsymbol{q}})-f_{m,\boldsymbol{k}+\boldsymbol{q}}(1-f_{n\boldsymbol{k}}) = f_{n\boldsymbol{k}}-f_{m,\boldsymbol{k}+\boldsymbol{q}}$, and the sum over the two spins gives the degeneracy factor $g_{\mathrm{deg}}=2$.

Independent-particle bubble of a crystal

$$R^0_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega) = \frac{g_{\mathrm{deg}}}{\mathscr{V}} \sum_{\boldsymbol{k}} \sum_{n,m} (f_{n\boldsymbol{k}}-f_{m,\boldsymbol{k}+\boldsymbol{q}})\, \frac{\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})^*\, \rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G}')}{\omega+\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}}+\mathrm{i}0^+}, \qquad (4.63)$$

with $\mathscr{V}=N\Omega$, the sum over the N crystal momenta of the first zone, and all pairs of bands. It is the density response at $\boldsymbol{q}+\boldsymbol{G}$ to a potential at $\boldsymbol{q}+\boldsymbol{G}'$ of electrons that move independently in the Kohn–Sham potential.

This is the result of Adler and Wiser [22, 23], and it is Eq. (E.38), written there with a finite broadening. For one band of plane waves, $\rho_{mn}\to 1$ at $\boldsymbol{G}=\boldsymbol{G}'=0$ and the matrix reduces to the bubble Eq. (4.55) of the gas. In a crystal the off-diagonal elements do not vanish: a potential at $\boldsymbol{q}$ moves electrons into states whose overlap with the initial ones is modulated on the scale of the cell, and the induced density has components at every $\boldsymbol{q}+\boldsymbol{G}$.

At small q the vertex at $\boldsymbol{G}=0$ between different bands is the optical one, Eq. (3.43), $\rho_{cv}(\boldsymbol{k},\boldsymbol{q}) = \boldsymbol{q}\cdot\boldsymbol{v}_{cv}(\boldsymbol{k})/(\varepsilon_{c\boldsymbol{k}}-\varepsilon_{v\boldsymbol{k}})$, while the vertices at $\boldsymbol{G}\neq 0$ tend to the finite $\rho_{cv}(\boldsymbol{k},\boldsymbol{G})$. In an insulator at $T=0$ only valence-to-conduction pairs contribute, with $f_{v\boldsymbol{k}}=1$ and $f_{c\boldsymbol{k}}=0$. Write $\Delta_{cv\boldsymbol{k}}=\varepsilon_{c\boldsymbol{k}}-\varepsilon_{v\boldsymbol{k}}$. The pairs $(n,m)=(v,c)$ and (c,v) of Eq. (4.63) give the head

$$R^0_{\boldsymbol{0}\boldsymbol{0}}(\boldsymbol{q}\to 0,\omega) = \frac{g_{\mathrm{deg}} q^2}{\mathscr{V}} \sum_{\boldsymbol{k}} \sum_{v,c} \frac{|\hat{\boldsymbol{q}}\cdot\boldsymbol{v}_{cv}(\boldsymbol{k})|^2}{\Delta^2_{cv\boldsymbol{k}}} \left[\frac{1}{\omega-\Delta_{cv\boldsymbol{k}}+\mathrm{i}0^+} - \frac{1}{\omega+\Delta_{cv\boldsymbol{k}}+\mathrm{i}0^+}\right], \qquad (4.64)$$

of order q^2, as Appendix D.7 requires of every proper response of an insulator. Its imaginary part, multiplied by $-v_{\mathrm{C}}(q) = -e^2/q^2$, is the absorptive part of the head of the dielectric matrix at positive frequency,

$$\varepsilon_2^{\mathrm{IP}}(\omega) = \frac{\pi g_{\mathrm{deg}} e^2}{\mathscr{V}} \sum_{\boldsymbol{k}} \sum_{v,c} \frac{|\hat{\boldsymbol{q}} \cdot \boldsymbol{v}_{cv}(\boldsymbol{k})|^2}{\Delta_{cv\boldsymbol{k}}^2} \, \delta(\omega - \Delta_{cv\boldsymbol{k}}), \tag{4.65}$$

where the label IP marks independent particles. It counts the vertical transitions at energy ω, each weighted by its squared velocity matrix element along the field: this is the golden-rule absorption of light, through $\sigma_1 = \omega\varepsilon_2$ of Eq. (4.50). The real part at $\omega = 0$ is

$$\varepsilon_1^{\mathrm{IP}}(0) = 1 + \frac{2 g_{\mathrm{deg}} e^2}{\mathscr{V}} \sum_{\boldsymbol{k}} \sum_{v,c} \frac{|\hat{\boldsymbol{q}} \cdot \boldsymbol{v}_{cv}(\boldsymbol{k})|^2}{\Delta_{cv\boldsymbol{k}}^3}, \tag{4.66}$$

since the bracket of Eq. (4.64) is $-2/\Delta_{cv\boldsymbol{k}}$ at $\omega = 0$. The third power of the transition energy makes the static screening sensitive to the smallest gaps. A Kohn–Sham gap that is too small (Sect. 3.4) therefore makes ε_∞ too large, as the PBE value 13.04 against the measured 11.7 shows. The scissor of Eq. (3.66) leaves ρ_{cv}, and so the ratio $\boldsymbol{v}_{cv}/\Delta_{cv\boldsymbol{k}}$, unchanged (Eq. (3.67)). It then shifts $\varepsilon_2^{\mathrm{IP}}$ rigidly to higher energy and lowers Eq. (4.66), whose terms go as $|\rho_{cv}/q|^2/\Delta_{cv\boldsymbol{k}}$.

In practice Eq. (4.63) is summed on a grid of $\boldsymbol{k}$ points, over the occupied bands and a finite number of empty ones, with a broadening in place of $\mathrm{i}0^+$ that must be larger than the spacing of the transition energies on the grid. The head and the wings, the elements with one index zero, are computed analytically in $\hat{\boldsymbol{q}}$ from velocity matrix elements, and the matrix is cut off at a finite number of reciprocal vectors. Appendix E.6 gives the cost of each of these parameters, how to test their convergence on the f-sum rule, and what each one does to a loss function at the momentum transfers of dark matter.

4.6.2 The random-phase approximation in a crystal

The random-phase approximation of Sect. 4.5 replaces the proper response by the bubble, now as a matrix,

$$\tilde{R}_{\boldsymbol{G}\boldsymbol{G}'} \simeq R^0_{\boldsymbol{G}\boldsymbol{G}'}, \qquad \varepsilon_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\mathrm{C}}(|\boldsymbol{q} + \boldsymbol{G}|)\, R^0_{\boldsymbol{G}\boldsymbol{G}'}, \tag{4.67}$$

with Eq. (4.43). Each electron responds as an independent particle to the total potential, external plus Hartree, at every $\boldsymbol{q} + \boldsymbol{G}$. The macroscopic function then comes at two levels. Keeping only the head gives $\varepsilon_{\mathrm{mac}} \simeq \varepsilon_{\boldsymbol{00}} = 1 - v_{\mathrm{C}}(q) R^0_{\boldsymbol{00}}$, the random-phase approximation without local fields. Inverting the whole matrix gives $\varepsilon_{\mathrm{mac}} = 1/(\varepsilon^{-1})_{\boldsymbol{00}}$, the random-phase approximation with local fields, which by Eq. (D.37) is

$$\varepsilon_{\mathrm{mac}} = 1 - v_{\mathrm{C}}(q)\, \bar{R}_{\boldsymbol{00}}, \qquad \bar{R} = R^0 + R^0\, \bar{v}\, \bar{R}, \tag{4.68}$$

with $\bar{v}$ the Coulomb matrix without its element at $\boldsymbol{G} = 0$. In this form the singular head of v_{C} never appears in the inversion, and the limit $\boldsymbol{q} \to 0$ is taken in R^0 alone (Appendix D.7).

We can now see what the head alone does and does not change. Since v_{C} is real, the imaginary part of $\varepsilon_{\boldsymbol{00}}$ is Eq. (4.65) exactly: without local fields the random-phase approximation has the absorption spectrum of independent particles. Screening enters through the real part, and so through the loss function $L = \varepsilon_2/|\varepsilon|^2$ of Sect. 4.4, which it suppresses by $|\varepsilon|^2$ wherever ε_1 is large. Going from independent particles to this level is a change of the rate, through $1/|\varepsilon|^2$, and no change of the optical spectrum. The local fields are the first correction that changes ε_2.

What they do has a classical picture. A capacitor filled with layers of different dielectric constants, stacked along the field, has a uniform D and a field $E = D/\varepsilon(x)$ that varies from layer to layer. The capacitance measures the average field, $\langle E\rangle = D\langle 1/\varepsilon\rangle$, so the effective constant is $1/\langle 1/\varepsilon\rangle$, the harmonic mean, which is never larger than the arithmetic mean $\langle\varepsilon\rangle$. The macroscopic function Eq. (4.44) has exactly this structure: it is the inverse of the average of the inverse, while the head $\varepsilon_{\mathbf{00}}$ is the average itself. In silicon the polarizable charge sits in the bonds. A uniform field pushes it along the bonds, the displaced charge forms dipoles on the scale of the cell, and their microscopic Hartree field, at $\boldsymbol{G} \neq 0$, points against the applied field in the regions where the polarizable electrons are. The electrons feel less than the average field, and the crystal screens less than the head alone predicts. The two would agree if the induced density were uniform, as in the gas, and they disagree most where the response is concentrated in a small part of the cell.

Fig. 4.11 shows the consequence for silicon in a model calculation with the empirical pseudopotential of Cohen and Bergstresser (Sect. 3.3), whose form factors were fitted to measured transition energies; no scissor is needed. The independent-particle spectrum, panel (a), has its upper peak at 4.26 eV, close to the measured 4.24 eV, with a height of 41 against the measured 46. At 3.42 eV, where the measured spectrum has its lower peak of height 36, the model has only 15, a shoulder. The local fields lower the upper peak to 34 and move it to 4.32 eV; at 3.42 eV they leave 13. Spectral weight moves up, as the f-sum rule with local fields (Appendix D.7) requires if the low-energy peaks lose some, and the agreement with experiment gets worse. The same happens in first-principles calculations on top of quasiparticle bands [28], and it was first found with empirical pseudopotential bands [29]. Since the static constant is the inverse moment $1 + (2/\pi)\int \varepsilon_2/\omega$ (Eq. (4.48)), weight moved up lowers it, and the local fields lower ε_∞ (Worked example 4.11).

Panel (b) shows what dark matter sees. Up to about 6 eV the loss function lies two to three orders of magnitude below the unscreened ε_2; this is the factor $1/|\varepsilon|^2$ that the head of the dielectric matrix already contains. On this scale the local fields are a correction of tens of percent. Between 3 and 12 eV they raise L by 10 to 35 percent, because they lower $|\varepsilon|$, and they cut the height of the plasmon peak from 7.6 to 3.4. A dark-matter rate integrates the loss function over $\boldsymbol{q}$ and over ω up to $qv_{\max}$, and for silicon and germanium the difference between rates computed with and without local fields is small [5]. Local fields matter more at momentum transfers of a few reciprocal vectors, where the off-diagonal elements of R^0 are as large as the diagonal ones, and at energies where semicore shells, localized near the nuclei, contribute (Appendix E.6).

Local fields have a second reading, which Sect. 4.9 uses. In the space of electron–hole pairs, the microscopic Hartree potential $\bar{v}$ is the bare Coulomb interaction between a pair and another pair at the same total momentum, with the long-range term at $\boldsymbol{G} = 0$ removed. This is the exchange kernel K^{x} of the Bethe–Salpeter equation. The random-phase approximation with local fields is that equation with the electron–hole attraction switched off, and it cannot bind an exciton.

4.6.3 Time-dependent density-functional theory

The bubble Eq. (4.63) is built from Kohn–Sham states, and until now that was a convenience: the states of some independent-particle model were needed, and these were at hand. Time-dependent density-functional theory makes the choice exact. Runge and Gross proved that the time-dependent density of an interacting system, evolving from its ground state, determines the time-dependent external potential up to a function of time alone [32]. It follows, as in the static theory of Sect. 3.4, that a system of independent electrons in a suitable time-dependent Kohn–Sham potential has the same density at all times. That potential is the external one

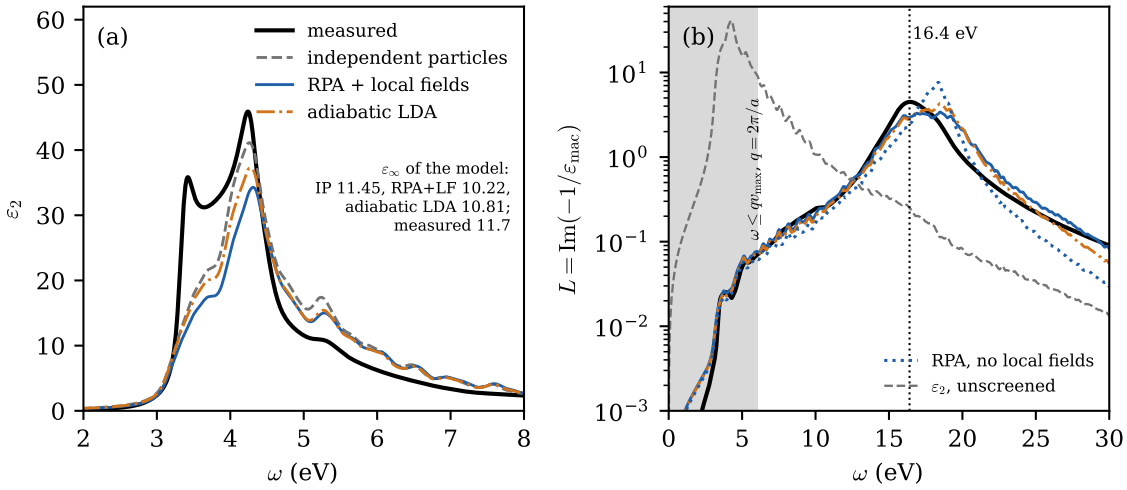


Figure 4.11: Three many-body approximations to the dielectric function of crystalline silicon at $\boldsymbol{q} \to 0$, in a model calculation. Band energies and Bloch states come from the local empirical pseudopotential of Cohen and Bergstresser [30], fitted to measured transition energies (no scissor shift). The sums run over a shifted 20^3 grid of $\boldsymbol{k}$ points over the full zone, with 4 valence and 26 conduction bands, a local-field matrix of the 59 shortest reciprocal vectors, and a Lorentzian broadening of 0.10 eV. The exchange–correlation kernel is the adiabatic local-density one, evaluated on the model's valence density with the Perdew–Zunger correlation energy [31]. Black: measured at 300 K [19]; panel (a) lists the static limits of the three approximations against the measured $\varepsilon_\infty = 11.7$ (Sect. 4.4). (a) Absorptive part ε_2: independent particles (dashed gray), which is also the random-phase approximation without local fields; the random-phase approximation with local fields (solid blue); adiabatic LDA (dash-dotted orange). The strength of the measured lower peak at 3.42 eV comes from the electron–hole attraction (Sect. 4.9), and all three underestimate it. (b) Loss function on a logarithmic scale: the random-phase approximation without local fields (dotted blue), with local fields (solid blue), and adiabatic LDA (dash-dotted orange). Dashed gray: the unscreened ε_2 of independent particles, which a rate without screening would use in place of L. The model's loss peak lies about 2 eV above the measured 16.4 eV (dotted line), because its high conduction bands are not fitted. Shaded: energies up to $qv_{\max} = 6.05$ eV at a momentum transfer of one reciprocal-lattice unit, $q = 2\pi/a$, the range that dark matter of any mass can reach there.

plus the Hartree potential plus an exchange–correlation potential $v_{\mathrm{xc}}(\boldsymbol{r},t)$, which now depends on the density at all earlier times.

Linear response follows in three lines. A weak external potential energy ϕ_{ext} changes the Kohn–Sham potential by

$$\delta V_{\mathrm{KS}}(\boldsymbol{r},t) = \phi_{\mathrm{ext}}(\boldsymbol{r},t) + \int \mathrm{d}^3r'\, v_{\mathrm{C}}(\boldsymbol{r}-\boldsymbol{r}')\,\delta n(\boldsymbol{r}',t) + \int \mathrm{d}^3r' \int \mathrm{d}t'\, f_{\mathrm{xc}}(\boldsymbol{r},\boldsymbol{r}',t-t')\,\delta n(\boldsymbol{r}',t'),$$

where the exchange–correlation kernel

$$f_{\mathrm{xc}}(\boldsymbol{r},\boldsymbol{r}',t-t') = \left.\frac{\delta v_{\mathrm{xc}}(\boldsymbol{r},t)}{\delta n(\boldsymbol{r}',t')}\right|_{n_{\mathrm{gs}}}, \tag{4.69}$$

is the functional derivative taken at the ground-state density n_{gs}. Since the Kohn–Sham electrons are independent, their density responds to their own potential through the bubble: $\delta n = R^0\,\delta V_{\mathrm{KS}}$, with R^0 of Eq. (4.63). And δn is, by construction, the density change of the interacting system, $\delta n = R\,\phi_{\mathrm{ext}}$. In reciprocal space, with $f_{\mathrm{Hxc}} = v_{\mathrm{C}} + f_{\mathrm{xc}}$ the matrix $(f_{\mathrm{Hxc}})_{\boldsymbol{G}\boldsymbol{G}'} = v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\delta_{\boldsymbol{G}\boldsymbol{G}'} + (f_{\mathrm{xc}})_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega)$, the two statements combine to $R\phi_{\mathrm{ext}} = R^0(\phi_{\mathrm{ext}} + f_{\mathrm{Hxc}}R\phi_{\mathrm{ext}})$ for every ϕ_{ext}.

Dyson equation of time-dependent density-functional theory

$$R = R^0 + R^0 f_{\mathrm{Hxc}} R, \qquad \tilde{R} = R^0 + R^0 f_{\mathrm{xc}} \tilde{R}, \tag{4.70}$$

as matrices in the reciprocal-lattice vectors at fixed $(\boldsymbol{q},\omega)$. With the exact kernel, R is the exact density response, although R^0 is built from Kohn–Sham states. The random-phase approximation is $f_{\mathrm{xc}} = 0$.

The proper response follows from the first form by splitting off the Coulomb kernel, since $R = \tilde{R} + \tilde{R}v_{\mathrm{C}}R$ (Eq. (4.40)) has the solution $\tilde{R} = (1 - R^0 f_{\mathrm{xc}})^{-1}R^0$. This is Eq. (E.39), and Petersilka, Gossmann and Gross used it to compute excitation energies as the poles of R [33]. For the macroscopic function, the kernel of Eq. (4.68) becomes $\bar{v} + f_{\mathrm{xc}}$, with the long-range Coulomb term still removed.

The exact kernel, however, is unknown, and it has to do a lot. The Kohn–Sham gap is smaller than the true one (Sect. 3.4), and the poles of R^0 sit at the wrong energies; the kernel must move them. It must also bind the electron and the hole into excitons, which takes an attraction of long range. Both are hard to get from a function of the density at one point.

The most widely used kernel is the *adiabatic local-density approximation* (ALDA). It takes the exchange–correlation potential of Eq. (3.63) at the instantaneous local density, so that the kernel is local in space and instantaneous in time [34],

$$f_{\mathrm{xc}}^{\mathrm{ALDA}}(\boldsymbol{r},\boldsymbol{r}',t-t') = \delta(\boldsymbol{r}-\boldsymbol{r}')\,\delta(t-t')\left.\frac{\mathrm{d}^2\left[n\,\varepsilon_{\mathrm{xc}}(n)\right]}{\mathrm{d}n^2}\right|_{n_{\mathrm{gs}}(\boldsymbol{r})}. \tag{4.71}$$

It is negative, since exchange and correlation make the potential more attractive where the density grows. In reciprocal space it is a function of $\boldsymbol{G}-\boldsymbol{G}'$ alone and does not depend on $\boldsymbol{q}$ or ω.

These properties fix what the kernel misses. Being instantaneous, it has no frequency dependence and cannot open the gap: the absorption edge stays at the Kohn–Sham gap, and a scissor or quasiparticle energies are still needed. Being local, its matrix is finite as $\boldsymbol{q}\to 0$. In the optical limit the head element enters $\tilde{R}_{\boldsymbol{0}\boldsymbol{0}}$ multiplied by $R^0_{\boldsymbol{0}\boldsymbol{0}} = O(q^2)$ and drops out, and the wings drop out with one power of q. The kernel acts only through the body of the matrix, at $\boldsymbol{G},\boldsymbol{G}' \neq 0$, alongside $\bar{v}$, which it partly cancels. An exciton, however, needs the

attraction between an electron and a hole at large separation, which in an insulator behaves as $-\alpha_{\mathrm{xc}}/|\boldsymbol{q}+\boldsymbol{G}|^2$ in reciprocal space [35]; a local kernel has no such term. In Fig. 4.11 the adiabatic LDA spectrum lies between the two random-phase ones and close to the one with local fields, with the upper peak at a height of 37, and its static constant lies between theirs (Worked example 4.11). Its lower peak is as weak as theirs. This is the general finding for semiconductors: for absorption the adiabatic LDA is little better than the random-phase approximation with local fields, and for energy-loss spectra it does well because the random-phase approximation already does [18, 36]. Kernels with a long-range part, and their cost, are the subject of Appendix E.6.

4.6.4 Local-field factor of the electron gas

In the homogeneous gas every matrix is diagonal, and Eq. (4.70) gives $\tilde{R} = R^0/(1 - f_{\mathrm{xc}}R^0)$ with the Lindhard function R^0 of Eq. (4.56). The many-body literature on the gas writes the same proper response with a static *local-field factor* $G_{\mathrm{LF}}(q)$. It is defined by

$$\varepsilon(q,\omega) = 1 - \frac{v_{\mathrm{C}}(q)\,R^0(q,\omega)}{1 + v_{\mathrm{C}}(q)\,G_{\mathrm{LF}}(q)\,R^0(q,\omega)}.$$

Its meaning is that an electron feels the Hartree potential of the induced density reduced by the factor $1 - G_{\mathrm{LF}}$, because the exchange–correlation hole keeps the other electrons away from it. The name is the gas analog of the local fields of a crystal: in both cases the field an electron actually feels differs from the average one. Comparing with $\varepsilon = 1 - v_{\mathrm{C}}R^0/(1 - f_{\mathrm{xc}}R^0)$,

$$f_{\mathrm{xc}}(q) = -v_{\mathrm{C}}(q)\,G_{\mathrm{LF}}(q) = -\frac{e^2}{q^2}\,G_{\mathrm{LF}}(q). \tag{4.72}$$

A static local-field factor is thus a static kernel, and every model of G_{LF} in the literature on the gas [2, Chap. 5] is a kernel of time-dependent density-functional theory for it.

The long-wavelength limit is fixed by thermodynamics. A static potential of long wavelength shifts the local chemical potential, as in the Thomas–Fermi argument of Sect. 4.4, and the proper response is $\tilde{R}(q \to 0, 0) = -\partial n_e/\partial\mu_{\mathrm{ch}}$, with μ_{ch} the chemical potential without the Hartree term. With $R^0(q \to 0, 0) = -2\mathcal{N}_{\mathrm{F}}$, the inverse of $\tilde{R} = R^0/(1 - f_{\mathrm{xc}}R^0)$ is

$$\frac{\partial\mu_{\mathrm{ch}}}{\partial n_e} = \frac{1}{2\mathcal{N}_{\mathrm{F}}} + f_{\mathrm{xc}}(q \to 0), \tag{4.73}$$

which is the compressibility sum rule: the kernel at long wavelength is the second derivative of the exchange–correlation energy density, $\mathrm{d}^2(n\varepsilon_{\mathrm{xc}})/\mathrm{d}n^2$. The adiabatic LDA satisfies it by construction. For exchange alone, $v_{\mathrm{x}} = \frac{4}{3}\varepsilon_{\mathrm{x}} \propto n^{1/3}$ with $\varepsilon_{\mathrm{x}} = -3e^2k_{\mathrm{F}}/16\pi^2$ (Sect. 3.4), so $f_{\mathrm{x}} = \mathrm{d}v_{\mathrm{x}}/\mathrm{d}n = v_{\mathrm{x}}/3n$, and with $n = k_{\mathrm{F}}^3/3\pi^2$,

$$f_{\mathrm{x}} = -\frac{e^2}{4k_{\mathrm{F}}^2}, \qquad G_{\mathrm{LF}}(q) = \frac{q^2}{4k_{\mathrm{F}}^2} \quad (q \ll k_{\mathrm{F}}).$$

Correlation adds to this. At $r_s = 2$, the Perdew–Zunger correlation energy [31] raises the coefficient from $\frac{1}{4}$ to 0.267, and Eq. (4.73) with $k_{\mathrm{TF}}^2/k_{\mathrm{F}}^2 = 1.33$ (Sect. 4.5) says that the gas is 1.55 times more compressible than free electrons of the same density. An electron surrounded by its exchange–correlation hole repels the others less, the gas yields more easily to a slow potential, and it screens more strongly than the random-phase approximation says.

At short wavelengths the adiabatic LDA fails, and this is where dark matter scatters. Its $G_{\mathrm{LF}} = 0.267\,q^2/k_{\mathrm{F}}^2$ grows without bound, while the factor of the real gas follows the q^2 law only for $q \ll k_{\mathrm{F}}$; beyond $q \sim k_{\mathrm{F}}$ it stops growing and stays of order one [2, Chap. 5]. At the

typical momentum transfer $q_0 = 3.73\,\mathrm{keV}$, which is $1.04\,k_\mathrm{F}$ at the silicon valence density, the local kernel is at the edge of its range of validity. There its factor is 0.29, and it raises the static Lindhard value $\varepsilon(q_0, 0) = 2.10$ of Sect. 4.5 to 2.62. The correction to the screening at q_0 is thus of order a quarter. Its sign is reliable, its size at $q \sim k_\mathrm{F}$ only to the extent that the local kernel is.

4.6.5 Many-body spread as an error measure

Independent particles, the random-phase approximation with the head alone, the same with local fields, and an exchange–correlation kernel on top form a hierarchy, which the Bethe–Salpeter equation of Sect. 4.9 completes. The hierarchy is not an expansion in a small number: each level adds a physical effect whose size depends on the material, the energy and the momentum, so no level bounds the error of the one below. What the hierarchy does give is a measure of how much a quantity depends on the many-body treatment. In silicon the static constant changes by about a tenth across the approximations of Fig. 4.11 (Worked example 4.11), and the screening factor $1/\varepsilon_\infty^2$ at small q by about a quarter. Near the gap, where excitons move weight, the spread is larger, and only the Bethe–Salpeter equation captures the shape of the spectrum there. Well above the gap, at the energies that hold most of the f-sum, the approximations agree to within the accuracy of the bands. This many-body spread, computed with the same bands, $\boldsymbol{k}$ grid and cutoffs in every approximation (Appendix E.6), is the many-body part of the theory uncertainty of a loss function. For heavier dark matter, whose rate is dominated by energies of several eV, it is small. For light dark matter, which deposits at most $\frac{1}{2} m_\chi v_\mathrm{max}^2$ (Eq. (1.24)) and so samples the first eV above the gap, it is the leading part.

Worked example 4.11: The local-field reduction of the dielectric constant of silicon

The calculation of Fig. 4.11 yields the static limit of each approximation at $\boldsymbol{q} \to 0$ from the same states, from Eq. (4.66) for independent particles and from Eqs. (4.68) and (4.70) at $\omega = 0$ for the others. Independent particles, and equally the random-phase approximation without local fields, give $\varepsilon_\infty = 11.45$, within 2 percent of the measured 11.7. The small difference is a property of the model, whose form factors were fitted to measured transition energies. With the local fields of the 59 shortest reciprocal vectors, $\varepsilon_\infty = 10.22$, unchanged to three digits with 113 vectors; the local fields lower the dielectric constant by 11 percent. The adiabatic LDA kernel raises it again to 10.81, 6 percent above the random-phase value with local fields, so that it cancels about half of the local-field reduction. All three are converged in the $\boldsymbol{k}$ grid to better than 0.1 percent.

Two comparisons place these numbers. The PBE value 13.04 of Sect. 3.4 is a static linear-response calculation with Kohn–Sham bands, local fields and the adiabatic PBE kernel: it is the third of our approximations, with Kohn–Sham energies in place of fitted ones. It lies above the measured value by 11 percent, and the reason is Eq. (4.66). The PBE gap is about half the measured one, and the static sum weights small transition energies by their inverse cube. The same representative PBE setting as Appendix E.1, at the measured lattice constant on a 20^3 grid, gives the other two approximations as well: 13.61 for independent particles and 12.20 with local fields, against 13.04 with the PBE kernel. The first-principles values follow the trend of the model values 11.45, 10.22 and 10.81: the local fields lower ε_∞ by 10 percent, and the kernel restores part of it. Independent particles with fitted energies happen to land close to experiment, but the adiabatic kernel with fitted energies does not: at 10.81 it is 8 percent below 11.7. What is

missing is the electron–hole attraction, which lowers the energies of the transitions near the gap and so raises ε_∞ (Sect. 4.9).
For a dark-matter rate at momentum transfers small enough that $L \simeq \varepsilon_2/\varepsilon_\infty^2$, the local-field reduction of ε_∞ alone changes the screening factor by $(11.45/10.22)^2 \simeq 1.25$. The kernel takes back part of it, and the exciton the rest and more. Across the three approximations, the factor varies by about a quarter, which is the size of the many-body uncertainty at the bottom of the spectrum, before any calculation of the excitonic weight near the gap.

The random-phase approximation with local fields is the standard for computed dark-matter loss functions, and the kernels of time-dependent density-functional theory change it little where the rate is large. The attraction that binds an electron to its hole, long-ranged, frequency dependent in principle and nonlocal in the pair of coordinates, is missing from both; Sect. 4.9 builds it from the screened interaction of Sect. 4.8 as the excitonic correction that completes the hierarchy.

4.7 Green's functions

The Bethe–Salpeter equation, which completes the hierarchy of Sect. 4.6, needs quasiparticle energies and a screened interaction, and both are properties of one added electron. Put one extra electron into the conduction band of silicon. The other electrons move away from it and screen its charge, the lattice relaxes around it, and if it sits high enough in the band it can lose energy by creating an electron–hole pair or by emitting a phonon. What propagates is an electron dressed by its surroundings, with an energy that differs from its Kohn–Sham band energy (Sect. 3.4) and, in general, with a finite lifetime. The density response R_{ee} of Sect. 4.1 does not follow this object, since the density operator moves an electron from one state to another and never changes their number. The function that does is the single-particle Green's function G: the amplitude that an electron added to the target at one time is still found in the same state at a later time. Its poles are the energies of the dressed electrons, the quasiparticles of the target, and the distance of each pole below the real axis is a decay rate. We will define G and its spectral function, write both in the exact eigenstates of the target, and see how an interaction shifts and broadens the poles through the self-energy. We will then move to imaginary time, where finite temperature costs nothing extra, and show that the density response of Sect. 4.1 is obtained there too. The treatment follows Refs. [3], [37, Chaps. 3 and 7] and [38, Chaps. 2 and 3].

4.7.1 The one-electron Green's function

The Green's function changes the number of electrons, so we work in the grand-canonical state Eq. (2.63), with $\hat{N}_e$ the electron number operator of Sect. 2.3. Every Heisenberg operator in this section evolves with $H_\mu = H - \mu_{\mathrm{ch}}\hat{N}_e$, so that $c(t) = \mathrm{e}^{\mathrm{i}H_\mu t} c\, \mathrm{e}^{-\mathrm{i}H_\mu t}$, and E_i denotes an eigenvalue of H_μ. Energies are then measured from the chemical potential: adding an electron to the N-electron state $|i\rangle$ to reach the $(N+1)$-electron state $|f\rangle$ costs $E_f - E_i$ in eigenvalues of H_μ, which is the change of the total energy minus μ_{ch}. Let $c_{\boldsymbol{k}}^\dagger$ create an electron in a single-particle state labeled $\boldsymbol{k}$, with the spin and, in a crystal, the band index included in the label. The retarded Green's function is

$$\mathsf{G}^{\mathrm{R}}(\boldsymbol{k}, t) = -\mathrm{i}\,\Theta(t)\,\big\langle \{c_{\boldsymbol{k}}(t),\, c_{\boldsymbol{k}}^\dagger(0)\} \big\rangle, \tag{4.74}$$

with the anticommutator $\{A, B\} = AB + BA$ and the average in the grand-canonical state. It has the form of the response function Eq. (4.3), with an anticommutator in place of the

commutator, and for good reason. The term $\langle c_{\boldsymbol{k}}(t) c_{\boldsymbol{k}}^\dagger \rangle$ is the overlap of the state with an electron added in $\boldsymbol{k}$ at time 0 with the same state at time t: the amplitude that the added electron is still there. The term $\langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}(t) \rangle$ is the same amplitude for a hole, an electron removed from $\boldsymbol{k}$. The anticommutator makes $\mathsf{G}^{\mathrm{R}}(\boldsymbol{k}, 0^+) = -\mathrm{i}$ whatever the state, since $\{c_{\boldsymbol{k}}, c_{\boldsymbol{k}}^\dagger\} = 1$ (Eq. (2.32)).

For independent electrons the amplitude is a pure phase. With $H_\mu = \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}}\, c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}$, where $\xi_{\boldsymbol{k}}$ is the energy of the orbital $\boldsymbol{k}$ of Sect. 2.3 measured from μ_{ch}, the Heisenberg operator is $c_{\boldsymbol{k}}(t) = c_{\boldsymbol{k}}\, \mathrm{e}^{-\mathrm{i}\xi_{\boldsymbol{k}} t}$ (Sect. 2.3). The anticommutator in Eq. (4.74) is then the number $\mathrm{e}^{-\mathrm{i}\xi_{\boldsymbol{k}} t}$, and with the transform of Appendix A.3 and $\omega \to \omega + \mathrm{i}0^+$, as for every retarded function (Sect. 4.1),

$$\mathsf{G}_0^{\mathrm{R}}(\boldsymbol{k}, t) = -\mathrm{i}\,\Theta(t)\, \mathrm{e}^{-\mathrm{i}\xi_{\boldsymbol{k}} t}, \qquad \mathsf{G}_0^{\mathrm{R}}(\boldsymbol{k}, \omega) = \frac{1}{\omega - \xi_{\boldsymbol{k}} + \mathrm{i}0^+}. \tag{4.75}$$

The subscript 0 marks the function without interactions. It has one pole, at the energy of the orbital, and it is independent of temperature and of which orbitals are occupied.

In an interacting target the same steps as in Sect. 4.1 lead to the eigenstates. Inserting a complete set of states between the operators, $\langle c_{\boldsymbol{k}}(t) c_{\boldsymbol{k}}^\dagger \rangle = \sum_{i,f} p_i\, |\langle f | c_{\boldsymbol{k}}^\dagger | i \rangle|^2\, \mathrm{e}^{-\mathrm{i}(E_f - E_i)t}$ and $\langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}(t) \rangle = \sum_{i,f} p_i\, |\langle f | c_{\boldsymbol{k}} | i \rangle|^2\, \mathrm{e}^{+\mathrm{i}(E_f - E_i)t}$, and the time integrals are those of Eq. (4.6). Since the two terms are added, not subtracted, both enter with a plus sign:

Lehmann representation of the Green's function

$$\begin{aligned} \mathsf{G}^{\mathrm{R}}(\boldsymbol{k}, \omega) &= \sum_{i,f} p_i \left[\frac{|\langle f | c_{\boldsymbol{k}}^\dagger | i \rangle|^2}{\omega - (E_f - E_i) + \mathrm{i}0^+} + \frac{|\langle f | c_{\boldsymbol{k}} | i \rangle|^2}{\omega + (E_f - E_i) + \mathrm{i}0^+} \right] \\ &= \int \mathrm{d}\omega'\, \frac{\mathsf{A}(\boldsymbol{k}, \omega')}{\omega - \omega' + \mathrm{i}0^+}, \end{aligned} \tag{4.76}$$

with the spectral function

$$\begin{aligned} \mathsf{A}(\boldsymbol{k}, \omega) &= -\frac{1}{\pi}\, \mathrm{Im}\, \mathsf{G}^{\mathrm{R}}(\boldsymbol{k}, \omega) \\ &= \sum_{i,f} p_i \Big[|\langle f | c_{\boldsymbol{k}}^\dagger | i \rangle|^2\, \delta\big(\omega - (E_f - E_i)\big) + |\langle f | c_{\boldsymbol{k}} | i \rangle|^2\, \delta\big(\omega + (E_f - E_i)\big) \Big]. \end{aligned} \tag{4.77}$$

The poles are at the energies needed to add an electron in $\boldsymbol{k}$, and at minus the energies released by removing one, each weighted by the probability that the added electron or hole is found in an exact eigenstate.

This is the representation of Lehmann [39], and it shows that A is positive, normalized, and tied to the occupations. It is positive since every weight in Eq. (4.77) is. Its integral over ω is $\sum_i p_i \langle i | c_{\boldsymbol{k}} c_{\boldsymbol{k}}^\dagger + c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}} | i \rangle = 1$, which makes $\mathsf{A}(\boldsymbol{k}, \omega)$ a probability density in energy: the probability that an electron or hole put into $\boldsymbol{k}$ has the energy ω. And the removal part alone counts the electrons already in $\boldsymbol{k}$. To see this, relabel $i \leftrightarrow f$ in the removal term; it becomes $\sum_{i,f} p_f\, |\langle f | c_{\boldsymbol{k}}^\dagger | i \rangle|^2\, \delta(\omega - (E_f - E_i))$, and on the energy shell $p_f = p_i\, \mathrm{e}^{-\beta\omega}$, the detailed-balance step of Eq. (2.68). The removal part is thus $\mathrm{e}^{-\beta\omega}$ times the addition part, the two add up to A, and the removal part is $\mathsf{A}\, \mathrm{e}^{-\beta\omega}/(1 + \mathrm{e}^{-\beta\omega}) = \mathsf{A}\, n_{\mathrm{F}}(\omega)$. Its integral is $\langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}} \rangle$:

$$\langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}} \rangle = \int \mathrm{d}\omega\, \mathsf{A}(\boldsymbol{k}, \omega)\, n_{\mathrm{F}}(\omega). \tag{4.78}$$

For independent electrons $\mathsf{A}_0(\boldsymbol{k}, \omega) = \delta(\omega - \xi_{\boldsymbol{k}})$ and Eq. (4.78) is the Fermi function of Eq. (2.69). With interactions the occupation of a plane-wave state is no longer $n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$, even

at $T = 0$, because A has weight at energies other than $\xi_{\boldsymbol{k}}$. Note that we normalize A to one. Many texts, Refs. [3, 38] among them, write $-2\,\mathrm{Im}\,\mathsf{G}^{\mathrm{R}}$ for the spectral function, which integrates to 2π.

The spectral function is close to a measured quantity. In photoemission a photon removes an electron from the crystal, and the rate at a given energy and momentum of the outgoing electron is, in the simplest approximation, proportional to the removal part $\mathsf{A}\,n_{\mathrm{F}}$; inverse photoemission measures the addition part. The analytic structure is that of Sect. 4.1. By Eq. (4.76), $\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},z) = \int \mathrm{d}\omega'\,\mathsf{A}(\boldsymbol{k},\omega')/(z-\omega')$ is analytic in the upper half plane, and the advanced function $\mathsf{G}^{\mathrm{A}}(\boldsymbol{k},\omega) = \mathsf{G}^{\mathrm{R}}(\boldsymbol{k},\omega)^*$, nonzero only for $t<0$, is its boundary value from below.

4.7.2 The time-ordered function

At $T = 0$ the perturbation theory of Sect. 4.8 produces a third function. The Dyson series Eq. (2.50) is a time-ordered exponential, and when it is used to expand a ground-state amplitude the operators come out in time order. The time-ordered Green's function is

$$\mathsf{G}(\boldsymbol{k},t) = -\mathrm{i}\,\langle 0|\mathrm{T}\,c_{\boldsymbol{k}}(t)\,c_{\boldsymbol{k}}^\dagger(0)|0\rangle = -\mathrm{i}\,\Theta(t)\,\langle 0|c_{\boldsymbol{k}}(t)\,c_{\boldsymbol{k}}^\dagger|0\rangle + \mathrm{i}\,\Theta(-t)\,\langle 0|c_{\boldsymbol{k}}^\dagger\,c_{\boldsymbol{k}}(t)|0\rangle, \tag{4.79}$$

where $|0\rangle$ is the ground state of H_μ and the time-ordering symbol T puts the operator with the later time on the left, with a minus sign when two fermion operators are exchanged. For $t>0$ it propagates an added electron forward in time, and for $t<0$ a hole. The integral over $t>0$ is that of the retarded function. The integral over $t<0$ needs a factor e^{+0^+t} to converge, $\mathrm{i}\int_{-\infty}^0 \mathrm{d}t\,\mathrm{e}^{\mathrm{i}(\omega+E_f-E_0)t+0^+t} = 1/(\omega+E_f-E_0-\mathrm{i}0^+)$, so

$$\mathsf{G}(\boldsymbol{k},\omega) = \sum_f \left[\frac{|\langle f|c_{\boldsymbol{k}}^\dagger|0\rangle|^2}{\omega-(E_f-E_0)+\mathrm{i}0^+} + \frac{|\langle f|c_{\boldsymbol{k}}|0\rangle|^2}{\omega+(E_f-E_0)-\mathrm{i}0^+}\right]. \tag{4.80}$$

The removal poles now lie above the real axis. At $T=0$ every $E_f-E_0\geq 0$, because $|0\rangle$ is the ground state of H_μ in all particle-number sectors. The addition poles therefore sit at $\omega>0$ and the removal poles at $\omega<0$. As a result, the time-ordered function equals G^{R} for $\omega>0$ and G^{A} for $\omega<0$, and both carry the same spectral function. In this section a G without superscript and with a real time or frequency is the time-ordered function. With an imaginary argument it is the Matsubara function introduced below, and with a complex argument z it is the function $\int \mathrm{d}\omega'\,\mathsf{A}(\boldsymbol{k},\omega')/(z-\omega')$, which unites them all.

4.7.3 Self-energy and the Dyson equation

How does an interaction change G^{R}? Split $H_\mu = \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}}\,c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}} + H_{\mathrm{int}}$, where H_{int} holds the Coulomb interaction between the electrons, their coupling to phonons, or any other term that is not a one-body energy. The Heisenberg operator obeys $\mathrm{i}\,\partial_t c_{\boldsymbol{k}}(t) = [c_{\boldsymbol{k}}, H_\mu](t) = \xi_{\boldsymbol{k}} c_{\boldsymbol{k}}(t) + [c_{\boldsymbol{k}}, H_{\mathrm{int}}](t)$, by Eq. (2.33). Differentiating Eq. (4.74), the derivative of the step function gives $\delta(t)\langle\{c_{\boldsymbol{k}}, c_{\boldsymbol{k}}^\dagger\}\rangle = \delta(t)$, and

$$\mathrm{i}\,\partial_t \mathsf{G}^{\mathrm{R}}(\boldsymbol{k},t) = \delta(t) + \xi_{\boldsymbol{k}}\,\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},t) + \mathsf{F}^{\mathrm{R}}(\boldsymbol{k},t), \qquad \mathsf{F}^{\mathrm{R}}(\boldsymbol{k},t) = -\mathrm{i}\,\Theta(t)\,\langle\{[c_{\boldsymbol{k}}, H_{\mathrm{int}}](t),\, c_{\boldsymbol{k}}^\dagger\}\rangle.$$

The function F^{R} is a retarded correlation of the same kind, built from the operator $[c_{\boldsymbol{k}}, H_{\mathrm{int}}]$, which removes an electron from $\boldsymbol{k}$ and scatters the others. With $\mathrm{e}^{\mathrm{i}\omega t}$, the derivative $\mathrm{i}\,\partial_t$ becomes ω, and $(\omega-\xi_{\boldsymbol{k}})\,\mathsf{G}^{\mathrm{R}} = 1 + \mathsf{F}^{\mathrm{R}}$. We define the self-energy $\Sigma^{\mathrm{R}}(\boldsymbol{k},\omega)$ as the ratio $\mathsf{F}^{\mathrm{R}}/\mathsf{G}^{\mathrm{R}}$. This is possible because $\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},z)$ has no zeros in the upper half plane: there $\mathrm{Im}\,\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},z) = -\,\mathrm{Im}\,z\int \mathrm{d}\omega'\,\mathsf{A}(\boldsymbol{k},\omega')/|z-\omega'|^2 < 0$ by the positivity of A. Multiplying by $\mathsf{G}_0^{\mathrm{R}}$ of Eq. (4.75) gives the Dyson equation.

Dyson equation

$$\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},\omega) = \mathsf{G}^{\mathrm{R}}_0(\boldsymbol{k},\omega) + \mathsf{G}^{\mathrm{R}}_0(\boldsymbol{k},\omega)\,\Sigma^{\mathrm{R}}(\boldsymbol{k},\omega)\,\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},\omega),$$
$$\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},\omega) = \frac{1}{\omega - \xi_{\boldsymbol{k}} - \Sigma^{\mathrm{R}}(\boldsymbol{k},\omega)}. \tag{4.81}$$

All the effects of the interaction on the propagation of one electron are in the self-energy Σ^{R}, a complex function of energy.

In a crystal, with $\boldsymbol{k}$ the crystal momentum and n the band, the same steps make G^{R} and Σ^{R} matrices in the band indices, and Eq. (4.81) a matrix equation of the type Eq. (C.59). In a translation-invariant target they are diagonal in $\boldsymbol{k}$, as we assume here.

So far Eq. (4.81) is a definition. Its content appears when it is iterated: $\mathsf{G}^{\mathrm{R}} = \mathsf{G}^{\mathrm{R}}_0 + \mathsf{G}^{\mathrm{R}}_0\Sigma^{\mathrm{R}}\mathsf{G}^{\mathrm{R}}_0 + \mathsf{G}^{\mathrm{R}}_0\Sigma^{\mathrm{R}}\mathsf{G}^{\mathrm{R}}_0\Sigma^{\mathrm{R}}\mathsf{G}^{\mathrm{R}}_0 + \cdots$. The electron propagates freely, is scattered out of $\boldsymbol{k}$ and back by the interaction, propagates freely again, and so on, and Σ^{R} is the amplitude of one such excursion. Sect. 4.8 computes Σ with diagrams and shows that it is the sum of the excursions that cannot be cut into two by removing a single free propagator. A self-energy known only to low order then yields a Green's function to all orders, and no finite order of the series has the shifted, broadened pole that the closed form has (Appendix C.7). This is what the width did for the intermediate state of Sect. 2.4, where summing the repeated emission and reabsorption turned $\mathrm{i}0^+$ into $\mathrm{i}\Gamma_{\mathrm{w}}/2$ in Eq. (2.57).

The sign of the width is fixed before any calculation. On the real axis, $\Sigma^{\mathrm{R}} = \omega - \xi_{\boldsymbol{k}} - 1/\mathsf{G}^{\mathrm{R}}$ has the imaginary part $\operatorname{Im}\mathsf{G}^{\mathrm{R}}/|\mathsf{G}^{\mathrm{R}}|^2$, so

$$\operatorname{Im}\Sigma^{\mathrm{R}}(\boldsymbol{k},\omega) = -\frac{\pi\,\mathsf{A}(\boldsymbol{k},\omega)}{\left|\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},\omega)\right|^2} \le 0. \tag{4.82}$$

The large-frequency limit is fixed as well. At large $|\omega|$, the expansion $1/(\omega-\omega') = \omega^{-1}+\omega'\omega^{-2}+\cdots$ in Eq. (4.76) gives $\mathsf{G}^{\mathrm{R}} = \omega^{-1} + m_1\omega^{-2} + O(\omega^{-3})$, with $m_1 = \int \mathrm{d}\omega\,\omega\,\mathsf{A} = \langle\{[c_{\boldsymbol{k}}, H_\mu], c^\dagger_{\boldsymbol{k}}\}\rangle$ by the same steps as Eq. (4.11). Then $1/\mathsf{G}^{\mathrm{R}} = \omega - m_1 + O(\omega^{-1})$, and $\Sigma^{\mathrm{R}}(\boldsymbol{k},\omega) \to m_1 - \xi_{\boldsymbol{k}} = \langle\{[c_{\boldsymbol{k}}, H_{\mathrm{int}}], c^\dagger_{\boldsymbol{k}}\}\rangle$. The self-energy tends to a static potential, linear in H_{int} and evaluated in the exact state. For the Coulomb interaction it is the mean-field (Hartree and exchange) potential of Sect. 3.4, built with the exact occupations. Everything that depends on frequency, the retardation of the screening and the decay into other excitations, sits in $\Sigma^{\mathrm{R}} - \Sigma^{\mathrm{R}}(\boldsymbol{k},\infty)$, which is analytic in the upper half plane and falls off at large $|\omega|$, and as a result obeys the Kramers–Kronig relations of Appendix C.5.

4.7.4 Quasiparticles

Suppose the self-energy is small and varies slowly with ω. The Green's function Eq. (4.81) is then large where its real denominator vanishes, at the quasiparticle energy $E_{\mathrm{qp}}(\boldsymbol{k})$ defined by

$$E_{\mathrm{qp}}(\boldsymbol{k}) - \xi_{\boldsymbol{k}} - \operatorname{Re}\Sigma^{\mathrm{R}}(\boldsymbol{k}, E_{\mathrm{qp}}(\boldsymbol{k})) = 0.$$

Near this energy we expand the real part to first order, $\omega - \xi_{\boldsymbol{k}} - \operatorname{Re}\Sigma^{\mathrm{R}} \simeq (\omega - E_{\mathrm{qp}})/z_{\mathrm{qp}}$, and keep the imaginary part at its value at E_{qp}.

Quasiparticle

$$\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},\omega) \simeq \frac{z_{\mathrm{qp}}(\boldsymbol{k})}{\omega - E_{\mathrm{qp}}(\boldsymbol{k}) + \mathrm{i}\Gamma_{\mathrm{qp}}(\boldsymbol{k})/2},$$
$$z_{\mathrm{qp}} = \Big[1 - \partial_\omega \operatorname{Re}\Sigma^{\mathrm{R}}(\boldsymbol{k},\omega)\big|_{E_{\mathrm{qp}}}\Big]^{-1}, \qquad \Gamma_{\mathrm{qp}} = -2 z_{\mathrm{qp}} \operatorname{Im}\Sigma^{\mathrm{R}}(\boldsymbol{k}, E_{\mathrm{qp}}). \tag{4.83}$$

The real part of the self-energy shifts the energy, its slope sets the residue z_{qp}, and its imaginary part sets the width Γ_{qp}.

In time, the pole at $E_{\mathrm{qp}} - \mathrm{i}\Gamma_{\mathrm{qp}}/2$, in the lower half plane of the continued function, gives $\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},t) \simeq -\mathrm{i} z_{\mathrm{qp}}\, \mathrm{e}^{-\mathrm{i}E_{\mathrm{qp}}t - \Gamma_{\mathrm{qp}}t/2}$: the added electron keeps the fraction z_{qp} of its amplitude in a state that oscillates at a definite energy and decays at the rate Γ_{qp}. It is a resonance pole of the kind met in Sect. 4.1, and $1/\Gamma_{\mathrm{qp}}$ is the quasiparticle lifetime. In the spectral function the pole is a Lorentzian of weight z_{qp} and full width Γ_{qp}. In an interacting system z_{qp} is typically below one, and the rest of the sum rule, $1 - z_{\mathrm{qp}}$, is spread over a broad incoherent background: states in which the added electron has created electron–hole pairs, plasmons or phonons. Since the background is broad, its contribution to $\mathsf{G}^{\mathrm{R}}(\boldsymbol{k},t)$ dephases within a time of order the inverse of its spread, and at later times the quasiparticle term is all that is left. The expansion requires Γ_{qp} to be small compared with the energy over which Σ^{R} changes, the condition of the Wigner–Weisskopf result of Sect. 2.4. The quasiparticle is a useful concept only if it also lives long compared with its own period, $\Gamma_{\mathrm{qp}} \ll |E_{\mathrm{qp}}|$.

Whether that holds is a matter of phase space, and an estimate shows why quasiparticles are sharp near the Fermi level of a metal. An electron added in a state of energy $\xi_{\boldsymbol{k}} > 0$ above μ_{ch} can decay by scattering off an electron of the Fermi sea, which leaves a hole below μ_{ch} and two electrons above it. At $T = 0$ the Pauli principle forces all three energies into a window of width $\xi_{\boldsymbol{k}}$: the energy ω' lost by the added electron lies between 0 and $\xi_{\boldsymbol{k}}$, and the partner must start between $-\omega'$ and 0. The allowed region in these two energy variables is a triangle of area $\xi_{\boldsymbol{k}}^2/2$, so $\Gamma_{\mathrm{qp}} \propto \xi_{\boldsymbol{k}}^2$ and $\Gamma_{\mathrm{qp}}/\xi_{\boldsymbol{k}} \to 0$ as $\xi_{\boldsymbol{k}} \to 0$. Luttinger showed that $\operatorname{Im}\Sigma^{\mathrm{R}}$ vanishes as $(\omega - \mu_{\mathrm{ch}})^2$ at every order of perturbation theory [40]. This is the basis of Landau's picture of a metal as a gas of weakly interacting quasiparticles. In a semiconductor the window is set by the gap instead. An electron at the bottom of the conduction band of silicon is in the lowest state with one extra electron, it has nothing to decay into, and its width is zero. Higher in the band, the electron–electron part of $\operatorname{Im}\Sigma^{\mathrm{R}}$ stays zero until the electron has enough energy to create an electron–hole pair, at least E_{g} above the band edge. Below that energy the width comes from phonon emission alone, which Sect. 6.3 computes from the same Eq. (4.83).

The quasiparticle energies of silicon are material quantities, and the dark-matter rate needs them: the band edges set the threshold for electron excitation. They are computed by starting from the Kohn–Sham Hamiltonian of Sect. 3.4 in place of the free one. Its eigenvalues $\varepsilon_{n\boldsymbol{k}}$ already contain the exchange–correlation potential v_{xc}, so $\Sigma - v_{\mathrm{xc}}$ takes the place of Σ in the Dyson equation. With the approximation that the Kohn–Sham states diagonalize Σ, the quasiparticle condition expanded to first order about $\varepsilon_{n\boldsymbol{k}}$ gives

$$E_{\mathrm{qp}}(n\boldsymbol{k}) \simeq \varepsilon_{n\boldsymbol{k}} + z_{\mathrm{qp}}\, \operatorname{Re}\langle\psi_{n\boldsymbol{k}}|\Sigma(\varepsilon_{n\boldsymbol{k}}) - v_{\mathrm{xc}}|\psi_{n\boldsymbol{k}}\rangle, \tag{4.84}$$

the form in which first-principles quasiparticle corrections are evaluated [41]. A PBE calculation of silicon puts the indirect gap at 0.602 eV, against the measured 1.17 eV at low temperature [42, Eq. (11)] (1.12 eV at 300 K; Appendix E.1); the self-energy in the *GW* approximation of Sect. 4.8 supplies most of the difference. Appendix E.3 describes the calculation and its convergence.

Worked example 4.12: A level coupled to a flat band

The simplest self-energy that can be computed exactly is that of a localized orbital d, of energy $E_{\rm lev}$, hybridized with a band of orbitals $\boldsymbol{k}$ of energies $\xi_{\boldsymbol{k}}$: an impurity level in a metal or an atomic orbital near a surface. The electrons do not interact with each other, and the model is Anderson's without his on-site repulsion [43]:

$$H_\mu = E_{\rm lev}\, d^\dagger d + \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}} + \sum_{\boldsymbol{k}} (V_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}} d + V^*_{\boldsymbol{k}}\, d^\dagger c_{\boldsymbol{k}}),$$

with energies measured from $\mu_{\rm ch}$ and d, $d^\dagger$ the fermion operators of the level. The commutators $[d, H_\mu] = E_{\rm lev} d + \sum_{\boldsymbol{k}} V^*_{\boldsymbol{k}} c_{\boldsymbol{k}}$ and $[c_{\boldsymbol{k}}, H_\mu] = \xi_{\boldsymbol{k}} c_{\boldsymbol{k}} + V_{\boldsymbol{k}} d$ follow from Eq. (2.33). Let $\mathsf{G}^{\rm R}_{dd}$ be defined by Eq. (4.74) with d for $c_{\boldsymbol{k}}$. Its equation of motion contains the mixed function $\mathsf{G}^{\rm R}_{\boldsymbol{k}d}(t) = -\mathrm{i}\Theta(t)\langle\{c_{\boldsymbol{k}}(t), d^\dagger\}\rangle$, whose own equation closes, since $\{c_{\boldsymbol{k}}, d^\dagger\} = 0$:

$$(\omega - E_{\rm lev})\, \mathsf{G}^{\rm R}_{dd} = 1 + \sum_{\boldsymbol{k}} V^*_{\boldsymbol{k}}\, \mathsf{G}^{\rm R}_{\boldsymbol{k}d}, \qquad (\omega - \xi_{\boldsymbol{k}})\, \mathsf{G}^{\rm R}_{\boldsymbol{k}d} = V_{\boldsymbol{k}}\, \mathsf{G}^{\rm R}_{dd}.$$

Eliminating $\mathsf{G}^{\rm R}_{\boldsymbol{k}d}$ gives Eq. (4.81) with

$$\Sigma^{\rm R}_d(\omega) = \sum_{\boldsymbol{k}} \frac{|V_{\boldsymbol{k}}|^2}{\omega - \xi_{\boldsymbol{k}} + \mathrm{i}0^+}, \tag{4.85}$$

exact at every order in $V_{\boldsymbol{k}}$, and the same at every temperature and filling of the band: the anticommutators are numbers, and the averages never enter. The electron leaves d for the band and returns, and Eq. (4.85) is the amplitude of one such excursion.

This is the level-shift function of the Wigner–Weisskopf problem, Eq. (D.11), at $z = \omega - E_{\rm lev} + \mathrm{i}0^+$, with $|V_{\boldsymbol{k}}|^2$ in place of $|H'_{fi}|^2$. Take the flat band of Worked example 2.1: states spread uniformly over a width $E_{\rm w} = 1\,\mathrm{eV}$ centered on $E_{\rm lev}$, all with the same $|V_{\boldsymbol{k}}|^2$, chosen so that the golden-rule rate $\Gamma = 2\pi|V_{\boldsymbol{k}}|^2\mathcal{D}_f$ is 10 meV. Appendix D.3 found $\Sigma^{\rm R}_d(E_{\rm lev}) = -\mathrm{i}\Gamma/2$ with no shift, and a slope $\partial_\omega \,\mathrm{Re}\,\Sigma^{\rm R}_d = 2\Gamma/(\pi E_{\rm w}) = 0.0064$ at $E_{\rm lev}$. By Eq. (4.83), $E_{\rm qp} = E_{\rm lev}$, $z_{\rm qp} = 1/(1 - 0.0064) = 1.0064$ and $\Gamma_{\rm qp} = z_{\rm qp}\Gamma$. The spectral function is a Lorentzian of full width Γ with peak height $2/(\pi\Gamma) = 63.7\,\mathrm{eV}^{-1}$, and the exact A_{dd} follows it to within 1.3 percent out to 10Γ from the peak (Fig. 4.12). This is the Lorentzian line of Eq. (2.56), now as the energy distribution of an electron put into d, and its width is $-2\,\mathrm{Im}\,\Sigma^{\rm R}_d$.

The residue above one is no paradox. The exact A_{dd} vanishes outside the band, where $\mathrm{Im}\,\Sigma^{\rm R}_d = 0$, while a Lorentzian of width Γ puts the weight $2\Gamma/(\pi E_{\rm w}) = 0.0064$ beyond the band edges. The pole carries that weight back to the center, and the sum rule $\int \mathsf{A}_{dd} = 1$ is satisfied by the exact function inside the band. The quasiparticle formulas also reproduce the corrections of Appendix D.3: there the decay rate was Γ divided by one minus the slope of the level-shift function, and the residue was the inverse of the same factor, which are $\Gamma_{\rm qp}$ and $z_{\rm qp}$. Indeed, in the state with the band empty, $\mathsf{G}^{\rm R}_{dd}(t) = -\mathrm{i}\Theta(t)\langle 0|d\,\mathrm{e}^{-\mathrm{i}H_\mu t} d^\dagger|0\rangle$ is $-\mathrm{i}$ times the amplitude that an electron put into d is still there, the $c_i(t)\,\mathrm{e}^{-\mathrm{i}E_{\rm lev}t}$ of Eq. (2.55), and $|\mathsf{G}^{\rm R}_{dd}(t)|^2$ is the survival probability. The Green's function contains the Wigner–Weisskopf result, with the condition $\Gamma \ll E_{\rm w}$ appearing as the slow variation of $\Sigma^{\rm R}_d$. Temperature enters only the occupation: at $T = 20\,\mathrm{meV}$ and $E_{\rm lev} = -30\,\mathrm{meV}$, Eq. (4.78) gives $\langle d^\dagger d\rangle = 0.787$, against $n_{\rm F}(E_{\rm lev}) = 0.818$ for the isolated level.

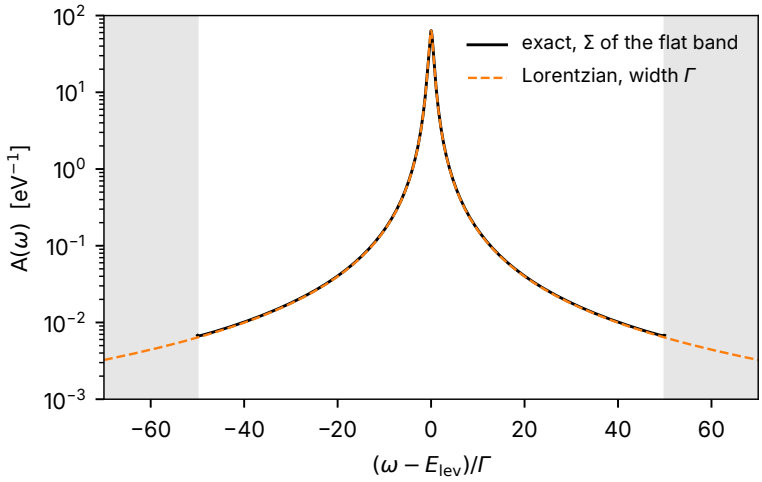


Figure 4.12: Spectral function of a level of energy E_{lev} coupled to a flat band of width $E_{\mathrm{w}} = 1\,\mathrm{eV}$ centered on E_{lev}, with golden-rule rate $\Gamma = 10\,\mathrm{meV}$ (the model of Worked example 4.12), on a logarithmic scale against $(\omega - E_{\mathrm{lev}})/\Gamma$. Solid: exact result, $\mathsf{A}_{dd} = -\pi^{-1}\,\mathrm{Im}[\omega - E_{\mathrm{lev}} - \Sigma_d^{\mathrm{R}}(\omega)]^{-1}$ with the self-energy of the band in the limit of a continuum, $\Sigma_d^{\mathrm{R}} = (\Gamma/2\pi)\ln[(\omega - E_{\mathrm{lev}} + E_{\mathrm{w}}/2)/(\omega - E_{\mathrm{lev}} - E_{\mathrm{w}}/2)]$ at $\omega + \mathrm{i}0^+$. Dashed: Lorentzian of full width Γ and unit weight. Shaded: outside the band, where the exact function vanishes (its two bound states lie exponentially close to the band edges and carry negligible weight) and the Lorentzian has the weight $2\Gamma/(\pi E_{\mathrm{w}}) = 0.0064$. Inside the band the exact function exceeds the Lorentzian by 1.0 percent at $|\omega - E_{\mathrm{lev}}| = \Gamma$ and by 1.3 percent at 10Γ, because its quasiparticle residue is 1.0064.

4.7.5 Imaginary time

At finite temperature two exponentials of the Hamiltonian appear in every average: $\mathrm{e}^{-\mathrm{i}H_\mu t}$ moves the operators in time, and $\mathrm{e}^{-\beta H_\mu}$ weights the states. The second is the first at the imaginary time $t = -\mathrm{i}\beta$. Matsubara used this to treat both on the same footing [44]. Define operators in imaginary time τ by

$$O(\tau) = \mathrm{e}^{\tau H_\mu}\, O\, \mathrm{e}^{-\tau H_\mu}, \qquad 0 \le \tau \le \beta,$$

which is the Heisenberg operator $O(t)$ at $t = -\mathrm{i}\tau$. Note that $c^\dagger(\tau)$ is not the adjoint of $c(\tau)$. The imaginary-time, or Matsubara, Green's function is

$$\mathsf{G}(\boldsymbol{k}, \tau) = -\langle \mathrm{T}_\tau\, c_{\boldsymbol{k}}(\tau)\, c^\dagger_{\boldsymbol{k}}(0)\rangle, \tag{4.86}$$

where T_τ puts the operator with the larger τ on the left, with a minus sign for an exchange of fermion operators. Its use is that a perturbation expansion of $\mathrm{e}^{-\beta H_\mu}$ and of $\mathrm{e}^{\pm\tau H_\mu}$ in H_{int} is one Dyson series along the imaginary-time axis, with the same structure as Eq. (2.50), and Wick's theorem then reduces every term to products of free Green's functions (Sect. 4.8). The real-time time-ordered function of Eq. (4.79), however, has no expansion of this single-exponential form at $T > 0$ (Appendix D.11), which is why the Matsubara function replaces it.

The cyclic property of the trace makes Eq. (4.86) antiperiodic, $\mathsf{G}(\boldsymbol{k}, \tau - \beta) = -\mathsf{G}(\boldsymbol{k}, \tau)$ for $0 < \tau < \beta$ (Appendix D.10). A function on $-\beta < \tau < \beta$ with this property is a Fourier series in the odd multiples of πT, the fermionic Matsubara frequencies $\omega_j^{\mathrm{F}} = (2j+1)\pi T$ with integer j:

$$\mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = \int_0^\beta \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}}\tau}\, \mathsf{G}(\boldsymbol{k}, \tau), \qquad \mathsf{G}(\boldsymbol{k}, \tau) = T\sum_j \mathrm{e}^{-\mathrm{i}\omega_j^{\mathrm{F}}\tau}\, \mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}). \tag{4.87}$$

A correlation of two operators that each change the electron number by an even amount, such as two densities, is periodic instead, and its series runs over the bosonic frequencies $\omega_j^{\mathrm{B}} = 2j\pi T$. As $T \to 0$ both sets become dense on the imaginary axis.

The Lehmann form follows as before. For $0 < \tau < \beta$ only the ordering $c_{\boldsymbol{k}}(\tau)c^\dagger_{\boldsymbol{k}}$ appears, and inserting states leads to $\mathsf{G}(\boldsymbol{k}, \tau) = -\sum_{i,f} p_i |\langle f|c^\dagger_{\boldsymbol{k}}|i\rangle|^2\, \mathrm{e}^{-(E_f - E_i)\tau}$. The integral in Eq. (4.87) is elementary, and $\mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}}\beta} = -1$:

$$\int_0^\beta \mathrm{d}\tau\, \mathrm{e}^{[\mathrm{i}\omega_j^{\mathrm{F}} - (E_f - E_i)]\tau} = -\frac{1 + \mathrm{e}^{-\beta(E_f - E_i)}}{\mathrm{i}\omega_j^{\mathrm{F}} - (E_f - E_i)}.$$

Since $p_i\, \mathrm{e}^{-\beta(E_f - E_i)} = p_f$, the weight becomes $(p_i + p_f)|\langle f|c^\dagger_{\boldsymbol{k}}|i\rangle|^2$, and the relabeling that led to Eq. (4.78) shows that this is the weight of the spectral function at $\omega = E_f - E_i$.

Matsubara Green's function and analytic continuation

$$\mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = \int \mathrm{d}\omega'\, \frac{\mathsf{A}(\boldsymbol{k}, \omega')}{\mathrm{i}\omega_j^{\mathrm{F}} - \omega'}, \qquad \mathsf{G}^{\mathrm{R}}(\boldsymbol{k}, \omega) = \mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}} \to \omega + \mathrm{i}0^+). \tag{4.88}$$

The Matsubara and retarded functions are values of one analytic function, $\int \mathrm{d}\omega'\, \mathsf{A}(\boldsymbol{k}, \omega')/(z - \omega')$, at the points $z = \mathrm{i}\omega_j^{\mathrm{F}}$ and just above the real axis.

In practice a Matsubara function computed in closed form is turned into the retarded one by the substitution $\mathrm{i}\omega_j^{\mathrm{F}} \to \omega + \mathrm{i}0^+$, made after every factor $\mathrm{e}^{\mathrm{i}\beta\omega_j^{\mathrm{F}}}$ has been replaced by -1.

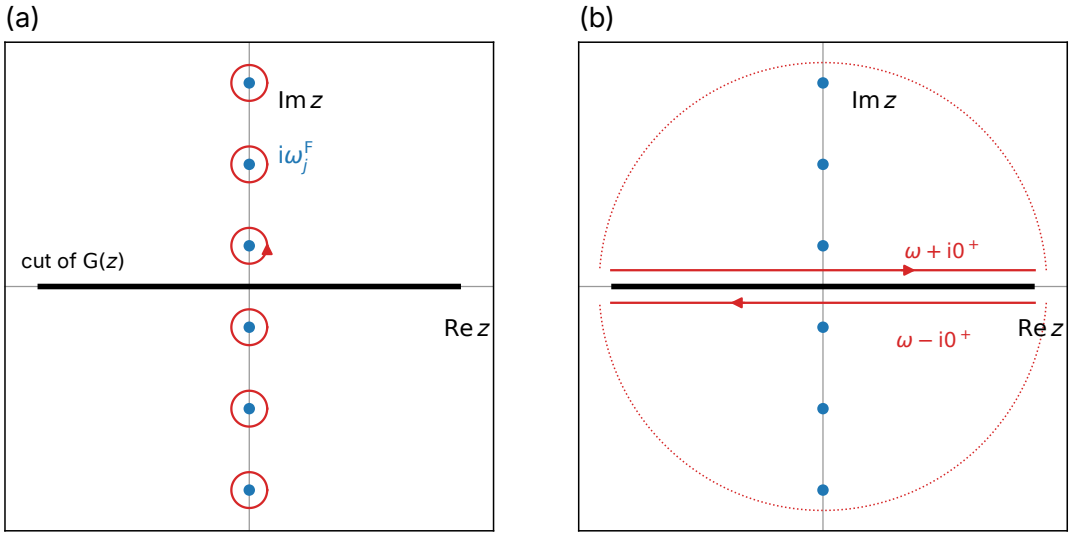


Figure 4.13: Frequency sums by contour integration in the complex frequency plane z. (a) The fermionic Matsubara frequencies $\mathrm{i}\omega_j^{\mathrm{F}} = (2j+1)\pi\mathrm{i}T$ (dots) are the poles of the Fermi function $n_{\mathrm{F}}(z)$, each with residue $-T$. A sum $T\sum_j g(\mathrm{i}\omega_j^{\mathrm{F}})$ is minus the integral of $n_{\mathrm{F}}(z)\,g(z)/2\pi\mathrm{i}$ around small circles that enclose them counterclockwise. For $g = \mathsf{G}(\boldsymbol{k}, z)$ the only other singularity is the cut on the real axis (thick line), where the spectral function is nonzero. (b) The circles are merged into the boundaries of the upper and lower half planes: a line just above the real axis, at $\omega + \mathrm{i}0^+$, run from left to right, and a line just below, at $\omega - \mathrm{i}0^+$, run from right to left. The large arcs (dotted) contribute nothing when $n_{\mathrm{F}}(z)\,g(z)$ falls off faster than $1/|z|$ in every direction. The two lines see the retarded and advanced functions, and their difference, $-2\pi\mathrm{i}\mathsf{A}$, turns the sum into an integral over real frequencies.

The continuation is unique, because the function falls off at large $|z|$ (Appendix C.5) [45]. The Dyson equation carries over with the same self-energy evaluated at $z = \mathrm{i}\omega_j^{\mathrm{F}}$, $\mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = [\mathrm{i}\omega_j^{\mathrm{F}} - \xi_{\boldsymbol{k}} - \Sigma(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}})]^{-1}$, and for free electrons $\mathsf{G}_0(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = 1/(\mathrm{i}\omega_j^{\mathrm{F}} - \xi_{\boldsymbol{k}})$.

Diagrams return their results as sums over Matsubara frequencies, and these are done by contour integration. The occupation is the simplest case. At $\tau = 0^-$ the time ordering puts $c_{\boldsymbol{k}}^\dagger$ on the left, $\mathsf{G}(\boldsymbol{k}, 0^-) = \langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}\rangle$, and by Eq. (4.87)

$$\langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}\rangle = T\sum_j \mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}} 0^+}\,\mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}). \tag{4.89}$$

The Fermi function $n_{\mathrm{F}}(z)$ has poles at the points $z = \mathrm{i}\omega_j^{\mathrm{F}}$, each with residue $-T$, and the sum is a contour integral of $n_{\mathrm{F}}(z)\,\mathsf{G}(\boldsymbol{k}, z)$ around small circles that enclose them (Fig. 4.13a). Opened up, the circles become two lines that run along the real axis, above and below the cut of $\mathsf{G}(\boldsymbol{k}, z)$ (Fig. 4.13b). The jump of G across the cut is $-2\pi\mathrm{i}\mathsf{A}$, by Eq. (4.76), and the sum becomes $\int \mathrm{d}\omega\,\mathsf{A}(\boldsymbol{k}, \omega)\,n_{\mathrm{F}}(\omega)$, the occupation Eq. (4.78). The factor $\mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}} 0^+}$, which records that $c^\dagger$ stands to the left, is what makes the arcs at infinity vanish. Appendix D.10 writes out the steps and the sum of two propagators that appears in every bubble diagram.

4.7.6 The density response in imaginary time

The density response obeys the same rules. For two operators O_a, O_b that conserve the electron number, define

$$R_{ab}(\boldsymbol{q},\tau) = -\frac{1}{\mathscr{V}}\langle \mathrm{T}_\tau\, O_a(\boldsymbol{q},\tau)^\dagger\, O_b(\boldsymbol{q},0)\rangle, \tag{4.90}$$

with $O_a(\boldsymbol{q},\tau)^\dagger$ meaning $\mathrm{e}^{\tau H_\mu} O_a(\boldsymbol{q})^\dagger \mathrm{e}^{-\tau H_\mu}$ and no sign on exchange, since the operators are bosonic. The function is periodic in τ with period β, and its coefficients are taken at $\mathrm{i}\omega_j^{\mathrm{B}}$. For $0<\tau<\beta$ it is $-\mathscr{V}^{-1}\sum_{i,f} p_i \langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle\, \mathrm{e}^{-(E_f-E_i)\tau}$. Integrated with $\mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{B}}\tau}$, now with $\mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{B}}\beta}=+1$, each term gives $[\mathrm{e}^{-\beta(E_f-E_i)}-1]/[\mathrm{i}\omega_j^{\mathrm{B}}-(E_f-E_i)]$ if the denominator is nonzero, and β if it is zero, which happens only for $j=0$ and $E_f=E_i$. With $p_i\,\mathrm{e}^{-\beta(E_f-E_i)}=p_f$,

$$R_{ab}(\boldsymbol{q},\mathrm{i}\omega_j^{\mathrm{B}}) = \frac{1}{\mathscr{V}}\sum_{i,f}^{E_f\neq E_i} \frac{(p_i-p_f)\,\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle}{\mathrm{i}\omega_j^{\mathrm{B}}-(E_f-E_i)} - \delta_{j0}\,\frac{\beta}{\mathscr{V}}\sum_{i,f}^{E_f=E_i} p_i\,\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle. \tag{4.91}$$

In the retarded Lehmann form Eq. (4.6), relabeling $i\leftrightarrow f$ in the antiresonant term turns its weight into $-p_f\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle$ and its denominator into $\omega-(E_f-E_i)+\mathrm{i}0^+$, and the two terms combine into (p_i-p_f) times one pole. That is the sum in Eq. (4.91) with $\mathrm{i}\omega_j^{\mathrm{B}}\to\omega+\mathrm{i}0^+$.

Response function from imaginary time

$$R_{ab}(\boldsymbol{q},\omega) = R_{ab}(\boldsymbol{q},\mathrm{i}\omega_j^{\mathrm{B}}\to\omega+\mathrm{i}0^+), \qquad j>0. \tag{4.92}$$

The retarded response function of Sect. 4.1 is the analytic continuation of the imaginary-time correlation function Eq. (4.90) from the positive bosonic Matsubara frequencies.

The term with δ_{j0} is not part of the continuation, which uses the points $j>0$ alone, and it has a physical meaning. It comes from matrix elements between states of equal energy, and it is the extra term proportional to β that distinguishes the isothermal static response from the limit $\omega\to 0$ of $R_{ab}(\boldsymbol{q},\omega)$, the isolated response of Sect. 4.1 [1]. The value at $\omega_0^{\mathrm{B}}=0$ is the isothermal one, since it is a derivative of an equilibrium average in which the field has been present long enough for the target to stay in equilibrium with the bath.

For independent electrons the imaginary-time route writes the response as a sum over one fermionic frequency of a product of two free Green's functions. With $O_a=O_b=n_e(\boldsymbol{q})$ from Eq. (2.38) and occupations $n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$, Appendix D.10 evaluates Eq. (4.90) directly and shows that it equals

$$R^0(\boldsymbol{q},\mathrm{i}\omega_j^{\mathrm{B}}) = \frac{1}{\mathscr{V}}\sum_{\boldsymbol{k}} T\sum_{j'} \mathsf{G}_0(\boldsymbol{k},\mathrm{i}\omega_{j'}^{\mathrm{F}})\,\mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q},\mathrm{i}\omega_{j'}^{\mathrm{F}}+\mathrm{i}\omega_j^{\mathrm{B}}) = \frac{1}{\mathscr{V}}\sum_{\boldsymbol{k}} \frac{n_{\mathrm{F}}(\xi_{\boldsymbol{k}})-n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}})}{\mathrm{i}\omega_j^{\mathrm{B}}+\xi_{\boldsymbol{k}}-\xi_{\boldsymbol{k}+\boldsymbol{q}}}, \tag{4.93}$$

with the sum over $\boldsymbol{k}$ including spin. Continued to $\omega+\mathrm{i}0^+$, it is the independent-electron response whose imaginary part is the transition sum Eq. (2.70), and Sect. 4.5 evaluates it for the electron gas. The product of two propagators, one carrying the electron from $\boldsymbol{k}$ to $\boldsymbol{k}+\boldsymbol{q}$ and one carrying the hole back, is the bubble diagram of Sect. 4.8. With interactions the same continuation holds term by term, and the diagrams of the next section compute Σ and R as Matsubara sums, which Eqs. (4.88) and (4.92) then carry to the real frequencies at which a dark-matter particle deposits its energy.

4.8 Feynman diagrams

An electron injected 5 eV above the conduction-band edge of silicon does not stay there for long. It knocks electrons across the gap and emits phonons, and its quasiparticle peak acquires the width $\Gamma_{\mathrm{qp}} = -2z_{\mathrm{qp}} \operatorname{Im} \Sigma^{\mathrm{R}}$ of Eq. (4.83). To get a number we need Σ, and the self-energy of an interacting system can only be computed as a series in the interaction. The same is true of the density response beyond the independent-electron bubble of Sect. 4.7. Feynman diagrams are the bookkeeping for these series: each term of the expansion is a picture, and a small set of rules turns the picture into an integral. We will state the rules, which Appendix D.11 derives from Wick's theorem. We then use them on the diagrams the rest of the book needs: the bubble chain behind screening, the boson self-energy behind phonon widths, the *GW* self-energy behind quasiparticle gaps and lifetimes, and the ladder behind excitons. We will also see why the imaginary part of every such diagram is a golden-rule rate. The treatment follows Refs. [3], [37, Chaps. 3 and 7] and [38, Chaps. 2 and 3].

4.8.1 Wick's theorem and the rules

Split $H_\mu = H_0 + H_{\mathrm{int}}$, with $H_0 = \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}}$ the independent electrons of Sect. 4.7. In imaginary time the Boltzmann factor and the Heisenberg operators can both be written with the ordered exponential $U_{\mathrm{I}}(\beta, 0) = \mathrm{T}_\tau \exp[-\int_0^\beta \mathrm{d}\tau\, H_{\mathrm{int,I}}(\tau)]$, where the subscript I marks operators that evolve with H_0 alone. The Matsubara Green's function Eq. (4.86) becomes the ratio Eq. (D.52),

$$\mathsf{G}(\boldsymbol{k}, \tau) = -\frac{\langle \mathrm{T}_\tau\, U_{\mathrm{I}}(\beta, 0)\, c_{\boldsymbol{k},\mathrm{I}}(\tau)\, c^\dagger_{\boldsymbol{k},\mathrm{I}}(0) \rangle_0}{\langle U_{\mathrm{I}}(\beta, 0) \rangle_0},$$

with $\langle \cdots \rangle_0$ the average in the free grand-canonical state. Expanding the exponential, the term of order n in H_{int} is an integral over n imaginary times of a free average of a time-ordered string of creation and annihilation operators. Nothing has been approximated so far.

Wick's theorem evaluates these averages [46]. In the free state, the time-ordered average of a string of fermion operators is a sum over all ways of grouping the operators into pairs. Each grouping contributes the product of its pair averages, with the sign of the permutation that brings the members of every pair next to each other. Only pairs of one creation and one annihilation operator of the same mode survive. Their time-ordered averages, the contractions, are free Green's functions:

$$\langle \mathrm{T}_\tau\, c_{\boldsymbol{k},\mathrm{I}}(\tau)\, c^\dagger_{\boldsymbol{k},\mathrm{I}}(\tau') \rangle_0 = -\mathsf{G}_0(\boldsymbol{k}, \tau - \tau'). \tag{4.94}$$

For bosons the same holds without signs, and a contraction of the field $d_{\boldsymbol{Q}} + d^\dagger_{-\boldsymbol{Q}}$ with itself is the free boson propagator. The theorem is the quantum version of the statement that every moment of a Gaussian distribution is a sum over pairings (Appendix C.6): the free grand-canonical state is the "Gaussian" state of the many-body problem. Appendix D.11 proves it with Gaudin's trace argument [47], which uses only the anticommutators and the cyclic property of the trace. At $T = 0$ the same theorem holds for averages in the Fermi sea $|\Phi_0\rangle$, the limit $\beta \to \infty$ of the free state, with the contraction $\mathrm{i}\mathsf{G}_0(\boldsymbol{k}, t - t')$ of the time-ordered function Eq. (4.79). The interacting ground state is reached from $|\Phi_0\rangle$ by switching on the interaction slowly, as long as no level crossing intervenes (the theorem of Gell-Mann and Low [48]), and the real-time expansion Eq. (D.53) has the same structure.

Each pairing is drawn as a picture. A contraction Eq. (4.94) is a solid line with an arrow, from the point where the electron is created to the point where it is annihilated. A factor H_{int} is a vertex, or for the Coulomb interaction a dashed line whose two ends are the two density operators of the pair interaction. Pairings that fall into a piece attached to the external

points and a piece that is not, the vacuum diagrams, cancel against the denominator, which is the linked-cluster theorem, and only connected diagrams remain. Relabeling the n identical interaction factors, and exchanging the two ends of each line, produces $n!\,2^n$ pairings with the same value, which cancel the $1/n!$ of the exponential and the $\frac{1}{2}$ of the pair interaction. Collecting the signs and the factors of T and $\mathcal{V}$, Appendix D.11 arrives at the following rules; a reader to whom the signs are new can follow their derivation there before going on.

Feynman rules in imaginary time

The Green's function $\mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}})$ is the sum of all connected, topologically distinct diagrams with two external points, each evaluated with the following factors.

1. Each solid line carrying $(\boldsymbol{p}, \mathrm{i}\omega^{\mathrm{F}})$ is $\mathsf{G}_0(\boldsymbol{p}, \mathrm{i}\omega^{\mathrm{F}}) = 1/(\mathrm{i}\omega^{\mathrm{F}} - \xi_{\boldsymbol{p}})$; spin is conserved along it.
2. Each Coulomb line carrying $(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{B}})$ is $-v_{\mathrm{C}}(k)/\mathcal{V}$. Each boson line is $-|g(\boldsymbol{Q})|^2 \mathsf{D}_0(\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{B}})/\mathcal{V}$, with $\mathsf{D}_0(\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{B}}) = 2\omega_{\boldsymbol{Q}}/[(\mathrm{i}\omega^{\mathrm{B}})^2 - \omega_{\boldsymbol{Q}}^2]$ for a boson of energy $\omega_{\boldsymbol{Q}}$ coupled through $H_{\mathrm{int}} = \mathcal{V}^{-1/2} \sum g(\boldsymbol{Q})\, c^\dagger_{\boldsymbol{k}+\boldsymbol{Q}} c_{\boldsymbol{k}} (d_{\boldsymbol{Q}} + d^\dagger_{-\boldsymbol{Q}})$.
3. Momentum and frequency are conserved at every vertex. Each independent internal momentum is summed, and each independent internal frequency carries $T\sum_j$.
4. Each closed fermion loop gives -1 and a sum over its spin.
5. A solid line that begins and ends on the same interaction carries $\mathrm{e}^{\mathrm{i}\omega^{\mathrm{F}} 0^+}$.

The response function Eq. (4.90) of two densities is $-1/\mathcal{V}$ times the sum of connected diagrams with two density vertices, at which $\boldsymbol{q}$ and $\mathrm{i}\omega_j^{\mathrm{B}}$ enter and leave, evaluated with the same rules. At $T = 0$ the rules for the time-ordered functions are the same with $\mathsf{G}_0(\boldsymbol{p}, \omega) = 1/(\omega - \xi_{\boldsymbol{p}} + \mathrm{i}0^+ \operatorname{sgn} \xi_{\boldsymbol{p}})$, each Coulomb line $+\mathrm{i}v_{\mathrm{C}}(k)/\mathcal{V}$, and each internal frequency $\int \mathrm{d}\omega/2\pi$, and the time-ordered response $-(\mathrm{i}/\mathcal{V})\langle \mathrm{T}\, O_a(\boldsymbol{q}, t)^\dagger O_b(\boldsymbol{q}, 0)\rangle$ carries $+\mathrm{i}/\mathcal{V}$ in place of $-1/\mathcal{V}$.

The sign of a loop is the one rule that has no counterpart in the old-fashioned perturbation theory of Sect. 2.4. It comes from the last contraction around the loop, which pairs the first creation operator with the last annihilation operator and costs an odd number of exchanges. The rule for the response function needs one line of its own. The definition Eq. (4.90) carries $-1/\mathcal{V}$. A response diagram has $2n + 2$ propagators at order n, an even number, so the -1 of each contraction drops out. The signs left over are those of the expansion, $(-1)^n$, which the rule puts into the lines, and of the loops. Appendix D.11 checks the rules for G at first order against the Hartree–Fock self-energy of Appendix D.4. The next subsection checks the rule for R against Eq. (4.93) and against the screening of Sect. 4.4.

In the frequency representation every line is a simple pole, every frequency sum is done by the residue formula Eq. (C.42), and the temperature enters only through these sums: G_0 and D_0 do not depend on it, and the occupation factors n_{F} and n_{B} appear as residues of $n_{\mathrm{F}}(z)$ and $n_{\mathrm{B}}(z)$.

4.8.2 The bubble and the random-phase chain

The density response of independent electrons is the simplest diagram there is: one fermion loop through the two density vertices (Fig. 4.14a). An electron is created in $\boldsymbol{k}+\boldsymbol{q}$ and a hole in $\boldsymbol{k}$ at one vertex, and both are annihilated at the other. By the rules, the loop contributes -1

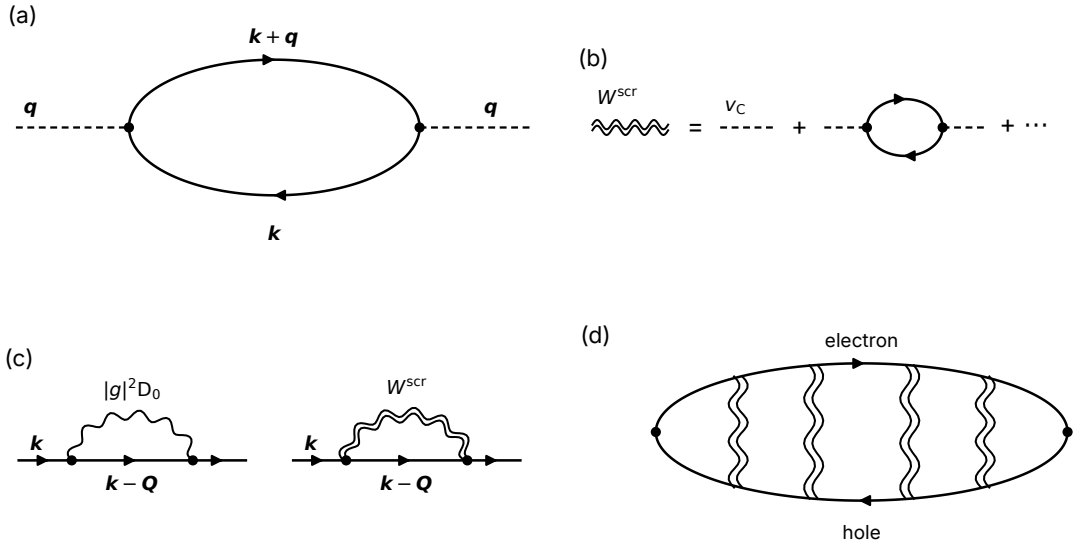


Figure 4.14: The diagrams of Sect. 4.8. Solid lines with arrows are free electron propagators G_0; an arrow against the direction of the momentum label is a hole. Dashed lines are the bare Coulomb interaction v_C, except the short dashed stubs in (a), which mark the external momentum $\boldsymbol{q}$ entering and leaving at the density vertices; single wavy lines a boson, double wavy lines the screened interaction W^scr. Dots are vertices. (a) The bubble: a density vertex creates an electron in $\boldsymbol{k}+\boldsymbol{q}$ and a hole in $\boldsymbol{k}$, and a second vertex annihilates them; it is the independent-electron response R^0. (b) The screened interaction as the random-phase chain of bubbles joined by Coulomb lines, $W^\mathrm{scr} = v_\mathrm{C} + v_\mathrm{C}R^0v_\mathrm{C} + \cdots$. (c) The self-energy of an electron that emits and reabsorbs a boson, with the line $|g|^2\mathsf{D}_0$, and the GW self-energy, the same diagram with the screened line. (d) The ladder: the electron and the hole of the bubble interact through screened interaction lines. Summed together with the chain of exchange lines, it becomes the Bethe–Salpeter equation of Sect. 4.9.

and a spin sum, the two lines give $\mathsf{G}_0(\boldsymbol{k},\mathrm{i}\omega^\mathrm{F}_{j'})\mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q},\mathrm{i}\omega^\mathrm{F}_{j'}+\mathrm{i}\omega^\mathrm{B}_j)$, and the internal momentum and frequency are summed. With the $-1/\mathscr{V}$ of the response,

$$R^0(\boldsymbol{q},\mathrm{i}\omega^\mathrm{B}_j) = \frac{1}{\mathscr{V}}\sum_{\boldsymbol{k}s} T\sum_{j'} \mathsf{G}_0(\boldsymbol{k},\mathrm{i}\omega^\mathrm{F}_{j'})\,\mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q},\mathrm{i}\omega^\mathrm{F}_{j'}+\mathrm{i}\omega^\mathrm{B}_j), \tag{4.95}$$

which is the first form of Eq. (4.93). The frequency sum is Eq. (D.51), and it produces the Pauli difference $[n_\mathrm{F}(\xi_{\boldsymbol{k}}) - n_\mathrm{F}(\xi_{\boldsymbol{k}+\boldsymbol{q}})]/(\mathrm{i}\omega^\mathrm{B}_j + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}})$. Continued to $\omega + \mathrm{i}0^+$ by Eq. (4.92), this is the independent-electron response Eq. (4.55), whose evaluation for the electron gas is the Lindhard function of Sect. 4.5. In a crystal the plane waves become Bloch states $\psi_{n\boldsymbol{k}}$, the density vertex $n_e(\boldsymbol{q}+\boldsymbol{G})$ acquires the matrix element $\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})$ between the two states it connects, and the same diagram is the crystal bubble Eq. (4.63) (Eqs. (D.60) and (D.61)). The sign of the rule passes its first test: the static bubble is negative, as a response must be.

At first order in v_C the bubble is corrected by three kinds of diagram, which differ in where the Coulomb line attaches. It can join two bubbles, the line carrying the momentum $\boldsymbol{q}$ from one to the other. It can run across the bubble, from the electron line to the hole line. Or it can begin and end on one propagator, which is a first-order self-energy inserted into that line. The

first kind has two loops, and by the rules its value is $-(1/\mathscr{V})(-1)^2(-v_{\mathrm{C}}(q)/\mathscr{V})\,(VR^0)(VR^0) = R^0v_{\mathrm{C}}(q)R^0$, with a plus sign, where each loop without its sign is $\sum_{\boldsymbol{k}s} T \sum_{j'} \mathsf{G}_0\mathsf{G}_0 = VR^0$. The second kind has the same propagators and the same line but only one loop, and the opposite sign. Repeating the first kind gives the chain $R^0 + R^0v_{\mathrm{C}}R^0 + R^0v_{\mathrm{C}}R^0v_{\mathrm{C}}R^0 + \cdots$, a geometric series with the sum

$$R = \frac{R^0}{1 - v_{\mathrm{C}}(q)R^0}. \tag{4.96}$$

This is the random-phase approximation Eq. (4.59), now as a definite selection of diagrams: the chain of bubbles, and nothing inside them. Its sign is the one Sect. 4.4 found from the classical argument, $\varepsilon = 1 - v_{\mathrm{C}}R^0 > 1$ in the static limit. The general statement is just as direct. A diagram for R either can or cannot be cut into two by cutting a single Coulomb line at the momentum $\boldsymbol{q}$. Call the sum of those that cannot the proper part $\tilde{R}$. Every diagram that can is a chain of proper pieces joined by such lines, and the sum of the chains is $R = \tilde{R} + \tilde{R}v_{\mathrm{C}}R$, the Dyson equation Eq. (4.40). The blobs of Fig. 4.6 are sums of Feynman diagrams, and the random-phase approximation keeps the bubble alone in each blob.

In a crystal the Coulomb line between two bubbles carries $\boldsymbol{q}+\boldsymbol{G}$ for every reciprocal-lattice vector $\boldsymbol{G}$, and the chain becomes the matrix equation of Sect. 4.6. The line with $\boldsymbol{G} = 0$ carries the macroscopic field. The lines with $\boldsymbol{G} \neq 0$ are the local fields, and in the language of transitions they form the exchange kernel K^{x} of Sect. 4.6, the Coulomb interaction without its $\boldsymbol{G} = 0$ term. The second kind of first-order diagram, the line across the bubble, is the attraction between the electron and the hole that the chain leaves out. Repeated, with each interaction line screened by its own chain of bubbles, it forms the ladder of Fig. 4.14d. The sum of ladders and exchange chains is the Bethe–Salpeter equation, the equation for an electron–hole pair that propagates while the two particles attract each other, and its bound states are the excitons of Sect. 4.9. Appendix D.12 derives it from these diagrams, with the direct kernel K^{d} built from the screened interaction line.

4.8.3 Reducible lines and the thermodynamic limit

The f-sum rule for ε_2 in Sect. 4.4 rested on one claim. Deleting the Coulomb interaction at the single wavevector $\boldsymbol{q}$ from the Hamiltonian leaves the equilibrium state unchanged in the thermodynamic limit, while it turns the full response into its proper part. Diagrams make both halves precise. Let $\bar{H}$ be H with the Coulomb terms at $\pm\boldsymbol{q}$ removed, $\boldsymbol{q} \neq 0$, and compare the diagrams of the two Hamiltonians.

Take first any equilibrium average: the Green's function at some $\boldsymbol{k}$, an occupation, or the free energy per volume. In its diagrams a Coulomb line has its momentum fixed by conservation only if cutting it separates the diagram into two pieces. The piece without external points then exchanges no momentum with the rest, and the line carries $\boldsymbol{k} = 0$, which is absent in a neutral system. Every other line carries a loop momentum, summed as $\mathscr{V}^{-1}\sum_{\boldsymbol{k}}$ with $v_{\mathrm{C}}(k)$ finite at $\boldsymbol{k} = \pm\boldsymbol{q}$. Deleting the two terms $\boldsymbol{k} = \pm\boldsymbol{q}$ from such a sum changes it by a relative amount of order $1/N$, with N the number of electrons, and it vanishes in the thermodynamic limit. The equilibrium state of $\bar{H}$ is therefore that of H, at every order.

Now take the response at $\boldsymbol{q}$. Here a line can carry exactly $\boldsymbol{q}$, and conservation forces it to when, and only when, the line is a bridge: cutting it splits the diagram into a piece attached to one density vertex and a piece attached to the other. These are the lines of the chain. In $\bar{H}$ they are gone, while the lines at $\pm\boldsymbol{q}$ inside loops lose a negligible term as before.

Removing one Fourier component

In the thermodynamic limit, the Hamiltonian $\bar{H}$ without the Coulomb terms at $\pm\boldsymbol{q}$ has the same equilibrium state as H, and its density response at $\boldsymbol{q}$ is the proper part of the response of H,

$$\bar{R}(\boldsymbol{q},\omega) = \tilde{R}(\boldsymbol{q},\omega). \tag{4.97}$$

In a crystal the same argument with the $\boldsymbol{G}=0$ term alone removed identifies $\bar{R}$ of Appendix D.7 with the response that is proper with respect to the macroscopic line only.

Because $\bar{R}$ is the response function of a Hamiltonian in its own equilibrium state, every exact property proved in Sects. 4.1 to 4.3 holds for it: the Lehmann form, positivity of the absorption and the f-sum rule with the same density $\boldsymbol{n}_e$. That is what Sect. 4.4 used for ε_2 and for the Ward identity of the proper current response. The argument is diagrammatic, and it holds at every order of perturbation theory for $\boldsymbol{q}\neq 0$. At $\boldsymbol{q}\to 0$ the deleted terms are at the edge of the sum and the limit is taken after the thermodynamic one, which is the order in which the optical limit of Appendix D.7 is defined.

4.8.4 An electron coupled to a boson

An electron in a crystal shakes the lattice as it moves: it emits a phonon, propagates in a different state, and absorbs the phonon again. The lowest self-energy diagram of this process is the Fock diagram with a boson line (Fig. 4.14c), second order in the coupling g. It carries every feature of finite-temperature perturbation theory in a few lines, and Sect. 6.3 builds the electron–phonon self-energy of silicon on it. By the rules, with the boson line carrying $(\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{B}}_{j'})$ and the electron line $(\boldsymbol{k}-\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{F}}_{j} - \mathrm{i}\omega^{\mathrm{B}}_{j'})$, and no loop,

$$\Sigma(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{F}}_j) = -\frac{1}{\mathscr{V}} \sum_{\boldsymbol{Q}} |g(\boldsymbol{Q})|^2 \, T \sum_{j'} \mathsf{D}_0(\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{B}}_{j'}) \, \mathsf{G}_0(\boldsymbol{k}-\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{F}}_j - \mathrm{i}\omega^{\mathrm{B}}_{j'}).$$

The sum runs over a bosonic frequency, and by Eq. (C.42) it is minus the sum of the residues of $n_{\mathrm{B}}(z)h(z)$ at the poles of

$$h(z) = \Big[\frac{1}{z-\omega_{\boldsymbol{Q}}} - \frac{1}{z+\omega_{\boldsymbol{Q}}}\Big] \frac{1}{\mathrm{i}\omega^{\mathrm{F}}_j - z - \xi'}, \qquad \xi' = \xi_{\boldsymbol{k}-\boldsymbol{Q}},$$

which falls off as $1/|z|^3$, so no convergence factor is needed. The pole at $z=\omega_{\boldsymbol{Q}}$ has the residue $n_{\mathrm{B}}(\omega_{\boldsymbol{Q}})/(\mathrm{i}\omega^{\mathrm{F}}_j - \xi' - \omega_{\boldsymbol{Q}})$. The pole at $z=-\omega_{\boldsymbol{Q}}$ has the residue $-n_{\mathrm{B}}(-\omega_{\boldsymbol{Q}})/(\mathrm{i}\omega^{\mathrm{F}}_j - \xi' + \omega_{\boldsymbol{Q}})$, and $n_{\mathrm{B}}(-x) = -1 - n_{\mathrm{B}}(x)$ turns its numerator into $1+n_{\mathrm{B}}$. The pole at $z=\mathrm{i}\omega^{\mathrm{F}}_j - \xi'$, where the electron line is singular, has the residue $-n_{\mathrm{B}}(\mathrm{i}\omega^{\mathrm{F}}_j - \xi')\,\mathsf{D}_0(\mathrm{i}\omega^{\mathrm{F}}_j - \xi')$. Before anything is continued, the Bose function at a fermionic frequency is simplified with $\mathrm{e}^{\mathrm{i}\beta\omega^{\mathrm{F}}_j} = -1$: $n_{\mathrm{B}}(\mathrm{i}\omega^{\mathrm{F}}_j - \xi') = 1/(-\mathrm{e}^{-\beta\xi'} - 1) = -[1 - n_{\mathrm{F}}(\xi')]$. This residue is then $[1-n_{\mathrm{F}}(\xi')][1/(\mathrm{i}\omega^{\mathrm{F}}_j - \xi' - \omega_{\boldsymbol{Q}}) - 1/(\mathrm{i}\omega^{\mathrm{F}}_j - \xi' + \omega_{\boldsymbol{Q}})]$. Adding the three residues, changing the sign, and multiplying by $-|g|^2/\mathscr{V}$ gives the self-energy.

Second-order self-energy with a boson

$$\Sigma(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{F}}_j) = \frac{1}{\mathscr{V}} \sum_{\boldsymbol{Q}} |g(\boldsymbol{Q})|^2 \left[\frac{n_{\mathrm{B}}(\omega_{\boldsymbol{Q}}) + 1 - n_{\mathrm{F}}(\xi_{\boldsymbol{k}-\boldsymbol{Q}})}{\mathrm{i}\omega^{\mathrm{F}}_j - \xi_{\boldsymbol{k}-\boldsymbol{Q}} - \omega_{\boldsymbol{Q}}} + \frac{n_{\mathrm{B}}(\omega_{\boldsymbol{Q}}) + n_{\mathrm{F}}(\xi_{\boldsymbol{k}-\boldsymbol{Q}})}{\mathrm{i}\omega^{\mathrm{F}}_j - \xi_{\boldsymbol{k}-\boldsymbol{Q}} + \omega_{\boldsymbol{Q}}} \right]. \tag{4.98}$$

The retarded self-energy is the same expression with $\mathrm{i}\omega_j^{\mathrm{F}} \to \omega + \mathrm{i}0^+$. The term with $-\omega_{\boldsymbol{Q}}$ in the denominator is the emission of a boson, the term with $+\omega_{\boldsymbol{Q}}$ its absorption.

The thermal factors have a direct reading. In the emission term, $n_{\mathrm{B}} + 1$ is the stimulated plus spontaneous emission of Worked example 2.7, and $-n_{\mathrm{F}}$ removes the intermediate states that are already occupied: the Pauli principle acts on the virtual state as well as on a real one. In the absorption term, n_{B} counts the thermal bosons available, and $+n_{\mathrm{F}}$ adds the process in which an electron already in $\boldsymbol{k} - \boldsymbol{Q}$ makes the transition instead. Eq. (4.103) below shows that the two factors are sums of a forward and a reverse rate. For an electron in an empty band at $T = 0$, $n_{\mathrm{B}} = n_{\mathrm{F}} = 0$, and the retarded self-energy is

$$\Sigma^{\mathrm{R}}(\boldsymbol{k}, \omega) = \frac{1}{\mathscr{V}} \sum_{\boldsymbol{Q}} \frac{|g(\boldsymbol{Q})|^2}{\omega - \xi_{\boldsymbol{k}-\boldsymbol{Q}} - \omega_{\boldsymbol{Q}} + \mathrm{i}0^+},$$

the second-order sum of Eq. (2.52) over intermediate states with one electron in $\boldsymbol{k} - \boldsymbol{Q}$ and one boson, with ω in place of the initial energy. It has the form of the self-energy Eq. (4.85) of the level coupled to a band, and the matrix element $g/\sqrt{\mathscr{V}}$ of H_{int} plays the role of $V_{\boldsymbol{k}}$.

Worked example 4.13: An electron coupled to an Einstein boson

Take the electron in an empty conduction band of a semiconductor, with μ_{ch} near midgap, 0.5 eV below the band bottom E_{c}. Then $n_{\mathrm{F}} \leq 4 \times 10^{-9}$ in the band at 300 K, and the n_{F} terms of Eq. (4.98) drop out. Let the boson be dispersionless, an Einstein mode with $\omega_{\boldsymbol{Q}} = \omega_{\mathrm{Ein}} = 60$ meV, a typical silicon phonon energy (Sect. 1.3), and let the coupling g be constant. The sum over $\boldsymbol{Q}$ then runs over all final electron states $\boldsymbol{k} - \boldsymbol{Q}$ with the same weight, and the self-energy depends on ω alone. For the band we take the flat continuum of Worked example 2.1: $\mathcal{D}_f$ states per unit energy, spread uniformly over a width $E_{\mathrm{w}} = 1$ eV above E_{c}. The golden-rule rate for emitting a boson from deep inside the band at $T = 0$ is $\Gamma_{\mathrm{em}} = 2\pi(|g|^2/\mathscr{V})\mathcal{D}_f$, and we choose the coupling so that $\Gamma_{\mathrm{em}} = 20$ meV. With $\sum_{\boldsymbol{Q}} \to \mathcal{D}_f \int \mathrm{d}\epsilon$ and the integral $\int_{E_{\mathrm{c}}}^{E_{\mathrm{c}}+E_{\mathrm{w}}} \mathrm{d}\epsilon/(x - \epsilon + \mathrm{i}0^+) = \ell_{\mathrm{b}}(x)$,

$$\Sigma^{\mathrm{R}}(\omega) = \frac{\Gamma_{\mathrm{em}}}{2\pi}\Big[(n_{\mathrm{B}}+1)\,\ell_{\mathrm{b}}(\omega-\omega_{\mathrm{Ein}}) + n_{\mathrm{B}}\,\ell_{\mathrm{b}}(\omega+\omega_{\mathrm{Ein}})\Big], \qquad \ell_{\mathrm{b}}(x) = \ln\frac{x - E_{\mathrm{c}} + \mathrm{i}0^+}{x - E_{\mathrm{c}} - E_{\mathrm{w}} + \mathrm{i}0^+},$$

with $n_{\mathrm{B}} = n_{\mathrm{B}}(\omega_{\mathrm{Ein}})$. The logarithm has the imaginary part $-\pi$ where its argument lies in the band and zero outside, and each term contributes a width only where its final electron state exists (Fig. 4.15).

At $T = 0$ an electron at the band bottom, $\omega = E_{\mathrm{c}}$, cannot emit: the final state would lie ω_{Ein} below the band. Its width is zero, as Sect. 4.7 argued for the conduction-band minimum. The virtual emission still shifts it, by $\mathrm{Re}\,\Sigma^{\mathrm{R}}(E_{\mathrm{c}}) = (\Gamma_{\mathrm{em}}/2\pi)\ln[\omega_{\mathrm{Ein}}/(E_{\mathrm{w}} + \omega_{\mathrm{Ein}})] = -9.14$ meV, and the slope $\partial_\omega \mathrm{Re}\,\Sigma^{\mathrm{R}} = -0.0500$ sets $z_{\mathrm{qp}} = 0.952$. The electron at the band bottom carries a cloud of virtual bosons, which lowers its energy and takes 5 percent of its weight into the incoherent background. Solving $E_{\mathrm{qp}} = E_{\mathrm{c}} + \mathrm{Re}\,\Sigma^{\mathrm{R}}(E_{\mathrm{qp}})$ instead of evaluating Σ at E_{c} moves the energy by -8.73 meV. An electron 0.3 eV up in the band emits freely, and $-2\,\mathrm{Im}\,\Sigma^{\mathrm{R}} = \Gamma_{\mathrm{em}}$ is the golden-rule rate.

At 300 K, $T = 25.85$ meV and $n_{\mathrm{B}} = 0.109$. The electron at the band bottom can now absorb a thermal boson, and its width is $\Gamma_{\mathrm{qp}} = z_{\mathrm{qp}}\Gamma_{\mathrm{em}} n_{\mathrm{B}} = 2.08$ meV, with $z_{\mathrm{qp}} = 0.953$: a lifetime of 0.32 ps. The shift grows to -11.09 meV. That is the zero-temperature shift times $2n_{\mathrm{B}} + 1$ to within 0.4 percent, since emission and absorption contribute nearly equal logarithms when $\omega_{\mathrm{Ein}} \ll E_{\mathrm{w}}$. Inside the band the width rises to $\Gamma_{\mathrm{em}}(2n_{\mathrm{B}} + 1) = 24.4$ meV. These two features, a shift that scales as $2n_{\mathrm{B}} + 1$ and a width at the band edge set by

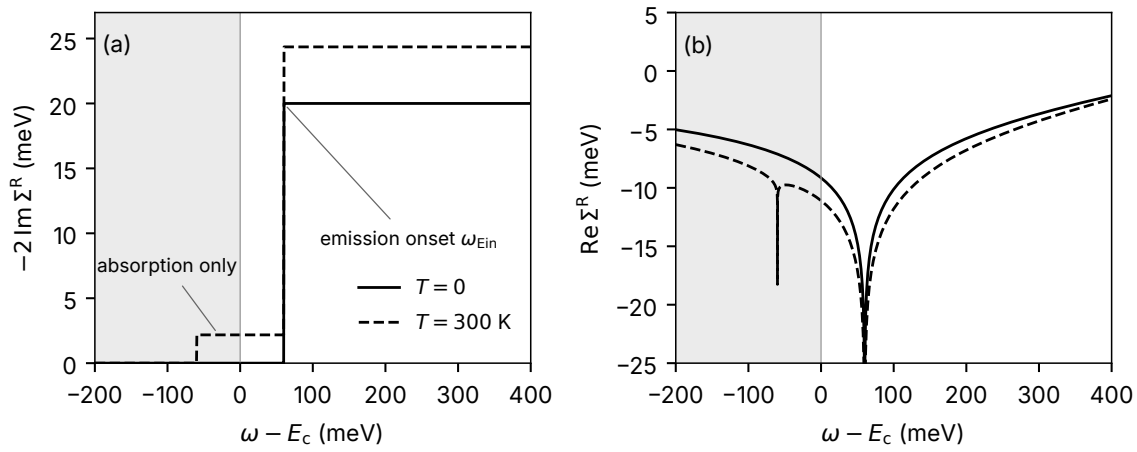


Figure 4.15: Self-energy of an electron coupled to a dispersionless boson of energy $\omega_{\mathrm{Ein}} = 60\,\mathrm{meV}$ (Worked example 4.13), against the energy ω measured from the bottom E_{c} of an empty band with a flat density of states over $E_{\mathrm{w}} = 1\,\mathrm{eV}$, for a constant coupling with the golden-rule emission rate $\Gamma_{\mathrm{em}} = 20\,\mathrm{meV}$; solid at $T = 0$, dashed at 300 K. The model and its parameters are illustrative. (a) The width $-2\,\mathrm{Im}\,\Sigma^{\mathrm{R}}$: zero below the emission threshold $E_{\mathrm{c}} + \omega_{\mathrm{Ein}}$ at $T = 0$; at 300 K, $\Gamma_{\mathrm{em}} n_{\mathrm{B}} = 2.2\,\mathrm{meV}$ from absorption above $E_{\mathrm{c}} - \omega_{\mathrm{Ein}}$, and $\Gamma_{\mathrm{em}}(2n_{\mathrm{B}} + 1)$ where both processes are allowed. (b) The shift $\mathrm{Re}\,\Sigma^{\mathrm{R}}$, $-9.14\,\mathrm{meV}$ at E_{c} for $T = 0$ and $-11.09\,\mathrm{meV}$ at 300 K, with logarithmic singularities at the thresholds caused by the sharp edge of the flat band. Shaded: below the band bottom.

absorption alone, are the temperature dependence of band energies that Sect. 6.3 computes for silicon. The logarithmic dip of $\mathrm{Re}\,\Sigma^{\mathrm{R}}$ at the emission threshold, $\omega = E_{\mathrm{c}} + \omega_{\mathrm{Ein}}$, comes from the step of the flat density of states at E_{c}; with the square-root density of states of a three-dimensional band edge it becomes a finite cusp.

4.8.5 The screened interaction and the GW self-energy

The Coulomb version of the same diagram, with a bare line, is the exchange self-energy Eq. (D.56), $\Sigma_{\mathrm{x}}(\boldsymbol{k}) = -\mathscr{V}^{-1}\sum_{\boldsymbol{p}} v_{\mathrm{C}}(|\boldsymbol{k}-\boldsymbol{p}|)\, n_{\mathrm{F}}(\xi_{\boldsymbol{p}})$, the Fock term of Sect. 3.4. It is static, it has no imaginary part, and in a metal it makes the group velocity of the electron gas infinite at the Fermi surface through a logarithm (Appendix D.4). What it misses is screening: while the electron propagates, the other electrons rearrange around it and weaken the interaction at long distances. The rules say how to put this in. Every chain of bubbles that can be inserted into the Coulomb line is a self-energy diagram, and summing the chains replaces the line, exactly as in the random-phase chain:

$$W^{\mathrm{scr}}(\boldsymbol{q}, \mathrm{i}\omega_j^{\mathrm{B}}) = v_{\mathrm{C}}(q) + v_{\mathrm{C}}(q)\, R(\boldsymbol{q}, \mathrm{i}\omega_j^{\mathrm{B}})\, v_{\mathrm{C}}(q) = \varepsilon^{-1}(\boldsymbol{q}, \mathrm{i}\omega_j^{\mathrm{B}})\, v_{\mathrm{C}}(q). \tag{4.99}$$

The last step is Eq. (4.41). With the bubble chain alone, R is the random-phase response Eq. (4.96) and ε the Lindhard function of Sect. 4.5 or its crystal form; with the proper blobs it is exact. The screened interaction is the Coulomb potential of a test charge seen through the dielectric response of the target, the same $\varepsilon^{-1} v_{\mathrm{C}}$ that Sect. 4.4 found for an external charge, but now at every frequency. In a crystal it is the matrix $W^{\mathrm{scr}}_{\boldsymbol{G}\boldsymbol{G}'} = (\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'} v_{\mathrm{C}}(|\boldsymbol{q} + \boldsymbol{G}'|)$, and the steps below hold element by element.

Replacing the bare line of the exchange diagram by W^{scr}, and the free propagator by the full one, is the *GW* approximation of Hedin [49]. In imaginary time,

$$\Sigma(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = -\frac{1}{\mathscr{V}} \sum_{\boldsymbol{q}} T \sum_{j'} W^{\mathrm{scr}}(\boldsymbol{q}, \mathrm{i}\omega_{j'}^{\mathrm{B}})\, \mathsf{G}(\boldsymbol{k}-\boldsymbol{q}, \mathrm{i}\omega_j^{\mathrm{F}} - \mathrm{i}\omega_{j'}^{\mathrm{B}}), \tag{4.100}$$

with the convergence factor of rule 5 on the static part v_{C}, which reproduces Σ_{x}. At $T = 0$ the real-time rules give the same diagram as

$$\Sigma(\boldsymbol{k}, \omega) = \frac{\mathrm{i}}{\mathscr{V}} \sum_{\boldsymbol{q}} \int \frac{\mathrm{d}\omega'}{2\pi}\, \mathsf{G}(\boldsymbol{k}-\boldsymbol{q}, \omega-\omega')\, W^{\mathrm{scr}}(\boldsymbol{q}, \omega'), \tag{4.101}$$

which in space and time is a product, $\Sigma(1,2) = \mathrm{i}\mathsf{G}(1,2)W^{\mathrm{scr}}(1,2)$, the form usually written $\Sigma = \mathrm{i}GW$ [50, Eqs. (60) and (61)]. Note that G and W^{scr} here are time-ordered functions: by the convention of Sect. 4.7, a function without superscript and with a real argument is time-ordered, and $W^{\mathrm{scr}}(\boldsymbol{q}, \omega')$ equals the retarded $\varepsilon^{-1}v_{\mathrm{C}}$ of Sect. 4.4 for $\omega' > 0$ and the advanced one for $\omega' < 0$. Lifetimes need the retarded self-energy, which the Matsubara form Eq. (4.100) delivers by continuation. The approximation keeps the screening to all orders and drops the vertex corrections, the diagrams in which a further line crosses the one of Fig. 4.14c; Hedin's exact equations contain them in a vertex function, which the *GW* approximation sets to one [50, Eqs. (43)–(46)]. With the Hartree term, which is the electrostatic potential of the density, *GW* is the self-energy behind the quasiparticle gap of silicon in Sects. 3.4 and 4.7.

The lifetime follows from the same frequency sum as for a boson. The difference $W^{\mathrm{scr}} - v_{\mathrm{C}} = v_{\mathrm{C}}^2 R$ is a response function, and like every response it has the spectral representation Eq. (4.19), $W^{\mathrm{scr}}(\boldsymbol{q}, z) - v_{\mathrm{C}}(q) = \int \mathrm{d}\omega'\, B(\boldsymbol{q}, \omega')/(z-\omega')$, with the weight $B = -\operatorname{Im} W^{\mathrm{scr,R}}/\pi = v_{\mathrm{C}}(q) L(\boldsymbol{q}, \omega')/\pi$, where L is the loss function Eq. (4.45). This is a superposition of bosons: the free boson line has $B = |g|^2[\delta(\omega' - \omega_{\boldsymbol{Q}}) - \delta(\omega' + \omega_{\boldsymbol{Q}})]$, and the screened Coulomb line is a continuum of them, the electron–hole pairs and plasmons of the target, weighted by the loss function. The residue sum that led to Eq. (4.98) then leads, with the free propagator in Eq. (4.100),

$$\Sigma(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = \Sigma_{\mathrm{x}}(\boldsymbol{k}) + \frac{1}{\mathscr{V}} \sum_{\boldsymbol{q}} \int \mathrm{d}\omega'\, B(\boldsymbol{q}, \omega')\, \frac{n_{\mathrm{B}}(\omega') + 1 - n_{\mathrm{F}}(\xi_{\boldsymbol{k}-\boldsymbol{q}})}{\mathrm{i}\omega_j^{\mathrm{F}} - \xi_{\boldsymbol{k}-\boldsymbol{q}} - \omega'},$$

and the imaginary part of its continuation, $\operatorname{Im}\Sigma^{\mathrm{R}}(\boldsymbol{k}, \omega) = -\mathscr{V}^{-1} \sum_{\boldsymbol{q}} v_{\mathrm{C}}(q) L(\boldsymbol{q}, \omega')[n_{\mathrm{B}}(\omega') + 1 - n_{\mathrm{F}}(\xi_{\boldsymbol{k}-\boldsymbol{q}})]$ with $\omega' = \omega - \xi_{\boldsymbol{k}-\boldsymbol{q}}$. On the shell $\omega = \xi_{\boldsymbol{k}}$, with $z_{\mathrm{qp}} \simeq 1$, the width is $\Gamma_{\mathrm{qp}} = -2 \operatorname{Im}\Sigma^{\mathrm{R}}$. The structure-factor relation $2L/v_{\mathrm{C}} = (1 - \mathrm{e}^{-\beta\omega'})S$ of Eq. (4.45), and the identity $(1 - \mathrm{e}^{-\beta\omega'})(n_{\mathrm{B}}(\omega') + 1) = 1$, turn it into a rate.

Electron–electron lifetime in the *GW* approximation

$$\Gamma_{\mathrm{qp}}(\boldsymbol{k}) = \int \frac{\mathrm{d}^3 q}{(2\pi)^3}\, v_{\mathrm{C}}(q)^2\, S(\boldsymbol{q}, \omega') \Big[1 - n_{\mathrm{F}}(\xi_{\boldsymbol{k}-\boldsymbol{q}}) + \mathrm{e}^{-\beta\omega'} n_{\mathrm{F}}(\xi_{\boldsymbol{k}-\boldsymbol{q}})\Big], \qquad \omega' = \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}-\boldsymbol{q}}. \tag{4.102}$$

The inverse lifetime of an electron is the rate Eq. (4.14) at which it would scatter off the target as a projectile with the potential $\tilde{V} = v_{\mathrm{C}}$, with the Pauli factor of its final state, plus the rate of the reverse process.

An injected electron is thus a dark-matter particle with a Coulomb coupling. It loses energy to exactly the excitations that the loss function describes, through the same structure factor, and Eq. (4.102) is a stopping-power calculation of the kind in Eq. (4.54), with a

quantum recoil and a Pauli-blocked final state. The reverse term, $\mathrm{e}^{-\beta\omega'} n_{\mathrm{F}}$, is the rate at which electrons in $\boldsymbol{k}-\boldsymbol{q}$ absorb energy from the target and refill $\boldsymbol{k}$, and it is negligible for an electron far above μ_{ch}. Near the Fermi level of a metal, $1-n_{\mathrm{F}}$ and the linear low-frequency loss function of Sect. 4.5 give the $\xi_{\boldsymbol{k}}^2$ of Sect. 4.7. In silicon, $S(\boldsymbol{q},\omega')$ vanishes for ω' below the gap, and an electron cannot decay by creating a pair until it sits at least E_{g} above the band edge. Higher up, the loss function peaks at the plasmon near 16.4 eV (Sect. 4.4), and an electron with tens of eV loses its energy mostly by plasmon emission. The exchange of the projectile with the electrons it excites, a vertex correction, is absent from Eq. (4.102).

In practice the *GW* quasiparticle energies and lifetimes of silicon are computed in the basis of Kohn–Sham states. The independent-electron response $R^0_{\boldsymbol{G}\boldsymbol{G}'}$ of Sect. 4.6 yields $\varepsilon^{-1}_{\boldsymbol{G}\boldsymbol{G}'}$ in the random-phase approximation, and Eq. (4.99) gives the matrix $W^{\mathrm{scr}}_{\boldsymbol{G}\boldsymbol{G}'}$. The matrix elements $\langle\psi_{n\boldsymbol{k}}|\Sigma(\omega)|\psi_{n\boldsymbol{k}}\rangle$ of Eq. (4.101), built from the same plane-wave matrix elements, enter the linearized quasiparticle equation Eq. (4.84). The frequency integral is the expensive step: a plasmon-pole model, which puts each element of $\operatorname{Im}\varepsilon^{-1}$ into a single delta function, is accurate for the energies; however, it produces no lifetimes, which need $\operatorname{Im}\varepsilon^{-1}$ resolved down to the gap. Appendix E.3 describes the method, its convergence parameters and the results for silicon.

4.8.6 The cutting rule

The imaginary parts found so far have all been golden-rule rates: the transition sums in Eq. (4.93), the emission and absorption rates of Worked example 4.13, the scattering rate Eq. (4.102). This is general, and it has a graphical form. After the frequency sums, a diagram at real frequency is a sum of terms, each a product of occupation factors divided by $\omega - E_{\mathrm{cs}} + \mathrm{i}0^+$, where E_{cs} is the total energy of a set of lines that together separate the two external points. Its imaginary part replaces each such denominator by $-\pi\delta(\omega - E_{\mathrm{cs}})$, by the Sokhotski–Plemelj formula Eq. (2.58). Drawing a line through the diagram across these propagators, the cut, puts them on shell. The pieces on either side of the cut are the amplitudes to reach the cut state and to return from it, and the imaginary part is π times their product summed over the cut states: a golden-rule rate. Cutkosky formulated this rule for relativistic amplitudes [51]. At finite temperature the occupation factors combine into rates in two ways, and Weldon gave the general form [52]; we show it for the two diagrams that the book uses.

For the boson self-energy Eq. (4.98), the cut runs through the electron line and the boson line. Write each thermal factor on its shell as a sum of products of occupations, $n_{\mathrm{B}} + 1 - n_{\mathrm{F}} = (1+n_{\mathrm{B}})(1-n_{\mathrm{F}}) + n_{\mathrm{B}} n_{\mathrm{F}}$ and $n_{\mathrm{B}} + n_{\mathrm{F}} = n_{\mathrm{B}}(1-n_{\mathrm{F}}) + (1+n_{\mathrm{B}})n_{\mathrm{F}}$. With $\xi' = \xi_{\boldsymbol{k}-\boldsymbol{Q}}$ and $n_{\mathrm{F}} = n_{\mathrm{F}}(\xi')$,

$$
\begin{aligned}
-2\operatorname{Im}\Sigma^{\mathrm{R}}(\boldsymbol{k},\omega) &= \Gamma_{\mathrm{out}}(\boldsymbol{k},\omega) + \Gamma_{\mathrm{in}}(\boldsymbol{k},\omega),\\
\Gamma_{\mathrm{out}} &= \frac{2\pi}{\mathscr{V}} \sum_{\boldsymbol{Q}} |g(\boldsymbol{Q})|^2 (1-n_{\mathrm{F}}) \Big[(1+n_{\mathrm{B}})\, \delta(\omega - \xi' - \omega_{\boldsymbol{Q}}) + n_{\mathrm{B}}\, \delta(\omega - \xi' + \omega_{\boldsymbol{Q}}) \Big],\\
\Gamma_{\mathrm{in}} &= \frac{2\pi}{\mathscr{V}} \sum_{\boldsymbol{Q}} |g(\boldsymbol{Q})|^2\, n_{\mathrm{F}} \Big[n_{\mathrm{B}}\, \delta(\omega - \xi' - \omega_{\boldsymbol{Q}}) + (1+n_{\mathrm{B}})\, \delta(\omega - \xi' + \omega_{\boldsymbol{Q}}) \Big].
\end{aligned}
\tag{4.103}
$$

Γ_{out} is the golden-rule rate Eq. (2.5) at which an electron in $\boldsymbol{k}$ with energy ω scatters out, by emission or absorption, into an empty state. Γ_{in} is the rate at which electrons scatter into an empty $\boldsymbol{k}$ from occupied states. On the shell, $n_{\mathrm{B}}/(1+n_{\mathrm{B}}) = \mathrm{e}^{-\beta\omega_{\boldsymbol{Q}}}$ and $n_{\mathrm{F}}/(1-n_{\mathrm{F}}) = \mathrm{e}^{-\beta\xi'}$, so that $\Gamma_{\mathrm{in}} = \mathrm{e}^{-\beta\omega}\Gamma_{\mathrm{out}}$, the fermion counterpart of detailed balance. The retarded Green's function is built from an anticommutator, and its width counts the decay of an added electron

and the filling of an added hole together. The density response is built from a commutator, and the corresponding combination is a difference.

Cutting rule

The imaginary part of the retarded continuation of a diagram is the sum over its cuts, each cut line on shell. For a fermion self-energy, $-2\,\mathrm{Im}\,\Sigma^{\mathrm{R}} = \Gamma_{\mathrm{out}}(1+\mathrm{e}^{-\beta\omega})$ with Γ_{out} the golden-rule rate out of the state; for a density response, $-2\,\mathrm{Im}\,R = (1-\mathrm{e}^{-\beta\omega})S$ with S the golden-rule sum into the cut states.

Worked example 4.14: The cut of the bubble

The bubble has a single cut, through its two propagators (Fig. 4.16a), and its cut state is one electron–hole pair. With orbitals γ in place of plane waves, the density vertex between γ and γ' is $\langle\gamma'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma\rangle$, and the frequency sum Eq. (D.51) gives

$$R^0(\boldsymbol{q},\mathrm{i}\omega_j^{\mathrm{B}}) = \frac{1}{\mathscr{V}}\sum_{\gamma\gamma'}\left|\langle\gamma'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma\rangle\right|^2\frac{n_\gamma - n_{\gamma'}}{\mathrm{i}\omega_j^{\mathrm{B}} + E_\gamma - E_{\gamma'}},$$

with $n_\gamma = n_{\mathrm{F}}(E_\gamma - \mu_{\mathrm{ch}})$; for plane waves the matrix element is $\delta_{\boldsymbol{k}',\boldsymbol{k}+\boldsymbol{q}}$ and this is Eq. (4.93). The denominator is $\omega - E_{\mathrm{cs}}$ with $E_{\mathrm{cs}} = E_{\gamma'} - E_\gamma$, the energy of the pair. Continuing to $\omega + \mathrm{i}0^+$ and cutting,

$$-2\,\mathrm{Im}\,R^0(\boldsymbol{q},\omega) = \frac{2\pi}{\mathscr{V}}\sum_{\gamma\gamma'}\left|\langle\gamma'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma\rangle\right|^2(n_\gamma - n_{\gamma'})\,\delta(\omega - E_{\gamma'} + E_\gamma).$$

The Pauli difference is a difference of two rates. On the shell, $n_{\gamma'}(1-n_\gamma) = \mathrm{e}^{-\beta\omega}n_\gamma(1-n_{\gamma'})$, and $n_\gamma - n_{\gamma'} = n_\gamma(1-n_{\gamma'}) - n_{\gamma'}(1-n_\gamma) = (1-\mathrm{e}^{-\beta\omega})\,n_\gamma(1-n_{\gamma'})$: an electron moves from occupied γ to empty γ', minus the reverse move. The factor $1-\mathrm{e}^{-\beta\omega}$ is that of the fluctuation–dissipation theorem Eq. (4.18), and what it multiplies is

$$S^0(\boldsymbol{q},\omega) = \frac{2\pi}{\mathscr{V}}\sum_{\gamma\gamma'}\left|\langle\gamma'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\gamma\rangle\right|^2 n_\gamma(1-n_{\gamma'})\,\delta(\omega - E_{\gamma'} + E_\gamma),$$

the golden-rule sum Eq. (4.21) over single electron–hole pairs, term by term. The terms with $\gamma' = \gamma$ have $n_\gamma - n_{\gamma'} = 0$ and drop out of the bubble; they are the elastic term of Eq. (4.21), which carries no dissipation. The diagram and the rate are the same object, read in two ways: the diagram assembles the pair from two propagators and a frequency sum, the golden rule counts the pairs directly.

For the electron gas the sum can be done by brute force. At the silicon valence density, $k_{\mathrm{F}} = 3.57\,\mathrm{keV}$, and at $q = q_0 = 3.73\,\mathrm{keV}$, we sample 4×10^6 occupied momenta $\boldsymbol{k}$, keep those with $\boldsymbol{k}+\boldsymbol{q}$ outside the Fermi sphere, and histogram their pair energies. The histogram agrees with the closed form Eq. (4.22), the cut of the bubble at $T=0$, to 0.6 percent rms across the continuum, which ends at $\omega_+ = 39.7\,\mathrm{eV}$ (Fig. 4.16b). Its total weight is the unblocked fraction $2\pi n_e(3q/4k_{\mathrm{F}} - q^3/16k_{\mathrm{F}}^3)$ of Worked example 4.3 to 10^{-4}.

The cutting rule is the reason a Feynman diagram can be trusted as a rate. Whatever is drawn, its imaginary part counts real final states with positive weights, and every approximation that is a selection of diagrams, with the cuts it allows, has a golden-rule reading. The random-phase chain cuts through electron–hole pairs and plasmons. The ladder, which Sect. 4.9 sums, cuts through pairs whose electron and hole are bound into excitons.

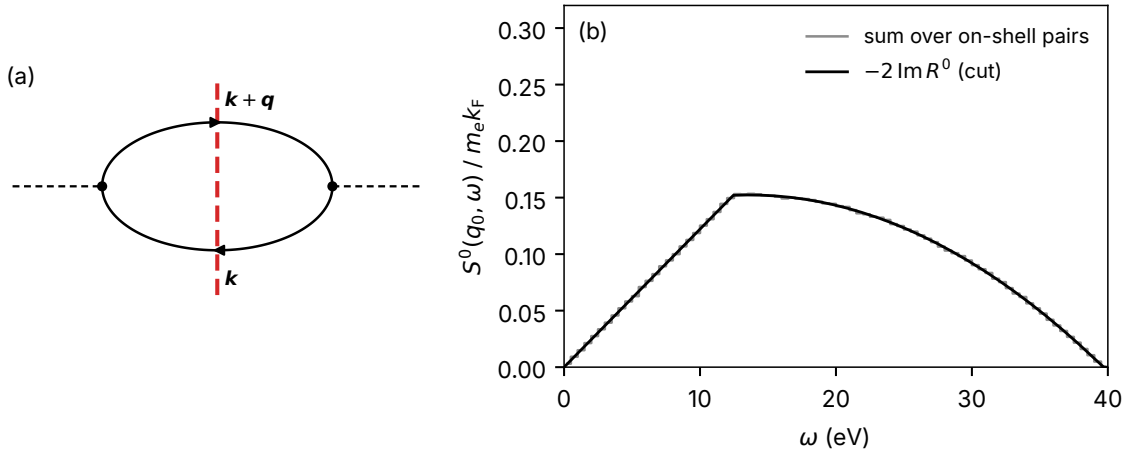


Figure 4.16: The cut of the bubble. (a) The dashed stubs mark the external momentum $\boldsymbol{q}$ at the two density vertices. The cut (dashed vertical line) passes through the electron line $\boldsymbol{k}+\boldsymbol{q}$ and the hole line $\boldsymbol{k}$ and puts both on shell, $\omega = \xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}}$; each half of the diagram is the amplitude for the density operator to create the pair. (b) The independent-electron structure factor of the free electron gas at $T = 0$ at the silicon valence density ($k_{\mathrm{F}} = 3.57\,\mathrm{keV}$, $E_{\mathrm{F}} = 12.5\,\mathrm{eV}$) and $q = q_0 = \alpha m_e = 3.73\,\mathrm{keV}$, in units of $m_e k_{\mathrm{F}}$. Gray: the golden-rule sum over on-shell electron–hole pairs, by Monte Carlo sampling of 4×10^6 momenta in the Fermi sphere, binned in 0.5 eV intervals. Black: $-2\,\mathrm{Im}\,R^0$, the cut of the bubble, in closed form, Eq. (4.22). The two agree to 0.6 percent rms; the continuum ends at $\omega_+ = qv_{\mathrm{F}} + q^2/2m_e = 39.7\,\mathrm{eV}$.

4.9 Excitons and the Bethe–Salpeter equation

The measured absorption spectrum of silicon has a peak of height 36 at 3.42 eV, and every approximation of Sect. 4.6 leaves only a shoulder of 13 to 15 in its place (Fig. 4.11). The bubble, the local fields and the adiabatic kernel all treat the electron lifted across the gap and the hole it leaves behind as independent. In fact the two carry opposite charges and attract, and the other electrons screen their attraction without cancelling it. In GaAs the attraction binds them into a hydrogen-like state 4.2 meV below the gap, which shows up as a sharp line at the absorption edge [53], and in silicon the bound pair lies 14.7 meV below the indirect gap [54]. Such correlated electron–hole pairs, bound or not, are excitons. For dark matter they matter exactly where light particles probe the target: in the first eV above the gap, where the loss function is smallest and the attraction moves spectral weight downward and adds lines below the continuum. We will write the attraction into an eigenvalue problem for electron–hole pairs, the Bethe–Salpeter equation, and see that a weakly bound pair is a hydrogen atom with a reduced mass and a screened charge. Its solutions then enter ε_2 and $S(\boldsymbol{q},\omega)$ at finite momentum transfer, and with them the hierarchy of approximations of Sect. 4.6 is complete: the differences between its levels become a number we can quote.

4.9.1 The electron–hole Hamiltonian

In the bubble of Sect. 4.8 one density vertex creates a pair, an electron in $(c, \boldsymbol{k}+\boldsymbol{q})$ and a hole in $(v, \boldsymbol{k})$, and the other vertex destroys it; such a pair is a transition, labeled $cv\boldsymbol{k}$ at fixed $\boldsymbol{q}$. Between the two vertices the pair propagates freely, and in an insulator at $T = 0$ each transition is an eigenstate of the problem with energy $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}}$. The ladder of Fig. 4.14(d) lets the electron and the hole interact any number of times on the way, and Appendix D.12

sums it with the rules of Appendix D.11. The two first-order insertions of Sect. 4.8 become two kernels, and with a statically screened interaction line the sum closes as a matrix equation in the space of transitions, which is an eigenvalue problem.

Bethe–Salpeter equation in the transition basis

At momentum transfer $\boldsymbol{q}$, the exciton energies $E_\lambda(\boldsymbol{q})$ and envelopes $A^\lambda_{cv\boldsymbol{k}}(\boldsymbol{q})$ solve

$$\sum_{c'v'\boldsymbol{k}'} H^{\mathrm{exc}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'}\, A^\lambda_{c'v'\boldsymbol{k}'} = E_\lambda\, A^\lambda_{cv\boldsymbol{k}},$$
$$H^{\mathrm{exc}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} = (\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}})\delta_{cc'}\delta_{vv'}\delta_{\boldsymbol{k}\boldsymbol{k}'} + g_{\mathrm{deg}} K^{\mathrm{x}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} - K^{\mathrm{d}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'}, \tag{4.104}$$

normalized to $\sum_{cv\boldsymbol{k}} |A^\lambda_{cv\boldsymbol{k}}|^2 = 1$, with $g_{\mathrm{deg}} = 2$ for the spin singlets that the density creates. The exchange and direct kernels are

$$K^{\mathrm{x}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} = \frac{1}{\mathcal{V}} \sum_{\boldsymbol{G}\neq 0} v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\, \rho_{cv}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{G})\, \rho_{c'v'}(\boldsymbol{k}', \boldsymbol{q}+\boldsymbol{G})^*,$$
$$K^{\mathrm{d}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} = \frac{1}{\mathcal{V}} \sum_{\boldsymbol{G}\boldsymbol{G}'} \rho_{cc'}(\boldsymbol{k}'+\boldsymbol{q}, \boldsymbol{p}+\boldsymbol{G})\, W^{\mathrm{scr}}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p}, \omega=0)\, \rho_{vv'}(\boldsymbol{k}', \boldsymbol{p}+\boldsymbol{G}')^*, \quad \boldsymbol{p} = \boldsymbol{k} - \boldsymbol{k}', \tag{4.105}$$

with the density vertices of Eq. (3.34), the screened interaction $W^{\mathrm{scr}}_{\boldsymbol{G}\boldsymbol{G}'} = (\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'} v_{\mathrm{C}}(|\boldsymbol{p}+\boldsymbol{G}'|)$ of Eq. (4.99), and quasiparticle band energies.

Eq. (4.104) is Eq. (D.66) with the kernels of Eq. (D.67), and Eq. (E.41) at finite $\boldsymbol{q}$. The exchange kernel is the one of Sects. 4.6 and 4.8, the Coulomb interaction without its $\boldsymbol{G} = 0$ term. For a momentum transfer beyond the first zone, $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$ in the unreduced indexing of Sect. 4.4, the kernels are written with $\boldsymbol{q}_r$ in place of $\boldsymbol{q}$, and the omitted term of K^{x} is then the one at $\boldsymbol{G} = \boldsymbol{G}_q$.

Each piece of H^{exc} has a plain meaning. The diagonal is the energy of the free pair. It must be a quasiparticle energy (Sect. 4.7 and Appendix E.3), because the self-energy insertions of the ladder are summed into the propagators, and with Kohn–Sham bands a scissor (Eq. (3.66)) takes their place. The direct kernel is the Coulomb interaction between the electron and the hole, screened by all the other electrons. The electron scatters from $(c, \boldsymbol{k}+\boldsymbol{q})$ to $(c', \boldsymbol{k}'+\boldsymbol{q})$ and the hole from $(v, \boldsymbol{k})$ to $(v', \boldsymbol{k}')$, so the pair keeps its total momentum $\boldsymbol{q}$ and its constituents exchange $\boldsymbol{p}$. Its diagonal element is the Coulomb energy of two positive densities, $|\psi_{c,\boldsymbol{k}+\boldsymbol{q}}|^2$ and $|\psi_{v\boldsymbol{k}}|^2$. Since that energy is positive, $-K^{\mathrm{d}}$ lowers the pair energy: it is the attraction. At large separation the screened interaction of two charges in an insulator is $e^2/(4\pi\varepsilon_\infty r)$, and this long-range tail is what binds the pairs of the next subsection. The exchange kernel describes something else. The pair annihilates at one point and a pair is re-created at another, and the Coulomb line between them is the microscopic Hartree potential of the induced density. It is the local-field kernel of Sect. 4.6, positive on the diagonal and repulsive. The long-range term $\boldsymbol{G} = 0$ is left out because it is the macroscopic field, which the definition of $\varepsilon_{\mathrm{mac}}$ already treats on its own (Eq. (D.70)); added back, it yields the full density response (Appendix D.12). Exchange is also what distinguishes the spin states. The density vertex couples to both spins alike, so K^{x} couples a transition of either spin to one of either spin and appears with g_{deg} in the singlet, while the triplet feels only $-K^{\mathrm{d}}$ and is not excited by a density coupling at all.

Behind Eq. (4.104) stand a kernel without the change of the screening, a static screened interaction and the Tamm–Dancoff approximation, and Appendix D.12 makes each one explicit. The kernel is the change of the Hartree potential plus the GW self-energy of Sect. 4.8 when the Green's function changes, with the change of W^{scr} itself neglected [55, 56]. The screened

interaction is static: W^{scr} is taken at zero frequency, which is good when the binding energies and the spread of transition energies that matter are small against the frequencies at which ε^{-1} changes, set in silicon by the plasma frequency of 16.6 eV. What is lost is the retarded response of the other electrons to the moving pair. And the coupling between the creation and the annihilation of a pair is dropped, the Tamm–Dancoff approximation, whose error on E_λ is of second order in the kernel divided by twice the gap. With these choices H^{exc} is Hermitian (Appendix D.12), and its eigenvectors are orthonormal.

We can now see how Eq. (4.104) contains what came before. Without kernels each transition is its own eigenstate and the response is the bubble of Eq. (4.63). With K^{d} switched off, $H^{\mathrm{exc}} = \mathbf{D} + g_{\mathrm{deg}}\mathbf{X}\mathbf{X}^\dagger$, where $\mathbf{D}$ is the diagonal matrix of transition energies and the column of $\mathbf{X}$ for each $\boldsymbol{G} \neq 0$ holds the vertices $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{G})$ times $[v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)/\mathscr{V}]^{1/2}$. Projected on the vertices, the resolvent $(\omega - H^{\mathrm{exc}})^{-1}$ becomes, by the Woodbury identity Eq. (C.60),

$$\mathbf{X}^\dagger(\omega - \mathbf{D} - g_{\mathrm{deg}}\mathbf{X}\mathbf{X}^\dagger)^{-1}\mathbf{X} = \mathbf{M}(\mathbf{1} - g_{\mathrm{deg}}\mathbf{M})^{-1}, \qquad \mathbf{M} = \mathbf{X}^\dagger(\omega - \mathbf{D})^{-1}\mathbf{X}.$$

Here $g_{\mathrm{deg}}\mathbf{M}$ is the resonant part of R^0 of Eq. (4.63) between the vertices at $\boldsymbol{G} \neq 0$, sandwiched between factors $\bar{v}^{1/2}$, and $g_{\mathrm{deg}}\mathbf{M}(\mathbf{1} - g_{\mathrm{deg}}\mathbf{M})^{-1}$, with the same factors, is the geometric series of $\bar{R} = R^0 + R^0\bar{v}\bar{R}$ on the block $\boldsymbol{G}, \boldsymbol{G}' \neq 0$; the head $\bar{R}_{\mathbf{00}}$ follows from the same equation with the row $\boldsymbol{G} = 0$ of R^0, which is Eq. (4.68). The random-phase approximation with local fields is thus the Bethe–Salpeter equation without the attraction, as Sect. 4.6 anticipated, and the attraction is the new physics of this level.

In practice the equation is set up from the same states as the random-phase response, with the screened interaction $\varepsilon^{-1}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p})$ computed beforehand for every difference $\boldsymbol{p}$ of two grid points. Its dimension is the number of transitions, $N_v N_c N_k$, and the $\boldsymbol{k}$ grid sets the cost: the kernel has $(N_v N_c N_k)^2$ elements, and a converged spectrum of silicon needs several thousand points in the zone. The spectrum then comes from an iterative (Lanczos–Haydock) solver, which needs products of H^{exc} with a vector and nothing else, and a full diagonalization is needed when individual envelopes are wanted. Appendix E.6 gives the sizes and costs, the choice of bands, and why symmetry is used only with care.

4.9.2 Wannier–Mott excitons

When the attraction is weak, the pair it binds is large, and a large pair samples just the bands near their extrema and the screening at long wavelengths. Eq. (4.104) then reduces to the hydrogen atom, a result of Wannier and of Mott [57, 58]. Take a direct gap at $\boldsymbol{k} = 0$, as in GaAs, with one isotropic conduction band of mass m^*, one valence band of hole mass m_{h}, and $\boldsymbol{q} \to 0$. Near the extrema the transition energy is

$$\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}} = E_{\mathrm{g}} + \frac{k^2}{2m^*} + \frac{k^2}{2m_{\mathrm{h}}} = E_{\mathrm{g}} + \frac{k^2}{2\mu_x}, \qquad \frac{1}{\mu_x} = \frac{1}{m^*} + \frac{1}{m_{\mathrm{h}}},$$

with μ_x the reduced mass of the exciton. Suppose the envelope $A_{\boldsymbol{k}}$ is confined to $|\boldsymbol{k}|$ small against the zone; we check this at the end. Then $\boldsymbol{p} = \boldsymbol{k} - \boldsymbol{k}'$ is small too, and in the direct kernel only $\boldsymbol{G} = \boldsymbol{G}' = 0$ survives: the terms with $\boldsymbol{G} \neq 0$ carry vertices between nearly equal states at momentum transfer $|\boldsymbol{p}+\boldsymbol{G}| \approx |\boldsymbol{G}|$, which are small, and a Coulomb factor that is finite instead of singular. The two vertices at $\boldsymbol{G} = 0$ are overlaps of the periodic parts of the same band at nearby $\boldsymbol{k}$, $\rho_{cc}(\boldsymbol{k}, \boldsymbol{p}) \simeq \rho_{vv}(\boldsymbol{k}, \boldsymbol{p}) \simeq 1$ (Eq. (3.34)), and the screened interaction at small $\boldsymbol{p}$ is the bare one divided by ε_∞. The direct kernel becomes

$$K^{\mathrm{d}}_{\boldsymbol{k}\boldsymbol{k}'} \simeq \frac{1}{\mathscr{V}} \frac{e^2}{\varepsilon_\infty |\boldsymbol{k} - \boldsymbol{k}'|^2}.$$

The exchange kernel, without its $\boldsymbol{G} = 0$ term, is however short-ranged in the relative coordinate: it acts only when the electron and the hole sit in the same cell. For a pair spread over many cells its effect is smaller than that of K^{d} by the probability of finding the two in one cell, and we drop it. It splits the singlet from the triplet by a small amount, the electron–hole exchange splitting. Eq. (4.104) now reads

$$\Big(E_{\mathrm{g}} + \frac{k^2}{2\mu_x}\Big) A_{\boldsymbol{k}} - \frac{1}{\mathscr{V}} \sum_{\boldsymbol{k}'} \frac{e^2}{\varepsilon_\infty |\boldsymbol{k} - \boldsymbol{k}'|^2} A_{\boldsymbol{k}'} = E\, A_{\boldsymbol{k}}.$$

Define the envelope function $F(\boldsymbol{r}) = \mathscr{V}^{-1/2} \sum_{\boldsymbol{k}} A_{\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$, the Fourier transform of the envelope in the relative coordinate $\boldsymbol{r}$ of the electron and the hole; by Parseval, $\int \mathrm{d}^3 r\, |F|^2 = \sum_{\boldsymbol{k}} |A_{\boldsymbol{k}}|^2 = 1$. Multiplying by $\mathscr{V}^{-1/2}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ and summing over $\boldsymbol{k}$ turns k^2 into $-\nabla^2$, and the kernel, a convolution in $\boldsymbol{k}$, into a product in $\boldsymbol{r}$ with the transform of $e^2/(\varepsilon_\infty k^2)$, which is $e^2/(4\pi\varepsilon_\infty r)$ by Eq. (A.3). The result is a Schrödinger equation for the relative motion,

$$\Big[-\frac{\nabla^2}{2\mu_x} - \frac{\alpha}{\varepsilon_\infty r}\Big] F(\boldsymbol{r}) = (E - E_{\mathrm{g}})\, F(\boldsymbol{r}), \tag{4.106}$$

the hydrogen atom with $m_e \to \mu_x$ and $\alpha \to \alpha/\varepsilon_\infty$. Its bound states follow from those of hydrogen by the same substitution.

Wannier–Mott excitons

Below the gap the pair has the levels $E_n = E_{\mathrm{g}} - E_x^b/n^2$, $n = 1, 2, \ldots$, with the binding energy and radius

$$E_x^b = \frac{\mu_x \alpha^2}{2\varepsilon_\infty^2} = \frac{\mu_x}{m_e} \frac{13.61\,\mathrm{eV}}{\varepsilon_\infty^2}, \qquad a_x = \frac{\varepsilon_\infty}{\alpha \mu_x} = \varepsilon_\infty \frac{m_e}{\mu_x}\, a_0, \tag{4.107}$$

and hydrogenic envelopes; the ground state is $F_{1s}(\boldsymbol{r}) = \mathrm{e}^{-r/a_x}/(\pi a_x^3)^{1/2}$, and $|F_{ns}(0)|^2 = 1/(\pi a_x^3 n^3)$. Above E_{g} the unbound solutions form the continuum of scattering states.

The approximation holds when the exciton is large compared with the lattice constant, $a_x \gg a$. Then the envelope occupies a region of size $1/a_x$ around the band extremum, where the bands are parabolic and the vertices close to one, and the electron and the hole are far enough apart for the macroscopic ε_∞ to screen them. Static screening needs a second condition: the internal motion of the pair, at frequencies $\sim E_x^b$, must be slow against the response that screens it. For the electrons, which respond at energies of the gap and above, this is the case whenever $E_x^b \ll E_{\mathrm{g}}$. The lattice of a polar crystal also screens, with the ions responding at the optical-phonon energies. If E_x^b is small against these, the ions follow the pair, and the static constant ε_0 with the lattice included (Sect. 5.5) takes the place of ε_∞; if it is large, the ions cannot follow and ε_∞ is right. In between, the interaction is not of the form $1/r$ at all.

In silicon the pair is indirect. Its lowest state has the electron in one of the conduction valleys near $\boldsymbol{k}_{\mathrm{c}}$ and the hole at the valence maximum at Γ, and its total crystal momentum is $\boldsymbol{k}_{\mathrm{c}}$ (Sect. 3.3). The relative motion is still governed by Eq. (4.106), now with the anisotropic valley mass and the degenerate, warped valence band. A spherical estimate brackets it. The kinetic energy of an s state averaged over directions involves $(1/3)(1/m_\parallel^* + 2/m_\perp^*)$, an electron mass of $0.26\, m_e$ from the masses of Table 3.1; with the heavy-hole mass between its values along [100] and [111], 0.28 and $0.72\, m_e$, Eq. (4.107) with $\varepsilon_\infty = 11.7$ gives 13 to 19 meV. The measured binding is 14.7 meV [54]. Read through Eq. (4.107), it corresponds to $\mu_x = 0.148\, m_e$ and to a radius $a_x = 4.2$ nm, nearly eight lattice constants, and the large-exciton picture

holds. In crystal momentum the envelope has a width $1/a_x = 0.024\,\text{Å}^{-1}$, about one percent of the zone, $|\boldsymbol{G}_{111}| = 2.0\,\text{Å}^{-1}$. A calculation of this state with Eq. (4.104) on a uniform grid would need hundreds of points per direction, which is why bound excitons near the gap are computed with grids refined around the band extrema, or in the effective-mass limit itself (Appendix E.6). Since silicon is not polar, $\varepsilon_0 = \varepsilon_\infty$ and the question of the lattice does not arise.

Worked example 4.15: The Wannier–Mott exciton of GaAs

The conduction band of GaAs at Γ has $m^* = 0.067\,m_e$, and the valence maximum is the fourfold level of heavy and light holes (Table 3.1). For a hydrogenic estimate the hole needs one mass. The two hole energies of Eq. (3.56) at the same $\boldsymbol{\kappa}$ differ by the sign of the square root, so their mean is $\gamma_1^{\text{Lu}}\kappa^2/2m_e$ in every direction: averaged over the two branches, the hole moves with the inverse mass γ_1^{Lu}/m_e. With $\gamma_1^{\text{Lu}} = 6.98$,

$$\frac{m_e}{\mu_x} = \frac{1}{0.067} + 6.98 = 21.9, \qquad \mu_x = 0.0457\,m_e.$$

With the measured electronic constant $\varepsilon_\infty = 10.88$ [59], Eq. (4.107) gives

$$E_x^b = 0.0457 \times \frac{13.61\,\text{eV}}{10.88^2} = 5.25\,\text{meV}, \qquad a_x = 10.88 \times \frac{0.529\,\text{Å}}{0.0457} = 12.6\,\text{nm},$$

22 lattice constants, which keeps the envelope well inside the parabolic region. The measured binding energy is 4.2 meV [53], and the estimate with ε_∞ overbinds by a quarter. The reason is the lattice. The optical phonons of GaAs lie near 35 meV [60, Fig. 2], eight times the binding energy, so the ions follow the slow relative motion of the pair and add their own screening. With the static constant $\varepsilon_0 = 12.9$ [60, Table I], the binding falls to $0.0457 \times 13.61\,\text{eV}/12.9^2 = 3.74\,\text{meV}$ and the radius grows to 14.9 nm. Most of the remaining difference from the measurement, 0.5 meV, comes from the part of the hole kinetic energy that the average over the two branches throws away. That part mixes heavy and light holes, has no first-order effect on the $1s$ state by symmetry, and lowers its energy in second order [61]. Either way the binding is a few meV, small against the gap of 1.52 eV, and the exciton is a line just below the absorption edge. Sect. 5.5 derives ε_0 and the lattice contribution to the screening.

At finite momentum transfer the pair moves as a whole. Take the transitions from $(v, \boldsymbol{k})$ to $(c, \boldsymbol{k}+\boldsymbol{q})$ with a conduction minimum at $\boldsymbol{k}_\text{c}$ (zero for GaAs) and $\boldsymbol{Q} = \boldsymbol{q} - \boldsymbol{k}_\text{c}$, small. The transition energy is $E_\text{g} + (\boldsymbol{k}+\boldsymbol{Q})^2/2m^* + k^2/2m_\text{h}$. With $\boldsymbol{k} = \boldsymbol{\kappa} - m_\text{h}\boldsymbol{Q}/(m^* + m_\text{h})$, the terms linear in $\boldsymbol{\kappa}$ cancel and it becomes $E_\text{g} + Q^2/[2(m^* + m_\text{h})] + \kappa^2/2\mu_x$. Since the direct kernel depends only on $\boldsymbol{k} - \boldsymbol{k}' = \boldsymbol{\kappa} - \boldsymbol{\kappa}'$, the relative motion obeys Eq. (4.106) unchanged, and

$$E_n(\boldsymbol{q}) = E_\text{g} - \frac{E_x^b}{n^2} + \frac{(\boldsymbol{q} - \boldsymbol{k}_\text{c})^2}{2(m^* + m_\text{h})}. \tag{4.108}$$

The exciton has the center-of-mass dispersion of a particle with the total mass of the pair, the same mass that governs the pair threshold $E_\text{min}(\boldsymbol{q})$ near a valley in Eq. (3.58). The bound levels follow the threshold, E_x^b/n^2 below it. In silicon the lowest exciton sits at $\boldsymbol{q} \simeq \boldsymbol{k}_\text{c}$, 14.7 meV below the indirect gap, at a crystal momentum that a photon cannot supply. Light reaches it only with a phonon (Sect. 6.5), while dark matter that transfers $\boldsymbol{q} \approx \boldsymbol{k}_\text{c}$ can create it on its own.

4.9.3 Excitonic absorption and scattering

The eigenpairs of Eq. (4.104) give the macroscopic dielectric function directly. In the resonant block the resolvent of the pair is $(\omega - H^{\mathrm{exc}} + \mathrm{i}0^+)^{-1} = \sum_\lambda A^\lambda A^{\lambda\dagger}/(\omega - E_\lambda + \mathrm{i}0^+)$, and projecting it on the density vertices at $\boldsymbol{q}$, as in Eq. (D.68), gives the response $\bar{R}_{\mathbf{00}}$ that is proper with respect to the long-range Coulomb term. With Eq. (4.68), $\varepsilon_{\mathrm{mac}} = 1 - v_{\mathrm{C}}(q)\bar{R}_{\mathbf{00}}$,

$$\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega) = 1 - \frac{g_{\mathrm{deg}}e^2}{Vq^2}\sum_\lambda \Big|\sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}\,\rho_{cv}(\boldsymbol{k},\boldsymbol{q})^*\Big|^2 \left[\frac{1}{\omega - E_\lambda + \mathrm{i}0^+} - \frac{1}{\omega + E_\lambda + \mathrm{i}0^+}\right], \quad (4.109)$$

where the antiresonant term, with its poles at $-E_\lambda$, is the mirror image that the Tamm–Dancoff approximation keeps. Its imaginary part at $\omega > 0$ and its static limit are

$$\begin{aligned} \varepsilon_2(\boldsymbol{q},\omega) &= \frac{\pi g_{\mathrm{deg}}e^2}{Vq^2}\sum_\lambda \Big|\sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}\,\rho_{cv}(\boldsymbol{k},\boldsymbol{q})^*\Big|^2 \delta(\omega - E_\lambda), \\ \varepsilon_{\mathrm{mac}}(\boldsymbol{q},0) &= 1 + \frac{2g_{\mathrm{deg}}e^2}{Vq^2}\sum_\lambda \frac{\big|\sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}\rho_{cv}(\boldsymbol{k},\boldsymbol{q})^*\big|^2}{E_\lambda}. \end{aligned} \quad (4.110)$$

At $\boldsymbol{q} \to 0$ the vertex is the optical one, $\rho_{cv} = \boldsymbol{q}\cdot\boldsymbol{v}_{cv}/(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}})$ (Eq. (3.43)), the factor q^2 cancels, and the first form is Eq. (D.71); the second is the static constant ε_∞. Without kernels each λ is a single transition, and the two reduce to Eqs. (4.65) and (4.66).

The new object is the exciton oscillator strength, the squared modulus of a coherent sum. Independent transitions contribute $|\rho_{cv}|^2$ each, whatever their phases. An exciton, however, contributes the square of the sum of the amplitudes of the transitions it is made of, weighted by A^λ. When the attraction lowers a state, the transitions in it add in phase, the low-lying excitons collect the oscillator strength of many transitions, and the weight moves to lower energy. Through the static limit in Eq. (4.110), with E_λ in the denominator, this also raises ε_∞. The f-sum rule of Sect. 4.3 fixes the total weight of the exact response, so the weight gained at low energies is lost higher up.

For a Wannier–Mott exciton the coherent sum has a closed form. The vertex varies little across the region of size $1/a_x$ that the envelope occupies, and with $\sum_{\boldsymbol{k}} A^\lambda_{\boldsymbol{k}} = \mathscr{V}^{1/2}F_\lambda(0)$, from the definition of F,

$$\Big|\sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}\,\rho_{cv}(\boldsymbol{k},\boldsymbol{q})^*\Big|^2 \simeq \mathscr{V}\,|F_\lambda(0)|^2\,|\rho_{cv}(\boldsymbol{k}_0,\boldsymbol{q})|^2, \quad (4.111)$$

with $\boldsymbol{k}_0$ the valence extremum. Only s states, whose envelope does not vanish where the electron and the hole coincide, are excited. Their strengths fall as $|F_{ns}(0)|^2 = 1/(\pi a_x^3 n^3)$, and since the levels crowd toward the gap as n^{-2}, the weight per unit energy just below E_{g} joins smoothly onto that of the continuum above it. The continuum itself is enhanced over the independent-particle one near the edge, since the attraction raises the probability of finding the electron at the hole. Both results are Elliott's [62]. Compared with independent particles, whose absorption at a direct edge rises from zero as $(\omega - E_{\mathrm{g}})^{1/2}$, the excitonic edge starts with a series of lines and a finite step.

Fig. 4.17 shows what the attraction does to silicon at $\boldsymbol{q} \to 0$, in a model calculation with the empirical pseudopotential states of Sect. 3.3. One property of these states decides how the figure is read. The bands were fitted to the measured positions of the absorption peaks, and those positions already contain the downward shift that the electron–hole attraction produces. Adding the attraction on top of the fitted bands counts that shift a second time, and every peak of the model lands too low. The figure therefore tests what the attraction does to the shape and the weight of the spectrum, not where the peaks lie. All three spectra come from

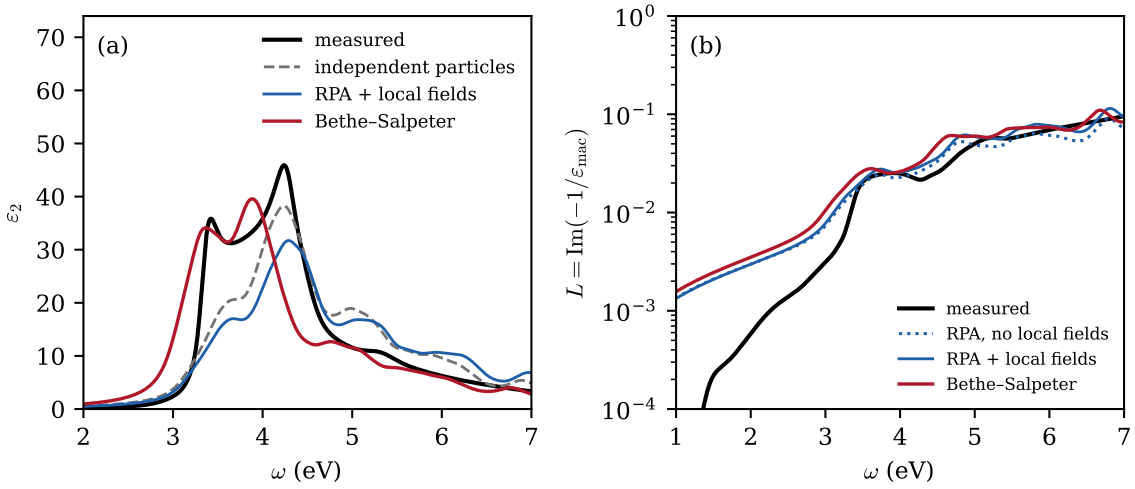


Figure 4.17: The electron–hole attraction in the optical spectrum of crystalline silicon at $\boldsymbol{q} \to 0$, in a model calculation. Band energies and states come from the local empirical pseudopotential of Cohen and Bergstresser [30], fitted to measured transition energies, as in Fig. 4.11. All curves come from the transition-basis Hamiltonian Eq. (4.104) in the Tamm–Dancoff approximation. The basis has 4 valence and 3 conduction bands on a shifted 12^3 grid of $\boldsymbol{k}$ points over the full zone (20,736 transitions); both kernels use the 59 shortest reciprocal vectors, and the spectra carry a Lorentzian broadening of 0.15 eV. The direct kernel uses the diagonal of the model's static inverse dielectric matrix in the random-phase approximation with local fields, computed with 26 conduction bands; spectra by the Lanczos–Haydock method. Black: measured at 300 K [19]. (a) ε_2: independent particles (dashed gray), the random-phase approximation with local fields, that is, exchange kernel only (blue), and the Bethe–Salpeter equation with both kernels (red). The attraction turns the shoulder near 3.4 eV into a peak; the whole spectrum shifts down because the fitted bands already contain part of the excitonic shift. (b) Loss function L on a logarithmic scale for the same three cases (dotted blue: no local fields, which has the independent-particle ε_2). Below about 3 eV the model curves are the Lorentzian tails of transitions that start at 3.13 eV on this grid, and lie above the measured L, which has no such tails.

one Hamiltonian, Eq. (4.104), on the same transitions: without kernels, with $g_{\mathrm{deg}} K^{\mathrm{x}}$ alone, and with $g_{\mathrm{deg}} K^{\mathrm{x}} - K^{\mathrm{d}}$. The screened interaction in K^{d} is the model's own random-phase one with local fields. On this grid the diagonal elements of K^{d} average 81 meV, a measure of the attraction between an electron and a hole spread over the crystal. The number depends on the grid, since the term at $\boldsymbol{p} \to 0$ grows with the volume of the zone per $\boldsymbol{k}$ point. The full random-phase result with local fields, computed from Eq. (4.68) on the same states without the Tamm–Dancoff approximation, differs from the spectrum with K^{x} alone by 3.4 percent of the peak, which is the Tamm–Dancoff error here.

The attraction restores the lower peak. At 3.42 eV independent particles give 15 and local fields 13; with the direct kernel the spectrum has a peak of height 34 at 3.36 eV, against the measured 36 at 3.42 eV. This peak, however, is not a bound state. The vertical transitions of silicon start near 3.1 eV, and the strength comes from many transitions along a region of the zone where conduction and valence bands run nearly parallel, which the attraction mixes coherently. The upper peak moves down to 3.88 eV, 0.36 eV below the measured 4.24 eV, which is the double counting described above. First-principles calculations avoid it by starting from quasiparticle energies of a *GW* calculation, which contain no excitonic shift. The Bethe–

Salpeter spectrum of silicon computed that way by Albrecht and coworkers places both main peaks at the measured energies with the measured ratio of their heights, and overestimates their absolute heights slightly [28]; Reining and coworkers show the same spectrum against the measured one [35, Fig. 1]. In the model, the static constant rises by 17 percent over the random-phase value with local fields in the same basis of bands. Part of that rise is again the double counting, since ε_∞ weights each exciton by $1/E_\lambda$, and the ratio is an upper estimate of what the attraction adds to the local-field value of Worked example 4.11.

For dark matter the loss function in panel (b) is the relevant quantity. At 3 eV the attraction raises L by 36 percent over the random-phase value with local fields; from 4 to 6 eV the two agree within 3.5 percent. The effect is concentrated in the first eV of the absorption, exactly where a slow particle deposits its energy.

At finite $\boldsymbol{q}$ the dark-matter rate needs the structure factor rather than ε_2. At $T = 0$ and $\omega > 0$, Eq. (4.45) gives $S = (2q^2/e^2)L$, with $L = \mathrm{Im}(-1/\varepsilon_{\mathrm{mac}}) = \varepsilon_2/|\varepsilon_{\mathrm{mac}}|^2$, and Eq. (4.110) then yields the exciton contribution

$$S(\boldsymbol{q},\omega) = \frac{2\pi g_{\mathrm{deg}}}{\mathscr{V}} \sum_\lambda \frac{\left|\sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}(\boldsymbol{q})\, \rho_{cv}(\boldsymbol{k},\boldsymbol{q})^*\right|^2}{|\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega)|^2}\, \delta\big(\omega - E_\lambda(\boldsymbol{q})\big). \tag{4.112}$$

Each exciton is a line at $E_\lambda(\boldsymbol{q})$ whose weight is its oscillator strength, screened by the factor $1/|\varepsilon_{\mathrm{mac}}|^2$ of Sect. 4.4. The exact statement is $S = (2q^2/e^2)\varepsilon_2/|\varepsilon_{\mathrm{mac}}|^2$, and the line form holds when the line's own contribution to $\varepsilon_{\mathrm{mac}}$ near its energy is small against the background, whose value then goes into the denominator. A line strong enough to change $\varepsilon_{\mathrm{mac}}$ near its own energy is moved up by the long-range exchange energy, since the pole of L sits at the zero of $\varepsilon_{\mathrm{mac}}$. This is the electronic analog of the splitting between the longitudinal optical (LO) and transverse optical (TO) phonons of a polar crystal, in which the macroscopic field of a longitudinal lattice vibration raises its frequency above that of the transverse one (Sect. 5.5). With the $\boldsymbol{G} = 0$ term kept in the kernel, the eigenvalues are these shifted energies and the weights are not divided by $|\varepsilon_{\mathrm{mac}}|^2$ (Eq. (D.69)). At the momentum transfers of a few keV that dark matter typically deposits, the Coulomb factor e^2/q^2 is small and the shift is negligible. Without kernels, Eq. (4.112) is the independent-particle structure factor Eq. (4.21) screened by $1/|\varepsilon_{\mathrm{mac}}|^2$, the random-phase result for the electrons.

For a Wannier–Mott exciton, Eq. (4.111) and Eq. (4.108) make the lines explicit,

$$S(\boldsymbol{q},\omega) = 2\pi g_{\mathrm{deg}}\, \frac{|\rho_{cv}(\boldsymbol{k}_0,\boldsymbol{q})|^2}{|\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega)|^2} \sum_n \frac{1}{\pi a_x^3 n^3}\, \delta\big(\omega - E_n(\boldsymbol{q})\big),$$

a weight per unit volume set by the probability $|F_{ns}(0)|^2$ of finding the electron at the hole, times the squared vertex between the two band extrema. In silicon the lowest line follows the indirect threshold of Sect. 3.3 at $\boldsymbol{q} \approx \boldsymbol{k}_{\mathrm{c}}$ and lies 14.7 meV below it. As a shift of the threshold this is 1.3 percent of the gap, small for the threshold mass of Sect. 3.3. As a redistribution of weight, however, it matters more: the edge of the zero-phonon response starts with a line, and the continuum above it is enhanced. Light dark matter samples exactly this region, and Sect. 10.5 computes the excitonic loss function at finite $\boldsymbol{q}$, while Sect. 13.1 places it against the phonon-assisted processes below the zero-phonon edge.

4.9.4 Levels of approximation

The approximations can now be listed with what each one adds and what it leaves out. *Independent particles*: the bubble Eq. (4.63) with quasiparticle or scissor-corrected energies; no screening, no interaction between the electron and the hole. *The random-phase approximation*: the head of the dielectric matrix screens the potential at $\boldsymbol{q}$ by the long-range Hartree field;

the absorption spectrum is unchanged, and the loss function is divided by $|\varepsilon|^2$. *Local fields*: the full Hartree matrix, which is $g_{\mathrm{deg}}K^{\mathrm{x}}$ in the transition basis; weight moves up in energy. *Time-dependent density-functional theory*: a kernel f_{xc} added to $\bar{v}$; the local adiabatic kernel changes little, and only a kernel with a long-range part can mimic the attraction [35]. *The Bethe–Salpeter equation*: the screened attraction $-K^{\mathrm{d}}$ on top of the local fields, with the energies of Sect. 4.7. Each approximation contains the previous one as a limit: $K^{\mathrm{d}} \to 0$ gives local fields, $\bar{v} \to 0$ the random-phase head, and the head alone at fixed energies the independent particles. What the Bethe–Salpeter equation in this form leaves out are the dynamical screening of the interaction, the coupling of pair creation and annihilation, and every process with a phonon, which Chap. 6 adds.

In silicon the approximations of Figs. 4.11 and 4.17 differ in characteristic ways. The static constant falls with the local fields and rises part of the way back with the adiabatic kernel (Worked example 4.11), and in the model the attraction raises it by 17 percent above the local-field value of 10.22, to about 11.9. The screening factor $1/\varepsilon_\infty^2$ at small q thus varies by up to a third across the hierarchy. At the lower absorption peak the spread is a factor of more than two, 13 to 34, and the loss function at 3 eV changes by a third between the adiabatic kernel and the Bethe–Salpeter equation alone. Above about 4 eV, which holds most of the f-sum, the approximations agree within the accuracy of the bands. As in Sect. 4.6, the spread is an estimate, not a bound. Light dark matter samples the first eV above the gap; for it the attraction is the leading correction, and the Bethe–Salpeter result is the reference against which the cheaper approximations are judged.

Summary

A weak field that couples to an operator O_b of a target in equilibrium changes the average of any other operator O_a through the retarded commutator, the Kubo formula $R_{ab}(\boldsymbol{q},t) = -(\mathrm{i}/\mathcal{V})\Theta(t)\langle[O_a(\boldsymbol{q},t)^\dagger, O_b(\boldsymbol{q},0)]\rangle$ (Sect. 4.1). It holds for any Hamiltonian and at any temperature. Its Lehmann representation has resonant and antiresonant poles at the excitation energies, its imaginary part is the power the target absorbs, and causality makes it analytic in the upper half of the frequency plane. Resonances, such as the plasmon and the quasiparticles, are poles continued below the real axis.

The rate of a dark-matter particle that couples through several operators is an integral over $\boldsymbol{q}$ of $\sum_{ab} \tilde{V}_a^* \tilde{V}_b S_{ab}(\boldsymbol{q}, \omega_{\boldsymbol{q}})$, with the dynamic structure factor S_{ab} the Fourier transform of an equilibrium correlation function (Sect. 4.2). For the density it is the transform of the Van Hove function, the probability of finding an electron at $\boldsymbol{r}$ at time t given one at the origin at time zero. The fluctuation–dissipation theorem, $(1 - \mathrm{e}^{-\beta\omega})S = -2\,\mathrm{Im}\,R$ on the diagonal, ties the rate to the response, and S_{ab} is a positive semidefinite matrix at every $(\boldsymbol{q},\omega)$, so that every rate is positive and every interference is bounded. Without interactions S is the sum over one-electron transitions of Sect. 2.5.

Exact constraints on any computed response follow from causality and from commutators (Sect. 4.3). The Kramers–Kronig relations rebuild the real part from the imaginary one, and the moments of S are equal-time commutators. The first is the f-sum rule, $\int \omega S$ fixed by the density and the mass whatever the interactions, which defines the plasma frequency, 16.6 eV for the valence electrons of silicon. The oscillator strengths of an atom obey the Thomas–Reiche–Kuhn rule, a check on tables and on ionization form factors.

The long-range Coulomb interaction is separated once and for all by the dielectric function (Sect. 4.4). The response to the total potential is the proper response $\tilde{R}$, $\varepsilon = 1 - v_{\mathrm{C}}\tilde{R}$ and $\varepsilon^{-1} = 1 + v_{\mathrm{C}}R$, and in a crystal both are matrices in the reciprocal-lattice vectors whose inverse defines the macroscopic $\varepsilon_{\mathrm{mac}} = 1/(\varepsilon^{-1})_{\mathbf{00}}$. The fluctuation–dissipation theorem then writes

the structure factor as the loss function, $S = (2q^2/e^2)L/(1 - \mathrm{e}^{-\beta\omega})$ with $L = \mathrm{Im}[-(\varepsilon^{-1})_{\mathbf{00}}]$, so that measured optical and energy-loss spectra enter the rate directly. Silicon has $\varepsilon_\infty = 11.7$, absorption peaks near 3.4 and 4.2 eV, and a loss peak of height 4.5 at 16.4 eV. Its static screening is metallic at a few keV and insulating below about 1 keV. The f-sum and inverse-moment rules fix the main loss peak, but not the low-energy side of L that slow particles, and dark matter, sample.

The homogeneous electron gas is the one system whose response is known in closed form (Sect. 4.5). The Lindhard function is its independent-particle bubble, the random-phase approximation screens it, and at long wavelength the gas oscillates collectively at ω_p with a dispersion rising as q^2, until the plasmon meets the pair continuum at $q_\mathrm{c} = 2.60$ keV and decays. A dark-matter particle reaches the plasmon energy only at 6.26 keV, beyond q_c, and the plasmon is kinematically out of reach. The static Lindhard screening at the typical momentum transfer q_0 is $\varepsilon = 2.10$. Mermin's relaxation-time form adds damping without destroying electrons.

In a crystal the bubble becomes the Adler–Wiser matrix of Bloch states (Sect. 4.6). The random-phase approximation with only the head leaves the absorption spectrum unchanged and screens the loss function. The full matrix adds local fields, the microscopic Hartree fields of the induced charge, and time-dependent density-functional theory adds an exchange–correlation kernel. In silicon the local fields lower ε_∞ by about a tenth, and the adiabatic local-density kernel restores part of it (Worked example 4.11). None of these approximations gives the lower absorption peak. Their spread is the many-body part of the uncertainty of a computed loss function, small for heavier dark matter and leading for light dark matter near the gap.

Green's functions describe one added electron or hole (Sect. 4.7). The spectral function is positive and normalized, the Dyson equation defines the self-energy Σ, and the poles of the Green's function near the real axis are quasiparticles with energy E_qp, residue z_qp and width Γ_qp. In imaginary time the thermal weight and the time evolution combine into one exponential, so that a single perturbation series serves both, and the analytic continuation from the Matsubara frequencies returns the retarded functions. The quasiparticle correction to Kohn–Sham energies repairs the gap, 0.602 eV with PBE against the measured 1.17 eV in silicon.

Feynman diagrams organize those series (Sect. 4.8). Wick's theorem reduces every term to products of free propagators, and the Feynman rules turn each diagram into an integral. The random-phase approximation is the chain of bubbles, the proper response the diagrams that one Coulomb line cannot cut, and the screened interaction $W^\mathrm{scr} = \varepsilon^{-1} v_\mathrm{C}$ replaces the bare one in the GW self-energy. The width of a hot electron is then a dark-matter-like rate with the Coulomb potential in place of $\tilde{V}$, and the cutting rule shows that every selection of diagrams has a golden-rule reading, $-2\,\mathrm{Im}\,R = (1 - \mathrm{e}^{-\beta\omega})S$.

The attraction between an excited electron and its hole is the last correction (Sect. 4.9). Summing the ladder diagrams with a statically screened interaction gives the Bethe–Salpeter equation, an eigenvalue problem in the space of transitions with a repulsive exchange kernel, the local fields, and an attractive direct kernel from W^scr. Near a band extremum it reduces to hydrogen with the reduced mass μ_x and the coupling $\alpha/\varepsilon_\infty$, with binding $E_x^b = \mu_x \alpha^2/(2\varepsilon_\infty^2)$. In silicon the measured binding, 14.7 meV, is not a prediction of the formula: it fixes $\mu_x = 0.15\, m_e$ and with it a radius of 4.2 nm. In GaAs, where the ions screen as well and ε_0 replaces ε_∞, the formula gives 3.7 meV against the measured 4.2 meV. Excitons enter ε_2 and the structure factor through coherent oscillator strengths, which move weight toward the gap, and as lines that follow the threshold at finite $\boldsymbol{q}$ with the center-of-mass dispersion of the pair. In the model calculation for silicon the attraction restores the lower absorption peak, raises ε_∞ toward the measured value and the loss function at 3 eV by a third; the spread across the

completed hierarchy is the many-body uncertainty a computed loss function can quote.

Throughout, the nuclei were held fixed. They move, and their motion adds a second set of excitations, phonons, with energies of tens of meV. It also adds a second contribution to screening in polar crystals and a coupling between electrons and the lattice. Chap. 5 builds the lattice dynamics on which all three rest.

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Chapter 5

Lattice dynamics

Dark matter lighter than 0.33 MeV cannot carry an electron of silicon across its 1.17 eV gap (Sects. 3.3 and 3.4), yet the crystal still has excitations within its reach. The nuclei vibrate, at energies up to 64 meV, and a quantum of that vibration is kinematically allowed for masses down to about 18 keV by the bound $2\omega/v_{\max}^2$ of Sect. 1.3; the acoustic vibrations reach lower still. Chaps. 3 and 4 held the nuclei fixed and computed the response of the electrons alone, so none of this appears in the dynamic structure factor they built. Letting the nuclei move raises questions that the static lattice could not ask. What potential do they move in, if the electrons that bind them also respond to their motion? How far does each nucleus stray from its site, even at a millikelvin, where zero-point motion is all there is? And what does a dark-matter particle that pushes on one nucleus set in motion in the whole crystal? The answers rest on one fact, the mass ratio $m_e/M \approx 2 \times 10^{-5}$, which separates the slow nuclei from the fast electrons and makes the vibrations of a crystal a nearly harmonic problem in a potential that density-functional theory computes. We will build from it, in the steps of Fig. 5.1, the phonons of silicon and of polar crystals, their quanta and their correlations, the quantities that every nuclear-recoil and phonon rate of the later chapters takes as input.

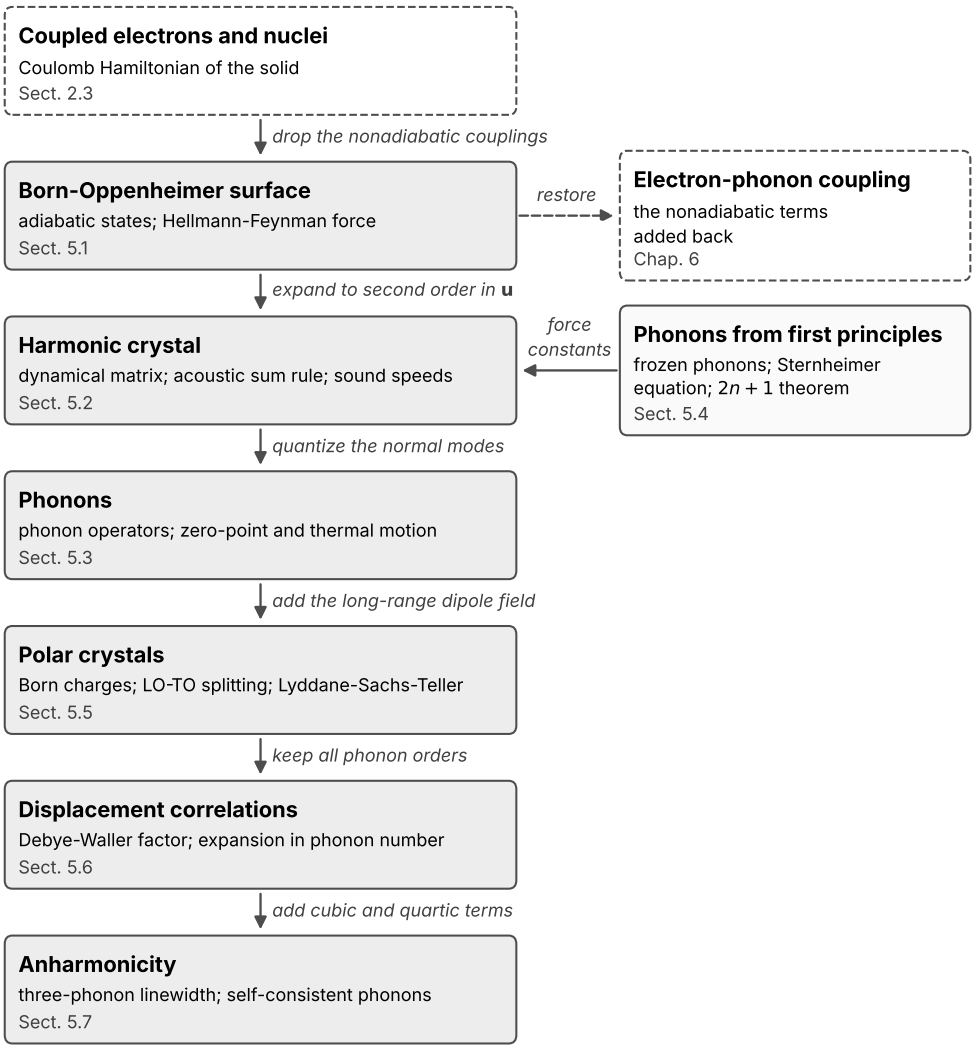


Figure 5.1: From coupled electrons and nuclei to the anharmonic crystal. Dropping the nonadiabatic couplings gives the Born–Oppenheimer surface (Sect. 5.1), which Chap. 6 restores; expanding it to second order in the displacements gives the harmonic crystal (Sect. 5.2), with force constants computed as a density response (Sect. 5.4); quantizing its normal modes gives phonons (Sect. 5.3). The long-range dipole field adds the Born charges and LO–TO splitting of polar crystals (Sect. 5.5); keeping all phonon orders gives the displacement correlations, the Debye–Waller factor and the multiphonon expansion (Sect. 5.6); cubic and quartic terms give linewidths and self-consistent phonons (Sect. 5.7).

5.1 The Born–Oppenheimer approximation

The optical phonon of silicon at the zone center vibrates at 15.5 THz [1], an energy of 64.1 meV and a period of 64.5 fs. An electron excited across the gap of 1.17 eV evolves on the time $2\pi/E_\mathrm{g} = 3.5\,\mathrm{fs}$, eighteen times faster. During one swing of the nuclei the electrons have time to settle, many times over, into the lowest state compatible with where the nuclei happen to be. This is the static lattice of Sect. 3.1 with the nuclei now allowed to move, slowly. We will turn the picture into an approximation with a definite small parameter, find the potential in which the nuclei move and the force that this potential exerts, and say what the approximation leaves out. The construction is due to Born and Oppenheimer [2], in the form that Born and Huang gave it for crystals [3, App. VIII]; the derivations are written out in Appendix D.13.

5.1.1 Adiabatic states

We start from the Coulomb Hamiltonian of electrons and point nuclei, Eq. (2.48). In a crystal it is natural to label the nucleus I of Sect. 2.3 by its cell and its place in the cell. The nucleus of species κ in the cell at the Bravais vector $\boldsymbol{R}$ sits at

$$\boldsymbol{X}_I = \boldsymbol{R} + \boldsymbol{\tau}_\kappa + \boldsymbol{u}_{\boldsymbol{R}\kappa}, \tag{5.1}$$

where $\boldsymbol{\tau}_\kappa$ is its basis position in the cell (Sect. 3.1) and $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ its displacement. We write u for the set of all displacements and $\nabla_{\boldsymbol{R}\kappa} = \partial/\partial\boldsymbol{u}_{\boldsymbol{R}\kappa}$. The equilibrium positions, about which u is measured, are fixed below. Of the terms of Eq. (2.48), only the kinetic energy of the nuclei contains $\nabla_{\boldsymbol{R}\kappa}$. We separate it from the rest,

$$H = -\sum_{\boldsymbol{R}\kappa} \frac{\nabla^2_{\boldsymbol{R}\kappa}}{2M_\kappa} + H_\mathrm{el}(u), \qquad H_\mathrm{el}(u) = T_e + V_{ee} + V_{eN}(u) + V_{NN}(u), \tag{5.2}$$

with T_e the kinetic energy of the electrons, the first term of Eq. (2.48), and V_{ee}, V_{eN} and V_{NN} the electron–electron, electron–nucleus and nucleus–nucleus terms of its bracket. At fixed u, $H_\mathrm{el}(u)$ acts on the electron coordinates alone, which we collect in r, and u enters it as a set of parameters. Its eigenstates are the *adiabatic states*,

$$H_\mathrm{el}(u)\,\Psi_i(r;u) = E_i(u)\,\Psi_i(r;u), \tag{5.3}$$

orthonormal at every u. Each is a state of all the electrons with the nuclei held at the positions u. In the absence of a magnetic field H_el is real and the Ψ_i can be chosen real, which we assume.

The adiabatic states are complete in r at each u, and an exact state of electrons and nuclei can be expanded in them with u-dependent coefficients,

$$\Psi(r,u) = \sum_i \Psi_i^\mathrm{nuc}(u)\,\Psi_i(r;u).$$

The coefficients Ψ_i^nuc are nuclear wavefunctions, one for each electronic state, and so far nothing has been approximated. Inserting the expansion into $H\Psi = E\Psi$ and projecting on Ψ_j at fixed u (Appendix D.13) leads to coupled equations for the nuclear wavefunctions,

$$\Big[-\sum_{\boldsymbol{R}\kappa} \frac{\nabla^2_{\boldsymbol{R}\kappa}}{2M_\kappa} + E_j(u)\Big]\Psi_j^\mathrm{nuc} + \sum_i \mathcal{N}_{ji}\,\Psi_i^\mathrm{nuc} = E\,\Psi_j^\mathrm{nuc}, \tag{5.4}$$

in which the electronic states talk to each other only through the operators

$$\mathcal{N}_{ji} = -\sum_{\boldsymbol{R}\kappa} \frac{1}{2M_\kappa}\Big[2\,\langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}\Psi_i\rangle\cdot\nabla_{\boldsymbol{R}\kappa} + \langle\Psi_j|\nabla^2_{\boldsymbol{R}\kappa}\Psi_i\rangle\Big]. \tag{5.5}$$

These are the *nonadiabatic couplings*: the first-derivative (or derivative) coupling, which acts on the nuclear wavefunction through $\nabla_{\boldsymbol{R}\kappa}$, and the second-derivative (or scalar) coupling. The brackets are integrals over r and functions of u. Both exist because the electronic state changes as the nuclei move, and both carry the factor $1/M_\kappa$.

The *Born–Oppenheimer approximation* drops them. The electrons then stay in one adiabatic state (for an insulator at low temperature, the ground state), and the nuclei move in its energy:

$$\Big[-\sum_{\boldsymbol{R}\kappa}\frac{\nabla^2_{\boldsymbol{R}\kappa}}{2M_\kappa}+E^{\mathrm{ad}}(u)\Big]\Psi_0^{\mathrm{nuc}}=E\,\Psi_0^{\mathrm{nuc}},\qquad E^{\mathrm{ad}}(u)\equiv E_0(u),\qquad \Psi\simeq\Psi_0^{\mathrm{nuc}}(u)\,\Psi_0(r;u). \quad (5.6)$$

The ground-state energy of the electrons at fixed nuclei, with the repulsion of the nuclei included, has become the potential energy of the nuclei. It is the *potential-energy surface*, or Born–Oppenheimer energy surface, $E^{\mathrm{ad}}(u)$, a function of the $3n_{\mathrm{at}}N$ coordinates of the nuclei of a crystal of N cells with n_{at} nuclei each. The picture is that of a slow nucleus pulling a cloud of electrons that rearranges instantly: the electrons follow the nuclei adiabatically, never get excited, and pass to the nuclei only the energy of their ground state. Everything the rest of this chapter says about phonons, from their frequencies to their anharmonic decay, is a statement about the shape of E^{ad} and about quantum mechanics in it. Born and Huang keep the diagonal term $\mathcal{N}_{00}$ as a small correction to E^{ad} and call the result the adiabatic approximation [3, App. VIII]. The field uses "adiabatic" and "Born–Oppenheimer" interchangeably, and so do we, since $\mathcal{N}_{00}$ turns out to be negligible (Appendix D.13). The off-diagonal couplings $\mathcal{N}_{j\neq0,0}$ are what let a moving nucleus excite the electrons; they are the subject of Sect. 6.1.

The surface is computed, not modeled. In a density-functional code, E^{ad} at a given set of nuclear positions is the Kohn–Sham total energy Eq. (D.22) plus the repulsion of the nuclei (evaluated as an Ewald sum), and every code prints it as its standard output. Evaluating it at displaced positions is the frozen-phonon method, and its derivatives at equilibrium are computed directly by density-functional perturbation theory (Sect. 5.4, Appendices E.1 and E.4). The Born–Oppenheimer approximation enters these calculations at one point only: the electrons are kept in their ground state at every configuration of the nuclei [4, Sect. III A].

5.1.2 Equilibrium positions and clamped nuclei

The nuclei of a crystal at rest sit at a minimum of the energy surface. We choose the equilibrium positions $\boldsymbol{R}+\boldsymbol{\tau}_\kappa$ so that $u=0$ is that minimum, $\nabla_{\boldsymbol{R}\kappa}E^{\mathrm{ad}}=0$ for every nucleus. In silicon symmetry does most of the work. Each atom of the diamond structure sits at a site of tetrahedral symmetry, and a vector left unchanged by every operation of the tetrahedral group is zero, so no atom feels a force at any lattice constant. The only free parameter is the lattice constant itself, and minimizing E^{ad} per cell with respect to it yields 5.401 Å in the local-density approximation and 5.477 Å with the PBE functional, against the measured 5.431 Å (Sects. 3.1 and 3.4). The minimum of E^{ad} is not exactly what a diffraction experiment measures, since the zero-point spread of the nuclei on an anharmonic surface shifts the mean lattice constant (Sect. 5.7).

Chaps. 3 and 4 went one step further than Eq. (5.6). They kept the electronic problem at the single configuration $u=0$, where $H_{\mathrm{el}}(0)$ has the full periodicity of the lattice, and gave the nuclei no motion at all. This is the *clamped-nuclei approximation*, or static lattice: the target is $H_{\mathrm{el}}(0)$, its states are $\Psi_i(r;0)$, and the nuclei enter only as the fixed periodic potential of the band structure (Sect. 3.2). The Born–Oppenheimer approximation lets the electronic state follow the nuclei as they move; the clamped-nuclei approximation does not let them

move. The difference is the electron–phonon coupling. Expanding the electronic Hamiltonian about the clamped one,

$$H_{\rm el}(u) = H_{\rm el}(0) + \sum_{\boldsymbol{R}\kappa} \boldsymbol{u}_{\boldsymbol{R}\kappa} \cdot \nabla_{\boldsymbol{R}\kappa} V_{eN}\Big|_{u=0} + \big[V_{NN}(u) - V_{NN}(0)\big] + O(u^2), \tag{5.7}$$

the linear term in V_{eN} is a one-body operator on the electrons times a displacement, the bare form of the electron–phonon Hamiltonian of Sect. 6.2 (the second-quantized form is Eq. (D.73)). The adiabatic states are the clamped-nuclei states dressed by this coupling to all orders in u, with the nuclei held still (Appendix D.13). The clamped-nuclei approximation therefore has no phonons, the only excitations below the gap. It has no phonon-assisted transitions, which let a slow dark-matter particle reach the indirect gap (Sect. 3.3), and no renormalization of the electronic energies by the zero-point motion of the nuclei, which in silicon closes the gap by about 60 meV (Appendix E.8). The dynamic structure factor of Chap. 4, computed at clamped nuclei, contains none of them.

5.1.3 The Born–Oppenheimer parameter

A nucleus and an electron in the same crystal feel forces of the same size, since the Coulomb interaction does not depend on mass. In the same potential, however, a heavy particle spreads much less than a light one, because confining it costs a kinetic energy that scales as $1/M$; the nuclei sit far more sharply at their sites than the electrons at theirs. Their zero-point spread, and the energy of their vibrations, are then set by a balance that contains the mass ratio m_e/M, and the question is which power of it.

Born and Oppenheimer measured lengths in units of the Bohr radius $a_0 = 1/(\alpha m_e) = 0.529\,\text{Å}$ and energies in units of $\alpha^2 m_e = 27.21\,\text{eV}$, the scales of the electronic problem. In these units $H_{\rm el}$ has no small parameter. Interatomic distances are a few a_0, electronic excitation energies are a fraction of $\alpha^2 m_e$, and $E^{\rm ad}$ changes by an amount of order $\alpha^2 m_e$ when a nucleus moves by a_0. Expanded about the minimum, the surface reads, schematically, $E^{\rm ad} = E_0^{\rm ad} + \frac{1}{2}\Phi^{(2)}u^2 + \frac{1}{6}\Phi^{(3)}u^3 + \cdots$, with force constants $\Phi^{(n)} \sim \alpha^2 m_e / a_0^n$ (Sect. 5.2). The nuclear kinetic energy is the only term that carries the mass ratio,

$$\frac{\nabla^2}{2M} = \alpha^2 m_e \, \frac{m_e}{M} \, \frac{a_0^2 \nabla^2}{2}.$$

A nuclear wavefunction spread over a length $\ell_{\rm zp}$ costs a kinetic energy $\alpha^2 m_e (m_e/M)(a_0/\ell_{\rm zp})^2$ and a potential energy $\alpha^2 m_e (\ell_{\rm zp}/a_0)^2$. The two balance at

$$\frac{\ell_{\rm zp}}{a_0} \sim \Big(\frac{m_e}{M}\Big)^{1/4} = \kappa_{\rm BO}, \qquad \omega_{\rm ph} \sim \alpha^2 m_e \, \kappa_{\rm BO}^2, \tag{5.8}$$

where the vibrational energy $\omega_{\rm ph}$ is the common size of the two terms. The fourth root is the point: the mass ratio of silicon is 2×10^{-5}, but the zero-point amplitude is a fraction $\kappa_{\rm BO} = 0.067$ of the electronic length, and a phonon a fraction $\kappa_{\rm BO}^2 = 4.4 \times 10^{-3}$ of the electronic energy.

With the scale of the displacements fixed, every other term of Eq. (5.4) is counted by the same substitution, $u \sim \kappa_{\rm BO} a_0$ and $\nabla \sim 1/(\kappa_{\rm BO} a_0)$ on a nuclear wavefunction, and Appendix D.13 carries the counting through term by term. The cubic term of the surface is $\sim \alpha^2 m_e \, \kappa_{\rm BO}^3$, one power of $\kappa_{\rm BO}$ below a phonon energy, the anharmonic small parameter of Sect. 5.7. Counted the same way, the derivative coupling is $\sim \alpha^2 m_e \, \kappa_{\rm BO}^3$ and the scalar one $\sim \alpha^2 m_e \, \kappa_{\rm BO}^4$, and since the derivative coupling connects the ground state to electronic states $\sim \alpha^2 m_e$ above it, the vibrational levels move first at $\alpha^2 m_e \, \kappa_{\rm BO}^6$.

Born–Oppenheimer hierarchy

With $\kappa_{\mathrm{BO}} = (m_e/M)^{1/4}$, the energies of electrons and nuclei are ordered as

$$\underbrace{\alpha^2 m_e}_{\text{electrons}} \;:\; \underbrace{\alpha^2 m_e\,\kappa_{\mathrm{BO}}^2}_{\text{phonons}} \;:\; \underbrace{\alpha^2 m_e\,\kappa_{\mathrm{BO}}^4}_{\text{anharmonic shifts}} \;:\; \underbrace{\alpha^2 m_e\,\kappa_{\mathrm{BO}}^6}_{\text{nonadiabatic shifts}} \;,$$

and the zero-point amplitude of the nuclei is $\ell_{\mathrm{zp}} \sim \kappa_{\mathrm{BO}} a_0$. The nonadiabatic correction to a vibrational energy is of relative order $\kappa_{\mathrm{BO}}^4 = m_e/M$.

This is the result of Born and Oppenheimer [2]. Appendix D.13 checks it on an exactly solvable model, one electron bound by a spring to a nucleus that is itself bound to a fixed point. There the exact squared vibrational frequency differs from the Born–Oppenheimer one by the relative amount m_e/M, and the frequency itself by $m_e/2M$: the electron is dragged along and adds its mass to the nucleus, and nothing else happens at that order.

The counting assumed that every electronic excitation costs an energy $\Delta E \sim \alpha^2 m_e$. When some cost less, the derivative coupling grows. Its exact form, derived below as Eq. (5.11), has ΔE in the denominator, and the amplitude with which a nuclear vibration mixes an excited electronic state j into the ground state is

$$\frac{\omega_{\mathrm{ph}}}{\Delta E}\,\frac{\ell_{\mathrm{zp}} F_{j0}}{\Delta E}, \tag{5.9}$$

with F_{j0} the matrix element of the force between the two states (Eq. (D.72)). The factor $\omega_{\mathrm{ph}}/\Delta E$ compares a phonon with an electronic excitation, and $\ell_{\mathrm{zp}} F_{j0}/\Delta E$ compares the work done by the force over a zero-point amplitude with the same excitation. At $\Delta E \sim \alpha^2 m_e$ they are κ_{BO}^2 and κ_{BO}, and we recover the κ_{BO}^3 of the derivative coupling. In an insulator the smallest ΔE is the gap, and the parameter that controls the adiabatic approximation is $\omega_{\mathrm{ph}}/E_{\mathrm{g}}$ rather than κ_{BO}^2. In a metal there are electronic excitations at every energy, the expansion in adiabatic states fails near the Fermi surface, and what survives is a weaker statement, Migdal's adiabatic theorem, with the small parameter $\omega_{\mathrm{ph}}/E_{\mathrm{F}}$ (Sect. 6.3). Two adiabatic levels that cross, as they can in a defect or during a violent recoil, make the denominator vanish, and there the approximation fails outright.

5.1.4 The Hellmann–Feynman force

The force on a nucleus is minus the gradient of the energy surface, and it takes a simple form. Write $E^{\mathrm{ad}} = \langle\Psi_0|H_{\mathrm{el}}|\Psi_0\rangle$ and differentiate with respect to the displacement of one nucleus. The derivative falls on the two states and on the Hamiltonian. Since $H_{\mathrm{el}}\Psi_0 = E^{\mathrm{ad}}\Psi_0$, the two terms in which it falls on a state add up to $E^{\mathrm{ad}}\,\nabla_{\boldsymbol{R}\kappa}\langle\Psi_0|\Psi_0\rangle$, which is zero because the state stays normalized. What is left is the *Hellmann–Feynman theorem* [5, 6],

$$\boldsymbol{F}_{\boldsymbol{R}\kappa} = -\nabla_{\boldsymbol{R}\kappa} E^{\mathrm{ad}} = -\big\langle\Psi_0\big|\nabla_{\boldsymbol{R}\kappa} H_{\mathrm{el}}\big|\Psi_0\big\rangle = -\big\langle\Psi_0\big|\nabla_{\boldsymbol{R}\kappa}(V_{eN} + V_{NN})\big|\Psi_0\big\rangle, \tag{5.10}$$

since only V_{eN} and V_{NN} depend on the positions of the nuclei. The change of the wavefunction does not enter. The operator V_{eN} is a sum of one-body terms, so its expectation value is fixed by the ground-state electron density $n(\boldsymbol{r})$, and the force is that of classical electrostatics. With Z_κ the charge number of the nucleus at $\boldsymbol{X}_I$,

$$\boldsymbol{F}_{\boldsymbol{R}\kappa} = Z_\kappa\, e^2\Big[\int \mathrm{d}^3 r\; n(\boldsymbol{r})\,\frac{\boldsymbol{r} - \boldsymbol{X}_I}{4\pi|\boldsymbol{r} - \boldsymbol{X}_I|^3} - \sum_{J\neq I} Z_{\kappa(J)}\,\frac{\boldsymbol{X}_J - \boldsymbol{X}_I}{4\pi|\boldsymbol{X}_J - \boldsymbol{X}_I|^3}\Big].$$

Each nucleus is pulled toward the electron cloud and pushed away from the other nuclei, and quantum mechanics enters only through the shape of the cloud. This is the electrostatic

theorem in the form Feynman stated it for molecules [6]. At equilibrium the two terms cancel. When a nucleus is displaced, the cloud is distorted, and the force that restores the nucleus comes from that distortion; computing it to first order in the displacement is exactly the linear-response problem of Sect. 5.4.

The same differentiation, projected on an excited state Ψ_j instead of Ψ_0, yields the derivative coupling of Eq. (5.5) (Appendix D.13):

$$\langle \Psi_j | \nabla_{\boldsymbol{R}\kappa} \Psi_i \rangle = \frac{\langle \Psi_j | \nabla_{\boldsymbol{R}\kappa} V_{eN} | \Psi_i \rangle}{E_i(u) - E_j(u)}, \qquad j \neq i. \tag{5.11}$$

The numerator is the matrix element of the force that the nucleus exerts on the electrons, the bare electron–phonon vertex, and the denominator is an electronic excitation energy. This is where the gap of the insulator enters the estimate Eq. (5.9).

In a periodic configuration the forces repeat from cell to cell. Displacing every nucleus of species κ by the same $\boldsymbol{u}_\kappa$ leaves the crystal periodic, and the force on each of those nuclei is the same in every cell. We write it as

$$\boldsymbol{F}_\kappa = -\frac{\partial E^{\mathrm{ad}}_{\mathrm{cell}}}{\partial \boldsymbol{u}_\kappa}, \tag{5.12}$$

with $E^{\mathrm{ad}}_{\mathrm{cell}}$ the energy surface per cell for that uniform displacement of the sublattice. This zone-center force is the one a calculation in the primitive cell returns, and the one whose response to a macroscopic electric field defines the Born effective charge of a polar crystal (Sect. 5.5).

The theorem holds for the Kohn–Sham energy as well, and this is what makes forces cheap. The Kohn–Sham energy is minimal with respect to the orbitals at self-consistency, so, as in the derivation above, the change of the orbitals drops out of its first derivative. The force is then the electrostatic expression above with the self-consistent Kohn–Sham density, or, with pseudopotentials, the derivative of the pseudopotential weighted by the density and the orbitals (Sect. 3.5). With a plane-wave basis, which does not move with the nuclei, there is no other term. A basis of atom-centered functions moves with the nuclei and adds a correction, the Pulay force [7]. A code computes the energy and every force at the cost of one self-consistent calculation (Appendix E.1).

Fig. 5.2(c) puts Eq. (5.10) to the test on silicon. A PBE calculation at the measured lattice constant moves the two atoms of the cell in opposite directions along [111], so that the bond pointing along [111] changes its length by u in every cell: the zone-center optical displacement. For $|u| \leq 0.2\,\text{Å}$, the Hellmann–Feynman force on the coordinate u agrees with $-\mathrm{d}E^{\mathrm{ad}}_{\mathrm{cell}}/\mathrm{d}u$, obtained by differentiating the computed energies, to better than 0.1 %. The curvature at the minimum is $\Phi_{\mathrm{opt}} = 13.6\,\text{eV}\,\text{Å}^{-2}$ per cell. Since the two atoms move in opposite directions, the coordinate u carries the reduced mass $M_\kappa/2$, and the optical frequency is $\omega_{\mathrm{opt}} = (2\Phi_{\mathrm{opt}}/M_\kappa)^{1/2} = 63.7\,\text{meV}$, against the measured 64.1 meV. The surface is visibly asymmetric: compressing the bond costs more energy than stretching it by the same amount. This is the cubic term of the expansion. It is allowed for the optical displacement along [111] and vanishes by symmetry along [100], and Worked example 5.1 checks its size against the counting.

Worked example 5.1: The small parameter of silicon

For natural silicon, $M = 26.16\,\text{GeV}$ and $m_e/M = 1.953 \times 10^{-5}$, so that

$$\kappa_{\mathrm{BO}} = 0.0665, \qquad \kappa^2_{\mathrm{BO}} = 4.42 \times 10^{-3}, \qquad \kappa^3_{\mathrm{BO}} = 2.94 \times 10^{-4}.$$

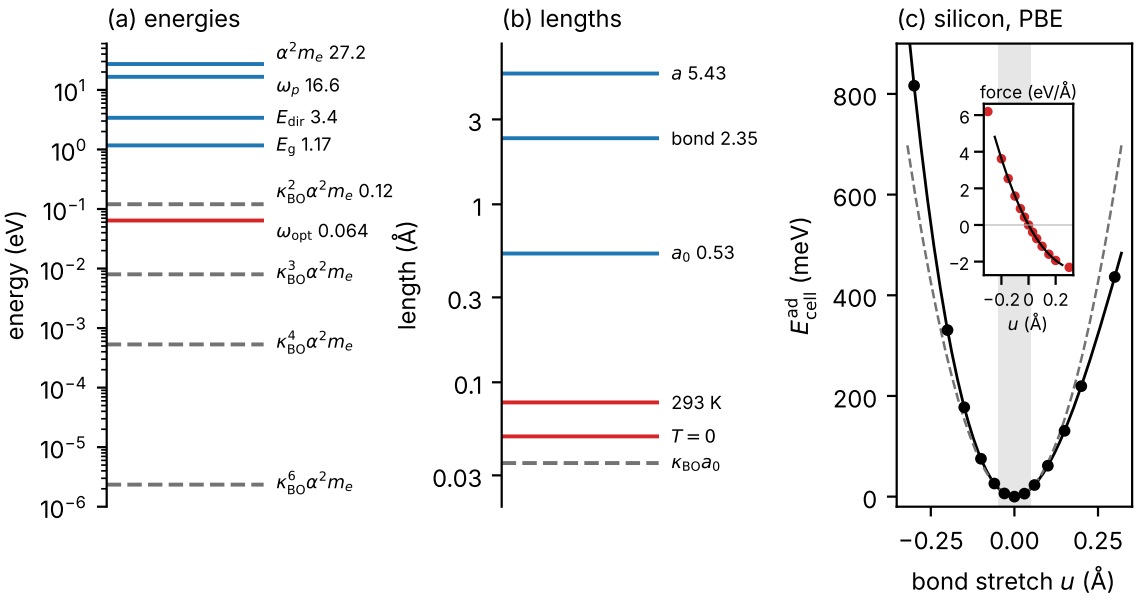


Figure 5.2: Scales of electrons and nuclei in silicon. (a) Energies on a logarithmic axis, in eV. Electronic (blue): the atomic unit $\alpha^2 m_e = 27.2\,\text{eV}$, the valence plasma energy $\omega_p = 16.6\,\text{eV}$, the direct gap $E_{\text{dir}} = 3.4\,\text{eV}$ and the indirect gap $E_{\text{g}} = 1.17\,\text{eV}$. Nuclear (red): the measured zone-center optical phonon, 64.1 meV (15.5 THz), the highest vibrational energy of the crystal. Gray dashed: the Born–Oppenheimer estimates $\kappa_{\text{BO}}^n \alpha^2 m_e$ for $n = 2$ (phonons), 3 (cubic anharmonicity), 4 (anharmonic level shifts) and 6 (nonadiabatic level shifts), with $\kappa_{\text{BO}} = (m_e/M)^{1/4} = 0.0665$ for ^{28}Si. (b) Lengths on a logarithmic axis, in Å: the lattice constant a, the bond length $\sqrt{3}a/4$ and the Bohr radius a_0 (blue). Red: the root-mean-square displacement of an atom along one Cartesian axis, $\langle u_x^2 \rangle^{1/2}$, at 293 K from a lattice-dynamical fit to neutron data and at $T = 0$ from the PBE phonon density of states of the calculation of panel (c) on a 4^3 grid of phonon wavevectors. Gray dashed: the estimate $\kappa_{\text{BO}} a_0$. (c) Energy per two-atom cell in a PBE calculation with an SG15 norm-conserving pseudopotential, a 50 Ry cutoff, an 8^3 grid of crystal momenta and $a = 5.431$ Å. The circles are computed at displacements in which the two atoms move in opposite directions along [111] and the [111] bond changes its length by u. The curves are a sixth-degree polynomial fit (solid) and the harmonic term $\frac{1}{2}\Phi_{\text{opt}} u^2$, $\Phi_{\text{opt}} = 13.6\,\text{eV}\,\text{Å}^{-2}$ (dashed), from a quartic fit over $|u| \le 0.1$ Å. Shaded: $|u| < \ell_{\text{zp}} = (M_\kappa \omega_{\text{opt}})^{-1/2} = 0.048$ Å, the zero-point length of the coordinate u. Inset: Hellmann–Feynman force on u (circles) and $-\text{d}E/\text{d}u$ from the fit (line).

The estimates Eq. (5.8) are $\kappa_{\mathrm{BO}} a_0 = 0.0352\,\text{Å}$ for the zero-point amplitude and $\kappa_{\mathrm{BO}}^2 \alpha^2 m_e = 0.120\,\mathrm{eV}$ for a phonon. We compare them with the crystal (Fig. 5.2).
Energies. The highest phonon of silicon, the optical mode at 64.1 meV, is $\omega_{\mathrm{opt}}/\alpha^2 m_e = 2.36 \times 10^{-3} = 0.53\,\kappa_{\mathrm{BO}}^2$: on the atomic scale for which it was built, the counting is right to a factor of two. Against the gap it does much worse. The ratio is $\omega_{\mathrm{opt}}/E_{\mathrm{g}} = 0.0548 = 12.4\,\kappa_{\mathrm{BO}}^2$, because the gap of silicon is 23 times smaller than $\alpha^2 m_e$. By Eq. (5.9), $\omega_{\mathrm{ph}}/E_{\mathrm{g}}$ is the ratio that controls the mixing of excited electronic states into the ground state, so the adiabatic approximation in silicon is an order of magnitude less safe than the mass ratio alone suggests. It is still safe: the gap is eighteen phonon energies away. A semiconductor whose gap approaches the phonon energies, however, or a metal, is a different matter (Sect. 6.3).
Lengths. The zero-point amplitude of one Cartesian component is $\langle u_x^2 \rangle^{1/2} = 0.050\,\text{Å}$ in the PBE calculation of Fig. 5.2, 1.4 times $\kappa_{\mathrm{BO}} a_0$. At 293 K, where thermal phonons add to the zero-point motion, a lattice-dynamical fit to neutron data gives 0.0771 Å, from $\langle u_x^2 \rangle = 0.005\,941\,\text{Å}^2$ [8], and the same calculation yields 0.081 Å, 5 % more, which suggests a similar accuracy for its zero-point value. Measured against the lattice constant, the zero-point amplitude is $\langle u_x^2 \rangle^{1/2}/a = 0.0091$, seven times smaller than κ_{BO} itself. The reason is the unit: the counting compares the amplitude with a_0, and the lattice constant of silicon is $10.3\,a_0$. As a fraction of the bond length, $4.4\,a_0$, the amplitude is 0.021.
Anharmonicity. The fit of Fig. 5.2(c) has the cubic coefficient $\Phi_{\mathrm{opt}}^{(3)} = -41.5\,\mathrm{eV\,Å}^{-3}$. At $u = \ell_{\mathrm{zp}} = 0.048\,\text{Å}$ the ratio of the cubic to the harmonic term of the energy is $|\Phi_{\mathrm{opt}}^{(3)}| \ell_{\mathrm{zp}} / 3\Phi_{\mathrm{opt}} = 0.049$, against the estimate $\kappa_{\mathrm{BO}} = 0.067$.
Nonadiabatic shifts. Their relative size is $\kappa_{\mathrm{BO}}^4 = m_e/M = 2 \times 10^{-5}$, about 1 µeV on the optical phonon. This is far below the anharmonic effects and below any resolution that matters for detection.
The upshot is that the powers of κ_{BO} order the effects correctly, with coefficients between one half and two when the comparison is made in atomic units. The coefficients become large when a ratio is taken to a scale of the crystal itself, such as the gap or the lattice constant, and those are the ratios that a calculation should quote.

The energy surface is the starting point for everything the nuclei do. Expanding it to second order about the equilibrium positions yields the force constants and the harmonic crystal of Sect. 5.2, and the cubic term visible in Fig. 5.2(c) returns as the anharmonic coupling of Sect. 5.7. The couplings $\mathcal{N}_{ji}$ dropped here are small in the counting but enhanced by a small gap. They come back in Sect. 6.1 as the electron–phonon interaction, and with them the processes in which a dark-matter particle that strikes a nucleus also excites the electrons, from phonon-assisted excitations to the Migdal effect.

5.2 The harmonic crystal

Neutron scattering finds that the vibrations of silicon fill a band from zero up to 64.1 meV, organized in six branches over the Brillouin zone. Three of them start at zero energy at the zone center and rise linearly, with slopes between 4.7 and $9.4\,\mathrm{km\,s^{-1}}$: these are sound waves. Three start at the optical frequency and stay near it. The density of vibrational states piles up in sharp peaks, near 17 meV, where the sound waves stop rising, and near 60 meV, and a dark-matter particle that deposits energy in single phonons sees these peaks directly in its spectrum. All of this follows from the shape of the energy surface $E^{\mathrm{ad}}(u)$ of Sect. 5.1 near its minimum. We will find these branches as its normal modes, see how translations and

rotations fix the acoustic branches at long wavelength, and end with the Debye model that the rest of the book uses for estimates.

5.2.1 Force constants and the dynamical matrix

The displacement of the nucleus κ in the cell $\boldsymbol{R}$ from its equilibrium site is $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ (Eq. (5.1)), and the crystal has n_{at} nuclei per cell, two for silicon (Sect. 3.1). Expanded about its minimum, the energy surface reads

$$\begin{aligned} E^{\mathrm{ad}}(u) &= E^{\mathrm{ad}}(0) + \frac{1}{2}\sum_{\boldsymbol{R}\kappa\alpha}\sum_{\boldsymbol{R}'\kappa'\beta} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},\boldsymbol{R}')\, u^{\alpha}_{\boldsymbol{R}\kappa} u^{\beta}_{\boldsymbol{R}'\kappa'} + O(u^3), \\ \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},\boldsymbol{R}') &= \left.\frac{\partial^2 E^{\mathrm{ad}}}{\partial u^{\alpha}_{\boldsymbol{R}\kappa}\,\partial u^{\beta}_{\boldsymbol{R}'\kappa'}}\right|_{u=0}, \end{aligned} \tag{5.13}$$

with Cartesian indices α, β. The linear term is absent, since every force vanishes at equilibrium. The coefficients $\Phi^{(2)}$ are the *force constants*: $-\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},\boldsymbol{R}')\, u^{\beta}_{\boldsymbol{R}'\kappa'}$ is the force along α on the nucleus $(\boldsymbol{R},\kappa)$ when the nucleus $(\boldsymbol{R}',\kappa')$ alone is displaced. As second derivatives they are symmetric under interchange of the two nuclei (the pair symmetry), $\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},\boldsymbol{R}') = \Phi^{(2),\beta\alpha}_{\kappa'\kappa}(\boldsymbol{R}',\boldsymbol{R})$, and real. A lattice translation moves every nucleus to an equivalent site and leaves the surface unchanged. As a result the force constants depend on the two cells only through their separation, $\Phi^{(2)}_{\kappa\kappa'}(\boldsymbol{R},\boldsymbol{R}') = \Phi^{(2)}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R}'-\boldsymbol{R})$. Dropping the terms beyond second order is the *harmonic approximation*. By the counting of Sect. 5.1, the cubic term is smaller than the quadratic one by a factor $\sim \kappa_{\mathrm{BO}}$ at the zero-point amplitude, about 5 % in silicon (Worked example 5.1), and we return to it in Sect. 5.7.

The nuclei move in this quadratic potential. We treat their motion classically here. For a Hamiltonian quadratic in the coordinates and momenta, the Heisenberg equations of motion of the displacement operators are the same linear equations as the classical ones, so the normal modes we find are also those of the quantum problem; Sect. 5.3 quantizes them. Newton's law for each nucleus is

$$M_\kappa\, \ddot{u}^{\alpha}_{\boldsymbol{R}\kappa} = -\frac{\partial E^{\mathrm{ad}}}{\partial u^{\alpha}_{\boldsymbol{R}\kappa}} = -\sum_{\boldsymbol{R}'\kappa'\beta} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},\boldsymbol{R}')\, u^{\beta}_{\boldsymbol{R}'\kappa'}, \tag{5.14}$$

$3n_{\mathrm{at}}N$ coupled linear equations for a crystal of N cells. Two steps reduce them to a small matrix problem: the mass weighting that works for any set of coupled oscillators, and the translation symmetry of the crystal. For a set of coordinates x_i with masses m_i and potential $\frac{1}{2}\sum_{ij} K_{ij} x_i x_j$, the mass-weighted coordinates $y_i = \sqrt{m_i}\, x_i$ obey $\ddot{y}_i = -\sum_j (K_{ij}/\sqrt{m_i m_j})\, y_j$. The matrix on the right is real and symmetric, so it has real eigenvalues ω^2 and orthonormal eigenvectors, and each eigenvector oscillates on its own as $\mathrm{e}^{-\mathrm{i}\omega t}$: a normal mode. The crystal is such a system, with one more property. Its matrix is invariant under lattice translations, and a translation-invariant matrix is diagonalized by plane waves in the cell index. We therefore try

$$u^{\alpha}_{\boldsymbol{R}\kappa}(t) = \frac{1}{\sqrt{M_\kappa}}\, e^{\alpha}_{\kappa}\, \mathrm{e}^{\mathrm{i}(\boldsymbol{Q}\cdot\boldsymbol{R}-\omega t)}, \tag{5.15}$$

with a wavevector $\boldsymbol{Q}$ and $3n_{\mathrm{at}}$ amplitudes e^{α}_{κ} to be found. The physical displacement is the real part, or the sum of the solutions at $\boldsymbol{Q}$ and $-\boldsymbol{Q}$. Inserting Eq. (5.15) into Eq. (5.14), dividing by $\sqrt{M_\kappa}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{Q}\cdot\boldsymbol{R}-\omega t)}$ and writing $\boldsymbol{R}' = \boldsymbol{R} + \boldsymbol{R}''$, the sum over $\boldsymbol{R}''$ no longer depends on $\boldsymbol{R}$. What is left is an eigenvalue problem in the $3n_{\mathrm{at}}$ amplitudes,

$$\sum_{\kappa'\beta} D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q})\, e^{\beta}_{\kappa'\nu}(\boldsymbol{Q}) = \omega^2_{\boldsymbol{Q}\nu}\, e^{\alpha}_{\kappa\nu}(\boldsymbol{Q}), \qquad D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q}) = \frac{1}{\sqrt{M_\kappa M_{\kappa'}}}\sum_{\boldsymbol{R}} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}, \tag{5.16}$$

the *dynamical matrix* $D^{\mathrm{dyn}}(\boldsymbol{Q})$, the Fourier transform of the mass-weighted force constants [3], [9, Chaps. 22 and 23], [4, Eq. (14)]; it is Eq. (A.23). Its eigenvalues are labeled by the *branch* index $\nu = 1, \ldots, 3n_{\mathrm{at}}$, and its eigenvectors $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$, with one three-vector for each nucleus of the cell, are the *polarization vectors*. The translation symmetry has turned $3n_{\mathrm{at}}N$ equations into N independent problems of size $3n_{\mathrm{at}}$, one for each $\boldsymbol{Q}$: six by six for silicon. The phase in Eq. (5.16) contains the cell vector $\boldsymbol{R}$ and not the position $\boldsymbol{R} + \boldsymbol{\tau}_\kappa$ of the nucleus. Some codes and texts use the full position; the two choices differ by the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{\tau}_\kappa}$ on each component of $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$ and give the same frequencies.

The properties of D^{dyn} follow from those of $\Phi^{(2)}$. By the pair symmetry of $\Phi^{(2)}$ and translation invariance, $\Phi^{(2),\beta\alpha}_{\kappa'\kappa}(\boldsymbol{0}, \boldsymbol{R}) = \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0}, -\boldsymbol{R})$, and replacing $\boldsymbol{R}$ by $-\boldsymbol{R}$ in the sum shows that $D^{\mathrm{dyn}}(\boldsymbol{Q})$ is Hermitian. Its eigenvalues $\omega^2_{\boldsymbol{Q}\nu}$ are therefore real and its eigenvectors orthonormal and complete,

$$\sum_{\kappa\alpha} e^{\alpha}_{\kappa\nu}(\boldsymbol{Q})^*\, e^{\alpha}_{\kappa\nu'}(\boldsymbol{Q}) = \delta_{\nu\nu'}, \qquad \sum_{\nu} e^{\alpha}_{\kappa\nu}(\boldsymbol{Q})\, e^{\beta}_{\kappa'\nu}(\boldsymbol{Q})^* = \delta_{\kappa\kappa'}\delta_{\alpha\beta}, \tag{5.17}$$

which contains the normalization $\sum_\kappa |\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2 = 1$ of Eq. (A.22). At a minimum of E^{ad} the quadratic form Eq. (5.13) is non-negative for every displacement, and so is $D^{\mathrm{dyn}}(\boldsymbol{Q})$ for every $\boldsymbol{Q}$; every $\omega^2_{\boldsymbol{Q}\nu}$ is then non-negative and $\omega_{\boldsymbol{Q}\nu}$ real. A computed $\omega^2_{\boldsymbol{Q}\nu} < 0$, usually printed as an imaginary frequency, signals that the structure is not at a minimum, or that the calculation is not converged. Since $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{R}} = 1$, $D^{\mathrm{dyn}}(\boldsymbol{Q} + \boldsymbol{G}) = D^{\mathrm{dyn}}(\boldsymbol{Q})$, and $\boldsymbol{Q}$ can be taken in the first Brillouin zone: a displacement wave of wavevector $\boldsymbol{Q} + \boldsymbol{G}$ moves the nuclei exactly as one of wavevector $\boldsymbol{Q}$, because the nuclei sit only at the lattice sites. Under Born–von Karman conditions there are N allowed $\boldsymbol{Q}$ in the zone (Sect. 3.1), and the crystal has $3n_{\mathrm{at}}N$ modes, one for each degree of freedom. Finally, since the force constants are real, $D^{\mathrm{dyn}}(-\boldsymbol{Q}) = D^{\mathrm{dyn}}(\boldsymbol{Q})^*$. As a result $\omega_{-\boldsymbol{Q}\nu} = \omega_{\boldsymbol{Q}\nu}$, and the phases can be chosen so that $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$, the convention of Eq. (A.22).

A quick estimate fixes the scale. The force constant between neighbors is $\Phi^{(2)} \sim \alpha^2 m_e / a_0^2 \sim 100\,\mathrm{eV\,\AA^{-2}}$ by the counting of Sect. 5.1, and the frequencies are $\omega \sim (\Phi^{(2)}/M_\kappa)^{1/2}$. For silicon the zone-center optical mode of Fig. 5.2 has $\Phi_{\mathrm{opt}} = 13.6\,\mathrm{eV\,\AA^{-2}}$ and $\omega = 63.7\,\mathrm{meV}$ in the same calculation: the estimate is high by a factor of seven in the force constant, and the frequencies are tens of meV.

The force constants are computed, not fitted. Each $\Phi^{(2)}$ is a second derivative of the Kohn–Sham total energy, and density-functional perturbation theory evaluates the whole matrix $D^{\mathrm{dyn}}(\boldsymbol{Q})$ at a given $\boldsymbol{Q}$ from the linear response of the electrons to a displacement wave (Sect. 5.4). A calculation computes it on a coarse uniform grid of $\boldsymbol{Q}$, transforms back to real-space force constants with Eq. (E.28), and evaluates Eq. (5.16) at any $\boldsymbol{Q}$ from them; Appendix E.4 describes the grids, the convergence tests and the cost. The phonons of silicon shown below come from such a calculation.

Worked example 5.2: The diatomic chain

The simplest crystal with both kinds of branch is a line of nuclei of masses M_1 and M_2, alternating, with period a_{c} and a spring of stiffness K between neighbors. The nucleus 1 of cell R is tied to the nucleus 2 of the same cell and to the nucleus 2 of the cell $R - a_{\mathrm{c}}$, and the energy is $\frac{1}{2}K\sum_R[(u_{R2} - u_{R1})^2 + (u_{R+a_{\mathrm{c}},1} - u_{R2})^2]$. Differentiating twice, the nonzero force constants from the cell at the origin are $\Phi^{(2)}_{11}(0,0) = \Phi^{(2)}_{22}(0,0) = 2K$, $\Phi^{(2)}_{12}(0,0) = \Phi^{(2)}_{12}(0,-a_{\mathrm{c}}) = -K$ and $\Phi^{(2)}_{21}(0,0) = \Phi^{(2)}_{21}(0,a_{\mathrm{c}}) = -K$. By Eq. (5.16),

$$D^{\mathrm{dyn}}(Q) = \begin{pmatrix} 2K/M_1 & -K(1+\mathrm{e}^{-\mathrm{i}Qa_{\mathrm{c}}})/\sqrt{M_1M_2} \\ -K(1+\mathrm{e}^{\mathrm{i}Qa_{\mathrm{c}}})/\sqrt{M_1M_2} & 2K/M_2 \end{pmatrix},$$

which is Hermitian, as it must be. With $|1+\mathrm{e}^{\mathrm{i}Qa_\mathrm{c}}|^2 = 4\cos^2(Qa_\mathrm{c}/2)$, the eigenvalues solve $\omega^4 - 2K(M_1^{-1}+M_2^{-1})\,\omega^2 + 4K^2\sin^2(Qa_\mathrm{c}/2)/(M_1M_2) = 0$:

$$\omega_{Q\pm}^2 = K\Big(\frac{1}{M_1}+\frac{1}{M_2}\Big) \pm K\Big[\Big(\frac{1}{M_1}+\frac{1}{M_2}\Big)^2 - \frac{4\sin^2(Qa_\mathrm{c}/2)}{M_1M_2}\Big]^{1/2}.$$

The lower branch vanishes at $Q=0$. For $Qa_\mathrm{c} \ll 1$, expanding the square root to first order in its small term, $\omega_{Q-}^2 = 2K\sin^2(Qa_\mathrm{c}/2)/(M_1+M_2) \simeq KQ^2a_\mathrm{c}^2/[2(M_1+M_2)]$: the frequency is linear in Q, a sound wave with speed $c = a_\mathrm{c}[K/2(M_1+M_2)]^{1/2}$, in which the chain moves as a medium of mass M_1+M_2 per length a_c. Its eigenvector at $Q=0$ is $\boldsymbol{e} \propto (\sqrt{M_1}, \sqrt{M_2})$, so that by Eq. (5.15) both nuclei move by the same amount: a rigid translation. This is the *acoustic* branch. The upper branch starts at $\omega_{0+}^2 = 2K/\mu_\mathrm{red}$, with $\mu_\mathrm{red} = M_1M_2/(M_1+M_2)$, and its eigenvector at $Q=0$ is $\boldsymbol{e} \propto (\sqrt{M_2}, -\sqrt{M_1})$: the two nuclei move against each other, $u_1/u_2 = -M_2/M_1$, and the center of mass of each cell stays at rest. This is the *optical* branch, so named because, when the two nuclei carry opposite charges, its oscillating dipole couples to light (Sect. 5.5). At the zone boundary, $Q = \pi/a_\mathrm{c}$, the off-diagonal elements vanish. Each mode then moves one species only, with $\omega^2 = 2K/M_1$ and $2K/M_2$, and between them lies a gap in which no wave propagates.
For equal masses the gap closes. Then $\omega_{Q\pm} = (2K/M)^{1/2}[1 \pm \cos(Qa_\mathrm{c}/2)]^{1/2}$, which is the single branch $\omega = 2(K/M)^{1/2}|\sin(Q'a_\mathrm{c}/4)|$ of a chain of identical nuclei with period $a_\mathrm{c}/2$, with $Q' = Q$ on the acoustic branch and $Q' = 2\pi/a_\mathrm{c} - Q$ on the optical one. The doubled cell has folded the zone in half, and the optical branch is the far half of the acoustic one. Silicon contains exactly this chain. For a wave along [100] with longitudinal polarization, every atomic plane perpendicular to [100] moves rigidly, and the planes, 1.358 Å ($a/4$) apart, alternate between the two sublattices of the diamond structure. The two kinds of plane are related by a translation by $a/4$ combined with a rotation by 90° about [100], and the rotation does not affect a longitudinal motion. The planes therefore act as identical masses, and the longitudinal branches along [100] form a folded chain with $a_\mathrm{c} = a/2$. Its zone boundary is $Q = 2\pi/a$, the point X. In the folded chain the longitudinal acoustic and optical branches must meet there, whatever the range of the interplanar forces, and in silicon they do, at 12.32 THz [1]. With springs between neighboring planes alone, fitted to the measured optical frequency at the zone center, $\omega_{0+} = 2(K/M)^{1/2} = 64.1\,\mathrm{meV}$, the model has $K = 6.90\,\mathrm{eV\,Å^{-2}}$ per atom and predicts

$$\omega_\mathrm{X} = \frac{\omega_{0+}}{\sqrt{2}} = 45.3\,\mathrm{meV}, \qquad c_\mathrm{LA} = \frac{a}{2}\Big(\frac{K}{4M}\Big)^{1/2} = \frac{a\,\omega_{0+}}{8} = 6.61\,\mathrm{km\,s^{-1}},$$

against the measured 50.95 meV at X and $8.43\,\mathrm{km\,s^{-1}}$ along [100] (Appendix C.18). The model gets the shape right; the numbers, however, are off by up to 22 %. The difference comes from forces between planes farther apart, which in silicon reach several bond lengths (Appendix E.4). For a chain of two different masses the same construction applies to GaAs, where the nearest-plane model opens a gap of $(M_\mathrm{As}/M_\mathrm{Ga})^{1/2} - 1 = 3.7\,\%$ at X between the two longitudinal branches.

5.2.2 The acoustic sum rule

The acoustic branch of the chain vanished at $Q=0$ because a rigid translation costs no energy. This holds for every crystal. Displacing all nuclei by the same vector $\boldsymbol{d}$ leaves the energy unchanged at any configuration, $E^\mathrm{ad}(u+d) = E^\mathrm{ad}(u)$, where $u+d$ means $\boldsymbol{u}_{\boldsymbol{R}\kappa} + \boldsymbol{d}$ for

every nucleus. Differentiating with respect to $u^{\alpha}_{\boldsymbol{R}\kappa}$ and to d^{β}, and setting $u = d = 0$, yields $\sum_{\boldsymbol{R}'\kappa'} \partial^2 E^{\mathrm{ad}} / \partial u^{\alpha}_{\boldsymbol{R}\kappa} \partial u^{\beta}_{\boldsymbol{R}'\kappa'} = 0$, that is

$$\sum_{\boldsymbol{R}\kappa'} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0}, \boldsymbol{R}) = 0 \qquad \text{for every } \kappa, \alpha, \beta. \tag{5.18}$$

This is the *acoustic sum rule*: the force on a nucleus when all others, and the nucleus itself, are moved together vanishes. Its consequence for the dynamical matrix follows at once. At $\boldsymbol{Q} = 0$, the vector $e^{\beta}_{\kappa'} = (M_{\kappa'}/M_{\mathrm{cell}})^{1/2} \hat{e}^{\beta}$, with $M_{\mathrm{cell}} = \sum_{\kappa} M_{\kappa}$ and any unit vector $\hat{\boldsymbol{e}}$, gives $\sum_{\kappa'\beta} D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0})\, e^{\beta}_{\kappa'} = (M_{\kappa} M_{\mathrm{cell}})^{-1/2} \sum_{\beta} \bigl[\sum_{\boldsymbol{R}\kappa'} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0}, \boldsymbol{R})\bigr] \hat{e}^{\beta} = 0$. Three independent choices of $\hat{\boldsymbol{e}}$ give three modes of zero frequency at the zone center, the *acoustic* branches, in which all nuclei move together. These are the polarization vectors of Appendix A.6, $\boldsymbol{e}_{\kappa\nu} = (M_{\kappa}/M_{\mathrm{cell}})^{1/2} \hat{\boldsymbol{e}}_{\nu}$. The other $3n_{\mathrm{at}} - 3$ branches, three in silicon, are the *optical* branches, finite at $\boldsymbol{Q} = 0$. In them the nuclei of a cell move against each other with their center of mass at rest: orthogonality to the three acoustic vectors in Eq. (5.17) is $\sum_{\kappa} \sqrt{M_{\kappa}}\, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{0}) = 0$, and the displacement $\boldsymbol{e}_{\kappa\nu}/\sqrt{M_{\kappa}}$ of Eq. (5.15) then has $\sum_{\kappa} M_{\kappa}\, \boldsymbol{u}_{\kappa} = 0$. In field-theory language the acoustic phonons are the Goldstone modes of the translation symmetry that the crystal breaks [10, p. 69]. A uniform translation is a different ground state at the same energy, and a translation that varies slowly in space costs an energy that vanishes as its wavelength grows.

A computed set of force constants violates Eq. (5.18) slightly, chiefly because the exchange–correlation potential is evaluated on a real-space grid that is not invariant under arbitrary translations. The acoustic frequencies at Γ then come out small but nonzero, 0.16 THz in the calculation of Fig. 5.3, and codes restore the rule by correcting the on-site constants (Appendix E.4).

5.2.3 Sound speeds

Near $\boldsymbol{Q} = 0$ the acoustic frequencies rise from zero, and we want the rate at which they do. Expanding the exponential in Eq. (5.16) in powers of $\boldsymbol{Q}$,

$$\begin{aligned} D^{\mathrm{dyn}}(\boldsymbol{Q}) &= D_0 + D_1(\boldsymbol{Q}) + D_2(\boldsymbol{Q}) + O(Q^3), \\ D^{\alpha\beta}_{n,\kappa\kappa'}(\boldsymbol{Q}) &= \frac{1}{\sqrt{M_{\kappa} M_{\kappa'}}} \sum_{\boldsymbol{R}} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0}, \boldsymbol{R})\, \frac{(\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R})^n}{n!}, \end{aligned} \tag{5.19}$$

and treating D_1 and D_2 as perturbations of $D_0 = D^{\mathrm{dyn}}(\boldsymbol{0})$, whose acoustic eigenvalue is triply degenerate at zero, leads to ω^2 as an expansion in Q. The acoustic frequencies are the eigenvalues of a 3×3 matrix in the space of the three acoustic vectors $\boldsymbol{a}_{\hat{\boldsymbol{e}}}$, with components $(M_{\kappa}/M_{\mathrm{cell}})^{1/2} \hat{\boldsymbol{e}}$.

The first-order term would make ω^2 linear in Q, and $\omega \propto Q^{1/2}$, which no crystal shows. Rotational invariance makes it vanish. Its matrix elements are

$$\boldsymbol{a}^{\dagger}_{\hat{\boldsymbol{e}}} D_1\, \boldsymbol{a}_{\hat{\boldsymbol{e}}'} = \frac{\mathrm{i}Q_{\gamma}}{M_{\mathrm{cell}}}\, \hat{e}^{\alpha} \hat{e}'^{\beta}\, T^{\alpha\beta\gamma}, \qquad T^{\alpha\beta\gamma} = \sum_{\kappa\kappa'\boldsymbol{R}} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0}, \boldsymbol{R})\, R^{\gamma},$$

with summation over repeated Cartesian indices. Two properties of T force it to zero. The first follows from the pair symmetry of $\Phi^{(2)}$. Writing $\Phi^{(2),\beta\alpha}_{\kappa\kappa'}(\boldsymbol{0}, \boldsymbol{R}) = \Phi^{(2),\alpha\beta}_{\kappa'\kappa}(\boldsymbol{0}, -\boldsymbol{R})$ and relabeling $\kappa \leftrightarrow \kappa'$ and $\boldsymbol{R} \to -\boldsymbol{R}$ in the sum shows $T^{\beta\alpha\gamma} = -T^{\alpha\beta\gamma}$. The second follows from rotations. A rigid infinitesimal rotation by the angle vector $\boldsymbol{\theta}$ moves every nucleus, at whatever position $\boldsymbol{X}_I$ it has, by $\boldsymbol{\theta} \times \boldsymbol{X}_I$ and leaves the energy unchanged. (For a crystal under Born–von Karman conditions a rotation is not a symmetry, but the force constants are those of a large free crystal, for which it is.) Hence $\sum_I \nabla_I E^{\mathrm{ad}} \cdot (\boldsymbol{\theta} \times \boldsymbol{X}_I) = 0$ at every configuration:

the total torque vanishes. Differentiating this identity with respect to the displacement of one nucleus and evaluating at equilibrium, where the forces vanish, leaves $\sum_I \Phi^{(2)}_{JI}\,(\boldsymbol{\theta}\times\boldsymbol{X}_I)=0$ for every J and every $\boldsymbol{\theta}$. In components, $\sum_{\boldsymbol{R}\kappa'}\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})\,X^{\gamma}_{\boldsymbol{R}\kappa'}$ is symmetric in $\beta\gamma$, with $\boldsymbol{X}_{\boldsymbol{R}\kappa'}=\boldsymbol{R}+\boldsymbol{\tau}_{\kappa'}$. Summed over κ, it is $T^{\alpha\beta\gamma}$ plus a term $\sum_{\kappa\kappa'}[\sum_{\boldsymbol{R}}\Phi^{(2)}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})]\tau^{\gamma}_{\kappa'}$, which vanishes because the sum rule Eq. (5.18) also holds with the roles of κ and κ' exchanged (by the same pair symmetry). So $T^{\alpha\beta\gamma}$ is symmetric in $\beta\gamma$ and antisymmetric in $\alpha\beta$, and a tensor with both properties is zero. Alternating the two properties six times returns to the start with a minus sign [3, Chap. V]:

$$T^{\alpha\beta\gamma}=-T^{\beta\alpha\gamma}=-T^{\beta\gamma\alpha}=T^{\gamma\beta\alpha}=T^{\gamma\alpha\beta}=-T^{\alpha\gamma\beta}=-T^{\alpha\beta\gamma}=0.$$

At second order two terms contribute. One is the expectation value of D_2 in the acoustic space. The other is D_1 taken twice, through the optical modes: D_1 does connect an acoustic vector to an optical one at $\boldsymbol{Q}=0$, with frequency ω_{o}, and second-order perturbation theory adds $-|\boldsymbol{o}^{\dagger}D_1\boldsymbol{a}_{\hat{\boldsymbol{e}}}|^2/\omega_{\mathrm{o}}^2$. Both are quadratic in $\boldsymbol{Q}$, so

$$\omega^2_{\boldsymbol{Q}\nu}=\Lambda^{\mathrm{ac}}_{\nu}(\hat{\boldsymbol{Q}})\,Q^2+O(Q^3),\qquad \omega_{\boldsymbol{Q}\nu}=c_{\nu}(\hat{\boldsymbol{Q}})\,Q,\qquad c_{\nu}(\hat{\boldsymbol{Q}})=\Lambda^{\mathrm{ac}}_{\nu}(\hat{\boldsymbol{Q}})^{1/2},\tag{5.20}$$

where $\Lambda^{\mathrm{ac}}_{\nu}(\hat{\boldsymbol{Q}})$ are the three eigenvalues of the 3×3 matrix

$$\Lambda^{\mathrm{ac},\alpha\beta}(\hat{\boldsymbol{Q}})\,Q^2=-\frac{1}{2M_{\mathrm{cell}}}\sum_{\kappa\kappa'\boldsymbol{R}}\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})\,(\boldsymbol{Q}\cdot\boldsymbol{R})^2-\sum_{\mathrm{o}}\frac{(\boldsymbol{a}^{\dagger}_{\alpha}D_1\boldsymbol{o})(\boldsymbol{o}^{\dagger}D_1\boldsymbol{a}_{\beta})}{\omega^2_{\mathrm{o}}},$$

with $\boldsymbol{a}_{\alpha}$ the acoustic vector polarized along the axis α and the sum over the optical modes at $\boldsymbol{Q}=0$. The frequency of an acoustic mode is linear in Q, with a sound speed $c_{\nu}(\hat{\boldsymbol{Q}})$ that depends on the direction of propagation and on the branch, and the polarizations $\hat{\boldsymbol{e}}_{\nu}(\hat{\boldsymbol{Q}})$ are the eigenvectors of Λ^{ac}. The first term is the stiffness of the crystal when every nucleus follows the long wave rigidly, the sum of the springs weighted by the squared lengths over which they are stretched. The second lowers it. In a crystal with more than one nucleus per cell, a strain pushes the sublattices against each other, and the optical coordinate relaxes in the long wave; this internal relaxation, which in the diamond structure reduces the shear stiffness, is what the second term describes [3, Chap. V]. In silicon the first-order coupling D_1 from acoustic to optical vectors vanishes for the longitudinal wave along [100], in which the planes of Worked example 5.2 move as identical masses and the two sublattices are not pushed against each other, but not for shear waves.

Since Λ^{ac} describes a long wave, it is a property of the crystal as an elastic continuum. Appendix C.18 derives the equation of motion of such a continuum, the Christoffel equation, with the elastic constants C_{ijkl} and the mass density ρ_T, and Eq. (5.20) is the same equation: $\rho_T\Lambda^{\mathrm{ac}}$, with $\rho_T=M_{\mathrm{cell}}/\Omega$, is the Christoffel matrix of the relaxed elastic constants. For a cubic crystal along [100] this reads $\rho_T c^2_{\mathrm{LA}}=C_{11}$ and $\rho_T c^2_{\mathrm{TA}}=C_{44}$ for the two degenerate transverse branches. The sound speeds of the dark-matter kinematics of Sect. 1.3 are these slopes, and the acoustic edge $m^{\mathrm{ac}}_{\mathrm{min}}$ of Sect. 9.2 uses them. The longitudinal speed is the largest, since compressing a medium costs more than shearing it. The scale is a vibrational frequency $(\Phi^{(2)}/M_{\kappa})^{1/2}$ times a distance of the order of the spacing between atomic planes: for silicon, the optical frequency times $a/4$ is $13\,\mathrm{km\,s^{-1}}$, against measured speeds of 4.7 to $9.4\,\mathrm{km\,s^{-1}}$. Note that $c/v_{\mathrm{max}}\sim10^{-2}$: sound is two orders of magnitude slower than the fastest dark matter, which is why an acoustic phonon takes only a small fraction of the kinetic energy (Sect. 1.3).

5.2.4 The phonons of silicon

Fig. 5.3(a) shows the six branches of silicon along the three lines of the zone on which they were measured: from Γ to X along [100], from X to Γ along [110] through K, and from Γ to

L along [111]. On these lines the symmetry of the crystal makes every polarization either parallel or perpendicular to $\boldsymbol{Q}$, and the branches are named accordingly: longitudinal and transverse acoustic (LA, TA), longitudinal and transverse optical (LO, TO). In a general direction the polarizations mix, and the names refer to the branch that is mostly longitudinal or mostly transverse. Along [100] and [111] the two transverse branches of each kind are degenerate by the symmetry of these axes; along [110] they split.

The measured frequencies at the high-symmetry points come from inelastic neutron scattering at room temperature [1, 11]. At Γ the three optical modes are degenerate, at 15.5 THz, or 64.1 meV. In a polar crystal the longitudinal one would be pushed up by the electric field it creates, the LO–TO splitting of Sect. 5.5; silicon has no Born charges and no splitting. At X the longitudinal acoustic and optical branches meet, at 12.32 THz (50.95 meV), as the folded chain of Worked example 5.2 requires. Near the zone boundary the transverse acoustic branches bend over and become almost flat, at 4.49 THz (18.6 meV) at X and 3.43 THz (14.2 meV) at L. The reason is the covalent bond. Stretching a bond is stiff, while changing the angle between two bonds is soft, and a short-wavelength transverse wave mostly bends bonds. Finally, the optical branches are narrow. From 15.5 THz at Γ they fall only to between 12.3 and 14.7 THz at the zone boundary. The reason is that the motion of the two atoms of a cell against each other is controlled by the stiff bond between them and depends only weakly on what the neighboring cells do.

The lines in Fig. 5.3(a) are a density-functional calculation with the PBE functional at the measured lattice constant, from force constants computed by density-functional perturbation theory (DFPT, Sect. 5.4) on a 6^3 grid of wavevectors. They follow the measurements closely. Of the eight measured frequencies at Γ, X and L, six agree to within 2 %, and the mean absolute difference over all eight is 2.6 %. The zone-center optical mode comes out at 15.40 THz (63.7 meV), 0.6 % below the measured 64.1 meV and the same as the frozen-phonon value of Fig. 5.2. The exceptions are the flat transverse acoustic modes, 8 % soft: 4.13 against 4.49 THz at X and 3.17 against 3.43 THz at L. PBE generally puts the phonons of silicon and germanium below the measured ones, a consequence of its underbinding [12, Sect. V]; in this calculation the deficit is largest on the branches set by bond bending, the weakest force constant of the crystal. The calculation describes a static lattice, while the measurement was made at room temperature, where the frequencies are lower by thermal expansion and by anharmonic shifts; these effects, and the widths $\gamma_{\boldsymbol{Q}\nu}$ that the harmonic crystal lacks entirely, are the subject of Sect. 5.7. A frequency also moves with the lattice constant at which it is computed, by an amount set by its Grüneisen parameter $\gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu} = -\partial \ln \omega_{\boldsymbol{Q}\nu} / \partial \ln \Omega$ (Appendix E.8). For most modes of silicon $\gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu} > 0$, and the PBE lattice constant, 0.85 % too large, would soften them; the flat transverse acoustic modes have $\gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu} < 0$ and would stiffen.

5.2.5 The density of states

For any quantity that sums over all phonons, such as a rate summed over final states or a thermal average, what matters is how many modes there are at each frequency. The phonon density of states per cell is

$$g_{\mathrm{ph}}(\omega) = \frac{1}{N} \sum_{\boldsymbol{Q}\nu} \delta(\omega - \omega_{\boldsymbol{Q}\nu}) = \frac{\Omega}{(2\pi)^3} \sum_{\nu} \int_{\mathrm{BZ}} \mathrm{d}^3 Q \; \delta(\omega - \omega_{\boldsymbol{Q}\nu}), \qquad \int_0^\infty \mathrm{d}\omega \, g_{\mathrm{ph}}(\omega) = 3 n_{\mathrm{at}}, \tag{5.21}$$

where the sum over the N wavevectors of the zone became an integral with the density $\mathscr{V}/(2\pi)^3$ of Sect. 3.1, $\mathscr{V} = N\Omega$. Its integral counts the $3n_{\mathrm{at}}$ modes per cell, six for silicon. A density of states per atom, normalized to three, is $g_{\mathrm{ph}}/n_{\mathrm{at}}$. In silicon, Fig. 5.3(b), g_{ph} has a sharp peak at 17 meV from the flat transverse acoustic branches near the zone boundary. Above it lies a

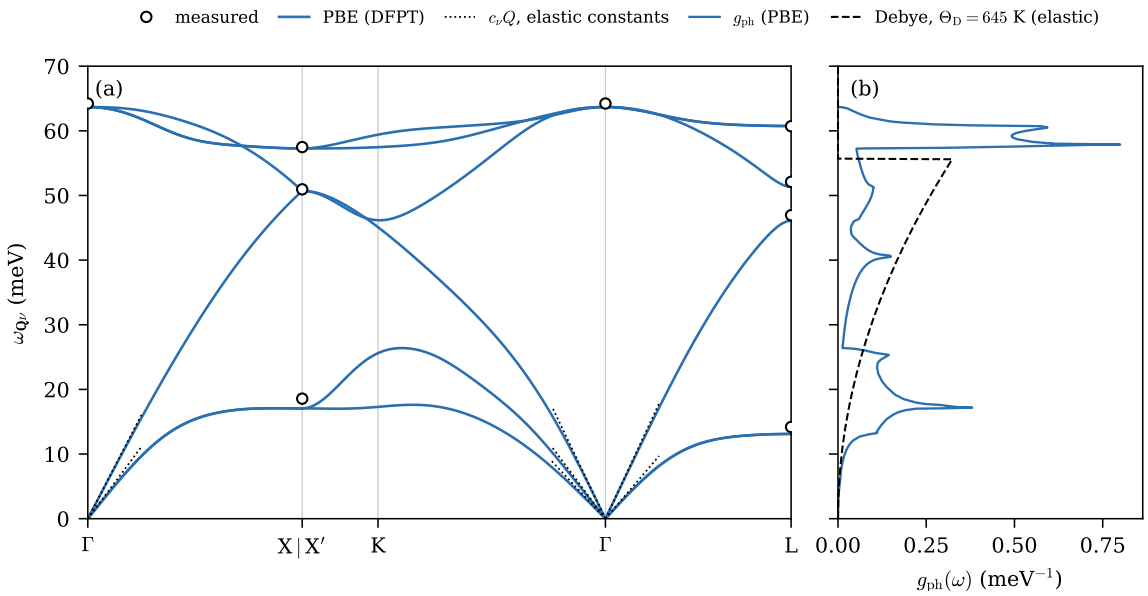


Figure 5.3: Phonons of silicon. (a) Dispersion along [100] from Γ to X, along [110] from X′ $= (1, 1, 0)2\pi/a$, which is equivalent to X, through K to Γ, and along [111] from Γ to L. Lines: density-functional perturbation theory with the PBE functional, an SG15 norm-conserving pseudopotential, a 60 Ry cutoff, a 12^3 grid of crystal momenta, the measured lattice constant $a = 5.431$ Å and the natural mass of silicon; force constants from a 6^3 grid of wavevectors, with the acoustic sum rule imposed, interpolated to the path. Circles: frequencies measured by inelastic neutron scattering at room temperature at Γ, X and L [1, 11], as tabulated in Ref. [13, Table I]. Dotted: $\omega = c_\nu Q$ with the sound speeds of the measured elastic constants (Worked example 5.3), drawn up to $Q = 0.25\,(2\pi/a)$. (b) Phonon density of states per two-atom cell, normalized to six modes, from the same force constants on a 48^3 grid (tetrahedron integration), and the Debye model Eq. (5.24) with $\bar{c}$ from the measured elastic constants, $\omega_{\mathrm{D}} = 55.6$ meV, in the convention of one wavevector per atom. One THz corresponds to 4.136 meV.

structured region between 25 and 52 meV from the longitudinal acoustic branch and the lower optical branches. The tallest peaks, at 58 and 61 meV, come from the flat transverse optical branches. The peaks are where the branches are flat. Near a frequency at which $\nabla_{\boldsymbol{Q}}\omega_{\boldsymbol{Q}\nu}$ vanishes, many wavevectors share nearly the same frequency, and in three dimensions the density of states has a kink with an infinite slope there, a Van Hove singularity [9, Chap. 23]. The large peak of the transverse acoustic branches is what makes silicon's low-energy phonons so numerous: a dark-matter particle that can deposit 20 meV in single phonons finds a fifth of all modes of the crystal, 22 % in the calculation, below that energy.

At low frequency only the acoustic branches contribute, with $\omega_{\boldsymbol{Q}\nu} = c_\nu(\hat{\boldsymbol{Q}})Q$. Writing $\mathrm{d}^3Q = Q^2\mathrm{d}Q\,\mathrm{d}\hat{\boldsymbol{Q}}$ and integrating over Q with $\int Q^2\mathrm{d}Q\,\delta(\omega - c_\nu Q) = \omega^2/c_\nu^3$,

$$g_{\mathrm{ph}}(\omega) \simeq \frac{\Omega\,\omega^2}{2\pi^2}\sum_\nu \langle c_\nu^{-3}\rangle, \qquad \langle c_\nu^{-3}\rangle = \int \frac{\mathrm{d}\hat{\boldsymbol{Q}}}{4\pi}\, c_\nu(\hat{\boldsymbol{Q}})^{-3}, \tag{5.22}$$

an ω^2 law with a coefficient fixed by the sound speeds alone, averaged over directions. It holds as long as the dispersion is linear, which in silicon means up to about 7 meV, where the computed g_{ph} exceeds the ω^2 law by 10 %.

5.2.6 The Debye model

Debye's model extends Eq. (5.22) to the whole spectrum. It replaces the $3n_{\mathrm{at}}$ branches by three, all with the linear dispersion $\omega = \bar{c}Q$ of a single averaged speed $\bar{c}$. It also replaces the zone by a sphere of radius Q_{D} that holds one wavevector per atom, $n_{\mathrm{at}}N$ in all, so that the three branches carry all $3n_{\mathrm{at}}N$ degrees of freedom [9, Chap. 23]. Counting with the density $\mathcal{V}/(2\pi)^3$, $(4\pi Q_{\mathrm{D}}^3/3)\,N\Omega/(2\pi)^3 = n_{\mathrm{at}}N$, so that

$$Q_{\mathrm{D}} = \Big(\frac{6\pi^2 n_{\mathrm{at}}}{\Omega}\Big)^{1/3}, \qquad \omega_{\mathrm{D}} = \bar{c}\,Q_{\mathrm{D}}, \qquad \Theta_{\mathrm{D}} = \omega_{\mathrm{D}}, \tag{5.23}$$

with the Debye frequency ω_{D} and the Debye temperature Θ_{D}, which in natural units are the same number. The density of states of the model follows from Eq. (5.22) with $c_\nu = \bar{c}$, cut off at ω_{D}:

$$g_{\mathrm{D}}(\omega) = \frac{3\Omega\,\omega^2}{2\pi^2\bar{c}^3}\,\Theta(\omega_{\mathrm{D}} - \omega) = \frac{9n_{\mathrm{at}}\,\omega^2}{\omega_{\mathrm{D}}^3}\,\Theta(\omega_{\mathrm{D}} - \omega), \tag{5.24}$$

where the second form uses Eq. (5.23); its integral is $3n_{\mathrm{at}}$, as it must be. Matching the low-frequency limit Eq. (5.22) exactly fixes the speed,

$$\frac{3}{\bar{c}^3} = \sum_\nu \langle c_\nu^{-3}\rangle. \tag{5.25}$$

This is the definition we use. It makes the model exact wherever only long acoustic waves matter, as in the heat capacity at low temperature, which then follows Debye's T^3 law with this Θ_{D}, and in a dark-matter rate dominated by the lowest phonon energies. Since c^{-3} is averaged, the slowest branches dominate $\bar{c}$. Note that the literature also uses a second convention. Some texts, particularly on phonon transport, put only the N acoustic wavevectors in the sphere, one per cell, which lowers Q_{D} and Θ_{D} by the factor $n_{\mathrm{at}}^{1/3}$, $2^{1/3} = 1.26$ for silicon, and treats the optical branches separately.

Above a few meV, however, the model is crude. It puts all modes below ω_{D}, and in silicon it misses both the peak of the flat transverse branches and the optical band at the top. Its strength is that one number fixes the low-energy spectrum, and any quantity that is an integral over g_{ph} weighted toward low frequencies, such as the mean-square displacement at room temperature (Sect. 5.3) or the Debye–Waller exponent (Sect. 5.6), comes out within tens of percent. Worked example 5.3 carries this through for silicon.

Worked example 5.3: Silicon sound speeds and Debye temperature

The elastic constants of silicon at room temperature are $C_{11} = 165.6\,\mathrm{GPa}$, $C_{12} = 63.9\,\mathrm{GPa}$ and $C_{44} = 79.5\,\mathrm{GPa}$, and its density is $\rho_T = 2.33\,\mathrm{g\,cm^{-3}}$ [14, Table I]. Along [100], $\rho_T c_{\mathrm{LA}}^2 = C_{11}$ and $\rho_T c_{\mathrm{TA}}^2 = C_{44}$, so

$$c_{\mathrm{LA}} = \Big(\frac{165.6\times10^9\,\mathrm{Pa}}{2330\,\mathrm{kg\,m^{-3}}}\Big)^{1/2} = 8.43\,\mathrm{km\,s^{-1}},$$

$$c_{\mathrm{TA}} = \Big(\frac{79.5\times10^9\,\mathrm{Pa}}{2330\,\mathrm{kg\,m^{-3}}}\Big)^{1/2} = 5.84\,\mathrm{km\,s^{-1}}\ \text{(twice)}.$$

In other directions the speeds are the eigenvalues of the Christoffel matrix (Appendix C.18): along [110], 9.13 for LA and 5.84 and $4.67\,\mathrm{km\,s^{-1}}$ for the two TA branches; along [111], 9.35 for LA and $5.09\,\mathrm{km\,s^{-1}}$ for both TA branches. The longitudinal speed varies by 11 % with direction and the transverse ones by 25 %. These are the slopes of the acoustic branches at Γ in Fig. 5.3(a). The PBE calculation of that figure, interpolated from its 6^3 grid to small Q, has 8.34 and $5.44\,\mathrm{km\,s^{-1}}$ along [100], 1 % and 7 % below; its transverse branches are soft at small Q as they are at X and L.

The Debye speed Eq. (5.25) needs the average of c_ν^{-3} over all directions, which we evaluate numerically by diagonalizing the Christoffel matrix on a fine grid of directions. Ordering the three branches by speed in each direction, the averages $\langle c_\nu^{-3}\rangle^{-1/3}$ are 9.00, 5.66 and $5.04\,\mathrm{km\,s^{-1}}$, and

$$\bar{c} = \Big[\frac{3}{\sum_\nu \langle c_\nu^{-3}\rangle}\Big]^{1/3} = 5.89\,\mathrm{km\,s^{-1}},$$

closer to the transverse speeds than to the longitudinal one: the longitudinal branch contributes only 9 % of $\sum_\nu \langle c_\nu^{-3}\rangle$. With $n_{\mathrm{at}}/\Omega = 8/a^3 = 4.994\times10^{22}\,\mathrm{cm^{-3}}$ (Sect. 3.1), Eq. (5.23) gives $Q_{\mathrm{D}} = 1.435\,\mathrm{Å}^{-1} = 1.24\,(2\pi/a)$, beyond the zone boundary since the sphere holds two wavevectors per cell, and

$$\omega_{\mathrm{D}} = \bar{c}Q_{\mathrm{D}} = 55.6\,\mathrm{meV}\ (13.4\,\mathrm{THz}), \qquad \Theta_{\mathrm{D}} = 645\,\mathrm{K}.$$

The Debye temperature measured from the heat capacity of silicon at low temperature is 645 K [15, Chap. 5, Table 1]. The agreement is expected: the heat capacity at low temperature measures the ω^2 coefficient of g_{ph}, which Eq. (5.25) takes from the elastic constants. In the other convention, with one wavevector per cell in the sphere, the same speed gives $\Theta_{\mathrm{D}} = 512\,\mathrm{K}$. The isotropic [100] model of Appendix C.18, which uses the [100] speeds $c_{\mathrm{LA}}^2 = C_{11}/\rho_T$ and $c_{\mathrm{TA}}^2 = C_{44}/\rho_T$, 8.43 and $5.84\,\mathrm{km\,s^{-1}}$, in every direction, overestimates $\bar{c}$ at $6.35\,\mathrm{km\,s^{-1}}$ and Θ_{D} at 696 K, 8 % high, because it misses the slow transverse wave along [110].

Against the full spectrum, $\omega_{\mathrm{D}} = 55.6\,\mathrm{meV}$ sits 13 % below the optical phonon at 64.1 meV. Fig. 5.3(b) shows how the Debye density of states differs from the computed one. Below about 7 meV both follow ω^2, but with different coefficients: the computed g_{ph} is 1.3 to 1.4 times g_{D} between 5 and 7 meV, because the PBE sound speeds are softer than the measured ones. Above 7 meV the computed g_{ph} rises far above the Debye curve toward the transverse acoustic peak at 17 meV, falls below it between 26 and 55 meV, and leaves 38 % of all modes, most of the optical band, above ω_{D}, where the model has none. The low-frequency coefficient of the computed g_{ph}, fitted to ω^2 between 1 and 6 meV, implies $\bar{c} = 5.42\,\mathrm{km\,s^{-1}}$ and $\Theta_{\mathrm{D}} = 594\,\mathrm{K}$. The 8 % deficit comes from the transverse branches, 7 % soft in the calculation, which dominate $\bar{c}$, and it enters the ω^2 coefficient cubed, $(5.89/5.42)^3 = 1.28$; for a Debye temperature, the measured elastic constants are the better input.

The harmonic crystal is now a set of $3n_{\mathrm{at}}N$ independent oscillators with known frequencies and polarizations, and in silicon we know them to a few percent, and to 8 % for the soft transverse acoustic branches, from a calculation that needs only the energy surface. What they are not yet is quantum. Sect. 5.3 quantizes each normal mode, which turns $\omega_{\boldsymbol{Q}\nu}$ into the energy of a phonon and $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$ into the pattern in which it displaces the nuclei. With the Debye model of this section it then fixes how far the nuclei of silicon move at zero temperature and at room temperature. A dark-matter particle that strikes one nucleus with momentum $\boldsymbol{q}$ couples to all of these modes at once, with a weight set by $\boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu}$, and the single-phonon rates of the later chapters are built from exactly these two ingredients.

5.3 Phonons

A silicon nucleus in a crystal cooled to a few millikelvin is not at rest. Its displacement along a cube axis has a root-mean-square value of 0.05 Å (Worked example 5.1), about a hundredth of the lattice constant, and at room temperature 0.077 Å. The classical normal modes of Sect. 5.2 cannot account for the first number, since at zero temperature a classical crystal sits motionless at the bottom of its energy surface. They also miss what matters most for detection: the energy of a mode comes in quanta. A dark-matter particle that scatters in a crystal at zero temperature cannot leave an arbitrary amount of energy in the lattice. It creates phonons, each with the energy $\omega_{\boldsymbol{Q}\nu}$ of one mode, out of a lattice in its ground state, and the amplitude for doing so is set by the zero-point motion of that mode. Both facts follow from quantizing the normal modes, which turns the crystal into a set of the harmonic oscillators of Sects. 2.3 and 2.5. We will derive the phonon expansion of the nuclear displacements (Appendix A.6) with its matrix elements for emitting and absorbing a phonon, and compute from it how far the nuclei of silicon move at zero and at room temperature.

5.3.1 Normal coordinates

In the harmonic approximation the nuclei move on the quadratic energy surface Eq. (5.13), and the nuclear Hamiltonian of Eq. (5.6) becomes

$$H_{\mathrm{N}} = \sum_{\boldsymbol{R}\kappa} \frac{\boldsymbol{p}_{\boldsymbol{R}\kappa}^2}{2M_\kappa} + \frac{1}{2}\sum_{\boldsymbol{R}\kappa\alpha}\sum_{\boldsymbol{R}'\kappa'\beta} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},\boldsymbol{R}')\, u^{\alpha}_{\boldsymbol{R}\kappa} u^{\beta}_{\boldsymbol{R}'\kappa'}, \qquad [u^{\alpha}_{\boldsymbol{R}\kappa}, p^{\beta}_{\boldsymbol{R}'\kappa'}] = \mathrm{i}\,\delta_{\boldsymbol{R}\boldsymbol{R}'}\delta_{\kappa\kappa'}\delta_{\alpha\beta}, \tag{5.26}$$

where $\boldsymbol{p}_{\boldsymbol{R}\kappa} = -\mathrm{i}\nabla_{\boldsymbol{R}\kappa}$ is the momentum of the nucleus and the constant $E^{\mathrm{ad}}(0)$ has been dropped. The displacements and momenta are now operators. The classical solution of Sect. 5.2 already tells us which combinations of them decouple: mass-weighted plane waves in the cell index, projected on the polarization vectors. We define, for each $\boldsymbol{Q}$ in the first zone and each branch ν,

$$u_{\boldsymbol{Q}\nu} = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{R}\kappa}\sqrt{M_\kappa}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\,\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}, \qquad p_{\boldsymbol{Q}\nu} = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{R}\kappa}\frac{1}{\sqrt{M_\kappa}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\,\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\cdot\boldsymbol{p}_{\boldsymbol{R}\kappa}, \tag{5.27}$$

the normal coordinate and the normal momentum of the mode $\boldsymbol{Q}\nu$. The transformation is inverted with the lattice sum Eq. (3.14), which gives $\sum_{\boldsymbol{R}}\mathrm{e}^{\mathrm{i}(\boldsymbol{Q}'-\boldsymbol{Q})\cdot\boldsymbol{R}} = N\delta_{\boldsymbol{Q}\boldsymbol{Q}'}$ for two wavevectors of the first zone, and with the completeness relation of Eq. (5.17):

$$\boldsymbol{u}_{\boldsymbol{R}\kappa} = \frac{1}{\sqrt{NM_\kappa}}\sum_{\boldsymbol{Q}\nu}\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\,u_{\boldsymbol{Q}\nu}, \qquad \boldsymbol{p}_{\boldsymbol{R}\kappa} = \sqrt{\frac{M_\kappa}{N}}\sum_{\boldsymbol{Q}\nu}\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*\,\mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\,p_{\boldsymbol{Q}\nu}. \tag{5.28}$$

Inserting Eq. (5.28) into Eq. (5.27) returns $u_{\boldsymbol{Q}\nu}$, by the lattice sum and the orthonormality of the polarization vectors. The normal coordinates are not Hermitian. Since $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ is, and $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$ (Eq. (A.22)), the adjoint of Eq. (5.27) is the coordinate at $-\boldsymbol{Q}$, and the same holds for the momenta:

$$u^\dagger_{\boldsymbol{Q}\nu} = u_{-\boldsymbol{Q}\nu}, \qquad p^\dagger_{\boldsymbol{Q}\nu} = p_{-\boldsymbol{Q}\nu}.$$

The commutator follows from that of Eq. (5.26). The masses cancel, the lattice sum sets $\boldsymbol{Q}' = \boldsymbol{Q}$, and orthonormality sets $\nu' = \nu$:

$$[u_{\boldsymbol{Q}\nu}, p_{\boldsymbol{Q}'\nu'}] = \frac{\mathrm{i}}{N}\sum_{\boldsymbol{R}\kappa} \mathrm{e}^{\mathrm{i}(\boldsymbol{Q}'-\boldsymbol{Q})\cdot\boldsymbol{R}}\, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^* \cdot \boldsymbol{e}_{\kappa\nu'}(\boldsymbol{Q}') = \mathrm{i}\,\delta_{\boldsymbol{Q}\boldsymbol{Q}'}\delta_{\nu\nu'},$$

while the coordinates commute among themselves, and so do the momenta. Each pair $(u_{\boldsymbol{Q}\nu}, p_{\boldsymbol{Q}\nu})$ is as canonical as the position and momentum of a single particle; the only novelty is that the pairs at $\boldsymbol{Q}$ and $-\boldsymbol{Q}$ are adjoints of each other. In field theory the momentum conjugate to a mode of wavevector $\boldsymbol{k}$ carries $-\boldsymbol{k}$; here the phase $\mathrm{e}^{+\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$ in the definition of $p_{\boldsymbol{Q}\nu}$ absorbs the sign, so that $p_{\boldsymbol{Q}\nu}$ is conjugate to $u_{\boldsymbol{Q}\nu}$ itself.

In these variables the Hamiltonian separates. In the kinetic term we write $\boldsymbol{p}^2_{\boldsymbol{R}\kappa} = \boldsymbol{p}^\dagger_{\boldsymbol{R}\kappa}\cdot\boldsymbol{p}_{\boldsymbol{R}\kappa}$ and insert Eq. (5.28); the sum over $\boldsymbol{R}$ sets the two wavevectors equal and the sum over κ the two branches. In the potential term we write the left displacement as $\boldsymbol{u}^\dagger_{\boldsymbol{R}\kappa}$ and expand both. The sum over $\boldsymbol{R}'$ of the force constants times $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}'\cdot\boldsymbol{R}'}$ is $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}'\cdot\boldsymbol{R}}\sqrt{M_\kappa M_{\kappa'}}\,D^{\mathrm{dyn}}_{\kappa\kappa'}(\boldsymbol{Q}')$, by translation invariance and Eq. (5.16). The dynamical matrix acting on its eigenvector returns $\omega^2_{\boldsymbol{Q}'\nu'}$, and the sums over $\boldsymbol{R}$ and κ again set $\boldsymbol{Q}' = \boldsymbol{Q}$ and $\nu' = \nu$. What remains is

$$H_{\mathrm{N}} = \frac{1}{2}\sum_{\boldsymbol{Q}\nu}\Big(p^\dagger_{\boldsymbol{Q}\nu}p_{\boldsymbol{Q}\nu} + \omega^2_{\boldsymbol{Q}\nu}\,u^\dagger_{\boldsymbol{Q}\nu}u_{\boldsymbol{Q}\nu}\Big). \tag{5.29}$$

The $3n_{\mathrm{at}}N$ coupled nuclei have become $3n_{\mathrm{at}}N$ independent oscillators of unit mass, one for each mode. With Eq. (5.29) the Heisenberg equations are $\dot{u}_{\boldsymbol{Q}\nu} = p^\dagger_{\boldsymbol{Q}\nu}$ and $\dot{p}^\dagger_{\boldsymbol{Q}\nu} = -\omega^2_{\boldsymbol{Q}\nu}u_{\boldsymbol{Q}\nu}$, so that $\ddot{u}_{\boldsymbol{Q}\nu} = -\omega^2_{\boldsymbol{Q}\nu}u_{\boldsymbol{Q}\nu}$: the classical equation of each normal mode, as Sect. 5.2 anticipated. Nothing quantum has happened yet. It enters through the commutator, which forbids $u_{\boldsymbol{Q}\nu}$ and $p_{\boldsymbol{Q}\nu}$ from vanishing together.

5.3.2 Phonon operators

For a single oscillator of unit mass the ladder operator is $(\omega/2)^{1/2}x + \mathrm{i}(2\omega)^{-1/2}p$. The normal modes need one change: since $u_{\boldsymbol{Q}\nu}$ and $p_{\boldsymbol{Q}\nu}$ are not Hermitian, the momentum enters through its adjoint,

$$b_{\boldsymbol{Q}\nu} = \Big(\frac{\omega_{\boldsymbol{Q}\nu}}{2}\Big)^{1/2}u_{\boldsymbol{Q}\nu} + \frac{\mathrm{i}}{(2\omega_{\boldsymbol{Q}\nu})^{1/2}}\,p^\dagger_{\boldsymbol{Q}\nu}, \qquad b^\dagger_{\boldsymbol{Q}\nu} = \Big(\frac{\omega_{\boldsymbol{Q}\nu}}{2}\Big)^{1/2}u_{-\boldsymbol{Q}\nu} - \frac{\mathrm{i}}{(2\omega_{\boldsymbol{Q}\nu})^{1/2}}\,p_{\boldsymbol{Q}\nu}. \tag{5.30}$$

With the commutator of the normal coordinates and its adjoint, $[u_{\boldsymbol{Q}\nu}, p^\dagger_{\boldsymbol{Q}'\nu'}] = \mathrm{i}\,\delta_{\boldsymbol{Q},-\boldsymbol{Q}'}\delta_{\nu\nu'}$, the two cross terms of $[b_{\boldsymbol{Q}\nu}, b^\dagger_{\boldsymbol{Q}'\nu'}]$ contribute $\frac{1}{2}\delta_{\boldsymbol{Q}\boldsymbol{Q}'}\delta_{\nu\nu'}$ each. The two cross terms of $[b_{\boldsymbol{Q}\nu}, b_{\boldsymbol{Q}'\nu'}]$ are $\mp\frac{1}{2}\delta_{\boldsymbol{Q},-\boldsymbol{Q}'}\delta_{\nu\nu'}$ and cancel, because $\omega_{-\boldsymbol{Q}\nu} = \omega_{\boldsymbol{Q}\nu}$. Hence

$$[b_{\boldsymbol{Q}\nu}, b^\dagger_{\boldsymbol{Q}'\nu'}] = \delta_{\boldsymbol{Q}\boldsymbol{Q}'}\delta_{\nu\nu'}, \qquad [b_{\boldsymbol{Q}\nu}, b_{\boldsymbol{Q}'\nu'}] = [b^\dagger_{\boldsymbol{Q}\nu}, b^\dagger_{\boldsymbol{Q}'\nu'}] = 0,$$

the boson algebra Eq. (2.35) with $\gamma = (\boldsymbol{Q}, \nu)$. Adding Eq. (5.30) at $\boldsymbol{Q}$ to the adjoint at $-\boldsymbol{Q}$, and subtracting, inverts the definition:

$$u_{\boldsymbol{Q}\nu} = \frac{b_{\boldsymbol{Q}\nu} + b^\dagger_{-\boldsymbol{Q}\nu}}{(2\omega_{\boldsymbol{Q}\nu})^{1/2}}, \qquad p^\dagger_{\boldsymbol{Q}\nu} = -\mathrm{i}\Big(\frac{\omega_{\boldsymbol{Q}\nu}}{2}\Big)^{1/2}(b_{\boldsymbol{Q}\nu} - b^\dagger_{-\boldsymbol{Q}\nu}).$$

A normal coordinate contains the annihilator of the mode $\boldsymbol{Q}$ and the creator of the mode $-\boldsymbol{Q}$. Inserting these forms into Eq. (5.29), one bracket is

$$p^\dagger_{\boldsymbol{Q}\nu} p_{\boldsymbol{Q}\nu} + \omega^2_{\boldsymbol{Q}\nu} u^\dagger_{\boldsymbol{Q}\nu} u_{\boldsymbol{Q}\nu} = \frac{\omega_{\boldsymbol{Q}\nu}}{2}\Big[(b_{\boldsymbol{Q}\nu} - b^\dagger_{-\boldsymbol{Q}\nu})(b^\dagger_{\boldsymbol{Q}\nu} - b_{-\boldsymbol{Q}\nu}) + (b^\dagger_{\boldsymbol{Q}\nu} + b_{-\boldsymbol{Q}\nu})(b_{\boldsymbol{Q}\nu} + b^\dagger_{-\boldsymbol{Q}\nu})\Big].$$

The products of two annihilators, $-b_{\boldsymbol{Q}\nu} b_{-\boldsymbol{Q}\nu} + b_{-\boldsymbol{Q}\nu} b_{\boldsymbol{Q}\nu}$, cancel since the two operators commute, and so do the products of two creators. The four remaining terms are $b_{\boldsymbol{Q}\nu} b^\dagger_{\boldsymbol{Q}\nu} + b^\dagger_{\boldsymbol{Q}\nu} b_{\boldsymbol{Q}\nu}$ and the same at $-\boldsymbol{Q}$, which in the sum over all $\boldsymbol{Q}$ repeats the first pair. With $bb^\dagger = b^\dagger b + 1$,

$$H_{\mathrm{N}} = \sum_{\boldsymbol{Q}\nu} \omega_{\boldsymbol{Q}\nu}\Big(b^\dagger_{\boldsymbol{Q}\nu} b_{\boldsymbol{Q}\nu} + \frac{1}{2}\Big). \tag{5.31}$$

We see that the spectrum is that of independent oscillators. The eigenstates are labeled by one occupation number $n_{\boldsymbol{Q}\nu} = 0, 1, 2, \ldots$ per mode, $|\{n_{\boldsymbol{Q}\nu}\}\rangle = \prod_{\boldsymbol{Q}\nu} (n_{\boldsymbol{Q}\nu}!)^{-1/2} (b^\dagger_{\boldsymbol{Q}\nu})^{n_{\boldsymbol{Q}\nu}} |0\rangle$, with energy $\sum_{\boldsymbol{Q}\nu} \omega_{\boldsymbol{Q}\nu}(n_{\boldsymbol{Q}\nu} + \frac{1}{2})$. A *phonon* is one quantum of a normal mode: the state $b^\dagger_{\boldsymbol{Q}\nu}|0\rangle$ lies $\omega_{\boldsymbol{Q}\nu}$ above the ground state, and $\omega_{\boldsymbol{Q}\nu}$ of Sect. 5.2 is its energy. Phonons are bosons, since any number of them can occupy a mode, and their number is not conserved, which is why their thermal occupation has no chemical potential (Sect. 2.5). Each phonon also carries a crystal momentum. A lattice translation $T_{\boldsymbol{R}}$, in the sense of Eq. (3.11), relabels the displacements, $\boldsymbol{u}_{\boldsymbol{R}'\kappa} \to \boldsymbol{u}_{\boldsymbol{R}'-\boldsymbol{R},\kappa}$. By Eqs. (5.27) and (5.30) this multiplies $b^\dagger_{\boldsymbol{Q}\nu}$ by $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$, while the ground state is invariant. A state with one phonon in the mode $\boldsymbol{Q}\nu$ has crystal momentum $\boldsymbol{Q}$, defined up to a reciprocal-lattice vector, exactly as an electron in a Bloch state has $\boldsymbol{k}$ (Sect. 3.1).

The ground state $|0\rangle$ holds no phonons, but its energy is not zero. Every mode keeps $\frac{1}{2}\omega_{\boldsymbol{Q}\nu}$, and in silicon the zero-point energy $\frac{1}{2}\int \mathrm{d}\omega\, g_{\mathrm{ph}}(\omega)\,\omega$ is 123 meV per cell for the computed density of states of Fig. 5.3: $\frac{1}{2} \times 40.9\,\mathrm{meV}$, the mean phonon energy, times six modes. It depends on the cell volume through the frequencies, which Sect. 5.7 takes up. In the Heisenberg picture the phonon operators of Eq. (5.31) evolve as $b_{\boldsymbol{Q}\nu}(t) = b_{\boldsymbol{Q}\nu} \mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{Q}\nu} t}$, and every displacement is a superposition of oscillations at the mode frequencies, as in the classical crystal.

The displacement of a nucleus follows from Eq. (5.28) once the normal coordinate is written with the phonon operators. For the momentum, the term $-\boldsymbol{Q}$ of the sum is relabeled with $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q})^* = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$. Both have the same structure:

Phonon expansion of the displacement

$$\boldsymbol{u}_{\boldsymbol{R}\kappa} = \sum_{\boldsymbol{Q}\nu} \frac{1}{\sqrt{2N M_\kappa \omega_{\boldsymbol{Q}\nu}}}\, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, B_{\boldsymbol{Q}\nu}, \qquad B_{\boldsymbol{Q}\nu} = b_{\boldsymbol{Q}\nu} + b^\dagger_{-\boldsymbol{Q}\nu}, \tag{5.32}$$

$$\boldsymbol{p}_{\boldsymbol{R}\kappa} = -\mathrm{i} \sum_{\boldsymbol{Q}\nu} \Big(\frac{M_\kappa \omega_{\boldsymbol{Q}\nu}}{2N}\Big)^{1/2} \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, (b_{\boldsymbol{Q}\nu} - b^\dagger_{-\boldsymbol{Q}\nu}),$$

with $[b_{\boldsymbol{Q}\nu}, b^\dagger_{\boldsymbol{Q}'\nu'}] = \delta_{\boldsymbol{Q}\boldsymbol{Q}'}\delta_{\nu\nu'}$, $\sum_\kappa |\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2 = 1$ and $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$. The harmonic Hamiltonian is Eq. (5.31).

This is Eq. (A.21), the convention used throughout, and that of Ashcroft and Mermin [9, Chap. 23] and of Giustino [4, Eqs. (16)–(21)], with the phonon wavevector written $\boldsymbol{Q}$. The *displacement field* $B_{\boldsymbol{Q}\nu}$ obeys $B^\dagger_{\boldsymbol{Q}\nu} = B_{-\boldsymbol{Q}\nu}$, so that the terms at $\boldsymbol{Q}$ and $-\boldsymbol{Q}$ of Eq. (5.32) are adjoints of each other and the displacement is Hermitian. It is the operator that every coupling linear in the displacements contains: the electron–phonon Hamiltonian Eq. (A.25) and the single-phonon response of a nucleus to dark matter.

5.3.3 The zero-point length

Each term of Eq. (5.32) has three factors: the pattern $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$ of Sect. 5.2, the operator $B_{\boldsymbol{Q}\nu}$, which changes the number of phonons by one, and a length that fixes how far one quantum moves the nuclei. We write that length as the *zero-point length* of the mode,

$$\ell_{\boldsymbol{Q}\nu} = \frac{1}{\sqrt{2M_{\mathrm{cell}}\,\omega_{\boldsymbol{Q}\nu}}}, \tag{5.33}$$

with the cell mass $M_{\mathrm{cell}} = \sum_\kappa M_\kappa$ and the factor $1/\sqrt{N}$ kept outside (Eq. (A.24)). For an acoustic mode at small $\boldsymbol{Q}$, where $\boldsymbol{e}_{\kappa\nu} = (M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$ (Sect. 5.2), the amplitude in Eq. (5.32) is $\ell_{\boldsymbol{Q}\nu}/\sqrt{N}$ for every nucleus: the whole cell moves as one particle of mass M_{cell}. For other modes $\ell_{\boldsymbol{Q}\nu}$ sets the scale, and the polarization vector distributes the motion over the nuclei of the cell. The name refers to the ground state, in which $\langle 0|B_{\boldsymbol{Q}\nu}B^\dagger_{\boldsymbol{Q}\nu}|0\rangle = \langle 0|b_{\boldsymbol{Q}\nu}b^\dagger_{\boldsymbol{Q}\nu}|0\rangle = 1$: $\ell_{\boldsymbol{Q}\nu}/\sqrt{N}$ is the zero-point amplitude that one mode contributes to each nucleus.

For silicon, with M_{cell} two silicon masses (52.33 GeV, Sect. 1.5), the zone-center optical phonon at 64.1 meV has $\ell_{\boldsymbol{Q}\nu} = 0.024\,\text{Å}$. This is half the zero-point length $\ell_{\mathrm{zp}} = 0.048\,\text{Å}$ of the bond coordinate in Fig. 5.2(c), and must be. That coordinate is the change of a bond length, and in the optical mode the two atoms of the cell move against each other with equal amplitude, $\boldsymbol{e}_{\kappa\nu} = \pm\hat{\boldsymbol{e}}/\sqrt{2}$, each by half of it. The transverse acoustic phonon at X, 18.6 meV, has $\ell_{\boldsymbol{Q}\nu} = 0.045\,\text{Å}$, and an acoustic phonon of 5 meV has 0.086 Å: soft modes move the nuclei farther, as $\omega_{\boldsymbol{Q}\nu}^{-1/2}$. The factor $1/\sqrt{N}$ makes the motion of any single mode unobservably small in a macroscopic crystal. A kilogram of silicon has $N = 1.07\times10^{25}$ cells (Sect. 3.1), and one optical phonon moves each of its nuclei by 7×10^{-15} Å. What is finite is the sum over all $3n_{\mathrm{at}}N$ modes, each contributing a share of order $\ell^2_{\boldsymbol{Q}\nu}/N$ to the mean-square displacement, and we compute that sum below.

5.3.4 Emission and absorption

The matrix elements of the displacement between phonon states follow from Eq. (5.32) and the boson relations Eq. (2.34). Only two terms of the sum connect a state with $n_{\boldsymbol{Q}\nu}$ phonons in the mode $\boldsymbol{Q}\nu$ to a state that differs from it in that mode alone. The term $\boldsymbol{Q}$, through $b_{\boldsymbol{Q}\nu}$, removes a phonon. The term $-\boldsymbol{Q}$, through $b^\dagger_{\boldsymbol{Q}\nu}$, adds one, with the pattern $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q})\mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*\mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$. Hence

$$\begin{aligned}
\langle n_{\boldsymbol{Q}\nu}+1|\,\boldsymbol{u}_{\boldsymbol{R}\kappa}\,|n_{\boldsymbol{Q}\nu}\rangle &= \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*\,\mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}}{\sqrt{2NM_\kappa\omega_{\boldsymbol{Q}\nu}}}\sqrt{n_{\boldsymbol{Q}\nu}+1},\\
\langle n_{\boldsymbol{Q}\nu}-1|\,\boldsymbol{u}_{\boldsymbol{R}\kappa}\,|n_{\boldsymbol{Q}\nu}\rangle &= \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}}{\sqrt{2NM_\kappa\omega_{\boldsymbol{Q}\nu}}}\sqrt{n_{\boldsymbol{Q}\nu}},
\end{aligned} \tag{5.34}$$

with the occupations of all other modes unchanged. In thermal equilibrium the squared matrix elements are averaged over the Gibbs state of Sect. 2.5, and the occupation becomes the mean *phonon occupation*

$$n_{\boldsymbol{Q}\nu} = \langle b^\dagger_{\boldsymbol{Q}\nu}b_{\boldsymbol{Q}\nu}\rangle = n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu}),$$

by Eq. (2.69). Emission then carries the weight $n_{\boldsymbol{Q}\nu}+1$ and absorption the weight $n_{\boldsymbol{Q}\nu}$, the factors $n_{\boldsymbol{Q}\nu}+\frac{1}{2}\pm\frac{1}{2}$ of Eq. (A.28), and their ratio $\mathrm{e}^{\beta\omega_{\boldsymbol{Q}\nu}}$ is detailed balance for the mode. Each mode of the crystal is the oscillator of Worked example 2.7, with the coupling λ_{osc} replaced by the projection of the dark-matter vertex on $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})/\sqrt{2NM_\kappa\omega_{\boldsymbol{Q}\nu}}$. The numbers of that example carry over. At room temperature the optical phonon of silicon has $n_{\boldsymbol{Q}\nu} = 0.09$ and the transverse acoustic phonon at X has 0.95. In a cryogenic detector every mode above a few

tens of μeV is empty, absorption is absent, and emission into an empty mode, with weight one, is the whole signal.

Consider an interaction of dark matter with the nuclei of one species κ that depends on their positions through $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$, the nuclear density $n_\kappa(\boldsymbol{q})$ of Sect. 2.3, with $\boldsymbol{q}$ the momentum transfer. With $\boldsymbol{X} = \boldsymbol{R} + \boldsymbol{\tau}_\kappa + \boldsymbol{u}_{\boldsymbol{R}\kappa}$ (Eq. (5.1)),

$$n_\kappa(\boldsymbol{q}) = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_\kappa)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}} \simeq \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_\kappa)}(1 + \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}).$$

The expansion holds as long as $\langle(\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa})^2\rangle = q^2\langle u_x^2\rangle \ll 1$, that is, for q well below 26 keV in silicon at room temperature (with the value of $\langle u_x^2\rangle$ given below); the full exponential is the subject of Sect. 5.6. The term 1 leaves the lattice unchanged. The term linear in $\boldsymbol{u}$, with Eq. (5.34) and the lattice sum Eq. (3.14), emits one phonon with the amplitude

$$\langle n_{\boldsymbol{Q}\nu} + 1|\, n_\kappa(\boldsymbol{q})\, |n_{\boldsymbol{Q}\nu}\rangle = \mathrm{i}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}\, \frac{\boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*}{\sqrt{2M_\kappa\omega_{\boldsymbol{Q}\nu}}}\, \sqrt{N(n_{\boldsymbol{Q}\nu}+1)}\, \sum_{\boldsymbol{G}} \delta_{\boldsymbol{q},\boldsymbol{Q}+\boldsymbol{G}}.$$

The lattice sum conserves crystal momentum, a feature of every later phonon rate: the emitted phonon carries the momentum transfer, up to a reciprocal-lattice vector, so that each $\boldsymbol{q}$ excites one wavevector $\boldsymbol{Q}$ in each branch. The squared amplitude is also proportional to N, as it must be for a target whose rate grows with its size. Per nucleus, the weight of the branch ν is $|\boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2(n_{\boldsymbol{Q}\nu}+1)/(2M_\kappa\omega_{\boldsymbol{Q}\nu})$: the projection of the momentum transfer on the polarization, which favors longitudinal modes at small $\boldsymbol{Q}$, divided by the energy of the mode, which favors soft ones. Absorption of a phonon $-\boldsymbol{q}$ follows in the same way with $n_{\boldsymbol{Q}\nu}$, and with several species the amplitudes of the nuclei of a cell add with their phases $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}$. Sects. 11.1 and 11.2 build the single-phonon rates from exactly this amplitude.

5.3.5 Zero-point and thermal motion

Each Cartesian component of a nuclear displacement behaves roughly like an oscillator of mass M_κ at a typical phonon frequency $\bar\omega$. In the ground state such an oscillator has $\langle x^2\rangle = 1/(2M_\kappa\bar\omega)$, and at a temperature well above $\bar\omega$ equipartition, $M_\kappa\bar\omega^2\langle x^2\rangle = T$, leads to $T/(M_\kappa\bar\omega^2)$. For silicon with $\bar\omega \sim 30$ meV the zero-point estimate is $\langle u_x^2\rangle^{1/2} \sim 0.05$ Å, the number of Worked example 5.1, and room temperature, where $T = 25.9$ meV is close to $\bar\omega$, should roughly double $\langle u_x^2\rangle$. The question is what $\bar\omega$ means for a crystal with a whole spectrum of modes.

The exact statement is a sum over modes. In the Gibbs state the occupations are independent, and the only nonzero averages of two displacement fields pair a mode with its partner at $-\boldsymbol{Q}$:

$$\langle B_{\boldsymbol{Q}\nu} B_{\boldsymbol{Q}'\nu'}\rangle = \delta_{\boldsymbol{Q}',-\boldsymbol{Q}}\delta_{\nu\nu'}\big(\langle b_{\boldsymbol{Q}\nu} b^\dagger_{\boldsymbol{Q}\nu}\rangle + \langle b^\dagger_{-\boldsymbol{Q}\nu} b_{-\boldsymbol{Q}\nu}\rangle\big) = \delta_{\boldsymbol{Q}',-\boldsymbol{Q}}\delta_{\nu\nu'}\,(2n_{\boldsymbol{Q}\nu}+1),$$

by Eq. (2.69) and $n_{-\boldsymbol{Q}\nu} = n_{\boldsymbol{Q}\nu}$. Inserting Eq. (5.32) twice, the phases $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$ cancel, $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$ supplies the conjugate, and the *displacement tensor* of the nucleus κ is

$$B_\kappa^{\alpha\beta} = \langle u^\alpha_{\boldsymbol{R}\kappa} u^\beta_{\boldsymbol{R}\kappa}\rangle = \frac{1}{N}\sum_{\boldsymbol{Q}\nu} \frac{2n_{\boldsymbol{Q}\nu}+1}{2M_\kappa\,\omega_{\boldsymbol{Q}\nu}}\, e^\alpha_{\kappa\nu}(\boldsymbol{Q})\, e^\beta_{\kappa\nu}(\boldsymbol{Q})^*, \qquad 2n_{\boldsymbol{Q}\nu}+1 = \coth\frac{\omega_{\boldsymbol{Q}\nu}}{2T}, \tag{5.35}$$

the same in every cell, real, symmetric and positive (Eq. (D.80)). The average is over the thermal state, and at $T = 0$ over the ground state: B_κ then measures the *zero-point motion*, the spread of the nuclear wavefunction that no cooling removes. The two limits are those of the estimate. At $T = 0$, $2n+1 = 1$, and each mode contributes $1/(2M_\kappa\omega_{\boldsymbol{Q}\nu})$ times the weight

$|\boldsymbol{e}_{\kappa\nu}|^2/N$. For T above every phonon energy, $2n+1 \to 2T/\omega_{\boldsymbol{Q}\nu}$, and each mode contributes its equipartition value $T/(M_\kappa \omega_{\boldsymbol{Q}\nu}^2)$, growing linearly in T. The acoustic modes have $\omega_{\boldsymbol{Q}\nu} \to 0$, but they are few: in three dimensions their number per unit frequency falls as ω^2 (Eq. (5.22)), and the sum stays finite at every temperature.

At a site of cubic symmetry, such as the tetrahedral sites of the diamond and zinc-blende structures, a symmetric second-rank tensor that is invariant under the site group is a multiple of the identity. There $B_\kappa = \langle u_x^2 \rangle \, \mathbf{1}$, and the mean-square displacement $\langle u^2 \rangle = \operatorname{tr} B_\kappa = 3\langle u_x^2 \rangle$; crystallographers quote $\langle u_x^2 \rangle$, or the isotropic displacement parameter $8\pi^2 \langle u_x^2 \rangle$. If, in addition, the n_{at} nuclei of the cell are of one species and related by symmetry, as in silicon and germanium, all of them have the same trace. We can then average the trace of Eq. (5.35) over κ, the normalization $\sum_\kappa |\boldsymbol{e}_{\kappa\nu}|^2 = 1$ removes the polarization vectors, and the sum over modes becomes an integral over the density of states Eq. (5.21):

$$\langle u^2 \rangle = \int_0^\infty \mathrm{d}\omega \, \frac{g_{\mathrm{ph}}(\omega)}{n_{\mathrm{at}}} \, \frac{\coth(\omega/2T)}{2M_\kappa \, \omega}. \tag{5.36}$$

The density of states enters per atom, $g_{\mathrm{ph}}/n_{\mathrm{at}}$, normalized to three. For a crystal of several species, such as GaAs, the same steps keep the weight of each nucleus separately, and $g_{\mathrm{ph}}/n_{\mathrm{at}}$ is replaced by the atom-projected density of states $g_{\mathrm{ph},\kappa}(\omega) = N^{-1} \sum_{\boldsymbol{Q}\nu} |\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2 \delta(\omega - \omega_{\boldsymbol{Q}\nu})$, also normalized to three. Eq. (5.36) makes the estimate precise: the typical frequency at $T = 0$ is the inverse of the mean of $1/\omega$ over the spectrum, $\langle u^2 \rangle = (3/2M_\kappa)\langle \omega^{-1} \rangle$, and at high temperature the inverse root of the mean of $1/\omega^2$, $\langle u^2 \rangle = (3T/M_\kappa)\langle \omega^{-2} \rangle$. Both means are weighted toward low frequencies, the second more strongly.

In the Debye model, $g_{\mathrm{ph}}/n_{\mathrm{at}} = 9\omega^2/\omega_{\mathrm{D}}^3$ below ω_{D} (Eq. (5.24)). Splitting $\coth = 1 + 2n_{\mathrm{B}}$ into its zero-point and thermal parts and writing $x = \omega/T$,

$$\langle u^2 \rangle = \frac{9}{4M_\kappa \omega_{\mathrm{D}}} \Big[1 + 4 \Big(\frac{T}{\omega_{\mathrm{D}}} \Big)^2 \int_0^{\omega_{\mathrm{D}}/T} \frac{x \, \mathrm{d}x}{\mathrm{e}^x - 1} \Big], \tag{5.37}$$

the result of Appendix D.15 for the Debye–Waller exponent, Eq. (D.84), written for the displacement itself. The zero-point value $9/(4M_\kappa \omega_{\mathrm{D}})$ corresponds to $\bar{\omega} = 2\omega_{\mathrm{D}}/3$ in the estimate. At high temperature the bracket becomes $4T/\omega_{\mathrm{D}} + \omega_{\mathrm{D}}/(9T) + \cdots$, and $\langle u^2 \rangle = (9T/M_\kappa \omega_{\mathrm{D}}^2)[1 + \omega_{\mathrm{D}}^2/36T^2 + \cdots]$, the equipartition value with the zero-point motion as its first correction. The thermal part equals the zero-point part when the bracket is two, at $T = 0.44\,\Theta_{\mathrm{D}}$. For silicon this is 284 K: at room temperature the nuclei of silicon move as much because of their zero-point motion as because of the heat.

For crystals of similar bonding, the zero-point value of Eq. (5.37) $\propto 1/(M_\kappa \omega_{\mathrm{D}})$ falls with the mass, since $\omega_{\mathrm{D}} \propto M_\kappa^{-1/2}$ at fixed force constants: heavier nuclei are better localized. The high-temperature value $\propto T/(M_\kappa \omega_{\mathrm{D}}^2)$ depends on the force constants alone, the classical statement that a spring stretched by thermal energy does not care what mass hangs on it. A heavier and softer crystal has less zero-point motion and, at room temperature, more thermal motion, as germanium shows in Table 5.1.

In practice B_κ is computed from DFPT phonons. The sum Eq. (5.35) runs over a uniform grid of wavevectors, with the frequencies and polarization vectors interpolated from the force constants of a coarser grid (Sect. 5.4 and Appendix E.4), or Eq. (5.36) is integrated over a density of states computed on a fine grid. The grid misses the small acoustic frequencies near Γ, whose terms grow as $1/\omega$ at $T = 0$ and $1/\omega^2$ at $T > 0$, and Eq. (E.30) adds them back analytically. For silicon the 4^3 grid of Fig. 5.2 yields $\langle u_x^2 \rangle^{1/2} = 0.050$ Å at zero temperature and 0.081 Å at 293 K, and the 48^3 density of states of Fig. 5.3 leads to 0.0495 Å at zero temperature, 0.0799 Å at 293 K and 0.0807 Å at 300 K. The two grids differ by 2 % at most, well inside the error that the functional itself makes (Worked example 5.4).

Worked example 5.4: Zero-point and room-temperature motion of silicon

We evaluate Eq. (5.37) for silicon with the natural isotope mixture, $M_\kappa = 28.0855\,\mathrm{u} = 26.16\,\mathrm{GeV}$. The Debye frequency is that of the measured elastic constants, $\omega_\mathrm{D} = 55.6\,\mathrm{meV}$ (Worked example 5.3). We convert with $\hbar c = 1973.27\,\mathrm{eV\,\AA}$.

Zero temperature. The product $M_\kappa\omega_\mathrm{D} = 1.455 \times 10^9\,\mathrm{eV}^2$, and

$$\langle u^2\rangle_{T=0} = \frac{9}{4M_\kappa\omega_\mathrm{D}} = 1.546 \times 10^{-9}\,\mathrm{eV}^{-2} = 6.02 \times 10^{-3}\,\mathrm{\AA}^2, \qquad \langle u_x^2\rangle_{T=0}^{1/2} = 0.0448\,\mathrm{\AA}.$$

Room temperature. At $T = 300\,\mathrm{K} = 25.85\,\mathrm{meV}$, $\omega_\mathrm{D}/T = 2.152$, and the integral of Eq. (5.37), evaluated numerically, is $\int_0^{2.152} x\,\mathrm{d}x/(\mathrm{e}^x - 1) = 1.259$. The bracket is $1 + 4 \times 0.4648^2 \times 1.259 = 2.088$, so that

$$\langle u^2\rangle_{300\,\mathrm{K}} = 1.257 \times 10^{-2}\,\mathrm{\AA}^2, \qquad \langle u_x^2\rangle_{300\,\mathrm{K}}^{1/2} = 0.0647\,\mathrm{\AA}.$$

Thermal phonons double the mean-square displacement, as the estimate suggested, and multiply the amplitude by $\sqrt{2.088} = 1.44$. Since T lies below $\omega_\mathrm{D}/2$, equipartition alone, $9T/(M_\kappa\omega_\mathrm{D}^2)$, falls 11 % short, at $\langle u_x^2\rangle = 3.73 \times 10^{-3}\,\mathrm{\AA}^2$; its first zero-point correction, the factor $1 + \omega_\mathrm{D}^2/36T^2 = 1.129$, recovers the full value to 0.5 %.

Against the full spectrum and the measurement. The value at 293 K from a lattice-dynamical fit to the measured phonons, the reference of Worked example 5.1, is $\langle u_x^2\rangle = 5.941 \times 10^{-3}\,\mathrm{\AA}^2$ [8]. The Debye model predicts $4.11 \times 10^{-3}\,\mathrm{\AA}^2$ at that temperature. Eq. (5.36) with the computed density of states of Fig. 5.3 gives $\langle u_x^2\rangle = 2.45 \times 10^{-3}\,\mathrm{\AA}^2$ at zero temperature and $6.38 \times 10^{-3}\,\mathrm{\AA}^2$ at 293 K, 7 % above the measurement, as expected from the transverse acoustic branches, which are 8 % soft in this calculation (Sect. 5.2) and enter as $1/\omega^2$. Against this spectrum the Debye model is low by 18 % at zero temperature and by about a third at room temperature, and the reason is the flat transverse acoustic branches. In the computed spectrum the modes below 20 meV are 22 % of all modes but carry 45 % of the zero-point $\langle u^2\rangle$, since they enter with the weight $1/\omega$. In the Debye spectrum the same window holds 5 % of the modes and 13 % of $\langle u^2\rangle$. The typical frequencies of the estimate make the same point. For the computed spectrum, $\langle\omega^{-1}\rangle^{-1} = 30.4\,\mathrm{meV}$ and $\langle\omega^{-2}\rangle^{-1/2} = 25.2\,\mathrm{meV}$, against $2\omega_\mathrm{D}/3 = 37.1\,\mathrm{meV}$ and $\omega_\mathrm{D}/\sqrt{3} = 32.1\,\mathrm{meV}$ in the model. Elastic constants fix the low-frequency density of states exactly, but the displacement is weighted toward the zone-boundary modes of the transverse branches, which the model pushes up to ω_D.

The verdict is that the Debye model reproduces the zero-point and room-temperature motion of silicon to 20 % to 35 %, which is enough for an estimate of where the Debye–Waller suppression sets in. A rate that depends on B_κ at the ten-percent level needs the full density of states, and at that level the softness of the computed transverse branches matters as well.

Table 5.1 puts silicon next to the two contrast materials of the book. Germanium has the same structure and bonding, a nucleus 2.6 times heavier and weaker bonds. Its optical phonon, at 37.6 meV, lies below the 39.9 meV that the mass alone would give, 64.1 meV scaled by the square root of the mass ratio, and its elastic constants imply $\Theta_\mathrm{D} = 373\,\mathrm{K}$. Its zero-point motion is smaller than that of silicon and its room-temperature motion larger, as the trends of Eq. (5.37) require. At 300 K, $T = 0.80\,\Theta_\mathrm{D}$, and the thermal part is 2.4 times the zero-point part. Its transverse acoustic branches are flat near the zone boundary as well, at 9.9 meV at X, a third of ω_D as in silicon, and the Debye values in the table should be read as low by a similar fraction. GaAs, with nuclei of nearly equal mass, sits close to germanium, but it is

Table 5.1: Phonon energies and root-mean-square displacements $\langle u_x^2 \rangle^{1/2}$ along one Cartesian axis, in Å, of silicon, germanium and GaAs. Phonon energies: measured by inelastic neutron scattering, silicon at room temperature [1, 11], germanium at 80 K [16] as tabulated in Ref. [12, Table IV], and GaAs [17] as tabulated in Ref. [18, Table I]; for GaAs the transverse (TO) and longitudinal (LO) optical energies at Γ. Debye temperatures from the room-temperature elastic constants and lattice constants with the angular average of Eq. (5.25): silicon from Worked example 5.3, germanium from $C_{11} = 131$, $C_{12} = 49$, $C_{44} = 68$ GPa [19] (as tabulated in Ref. [12, Table III]) and the room-temperature $a = 5.658$ Å of Table 3.1. Debye columns: Eq. (5.37) with the natural isotope masses. DFPT columns: Eq. (5.36) at 0 and 300 K with the PBE density of states of Fig. 5.3 for silicon, on a 48^3 grid of wavevectors. Germanium has no DFPT entry because PBE leaves it nearly metallic at the measured lattice constant, with a direct gap of 0.05 eV and the wrong band order, and metallic at its own (Appendix E.1). Entries marked † are computed for GaAs with the same method (PBE, $a = 5.638$ Å, a 4^3 grid of wavevectors and the dipole term of Sect. 5.5): the transverse acoustic energy at X, and the displacements of gallium (first) and arsenic (second) from the atom-projected densities of states $g_{\mathrm{ph},\kappa}$ on a 32^3 grid. Reference: lattice-dynamical fit to neutron data at 293 K [8]. The germanium energies are tabulated to 0.1 THz, 0.4 meV. A dash marks a value not given here.

	Phonon energy (meV)			Debye		DFPT		Reference
	optical at Γ	TA at X	Θ_D (K)	0 K	300 K	0 K	300 K	293 K
Si	64.1	18.6	645	0.045	0.065	0.049	0.081	0.077
Ge	37.6	9.9	373	0.037	0.067	–	–	–
GaAs	33.6, 36.4	10.5†	–	–	–	0.040, 0.039†	0.080, 0.078†	–

polar: its longitudinal optical phonon at Γ lies 2.7 meV above the transverse one, the LO–TO splitting of Sect. 5.5.

With the phonon operators, the displacement field $B_{\boldsymbol{Q}\nu}$ and the tensor B_κ, the lattice is ready for the questions of the rest of the chapter. The tensor sets the scale beyond which a nucleus no longer absorbs a momentum transfer as a single phonon. With the reference value for silicon at room temperature, $\boldsymbol{q} \cdot B_\kappa \cdot \boldsymbol{q} = q^2 \langle u_x^2 \rangle$ reaches one at $q = 26$ keV, and Sect. 5.6 turns this into the Debye–Waller factor and the multiphonon response. The displacement field, with the normalization of Eq. (5.32), is what the electron–phonon vertex of Sect. 6.2 multiplies, and the matrix elements Eq. (5.34), with the $n_{\boldsymbol{Q}\nu} + 1$ that survives in a cold crystal, are where every single-phonon rate of the later chapters starts. All of this needs $\omega_{\boldsymbol{Q}\nu}$ and $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$ on fine grids of wavevectors, and Sect. 5.4 shows how density-functional theory supplies them.

5.4 Phonons from first principles

The dispersion of silicon in Fig. 5.3 came from sixteen calculations, each in the two-atom primitive cell, one for each inequivalent wavevector of a $6 \times 6 \times 6$ grid. The same force constants could have been obtained by moving atoms in a supercell of $6^3 = 216$ primitive cells, 432 atoms, and recording the forces. The cost of a density-functional calculation grows as the cube of the number of atoms N_at in the cell times the number of $\boldsymbol{k}$ points. A cell n times larger needs n times fewer $\boldsymbol{k}$ points for the same sampling, as long as more than one is left. Each self-consistent calculation of that supercell would therefore cost $216^3/216 = 216^2 \approx 5 \times 10^4$ times one of the primitive cell. The two routes measure the same thing, the second derivatives of the energy surface $E^\mathrm{ad}(u)$ of Sect. 5.1, from finite differences of forces or from the linear response of the electrons to a displacement wave. We will derive both, the Sternheimer

equation that makes the linear-response route, density-functional perturbation theory, cheap, and the Fourier interpolation that turns a few dynamical matrices into the whole dispersion. For dark matter the result matters at both ends of the spectrum: the sound speeds that set the reach of acoustic-phonon searches (Sect. 9.2) and the optical branches of polar crystals (Sect. 5.5) all come out of density-functional perturbation theory. Our account follows Baroni and coworkers [20] and Giustino [4].

5.4.1 Frozen phonons

The most direct way to get a force constant is to displace an atom and measure the forces on all the others. The force on the nucleus $(\boldsymbol{R},\kappa')$ along α when only the nucleus $(\boldsymbol{0},\kappa)$ is displaced by h along β is, by Eq. (5.13) carried one order further,

$$F^{\alpha}_{\boldsymbol{R}\kappa'}(h) = -\Phi^{(2),\alpha\beta}_{\kappa'\kappa}(\boldsymbol{R},\boldsymbol{0})\,h - \tfrac{1}{2}\Phi^{(3),\alpha\beta\beta}_{\kappa'\kappa\kappa}(\boldsymbol{R},\boldsymbol{0},\boldsymbol{0})\,h^2 - \tfrac{1}{6}\Phi^{(4),\alpha\beta\beta\beta}_{\kappa'\kappa\kappa\kappa}(\boldsymbol{R},\boldsymbol{0},\boldsymbol{0},\boldsymbol{0})\,h^3 + O(h^4),$$

where $\Phi^{(3)}$ and $\Phi^{(4)}$ are the third and fourth derivatives of E^{ad} (Sect. 5.7). The forces are the Hellmann–Feynman forces of Eq. (5.10), which a code returns at the cost of one self-consistent calculation. The quotient $-F/h$ has an error of first order in h, from the cubic term. Displacing by $+h$ and by $-h$ and taking the difference cancels all even powers of h:

$$\Phi^{(2),\alpha\beta}_{\kappa'\kappa}(\boldsymbol{R},\boldsymbol{0}) = -\frac{F^{\alpha}_{\boldsymbol{R}\kappa'}(h) - F^{\alpha}_{\boldsymbol{R}\kappa'}(-h)}{2h} + O(h^2). \tag{5.38}$$

This is the *frozen-phonon* method in its finite-displacement form [20, Sect. IV.B]. Because each calculation returns the force on every nucleus, one displacement gives a whole column of the force-constant matrix, $3N_{\mathrm{at}}$ numbers for a cell of N_{at} atoms; an energy alone would give one number. Symmetry reduces the work further. In silicon both atoms of the cell are equivalent under the space group, and the site symmetry T_d contains the threefold rotation about [111] that turns x into y and z. A single displacement of one atom along x, made with both signs, therefore fixes every force constant the supercell can hold. The step h is a compromise: large enough that the force exceeds the numerical noise of the calculation, typically 0.01 Å to 0.03 Å, and small enough that the $O(h^2)$ term of Eq. (5.38), set by $\Phi^{(4)}$, is negligible.

A displaced atom in a periodic calculation is not alone. A supercell built from $N_Q\times N_Q\times N_Q$ primitive cells repeats with the lattice of supercell translations $\boldsymbol{T}$, and moving the nucleus $(\boldsymbol{0},\kappa)$ moves all its images $(\boldsymbol{T},\kappa)$ with it. The force recorded on $(\boldsymbol{R},\kappa')$ is then the sum over images. With $\Phi^{(2)}(\boldsymbol{R},\boldsymbol{T}) = \Phi^{(2)}(\boldsymbol{R}-\boldsymbol{T},\boldsymbol{0})$ by translation invariance (Sect. 5.2), the supercell measures

$$\Phi^{\mathrm{SC},\alpha\beta}_{\kappa'\kappa}(\boldsymbol{R}) = \sum_{\boldsymbol{T}} \Phi^{(2),\alpha\beta}_{\kappa'\kappa}(\boldsymbol{R}+\boldsymbol{T},\boldsymbol{0}), \tag{5.39}$$

a sum of the true constant and its copies displaced by every supercell translation. The true $\Phi^{(2)}$ cannot be recovered from Φ^{SC} in general, but the dynamical matrix can, at special wavevectors. Write each cell vector of Eq. (5.16) as $\boldsymbol{R} = \boldsymbol{R}_s + \boldsymbol{T}$, with $\boldsymbol{R}_s$ one of the N_Q^3 cells of the supercell. If $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{T}} = 1$ for every $\boldsymbol{T}$, the phase depends only on $\boldsymbol{R}_s$, the sum over $\boldsymbol{T}$ acts on $\Phi^{(2)}$ alone, and

$$D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q}) = \frac{1}{\sqrt{M_\kappa M_{\kappa'}}} \sum_{\boldsymbol{R}_s} \Phi^{\mathrm{SC},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R}_s)\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}_s}$$

holds exactly, with $\Phi^{\mathrm{SC}}_{\kappa\kappa'}(\boldsymbol{R}_s)$ read off the supercell with the roles of the two nuclei exchanged by pair symmetry. The wavevectors with $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{T}} = 1$ are the N_Q^3 points of a uniform grid in the zone, the zone center among them. A frozen-phonon calculation in an N_Q^3 supercell therefore gives the exact phonons on an N_Q^3 grid of wavevectors and nothing more. At the zone center

the primitive cell is the supercell. The zone-boundary point X of silicon, $\boldsymbol{Q} = (0,0,1)\,2\pi/a$, has $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} = -1$ for half of the cell vectors and needs a cell twice as large, as the doubled period of the folded chain in Worked example 5.2 already showed. Long wavelengths need large cells. Force constants of range L need a supercell of side $\sim L$, with $N_{\mathrm{at}} \propto L^3$ atoms. By the rule of the opening paragraph each calculation then costs $\propto N_{\mathrm{at}}^3$ times the number of $\boldsymbol{k}$ points: $\propto L^6$ while the $\boldsymbol{k}$ grid can still be thinned, and $\propto L^9$ once a single $\boldsymbol{k}$ point suffices, the regime of the ninth-power scaling quoted by Baroni and coworkers [20, Sect. IV.B] and worked out in Appendix E.4. A dark-matter calculation needs the acoustic branches precisely near the zone center, where this route is most expensive.

The energy itself, rather than the force, yields a frequency in one stroke when the pattern of the displacement is fixed by symmetry. This is the original frozen-phonon method [20, Sect. IV.B]: the atoms are displaced along a known normal mode, and the energy is fitted as a function of its amplitude. The zone-center optical mode of silicon is such a mode.

Worked example 5.5: The optical frequency of silicon from a frozen phonon

At the zone center the two atoms of the cell move against each other, and by symmetry the polarization is fixed up to its direction. For a displacement along [111], atom 1 moves by $-(u/2)\hat{\boldsymbol{n}}$ and atom 2 by $+(u/2)\hat{\boldsymbol{n}}$, with $\hat{\boldsymbol{n}} = (1,1,1)/\sqrt{3}$, so that the bond along [111] changes its length by u. Every cell moves in the same way, and the primitive cell is the whole calculation. The energy per cell is $E_{\mathrm{cell}}^{\mathrm{ad}}(u) = \frac{1}{2}\Phi_{\mathrm{opt}}u^2 + \frac{1}{6}\Phi_{\mathrm{opt}}^{(3)}u^3 + \frac{1}{24}\Phi_{\mathrm{opt}}^{(4)}u^4 + \ldots$, measured from the minimum. In the PBE calculation of Fig. 5.2, $E_{\mathrm{cell}}^{\mathrm{ad}} = 5.943\,\mathrm{meV}$ at $u = +0.03\,\text{Å}$ and $6.316\,\mathrm{meV}$ at $u = -0.03\,\text{Å}$. The bond resists compression more than stretching.

From a single energy, $2E_{\mathrm{cell}}^{\mathrm{ad}}(u)/u^2 = \Phi_{\mathrm{opt}} + \Phi_{\mathrm{opt}}^{(3)}u/3 + \ldots$, we find $\Phi_{\mathrm{opt}} = 13.21\,\mathrm{eV\,Å^{-2}}$ from the stretched bond and $14.04\,\mathrm{eV\,Å^{-2}}$ from the compressed one. The spread is the cubic term: half the difference, $0.415\,\mathrm{eV\,Å^{-2}} = |\Phi_{\mathrm{opt}}^{(3)}|u/3$, puts $\Phi_{\mathrm{opt}}^{(3)}$ at $-41.5\,\mathrm{eV\,Å^{-3}}$, the value of Sect. 5.1. The symmetric combination removes it,

$$\frac{E_{\mathrm{cell}}^{\mathrm{ad}}(u) + E_{\mathrm{cell}}^{\mathrm{ad}}(-u) - 2E_{\mathrm{cell}}^{\mathrm{ad}}(0)}{u^2} = \Phi_{\mathrm{opt}} + \frac{\Phi_{\mathrm{opt}}^{(4)}u^2}{12} + \cdots = 13.62\,\mathrm{eV\,Å^{-2}},$$

and the same combination at $u = 0.06\,\text{Å}$ is $13.64\,\mathrm{eV\,Å^{-2}}$. The two symmetric estimates differ by $0.02\,\mathrm{eV\,Å^{-2}}$, within the precision of the computed energies, so the quartic term has to come from a fit to the whole curve. Polynomial fits to all the computed energies and forces, over ranges of $|u|$ from 0.1 Å to 0.3 Å, put the quartic constant $\Phi_{\mathrm{opt}}^{(4)}$ at $35\,\mathrm{eV\,Å^{-4}}$ to $42\,\mathrm{eV\,Å^{-4}}$, about $40\,\mathrm{eV\,Å^{-4}}$, so the quartic correction $\Phi_{\mathrm{opt}}^{(4)}u^2/12$ is $0.003\,\mathrm{eV\,Å^{-2}}$ at 0.03 Å, 0.02 %. The coordinate u carries the reduced mass $M_\kappa/2$, since each atom moves by $u/2$ and the kinetic energy is $\frac{1}{2}(M_\kappa/2)\dot{u}^2$. With $M_\kappa = 28.0855\,\mathrm{u}$,

$$\omega_{\mathrm{opt}} = \left(\frac{2\Phi_{\mathrm{opt}}}{M_\kappa}\right)^{1/2} = 63.68\,\mathrm{meV}.$$

The central difference of the Hellmann–Feynman forces, Eq. (5.38) applied to the coordinate u, yields $13.63\,\mathrm{eV\,Å^{-2}}$ and 63.71 meV from the same two calculations. DFPT at the same settings, the 4^3 calculation of Fig. 5.2(b), finds 63.7 meV, and the calculation of Fig. 5.3, with a higher cutoff and a finer grid, 63.69 meV. Neutron scattering finds 64.1 meV [1], 0.6 % higher, a difference set by the functional and the static lattice, not by the method. The one-sided estimates, by contrast, miss by 1.5 % at the smallest step and by 3 % at 0.06 Å. A finite-difference force constant is accurate only when it is symmetric.

5.4.2 Force constants as a density response

A second derivative of the energy needs only the first-order change of the electron density. Differentiate the Hellmann–Feynman force Eq. (5.10) once more. Its electrostatic form contains the nuclear positions in two places, the electron–nucleus potential and the nucleus–nucleus repulsion, and the electron density, which also changes when a nucleus moves. In a Kohn–Sham calculation the electron–nucleus term is the external potential $V_{\text{ext}}(\boldsymbol{r})$ of the electrons, the potential U_1 of Eq. (3.62) and the operator V_{eN} of Eq. (5.2). Writing $E^{\text{ad}} = E_{NN} + E_{\text{el}}$, with E_{NN} the repulsion of the nuclei and $\partial E_{\text{el}}/\partial u_i = \int \mathrm{d}^3 r\, n(\boldsymbol{r})\, \partial V_{\text{ext}}(\boldsymbol{r})/\partial u_i$ by the theorem, for any two displacement coordinates u_i, u_j [20, Eq. (10)],

$$\frac{\partial^2 E^{\text{ad}}}{\partial u_i\, \partial u_j} = \int \mathrm{d}^3 r\, \frac{\partial n(\boldsymbol{r})}{\partial u_j}\, \frac{\partial V_{\text{ext}}(\boldsymbol{r})}{\partial u_i} + \int \mathrm{d}^3 r\, n(\boldsymbol{r})\, \frac{\partial^2 V_{\text{ext}}(\boldsymbol{r})}{\partial u_i\, \partial u_j} + \frac{\partial^2 E_{NN}}{\partial u_i\, \partial u_j}. \tag{5.40}$$

The terms with n and with E_{NN} need only the ground state: the density at equilibrium and the positions of the nuclei, whose repulsion is an Ewald sum. The term with $\partial n/\partial u_j$ needs the change of the density to first order in a displacement, and nothing else. With a nonlocal pseudopotential (Sect. 3.5) the integrals of n against derivatives of V_{ext} become sums $2\sum_v \langle\psi_v|\partial V_{\text{ext}}|\psi_v\rangle$ over occupied states, and the structure is unchanged. This is the statement of Sect. 5.1 made precise: the restoring force on a displaced nucleus comes from the distortion of the electron cloud, and the force constant needs that distortion to first order.

A crystal is perturbed one wavevector at a time. Let the nuclei of species κ move along α in a wave, $u^{\beta}_{\boldsymbol{R}\kappa'} = \delta_{\kappa\kappa'}\delta_{\alpha\beta}\,\lambda\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$. The derivative with respect to its amplitude is

$$\partial_{\kappa\alpha,\boldsymbol{Q}} = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, \frac{\partial}{\partial u^{\alpha}_{\boldsymbol{R}\kappa}}.$$

Applied to the bare potential, with v_κ the potential of one nucleus (or pseudo-ion) of species κ, it yields $\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\text{ext}}(\boldsymbol{r}) = -\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, \partial_\alpha v_\kappa(\boldsymbol{r} - \boldsymbol{R} - \boldsymbol{\tau}_\kappa)$, which is a lattice-periodic function times $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}}$. The double sum $\sum_{\boldsymbol{R}\boldsymbol{R}'} \mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}'} \Phi^{(2)}(\boldsymbol{R}, \boldsymbol{R}')$ collapses by translation invariance to $N\sum_{\boldsymbol{R}} \Phi^{(2)}(\boldsymbol{0}, \boldsymbol{R})\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$, so the dynamical matrix Eq. (5.16) is a second derivative with respect to two such waves [20, Eq. (82)],

$$D^{\text{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q}) = \frac{1}{N\sqrt{M_\kappa M_{\kappa'}}}\, \partial_{\kappa\alpha,-\boldsymbol{Q}}\, \partial_{\kappa'\beta,\boldsymbol{Q}}\, E^{\text{ad}}. \tag{5.41}$$

Inserted in Eq. (5.40), its electronic part contains the integral of $[\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\text{ext}}]^*\, \partial_{\kappa'\beta,\boldsymbol{Q}} n$ over the crystal [20, Eq. (86)], with $[\partial_{\boldsymbol{Q}} V]^* = \partial_{-\boldsymbol{Q}} V$ since the potential is real. The density response to the wave at $\boldsymbol{Q}$ carries the same crystal momentum: the perturbed crystal is invariant under a lattice translation combined with the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$, and so is its first-order density. Different $\boldsymbol{Q}$ decouple, and each is a problem in the primitive cell.

The density response is the static bubble of Sect. 4.6. Kohn–Sham electrons move in V_{KS}, and a change of it changes their density by $\delta n = R^0\, \delta V_{\text{KS}}$, with R^0 the independent-particle response Eq. (4.63) at $\omega = 0$. The potential itself responds to the density through the Hartree and exchange–correlation terms,

$$\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\text{KS}}(\boldsymbol{r}) = \partial_{\kappa\alpha,\boldsymbol{Q}} V_{\text{ext}}(\boldsymbol{r}) + \int \mathrm{d}^3 r'\, \left[v_{\text{C}}(\boldsymbol{r} - \boldsymbol{r}') + f_{\text{xc}}(\boldsymbol{r}, \boldsymbol{r}')\right]\, \partial_{\kappa\alpha,\boldsymbol{Q}} n(\boldsymbol{r}'), \tag{5.42}$$

where f_{xc} is the kernel Eq. (4.69) in its static form, the second functional derivative $\delta^2 E_{\text{xc}}/\delta n(\boldsymbol{r})\delta n(\boldsymbol{r}')$ of the ground-state functional [4, Eqs. (42)–(46)]. This is Eq. (E.21). With $f_{\text{Hxc}} = v_{\text{C}} + f_{\text{xc}}$, solving $\delta n = R^0(\delta V_{\text{ext}} + f_{\text{Hxc}}\delta n)$ for δn yields $\delta n = R\, \delta V_{\text{ext}}$, with

$R = R^0 + R^0 f_{\mathrm{Hxc}} R$ the Dyson equation Eq. (4.70) at zero frequency. Inserting $\partial\boldsymbol{n} = R\,\partial V_{\mathrm{ext}}$ into the first term of Eq. (5.40), the electronic force constant is

$$N\sqrt{M_\kappa M_{\kappa'}}\, D^{\mathrm{dyn,el},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q}) = \iint \mathrm{d}^3r\,\mathrm{d}^3r'\,\left[\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\mathrm{ext}}(\boldsymbol{r})\right]^* R(\boldsymbol{r},\boldsymbol{r}';\omega=0)\,\partial_{\kappa'\beta,\boldsymbol{Q}} V_{\mathrm{ext}}(\boldsymbol{r}') + \int \mathrm{d}^3r\,\boldsymbol{n}(\boldsymbol{r})\,\partial_{\kappa\alpha,-\boldsymbol{Q}}\partial_{\kappa'\beta,\boldsymbol{Q}} V_{\mathrm{ext}}(\boldsymbol{r}),$$

the microscopic theory of force constants of Pick, Cohen and Martin [21]. The bare nuclei and the rigid ground-state density give a stiff lattice; the static response R is negative, since electrons move away from where their potential energy rises, and the electrons that flow to screen the displaced nuclei soften it. In a metal the screening is strong enough at long wavelength to turn the bare ionic plasma frequency into a sound wave. Within density-functional theory the static kernel is exactly $\delta^2 E_{\mathrm{xc}}/\delta\boldsymbol{n}\delta\boldsymbol{n}$, so a DFPT phonon contains the full static electronic response of the functional, and its error is the error of the functional (Appendix E.4). And since the electrons are relaxed at every frozen configuration, the adiabatic phonon frequency already includes the static part of the electron–phonon self-energy; Appendix D.17 uses this to avoid counting it twice.

5.4.3 The Sternheimer equation

The bubble Eq. (4.63) is a sum over pairs of occupied and empty states, and a converged sum over empty states is expensive. DFPT avoids it. Start from the change of one occupied Kohn–Sham state. Linearizing $(H_{\mathrm{KS}} - \varepsilon_v)\psi_v = 0$ leads to [20, Eq. (25)]

$$(H_{\mathrm{KS}} - \varepsilon_v)\,|\delta\psi_v\rangle = -(\delta V_{\mathrm{KS}} - \delta\varepsilon_v)\,|\psi_v\rangle, \qquad \delta\varepsilon_v = \langle\psi_v|\delta V_{\mathrm{KS}}|\psi_v\rangle. \tag{5.43}$$

Solved by expanding in eigenstates it is ordinary first-order perturbation theory, $\delta\psi_v = \sum_{m\neq v}\psi_m\langle\psi_m|\delta V_{\mathrm{KS}}|\psi_v\rangle/(\varepsilon_v - \varepsilon_m)$. In the density, $\delta\boldsymbol{n} = 2\sum_v(\psi_v^*\delta\psi_v + \psi_v\delta\psi_v^*)$ with the factor two for spin, the terms in which m is another occupied state v' cancel. The pair (v, v') contributes $\psi_v^*\psi_{v'}\langle\psi_{v'}|\delta V_{\mathrm{KS}}|\psi_v\rangle/(\varepsilon_v - \varepsilon_{v'})$; the complex conjugate of the pair (v', v) has the same numerator, since δV_{KS} is Hermitian, and the opposite denominator. A rotation among occupied states does not change the density, and only the component of $\delta\psi_v$ in the empty manifold matters [20, Eq. (29)]. Call it $\hat{P}^{\mathrm{c}}\delta\psi_v = \sum_c \psi_c\langle\psi_c|\delta V_{\mathrm{KS}}|\psi_v\rangle/(\varepsilon_v - \varepsilon_c)$, with $\hat{P}^{\mathrm{c}} = 1 - \hat{P}^{\mathrm{occ}}$ the projector on the empty states. Acting on it with $H_{\mathrm{KS}} - \varepsilon_v$ turns each denominator into -1:

$$(H_{\mathrm{KS}} - \varepsilon_v)\,\hat{P}^{\mathrm{c}}|\delta\psi_v\rangle = -\sum_c |\psi_c\rangle\langle\psi_c|\delta V_{\mathrm{KS}}|\psi_v\rangle = -\hat{P}^{\mathrm{c}}\,\delta V_{\mathrm{KS}}\,|\psi_v\rangle.$$

This is a linear equation for $\hat{P}^{\mathrm{c}}\delta\psi_v$ whose right-hand side contains only the occupied state ψ_v. The empty states have disappeared, and so has $\delta\varepsilon_v$, which multiplies ψ_v and is removed by the projector. The operator $H_{\mathrm{KS}} - \varepsilon_v$ is positive on the empty manifold of an insulator, where every eigenvalue exceeds ε_v, but singular or negative on the occupied one. Adding $a_{\mathrm{sh}}\hat{P}^{\mathrm{occ}}$ changes nothing for a solution that lies in the empty manifold, and with a_{sh} larger than the valence bandwidth every eigenvalue $\varepsilon_{v'} + a_{\mathrm{sh}} - \varepsilon_v$ on the occupied manifold becomes positive too. The operator is then positive definite, and the equation can be solved by conjugate gradients with the cost of a few applications of H_{KS} [20, Eq. (30)].

For a displacement wave the bookkeeping is in crystal momentum. The perturbation $\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\mathrm{KS}}$ carries $\boldsymbol{Q}$, so its matrix element between $\psi_{v\boldsymbol{k}}$ and a state of crystal momentum $\boldsymbol{k}'$ vanishes unless $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{Q}$ (Sect. 3.2). The change of $\psi_{v\boldsymbol{k}}$ lives at $\boldsymbol{k} + \boldsymbol{Q}$, the projector is the one on the occupied states at $\boldsymbol{k} + \boldsymbol{Q}$, and $\delta\varepsilon_{v\boldsymbol{k}}$ vanishes unless $\boldsymbol{Q} = 0$, where the projector removes it.

Sternheimer equation of density-functional perturbation theory

For each occupied state $\psi_{v\boldsymbol{k}}$ and each perturbation $(\kappa, \alpha, \boldsymbol{Q})$, the change $|\partial\psi_{v\boldsymbol{k}}\rangle$, of crystal momentum $\boldsymbol{k}+\boldsymbol{Q}$ and orthogonal to the occupied states, solves

$$(H_{\mathrm{KS}} + a_{\mathrm{sh}}\hat{P}^{\mathrm{occ}}_{\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}})\,|\partial\psi_{v\boldsymbol{k}}\rangle = -(1-\hat{P}^{\mathrm{occ}}_{\boldsymbol{k}+\boldsymbol{Q}})\,\partial_{\kappa\alpha,\boldsymbol{Q}}V_{\mathrm{KS}}\,|\psi_{v\boldsymbol{k}}\rangle, \tag{5.44}$$

with $\hat{P}^{\mathrm{occ}}_{\boldsymbol{k}+\boldsymbol{Q}} = \sum_{v'}|\psi_{v',\boldsymbol{k}+\boldsymbol{Q}}\rangle\langle\psi_{v',\boldsymbol{k}+\boldsymbol{Q}}|$ and a_{sh} larger than the valence bandwidth. The first-order density is

$$\partial_{\kappa\alpha,\boldsymbol{Q}}n(\boldsymbol{r}) = 4\sum_{v\boldsymbol{k}}\psi^*_{v\boldsymbol{k}}(\boldsymbol{r})\,\partial\psi_{v\boldsymbol{k}}(\boldsymbol{r}), \tag{5.45}$$

and Eqs. (5.42), (5.44) and (5.45) are solved to self-consistency.

Eq. (5.44) is the equation of Baroni and coworkers [20, Eq. (31)] and of Giustino [4, Eq. (48)], and the one that Appendix E.4 solves, Eq. (E.20). The factor four in Eq. (5.45) is two for spin and two from time reversal: the term $\psi_{v\boldsymbol{k}}\,(\partial_{-\boldsymbol{Q}}\psi_{v\boldsymbol{k}})^*$ of the density equals $\psi^*_{v,-\boldsymbol{k}}\,\partial_{\boldsymbol{Q}}\psi_{v,-\boldsymbol{k}}$ and doubles the sum over $\boldsymbol{k}$ [20, Eq. (35)]. Writing $\partial\psi_{v\boldsymbol{k}} = \mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{Q})\cdot\boldsymbol{r}}\,\partial u_{v\boldsymbol{k}}$ turns every quantity into a lattice-periodic function, and the calculation runs in the primitive cell at any $\boldsymbol{Q}$, commensurate or not, with a cost per perturbation comparable to that of the ground state. This is what the supercell could not do.

The equation is old. Sternheimer used it in 1954 to compute the polarizabilities of ions without summing over excited states [22]. In solids its self-consistent form, with the projector that makes it well posed, is due to Baroni, Giannozzi and Testa [23] and to Giannozzi and coworkers, who computed the full dispersion of silicon and other semiconductors with it [24]. The self-consistency loop has the structure of a ground-state calculation: Eq. (5.44) yields $\partial\psi$ for a trial potential, Eq. (5.45) the density, Eq. (5.42) a new potential. The loop starts from the bare $\partial V_{\mathrm{ext}}$ and converges to the screened ∂V_{KS}, which is also the electron–phonon vertex of Sect. 6.2. Appendix E.4 describes what a run returns, its convergence parameters and its cost.

5.4.4 The $2n+1$ theorem

The Hellmann–Feynman theorem gets a first derivative of the energy from the unperturbed state alone, and Eq. (5.40) gets a second derivative from the first-order density. Both are the lowest cases of one statement. Let a perturbation of strength λ change the energy functional to $E[\psi;\lambda]$, where ψ stands for all the occupied orbitals, and let $\psi(\lambda)$ be its minimum, so that the ground-state energy is $E_0(\lambda) = E[\psi(\lambda);\lambda]$. Expand $\psi(\lambda) = \psi^{(0)} + \lambda\psi^{(1)} + \lambda^2\psi^{(2)} + \dots$ and $E_0(\lambda) = \sum_l \lambda^l E^{(l)}$.

$2n+1$ theorem

The derivatives of the wavefunctions up to order n determine the derivatives of the energy up to order $2n+1$ [25, 26], [20, Sect. II.F.1].

The proof uses only stationarity. Truncate the expansion of the orbitals at order n, $\tilde{\psi} = \sum_{l\le n}\lambda^l\psi^{(l)}$, so that $\tilde{\psi} - \psi(\lambda) = O(\lambda^{n+1})$. Since $E[\,\cdot\,;\lambda]$ is stationary at $\psi(\lambda)$, its first variation there vanishes and a change of the orbitals by $\delta\psi$ changes it only at second order, $E[\psi(\lambda)+\delta\psi;\lambda] - E_0(\lambda) = O(\delta\psi^2)$. With $\delta\psi = O(\lambda^{n+1})$ the error is $O(\lambda^{2n+2})$, and $E[\tilde{\psi};\lambda]$, expanded in λ, agrees with $E_0(\lambda)$ through order λ^{2n+1}. The orbitals must stay orthonormal, which the truncated $\tilde{\psi}$ does only to order n. The argument goes through for a functional written for non-orthonormal orbitals, or with Lagrange multipliers expanded in the same way;

this is the part Gonze worked out for density-functional theory at arbitrary order [25].

For $n = 0$ the theorem is Eq. (5.10): the unperturbed orbitals fix the energy to first order. For $n = 1$ the first-order orbitals of the Sternheimer equation fix the energy to third order. The second derivatives are the force constants, and the third are the cubic constants $\Phi^{(3)}$ that make phonons decay (Sect. 5.7); one DFPT run per wavevector thus contains, in principle, the anharmonic couplings as well, without a second-order Sternheimer equation [20, Eq. (123)]. For two different perturbations the same reasoning shows that a mixed second derivative needs the first-order response to only one of them, as Eq. (5.40) needed $\partial \boldsymbol{n}/\partial u_j$ but not $\partial \boldsymbol{n}/\partial u_i$. The force on a nucleus per unit electric field, the Born charge of Sect. 5.5, can therefore be computed from the response to the displacement or from the response to the field, whichever is cheaper (Appendix E.4).

Stationarity also controls the numerical error. The second-order energy can be written as a functional of the trial $\delta\psi$ that is stationary at the solution of Eq. (5.44), and its error is then quadratic in the error of $\delta\psi$ [20, Sect. II.C.3]. The expression Eq. (5.40), which codes usually evaluate, is not of this form: it is linear in $\partial \boldsymbol{n}$, and an error in the first-order density enters the force constant linearly. That is why the self-consistency of a phonon calculation is converged far more tightly than that of a ground state (Appendix E.4).

5.4.5 Interpolation and the sum rule

A dispersion curve, a density of states or a Debye–Waller sum needs $D^{\mathrm{dyn}}(\boldsymbol{Q})$ at thousands of wavevectors, and a Sternheimer calculation at each is out of the question. The way out is Eq. (5.39) read backward. Compute D^{dyn} by DFPT on the uniform N_Q^3 grid, and transform back with Eq. (E.28). Inserting Eq. (5.16) and using $N_Q^{-3}\sum_{\boldsymbol{Q}\in\mathrm{grid}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot(\boldsymbol{R}'-\boldsymbol{R})} = \sum_{\boldsymbol{T}} \delta_{\boldsymbol{R}',\boldsymbol{R}+\boldsymbol{T}}$, where $\boldsymbol{T}$ runs over the translations of the N_Q^3 supercell, the transform returns

$$\frac{\sqrt{M_\kappa M_{\kappa'}}}{N_Q^3} \sum_{\boldsymbol{Q}\in\mathrm{grid}} D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} = \sum_{\boldsymbol{T}} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0}, \boldsymbol{R}+\boldsymbol{T}),$$

the aliased force constants Eq. (5.39) of a frozen-phonon calculation in the same supercell. DFPT on an N_Q^3 grid and finite displacements in an N_Q^3 supercell contain exactly the same information. They differ in cost, and DFPT can add any other $\boldsymbol{Q}$ at will. To interpolate, each aliased constant is assigned to the image $\boldsymbol{R}$ that lies in the Wigner–Seitz cell of the supercell, with equal weights on images at equal distance, and Eq. (5.16) is summed with these constants at any $\boldsymbol{Q}$ [20, Sect. II.D.3]. The result is exact on the grid. Off the grid its error is the part of $\Phi^{(2)}$ that reaches beyond the Wigner–Seitz cell, which the aliasing has folded back onto shorter vectors. The interpolation therefore works as well as the force constants are short ranged, and their range is the smoothness of $D^{\mathrm{dyn}}(\boldsymbol{Q})$ seen from the other side of the Fourier transform. In a nonpolar insulator, where D^{dyn} is a smooth function of $\boldsymbol{Q}$, the constants fall off rapidly with distance. In a metal the Kohn anomalies, kinks of D^{dyn} at wavevectors that connect two parallel pieces of the Fermi surface, make them oscillate and decay slowly. In a polar crystal the dipoles of the displaced charges make them fall as the inverse cube of the distance, and D^{dyn} is nonanalytic at $\boldsymbol{Q} = 0$. The remedy there is to subtract the dipole part, whose form is known exactly, interpolate the short-ranged remainder, and add the dipole part back (Sect. 5.5, Eq. (E.24)).

Fig. 5.4 tests this on silicon, with grids cut out of the single 6^3 calculation of Fig. 5.3. The 2^3 grid has three inequivalent points, Γ, X and L. At those points its frequencies agree with the 6^3 ones to the last digit, 17.06 meV for the transverse acoustic mode at X, as exactness on the grid requires. Between them it fails: at the three test points off all grids, its frequencies differ from direct DFPT by up to 4.0 meV, a quarter of the transverse acoustic frequency.

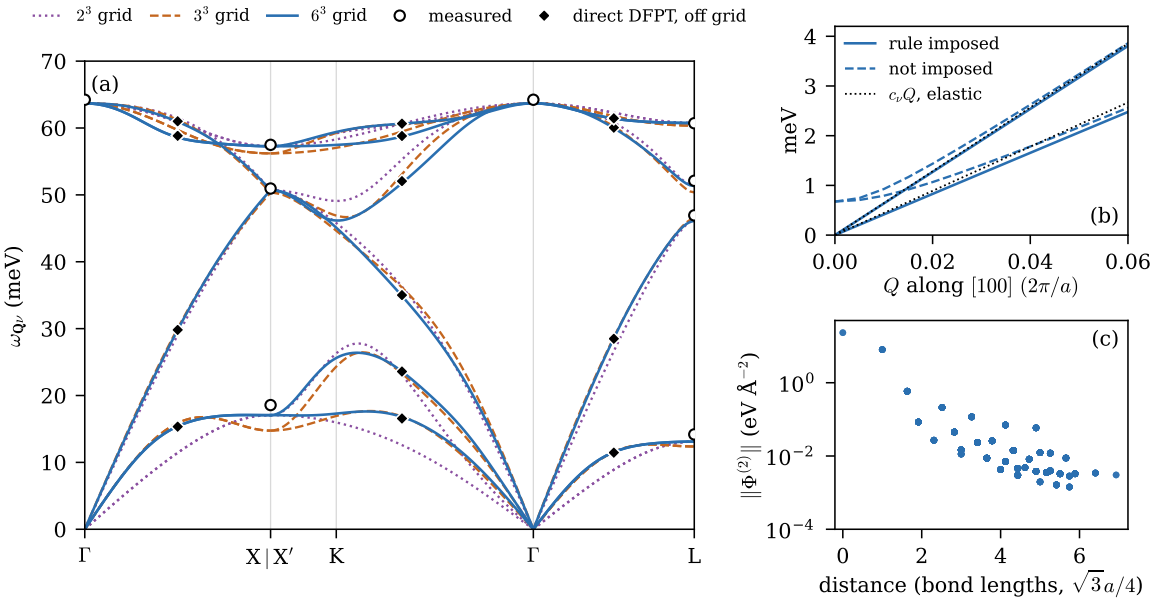


Figure 5.4: Fourier interpolation of the force constants of silicon. All curves come from one set of dynamical matrices: density-functional perturbation theory with the PBE functional, an SG15 norm-conserving pseudopotential, a 60 Ry cutoff, a 12^3 grid of crystal momenta, $a = 5.431$ Å and the natural mass of silicon, computed on a 6^3 grid of wavevectors (the calculation of Fig. 5.3). (a) Dispersion interpolated from the full 6^3 grid (solid), from its 3^3 subgrid (dashed) and from its 2^3 subgrid (dotted), which contains only Γ, X and L; acoustic sum rule imposed in each. Diamonds: frequencies from direct calculations, with the same settings, at three wavevectors that lie on none of the grids, $(0, 0, \frac{1}{2})$, $(\frac{1}{2}, \frac{1}{2}, 0)$ and $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ in units of $2\pi/a$. Circles: frequencies measured by inelastic neutron scattering at room temperature [1, 11], as tabulated in Ref. [13, Table I]. (b) Acoustic branches along [100] near Γ from the 6^3 grid, with the sum rule imposed (solid) and not imposed (dashed); dotted, $\omega = c_\nu Q$ with the measured sound speeds 8.43 km s^{-1} (longitudinal) and 5.84 km s^{-1} (transverse). (c) Norm of the 3×3 blocks $\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\mathbf{0}, \boldsymbol{R})$ of the 6^3 grid against the distance $|\boldsymbol{R} + \boldsymbol{\tau}_{\kappa'} - \boldsymbol{\tau}_\kappa|$. The distance is that of the shortest image in the supercell, in units of the bond length $\sqrt{3}a/4 = 2.35$ Å.

The 3^3 grid, with four inequivalent points, misses X and puts the transverse acoustic mode there 14 % low, at 14.75 meV; at the test points, however, it is within 1.2 meV. The 6^3 grid reproduces them to 0.3 meV, 1.5 % at worst, which is below the 2.6 % mean difference between this functional and experiment (Sect. 5.2). Panel (c) shows why. The on-site block is $23.3\,\mathrm{eV\,\AA^{-2}}$ and the nearest-neighbor block $8.0\,\mathrm{eV\,\AA^{-2}}$. Beyond the second neighbors most blocks fall below $0.1\,\mathrm{eV\,\AA^{-2}}$, with one family of exceptions: the neighbors along the $\langle 110\rangle$ zigzag chains of bonds, $0.59\,\mathrm{eV\,\AA^{-2}}$ at $a/\sqrt{2} = 3.84\,\text{Å}$, $0.21\,\mathrm{eV\,\AA^{-2}}$ at $\sqrt{19}\,a/4 = 5.92\,\text{Å}$ and still $0.12\,\mathrm{eV\,\AA^{-2}}$ at $\sqrt{2}a = 7.68\,\text{Å}$, 1.5 % of the nearest-neighbor value at more than three bond lengths. These long-range constants along $\langle 110\rangle$ are a known feature of the diamond and zincblende structures [20, Sect. V.A.1], and they are what the 2^3 and 3^3 supercells cannot hold. They control the flat transverse acoustic branch, which is where the coarse grids fail most.

The acoustic sum rule Eq. (5.18) is exact for E^{ad} but not for its computed approximation. The violation comes chiefly from the exchange–correlation energy, which is integrated on a real-space grid that is fixed in space. A rigid translation of the crystal relative to the grid changes the energy by a small periodic amount, and the computed $\sum_{\boldsymbol{R}\kappa'} \Phi^{(2)}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})$ is a small matrix $\Delta^{\mathrm{ASR}}_{\kappa}$ instead of zero. At $\boldsymbol{Q} = 0$ the acoustic modes then have $\omega^2 \sim \Delta^{\mathrm{ASR}}_{\kappa}/M_\kappa$ instead of zero, 0.675 meV (0.163 THz) in Fig. 5.4(b). The simplest repair subtracts the violation from the on-site constants,

$$\Phi^{(2),\alpha\beta}_{\kappa\kappa}(\boldsymbol{0},\boldsymbol{0}) \;\to\; \Phi^{(2),\alpha\beta}_{\kappa\kappa}(\boldsymbol{0},\boldsymbol{0}) - \sum_{\boldsymbol{R}\kappa'} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R}), \tag{5.46}$$

after which Eq. (5.18) holds exactly. The repair is not unique. When $\Delta^{\mathrm{ASR}}_{\kappa}$ is not symmetric in $\alpha\beta$, Eq. (5.46) breaks the pair symmetry of Sect. 5.2. Codes therefore also offer the smallest change of all the constants that restores both rules; in silicon $\Delta^{\mathrm{ASR}}_{\kappa}$ is a multiple of the identity by the cubic site symmetry, and the on-site repair keeps it. Neither repair imposes the rotational invariance of Sect. 5.2.

The repair matters most where dark matter looks. Near Γ the unrepaired branches follow $\omega^2 \approx \omega^2_{\mathrm{ASR}} + c_\nu^2 Q^2$, with $\omega_{\mathrm{ASR}} = 0.675\,\mathrm{meV}$, and they meet the straight lines only for $\omega \gg \omega_{\mathrm{ASR}}$. At $Q = 0.02\,(2\pi/a)$ the transverse acoustic frequency is 1.07 meV instead of 0.83 meV, 29 % high; at $Q = 0.1\,(2\pi/a)$ the difference has dropped to 1 %. With the rule imposed, the slopes along [100] are 5.44 and $8.34\,\mathrm{km\,s^{-1}}$, the PBE sound speeds of Sect. 5.2. An acoustic-phonon search with a meV threshold samples exactly the region that the unrepaired constants get wrong (Sect. 9.2). The size of ω_{ASR} before the repair is also a cheap diagnostic of the real-space grid, which Appendix E.4 uses.

The same linear-response calculation, with an electric field in place of a displacement, also yields the response of a polar crystal to the macroscopic fields that its optical phonons create, and the long-range part of D^{dyn} that the interpolation had to set aside. Sect. 5.5 turns to it.

5.5 Polar crystals

At the center of the zone the three optical phonons of silicon share one energy, 64.1 meV. In GaAs they do not. Two transverse optical (TO) modes sit at 33.6 meV and the longitudinal one (LO) at 36.4 meV [17], although the three move the nuclei in the same way, gallium against arsenic, and differ only in the direction of that motion relative to $\boldsymbol{Q}$. The reason is that gallium and arsenic are not equivalent. When the two sublattices move against each other, every cell acquires an electric dipole. In a longitudinal wave the dipoles add up to a macroscopic polarization, whose field pulls the ions back and stiffens the mode; in a transverse wave they produce no such field. For dark matter this dipole is what makes a polar crystal

a useful target. A dark photon, or a dark-matter particle that couples to charge through a light dark photon, couples to it coherently, cell by cell, and polar crystals such as GaAs were proposed as targets for sub-MeV dark matter for this reason [27]. The charge that sets the size of the dipole is the Born effective charge. We will see how it splits LO from TO, how the Lyddane–Sachs–Teller relation ties the splitting to the two dielectric constants, and how a polar mode couples to a uniform field. The derivations, for the two-atom cubic crystal and for any number of polar modes, are in Appendix D.14.

5.5.1 Born effective charges

The fields that matter here vary over many cells. The microscopic field $\boldsymbol{E}$ of Sect. 1.4 oscillates within each cell, and what a long wave sees is its average over a cell, the *macroscopic electric field* $\boldsymbol{\mathcal{E}}$, which is the $\boldsymbol{G} = 0$ component of the field in the language of Sect. 4.4. The matching density of dipole moment is the *polarization* $\boldsymbol{P}$, and in the Heaviside–Lorentz units of the book the displacement field is $\boldsymbol{D} = \boldsymbol{\mathcal{E}} + \boldsymbol{P}$ (Appendix A.2), so that the electronic dielectric constant of Sect. 4.4 is $\varepsilon_\infty = 1 + \partial P/\partial \mathcal{E}$ at clamped nuclei.

Now let each sublattice κ of the crystal move rigidly by $\boldsymbol{u}_\kappa$, the zone-center displacement of Eq. (5.12), in a uniform field $\boldsymbol{\mathcal{E}}$. The electrons follow both adiabatically. The energy per cell at fixed field, $E^{\mathrm{ad}}_{\mathcal{E}}$, reduces to the energy surface $E^{\mathrm{ad}}_{\mathrm{cell}}$ of Sect. 5.1 at $\boldsymbol{\mathcal{E}} = 0$. To second order in both variables it is

$$E^{\mathrm{ad}}_{\mathcal{E}}(u, \boldsymbol{\mathcal{E}}) = \frac{1}{2} \sum_{\kappa\kappa'} \boldsymbol{u}_\kappa \cdot C^{\mathrm{an}}_{\kappa\kappa'} \cdot \boldsymbol{u}_{\kappa'} - e \sum_{\kappa} \boldsymbol{u}_\kappa \cdot Z^*_\kappa \cdot \boldsymbol{\mathcal{E}} - \frac{\Omega}{2}\, \boldsymbol{\mathcal{E}} \cdot (\varepsilon_\infty - 1) \cdot \boldsymbol{\mathcal{E}}, \tag{5.47}$$

with Ω the cell volume. Each coefficient is a second derivative, and this defines it. $C^{\mathrm{an}}_{\kappa\kappa'} = \sum_{\boldsymbol{R}} \Phi^{(2)}_{\kappa\kappa'}(\boldsymbol{0}, \boldsymbol{R})$ is the zone-center block of the force constants of Sect. 5.2, taken at zero macroscopic field; ε_∞ is the electronic dielectric tensor; and the mixed derivative defines the *Born effective charge* tensor Z^*_κ of each sublattice. The first derivatives are the force on a nucleus and the polarization, $\boldsymbol{F}_\kappa = -\partial E^{\mathrm{ad}}_{\mathcal{E}}/\partial \boldsymbol{u}_\kappa$ and $\boldsymbol{P} = -\Omega^{-1} \partial E^{\mathrm{ad}}_{\mathcal{E}}/\partial \boldsymbol{\mathcal{E}}$:

$$\boldsymbol{F}_\kappa = -\sum_{\kappa'} C^{\mathrm{an}}_{\kappa\kappa'} \cdot \boldsymbol{u}_{\kappa'} + e\, Z^*_\kappa \cdot \boldsymbol{\mathcal{E}}, \qquad \boldsymbol{P} = \frac{e}{\Omega} \sum_{\kappa} \boldsymbol{u}_\kappa \cdot Z^*_\kappa + (\varepsilon_\infty - 1) \cdot \boldsymbol{\mathcal{E}}. \tag{5.48}$$

The force identifies the Born charge as the force on nucleus κ per unit uniform field. The polarization identifies the same tensor as the dipole per cell created by a unit displacement:

$$e\, Z^{*,\alpha\beta}_\kappa = \left.\frac{\partial F^\alpha_\kappa}{\partial \mathcal{E}^\beta}\right|_{\boldsymbol{u}=0} = \Omega \left.\frac{\partial P^\beta}{\partial u^\alpha_\kappa}\right|_{\boldsymbol{\mathcal{E}}=0}. \tag{5.49}$$

The two derivatives are equal because both are the same mixed second derivative of $E^{\mathrm{ad}}_{\mathcal{E}}$, a Maxwell relation. The first Cartesian index of Z^*_κ belongs to the force, or to the displacement, and the second to the field, or to the polarization; this is the convention of Eq. (E.23). Some codes and papers store the transpose [20, Eqs. (89)–(91)], which matters only at sites of low symmetry, since at a cubic site Z^*_κ is a multiple of the identity.

The Born charge measures the charge that moves with a nucleus. A field acts on the bare nucleus with the force $Z_\kappa e \boldsymbol{\mathcal{E}}$, and on its electrons, which are pulled the other way and drag on the nucleus through the forces of Eq. (5.10). If the electrons of each atom followed their nucleus rigidly, Z^*_κ would be the net charge of the atom. They do not: as the sublattices move, the bonds change length and electronic charge flows along them, and this flow adds to the dipole. A Born charge is therefore a dynamical quantity. It need not be an integer, and it differs from any charge one might assign to an atom by dividing up the static density, a division that has no unique answer.

The Born charges of a neutral crystal add up to zero. Displacing all sublattices together by the same $\boldsymbol{u}$, with the electrons following, translates the charge distribution of every cell. The dipole moment of a neutral distribution does not depend on where it sits, and $\boldsymbol{P}$ does not change. By Eq. (5.48) this requires $\boldsymbol{u} \cdot \sum_\kappa Z^*_\kappa = 0$ for every $\boldsymbol{u}$, that is

$$\sum_\kappa Z^*_\kappa = 0, \tag{5.50}$$

the charge neutrality of the Born charges, also called their acoustic sum rule. In GaAs each site has tetrahedral symmetry, both tensors are multiples of the identity, and neutrality leaves one number, $Z^*_{\mathrm{Ga}} = -Z^*_{\mathrm{As}} \equiv Z^*$. In silicon the two atoms are exchanged by inversion through the bond center, which forces $Z^*_1 = Z^*_2$, and neutrality then forces both to vanish (Appendix E.4). Silicon has no dipole to offer.

The polarization in Eq. (5.49) needs a definition. For a molecule the dipole moment is $\int \mathrm{d}^3 r\, \boldsymbol{r}\, \rho(\boldsymbol{r})$ with ρ the charge density. For a crystal the same integral over one cell depends on where the cell boundaries are drawn, and with the periodic boundary conditions of Sect. 3.1 the position operator is not even well defined. What is measurable is a change of polarization, the charge per unit area that flows through the crystal while the nuclei are displaced slowly. King-Smith and Vanderbilt showed that this change is a Berry phase of the occupied Bloch states, accumulated as $\boldsymbol{k}$ crosses the zone [28]. Equivalently, the electronic polarization is $-e g_{\mathrm{deg}}/\Omega$ times the sum of the centers of the occupied Wannier functions of one cell (Sect. 3.6), with $g_{\mathrm{deg}} = 2$ for spin. This *Berry-phase theory* of polarization defines $\boldsymbol{P}$ only up to a quantum, a lattice vector times $e g_{\mathrm{deg}}/\Omega$, because a Wannier center can be assigned to any cell; the Born charge, a derivative, is free of that ambiguity. The formula and its evaluation on a $\boldsymbol{k}$ grid are Eq. (E.26).

The density-functional perturbation theory that yields the phonons (Sect. 5.4) also yields ε_∞ and Z^*_κ. A uniform field enters the Sternheimer equation as one more perturbation, through the matrix elements of $\boldsymbol{r}$ between occupied and empty states (Eq. (E.22)), and its self-consistent solution is the induced polarization. Either side of Eq. (5.49) can then be evaluated, and Appendix E.4 describes the practice. The sum rule Eq. (5.50) is a cheap convergence test. In the PBE calculation behind Fig. 5.5, the two charges of GaAs miss neutrality by 12 % on the 8^3 grid of crystal momenta used for the phonons. Only near a 28^3 grid does their sum vanish, with $Z^* = 2.094$, within 1 % of the value 2.11 that the measured LO–TO splitting implies (Worked example 5.6).

The dielectric constant converges more slowly, from 26.0 on the 8^3 grid to 13.7 on the 28^3 grid, where successive grids differ by less than one percent. In this calculation PBE puts the direct gap of GaAs at Γ at 0.57 eV, against the measured 1.52 eV (Table E.1). The response to a field, built from the energy denominators of Eq. (E.22), is then dominated by the small region around Γ where the gap is smallest, and a coarse grid samples it poorly. The converged value is still 26 % above the measured $\varepsilon_\infty = 10.88$ [29]. That error belongs to the functional: a gap that is too small makes the electrons too polarizable (Eq. (E.27)), and no numerical parameter removes it. The Born charge, as we saw, comes through almost untouched.

5.5.2 LO–TO splitting

In Eq. (5.48) the field is still an external parameter. In a lattice wave no one applies the field: the displaced ions create it, and it acts back on them. Let the displacements vary as $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}}$, with $|\boldsymbol{Q}|$ small compared with the reciprocal-lattice vectors, so that within many cells the displacement is the uniform $\boldsymbol{u}_\kappa$ of Eq. (5.47). We also take $|\boldsymbol{Q}|$ large compared with ω/c, which lets us neglect retardation and use electrostatics; we will come back to this condition. Two equations then fix the field. A static field has no curl, so the field of a wave is longitudinal,

$\boldsymbol{\mathcal{E}} = \hat{\boldsymbol{Q}}\,\mathcal{E}_\parallel$, and there is no free charge, so $\hat{\boldsymbol{Q}} \cdot \boldsymbol{D} = 0$. With Eq. (5.48) the second condition fixes $\mathcal{E}_\parallel$ by the displacements. Inserted into the force, the field adds an extra force-constant matrix (Appendix D.14):

Nonanalytic term of the dynamical matrix

$$D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q} \to 0) = \frac{1}{\sqrt{M_\kappa M_{\kappa'}}}\Big[C^{\mathrm{an},\alpha\beta}_{\kappa\kappa'} + \frac{e^2}{\Omega}\,\frac{(Z^*_\kappa \cdot \hat{\boldsymbol{Q}})^\alpha\,(Z^*_{\kappa'} \cdot \hat{\boldsymbol{Q}})^\beta}{\hat{\boldsymbol{Q}} \cdot \varepsilon_\infty \cdot \hat{\boldsymbol{Q}}}\Big], \tag{5.51}$$

with $(Z^*_\kappa \cdot \hat{\boldsymbol{Q}})^\alpha = \sum_\beta Z^{*,\alpha\beta}_\kappa \hat{Q}^\beta$, the analytic part C^{an} computed at zero macroscopic field, and the masses as in Eq. (5.16).

This is C^{na} of Eq. (D.75) with the masses divided out, the Heaviside–Lorentz form of the result of Gonze and Lee [30] and of Ref. [20, Eq. (98)], and the term that Eq. (E.24) adds in a calculation. The second term is the *nonanalytic term*. The reason for the name is that it depends on the direction $\hat{\boldsymbol{Q}}$ from which the zone center is approached but not on $|\boldsymbol{Q}|$. The macroscopic field is the $\boldsymbol{G} = 0$ Coulomb interaction e^2/Q^2 acting between two dipole densities, each proportional to $\boldsymbol{Q}$, and the ratio stays finite as $Q \to 0$ without having a limit. No Taylor series in $\boldsymbol{Q}$ can represent such a function, and the dynamical matrix of a polar crystal at $\boldsymbol{Q} = 0$ is not one matrix but a family labeled by $\hat{\boldsymbol{Q}}$.

In a cubic crystal with two atoms per cell the consequences are explicit. There $Z^*_1 = -Z^*_2 = Z^*$ times the identity, ε_∞ is a scalar, and the acoustic sum rule of Sect. 5.2 leaves one zone-center constant, $C^{\mathrm{an}}_{11} = C^{\mathrm{an}}_{22} = -C^{\mathrm{an}}_{12} = C_0$ times the identity. The optical modes move only the relative coordinate $\boldsymbol{u}_{\mathrm{rel}} = \boldsymbol{u}_1 - \boldsymbol{u}_2$, an oscillator with the reduced mass $\mu_{\mathrm{red}} = M_1M_2/(M_1 + M_2)$ of the pair, as in the diatomic chain of Worked example 5.2, driven by the force $eZ^*\boldsymbol{\mathcal{E}}$ (Appendix D.14). In a transverse optical mode, $\boldsymbol{u}_{\mathrm{rel}} \perp \hat{\boldsymbol{Q}}$, the dipoles create no longitudinal field, and the frequency is that of the springs alone, $\omega^2_{\mathrm{T}} = C_0/\mu_{\mathrm{red}}$. In the longitudinal mode, $\boldsymbol{u}_{\mathrm{rel}} \parallel \hat{\boldsymbol{Q}}$, the condition $\hat{\boldsymbol{Q}} \cdot \boldsymbol{D} = 0$ makes the field $\boldsymbol{\mathcal{E}} = -eZ^*\boldsymbol{u}_{\mathrm{rel}}/(\Omega\varepsilon_\infty)$, which opposes the displacement. The longitudinal frequency ω_{L} is then higher,

$$\omega^2_{\mathrm{L}} = \omega^2_{\mathrm{T}} + \frac{e^2 Z^{*2}}{\Omega\,\varepsilon_\infty\,\mu_{\mathrm{red}}}, \tag{5.52}$$

which is Eq. (E.25). The difference $\omega_{\mathrm{L}} - \omega_{\mathrm{T}}$ is the *LO–TO splitting*. The longitudinal mode carries the electrostatic energy of its own polarization, screened by the electrons, and the Born charge enters squared, divided by ε_∞. A quick estimate shows the size. With $Z^* = 2$, $\Omega = 45\,\text{Å}^3$, $\varepsilon_\infty = 11$ and $\mu_{\mathrm{red}} = 36\,\mathrm{u}$, the added term is $\sim 2 \times 10^{-4}\,\mathrm{eV}^2$, small against $\omega^2_{\mathrm{T}} \approx 1.1 \times 10^{-3}\,\mathrm{eV}^2$; the splitting is a few meV, less than a tenth of the optical energy.

Fig. 5.5 shows the computed phonons of GaAs. The transverse optical energy at Γ comes out at 34.22 meV, 1.8 % above the measured 33.62 meV. PBE usually makes phonons slightly soft, but here the measured lattice constant compresses the crystal relative to the PBE minimum at 5.746 Å (Table E.1). The longitudinal branch meets the zone center at 36.34 meV from every direction. In a cubic crystal $Z^*_\kappa \cdot \hat{\boldsymbol{Q}} = Z^*_\kappa \hat{\boldsymbol{Q}}$ and $\hat{\boldsymbol{Q}} \cdot \varepsilon_\infty \cdot \hat{\boldsymbol{Q}} = \varepsilon_\infty$ for every $\hat{\boldsymbol{Q}}$, and the nonanalytic term does not depend on the direction; in an anisotropic crystal such as sapphire it does, and so does the LO frequency at Γ [31]. The computed splitting is 2.12 meV against the measured 2.73 meV, 22 % short, and Eq. (5.52) names the culprit. The added term is $e^2Z^{*2}/(\Omega\varepsilon_\infty\mu_{\mathrm{red}})$, the Born charge is right to a percent, and ε_∞ is 26 % too large. With the measured ε_∞ and the computed Z^* and ω_{T}, the same formula yields a splitting of 2.65 meV, within 3 % of the measurement (the diamond in Fig. 5.5(b)). The computed LO energy itself lands on the measured one, but by accident: the transverse energy is high and the splitting short by nearly the same amount. The near-equal masses of gallium and arsenic

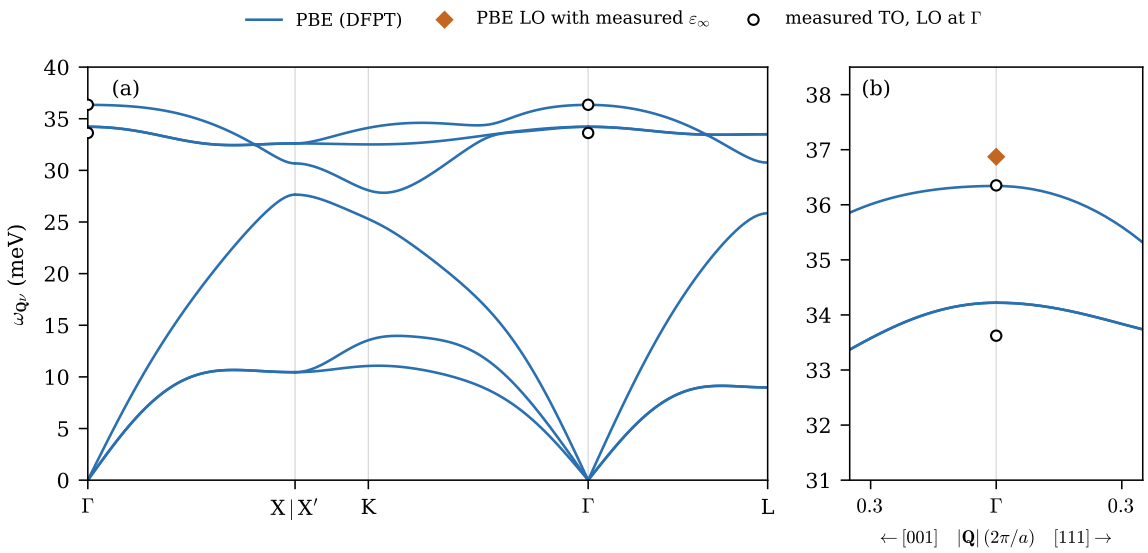


Figure 5.5: Phonon dispersion of GaAs. (a) Computed along Γ–X–K–Γ–L with density-functional perturbation theory: PBE functional, SG15 norm-conserving pseudopotentials with the gallium $3d$ shell in the valence, a 60 Ry cutoff, an 8^3 grid of crystal momenta, the static-lattice constant $a = 5.638$ Å of Table E.1 (the measured value extrapolated to zero temperature with the zero-point expansion removed) and natural masses, on a 4^3 grid of wavevectors. The force constants are Fourier interpolated with the dipole term of Eq. (5.51) subtracted and added back, using $\varepsilon_\infty = 13.7$ and $Z^* = 2.094$ computed on a 28^3 grid of crystal momenta; acoustic sum rules imposed on the force constants and on the Born charges. Circles: measured TO and LO energies at Γ, 8.13 and 8.79 THz [17], as tabulated in Ref. [18, Table I]. (b) Optical branches near Γ along [001] (left) and [111] (right), with $|\boldsymbol{Q}|$ in units of $2\pi/a$. Diamond: LO energy at Γ from Eq. (5.52) with the computed TO energy and Born charge and the measured $\varepsilon_\infty = 10.88$ [29].

leave almost no gap between the acoustic and optical bands: the longitudinal acoustic branch ends at 27.65 meV at X, and the optical band begins 0.2 meV higher.

The nonanalytic term is also why the interpolation of Sect. 5.4 needs an extra step in a polar crystal. In real space the dipole–dipole interaction behind Eq. (5.51) falls off as the inverse cube of the distance, too slowly for any grid of wavevectors to capture, and the computed $D^{\mathrm{dyn}}(\boldsymbol{Q})$ near Γ is not smooth. Since the term is fixed by Z^*_κ and ε_∞ alone, it can be subtracted from the computed matrices in an Ewald-summed form that is periodic in $\boldsymbol{Q}$, the short-ranged remainder interpolated, and the term added back analytically at each target $\boldsymbol{Q}$ [30]. Fig. 5.5 was interpolated in this way from a 4^3 grid (Appendix E.4).

The electrostatic treatment assumed $|\boldsymbol{Q}| \gg \omega/c$. For wavevectors comparable with ω/c, the transverse optical phonon mixes with the photon of the same wavevector and frequency, and the two form the coupled modes called *phonon polaritons*. The crossover happens near $Q \sim \varepsilon_0^{1/2}\omega_{\mathrm{T}}$ in natural units, where the photon line $\omega = Q/\varepsilon_0^{1/2}$ meets the phonon, about 0.1 eV for $\omega_{\mathrm{T}} \sim 30$ meV, less than 10^{-4} of the size of the zone. Dark-matter scattering transfers $q \sim m_\chi v \gtrsim 1$ eV for any mass above a keV, and for the phonons it excites the electrostatic result is exact. Absorption is different. A boson of mass m from the halo, moving at $v \sim 10^{-3}$, deposits $\omega \approx m$ with a momentum of only $mv \sim 10^{-3}\omega$, deep inside the polariton region, where the crystal sees a field that is uniform in space, the case of the coupling below.

5.5.3 The Lyddane–Sachs–Teller relation

A uniform field oscillating at frequency ω, $\boldsymbol{\mathcal{E}}\,\mathrm{e}^{-\mathrm{i}\omega t}$, is the opposite limit, $Q \to 0$ at fixed ω, and it measures the dielectric function of the crystal with the lattice included. The field now drives the relative displacement without any longitudinal constraint, as an oscillator with its natural frequency at ω_{T}, and the electrons respond with ε_∞ as long as ω lies far below the gap, the adiabatic condition of Sect. 5.1. The dielectric function $\varepsilon(\omega) = D/\mathcal{E}$ is then (Appendix D.14)

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{e^2 Z^{*2}}{\Omega\,\mu_{\mathrm{red}}\,(\omega_{\mathrm{T}}^2 - \omega^2)} = \varepsilon_\infty\,\frac{\omega_{\mathrm{L}}^2 - \omega^2}{\omega_{\mathrm{T}}^2 - \omega^2}, \tag{5.53}$$

where the second form uses Eq. (5.52) to replace $e^2Z^{*2}/(\Omega\mu_{\mathrm{red}})$ by $\varepsilon_\infty(\omega_{\mathrm{L}}^2 - \omega_{\mathrm{T}}^2)$. At frequencies well above ω_{T} the ions cannot follow, and $\varepsilon \to \varepsilon_\infty$, which is why the electronic constant is also called the high-frequency one (Sect. 4.4). At $\omega = 0$ the ions follow fully and add their polarization. The *static dielectric constant* $\varepsilon_0 = \varepsilon(0)$, lattice included, is $\varepsilon_\infty + e^2Z^{*2}/(\Omega\mu_{\mathrm{red}}\omega_{\mathrm{T}}^2)$, and the second form of Eq. (5.53) turns this into a relation between measurable quantities alone:

Lyddane–Sachs–Teller relation

$$\frac{\varepsilon_0}{\varepsilon_\infty} = \frac{\omega_{\mathrm{L}}^2}{\omega_{\mathrm{T}}^2}. \tag{5.54}$$

For several polar modes, $\varepsilon_0/\varepsilon_\infty = \prod_\nu \omega_{\mathrm{L}\nu}^2/\omega_{\mathrm{T}\nu}^2$ (Eq. (D.76)).

This is the relation of Lyddane, Sachs and Teller [32]. The Born charge, the masses and the cell volume have dropped out: the ratio of the two dielectric constants is fixed by the ratio of two frequencies. The general form, with one factor per polar mode, is due to Cochran and Cowley [33], and Appendix D.14 derives it from the interlacing of the poles and zeros of $\varepsilon(\omega)$.

Those poles and zeros are the content of Eq. (5.53), and they hold beyond the model. The transverse frequency is a pole: a transverse mode creates no field, so it needs none, and it shows up where the response to a field diverges. The longitudinal frequency is a zero. A longitudinal mode has $\boldsymbol{\mathcal{E}} \parallel \hat{\boldsymbol{Q}}$ and needs $\hat{\boldsymbol{Q}} \cdot \boldsymbol{D} = \varepsilon(\omega)\,\mathcal{E}_\parallel = 0$, which allows a nonzero field

only where $\varepsilon(\omega)$ vanishes. This is the same condition that fixes the plasmon of Sect. 4.5, a self-sustained longitudinal oscillation of the electrons; the LO phonon is a self-sustained longitudinal oscillation of the ions. Between ω_T and ω_L, Eq. (5.53) is negative, no transverse wave propagates, and light in this band, the reststrahlen band, is reflected [9, Chap. 27]. The loss function $\mathrm{Im}[-1/\varepsilon(\omega)]$ of Sect. 4.4 peaks at ω_L, and the absorption ε_2 at ω_T, once the phonons are given a width.

Worked example 5.6: The Lyddane–Sachs–Teller relation for GaAs

We test Eq. (5.54) on measured numbers. Neutron scattering puts the optical phonons of GaAs at $\omega_\mathrm{T} = 8.13\,\mathrm{THz} = 33.62\,\mathrm{meV}$ and $\omega_\mathrm{L} = 8.79\,\mathrm{THz} = 36.35\,\mathrm{meV}$ [17], as tabulated in Ref. [18, Table I], and infrared spectroscopy measured $\varepsilon_0 = 12.9$ and $\varepsilon_\infty = 10.88$ [29], as quoted in Ref. [27].

The relation. The two sides are

$$\frac{\varepsilon_0}{\varepsilon_\infty} = \frac{12.9}{10.88} = 1.186, \qquad \frac{\omega_\mathrm{L}^2}{\omega_\mathrm{T}^2} = \left(\frac{8.79}{8.13}\right)^2 = 1.169,$$

equal to 1.4 %. Read as a prediction, the relation puts the longitudinal phonon at $\omega_\mathrm{L} = \omega_\mathrm{T}(\varepsilon_0/\varepsilon_\infty)^{1/2}$, or 36.61 meV, 0.26 meV or 0.7 % above the measurement. The splitting itself comes out as 2.99 meV against 2.73 meV, 9 % high. The relative error grows because the ratio is close to one: $\omega_\mathrm{L}^2 - \omega_\mathrm{T}^2$ is $0.17\,\omega_\mathrm{T}^2$, and a 1.4 % error in the ratio is a 10 % error in that difference. For a harmonic crystal with adiabatic electrons the relation is exact. The residual therefore belongs to the inputs. The four numbers come from different measurements on different samples, and the dielectric constants of GaAs change at the percent level between room temperature and liquid-helium temperature [31].

The Born charge. Solved for the Born charge, Eq. (5.52) reads $Z^{*2} = \Omega\varepsilon_\infty\mu_\mathrm{red}(\omega_\mathrm{L}^2 - \omega_\mathrm{T}^2)/e^2$. With $a = 5.638\,\text{Å}$ (Table E.1), $\Omega = a^3/4 = 44.80\,\text{Å}^3 = 5.831 \times 10^{-9}\,\mathrm{eV}^{-3}$ in natural units, the masses 69.723 and 74.922 u, so that $\mu_\mathrm{red} = 36.11\,\mathrm{u} = 33.64\,\mathrm{GeV}$, $e^2 = 4\pi\alpha = 0.0917$ and $\omega_\mathrm{L}^2 - \omega_\mathrm{T}^2 = 1.910 \times 10^{-4}\,\mathrm{eV}^2$,

$$Z^* = \left[\frac{(5.831 \times 10^{-9}) \times 10.88 \times (3.364 \times 10^{10}) \times (1.910 \times 10^{-4})}{0.0917}\right]^{1/2} = 2.11.$$

The static constant offers a second route, since Eq. (5.53) at $\omega = 0$ gives $\varepsilon_0 - \varepsilon_\infty = e^2 Z^{*2}/(\Omega\mu_\mathrm{red}\omega_\mathrm{T}^2)$. It yields 2.21, 5 % more, the same 1.4 % inconsistency of the inputs magnified by the difference $\varepsilon_0 - \varepsilon_\infty = 2.02$. The splitting, measured directly by one experiment, is the better input.

Against the calculation. The converged PBE Born charge, 2.094, lies 0.7 % below the value just extracted, while the PBE ε_∞, 13.7, lies 26 % above the measurement, which is why the computed splitting of Fig. 5.5 falls short. Wherever a rate needs the combination $Z^{*2}/\varepsilon_\infty$, the measured splitting, or the computed Born charge with the measured ε_∞, is the input to use (Appendix E.4).

5.5.4 Coupling to a uniform field

The middle term of Eq. (5.47), $-e\sum_\kappa \boldsymbol{u}_\kappa \cdot Z_\kappa^* \cdot \boldsymbol{\mathcal{E}}$, is the energy of the dipoles of the displaced sublattices in the field. For a field uniform over the crystal and slow compared with the electrons, it adds a term to the nuclear Hamiltonian of Eq. (5.26):

$$H_\mathcal{E}(t) = -e\sum_{\boldsymbol{R}\kappa} \boldsymbol{u}_{\boldsymbol{R}\kappa} \cdot Z_\kappa^* \cdot \boldsymbol{\mathcal{E}}(t).$$

Inserting the phonon expansion Eq. (5.32), the lattice sum $\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} = N\delta_{\boldsymbol{Q},0}$ keeps only the modes at the zone center:

$$H_{\mathcal{E}}(t) = -e\sqrt{N}\sum_{\nu} \frac{\boldsymbol{z}_\nu \cdot \boldsymbol{\mathcal{E}}(t)}{\sqrt{2\omega_{0\nu}}}\, B_{0\nu}, \qquad \boldsymbol{z}_\nu = \sum_{\kappa} \frac{\boldsymbol{e}_{\kappa\nu}(0)\cdot Z^*_\kappa}{\sqrt{M_\kappa}}, \tag{5.55}$$

with $\boldsymbol{z}_\nu$ the mode charge vector of Appendix D.14 and $\omega_{0\nu}$ the frequency of branch ν at Γ. The amplitude to create one phonon from the ground state, by Eq. (5.34), is $-e\sqrt{N}\,\boldsymbol{z}_\nu\cdot\boldsymbol{\mathcal{E}}/\sqrt{2\omega_{0\nu}}$.

In Eq. (5.55) the acoustic modes do not couple. At Γ their polarization vectors are $(M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}$ (Sect. 5.2), so that $\boldsymbol{z}_\nu = \hat{\boldsymbol{e}}\cdot\sum_\kappa Z^*_\kappa/\sqrt{M_{\mathrm{cell}}}$, which vanishes by the charge neutrality Eq. (5.50): a uniform field cannot push a neutral crystal. In GaAs the optical modes at Γ have $\boldsymbol{e}_1 = (M_2/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}$ and $\boldsymbol{e}_2 = -(M_1/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}$, and the same algebra as in Appendix D.14 leads to $\boldsymbol{z}_\nu = Z^*\hat{\boldsymbol{e}}/\sqrt{\mu_{\mathrm{red}}}$: the field couples to the optical mode polarized along it, through the Born charge and the reduced mass. The squared amplitude is proportional to N. All cells are driven in phase, the whole crystal absorbs the quantum, and the rate grows with the target mass, as for the single-phonon amplitude of Sect. 5.3, without the momentum transfer that a scattering process has to supply.

Which resonance a probe excites depends on what it holds fixed. A field $\boldsymbol{\mathcal{E}}$ imposed inside the crystal drives the transverse resonance, the pole of ε at ω_{T}. A probe that fixes the source of $\boldsymbol{D}$, as a moving charge does, sees $1/\varepsilon$ and excites the longitudinal mode at its zero, ω_{L}. A dark photon couples through its in-medium mixing with the photon. For a dark photon at rest, of mass $m = \omega$, the medium screens that mixing by $1/\varepsilon(\omega)$: the absorption rate is the photon absorption $\omega\,\mathrm{Im}\,\varepsilon$ times the screened coupling $1/|\varepsilon|^2$, and so proportional to the loss function $\mathrm{Im}[-1/\varepsilon(\omega)]$ [27, Eqs. (2) and (3)], which peaks at the zero of ε, ω_{L}. The absorption of dark-photon dark matter in GaAs therefore peaks at the LO energy, and Chap. 11 works out the rate. The longitudinal field also acts on the electrons. Its potential grows as $1/Q$ at small Q, the Fröhlich coupling, with a strength set by ω_{L} and $1/\varepsilon_\infty - 1/\varepsilon_0$ (Eq. (D.78)); Sect. 6.2 takes it up.

The five numbers of this section, Z^*, ε_∞, ε_0, ω_{T} and ω_{L}, obey two relations, Eqs. (5.52) and (5.54), so that any three fix the other two. The polar single-phonon rate of Sect. 11.2 and the Fröhlich vertex of Sect. 6.2 need $Z^{*2}/\varepsilon_\infty$, which the LO–TO splitting measures directly, and for GaAs the splitting is a safer input than a semilocal ε_∞. So far each phonon has been probed one quantum at a time. How the displacements of a harmonic crystal are correlated at finite temperature, which controls the response at large momentum transfer, is the question of Sect. 5.6.

5.6 Displacement correlations

A dark-matter particle that transfers 50 keV of momentum to a silicon nucleus in a cold crystal pushes on a particle that is spread over 0.05 Å by its zero-point motion. The phase $\boldsymbol{q}\cdot\boldsymbol{u}$ that the push imprints then varies by more than a radian across the nuclear wavefunction, and the expansion of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}}$ to first order, which gave the single-phonon amplitude of Sect. 5.3, no longer holds. In some events the nucleus takes up the momentum and the crystal is left in its ground state. Otherwise it emits one phonon, or two, or more, and at large enough momentum transfer the many-phonon sum turns into the free recoil of a lone nucleus. To find out how often each happens, we need the thermal averages of exponentials of displacements, at one time and at two. For a harmonic crystal both are exponentials of quadratic forms in $\boldsymbol{q}$, the Debye–Waller exponent $2W_\kappa(\boldsymbol{q})$ and the displacement correlation $\mathcal{C}_{\kappa\kappa'}(t)$, and both follow from the tensor B_κ and the phonon modes of Sect. 5.3. The same algebra describes the optical spectrum of a localized electron whose transition moves the nuclei around it, where it

leads to the Franck–Condon factors and the Huang–Rhys factor. We will compute the elastic weight and the phonon-number distribution of a silicon nucleus at 10 keV and 50 keV, and the Franck–Condon progression of a displaced oscillator.

5.6.1 The Bloch identity

The operator through which a momentum transfer $\boldsymbol{q}$ acts on the nucleus κ in cell $\boldsymbol{R}$ is $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}$, with $\boldsymbol{X} = \boldsymbol{R} + \boldsymbol{\tau}_\kappa + \boldsymbol{u}_{\boldsymbol{R}\kappa}$ (Eq. (5.1)). The lattice position contributes a phase, and the displacement contributes the operator $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}}$. By Eq. (5.32), $\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}$ is a linear combination of the phonon operators $b_{\boldsymbol{Q}\nu}$ and $b^\dagger_{\boldsymbol{Q}\nu}$. Its exponential contains every power of them and connects the thermal state to states with any number of extra phonons. The first question is its average.

For a classical Gaussian variable x with zero mean, the average of $\mathrm{e}^{\mathrm{i}qx}$ is $\mathrm{e}^{-q^2\langle x^2\rangle/2}$ (Eq. (C.45)). A harmonic crystal in thermal equilibrium makes the same statement for operators. Its density matrix $\mathrm{e}^{-\beta H_\mathrm{N}}/\mathcal{Z}$ is Gaussian in the normal coordinates and momenta, because H_N of Eq. (5.31) is quadratic in them. For any operator $\hat{A}$ linear in the phonon operators,

$$\langle \mathrm{e}^{\hat{A}}\rangle = \mathrm{e}^{\langle \hat{A}^2\rangle/2},$$

which is Bloch's identity, Eq. (C.53). The proof in Appendix D.15 differentiates $\langle \mathrm{e}^{\theta\hat{A}}\rangle$ with respect to θ. Moving each ladder operator around the trace and past $\mathrm{e}^{\theta\hat{A}}$ costs only numbers, because the commutators of $\hat{A}$ with b and $b^\dagger$ are numbers and $b\,\mathrm{e}^{-\beta H_\mathrm{N}} = \mathrm{e}^{-\beta\omega}\mathrm{e}^{-\beta H_\mathrm{N}} b$ for each mode. The result is $\mathrm{d}\langle \mathrm{e}^{\theta\hat{A}}\rangle/\mathrm{d}\theta = \theta\langle\hat{A}^2\rangle\langle \mathrm{e}^{\theta\hat{A}}\rangle$, whose solution at $\theta = 1$ is the identity. The identity was first proved by Bloch, and Mermin gave a short derivation by a different route [34]. Expanded in powers of $\hat{A}$, it is Wick's theorem: the average of $\hat{A}^{2n}$ is the sum over the $(2n)!/(2^n n!)$ ways of pairing the $2n$ factors, each pair worth $\langle\hat{A}^2\rangle$.

With $\hat{A} = \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}$, $\langle\hat{A}^2\rangle = -\langle(\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa})^2\rangle = -\sum_{\alpha\beta} q^\alpha B_\kappa^{\alpha\beta} q^\beta$, with the displacement tensor B_κ of Eq. (5.35).

Debye–Waller factor

For a nucleus κ of a harmonic crystal in thermal equilibrium,

$$\left\langle \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}}\right\rangle = \mathrm{e}^{-W_\kappa(\boldsymbol{q})}, \qquad 2W_\kappa(\boldsymbol{q}) = \boldsymbol{q}\cdot B_\kappa\cdot\boldsymbol{q} = \frac{1}{N}\sum_{\boldsymbol{Q}\nu}\frac{|\boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2}{2M_\kappa\,\omega_{\boldsymbol{Q}\nu}}\coth\frac{\omega_{\boldsymbol{Q}\nu}}{2T}, \quad (5.56)$$

the same in every cell (Eq. (D.81)). At a site of cubic symmetry, $2W_\kappa(\boldsymbol{q}) = q^2\langle u_x^2\rangle$.

The factor e^{-W_κ} is an amplitude: the probability amplitude for the nucleus to absorb $\boldsymbol{q}$ and leave the lattice in the state it found it in, averaged over that state. The probability is its square, the *Debye–Waller factor* $\mathrm{e}^{-2W_\kappa(\boldsymbol{q})}$, and $W_\kappa(\boldsymbol{q})$ is the Debye–Waller exponent. In X-ray and neutron diffraction the factor multiplies the intensity of every Bragg peak, which is how B_κ is measured; in silicon at 293 K, with the reference value $\langle u_x^2\rangle = 5.94\times10^{-3}\,\text{Å}^2$ of Worked example 5.4, it reduces the (111) reflection, at $|\boldsymbol{G}_{111}| = \sqrt{3}\,2\pi/a = 3.95$ keV, by only 2.4 %. Note that the electron–phonon literature also uses the name "Debye–Waller" for the term of second order in the displacements of the electron–phonon interaction. That term shifts electronic energies and is a different object, the Allen–Heine term of Sect. 6.3 (Appendix A.6); in this book the name means Eq. (5.56) only.

Eq. (5.56) inherits the structure of B_κ. The exponent grows as q^2, and it is largest along directions in which the nucleus is soft. Each mode contributes in proportion to the projection of $\boldsymbol{q}$ on its polarization, divided by its energy, and weighted by $2n_{\boldsymbol{Q}\nu}+1$, which grows from

one at $T = 0$ to $2T/\omega_{\boldsymbol{Q}\nu}$ at high temperature. For one species at a cubic site in the Debye model, the exponent is Eq. (D.84). At $T = 0$ it is

$$2W_\kappa(\boldsymbol{q}) = \frac{3q^2}{4M_\kappa\omega_\mathrm{D}} = \frac{3}{2}\frac{E_R}{\omega_\mathrm{D}}, \qquad E_R = \frac{q^2}{2M_\kappa},$$

with E_R the recoil energy that a free nucleus would take (Sect. 1.3). This is the scale that controls everything below: the lattice stays in its ground state with high probability as long as the recoil energy is small compared with typical phonon energies, and almost never once it is large. For numbers we use the computed spectrum, since the Debye model misses the flat transverse acoustic branches (Worked example 5.4).

5.6.2 The displacement correlation

A scattering rate needs more than an average. By Eq. (4.13), the dynamic structure factor of the nuclei is the Fourier transform in time of the correlation of the density operator at time t with its adjoint at time zero. For one nucleus the density is $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}$, and the static phases cancel between the two factors, leaving

$$\big\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(t)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(0)} \big\rangle,$$

with the Heisenberg operators $b_{\boldsymbol{Q}\nu}(t) = b_{\boldsymbol{Q}\nu}\mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}$ of Sect. 5.3. The two exponents are still linear in the phonon operators, but they do not commute, and Bloch's identity has to be combined with the Baker–Campbell–Hausdorff formula. For two operators $\hat{A}$ and $\hat{B}$ linear in the phonon operators, the commutator is a number, $\mathrm{e}^{\hat{A}}\mathrm{e}^{\hat{B}} = \mathrm{e}^{\hat{A}+\hat{B}}\mathrm{e}^{[\hat{A},\hat{B}]/2}$ (Eq. (C.49)), and Bloch's identity applied to $\hat{A}+\hat{B}$ leads to

$$\langle \mathrm{e}^{\hat{A}}\mathrm{e}^{\hat{B}} \rangle = \exp\Big[\tfrac{1}{2}\langle\hat{A}^2\rangle + \tfrac{1}{2}\langle\hat{B}^2\rangle + \langle\hat{A}\hat{B}\rangle\Big],$$

Eq. (C.54). With $\hat{A} = -\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(t)$ and $\hat{B} = \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(0)$, the first two terms are $-W_\kappa(\boldsymbol{q})$ each, since the thermal state is stationary and the average of a square does not depend on time. The third term is a correlation of two displacements at different times. Inserting Eq. (5.32) twice, the averages $\langle b_{\boldsymbol{Q}\nu}b^\dagger_{\boldsymbol{Q}\nu}\rangle = n_{\boldsymbol{Q}\nu}+1$ and $\langle b^\dagger_{\boldsymbol{Q}\nu}b_{\boldsymbol{Q}\nu}\rangle = n_{\boldsymbol{Q}\nu}$ now come with the phases $\mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}$ and $\mathrm{e}^{+\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}$, and the *displacement correlation* of the nuclei κ and κ' is

$$\mathcal{C}^{\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},t) = \big\langle u^\alpha_{\boldsymbol{R}\kappa}(t)\, u^\beta_{\boldsymbol{0}\kappa'}(0)\big\rangle = \frac{1}{N}\sum_{\boldsymbol{Q}\nu} \frac{e^\alpha_{\kappa\nu}(\boldsymbol{Q})\, e^\beta_{\kappa'\nu}(\boldsymbol{Q})^*\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}}{2\omega_{\boldsymbol{Q}\nu}\sqrt{M_\kappa M_{\kappa'}}} \Big[(n_{\boldsymbol{Q}\nu}+1)\,\mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{Q}\nu}t} + n_{\boldsymbol{Q}\nu}\,\mathrm{e}^{\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}\Big], \tag{5.57}$$

Eq. (D.82). For one nucleus, $\boldsymbol{R} = \boldsymbol{0}$ and $\kappa' = \kappa$, we write $\mathcal{C}_{\kappa\kappa}(t)$. Its value at $t = 0$ is the displacement tensor, $\mathcal{C}_{\kappa\kappa}(0) = B_\kappa$. The term with $n_{\boldsymbol{Q}\nu}+1$ creates a phonon at time zero and removes it at time t, and oscillates at the positive frequency $\omega_{\boldsymbol{Q}\nu}$; the term with $n_{\boldsymbol{Q}\nu}$ absorbs a thermal phonon first, and it vanishes at $T = 0$. Unlike B_κ, the correlation is complex, because $\boldsymbol{u}(t)$ and $\boldsymbol{u}(0)$ do not commute. Its imaginary part, $\mathrm{Im}\,\mathcal{C}_{\kappa\kappa}(t) = -\sum_{\boldsymbol{Q}\nu}\boldsymbol{e}_{\kappa\nu}\boldsymbol{e}^*_{\kappa\nu}\sin(\omega_{\boldsymbol{Q}\nu}t)/(2NM_\kappa\omega_{\boldsymbol{Q}\nu})$, does not depend on temperature; we will see that it carries the recoil energy.

Generating formula

For a nucleus κ of a harmonic crystal in thermal equilibrium,

$$\big\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(t)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(0)} \big\rangle = \mathrm{e}^{-2W_\kappa(\boldsymbol{q})}\, \mathrm{e}^{\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa}(t)\cdot\boldsymbol{q}}, \tag{5.58}$$

with $2W_\kappa$ of Eq. (5.56) and $\mathcal{C}_{\kappa\kappa}(t)$ of Eq. (5.57). For two nuclei in different cells or sublattices, the prefactor is $\mathrm{e}^{-W_\kappa - W_{\kappa'}}$ and the correlation is $\mathcal{C}_{\kappa\kappa'}(\boldsymbol{R}, t)$ (Eq. (D.83)).

This is the starting point of the theory of neutron scattering from crystals [9, App. N], [35, Chaps. 3 and 4]. The left side equals one at $t = 0$, and so does the right side, since $\boldsymbol{q} \cdot \mathcal{C}_{\kappa\kappa}(0) \cdot \boldsymbol{q} = 2W_\kappa$. The single-nucleus form, Eq. (5.58), is the *self* part of the nuclear response, in which the same nucleus scatters at both times. The full structure factor of the crystal adds the pairs of different nuclei, whose phases $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_\kappa-\boldsymbol{\tau}_{\kappa'})}$ produce the Bragg peaks and the crystal-momentum conservation of the single-phonon amplitude of Sect. 5.3; Sect. 11.3 builds the complete response from both.

Both W_κ and $\mathcal{C}_{\kappa\kappa'}(t)$ are sums over all phonons of the zone, computed after the phonon calculation from the frequencies and polarization vectors that DFPT yields on a fine grid of wavevectors, by Fourier interpolation of the force constants (Sect. 5.4 and Appendix E.4). The cost of the sums is negligible next to the phonon calculation itself. The acoustic modes near Γ make the summand of $2W_\kappa$ diverge as $1/\omega$ or $1/\omega^2$, and a uniform grid needs the analytic correction Eq. (E.30). The same small frequencies control $\mathcal{C}_{\kappa\kappa'}(t)$ at long times, which set the shape of the spectrum at small energies.

5.6.3 Expansion in phonon number

Expanding the second exponential of Eq. (5.58) in powers of $\boldsymbol{q} \cdot \mathcal{C}_{\kappa\kappa}(t) \cdot \boldsymbol{q}$ sorts the response by the number of phonons involved. The nth power is a sum of products of n factors $\mathrm{e}^{\mp\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}$, one per phonon created (upper sign) or absorbed (lower sign), and its Fourier transform puts weight at the sums of n phonon energies with these signs. The term $n = 0$ is time independent and is the elastic weight e^{-2W_κ} at $\omega = 0$. The term $n = 1$ is the single-phonon response of Sect. 5.3, now multiplied by e^{-2W_κ}.

The bookkeeping is simplest at a site of cubic symmetry with one species of nucleus, as in silicon. There $\mathcal{C}_{\kappa\kappa}(t)$ is a multiple of the identity, like B_κ. Its trace reduces to an integral over the density of states as in Eq. (5.36):

$$\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa}(t)\cdot\boldsymbol{q} = \frac{q^2}{3}\int_0^\infty \mathrm{d}\omega\, \frac{g_{\mathrm{ph}}(\omega)}{n_{\mathrm{at}}}\, \frac{(n_{\mathrm{B}}(\omega)+1)\,\mathrm{e}^{-\mathrm{i}\omega t} + n_{\mathrm{B}}(\omega)\,\mathrm{e}^{\mathrm{i}\omega t}}{2M_\kappa\,\omega} = 2W_\kappa(\boldsymbol{q})\int_{-\infty}^\infty \mathrm{d}\omega\; P_1(\omega)\,\mathrm{e}^{-\mathrm{i}\omega t}.$$

Here $P_1(\omega)$ is the single-phonon distribution, proportional to $g_{\mathrm{ph}}(|\omega|)[n_{\mathrm{B}}(\omega)+1]/\omega$ for both signs of ω and normalized to one; at negative ω, $n_{\mathrm{B}}(\omega)+1 = -n_{\mathrm{B}}(|\omega|)$ gives the absorption weight. The prefactor is $2W_\kappa$, since at $t = 0$ the left side is $\boldsymbol{q}\cdot B_\kappa\cdot\boldsymbol{q}$. Each power of the exponential is then a convolution in frequency, and the Fourier transform of Eq. (5.58) becomes

$$\int_{-\infty}^\infty \mathrm{d}t\; \mathrm{e}^{\mathrm{i}\omega t}\,\mathrm{e}^{-2W_\kappa}\,\mathrm{e}^{\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa}(t)\cdot\boldsymbol{q}} = 2\pi\,\mathrm{e}^{-2W_\kappa}\sum_{n=0}^\infty \frac{(2W_\kappa)^n}{n!}\,P_n(\omega), \tag{5.59}$$

with $P_0(\omega) = \delta(\omega)$ and P_n the n-fold convolution of P_1 with itself, normalized to one. The shape of each term comes from the phonon spectrum; its weight does not. The probability that a momentum transfer $\boldsymbol{q}$ to one nucleus involves n phonons is the Poisson distribution

$$p_n^{\mathrm{ph}} = \mathrm{e}^{-2W_\kappa(\boldsymbol{q})}\,\frac{[2W_\kappa(\boldsymbol{q})]^n}{n!}, \tag{5.60}$$

with mean $2W_\kappa$. At $T = 0$ every phonon is created, P_1 lives at positive ω, and n is the number of phonons emitted. At $T > 0$, n counts creations and absorptions together, and a term $n = 2$ that absorbs one phonon and emits another lands near $\omega = 0$. Eq. (5.60) is the oscillator result Eq. (C.51) for a momentum kick, distributed over all the modes of the crystal: a kick is

a displacement in momentum, and the number of quanta it creates out of a Gaussian state is Poisson distributed.

The mean energy deposited takes no calculation at all. The first-moment identity Eq. (4.28) fixes it by a double commutator, and for $\hat{O} = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}$ only the kinetic energy $\boldsymbol{p}^2/2M_\kappa$ of the nucleus fails to commute with $\hat{O}$. With $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}\,\boldsymbol{p}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}} = \boldsymbol{p}-\boldsymbol{q}$, the commutator is $[\boldsymbol{p}^2/2M_\kappa, \hat{O}] = \hat{O}\,(\boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa + q^2/2M_\kappa)$, and the double commutator is the number $[\hat{O}^\dagger, [H, \hat{O}]] = q^2/M_\kappa$. The self response is even in $\boldsymbol{q}$, since it depends on $\boldsymbol{q}$ only through quadratic forms, so the two terms on the right of Eq. (4.28) are equal. Since the weights of Eq. (5.59) add up to one, the mean energy deposited is

$$\mathrm{e}^{-2W_\kappa} \sum_{n=0}^{\infty} \frac{(2W_\kappa)^n}{n!} \int_{-\infty}^{\infty} \mathrm{d}\omega\; \omega\, P_n(\omega) = \frac{q^2}{2M_\kappa} = E_R. \tag{5.61}$$

The average energy left in the lattice is the free recoil energy, at every q and every temperature, whatever the force constants. The derivation used only the kinetic energy of the nucleus, not the harmonic approximation or the phonon modes, so the identity holds on any potential-energy surface, anharmonic ones included. In Eq. (5.58) it appears as the slope at $t = 0$. The polarization vectors are complete at each $\boldsymbol{Q}$ (Eq. (5.17)), $\sum_\nu e^\alpha_{\kappa\nu}(\boldsymbol{Q}) e^\beta_{\kappa\nu}(\boldsymbol{Q})^* = \delta_{\alpha\beta}$, and the derivative of Eq. (5.57) at $t = 0$ is therefore $-\mathrm{i}\,\mathbf{1}/2M_\kappa$. The thermal factors cancel between creation and absorption, as for the single oscillator of Worked example 2.7, and $\mathrm{i}\,\mathrm{d}/\mathrm{d}t$ of Eq. (5.58) at $t = 0$ is $q^2/2M_\kappa$. The elastic weight e^{-2W_κ} and the first moment behave oppositely: heating the crystal shrinks the elastic weight, since $2W_\kappa$ grows with $\langle u_x^2\rangle$, but leaves the first moment untouched, since the weight that leaves $\omega = 0$ goes to both signs of ω.

The Poisson weights and the sum rule together lay out the regimes of the nuclear response. While $2W_\kappa \ll 1$, the elastic term dominates the total weight, the single-phonon term carries nearly all the inelastic weight, and the first-order expansion of Sect. 5.3 holds. Once $2W_\kappa$ approaches one, two- and three-phonon terms carry a comparable share. For $2W_\kappa \gg 1$, the Poisson distribution peaks at $n \approx 2W_\kappa$ with relative width $(2W_\kappa)^{-1/2}$, the elastic weight is exponentially small, and the spectrum is a broad peak centered at E_R, the free recoil of the standard nuclear-recoil rate (Sect. 11.4). At $T = 0$ the mean energy per phonon, $E_R/2W_\kappa = 1/(2M_\kappa\langle u_x^2\rangle)$, is the inverse of the mean of $1/\omega$ over the spectrum, the typical frequency $\langle\omega^{-1}\rangle^{-1}$ of Sect. 5.3. The ratio of the recoil energy to that typical phonon energy is the zero-temperature $2W_\kappa$ itself, and it decides which regime a given momentum transfer is in.

Worked example 5.7: The Debye–Waller exponent of silicon

We take the natural mass of silicon, $M_\kappa = 26.16\,\mathrm{GeV}$, and the values of $\langle u_x^2\rangle$ from Worked example 5.4. The computed density of states gives $2.45 \times 10^{-3}\,\text{Å}^2$ at $T = 0$ and $6.38 \times 10^{-3}\,\text{Å}^2$ at 293 K; the lattice-dynamical fit to the measured phonons gives $5.94 \times 10^{-3}\,\text{Å}^2$ at 293 K [8]. With $\hbar c = 1973.27\,\mathrm{eV\,Å}$, $1\,\text{Å}^2$ is $2.568 \times 10^{-7}\,\mathrm{eV}^{-2}$.

$q = 10\,\mathrm{keV}$. The recoil energy is $E_R = q^2/2M_\kappa = 1.91\,\mathrm{meV}$, far below the phonon energies. At $T = 0$, $2W_\kappa = q^2\langle u_x^2\rangle = 1 \times 10^8 \times 2.45 \times 10^{-3} \times 2.568 \times 10^{-7} = 0.063$, the elastic weight is 0.939, and by Eq. (5.60) one phonon is created with probability 0.059 and two with probability 0.0019. At 293 K, $2W_\kappa = 0.164$ (0.153 with the reference value) and $\mathrm{e}^{-2W_\kappa} = 0.85$. A momentum transfer of 10 keV is in the single-phonon regime, and the expansion of Sect. 5.3 is accurate to the size of the two-phonon term, a few percent of the inelastic weight.

$q = 50\,\mathrm{keV}$. The recoil energy is 47.8 meV, comparable to the optical phonon. At $T = 0$, $2W_\kappa = 1.57$, and the Poisson weights for zero, one, two, three and four phonons are 0.21,

0.33, 0.26, 0.13 and 0.05: almost half of all scatterings, 0.47, create two or more phonons. The mean energy per phonon is $E_R/2W_\kappa = 30.4\,\mathrm{meV}$, the typical frequency $\langle\omega^{-1}\rangle^{-1}$ of the computed spectrum in Worked example 5.4. At 293 K, $2W_\kappa = 4.10$ (3.81 with the reference value), and the elastic weight has dropped to 0.017.
Where $2W_\kappa = 1$. The crossover from the single-phonon to the multiphonon regime sits at $q = \langle u_x^2\rangle^{-1/2}$: 40 keV at $T = 0$ and 26 keV at 293 K with the reference value, the scale anticipated in Sect. 5.3. For a cryogenic detector the first number applies. It is smaller than the 56 keV of the single 60 meV oscillator of Appendix C.6 because the crystal's zero-point motion is set by $\langle\omega^{-1}\rangle^{-1} = 30.4\,\mathrm{meV}$, half the optical energy: the soft transverse acoustic modes move the nucleus most. Indeed, a single oscillator of frequency 30.4 meV reproduces the zero-temperature $2W_\kappa$ exactly, since its $|\alpha_\mathrm{d}|^2 = E_R/\omega_0$ is then $1/(2M_\kappa\omega_0) \times q^2 = q^2\langle u_x^2\rangle$; it misses only the shape of the spectrum.
The Debye model. With $\omega_\mathrm{D} = 55.6\,\mathrm{meV}$, Eq. (D.84) gives $2W_\kappa = 1.29$ at 50 keV and $T = 0$, and 2.64 at 293 K, low by 18 % and 35 %. The elastic weight is correspondingly too large, 0.28 instead of 0.21 at $T = 0$, and at room temperature 0.071 instead of 0.017, a factor of four. A Debye estimate locates the crossover to within 10 % in q, but a rate that depends on the elastic or single-phonon weight at $2W_\kappa \gtrsim 1$ needs the full spectrum.

5.6.4 The displaced oscillator

An electron bound to a defect in silicon, or to the nitrogen–vacancy center of diamond, absorbs light at one energy and emits it at a lower one. Both spectra show a sharp line with a train of broader replicas spaced by a phonon energy. The reason is that the electronic transition changes the forces on the nearby nuclei. In the language of Sect. 5.1, each electronic state has its own potential-energy surface, and the minima of the two surfaces sit at different nuclear positions. The transition happens at fixed nuclei, and afterwards the nuclei relax to the new minimum and shed the difference as phonons. The momentum kick of Eq. (5.60) displaces the nuclear momentum; the electronic transition displaces the equilibrium position. In both cases the lattice ends up in a displaced Gaussian state, and the same Poisson statistics follow.

The simplest model keeps one vibrational mode, of mass M and frequency ω_0, with the same frequency in both electronic states. In the ground electronic state g and the excited state e the nuclear Hamiltonians are

$$H_\mathrm{g} = \frac{p^2}{2M} + \frac{1}{2}M\omega_0^2 x^2, \qquad H_\mathrm{e} = E_\mathrm{ZPL} + \frac{p^2}{2M} + \frac{1}{2}M\omega_0^2 (x - \Delta x)^2, \tag{5.62}$$

where Δx is the shift of the equilibrium position and E_ZPL the energy between the two minima, the zero-point energies being equal. The difference $H_\mathrm{e} - H_\mathrm{g}$ is linear in x, the linear coupling of the electron to the mode. In the Born–Oppenheimer approximation of Sect. 5.1, the vibronic states are products of an electronic state and an eigenstate of the oscillator that belongs to it. With $x = \ell_0(d + d^\dagger)$ and $\ell_0 = (2M\omega_0)^{-1/2}$ as in Appendix C.6, the displacement operator $\hat{D}(\alpha_\mathrm{d})$ of Eq. (C.50) with real $\alpha_\mathrm{d} = \Delta x/2\ell_0$ shifts the coordinate by Δx. Hence $H_\mathrm{e} = E_\mathrm{ZPL} + \hat{D} H_\mathrm{g} \hat{D}^\dagger$, and the oscillator states of the excited surface are $|m_\mathrm{e}\rangle = \hat{D}|m\rangle$, with energies $E_\mathrm{ZPL} + m\omega_0$ above the ground-state energy of the lower surface (we drop the common $\frac{1}{2}\omega_0$).

An optical transition is driven by the electronic dipole, whose matrix element between g and e depends weakly on the nuclear positions. The Condon approximation takes it constant, and the transition from $|n_\mathrm{g}\rangle$ to $|m_\mathrm{e}\rangle$ then has a probability proportional to the *Franck–Condon factor* $|\langle m_\mathrm{e}|n_\mathrm{g}\rangle|^2$, the squared overlap of the two vibrational wavefunctions.

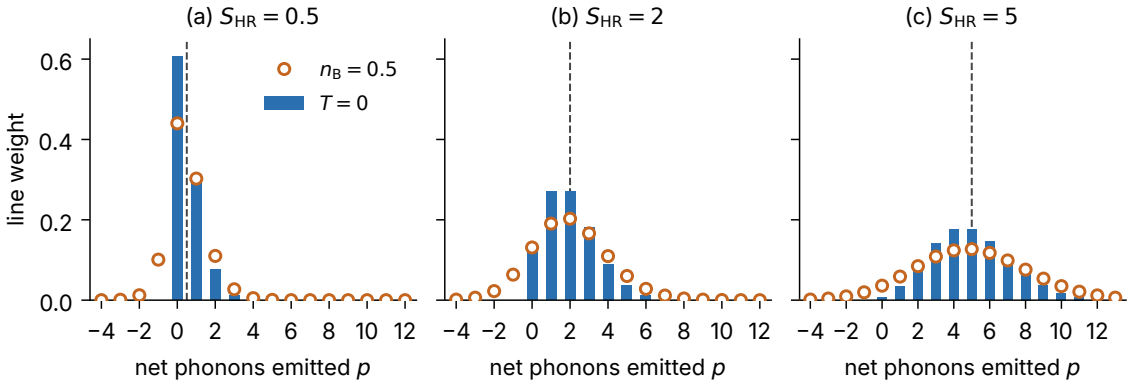


Figure 5.6: Franck–Condon progressions of a displaced oscillator, for Huang–Rhys factors $S_{\mathrm{HR}} = 0.5$, 2 and 5. Each panel shows the weights of the absorption lines at $E_{\mathrm{ZPL}} + p\,\omega_0$ against the net number p of phonons emitted, for one mode of frequency ω_0 that is the same in both electronic states, in the Condon approximation (dipole independent of the nuclear coordinate). Bars: zero temperature, Poisson weights $\mathrm{e}^{-S_{\mathrm{HR}}} S_{\mathrm{HR}}^{p}/p!$; the bar at $p = 0$ is the zero-phonon line. Open circles: mean occupation $n_{\mathrm{B}}(\omega_0) = 0.5$, that is $T = 0.91\,\omega_0$, from the Bessel-function weights of Eq. (5.65); the lines at $p < 0$ are anti-Stokes lines, which start from thermally excited vibrational states. Dashed line: the mean $p = S_{\mathrm{HR}}$, the same at both temperatures. The zero-phonon fractions at $T = 0$ are 0.61, 0.14 and 0.007.

From the vibrational ground state, $\langle m_{\mathrm{e}}|0\rangle = \langle m|\hat{D}^{\dagger}|0\rangle = \langle m|\hat{D}(-\alpha_{\mathrm{d}})|0\rangle$, and by Eq. (C.51)

$$|\langle m_{\mathrm{e}}|0_{\mathrm{g}}\rangle|^2 = \mathrm{e}^{-S_{\mathrm{HR}}}\,\frac{S_{\mathrm{HR}}^{m}}{m!}, \qquad S_{\mathrm{HR}} = \alpha_{\mathrm{d}}^2 = \frac{\Delta x^2}{4\ell_0^2} = \frac{M\omega_0\,\Delta x^2}{2} = \frac{\omega_0\,\Delta Q^2}{2}. \tag{5.63}$$

The *Huang–Rhys factor* S_{HR} [36] is written last with the mass-weighted displacement $\Delta Q = M^{1/2}\Delta x$, the form in which a first-principles calculation supplies it for each mode (Appendix E.10). It has a direct meaning. The relaxation energy that the lattice releases after a vertical transition, $\frac{1}{2}M\omega_0^2\Delta x^2$, equals $S_{\mathrm{HR}}\,\omega_0$: S_{HR} is the mean number of phonons emitted in the transition. At $T = 0$ the absorption spectrum is a progression of lines at $E_{\mathrm{ZPL}} + m\omega_0$ with Poisson weights (Fig. 5.6). The line $m = 0$ is the *zero-phonon line*, with weight $\mathrm{e}^{-S_{\mathrm{HR}}}$; the lines $m \geq 1$ form the *phonon sideband*. Emission from the relaxed excited state is the mirror image, with lines at $E_{\mathrm{ZPL}} - m\omega_0$ and the same weights. The means of the two spectra lie at $E_{\mathrm{ZPL}} \pm S_{\mathrm{HR}}\omega_0$, and their separation $2S_{\mathrm{HR}}\omega_0$ is the Stokes shift. For $S_{\mathrm{HR}} \ll 1$ the zero-phonon line carries most of the weight; for $S_{\mathrm{HR}} \gg 1$ it is exponentially weak, and the band is a near-Gaussian of width $S_{\mathrm{HR}}^{1/2}\omega_0$ around its mean. Because the weight $\mathrm{e}^{-S_{\mathrm{HR}}}$ is the fraction of transitions that leave the lattice untouched, the color-center literature calls it the Debye–Waller factor of the line, by analogy with diffraction [37, Sect. 2.2].

At finite temperature the initial vibrational state is thermal, and the overlaps $|\langle m_{\mathrm{e}}|n_{\mathrm{g}}\rangle|^2$ for $n > 0$ are needed as well. Summing them with Gibbs weights is a job for the generating function. The absorption line shape is

$$\begin{aligned} A(E) &= \sum_n p_n \sum_m |\langle m_{\mathrm{e}}|n_{\mathrm{g}}\rangle|^2\,\delta\big(E - E_{\mathrm{ZPL}} - (m-n)\,\omega_0\big) \\ &= \int_{-\infty}^{\infty} \frac{\mathrm{d}t}{2\pi}\,\mathrm{e}^{\mathrm{i}(E-E_{\mathrm{ZPL}})t}\,\big\langle \mathrm{e}^{\mathrm{i}H_{\mathrm{g}}t}\hat{D}\,\mathrm{e}^{-\mathrm{i}H_{\mathrm{g}}t}\hat{D}^{\dagger}\big\rangle, \end{aligned}$$

where p_n are the Gibbs weights of the ground-state oscillator (Sect. 2.5). The second form writes each δ-function as the time integral of $\mathrm{e}^{\mathrm{i}(E-E_{\mathrm{ZPL}}-m\omega_0+n\omega_0)t}$, uses $\langle m_{\mathrm{e}}|n_{\mathrm{g}}\rangle = \langle m|\hat{D}^{\dagger}|n\rangle$,

and lets $\mathrm{e}^{\pm \mathrm{i}H_\mathrm{g}t}$ supply the phases $\mathrm{e}^{\mathrm{i}n\omega_0 t}$ and $\mathrm{e}^{-\mathrm{i}m\omega_0 t}$; the sum over m is then a completeness sum. The operator $\mathrm{e}^{\mathrm{i}H_\mathrm{g}t}\hat{D}\mathrm{e}^{-\mathrm{i}H_\mathrm{g}t}$ is $\hat{D}$ with $d \to d\,\mathrm{e}^{-\mathrm{i}\omega_0 t}$, so both exponents are linear in d and $d^\dagger$, and Eq. (C.54) applies, with $\hat{A} = \alpha_\mathrm{d}(d^\dagger \mathrm{e}^{\mathrm{i}\omega_0 t} - d\,\mathrm{e}^{-\mathrm{i}\omega_0 t})$ and $\hat{B} = -\alpha_\mathrm{d}(d^\dagger - d)$. The averages are $\langle \hat{A}^2\rangle = \langle \hat{B}^2\rangle = -S_\mathrm{HR}(2n_\mathrm{B}+1)$ and $\langle \hat{A}\hat{B}\rangle = S_\mathrm{HR}[(n_\mathrm{B}+1)\mathrm{e}^{-\mathrm{i}\omega_0 t} + n_\mathrm{B}\mathrm{e}^{\mathrm{i}\omega_0 t}]$, with $n_\mathrm{B} = n_\mathrm{B}(\omega_0)$, so that

$$\langle \mathrm{e}^{\mathrm{i}H_\mathrm{g}t}\hat{D}\,\mathrm{e}^{-\mathrm{i}H_\mathrm{g}t}\hat{D}^\dagger\rangle = \mathrm{e}^{-S_\mathrm{HR}(2n_\mathrm{B}+1)}\,\exp\Big\{S_\mathrm{HR}\big[(n_\mathrm{B}+1)\,\mathrm{e}^{-\mathrm{i}\omega_0 t} + n_\mathrm{B}\,\mathrm{e}^{\mathrm{i}\omega_0 t}\big]\Big\}. \tag{5.64}$$

It has the form of the generating formula Eq. (5.58) for a crystal with a single mode: $2W_\kappa$ becomes $S_\mathrm{HR}(2n_\mathrm{B}+1)$, and $\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa}(t)\cdot\boldsymbol{q}$ becomes the second exponent. A kick of momentum q to a single oscillator is the displacement $\hat{D}(\mathrm{i}q\ell_0)$ (Appendix C.6), and it leads to the same expression with $S_\mathrm{HR} \to E_R/\omega_0$. The second exponential expands with the generating function of the modified Bessel functions, $\mathrm{e}^{(z/2)(s+1/s)} = \sum_p s^p I_p(z)$, taken at $s = [(n_\mathrm{B}+1)/n_\mathrm{B}]^{1/2}\mathrm{e}^{-\mathrm{i}\omega_0 t}$. The coefficient of $\mathrm{e}^{-\mathrm{i}p\omega_0 t}$ is the weight of the line at $E_\mathrm{ZPL} + p\,\omega_0$, for every integer p:

$$w_p = \mathrm{e}^{-S_\mathrm{HR}(2n_\mathrm{B}+1)}\Big(\frac{n_\mathrm{B}+1}{n_\mathrm{B}}\Big)^{p/2} I_p\big(2S_\mathrm{HR}\sqrt{n_\mathrm{B}(n_\mathrm{B}+1)}\big). \tag{5.65}$$

At $T = 0$ only $p \geq 0$ survives, and w_p returns to the Poisson weights of Eq. (5.63). At $T > 0$ the lines with $p < 0$ appear, and since $I_{-p} = I_p$, $w_{-p}/w_p = \mathrm{e}^{-p\beta\omega_0}$, detailed balance between the Stokes and anti-Stokes sides. The mean of p is S_HR at every temperature, the slope of Eq. (5.64) at $t = 0$, as the mean recoil energy was in Eq. (5.61).

A real center couples to many modes k, each with its own factor $S_k = \omega_k \Delta Q_k^2/2$, and the generating function is the product of factors Eq. (5.64), one per mode. The total $S_\mathrm{HR} = \sum_k S_k$ is the mean number of phonons emitted, and at $T = 0$ the zero-phonon weight is $\mathrm{e}^{-S_\mathrm{HR}}$. With a continuum of modes, the replicas merge into a continuous sideband with the shape of the convolutions of Eq. (5.59), while the zero-phonon line stays sharp, since it involves no phonon. Huang and Rhys derived this structure for the F center of alkali halides, where S_HR is large and the band is broad [36]. Sect. 7.4 applies it to the color centers of silicon and diamond, with their measured Huang–Rhys factors.

Worked example 5.8: A displaced oscillator

As a toy model we take the bond coordinate of silicon from Sect. 5.1, an oscillator of frequency $\omega_0 = 64.1\,\mathrm{meV}$ whose zero-point length is $\ell_0 = \ell_\mathrm{zp} = 0.048\,\text{Å}$ (with the reduced mass $M/2$ of the pair of nuclei, $\ell_0 = (2\cdot\frac{M}{2}\cdot\omega_0)^{-1/2}$ is the ℓ_zp of Fig. 5.2). Suppose a localized electronic excitation lengthens the equilibrium bond by Δx.

The size of the shift. A Huang–Rhys factor of two needs $\Delta x = 2\ell_0\sqrt{S_\mathrm{HR}} = 0.136\,\text{Å}$, 5.8 % of the bond length $\sqrt{3}a/4 = 2.35\,\text{Å}$. A shift of a few times the zero-point length is enough to make the zero-phonon line a minority: the relaxation energy is $S_\mathrm{HR}\omega_0 = 128\,\mathrm{meV}$, and the Stokes shift between the means of absorption and emission is 256 meV.

The overlap directly. The ground-state wavefunction is $\psi_0(x) = (2\pi\ell_0^2)^{-1/4}\mathrm{e}^{-x^2/4\ell_0^2}$, and the overlap with the same Gaussian centered at Δx is

$$\langle 0_\mathrm{e}|0_\mathrm{g}\rangle = \frac{1}{\sqrt{2\pi}\,\ell_0}\int \mathrm{d}x\;\mathrm{e}^{-[x^2+(x-\Delta x)^2]/4\ell_0^2} = \frac{\mathrm{e}^{-\Delta x^2/4\ell_0^2}}{\sqrt{2\pi}\,\ell_0}\int \mathrm{d}x\;\mathrm{e}^{-x^2/2\ell_0^2 + x\Delta x/2\ell_0^2} = \mathrm{e}^{-\Delta x^2/8\ell_0^2},$$

by Eq. (C.43) with $\mu = 1/2\ell_0^2$ and $\eta = \Delta x/2\ell_0^2$, which contributes $\sqrt{2\pi}\,\ell_0\,\mathrm{e}^{\Delta x^2/8\ell_0^2}$. The squared overlap is $\mathrm{e}^{-\Delta x^2/4\ell_0^2} = \mathrm{e}^{-S_\mathrm{HR}}$, the zero-phonon weight of Eq. (5.63). For the first replica, $\psi_1(x) = (x/\ell_0)\psi_0(x)$, and the same integral with an extra factor $(x-\Delta x)/\ell_0$ gives $|\langle 1_\mathrm{e}|0_\mathrm{g}\rangle|^2 = S_\mathrm{HR}\mathrm{e}^{-S_\mathrm{HR}}$.

The progression. For $S_\mathrm{HR} = 2$ the weights of the lines with $p = 0$ to 5 phonons are 0.135,

0.271, 0.271, 0.180, 0.090 and 0.036, the middle panel of Fig. 5.6. The zero-phonon line carries 14 % of the absorption, and the strongest replicas are those with one and two phonons, which for an integer S_{HR} are equal.
Room temperature. At 300 K, $n_{\mathrm{B}}(\omega_0) = 0.091$, the value of Sect. 5.3 for the optical phonon. By Eq. (5.65), the first anti-Stokes line, at $E_{\mathrm{ZPL}} - \omega_0$, has weight 0.021, and the first Stokes replica 0.249; their ratio is $\mathrm{e}^{-\beta\omega_0} = 0.084$. The thermal population thus adds a weak line on the low-energy side of the zero-phonon line, while the mean of the spectrum stays at $E_{\mathrm{ZPL}} + 2\omega_0$.

The two exponents of this section, $2W_\kappa(\boldsymbol{q})$ for a momentum kick and S_{HR} for a shift of equilibrium, both count the mean number of phonons that a sudden change of the nuclear state creates, and both enter as Poisson weights. With B_κ and $\mathcal{C}_{\kappa\kappa'}(t)$ in hand, Sect. 11.3 builds the complete multiphonon response of a crystal, including the pairs of different nuclei, and Sect. 11.4 follows it to the free recoil of a nucleus at large q. Both rest on the harmonic crystal, whose thermal state is Gaussian; Sect. 5.7 asks what the anharmonic terms of the potential do to that picture and to the phonons themselves.

5.7 Anharmonicity

The Raman line of the zone-center optical phonon of silicon is not infinitely sharp. Near zero temperature its full width at half maximum is $1.45\,\mathrm{cm}^{-1}$, or 0.18 meV [38], so that a phonon created at 64.1 meV survives for about 3.7 ps, some sixty oscillations, before it disappears. The harmonic crystal of Sects. 5.2 and 5.3 cannot account for this. Its phonons are exact eigenstates of Eq. (5.31) and have infinite lifetime. What limits their lifetime is the part of the energy surface that the harmonic approximation dropped, the cubic term already visible in Fig. 5.2(c), which lets one phonon turn into two. For detection this matters in two ways. An optical phonon that a dark-matter particle creates in a cryogenic crystal does not stay where it was made: within picoseconds it splits into acoustic phonons, which carry the energy to the sensors on the surface (Sect. 20.2). And every lattice response computed in this chapter is the leading term of an expansion in the anharmonic terms, whose size we need to know before trusting it. We will derive the three-phonon linewidth from the golden rule, estimate the small parameter that controls it in powers of κ_{BO}, and see what the three standard effective harmonic theories fit when that parameter is not small.

5.7.1 Cubic and quartic force constants

The expansion Eq. (5.13) of the energy surface continues beyond second order. With the compact label $i = (\boldsymbol{R}_i\kappa_i\alpha_i)$ for a nucleus and a Cartesian direction, so that $u_i = u^{\alpha_i}_{\boldsymbol{R}_i\kappa_i}$,

$$\begin{aligned} E^{\mathrm{ad}}(u) = E^{\mathrm{ad}}(0) &+ \frac{1}{2!}\sum_{ij}\Phi^{(2)}_{ij}\,u_iu_j + \frac{1}{3!}\sum_{ijk}\Phi^{(3)}_{ijk}\,u_iu_ju_k \\ &+ \frac{1}{4!}\sum_{ijkl}\Phi^{(4)}_{ijkl}\,u_iu_ju_ku_l + \cdots , \end{aligned} \tag{5.66}$$

$$\Phi^{(n)}_{i_1\ldots i_n} = \left.\frac{\partial^n E^{\mathrm{ad}}}{\partial u_{i_1}\cdots\partial u_{i_n}}\right|_{u=0}.$$

The *third- and fourth-order force constants* $\Phi^{(3)}$ and $\Phi^{(4)}$ share the properties of $\Phi^{(2)}$ that Sect. 5.2 derived. As derivatives they are real and symmetric under any permutation of their labels. Since a lattice translation leaves the surface unchanged, they depend on the cells

only through differences, $\Phi^{(3)}(\boldsymbol{R},\boldsymbol{R}',\boldsymbol{R}'') = \Phi^{(3)}(\boldsymbol{0},\boldsymbol{R}'-\boldsymbol{R},\boldsymbol{R}''-\boldsymbol{R})$. A rigid displacement of the whole crystal, $u_i \to u_i + d^{\alpha_i}$, also leaves it unchanged, which means that the total force $\sum_{\boldsymbol{R}\kappa} \partial E^{\mathrm{ad}}/\partial u^{\gamma}_{\boldsymbol{R}\kappa}$ vanishes at every configuration. Differentiating this identity $n-1$ times at $u=0$ gives the sum rules

$$\sum_{\boldsymbol{R}''\kappa''} \Phi^{(n),\alpha\beta\cdots\gamma}_{\kappa\kappa'\cdots\kappa''}(\boldsymbol{R},\boldsymbol{R}',\ldots,\boldsymbol{R}'') = 0, \tag{5.67}$$

of which Eq. (5.18) is the case $n=2$. Physically, $\Phi^{(3)}$ says how the force constants change when the nuclei move, $\Phi^{(2)}_{ij}(u) = \Phi^{(2)}_{ij} + \sum_k \Phi^{(3)}_{ijk} u_k + \cdots$, and $\Phi^{(4)}$ how that change changes in turn. The asymmetric curve of Fig. 5.2(c) is a direct measurement of one combination: along the [111] bond coordinate of silicon the cubic coefficient is $\Phi^{(3)}_{\mathrm{opt}} = -41.5\,\mathrm{eV\,\AA^{-3}}$ (Worked example 5.1), and the quartic one is about $40\,\mathrm{eV\,\AA^{-4}}$ (Worked example 5.5). A bond is stiffer under compression than under stretching: its force constant falls as it lengthens.

In practice the cubic constants come from the same calculations as the harmonic ones. By the $2n+1$ theorem of Sect. 5.4, the first-order density responses of DFPT fix the third derivatives of the energy, so that one set of linear-response calculations on a grid of wavevectors contains $\Phi^{(3)}$ as well [20, Eq. (123)]. The alternative is finite differences of forces in a supercell, with one calculation for each symmetry-distinct pair of displaced atoms and a cutoff on the range of the triplets, or a fit to forces sampled at temperature T. The fourth-order constants are rarely computed term by term; they enter through the effective harmonic theories at the end of this section. Appendix E.8 describes the practice.

To see what the cubic term does to phonons, we write it in the phonon operators of Sect. 5.3. Inserting the phonon expansion Eq. (5.32) three times, each displacement brings a factor $(2NM_\kappa\omega)^{-1/2}$, a polarization vector, a phase and a displacement field. The sum over the first cell $\boldsymbol{R}$ of the three phases, at fixed relative positions, is $\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}(\boldsymbol{Q}_1+\boldsymbol{Q}_2+\boldsymbol{Q}_3)\cdot\boldsymbol{R}} = N\sum_{\boldsymbol{G}} \delta_{\boldsymbol{Q}_1+\boldsymbol{Q}_2+\boldsymbol{Q}_3,\boldsymbol{G}}$, by the lattice sum Eq. (3.14). Writing $1 = (\boldsymbol{Q}_1\nu_1)$ and so on for the phonon labels,

$$H_3 = \frac{1}{3!}\frac{1}{\sqrt{N}} \sum_{123} \sum_{\boldsymbol{G}} \delta_{\boldsymbol{Q}_1+\boldsymbol{Q}_2+\boldsymbol{Q}_3,\boldsymbol{G}}\, V^{(3)}(1,2,3)\, B_1 B_2 B_3, \tag{5.68}$$

with the *three-phonon vertex*

$$\begin{aligned} V^{(3)}(1,2,3) = \frac{1}{\sqrt{8\,\omega_1\omega_2\omega_3}} \sum_{\kappa} \sum_{\boldsymbol{R}'\kappa'} \sum_{\boldsymbol{R}''\kappa''} \mathrm{e}^{\mathrm{i}(\boldsymbol{Q}_2\cdot\boldsymbol{R}'+\boldsymbol{Q}_3\cdot\boldsymbol{R}'')} \\ \times \frac{\Phi^{(3),\alpha\beta\gamma}_{\kappa\kappa'\kappa''}(\boldsymbol{0},\boldsymbol{R}',\boldsymbol{R}'')\, e^{\alpha}_{\kappa\nu_1}(\boldsymbol{Q}_1)\, e^{\beta}_{\kappa'\nu_2}(\boldsymbol{Q}_2)\, e^{\gamma}_{\kappa''\nu_3}(\boldsymbol{Q}_3)}{\sqrt{M_\kappa M_{\kappa'} M_{\kappa''}}}, \end{aligned} \tag{5.69}$$

summed over Cartesian indices, and $\omega_i = \omega_{\boldsymbol{Q}_i\nu_i}$. The vertex has the dimension of an energy and is symmetric in its three labels, and $V^{(3)}(-1,-2,-3) = V^{(3)}(1,2,3)^*$ by Eq. (A.22). The factor $1/\sqrt{N}$ is the price of spreading three phonons over the whole crystal, and the Kronecker delta is the conservation of crystal momentum: three phonons can meet only if their wavevectors add up to a reciprocal-lattice vector. Since each B_i creates or destroys one phonon, H_3 turns one phonon into two, two into one, or creates or destroys three at once. The quartic term, written in the same way, has four fields and the factor $1/N$.

5.7.2 Three-phonon decay and the linewidth

Consider a crystal in thermal equilibrium, except for a small excess population in one mode $\boldsymbol{Q}\nu$. Since $\omega_{\boldsymbol{Q}\nu} > 0$, the processes of H_3 that conserve energy are of two kinds. The phonon can decay into two others, $\boldsymbol{Q}\nu \to 1+2$, or merge with a thermal phonon into a third,

$\boldsymbol{Q}\nu + 1 \to 2$. For the decay, the matrix element of Eq. (5.68) between $|n_{\boldsymbol{Q}\nu}, n_1, n_2\rangle$ and $|n_{\boldsymbol{Q}\nu} - 1, n_1 + 1, n_2 + 1\rangle$ comes from the terms that contain $b_{\boldsymbol{Q}\nu}$, $b_1^\dagger$ and $b_2^\dagger$. These sit in $B_{\boldsymbol{Q}\nu}$, B_{-1} and B_{-2}, with $-1 = (-\boldsymbol{Q}_1\nu_1)$, and there are 3! ways of assigning the three fields to the three slots of H_3, all equal by the symmetry of the vertex. The 3! cancels, and the boson factors of Eq. (2.34) give

$$\langle n_{\boldsymbol{Q}\nu} - 1, n_1 + 1, n_2 + 1 | H_3 | n_{\boldsymbol{Q}\nu}, n_1, n_2\rangle = \frac{V^{(3)}(\boldsymbol{Q}\nu, -1, -2)}{\sqrt{N}} \sqrt{n_{\boldsymbol{Q}\nu}(n_1+1)(n_2+1)},$$

with $\boldsymbol{Q}_1 + \boldsymbol{Q}_2 = \boldsymbol{Q}$ up to a reciprocal-lattice vector. By the golden rule Eq. (2.5), the rate at which the mode loses phonons this way is $(2\pi/N)\sum |V^{(3)}|^2 n_{\boldsymbol{Q}\nu}(n_1+1)(n_2+1)\,\delta(\omega_{\boldsymbol{Q}\nu} - \omega_1 - \omega_2)$, summed over distinct final pairs, which is half the sum over $\boldsymbol{Q}_1$, ν_1 and ν_2 with $\boldsymbol{Q}_2$ fixed. The inverse process, in which the pair merges back, feeds the mode at the rate with $(n_{\boldsymbol{Q}\nu}+1)n_1 n_2$ in place of $n_{\boldsymbol{Q}\nu}(n_1+1)(n_2+1)$. The difference of the two occupation factors is $n_{\boldsymbol{Q}\nu}(1 + n_1 + n_2) - n_1 n_2$. At equilibrium it vanishes, since $n_{\rm B}(\omega_1+\omega_2)(1+n_1+n_2) = n_1 n_2$ for Bose functions, which is detailed balance for the process (Sect. 2.5). For an excess δn in the mode it is $(1+n_1+n_2)\,\delta n$, so the excess decays exponentially. The merging processes contribute in the same way, with the occupation difference $n_{\boldsymbol{Q}\nu}n_1(1+n_2) - (n_{\boldsymbol{Q}\nu}+1)(n_1+1)n_2 = (n_1 - n_2)\,\delta n$, and with no factor one half, since the two partners now play different roles. Relabeling $\boldsymbol{Q}_i \to -\boldsymbol{Q}_i$, which leaves ω_i and n_i unchanged, and adding both, the excess decays as $\mathrm{e}^{-\gamma^{\rm anh}_{\boldsymbol{Q}\nu} t}$ with

Three-phonon linewidth

$$\begin{aligned}\gamma^{\rm anh}_{\boldsymbol{Q}\nu} = \frac{\pi}{N} \sum_{\boldsymbol{Q}_1\nu_1\nu_2} & \left|V^{(3)}(\boldsymbol{Q}\nu, \boldsymbol{Q}_1\nu_1, \boldsymbol{Q}_2\nu_2)\right|^2 \\ & \times \Big[(1+n_1+n_2)\,\delta(\omega_{\boldsymbol{Q}\nu} - \omega_1 - \omega_2) + 2(n_1 - n_2)\,\delta(\omega_{\boldsymbol{Q}\nu} + \omega_1 - \omega_2)\Big],\end{aligned} \tag{5.70}$$

with $\boldsymbol{Q}_2 = -\boldsymbol{Q} - \boldsymbol{Q}_1$ up to a reciprocal-lattice vector, $\omega_i = \omega_{\boldsymbol{Q}_i\nu_i}$ and $n_i = n_{\rm B}(\omega_i)$. The first term is the decay of the phonon into the pair $(-\boldsymbol{Q}_1\nu_1, -\boldsymbol{Q}_2\nu_2)$, the second its merging with $\boldsymbol{Q}_1\nu_1$ into $-\boldsymbol{Q}_2\nu_2$.

This is the result of Maradudin and Fein [39, 40, Eq. (6)]. The rate at which an excess population relaxes is also the width of the line. A state that decays with probability $\mathrm{e}^{-\gamma t}$ has an amplitude that decays as $\mathrm{e}^{-\gamma t/2}$, and its spectrum is a Lorentzian with full width γ at half maximum. The *phonon linewidth* $\gamma_{\boldsymbol{Q}\nu}$ is this full width, the inverse lifetime of the phonon, and $\gamma^{\rm anh}_{\boldsymbol{Q}\nu}$ is its anharmonic part; in the language of the phonon self-energy of Appendix D.17, defined there in units of a squared frequency, it is $-\operatorname{Im}\Pi_\nu/\omega_{\boldsymbol{Q}\nu}$ for the bubble with two cubic vertices. Other mechanisms add to it: the creation of electron–hole pairs, $\gamma^{\rm ep}_{\boldsymbol{Q}\nu}$, which vanishes in an insulator at zero temperature (Sect. 6.4), and the scattering by isotopes and defects (Sect. 11.6).

Eq. (5.70) has two ingredients, the strength of the vertex and the number of pairs that conserve energy and crystal momentum, and the second is often the more restrictive. Temperature enters only through the occupations. At $T = 0$ only spontaneous decay survives, with weight one: a phonon of a cryogenic crystal can split, but it finds no thermal phonon to merge with. At high temperature $n_i \approx T/\omega_i$, and the width grows linearly with T.

The zone-center optical phonon of silicon makes this concrete. Since its wavevector is zero, it decays into pairs with opposite wavevectors, $(\boldsymbol{Q}_1, -\boldsymbol{Q}_1)$, and their energies must add to $\omega_{\rm opt} = \omega_{\Gamma,\rm opt}$. It is also the highest phonon of the crystal and has no partner to merge with, and only the decay term of Eq. (5.70) contributes, at any temperature. Fig. 5.7(a) shows

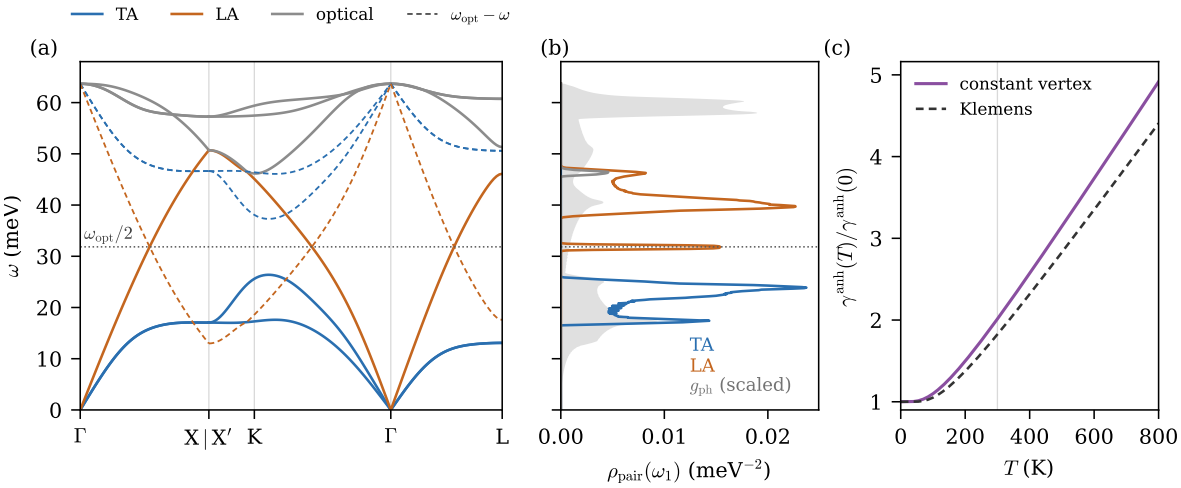


Figure 5.7: Three-phonon decay of the zone-center optical phonon of silicon, $\omega_{\mathrm{opt}} = 63.7\,\mathrm{meV}$ in this calculation. Phonons from the PBE force constants of Fig. 5.3 (DFPT on a 6^3 grid of wavevectors, $a = 5.431\,\text{Å}$), Fourier-interpolated; branches are labeled by energy order, the lowest two transverse acoustic (TA, blue), the third longitudinal acoustic (LA, orange), the upper three optical (gray). (a) Dispersion (solid) and the mirror curves $\omega_{\mathrm{opt}} - \omega_{\boldsymbol{Q}\nu}$ of the acoustic branches (dashed). A solid branch crossing a dashed one marks a pair $(\boldsymbol{Q}, -\boldsymbol{Q})$ whose energies add to ω_{opt}; dotted: $\omega_{\mathrm{opt}}/2$. (b) Two-phonon density of final states resolved by the energy ω_1 of one member of the pair and by its branch, $\rho_{\mathrm{pair}}(\omega_1) = N^{-1} \sum_{\boldsymbol{Q}\nu_1\nu_2} \delta(\omega_1 - \omega_{\boldsymbol{Q}\nu_1})\, \delta(\omega_{\mathrm{opt}} - \omega_{\boldsymbol{Q}\nu_1} - \omega_{\boldsymbol{Q}\nu_2})$, on an 80^3 grid with Gaussians of width 0.25 meV and a 0.9 meV running mean; it is symmetric about $\omega_{\mathrm{opt}}/2$, each TA phonon (blue) pairing with an LA phonon (orange) at the mirror energy. Shaded: the one-phonon density of states g_{ph}, scaled. LA+TA pairs make up 89 % of the final states, LA+LA 7 %, and TA+TA none; the remaining 4 % are optical+TA pairs from a small region near K. (c) Temperature dependence of the width relative to $T = 0$: Klemens's law Eq. (5.71) (dashed) and the average of $1 + n_1 + n_2$ over the final states of panel (b) (solid), which assumes a constant vertex; the vertical line marks 300 K.

the allowed pairs on the computed dispersion as crossings between a branch and the mirror image $\omega_{\mathrm{opt}} - \omega_{\boldsymbol{Q}\nu}$ of another. Klemens proposed the simplest channel: two acoustic phonons of the same branch, each with energy $\omega_{\mathrm{opt}}/2$ [41]. With $\omega_1 = \omega_2 = \omega_{\mathrm{opt}}/2$ in Eq. (5.70) the occupation factor is the same for every pair, and the width follows the *Klemens law*

$$\gamma^{\mathrm{anh}}_{\Gamma,\mathrm{opt}}(T) = \gamma^{\mathrm{anh}}_{\Gamma,\mathrm{opt}}(0) \left[1 + 2n_{\mathrm{B}}\Big(\frac{\omega_{\mathrm{opt}}}{2}\Big)\right]. \tag{5.71}$$

In silicon the channel is only partly open. The transverse acoustic branches never reach $\omega_{\mathrm{opt}}/2$: in the PBE calculation of Fig. 5.7 they top out at 26.4 meV, below $\omega_{\mathrm{opt}}/2 = 31.8\,\mathrm{meV}$, and two transverse phonons cannot share the energy at all. Two longitudinal acoustic phonons can. However, few modes sit near $\omega_{\mathrm{opt}}/2$, where the phonon density of states is small. Most of the phase space is in asymmetric pairs, a transverse acoustic phonon near 22 meV with a longitudinal one near 42 meV, which make up 89 % of the density of final states in Fig. 5.7(b), against 7 % for the pairs of longitudinal phonons. A full DFPT calculation of the vertex finds the same ordering, 94.3 % and 5.7 % [42, Table 1]. For asymmetric pairs the occupation factor $1 + n_1 + n_2$ is larger than Klemens's at the same temperature, because the softer partner is populated first (Fig. 5.7(c)).

Worked example 5.9: Klemens decay of the optical phonon of silicon

We compare the width of the zone-center optical phonon with the Klemens law Eq. (5.71), using the measured $\omega_{\text{opt}} = 64.1\,\text{meV}$.

The measured width. At low temperature the Raman line of the mode has a full width of $1.45\,\text{cm}^{-1}$ [38], as tabulated in Ref. [42, Table 1]. With $1\,\text{cm}^{-1} = 0.123\,98\,\text{meV}$, this is $\gamma_{\Gamma,\text{opt}} = 0.180\,\text{meV}$, and

$$\frac{\gamma_{\Gamma,\text{opt}}}{\omega_{\text{opt}}} = 2.80 \times 10^{-3}, \qquad \frac{1}{\gamma_{\Gamma,\text{opt}}} = 3.66\,\text{ps},$$

57 periods of 64.5 fs. Other measurements range from 1.24 to $2.1\,\text{cm}^{-1}$ [42], a spread that reflects sample quality and spectral resolution. Silicon is an insulator, with $\gamma^{\text{ep}}_{\boldsymbol{Q}\nu} = 0$ at low temperature, and isotope scattering is weak for a mode at the top of the spectrum. The width is anharmonic. A DFPT calculation of the full vertex Eq. (5.69), inserted in Eq. (5.70), yields $1.43\,\text{cm}^{-1}$ [42, Table 1], 99 % of the measured value, so three-phonon decay accounts for the whole width.

The temperature dependence. At the 10 mK of a detector, $\omega_{\text{opt}}/2T = 3.7 \times 10^4$, the occupation $n_{\text{B}}(\omega_{\text{opt}}/2)$ is e^{-37000}, and the width is its zero-temperature value: the optical phonon decays spontaneously, with no help from thermal phonons. At 77 K, $n_{\text{B}}(\omega_{\text{opt}}/2) = 0.008$ and the Klemens factor is 1.016. At 300 K, $n_{\text{B}}(\omega_{\text{opt}}/2) = 0.407$, and Eq. (5.71) predicts

$$\gamma_{\Gamma,\text{opt}}(300\,\text{K}) = 1.45\,\text{cm}^{-1} \times (1 + 2 \times 0.407) = 2.63\,\text{cm}^{-1}.$$

The average of $1 + n_1 + n_2$ over the asymmetric pairs that actually dominate, with a constant vertex, leads to a factor 2.01 instead of 1.82 and a width of $2.92\,\text{cm}^{-1}$ (Fig. 5.7(c)). The Klemens law thus gets the form of the temperature dependence right: a constant plus a term that grows as the Bose function of the daughters. Its single daughter energy, $\omega_{\text{opt}}/2$, misses the true pairs of silicon, and at room temperature it gives a width about 10 % below the estimate from those pairs. With the full vertex and the true daughters, Eq. (5.70) follows the measured widths up to temperatures of the order of $\omega_{\text{opt}} \approx 740\,\text{K}$ [42].

The Klemens law is a good parametrization of the temperature dependence once its zero-temperature width is taken from experiment or from a calculation. It does not predict that width, and its picture of the channel is wrong in silicon, where two transverse phonons cannot share the energy at all.

5.7.3 The anharmonic small parameter

How large are the anharmonic terms compared with the harmonic ones? The natural measure is the ratio of the cubic to the harmonic energy at the zero-point amplitude of a mode. Take a bond-like coordinate with curvature $\Phi^{(2)}$, cubic coefficient $\Phi^{(3)}$ and zero-point length ℓ_{zp}. The ratio of $\frac{1}{6}\Phi^{(3)}\ell_{\text{zp}}^3$ to $\frac{1}{2}\Phi^{(2)}\ell_{\text{zp}}^2$ defines the *anharmonic small parameter*

$$\eta_{\text{anh}} = \frac{|\Phi^{(3)}|\,\ell_{\text{zp}}}{3\,\Phi^{(2)}}. \tag{5.72}$$

Equivalently, η_{anh} is the zero-point amplitude measured in units of the length $3\Phi^{(2)}/|\Phi^{(3)}|$ over which the force constants change by their own size. The counting of Sect. 5.1 estimates it at once. Force constants of every order scale as $\Phi^{(n)} \sim \alpha^2 m_e / a_0^n$, and the zero-point amplitude as $\ell_{\text{zp}} \sim \kappa_{\text{BO}} a_0$, so

$$\eta_{\text{anh}} \sim \frac{\alpha^2 m_e / a_0^3}{\alpha^2 m_e / a_0^2}\,\kappa_{\text{BO}} a_0 = \kappa_{\text{BO}}.$$

The anharmonic parameter is the Born–Oppenheimer parameter itself: the cubic term is smaller than a phonon energy by one power of κ_{BO}, as Sect. 5.1 found. The quartic term, $\frac{1}{24}\Phi^{(4)}\ell_{\mathrm{zp}}^4$, is smaller than the harmonic one by η_{anh}^2, since $\Phi^{(4)}\ell_{\mathrm{zp}}^2/\Phi^{(2)} \sim (\ell_{\mathrm{zp}}/a_0)^2$, and each further order brings one more power.

The physical effects follow by counting powers of η_{anh}. The cubic term changes no energy at first order, since its average over any harmonic state, a product of an odd number of displacements, vanishes. At second order it shifts the levels by $(\eta_{\mathrm{anh}}\omega_{\mathrm{ph}})^2/\omega_{\mathrm{ph}} = \eta_{\mathrm{anh}}^2\omega_{\mathrm{ph}}$, since the energy denominators are themselves phonon energies. The quartic term shifts them at first order by the same amount, $\eta_{\mathrm{anh}}^2\omega_{\mathrm{ph}}$. The decay rate Eq. (5.70) is the square of a cubic matrix element of size $\eta_{\mathrm{anh}}\omega_{\mathrm{ph}}$ times a density of final states of order $1/\omega_{\mathrm{ph}}$, again $\eta_{\mathrm{anh}}^2\omega_{\mathrm{ph}}$. The three effects are of one order:

$$\frac{\gamma_{\boldsymbol{Q}\nu}^{\mathrm{anh}}}{\omega_{\boldsymbol{Q}\nu}} \sim \frac{\Delta\omega_{\boldsymbol{Q}\nu}^{\mathrm{anh}}}{\omega_{\boldsymbol{Q}\nu}} \sim \eta_{\mathrm{anh}}^2 \sim \kappa_{\mathrm{BO}}^2, \tag{5.73}$$

where $\Delta\omega_{\boldsymbol{Q}\nu}^{\mathrm{anh}}$ is the anharmonic shift of the frequency. In absolute terms this is $\alpha^2 m_e\, \kappa_{\mathrm{BO}}^4$, the anharmonic level of the Born–Oppenheimer hierarchy of Sect. 5.1. The three-phonon width, with two cubic vertices, and the first-order quartic shift, with one quartic vertex, are thus of one order. In the label of Sect. 8.5 it is anharmonic order $n_A = 2$, where n_A counts powers of the anharmonic expansion parameter in the response rather than vertices, and the leading anharmonic phonon self-energy sits at that order. In thermal equilibrium the zero-point length is replaced by the thermal amplitude, $\ell_{\mathrm{zp}}^2 \to \ell_{\mathrm{zp}}^2(2n_{\mathrm{B}}+1)$, as in Eq. (5.35), and at high temperature η_{anh}^2 grows in proportion to T, as the widths do.

Silicon bears this out. Along the bond coordinate of Fig. 5.2(c), $\Phi^{(2)} = 13.6\,\mathrm{eV\,\AA^{-2}}$, $\Phi^{(3)} = -41.5\,\mathrm{eV\,\AA^{-3}}$ and $\ell_{\mathrm{zp}} = 0.048\,\mathrm{\AA}$ give $\eta_{\mathrm{anh}} = 0.049$, against $\kappa_{\mathrm{BO}} = 0.067$; the length over which the force constants change is $3\Phi^{(2)}/|\Phi^{(3)}| = 0.98\,\mathrm{\AA}$, about 40 % of the bond. Then $\eta_{\mathrm{anh}}^2 = 2.4\times10^{-3}$, while the quartic-to-harmonic ratio at the same amplitude, $\Phi^{(4)}\ell_{\mathrm{zp}}^2/12\Phi^{(2)}$ with $\Phi^{(4)} \approx 40\,\mathrm{eV\,\AA^{-4}}$, is 0.6×10^{-3}, a quarter of η_{anh}^2, and the measured width of Worked example 5.9 is $\gamma/\omega_{\mathrm{opt}} = 2.8\times10^{-3}$, 1.2 times η_{anh}^2. The coefficients of the counting are close to one, between a quarter and two here and in the Born–Oppenheimer hierarchy of Worked example 5.1. Silicon at detector temperatures is a weakly anharmonic crystal, and its harmonic phonons are correct to a few parts in a thousand, well below the few-percent error of the functional (Sect. 5.2).

The parameter is not always small, and it grows with ℓ_{zp}. Light atoms have larger amplitudes, since $\ell_{\mathrm{zp}} \propto M^{-1/4}$ at fixed force constants, and hydrogen has $\kappa_{\mathrm{BO}} = 0.15$. The same happens for soft modes near a structural instability, where $\Phi^{(2)} \to 0$ and the harmonic expansion about the minimum loses its meaning. It happens in quantum crystals such as solid helium, where the zero-point motion is a sizable fraction of the spacing, and in any crystal near melting. At low energy phonon decay is controlled by phase space rather than by η_{anh}. The width of a long-wavelength acoustic phonon at $T = 0$ scales as the fifth power of its energy, which sets the pace of the down-conversion of the phonons that a dark-matter event leaves behind (Sect. 20.2 and Appendix E.12). For the nuclear response to dark matter, η_{anh} measures how far the harmonic results of this chapter can be trusted, and Sect. 11.5 turns it into a correction.

5.7.4 Effective harmonic theories

When temperature, light atoms or soft modes make the expansion Eq. (5.66) converge slowly, the usual remedy is not to add orders but to find a better harmonic problem. Three such theories are in common use, and they differ in what the harmonic force constants are fitted to.

The *quasi-harmonic approximation* fits them to the volume. It keeps every mode harmonic but computes the frequencies at the cell volume Ω that minimizes the free energy at temperature T, Eq. (E.51). The volume dependence is itself a cubic effect. A homogeneous strain ϵ displaces every nucleus by $u_i = \sum_\beta \epsilon^{\alpha_i\beta} X_i^\beta$, with $\boldsymbol{X}_i$ its equilibrium position. The force constants about the strained configuration then change, to first order, by

$$\Phi^{(2)}_{ij}(\epsilon) - \Phi^{(2)}_{ij} = \sum_k \Phi^{(3)}_{ijk} \sum_\beta \epsilon^{\alpha_k\beta} X_k^\beta. \tag{5.74}$$

The Grüneisen parameter $\gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu} = -\partial \ln \omega_{\boldsymbol{Q}\nu} / \partial \ln \Omega$ of Sect. 5.2 is therefore a projection of $\Phi^{(3)}$, and a harmonic crystal, with $\Phi^{(3)} = 0$, neither expands when heated nor shifts its phonons under pressure. The zero-point energy of Sect. 5.3 also depends on the volume, and its pressure expands the lattice at $T = 0$ beyond the static minimum of Sect. 5.1 (Appendix E.8). What the approximation keeps is the cubic term at the level of the volume. It drops the shifts at fixed volume, which Eq. (5.73) puts at the same order, and it has no widths at all.

The *self-consistent harmonic approximation* (SCHA) fits the force constants to the distribution of the nuclei, by a variational principle [43]. For any trial density matrix $\tilde{\varrho}$, the functional $\mathcal{F}[\tilde{\varrho}] = \langle H \rangle_{\tilde{\varrho}} - T S_{\mathrm{ent}}[\tilde{\varrho}]$ is an upper bound on the free energy, the Gibbs–Bogoliubov inequality. The SCHA takes for $\tilde{\varrho}$ the thermal state of a harmonic Hamiltonian H_{trial} with centroids $\bar{X}_a$ and auxiliary force constants $\Phi^{(2)}_{\mathrm{aux}}$, Eq. (E.53), with a running over the Cartesian coordinates X_a of all nuclei and $\delta X_a = X_a - \bar{X}_a$. Its distribution of positions is a Gaussian centered at $\bar{X}$. Writing $\mathcal{F}_{\mathrm{trial}}$ for the free energy of H_{trial} and $V_{\mathrm{trial}} = \frac{1}{2}\sum_{ab} \delta X_a \Phi^{(2)}_{\mathrm{aux},ab} \delta X_b$ for its potential, the bound reads

$$\mathcal{F}[\tilde{\varrho}] = \mathcal{F}_{\mathrm{trial}} + \big\langle E^{\mathrm{ad}} - V_{\mathrm{trial}} \big\rangle_{\tilde{\varrho}}.$$

We minimize it. A change of $\Phi^{(2)}_{\mathrm{aux},ab}$ changes $\mathcal{F}_{\mathrm{trial}}$ by $\frac{1}{2}\langle \delta X_a \delta X_b \rangle\, \delta\Phi^{(2)}_{\mathrm{aux},ab}$, the thermal form of the Hellmann–Feynman theorem of Sect. 5.1, and the average of V_{trial} by the same amount plus $\frac{1}{2}\sum_{cd} \Phi^{(2)}_{\mathrm{aux},cd}\, \delta\langle \delta X_c \delta X_d \rangle$. The first contributions cancel. The average of E^{ad} over a Gaussian depends on the force constants only through the covariance $\langle \delta X_c \delta X_d \rangle$, and a Gaussian obeys the diffusion equation in its covariance, so that $\delta\langle E^{\mathrm{ad}} \rangle = \frac{1}{2}\sum_{cd} \langle \partial_c \partial_d E^{\mathrm{ad}} \rangle\, \delta\langle \delta X_c \delta X_d \rangle$. Altogether

$$\delta\mathcal{F} = \frac{1}{2} \sum_{cd} \Big[\big\langle \partial_c \partial_d E^{\mathrm{ad}} \big\rangle_{\tilde{\varrho}} - \Phi^{(2)}_{\mathrm{aux},cd} \Big]\, \delta\langle \delta X_c \delta X_d \rangle,$$

and the centroids enter only through $\langle E^{\mathrm{ad}} \rangle$, whose derivative is the average force. Both variations vanish at the minimum:

Self-consistent harmonic approximation

$$\Phi^{(2)}_{\mathrm{aux},ab} = \Big\langle \frac{\partial^2 E^{\mathrm{ad}}}{\partial X_a\, \partial X_b} \Big\rangle_{\tilde{\varrho}}, \qquad \Big\langle \frac{\partial E^{\mathrm{ad}}}{\partial X_a} \Big\rangle_{\tilde{\varrho}} = 0. \tag{5.75}$$

The auxiliary force constants are the curvature of the true energy surface averaged over the Gaussian distribution that they themselves generate, and the centroids sit where the average force vanishes.

These are the stationarity conditions of Ref. [43, Eqs. (21) and (22)], which writes them as averages of forces. The condition is self-consistent because the width of the distribution, $\langle \delta X_a \delta X_b \rangle$, is computed from the auxiliary frequencies with the zero-point and thermal factors of Eq. (5.35). Expanded in the force constants, the average curvature is $\Phi^{(2)} + \frac{1}{2}\sum_{cd} \Phi^{(4)}_{abcd} \langle \delta X_c \delta X_d \rangle + \cdots$: its first correction is the first-order quartic shift of Eq. (5.73),

which the SCHA sums to all orders together with its higher even companions. The cubic term enters only through the shift of the centroids and, in the physical phonon frequencies, through an added bubble term (Appendix E.8). For soft modes the SCHA is the right starting point even when the harmonic frequencies are imaginary, since the average curvature can be positive where the curvature at the minimum is not.

The *temperature-dependent effective potential* fits the force constants to forces. It samples nuclear configurations at temperature T, by molecular dynamics, computes the forces in each, and finds the effective constants $\tilde{\Phi}^{(2)}$ that reproduce them best in the least-squares sense, Eq. (E.54) [44]. For a Gaussian sample the least-squares fit returns exactly the average curvature of Eq. (5.75). The reason is that a Gaussian average of the force times a displacement is the average of the force's derivative times the covariance. The two methods differ in the distribution they sample, a classical one in molecular dynamics and a quantum Gaussian in the SCHA. That difference matters at detector temperatures, where a classical sample sees an almost static lattice and misses the zero-point motion. The same fit extends to effective third-order constants, from which Eq. (5.70) gives temperature-dependent widths [45]. Appendix E.8 gives the practice of all three methods, their cost and their pitfalls.

For silicon at 10 mK, all three reduce to the harmonic phonons of Sect. 5.4 with corrections of a few parts in a thousand, which is why the rest of the book can work with them.

Summary

Electrons are lighter than nuclei by a factor of about 2×10^{-5} in silicon, and they settle into their ground state many times during one vibration of the lattice (Sect. 5.1). The Born–Oppenheimer approximation turns their ground-state energy at fixed nuclei, with the repulsion of the nuclei added, into the potential-energy surface $E^{\mathrm{ad}}(u)$ on which the nuclei move. The small parameter is the fourth root of the mass ratio, $\kappa_{\mathrm{BO}} = (m_e/M)^{1/4} = 0.067$: zero-point amplitudes are $\sim \kappa_{\mathrm{BO}} a_0$, and phonon energies, anharmonic shifts and nonadiabatic shifts are $\sim \kappa_{\mathrm{BO}}^2$, κ_{BO}^4 and κ_{BO}^6 times the electronic scale. In an insulator the ratio that controls the approximation is the phonon energy over the gap, 0.0548 in silicon. The clamped-nuclei approximation of Chaps. 3 and 4 differs from the Born–Oppenheimer one by the electron–phonon coupling. Forces on the nuclei follow from the Hellmann–Feynman theorem as electrostatic forces of the ground-state density, at the cost of one self-consistent calculation.

Expanded to second order about its minimum, the energy surface defines the force constants $\Phi^{(2)}$ and the dynamical matrix $D^{\mathrm{dyn}}(\boldsymbol{Q})$, whose eigenvalues are the squared phonon frequencies $\omega^2_{\boldsymbol{Q}\nu}$ and whose eigenvectors are the polarizations $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$ (Sect. 5.2). Translation invariance implies the acoustic sum rule and three acoustic branches that start at zero, and rotation invariance makes them linear at long wavelengths, with the sound speeds of the elastic constants. Silicon has six branches up to 64.1 meV, flat transverse acoustic branches that pile up states near 17 meV, and no LO–TO splitting. A PBE calculation reproduces six of its eight measured frequencies at Γ, X and L to within 2 %, the transverse acoustic modes being 8 % soft. The Debye model, fitted to the elastic constants, has $\omega_{\mathrm{D}} = 55.6\,\mathrm{meV}$ and $\Theta_{\mathrm{D}} = 645\,\mathrm{K}$.

Quantized, each normal mode is an oscillator and a phonon is one of its quanta (Sect. 5.3). The displacement $\boldsymbol{u}_{\boldsymbol{R}\kappa} = \sum_{\boldsymbol{Q}\nu} (2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{-1/2} \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} B_{\boldsymbol{Q}\nu}$ is the convention used throughout. Emitting a phonon carries the weight $n_{\boldsymbol{Q}\nu} + 1$ and absorbing one the weight $n_{\boldsymbol{Q}\nu}$, and in a cryogenic detector every mode is empty, so emission is the whole signal. A nuclear density at momentum transfer $\boldsymbol{q}$ creates a single phonon with crystal momentum conserved up to a reciprocal-lattice vector and a rate proportional to the number of cells. Zero-point motion alone spreads a silicon nucleus over 0.05 Å per Cartesian component, and thermal motion raises this to 0.077 Å at room temperature; the Debye model underestimates both,

because the transverse acoustic branches are softer than it assumes.

The force constants come from density-functional theory (Sect. 5.4). Finite displacements in a supercell give them directly but alias every constant beyond the supercell; density-functional perturbation theory computes the response to a displacement wave of any $\boldsymbol{Q}$ in the primitive cell, from the first-order Sternheimer equation. By the $2n + 1$ theorem the same first-order responses fix the third derivatives as well. A coarse grid of wavevectors and a Fourier interpolation of the force constants then give the phonons everywhere, to within 0.3 meV in silicon on a 6^3 grid, once the acoustic sum rule, which every computed set violates slightly, has been restored.

In a polar crystal the sublattices carry Born effective charges Z^*_κ, the force on a nucleus per unit macroscopic field, which add up to zero (Sect. 5.5). The long-range field of a longitudinal optical wave adds a nonanalytic term to the dynamical matrix that depends on the direction of $\boldsymbol{Q}$ only, and splits the longitudinal from the transverse modes: in GaAs, 36.4 meV against 33.6 meV. The Lyddane–Sachs–Teller relation $\varepsilon_0/\varepsilon_\infty = \omega_\mathrm{L}^2/\omega_\mathrm{T}^2$ ties the splitting to the two dielectric constants, and the measured values of GaAs satisfy it to 1.4 %; the splitting gives $Z^* = 2.11$. PBE gets Z^* right to a percent but ε_∞ 26 % too high, so wherever a rate needs $Z^{*2}/\varepsilon_\infty$ the measured splitting, or the computed Z^* with the measured ε_∞, is the input to use, not the PBE ε_∞. A uniform field couples only to the polar optical modes at the zone center, through Z^*_κ, coherently in every cell.

At finite momentum transfer the response of a nucleus is controlled by thermal averages of exponentials of displacements (Sect. 5.6). For a harmonic crystal the probability that the lattice absorbs $\boldsymbol{q}$ and stays in its state is the Debye–Waller factor e^{-2W_κ}, with $2W_\kappa = \boldsymbol{q} \cdot B_\kappa \cdot \boldsymbol{q}$, and the time-dependent version brings in the displacement correlation $\mathcal{C}_{\kappa\kappa'}(t)$. Expanded in $\mathcal{C}$, it yields the number of phonons created as a Poisson distribution with mean $2W_\kappa$. In silicon at zero temperature $2W_\kappa = 1$ at $q = 40$ keV, the crossover from single phonons to multiphonon emission. The same algebra applied to a sudden shift of the equilibrium positions gives the Franck–Condon factors and the Huang–Rhys factor S_HR of a localized electronic transition.

The cubic and quartic terms of the energy surface make phonons interact (Sect. 5.7). The cubic term lets a phonon decay into two, with a full width $\gamma^\mathrm{anh}_{\boldsymbol{Q}\nu}$ that the golden rule expresses through the third-order force constants, the energy and momentum balance of the pairs and the factor $1 + n_1 + n_2$. The zone-center optical phonon of silicon has a measured width of 1.45 cm^{-1} at low temperature, a lifetime of 3.7 ps, and decays mainly into a transverse and a longitudinal acoustic phonon of unequal energies. The anharmonic small parameter η_anh, the ratio of the cubic to the harmonic energy at the zero-point amplitude, is of order κ_BO, 0.049 in silicon, and widths and anharmonic shifts are both of relative order η^2_anh. Where the expansion fails, the quasi-harmonic approximation fits harmonic frequencies to the volume, the self-consistent harmonic approximation to the average curvature of the energy surface over the distribution of the nuclei, and the temperature-dependent effective potential to sampled forces.

The chapter has produced the phonons of a crystal, their quanta, their correlations and their lifetimes, all on the energy surface of electrons that never leave their ground state. The couplings that the Born–Oppenheimer approximation dropped let a moving nucleus excite the electrons, and a passing electron excite the lattice; Chap. 6 derives them.

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Chapter 6

Electron–phonon coupling

The band gap of silicon is 1.17 eV at low temperature and 1.12 eV at room temperature (Sect. 3.3), and about 60 meV of it is already lost to the zero-point motion of the nuclei (Sect. 6.3). The static lattice of Chaps. 3 and 4 has a gap that changes with neither the temperature nor the nuclear mass, and the harmonic crystal of Chap. 5 treats the phonons as a system of their own. The shift and its temperature dependence come from the coupling between electrons and nuclei: a displaced nucleus changes the potential the electrons feel, and an electron in turn changes the forces between nuclei. For dark-matter detection this coupling matters well beyond the gap. A phonon can supply the crystal momentum that a slow particle lacks to cross the indirect gap. The same coupling limits how long an excited carrier keeps its energy before it ends up as heat in the lattice. And when a dark-matter particle strikes a nucleus instead of an electron, the coupling decides whether the electrons stay behind, which is the Migdal effect. All of this follows from the Born–Oppenheimer approximation of Chap. 5 by adding back what it dropped. We will start from the couplings it neglected and end, along Fig. 6.1, with the absorption of light, and of dark matter, at the indirect gap of silicon.

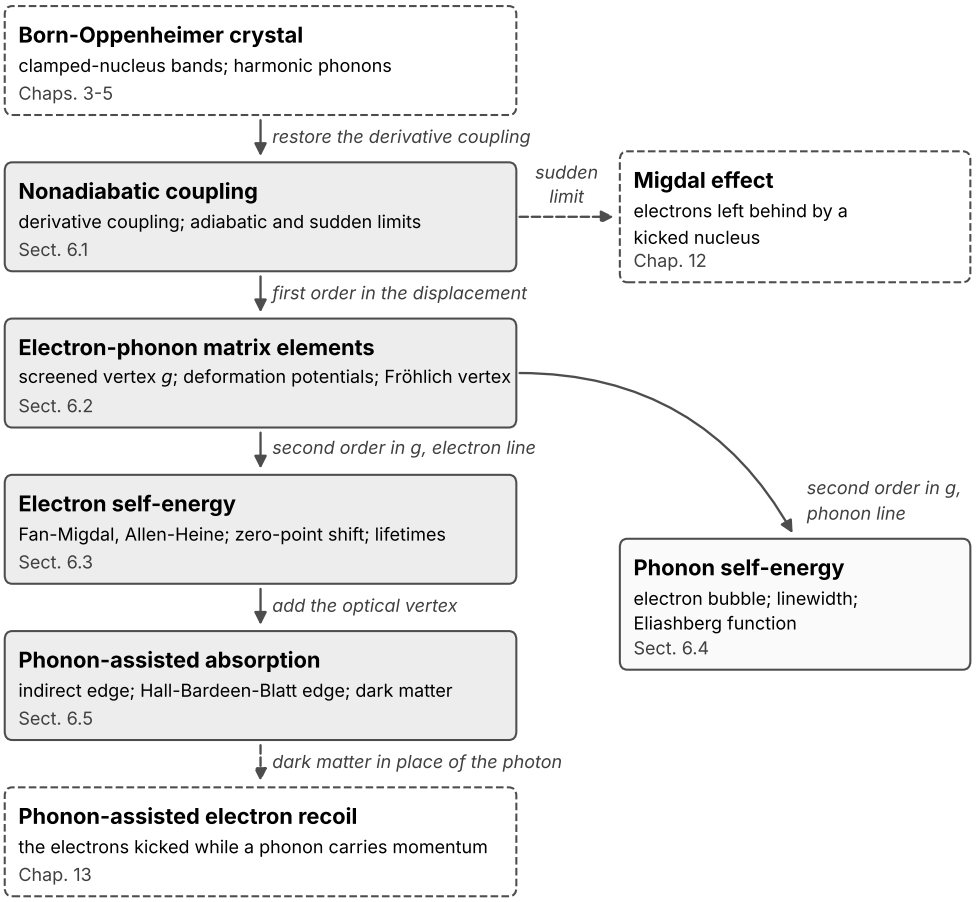


Figure 6.1: Adding back to the Born–Oppenheimer crystal (Chaps. 3 and 5) what it dropped. The derivative coupling, with its adiabatic and sudden limits (Sect. 6.1), gives at first order in the displacement the electron–phonon vertex g (Sect. 6.2). At second order in g it gives the electron self-energy, with the zero-point shift of the gap and carrier lifetimes (Sect. 6.3), and the phonon self-energy and Eliashberg function (Sect. 6.4). With an optical vertex it gives phonon-assisted absorption at the indirect gap (Sect. 6.5). Dashed boxes: the sudden limit of a kicked nucleus becomes the Migdal effect (Chap. 12), and dark matter in place of the photon becomes phonon-assisted electron recoil (Chap. 13).

6.1 Nonadiabatic coupling

A dark-matter particle of 1 GeV that scatters off a silicon nucleus at the maximum momentum transfer, $q = 2\mu_{\chi N} v_{\max} = 5.10\,\mathrm{MeV}$ (Sect. 1.3), sends the nucleus off at $v_N = q/M = 1.95 \times 10^{-4}$, or $58\,\mathrm{km\,s^{-1}}$. The outer electrons of the atom move at typical speeds $\sim \alpha = 7.3 \times 10^{-3}$, 37 times faster, and the Born–Oppenheimer picture of Sect. 5.1 says that they follow the nucleus. They follow it closely but not exactly. Seen from the recoiling nucleus, every electron has suddenly acquired the velocity $-v_N$ and the momentum $m_e v_N = 100\,\mathrm{eV}$, and a cloud bound with momenta $\sim \alpha m_e = 3.7\,\mathrm{keV}$ absorbs that small kick elastically most of the time but not always. For an electron bound as in hydrogen, the probability that it is left excited or ionized is $(v_N/\alpha)^2 = 7 \times 10^{-4}$. Small as it is, this is the origin of the Migdal effect, which turns a nuclear recoil too small to detect into an electronic signal (Chap. 12). The same couplings, at the slow speeds of thermal and zero-point motion, are what the Born–Oppenheimer approximation drops.

We will make this quantitative. The nonadiabatic couplings of Eq. (5.5) turn out to be a single object, the derivative coupling, which enters multiplied by the nuclear velocity. Expanding the electronic potential in the displacements instead yields the electron–phonon Hamiltonian, the starting point of the rest of this chapter. And a nucleus that is set in motion excites the electrons with an amplitude set by how fast it is accelerated: slowly, and the electrons follow (the adiabatic limit); abruptly, and they are left behind (the sudden limit). The derivations of the Born–Oppenheimer expansion that we build on are in Appendix D.13, following Born and Huang [1, App. VIII].

6.1.1 Derivative coupling

The two brackets of Eq. (5.5) are not independent. Both come from the change of the adiabatic states $\Psi_i(r;u)$ with the displacements. Both can be written through the *derivative-coupling vector*

$$\boldsymbol{d}_{ji}^{\boldsymbol{R}\kappa}(u) = \langle \Psi_j | \nabla_{\boldsymbol{R}\kappa} \Psi_i \rangle, \qquad \boldsymbol{d}_{ij}^{\boldsymbol{R}\kappa} = -\boldsymbol{d}_{ji}^{\boldsymbol{R}\kappa}. \tag{6.1}$$

The antisymmetry follows from differentiating the orthonormality $\langle \Psi_j | \Psi_i \rangle = \delta_{ji}$ with respect to a displacement, $\langle \nabla\Psi_j | \Psi_i \rangle + \langle \Psi_j | \nabla \Psi_i \rangle = 0$, together with the reality of the states; in particular $\boldsymbol{d}_{ii} = 0$. For the second derivative, we move one gradient to the left. Then we insert the completeness of the adiabatic states at fixed u, $\sum_k |\Psi_k\rangle\langle\Psi_k| = 1$:

$$\langle \Psi_j | \nabla^2 \Psi_i \rangle = \nabla \cdot \langle \Psi_j | \nabla \Psi_i \rangle - \langle \nabla\Psi_j | \nabla \Psi_i \rangle = \nabla \cdot \boldsymbol{d}_{ji} - \sum_k \boldsymbol{d}_{kj} \cdot \boldsymbol{d}_{ki} = \nabla \cdot \boldsymbol{d}_{ji} + \sum_k \boldsymbol{d}_{jk} \cdot \boldsymbol{d}_{ki},$$

where we dropped the label $\boldsymbol{R}\kappa$, which every gradient and every $\boldsymbol{d}$ carries, and used the antisymmetry in the last step. Inserted in Eq. (5.5), the scalar coupling contributes $\nabla \cdot \boldsymbol{d}_{ji} + (\boldsymbol{d} \cdot \boldsymbol{d})_{ji}$, a product of matrices in the state labels, and the derivative coupling $2\boldsymbol{d}_{ji} \cdot \nabla$. These are exactly the cross terms of a square. Writing the nuclear wavefunctions Ψ_i^{nuc} as a column, the nuclear kinetic energy and the couplings of Eq. (5.4) combine into

$$-\sum_{\boldsymbol{R}\kappa} \frac{\nabla_{\boldsymbol{R}\kappa}^2}{2M_\kappa}\,\delta_{ji} + \mathcal{N}_{ji} = \sum_{\boldsymbol{R}\kappa} \frac{1}{2M_\kappa} \Big[\big(-\mathrm{i}\nabla_{\boldsymbol{R}\kappa} - \mathrm{i}\boldsymbol{d}^{\boldsymbol{R}\kappa}\big)^2 \Big]_{ji}, \tag{6.2}$$

since $(\nabla + \boldsymbol{d})^2 = \nabla^2 + 2\boldsymbol{d}\cdot\nabla + (\nabla\cdot\boldsymbol{d}) + \boldsymbol{d}\cdot\boldsymbol{d}$, with ∇ acting on everything to its right. In the adiabatic basis the momentum of a nucleus, $\boldsymbol{p}_{\boldsymbol{R}\kappa} = -\mathrm{i}\nabla_{\boldsymbol{R}\kappa}$ of Sect. 5.3, becomes the matrix $-\mathrm{i}(\nabla_{\boldsymbol{R}\kappa} + \boldsymbol{d}^{\boldsymbol{R}\kappa})$. The nucleus carries its electronic state along, and the derivative-coupling vector measures how much of the electronic state changes per unit of displacement, much as a vector potential enters the momentum of a charge. It is a Berry connection, in the space of

nuclear displacements instead of crystal momenta. The Born–Oppenheimer approximation keeps only the diagonal part of the matrix, in which $\boldsymbol{d}_{00} = 0$ and only the small potential $\mathcal{N}_{00}$ of Appendix D.13 survives.

The derivative coupling has a simple reading if we take $\boldsymbol{p}/M_\kappa$ to be a velocity, heuristically at first. The off-diagonal, first-order part of Eq. (6.2) is $-(\mathrm{i}/M_\kappa)\,\boldsymbol{d}_{ji}\cdot\boldsymbol{p}_{\boldsymbol{R}\kappa}$, and on a nuclear wavefunction moving with velocity $\dot{\boldsymbol{u}}_{\boldsymbol{R}\kappa}$ it acts as $-\mathrm{i}\,\dot{\boldsymbol{u}}_{\boldsymbol{R}\kappa}\cdot\boldsymbol{d}_{ji}$. The same coupling appears when the nuclei are treated as classical particles that follow a trajectory $u(t)$. The electrons then obey $\mathrm{i}\partial_t\Psi = H_{\mathrm{el}}(u(t))\Psi$, and we expand $\Psi(t) = \sum_i c_i(t)\,\Psi_i(r;u(t))$ in the adiabatic states at the instantaneous positions. The time derivative falls on the coefficients and, through $u(t)$, on the states, $\partial_t\Psi_i = \sum_{\boldsymbol{R}\kappa}\dot{\boldsymbol{u}}_{\boldsymbol{R}\kappa}\cdot\nabla_{\boldsymbol{R}\kappa}\Psi_i$. Projecting on Ψ_j,

$$\mathrm{i}\dot{c}_j = E_j(u(t))\,c_j - \mathrm{i}\sum_i\sum_{\boldsymbol{R}\kappa}\dot{\boldsymbol{u}}_{\boldsymbol{R}\kappa}\cdot\boldsymbol{d}_{ji}^{\boldsymbol{R}\kappa}\,c_i. \tag{6.3}$$

The electrons are driven out of their adiabatic state by the velocity of the nuclei times the derivative coupling, and by nothing else. By Eq. (5.11) the coupling is $\dot{\boldsymbol{u}}\cdot\boldsymbol{F}_{ji}/(E_i - E_j)$, with $\boldsymbol{F}_{ji} = \langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}V_{eN}|\Psi_i\rangle$ the matrix element of the force: the power delivered by the nucleus to the transition, divided by the energy the transition costs.

The scalar coupling has no counterpart in Eq. (6.3). Roughly speaking, it is the quantum correction to a classical trajectory, and compared with the derivative coupling it is suppressed by $|\boldsymbol{d}|/(Mv)$, with $|\boldsymbol{d}| \sim 1/a_0$ for an electronic state of size a_0. A nucleus in its zero-point motion has a momentum $Mv \sim 1/\ell_{\mathrm{zp}}$, and the ratio is $\ell_{\mathrm{zp}}/a_0 \sim \kappa_{\mathrm{BO}}$, which is the extra power of κ_{BO} that separates the scalar from the derivative coupling in Sect. 5.1. A nucleus kicked by dark matter has $Mv_N = q$, and the ratio is $1/(qa_0)$, small for every momentum transfer well above $1/a_0 = 3.7\,\mathrm{keV}$: such a nucleus is a classical projectile as far as its electrons are concerned.

The velocity form also makes the counting of Sect. 5.1 transparent. In zero-point motion a nucleus moves at $v \sim \omega_{\mathrm{ph}}\ell_{\mathrm{zp}} \sim (\alpha^2 m_e\kappa_{\mathrm{BO}}^2)(\kappa_{\mathrm{BO}}a_0) = \alpha\,\kappa_{\mathrm{BO}}^3$, while the electrons move at $\sim\alpha$. The nuclei are slower than the electrons by $\kappa_{\mathrm{BO}}^3 = (m_e/M)^{3/4}$, which is 2.9×10^{-4} in silicon, and the derivative coupling $v\,|\boldsymbol{d}| \sim \alpha^2 m_e\,\kappa_{\mathrm{BO}}^3$ is the κ_{BO}^3 of the Born–Oppenheimer hierarchy. Compared with the excitation energy $\Delta E \sim \alpha^2 m_e$ it mixes in excited states with an amplitude κ_{BO}^3; at second order it shifts the vibrational levels by $\alpha^2 m_e\,\kappa_{\mathrm{BO}}^6$, the nonadiabatic shift of the hierarchy, a relative correction $\kappa_{\mathrm{BO}}^4 = m_e/M$ to a phonon energy. The kicked nucleus of the opening is 91 times faster than this zero-point estimate, and correspondingly more disruptive.

The small parameter is not always κ_{BO}. With an excitation energy ΔE in place of $\alpha^2 m_e$, the mixing amplitude becomes Eq. (5.9), $(\omega_{\mathrm{ph}}/\Delta E)(\ell_{\mathrm{zp}}F_{j0}/\Delta E)$, which in an insulator is controlled by $\omega_{\mathrm{ph}}/E_{\mathrm{g}}$, 0.055 for the optical phonon of silicon. In a metal ΔE reaches zero at the Fermi surface and the expansion in adiabatic states breaks down. What survives is a statement about the electrons that do respond: only those within ω_{ph} of the Fermi energy cannot follow the nuclei, a fraction $\omega_{\mathrm{ph}}/E_{\mathrm{F}}$ of the conduction electrons. Since $E_{\mathrm{F}} \sim \alpha^2 m_e$ in a typical metal, this fraction is $\sim \kappa_{\mathrm{BO}}^2 = (m_e/M)^{1/2}$, between 10^{-3} and 10^{-2}. The corrections to the electron–phonon interaction that come from these electrons, the vertex corrections, are suppressed by the same ratio. This is Migdal's adiabatic theorem [2]; its precise statement with diagrams, and its form for insulators, are given in Sect. 6.3.

6.1.2 The electron–phonon Hamiltonian

The adiabatic basis follows the nuclei and pays with the derivative couplings. The basis of clamped-nuclei states $\Psi_i(r;0)$ does not move, and the coupling sits in the Hamiltonian instead, as the change $V_{eN}(u) - V_{eN}(0)$ of the electron–nucleus potential (Eq. (5.7)). This is the form in which the coupling of electrons to phonons is computed and used. Let $v_\kappa(\boldsymbol{r})$ be the

potential energy of an electron at $\boldsymbol{r}$ due to a nucleus of species κ at the origin: $-Z_\kappa e^2/(4\pi r)$ for a bare nucleus, or a pseudopotential (Sect. 3.5). With the electron density operator $\hat{n}_e(\boldsymbol{r}) = \sum_e \delta(\boldsymbol{r} - \boldsymbol{r}_e)$, whose Fourier transform is Eq. (2.38),

$$V_{eN}(u) = \int \mathrm{d}^3 r \; \hat{n}_e(\boldsymbol{r}) \sum_{\boldsymbol{R}\kappa} v_\kappa(\boldsymbol{r} - \boldsymbol{R} - \boldsymbol{\tau}_\kappa - \boldsymbol{u}_{\boldsymbol{R}\kappa}).$$

To first order in the displacements, $V_{eN}(u) - V_{eN}(0) = \sum_{\boldsymbol{R}\kappa} \boldsymbol{u}_{\boldsymbol{R}\kappa} \cdot \int \mathrm{d}^3 r \, \hat{n}_e(\boldsymbol{r}) \, \nabla_{\boldsymbol{R}\kappa} V$, with $\nabla_{\boldsymbol{R}\kappa} V = -\nabla v_\kappa(\boldsymbol{r} - \boldsymbol{R} - \boldsymbol{\tau}_\kappa)$, the bare force field of one nucleus. We now insert the phonon expansion of the displacement, Eq. (5.32),

$$\boldsymbol{u}_{\boldsymbol{R}\kappa} = \sum_{\boldsymbol{Q}\nu} \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) \, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}}{\sqrt{2NM_\kappa \omega_{\boldsymbol{Q}\nu}}} \, B_{\boldsymbol{Q}\nu}.$$

The sum over cells collects the phase factors into a displacement wave of one sublattice,

$$\partial_{\kappa,\boldsymbol{Q}} V(\boldsymbol{r}) = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} \, \nabla_{\boldsymbol{R}\kappa} V(\boldsymbol{r}), \qquad \partial_{\kappa,\boldsymbol{Q}} V(\boldsymbol{r} + \boldsymbol{R}') = \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}'} \, \partial_{\kappa,\boldsymbol{Q}} V(\boldsymbol{r}),$$

where the second relation follows from relabeling $\boldsymbol{R} - \boldsymbol{R}' \to \boldsymbol{R}$ in the sum. The potential of a displacement wave with wavevector $\boldsymbol{Q}$ is a Bloch function with crystal momentum $\boldsymbol{Q}$. Its matrix element between Bloch states $\psi_{m\boldsymbol{k}'}$ and $\psi_{n\boldsymbol{k}}$ (normalized as in Eq. (A.15)) is therefore zero unless $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{Q}$ up to a reciprocal-lattice vector: a lattice translation by $\boldsymbol{R}'$ multiplies the integrand by $\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{Q}-\boldsymbol{k}')\cdot\boldsymbol{R}'}$, and the integral cannot change. Writing the one-body operator in second quantization, Eq. (2.36), we arrive at the *electron–phonon Hamiltonian*,

$$\begin{aligned} H_{\mathrm{ep}} &= \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}\boldsymbol{Q}} \sum_{mn\nu} g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q}) \, c^\dagger_{m,\boldsymbol{k}+\boldsymbol{Q}} c_{n\boldsymbol{k}} \, B_{\boldsymbol{Q}\nu}, \\ g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q}) &= \frac{1}{\sqrt{2\omega_{\boldsymbol{Q}\nu}}} \sum_\kappa \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})}{\sqrt{M_\kappa}} \cdot \langle \psi_{m,\boldsymbol{k}+\boldsymbol{Q}} | \partial_{\kappa,\boldsymbol{Q}} V | \psi_{n\boldsymbol{k}} \rangle, \end{aligned} \tag{6.4}$$

with the spin carried in the band labels. This is Eqs. (A.25) and (A.26), with the bare potential in place of the Kohn–Sham one [3, Eqs. (31)–(38)]. It has a simple reading. An electron in state $n\boldsymbol{k}$ is scattered to $m, \boldsymbol{k} + \boldsymbol{Q}$ while a phonon of crystal momentum $\boldsymbol{Q}$ is absorbed, or one of $-\boldsymbol{Q}$ is emitted, through $B_{\boldsymbol{Q}\nu} = b_{\boldsymbol{Q}\nu} + b^\dagger_{-\boldsymbol{Q}\nu}$. The amplitude is the force of the displacement wave on the electron, weighted by the zero-point amplitude $(2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{-1/2}$ of each nucleus in the mode.

Three things are left out of Eq. (6.4). The second order in u gives a vertex quadratic in B, the Allen–Heine vertex g^{AH}, which matters as much as the first order squared for the shifts of electronic levels (Sect. 6.3). The change of V_{NN} acts on the nuclei only and belongs to the energy surface E^{ad} that already fixes the phonons. Most important, the bare vertex ignores the response of the other electrons. When a nucleus moves, the electron cloud around it moves too and screens most of its force field, and the potential an electron actually feels is the change of the self-consistent potential. In density-functional theory, $\partial_{\kappa,\boldsymbol{Q}} V$ is replaced by $\partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}}$, computed by density-functional perturbation theory at the same time as the phonons themselves (Sect. 5.4), in calculations of the kind described in Appendix E.4.

The electron–phonon Hamiltonian and the nonadiabatic couplings describe one interaction in two bases. To first order, Eq. (6.4) mixes the clamped states with an amplitude $\boldsymbol{u} \cdot \boldsymbol{F}_{ji}/(E_i - E_j)$, proportional to the displacement, which by Eq. (5.11) is just $\boldsymbol{u} \cdot \boldsymbol{d}_{ji}$, the first term in the Taylor series of the adiabatic state $\Psi_i(r; u)$ (Appendix D.13). This part of the mixing is the electron cloud deforming with the lattice, and it costs nothing; it is already in the adiabatic

states. Real transitions need more: Eq. (6.3) shows that they are driven by the velocity, and their amplitude carries one more factor $\omega_{\mathrm{ph}}/\Delta E$. That is why a phonon, far below every electronic excitation in energy, cannot excite the electrons of an insulator on its own. The same vertex readily scatters electrons that are already excited, or takes part in a second-order process that supplies the energy (Sect. 6.5).

6.1.3 Adiabatic and sudden limits

Whether a moving nucleus leaves its electrons excited depends on how its velocity changes in time. Take one nucleus, or a whole atom, that is set in motion along a prescribed path $\boldsymbol{X}(t)$, from rest to the final velocity $\boldsymbol{v}_N$, and let its electrons start in the ground state. Translation invariance makes the electronic energies and the derivative couplings independent of the position of the nucleus. Let j be an excited state at $\Delta E_j = E_j - E_0$ above the ground state. In the interaction picture of Eq. (2.49), Eq. (6.3) gives its amplitude to first order,

$$c_j(t) = -\int_{-\infty}^{t} \mathrm{d}t'\; \dot{\boldsymbol{X}}(t') \cdot \boldsymbol{d}_{j0}\, \mathrm{e}^{\mathrm{i}\Delta E_j t'}.$$

An integration by parts separates two kinds of response,

$$c_j(t) = -\frac{\dot{\boldsymbol{X}}(t)\cdot\boldsymbol{d}_{j0}}{\mathrm{i}\Delta E_j}\,\mathrm{e}^{\mathrm{i}\Delta E_j t} + \frac{\boldsymbol{d}_{j0}}{\mathrm{i}\Delta E_j}\cdot\int_{-\infty}^{t}\mathrm{d}t'\;\ddot{\boldsymbol{X}}(t')\,\mathrm{e}^{\mathrm{i}\Delta E_j t'}. \tag{6.5}$$

The first term follows the instantaneous velocity and disappears when the nucleus stops. It describes the electron cloud of a uniformly moving atom, which carries the momentum of its electrons along and is, for that reason, not the ground state of the atom at rest. The second term is what remains after the motion: a real excitation, driven by the acceleration and weighted by its Fourier component at the excitation energy. Two limits follow at once. If the nucleus is accelerated over a time τ_{acc} long compared with $1/\Delta E_j$, the integrand oscillates many times during the acceleration and the Fourier component is small; for a smooth velocity profile it falls faster than any power of $\Delta E_j \tau_{\mathrm{acc}}$. This is the *adiabatic limit*, in which the electrons follow the nuclei and end in the ground state. If instead $\Delta E_j \tau_{\mathrm{acc}} \ll 1$, the phase stays fixed while the nucleus accelerates. The integral is then just $\boldsymbol{v}_N$, and

$$c_j = \frac{\boldsymbol{v}_N\cdot\boldsymbol{d}_{j0}}{\mathrm{i}\Delta E_j}. \tag{6.6}$$

This is the *sudden limit*: the electrons have no time to react while the nucleus is accelerated, and they find themselves in a state that is no longer an eigenstate around the moving nucleus. The nucleus has been given a velocity, but its electrons have not.

A model that can be solved exactly shows the crossover (Fig. 6.2). One electron of mass m_e is bound to a nucleus by a harmonic force of frequency ω_{e}, the electronic oscillator of the solvable model of Appendix D.13, and the nucleus follows the velocity profile $\dot{X}(t) = \frac{1}{2}v_N[1+\tanh(t/\tau_{\mathrm{acc}})]$. Let $y = x - X(t)$ be the position of the electron relative to the nucleus. In the frame of the nucleus the electron is an oscillator pushed by the inertial force $-m_e\ddot{X}(t)$,

$$H = \frac{p_y^2}{2m_e} + \frac{1}{2}m_e\omega_{\mathrm{e}}^2 y^2 + m_e\ddot{X}(t)\,y.$$

A linear force acting on an oscillator in its ground state leaves it in a coherent state, the displaced ground state $\hat{D}(\alpha_{\mathrm{d}})|0\rangle$ of Eq. (C.50). The amplitude follows from the classical equation of motion, or from the exact form of Eq. (6.5) for this model, as $\alpha_{\mathrm{d}} = -\mathrm{i}(m_e/2\omega_{\mathrm{e}})^{1/2}\int \mathrm{d}t\,\ddot{X}\,\mathrm{e}^{\mathrm{i}\omega_{\mathrm{e}}t}$, and the number of quanta is Poisson distributed (Eq. (C.51)). The electron stays in the

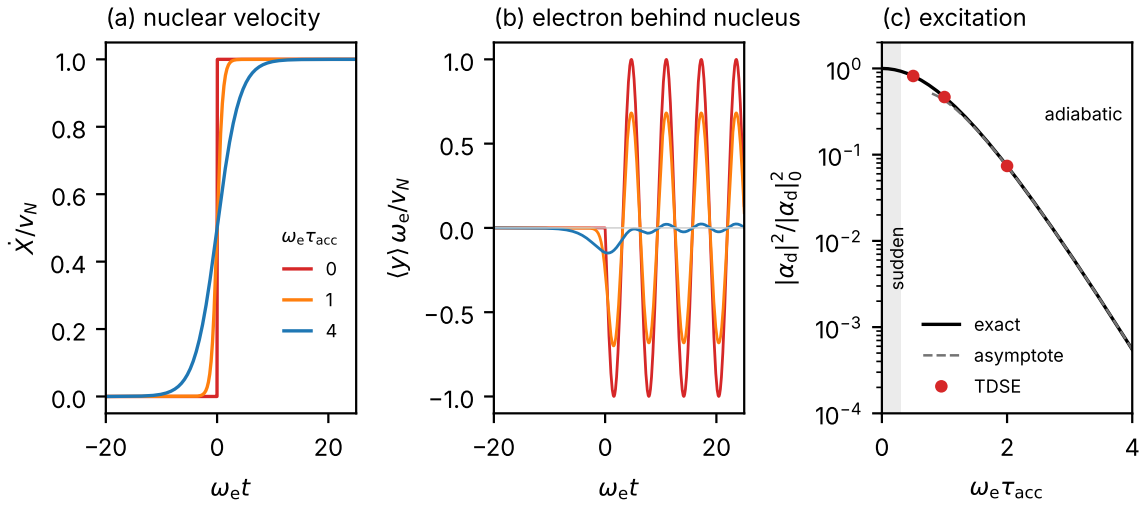


Figure 6.2: Adiabatic and sudden responses of an electron to a moving nucleus, in a model in which one electron is bound to the nucleus by a harmonic force of frequency ω_e and the nucleus follows a prescribed classical path. (a) The velocity of the nucleus, $\dot{X} = \frac{1}{2}v_N[1+\tanh(t/\tau_{acc})]$, for acceleration times $\omega_e\tau_{acc} = 0$ (an instantaneous kick), 1 and 4. (b) The mean position $\langle y\rangle$ of the electron relative to the nucleus, in units of v_N/ω_e, for the same three cases. After a kick the electron oscillates about the nucleus with the full amplitude; after a slow acceleration it lags behind while the force acts and returns to rest. (c) The excitation $|\alpha_d|^2 = -\ln P_0$, with P_0 the probability that the electron ends in the ground state, divided by its value $|\alpha_d|_0^2 = m_e v_N^2/2\omega_e$ for an instantaneous kick, against $\omega_e\tau_{acc}$. Solid: the exact result Eq. (6.7). Dashed: its adiabatic asymptote $\pi^2(\omega_e\tau_{acc})^2 e^{-\pi\omega_e\tau_{acc}}$. Dots: $-\ln P_0$ from a split-operator solution of the time-dependent Schrödinger equation for $m_e v_N^2/2\omega_e = 1/2$. Shaded: the sudden region $\omega_e\tau_{acc} < 0.3$, where the excitation is within 10 % of its sudden value.

ground state with probability $P_0 = e^{-|\alpha_d|^2}$. With $\int ds\,\mathrm{sech}^2 s\; e^{iks} = \pi k/\sinh(\pi k/2)$ the Fourier integral is elementary, and

$$|\alpha_d|^2 = \frac{m_e v_N^2}{2\omega_e}\left[\frac{\pi\omega_e\tau_{acc}/2}{\sinh(\pi\omega_e\tau_{acc}/2)}\right]^2. \tag{6.7}$$

The prefactor is the sudden value. It is the kinetic energy $\frac{1}{2}m_e v_N^2$ that the electron lacks after the kick, measured in quanta of the oscillator. The bracket is the adiabatic suppression: it is 0.47 at $\omega_e\tau_{acc} = 1$ and 0.074 at $\omega_e\tau_{acc} = 2$, and beyond that it is close to $\pi^2(\omega_e\tau_{acc})^2\, e^{-\pi\omega_e\tau_{acc}}$. The dots in Fig. 6.2(c) come from a direct solution of the time-dependent Schrödinger equation. Panel (b) shows the two behaviors in time. A sudden kick leaves the electron oscillating about the nucleus with the amplitude v_N/ω_e; a slow acceleration displaces it behind the nucleus by $\ddot{X}/\omega_e^2$ while the force acts, the first term of Eq. (6.5), and then lets it return to rest at the center.

For a nucleus in a crystal two times matter: how long the kick lasts, and how soon the neighbors start pulling the nucleus back. The kick itself is fast. A classical estimate of its duration, a heuristic only, is the time the dark matter takes to cross the region where it interacts with the nucleus. For a contact interaction and a silicon nucleus of radius 3.5 fm this is 4×10^{-21} s at v_{max}, while $1/E_g = 0.56$ fs. The deceleration is slow. After the kick the nucleus is pulled back by the lattice over a time $\sim 1/\omega_{ph}$, which is the adiabatic regime for every excitation across the gap. In the model, take a nucleus that is kicked at $t = 0$

and then oscillates at $\omega_{\rm ph}$ in a harmonic well, $X(t) = (v_N/\omega_{\rm ph})\sin\omega_{\rm ph}t$ for $t > 0$. Its acceleration is $\ddot{X} = v_N\delta(t) - v_N\omega_{\rm ph}\sin(\omega_{\rm ph}t)\,\Theta(t)$, and $\int \mathrm{d}t\,\ddot{X}\,\mathrm{e}^{\mathrm{i}\omega_e t} = v_N/[1-(\omega_{\rm ph}/\omega_{\rm e})^2]$, so $|\alpha_{\rm d}|^2$ exceeds the free-recoil value by the factor $[1-(\omega_{\rm ph}/\omega_{\rm e})^2]^{-2}$. With $\omega_{\rm ph}/\omega_{\rm e}$ set to the ratio $\omega_{\rm opt}/E_{\rm g} = 0.055$ of the zone-center optical phonon of silicon, $\omega_{\rm opt} = 64.1\,\mathrm{meV}$, to its gap, the factor is 1.006. A kicked nucleus in an insulator therefore excites its electrons as if it were free, with corrections of relative order $(\omega_{\rm ph}/\Delta E)^2$, the square of the small ratio that controls the adiabatic approximation. The sudden picture fails where that ratio is not small, for excitation energies comparable to phonon energies, as at the Fermi surface of a metal, and where the classical path itself fails, for recoil energies $q^2/2M$ that do not exceed a phonon energy (Sect. 5.6). There the nucleus does not recoil, the kick creates a few phonons, and the electron–lattice response of Sect. 12.1 replaces the trajectory.

6.1.4 The sudden kick

In the sudden limit the result does not need perturbation theory. Immediately after the kick the electronic wavefunction is what it was before, $\Psi_0(r;u)$, but the nucleus moves with velocity $\boldsymbol{v}_N$. In the frame that moves with the nucleus, the electrons have all acquired the velocity $-\boldsymbol{v}_N$. A Galilean boost of every electron by $-\boldsymbol{v}_N$ multiplies the wavefunction by a plane wave:

$$\Psi_0 \;\to\; \exp\Big(-\mathrm{i}m_e\boldsymbol{v}_N\cdot\sum_e(\boldsymbol{r}_e-\boldsymbol{X})\Big)\Psi_0, \qquad P_0 = \Big|\Big\langle\Psi_0\Big|\exp\Big(-\mathrm{i}m_e\boldsymbol{v}_N\cdot\sum_e\boldsymbol{r}_e\Big)\Big|\Psi_0\Big\rangle\Big|^2, \quad (6.8)$$

where in P_0, the probability that the electrons are found in the ground state around the moving nucleus, the phase $\mathrm{e}^{\mathrm{i}m_e\boldsymbol{v}_N\cdot\boldsymbol{X}}$ drops out. This is Migdal's formulation [4, 5, Eqs. (1) and (2)]. To first order in v_N the amplitude of an excited state is $-\mathrm{i}m_e\boldsymbol{v}_N\cdot\langle\Psi_j|\sum_e\boldsymbol{r}_e|\Psi_0\rangle$, a dipole transition driven by the kick. It agrees with Eq. (6.6). For an atom the adiabatic states depend on the nuclear position only through $\boldsymbol{r}_e-\boldsymbol{X}$, so that $\nabla_{\boldsymbol{X}}\Psi_0 = -\sum_e\nabla_e\Psi_0$, and the total electron momentum is $-\mathrm{i}\sum_e\nabla_e = \mathrm{i}m_e[H_{\rm el},\sum_e\boldsymbol{r}_e]$. Taking the matrix element between Ψ_j and Ψ_0, $\boldsymbol{d}_{j0} = m_e\Delta E_j\,\langle\Psi_j|\sum_e\boldsymbol{r}_e|\Psi_0\rangle$, and Eq. (6.6) becomes the dipole amplitude. The derivative coupling of an atom is its dipole matrix element times the excitation energy and the electron mass.

For one electron in hydrogen the sudden result is exact. The Hamiltonian $p_e^2/2m_e + p_p^2/2m_p - \alpha/|\boldsymbol{r}_e-\boldsymbol{X}|$ separates into the motion of the center of mass $\boldsymbol{R}_{\rm cm} = (m_p\boldsymbol{X}+m_e\boldsymbol{r}_e)/M_{\rm H}$, with $M_{\rm H} = m_p+m_e$, and the relative coordinate $\boldsymbol{\rho} = \boldsymbol{r}_e-\boldsymbol{X}$, whose ground state is the 1s state with the reduced mass $\mu_{ep} = m_em_p/M_{\rm H}$ and the Bohr radius $a_{ep} = 1/(\alpha\mu_{ep})$. Dark matter that couples to the proton acts on the atom, to first order in its coupling, through $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}$ (Sect. 2.2); a coupling of the dark matter to the electron itself is a separate process, which we leave aside. Since $\boldsymbol{X} = \boldsymbol{R}_{\rm cm} - (m_e/M_{\rm H})\boldsymbol{\rho}$,

$$\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}} = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_{\rm cm}}\;\mathrm{e}^{-\mathrm{i}(m_e/M_{\rm H})\boldsymbol{q}\cdot\boldsymbol{\rho}}.$$

The first factor gives the whole atom the momentum $\boldsymbol{q}$. The second acts on the internal state like a momentum transfer $-(m_e/M_{\rm H})\boldsymbol{q}$ to the electron, and its ground-state expectation value is the elastic form factor Eq. (2.21) with a_{ep} in place of a_0. Its argument is $(m_eq/M_{\rm H})\,a_{ep} = q/(\alpha m_p) = v_N/\alpha$, where $v_N = q/m_p$ is the velocity given to the proton, and the reduced-mass corrections cancel. With $a_0 = 1/(\alpha m_e)$ we may write the argument as $m_ev_Na_0$:

Sudden kick of a hydrogen atom

A hydrogen atom in its ground state whose nucleus is given the velocity $v_N = q/m_p$

remains in its ground state with probability

$$P_0 = \left|F_{1s}(m_e v_N)\right|^2 = \left[\frac{16}{(4+v_N^2/\alpha^2)^2}\right]^2 \simeq 1 - \frac{v_N^2}{\alpha^2} \quad (v_N \ll \alpha), \tag{6.9}$$

and is excited or ionized with probability $1 - P_0$. The controlling ratio is that of the nuclear velocity to the orbital velocity α of the electron.

The derivation used no assumption about time. In a first-order amplitude the momentum transfer is a single act, and the dark matter couples to the nuclear coordinate only. Nothing in a free atom decelerates the nucleus afterward, and the separation of the center of mass does the work that suddenness did in Eq. (6.8). Ibe et al. derive the general atom in this way, with the amplitude of every electronic transition given by Eq. (6.8) and the energy of the atom conserved overall [5, Eqs. (55)–(58)]. For more than one electron the separation is not exact, and the argument $m_e v_N$ is corrected at relative order m_e/M; Sect. 12.2 makes this precise.

The fraction $1 - P_0$ of nuclear recoils that leave the atom excited or ionized is the *Migdal effect*. Its interest for dark-matter searches is the energy it adds. A recoil of a few hundred eV in silicon may be below the threshold of a phonon or charge readout. An electron–hole pair from the same event, carrying at least the gap, is easy to count. Chap. 12 computes the effect in crystals, where the electrons are shared between atoms and the excitation is an electron–hole pair rather than an atomic transition.

Worked example 6.1: Ground-state survival of a kicked hydrogen atom

Dark matter of $m_\chi = 1\,\mathrm{GeV}$ scatters elastically off a hydrogen atom at rest, at the largest laboratory speed $v_{\max} = 2.650 \times 10^{-3}$ (Eq. (1.9)). The atom has the mass $M_\mathrm{H} = 938.783\,\mathrm{MeV}$, the reduced mass is $\mu_{\chi\mathrm{H}} = (1000 \times 938.783/1938.783)\,\mathrm{MeV} = 484.2\,\mathrm{MeV}$, and the largest momentum transfer is $q = 2\mu_{\chi\mathrm{H}} v_{\max} = 2.566\,\mathrm{MeV}$, at which the atom recoils with $q^2/2M_\mathrm{H} = 3.51\,\mathrm{keV}$.

The proton, of mass $m_p = 938.272\,\mathrm{MeV}$, is kicked to $v_N = q/m_p = 2.735 \times 10^{-3}$, or $820\,\mathrm{km\,s^{-1}}$; dark matter as heavy as the target bounces it at about its own speed. In units of the orbital velocity, $v_N/\alpha = 0.375$. Then $F_{1s}(m_e v_N) = 16/(4 + 0.1405)^2 = 0.933$, and Eq. (6.9) gives

$$P_0 = 0.871, \qquad 1 - P_0 = 0.129.$$

One such recoil in eight leaves the atom excited or ionized, with at least the 10.2 eV of the $1s \to 2p$ transition in its electrons. The small-v_N form, $v_N^2/\alpha^2 = 0.140$, is 9 % high at this velocity. The atom survives with probability one half at $v_N/\alpha = 0.87$, where $F_{1s}^2 = \frac{1}{2}$ (Worked example 2.2): $v_N = 6.35 \times 10^{-3}$, $q = 5.96\,\mathrm{MeV}$ and a recoil of 18.9 keV, beyond the reach of dark matter at this mass.

Is the kick sudden? The excitation energies start at $\Delta E = \frac{3}{8}\alpha^2 m_e = 10.2\,\mathrm{eV}$, so the electron responds on the time $1/\Delta E = 0.065\,\mathrm{fs}$. Dark matter with a contact interaction crosses the proton, of charge radius 0.84 fm [6], in $1.1 \times 10^{-21}\,\mathrm{s}$, and $\Delta E\,\tau_\mathrm{acc} = 1.6 \times 10^{-5}$: on the scale of Fig. 6.2(c) the kick is instantaneous. A free atom has nothing that decelerates the nucleus afterward, and Eq. (6.9) is exact to first order in the dark-matter coupling.

For the kick of the opening, $v_N = 1.95 \times 10^{-4}$ for a silicon nucleus, the same formula gives $1 - P_0 = 7.1 \times 10^{-4}$ for an electron bound as in hydrogen. The outer electrons of silicon are bound more weakly and the inner ones far more strongly, so this is an order-of-magnitude statement; Sect. 12.2 computes the atom.

Two different results carry Migdal's name, and they sit at opposite ends of this section. In 1941 Migdal computed the ionization of an atom whose nucleus is suddenly set in motion, as in α decay, or suddenly changes its charge, as in β decay [4]. That is the sudden limit of Eq. (6.8), and the "Migdal effect" of the dark-matter literature is its application to nuclear recoils (Chap. 12). In 1958 he showed that the vertex corrections in a metal are small by the ratio $\omega_{\mathrm{ph}}/E_{\mathrm{F}}$ of the counting above, which is why Eq. (6.4) with a single vertex is enough for most of this chapter (Sect. 6.3) [2]. The first result is about electrons left behind by a fast nucleus; the second is about electrons that keep up with slow ones.

The vertex g of Eq. (6.4) is now the central object. With the bare potential it overestimates the coupling, since the other electrons screen most of the force of a displaced nucleus. The screened vertex, its forms for acoustic and polar optical phonons and its first-principles calculation are the subject of Sect. 6.2.

6.2 Electron–phonon matrix elements

An electron at the bottom of the conduction band of GaAs absorbs a longitudinal optical phonon of wavevector $0.01\,\text{Å}^{-1}$ with a vertex $|g| = 3.3\,\text{eV}$. An electron in a conduction valley of silicon absorbs a longitudinal sound wave of the same wavevector, running along the valley axis, with $|g| = 26\,\text{meV}$, more than a hundred times less. Ten times longer wavelengths make the GaAs vertex ten times larger and the silicon one $\sqrt{10}$ times smaller. Both numbers follow from the definition Eq. (6.4) once the bare potential of a displaced nucleus is replaced by the potential that the electrons actually feel, screened by all the other electrons. The coupling to long-wavelength sound waves controls how an excited carrier in silicon sheds its last few meV. The Fröhlich coupling of GaAs is the electron's view of the same macroscopic field through which a dark photon drives the optical phonons of a polar crystal (Sect. 5.5). The phonon that lets a slow dark-matter particle cross the indirect gap of silicon enters through the same vertex, at large wavevector (Sect. 6.5).

We will derive the screened vertex in the form that density-functional perturbation theory computes, and then its three long-wavelength limits, which scale as $Q^{1/2}$, Q^0 and Q^{-1} and carry the classic deformation potentials and Fröhlich's constant; for GaAs that constant comes out as 0.07. The same analysis explains why a vertex computed on a few hundred phonon wavevectors can be interpolated to the billions that a rate requires, and why the Fröhlich part must be taken out before the interpolation and put back after it.

6.2.1 The screened vertex

The bare potential of a displaced nucleus is long-ranged. A pseudo-ion of valence charge Z_κ has the potential energy $v_\kappa(k) \to -Z_\kappa e^2/k^2$ for an electron at small wavevector k, the Coulomb tail of Eq. (A.3). Its gradient, the force field of Sect. 6.1, has the Fourier components $-\mathrm{i}\boldsymbol{k}\, v_\kappa(k)$. A displacement wave of wavevector $\boldsymbol{Q}$ therefore has a bare potential whose component at $\boldsymbol{Q}$ grows as $Z_\kappa e^2/Q$, for every crystal. The electrons undo most of this. They flow toward a displaced positive ion and away from the region it left, and what an electron near the ion feels is the change of the self-consistent potential, $\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\mathrm{KS}}$ of Eq. (5.42), not the change of the bare one.

The screening can be written in one line. The first-order density of Eq. (5.45) is the static Kohn–Sham response to the screened potential, $\partial n = R^0\, \partial V_{\mathrm{KS}}$, with R^0 the independent-particle response of Sect. 4.6 at $\omega = 0$, and Eq. (5.42) adds back the Hartree and exchange–correlation potentials of that density, $\partial V_{\mathrm{KS}} = \partial V_{\mathrm{ext}} + f_{\mathrm{Hxc}}\, \partial n$ with $f_{\mathrm{Hxc}} = v_{\mathrm{C}} + f_{\mathrm{xc}}$. Eliminating ∂n,

$$\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\mathrm{KS}} = \left(1 - f_{\mathrm{Hxc}} R^0\right)^{-1} \partial_{\kappa\alpha,\boldsymbol{Q}} V_{\mathrm{ext}} = \left(1 + f_{\mathrm{Hxc}} R\right) \partial_{\kappa\alpha,\boldsymbol{Q}} V_{\mathrm{ext}}, \tag{6.10}$$

where products are integral operators in space, or matrices in the reciprocal-lattice vectors $\boldsymbol{Q}+\boldsymbol{G}$ at fixed $\boldsymbol{Q}$, and $R = R^0(1 - f_{\mathrm{Hxc}}R^0)^{-1}$ is the full static response of Eq. (4.70); the form with $1 + f_{\mathrm{Hxc}}R$ follows from expanding the inverse in powers of $f_{\mathrm{Hxc}}R^0$. With f_{xc} dropped, $1 + v_{\mathrm{C}}R$ is the inverse dielectric matrix $\varepsilon^{-1}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{Q}, 0)$ of Eq. (4.43). With it, the operator is the inverse dielectric function of a *test electron*: the potential that a second electron feels includes the exchange–correlation hole that the response builds around it, while the test charge of Sect. 4.4 feels only the Hartree part. The screening is static. A phonon of tens of meV is slow compared with the electronic response of an insulator, whose scale is the gap, and the static limit is the adiabatic approximation of Sect. 6.1. Below the gap a response function is even in the frequency, so the static screening is accurate to relative order $(\omega_{\mathrm{ph}}/E_{\mathrm{g}})^2$, as are the phonon frequencies (Sect. 6.4). The first power of $\omega_{\mathrm{ph}}/E_{\mathrm{g}}$ in Sect. 6.1 measures the amplitude for exciting the electrons, not the change of their response. Screening of the ionic potential by the static Lindhard function goes back to Bardeen, as quoted by Giustino [3, Eqs. (3)–(5)]. In a metal, whose response has no gap, the same structure appears with the response evaluated at the phonon frequency, the dynamically screened vertex of Bardeen and Pines [7].

How much is left at long wavelength depends on the material. In a metal, the Thomas–Fermi limit of Eq. (4.53) divides the bare $Z_\kappa e^2/Q$ by $\varepsilon_{\mathrm{TF}}(Q) = 1 + k^2_{\mathrm{TF}}/Q^2$, and the screened vertex becomes $Z_\kappa e^2 Q/(Q^2 + k^2_{\mathrm{TF}})$: the divergence has turned into a coupling that vanishes linearly in Q. The electrons have neutralized the displaced ion completely. In an insulator the head of the dielectric matrix tends to ε_∞, which reduces the bare $1/Q$ tail by a factor of 11.7 in silicon but does not remove it. The tail is removed by the local fields of Sect. 4.4, the off-diagonal elements $\varepsilon^{-1}_{\boldsymbol{0}\boldsymbol{G}}$ that describe electrons moving with the ion on the scale of a bond. Their effect is to replace the ionic charge by the Born effective charge of Sect. 5.5. A displaced ion drags part of its bond charge along, and the induced density at the wavevectors $\boldsymbol{Q}+\boldsymbol{G}$ with $\boldsymbol{G} \neq 0$ is this comoving cloud. Its dipole per unit displacement is $-e(Z_\kappa - Z^*_\kappa)$, so that ion and cloud together carry eZ^*_κ. At long wavelength a displaced, screened ion is a dipole $eZ^*_\kappa \cdot \boldsymbol{u}$ in a medium of dielectric constant ε_∞, as Pick, Cohen and Martin showed from the structure of the dielectric matrix [8]. In silicon $Z^* = 0$. The electron cloud follows each nucleus so well that no dipole is left, and the long-range part of the screened potential starts at the quadrupole. In GaAs $Z^* = 2.1$, the dipole survives, and its field gives the $1/Q$ vertex of the opening, the Fröhlich coupling.

Eq. (6.4) with the screened potential is the vertex that density-functional perturbation theory computes. Since $\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\mathrm{KS}}$ is a Bloch function of crystal momentum $\boldsymbol{Q}$, we write it as $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}}\, \partial_{\kappa\alpha,\boldsymbol{Q}} v_{\mathrm{KS}}(\boldsymbol{r})$, with $\partial_{\kappa\alpha,\boldsymbol{Q}} v_{\mathrm{KS}}$ periodic in the lattice. With the Bloch states of Eq. (A.15), the phases $\mathrm{e}^{-\mathrm{i}(\boldsymbol{k}+\boldsymbol{Q})\cdot\boldsymbol{r}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ cancel, as they did for the density vertex Eq. (A.17), the factor $1/N$ cancels the N cells, and

$$g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q}) = \ell_{\boldsymbol{Q}\nu} \left\langle u_{m,\boldsymbol{k}+\boldsymbol{Q}} \right| \Delta_{\boldsymbol{Q}\nu} v_{\mathrm{KS}} \left| u_{n\boldsymbol{k}} \right\rangle_\Omega, \qquad \Delta_{\boldsymbol{Q}\nu} v_{\mathrm{KS}} = \sum_{\kappa\alpha} \Big(\frac{M_{\mathrm{cell}}}{M_\kappa} \Big)^{1/2} e^\alpha_{\kappa\nu}(\boldsymbol{Q})\, \partial_{\kappa\alpha,\boldsymbol{Q}} v_{\mathrm{KS}}, \tag{6.11}$$

where the bracket is an integral over one cell. We have written the prefactor $(2M_\kappa \omega_{\boldsymbol{Q}\nu})^{-1/2}$ of Eq. (A.26) as the zero-point length $\ell_{\boldsymbol{Q}\nu}$ of Eq. (A.24) times $(M_{\mathrm{cell}}/M_\kappa)^{1/2}$. This is the normalization of Refs. [3, Eqs. (33)–(38)] and [9, Eq. (2)]. The zero-point length, typically a few hundredths of an Å for an optical phonon, carries all the mass and frequency dependence. What is left is how much the potential changes when the atoms move, the matrix element of $\Delta_{\boldsymbol{Q}\nu} v_{\mathrm{KS}}$ per unit mass-weighted displacement, in eV per Å. For an acoustic mode at small Q, where $(M_{\mathrm{cell}}/M_\kappa)^{1/2} \boldsymbol{e}_{\kappa\nu} = \hat{\boldsymbol{e}}_\nu$ for every κ (Sect. 5.3), it is the potential change per unit displacement of the whole crystal. The vertex itself has the dimension of energy, and $|g|^2/N$ is the squared amplitude per mode. The vertex depends on the phases of the two Bloch states, as ρ_{mn} does in Eq. (A.19); only sums of $|g|^2$ over degenerate partners can be compared

between calculations.

In practice one run of density-functional perturbation theory per phonon wavevector solves the Sternheimer equation Eq. (5.44) for the three displacement directions of every atom and returns $\partial_{\kappa\alpha,\boldsymbol{Q}} v_{\mathrm{KS}}$ together with the phonon frequencies and polarization vectors. The vertex for any pair of bands and any $\boldsymbol{k}$ then costs just one matrix element, Eq. (6.11), one application of a known potential to a known state. The run does not change for the vertex: the screened potential is the quantity whose self-consistency already produced the phonon. Appendix E.4 describes such a run and its convergence parameters, and Appendix E.5 the step from a coarse set of such vertices to the dense grids of a rate, whose principle closes this section.

6.2.2 Acoustic phonons and deformation potentials

Moving the whole crystal changes no band energy. If every nucleus is displaced by the same vector $\boldsymbol{t}$ and the electrons follow, the self-consistent potential is translated with them, $V_{\mathrm{KS}}(\boldsymbol{r}; u + t) = V_{\mathrm{KS}}(\boldsymbol{r} - \boldsymbol{t}; u)$, or $\hat{T}_{\boldsymbol{t}} V_{\mathrm{KS}} \hat{T}_{\boldsymbol{t}}^{\dagger}$ with the translation operator $\hat{T}_{\boldsymbol{t}} = \mathrm{e}^{-\mathrm{i}\boldsymbol{t}\cdot\hat{\boldsymbol{p}}}$ when the potential is a nonlocal pseudopotential. Differentiating at $\boldsymbol{t} = 0$ collects the derivatives with respect to all nuclei into the derivative of the potential with respect to the electron coordinate. The sum over all cells and all species is the displacement wave at $\boldsymbol{Q} = 0$ summed over κ, so $\sum_\kappa \partial_{\kappa\alpha,\boldsymbol{0}} V_{\mathrm{KS}} = -\mathrm{i}[\hat{p}_\alpha, V_{\mathrm{KS}}] = -\mathrm{i}[\hat{p}_\alpha, H_{\mathrm{KS}}]$, where the last step uses that the kinetic energy commutes with the momentum. Between two Bloch states at the same $\boldsymbol{k}$, eigenstates of H_{KS},

$$\sum_{\kappa} \langle \psi_{m\boldsymbol{k}} |\, \partial_{\kappa\alpha,\boldsymbol{0}} V_{\mathrm{KS}}\, | \psi_{n\boldsymbol{k}} \rangle = -\mathrm{i}\,(\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}})\, p^{\alpha}_{mn}(\boldsymbol{k}), \tag{6.12}$$

with p^{α}_{mn} the momentum matrix element of Sect. 3.2. The diagonal element vanishes: a rigid translation does not shift band n at first order. This is the acoustic sum rule of the vertex, the counterpart of Eq. (5.18) for the force constants, and like it, it follows from translation invariance alone.

An acoustic phonon at small Q approaches this rigid translation. Its polarization vectors tend to $(M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$ (Sect. 5.3), so that $\Delta_{\boldsymbol{Q}\nu} v_{\mathrm{KS}}$ in Eq. (6.11) becomes $\hat{\boldsymbol{e}}_\nu \cdot \sum_\kappa \partial_{\kappa,\boldsymbol{Q}} v_{\mathrm{KS}}$, and Eq. (6.12) makes its diagonal matrix element vanish as $Q \to 0$. What survives is the next order in Q, multiplied by a zero-point length that grows as $\ell_{\boldsymbol{Q}\nu} = (2M_{\mathrm{cell}} c_\nu Q)^{-1/2}$. The intraband vertex therefore vanishes as

$$g_{nn\nu}(\boldsymbol{k}, \boldsymbol{Q}) \propto Q\, \ell_{\boldsymbol{Q}\nu} \propto Q^{1/2} \qquad (\nu \text{ acoustic},\ Q \to 0),$$

the statement that $g_{nn\nu}(\boldsymbol{k}, \boldsymbol{0}) = 0$ for acoustic modes. The interband elements of Eq. (6.12) do not vanish. They describe the rigid translation of the electronic state itself: shifted by $\ell_{\boldsymbol{Q}\nu}\hat{\boldsymbol{e}}_\nu$, a state changes by $-\ell_{\boldsymbol{Q}\nu}\hat{\boldsymbol{e}}_\nu \cdot \nabla\psi_{n\boldsymbol{k}}$, which written in the other bands at the same $\boldsymbol{k}$ is exactly this matrix element divided by $\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}}$. No energy is exchanged in such a change, and in the energy shifts of Sect. 6.3 it cancels against the second-order Allen–Heine term, as translation invariance demands. Codes therefore set the vertices of the three acoustic modes at $\boldsymbol{Q} = 0$ to zero [3, below Eq. (38)].

The term of first order in Q is the deformation potential. Consider a region short compared with the wavelength $2\pi/Q$ but containing many cells. There one long acoustic mode displaces every nucleus by the same amount, $\boldsymbol{u}(\boldsymbol{r}) = (\ell_{\boldsymbol{Q}\nu}/\sqrt{N})\,\hat{\boldsymbol{e}}_\nu\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}} B_{\boldsymbol{Q}\nu}$, and the crystal is locally translated and uniformly strained, with strain tensor

$$S_{\alpha\beta}(\boldsymbol{r}) = \tfrac{1}{2}(\partial_\alpha u_\beta + \partial_\beta u_\alpha) = \frac{\mathrm{i}}{2}(Q_\alpha \hat{e}_{\nu\beta} + Q_\beta \hat{e}_{\nu\alpha})\, \frac{\ell_{\boldsymbol{Q}\nu}}{\sqrt{N}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}} B_{\boldsymbol{Q}\nu}.$$

The translation does nothing, by Eq. (6.12). A uniform strain shifts the band energies at first order, $\delta\varepsilon_{n\boldsymbol{k}} = \sum_{\alpha\beta} D^n_{\alpha\beta} S_{\alpha\beta}$, and this defines the *deformation-potential tensor* $D^n_{\alpha\beta}$ of band n

near the extremum of interest. It is symmetric, and it is taken with the nuclei inside each cell relaxed, since a long acoustic wave carries the internal relaxation of Sect. 5.2 with it. An electron near the band edge, in a wavepacket much longer than the cell but short compared with $2\pi/Q$, feels the slowly varying shift $\delta\varepsilon_n(\boldsymbol{r})$ as a potential energy. This is the argument of Bardeen and Shockley [10]. The matrix element of that potential between $n\boldsymbol{k}$ and $n, \boldsymbol{k}+\boldsymbol{Q}$ is the amplitude of the shift times $\rho_{nn}(\boldsymbol{k},\boldsymbol{Q}) \simeq 1$. Comparison with Eq. (6.4) gives

$$g_{nn\nu}(\boldsymbol{k},\boldsymbol{Q}) = \mathrm{i}\,\ell_{\boldsymbol{Q}\nu} \sum_{\alpha\beta} Q_\alpha D^n_{\alpha\beta}\,\hat{e}_{\nu\beta} \qquad (\nu \text{ acoustic},\ Q \to 0). \tag{6.13}$$

For a band extremum at Γ in a cubic crystal the tensor is a multiple of the identity, $D^n_{\alpha\beta} = E_1\delta_{\alpha\beta}$, and $E_1 = \Omega\,\partial\varepsilon_n/\partial\Omega$, since the trace of the strain is the relative change of volume [3, Eq. (6)]. Only the longitudinal mode couples, through $\boldsymbol{Q}\cdot\hat{\boldsymbol{e}}_\nu$: a transverse wave shears the crystal without changing its volume. With $\omega_{\boldsymbol{Q}\nu} = c_{\mathrm{LA}}Q$, $|g|^2 = E_1^2 Q/(2M_{\mathrm{cell}}c_{\mathrm{LA}})$. Bardeen and Shockley obtained E_1 for the band edges of silicon by fitting measured mobilities to this form [10].

The conduction valleys of silicon are not at Γ. Each is symmetric only about its own axis $\hat{\boldsymbol{k}}_{\mathrm{c}}$ (Sect. 3.3), and the most general symmetric tensor with that symmetry has two constants, $D_{\alpha\beta} = \Xi_d\,\delta_{\alpha\beta} + \Xi_u\,\hat{k}_{\mathrm{c}\alpha}\hat{k}_{\mathrm{c}\beta}$, the dilatation and uniaxial deformation potentials of Herring and Vogt [11]. With θ the angle between $\boldsymbol{Q}$ and the valley axis, Eq. (6.13) gives the effective constants $\Xi_d + \Xi_u\cos^2\theta$ for the longitudinal mode and $\Xi_u \sin\theta\cos\theta$ for the transverse mode polarized in the plane of $\boldsymbol{Q}$ and $\hat{\boldsymbol{k}}_{\mathrm{c}}$; the other transverse mode does not couple [12, Eqs. (13) and (14)]. Li, Graziosi and Neophytou extracted the two constants from the slopes of DFPT vertices of silicon at small Q, and found $\Xi_d = 1.01\,\mathrm{eV}$ and $\Xi_u = 8.84\,\mathrm{eV}$ [12]. A longitudinal phonon of $Q = 0.01\,\mathrm{\AA}^{-1}$ along the valley axis has 0.55 meV and a zero-point length $\ell_{\boldsymbol{Q}\nu} = 0.26\,\mathrm{\AA}$, and Eq. (6.13) yields $|g| = 9.85 \times 0.01 \times 0.26\,\mathrm{eV} = 26\,\mathrm{meV}$, the number of the opening. Across the valley axis the coupling is almost ten times weaker, 2.6 meV. Fig. 6.3 shows both against Q. The form holds for intraband scattering near a band edge, for Q small compared with the zone; the computed matrix elements are linear in Q over the whole computed range, up to $0.1\,\mathrm{\AA}^{-1}$ [12, Fig. 5].

In a crystal without inversion symmetry, such as GaAs, a strain also polarizes the cell, $P_\alpha = \sum_{\beta\gamma} e^{\mathrm{pz}}_{\alpha\beta\gamma} S_{\beta\gamma}$ with the piezoelectric tensor e^{pz}, and an acoustic wave then carries a macroscopic field. The steps of Appendix D.14 that turn the polarization of a longitudinal optical wave into the Fröhlich vertex apply unchanged. The field obeys $\hat{\boldsymbol{Q}}\cdot(\varepsilon_\infty\boldsymbol{\mathcal{E}} + \boldsymbol{P}) = 0$, an electron feels the potential energy $V = e\hat{\boldsymbol{Q}}\cdot\boldsymbol{\mathcal{E}}/(\mathrm{i}Q)$, and with the strain above

$$g^{\mathrm{pz}}_{mn\nu}(\boldsymbol{k},\boldsymbol{Q}) = -\frac{e}{\varepsilon_\infty}\,\ell_{\boldsymbol{Q}\nu} \sum_{\alpha\beta\gamma} \hat{Q}_\alpha\, e^{\mathrm{pz}}_{\alpha\beta\gamma}\, \hat{Q}_\beta\, \hat{e}_{\nu\gamma}\, \rho_{mn}(\boldsymbol{k},\boldsymbol{Q}),$$

for a cubic crystal. This is the *piezoelectric coupling*. The factor Q of the strain cancels the $1/Q$ of the field, and the vertex depends on Q only through $\ell_{\boldsymbol{Q}\nu} \propto Q^{-1/2}$ and on the direction of $\boldsymbol{Q}$. It grows at long wavelength, more gently than the Fröhlich vertex. In a first-principles calculation it comes from two terms. One is the dynamical quadrupoles of the displaced atoms, the next term after the Born charge in the long-wavelength expansion of the charge that a displaced atom induces. The other is the Born charges of the relative sublattice displacement that the strain induces. In GaAs it dominates the scattering of electrons by acoustic phonons [13]; Appendix E.5 treats it with the other long-range terms.

6.2.3 Optical deformation potential

An optical phonon at $\boldsymbol{Q} = 0$ moves the sublattices against each other. Nothing in the translation argument forces its vertex to vanish, and in general it does not: bond lengths and

angles inside the cell change at first order in the displacement, and so do the band energies. The vertex is finite at $\boldsymbol{Q} = 0$, and it is conventionally written as

$$g_{nn\nu}(\boldsymbol{k}, \boldsymbol{0}) = \ell_{\boldsymbol{0}\nu}\, d^n_\nu, \qquad d^n_\nu = \langle \psi_{n\boldsymbol{k}} | \sum_\kappa \Big(\frac{M_{\text{cell}}}{M_\kappa} \Big)^{1/2} \boldsymbol{e}_{\kappa\nu}(\boldsymbol{0}) \cdot \partial_{\kappa,\boldsymbol{0}} V_{\text{KS}} | \psi_{n\boldsymbol{k}} \rangle, \tag{6.14}$$

with d^n_ν the *optical deformation potential* of band n, in eV per Å. Since $\partial_{\kappa,\boldsymbol{0}} V_{\text{KS}}$ displaces a whole sublattice, d^n_ν is, by the Hellmann–Feynman theorem of Sect. 5.1, the shift of $\varepsilon_{n\boldsymbol{k}}$ per unit mass-weighted displacement of the sublattices in the pattern of mode ν. In silicon the two atoms have equal masses, $\boldsymbol{e}_{1\nu} = -\boldsymbol{e}_{2\nu} = \hat{\boldsymbol{e}}/\sqrt{2}$, and the pattern moves the two atoms of every cell by $+u\hat{\boldsymbol{e}}$ and $-u\hat{\boldsymbol{e}}$; d^n_ν is then $\mathrm{d}\varepsilon_{n\boldsymbol{k}}/\mathrm{d}u$. Compared with the acoustic constant D^n, which enters with an extra factor Q, d^n_ν is of zeroth order in Q, and the two are often called the first-order and zero-order deformation potentials.

Symmetry can still make d^n_ν vanish, and for the conduction valleys of silicon it does: within a valley the electrons scatter off acoustic phonons only, and DFPT finds the intravalley optical vertices small near $\boldsymbol{Q} = 0$ [12]. The valence maximum is threefold degenerate at Γ (Sect. 3.3), and the optical phonon splits it. For a degenerate level the coupling is a matrix in the partners; Li et al. combine its elements into a single value, $10.45\,\text{eV}\,\text{Å}^{-1}$, the mean of two such combinations [12, Eq. (12) and the text below it]. With $\omega_{\boldsymbol{0}\nu} = 64.1\,\text{meV}$ the zero-point length of the optical mode is $\ell_{\boldsymbol{0}\nu} = 0.024\,\text{Å}$. That is half the 0.048 Å of the bond coordinate in Sect. 5.1, because ℓ refers the motion to the mass of the whole cell, and the vertex of the holes is $|g| \simeq 0.25\,\text{eV}$, independent of Q near the zone center (Fig. 6.3). It is ten times the acoustic vertex of the electrons at $Q = 0.01\,\text{Å}^{-1}$, but the phonon it emits costs 64 meV, which a hole near the band edge does not have. Electrons in silicon meet the same zero-order coupling when they scatter between valleys. The crystal momentum between two valleys is a sizable fraction of the zone, the phonons that supply it are near the zone boundary, and the vertex is finite there; such intervalley processes cool hot carriers in a detector (Appendix E.12). In a polar crystal the optical deformation potential is present as well, but the longitudinal optical mode carries in addition the macroscopic field, to which we turn now.

6.2.4 The Fröhlich vertex

In a longitudinal optical wave of GaAs the gallium and arsenic sublattices move against each other along $\boldsymbol{Q}$, every cell acquires a dipole $eZ^*\boldsymbol{u}_{\text{rel}}$, with $\boldsymbol{u}_{\text{rel}} = \boldsymbol{u}_1 - \boldsymbol{u}_2$ the relative displacement of the two sublattices (Appendix D.14), and the dipoles of a long wave add up to a macroscopic polarization. Its field reaches every electron in the crystal, however far from the displaced atoms. Appendix D.14 computes that field from the condition $\hat{\boldsymbol{Q}} \cdot \boldsymbol{D} = 0$ of electrostatics, in the same way as the LO–TO splitting of Sect. 5.5, and the potential energy of an electron in it. Written as a vertex in the normalization of Eq. (6.4), it is the *long-range vertex*

$$g^{\text{L}}_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q}) = \mathrm{i}\, \frac{e^2}{\Omega} \sum_\kappa \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) \cdot Z^*_\kappa \cdot \boldsymbol{Q}}{(2M_\kappa \omega_{\boldsymbol{Q}\nu})^{1/2}\, \boldsymbol{Q} \cdot \varepsilon_\infty \cdot \boldsymbol{Q}}\, \rho_{mn}(\boldsymbol{k}, \boldsymbol{Q}),$$

Eq. (D.77), the generalization of Fröhlich's coupling to any crystal and any number of polar modes due to Verdi and Giustino [14, Eq. (4)]. Only longitudinal modes couple, since only they have $Z^*_\kappa \cdot \boldsymbol{Q}$ along their displacement, and the vertex grows as $1/Q$, the Fourier transform of a field that falls off as the inverse square of the distance from a dipole. The density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{Q})$ tends to δ_{mn} as $Q \to 0$, so the divergence is intraband. Between different bands ρ_{mn} vanishes linearly in Q, by Eq. (A.18), and the interband long-range vertex tends to a finite limit that depends on the direction of $\boldsymbol{Q}$.

For a cubic crystal with two atoms per cell the Born charges and the masses drop out. The longitudinal polarization vectors, the reduced mass and the LO–TO relation Eq. (5.52)

combine with the Lyddane–Sachs–Teller relation Eq. (5.54) into a vertex that contains only macroscopic, measurable constants (Appendix D.14).

Fröhlich vertex

In a cubic polar crystal with two atoms per cell, the longitudinal optical phonon of small wavevector $\boldsymbol{Q}$ couples to the electrons with

$$g^{\mathrm{L}}_{mn}(\boldsymbol{k},\boldsymbol{Q}) = \frac{\mathrm{i}\,C_{\mathrm{F}}}{Q}\,\rho_{mn}(\boldsymbol{k},\boldsymbol{Q}), \qquad C_{\mathrm{F}}^2 = \frac{e^2\omega_{\mathrm{L}}}{2\Omega}\Big(\frac{1}{\varepsilon_\infty}-\frac{1}{\varepsilon_0}\Big), \tag{6.15}$$

with ω_{L} the longitudinal optical frequency at Γ and ε_∞ and ε_0 the electronic and static dielectric constants. The sign is that of the Born charge of the first atom; the coupling constant C_{F} has the dimension of energy squared.

This is Fröhlich's vertex [15], generalized to the density vertex ρ_{mn} in place of the single-band overlap [14, Eq. (1)]. The combination $1/\varepsilon_\infty - 1/\varepsilon_0$ measures the part of the screening that comes from the ions. A static charge in the crystal is screened by electrons and ions together, through ε_0. The field of the displaced ions, as seen by the electrons, is screened by the electrons alone, through ε_∞. The difference of the two inverse constants is the field that the ions contribute. In silicon $\varepsilon_0 = \varepsilon_\infty$ and $C_{\mathrm{F}} = 0$. The vertex is the Coulomb propagator $1/Q^2$ acting on a dipole source proportional to Q, the bound charges of the polarization wave, and its square falls as $1/Q^2$. A light mediator weights a rate toward small momentum transfers more steeply, as $1/q^4$, the squared propagator acting on a point charge (Sect. 1.4).

The strength of the Fröhlich coupling is measured by a dimensionless number. Take an electron at rest at the bottom of an isotropic parabolic band of mass m^*, at $T = 0$. At second order in g^{L} it emits a phonon $-\boldsymbol{Q}$ and moves to the state $\boldsymbol{Q}$, at an energy $\omega_{\mathrm{L}} + Q^2/2m^*$ above its own, and comes back. The energy is lowered by

$$\Delta E = -\frac{1}{N}\sum_{\boldsymbol{Q}} \frac{|g^{\mathrm{L}}|^2}{\omega_{\mathrm{L}} + Q^2/2m^*} = -\Omega\int \frac{\mathrm{d}^3Q}{(2\pi)^3}\,\frac{C_{\mathrm{F}}^2/Q^2}{\omega_{\mathrm{L}} + Q^2/2m^*} = -\frac{\Omega C_{\mathrm{F}}^2}{2\pi^2}\,\frac{\pi}{2}\Big(\frac{2m^*}{\omega_{\mathrm{L}}}\Big)^{1/2},$$

where we used $\rho_{nn} = 1$, replaced the sum over the zone by an integral (Appendix A.3) and extended it to infinity, and integrated $\int_0^\infty \mathrm{d}Q\,(\omega_{\mathrm{L}} + Q^2/2m^*)^{-1} = (\pi/2)(2m^*/\omega_{\mathrm{L}})^{1/2}$. The integral is dominated by $Q \sim (2m^*\omega_{\mathrm{L}})^{1/2}$, the wavevector at which the kinetic energy of the electron equals the phonon energy. Writing $\Delta E = -\alpha_{\mathrm{F}}\,\omega_{\mathrm{L}}$ and inserting C_{F}^2 from Eq. (6.15), with $e^2 = 4\pi\alpha$,

$$\alpha_{\mathrm{F}} = \frac{\Omega\,C_{\mathrm{F}}^2}{4\pi\,\omega_{\mathrm{L}}^2}\,(2m^*\omega_{\mathrm{L}})^{1/2} = \alpha\,\Big(\frac{1}{\varepsilon_\infty}-\frac{1}{\varepsilon_0}\Big)\Big(\frac{m^*}{2\omega_{\mathrm{L}}}\Big)^{1/2}, \tag{6.16}$$

the *Fröhlich coupling constant* [15]. It is the fine-structure constant, reduced by the ionic part of the screening and enhanced by the square root of the ratio of the band mass to twice the phonon energy. For $\alpha_{\mathrm{F}} \ll 1$ the electron carries a thin cloud of virtual phonons and the second order is accurate; the electron dressed by this cloud is the Fröhlich polaron, and Sect. 6.3 treats the energy shift as part of the electron self-energy. The cloud extends over the length $(2m^*\omega_{\mathrm{L}})^{-1/2}$, many cells for a light electron, which is why the continuum description of the crystal, ε_∞ and ε_0 in place of the atoms, is adequate.

Worked example 6.2: The Fröhlich constant of GaAs

We take the measured constants of Worked example 5.6: $\omega_{\mathrm{L}} = 8.79\,\mathrm{THz} = 36.35\,\mathrm{meV}$ [16], $\varepsilon_\infty = 10.88$ and $\varepsilon_0 = 12.9$ [17], the cell volume $\Omega = a^3/4 = 44.80\,\text{Å}^3 = 5.831\times10^{-9}\,\mathrm{eV}^{-3}$ with the static-lattice $a = 5.638\,\text{Å}$ of Table E.1, and the conduction mass $m^* = 0.067\,m_e =$

34.24 keV (Sect. 3.3) [18]. The ionic part of the screening is

$$\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0} = 0.09191 - 0.07752 = 0.01439,$$

a small difference of two larger numbers. With $e^2 = 0.0917$, Eq. (6.15) gives $C_\mathrm{F}^2 = 0.0917 \times 0.036\,35\,\mathrm{eV} \times 0.01439/(2 \times 5.831 \times 10^{-9}\,\mathrm{eV}^{-3}) = 4.11 \times 10^3\,\mathrm{eV}^4$, or $C_\mathrm{F} = 64.1\,\mathrm{eV}^2$. A phonon of $Q = 0.01\,\text{Å}^{-1} = 19.7\,\mathrm{eV}$ couples to an electron at the band bottom with $|g^\mathrm{L}| = C_\mathrm{F}/Q = 3.25\,\mathrm{eV}$, the number of the opening; at $0.1\,\text{Å}^{-1}$ the vertex has fallen to 0.33 eV.

The constant follows from Eq. (6.16). The square root is $(34{,}240/(2 \times 0.03635))^{1/2} = 686$, and

$$\alpha_\mathrm{F} = \frac{0.01439 \times 686}{137.036} = 0.072.$$

The coupling is weak. An electron at the bottom of the band is lowered by $\alpha_\mathrm{F}\omega_\mathrm{L} = 2.6\,\mathrm{meV}$, and its phonon cloud extends over $(2m^*\omega_\mathrm{L})^{-1/2} = 39.6\,\text{Å}$, seven lattice constants. The wavevectors that matter are near $(2m^*\omega_\mathrm{L})^{1/2} = 0.025\,\text{Å}^{-1}$, about a fortieth of the distance to the zone boundary, where $|g^\mathrm{L}| = 1.29\,\mathrm{eV}$.

How well is α_F known? The measured constants obey the Lyddane–Sachs–Teller relation only to 1.4 % (Worked example 5.6). If we eliminate ε_0 with that relation, $1/\varepsilon_\infty - 1/\varepsilon_0 = (1 - \omega_\mathrm{T}^2/\omega_\mathrm{L}^2)/\varepsilon_\infty$, the measured $\omega_\mathrm{T} = 33.62\,\mathrm{meV}$ yields 0.01328, 8 % less, and $\alpha_\mathrm{F} = 0.067$. The small difference amplifies the residual of the inputs fivefold. For GaAs the Fröhlich constant is therefore 0.07 to within several percent. The same amplification is why Eq. (6.15) needs dielectric constants from one consistent source: the PBE value $\varepsilon_\infty = 13.7$ of Sect. 5.5, combined with the measured ε_0, would even reverse the sign of the difference.

6.2.5 Wannier interpolation of the vertex

A phonon-assisted rate in silicon with an energy resolution of 14 meV needs the vertex at about 4×10^9 pairs $(\boldsymbol{k}, \boldsymbol{Q})$, for every pair of bands and every branch (Appendix E.5). Density-functional perturbation theory costs about one ground-state calculation per phonon wavevector, and a few hundred wavevectors, a 6^3 or 8^3 grid, are what a calculation can afford. The step from such a *coarse grid* to the *fine grids* of a rate is an interpolation, and it works for the same reason as the interpolation of bands in Sect. 3.6: in a suitable real-space representation the vertex is short-ranged.

The representation uses Wannier functions for the electrons and single displaced atoms for the phonons. We insert the Bloch sums of Wannier functions of one gauge, $\psi_{n\boldsymbol{k}} = N^{-1/2} \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} \sum_{n'} w_{n'}(\boldsymbol{r} - \boldsymbol{R})\, U^\dagger_{n'n}(\boldsymbol{k})$ (Eq. (A.20) inverted), into the matrix element of $\partial_{\kappa\alpha,\boldsymbol{Q}} V_\mathrm{KS} = \sum_{\boldsymbol{R}_p} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}_p}\, \partial V_\mathrm{KS}/\partial u^\alpha_{\boldsymbol{R}_p\kappa}$. We then translate every term by the cell of the bra and sum over that cell, which cancels the factor $1/N$. What is left is

$$\langle \psi_{m,\boldsymbol{k}+\boldsymbol{Q}} | \partial_{\kappa\alpha,\boldsymbol{Q}} V_\mathrm{KS} | \psi_{n\boldsymbol{k}} \rangle = \sum_{\boldsymbol{R}_e \boldsymbol{R}_p} \mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{R}_e + \boldsymbol{Q}\cdot\boldsymbol{R}_p)} \left[U(\boldsymbol{k}+\boldsymbol{Q})\, g^W_{\kappa\alpha}(\boldsymbol{R}_e, \boldsymbol{R}_p)\, U^\dagger(\boldsymbol{k}) \right]_{mn},$$

$$g^W_{m'n'\kappa\alpha}(\boldsymbol{R}_e, \boldsymbol{R}_p) = \Big\langle w_{m'}(\boldsymbol{r}) \Big| \frac{\partial V_\mathrm{KS}}{\partial u^\alpha_{\boldsymbol{R}_p\kappa}} \Big| w_{n'}(\boldsymbol{r} - \boldsymbol{R}_e) \Big\rangle. \tag{6.17}$$

The real-space vertex g^W is the matrix element, between a Wannier function at the origin and one in cell $\boldsymbol{R}_e$, of the screened potential change when a single atom in cell $\boldsymbol{R}_p$ moves. Multiplying by the polarization factors of Eq. (A.26) gives Eq. (E.36), and on the coarse grids

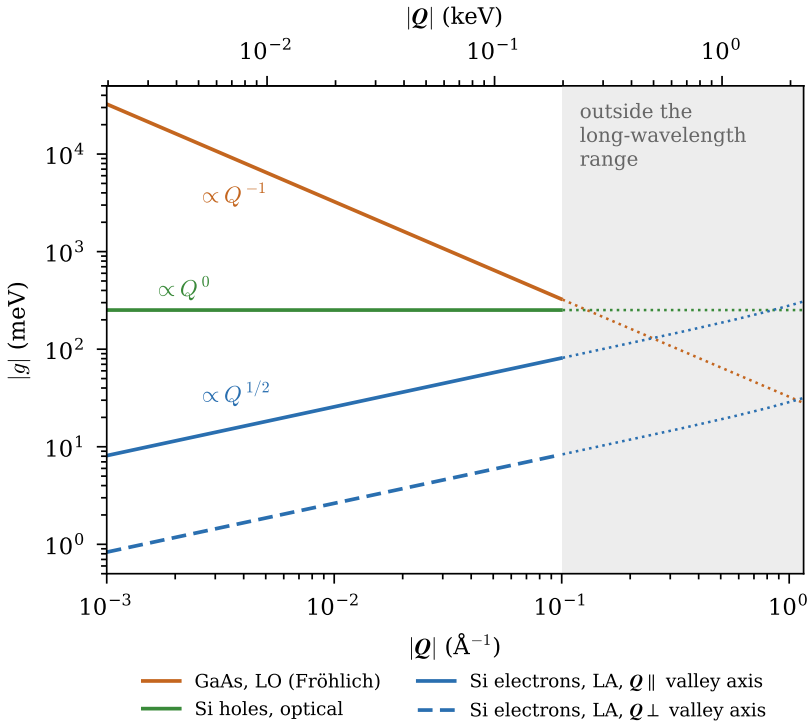


Figure 6.3: Magnitude of the electron–phonon vertex $|g|$, in the normalization in which H_{ep} carries $N^{-1/2}$, against the phonon wavevector $|\boldsymbol{Q}|$ (bottom axis in Å^{-1}, top axis in keV), for the three long-wavelength forms of this section. Orange: Fröhlich vertex Eq. (6.15) of GaAs for an electron at the conduction-band minimum, with $\rho_{nn} = 1$ and the measured $\omega_{\text{L}} = 36.35\,\text{meV}$, $\varepsilon_\infty = 10.88$, $\varepsilon_0 = 12.9$. Green: optical deformation potential Eq. (6.14) for the holes at the valence maximum of silicon, with the DFPT value $10.45\,\text{eV}\,\text{Å}^{-1}$ combined over the degenerate bands [12] and $\omega = 64.1\,\text{meV}$. Blue: acoustic deformation potential Eq. (6.13) for an electron in a conduction valley of silicon and the longitudinal acoustic phonon, with $\boldsymbol{Q}$ along the valley axis (solid, $\Xi_d + \Xi_u = 9.85\,\text{eV}$) and perpendicular to it (dashed, $\Xi_d = 1.01\,\text{eV}$). These use the DFPT constants of Ref. [12] and the DFPT longitudinal acoustic branch along Γ–X of Fig. 5.3, whose small-Q slope is $8.34\,\text{km}\,\text{s}^{-1}$ (measured: $8.43\,\text{km}\,\text{s}^{-1}$). The curves are these limiting forms, not vertices computed point by point. They are drawn solid up to $0.1\,\text{Å}^{-1}$, the end of the range over which the acoustic vertices of silicon were computed and found linear in Q [12, Fig. 5]. Beyond it they are dotted, in the shaded region up to the zone boundary of silicon at $2\pi/a = 1.157\,\text{Å}^{-1}$. On logarithmic axes the slopes -1, 0 and $\frac{1}{2}$ distinguish the polar, optical and acoustic couplings.

the relation is inverted by a discrete Fourier transform, as for $H^W(\boldsymbol{R})$ in Eq. (3.86). This is the scheme of Giustino, Cohen and Louie [19], reviewed in Ref. [3, Eqs. (177)–(179)] and implemented in the Electron–Phonon Wannier (EPW) code [9].

The real-space vertex decays in both of its lattice vectors. In $\boldsymbol{R}_e$ it decays as the overlap of two Wannier functions does, exponentially for the smooth gauges of Sect. 3.6, because both must reach the region where the potential changes. In $\boldsymbol{R}_p$ it decays as the screened potential of one displaced atom, seen from the Wannier function at the origin. In a nonpolar insulator this potential has neither a monopole, since the displaced atom is screened to neutrality, nor a dipole, since $Z^* = 0$. It still has a quadrupole, and the vertex falls off as R_p^{-3}, a tail of no finite range whose Fourier transform carries a term of order Q^0 that depends on the direction of $\boldsymbol{Q}$. In silicon this quadrupolar part is small, and it can be subtracted like the dipole of a polar crystal below. By the argument that led to Eq. (3.89), a coarse grid whose supercell contains the short-ranged remainder of g^W reproduces the vertex everywhere, and the tails beyond it set the error. For diamond and silicon a 6^3 or 8^3 coarse grid of phonon wavevectors is enough (Appendix E.5).

A polar crystal breaks the argument. The displaced atom carries the dipole $eZ^*_\kappa \cdot \boldsymbol{u}$ screened by ε_∞, its potential falls off as R_p^{-2}, and the Fourier transform of that tail is the $1/Q$ singularity of the Fröhlich vertex. No finite sum of Fourier components reproduces a singularity. The remedy is the one that Sect. 5.5 used for the nonanalytic part of the dynamical matrix. The long-range vertex is known in closed form from Z^*_κ and ε_∞ alone, Eq. (D.77) or its version with the sum over reciprocal-lattice vectors, Eq. (E.37). One subtracts it from the coarse-grid vertices, interpolates the short-range remainder $g^{\mathrm{S}} = g - g^{\mathrm{L}}$, which is as smooth as the vertex of a nonpolar crystal, and adds g^{L} back analytically at every fine-grid point [14]. The split $g = g^{\mathrm{S}} + g^{\mathrm{L}}$ is not unique, since any smooth function can be moved from one part to the other, but every choice puts the whole singularity into g^{L}, and that is all the interpolation needs. The density vertex in g^{L} is needed at the fine-grid points too, and at small $\boldsymbol{Q}$ it is $[U(\boldsymbol{k}+\boldsymbol{Q})U^\dagger(\boldsymbol{k})]_{mn}$ in the Wannier gauge, so that g^{S} and g^{L} carry the same phases (Appendix A.5). The quadrupolar terms, present in silicon as in GaAs and, together with the Born charges, the origin of the piezoelectric vertex of GaAs, are long-ranged in the same way and are subtracted with the dipole where the code allows it (Appendix E.5).

Fig. 6.4 summarizes the chain. Its pitfalls are practical: a Wannier manifold that misses the states a rate needs, a coarse grid that misplaces the conduction minimum of silicon, a gauge that differs between g and ρ. Appendix E.5 discusses each, with the grid sizes, the counts of matrix elements and the checks against direct DFPT at points off the coarse grid. The result is a vertex at any $(\boldsymbol{k}, \boldsymbol{Q})$ for about the cost of a few hundred phonon calculations.

With the vertex in hand, the rest of the chapter takes sums of $|g|^2$. Summed over all phonons, with the energies of the intermediate states, $|g|^2$ shifts the bands and gives the electrons a finite lifetime (Sect. 6.3).

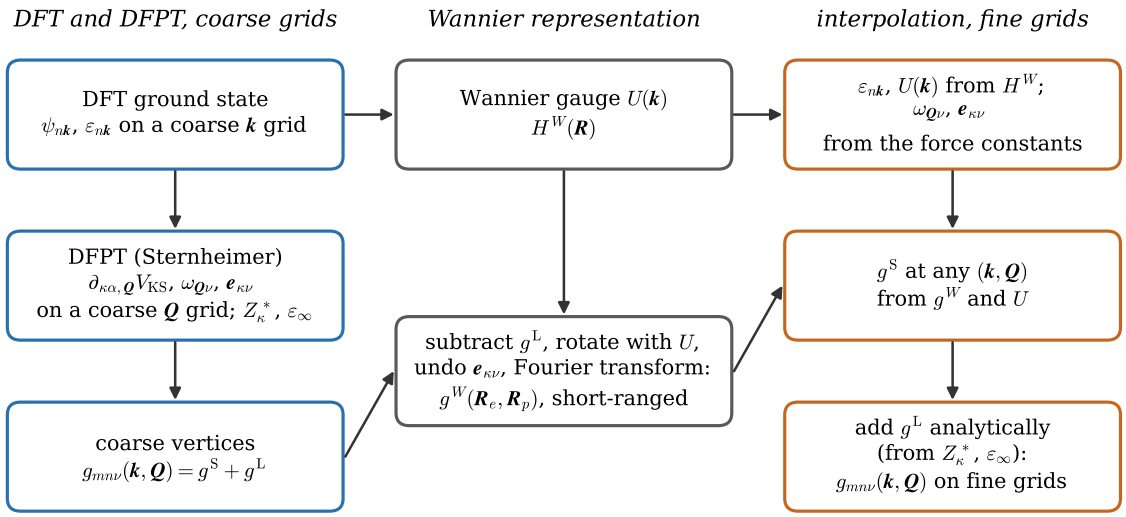


Figure 6.4: From first principles to interpolated electron–phonon vertices. Left: a density-functional calculation on a coarse grid of crystal momenta gives the Bloch states and energies. Density-functional perturbation theory on a coarse grid of phonon wavevectors gives the screened potential changes $\partial_{\kappa\alpha,\boldsymbol{Q}}V_{\mathrm{KS}}$, the phonons, and in a polar crystal Z^*_κ and ε_∞. Their matrix elements are the coarse vertices, Eq. (6.11). Middle: the Bloch states define the Wannier gauge $U(\boldsymbol{k})$; the long-range part g^{L} is subtracted, and the rotated remainder is transformed to the real-space vertex $g^W(\boldsymbol{R}_e,\boldsymbol{R}_p)$ of Eq. (6.17), short-ranged in both lattice vectors. Right: the real-space Hamiltonian and force constants give bands, gauge matrices and phonons at any point; g^W is transformed back, Eq. (E.36), and g^{L}, Eq. (E.37), is added analytically. Every quantity in one amplitude uses the same $U(\boldsymbol{k})$.

6.3 Electron self-energy from phonons

At 10 mK a silicon crystal holds no thermal phonons, yet its band gap is about 60 meV smaller than the gap of the same crystal with its nuclei frozen at their equilibrium positions. The zero-point motion of Sect. 5.3, with a root-mean-square amplitude of 0.05 Å along each axis, moves the bands by an amount that matters near threshold: the lightest dark-matter mass a detector can see scales with the gap (Eq. (1.25)). The same coupling decides how long a conduction electron with a few meV of excess energy keeps that energy before it hands it to the lattice. Both effects are the electron self-energy of Sect. 4.7, with phonons as the boson. We will find that it has two terms of the same order, that translation invariance ties them together, and that an Einstein model with one silicon phonon energy already reproduces the measured temperature dependence of the gap to about 10 %. Along the way we will see why the lowest diagrams are enough, which is Migdal's adiabatic theorem.

6.3.1 Fan–Migdal and Allen–Heine terms

A displaced nucleus changes the potential that the electrons feel, at first and at second order in the displacement. Expanding the self-consistent potential V_{KS} about equilibrium,

$$V_{\mathrm{KS}} = V_{\mathrm{KS}}^{(0)} + \sum_a \frac{\partial V_{\mathrm{KS}}}{\partial u_a}\, u_a + \frac{1}{2}\sum_{ab} \frac{\partial^2 V_{\mathrm{KS}}}{\partial u_a\, \partial u_b}\, u_a u_b,$$

with $a = (\boldsymbol{R}\kappa\alpha)$ running over cells, species and Cartesian directions. The linear term is the electron–phonon Hamiltonian H_{ep} of Eq. (6.4), with the screened vertex $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})$ of Sect. 6.2. The quadratic term, with each displacement expanded in normal modes by Eq. (A.21), is a vertex with two phonon operators,

$$\begin{aligned} H_{\mathrm{ep}}^{(2)} &= \frac{1}{N}\sum_{\boldsymbol{k}\boldsymbol{Q}\boldsymbol{Q}'}\sum_{mn\nu\nu'} g^{\mathrm{AH}}_{mn\nu\nu'}(\boldsymbol{k},\boldsymbol{Q},\boldsymbol{Q}')\, c^\dagger_{m,\boldsymbol{k}+\boldsymbol{Q}+\boldsymbol{Q}'} c_{n\boldsymbol{k}}\, B_{\boldsymbol{Q}\nu} B_{\boldsymbol{Q}'\nu'}, \\ g^{\mathrm{AH}}_{mn\nu\nu'}(\boldsymbol{k},\boldsymbol{Q},\boldsymbol{Q}') &= \frac{1}{2}\sum_{\kappa\alpha,\kappa'\beta} \frac{e^\alpha_{\kappa\nu}(\boldsymbol{Q})\, e^\beta_{\kappa'\nu'}(\boldsymbol{Q}')}{\sqrt{2M_\kappa\omega_{\boldsymbol{Q}\nu}\, 2M_{\kappa'}\omega_{\boldsymbol{Q}'\nu'}}} \langle \psi_{m,\boldsymbol{k}+\boldsymbol{Q}+\boldsymbol{Q}'} | \partial^2_{\kappa\alpha\boldsymbol{Q},\kappa'\beta\boldsymbol{Q}'} V_{\mathrm{KS}} | \psi_{n\boldsymbol{k}} \rangle, \end{aligned} \tag{6.18}$$

where $\partial^2_{\kappa\alpha\boldsymbol{Q},\kappa'\beta\boldsymbol{Q}'} V_{\mathrm{KS}}$ is the second derivative with respect to two displacement waves, the sum over $\boldsymbol{R}$ and $\boldsymbol{R}'$ of $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}+\mathrm{i}\boldsymbol{Q}'\cdot\boldsymbol{R}'}$ times the second derivative with respect to $u^\alpha_{\boldsymbol{R}\kappa}$ and $u^\beta_{\boldsymbol{R}'\kappa'}$ (Appendix D.16). This is the Allen–Heine vertex g^{AH} (Appendix A.6). In the energy of an electron, H_{ep} enters at second order and $H_{\mathrm{ep}}^{(2)}$ at first order, and both carry two factors of the zero-point length $\ell_{\boldsymbol{Q}\nu}$: they are of the same order, $\langle u^2\rangle$, and a consistent theory keeps both. We will see below that dropping either violates translation invariance.

The first term is the boson-exchange diagram of Sect. 4.8 (Fig. 4.14c): the electron emits or absorbs a phonon, propagates in another band and wavevector, and reabsorbs or re-emits it. Compared with Eq. (4.98), the vertex carries $1/\sqrt{N}$ instead of $1/\sqrt{\mathcal{V}}$ and band indices, and the intermediate electron can be in any band m. The second term needs no propagator at all. Its two phonon operators are contracted with each other, a phonon line that leaves and returns to the same point of the electron line, and the pair is replaced by its thermal average $\langle B_{\boldsymbol{Q}\nu} B_{-\boldsymbol{Q}\nu}\rangle = 2n_{\boldsymbol{Q}\nu} + 1$. Appendix D.16 carries out the Matsubara sum and the continuation for both diagrams.

Fan–Migdal and Allen–Heine self-energies

For a nondegenerate band n at wavevector $\boldsymbol{k}$, with energies ω and $\xi_{m\boldsymbol{p}} = \varepsilon_{m\boldsymbol{p}} - \mu_{\mathrm{ch}}$ measured from the chemical potential, $f_{m\boldsymbol{p}} = n_{\mathrm{F}}(\xi_{m\boldsymbol{p}})$ and $n_{\boldsymbol{Q}\nu} = n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$,

$$\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\omega) = \frac{1}{N}\sum_{\boldsymbol{Q}\nu m} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2 \Big[\frac{n_{\boldsymbol{Q}\nu} + 1 - f_{m,\boldsymbol{k}+\boldsymbol{Q}}}{\omega - \xi_{m,\boldsymbol{k}+\boldsymbol{Q}} - \omega_{\boldsymbol{Q}\nu} + \mathrm{i}0^+} + \frac{n_{\boldsymbol{Q}\nu} + f_{m,\boldsymbol{k}+\boldsymbol{Q}}}{\omega - \xi_{m,\boldsymbol{k}+\boldsymbol{Q}} + \omega_{\boldsymbol{Q}\nu} + \mathrm{i}0^+} \Big], \tag{6.19}$$

$$\Sigma^{\mathrm{AH}}_{n\boldsymbol{k}} = \frac{1}{N}\sum_{\boldsymbol{Q}\nu} g^{\mathrm{AH}}_{nn\nu\nu}(\boldsymbol{k},\boldsymbol{Q},-\boldsymbol{Q})\,(2n_{\boldsymbol{Q}\nu}+1). \tag{6.20}$$

The Fan–Migdal term depends on frequency and has an imaginary part; the Allen–Heine term is static and real.

Eq. (6.19) is the second-order self-energy of Fan [20], called the Migdal self-energy in the theory of metals [2], and Eq. (6.20) is the term added by Allen and Heine [21]; together they are Ref. [3, Eqs. (157), (158), (160) and (161)]. The thermal factors read as in Sect. 4.8. In the emission term, $n+1$ is stimulated plus spontaneous emission and $-f$ blocks an occupied intermediate state. In the absorption term, n counts the phonons present, and f adds the process in which an electron already in $(m, \boldsymbol{k}+\boldsymbol{Q})$ emits a phonon and drops into the state. The sum over m now runs over occupied bands as well. For a degenerate level, such as the threefold valence maximum of silicon (Sect. 3.3), the self-energy is a matrix in the partners, and the terms between partners, whose denominators vanish, are left out as the term $m = n$ at $\boldsymbol{Q} = 0$ is [22, Sect. III A]. Summed over the whole zone, the matrix commutes with the symmetry operations that leave $\boldsymbol{k}$ unchanged. On a level whose partners form one irreducible representation it is therefore a multiple of the identity, and every partner shifts by the average of the diagonal elements over the level. This is how the shifts of degenerate band edges below are defined. At $T = 0$ the zero-point motion enters Σ^{FM} through the 1 of $n+1$ and through f, and Σ^{AH} through the 1 of $2n+1$. One more diagram of the same order, a phonon line at zero wavevector ending on a closed electron loop, would describe a static shift of the equilibrium positions; it vanishes because the expansion is made about the minimum of the Born–Oppenheimer surface.

The pole of the Dyson equation Eq. (4.81) then sits at the renormalized band energy

$$\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}} + \mathrm{Re}\,\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}}) + \Sigma^{\mathrm{AH}}_{n\boldsymbol{k}}, \tag{6.21}$$

with ω now measured from the same origin as ε, the quasiparticle condition of Eq. (4.83). Its value at $T = 0$, less $\varepsilon_{n\boldsymbol{k}}$, is the *zero-point renormalization* (ZPR) of the state, and the ZPR of a gap is that of the conduction state minus that of the valence state. The theory built on the two terms, usually in the adiabatic form below and with the rigid-ion approximation of Allen and Cardona [23], is called Allen–Heine–Cardona (AHC) theory. In practice Σ^{FM} is often evaluated at $\omega = \varepsilon_{n\boldsymbol{k}}$ rather than at the solution of Eq. (6.21); for the Einstein boson of Worked example 4.13 the two choices yield -9.14 meV and -8.73 meV, a difference set by $1 - z_{\mathrm{qp}}$.

6.3.2 The adiabatic limit

Where the phonon energies are small next to the electronic energy differences that dominate the sum over m, they can be dropped from the denominators of Eq. (6.19). Setting also $\omega = \varepsilon_{n\boldsymbol{k}}$, the two terms acquire the same denominator, and their numerators add to $(n+1-f)+(n+f) =$

$2n+1$: the occupations of the intermediate states drop out. The result is the formula of Allen and Heine [21],

$$\varepsilon_{n\boldsymbol{k}}^{\mathrm{ren}} - \varepsilon_{n\boldsymbol{k}} = \frac{1}{N}\sum_{\boldsymbol{Q}\nu}(2n_{\boldsymbol{Q}\nu}+1)\left[\sum_m \frac{|g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}} + g^{\mathrm{AH}}_{nn\nu\nu}(\boldsymbol{k},\boldsymbol{Q},-\boldsymbol{Q})\right], \tag{6.22}$$

Ref. [3, Eq. (167)]. Every mode now enters with the weight $2n_{\boldsymbol{Q}\nu}+1$ of its mean-square displacement (Eq. (5.35)). That is no accident. Allen and Heine derived Eq. (6.22) without diagrams. With the nuclei frozen at displacements u_a, a band energy is a function $\varepsilon_{n\boldsymbol{k}}(\{u\})$, and second-order perturbation theory gives its curvature: the Fan sum is the second-order term of H_{ep}, the Allen–Heine vertex the first-order term of the quadratic potential (Eq. (D.91)). The nuclei are not frozen, however, but spread over a Gaussian distribution of width $\langle u_a u_b\rangle$. Averaged over it, the linear term vanishes and the energy shifts by half the curvature times the width, $\frac{1}{2}\sum_{ab}\langle u_a u_b\rangle\, \partial^2\varepsilon_{n\boldsymbol{k}}/\partial u_a \partial u_b$. In the adiabatic limit the AHC shift is the average of the band energy over the zero-point and thermal smearing of the lattice.

The adiabatic form fails where the dropped phonon energies are not small. That happens for $m = n$ at small Q, where $\varepsilon_{n\boldsymbol{k}} - \varepsilon_{n,\boldsymbol{k}+\boldsymbol{Q}}$ vanishes with Q, and wherever another band is degenerate with the state. Both occur at band edges, the states that set a detector's threshold. The Fröhlich vertex of a polar crystal shows how badly it can fail. Take an electron at the bottom of a parabolic band of mass m^* at $T = 0$, coupled to the longitudinal optical phonon by $|g^{\mathrm{L}}|^2 = C_{\mathrm{F}}^2/Q^2$ (Eq. (6.15)). In the adiabatic form its intraband Fan term is

$$\frac{1}{N}\sum_{\boldsymbol{Q}} \frac{|g^{\mathrm{L}}|^2}{-Q^2/2m^*} = -\Omega\int \frac{\mathrm{d}^3Q}{(2\pi)^3}\,\frac{2m^*C_{\mathrm{F}}^2}{Q^4} = -\frac{\Omega\, m^* C_{\mathrm{F}}^2}{\pi^2}\int_0 \frac{\mathrm{d}Q}{Q^2},$$

which diverges at small Q. The divergence is an artifact. With the phonon energy kept, only the emission term of Eq. (6.19) survives for an empty band at $T = 0$, its denominator at $\omega = \varepsilon_{n\boldsymbol{k}}$ is $-Q^2/2m^* - \omega_{\mathrm{L}}$, and $\mathrm{Re}\,\Sigma^{\mathrm{FM}}$ is the finite second-order shift of Sect. 6.2, $-\alpha_{\mathrm{F}}\omega_{\mathrm{L}} = -2.6\,\mathrm{meV}$ for the conduction band of GaAs (Worked example 6.2). The Fröhlich polaron shift is the Fan–Migdal self-energy at the band bottom. Every infrared-active crystal has this $1/Q^2$ coupling. For all of them the adiabatic ZPR is infinite, and the nonadiabatic form, with the phonon energies in the denominators, is needed to get a number [22]. In silicon the vertex has no $1/Q$ term (Sect. 6.2), both forms converge, and they differ by 8 meV: $-64.3\,\mathrm{meV}$ adiabatic and $-56.2\,\mathrm{meV}$ nonadiabatic for the indirect gap in the local-density approximation [22, Table VI].

6.3.3 Translation invariance and the rigid-ion approximation

Moving the whole crystal by a constant vector shifts no band energy. For long-wavelength acoustic phonons this ties the Fan and Allen–Heine terms together. As $Q \to 0$ an acoustic mode becomes a rigid translation, with $\boldsymbol{e}_{\kappa\nu} \to (M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$ and every nucleus displaced by $\ell_{\boldsymbol{Q}\nu}\hat{\boldsymbol{e}}_\nu/\sqrt{N}$ per unit of $B_{\boldsymbol{Q}\nu}$ (Appendix A.6). Its vertex then follows from Eq. (6.11) and the acoustic sum rule Eq. (6.12),

$$g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q}) \to \ell_{\boldsymbol{Q}\nu}\sum_\kappa \hat{\boldsymbol{e}}_\nu \cdot \langle\psi_{m\boldsymbol{k}}|\partial_{\kappa,\boldsymbol{0}}V_{\mathrm{KS}}|\psi_{n\boldsymbol{k}}\rangle = -\mathrm{i}\,\ell_{\boldsymbol{Q}\nu}\,(\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}})\,\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{p}_{mn}(\boldsymbol{k}).$$

The intraband element vanishes, as Sect. 6.2 showed; the interband elements do not. In the Fan sum of Eq. (6.22) the energy difference is squared in the numerator and appears once in the denominator. What is left is

$$\sum_{m\neq n}\frac{|g_{mn\nu}|^2}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}}} \to \ell_{\boldsymbol{Q}\nu}^2\sum_m(\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}})\,|\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{p}_{mn}|^2.$$

For the Allen–Heine vertex of the same mode, the same polarization vectors turn the prefactor of Eq. (6.18) into $\ell^2_{\boldsymbol{Q}\nu}\hat{e}_{\nu\alpha}\hat{e}_{\nu\beta}$, and the second derivative with respect to two displacement waves of zero wavevector, summed over κ and κ', becomes the second derivative with respect to a uniform translation $\boldsymbol{t}$ of the whole crystal. The self-consistent potential moves with the crystal, $V_{\mathrm{KS}} \to \hat{T}_{\boldsymbol{t}} V_{\mathrm{KS}} \hat{T}_{\boldsymbol{t}}^\dagger$ with $\hat{T}_{\boldsymbol{t}} = \mathrm{e}^{-\mathrm{i}\boldsymbol{t}\cdot\hat{\boldsymbol{p}}}$ (Sect. 6.2). Differentiating twice along $\hat{\boldsymbol{e}}_\nu$ gives $-[\hat{p}_\nu,[\hat{p}_\nu,V_{\mathrm{KS}}]]$ with $\hat{p}_\nu = \hat{\boldsymbol{e}}_\nu\cdot\hat{\boldsymbol{p}}$, and since the kinetic energy commutes with $\hat{p}_\nu$, V_{KS} may be replaced by H_{KS}. Expanding the double commutator in the eigenstates of H_{KS} at $\boldsymbol{k}$, the states that $\hat{p}_\nu$ connects to $\psi_{n\boldsymbol{k}}$,

$$g^{\mathrm{AH}}_{nn\nu\nu}(\boldsymbol{k},\boldsymbol{Q},-\boldsymbol{Q}) \to -\frac{\ell^2_{\boldsymbol{Q}\nu}}{2}\langle\psi_{n\boldsymbol{k}}|[\hat{p}_\nu,[\hat{p}_\nu,H_{\mathrm{KS}}]]|\psi_{n\boldsymbol{k}}\rangle = -\ell^2_{\boldsymbol{Q}\nu}\sum_m(\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m\boldsymbol{k}})\,|\hat{\boldsymbol{e}}_\nu\cdot\boldsymbol{p}_{mn}|^2.$$

The two contributions cancel term by term:

$$\sum_{m\neq n}\frac{|g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2}{\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}} + g^{\mathrm{AH}}_{nn\nu\nu}(\boldsymbol{k},\boldsymbol{Q},-\boldsymbol{Q}) \to 0 \qquad (\nu \text{ acoustic},\ Q\to 0). \tag{6.23}$$

The picture is the one of Sect. 6.2. The interband vertex translates the electronic state along with the crystal, and at second order it shifts the energy by $\ell^2_{\boldsymbol{Q}\nu}\sum_m(\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m\boldsymbol{k}})|\hat{\boldsymbol{e}}_\nu\cdot\boldsymbol{p}_{mn}|^2$, a sum of either sign: bands above n push it down, bands below push it up. The Allen–Heine term measures the curvature of the potential under the same translation, and it shifts the energy back by exactly as much. Neither term means anything alone, and a calculation must treat both on the same grid of wavevectors. In the language of scalar quantum electrodynamics (QED) the two are the rainbow and the seagull diagrams, and their cancellation for a rigid translation is the analogue of the one that makes the scattering of low-frequency light by a bound electron vanish. The same double commutator underlies the f-sum rule (Eq. (4.28)). The three translational modes at exactly $\boldsymbol{Q}=0$, with zero frequency, are left out of both sums; Eq. (6.23) states that the integrand is continuous there. For small Q the first correction to the cancellation is linear in $\boldsymbol{Q}$ and drops out of the sum over directions. The interband remainder is then smaller by a factor $\sim Q^2$, and long-wavelength acoustic phonons, which carry a large share of the thermal $\langle u^2\rangle$ of silicon (Worked example 5.4), barely shift the bands. What survives at small Q is the intraband term, $m=n$, whose denominator vanishes with Q, the term that needs the phonon energy kept.

The Allen–Heine vertex needs second derivatives of the self-consistent potential, a second-order perturbation calculation that few codes perform. Translation invariance offers a way around it. A rigid translation leaves every band energy unchanged at every configuration of the nuclei, so the sum over all nuclei of $\partial\varepsilon_{n\boldsymbol{k}}/\partial u_{\boldsymbol{R}\kappa\alpha}$ vanishes everywhere, and so does its derivative. At equilibrium this is an exact sum rule, Eq. (D.93): the diagonal element of the second derivative, summed over the sites of the whole crystal, equals a product of first derivatives at $\boldsymbol{Q}=0$. The sum rule fixes a sum over sites, while g^{AH} needs the on-site element. The *rigid-ion approximation* supplies the missing step. It writes the potential as a sum of atom-centered terms that move rigidly with their nuclei, $V_{\mathrm{KS}} = \sum_{\boldsymbol{R}\kappa} v_\kappa(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\kappa-\boldsymbol{u}_{\boldsymbol{R}\kappa})$. Second derivatives with respect to two different atoms then vanish, the sum rule determines the on-site element (Eq. (D.94)), and the Allen–Heine term becomes

$$\Sigma^{\mathrm{AH}}_{n\boldsymbol{k}} = -\frac{1}{2}\sum_{\kappa\kappa'}\sum_{\alpha\beta}(B^{\alpha\beta}_\kappa + B^{\alpha\beta}_{\kappa'})\ \mathrm{Re}\sum_m{}' \frac{\langle\psi_{n\boldsymbol{k}}|\partial_{\kappa\alpha,\boldsymbol{0}}V_{\mathrm{KS}}|\psi_{m\boldsymbol{k}}\rangle\,\langle\psi_{m\boldsymbol{k}}|\partial_{\kappa'\beta,\boldsymbol{0}}V_{\mathrm{KS}}|\psi_{n\boldsymbol{k}}\rangle}{\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m\boldsymbol{k}}}, \tag{6.24}$$

where $B_\kappa = \langle\boldsymbol{u}_\kappa\boldsymbol{u}_\kappa\rangle$ is the displacement tensor of Eq. (5.35) at temperature T and the prime excludes the states degenerate with $(n,\boldsymbol{k})$ (Appendix D.16). This is the form of Allen and Cardona [23], Ref. [3, Eqs. (195)–(198)]. It needs just the first-order potentials at $\boldsymbol{Q}=0$, which

every phonon calculation already supplies. Its structure is a Fan term at zero wavevector with the opposite sign, weighted by the mean-square displacement of each atom; the cancellation of Eq. (6.23) is built in. What it neglects is the change of screening when two atoms move together rather than one at a time. Poncé et al. find that correction large in molecules [22]; for crystals the rigid-ion form is the standard choice.

In practice the AHC shift is computed from DFPT vertices on a grid of phonon wavevectors. The sum over bands is split into an active window, summed explicitly, and the rest, obtained from a Sternheimer equation as in Sect. 5.4. The sum over $\boldsymbol{Q}$ converges slowly near the vanishing intraband denominators and needs grids of 10^5 to 10^6 points, or Wannier interpolation of the vertex (Sect. 6.2), with a small imaginary part in the denominators extrapolated to zero. Appendix E.8 describes the calculation, its convergence parameters and its pitfalls, among them the small residual Born charges of a nonpolar DFPT calculation, which must be removed before the sum or the integral diverges at a band extremum.

6.3.4 Smeared ionic potentials

The literature calls Σ^{AH} the Debye–Waller self-energy, and in the rigid-ion approximation the name has a reason. Average the rigid-ion potential over the Gaussian distribution of the displacements of each nucleus. A Fourier component $v_\kappa(\boldsymbol{G})\,\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\kappa)}$ acquires the factor $\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{u}_\kappa}\rangle = \mathrm{e}^{-\boldsymbol{G}\cdot B_\kappa\cdot\boldsymbol{G}/2}$, and to first order in B_κ

$$\big\langle v_\kappa(\boldsymbol{r}-\boldsymbol{u}_\kappa)\big\rangle - v_\kappa(\boldsymbol{r}) = \frac{1}{2}\sum_{\alpha\beta} B_\kappa^{\alpha\beta}\,\partial_\alpha\partial_\beta v_\kappa(\boldsymbol{r}) = -\sum_{\boldsymbol{G}} W_\kappa(\boldsymbol{G})\,v_\kappa(\boldsymbol{G})\,\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}, \tag{6.25}$$

for the atom at the origin, with $2W_\kappa(\boldsymbol{G}) = \boldsymbol{G}\cdot B_\kappa\cdot\boldsymbol{G}$, the exponent of Eq. (5.56) evaluated at a reciprocal-lattice vector. The diagonal matrix element of the middle expression, summed over all the sites of every species, is the rigid-ion Allen–Heine term (Appendix D.16). So Σ^{AH} is the first-order change of a band energy when each ionic potential is smeared out by the zero-point and thermal motion of its nucleus, the "fuzziness" of the nuclear charge described in Ref. [3, below Eq. (154)].

It is nonetheless a different object from the Debye–Waller factor $\mathrm{e}^{-2W_\kappa(\boldsymbol{q})}$ of Sect. 5.6, to which we reserve the name (Appendix A.6). The factor of Sect. 5.6 is a probability: the chance that a nucleus absorbs a momentum transfer $\boldsymbol{q}$ from outside and leaves the lattice in the state it found it in. It multiplies a rate, at the momentum transfer of the scattering. The smearing of Eq. (6.25) involves no momentum transfer and no rate. It is evaluated at the reciprocal-lattice vectors of the static potential, enters an energy at first order in B_κ, and is one of the two terms of the second-order shift. The Fan term is of the same order and partly cancels it, exactly so for acoustic phonons at small Q (Eq. (6.23)). Allen and Heine showed that a theory of the band shifts with the smeared potential alone is incomplete [21]. The two share the displacement tensor B_κ, and nothing else.

6.3.5 Zero-point renormalization

For the indirect gap of silicon, AHC theory in the local-density approximation gives a zero-point renormalization of $-56.2\,\mathrm{meV}$ in the nonadiabatic form [22, Table VI]. The valence maximum rises and the conduction minimum drops, by 35 meV and 22 meV in a calculation by a different method [24, Sect. VII B], and the gap closes from both sides. That calculation used *special displacements*, a non-perturbative route that avoids vertices and sums over wavevectors altogether. Since the adiabatic shift is the average of the band energy over the Gaussian distribution of the nuclei, it can be computed directly: displace the nuclei of a large supercell and compute the bands. Zacharias and Giustino showed that a single configuration suffices,

one in which every normal coordinate has its root-mean-square amplitude and the signs alternate between modes so that the cross terms cancel [25]. In an $8 \times 8 \times 8$ supercell of 1024 atoms it yields $-57\,\mathrm{meV}$ [24, Sect. VII B]. The method includes all orders in the displacements but is adiabatic by construction; Appendix E.8 describes it and compares the costs of the two routes (Worked example E.7).

The renormalization extracted from experiment is $-62\,\mathrm{meV}$ or $-64\,\mathrm{meV}$, depending on the route [26], as tabulated in Ref. [22, Table VI]. A ZPR cannot be measured directly, since no experiment freezes the nuclei. Both routes rest on how the shift depends on temperature and on the nuclear mass, which the simplest model makes explicit.

Suppose every phonon has the same energy ω_{Ein}, an Einstein crystal. The weight $2n_{\boldsymbol{Q}\nu}+1 = \coth(\omega_{\mathrm{Ein}}/2T)$ is then the same for every mode and comes out of the sum in Eq. (6.22). The shift of any band at temperature T is its zero-point value times one function of temperature,

$$\Delta\varepsilon_{n\boldsymbol{k}}(T) = \Delta\varepsilon_{n\boldsymbol{k}}(0)\,\coth\frac{\omega_{\mathrm{Ein}}}{2T}. \tag{6.26}$$

At $T \gg \omega_{\mathrm{Ein}}$, $\coth x \simeq 1/x + x/3$, and the shift becomes linear in T with the slope $2\Delta\varepsilon(0)/\omega_{\mathrm{Ein}}$. Since this line extrapolates to zero shift at $T = 0$, extrapolating a measured gap back from high temperature lands on the gap of the clamped lattice, and the low-temperature gap sits below that intercept by the ZPR. This is one of the two extractions. The other uses the nuclear mass. At fixed force constants $\omega_{\mathrm{Ein}} \propto M^{-1/2}$, the squared vertex and the Allen–Heine term are both proportional to $\ell^2 \propto 1/(M\omega_{\mathrm{Ein}}) \propto M^{-1/2}$, and so is the ZPR. Then $\mathrm{ZPR} = -2M\,\partial E_{\mathrm{g}}/\partial M$, and the gaps of isotopically enriched crystals measure it.

Worked example 6.3: The Einstein model of the silicon gap

The Einstein boson of Worked example 4.13 shifted the band bottom by $-9.14\,\mathrm{meV}$ at $T = 0$ and by $-11.09\,\mathrm{meV}$ at 300 K, a ratio of 1.213 against $2n_{\mathrm{B}} + 1 = 1.218$. The small difference came from the boson energy in the denominators. In the adiabatic limit Eq. (6.26) makes the ratio exact, and the model reduces to two numbers: the ZPR and ω_{Ein}.

For silicon we choose ω_{Ein} by the zero-point motion. At $T = 0$ every term of the ZPR carries a factor $1/\omega_{\boldsymbol{Q}\nu}$ from $\ell^2_{\boldsymbol{Q}\nu}$, as the mean-square displacement does, and the Einstein energy that reproduces the zero-point $\langle u^2\rangle$ of the computed phonon spectrum is $\langle\omega^{-1}\rangle^{-1} = 30.4\,\mathrm{meV}$, or 353 K (Worked example 5.4). For the ZPR we take the nonadiabatic AHC value, $-56.2\,\mathrm{meV}$. Applying the adiabatic weight of Eq. (6.26) to a nonadiabatic value is an approximation; the two forms differ by 8 meV at $T = 0$.

At 300 K, $T = 25.85\,\mathrm{meV}$, $\omega_{\mathrm{Ein}}/2T = 0.588$ and $\coth(0.588) = 1.891$. The model predicts that the gap closes between zero and room temperature by

$$E_{\mathrm{g}}(300\,\mathrm{K}) - E_{\mathrm{g}}(0) = -56.2\,\mathrm{meV} \times (1.891 - 1) = -50.1\,\mathrm{meV}.$$

The measured gap, $E_{\mathrm{g}}(T) = E_{\mathrm{g}}(0) - aT^2/(T + b)$ with $E_{\mathrm{g}}(0) = 1.1692\,\mathrm{eV}$, $a = 4.9 \times 10^{-4}\,\mathrm{eV\,K^{-1}}$ and $b = 655\,\mathrm{K}$, a fit to photoluminescence up to 1000 K [27, 28, Eq. (11)], falls by 46.2 meV, from 1.1692 eV to 1.1230 eV. The model is 8 % high; with the adiabatic ZPR, $-64.3\,\mathrm{meV}$, it would predict $-57.3\,\mathrm{meV}$. Its high-temperature slope is $2 \times 56.2\,\mathrm{meV}/353\,\mathrm{K} = 0.318\,\mathrm{meV\,K^{-1}}$, between the slopes of the measured curve at 300 K and 500 K, 0.26 and $0.33\,\mathrm{meV\,K^{-1}}$.

The choice of ω_{Ein} carries physics. Had we assumed that the gap follows the full mean-square displacement of the computed spectrum, the shift would have scaled with $\langle u^2\rangle(T)/\langle u^2\rangle(0) = 2.661$, and the gap would have closed by $56.2\,\mathrm{meV} \times 1.661 = 93.3\,\mathrm{meV}$ at 300 K, twice the measured value. That assumption gives every mode a shift in proportion to its mean-square displacement, and the thermal $\langle u^2\rangle$ is weighted toward the

softest modes, which enter with $1/\omega^2$: the long-wavelength acoustic phonons and the flat transverse acoustic branches near the zone boundary (Worked example 5.4). The measured curve says that these modes move the gap less than their share of $\langle u^2 \rangle$, and for the acoustic modes at small Q Eq. (6.23) explains why: they are nearly rigid translations. The gap responds to the displacements of atoms relative to their neighbors, not to their displacement as such.
The model also runs the other way, which is how a ZPR is extracted from measurements. With ω_{Ein} fixed, the measured fall of 46.2 meV between zero and room temperature implies ZPR $= -46.2\,\mathrm{meV}/0.891 = -51.8\,\mathrm{meV}$, and the measured slope at 500 K, $0.33\,\mathrm{meV\,K^{-1}}$, implies $-(353\,\mathrm{K}/2) \times 0.33\,\mathrm{meV\,K^{-1}} = -58\,\mathrm{meV}$. The two estimates bracket the nonadiabatic AHC value, −56.2 meV.
Two further numbers follow. The model's clamped-lattice gap, the intercept of its high-temperature line, is $1.1692\,\mathrm{eV} + 0.0562\,\mathrm{eV} = 1.2254\,\mathrm{eV}$; the model is anchored at the measured $E_{\mathrm{g}}(0)$, which contains the zero-point expansion of the lattice, and it leaves out thermal expansion. And replacing natural silicon, $M = 28.09\,\mathrm{u}$, by ^{30}Si scales the ZPR by $(28.09/29.97)^{1/2}$ and raises the gap by 1.8 meV. At detector temperatures nothing of the thermal part survives. At 1 K, $\omega_{\mathrm{Ein}}/T = 353$ and $2n_{\mathrm{B}}(\omega_{\mathrm{Ein}}) \sim 10^{-153}$; the only phonons still occupied are acoustic modes at small Q, and their contribution vanishes by Eq. (6.23) and with their phase space. A detector at 10 mK therefore sees $E_{\mathrm{g}}(0)$.

Fig. 6.5 puts the pieces together. The Einstein model with a computed ZPR and an Einstein energy fixed by the zero-point motion follows the measured gap to within about 5 meV from zero to 600 K. Below 150 K it falls slightly too slowly, since a single energy of 30 meV cannot represent the softer modes that the real spectrum starts to populate first. The curve that follows $\langle u^2 \rangle$ fails from the start.

Three sets of numbers for the silicon ZPR have now appeared, and they measure different things. The inversions of Worked example 6.3, −52 meV and −58 meV, assume $\omega_{\mathrm{Ein}} = 353\,\mathrm{K}$ and read the measured curve at one temperature. The slope of the measured gap still grows above 500 K, toward $a = 0.49\,\mathrm{meV\,K^{-1}}$, so these inversions test whether the computed ZPR and the chosen ω_{Ein} are consistent with the curve; they do not measure the ZPR. The values −62 meV and −64 meV extracted from experiment come from the isotope dependence of the gap and from the linear extrapolation of its high-temperature part [26], and they need no Einstein energy. They contain the whole zero-point shift of the gap, including a part that is not electron–phonon coupling at all: the zero-point motion also expands the lattice, which by itself opens the gap by 9 meV [29, Table S2]. The electron–phonon part implied by experiment is therefore −71 to −73 meV. That is the number to compare with the computed −56.2 meV, a fixed-volume electron–phonon value, and the calculation falls about 20 % short of it, as does the special-displacement value of −57 meV. The shortfall is consistent with the overscreening of the vertex by the small Kohn–Sham gap: a calculation of the same kind on *GW* bands gives 66 meV [24, Sect. VII B].

6.3.6 Phonon-limited lifetimes

The imaginary part of the self-energy is a decay rate. The Allen–Heine term is real and contributes nothing to it. For the Fan–Migdal term, the Sokhotski–Plemelj formula Eq. (2.58)

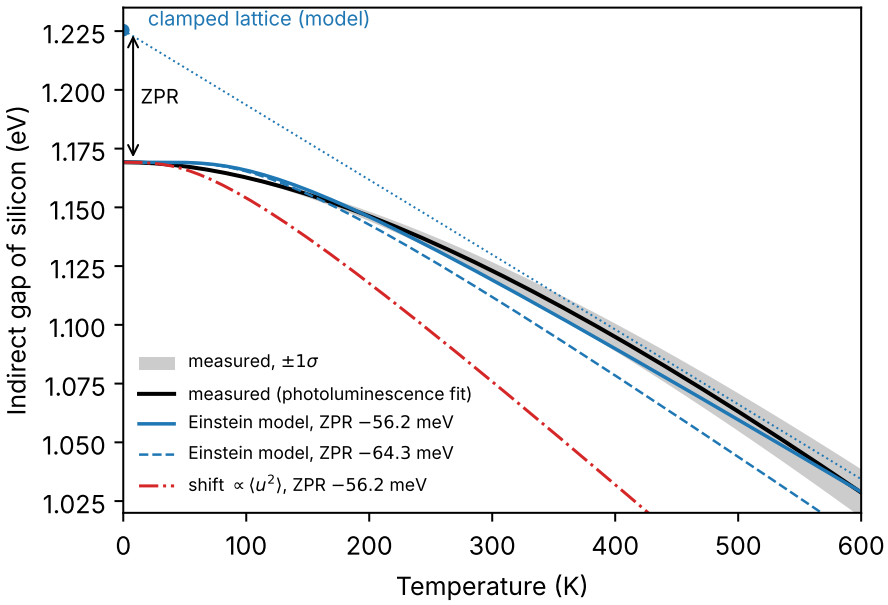


Figure 6.5: The indirect gap of silicon against temperature. Black: measured, the fit $E_g(T) = 1.1692\,\text{eV} - aT^2/(T+b)$ with $a = 4.9 \times 10^{-4}\,\text{eV K}^{-1}$ and $b = 655\,\text{K}$ to photoluminescence data [27, 28]; the gray band is the envelope of the fits with a and b varied independently by their quoted errors of $0.2 \times 10^{-4}\,\text{eV K}^{-1}$ and 40 K. Blue: Einstein model $E_g(0) + \text{ZPR}\,[\coth(\omega_{\text{Ein}}/2T) - 1]$ with $\omega_{\text{Ein}} = 30.4\,\text{meV}$, the Einstein energy that reproduces the zero-point mean-square displacement of the computed phonon spectrum, and the computed zero-point renormalization, −56.2 meV (solid, nonadiabatic) or −64.3 meV (dashed, adiabatic) [22]. Red, dash-dotted: a shift proportional to the full mean-square displacement $\langle u^2 \rangle(T)$ of the computed spectrum, with the same zero-point value. Blue dotted line and dot: the high-temperature asymptote of the solid model, which extrapolates to the model's clamped-lattice gap, 1.2254 eV, at $T = 0$; the arrow marks the ZPR. All model curves are anchored at the measured gap at $T = 0$ and leave out thermal expansion.

turns each denominator of Eq. (6.19) into a delta function,

$$\operatorname{Im}\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\omega) = -\frac{\pi}{N}\sum_{\boldsymbol{Q}\nu m} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2 \Big[(n_{\boldsymbol{Q}\nu} + 1 - f_{m,\boldsymbol{k}+\boldsymbol{Q}})\,\delta(\omega - \xi_{m,\boldsymbol{k}+\boldsymbol{Q}} - \omega_{\boldsymbol{Q}\nu}) + (n_{\boldsymbol{Q}\nu} + f_{m,\boldsymbol{k}+\boldsymbol{Q}})\,\delta(\omega - \xi_{m,\boldsymbol{k}+\boldsymbol{Q}} + \omega_{\boldsymbol{Q}\nu}) \Big], \tag{6.27}$$

and by Eq. (4.83) the state decays at the rate $\Gamma_{n\boldsymbol{k}} = -2 z_{\mathrm{qp}} \operatorname{Im}\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\xi_{n\boldsymbol{k}})$, its *phonon-limited lifetime* being $1/\Gamma_{n\boldsymbol{k}}$. The cutting rule of Sect. 4.8 reads Eq. (6.27) as golden-rule rates: the cut runs through the intermediate electron and the phonon, and puts both on shell. For an electron in an empty band the out-scattering rates are all that remain, emission of a phonon with the weight $n+1$ and absorption with the weight n of Eq. (A.28).

At $T = 0$ only emission is left, and an electron can emit only if a state of lower energy is available. An electron at the bottom of the conduction band of an insulator has none, and its width is zero: band edges are sharp at zero temperature, as Sect. 4.7 anticipated. A width found at a band extremum at $T = 0$ in a calculation is the imaginary part added to the denominators for numerical reasons (Appendix E.8), not a lifetime. At 10 mK the absorption term is absent for every phonon above about 10 µeV, and a hot carrier in a detector loses its energy by spontaneous emission only. How fast, and down to what energy?

For a conduction electron near a band minimum the answer is set by acoustic phonons. Take a parabolic band of mass m^*, an electron of wavevector $\boldsymbol{k}$ measured from the minimum, and the deformation-potential vertex $|g|^2 = \Xi^2 Q/(2M_{\mathrm{cell}}c)$ of Eq. (6.13), for a branch of sound speed c and an effective constant Ξ. At $T = 0$, with $(1/N)\sum_{\boldsymbol{Q}} \to \Omega \int \mathrm{d}^3Q/(2\pi)^3$,

$$\Gamma(\boldsymbol{k}) = 2\pi\,\Omega \int \frac{\mathrm{d}^3Q}{(2\pi)^3}\,\frac{\Xi^2 Q}{2M_{\mathrm{cell}}c}\,\delta\Big(\frac{k^2 - |\boldsymbol{k}+\boldsymbol{Q}|^2}{2m^*} - cQ\Big).$$

The delta function fixes the angle θ between $\boldsymbol{Q}$ and $\boldsymbol{k}$ at $\cos\theta = -(Q/2 + m^*c)/k$, and the integral over $\cos\theta$ gives $m^*/(kQ)$. A solution exists only for $Q \le Q_{\max} = 2(k - m^*c)$, which requires $k > m^*c$: the electron must move faster than sound. The emission is Cherenkov radiation of sound, and the phonons go out on a cone about the direction of motion. The remaining integral over Q gives

$$\Gamma(\boldsymbol{k}) = \frac{\Omega\,\Xi^2\,m^*\,Q_{\max}^3}{12\pi M_{\mathrm{cell}}\,c\,k}, \qquad Q_{\max} = 2(k - m^*c), \tag{6.28}$$

which grows linearly with the kinetic energy $k^2/2m^*$ far above the threshold.

For silicon we estimate the rate with three simplifications. The valley is taken spherical, with the density-of-states mass $m^* = m^*_{\mathrm{d}} = (m^*_{\parallel} m^{*2}_{\perp})^{1/3} = 0.322\,m_e$ from the valley masses of Sect. 3.3. The valley constants of Sect. 6.2, $\Xi_d = 1.01\,\mathrm{eV}$ and $\Xi_u = 8.84\,\mathrm{eV}$, enter with their angular factors averaged over the angle between $\boldsymbol{Q}$ and the valley axis, as for valleys of random orientation: $\Xi^2 = \Xi_d^2 + \frac{2}{3}\Xi_d\Xi_u + \frac{1}{5}\Xi_u^2 = (4.75\,\mathrm{eV})^2$ for the longitudinal branch and $\frac{2}{15}\Xi_u^2 = (3.23\,\mathrm{eV})^2$ for the one transverse branch that couples. And the sound speeds are the measured ones along [100], 8.43 and $5.84\,\mathrm{km\,s^{-1}}$ (Sect. 5.2). The thresholds $m^*c^2/2$ are 0.065 meV for longitudinal and 0.031 meV for transverse emission. At 10 meV of kinetic energy the two branches give $\Gamma = 7.2 + 5.2\,\mu\mathrm{eV} = 12.3\,\mu\mathrm{eV}$, a lifetime of 53 ps; at 1 meV the lifetime is 0.9 ns, and at 30 meV 16 ps. Above the optical phonon energy, 64 meV, and through intervalley scattering (Sect. 6.2) the rates are far higher. The estimate fixes the scale. An electron created with a few meV in a silicon detector sheds it as acoustic phonons over a fraction of a nanosecond, and then stops emitting once its kinetic energy falls below a few tens of µeV, under 1 K. In a biased detector the field keeps the carriers above that threshold, and the phonons emitted along the drift are the Luke phonons of Appendix E.12. A quantitative rate uses Eq. (6.27) with DFPT vertices on fine grids and the real anisotropic valleys (Appendix E.5).

6.3.7 Migdal's adiabatic theorem

The Fan–Migdal self-energy keeps the vertex g bare at both ends and the propagators free. The exact self-energy has the same structure with every piece dressed: the full electron propagator, the full phonon propagator, and a vertex function in place of one of the two g's [3, Eq. (156)]. At fourth order in g two new diagrams appear. In one, a second phonon line is emitted and reabsorbed inside the first, on the intermediate electron line; it is a self-energy insertion, and replacing the free propagator in Eq. (6.19) by the full one sums all such insertions. In the other, the two phonon lines cross. This is a vertex correction, and no self-consistency of the propagators includes it. Migdal's question was how large it is [2].

In a metal, the counting of Sect. 6.1 answers it. Consider an electron at the Fermi surface. In the self-energy insertion the inner phonon carries a frequency of at most ω_{D}. The intermediate electron can sit anywhere on the Fermi surface, within ω_{D} of it in energy. The insertion is then as large as the self-energy itself, a fraction λ_{ep} of ω_{D}, with λ_{ep} the dimensionless electron–phonon coupling, defined so that the mass of an electron at the Fermi surface is enhanced by $1 + \lambda_{\mathrm{ep}}$ (Sect. 6.4). In the crossed diagram, two intermediate electron lines carry momenta that differ by the momentum of the outer phonon. Both lie within ω_{D} of the Fermi surface only where two thin shells, of thickness $\omega_{\mathrm{D}}/v_{\mathrm{F}}$ with v_{F} the Fermi velocity and shifted by that momentum, intersect, a fraction $\sim \omega_{\mathrm{D}}/E_{\mathrm{F}}$ of the momenta summed over. Elsewhere one of the two lines is off shell by an energy of order E_{F}.

Migdal's adiabatic theorem

In a metal, the vertex corrections to the electron–phonon self-energy are of relative order $\lambda_{\mathrm{ep}}\,\omega_{\mathrm{D}}/E_{\mathrm{F}}$. To that accuracy, the self-energy is the Fan–Migdal diagram with the screened vertex g, and the electron propagator inside it may be made self-consistent when λ_{ep} is not small.

The small ratio is the one of Sect. 6.1: $\omega_{\mathrm{D}}/E_{\mathrm{F}} \sim \kappa_{\mathrm{BO}}^2 = (m_e/M)^{1/2}$, 10^{-3} to 10^{-2}, the fraction of the conduction electrons that cannot follow the nuclei. The theorem does not require weak coupling. It allows λ_{ep} of order one, and that is why the theory of the electron–phonon coupling in metals and superconductors is built on the single diagram Eq. (6.19) and its phonon counterpart (Sect. 6.4 and Appendix D.17). It fails where its counting fails. Small Fermi surfaces, as in a doped semiconductor, bring E_{F} down toward the phonon energy. In nearly forward scattering the two shells are parallel, and the counting breaks down for $v_{\mathrm{F}}Q \lesssim \omega_{\boldsymbol{Q}\nu}$. Flat bands fail for the same reason, and so does strong coupling once the lattice distorts around each electron.

An insulator has no Fermi surface, and the role of E_{F} passes to the gap. Every virtual excitation of the filled bands costs at least E_{g}, and these electrons follow the nuclei. Their response to a phonon is the static screening of the vertex (Sect. 6.2) together with the Allen–Heine term, both computed with the nuclei at rest, and the vertex corrections that run through interband excitations are of this kind. The theorem for insulators states that the electron–phonon vertex may be computed statically, by DFPT with frozen nuclei, with corrections of relative order $(\omega_{\mathrm{ph}}/E_{\mathrm{g}})^2$, 0.003 for the optical phonon of silicon, since the screening is even in the frequency (Sect. 6.2). It says nothing about a carrier moving within its own band near an edge. There the electron has kinetic energies comparable to ω_{ph}, the ratio $\omega_{\mathrm{ph}}/E_{\mathrm{F}}$ is not small, and the only expansion parameter is the coupling itself. For the Fröhlich polaron we estimate the crossed diagram, by the same counting with the coupling as the only parameter, to be smaller than the Fan–Migdal one by a relative order $\sim \alpha_{\mathrm{F}}$, a few percent for GaAs with $\alpha_{\mathrm{F}} = 0.072$ (Worked example 6.2). In strongly polar crystals with α_{F} of order one the carrier becomes a polaron that no second-order self-energy describes.

The gap at 10 mK is the measured low-temperature gap, which already contains the full zero-point renormalization, so a calculation that starts from it must not add a computed renormalization again (Worked example E.7). The same two vertices, joined the other way into an electron loop, shift and broaden the phonons, the subject of Sect. 6.4.

6.4 Phonon self-energy and the Eliashberg function

A longitudinal phonon of 35 meV in aluminum lasts a few tens of periods before it disappears into an electron–hole pair at the Fermi surface. The zone-center optical phonon of silicon, at 64.1 meV, cannot decay that way at all at low temperature: the cheapest pair costs the 1.17 eV gap, and the phonon lives until anharmonicity breaks it into two, after about 57 periods (Sect. 5.7). The vertex g of Sect. 6.2 is at work in both crystals. What differs is whether real pairs exist at the phonon's energy. The same pairs, created virtually, also shift the phonon frequency, and in a metal they leave a kink in the dispersion wherever the Fermi surface can just span the phonon wavevector. For dark-matter detection these effects matter in two places. The superconducting sensors of Chap. 7 owe their gap to the phonon-mediated attraction, whose strength is a single number λ_{ep} built from the same widths. And every phonon energy used in a scattering rate is a frequency from density-functional perturbation theory (Sect. 5.4), and adding a self-energy correction to it counts the static response twice unless that part is removed (Eq. (6.31)).

We will derive the phonon self-energy as the electron bubble, read its imaginary part as a golden-rule decay rate, and show why it vanishes in an undoped insulator. In a metal the width leads to Allen's formula, the Eliashberg function $\alpha^2F(\omega)$ and the coupling constant λ_{ep}, which for aluminum comes out as 0.44 from a model with one adjustable length.

6.4.1 The electron bubble

A phonon $(\boldsymbol{Q},\nu)$ can be absorbed by an electron in $(n,\boldsymbol{k})$, which jumps to $(m,\boldsymbol{k}+\boldsymbol{Q})$ and leaves a hole behind; the pair can later recombine and emit the phonon again. At second order in H_{ep} of Eq. (A.25) this is the only correction to the propagation of a phonon, the bubble of Fig. 6.6(b), with the vertex $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})$ where the pair is created and its complex conjugate where it is destroyed. The electron self-energy of Sect. 6.3, Fig. 6.6(a), is the same pair of vertices joined the other way: there the electron line runs through and the phonon line closes on itself. The bubble is the density bubble of Eq. (4.93) with the density vertex replaced by g, which is why a phonon is screened by the same transitions that screen a Coulomb potential.

Appendix D.17 evaluates the bubble with the diagram rules of Appendix D.11: two vertices with the factor $1/N$, a closed electron loop with its sign and its sum over the two spin orientations, and the frequency sum of Eq. (D.51). Repeating the bubble along the phonon line, we obtain the geometric series of Fig. 6.6(c), the free propagator D_0 with any number of bubbles inserted, which sums into a Dyson equation like Eq. (4.81) for electrons. Since the free propagator Eq. (D.57) has $\mathsf{D}_0^{-1} = (\omega^2-\omega_{\boldsymbol{Q}\nu}^2)/2\omega_{\boldsymbol{Q}\nu}$, it is natural to measure the self-energy in units of a squared frequency, and after continuation to real frequency,

Phonon self-energy and Dyson equation

$$\mathsf{D}_\nu(\boldsymbol{Q},\omega) = \frac{2\omega_{\boldsymbol{Q}\nu}}{\omega^2-\omega_{\boldsymbol{Q}\nu}^2-\Pi_\nu(\boldsymbol{Q},\omega)},$$

$$\Pi_\nu(\boldsymbol{Q},\omega) = \frac{4\omega_{\boldsymbol{Q}\nu}}{N}\sum_{\boldsymbol{k}mn}|g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2\,\frac{f_{n\boldsymbol{k}}-f_{m,\boldsymbol{k}+\boldsymbol{Q}}}{\omega+\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}+\mathrm{i}0^+}, \tag{6.29}$$

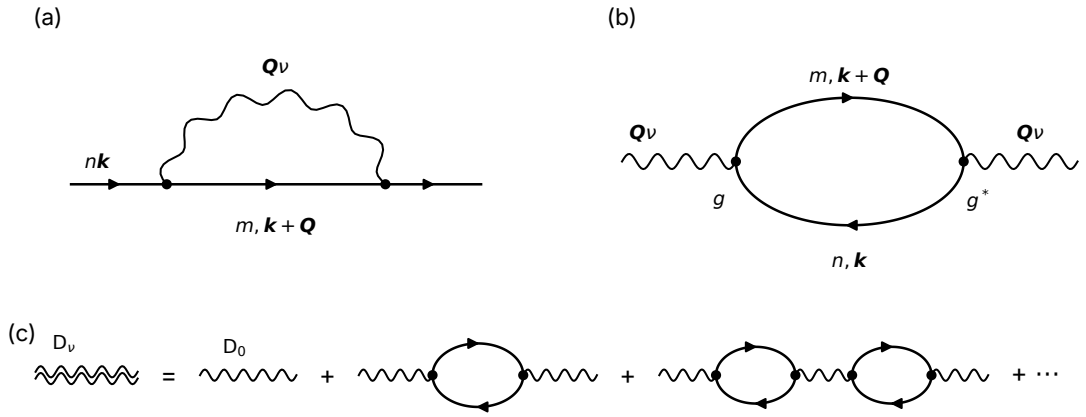


Figure 6.6: The two second-order diagrams of the electron–phonon interaction. Solid lines with arrows are electrons, wavy lines phonons, dots the vertex g. (a) The electron self-energy: an electron in $(n, \boldsymbol{k})$ exchanges a phonon $(\boldsymbol{Q}, \nu)$ with the intermediate state $(m, \boldsymbol{k}+\boldsymbol{Q})$. (b) The phonon self-energy: a phonon creates an electron in $(m, \boldsymbol{k}+\boldsymbol{Q})$ and a hole in $(n, \boldsymbol{k})$ through g, and the pair recombines through g^*. (c) The Dyson series for the dressed phonon propagator D_ν (double wavy line): the free propagator D_0 with any number of bubbles inserted.

with $f_{n\boldsymbol{k}} = n_\mathrm{F}(\xi_{n\boldsymbol{k}})$ the occupations and Π_ν in eV^2. This is Eq. (D.96). The factor 4 is the 2 of D_0^{-1} times the two spin orientations of the loop. Note that we have put the screened vertex g at both ends, with noninteracting electrons between them. Strictly, one end should carry the bare vertex [3, Eqs. (145) and (146)], and we will see below why this matters for the static part only. We have dropped the bubbles that connect two branches $\nu \neq \nu'$ at the same $\boldsymbol{Q}$. They mix the modes, and they matter where two branches are nearly degenerate.

The poles of D_ν are the dressed phonons, $\omega^2 = \omega_{\boldsymbol{Q}\nu}^2 + \Pi_\nu(\boldsymbol{Q}, \omega)$. When $|\Pi_\nu| \ll \omega_{\boldsymbol{Q}\nu}^2$ the root lies close to $\omega_{\boldsymbol{Q}\nu}$. Expanding the denominator there, $\omega^2 - \omega_{\boldsymbol{Q}\nu}^2 \simeq 2\omega_{\boldsymbol{Q}\nu}(\omega - \omega_{\boldsymbol{Q}\nu})$, turns D_ν into a Lorentzian, $1/(\omega - \tilde{\omega}_{\boldsymbol{Q}\nu} + \mathrm{i}\gamma_{\boldsymbol{Q}\nu}/2)$, with

$$\tilde{\omega}_{\boldsymbol{Q}\nu} = \omega_{\boldsymbol{Q}\nu} + \frac{\mathrm{Re}\,\Pi_\nu(\boldsymbol{Q}, \omega_{\boldsymbol{Q}\nu})}{2\omega_{\boldsymbol{Q}\nu}}, \qquad \gamma_{\boldsymbol{Q}\nu} = -\frac{\mathrm{Im}\,\Pi_\nu(\boldsymbol{Q}, \omega_{\boldsymbol{Q}\nu})}{\omega_{\boldsymbol{Q}\nu}}. \tag{6.30}$$

The real part shifts the phonon, and the imaginary part gives it a full width at half maximum $\gamma_{\boldsymbol{Q}\nu}$, the quantity of Sect. 5.7. Its electron–phonon part is $\gamma_{\boldsymbol{Q}\nu}^{\mathrm{ep}}$, and the anharmonic bubble of Sect. 5.7 adds $\gamma_{\boldsymbol{Q}\nu}^{\mathrm{anh}}$ to it.

The sign of the static part is fixed. In an insulator at $T = 0$, $f_{n\boldsymbol{k}} - f_{m,\boldsymbol{k}+\boldsymbol{Q}} = +1$ for n occupied and m empty, and then $\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} < 0$; the term with the roles exchanged has both signs reversed. Every term of $\Pi_\nu(\boldsymbol{Q}, 0)$ is therefore negative. The virtual pairs soften the lattice, as second-order perturbation theory always lowers a ground-state energy. In a metal the static bubble inherits the structure of the Lindhard function (Sect. 4.5). For a spherical Fermi surface the density of pairs with $\varepsilon_{n\boldsymbol{k}} = \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} = E_\mathrm{F}$ changes abruptly when Q passes $2k_\mathrm{F}$, the largest wavevector that joins two points of the surface, and the derivative of the Lindhard function diverges there logarithmically (Appendix D.8). The phonon frequencies carry the singularity as a kink at $Q = 2k_\mathrm{F}$, or at whatever wavevectors span flat, parallel pieces of a real Fermi surface. This is the Kohn anomaly [30], and its position in a measured dispersion is an image of the Fermi surface.

6.4.2 Adiabatic and nonadiabatic phonons

The static part of Π_ν is already in the phonons we start from. Density-functional perturbation theory computes second derivatives of the Born–Oppenheimer energy with the electrons relaxed to their ground state at every nuclear configuration (Sect. 5.4). The relaxation is the electronic response at zero frequency, which at this order is $\Pi_\nu(\boldsymbol{Q},0)$. The first-order density of Eq. (5.45) is precisely that response, and the Hartree and exchange–correlation potential it generates is what turns the bare vertex at one end of the bubble into the screened one. Starting from the frequencies of such a calculation and adding the full $\Pi_\nu(\boldsymbol{Q},\omega)$ would count the static response twice. Only the difference between the dynamic and the static response may be added [31, Eqs. (22) and (23)],

$$\left(\omega^{\mathrm{ren}}_{\boldsymbol{Q}\nu}\right)^2 = \omega^2_{\boldsymbol{Q}\nu} + \mathrm{Re}\left[\Pi_\nu(\boldsymbol{Q},\omega^{\mathrm{ren}}_{\boldsymbol{Q}\nu}) - \Pi_\nu(\boldsymbol{Q},0)\right], \tag{6.31}$$

where $\omega_{\boldsymbol{Q}\nu}$ is now the adiabatic frequency of density-functional perturbation theory and $\omega^{\mathrm{ren}}_{\boldsymbol{Q}\nu}$ the nonadiabatic one (Eq. (D.97)). The imaginary part needs no subtraction, since $\Pi_\nu(\boldsymbol{Q},0)$ is real.

How large is the remainder? In an insulator every denominator of Eq. (6.29) is at least E_g in magnitude, and the pair of terms with roles exchanged makes Π_ν even in ω. The difference $\Pi_\nu(\omega) - \Pi_\nu(0)$ is then of relative order $(\omega_{\boldsymbol{Q}\nu}/E_\mathrm{g})^2$, which is 0.003 for the optical phonon of silicon. The adiabatic phonons of an insulator are accurate, and this is the phonon-side statement of the counting of Sect. 6.1, where $\omega_\mathrm{ph}/E_\mathrm{g}$ was the small parameter. A metal has no gap. Its intraband pairs with $|\varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{n\boldsymbol{k}}| \lesssim \omega_{\boldsymbol{Q}\nu}$ respond differently at $\omega = 0$ and at $\omega = \omega_{\boldsymbol{Q}\nu}$, and for them the difference is of order one. Such pairs dominate the bubble at small wavevector. For a band with Fermi velocity v_F, the energy difference of a pair is $v_\mathrm{F} Q\cos\theta$, and the adiabatic and nonadiabatic frequencies differ appreciably only for $Q \lesssim \omega_{\boldsymbol{Q}\nu}/v_\mathrm{F}$ [31]. In aluminum, with $v_\mathrm{F} = 2.0\times10^6\,\mathrm{m\,s^{-1}}$, that is $Q \lesssim 2\times10^{-3}\,\mathrm{Å}^{-1}$ for a 30 meV phonon, a thousandth of the zone. Near the zone center of metals with few carriers and stiff optical modes the correction can, however, be large. Calandra, Profeta and Mauri note corrections of up to 30 % of the phonon frequency there [31]; graphite intercalation compounds and doped graphene are the standard examples. Everywhere else the adiabatic frequencies are the right starting point.

The bubble with undressed vertices is itself an approximation. The counting of Migdal's theorem (Sect. 6.3) makes the vertex corrections to it of relative order $\lambda_\mathrm{ep}\omega_\mathrm{D}/E_\mathrm{F}$, with the coupling constant λ_ep defined below; for aluminum, with phonon energies up to $\omega_\mathrm{max} = 40\,\mathrm{meV}$ and $E_\mathrm{F} = 11.7\,\mathrm{eV}$, that is at most 0.0015.

6.4.3 The electron–phonon linewidth

The width follows from the imaginary part of Eq. (6.29). With $1/(x+\mathrm{i}0^+) = \mathrm{PV}(1/x) - \mathrm{i}\pi\delta(x)$ from Eq. (2.58) in Eq. (6.30),

Electron–phonon linewidth

$$\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu} = -\frac{\mathrm{Im}\,\Pi_\nu(\boldsymbol{Q},\omega_{\boldsymbol{Q}\nu})}{\omega_{\boldsymbol{Q}\nu}} = \frac{4\pi}{N}\sum_{\boldsymbol{k}mn} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2 \left(f_{n\boldsymbol{k}} - f_{m,\boldsymbol{k}+\boldsymbol{Q}}\right)\delta\left(\varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{n\boldsymbol{k}} - \omega_{\boldsymbol{Q}\nu}\right), \tag{6.32}$$

a full width, with the spin sum in the factor $4\pi = 2\times2\pi$ (Eq. (D.98)). The cutting rule of Sect. 4.8 tells us how to read it. The delta function puts the pair on shell with the energy of the phonon, and $2\pi|g|^2/N$ times it is the golden-rule rate Eq. (2.67) for absorbing the

phonon into the pair $(n\boldsymbol{k} \to m, \boldsymbol{k}+\boldsymbol{Q})$. The occupation factor is the rate of that process, $f_n(1-f_m)$, minus the rate of the inverse one, $f_m(1-f_n)$, in which a pair recombines and emits the phonon; the difference is $f_n - f_m$.

In an undoped insulator at $T = 0$ the width vanishes below the gap. The delta function needs an occupied state n and an empty state m with $\varepsilon_m - \varepsilon_n = \omega_{\boldsymbol{Q}\nu}$, and every such difference is at least E_g, while the largest phonon energy of silicon is 64 meV. No term survives. The phonons of an insulator are, however, still broadened, by anharmonicity (Sect. 5.7) and by isotopes and defects (Sect. 11.6), which is why the measured 0.18 meV width of the silicon Raman line could be compared with the three-phonon formula alone. At $T > 0$ thermally excited electrons and holes absorb phonons by intraband transitions, with a width proportional to their density, which in an intrinsic crystal is suppressed by $\mathrm{e}^{-E_\mathrm{g}/2T}$: 4×10^{-10} in silicon at room temperature. Doping supplies such carriers without the exponential, and a doped semiconductor behaves for this purpose like a dilute metal.

In a metal the electronic states near the Fermi level change on scales of eV, far larger than $\omega_{\boldsymbol{Q}\nu}$. For $T \ll E_\mathrm{F}$ the occupation difference at $\varepsilon_m = \varepsilon_n + \omega_{\boldsymbol{Q}\nu}$ is $f(\varepsilon_n) - f(\varepsilon_n + \omega_{\boldsymbol{Q}\nu}) \simeq \omega_{\boldsymbol{Q}\nu}\delta(\xi_{n\boldsymbol{k}})$, and dropping $\omega_{\boldsymbol{Q}\nu}$ also in the remaining delta function, we arrive at Allen's formula [32],

$$\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu} \simeq \frac{4\pi\omega_{\boldsymbol{Q}\nu}}{N} \sum_{\boldsymbol{k}mn} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2\, \delta(\xi_{n\boldsymbol{k}})\, \delta(\xi_{m,\boldsymbol{k}+\boldsymbol{Q}}), \tag{6.33}$$

which is Eq. (D.99). Both electrons now sit on the Fermi surface, and the width is proportional to the phonon frequency. It no longer depends on temperature: thermal smearing removes states from below the Fermi level and adds them above, and the net number of pairs that can absorb $\omega_{\boldsymbol{Q}\nu}$ stays the same. The approximation requires bands that are smooth on the scale of $\omega_{\boldsymbol{Q}\nu}$ and a temperature small on the electronic scale. It fails for a band edge within $\omega_{\boldsymbol{Q}\nu}$ of the Fermi level, and for $Q \lesssim \omega_{\boldsymbol{Q}\nu}/v_\mathrm{F}$, the same small-$Q$ region where the nonadiabatic correction lives. Appendix D.17 checks Eq. (6.33) against Eq. (6.32) for the electron gas, where both give $m_e^2\omega/(4\pi^2 Q)$ per spin and volume for $\omega/v_\mathrm{F} \ll Q < 2k_\mathrm{F}$.

6.4.4 The Eliashberg function

The double sum in Eq. (6.33) is a Fermi-surface average of $|g|^2$ for one phonon. Superconductivity, the electron mass enhancement and the resistivity of a metal all involve the same average, weighted over all phonons at a given energy. This weighted average is the Eliashberg function [33],

$$\alpha^2F(\omega) = \frac{1}{\mathcal{N}_\mathrm{F}\Omega\, N^2} \sum_{\boldsymbol{k}\boldsymbol{Q}} \sum_{mn\nu} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2\, \delta(\xi_{n\boldsymbol{k}})\, \delta(\xi_{m,\boldsymbol{k}+\boldsymbol{Q}})\, \delta(\omega - \omega_{\boldsymbol{Q}\nu}), \tag{6.34}$$

with $\mathcal{N}_\mathrm{F}$ the density of states per spin and volume at the Fermi level (Sect. 2.5), so that $\mathcal{N}_\mathrm{F}\Omega$, per spin and per cell, satisfies $N\mathcal{N}_\mathrm{F}\Omega = \sum_{n\boldsymbol{k}} \delta(\xi_{n\boldsymbol{k}})$. The prefactor makes α^2F an average over initial states on the Fermi surface and a sum over final states, and it is dimensionless [3, Eq. (205)]. The symbol is read as one. If $|g|^2$ were the same for every pair on the Fermi surface, α^2F would be that constant times $\mathcal{N}_\mathrm{F}\Omega$ times the phonon density of states $g_\mathrm{ph}(\omega)$ of Eq. (5.21). The historical notation splits it this way into a coupling $\alpha^2(\omega)$ and a density $F(\omega)$. Comparing Eqs. (6.33) and (6.34) term by term turns the Eliashberg function into a sum of linewidths, and it defines the coupling constant as its inverse moment [3, Eq. (206)]:

Eliashberg function, coupling constant and linewidths

$$\alpha^2F(\omega) = \frac{1}{4\pi\mathcal{N}_\mathrm{F}\Omega\,N}\sum_{\boldsymbol{Q}\nu}\frac{\gamma^\mathrm{ep}_{\boldsymbol{Q}\nu}}{\omega_{\boldsymbol{Q}\nu}}\,\delta(\omega-\omega_{\boldsymbol{Q}\nu}),$$
$$\lambda_\mathrm{ep} = 2\int_0^\infty \mathrm{d}\omega\,\frac{\alpha^2F(\omega)}{\omega} = \frac{1}{2\pi\mathcal{N}_\mathrm{F}\Omega\,N}\sum_{\boldsymbol{Q}\nu}\frac{\gamma^\mathrm{ep}_{\boldsymbol{Q}\nu}}{\omega^2_{\boldsymbol{Q}\nu}}. \tag{6.35}$$

With a full width and a density of states per spin the factors are as written; a half width [3, Eq. (185)] or a two-spin $\mathcal{N}_\mathrm{F}$ changes them by factors of two (Appendix D.17). The coupling constant is a zone average of mode contributions $\lambda_{\boldsymbol{Q}\nu} = \gamma^\mathrm{ep}_{\boldsymbol{Q}\nu}/(2\pi\mathcal{N}_\mathrm{F}\Omega\,\omega^2_{\boldsymbol{Q}\nu})$. We can therefore read the coupling of a mode off its measured linewidth, which was the point of Allen's relation [32], and the factor $1/\omega^2$ makes soft modes with large widths dominate.

The name "coupling constant" comes from the electrons. Averaging the Fan–Migdal self-energy of Sect. 6.3 over the Fermi surface leaves an integral over α^2F alone, and its slope at the Fermi level is (Appendix D.17)

$$-\frac{\partial\,\mathrm{Re}\langle\Sigma^\mathrm{FM}\rangle_\mathrm{FS}}{\partial\omega}\bigg|_{\omega=0} = \lambda_\mathrm{ep}. \tag{6.36}$$

By Eq. (4.83) the quasiparticles within about ω_D of the Fermi level have the residue $z_\mathrm{qp} = 1/(1+\lambda_\mathrm{ep})$, their velocity is reduced by the same factor, and their density of states, and with it the electronic specific heat, is enhanced by $1+\lambda_\mathrm{ep}$. The free electron gas has the specific heat $(\pi^2/3)\,2\mathcal{N}_\mathrm{F}\,T$ per volume [34, Chap. 2]. For aluminum, for example, a density of states $2\mathcal{N}_\mathrm{F}\Omega = 5.49$ states per Ry and cell for both spins, computed from first principles [35, Table II], corresponds to $0.951\,\mathrm{mJ\,K^{-2}\,mol^{-1}}$, while the measured coefficient is $1.36\,\mathrm{mJ\,K^{-2}\,mol^{-1}}$ [35, Table II]. The ratio is $1+\lambda_\mathrm{ep} = 1.43$. The specific heat alone measures $\lambda_\mathrm{ep} = 0.43$, without any phonon calculation.

The Eliashberg function has a different shape from the phonon density of states, and Fig. 6.7 shows by how much for aluminum. The model behind it is the simplest that still has the right ingredients. The phonons come from force constants between first and second neighbors of the fcc lattice, with $a = 4.05\,\text{Å}$ and $M = 26.98\,\mathrm{u}$. A neighbor at the lattice vector $\boldsymbol{R}$ contributes its spring tensor times $1-\cos(\boldsymbol{Q}\cdot\boldsymbol{R})$ to the dynamical matrix, divided by M. The tensor of the first neighbor at $(a/2)(1,1,0)$ has the elements 0.648 ($xx = yy$), -0.187 (zz) and 0.574 (xy), and that of the second neighbor at $a(1,0,0)$ has 0.038 (xx) and -0.052 ($yy = zz$), all in $\mathrm{eV\,Å^{-2}}$. These five numbers are a least-squares fit to the measured zone-boundary frequencies, 9.69 and 5.78 THz (longitudinal and transverse) at X and 9.69 and 4.19 THz at L [35, Table I], and to the elastic constants $C_{11} = 106.8$, $C_{12} = 60.7$ and $C_{44} = 28.2\,\mathrm{GPa}$ at a density of $2.70\,\mathrm{g\,cm^{-3}}$ [36], the room-temperature values of that source. They reproduce the transverse frequencies at X and L to 2 % and the longitudinal one at L to 8 %, and the largest phonon energy is 40 meV.

The electrons are a free gas of three electrons per atom, with $k_\mathrm{F} = 1.75\,\mathrm{Å^{-1}}$ and $E_\mathrm{F} = 11.7\,\mathrm{eV}$. The potential of each ion is a Coulomb potential $-3e^2/(4\pi r)$ outside a core radius r_c and zero inside, screened by the Lindhard function of Sect. 4.5. Its Fourier transform per cell is $w(q) = -3e^2\cos(qr_\mathrm{c})/[\Omega q^2\varepsilon_\mathrm{Lin}(q,0)]$. For plane waves and one atom per cell, Eq. (A.26) becomes $g_\nu = (2M\omega_{\boldsymbol{Q}\nu})^{-1/2}(\boldsymbol{q}\cdot\boldsymbol{e}_\nu)\,w(q)$, where $\boldsymbol{q} = \boldsymbol{Q}+\boldsymbol{G}$ is the full momentum transfer, umklapp included. The sums over $\boldsymbol{k}$ and over the reciprocal-lattice vectors become integrals, $N^{-1}\sum_{\boldsymbol{k}} \to \Omega\int \mathrm{d}^3k/(2\pi)^3$ and $N^{-1}\sum_{\boldsymbol{Q}}\sum_{\boldsymbol{G}} \to \Omega\int \mathrm{d}^3q/(2\pi)^3$. The double delta function integrates to $\int \mathrm{d}^3k\,(2\pi)^{-3}\delta(\xi_{\boldsymbol{k}})\delta(\xi_{\boldsymbol{k}+\boldsymbol{q}}) = m_e^2/(4\pi^2 q)$ for $q < 2k_\mathrm{F}$ (Appendix D.17), and with

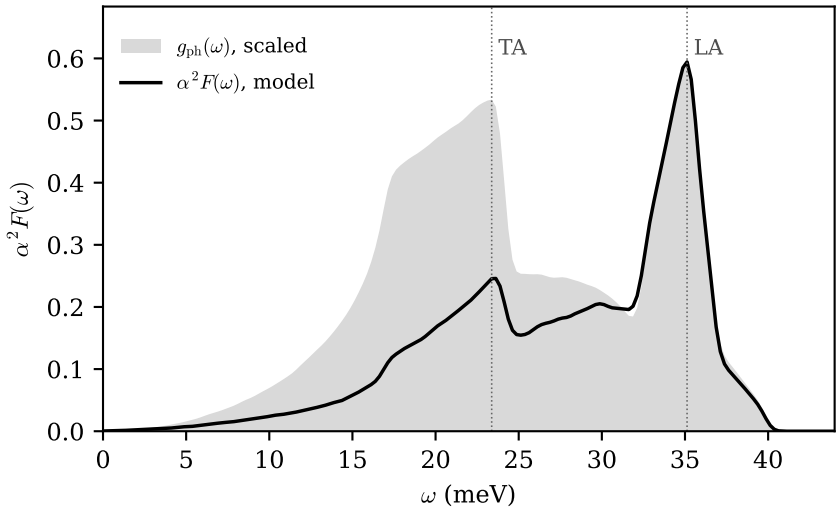


Figure 6.7: Eliashberg function of aluminum (black line) in a model of free electrons scattered by screened empty-core ions, Eq. (6.37), with the core radius, 0.56 Å, taken where λ_{ep} is stationary in it (between 0.55 and 0.59 Å, λ_{ep} stays within 0.44–0.46), with phonons from force constants between first and second neighbors fitted to measured zone-boundary frequencies and elastic constants. The gray area is the phonon density of states $g_{\mathrm{ph}}(\omega)$ of the same force constants, scaled to the height of the α^2F peak. Dotted lines mark the transverse (TA) and longitudinal (LA) maxima of g_{ph}. In α^2F the transverse peak is suppressed, although three quarters of the modes lie below 32 meV, because a transverse displacement couples to an electron only when the momentum transfer leaves the first zone. The model gives $\lambda_{\mathrm{ep}} = 0.436$ and a logarithmic average frequency $\omega_{\log} = 250$ K, against 0.44 and 270 K from linear-response calculations and 0.42 from tunneling [35]. Both curves are histograms in bins of 0.25 meV smoothed by a Gaussian of 0.3 meV.

$\mathcal{N}_{\mathrm{F}} = m_e k_{\mathrm{F}}/2\pi^2$, Eq. (6.34) becomes

$$\alpha^2F(\omega) = \frac{m_e\Omega}{2k_{\mathrm{F}}} \int_{q<2k_{\mathrm{F}}} \frac{\mathrm{d}^3q}{(2\pi)^3} \sum_{\nu} \frac{(\hat{\boldsymbol{q}}\cdot\boldsymbol{e}_\nu)^2\, q\,|w(q)|^2}{2M\omega_{\boldsymbol{q}\nu}}\, \delta(\omega-\omega_{\boldsymbol{q}\nu}). \tag{6.37}$$

The core radius is the only free parameter, a choice of the model. As a function of r_{c}, λ_{ep} has a flat minimum at $r_{\mathrm{c}} = 0.56$ Å, where the zero of $\cos(qr_{\mathrm{c}})$ sits among the umklapp momentum transfers between $|\boldsymbol{G}_{111}| = 2.69\ \text{Å}^{-1}$ and $2k_{\mathrm{F}} = 3.50\ \text{Å}^{-1}$. We take r_{c} there, where the result depends least on it; between 0.55 and 0.59 Å the coupling stays between 0.44 and 0.46. The model uses no measured coupling.

The result is $\lambda_{\mathrm{ep}} = 0.436$, against 0.44 from a linear-response calculation and 0.42 from tunneling spectra [35, Table II], and 0.43 from the specific heat above. Savrasov and Savrasov found their first-principles α^2F practically identical to an empirical rigid-ion pseudopotential result of this kind [35], and aluminum is the metal where the free-electron picture is expected to work. The shape carries more information than the number. Compared to the density of states, the transverse peak near 23 meV is strongly suppressed. The reason is the factor $(\hat{\boldsymbol{q}}\cdot\boldsymbol{e}_\nu)^2$: within the first zone a transverse wave moves the ions perpendicular to $\boldsymbol{q}$, and an electron feels it only through an umklapp process, where $\boldsymbol{q} = \boldsymbol{Q}+\boldsymbol{G}$ has a component along the displacement. This is the suppression of the low-frequency peak by $\alpha^2(\omega)$ that Savrasov and Savrasov note for aluminum [35]. The logarithmic average frequency, $\omega_{\log} = \exp[(2/\lambda_{\mathrm{ep}}) \int \mathrm{d}\omega\, \alpha^2F(\omega) \ln\omega/\omega]$, which sets the prefactor of the critical temperature in Appendix E.9, comes out as 250 K,

against 270 K from the linear-response calculation [35, Table III]. With the zero-temperature elastic constants of Appendix C.18, $C_{11} = 114.3$, $C_{12} = 61.9$ and $C_{44} = 31.6$ GPa, in place of the room-temperature ones, the fit gives $\lambda_{\mathrm{ep}} = 0.415$ and $\omega_{\log} = 259$ K, and the two-peak model of Worked example 6.4 gives 0.385 in place of 0.396. A constant α^2 with the same total weight would give 0.50 and 228 K. Summed over the zone with Eq. (6.35), the model widths average $\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}/\omega_{\boldsymbol{Q}\nu} = 4.4 \times 10^{-3}$ over all modes, or 0.12 meV: a phonon in aluminum decays into electron–hole pairs after about 36 periods, much as the optical phonon of silicon decays anharmonically.

Worked example 6.4: Coupling constant of aluminum from two Einstein peaks

Replace the model α^2F of Fig. 6.7 by two delta functions at the maxima of the phonon density of states, $\omega_{\mathrm{TA}} = 23.4$ meV and $\omega_{\mathrm{LA}} = 35.1$ meV, each carrying the weight of α^2F on its side of the minimum of g_{ph} at 31.9 meV. The weights are $I_{\mathrm{TA}} = \int_0^{31.9} \mathrm{d}\omega\, \alpha^2F = 3.13$ meV and $I_{\mathrm{LA}} = 2.26$ meV. Below the minimum lie 2.24 of the three modes per atom and above it 0.76, a ratio of 2.9; the weights of α^2F stand in the ratio 1.4, so each transverse mode couples half as strongly as a longitudinal one.
With $\alpha^2F = I_{\mathrm{TA}}\delta(\omega - \omega_{\mathrm{TA}}) + I_{\mathrm{LA}}\delta(\omega - \omega_{\mathrm{LA}})$, Eq. (6.35) gives

$$\lambda_{\mathrm{ep}} = 2\Big(\frac{I_{\mathrm{TA}}}{\omega_{\mathrm{TA}}} + \frac{I_{\mathrm{LA}}}{\omega_{\mathrm{LA}}}\Big) = 2\,(0.134 + 0.064) = 0.396,$$

9 % below the full model's 0.436 and 6 to 8 % below the measured 0.42–0.43. The deficit is all in the transverse peak. The same weight spread over the low-frequency tail of Fig. 6.7 counts for more, since $1/\omega$ is convex, and the full transverse part is 0.306 instead of $2 \times 0.134 = 0.268$; the longitudinal part is 0.130 in both. Although the longitudinal peak is the taller, the transverse branches carry 70 % of the coupling. For the two-peak model $\omega_{\log} = \exp[(2/\lambda_{\mathrm{ep}})(I_{\mathrm{TA}} \ln \omega_{\mathrm{TA}}/\omega_{\mathrm{TA}} + I_{\mathrm{LA}} \ln \omega_{\mathrm{LA}}/\omega_{\mathrm{LA}})] = 26.7$ meV $= 310$ K, too high for the same reason. With $\lambda_{\mathrm{ep}} = 0.44$, the mass enhancement $1 + \lambda_{\mathrm{ep}} = 1.44$ is what the specific heat measures.

In practice $\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}$ and α^2F are computed from Eqs. (6.32) and (6.34) with the vertices of density-functional perturbation theory, Wannier-interpolated to fine grids of $\boldsymbol{k}$ and $\boldsymbol{Q}$ as described in Sect. 6.2. The two delta functions that pin both electrons to the Fermi surface make these the slowest-converging integrals of electron–phonon theory, and Appendix E.5 lists the grids and smearings they need. Appendix E.9 takes the resulting α^2F into the Eliashberg equations for the gap of aluminum, which come out at the measured 0.18 meV once the Coulomb repulsion between paired electrons is fixed by the measured critical temperature.

The phonon-mediated attraction that binds those pairs is the exchange of the phonons whose widths we have just summed. Sect. 7.2 reduces it to the model interaction of Bardeen, Cooper and Schrieffer, with λ_{ep} as its strength and ω_{D} as its range in energy, and finds the gap that sets the threshold of a superconducting detector.

6.5 Phonon-assisted optical absorption

Silicon solar cells absorb the whole visible spectrum, from about 1.6 to 3.3 eV, and silicon photodiodes respond down to its gap of 1.17 eV. Yet the smallest vertical gap of silicon, E_{dir}, is 3.4 eV (Sect. 3.3). A photon carries almost no crystal momentum, and on its own it can lift an electron only vertically, from $(v, \boldsymbol{k})$ to $(c, \boldsymbol{k})$. Every photon that silicon absorbs between 1.17 and 3.4 eV is therefore absorbed together with a phonon, which carries the crystal momentum

$\boldsymbol{k}_\mathrm{c}$ from the valence maximum at Γ to a conduction valley. The phonon costs or returns a few tens of meV, and it can be created or destroyed. It is the simplest composite process: two vertices, the photon's and the electron–phonon vertex g of Sect. 6.2, joined through virtual intermediate states with the energy denominators of Sect. 2.4.

Hall, Bardeen and Blatt worked out the theory in 1954 for germanium [37]. Near the edge the absorption coefficient grows as the square of the photon energy above a threshold, and each phonon branch that couples adds two thresholds, $E_\mathrm{g} - \omega$ for phonon absorption and $E_\mathrm{g} + \omega$ for phonon emission. Four years later Macfarlane and co-workers read the phonon energies of silicon off its absorption edge [38]. The process matters for dark matter twice. An absorbed dark-matter boson has even less momentum for its energy than a photon, and between the indirect and the direct gap it is absorbed only with a phonon (Chap. 8). A scattering dark-matter particle just above its threshold mass delivers its energy at momentum transfers too small to reach $\boldsymbol{k}_\mathrm{c}$, and a phonon again makes up the difference (Sect. 13.1). We will derive the rate in a form that applies to both, compute its edge, and read the silicon thresholds off it.

6.5.1 The optical vertex

A uniform optical field $\boldsymbol{\mathcal{E}}(t) = -\mathcal{E}\,\hat{\boldsymbol{e}}_\gamma \cos\omega t$, with unit polarization vector $\hat{\boldsymbol{e}}_\gamma$, is described by the vector potential $\boldsymbol{A}(t) = (\mathcal{E}/\omega)\,\hat{\boldsymbol{e}}_\gamma \sin\omega t$. To first order in $\boldsymbol{A}$ it couples to the electrons through

$$H_A = e\,\boldsymbol{A}\cdot\hat{\boldsymbol{V}}, \qquad \hat{\boldsymbol{V}} = \sum_{\boldsymbol{k}}\sum_{mn} \boldsymbol{v}_{mn}(\boldsymbol{k})\, c^\dagger_{m\boldsymbol{k}} c_{n\boldsymbol{k}}, \tag{6.38}$$

where $\hat{\boldsymbol{V}}$ is the total velocity of the electrons, written over Bloch states with the velocity matrix elements $\boldsymbol{v}_{mn}(\boldsymbol{k})$ of Eq. (3.38), and the spin is summed implicitly. For a local potential $\hat{\boldsymbol{V}} = \sum_j \hat{\boldsymbol{p}}_j/m_e$. The operator conserves $\boldsymbol{k}$, because the velocity commutes with lattice translations. The part of H_A that oscillates as $\mathrm{e}^{-\mathrm{i}\omega t}$ absorbs the energy ω, with the amplitude $(e\mathcal{E}/2\omega)\,\hat{\boldsymbol{e}}_\gamma\cdot\hat{\boldsymbol{V}}$ up to a phase (Appendix D.18).

Eq. (6.38) is the *velocity form* of the optical vertex. Its alternative, the length form $e\boldsymbol{\mathcal{E}}\cdot\sum_j \boldsymbol{r}_j$, is awkward in a crystal, since the position has no matrix elements between extended Bloch states except through the commutator of Eq. (3.39). The velocity form needs just $\boldsymbol{v} = \mathrm{i}[H_\mathrm{e}, \boldsymbol{r}]$, which stays well defined when the potential is nonlocal (Eq. (3.40)). It is also the vertex that dark matter shares with light. By Eq. (3.43), the density vertex at small momentum transfer is $\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) \to \boldsymbol{q}\cdot\boldsymbol{v}_{mn}/(\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}})$. On the energy shell the denominator is ω, and a longitudinal field coupled through the density and a transverse field coupled through the velocity carry the same matrix element, $\hat{\boldsymbol{e}}_\gamma\cdot\boldsymbol{v}_{mn}/\omega$. Replacing the photon by dark matter replaces this optical limit by the full density vertex, and nothing else in the structure changes.

At first order the golden rule with H_A gives the direct absorption of Sect. 4.4. The absorbed power per unit volume is $\sigma_1\mathcal{E}^2/2$, and with $\sigma_1 = \omega\varepsilon_2$ (Eq. (4.50)) the rate becomes the familiar sum over vertical transitions,

$$\varepsilon_2(\omega) = \frac{g_\mathrm{deg}\,\pi e^2}{\omega^2 \mathcal{V}} \sum_{\boldsymbol{k}}\sum_{cv} \left|\hat{\boldsymbol{e}}_\gamma\cdot\boldsymbol{v}_{cv}(\boldsymbol{k})\right|^2 \delta(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}} - \omega), \tag{6.39}$$

for $T \ll E_\mathrm{g}$, with $g_\mathrm{deg} = 2$ for the spin [39, Chap. 6]. In a frozen lattice this vanishes below E_dir. Below it, light needs something else to supply the crystal momentum. A phonon can, and so can an impurity or a free carrier; in a pure, cold crystal the phonon is all there is. The absorption coefficient, the inverse length over which the intensity falls by e, follows from ε_2

and the refractive index $n_r(\omega)$ as

$$\alpha_{\text{abs}}(\omega) = \frac{\omega\, \varepsilon_2(\omega)}{n_r(\omega)}, \tag{6.40}$$

since the intensity decays as $\mathrm{e}^{-2\omega\kappa_{\text{ext}}x}$ with the extinction coefficient $\kappa_{\text{ext}} = \varepsilon_2/2n_r$ (Appendix D.18).

6.5.2 The phonon-assisted amplitude

Add the electron–phonon Hamiltonian H_{ep} of Eq. (A.25) and count the phonons in the states. The initial state $|i\rangle$ has filled valence bands, empty conduction bands and $n_{\boldsymbol{Q}\nu}$ phonons in each mode. The final state has an electron in $(c, \boldsymbol{k}+\boldsymbol{Q})$, a hole in $(v, \boldsymbol{k})$, and one phonon fewer in the mode $(\boldsymbol{Q}, \nu)$ or one more in $(-\boldsymbol{Q}, \nu)$. No single vertex connects the two. H_A conserves $\boldsymbol{k}$ and does not touch the phonons, and H_{ep} changes $\boldsymbol{k}$ but cannot create a pair across a gap of an electronvolt with tens of meV. The second-order amplitude of Eq. (2.51), linear in H_A and linear in H_{ep}, does both. Its long-time limit is the rate $2\pi|T_{fi}|^2\delta(E_f - E_i - \omega)$ of Eq. (2.52), and T_{fi} has the two time orderings of Eq. (2.54): the photon is absorbed first and the phonon acts second, or the other way round.

Fig. 6.8 draws the two orderings for silicon in the one-electron picture. If the photon acts first, it lifts the electron vertically at $\boldsymbol{k}$ into some empty band p, and the phonon then scatters it to $(c, \boldsymbol{k}+\boldsymbol{Q})$. The intermediate state lies above $E_i + \omega$ by $\varepsilon_{p\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}} - \omega$. Near the edge of silicon, with $\omega \approx E_{\text{g}}$ and the lowest vertical transition at 3.4 eV, the electron is off the energy shell by about 2 eV. If the phonon acts first, it scatters the valence electron to a state $(p, \boldsymbol{k}+\boldsymbol{Q})$, and the photon then lifts it vertically to $(c, \boldsymbol{k}+\boldsymbol{Q})$. The photon's energy has not yet been absorbed, and the intermediate state lies above E_i by $\varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$, with the upper sign for emission.

The one-electron picture ignores Pauli blocking, which Sect. 2.4 anticipated: the electron cannot enter a band p that is already full. In silicon the natural intermediate state of the phonon-first ordering is a valence state at $\boldsymbol{k}_{\text{c}}$, which is occupied. The many-electron amplitude has, in its place, two hole orderings, the Z graphs of old-fashioned perturbation theory. In one, the photon first creates the electron in $(c, \boldsymbol{k}+\boldsymbol{Q})$ from an occupied $(p, \boldsymbol{k}+\boldsymbol{Q})$, and the phonon then moves an electron from $(v, \boldsymbol{k})$ into the hole left behind. In the other, the phonon creates the electron in $(c, \boldsymbol{k}+\boldsymbol{Q})$ from an occupied $(p, \boldsymbol{k})$, and the photon then fills that hole from $(v, \boldsymbol{k})$. Reordering the creation and annihilation operators attaches a minus sign to each. On the energy shell their denominators turn into those of the electron orderings with the band p occupied, and the sign is what makes them fit (Appendix D.18). The four paths combine into a single one-electron amplitude, summed over every band p, occupied or empty:

$$M^{\pm}_{cv\nu}(\boldsymbol{k}, \boldsymbol{Q}) = \sum_p \left[\frac{g_{cp\nu}(\boldsymbol{k}, \boldsymbol{Q})\; \hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{pv}(\boldsymbol{k})}{\omega - \varepsilon_{p\boldsymbol{k}} + \varepsilon_{v\boldsymbol{k}}} + \frac{\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{cp}(\boldsymbol{k}+\boldsymbol{Q})\; g_{pv\nu}(\boldsymbol{k}, \boldsymbol{Q})}{\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} \mp \omega_{\boldsymbol{Q}\nu}} \right]. \tag{6.41}$$

The common factors $e\mathcal{E}/2\omega$, $N^{-1/2}$ from Eq. (A.25) and the phonon matrix element $(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})^{1/2}$ of Eq. (A.28) have been taken out. The sum includes $p = v$ and $p = c$, in which the photon acts as an intraband velocity and the phonon does all the interband work. Refs. [3, Eq. (200)] and [40, Eqs. (2) and (3)] write the same sum with both denominators of opposite sign, which leaves $|M^{\pm}|^2$ unchanged. Note that the phonon energy enters twice: in the denominator of the phonon-first term, where against electronic energies of electronvolts it is usually negligible, and in the energy balance $\omega = \varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$, where it shifts the threshold.

A rough size follows from Eq. (6.41) before any calculation. Compared to a direct transition with a velocity matrix element of similar size, the phonon-assisted amplitude carries an extra

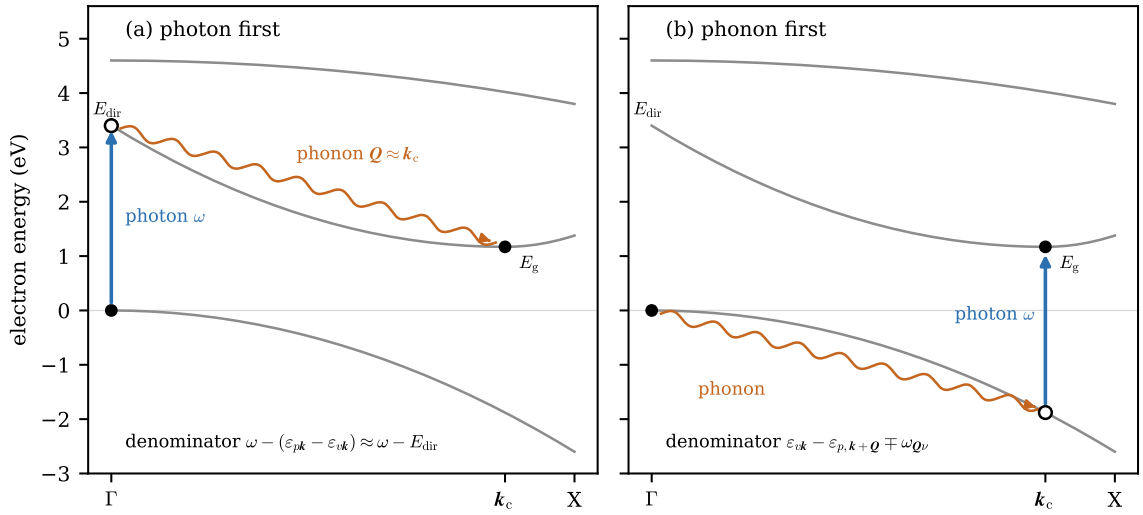


Figure 6.8: The two time orderings of phonon-assisted absorption across the indirect gap of silicon, on schematic bands along Γ–X. Filled dots: the initial electron at the valence maximum at Γ and the final electron at the conduction minimum near $\boldsymbol{k}_{\mathrm{c}} = 0.85\,(2\pi/a)$, 1.17 eV higher. Open dots: virtual intermediate states. Straight arrows: the vertical optical vertex $\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}$; wavy arrows: the electron–phonon vertex g, which changes the crystal momentum by $\boldsymbol{Q} \approx \boldsymbol{k}_{\mathrm{c}}$ and the energy by just $\pm\omega_{\boldsymbol{Q}\nu}$, tens of meV. Energy is not conserved at the intermediate state; each panel gives the energy denominator with which its amplitude enters Eq. (6.41). (a) Photon first: the intermediate electron sits at Γ, and the denominator vanishes when ω reaches the vertical gap $E_{\mathrm{dir}} = 3.4\,\mathrm{eV}$. (b) Phonon first: the intermediate state is a valence state at $\boldsymbol{k}_{\mathrm{c}}$, which is occupied. The process then runs through hole orderings, in which an electron from an occupied band is promoted first and the valence electron fills the hole it leaves (sign -1), and its denominator never vanishes. Band curves drawn by hand; only E_{g}, E_{dir} and $\boldsymbol{k}_{\mathrm{c}}$ are measured values.

factor of the vertex over a denominator. Typical optical-phonon vertices in silicon are a few tenths of an eV (Sect. 6.2), and the denominators near the silicon edge are 2 to 3 eV, which puts $|M^{\pm}|^2$ at $\sim 10^{-2}$ of the squared optical vertex for each phonon mode that couples. The phonon-assisted ε_2 near the indirect edge is then $\sim 10^{-2}$ of the direct one above E_{dir}, before phase space and occupations are counted. This is why silicon absorbs weakly between its two gaps, and why, for dark matter, the phonon-assisted rate leads only where the zero-phonon rate is kinematically forbidden.

The photon-first denominator vanishes when $\omega = \varepsilon_{p\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}$, that is, when the photon alone could create a real pair at $\boldsymbol{k}$. In silicon this happens first at E_{dir}. There the intermediate electron becomes real, and Sect. 2.4 gives the treatment. A real electron decays, by phonon emission among other channels, with a width Γ_{w} (Sect. 6.3), and the denominator acquires $+\mathrm{i}\Gamma_{\mathrm{w}}/2$ (Eq. (2.57)). By Eq. (2.60) the squared resonant term then splits into an on-shell part and a finite part. The on-shell part is a cascade: direct absorption at $\boldsymbol{k}$, followed by the scattering of the real electron by a phonon, weighted by its branching ratio. Direct absorption already counts this process once, in Eq. (6.39) with a broadened delta function. Adding the full phonon-assisted ε_2 to the direct one above E_{dir} counts the cascade twice. Below E_{dir} the problem does not arise: every intermediate state is off shell, by electronvolts at the indirect edge. First-principles calculations in silicon give the intermediate states a constant width of 100 meV and find the spectrum between the gaps insensitive to its value [40], which is the numerical form of the same statement.

6.5.3 The phonon-assisted dielectric function

The rest is bookkeeping, done as for Eq. (6.39): T_{fi} goes into the rate, the rate into the absorbed power, and the power into ε_2. Summing over the final states, labeled by $\boldsymbol{k}$, $\boldsymbol{Q}$, ν, c, v, the spin and the sign, we obtain (Appendix D.18)

Phonon-assisted dielectric function

$$\varepsilon_2^{\mathrm{PA}}(\omega) = \frac{g_{\mathrm{deg}}\,\pi e^2}{\omega^2\,\Omega} \left\langle\!\!\left\langle \sum_{\nu} \sum_{cv} \sum_{\pm} \left(n_{\boldsymbol{Q}\nu} + \tfrac{1}{2} \pm \tfrac{1}{2}\right) \left|M^{\pm}_{cv\nu}(\boldsymbol{k},\boldsymbol{Q})\right|^2 \delta\left(\varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu} - \omega\right) \right\rangle\!\!\right\rangle, \tag{6.42}$$

where $\langle\!\langle \cdots \rangle\!\rangle$ is the average over the electron wavevector $\boldsymbol{k}$ and the phonon wavevector $\boldsymbol{Q}$ in the zone of Eq. (A.11), and $g_{\mathrm{deg}} = 2$ counts the spin. The formula holds for $T \ll E_{\mathrm{g}}$, where thermally excited pairs, which would run the process backwards, are absent. With Eq. (6.40) it is the absorption coefficient used in first-principles work, written there in Gaussian units, $e_{\mathrm{G}}^2 = e^2/4\pi$ (Appendix A.2), with the spin factor explicit [40, Eq. (1)], and in Ref. [3, Eq. (200)] with the spin sum implicit. Eq. (6.42) adds to the direct ε_2 of Eq. (6.39), and below E_{dir} it is all there is.

The formula contains three sums, and each sets a cost. The sum over intermediate bands p, inside $M^{\pm}$, runs over occupied and empty bands alike, and it converges no faster than its denominators grow; a calculation must include bands several electronvolts above the conduction minimum. The sum over phonon branches is short, six in silicon. The average over $\boldsymbol{k}$ and $\boldsymbol{Q}$ is the expensive one. The two wavevectors range independently over the whole zone, linked by the energy delta function alone, and the onset region needs both on fine grids. An energy resolution of 14 meV in silicon took 40^3 points for each, 4×10^9 pairs, each with its own vertices for every pair of bands [40].

In practice no calculation computes so many vertices directly. The energies, velocities and vertices are computed by density-functional perturbation theory on coarse grids, rotated

to Wannier functions, and interpolated to the fine grids, with every quantity in one gauge (Sects. 3.6 and 6.2); Appendix E.5 describes the procedure and its convergence. Two corrections enter the optical vertex. With a nonlocal pseudopotential the velocity includes the commutator of the nonlocal part (Eq. (3.40) and Appendix E.7), and when the gap is corrected the velocities are rescaled by the ratio of corrected to Kohn–Sham transition energies (Eq. (E.19)). A different route avoids the double sum altogether. A supercell whose nuclei are frozen in a special displaced configuration breaks translation symmetry, the photon then connects states of different crystal momentum, and one static calculation of ε_2 in that supercell contains the phonon-assisted absorption at a given temperature (Appendix E.8). Being adiabatic, it does not distinguish the emission of a phonon from its absorption, and it cannot resolve the two thresholds $E_\mathrm{g} \pm \omega_{\boldsymbol{Q}\nu}$ that the next subsection derives.

6.5.4 The Hall–Bardeen–Blatt edge

Just above the threshold, Eq. (6.42) simplifies enough to be done by hand. The final hole lies near the valence maximum at $\boldsymbol{k} = 0$, and the final electron near one of the conduction minima $\boldsymbol{k}_\mathrm{c}$. We measure their crystal momenta from the extrema, $\boldsymbol{k} = \boldsymbol{\kappa}_\mathrm{h}$ and $\boldsymbol{k} + \boldsymbol{Q} = \boldsymbol{k}_\mathrm{c} + \boldsymbol{\kappa}_\mathrm{e}$, and take the bands as parabolic, with kinetic energies $E_\mathrm{h} = \kappa_\mathrm{h}^2/2m_\mathrm{h}$ and $E_\mathrm{e} = \kappa_\mathrm{e}^2/2m_\mathrm{d}^*$. Here m_h is the mass of one valence band and $m_\mathrm{d}^* = (m_\parallel^* m_\perp^{*2})^{1/3}$ the density-of-states mass of one valley, $0.322\,m_e$ in silicon with the valley masses of Table 3.1. Two approximations make the sums elementary. First, the phonon wavevector $\boldsymbol{Q} = \boldsymbol{k}_\mathrm{c} + \boldsymbol{\kappa}_\mathrm{e} - \boldsymbol{\kappa}_\mathrm{h}$ stays within the small pockets around $\boldsymbol{k}_\mathrm{c}$, and the phonon energies and vertices may be taken at $\boldsymbol{Q} = \boldsymbol{k}_\mathrm{c}$; we write $\omega_\nu \equiv \omega_{\boldsymbol{k}_\mathrm{c}\nu}$ and $n_\nu = n_\mathrm{B}(\omega_\nu)$. Second, the denominators of Eq. (6.41) are electronic energy differences of an electronvolt or more, far from any resonance, and they barely change across the pockets. The amplitude $|M^\pm|^2$ is then a constant, its value $|M_\nu^\pm|^2$ at the extrema.

What remains is phase space. The averages over $\boldsymbol{k}$ and $\boldsymbol{Q}$ become independent integrals over $\boldsymbol{\kappa}_\mathrm{h}$ and $\boldsymbol{\kappa}_\mathrm{e}$, each of which turns into an energy integral with the density of states of a parabolic band, $\int \mathrm{d}^3\kappa\,(2\pi)^{-3} = \int \mathrm{d}E\,(2m)^{3/2}E^{1/2}/4\pi^2$. The delta function fixes $E_\mathrm{e} + E_\mathrm{h} = x$, the energy above the threshold, and

$$\begin{aligned}\Omega^2 \int \frac{\mathrm{d}^3\kappa_\mathrm{h}}{(2\pi)^3}\frac{\mathrm{d}^3\kappa_\mathrm{e}}{(2\pi)^3}\,\delta(x - E_\mathrm{e} - E_\mathrm{h}) &= \frac{\Omega^2\,(2m_\mathrm{d}^*)^{3/2}(2m_\mathrm{h})^{3/2}}{16\pi^4}\int_0^x \mathrm{d}E_\mathrm{e}\,E_\mathrm{e}^{1/2}(x - E_\mathrm{e})^{1/2} \\ &= \frac{\Omega^2\,(m_\mathrm{d}^* m_\mathrm{h})^{3/2}}{16\pi^3}\,x^2,\end{aligned} \tag{6.43}$$

for $x \geq 0$, since the last integral is $\pi x^2/8$. Two densities of states that each grow as $\varepsilon^{1/2}$, convolved, grow as x^2. A direct edge has one free momentum, because the photon ties the electron to the hole, and its joint density of states grows as $x^{1/2}$; the phonon frees the second momentum and adds a power 3/2. Collecting the factors, each valley, valence band and branch adds a term

Hall–Bardeen–Blatt edge

$$\begin{aligned}\alpha_\mathrm{abs}(\omega) \simeq \sum_\nu A_\nu \Big[& n_\nu\,(\omega - E_\mathrm{g} + \omega_\nu)^2\,\Theta(\omega - E_\mathrm{g} + \omega_\nu) \\ & + (n_\nu + 1)\,(\omega - E_\mathrm{g} - \omega_\nu)^2\,\Theta(\omega - E_\mathrm{g} - \omega_\nu)\Big],\end{aligned} \tag{6.44}$$

with

$$A_\nu = \frac{g_\mathrm{deg}\,e^2\,\Omega\,(m_\mathrm{d}^* m_\mathrm{h})^{3/2}}{16\pi^2\,\omega\,n_r} \sum_{\text{valleys}} |M_\nu|^2,$$

constant across the edge. In A_ν we have set $|M_\nu^+| = |M_\nu^-| \equiv |M_\nu|$, which holds when ω_ν is negligible in the denominators, and treated ω and n_r as constant over the width of the edge, a few times ω_ν or about 0.1 eV. Heavy and light holes enter as separate valence bands, each with its own m_h. This is the form of Hall, Bardeen and Blatt [37], derived in Appendix D.18.

Eq. (6.44) turns the edge into a spectroscopy. Each branch that couples contributes two components with the same coefficient A_ν: one starting at $E_\mathrm{g} - \omega_\nu$, where the photon is helped by a phonon absorbed from the lattice, and one at $E_\mathrm{g} + \omega_\nu$, where the photon pays for a phonon emitted. The midpoint of each pair of thresholds is E_g, and their separation is $2\omega_\nu$. Each component is a straight line in a plot of $\sqrt{\alpha_\mathrm{abs}}$ against ω, with slope $\sqrt{A_\nu n_\nu}$ or $\sqrt{A_\nu(n_\nu+1)}$. The sum of several components is not a straight line, and the measured curve is decomposed by subtracting, at each threshold, the extrapolation of the components that started below it. This is how the phonon energies of silicon were first extracted from optical data.

The quadratic law has two caveats. It assumes that M_ν is finite at the extrema, that is, that the transition is *allowed* there. If symmetry forces M_ν to vanish at the extrema, it grows linearly with κ_e or κ_h, an extra power of energy enters the integrand of Eq. (6.43), and the forbidden edge grows as x^3. And it neglects the attraction between the electron and the hole. In silicon the bound pair lies $E_x^b = 14.7\,\mathrm{meV}$ below the gap at total crystal momentum $\boldsymbol{k}_\mathrm{c}$ (Sect. 4.9), and a photon reaches it with a phonon just as it reaches a free pair. Each threshold of Eq. (6.44) therefore has an excitonic companion E_x^b lower, with its own onset shape, and the kinks of a measured edge are exciton thresholds at $E_\mathrm{g} - E_x^b \pm \omega_\nu$.

6.5.5 Temperature dependence

All of the temperature dependence of Eq. (6.44) at a fixed gap sits in the occupations. The component that absorbs a phonon is proportional to n_ν, the number of phonons available; the one that emits is proportional to $n_\nu + 1$, spontaneous plus stimulated emission. Their ratio,

$$\frac{n_\nu}{n_\nu + 1} = \mathrm{e}^{-\omega_\nu/T}, \tag{6.45}$$

is detailed balance (Eq. (2.68)) for the phonon system alone: the two components have the same $|M_\nu|^2$, and the lattice gives up an energy ω_ν with a Boltzmann penalty relative to taking it. The ratio of the slopes of the two components in $\sqrt{\alpha_\mathrm{abs}}$ is $\mathrm{e}^{-\omega_\nu/2T}$, and the edge is a thermometer of the lattice that needs no knowledge of A_ν. At high temperature both components grow as T/ω_ν, and the total weight of a branch, proportional to $2n_\nu + 1$, rises with it.

As $T \to 0$ the absorption components vanish. A cold crystal has no phonons to give, and every photon absorbed at the indirect edge must create one. The edge then sits at $E_\mathrm{g} + \omega_\nu$ for the softest branch that couples, and the energy ω_ν of the emitted phonon goes into the lattice, not into the pair. At the 10 mK of a dark-matter detector this is exact for every practical purpose: the transverse-acoustic occupation in silicon is vanishingly small (Worked example 6.5).

The gap itself also moves with temperature. The occupations above are what the phonon-assisted formula contains explicitly; the bands in Eq. (6.42) are the bands the electrons actually occupy, renormalized by the same electron–phonon coupling (Sect. 6.3). In silicon the gap shrinks from 1.17 eV at low temperature to 1.12 eV at 300 K (Sect. 3.3), and all four thresholds of each branch move down with it. First-principles spectra that include the occupations alone must be shifted rigidly to match the measured onsets, by 0.15 to 0.23 eV in silicon, a shift that absorbs both the remaining error of the computed gap and its electron–phonon renormalization [40].

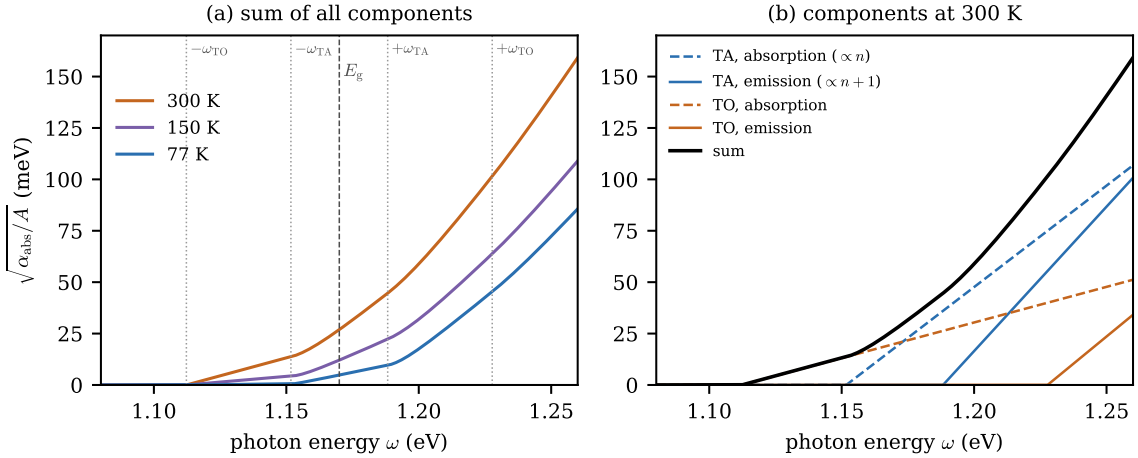


Figure 6.9: The Hall–Bardeen–Blatt edge of silicon, Eq. (6.44), plotted as $\sqrt{\alpha_{\text{abs}}/A}$ against the photon energy, with $E_{\text{g}} = 1.17\,\text{eV}$ and the phonon energies $\omega_{\text{TA}} = 18.27\,\text{meV}$ and $\omega_{\text{TO}} = 57.73\,\text{meV}$ measured from the optical edge [38]. Dotted lines mark the free-pair thresholds $E_{\text{g}} \mp \omega_{\text{TO}}$ and $E_{\text{g}} \mp \omega_{\text{TA}}$; the dashed line marks E_{g}, their common midpoint. (a) Sum of the four components at 300 K, 150 K and 77 K. (b) The components at 300 K: dashed, phonon absorption, proportional to n_ν; solid, phonon emission, proportional to $n_\nu + 1$. Each component is a straight line; the sum (black) is not. The two branches are given equal coefficients $A_{\text{TA}} = A_{\text{TO}} = A$, so their relative heights are illustrative. The gap is held at its low-temperature value at every temperature, to isolate the phonon occupations (Sect. 6.3 treats its shift). The excitonic thresholds, 14.7 meV lower, are omitted.

6.5.6 The silicon edge

Macfarlane, McLean, Quarrington and Roberts measured the absorption edge of silicon between 4.2 and 415 K, with enough resolution to see the thresholds of Eq. (6.44) [38]. They found four series of thresholds and quoted the phonon energies as temperatures θ, which in natural units are the energies themselves. Two of them, with $\theta = 212\,\text{K}$ and 670 K, are the transverse-acoustic and transverse-optical phonons at $\boldsymbol{k}_{\text{c}}$ [39, Chap. 6]. With $1\,\text{K} = 86.17\,\mu\text{eV}$ (Sect. 2.5) they are

$$\omega_{\text{TA}} = 18.27\,\text{meV}, \qquad \omega_{\text{TO}} = 57.73\,\text{meV}.$$

Neutron scattering measures the same branches at the zone boundary X, at 4.49 THz and 13.90 THz, or 18.57 meV and 57.49 meV (Sect. 5.2). Both branches are flat between $\boldsymbol{k}_{\text{c}} = 0.85\,\text{X}$ and X: in the computed dispersion of Sect. 5.2 they change by 0.1 % (TA) and 0.3 % (TO) over that interval. The optical and the neutron values measure the same phonons, and they agree to 2 % for the transverse-acoustic branch and to 0.4 % for the transverse-optical one. The measured thresholds were exciton thresholds, a binding energy below the free-pair ones, as Sect. 4.9 anticipated.

Fig. 6.9 shows Eq. (6.44) with these energies. Panel (b) separates the four components at 300 K and shows that each is a straight line in $\sqrt{\alpha_{\text{abs}}}$ with its own threshold, while their sum bends at every threshold. Panel (a) follows the sum down in temperature. The components that absorb a phonon fade first, the transverse-optical one long before the transverse-acoustic one, and at 77 K the edge is already dominated by emission.

Worked example 6.5: The phonon thresholds of silicon

With $E_\mathrm{g} = 1.17\,\mathrm{eV}$ and the phonon energies above, the four free-pair thresholds of Eq. (6.44) are

$$E_\mathrm{g} - \omega_\mathrm{TO} = 1.1123\,\mathrm{eV}, \qquad E_\mathrm{g} - \omega_\mathrm{TA} = 1.1517\,\mathrm{eV},$$
$$E_\mathrm{g} + \omega_\mathrm{TA} = 1.1883\,\mathrm{eV}, \qquad E_\mathrm{g} + \omega_\mathrm{TO} = 1.2277\,\mathrm{eV}.$$

The exciton binding $E_x^b = 14.7\,\mathrm{meV}$ moves each of them down by that amount, and a measured edge shows its kinks there. The pairs of thresholds are symmetric about E_g, separated by $2\omega_\mathrm{TA} = 36.5\,\mathrm{meV}$ and $2\omega_\mathrm{TO} = 115.5\,\mathrm{meV}$.
The slopes depend on the occupations alone, not on the gap. At 300 K, $T = 25.85\,\mathrm{meV}$, and

$$n_\mathrm{TA} = \frac{1}{\mathrm{e}^{18.27/25.85} - 1} = 0.973, \qquad n_\mathrm{TO} = \frac{1}{\mathrm{e}^{57.73/25.85} - 1} = 0.120.$$

In units of $\sqrt{A_\nu}$, the transverse-acoustic components of $\sqrt{\alpha_\mathrm{abs}}$ rise with slopes $\sqrt{n_\mathrm{TA}} = 0.987$ (absorption) and $\sqrt{n_\mathrm{TA}+1} = 1.405$ (emission), the transverse-optical ones with 0.346 and 1.058. The ratios of the slopes in each pair, 0.702 and 0.327, are $\mathrm{e}^{-\omega_\nu/2T}$ for the two branches, independent of A_ν. Read backwards, $-2T\ln 0.702 = 18.27\,\mathrm{meV}$: the slope ratio of one branch returns its phonon energy if the temperature is known, or the temperature if the phonon energy is. At 77 K, $T = 6.635\,\mathrm{meV}$, the occupations fall to $n_\mathrm{TA} = 0.068$ and $n_\mathrm{TO} = 1.7 \times 10^{-4}$, and the slope ratios to 0.252 and 0.013. At 20 K the transverse-acoustic absorption component has $n_\mathrm{TA} = 2.5 \times 10^{-5}$, and at the 10 mK of a cryogenic detector $\omega_\mathrm{TA}/T = 2.1 \times 10^4$: the occupation is e^{-21200}, and only the emission thresholds exist. The lowest of them is the true optical edge of cold silicon, 18 meV above the gap. With the measured low-temperature gap $E_\mathrm{g}(0) = 1.1692\,\mathrm{eV}$ of Sect. 6.3 in place of the rounded value it lies at $E_\mathrm{g}(0) + \omega_\mathrm{TA} = 1.1875\,\mathrm{eV}$ for a free pair, and the excitonic one at 1.1728 eV, since ω_TA and E_x^b nearly cancel.

6.5.7 Phonon-assisted dark-matter scattering

The derivation of Eq. (6.41) used only two properties of the photon: it couples to the electrons through a one-body operator, and it carries negligible momentum. A dark-matter particle that interacts with the electron density keeps the first fact and drops the second. Its vertex is the matrix element of $n_e(\boldsymbol{q}) = \int \mathrm{d}^3 r\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\, \hat{n}_e(\boldsymbol{r})$, the Fourier component of the density operator, which between Bloch states is the density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ of Eq. (3.34), multiplied by the dark-matter potential $\tilde{V}(\boldsymbol{q})$. The optical vertex $\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{pv}(\boldsymbol{k})$ is replaced by $\rho_{pv}(\boldsymbol{k},\boldsymbol{q})$, which is no longer vertical: it takes the electron from $\boldsymbol{k}$ to $\boldsymbol{k}+\boldsymbol{q}$. The phonon then supplies whatever crystal momentum is still missing, $\boldsymbol{Q} = \boldsymbol{k}_\mathrm{c} - \boldsymbol{q}$ up to a reciprocal-lattice vector for a transition from the valence maximum to a valley. The amplitude has the same orderings and the same hole paths, but its vertices and energies sit at three crystal momenta, $\boldsymbol{k}$, $\boldsymbol{k}+\boldsymbol{q}$ and $\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}$, instead of two.

The energy denominators take the energy deposited ω in place of the photon energy, and they resonate wherever ω matches a real pair at momentum transfer $\boldsymbol{q}$, with the same risk of double counting as at E_dir. Since ρ_{mn} reduces to the optical vertex as $q \to 0$ (Eq. (3.43)), the measured silicon edge is the zero-momentum limit of the dark-matter amplitude and a test of any calculation of it.

Two kinds of dark matter meet this amplitude in different limits. An absorbed boson, a kinetically mixed dark photon for instance, deposits its whole mass with a momentum $m_\phi v \approx 10^{-3} m_\phi$, a smaller momentum for its energy than a photon carries. Its absorption between E_g and E_dir is Eq. (6.42) with the photon's coupling replaced by its own and

screened by the medium (Chap. 8). A scattering particle transfers $\boldsymbol{q}$ and ω together, with $\omega \leq q v_{\max} - q^2/2m_\chi$ (Eq. (1.26)). Far above its threshold mass it reaches $\boldsymbol{k}_\mathrm{c}$ at first order, but just above it the energy needed to cross the gap is available at momentum transfers well below k_c, where the zero-phonon transition is forbidden (Sect. 3.3). At the threshold mass, $\omega = E_\mathrm{g}$ is reached just at $q = 2E_\mathrm{g}/v_{\max} = 0.88\,\mathrm{keV}$, and for heavier dark matter the smallest such q falls toward $E_\mathrm{g}/v_{\max} = 0.44\,\mathrm{keV}$; both lie well below $k_\mathrm{c} = 0.85\,(2\pi/a) = 1.9\,\mathrm{keV}$. There, as for light below E_dir, the phonon-assisted amplitude is the leading process, and it opens a window of masses in which silicon sees dark matter only through it (Sect. 13.1). In a cold detector, however, the emission branch is all there is: the phonon takes part of the energy deposited, and the pair receives $\omega - \omega_{\boldsymbol{Q}\nu}$. The rate in that window, the four-path amplitude at finite $\boldsymbol{q}$, and the screening it needs are the subject of Chap. 13; the optical edge measured in 1958 is the limit that every such calculation has to reproduce.

Summary

The Born–Oppenheimer approximation keeps only the diagonal of the coupling between electrons and nuclei, and the rest follows from the derivative coupling (Sect. 6.1). Both nonadiabatic couplings follow from the derivative-coupling vector $\boldsymbol{d}_{ji} = \langle \Psi_j | \nabla \Psi_i \rangle$, and in the adiabatic basis the nuclear momentum becomes $-\mathrm{i}(\nabla + \boldsymbol{d})$, a vector potential in the space of electronic states. Written with the nuclear velocity, the coupling makes the hierarchy of Chap. 5 explicit: nuclei in zero-point motion are slower than electrons by κ_BO^3, and the corrections to the adiabatic energies are of relative order m_e/M. To first order in the displacements the same physics is the electron–phonon Hamiltonian H_ep, in which a phonon of wavevector $\boldsymbol{Q}$ moves an electron from $\boldsymbol{k}$ to $\boldsymbol{k} + \boldsymbol{Q}$ with the vertex $g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q})$. Whether the electrons follow a moving nucleus depends on how fast it is accelerated: slowly compared with $1/\Delta E$, they stay in their ground state. Accelerated suddenly, they keep their state while the nucleus acquires a velocity v_N, and the overlap of the boosted ground state with the original one gives the probability that nothing happens to them. For a hydrogen atom struck by dark matter this is exact, and its complement is the Migdal effect, the subject of Chap. 12.

The vertex that density-functional perturbation theory computes is screened (Sect. 6.2). The bare potential of a displaced ion has a $1/Q$ tail in every crystal; the electrons remove it in a metal, reduce it by ε_∞ in an insulator, and through local fields replace the ionic charge by the Born effective charge. Three long-wavelength limits follow. Acoustic phonons couple through deformation potentials, with $g \propto Q^{1/2}$, a consequence of translation invariance. Optical phonons in nonpolar crystals couple through optical deformation potentials, with a vertex that tends to a constant. Longitudinal optical phonons in polar crystals couple through the Fröhlich vertex $g^\mathrm{L} = \mathrm{i}C_\mathrm{F}\rho_{mn}/Q$, which in GaAs gives a polaron constant $\alpha_\mathrm{F} = 0.072$. Since the screened vertex is short-ranged in a Wannier basis, a few hundred computed wavevectors interpolate to the billions a rate needs, once the long-range Fröhlich part has been subtracted before the interpolation and restored after it.

At 10 mK the gap of silicon is about 60 meV smaller than that of the same crystal with its nuclei clamped, because of zero-point motion (Sect. 6.3). Second order in the displacements yields two terms of the electron self-energy of the same size: the Fan–Migdal term, built from two first-order vertices with the phonon energies and occupations in its denominators, and the Allen–Heine term from the vertex quadratic in the displacements. In the adiabatic limit they combine into the average of the band energy over the zero-point and thermal smearing of the lattice, each mode weighted by $2n_{\boldsymbol{Q}\nu} + 1$. Translation invariance makes the two terms cancel for long-wavelength acoustic phonons, and the rigid-ion approximation expresses the Allen–Heine term through first-order vertices at $\boldsymbol{Q} = 0$. The adiabatic form diverges in polar

crystals, where the nonadiabatic Fan–Migdal term at a band bottom is the Fröhlich polaron shift of Sect. 6.2. For the indirect gap of silicon the computed zero-point renormalization is −56.2 meV. Experiment yields −62 to −64 meV for the whole zero-point shift, which includes a 9 meV opening by the zero-point expansion of the lattice; the electron–phonon part, −71 to −73 meV, is about 20 % larger than the computed value. Einstein-model inversions of the measured curve, −52 and −58 meV, test the consistency of the computed value with that curve rather than measure it. An Einstein model with its energy fixed by the zero-point motion follows the measured gap to within a few meV up to room temperature, and a detector at 10 mK sees the low-temperature gap, which already contains the full renormalization.

A phonon decays into electron–hole pairs only where real pairs exist at its energy (Sect. 6.4). Its self-energy is the electron bubble. The static part of the bubble is already contained in the frequencies of density-functional perturbation theory, and only the difference between the dynamic and the static bubble may be added; in an insulator that difference is of relative order $(\omega_{\boldsymbol{Q}\nu}/E_\mathrm{g})^2$. In a metal the static bubble leaves Kohn anomalies in the dispersion. The imaginary part is a golden-rule linewidth. It vanishes in an undoped insulator, where the phonons of silicon are broadened by anharmonicity and disorder instead. In a metal it leads to Allen's formula, the Eliashberg function $\alpha^2F(\omega)$ and the coupling constant λ_ep, which also enhances the electron mass by $1 + \lambda_\mathrm{ep}$. For aluminum a free-electron model with one core radius yields $\lambda_\mathrm{ep} = 0.436$, against 0.42 to 0.43 from tunneling and the specific heat, and this number sets the strength of the pairing in the superconducting sensors of Chap. 7.

Photons between the indirect gap of silicon and its direct gap of 3.4 eV are absorbed only with a phonon (Sect. 6.5). The amplitude has two time orderings, photon first or phonon first, and the Pauli principle replaces the forbidden intermediate states by hole orderings, which combine with the electron ones into a single sum over all bands. Summed over electron and phonon wavevectors in the whole zone, the squared amplitude gives the phonon-assisted dielectric function $\varepsilon_2^\mathrm{PA}$, computed from first principles with Wannier-interpolated vertices. Near the edge it reduces to the Hall–Bardeen–Blatt law: each phonon branch adds components growing as $(\omega - E_\mathrm{g} \pm \omega_\nu)^2$, weighted by n_ν for phonon absorption and $n_\nu + 1$ for emission. Their ratio is detailed balance, and in a cold detector only emission survives. The silicon edge measured in 1958 yields transverse-acoustic and transverse-optical phonon energies at $\boldsymbol{k}_\mathrm{c}$ that agree with neutron data. The optical vertex is the zero-momentum limit of the density vertex. Replacing one by the other turns phonon-assisted absorption of light into the phonon-assisted response to dark matter, which governs the absorption of light bosons (Chap. 8) and the threshold window of electron recoils in silicon (Chap. 13).

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Chapter 7

Magnets, superconductors, Dirac materials and defects

Silicon offers dark matter two kinds of excitation: an electron lifted across the 1.17 eV gap, and phonons up to 64 meV. A particle that couples to the spin of the electron has no cheaper excitation in silicon, since every spin is paired in a bond and flipping one costs the gap. Other solids have excitations below these energies or of other kinds. In NiO the spins of the nickel ions order, and their precession, a magnon, can take up anything from a few meV to 117 meV for a spin-coupled particle. In aluminum below 1.18 K the electrons at the Fermi surface bind into pairs, and breaking a pair costs only 0.36 meV, which by the bound $2E_{\mathrm{th}}/v_{\mathrm{max}}^2$ of Sect. 1.3 brings dark matter of 0.1 keV within kinematic reach. Dirac materials, whose electrons obey a Dirac equation with a velocity far below that of light, offer pair excitations from a few to tens of meV. Donors and color centers put bound levels inside the gap of silicon itself, shallow for a donor and deep for a color center, and a struck atom can leave a defect that stores energy. Each of these targets is built largely from pieces already in hand: the exchange term of the Coulomb interaction (Sects. 2.3 and 3.4), the electron–phonon coupling (Chap. 6), the $\boldsymbol{k}\cdot\boldsymbol{p}$ expansion (Sect. 3.3) and the hydrogenic envelope (Sect. 4.9). What is new is physics that silicon does not have, and above all spontaneously broken symmetry, which brings with it collective modes and quasiparticles whose energies are set by an interaction rather than by a band gap. We will derive, for each target, the excitation spectrum and the response that the rate formulas of the later chapters take as input.

7.1 Magnetic order and spin waves

Each nickel ion in NiO carries a spin, and below the Néel temperature of 523 K, the measured value as tabulated by Fischer et al. [1, Table IV], the ordering temperature of an antiferromagnet, the spins order: those in one (111) plane of the crystal point one way, those in the next plane the opposite way. A disturbance of that order travels through the crystal as a spin wave, and its quanta, the *magnons*, have energies up to ∼117 meV. Inelastic neutron scattering at 78 K measured their dispersion, with an initial slope of ∼250 meV Å [2]. In an ordered magnet a particle that couples to the spin of the electron can create a magnon, as Trickle, Zhang and Zurek pointed out [3]. The energy deposits then run from a few meV to about a tenth of an eV. To compute such a rate we need three things: the interaction that orders the spins, the spectrum of the waves, and the operator through which a probe creates them.

7.1.1 Exchange and the Heisenberg model

The Coulomb interaction does not act on spin, yet it makes the energy of two electrons depend on how their spins are aligned. Put one electron in each of two orthogonal orbitals φ_a and φ_b. In the spin triplet the spatial wavefunction is antisymmetric, $[\varphi_a(\boldsymbol{r})\varphi_b(\boldsymbol{r}') - \varphi_b(\boldsymbol{r})\varphi_a(\boldsymbol{r}')]/\sqrt{2}$, and in the singlet it is symmetric. The expectation value of the Coulomb repulsion $v_{\rm C}(\boldsymbol{r}-\boldsymbol{r}')$ is then $C \mp K_{\rm ex}$, upper sign for the triplet, with the direct integral C and the exchange integral

$$K_{\rm ex} = \int \mathrm{d}^3r\, \mathrm{d}^3r'\, \varphi_a^*(\boldsymbol{r})\varphi_b^*(\boldsymbol{r}')\, v_{\rm C}(\boldsymbol{r}-\boldsymbol{r}')\, \varphi_b(\boldsymbol{r})\varphi_a(\boldsymbol{r}'). \tag{7.1}$$

This is the exchange term of Eq. (2.46), written for localized orbitals instead of plane waves. It is the Coulomb self-energy of the charge distribution $\varphi_a^*\varphi_b$, and since the Fourier transform $v_{\rm C}(k) = e^2/k^2$ is positive, $K_{\rm ex} > 0$. The triplet therefore lies $2K_{\rm ex}$ below the singlet: parallel spins keep the electrons apart and save Coulomb energy. This is Hund's rule, and within one Ni^{2+} ion it aligns the two holes of the $3d$ shell into a spin $|\boldsymbol{S}| = 1$.

The four spin states can be described without any reference to orbitals. With $\boldsymbol{S}_1 \cdot \boldsymbol{S}_2 = \frac{1}{2}[(\boldsymbol{S}_1+\boldsymbol{S}_2)^2 - \boldsymbol{S}_1^2 - \boldsymbol{S}_2^2]$, which is $-\frac{3}{4}$ in the singlet and $\frac{1}{4}$ in the triplet, the operator

$$H_{12} = E_{\rm orb} + C - \tfrac{1}{2}K_{\rm ex} - 2K_{\rm ex}\, \boldsymbol{S}_1 \cdot \boldsymbol{S}_2 \tag{7.2}$$

has the singlet energy $E_{\rm orb} + C + K_{\rm ex}$ and the triplet energy $E_{\rm orb} + C - K_{\rm ex}$, with $E_{\rm orb}$ the sum of the two one-electron orbital energies. An interaction that depends only on the positions of the electrons has become a coupling between their spins, of strength $-2K_{\rm ex}$ and ferromagnetic sign.

Kinetic exchange. Between electrons on neighboring ions of an insulator, however, a second mechanism usually dominates, and it has the opposite sign. Take two sites with one orbital each and the hopping $t_{\rm hop} > 0$ of the tight-binding model of Sect. 3.2. Putting two electrons on the same site costs the on-site or Hubbard repulsion $U_{\rm H}$:

$$H = -t_{\rm hop} \sum_s (c_{1s}^\dagger c_{2s} + c_{2s}^\dagger c_{1s}) + U_{\rm H} \sum_{i=1,2} c_{i\uparrow}^\dagger c_{i\uparrow}\, c_{i\downarrow}^\dagger c_{i\downarrow}. \tag{7.3}$$

With two electrons and $t_{\rm hop} = 0$, the four states with one electron on each site are degenerate at zero energy, and the two doubly occupied states lie at $U_{\rm H}$. The hopping connects the two groups. A triplet cannot hop, since the Pauli principle forbids two parallel spins in one orbital: for the state $c_{1\uparrow}^\dagger c_{2\uparrow}^\dagger|0\rangle$ this is evident, and the other two members of the triplet follow by rotational symmetry. The triplet stays at zero energy to all orders. The singlet couples to the symmetric combination of the doubly occupied states, through two hops (from site 1 or from site 2) that add, with a matrix element of magnitude $2t_{\rm hop}$. To second order in $t_{\rm hop}/U_{\rm H}$ its energy is $-(2t_{\rm hop})^2/U_{\rm H}$. As Eq. (7.2) shows, the coefficient of $\boldsymbol{S}_1 \cdot \boldsymbol{S}_2$ is the triplet energy minus the singlet energy. With both mechanisms present it is

$$J^{\rm ex} = \frac{4t_{\rm hop}^2}{U_{\rm H}} - 2K_{\rm ex}. \tag{7.4}$$

The picture behind the first term is that an electron lowers its kinetic energy by a virtual hop onto its neighbor and back, and it can do so only if the neighbor's spin is antiparallel. The hop costs $U_{\rm H}$, so the gain is second order in $t_{\rm hop}$. The expansion holds as long as $t_{\rm hop} \ll U_{\rm H}$, which is also the condition for the charge to stay frozen, one electron per site: a *Mott insulator*, in this two-site picture. The corrections are of relative order $(t_{\rm hop}/U_{\rm H})^2$.

In NiO the nickel ions do not touch; between two of them sits an oxygen ion. Anderson's mechanism of *superexchange* [4] applies Eq. (7.4) with an effective hopping that runs through

the oxygen p orbital, so that t_{hop} is itself second order in the nickel–oxygen hopping. The geometry then decides the sign and the size. Next-nearest nickel neighbors sit on a straight line through one oxygen, whose p orbital along that line overlaps strongly with a $3d$ orbital of each, and their coupling is large and antiferromagnetic. Nearest neighbors see each other at 90° through two different p orbitals, and their coupling is weak and ferromagnetic. The measured values are [2]

$$J_2 = 19.01\,\mathrm{meV}, \qquad J_1 = -1.37\,\mathrm{meV}, \tag{7.5}$$

in the convention of Eq. (7.6) below. With several $3d$ orbitals and $|\boldsymbol{S}| = 1$, Eq. (7.4) acquires factors of order one, and we use it as an estimate, $J^{\mathrm{ex}} \sim t_{\mathrm{hop}}^2/U_{\mathrm{H}}$. The Hubbard repulsion of the nickel $3d$ shell is several eV; the linear-response values of Appendix E.9 lie between 4.6 and 8.2 eV. Inverting the estimate, J_2 requires an effective hopping $(J_2 U_{\mathrm{H}})^{1/2} \approx 0.3$ to 0.4 eV, so that $t_{\mathrm{hop}}/U_{\mathrm{H}} \approx 0.05$ to 0.06. The electrons of NiO hop little compared with what a double occupancy costs, which is why only their spins remain free at low energy. The estimate is rough in one more respect. NiO is a *charge-transfer insulator*: the cheapest charge excitation moves an electron from oxygen to nickel, at the charge-transfer energy Δ_{CT}, which is smaller than U_{H}, and superexchange through the oxygen depends on both energies. The effective hopping above absorbs this into one number.

The Heisenberg model. For a crystal of such ions, each with a spin of length $|\boldsymbol{S}|$ fixed by Hund's rule and the charge frozen, the low-energy states are the spin configurations. To leading order in $t_{\mathrm{hop}}/U_{\mathrm{H}}$ the coupling is a sum over pairs, and the only coupling of two spins that is invariant under a common rotation and bilinear in each is their scalar product. The result is the *Heisenberg model*,

$$H_{\mathrm{H}} = \sum_{\langle ij \rangle} J_{ij}^{\mathrm{ex}}\, \boldsymbol{S}_i \cdot \boldsymbol{S}_j, \tag{7.6}$$

with each pair of ions counted once and the *exchange constant* $J_{ij}^{\mathrm{ex}} > 0$ antiferromagnetic. The ionic spins obey $[S_i^a, S_j^b] = \mathrm{i}\delta_{ij}\epsilon_{abc}S_i^c$ and $\boldsymbol{S}_i^2 = |\boldsymbol{S}|(|\boldsymbol{S}|+1)$. Note that the literature uses every combination of sign, of single or double counting and of spin length, and a quoted J^{ex} is meaningful only together with its Hamiltonian; Appendix E.9 lists the conversions.

If every $J_{ij}^{\mathrm{ex}} < 0$, the state with all spins parallel is the ground state, a *ferromagnet*. If the antiferromagnetic bonds connect two sublattices, the classical ground state has the spins of one sublattice up and those of the other down, an *antiferromagnet*. In NiO, J_2 connects the ions of adjacent (111) planes, and the order is the alternation of ferromagnetic (111) planes described above, called type II (Fig. 7.1a). Eq. (7.6) leaves out the coupling of the spins to the lattice through spin–orbit coupling, which fixes the direction of the moments in the crystal and opens small gaps in the magnon spectrum, below 5 meV in NiO [5, Sect. III]. We neglect it.

The exchange constants of a real magnet are computed from the electronic structure in the density-functional theory (DFT) of Sect. 3.4 with a Hubbard correction on the $3d$ shell, DFT $+U$ (Appendix E.9). Three routes are common: total energies of several collinear orders mapped onto Eq. (7.6), the energies of spin spirals, and a Green's-function formula for small rotations of two moments. All three inherit the dependence of superexchange on U_{H}. For NiO the density-functional values of J_2 in one comparison range from 11 to 27 meV [1, Table III] (converted to our convention in Appendix E.9), and linear spin-wave theory with the best of them overestimates the measured maximum magnon energy by about 14 % [5, Sect. III]. For a target without neutron data, this spread is the theory uncertainty of its magnon spectrum.

7.1.2 Spin waves of a ferromagnet

The ferromagnetic ground state is known exactly, and its excitations are the simplest place to see what a magnon is. Order the spins along $+\hat{\boldsymbol{z}}$. A state with one unit of spin removed from site j is not an eigenstate, because the transverse part of $\boldsymbol{S}_i \cdot \boldsymbol{S}_j = S_i^z S_j^z + \frac{1}{2}(S_i^+ S_j^- + S_i^- S_j^+)$ moves the deviation to the neighbors. The eigenstates are deviations spread over all sites with a definite crystal momentum: spin waves.

Holstein and Primakoff turned this into a boson problem [6]. On each site write

$$S_j^z = |\boldsymbol{S}| - d_j^\dagger d_j, \qquad S_j^+ = (2|\boldsymbol{S}| - d_j^\dagger d_j)^{1/2} d_j, \qquad S_j^- = d_j^\dagger (2|\boldsymbol{S}| - d_j^\dagger d_j)^{1/2}, \tag{7.7}$$

with boson operators $[d_i, d_j^\dagger] = \delta_{ij}$ of the kind introduced in Sect. 2.3. The number $n_j = d_j^\dagger d_j$ counts the deviation of the site from full polarization. The spin algebra holds on the physical states $n_j \le 2|\boldsymbol{S}|$. To check $[S^+, S^-] = 2S^z$ on one site, use $f(n)d = d\,f(n-1)$ for any function f. Then $S^+S^- = (2|\boldsymbol{S}| - n)(n+1)$ and $S^-S^+ = d^\dagger(2|\boldsymbol{S}| - n)d = n(2|\boldsymbol{S}| - n + 1)$, whose difference is $2|\boldsymbol{S}| - 2n = 2S^z$; the relations $[S^z, S^\pm] = \pm S^\pm$ follow since S^+ lowers n by one. The square roots make Eq. (7.7) exact and nonlinear. *Linear spin-wave theory* expands in the small number of deviations per site compared with $2|\boldsymbol{S}|$, that is in $1/|\boldsymbol{S}|$, replaces the square roots by $(2|\boldsymbol{S}|)^{1/2}$ and keeps the Hamiltonian to second order in the bosons. For one bond,

$$\boldsymbol{S}_i \cdot \boldsymbol{S}_j \simeq |\boldsymbol{S}|^2 - |\boldsymbol{S}|(n_i + n_j) + |\boldsymbol{S}|(d_i^\dagger d_j + d_j^\dagger d_i),$$

where the dropped quartic term $n_i n_j$ comes from $S_i^z S_j^z$. The deviations hop from site to site with amplitude $J_{ij}^{\mathrm{ex}}|\boldsymbol{S}|$.

On a Bravais lattice of N magnetic ions at the sites $\boldsymbol{R}_j$, the hopping is diagonalized by plane waves,

$$d_j = N^{-1/2} \sum_{\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_j}\, a_{\boldsymbol{k}},$$

with $\boldsymbol{k}$ in the Brillouin zone. The operators $a_{\boldsymbol{k}}$ are bosons, the *magnon operators*; with several magnetic ions per cell they carry a branch index, $a_{\boldsymbol{k}\nu}$, which we drop for a single branch. Summing over pairs, each site has the bonds $\sum_{\boldsymbol{R}\neq 0} J_{0\boldsymbol{R}}^{\mathrm{ex}}$, which multiply $-|\boldsymbol{S}|n_j$, and the hopping terms collect into $|\boldsymbol{S}| \sum_{\boldsymbol{k}} J^{\mathrm{ex}}(\boldsymbol{k})\, a_{\boldsymbol{k}}^\dagger a_{\boldsymbol{k}}$ with the lattice Fourier transform of the exchange constants. The result is a set of independent oscillators:

Linear spin waves of a ferromagnet

$$H_{\mathrm{H}} \simeq E_{\mathrm{cl}} + \sum_{\boldsymbol{k}} \omega_{\boldsymbol{k}}^{\mathrm{M}}\, a_{\boldsymbol{k}}^\dagger a_{\boldsymbol{k}}, \qquad \omega_{\boldsymbol{k}}^{\mathrm{M}} = |\boldsymbol{S}|\big[J^{\mathrm{ex}}(\boldsymbol{k}) - J^{\mathrm{ex}}(0)\big], \qquad J^{\mathrm{ex}}(\boldsymbol{k}) = \sum_{\boldsymbol{R}\neq 0} J_{0\boldsymbol{R}}^{\mathrm{ex}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}, \tag{7.8}$$

with the classical energy $E_{\mathrm{cl}} = \frac{1}{2}N|\boldsymbol{S}|^2 J^{\mathrm{ex}}(0)$, each pair counted once.

For inversion-symmetric lattices $J^{\mathrm{ex}}(\boldsymbol{k})$ is real and even. With all $J_{0\boldsymbol{R}}^{\mathrm{ex}} < 0$, $J^{\mathrm{ex}}(0)$ is the minimum of $J^{\mathrm{ex}}(\boldsymbol{k})$ and every $\omega_{\boldsymbol{k}}^{\mathrm{M}} \ge 0$. The mode $\omega_0^{\mathrm{M}} = 0$ tilts all spins together, which costs nothing because Eq. (7.6) is invariant under a common rotation; it is the Goldstone mode of the broken rotational symmetry, the analog of the acoustic phonon of a crystal. A particle physicist would count two Goldstone modes for the two broken rotations about $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$. The ferromagnet has one, with a quadratic dispersion. Its two broken generators do not commute in the ground state, $\langle [S_{\mathrm{tot}}^x, S_{\mathrm{tot}}^y] \rangle = \mathrm{i}\langle S_{\mathrm{tot}}^z \rangle \neq 0$, so they act as a coordinate and its conjugate momentum and together describe a single mode. In the antiferromagnet $\langle S_{\mathrm{tot}}^z \rangle = 0$, and the two modes of the count appear, both linear. At small k, expanding the cosine,

$\omega_{\boldsymbol{k}}^{\mathrm{M}} \simeq \frac{1}{2}|\boldsymbol{S}| \sum_{\boldsymbol{R}} (-J_{0\boldsymbol{R}}^{\mathrm{ex}})(\boldsymbol{k}\cdot\boldsymbol{R})^2$, quadratic in k like the energy of a free particle. Physically, the magnon $a_{\boldsymbol{k}}^{\dagger}|0\rangle$ is one unit of spin deviation shared by all ions: each spin precesses on a narrow cone about $\hat{\boldsymbol{z}}$, and the phase of the precession advances by $\boldsymbol{k}\cdot\boldsymbol{R}$ from ion to ion. Eq. (7.8) agrees with the frozen-magnon relation of Appendix E.9, which reads $\omega_{\boldsymbol{k}}^{\mathrm{M}}$ off the energy of a static spin spiral.

Worked example 7.1: The ferromagnetic chain

Take a ring of N spins at spacing a with ferromagnetic nearest-neighbor exchange $J^{\mathrm{ex}} < 0$. Then $J^{\mathrm{ex}}(k) = 2J^{\mathrm{ex}} \cos ka$, and Eq. (7.8) gives

$$\omega_k^{\mathrm{M}} = 2|J^{\mathrm{ex}}||\boldsymbol{S}|\,(1-\cos ka),$$

which rises as $|J^{\mathrm{ex}}||\boldsymbol{S}|a^2k^2$ from $k=0$ and reaches $4|J^{\mathrm{ex}}||\boldsymbol{S}|$ at the zone boundary $k=\pi/a$. For one magnon this linear result is exact. Let $|F\rangle$ be the fully polarized state, of energy $E_F = NJ^{\mathrm{ex}}|\boldsymbol{S}|^2$, and $|j\rangle = (2|\boldsymbol{S}|)^{-1/2} S_j^- |F\rangle$ the normalized state with one unit of deviation on site j. The bonds that do not touch j keep their energy $J^{\mathrm{ex}}|\boldsymbol{S}|^2$. The two bonds that do touch it have $S^zS^z = |\boldsymbol{S}|(|\boldsymbol{S}|-1)$, which raises $\langle j|H_{\mathrm{H}}|j\rangle$ to $E_F - 2J^{\mathrm{ex}}|\boldsymbol{S}|$, above E_F since $J^{\mathrm{ex}} < 0$. The transverse term $\frac{1}{2}J^{\mathrm{ex}} S_j^+ S_{j\pm1}^-$ moves the deviation to a neighbor with amplitude $\frac{1}{2}J^{\mathrm{ex}} \cdot 2|\boldsymbol{S}| = J^{\mathrm{ex}}|\boldsymbol{S}|$, since $S_{j\pm1}^-$ acting on a full site and S_j^+ acting on a site with one deviation each contribute $(2|\boldsymbol{S}|)^{1/2}$. Hence

$$H_{\mathrm{H}}|j\rangle = (E_F - 2J^{\mathrm{ex}}|\boldsymbol{S}|)\,|j\rangle + J^{\mathrm{ex}}|\boldsymbol{S}|\,(|j+1\rangle + |j-1\rangle),$$

an exact statement for any $|\boldsymbol{S}|$, because the states with one deviation are closed under H_{H}. The Bloch combination $N^{-1/2}\sum_j \mathrm{e}^{\mathrm{i}kja}|j\rangle$ is an eigenstate with energy $E_F - 2J^{\mathrm{ex}}|\boldsymbol{S}| + 2J^{\mathrm{ex}}|\boldsymbol{S}|\cos ka = E_F + \omega_k^{\mathrm{M}}$. The quartic terms dropped in linear spin-wave theory act only on states with two or more deviations. With two deviations, however, linear spin-wave theory is no longer exact, as a numerical diagonalization of a ring of 10 spins $\frac{1}{2}$ shows. Among the two-deviation states we leave out those that are rotated copies of states with fewer deviations, the states $S_{\mathrm{tot}}^-|F\rangle$ and S_{tot}^- acting on one-magnon states, where $S_{\mathrm{tot}}^- = \sum_j S_j^-$ rotates all spins together. The lowest remaining level lies at $0.328|J^{\mathrm{ex}}|$, while the cheapest pair of magnons with nonzero wavevectors, $k = \pm 2\pi/10a$, costs $0.382|J^{\mathrm{ex}}|$ in linear spin-wave theory. Magnons attract each other, an effect of relative order $1/|\boldsymbol{S}|$ that linear spin-wave theory misses. For $|\boldsymbol{S}| = 1$ on a ring of 8 sites the same comparison is 1.083 against $1.172|J^{\mathrm{ex}}|$.

For dark matter the quadratic dispersion matters. A magnon of energy ω has $k = [\omega/(|J^{\mathrm{ex}}||\boldsymbol{S}|a^2)]^{1/2}$, and its group velocity $2(\omega|J^{\mathrm{ex}}||\boldsymbol{S}|)^{1/2}a$ vanishes at small ω. At low energy a ferromagnet offers excitations of small momentum and small velocity, a different kinematic match from the linear branches of phonons and of the antiferromagnet below.

7.1.3 Spin waves of an antiferromagnet

In an antiferromagnet, however, the classical state, with sublattice A up and B down, is not an eigenstate of Eq. (7.6), and the reason fixes the form of its spin waves. For an antiparallel pair the transverse term $S_i^+S_j^-$, with i on A and j on B, lowers a spin that already points down and raises one that points up: it creates two deviations at once. Spin waves are then mixtures of deviations and their absence, and the Bogoliubov transformation of Appendix C.7 diagonalizes them. Appendix D.19 carries the derivation for exchange of any range.

On B the reference spin points along $-\hat{\boldsymbol{z}}$, so the roles of raising and lowering are exchanged: $S^z_j = -|\boldsymbol{S}| + d^\dagger_{\mathrm{B}j} d_{\mathrm{B}j}$ and $S^-_j \simeq (2|\boldsymbol{S}|)^{1/2} d_{\mathrm{B}j}$, while A uses Eq. (7.7) with bosons $d_{\mathrm{A}i}$. At linear order an antiparallel bond becomes

$$\boldsymbol{S}_i \cdot \boldsymbol{S}_j \simeq -|\boldsymbol{S}|^2 + |\boldsymbol{S}|(n_{\mathrm{A}i} + n_{\mathrm{B}j}) + |\boldsymbol{S}|(d_{\mathrm{A}i} d_{\mathrm{B}j} + d^\dagger_{\mathrm{A}i} d^\dagger_{\mathrm{B}j}),$$

with $n_{\mathrm{A}i} = d^\dagger_{\mathrm{A}i} d_{\mathrm{A}i}$ and $n_{\mathrm{B}j}$ likewise. Each deviation raises the energy of the bond, and the pair term $d_{\mathrm{A}i} d_{\mathrm{B}j} + \text{h.c.}$ creates or destroys deviations in pairs. Bonds within a sublattice keep the ferromagnetic form. Exchange enters through two lattice sums over the neighbors $\boldsymbol{\delta}$ of a site, on its own sublattice and on the other, $J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{k}) = \sum_{\boldsymbol{\delta}\in\mathrm{A},\,\boldsymbol{\delta}\neq 0} J^{\mathrm{ex}}_{0\boldsymbol{\delta}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}}$ and $J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{k}) = \sum_{\boldsymbol{\delta}\in\mathrm{B}} J^{\mathrm{ex}}_{0\boldsymbol{\delta}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}}$ (Eq. (D.109)). After a Fourier transform on each of the two sublattices of N_{s} sites, with the opposite phase on B so that the pair terms couple equal wavevectors, the Hamiltonian is a set of two-mode problems,

$$\begin{gathered} H_{\mathrm{H}} \simeq E_{\mathrm{cl}} + |\boldsymbol{S}| \sum_{\boldsymbol{k}} \Big[A_{\boldsymbol{k}} (d^\dagger_{\mathrm{A}\boldsymbol{k}} d_{\mathrm{A}\boldsymbol{k}} + d^\dagger_{\mathrm{B}\boldsymbol{k}} d_{\mathrm{B}\boldsymbol{k}}) + B_{\boldsymbol{k}} (d_{\mathrm{A}\boldsymbol{k}} d_{\mathrm{B}\boldsymbol{k}} + d^\dagger_{\mathrm{B}\boldsymbol{k}} d^\dagger_{\mathrm{A}\boldsymbol{k}}) \Big], \\ A_{\boldsymbol{k}} = J^{\mathrm{ex}}_{\mathrm{AB}}(0) - J^{\mathrm{ex}}_{\mathrm{AA}}(0) + J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{k}), \qquad B_{\boldsymbol{k}} = J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{k}), \end{gathered} \tag{7.9}$$

which is Eq. (D.111). The coefficient $A_{\boldsymbol{k}}$ is the cost of one deviation: the exchange field of the antiparallel neighbors minus that of the parallel ones, plus the hopping on its own sublattice. Without $B_{\boldsymbol{k}}$ this would be a ferromagnet on each sublattice. At each $\boldsymbol{k}$, Eq. (7.9) is the two-mode Hamiltonian Eq. (C.61), with the diagonal energy $|\boldsymbol{S}|A_{\boldsymbol{k}}$ and the pairing amplitude $g_0 = |\boldsymbol{S}|B_{\boldsymbol{k}}$. The new operators

$$a_{\boldsymbol{k}1} = u^{\mathrm{M}}_{\boldsymbol{k}}\, d_{\mathrm{A}\boldsymbol{k}} + v^{\mathrm{M}}_{\boldsymbol{k}}\, d^\dagger_{\mathrm{B}\boldsymbol{k}}, \qquad a_{\boldsymbol{k}2} = u^{\mathrm{M}}_{\boldsymbol{k}}\, d_{\mathrm{B}\boldsymbol{k}} + v^{\mathrm{M}}_{\boldsymbol{k}}\, d^\dagger_{\mathrm{A}\boldsymbol{k}}, \tag{7.10}$$

with $u^{\mathrm{M}}_{\boldsymbol{k}} = \cosh\theta_{\boldsymbol{k}}$ and $v^{\mathrm{M}}_{\boldsymbol{k}} = \sinh\theta_{\boldsymbol{k}}$, are bosons because $(u^{\mathrm{M}}_{\boldsymbol{k}})^2 - (v^{\mathrm{M}}_{\boldsymbol{k}})^2 = 1$, and the pair terms vanish for $\tanh 2\theta_{\boldsymbol{k}} = B_{\boldsymbol{k}}/A_{\boldsymbol{k}}$. These are the *magnon Bogoliubov coefficients*. Eq. (C.62) then yields:

Linear spin waves of a two-sublattice antiferromagnet

$$H_{\mathrm{H}} \simeq E_{\mathrm{gs}} + \sum_{\boldsymbol{k}} \sum_{\nu=1,2} \omega^{\mathrm{M}}_{\boldsymbol{k}}\, a^\dagger_{\boldsymbol{k}\nu} a_{\boldsymbol{k}\nu}, \qquad \omega^{\mathrm{M}}_{\boldsymbol{k}} = |\boldsymbol{S}|(A^2_{\boldsymbol{k}} - B^2_{\boldsymbol{k}})^{1/2}, \tag{7.11}$$

with the ground-state energy $E_{\mathrm{gs}} = E_{\mathrm{cl}} + \sum_{\boldsymbol{k}} (\omega^{\mathrm{M}}_{\boldsymbol{k}} - |\boldsymbol{S}|A_{\boldsymbol{k}})$ and $E_{\mathrm{cl}} = N_{\mathrm{s}}|\boldsymbol{S}|^2[J^{\mathrm{ex}}_{\mathrm{AA}}(0) - J^{\mathrm{ex}}_{\mathrm{AB}}(0)]$.

The two branches are degenerate without anisotropy, and they differ in the spin they carry: $a^\dagger_{\boldsymbol{k}1}$ lowers the total S^z by one and $a^\dagger_{\boldsymbol{k}2}$ raises it. Since $\omega^{\mathrm{M}}_{\boldsymbol{k}} \leq |\boldsymbol{S}|A_{\boldsymbol{k}}$, the ground state lies below the classical Néel energy: in the true ground state each site of A carries the zero-point deviation $(v^{\mathrm{M}}_{\boldsymbol{k}})^2$ per mode, which reduces the ordered moment (Sect. 14.1 evaluates it for NiO). The theory is consistent only if $A_{\boldsymbol{k}} \geq |B_{\boldsymbol{k}}|$ at every $\boldsymbol{k}$; otherwise $\omega^{\mathrm{M}}_{\boldsymbol{k}}$ is imaginary and the assumed order is unstable.

The antiferromagnet differs from the ferromagnet at long wavelengths. At $\boldsymbol{k} = 0$, $A_0 = B_0 = J^{\mathrm{ex}}_{\mathrm{AB}}(0)$ and $\omega^{\mathrm{M}}_0 = 0$, the Goldstone mode again. Near it, $A_{\boldsymbol{k}} + B_{\boldsymbol{k}} \to 2J^{\mathrm{ex}}_{\mathrm{AB}}(0)$ stays finite while $A_{\boldsymbol{k}} - B_{\boldsymbol{k}}$ is quadratic in k, so $\omega^{\mathrm{M}}_{\boldsymbol{k}} = |\boldsymbol{S}|[(A_{\boldsymbol{k}} - B_{\boldsymbol{k}})(A_{\boldsymbol{k}} + B_{\boldsymbol{k}})]^{1/2}$ is linear in k (Eq. (D.114)). The velocity is the geometric mean of the large exchange field $|\boldsymbol{S}|J^{\mathrm{ex}}_{\mathrm{AB}}(0)$ and the small stiffness in $A_{\boldsymbol{k}} - B_{\boldsymbol{k}}$. Physically, a long-wavelength deviation on one sublattice drags the other along through the pair term, and every cycle of the precession works against the full exchange field. For type-II order on the fcc lattice with J_2 alone, the 6 next-nearest

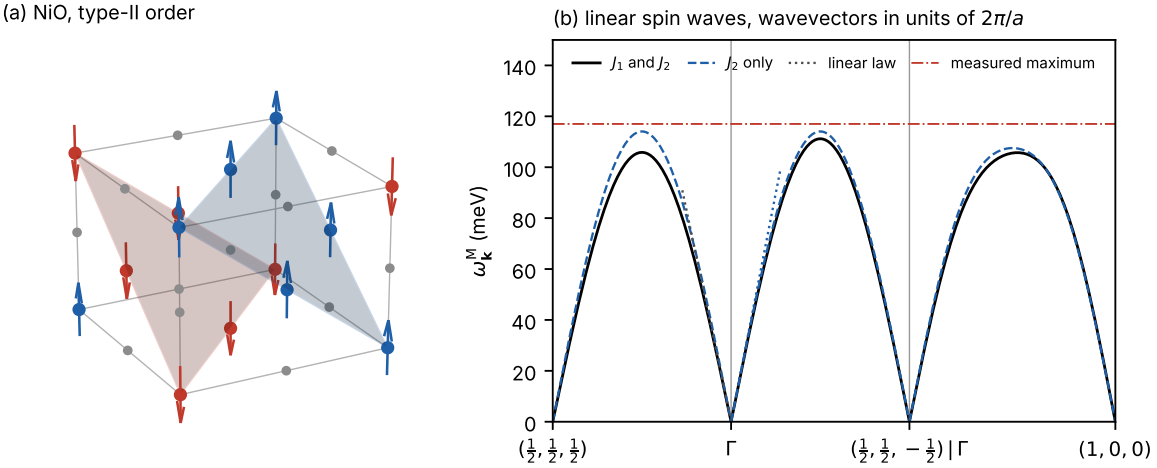


Figure 7.1: Magnetic order and spin waves of NiO. (a) Conventional cubic cell of the rock-salt structure: nickel ions (colored) on the fcc sites, oxygen ions (gray) between them. In the type-II order the spins of each (111) plane are parallel and alternate in sign from plane to plane; two such planes are shaded. The arrows show only the sign of each moment; in NiO the moments lie within the planes, along a direction set by the weak anisotropy. (b) Magnon energies $\omega^{\mathrm{M}}_{\boldsymbol{k}}$ (meV) from linear spin-wave theory, Eq. (7.11), along three lines from the zone center Γ, with wavevectors in units of $2\pi/a$ and $a = 4.17\,\text{Å}$. Model: undistorted fcc lattice, Heisenberg exchange with $J_1 = -1.37\,\text{meV}$ and $J_2 = 19.01\,\text{meV}$ (each pair counted once, positive antiferromagnetic) measured by neutron scattering, $|\boldsymbol{S}| = 1$, no anisotropy. Solid: J_1 and J_2; dashed: J_2 alone, Eq. (7.12). Dotted: the linear laws $2\sqrt{3}(J_1 + J_2)|\boldsymbol{S}|ak$ along [111] and $2\sqrt{3}J_2|\boldsymbol{S}|ak$. Dash-dotted: the measured maximum, ∼117 meV. The point $(\frac{1}{2},\frac{1}{2},\frac{1}{2})$, along the stacking direction of the planes, is equivalent to Γ in the magnetic structure, and the energy returns to zero there. The zeros at $(\frac{1}{2},\frac{1}{2},-\frac{1}{2})$ and $(1,0,0)$ are soft points of the undistorted model, which the small rhombohedral distortion of real NiO lifts to a few meV. The largest energy in the zone, $|\boldsymbol{S}|(6J_2 - 3J_1) = 118.2\,\text{meV}$, lies off these lines.

neighbors at $\pm a\hat{\boldsymbol{x}}$, $\pm a\hat{\boldsymbol{y}}$, $\pm a\hat{\boldsymbol{z}}$ are all antiparallel, $J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{k}) = 2J_2(\cos k_x a + \cos k_y a + \cos k_z a)$, and $J^{\mathrm{ex}}_{\mathrm{AA}} = 0$. Then

$$\omega^{\mathrm{M}}_{\boldsymbol{k}} = 6J_2|\boldsymbol{S}|(1 - \zeta^2_{\boldsymbol{k}})^{1/2}, \qquad \zeta_{\boldsymbol{k}} = \tfrac{1}{3}(\cos k_x a + \cos k_y a + \cos k_z a), \tag{7.12}$$

with the maximum $6J_2|\boldsymbol{S}|$ where $\zeta_{\boldsymbol{k}} = 0$ and the small-k law $\omega^{\mathrm{M}}_{\boldsymbol{k}} \simeq 2\sqrt{3}\,J_2|\boldsymbol{S}|\,ak$, the same in every direction. The nearest-neighbor J_1 adds 6 parallel bonds, in the same (111) plane, and 6 antiparallel ones, in the adjacent planes, and makes the velocity depend on direction. Fig. 7.1 shows the result for NiO.

Worked example 7.2: The magnon velocity of NiO

With the exchange constants Eq. (7.5), $|\boldsymbol{S}| = 1$ for Ni^{2+}, and the measured lattice constant $a = 4.17\,\text{Å}$ as tabulated by Fischer et al. [1, Table I], Eq. (7.12) gives the velocity

$$2\sqrt{3}\,J_2|\boldsymbol{S}|\,a = 2\sqrt{3} \times 19.01\,\text{meV} \times 4.17\,\text{Å} = 274.6\,\text{meV Å}.$$

Dividing by $\hbar c = 1973.27\,\text{eV Å}$ turns this into 1.39×10^{-4} in natural units, or 41.7 km/s. The nearest-neighbor exchange modifies it. Along [111], the direction normal to the ferromagnetic planes, the 6 parallel neighbors lie in a plane perpendicular to $\boldsymbol{k}$ and drop

out of Eq. (D.114). Each of the 6 antiparallel ones has $(\boldsymbol{k}\cdot\boldsymbol{\delta})^2 = a^2k^2/3$, the same as each next-nearest neighbor contributes on average, and $J^{\rm ex}_{\rm AB}(0) = 6(J_1 + J_2)$. The bracket of Eq. (D.114) is $2a^2k^2(J_1 + J_2)$, and

$$\omega^{\rm M}_{\boldsymbol{k}} \simeq 2\sqrt{3}\,(J_1 + J_2)|\boldsymbol{S}|\,ak = 254.8\,\text{meV Å} \times k \qquad (\boldsymbol{k} \parallel [111]).$$

In other directions the parallel bonds no longer drop out, and evaluating Eq. (7.11) numerically at small k gives slopes between 255 and 269 meV Å. The measured initial slope is ∼250 meV Å.
The largest energy follows from a bound. With $J_1 < 0$, the in-plane sum $J^{\rm ex}_{\rm AA}(\boldsymbol{k}) = J_1 \sum_{\boldsymbol{\delta}} \cos \boldsymbol{k}\cdot\boldsymbol{\delta}$ over the 6 parallel neighbors is largest when the cosines sum to their minimum, −3, the value of a triangular lattice at the corner of its zone. Then $A_{\boldsymbol{k}} \le J^{\rm ex}_{\rm AB}(0) - J^{\rm ex}_{\rm AA}(0) - 3J_1 = 6J_2 - 3J_1$, and $\omega^{\rm M}_{\boldsymbol{k}} \le |\boldsymbol{S}|A_{\boldsymbol{k}}$ with equality where $B_{\boldsymbol{k}} = 0$. Both conditions hold at once, at $\omega^{\rm M}_{\rm max} = |\boldsymbol{S}|(6J_2 - 3J_1) = 118.2$ meV, against $6J_2|\boldsymbol{S}| = 114.1$ meV for J_2 alone and a measured maximum of ∼117 meV. Linear spin-wave theory with two exchange constants thus reproduces both ends of the measured spectrum to a few percent. The agreement is partly by construction, since the constants were fitted to the same data, and it does not test the $1/|\boldsymbol{S}|$ corrections, which a fit absorbs into J_1 and J_2.
The magnon velocity, 1.39×10^{-4}, is 19 times smaller than the fastest dark-matter speed in the laboratory, $v_{\rm max} = 2.650 \times 10^{-3}$ (Sect. 1.3), and five times larger than the longitudinal sound speed of silicon, 8.43 km/s along [100] (Appendix C.18). A dark-matter particle moves faster than the magnons it can emit. Emitting a magnon near the zone center is kinematically allowed in the same way as emitting an acoustic phonon, and at the same momentum transfer it deposits about five times more energy.

7.1.4 Spin-density response

A probe that couples to the spin of the electron, such as a slow neutron or a dark-matter particle with a spin-dependent interaction, couples to the *electron spin density*

$$\boldsymbol{s}_e(\boldsymbol{q}) = \sum_i \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}\,\frac{\boldsymbol{\sigma}_i}{2}, \tag{7.13}$$

the spin analog of the density $n_e(\boldsymbol{q})$ of Sect. 2.3, with the Pauli matrices $\boldsymbol{\sigma}_i$ of electron i. Like $n_e(\boldsymbol{q})$, it adds the momentum $\boldsymbol{q}$ to the target. For dark matter of spin $\frac{1}{2}$ with a spin-dependent coupling, the interaction takes the form $-\sum_a \hat{O}_{\chi a}(\boldsymbol{q})\, s_{e,a}(\boldsymbol{q})$ [3, Eq. (4)], with an operator $\hat{O}_{\chi a}$ that acts on the spin of the dark-matter particle; Sect. 8.1 lists and derives these operators. The three components $s_{e,a}(\boldsymbol{q})$ are then target operators O_a of the response matrix S_{ab} of Eq. (4.13), as the density was for a charge coupling.

In a magnetic insulator the spin density sits on the ions. The filled shells carry no spin, and the unpaired electrons of ion j occupy an open shell whose spin, at low energy, is $\boldsymbol{S}_j$. Within the ground multiplet of the ion every vector operator is proportional to $\boldsymbol{S}_j$, by the Wigner–Eckart theorem (the matrix elements of any vector operator within a multiplet of fixed total spin are those of the spin itself times one constant), so the spin density near ion j is $\boldsymbol{S}_j\,\varrho_{\rm s}(\boldsymbol{r} - \boldsymbol{R}_j)$, with $\varrho_{\rm s}$ the normalized density of the unpaired electrons. Taking the Fourier transform,

$$\boldsymbol{s}_e(\boldsymbol{q}) = F^{\rm mag}(q) \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_j}\,\boldsymbol{S}_j, \qquad F^{\rm mag}(q) = \int \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,\varrho_{\rm s}(\boldsymbol{r}), \tag{7.14}$$

where $F^{\rm mag}(q)$, the *magnetic form factor* of the ion, is spherically averaged and normalized to $F^{\rm mag}(0) = 1$. It plays for the spin the role that the atomic form factor of Sect. 2.2 plays

for the charge. For a $3d$ shell it stays close to one up to the inverse shell radius, a few Å^{-1} or several keV (Appendix C.19), above most momentum transfers of sub-MeV dark matter. Appendix C.19 adds the orbital moment of shells that have one and derives the neutron cross-section, which measures the same S_{ab}. Eq. (7.14) reduces the response of the electrons to that of the ionic spins, which linear spin-wave theory supplies.

For the ferromagnet, at zero temperature, the transverse spin operators create single magnons. Define $S^{\pm}(\boldsymbol{q}) = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_j} S_j^{\pm}$. At linear order $S_j^- \simeq (2|\boldsymbol{S}|)^{1/2} d_j^\dagger$, and with $d_j^\dagger = N^{-1/2}\sum_{\boldsymbol{k}} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_j} a_{\boldsymbol{k}}^\dagger$ and the lattice sum Eq. (3.14),

$$S^-(\boldsymbol{q}) \simeq (2|\boldsymbol{S}|N)^{1/2}\, a_{\boldsymbol{q}_r}^\dagger,$$

where $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_{\boldsymbol{q}}$ is the split of Eq. (3.16) into a wavevector in the zone and a reciprocal-lattice vector. Crystal momentum is conserved: the magnon carries $\boldsymbol{q}_r$, and the lattice takes $\boldsymbol{G}_{\boldsymbol{q}}$. The operator $S^+(\boldsymbol{q})$ annihilates the ground state, which is fully polarized, and S^z is quadratic in the bosons and creates no single magnon. With $s_{e,x} = F^{\mathrm{mag}}(S^+ + S^-)/2$ and $s_{e,y} = F^{\mathrm{mag}}(S^+ - S^-)/2\mathrm{i}$, the one-magnon matrix elements are $\langle \boldsymbol{q}_r | s_{e,x} | 0\rangle = F^{\mathrm{mag}}(2|\boldsymbol{S}|N)^{1/2}/2$ and $\langle \boldsymbol{q}_r | s_{e,y} | 0\rangle = \mathrm{i}F^{\mathrm{mag}}(2|\boldsymbol{S}|N)^{1/2}/2$. Inserting them into Eq. (4.13) with $\mathscr{V} = N\Omega$, where Ω is the volume per magnetic ion:

One-magnon spin response of a ferromagnet

$$S_{ab}(\boldsymbol{q},\omega) = \frac{\pi|\boldsymbol{S}|}{\Omega}\,|F^{\mathrm{mag}}(q)|^2\,(\delta_{ab} - \hat{z}_a\hat{z}_b + \mathrm{i}\epsilon_{abc}\hat{z}_c)\,\delta(\omega - \omega^{\mathrm{M}}_{\boldsymbol{q}_r}), \tag{7.15}$$

for $T = 0$, $\omega > 0$, and spins ordered along $\hat{\boldsymbol{z}}$, with a, b the Cartesian components of $\boldsymbol{s}_e$.

The matrix in brackets has the eigenvalues 2, for the circular combination $s_{e,x} - \mathrm{i}s_{e,y}$ that lowers the spin, and 0 for the other two: Eq. (7.15) is Hermitian and positive semidefinite, as Eq. (4.20) requires. A ferromagnet absorbs one sense of rotation only, the one in which its spins precess; this is the polarization vector $(|\boldsymbol{S}|/2)^{1/2}(1, \mathrm{i}, 0)$ of Ref. [3, Eq. (6)]. Apart from $|F^{\mathrm{mag}}|^2$ the weight does not depend on $\boldsymbol{q}$: every momentum transfer finds its magnon with the same strength. Through the fluctuation–dissipation theorem Eq. (4.18), the same delta function is a pole of the transverse spin response R_{ab}, the susceptibility that magnetic resonance and neutron scattering measure.

In the antiferromagnet a transverse spin operator again creates one magnon, now with the amplitudes of Eq. (7.10). Inverting it, $d_{\mathrm{A}\boldsymbol{k}} = u^{\mathrm{M}}_{\boldsymbol{k}} a_{\boldsymbol{k}1} - v^{\mathrm{M}}_{\boldsymbol{k}} a^\dagger_{\boldsymbol{k}2}$. Appendix D.19 shows that the transverse spin of both sublattices together creates a magnon near the zone center with the amplitude $(2|\boldsymbol{S}|N_{\mathrm{s}})^{1/2}(u^{\mathrm{M}}_{\boldsymbol{k}} - v^{\mathrm{M}}_{\boldsymbol{k}})$. Near the ordering wavevector, where the two sublattices enter with opposite phases, the amplitude is $(2|\boldsymbol{S}|N_{\mathrm{s}})^{1/2}(u^{\mathrm{M}}_{\boldsymbol{k}} + v^{\mathrm{M}}_{\boldsymbol{k}})$. From the expressions of $\cosh 2\theta_{\boldsymbol{k}}$ and $\sinh 2\theta_{\boldsymbol{k}}$ in Appendix C.7,

$$(u^{\mathrm{M}}_{\boldsymbol{k}} \mp v^{\mathrm{M}}_{\boldsymbol{k}})^2 = \frac{A_{\boldsymbol{k}} \mp B_{\boldsymbol{k}}}{(A^2_{\boldsymbol{k}} - B^2_{\boldsymbol{k}})^{1/2}} = \frac{|\boldsymbol{S}|(A_{\boldsymbol{k}} \mp B_{\boldsymbol{k}})}{\omega^{\mathrm{M}}_{\boldsymbol{k}}}. \tag{7.16}$$

Near Γ, $A_{\boldsymbol{k}} - B_{\boldsymbol{k}} \propto k^2$ while $\omega^{\mathrm{M}}_{\boldsymbol{k}} \propto k$, so the weight vanishes linearly in k: a uniform field cannot tilt two opposite sublattices without paying the exchange field. Near the ordering wavevector the weight grows as the inverse distance from it. Unlike the ferromagnet, the antiferromagnet thus distributes its one-magnon weight unevenly: at the same $\boldsymbol{k}$ the two regions differ by the factor $(A_{\boldsymbol{k}} + B_{\boldsymbol{k}})/(A_{\boldsymbol{k}} - B_{\boldsymbol{k}})$. Dark matter samples the suppressed part near Γ unless its momentum transfer reaches the ordering wavevector, $(\pi/a)(1,1,1)$ in NiO, of length $\sqrt{3}\pi/a = 1.30\,\text{Å}^{-1} = 2.6\,\mathrm{keV}$. Since $q \le 2m_\chi v_{\max}$, that takes $m_\chi \gtrsim 0.5\,\mathrm{MeV}$. The longitudinal component S^z creates magnons in pairs. Sect. 14.1 assembles these weights into the one-magnon rate, and Sect. 14.2 treats the pairs.

The lower end of the magnon spectrum, a few meV, is set by the anisotropy that cuts off the Goldstone mode. The spin coupling itself is the subject of Sect. 8.1.

7.2 Superconductivity

Aluminum loses its electrical resistance below 1.18 K, and at the lowest temperatures tunneling measures an energy gap of 0.18 meV at its Fermi surface [7, Table III]. The gap is not a band gap. It opens because the electrons within a few tens of meV of the Fermi surface bind into pairs, and an excitation that conserves the number of electrons must break a pair, at a cost of $2\Delta_{\mathrm{BCS}} \approx 0.36\,\mathrm{meV}$. This is the smallest quantum of Table 1.4, and by the bound $m_{\min} = 2E_{\mathrm{th}}/v_{\max}^2$ of Sect. 1.3 it would let dark matter far below a keV deposit detectable energy (Worked example 7.3), as Hochberg and collaborators proposed [8]. Superconducting films are also the sensors that read out most phonon-based detectors (Sect. 1.5). Where does the attraction that binds the pairs come from, and how strong is it? What are the excitations of the paired state, and how do the density and the spin density of the electrons create them? And does the answer respect charge conservation, which the simplest treatment of the paired state does not? We answer them in turn, starting from the electron–phonon vertex of Sect. 6.2.

7.2.1 The phonon-mediated attraction

Two electrons repel each other through the Coulomb interaction, yet in a metal they can attract each other by exchanging a phonon. Appendix D.11 showed how. In the diagram rules, a phonon line replaces the Coulomb line, and with the normalization of Eq. (A.25), in which the vertex g comes with $1/\sqrt{N}$, the rule Eq. (D.58) reads $-v_{\mathrm{C}}/\mathcal{V} \to -|g|^2 \mathsf{D}_0(\boldsymbol{Q},\omega)/N$. Two electrons that exchange energy ω through a phonon $(\boldsymbol{Q},\nu)$ therefore feel the interaction

$$\frac{1}{N}\,|g_\nu(\boldsymbol{k},\boldsymbol{Q})|^2\,\mathsf{D}_0(\boldsymbol{Q}\nu,\omega) = \frac{1}{N}\,\frac{2\omega_{\boldsymbol{Q}\nu}\,|g_\nu(\boldsymbol{k},\boldsymbol{Q})|^2}{\omega^2-\omega_{\boldsymbol{Q}\nu}^2}, \tag{7.17}$$

with g_ν the vertex of a single band at the Fermi level, whose band indices we drop. For $|\omega| < \omega_{\boldsymbol{Q}\nu}$ it is negative, an attraction. The picture is one of timing. An electron moving through the crystal pulls the ions toward its path, and the ions, which respond on the scale $1/\omega_{\boldsymbol{Q}\nu}$, are still displaced after the electron has left. A second electron that arrives within that time is attracted by the positive charge the first one left behind. The screened Coulomb repulsion between the two acts only when they are close at the same moment, and by then the first electron is far away. Bardeen and Pines combined the two interactions in the electron gas and found the net interaction attractive for energy transfers below the phonon energy [9]. In a scattering of two electrons near the Fermi surface the energy transfer is $\omega = \xi_{\boldsymbol{k}'} - \xi_{\boldsymbol{k}}$, with $\xi_{\boldsymbol{k}}$ the band energy measured from the Fermi level. The attraction acts between electrons whose energies lie within about the Debye frequency ω_{D} of the Fermi surface, the scale of the largest phonon energies (Sect. 5.2).

The pairing strength. Take two electrons of opposite momentum and spin, $(\boldsymbol{k}\uparrow, -\boldsymbol{k}\downarrow)$, and let them scatter into $(\boldsymbol{k}'\uparrow, -\boldsymbol{k}'\downarrow)$ with $\boldsymbol{Q} = \boldsymbol{k}' - \boldsymbol{k}$. For energy transfers small compared with the phonon energy, Eq. (7.17) is its static value $-2|g_\nu|^2/(N\omega_{\boldsymbol{Q}\nu})$. In the interaction Hamiltonian $(1/2N)\sum_{\boldsymbol{Q}} v(\boldsymbol{Q})\, n_e(\boldsymbol{Q}) n_e(-\boldsymbol{Q})$ the pair process appears twice, once with the spin-up electron in $n_e(\boldsymbol{Q})$ and once with the spin-down electron there, which cancels the $\frac{1}{2}$. Time reversal makes the vertex of the spin-down electron, $g_\nu(-\boldsymbol{k},-\boldsymbol{Q})$, the complex conjugate of $g_\nu(\boldsymbol{k},\boldsymbol{Q})$, so the product of the two vertices is $|g_\nu(\boldsymbol{k},\boldsymbol{Q})|^2$. The pair term is therefore

$$-\frac{2}{N}\sum_\nu \frac{|g_\nu(\boldsymbol{k},\boldsymbol{Q})|^2}{\omega_{\boldsymbol{Q}\nu}}\; c^\dagger_{\boldsymbol{k}'\uparrow} c^\dagger_{-\boldsymbol{k}'\downarrow} c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow}.$$

The model of Bardeen, Cooper and Schrieffer [10] replaces the coefficient by a constant $-V_{\rm BCS}/\mathcal{V}$, with $\mathcal{V} = N\Omega$ the volume of the crystal and Ω the unit-cell volume, for all $\boldsymbol{k}$ and $\boldsymbol{k}'$ within $\omega_{\rm D}$ of the Fermi surface, and by zero outside that shell. The natural constant is the Fermi-surface average, $V_{\rm BCS} = 2\Omega\langle\sum_\nu |g_\nu|^2/\omega_{\boldsymbol{Q}\nu}\rangle_{\rm FS}$, where $\langle X\rangle_{\rm FS} = \sum_{\boldsymbol{k}\boldsymbol{k}'} X\,\delta(\xi_{\boldsymbol{k}})\delta(\xi_{\boldsymbol{k}'})/[\sum_{\boldsymbol{k}}\delta(\xi_{\boldsymbol{k}})]^2$ averages over both electrons on the Fermi surface. With $\sum_{\boldsymbol{k}}\delta(\xi_{\boldsymbol{k}}) = N\Omega\mathcal{N}_{\rm F}$ for one spin orientation and the Eliashberg function Eq. (6.34),

$$\mathcal{N}_{\rm F}V_{\rm BCS} = \frac{2}{\mathcal{N}_{\rm F}\Omega\,N^2}\sum_{\boldsymbol{k}\boldsymbol{Q}\nu}\frac{|g_\nu(\boldsymbol{k},\boldsymbol{Q})|^2}{\omega_{\boldsymbol{Q}\nu}}\,\delta(\xi_{\boldsymbol{k}})\,\delta(\xi_{\boldsymbol{k}+\boldsymbol{Q}}) = 2\int_0^\infty {\rm d}\omega\,\frac{\alpha^2F(\omega)}{\omega}.$$

The right side is the coupling constant of Eq. (6.35):

Phonon-mediated pairing strength

$$\mathcal{N}_{\rm F}V_{\rm BCS} = \lambda_{\rm ep}. \tag{7.18}$$

The dimensionless strength of the attraction is the same number that enhances the electron mass and that the phonon linewidths of Sect. 6.4 measure.

Eq. (7.18) leaves out the Coulomb repulsion and the quasiparticle renormalization, both of the same order. The screened Coulomb repulsion acts at all energy transfers, while the attraction acts only within $\omega_{\rm D}$ of the Fermi surface. Its Fermi-surface average is reduced by this mismatch of time scales to the Coulomb pseudopotential $\mu^*_{\rm C}$ of Appendix E.9, of order 0.1, which subtracts from $\lambda_{\rm ep}$. The pairing electrons are also not bare electrons. They are the quasiparticles of Sect. 6.4, with the residue $1/(1+\lambda_{\rm ep})$. The Eliashberg equations of Appendix E.9 contain both effects. At low frequencies the renormalization function of Eq. (E.66) is $1+\lambda_{\rm ep}$, the mass enhancement. If we go further and cut the frequency-dependent coupling Eq. (E.68) off at about $\omega_{\rm D}$, replacing it by $\lambda_{\rm ep}$ below and by zero above, the Eliashberg gap equation Eq. (E.67) reduces to the BCS one derived below, Eq. (7.26), written in Matsubara form, with the effective coupling

$$\mathcal{N}_{\rm F}V_{\rm BCS} \simeq \frac{\lambda_{\rm ep}-\mu^*_{\rm C}}{1+\lambda_{\rm ep}}. \tag{7.19}$$

For aluminum, with $\lambda_{\rm ep} = 0.44$ and $\mu^*_{\rm C} = 0.12$, this is 0.22, half of Eq. (7.18). In practice the gap of a real superconductor is computed by solving the Eliashberg equations with the α^2F of Sect. 6.4, which comes from density-functional perturbation theory and Wannier interpolation (Appendices E.5 and E.9). The model interaction keeps only what the gap needs: a constant attraction in a thin shell.

The reduced Hamiltonian. The attraction scatters pairs of any total momentum, but pairs of zero total momentum are special. For every $\boldsymbol{k}$ in the shell, every $\boldsymbol{k}'$ in the shell is a possible final state of the pair $(\boldsymbol{k}\uparrow, -\boldsymbol{k}\downarrow)$. A pair with total momentum $\boldsymbol{P} \neq 0$, by contrast, needs both $\boldsymbol{k}$ and $\boldsymbol{P}-\boldsymbol{k}$ in the shell, a condition that removes most of the phase space once $|\boldsymbol{P}|$ exceeds $\omega_{\rm D}/v_{\rm F}$. The *reduced* BCS Hamiltonian keeps only the zero-momentum pairs:

$$H_{\rm BCS} = \sum_{\boldsymbol{k}\sigma}\xi_{\boldsymbol{k}}\,c^\dagger_{\boldsymbol{k}\sigma}c_{\boldsymbol{k}\sigma} - \frac{V_{\rm BCS}}{\mathcal{V}}{\sum_{\boldsymbol{k}\boldsymbol{k}'}}' c^\dagger_{\boldsymbol{k}'\uparrow}c^\dagger_{-\boldsymbol{k}\downarrow}c_{-\boldsymbol{k}'\downarrow}c_{\boldsymbol{k}'\uparrow}, \tag{7.20}$$

with the prime restricting $|\xi_{\boldsymbol{k}}|$ and $|\xi_{\boldsymbol{k}'}|$ to less than $\omega_{\rm D}$. This is Eq. (D.116). The pairs of finite momentum return when we ask how the paired state responds to a probe at wavevector $\boldsymbol{q}$.

7.2.2 Cooper pairs and the BCS state

A weak attraction does not bind two particles in three-dimensional vacuum, but it does bind two electrons on top of a filled Fermi sea. Cooper showed this with a two-electron problem [11]. Add two electrons to a Fermi sea that is held fixed, in the state $\sum'_{\boldsymbol{k}} a_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow}|\mathrm{FS}\rangle$ with $0 < \xi_{\boldsymbol{k}} < \omega_\mathrm{D}$, the states below the Fermi level being occupied. Eq. (7.20) leads to $(2\xi_{\boldsymbol{k}} - E)\, a_{\boldsymbol{k}} = (V_\mathrm{BCS}/\mathcal{V}) \sum'_{\boldsymbol{k}'} a_{\boldsymbol{k}'}$ for the energy E of the pair above the Fermi level. Since the right side does not depend on $\boldsymbol{k}$, $a_{\boldsymbol{k}} \propto 1/(2\xi_{\boldsymbol{k}} - E)$, and summing both sides over $\boldsymbol{k}$ gives the condition

$$1 = \frac{V_\mathrm{BCS}}{\mathcal{V}} \sum_{\boldsymbol{k}}{}' \frac{1}{2\xi_{\boldsymbol{k}} + E_\mathrm{bind}} = \mathcal{N}_\mathrm{F} V_\mathrm{BCS} \int_0^{\omega_\mathrm{D}} \frac{\mathrm{d}\xi}{2\xi + E_\mathrm{bind}} = \frac{\mathcal{N}_\mathrm{F} V_\mathrm{BCS}}{2} \ln\Bigl(1 + \frac{2\omega_\mathrm{D}}{E_\mathrm{bind}}\Bigr),$$

for a bound state at $E = -E_\mathrm{bind}$. It has a solution for every $V_\mathrm{BCS} > 0$, $E_\mathrm{bind} \simeq 2\omega_\mathrm{D} \mathrm{e}^{-2/\mathcal{N}_\mathrm{F} V_\mathrm{BCS}}$ at weak coupling, because the density of pair states stays finite down to zero energy above a Fermi surface, while in vacuum it vanishes as $\xi^{1/2}$ and the integral converges. The bound pair is a *Cooper pair*. Its binding is exponentially small and nonanalytic in the coupling. For aluminum, with $\mathcal{N}_\mathrm{F} V_\mathrm{BCS} = 0.164$ and $\omega_\mathrm{D} = 40\,\mathrm{meV}$ (Worked example 7.3), it would be $0.40\,\mu\mathrm{eV}$, some 900 times smaller than the measured $2\Delta_\mathrm{BCS}$. The difference is in the exponent, $2/\mathcal{N}_\mathrm{F} V_\mathrm{BCS}$ here against $1/\mathcal{N}_\mathrm{F} V_\mathrm{BCS}$ below. The Cooper problem freezes the Fermi sea, whereas every electron near the Fermi surface pairs at once, and the pairs bind each other more strongly than a single pair is bound.

Mean field. The many-pair problem is solved by the mean-field theory of Appendix D.20, which we summarize. Define the pair amplitude $b_{\boldsymbol{k}} = \langle c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow}\rangle$, which is zero in any state with a definite number of electrons, and the gap

$$\Delta_\mathrm{BCS} = \frac{V_\mathrm{BCS}}{\mathcal{V}} \sum_{\boldsymbol{k}}{}' b_{\boldsymbol{k}}, \tag{7.21}$$

real by the choice of an overall phase. In Eq. (7.20) each pair operator interacts with all $\sim N$ others with a strength of order $1/\mathcal{V}$, and replacing their sum by its average, Eq. (7.21), makes an error that vanishes as $\mathcal{V} \to \infty$. The interaction then becomes the pair field $-\Delta_\mathrm{BCS}(c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow} + c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow})$ acting on each pair of modes, and for each $\boldsymbol{k}$ the mean-field Hamiltonian H_MF is the two-mode fermion problem Eq. (C.63) with $\xi_0 = \xi_{\boldsymbol{k}}$ and $g_0 = \Delta_\mathrm{BCS}$. It is diagonalized by the fermionic Bogoliubov transformation of Appendix C.7, the counterpart of the bosonic one that diagonalized the antiferromagnet in Sect. 7.1, with sines and cosines in place of hyperbolic functions. The quasiparticle operators are

$$\gamma_{\boldsymbol{k}1} = u^\mathrm{B}_{\boldsymbol{k}}\, c_{\boldsymbol{k}\uparrow} - v^\mathrm{B}_{\boldsymbol{k}}\, c^\dagger_{-\boldsymbol{k}\downarrow}, \qquad \gamma_{\boldsymbol{k}2} = u^\mathrm{B}_{\boldsymbol{k}}\, c_{-\boldsymbol{k}\downarrow} + v^\mathrm{B}_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}\uparrow}, \tag{7.22}$$

and by Eq. (C.64),

Bogoliubov quasiparticles

$$\begin{gathered} H_\mathrm{MF} = \sum_{\boldsymbol{k}}{}' E_{\boldsymbol{k}} (\gamma^\dagger_{\boldsymbol{k}1}\gamma_{\boldsymbol{k}1} + \gamma^\dagger_{\boldsymbol{k}2}\gamma_{\boldsymbol{k}2}) + \mathrm{const}, \qquad E_{\boldsymbol{k}} = \bigl(\xi^2_{\boldsymbol{k}} + \Delta^2_\mathrm{BCS}\bigr)^{1/2}, \\ (u^\mathrm{B}_{\boldsymbol{k}})^2 = \frac{1}{2}\Bigl(1 + \frac{\xi_{\boldsymbol{k}}}{E_{\boldsymbol{k}}}\Bigr), \qquad (v^\mathrm{B}_{\boldsymbol{k}})^2 = \frac{1}{2}\Bigl(1 - \frac{\xi_{\boldsymbol{k}}}{E_{\boldsymbol{k}}}\Bigr), \qquad 2u^\mathrm{B}_{\boldsymbol{k}} v^\mathrm{B}_{\boldsymbol{k}} = \frac{\Delta_\mathrm{BCS}}{E_{\boldsymbol{k}}}. \end{gathered} \tag{7.23}$$

The coefficients $u^\mathrm{B}_{\boldsymbol{k}}$ and $v^\mathrm{B}_{\boldsymbol{k}}$ are the *coherence factors* of the BCS state. The quasiparticle $\gamma^\dagger_{\boldsymbol{k}1}$ carries momentum $\boldsymbol{k}$ and spin up. Far above the Fermi surface $u^\mathrm{B}_{\boldsymbol{k}} \to 1$ and it is an added

electron; far below, $v_{\boldsymbol{k}}^{\mathrm{B}} \to 1$ and it is a removed electron of momentum $-\boldsymbol{k}$ and spin down, a hole; at the Fermi surface it is half of each, with energy Δ_{BCS}. The ground state, annihilated by every γ, is the BCS state

$$|\mathrm{BCS}\rangle = \prod_{\boldsymbol{k}} (u_{\boldsymbol{k}}^{\mathrm{B}} + v_{\boldsymbol{k}}^{\mathrm{B}}\, c_{\boldsymbol{k}\uparrow}^{\dagger} c_{-\boldsymbol{k}\downarrow}^{\dagger})|0\rangle, \tag{7.24}$$

in which each pair state $(\boldsymbol{k}\uparrow, -\boldsymbol{k}\downarrow)$ is occupied with probability $(v_{\boldsymbol{k}}^{\mathrm{B}})^2$. The step of the Fermi sea is smeared over an energy Δ_{BCS}, and the coherent superposition of empty and full pair states is what makes $b_{\boldsymbol{k}} = u_{\boldsymbol{k}}^{\mathrm{B}} v_{\boldsymbol{k}}^{\mathrm{B}}$ nonzero. Note that $|\mathrm{BCS}\rangle$ has no definite electron number. Its relative number fluctuations are of order $N^{-1/2}$. They do not matter for energies, but they spoil the density response, which the last subsection repairs.

Quasiparticle density of states. Since $\mathrm{d}E_{\boldsymbol{k}}/\mathrm{d}\xi_{\boldsymbol{k}} = \xi_{\boldsymbol{k}}/E_{\boldsymbol{k}}$, each energy $E > \Delta_{\mathrm{BCS}}$ is reached from two values of ξ, an electronlike state above the Fermi level and a holelike one below. With $\mathcal{N}_{\mathrm{F}}$ constant over the shell, the quasiparticles of one spin orientation have the density of states per volume

$$\mathcal{N}_{\mathrm{qp}}(E) = 2\mathcal{N}_{\mathrm{F}} \frac{E}{\sqrt{E^2 - \Delta_{\mathrm{BCS}}^2}}\, \Theta(E - \Delta_{\mathrm{BCS}}), \tag{7.25}$$

which is Eq. (D.120). The factor 2 counts the two branches, not the spin; both spin orientations together have $g_{\mathrm{deg}}\mathcal{N}_{\mathrm{qp}}$, with $g_{\mathrm{deg}} = 2$. In the normal metal, $\mathcal{N}_{\mathrm{qp}} \to 2\mathcal{N}_{\mathrm{F}}$, the density of electron and hole excitations at energy E above the ground state. Pairing removes the states from $|E| < \Delta_{\mathrm{BCS}}$ and piles them up at the edge, where the density diverges as $(E - \Delta_{\mathrm{BCS}})^{-1/2}$ (Fig. 7.2a). The number of states is conserved: $\int_{\Delta_{\mathrm{BCS}}}^{\omega} \mathrm{d}E\, \mathcal{N}_{\mathrm{qp}} = 2\mathcal{N}_{\mathrm{F}}(\omega^2 - \Delta_{\mathrm{BCS}}^2)^{1/2}$, which approaches the normal-state count $2\mathcal{N}_{\mathrm{F}}\omega$ as ω grows.

The gap equation. The gap must reproduce itself through Eq. (7.21). In thermal equilibrium the quasiparticles are independent fermions, occupied with $n_{\mathrm{F}}(E_{\boldsymbol{k}})$, and inverting Eq. (7.22) yields $b_{\boldsymbol{k}} = u_{\boldsymbol{k}}^{\mathrm{B}} v_{\boldsymbol{k}}^{\mathrm{B}} [1 - 2n_{\mathrm{F}}(E_{\boldsymbol{k}})] = (\Delta_{\mathrm{BCS}}/2E_{\boldsymbol{k}}) \tanh(E_{\boldsymbol{k}}/2T)$. Inserting this, dividing by Δ_{BCS} and replacing $\mathcal{V}^{-1}\sum_{\boldsymbol{k}}$ by $\mathcal{N}_{\mathrm{F}} \int \mathrm{d}\xi$,

BCS gap equation and its weak-coupling solutions

$$\begin{gathered}
\frac{1}{\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}}} = \int_0^{\omega_{\mathrm{D}}} \mathrm{d}\xi\, \frac{\tanh(\sqrt{\xi^2 + \Delta_{\mathrm{BCS}}^2}/2T)}{\sqrt{\xi^2 + \Delta_{\mathrm{BCS}}^2}}, \\
\Delta_{\mathrm{BCS}}(0) \simeq 2\omega_{\mathrm{D}}\, \mathrm{e}^{-1/\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}}}, \qquad T_c = \frac{2\mathrm{e}^{\gamma_{\mathrm{E}}}}{\pi}\, \omega_{\mathrm{D}}\, \mathrm{e}^{-1/\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}}}, \\
\frac{\Delta_{\mathrm{BCS}}(0)}{T_c} = \pi \mathrm{e}^{-\gamma_{\mathrm{E}}} = 1.764,
\end{gathered} \tag{7.26}$$

with $\gamma_{\mathrm{E}} = 0.5772$ Euler's constant (Eqs. (D.121) to (D.124)). The right side of the gap equation diverges logarithmically as Δ_{BCS} and T go to zero, which is why a solution exists for any attraction, as in the Cooper problem. At $T = 0$ the integral is $\operatorname{arsinh}(\omega_{\mathrm{D}}/\Delta_{\mathrm{BCS}})$, and the gap follows; at T_c the gap closes, and the integral, evaluated in Appendix D.20 by expanding the hyperbolic tangent over Matsubara frequencies, fixes the critical temperature. The gap, like the Cooper binding, is nonanalytic in the coupling and invisible at any finite order of perturbation theory. Both $\Delta_{\mathrm{BCS}}(0)$ and T_c depend exponentially on $\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}}$ and linearly on the cutoff, but their ratio depends on neither, and it is the same for every material

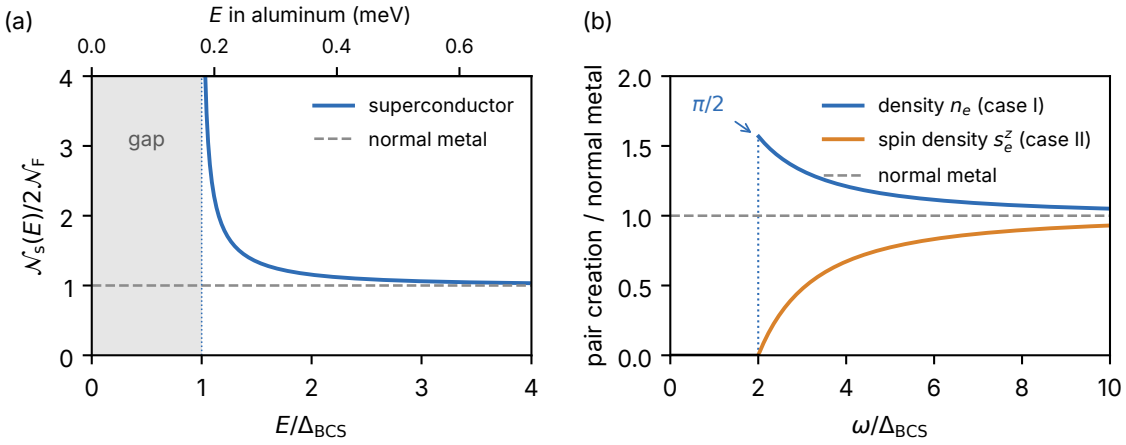


Figure 7.2: Quasiparticles of a BCS superconductor at $T = 0$. (a) Density of states per spin orientation, Eq. (7.25), in units of its normal-state value $2\mathcal{N}_F$: no states below Δ_{BCS} (gray), an integrable divergence at the edge. The top axis shows the energy for aluminum, $\Delta_{BCS} = 1.764\,T_c = 0.179\,\text{meV}$ with the measured $T_c = 1.18\,\text{K}$. (b) Rate of pair creation by a probe of energy ω coupled to the electron density (case I) or to the z component of the spin density (case II), divided by its normal-state value, Eq. (7.30). Both are bare quasiparticle bubbles for a linearized band with $q \ll k_F$ and $qv_F \geq \omega$, the regime of dark-matter scattering in aluminum, where the ratio does not depend on q. The density curve jumps to $\pi/2$ at the threshold $2\Delta_{BCS}$; the spin curve starts from zero and is the zero-temperature conductivity ratio of Mattis and Bardeen. Both tend to one for $\omega \gg \Delta_{BCS}$.

within the model. Deviations from 1.764 measure the strength and the retardation of the real interaction. Aluminum, with the measured 0.18 meV and 1.18 K, has $2\Delta_{BCS}/T_c = 3.54$, against $2\pi e^{-\gamma_E} = 3.53$.

The pair-breaking threshold. A dark-matter particle or a phonon couples to the electrons through operators that conserve their number, such as the density $n_e(\boldsymbol{q})$ and the spin density. Expressed through Eq. (7.22), such an operator either scatters a quasiparticle that is already present or creates two at once, one from each branch; it never creates a single one, which would change the electron number by one. At $T = 0$ no quasiparticles are present, and the lowest excitation that a probe can make is a pair, at

$$\omega \geq E_{\boldsymbol{k}} + E_{\boldsymbol{k}+\boldsymbol{q}} \geq 2\Delta_{BCS}, \tag{7.27}$$

the *pair-breaking threshold*. It is reached when both quasiparticles sit at the Fermi surface, which for $q < 2k_F$ is always possible. In the language of Sect. 1.3, $E_{th} = 2\Delta_{BCS}$. At $T > 0$ thermally excited quasiparticles can absorb energy in smaller amounts, but their number is suppressed by $e^{-\Delta_{BCS}/T}$, and in aluminum at 10 mK the exponent is -208.

7.2.3 Coherence factors

The amplitude to break a pair depends on what the probe couples to, because each quasiparticle is part electron and part hole and the two parts can interfere. Take the density, $n_e(\boldsymbol{q}) = \sum_{\boldsymbol{k}\sigma} c^\dagger_{\boldsymbol{k}+\boldsymbol{q},\sigma} c_{\boldsymbol{k}\sigma}$, and the z component of the spin density of Eq. (7.13), $s^z_e(\boldsymbol{q}) = \frac{1}{2}\sum_{\boldsymbol{k}}(c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\uparrow} c_{\boldsymbol{k}\uparrow} - c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\downarrow} c_{\boldsymbol{k}\downarrow})$. Write $u = u^B_{\boldsymbol{k}}$, $v = v^B_{\boldsymbol{k}}$, and primes for $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{q}$. Substituting the inverse of

Eq. (7.22) into the spin-up term gives $c^\dagger_{\boldsymbol{k}'\uparrow}c_{\boldsymbol{k}\uparrow} = u'u\,\gamma^\dagger_{\boldsymbol{k}'1}\gamma_{\boldsymbol{k}1} + u'v\,\gamma^\dagger_{\boldsymbol{k}'1}\gamma^\dagger_{\boldsymbol{k}2} + \dots$, and the spin-down term, after relabeling $\boldsymbol{k} \to -\boldsymbol{k}-\boldsymbol{q}$, contributes to the same two processes with $-vv'$ and uv' (Appendix D.20). The density adds the spin-up and spin-down terms and the spin density subtracts them, so that the amplitudes are

$$\begin{aligned} n_e(\boldsymbol{q}):&\quad \text{pair creation } u'v + v'u, &&\quad \text{quasiparticle scattering } uu' - vv',\\ s^z_e(\boldsymbol{q}):&\quad \text{pair creation } \tfrac{1}{2}(u'v - v'u), &&\quad \text{quasiparticle scattering } \tfrac{1}{2}(uu' + vv'). \end{aligned} \tag{7.28}$$

Squared and expressed through the energies with Eq. (7.23),

Coherence factors of pair creation and quasiparticle scattering

$$(uv' \pm vu')^2 = \frac{1}{2}\Big(1 - \frac{\xi\xi' \mp \Delta^2_{\mathrm{BCS}}}{EE'}\Big), \qquad (uu' \mp vv')^2 = \frac{1}{2}\Big(1 + \frac{\xi\xi' \mp \Delta^2_{\mathrm{BCS}}}{EE'}\Big), \tag{7.29}$$

with the upper signs for the density and the lower signs for the spin density (Eq. (D.126)). The two operators differ in their behavior under time reversal: the density is even and the spin density odd. These are the two classes that Tinkham calls case I and case II [12]. Every one-body probe is a sum of a part even and a part odd under time reversal, one from each class; the current, which drives the absorption of light, is odd and belongs to case II with the spin.

At the threshold both quasiparticles sit at the Fermi surface, $\xi = \xi' = 0$, and the creation factor is $\frac{1}{2}(1 \pm 1)$: one for the density, zero for the spin density. In the density the electron and hole parts of the pair add; in the spin density they cancel. Away from the Fermi surface the factors depend on the signs of ξ and ξ', and the dark-matter kinematics decide how these are sampled. For a linearized band, $\xi' = \xi + v_{\mathrm{F}}q\cos\theta$ with θ the angle between $\boldsymbol{k}$ and $\boldsymbol{q}$. Two quasiparticles with $E + E' = \omega$ have $|\xi' - \xi| \le |\xi| + |\xi'| < \omega$, so as long as $v_{\mathrm{F}}q \ge \omega$ every pair of values (ξ, ξ') in that range is reached at some angle. Since $\cos\theta$ is uniformly distributed over the Fermi sphere, each pair is reached with the same weight.

This is the regime of dark-matter scattering in a metal. A dark-matter particle that deposits ω transfers at least $q = \omega/v_{\max}$ (Sect. 1.3), and in aluminum $v_{\mathrm{F}} = 6.75 \times 10^{-3}$, the free-electron value of Sect. 6.4, is 2.5 times $v_{\max}$. Then ξ and ξ' are independent and uniformly distributed, and the terms odd in either one average to zero. The mean creation factor of the density becomes $\frac{1}{2}(1 + \Delta^2_{\mathrm{BCS}}/EE')$, and that of the spin density $\frac{1}{2}(1 - \Delta^2_{\mathrm{BCS}}/EE')$.

The pair-creation spectrum. With these factors the absorption spectrum of the paired state follows in closed form. At $T = 0$ the dynamic structure factor of the density is the golden-rule sum over pairs, $S(\boldsymbol{q},\omega) = (2\pi/\mathcal{V})\sum_{\boldsymbol{k}}(uv' + vu')^2\,\delta(\omega - E_{\boldsymbol{k}} - E_{\boldsymbol{k}+\boldsymbol{q}})$; the sum over $\boldsymbol{k}$ with both branches covers both spin orientations, as the normal limit below confirms. Replace $\mathcal{V}^{-1}\sum_{\boldsymbol{k}}$ by $\mathcal{N}_{\mathrm{F}}\int \mathrm{d}\xi \int_{-1}^{1} \mathrm{d}\cos\theta/2$, trade the angle for ξ' at $\mathrm{d}\xi'/2v_{\mathrm{F}}q$, and change variables from ξ and ξ' to E and E' with the density of states Eq. (7.25) for each, which absorbs the sum over the four sign combinations. In the normal state, $\Delta_{\mathrm{BCS}} = 0$, the result is $S_{\mathrm{n}}(\boldsymbol{q},\omega) = 2\pi\mathcal{N}_{\mathrm{F}}\,\omega/v_{\mathrm{F}}q = m_e^2\omega/\pi q$, the electron gas of Sect. 4.5 below its curve $\omega = qv_{\mathrm{F}} - q^2/2m_e$. In the superconductor,

Bare pair-creation spectra for $qv_{\mathrm{F}} \ge \omega$

$$\frac{S_{\mathrm{I,II}}(\boldsymbol{q},\omega)}{S_{\mathrm{n}}(\boldsymbol{q},\omega)} = \frac{1}{\omega}\int_{\Delta_{\mathrm{BCS}}}^{\omega-\Delta_{\mathrm{BCS}}} \mathrm{d}E\, \frac{E(\omega - E) \pm \Delta^2_{\mathrm{BCS}}}{\sqrt{E^2 - \Delta^2_{\mathrm{BCS}}}\,\sqrt{(\omega - E)^2 - \Delta^2_{\mathrm{BCS}}}}\,\Theta(\omega - 2\Delta_{\mathrm{BCS}}), \tag{7.30}$$

with the upper sign for the density (case I) and the lower sign for the spin density (case II), each divided by its own normal-state value. Fig. 7.2(b) shows both. Near the threshold $E(\omega - E) \to \Delta_{\mathrm{BCS}}^2$, so the numerator of the integrand tends to $\Delta_{\mathrm{BCS}}^2 \pm \Delta_{\mathrm{BCS}}^2$, that is $2\Delta_{\mathrm{BCS}}^2$ for the density and zero for the spin density, while the denominator tends to $2\Delta_{\mathrm{BCS}}$ times the product of the two edge divergences of Eq. (7.25). The integral of that product over the shrinking interval stays finite, $\int \mathrm{d}E\,[(E-\Delta_{\mathrm{BCS}})(\omega-\Delta_{\mathrm{BCS}}-E)]^{-1/2} = \pi$. For the density the integral is therefore $\Delta_{\mathrm{BCS}}\pi$, and with the prefactor $1/\omega = 1/2\Delta_{\mathrm{BCS}}$ the spectrum jumps at $\omega = 2\Delta_{\mathrm{BCS}}$ to $\pi/2$ times the normal value. Since $E(\omega - E) - \Delta_{\mathrm{BCS}}^2$ vanishes at the threshold, the spin spectrum instead rises continuously from zero. Far above the gap both approach the normal metal: at $\omega = 4\Delta_{\mathrm{BCS}}$ the ratios are 1.21 and 0.67. The case-II integral is the one that Mattis and Bardeen found for the absorption of light by a superconductor at zero temperature [13], in the limit of short mean free path, where ξ and ξ' are again independent, and it reproduces their closed form in elliptic integrals. For a dark-matter particle that couples to the electron density, the gap therefore enhances the bare rate just above threshold instead of suppressing it. The pile-up of states at the edge outweighs the narrow window of energies.

Quasiparticle scattering, the second process in Eq. (7.28), needs quasiparticles in the initial state. It is weighted by $n_{\mathrm{F}}(E)[1 - n_{\mathrm{F}}(E')]$ and absorbs energies below $2\Delta_{\mathrm{BCS}}$. Its coherence factors are the mirror images of the creation factors: the density scattering factor averages to $\frac{1}{2}(1 - \Delta_{\mathrm{BCS}}^2/EE')$ and vanishes at the edge, which is why the attenuation of sound in a superconductor falls with temperature as $2n_{\mathrm{F}}(\Delta_{\mathrm{BCS}})$ instead of showing the pile-up [12]. In a detector at millikelvin temperatures this channel is closed by the factor $\mathrm{e}^{-\Delta_{\mathrm{BCS}}/T}$.

Eq. (7.30) for the density is the bare quasiparticle bubble, the response of independent Bogoliubov quasiparticles. It is not yet the density response of the superconductor, and the next subsection shows why.

7.2.4 Charge conservation and the Nambu formalism

At $\boldsymbol{q} = 0$ the density is the total electron number $\hat{N}_e$, which commutes with Eq. (7.20) and cannot excite anything. Yet by Eq. (7.28) its pair-creation amplitude at $\boldsymbol{q} = 0$ is $2uv = \Delta_{\mathrm{BCS}}/E_{\boldsymbol{k}}$, not zero. The culprit is the mean field. H_{MF} creates and destroys pairs, it does not commute with $\hat{N}_e$, and its eigenstates have no definite electron number. A response computed from independent quasiparticles then puts spectral weight $\Delta_{\mathrm{BCS}}^2/E_{\boldsymbol{k}}^2$ at $\omega = 2E_{\boldsymbol{k}}$ into the uniform density response. Its first moment, by the gap equation at $T = 0$, $\mathscr{V}^{-1}\sum_{\boldsymbol{k}}' 1/E_{\boldsymbol{k}} = 2/V_{\mathrm{BCS}}$, is

$$\int_0^\infty \mathrm{d}\omega\,\omega\,S^0(0,\omega) = \frac{4\pi}{\mathscr{V}}\sum_{\boldsymbol{k}}{}' \frac{\Delta_{\mathrm{BCS}}^2}{E_{\boldsymbol{k}}} = \frac{8\pi\Delta_{\mathrm{BCS}}^2}{V_{\mathrm{BCS}}}, \tag{7.31}$$

which is Eq. (D.129). All bubbles here carry the restriction of the pair states to the shell, as in Eq. (7.20). The f-sum rule Eq. (4.30) requires $\pi n_e q^2/m_e$, which vanishes at $q = 0$. The bare bubble violates it. The violation is not confined to $q = 0$: as long as $v_{\mathrm{F}}q \ll \omega_{\mathrm{D}}$ the spurious first moment keeps the value Eq. (7.31), since it comes from pairs spread over the whole shell, at frequencies up to $2\omega_{\mathrm{D}}$. For a linearized band the moment stays within 0.2 % of this value up to $v_{\mathrm{F}}q = 20\Delta_{\mathrm{BCS}}$. With $n_e = \frac{4}{3}\mathcal{N}_{\mathrm{F}}E_{\mathrm{F}}$ for free electrons, the spurious moment relative to the f-sum is $12(\Delta_{\mathrm{BCS}}/v_{\mathrm{F}}q)^2/\mathcal{N}_{\mathrm{F}}V_{\mathrm{BCS}}$, which exceeds one for $v_{\mathrm{F}}q < 8.6\,\Delta_{\mathrm{BCS}}$ at the aluminum coupling $\mathcal{N}_{\mathrm{F}}V_{\mathrm{BCS}} = 0.164$ of Worked example 7.3, obtained with a 40 meV cutoff. The bare bubble is a poor guide to the total spectral weight even at the momenta of dark-matter scattering.

Nambu form. The cure is easiest to see in the notation of Nambu [14], which treats the two operators that pairing mixes as one spinor, $\Psi_{\boldsymbol{k}} = (c_{\boldsymbol{k}\uparrow}, c_{-\boldsymbol{k}\downarrow}^\dagger)^{\mathrm{T}}$. Bilinears $O_A(\boldsymbol{q}) =$

$\sum_{\boldsymbol{k}} \Psi^\dagger_{\boldsymbol{k}+\boldsymbol{q}} A\, \Psi_{\boldsymbol{k}}$ with the 2×2 matrices

$$\boldsymbol{\eta} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \mathsf{s}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathsf{s}_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix} \tag{7.32}$$

are the physical operators: $O_{\boldsymbol{\eta}}(\boldsymbol{q}) = n_e(\boldsymbol{q})$ for $\boldsymbol{q} \neq 0$, after anticommuting the spin-down term, while O_{s_1} and O_{s_2} create and destroy pairs of total momentum $\boldsymbol{q}$ (Appendix D.21). For a real gap, s_1 changes the magnitude of the pair field and s_2 its phase. The matrix $\boldsymbol{\eta}$ is the one of Appendix C.7; the three matrices obey the algebra of the Pauli matrices, which we do not call σ_i because those act on the electron spin. In this notation $H_{\mathrm{MF}} = \sum_{\boldsymbol{k}} \Psi^\dagger_{\boldsymbol{k}} h_{\boldsymbol{k}} \Psi_{\boldsymbol{k}} + \mathrm{const}$ with $h_{\boldsymbol{k}} = \xi_{\boldsymbol{k}} \boldsymbol{\eta} - \Delta_{\mathrm{BCS}} \mathsf{s}_1$, whose eigenvalues are $\pm E_{\boldsymbol{k}}$, and the Matsubara Green's function of Sect. 4.7 becomes the 2×2 matrix $\mathsf{G}_0(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{F}}) = (\mathrm{i}\omega^{\mathrm{F}} - h_{\boldsymbol{k}})^{-1}$, whose off-diagonal elements are the pair amplitudes. The bare bubble with vertices A and B is the loop of Sect. 4.8 with a trace over the Nambu index, and the trace for $A = B = \boldsymbol{\eta}$ reproduces the case-I factor of Eq. (7.29).

The violation has a simple origin in this language. Since $[\mathsf{s}_1, \boldsymbol{\eta}] = -2\mathrm{i}\mathsf{s}_2$,

$$[H_{\mathrm{MF}}, n_e(\boldsymbol{q})] = \boldsymbol{q} \cdot \boldsymbol{j}_e(\boldsymbol{q}) + 2\mathrm{i}\Delta_{\mathrm{BCS}}\, O_{\mathsf{s}_2}(\boldsymbol{q}). \tag{7.33}$$

The kinetic term gives the current of the continuity equation Eq. (4.29), and the pair field adds a term that turns density into fluctuations of the pair phase. Under H_{MF}, charge leaks into the condensate and does not come back. The full Hamiltonian has no such term, because the attraction $-V_{\mathrm{BCS}} \psi^\dagger_\uparrow \psi^\dagger_\downarrow \psi_\downarrow \psi_\uparrow$ is a function of the local densities and commutes with $n_e(\boldsymbol{q})$, up to corrections of relative order $v_{\mathrm{F}} q/\omega_{\mathrm{D}}$ from the shell (Appendix D.21).

The conserving response. What the bare bubble misses is that the pair field is itself made of electrons. A density perturbation at wavevector $\boldsymbol{q}$ changes the pair amplitudes, and the changed amplitudes act back on the quasiparticles as a pair field oscillating at $\boldsymbol{q}$ and ω. Anderson [15] included this as the linear response of the time-dependent mean field. To let the pair field respond at $\boldsymbol{q}$, the attraction must scatter pairs of every total momentum, not only the zero-momentum pairs of Eq. (7.20); written in the pair channels it is $-(V_{\mathrm{BCS}}/4V) \sum_{\boldsymbol{P}} \sum_{a=1,2} O_{\mathsf{s}_a}(\boldsymbol{P}) O_{\mathsf{s}_a}(-\boldsymbol{P})$ plus one-body terms. Its mean field at wavevector $\boldsymbol{q}$ couples the induced $\delta\langle O_{\mathsf{s}_a}\rangle$ back to O_{s_a}. The response is then the random-phase approximation of Sect. 4.5, with the pair channels $(\boldsymbol{\eta}, \mathsf{s}_1, \mathsf{s}_2)$ in place of the Coulomb channel and the kernel $K = \mathrm{diag}(0, -V_{\mathrm{BCS}}/2, -V_{\mathrm{BCS}}/2)$:

Conserving density response

$$R = R^0 + R^0 K R, \qquad R_{\boldsymbol{\eta}\boldsymbol{\eta}} = R^0_{\boldsymbol{\eta}\boldsymbol{\eta}} + \sum_{a,b=1,2} R^0_{\boldsymbol{\eta}\mathsf{s}_a} \Big[(1 - K R^0)^{-1} K \Big]_{ab} R^0_{\mathsf{s}_b \boldsymbol{\eta}}, \tag{7.34}$$

with the bare Nambu bubbles R^0_{AB} and the inverse taken in the block of the two pair channels (Eq. (D.130)). In diagrams the correction is the bare bubble with a ladder of pair propagators between its two vertices. That it conserves charge follows from a general property of mean-field dynamics. In a state for which Wick's theorem holds, the time derivative of any one-body average computed with the self-consistent mean field equals the one computed with the full Hamiltonian, and the full Hamiltonian obeys the continuity equation. The term $2\mathrm{i}\Delta_{\mathrm{BCS}}\, \delta\langle O_{\mathsf{s}_2}\rangle$ that Eq. (7.33) produces is cancelled by the commutator of n_e with the induced pair field. A response that obeys the continuity equation satisfies the f-sum rule by the double-commutator argument of Sect. 4.3.

At $\boldsymbol{q} = 0$ the cancellation can be seen in closed form. If the density of states is symmetric about the Fermi level over the shell, the bubbles that connect s_1 to $\boldsymbol{\eta}$ or to s_2 are odd in ξ and vanish, and the magnitude of the pair field decouples. We assume this particle–hole symmetry here; real metals break it weakly, and the general argument above does not need it. The remaining bubbles are, with $F(\omega) = \mathscr{V}^{-1} \sum_{\boldsymbol{k}}' [E_{\boldsymbol{k}}(\omega^2 - 4E_{\boldsymbol{k}}^2)]^{-1}$,

$$R^0_{\boldsymbol{\eta}\boldsymbol{\eta}} = 4\Delta_{\mathrm{BCS}}^2 F, \qquad R^0_{\mathsf{s}_2\mathsf{s}_2} = \omega^2 F - \frac{2}{V_{\mathrm{BCS}}}, \qquad R^0_{\mathsf{s}_2\boldsymbol{\eta}} = -R^0_{\boldsymbol{\eta}\mathsf{s}_2} = 2\mathrm{i}\Delta_{\mathrm{BCS}}\, \omega F,$$

where the gap equation was used in $R^0_{\mathsf{s}_2\mathsf{s}_2}$ (Eq. (D.128)). The phase channel has the denominator $1 + \frac{1}{2} V_{\mathrm{BCS}} R^0_{\mathsf{s}_2\mathsf{s}_2} = \frac{1}{2} V_{\mathrm{BCS}} \omega^2 F$, and Eq. (7.34) gives

$$R_{\boldsymbol{\eta}\boldsymbol{\eta}}(0,\omega) = 4\Delta_{\mathrm{BCS}}^2 F + \frac{(-2\mathrm{i}\Delta_{\mathrm{BCS}}\omega F)(-\frac{1}{2}V_{\mathrm{BCS}})(2\mathrm{i}\Delta_{\mathrm{BCS}}\omega F)}{\frac{1}{2}V_{\mathrm{BCS}}\, \omega^2 F} = 4\Delta_{\mathrm{BCS}}^2 F - 4\Delta_{\mathrm{BCS}}^2 F = 0 \tag{7.35}$$

at every frequency. The phase fluctuations of the condensate absorb exactly the spurious pair-breaking weight. The cancellation used the gap equation twice, once in $R^0_{\mathsf{s}_2\mathsf{s}_2}$ and once through the coefficient V_{BCS} of the kernel: a response built on a gap that does not solve the gap equation does not conserve charge.

The Goldstone mode. The same denominator vanishes at $\omega = 0$ for $\boldsymbol{q} = 0$, because rotating the phase of Δ_{BCS} uniformly costs no energy, just as rotating all spins of the ferromagnet of Sect. 7.1 cost none. At small $\boldsymbol{q}$ the zero moves to a pole at $\omega = c_{\mathrm{AB}} q$, a sound wave of the condensate phase, the Anderson–Bogoliubov mode. Anderson found $c_{\mathrm{AB}} = v_{\mathrm{F}}/\sqrt{3}$ [15], and a numerical zero of $\det(1 - KR^0)$ for a parabolic band lies at $0.579\, v_{\mathrm{F}}$ against $1/\sqrt{3} = 0.577$ (Appendix D.21). In a neutral superconductor this mode carries the density response below $2\Delta_{\mathrm{BCS}}$ and exhausts the f-sum rule as $q \to 0$. Electrons are charged, however. Adding the Coulomb interaction to Eq. (7.34) as the random-phase kernel $v_{\mathrm{C}}(q)$ in the $\boldsymbol{\eta}$ channel pushes the mode up to the plasma frequency of Sect. 4.5, and the density response of a superconductor at low frequency is screened as in the normal metal [15]. In the language of field theory this is the Anderson–Higgs mechanism, named after Anderson, who saw it in the superconductor, and Higgs, who gave its relativistic form: a Goldstone mode coupled to a long-range gauge field becomes massive. Since a density–density interaction commutes with the density, the response stays conserving. The magnitude channel s_1, which decoupled above, carries a second collective mode at the edge $2\Delta_{\mathrm{BCS}}$, the amplitude or Higgs mode of the condensate, which the density does not excite in a particle–hole-symmetric metal.

The Ward identity. Charge conservation also constrains the building blocks of diagrams, the *vertices*: the amplitudes with which the density or the current scatters an electron from $(\boldsymbol{k}, \nu)$ to $(\boldsymbol{k} + \boldsymbol{q}, \nu + \omega)$, with ν the frequency of the incoming electron. The bare density vertex is the matrix $\boldsymbol{\eta}$ of the operator $O_{\boldsymbol{\eta}}$; the interactions dress it, as the ladder of Eq. (7.34) dresses the bubble. To find the constraint, start from the correlation function of the density with the two Nambu fields of the incoming and outgoing electron. Its time derivative turns the density into the divergence of the current, by the continuity equation, and leaves two extra terms from the equal-time commutators of the density with the two fields, each with a single Green's function. Multiplying by G^{-1} on the left and on the right then removes the propagation of the incoming and the outgoing electron and leaves the dressed vertices alone, the dressed density vertex $\mathsf{V}_{\boldsymbol{\eta}}$ and current vertex $\mathbf{V}_{\boldsymbol{j}}$ (Appendix D.21). They obey

Density Ward identity in Nambu form

$$\boldsymbol{q}\cdot\mathbf{V}_{\boldsymbol{j}}(\boldsymbol{k}+\boldsymbol{q},\boldsymbol{k})-\omega\,\mathsf{V}_{\boldsymbol{\eta}}(\boldsymbol{k}+\boldsymbol{q},\boldsymbol{k})=\boldsymbol{\eta}\,\mathsf{G}^{-1}(\boldsymbol{k},\nu)-\mathsf{G}^{-1}(\boldsymbol{k}+\boldsymbol{q},\nu+\omega)\,\boldsymbol{\eta}, \tag{7.36}$$

which is Eq. (D.131). In words, the combination of vertices on the left, the one that the continuity equation tests, is fixed by the Green's functions of the two electrons. In the normal state, $\mathsf{G}^{-1}=\nu-\xi_{\boldsymbol{k}}\boldsymbol{\eta}$, and the bare vertices $\mathsf{V}_{\boldsymbol{\eta}}=\boldsymbol{\eta}$ and $\mathbf{V}_{\boldsymbol{j}}=(\boldsymbol{k}+\boldsymbol{q}/2)/m_e$ satisfy it for a parabolic band. In the superconductor, $\mathsf{G}^{-1}=\nu-h_{\boldsymbol{k}}$ adds $\Delta_{\mathrm{BCS}}(\boldsymbol{\eta}\mathsf{s}_1-\mathsf{s}_1\boldsymbol{\eta})=2\mathrm{i}\Delta_{\mathrm{BCS}}\mathsf{s}_2$ to the right side, which no bare vertex supplies; this is the identity's version of Eq. (7.33). The ladder of Eq. (7.34) supplies it. Its dressed density vertex is $\mathsf{V}_{\boldsymbol{\eta}}=\boldsymbol{\eta}+\mathsf{c}_2\mathsf{s}_2$, and at $\boldsymbol{q}=0$ the phase-channel coefficient is $\mathsf{c}_2=-2\mathrm{i}\Delta_{\mathrm{BCS}}/\omega$, so that $-\omega\mathsf{V}_{\boldsymbol{\eta}}=-\omega\boldsymbol{\eta}+2\mathrm{i}\Delta_{\mathrm{BCS}}\mathsf{s}_2$ is exactly the right side. The pole of c_2 at $\omega=0$ is the Goldstone mode.

The spin density needs no such correction. Its Nambu vertex is the unit matrix, which commutes with $h_{\boldsymbol{k}}$. Total S^z commutes with H_{MF}, and the bare quasiparticle response to a spin probe already conserves spin. The case-II curve of Fig. 7.2(b) is therefore final at this level. Eq. (7.36) is the identity for the density alone, local in the electron's momentum and frequency. The identities that translation invariance imposes on every dark-matter operator are the subject of Sect. 8.3.

For dark matter the size of the correction is set by $\Delta_{\mathrm{BCS}}/v_{\mathrm{F}}q$, which is $1/\pi$ times the ratio of the probed wavelength $1/q$ to the *coherence length* $v_{\mathrm{F}}/\pi\Delta_{\mathrm{BCS}}$, the size of a pair. Since a deposit $\omega\geq 2\Delta_{\mathrm{BCS}}$ needs $q\geq\omega/v_{\mathrm{max}}$, the ratio is at most $v_{\mathrm{max}}/2v_{\mathrm{F}}=0.20$ in aluminum. That bound is approached only for heavy particles, which can deposit $2\Delta_{\mathrm{BCS}}$ at the smallest momentum; the typical momentum transfer, $q\sim m_\chi v$, grows with the mass, and the typical ratio falls. The conserving response of Eq. (7.34), screened by the Coulomb interaction, is what the rate of pair breaking uses, and Sects. 15.1 and 15.2 evaluate how it modifies Eq. (7.30) near the edge.

Worked example 7.3: The aluminum gap and the threshold mass

The measured critical temperature of aluminum, $T_c=1.18\,\mathrm{K}$ [7, Table III], is 0.1017 meV in energy units. The weak-coupling ratio of Eq. (7.26) yields

$$\Delta_{\mathrm{BCS}}=1.764\times 0.1017\,\mathrm{meV}=0.179\,\mathrm{meV},\qquad 2\Delta_{\mathrm{BCS}}=0.36\,\mathrm{meV},$$

in agreement with the measured gap of 0.18 meV [7, Table III]. Aluminum is a weak-coupling superconductor, and the ratio needs neither the cutoff nor the coupling.
The threshold mass follows from Eq. (1.25) with $E_{\mathrm{th}}=2\Delta_{\mathrm{BCS}}$ and $v_{\mathrm{max}}=2.650\times10^{-3}$:

$$m_{\mathrm{min}}=\frac{2\times 0.36\,\mathrm{meV}}{(2.650\times10^{-3})^2}=0.10\,\mathrm{keV}.$$

This is the bulk value. Thin aluminum films have higher critical temperatures, and dark-matter projections adopt the film convention $2\Delta_{\mathrm{BCS}}=0.6\,\mathrm{meV}$ (Sect. 1.5), which moves the threshold to 0.17 keV. At the threshold mass the particle moves at v_{max} and gives up its whole kinetic energy, with the momentum transfer $q=m_{\mathrm{min}}v_{\mathrm{max}}=2\times(2\Delta_{\mathrm{BCS}})/v_{\mathrm{max}}=0.27\,\mathrm{eV}$, so $1/q=0.73\,\mu\mathrm{m}$. With the free-electron $v_{\mathrm{F}}=6.75\times10^{-3}$, the coherence length, the size of a pair, is $v_{\mathrm{F}}/\pi\Delta_{\mathrm{BCS}}=2.4\,\mu\mathrm{m}$, three times larger, and $v_{\mathrm{F}}q=10\,\Delta_{\mathrm{BCS}}$. The condition $v_{\mathrm{F}}q\geq\omega$ of Eq. (7.30) holds with room to spare, and the ratio $\Delta_{\mathrm{BCS}}/v_{\mathrm{F}}q=0.10$ that controls the conserving correction is small but not negligible. The coupling of Sect. 6.4 does much worse on its own. The gap formula of Eq. (7.26) needs a cutoff, and with ω_{D} replaced by the largest phonon energy of aluminum, 40 meV,

the measured gap requires $\mathcal{N}_F V_{BCS} = 1/\operatorname{arsinh}(\omega_D/\Delta_{BCS}) = 0.164$; with the logarithmic average phonon energy of Sect. 6.4, $\omega_{log} = 270\,\mathrm{K} = 23.3\,\mathrm{meV}$ [7, Table III], as the cutoff it requires 0.18. The bare identification Eq. (7.18), $\mathcal{N}_F V_{BCS} = \lambda_{ep} = 0.44$, would predict $T_c = 54\,\mathrm{K}$. The quasiparticle residue reduces the coupling to $\lambda_{ep}/(1+\lambda_{ep}) = 0.31$ and T_c to 20 K; the Coulomb pseudopotential, Eq. (7.19) with $\mu_C^* = 0.12$, brings them to 0.22 and 5.8 K, still a factor 4.9 above the measured value. Since $\mathrm{d}\ln T_c/\mathrm{d}(\mathcal{N}_F V_{BCS}) = (\mathcal{N}_F V_{BCS})^{-2}$, a coupling a third too large makes that factor, $\exp(1/0.164 - 1/0.222) = 4.9$, and the choice of cutoff between 40 meV and ω_{log} changes the prefactor $1.134\,\omega_D$ by another factor of 1.7. The Eliashberg equations with the full α^2F reproduce the measured T_c only after μ_C^* has been fixed to it (Worked example E.8). For a dark-matter target the gap is taken from measurement, and theory supplies its relation to T_c.

The pair-breaking edge of aluminum lies more than three thousand times below the gap of silicon (Worked example 7.3). The f-sum rule is the check that a computed response above it has been built correctly. Chap. 15 puts the conserving and screened response into the rate of pair breaking, and Sect. 20.4 follows the quasiparticles it creates to the sensor.

7.3 Dirac and narrow-gap materials

The pentatelluride $ZrTe_5$ has a band gap of about 24 meV, and near the gap its electrons move at about 10^{-3} of the speed of light whatever their energy. Dark matter of about 8 keV can carry an electron across this gap, about forty times lighter than the 0.333 MeV that silicon requires (Sect. 1.3). Materials of this kind are called Dirac materials, because near the gap the electrons obey the Dirac equation of a relativistic particle, with a velocity v_D in place of the speed of light and half the gap in place of the mass. They were proposed as targets for sub-MeV dark matter [16]. As in the superconductor of Sect. 7.2, the cheapest excitation is a pair whose energy starts at twice a half-gap, here an electron and a hole on either side of a band gap. Pairs are cheap in momentum too: the bands are linear, and slow enough, $v_D < v_{max}$, for a dark-matter particle to outrun them. And the same electrons screen the interaction in a way that is fixed by two numbers, v_D and the dielectric constant of everything else in the crystal.

7.3.1 The Dirac Hamiltonian from $\boldsymbol{k}\cdot\boldsymbol{p}$ theory

The $\boldsymbol{k}\cdot\boldsymbol{p}$ expansion of Sect. 3.3 assumed a band well separated from all others. Near a point where two bands nearly touch, the second-order formula Eq. (3.50) breaks down, because its energy denominator becomes small, and the two bands have to be treated together. Take a time-reversal-invariant momentum $\boldsymbol{k}_0$, such as Γ, in a crystal with inversion symmetry. Time reversal combined with inversion maps a Bloch state at $\boldsymbol{k}$ onto a different state at the same $\boldsymbol{k}$, and because time reversal squares to -1 for electrons, every band is then doubly degenerate at every $\boldsymbol{k}$: the Kramers degeneracy. Suppose that at $\boldsymbol{k}_0$ two such doublets lie close to each other and far from all other bands: a conduction doublet at energy $+\Delta_D$ and a valence doublet at $-\Delta_D$, measured from the middle of the gap, with opposite parity under inversion. We write Δ_D for this *half-gap*. Keep these four states exactly and drop the remote bands. At first order in $\delta\boldsymbol{k} = \boldsymbol{k} - \boldsymbol{k}_0$, the Bloch Hamiltonian of Sect. 3.3 acts in the four-state space through the matrix elements of $\delta\boldsymbol{k}\cdot\boldsymbol{v}$, with $\boldsymbol{v}$ the velocity operator of Eq. (3.40). The velocity is odd under inversion, and its matrix elements between two states of the same parity vanish. Only

the block between the doublets survives:

$$H_{\boldsymbol{k}_0+\delta\boldsymbol{k}} = \begin{pmatrix} \Delta_{\mathrm{D}}\,\mathbb{1} & M(\delta\boldsymbol{k}) \\ M(\delta\boldsymbol{k})^\dagger & -\Delta_{\mathrm{D}}\,\mathbb{1} \end{pmatrix}, \qquad M(\delta\boldsymbol{k}) = \sum_{i=1}^{3} \delta k_i M_i, \tag{7.37}$$

where $\mathbb{1}$ is the 2×2 unit matrix and M_i is the 2×2 matrix of the velocity component v^i between the conduction and the valence doublet. The free-electron term $\delta k^2/2m_e$ and the second-order couplings to the remote bands are of order δk^2; they bend the bands at large δk, and we return to them when we fix the cutoff.

The block $M(\delta\boldsymbol{k})$ is not arbitrary, and the Kramers degeneracy is what pins it down. Square Eq. (7.37). The off-diagonal blocks of H^2 are $\Delta_{\mathrm{D}}M - M\Delta_{\mathrm{D}}$, which cancel because Δ_{D} multiplies the unit matrix, and H^2 is block diagonal, with the blocks $\Delta_{\mathrm{D}}^2 + MM^\dagger$ and $\Delta_{\mathrm{D}}^2 + M^\dagger M$. The eigenvalues of H come in pairs $\pm E$. If each is doubly degenerate at every $\delta\boldsymbol{k}$, as the Kramers degeneracy requires, then E^2 is a fourfold eigenvalue of H^2, and the 2×2 matrix $MM^\dagger$ must be a multiple of the unit matrix for every direction of $\delta\boldsymbol{k}$. This is a strong condition on the three matrices M_i. A general complex 2×2 matrix is a combination of the unit matrix and the three Pauli matrices, with eight real coefficients. The Pauli matrices here act within a doublet, not on the electron spin of Eq. (7.13), and we write them σ_i^{d}, with d for doublet. Appendix D.22 shows step by step what the condition leaves. Up to a common unitary factor, which a choice of basis in the valence doublet absorbs, each M_i is a real combination of the four matrices $\mathbb{1}$, $\mathrm{i}\sigma_1^{\mathrm{d}}$, $\mathrm{i}\sigma_2^{\mathrm{d}}$ and $\mathrm{i}\sigma_3^{\mathrm{d}}$: four real numbers per direction instead of eight. A unitary change of basis within each doublet turns these three four-vectors rigidly, by any rotation of four-dimensional space, and a suitable rotation brings M to the form $M = \sum_i v_i\,\delta k_i\,\sigma_i^{\mathrm{d}}$ with three real velocities v_i along three principal axes. In that basis we define the 4×4 Dirac matrices

$$\alpha_{\mathrm{D}}^i = \begin{pmatrix} 0 & \sigma_i^{\mathrm{d}} \\ \sigma_i^{\mathrm{d}} & 0 \end{pmatrix}, \qquad \beta_{\mathrm{D}} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix},$$

the matrices of Appendix C.15 with the subscript D that keeps them apart from the fine-structure constant and the inverse temperature. They anticommute pairwise and square to one. The four-state Hamiltonian then takes the Dirac form.

Dirac Hamiltonian

Near a time-reversal-invariant momentum $\boldsymbol{k}_0$ of a crystal with inversion symmetry, two Kramers doublets of opposite parity at $\pm\Delta_{\mathrm{D}}$ are described, to first order in $\delta\boldsymbol{k} = \boldsymbol{k} - \boldsymbol{k}_0$, by

$$H_{\boldsymbol{k}_0+\delta\boldsymbol{k}} = \sum_{i=1}^{3} v_i\,\delta k_i\,\alpha_{\mathrm{D}}^i + \Delta_{\mathrm{D}}\,\beta_{\mathrm{D}}, \qquad \varepsilon_{\pm,\delta\boldsymbol{k}} = \pm\Big(\sum_i v_i^2 \delta k_i^2 + \Delta_{\mathrm{D}}^2\Big)^{1/2}, \tag{7.38}$$

with each band doubly degenerate. The velocities v_i are the singular values of the interband velocity matrix elements, and Δ_{D} is half the energy separation of the two doublets at $\boldsymbol{k}_0$.

Eq. (7.38) is the Dirac Hamiltonian of a particle of mass Δ_{D}, with the speed of light replaced by v_i along the principal axis i; it is the form used for $ZrTe_5$ in Refs. [17, Eq. (1.1)] and [16]. When the crystal symmetry makes the three velocities equal, $v_i = v_{\mathrm{D}}$, the *Dirac velocity*, and $H = v_{\mathrm{D}}\boldsymbol{\alpha}_{\mathrm{D}}\cdot\delta\boldsymbol{k} + \Delta_{\mathrm{D}}\beta_{\mathrm{D}}$ with $E_{\boldsymbol{k}} = (v_{\mathrm{D}}^2\delta k^2 + \Delta_{\mathrm{D}}^2)^{1/2}$, the notation of Appendix D.22. The literature usually writes v_{F} for the Dirac velocity; in this book v_{F} is the Fermi velocity of the electron gas. One of the v_i may come out negative in the last rotation (Appendix D.22). Its

sign does not enter the spectrum or anything else in this section, which depends on v_i^2, and we take all three positive.

The two-level Kane model of Sect. 3.3 is the same physics in a familiar setting. There the conduction state of GaAs couples to the valence state $|X\rangle$ through $\delta k_x v_{cX}^x$ across the gap E_g, and the two levels repel as $\pm[(E_\mathrm{g}/2)^2 + |v_{cX}^x|^2 \delta k_x^2]^{1/2}$, which is Eq. (7.38) with $\Delta_\mathrm{D} = E_\mathrm{g}/2$ and $v_\mathrm{D} = |v_{cX}^x|$. Near its bottom the band is parabolic, with mass $\Delta_\mathrm{D}/v_\mathrm{D}^2 = m_e E_\mathrm{g}/E_P$: the Kane mass Eq. (3.52) without the free-electron term. With $E_P = 22.9\,\mathrm{eV}$ ($|v_{cX}^x| = 4.7 \times 10^{-3}$) and the GaAs gap of 1.52 eV, this is $0.066\,m_e$, within 2 % of the measured $0.067\,m_e$. A narrow-gap semiconductor is thus a massive Dirac material, and since the Kane energy changes little from one III–V compound to the next, its Dirac velocity stays near 4.7×10^{-3} as the gap closes. That is $1.8\,v_\mathrm{max}$, and we will see below that this is too fast for dark-matter scattering.

At $\Delta_\mathrm{D} = 0$ the conduction and valence doublets are degenerate at $\boldsymbol{k}_0$, and Eq. (7.38) separates. In the basis of the sums and differences of the conduction and valence states, it becomes two 2×2 blocks, $\pm \sum_i v_i\, \delta k_i\, \sigma_i^\mathrm{d}$. Each block is a *Weyl node*, a crossing of two nondegenerate bands that uses all three Pauli matrices, so that no small perturbation of the block alone can open a gap. The two nodes have opposite chirality, the sign in front of the block, and only a perturbation that couples them, such as the mass term $\Delta_\mathrm{D}\beta_\mathrm{D}$, gaps the spectrum. A crystal in which the nodes sit at the same momentum and symmetry keeps $\Delta_\mathrm{D} = 0$ is a Dirac semimetal. Na_3Bi and Cd_3As_2 are examples, although their nodes sit on a rotation axis away from the time-reversal-invariant momenta, where a crystal rotation protects them. If inversion or time reversal is broken, the Kramers degeneracy is lifted, the nodes can move apart in momentum, and the material is a Weyl semimetal [18]. The sign of Δ_D distinguishes an ordinary from an inverted band order, and pressure, strain or composition can tune it through zero. $ZrTe_5$ sits close to this transition, which is why its gap is small and why first-principles values disagree on its size: they range from 24 to 35 meV [17, Table 1] and [16, Table 1], and the gap is sensitive to the lattice constant used.

The linear form Eq. (7.38) holds while the neglected terms of order δk^2 are small compared with $v_i \delta k_i$. Beyond some momentum the bands bend away, and the states there behave like those of an ordinary insulator. We model this by a hard *momentum cutoff* Λ_D on $|\delta\boldsymbol{k}|$, inside which the Dirac form is used and outside which the states are counted with the remote bands. The value of Λ_D is read off a computed band structure, as the momentum at which the bands deviate from linear; for $ZrTe_5$ it is 0.1 to $0.23\,\text{Å}^{-1}$ depending on the direction, or 200 to 450 eV. Finally, let g_deg count the degenerate states per band and wavevector, summed over all Dirac points in the zone: $g_\mathrm{deg} = 2$ for one four-component cone, and twice the number of cones in general.

7.3.2 Density of states

How many states does a Dirac cone offer at energy E? For the isotropic cone the number of states per volume in one band with energy below E is g_deg times the volume of a sphere of radius $\delta k = (E^2 - \Delta_\mathrm{D}^2)^{1/2}/v_\mathrm{D}$, divided by $(2\pi)^3$. Differentiating with $\delta k\, \mathrm{d}(\delta k) = E\, \mathrm{d}E/v_\mathrm{D}^2$ gives the density of states per volume of the conduction band,

$$\mathcal{N}_\mathrm{D}(E) = \frac{g_\mathrm{deg} \delta k^2}{2\pi^2} \frac{\mathrm{d}(\delta k)}{\mathrm{d}E} = \frac{g_\mathrm{deg}}{2\pi^2 v_\mathrm{D}^3}\, E\, (E^2 - \Delta_\mathrm{D}^2)^{1/2}\, \Theta(E - \Delta_\mathrm{D}), \tag{7.39}$$

and the valence band mirrors it at $-E$. For an anisotropic cone, the substitution $\delta k_i \to \delta k_i v_\mathrm{D}/v_i$ shows that v_D^3 is replaced by $v_1 v_2 v_3$. Just above the edge, $E - \Delta_\mathrm{D} \ll \Delta_\mathrm{D}$, the band is parabolic with mass $\Delta_\mathrm{D}/v_\mathrm{D}^2$ and $\mathcal{N}_\mathrm{D}$ grows as $(E - \Delta_\mathrm{D})^{1/2}$, as at any band edge. Far above it, or at any energy in a gapless cone, $\mathcal{N}_\mathrm{D} = g_\mathrm{deg} E^2/(2\pi^2 v_\mathrm{D}^3)$, the density of states of massless particles, which vanishes at the node.

The vanishing density of states at $E = 0$ means that an undoped gapless cone has no Fermi surface. Thomas–Fermi screening (Eq. (4.53)), which needs a finite density of states at the Fermi level, is then absent, and whatever screening the Dirac electrons provide comes from interband pairs, which we compute below. The factor $1/v_{\mathrm{D}}^3$ means that slow cones have many states: at fixed E, halving the velocity multiplies the states by eight. The thermal carriers follow from the parabolic edge. The symmetry of the two bands puts the chemical potential in the middle of the gap. The conduction-electron density is then the intrinsic density Eq. (3.48) with both density-of-states masses equal to $\Delta_{\mathrm{D}}/v_{\mathrm{D}}^2$, the spin factor 2 replaced by g_{deg} and $E_{\mathrm{g}} = 2\Delta_{\mathrm{D}}$:

$$n_{\mathrm{i}} = g_{\mathrm{deg}} \Big(\frac{\Delta_{\mathrm{D}} T}{2\pi v_{\mathrm{D}}^2} \Big)^{3/2} \mathrm{e}^{-\Delta_{\mathrm{D}}/T}, \qquad T \ll \Delta_{\mathrm{D}}. \tag{7.40}$$

For the $ZrTe_5$-like cone of Worked example 7.4, with $\Delta_{\mathrm{D}} = 11.8\,\mathrm{meV}$, $v_{\mathrm{D}} = 7.9 \times 10^{-4}$ and $g_{\mathrm{deg}} = 2$, the edge mass is $0.037\,m_e$. At room temperature $\Delta_{\mathrm{D}}/T = 0.46$ and the material is full of carriers. At 4 K, $\Delta_{\mathrm{D}}/T = 34$ and Eq. (7.40) gives $n_{\mathrm{i}} \approx 0.4\,\mathrm{cm}^{-3}$, within 5 % of the exact integral over Eq. (7.39); at the millikelvin temperatures of a phonon or charge readout the thermal population is zero for all purposes. A gap of tens of meV is the right size: small enough for light dark matter, large enough to freeze out the carriers.

7.3.3 Interband pair threshold

At $T = 0$, with the chemical potential in the gap, the lower band is full and the upper band empty. The excitations are pairs: an electron at $\boldsymbol{k} + \boldsymbol{q}$ in the upper band and a hole at $\boldsymbol{k}$ in the lower one, with energy $\omega = E_{\boldsymbol{k}} + E_{\boldsymbol{k}+\boldsymbol{q}}$. For given $\boldsymbol{q}$, the cheapest pair has $\boldsymbol{k} = -\boldsymbol{q}/2$. The reason is that $E_{\boldsymbol{k}}$ is even and convex in $\boldsymbol{k}$, so that $E_{\boldsymbol{k}} + E_{\boldsymbol{k}+\boldsymbol{q}} = E_{-\boldsymbol{k}} + E_{\boldsymbol{k}+\boldsymbol{q}} \geq 2E_{\boldsymbol{q}/2}$, with equality at $\boldsymbol{k} = -\boldsymbol{q}/2$. The pair continuum is therefore

$$\omega \geq (v_{\mathrm{D}}^2 q^2 + 4\Delta_{\mathrm{D}}^2)^{1/2}, \tag{7.41}$$

the shaded region of Fig. 7.3b. The edge is the dispersion of a relativistic particle of mass $2\Delta_{\mathrm{D}}$ and momentum q, with v_{D} in place of the speed of light. In a gapless cone the edge is the line $\omega = v_{\mathrm{D}} q$. Compare the electron gas of Sect. 4.5, whose continuum reaches down to $\omega = 0$ at every $q < 2k_{\mathrm{F}}$: pairs in a Fermi sea can be made at no energy cost, while in a Dirac cone every pair costs at least $v_{\mathrm{D}} q$.

Dark matter of mass m_χ can deposit at most $\omega = q v_{\max} - q^2/2m_\chi$ at momentum q (Eq. (1.26)), and a pair is kinematically allowed where this curve enters the continuum. Fig. 7.3b shows that the velocity decides whether a cone can be reached at all, and the gap decides at which mass. The dark-matter curve lies below the line $\omega = v_{\max} q$, and the continuum edge lies above the line $\omega = v_{\mathrm{D}} q$, so that a cone with $v_{\mathrm{D}} > v_{\max}$ is out of reach of scattering at any mass, with or without a gap. This rules out the III–V narrow-gap semiconductors, with $v_{\mathrm{D}} \approx 1.8\, v_{\max}$, as scattering targets (absorption at $q \approx 0$ is another matter), and it is why the slow cones of $ZrTe_5$, at $v_{\mathrm{D}} = 0.30\, v_{\max}$, are of interest. In the limit $v_{\mathrm{D}} \ll v_{\max}$ the edge is flat at $2\Delta_{\mathrm{D}}$, and the threshold mass is that of Eq. (1.25) with $E_{\mathrm{th}} = 2\Delta_{\mathrm{D}}$, $m_{\min} = 4\Delta_{\mathrm{D}}/v_{\max}^2 = 6.7\,\mathrm{keV}$ for $\Delta_{\mathrm{D}} = 11.8\,\mathrm{meV}$. The finite velocity raises it: the smallest m_χ for which $q v_{\max} - q^2/2m_\chi \geq (v_{\mathrm{D}}^2 q^2 + 4\Delta_{\mathrm{D}}^2)^{1/2}$ at some q, found numerically, is 8.0 keV at $v_{\mathrm{D}} = 7.9 \times 10^{-4}$. In an anisotropic cone, v_{D} in Eq. (7.41) is replaced by $(\sum_i v_i^2 \hat{q}_i^2)^{1/2}$ for $\boldsymbol{q}$ along the unit vector $\hat{\boldsymbol{q}}$; for $ZrTe_5$ the threshold is 7.1 keV along its slowest axis and 9.6 keV along its fastest. The dark-matter wind arrives from a fixed direction in the Galaxy, which turns once a day in the laboratory with the rotation of the Earth, so this direction dependence is a source of daily modulation [17].

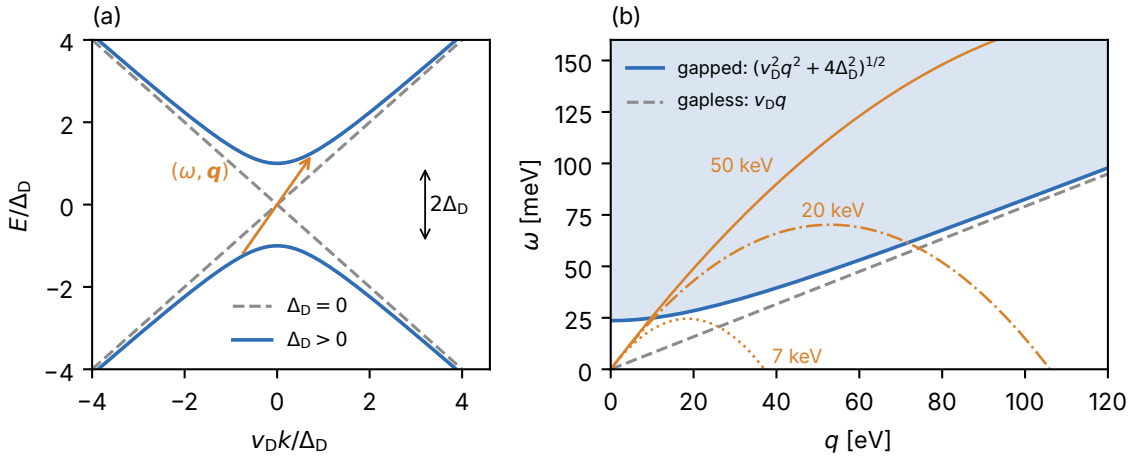


Figure 7.3: Dirac cones and their pairs. (a) Bands $\pm(v_\mathrm{D}^2 k^2 + \Delta_\mathrm{D}^2)^{1/2}$ of a gapped cone (solid) and of a gapless one (dashed), in units of the half-gap Δ_D; the arrow is an interband pair with energy ω and momentum $\boldsymbol{q}$. (b) Energy ω against momentum q of the pairs of an isotropic cone with the $ZrTe_5$-like parameters $v_\mathrm{D} = 7.9 \times 10^{-4}$ (geometric mean of the three principal velocities) and $\Delta_\mathrm{D} = 11.8\,\mathrm{meV}$ of Ref. [17, Table 1 and text below it]. Shaded: the pair continuum $\omega \geq (v_\mathrm{D}^2 q^2 + 4\Delta_\mathrm{D}^2)^{1/2}$; dashed: its edge $\omega = v_\mathrm{D} q$ for $\Delta_\mathrm{D} = 0$. Orange: the largest energy that dark matter of mass 7, 20 and 50 keV can deposit at momentum q, $\omega = q v_\mathrm{max} - q^2/2m_\chi$ with $v_\mathrm{max} = 2.650 \times 10^{-3}$. The 7 keV curve stays just below the continuum; the threshold mass is 8.0 keV.

7.3.4 Screening by Dirac electrons

The remote bands that we dropped from Eq. (7.37) do not disappear from the response. They are the core and the bonding states of the crystal, separated from the Dirac bands by an eV or more, and at the energies $\omega \lesssim 0.1\,\mathrm{eV}$ and momenta $q \lesssim \Lambda_\mathrm{D}$ that matter here they respond instantly and uniformly, like a dielectric medium. We describe them by a constant, the *background dielectric constant* ε_bg: the electronic dielectric constant ε_∞ of Sect. 4.4 that the crystal would have without its Dirac electrons. In $ZrTe_5$ the remote interband transitions and the polar lattice make it large, between about 20 and 300 depending on the direction. Two charges among the Dirac electrons then interact through $e^2/(\varepsilon_\mathrm{bg} q^2)$, not e^2/q^2, and the Dirac electrons screen on top of the background. In the random-phase approximation of Sect. 4.5, with R_D^0 the independent-pair response of the Dirac electrons,

$$\varepsilon(\boldsymbol{q},\omega) = \varepsilon_\mathrm{bg} - v_\mathrm{C}(q)\, R_\mathrm{D}^0(\boldsymbol{q},\omega) = \varepsilon_\mathrm{bg}\Big[1 - \frac{e^2}{\varepsilon_\mathrm{bg} q^2} R_\mathrm{D}^0(\boldsymbol{q},\omega)\Big]. \tag{7.42}$$

The bare Coulomb interaction $v_\mathrm{C} = e^2/q^2$ multiplies the Dirac response, and the background enters as an additive constant. Inside the bracket the Dirac electrons see the coupling $e^2/\varepsilon_\mathrm{bg}$. This placement of ε_bg fixes the overall size of every rate in a Dirac target, and Sect. 16.1 returns to it.

How strong is that coupling? A pair of Dirac electrons at distance r has Coulomb energy $\alpha/(\varepsilon_\mathrm{bg} r)$ and kinetic energy of order v_D/r, since a particle localized within r has momentum $\sim 1/r$ and energy v_D/r. Their ratio,

$$\tilde{\alpha} = \frac{\alpha}{\varepsilon_\mathrm{bg} v_\mathrm{D}}, \tag{7.43}$$

is the same at every distance. The *effective coupling* $\tilde{\alpha}$ plays the role that the fine-structure constant plays in quantum electrodynamics, where $v_\mathrm{D} = 1$ and $\varepsilon_\mathrm{bg} = 1$. A small velocity makes it large, and a large background constant makes it small again. For $ZrTe_5$ the two compete: $1/v_\mathrm{D} \approx 1300$ and $\varepsilon_\mathrm{bg} \approx 50$ on average, and $\tilde{\alpha} = 0.18$ to 0.20 with $g_\mathrm{deg} = 2$ (Worked example 7.4). Part of the literature folds the degeneracy into the coupling and writes $g_\mathrm{deg}\alpha/(\varepsilon_\mathrm{bg} v_\mathrm{D})$ for what we write $g_\mathrm{deg}\tilde{\alpha}$ [16, Eq. (2.5)].

Appendix D.22 evaluates R^0_D for one isotropic cone at $T = 0$ with the chemical potential in the gap. Only interband pairs contribute, and each enters with the overlap of the band spinors, Eq. (D.133), which vanishes at $q = 0$, because the two bands are orthogonal, and grows as q^2. That factor q^2 cancels the $1/q^2$ of the Coulomb interaction. In a metal the Thomas–Fermi term k^2_TF/q^2 diverges as $q \to 0$; in a Dirac material the Dirac contribution to ε stays finite, as it does in an insulator. The sum over $\boldsymbol{k}$ diverges logarithmically at large $\boldsymbol{k}$, for the same reason as the vacuum polarization of quantum electrodynamics, and the cutoff $|\boldsymbol{k}| < \Lambda_\mathrm{D}$ makes it finite. The result, Eq. (D.136), is

$$\frac{\varepsilon(\boldsymbol{q},\omega)}{\varepsilon_\mathrm{bg}} = 1 + \frac{g_\mathrm{deg}\tilde{\alpha}}{6\pi}\Big[\ln\frac{4v_\mathrm{D}^2\Lambda_\mathrm{D}^2}{\Delta_\mathrm{D}^2} - \frac{5}{3} - 6\int_0^1 \mathrm{d}x\, x(1-x)\ln\Big(1 - x(1-x)\frac{\omega^2 - v_\mathrm{D}^2 q^2}{\Delta_\mathrm{D}^2} - \mathrm{i}0^+\,\mathrm{sgn}\,\omega\Big)\Big], \tag{7.44}$$

valid for q, ω/v_D and $\Delta_\mathrm{D}/v_\mathrm{D}$ small compared with Λ_D. The response depends on $\boldsymbol{q}$ and ω only through $\omega^2 - v_\mathrm{D}^2 q^2$, the invariant of a Lorentz transformation with v_D in place of the speed of light. It is the one-loop vacuum polarization of quantum electrodynamics with an electron of mass Δ_D, which is how Ref. [16, Eq. (2.6)] obtained it. That reference fixes the constant inside the bracket by a subtraction at the cutoff instead of a hard cutoff, and its cutoff corresponds to $\mathrm{e}^{-5/6}\Lambda_\mathrm{D} = 0.43\,\Lambda_\mathrm{D}$ in ours.

The imaginary part, which counts the pairs, needs no cutoff. For $\omega > 0$ the logarithm in Eq. (7.44) has a negative argument inside the continuum Eq. (7.41), and with $s = \omega^2 - v_\mathrm{D}^2 q^2$,

$$\frac{\varepsilon_2(\boldsymbol{q},\omega)}{\varepsilon_\mathrm{bg}} = \frac{g_\mathrm{deg}\tilde{\alpha}}{6}\Big(1 - \frac{4\Delta_\mathrm{D}^2}{s}\Big)^{1/2}\Big(1 + \frac{2\Delta_\mathrm{D}^2}{s}\Big)\Theta(s - 4\Delta_\mathrm{D}^2), \tag{7.45}$$

which is Eq. (D.135). It rises from zero at the edge as the square root of the distance to it, like the density of states of a band edge, and for $s \gg \Delta_\mathrm{D}^2$ it tends to the constant $\varepsilon_\mathrm{bg} g_\mathrm{deg}\tilde{\alpha}/6$, the same at every q and ω. A Dirac material absorbs and scatters with a strength that, deep in the continuum, does not depend on how much energy or momentum is deposited. The flat ε_2 of a gapless cone, $\varepsilon_\mathrm{bg} g_\mathrm{deg}\tilde{\alpha}/6$ above $\omega = v_\mathrm{D} q$, is a signature of linear bands; optical measurements on the Dirac semimetal Na_3Bi agree with it to about 10 % [16].

The real part carries the screening. At $q \to 0$ and $\omega = 0$ the integral in Eq. (7.44) vanishes. The static dielectric constant of a gapped Dirac material is then Eq. (D.137),

$$\varepsilon(0,0) = \varepsilon_\mathrm{bg}\Big[1 + \frac{g_\mathrm{deg}\tilde{\alpha}}{3\pi}\Big(\ln\frac{2v_\mathrm{D}\Lambda_\mathrm{D}}{\Delta_\mathrm{D}} - \frac{5}{6}\Big)\Big]. \tag{7.46}$$

The Dirac electrons add to the background a term proportional to $g_\mathrm{deg}\tilde{\alpha}$ times a logarithm of the ratio of the band width $v_\mathrm{D}\Lambda_\mathrm{D}$ to the gap. Pairs of every energy between $2\Delta_\mathrm{D}$ and $2v_\mathrm{D}\Lambda_\mathrm{D}$ contribute equally per logarithmic interval, which is where the logarithm comes from. The dependence on the cutoff is mild: doubling Λ_D changes $\varepsilon(0,0)/\varepsilon_\mathrm{bg}$ by $g_\mathrm{deg}\tilde{\alpha}\ln 2/3\pi$, under 3 % for $ZrTe_5$. And the split between ε_bg and the Dirac term is not unique. States beyond Λ_D belong to the background, so a larger cutoff moves weight from ε_bg into the logarithm, and only the total ε is measurable.

As q grows at $\omega = 0$, the argument of the logarithm in Eq. (7.44) grows and the screening weakens. Once $v_{\mathrm{D}} q \gg 2\Delta_{\mathrm{D}}$ the gap no longer matters, and the result approaches that of a gapless cone.

Limit Check 7.1. Dielectric function of a gapless Dirac material

For $\Delta_{\mathrm{D}} \to 0$, with a hard momentum cutoff Λ_{D}, $\varepsilon/\varepsilon_{\mathrm{bg}} = 1 + (g_{\mathrm{deg}}\tilde{\alpha}/6\pi)\ln[4\Lambda_{\mathrm{D}}^2/|q^2 - \omega^2/v_{\mathrm{D}}^2|] + \mathrm{i}(g_{\mathrm{deg}}\tilde{\alpha}/6)\,\Theta(\omega - v_{\mathrm{D}}q)$, after [16, Eq. (2.7)].

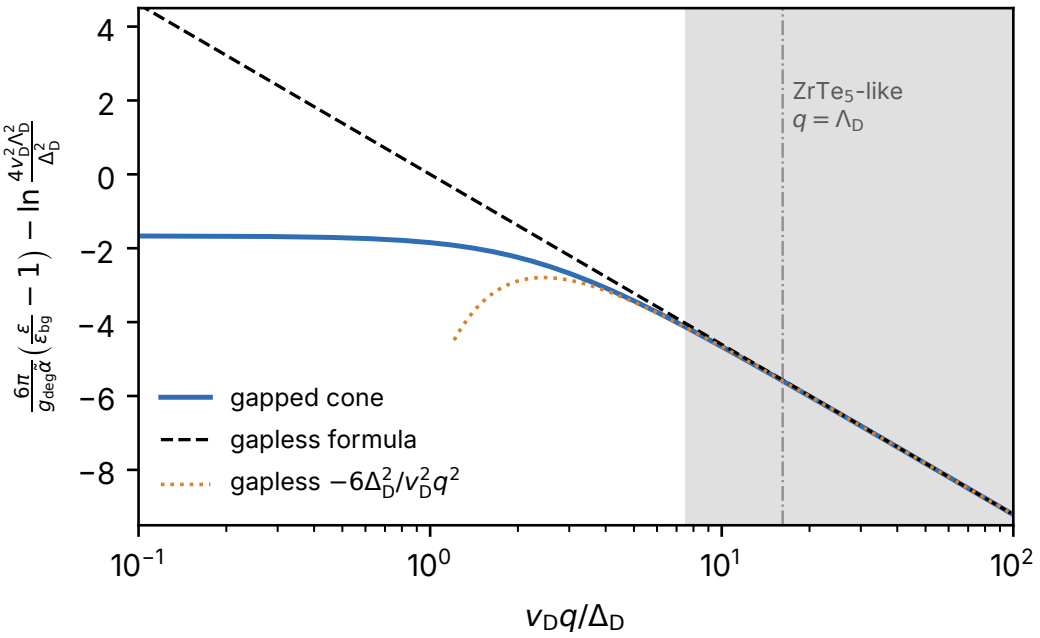


Figure 7.4: Static dielectric function of a gapped Dirac cone against the gapless formula. Plotted is the induced part $\varepsilon(q,0)/\varepsilon_{\mathrm{bg}} - 1$ in units of $g_{\mathrm{deg}}\tilde{\alpha}/6\pi$, less the cutoff term $\ln(4v_{\mathrm{D}}^2\Lambda_{\mathrm{D}}^2/\Delta_{\mathrm{D}}^2)$, which makes the curves independent of the cutoff (hard cutoff on $|\boldsymbol{k}|$). Solid: Eq. (7.44) at $\omega = 0$; dashed: the gapless formula, $-2\ln(v_{\mathrm{D}}q/\Delta_{\mathrm{D}})$ on this scale; dotted: the gapless formula with its leading correction $-6\Delta_{\mathrm{D}}^2/v_{\mathrm{D}}^2q^2$. Gray: $v_{\mathrm{D}}q > 7.5\,\Delta_{\mathrm{D}}$, where the gapless formula is within 0.1 of the full result; with the correction this holds from $3.7\,\Delta_{\mathrm{D}}$, and at $v_{\mathrm{D}}q = 10\,\Delta_{\mathrm{D}}$ the residual is 3×10^{-3}. Dash-dotted: $q = \Lambda_{\mathrm{D}}$ for the ZrTe$_5$-like cone of Worked example 7.4, where $v_{\mathrm{D}}\Lambda_{\mathrm{D}} = 16.2\,\Delta_{\mathrm{D}}$. The Dirac region ends there.

Below $v_{\mathrm{D}}q \approx 2\Delta_{\mathrm{D}}$ the gapless formula diverges logarithmically, while the gapped cone saturates at Eq. (7.46): the gap cuts off the pairs that would screen at long wavelength.

The gapless formula follows from Eq. (7.44) by expanding the logarithm for $|s| \gg \Delta_{\mathrm{D}}^2$: $\ln[1 - x(1-x)s/\Delta_{\mathrm{D}}^2] \to \ln[-x(1-x)s/\Delta_{\mathrm{D}}^2] + \Delta_{\mathrm{D}}^2/[-x(1-x)s]$, and with $\int_0^1 x(1-x)\ln[x(1-x)]\,\mathrm{d}x = -5/18$ the constants $-\frac{5}{3}$ cancel and Δ_{D} drops out of the logarithm (Eq. (D.138)). The next term contributes $-6\Delta_{\mathrm{D}}^2/(v_{\mathrm{D}}^2q^2 - \omega^2)$ to the bracket. Static screening in a gapless cone, $\varepsilon(q,0) = \varepsilon_{\mathrm{bg}}[1 + (g_{\mathrm{deg}}\tilde{\alpha}/3\pi)\ln(2\Lambda_{\mathrm{D}}/q)]$, grows without bound as $q \to 0$, but only logarithmically. There is no screening length, since the density of states vanishes at the node; the electron gas screens a charge completely beyond $1/k_{\mathrm{TF}}$, whereas a gapless Dirac material only reduces it by a slowly growing factor.

The logarithms also set the validity of the calculation. The random-phase approximation keeps the first term in $\tilde{\alpha}$ of the polarization, and the induced part of Eq. (7.44) is of order

$g_{\mathrm{deg}}\tilde{\alpha}\ln(v_{\mathrm{D}}\Lambda_{\mathrm{D}}/\max(\Delta_{\mathrm{D}}, v_{\mathrm{D}}q))$, up to the numerical factor $1/3\pi$. We take this product as the expansion parameter: perturbation theory is reliable when it is small compared with one. Reading the screened interaction $e^2/\varepsilon q^2$ as a coupling that depends on q, the screening makes it weaker at smaller q, the counterpart of the running of the fine-structure constant in quantum electrodynamics, and the theory becomes weakly coupled at long wavelength [16]. For $\tilde{\alpha} \gtrsim 1$ the one-loop result is an estimate, and what replaces it is an open question that Sect. 16.1 takes up.

Anisotropy enters through a rescaling. With $\delta k_i \to \delta k_i/v_i$, the Hamiltonian Eq. (7.38) becomes isotropic with unit velocity, the sum over $\boldsymbol{k}$ acquires the Jacobian $1/(v_1v_2v_3)$, and q^2 in the overlap becomes $\sum_i v_i^2 q_i^2$ (Appendix D.22). The cutoff becomes an ellipsoid in $\boldsymbol{k}$, a sphere of radius $\Lambda_{\mathrm{D}}v_{\mathrm{D}}$ in energy units for the rescaled momenta. For $\boldsymbol{q}$ along the principal axis i, the Coulomb factor e^2/q^2 then multiplies $v_i^2/(v_1v_2v_3)$ in place of $1/v_{\mathrm{D}}$, and the background constant becomes its component $\varepsilon_{\mathrm{bg},ii}$ along that axis, assuming the background and the cone share principal axes. The static constant along i is

$$\varepsilon_{ii}(0,0) = \varepsilon_{\mathrm{bg},ii} + \frac{g_{\mathrm{deg}}\alpha}{3\pi}\,\frac{v_i^2}{v_1v_2v_3}\Big(\ln\frac{2\tilde{\Lambda}}{\Delta_{\mathrm{D}}} - \frac{5}{6}\Big), \tag{7.47}$$

with $\tilde{\Lambda}$ the cutoff on the energy $(\sum_i v_i^2\delta k_i^2)^{1/2}$. Along each axis the effective coupling is $\alpha v_i^2/(\varepsilon_{\mathrm{bg},ii}v_1v_2v_3)$, which reduces to Eq. (7.43) for an isotropic cone.

The four inputs of Eq. (7.44), v_i, Δ_{D}, Λ_{D} and $\varepsilon_{\mathrm{bg}}$, come from first-principles calculations. The velocities and the cutoff are fitted to a computed band structure near the node (Appendix E.1). The half-gap is the delicate one: semilocal functionals misplace small gaps and can invert the band order, and quasiparticle corrections (Appendix E.3) or the measured gap are needed. The background constant comes from density-functional perturbation theory (Appendix E.4) or from the random-phase sum over bands (Appendix E.6). Both compute the dielectric constant of the whole crystal, which includes the Dirac pairs, and they do so at the calculated gap, to which the Dirac term is sensitive through $\ln(1/\Delta_{\mathrm{D}})$. A consistent $\varepsilon_{\mathrm{bg}}$ leaves the Dirac bands out of the sum over transitions, or subtracts Eq. (7.46) from the computed total. The parameters of $ZrTe_5$ used in rate calculations are specified in Appendix E.20.

Worked example 7.4: Effective coupling and static screening of $ZrTe_5$

$ZrTe_5$ has one gapped cone at Γ, $g_{\mathrm{deg}} = 2$. We take the first-principles values for the measured crystal structure [17, Table 1 and text below it]. They are the principal velocities $v_x, v_y, v_z = 1.1\times10^{-3}$, 4.5×10^{-4}, 1.0×10^{-3}, a gap of 23.6 meV ($\Delta_{\mathrm{D}} = 11.8$ meV), and cutoffs $\Lambda_x, \Lambda_y, \Lambda_z = 0.23$, 0.215, 0.1 Å^{-1}. A relaxed cell volume gives the same v_x, slower $v_y = 4.4\times10^{-4}$ and $v_z = 9.1\times10^{-4}$, and a gap of 31.2 meV ($\Delta_{\mathrm{D}} = 15.6$ meV); the reference implements that row in its rates. The dielectric tensor from density-functional perturbation theory has the diagonal components 308.4, 20.75 and 126.1 [17, Table 2].

Cutoff. With 1 Å^{-1} = 1.973 keV, the band energies at the three cutoffs are $v_i\Lambda_i = 0.499$, 0.191 and 0.197 eV. The cone is linear to a much higher energy along x than along y. We keep the states with $(\sum_i v_i^2\delta k_i^2)^{1/2} < \tilde{\Lambda} = 0.191$ eV, the smallest of the three [17, Eq. (2.21)], a conservative choice that a better fit of the bands would replace by a larger ellipsoid. Their dielectric function, Ref. [17, Eq. (3.10)], carries the same constant $-\frac{5}{3}$ as Eq. (7.44), so that $\tilde{\Lambda}$ is a hard cutoff in our sense, $\tilde{\Lambda} = v_{\mathrm{D}}\Lambda_{\mathrm{D}}$.

Isotropic estimate. To get a single number, replace the cone by an isotropic one with the same density of states, $v_{\mathrm{D}} = (v_xv_yv_z)^{1/3} = 7.9\times10^{-4}$, and the tensor by the harmonic mean of its components, $\varepsilon_{\mathrm{bg}} = 3/(\sum_i 1/\varepsilon_{\mathrm{bg},ii}) = 50.5$, the average used in Ref. [16]. Then Eq. (7.43) gives $\tilde{\alpha} = 0.18$ and $g_{\mathrm{deg}}\tilde{\alpha} = 0.37$. The cutoff in momentum

is $\Lambda_D = \tilde{\Lambda}/v_D = 241\,\text{eV} = 0.12\,\text{Å}^{-1}$. An earlier calculation [16, Table 1], with faster velocities (2.9×10^{-3}, 5.0×10^{-4}, 2.1×10^{-3}) and smaller dielectric constants (187.5, 9.8 and 90.9, harmonic mean 25.3), yields a similar $\tilde{\alpha} = 0.20$, quoted there as $g_{\text{deg}}\tilde{\alpha} = 0.80$ with $g_{\text{deg}} = 4$ states per band.

Static screening. With $\ln(2\tilde{\Lambda}/\Delta_D) = \ln 32.4 = 3.48$, Eq. (7.46) yields

$$\frac{\varepsilon(0,0)}{\varepsilon_{\text{bg}}} - 1 = \frac{0.37}{3\pi}\,(3.48 - 0.83) = 0.102,$$

so that $\varepsilon(0,0) = 55.7$. The Dirac electrons add 10 % to the screening of the background. At finite q, the integral in Eq. (7.44) reduces this to 0.091 at $q = 2\Delta_D/v_D = 30\,\text{eV}$ and to 0.052 at $q = \Lambda_D/2 = 121\,\text{eV}$, momenta at which sub-MeV dark matter scatters (Fig. 7.3b). The expansion parameter at long wavelength, $g_{\text{deg}}\tilde{\alpha}\ln(v_D\Lambda_D/\Delta_D) = 0.37\ln 16.2 = 1.0$, is about one. The induced term itself, 0.102, is small mainly through the factor $1/3\pi$, and the one-loop result is a fair estimate of it. Since $v_D\Lambda_D = 16.2\,\Delta_D$, $ZrTe_5$ reaches the gapless regime of Fig. 7.4 only above $v_D q = 7.5\,\Delta_D$, or $q = 112\,\text{eV}$: below that momentum the gap matters.

Anisotropy. Along the principal axes, Eq. (7.47) with $\ln(2\tilde{\Lambda}/\Delta_D) - \frac{5}{6} = 2.64$ adds 10.0, 1.67 and 8.3 to the background components 308.4, 20.75 and 126.1, that is 3.2, 8.1 and 6.6 %. The effective couplings along the three axes, $\alpha v_i^2/(\varepsilon_{\text{bg},ii} v_x v_y v_z)$, are 0.058, 0.144 and 0.117: the slow direction y has the smallest background constant as well, and it is the most strongly coupled. The averaged $\tilde{\alpha} = 0.18$ is a fair summary, though it overstates the coupling along x threefold.

Size of the Dirac screening. The Dirac contribution is 3 to 8 % of the dielectric constant, smaller than the spread between the two sets of computed parameters. Both sets of dielectric constants are totals for the whole crystal, computed at a gap of 24 to 35 meV, and already contain a Dirac term of this size, which is harmless at the few-percent level. The coupling that matters for a rate is the total ε, dominated here by the background: with $\varepsilon \approx 56$, every Coulomb-like interaction inside $ZrTe_5$ is reduced about fifty-fold.

Dirac materials screen the interaction mostly through their large background dielectric constant, with the Dirac electrons adding a logarithm of order $g_{\text{deg}}\tilde{\alpha}$. The loss function that follows from Eq. (7.44), and the rates of scattering and absorption with the background placed in the total ε as in Eq. (7.42), are the subject of Sect. 16.1.

7.4 Defects and dopants

A phosphorus atom in silicon binds its fifth electron by 45.6 meV, a twenty-fifth of the gap. Dark matter of 13 keV carries enough kinetic energy to set that electron free, where the gap of the perfect crystal needs 0.333 MeV (Sect. 1.3). A carbon–hydrogen complex in the same crystal, the T center, emits light at 935 meV, and a quarter of its photons leave without exciting a single phonon. And a silicon atom struck with a few tens of eV leaves its site for good, storing several eV in the vacancy and the interstitial it leaves behind. Defects therefore enter a dark-matter experiment in two roles: as targets with their own low thresholds, and as final states or readouts of an energy deposit. Each is built from formalism already in hand. The shallow donor is the hydrogenic envelope of Sect. 4.9 with the valley masses of Sect. 3.3, the color center is the displaced oscillator of Sect. 5.6, and the displacement threshold is a property of the energy surface of Sect. 5.1.

7.4.1 Shallow donors

An atom from group V on a silicon site, such as phosphorus, arsenic or antimony, uses four of its five valence electrons in the bonds. The fifth is bound to the ion core and its one extra positive charge, in a hydrogenic donor level just below the conduction band, built from conduction-band states. An impurity of this kind is a *donor*; a group-III atom such as boron, one electron short, binds a hole instead and is an *acceptor*. Both bind their carrier weakly, because the host screens the Coulomb attraction by the static dielectric constant ε_0 and the carrier moves with the band mass m^* rather than m_e.

The hydrogen atom with these two substitutions already sets the scale. Appendix D.23 derives the envelope equation of Kohn and Luttinger [19] and [20, Chap. 4]. The donor potential varies slowly on the scale of the lattice and mixes only conduction states within $\sim 1/a^*$ of a band minimum $\boldsymbol{k}_j$. The donor state is then the Bloch function at the minimum modulated by an envelope,

$$\psi(\boldsymbol{r}) \simeq F_j(\boldsymbol{r})\, \psi_{c\boldsymbol{k}_j}(\boldsymbol{r}), \qquad \Big[-\frac{\partial_x^2+\partial_y^2}{2m_\perp^*} - \frac{\partial_z^2}{2m_\parallel^*} - \frac{\alpha}{\varepsilon_0 r}\Big] F_j(\boldsymbol{r}) = -E_{\mathrm{b}}^{\mathrm{D}}\, F_j(\boldsymbol{r}), \tag{7.48}$$

with z along the axis of the valley and $E_{\mathrm{b}}^{\mathrm{D}}$ the *donor binding energy*, measured down from the conduction-band minimum. This is *effective-mass theory*. It is the Wannier–Mott equation Eq. (4.106) with two changes. The electron is bound to a fixed ion rather than to a hole, which makes the mass the band mass and not a reduced mass. And the mass is a tensor, because the valleys of silicon are ellipsoids (Sect. 3.3).

Effective-mass donor levels

For an isotropic mass m^* the solution of Eq. (7.48) is hydrogenic, with the *effective Bohr radius* a^*,

$$E_{\mathrm{b}}^{\mathrm{D}} = \frac{m^*\alpha^2}{2\varepsilon_0^2} = \frac{m^*/m_e}{\varepsilon_0^2}\,\mathrm{Ry}, \qquad a^* = \frac{\varepsilon_0}{m^*\alpha} = \varepsilon_0\,\frac{m_e}{m^*}\,a_0, \tag{7.49}$$

and excited levels at $E_{\mathrm{b}}^{\mathrm{D}}/n^2$ below the band edge (Eq. (D.140)).

With $\varepsilon_0 = 11.7$ and masses of a few tenths of m_e, the binding is tens of meV and the orbit tens of Å, several lattice constants. The assumption behind Eq. (7.48), an envelope spread over many cells, is then satisfied, which makes the estimate self-consistent. For silicon the estimate brackets the answer without giving it: the transverse mass $m_\perp^* = 0.191\, m_e$ puts the binding at 19.0 meV and $a^* = 32.4$ Å, the longitudinal mass $m_\parallel^* = 0.916\, m_e$ at 91.0 meV and 6.8 Å.

The anisotropic problem has one parameter, the mass ratio $\gamma_m = m_\perp^*/m_\parallel^* = 0.209$. In units of the transverse Rydberg $\mathrm{Ry}_\perp^* = m_\perp^*\alpha^2/2\varepsilon_0^2 = 18.98$ meV and radius $a_\perp^* = 32.4$ Å, the variational envelope of Kohn and Luttinger, a hydrogen $1s$ function squeezed along the heavy axis, binds the electron by $E_{\mathrm{b}}^{\mathrm{D}} = 1.564\,\mathrm{Ry}_\perp^* = 29.7$ meV, with the radii $r_\perp = 24.3$ Å across the valley axis and $r_\parallel = 14.0$ Å along it (Eq. (D.142)). The orbit is an oblate spheroid, flattened along the heavy direction, where the electron is less mobile, and the binding lies between the two isotropic values, closer to the light one, because two of the three directions are light. The full spectrum of Eq. (7.48) needs a numerical solution. Scaling z by $\gamma_m^{1/2}$ makes the kinetic energy isotropic and turns the Coulomb potential into $-2/[r\,(1-(1-\gamma_m)\cos^2\theta)^{1/2}]$ in these units. Expanding that angular factor in Legendre polynomials, and the envelope in spherical harmonics of fixed m and parity, leaves a set of coupled radial equations, which we solve on

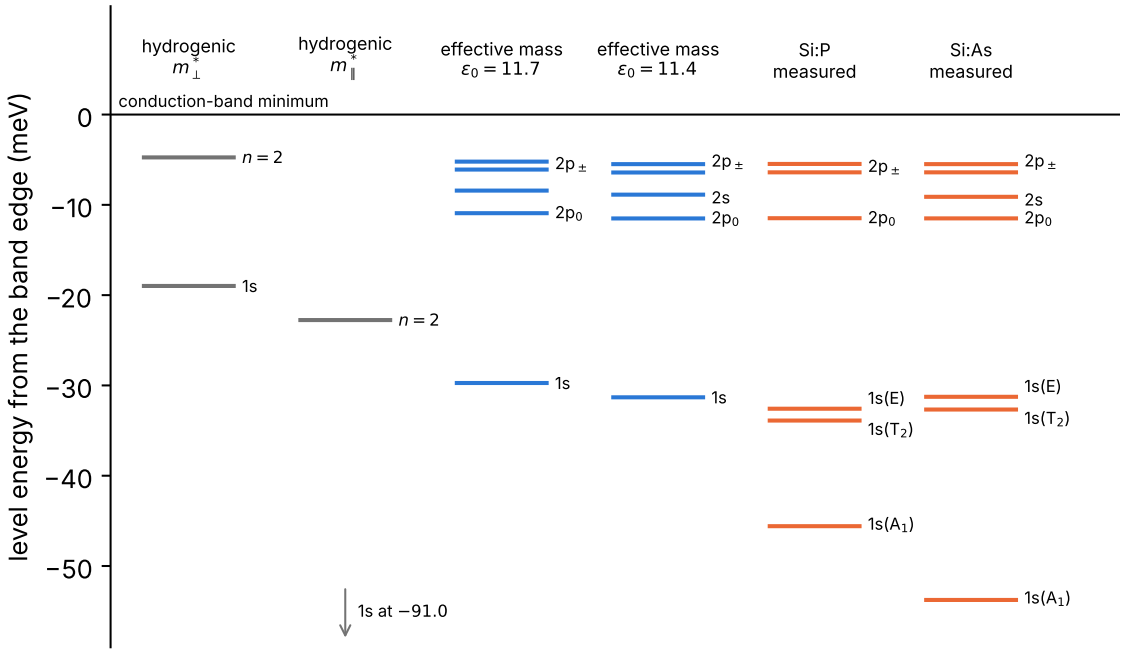


Figure 7.5: Bound levels of a shallow donor in silicon, measured from the conduction-band minimum (meV). Gray: the hydrogenic model, Eq. (7.49), with $\varepsilon_0 = 11.7$ and the transverse mass $0.191\,m_e$ ($1s$ at 19.0 meV) or the longitudinal mass $0.916\,m_e$ ($1s$ at 91.0 meV, off the scale), each with its $n = 2$ level. Blue: the effective-mass equation Eq. (7.48) with both masses, solved numerically (coupled radial equations in spherical harmonics after scaling z by $\gamma_m^{1/2}$, $l \le 14$), for $\varepsilon_0 = 11.7$ (room temperature) and 11.4 (liquid-helium temperature); from the top, $3p_0$, $2p_\pm$, $2s$, $2p_0$ and $1s$. Orange: measured levels of phosphorus and arsenic donors [21, Fig. 1]. The p levels agree with effective-mass theory at $\varepsilon_0 = 11.4$ to 0.4 %; the $1s$ level splits into A_1, T_2 and E states, and the A_1 ground state lies below the effective-mass value by 14 meV for phosphorus and 22 meV for arsenic.

a grid. The ground state comes out at $1.566\,\mathrm{Ry}_\perp^* = 29.7$ meV, so the variational envelope is within $0.002\ \mathrm{Ry}_\perp^*$ of the exact effective-mass ground state. The excited states are labeled by their hydrogenic parents; the anisotropy splits each np level into np_0, with the node across the valley axis, and $np_\pm$. Fig. 7.5 shows them.

The measured spectra test the theory state by state [21, Fig. 1]. The p levels of phosphorus and arsenic agree with each other to 0.02 meV: $2p_0$ at 11.48 and 11.50 meV, $2p_\pm$ at 6.40 meV for both. They do not depend on the chemistry of the donor, as effective-mass theory says they should. With $\varepsilon_0 = 11.7$, however, the theory puts them 5 % too shallow, at 10.92 and 6.09 meV. Since every level of Eq. (7.48) scales as $1/\varepsilon_0^2$, the measured $2p_\pm$ fixes $\varepsilon_0 = 11.7 \times (6.09/6.40)^{1/2} = 11.4$. It is the value of the static constant at liquid-helium temperature, where donor spectra are taken, that calculations of donor levels use [22, 23], 2.6 % below the room-temperature 11.7. With 11.4, effective-mass theory puts $2p_0$, $2p_\pm$ and $3p_0$ at 11.51, 6.42 and 5.49 meV, within 0.4 % of the measured 11.48, 6.40 and 5.47 meV, and the $1s$ state at 31.3 meV, the value Faulkner obtained [22, 23]. The p envelopes vanish at the donor site and never see the ion core. For them, effective-mass theory is as good as the hydrogen atom.

The ground states are another matter. Phosphorus binds its electron by 45.59 meV [24], arsenic by 53.76 meV [21, Fig. 1], and antimony by 42.7 meV [25, Table 3], all deeper than

31.3 meV and different from one another. The difference lies near the core. Within a lattice constant of the donor the screening by ε_0 is incomplete, and the potential carries the chemistry of the ion; the envelope equation replaces both by $-\alpha/\varepsilon_0 r$. The $1s$ envelope is largest exactly there, and the potential that Eq. (7.48) leaves out shifts it. This shift is the *central-cell correction*.

The central cell also lifts a degeneracy that the effective-mass equation leaves intact. Silicon has six valleys, on the $\langle 100 \rangle$ axes, and each carries its own envelope, rotated with its axis; at the order of Eq. (7.48), any combination $\sum_j c_j F_j \psi_{c\boldsymbol{k}_j}$ is a ground state, sixfold degenerate without spin. The short-range part of the donor potential couples the valleys, because a potential concentrated in one cell has Fourier components at the large momenta $\boldsymbol{k}_j - \boldsymbol{k}_{j'}$ between them. The donor site has the symmetry of a regular tetrahedron, the point group T_d, which leaves three matrix elements, Δ_0 within a valley, Δ_g between opposite valleys and Δ_f between perpendicular ones. The six states split into a singlet A_1, a doublet E and a triplet T_2. These are the standard names of the irreducible representations of T_d: each label says how a set of states transforms under the rotations and reflections of the tetrahedron, and here it fixes the degeneracy, one, two or three. The levels lie at

$$E_{A_1} = \Delta_0 + \Delta_\mathrm{g} + 4\Delta_\mathrm{f}, \qquad E_E = \Delta_0 + \Delta_\mathrm{g} - 2\Delta_\mathrm{f}, \qquad E_{T_2} = \Delta_0 - \Delta_\mathrm{g}, \tag{7.50}$$

relative to the effective-mass level (Eq. (D.143)). In the A_1 state all six valleys add in phase at the donor site; it has the largest density on the core and an attractive central cell lowers it most. This *valley-orbit splitting* is what the measured spectra show: the phosphorus $1s$ level is a singlet at 45.59 meV, with the triplet T_2 at 33.89 meV and the doublet E at 32.58 meV, close to the effective-mass value [21, Fig. 1]. The central-cell correction is a property of the s-like ground state, and a large one, a third of the binding of phosphorus.

The donor's size sets the density at which it stays a single donor. The *defect density* n_D, the number of donors (or of any defect) per unit volume, has to be small enough that orbits do not overlap, $n_\mathrm{D} a^{*3} \ll 1$. The spheroid of radii $r_\perp$, $r_\perp$ and $r_\parallel$ has a volume of $3.5 \times 10^{-20}\,\mathrm{cm}^3$, and one donor per orbit means $n_\mathrm{D} \approx 3 \times 10^{19}\,\mathrm{cm}^{-3}$. In a lightly doped crystal, with $n_\mathrm{D} = 10^{15}\,\mathrm{cm}^{-3}$, donors sit 100 nm apart and each is an isolated hydrogen atom of the crystal. At higher densities the electrons hop between donors and form an impurity band, and the isolated-donor picture fails before the orbits touch, at the Mott metal–insulator transition (Sect. 16.5).

Acceptors are bound by similar amounts, tens of meV in silicon, but their effective-mass problem is harder. The valence maximum is degenerate, with heavy and light holes and a split-off band (Sect. 3.3), and Eq. (7.48) becomes a set of coupled equations for several envelopes, the same complication that the heavy–light mixing brings to excitons (Sect. 4.9). Donors are the cleaner case, and we keep to them.

Effective-mass theory is the computational method of choice for shallow donors, because no supercell is large enough to hold the envelope. An orbit of radius 2.4 nm spans thousands of atoms, and a supercell calculation of the donor (Appendix E.10) has to extrapolate in its size. It also inherits the error of its functional, magnified: a semilocal functional spreads the donor electron too far and binds phosphorus by only 19.2 meV after extrapolation, against the measured 45.6 meV [25, Table 3]. What a supercell supplies well is the potential within a cell or two of the core. Combined with the envelope equation, in multivalley effective-mass theory, that potential gives the central-cell correction and the valley-orbit splitting, which is the division of labor that Appendix D.23 sets up.

Worked example 7.5: The binding of phosphorus in silicon

We compare the effective-mass binding of a phosphorus donor with the measured 45.59 meV and extract the valley-orbit matrix elements, with the masses $m_\perp^* = 0.191\,m_e$ and $m_\parallel^* = 0.916\,m_e$ of Table 3.1.

Units. With $\varepsilon_0 = 11.7$, $\mathrm{Ry}_\perp^* = 0.191 \times 13\,606\,\mathrm{meV}/11.7^2 = 18.98\,\mathrm{meV}$, $a_\perp^* = 11.7 \times 0.529\,\text{Å}/0.191 = 32.4\,\text{Å}$ and $\gamma_m = 0.209$.

Effective-mass ground state. The variational envelope of Appendix D.23 binds the electron by 29.7 meV, and the numerical solution of Eq. (7.48) gives $1.566\,\mathrm{Ry}_\perp^*$, also 29.7 meV. At the liquid-helium value $\varepsilon_0 = 11.4$, which the p levels require, $\mathrm{Ry}_\perp^* = 20.00\,\mathrm{meV}$ and the effective-mass $1s$ level lies at 31.3 meV.

Central-cell correction. The measured ground state is $45.59\,\mathrm{meV} - 31.3\,\mathrm{meV} = 14.3\,\mathrm{meV}$ deeper, 31 % of the binding. (At $\varepsilon_0 = 11.7$ the difference would read 15.9 meV, with 1.6 meV of it due to the dielectric constant rather than to the core.) The measured binding itself leans on the theory: the spectroscopy measures transitions, and the absolute ground-state energy follows by adding the effective-mass binding of a high p state, 3.12 meV for $3p_\pm$, to the measured $1s(A_1) \to 3p_\pm$ line [23]. The p levels can bear that weight, as Fig. 7.5 shows.

Valley-orbit matrix elements. Taking the effective-mass level -31.3 meV as the reference, the three measured $1s$ levels of phosphorus, A_1 at -45.59, T_2 at -33.89 and E at -32.58 meV, sit at -14.29, -2.59 and -1.28 meV. Eq. (7.50) gives $E_{A_1} - E_E = 6\Delta_\mathrm{f}$, so $\Delta_\mathrm{f} = -13.01/6 = -2.17$ meV; then $\Delta_0 + \Delta_\mathrm{g} = E_E + 2\Delta_\mathrm{f} = -5.62$ meV and $\Delta_0 - \Delta_\mathrm{g} = E_{T_2} = -2.59$ meV, so that $\Delta_0 = -4.10$ meV and $\Delta_\mathrm{g} = -1.51$ meV. All three are negative, as an attractive core requires, and the coupling between perpendicular valleys is the larger of the two intervalley terms. The E and T_2 states lie only 1.3 and 2.6 meV below the effective-mass value, and the A_1 state takes 14.3 meV, because only A_1 collects all six valleys in phase at the core.

Size. The variational envelope of Appendix D.23 has $|F(0)|^2 = 1/(\pi r_\perp^2 r_\parallel)$, with the radii $r_\perp = 24.3\,\text{Å}$ and $r_\parallel = 14.0\,\text{Å}$ found above. The volume of one silicon atom is $a^3/8$, with the lattice constant $a = 5.43\,\text{Å}$, and the probability of finding the electron within it around the donor is $|F(0)|^2 a^3/8 = (5.43\,\text{Å})^3/(8\pi \times 24.3^2 \times 14.0\,\text{Å}^3) = 7.7 \times 10^{-4}$ per valley envelope. The central cell is a small region of the orbit, and still it moves the ground state by a third: the potential there is that of an unscreened ion, many eV deep.

Kinematics. Dark matter can ionize the donor if $\frac{1}{2} m_\chi v_\mathrm{max}^2 \geq E_\mathrm{b}^\mathrm{D}$, that is for $m_\chi \geq 2 \times 45.59\,\mathrm{meV}/7.023 \times 10^{-6} = 13.0\,\mathrm{keV}$, twenty-six times lighter than the 0.333 MeV of the silicon gap. At threshold the momentum transfer is $m_\chi v_\mathrm{max} = 34.4\,\mathrm{eV}$. The envelope spreads over momenta up to $1/r_\perp = 81\,\mathrm{eV}$ across the valley axis and $1/r_\parallel = 141\,\mathrm{eV}$ along it, so $q r_\perp \approx 0.42$: the dark matter that can just ionize the donor transfers momenta that the bound electron has. In the language of Sect. 1.3, the donor is a good kinematic match to dark matter of a few tens of keV.

7.4.2 Color centers

The T center of silicon emits at 935 meV, 1326 nm in the telecommunications O band, and the nitrogen–vacancy center of diamond at 1.945 eV, 637 nm. Both lines are sharp, and both sit on a broad band of lower-energy light. Defects of this kind are called *color centers*, after the F centers that color the alkali halides. Their levels lie deep in the gap, and their electrons are confined within a cell or two of the defect, unlike the donor electron of the previous subsection. A localized electron that changes its state exerts large forces on the few atoms around it.

The nuclei settle at different positions in the ground and the excited state, and an optical transition, which is fast compared with the nuclear motion, leaves them displaced from their new equilibrium. This is the displaced oscillator of Sect. 5.6, and color centers are where its parameters are measured.

The *configuration-coordinate diagram* draws the two Born–Oppenheimer surfaces of Sect. 5.1, one for each electronic state, along the single coordinate Q that connects their minima. With Q mass-weighted, as $\Delta Q = M^{1/2}\Delta x$ in Sect. 5.6, and one effective mode of energy ω_0 that is the same in both states,

$$E_{\mathrm{g}}^{\mathrm{ad}}(Q) = \tfrac{1}{2}\omega_0^2 Q^2, \qquad E_{\mathrm{e}}^{\mathrm{ad}}(Q) = E_{\mathrm{ZPL}} + \tfrac{1}{2}\omega_0^2 (Q-\Delta Q)^2, \qquad S_{\mathrm{HR}} = \tfrac{1}{2}\omega_0 \Delta Q^2. \tag{7.51}$$

The *zero-phonon-line energy* E_{ZPL} is the difference between the two minima, the energy of the transition that leaves the vibrational state unchanged, and the Huang–Rhys factor S_{HR} measures the displacement in units of the zero-point spread of the mode. A vertical transition from the ground minimum lands $\frac{1}{2}\omega_0^2\Delta Q^2 = S_{\mathrm{HR}}\omega_0$ above the excited minimum, so absorption peaks near $E_{\mathrm{ZPL}} + S_{\mathrm{HR}}\omega_0$; emission from the relaxed excited state peaks near $E_{\mathrm{ZPL}} - S_{\mathrm{HR}}\omega_0$, and the lattice takes up the relaxation energy $S_{\mathrm{HR}}\omega_0$ each way. At $T = 0$ the emission lines at $E_{\mathrm{ZPL}} - p\,\omega_0$ carry the Poisson weights $\mathrm{e}^{-S_{\mathrm{HR}}} S_{\mathrm{HR}}^p / p!$ of Eq. (5.63), and the zero-phonon line carries

$$w_{\mathrm{ZPL}} = \mathrm{e}^{-S_{\mathrm{HR}}}. \tag{7.52}$$

A real center couples to a continuum of crystal modes and to a few local modes of its own atoms; S_{HR} is then the sum of the factors of all the modes, the replicas merge into a sideband, and the zero-phonon weight is still $\mathrm{e}^{-S_{\mathrm{HR}}}$ (Sect. 5.6). The color-center literature calls this weight the Debye–Waller factor of the line; in this book that name belongs to e^{-2W} of Eq. (5.56), and we write *zero-phonon fraction*.

The *T center* is a pair of carbon atoms sharing one silicon site, with a hydrogen atom on one of them [26]. Its ground state has one unpaired electron. Its luminescent state is a bound exciton: a second electron pairs with the first into a singlet, and a hole is bound to the negatively charged complex by only (34.78 ± 0.15) meV, out of a total pair binding of about 235 meV. The hole binding comes from the series of excited hole levels seen in photoluminescence excitation, extrapolated to the ionization limit of the hole [26, Sect. IV]; earlier estimates, quoted there, had put the hole binding at 22 to 32 meV. The loosely bound hole is a shallow, effective-mass-like particle, while the electrons are deep and localized. Measured on an ensemble in isotopically pure ^{28}Si between 1.4 and 4.2 K, the exciton lives 0.94 µs, and 23 % of the emitted photons are in the zero-phonon line [27].

The *nitrogen–vacancy center* of diamond is a substitutional nitrogen next to a vacant site along [111]. In its negative charge state six electrons occupy orbitals built from the dangling bonds around the vacancy, and its spin-triplet ground state is optically read out at room temperature [28, Sect. 1.1]. A fit of its measured emission band at 8 K returns $S_{\mathrm{HR}} = 3.85 \pm 0.05$ with an effective phonon energy of 64 meV [29, Sects. 3 and 4.3].

Fig. 7.6 draws both centers in units of their own phonon energy, where the diagram depends on S_{HR} alone. The T center's minima are displaced by $(2S_{\mathrm{HR}})^{1/2} = 1.7$ zero-point units, the nitrogen–vacancy center's by 2.8. That difference becomes a factor of ten in the zero-phonon weight. The measurements are taken at temperatures where the coupled modes are empty: at 4.2 K, a phonon of 10 meV has a thermal occupation of $\mathrm{e}^{-\omega/T} = 1 \times 10^{-12}$, and the zero-temperature weights of Eq. (7.52) apply.

Comparing Eq. (7.52) with a luminescence spectrum needs one correction. The rate of spontaneous emission into a line of energy E grows as $E^3|\boldsymbol{d}|^2$, with $\boldsymbol{d}$ the transition dipole, so a luminescence spectrum counts photons with the Franck–Condon weight times E^3 [29, Eq. (2)]. The sideband lies at lower energy and is suppressed by this factor, and the zero-phonon line

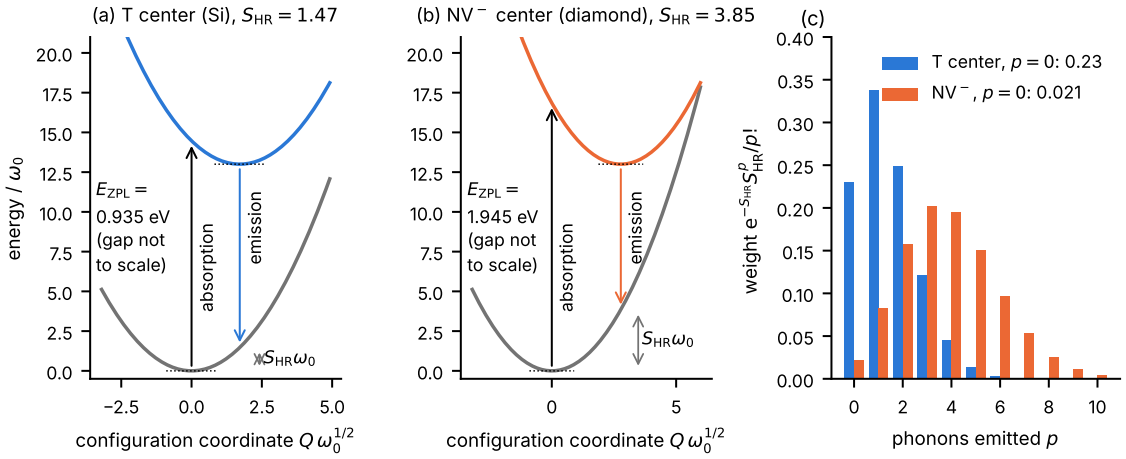


Figure 7.6: Configuration-coordinate diagrams of (a) the T center in silicon, $E_{ZPL} = 0.935\,\text{eV}$, $S_{HR} = 1.47$ (from the measured zero-phonon fraction 0.23 [27]), and (b) the negatively charged nitrogen–vacancy center in diamond, $E_{ZPL} = 1.945\,\text{eV}$, $S_{HR} = 3.85$ [29]. Each center is modeled by one effective mode of energy ω_0, the same in both electronic states, in the Condon approximation; energies are in units of ω_0 (64 meV for the nitrogen–vacancy center) and the coordinate in units of $\omega_0^{-1/2}$, Eq. (7.51). Gray: ground-state surface; colored: excited state, drawn with a reduced gap E_{ZPL} (not to scale). Arrows: vertical absorption from the ground minimum and emission from the excited minimum; the double arrow is the relaxation energy $S_{HR}\omega_0$. (c) Zero-temperature weights of the emission lines with p phonons emitted, $e^{-S_{HR}} S_{HR}^p/p!$; the zero-phonon line is $p = 0$.

takes a larger share of the photons than of the Franck–Condon weight:

$$\frac{\text{photons in the zero-phonon line}}{\text{all photons}} = \frac{e^{-S_{HR}} E_{ZPL}^3}{\sum_p e^{-S_{HR}} S_{HR}^p (E_{ZPL} - p\,\omega_0)^3/p!}, \tag{7.53}$$

in the single-mode model. The correction is of order $3S_{HR}\omega_0/E_{ZPL}$, large when the relaxation energy is a sizable fraction of the line energy.

Computing these numbers is a supercell problem (Appendix E.10). E_{ZPL} is the difference of two total energies, each at its own relaxed geometry, with the excited state obtained by constraining the occupations of the defect orbitals; S_{HR} follows from the displacement between the two geometries projected on the vibrational modes, $S_{HR} = \sum_k \omega_k \Delta Q_k^2/2$. A hybrid functional in a 512-atom cell puts the T-center line at 985 meV, 50 meV above the measurement [30], and the converged Huang–Rhys factor of the nitrogen–vacancy center is 3.67 against the measured 3.85 [29, Sects. 4.2 and 4.3].

Worked example 7.6: Zero-phonon fractions

We compute the zero-phonon fractions of the T center and the nitrogen–vacancy center and check the second against its measured photon count.

T center. Through Eq. (7.52), the measured photon fraction 0.23 means $S_{HR} = -\ln 0.23 = 1.47$. The weights of the lines with 0, 1, 2 and 3 phonons are 0.23, 0.34, 0.25 and 0.12 (Fig. 7.6c): one photon in four is in the zero-phonon line, and one in three has emitted exactly one phonon. Read through Eq. (7.53), the photon fraction slightly overstates the Franck–Condon weight. The bound on the correction, $3S_{HR}\omega_0/E_{ZPL}$ with ω_0 at most the

64 meV of the silicon optical phonon (Sect. 5.3), is $3 \times 1.47 \times 0.064/0.935 = 0.30$, so the photon count shifts S_{HR} by at most 0.30. The true Huang–Rhys factor lies between 1.47 and about 1.8, and the zero-phonon weight between 0.23 and 0.17.
Nitrogen–vacancy center. With $S_{\mathrm{HR}} = 3.85$, Eq. (7.52) gives $w_{\mathrm{ZPL}} = \mathrm{e}^{-3.85} = 0.021$, ten times less than the T center. The line with three phonons is the strongest, at 0.20. The relaxation energy is $S_{\mathrm{HR}}\omega_0 = 3.85 \times 64\,\mathrm{meV} = 0.246\,\mathrm{eV}$, so absorption peaks near 2.19 eV and emission near 1.70 eV, a Stokes shift of 0.49 eV.
The photon check. In Eq. (7.53), with $\omega_0/E_{\mathrm{ZPL}} = 0.064/1.945 = 0.0329$, the Poisson average of $(1 - 0.0329\,p)^3$ is

$$1 - 3(0.0329)S_{\mathrm{HR}} + 3(0.0329)^2(S_{\mathrm{HR}} + S_{\mathrm{HR}}^2) - (0.0329)^3(S_{\mathrm{HR}} + 3S_{\mathrm{HR}}^2 + S_{\mathrm{HR}}^3) = 0.677,$$

from the first three moments of the Poisson distribution. The photon fraction is then $0.0213/0.677 = 0.031$. The measured share of the zero-phonon line in the emission is 0.032 [29, Sect. 3]. Taken at face value, $-\ln 0.032$ would give $S_{\mathrm{HR}} = 3.44$, an underestimate by 0.4 that the single-mode model attributes entirely to the E^3 weighting.
Detection fraction. A detector that reads a color center through its zero-phonon line collects a quarter of the T center's photons and three percent of the nitrogen–vacancy center's. The rest come out in the sideband, spread over several phonon energies below the line, where a narrow filter rejects them together with the background.

7.4.3 Displacement thresholds and Frenkel pairs

A silicon atom struck along ⟨111⟩ with 12.5 eV can leave its site for good; struck along ⟨100⟩, it needs 20 eV [31, Table IV]. Below these energies the atom swings out, hands its energy to its neighbors and returns. The difference between the two outcomes is a statement about the energy surface $E^{\mathrm{ad}}(u)$ of Sect. 5.1. The perfect crystal is its lowest minimum, but the surface has many other local minima, separated from it by barriers of several eV. A *vacancy* is an empty lattice site. An *interstitial* is an extra atom between the sites; in silicon it can sit at the tetrahedral or the hexagonal interstitial site or share a site with a lattice atom as a split ⟨110⟩ dumbbell. An atom moved from its site into an interstitial position leaves a vacancy behind, and the two together are a *Frenkel pair*. Each configuration costs a formation energy, the height of its minimum above that of the perfect crystal. Density-functional calculations put the isolated vacancy at 3.4 to 3.9 eV, the interstitials at 4.7 to 5.8 eV, and the most stable close Frenkel pair, a tetrahedral interstitial 4.7 Å from its vacancy, at 6.8 to 7.5 eV [31, Table III]. A cheaper minimum exists without any vacancy: the *bond defect*, a local rearrangement of bonds, at 2.6 to 3.0 eV.

The *displacement threshold energy* $E_{\mathrm{d}}(\hat{\boldsymbol{r}})$ is the smallest kinetic energy that, given to one atom in the direction $\hat{\boldsymbol{r}}$, leaves the crystal in a minimum other than the perfect one once the motion has died out,

$$E_{\mathrm{d}}(\hat{\boldsymbol{r}}) = \min\{E_R : \text{ the trajectory on } E^{\mathrm{ad}} \text{ from } (E_R, \hat{\boldsymbol{r}}) \text{ ends in a defect}\}. \tag{7.54}$$

It is not the formation energy of the pair it creates, and it is always larger. The struck atom must squeeze between its neighbors, over a barrier that lies on its path and not at the end points. It loses energy to every atom it pushes. And if it stops too close to its own vacancy, the pair recombines and the crystal returns to its minimum. E_{d} therefore depends on the direction, through the open and closed channels of the lattice, and on the path of the whole collision. It is computed by molecular dynamics: a single trajectory on E^{ad} for each direction and energy, repeated over many directions (Appendices E.10 and E.11).

For silicon, density-functional molecular dynamics finds the lowest threshold, 12.5 eV, along $\langle 111 \rangle$, where the struck atom pushes its neighbor into the tetrahedral site in a short replacement sequence; 20 eV along $\langle 100 \rangle$; and 36 eV for a Frenkel pair averaged over all directions [31, Table IV]. Electron-irradiation experiments see damage set in at 12.9 ± 0.6 eV [32], in agreement with the minimum. In most directions a bond defect forms at a lower energy than a Frenkel pair, and the average threshold for a defect of either kind is 24 eV [31, Table IV]. Which number is quoted depends on what is counted: a count of vacant and doubly occupied lattice sites sees the Frenkel pair but not the bond defect. The classical Stillinger–Weber potential, used for the large statistical scans, overshoots the density-functional thresholds along $\langle 111 \rangle$ by 4 to 6 eV [31, Sect. IV], and its thresholds range from 17.5 to 63.5 eV over all directions (Appendix E.10).

The threshold is not the energy that stays in the crystal. A close Frenkel pair stores 6.8 eV to 7.5 eV, about half of the minimum threshold and a fifth of the average one; the rest of the recoil energy becomes heat within picoseconds. The stored energy is invisible to a calorimeter that measures the heat. It is a missing part of the deposit, and Sect. 19.4 turns this into the defect fraction of a recoil and the bounds that calibration data put on it. For dark matter, only heavy enough particles can reach these thresholds at all. By Eq. (1.21), with the silicon nuclear mass $M = 26.16$ GeV and $v_{\max} = 2.650 \times 10^{-3}$, a recoil of 12.5 eV needs $m_\chi \geq 0.153$ GeV, and one of 36 eV needs 0.26 GeV. Since E_d depends on the direction of the recoil relative to the crystal axes, the rate of defect creation near threshold carries the direction of the dark-matter wind, a handle for directional detection [33].

7.4.4 Defects as targets and readouts

A shallow donor is a target in its own right. Its threshold of 45.6 meV for phosphorus opens silicon to dark matter of 13 keV, and its envelope covers the momenta that such particles carry (Worked example 7.5). Its weakness is the number of electrons. A donor density of $10^{15}\,\mathrm{cm}^{-3}$ is five parts in 10^9 of the valence electrons of silicon, $2.0 \times 10^{23}\,\mathrm{cm}^{-3}$, and the interaction with the bound electron is screened by the same ε_0 that makes it shallow. The rate requires the response of the donor to the dark-matter operators, with its valleys and central cell, which Sect. 16.5 derives.

A color center is a target with a level inside the gap and a readout with an optical line. As a target, the T center needs $m_\chi \geq 2 \times 0.935\,\mathrm{eV}/v_{\max}^2 = 0.266$ MeV, below the 0.333 MeV of the silicon gap, because its excitation sits 0.24 eV below the band edge. The momentum at threshold is $2E_{\mathrm{ZPL}}/v_{\max} = 0.71$ keV, and a transition at that momentum needs states localized to about $1/0.71\,\mathrm{keV} = 2.8$ Å, which the deep levels of the center are. The nitrogen–vacancy center needs 0.554 MeV and 1.47 keV. As a readout, the zero-phonon fraction sets how many of a center's photons survive a narrow filter (Worked example 7.6), and the spins of both centers can be polarized and read optically. As a final state, a color center can be created by a nuclear recoil, by a displacement or by the capture of a vacancy, and the threshold E_d sets where that begins (Sect. 16.6).

A displaced atom, finally, is a long-lived final state. The Frenkel pair or the bond defect left by a recoil holds several eV that the phonon and charge channels never see, and it persists as long as its barriers allow. At 10 mK the Boltzmann factor for crossing a barrier of 0.1 eV is $\mathrm{e}^{-1.2\times10^5}$, and only tunneling or the phonons of a later event move the defect (Appendix E.11). Chap. 19 follows the energy of a recoil into heat, charge and defects, and the numbers of this section, E_d and the formation energies, are the inputs to that energy balance.

Summary

The Coulomb interaction does not act on spin, yet it couples spins, and in NiO it orders them below 523 K (Sect. 7.1). For two electrons in orthogonal orbitals the exchange integral favors parallel spins, which is Hund's rule. Between neighboring ions of a Mott insulator a virtual hop is allowed only between antiparallel spins and produces the antiferromagnetic coupling $4t_{\mathrm{hop}}^2/U_{\mathrm{H}}$, which superexchange carries through the oxygen ion. At low energy only the spins remain, coupled by the Heisenberg model, and in NiO one exchange constant, the measured $J_2 = 19.01\,\mathrm{meV}$ between next-nearest nickel neighbors, sets the scale. The Holstein–Primakoff transformation turns spin deviations into bosons, and linear spin-wave theory, the leading order in $1/|\boldsymbol{S}|$, makes them independent magnons. In a ferromagnet the magnon energy rises quadratically from zero. In an antiferromagnet the exchange creates spin deviations in pairs, a bosonic Bogoliubov transformation removes the pair terms, and the dispersion becomes linear, with the measured initial slope of NiO, $\sim$250 meV Å, and a maximum of $\sim$117 meV. Density-functional values of J_2 spread from 11 to 27 meV, and that spread is the theory uncertainty for a magnet without neutron data. A dark-matter particle that couples to the electron spin acts through the spin density $\boldsymbol{s}_e(\boldsymbol{q})$, which for localized moments is the magnetic form factor times the ionic spins, and its one-magnon response follows from the Bogoliubov coefficients.

Phonon exchange makes electrons near the Fermi surface attract each other, with the dimensionless strength $\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}} = \lambda_{\mathrm{ep}}$, reduced by the quasiparticle residue and the Coulomb pseudopotential (Sect. 7.2). On a Fermi sea any attraction binds a Cooper pair, and the electrons near the Fermi surface pair together into the BCS state. Its excitations are Bogoliubov quasiparticles of energy $(\xi_{\boldsymbol{k}}^2 + \Delta_{\mathrm{BCS}}^2)^{1/2}$, mixtures of an electron and a hole, whose density of states piles up at the gap edge. The gap equation fixes Δ_{BCS} and T_c, both exponentially sensitive to the coupling, and their ratio 1.764, which is not. A probe that conserves the electron number creates quasiparticles in pairs, so the threshold is $2\Delta_{\mathrm{BCS}}$. The coherence factors decide what happens above it: the response to the density jumps to $\pi/2$ times the normal-metal value at the edge, and the response to the spin density rises continuously from zero. Independent quasiparticles, however, do not conserve charge, and their density response misplaces a first moment $8\pi\Delta_{\mathrm{BCS}}^2/V_{\mathrm{BCS}}$ of the f-sum rule. In the Nambu formalism the defect is a pair-field term in the continuity equation, and the conserving response, which lets the pair field respond, removes it through the phase mode of the condensate and satisfies the density Ward identity. For aluminum the measured critical temperature fixes $\Delta_{\mathrm{BCS}} = 0.179\,\mathrm{meV}$, a threshold of 0.36 meV and a threshold mass of 0.10 keV.

Near a point where two bands nearly touch, the $\boldsymbol{k}\cdot\boldsymbol{p}$ expansion becomes a Dirac Hamiltonian, with the velocity v_{D} in place of the speed of light and the half-gap Δ_{D} in place of the mass (Sect. 7.3). The Kane model of a direct-gap semiconductor is such a cone, and it reproduces the GaAs effective mass, $0.066\,m_e$. Since the edge of the pair continuum never lies below the line $\omega = v_{\mathrm{D}} q$, a cone with $v_{\mathrm{D}} > v_{\max}$ cannot be excited by scattering at any mass, while a slow cone opens the target to light particles: 8.0 keV for $ZrTe_5$-like parameters. The Dirac electrons screen on top of a background dielectric constant $\varepsilon_{\mathrm{bg}}$, with the effective coupling $\tilde{\alpha} = \alpha/(\varepsilon_{\mathrm{bg}} v_{\mathrm{D}})$, 0.18 to 0.20 for $ZrTe_5$ with $g_{\mathrm{deg}} = 2$. Their random-phase polarization grows logarithmically up to the momentum cutoff, and a gapless cone has no screening length. For $ZrTe_5$-like parameters the Dirac electrons raise the static dielectric constant above $\varepsilon_{\mathrm{bg}}$ by a fraction 0.102.

Defects put levels inside the gap of silicon itself (Sect. 7.4). A shallow donor binds its extra electron in a hydrogenic envelope set by the anisotropic conduction-band masses and the dielectric constant. Effective-mass theory binds every donor by 31.3 meV at the liquid-helium dielectric constant 11.4 (29.7 meV at 11.7), and the central-cell correction of the s-like ground

state, 14.3 meV, brings phosphorus to its measured 45.6 meV, which opens silicon to dark matter of 13 keV. A color center couples its electronic transition to local vibrations, and the configuration-coordinate diagram turns the Huang–Rhys factor into the zero-phonon fraction $e^{-S_{HR}}$. The measured zero-phonon fraction 0.23 of the T center of silicon means $S_{HR} = 1.47$, while the nitrogen–vacancy center of diamond, with $S_{HR} = 3.85$, keeps only $e^{-3.85} = 0.021$ of its weight in the line. A recoiling atom that receives more than the displacement threshold, 12.5 eV along $\langle 111 \rangle$ in silicon and 36 eV for a Frenkel pair averaged over directions, leaves a Frenkel pair or a bond defect that stores energy no calorimeter sees. Chap. 8 gathers the responses of all these targets, and those of the earlier chapters, into one framework for the rate.

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Chapter 8

The response framework

A particle with a magnetic moment couples to the spins and currents of a solid as well as to its charge, and a fermion that is absorbed leaves a light partner behind; neither fits the single product of Eq. (1.1), a halo factor times a coupling $|\tilde{V}|^2$ times one dynamic structure factor. So far Chap. 4 took the electron density of silicon, Chap. 5 the nuclei of a harmonic crystal, Chap. 6 electrons and lattice together, and Chap. 7 the spins of NiO and the paired electrons of aluminum. A reader who meets a new model or a new target should not have to start again. What do these calculations have in common, and what does each of them leave out?

They share one structure. Every model collapses into a tensor that weights the response matrix of a few target operators, and every target into that one matrix. A new model then needs a new tensor, and a new target a new matrix, but not a new calculation of the other. Because the matrix belongs to the target alone, what is exact about it, how the medium screens it and where a calculation truncates it can be settled once, for every coupling at the same time. What a calculation leaves out then becomes a named step with a size, not a guess. We will build this framework (Fig. 8.1) from the separate results of Chaps. 1 to 7 and carry it to the point where the detector chain of Chap. 18 takes over, the final state that the dark matter leaves in the target.

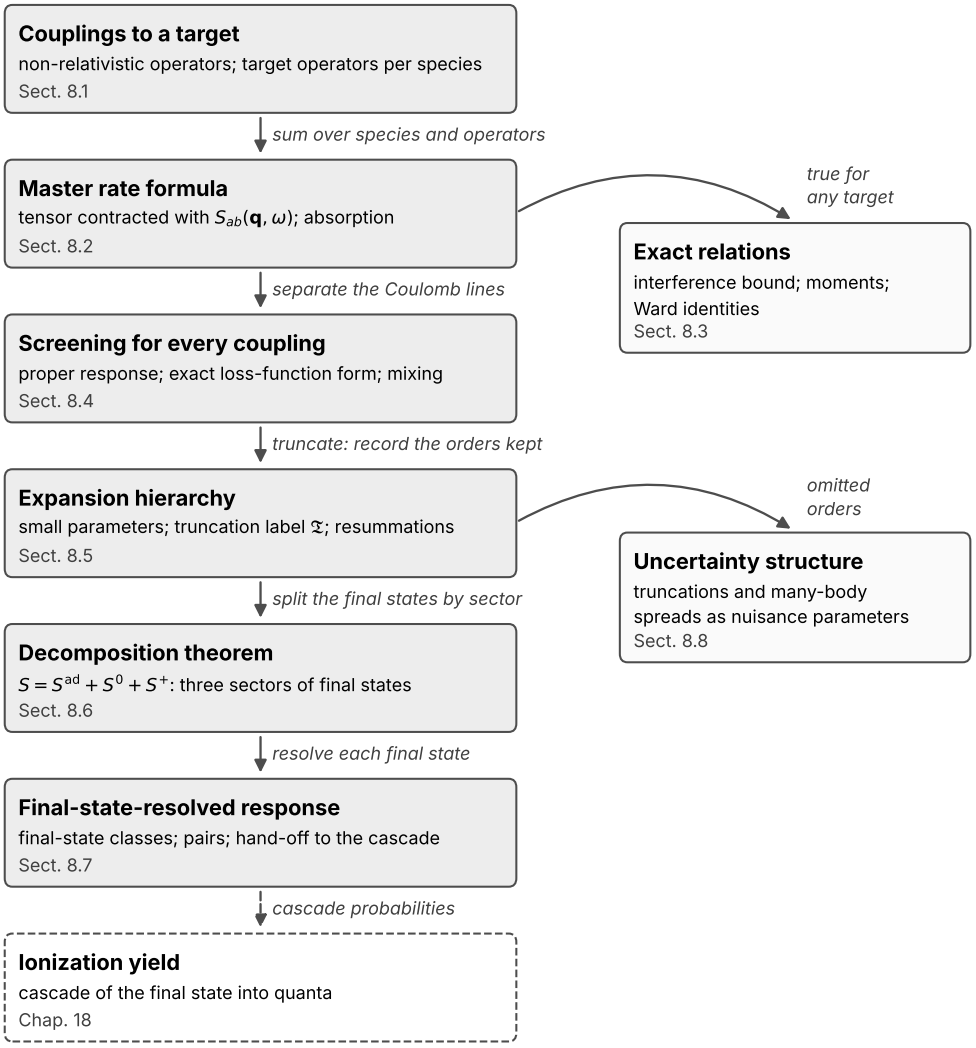


Figure 8.1: The response framework from couplings to final states. The non-relativistic operators of every model reduce to target operators per species (Sect. 8.1), and the master rate formula contracts a dark-matter tensor with the response matrix $S_{ab}(\boldsymbol{q}, \omega)$ (Sect. 8.2), which obeys exact relations for any target (Sect. 8.3). Separating the Coulomb lines screens every coupling (Sect. 8.4); the truncation label records the orders a calculation keeps (Sect. 8.5), and its omitted orders and many-body spreads become nuisance parameters (Sect. 8.8). The decomposition theorem splits the final states into three sectors (Sect. 8.6), and the final-state-resolved response hands each final state to the ionization cascade (Sect. 8.7; Chap. 18).

8.1 Dark-matter couplings to a target

A dark photon couples to a solid through its charge alone. In the language of Sect. 4.2 it needs one target operator per species, the electron density $n_e(\boldsymbol{q})$ and the density $n_\kappa(\boldsymbol{q})$ of each kind of nucleus, with potentials of opposite sign. A dark-matter particle with a magnetic dipole moment does more: it feels the spins of the electrons, and, since a charge moving past a magnetic moment feels a force, their velocities as well [1, Table II]. Particle physics offers many such models, and at first sight each would need its own calculation of the target. None does. A slow dark-matter particle sees an electron or a nucleus through three things only: the momentum it hands over, the velocity of the two particles relative to each other, and their spins. As a result every model reduces to the same short list of target operators per species: the density, the spin density, the current, the spin current and a kinetic-energy density. What remains of the model is a tensor on the dark-matter side that weights the response matrix of these operators. We will build the list, reduce each non-relativistic interaction to it, and add the operators through which bosonic and fermionic dark matter are absorbed.

8.1.1 The non-relativistic operators

Take a dark-matter particle of mass m_χ and spin operator $\boldsymbol{S}_\chi$ that scatters on a free electron, with spin $\boldsymbol{S}_e = \boldsymbol{\sigma}/2$. The dark matter goes from $\boldsymbol{p}$ to $\boldsymbol{p}' = \boldsymbol{p} - \boldsymbol{q}$ and the electron from $\boldsymbol{k}$ to $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{q}$, with $\boldsymbol{q}$ the momentum transfer of Sect. 1.3. Momentum conservation leaves three of the four momenta independent, and Galilean invariance, the freedom to add a common velocity to both particles, allows an interaction to depend on them only through $\boldsymbol{q}$ and one velocity difference [2, Sect. 2.2]. The difference that is Hermitian, symmetric between the incoming and outgoing particles, is

$$\boldsymbol{v}^\perp = \bar{\boldsymbol{v}}_\chi - \frac{\boldsymbol{k}+\boldsymbol{k}'}{2m_e}, \qquad \bar{\boldsymbol{v}}_\chi \equiv \frac{\boldsymbol{p}+\boldsymbol{p}'}{2m_\chi} = \boldsymbol{v} - \frac{\boldsymbol{q}}{2m_\chi}, \tag{8.1}$$

the mean velocity of the dark matter minus that of the electron, with $\boldsymbol{v} = \boldsymbol{p}/m_\chi$ the incoming velocity. For an electron initially at rest, $\boldsymbol{v}^\perp = \boldsymbol{v} - \boldsymbol{q}/2\mu_{\chi e}$, and energy conservation in the form $\boldsymbol{q}\cdot\boldsymbol{v} = q^2/2\mu_{\chi e}$ (Sect. 1.3) makes $\boldsymbol{q}\cdot\boldsymbol{v}^\perp = 0$. This transversality is the origin of the name [2, Eq. (14)], and Appendix D.24 shows that it holds for any free target. The momentum enters as $\mathrm{i}\boldsymbol{q}$. Hermitian conjugation exchanges the incoming and outgoing particles, which reverses $\boldsymbol{q}$ and leaves $\boldsymbol{v}^\perp$ unchanged, so that $\mathrm{i}\boldsymbol{q}$, $\boldsymbol{v}^\perp$ and the spins are all Hermitian.

An interaction is then a scalar built from $\mathrm{i}\boldsymbol{q}$, $\boldsymbol{v}^\perp$, $\boldsymbol{S}_\chi$ and $\boldsymbol{S}_e$. Powers of q^2 can be absorbed into a momentum-dependent coefficient, as the mediator propagator of Eq. (1.32) already is, and products of more than one spin of the same particle reduce to at most one by the algebra of the Pauli matrices [2, Sect. 2.2]. What remains is a finite list of operators, which Fitzpatrick and collaborators wrote down and Anand, Fitzpatrick and Haxton completed [2, Eqs. (19)–(24) and (45), 3, Eqs. (12) and (13)]. They are listed in Table 8.1, which writes $\boldsymbol{q}/m_e$ in place of $\boldsymbol{q}$, so that each operator is dimensionless. The list is complete for dark matter of spin $\frac{1}{2}$, whose spin products the Pauli algebra reduces. For spin 1 a product of two dark-matter spins leaves a symmetric tensor built from the polarization vectors, and two more operators appear, $\boldsymbol{q}$ contracted through that tensor with $\boldsymbol{v}^\perp$ or with $\boldsymbol{S}_e$ [4, Eq. (18)]; for spin 0 only the operators free of $\boldsymbol{S}_\chi$ remain. The target side does not change: the new operators are linear in the electron's velocity or spin and reduce to the same rows. The potential between the dark matter and the electron is a sum

$$\tilde{V}(\boldsymbol{q}) = \sum_i c_i^{(e)}(q)\, \mathcal{O}_i^{(e)}, \tag{8.2}$$

an operator in the spin spaces of both particles, whose matrix element between plane waves is the transition amplitude in the first Born approximation (Sect. 2.2). The coefficients $c_i^{(e)}(q)$ carry the dimension eV^{-2} of $\tilde{V}$ and the momentum dependence of the mediator. For the Yukawa potential of Sect. 1.4, $c_1^{(e)}(q) = \tilde{V}(q) = y_\chi y_e/(q^2 + m_V^2)$, and every other coefficient vanishes. Papers that normalize their states relativistically quote $4m_\chi m_e\, c_i^{(e)}$ instead, the factor of Eq. (A.47). Refs. [2, 3] define $\boldsymbol{q}$ as the momentum the dark matter gains, the opposite of the $\boldsymbol{q}$ of this book, so a coefficient quoted there changes sign for the operators odd in $\boldsymbol{q}$, $i = 3, 5, 9, 10, 11, 13, 14$; $\boldsymbol{v}^\perp$ is the same in both conventions.

Which operators a model produces follows from the non-relativistic limit of its couplings. The limits of the four Dirac bilinears are derived in Appendix C.15 and, between plane waves, in Appendix D.24. The scalar $\bar{\psi}\psi$ and the time component of the vector $\bar{\psi}\gamma^\mu\psi$ both reduce to the density, 1. The spatial vector reduces to the velocity plus a spin term, $(\boldsymbol{k} + \boldsymbol{k}')/2m_e - \mathrm{i}\boldsymbol{q} \times \boldsymbol{S}_e/m_e$, the spatial axial vector $\bar{\psi}\gamma^\mu\gamma^5\psi$ to twice the spin, $2\boldsymbol{S}_e$, and the pseudoscalar $\bar{\psi}\mathrm{i}\gamma^5\psi$ to $-\mathrm{i}\boldsymbol{q} \cdot \boldsymbol{S}_e/m_e$. Multiplying a dark-matter bilinear by an electron bilinear and by the mediator propagator gives the coefficients. A vector mediator coupled to both vector currents leads to $\mathcal{O}_1$, and an axial-vector mediator coupled to both axial currents to $\mathcal{O}_4$, the standard spin-independent and spin-dependent interactions. A scalar mediator with a pseudoscalar coupling to the electron produces $\mathcal{O}_{10}$, and with pseudoscalar couplings on both sides $\mathcal{O}_6$. Dark matter with an electric dipole moment that couples through a photon-like mediator produces $\mathcal{O}_{11}$; a magnetic dipole moment, $\mathcal{O}_1$, $\mathcal{O}_4$, $\mathcal{O}_5$ and $\mathcal{O}_6$; and an anapole moment, $\mathcal{O}_8$ and $\mathcal{O}_9$ [1, Table II]. The dipole and anapole couplings bring extra powers of q, in the operator or in the coefficient, and those coupled through a light mediator are strongly weighted toward small q. Operator $\mathcal{O}_2 = (\boldsymbol{v}^\perp)^2$ arises in none of these models at leading order [3, Sect. 2.1]. It appears as a correction, of relative size v^2, to $\mathcal{O}_1$, and many lists omit it [5, Table I]; we keep it, because its reduction needs a row that the scalar absorption of Sect. 8.1.5 needs anyway.

8.1.2 Target operators per species

Inside a crystal the electron that the dark matter strikes is one of 10^{23}, bound by the nuclei and by the other electrons, and its bare momentum $\boldsymbol{k}$ is not conserved; only its crystal momentum is (Chap. 3). What survives is the structure of Eq. (8.2). Each operator acts on the target particle through three ingredients, the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ that adds the momentum $\boldsymbol{q}$, the velocity $(\boldsymbol{k} + \boldsymbol{k}')/2m_e$ inside $\boldsymbol{v}^\perp$, and the spin $\boldsymbol{S}_e$, and each has a one-body operator whose matrix elements between plane waves reproduce it. The phase is $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ itself. The velocity is the symmetrized product $\frac{1}{2}\{\boldsymbol{p}/m_e, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\}$, whose element from $\boldsymbol{k}$ to $\boldsymbol{k}'$ is $(\boldsymbol{k} + \boldsymbol{k}')/2m_e$, since $\boldsymbol{p}$ acting to the left gives $\boldsymbol{k}'$ and to the right gives $\boldsymbol{k}$. The square of the velocity is the doubly symmetrized $\{\boldsymbol{p}, \{\boldsymbol{p}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\}\}/4m_e^2$. Summing over the particles, as in Eq. (2.36), leads to five operators. Let s be a species, either the electrons ($s = e$) or the nuclei of species κ ($s = \kappa$), with mass m_s. Its particles j have positions $\boldsymbol{r}_j$, momenta $\boldsymbol{p}_j$ and spins $\boldsymbol{S}_j$, and

Target operators per species

$$n_s(\boldsymbol{q}) = \sum_{j\in s} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}, \qquad \boldsymbol{s}_s(\boldsymbol{q}) = \sum_{j\in s} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\,\boldsymbol{S}_j,$$
$$\boldsymbol{j}_s(\boldsymbol{q}) = \sum_{j\in s} \frac{1}{2}\Big\{\frac{\boldsymbol{p}_j}{m_s}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\Big\}, \quad \mathcal{J}_s^{\alpha\beta}(\boldsymbol{q}) = \sum_{j\in s} \frac{1}{2}\Big\{\frac{p_j^\alpha}{m_s}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\Big\}\, S_j^\beta, \tag{8.3}$$
$$\mathcal{T}_s(\boldsymbol{q}) = \sum_{j\in s} \frac{1}{8m_s}\{\boldsymbol{p}_j, \{\boldsymbol{p}_j, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\}\} = \sum_{j\in s} \frac{\boldsymbol{p}_j\cdot \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\,\boldsymbol{p}_j}{2m_s} + \frac{q^2}{8m_s}\, n_s(\boldsymbol{q}).$$

These are the density, the spin density, the paramagnetic current, the spin current (velocity index α, spin index β) and the kinetic-energy density. Their matrix elements from a plane wave $\boldsymbol{k}$ to $\boldsymbol{k}+\boldsymbol{q}$ are 1, $\boldsymbol{S}$, $(\boldsymbol{k}+\boldsymbol{k}')/2m_s$, $(k^\alpha+k'^\alpha)S^\beta/2m_s$ and $(\boldsymbol{k}+\boldsymbol{k}')^2/8m_s$.

For the electrons they are old acquaintances. The density $n_e(\boldsymbol{q})$ is Eq. (2.38), the spin density $\boldsymbol{s}_e(\boldsymbol{q})$ is Eq. (7.13), and the current $\boldsymbol{j}_e(\boldsymbol{q})$ is that of the continuity equation Eq. (4.29). The spin current and the kinetic-energy density are new. At $\boldsymbol{q}=0$ the kinetic-energy density is the kinetic energy $\sum_j p_j^2/2m_s$ of the species, and at finite $\boldsymbol{q}$ it is the kinetic energy of the mean of the incoming and outgoing momenta, which is why the term $q^2/8m_s$ appears beside the familiar $\boldsymbol{p}\cdot\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\boldsymbol{p}/2m_s$. For nuclei the spin $\boldsymbol{S}_j$ is the nuclear spin, and Sect. 8.1.3 turns couplings to nucleons into couplings to these rows.

With the rows in hand the reduction is mechanical. Substitute $\boldsymbol{v}^\perp = \bar{\boldsymbol{v}}_\chi - (\boldsymbol{k}+\boldsymbol{k}')/2m_e$ in each operator, expand, and replace every monomial in the electron's velocity and spin by its row: $1 \to n_e$, $\boldsymbol{S}_e \to \boldsymbol{s}_e$, the velocity $\to \boldsymbol{j}_e$, the velocity times the spin $\to \mathcal{J}_e$, and the square of the velocity $\to (2/m_e)\mathcal{T}_e$. The dark-matter velocity $\bar{\boldsymbol{v}}_\chi$ and spin $\boldsymbol{S}_\chi$ are not target operators; they stay in the coefficients. For instance,

$$\mathcal{O}_8 = \boldsymbol{S}_\chi\cdot\boldsymbol{v}^\perp \;\to\; (\boldsymbol{S}_\chi\cdot\bar{\boldsymbol{v}}_\chi)\, n_e(\boldsymbol{q}) - \boldsymbol{S}_\chi\cdot\boldsymbol{j}_e(\boldsymbol{q}), \qquad \mathcal{O}_7 = \boldsymbol{S}_e\cdot\boldsymbol{v}^\perp \;\to\; \bar{\boldsymbol{v}}_\chi\cdot\boldsymbol{s}_e(\boldsymbol{q}) - \mathcal{J}_e^{\alpha\alpha}(\boldsymbol{q}), \tag{8.4}$$

with a sum over the repeated Cartesian index. Table 8.1 lists all fifteen reductions, and Appendix D.24 derives them. Every operator except $\mathcal{O}_2$ is at most linear in the electron's velocity and in its spin, and needs at most the four rows n_e, $\boldsymbol{s}_e$, $\boldsymbol{j}_e$ and $\mathcal{J}_e$ [1, Table III]; $\mathcal{O}_2$ adds $\mathcal{T}_e$.

In a solid the electron moves too. The velocity-dependent operators split into a piece in which the dark-matter velocity multiplies a velocity-independent row and a piece that needs the target's own velocity. For a free electron at rest the second piece vanishes, but in a solid it does not, and it is often the larger: the valence electrons of silicon, taken as a free gas at their density, move at $v_\mathrm{F} = k_\mathrm{F}/m_e = 7.0\times10^{-3}$ (Sect. 4.5), faster than any dark-matter particle in the laboratory. In the language of Ref. [1, Table III], the target velocity turns a coupling to the charge or spin of the electron into a coupling to its orbital motion. The rows also overlap, in the longitudinal direction. The longitudinal current is not independent of the density: by the continuity equation Eq. (4.29), $\langle f|\boldsymbol{q}\cdot\boldsymbol{j}_e|i\rangle = (E_f-E_i)\langle f|n_e|i\rangle$ between eigenstates of any Hamiltonian whose potentials are local. On the dark-matter side, $\boldsymbol{q}\cdot\bar{\boldsymbol{v}}_\chi = \boldsymbol{q}\cdot\boldsymbol{v} - q^2/2m_\chi = \omega + \Delta_\chi$ by Eq. (1.22). In every matrix element that enters a rate, therefore,

$$\boldsymbol{q}\cdot\boldsymbol{v}^\perp \;\to\; \Delta_\chi\, n_e(\boldsymbol{q}), \tag{8.5}$$

which vanishes for elastic dark matter (Appendix D.24). The transversality of $\boldsymbol{v}^\perp$ is thus not a property of free particles alone: as long as the target's potentials are local, only the components of $\bar{\boldsymbol{v}}_\chi$ and $\boldsymbol{j}_e$ transverse to $\boldsymbol{q}$ enter the rate of an elastic process. The same argument applies to the spin current, $q^\alpha\mathcal{J}_e^{\alpha\beta} \to \omega\, s_e^\beta$, provided the Hamiltonian conserves each electron's spin, which spin–orbit coupling and the hyperfine interaction violate.

Table 8.1: Non-relativistic operators of dark-matter scattering and absorption, and the target operators each needs. Upper block: the operators $\mathcal{O}_i$ of Refs. [2, Eqs. (19)–(24) and (45)] and [3, Eqs. (12) and (13)], written for an electron with $\boldsymbol{q}$ the momentum the dark matter loses; $\boldsymbol{v}^\perp$ is Eq. (8.1), $\bar{\boldsymbol{v}} \equiv \bar{\boldsymbol{v}}_\chi = \boldsymbol{v} - \boldsymbol{q}/2m_\chi$, and $m = m_e$. The rows are those of Eq. (8.3) for $s = e$, all at the same $\boldsymbol{q}$; repeated Cartesian indices are summed, and $\epsilon_{\alpha\beta\gamma}$ is the Levi-Civita symbol. For nucleons, replace e by N and m_e by the nucleon mass; the nuclear rows follow from Eq. (8.8). Lower block: absorption of a boson ϕ of mass m_ϕ at $\boldsymbol{q} = \boldsymbol{p}_\phi$, and fermionic absorption with an outgoing fermion $\tilde{\nu}$ (Sect. 8.1.5). Last column: what each row excites in an insulator. Through the nuclei the density row also excites phonons, and in an ordered magnet the spin rows excite magnons (Sect. 7.1).

	Operator	Target operators	Excites
$\mathcal{O}_1$	1	n	pairs, plasmons
$\mathcal{O}_2$	$(\boldsymbol{v}^\perp)^2$	$\bar{v}^2 n - 2\bar{\boldsymbol{v}}\cdot\boldsymbol{j} + (2/m)\mathcal{T}$	pairs
$\mathcal{O}_3$	$\mathrm{i}\boldsymbol{S}_e\cdot(\boldsymbol{q}/m\times\boldsymbol{v}^\perp)$	$(\mathrm{i}/m)\big[(\boldsymbol{q}\times\bar{\boldsymbol{v}})\cdot\boldsymbol{s} - \epsilon_{\alpha\beta\gamma}q^\beta\mathcal{J}^{\gamma\alpha}\big]$	spin flips
$\mathcal{O}_4$	$\boldsymbol{S}_\chi\cdot\boldsymbol{S}_e$	$\boldsymbol{S}_\chi\cdot\boldsymbol{s}$	spin flips
$\mathcal{O}_5$	$\mathrm{i}\boldsymbol{S}_\chi\cdot(\boldsymbol{q}/m\times\boldsymbol{v}^\perp)$	$(\mathrm{i}/m)\,\boldsymbol{S}_\chi\cdot\big[(\boldsymbol{q}\times\bar{\boldsymbol{v}})\,n - \boldsymbol{q}\times\boldsymbol{j}\big]$	pairs
$\mathcal{O}_6$	$(\boldsymbol{S}_\chi\cdot\boldsymbol{q}/m)(\boldsymbol{S}_e\cdot\boldsymbol{q}/m)$	$(\boldsymbol{S}_\chi\cdot\boldsymbol{q})(\boldsymbol{q}\cdot\boldsymbol{s})/m^2$	spin flips
$\mathcal{O}_7$	$\boldsymbol{S}_e\cdot\boldsymbol{v}^\perp$	$\bar{\boldsymbol{v}}\cdot\boldsymbol{s} - \mathcal{J}^{\alpha\alpha}$	spin flips
$\mathcal{O}_8$	$\boldsymbol{S}_\chi\cdot\boldsymbol{v}^\perp$	$(\boldsymbol{S}_\chi\cdot\bar{\boldsymbol{v}})\,n - \boldsymbol{S}_\chi\cdot\boldsymbol{j}$	pairs
$\mathcal{O}_9$	$\mathrm{i}\boldsymbol{S}_\chi\cdot(\boldsymbol{S}_e\times\boldsymbol{q}/m)$	$(\mathrm{i}/m)\,\boldsymbol{S}_\chi\cdot(\boldsymbol{s}\times\boldsymbol{q})$	spin flips
$\mathcal{O}_{10}$	$\mathrm{i}\boldsymbol{S}_e\cdot\boldsymbol{q}/m$	$(\mathrm{i}/m)\,\boldsymbol{q}\cdot\boldsymbol{s}$	spin flips
$\mathcal{O}_{11}$	$\mathrm{i}\boldsymbol{S}_\chi\cdot\boldsymbol{q}/m$	$(\mathrm{i}/m)(\boldsymbol{S}_\chi\cdot\boldsymbol{q})\,n$	pairs
$\mathcal{O}_{12}$	$\boldsymbol{S}_\chi\cdot(\boldsymbol{S}_e\times\boldsymbol{v}^\perp)$	$\boldsymbol{S}_\chi\cdot(\boldsymbol{s}\times\bar{\boldsymbol{v}}) - \epsilon_{\alpha\beta\gamma}S_\chi^\alpha\mathcal{J}^{\gamma\beta}$	spin flips
$\mathcal{O}_{13}$	$\mathrm{i}(\boldsymbol{S}_\chi\cdot\boldsymbol{v}^\perp)(\boldsymbol{S}_e\cdot\boldsymbol{q}/m)$	$(\mathrm{i}/m)\big[(\boldsymbol{S}_\chi\cdot\bar{\boldsymbol{v}})(\boldsymbol{q}\cdot\boldsymbol{s}) - S_\chi^\alpha q^\beta\mathcal{J}^{\alpha\beta}\big]$	spin flips
$\mathcal{O}_{14}$	$\mathrm{i}(\boldsymbol{S}_\chi\cdot\boldsymbol{q}/m)(\boldsymbol{S}_e\cdot\boldsymbol{v}^\perp)$	$(\mathrm{i}/m)(\boldsymbol{S}_\chi\cdot\boldsymbol{q})\big(\bar{\boldsymbol{v}}\cdot\boldsymbol{s} - \mathcal{J}^{\alpha\alpha}\big)$	spin flips
$\mathcal{O}_{15}$	$-(\boldsymbol{S}_\chi\cdot\boldsymbol{q}/m)\big[(\boldsymbol{S}_e\times\boldsymbol{v}^\perp)\cdot\boldsymbol{q}/m\big]$	$-(\boldsymbol{S}_\chi\cdot\boldsymbol{q})\big[\boldsymbol{q}\cdot(\boldsymbol{s}\times\bar{\boldsymbol{v}}) - \epsilon_{\alpha\beta\gamma}q^\alpha\mathcal{J}^{\gamma\beta}\big]/m^2$	spin flips
Dark photon	$-\zeta e\,A'_\mu J^\mu_{\mathrm{em}}$	$-\boldsymbol{j}_e + \sum_\kappa Z_\kappa\boldsymbol{j}_\kappa$	pairs; polar phonons
ALP	$g_{aee}\,\phi\,\bar{\psi}\mathrm{i}\gamma^5\psi$	$\boldsymbol{s}_e$ ($\propto|\boldsymbol{p}_\phi|$); $\mathcal{J}_e^{\alpha\alpha}$ ($\propto m_\phi$)	pairs
Scalar	$g_{\phi ee}\,\phi\,\bar{\psi}\psi$	$n_e - \mathcal{T}_e/m_e$	pairs
Fermionic	$\Lambda_F^{-2}(\bar{\tilde{\nu}}\Gamma\chi)(\bar{\psi}\Gamma'\psi)$	n, $\boldsymbol{j}$, $\boldsymbol{s}$, $\mathcal{J}^{\alpha\alpha}$ or $n - \mathcal{T}/m$, by Γ'	pairs; nuclear recoils

Collecting the coefficients, the interaction between a dark-matter particle and the target, from dark-matter momentum $\boldsymbol{p}$ to $\boldsymbol{p}-\boldsymbol{q}$, takes the form assumed in Sect. 4.2,

$$\langle \boldsymbol{p}-\boldsymbol{q}|H_{\mathrm{int}}|\boldsymbol{p}\rangle = \frac{1}{\mathscr{V}}\sum_a \tilde{V}_a(\boldsymbol{q},\bar{\boldsymbol{v}}_\chi)\,O_a(\boldsymbol{q}), \tag{8.6}$$

with the index a running over the rows of every species. The only change from Sect. 4.2 is that the potentials $\tilde{V}_a$ now depend on the dark-matter velocity and are matrices in the spin space of the dark matter. For $\mathcal{O}_8$, for instance, $\tilde{V}_{n_e} = c_8^{(e)}\,\boldsymbol{S}_\chi\cdot\bar{\boldsymbol{v}}_\chi$ and $\tilde{V}_{j_e^\alpha} = -c_8^{(e)}S_\chi^\alpha$. An unpolarized beam averages over the initial spin of the dark matter and sums over its final spin, and the square of the amplitude $\mathscr{V}^{-1}\sum_b\langle f|O_b|i\rangle\tilde{V}_b$ becomes a trace over that spin space:

Dark-matter-side tensor

$$\mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v}) = \frac{1}{2|\boldsymbol{S}_\chi|+1}\,\mathrm{Tr}_\chi[\tilde{V}_a(\boldsymbol{q},\bar{\boldsymbol{v}}_\chi)^\dagger\,\tilde{V}_b(\boldsymbol{q},\bar{\boldsymbol{v}}_\chi)], \tag{8.7}$$

with $|\boldsymbol{S}_\chi|$ the spin quantum number of the dark matter. The rate of Eq. (4.14) holds with $\tilde{V}_a^*\tilde{V}_b \to \mathfrak{D}_{ab}$. The tensor is Hermitian and positive semidefinite, and for a coupling that does not involve $\boldsymbol{S}_\chi$ it is $\tilde{V}_a^*\tilde{V}_b$.

Since $\mathrm{Tr}_\chi\,\boldsymbol{S}_\chi = 0$, a coefficient linear in $\boldsymbol{S}_\chi$ never interferes with one that is independent of it, and the operators split into two classes that add in the rate. For spin $\frac{1}{2}$, $\mathrm{Tr}_\chi(S_\chi^\alpha S_\chi^\beta)/2 = \delta_{\alpha\beta}/4$. Every model enters the rest of the book through $\mathfrak{D}_{ab}$, and every target through the response matrix S_{ab} of the rows; Sect. 8.2 derives the rate formula that contracts them.

Responses of the density are what Chap. 4 computed. The other rows need the same calculation with other vertices. For the spin density and the spin current, the vertex is that of the density or of the current multiplied by a Pauli matrix. For the current and the kinetic-energy density it needs the momentum operator, whose matrix elements change when the pseudopotential is nonlocal and must be corrected for it. In practice a calculation computes the independent-particle response matrix of all the electron rows at once and screens it with one Dyson equation (Sect. 8.4); Appendix E.16 describes the tables it exports.

8.1.3 Nuclei

A model specifies its couplings to protons and neutrons, $c_i^{(p)}$ and $c_i^{(n)}$, with operators $\mathcal{O}_i^{(N)}$ built from the nucleon spin $\boldsymbol{S}_N$, the nucleon mass m_N and the nucleon velocity. The solid, however, contains nuclei. Nucleus I of species κ sits at $\boldsymbol{X}_I$, and its A_κ nucleons sit at $\boldsymbol{X}_I+\boldsymbol{y}_j$, within a radius of a few fm. The small size of the nucleus and the small energy deposited reduce the nucleon operators to nuclear rows. Light dark matter transfers at most $2m_\chi v_{\max}\approx 5\,\mathrm{MeV}$, against $1/R\approx 57\,\mathrm{MeV}$ for silicon (Sect. 1.4), so $\boldsymbol{q}\cdot\boldsymbol{y}_j$ is small. And the energy it deposits on a nucleus, at most 0.50 keV on silicon at $m_\chi = 1\,\mathrm{GeV}$ (Sect. 1.3), lies far below the excitation energies of nuclei [2, Sect. 2.2], so the nucleus stays in its ground multiplet. The small value of $\boldsymbol{q}\cdot\boldsymbol{y}_j$ turns the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{X}_I+\boldsymbol{y}_j)}$ into $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$ times the nuclear form factor $F_N(q)$ of Eq. (2.26), and the nucleon couplings add coherently, as in Sect. 1.4. The nucleon velocity is the velocity of the nucleus, $\boldsymbol{P}_I/M_\kappa$, plus an internal velocity; the internal part averages to zero in the ground state and contributes at higher order in $\boldsymbol{q}\cdot\boldsymbol{y}_j$ [2, Eqs. (26) and (27)], and we drop it. The sum of the nucleon spins, weighted by their couplings, is a vector operator, and since the nucleus stays in its ground multiplet, the Wigner–Eckart theorem makes it proportional to the nuclear spin $\boldsymbol{J}_I^{\mathrm{nuc}}$ within the ground multiplet of spin J_{nuc}. The constant follows from the state with $M = J_{\mathrm{nuc}}$, in which the proton and neutron spins have the expectation values $\langle S_p\rangle$ and $\langle S_n\rangle$ along the nuclear spin. Together these give, for nucleus species κ (and, for spin

couplings, a single isotope),

$$c_i^{(\kappa)}(q) = [Z_\kappa c_i^{(p)} + (A_\kappa - Z_\kappa)\, c_i^{(n)}] F_N(q) \quad \text{or} \quad c_i^{(\kappa)}(q) = \frac{c_i^{(p)}\langle S_p\rangle + c_i^{(n)}\langle S_n\rangle}{J_{\text{nuc}}}, \tag{8.8}$$

the first for operators without $\boldsymbol{S}_N$ and the second, with $\boldsymbol{S}_N \to \boldsymbol{J}_I^{\text{nuc}}$, for those linear in it. The nuclear operators then reduce as in Table 8.1, to the rows of Eq. (8.3) with $s = \kappa$, $m_s = M_\kappa$ and $\boldsymbol{S}_j = \boldsymbol{J}_I^{\text{nuc}}$; in particular the nuclear spin density is $\boldsymbol{s}_\kappa(\boldsymbol{q}) = \sum_{I\in\kappa} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I} \boldsymbol{J}_I^{\text{nuc}}$. The operators keep $\boldsymbol{q}/m_N$ with the nucleon mass, since that is how the coefficients are defined.

The spin coupling is far weaker than the coherent one. Silicon has one isotope with a nuclear spin, ^{29}Si, with $J_{\text{nuc}} = \frac{1}{2}$ [6, Fig. 5] and an abundance of 4.685 % in natural silicon [7]. Shell-model calculations give $\langle S_n\rangle = 0.156$ and $\langle S_p\rangle = 0.016$ [6, Table III]. For a coupling to neutron spins alone, Eq. (8.8) gives $c_4^{(\kappa)} = 0.312\, c_4^{(n)}$. Averaged over the initial orientations of the nucleus and summed over the final ones, the squared spin operator is $J_{\text{nuc}}(J_{\text{nuc}}+1) = \frac{3}{4}$. The strength is then $(0.312)^2 \times \frac{3}{4} = 0.073$ in units of $|c_4^{(n)}|^2$, against $\frac{3}{4}$ for a free neutron. A ^{29}Si nucleus thus responds with 10 % of the strength of one neutron, whereas its spin-independent coupling is $A^2 = 841$ times that of one nucleon, and in natural silicon one nucleus in 21 carries the spin. A sample of unpolarized nuclear spins has no coherent spin response, because the spins of two different nuclei are uncorrelated; only the incoherent part of $S_{\boldsymbol{s}_\kappa \boldsymbol{s}_\kappa}$ survives (Sect. 11.6).

Worked example 8.1: A heavy dark photon reduces to the density rows

Take the thermal benchmark of Worked example 1.5: $m_\chi = 10\,\text{MeV}$, $m_V = 30\,\text{MeV}$, $y_\chi = 0.3$ and $\zeta = 10^{-4}$. The dark photon couples to the dark-matter current with strength y_χ and to J^μ_{em} with strength ζe (Sect. 1.4). Exchanging it between a dark-matter particle and a target fermion with the currents J^μ_χ and J^μ gives, for $\omega \ll q$, the potential $y_\chi y\,(J^0_\chi J^0 - \boldsymbol{J}_\chi\cdot\boldsymbol{J})/(q^2+m_V^2)$, with y the coupling of the target fermion (Appendix D.24). The time components reduce to 1 on both sides, and the leading term is $\mathcal{O}_1$ with

$$c_1^{(e)} = \frac{y_\chi y_e}{q^2+m_V^2} = \tilde{V}(q), \qquad c_1^{(p)} = -c_1^{(e)}, \qquad c_1^{(n)} = 0,$$

since $y_e = -\zeta e$ and a proton has $+\zeta e$. By Eq. (8.8), $c_1^{(\kappa)} = -Z_\kappa \tilde{V} F_N(q)$, and with $F_N = 1$ to better than 0.1 % at these momenta the interaction is $\mathscr{V}^{-1}\tilde{V}(q)\,[n_e(\boldsymbol{q}) - \sum_\kappa Z_\kappa n_\kappa(\boldsymbol{q})] = -\mathscr{V}^{-1}\tilde{V}(q)\, n_Z(\boldsymbol{q})$, with n_Z the charge-number density of Sect. 2.3. The tensor is $\mathfrak{D}_{ab} = \tilde{V}_a^* \tilde{V}_b$ over the rows n_e and n_κ, and the rate is $|\tilde{V}|^2$ times the structure factor of the total charge. Its electron block is the S of Eq. (1.1); the nuclear block and the cross terms, of opposite sign, matter where the nuclei move, at phonon energies (Sect. 8.4). What is dropped? At $m_V = 30\,\text{MeV}$ the mediator is heavy at every momentum transfer, $q \le 2m_\chi v_{\max} = 53\,\text{keV}$, and the propagator is constant to $(q/m_V)^2 \le 3\times10^{-6}$. The largest correction to the density vertex is the factor $1 - q^2/8m_e^2$ of Eq. (C.134), which is 7×10^{-6} at q_0 and 1.3×10^{-3} at 53 keV. The spatial currents, $\boldsymbol{J}_\chi\cdot\boldsymbol{J}$, are products of velocities and spin terms. Their spin-independent piece, $\bar{\boldsymbol{v}}_\chi\cdot(\boldsymbol{k}+\boldsymbol{k}')/2m_e$, is at most $v_{\max}v_{\text{F}} = 1.9\times10^{-5}$ for valence electrons. The pieces linear in one spin do not interfere with the density coupling in a nonmagnetic target and enter the rate squared, and the product of the two spin terms is at most $q^2/(4m_\chi m_e) = 1.4\times10^{-4}$. To a part in 10^3, therefore, a heavy dark photon couples to the density rows and nothing else.

8.1.4 The velocity-squared row

The operator $\mathcal{O}_2 = (\boldsymbol{v}^\perp)^2$ is the only one that is quadratic in a velocity. Expanding the square,

$$(\boldsymbol{v}^\perp)^2 = \bar{v}_\chi^2 - 2\bar{\boldsymbol{v}}_\chi \cdot \frac{\boldsymbol{k}+\boldsymbol{k}'}{2m_e} + \frac{(\boldsymbol{k}+\boldsymbol{k}')^2}{4m_e^2} \;\to\; \bar{v}_\chi^2\, n_e(\boldsymbol{q}) - 2\bar{\boldsymbol{v}}_\chi \cdot \boldsymbol{j}_e(\boldsymbol{q}) + \frac{2}{m_e}\,\mathcal{T}_e(\boldsymbol{q}). \tag{8.9}$$

The first two terms use the density and current rows. The third needs the square of the electron's velocity, and since $\mathcal{O}_2$ is a scalar, only the trace of the velocity tensor $v^\alpha v^\beta$ appears, which is $2/m_e$ times the kinetic-energy density of Eq. (8.3). One row per species therefore suffices.

How large is each term? A naive estimate sets $\boldsymbol{v}^\perp \sim \boldsymbol{v} \sim 10^{-3}$ and makes $\mathcal{O}_2$ a correction of about 10^{-6} to $\mathcal{O}_1$. That estimate, however, treats the electron as being at rest, and Eq. (8.9) shows what it misses. By Eq. (8.5), the components along $\boldsymbol{q}$ of the dark-matter and electron velocities cancel for elastic scattering, and what remains of $\boldsymbol{v}^\perp$ is the difference of their transverse parts. The dark matter's transverse velocity obeys $v_\mathrm{T}^2 = v^2 - v_\mathrm{min}^2 \le v_\mathrm{max}^2$, since its longitudinal part is v_min at $\Delta_\chi = 0$. The electron's is set by the motion of the electrons in the solid, about v_F for valence states and larger for the core.

Worked example 8.2: The velocity-squared operator in silicon

We model the valence electrons of silicon as the free electron gas of Sect. 4.5, at $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$, with $k_\mathrm{F} = 3.57\,\mathrm{keV}$ and $v_\mathrm{F} = 6.99 \times 10^{-3}$, at $T = 0$. Take dark matter of mass 100 MeV moving at $v = 2.0 \times 10^{-3}$ that transfers $q = q_0 = 3.73\,\mathrm{keV}$ and deposits $\omega = 5\,\mathrm{eV}$. A pair takes an electron from $\boldsymbol{k}$ inside the Fermi sphere to $\boldsymbol{k}+\boldsymbol{q}$ outside it, with $\omega = \boldsymbol{q}\cdot(2\boldsymbol{k}+\boldsymbol{q})/2m_e$. This fixes the component of $\boldsymbol{k}$ along $\boldsymbol{q}$ at $m_e\omega/q - q/2 = -1.18\,\mathrm{keV}$. The two Pauli conditions then confine the transverse momentum k_T to an annulus, $k_\mathrm{F}^2 - (2.55\,\mathrm{keV})^2 < k_\mathrm{T}^2 < k_\mathrm{F}^2 - (1.18\,\mathrm{keV})^2$, with 2.55 keV the longitudinal momentum after the transfer, that is $2.50\,\mathrm{keV} < k_\mathrm{T} < 3.37\,\mathrm{keV}$, which every pair of this $(\boldsymbol{q},\omega)$ fills uniformly in area. The electron's transverse velocity, $(\boldsymbol{k}+\boldsymbol{k}')_\mathrm{T}/2m_e = \boldsymbol{k}_\mathrm{T}/m_e$, has $\langle v_{e\mathrm{T}}^2\rangle = 3.37 \times 10^{-5}$ and $\langle v_{e\mathrm{T}}^4\rangle = 1.17 \times 10^{-9}$ over the annulus.
On the dark-matter side, $v_\mathrm{min} = \omega/q + q/2m_\chi = 1.36 \times 10^{-3}$ and $v_\mathrm{T}^2 = v^2 - v_\mathrm{min}^2 = 2.15 \times 10^{-6}$, smaller than the electron's by a factor of 16; at $v = v_\mathrm{max} = 2.650 \times 10^{-3}$ the factor is still 6.5. Per pair, $(\boldsymbol{v}^\perp)^2 = |\boldsymbol{v}_\mathrm{T} - \boldsymbol{v}_{e\mathrm{T}}|^2$, and the azimuth of $\boldsymbol{v}_{e\mathrm{T}}$ about $\boldsymbol{q}$ is uniform. Averaging its square, with $A = v_\mathrm{T}^2$ and $B = v_{e\mathrm{T}}^2$, gives $(A+B)^2 + 2AB$, and over the annulus

$$\langle(\boldsymbol{v}^\perp)^4\rangle = A^2 + 4A\langle B\rangle + \langle B^2\rangle = (0.0046 + 0.290 + 1.167) \times 10^{-9} = 1.46 \times 10^{-9}.$$

The term that the naive estimate keeps, A^2, is 0.3 % of the total. The fourth power of the electron's transverse velocity, $\langle B^2\rangle$, carries 80 %, and the cross term $4A\langle B\rangle$, in which the two transverse velocities meet, the remaining 20 %. These are shares of velocities, not of rows: every row of Eq. (8.9) also carries longitudinal velocity, which cancels in their sum. At equal coefficients the rate of $\mathcal{O}_2$ is 1.5×10^{-9} times that of $\mathcal{O}_1$ at this $(\boldsymbol{q},\omega)$, about 90 times the naive $v^4 = 1.6 \times 10^{-11}$.

The electron gas understates the effect in a real crystal. The valence wavefunctions of silicon have Fourier components at the reciprocal-lattice vectors (Sect. 3.2), so the electron velocities spread beyond v_F, and the $1s$ electrons, whose absorption edge lies at 1.84 keV (Sect. 4.3), move at $Z\alpha \approx 0.10$, where $(\boldsymbol{v}^\perp)^2 \approx 10^{-2}$ (Appendix C.15). A velocity-dependent coupling therefore weights the core and the high-momentum tails of the valence states far

more than the density coupling does. In practice, it is the target term $\mathcal{T}_e$ that a calculation must get right.

8.1.5 Absorption operators

A bosonic dark-matter field ϕ of mass m_ϕ oscillates as $\mathrm{e}^{-\mathrm{i}m_\phi t+\mathrm{i}\boldsymbol{p}_\phi\cdot\boldsymbol{x}}$, and absorbing one quantum deposits $\omega = m_\phi$ and $\boldsymbol{q} = \boldsymbol{p}_\phi$ (Eq. (1.38)). Its coupling to the target is a bilinear of the target fields multiplied by ϕ or a derivative of ϕ, and the spatial integral over $\mathrm{e}^{\mathrm{i}\boldsymbol{p}_\phi\cdot\boldsymbol{x}}$ turns each bilinear into a row of Eq. (8.3) at $\boldsymbol{q} = \boldsymbol{p}_\phi$. The rows are the same as for scattering. What differs is the point $(\boldsymbol{q},\omega)$ at which their response is needed, $|\boldsymbol{p}_\phi| \approx 10^{-3}\omega$ instead of $\omega \le qv$. The three bosons of Sect. 1.4 need different rows (Appendix D.24).

The dark photon couples to J^μ_{em} like a photon, with $e \to \zeta e$. For a dark photon A'_μ of mass $m_V = m_\phi$ at rest, the time component is smaller than the spatial ones by $|\boldsymbol{p}_\phi|/m_\phi$, by the Lorenz condition, and the coupling is to the spatial current of all charges, $-\boldsymbol{j}_e(\boldsymbol{p}_\phi) + \sum_\kappa Z_\kappa \boldsymbol{j}_\kappa(\boldsymbol{p}_\phi)$. At $\boldsymbol{p}_\phi \to 0$ its response is the optical conductivity (Sect. 8.2.3). The nuclear currents carry the absorption into polar optical phonons below the electronic gap.

The axion-like particle couples through Eq. (1.40), in which we write ϕ, $\nabla\phi$ and $\partial_t\phi$ for the amplitudes of the plane wave. Its leading term is $(g_{aee}/m_e)(\nabla\phi)\cdot\boldsymbol{s}_e$, the spin-density row with a coefficient proportional to $|\boldsymbol{p}_\phi|$. Its second term contains $\hat\psi^\dagger\boldsymbol{\sigma}\cdot\overleftrightarrow{\nabla}\hat\psi$, whose matrix element between plane waves is $\mathrm{i}\boldsymbol{\sigma}\cdot(\boldsymbol{k}+\boldsymbol{k}')$; it is the trace of the spin current, and the interaction becomes

$$H_{aee} = \frac{g_{aee}}{m_e}\Big[(\nabla\phi)\cdot\boldsymbol{s}_e(\boldsymbol{p}_\phi) + (\partial_t\phi)\,\mathcal{J}^{\alpha\alpha}_e(\boldsymbol{p}_\phi)\Big]. \tag{8.10}$$

The trace couples $\boldsymbol{S}_j$ to the velocity of the same electron. With $|\partial_t\phi| = m_\phi|\phi|$ and $|\nabla\phi| = |\boldsymbol{p}_\phi||\phi|$, the second term exceeds the first by the electron velocity over the dark-matter velocity, as Sect. 1.4 found. When spin–orbit coupling is negligible the spin and the orbit of each electron separate, and the response of $\mathcal{J}^{\alpha\alpha}_e$ reduces to that of the current, which is why the rate again follows from the conductivity [8, Sect. III B].

The scalar couples through Eq. (1.41). The operator $\hat\psi^\dagger\overleftrightarrow{\nabla}^2\hat\psi$ has the plane-wave matrix element $-(\boldsymbol{k}+\boldsymbol{k}')^2$, so $(1/8m_e^2)\hat\psi^\dagger\overleftrightarrow{\nabla}^2\hat\psi$ is $-\mathcal{T}_e/m_e$, and

Scalar absorption through the kinetic-energy density

$$H_{\phi ee} = -g_{\phi ee}\,\phi\Big[n_e(\boldsymbol{p}_\phi) - \frac{\mathcal{T}_e(\boldsymbol{p}_\phi)}{m_e}\Big]. \tag{8.11}$$

The bracket is the non-relativistic limit of $\bar\psi\psi$: the number density times $1 - v^2/2$, the factor $(1-v^2)^{1/2}$ by which a moving particle's scalar density falls below its number density. Matrix elements of $n_e(\boldsymbol{p}_\phi)$ between different states vanish as $\boldsymbol{p}_\phi \to 0$, and those of $\mathcal{T}_e$ do not, so the kinetic-energy term dominates absorption [8, Eq. (18) and Sect. III C].

The kinetic-energy row is the one that $\mathcal{O}_2$ needed, so a single new row per species covers both. The density term is suppressed twice: by orthogonality, since $\langle f|n_e(\boldsymbol{p}_\phi)|i\rangle \simeq \mathrm{i}\boldsymbol{p}_\phi\cdot\langle f|\sum_j \boldsymbol{r}_j|i\rangle$, and by the screening of a density coupling in the medium [8, Sect. III C]. The kinetic energy does not commute with the crystal Hamiltonian and has matrix elements between different states already at $\boldsymbol{p}_\phi = 0$. Its rate, unlike those of the dark photon and the axion-like particle, is not fixed by the optical conductivity, and Sect. 10.7 computes it.

Fermionic dark matter can be absorbed too. In the operators of Refs. [9, Eq. (1), 10, Eqs. (1) and (2)], $\Lambda_F^{-2}(\bar\nu\Gamma\chi)(\bar\psi\Gamma'\psi)$ with a heavy scale Λ_F and Dirac matrices Γ, Γ', the

dark-matter fermion χ is destroyed and a light fermion $\tilde{\nu}$ of mass $m_{\tilde{\nu}} \ll m_\chi$ is emitted. The target bilinear $\bar{\psi}\Gamma'\psi$ reduces as for scattering: the vector to the density and current rows, the axial vector to the spin density and the trace of the spin current, the scalar to the bracket of Eq. (8.11). What is new is the dark-matter side, which Sect. 8.2.3 builds into the rate, and the kinematics of absorption, which fixes $\omega \simeq m_\phi$ at $q = m_\phi v$ (Sect. 8.2.3).

8.1.6 Energy deposited

In every process of this section the target receives the energy that the dark-matter side gives up. A particle of mass m_χ and momentum $\boldsymbol{p}$ enters, and either nothing or a particle of momentum $\boldsymbol{p} - \boldsymbol{q}$ leaves, so that

$$\omega = E_{\text{in}}(\boldsymbol{p}) - E_{\text{out}}(\boldsymbol{p} - \boldsymbol{q}). \tag{8.12}$$

For elastic scattering, E_{out} is the kinetic energy of the same particle and Eq. (8.12) is $\omega_{\boldsymbol{q}}$ of Eq. (1.22) at $\Delta_\chi = 0$. For inelastic dark matter the outgoing state χ_2 has mass $m_\chi + \Delta_\chi$, and expanding its energy to first order in $(\boldsymbol{p} - \boldsymbol{q})^2$,

$$\omega = \omega_{\boldsymbol{q}} + \frac{\Delta_\chi}{m_\chi + \Delta_\chi} \frac{(\boldsymbol{p} - \boldsymbol{q})^2}{2m_\chi}, \tag{8.13}$$

with $\omega_{\boldsymbol{q}}$ including the term $-\Delta_\chi$ (Appendix D.24). The second term corrects the recoil energy by the fraction Δ_χ/m_χ, which Eq. (1.22) drops; endothermic and exothermic scattering are then exactly the cases of Sect. 1.3. A light outgoing fermion changes the picture. For $m_{\tilde{\nu}} \to 0$ and $v \to 0$, $E_{\text{in}} = m_\chi$ and $E_{\text{out}} = |\boldsymbol{q}|$, so

$$\omega = m_\chi - q, \tag{8.14}$$

and nearly all of the rest energy goes to $\tilde{\nu}$: the target is probed along a line in the (q, ω) plane at $q \approx m_\chi$, far above the $q \leq 2m_\chi v$ of scattering. On a free nucleus of mass M the recoil is then $\omega \approx m_\chi^2/2M$ [9, Eq. (2)], larger than an elastic recoil by $1/v^2$. Bosonic absorption is the limit in which nothing leaves, with $\omega = m_\phi$ at $\boldsymbol{q} = \boldsymbol{p}_\phi$. Scattering fills a region of the plane, fermionic absorption a line and bosonic absorption a point, and Sect. 8.2 shows that one rate formula, with the rows of Eq. (8.3) and the tensor of Eq. (8.7), covers all three.

8.2 The master rate formula

A dark-matter particle that scatters in silicon, a dark photon of a few eV that is absorbed in it, and an excited dark-matter state that drops to its ground state on the way through all probe one property of the crystal: how readily it takes up a momentum $\boldsymbol{q}$ together with an energy ω. For scattering through several target operators at once, Eq. (4.14) already answers it, for one dark-matter particle of fixed velocity and for potentials that are plain numbers. Three things are missing before it can serve every model of Sect. 8.1. The potentials are matrices in the spin space of the dark matter, and the rate needs the trace $\mathfrak{D}_{ab}$ of Eq. (8.7). The halo supplies a distribution of velocities, and experiments quote rates per unit target mass. And absorption, in which the dark matter disappears, and the transitions in which the target gives energy away, look at first sight like different processes. We will see that all of them follow from one golden-rule calculation and land in different regions of the $(\boldsymbol{q}, \omega)$ plane (Fig. 8.3). The formula is exact in the target and first order in the coupling to the dark matter.

8.2.1 Scattering

Take a dark-matter particle in the plane wave $\boldsymbol{p} = m_\chi \boldsymbol{v}$, in spin state s, and a target in the eigenstate $|i\rangle$. By Eq. (8.6) the amplitude to reach $\boldsymbol{p} - \boldsymbol{q}$, spin s' and $|f\rangle$ is $\mathcal{V}^{-1}\sum_b \langle s'|\tilde{V}_b|s\rangle\langle f|O_b(\boldsymbol{q})|i\rangle$. The golden rule squares it, the spin sums turn the product of two potentials into the trace of Eq. (8.7), the thermal average weights $|i\rangle$ with p_i, and the sum over final dark-matter momenta becomes $\mathcal{V}\int \mathrm{d}^3q/(2\pi)^3$, as in Sect. 2.5. Each step is written out in Appendix D.25. The target then enters through the response matrix Eq. (4.13), and the dark matter through $\mathfrak{D}_{ab}$:

Master rate formula: scattering

$$\Gamma(\boldsymbol{v}) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} \sum_{a,b} \underbrace{\mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})}_{\text{coupling}} \underbrace{S_{ab}(\boldsymbol{q},\omega_{\boldsymbol{q}})}_{\text{target}}, \tag{8.15}$$

the rate of one dark-matter particle of velocity $\boldsymbol{v}$, with $\omega_{\boldsymbol{q}}$ of Eq. (1.22), and per unit target mass

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{1}{\rho_T} \underbrace{\frac{\rho_\chi}{m_\chi}\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})}_{\text{halo}} \int \frac{\mathrm{d}^3q}{(2\pi)^3} \sum_{a,b} \underbrace{\mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})}_{\text{coupling}} \times \underbrace{S_{ab}(\boldsymbol{q},\omega)}_{\text{target}} \underbrace{\delta(\omega-\omega_{\boldsymbol{q}})}_{\text{kinematics}}. \tag{8.16}$$

The rows a, b run over the target operators of every species (Eq. (8.3)), the target is in thermal equilibrium at temperature T, and ω takes either sign.

The integrand is a contraction of two Hermitian positive semidefinite matrices, $\mathfrak{D}$ from the dark-matter side and S from the target, and is therefore never negative. For potentials without dark-matter spin it is the rate Eq. (4.14), which Sect. 4.2 derived and we do not repeat, and for the density row alone it is Eq. (2.67). In the same form, Trickle, Zhang and Zurek write the scattering rate with an operator-valued potential [1, Eq. (1)], and Trickle and collaborators write the total rate per unit target mass [11, Eqs. (14)–(16)], both for a target that starts in one eigenstate.

The order of the indices matters once both matrices are complex. Since the golden rule pairs the conjugated potential $\tilde{V}_a^\dagger$ with the conjugated operator $O_a^\dagger$, Eq. (8.15) contracts $\mathfrak{D}_{ab}$ with S_{ab}, element by element, which is the trace $\mathrm{Tr}[\mathfrak{D}S^{\mathrm{T}}]$. Splitting each Hermitian matrix into a real symmetric and a real antisymmetric part, $\mathfrak{D} = \mathfrak{D}_{\mathrm{R}} + \mathrm{i}\mathfrak{D}_{\mathrm{I}}$ and $S = S_{\mathrm{R}} + \mathrm{i}S_{\mathrm{I}}$, the other order, $\mathrm{Tr}[\mathfrak{D}S]$, differs from it by $2\,\mathrm{Tr}[\mathfrak{D}_{\mathrm{I}}S_{\mathrm{I}}]$ (Appendix D.25). The two agree whenever one of the matrices is real. They differ when both are complex: for instance, when an operator with an explicit i, such as $\mathcal{O}_3$ of Table 8.1, meets a target whose current–spin response has an imaginary antisymmetric part, as spin–orbit coupling produces.

Eq. (8.16) reduces exactly to the rates of Chap. 1. For an interaction through the electron density alone, $\mathfrak{D}_{ab} = |\tilde{V}|^2$ for $a = b = n_e$ and zero otherwise, and the contraction is $|\tilde{V}(\boldsymbol{q})|^2 S(\boldsymbol{q},\omega)$. With $\Delta_\chi = 0$ the delta function is that of Eq. (1.1), and the two formulas coincide term by term, including the normalization $\rho_\chi/m_\chi\rho_T$; this is Eq. (A.43). For a structure factor that depends only on q, or after averaging it over the directions of $\boldsymbol{q}$, the velocity integral and the angular integral act on the delta function alone. They give $\eta(v_{\min})/2q$ per unit $\mathrm{d}^3q/(2\pi)^3$ (Eq. (A.42)), and with $|\tilde{V}|^2 = \pi\bar{\sigma}_e F_{\mathrm{DM}}^2/\mu_{\chi e}^2$ the result is Eq. (1.47). For a crystal S_{ab} depends on the direction of $\boldsymbol{q}$, and the full velocity kernel is kept, as Appendix A.8 describes for the daily modulation.

8.2.2 Assumptions of the formula

The derivation uses four assumptions, and each can fail in a known place.

First Born approximation. The interaction with the dark matter acts once. The next term in the amplitude passes through intermediate dark-matter states and is smaller by roughly the ratio of the scattering amplitude $|f| = \mu_{\chi T}|\tilde{V}|/2\pi$ (Sect. 2.2) to the range of the potential [12, Chap. 6]. For the heavy dark photon of Worked example 8.1, at $m_\chi = 10\,\mathrm{MeV}$ and $m_V = 30\,\mathrm{MeV}$, $|f|\,m_V = 2 \times 10^{-8}$. At low energy the estimate is exact: an attractive potential raises the cross-section by the fraction $|f|\,m_V$ and a repulsive one lowers it by as much. Once $p_\mathrm{rel} = \mu_{\chi T} v$ exceeds a few m_V, the correction stays below the Sommerfeld parameter $y_\chi y_e/4\pi v$ of Sect. 2.2, which the relative momentum sets, not q (Limit Check 8.1). At the cross-sections used in this book the correction is below 10^{-4} for m_χ from 1 keV to 1 GeV and m_V up to 10 GeV, and, with couplings below 4π, 10 % requires $\bar{\sigma}_e \gtrsim 5 \times 10^{-25}\,\mathrm{cm}^2$ or $\sigma_p \gtrsim 1.6 \times 10^{-34}\,\mathrm{cm}^2$. The interactions inside the target are not expanded at all: S_{ab} is exact, and any approximation in it is an approximation to the target. Sect. 8.5 records such approximations in a truncation label $\mathfrak{T}$, the list of orders that a calculation keeps in each small parameter of the target. No truncation of the target changes the master formula; it changes only S_{ab}.

Plane waves. The halo is an incoherent mixture of momenta, distributed as f_χ, and the final dark-matter state is not observed. Averaging plane-wave rates over f_χ is then exact. For bosons light enough that the halo is a classical wave, the ensemble average still holds once the observation is longer than the shorter of the coherence time of the wave and the inverse width of the target's response, as we show below; a single record matches that average only once it spans many coherence times. Shorter observations are the coherent regime of Sect. 14.6.

Gibbs target. The target is in thermal equilibrium before each event, with the weights $p_i = \mathrm{e}^{-\beta E_i}/\mathcal{Z}$. A detector qualifies when events are rare compared with its relaxation. A target prepared out of equilibrium, such as an ensemble of polarized nuclear spins, still obeys Eq. (8.15) with its own populations in S_{ab}, provided its density operator is diagonal in energy, but detailed balance and the fluctuation–dissipation theorem then fail. The extension to a density operator with coherences between energy eigenstates, or to a target driven during the exposure, is open.

Unpolarized dark matter. The halo has no preferred spin direction, which is what the factor $1/(2|\boldsymbol{S}_\chi| + 1)$ in $\mathfrak{D}_{ab}$ assumes. A polarized flux, with spin density matrix $\hat{\varrho}_\chi$ of unit trace, has

$$\mathfrak{D}_{ab}(\boldsymbol{q}, \boldsymbol{v}) = \mathrm{Tr}_\chi\big[\hat{\varrho}_\chi\, \tilde{V}_a(\boldsymbol{q}, \bar{\boldsymbol{v}}_\chi)^\dagger\, \tilde{V}_b(\boldsymbol{q}, \bar{\boldsymbol{v}}_\chi)\big], \tag{8.17}$$

which is Eq. (8.7) for $\hat{\varrho}_\chi = 1/(2|\boldsymbol{S}_\chi| + 1)$.

Limit Check 8.1. First Born approximation for a Yukawa potential

For a Yukawa potential of strength $y_\chi y_T$, with $y_T = y_e$ for an electron and $A y_N$ for a nucleus, and range $1/m_V$, the second Born term changes the cross-section by the fraction

$$\delta_\mathrm{Born} \simeq \pm\frac{\mu_{\chi T}\, y_\chi y_T}{2\pi(m_V + p_\mathrm{rel})} = \pm|f(0)|\, m_V\, \frac{m_V}{m_V + p_\mathrm{rel}}, \tag{8.18}$$

upper sign for attraction [12, Chap. 6]. At $p_\mathrm{rel} \ll m_V$ it is exact: the zero-energy radial equation gives the scattering length $a = a_\mathrm{Born}(1 \pm |f(0)|m_V/2)$. At $p_\mathrm{rel} \gg m_V$, partial waves give a correction below $y_\chi y_T/4\pi v$ at every $q \le 2p_\mathrm{rel}$ (0.28 of it at $q = 2p_\mathrm{rel} = 20\,m_V$), falling as $m_V/p_\mathrm{rel} \to 0$, where the Born cross-section of the Coulomb potential is exact; Eq. (8.18) then overestimates it.

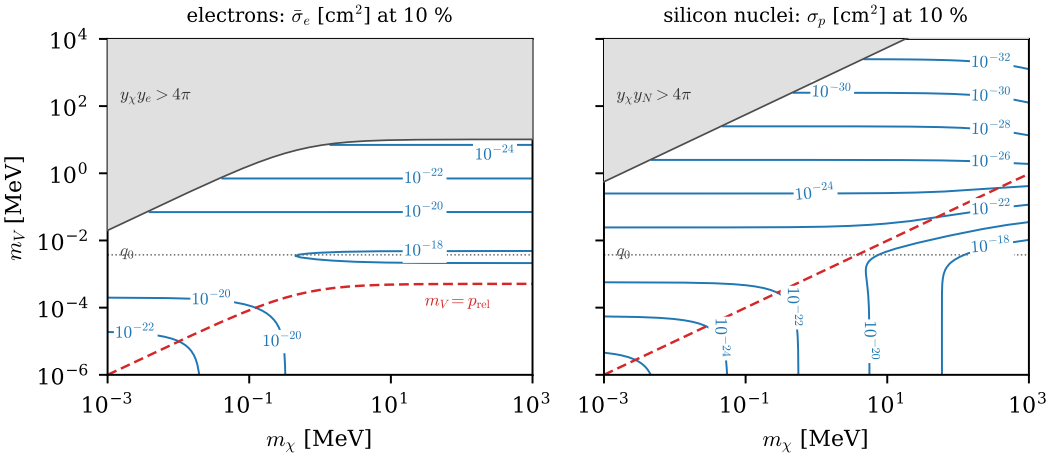


Figure 8.2: Cross-section at which $|\delta_{\rm Born}|$ of Eq. (8.18) reaches 0.1, against m_χ and m_V, at $v = 10^{-3}$. Left: electrons, $\bar{\sigma}_e$ with the form factor normalized at $q_0 = \alpha m_e$ (Eq. (1.35)). Right: silicon nuclei, coherent ($A = 28$), σ_p normalized in the same way. With the coupling fixed by the cross-section, m_χ enters only through $p_{\rm rel} = \mu_{\chi T} v$ (dashed, $m_V = p_{\rm rel}$; dotted, $m_V = q_0$). Gray: the 10 % point needs $y_\chi y_e$ or $y_\chi y_N$ above 4π, which happens for $m_V + p_{\rm rel} > 20 A \mu_{\chi T}$ ($A = 1$ for electrons). At $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$ and $\sigma_p = 10^{-40}\,\mathrm{cm}^2$ the largest correction in the plot is 4.5×10^{-5} and 8.0×10^{-5}, at $m_V = 10\,\mathrm{GeV}$. With perturbative couplings, 10 % needs $\bar{\sigma}_e \geq 4.9 \times 10^{-25}\,\mathrm{cm}^2$ (at $m_V \approx 10\,\mathrm{MeV}$) and $\sigma_p \geq 1.6 \times 10^{-34}\,\mathrm{cm}^2$ (at $m_\chi = 1\,\mathrm{GeV}$, $m_V = 10\,\mathrm{GeV}$).

A correction of order one needs couplings near 4π with $m_V \lesssim 20 A \mu_{\chi T}$, where the attractive potential binds and the cross-section has resonances that only the full partial-wave solution describes.

8.2.3 Absorption of a bosonic field

A boson ϕ of mass m_ϕ that is absorbed leaves nothing behind, and its whole energy and momentum go into the target. By Sect. 8.1.5 its coupling is linear in the field and turns each target bilinear into a row at $\boldsymbol{q} = \boldsymbol{p}_\phi$. We write the coefficient of row a per unit field amplitude as g_a^ϕ, so that for the scalar of Eq. (8.11), $g_{n_e}^\phi = -g_{\phi ee}$ and $g_{\mathcal{T}_e}^\phi = g_{\phi ee}/m_e$. For a vector field the polarization vector of the absorbed quantum sits inside g_a^ϕ. Quantize the field in the box, with quanta of energy $\omega_\phi = m_\phi(1 + v^2/2)$. The matrix element that removes one quantum from a mode with N_ϕ quanta is $\sqrt{N_\phi/2\omega_\phi \mathcal{V}}\ \sum_b g_b^\phi \langle f|O_b(\boldsymbol{p}_\phi)|i\rangle$, and the golden rule, summed over f and averaged over i, gives the response matrix again (Appendix D.25). Per absorbed particle,

Master rate formula: absorption

$$\Gamma_{\rm abs} = \frac{1}{2\omega_\phi} \sum_{a,b} \underbrace{g_a^{\phi *}\, g_b^\phi}_{\text{coupling}}\ \underbrace{S_{ab}(\boldsymbol{p}_\phi, \omega_\phi)}_{\text{target}},$$
$$\Gamma_{\rm net} = (1 - \mathrm{e}^{-\beta\omega_\phi})\,\Gamma_{\rm abs} = -\frac{1}{\omega_\phi}\ \mathrm{Im} \sum_{a,b} g_a^{\phi *}\, g_b^\phi\, R_{ab}(\boldsymbol{p}_\phi, \omega_\phi), \tag{8.19}$$

with $\omega_\phi \simeq m_\phi$. The event rate per unit target mass is $(\rho_\chi/m_\phi\rho_T)\langle\Gamma_{\rm abs}\rangle$, averaged over the directions and magnitudes of $\boldsymbol{p}_\phi = m_\phi\boldsymbol{v}$ with f_χ.

The absorption rate is the scattering formula evaluated at a single point, with $\mathfrak{D}_{ab}$ replaced by $g_a^{\phi\,*}g_b^\phi/2\omega_\phi$. The factor $1/2\omega_\phi$ is the relativistic normalization of one quantum. The number of quanta per unit volume, ρ_χ/m_ϕ, sets the event rate, not their flux: the speed of the dark matter enters only through the small momentum $\boldsymbol{p}_\phi$, as Sect. 2.1 found for a classical field.

The two rates in Eq. (8.19) answer different questions. $\Gamma_{\rm abs}$ counts the quanta the target absorbs, each depositing ω_ϕ, and it is what a detector records. The field also stimulates the target to emit quanta into the occupied modes, at the rate $\mathrm{e}^{-\beta\omega_\phi}\Gamma_{\rm abs}$ by detailed balance, and such an event takes energy out of the target. $\Gamma_{\rm net}$ is the difference, the rate at which the field loses quanta, and the fluctuation–dissipation theorem Eq. (4.18) writes it through the imaginary part of the response. That is the form in which Mitridate and collaborators state absorption, as the imaginary part of the self-energy of ϕ, $\Gamma = -\operatorname{Im}\Pi_{\phi\phi}/\omega$ [8, Eqs. (20)–(22)], with $\Pi_{\phi\phi} = \sum_{ab} g_a^{\phi\,*}g_b^\phi R_{ab}$ in our normalization. For ω_ϕ of an eV in a detector at 10 mK, however, $\mathrm{e}^{-\beta\omega_\phi}$ is zero for every purpose, and the two rates coincide.

The number of quanta in a volume $(1/m_\phi v)^3$ is $\rho_\chi/(m_\phi^4 v^3)$, which exceeds one for m_ϕ below about 7 eV at $v = 10^{-3}$, and grows as m_ϕ^{-4} below. Lighter bosons form a classical field rather than a gas of particles. The quantized calculation still applies, because the rate is linear in N_ϕ and the difference between N_ϕ and $N_\phi + 1$ in the emission matrix element, the spontaneous emission that a classical field lacks (Sect. 2.1), is negligible. The same rate also follows from the classical field directly. Its amplitude is a sum of many modes with random phases, and by the central limit theorem it is a Gaussian random field. Averaging the transition probability over this Gaussian ensemble, the *Gaussian field average*, removes the cross terms between different modes and leaves $\langle|\phi_{\boldsymbol{k}}|^2\rangle$ for each mode, which the energy density fixes; Appendix D.25 shows that the result is Eq. (8.19) term by term.

The ensemble average needs an observation time long compared with the shorter of the coherence time $t_{\rm coh}$ of the field, of order $1/m_\phi v_0^2$, about 1 ns at $m_\phi = 1$ eV, and the inverse width of the target's response; a single record reproduces it only when it is also long compared with $t_{\rm coh}$. When both times exceed the record, for a field of neV mass or below acting on a sharp resonance such as a nuclear spin, the field drives the target coherently. The transition amplitude rather than the probability then grows linearly in time, and the golden rule gives way to the coherent classical-field treatment of Sect. 14.6.

Eq. (8.19) reduces exactly to the dark-photon rate Eq. (1.39). A dark photon at rest couples to the current of the electrons with strength ζe times its polarization $\hat{\boldsymbol{e}}$ (Sect. 8.1.5), and the only row is $\hat{\boldsymbol{e}}\cdot\boldsymbol{j}_e$, with $g^\phi = \zeta e$. With the proper current response $\tilde{R}^{\rm T}_{jj}$ of Sect. 4.4, Eq. (8.19) gives $\Gamma_{\rm net} = -(\zeta^2 e^2/m_\phi)\operatorname{Im}\tilde{R}^{\rm T}_{jj}(m_\phi)$. The diamagnetic term of the current is real and does not contribute. By the transverse dielectric function of Sect. 4.4, $\varepsilon_2 = -(e^2/\omega^2)\operatorname{Im}\tilde{R}^{\rm T}_{jj}$, and with $\sigma_1 = \omega\varepsilon_2$ (Eq. (4.50)) the rate is $\zeta^2\sigma_1(m_\phi)$, the same for every polarization in a cubic crystal (Eq. (4.52)). The full response also contains the chain in which the induced current radiates an ordinary photon that the target absorbs again. Summing it, Appendix D.25 shows that it turns ζ^2 into $\zeta^2_{\rm eff}$ of Sect. 1.4. Multiplying by $\rho_\chi/m_\phi\rho_T$ gives Eq. (1.39).

Fermionic absorption (Sect. 8.1.5) fits the same formula. The dark matter disappears and a light fermion $\tilde{\nu}$ leaves with momentum $\boldsymbol{p}-\boldsymbol{q}$. The sum over final dark-matter momenta that produced Eq. (8.15) becomes the sum over the momenta of $\tilde{\nu}$, and energy conservation sets ω to Eq. (8.12), which is $m_\chi - q$ for a massless $\tilde{\nu}$. The rows and S_{ab} are those of scattering, now sampled along a line at $q \approx m_\chi$. Only $\mathfrak{D}_{ab}$ changes: the spinors of the relativistic $\tilde{\nu}$ enter it with relativistic normalization, which Sect. 10.9 constructs.

8.2.4 Negative energies and exothermic dark matter

The master formula holds for either sign of ω. The kinematics of scattering allows it to be negative: a dark-matter particle that moves against $\boldsymbol{q}$, or that recoils with more kinetic energy than it brings, gains energy, $\omega_{\boldsymbol{q}} < 0$. Whether such a transfer happens is decided by the target alone. At $T = 0$ the target starts in its ground state, $S_{ab}(\boldsymbol{q}, \omega)$ vanishes for $\omega < 0$, and the half-plane below the axis in Fig. 8.3 is closed. A warm target opens it, with the weight fixed by detailed balance. In matrix form, $S_{ab}(-\boldsymbol{q}, -\omega) = \mathrm{e}^{-\beta\omega} S_{ba}(\boldsymbol{q}, \omega)$ (Sect. 4.2), so a transfer of $\omega < 0$ is suppressed relative to the reverse one by $\mathrm{e}^{-\beta|\omega|}$, the Boltzmann factor of the excitation the target gives up. For an electronic excitation across the gap of silicon, this factor is smaller than any power of the temperature, and an expansion in the phonon occupations (the thermal order n_T of Sect. 8.5) drops it at every order. For a phonon it is the occupation $n_{\boldsymbol{Q}\nu}$ itself: the negative-ω part of a one-phonon rate is proportional to $n_{\boldsymbol{Q}\nu}$, of first order in the occupations. The same holds for absorption, whose emission partner sits at the point $(-\boldsymbol{p}_\phi, -\omega_\phi)$.

Exothermic dark matter does something different: it does not reweight the band, it moves it. A particle that arrives in an excited state and leaves in the lower one, with $\Delta_\chi < 0$, adds $|\Delta_\chi|$ to every $\omega_{\boldsymbol{q}}$ of Eq. (1.22). The band of allowed $(\boldsymbol{q}, \omega)$ shifts up by $|\Delta_\chi|$, reaches $q \to 0$ at $\omega = |\Delta_\chi|$, and there looks like the absorption of a boson of mass $|\Delta_\chi|$, with $\boldsymbol{q}$ set by the velocity rather than fixed at $m_\phi \boldsymbol{v}$ (Fig. 8.3). Since the energy now comes from the dark matter, the target needs no temperature, and the response is sampled at positive ω even for the slowest particles. An endothermic splitting, $\Delta_\chi > 0$, shifts the band down, and part of it can then be reached only with energy taken from a warm target.

Detailed balance ties the two directions of each transition, for the dark matter as well as for the target. Let $\boldsymbol{p}$ and $\boldsymbol{p}'$ be two dark-matter states, of any masses, and $\omega = E(\boldsymbol{p}) - E(\boldsymbol{p}')$ the energy that the transition $\boldsymbol{p} \to \boldsymbol{p}'$ deposits. Then, for unpolarized dark-matter states of equal spin,

$$\varGamma(\boldsymbol{p}' \to \boldsymbol{p}) = \mathrm{e}^{-\beta\omega}\, \varGamma(\boldsymbol{p} \to \boldsymbol{p}'), \tag{8.20}$$

for the rate into one final momentum (Appendix D.25). The reverse transition is driven by the conjugated interaction, which exchanges $\mathfrak{D}_{ab}$ for $\mathfrak{D}_{ba}$ and O_b for $O_b^\dagger$, and the two exchanges cancel in the contraction; what remains is the ratio of the Gibbs weights of the two target states. The down-scattering of an exothermic particle, $\chi_2 \to \chi_1$, and the up-scattering $\chi_1 \to \chi_2$ that a warm target can drive are therefore one process read in two directions. A population of dark matter in contact with the target for long enough would relax to the temperature of the target.

8.2.5 Conditions for off-diagonal terms

The off-diagonal terms of the contraction are the part of a rate that no single-operator calculation contains. They need two things at once. The dark-matter side must couple two rows with a nonzero product $\mathfrak{D}_{ab}$, and the target must correlate the same two rows, $S_{ab} \neq 0$. Each factor vanishes for a reason of symmetry in the commonest cases. On the dark-matter side, a coefficient linear in $\boldsymbol{S}_\chi$ never interferes with one that is independent of it (Sect. 8.1). On the target side, the density and the spin density of a target without magnetic order are uncorrelated, since $\mathrm{Tr}\,\boldsymbol{\sigma} = 0$. The worked example takes a target with magnetic order and a coupling that survives the trace over the dark-matter spin.

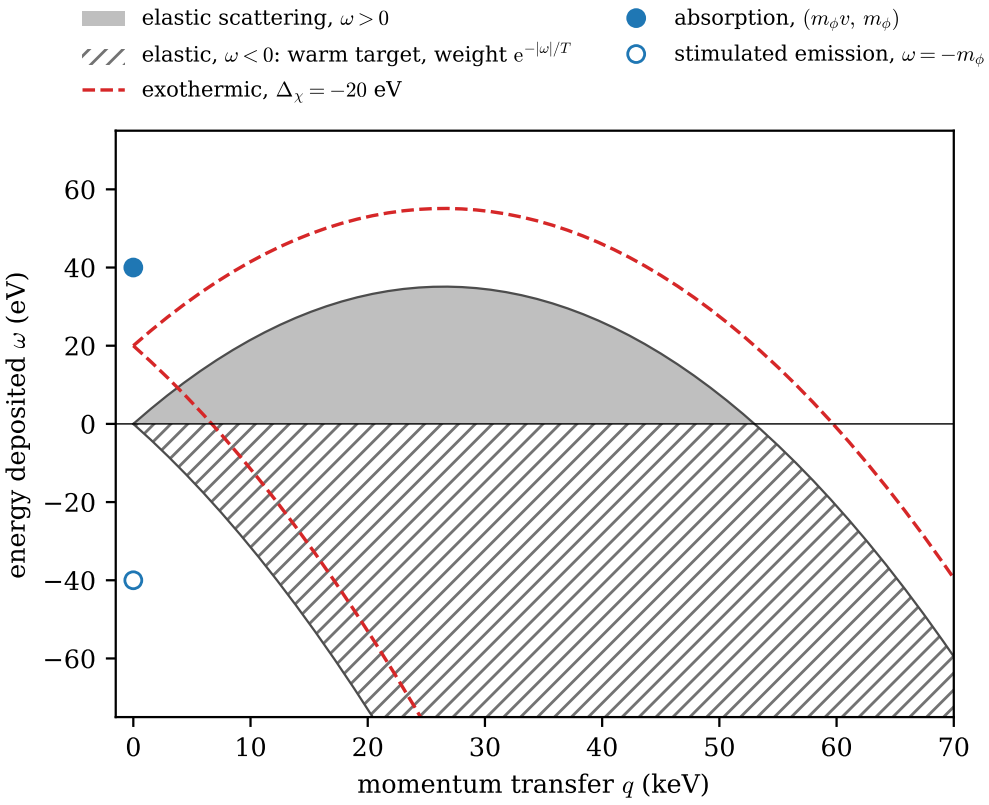


Figure 8.3: Where the master rate formula samples the response, in the plane of momentum transfer q and energy deposited ω, for $m_\chi = 10\,\mathrm{MeV}$ and speeds up to $v_{\max} = 2.650 \times 10^{-3}$. Elastic scattering reaches the band $-qv_{\max} - q^2/2m_\chi \le \omega \le qv_{\max} - q^2/2m_\chi$, the extreme values of $\omega_{\boldsymbol{q}}$ over all directions of $\boldsymbol{v}$. Its upper part (gray, $\omega > 0$, maximum 35 eV at $q = m_\chi v_{\max} = 26.5\,\mathrm{keV}$) is open at any temperature. Its lower part (hatched, $\omega < 0$) is open only for a target at $T > 0$, with weight $e^{-|\omega|/T}$ relative to the reverse transfer. Exothermic dark matter with $\Delta_\chi = -20\,\mathrm{eV}$ shifts the band up by $|\Delta_\chi|$ (dashed). Absorption of a boson of mass $m_\phi = 40\,\mathrm{eV}$ is a point at $q = m_\phi v \approx 40\,\mathrm{meV}$, $\omega = m_\phi$ (filled circle); stimulated emission into the occupied field is the point $\omega = -m_\phi$ (open circle). Fermionic absorption with a massless outgoing fermion, $\omega = m_\chi - q$, is a line at $q \approx m_\chi$, far to the right.

Worked example 8.3: Density and spin couplings in a polarized electron gas

Take the free electron gas at the valence density of silicon, $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$, fully polarized along $\hat{\boldsymbol{z}}$ by a uniform exchange field strong enough to lift the minority band above the Fermi level, a toy half-metal. All electrons have $S_z = +\frac{1}{2}$ and fill one Fermi sphere of radius $k_{\mathrm{F}\uparrow} = 2^{1/3} k_{\mathrm{F}} = 4.50\,\mathrm{keV}$, with $v_{\mathrm{F}\uparrow} = 8.80 \times 10^{-3}$ and $E_{\mathrm{F}\uparrow} = 19.8\,\mathrm{eV}$; the exchange splitting must exceed $E_{\mathrm{F}\uparrow}$. We work at $T = 0$ with two rows, the density n_e and the spin density s_e^z of Eq. (8.3).

The Hamiltonian conserves the spin of each electron, so every state that n_e or s_e^z reaches from the ground state has all spins up, and on those states $s_e^z(\boldsymbol{q}) = \frac{1}{2} n_e(\boldsymbol{q})$ exactly. The response matrix is then

$$\begin{pmatrix} S_{nn} & S_{ns} \\ S_{sn} & S_{ss} \end{pmatrix} = S_\uparrow(\boldsymbol{q}, \omega) \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} \end{pmatrix},$$

with $S_\uparrow$ the structure factor of one spin, half of Eq. (4.22) with $k_{\mathrm{F}} \to k_{\mathrm{F}\uparrow}$. Its determinant vanishes: the two rows are one operator, and the matrix saturates the bound Eq. (4.20). At $q = q_0 = 3.73\,\mathrm{keV}$ and $\omega = 5\,\mathrm{eV}$ we are in the linear region of the sphere, below $q v_{\mathrm{F}\uparrow} - q^2/2m_e = 19.2\,\mathrm{eV}$, and $S_\uparrow = m_e^2 \omega / 2\pi q = 5.57 \times 10^7\,\mathrm{eV}^2$, half of the unpolarized gas at the same density.

For potentials $\tilde{V}_n$ and $\tilde{V}_s$ on the two rows, real and without dark-matter spin, the contraction is

$$\tilde{V}_n^2 S_{nn} + 2\tilde{V}_n \tilde{V}_s S_{ns} + \tilde{V}_s^2 S_{ss} = S_\uparrow \big(\tilde{V}_n + \tfrac{1}{2}\tilde{V}_s\big)^2 .$$

Every electron carries $S_z = \frac{1}{2}$, so a coupling to the spin acts as an extra density potential and adds to it in the amplitude. The sum of the two single-operator rates is $S_\uparrow(\tilde{V}_n^2 + \tilde{V}_s^2/4)$. It misses the term $\tilde{V}_n \tilde{V}_s S_\uparrow$, which is first order in the weaker coupling, and at $\tilde{V}_s = -2\tilde{V}_n$, where the true rate vanishes, it is wrong by the whole of $2\tilde{V}_n^2 S_\uparrow$.

Which models produce such a $\tilde{V}_s$? The standard pair $\mathcal{O}_1$ and $\mathcal{O}_4 = \boldsymbol{S}_\chi \cdot \boldsymbol{S}_e$ does not: $\tilde{V}_s = c_4^{(e)} \boldsymbol{S}_\chi$, and the trace over the unpolarized dark-matter spin removes the cross term. A spin coupling free of $\boldsymbol{S}_\chi$ is $\mathcal{O}_7 = \boldsymbol{S}_e \cdot \boldsymbol{v}^\perp$, which reduces to $\bar{\boldsymbol{v}}_\chi \cdot \boldsymbol{s}_e - \mathcal{J}_e^{\alpha\alpha}$ (Table 8.1). No standard model produces $\mathcal{O}_7$ at leading order (Sect. 8.1); we take it as the simplest spin coupling that survives the trace. Take it together with $\mathcal{O}_1$, with coefficients c_1 and c_7. The rows s_e^x, s_e^y, $\mathcal{J}_e^{xx}$ and $\mathcal{J}_e^{yy}$ flip the spin and reach states orthogonal to those of n_e, so the terms of first order in c_7 are

$$2 c_1 c_7 \big[\bar{v}_{\chi,z}\, S_{ns} - S_{n\mathcal{J}}\big], \qquad S_{n\mathcal{J}} = \tfrac{1}{2} \Big\langle \frac{(2\boldsymbol{k} + \boldsymbol{q})_z}{2m_e} \Big\rangle S_\uparrow ,$$

since the element of $\mathcal{J}_e^{zz}$ between $\boldsymbol{k}$ and $\boldsymbol{k} + \boldsymbol{q}$ is $\frac{1}{2}(2\boldsymbol{k} + \boldsymbol{q})_z / 2m_e$, and the brackets average over the pairs at this $(\boldsymbol{q}, \omega)$. Since the pairs fill an annulus symmetric about $\boldsymbol{q}$ (Worked example 8.2), the transverse part of $\boldsymbol{k}$ averages away, and $(2\boldsymbol{k} + \boldsymbol{q}) \cdot \hat{\boldsymbol{q}} / 2m_e = \omega/q$ for every pair. As a result, $S_{n\mathcal{J}} = \frac{1}{2}(\omega/q) \hat{q}_z S_\uparrow$. On the dark-matter side, $\hat{\boldsymbol{q}} \cdot \bar{\boldsymbol{v}}_\chi = \omega/q$ for elastic scattering, so $\bar{v}_{\chi,z} = (\omega/q)\hat{q}_z + \boldsymbol{v}_{\mathrm{T}} \cdot \hat{\boldsymbol{z}}$, with $\boldsymbol{v}_{\mathrm{T}}$ the part of $\boldsymbol{v}$ transverse to $\boldsymbol{q}$. The longitudinal parts cancel, as Eq. (8.5) requires, and the integrand of Eq. (8.15) becomes

$$c_1^2\, S_\uparrow + c_1 c_7\, (\boldsymbol{v}_{\mathrm{T}} \cdot \hat{\boldsymbol{z}})\, S_\uparrow + O(c_7^2).$$

For the dark matter of Worked example 8.2, $m_\chi = 100\,\mathrm{MeV}$ and $v = 2.0 \times 10^{-3}$, $v_{\min} = 1.36 \times 10^{-3}$ and $v_{\mathrm{T}} = 1.47 \times 10^{-3}$. With $\boldsymbol{v}_{\mathrm{T}}$ along the magnetization, the cross term is $1.5 \times 10^{-3}\, c_7/c_1$ of the $\mathcal{O}_1$ rate. The rate of $\mathcal{O}_7$ alone is second order in c_7 and quadratic in the velocities. Its squared matrix elements, summed over the final spin, are

$|\boldsymbol{v}^\perp|^2/4 \le (v_{\max} + v_{\mathrm{F}\uparrow} + q/2m_e)^2/4 = 5.7 \times 10^{-5}$, and its structure factors are at most $S_\uparrow$ and $m_e k_{\mathrm{F}\uparrow}^2/4\pi q = 4.0\, S_\uparrow$, the second for spin flips into the empty minority band, which an exchange splitting above $E_{\mathrm{F}\uparrow} + \omega$ closes; the bound is generous. At $c_7 = c_1$ the cross term is therefore at least five times the rate of $\mathcal{O}_7$ alone.
The cross term is odd in $\boldsymbol{v}_\mathrm{T}$. At fixed $\boldsymbol{q}$ the halo integral runs over the velocities with $\hat{\boldsymbol{q}} \cdot \boldsymbol{v} = v_{\min}$, and for the laboratory Maxwellian $f_\chi \propto \mathrm{e}^{-|\boldsymbol{v}+\boldsymbol{v}_\mathrm{E}|^2/v_0^2}$ of Appendix A.8 the exponent splits into a longitudinal and a transverse Gaussian. Apart from the escape cut, the transverse Gaussian is centered on $-\boldsymbol{v}_{\mathrm{E,T}}$, the part of $-\boldsymbol{v}_\mathrm{E}$ transverse to $\boldsymbol{q}$, and the averaged cross term is $-c_1 c_7\, (\boldsymbol{v}_{\mathrm{E,T}} \cdot \hat{\boldsymbol{z}})\, S_\uparrow$, at most $8.4 \times 10^{-4}\, c_7/c_1$ of the $\mathcal{O}_1$ rate for $|\boldsymbol{v}_\mathrm{E}| = 250.5\,\mathrm{km\,s^{-1}}$. It vanishes for an isotropic halo, follows the angle between the magnetization and the dark-matter wind, and changes sign when the magnetization is reversed.
A partially polarized gas has two spheres and $S_{ns} = \frac{1}{2}(S_\uparrow - S_\downarrow)$. In the linear region, however, the structure factor of one spin, $m_e^2\omega/2\pi q$, does not depend on the radius of its sphere, and S_{ns} vanishes there even though the gas is magnetized. At a polarization of 0.5 and $q = q_0$, $S_{ns} = 0$ at 5 eV; at 10 eV, where the minority sphere has left its linear region (its edge is at 7.07 eV), $S_{ns}/S_{nn} = 0.068$.

The example shows the general pattern. A cross term needs a coupling and a target that break the same symmetry, here time reversal, and when both do, it enters the rate at first order in the weaker coupling, where each single-operator rate enters at second order. How large it can be is fixed by positivity, and the polarized gas reaches the limit; Sect. 8.3 derives the bound together with the other relations that hold for every target.

8.3 Exact relations

A computed response of silicon can be wrong in ways that no convergence test reveals. Take an electron–nucleus interference spectrum that keeps one sign, an energy-weighted spectrum that counts thirteen electrons per atom, or a coupling proportional to mass that excites optical phonons at vanishing momentum transfer. Each is an error, whatever the method and however well converged, because each violates a relation that holds for every target. Sects. 4.2 and 4.3 derived the first of these relations: detailed balance, the positivity of the response matrix, the fluctuation–dissipation theorem, the Kramers–Kronig relations and the f-sum rule of every species. The multi-operator rate of Sect. 8.2 needs three more: an interference bound, higher moments and a Ward identity. Positivity limits how strongly two couplings can interfere. The higher frequency moments of the response matrix are fixed by commutators, and the third moment of the electron–nucleus interference turns out to be set by the electron density at the nuclei. Translation invariance links the densities of all species to the momentum density, and as a result a coupling proportional to mass can only make sound at long wavelengths. None of these relations depends on an approximation to the target, so a response that violates one is wrong, not merely truncated, and Table 8.2 lists the sections that use them as checks.

8.3.1 Interference of two couplings

In the polarized electron gas of Worked example 8.3 the rate of a density and a spin coupling together was $S_\uparrow(\tilde{V}_n + \frac{1}{2}\tilde{V}_s)^2$, which can be anything from zero to twice the sum of the two single-operator rates. How far can a cross term go in general? Split the rows of Eq. (8.15) into two groups, A and B, for instance the electron rows and the nuclear rows of a dark photon,

or the density and the current rows of Sect. 8.1. The rate is then

$$\Gamma = \Gamma_A + \Gamma_{AB} + \Gamma_B,$$

where Γ_A and Γ_B keep only the terms with both indices in one group, the rates of each group alone, and the cross term Γ_{AB} collects the terms with one index in each, $2\,\mathrm{Re}\sum_{a\in A,b\in B}\mathfrak{D}_{ab}S_{ab}$ under the integral.

The bound follows from a family of models. Multiply every coupling of group B by a real number λ. Each element $\mathfrak{D}_{ab}$ is multiplied by λ once for every index in B, which is $\mathfrak{D}$ multiplied on both sides by a real diagonal matrix, so the tensor is still Hermitian and positive semidefinite. Its contraction with the positive semidefinite S_{ab} of Eq. (4.20) is therefore never negative, at every $(\boldsymbol{q},\omega)$ and for every velocity. So is the rate after any integration over $\boldsymbol{q}$, ω and the halo:

$$\Gamma(\lambda) = \Gamma_A + \lambda\,\Gamma_{AB} + \lambda^2\,\Gamma_B \geq 0 \qquad \text{for every real } \lambda.$$

A quadratic that never goes negative has a discriminant that is not positive, $\Gamma_{AB}^2 \leq 4\Gamma_A\Gamma_B$.

Interference bound

For any two groups of rows, any target and any dark-matter model,

$$|\Gamma_{AB}| \leq 2\,(\Gamma_A\Gamma_B)^{1/2}, \qquad \text{equivalently} \qquad \left(\Gamma_A^{1/2}-\Gamma_B^{1/2}\right)^2 \leq \Gamma \leq \left(\Gamma_A^{1/2}+\Gamma_B^{1/2}\right)^2. \tag{8.21}$$

The cross term is at most twice the geometric mean of the separate rates. The bound holds pointwise in $(\boldsymbol{q},\omega)$, for the spectrum $\mathrm{d}\mathcal{R}/\mathrm{d}\omega$ in any bin, and for the total rate.

For one row in each group the pointwise statement is the Cauchy–Schwarz inequality $|S_{ab}|^2 \leq S_{aa}S_{bb}$ of Eq. (4.20) combined with $|\mathfrak{D}_{ab}|^2 \leq \mathfrak{D}_{aa}\mathfrak{D}_{bb}$ from Eq. (8.7), and the integrated statement then follows from the Cauchy–Schwarz inequality for integrals, $\int(xy)^{1/2} \leq (\int x \int y)^{1/2}$. The argument through λ gives both at once, and for blocks of any size.

The bound is reached when both matrices have rank one on the two groups and their phases line up, as in the polarized gas: there $S_{ns}^2 = S_{nn}S_{ss}$, and at $\tilde{V}_s = -2\tilde{V}_n$ the rate sits on the lower edge of Eq. (8.21), zero. The bound is not an estimate. For an unpolarized target and couplings that are blind to the dark-matter spin, the cross terms between density and spin vanish by symmetry (Sect. 8.2.5), far inside the band. Its use is to put a rigorous band on a rate whose interference term has not been computed: writing $r = \Gamma_B/\Gamma_A$, the total lies between $(\sqrt{r}-1)^2$ and $(\sqrt{r}+1)^2$ times Γ_A, which is how Sect. 10.4 treats the mixed density–current blocks of the electronic response. The same inequality bounds the Migdal interference of Chap. 12 and Sect. 17.4, $|S_{e\kappa}|^2 \leq S_{ee}S_{\kappa\kappa}$, and enters the nuisance parameters of Sect. 8.8 as a constraint.

8.3.2 Moment matrices

The f-sum rule fixed the first frequency moment of every density by a double commutator (Eqs. (4.28) and (4.34)). Higher moments weight the spectrum toward high energies, and they are where an approximation to the core electrons or to the electron–nucleus interaction shows up. We collect them into matrices,

$$\mathscr{M}_{ab}^{k}(\boldsymbol{q}) \equiv \int_{-\infty}^{\infty}\frac{\mathrm{d}\omega}{2\pi}\,\omega^k\left[S_{ab}(\boldsymbol{q},\omega) + S_{ba}(-\boldsymbol{q},\omega)\right] = \frac{1}{\mathscr{V}}\Big\langle\left[O_a(\boldsymbol{q})^\dagger,\,[H,O_b(\boldsymbol{q})]_k\right]_{\mp}\Big\rangle, \tag{8.22}$$

for operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$ and any stationary state, where $[H,O]_k$ is O commuted k times with H, as in Sect. 4.3, and $[X,Y]_{\mp}$ is the commutator $XY - YX$ for odd k and the

anticommutator $XY + YX$ for even k. The derivation of Eq. (4.27) goes through unchanged. The term $O_a^\dagger [H, O_b]_k$ gives the k-th moment of $VS_{ab}(\boldsymbol{q}, \omega)/2\pi$, the term $[H, O_b]_k O_a^\dagger$ gives $(-1)^k$ times that of $VS_{ba}(-\boldsymbol{q}, \omega)/2\pi$, and the sign of the bracket makes the two add for every k. For $k = 1$ this is Eq. (4.28), so $\mathscr{M}^1$ is the first-moment matrix of Sect. 4.3. Since S_{ab} has the units of eV^2 for densities, $\mathscr{M}^k$ has those of eV^{k+3}.

The odd and the even moments carry different information. An odd moment is a commutator. It needs only the algebra of H and equal-time averages of simpler operators, and for the densities of electrons and nuclei the first one is a number, Eq. (4.34). An even moment is an equal-time correlation function. On the diagonal, with $X = [H, O_a]_{k/2}$, it reads $\mathscr{V}^{-1}\langle X^\dagger X + X X^\dagger \rangle$; for instance $\mathscr{M}^0_{ee} = 2\boldsymbol{n}_e S^{\mathrm{st}}$ for a target with inversion symmetry, by Eq. (4.17). It depends on the state, its temperature and its correlations, but no dynamics is needed to compute it, which is why the moments of the nuclear self response can be taken from a static path-integral simulation (Appendix E.19). In the same way as for $k = 1$ in Sect. 4.3, an odd $\mathscr{M}^k$ is the coefficient of $z^{-(k+1)}$ in the expansion of $R_{ab}(\boldsymbol{q}, z)$ at large z (Appendix D.26).

Positivity constrains the moments among themselves. The spectral weight of one row, summed over $\pm\boldsymbol{q}$, is never negative. It defines a positive measure on the real line, and its moments $\mathscr{M}^k_{aa}$ form a Hankel matrix, with elements $\mathscr{M}^{i+j}_{aa}$, that is positive semidefinite:

$$\sum_{i,j=0}^{n} x_i\, x_j\, \mathscr{M}^{i+j}_{aa} = \int \frac{\mathrm{d}\omega}{2\pi} \Big(\sum_i x_i \omega^i \Big)^2 \big[S_{aa}(\boldsymbol{q}, \omega) + S_{aa}(-\boldsymbol{q}, \omega) \big] \geq 0. \tag{8.23}$$

The smallest case is $(\mathscr{M}^1_{aa})^2 \leq \mathscr{M}^0_{aa} \mathscr{M}^2_{aa}$. At $T = 0$ the measure lives on $\omega \geq 0$, and the shifted matrix $\mathscr{M}^{i+j+1}_{aa}$ is positive semidefinite too, which gives for instance $(\mathscr{M}^2_{aa})^2 \leq \mathscr{M}^1_{aa} \mathscr{M}^3_{aa}$. A spectrum is determined, within a band that these inequalities make precise, by its first few moments, the Markov–Krein band of Sect. 11.5.

How many moments exist? For electrons and point nuclei with Coulomb interactions, $\mathscr{M}^3$ is finite and $\mathscr{M}^4$ is not. The fourth moment of a density is the equal-time average of $|[H, [H, n_s]]|^2$, and $[H, [H, n_s]]$ contains the force on each particle (Appendix D.26). The mean square of the Coulomb force between an electron and a nucleus, $\int \mathrm{d}^3 y\, \boldsymbol{n}(\boldsymbol{y})\, (Z_\kappa \alpha / y^2)^2$ with $\boldsymbol{n}$ finite at the nucleus, diverges as $\int \mathrm{d}y / y^2$. The spectrum shows the same thing at large ω. The oscillator strength of the hydrogen continuum falls as $\omega^{-7/2}$ (Worked example 4.5), and at small q, where $S \propto q^2\, \omega^{-1} \mathrm{d}f^{\mathrm{osc}}/\mathrm{d}\omega$, the structure factor falls as $\omega^{-9/2}$: $\omega^3 S$ is integrable and $\omega^4 S$ is not. In the Born–Oppenheimer approximation, the order $n_B = 0$ of the expansion in the derivative coupling of Sect. 8.5, the nuclei move on a smooth energy surface, and every moment of their self response is finite. This is the setting in which the five moments $k = 0, \ldots, 4$ of Appendix E.19 are used.

8.3.3 The mixed third moment

The first moment of the electron–nucleus interference vanishes, Eq. (4.34), and at $T = 0$ the real part of $S_{e\kappa}$ must change sign along the energy axis. Which sign wins at high energy, where the Migdal effect and core excitations live? The third moment weights the spectrum by ω^3 and answers the question.

The calculation takes three steps, written out in Appendix D.26. First, in a stationary state the average of $[X, [H, Y]]$ equals that of $[[X, H], Y]$, since the average of a commutator with H vanishes; this moves one commutator onto the electron density, $\mathscr{M}^3_{e\kappa} = \mathscr{V}^{-1}\langle [[n_e^\dagger, H], [H, [H, n_\kappa]]] \rangle$. Second, the continuity equations of the two species, Eq. (4.29) and its nuclear analogue, turn the two inner commutators into currents, $[n_e(\boldsymbol{q})^\dagger, H] = \boldsymbol{q} \cdot \boldsymbol{j}_e(\boldsymbol{q})^\dagger$ and $[H, n_\kappa(\boldsymbol{q})] = \boldsymbol{q} \cdot \boldsymbol{j}_\kappa(\boldsymbol{q})$. Third, the commutator of H with the nuclear current contains

the kinetic energy of the nuclei and the gradient of the potential energy at each nucleus, $\mathrm{i}M_\kappa^{-1}\sum_{I\in\kappa}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}\,\boldsymbol{q}\cdot\nabla_I U$, with U the potential energy. The electron current commutes with every nuclear operator. It fails to commute only with the electron positions inside the force, and only the electron–nucleus Coulomb energy $\sum_{j,I} w_\kappa(\boldsymbol{r}_j-\boldsymbol{X}_I)$, $w_\kappa(\boldsymbol{y})=-Z_\kappa\alpha/y$, depends on both. The electron–electron and nucleus–nucleus interactions, and every kinetic term, drop out. What is left involves one function, the mean electron density $n_e^{(\kappa)}(\boldsymbol{y})$ at the displacement $\boldsymbol{y}$ from a nucleus of species κ, averaged over the nuclei of that species and over their motion. Its value at $\boldsymbol{y}=0$ is the contact density $n_e^{(\kappa)}$, the electron density at the nucleus, and far from the nucleus it averages to the mean density n_e.

Mixed electron–nucleus third moment

For point nuclei, nonrelativistic electrons and local potentials, at any $\boldsymbol{q}$, temperature and coupling strength,

$$\mathscr{M}^3_{e\kappa}(\boldsymbol{q})=-\frac{n_\kappa}{m_e M_\kappa}\int \mathrm{d}^3y\; n_e^{(\kappa)}(\boldsymbol{y})\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{y}}\,(\boldsymbol{q}\cdot\nabla)^2 w_\kappa(\boldsymbol{y}). \tag{8.24}$$

At sites of cubic symmetry, for q small compared with the shortest reciprocal-lattice vector $G_{\min}$ but large compared with the inverse size of the sample,

$$\mathscr{M}^3_{e\kappa}(\boldsymbol{q})=-\frac{Z_\kappa e^2 q^2\, n_\kappa}{3\, m_e M_\kappa}\big[n_e^{(\kappa)}+2n_e\big]. \tag{8.25}$$

The high-frequency interference of electrons and nuclei is negative, and its weight is fixed by the electron density at the nuclei.

We derive this mixed rule here; the third moment of a single species is classical, and Puff wrote it for an interacting Bose fluid [13]. The two terms of the bracket have different origins, which the identity

$$(\boldsymbol{q}\cdot\nabla)^2\frac{1}{y}=-\frac{4\pi}{3}\,q^2\,\delta^3(\boldsymbol{y})+q^2\,\frac{3(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{y}})^2-1}{y^3}$$

makes visible. The contact part picks out the density at the nucleus and gives $n_e^{(\kappa)}/3$ at every q, with the phase $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{y}}$ equal to one on its support. The quadrupolar part averages to zero against any density with cubic symmetry about the nucleus that varies on scales short compared with $1/q$, the core and the valence bonds alike. Against the uniform part of the density, however, it does not vanish. That integral is dominated by distances $y\sim 1/q$, where the phase matters, and since the kernel scales as $1/y^3$ it is independent of q; it contributes $2n_e/3$. It is the Coulomb field of the bare nucleus acting on the electrons of the whole crystal at distances $\sim 1/q$. The order of limits is the physical one: the sample is infinite at fixed q, and only then is q small.

The sign has a simple reading. For an electron bound to a nucleus by a spring of constant K, Eq. (8.24) gives $\mathscr{M}^3_{e\kappa}=-Kq^2/(m_eM_\kappa\mathscr{V})$ at small q. The relative coordinate makes the sign obvious: in the mode in which the two particles oscillate against each other, at $\omega^2=K(m_e^{-1}+M_\kappa^{-1})$, a displacement of one is accompanied by an opposite displacement of the other (Appendix D.26). The Coulomb attraction is a spring whose stiffness, the curvature $\nabla^2 w_\kappa=Z_\kappa e^2\delta^3(\boldsymbol{y})$, sits at the nucleus, and the contact density measures how much of the electron cloud feels it. Together with the vanishing first moment, the negative third moment says that the interference is positive at low energies, where the electrons follow the nucleus (Sect. 8.6), and negative at high energies, where they are left behind. The positive part sits in the phonon final states, on the acoustic branch, at energies that vanish with q, and on the optical phonons, wherever the electrons move with the nuclei (Sect. 8.6). Every electronic final

state carries the negative interference $-(m_e/M_\kappa)|\langle f|n_e|0\rangle|^2$ at small q, as Eq. (8.32) below shows.

Eq. (8.25) has a limited range in q, and three regimes follow from the exact form, which Appendix D.26 writes as a sum over reciprocal-lattice vectors. For $q \ll G_{\min}$, Eq. (8.25) holds. Its real part is corrected at relative order $(q/G_{\min})^2$ at every cubic site. At sites without inversion symmetry, such as the tetrahedral sites of silicon, there is also a correction of first order in $q/G_{\min}$. It is purely imaginary. It comes from the part of the density odd about the nucleus, whose Fourier components are imaginary, and it enters $\operatorname{Im}\mathscr{M}^3_{e\kappa}$ alone. In silicon $G_{\min} = |\boldsymbol{G}_{111}| = 3.95\,\mathrm{keV}$ (Sect. 3.1). Once q passes $G_{\min}$ the valence structure starts to matter, and the bracket depends on q, although the contact term still dominates. Once q reaches the inverse size $1/a_{\mathrm{core}}$ of the innermost shell, the core itself is resolved: the quadrupolar kernel then acts on the density near the nucleus as on a uniform one, and the bracket climbs from $n_e^{(\kappa)}/3$ toward $n_e^{(\kappa)}$. For a density with spherical symmetry about the nucleus, which is exact for a free atom and a good approximation for the core in a cubic crystal, the angular integral of Eq. (8.24) can be done once and for all (Appendix D.26):

$$\mathscr{M}^3_{e\kappa}(\boldsymbol{q}) = -\frac{Z_\kappa e^2 q^2\, n_\kappa}{m_e M_\kappa}\Big[\frac{n_e^{(\kappa)}}{3} + 2\int_0^\infty \mathrm{d}y\, \frac{\bar{n}_e^{(\kappa)}(y)\, j_2(qy)}{y}\Big], \tag{8.26}$$

with $\bar{n}_e^{(\kappa)}(y)$ the spherically averaged density at distance y from the nucleus and j_2 the spherical Bessel function of order two. Since $\int_0^\infty \mathrm{d}x\, j_2(x)/x = 1/3$, a density that is uniform over the range $y \sim 1/q$ contributes $2/3$ of its value through the integral. This is the origin of the $2n_e/3$ of the small-q limit, and at large q, where the whole atom looks uniform on the scale $1/q$ near the nucleus, the square bracket of Eq. (8.26) rises from $n_e^{(\kappa)}/3$ to $n_e^{(\kappa)}$.

The contact density is a material quantity in its own right, and it is computed from all-electron densities. Pseudopotential calculations replace the cores and cannot supply it; the projector-augmented-wave reconstruction of Appendix E.2 or a full-potential code restores the density near the nucleus. The spherical atomic part comes from a calculation of the free atom, and the crystal adds a small anisotropy of the valence; Appendix E.17 specifies both. Relativity changes the density at the nucleus more than anything else. For a finite nucleus it raises the contact density by about 8 % in silicon, an effect that grows roughly as $(Z\alpha)^2$ in heavier atoms and that Sect. 12.5 adds to Eq. (8.25).

Worked example 8.4: The mixed third moment of silicon

We evaluate Eq. (8.26) for silicon at $q = 10\,\mathrm{keV}$, a typical momentum transfer for the Migdal effect. For the density we take the free atom in its ground state from an all-electron Hartree–Fock calculation, nonrelativistic and with a point nucleus. Its total energy, -288.854 hartree, is the Roothaan–Hartree–Fock value quoted in Sect. 4.3 [14]. The contact density is $n_e^{(\kappa)} = 1765\, a_0^{-3} = 1.19 \times 10^{28}\,\mathrm{cm}^{-3}$, converged to 0.1 % in the Gaussian basis, of which the ten core electrons contribute $1757.7\, a_0^{-3}$ and the four valence electrons, through their 3s orbital, $7.6\, a_0^{-3}$, or 0.43 %. Against it, the mean density of all fourteen electrons, $7.00 \times 10^{23}\,\mathrm{cm}^{-3}$, is tiny: $2n_e/n_e^{(\kappa)} = 1.2 \times 10^{-4}$. In silicon the term that comes from the uniform density is a correction at the level of 10^{-4}.
At $q = 10\,\mathrm{keV} = 2.68\, a_0^{-1}$ the kernel $j_2(qy)/y$ samples distances $y \sim 1/q = 0.20\,\text{Å}$. This is above $G_{\min} = 3.95\,\mathrm{keV}$, in the middle regime, and below the inverse size $\sim 14/a_0 = 52\,\mathrm{keV}$ of the 1s shell. The bracket of Eq. (8.26) comes out at $0.3353\, n_e^{(\kappa)}$, against $1/3$ in the limit $q \to 0$: Eq. (8.25) is 0.6 % low at 10 keV. The core supplies 99.56 % of the bracket and the valence 0.44 %, almost all of it the 3s density at the nucleus. The crystal valence, which the free atom misses, enters at most through the mean density, 1.8×10^{-4} of the bracket.

As q grows the 1s shell is resolved; the bracket is $0.347\,n_e^{(\kappa)}$ at 30 keV and $0.426\,n_e^{(\kappa)}$ at 100 keV.
The mixed moment is best read against the nuclear f-sum $\mathscr{M}^1_{\kappa\kappa} = q^2 n_\kappa / M_\kappa$. Their ratio is a squared energy,

$$-\frac{\mathscr{M}^3_{e\kappa}}{\mathscr{M}^1_{\kappa\kappa}} = \frac{Z_\kappa e^2}{m_e} \times (\text{bracket}) = (8.78\,\text{keV})^2,$$

with $(8.76\,\text{keV})^2$ in the limit $q \to 0$. The scale is that of the K shell: the 1s orbital energy of the same calculation is 1.87 keV, and the 1s electrons move with kinetic energies near 2.7 keV (Sect. 4.3). The third moment of the electron–nucleus interference is a property of the core electrons of silicon, to better than a percent, and a calculation that freezes the core or replaces it by a pseudopotential gets it wrong by construction.

The diagonal third moments follow from the same algebra. For species s of mass m_s,

$$\mathscr{M}^3_{ss}(\boldsymbol{q}) = \frac{n_s q^2}{m_s}\Big[\Big(\frac{q^2}{2m_s}\Big)^2 + \frac{3q^2\langle(\hat{\boldsymbol{q}}\cdot\boldsymbol{p})^2\rangle_s}{m_s^2}\Big] + \frac{1}{m_s^2 \mathscr{V}}\Big\langle \sum_{i,j\in s} \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}_i-\boldsymbol{r}_j)}(\boldsymbol{q}\cdot\nabla_i)(\boldsymbol{q}\cdot\nabla_j)\,U\Big\rangle, \quad (8.27)$$

with $\langle(\hat{\boldsymbol{q}}\cdot\boldsymbol{p})^2\rangle_s$ the mean square of one particle's momentum along $\hat{\boldsymbol{q}}$, $2m_s\bar{K}/3$ for electrons in a cubic crystal, with the $\bar{K}$ of Eq. (4.33) (Appendix D.26). The kinetic bracket is the free recoil, broadened by the motion of the particles. The potential term measures the curvature of the energy surface on which the particles move, and for electrons in jellium it reduces to $n_e q^2 \omega_p^2 / m_e$: the third moment of the electron gas is the plasma frequency squared times the f-sum. In a crystal with mobile nuclei, however, the electron–nucleus curvature at the nuclei dominates it, and the next subsection shows that it is tied to Eq. (8.25) exactly.

8.3.4 Translation Ward identities

A dark-matter coupling that is the same per unit mass for every particle, a heavy scalar coupled to nucleons in a crystal of one element for instance, pushes every part of the crystal in the same proportion. At long wavelength the push is uniform, like a gravitational field, and by the equivalence principle it accelerates every particle equally: it can move the crystal, or at finite q launch a sound wave, but it cannot excite the crystal's internal motion. The identities that make this precise follow from translation invariance, which in quantum mechanics is the conservation of total momentum.

Add the continuity equations of all species, each weighted by its mass. The mass-density operator $n_{\mathrm{M}}(\boldsymbol{q}) = \sum_s m_s n_s(\boldsymbol{q})$, with s running over the electrons and every nuclear species, then obeys

$$[H, n_{\mathrm{M}}(\boldsymbol{q})] = \boldsymbol{q}\cdot\boldsymbol{g}(\boldsymbol{q}), \qquad \boldsymbol{g}(\boldsymbol{q}) = \sum_s m_s \boldsymbol{j}_s(\boldsymbol{q}) = \sum_i \frac{1}{2}\{\boldsymbol{p}_i,\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}\}, \quad (8.28)$$

where the last sum runs over all particles, electrons and nuclei, with their positions and momenta. The operator $\boldsymbol{g}(\boldsymbol{q})$ is the momentum density, and at $\boldsymbol{q} = 0$ it is the total momentum. Between eigenstates, $[H, X]$ becomes $\omega_{fi} X$, so $\omega_{fi}\langle f|n_{\mathrm{M}}(\boldsymbol{q})|i\rangle = \boldsymbol{q}\cdot\langle f|\boldsymbol{g}(\boldsymbol{q})|i\rangle$, and inserted into Eq. (4.13) this is a statement about every response that involves the mass density.

Translation Ward identity

For any operator O_B and $\omega \neq 0$, with S_{Bs} the response matrix of O_B and n_s,

$$\sum_s m_s\, S_{Bs}(\boldsymbol{q},\omega) = \frac{\boldsymbol{q}}{\omega} \cdot S_{B\boldsymbol{g}}(\boldsymbol{q},\omega). \tag{8.29}$$

In amplitudes, $\omega_{fi} \sum_s m_s \langle f|n_s(\boldsymbol{q})|i\rangle = \boldsymbol{q} \cdot \langle f|\boldsymbol{g}(\boldsymbol{q})|i\rangle$ for every pair of states.

The name follows field theory, where a Ward identity is a conservation law written for matrix elements. Those of quantum electrodynamics, and the density identity Eq. (7.36) of the superconductor, express the conservation of charge; the identities here express the conservation of momentum. Eq. (8.28) itself is the standard momentum-conservation identity of a fluid of several components. In the form Eq. (8.29), for the response matrix that a dark-matter rate contracts, we derive it here, under the hypothesis of the f-sum rule, that every interaction depends on positions and spins only.

So far the identity only adds up continuity equations. Its content comes from the next commutator. The total momentum $\boldsymbol{g}(\boldsymbol{0})$ commutes with H, since the crystal holds itself together and no external potential acts. For interactions between pairs of particles, the Coulomb interactions among them, $[H, \boldsymbol{g}(\boldsymbol{q})]$ is then not merely small at small q but proportional to $\boldsymbol{q}$ (Appendix D.26):

$$[H, g^\alpha(\boldsymbol{q})] = q_\beta\, \mathsf{T}^{\alpha\beta}(\boldsymbol{q}), \tag{8.30}$$

with $\mathsf{T}^{\alpha\beta}$ the momentum-flux tensor, the kinetic flux of momentum plus the stress carried by the forces between pairs. This is Newton's third law in operator form: the forces inside the crystal add up to zero, and momentum can only flow from one region to the next. Combined with Eq. (8.29), it gives each factor of $\boldsymbol{q}/\omega$ a partner.

Soft acoustic limit

For pairwise, translation-invariant interactions and $\omega \neq 0$,

$$\sum_s m_s\, S_{Bs}(\boldsymbol{q},\omega) = \frac{q_\alpha q_\beta}{\omega^2}\, S_{B\mathsf{T}^{\alpha\beta}}(\boldsymbol{q},\omega), \qquad S_{n_{\mathrm{M}} n_{\mathrm{M}}}(\boldsymbol{q},\omega) = \frac{q_\alpha q_\beta q_\gamma q_\delta}{\omega^4}\, S_{\mathsf{T}^{\alpha\beta}\mathsf{T}^{\gamma\delta}}(\boldsymbol{q},\omega). \tag{8.31}$$

At fixed $\omega \neq 0$ the response of the mass density vanishes as q^4, two powers faster than that of any single species. As $q \to 0$ its weight collects on the acoustic branches, where $\omega = c_\nu q$ vanishes with q.

On the acoustic branch $q^4/\omega^4 = c_\nu^{-4}$ stays finite, and the mass density makes sound. Everywhere else, at an optical phonon, a pair of phonons with a finite total energy, or an electronic excitation, its amplitude is $O(q^2)$, against $O(q)$ for the density of one species. Campbell-Deem and collaborators found this scaling in a crystal with a coupling proportional to mass: the single optical phonon, the two-phonon final states and the anharmonic channel have structure factors $\propto q^4$, and the single acoustic phonon $\propto q$ [15, Table I and Sect. V B]. Eq. (8.31) shows that the scaling is exact, anharmonic terms included. In the harmonic lattice it is the acoustic sum rule Eq. (5.18): a rigid displacement exerts no force, which is the statement that the potential part of $[H, \boldsymbol{g}(\boldsymbol{q})]$ vanishes at $\boldsymbol{q} = 0$. Sect. 13.6 applies the same identity to the amplitude of phonon-assisted electron scattering, where it fixes the limit of a soft acoustic phonon.

The identity sorts any coupling to the densities. Write $\sum_s c_s n_s = \lambda n_{\mathrm{M}} + \sum_s (c_s - \lambda m_s) n_s$ for any λ. The first term reaches internal excitations at order q^2 alone. At long wavelengths,

as a result, only the part of a coupling that is not proportional to mass shakes the inside of the crystal. A coupling to nucleon number differs from one to mass only by the electron masses and the nuclear binding energies, and for it the optical phonons of a polar crystal are suppressed accordingly at small q.

The electron masses give the most instructive case. In a crystal of one element, the density of all nuclei is $\sum_\kappa n_\kappa = (n_\mathrm{M} - m_e n_e)/M_\kappa$. For every final state f at a finite energy ω_{f0} above the ground state, Eq. (8.31) then makes the mass density drop out to the order that matters:

$$\Big\langle f \Big| \sum_\kappa n_\kappa(\boldsymbol{q}) \Big| 0 \Big\rangle = -\frac{m_e}{M_\kappa} \langle f | n_e(\boldsymbol{q}) | 0 \rangle + O(q^2). \tag{8.32}$$

A nucleus that is kicked excites the electrons as if they had been kicked with the momentum $m_e \boldsymbol{q}/M_\kappa$, in the opposite direction. This is the boost of the Migdal effect, Eq. (8.53) summed over all nuclei, now as an exact consequence of momentum conservation (Appendix D.26). At small q the electronic part of the nuclear response is $(m_e/M_\kappa)^2 S_{ee}$, 3.8×10^{-10} times the electron structure factor in silicon, and the interference $S_{e\kappa}$ summed over the nuclei is $-(m_e/M_\kappa) S_{ee}$ there: negative, as the third moment requires.

The third moments obey the same identity. Multiplying Eq. (8.29) by ω^3 and integrating leaves one commutator with $n_e^\dagger$ acting on $q_\alpha q_\beta \mathsf{T}^{\alpha\beta}$, so $m_e \mathscr{M}^3_{ee} + \sum_\kappa M_\kappa \mathscr{M}^3_{e\kappa}$ is $O(q^3)$ at least. The kinetic part of Eq. (8.27) is of order q^4, and the terms of order q^2 must cancel between the electron self moment and the mixed moments. They do (Appendix D.26). The potential term of Eq. (8.27) for electrons is, at small q and cubic sites,

$$\mathscr{M}^3_{ee}(\boldsymbol{q}) \to \frac{e^2 q^2}{3 m_e^2} \Big[\sum_\kappa Z_\kappa \boldsymbol{n}_\kappa \boldsymbol{n}_e^{(\kappa)} + 2 \boldsymbol{n}_e^2 \Big], \tag{8.33}$$

which by charge neutrality, $\sum_\kappa Z_\kappa \boldsymbol{n}_\kappa = \boldsymbol{n}_e$, is exactly $-\sum_\kappa (M_\kappa/m_e)$ times Eq. (8.25). The electron self moment carries the same contact densities. In jellium, where the nuclei are replaced by a fixed uniform background and translation invariance is lost, the contact density is replaced by $\boldsymbol{n}_e$, and the bracket $3\boldsymbol{n}_e^2$ returns Puff's $\boldsymbol{n}_e q^2 \omega_p^2/m_e$. For silicon at small q the potential term divided by the f-sum is $e^2[\boldsymbol{n}_e^{(\kappa)} + 2\boldsymbol{n}_e]/3m_e = (2.34\,\mathrm{keV})^2$, far above the kinetic terms $(q^2/2m_e)^2 = (98\,\mathrm{eV})^2$ and $2q^2\bar{K}/m_e = (469\,\mathrm{eV})^2$ at $q = 10\,\mathrm{keV}$.

Worked example 8.5: The Ward identity of a harmonic chain

Take a chain of identical nuclei of mass M, joined by springs of constant K at spacing d, with positions $x_l = ld + u_l$, momenta P_l and periodic boundary conditions over N sites. Its single branch is $\omega_k = 2(K/M)^{1/2} |\sin(kd/2)|$. For numbers we use the longitudinal [100] chain of silicon planes of Sect. 5.2: $M = 26.16\,\mathrm{GeV}$, $d = a/4 = 1.358\,\text{Å}$ and a maximum frequency $2(K/M)^{1/2} = 64.1\,\mathrm{meV}$, which give the sound speed $c = d(K/M)^{1/2} = 6.61\,\mathrm{km\,s^{-1}}$. The two operators of Eq. (8.29) are $n(q) = \sum_l \mathrm{e}^{\mathrm{i}qx_l}$ and $g(q) = \sum_l \frac{1}{2}\{P_l, \mathrm{e}^{\mathrm{i}qx_l}\}$, and the identity reads $M\omega_{fi}\langle f|n|i\rangle = q\langle f|g|i\rangle$.

One phonon. With $u_l = \sum_k (2NM\omega_k)^{-1/2}(a_k \mathrm{e}^{\mathrm{i}kld} + a_k^\dagger \mathrm{e}^{-\mathrm{i}kld})$, the term of first order in u_l creates the phonon $k = q$ from the ground state with $\langle 1_q|n|0\rangle = \mathrm{i}q\,(N/2M\omega_q)^{1/2}$. The momentum density to the same order is $\sum_l \mathrm{e}^{\mathrm{i}qld} P_l$, and $\langle 1_q|P_l|0\rangle = \mathrm{i}M\omega_q \langle 1_q|u_l|0\rangle$ gives $\langle 1_q|g|0\rangle = \mathrm{i}(NM\omega_q/2)^{1/2}$. The ratio is $q/M\omega_q$, as the identity demands. At $q = 1\,\mathrm{keV} = 0.507\,\text{Å}^{-1}$, $qd = 0.688$ and $\omega_q = 21.6\,\mathrm{meV}$.

Two phonons. The second-order terms are $-\frac{1}{2}q^2 \sum_l \mathrm{e}^{\mathrm{i}qld} u_l^2$ in n and $\frac{1}{2}\mathrm{i}q \sum_l \mathrm{e}^{\mathrm{i}qld}\{P_l, u_l\}$ in

g. For the pair $|k, q-k\rangle$, with $k \neq q-k$, they give

$$\langle k, q-k|n|0\rangle = -\frac{q^2}{2M(\omega_k\omega_{q-k})^{1/2}}, \qquad \langle k, q-k|g|0\rangle = -\frac{q\,(\omega_k+\omega_{q-k})}{2(\omega_k\omega_{q-k})^{1/2}},$$

and $M(\omega_k+\omega_{q-k})$ times the first is q times the second. The pair has a finite energy as $q \to 0$, near $2\omega_k$, and its amplitude is of order q^2: the soft acoustic limit, here visible term by term. Measured against the one-phonon weight, the two-phonon weight carries a factor of order $q^2/(2M\omega)$, the recoil energy over a phonon energy, 8.8×10^{-4} at 1 keV.

The flux. The forces enter through $[H, g(q)]$. Each spring pulls its two ends with opposite forces, $\pm K\Delta_l$ with $\Delta_l = u_{l+1} - u_l$, and the potential part of the commutator is $\mathrm{i}K\sum_l \Delta_l(\mathrm{e}^{\mathrm{i}qx_{l+1}} - \mathrm{e}^{\mathrm{i}qx_l})$. The difference of the phases, $\mathrm{e}^{\mathrm{i}qld}(\mathrm{e}^{\mathrm{i}qd}-1)$ at lowest order, carries the factor q of Eq. (8.30): Newton's third law makes the force on the chain as a whole vanish, the one-dimensional form of Eq. (5.18). Its one-phonon element is $4\mathrm{i}K\sin^2(qd/2)\,(N/2M\omega_q)^{1/2}$, and with $\omega_q\langle 1_q|g|0\rangle = \langle 1_q|[H,g]|0\rangle$ we recover $M\omega_q^2 = 4K\sin^2(qd/2)$, the dispersion of the chain, up to terms of relative order $q^2/(M\omega_q)$ from the kinetic part of the flux. The Ward identity, read at one phonon, is the equation of motion.

A cubic term in the springs changes none of this. It adds to the two-phonon amplitude a path through the one-phonon state $|1_q\rangle$, whose amplitude from n is of order q. The anharmonic path alone is of order q times the cubic coupling. The flux Eq. (8.30) still carries its factor q, because the cubic springs are still forces between pairs, and the total amplitude to the pair therefore remains of order q^2: the terms of order q cancel between the paths.

8.3.5 Table of exact relations

Table 8.2 collects the exact relations of Sects. 4.2 and 4.3 and of this section with their hypotheses and the places where later chapters use them. Each follows from stationarity, the positivity of probabilities, causality, or the conservation of particle number or momentum, and together they constrain the nuisance parameters with which Sect. 8.8 turns truncations into uncertainties.

None of these relations involves the coupling to the dark matter beyond the positivity of $\mathfrak{D}_{ab}$, and none involves screening. Which rows of the response matrix the medium screens, and how the Coulomb lines are counted when electrons and nuclei respond together, is the question of Sect. 8.4.

Table 8.2: Relations that hold for every target, with the hypotheses they need and the sections that use them as checks or constraints. "Stationary" means any state diagonal in the energy basis, the Gibbs state at any temperature included; "local" means that every interaction depends on positions and spins only; "translation invariant" means pairwise interactions, no external potential.

Relation	Statement	Hypotheses	Used in
Detailed balance	$S_{ab}(-\boldsymbol{q},-\omega) = \mathrm{e}^{-\beta\omega} S_{ba}(\boldsymbol{q},\omega)$, Eq. (A.34)	Gibbs state	Sects. 8.2 and 13.7
Fluctuation–dissipation	S from $\operatorname{Im} R$, Eq. (4.18)	Gibbs state	Sect. 4.6 and Appendix E.6
Kramers–Kronig	$\operatorname{Re} R$ from $\operatorname{Im} R$, Eq. (4.24)	causality	Sect. 4.4 and Appendix E.6
Positivity	S_{ab} positive semidefinite, Eq. (4.20)	stationary	Sects. 8.2 and 14.4
Interference bound	$\lvert\Gamma_{AB}\rvert \le 2(\Gamma_A\Gamma_B)^{1/2}$, Eq. (8.21)	stationary	Sects. 8.8, 10.4 and 17.4
Species f-sum	$\mathscr{M}^1_{ss'} = \delta_{ss'} q^2 \boldsymbol{n}_s / m_s$, Eq. (4.34)	stationary, local	Worked example 4.7, Sect. 8.6, and Chap. 22
Moment matrices	Eq. (8.22), Hankel positivity Eq. (8.23)	stationary	Sect. 11.5 and Appendix E.19
Mixed third moment	$\mathscr{M}^3_{e\kappa}$, Eqs. (8.24) and (8.25)	stationary, local, point nuclei	Sect. 12.5
Ward identity	$\sum_s m_s S_{Bs} = (\boldsymbol{q}/\omega)\cdot S_{B\boldsymbol{g}}$, Eq. (8.29)	stationary, local	Sects. 8.6 and 13.6
Soft acoustic limit	$S_{n_\mathrm{M} n_\mathrm{M}} = O(q^4)$ at $\omega \neq 0$, Eq. (8.31)	translation invariant	Sects. 11.3 and 13.6

8.4 Screening for every coupling

In GaAs the dielectric function vanishes at 36.35 meV, the energy of the longitudinal optical phonon (Sect. 5.5). A rate written as the unscreened rate divided by $|\varepsilon|^2$, the form Sect. 4.4 derived for a probe of the electron density, would be infinite there. For a scalar that couples to nucleons it is finite, and the phonon it excites sits at 36.35 meV rather than at the 33.62 meV of the transverse mode, because the Born charges shift the pole and leave the coupling alone (Worked example 8.6). The rule $1/|\varepsilon|^2$, and the loss function behind it, belong to couplings that see the charge. A dark photon sees the total charge of electrons and nuclei, an electrophilic mediator the electrons alone, a hadrophilic scalar the nucleons, and a spin coupling no charge at all, and each is screened differently. We will write one screened response that covers all of them and prove when the loss function gives the rate exactly. We then sort the Coulomb lines of a process that involves both electrons and phonons, so that none is counted twice, and read the in-medium mixing of a dark photon as the same construction.

8.4.1 The proper response of a channel

Electrons and nuclei interact through one thing, the Coulomb energy H_C of Eq. (2.48), which couples them only through the charge-number density $n_Z = -n_e + \sum_\kappa Z_\kappa n_\kappa$ of Sect. 2.3. Phonons, electron–phonon vertices and screening are consequences of it. A dark-matter particle that transfers $\boldsymbol{q}$ couples to the rows of Eq. (8.3), and in a crystal the rows it excites induce charge at $\boldsymbol{q}+\boldsymbol{G}$ for every reciprocal-lattice vector (Sect. 4.4), with $\boldsymbol{q}$ in the first zone and the dark-matter vertex at the row $\boldsymbol{G}_q$ of Eq. (A.13). The set of momenta $\{\boldsymbol{q}+\boldsymbol{G}\}$ is the *channel* of the external field. The Coulomb lines that carry a momentum of the channel are

the ones that screen the dark-matter vertex. All others, at $\boldsymbol{q}' \neq \boldsymbol{q}+\boldsymbol{G}$, belong to the material: they bind the electrons, make the phonons and dress the electron–phonon vertex.

Follow the argument of Sect. 4.4 with every species at once. Let the external field couple to row b with amplitude ϕ_b, and write $\delta O_a = \delta\langle O_a(\boldsymbol{q})^\dagger\rangle/\mathcal{V}$ for the induced change of row a, so that the Kubo formula Eq. (4.3) reads $\delta O_a = R_{ab}\phi_b$. The induced charge-number density at $\boldsymbol{q}+\boldsymbol{G}$, $\delta n_{Z,\boldsymbol{G}}$, produces the Hartree potential energy $v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\,\delta n_{Z,\boldsymbol{G}}$ on a unit charge, and on the target it acts as an extra field on the row $n_Z(\boldsymbol{q}+\boldsymbol{G})$. With electrons alone, $n_Z = -n_e$ and the two signs cancel, which is Eq. (4.38). The *proper response of the channel*, $\tilde{R}$, is the response to the external and Hartree fields together,

$$\delta O_a = \sum_b \tilde{R}_{ab}\,\phi_b + \sum_{\boldsymbol{G}} \tilde{R}_{aZ_{\boldsymbol{G}}}\,v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\,\delta n_{Z,\boldsymbol{G}}, \tag{8.34}$$

where the subscript $Z_{\boldsymbol{G}}$ stands for the row $n_Z(\boldsymbol{q}+\boldsymbol{G})$, so that $\tilde{R}_{aZ_{\boldsymbol{G}}} = -\tilde{R}_{ae_{\boldsymbol{G}}} + \sum_\kappa Z_\kappa \tilde{R}_{a\kappa_{\boldsymbol{G}}}$. Electrons and nuclei enter on the same footing, as two kinds of charge. In the language of Sect. 4.8, $\tilde{R}$ is the sum of the diagrams that cannot be cut by removing a single Coulomb line of the channel. By the argument of Eq. (4.97) it is also the response of the Hamiltonian with the Coulomb terms of the channel deleted, which has the same equilibrium state in the thermodynamic limit. Every exact property of Chap. 4 therefore holds for $\tilde{R}$, the Lehmann form and positivity included.

Taking $a = Z_{\boldsymbol{G}}$ in Eq. (8.34) closes the equations for the induced charge, and the result for any pair of rows is (Appendix D.27)

Screened response for any coupling

$$\begin{aligned} R_{ab}(\boldsymbol{q},\omega) &= \tilde{R}_{ab} + \sum_{\boldsymbol{G},\boldsymbol{G}'} \tilde{R}_{aZ_{\boldsymbol{G}}}\,(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'}\,v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}'|)\,\tilde{R}_{Z_{\boldsymbol{G}'}b}, \\ \varepsilon_{\boldsymbol{G}\boldsymbol{G}'} &= \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\,\tilde{R}_{Z_{\boldsymbol{G}}Z_{\boldsymbol{G}'}}, \end{aligned} \tag{8.35}$$

with every function at $(\boldsymbol{q},\omega)$. The matrix ε is the total dielectric matrix of electrons and nuclei. A row is screened through its proper correlation with the charge, $\tilde{R}_{aZ}$, and through nothing else.

Without local fields the sums reduce to $\boldsymbol{G} = 0$, and

$$R_{ab} = \tilde{R}_{ab} + v_{\mathrm{C}}(q)\,\frac{\tilde{R}_{aZ}\,\tilde{R}_{Zb}}{\varepsilon}, \qquad \varepsilon = 1 - v_{\mathrm{C}}(q)\,\tilde{R}_{ZZ}. \tag{8.36}$$

The reading is the one of Sect. 4.4, with the vertex made explicit. The dark matter acts on row b; the target answers with a proper charge $\tilde{R}_{Zb}$; its field, screened by the whole medium, $\varepsilon^{-1}v_{\mathrm{C}}$, acts back on row a through $\tilde{R}_{aZ}$. The dielectric function contains the polarization of the nuclei as well as that of the electrons, the static ε_0 of Sect. 5.5 at low frequency and ε_∞ between the phonons and the gap. With the nuclei clamped it reduces to the electronic ε of Sect. 4.4. For time-dependent density-functional theory the same algebra holds with f_{xc} inside the proper part (Sect. 4.6).

The proper cross responses are material quantities. The charge–charge block is the dielectric matrix that codes compute in the random-phase approximation with local fields or with an exchange–correlation kernel (Appendix E.6), and its ionic part at long wavelength is fixed by the Born charges and phonons of density-functional perturbation theory (Appendix E.4). The blocks $\tilde{R}_{aZ}$ for a current, spin or kinetic-energy row are computed from the same Kohn–Sham states as the response of row a to a scalar potential.

8.4.2 Four couplings

The rows of a density coupling are the densities of the species, weighted by the coefficients $c_1^{(s)}$ of Sect. 8.1: the dark matter acts through $\sum_s c_1^{(s)} n_s(\boldsymbol{q})$, with $c_1^{(\kappa)} = [Z_\kappa c_1^{(p)} + (A_\kappa - Z_\kappa) c_1^{(n)}] F_N$ from Eq. (8.8). Three choices of the coefficients cover the standard density couplings, a fourth coupling, to the spin, carries no charge at all, and Eq. (8.35) treats each in one line.

Total charge. A dark photon, heavy or light, couples to $-\tilde{V} n_Z$ (Worked example 8.1), and so does any millicharged particle. With $a = b = Z$, Eq. (8.35) collapses to $v_\mathrm{C} R_{ZZ} = \varepsilon^{-1} - 1$ (Appendix D.27), the relation Sect. 4.4 stated, now with the nuclei moving. The fluctuation–dissipation theorem then gives $S_{ZZ} = (2q^2/e^2)\, L/(1 - \mathrm{e}^{-\beta\omega})$, with $L = \mathrm{Im}[-(\varepsilon^{-1})_{00}]$ the loss function of the total dielectric matrix, exactly and at every energy. Below the gap L contains the longitudinal optical phonons of a polar crystal, which Sect. 11.2 turns into a rate.

Electron density. A scalar or vector mediator that couples to electrons and not to nucleons acts through $c_1^{(e)} n_e$, and Eq. (8.35) gives

$$R_{ee} = \tilde{R}_{ee} + \tilde{R}_{eZ}\, \varepsilon^{-1} v_\mathrm{C}\, \tilde{R}_{Ze}, \qquad \tilde{R}_{eZ} = -\tilde{R}_{ee} + \sum_\kappa Z_\kappa \tilde{R}_{e\kappa}. \tag{8.37}$$

With the nuclei clamped, n_κ is a static Bragg sum, $\tilde{R}_{e\kappa} = 0$, and this is Eq. (4.41) again. With moving nuclei the second term of $\tilde{R}_{eZ}$ is the electron density that follows a nucleus, and it is large at phonon energies, where the electrons of an atom move with their nucleus (Sect. 8.6). The exact statement for this coupling is the electron-channel form of Appendix A.7, $S_{ee} = (2q^2/e^2)\, L_e/(1 - \mathrm{e}^{-\beta\omega})$ with $\varepsilon_e^{-1} = 1 + v_\mathrm{C} R_{ee}$, but ε_e is not the dielectric function of any experiment. Its inverse has a pole at every acoustic phonon, weighted by the electrons that follow the nuclei, while the total ε^{-1} at small q has none, since a neutral cell that moves carries no charge.

Nucleon number. A scalar that couples to protons and neutrons equally acts through $\sum_\kappa A_\kappa n_\kappa$, times its coupling per nucleon. Its vertex has no electronic part, and its screening runs entirely through $\tilde{R}_{NZ}$, the proper charge that moves when the nucleon density does. At phonon energies that charge is the Born charge of each sublattice; at electronic energies the nuclei do not move and the coupling is not screened at all. The Born charge never enters the vertex itself. It enters the pole: the proper response has its optical pole at the transverse frequency, the screened one at the longitudinal frequency, with the same f-sum weight (Worked example 8.6).

Spin. The spin density $\boldsymbol{s}_e$ of a target without spin–orbit coupling and without magnetic order is not screened. Its Hamiltonian commutes with the total spin, the charge density is invariant under spin rotations and $\boldsymbol{s}_e$ is a vector under them, and the equilibrium state is invariant too. The cross response $\tilde{R}_{sZ}$ then vanishes, and $R_{ss} = \tilde{R}_{ss}$. Spin–orbit coupling or magnetic order breaks the argument; the polarized gas of Worked example 8.3 has $S_{ns} \neq 0$ and screens its spin rows with its density. The nuclear spin density $\boldsymbol{s}_\kappa$ is unscreened for the same reason. Beyond independent particles a spin-flip pair still feels the attraction between its electron and hole, the direct kernel of Sect. 4.9, but not the exchange kernel that carries the local fields: it has no charge to exchange.

The rows that are not densities follow the same rule. The longitudinal current is screened as the density. The continuity equation in the form of Eq. (8.5) makes $\hat{\boldsymbol{q}} \cdot \boldsymbol{j}_s$ act as $(\omega/q)\, n_s$ on the energy shell, and the Ward identity Eq. (4.51) holds for proper and full responses alike. The transverse current and the kinetic-energy density $\mathcal{T}_e$ are screened where their proper correlation with the charge is nonzero; the spin current, a vector under spin rotations like $\boldsymbol{s}_e$, has none without spin–orbit coupling or magnetic order and is unscreened. The kinetic-energy density is a scalar under spin rotations and the point group and correlates with the charge already at $\boldsymbol{G} = 0$, so it is screened at the head, by a term of order q^2 at small q in a crystal

without a polar axis. In a cubic crystal, with $\boldsymbol{q}$ along a symmetry axis, $\tilde{R}_{j_\mathrm{T}Z}$ vanishes at $\boldsymbol{G}=0$ by symmetry, and the transverse current is screened through the local fields at $\boldsymbol{G}\neq 0$. In a crystal of lower symmetry, or off the symmetry axes, longitudinal and transverse parts mix already at $\boldsymbol{G}=0$. Exchange of a transverse photon between currents is a separate and smaller effect at the momentum transfers of scattering: its propagator $1/(\omega^2-q^2-\Pi_\mathrm{em})$, with Π_em the photon self-energy of the medium (Sect. 8.4.5), differs from $-1/q^2$ by $\Pi_\mathrm{em}/q^2\sim\omega_p^2/q^2$, which is 2×10^{-5} in silicon at q_0. The working rule is short. Every row with $\tilde{R}_{aZ}\neq 0$ is screened through Eq. (8.35), and a rate that contracts Eq. (8.15) with the proper response $\tilde{R}$ alone, or with a factor $1/|\varepsilon|^2$ in place of the cross terms, is wrong for every coupling except the charge.

8.4.3 Exactness of the loss function

The loss function is measured for many materials, optically at $q\to 0$ and by electron energy loss at finite q, and it is the quantity that a computed response is checked against. Hochberg and collaborators pointed out that every coupling to the electron density is screened, and that couplings that treat nucleons and electrons differently are screened identically as long as the ions contribute negligibly to the loss function [16, p. 2, discussion of Eq. (1)]. The screened response turns this into a sharp statement.

Exactness of the loss-function form

Let a dark-matter particle couple to $\sum_s c_1^{(s)} n_s(\boldsymbol{q})$ in a crystal, a target whose long-wavelength longitudinal acoustic phonon moves it locally as a whole. Let nuclei of different atomic number have different masses, and set the nuclear form factors to one. Then

$$S_{cc}(\boldsymbol{q},\omega)=\lambda_Z^2\,\frac{2q^2}{e^2}\,\frac{L(\boldsymbol{q},\omega)}{1-\mathrm{e}^{-\beta\omega}}\quad\text{at every }\boldsymbol{q}\text{ and every }\omega\neq 0, \tag{8.38}$$

with S_{cc} the structure factor of the coupling and L the loss function of the total dielectric matrix. For some constant λ_Z it holds if and only if the coupling is proportional to the charge, $c_1^{(e)}=-\lambda_Z$ and $c_1^{(\kappa)}=\lambda_Z Z_\kappa$, or the nuclei are clamped and $c_1^{(e)}=\pm\lambda_Z$. A coupling to the electron density is exact only at a static lattice.

Sufficiency is immediate. If the coupling differs from $\lambda_Z n_Z$ by an operator with no dynamics, a sum of static Bragg terms when the nuclei are clamped, the two have the same transitions and $S_{cc}=\lambda_Z^2 S_{ZZ}$ at every $\omega\neq 0$, which is Eq. (8.38) by the total-charge relation above.

The converse takes two limits in which electrons and nuclei come apart. At large momentum transfer, above the inverse size of every electron orbital and in the regime where a nucleus recoils freely from its site (the impulse regime of Sect. 8.5), each particle recoils on its own. The structure factor of the coupling is then a sum of recoil peaks, at $q^2/2m_e$ for the electrons and $q^2/2M_\kappa$ for each nuclear species, with weights $(c_1^{(s)})^2\boldsymbol{n}_s$. The cross terms vanish, because the electrons no longer follow the nuclei and different nuclei are uncorrelated at such q. The peaks of different masses separate, since their distance grows as q^2 and their widths as q. The same holds for S_{ZZ} with weights $\boldsymbol{n}_e$ and $Z_\kappa^2\boldsymbol{n}_\kappa$. Equality peak by peak requires $|c_1^{(e)}|=|\lambda_Z|$ and $|c_1^{(\kappa)}|=|\lambda_Z|Z_\kappa$ for every species (Appendix D.27 treats species of equal mass): the magnitudes are those of the charge, but the signs are still free.

The signs are fixed at the opposite end, $\boldsymbol{q}\to 0$ at the frequency of a longitudinal acoustic phonon. There the crystal moves as a whole, electrons included, over a region much larger than a cell, and every density row picks up the factor $1+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{U}$ with $\boldsymbol{U}$ the local displacement. The amplitude to excite the phonon is therefore proportional to the coupling carried by a

unit volume, $\sum_s c_1^{(s)} n_s$, and its weight in S_{cc} to q^2 times its square. For the charge this sum is the net charge density, zero, and the acoustic weight of S_{ZZ} starts at order q^4, from the charge that a strain induces. Equality at small q needs $c_1^{(e)} n_e + \sum_\kappa c_1^{(\kappa)} n_\kappa = 0$, and with the magnitudes already fixed this is possible only if every nuclear coefficient has the sign opposite to the electron coefficient: a sum of positive terms $|\lambda_Z| Z_\kappa n_\kappa$ with mixed signs cannot equal $|\lambda_Z| n_e = |\lambda_Z| \sum_\kappa Z_\kappa n_\kappa$. The coupling is then $\pm\lambda_Z n_Z$. With the nuclei clamped, the nuclear peaks and the acoustic phonon are absent, and only $|c_1^{(e)}| = |\lambda_Z|$ survives.

The theorem leaves no exact use of the loss function for an electrophilic mediator once the lattice moves, and none at all for a nucleon or isospin-violating coupling. Its practical content is a bound on where the error sits. For the electron density, the difference between S_{ee} and S_{ZZ} comes from the nuclear rows of Eq. (8.37). Above the phonon band, however, the nuclei reach the electronic final states only through the Migdal amplitudes, by which a moving nucleus excites the electrons (Sect. 8.6), suppressed by powers of κ_{BO}, and the loss function gives the electron-only rate to that accuracy, the statement of Sect. 4.4. Inside the phonon band the two differ at leading order: S_{ee} has an acoustic peak carrying the electrons that follow the nuclei, a fraction $Z_\kappa m_e/M_\kappa$ of its f-sum (Eq. (8.54)), while S_{ZZ} has none at small q. Sect. 10.2 uses the theorem for the electronic rate, and the relaxed-lattice correction to its sum-rule bound comes from this acoustic peak.

Table 8.3 collects the couplings with the screening each takes.

Worked example 8.6: The nucleon-number coupling of a heavy scalar in GaAs

Take a heavy scalar that couples to protons and neutrons with the same coefficient, $c_1^{(p)} = c_1^{(n)}$, and to no electron. By Eq. (8.8) its row is $c_1^{(p)} \sum_\kappa A_\kappa n_\kappa(\boldsymbol{q})$, with $F_N = 1$ at the momenta of phonon scattering. We use the measured GaAs numbers of Worked example 5.6: $\omega_{\mathrm{T}} = 33.62\,\mathrm{meV}$, $\omega_{\mathrm{L}} = 36.35\,\mathrm{meV}$, $\varepsilon_\infty = 10.88$, $\Omega = 44.80\,\text{Å}^3$, $\mu_{\mathrm{red}} = 36.11\,\mathrm{u}$ and $Z^* = 2.11$. Take $\boldsymbol{q}$ small against the zone and ω near the optical phonons, where the macroscopic field is the whole of the long-range Coulomb line and Eq. (8.36) applies.

Vertices. In an optical mode at small $\boldsymbol{q}$ the two sublattices move against each other with the center of mass at rest, $\boldsymbol{u}_{\mathrm{Ga}} = (\mu_{\mathrm{red}}/M_{\mathrm{Ga}})\boldsymbol{u}_{\mathrm{rel}}$ and $\boldsymbol{u}_{\mathrm{As}} = -(\mu_{\mathrm{red}}/M_{\mathrm{As}})\boldsymbol{u}_{\mathrm{rel}}$, with $\boldsymbol{u}_{\mathrm{rel}} = \boldsymbol{u}_{\mathrm{Ga}} - \boldsymbol{u}_{\mathrm{As}}$ the relative coordinate of Sect. 5.5. Expanding $n_\kappa(\boldsymbol{q}) = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_\kappa+\boldsymbol{u}_{\boldsymbol{R}\kappa})}$ to first order in the displacements, the nucleon row creates the mode through $\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\mathrm{rel}}$ times an amplitude per cell, which at leading order in $\boldsymbol{q}$ is

$$a_N = \mu_{\mathrm{red}}\Big(\frac{A_{\mathrm{Ga}}}{M_{\mathrm{Ga}}} - \frac{A_{\mathrm{As}}}{M_{\mathrm{As}}}\Big).$$

At finite $\boldsymbol{q}$ the site phases $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}$ of n_κ, combined with the zone-center displacements, give a vertex that changes when another arsenic atom is assigned to the cell. Only with the eigenvector of the mode at $\boldsymbol{q}$ is the vertex independent of that choice, $a_N(\boldsymbol{q}) = \mu_{\mathrm{red}}^{1/2} \sum_\kappa A_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}\, \hat{\boldsymbol{q}}\cdot\boldsymbol{e}^*_{\kappa\mathrm{L}}(\boldsymbol{q})/M_\kappa^{1/2}$, with the conventions of Eq. (11.1). The charge row creates it through the Born charges, $\sum_\kappa Z^*_\kappa\, \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_\kappa = Z^*\,\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\mathrm{rel}}$ at leading order in $\boldsymbol{q}$, since $\mu_{\mathrm{red}}(1/M_{\mathrm{Ga}} + 1/M_{\mathrm{As}}) = 1$. The Born charge is the right vertex for the proper response: Sect. 5.5 defines it at zero macroscopic field, which is what removing the Coulomb line at $\boldsymbol{q}$ imposes. For the same reason the relative coordinate of the proper response oscillates at ω_{T}, for longitudinal as well as transverse motion, with the response $1/[\mu_{\mathrm{red}}(\omega^2 - \omega_{\mathrm{T}}^2)]$ per cell. With $N/\mathcal{V} = 1/\Omega$,

$$\tilde{R}_{NN} = \frac{q^2|a_N|^2}{\mu_{\mathrm{red}}\Omega\,(\omega^2 - \omega_{\mathrm{T}}^2)}, \qquad \tilde{R}_{NZ} = \frac{q^2 a_N Z^*}{\mu_{\mathrm{red}}\Omega\,(\omega^2 - \omega_{\mathrm{T}}^2)}, \qquad \varepsilon = \varepsilon_\infty + \frac{e^2 Z^{*2}}{\Omega\,\mu_{\mathrm{red}}(\omega_{\mathrm{T}}^2 - \omega^2)},$$

Table 8.3: External couplings of dark matter and the screening each takes, from the screened response Eq. (8.35). "Exact L" states whether the structure factor of the coupling equals $(2q^2/e^2)L/(1-\mathrm{e}^{-\beta\omega})$, with L the loss function of the total dielectric matrix of electrons and nuclei, at every momentum transfer and energy (Eq. (8.38)). SOC: spin–orbit coupling.

Coupling	Example	Screened by	Exact L
Total charge n_Z	dark photon, heavy or light; millicharge	ε^{-1} of electrons and nuclei; $v_\mathrm{C}R_{ZZ} = \varepsilon^{-1} - 1$	yes, any lattice
Electron density n_e	electrophilic scalar or vector mediator	$\tilde{R}_{eZ}\varepsilon^{-1}v_\mathrm{C}\tilde{R}_{Ze}$: electrons, and the nuclei they follow	clamped nuclei, $\omega \neq 0$; above the phonon band to Migdal order
Nucleon number $\sum_\kappa A_\kappa n_\kappa$	hadrophilic scalar	ionic polarization through the Born charges; it moves the poles (TO to LO), not the vertex	no
Other density mixtures	isospin-violating couplings	$\tilde{R}_{cZ}\varepsilon^{-1}v_\mathrm{C}\tilde{R}_{Zc}$	only if proportional to n_Z
Spin density $\boldsymbol{s}_e$, $\boldsymbol{s}_\kappa$	$\mathcal{O}_4$, $\mathcal{O}_6$, $\mathcal{O}_{10}$	none without SOC or magnetic order; the triplet exchange–correlation kernel remains	no
Longitudinal current $\hat{\boldsymbol{q}}\cdot\boldsymbol{j}$	$\mathcal{O}_8$, $\mathcal{O}_2$ (parts along $\boldsymbol{q}$)	as the density, (ω/q) times its row	as the density
Transverse current $\boldsymbol{j}_\mathrm{T}$	$\mathcal{O}_5$, $\mathcal{O}_8$ (part across $\boldsymbol{q}$)	only through local fields ($\boldsymbol{G} \neq 0$) on symmetry axes; at $\boldsymbol{G} = 0$ only through longitudinal–transverse mixing	no
Spin current $\mathcal{J}_e^{\alpha\beta}$	$\mathcal{O}_3$, $\mathcal{O}_7$, $\mathcal{O}_{12}$–$\mathcal{O}_{15}$; ALP absorption	none without SOC or magnetic order, as the spin density	no
Kinetic-energy density $\mathcal{T}_e$	$\mathcal{O}_2$ (part in $(\boldsymbol{v}^\perp)^2$); scalar absorption	head and local fields; head term $O(q^2)$ at small q in crystals without a polar axis	no

where ε is Eq. (5.53): the electrons give ε_∞, and the ionic part of $\tilde{R}_{ZZ}$ the rest.
Screening. Insert these into Eq. (8.36). By Eqs. (5.52) and (5.53), $\varepsilon = \varepsilon_\infty(\omega_\mathrm{L}^2 - \omega^2)/(\omega_\mathrm{T}^2 - \omega^2)$ and $e^2 Z^{*2}/(\Omega\mu_\mathrm{red}\varepsilon_\infty) = \omega_\mathrm{L}^2 - \omega_\mathrm{T}^2$, and the screening term is

$$v_\mathrm{C}\frac{\tilde{R}_{NZ}\tilde{R}_{ZN}}{\varepsilon} = \frac{q^2|a_N|^2}{\mu_\mathrm{red}\Omega}\,\frac{\omega_\mathrm{L}^2 - \omega_\mathrm{T}^2}{(\omega^2 - \omega_\mathrm{T}^2)(\omega^2 - \omega_\mathrm{L}^2)},$$

and added to $\tilde{R}_{NN}$ it cancels the transverse pole:

$$R_{NN}(\boldsymbol{q},\omega) = \frac{q^2|a_N|^2}{\mu_\mathrm{red}\Omega\,(\omega^2 - \omega_\mathrm{L}^2)}, \qquad S_{NN}(\boldsymbol{q},\omega) = \frac{\pi q^2|a_N|^2}{\mu_\mathrm{red}\Omega\,\omega_\mathrm{L}}\,\delta(\omega - \omega_\mathrm{L}),$$

at $T = 0$ and $\omega > 0$. The screened response is the proper one with its pole moved from ω_T to ω_L, and its f-sum weight $\int \mathrm{d}\omega\,\omega S_{NN}/2\pi$ is unchanged; the area of the peak, $\int \mathrm{d}\omega\, S_{NN}/2\pi$, falls by $\omega_\mathrm{T}/\omega_\mathrm{L} = 0.925$. The Born charges and ε_∞ enter through ω_L and nowhere else; the vertex is a_N, a property of the nuclei and their masses. The rule $1/|\varepsilon|^2$ gets this wrong in an instructive way. The proper structure factor has its weight at ω_T, where ε has a pole and $1/|\varepsilon|^2 = 0$, so the naive rate would vanish; the true rate sits at ω_L, where $1/|\varepsilon|^2$ is infinite. The cross term of Eq. (8.36), which the naive rule drops, resolves both.
The size of the vertex. At $\boldsymbol{q} \to 0$ the phases drop out and $a_N \to \mu_\mathrm{red}(A_\mathrm{Ga}/M_\mathrm{Ga} - A_\mathrm{As}/M_\mathrm{As})$, which would vanish if atomic masses were proportional to nucleon numbers. They nearly are. With the atomic masses of the isotopes [17] and the natural abundances of gallium, 60.108 % ^{69}Ga and 39.892 % ^{71}Ga [7], the mean nucleon number per mass is 1.001072 per u for gallium and 1.001046 per u for arsenic, and $a_N = 36.11 \times 2.59 \times 10^{-5} = 9.4 \times 10^{-4}$. A nucleon-number coupling barely moves the center of nucleon number when the center of mass stays put. At finite $\boldsymbol{q}$ the center of mass no longer stays put. In zincblende a shear strain displaces the two sublattices against each other, the internal strain behind the piezoelectric charge of Sect. 6.2, and by the same coupling an optical wave drags the center of mass along wherever a longitudinal wave contains shear. The nucleon row then sees the mass density of the cell, $A_\kappa/M_\kappa \approx 1$ per u, and a coupling to the mass density alone gives nearly the same vertex. Along [111], the direction of the Ga–As bond, a longitudinal wave contains shear, and the drag grows linearly in q and overtakes the mass term near $q \approx 0.05\,\mathrm{eV}$. Scattering reaches ω_L only for $q \geq \omega_\mathrm{L}/v_\mathrm{max} = 13.7\,\mathrm{eV}$, where, with the phonons of Fig. 5.5 at that $\boldsymbol{q}$ (Sect. 11.2), $|a_N| = 0.24$, 250 times the long-wavelength value; at 100 eV, $|a_N| = 1.74$. Along [100] a longitudinal wave contains no shear, the drag is absent at first order in q, and $|a_N| = 2.1 \times 10^{-3}$ at 13.7 eV, close to its value at $q \to 0$. Compared with the longitudinal acoustic phonon, whose vertex $\sum_\kappa A_\kappa = 144.8$ per cell is not screened at leading order because $\sum_\kappa Z_\kappa^* = 0$, the energy-weighted optical share, $|a_N|^2 M_\mathrm{cell}/\mu_\mathrm{red}$ divided by the squared acoustic vertex, grows as q^2; along [111] it is 1.1×10^{-5} at 13.7 eV and 5.8×10^{-4} at 100 eV. This is the suppression of optical phonons by destructive interference within the cell that Griffin and collaborators found for couplings proportional to the mass number [18, Eqs. (30) and (33)]. For a dark photon the same mode couples through $Z^*/\varepsilon_\infty = 0.19$ at any small q, and it is the polar phonon of Sect. 11.2.

8.4.4 Coulomb-line classes

Below the direct gap of silicon, dark matter can excite an electron–hole pair only together with a phonon (Sect. 6.5). The amplitude contains three objects that the medium screens. The dark-matter vertex sits at $(\boldsymbol{q}, \omega)$. The electron–phonon vertex sits at the phonon wavevector $\boldsymbol{Q}$, and density-functional perturbation theory returns it already screened (Sect. 6.2). The pair that leaves has momentum $\boldsymbol{q} + \boldsymbol{Q}$ and energy $\omega \mp \omega_{\boldsymbol{Q}\nu}$, upper sign for emission. A calculation that divides the dark-matter vertex by $\varepsilon(\boldsymbol{q}, \omega)$ and uses the computed vertex g might count one Coulomb line twice, or miss one. The question is which Coulomb lines each factor contains, and the diagrams of Sect. 4.8 answer it.

Work at the random-phase level, where every Coulomb line joins two different fermion loops and nothing runs inside a loop, at second order in the electron–phonon vertex g in the rate, with one phonon in the final state. Draw a diagram of the response as loops joined by Coulomb lines, with one phonon line between two ion vertices and the two dark-matter vertices on loops.

Coulomb-line classes

In such a diagram every Coulomb line belongs to exactly one of three classes.

- *External*, at $(\boldsymbol{q} + \boldsymbol{G}, \omega)$: a line whose removal cuts the diagram in two. These lines form the chains that Eq. (8.35) resums into $(\varepsilon^{-1})_{\boldsymbol{G}0}(\boldsymbol{q}, \omega)$ on the dark-matter vertex, local fields included.
- *Vertex*, at $\boldsymbol{Q} + \boldsymbol{G}$ and frequencies of order $\omega_{\boldsymbol{Q}\nu}$: lines between an ion vertex and the first loop that carries a dark-matter vertex. They make the screened electron–phonon vertex g.
- *Pair*, at $(\boldsymbol{q} + \boldsymbol{Q} + \boldsymbol{G}, \omega \mp \omega_{\boldsymbol{Q}\nu})$: lines between the two loops that carry the dark-matter vertices. They resum into the screened interaction $W^{\mathrm{scr}}(\boldsymbol{q} + \boldsymbol{Q}, \omega \mp \omega_{\boldsymbol{Q}\nu})$ of Sect. 4.8, acting on the final pair.

A phonon line that itself cuts the diagram in two is not a Coulomb line. It is the ionic part of the total dielectric matrix, included by using the ε of Eq. (8.35) in the external and pair chains.

The proof is by the topology of the diagram (Appendix D.27). With the phonon line removed, the loops and Coulomb lines form a tree, since at the random-phase level every Coulomb line is a bridge; a closed cycle of Coulomb lines would need a line inside a loop. Putting the phonon line back closes exactly one cycle. Lines off the cycle remain bridges, and momentum conservation sets them to $\boldsymbol{q} + \boldsymbol{G}$; these are the external lines. On the cycle, the segments next to the ion vertices carry $\boldsymbol{Q} + \boldsymbol{G}$, and the segment between the two loops with dark-matter vertices carries $\boldsymbol{q} + \boldsymbol{Q} + \boldsymbol{G}$. When both dark-matter vertices sit on one loop, the pair segment is absent. The class of a line follows from its position, not from its momentum label, and the classes stay distinct where the labels coincide. At $\boldsymbol{q} \to 0$ the pair segment carries the momentum $\boldsymbol{Q}$ of the vertex segments but an energy of order eV against theirs of order $\omega_{\boldsymbol{Q}\nu}$, and at $\boldsymbol{Q} \to 0$ the pair segment lies on the cycle while the external lines do not.

Three consequences follow. First, the screened vertex of density-functional perturbation theory, an external $\varepsilon^{-1}(\boldsymbol{q}, \omega)$ and the pair interaction W^{scr} together count every random-phase diagram once. An external $1/\varepsilon(\boldsymbol{Q})$ on top of the computed g would count the vertex lines twice; leaving out W^{scr} is an omission, not a cure for a double count. Second, the vertex lines are evaluated by density-functional perturbation theory at zero frequency, while they carry

frequencies of order $\omega_{\boldsymbol{Q}\nu}$. Below the gap, time reversal makes $\varepsilon(\boldsymbol{Q},\omega')$ an even and analytic function of the frequency ω' for $|\omega'| < E_\mathrm{g}$ (Appendix D.27). The error of the static vertex is then of order $(\omega_{\boldsymbol{Q}\nu}/E_\mathrm{g})^2$, about 3×10^{-3} in silicon, the number of Worked example 8.7; in a magnetic insulator, without time reversal, it can be of order $\omega_{\boldsymbol{Q}\nu}/E_\mathrm{g}$. Third, the ionic insertion is small at electronic energies. Far above the optical phonons, Eq. (5.53) gives an ionic part $\varepsilon-\varepsilon_\infty \approx -(\varepsilon_0-\varepsilon_\infty)\,\omega_\mathrm{T}^2/\omega^2$, which is -0.9×10^{-3} in GaAs at 1.5 eV, 10^{-4} of ε_∞; in silicon, whose Born charges vanish, it is smaller still.

The pair lines are the new element. They screen the final pair at its own momentum and energy: once the phonon has taken $\boldsymbol{Q}$, the charge that the dark-matter vertex induced oscillates at $\boldsymbol{q}+\boldsymbol{Q}$, and the medium screens it there, not at $\boldsymbol{q}$. The final pair is on shell, the screened interaction at its momentum and energy is complex, and it acts between the two halves of the amplitude, which Sect. 13.4 writes out with its size in silicon (Appendix D.36). Lines inside a loop, the electron–hole ladder of Sect. 4.9, lie outside the classification; an adiabatic exchange–correlation kernel adds a source term of its own (Sect. 13.4). A frozen-phonon supercell calculation with local fields in the supercell basis contains all three classes at once, since every $\boldsymbol{Q}+\boldsymbol{G}$ is then a reciprocal vector of the supercell (Appendix E.16).

8.4.5 In-medium mixing

At 3 eV silicon has $\varepsilon_2 = 2.4$ and a loss function $L = 3.1\times10^{-3}$ (Sect. 4.4), so $|\varepsilon|^2 = \varepsilon_2/L = 761$. A dark photon absorbed there couples as if its squared mixing were $\zeta^2/761$ rather than ζ^2. The factor is the in-medium mixing ζ_eff of Sect. 1.4, and it comes from the photon-reducible part of the response. A dark photon couples to the electromagnetic current with strength ζe (Sect. 1.4). The current it induces radiates an ordinary photon, which the medium absorbs again, and the chain repeats. Split the current response as the density response was split: a part $\tilde{R}$ that is proper with respect to one photon line, longitudinal or transverse, and the photon-reducible chains. For absorption at rest, Appendix D.25 summed the chain and found that it turns ζ^2 into

$$\zeta_\mathrm{eff}^2 = \frac{\zeta^2 m_\phi^4}{|m_\phi^2 - \Pi_\mathrm{em}|^2}, \tag{8.39}$$

with Π_em the photon self-energy, the proper current response times e^2, diamagnetic term included. In the language of Mitridate and collaborators, the dark-matter self-energy acquires a mixing term $\Pi_{\phi A}\Pi_{A\phi}/(m_\phi^2-\Pi_{AA})$ beside its proper part [8, Eqs. (20)–(22)]; that term is the photon-reducible response.

Both polarizations follow from one relation. The transverse self-energy is $\Pi_\mathrm{T} = \omega^2(1-\varepsilon^\mathrm{T})$ (Sect. 4.4), and the longitudinal one is $\Pi_\mathrm{L} = (\omega^2-k^2)(1-\varepsilon)$ [8, Eq. (34)], with k the momentum of the dark photon and ε the longitudinal function of Eq. (4.41). On shell, $\omega^2-k^2=m_\phi^2$, and $m_\phi^2-\Pi_\mathrm{L} = m_\phi^2\varepsilon$: a longitudinal dark photon mixes with $\zeta_\mathrm{eff} = \zeta/\varepsilon$ exactly. A transverse one does so at $k\to0$, where $\varepsilon^\mathrm{T}=\varepsilon$ by Eq. (4.52) and $\omega = m_\phi$. The net rate per quantum, Eq. (8.19) with the full response, is then

$$\Gamma_\mathrm{net} = -\frac{\zeta^2 m_\phi^2}{\omega}\,\mathrm{Im}\,\frac{\Pi_\mathrm{em}}{m_\phi^2-\Pi_\mathrm{em}} = -\zeta^2 m_\phi\,\mathrm{Im}\Big(\frac{1}{\varepsilon}-1\Big) = \zeta^2 m_\phi\, L(m_\phi),$$

which is $\zeta_\mathrm{eff}^2\sigma_1$ with $\sigma_1 = \omega\varepsilon_2$ [8, Eq. (40)]. A dark photon at rest measures the loss function of the target, not its absorption. This is why its absorption in GaAs peaks at the longitudinal phonon (Sect. 5.5), and why it is suppressed by 761 at 3 eV in silicon, where photons are strongly absorbed.

Scattering through a dark-photon mediator is the same construction at $\omega \ll q$. The interaction of Worked example 8.1 uses the potential of the dark matter in vacuum, $\tilde{V}$,

and the full response R_{ZZ}, which carries the chain through ε^{-1}. We can instead put the chain into the mediator: the dark photon then mixes with the longitudinal photon of the medium, its propagator acquires $1/\varepsilon(\boldsymbol{q},\omega)$, and it acts on the proper response. The two bases agree [19, Eq. (8) and the text below it]: with $\varepsilon = 1 - v_C\tilde{R}_{ZZ}$, $\tilde{R}_{ZZ}/\varepsilon = \tilde{R}_{ZZ}\varepsilon^*/|\varepsilon|^2$, and $\mathrm{Im}(\tilde{R}_{ZZ}\varepsilon^*) = \mathrm{Im}\,\tilde{R}_{ZZ}$ because $v_C|\tilde{R}_{ZZ}|^2$ is real, so that $\mathrm{Im}\,R_{ZZ} = \mathrm{Im}\,\tilde{R}_{ZZ}/|\varepsilon|^2$. In scattering, the in-medium mixing is the factor $1/\varepsilon(\boldsymbol{q},\omega)$, and it is the screening of the total-charge row of Table 8.3.

The bookkeeping rule is the one of the Coulomb lines. Either the vacuum mixing ζ multiplies the full response, or ζ_{eff} multiplies the proper one. The loss function already contains the screening, and a factor $\zeta_{\text{eff}}^2/\zeta^2$ applied to a loss-function rate, such as that of Sect. 10.2, counts it twice. The optical conductivity σ_1 is a proper response, and Eq. (1.39) multiplies it by ζ_{eff}^2 once. Other bosons mix in the same way through their own rows. The scalar of Eq. (8.11) couples to n_e and $\mathcal{T}_e$, both of which correlate with the charge, and its absorption carries a photon-reducible part through $\tilde{R}_{n_eZ}$ and $\tilde{R}_{\mathcal{T}_eZ}$. An axion-like particle in a static field $\boldsymbol{B}_0$ acts as a source current along $\boldsymbol{B}_0$ and mixes as a dark photon does; the spin rows of the axioelectric coupling do not mix at all without spin–orbit coupling. Sect. 10.7 works out these rates.

With Eq. (8.35), every row of the master formula has its medium correction in one place: the proper part of its response, its proper correlation with the charge, and the total dielectric matrix of electrons and nuclei. The proper response is exact in the target, and what a calculation keeps of it is a truncation that Sect. 8.5 labels.

8.5 The expansion hierarchy

The phonon-assisted absorption of silicon, Eq. (6.42), accounts for the measured indirect edge so well that the phonon energies read off its thresholds agree with the neutron values to 2 % (Sect. 6.5), and yet it is a truncated formula. It keeps the electron–phonon vertex to second order and drops the fourth. It takes the phonons as harmonic, the electrons as independent particles, the thermal occupations to first order, and the nuclei as slow. Each of these choices drops the next term of an expansion in some small number. In a silicon detector at 10 mK each number has a value: the Born–Oppenheimer parameter is 0.067, the anharmonic parameter 0.049, the electron–phonon parameter at the edge $\sim 10^{-2}$, the ratio of temperature to phonon energy $\sim 10^{-5}$. Every rate in this book is built in the same way, and so every rate can be labeled by the orders it keeps. We will collect the small parameters with their silicon values, attach to each the order that a formula keeps, and assemble the orders into one label. The label says where the next correction comes from and, with the small parameter, roughly how large it is.

8.5.1 Small parameters

Nuclear mass. The Born–Oppenheimer parameter $\kappa_{\text{BO}} = (m_e/M)^{1/4}$ of Sect. 5.1 is 0.0665 for natural silicon (Worked example 5.1). Through the hierarchy Eq. (5.8) it sets the zero-point amplitude in units of the Bohr radius, the phonon energy in units of $\alpha^2 m_e$ (as κ_{BO}^2), the anharmonic level shifts (κ_{BO}^4) and the nonadiabatic ones (κ_{BO}^6). What the Born–Oppenheimer approximation drops are the non-adiabatic couplings $\mathcal{N}_{ji}$ of Eq. (5.5), both of which are built from the derivative coupling $\boldsymbol{d}_{ji}^{\boldsymbol{R}\kappa}$ of Eq. (6.1) (Eq. (6.2)). How much a given process feels them depends on how fast the nuclei move. In zero-point motion a nucleus mixes excited electronic states into the ground state with an amplitude $\kappa_{\text{BO}}^3 = 2.9\times10^{-4}$, and in an insulator the ratio that controls the mixing is $\omega_{\text{ph}}/\Delta E$, which is $\omega_{\text{opt}}/E_{\text{g}} = 0.055 \approx 12\,\kappa_{\text{BO}}^2$ in silicon (Sects. 5.1 and 6.1). A nucleus kicked by dark matter moves far faster, and the same coupling then

produces the sudden amplitude Eq. (6.6), which is the Migdal effect. The parameter enters the rates through Chaps. 12 and 13.

Electron–phonon coupling. A second-order amplitude is smaller than a first-order one by the coupling that the second route adds, divided by the off-shell distance of the intermediate state (Sect. 2.4). For the electron–phonon coupling the square of that ratio is the *off-shell parameter*

$$\theta_{\mathrm{ep}} = \frac{\langle |g|^2 \rangle}{\delta_{\mathrm{off}}^2}, \tag{8.40}$$

where δ_{off} is the distance of the intermediate electronic state from the energy shell and $\langle |g|^2 \rangle$ is the mean squared vertex $g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q})$ of Eq. (6.11) over the phonon modes and intermediate states that the amplitude visits. Each pair of vertices added to a rate costs one power of θ_{ep}. In silicon the optical vertices of the holes are about 0.25 eV (Sect. 6.2), and at the absorption edge the intermediate states lie 2 to 2.5 eV off shell (Sect. 6.5). These give $\theta_{\mathrm{ep}} = 0.010$ to 0.016, the "$\sim 10^{-2}$ per mode" that Sect. 6.5 estimated for the phonon-assisted amplitude. Acoustic vertices near the zone center are ten times smaller (26 meV at $Q = 0.01\,\text{Å}^{-1}$) and their θ_{ep} a hundred times smaller still. The parameter is not small everywhere. When an intermediate state approaches the shell, at the direct gap for light or near the edge $E_{\mathrm{min}}(q)$ for dark-matter scattering, δ_{off} falls toward $|g|$ and θ_{ep} reaches one at $\delta_{\mathrm{off}} \approx 0.25$ eV. Closer still the expansion fails, the widths of Eq. (2.57) enter the denominators, and the resonant part becomes a cascade of real transitions (Eq. (2.60)). Away from the shell, where $\theta_{\mathrm{ep}} \ll 1$, lies the off-shell window in which Sect. 8.6 separates the zero-phonon from the phonon-dressed final states. Its edge is not where θ_{ep} reaches one: at the margin of 0.3 eV that Sect. 13.6 fixes for silicon, $\theta_{\mathrm{ep}} \approx 0.7$ bounds the correction from the next pair of vertices, whose size there is about 0.2. The parameter of Sect. 6.2 becomes a number in Chap. 13.

Anharmonicity. The anharmonic parameter $\eta_{\mathrm{anh}} = |\Phi^{(3)}|\ell_{\mathrm{zp}}/3\Phi^{(2)}$ of Eq. (5.72) measures the zero-point amplitude against the length over which the force constants change, $\eta_{\mathrm{anh}} \sim \langle u \rangle / a$. By the counting of Sect. 5.7 it is of the size of κ_{BO}, and in silicon it is 0.049. The cubic force constants are of first order in it and the quartic ones of second, so the three-phonon width and the first-order quartic shift are both of relative size $\eta_{\mathrm{anh}}^2 = 2.4 \times 10^{-3}$ (Eq. (5.73)). The measured width of the silicon Raman line, $\gamma/\omega_{\mathrm{opt}} = 2.8 \times 10^{-3}$, confirms the size. Sect. 11.5 turns η_{anh} into a correction to the nuclear response.

Temperature. A detector at 10 mK has $T = 0.862\,\mu\text{eV}$. Against the transverse acoustic phonon at X, $\omega_{\mathrm{TA}} = 18.57$ meV, the softest phonon that assists the silicon edge, $T/\omega_{\boldsymbol{Q}\nu} = 4.6 \times 10^{-5}$ and $n_{\boldsymbol{Q}\nu} = \mathrm{e}^{-21550}$; against the gap, $T/E_{\mathrm{g}} = 7.4 \times 10^{-7}$. The ratio $T/\omega_{\boldsymbol{Q}\nu}$ is not small for every mode, since acoustic phonons near the zone center have energies below T. These are, however, a fraction $(T/\omega_{\mathrm{D}})^3 = 4 \times 10^{-15}$ of the modes, with $\omega_{\mathrm{D}} = 55.6$ meV (Worked example 5.3), and the expansion is in the occupations $n_{\boldsymbol{Q}\nu} = n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$ that weight each mode, not in T itself. The occupations enter every thermal formula of Sect. 2.5, and they matter wherever a measured input was taken warm: the absorption edge of Worked example 6.5 at 300 K, the gap at room temperature, the phonon widths at 77 K. Sect. 13.7 follows their effect on the rate.

Recoil. A nucleus given the momentum $\boldsymbol{q}$ would recoil freely with the energy $E_R = q^2/2M_\kappa$, and the *recoil parameter*

$$x_R = \frac{q^2}{2M_\kappa \langle \omega \rangle} = \frac{E_R}{\langle \omega \rangle}, \tag{8.41}$$

compares that energy with the mean phonon energy $\langle \omega \rangle$ of the atom. The mean is taken over the phonon density of states per atom, normalized to three,

$$\langle \omega^k \rangle = \frac{1}{3} \int_0^\infty \mathrm{d}\omega\, \omega^k\, \frac{g_{\mathrm{ph}}(\omega)}{n_{\mathrm{at}}},$$

and for the density of states of Sect. 5.2 it is $\langle\omega\rangle = 40.9\,\text{meV}$. At $q = 10\,\text{keV}$, $E_R = 1.91\,\text{meV}$ and $x_R = 0.047$; x_R reaches one at 46 keV. Together with the Debye–Waller exponent $2W_\kappa$ of Eq. (5.56), x_R decides whether a momentum transfer creates one phonon, several, or a free recoil. The next subsection derives how. The physics is that of Sect. 5.6, and Sects. 11.3 and 11.4 turn it into rates.

Suddenness. A nucleus accelerated over a time τ_{acc} excites an electronic state of energy ΔE with the sudden amplitude when $\Delta E\,\tau_{\text{acc}} \ll 1$, and with an amplitude that falls faster than any power of $\Delta E\,\tau_{\text{acc}}$ when that product is large (Sect. 6.1). The *suddenness ratio* $\Delta E\,\tau_{\text{acc}}$ is the small parameter of the sudden limit. A silicon nucleus is kicked in about $4\times10^{-21}\,\text{s}$, while $1/E_{\text{g}} = 0.56\,\text{fs}$, so that $E_{\text{g}}\tau_{\text{acc}} = 7\times10^{-6}$ across the gap. The lattice pulls the nucleus back on the slower time $1/\omega_{\text{ph}}$, and Sect. 6.1 found that this changes the excitation probability by the factor $[1-(\omega_{\text{ph}}/\Delta E)^2]^{-2}$, 1.006 for silicon: the deceleration enters through the square of the adiabatic ratio $\omega_{\text{ph}}/\Delta E$ of the first paragraph. Both numbers become rates in Chap. 12.

Multipoles. The density vertex $e^{i\boldsymbol{q}_e\cdot\boldsymbol{r}}$ that excites an electron can be expanded in powers of the *multipole parameter* $\xi_D = q_e r_{\text{orb}}$, the momentum given to the electron times the size r_{orb} of the orbital it leaves or enters. The first nonvanishing term between orthogonal states is the dipole term of Sect. 2.2. Three dark-matter processes give the electrons momenta that span six orders of magnitude. In electron scattering $q_e = q$, and at the typical $q_0 = \alpha m_e = 1.89\,\text{Å}^{-1}$ an orbital of 1 Å gives $\xi_D \approx 1.9$: there is no multipole expansion, and the crystal form factors of Chap. 10 keep the plane wave whole (Appendix E.7). In the Migdal effect the electrons receive only the boost $m_e v_N$, 100 eV for a silicon nucleus kicked by 1 GeV dark matter at the largest momentum transfer (Sect. 6.1), and $\xi_D \approx 0.05$. An absorbed boson of mass m_ϕ brings $q_e = m_\phi v \approx 10^{-3} m_\phi$, and $\xi_D = 5\times10^{-6}$ at $m_\phi = 10\,\text{eV}$. The parameter introduced with the hydrogen form factor of Sect. 2.2 is used again in Sects. 10.6 and 12.2.

Table 8.4 collects the seven parameters. Each is an estimate with a coefficient of order one in front: the counting of Sect. 5.7 found coefficients between a quarter and two in silicon, and nothing better should be expected of the others. Each is also small for silicon at the conditions of a dark-matter search, with three exceptions that the next chapters live with: θ_{ep} near resonances, x_R at large momentum transfer, and ξ_D in electron scattering. Where a parameter is of order one the corresponding expansion is not used, and the label below records that the quantity is computed to all orders in it.

8.5.2 Lattice regimes

The multiphonon expansion Eq. (5.59) has a simple statistical reading. The number of phonons is Poisson distributed with mean $2W_\kappa$, and each phonon carries an energy drawn from the single-phonon distribution $P_1(\omega)$. For such a compound Poisson distribution the cumulants of the energy deposited are $2W_\kappa$ times the moments of P_1, $c_k = 2W_\kappa \int \mathrm{d}\omega\, \omega^k P_1(\omega)$. At $T = 0$ and on a cubic site, $P_1(\omega) \propto g_{\text{ph}}(\omega)/\omega$ for $\omega > 0$, and $2W_\kappa = E_R\langle\omega^{-1}\rangle$ from Eq. (5.56). The factors of $\langle\omega^{-1}\rangle$ cancel, and

$$c_k = E_R\,\langle\omega^{k-1}\rangle, \qquad c_1 = E_R, \qquad c_2 = E_R\langle\omega\rangle. \tag{8.42}$$

The mean deposit is the free recoil energy, the sum rule Eq. (5.61). The variance is $E_R\langle\omega\rangle$. The width of the spectrum relative to its mean is

$$\frac{c_2^{1/2}}{c_1} = \Big(\frac{\langle\omega\rangle}{E_R}\Big)^{1/2} = x_R^{-1/2}. \tag{8.43}$$

This is why x_R is defined with the arithmetic mean $\langle\omega\rangle$, while the Debye–Waller exponent carries the inverse mean: $2W_\kappa = x_R\,\langle\omega\rangle\langle\omega^{-1}\rangle \geq x_R$, with $\langle\omega\rangle\langle\omega^{-1}\rangle = 1.35$ in silicon

Table 8.4: Small parameters of the expansion hierarchy, with their values for silicon in a detector at 10 mK. Each controls one component of the truncation label $\mathfrak{T}$ (Eq. (8.44)). "Introduced" is the section where the physics first appears; "Number" is where it becomes a correction to a dark-matter rate. The values are estimates, with coefficients of order one. The inputs are $M = 26.16\,\mathrm{GeV}$ (natural silicon) for κ_{BO}, the optical hole vertex of 0.25 eV and off-shell distances of 2 to 2.5 eV at the absorption edge for θ_{ep}, the bond coordinate of Sect. 5.1 for η_{anh}, the transverse acoustic phonon at X (18.57 meV) for $T/\omega_{\boldsymbol{Q}\nu}$, the density of states of Sect. 5.2 ($\langle\omega\rangle = 40.9\,\mathrm{meV}$) and the natural isotope mixture for x_R, a kick of $4 \times 10^{-21}\,\mathrm{s}$ across the 1.17 eV gap for $\Delta E\,\tau_{\mathrm{acc}}$, and an orbital of 1 Å for ξ_D.

Parameter	Definition	Silicon	Component	Introduced	Number
κ_{BO}	$(m_e/M)^{1/4}$	0.0665; $\omega_{\mathrm{opt}}/E_{\mathrm{g}} = 0.055$, so $(\omega_{\mathrm{opt}}/E_{\mathrm{g}})^2 = 3{\times}10^{-3}$ per step of n_B in a lattice at rest	n_B	Sect. 5.1	Chaps. 12 and 13
θ_{ep}	$\langle\lvert g\rvert^2\rangle/\delta_{\mathrm{off}}^2$, Eq. (8.40)	0.010–0.016 at the edge; 1 at $\delta_{\mathrm{off}} \approx 0.25\,\mathrm{eV}$	n_g	Sect. 6.2	Chap. 13
η_{anh}	$\lvert\Phi^{(3)}\rvert\ell_{\mathrm{zp}}/3\Phi^{(2)}$, Eq. (5.72)	0.049 ($\eta_{\mathrm{anh}}^2 = 2.4 \times 10^{-3}$)	n_A	Sect. 5.7	Sect. 11.5
$T/\omega_{\boldsymbol{Q}\nu}$	temperature over phonon energy	4.6×10^{-5}; $n_{\boldsymbol{Q}\nu} = \mathrm{e}^{-21550}$	n_T	Sect. 2.5	Sect. 13.7
x_R	$q^2/2M_\kappa\langle\omega\rangle$, Eq. (8.41)	0.047 at 10 keV; 1 at 46 keV	$\mathcal{K}$	Sect. 5.6	Sects. 11.3 and 11.4
$\Delta E\,\tau_{\mathrm{acc}}$	excitation energy times kick duration	7×10^{-6}	$\mathfrak{M}$	Sect. 6.1	Chap. 12
ξ_D	$q_e r_{\mathrm{orb}}$	1.9 (electron scattering at q_0); 0.05 (Migdal); 5×10^{-6} (absorption at 10 eV)	n_D	Sect. 2.2	Sects. 10.6 and 12.2

($\langle\omega^{-1}\rangle^{-1} = 30.4\,\mathrm{meV}$, Worked example 5.7). The two numbers answer different questions. The exponent $2W_\kappa$ counts phonons; the recoil parameter says whether the deposit is concentrated near E_R.

The exponent and the recoil parameter, read together, fix the lattice regime. For $2W_\kappa \ll 1$, and therefore $x_R \ll 1$, most scatterings leave the lattice unchanged and a fraction $2W_\kappa$ creates one phonon, which carries an energy of the order of $\langle\omega\rangle$, far above E_R. This is the *one-phonon* regime, in which the coherent response of the crystal (Sect. 11.3) decides the rate. In silicon this holds below about 10 keV, where $2W_\kappa \leq 0.063$ and $x_R \leq 0.047$. For $2W_\kappa$ of order one, several terms of Eq. (5.59) compete and the deposit comes in a few whole phonons: the *multiphonon* regime, with $2W_\kappa = 1$ at 40 keV and $x_R = 1$ at 46 keV. For $x_R \gg 1$ the spectrum narrows around E_R, and by the central limit theorem it approaches a Gaussian of width $(E_R\langle\omega\rangle)^{1/2}$: the free recoil, broadened by the motion of the nucleus in its well. This is the *impulse* regime, and its formula is the elastic recoil of Sect. 1.3 with the width as its leading correction [20, Sect. IV A, Eqs. (90)–(92)]. The width falls below $E_R/3$ for $x_R \geq 9$, at $q \geq 139\,\mathrm{keV}$ or $E_R \geq 0.37\,\mathrm{eV}$ in silicon. The Einstein oscillator of Sect. 1.3, at 60 meV, puts the same boundary at 0.54 eV; the computed density of states, whose mean phonon energy is lower, moves it down by a third.

Temperature acts on the two numbers differently. Thermal phonons add the factor $\coth(\omega/2T)$ to $2W_\kappa$ (Eq. (5.56)), raising the number of phonons absorbed and emitted, while the mean deposit stays at E_R at any temperature. At 10 mK the factor is one for every mode that matters, and the regimes are those of $T = 0$. The boundaries are, of course, gradual, and a formula is placed in a regime by what it assumes, not by the momentum at which it is evaluated.

8.5.3 The truncation label

Each small parameter has an order to which a formula keeps it, and the orders, with the three labels that are not orders, form the *truncation label*

$$\mathfrak{T} = (n_B,\, n_g,\, n_A,\, \mathcal{X},\, n_T,\, \mathcal{K},\, \mathfrak{M},\, n_D). \tag{8.44}$$

The five integers are the Born–Oppenheimer, electron–phonon, anharmonic, thermal and multipole orders. One rule fixes them all: *an order is the power of its expansion parameter carried by the quantity being labeled*, counted in the response or the rate, not in the number of vertices. The rule generalizes the counting of Sect. 5.7, where a width with two cubic vertices and a shift with one quartic vertex are of one order. The three remaining components are labels: the many-body approximation $\mathcal{X}$, the lattice regime $\mathcal{K}$ and the Migdal regime $\mathfrak{M}$. The Born–Oppenheimer order n_B always appears with its value, as in $n_B = 0$, which distinguishes it from the Bose function $n_\mathrm{B}(\omega)$.

Each component is best seen at its two lowest values, and the earlier chapters have met both.

Born–Oppenheimer order n_B, the power of the derivative coupling $\boldsymbol{d}^{\boldsymbol{R}\kappa}_{ji}$ in the quantity. At $n_B = 0$ the electrons are in the adiabatic state of the instantaneous nuclear configuration. This covers the clamped-nucleus band structures of Chap. 3, the response functions of Chap. 4, and the phonons on the Born–Oppenheimer surface of Chap. 5, all of which are evaluated without any derivative coupling. The sudden amplitude Eq. (6.6) is linear in $\boldsymbol{d}$, so the probability $1 - P_0$ that a kick excites the electrons, and with it the Migdal rate, is at $n_B = 2$. The odd order $n_B = 1$ appears only in interference, between an amplitude that contains the coupling and one that does not; Sect. 8.6 meets such a term. The cost of a step depends on how fast the nuclei move, because each power of $\boldsymbol{d}$ enters with the nuclear velocity, as $\boldsymbol{v}_N \cdot \boldsymbol{d}/\Delta E$ (Eq. (6.6)). For a lattice in its vibrational ground state the velocity is that of zero-point

motion. The step $n_B \to n_B + 2$ then multiplies an adiabatic quantity by $1 + O((\omega_{\mathrm{ph}}/\Delta E)^2)$, which is 3×10^{-3} for the optical phonon across the gap of silicon. The amplitude κ_{BO}^3 above is the same ratio in the atomic counting, with $\Delta E \sim \alpha^2 m_e$. For a nucleus kicked by dark matter the velocity is $\boldsymbol{q}/M_\kappa$, and the step from $n_B = 0$ to $n_B = 2$ is the sudden probability itself. The share $\kappa_{\mathrm{BO}}^4 = m_e/M$ of an f-sum that the electrons following a nucleus carry (Sect. 8.6) is fixed by the masses alone and is not a step of this kind.

Electron–phonon order n_g, the power of the vertex g. Direct absorption, Eq. (6.39), is at $n_g = 0$; phonon-assisted absorption, Eq. (6.42), with one vertex in the amplitude, is at $n_g = 2$, as are the one-phonon emission rates and the Fan–Migdal widths of Sect. 6.3. The vertex itself, a quantity of first order in the displacement, is at $n_g = 1$. Raising n_g by two costs one power of θ_{ep}.

Anharmonic order n_A, the power of η_{anh}. The harmonic crystal of Sects. 5.2, 5.3 and 5.6, and with it the multiphonon formula, is at $n_A = 0$. The three-phonon width Eq. (5.70) and the first-order quartic shift are both at $n_A = 2$, and the leading anharmonic phonon self-energy sits there; at $n_A = 1$ the energies do not change, since the average of the cubic term over a harmonic state vanishes. First-principles electron–phonon theory starts from the point $n_B = 0$, $n_A = 0$: its lattice Hamiltonian rests on the harmonic and the Born–Oppenheimer approximations and on nothing else [21, Sect. III A, text after Eq. (19)].

Many-body approximation $\mathcal{X}$, one of five: the independent-particle bubble Eq. (4.63), IP (the Lindhard function of Sect. 4.5 in the electron gas); the random-phase approximation (RPA) Eq. (4.67) with the head alone, or with local fields (LF) Eq. (4.68), written RPA + LF; an exchange–correlation kernel of time-dependent density-functional theory (TDDFT) Eq. (4.70); and the Bethe–Salpeter equation (BSE) Eq. (4.104). $\mathcal{X}$ is not an order, and the next subsection says why. Inputs from density-functional perturbation theory, the phonons, Born charges and ε_∞, are static self-consistent responses with the Hartree and exchange–correlation kernel of the functional, and they carry $\mathcal{X} =$ TDDFT at $\omega = 0$, whatever the approximation of the electronic response they are combined with.

Thermal order n_T, the power of the phonon occupations $n_{\boldsymbol{Q}\nu}$. At $n_T = 0$ there is no thermal occupation, though zero-point motion is kept wherever phonons enter; at $n_T = 1$ a rate is linear in the occupations, as Eq. (6.42) is through $n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2}$. The thermal occupation of electrons across a gap, $\mathrm{e}^{-E_{\mathrm{g}}/T}$, is smaller than any power and is dropped at every order.

Lattice regime $\mathcal{K}$, one of one-phonon, multiphonon and impulse: the phonon content of the final state. For a coupling to the nuclei it is read from x_R and $2W_\kappa$ as above. The single term $n = 1$ of Eq. (5.59) is one-phonon; the full sum is multiphonon; its Gaussian limit around E_R is impulse. A process that creates its phonon through one electron–phonon vertex in the amplitude, as phonon-assisted absorption does, is one-phonon as well. Without phonons in the final state the label does not apply.

Migdal regime $\mathfrak{M}$, sudden or adiabatic. The boost Eq. (6.8) and the hydrogen survival probability Eq. (6.9) are sudden; the adiabatic limit of Sect. 6.1, in which the electrons follow the nucleus and end in their ground state, is the other value. The label applies to a formula that treats the struck nucleus as a classical trajectory. Where the phonon final states are kept, at small x_R, the same physics is carried by n_B and $\mathcal{K}$ instead (Sect. 12.1).

Multipole order n_D, the power of ξ_D. The dipole transition $1s \to 2p$ of hydrogen at small momentum transfer, $|f|^2 \simeq 0.555\, q^2 a_0^2$ (Sect. 2.2), is at $n_D = 2$, the lowest order at which a transition between orthogonal states occurs. The full plane wave, kept whole, has no multipole order.

The *base point* of the label is

$$\mathfrak{T}_0 = (n_B = 0,\, n_g = 0,\, n_A = 0,\, \mathcal{X} = \mathrm{IP},\, n_T = 0;\ \mathcal{K},\, \mathfrak{M},\, n_D \text{ not applicable}) : \tag{8.45}$$

clamped nuclei, no electron–phonon vertex, independent particles, zero temperature, and no

expansion of the plane wave, with no phonon in the final state and no nuclear recoil to classify. The unscreened crystal form factor of Chap. 10 sits there (Appendix E.7). A formula that takes its phonons or Born charges from density-functional perturbation theory keeps $\mathcal{X}$ for its electronic response and names those inputs as static TDDFT. A formula elsewhere on the label is written by listing only the components that differ from $\mathfrak{T}_0$. The direct dielectric function Eq. (6.39) is $\mathfrak{T}_0$ in this notation, a three-phonon linewidth is "$n_A = 2$", and the screened loss function with local fields is "$\mathcal{X} = \mathrm{RPA} + \mathrm{LF}$". Qualifiers that are not orders, such as "unscreened" or "with spin–orbit coupling", are added in words. Appendix F gives each literature formula in this form, and Sect. 9.4 places them.

Several approximations are deliberately left off the label, because none of them is an expansion in a small number with a next order that could be computed. The exchange–correlation functional, the frozen core of a pseudopotential, the *GW* quasiparticle energies, and the numerical parameters of a calculation (grids, cutoffs, Wannier interpolation) are of this kind; their errors are measured by tests and convergence, as Appendices E.1 to E.5 describe. Methods that sum an expansion to all orders sit off the finite values of an axis and are described in words. The self-consistent harmonic approximation of Sect. 5.7 resums the anharmonic expansion, molecular dynamics on the full Born–Oppenheimer surface (Appendix E.11) is exact in n_A at $n_B = 0$, and a classical cascade calculation is off the perturbative axes altogether.

8.5.4 Many-body approximations as resummations

The approximations of Sects. 4.6 and 4.9 are ordered by what they include, and each is a limit of the next, but they are not successive orders in a small number. Each is a resummation. The random-phase approximation already sums the Coulomb interaction to all orders through the Dyson equation, the kernel of time-dependent density-functional theory adds a correction inside that sum, and the Bethe–Salpeter equation adds a ladder of electron–hole attractions. The quantity that would have to be small, the kernel measured against the Coulomb interaction it corrects, is not. In silicon the local fields lower ε_∞ by 11 %, from 11.45 to 10.22; the adiabatic kernel restores about half of that, to 10.81; and the attraction, added to the local fields, raises the local-field value by 17 %, to ≈ 11.9 (Worked example 4.11 and Sect. 4.9). Each correction moves ε_∞ by about as much as the last, and not always in the same direction, so there is nothing to expand in. At the lower absorption peak the approximations disagree by a factor of more than two, while above 4 eV they agree within the accuracy of the bands. Nothing like η_{anh}^2 predicts the size of the next step.

The error of a many-body approximation is therefore estimated by computing the next one, not by a power. The difference between two computed approximations is the many-body uncertainty of a rate: for the screening factor $1/\varepsilon_\infty^2$ of silicon it is up to a third across the hierarchy, and for light dark matter near the gap it is the leading uncertainty. As Sect. 4.9 stressed, the spread is an estimate, not a bound, since a further correction could in principle move the result outside it. Sect. 8.8 turns such computed differences, and the next-order terms of the other components, into the nuisance parameters of a rate.

Worked example 8.7: The label of phonon-assisted absorption

We read the label of the phonon-assisted dielectric function Eq. (6.42) off its assumptions, component by component, and estimate the size of each first omitted term in silicon at the indirect edge.

The phonon frequencies and vertices come from density-functional perturbation theory on the Born–Oppenheimer surface, with the electrons responding statically, so the formula is at $n_B = 0$, and these inputs sit at $\mathcal{X} = \mathrm{TDDFT}$ in its static limit. The static screening of

the vertex misses corrections of relative order $(\omega_{\mathrm{ph}}/E_{\mathrm{g}})^2$ (Sect. 6.2), 3×10^{-3} for the optical phonon. The phonons are harmonic, $n_A = 0$; their anharmonic widths, $\gamma/\omega = 2.8 \times 10^{-3}$ for the optical phonon, would broaden each threshold by a fraction of a meV.
The amplitude $M^{\pm}_{cv\nu}$ contains one optical vertex and one electron–phonon vertex, so the rate is quadratic in g: $n_g = 2$. The first omitted order, $n_g = 4$, adds processes with two phonons and the dressing of the one-phonon process by a second. It is smaller by $\theta_{\mathrm{ep}} \approx 10^{-2}$ below E_{dir}, where every intermediate state lies electronvolts off shell, and it is not small near E_{dir}, where the intermediate electron becomes real and Sect. 6.5 replaced the expansion by the width Γ_{w}.
The occupation factor $n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2}$ is linear in $n_{\boldsymbol{Q}\nu}$, which makes the formula $n_T = 1$. Linearity is exact at $n_g = 2$ for harmonic phonons: a term quadratic in the occupations needs two phonons and therefore $n_g = 4$. At 10 mK the occupations are e^{-21550} and smaller, and only the emission branch survives; the label still records $n_T = 1$, because the formula holds at the 300 K and 77 K of the measured edge in Worked example 6.5. The electrons are thermally unoccupied across the gap, $T \ll E_{\mathrm{g}}$, as at every order.
The bands are independent-particle bands with a rigid shift, and the electron and hole do not attract each other: $\mathcal{X} = \mathrm{IP}$. Each event creates or absorbs exactly one phonon, through the vertex g, so $\mathcal{K}$ = one-phonon; the photon vertex acts on the electrons and no nucleus recoils, so $\mathfrak{M}$ does not apply. The dielectric function ε_2 is defined in the limit $\boldsymbol{q} \to 0$, where no multipole expansion is made, so n_D does not apply either. In full,

$$\mathfrak{T}[\varepsilon_2^{\mathrm{PA}}] = (n_B = 0,\, n_g = 2,\, n_A = 0,\, \mathcal{X} = \mathrm{IP},\, n_T = 1,\, \mathcal{K} = \text{one-phonon};\ \mathfrak{M},\, n_D \text{ not applicable}),$$

written as "$n_g = 2$, $n_T = 1$, $\mathcal{K}$ = one-phonon" against the base point, with $\mathcal{X}$ at its base value IP. The same formula with the absorption branch dropped, as a calculation for a detector at millikelvin temperatures does, is at $n_T = 0$; the phonon-assisted loss function of Appendix E.14 is labeled that way.
The label also ranks the corrections. The next electron–phonon order costs 10^{-2}, the static screening of the vertex a few 10^{-3}, and the anharmonic widths less. The excitonic correction, however, is not a percent effect at the edge. The electron–hole attraction moves every threshold down by the exciton binding energy, 14.7 meV (Sect. 4.9), comparable to the energy of the transverse acoustic phonon that assists the transition, and the measured kinks of Worked example 6.5 are exciton thresholds. Near the edge, then, the largest error of Eq. (6.42) lies on the axis that has no small parameter, $\mathcal{X} = \mathrm{IP} \to \mathrm{BSE}$, and it is estimated by solving the Bethe–Salpeter equation, not by a power counting.

The same reading applies to every rate of the following chapters. A formula sits at a point of $\mathfrak{T}$, its first omitted term is one step along an axis, and Table 8.4 gives the size of that step for silicon wherever the axis has a small parameter. Where it has none, as on the many-body axis, the next approximation is computed. The decomposition of the next section needs one of these numbers in particular: the final states can be split into zero-phonon and phonon-dressed electronic excitations only where $\theta_{\mathrm{ep}} \ll 1$.

8.6 The decomposition theorem

A dark-matter particle of 1 GeV that kicks a silicon nucleus at the maximum momentum transfer, 5.10 MeV, sends it off at $v_N = 1.95 \times 10^{-4}$ (Sect. 6.1). Most of the time the electrons follow, and the recoil energy ends up in the lattice as phonons. In a fraction of events that

Sect. 6.1 estimated at 7×10^{-4}, for an electron bound as in hydrogen, they do not, and the same kick leaves an electron–hole pair behind. At 50 keV the nucleus does not recoil freely at all: it leaves the lattice untouched in 21 % of the events and creates two or more phonons in 47 % (Worked example 5.7). The electron density offers the same variety. A slow particle that couples to the electrons excites a pair with no phonon (Chap. 4), a pair with a phonon below the direct gap (Sect. 6.5), or no pair at all, since the electrons that follow a vibrating nucleus scatter as well (Sect. 5.5). The literature computes each of these final states with its own formula and its own code. Their sum is the rate only if no final state is counted twice and none is left out. Worked example 2.6 and Sect. 6.5 have already met the danger: above the direct gap, the phonon-assisted absorption of light contains a cascade that direct absorption counts once more.

To make the sum count each final state once we split the final states of every response into three sectors, defined below. We will find the leading operator that reaches each sector, the share each sector takes of the sum rules of Sects. 4.3 and 8.3, and the convention that makes the split unique where resonances blur it. The construction needs a gap, and we end with what replaces it in metals, superconductors and semimetals. The derivations are in Appendix D.28.

8.6.1 Three sectors of final states

The Born–Oppenheimer approximation of Sect. 5.1 already suggests the first cut. Every state of the form $\Psi^{\mathrm{nuc}}(u)\,\Psi_0(r;u)$, with the electrons in the adiabatic ground state at each configuration u of the nuclei and any nuclear wavefunction, lies in the range of the projector

$$\mathcal{P}_{\mathrm{BO}} = \int \mathrm{d}u\; |\Psi_0(u)\rangle\langle\Psi_0(u)| \otimes |u\rangle\langle u|, \tag{8.46}$$

where $|\Psi_0(u)\rangle$ acts on the electrons and $|u\rangle$ on the nuclei. Phonons, multiphonon states and a freely recoiling nucleus are all in this subspace; electron–hole pairs are not. The subspace is not exactly invariant under the Hamiltonian, however. The nuclear kinetic energy differentiates $\Psi_0(r;u)$ with respect to u, and the nonadiabatic couplings Eq. (5.5) carry part of the state into the excited adiabatic states. For zero-point motion the leak is the derivative coupling $\alpha^2 m_e\,\kappa_{\mathrm{BO}}^3$ of the Born–Oppenheimer hierarchy (Sect. 6.1).

A final state must have a definite sector, and a projector that leaks does not give it one. Panati, Spohn and Teufel showed how to repair this order by order [22, Thm. 2.1]. Their small parameter is $(m_e/M)^{1/2} = \kappa_{\mathrm{BO}}^2$, the ratio of a nuclear to an electronic velocity at equal kinetic energy. A unitary U_n, built as a power series in it, rotates $\mathcal{P}_{\mathrm{BO}}$ into a *superadiabatic* projector

$$\mathcal{P}_{\mathrm{ad}} = U_n\,\mathcal{P}_{\mathrm{BO}}\,U_n^{\dagger}, \qquad \big\|(1-\mathcal{P}_{\mathrm{ad}})\,H\,\mathcal{P}_{\mathrm{ad}}\big\| = O\big(\kappa_{\mathrm{BO}}^{2(n+1)}\big)\,\alpha^2 m_e, \tag{8.47}$$

on states below any fixed energy [22, Sect. 4.5]. The order $n = 0$ is Eq. (8.46) itself, and its bound κ_{BO}^2 lies above the κ_{BO}^3 of the counting, as an upper bound should. Each further order removes one more power of κ_{BO}^2. The first correction tilts the electronic state in the direction of the nuclear velocity, $\Psi_0 \to \Psi_0 + \sum_{j\neq 0} \Psi_j\,\mathrm{i}\,\boldsymbol{d}_{j0}\cdot\boldsymbol{p}/[M_\kappa(E_j - E_0)]$, with $\boldsymbol{d}_{j0}$ the derivative-coupling vector Eq. (6.1) and $\boldsymbol{p}$ the momentum of the nucleus: the electron cloud of a moving nucleus lags behind it and is slightly polarized. Appendix D.28 gives the construction and its hypotheses.

The range of $\mathcal{P}_{\mathrm{ad}}$ is the *electron-following sector*. Its complement $1 - \mathcal{P}_{\mathrm{ad}}$ contains every state with an electronic excitation, and it splits once more by what happens to the lattice. Measured from the initial state, the number of phonons can stay the same or change. We write

$$1 = \mathcal{P}_{\mathrm{ad}} + \mathcal{P}_0 + \mathcal{P}_+, \tag{8.48}$$

with $\mathcal{P}_0$ the *zero-phonon electronic sector*, an electronic excitation with no net change of the phonon number, and $\mathcal{P}_+$ the *phonon-dressed electronic sector*, an electronic excitation with a nonzero net change of the phonon number. At $T = 0$ phonons can only be emitted, and the change is at least one phonon created; a warm target can also give up phonons, as at the absorption thresholds of Sect. 6.5. The phonon number here is that of the harmonic reference lattice in which the electronic excitation sits. The excited surfaces are displaced, and a zero-phonon state of an excited surface is a coherent superposition of phonons of the ground one, as the displaced oscillator of Sect. 5.6 showed. At $T = 0$, $\mathcal{P}_0$ holds the zero-phonon line of every electronic transition and $\mathcal{P}_+$ its sideband. The three projectors are orthogonal and sum to one, so every final state has one label, and Table 8.5 below lists what each sector holds. The lattice regime $\mathcal{K}$ of Sect. 8.5 labels the phonon content within a sector, one phonon, several or a free recoil inside $\mathcal{P}_{\mathrm{ad}}$, and the phonons that dress the excitation inside $\mathcal{P}_+$; the two labels are nested, not competing.

The response of each sector follows from Eq. (4.13) by inserting the projector on both sides of the energy delta function,

$$S^A_{ab}(\boldsymbol{q},\omega) = \frac{2\pi}{\mathscr{V}} \sum_i p_i \, \langle i| O_a(\boldsymbol{q})^\dagger \, \mathcal{P}_A \, \delta(\omega + E_i - H) \, \mathcal{P}_A \, O_b(\boldsymbol{q}) |i\rangle, \qquad A \in \{\mathrm{ad}, 0, +\}, \quad (8.49)$$

with the initial states in the electron-following sector, as they are at $T \ll E_{\mathrm{g}}$. If $\mathcal{P}_A$ commutes with H, the delta function can be written in eigenstates that each lie in one sector, S^A_{ab} is the sum over the final states of sector A, and $\sum_A S^A_{ab} = S_{ab}$ exactly. Each S^A_{ab} is then a positive semidefinite matrix by the argument of Sect. 4.2, because it is again a sum of outer products with positive weights. If $\mathcal{P}_A$ commutes with H approximately, the cross terms $\mathcal{P}_A \, \delta(\omega + E_i - H) \, \mathcal{P}_B$ with $A \neq B$ do not vanish. They are small on energy bins wider than the leak, and not on narrower ones. For a bin of width $\Delta\omega_{\mathrm{bin}}$ they are bounded by the leakage norm of Eq. (8.47) divided by $\Delta\omega^2_{\mathrm{bin}}$, times the norms of $O_a|i\rangle$ and $O_b|i\rangle$ (Eq. (D.164)). This is the precise sense in which the sectors partition a spectrum.

Decomposition theorem

For a target with a gap, with the nuclear charges smeared over the finite size of the nuclei, at temperatures $T \ll E_{\mathrm{g}}$, and with the electronic ground state nondegenerate at every nuclear configuration that the vibrational state visits,

$$S_{ab}(\boldsymbol{q},\omega) = S^{\mathrm{ad}}_{ab}(\boldsymbol{q},\omega) + S^0_{ab}(\boldsymbol{q},\omega) + S^+_{ab}(\boldsymbol{q},\omega) + \delta S_{ab}, \qquad \frac{|\delta S_{ab}|}{S^{\mathrm{bin}}} = O\Big(\kappa_{\mathrm{BO}}^{2(n+1)} \, \frac{\alpha^2 m_e}{\Delta\omega_{\mathrm{bin}}}\Big), \quad (8.50)$$

for every pair of target operators, with S^{bin} the response averaged over an energy bin of width $\Delta\omega_{\mathrm{bin}}$ and n the order of the superadiabatic projector. The electron-following sector and the sum of the two electronic sectors are positive semidefinite matrices. The split between S^0 and S^+ is unique to relative order θ_{ep} inside the off-shell window, where each is positive; outside it the split is fixed by the Dyson convention of Eq. (8.59), and near a resonance only the sum of the two is positive.

For bins of 1 meV, the scale of phonon spectra, the factor $\alpha^2 m_e/\Delta\omega_{\mathrm{bin}}$ is 2.7×10^4, and the bound becomes useful from $n = 2$ on, where it is 2×10^{-3}. The hypotheses are physical. The finite size of the nuclei keeps the electronic Hamiltonian a smooth function of the nuclear positions, which the adiabatic construction requires and which point charges violate at coincidence [23, Sect. 2]; the nuclear charge radius changes nothing else in this book. The gap excludes metals and level crossings, and the temperature condition keeps the initial state in the electron-following sector. The theorem is proved for a finite crystal. Its extension

to the infinite crystal, with constants that do not grow with the number of nuclei, is open and is stated with the other open problems at the end of this section. The theorem does not depend on which operators O_a appear: the final state decides the sector, not the coupling.

8.6.2 Densities in each sector

Inside the electron-following sector an electronic operator acts on the nuclei alone. The electrons sit in $\Psi_0(r;u)$, and the electron density reaches the sector only through its expectation value at each configuration,

$$\mathcal{P}_{\mathrm{ad}}\, n_e(\boldsymbol{q})\, \mathcal{P}_{\mathrm{ad}} \simeq \bar{\rho}_e(\boldsymbol{q};u) \equiv \langle \Psi_0(u) | n_e(\boldsymbol{q}) | \Psi_0(u) \rangle, \tag{8.51}$$

up to corrections of relative order κ_{BO}^2 from the tilt of the state. The function $\bar{\rho}_e(\boldsymbol{q};u)$ is the Fourier component of the ground-state electron density of the crystal frozen in configuration u, an operator on nuclear wavefunctions like the nuclear density $n_\kappa(\boldsymbol{q})$ itself. At $u=0$ it is the static density, nonzero only at reciprocal-lattice vectors: Bragg scattering. Its term linear in u creates one phonon, and its higher terms several. A dark-matter particle that couples only to electrons therefore excites phonons, through the electrons that are carried along by the vibrating nuclei. If each nucleus dragged a rigid cloud with form factor $f_\kappa(\boldsymbol{q})$, we would have $\bar{\rho}_e = \sum_I f_{\kappa(I)}(\boldsymbol{q})\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$, and the electron channel would see the lattice with the weight $|f_\kappa|^2$ of the nuclear one. In a real crystal the cloud deforms. At long wavelength and zero macroscopic field, the electrons that follow nucleus κ carry the charge $Z_\kappa \mathbf{1} - Z_\kappa^*$, a tensor, so that nucleus and electrons together carry the Born effective charge of Eq. (5.49). In silicon $Z_\kappa^* = 0$ and all fourteen electrons follow; in GaAs a fraction is left behind, and the lattice is seen by a dark photon through Z_κ^*, screened by ε_∞ (Sects. 8.4 and 11.2).

The electronic sectors are reached through the transition densities between adiabatic states. A final state $\Psi^{\mathrm{nuc}\prime}\Psi_j$ on the excited surface j, with $\Psi^{\mathrm{nuc}\prime}$ one of its vibrational states, is reached from $\Psi^{\mathrm{nuc}}\Psi_0$ with the amplitude

$$\begin{aligned} \langle \Psi^{\mathrm{nuc}\prime}\Psi_j | n_e(\boldsymbol{q}) | \Psi^{\mathrm{nuc}}\Psi_0 \rangle &= \langle \Psi^{\mathrm{nuc}\prime} | \rho_{j0}(\boldsymbol{q};u) | \Psi^{\mathrm{nuc}} \rangle, \\ \rho_{j0}(\boldsymbol{q};u) &= \rho_{j0}(\boldsymbol{q};0) + \sum_{\boldsymbol{R}\kappa} \boldsymbol{u}_{\boldsymbol{R}\kappa} \cdot \nabla_{\boldsymbol{R}\kappa} \rho_{j0} + \cdots , \end{aligned} \tag{8.52}$$

where $\rho_{j0}(\boldsymbol{q};u) = \langle \Psi_j(u) | n_e(\boldsymbol{q}) | \Psi_0(u) \rangle$ is the transition density of the frozen crystal. The constant term is the Condon approximation of Sect. 5.6. It weights the vibrational states by the Franck–Condon overlaps $|\langle \Psi^{\mathrm{nuc}\prime} | \Psi^{\mathrm{nuc}} \rangle|^2$: the zero-phonon line, in $\mathcal{P}_0$, with the weight $\mathrm{e}^{-S_{\mathrm{HR}}}$ of Eq. (5.63), and the replicas, in $\mathcal{P}_+$. For an electron–hole pair in a crystal the surfaces of the ground and excited states are displaced by almost nothing, since one pair spread over many cells exerts a negligible force on each nucleus, and the zero-phonon line carries nearly all of the Condon term. The linear term, the Herzberg–Teller term of molecular spectroscopy, creates a phonon even without a displacement. In a crystal it does more: at $u \neq 0$ the frozen lattice is not periodic, and ρ_{j0} connects pairs of different crystal momentum that the static lattice keeps apart. This is phonon-assisted electron excitation in adiabatic form, the term that the special-displacement supercell of Sect. 6.5 computes. Being adiabatic, it misses the phonon energy in the final state, a correction of relative order $\omega_{\boldsymbol{Q}\nu}/\delta_{\mathrm{off}}$ that the electron–phonon form Eq. (6.42) restores.

The nuclear density cannot reach the electronic sectors at all in the Born–Oppenheimer approximation. Its amplitude between the same two states is $\int \mathrm{d}u\, \Psi^{\mathrm{nuc}\prime *}(u)\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}\, \Psi^{\mathrm{nuc}}(u)$ $\langle \Psi_j(u) | \Psi_0(u) \rangle$, and the electronic overlap vanishes at every u because the adiabatic states are orthogonal. The amplitude comes from the tilt of the initial and final states by the

Table 8.5: The three final-state sectors for the densities of electrons and nuclei and their interference, with the physics in each cell, the literature formula it recovers, and the section that treats it. Electron-following: electrons in the adiabatic ground state, $\mathcal{P}_{\mathrm{ad}}$. Zero-phonon electronic: an electronic excitation with no net phonon, $\mathcal{P}_0$. Phonon-dressed electronic: an electronic excitation with a nonzero net change of the phonon number, or a free recoil, $\mathcal{P}_+$.

	Electron-following	Zero-phonon electronic	Phonon-dressed electronic
S_{ee}	electrons carried by the vibrating nuclei: phonon peaks in the electron channel; dark-photon single-phonon rate with Z^*_κ and ε_∞ [18, Eqs. (19)–(23)] (Sect. 11.2)	zero-phonon line: the clamped-nuclei response with renormalized bands; crystal-form-factor rate [24, Eq. (3.13)] and loss-function rate [19, Eqs. (6) and (13)] (Sects. 10.1 and 10.2)	phonon sideband: phonon-assisted electron recoil and absorption, exciton–phonon replicas; Eq. (6.42) at finite $\boldsymbol{q}$ (Chap. 13)
$S_{\kappa\kappa}$	single phonons, multi-phonons, free nuclear recoil [15, Eqs. (5)–(8)] (Sects. 11.1, 11.3 and 11.4)	zero-phonon Migdal: the electrons excited, the lattice left in its state (Chap. 12)	Migdal with phonons or free recoil; atomic boost [25, Eqs. (55)–(58)], crystal form [26, Eqs. (3) and (6)] (Sects. 12.2 to 12.4)
$S_{e\kappa}$	Born-charge interference: electrons and nuclei moving together in a phonon (Sects. 8.4 and 11.2)	interference of zero-phonon electron recoil and zero-phonon Migdal, for couplings to both species (Sect. 12.5)	interference of phonon-assisted electron recoil and Migdal with phonons, for couplings to both species (Sects. 12.5 and 17.4)

superadiabatic correction. At its first order (Appendix D.28),

$$\langle \widetilde{\Psi^{\mathrm{nuc}\prime}\Psi}_j | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I} | \widetilde{\Psi^{\mathrm{nuc}}\Psi}_0 \rangle = -\mathrm{i}\,\frac{\boldsymbol{q}\cdot\boldsymbol{d}^I_{j0}}{M_\kappa(E_j - E_0)}\,\langle \Psi^{\mathrm{nuc}\prime} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I} | \Psi^{\mathrm{nuc}} \rangle, \tag{8.53}$$

with $\boldsymbol{d}^I_{j0}$ the derivative coupling with respect to the position of nucleus I of species κ, and the tildes marking the tilted states. Two terms proportional to the gradient of the nuclear wavefunctions cancel between the initial and the final tilt, and only the momentum $\boldsymbol{q}$ of the kick survives. The electronic factor is the sudden amplitude $\boldsymbol{v}_N \cdot \boldsymbol{d}_{j0}/(\mathrm{i}\Delta E_j)$ of Eq. (6.6), with $\boldsymbol{v}_N = \boldsymbol{q}/M_\kappa$: the electrons see a nucleus that suddenly moves. The nuclear factor is the kick of Sect. 5.6, with its Debye–Waller and multiphonon structure. The Migdal effect is purely nonadiabatic. Its zero-phonon term lies in $\mathcal{P}_0$, and the terms in which the kick also creates phonons, or a free recoil, lie in $\mathcal{P}_+$. In the truncation label of Sect. 8.5, which counts powers in the rate, a Migdal rate carries the derivative coupling squared and sits at $n_B = 2$; its interference with an electron recoil to the same final state is linear in it, at $n_B = 1$.

Table 8.5 collects the cells. Each literature formula of the table computes one cell, and the rates of Part II are added cell by cell. Two cells of the bottom row have no standard formula. They are nonzero whenever the dark matter couples to electrons and nuclei at once, as a dark photon does, and the next subsection shows that they cannot be dropped as small without violating a sum rule.

In practice each sector comes from a different chain of codes (Appendices E.4 to E.8), and the three must use one set of phonon energies, vertices and band energies, or their sum is the response of no single Hamiltonian.

8.6.3 Partial sum rules

The first-moment identity Eq. (4.28) holds sector by sector, because a projector that commutes with H can be carried through the double commutator. Restricted to the electron-following sector, both densities are functions of the nuclear positions, and the only term of the Hamiltonian that fails to commute with them is the kinetic energy of the nuclei, exactly as for n_κ in Sect. 4.3. For the electron density, Eq. (8.51) turns the double commutator into a sum over nuclei of squared gradients (Appendix D.28),

$$\mathscr{M}^1_{ee}\big|_{\mathrm{ad}}(\boldsymbol{q}) = \frac{1}{\mathscr{V}}\Big\langle \sum_{\boldsymbol{R}\kappa} \frac{|\nabla_{\boldsymbol{R}\kappa}\bar{\rho}_e(\boldsymbol{q};u)|^2}{M_\kappa}\Big\rangle [1+O(\kappa_{\mathrm{BO}}^2)] \;\simeq\; q^2 \sum_\kappa \frac{n_\kappa\,|f_\kappa(\boldsymbol{q})|^2}{M_\kappa}, \tag{8.54}$$

where the second form holds for rigid clouds. Measured against the f-sum $n_e q^2/m_e$, it is of relative size $Z_\kappa m_e/M_\kappa$ at small q, where the whole cloud follows. This is 2.73×10^{-4} for silicon with all fourteen electrons, the number that Sect. 4.3 found for the nuclear share of the f-sum of the total charge, and for the same reason: either way, Z_κ units of charge are attached to a mass M_κ. In powers of the small parameter it is $\kappa_{\mathrm{BO}}^4 = m_e/M$. The two electronic sectors together therefore carry the f-sum up to a part in 10^3, and the f-sum test of a clamped-nuclei calculation (Sect. 4.3) applies to them unchanged.

For the nuclear density the electronic sectors carry the Migdal events, and their share follows from Eq. (8.53). Summed over the vibrational states $\Psi^{\mathrm{nuc}\prime}$, which form a complete set, the probability that a kick of nucleus I leaves the electrons excited is

$$P_{\mathrm{Mig}}(\boldsymbol{q}) = \sum_{j\neq0} \frac{|\boldsymbol{q}\cdot\boldsymbol{d}^I_{j0}|^2}{M_\kappa^2 (E_j-E_0)^2}, \tag{8.55}$$

at leading order, the sudden probability of Sect. 6.1 with $\boldsymbol{v}_N = \boldsymbol{q}/M_\kappa$. Each such event deposits the electronic energy $E_j - E_0$ plus, on average, the recoil energy $E_R = q^2/2M_\kappa$ of Eq. (5.61). With $\boldsymbol{d}_{j0} = \boldsymbol{F}_{j0}/(E_0-E_j)$ from Eq. (5.11), the electronic part adds up to $\boldsymbol{q}\cdot\delta M_\kappa\cdot\boldsymbol{q}/2M_\kappa^2$, and the partial sum rules per nucleus read (Appendix D.28)

$$\begin{aligned}
&\mathscr{M}^1_{\kappa\kappa}\big|_{0+} = n_\kappa\Big[P_{\mathrm{Mig}}\,\frac{q^2}{M_\kappa} + \frac{\boldsymbol{q}\cdot\delta M_\kappa\cdot\boldsymbol{q}}{M_\kappa^2}\Big], \qquad \mathscr{M}^1_{\kappa\kappa}\big|_{\mathrm{ad}} = n_\kappa\,(1-P_{\mathrm{Mig}})\;\boldsymbol{q}\cdot(M_\kappa+\delta M_\kappa)^{-1}\cdot\boldsymbol{q},\\
&\delta M_\kappa = 2\sum_{j\neq0}\frac{\mathrm{Re}[\boldsymbol{F}_{0j}\,\boldsymbol{F}_{j0}]}{(E_j-E_0)^3}, \qquad \boldsymbol{F}_{j0} = \langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}V_{eN}|\Psi_0\rangle,
\end{aligned} \tag{8.56}$$

to first order in P_{Mig} and $\delta M_\kappa/M_\kappa$, with 0+ marking the two electronic sectors together. The tensor δM_κ is the *electron drag mass*: a nucleus that moves slowly drags part of its electron cloud along and vibrates as if it were heavier, the effect that the solvable model of Appendix D.13 found at order m_e/M. The same term appears in the second-order Born–Oppenheimer Hamiltonian of Panati, Spohn and Teufel [22, Eq. (40), last term]. For an isolated atom the sum is fixed by the Thomas–Reiche–Kuhn rule Eq. (4.37), and $\delta M = N_e m_e$: the atom recoils with its electrons, as it must (Appendix D.28). In a crystal the f-sum rule fixes the sum of the drag masses over all nuclei, exactly $14\,m_e$ per silicon atom (Eq. (12.42)), which gives $\delta M_\kappa/M_\kappa \approx 2.7\times10^{-4}$. A kick on one nucleus probes instead the on-site element, which no sum rule fixes: Sect. 12.5.2 evaluates its valence part as $18.6\,m_e$, between 11.8 and $24.0\,m_e$ in the approximations behind it.

Eq. (8.56) refines Eq. (5.61). That rule says that the mean energy given to the lattice is E_R on any potential-energy surface, and it is a statement within the Born–Oppenheimer approximation. Beyond it the total first moment is still E_R per nucleus, by Eq. (4.34), but the lattice receives less. The electrons take $P_{\mathrm{Mig}}E_R + \boldsymbol{q}\cdot\delta M_\kappa\cdot\boldsymbol{q}/2M_\kappa^2$, and the lattice the

remainder, which is the recoil of a nucleus of mass $M_\kappa + \delta M_\kappa$ in the events that leave the electrons alone. In silicon both corrections are below 10^{-3} of E_R for recoil energies up to about 16 eV, which covers the phonon and multiphonon regime, far below the accuracy of any multiphonon calculation; at 500 eV the weight $1 - P_{\mathrm{Mig}}$ is a correction of 3 %. The bookkeeping, however, is exact, and Sect. 12.5 carries it through.

The interference of electrons and nuclei is the sharpest case. Its total first moment vanishes, Eq. (4.34), but its sectors do not. In the electron-following sector the same steps that led to Eq. (8.54) give

$$\mathscr{M}^1_{e\kappa}\big|_{\mathrm{ad}}(\boldsymbol{q}) = \frac{1}{\mathscr{V}}\Big\langle \sum_{I\in\kappa} \frac{\nabla_I \bar{\rho}_e(\boldsymbol{q};u)^\dagger \cdot \mathrm{i}\boldsymbol{q}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}}{M_\kappa} \Big\rangle \simeq \frac{q^2\, \boldsymbol{n}_\kappa\, f_\kappa(\boldsymbol{q})^*}{M_\kappa}, \qquad \mathscr{M}^1_{e\kappa}\big|_{0+} = -\,\mathscr{M}^1_{e\kappa}\big|_{\mathrm{ad}}, \tag{8.57}$$

to leading order in κ_{BO}. Electrons that follow the nucleus move in phase with it and carry a positive energy-weighted interference. The electronic sectors carry the same weight with the opposite sign. This is the sign change that Sect. 4.3 required of the interference spectrum, now located: the electron-following and the electronic parts of the bottom row of Table 8.5 are mirror images in their first moment. A model that keeps the Born-charge interference and drops the electronic cells of that row violates the sum rule by the full Born-charge term.

These rules turn into four prescriptions for adding rates. A multiphonon $S_{\kappa\kappa}$ normalized to the full nuclear f-sum, added to a Migdal rate, counts the electronic share twice; the electron-following part must be depleted by P_{Mig} and computed with the mass $M_\kappa + \delta M_\kappa$. A resonant sideband is never added to a zero-phonon line that already contains its cascade (Sects. 2.4 and 6.5); the next subsection makes this precise. A measured loss function or optical conductivity contains both electronic sectors, and a computed sideband is never added to it. And when the dark matter couples to both species, the electron and nuclear amplitudes to the same final state interfere in every sector, with the interference bounded by the positivity of Eq. (4.20).

8.6.4 Zero-phonon line and phonon sideband

The superadiabatic projector commutes with H to the order of Eq. (8.47) because a gap separates the electronic states. Nothing comparable, however, separates $\mathcal{P}_0$ from $\mathcal{P}_+$. A zero-phonon pair at energy E and a one-phonon state at the same energy are both electronic excitations, and the electron–phonon coupling H_{ep} of Sect. 6.2 turns one into the other. Whether the split means anything depends on where the energy deposited ω lies relative to the zero-phonon states at the same momentum transfer.

Below them the answer is simple. Take ω inside the off-shell window, below the bottom $E_{\mathrm{min}}(q)$ of the zero-phonon pair continuum by a margin δ_{w} large compared with the phonon energies, the widths and $|g|$ (the window is made precise in Sect. 13.6). No zero-phonon state exists at that energy, and every final state contains a phonon. An exact final state $|\tilde{f}\rangle$ is a one-phonon state $|f\rangle$ dressed by a virtual admixture of zero-phonon pairs m, each off shell by at least δ_{w},

$$|\tilde{f}\rangle = |f\rangle + \sum_m |m\rangle\, \frac{\langle m|H_{\mathrm{ep}}|f\rangle}{E_f - E_m} + \cdots, \qquad \big\|\mathcal{P}_0^{(0)}|\tilde{f}\rangle\big\|^2 = O(\theta_{\mathrm{ep}}),$$

with $\mathcal{P}_0^{(0)}$ the zero-phonon projector of the uncoupled reference and θ_{ep} the off-shell parameter Eq. (8.40). The perturbation series converges, since no denominator comes close to zero, and the state can be followed continuously from $g = 0$. Every reasonable definition of its label agrees: the phonon number of its leading component, the number obtained after a Schrieffer–Wolff transformation that removes H_{ep} to a given order, and the number of phonons

that eventually leave. They differ by the weight $O(\theta_{\mathrm{ep}})$ of the virtual admixture, and the whole response at ω belongs to $\mathcal{P}_+$ to that accuracy (Appendix D.28). In silicon, at the absorption edge, $\theta_{\mathrm{ep}} = 0.010$ to 0.016 (Sect. 8.5). Note that the bare projector would not do. Inserted in Eq. (8.49), $\mathcal{P}_0^{(0)}$ would pick out the virtual admixture through which the whole phonon-assisted amplitude passes, and assign to the zero-phonon sector a weight as large as the rate itself. Sector projectors must commute with H, and here this means that they are dressed.

Above $E_{\min}(q)$ the zero-phonon pair is real, and the label becomes a matter of time. A pair m created at energy ω lies inside the continuum of one-phonon states f and decays into it by emitting a phonon. Its amplitude follows from the level-and-band model of Worked example 4.12, with the zero-phonon pair as the level and the one-phonon states as the band. Let $\rho_m = \langle m|O_b|i\rangle$ be the zero-phonon vertex, $g_{fm} = \langle f|H_{\mathrm{ep}}|m\rangle$ the electron–phonon matrix element built from the $g_{mn\nu}$ of Eq. (6.11), and $g_{fm}\mathcal{A}_f^{\mathrm{off}}$ the amplitude with which the probe reaches f by every path that does not pass through m on shell: the phonon emitted first, hole orderings, and nonresonant pairs. The amplitude into f at $\omega = E_f$ is then exactly (Appendix D.28)

$$A_f(\omega) = g_{fm}\Big[\mathcal{A}_f^{\mathrm{off}} + \mathsf{G}_m(\omega)\big(\rho_m + \Upsilon_m(\omega)\big)\Big], \qquad \mathsf{G}_m(\omega) = \frac{1}{\omega - E_m - \Sigma_m(\omega)}, \tag{8.58}$$

with the self-energy of the pair $\Sigma_m(\omega) = \sum_f |g_{fm}|^2/(\omega - E_f + \mathrm{i}0^+)$ and the vertex correction $\Upsilon_m(\omega) = \sum_f |g_{fm}|^2\mathcal{A}_f^{\mathrm{off}}/(\omega - E_f + \mathrm{i}0^+)$. The width of the pair is $\Gamma_m(\omega) = -2\,\mathrm{Im}\,\Sigma_m(\omega)$, its part due to phonon branch ν is $\Gamma_m^{(\nu)}$, and the quasiparticle energy $\tilde{E}_m$ and residue $z_{\mathrm{qp}}(m)$ follow from Σ_m as in Eq. (4.83). The response is $(2\pi/\mathcal{V})\sum_f |A_f|^2\delta(\omega - E_f)$. Near $\tilde{E}_m$ the squared propagator $|\mathsf{G}_m|^2$ is a Lorentzian of width Γ_m, and Eq. (2.60) splits it into a delta function and a finite part. Assigning the delta function to the zero-phonon sector and the rest to the phonon-dressed one is the convention we adopt.

Dyson convention

For Γ_m small compared with the detuning δ_{F}, the distance from $\tilde{E}_m$ to the nearest structure of the one-phonon continuum,

$$\begin{aligned} S^0(\boldsymbol{q},\omega) &= \frac{2\pi}{\mathcal{V}}\sum_m z_{\mathrm{qp}}(m)\,|\tilde{\rho}_m|^2\,\delta(\omega - \tilde{E}_m), \\ S^+(\boldsymbol{q},\omega) &= \frac{2\pi}{\mathcal{V}}\sum_m\sum_f |g_{fm}|^2\,\delta(\omega - E_f)\Big[|\rho_m|^2\,\mathrm{fp}\,\frac{1}{(\omega - \tilde{E}_m)^2} \\ &\qquad + 2\,\mathrm{Re}(\mathcal{A}_f^{\mathrm{off}*}\rho_m)\,\mathrm{PV}\,\frac{1}{\omega - \tilde{E}_m} + |\mathcal{A}_f^{\mathrm{off}}|^2\Big], \end{aligned} \tag{8.59}$$

where $|\tilde{\rho}_m|^2 = |\rho_m|^2 + 2\,\mathrm{Re}[\rho_m^*\,\mathrm{PV}\,\Upsilon_m(\tilde{E}_m)]$ contains the virtual-phonon correction of the vertex. Inside the off-shell window no $\tilde{E}_m$ lies at ω, the delta term is absent, and S^+ is the ordinary second-order rate.

The delta part is a real zero-phonon pair, handed as such to the cascade of Part III, which lets it emit its first phonon (Sects. 8.7 and 18.1). The finite-part and principal-value terms are the genuine sideband, a pair and a phonon created together. The finite part is negative against a narrow positive test function, as Sect. 2.4 found, so S^+ alone is not a positive spectrum near $\tilde{E}_m$. Near a resonance the two sectors need not be separately positive; their sum is the response, which is.

The sum is what the convention protects. Against any test function smooth on the scale δ_{F}, a detector resolution or the integral over $\boldsymbol{q}$ and the velocity distribution, $S^0 + S^+$ reproduces the exact response up to terms of order g^4 and $\Gamma_m^2/\delta_{\mathrm{F}}^2$ (Appendix D.28). The step that makes the weights add up is a per-pair zeroth-moment identity. Differentiating the Kramers–Kronig form of $\mathrm{Re}\,\Sigma_m$ with Eq. (C.31) gives $\partial_\omega \mathrm{Re}\,\Sigma_m = -(2\pi)^{-1}\,\mathrm{fp}\!\int \mathrm{d}\omega'\,\Gamma_m(\omega')/(\omega'-\omega)^2$, and since $z_{\mathrm{qp}} = 1 + \partial_\omega \mathrm{Re}\,\Sigma_m + O(g^4)$,

$$z_{\mathrm{qp}}(m) + \frac{1}{2\pi}\,\mathrm{fp}\!\int \mathrm{d}\omega'\,\frac{\Gamma_m(\omega')}{(\omega' - \tilde{E}_m)^2} = 1 + O(g^4). \tag{8.60}$$

The weight that the zero-phonon line loses to its residue is exactly the finite-part weight of the sideband, the first term of S^+ integrated over ω. The level of Worked example 4.12 shows that the residue is not a probability. There $z_{\mathrm{qp}} = 1.0064$, and the finite part of the flat band is $-2\Gamma/(\pi E_{\mathrm{w}}) = -0.0064$: the line carries more than the full weight and the off-shell part returns the excess. Inside a continuum whose width grows with energy, the same happens, and a negative finite-part weight in a computed sideband is no error. The identity concerns the self-energy only; the vertex correction in $|\tilde{\rho}_m|^2$ redistributes weight that no zeroth-moment rule fixes, and there is no global zeroth-moment rule for n_e.

The split itself is a convention, and its ambiguity has a size. Three choices are natural. The first is the power of the residue, which a time-domain definition by the surviving amplitude raises from z_{qp} to z_{qp}^2. The second is the shape of the line, a Lorentzian or any normalized kernel no wider than Γ_m in place of the delta function. The third is a window $|\omega - \tilde{E}_m| < x_{\mathrm{w}}$, with $\Gamma_m \ll x_{\mathrm{w}} \le \delta_{\mathrm{F}}$, all of whose weight is assigned to the line. Any two conventions that differ from Eq. (8.59) only in these ways assign zero-phonon weights w_0 in a bin around $\tilde{E}_m$ that differ by (Appendix D.28)

$$\frac{|\Delta w_0|}{w_0} \le \frac{5}{2\pi}\,\frac{\hat{\Gamma}_m}{\delta_{\mathrm{F}}} + \frac{\hat{\Gamma}_m}{\pi x_{\mathrm{w}}} + \Big(1 + \frac{|\mathcal{A}^{\mathrm{off}}|\,\delta_{\mathrm{F}}}{|\rho_m|}\Big)^2 \frac{\hat{\Gamma}_m x_{\mathrm{w}}}{\delta_{\mathrm{F}}^2}, \tag{8.61}$$

with $\hat{\Gamma}_m$ the largest width within δ_{F} of the line. At $x_{\mathrm{w}} \approx \delta_{\mathrm{F}}$ the first two terms add up to $1.1\,\hat{\Gamma}_m/\delta_{\mathrm{F}}$, and the bound is of order $\hat{\Gamma}_m/\delta_{\mathrm{F}}$ times a number of order one when the off-shell amplitude is no larger than $|\rho_m|/\delta_{\mathrm{F}}$. The split is sharp where the pair lives long compared with $1/\delta_{\mathrm{F}}$ and becomes meaningless as Γ_m approaches δ_{F}. The bound does not extend to every convention. Defined by the number of phonons present after an infinite time, every pair inside the continuum eventually decays, the zero-phonon sector is empty, and the split moves by an amount of order one. The sector label belongs to a time window between $1/\delta_{\mathrm{F}}$ and $1/\Gamma_m$, and the Dyson convention is the choice that makes the totals and the cascade consistent.

Two mixed conventions are common and wrong. A Lorentzian zero-phonon line added to the finite-part sideband counts the off-shell weight $(\Gamma_m/2\pi)|\rho_m|^2\,\mathrm{fp}(1/x^2)$ twice, an error of order g^2. A Lorentzian line added to the resummed sideband $|x + \mathrm{i}\Gamma_m/2|^{-2}$ counts the delta function twice, an error of order one; it is the double counting of the cascade in Sect. 2.4. Within a few widths of $E_{\mathrm{min}}(q)$, or of a singularity of the joint density of states, no test function is smooth on the scale δ_{F}, and only totals computed from the unexpanded amplitude Eq. (8.58) are quoted (Sect. 13.6).

There is one more term, and it changes no total. When the pair decays, its first decay product can interfere with the same product made directly, and that shifts which channel the first decay goes into. The delta function receives, besides $z_{\mathrm{qp}}|\tilde{\rho}_m|^2$, a contribution from the interference between the off-shell path $\mathcal{A}_f^{\mathrm{off}}$ and the decay of the real pair. Summed over the final states of one decay channel ν, the zero-phonon events that decay first into channel ν

occur with the probability (Appendix D.28)

$$\frac{\Gamma_m^{(\nu)}}{\Gamma_m^{\mathrm{tot}}}(1+\bar{c}^{\mathrm{F}}-c_\nu^{\mathrm{F}}), \qquad c_\nu^{\mathrm{F}}=\frac{\Gamma_m^{\mathrm{tot}}\,\mathrm{Im}(\rho_m^*\mathcal{A}_\nu^{\mathrm{off}})}{|\rho_m|^2}, \qquad \bar{c}^{\mathrm{F}}=\sum_\nu \frac{\Gamma_m^{(\nu)}}{\Gamma_m^{\mathrm{tot}}}\,c_\nu^{\mathrm{F}}, \tag{8.62}$$

with ν running over every decay channel of the pair, each phonon branch and the others, such as impact ionization, $\mathcal{A}_\nu^{\mathrm{off}}$ the off-shell amplitude on the shell of channel ν, and $\Gamma_m^{\mathrm{tot}}=\sum_\nu \Gamma_m^{(\nu)}$ the total width. The vertex correction Υ_m then sums over all channels as well. The deviations $\bar{c}^{\mathrm{F}}-c_\nu^{\mathrm{F}}$ of the *Fano coefficients* from their mean average to zero with the weights $\Gamma_m^{(\nu)}$, because ν runs over all channels, so they change no total and no number of zero-phonon events; the coefficients themselves need not. They change the branching of the first decay, at relative order $\Gamma_m|\mathcal{A}^{\mathrm{off}}|/|\rho_m|\sim\Gamma_m/\delta_{\mathrm{F}}$, which is the order of the cascade itself. This is the interference of a discrete state with a continuum that Fano analyzed for atomic autoionization [27]. It has nothing to do with the Fano factor of the number of electron–hole pairs (Sect. 18.2), which measures the variance of a count. An incoherent cascade cannot generate the term, since it is a coherence between production and decay, and the response calculation must supply it.

In many crystals the Fano term vanishes. Let the target be invariant under inversion combined with time reversal, and let its number of electrons be even. The antiunitary operator PT then squares to one, and every state of the electrons and phonons can be chosen invariant under it, with the phases of the phonon polarization vectors fixed to match. The density $n_e(\boldsymbol{q})$ and H_{ep} are invariant too, so every matrix element and every off-shell amplitude is real, and $\mathrm{Im}(\rho_m^*\mathcal{A}_\nu^{\mathrm{off}})=0$ (Appendix D.28). Silicon and germanium satisfy the hypothesis, with or without spin–orbit coupling, at the level of one-body self-energies, whose level-shift matrix is a multiple of the identity on each degenerate zero-phonon manifold, a Kramers pair or an irreducible representation, so that the invariant basis also diagonalizes it. Level shifts from the electron–hole vertex can split such a manifold in a basis that is not invariant; then only the sum of the Fano terms over the branches vanishes, not each one. GaAs has no center of inversion, and its Fano coefficients are generically nonzero; Appendix E.14 specifies the calculation that sizes them.

8.6.5 Targets without a gap

Aluminum has electron–hole pairs at every energy down to zero, and its phonons, up to 40 meV (Sect. 6.4), sit inside that continuum. The adiabatic construction of Eq. (8.47) needs a gap between the electronic ground state and the rest at every nuclear configuration, and a metal has none: as Sect. 5.1 found, the expansion in adiabatic states fails at the Fermi surface. A superconductor has a gap, but $2\Delta_{\mathrm{BCS}}=0.36\,\mathrm{meV}$ in aluminum lies far below the phonon energies, and the adiabatic hierarchy is reversed: the electrons are the slow degrees of freedom near the Fermi surface. A Dirac semimetal is gapless at its nodes. Magnetic insulators are a fourth case. Their charge excitations are gapped, but at clamped nuclei the magnons are electronic excitations whose gaps, set by the magnetic anisotropy, lie below most phonon energies, so that no gap separates them from the lattice. The construction then goes through with the whole low-energy spin manifold in place of the electronic ground state, so that magnons and phonons share the electron-following sector (the magnetic targets of Chap. 14).

For the gapless targets the sectors must count quasiparticles instead. The low-energy excitations of a metal are the quasiparticles of Sect. 4.7 and the phonons, and Migdal's theorem (Sect. 6.3) separates them with the small parameter $\omega_{\mathrm{D}}/E_{\mathrm{F}}\sim\kappa_{\mathrm{BO}}^2$. Suppose a dressing unitary U_{qp}, of the kind that Landau's theory and the Migdal–Eliashberg equations construct order by order, maps bare electrons and phonons onto quasiparticles and dressed phonons with number operators $\hat{N}_{\mathrm{qp}}$ and $\hat{N}_{\mathrm{ph}}$. The joint spectral projectors $\mathcal{P}_{n,m}^{\mathrm{qp}}$ of $U_{\mathrm{qp}}\hat{N}_{\mathrm{qp}}U_{\mathrm{qp}}^\dagger$ and $U_{\mathrm{qp}}\hat{N}_{\mathrm{ph}}U_{\mathrm{qp}}^\dagger$,

with n and m the net numbers of quasiparticles and phonons created, are orthogonal and sum to one. They do not commute with H, since quasiparticles have a width Γ_{qp} and phonons decay into pairs. Split $H = H_{\mathrm{d}} + H_{\mathrm{off}}$, with $H_{\mathrm{d}} = \sum_{n,m} \mathcal{P}^{\mathrm{qp}}_{n,m} H \mathcal{P}^{\mathrm{qp}}_{n,m}$, and define the sector responses with H_{d}, which commutes with every $\mathcal{P}^{\mathrm{qp}}_{n,m}$. Each is then positive, and they sum exactly to the response of H_{d}. On a Gaussian bin of standard deviation $\Delta\omega_{\mathrm{bin}}$, the response of H from one initial state $|i\rangle$, without the factor $2\pi/\mathcal{V}$, differs from it by at most (Appendix D.28)

$$|C^{\mathrm{cross}}_{ab}| \le \frac{\mathfrak{h}_{\mathrm{off}}}{\pi\,\Delta\omega^2_{\mathrm{bin}}}\,\big\|O_a|i\rangle\big\|\,\big\|O_b|i\rangle\big\|, \qquad \mathfrak{h}_{\mathrm{off}} = \sup_{t'} \frac{\big\|H_{\mathrm{off}}\,\mathrm{e}^{-\mathrm{i}H_{\mathrm{d}}t'}\,O_b|i\rangle\big\|}{\big\|O_b|i\rangle\big\|}. \tag{8.63}$$

With the dressing in place, the bound needs only the few-quasiparticle state that the probe creates, not the whole crystal, and it holds for any orthogonal sectors; with the leakage of Eq. (8.47) in place of $\mathfrak{h}_{\mathrm{off}}$ it is the remainder of the decomposition theorem. Relative to a typical binned value, $\|O|i\rangle\|^2/\Delta\omega_{\mathrm{bin}}$, the cross terms are of order $\mathfrak{h}_{\mathrm{off}}/\Delta\omega_{\mathrm{bin}}$. For a state that decays by the golden rule into a continuum of width ΔE_{c}, $\mathfrak{h}^2_{\mathrm{off}} \approx \Gamma\,\Delta E_{\mathrm{c}}/2\pi$, which is much larger than Γ^2. The bound is therefore sufficient but loose: it guarantees a clean split on bins wide compared with $(\Gamma\,\Delta E_{\mathrm{c}})^{1/2}$, while the natural expectation, that bins wider than the width $\Gamma_{\mathrm{qp}} + \Gamma_{\mathrm{ph}}$ of the sector label suffice, is not proved. Without any dressing, $U_{\mathrm{qp}} = 1$, the bound fails: H_{off} acts on the filled Fermi sea as well as on the quasiparticles the probe creates, and this disconnected part makes $\mathfrak{h}_{\mathrm{off}}$ grow with the size of the crystal (Appendix D.28). What a bound has to control is the connected part, in which H_{off} acts on the probe's excitations. Its square is at most the largest eigenvalue of the band matrix of zeroth moments of the Fan–Migdal spectral function, whose diagonal entries are the sums $N^{-1}\sum |g_{mn\nu}|^2(2n_{\boldsymbol{Q}\nu}+1)$ of Sect. 6.3 without their energy delta function, and in aluminum it is estimated at about $0.2\,\mathrm{eV}$.

The sectors of a superconductor follow the counting of Sect. 7.2. A density or spin operator acting on the BCS state creates Bogoliubov quasiparticles in pairs, so $n \in \{0, 2\}$ at leading order. Phonons with no broken pair, $(0, m)$, are the analogue of the electron-following sector, and they are stable only below $2\Delta_{\mathrm{BCS}}$, since a phonon above it breaks a pair. Pair breaking, $(2, 0)$, computed with the conserving response of Sect. 7.2, is the analogue of the zero-phonon sector, and phonon-assisted pair breaking, $(2, m \ge 1)$, that of the phonon-dressed one. The collective modes leave no sub-gap line in the density response: the phase mode is pushed to the plasma frequency, and the amplitude mode is not excited by the density in a particle–hole-symmetric metal (Sect. 7.2). Assigning these pair composites to a sector is a convention, not a consequence. In a normal metal n counts electron–hole pairs, and a phonon decays into pairs at the rate $\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}$ of Eq. (6.32); the sectors $(0, m)$ and $(2, m-1)$ mix within that width. In a Dirac semimetal the spectrum is linear and the effective fine-structure constant $\tilde{\alpha}$ of Sect. 7.3 is dimensionless, so the only energy scale near the node is ω itself. By dimensional analysis the quasiparticle width is then ω times a function of $\tilde{\alpha}$, which starts at $\tilde{\alpha}^2$ because a decay needs two Coulomb vertices. The split is therefore meaningful only for $\tilde{\alpha}^2 \ll 1$, and near $\tilde{\alpha} \sim 1$ only totals are quoted. The totals are safe in every case if the response is conserving. The f-sum then holds for the sum of the sectors, while the share of each sector is not fixed, and a nonconserving pair-breaking response, such as the bare bubble that violates the f-sum in Sect. 7.2, cannot be paired with any phonon sector.

Worked example 8.8: A two-level electron on a vibrating bond

We take the bond coordinate x of silicon from Worked example 5.8, an oscillator of frequency $\omega_0 = 64.1\,\mathrm{meV}$, zero-point length $\ell_0 = 0.048\,\text{Å}$ and mass $M/2$, and attach to it an electron with two levels, a lower one $|\mathrm{lo}\rangle$ and an upper one $|\mathrm{up}\rangle$, split by $\Delta_{\mathrm{tl}} = 1.17\,\mathrm{eV}$,

the gap of silicon. With $x = \ell_0(d + d^\dagger)$,

$$H = \omega_0\, d^\dagger d + \left[\Delta_{\mathrm{tl}} + g_{\mathrm{d}}(d + d^\dagger)\right] |\mathrm{up}\rangle\langle\mathrm{up}| + g_{\mathrm{n}}(d + d^\dagger)\left(|\mathrm{up}\rangle\langle\mathrm{lo}| + |\mathrm{lo}\rangle\langle\mathrm{up}|\right).$$

The diagonal coupling $g_{\mathrm{d}} = -\sqrt{2}\,\omega_0$ displaces the excited surface as in Worked example 5.8, with $S_{\mathrm{HR}} = 2$. The off-diagonal coupling $g_{\mathrm{n}} = 20\,\mathrm{meV}$, a model choice, mixes the two levels as the bond moves; it is the only source of nonadiabatic coupling, with the mixing $g_{\mathrm{n}}/\Delta_{\mathrm{tl}} = 0.017$ at the zero-point amplitude. The probes are the electronic transition operator $\sigma_x = |\mathrm{up}\rangle\langle\mathrm{lo}| + |\mathrm{lo}\rangle\langle\mathrm{up}|$ and a kick $\mathrm{e}^{\mathrm{i}qx}$ with $q\ell_0 = 1$, that is $q = 41.1\,\mathrm{keV}$, whose recoil energy on the coordinate, q^2/M, equals ω_0. We diagonalize H exactly in 160 oscillator states; 90 give the same numbers to the digits quoted.

Labels. The two probes reach forty eigenstates with a weight above 10^{-10}. Each has at least 0.996 of its norm on a single Born–Oppenheimer product state, $\Psi^{\mathrm{nuc}}_{0,m}(x)\Psi_0(x)$ on the ground surface or $\Psi^{\mathrm{nuc}}_{1,m}(x)\Psi_1(x)$ on the excited one, with m the vibrational level. We assign the ground surface to the electron-following sector, $m = 0$ on the excited surface to the zero-phonon sector and $m \geq 1$ there to the phonon-dressed one. The sector projectors are sums of exact eigenprojectors, they commute with H, and the split is exact.

The electron probe. The weight of σ_x, one in total, splits as 1.25×10^{-3}, 0.152 and 0.847 among the three sectors. The electron-following part is a line at 63.4 meV: the bond's ground-state surface, softened by the mixing, carries an electronic dipole $\langle\sigma_x\rangle_0 \simeq -2g_{\mathrm{n}}x/(\ell_0\Delta_{\mathrm{tl}})$ that follows the bond, with the one-phonon weight $(2g_{\mathrm{n}}/\Delta_{\mathrm{tl}})^2 = 1.17\times10^{-3}$ at leading order. The zero-phonon line, at 1.0463 eV, has the weight 0.152, the Franck–Condon factor of the two computed surfaces (0.1519 with the dependence of the transition dipole on x). It exceeds the $\mathrm{e}^{-2} = 0.135$ of Worked example 5.8 because the mixing softens the ground surface to 63.4 meV and stiffens the excited one to 65.5 meV. The first moment of σ_x is fixed by its double commutator at $-\langle(\Delta_{\mathrm{tl}} + g_{\mathrm{d}}x/\ell_0)\sigma_z\rangle = 1.169\,15\,\mathrm{eV}$, with $\sigma_z = |\mathrm{up}\rangle\langle\mathrm{up}| - |\mathrm{lo}\rangle\langle\mathrm{lo}|$, and the electron-following sector takes $8.32 \times 10^{-5}\,\mathrm{eV}$ of it, a fraction 7.1×10^{-5}. The leading-order estimate of that fraction, $4g_{\mathrm{n}}^2\omega_0/\Delta_{\mathrm{tl}}^3 = 6.4 \times 10^{-5}$, is also the drag ratio $\delta M/(M/2)$ of Eq. (8.56) for this model, for the reason that made $Z_\kappa m_e/M_\kappa$ appear twice in silicon.

The kick. The electrons are left excited with probability $P_{\mathrm{Mig}} = 3.94 \times 10^{-6}$, of which 0.38×10^{-6} is zero-phonon and 3.56×10^{-6} phonon-dressed. The boost formula Eq. (8.55), evaluated at $x = 0$ with the vertical gap, gives $(2\omega_0 q\ell_0 g_{\mathrm{n}}/\Delta_{\mathrm{tl}}^2)^2 = 3.51\times10^{-6}$; the difference, 12 %, is of the order $S_{\mathrm{HR}}\omega_0/\Delta_{\mathrm{tl}} = 0.11$ of the displacement of the excited surface, which the leading formula ignores. As $g_{\mathrm{n}} \to 0$ the probability falls as g_{n}^2: the kick reaches the electronic sectors through the nonadiabatic coupling alone. The electronic sectors take 4.72 µeV of the mean recoil energy of 64.1 meV, against $P_{\mathrm{Mig}}\, q^2/M + (q^2/M)\,\delta M/(M/2) = 0.25 + 4.10\,\mu\mathrm{eV}$ from Eq. (8.56), and the lattice keeps the rest. The difference of 8 % has the origin of the 12 % in P_{Mig}: with the displacement removed, $g_{\mathrm{d}} = 0$, the exact values are 3.52×10^{-6} and 4.34 µeV, within 0.3 % of the leading formulas. This is the correction to Eq. (5.61) in its smallest form.

The interference. The energy-weighted interference of the two probes is $(2.17 - 13.22\,\mathrm{i}) \times 10^{-4}\,\mathrm{eV}$ in the electron-following sector and exactly the opposite in the two electronic sectors together. Its total vanishes because σ_x commutes with $[H, \mathrm{e}^{\mathrm{i}qx}]$, the toy form of Eq. (8.57).

In this model every zero-phonon level is discrete and nothing decays, so the labels are exact and no convention is needed. Couple the excited level to a continuum of phonons, as the band of Worked example 4.12 couples to its level, and the zero-phonon line acquires a width; the split then follows Eq. (8.59), with the residue 1.0064 and the finite-part weight −0.0064 of that example.

The decomposition rests on three statements that are not yet theorems. First, the split between the zero-phonon and the phonon-dressed sectors outside the off-shell window, and the convention independence of the totals, are derived at the level of perturbation theory in g with distributional limits. A proof needs a limiting-absorption principle for the coupled electron–phonon Hamiltonian at fixed total crystal momentum, on the energies outside the window, uniform in the coupling and in the number of cells. The same estimate would sharpen the gapless bound Eq. (8.63) from $(\Gamma\,\Delta E_{\mathrm{c}})^{1/2}$ to the width Γ. Second, the constants of the superadiabatic construction grow with the number of nuclei, so Eq. (8.47) is proved for a finite crystal. The missing step is a superadiabatic generator built from exponentially localized terms, so that the leak per unit volume stays finite in the infinite crystal. Third, for gapless targets the existence of the dressing U_{qp} beyond perturbation theory, with bounds, is open, since without a gap there is no adiabatic theorem to provide it. The bin bound Eq. (8.63) needs this dressed vacuum and, with it, a bound on the connected part of $\mathfrak{h}_{\mathrm{off}}$ through a commutator estimate uniform in the number of cells. Until these are closed, the decomposition is exact for a finite insulator and controlled, with the stated error terms, for everything else; the rates of Part II are added cell by cell under exactly these conditions.

8.7 Final-state-resolved response

A dark-matter particle that deposits 10 eV in silicon leaves an electron and a hole with about 9 eV of kinetic energy between them. How many further pairs the two carriers make on their way down depends on how they share it. The rate of impact ionization rises steeply with the energy of a carrier, so a lopsided split makes more pairs than an even one, and an even split is the one that minimizes impact ionization [28, Sect. II A]. The step model of Eq. (1.43) assigns three pairs to every 10 eV deposit, and the rate Eq. (8.16) cannot do better, because the structure factor sums over all final states with the same ω and keeps no record of which one was made. The detector, however, counts the products of a particular final state. What must the response keep, so that the cascade of Chaps. 18 to 20 receives the state the dark matter actually produced, and nothing is counted twice or lost on the way?

The answer is a response matrix resolved by the class of the final state, $S_{ab}(\boldsymbol{q}, \omega; \Lambda)$, together with a rule for handing it over. We will define the classes, compute the resolved response of an insulator with independent particles and with screening, and find that screening, which changes the rate of silicon at a few keV by a factor of three, hardly changes which pairs are made. The hand-off itself has one condition that is easy to violate: the momentum transfer, the velocity of the dark matter and the operator structure of the coupling must all stay inside the sum over final states.

8.7.1 Final-state classes

The cascade needs more than the sector of Sect. 8.6 in which a final state lies. It needs the state itself, at the moment the dark matter has finished acting and before the target starts to relax. Two time scales bracket that moment. The excitation is dressed by the electrons and phonons around it within a time of order $1/\delta_{\mathrm{F}}$, with δ_{F} the distance from the state to the nearest structure of the continuum it can decay into, about a femtosecond for a pair in silicon. It decays, by emitting its first phonon or by impact ionization, within $1/\Gamma_m$, with Γ_m the width of Sect. 8.6. Between the two, the state is a well-defined *prompt* final state. This is the time window in which Sect. 8.6 found the split between $\mathcal{P}_0$ and $\mathcal{P}_+$ meaningful, and the Dyson convention Eq. (8.59) is the rule that labels a state inside it.

We group the prompt final states into *final-state classes* Λ, each described by the quantum numbers the cascade needs. The commonest are these. An electron–hole pair is labeled by

the conduction band c and crystal momentum $\boldsymbol{k}+\boldsymbol{q}$ of the electron and the valence band v and crystal momentum $\boldsymbol{k}$ of the hole. These fix the valleys the carriers occupy and their kinetic energies E_e and E_h, measured from the band edges as in Sect. 6.5. A spin label is added when the coupling acts on the spin. A pair with phonons carries in addition the branch, momentum and energy of each phonon. A core hole is labeled by its shell, since it relaxes by emitting an Auger electron (Sect. 10.6). In the electron-following sector the class is the phonon content, or, for a nucleus that recoils freely, its energy E_R and direction. Each class lies in one sector, so the classes refine the partition Eq. (8.48) rather than cut across it.

The plasmon of silicon is not a class of its own. At the zero of ε_1 the measured ε_2 is 0.224 (Worked example 4.10), interband pairs exist at the plasmon energy, and the collective oscillation decays into them. We will see in Sect. 8.7.2 that, in the random-phase approximation, the screened pair weights already contain the plasmon peak of the loss function, distributed over the pairs it decays into. A plasmon needs a label only where it cannot decay into pairs, below the cutoff q_c of the electron gas of Sect. 4.5, or where it can leave the target or convert into another excitation, as at the surfaces of films (Sect. 16.2). It then has its own decay, which the cascade must treat.

For each class we write P_Λ for the projector on its states. The classes are orthogonal and exhaust the final states, $P_\Lambda P_{\Lambda'} = \delta_{\Lambda\Lambda'} P_\Lambda$ and $\sum_\Lambda P_\Lambda = 1$. Inserting P_Λ around the energy delta function of Eq. (4.13), as Eq. (8.49) did for the sectors, gives the response into one class.

Final-state-resolved response

$$S_{ab}(\boldsymbol{q},\omega;\Lambda) = \frac{2\pi}{\mathcal{V}} \sum_i p_i \,\langle i|O_a(\boldsymbol{q})^\dagger\, P_\Lambda\, \delta(\omega + E_i - H)\, P_\Lambda\, O_b(\boldsymbol{q})|i\rangle,$$
$$\sum_\Lambda S_{ab}(\boldsymbol{q},\omega;\Lambda) = S_{ab}(\boldsymbol{q},\omega). \tag{8.64}$$

The sum rule holds exactly when each P_Λ commutes with H, and each $S_{ab}(\boldsymbol{q},\omega;\Lambda)$ is then a positive semidefinite matrix.

Both statements follow as for the sectors. When P_Λ commutes with H, the delta function can be written in eigenstates of H that each lie in one class, the cross terms between classes vanish, and each $S_{ab}(\boldsymbol{q},\omega;\Lambda)$ is a sum of outer products with positive weights. Since the prompt classes are defined on the window between $1/\delta_\mathrm{F}$ and $1/\Gamma_m$, they commute with H only to the accuracy of that window. Inside the off-shell window of Sect. 8.6, where the labels are unique to relative order θ_ep, each resolved response is positive. Near a zero-phonon resonance, the classes of $\mathcal{P}_+$ are not separately positive, and only their sum with the zero-phonon pair is, as Eq. (8.59) showed for the sectors. The delta part of the Dyson convention is a zero-phonon pair, weighted by its residue $z_\mathrm{qp}(m)$, and it enters $S_{ab}(\boldsymbol{q},\omega;\Lambda)$ as such. That this pair later emits a phonon is the first step of the cascade, not a second final state.

The rate into one class follows from the steps that gave Eq. (8.15), with P_Λ inserted before and after the delta function. For one dark-matter particle of velocity $\boldsymbol{v}$,

$$\Gamma(\boldsymbol{v};\Lambda) = \int \frac{\mathrm{d}^3 q}{(2\pi)^3} \sum_{a,b} \mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})\, S_{ab}(\boldsymbol{q},\omega_{\boldsymbol{q}};\Lambda), \tag{8.65}$$

and $\sum_\Lambda \Gamma(\boldsymbol{v};\Lambda) = \Gamma(\boldsymbol{v})$. For absorption, $\mathfrak{D}_{ab}$ is replaced by $g_a^{\phi\,*} g_b^\phi / 2\omega_\phi$ at the single point $(\boldsymbol{p}_\phi, \omega_\phi)$, as in Eq. (8.19). In a crystal the resolved response depends on the direction of $\boldsymbol{q}$, as the total does, and so does the mix of classes a given particle produces.

8.7.2 Pairs in an insulator

For independent electrons the classes of the zero-phonon sector are single pairs. The ground state of a clamped insulator at $T = 0$ is the Slater determinant with every valence state filled. The density $n_e(\boldsymbol{q})$ of Eq. (3.37) moves one electron from $v\boldsymbol{k}$ to $c, \boldsymbol{k}+\boldsymbol{q}$, with the amplitude $\rho_{cv}(\boldsymbol{k},\boldsymbol{q})$ of Eq. (3.34) and the energy $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}}$. Each final state of Eq. (4.13) is therefore one pair. Summed over the two spins of the pair,

$$S^0(\boldsymbol{q},\omega; cv\boldsymbol{k}) = \frac{4\pi}{\mathscr{V}} \left|\rho_{cv}(\boldsymbol{k},\boldsymbol{q})\right|^2 \delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}). \tag{8.66}$$

This is the term of the independent-particle structure factor that belongs to one pair; summed over c, v and $\boldsymbol{k}$ it is the crystal form of Eq. (2.67). Here $\boldsymbol{q}$ is not reduced to the first zone, and the folding vector $\boldsymbol{G}_\mathrm{U}$ of the final state stays in the vertex.

Screening changes the picture. In the random-phase approximation, or in time-dependent density-functional theory, the external potential at $\boldsymbol{q}$ induces a density, and the density adds its own Hartree and exchange–correlation potentials at every $\boldsymbol{q}+\boldsymbol{G}$. The electrons, independent in the Kohn–Sham sense, respond to the total. That total is the external potential times the column

$$(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q},\omega) = \delta_{\boldsymbol{G}0} + \sum_{\boldsymbol{G}'} (f_\mathrm{Hxc})_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega)\, R_{\boldsymbol{G}'0}(\boldsymbol{q},\omega),$$

the inverse dielectric matrix of a test electron, which Eq. (6.10) met for the static potential of a phonon. Without the exchange–correlation kernel it is the inverse dielectric matrix Eq. (4.43). The index 0 is the row $n_e(\boldsymbol{q})$ itself, and R is the retarded response. It is natural to guess that the screened pair weight replaces $\rho_{cv}(\boldsymbol{k},\boldsymbol{q})$ in Eq. (8.66) by the vertex of the total potential, $\sum_{\boldsymbol{G}} (\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}\, \rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})$. The guess is right, and the algebra that confirms it admits a second answer.

Let $f = f_\mathrm{Hxc}$ be Hermitian as a matrix in the reciprocal-lattice vectors, as it is in the random-phase approximation and for any real kernel without frequency dependence, such as the adiabatic local-density kernel of Sect. 4.6. Write $R^\dagger$ for the Hermitian conjugate matrix, the advanced response. The Dyson equation Eq. (4.70) reads $R = R^0 + R^0 f R$, and its conjugate $R^\dagger = R^{0\dagger} + R^\dagger f R^{0\dagger}$. The first says $R^0(1+fR) = R$, the second $(1+R^\dagger f)R^{0\dagger} = R^\dagger$, and together

$$(1+R^\dagger f)\,(R^0 - R^{0\dagger})\,(1+fR) = (R + R^\dagger f R) - (R^\dagger + R^\dagger f R) = R - R^\dagger, \tag{8.67}$$

for every target. The fluctuation–dissipation theorem Eq. (4.18) turns both differences into structure factors: at $T = 0$ and $\omega > 0$, $S_{\boldsymbol{G}\boldsymbol{G}'} = \mathrm{i}(R - R^\dagger)_{\boldsymbol{G}\boldsymbol{G}'}$, and the same holds for R^0. In the bubble Eq. (4.63) only the resonant term has an imaginary part at $\omega > 0$, and $\mathrm{i}[1/(x+\mathrm{i}0^+) - 1/(x-\mathrm{i}0^+)] = 2\pi\delta(x)$, so

$$\mathrm{i}(R^0 - R^{0\dagger})_{\boldsymbol{G}\boldsymbol{G}'} = \frac{4\pi}{\mathscr{V}} \sum_{cv\boldsymbol{k}} \rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})^*\, \rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G}')\, \delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}).$$

Take the element 00 of Eq. (8.67). Since f is Hermitian, $(1+R^\dagger f)_{0\boldsymbol{G}} = [(1+fR)_{\boldsymbol{G}0}]^* = [(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}]^*$, and the double sum over $\boldsymbol{G}$ and $\boldsymbol{G}'$ closes into a squared modulus for each pair.

Screened pair-resolved response

$$S(\boldsymbol{q},\omega; cv\boldsymbol{k}) = \frac{4\pi}{\mathscr{V}} \left|\sum_{\boldsymbol{G}} (\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q},\omega)\, \rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})\right|^2 \delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}), \tag{8.68}$$

with the retarded $\tilde{\varepsilon}^{-1}$, in the random-phase approximation and in time-dependent

density-functional theory with a Hermitian kernel. The sum over c, v and $\boldsymbol{k}$ is the screened structure factor $S(\boldsymbol{q},\omega) = (q^2/2\pi\alpha)L(\boldsymbol{q},\omega)$ of Eq. (4.45), local fields included.

Each term is positive, the terms add up to the full screened response, and the plasmon peak of L is in the sum, spread over the pairs into which it decays. The weight is the bare vertex of the total potential, evaluated at the energy deposited. In the deviation notation of Sect. 8.5, Eq. (8.66) sits at $\mathcal{X} = \text{IP}$ and Eq. (8.68) at $\mathcal{X} = \text{RPA} + \text{LF}$ or TDDFT; with the BSE the final states are eigenstates of the pair Hamiltonian, excitons among them, and these become the classes (Sect. 10.5).

The second answer comes from the same algebra in the other order. The Dyson equation can equally be written $R = R^0 + RfR^0$, with conjugate $R^\dagger = R^{0\dagger} + R^{0\dagger}fR^\dagger$, and then the identity $(1 + Rf)(R^0 - R^{0\dagger})(1 + fR^\dagger) = R - R^\dagger$ holds as well, and it distributes the same total over the pairs with the advanced weight $\delta_{\boldsymbol{G}0} + \sum_{\boldsymbol{G}'} f_{\boldsymbol{G}\boldsymbol{G}'}R^*_{0\boldsymbol{G}'}$ in place of $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}$. The two distributions have the same sum for every target, and pair by pair they need not agree. Which one labels the final state is a question about time. The pair that the cascade receives is the state found at late times, an *out-state* of scattering theory, and the overlap of a free final state with the evolving one brings in the retarded resolvent: in Lippmann–Schwinger form, $\langle\psi_f^{(-)}| = \langle\phi_f|(1 + VG^{(+)})$ [12, Chap. 6]. In the time-dependent mean-field picture this is plain causality. The external potential induces, after it, the total potential $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}\phi_{\text{ext}}$, which creates bare pairs, and the bare pairs are the asymptotic states of the mean-field evolution. The advanced form labels each pair by the state it is evolved from, an in-state, and it is not the label the cascade needs.

The two labelings agree pair by pair whenever $R(\boldsymbol{q},\omega)$ is Hermitian, since then the two weights coincide. This is the case inside the off-shell window, where the zero-phonon pairs are closed and the anti-Hermitian part of the external response comes from the phonon-assisted absorption itself, of order g^2; the difference in a pair-resolved rate is then of order g^4. They also agree in every target invariant under the product PT of inversion and time reversal with an even number of electrons, for the reason that made the Fano coefficients of Eq. (8.62) vanish. This needs a kernel that is real symmetric, $f_{\boldsymbol{G}\boldsymbol{G}'} = f_{\boldsymbol{G}'\boldsymbol{G}} = f^*_{\boldsymbol{G}\boldsymbol{G}'}$, as the random-phase and adiabatic local-density kernels are; a kernel that is merely Hermitian does not suffice. Every state can then be chosen invariant under PT, and with the origin at an inversion center $n_e(\boldsymbol{q}+\boldsymbol{G})$ is invariant too, since inversion reverses $\boldsymbol{q}+\boldsymbol{G}$ and complex conjugation restores it. Every amplitude $\langle f|n_e(\boldsymbol{q}+\boldsymbol{G})|i\rangle$ is then real, $R_{\boldsymbol{G}\boldsymbol{G}'} = R_{\boldsymbol{G}'\boldsymbol{G}}$, and the advanced weight is the complex conjugate of the retarded one. The symmetry of R depends on that choice of origin, since a shift multiplies $R_{\boldsymbol{G}\boldsymbol{G}'}$ by phases; the pair weights do not. A real vertex sum has the same modulus with either, so each pair, or each set of degenerate pairs, gets the same weight. Silicon and germanium satisfy the hypothesis. GaAs has no center of inversion, and there the choice of the out-state matters.

The pair weights need nothing beyond the ingredients of the structure factor itself. The vertices $\rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})$ come from the plane-wave coefficients of the Bloch states (Appendix E.7), and the column $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}$ from a dielectric-matrix calculation at the same $(\boldsymbol{q},\omega)$ (Appendix E.6). The difference is in what is kept. A code that returns $S(\boldsymbol{q},\omega)$ sums the pairs and discards them. The resolved response stores their weights, binned in the carrier energies and valleys, for each $\boldsymbol{q}$, ω and coupling row, and a response library must say which binning it keeps (Sect. 21.6).

Worked example 8.9: Pairs made at one momentum transfer in silicon

Which pairs does a dark-matter kick make, and how does that differ from what a photon of the same energy makes? We take the empirical pseudopotential of Cohen and Bergstresser for silicon [29], the model of Sects. 3.3 and 4.6: three form factors fitted to measured optical transitions, $a = 5.43\,\text{Å}$, 113 plane waves and no spin–orbit coupling. Its indirect gap is 0.82 eV, below the measured 1.17 eV, with the conduction minimum at 0.853 of the way to X. The momentum transfer is $q_0 = \alpha m_e = 3.73\,\text{keV}$ along [100], a typical value for electron scattering, which is $1.633\,(2\pi/a)$ and lies beyond the first zone; its Umklapp partner $\boldsymbol{q} + \boldsymbol{G}$ with $\boldsymbol{G} = (-2, 0, 0)\,2\pi/a$ has $|\boldsymbol{q} + \boldsymbol{G}| = 0.84\,\text{keV}$. The energy deposited is $\omega = 10\,\text{eV}$. The sums run over a shifted 24^3 grid of $\boldsymbol{k}$ over the full zone, with 4 valence and 26 conduction bands. The local-field matrix has the 51 vectors with $|\boldsymbol{q} + \boldsymbol{G}|^2 \leq 12\,(2\pi/a)^2$, and the $\mathrm{i}0^+$ of the bubble is a Lorentzian of half-width 0.10 eV. The energy delta function of each pair is a Gaussian of standard deviation 0.10 eV instead. A Lorentzian has long tails, and in the optical limit below, whose weights carry $1/(\varepsilon_c - \varepsilon_v)^2$, the tails would admit strong transitions several eV below ω and break the energy balance of the pairs. The numbers quoted change by at most 3 % between 20^3 and 24^3 points and between Gaussians of 0.10 and 0.05 eV.

Screening. The independent-particle structure factor is $S^{\text{IP}}(\boldsymbol{q}_0, \omega) = 1.7 \times 10^8\,\text{eV}^2$. The random-phase approximation with local fields gives $(\varepsilon^{-1})_{00} = 0.55 - 0.16\,\mathrm{i}$, and the screened structure factor, Eq. (8.68) summed over pairs, is $4.9 \times 10^7\,\text{eV}^2$, a factor of 0.29 of the unscreened one. The head of the column does most of this, since $|(\varepsilon^{-1})_{00}|^2 = 0.33$; the other 50 elements are at most 0.05 each, the Umklapp partner among the small ones, and together they lower the rate by a further 12 %. For comparison, the free gas at the valence density gives $m_e^2\omega/\pi q_0 = 2.2 \times 10^8\,\text{eV}^2$ at the same point, the unpolarized form of the one-spin structure factor of Worked example 8.3.

Which pairs. Every pair at this ω shares $\omega - E_{\text{g}} = 9.2\,\text{eV}$ between the kinetic energies E_{e} and E_{h} of its carriers. With the screened weights, the mean electron energy is 4.8 eV and the mean hole energy 4.4 eV; the electron takes more than half in 64 % of the weight, and one carrier takes more than three quarters in 45 %. The lowest valence band supplies 4 % of the holes and the other three about a third each. The independent-particle weights give the same picture, each of these fractions within three percentage points. Screening changes how many pairs are made by a factor of three, and which pairs hardly at all.

Other seeds at the same energy. A photon of 10 eV, or a dark photon, makes vertical pairs with the weights $|\hat{\boldsymbol{e}} \cdot \boldsymbol{v}_{cv}|^2/(\varepsilon_c - \varepsilon_v)^2$ of the optical limit Eq. (3.43). Its electrons are more energetic, 5.7 eV on average, and its holes less, 3.5 eV, about 0.9 eV below those of the dark-matter kick; the electron takes more than half in 85 % of the weight. The transverse current $\hat{\boldsymbol{y}} \cdot \boldsymbol{j}_e(\boldsymbol{q}_0)$ at the same momentum transfer lies in between, with 5.3 eV and 3.8 eV, and puts half its weight on lopsided pairs with one carrier above three quarters. Every seed shares the same 9.2 eV, and yet the three hand the cascade three different starting distributions.

In-states and out-states. In this model of silicon, with the origin at the bond center, an inversion center, the two labelings give every pair the same weight, and $R_{\boldsymbol{G}\boldsymbol{G}'} = R_{\boldsymbol{G}'\boldsymbol{G}}$, as the PT argument requires for the real symmetric kernel of the random-phase approximation. The same model of GaAs has no center of inversion. At the same q_0 and 10 eV, its pair-by-pair differences add up, in absolute value, to 15 % of the screened total, although the mean carrier energies of the two labelings differ by 0.02 eV at most. The choice of the out-state matters for momentum- and valley-resolved seeds, and much less for the energies alone.

The example makes the point of the next subsection concrete. At a fixed energy deposited, the pairs that are made depend on the momentum transfer and on the operator that makes them, by about 0.9 eV in the mean energy of the hole. That is close to the gap, the energy scale on which impact ionization switches on.

8.7.3 Cascade probabilities of a final state

Once the dark matter has acted, the target evolves on its own. A pair of class Λ emits phonons, ionizes further pairs and slows down, and the event ends with some number N_{eh} of pairs, some energy in phonons and some energy stored in defects. We write $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ for the probability that the cascade started by a final state of class Λ ends with N_{eh} pairs; Sect. 18.3 computes it from the branching process of Sect. 18.1. The step that turns the prompt state into the initial condition of the cascade, *seeding*, is made precise in Sect. 18.3. The one assumption here is that the class carries everything the later evolution depends on. The prompt state is reached before the first decay, so the history of how it was made can enter only through coherences between production and decay. These are the Fano coefficients of Eq. (8.62), and they vanish in silicon and germanium.

Under that assumption, the probability of an event with N_{eh} pairs is the probability of each class times the probability that its cascade ends with N_{eh} pairs, summed over the classes. With the rate into a class, Eq. (8.65), and the halo average of Eq. (8.16):

Hand-off to the cascade

$$\mathcal{R}_{N_{\mathrm{eh}}} = \frac{\rho_\chi}{m_\chi\,\rho_T} \int \mathrm{d}^3v\, f_\chi(\boldsymbol{v}) \int \frac{\mathrm{d}^3q}{(2\pi)^3} \sum_{\Lambda} \sum_{a,b} \mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})\, S_{ab}(\boldsymbol{q},\omega_{\boldsymbol{q}};\Lambda)\, P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda), \quad (8.69)$$

the rate per unit target mass of events with N_{eh} pairs. The coupling tensor is contracted inside the sum over classes, and the sum over classes sits inside the integrals over $\boldsymbol{q}$ and $\boldsymbol{v}$. Equivalently, at fixed $(\boldsymbol{q},\omega,\boldsymbol{v})$ the pair number is distributed as

$$P(N_{\mathrm{eh}}|\boldsymbol{q},\omega,\boldsymbol{v}) = \frac{\sum_\Lambda \sum_{a,b} \mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})\, S_{ab}(\boldsymbol{q},\omega;\Lambda)\, P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)}{\sum_{a,b} \mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})\, S_{ab}(\boldsymbol{q},\omega)}. \quad (8.70)$$

The same form holds for any other count of the cascade, the phonon energy or the energy stored in defects, with P_{casc} replaced by the distribution of that quantity, and for absorption, with $\mathfrak{D}_{ab}$ replaced as in Eq. (8.65). It contains the pair bins of Eq. (1.46) as a special case. If $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ depended on the class only through its energy ω, Eq. (8.70) would reduce to $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\omega)$, the sum over Λ would close into $S_{ab}(\boldsymbol{q},\omega)$ by Eq. (8.64), and the integrals would collapse to $\int \mathrm{d}\omega\, (\mathrm{d}\mathcal{R}/\mathrm{d}\omega)\, P_{\mathrm{casc}}(N_{\mathrm{eh}}|\omega)$. The step model is the further choice of a distribution that puts all its weight on $1 + \lfloor(\omega - E_{\mathrm{g}})/\varepsilon_{\mathrm{eh}}\rfloor$.

That reduction is the common shortcut, and Eq. (8.69) says when it fails. Three conditions are easy to break.

The momentum transfer stays inside. At one ω, different $\boldsymbol{q}$ make different pairs. Worked example 8.9 shows that at 10 eV in silicon the hole carries about 0.9 eV more, on average, for a dark-matter kick at q_0 than for an absorbed photon. Integrating the spectrum over $\boldsymbol{q}$ first and seeding the cascade afterwards with a distribution of pairs chosen at fixed ω attaches the pairs of one momentum to the rate of another. The resolved response depends on the direction of $\boldsymbol{q}$ as well, so in a crystal the mix of seeds follows the dark-matter wind through the day, and a directional analysis (Sects. 11.7 and 21.2) needs the seeds at every direction it uses.

The velocity and the model stay inside. With one coupling row a, the weights of the classes in Eq. (8.70) are $S_{aa}(\boldsymbol{q},\omega;\Lambda)/S_{aa}(\boldsymbol{q},\omega)$, independent of $\boldsymbol{v}$ and of the model, and a table of $P(N_{\mathrm{eh}}|\boldsymbol{q},\omega)$ per row serves every model that couples through that row. With several rows it does not. The weights then depend on $\mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})$, so on the coefficients of the model and on the velocity, and the off-diagonal elements $S_{ab}(\boldsymbol{q},\omega;\Lambda)$ populate the classes too. Mixing single-row tables in proportion to the single-row rates misses the interference of Sect. 8.2.5 class by class, even where it cancels in the total.

The operator structure stays inside. Rows that probe different parts of the electron's state make different pairs at the same $(\boldsymbol{q},\omega)$. A spin row reaches spin-flipped pairs that the density cannot. A current row weights each pair by the velocity of its carriers, not by the overlap of their periodic parts. The longitudinal current is the exception: the continuity equation Eq. (4.29) gives $\langle f|\hat{\boldsymbol{q}}\cdot\boldsymbol{j}_e(\boldsymbol{q})|i\rangle = (\omega/q)\langle f|n_e(\boldsymbol{q})|i\rangle$ for every pair of eigenstates, so the longitudinal current makes exactly the pairs the density makes. The transverse current, the spin density and the spin current make their own, and the velocity-dependent operators of Table 8.1 reach the target through them.

The cascade depends on $\boldsymbol{q}$ mainly through its first generation. The electron and hole of the prompt pair carry the crystal momenta $\boldsymbol{k}+\boldsymbol{q}$ and $\boldsymbol{k}$. Where collisions randomize the momentum of a carrier faster than they change its energy, its later history depends on its energy alone, and the secondary carriers inherit nothing of $\boldsymbol{q}$; Sect. 18.1 states this reduction and the energies at which it fails. The difference between dark-matter and photon seeds then enters once, through the distribution of $(E_{\mathrm{e}},E_{\mathrm{h}})$ and of the valleys at the start. That is enough to change the pair count at the few-pair level, where a single impact ionization more or less is a whole bin. The ionization model of Ramanathan and Kurinsky describes the split of the energy between the carriers by one distribution of the electron's energy at each total, fitted to the ionization yields of photon absorption [28, Eqs. (5)–(7)]. It is the $\boldsymbol{q}\to 0$ marginal of the seeds of Eq. (8.70), and using it for dark-matter scattering replaces the seeds of one momentum by those of another; Sect. 18.3 sizes the error. For absorption the momentum is $p_\phi = m_\phi v \approx 10^{-3} m_\phi$, the transitions are vertical, and a dark photon is absorbed through the same current as a photon. Its seeds are the photon's, and a photon calibration applies to it exactly.

8.7.4 Inputs to the measurement chain

Fig. 8.4 follows the hand-off from the resolved response to a limit. Each sector of Sect. 8.6 hands over a different set of numbers, and Eq. (8.69) fixes the form: for every class, a weight at each $(\boldsymbol{q},\omega)$ and for each pair of coupling rows.

Zero-phonon electronic sector. For each pair of rows and each $(\boldsymbol{q},\omega)$, the cascade needs the joint distribution of the electron and hole over bands, crystal momenta and valleys, from Eq. (8.68) or its excitonic counterpart. A pair that lies inside the one-phonon continuum carries its Dyson weight $z_{\mathrm{qp}}(m)$. The cascade of Chap. 18 lets that pair emit its first phonon. In a target without PT symmetry, such as GaAs, the response must also supply the Fano coefficients c_ν^{F} that redistribute its first decay among the phonon branches and the other decay channels (Eq. (8.62)), since an incoherent cascade cannot generate them. A core hole is handed over with its shell, and the cascade starts from its Auger electron (Sect. 10.6).

Phonon-dressed electronic sector. The pair as above, together with the branch, momentum and energy of each phonon created with it, from the phonon-assisted response of Chap. 13 and the joint electron–lattice response of Sect. 12.1. The phonon energy goes to the phonon channel once and the pair to the carrier cascade; the balance law of Sect. 19.5 books each share once. Inside the off-shell window every final state is in this sector, and the pair-resolved weights there carry the extra screening of the pair channel (Sect. 13.4).

Electron-following sector. Single phonons and multiphonon states go to the phonon readout of Chap. 20 as they are, with their branches and momenta, which set how fast they decay and how they reach a sensor. A nucleus recoiling freely, in the impulse regime of Sect. 8.5, is handed over with its energy E_R and direction from the nuclear response of Sect. 11.4, and Sect. 19.1 follows it through the lattice. The direction matters there, because the energy needed to displace an atom depends on it (Sect. 7.4).

Targets without a gap. Superconductors and magnetic targets hand over quasiparticle and magnon numbers, the sectors $\mathcal{P}^{\mathrm{qp}}_{n,m}$ of Sect. 8.6, to the quasiparticle readout of Sect. 20.4 and the spin chapters (Chaps. 14 and 15).

What the chain does not need is as clear. Part III never asks for the total $S_{ab}(\boldsymbol{q},\omega)$ on its own: every quantity it computes is a sum over classes weighted by what each class does next. A response computed to high accuracy and then integrated over $\boldsymbol{q}$ before seeding has lost the information that the few-pair bins depend on, and no later step can restore it. Conversely, an error in the resolved response that leaves the class weights unchanged, such as the overall screening factor of Worked example 8.9, passes through to the rate but not to the shape of the pair-number distribution. Sect. 8.8 turns these errors into nuisance parameters, one per truncation, each acting coherently across every momentum, energy and target that share the physics.

Two questions remain open. First, the identification of the prompt pair with the out-state of Eq. (8.68) is derived only within time-dependent mean-field linear response, where the bare pair is the asymptotic state of the evolution. The exact final states of an interacting insulator are not labeled by pairs, and whether the retarded weights give the distribution of the carriers found at late times beyond mean field is not established. Second, the assumption behind Eq. (8.69), that the class fixes the statistics of the cascade, holds when the only coherence between production and decay is the one the Fano coefficients carry; a bound on what else the history of an event can contribute, set by the widths of the prompt states, is not known.

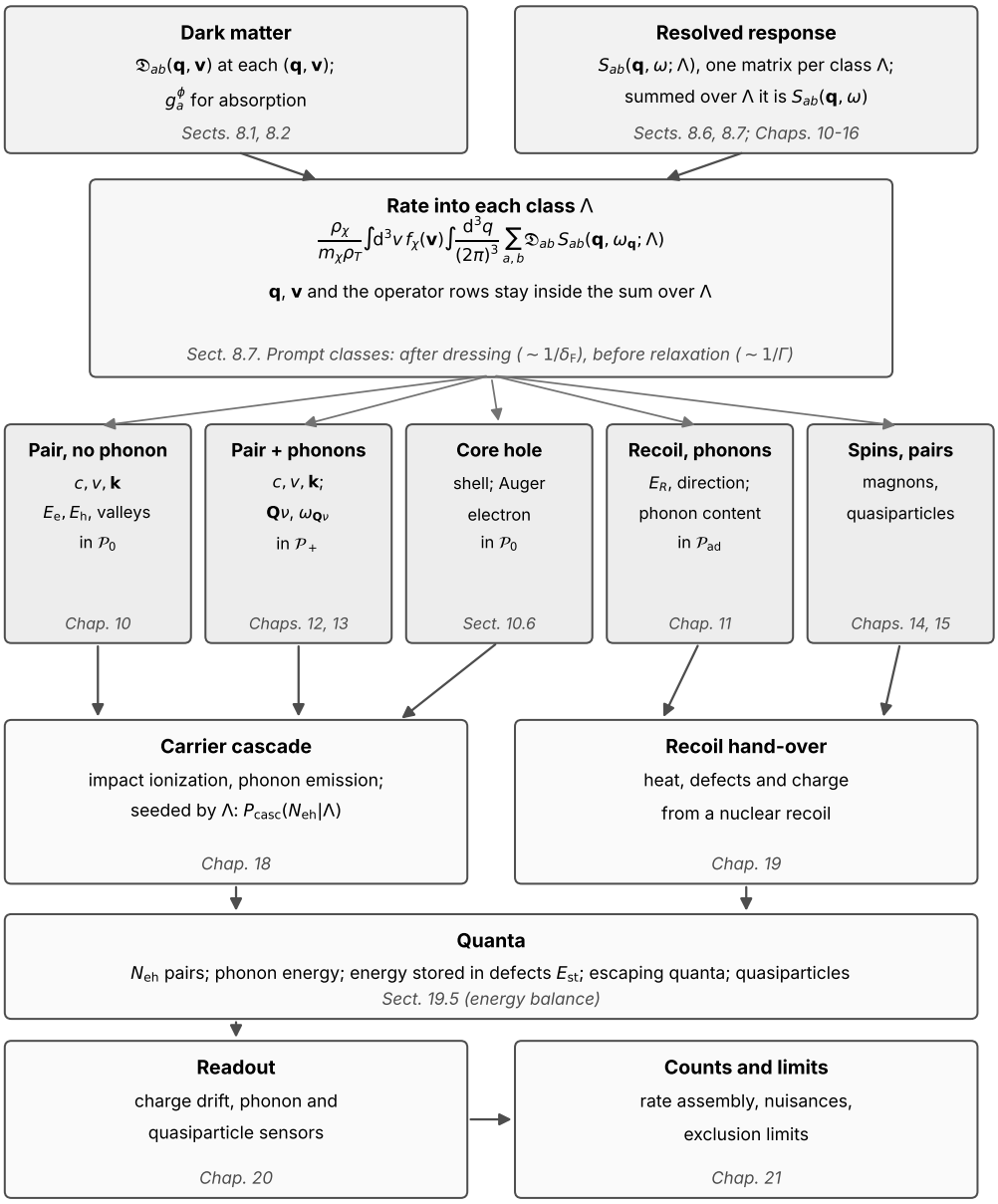


Figure 8.4: From the resolved response to a limit. The dark-matter side is a coupling tensor $\mathfrak{D}_{ab}(\boldsymbol{q}, \boldsymbol{v})$ for scattering or couplings g_a^ϕ for absorption (Sects. 8.1 and 8.2). Contracted with the final-state-resolved response $S_{ab}(\boldsymbol{q}, \omega; \Lambda)$ (Sects. 8.6 and 8.7; computed for each channel in Chaps. 10 to 16), it gives the rate into each prompt class Λ, a state after its dressing and before its first decay. The classes shown lie in the sectors of Sect. 8.6: a pair with no phonon (Chap. 10), a pair with phonons (Chaps. 12 and 13), a core hole (Sect. 10.6), a recoil or phonons (Chap. 11), and the magnons and quasiparticles of gapless targets (Chaps. 14 and 15). Each class seeds a cascade, of hot carriers (Chap. 18) or of a recoiling nucleus (Chap. 19); the energy ends as pairs, phonons, stored energy and escaping quanta, booked once by the balance law of Sect. 19.5, and is read out (Chap. 20) and turned into a limit (Chap. 21). The momentum transfer, the velocity and the coupling rows stay inside the sum over classes (Eq. (8.69)).

8.8 Uncertainty structure

The local fields of silicon raise its screening factor at small momentum transfer by a quarter (Sect. 4.6), and at a momentum transfer of 7.5 keV they lower the screened structure factor by 22 % to 44 % between 2 and 10 eV (Worked example 8.10). A limit computed with the random-phase approximation alone carries that difference whether or not anyone states it. Sect. 1.5 called such errors the theory uncertainty of a limit: they come from approximations, and a longer exposure does not reduce them. Sect. 8.5 gave every approximation of the response a place on the truncation label $\mathfrak{T}$, and with it a next term and, where the axis has a small parameter, a size. What remains is to turn each truncation into a number that a likelihood can use. We will define the nuisance parameter of a truncation as the step to its next order, impose on it the exact relations of Sect. 8.3, sort the nuisances by how they deform a spectrum, and write the likelihood of one exposure with the nuisances in place. A theory uncertainty built this way is computed from the physics, and two analyses that use the same truncation share it.

8.8.1 The nuisance of a truncation

A formula sits at a point $\mathfrak{T}$ of Eq. (8.44). Let $\mathfrak{T}_k^+$ be the label one step further along the axis k and nowhere else. On the integer axes the step goes to the next order at which the labeled quantity changes, by the counting rule of Sect. 8.5: $n_g = 0 \to 2$, $n_B = 0 \to 2$, $n_A = 0 \to 2$ and $n_T = 0 \to 1$. On the many-body axis it is the next approximation, from IP to RPA, to RPA + LF, to TDDFT or BSE. The lattice regime is not an order, and its step is the next term of the multiphonon series Eq. (5.59), which is known exactly. The difference of the two responses and its size relative to the first are

$$\Delta S_k(\boldsymbol{q},\omega) = S^{\mathfrak{T}_k^+}(\boldsymbol{q},\omega) - S^{\mathfrak{T}}(\boldsymbol{q},\omega), \qquad \delta_k(\boldsymbol{q},\omega) = \frac{\Delta S_k(\boldsymbol{q},\omega)}{S^{\mathfrak{T}}(\boldsymbol{q},\omega)}. \tag{8.71}$$

The nuisance parameter ϑ_k interpolates between the two. For a single response, such as S_{ee} or the contraction $\sum_{ab} \mathfrak{D}_{ab} S_{ab}$ of one model,

$$S(\boldsymbol{q},\omega;\vartheta) = S^{\mathfrak{T}}(\boldsymbol{q},\omega) \prod_k \left[1 + \delta_k(\boldsymbol{q},\omega)\right]^{\vartheta_k}, \tag{8.72}$$

so that $\vartheta_k = 0$ is the formula, $\vartheta_k = 1$ is its next order, and ϑ_k is given a standard normal constraint in the likelihood. To first order, $S(\vartheta) = S^{\mathfrak{T}} + \sum_k \vartheta_k \ln(1+\delta_k)\, S^{\mathfrak{T}}$, the familiar additive form. The power form differs from it at second order, and the difference matters: $S^{\mathfrak{T}}(1 + \vartheta_k \delta_k)$ turns negative at $\vartheta_k = -1/\delta_k$, two standard deviations out for $\delta_k = 0.5$, while Eq. (8.72) is positive for every ϑ.

A nuisance must not vary from bin to bin. A truncation is a property of the calculation, not of the bin, and the same ϑ_k acts at every $\boldsymbol{q}$ and ω, in every bin and on every final-state class Λ of Sect. 8.7. It is also shared by every target computed with the same approximation and by every experiment that uses the result. Only the template $\delta_k(\boldsymbol{q},\omega)$ varies. This is what distinguishes a nuisance parameter from an error bar attached to each bin. A truncation that lowers the spectrum near the gap and raises it at higher energy is one parameter, and an analysis that treats the two regions as independent errors would let the data fit a shape that no calculation can produce.

When is ΔS_k computable? The next order may have been computed, it may be estimated from a small parameter, or neither may be possible. In the first case the next order is known, as for the next many-body approximation of Sect. 4.6, the zero-point renormalization of a gap at $n_g = 2$ (Sect. 6.3) or the next term of Eq. (5.59). The template is then a number at every

point, and the better result usually becomes the central value, with the nuisance carried by the step after it. For the many-body approximations that step is again a computed difference, and like every many-body spread it is an estimate, not a bound (Sect. 4.9). In the second case the next order has not been computed, but the axis has a small parameter whose size Table 8.4 gives. Then δ_k is the parameter raised to the power of the step, times a coefficient of order one, between a quarter and two in the silicon counting of Sect. 5.7. The nuisance is then an estimate and is labeled as such. In the third case there is neither: no computed next order and no small parameter. The size of the truncation is then not known, and no width is assigned to it. The pair-channel screening of the phonon-assisted sideband is the example of this chapter. Sect. 8.4 showed that dropping it is an omission, not a choice of convention, and Sect. 13.4 gives only its two solvable limits, which bracket it loosely and do not fix its sign. The calculation that would size it is specified in Appendix E.16 and has not been done. Until it is, a rate that omits it is reported at its nominal value with the omission named, because a width invented for it would hide the one fact a reader needs.

Approximations off the label have nuisances too. The exchange–correlation functional, the frozen core and the numerical parameters of a calculation have no next order, and their sizes come from the spreads and convergence tests of Appendices E.1 to E.5. They enter the likelihood in the same way, with the spread in place of ΔS_k.

For a response matrix S_{ab} the power form needs care. Raising each element to its own power leaves the cone of positive semidefinite matrices, and $\delta_{k,ab}$ is undefined wherever $S^{\mathfrak{T}}_{ab} = 0$. The matrix analog of Eq. (8.72) is

$$S(\vartheta_k) = S^{1/2}\big(S^{-1/2}S^{+}S^{-1/2}\big)^{\vartheta_k}S^{1/2}, \qquad S \equiv S^{\mathfrak{T}}(\boldsymbol{q},\omega),\ S^{+} \equiv S^{\mathfrak{T}^{+}_{k}}(\boldsymbol{q},\omega), \tag{8.73}$$

which reduces to Eq. (8.72) for one row. It equals S at $\vartheta_k = 0$ and S^{+} at $\vartheta_k = 1$. It is positive semidefinite for every $\vartheta_k \geq 0$, since $S^{-1/2}S^{+}S^{-1/2}$ is positive semidefinite, so is any nonnegative power of it, and the congruence $X \to S^{1/2}XS^{1/2}$ preserves the property. The inverse square root is taken on the subspace where S is nonzero, which requires S^{+} to have no weight outside it. A profile explores negative ϑ_k as well, and a negative power exists only if S^{+} is positive definite on that subspace: the two truncations must have the same support, $\operatorname{supp} S^{+} = \operatorname{supp} S$, and then Eq. (8.73) is positive semidefinite for every real ϑ_k. Where the next order opens new support, Eq. (8.75) below applies instead. Several truncations are applied one after another; their order matters only at second order in the δ_k.

8.8.2 Scale, tilt and edge shifts

A nuisance scales a spectrum, tilts it or moves its edge, and each moves a limit differently.

A *normalization* nuisance has a template δ_k that is constant over the energies a rate samples at fixed $\boldsymbol{q}$. It scales the spectrum and leaves its shape alone, and by Eq. (1.54) it moves the limit by the same fraction. It also leaves the weights of the final-state classes unchanged, so it passes to the pair-number spectrum of Eq. (8.70) as an overall factor (Sect. 8.7).

A *tilt* has a template that varies across the energies a rate samples. It moves weight between bins, and its effect on a limit depends on which bins the dark-matter spectrum fills, so on the mass and the mediator. A tilt changes the mix of final-state classes as well, and with it the pair-number spectrum.

A *threshold* nuisance moves an edge,

$$S(\boldsymbol{q},\omega;\vartheta_k) = S^{\mathfrak{T}}(\boldsymbol{q},\omega - \vartheta_k\,\delta E_k), \tag{8.74}$$

with δE_k the shift that the next order produces. A shift acts on the limit through the edge of Sect. 1.3.5, $\delta \ln m_{\min} = \delta \ln E_{\mathrm{th}}$, and near the edge it moves the limit by a far larger factor than its fractional size, because $\eta(v_{\min})$ falls steeply there (Sect. 1.5.5).

Table 8.6: Nuisance types, with one truncation of each type in each sector of the decomposition Eq. (8.48): $\mathcal{P}_{\mathrm{ad}}$ (electrons following the nuclei; phonons and recoils), $\mathcal{P}_0$ (zero-phonon electronic excitations) and $\mathcal{P}_+$ (electronic excitations with phonons). The axis of the truncation label is given in parentheses. Sizes are for silicon, and the mark "est." denotes an estimate from a small parameter of Table 8.4. The residue entry in $\mathcal{P}_0$ and the finite-part entry in $\mathcal{P}_+$ are one nuisance, tied by Eq. (8.60); so are the two $\mathcal{K}$ entries of $\mathcal{P}_{\mathrm{ad}}$, the two parts of one step of Eq. (5.59).

Type	Form	$\mathcal{P}_{\mathrm{ad}}$	$\mathcal{P}_0$	$\mathcal{P}_+$
Normalization	Eq. (8.72), δ_k constant in ω	Debye–Waller factor e^{-2W_κ} dropped from a one-phonon rate ($\mathcal{K}$): $\delta = -0.061$ at 10 keV	residue z_{qp} of the pair ($n_g = 2$); est. of order $\theta_{\mathrm{ep}} \sim 10^{-2}$	static screening of the vertex (n_B): est. $(\omega_{\mathrm{ph}}/E_{\mathrm{g}})^2 = 3 \times 10^{-3}$; finite part tied to the residue
Tilt	Eq. (8.72), $\delta_k(\omega)$	anharmonic reshaping of the recoil spectrum at fixed area and mean ($n_A = 2$): Eq. (8.77)	local-field correction, RPA $\to$ RPA + LF ($\mathcal{X}$): -0.44 to $+0.25$ across q and ω, Worked example 8.10	two more vertices ($n_g = 4$): est. $\theta_{\mathrm{ep}} \approx 10^{-2}$ below E_{dir}, rising to 1 near it
Threshold	Eq. (8.74)	anharmonic shift of the optical phonon ($n_A = 2$): est. $\eta^2_{\mathrm{anh}}\omega_{\mathrm{opt}} = 0.15$ meV	zero-point renormalization of the gap ($n_g = 2$): $\delta E = -56.2$ meV; adiabatic against nonadiabatic 8 meV	exciton binding ($\mathcal{X} = \mathrm{IP} \to \mathrm{BSE}$): every phonon-assisted threshold 14.7 meV lower
New support	Eq. (8.75)	two-phonon response above the one-phonon band ($\mathcal{K}$)	exciton lines below the pair continuum at $\boldsymbol{q} \approx \boldsymbol{k}_{\mathrm{c}}$ ($\mathcal{X} = \mathrm{BSE}$)	phonon-absorption branch ($n_T = 1$): weight e^{-21550} at 10 mK

Some truncations do more than deform the spectrum: their next order puts weight where the truncated formula has none. The exciton lines of silicon below the pair continuum, which dark matter with $\boldsymbol{q} \approx \boldsymbol{k}_{\mathrm{c}}$ can create directly (Sect. 4.9), are an example, and so is the two-phonon response above the top of the one-phonon band. No power of $1 + \delta_k$ can create weight from zero. A positive piece that the formula drops is an error of known sign, so once computed it belongs in the central value, and its nuisance is on its own size,

$$S(\boldsymbol{q}, \omega; \vartheta_k) = S^{\mathfrak{T}}(\boldsymbol{q}, \omega) + \Delta S_k(\boldsymbol{q}, \omega)\,(1 + \delta'_k)^{\vartheta_k}, \tag{8.75}$$

where δ'_k is the fractional size of the next correction to the new piece. The rate stays positive for every ϑ_k as long as ΔS_k is itself a positive spectrum, which holds for new lines and for a sideband inside the off-shell window of Sect. 8.6. Outside the window it fails, since the finite-part terms of Eq. (8.59) are not positive on their own, and there the nuisance acts on the zero-phonon and phonon-dressed sectors together.

Table 8.6 gives an example of each type in each sector of Eq. (8.48), for silicon wherever the size is known. Two entries share a nuisance. The quasiparticle residue z_{qp} lowers the zero-phonon pair weight, and Eq. (8.60) returns exactly that weight to the finite part of the sideband. One parameter acts on both sectors with opposite signs, and their sum does not move. Treated as two independent nuisances, the residue and the sideband would double the uncertainty of a quantity that the identity holds fixed.

The Debye–Waller entry uses $2W_\kappa = 0.063$ at 10 keV (Sect. 8.5): the correct one-phonon rate is lower by $1 - \mathrm{e}^{-0.063} = 6.1\,\%$ than one that sets $\mathrm{e}^{-2W_\kappa} = 1$, at that momentum and

uniformly in ω. The template is exact, and its value grows with q. The two-phonon term of the new-support row is the other half of the same step of Eq. (5.59): the weight that e^{-2W_κ} removes from the one-phonon term goes to it and to the elastic line, and the two entries carry one ϑ. The gap entry is the one in which the central value moves. A clamped-nucleus gap is at $n_g = 0$, and the zero-point renormalization, $-56.2\,\mathrm{meV}$ in the nonadiabatic form (Sect. 6.3), is its computed next order. Once it is included, the nuisance that remains is the step after it, of which the 8 meV between the adiabatic and nonadiabatic forms and the difference to the measured $-62\,\mathrm{meV}$ to $-64\,\mathrm{meV}$ give the size. The anharmonic shift uses the counting $\eta^2_{\mathrm{anh}}\omega_{\mathrm{ph}}$ of Sect. 5.7, with $\eta^2_{\mathrm{anh}} = 2.4 \times 10^{-3}$ and $\omega_{\mathrm{opt}} = 64.1\,\mathrm{meV}$. The thermal entry is a nuisance whose template has been computed and found to vanish: at 10 mK the occupations of Sect. 8.5 make it smaller than any number a likelihood can resolve.

8.8.3 Exact relations as constraints

The relations of Sect. 8.3 hold for the exact response and for every truncation that respects them. A nuisance that breaks one of them describes no possible calculation, and the relations are therefore constraints on the nuisances, not further sources of uncertainty. Positivity, the interference bound, the sum rules, detailed balance and the moments of the recoil spectrum do the work.

Positivity. The response matrix is positive semidefinite at every $(\boldsymbol{q}, \omega)$ (Eq. (4.20)), and $S(\vartheta)$ must be so for every ϑ that the likelihood explores. The power form Eq. (8.72), the matrix form Eq. (8.73) and the new-support form Eq. (8.75) with a positive ΔS_k satisfy this by construction. The additive form $S^{\mathfrak{T}} + \vartheta_k \Delta S_k$ does not, wherever $|\Delta S_k|$ is comparable to $S^{\mathfrak{T}}$, which is common near edges.

The interference bound. The cross term of two groups of rows can never exceed twice the geometric mean of their separate rates, Eq. (8.21). A nuisance on the cross term, however, is not free. If a truncation scales the separate rates by $(1+\delta_A)^\vartheta$ and $(1+\delta_B)^\vartheta$, the largest cross term allowed scales by $[(1+\delta_A)(1+\delta_B)]^{\vartheta/2}$, and a nuisance that grows the cross term faster than the separate rates describes no target. In practice nuisances are assigned to the matrix, through Eq. (8.73), and never to an interference term on its own; the bound is then kept automatically. The same holds for the mixed electron–nucleus response of the Migdal effect, where $|S_{e\kappa}|^2 \le S_{ee} S_{\kappa\kappa}$.

Sum rules. The first moment of the density response is fixed by the f-sum rule (Eq. (4.28)) for any approximation that conserves particle number with local potentials. Every approximation of Sect. 4.6 is of that kind: the bubble, the random-phase approximation with or without local fields, and the adiabatic kernel all add static, local terms that leave the tail coefficient of the response unchanged. For these truncations

$$\int_{-\infty}^{\infty} \mathrm{d}\omega\, \omega\, \Delta S_k(\boldsymbol{q}, \omega) = 0. \tag{8.76}$$

A many-body nuisance moves weight in energy but cannot add or remove first moment, and its template must change sign: what it lowers in one range it raises in another. A normalization nuisance on the whole spectrum would violate Eq. (8.76), which is why the many-body nuisances are tilts. The relation is not vacuous for a rate, which samples only energies below $\frac{1}{2} m_\chi v_{\mathrm{max}}^2$. Within that window a many-body nuisance may look like a normalization, and Eq. (8.76) says the compensating weight lies above it. Sect. 4.6 saw the effect in the optical spectrum, where the local fields push weight up. Some truncations do not conserve the f-sum: a cut in the band sum or in q, a nonlocal pseudopotential, a scissor shift of the conduction bands. For them the violation of the sum rule measures the error, and the template can be calibrated against it. The partial sum rules of Sect. 8.6 act in the same way between sectors, and the residue identity Eq. (8.60) is one of them.

Detailed balance. In a warm target the response at negative energies is the Boltzmann-weighted image of the response at positive energies (Eq. (2.68)), and the dark-matter rates in the two directions are tied by Eq. (8.20). A nuisance applied to $\omega > 0$ alone would break the tie. The same ϑ_k must act on both signs of ω, with the template at $(-\boldsymbol{q}, -\omega)$ fixed by the one at $(\boldsymbol{q}, \omega)$.

Moments of the recoil spectrum. The self response of a nucleus has a fixed zeroth moment, the norm $\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}} \rangle = 1$, which no Hamiltonian changes, and a fixed first moment, the recoil energy E_R of Eq. (5.61). The derivation of Eq. (5.61) used only the kinetic energy of the nucleus, so both moments hold on any energy surface, anharmonic ones included. The anharmonic truncation $n_A = 0 \to 2$ can only reshape the recoil spectrum at fixed area and fixed mean. The power form keeps both moments only at $\vartheta = 0$ and 1. A *constrained nuisance* keeps them at every ϑ. With $y = \omega - E_R$ and $\psi = \ln(1 + \delta_A)$, the template of the anharmonic step, it reads

$$S(\boldsymbol{q}, \omega; \vartheta) = S^{\mathfrak{T}}(\boldsymbol{q}, \omega)\, \exp\big[\vartheta\, \psi(\omega) + \lambda_0(\vartheta) + \lambda_1(\vartheta)\, y\big], \tag{8.77}$$

where λ_0 and λ_1 are fixed at each ϑ by requiring the zeroth and first moments to equal those of $S^{\mathfrak{T}}$. Positivity is manifest. That λ_0 and λ_1 exist and are unique follows from one convex function. Write $m_n(\vartheta) = \int \mathrm{d}\omega\, y^n S(\boldsymbol{q}, \omega; \vartheta)$ for the moments in the variable y, and let $F(\lambda_0, \lambda_1) = \int \mathrm{d}\omega\, S^{\mathfrak{T}} \exp(\vartheta\psi + \lambda_0 + \lambda_1 y) - \lambda_0 m_0(0) - \lambda_1 m_1(0)$. Its gradient vanishes exactly where $m_0(\vartheta) = m_0(0)$ and $m_1(\vartheta) = m_1(0)$, and its matrix of second derivatives is the moment matrix $\left(\begin{smallmatrix} m_0 & m_1 \\ m_1 & m_2 \end{smallmatrix}\right)$ of the deformed spectrum. This is the Hankel matrix of Eq. (8.23), positive definite for any spectrum with more than one point of support. As a result F is strictly convex and has at most one stationary point, which exists when the prescribed mean lies inside the support, as the free-recoil energy does. Since $S^{\mathfrak{T}_A^+}$ obeys the same two moment conditions, the solution at $\vartheta = 1$ is $\lambda = 0$, and Eq. (8.77) passes through the next order exactly. The second moment is not protected. It contains the mean kinetic energy of the nucleus along $\boldsymbol{q}$, which anharmonicity changes, and its change is a separate nuisance with its own computed size. The tilt of Eq. (8.77) is not limited to the nuclear self response: any truncation that must preserve known linear moments uses it, with one λ per moment.

8.8.4 Nuisances in the likelihood

A search records counts $N_{\mathrm{obs},j}$ in bins j of the measured quantity, pair number for a charge detector, phonon energy for a calorimeter. Its expected signal in bin j is $N_{\mathrm{sig},j} = \bar{\sigma}_e\, M_{\mathrm{det}} t_{\mathrm{exp}}\, r_j(m_\chi; \vartheta)$, with r_j the rate per unit cross-section into the bin, computed from Eq. (8.69) with the response $S(\vartheta)$ in place of S. It generalizes the $r(m_\chi)$ of Sect. 1.5 to one bin and to nonzero nuisances. With a known background $N_{\mathrm{bkg},j}$, the Poisson probabilities of Eq. (1.51) and a standard normal constraint on each ϑ_k, the likelihood of one exposure is

$$\mathsf{L}(\bar{\sigma}_e, \vartheta) = \prod_j P\big(N_{\mathrm{obs},j} \,\big|\, N_{\mathrm{sig},j}(\bar{\sigma}_e, \vartheta) + N_{\mathrm{bkg},j}\big) \prod_k \mathrm{e}^{-\vartheta_k^2/2}. \tag{8.78}$$

The halo enters r_j at the fixed values of Sect. 1.2, by the convention of Sect. 1.5.5, and the detector's own nuisances, efficiency and background, enter in the same way (Sect. 21.4). The nuisances are removed by profiling [30, Eq. (7)], Eq. (C.122) of Appendix C.14: at each $\bar{\sigma}_e$ every ϑ_k takes the value that maximizes L. To first order in ϑ, a normalization nuisance $(1 + \delta_k)^{\vartheta_k} \simeq 1 + \vartheta_k \ln(1 + \delta_k)$ is the Gaussian offset of Appendix C.14 with the width $\sigma_\vartheta = \ln(1 + \delta_k)$, and the widths add in quadrature as in Eq. (C.123). The power form differs only where δ_k is not small, and there it is the one that stays positive. The general treatment of nuisances in a profile likelihood, and the limit built from it, are the subject of Appendix D.47 and Sect. 21.4.

How much does a theory nuisance cost? One bin with no events and no background shows the structure. With $N(\vartheta) = N_{\mathrm{nom}}(1+\delta)^{\vartheta}$, Eq. (8.78) gives $-2\ln \mathsf{L} = 2N(\vartheta) + \vartheta^2$ up to a constant. The best fit, at $N_{\mathrm{nom}} = 0$ and $\vartheta = 0$, has the value zero. At fixed N_{nom} the profile condition $2kN(\hat{\hat{\vartheta}}) + 2\hat{\hat{\vartheta}} = 0$, with $k = \ln(1+\delta)$, gives $\hat{\hat{\vartheta}} = -kN(\hat{\hat{\vartheta}})$. The nuisance is pulled toward less signal, since no events were seen, and the profiled statistic is $\mathsf{t}_{\mathrm{p}} = 2u + k^2u^2$, with $u = N(\hat{\hat{\vartheta}})$. The asymptotic thresholds of Appendix C.14 fail at zero counts, and for this illustration we set the threshold to the value $\mathsf{t}^* = 2N_{90} = 4.61$ at which the statistic without nuisance reproduces the exact limit $N_{90} = 2.303$ of Eq. (1.52). Solving $\mathsf{t}_{\mathrm{p}} = \mathsf{t}^*$ for u, and $N_{\mathrm{nom}} = u\,\mathrm{e}^{k^2u}$, the limit grows by the factor

$$\frac{N_{\mathrm{nom}}^{\mathrm{lim}}}{N_{90}} = \frac{2w\,\mathrm{e}^{w}}{k^2\mathsf{t}^*} = 1 + \frac{k^2\mathsf{t}^*}{4} + O(k^6), \qquad w = \left(1 + k^2\mathsf{t}^*\right)^{1/2} - 1. \tag{8.79}$$

The cost is of second order in the size of the truncation: 1.0 % for $\delta = 0.1$, 8 % for $\delta = 0.3$ and 19 % for $\delta = 0.5$. Quoting instead the worst case within one standard deviation, $\vartheta = -1$, costs $1 + \delta$, of first order: 10 %, 30 % and 50 %. The two are different statements, a likelihood and an envelope, and a result must say which it reports. The power form matters here as well. A linear nuisance $1 + \vartheta\delta$ can switch the signal off entirely at $\vartheta = -1/\delta$, at the price $\vartheta^2 = 1/\delta^2$ in the likelihood, and for $\delta > (\mathsf{t}^*)^{-1/2} = 0.47$ that price is below the threshold: the profile then excludes nothing at all.

With several bins the data begin to constrain the nuisances themselves. A normalization nuisance is degenerate with $\bar{\sigma}_e$ and is never constrained by the counts. A tilt or a threshold shift changes the shape of the expected spectrum, and a measured spectrum with enough events can prefer one value of ϑ_k over another, which an envelope of fixed width cannot express. Since ϑ_k is shared by every target and experiment that uses the same truncation, a combination of searches fits one value for it, not one per experiment (Sect. 21.5).

Worked example 8.10: The local-field nuisance of the random-phase rate in silicon

A rate computed with the random-phase approximation of the head element alone, Eq. (4.67), sits at $\mathcal{X} = \mathrm{RPA}$. The next approximation adds the local fields, $\mathcal{X} = \mathrm{RPA} + \mathrm{LF}$ of Eq. (4.68), and the template of its nuisance is $\delta_{\mathcal{X}} = S^{\mathrm{RPA+LF}}/S^{\mathrm{RPA}} - 1$ for the density structure factor. We compute both approximations in the empirical pseudopotential model of silicon used in Worked example 8.9, with the same bands, the same 51 local-field vectors and the same Lorentzian half-width of 0.10 eV, at $q_0 = 3.73$ keV and $2q_0 = 7.46$ keV along [100] and at energies from 0.5 to 100 eV. The model gap is 0.82 eV, and we quote energies from 2 eV up. The quoted values change by 0.01 or less between 16^3 and 20^3 points of $\boldsymbol{k}$. The comparison differs from the 12 % of Worked example 8.9, which compared the head of the inverse dielectric matrix with the full column inside one approximation; here two approximations are compared.

At q_0 the template is small and has structure. It is -0.06 at 2 and 3 eV, passes through zero at 4 eV, is -0.03 at 5 to 7 eV and -0.06 at 10 eV, and reaches its minimum, -0.15, near 17.5 eV. It crosses zero at 25.6 eV and is $+0.02$ at 30 eV. At $2q_0$ the local fields matter more: $\delta_{\mathcal{X}} = -0.44$ at 2 eV, -0.30 at 4 eV, -0.22 at 10 eV and -0.06 at 20 eV, with the zero near 29 eV. At small q, finally, the local-field reduction of ε_∞ alone raises the screening factor $1/\varepsilon_\infty^2$ by $(11.45/10.22)^2 \simeq 1.25$ (Sect. 4.6), so that $\delta_{\mathcal{X}} \approx +0.25$ at the bottom of the spectrum.

The nuisance is a tilt, in ω and in q. Weighted with the random-phase spectrum between 1.5 and 30 eV, its mean is -0.06 at q_0 and -0.09 at $2q_0$. A normalization nuisance of that size would understate the template near the gap at $2q_0$ by a factor of about five and would

have the wrong sign at small q. The f-sum rule explains the change of sign. Over 0.5 to 100 eV the first moments of the two approximations agree to 1×10^{-4} at q_0 and 2×10^{-4} at $2q_0$, the residue that the Lorentzian tails and the finite grid leave, while the zeroth moments fall by 2.1 % and 3.0 %. The local fields move weight upward, as Eq. (8.76) demands and as Sect. 4.6 found for the optical spectrum, and what they remove below 25 eV reappears above it. A particle of mass m_χ deposits at most $\frac{1}{2} m_\chi v_{\max}^2 = 3.5\,\mathrm{eV}$ per MeV, so dark matter lighter than about 7 MeV samples only energies where $\delta_\mathcal{X} \le 0$ at these momenta and never meets the compensating weight.
In the likelihood a single $\vartheta_\mathcal{X}$ multiplies $\ln(1 + \delta_\mathcal{X})$ at every point. At $\vartheta_\mathcal{X} = 1$ the rate is that of the random-phase approximation with local fields. Once that approximation is computed it becomes the central value, and the nuisance moves to the step to the adiabatic kernel or to the Bethe–Salpeter equation. The adiabatic kernel restores about half of the local-field reduction of ε_∞ (Sect. 4.6), so its template has the opposite sign at small q, and near the gap the Bethe–Salpeter step is the larger one (Worked example 8.7).

The uncertainty of a rate then rests on computed steps wherever they exist, and it names its omissions where they do not. The likelihood carries the templates to the limit, and Sect. 21.4 sets it. Each channel of the following chapters closes with a table of its nuisances in this form.

Summary

A slow dark-matter particle sees an electron or a nucleus through the momentum it hands over, their relative velocity and their spins, and nothing else (Sect. 8.1). Every nonrelativistic interaction therefore reduces to five target operators per species: the density, the spin density, the current, the spin current and a kinetic-energy density. The model survives only in the dark-matter-side tensor $\mathfrak{D}_{ab}$, a Hermitian positive semidefinite matrix that weights the response matrix of these operators. A heavy dark photon reduces to the density rows of electrons and nuclei; a spin coupling to ^{29}Si responds with 10 % of the strength of one neutron, against the $A^2 = 841$ of the spin-independent coupling. Absorbed bosons and fermions use the same operators at the momentum of the absorbed particle, and the scalar couples through the kinetic-energy density as well as the density.

The contraction of $\mathfrak{D}_{ab}$ with S_{ab}, integrated over the momentum transfer and the halo, is the master rate formula (Sect. 8.2). It holds at any temperature and for energies of either sign, and its integrand is never negative. Scattering fills a region of the (q, ω) plane, absorption a point, the absorption of a fermion a line, and an exothermic particle shifts the scattering band upward. Detailed balance makes the down-scattering of an exothermic particle and the up-scattering driven by a warm target one process. Cross terms between rows need both a dark-matter coefficient and a target correlation, and a polarized electron gas shows a destructive one that the sum of single-operator rates misses.

Exact relations constrain every target (Sect. 8.3). Positivity of the response bounds any interference term by twice the geometric mean of the separate rates. The frequency moments of the response form matrices, the odd ones fixed by commutators, the even ones by equal-time correlations, and all of them obey Hankel positivity. The mixed third moment of electrons and nuclei is set by the electron density at the nucleus, $1765\,a_0^{-3}$ for a silicon atom. Momentum conservation gives a Ward identity that ties the mass-density response to the momentum flux, so that a coupling proportional to mass shakes the interior of a crystal only at order q^4.

Screening is a property of the coupling, not of the target alone (Sect. 8.4). The response to any row is its proper response plus a term in which the total dielectric matrix of electrons

and nuclei carries the induced charge back to the row. Only a coupling to the total charge reproduces the loss function at every $(\boldsymbol{q}, \omega)$, and an electron-density coupling does so only at a static lattice. A scalar that couples to the nucleons of GaAs excites the phonon at the longitudinal energy 36.35 meV, not at the transverse 33.62 meV, with its f-sum weight unchanged, and an absorbed dark photon in silicon at 3 eV is suppressed by the factor 770 of in-medium mixing. In a phonon-assisted rate each Coulomb line belongs to exactly one of three classes, which fixes where each screening factor goes and counts each once.

Every rate is a truncation of one expansion (Sect. 8.5). Seven small parameters control it. Their silicon values at 10 mK are the Born–Oppenheimer parameter 0.0665, the off-shell parameter 0.010 to 0.016 at the edge, the anharmonic parameter 0.049, the thermal ratio 4.6×10^{-5}, the recoil parameter 0.047 at 10 keV, the suddenness ratio 7×10^{-6}, and the multipole parameter, 1.9 in electron scattering and 0.05 in the Migdal effect. The truncation label $\mathfrak{T}$ records the order a formula keeps in each and the many-body approximation, lattice regime and Migdal regime it uses. The many-body approximation is not an order: its error is estimated by computing the next one.

In a target with a gap the final states split into three sectors (Sect. 8.6): electrons following the nuclei, electronic excitations without phonons and electronic excitations with phonons. The split is unique to the order of the off-shell parameter inside the off-shell window, and the Dyson convention extends it near resonances so that totals and cascade stay consistent. The electrons that follow a silicon nucleus carry a share 2.73×10^{-4} of the f-sum, and the Fano coefficients that couple production and decay vanish in silicon and germanium.

What the detector counts depends on which final state was made (Sect. 8.7). The final-state-resolved response sums to the full response, and the screened pair weights follow from a factorization of the random-phase response. At q_0 and 10 eV in silicon, dark matter leaves the electron with 4.8 eV and the hole with 4.4 eV on average, while a photon of the same energy gives 5.7 eV and 3.5 eV, with about 0.9 eV less in the hole. The cascade must therefore start from the pair distribution at the momentum transfer that made it, and the hand-off to the cascade is a probability of each pair number at each $(\boldsymbol{q}, \omega)$.

Each truncation becomes a nuisance parameter whose size is the computed step to its next order (Sect. 8.8). One parameter acts coherently at every momentum, energy, final-state class, target and experiment that share the approximation. Nuisances act as normalizations, tilts or threshold shifts, or open new support, and the exact relations constrain them: positivity, the interference bound, the f-sum rule that makes every many-body nuisance a tilt, detailed balance and the moments of the recoil spectrum. In the likelihood of one exposure a profiled nuisance costs a limit at second order in its size, an envelope at first order. In silicon the step from the random-phase approximation to local fields is such a tilt, from -0.44 to $+0.25$ across momentum and energy. Chap. 9 begins to apply the framework, by mapping each excitation of a solid and each coupling onto the sections that treat them.

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Part II

Channels

Chapter 9

Channel atlas

A dark-matter particle of 1 MeV brings at most 3.5 eV of kinetic energy into a silicon crystal (Sect. 1.3). That is enough to lift an electron across the gap of 1.17 eV, or to create an optical phonon of 64.1 meV, or, if the particle couples to nucleons, to make a nucleus recoil, which now and then leaves its electrons excited (the Migdal effect). Which of these happens depends on what the particle couples to and on its mass. GaAs has polar phonons that a dark photon sees, an antiferromagnet has spin waves that only a spin coupling creates, and aluminum has broken Cooper pairs at 0.36 meV. Chap. 8 gave the rate for all of them: every coupling reduces to a few target operators, the rate contracts their response with a tensor of the dark-matter model, and the final states split into sectors so that none is counted twice. The chapters that follow apply it to two dozen kinds of excitation and six classes of coupling. At that size the hazard is counting a process twice or not at all. The literature computes many channels with formulas of its own, and a process can enter two of them: phonon-assisted absorption above the direct gap contains a cascade that direct absorption counts again (Sect. 6.5). A process can also enter none, as the spin flip of a core electron by a spin coupling enters none of the rates of the following chapters.

A dark photon reaches the electrons and the polar phonons of GaAs through one charge response, and adding its electron-recoil rate to its phonon rate counts the same final states twice unless the sectors of Sect. 8.6 keep them apart. A detector that counts pairs in silicon reaches masses from 0.333 to 0.405 MeV through pairs created together with a phonon, which a zero-phonon rate does not contain. And a published formula that is accurate for one material and one range of momentum transfer is used outside both, unless something records where it fails. Before any rate is computed, we therefore fix for each pairing of an excitation with a coupling the section that computes it, the lightest dark-matter mass it reaches and where its final states lie on the plane of momentum transfer and energy deposit.

9.1 The excitation–interaction matrix

Table 9.1 has 24 rows, one for each kind of excitation that this book treats, and six columns, one for each class of dark-matter coupling. Of its 144 cells, 57 hold a channel, and each of those names the one section that computes its rate. Eight more are allowed at leading order but computed in no section. The remaining 79 are empty at leading order, for reasons of operator content, kinematics or order in the coupling that we work out below. A channel is a pair of row and column, so the table is also a check that no channel is computed twice: where two sections seem to treat the same process, they treat different cells, or different parts of one response.

9.1.1 Excitation and interaction axes

The columns come from the reduction of Sect. 8.1. Every slow interaction, written in the operator basis $\mathcal{O}_1, \dots, \mathcal{O}_{15}$, reduces to a few target operators per species s: the number density n_s, the spin density $\boldsymbol{s}_s$, the paramagnetic current $\boldsymbol{j}_s$, the spin current $\mathcal{J}_s^{\alpha\beta}$ and the kinetic-energy density $\mathcal{T}_s$ (Eq. (8.3) and Table 8.1). Three distinctions separate the columns. The first is scattering against absorption. A scattered particle deposits ω over a region of the $(\boldsymbol{q}, \omega)$ plane bounded by the kinematics of Sect. 1.3, while an absorbed boson deposits its mass at the single point $\boldsymbol{q} = \boldsymbol{p}_\phi = m_\phi \boldsymbol{v}$, $\omega = m_\phi$ (Sects. 8.1.6 and 8.2.3). The second is the species: the electron rows and the nuclear rows reach different final states, as Sect. 8.6 showed. The third is the operator: a density cannot flip a spin, and a spin density carries no charge.

The first column is the one most models use. *Electron density* holds every scattering coupling whose leading target operator is n_e, for a heavy or a light mediator: $\mathcal{O}_1$ with the coefficient $c_1^{(e)} = \tilde{V}$, and the density parts of $\mathcal{O}_5$, $\mathcal{O}_8$ and $\mathcal{O}_{11}$. *Nuclei* holds the couplings to n_κ, with the coherent weight $Z c_1^{(p)} + (A - Z) c_1^{(n)}$ of Eq. (8.8). *Spin and current* holds the couplings to $\boldsymbol{s}_s$, $\boldsymbol{j}_s$ and $\mathcal{J}_s^{\alpha\beta}$, of electrons or of nuclei: $\mathcal{O}_4$ and the other operators of Table 8.1 marked as flipping spins, and the current parts of $\mathcal{O}_2$, $\mathcal{O}_5$ and $\mathcal{O}_8$. In the lattice rows its cells are those of the nuclear spin. *Vector absorption* is the absorption of a dark photon, whose row $-\boldsymbol{j}_e + \sum_\kappa Z_\kappa \boldsymbol{j}_\kappa$ is the current of all charges, or of another vector boson. *Axion-like and scalar absorption* uses $\boldsymbol{s}_e$ and the trace $\mathcal{J}_e^{\alpha\alpha}$ for the axion-like particle (Eq. (8.10)) and $n_e - \mathcal{T}_e / m_e$ for the scalar (Eq. (8.11)), or their nucleon analogues.

The last column keeps the target rows and changes the kinematics. It collects inelastic dark matter with a mass splitting Δ_χ (Eq. (8.13)) and its exothermic version, which deposits more than its kinetic energy (Sect. 8.2.4). It also collects dark matter boosted to relativistic speed, and fermionic absorption, in which the dark matter turns into a light fermion and deposits energy along the line $\omega = m_\chi - q$ (Eq. (8.14)). A coupling to the electron density thus belongs to the first column when the dark matter scatters elastically and to the last when it scatters inelastically. The last column, however, holds a cell only where the change of kinematics changes the physics (Sect. 9.1.3).

The rows follow the solid-state sectors of the earlier chapters, in the order in which the channel chapters take them up. Electronic excitations at clamped nuclei come first, then lattice excitations, then the mixed processes in which a phonon and an electronic excitation are created together, and last the magnetic, superconducting and other targets. Most rows are excitations of Chaps. 3 to 7, and Table 9.1 cites their defining sections. One row is new. *Phonon-assisted electron recoil* is the creation of an electron–hole pair together with a phonon, which supplies crystal momentum that the dark matter cannot. It is the scattering counterpart of the phonon-assisted absorption of light in Sect. 6.5, and Sect. 13.2, which derives its second-order amplitude, makes it precise. In silicon it is the only way that dark matter scattering on electrons can create a pair at masses just below the zero-phonon threshold, because the gap is indirect (Sect. 3.3). Rows that no earlier chapter introduces carry a one-line description.

Table 9.1 is read in two directions. Along a row it lists the couplings that can create a given excitation, which is the question of a detector built to read out that excitation. Down a column it lists the excitations that a given coupling can create, which is the question of a model builder, and of Chap. 17. Some entries stand for a short run of sections, and the table names the one that computes the rate. The density cell of the pair row, for instance, is the crystal form factor of independent electrons in Sect. 10.1, and Sect. 10.2 screens it by the medium and writes it through the loss function. The nuclear cell of the Migdal row is the probability of Sect. 12.3, written through the loss function of the crystal, which Sect. 12.2 derives in the atomic limit and Sect. 12.4 extends to phonon final states.

Table 9.1: The channel atlas. Rows: excitations of a solid, with their defining section or a short description. Sector: the final-state sector of Eq. (8.48) (Sect. 8.6). ad: electron-following; 0: zero-phonon electronic; +: phonon-dressed electronic; ad^{s}: electron-following on the spin manifold of a magnet; qp: the quasiparticle-number sectors of a gapless target; $(2,0)$: pair breaking, and $(2,m)$ pair breaking with m phonons. Columns: classes of coupling, with the target operators of Sect. 8.1; ALP: axion-like particle; abs.: absorption. Entry: the one section that computes the rate; ∘: allowed at leading order, computed in no section; —: no channel at leading order (Sect. 9.1.3). The density cell of single phonons is the coupling to charge through the Born charges; for a coupling to electrons alone, the acoustic peak of the electrons that follow the nuclei is computed in Sect. 10.2. The absorption cells of excitons rest on measured optical data.

Excitation	Sector	Electron density n_e	Nuclei n_κ	Spin, current $\boldsymbol{s}$, $\boldsymbol{j}$, $\mathcal{J}$	Vector abs. $\boldsymbol{j}_e$, $\boldsymbol{j}_\kappa$	ALP, scalar abs. $\boldsymbol{s}_e$, $\mathcal{J}_e^{\alpha\alpha}$, $\mathcal{T}_e$	Inelastic, other rows of $\boldsymbol{q}$, Δ_χ
Electron–hole pairs, band to band (3.2, 4.2)	0	10.1	—	10.4	10.7	10.7	10.9
Electron-gas continua and plasmons (4.5)	0; qp	10.3	—	—	10.7	—	17.3
Excitons (4.9)	0	10.5	—	10.5	10.7	10.7	—
Core and semicore electrons (3.5)	0	10.6	—	∘	10.7	—	—
Single phonons (5.3, 5.5)	ad	11.2	11.1	11.6	11.8	11.8	17.3
Multiphonons (5.6)	ad	11.3	11.3	11.6	—	—	—
Free nuclear recoil (1.3, 5.6)	ad	∘	11.4	11.6	—	—	11.4
Quantum crystal: solid helium	ad	—	11.5	—	—	—	—
Migdal effect: electrons with a nuclear recoil (6.1)	0, +	—	12.3	∘	—	12.6	—
Phonon-assisted electron recoil (6.5)	+	13.3	—	∘	13.8	13.8	—
Magnons (7.1)	ad^{s}	—	—	14.1	14.3	14.3	17.3
Two-magnon continua, magnon–phonon modes	ad^{s}	—	—	14.2	—	—	—
Chiral and nonreciprocal modes: crystals without inversion or time reversal	ad, ad^{s}	—	—	14.5	14.5	14.5	—
Precessing nuclear spins in a coherent field	ad	—	—	—	—	14.6	—
Superconducting quasiparticles (7.2)	$(2,0)$, $(2,m)$	15.1	—	∘	15.4	15.4	—
Dirac and narrow-gap pairs (7.3)	qp; 0	16.1	—	∘	16.1	—	—
Plasmons of sheets, wires and layered stacks	0; qp	16.2	—	—	—	—	—
Graphene, nanotubes, quantum dots: confined electrons	0; qp	16.3	—	—	16.3	—	—
Frenkel excitons, localized on one molecule	0	16.4	—	∘	—	—	—
Dopant transitions (7.4)	0	16.5	—	∘	16.5	—	—
Color centers and broken bonds (7.4)	ad	—	16.6	—	—	—	—
Two-level systems: tunneling atoms of a glass	ad	16.7	—	—	16.7	—	—
Mobile defects: defectons and crowdions, kinetics only (19.6)	ad	—	—	—	—	—	—
Acoustic modes of the whole body	ad	—	—	—	11.8	11.8	—

Coherent scattering off a whole random medium and the absorption of a dark photon by the vibrations of a glass, where no crystal momentum selects one optical phonon, are not single excitations and have no row; Sect. 16.7 treats them with the two-level systems. Processes second order in the coupling have no row either (Sect. 9.1.3).

The three entries of Sect. 17.3 in the last column are one formula, for exothermic or inelastic scattering into a collective mode, that serves phonons, magnons and plasmons alike, which is why the comparison chapter is their home. Couplings to the nuclear spin, and to baryon-minus-lepton number, which differs between isotopes through their neutron number, change from site to site. They reach the lattice through the random couplings of Sect. 11.6. That section holds the spin entries of the lattice rows, from one phonon to a free recoil, where its self response at large q becomes the impulse limit, and also the absorption of such bosons by phonons.

9.1.2 Operators and sectors of the rows

A row of Table 9.1 is a set of final states, and the decomposition theorem of Sect. 8.6 assigns every final state of a target with a gap to exactly one of three sectors. The sector of a row says which partial sum rule its rate shares and which other rows it can be added to. Two rows in different sectors add, apart from the interference cells of Table 8.5; two cells in the same sector and the same column are pieces of one response, and their sections compute different parts of it, not terms of a sum.

The rows built on electronic excitations of an insulator sit in the zero-phonon electronic sector, $\mathcal{P}_0$. A band-to-band pair, an exciton, a core hole, a pair on a dopant and a Frenkel exciton all change the electronic state and leave the phonon number of the reference lattice unchanged at their zero-phonon line. The four electronic rows of the density column, pairs, plasmons, excitons and core electrons, are thus one response, S^0_{ee}, divided by the region of the $(\boldsymbol{q}, \omega)$ plane and by the approximation that is accurate there. The exciton row is the part of the spectrum just above $E_{\min}(\boldsymbol{q})$ that the electron–hole attraction of Sect. 4.9 reshapes, the plasmon row the collective pole of Sect. 4.5, and the core row the shells that open at large ω and q. The rate is the integral of S^0_{ee} once, and the f-sum rule of Sect. 4.3 constrains the four cells together.

Phonons, multiphonons and a freely recoiling nucleus are all in the electron-following sector, $\mathcal{P}_{\mathrm{ad}}$: the electrons stay in their adiabatic ground state and the nuclei carry the energy. These three rows differ in the lattice regime alone of Sect. 8.5, one phonon, several or a recoil, which grows with q through the Debye–Waller exponent of Sect. 5.6. The quantum crystal, the color centers created by a recoil, the two-level systems of a glass and the acoustic modes of a finite body are lattice states as well. Precessing nuclear spins belong to the same sector, although they are not a lattice state: the electrons stay in their ground state and the nuclear moments carry the energy. A color center is a different arrangement of the nuclei with the electrons again in their ground state; the electronic excitation that reads it out comes later and is not part of the dark-matter vertex. The electron-following sector is also how an electron coupling reaches the lattice. Through the density $\bar{\rho}_e(\boldsymbol{q}; u)$ of electrons that follow the nuclei, Eq. (8.51), the electron-density column has entries in the single-phonon and multiphonon rows. In a polar crystal they enter through the electronic part of the Born effective charges, and at large momentum through the part of the electron cloud that a displaced atom carries along.

Two rows fill the phonon-dressed electronic sector, $\mathcal{P}_+$. The phonon-assisted row is the sideband of the electronic transitions, an electronic excitation with a net change of phonon number. The Migdal row straddles two sectors. The nuclear density reaches an electronic excitation only through the nonadiabatic boost of Eq. (8.53); its zero-phonon term lies in

$\mathcal{P}_0$, and the terms in which the kick also creates phonons, or a free recoil, lie in $\mathcal{P}_+$. The interference of a Migdal amplitude with an electron recoil to the same final state is an off-diagonal block $S_{e\kappa}$ in these two sectors, nonzero for a coupling to both species (Table 8.5). Sect. 12.5 computes its zero-phonon block, and, with Sect. 17.4, which treats the interference between channels, its phonon-dressed block.

The decomposition theorem needs a gap, and four kinds of row lack one in the sense of Sect. 8.6. For a magnetic insulator the construction goes through with the low-energy spin manifold in place of the electronic ground state, so magnons and phonons share an electron-following sector, ad^{s} in the table. Superconducting pairs are counted by the number of Bogoliubov quasiparticles: pair breaking is the sector $(2, 0)$, the analogue of $\mathcal{P}_0$, and pair breaking with a phonon, $(2, m \geq 1)$, the analogue of $\mathcal{P}_+$. For a metal, a Dirac semimetal and a gapless sheet such as graphene the quasiparticle-number sectors replace the three projectors, with the caveats of Eq. (8.63). A narrow-gap semiconductor and a semiconducting nanotube have a gap and use $\mathcal{P}_0$, which is why several rows carry two entries.

9.1.3 Empty cells and selection rules

A cell is empty when the target operator of its column has no matrix element into the final states of its row at leading order. Leading order means first order in the coupling, one insertion of the interaction as in the master rate Eq. (8.15), and the lowest order of the expansion of Sect. 8.5 at which the process occurs. Operator content, the kinematics of absorption, the order in the coupling and the absence of a new response account for nearly all of the 79 empty cells.

Operator content. The nuclear density cannot create an electronic excitation in the Born–Oppenheimer approximation. Its matrix element between the adiabatic ground state and an excited adiabatic state carries the electronic overlap $\langle \Psi_j(u) | \Psi_0(u) \rangle$, which vanishes at every nuclear configuration (Sect. 8.6). The nuclear column is empty in every electronic row, and the nonadiabatic remainder, which does create pairs, is the Migdal row. In the same way the electron density cannot create a magnon. The density $n_e(\boldsymbol{q}) = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$ acts on positions and commutes with the total electron spin, while a one-magnon state differs from the ordered ground state by one unit of the spin component along the order (Sect. 7.1). Without spin–orbit coupling the matrix element vanishes, and the density column is empty in every magnetic row. The converse rule empties the spin column of the plasmon row. A spin density creates electron–hole pairs in the spin triplet, which carry no charge and produce no Coulomb field, and the plasmon is a pole that exists only because the induced charge feeds back through the long-range Coulomb interaction (Sect. 4.5). The pair row has a spin entry, the plasmon row none.

Kinematics of absorption. An absorbed boson brings $\omega = m_\phi$ at $q = m_\phi v \approx 10^{-3} m_\phi$. An acoustic phonon of that momentum has the energy $c_\nu m_\phi v$, with c_ν between 1.6×10^{-5} and 3.1×10^{-5} in silicon (Sect. 5.2), some 10^{-8} of the boson mass. Since no acoustic phonon of an infinite crystal can absorb a boson, the single-phonon absorption entries are optical phonons at the zone center. A freely recoiling nucleus fares worse. Taking the momentum $m_\phi v$, it gains the energy $(m_\phi v)^2/2M$, a fraction $m_\phi v^2/2M$ of the boson mass. Exactly, a free particle at rest absorbs a boson of energy E_ϕ and momentum $\boldsymbol{p}_\phi$ only if $(E_\phi + M)^2 - p_\phi^2 = M^2$, that is $m_\phi^2 + 2E_\phi M = 0$, which no massive boson satisfies. A vector or axion-like coupling at $\boldsymbol{q} \to 0$ is linear in the nuclear velocities, so in a harmonic crystal it creates exactly one phonon. Two phonons need the nonlinear dependence of the dipole on the displacements, or anharmonicity, both higher orders of Sect. 8.5, and the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{p}_\phi \cdot \boldsymbol{u}}$ adds a correction of order $p_\phi^2 \langle u^2 \rangle$. The multiphonon and recoil rows are thus empty in the absorption columns. Two rows exist because these rules can be evaded. A body smaller than the wavelength of the boson

has discrete acoustic modes, at MHz in a millimeter crystal, that a coherent field drives as a whole (Sect. 11.8, which treats absorption on phonons and on the acoustic modes of a finite body). And a coupling that changes at random from site to site, such as the coupling to the nuclear spin of ^{29}Si, does not conserve crystal momentum. A boson of negligible momentum can then create a phonon of any wavevector (Sect. 11.6, which treats such random couplings).

Order in the coupling. Two processes act on the target with two vertices: the absorption of two dark-matter particles at once, and dark Compton scattering, in which an absorbed boson leaves as a photon, $\phi e \to \gamma e$, with one dark and one electromagnetic vertex. They belong to no single cell and have no row (Sect. 10.8, which builds their second-order transition operator).

No new response. Inelastic or boosted dark matter changes where on the $(\boldsymbol{q}, \omega)$ plane a target response is sampled, not the response. For most rows the rate of the last column follows from the elastic cell of the same operator with the kinematics of Eq. (8.13), and the cell is left empty. The column has entries where the change of kinematics changes the physics. For pairs these are the endothermic threshold and the boosted and fermionic cases (Sect. 10.9, which treats inelastic and relativistic dark matter). For nuclei it is fermionic absorption, where the outgoing fermion carries $q \simeq m_\chi$ and the nucleus recoils with $m_\chi^2/2M$ (Sect. 11.4). For collective modes it is exothermic down-scattering at a fixed energy (Sect. 17.3).

The row of mobile defects is empty by construction. Dark matter does not create defectons or crowdions at leading order; the row records that they release the energy of earlier excitations late, through the defect kinetics of Sect. 19.6.

The eight cells marked ∘ are allowed at leading order, but no section computes their rate. In seven of them a spin coupling has a matrix element into the final states of the row. It can flip the spin of a core electron, break a Cooper pair or accompany a Migdal transition; it can create a Dirac pair or a phonon-assisted pair in the spin triplet; and it can flip the spin of a dopant electron or excite a molecule into its triplet state. The eighth is the electron density acting on a freely recoiling atom through the electrons that move with it (Eq. (8.51)), whose rigid-atom density $f_\kappa(q)$ enters the multiphonon vertex $Z_\kappa - f_\kappa(q)$ of Sect. 11.3. Sect. 22.2, which collects the open items of the book, lists the eight as one gap. A populated cell can also be empty for a particular material. Silicon has $Z_\kappa^* = 0$ (Sect. 5.5): at long wavelength the electrons that follow each nucleus cancel its charge, so a dark photon couples to no single optical phonon of silicon, while in GaAs it couples through Z_κ^*, screened by ε_∞. And a clean metal or superconductor, described as a free-electron gas, absorbs no dark photon below its interband transitions. The current at zero momentum is then the total momentum of the electrons divided by m_e, which the electron–electron interaction conserves, so absorption needs a third body, an impurity or a phonon; Sect. 15.4, on absorption in superconductors and metals, computes both.

9.1.4 Cells of a dark-matter model

A heavy scalar mediator coupled to electrons enters only the density column: the electronic rows from pairs to core shells, the lattice through the electrons that follow the nuclei, the phonon-assisted row, and the superconductors and novel targets of Chaps. 15 and 16. The same mediator coupled to nucleons enters only the nuclear column. It runs from single phonons through multiphonons to a free recoil, adds the Migdal row and creates color centers and broken bonds, and in silicon its lowest-mass coherent channel is the single acoustic phonon (Sect. 11.1, which treats acoustic phonons). If its couplings to protons and neutrons differ, the isotopes of the crystal make it a random coupling, which reaches lighter masses incoherently (Sect. 11.6). A spin-dependent coupling of the form $\mathcal{O}_4 = \boldsymbol{S}_\chi \cdot \boldsymbol{S}_e$ reaches the triplet pairs and excitons of a nonmagnetic crystal, and in a magnet the magnons and their continua.

Coupled to nuclear spins instead, it reaches the lattice of an unpolarized crystal through the random couplings of Sect. 11.6. Each of these models leaves most of Table 9.1 empty. The dark photon, which couples to every charge, does not.

Worked example 9.1: The dark photon along its column

A dark photon couples to the electromagnetic current with strength ζe (Sect. 1.4), and Table 8.1 reduces it to the current of all charges, $-\boldsymbol{j}_e + \sum_\kappa Z_\kappa \boldsymbol{j}_\kappa$. We follow it through Table 9.1, first as the mediator of dark-matter scattering and then as the dark matter itself.

Scattering. For a heavy dark photon the interaction is $-\mathscr{V}^{-1}\tilde{V}(q)\, n_Z(\boldsymbol{q})$, with $n_Z = -n_e + \sum_\kappa Z_\kappa n_\kappa$ the charge-number density (Worked example 8.1), so the dark photon enters the density and nuclear columns at once, with opposite signs. In the medium it mixes with the longitudinal photon, and the rate becomes an integral of the loss function of the total charge (Sect. 8.4). That fixes its cells. All four electronic rows of the density column are open: the screened pair rate of Sect. 10.2, which writes the electronic rate through the loss function, with its plasmon, exciton and core parts in Sects. 10.3, 10.5 and 10.6. In the lattice rows the electron and nuclear blocks combine into the charge carried by a vibration. An acoustic wave of long wavelength moves neutral cells and carries no charge, so the dark photon creates no acoustic phonon at small q, whereas an electron-only coupling does. An optical phonon of a polar crystal carries the Born charge, and this is the polar single-phonon rate of Sect. 11.2; in silicon, with $Z^*_\kappa = 0$, the cell is empty. Above the zone the charge carried by a displaced atom is the nuclear charge minus the part of its cloud that follows, $Z_\kappa - f_\kappa(q)$, the charge vertex of the multiphonon response in Sect. 11.3. Once q exceeds the momenta of the core electrons the cloud is left behind, the vertex tends to Z_κ, and the dark photon makes the nucleus recoil freely with the weight Z_κ^2 (Sect. 11.4). The same recoil creates color centers (Sect. 16.6), and the same vertex, with the two electrons of helium, reaches the quantum crystal (Sect. 11.5). Because the dark photon couples to both species, the Migdal row opens, together with its interference with electron recoil, the cross cells of Table 8.5. The phonon-assisted row opens through its electron vertex (Sect. 13.3, which builds the phonon-assisted structure factor), and so do the superconducting, Dirac, layered, confined, molecular, dopant and glass rows of the density column.

Absorption. A dark photon of mass m_ϕ is absorbed at $q = m_\phi v$ and $\omega = m_\phi$, and with the current row it measures the loss function of the target at that point. The vector column of Table 9.1 lists where this is nonzero. On electrons it is the optical response at $\omega = m_\phi$ (Sect. 10.7), up through the L and K edges of the core shells; near the plasma frequency the dark photon converts into the plasmon. In silicon between the indirect gap of 1.17 eV and the direct gap of 3.4 eV no vertical transition exists (Sect. 3.3), and absorption needs a phonon, the phonon-assisted cell of Sect. 13.8. On the lattice the absorber is the zone-center longitudinal optical phonon of a polar crystal, at $m_\phi = \omega_\mathrm{L} = 36.35\,\mathrm{meV}$ in GaAs (Worked example 5.6), and nothing in silicon; a finite piezoelectric body adds its acoustic modes, which the field inside it drives through the polarization of a strain (Sect. 11.8). In a magnet the dark photon reaches the spins through its magnetic field, $\boldsymbol{B}' = \mathrm{i}\boldsymbol{p}_\phi \times \hat{\boldsymbol{e}}\, A'$, with $\hat{\boldsymbol{e}}$ the polarization and A' the field amplitude. That field is smaller than the electric one by the factor v (Sect. 14.3, which treats absorption into magnons), and single-molecule magnets absorb in the same way (Sect. 14.5). A superconductor needs an impurity or a phonon to absorb (Sect. 15.4), and the remaining entries, Dirac materials, aligned nanotubes, dopants and two-level systems, each measure their own loss function (Sects. 16.1, 16.3, 16.5 and 16.7).

The last column. Inelastic dark matter that scatters through a dark photon uses the same charge density with the kinematics of Eq. (8.13). Its endothermic and exothermic pair rates are in Sect. 10.9, and its exothermic down-scattering into plasmons and optical phonons in Sect. 17.3.
Altogether the dark photon populates all 14 cells of the density column, all six of the nuclear column, all 14 of the vector column and three of the last: 37 of the 57 channels of the atlas. It is the model that tests the atlas most, and the one for which double counting is easiest: its loss-function rate on electrons and its polar-phonon rate are two sectors of one charge response, and only the sector split of Sect. 8.6 keeps them apart.

9.2 The kinematic map

An aluminum film breaks Cooper pairs at $2\Delta_{\rm BCS} = 0.36\,\text{meV}$, and dark matter of 0.10 keV can break them (Worked example 7.3). A silicon crystal read out through its acoustic phonons at a threshold of 1 meV needs 6.1 keV on its fastest longitudinal branch, sixty times more, although its threshold is less than three times higher. Its electron–hole pairs need 0.333 MeV, and a free silicon nucleus receives the 1.17 eV of the gap from 46.8 MeV. A dark photon, by contrast, is absorbed in the same crystal from a mass of 1.17 eV. These lightest masses span almost eight decades, and three things fix them: the largest laboratory speed of the halo, the energy threshold, and the shape of the dispersion of the excitation. We derive the lightest mass of a channel from these three, place every channel of Table 9.1 on one map of mass against threshold (Fig. 9.1), and list the edges of silicon, germanium, GaAs, aluminum and the Dirac material $ZrTe_5$ in Table 9.2.

9.2.1 The lightest mass of a channel

A particle of mass m_χ at the largest laboratory speed $v_{\rm max}$ deposits at most $qv_{\rm max} - q^2/2m_\chi$ at momentum transfer q (Eq. (1.26)). Let $E_{\rm ch}(q)$ be the lowest energy, at or above the threshold $\omega_{\rm th}$, of a final state of the channel that carries momentum q; we write $\omega_{\rm th}$ for the threshold $E_{\rm th}$ of Sect. 1.3. The channel is open to m_χ if $qv_{\rm max} - q^2/2m_\chi \geq E_{\rm ch}(q)$ at some q. Solving for m_χ and taking the most favorable q gives the lightest mass of the channel,

$$m_{\rm min}^{\rm ch} = \min_q \frac{q^2}{2\big[qv_{\rm max} - E_{\rm ch}(q)\big]}, \qquad qv_{\rm max} > E_{\rm ch}(q), \tag{9.1}$$

which is the condition Eq. (3.59) read as a bound on the mass. Let $\bar{q}$ be the momentum at which the minimum is reached. Since $E_{\rm ch}(q) \geq \omega_{\rm th}$, the bracket never exceeds $qv_{\rm max} - \omega_{\rm th}$, and the minimum over q of $q^2/2(qv_{\rm max} - \omega_{\rm th})$ is $2\omega_{\rm th}/v_{\rm max}^2$. Every channel thus lies on or above the threshold mass $m_{\rm min}$ of Eq. (1.25), the floor of the map, and it reaches the floor only if it has a state of energy $\omega_{\rm th}$ at $q = 2\omega_{\rm th}/v_{\rm max}$. Four shapes of $E_{\rm ch}(q)$ cover the channels of Table 9.1, and absorption adds a fifth case.

Flat onset. If $E_{\rm ch}(q) = E_0$ near $q = 2E_0/v_{\rm max}$, the minimum lies there and $m_{\rm min}^{\rm ch} = 2E_0/v_{\rm max}^2$, the floor with $\omega_{\rm th} \to E_0$. Optical phonons are nearly flat across the zone [1, Sect. IV B, Eq. (94)]: the longitudinal optical phonon of GaAs at $\omega_{\rm L} = 36.35\,\text{meV}$ (Worked example 5.6) is reached from 10.35 keV, and the optical phonon of silicon at 64.1 meV (Sect. 5.2) from 18.3 keV. Pair breaking in a superconductor costs $2\Delta_{\rm BCS}$ at every momentum the halo delivers, because $v_{\rm F}q \geq \omega$ there (Sect. 7.2), and aluminum sits on the floor at 0.10 keV. A core shell costs its binding energy plus the gap. The $3d$ doublet of germanium, bound by 29.2 and

29.8 eV, opens at 8.5 and 8.7 MeV, and the $2p$ shell of silicon, which costs $E_0 = 99.4 + 1.17$ eV, at $2E_0/v_{\max}^2 = 28.6$ MeV (Sect. 10.6, which treats ionization of the core and semicore shells). The floor is twice the least momentum at which the shell absorbs, $E_0/v_{\max} = 38$ keV (Sect. 3.5), divided by $v_{\max}$. Two channels of gapless excitations also sit on the floor, because they do not tie the energy to one momentum: a coupling that changes at random from site to site creates a phonon of any wavevector (Sect. 11.6), and two magnons share the momentum between them (Sect. 14.2). At a threshold of 1 meV both open at 0.285 keV, 21.5 times below the acoustic edge of silicon and 5.1 times below the one-magnon edge of NiO; the map draws them as the floor line.

Gapless linear branch. If the lowest state at momentum q has energy $c_{\rm ch}q$, a state above threshold needs $q \geq \omega_{\rm th}/c_{\rm ch}$. The quantity minimized in Eq. (9.1) is then $q/2(v_{\max} - c_{\rm ch})$, which grows with q, so the minimum sits at $\bar{q} = \omega_{\rm th}/c_{\rm ch}$ and

$$m_{\rm min}^{\rm ch} = \frac{\omega_{\rm th}}{2c_{\rm ch}(v_{\max} - c_{\rm ch})} \simeq \frac{\omega_{\rm th}}{2c_{\rm ch}v_{\max}},$$

for $c_{\rm ch} < v_{\max}$. For phonons this is the *acoustic edge* $m_{\rm min}^{\rm ac}$, with $c_{\rm ch} = c_\nu$ the speed of the fastest branch that has a longitudinal component, since a density changes only under the component of the displacement along $\boldsymbol{q}$. It is the bound $\omega \lesssim 2c_\nu m_\chi v$ of Sect. 1.3 [1, Eq. (93)] solved for the mass. Every acoustic edge of this chapter uses this convention; Sect. 11.1 also quotes the edge for the polarization-weighted inverse speed of the rate, 6.39 keV in silicon at 1 meV. Sect. 11.1, which computes the rate of single acoustic phonons, writes the edge as Eq. (11.5) and finds how the rate approaches it. Compared with a flat onset of the same energy the edge is higher by $v_{\max}^2/4c_\nu(v_{\max} - c_\nu) \approx v_{\max}/4c_\nu$, a factor 21.5 for the longitudinal speed 9.35 km s^{-1} of silicon along [111] (Worked example 5.3). The reason is momentum. A phonon of energy $\omega_{\rm th}$ carries $\omega_{\rm th}/c_\nu$, and the dark matter hands over that much momentum by backscattering, at $q = 2m_\chi(v_{\max} - c_\nu)$, where it gives up a fraction $4c_\nu/v_{\max}$ of its kinetic energy. The magnons of an antiferromagnet are linear near their gapless points, with a velocity $c_{\rm M}$ near 10^{-4} (Worked example 7.2), some four times the fastest sound speed of silicon. In NiO the fastest magnon velocity is $c_{\rm M} = 1.36 \times 10^{-4}$, and the edge at a 5 meV threshold is 7.3 keV (Sect. 14.1, which computes the one-magnon rate of NiO).

If $c_{\rm ch} \geq v_{\max}$, no mass reaches the channel: the deposit $qv_{\max} - q^2/2m_\chi$ lies below $c_{\rm ch}q$ at every q. Two channels inside populated cells of Table 9.1 are closed this way to elastic scattering. The bulk plasmon of silicon, with the quadratic dispersion $\omega_{\rm pl}^2 = \omega_p^2 + \frac{3}{5}q^2v_{\rm F}^2$ of Sect. 4.5, has a phase velocity $\omega_{\rm pl}(q)/q$ that falls to 9.2×10^{-3}, or $3.5\,v_{\max}$, where it meets the pair continuum at 2.24 keV (Sect. 10.3, electron-gas models and the plasmon), so halo dark matter never excites it. The narrow-gap III–V semiconductors, however, are massive Dirac cones with $v_{\rm D} \approx 1.8\,v_{\max}$ (Sect. 7.3), and their pairs are closed to scattering at every mass, while absorption stays open.

Gapped linear branch. A branch $E_{\rm ch}(q) = (c_{\rm ch}^2q^2 + E_0^2)^{1/2}$ interpolates between the two shapes: flat for $c_{\rm ch} \ll v_{\max}$, closed for $c_{\rm ch} \geq v_{\max}$. The pair continuum of a Dirac material has this form with $c_{\rm ch} = v_{\rm D}$ and $E_0 = 2\Delta_{\rm D}$ (Eq. (7.41)). For $ZrTe_5$, with the half-gap $\Delta_{\rm D} = 11.8$ meV of the measured crystal structure (the relaxed cell of Worked example 7.4 gives 15.6 meV) and $v_{\rm D}/v_{\max}$ between 0.17 and 0.42 along its principal axes (Sect. 16.1), the lightest mass is 7.1 keV, with $\boldsymbol{q}$ along the slow axis; along the other two axes it is 9.0 and 9.6 keV. A flat continuum would give $4\Delta_{\rm D}/v_{\max}^2 = 6.72$ keV, and the isotropic cone with the geometric-mean velocity 7.9×10^{-4}, which Sect. 16.1 uses for the average over orientations, gives 8.0 keV. A magnon with a gap has the same shape. Single-ion anisotropy lifts the gapless mode of an antiferromagnet to the antiferromagnetic resonance, the geometric mean of the

anisotropy energy and twice the exchange energy of a spin (Eq. (14.26)). It lies at 1.1 meV in MnF_2 (Sect. 14.3, absorption into magnons). In NiO the anisotropy gaps the two branches differently: far-infrared absorption places one zone-center magnon at 4.54 meV, and the other lies near 0.6 meV (Sect. 14.1). Since $c_M \ll v_{max}$, a gapped magnon sits within a percent of the flat value: 1.30 keV for the mode at 4.54 meV.

Offset minimum. In an indirect-gap crystal $E_{ch}(q)$ comes down to E_g only near the valley momentum k_c (Eq. (3.57)), while the floor needs E_g at $q = 2E_g/v_{max}$, 0.88 keV in silicon against $k_c = 1.94$ keV. The minimum of Eq. (9.1) lies between the floor and the mass that deposits E_g at exactly k_c: between 0.333 and 0.474 MeV in silicon, and between 0.212 and 0.420 MeV in germanium (Worked examples 3.5 and 3.6). A phonon can take up the mismatch of momentum cheaply. At the floor the dark matter delivers $\bar{q} \approx 0.89$ keV, and the phonon supplies the rest of k_c, 1.06 keV, where a transverse acoustic phonon costs 14.8 meV. Absorption brings no momentum and needs a phonon at k_c itself, at 18.3 meV; that is why its threshold, 1.1728 eV, lies above the bound pair plus 15 meV. This places the phonon-assisted onset just above the floor: 0.3332 MeV for a pair bound to an exciton and 0.3374 MeV for a free pair in silicon, and 0.2119 to 0.2141 MeV in germanium (Sect. 13.1, which derives both thresholds of an indirect gap). Without a phonon the edge of silicon lies at 0.405 to 0.408 MeV (Sect. 13.9, which collects the silicon window), and between the two edges dark matter creates pairs together with a phonon or not at all. For germanium the zero-phonon edge is an estimate. A parabolic pair on the L axis gives 0.32 MeV. The minimum need not lie on the axis, however, and the heavy hole is warped and mixed by spin–orbit coupling, so the value bounds the edge in neither direction (Sect. 13.1); Appendix E.14 specifies the band calculation that would fix it. A direct gap does not remove the problem. GaAs has both band edges at Γ, but a pair there that absorbs the 1.15 keV at which $E_g = 1.52$ eV is deposited gains a kinetic energy comparable to the gap. Its phonon-assisted onset, 0.433 to 0.436 MeV, sits at the floor, and its first zero-phonon states are those of the L valleys at 1.815 eV, which open between 0.517 and 0.560 MeV (Sect. 13.1).

Quadratic branch. A free nucleus recoils with $E = q^2/2M$, and Eq. (9.1) reproduces the largest recoil energy $E_R^{max} = 2\mu_{\chi N}^2 v_{max}^2/M$ of Eq. (1.21). The lightest mass that deposits ω_{th} solves $E_R^{max}(m_\chi) = \omega_{th}$, which for $m_\chi \ll M$ is $m_\chi \simeq (M\omega_{th}/2)^{1/2}/v_{max}$: it grows as the square root of the threshold. The recoil is free once it exceeds several phonon energies, and how many depends on the quantity compared. Sect. 11.4, which treats nuclear recoil in the impulse approximation, replaces the sum over phonon numbers by the impulse Gaussian at four mean phonon energies, 0.164 eV in silicon, reached from 17.5 MeV. For germanium the estimate is 21.3 MeV (Sect. 17.1, which compares the scattering channels of silicon, germanium and GaAs). The map and the plane of Sect. 9.3 use this onset. The spectrum at one momentum transfer needs nine phonon energies before its width falls below a third of the recoil energy (Limit Check 1.1), while a rate summed over momentum transfers agrees with the free form within 10 % from 2.3 (Worked example 9.4). Below the onset mass a nucleon coupling excites several phonons or one, whose kinematics are the shapes above. The Migdal effect, in which a nuclear recoil also excites the electrons, lets the crystal absorb the momentum while the electrons take the energy, so it opens at the floor of the gap, 0.333 MeV in silicon (Sect. 17.1).

Absorption. A boson of mass m_ϕ brings the energy m_ϕ and the momentum $m_\phi v \leq 2.65 \times 10^{-3}\, m_\phi$ (Sect. 8.2.3). It is absorbed by an excitation at $q \simeq 0$ with $\omega = m_\phi$. The lightest mass a target absorbs is the lowest energy of such an excitation that the coupling

reaches:

$$m_\phi = \omega, \qquad q = m_\phi v \simeq 0. \tag{9.2}$$

At equal deposited energy, absorption reaches masses $2/v_{\max}^2 = 2.85 \times 10^5$ times lighter than scattering, five and a half decades. The condition $q \simeq 0$ also selects the excitations: no acoustic phonon absorbs (Sect. 9.1.3), and across the indirect gap of silicon or germanium a phonon must again supply the momentum (Sect. 3.3). Electronic absorption in cold silicon begins at 1.1728 eV, where a pair bound by 14.7 meV is created together with a transverse acoustic phonon (Sect. 13.8, exciton replicas and bosonic absorption). No single phonon of silicon absorbs at all (Sect. 9.1.3). GaAs absorbs a dark photon in its longitudinal optical phonon at 36.35 meV (Sect. 11.8), aluminum from $2\Delta_{\mathrm{BCS}}$ (Sect. 15.4, absorption in superconductors and metals), and $ZrTe_5$ from $2\Delta_{\mathrm{D}} = 23.6$ meV (Sect. 16.1). In a ferromagnet the uniform precession of the spins absorbs at a frequency tuned by an applied field at 116 µeV per tesla, from a few µeV to about a meV (Sect. 14.3). The normal modes of the detector body, at MHz in a millimeter crystal, continue the line down to the neV range (Sect. 11.8).

9.2.2 Edges of five targets

Fig. 9.1a draws the lightest mass of every channel against its threshold, and Table 9.2 lists the edges of the five targets. On logarithmic axes the absorption line $m_\phi = \omega$ and the floor $2\omega_{\mathrm{th}}/v_{\max}^2$ are parallel. The acoustic edges and the magnon edge run parallel to both, above the floor by $v_{\max}/4c_{\mathrm{ch}}$, while the free-recoil curves rise with half the slope; they would meet the floor only at $m_\chi = M$, where an elastic collision can transfer the whole kinetic energy. The scattering channels fall into three groups of mass. Collective modes and the pairs of small-gap materials, with energies from a fraction of a meV to tens of meV, open between 0.1 keV and 20 keV. Electron–hole pairs across gaps near an eV open between 0.2 and 0.6 MeV, the region enlarged in Fig. 9.1b. Free nuclear recoils and core shells need about 10 MeV or more.

Inside the first group the map orders the targets by stiffness and by gap. At a common threshold of 1 meV the acoustic edge falls as the longitudinal sound speed rises. It is 6.1 keV in silicon, 8.6 keV in aluminum, 10.2 keV in germanium and 10.3 to 12.3 keV in GaAs, whose elastic tensor is a computed one (Sect. 17.6, which ranks the targets when their inputs are uncertain). Diamond reaches 3.1 keV (Sect. 17.5, figures of merit of the channels). A flat mode competes with an acoustic one read out at a much lower threshold. The optical phonon of GaAs at 36.35 meV opens at 10.35 keV, the acoustic edge that silicon has at a threshold of 1.7 meV. The pairs of $ZrTe_5$ across 23.6 meV open at 7.1 keV, the silicon edge at 1.16 meV; and the NiO magnon at 5 meV opens at 7.3 keV, the silicon edge at 1.2 meV.

In the second group the gap sets the order, germanium at 0.212 MeV, silicon at 0.333 and GaAs at 0.433 MeV, and every crystal of the three has a band above its floor in which pairs need a phonon. In germanium the upper end of that band is not known, and in GaAs it ends where the L valleys open. A detector that counts pairs reaches its floor through phonon-assisted scattering, the process that Chap. 13 treats. In the third group the square-root growth of the free-recoil mass makes the nuclear edge of germanium higher than that of silicon by only $(M_{\mathrm{Ge}}/M_{\mathrm{Si}})^{1/2}$ at equal recoil energy.

The map ranks reach, not rate. At the edge itself only particles at $v_{\max}$, moving along the right direction, contribute, and the rate vanishes (Sect. 1.3.5); a reach curve turns up steeply as the mass approaches the edge from above. For one acoustic phonon, Sect. 11.1 finds that the rate comes within 10 % of its value for heavy particles only above 44 times the edge. Comparing two targets near their edges therefore needs the rates of the chapters that follow.

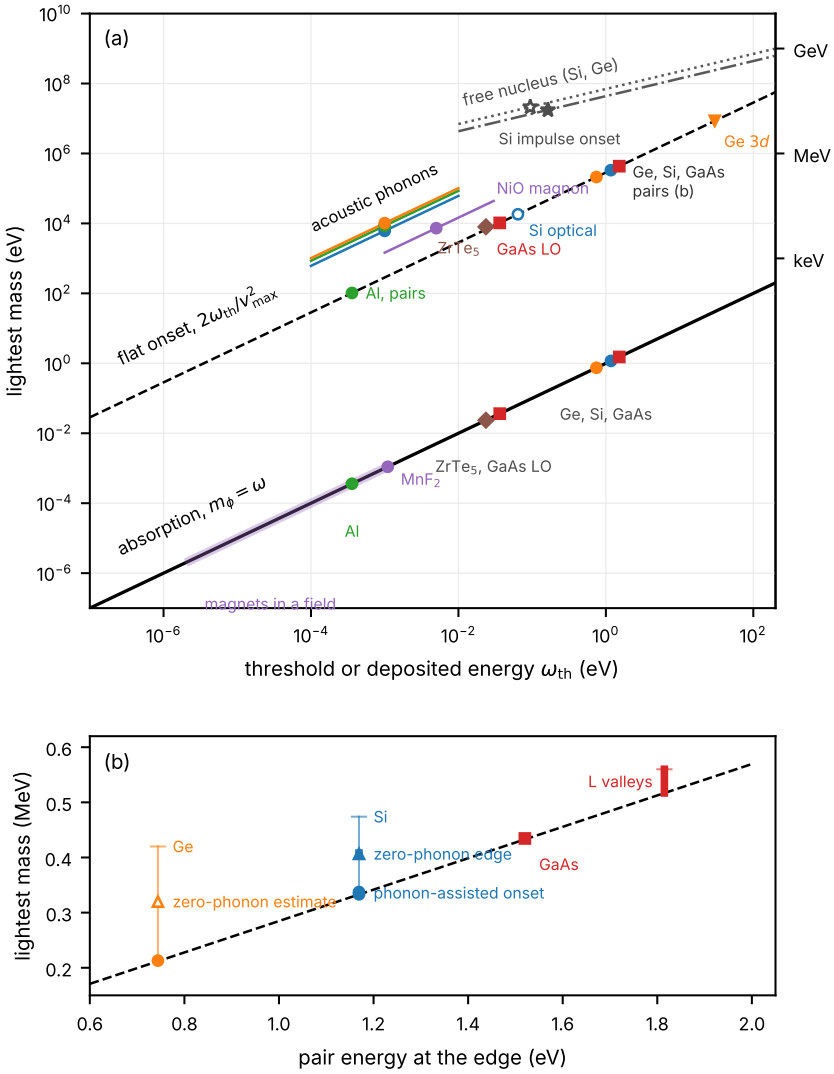


Figure 9.1: The kinematic map at $v_{\text{max}} = 794.5\,\text{km}\,\text{s}^{-1}$ (Table 1.1), elastic scattering. (a) Lightest dark-matter mass of each scattering channel and lightest absorbed boson mass, against the threshold or the energy of the excitation. Solid black: absorption, $m_\phi = \omega$. Dashed: the floor $2\omega_{\text{th}}/v_{\text{max}}^2$, reached by flat onsets. Colored lines: acoustic edges $\omega_{\text{th}}/[2c_\nu(v_{\text{max}} - c_\nu)]$ for the fastest longitudinal speeds of silicon ($9.35\,\text{km}\,\text{s}^{-1}$, lowest), aluminum ($6.67\,\text{km}\,\text{s}^{-1}$, along [111]) and germanium ($5.60\,\text{km}\,\text{s}^{-1}$), drawn from 0.1 to 10 meV, with their values at 1 meV (dots); purple, the magnon edge of NiO with $c_{\text{M}} = 1.36 \times 10^{-4}$, with its value at 5 meV. Gray: largest free recoil of a silicon (dash-dotted) and germanium (dotted) nucleus, with the impulse onsets (stars; open: estimate). Points on the floor: aluminum pair breaking, the optical phonons of GaAs (square) and silicon (open circle), the germanium $3d$ shell; diamond and bar: $ZrTe_5$, isotropic cone and principal axes (the lowest end, 7.1 keV, is its lightest mass). Points on the absorption line: aluminum, MnF_2, $ZrTe_5$, GaAs (optical phonon), germanium, silicon and GaAs (electrons); band: the uniform mode of a magnet tuned by a field. (b) Electron–hole pairs. Filled circles and squares: phonon-assisted onsets; thick bars and filled triangle: zero-phonon edges (silicon, GaAs L valleys); open triangle: germanium zero-phonon estimate, which bounds the edge in neither direction; thin lines: from the floor to the mass that deposits the gap at exactly the valley momentum. Values and their sources in Table 9.2.

Table 9.2: Channel edges of five targets: the lightest mass that reaches each channel in elastic scattering, and the lightest absorbed boson mass, at $v_{\max} = 794.5\,\mathrm{km\,s^{-1}}$. Masses in MeV unless marked. "–": no such channel; "n.e.": not evaluated in this book. The last column gives the section where each value is computed. Entries without a section there are computed here from Eq. (9.1): the aluminum acoustic edge, with the speed $6.67\,\mathrm{km\,s^{-1}}$ along [111] from the zero-temperature elastic constants and density of Sect. 20.3; the floor of $ZrTe_5$; and the silicon $2p$ edge. Estimates are marked.

Channel	Si	Ge	GaAs	Al	$ZrTe_5$	Section
Floor of the gap, $2E_g/v_{\max}^2$	0.333	0.212	0.433	–	6.72 keV ($4\Delta_D/v_{\max}^2$)	Sect. 1.3
Pairs with a phonon (onset)	0.3332 (exciton), 0.3374 (free pair)	0.2119–0.2141	0.433–0.436	–	n.e.	Sect. 13.1
Pairs without a phonon	0.405–0.408	≈ 0.32; estimate	0.517–0.560 (L valleys)	–	7.1 keV (slow axis; 9.0, 9.6 keV along the others)	Sects. 13.1, 13.9 and 16.1
Pair breaking, $2\Delta_{BCS}$	–	–	–	0.10 keV	–	Sect. 7.2
One acoustic phonon, $\omega_{th} = 1$ meV	6.1 keV	10.2 keV	10.3–12.3 keV (computed tensors)	8.6 keV	n.e.	Sects. 11.1, 17.1 and 17.6
One optical phonon	18.3 keV	n.e.	10.35 keV	–	n.e.	Sects. 11.2 and 17.3
Core shell	28.6 ($2p$)	8.5–8.7 ($3d$)	n.e.	n.e.	n.e.	Sect. 10.6
Free nuclear recoil (impulse onset)	17.5	21.3; estimate	n.e.	n.e.	n.e.	Sects. 11.4 and 17.1
Absorption, lowest m_ϕ	1.1728 eV (pairs with a phonon)	≈ 0.74 eV with a phonon, 0.89 eV without	36.35 meV (optical phonon); 1.52 eV (pairs)	0.36 meV	23.6 meV	Sects. 3.3, 11.8, 13.8, 15.4 and 16.1

Worked example 9.2: The acoustic edge of silicon at a threshold of 1 meV

The fastest longitudinal branch of silicon runs along [111], with $c_{\mathrm{LA}} = 9.35\,\mathrm{km\,s^{-1}}$ from the measured elastic constants (Worked example 5.3). In units of the speed of light, $c_{\mathrm{LA}} = 9.35/299792 = 3.119 \times 10^{-5}$, and $v_{\mathrm{max}} = 794.5/299792 = 2.650 \times 10^{-3}$. A phonon of energy $\omega_{\mathrm{th}} = 1\,\mathrm{meV}$ on this branch carries

$$\bar{q} = \frac{\omega_{\mathrm{th}}}{c_{\mathrm{LA}}} = \frac{10^{-3}\,\mathrm{eV}}{3.119 \times 10^{-5}} = 32.1\,\mathrm{eV}.$$

This is $0.014\,(2\pi/a)$, with $2\pi/a = 2.28\,\mathrm{keV}$, deep in the linear part of the branch, where the edge needs slopes that obey the acoustic sum rule (Sect. 5.4). With $v_{\mathrm{max}} - c_{\mathrm{LA}} = 2.619 \times 10^{-3}$,

$$m^{\mathrm{ac}}_{\mathrm{min}} = \frac{10^{-3}\,\mathrm{eV}}{2 \times 3.119 \times 10^{-5} \times 2.619 \times 10^{-3}} = 6.12\,\mathrm{keV}.$$

The floor at the same threshold is $2\omega_{\mathrm{th}}/v^2_{\mathrm{max}} = 0.285\,\mathrm{keV}$. A particle of 6.12 keV at v_{max} has 21.5 meV of kinetic energy and could give all of it to a target that absorbs the momentum $m_\chi v_{\mathrm{max}} = 16.2\,\mathrm{eV}$, but a phonon with that momentum carries only 0.51 meV. To create the 1 meV phonon the particle must reverse its direction, giving $2m_\chi(v_{\mathrm{max}} - c_{\mathrm{LA}}) = 32.1\,\mathrm{eV}$, and it keeps 20.5 meV of its energy. Along [100] the transverse speed $5.84\,\mathrm{km\,s^{-1}}$ of Worked example 5.3 would give 9.76 keV, but a transverse phonon along a symmetry axis carries no density and is not excited by a density coupling. The edge scales linearly with the threshold, 0.61 keV at 0.1 meV and 61 keV at 10 meV, as long as $\bar{q}$ stays in the linear part of the branch; at 10 meV it is already $0.14\,(2\pi/a)$. Sect. 11.1 evaluates the rate near this edge with all branches and directions.

9.2.3 Dependence on the halo

Every scattering edge moves with $v_{\mathrm{max}} = v_{\mathrm{esc}} + v^{\mathrm{ref}}_{\mathrm{E}}$, but not by the same amount. At the minimum of Eq. (9.1) either the derivative with respect to q vanishes or q sits on a constraint that does not depend on v_{max}. Differentiating then leaves the explicit dependence alone:

$$\frac{\mathrm{d}\ln m^{\mathrm{ch}}_{\mathrm{min}}}{\mathrm{d}\ln v_{\mathrm{max}}} = -\frac{\bar{q}v_{\mathrm{max}}}{\bar{q}v_{\mathrm{max}} - E_{\mathrm{ch}}(\bar{q})}. \tag{9.3}$$

The ratio on the right compares the largest energy the particle could deposit at $\bar{q}$ with what is left after the excitation is paid. For a flat onset $\bar{q} = 2E_0/v_{\mathrm{max}}$, half of $\bar{q}v_{\mathrm{max}}$ goes into the excitation, and the exponent is -2, the halo dependence of the floor in Sect. 1.3. For a gapless linear branch $\bar{q} = \omega_{\mathrm{th}}/c_{\mathrm{ch}}$ and the exponent is $-v_{\mathrm{max}}/(v_{\mathrm{max}} - c_{\mathrm{ch}})$, which is -1.012 for silicon: the acoustic edge scales as $1/v_{\mathrm{max}}$, not $1/v^2_{\mathrm{max}}$, because the excitation takes the small fraction c_ν/v_{max} of $\bar{q}v_{\mathrm{max}}$. A free nucleus gives -1 for $m_\chi \ll M$, from $m_\chi \simeq (M\omega_{\mathrm{th}}/2)^{1/2}/v_{\mathrm{max}}$. At a stationary minimum, where the derivative with respect to q vanishes, $\bar{q}v_{\mathrm{max}} = 2E_{\mathrm{ch}} - \bar{q}E'_{\mathrm{ch}}$. A branch that rises there takes more than half of $\bar{q}v_{\mathrm{max}}$, and the exponent exceeds 2 in magnitude: -2.37 for the isotropic cone of $ZrTe_5$, and -2.12 to -2.74 along its axes. A branch that falls there, as on the way to an indirect valley, gives less than 2. The gapless linear branch also rises, but its minimum sits on the constraint $q \geq \omega_{\mathrm{th}}/c_{\mathrm{ch}}$, not at a stationary point. Absorption, Eq. (9.2), does not depend on the halo at all.

The halo matters at the level of a few percent. A change of the escape speed by $25\,\mathrm{km\,s^{-1}}$, its uncertainty (Sect. 1.2), changes v_{max} by 3.15 %. To first order it moves the floor and every flat onset by 6.3 %, the acoustic edge of silicon by 3.2 %, the free-recoil edges by 3.1 %, and

the Dirac edge of $ZrTe_5$ by 7.4 %. The exact shifts of the floor for $\pm 25\,\mathrm{km\,s^{-1}}$ are −6.0 % and 6.6 %. Over the year $v_{\max}$ moves between 781.4 and 810.4 $\mathrm{km\,s^{-1}}$ (Sect. 1.2); the floor moves by −3.9 % in June and 3.4 % in December (Sect. 1.3), and the acoustic edge of silicon by −2.0 % and 1.7 %. A different halo thus shifts the map without reordering channels of the same shape, but it changes the ratio between a flat and a linear edge by about $\delta \ln v_{\max}$. An endothermic splitting $\Delta_\chi > 0$ raises every edge, by adding Δ_χ to the energy the particle must give up. Sect. 17.3, which treats inelastic scattering into collective modes, finds that a splitting of 10 meV moves the phonon and magnon edges by 12 to 28 % and the electronic edge of silicon by 1 % (Table 17.4). An exothermic splitting larger than the excitation energy removes the edge altogether, as Sect. 1.3 found for the threshold mass.

9.3 Excitations on the momentum–energy plane

At $q = 50\,\mathrm{keV}$ and $\omega = 50\,\mathrm{meV}$ silicon has phonon final states and nothing else. At 1 keV and 1.3 eV its one final state is an electron–hole pair together with a phonon; at 1 keV and 3 eV a pair without a phonon is open as well. Each excitation in the atlas of Table 9.1 has a *support*, the region of the (q, ω) plane where its response $S(\boldsymbol{q}, \omega)$ of Sect. 4.2 carries weight. Where the support is formally unbounded, as for the multiphonon tails of Sect. 5.6, we draw the region that holds 90 % of the weight. Fig. 1.4 showed the region that halo dark matter can reach. Here we draw the supports of silicon on the same axes, label each region by its sector of Eq. (8.48), and find the places where two regions overlap. Those are the places where a rate can count a final state twice, or drop an interference term, and the decomposition theorem of Sect. 8.6 decides which of the two dangers is real.

9.3.1 The silicon plane by sector

In Fig. 9.2 the three colors are the three sectors: green for the electron-following sector $\mathcal{P}_{\mathrm{ad}}$, blue for the zero-phonon electronic sector $\mathcal{P}_0$, orange for the phonon-dressed electronic sector $\mathcal{P}_+$. The axes and the reach curves are those of Fig. 1.4.

Electron-following sector. Inside the zone the support is a set of lines, the phonon branches of Sect. 5.2. The acoustic branches rise as $\omega = c_\nu q$ from the origin and flatten at the zone boundary, at 17 meV (transverse) and 51 meV (longitudinal) at X in the PBE calculation; the optical branch stays near 64 meV. Beyond the zone nothing new appears in energy. The momentum transfer is never folded (Sect. 1.3); the nucleus keeps $\boldsymbol{q}$, and the phonon carries $\boldsymbol{q}$ minus a reciprocal-lattice vector. A single phonon therefore has energies between zero and 64 meV at every q, and its weight changes as the Debye–Waller exponent $2W_\kappa = q^2\langle u_x^2\rangle$ of Eq. (5.56) grows. The PBE density of states of Sect. 5.2 gives $\langle u_x^2\rangle = 2.45 \times 10^{-3}\,\text{Å}^2$ at $T = 0$, so $2W_\kappa = 0.016$ at 5 keV, 0.063 at 10 keV and 1.0 at 40 keV. Two and more phonons fill the band as $2W_\kappa$ approaches one. In the incoherent approximation the central 90 % of the weight that creates phonons lies between 11 and 60 meV at the zone radius $q_{\mathrm{BZ}} = 2.25\,\mathrm{keV}$, still the one-phonon band. It lies between 13 and 129 meV at 46 keV, where the recoil parameter x_R of Eq. (8.41) is one, and between 46 and 311 meV at 92.5 keV. At that momentum the impulse approximation of Sect. 11.4 takes over from the sum over phonon numbers. It treats the nucleus as kicked so hard that its spectrum is a Gaussian of width $(E_R\langle\omega\rangle)^{1/2}$ about the free recoil energy $E_R = q^2/2M$, with $\langle\omega\rangle = 40.9\,\mathrm{meV}$ the mean phonon energy. From there on the support is a band that follows $q^2/2M$ and narrows in relative width as $x_R^{-1/2}$.

The free recoil line crosses the gap at $q = (2ME_{\mathrm{g}})^{1/2} = 247\,\mathrm{keV}$, and the upper edge of the band does so at 212 keV. Above these momenta electron-following final states sit at the

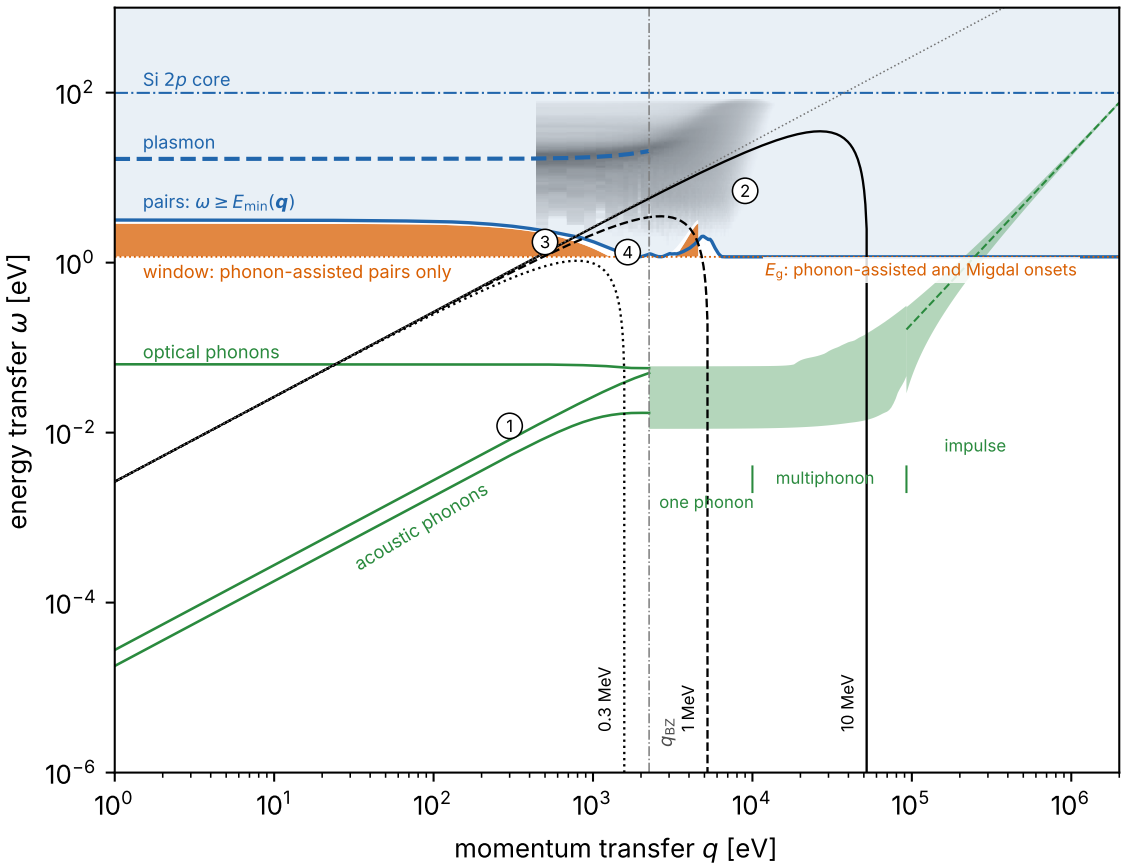


Figure 9.2: The excitations of silicon on the plane of momentum transfer q and energy deposited ω, on the axes of Fig. 1.4. Green, electron-following sector: the transverse and longitudinal acoustic branches and the optical branch along [100] inside the zone, from the PBE force constants of Sect. 5.2; beyond the zone radius $q_{\mathrm{BZ}} = 2.25\,\mathrm{keV}$ (dash-dotted), the band holding the central 90 % of the weight that creates phonons (5 and 95 % quantiles), in the incoherent harmonic response at $T = 0$ with the PBE density of states, and above 92.5 keV the same quantiles of the impulse Gaussian about the free recoil $q^2/2M$ (dashed); ticks at 10 keV and 92.5 keV separate the one-phonon, multiphonon and impulse regimes. Blue, zero-phonon electronic sector: the pair continuum above $E_{\mathrm{min}}(\boldsymbol{q})$, drawn as the lowest of the PBE edges along [100], [110] and [111] with the conduction bands shifted to the measured gap $E_{\mathrm{g}} = 1.17\,\mathrm{eV}$, and as E_{g} beyond 6.5 keV; gray scale, the random-phase loss function with local fields of the tables of Knapen, Kozaczuk and Lin [2] (scissor to the gap of 1.17 eV, directions averaged, 10^{-3} to 4 on a logarithmic scale); dashed, the plasmon of the valence electron gas, $\omega_{\mathrm{pl}}^2 = \omega_p^2 + \frac{3}{5}q^2 v_{\mathrm{F}}^2$ with $\omega_p = 16.6\,\mathrm{eV}$, up to 2.24 keV, where this dispersion meets the pair continuum; dash-dotted, the $2p$ core level at 99.4 eV. Orange, phonon-dressed electronic sector: the off-shell window along [100], from the phonon-assisted onset $E_{\mathrm{g}} + \omega_{\mathrm{TA}}$ up to $E_{\mathrm{min}} - \delta_{\mathrm{w}}$ with $\delta_{\mathrm{w}} = 0.3\,\mathrm{eV}$; white above it, the margin of width δ_{w} below the edge; dotted, E_{g}, above which phonon-assisted pairs and Migdal events exist at every q. Black: the largest energy deposited, $qv_{\mathrm{max}} - q^2/2m_\chi$, for $m_\chi = 0.3$ (dotted), 1 (dashed) and 10 MeV (solid), and qv_{max} (gray dotted), with $v_{\mathrm{max}} = 2.650 \times 10^{-3}$. Circled numbers mark the interference terms of Table 9.3.

same (q, ω) as electronic ones. A particle must be heavier than 46.8 MeV to supply 247 keV.

Zero-phonon electronic sector. An electron–hole pair with no net phonon must take the whole momentum transfer itself, and its support is the continuum $\omega \geq E_{\min}(\boldsymbol{q})$ of Eq. (3.57). With the bands of Fig. 3.7, shifted to the measured indirect gap, the edge lies at 3.17 eV for $q \to 0$, at 1.77 eV at 1 keV along the lowest of the three symmetry directions, and reaches $E_{\mathrm{g}} = 1.17\,\mathrm{eV}$ at the valley momentum $k_{\mathrm{c}} = 1.94\,\mathrm{keV}$. The value at $q \to 0$ lies below the measured direct gap of 3.4 eV, because a shift fitted to the indirect gap does not correct the vertical gap at Γ (Sect. 3.3). At larger $|\boldsymbol{q}|$ the edge depends on the direction of $\boldsymbol{q}$. It returns to E_{g} whenever $\boldsymbol{q}$ equals a valley momentum plus a reciprocal-lattice vector, $\boldsymbol{k}_{\mathrm{c}} + \boldsymbol{G}$, whose lengths are 1.94, 2.63, 3.25, 4.96, 5.27, 5.32 keV and so on. Above 6.4 keV they leave no gap wider than 0.72 keV, and half that distance costs a pair of mass $1.19\,m_e$ about 0.11 eV above E_{g}. There the onset of the direction average that enters an isotropic rate stays within about a tenth of an eV of the gap. Below, the widest gap lies between 3.25 and 4.96 keV; there the lowest of the three symmetry-line edges rises to 2.07 eV, an upper value for the onset over all directions. Upward the continuum is unbounded. At small q its weight sits at the absorption peaks of 3.4 and 4.2 eV and in the plasmon; at large q it concentrates along the free-electron recoil $q^2/2m_e$, the ridge of the gray scale in Fig. 9.2.

The plasmon, the exciton and the core shells lie inside this region and have homes of their own. The plasmon of Sect. 4.5 starts at the measured loss peak of 16.4 eV, 0.2 eV below the $\omega_p = 16.6\,\mathrm{eV}$ of the valence electron gas, and disperses upward. With the quadratic dispersion of the gas it is a line up to 2.24 keV, where it enters the pair continuum and Landau damping removes its weight; the exact zero of the Lindhard function enters at $q_{\mathrm{c}} = 2.60\,\mathrm{keV}$ (Sect. 4.5). In the loss-function tables that Sect. 10.2 uses, the peak moves from 18.9 eV at 0.5 keV to 22.3 eV at 2.45 keV; the small-q peak lies 2.5 eV above the measured 16.4 eV because of the shift that sets their gap (Sect. 10.3, which computes the plasmon's share of the rate). The bound pair, the exciton of Sect. 4.9, lies 14.7 meV below the edge near $q = k_{\mathrm{c}}$; on the scale of the figure it coincides with the edge (Sect. 10.5, the exciton lines at finite momentum). The core shells start at fixed energies, 99.4 eV for $2p$ and 150 eV for $2s$ (Sect. 3.5), and carry weight at momentum transfers of tens of keV, the momenta of the electrons in them (Sect. 10.6).

Phonon-dressed electronic sector. A pair created together with a phonon needs no particular momentum, because the phonon carries whatever crystal momentum the pair lacks. Its support starts just above the gap at every q, at the onset $E^{\mathrm{PA}}_{\min}(\boldsymbol{q}) = E_{\mathrm{g}} + \omega_{\mathrm{TA}}(\boldsymbol{k}_{\mathrm{c}} - \boldsymbol{q})$ of Eq. (13.2), and never above $E_{\mathrm{g}} + 64\,\mathrm{meV} = 1.23\,\mathrm{eV}$. With the PBE force constants the onset is 1.187 eV at $q \to 0$ and 1.184 eV at 1 keV. Between this onset and the zero-phonon edge lies the *off-shell window* of Sect. 8.6, the region below $E_{\min}(\boldsymbol{q}) - \delta_{\mathrm{w}}$ where no zero-phonon pair exists and every electronic final state carries a phonon. Along the valley axis, with $\delta_{\mathrm{w}} = 0.3\,\mathrm{eV}$, it spans 1.184 to 1.467 eV at 1 keV and closes at 1.27 keV with the computed bands. It reopens beyond 3.30 keV, on the far side of the Umklapp image of the opposite valley (Sect. 13.6, which fixes the margin and the window edges). Above the zero-phonon edge the same sector is the phonon sideband of the zero-phonon continuum, and there it is a correction to the zero-phonon rate (Sect. 13.9 compares the two).

The Migdal effect of a nucleon coupling, which straddles the two electronic sectors (Sect. 9.1.2), has the same support. The struck nucleus absorbs the momentum and the electrons take an energy $\omega_e \geq E_{\mathrm{g}}$, at every q. What separates it from a pair made through the electron density is the operator, not the region of the plane.

Table 9.3: Where amplitudes interfere on the silicon plane of Fig. 9.2. The first three rows are the cross terms $S^A_{e\kappa}$ of Table 8.5, nonzero only for a coupling to electrons and nuclei at once; the fourth is the boundary between the two electronic sectors. Sizes are for silicon at momentum transfers well below the zone radius, from the sections listed. A dark photon couples with $\tilde{V}_\kappa = -Z_\kappa \tilde{V}_e$ to nucleus κ of charge Z_κ and mass M_κ.

	Region	Term	Size in silicon	Home
1	phonon branches and band, $\mathcal{P}_{\text{ad}}$	nucleus and the electrons that follow it, $S^{\text{ad}}_{e\kappa}$	at long wavelength the following cloud cancels the nuclear charge, $14 - 14 = 0$: no dark-photon single-phonon rate at leading order	Sects. 11.2 and 17.4
2	pair continuum, $\mathcal{P}_0$	zero-phonon electron recoil with zero-phonon Migdal, $S^0_{e\kappa}$	dark photon: $(1 + Z_\kappa m_e/M_\kappa)^2 - 1 = 5.47 \times 10^{-4}$ of the zero-phonon rate	Sects. 12.5 and 17.4
3	off-shell window, $\mathcal{P}_+$	phonon-assisted recoil with Migdal with one phonon, $S^+_{e\kappa}$	the same fraction, 5.47×10^{-4} for a dark photon; the whole electronic rate vanishes at leading order for a coupling in proportion to mass	Sect. 17.4
4	margin and shell at $E_{\text{min}}(\boldsymbol{q})$, $\mathcal{P}_0 + \mathcal{P}_+$	production and decay of a zero-phonon pair, Fano coefficients c^{F}_ν	zero for every one-body self-energy: inversion with time reversal, with an even electron number, makes every amplitude real; nonzero in GaAs, estimated to move a few percent of first emissions	Sects. 8.6 and 13.6

9.3.2 Overlap regions and interference

One phonon and two phonons share the band below 64 meV. The impulse band and the pair continuum share the region above 212 keV and 1.17 eV, and the phonon sideband and the zero-phonon continuum share everything above $E_{\text{min}}(\boldsymbol{q})$. An overlap in the plane is not by itself an interference. Sectors add (Sect. 9.1.2), to the accuracy of the leak in Eq. (8.50), and within one sector so do different phonon numbers: one and two phonons are orthogonal final states, and so are a nucleus recoiling with the electrons in their ground state and a pair. Interference needs two amplitudes into the same final state, and on the silicon plane that happens in two ways.

The first is a coupling to both species. When the dark matter couples to electrons and to nuclei, as a dark photon does, both amplitudes reach the same final states of every sector, and the bottom row of Table 8.5 holds their cross terms $S^A_{e\kappa}$. Each is bounded by the positivity of Eq. (4.20), and each lives where its sector lives. The second is the boundary between the two electronic sectors. Within δ_{w} below the zero-phonon edge and a few pair widths above it, the label of a final state is not sharp. Only the sum $S^0 + S^+$ is defined there, and the Dyson convention Eq. (8.59) splits it. The decay of a zero-phonon pair there can interfere with the same one-phonon state made directly, the Fano term Eq. (8.62). Table 9.3 lists the four regions, numbered as in Fig. 9.2, with what is known of each term in silicon.

In region 1 alone the cross term decides a rate at leading order. There the electrons that follow a nucleus are part of the vertex: a dark photon sees the nucleus and its cloud together, and in a nonpolar crystal their charges cancel at long wavelength. This is why Sect. 11.2, on optical phonons in polar crystals, finds a channel in GaAs and none in silicon.

Omitting the cross term there would make a vanishing rate look like the sum of two large ones, $196 + 196$ in units of the squared electron coupling. In regions 2 and 3 the nuclear amplitude is suppressed by m_e/M_κ, and the cross term is a correction of a few parts in 10^4 for a dark photon. It competes with the direct term for couplings that favor nuclei over electrons by about $M_\kappa/m_e = 5.1 \times 10^4$. For a coupling exactly in proportion to mass it cancels the other two, and the electronic final states are reached only at the next order in q (Sect. 17.4, which derives these relations and their limits of validity). Region 4 is the narrowest. The Fano coefficients change no total and no number of zero-phonon events, only which decay channel, a phonon branch or another, the first decay goes into. In silicon and germanium they vanish at the level of one-body self-energies. In GaAs, which lacks a center of inversion, Sect. 13.6 estimates that they move a few percent at most of the first emissions from one branch to another, and Appendix E.14 specifies the calculation that sizes them. The practical consequence of region 4 is a rule for the plane: inside the margin and the shell above it, only totals of the two electronic sectors are quoted (Sect. 13.6).

9.3.3 The kinematic reach on the plane

Particles of mass m_χ reach the region below $\omega = q v_{\max} - q^2/2m_\chi$ (Eq. (1.26)), and the black curves of Fig. 9.2 show it for three masses. A particle of 0.3 MeV deposits at most 1.05 eV, at $q = 0.80$ keV, and transfers at most 1.59 keV. Its region lies below the gap and inside the zone, so it meets single acoustic and optical phonons and nothing else, with $2W_\kappa \leq 0.002$. A particle of 1 MeV deposits at most 3.51 eV and crosses the gap for momentum transfers between 0.49 and 4.81 keV. It enters the off-shell window, where every pair it makes carries a phonon, and the zero-phonon continuum above the edge, while its lattice final states are still single phonons ($2W_\kappa \leq 0.018$ at its largest q, 5.3 keV). A particle of 10 MeV transfers up to 53 keV, where $2W_\kappa = 1.8$, and reaches deep into the multiphonon band. In energy it reaches 35 eV, across the region of the plasmon, but not the plasmon line, which ends at 2.24 keV, where this particle deposits no more than 5.9 eV; nor the $2p$ shell at 99.4 eV, nor the impulse band, which opens at 17.5 MeV. The lightest mass whose curve touches a region is the lightest mass of that channel (Sect. 9.2).

Worked example 9.3: The excitations open at 5 keV and 2 eV in silicon

Take $q = |\boldsymbol{q}| = 5$ keV and $\omega = 2$ eV. The point is reachable if $q v_{\max} - q^2/2m_\chi \geq \omega$, with $q v_{\max} = 13.25$ eV, that is for $m_\chi \geq q^2/[2(q v_{\max} - \omega)] = 1.11$ MeV; at 10 MeV the minimum speed of Eq. (1.23) is $v_{\min} = \omega/q + q/2m_\chi = 6.50 \times 10^{-4}$, or 195 km s^{-1}, well inside the halo.

Electron-following. At 5 keV, $2W_\kappa = 0.016$ and $E_R = 0.48$ meV. Two electronvolts would take at least 32 phonons of 64 meV, each costing a factor of order $2W_\kappa$ in weight. The sector is closed.

Zero-phonon electronic. The edge depends on the direction of $\boldsymbol{q}$. Along [100], [110] and [111] it lies at 2.59, 2.02 and 2.31 eV, all above 2 eV: along the symmetry lines no pair without a phonon exists at this point. But a valley image $\boldsymbol{k}_c + \boldsymbol{G}$ has length 4.96 keV, and for $\boldsymbol{q}$ near that direction the pair costs E_g plus about 1 meV. Averaged over directions the zero-phonon response is thus open, but small: the direction-averaged loss function of the tables of Sect. 10.2 at this q is 8.8×10^{-4} at 2 eV, against its maximum of 0.33 at 35.7 eV. The plasmon line has ended at 2.24 keV, the exciton lies 14.7 meV below the edge, and the $2p$ shell needs 99.4 eV: none of them is open.

Phonon-dressed electronic. The phonon-assisted onset lies below 1.23 eV for every $\boldsymbol{q}$, so phonon-assisted pairs are open. Where the point sits relative to the zero-phonon edge

again depends on direction. Along [100] it lies below $E_{\min} - \delta_{\mathrm{w}} = 2.29\,\mathrm{eV}$, inside the window, and phonon-assisted pairs are the one electronic final state there. Along [111] it lies inside the window by 11 meV ($E_{\min} - \delta_{\mathrm{w}} = 2.011\,\mathrm{eV}$), less than the exciton binding and less than the difference between a symmetry-line edge and the minimum over all crystal momenta. Along [110] it lies in the margin, between 1.72 and 2.02 eV, where only the total of the two electronic sectors is defined. A nucleon coupling reaches the same final states through the Migdal effect, with a probability per scattering of order $(E_R/\langle\omega_e\rangle)(\delta M_\kappa/M_\kappa) = 2 \times 10^{-8}$ by the moment estimate of Sect. 12.1. Here $\langle\omega_e\rangle \approx 6\,\mathrm{eV}$ is the mean electronic excitation energy, and $\delta M_\kappa/M_\kappa = Z_\kappa m_e/M_\kappa = 2.7 \times 10^{-4}$ the share of the atom's mass carried by the electrons that move with the nucleus (Sect. 8.6).
Interference. For a dark photon the cross cells of regions 2 and 3 of Table 9.3 are both present, each a few parts in 10^4 of its sector, and for directions in the margin the Fano term of region 4 vanishes in silicon. The point is thus served by two sectors in different directions of $\boldsymbol{q}$, which is why a calculation applies the window bin by bin in $(|\boldsymbol{q}|, \hat{\boldsymbol{q}})$ and not to the direction average (Sect. 13.6).

9.3.4 Supports in other materials

The remaining rows of the atlas belong to other materials, and their supports reduce to a few shapes: a dispersing line, a flat line at a fixed energy, a continuum above a threshold, or a point at $q \to 0$. Table 9.4 gives the shape and one number for each, with the section that treats it. Absorption samples the single point $\omega = m_\phi$, $q = m_\phi v \approx 10^{-3}\omega$, far to the left of every scattering curve, and a support absorbs if it reaches $q \to 0$ at that energy (Sect. 9.1.3).

For superconductors and Dirac materials the sectors are those of quasiparticle number (Sect. 9.1.2), so the rules of Sect. 9.3.2 for which overlaps add and which interfere hold only as far as those sectors are defined, which Sect. 8.6 bounds but does not settle.

Table 9.4: Supports in the (q, ω) plane of the excitations of the atlas that silicon does not have, with one characteristic number for the material that the treating section uses. "Line" means one energy for each q; "continuum" a band of energies at each q. g_{33}: component of the gyration tensor along the optic axis; g_c: Luttinger charge parameter of a one-dimensional metal.

Excitation	Support	Scale	Home
Optical phonons of polar crystals	line at ω_{L} as $q \to 0$, reached by a coupling to charge	GaAs 36.35 meV	Sect. 11.2
Phonons of a quantum crystal	acoustic lines and multiphonon band of a light, strongly anharmonic lattice	solid helium, below 3 meV at low pressure	Sect. 11.5
Magnons	line from an anisotropy gap at the zone center to the top of the band	NiO, gap 4.54 meV (measured; anisotropy added to the model), top 118 meV	Sect. 14.1
Two-magnon continua	continuum from the one-magnon energy to twice the band top at small q	NiO, up to 236 meV	Sect. 14.2
Chiral and nonreciprocal modes	the supports above; the response acquires a part odd in $\boldsymbol{q}$	α-quartz, $q\, g_{33} = 7 \times 10^{-5}$ at 2.1 eV	Sect. 14.5
Precessing nuclear spins	point at $q \to 0$, ω at the Larmor frequency	^{207}Pb in 26.8 mT: 241.8 kHz (1 neV)	Sect. 14.6
Superconducting quasiparticle pairs	continuum $\omega \geq 2\Delta_{\mathrm{BCS}}$ inside the particle–hole band $\omega \leq q v_{\mathrm{F}} + q^2/2m_e$ of the metal	aluminum, $2\Delta_{\mathrm{BCS}} = 0.36$ meV	Sect. 15.1
Dirac and narrow-gap pairs	continuum $\omega \geq (4\Delta_{\mathrm{D}}^2 + v_{\mathrm{D}}^2 q^2)^{1/2}$	$ZrTe_5$, $\Delta_{\mathrm{D}} = 11.8$ meV	Sect. 16.1
Low-dimensional plasmons	line $\omega \propto q_{\parallel}$, faster than the halo, reached by oblique $\boldsymbol{q}$	metallic nanotube, $\omega = v_{\mathrm{D}} q_{\parallel}/g_c$, 3.4 to 6.9 times $v_{\max}$	Sect. 16.2
Graphene, nanotubes, quantum dots	continuum above a confinement gap; ejection above the work function	PbS dot 1.2 eV; graphene 4.3 eV	Sect. 16.3
Molecular (Frenkel) excitons	nearly flat lines at molecular energies	anthracene 3.1 eV	Sect. 16.4
Dopant transitions	flat lines and an ionization continuum	Si:P, binding 45.59 meV	Sect. 16.5
Color centers, broken bonds	reached through a nuclear recoil above a displacement or bond energy	H_2 bond 4.75 eV; displacement tens of eV	Sect. 16.6
Two-level systems in glasses	flat continuum of splittings, independent of q	down to μeV; $10^{20}\,\mathrm{eV^{-1}\,cm^{-3}}$	Sect. 16.7
Bulk acoustic modes of a body	points at $q \to 0$, the normal modes of the whole detector	MHz (mm crystal) to kHz (ton-scale bar)	Sect. 11.8

9.4 Limit Checks and the truncation label

The free-recoil formula $\omega = q^2/2M$ holds for a silicon nucleus once the recoil energy exceeds nine phonon energies, 0.54 eV for a nucleus in an Einstein well of 60 meV. Limit Check 1.1 showed this with one plot: the formula against the bound nucleus, the recoil energy on the horizontal axis, and a gray band where the two agree. The channel chapters recover 90 published formulas as limits of their own results, and test 80 of them in 75 boxes of the same kind. Each box answers one question: over what range of a single parameter does the published formula agree with a fuller calculation, and what fails outside that range? We build such a box, show that its horizontal axis follows from the place of the formula on the truncation label $\mathfrak{T}$ of Sect. 8.5, and place every formula of Chaps. 10 to 17 on that label.

9.4.1 The comparison plot

A Limit Check has three parts: the formula, a plot, and one sentence. The formula is written in the book's notation, with the symbols of Chap. 8, and a one-line citation names the equation of the source where it was first given.

The plot compares the formula with a fuller result for the same quantity. In the language of Sect. 8.8, the formula is a quantity evaluated at a point $\mathfrak{T}$ of the label and the fuller result is the same quantity one step further along one axis, at $\mathfrak{T}_k^+$; the plot shows their relative difference δ_k of Eq. (8.71). Most boxes plot the ratio of the two, $1 + \delta_k$ or its inverse, so that the formula is the horizontal line at one. Some plot both curves, when their shapes matter more than their ratio. The fuller result is computed in the book, from a crystal calculation or from a model that keeps the physics the formula drops, or it is measured. In Limit Check 1.1 it is the nucleus bound in an Einstein potential, whose spectrum is a series of phonon lines.

The horizontal axis carries the *controlling parameter*, the variable along which δ_k runs from negligible at one end of the axis to order one at the other. In Limit Check 1.1 it is E_R/ω_0, the recoil energy in units of the phonon energy. The *leading correction*, the first term of δ_k in the controlling parameter, is drawn as a third curve, usually dotted or dashed. In Limit Check 1.1 it is the rms width $(E_R\omega_0)^{1/2}$ of the deposit, which falls relative to E_R as $(\omega_0/E_R)^{1/2}$. The gray band marks the values of the controlling parameter at which the formula agrees with the fuller result. The tolerance is stated in the caption: usually 10 %, sometimes 1 % or 15 %, or a fixed fraction of a width, as the rms width below $E_R/3$ that bounds the band of Limit Check 1.1. The caption prints the edge of the band.

The sentence under the plot says what breaks down outside the band, in physical terms: in Limit Check 1.1, that below a few phonon energies the deposit arrives in whole phonons and spreads as widely as E_R itself.

The *approximation chain*, the sequence of steps that leads from the book's general result to the formula, lives in the running text next to the box, where each step is derived or cited to a specific equation. The formal record, with the chain, the conditions, the specification of the plot and the size of the leading correction, is the row of the formula in Table F.1 of Appendix F. How far the book has established the formula is read from what the plot compares with.

Inside the gray band the formula is established to the stated tolerance, relative to the fuller result. That statement is as strong as the fuller result. Where the fuller result is a crystal calculation at a known point of $\mathfrak{T}$, such as the random-phase response with local fields, the band says that the formula agrees with that calculation; the calculation has its own next step, which the label names. Where the fuller result is a model, as the Einstein well of Limit Check 1.1 is a model of the silicon lattice, the edge of the band is an estimate of where the formula fails, not a bound. Where nothing is shaded, the formula fails over the whole plotted

range, and the box gives the size of the failure instead of a range of validity. Some boxes compare two published forms of one result, or test a sum rule; they are read in the same way, with the exact relation as the fuller result.

9.4.2 Placement on the truncation label

A formula is placed on $\mathfrak{T}$ by reading its assumptions, one component at a time, as Worked example 8.7 did for phonon-assisted absorption. The eight components of Eq. (8.44) are five orders and three labels. The orders are those of the Born–Oppenheimer coupling (n_B), the electron–phonon vertex (n_g), the anharmonicity (n_A), the temperature (n_T) and the multipole expansion (n_D); the labels are the many-body approximation $\mathcal{X}$, the lattice regime $\mathcal{K}$ and the Migdal regime $\mathfrak{M}$. Clamped nuclei put a formula at $n_B = 0$; independent particles at $\mathcal{X} = \text{IP}$; a free nucleus that recoils as a whole at $\mathcal{K} = \text{impulse}$; a kick that the electrons see as instantaneous at $\mathfrak{M} = \text{sudden}$. The deviation notation of Sect. 8.5 then lists only the components that differ from the base point $\mathfrak{T}_0$ of Eq. (8.45).

The placement fixes the plot. The leading correction of a formula is the largest of its steps $\mathfrak{T} \to \mathfrak{T}_k^+$, and its controlling parameter is whatever makes that step grow. On an axis with a small parameter, the step is a power of the parameter of Table 8.4, and the controlling parameter is that small parameter or a variable it grows with. A formula in the impulse regime has the Doppler width as its leading correction, of relative size $x_R^{-1/2}$ by Eq. (8.43), so its plot runs against the recoil parameter x_R. A Migdal probability at dipole order, $n_D = 2$, misses the higher multipoles, and its plot runs against ξ_D. On the many-body axis there is no small parameter (Sect. 8.5). The fuller result is then the next approximation, computed, and the controlling parameter is the variable along which screening changes: the momentum transfer, the energy deposit or the mass of an absorbed boson.

Many formulas truncate an expansion that is not one of the eight components. A long-wavelength form keeps the leading power of q against a lattice momentum, and a light-mediator rate assumes a dark-matter mass far above the kinematic edge. A spin-wave rate keeps the leading order in $1/|\boldsymbol{S}|$, and a dopant rate the first order in the donor density. Sect. 8.5 records such approximations as qualifiers, in words, and their controlling parameter is the expansion parameter itself. We sort them into four kinds: long wavelength, as in the plasmon pole of the electron gas; the kinematics of the halo and of the signal, as in a light-mediator rate far above its edge or the coherence time of a classical field; the dark-matter side and the order in the coupling, as in the speed of boosted dark matter; and the model or method used for the target, as in the spin-wave order of a magnon rate or the density of a dopant.

Fig. 9.3 places the 75 Limit Checks of Chaps. 10 to 17 by the axis of their leading correction, and Table 9.5 lists each with its controlling parameter and its step. Twenty-five lie on a component of the label. Eleven are on the many-body axis. Among them are the crystal form-factor rate of Sect. 10.1, which computes the rate of creating electron–hole pairs from independent-particle bands, and the Migdal rate through the loss function of Sect. 12.3. The loss function of a superconductor in Sect. 15.1 is there too: its box compares the bare quasiparticle response with a conserving one, built so that it obeys charge conservation. Six are on the lattice regime, from the clamped lattice to the impulse limit, two on the anharmonic order, two on the Migdal regime, two on the multipole order, one on the thermal order and one on the Born–Oppenheimer order. Every Migdal scattering formula sits at $n_B = 2$, and the rate with phonon final states of Sect. 12.4 takes its next step there: the phonon energies in the electronic denominators give $2\langle\omega_{\text{ph}}^2\rangle/\omega_e^2 = 2.9 \times 10^{-3}$ at the silicon gap, one step of n_B in the counting of Table 8.4. None of the boxes has its leading correction on n_g. The one formula whose next step is $n_g = 4$, the phonon-assisted electron-recoil rate of Sect. 13.3, has no box, and its correction is the off-shell parameter θ_{ep} of Table 8.4.

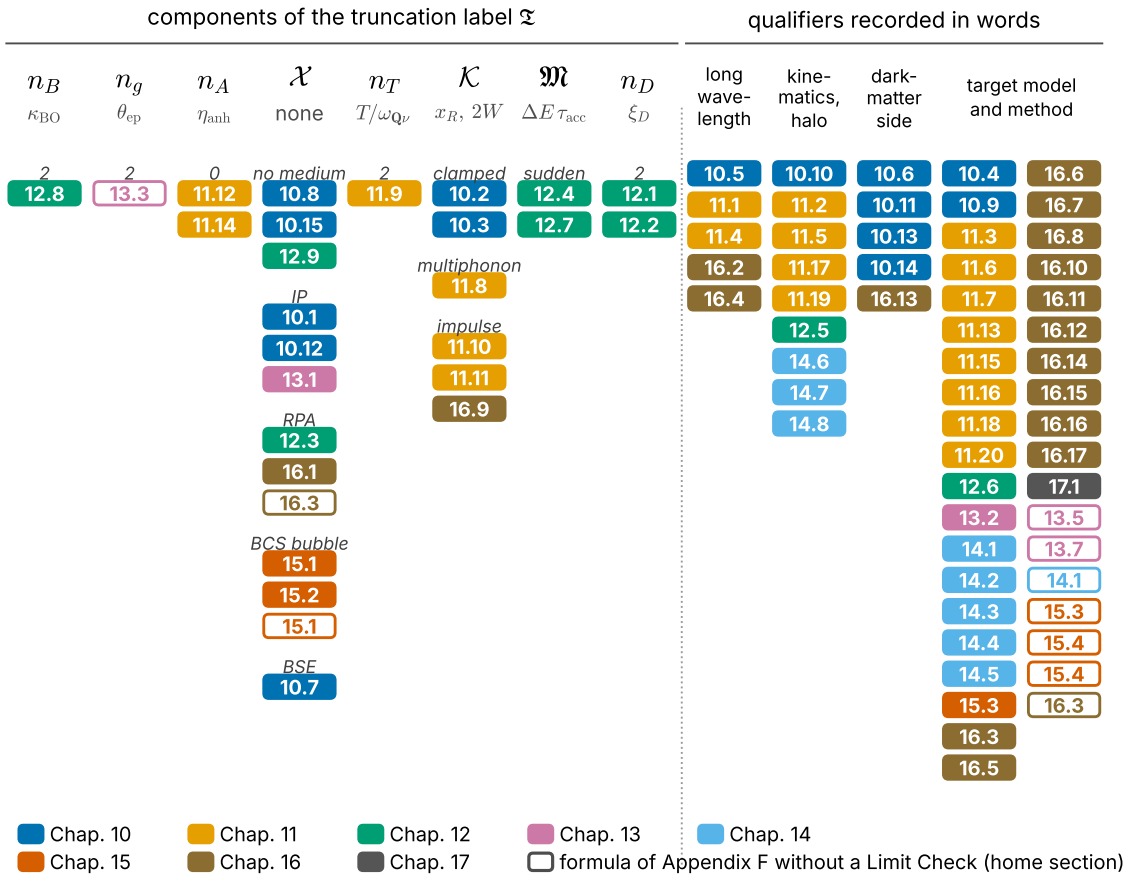


Figure 9.3: The Limit Checks of Chaps. 10 to 17 placed by the axis of their leading correction. Left: the eight components of the truncation label $\mathfrak{T}$, each with its small parameter (Table 8.4); under each, the boxes whose comparison result lies one step from the formula along that component, grouped by the formula's value of that component, printed in italic (for example IP or impulse). Right: boxes whose leading correction is an expansion in a parameter that the label records only in words, sorted into four kinds. Filled chips: Limit Checks, colored by chapter. Hollow chips: formulas of Table F.1 without a box, labeled by the number of their home section, not of a box. The step of each box and its controlling parameter are listed in Table 9.5.

The other 50 boxes expand in a qualifier: 5 in a long wavelength, 9 in the kinematics of the halo or the signal, 5 on the dark-matter side, and 31 in the model of the target. Two of the last are close to the label. Before any change of $\mathcal{X}$, the rate of the electron gas in Sect. 10.3 misses the gap and the interband peaks of the crystal, a change of target model. The band of Sect. 11.5 built from exact moments misses the skewness of the moment problem, not an anharmonic order. The split follows the chapters. In Chap. 12 seven of nine boxes lie on the label, in Chap. 10 seven of fifteen and in Chap. 11 six of twenty. In Chap. 14, on magnetic excitations, none of the eight does, and in Chap. 16, on novel targets such as Dirac materials, quantum dots and glasses, two of seventeen do. The label was built from the expansion of a crystal of electrons and nuclei in its small parameters, and the electronic, lattice and Migdal channels truncate along it. A magnon rate truncates in $1/|\boldsymbol{S}|$, a tunneling defect in its coupling to the bath, and a dopant in its density, and for these targets the controlling parameter of the box is the target's own.

Worked example 9.4: Three literature rates on the truncation label

We place three rates on $\mathfrak{T}$ and read from each placement the plot that its Limit Check draws. The rates are derived in Sects. 10.1, 10.2 and 11.4; here we use only their assumptions.

Crystal form-factor rate. The electron-recoil rate of Essig and collaborators [3, Eqs. (3.13) and (3.17)] squares the matrix elements of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ between Bloch states of independent electrons (Chap. 3), with the valence bands filled at $T = 0$ and the nuclei clamped. Each assumption is a base value: no derivative coupling, no electron–phonon vertex, independent particles, no thermal occupation, and no multipole expansion, since the plane wave is kept whole. The rate sits at $\mathfrak{T}_0$, with the qualifier "unscreened". Every integer step costs a small parameter of Table 8.4, and the many-body step costs an amount that no parameter predicts, so the many-body step is computed: IP to RPA to RPA with local fields. Screening is strongest at small q, where the long-range Coulomb field of the induced charge acts, and fades at large q. The controlling parameter is therefore the momentum transfer at fixed energy, and Limit Check 10.1 plots the rate weight against q/q_0, with $q_0 = \alpha m_e$, for the three approximations. The screened response is 0.05 of the unscreened one at the kinematic edge and 0.22 at q_0, and the q-integrated rate at 5 eV is 0.47 of it. Nothing is shaded. The status follows: the formula is a definite point of the label, but its leading correction is of order one across the whole range, and Sect. 10.2 computes the rate with screening.

Loss-function rate. The rate of Knapen and collaborators and of Hochberg and collaborators [2, Eqs. (6) and (13), 4, Eq. (1)] writes the response through the loss function $\mathrm{Im}[-1/\varepsilon(\boldsymbol{q},\omega)]$. The identity Eq. (8.38) makes this form exact for a coupling to the total charge at any lattice and for a coupling to the electron density with the nuclei clamped. The many-body axis is then not truncated by the formula; $\mathcal{X}$ is whatever approximation the loss function is computed in, RPA with local fields in the tables of Sect. 10.2. The truncation is the clamped lattice, for a coupling to electrons alone. Releasing the nuclei is a step of the lattice regime, from no phonon in the final state to one phonon, and it adds the electron-following sector of Sect. 8.6: the electrons that move with a nucleus carry an acoustic peak inside the phonon band. The controlling parameter is the energy deposit against the top of the phonon band, $\omega/\omega_{\max}$, and Limit Check 10.2 shades the energies above the band, where the two couplings share the formula. Below it, for dark matter of 1 MeV and in a model of the acoustic peak, the electron-only coupling has more than 10^4 times the rate that electronic excitations give at the same energies. The formula is established above the phonon band; inside it, for an electron-only coupling, the size of the correction rests on a model.

Free nuclear recoil. The recoil rate of Lewin and Smith with the Helm form factor [5], Eq. (11.35) with Eq. (2.28), assumes a nucleus that is free, at rest and recoils as a whole. In the label this is $\mathcal{K}$ = impulse at zero width, $n_B = 0$, the same placement as the elastic recoil of Limit Check 1.1. The mean deposit E_R is exact at every anharmonic and thermal order, since it is fixed by the sum rule Eq. (5.61). The leading correction is the next term of the impulse expansion, the Doppler width of the bound nucleus, of relative size $x_R^{-1/2}$. The controlling parameter is x_R, and Limit Check 11.10 plots the crystal rate over the free-recoil rate against it, with the multiphonon response of silicon as the fuller result and the Gaussian of the impulse approximation as the leading correction. For 1 GeV dark matter the two agree within 10 % for $x_R \geq 2.3$, at $q \geq 70\,\mathrm{keV}$.
The two boxes on the impulse axis draw their bands at different values of one parameter. Limit Check 1.1 asks of the spectrum at one momentum transfer that its rms width be below a third of E_R, which needs $x_R \geq 9$ for its oscillator. Limit Check 11.10 asks of the rate at a given energy, which sums the response over momentum transfers with a smooth halo weight, that it agree within 10 %. That is met at smaller x_R: with the mean deposit exact, the width changes such a sum only at second order. The placement fixes the axis and the leading correction. Where the band starts depends on the quantity compared and on the tolerance, and each caption states both.

9.4.3 The formal record of the limits

Table F.1 of Appendix F has one row for each formula that the book recovers as a limit, in the order of the chapters. Its first column names the formula, its equation in the book and its Limit Check if it has one. The leading-correction column opens with the size of the correction in bold, taken from the text, the box or its figure; the word *unsized* in its place marks a correction that the book identifies but to which it assigns no number.

For the channel chapters the appendix has 90 rows. Eighty belong to the 75 Limit Checks, since a box may compare two formulas in its panels. Ten have no box; they are marked so in the first column and appear as hollow chips in Fig. 9.3. Six of the 90 rows have an unsized leading correction. A formula of this kind is established at its own point of $\mathfrak{T}$ and open beyond it; Chap. 22 collects the open items of the book, each with the missing step that would close it. To find a formula, a reader goes from the channel in Table 9.1 to its home section, and from there to its row in Table F.1; Table 9.5 gives the same entry point by Limit Check.

Table 9.5: The Limit Checks of Chaps. 10 to 17, by chapter. Controlling parameter: the variable on the horizontal axis of the box. Leading correction: the component of the truncation label $\mathfrak{T}$ of Eq. (8.44) along which the comparison result lies one step from the formula, with the step; where the step is not along a component, the kind of qualifier it expands in. Formulas recovered as limits without a box are listed in Table F.1.

Limit Check	Formula	Controlling parameter	Leading correction
Chap. 10			
10.1	Crystal form-factor rate	momentum transfer q/q_0 at fixed ω	$\mathcal{X}$: IP → RPA + LF
10.2	Loss-function rate	energy over the top of the phonon band	$\mathcal{K}$: clamped → one-phonon

Limit Check	Formula	Controlling parameter	Leading correction
10.3	Sum-rule bounds on the loss function	momentum transfer q	$\mathcal{K}$: clamped $\to$ one-phonon
10.4	Lindhard and Mermin loss-function rates	energy over the gap, ω/E_{g}	target model and method
10.5	Plasmon production from the pole of $1/\varepsilon$	momentum over the plasmon cutoff	long wavelength
10.6	Crystal responses of the effective operators	current-row rate over density-row rate	dark-matter side and coupling
10.7	Bethe–Salpeter exciton rate	momentum transfer $\lvert\boldsymbol{q}\rvert$	$\mathcal{X}$: BSE, one time ordering $\to$ both
10.8	Absorption with in-medium mixing	boson mass m_ϕ	$\mathcal{X}$: no medium $\to$ measured ε
10.9	Scalar absorption through the kinetic-energy row	upper cutoff of the spectral sum	target model and method
10.10	Pair absorption	line width over the kinetic energy of the pair	kinematics and halo
10.11	Dark Compton scattering	boson mass over electron mass	dark-matter side and coupling
10.12	Inelastic electron recoil in crystals	mass splitting Δ_χ	$\mathcal{X}$: IP $\to$ RPA + LF
10.13	Boosted dark matter: the density row	dark-matter speed v	dark-matter side and coupling
10.14	Fermionic absorption on electrons	dark-matter mass over the binding energy	dark-matter side and coupling
10.15	In-medium decay of an excited state	splitting over the valence plasma frequency	$\mathcal{X}$: no medium $\to$ measured ε
Chap. 11			
11.1	Long-wavelength acoustic structure factor	q/q_{BZ}	long wavelength
11.2	Light-mediator single-phonon rate	dark-matter mass over the acoustic edge	kinematics and halo
11.3	Sphere-averaged longitudinal speed	elastic anisotropy (Zener ratio)	target model and method
11.4	Born-charge single-phonon rate	momentum transfer q in three directions	long wavelength
11.5	Polar matrix element and band integral	dark-matter mass m_χ	kinematics and halo
11.6	Static dielectric constant of one polar branch	share of one polar branch	target model and method
11.7	Born charge over ε_∞ from the LO–TO splitting	misfit of the inputs to the Lyddane–Sachs–Teller relation	target model and method
11.8	Multiphonon incoherent approximation	q/q_{BZ}	$\mathcal{K}$: incoherent $\to$ coherent multiphonon
11.9	Second-order thermal diffuse scattering	distance from a reciprocal-lattice vector	n_T: classical $\to$ quantum occupations
11.10	Free nuclear recoil with the Helm form factor	recoil parameter x_R	$\mathcal{K}$: impulse $\to$ multiphonon
11.11	Fermionic absorption on nuclei	dark-matter mass m_χ	$\mathcal{K}$: impulse $\to$ multiphonon
11.12	Anharmonic corrections to multiphonon weights	momentum in units of the zero-point momentum	n_A: $0 \to 2$

Limit Check	Formula	Controlling parameter	Leading correction
11.13	The Markov–Krein band	threshold in standard deviations	target model and method
11.14	Solid helium: quasi-harmonic and self-consistent phonons	Lindemann ratio	n_A: $0 \to$ all even orders
11.15	Incoherent spin absorption	boson mass m_ϕ	target model and method
11.16	Lifshitz mass-defect Green function	phonon frequency, at three mass defects	target model and method
11.17	Daily modulation from the anisotropic response	momentum transfer q	kinematics and halo
11.18	Single-phonon absorption	boson mass m_ϕ near the polar phonon	target model and method
11.19	Bulk acoustic resonator	quality factor of the mode over that of the halo field	kinematics and halo
11.20	Resonant-mass drive of a scalar	frequency over the lowest mode of the body	target model and method
Chap. 12			
12.1	Atomic Migdal effect at dipole order	multipole parameter ξ_D	n_D: $2 \to$ all orders
12.2	Migdal excitation of bound final states	multipole parameter ξ_D	n_D: $2 \to$ all orders
12.3	Crystal Migdal effect through the loss function	momentum given to the electrons, over q_{BZ}	$\mathcal{X}$: RPA $\to$ RPA + LF (diagonal)
12.4	Crystal Migdal rate with free recoil	dark-matter mass (largest recoil energy)	$\mathfrak{M}$: sudden $\to$ moving nucleus
12.5	Per-mass Migdal rate with the valence ionization factor	dark-matter mass m_χ	kinematics and halo
12.6	The Wannier picture of the boost	valence band width over the gap	target model and method
12.7	Migdal effect with phonon final states	Doppler shift over the electronic energy	$\mathfrak{M}$: sudden $\to$ exact denominators
12.8	Phonon-mediated Migdal effect	electronic energy against the phonon band	n_B: $2 \to 4$
12.9	Migdal-type absorption of a scalar	boson mass m_ϕ	$\mathcal{X}$: atomic $\to$ crystal loss function
Chap. 13			
13.1	Phonon-assisted absorption as the optical limit	photon energy above the gap	$\mathcal{X}$: IP $\to$ BSE
13.2	Zero-point renormalization by special displacements	supercell size	target model and method
Chap. 14			
14.1	One-magnon rate at linear spin-wave order	inverse spin, $1/2\lvert\boldsymbol{S}\rvert$	target model and method
14.2	Absorption into magnons	applied over demagnetizing field; anisotropy over exchange	target model and method
14.3	Data-driven spin response	orbital shift of the ion's g factor	target model and method
14.4	Chiral phonons and molecular magnets	dark-matter mass; level spacing over temperature	target model and method
14.5	Multiferroic cross response	temperature over the ordering temperature	target model and method

Limit Check	Formula	Controlling parameter	Leading correction
14.6	Transverse magnetization of a driven nuclear-spin ensemble	time over the coherence time of the field	kinematics and halo
14.7	Gaussian-field signal and its power spectrum	record length over the coherence time	kinematics and halo
14.8	Sensitivity scalings of the exponential-bin likelihood	record length over the coherence time	kinematics and halo
Chap. 15			
15.1	Loss function of a superconductor near the pair-breaking edge	energy over the gap, on the kinematic edge	$\mathcal{X}$: bare BCS bubble $\to$ conserving
15.2	Resummed loss function on the kinematic edge	energy over the gap, on the kinematic edge	$\mathcal{X}$: bare BCS bubble $\to$ conserving
15.3	Phonon-assisted absorption: the normal-metal limit	energy over the gap	target model and method
Chap. 16			
16.1	Background screening of Dirac-material rates	background dielectric constant	$\mathcal{X}$: Dirac bands $\to$ with remote bands
16.2	One-dimensional plasmons of linear channels	momentum along the channel over k_{F}	long wavelength
16.3	Acoustic plasmon of two components and its Landau width	plasmon speed over the Fermi speed of the light carriers	target model and method
16.4	Effective medium of a layered stack	wave number times layer period	long wavelength
16.5	Vacuum ejection with plane-wave final states	kinetic energy of the ejected electron	target model and method
16.6	Quantum-dot rate and its bulk limit	energy above the dot gap, in confinement energies	target model and method
16.7	Binned Frenkel rate	dark-matter mass m_χ	target model and method
16.8	Dopant absorption	donor density	target model and method
16.9	Color-center creation and bond breaking	largest recoil energy over the threshold	$\mathcal{K}$: impulse $\to$ bound atom
16.10	Two-level response to a classical strain or field	temperature	target model and method
16.11	Dark-photon absorption on two-level systems	boson mass m_ϕ	target model and method
16.12	Glass absorption below the Ioffe–Regel frequency	boson mass against the Ioffe–Regel frequency	target model and method
16.13	Random media and the Born limit	medium potential over the dark-matter energy	dark-matter side and coupling
16.14	Two-level relaxation by phonons and electrons	splitting over temperature; coupling to the electrons	target model and method
16.15	The ohmic scaling limit	dimensionless ohmic coupling	target model and method
16.16	Hopping and tunneling through one barrier	temperature over the tunneling crossover	target model and method
16.17	Heat release of a cooled glass	time after cooling	target model and method
Chap. 17			

Limit Check	Formula	Controlling parameter	Leading correction
17.1	Partial f-sum of the valence electrons	upper limit of the f-sum integral	target model and method

Summary

Table 9.1 pairs the 24 kinds of excitation that this book treats with six classes of dark-matter coupling: electron density, nuclei, spin and current, vector absorption, axion-like and scalar absorption, and inelastic and other kinematics. The columns come from the target operators of Sect. 8.1, the rows from the solid-state chapters. Of the 144 cells, 57 hold a channel, each with one home section; eight more are allowed at leading order but computed in no section, and 79 are empty at leading order. Four reasons account for nearly all of the empty ones. The first is operator content: the nuclear density creates no electronic excitation in the Born–Oppenheimer approximation, and the electron density no magnon without spin–orbit coupling. The others are the kinematics of absorption at $q = m_\phi v \simeq 0$, which no acoustic phonon or free nucleus can meet, the order in the coupling, and the absence of a new response for inelastic kinematics. Each row sits in a sector of the decomposition theorem. The four electronic rows of the density column are parts of one zero-phonon response; phonons, multiphonons and free recoil are electron-following and differ only in their lattice regime; the phonon-assisted row and part of the Migdal row are phonon-dressed. The dark photon, which couples to every charge, populates more of the atlas than any other model, and its loss-function rate on electrons and its polar-phonon rate are two sectors of one charge response.

The lightest mass that reaches a channel, Eq. (9.1), depends on the largest laboratory speed of the halo, the threshold and the shape of the dispersion. No channel lies below the floor $2\omega_{\rm th}/v_{\rm max}^2$, which a flat onset reaches. A gapless linear branch of speed $c_{\rm ch}$ lies above the floor by about $v_{\rm max}/4c_{\rm ch}$, because the particle must reverse its direction to hand over the momentum of the excitation: the acoustic edge of silicon at a threshold of 1 meV is 6.12 keV (Worked example 9.2). A branch faster than $v_{\rm max}$, such as the bulk plasmon of silicon, is closed to elastic scattering. An offset minimum puts the electron–hole pairs of silicon between the phonon-assisted onset just above the floor, 0.333 MeV, and the zero-phonon edge at 0.405 to 0.408 MeV. A free nucleus needs a mass that grows as the square root of the threshold; a free silicon nucleus takes up the gap from 46.8 MeV. Absorption reaches $m_\phi = \omega$, five and a half decades below scattering at equal energy. Fig. 9.1 and Table 9.2 place silicon, germanium, GaAs, aluminum and $ZrTe_5$ on one map, which ranks reach and not rate. The edges move with $v_{\rm max}$ as Eq. (9.3) states: as $v_{\rm max}^{-2}$ for a flat onset, nearly as $v_{\rm max}^{-1}$ for an acoustic edge.

On the plane of momentum transfer and energy deposit, Fig. 9.2 draws the support of every excitation of silicon, colored by sector. A pair without a phonon must take the whole momentum transfer and lives above $E_{\rm min}(\boldsymbol{q})$; a pair with a phonon starts just above the gap at every q, and between the two lies the off-shell window. Overlap on the plane is not interference. Sectors add, and different phonon numbers add; amplitudes interfere only for a coupling to electrons and nuclei at once and at the boundary between the two electronic sectors (Table 9.3). The only place where the cross term decides a rate at leading order is the electron-following sector, where the cloud that follows a silicon nucleus cancels its charge at long wavelength and leaves a dark photon without a single-phonon channel.

A Limit Check compares a published formula, written in the book's notation, with a fuller result one step further along one axis of the truncation label, or in a parameter that the label records in words, against a controlling parameter, and shades where the two agree to a stated tolerance. The axis of the step fixes the leading correction and the controlling parameter: x_R

for an impulse formula, ξ_D for a Migdal probability at dipole order, and the momentum or energy along which screening changes for a formula on the many-body axis. Of the 75 Limit Checks of the channel chapters, 25 have their leading correction on a component of the label (Fig. 9.3, Table 9.5), most of them in the electronic, lattice and Migdal chapters. The other 50, which include every magnetic one, expand in a parameter of their target, kinematics or coupling that the label records in words. Appendix F records each formula's chain, validity and leading correction. Of its 90 rows for these chapters, 80 are tested in the 75 boxes, and six carry a correction that the book does not size.

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Chapter 10

Electronic excitations

An electron–hole pair is the signal most searches for sub-GeV dark matter look for. A single pair in silicon costs at least the gap, 1.17 eV, and a pair is easy to count: the charge-coupled devices of Sect. 1.5 read out one electron at a time. Dark matter of 1 MeV carries at most 3.5 eV of kinetic energy, enough to cross the gap, and this is why electron scattering in semiconductors reaches masses that nuclear recoils cannot. Chap. 8 wrote the rate for any coupling as a contraction of the response matrix with a tensor from the dark-matter side, and Chap. 9 found where each electronic excitation sits on the map of channels. What remains is the response itself. How faithfully can the rate of making pairs be computed in a real crystal? The band structure alone gives the crystal form factor on which most published limits rest. The answer, however, turns on what the band structure leaves out: other electrons that screen the potential and bind the pair, couplings to spin and current rather than number, and particles that are absorbed, split or fast rather than slow and elastic. Throughout, the nuclei stay clamped. This is the electronic sector of Sect. 8.6, and the lattice enters only in Chaps. 11 to 13, while the pairs made here are followed into the detector in Chap. 18. We start from independent electrons and add, one at a time, the pieces of the response that they leave out (Fig. 10.1).

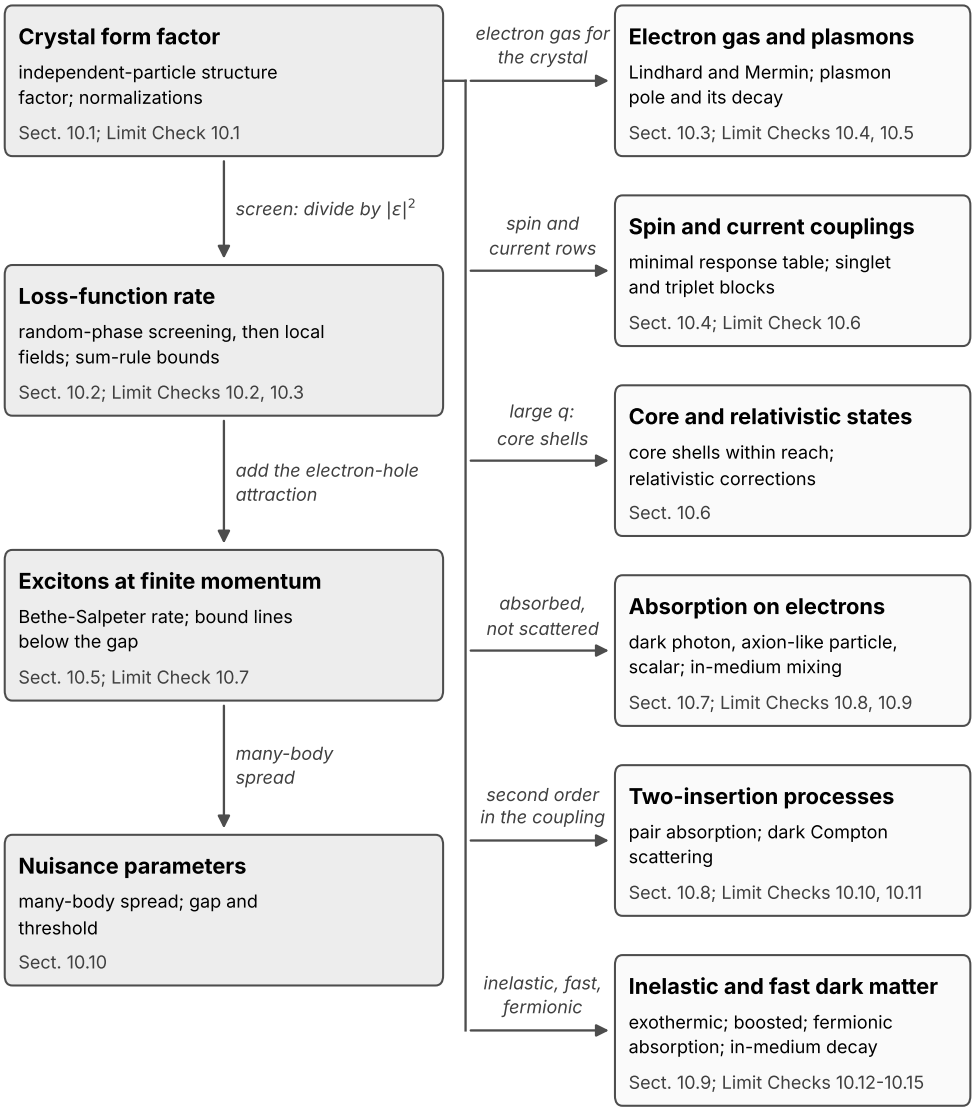


Figure 10.1: The electronic response and its extensions. Left, the hierarchy of approximations to the density response: the independent-particle crystal form factor (Sect. 10.1; Limit Check 10.1), screened by $|\varepsilon|^2$ in the random-phase approximation and then with local fields (Sect. 10.2; Limit Checks 10.2 and 10.3), with the electron–hole attraction added (Sect. 10.5; Limit Check 10.7); the spread across these approximations is a nuisance of the rate (Sect. 10.10). Right, the response classes beyond slow, elastic scattering on the density: the electron gas and its plasmon (Sect. 10.3; Limit Checks 10.4 and 10.5), spin and current rows (Sect. 10.4; Limit Check 10.6), core shells at large q (Sect. 10.6), absorption (Sect. 10.7; Limit Checks 10.8 and 10.9), processes second order in the coupling (Sect. 10.8; Limit Checks 10.10 and 10.11), and inelastic, boosted or fermionic dark matter (Sect. 10.9; Limit Checks 10.12 to 10.15).

10.1 The crystal form factor

A dark-matter particle of 10 MeV that crosses silicon can transfer up to 53 keV of momentum and deposit up to 35 eV (Table 1.2). Whether it leaves an electron–hole pair behind depends, for each pair $(\boldsymbol{q}, \omega)$, on how strongly the filled bands respond at that momentum and energy. In the simplest description of the crystal, electrons that move independently in the potential of clamped nuclei, that response is a sum over transitions from one valence state to one conduction state, each weighted by the square of the density vertex of Sect. 3.2. Experiments quote it as the *crystal form factor* of Essig and collaborators [1], and nearly every projected or published limit on electron scattering in silicon and germanium rests on a table of it. We will derive the rate from the master formula, define the form factor and its normalizations, read the silicon form factor in the (q, ω) plane, and compute the rate at 10 MeV from it. The rest of the chapter adds, one at a time, what such a calculation leaves out.

10.1.1 The independent-particle structure factor

Electron scattering through a mediator that couples to the electron number uses one row of the master formula, the density $n_e(\boldsymbol{q})$, with $\mathfrak{D}_{ee} = |\tilde{V}(\boldsymbol{q})|^2$ and all other entries zero (Sect. 8.2). The spectrum per unit target mass is then Eq. (8.16) with the dynamic structure factor of the electrons in place of the contraction, which is Eq. (A.43). Everything about the crystal sits in $S(\boldsymbol{q}, \omega)$, and the question is how to compute it.

At the base point $\mathfrak{T}_0$ of the expansion hierarchy, Eq. (8.45), the nuclei are clamped, the temperature is zero and the electrons are independent particles. The ground state is then a single Slater determinant in which the four valence bands of silicon are filled with both spins and the conduction bands are empty. The one-electron states are Kohn–Sham states, and the conduction energies are raised by the scissor shift of Sect. 3.4, which leaves the states themselves unchanged. In this ground state the density operator Eq. (3.37) can do only one thing: move one electron from a valence state $v\boldsymbol{k}$ to the conduction state $c, \boldsymbol{k}+\boldsymbol{q}$ of the same spin, with amplitude $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q})$. Every other term either removes an electron that is not there or adds one to a state that is full. The final states are single electron–hole pairs, each costs $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}}$, and they are orthonormal. Inserting them into Eq. (4.13), the two spins contribute equally, and the sum over the N crystal momenta of the zone becomes $\mathscr{V} \int_{\mathrm{BZ}} \mathrm{d}^3k/(2\pi)^3$ with $\mathscr{V} = N\Omega$. The volume cancels against the $1/\mathscr{V}$ in the definition of S, and (Appendix D.29)

Independent-particle structure factor of a semiconductor

$$S^0(\boldsymbol{q}, \omega) = 4\pi \sum_{v,c} \int_{\mathrm{BZ}} \frac{\mathrm{d}^3k}{(2\pi)^3} \left|\rho_{cv}(\boldsymbol{k}, \boldsymbol{q})\right|^2 \delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}), \tag{10.1}$$

for $\omega > 0$ at $T = 0$, with v over the valence and c over the conduction bands. The electron-recoil rate at $\mathfrak{T}_0$ is Eq. (8.16) with $\mathfrak{D}_{ee} = \pi\bar{\sigma}_e F_{\mathrm{DM}}(q)^2/\mu_{\chi e}^2$ (Eq. (1.34)) and $S = S^0$.

The factor 4π in S^0 is the 2π of Eq. (4.13) times 2 for spin. Each transition counts once, with the square of an amplitude of order one, and the rate is a sum of $|\rho_{cv}|^2$ over pairs, weighted by energy conservation. The same expression is $-2\,\mathrm{Im}\, R^0_{\boldsymbol{00}}(\boldsymbol{q}, \omega)$, the head of the independent-particle bubble Eq. (4.63) at $T = 0$: in the bubble, a pair costs $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}}$ and its resonant denominator carries the imaginary part $-\pi\delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}})$. So the codes that compute dielectric functions compute S^0 along the way, and the screened response of Sect. 10.2 starts from the same object.

As in Chap. 3, S^0 vanishes below the smallest pair energy $E_{\min}(\boldsymbol{q})$ of Eq. (3.57), and a mass is kinematically allowed only where Eq. (3.59) holds. Its total weight is fixed by the closure relation Eq. (3.36), each valence state distributes a total weight of one over all bands at $\boldsymbol{k}+\boldsymbol{q}$, and only the part that lands in conduction bands counts. At small q most of it stays in the same band and is lost to the rate; at large q it spreads over conduction bands far above the gap. Its energy-weighted sum is fixed as well: for a local one-electron potential the first moment of Eq. (10.1) equals the f-sum value $\pi n_e q^2/m_e$ with $n_e = 8/\Omega$ (Appendix D.29), which makes it the standard test of a computed S^0, with the caveats of Sect. 4.3 for nonlocal potentials.

Note that $\boldsymbol{q}$ in Eq. (10.1) is the physical momentum transfer and is never folded. The crystal momentum of the final electron is $\boldsymbol{k}+\boldsymbol{q}$ reduced to the first zone, but the vertex ρ_{cv} depends on the whole $\boldsymbol{q}$ through the Umklapp vector that stays in the plane wave (Eq. (3.34)), and so do $v_{\min}$ and F_{DM} on the dark-matter side (Sect. 1.3). Two momentum transfers that differ by a reciprocal-lattice vector reach the same pairs with different amplitudes and different kinematic weights.

The structure factor of a crystal depends on the direction of $\boldsymbol{q}$, and the full velocity kernel of Eq. (8.16) gives a rate that changes as the Earth turns the crystal through the dark-matter wind, the daily modulation of Sect. 11.7. For the time-averaged rate of a target whose orientation is not resolved, the literature replaces S^0 by its average $\bar{S}^0(q,\omega)$ over the directions of $\boldsymbol{q}$. The velocity and angular integrals then reduce to the mean inverse speed, Eq. (A.42). The rate takes the isotropic form Eq. (A.44),

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{\rho_\chi\,\bar{\sigma}_e}{4\pi\, m_\chi\, \rho_T\, \mu_{\chi e}^2} \int q\,\mathrm{d}q\; F_{\mathrm{DM}}(q)^2\, \eta\big(v_{\min}(q,\omega)\big)\, \bar{S}^0(q,\omega). \tag{10.2}$$

Appendix A.8 explains why this is not the same as averaging the directional rate over a day: a detector at fixed latitude does not sample all orientations equally.

10.1.2 Form-factor normalizations

Essig, Fernandez-Serra, Mardon, Soto, Volansky and Yu wrote the same rate per target, with the target information collected into a dimensionless function [1, Eqs. (3.13) and (3.17)]. In our notation their crystal form factor is

$$|f_{\mathrm{crys}}(q,\omega)|^2 = \frac{q^3\,\Omega}{4\pi\alpha\, m_e^2}\, \bar{S}^0(q,\omega), \tag{10.3}$$

which is Eq. (A.45) at $\mathfrak{T}_0$. The factor $q^3/\alpha m_e^2$ is chosen so that $|f_{\mathrm{crys}}|^2$ is close to one where the rate is made. Written out with Eq. (10.1), it is a double integral over the initial and final crystal momenta $\boldsymbol{k}$ and $\boldsymbol{k}'$ and a sum over the Umklapp vectors $\boldsymbol{G}'$ that connect them. The angle average is done by a delta function $\delta(q-|\boldsymbol{k}'-\boldsymbol{k}+\boldsymbol{G}'|)$. Ref. [1] defines f_{crys} in this form, which Eq. (D.172) derives. We substitute $\Omega\bar{S}^0 = 4\pi\alpha m_e^2 |f_{\mathrm{crys}}|^2/q^3$ into Eq. (10.2) and use $\rho_T\Omega = M_{\mathrm{cell}}$, the cell mass. Of the crystal, only the cell mass is left outside the form factor:

Limit Check 10.1. Crystal form-factor rate

At independent-particle level, unscreened, at zero temperature and with clamped nuclei, the electron-recoil spectrum per unit target mass is

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\ln\omega} = \frac{\rho_\chi\,\bar{\sigma}_e\,\alpha\,m_e^2}{m_\chi\,M_{\mathrm{cell}}\,\mu_{\chi e}^2} \times \int \mathrm{d}\ln q\,\frac{\omega}{q}\,\eta(v_{\mathrm{min}}(q,\omega))\,F_{\mathrm{DM}}(q)^2\,|f_{\mathrm{crys}}(q,\omega)|^2,$$

after Ref. [1, Eqs. (3.13) and (3.17)], divided by the target mass.

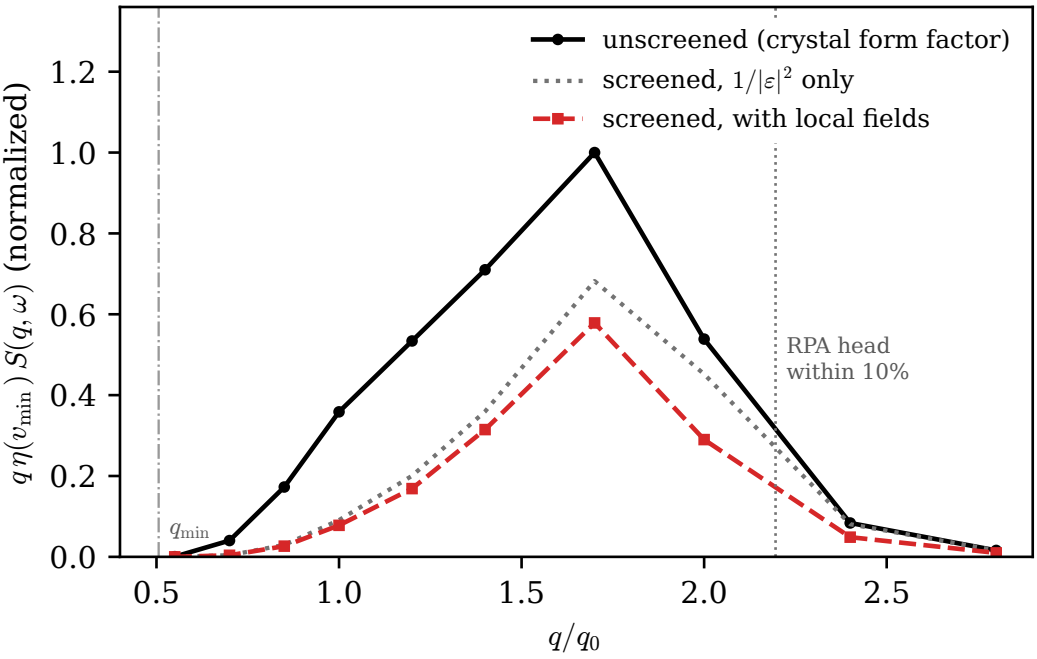


Figure 10.2: Unscreened rate against screened rates in silicon at $\omega = 5\,\mathrm{eV}$, for $m_\chi = 10\,\mathrm{MeV}$ and a heavy mediator: the rate weight $q\,\eta(v_{\mathrm{min}})\,S(q,\omega)$ of Eq. (10.2) against q, normalized to the maximum of the unscreened curve. All three curves come from one model, the local empirical pseudopotential of Worked example 8.10 (model gap 0.82 eV), with $\boldsymbol{q}$ along the direction $(5,3,1)$, chosen so that no $\boldsymbol{q}+\boldsymbol{G}$ comes close to zero, and spectra averaged over 4.5 to 5.5 eV. Solid: independent particles, the crystal form factor of the box. Dotted: random-phase screening by the macroscopic $1/|\varepsilon|^2$ alone, the leading correction. Dashed: random-phase screening with local fields over the 51 shortest reciprocal-lattice vectors, the fuller result. Dash-dotted: the smallest q the halo reaches at 5 eV. Dotted vertical: above $2.2\,q_0 = 8.2\,\mathrm{keV}$, with $q_0 = \alpha m_e$, the macroscopic screening changes the rate by less than 10 percent. The model has no momentum components above $2.8\,q_0$, where the plot ends. Nothing is shaded. No q in the plotted range brings the fuller result within 10 percent of the unscreened one.

Near the kinematic edge the unscreened formula overstates the response twentyfold, at q_0 fourfold; macroscopic screening fades above about 8 keV, but local fields keep the screened response near 0.6 of the unscreened one up to the largest q this model can describe.

Appendix D.29 writes out each step from the master formula to the box. Its leading correction lies on the many-body axis $\mathcal{X}$ of Sect. 8.5: the polarization of the valence electrons screens the dark matter's potential and multiplies the response at small q by roughly $1/|\varepsilon(q,\omega)|^2$, and the local fields of the crystal modify that factor. Sect. 10.2 adds both. In silicon at 5 eV the correction is not small (Fig. 10.2). The kinematic edge at that energy lies at $0.525\,q_0$, with $q_0 = \alpha m_e = 3.73$ keV; just above it, at $q = 0.55\,q_0$, screening cuts the response to 0.05 of the unscreened value, and at q_0 to 0.22. The macroscopic factor alone fades with q and is within 10 percent of one above $2.2\,q_0$. The local fields, however, do not fade in the same way: between 2 and $2.8\,q_0$ they hold the response at 0.54 to 0.61 of the unscreened value, with an uncertainty of about 10 percent from the size of the set of reciprocal-lattice vectors. Integrated over q, the screened rate at 5 eV is 0.47 of the unscreened one. The model is cruder than the tabulated form factor, which it matches to within 25 percent at q_0, and Sect. 10.2 computes the same ratios from first-principles tables.

Since a tabulated form factor is used far from the paper that produced it, its normalization is where errors enter. The literature writes the same rate per target or per unit mass, as a form factor, a structure factor or an unscreened ε_2, with the spin factor outside a sum over orbitals or inside the weights of a code, and with bin averages or bin sums; Appendix D.29 converts each to $\bar{S}^0$. A factor of two or ten in a rate is easy to make in these steps and hard to see in a plot. The f-sum rule catches it. At $q \approx q_0$ the free recoil energy, 13.6 eV, lies well inside any tabulated range, and the first moment $\int \mathrm{d}\omega\, \omega\, \bar{S}^0$ must then be close to $\pi n_e q^2/m_e$.

10.1.3 The silicon form factor

Fig. 10.3 shows the crystal form factor of silicon as Ref. [1] tabulated it, from PBE bands with the gap set by a scissor shift, four valence and 52 conduction bands, and a plane-wave cutoff of 70 Ry [1, Sects. 4.1 and 4.2]. The scissor puts the gap at 1.11 eV, close to the room-temperature gap of 1.12 eV (Sect. 10.7.2); the gap at zero temperature is 1.17 eV (Sect. 3.3), and a cryogenic detector sees the difference as a shift of the onset. The table begins at 1.2 eV, its first bin above the gap, and ends at 50 eV, where the highest conduction band runs out.

Before reading the figure we check its normalization, by the test just described. Divided by $\pi n_e q^2/m_e$ with the valence density $n_e = 8/\Omega$, the first moment of the tabulated $\bar{S}^0$ over its 50 eV is 1.03 at q_0, and 1.1 to 1.2 between $0.3\,q_0$ and $0.7\,q_0$, as means over windows of $0.2\,q_0$ with the bins the grid does not reach counted as zero. The scissor shift raises the first moment, since it moves every transition up by about 0.55 eV without changing its weight; that is the difference between the PBE gap of silicon (Sect. 3.4) and 1.11 eV. At small q, where the mean energy deposited is a few eV, the shift accounts for about ten percent; the rest of the excess has no identified cause (Sect. 10.10.1). Beyond $1.5\,q_0$ the ratio falls, to 0.83 at $1.5\,q_0$ and 0.33 at $2\,q_0$: the free recoil energy is then 31 and 54 eV, and the weight has moved past the last tabulated band. As a second check, the tabulated $\bar{S}^0$ at q_0 lies within 25 percent of the independent-particle structure factor of the local empirical pseudopotential of Worked example 8.10, along [100], between 5 and 20 eV. The two calculations share no input other than the crystal structure.

The form factor varies with q on two scales that silicon happens to have in common, the zone scale and q_0 (Sect. 1.3). At momentum transfers small compared with the zone, $2\pi/a = 0.61\,q_0$, the vertex reduces to its optical limit, $\rho_{cv} \simeq \boldsymbol{q}\cdot\boldsymbol{v}_{cv}/(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}})$ (Sect. 3.2), so $\bar{S}^0 \propto q^2$ at fixed ω, and by Eq. (A.45) $|f_{\mathrm{crys}}|^2 \simeq q^5\Omega\varepsilon_2(\omega)/8\pi^2\alpha^2 m_e^2$ with the optical ε_2. Integrated over energy, the tabulated form factor grows as $q^{4.8}$ between $0.3\,q_0$ and $0.75\,q_0$, close to this dipole scaling even where q is no longer small compared with the zone. Near $|\boldsymbol{G}_{111}| = 1.06\,q_0$ the momentum transfer resolves the bond, the optical limit fails, and the vertex is set by the plane-wave content of the valence states at momenta of order q (Eq. (3.35)).

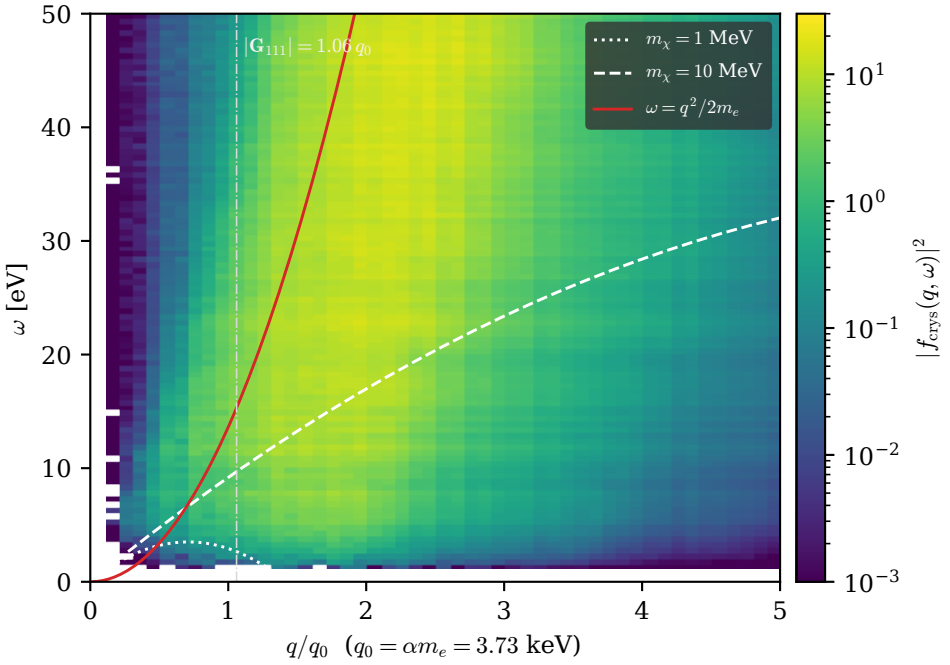


Figure 10.3: Crystal form factor $|f_{\mathrm{crys}}(q,\omega)|^2$ of silicon, Eq. (10.3), from the tabulation of Ref. [1]: PBE bands with the gap set to 1.11 eV by a scissor shift, valence to conduction, unscreened, averaged over the directions of $\boldsymbol{q}$. For display, bins are merged to $0.1\,q_0$ in q and 0.5 eV in ω; white cells below $0.3\,q_0$ are bins that the grid of crystal momenta of the calculation does not reach, and white cells in the lowest row are bins next to the gap that hold no weight. White curves: the largest energy that dark matter of 1 and 10 MeV can deposit at each q, $qv_{\mathrm{max}} - q^2/2m_\chi$ with $v_{\mathrm{max}} = 794.5\,\mathrm{km\,s^{-1}}$ (Eq. (1.26)); everything below a curve is kinematically allowed. Red: the free-electron recoil $\omega = q^2/2m_e$, along which the weight concentrates at large q. Dash-dotted: $|\boldsymbol{G}_{111}| = \sqrt{3}\,2\pi/a = 1.06\,q_0$, the shortest reciprocal-lattice vector. The color scale is logarithmic.

At fixed ω between 3 and 10 eV, $|f_{\mathrm{crys}}|^2$ peaks at 1.7 to $1.8\,q_0$ and at $2\,q_0$ it is still 2.1 to 2.5 times its value at q_0; by $3\,q_0$ it has fallen to a quarter of its value at q_0. The factor q^3 in Eq. (10.3) pushes this peak outward. The rate weight of Eq. (10.2), $q\bar{S}^0$, peaks earlier, at 1.1 to $1.4\,q_0$.

At large q the figure shows where the weight goes. Along the red curve, $\omega = q^2/2m_e$, a dark-matter particle would give a free electron at rest its recoil energy, and at large q the response of bound electrons gathers around it, broadened by their momenta, as in the impulse approximation of Worked example 4.3. At a fixed small energy deposited, a large q needs an electron that already moves against the momentum transfer with a momentum of about $q/2$, and the bound valence states have little weight at momenta much above q_0. That is why, integrated over the tabulated energies, $|f_{\mathrm{crys}}|^2$ falls as q^{-5} between 2 and $4\,q_0$. The white curves show which part of the plane the halo can reach. Dark matter of 1 MeV reaches only a thin sliver above the gap at $q < 1.4\,q_0$, near the bottom of the figure; at 10 MeV it reaches deposits up to 35 eV and momentum transfers well past the peak of $|f_{\mathrm{crys}}|^2$.

Worked example 10.1: The silicon rate at 10 MeV

Take $m_\chi = 10\,\mathrm{MeV}$, $\bar\sigma_e = 10^{-37}\,\mathrm{cm}^2$ and the standard halo of Appendix A.8: $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$, $v_0 = 238\,\mathrm{km\,s^{-1}}$, $v_\mathrm{esc} = 544\,\mathrm{km\,s^{-1}}$ and $v_\mathrm{E}^\mathrm{ref} = 250.5\,\mathrm{km\,s^{-1}}$. The reduced mass is $\mu_{\chi e} = 486.2\,\mathrm{keV}$, so $m_e^2/\mu_{\chi e}^2 = 1.105$, and the number density is $\rho_\chi/m_\chi = 30\,\mathrm{cm^{-3}}$. The prefactor of the box, with $M_\mathrm{cell} = 52.33\,\mathrm{GeV}$, is

$$\frac{\rho_\chi\,\bar\sigma_e\,\alpha\,m_e^2}{m_\chi\,M_\mathrm{cell}\,\mu_{\chi e}^2} = \frac{30\,\mathrm{cm^{-3}}\times 10^{-37}\,\mathrm{cm^2}\times c\times 1.105}{137.036\times M_\mathrm{cell}} = 2.45\times 10^5\,\mathrm{kg^{-1}\,yr^{-1}},$$

where the factor c turns $\rho_\chi\bar\sigma_e/m_\chi$ into a rate and η is then a pure number, $\langle 1/v\rangle = 1041$ in units of $1/c$. Everything else is the double integral of $\eta F_\mathrm{DM}^2|f_\mathrm{crys}|^2\,\mathrm{d}q\,\mathrm{d}\omega/q^2$ over the tabulated bins, with $v_\mathrm{min} = \omega/q + q/2m_\chi$ and η in the closed form of Appendix D.1.
For a heavy mediator, $F_\mathrm{DM} = 1$, the double integral is 0.652, and the rate above the gap is

$$\mathcal{R} = 2.45\times 10^5\,\mathrm{kg^{-1}\,yr^{-1}}\times 0.652 = 1.60\times 10^5\,\mathrm{kg^{-1}\,yr^{-1}},$$

about 440 events per kilogram and day. The spectrum $\mathrm{d}\mathcal{R}/\mathrm{d}\omega$ peaks at 6.5 eV, its mean is 6.0 eV, 38 percent of the events deposit at most 4.9 eV, and it ends at 35 eV, the largest energy a particle at v_max can deposit. At the peak energy the integrand in q is flat between about $1.7\,q_0$ and $1.9\,q_0$ (6.5 to 7.09 keV); only 5 percent of the rate comes from $q < q_0$, and 27 percent from $q > 2\,q_0$.
For a light mediator, $F_\mathrm{DM} = (q_0/q)^2$, the integral is 0.193, and $\mathcal{R} = 4.73\times 10^4\,\mathrm{kg^{-1}\,yr^{-1}}$, about 130 per kilogram and day. The factor $(q_0/q)^4$ moves the rate to small q and low energy: the spectrum peaks at 3.3 eV, its mean is 4.5 eV, 66 percent of the events deposit at most 4.9 eV, and the q integrand at the peak is largest near $0.6\,q_0 = 2.2\,\mathrm{keV}$, near the kinematic edge. Now 44 percent of the rate comes from $q < q_0$.
The two mediators probe different parts of Fig. 10.3. The heavy mediator samples the form factor near its maximum, at twice q_0 and a few eV, where the bands and the plane-wave content of the valence states set the answer. The light mediator samples its small-q edge, where the form factor is smallest and where, as the limit check shows, screening matters most. The scaling of Eq. (1.50) gives the rest: at fixed $\bar\sigma_e$ the rate falls as $1/m_\chi$ once m_χ is well above the threshold and the reach in q covers the peak. Published rates differ from these numbers mainly through the halo: Ref. [1] uses $\rho_\chi = 0.4\,\mathrm{GeV\,cm^{-3}}$, $v_0 = 230\,\mathrm{km\,s^{-1}}$, $v_\mathrm{esc} = 600\,\mathrm{km\,s^{-1}}$ and an Earth speed of $240\,\mathrm{km\,s^{-1}}$ [1, Sect. 6].

10.1.4 Convergence parameters and omitted physics

A crystal form factor is computed from Kohn–Sham bands and plane-wave vertices. The bands come from a density-functional calculation with a pseudopotential (Sects. 3.4 to 3.5), with the gap set by a scissor shift or by quasiparticle corrections (Appendices E.1 and E.3). The vertex is the plane-wave overlap Eq. (3.35), evaluated for every valence state, every conduction state and every Umklapp vector that a given $\boldsymbol{q}$ requires, with $\boldsymbol{q}$ kept unfolded. Appendix E.7 sets out the calculation and its cost, about 5×10^{11} vertices for silicon with 52 conduction bands and momentum transfers up to 30 keV. The plane-wave cutoff limits the largest q a vertex can carry, the number of conduction bands the largest ω, and the grid of crystal momenta the resolution at small q and near the onset; the white cells of Fig. 10.3 are this last limit. At large q the pseudopotential matters as well. A pseudo-wavefunction is smooth inside the core region by construction, and above a few q_0 the vertex needs the nodes of the true valence states near the nucleus. The all-electron reconstruction of Appendix E.2 restores them; Griffin

and collaborators found that it raises the valence-to-conduction rate of germanium at energies above about 15 eV [2, Sect. 1]. For silicon at 10 MeV, Worked example 10.1 draws a quarter of the heavy-mediator rate from $q > 2\,q_0$, and that part rests on the same correction.

Its place in the expansion hierarchy, $\mathfrak{T}_0$, fixes what the calculation leaves out. It has no screening: the dark matter's potential acts on independent electrons, while in the crystal the other electrons polarize and reduce it, most strongly at small q, where the light mediator and the lightest masses collect their rate. Sect. 10.2 adds the random-phase screening and the local fields, and shows when the loss function gives the rate exactly. It has no electron–hole attraction: an exciton moves weight from the continuum toward the gap, which matters for the lowest thresholds, and Sect. 10.5 treats excitons at finite momentum. It has no lattice: the nuclei are clamped, so the phonon-assisted transitions that open the indirect gap below the zero-phonon edge (Sect. 13.1) and the Migdal effect (Chap. 12) are absent. It holds only valence-to-conduction transitions: the $2p$ shell of silicon, 99 eV below the valence maximum, and the $3d$ shell of germanium, about 29 eV below it, are the subject of Sect. 10.6, and the table ends at the highest conduction band it includes. And it is an angle average at zero temperature, which drops the daily modulation and the thermal shift of the gap.

Each of these corrections is the next order of one entry of $\mathfrak{T}$ and has its own section. Screening, the largest of them at small momentum transfer, comes next.

10.2 Screening and the loss-function rate

At a momentum transfer of 2 keV the static dielectric function of silicon is about 2.9, and a potential that varies on that scale reaches the valence electrons weakened by $1/|\varepsilon|^2 \approx 0.12$. Yet the screened rate of a 10 MeV particle with a heavy mediator is only 2.3 times smaller than the unscreened rate of Sect. 10.1. The two numbers measure different things. The first is the screening of one Fourier component of the potential at zero frequency. The second is an integral over all the momentum transfers and energies that kinematics allows, most of which lie where the medium screens little, and over a spectrum whose total weight screening cannot change. We derive the screened rate of a density coupling from the master formula, as an integral of the loss function along the kinematic curve, and compute it in silicon in three approximations of Sect. 4.6: independent particles, the random-phase approximation, and the random-phase approximation with local fields. Two sum rules of the loss function then bound it from above, with only the plasma frequency and the static dielectric function as input. The bound on the static step fails for a mediator that couples to electrons alone once the lattice is free to move, because the electrons that follow the nuclei add an acoustic peak.

10.2.1 The loss-function rate

A mediator that couples to the electron number acts on one row of the master formula, the density $n_e(\boldsymbol{q})$, with $\mathfrak{D}_{ee} = |\tilde{V}(\boldsymbol{q})|^2 = \pi\bar{\sigma}_e F_{\mathrm{DM}}(q)^2/\mu_{\chi e}^2$ (Eq. (1.34)), and the spectrum Eq. (8.16) needs only the structure factor S_{ee}. Sect. 10.1 computed it for independent electrons. The exact structure factor follows from the fluctuation–dissipation theorem as $S = (2q^2/e^2)\,L/(1 - \mathrm{e}^{-\beta\omega})$, with $L(\boldsymbol{q},\omega) = \mathrm{Im}[-(\varepsilon^{-1})_{\mathbf{00}}(\boldsymbol{q},\omega)]$ the loss function of Eq. (4.45), and the exactness theorem Eq. (8.38) says for which couplings this holds. For a dark photon, or any mediator that couples to the total charge n_Z, it holds at every $(\boldsymbol{q},\omega)$ with the loss function of the total dielectric matrix of electrons and nuclei. For a mediator that couples to electrons and not to nucleons it holds exactly only with the nuclei clamped. Above the phonon band the nuclei reach the electronic final states through the Migdal amplitudes alone, suppressed by powers of κ_{BO}, and the loss function gives the electron-only rate to that accuracy (Sect. 8.4.3). An electron–hole pair in silicon costs at least 1.17 eV, eighteen times the top of the phonon

band, so for the electronic rate the two couplings share one formula. In silicon the Born charges vanish, and the ionic part of the total dielectric function at these energies is negligible.

Inserting S into Eq. (8.16), with $e^2 = 4\pi\alpha$, gives $|\tilde{V}|^2 S = \bar{\sigma}_e F_{\rm DM}^2 q^2 L/[2\alpha\mu_{\chi e}^2(1-\mathrm{e}^{-\beta\omega})]$ under the integrals [3, Eq. (13), 4, Eq. (1)]. For an isotropic target, or a crystal whose loss function is averaged over the directions of $\boldsymbol{q}$, the velocity integral and the angular integral act on the delta function alone and produce the measure $q\,\mathrm{d}q\,\eta(v_{\rm min})/4\pi^2$ of Eq. (A.42), with $v_{\rm min} = \omega/q + q/2m_\chi$ from Eq. (1.23). Collecting the factors,

Loss-function rate

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{\rho_\chi\,\bar{\sigma}_e}{8\pi^2\alpha\,\rho_T\,m_\chi\,\mu_{\chi e}^2}\int \mathrm{d}q\; q^3\,\eta(v_{\rm min}(q,\omega))\,F_{\rm DM}(q)^2\,\frac{\bar{L}(q,\omega)}{1-\mathrm{e}^{-\beta\omega}}, \tag{10.4}$$

with $\bar{L}$ the loss function averaged over the directions of $\boldsymbol{q}$. It is exact for a coupling to the total charge, and for a coupling to the electron density at clamped nuclei or above the phonon band to Migdal order.

At the temperatures of a detector $\beta E_{\rm g}$ is of order 10^4 or more, and the factor $1/(1-\mathrm{e}^{-\beta\omega})$ is one. All of the medium's physics now sits in one measurable function. Screening is in it through the inverse of the dielectric matrix, and so are the local fields, because L is a diagonal element of ε^{-1} and not the inverse of a diagonal element of ε. For a momentum transfer beyond the first zone the element sits at the reciprocal vector $\boldsymbol{G}_q$ of the unreduced $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$, $L = \mathrm{Im}[-(\varepsilon^{-1})_{\boldsymbol{G}_q\boldsymbol{G}_q}(\boldsymbol{q}_r,\omega)]$ (Eq. (E.40)). A crystal has a loss function that depends on the direction of $\boldsymbol{q}$, and a rate at a fixed time keeps the full kernel of Appendix A.8 in place of $\eta/2q$. In a cubic crystal the first anisotropic term of L has angular momentum four, and it is the source of the daily modulation of Sect. 11.7.

The rate of Sect. 10.1 is the same formula with L replaced by ε_2 of the independent-particle head: since $S^{\rm IP} = -2\,\mathrm{Im}\,R^0_{\mathbf{00}}$ (Sect. 10.1) and $\varepsilon_2 = -v_{\rm C}(q)\,\mathrm{Im}\,R^0_{\mathbf{00}}$ (Eq. (4.63)), $S^{\rm IP} = (2q^2/e^2)\,\varepsilon_2^{\rm IP}$. The prefactor is the one of Eq. (10.4), and Eq. (10.3) is this relation rewritten as a form factor. Each many-body approximation of Sect. 4.6 is then a choice of what to put in place of $\bar{L}$. Independent particles put $\varepsilon_2^{\rm IP}$. The random-phase approximation without local fields puts $\varepsilon_2^{\rm IP}/|\varepsilon_{\mathbf{00}}|^2$, in which the head $\varepsilon_{\mathbf{00}} = 1 - v_{\rm C}(q)R^0_{\mathbf{00}}$ is inverted as a number. With local fields it is $\mathrm{Im}[-(\varepsilon^{-1})_{\mathbf{00}}]$ with the full matrix of Eq. (4.68). Exchange–correlation kernels and the electron–hole attraction change the proper response inside the same inverse (Sects. 4.6 and 4.9). The three rates differ only in the many-body approximation $\mathcal{X}$ of Sect. 8.5, IP, RPA and RPA + LF, with every other entry of $\mathfrak{T}$ at its base value.

For a dark photon the formula needs one remark. Eq. (10.4) uses the reference cross section with the vacuum mixing ζ, and the loss function already contains the in-medium mixing. In scattering the dark photon mixes with the longitudinal photon of the medium, its propagator acquires the factor $1/\varepsilon(\boldsymbol{q},\omega)$, and that factor is the screening of the total-charge row (Sect. 8.4.5). The vacuum mixing goes with the full response and the in-medium mixing with the proper one; multiplying Eq. (10.4) by $\zeta_{\rm eff}^2/\zeta^2$ would count the screening twice.

The loss function is computed from Kohn–Sham states, the random-phase dielectric matrix with local fields being inverted at each $(\boldsymbol{q}_r,\omega)$ and the diagonal element read off at $\boldsymbol{G}_q$ (Appendix E.6). Two kinds of measurement test it. Optical constants give L at $q \to 0$, the curve of Fig. 4.7, and electron energy-loss and inelastic x-ray scattering give it at finite q. The silicon tables used below are a published calculation of this kind [3], distributed with the DarkELF code [5], whose settings Worked example E.5 checks. They use Kohn–Sham states with the gap fixed by a scissor shift, a $\boldsymbol{k}$ grid of 8^3 and 70 bands, and give the dielectric matrix with and without local fields. They extend to $q = 22.5\,\mathrm{keV}$ and $\omega = 89\,\mathrm{eV}$. Kinematics keeps the rate inside them, since $\omega \le q v_{\rm max}$ is below 60 eV for every momentum of the table.

Limit Check 10.2. Loss-function rate

For a coupling to the electron density or to the total charge, the spectrum per unit target mass is $\mathrm{d}\mathcal{R}/\mathrm{d}\omega = [\rho_\chi \bar{\sigma}_e/(8\pi^2 \alpha \rho_T m_\chi \mu_{\chi e}^2)] \int \mathrm{d}q\, q^3 \eta(v_{\min}) F_{\mathrm{DM}}^2\, \mathrm{Im}[-1/\varepsilon(q,\omega)]/(1-\mathrm{e}^{-\beta\omega})$ [3, Eqs. (6) and (13), 4, Eq. (1)].

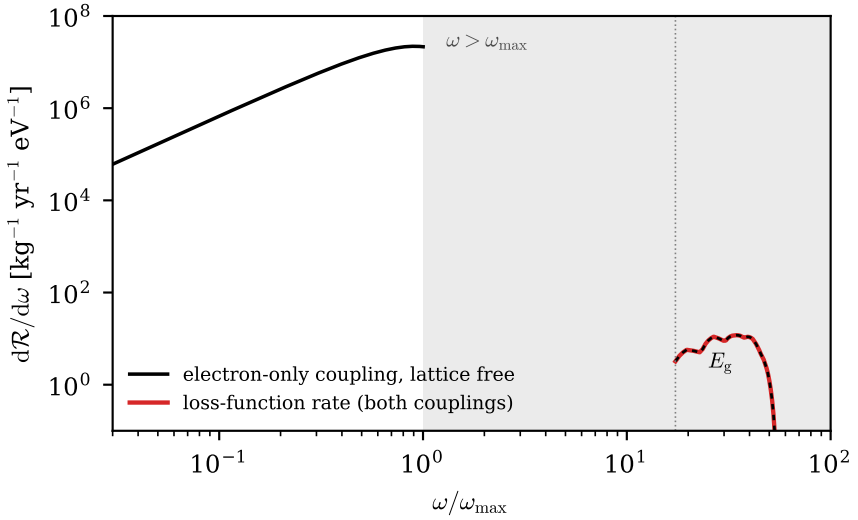


Figure 10.4: Spectrum of silicon at $m_\chi = 1\,\mathrm{MeV}$, heavy mediator, $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$, against $\omega/\omega_{\max}$, with $\omega_{\max} = 64.1\,\mathrm{meV}$ the top of the phonon band. Red: the loss-function rate with the random-phase loss function with local fields; it is the whole rate of a dark photon, since the Born charges of silicon vanish. Black: an electron-only coupling with the lattice free to move, which adds the acoustic peak carried by the electrons that follow the nuclei. Model of the peak: isotropic linear longitudinal branch with $c_{\mathrm{LA}} = 8.43\,\mathrm{km/s}$ up to $|\boldsymbol{q}| = 2\pi/a$, 14 electrons per atom following rigidly, no Debye–Waller factor, no Umklapp, transverse or multiphonon terms. Gray: $\omega > \omega_{\max}$, where the two couplings share the loss-function rate; dotted, the gap.

Inside the phonon band an electron-only coupling excites acoustic phonons through the electrons that move with the nuclei, a peak that the loss function of the total charge does not have and that, in this model, carries more than 10^4 times the electronic rate.

10.2.2 Screening corrections in silicon

Fig. 10.5 shows the silicon rate in the three approximations. At 10 MeV with a heavy mediator the spectrum peaks near 5 eV in each of them, and only its height changes. Per kilogram-year at $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$ the total above the gap is 1.36×10^5 events for independent particles, from the independent-particle head of these tables and not from the form-factor table of Sect. 10.1, which gives 1.60×10^5 (Sect. 10.10.1 traces the difference), 8.6×10^4 in the random-phase approximation without local fields and 5.9×10^4 with them. The ratio of the unscreened rate to the screened one with local fields is 2.3 for a heavy mediator and 3.7 for a light one, and above about 10 MeV it hardly depends on the mass. Below a few MeV it rises steeply, to 9 and 13 at 1 MeV. The local fields alone lower the rate by a factor 1.46 for a heavy mediator and 1.27 for a light one at 10 MeV, and by about 1.1 at 1 MeV.

A pointwise screening factor would suggest a larger effect. The static values of the same tables give $\varepsilon(q,0) = 2.87$ at 2 keV and 1.76 at $q_0 = 3.73\,\mathrm{keV}$, so $1/\varepsilon^2$ is 0.12 and 0.32; a widely used model dielectric function of semiconductors [6], with the silicon values $\varepsilon_\infty = 11.7$ and

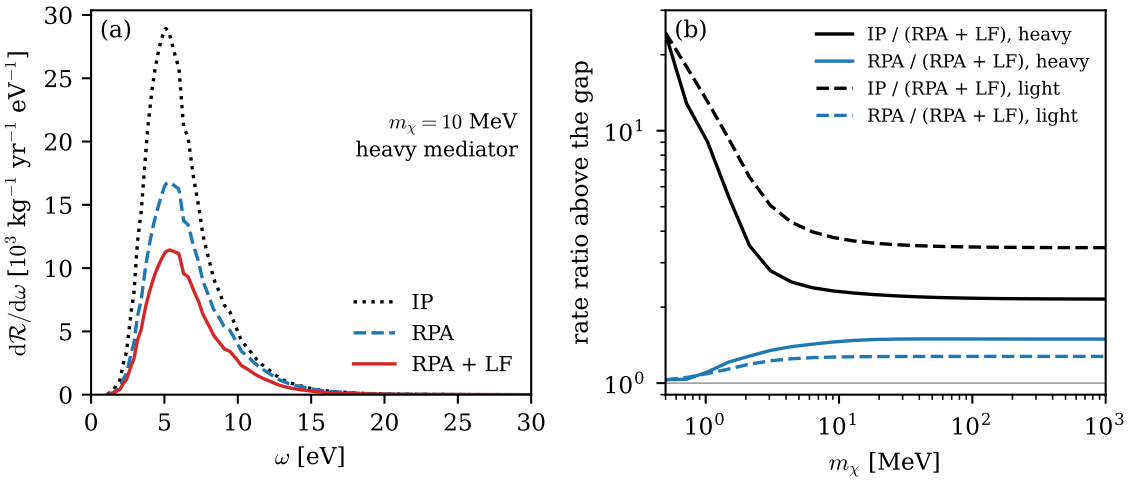


Figure 10.5: Electron-recoil rate of silicon in three approximations, independent particles, RPA, and RPA with local fields, from the random-phase dielectric matrices of Knapen, Kozaczuk and Lin [3], with and without local fields, with $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$, the standard halo of Sect. 1.2, $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$ and threshold 1.11 eV, the step-model gap of pair creation that accompanies the tables (their scissor puts the gap at 1.17 eV). IP: ε_2 of the independent-particle head of the same tables in place of L, the unscreened rate of Sect. 10.1 from a different tabulation; RPA: $\varepsilon_2/|\varepsilon_{\mathbf{00}}|^2$; RPA + LF: $\mathrm{Im}[-(\varepsilon^{-1})_{\mathbf{00}}]$. The tabulated functions are averaged over the directions of $\boldsymbol{q}$ in bins of 100 eV in $|\boldsymbol{q}|$, a choice of this figure, and end at $|\boldsymbol{q}| = 22.5\,\mathrm{keV}$; momenta above 15 keV already carry less than 2 % of the rate at 10 MeV. (a) Spectra at $m_\chi = 10\,\mathrm{MeV}$, heavy mediator. (b) Ratio of the rate above the gap to the rate with local fields, against m_χ; solid, heavy mediator, $F_\mathrm{DM} = 1$; dashed, light mediator, $F_\mathrm{DM} = (q_0/q)^2$.

$k_{\mathrm{TF}} = 4.12\,\mathrm{keV}$ of Worked example 4.8, gives 0.12 and 0.45. At finite frequency the factor is stronger still: at q_0 the ratio $\varepsilon_2^{\mathrm{IP}}/L$ is 4.7 at 3 eV, 5.2 at 5 eV and 4.4 at 10 eV. A pointwise factor leaves out where in q the rate comes from, the f-sum rule, and the local fields, and these set the rate ratio of 2.3.

Consider first where the rate lives. With a heavy mediator, $q^3\eta(v_{\mathrm{min}})$ weights large momentum transfers until the kinematic edge closes them, and at 10 MeV half of the screened rate comes from $q > 6.7\,\mathrm{keV}$ and 29 % from $q > 2q_0$. There the valence electrons barely screen. At 8 keV the static factor is 0.87, and the head screening, the ratio of the independent-particle rate to the random-phase rate without local fields at that momentum, is 1.18, against 13 at 2 keV and 4.2 at q_0. A light mediator weights the small momenta, where screening is strong, and its ratio is correspondingly larger.

The f-sum rule is the second. Screening does not remove spectral weight, since $\int \mathrm{d}\omega\, \omega L$ and $\int \mathrm{d}\omega\, \omega\varepsilon_2$ are both $(\pi/2)\omega_p^2$ at every $\boldsymbol{q}$ (Eqs. (4.46) and (4.47)); it moves the weight from the transition energies up toward the plasmon. At q_0 the peak of $\varepsilon_2^{\mathrm{IP}}$ sits at 14 eV and that of L at 26 eV. Whether the moved weight is lost depends on the largest energy the dark matter can deposit, $m_\chi v_{\mathrm{max}}^2/2$. At 10 MeV this is 35 eV, the moved weight stays mostly in reach, and the mean energy of the heavy-mediator spectrum rises only from 6.2 to 6.6 eV. At 1 MeV it is 3.5 eV, the rate samples only the depleted low-energy side of L, and the ratio grows to 9. This is why the curves of Fig. 10.5b turn up at small mass.

The local fields are the third, and at the momenta that carry the heavy-mediator rate they are larger than the head screening. Per unit momentum transfer, the rate without local fields exceeds the rate with them by 1.17 at q_0, 1.67 at 8 keV and 1.83 at 10 keV, two to three times the shortest reciprocal-lattice vector $|\boldsymbol{G}_{111}| = 3.95\,\mathrm{keV}$. The reason is the crystal potential. A density wave at $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$ induces, through the Bloch states, charge at every $\boldsymbol{q}_r + \boldsymbol{G}$, including the long-wavelength component at $\boldsymbol{q}_r$ itself, whose Coulomb field $v_{\mathrm{C}}(|\boldsymbol{q}_r|)$ is large and acts back. The head-only random-phase approximation keeps only the diagonal element and misses this field; the inverse of the full matrix contains it. An all-electron calculation with local fields finds the same suppression of the structure factor above about $3\alpha m_e$, and a reduction of the projected reach by 20 to 50 % for masses above a few MeV [7]. The size is not settled. At 10 keV the energies below 89 eV hold only 17 % of the f-sum, the rest lying near the free recoil at $q^2/2m_e = 98\,\mathrm{eV}$ and above. The rate samples the far low-energy tail of the spectrum, and the tail is sensitive to the band count and the pseudopotential (Appendix E.6). The spread between published calculations at these momenta is part of the many-body uncertainty of the rate (Sect. 8.8).

The ratio quoted in the literature compares yet another pair of approximations. Knapen, Kozaczuk and Lin kept local fields in both the unscreened and the screened response, comparing ε_2 of the macroscopic ε with L. They found a suppression by about 1.4 for a heavy mediator and 2.5 for a light one above 10 MeV [3]. Their tables give the same comparison as 1.56 and 2.73 at 10 MeV and 1.45 and 2.54 at 100 MeV with the halo of Sect. 1.2. "The screening factor" is therefore not one number, and a quoted value has to say which two approximations it compares. The independent-particle rate, the one most published limits rest on, overestimates the screened rate by 2.2 to 2.3 for a heavy mediator and 3.4 to 3.7 for a light one above 10 MeV.

The further corrections, an exchange–correlation kernel and the electron–hole attraction, change the picture mainly near the gap. An adiabatic local kernel changes the loss function by a few percent at energies of several eV, since it cannot bind an electron to a hole (Sect. 4.6). The electron–hole attraction raises the spectrum within about an eV of the gap and adds bound excitons below it (Sect. 4.9); Sect. 10.5 computes its effect on the rate. The optical limit of the tables shows what is missing at small q: $\varepsilon(0,0)$ is 10.2 without local fields and 9.1 with them, against the measured 11.7, the same trend as Worked example 4.11. A rate dominated

by the region just above the gap, as for masses of a few MeV, inherits that uncertainty, and a heavier one is less sensitive to it.

10.2.3 Sum-rule bounds on the rate

Silicon has 28 electrons per cell, and the f-sum rule fixes $\int_0^\infty \mathrm{d}\omega\, \omega L$ at $(\pi/2)\omega_p^2$ with $\omega_p = 31.1\,\mathrm{eV}$ for all of them (Sect. 4.3), whatever the bands, the screening or the many-body approximation. The inverse moment is fixed by the static dielectric function alone. These two integrals bound the rate of Eq. (10.4) from above, for any material, with no calculation of the spectrum [8, 9]. Define the moments of the loss function at fixed $\boldsymbol{q}$,

$$K_n(\boldsymbol{q}) = \int_0^\infty \mathrm{d}\omega\, \omega^n L(\boldsymbol{q},\omega), \qquad K_1 = \frac{\pi}{2}\,\omega_p^2, \qquad K_{-1} = \frac{\pi}{2}\big[1 - \varepsilon^{-1}(\boldsymbol{q},0)\big], \tag{10.5}$$

where K_1 is Eq. (4.46) and K_{-1} is the Kramers–Kronig relation Eq. (C.40) at zero frequency, at which ε^{-1} is real. At $\boldsymbol{q} \to 0$ in an insulator the second is the inverse moment Eq. (4.48); at finite q it is the static step $1 - \varepsilon^{-1}(\boldsymbol{q},0)$, which falls from 0.87 at 0.5 keV to 0.43 at q_0 and 0.28 at 5 keV in the silicon tables.

The intermediate moments follow by Hölder's inequality. For $-1 \le n \le 1$ write $\omega^n L = (\omega L)^{(1+n)/2}(L/\omega)^{(1-n)/2}$, a product of two nonnegative functions, and apply $\int fg \le (\int f^r)^{1/r}(\int g^s)^{1/s}$ with $r = 2/(1+n)$ and $s = 2/(1-n)$, so that $1/r + 1/s = 1$. Then $K_n \le K_1^{(1+n)/2} K_{-1}^{(1-n)/2}$, or

$$K_n(\boldsymbol{q}) \le \frac{\pi}{2}\,\omega_p^{n+1}\big[1 - \varepsilon^{-1}(\boldsymbol{q},0)\big]^{(1-n)/2}, \qquad -1 \le n \le 1, \tag{10.6}$$

which is Ref. [9, Eq. (8)]; the end points are the two sum rules themselves, and $n = 0$ is the Cauchy–Schwarz inequality. In the rate, the halo factor $\eta(v_{\min})$ is the only other function of ω. At $T = 0$ and for energies above a threshold ω_{th}, write $\eta \le \omega^n \eta_n^{\max}(q)$ with $\eta_n^{\max}(q) = \max_{\omega \ge \omega_{\mathrm{th}}} \omega^{-n}\eta(v_{\min}(q,\omega))$, extend the energy integral down to zero, which only adds positive terms since $L \ge 0$ for $\omega > 0$, and use Eq. (10.6):

$$\mathcal{R} \le \frac{\rho_\chi\, \bar{\sigma}_e\, \omega_p^{n+1}}{16\pi\alpha\, \rho_T\, m_\chi\, \mu_{\chi e}^2} \int \mathrm{d}q\, q^3\, F_{\mathrm{DM}}(q)^2\, \eta_n^{\max}(q)\, \big[1 - \varepsilon^{-1}(q,0)\big]^{(1-n)/2}, \tag{10.7}$$

Ref. [9, Eq. (9)]. The case $n = -1$ uses the static step alone and is the bound of Lasenby and Prabhu [8, Eqs. (8) and (9)]. Every n gives a valid bound, and the best one at a given mass is the smallest.

The useful form of the inverse moment needs $0 \le 1 - \varepsilon^{-1}(q,0) \le 1$. The lower limit is positivity: $1 - \varepsilon^{-1} = -v_{\mathrm{C}} R(\boldsymbol{q},0)$, and a static density response is negative by its Lehmann form (Eq. (4.20)). The upper limit, $\varepsilon^{-1}(\boldsymbol{q},0) \ge 0$, is not guaranteed in general, although we can show that it holds in the random-phase approximation with local fields for any insulator. The static independent-particle matrix $R^0_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},0)$ is Hermitian and negative semidefinite, each pair of states adding a positive rank-one matrix divided by minus its excitation energy, and so is $X = v^{1/2} R^0 v^{1/2}$, with v the diagonal matrix of $v_{\mathrm{C}}(|\boldsymbol{q} + \boldsymbol{G}|)$. The random-phase response gives $v^{1/2} R v^{1/2} = X(1-X)^{-1}$, whose eigenvalues $\lambda/(1-\lambda)$, with $\lambda \le 0$, lie in $(-1, 0]$. A diagonal element of a Hermitian matrix lies between its smallest and largest eigenvalue, so $(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}} = 1 + v_{\mathrm{C}} R_{\boldsymbol{G}\boldsymbol{G}}$ lies in $(0, 1]$ at every $\boldsymbol{q}$, local fields included. Beyond the random-phase approximation the same argument goes through with R^0 replaced by the proper response $\tilde{R} = (1 - R^0 f_{\mathrm{xc}})^{-1} R^0$ of Eq. (4.70), provided $\tilde{R}$ stays negative definite, that is $(-R^0)^{-1} + f_{\mathrm{xc}} > 0$. At long wavelength this is the positive inverse compressibility of Eq. (4.73), which the electron gas loses only near $r_s \approx 5.25$ [10], far from the $r_s \approx 2$ of the

silicon valence electrons. For an arbitrary material a negative static dielectric function at finite q is not excluded [11], and the conservative replacement $1-\varepsilon^{-1}\to 1$ that makes Eq. (10.7) material-independent is an assumption there, not a theorem.

The bound applies to the total charge as well. For a dark photon the loss function and the static step are those of the total dielectric matrix of Sect. 8.4, and the f-sum includes the nuclei, $\omega_p^2+\sum_\kappa Z_\kappa^2 e^2 n_\kappa/M_\kappa$. In silicon the nuclear share is 2.73×10^{-4} (Sect. 4.3) and the Born charges vanish, so the numbers are those of the electrons. The inverse moment also gives a check on calculations, which is how Lasenby and Prabhu first used it. The unscreened $\varepsilon_2^{\rm IP}$ has $(2/\pi)\int \mathrm{d}\omega\,\varepsilon_2/\omega=\varepsilon_{\mathbf{00}}(\boldsymbol{q},0)-1$, by the Kramers–Kronig relation for $\varepsilon-1$ (Eq. (C.41)), and this exceeds the bound $1-1/\varepsilon_{\mathbf{00}}$ on the screened L by the factor $\varepsilon_{\mathbf{00}}(\boldsymbol{q},0)$, which is 6.3 at 1 keV in silicon. Catena and Iglicki extended the inverse-moment bound to couplings to the spin and the current, through the Kramers–Kronig relations of the corresponding responses [12, Eqs. (4.16)–(4.20)]; Sect. 10.4 treats those couplings.

Worked example 10.2: The f-sum bound on the silicon rate at 1 MeV

Take $m_\chi=1\,\mathrm{MeV}$, a heavy mediator, $\bar\sigma_e=10^{-37}\,\mathrm{cm}^2$, $\rho_\chi=0.3\,\mathrm{GeV\,cm^{-3}}$ and the threshold $\omega_{\rm th}=1.11\,\mathrm{eV}$ of a charge detector, the step-model gap that accompanies the tables. Pairs are kinematically open for $\omega_{\rm th}/v_{\rm max}=0.42\,\mathrm{keV}\le q\le 2m_\chi v_{\rm max}=5.30\,\mathrm{keV}$ and $\omega\le m_\chi v_{\rm max}^2/2=3.51\,\mathrm{eV}$. The screened rate of Eq. (10.4) with local fields is 16.7 events per kilogram-year (18.5 without local fields, 157 for independent particles).

For $n=1$, $\eta(v_{\rm min})/\omega$ decreases with ω at fixed q, since $v_{\rm min}$ grows with ω, so its maximum is at the threshold, $\eta_1^{\rm max}(q)=\eta(v_{\rm min}(q,\omega_{\rm th}))/\omega_{\rm th}$. The bound Eq. (10.7) then needs only ω_p. With all 14 electrons per atom, $\omega_p=31.1\,\mathrm{eV}$, it gives 7.3×10^7 events per kilogram-year, 4×10^6 times the computed rate. The valence plasma frequency, $\omega_p^{\rm val}=16.6\,\mathrm{eV}$, would give 2.1×10^7, but it is not a bound, since the core electrons hold the rest of the f-sum and the inequality runs over the whole spectrum. The $n=0$ bound with the static step of the tables gives 2.0×10^6, and $n=-1$ gives 5.5×10^4, 3300 times the computed rate; with the step replaced by one it is 9.3×10^4. The tables contain only the valence electrons, whose step is slightly smaller than the all-electron one, so the $n=-1$ figure is a little low as a bound.

The bounds are far above the rate because they place all of the weight of L at the most favorable energy, while silicon puts it where a 1 MeV particle cannot go. The f-sum weight sits near the plasmon, around 17 eV at small q, and in the cores at hundreds of eV; the window ends at 3.5 eV, where the gap and the screening keep L small. The inverse moment weights low energies and comes closest. At 10 MeV, with a window reaching 35 eV, the $n=-1$ bound is 16 times the computed rate and the f-sum bound 8.5×10^5 times, both evaluated, like the rate, for $q\le 22.5\,\mathrm{keV}$, which is valid because the bound holds at each q. A sum-rule bound is a ceiling for any target, not an estimate for a given one. It comes close only for a material whose loss function concentrates its weight near the kinematic edge, a plasmon at a few eV, which is what the electron gas of Sect. 10.3 and the low-dimensional plasmons of Sect. 16.2 provide.

10.2.4 Static step of a relaxing lattice

For a mediator that couples to electrons alone, the exact relation is $S_{ee}=(2q^2/e^2)L_e/(1-\mathrm{e}^{-\beta\omega})$ with the electron-channel function $\varepsilon_e^{-1}=1+v_C R_{ee}$ (Appendix A.7), and both sum rules hold for L_e. The f-sum is unchanged: the double commutator of n_e with the Hamiltonian involves only the kinetic energy of the electrons (Sect. 4.3), so $\int \mathrm{d}\omega\,\omega L_e=(\pi/2)\omega_p^2$ with

the nuclei moving, and the acoustic peak of the electron-following sector takes the share $Z_\kappa m_e/M_\kappa = 2.73 \times 10^{-4}$ of it in silicon (Eq. (8.54)). The static step is a different matter. With the nuclei free to move, an electron-only probe at small q compresses neutral matter: the electrons it displaces drag their nuclei along, and the crystal responds as an elastic medium.

The size follows from the acoustic peak. At small q a longitudinal acoustic phonon moves each cell rigidly by $\boldsymbol{U_R}$, and the electrons that follow the nuclei, all $n_e\Omega = 28$ per cell in silicon since the Born charges vanish, give the density row the term $-\mathrm{i}\, n_e\Omega \sum_{\boldsymbol{R}} \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}}\, \boldsymbol{q}\cdot\boldsymbol{U_R}$ (Sect. 8.4.3). One phonon of wavevector $\boldsymbol{q}$ displaces each cell by $(2M_{\mathrm{cell}}N\omega_{\boldsymbol{q}})^{-1/2}$ along its polarization, with N cells in the volume $\mathcal{V} = N\Omega$. The squared matrix element is then $N(n_e\Omega)^2 q^2/(2M_{\mathrm{cell}}\omega_{\boldsymbol{q}})$, and

$$S^{\mathrm{ac}}_{ee}(\boldsymbol{q},\omega) = \frac{\pi n_e^2 \Omega\, q^2}{M_{\mathrm{cell}}\,\omega_{\boldsymbol{q}}}\,\delta(\omega-\omega_{\boldsymbol{q}}), \qquad \omega_{\boldsymbol{q}} = c_{\mathrm{LA}}q,$$

at $T = 0$, with c_{LA} the longitudinal sound speed, the c_ν of Eq. (5.20) for the longitudinal branch. Its first moment, $\int \mathrm{d}\omega\, \omega S/2\pi = n_e^2\Omega q^2/2M_{\mathrm{cell}}$, is $n_e\Omega\, m_e/M_{\mathrm{cell}}$ times the electronic f-sum $n_e q^2/2m_e$, the share quoted above. Its inverse moment is not small. With $L_e = (e^2/2q^2)(1-\mathrm{e}^{-\beta\omega})S_{ee}$, the peak adds $(2/\pi)\int \mathrm{d}\omega\, L^{\mathrm{ac}}_e/\omega = e^2 n_e^2\Omega/(M_{\mathrm{cell}}\,\omega_{\boldsymbol{q}}^2)$ to the static step, and

Static step of a relaxing lattice

$$1-\varepsilon_e^{-1}(\boldsymbol{q},0) = [1-\varepsilon_{\mathrm{cl}}^{-1}(\boldsymbol{q})] + \mathcal{G}_{\mathrm{rel}}(\boldsymbol{q}), \qquad \mathcal{G}_{\mathrm{rel}}(\boldsymbol{q}) \to \frac{\Omega_{P,e}^2}{(c_{\mathrm{LA}}q)^2}, \quad \Omega_{P,e}^2 = \frac{e^2 n_e^2}{\rho_T}, \tag{10.8}$$

as $q \to 0$, with $\varepsilon_{\mathrm{cl}}$ the dielectric function at clamped nuclei, n_e the density of all the electrons and $\rho_T = M_{\mathrm{cell}}/\Omega$. The lattice term grows as $(c_{\mathrm{LA}}q)^{-2}$ in every solid, polar or not. Its general form, a sum over branches with the electron-following charge $Z^{(e)}_\kappa(\boldsymbol{q})$ of each atom, is Eq. (E.93).

For silicon, $n_e = 7.00 \times 10^{23}\,\mathrm{cm}^{-3}$ and $\rho_T = 2.33\,\mathrm{g\,cm}^{-3}$ give $\Omega_{P,e} = 0.51\,\mathrm{eV}$, and with $c_{\mathrm{LA}} = 8.43\,\mathrm{km/s}$ along [100] (Appendix C.18) the lattice term equals one at $q = 18\,\mathrm{keV}$, far beyond the zone boundary. The limiting form is derived only as $q \to 0$. At finite q the dispersion bends over and the electron-following charge falls below 14 per atom, at relative order $(qa)^2$ and in opposite directions; at 1 keV the limiting form gives 3×10^2, and the true value there is only estimated to exceed 10^2 (Appendix E.16). Either way the step exceeds one by orders of magnitude throughout the zone, and the bound Eq. (10.7) with $n = -1$, and with every $n < 1$ through the power $(1-n)/2$, fails for an electron-only coupling once the lattice relaxes. Lasenby and Prabhu noted that phonon rates of electrophilic mediators exceed the sum-rule bounds of charge couplings [8, Sect. VII]; Eq. (10.8) gives the term responsible. It is the acoustic peak of S_{ee} that the exactness theorem of Sect. 8.4 attributed to the electrons that follow the nuclei. Its amplitude is the coefficient of $\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{U}$ in the density row, the coupling per unit volume $c_1^{(e)} n_e$ of the electrons that move with the cells; for the charge this coefficient cancels between electrons and nuclei, and the peak is absent.

The clamped bound holds in every many-body approximation, with the proviso on $\tilde{R}$ above; the lattice term is harmonic and adiabatic, with one phonon in the final state. For the electron-only coupling the f-sum bound and a restricted static-step bound survive. The f-sum bound holds as it stands. And for a threshold above the phonon band the electron-following sector contributes nothing to $\int_{\omega_{\mathrm{th}}}^{\infty} \mathrm{d}\omega\, L_e/\omega$ except through Migdal amplitudes, so the clamped step bounds that integral up to corrections in powers of κ_{BO} (Sect. 8.6). A

detector with an electronic threshold can use the clamped bound for either coupling; a calorimeter with a threshold in the phonon band cannot use it for an electrophilic mediator. Appendix E.16 specifies the calculation of $\mathcal{G}_{\rm rel}$ at finite q, with the electron-following charges from frozen-phonon supercells.

Limit Check 10.3. Sum-rule bounds on the loss function

For any target in its ground state, $\int_0^\infty \mathrm{d}\omega\, \omega L = (\pi/2)\omega_p^2$ and $\int_0^\infty \mathrm{d}\omega\, L/\omega = (\pi/2)[1 - \varepsilon^{-1}(q,0)] \le \pi/2$, and with $\eta \le \omega^n \eta_n^{\max}$ they bound the rate as in Eq. (10.7) [8, Eqs. (1), (8) and (9), 9, Eqs. (1), (2), (8) and (9)].

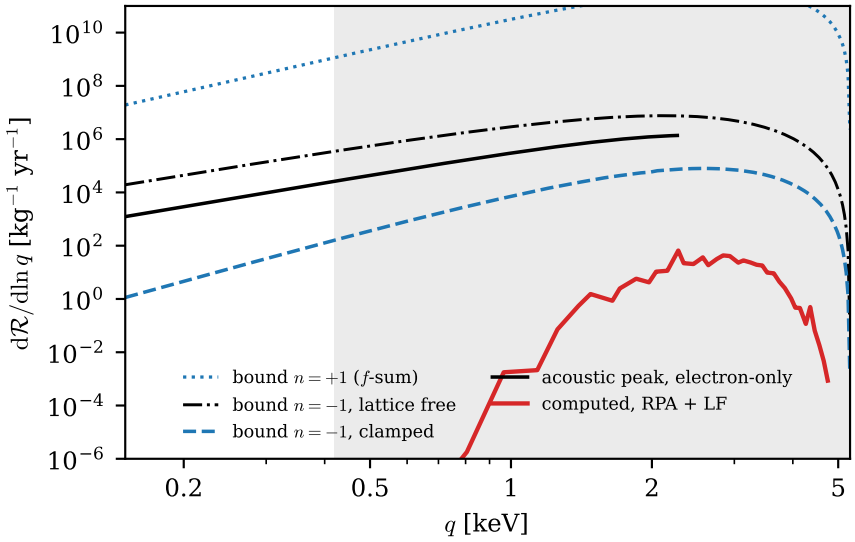


Figure 10.6: Rate per unit $\ln q$ in silicon at $m_\chi = 1\,\mathrm{MeV}$, heavy mediator, $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$, threshold 1 meV, against the bounds of Eq. (10.7). Red: the computed electronic rate (random-phase loss function with local fields). Dashed: the clamped bound with $n = -1$ and the static step of the same tables; dotted: the f-sum bound, $n = 1$, $\omega_p = 31.1\,\mathrm{eV}$. Black: the acoustic peak of an electron-only coupling, in the model of Fig. 10.4; dash-dotted: the $n = -1$ bound with the lattice term Eq. (10.8) added, in its limiting form $\Omega_{P,e}^2/(c_{\rm LA}q)^2$ at every q shown. Gray: momenta at which electron–hole pairs are open, $q \ge E_{\rm g}/v_{\max}$; there the clamped bound holds for both couplings when the threshold lies above the phonon band. The acoustic peak exceeds the clamped bound by 120 at 0.5 keV, 42 at 1 keV and 21 at 2 keV.

For an electron-only coupling with the lattice free to move, the static step grows as $(c_{\rm LA}q)^{-2}$ and exceeds one throughout the zone, and only the f-sum bound survives.

Sect. 10.3 takes the loss function of the electron gas, where the plasmon that would saturate these bounds is explicit.

10.3 Electron-gas models and plasmons

Put the Lindhard loss function of the silicon valence electrons into the rate of Sect. 10.2, and a 100 MeV particle with a heavy mediator makes 1.31×10^4 events per kilogram-year above 1.11 eV at $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$. The crystal tables of Sect. 10.2, with bands, local fields and a gap, give 1.22×10^4. A model with one parameter, the electron density, comes within seven percent of the crystal rate. At 1 MeV the same model gives 45 times the crystal rate. The gas

also carries the plasmon, which at small momentum transfer holds almost the whole f-sum, and it is natural to ask whether dark matter can excite it. We derive the electron-gas rate and find where it agrees with silicon and why, then isolate the plasmon pole in the rate and show that halo dark matter falls short of it by about ω_p in energy and a factor of 3.5 in speed.

10.3.1 The Lindhard loss-function rate

The electron gas of Sect. 4.5 is isotropic, so the direction average in Eq. (10.4) does nothing, and its loss function in the random-phase approximation is $\mathrm{Im}[-1/\varepsilon_{\mathrm{Lin}}(q,\omega)]$ with the Lindhard dielectric function of Eqs. (4.56) and (4.59). Inserting it,

Electron-gas loss-function rate

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{\rho_\chi\,\bar\sigma_e}{8\pi^2\alpha\,\rho_T\,m_\chi\,\mu_{\chi e}^2}\int \mathrm{d}q\; q^3\,\eta\big(v_{\min}(q,\omega)\big)\,F_{\mathrm{DM}}(q)^2\;\mathrm{Im}\!\left[\frac{-1}{\varepsilon_{\mathrm{Lin}}(q,\omega)}\right], \tag{10.9}$$

at $T=0$, with $\varepsilon_{\mathrm{Lin}}$ fixed by the electron density alone through k_{F} and ω_p. The Mermin form replaces $\varepsilon_{\mathrm{Lin}}$ by $\varepsilon_{\mathrm{Mer}}$ of Eq. (4.62).

This is the rate of Knapen, Kozaczuk and Lin with their Lindhard and Mermin functions [3, Eqs. (13), (17) and (19)]. Their prefactor $3\omega_p^2/q^2v_{\mathrm{F}}^2$ is the k_{TF}^2/q^2 of the static Lindhard screening $\varepsilon = 1 + (k_{\mathrm{TF}}^2/q^2)F_{\mathrm{L}}$ (Eq. (4.57)), since $k_{\mathrm{TF}}^2 = 2e^2\mathcal{N}_{\mathrm{F}} = e^2m_ek_{\mathrm{F}}/\pi^2$ and $\omega_p^2 = e^2k_{\mathrm{F}}^3/3\pi^2m_e$ give $k_{\mathrm{TF}}^2 = 3\omega_p^2/v_{\mathrm{F}}^2$, and their bracket $f_1^{\mathrm{L}}+\mathrm{i}f_2^{\mathrm{L}}$ is $-R^0/2\mathcal{N}_{\mathrm{F}}$, the braced expression of Eq. (4.56). For silicon the density is that of the four valence electrons per atom, $r_s = 2.0$, $k_{\mathrm{F}} = 3.57\,\mathrm{keV}$ and $\omega_p = 16.6\,\mathrm{eV}$ (Worked example 4.10). Below q_{c} the gas also has the undamped plasmon of Eq. (4.61), a delta function in ω; we return to it in Sect. 10.3.2 and leave it out of Eq. (10.9) for now, since halo dark matter cannot reach it.

The comparison with silicon uses the random-phase tables with local fields of Sect. 10.2. Both rates are integrated on the momenta of those tables, from 0.49 to 22.5 keV, above the threshold $E_{\mathrm{g}} = 1.11\,\mathrm{eV}$, the step-model gap that accompanies the tables; their scissor puts the gap at the physical low-temperature value, 1.17 eV. Per kilogram-year at $\bar\sigma_e = 10^{-37}\,\mathrm{cm}^2$:

	1 MeV		10 MeV		100 MeV	
	heavy	light	heavy	light	heavy	light
Silicon, RPA + LF	16.7	90	5.89×10^4	9.8×10^3	1.22×10^4	1.62×10^3
Lindhard / silicon	45	70	1.42	2.6	1.07	2.1
Mermin / silicon	54	94	1.57	2.9	1.21	2.4

The Mermin rows use one term with $\gamma_{\mathrm{rel}} = 3.66\,\mathrm{eV}$, the width of the measured silicon loss peak, which is a fit parameter and not a collision rate (Sect. 4.5).

The gas lacks the gap of silicon and the high-momentum tail of its bound electrons, two errors of opposite sign that set these ratios and that the spectrum of Fig. 10.7 shows. The first is the gap. The gas is a metal. Its continuum reaches down to $\omega = 0$, where $L \simeq e^2m_e^2q\omega/2\pi k_{\mathrm{TF}}^4$ (Sect. 4.5), while silicon has no pairs below its gap and weak absorption up to its peaks at 3.4 and 4.2 eV (Fig. 4.7). At 100 MeV with a heavy mediator the gas spectrum exceeds the crystal one by a factor 24 at 2 eV and 4.2 at 3 eV. A particle of 1 MeV deposits at most $m_\chi v_{\max}^2/2 = 3.5\,\mathrm{eV}$ (Eq. (1.24)), so it samples only this region, and the gas overestimates its rate by a factor 45.

The second error is the momentum distribution of the electrons. The halo deposits at most $\omega = qv_{\max}$ at momentum q. In the gas the pairs at $q > 2k_{\mathrm{F}}$ lie in the band $\omega_- < \omega < \omega_+$ around the free recoil $q^2/2m_e$, with $\omega_- = q^2/2m_e - qv_{\mathrm{F}}$, because no electron of the Fermi sea

moves faster than v_F. The lower edge passes the halo line at $q = 2m_e(v_F + v_{max}) = 9.85\,\mathrm{keV}$, and beyond it the gas has nothing the halo can reach. The valence electrons of silicon are bound, their orbitals contain momenta above k_F, and their loss function keeps weight far below the free recoil. An energy of 20 eV needs $q \geq 7.5\,\mathrm{keV}$, and there the gas spectrum is 3 percent of the crystal one; at 15 eV it is 27 percent. Between these two regions, from the upper absorption peak at 4.2 eV to $k_F v_{max} = 9.5\,\mathrm{eV}$, the largest energy deposited at $q = k_F$, the valence electrons respond nearly as free electrons do and the gas is within a factor 1.5 of the crystal. That band holds 62 percent of the silicon rate at 100 MeV. The integrated agreement at that mass, seven percent, is the excess below the band cancelling the deficit above it. A light mediator weights small q and therefore small ω, where the gas has its excess, and its ratio stays near 2.

The Mermin function does not repair either error. Its relaxation rate spreads the plasmon and gives the loss function a tail $\gamma_{rel}\omega/\omega_p^2$ below it (Sect. 4.5), which adds 10 to 35 percent to the rates in the table above; the function still has no gap. The Mermin oscillator method of Sect. 4.5, however, does better, because it is a fit. Its terms reproduce the measured optical loss function, gap and absorption peaks included, and each step function removes weight below its onset. The Mermin form only carries the fit to finite q, keeping each term's share of the f-sum. For silicon the method agrees with the first-principles tables to within 10 percent in the rate above a two-electron threshold, and the uncertainty grows for masses whose rate lies near the gap [3, App. C]. The momentum distribution is still that of a gas, and the two methods differ most at large ω, which carries a small share of the rate above that threshold.

Compared to silicon, the gas is the random-phase approximation for a homogeneous medium; the gap, the local fields and the bound momentum distribution are what the crystal adds. The computed crystal loss function that replaces ε_{Lin} is the one of Appendix E.6. The gas needs only n_e, which follows from the cell volume and the count of valence electrons (Appendix A.2), and it serves as a test of the codes that compute the crystal function (Sect. 4.5).

Limit Check 10.4. Lindhard and Mermin loss-function rates

For a homogeneous electron gas the rate is Eq. (10.9), with $\varepsilon_{Lin}(q,\omega) = 1 + (3\omega_p^2/q^2 v_F^2)[f_1^L + \mathrm{i} f_2^L]$ and $f_1^L + \mathrm{i} f_2^L$ the braced expression of Eq. (4.56), or the Mermin function ε_{Mer} with a relaxation rate γ_{rel} [3, Eqs. (13), (17)–(19)].

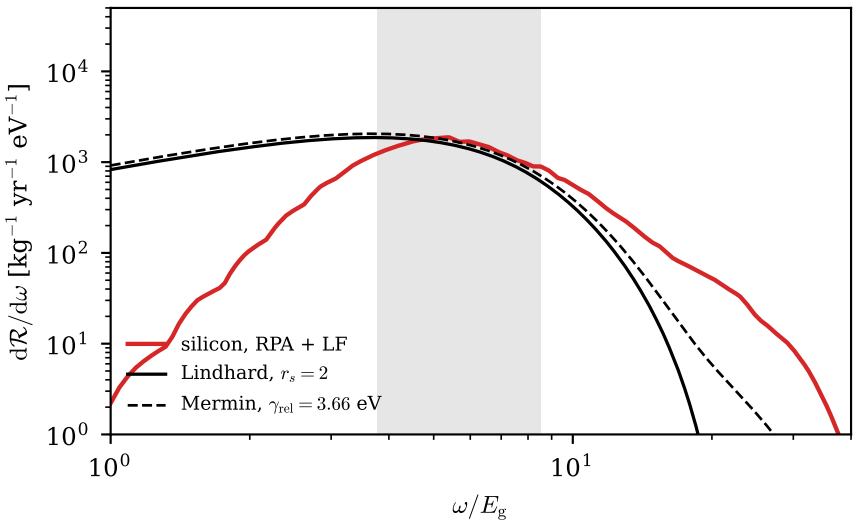


Figure 10.7: Spectrum of silicon at $m_\chi = 100\,\mathrm{MeV}$, heavy mediator, $\bar\sigma_e = 10^{-37}\,\mathrm{cm}^2$, $\rho_\chi = 0.3\,\mathrm{GeV\,cm}^{-3}$ and the standard halo, against ω/E_g with $E_\mathrm{g} = 1.11\,\mathrm{eV}$, the step-model gap that accompanies the tables. Red: the random-phase loss function of silicon with local fields (the tables of Fig. 10.5). Black: the Lindhard loss function of the electron gas at the valence density, $r_s = 2$; dashed, the Mermin function with $\gamma_\mathrm{rel} = 3.66\,\mathrm{eV}$, a fit parameter. All three are integrated over the same momenta, from 0.49 to 22.5 keV. Gray: from the upper absorption peak of silicon, 4.2 eV, to $k_\mathrm{F} v_\mathrm{max} = 9.5\,\mathrm{eV}$, a band chosen for this figure. There the Lindhard spectrum is between 0.70 and 1.48 times the crystal one.

Below the band the gas has no gap and overestimates the rate, by 24 at 2 eV; above it the gas has no electrons faster than v_F and loses the high-energy tail, and the integrated agreement at this mass comes from the two errors cancelling.

10.3.2 Plasmon pole and residue

Below q_c the loss function of the gas is the particle–hole continuum plus a line, Eq. (4.61), whose weight is π divided by the slope of ε_1 at the zero. At small q the slope follows from the high-frequency expansion of Sect. 4.5 kept to order q^2 [13, Eq. (5)],

$$\varepsilon_1(q,\omega) \simeq 1 - \frac{\omega_p^2}{\omega^2}\Big(1 + \frac{3}{5}\,\frac{q^2 v_\mathrm{F}^2}{\omega^2}\Big).$$

Its zero is $\omega_\mathrm{pl}^2 = \omega_p^2 + \frac{3}{5} q^2 v_\mathrm{F}^2$, the first two terms of Eq. (4.60). With $x = \frac{3}{5} q^2 v_\mathrm{F}^2/\omega_\mathrm{pl}^2$ the zero condition reads $\omega_p^2(1+x) = \omega_\mathrm{pl}^2$, and differentiating,

$$\left.\frac{\partial\varepsilon_1}{\partial\omega}\right|_{\omega_\mathrm{pl}} = \frac{2\omega_p^2}{\omega_\mathrm{pl}^3}\,(1+2x) = \frac{2}{\omega_\mathrm{pl}}\,\frac{1+2x}{1+x},$$

$$\frac{\pi}{\partial\varepsilon_1/\partial\omega} = \frac{\pi\omega_\mathrm{pl}}{2}\,Z_L(q), \qquad Z_L(q) = \frac{1+x}{1+2x} = 1 - \frac{3}{5}\,\frac{q^2 v_\mathrm{F}^2}{\omega_p^2} + O(q^4).$$

The factor Z_L is the residue of the longitudinal photon propagator at the plasmon pole that Kozaczuk and Lin use for the plasmon's polarization vector [13, Eq. (7)]; here it is the weight of the line in L. The first moment of the line, $\omega_\mathrm{pl}\cdot(\pi\omega_\mathrm{pl}/2)Z_L = (\pi/2)\omega_p^2(1+x)(1-x)$, is the whole f-sum to order q^2: at small q the plasmon carries all of the oscillator strength, and the continuum none. Integrating Eq. (10.4) over the delta function gives

Plasmon production rate

$$\mathcal{R}_\mathrm{pl} = \frac{\rho_\chi\,\bar\sigma_e}{8\pi^2\alpha\,\rho_T\,m_\chi\,\mu_{\chi e}^2}\int_0^{q_\mathrm{c}} \mathrm{d}q\; q^3\,F_\mathrm{DM}(q)^2\,\eta\big(v_\mathrm{min}(q,\omega_\mathrm{pl}(q))\big)\,\frac{\pi\omega_\mathrm{pl}(q)}{2}\,Z_L(q), \qquad (10.10)$$

with v_min of Eq. (1.23), the splitting Δ_χ included. Beyond the small-q forms, ω_pl is the exact zero of ε_1 and $(\pi\omega_\mathrm{pl}/2)Z_L$ is replaced by $\pi/(\partial\varepsilon_1/\partial\omega)$ there.

The small-q residue, with Z_L expanded to $1 - \frac{3}{5}q^2v_\mathrm{F}^2/\omega_p^2$ as in the limit check below, agrees with the exact one of the Lindhard function to 10^{-4} at $q = 0.1\,k_\mathrm{F}$ and 8×10^{-3} at $0.3\,k_\mathrm{F}$, the size of the dropped q^4 terms. Near the cutoff the two part. The exact line carries 0.99 of the f-sum at $q_\mathrm{c}/2$, 0.77 at $0.9\,q_\mathrm{c}$, and nothing at q_c. At the continuum edge ω_+ the real part of the Lindhard function contains $(1-\nu_-^2)\ln|(\nu_-+1)/(\nu_--1)|$ with $\nu_-\to1$ (Eq. (4.56)), and

its derivative in ω diverges logarithmically. The slope at the zero therefore grows without bound, and the weight of the line goes to zero as the line touches the continuum. The weight it loses appears in the continuum just below it; this is Landau damping seen in the residue.

The cutoff itself depends on how the dispersion is treated, and the literature uses several values. The exact zero of $\varepsilon_{\mathrm{Lin}}$ meets the edge $\omega_+ = qv_{\mathrm{F}} + q^2/2m_e$ at $q_{\mathrm{c}} = 2.60\,\mathrm{keV}$ and $24.8\,\mathrm{eV}$ (Sect. 4.5). The quadratic dispersion, with the q^4 terms dropped, meets it at $2.24\,\mathrm{keV}$ and $20.6\,\mathrm{eV}$, a dispersion fixed at ω_p at $1.88\,\mathrm{keV}$, and the estimate ω_p/v_{F} is $2.38\,\mathrm{keV}$. The spread is large because the correction is not small near the cutoff: at $2.24\,\mathrm{keV}$, $\frac{3}{5}q^2v_{\mathrm{F}}^2/\omega_p^2 = 0.53$. All four values lie close to the zone scale $2\pi/a = 2.28\,\mathrm{keV}$ of silicon. The crystal shows the same onset (Fig. 10.9). In the silicon tables the loss peak follows the gas dispersion from $18.9\,\mathrm{eV}$ at $0.5\,\mathrm{keV}$ to $22.3\,\mathrm{eV}$ at $2.45\,\mathrm{keV}$, with a half-maximum width of about $5\,\mathrm{eV}$. Beyond that it broadens fast: at $3.5\,\mathrm{keV}$ the half maximum spans 17.7 to $32.2\,\mathrm{eV}$, and at $4.6\,\mathrm{keV}$ 14.9 to $44\,\mathrm{eV}$. The tables place the small-q peak $2.5\,\mathrm{eV}$ above the measured $16.4\,\mathrm{eV}$, because the scissor that fixes the gap also raises the interband energies that push the plasmon up (Worked example 4.10).

Kurinsky, Baxter, Kahn and Krnjaic wrote the plasmon rate of a fast component of dark matter with a light mediator as the optical line shape times a logarithm [14, Eqs. (13) and (14)]. It follows from Eq. (10.4) in two steps. A light mediator has $F_{\mathrm{DM}} = (q_0/q)^2$, so $q^3F_{\mathrm{DM}}^2 = q_0^4/q$. Replacing $L(q,\omega)$ by its optical value $L(0,\omega)$ for all q below a cutoff then gives

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{\rho_\chi\,\bar{\sigma}_e\,q_0^4}{8\pi^2\alpha\,\rho_T\,m_\chi\,\mu_{\chi e}^2}\,L(0,\omega)\int_0^{q_{\mathrm{c}}}\frac{\mathrm{d}q}{q}\,\eta(v_{\mathrm{min}}(q,\omega)),$$

which is Ref. [14, Eq. (14)] in the book's normalization. They set the cutoff at $2\pi/a \sim 5\,\mathrm{keV}$ with a the lattice spacing [14, Eq. (4)]. For silicon, with $a = 5.43\,\text{Å}$, $2\pi/a$ is $2.28\,\mathrm{keV}$; $5\,\mathrm{keV}$ corresponds to a length of $2.5\,\text{Å}$, close to the bond length. The error this makes is logarithmic, because the integrand is $1/q$. For a particle of speed $v = 0.1$ the plasmon opens at $q_{\mathrm{min}} = \omega_p/v = 0.17\,\mathrm{keV}$, and moving the cutoff from $q_{\mathrm{c}} = 2.60\,\mathrm{keV}$ to $5\,\mathrm{keV}$ raises the integral by $\ln(5/0.166)/\ln(2.60/0.166) = 1.24$, or 1.31 from $2.24\,\mathrm{keV}$. A measure growing like q would have given $(5/2.60)^2 = 3.7$. The constant optical line shape also ignores the weight the line loses near q_{c}, which removes a further 3 percent of the logarithmic integral. For a fast component the choice of cutoff therefore moves the optical form by 25 to 30 percent, not by a factor of several.

Limit Check 10.5. Plasmon production from the pole of $1/\varepsilon$

Below the cutoff the plasmon is the line $L \simeq (\pi\omega_{\mathrm{pl}}/2)Z_L\,\delta(\omega - \omega_{\mathrm{pl}}(q))$, with $\omega_{\mathrm{pl}}^2 \simeq \omega_p^2 + \frac{3}{5}q^2v_{\mathrm{F}}^2$ and $Z_L \simeq 1 - \frac{3}{5}q^2v_{\mathrm{F}}^2/\omega_p^2$ [13, Eqs. (5) and (7)]. Kurinsky et al. write the light-mediator rate with L taken at $q = 0$ up to the cutoff [14, Eqs. (4), (13) and (14)].

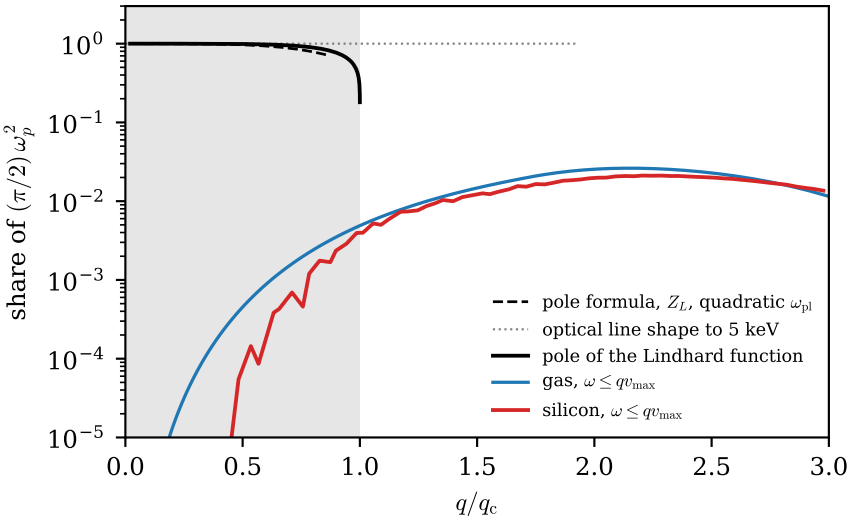


Figure 10.8: Share of the valence f-sum $(\pi/2)\omega_p^2$ of silicon, $\omega_p = 16.6\,\mathrm{eV}$, against q/q_c with $q_c = 2.60\,\mathrm{keV}$. Dashed: the first moment of the pole formula, $\omega_{pl}^2 Z_L/\omega_p^2$ with the quadratic dispersion, up to its own cutoff at 2.24 keV; dotted: the optical line shape continued to 5 keV. Solid black: the pole of the Lindhard function at $r_s = 2$, whose weight falls to zero at q_c. Blue: the weight of the Lindhard continuum below $\omega = qv_{max}$, the most a halo particle can deposit; red: the same for the random-phase loss function of silicon with local fields, from 0.49 keV. Gray: $q < q_c$, where the pole exists.

Near q_c the small-q residue misstates the line, whose weight goes to zero at the continuum edge, and beyond q_c there is no line at all. The weight within reach of the halo is never more than about 3 percent of the f-sum and lies in the pair continuum.

10.3.3 Plasmon threshold speed

A particle of speed v makes a plasmon of momentum q only if $v \geq v_{min}(q, \omega_{pl}(q)) = \omega_{pl}(q)/q + q/2m_\chi$ for elastic scattering, and Eq. (10.10) is nonzero only if this holds for some $q < q_c$. The phase velocity ω_{pl}/q is infinite at $q \to 0$ and falls as q grows. At q_c the line meets the continuum edge, where $\omega_{pl} = \omega_+$ and $\omega_+/q = v_F + q/2m_e$, which rises with q; in the Lindhard function the phase velocity reaches its smallest value at $0.989\,q_c$, within 10^{-3} of its value at q_c. For a heavy particle the slowest one that can make a plasmon therefore moves at

$$v_{th} \simeq \frac{\omega_+(q_c)}{q_c} = v_F + \frac{q_c}{2m_e}, \tag{10.11}$$

plus $q_c/2m_\chi$ for a finite mass. The threshold is set by the electrons, not by the dark matter. The plasmon is undamped only while its phase velocity exceeds ω_+/q, the largest ratio of energy to momentum that a single pair can have, so the slowest plasmon is as fast as the fastest pair. At the silicon valence density $v_{th} = 9.54 \times 10^{-3}$, or $2860\,\mathrm{km\,s^{-1}}$, and with the quadratic dispersion and its cutoff at 2.24 keV, 9.18×10^{-3} or $2750\,\mathrm{km\,s^{-1}}$. Either is 3.5 to 3.6 times the largest speed of the halo, $v_{max} = 794.5\,\mathrm{km\,s^{-1}}$ (Eq. (1.9)), and agrees with the estimate $v \gtrsim 10^{-2}$ of Kurinsky et al. [14, Eq. (7)]. Worked example 4.10 reached the same conclusion from the other side: the line $\omega = qv_{max}$ reaches ω_p only at $q = 6.27\,\mathrm{keV}$, more than twice q_c.

In the standard halo the speed distribution ends at v_{max}, and $\mathcal{R}_{pl}$ is zero at every mass. The suppression is not a power of a small ratio. Remove the escape cut and keep the Maxwellian

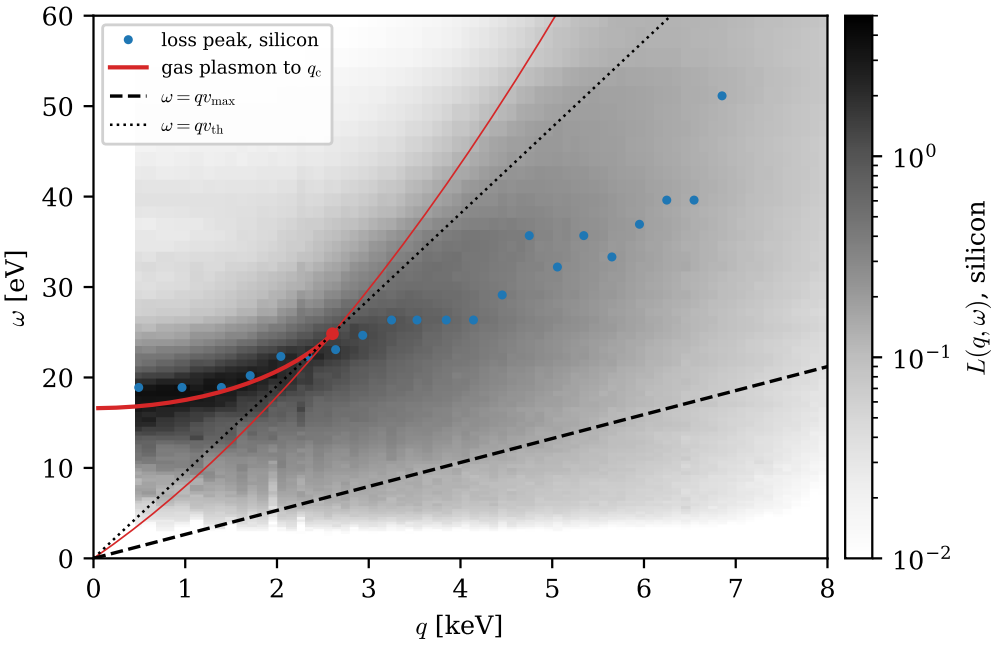


Figure 10.9: The silicon plasmon against the reach of halo dark matter. Gray scale: the random-phase loss function of silicon with local fields, $L(q,\omega)$, from the tables of Knapen, Kozaczuk and Lin [3], averaged over the directions of $\boldsymbol{q}$ in bins of 100 eV in $|\boldsymbol{q}|$ starting at 0.49 keV, logarithmic scale. Blue dots: the maximum of L in every third bin. Red: the undamped plasmon of the electron gas at the valence density, the exact zero of $\varepsilon_{\rm Lin}$, up to $q_c = 2.60\,\text{keV}$ (dot), and the continuum edge $\omega_+ = qv_F + q^2/2m_e$ (thin). Dashed: $\omega = qv_{max}$, the largest energy a halo particle deposits at momentum q. Dotted: $\omega = qv_{th}$ with $v_{th} = 9.54 \times 10^{-3}$ of Eq. (10.11), the slowest particle that reaches the plasmon of the gas. The tables place the small-q loss peak at 18.9 eV, against the measured 16.4 eV, because of the scissor shift that fixes their gap.

tail of Sect. 1.2: the fraction of particles faster than $2750\,\mathrm{km\,s^{-1}}$ in the laboratory is about 10^{-49}, and about 10^{-53} above $2860\,\mathrm{km\,s^{-1}}$. A real halo is not Maxwellian that far out, and these numbers only measure how far the plasmon lies from the bulk of the distribution, 10.5 to 11 times v_0 beyond the speed of the Earth.

Worked example 10.3: The plasmon rate of silicon at the highest halo speed

Take a particle at $v = v_{\max} = 2.650 \times 10^{-3}$, heavy enough that $q/2m_\chi$ is negligible; at 100 MeV it raises v_{th} by 0.14 percent. At momentum q it deposits at most $qv_{\max}$: 5.94 eV at the quadratic cutoff 2.24 keV, 6.90 eV at $q_{\mathrm{c}} = 2.60$ keV. The plasmon of the gas sits at 20.6 and 24.8 eV there. The distance between the plasmon and the reach, $\omega_{\mathrm{pl}}(q) - qv_{\max}$, is smallest at $q = 1.34$ keV, where it is 14.7 eV for the exact zero of the Lindhard function (the quadratic dispersion gives 14.5 eV at 1.72 keV). The particle is short of the plasmon by about ω_p, at every momentum where the plasmon exists, and Eq. (10.10) gives $\mathcal{R}_{\mathrm{pl}} = 0$. The silicon plasmon is not a sharp line: interband pairs give it a width of 3.7 eV already at $q \to 0$ (Worked example 4.10). The tables show what the particle meets instead. In the bin at 2.28 keV the loss peak sits at 22.3 eV, and at the reach of the particle, 6.0 eV, L is 6.9 percent of its peak value; the energies the particle can deposit hold 1.7×10^{-3} of the valence f-sum. At 2.57 keV the numbers are 8.5 percent and 4.0×10^{-3}. These are pairs on the low-energy side of the resonance, the same pairs that the rate of Sect. 10.2 already counts. They also carry little of that rate. At 100 MeV, momenta below 2.60 keV give 4.9×10^{-4} of the heavy-mediator rate (9.6×10^{-5} below 2.24 keV) and 2.5 percent of the light-mediator rate.
The answer is that the plasmon rate of silicon is zero for halo dark matter, not small: the pole needs $2750\,\mathrm{km\,s^{-1}}$ or more, and the halo stops at $794.5\,\mathrm{km\,s^{-1}}$. A loss-function rate that seems to contain "plasmon" events at 15 to 25 eV draws them from $q > 5.7$ keV, where Fig. 10.9 shows only the broad pair continuum.

The pole opens for a fast component of dark matter, for an exothermic splitting, or when a third body takes the momentum, and each case belongs to another section. A fast component, accelerated or boosted, can exceed v_{th}, and Eq. (10.10) applies with its speed distribution in η (Sect. 10.9). An exothermic transition lowers $v_{\min}$ through $\Delta_\chi < 0$ in Eq. (1.23), and a particle at $v_{\max}$ reaches the line once $|\Delta_\chi| \gtrsim 14.7$ eV, the closest approach of the worked example; Sect. 17.3 treats such down-scattering into collective modes. When the dark matter scatters off a nucleus, the recoiling ion can emit a plasmon whose momentum is small, as in bremsstrahlung. Kozaczuk and Lin computed this rate in the electron-gas picture and found it four to five orders of magnitude below the elastic nuclear rate for masses from 10 MeV to 1 GeV [13]; it is a process of the joint electron–lattice response of Chap. 12.

10.3.4 Plasmon decay into pairs and phonons

With the nuclei clamped, a plasmon of silicon decays into electron–hole pairs and nothing else. At the measured loss peak, 16.4 eV, vertical interband transitions are open, $\varepsilon_2 = 0.22$ where ε_1 vanishes, and the width of the loss peak follows from that value (Worked example 4.10). The branching ratio into pairs is one, and Sect. 8.7 showed that the screened pair weights of Eq. (8.68) already contain the plasmon peak, so that a plasmon is not a separate final state of silicon. Its energy is shared by the pairs, and the step model Eq. (1.43) turns 16.4 eV into five; the cascade that produces them is the subject of Sect. 18.1.

With the lattice free to move, part of the decay goes into pairs plus phonons. By the decomposition theorem Eq. (8.50), the absorptive part of the response at $(\boldsymbol{q}, \omega_{\mathrm{pl}})$ splits into a

zero-phonon sector S^0 and a phonon-dressed sector S^+. Since $L = \varepsilon_2/|\varepsilon|^2$ and the factor $|\varepsilon|^2$ is common to both sectors, the phonon branching of the plasmon is

$$B^+_{\mathrm{pl}}(\boldsymbol{q}) = \frac{S^+(\boldsymbol{q},\omega_{\mathrm{pl}})}{S^0(\boldsymbol{q},\omega_{\mathrm{pl}}) + S^+(\boldsymbol{q},\omega_{\mathrm{pl}})} = \frac{\varepsilon_2^+(\boldsymbol{q},\omega_{\mathrm{pl}})}{\varepsilon_2(\boldsymbol{q},\omega_{\mathrm{pl}})}, \tag{10.12}$$

the share of phonon-assisted absorption in ε_2 at the plasmon energy. At 16 eV direct transitions are allowed. A phonon-assisted transition needs one electron–phonon vertex besides the density vertex, and its amplitude carries the vertex over an energy denominator. With optical-phonon vertices of a few tenths of an eV and denominators of a few eV, Sect. 6.5 found the phonon-assisted absorption to be about 10^{-2} of the direct one per coupled mode. At the plasmon energy the direct transitions are strong, and the same estimate puts B^+_{pl} between 10^{-3} and 10^{-2}; it has not been computed. The phonons counted by Eq. (10.12) always come with a pair. A decay into phonons alone would have to pass 16 eV to modes of at most 64 meV, some 260 of them, with no electronic final state, and it is not part of this estimate. Kurinsky et al. needed plasmons that decay mostly into phonons to explain low-energy excess events [14]; the estimate here gives a phonon share of at most a percent.

Sect. 10.4 turns from the number of the electrons to their spin and their current.

10.4 Spin and current couplings

Dark matter with an anapole moment couples to electrons partly through $\mathcal{O}_8 = \boldsymbol{S}_\chi \cdot \boldsymbol{v}^\perp$, the dark-matter spin times the relative velocity of the two particles (Sect. 8.1). For an electron at rest the relative velocity is that of the dark matter, typically 10^{-3}, and the rate is the density rate of Sect. 10.2 times $|\bar{\boldsymbol{v}}_{\mathrm{T}}|^2/4$, with $\bar{\boldsymbol{v}}_{\mathrm{T}}$ the part of the dark matter's mean velocity across $\boldsymbol{q}$. A recast of silicon and superconducting-nanowire data uses exactly this form, with the loss function as the only material input [15, Eq. (6)]. The valence electrons of silicon, however, are not at rest. Taken as a free gas at their density they move at $v_{\mathrm{F}} = 7.0 \times 10^{-3}$, nine times the dispersion v_0 of the halo, and the velocity in $\mathcal{O}_8$ is the difference of the two. The electron's own motion couples through its current, a row of the response matrix that the density does not contain and that the medium screens differently. Every coupling beyond the density needs new responses of the crystal, screened in its own way, and an interference term between rows that has not been computed. We will list the responses for all the operators of Sect. 8.1 in one table, screen them row by row, and find that the ratio of the current-row rate to the density-row rate bounds the interference term.

10.4.1 The minimal response table

The rows of Eq. (8.3) for the electrons are the density n_e, the spin density $\boldsymbol{s}_e$, the current $\boldsymbol{j}_e$, the spin current $\mathcal{J}_e^{\alpha\beta}$ and the kinetic-energy density $\mathcal{T}_e$, three of them vectors and one a tensor. Counted by components they are 17 operators, and the response matrix has 289 elements at every $(\boldsymbol{q},\omega)$. Charge conservation, spin-rotation symmetry and the combined symmetry of inversion and time reversal reduce them to a few functions, and Appendix D.30 derives each step.

Charge conservation comes first. For a Hamiltonian with local potentials the current along $\boldsymbol{q}$, $j_{\mathrm{L}} = \hat{\boldsymbol{q}} \cdot \boldsymbol{j}_e$, has the matrix elements of $(\omega/q)\, n_e$ between eigenstates (Eq. (4.29)), and every element of the response matrix that contains it is a multiple of one that does not (Eq. (D.174)). On the dark-matter side the same holds for the velocity, $\hat{\boldsymbol{q}} \cdot \bar{\boldsymbol{v}}_\chi = (\omega + \Delta_\chi)/q$. For elastic dark matter the two longitudinal parts cancel (Eq. (8.5)), and only the transverse current survives, $j_{\mathrm{T}i} = \hat{\boldsymbol{e}}_i \cdot \boldsymbol{j}_e$ along two unit vectors $\hat{\boldsymbol{e}}_{1,2}$ perpendicular to $\boldsymbol{q}$. If the electron

Table 10.1: The minimal response table: the rows of the electronic response matrix that each non-relativistic operator of Table 8.1 needs, and their screening. The longitudinal current and spin current have been eliminated by continuity (elastic scattering, local potentials, electron spin conserved). Screening is for a cubic crystal without spin–orbit coupling or magnetic order. Singlet rows make spin-singlet electron–hole pairs and triplet rows spin-triplet pairs; the two blocks do not interfere. Notation: j_T, current across $\boldsymbol{q}$; $\mathcal{J}_\mathrm{T}$: spin current with its velocity index across $\boldsymbol{q}$. With inversion and time reversal, the singlet block on $\{n_e, j_{\mathrm{T}1}, j_{\mathrm{T}2}\}$ holds six real functions per $(\boldsymbol{q},\omega)$. For independent particles the triplet block is one quarter of it (Eq. (D.178)).

Operators	Rows	Block	Screening
$\mathcal{O}_1$, $\mathcal{O}_{11}$	n_e	singlet	head and local fields (Sect. 10.2)
$\mathcal{O}_5$, $\mathcal{O}_8$	n_e, j_T	singlet	n_e as above; j_T unscreened at the head on symmetry axes, local fields only (about 10 %)
$\mathcal{O}_2$	n_e, j_T, $\mathcal{T}_e$	singlet	n_e, j_T as above; $\mathcal{T}_e$ at the head and through local fields
$\mathcal{O}_4$, $\mathcal{O}_6$, $\mathcal{O}_9$, $\mathcal{O}_{10}$	$\boldsymbol{s}_e$	triplet	none; electron–hole attraction beyond independent particles
$\mathcal{O}_3$, $\mathcal{O}_7$, $\mathcal{O}_{12}$–$\mathcal{O}_{15}$	$\boldsymbol{s}_e$, $\mathcal{J}_\mathrm{T}$	triplet	none; as above

spin is conserved, the spin current obeys its own continuity equation, and only its transverse velocity components $\mathcal{J}_e^{\mathrm{T}i,\beta} = \hat{e}_i^\alpha \mathcal{J}_e^{\alpha\beta}$ survive.

Without spin–orbit coupling and without magnetic order the Hamiltonian is unchanged by a rotation of all spins, and the ground state of silicon is a spin singlet. The density, the current and the kinetic-energy density do not act on spin and make singlet pairs; the spin density and the spin current make triplets. The two never interfere, and the response matrix splits into a singlet block on $\{n_e, j_{\mathrm{T}1}, j_{\mathrm{T}2}, \mathcal{T}_e\}$ and a triplet block on $\{s_e^z, \mathcal{J}_e^{\mathrm{T}1,z}, \mathcal{J}_e^{\mathrm{T}2,z}\}$, the same for every spin component. For independent electrons the triplet block is a quarter of the singlet one (Eq. (D.178)): the spin vertex $\sigma^z/2$ has $(\pm\frac{1}{2})^2 = \frac{1}{4}$ where the density vertex has 1, and the spatial sums are the same. This is why the unscreened crystal form factor of Sect. 10.1 also gives the spin-dependent rate of $\mathcal{O}_4$ at that level, up to the spin traces of the two particles. Beyond independent particles the factor $\frac{1}{4}$ fails, for a reason that belongs to the next subsection.

Silicon and germanium are also symmetric under inversion combined with time reversal, which makes every element of both blocks real. The singlet block on $\{n_e, j_{\mathrm{T}1}, j_{\mathrm{T}2}\}$ is then a real symmetric 3×3 matrix: six real functions of $(\boldsymbol{q},\omega)$, namely S_{nn}, two density–current elements and three current–current elements. A complete table holds them twice, screened and for independent particles, and the independent-particle set fixes the triplet block at that level. That makes twelve functions. Operator $\mathcal{O}_2$ alone needs more: its velocity squared brings in $\mathcal{T}_e$, with four further elements in each version. On a two-, three- or fourfold axis of the cubic crystal, a rotation about $\hat{\boldsymbol{q}}$ leaves the density unchanged and turns the transverse current, and the density–current elements vanish.

Table 10.1 sorts the operators of Table 8.1 by block. Its first lines are all that a density coupling needs; the rest is what the effective operators add.

The published crystal responses are this table in other variables. Catena and collaborators write the general rate with a density vertex and a vector vertex built on $\mathrm{i}\nabla/m_e$, and five response functions $\overline{W}_1$ to $\overline{W}_5$ made from their products [16, Eqs. (36)–(42)]. Their vector vertex is the current vertex minus $\boldsymbol{q}/2m_e$ times the density vertex, and continuity fixes its

component along $\boldsymbol{q}$ (Eq. (D.179)). Two of the five responses, $\overline{W}_2$ and $\overline{W}_4$, are then the density response times known functions of ω and q. The combination $\overline{W}_3 - (m_e^2/q^2)\overline{W}_4$ is the transverse current response, $(2\Omega\omega/\pi)\,\mathrm{Tr}\,\bar{S}_{\boldsymbol{j}_\mathrm{T}\boldsymbol{j}_\mathrm{T}}$; and $\overline{W}_5$ and $\mathrm{Im}\,\overline{W}_2$ vanish exactly in silicon and germanium for any Hamiltonian, nonlocal pseudopotentials included, because inversion with time reversal makes the products of vertices real (Appendix D.30). Of the five, two are independent in these crystals, $\bar{S}_{nn}$ and $\mathrm{Tr}\,\bar{S}_{\boldsymbol{j}_\mathrm{T}\boldsymbol{j}_\mathrm{T}}$. The small nonzero $\overline{W}_5$ and $\mathrm{Im}\,\overline{W}_2$ of the published calculation, which its authors attribute to departures from axial symmetry, are therefore numerical. Trickle, Zhang and Zurek list the same operator content in their reduction of the effective operators to target operators [17, Table III].

The current and spin-current rows are computed from the same Kohn–Sham states as the density, with one extra step: a nonlocal pseudopotential adds a commutator to the current vertex (Eq. (E.46)), which in silicon changes the transverse current by about 10 % if it is left out. Appendix E.16 specifies the calculation of the twelve functions, and of the full independent-particle matrix to which other kernels can be applied.

10.4.2 Screening of non-density rows

A row is screened to the extent that it correlates with the charge (Sect. 8.4). In the reciprocal-lattice basis of Sect. 4.6 the rule takes the form of Eq. (D.180). The Coulomb interaction acts on the densities $n_e(\boldsymbol{q}+\boldsymbol{G})$, and any other row enters as an external vertex. Its screened response differs from the proper one by a term built from its proper correlation with those densities. We go down Table 10.1 row by row.

The density is screened at the head and through local fields, the subject of Sect. 10.2. In silicon the screened rate at 10 MeV is 2.3 times smaller than the independent-particle rate for a heavy mediator and 3.7 times for a light one. The longitudinal current is screened exactly as the density, since it is the density times ω/q.

The transverse current cannot drive the long-wavelength density when $\hat{\boldsymbol{q}}$ lies on a symmetry axis: the rotation that leaves $\boldsymbol{q}$ fixed turns j_T and leaves $n_e(\boldsymbol{q})$ alone, so their correlation at $\boldsymbol{G}=0$ vanishes. The head, which dominates the screening of the density at small q, then drops out, and the transverse current feels only the local fields, through its correlation with $n_e(\boldsymbol{q}+\boldsymbol{G})$ at $\boldsymbol{G}\neq 0$. Off the axes the head returns through the mixing of longitudinal and transverse components, but weakly. In the empirical-pseudopotential model of silicon used for the box of Sect. 10.1, with $\boldsymbol{q}$ along the generic direction $(5,3,1)$, the head changes $\mathrm{Tr}\,S_{\boldsymbol{j}_\mathrm{T}\boldsymbol{j}_\mathrm{T}}$ by less than 10^{-3} at q_0 and 5 eV, while it lowers S_{nn} by a factor 3.9. With local fields the current response falls by 15 % at q_0 and 5 eV, by 11 at 10 eV, and by 8 to 9 % at $2q_0$; along [100] at q_0 and 5 eV it falls by 4 %. The density response at the same point falls to 0.22 of its independent-particle value. The transverse current is, to this accuracy, unscreened at the head, with a local-field correction of about ten percent.

The kinetic-energy density is a scalar under the point group, like the density itself, and correlates with the charge already at $\boldsymbol{G}=0$. It is screened at the head. The same correlation gives the absorption of a scalar its photon-reducible part (Sect. 8.4.5), which Sect. 10.7 computes.

The spin density and the spin current are not screened at all. The Coulomb interaction acts on the total density of both spins and not on their difference, so a triplet row has no proper correlation with the charge, and Eq. (D.180) leaves it as it is. EXCEED-DM (Extended Calculation of Electronic Excitations for Direct detection of Dark Matter), a code for electronic dark-matter rates, applies the same rule: it screens the density row of a dark photon with $1/(\hat{\boldsymbol{q}}\cdot\varepsilon\cdot\hat{\boldsymbol{q}})$ and leaves the spin current of an axial-vector coupling unscreened [18, Eq. (23) and App. B]. What the triplet rows do feel, beyond independent particles, is the attraction between the electron and the hole. In the language of Sect. 4.9 the triplet pair keeps the

direct kernel and loses the exchange kernel, the part that carries the local fields. Sect. 10.5.5 computes what this does to the triplet rows. Spin–orbit coupling mixes the two blocks, by amounts that Sect. 10.6 estimates near the band edges.

The difference in screening has a clean form at long wavelength. There the transverse current response is fixed by the transverse dielectric function and the density response by the loss function (Eq. (D.181)),

$$\frac{\operatorname{Tr} S_{\boldsymbol{j}_\mathrm{T}\boldsymbol{j}_\mathrm{T}}(\boldsymbol{q},\omega)}{S_{nn}(\boldsymbol{q},\omega)} \;\to\; \frac{2\omega^2}{q^2}\,|\varepsilon(\omega)|^2 \qquad (q\to 0), \tag{10.13}$$

and for independent particles the same ratio is $2\omega^2/q^2$ without the factor $|\varepsilon|^2$. At 3 eV silicon has $|\varepsilon|^2 = 761$ (Sect. 10.7.1). At small momentum transfer the medium raises the weight of the current row against the density row by orders of magnitude. At keV momenta the head screening weakens, to a factor 4.2 at q_0 and 1.2 at 8 keV for the density row (Sect. 10.2.2), and the enhancement with it.

Published treatments sit on different rows of this list. Catena and collaborators compute all five responses for independent particles, without screening [16]. Hochberg and collaborators write every rate through three measurable responses, the density response, the spin susceptibility and their cross response, and assign $\mathcal{O}_1$, $\mathcal{O}_2$, $\mathcal{O}_5$, $\mathcal{O}_8$ and $\mathcal{O}_{11}$ to the density response alone [15, Eq. (5) and Sect. III]. For $\mathcal{O}_1$ and $\mathcal{O}_{11}$ this is exact, with the screening of the density row included, and for $\mathcal{O}_4$, $\mathcal{O}_6$, $\mathcal{O}_9$ and $\mathcal{O}_{10}$ their spin susceptibility is the triplet row, unscreened, as it should be.

The operators that contain $\boldsymbol{v}^\perp$ are treated as for an electron at rest. Their $\boldsymbol{v}_\perp$ is the dark-matter velocity across $\boldsymbol{q}$, the part $\bar{\boldsymbol{v}}_\mathrm{T}$ of the book's $\boldsymbol{v}^\perp$. The target part of $\boldsymbol{v}^\perp$ is absent: it brings in the current row for $\mathcal{O}_2$, $\mathcal{O}_5$ and $\mathcal{O}_8$, and the spin current for $\mathcal{O}_3$, $\mathcal{O}_7$ and $\mathcal{O}_{12}$–$\mathcal{O}_{15}$. For independent particles the missing spin-current block is a quarter of the missing current block, as the spin block is of the density block, and for $\mathcal{O}_7$ the two terms stand in the same ratio as for $\mathcal{O}_8$. EXCEED-DM makes the opposite choice for its axial-vector model: it keeps the electron-velocity term and drops the dark-matter-velocity terms, smaller by the ratio of the two speeds [18, Eqs. (B16), (B17)].

The sum-rule bounds of Sect. 10.2 have been extended to the same operators by Catena and Iglicki, who bound each susceptibility by its static value [12, Eqs. (4.16)–(4.21)]. Two of their steps fail for rows other than the density, as they note. A combination of responses that enters the rate of a non-density coupling need not be positive at each energy, and they find one that is negative at every energy, the current–current combination of anapole dark matter at 15 keV in silicon. And the static value of an unscreened row is not limited by the Coulomb kernel, which is what bounds the static step of the density by one; for the spin coupling they find it exceeded by a factor of about 20 at 500 eV. The structure of Table 10.1 avoids both problems: each diagonal block obeys its own sum rules, and the cross terms are bounded by positivity, as the next subsection shows.

Limit Check 10.6. Crystal responses of the effective operators

For $\mathcal{O}_8 = \boldsymbol{S}_\chi \cdot \boldsymbol{v}^\perp$ with the electron at rest, the rate per unit volume of momentum transfer is that of the density coupling, screened through the loss function, times $|\bar{\boldsymbol{v}}_\mathrm{T}|^2/4$: $\mathrm{d}\Gamma_8/\mathrm{d}^3q = (|\bar{\boldsymbol{v}}_\mathrm{T}|^2/4)\,\mathrm{d}\Gamma_1/\mathrm{d}^3q$ [15, Eq. (6)]. The crystal responses of Refs. [16, 18] keep the current row, without its screening.

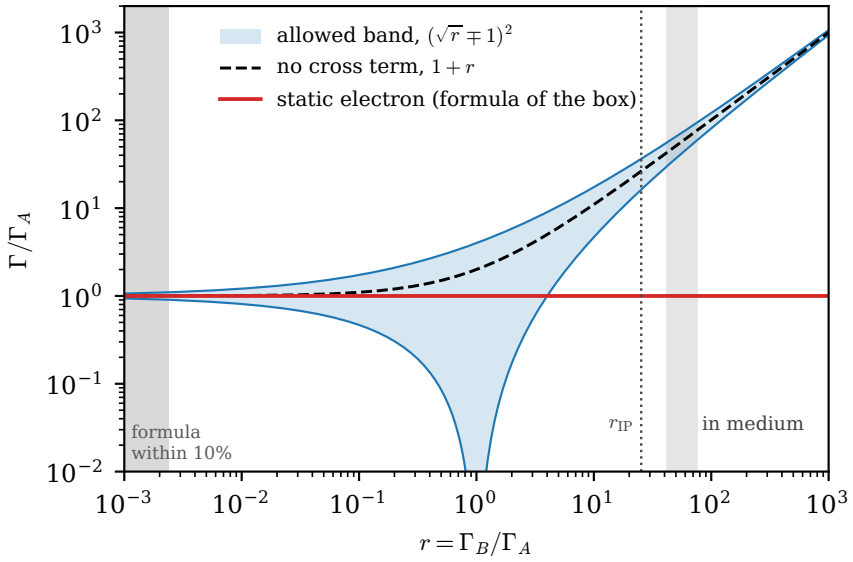


Figure 10.10: Rate of $\mathcal{O}_8 = \boldsymbol{S}_\chi \cdot \boldsymbol{v}^\perp$ (spin-$\frac{1}{2}$ dark matter, heavy mediator, elastic) relative to the static-electron formula of the box, Γ_A, against the ratio $r = \Gamma_B/\Gamma_A$ of the current-row rate to it. Blue: the band that positivity allows when the density–current cross term is not computed, $(\sqrt{r}\mp1)^2$ (Eq. (10.15)). Dashed: $1+r$, exact for responses averaged over the directions of $\boldsymbol{q}$, for which the cross term vanishes. Red: the formula. Gray at left: $r \le 2.4\times10^{-3}$, where every value in the band lies within 10 % of the formula, $(\sqrt{r}+1)^2 \le 1.1$. Dotted: silicon at $m_\chi = 10\,\mathrm{MeV}$ for independent particles, $r_{\mathrm{IP}} = 25.3$, from the empirical pseudopotential of Cohen and Bergstresser (model gap 0.82 eV, 10^3 $\boldsymbol{k}$ points, 36 conduction bands, bins of 0.1 eV, 26-point average over directions, threshold 1.11 eV, standard halo of Sect. 1.2). Gray at right: the same ratio with the density row screened, $r_{\mathrm{IP}}\times2.31\times(1\pm0.3)$. The factor 2.31 is the screening of the density rate in the random-phase tables with local fields of Sect. 10.2, and ±0.3 the range allowed for the local-field change of the current row.

The formula leaves out the electron's own transverse motion, which in silicon carries about 25 times the rate it keeps, so the true rate is at least 16 times the formula for independent particles and at least 29 times with the density row screened.

10.4.3 Interference bound on a mixed block

The rate of $\mathcal{O}_8$ shows how the rows combine. With the potentials of Eq. (8.4) on the density and the current, the contraction Eq. (8.15) has a density term, a current term and their interference. Continuity removes the parts along $\boldsymbol{q}$ on both sides (Eq. (D.175)). For elastic dark matter, and with the responses averaged over the directions of $\boldsymbol{q}$, the velocity integral is the one of Appendix A.8 with one change. The density term carries the dark matter's transverse velocity squared, $|\bar{\boldsymbol{v}}_{\mathrm{T}}|^2 = v^2 - v_{\mathrm{min}}^2$ on the energy shell, and its halo average is the second speed moment $\eta_2(v_{\mathrm{min}}) = 2\int_{v_{\mathrm{min}}}^{\infty} \mathrm{d}v'\, v'\eta(v')$ in place of η (Eq. (D.176)).

Rate of a velocity coupling

For $\mathcal{O}_8$ with coefficient $c_8^{(e)}(q)$ and spin-$\frac{1}{2}$ dark matter, the spectrum per unit target mass is

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{\rho_\chi}{16\pi^2 m_\chi \rho_T} \int \mathrm{d}q\, q\, |c_8^{(e)}(q)|^2 \Big[\underbrace{\eta_2(v_{\min})\, \bar{S}_{nn}(q,\omega)}_{\text{electron at rest}} + \underbrace{\eta(v_{\min})\ \mathrm{Tr}\, \bar{S}_{\boldsymbol{j}_\mathrm{T}\boldsymbol{j}_\mathrm{T}}(q,\omega)}_{\text{electron motion}} \Big], \tag{10.14}$$

with $\bar{S}$ averaged over the directions of $\boldsymbol{q}$, for which the density–current cross term vanishes. The first term is the rate of an electron at rest; the second is the electron's own transverse motion.

The prefactor follows from Eq. (8.16) with the kernel $\eta/(2q)$ per unit $\mathrm{d}^3q/(2\pi)^3$ of Eq. (A.42) and the factor $\frac{1}{4}$ of the spin trace. For $c_8^{(e)} = \tilde{V}$ and $\mathrm{Tr}\, \bar{S}_{\boldsymbol{j}_\mathrm{T}\boldsymbol{j}_\mathrm{T}} = 0$ it is the density rate with $\eta \to \eta_2/4$, the form of the box.

We can estimate the two terms before computing them. On the dark-matter side, the ratio η_2/η is the mean of $|\bar{\boldsymbol{v}}_\mathrm{T}|^2$ over the particles fast enough to deposit ω. It is 1.1×10^{-6} at $v_{\min} = 0$, 1.0×10^{-6} at 10^{-3} and 0.7×10^{-6} at 2×10^{-3}, and it vanishes at the kinematic edge, where the only particles left move along $\boldsymbol{q}$. On the target side, the ratio $\mathrm{Tr}\, \bar{S}_{\boldsymbol{j}_\mathrm{T}\boldsymbol{j}_\mathrm{T}}/\bar{S}_{nn}$, integrated over ω, is the mean transverse velocity squared of the electrons that make the pairs. In the empirical-pseudopotential model of silicon it is 3.2×10^{-5} at q_0 and 2.1×10^{-5} at $2q_0$, close to the free-gas value $\frac{2}{5}v_\mathrm{F}^2 = 2.0 \times 10^{-5}$. At small q it grows as $2\langle\omega^2\rangle/q^2$, the independent-particle form of Eq. (10.13): at $0.12\, q_0$ the model gives 4.1×10^{-4} against $2\langle\omega^2\rangle/q^2 = 4.4 \times 10^{-4}$. We expect the electron term to exceed the dark-matter term by a factor of about 20 to 30, and more near the kinematic edge.

Writing $\mathit{\Gamma}_A$ for the rate of the density term and $\mathit{\Gamma}_B$ for that of the current term, the interference bound of Sect. 8.3 limits the full rate whatever the cross term is.

Band on a mixed block

For any target, halo and binning,

$$\left(\sqrt{r} - 1\right)^2 \;\le\; \frac{\mathit{\Gamma}}{\mathit{\Gamma}_A} \;\le\; \left(\sqrt{r} + 1\right)^2, \qquad r = \frac{\mathit{\Gamma}_B}{\mathit{\Gamma}_A}, \tag{10.15}$$

and with responses averaged over directions $\mathit{\Gamma} = \mathit{\Gamma}_A + \mathit{\Gamma}_B$. For independent particles r is written r_IP.

One ratio decides how much the cross term can matter. The half-width of the band relative to $1 + r$ is $2\sqrt{r}/(1 + r)$, below 10 % only for $r < 2.5 \times 10^{-3}$ or $r > 4.0 \times 10^2$. For r above 4 the lower edge exceeds one: the rate of an electron at rest is then an underestimate whatever the phase of the cross term. Positivity also says when the band is reached. Both edges need the density and current rows to be proportional on every final state the rate samples, and in an unpolarized crystal they are far from it. In the model of silicon, along the generic direction $(5, 3, 1)$, the ω-integrated cross block $|\boldsymbol{S}_{n\boldsymbol{j}_\mathrm{T}}|$ stays below 0.012 of $(S_{nn}\, \mathrm{Tr}\, S_{\boldsymbol{j}_\mathrm{T}\boldsymbol{j}_\mathrm{T}})^{1/2}$ up to $2.4\, q_0$, and on the two-, three- and fourfold axes it vanishes by symmetry. A crystal at fixed orientation keeps a cross term through the anisotropy of its response, the same anisotropy that makes the daily modulation of Sect. 11.7.

In the medium the two terms of Eq. (10.14) are screened differently. For a heavy mediator the density rate weighted with η_2 loses a factor 2.2 at 30 MeV, 2.3 at 10 MeV and 9.5 at 1 MeV in the tables of Sect. 10.2, while the current row changes only through local fields. The ratio r grows accordingly (Eq. (D.183)), and with it the lower edge of the band, as long as the

screening of the density rate outweighs the local-field change of the current.

Worked example 10.4: The band for a velocity coupling in silicon

Take $\mathcal{O}_8$ with a heavy mediator, $m_\chi = 10\,\mathrm{MeV}$, the halo of Sect. 1.2 and the threshold $E_\mathrm{g} = 1.11\,\mathrm{eV}$ of the tables of Sect. 10.2. The empirical-pseudopotential model, averaged over directions with a 26-point rule, provides the two integrals of Eq. (10.14) for independent particles. Their ratio is

$$r_\mathrm{IP} = \frac{\int \mathrm{d}\omega \int \mathrm{d}q\, q\, \eta\ \mathrm{Tr}\, \bar{S}^0_{\boldsymbol{\jmath}_\mathrm{T}\boldsymbol{\jmath}_\mathrm{T}}}{\int \mathrm{d}\omega \int \mathrm{d}q\, q\, \eta_2\, \bar{S}^0_{nn}} = 25.3.$$

The model's gap is 0.82 eV; shifting all its pair energies up to the gap of 1.11 eV changes r_IP by 2 %. With $\sqrt{r_\mathrm{IP}} = 5.03$, Eq. (10.15) puts the rate between $(5.03-1)^2 = 16.2$ and $(5.03+1)^2 = 36.4$ times the rate of an electron at rest, and the direction-averaged value is $1 + r_\mathrm{IP} = 26.3$. The rate of the density row alone is low by more than an order of magnitude.

Now screen. With the random-phase tables with local fields of Sect. 10.2, the density rate weighted with η_2 falls by a factor 2.31 at this mass, the same as with η (2.30). The local-field change of the current row is not known for silicon; the model finds -4 to -15 percent at q_0 and $2q_0$, and we allow ± 30 percent. Then $r_\mathrm{med} = 25.3 \times 2.31 \times (1 \pm 0.3)$ lies between 41 and 76, the lower edge of the band between 29 and 60, and the direction-averaged rate is 42 to 77 times the static-electron formula with the same screened density row. The two models in this estimate differ: the current comes from the empirical pseudopotential, the screening of the density from a Kohn–Sham calculation. At q_0 and 5 eV the model itself lowers S_{nn} to 0.22 of its unscreened value, in line with the tables.

The ratio grows toward small masses, since the dark matter's transverse velocity vanishes at the kinematic edge: $r_\mathrm{IP} = 22$ at 30 MeV, 33 at 5 MeV, 56 at 3 MeV and about 150 at 2 MeV. The model's plane waves end at $2.8\,q_0$, while 27 % of the heavy-mediator density rate of silicon at 10 MeV comes from $q > 2q_0$ (Worked example 10.1). At large q the ratio of the two responses tends to the mean transverse velocity squared of the valence electrons, which the model puts at 2×10^{-5} to 3×10^{-5} and which the core-region oscillations missing from pseudo-wavefunctions can only raise. The value of r_IP above is, if anything, low.

The local-field screening of the transverse current in silicon is not known. Estimates put it at 10 to 30 % (Appendix E.16), the model above at 4 to 15 %, both of the sign that lowers the current; Appendix E.16 sets out its calculation from the mixed blocks of Eq. (D.180), and with it the in-medium ratio r_med. That ratio is open, and with it the factor between the true rate of $\mathcal{O}_8$, $\mathcal{O}_5$ or $\mathcal{O}_2$, or of the spin operators that carry $\boldsymbol{v}^\perp$, and the static-electron formula. Eq. (10.15) bounds the factor from below, by 16 at 10 MeV for independent particles and by about 29 with the density row screened, but it does not fix it. Until it is computed, limits set on these operators with the density response alone are conservative by at least an order of magnitude, and they need a material input beyond the density response.

10.5 Excitons at finite momentum

At a momentum transfer equal to the valley momentum $\boldsymbol{k}_\mathrm{c}$, the lowest final state of silicon is not a free electron and hole at the indirect gap of 1.17 eV. It is the pair bound by its own attraction, 14.7 meV lower (Sect. 4.9), and for the density coupling of Sect. 10.2 this attraction is the first correction beyond the random-phase approximation with local fields. Whether a

given coupling can reach the bound pair is a question of symmetry. Besides adding a line below the threshold, the attraction raises the probability of finding the electron at the hole in every pair of the continuum. The continuum just above $E_{\min}(\boldsymbol{q})$ grows at the expense of higher energies, and the change is largest in the first eV above the gap, where most of the rate of light dark matter lies. Sect. 10.2 wrote the screened rate as an integral of the loss function, and Sect. 4.9 gave the loss function of the Bethe–Salpeter equation. Put together at finite $\boldsymbol{q}$, they give the rate as a sum over exciton lines. We will find where its onset lies, which lines a density coupling can reach at the zone boundary of silicon, and how much a resonant-only solution of the equation costs. The spin-triplet excitons, which only spin couplings reach, close the section.

10.5.1 The exciton rate

The rate of Eq. (10.4) holds for any loss function, so the electron–hole attraction enters it through L alone. With the eigenpairs $E_\lambda(\boldsymbol{q})$ and $A^\lambda_{cv\boldsymbol{k}}(\boldsymbol{q})$ of Eq. (4.104) the structure factor is Eq. (4.112), a sum of lines, one per exciton. Each line carries the squared matrix element of the density between the ground state and the singlet exciton λ. The singlet pair state $cv\boldsymbol{k}$ has amplitude $\sqrt{g_{\deg}}\,\rho_{cv}(\boldsymbol{k},\boldsymbol{q})$ to be created by $n_e(\boldsymbol{q})$, and the exciton is the superposition $\sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}|cv\boldsymbol{k}\rangle$, so $\langle\lambda|n_e(\boldsymbol{q})|0\rangle = \sqrt{g_{\deg}}\,\rho_{\lambda 0}(\boldsymbol{q})$ with

$$\rho_{\lambda 0}(\boldsymbol{q}) = \sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}(\boldsymbol{q})^*\,\rho_{cv}(\boldsymbol{k},\boldsymbol{q}), \tag{10.16}$$

the exciton form factor. It is the coherent sum of Sect. 4.9: the vertices of all the transitions in the exciton add with the amplitudes of its envelope. At $T=0$ and $\omega>0$, $L=(e^2/2q^2)S$ by Eq. (4.45), and Eq. (4.112) gives $L=(4\pi^2\alpha g_{\deg}/Vq^2)\sum_\lambda|\rho_{\lambda 0}|^2\delta(\omega-E_\lambda)/|\varepsilon_{\mathrm{mac}}|^2$. In Eq. (10.4) the factor $4\pi^2\alpha$ cancels against $8\pi^2\alpha$ and q^3/q^2 leaves one power of q.

Exciton rate at finite momentum

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{\rho_\chi\,\bar\sigma_e\,g_{\deg}}{2\rho_T\,m_\chi\,\mu^2_{\chi e}}\int \mathrm{d}q\;q\,\eta(v_{\min}(q,\omega))\,F_{\mathrm{DM}}(q)^2\left\langle\frac{1}{\mathscr{V}}\sum_\lambda\frac{|\rho_{\lambda 0}(\boldsymbol{q})|^2}{|\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega)|^2}\,\delta(\omega-E_\lambda(\boldsymbol{q}))\right\rangle_{\hat{\boldsymbol{q}}}, \tag{10.17}$$

with $\langle\cdots\rangle_{\hat{\boldsymbol{q}}}$ the average over the directions of $\boldsymbol{q}$, the exciton energies and form factors of Eqs. (4.104) and (10.16) at the momentum transfer $\boldsymbol{q}$ (reduced to the first zone in the kernels, Sect. 4.9), and $g_{\deg}=2$.

Without the kernels each λ is one transition, $\rho_{\lambda 0}=\rho_{cv}$, $E_\lambda=\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}}-\varepsilon_{v\boldsymbol{k}}$, and Eq. (10.17) is the random-phase rate of Sect. 10.2; without $1/|\varepsilon_{\mathrm{mac}}|^2$ as well, it is the crystal form-factor rate of Sect. 10.1. With the attraction on, the energies $E_\lambda(\boldsymbol{q})$ start below $E_{\min}(\boldsymbol{q})$, so the support of the spectrum moves down, and the weights $|\rho_{\lambda 0}|^2$ are coherent sums, so the spectrum changes shape at fixed total weight. The f-sum rule of Sect. 4.3 holds for the exact L at every $\boldsymbol{q}$, and what the attraction adds at low energy it removes higher up. For a rate this redistribution is not neutral: $\eta(v_{\min})$ falls with ω at fixed q, and a particle that cannot deposit more than $\frac{1}{2}m_\chi v^2_{\max}$ sees only the low side. The line form with $|\varepsilon_{\mathrm{mac}}|^2$ in the denominator holds when no single line changes $\varepsilon_{\mathrm{mac}}$ near its own energy, which at keV momentum transfers, where the Coulomb factor e^2/q^2 is small, is always the case (Sect. 4.9).

The equation is solved as in Sect. 4.9, now at the reduced momentum $\boldsymbol{q}_r$ of each $\boldsymbol{q}$. It needs pairs from $(v,\boldsymbol{k})$ to $(c,\boldsymbol{k}+\boldsymbol{q}_r)$ on a grid of crystal momenta, quasiparticle energies or a scissor on Kohn–Sham bands, the screened interaction W^{scr} for all grid differences, and the head of the exchange kernel at the unreduced $\boldsymbol{G}_q$ left out (Appendix E.6). The first such

calculations for dark matter were made for GaAs and NaI, with the scissor set to the measured gaps, a 6^3 grid, 10 conduction bands and momenta along [100] up to 14.7 keV [19]. In both materials the loss function with the attraction lies above the random-phase one with local fields at low energy, at all the momenta shown. In GaAs the rate changes little. In NaI, whose lowest optical exciton lies about 0.3 eV below the gap (an absorption peak at 5.6 eV against a gap of 5.9 eV [19]), some seventy times deeper than the 4.2 meV of GaAs (Sect. 4.9), the rate at energies near the gap rises enough to improve the projected reach above a few MeV by close to an order of magnitude [19]. No such calculation exists yet for silicon or germanium. Appendix E.15 specifies one, with the coupling between pair creation and annihilation that the published calculations drop, and with the bound lines added from the effective-mass model below, since a 6^3 or even a 24^3 grid cannot resolve an exciton of radius 4.2 nm.

What the attraction does at a dark-matter momentum transfer can be seen in silicon itself, in the model of Fig. 4.17 carried to finite $\boldsymbol{q}$ (Fig. 10.11). At $|\boldsymbol{q}| = 0.91$ keV along [100], halfway to the valley, the lowest pair of the model costs 1.69 eV. The attraction changes the proper absorption ε_2 the way it did at $q \to 0$. The main peak moves from 5.3 eV to 4.5 eV and grows from 16.5 to 27.5. The weight below 3 eV grows by a factor of 2.2 over the random-phase value with local fields. The loss function follows with the screening of the same model, whose static value rises from 6.56 to 7.80. Between the onset and 3.5 eV the attraction raises L by a nearly constant 27 to 29 percent, by 49 percent at 4 eV, and by about 30 percent above. The rise at high energies is partly an artifact: four conduction bands and the Tamm–Dancoff form do not hold the f-sum fixed, and the first moment of ε_2 below 10 eV falls by 12 percent. The grid, ten points per direction, resolves no bound line, and the lines of Sect. 10.5.2 lie below the gray region of the figure. For a rate the near-edge factor is what counts. Dark matter below a few MeV reaches only the first two eV above $E_{\min}(\boldsymbol{q})$, and there the attraction adds about a quarter to the screened loss function, on top of whatever bound lines the grid misses.

10.5.2 Elliott onset and bound exciton lines

Near the bottom of the pair continuum at a given $\boldsymbol{q}$ the exciton form factor has a closed form, and with it the onset of Eq. (10.17). At fixed $\boldsymbol{q}$ the pair energy $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}}$ has its minimum $E_{\min}(\boldsymbol{q})$ of Eq. (3.57) at some $\boldsymbol{k}_0(\boldsymbol{q})$, and near it the pair energy rises quadratically in $\boldsymbol{k} - \boldsymbol{k}_0$, with a reduced mass set by the curvatures of the two bands at $\boldsymbol{k}_0$ and $\boldsymbol{k}_0 + \boldsymbol{q}$. This is the situation of Eq. (4.108), with the extremum of the pair energy in place of the band extrema. The relative motion obeys the hydrogenic equation Eq. (4.106), and the bound states sit at $E_{\min}(\boldsymbol{q}) - E_x^b/n^2$. In silicon at $\boldsymbol{q} = \boldsymbol{k}_{\mathrm{c}}$ the pair minimum joins the two band extrema and E_x^b is the measured 14.7 meV. Away from the valley, $\boldsymbol{k}_0$ moves off the band extrema, the reduced mass changes, and E_x^b with it (Eq. (E.90)); we keep the symbols E_x^b, a_x and μ_x for the values at the $\boldsymbol{q}$ in question.

When $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q})$ is smooth and nonzero at $\boldsymbol{k}_0$, the Elliott form Eq. (4.111) applies: $|\rho_{\lambda 0}|^2 = \mathcal{V}|F_\lambda(0)|^2|\rho_{cv}(\boldsymbol{k}_0, \boldsymbol{q})|^2$. For the bound s states, $|F_{ns}(0)|^2 = 1/(\pi a_x^3 n^3)$. For the unbound states of relative momentum κ, normalized in the volume $\mathcal{V}$, $\mathcal{V}|F_\kappa(0)|^2$ is its free value, one, times the factor by which the attraction raises the probability of finding the electron at the hole. That factor is the Fermi factor Eq. (2.25) of an electron leaving a charge, with the coupling $\alpha/\varepsilon_\infty$ and the mass μ_x in place of α and m_e, so that $\xi_{\mathrm{C}} = 1/(\kappa a_x)$. With $\kappa^2 = 2\mu_x(\omega - E_{\min})$

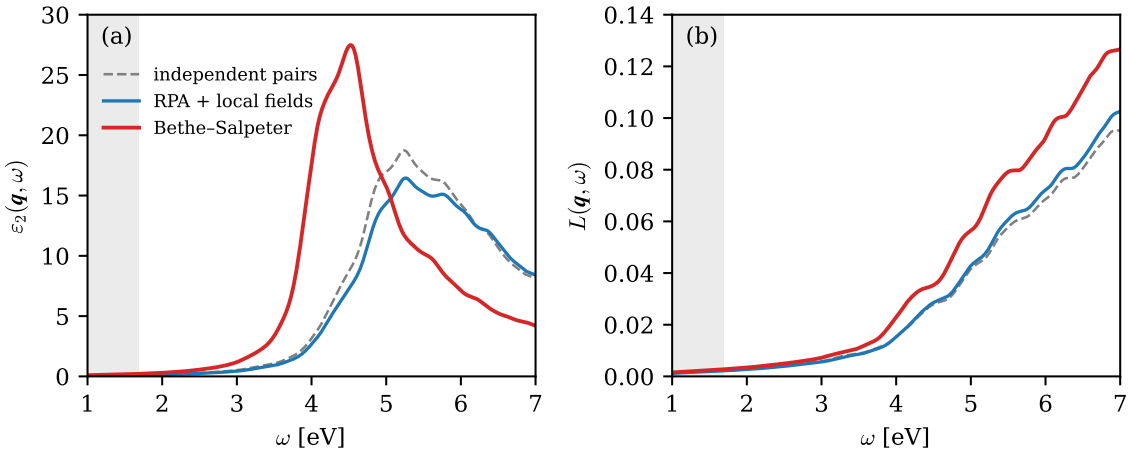


Figure 10.11: The electron–hole attraction at finite momentum transfer in silicon, in a model calculation: $|\boldsymbol{q}| = 0.91\,\mathrm{keV}$ $(0.4 \times 2\pi/a)$ along [100]. States from the empirical pseudopotential of Cohen and Bergstresser [20], fitted to measured transition energies (model indirect gap 0.82 eV; no scissor); 4 valence and 4 conduction bands on a shifted 10^3 grid over the full zone (16,000 pairs); the 59 shortest reciprocal vectors in both kernels; direct kernel with the model dielectric function of Cappellini and coworkers [6] with $\varepsilon_\infty = 11.7$; Tamm–Dancoff resonant block with both time orderings, Eq. (4.109); Lorentzian broadening 0.15 eV. On this grid the independent-pair spectrum agrees with a 15^3 grid within 8 percent below 6 eV. Gray: below the lowest pair of the grid, 1.69 eV, where the curves are broadening tails; the bound lines are not resolved. Dashed: independent pairs (with the head screening only, in panel b); blue: random-phase approximation with local fields; red: Bethe–Salpeter equation. (a) Proper absorption $\varepsilon_2(\boldsymbol{q},\omega)$. (b) Loss function. The fitted bands already contain part of the excitonic shift, which the attraction then counts again (Sect. 4.9).

and $2\mu_x E_x^b a_x^2 = 1$ from Eq. (4.107), the line sum in Eq. (10.17) becomes

$$\frac{1}{\mathcal{V}}\sum_\lambda |\rho_{\lambda 0}|^2 \delta(\omega - E_\lambda) = |\rho_{cv}(\boldsymbol{k}_0, \boldsymbol{q})|^2 \Big[\sum_{n\geq 1} \frac{\delta(\omega - E_{\min} + E_x^b/n^2)}{\pi a_x^3 n^3} + \Theta(\omega - E_{\min}) \frac{(2\mu_x)^{3/2}(\omega - E_{\min})^{1/2}}{4\pi^2} \frac{2\pi\xi_{\mathrm{C}}}{1 - \mathrm{e}^{-2\pi\xi_{\mathrm{C}}}} \Big], \tag{10.18}$$

with $\xi_{\mathrm{C}} = [E_x^b/(\omega - E_{\min})]^{1/2}$. The density of free pairs, $\int \mathrm{d}^3\kappa\,(2\pi)^{-3}\delta(\omega - E_{\min} - \kappa^2/2\mu_x)$, supplies the square root. The two parts meet smoothly. The lines crowd toward $E_{\min}$ with spacing $2E_x^b/n^3$, so their weight per unit energy tends to $1/(2\pi a_x^3 E_x^b)$, and as $\omega \to E_{\min}^+$ the continuum tends to $(2\mu_x)^{3/2}(E_x^b)^{1/2}/2\pi$, which is the same number. The edge of the excitonic spectrum at fixed $\boldsymbol{q}$ is therefore a series of lines followed by a finite step, where independent pairs start from zero as $(\omega - E_{\min})^{1/2}$.

How much weight the lines take is a short calculation. Their total is $\sum_n 1/(\pi a_x^3 n^3) = \zeta(3)/(\pi a_x^3)$. The free continuum without the Fermi factor holds $(2\mu_x)^{3/2}\epsilon^{3/2}/6\pi^2$ between $E_{\min}$ and $E_{\min} + \epsilon$. With $(2\mu_x)^{3/2} = 1/(a_x^3 (E_x^b)^{3/2})$ the two are equal at $\epsilon = (6\pi\zeta(3))^{2/3} E_x^b = 8.0\, E_x^b$: the bound lines together carry as much weight as the free pairs within eight binding energies of the edge would have carried. The attraction does not stop there. Above the edge the Fermi factor is 6.3 at $\omega - E_{\min} = E_x^b$, falls to 2.3 at $10\, E_x^b$, and is still $1 + \pi\xi_{\mathrm{C}} + \ldots$ far above, a slow approach to one. The hydrogenic form of that tail is not to be trusted beyond a few tenths of an eV, where the bands stop being parabolic and the pair samples the screening at shorter distances, and the sign is all it fixes there. The spectrum it describes comes from the region near the pair minimum only. In silicon at $\boldsymbol{q} = \boldsymbol{k}_{\mathrm{c}}$ the heavy and light holes at the valence maximum each add their own copy of Eq. (10.18), with their own masses and vertices.

Worked example 10.5: The exciton line of silicon below the pair onset

Take a momentum transfer close to $\boldsymbol{k}_{\mathrm{c}}$ in silicon, where $E_{\min}$ is the gap. With the low-temperature gap of 1.17 eV and $E_x^b = 14.7$ meV, the $1s$ line sits at 1.155 eV and the $2s$ line at 1.17 eV $- E_x^b/4 = 1.166$ eV; the radius is $a_x = 4.2$ nm (Sect. 4.9). For a vertex that is smooth at the band extrema, Eq. (10.18) gives the $1s$ line the weight $|\rho_{cv}|^2/(\pi a_x^3) = |\rho_{cv}|^2 \times 4.3 \times 10^{-6}\ \text{Å}^{-3}$ per unit volume. It is $1/\zeta(3) = 0.83$ of all the lines, and it equals the free-pair weight within $(6\pi)^{2/3} E_x^b = 7.1\, E_x^b = 0.10$ eV of the edge. A particle that deposits ω between 1.155 eV and 1.17 eV at this $\boldsymbol{q}$ has no free pair to make and makes the exciton; one that deposits 1.27 eV finds the free-pair density raised by the factor 2.6. For the rate integrated over ω, a line 14.7 meV below the edge changes little; for the threshold mass it changes even less. The kinematic floor of Eq. (1.25), $2E_{\mathrm{g}}/v_{\max}^2 = 0.333$ MeV, becomes $2(E_{\mathrm{g}} - E_x^b)/v_{\max}^2 = 0.329$ MeV, a shift of 1.3 percent, and Sect. 3.3 showed that this floor is not reached at all, since the momentum at which a particle that light deposits most energy falls short of the valley. Where the zero-phonon edge of silicon lies, and that its first line is weak, are the subjects of Sects. 13.1 and 13.9. The line is weak because on the valley axis the vertex at the extrema vanishes, and the $1s$ line is dark.

10.5.3 Selection rules at the zone boundary

Spin decides first which excitons a dark-matter coupling can make in silicon. The ground state is a singlet, the density does not act on spin, and $n_e(\boldsymbol{q})$ creates only singlet excitons, the ones of Eq. (4.104) with $g_{\mathrm{deg}} K^{\mathrm{x}}$. The triplets, in which the electron and hole spins combine

to one, are reached by the spin density and the spin current of Table 10.1, and Sect. 10.5.5 treats them. Crystal momentum decides next. The lowest exciton of silicon has the electron in a valley and the hole at Γ, so it carries $\boldsymbol{k}_\mathrm{c}$ up to a reciprocal-lattice vector, and a coupling reaches it at first order only through $\boldsymbol{q} = \boldsymbol{k}_\mathrm{c} + \boldsymbol{G}$. Light, with $q \approx 0$, needs a phonon (Sect. 6.5). Dark matter of a few hundred keV and more carries $|\boldsymbol{k}_\mathrm{c}| = 1.94\,\mathrm{keV}$ and needs none.

The symmetry of the states at the extrema decides last, and it is what makes the first line of silicon weak. Take $\boldsymbol{q} = \boldsymbol{k}_\mathrm{c}$ along [100], the x axis, with no Umklapp vector. Consider the operations of the cubic group that leave this axis fixed: the rotations by multiples of 90° about it and the mirrors that contain it. The conduction state at the valley minimum is invariant under all of them, and so is the operator $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$. The three valence states at the maximum transform like the products yz, zx and xy (in the notation of the band structure, $\Gamma_{25'}$). Under the quarter turn about x, which takes $y \to z$ and $z \to -y$, yz changes sign. The pair (zx, xy) transforms like the two components (z, y) of a vector across the axis and contains no invariant combination. A matrix element $\langle c, \boldsymbol{k}_\mathrm{c}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|v, \Gamma\rangle$ between an invariant state, through an invariant operator, to a state with no invariant part vanishes. Half of these operations carry the fractional translation of the diamond structure (Appendix C.17), which multiplies the Bloch states and the operator by phases; for $\boldsymbol{q} = \boldsymbol{k}_\mathrm{c}$ exactly, the phase of the operator, $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_2}$, cancels that of the conduction state, $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}_\mathrm{c}\cdot\boldsymbol{\tau}_2}$, and the argument stands. In germanium the valleys lie along [111]. The rotations by 120° about that axis and the mirrors containing it leave the combination $yz + zx + xy$ invariant, and the vertex at the extrema does not vanish.

A vertex that vanishes at the extrema grows linearly away from them: $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q}) \simeq (\boldsymbol{k} - \boldsymbol{k}_0)\cdot\boldsymbol{g}$ with $\boldsymbol{g} = \nabla_{\boldsymbol{k}}\rho_{cv}$, and $\boldsymbol{q}$ itself off the axis adds a term proportional to its distance $\delta q_\perp$ from it. Transitions with a vertex of this kind are called second class [21]. Their exciton form factor picks out the gradient of the envelope where the electron meets the hole. From $A_{\boldsymbol{k}} = \mathscr{V}^{-1/2}\int \mathrm{d}^3r\, F(\boldsymbol{r})\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$, $\sum_{\boldsymbol{k}} A_{\boldsymbol{k}}\,\boldsymbol{k}\cdot\boldsymbol{g} = -\mathrm{i}\mathscr{V}^{1/2}\,\boldsymbol{g}\cdot\nabla F(0)$, which vanishes for s states and is nonzero for the p state oriented along $\boldsymbol{g}$. Its envelope is $R_{n1}(r)\,(3/4\pi)^{1/2}\cos\theta$, with θ measured from $\boldsymbol{g}$, and the hydrogen radial function starts as $R_{n1} \to \frac{2}{3}[(n^2-1)/n^5]^{1/2} r/a_x^{5/2}$; for $n = 2$ this is $R_{21} = r\,\mathrm{e}^{-r/2a_x}/(2\sqrt{6}\,a_x^{5/2})$. So $|\boldsymbol{g}\cdot\nabla F_{np}(0)|^2 = |\boldsymbol{g}|^2(n^2-1)/(3\pi a_x^5 n^5)$, and in place of the s lines of Eq. (10.18),

$$\frac{1}{\mathscr{V}}\sum_{\lambda\in\mathrm{lines}} |\rho_{\lambda 0}|^2\delta(\omega - E_\lambda) = |\nabla_{\boldsymbol{k}}\rho_{cv}|^2 \sum_{n\geq 2} \frac{n^2-1}{3\pi a_x^5 n^5}\,\delta(\omega - E_\mathrm{min} + E_x^b/n^2) \qquad \text{(second class)}. \tag{10.19}$$

There is no $1s$ line. The first line is $2p$, at $E_x^b/4 = 3.7\,\mathrm{meV}$ below the continuum. Against an allowed $1s$ line with a vertex of size $|\rho|$ it is weaker by $(|\nabla_{\boldsymbol{k}}\rho_{cv}|/|\rho|)^2/(32a_x^2)$, a factor of about 10^{-3} for a vertex that changes by order one across a tenth of the zone ($|\nabla_{\boldsymbol{k}}\rho_{cv}|/|\rho| \approx 9\,\text{Å}$). The continuum changes the same way. Its vertex is $|\boldsymbol{g}\cdot\boldsymbol{\kappa}|^2$, or $|\boldsymbol{g}|^2\kappa^2/3$ averaged over directions, so the free pairs start as $(\omega - E_\mathrm{min})^{3/2}$. The p-wave Coulomb function at the origin carries the extra factor $1 + \xi_\mathrm{C}^2$, from $|\Gamma(2 + \mathrm{i}\xi_\mathrm{C})|^2 = (1+\xi_\mathrm{C}^2)|\Gamma(1+\mathrm{i}\xi_\mathrm{C})|^2$, which keeps the step at the edge finite. Off the axis the $1s$ line returns with a weight proportional to $\delta q_\perp^2$, and in a rate averaged over directions it enters at the energy of the $1s$ line with a weight that grows from zero.

The rule holds only for $\boldsymbol{q} = \boldsymbol{k}_\mathrm{c}$ without an Umklapp vector. With $\boldsymbol{q} = \boldsymbol{k}_\mathrm{c} + \boldsymbol{G}$ the phases of the shifted operations no longer cancel, $\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_2}$ can be -1, and transitions that are forbidden at $\boldsymbol{G} = 0$ become allowed. Which classes open, at what momenta and with what strength, follows from the glide symmetry of Appendix C.17 and is worked out in Sect. 13.9, where it decides the zero-phonon edge of silicon. Spin–orbit coupling lifts the zero on the axis as well, by an amount too small to matter (Sect. 13.9).

10.5.4 Time orderings and the coupling block

A dark-matter calculation with excitons must give the loss function, not only the absorption ε_2, and the loss function needs the real part of the dielectric function as well. Here the way the equation is solved matters more than one might expect. Eq. (4.109) has two terms for each exciton, a resonant pole at $+E_\lambda$ and an antiresonant one at $-E_\lambda$. The second is the same process with the opposite time ordering, the pair annihilated before it is created, and it is the mirror image that causality requires: with both terms, $\varepsilon_{\rm mac}(\boldsymbol{q},\omega)$ is analytic in the upper half plane and obeys the Kramers–Kronig relations of Appendix C.5. A program that solves the resonant block alone, as the Tamm–Dancoff approximation does, can still form the response with both orderings; this is the retarded Tamm–Dancoff form, and Eq. (4.109) is written that way. If instead the antiresonant term is dropped altogether, the response is resonant-only.

At $\omega > 0$ the antiresonant term has no imaginary part, so the two forms have the same ε_2. Their real parts, however, differ. With $\varepsilon_2 = \pi C \sum_\lambda w_\lambda \delta(\omega - E_\lambda)$ for the weights w_λ and the positive constant C of Eq. (4.109), the two orderings give $\varepsilon_1 - 1 = C\sum_\lambda w_\lambda[1/(E_\lambda-\omega) + 1/(E_\lambda+\omega)]$, and the resonant term alone gives $\varepsilon_1^{\rm res} - 1 = C\sum_\lambda w_\lambda/(E_\lambda-\omega)$. Written with ε_2,

$$\varepsilon_1 - 1 = \frac{2}{\pi}\,\mathrm{P}\int_0^\infty \mathrm{d}\omega'\,\frac{\omega'\varepsilon_2(\omega')}{\omega'^2-\omega^2}, \qquad \varepsilon_1^{\rm res} - 1 = \frac{1}{\pi}\,\mathrm{P}\int_0^\infty \mathrm{d}\omega'\,\frac{\varepsilon_2(\omega')}{\omega'-\omega}, \tag{10.20}$$

the first being Eq. (C.41). For a single oscillator at E_0 the ratio is

$$\frac{\varepsilon_1^{\rm res}-1}{\varepsilon_1-1} = \frac{E_0+\omega}{2E_0}, \qquad \frac{L^{\rm res}}{L} = \frac{\varepsilon_1^2+\varepsilon_2^2}{(\varepsilon_1^{\rm res})^2+\varepsilon_2^2} \simeq \Big(\frac{\varepsilon_1}{\varepsilon_1^{\rm res}}\Big)^2 \quad (\varepsilon_2 \ll \varepsilon_1). \tag{10.21}$$

At $\omega \to 0$ the resonant-only form keeps half of $\varepsilon_1 - 1$. In silicon at small q, with $\varepsilon_1 = 11.7$, it overstates the loss function by $[2\varepsilon_1/(\varepsilon_1+1)]^2 = 3.4$ as $\omega \to 0$ (at 1.3 eV and the smallest momentum of the tables below, 0.49 keV, the factor is 2.2), and wherever ε_1 is large and ε_2 small, as below and just above the gap, the error is typically a factor of two. It vanishes as $q \to \infty$, where $\varepsilon_1 \to 1$ and screening disappears.

The silicon tables of Sect. 10.2 put numbers on this. From their ε_2 with local fields, Eq. (10.20) reproduces the tabulated static $\varepsilon(\boldsymbol{q},0)$ to within one percent at every q, which shows that the tables hold enough of the spectrum for the transform. At $|\boldsymbol{q}| = 1.39\,\mathrm{keV}$ ($0.71\,\text{Å}^{-1}$) and $\omega = 1.3\,\mathrm{eV}$, below the pair continuum at that momentum, $\varepsilon_1 = 4.45$ and $\varepsilon_1^{\rm res} = 3.05$, and the resonant-only form overstates L by a factor of 2.12. A single oscillator that carries the whole valence f-sum and reproduces the static value 4.31, $E_0 = \omega_p^{\rm val}/[\varepsilon(\boldsymbol{q},0)-1]^{1/2} = 9.1\,\mathrm{eV}$ with $\omega_p^{\rm val} = 16.6\,\mathrm{eV}$, gives 2.23 from Eq. (10.21), and the model dielectric function of Sect. 10.2 gives 2.25. These momenta and energies are those at which the phonon-assisted pairs of Sect. 13.3 are made, and that is where the factor of two would land. It falls to 1.53 at $q_0 = 3.73\,\mathrm{keV}$ and to 1.07 at 8 keV, and it is within 15 percent of one above 6.7 keV. A resonant-only evaluation is harmless for heavy mediators at masses where large q dominates, and wrong by a factor of two for light mediators and light dark matter.

The retarded form has no such error, but it still drops something: the coupling block of Eq. (D.65), which mixes the creation of one pair with the annihilation of another. Its effect follows from a single pair of transition energy Δ coupled to itself by a kernel K in both blocks. With coupling the poles are at $\pm E_{\rm cp}$, $E_{\rm cp}^2 = (\Delta+K)^2 - K^2 = \Delta(\Delta+2K)$, so $E_{\rm cp} = \Delta + K - K^2/2\Delta + \ldots$; without it they are at $\pm(\Delta+K)$. The energy is wrong only at second order, the statement of Sect. 4.9. The weight is not. With $R^0(0) = -2|\rho|^2/\Delta$ for the free pair, the static response of the coupled pair is $R(0) = R^0/(1 - KR^0/|\rho|^2) = -2|\rho|^2/(\Delta+2K)$. The retarded resonant block gives $R^{\rm TD}(0) = -2|\rho|^2/(\Delta+K)$, and

$$\frac{R^{\rm TD}(0)}{R(0)} = \frac{\Delta+2K}{\Delta+K} = 1 + \frac{K}{\Delta+K}, \tag{10.22}$$

an error of first order in the kernel, here written for an exchange-type K that enters the coupling block with the same sign as the resonant one. Which kernel remains in the coupling block depends on how the long-range part is handled. The head of the exchange kernel, the term at $\boldsymbol{G}_q$, is the macroscopic field, and it turns the proper response into the screened one (Sect. 4.9). If it is kept in the Bethe–Salpeter Hamiltonian and the coupling block is dropped, the screening itself is computed in the Tamm–Dancoff approximation, and for the screened L the error is set by $v_C(q)$, large at small q. The safe arrangement is the one of Appendix E.15: the Tamm–Dancoff equation, without the head, for the proper ε_2 with its lines, and the screening from the full response, with both orderings, through $L = \varepsilon_2/|\varepsilon_{\text{mac}}|^2$ and Eq. (10.20). The coupling block then still holds the local-field exchange and the small interband part of the direct kernel. Since the local fields change ε_2 of silicon by about ten percent (Sect. 4.6), the remaining error is of that size, about 10 % on the proper ε_2. In the model of Fig. 10.11, where the coupling block holds the local-field exchange alone, the full random-phase response with local fields differs from its retarded Tamm–Dancoff form by 2.6 percent of the peak of ε_2. The 10 % is an estimate; the full coupled calculation, which also restores the f-sum rule that the Tamm–Dancoff form strains, is part of Appendix E.15.

The static screened interaction is the other approximation in the equation. It holds when the binding energies and the spread of transition energies that matter are small against the frequencies at which ε^{-1} changes, the plasma frequency of 16.6 eV in silicon (Sect. 4.9). For bound lines of 14.7 meV and pairs within a few eV of the gap this holds, and the dynamical screening of the interaction, left for the full calculation of Appendix E.15, is not expected to rival the two errors above.

Limit Check 10.7. Bethe–Salpeter exciton rate

With the loss function of the Bethe–Salpeter equation in the Tamm–Dancoff approximation, the rate is Eq. (10.4) with $\bar{L} = \text{Im}[-1/\varepsilon_{\text{mac}}^{\text{BSE}}(q,\omega)]$, or Eq. (10.17) line by line [19, Eqs. (1) and (2)].

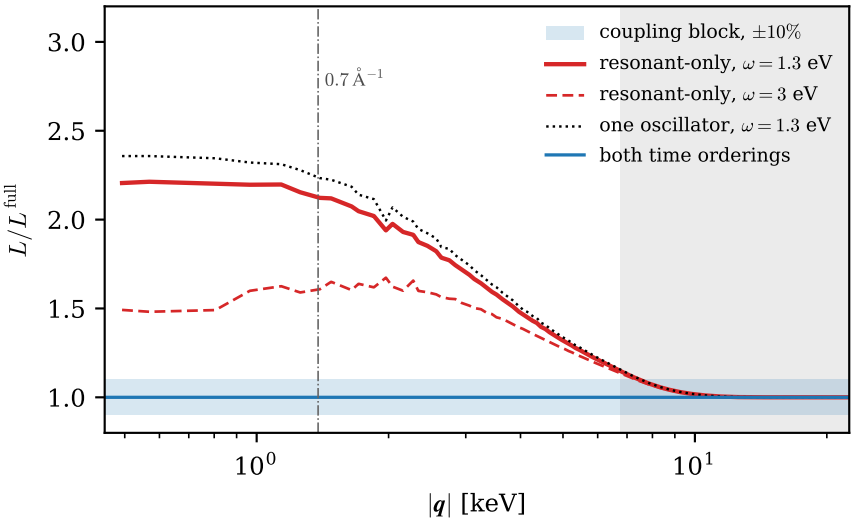


Figure 10.12: Loss function of silicon in two evaluations of the same response, divided by the evaluation with both time orderings, against $|\boldsymbol{q}|$. Red: resonant-only real part, Eq. (10.20), at $\omega = 1.3$ eV (solid) and 3 eV (dashed), from ε_2 of the random-phase tables with local fields of Sect. 10.2 (threshold 1.11 eV, the step-model gap that accompanies them; directions averaged in bins of 100 eV). Dotted: a single oscillator with the valence

f-sum and the tabulated static $\varepsilon(q,0)$, Eq. (10.21), at 1.3 eV. Blue line and band: both time orderings, with the estimated $\pm 10\,\%$ error of the coupling block, Eq. (10.22). Dash-dotted: $|\boldsymbol{q}| = 0.7\,\text{Å}^{-1}$ (1.39 keV), the momentum of the numbers quoted in the text. Gray: momenta at which the resonant-only form is within 15 %, $|\boldsymbol{q}| \geq 6.7$ keV.

Below about 7 keV, where screening is strong, a resonant-only real part overstates the loss function by up to a factor of two or more; with both time orderings the remaining error is that of the coupling block.

10.5.5 Triplet excitons

A coupling to the electron spin, such as $\mathcal{O}_4$ of Table 8.1, creates the electron–hole pair with its spins combined to one, a triplet, and the triplet rows of Table 10.1 are its responses. Without interaction they are a quarter of the density response (Eq. (D.178)), and they are not screened (Sect. 10.4). With the interaction the triplet pair has its own Bethe–Salpeter equation. The exchange kernel comes from a pair that annihilates and is re-created, and a spin-one pair cannot annihilate into the spinless density that the Coulomb line couples to, so its contribution cancels between the two spin projections. The direct kernel, the attraction between the electron and the hole, acts in every spin state alike. The triplet Hamiltonian is therefore

$$H_{\mathrm{T}}^{\mathrm{exc}} = \Delta - K^{\mathrm{d}}, \tag{10.23}$$

Eq. (4.104) with $g_{\mathrm{deg}} K^{\mathrm{x}}$ removed, and the spin structure factor follows from its eigenpairs as the density one does, with the vertex $\sigma^z/2$ in place of one and no $1/|\varepsilon_{\mathrm{mac}}|^2$.

The bound lines of the triplet sit where the singlet ones do, up to the electron–hole exchange splitting. That splitting is the short-range exchange of a pair whose electron and hole meet in one cell with probability of order $(a/a_x)^3$, 2×10^{-3} in silicon, so it is a fraction of order 10^{-3} of the binding, well below a meV. And the Wannier analysis above, which used only K^{d}, applies unchanged: the triplet edge has the same lines, the same Fermi factor and the same selection rules of point symmetry as the singlet one, with the spin vertex in place of the density vertex. What separates the two channels is the rest of K^{x}, the local fields, which raise the singlet transitions and move weight up (Sect. 4.6) and which the triplet lacks. In the model of Fig. 10.11, the triplet spectrum below 3 eV carries 2.2 times the independent-pair weight, and between 2.5 and 4 eV five times, while above 4 eV the two agree within 4 percent. The attraction alone moves the triplet weight down more than the singlet's, since no local fields push back.

What is not known is the triplet response at keV momentum transfers beyond independent particles: the ladder with the screened attraction over the whole spectrum, together with the exchange–correlation kernel of the spin channel, which is not the one of the density. As an estimate, near the edge the spin rows of Table 10.1 inherit the excitonic enhancement of the singlet ones, without their local fields and without screening. Appendix E.16 specifies the calculation, as the Bethe–Salpeter run of Appendix E.15 with the exchange kernel switched off; its control is the limit without interaction, which must return a quarter of the independent-particle density response.

10.5.6 Open problems

For silicon and germanium no finite-momentum Bethe–Salpeter calculation exists, with or without the coupling block, and no calculation resolves the bound lines of an indirect gap. The lines need grids around the pair minimum far finer than a uniform grid can afford, and are therefore to be added from Eqs. (10.18) and (10.19) with computed masses, vertices and

vertex gradients. The full coupled calculation would settle the 10 % of Eq. (10.22), test the factor of two of Eq. (10.21) directly, and check the f-sum rule to a percent (Appendix E.15). The triplet response beyond independent particles is open as stated above.

So is the response of crystals in which the exciton is not large. In NaI the lowest excitons are compact, of the size of an ion pair, and the hole localizes on two anions while the lattice relaxes around it: the exciton becomes self-trapped. The Wannier picture fails, and the line that dark matter creates is dressed by the phonons that the relaxation emits. Take linear coupling, a harmonic lattice and a vertex independent of the lattice coordinates. The line at E_λ then becomes a Poisson series of replicas spaced by a mean phonon energy $\bar{\omega}$, with the weight $\mathrm{e}^{-S_{\mathrm{HR}}} S_{\mathrm{HR}}^n/n!$ in the nth (Eq. (E.91)), where the Huang–Rhys factor S_{HR} is the mean number of phonons emitted, the relaxation energy divided by $\bar{\omega}$. For NaI none of S_{HR}, $\bar{\omega}$ or the form factor $\rho_{\lambda 0}(\boldsymbol{q})$ is known at keV momenta, and Appendix E.15 specifies their calculation. The localized (Frenkel) exciton itself, the limit of a pair confined to one site, is made precise in Sect. 16.4. The spread between the random-phase and the excitonic rate near the gap, the factor that a resonant-only evaluation can introduce, and the coupling error enter the many-body uncertainty of Sect. 8.8, collected for this chapter in Sect. 10.10.

10.6 Core and relativistic states

A germanium atom has 32 electrons, and the band structure of Sect. 10.1 follows four of them. Ten more sit in the $3d$ shell, 29.4 eV below the valence-band maximum, close enough for light dark matter to excite them, and the deeper shells follow at 122 eV and beyond. Sect. 3.5 found where each shell becomes kinematically accessible and why the momenta it needs are large: an electron that absorbs ω must receive at least $q = \omega/v_{\max}$, 11 keV for germanium $3d$ and 38 keV for silicon $2p$. It left two questions for the rate. At which momentum transfer does a core shell actually take over from the valence electrons, given that both are probed far out in the tails of their momentum distributions? And how much do the relativistic corrections of order $(Z\alpha)^2$, which grow with the depth of the shell, change the response there? We answer both for silicon and germanium, and find the three hypotheses under which the response of the core electrons separates from that of the valence electrons: a filled shell, a flat band and orthogonal site orbitals.

10.6.1 Core shells within kinematic reach

The structure factor of a crystal with core shells is the sum of the valence term of Eq. (10.1) and one term per core shell. The core states of atom κ in the cell are the Bloch sums $\psi_{\kappa nlm,\boldsymbol{k}}$ of Eq. (E.48), with the energy $-E_{\mathrm{core}}$ measured from the valence maximum, and their vertex with a conduction state is $\rho_{c,\kappa nlm}(\boldsymbol{k},\boldsymbol{q}) = \langle \psi_{c,\boldsymbol{k}+\boldsymbol{q}} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \psi_{\kappa nlm,\boldsymbol{k}} \rangle$. Following the steps of Eq. (10.1) with these states in place of the valence bands,

$$S^0_{\mathrm{core}}(\boldsymbol{q},\omega) = 4\pi \sum_{\kappa nlm} \sum_{c} \int_{\mathrm{BZ}} \frac{\mathrm{d}^3k}{(2\pi)^3} \, |\rho_{c,\kappa nlm}(\boldsymbol{k},\boldsymbol{q})|^2 \, \delta\big(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - E_{\mathrm{core},\kappa nl}\big). \tag{10.24}$$

Both states in the vertex must be all-electron states. The core orbital is all-electron by construction. The conduction state must carry its core-region oscillations, which the reconstruction Eq. (3.74) restores, because the overlap with a core orbital is taken inside the sphere where a pseudo-wavefunction is wrong (Appendix E.2). A semicore shell kept in the valence of the pseudopotential, such as germanium $3d$, enters Eq. (10.1) instead, with reconstructed vertices on both sides. Griffin and coworkers computed the germanium $3d$ contribution both ways and found that the atomic orbitals reproduce the reconstructed crystal states up to the largest momentum the crystal calculation resolves, about 45 keV [2, Sects. 2.2 and 3.3].

How many core electrons can dark matter reach at a given q? A momentum transfer q deposits at most $qv_{\max}$, whatever the mass (Eq. (1.23)), so the relevant measure is the part of the structure factor below that energy. Per atom and per shell we define

$$N_{nl}^{\rm acc}(q) = \int_{E_{\rm g}+E_{\rm core}}^{qv_{\max}} \mathrm{d}\omega \; S_{nl}^{\rm at}(q,\omega), \tag{10.25}$$

where $S_{nl}^{\rm at}(q,\omega)$ is the response of one atom's shell, normalized so that its integral over all ω is the occupation N_{nl} when q is large (closure, Sect. 4.3). The structure factor of Eq. (10.24) is $(2\pi/\Omega)\sum_\kappa S_{\kappa nl}^{\rm at}$, as the next subsection shows, so $N_{nl}^{\rm acc}$ counts the electrons of a shell, per atom, that the halo can excite at momentum transfer q. With plane-wave final states the shell response follows from the ionization form factor of Sect. 2.2. The number of final states per unit ejected energy is $|f_{\rm ion}|^2/4E_{\rm ej}$, and Eq. (2.24), with the radial momentum orbital $\phi_{nl}(k)$ of Sect. 3.5 normalized to $\int k^2\phi_{nl}^2\,\mathrm{d}k = 1$, gives

$$S_{nl}^{\rm at}(q,\omega) = \frac{N_{nl}}{2}\,\frac{m_e}{q}\int_{|p-q|}^{p+q} k\,\phi_{nl}(k)^2\,\mathrm{d}k, \qquad p = [2m_e(\omega - E_{\rm core})]^{1/2}, \tag{10.26}$$

for a full shell, $N_{nl} = 2(2l+1)$, and for the spherical average of a partly filled valence shell. The final kinetic energy is counted from the valence maximum, as for the free final states of Griffin and coworkers [2, Eq. (3.21)]. Integrating over ω, with $\mathrm{d}\omega = p\,\mathrm{d}p/m_e$ and the order of integration exchanged, each k collects $\int_{|k-q|}^{k+q} p\,\mathrm{d}p = 2kq$, and the integral is $N_{nl}\int k^2\phi_{nl}^2\,\mathrm{d}k = N_{nl}$: closure holds.

Eq. (10.26) explains when a shell takes over. Near its opening the ejected momentum p is much smaller than q, the integral is $2pq\,\phi_{nl}(q)^2$, and

$$N_{nl}^{\rm acc}(q) \simeq N_{nl}\,m_e\,\phi_{nl}(q)^2\int_{E_{\rm core}}^{qv_{\max}} p\,\mathrm{d}\omega = \frac{2\sqrt{2}}{3}\,N_{nl}\,m_e^{3/2}\,\phi_{nl}(q)^2\,(qv_{\max} - E_{\rm core})^{3/2},$$

where the gap has been dropped from the lower limit. Two factors decide. The first is the number of electrons of the shell per unit volume of momentum at $p = q$, $N_{nl}\phi_{nl}(q)^2$, in which a core shell beats the valence above about 5 keV (Fig. 3.11). The second is the phase space $(qv_{\max} - E_{\rm core})^{3/2}$, which is zero at the opening and approaches that of the valence, $(qv_{\max})^{3/2}$, only once $qv_{\max}$ is several times $E_{\rm core}$. A core shell therefore overtakes the valence a little above its opening, at the momentum where its larger density outweighs the reduced phase space.

Fig. 10.13 evaluates Eq. (10.25) for germanium. The core and semicore shells use the free-atom orbitals of Fig. 3.11. The valence uses the crystal: the independent-particle ε_2 of Knapen, Kozaczuk and Lin [3, Sect. III B], from which $S = (2q^2/e^2)\varepsilon_2$ (Sect. 10.2) and, per atom of the two-atom cell, $S^{\rm at} = q^2\Omega\varepsilon_2/8\pi^2\alpha$. That table holds only the four valence electrons, with the $3d$ shell frozen, on bands scissored to the measured gap at zero temperature. It ends at 22.5 keV, where its plane-wave basis ends too, and we use it up to 15 keV and Eq. (10.26) with the atomic $4s$ and $4p$ orbitals above. At the switch the two differ by 27 percent. Between 5 and 15 keV the atomic model and the crystal tables of silicon and germanium differ by up to a factor of six in either direction, since the low-energy corner $\omega \le qv_{\max}$ is where the band structure matters most. That is the accuracy of the valence curve beyond the table.

The figure answers the first question for germanium. Below 11.3 keV only the valence is reachable, and its count falls from 0.44 electrons per atom at 5 keV to 0.03 at 10 keV as the window $\omega \le qv_{\max}$ moves into the low-energy tail of a spectrum that is centered on $q^2/2m_e$. The $3d$ shell opens at 11.3 keV and overtakes the valence at 12.5 keV. By 20 keV it holds 0.036 reachable electrons per atom against 0.002 for the valence, and it keeps more than 90 percent of the total until the $3p$ and $3s$ shells, opening at 46 and 68 keV, take over near 69 keV. For

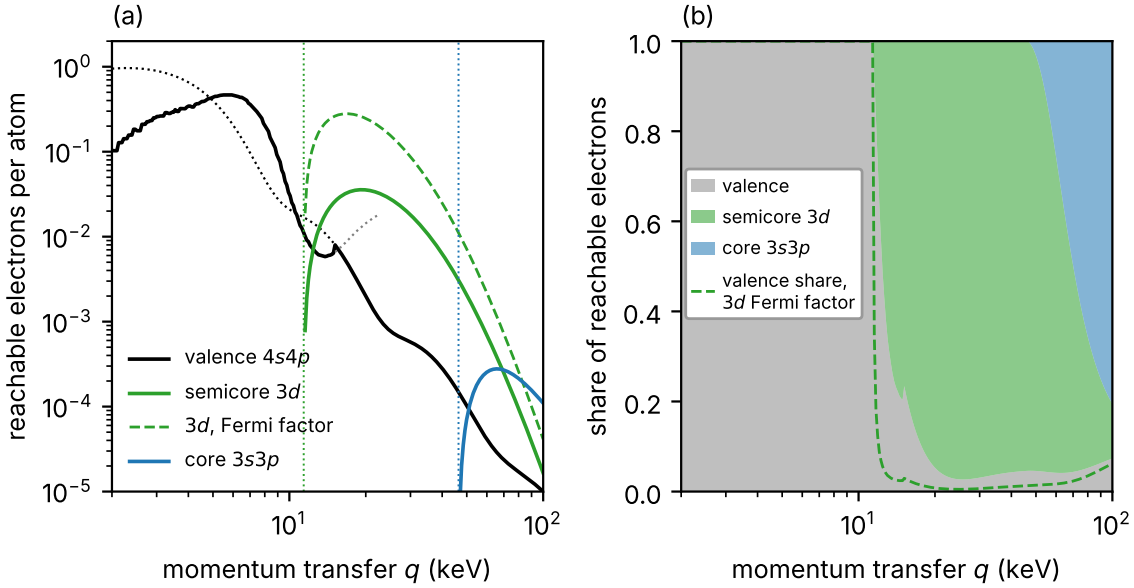


Figure 10.13: Germanium electrons that dark matter of any mass can excite at momentum transfer q, that is, with energy deposits between the threshold of the shell and $qv_{\max}$, $v_{\max} = 2.650 \times 10^{-3}$. (a) Electrons per atom, $N_{nl}^{\mathrm{acc}}(q)$ of Eq. (10.25), for the valence shell (black), the semicore $3d$ shell (green) and the core $3s$ and $3p$ shells (blue). Valence: crystal independent-particle response of Knapen, Kozaczuk and Lin (PAW, $3d$ frozen) below 15 keV; above, the plane-wave model of Eq. (10.26) with the free-atom $4s$, $4p$ orbitals (dotted lines continue each curve past the switch). Core and semicore: Eq. (10.26) with free-atom, scalar-relativistic PBE orbitals and plane-wave final states whose kinetic energy is counted from the valence maximum; dashed, the $3d$ shell with the Fermi factor of Eq. (2.25) at $Z_{\mathrm{eff}} = 1$. Dotted vertical lines: the openings $(E_{\mathrm{g}} + E_{\mathrm{core}})/v_{\max}$ of the $3d$ and $3p$ shells, with centroid binding energies 29.4 and 122.2 eV and $E_{\mathrm{g}} = 0.744$ eV. Germanium $2s$, $2p$ and $1s$ open above 460 keV. (b) Shares of the reachable electrons: valence (gray), $3d$ (green), $3s3p$ (blue); dashed, the valence share when the $3d$ shell carries the Fermi factor. The $3d$ shell overtakes the valence at 12.5 keV (11.5 keV with the Fermi factor) and holds more than 90 percent of the reachable electrons from 20 to 50 keV.

silicon the same model puts the opening of $2p$ at 38 keV and its crossing with the valence at 46 keV, and the $2s$ shell overtakes $2p$ near 77 keV. Since the silicon valence count there comes from the atomic model, uncertain by a factor of a few, and the $2p$ count rises steeply above its opening, the crossing is uncertain by a few keV. Above about 45 keV in silicon and 12 keV in germanium, the response that dark matter samples is that of core electrons.

Two features of the model need checking, the non-orthogonality of plane-wave final states and the missing Coulomb attraction of the ion. A plane wave is not orthogonal to the bound orbital. Its projection onto the initial orbital adds a spurious term of relative size $|F_{nl}(q)|^2\phi_{nl}(p)^2/\langle\phi_{nl}(|\boldsymbol{p}-\boldsymbol{q}|)^2\rangle$, with $F_{nl}(q)$ the elastic form factor of the shell; for germanium $3d$ between 12 and 30 keV it stays below 6 percent. The final state is the larger issue. A slow electron leaving a core region is pulled toward the ion, and the Fermi factor, 12 at a final momentum of 2 keV and 5 at 5 keV for $Z_{\rm eff}=1$, raises the $3d$ count by about that much near the edge (dashed curves). This moves the crossing only from 12.5 to 11.5 keV, because the count rises so steeply above the opening, but it changes the size of the $3d$ contribution to a rate by an order of magnitude. Crystal conduction states, as in Griffin's calculation, are the correct final states, and the Fermi factor only brackets them.

Worked example 10.6: The germanium $3d$ onset in q

The $3d$ shell is a doublet, $3d_{5/2}$ at 29.2 eV and $3d_{3/2}$ at 29.8 eV below the valence maximum [22, Table 1-1]. With $E_{\rm g}=0.744\,$eV the thresholds are 29.94 and 30.54 eV, and the openings $(E_{\rm g}+E_{\rm core})/v_{\rm max}$ are 11.3 keV and 11.5 keV. The threshold masses $2(E_{\rm g}+E_{\rm core})/v_{\rm max}^2$ are 8.5 and 8.7 MeV; the rounded values of Sect. 3.5, 11 keV and 8.5 MeV, are those of a single level at 29 eV.

Opening is not taking over. Just above 11.3 keV the window $qv_{\rm max}-E_{\rm core}$ is a fraction of an eV wide, and the $3d$ count of Eq. (10.25) starts from zero. From Fig. 10.13a, at 12 keV the $3d$ shell offers 0.0040 reachable electrons per atom against 0.0084 for the valence, at 15 keV 0.025 against 0.0065. The two are equal at 12.5 keV with the crystal valence. With the atomic valence the crossing is at 13.1 keV, and with the Fermi factor on the $3d$ final states at 11.5 keV. The onset in q is therefore

$$q_{3d}\simeq(12.5\pm1)\,\text{keV}=(3.3\pm0.3)\,q_0,$$

with $q_0=\alpha m_e=3.73\,$keV. Three quarters of the $3d$ electrons have momenta above 11 keV (Sect. 3.5), so the shell is probed near the peak of its distribution, while the valence is probed far in its tail.

Which masses use this momentum? A momentum transfer q with energy deposited ω needs $v_{\rm min}=\omega/q+q/2m_\chi\le v_{\rm max}$ (Eq. (1.23)), so at the $3d_{5/2}$ threshold, $\omega=29.94\,$eV,

$$m_\chi\ge\frac{q}{2(v_{\rm max}-\omega/q)}=\frac{12.5\,\text{keV}}{2\,(2.650-2.395)\times10^{-3}}=25\,\text{MeV},$$

and 19 MeV for $q=13\,$keV. A particle of 10 MeV reaches the $3d_{5/2}$ edge only for q between 16 and 37 keV, where the $3d$ shell dominates the reachable electrons but only the fastest particles of the halo contribute, through the tail of $\eta(v_{\rm min})$. Weighting each (q,ω) with $\eta(v_{\rm min})$ as in the heavy-mediator rate Eq. (1.47), with a one-pair threshold and the valence from the crystal table below 22.5 keV and the atomic model above, gives the ratio of the $3d$ rate to the valence rate. In the plane-wave model it is 6×10^{-5} at 10 MeV, 0.07 at 30 MeV, 0.27 at 100 MeV and 0.34 at 1 GeV. With the Fermi factor it is 1×10^{-3}, 0.69, 2.1 and 3.4. Griffin and coworkers, with crystal conduction states, find the $3d$ term dominant for a heavy mediator above about 30 MeV at a one-pair threshold, and at lower masses

for higher thresholds [2, Sect. 4.1]. The onset in q is robust to the final state; the mass at which the $3d$ shell dominates the rate is not, and only a calculation with crystal final states fixes it.

The computational connection is the one of Sect. 10.1, extended inward. The valence and semicore vertices need the all-electron reconstruction of Appendix E.2, with a cutoff fixed by the largest q as in Worked example 3.8. The deeper shells are atomic orbitals in Bloch sums (Appendix E.7), with conduction states from the crystal calculation and plane waves above its highest band.

10.6.2 Separation of core and valence responses

The core term Eq. (10.24) is a double sum over $\boldsymbol{k}$ and the orbitals m of every atom, and one might expect interference between the orbitals of neighboring atoms, with phases $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{\tau}_\kappa-\boldsymbol{\tau}_{\kappa'})}$. There is none, under three conditions. First, the shell is completely filled. Second, its band is flat on the scale of the energies that the response resolves, so that all its states share one energy $-E_{\mathrm{core}}$. Third, the atomic orbitals of different sites are orthogonal. For a final state $|f\rangle$ the sum over the initial states of the shell is then $\sum_i |\langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle|^2 = \langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}P_{nl}\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|f\rangle$, with P_{nl} the projector on the shell. A projector does not depend on the basis that spans it, and in the basis of the atomic orbitals $\psi^{\mathrm{at}}_{\boldsymbol{R}\kappa nlm}$ of Eq. (E.48), centered on the atom κ of cell $\boldsymbol{R}$, it is $\sum_{\boldsymbol{R}\kappa m\sigma} |\psi^{\mathrm{at}}_{\boldsymbol{R}\kappa nlm}\sigma\rangle\langle\psi^{\mathrm{at}}_{\boldsymbol{R}\kappa nlm}\sigma|$. The sum over initial states is a sum of squares, one per site, with no cross terms. Every cell gives the same contribution by translation symmetry, and dividing by the volume,

Core response of a crystal

For a filled, flat core shell with orthogonal site orbitals,

$$
\begin{aligned}
S^0_{\mathrm{core}}(\boldsymbol{q},\omega) &= \frac{2\pi}{\Omega}\sum_\kappa S^{\mathrm{at}}_{\kappa nl}(\boldsymbol{q},\omega),\\
S^{\mathrm{at}}_{\kappa nl}(\boldsymbol{q},\omega) &= \sum_{m,\sigma}\sum_f \left|\langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\psi^{\mathrm{at}}_{\kappa nlm}\sigma\rangle\right|^2 \delta(\omega-\varepsilon_f-E_{\mathrm{core}}),
\end{aligned}
\tag{10.27}
$$

the response of one site's shell to final states $|f\rangle$ of the crystal, at energies ε_f above the valence maximum. With plane-wave final states it is Eq. (10.26).

The core response of the crystal is the density of atoms times a single-site response, and adds to the valence response without interference, since a core hole and a valence hole are different final states. This separation is what lets a calculation take the valence from the crystal and the core shells from the atom.

The three conditions hold well for the shells the halo reaches. The germanium $3d$ orbitals have a mean radius of 0.39 Å, against a bond length of 2.45 Å, and the crystal $3d$ bands computed by Griffin and coworkers are flat to within the spread they test [2, Sect. 2.2]. Flatness must be judged per spin–orbit component. The two components of the doublet lie 0.6 eV apart, so Eq. (10.27) applies to each j separately with its own E_{core}, and treating the shell as one level at the centroid misplaces the edge by 0.3 eV.

The separation fails through the final state, the core hole, screening and shells that are not flat, each outside the independent-particle response. The final state comes first, where Eq. (10.27) keeps crystal states but every practical model replaces them: the factor of ten between free and Coulomb-attracted final states found above is the largest uncertainty in the core rate. Then the core hole. The excited electron and the hole left in the $3d$ shell attract

each other and the valence electrons relax around the hole. Both shift weight toward the edge, the same physics as the valence excitons of Sect. 10.5, and a core-level Bethe–Salpeter calculation at finite q is specified in Appendix E.15. Screening is the third. The valence electrons screen the potential that excites a core electron, by $1/|\varepsilon(\boldsymbol{q},\omega)|^2$. At large q the f-sum rule puts the valence oscillator strength near the free recoil energy $q^2/2m_e$, so that $\varepsilon - 1 \approx \omega_p^2/[(q^2/2m_e)^2 - \omega^2]$, with $\omega_p^{\rm val} = 15.6\,\text{eV}$ for the valence electrons of germanium. At 12 keV, where $q^2/2m_e = 141\,\text{eV}$, this is 0.012, and macroscopic screening changes the core response by about 2 percent. Local fields are less certain. For the valence response Sect. 10.2 found a suppression that persists at two to three times $|\boldsymbol{G}_{111}|$, and nothing comparable has been computed for core transitions. That uncertainty belongs with the many-body nuisances of Sect. 8.8. The last is a shell that is not flat. A semicore shell kept in the valence hybridizes with the valence bands, and its contribution is then part of Eq. (10.1) rather than a separate term; the separation still holds for the total, but the split between "valence" and "semicore" becomes a matter of bookkeeping.

A second kind of factorization concerns the dark matter. The master formula Eq. (8.15) separates the dark-matter side $\mathfrak{D}_{ab}$ from the target response S_{ab} exactly, at first Born order, for every set of target operators. What does not hold in general is the older form in which the rate is the cross section of a free electron times a single form factor. Ge, Sheng and Xing showed that, with an electron that moves, the free-electron squared amplitude, evaluated at the kinematics of a bound electron, can turn negative, and that a consistent form puts the electron at rest in the spinor part while its motion stays in the form factor [23, Sect. 4.2]. In the language of Sect. 8.1 this is the statement that only the density row n_e at leading order in the electron velocity reduces to a free amplitude times a scalar form factor. Every correction in the electron velocity is a separate row, the current, the spin current or the kinetic-energy density of Eq. (8.3), with its own response function.

10.6.3 Relativistic corrections to the core response

A target described by the Dirac equation couples to dark matter through bilinears $\bar\psi\Gamma\psi\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$. The Foldy–Wouthuysen transformation of Appendix C.15 turns each into nonrelativistic operators plus corrections in powers of $1/m_e$, acting on the positive-energy states. Two kinds of correction follow. The operators change, which is the picture change of the vertex. And the states change, through the mass–velocity, Darwin and spin–orbit terms of Eq. (3.75).

The density vertex is the time component $\bar\psi\gamma^0\psi = \psi^\dagger\psi$, and to order $1/m_e^2$ its transform is Eq. (C.134), $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}[1 - q^2/8m_e^2 + (\mathrm{i}/4m_e^2)\boldsymbol{\sigma}\cdot(\boldsymbol{q}\times\boldsymbol{p})]$. The scalar factor multiplies every matrix element and changes $|\rho|^2$ by $-q^2/4m_e^2$: 1.2×10^{-4} at the germanium $3d$ opening, 8.6×10^{-4} at 30 keV, 2.0×10^{-3} at the silicon $2p$ crossing at 46 keV. The spin term has amplitude $qp/4m_e^2$. With the root-mean-square momentum of the germanium $3d$ shell, $p = 20.4\,\text{keV}$, it is 5.9×10^{-4} at 30 keV. It is odd in the spin, so for a shell summed over both spin states without spin–orbit coupling its interference with the leading term vanishes and only its square, 3×10^{-7}, remains. With spin–orbit coupling the j-resolved states carry a correlation between spin and orbital motion, and the interference survives at most at the size of the amplitude itself, below 10^{-3}.

The scalar coupling $\bar\psi\psi = \psi_{\rm A}^\dagger\psi_{\rm A} - \psi_{\rm B}^\dagger\psi_{\rm B}$ differs from the density by twice the small component squared (Eq. (C.128)). With $\psi_{\rm B} \simeq (\boldsymbol{\sigma}\cdot\boldsymbol{p}/2m_e)\psi_{\rm A}$ this is the kinetic-energy row,

$$\bar\psi\psi\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \;\to\; n_e(\boldsymbol{q}) - \frac{\mathcal{T}_e(\boldsymbol{q})}{m_e}, \tag{10.28}$$

the bracket of Eq. (8.11), plus a spin term of the same form and size as the one in Eq. (C.134). The kinetic-energy part has the relative size $\langle v^2\rangle/2$. This is the scalar case that Ge, Sheng

and Xing computed with Dirac states. Their 30 to 50 percent reduction of the xenon response arises for inner shells with an effective charge near 50, where $(Z_{\text{eff}}\alpha)^2 = 0.13$, and for ejected electrons of 10 to 50 keV, a recoil of 0.02 to 0.1 of m_e [23, Sects. 1 and 5.2]. Halo dark matter deposits at most a few hundred eV in the shells of silicon and germanium it reaches, so the final states move with kinetic energies below $2 \times 10^{-4}\, m_e$ at 100 eV, and neither parameter is large.

The state corrections are measured by the mean squared velocity of a shell, $\langle v^2 \rangle = \langle p^2 \rangle / m_e^2$, which we take directly from the momentum distributions of Fig. 3.11. It is 1.3×10^{-3} for silicon $2p$, 1.6×10^{-3} for germanium $3d$ and about 10^{-4} for the valence shells. For germanium $3d$, Worked example 3.9 found 1.2×10^{-3} from the radius of the shell through a hydrogen-like formula; the direct second moment is larger because it weights the high-momentum tail. The mass–velocity and Darwin terms change the states at this relative order, and a scalar-relativistic calculation already contains them (Sect. 3.5). What it omits is spin–orbit coupling, whose largest visible effect in the shells within reach is the splitting of the $3d$ doublet, 0.6 eV of a binding energy of 29.4 eV, or 2.0 percent.

The next order of the expansion is the square of the first. For germanium $3d$, $\langle v^2 \rangle^2 = 2.5 \times 10^{-6}$, and for silicon $2p$ it is 1.7×10^{-6}. The deep s shells, whose Dirac wavefunctions behave as $r^{\gamma_D - 1}$ at the nucleus (Appendix C.15), gain a factor $(q/Z\alpha m_e)^{(Z\alpha)^2}$ in their momentum density at large q. At the opening of silicon $2s$, 57 keV, this factor is 1.0009. For germanium $1s$ at $q = 10 Z\alpha m_e = 1.2$ MeV it would be 1.13, but that shell opens only above 4 MeV.

Relativistic corrections to the core response

For the silicon and germanium shells that halo dark matter reaches, every correction of order $1/m_e^2$ to the response is of relative size $\max(q^2/4m_e^2, \langle v^2 \rangle) \approx 10^{-3}$ at $q \leq 30$ keV. The largest, the spin–orbit structure of the germanium $3d$ edge, is bounded by the ratio of the doublet splitting to the binding energy, $\leq 2\,\%$. The next order of the Foldy–Wouthuysen expansion is $\langle v^2 \rangle^2 \approx 3 \times 10^{-6}$.

A two-component calculation, scalar-relativistic with spin–orbit coupling and the vertex of Eq. (C.134), therefore misses only that next order and the error of projecting onto positive-energy states. That error reaches order $(Z\alpha)^2$ only for shells that halo dark matter cannot reach, 0.055 for germanium $1s$.

10.6.4 Relativistic shifts of the f-sum rule

The f-sum rule is the standard test of a computed response, and at the precision a core calculation can reach it is no longer exact. Sect. 4.3 showed that the Darwin and spin–orbit terms drop out of the double commutator and that the mass–velocity term reduces it by $\delta_{\text{mv}} = -(5/3)\bar{K}/m_e - q^2/4m_e^2$ (Eq. (4.33)). For silicon, $\bar{K} = 561$ eV and the first term is -1.83×10^{-3}. For germanium the kinetic energy of the all-electron atom, 2075.3 hartree, gives $\bar{K} = 1765$ eV and $\delta_{\text{mv}} = -5.76 \times 10^{-3}$ at small q.

The interaction between the electrons has a correction of the same order. Moving charges attract each other magnetically, and in the Pauli approximation to the Breit interaction this is the orbit–orbit term [24, Sect. 39]

$$H_{\text{oo}} = -\frac{\alpha}{2m_e^2} \sum_{i<j} \frac{1}{r_{ij}}\, \boldsymbol{p}_i \cdot (\mathbb{1} + \hat{\boldsymbol{r}}_{ij}\hat{\boldsymbol{r}}_{ij}) \cdot \boldsymbol{p}_j,$$

with $\boldsymbol{r}_{ij} = \boldsymbol{r}_i - \boldsymbol{r}_j$. Each term is linear in the momentum of each of two electrons, and its double commutator with the density does not vanish. Write $T_{ij} = (\mathbb{1} + \hat{\boldsymbol{r}}_{ij}\hat{\boldsymbol{r}}_{ij})/r_{ij}$, a

function of positions. In $[H_{\rm oo}, n_e(\boldsymbol{q})]$ only the plane waves of electrons i and j act, through $[\boldsymbol{p}_i, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}] = \boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$, which gives $\boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}\cdot T_{ij}\cdot\boldsymbol{p}_j$ and the same with $i \leftrightarrow j$. The outer commutator with $n_e(\boldsymbol{q})^\dagger$ removes the remaining momentum through $[\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}, \boldsymbol{p}_j] = \boldsymbol{q}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$ and leaves the number $\boldsymbol{q}\cdot T_{ij}\cdot\boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_{ij}}$. The term with $i \leftrightarrow j$ gives the complex conjugate. With $\boldsymbol{q}\cdot T_{ij}\cdot\boldsymbol{q} = [q^2 + (\boldsymbol{q}\cdot\hat{\boldsymbol{r}}_{ij})^2]/r_{ij}$ and Eq. (4.30),

$$\delta_{\rm oo}(\boldsymbol{q}) = -\frac{\alpha}{N_e m_e q^2}\Big\langle \sum_{i<j} \frac{q^2 + (\boldsymbol{q}\cdot\hat{\boldsymbol{r}}_{ij})^2}{r_{ij}} \cos(\boldsymbol{q}\cdot\boldsymbol{r}_{ij}) \Big\rangle \tag{10.29}$$

is the relative shift of the first moment $\mathscr{M}^1_{ee}$ of Eq. (4.30). It has the sign of the mass–velocity shift. A uniform pair density contributes nothing, since the Fourier transform of T_{ij}, $(8\pi/k^2)(\mathbb{1} - \hat{\boldsymbol{k}}\hat{\boldsymbol{k}})$, is transverse and vanishes when contracted with $\boldsymbol{q}$ at $\boldsymbol{k} = \boldsymbol{q}$; only the correlated part of the pair density, the part within an atom, counts. For pairs with $qr_{ij} \ll 1$ the cosine is one and the angular average of $(\boldsymbol{q}\cdot\hat{\boldsymbol{r}})^2$ is $q^2/3$, so that

$$\delta_{\rm oo} \simeq -\frac{4}{3}\frac{E_{\rm ee}}{N_e m_e}, \qquad E_{\rm ee} = \alpha\Big\langle \sum_{i<j} \frac{1}{r_{ij}} \Big\rangle .$$

We estimate the electron repulsion energy of the atoms as the Hartree plus exchange–correlation energy of the same all-electron calculations: 111.3 hartree for silicon and 817.5 hartree for germanium. Then $\delta_{\rm oo} = -5.7 \times 10^{-4}$ and -1.8×10^{-3}, about 0.31 of the mass–velocity shift in both atoms, and the total relativistic shift at small q is

$$\delta_{\rm mv} + \delta_{\rm oo} = -0.24\,\%\ \text{(silicon)}, \qquad -0.76\,\%\ \text{(germanium)}.$$

A Thomas–Fermi estimate of the same terms gives -0.23 and $-0.73\,\%$ (Appendix E.16). The limit $qr_{ij} \ll 1$ holds for the core pairs only below a few keV. At the momenta where the core shells open, 11 to 46 keV, the cosine averages over many core pairs, the orbit–orbit shift is smaller than this estimate, and its sign is not fixed by a positivity argument, while the mass–velocity term $-q^2/4m_e^2$ grows. Appendix E.16 specifies the calculation of $\delta_{\rm oo}(q)$ from 1 to 100 keV from the all-electron static structure factor.

These shifts set the precision of an f-sum test. A calculation with positive-energy relativistic states cannot reproduce the nonrelativistic rule for germanium to better than 0.8 percent, and a four-component response (Appendix E.15) should be checked against the shifted value. A calculation that keeps the negative-energy states without projection fails the rule altogether, since the double commutator of the Dirac Hamiltonian, linear in the momentum, vanishes and the strength moves to pair creation near $2m_e$ (Sect. 4.3).

10.6.5 Open problems

No crystal code computes the finite-q response with four-component states projected onto positive energies. The bounds of Sect. 10.6.3 say how little such a calculation can change for the shells within reach: at most 2 percent for germanium $3d$, with a central value near 10^{-3}, and terms of order 10^{-6} beyond a two-component treatment. What is missing is a calculation that turns the bound into a number, with the $3d$ binding energy itself taken as a nuisance across functionals and many-body methods; Appendix E.15 specifies it, with the relativistic f-sum shift of germanium as its check.

Spin–orbit coupling also mixes spin into the states at the band edges and can lift the symmetry zeros of the valence density vertex. In silicon the effect is computed and small (Appendix E.15 and Sect. 13.9). In germanium the mixing near the conduction-band minimum at L is estimated at up to 10 percent in amplitude. It has not been computed, because the

PBE bands with spin–orbit coupling come out in the wrong order, and so do bands of HSE06, the 2006 parametrization of the hybrid functional of Heyd, Scuseria and Ernzerhof, with spin–orbit coupling, which put Γ about 0.1 eV below L (Appendix E.14); Appendix E.15 specifies the calculation for bands that pass the check of order, gaps and masses there. Up to these open items, the response of every shell that halo dark matter scatters off is in hand. A particle that is absorbed asks for the response at a single point, $\boldsymbol{q} \simeq 0$.

10.7 Absorption on electrons

A dark photon of 10 eV that is absorbed in silicon deposits all of its energy, but at a momentum of only about 10 meV, the momentum $\boldsymbol{p}_\phi = m_\phi \boldsymbol{v}$ it brings from the halo. A scattering particle that deposits the same 10 eV needs at least $\omega/v_{\max} \approx 4\,\mathrm{keV}$. Absorption therefore asks the crystal for its response at a single point, $\boldsymbol{q} \simeq 0$ and $\omega = m_\phi$, the point at which light probes it too (Sects. 1.4 and 8.2.3). The three standard bosons ask for three different functions at that point. The dark photon couples to the current of the electrons, and through its mixing with the photon of the medium it measures the loss function. The axion-like particle couples to their spin current, which without spin–orbit coupling has the response of the current itself but no mixing, so it measures the optical conductivity. In a laboratory magnetic field it also couples through the photon, and then measures the loss function along the field. The scalar couples to the kinetic energy of the electrons, a response that no optical measurement provides. We will derive each rate from Eq. (8.19), compute it for silicon from measured optical constants where they suffice and from a model where they do not, and find for each formula where it stops holding.

10.7.1 Dark-photon absorption and in-medium mixing

Sect. 8.4.5 found that a dark photon at rest is absorbed at the net rate $\Gamma_{\mathrm{net}} = \zeta^2 m_\phi L(m_\phi)$, the loss function at $\boldsymbol{q} \to 0$, and that this is the familiar $\zeta_{\mathrm{eff}}^2 \sigma_1$ with the in-medium mixing $\zeta_{\mathrm{eff}} = \zeta/\varepsilon$. A dark photon of the halo is not at rest, and it has three polarizations: two transverse to $\boldsymbol{p}_\phi$, which mix with the transverse photon of the medium, and one longitudinal, which mixes with the longitudinal one. Each mixes through its own self-energy, $\Pi_{\mathrm{T}} = \omega^2(1 - \varepsilon^{\mathrm{T}})$ and $\Pi_{\mathrm{L}} = (\omega^2 - p_\phi^2)(1 - \varepsilon)$ (Sect. 8.4.5), both evaluated at $(\boldsymbol{p}_\phi, \omega)$ with $\omega^2 = m_\phi^2 + p_\phi^2$. The photon-reducible chain of Appendix D.25 applies to each polarization separately and replaces ζ^2 by $\zeta^2 m_\phi^4/|m_\phi^2 - \Pi_{\mathrm{T,L}}|^2$, and the net rate of each is $-1/\omega$ times the imaginary part of its proper self-energy times that factor [25, Eqs. (2) and (18), 26, Eqs. (38)–(40)]. For the longitudinal mode $m_\phi^2 - \Pi_{\mathrm{L}} = m_\phi^2 \varepsilon$ exactly, as Sect. 8.4.5 found, and $\mathrm{Im}\,\Pi_{\mathrm{L}} = -m_\phi^2 \varepsilon_2$. For a transverse mode $m_\phi^2 - \Pi_{\mathrm{T}} = \omega^2 \varepsilon^{\mathrm{T}} - p_\phi^2$ and $\mathrm{Im}\,\Pi_{\mathrm{T}} = -\omega^2 \varepsilon_2^{\mathrm{T}}$. With $\sigma_1 = \omega \varepsilon_2$,

$$\Gamma_{\mathrm{L}} = \zeta^2 \frac{m_\phi^2}{\omega^2} \frac{\sigma_1}{|\varepsilon|^2}, \qquad \Gamma_{\mathrm{T}} = \zeta^2 \frac{m_\phi^4}{|\omega^2 \varepsilon^{\mathrm{T}} - p_\phi^2|^2} \sigma_1^{\mathrm{T}}, \tag{10.30}$$

with ε, ε^{T} and the conductivities at $(\boldsymbol{p}_\phi, \omega)$, and $\sigma_1^{\mathrm{T}} = \omega \varepsilon_2^{\mathrm{T}}$.

The two polarizations differ through two small numbers. One is the momentum against the size of the cell: ε^{T} and ε both tend to the same $\varepsilon(\omega)$ as $\boldsymbol{p}_\phi \to 0$ (Eq. (4.52)), and their dependence on $\boldsymbol{p}_\phi$ is of order $(p_\phi a)^2$, about 10^{-9} at 10 eV. The other is $p_\phi^2/\omega^2 \simeq v^2 \approx 10^{-6}$. Expanding Eq. (10.30) to first order in it, $\Gamma_{\mathrm{T}}/\Gamma_{\mathrm{L}} = 1 + v^2[2\,\mathrm{Re}(1/\varepsilon) - 1]$. The transverse factor $|\omega^2 \varepsilon - p_\phi^2|$ could single out the transverse modes only where $|\varepsilon|$ itself falls to v^2, which would need a medium whose ε_2 at the plasma frequency is below 10^{-6}. Silicon has $\varepsilon_2 \approx 0.2$ there, and even a clean metal has orders of magnitude more than 10^{-6}. Averaged over the three polarizations of an unpolarized field, and with $\omega_\phi \simeq m_\phi$, the rate per particle is therefore

$\zeta^2 m_\phi L(m_\phi)$ up to corrections of order v^2. In a cubic crystal $\varepsilon(\omega)$ is a scalar at $\boldsymbol{q} \to 0$, and the orientation of the polarization relative to the crystal does not matter either. Multiplying by the number density ρ_χ/m_ϕ and dividing by ρ_T (Eq. (8.19)),

Dark-photon absorption

$$\mathcal{R} = \frac{\rho_\chi}{\rho_T\, m_\phi}\, \zeta_{\text{eff}}^2\, \sigma_1(m_\phi) = \frac{\rho_\chi}{\rho_T}\, \zeta^2\, L(m_\phi)\, [1 + O(v^2)], \tag{10.31}$$

with $\zeta_{\text{eff}} = \zeta/\varepsilon(m_\phi)$ and $L = \varepsilon_2/|\varepsilon|^2$ at $\boldsymbol{q} \to 0$ [27, Eq. (7), 28, Eq. (3)]. The mass of the dark photon cancels between the number density and the rate per particle; the rate per unit mass follows the loss function of the target.

Where the target absorbs light strongly it also screens strongly, and the rate is small: at 3 eV silicon has $|\varepsilon|^2 = 761$, and the dark photon is absorbed 761 times more weakly than the vacuum mixing ζ would suggest. Near the plasma frequency the reverse happens. At 16.3 eV ε_1 crosses zero, and at the peak of the loss function, 16.4 eV (Sect. 4.4), $|\varepsilon|^2 = 0.049$ and the in-medium factor $1/|\varepsilon|^2 = 20$ enhances the rate; the dark photon converts into the plasmon. Fig. 10.14a shows both, against the vacuum-mixing rate $\zeta^2\sigma_1$ that the medium would produce without the photon-reducible chain. Well above the valence plasma frequency, 16.6 eV (Sect. 4.3), $|1 - \varepsilon|$ is small, and it stays small through the L and K edges, since the core electrons add little to it; the two rates meet. In silicon they agree to 10 % for every mass above 63 eV.

The nuclei add a current of their own, $\sum_\kappa Z_\kappa \boldsymbol{j}_\kappa$ in the row of Sect. 8.1.5. In silicon it does not reach the electronic energies: the Born charges vanish, and the ionic part of ε consists of weak two-phonon absorption far below the gap. In a polar crystal the same Eq. (10.31) holds with the total dielectric function, and below the gap the rate peaks at the longitudinal optical phonon (Sects. 5.5 and 8.4.5).

The function σ_1 that enters Eq. (10.31) is the optical conductivity of the material, the proper transverse current response of Sect. 4.4, and it is measured. Ellipsometry, transmission and reflectance give ε from the infrared to the vacuum ultraviolet, and x-ray absorption gives it above about 30 eV through the atomic scattering factors $f_1 + \mathrm{i} f_2$, with $\varepsilon_1 = 1 - (4\pi\alpha n_{\text{at}}/m_e\omega^2) f_1$ and $\varepsilon_2 = (4\pi\alpha n_{\text{at}}/m_e\omega^2) f_2$ for an atomic density n_{at}. For silicon we use the optical constants of float-zone silicon at 300 K from 4 meV to 40 eV [29], the data of Fig. 4.7, and the tables of Henke, Gullikson and Davis above [30]; at 40 eV the two give $\varepsilon = 0.840 + 0.019\,\mathrm{i}$ and $0.852 + 0.019\,\mathrm{i}$. A calculation of σ_1 at $\boldsymbol{q} \to 0$ needs the electron–hole attraction near the gap, which is the Bethe–Salpeter equation of Sect. 4.9, and the local fields above it (Appendix E.6). For the dark photon and the axion-like particle, measured data are the better input wherever they exist at the right temperature, and that proviso is the subject of Sect. 10.7.2.

Worked example 10.7: Dark-photon absorption at 10 eV

At 10 eV the measured dielectric function of silicon is $\varepsilon = -1.879 + 1.268\,\mathrm{i}$: the frequency lies above the main interband peaks, where ε_1 has turned negative, and below the plasma frequency, where it turns positive again. Then $|\varepsilon|^2 = 5.139$, $L = \varepsilon_2/|\varepsilon|^2 = 0.2467$ and $\sigma_1 = \omega\varepsilon_2 = 12.68\,\text{eV}$. The in-medium mixing is $\zeta_{\text{eff}}/\zeta = 1/|\varepsilon| = 0.441$, and the rate is suppressed by $1/|\varepsilon|^2 = 0.195$, a factor of 5.1. The photon self-energy is $\Pi_{\text{em}} = m_\phi^2(1-\varepsilon) = (288 - 127\,\mathrm{i})\,\text{eV}^2$, larger than $m_\phi^2 = 100\,\text{eV}^2$. An expansion in Π_{em}/m_ϕ^2 is useless here: to first order it would predict $1 + 2(1 - \varepsilon_1) = 6.76$, an enhancement, where the true factor is a suppression by five.

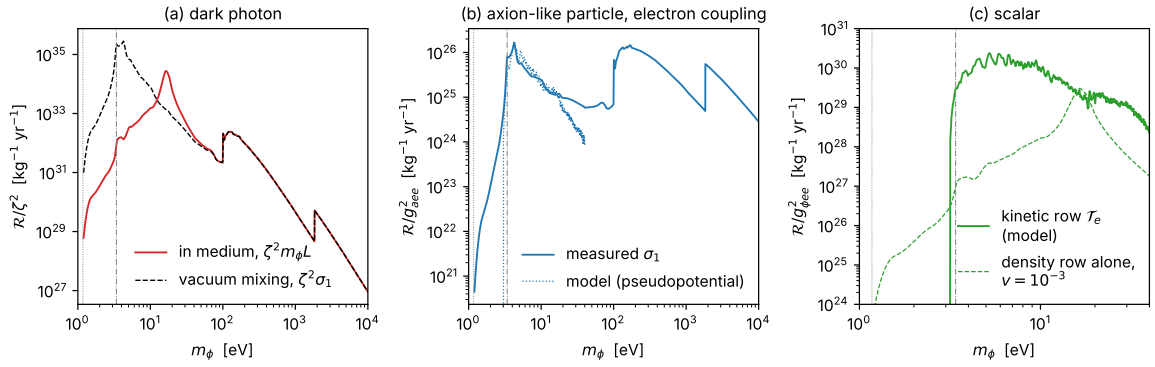


Figure 10.14: Absorption rates of bosonic dark matter on the electrons of silicon, per kilogram-year and per unit squared coupling, against the boson mass m_ϕ, with $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$ and $\rho_T = 2.33\,\mathrm{g\,cm^{-3}}$. Measured optical constants of float-zone silicon at 300 K [29] below 40 eV, and ε from the atomic scattering factors of Ref. [30] above, spliced at 40 eV. The data are used without transfer to millikelvin temperatures and are wrong within a few tens of meV of the edge (Sect. 10.7.2). Dotted line: the excitonic edge of cold silicon, 1.1728 eV with $E_\mathrm{g}(0) = 1.1692\,\mathrm{eV}$; dash-dotted: the direct gap, 3.4 eV. (a) Dark photon: in-medium rate $\zeta^2(\rho_\chi/\rho_T)L$, Eq. (10.31) (red), and the vacuum-mixing rate $\zeta^2(\rho_\chi/\rho_T)\sigma_1/m_\phi$ (dashed). The same curve gives the axion-like particle in a field $\boldsymbol{B}_0$ with $\zeta \to g_{a\gamma\gamma}B_0/m_\phi$, Eq. (10.33). (b) Axion-like particle with the electron coupling, Eq. (10.32), from the measured σ_1 (solid) and from the empirical-pseudopotential model of Sect. 4.6 (dotted; independent particles, no spin–orbit coupling, 16^3 $\boldsymbol{k}$ points). (c) Scalar, Eq. (10.35): kinetic-energy row in the same model, which has pseudo-wavefunctions without core oscillations and its own vertical gap of 3.12 eV (solid; histogram of 0.05 eV bins, smoothed over 0.25 eV); and the screened density row alone at $|\boldsymbol{p}_\phi| = 10^{-3}m_\phi$ with the measured loss function (dashed). The model curves show the size and shape of the kinetic and spin responses, not predictions for silicon.

With $\rho_\chi/m_\phi = 3.0\times10^7\,\mathrm{cm}^{-3}$, Eq. (10.31) gives $\mathcal{R} = 1.52\times10^{33}\,\zeta^2$ per kilogram-year, against $7.82\times10^{33}\,\zeta^2$ with the vacuum mixing. Three events in a kilogram-year, the background-free reach, need $\zeta = 4.4\times10^{-17}$; the vacuum mixing would claim 2.0×10^{-17}, a reach better by $|\varepsilon| = 2.27$ than the true one. For a halo particle with $v = 10^{-3}$ the transverse and longitudinal rates differ by $v^2[2\,\mathrm{Re}(1/\varepsilon) - 1] = -1.7\times10^{-6}$, and each differs from Eq. (10.31) at the level of 10^{-6}. At 10 eV the in-medium factor is the only correction that matters, and it is a factor of five.

10.7.2 Optical data near the silicon edge

Optical constants are measured at room temperature or at 77 K, and a detector runs at 10 mK. Above the direct gap of silicon, 3.4 eV, the difference hardly matters: the transitions are vertical, the bands shift by tens of meV, and σ_1 changes little. Between the indirect gap and 3.4 eV it matters a great deal. There every absorbed photon, and every absorbed dark-matter boson, needs a phonon to supply the crystal momentum $\boldsymbol{k}_\mathrm{c}$ of the conduction valley (Sect. 6.5). The measured σ_1 in that window is the phonon-assisted $\omega\varepsilon_2^{\mathrm{PA}}$, second order in the electron–phonon vertex and first order in the phonon occupations, and both of its temperature dependences act at the edge. At 300 K the gap is 1.12 eV instead of 1.17 eV, and the transverse-acoustic phonon at $\boldsymbol{k}_\mathrm{c}$ has the occupation 0.97, so that a photon can be absorbed together with a phonon taken from the lattice, from $E_\mathrm{g} - \omega_\mathrm{TA}$ up. At 10 mK no phonons are present to be absorbed, and the edge of cold silicon lies at $E_\mathrm{g}(0) + \omega_\mathrm{TA} - E_x^b = 1.1728$ eV with $E_\mathrm{g}(0) = 1.1692$ eV, the excitonic threshold of phonon emission (Sect. 6.5).

The consequence for a rate is direct. The 300 K data give $\sigma_1 = 4.9\times10^{-4}$ eV at 1.15 eV, where a cold crystal absorbs nothing at all, and Fig. 10.14 inherits the warm edge. The 77 K data, which earlier projections used near the gap [27], are closer, but the occupation of the transverse-acoustic phonon is still 0.068 there, and the gap is 1.165 eV against 1.169 eV at millikelvin temperatures, from the measured temperature dependence of the gap [31] (Sect. 6.3). Measured σ_1 near the edge must therefore be transferred to the temperature of the detector before it enters Eq. (10.31) or the axion-like rate below. The transfer runs the phonon-assisted formula backward: each branch is separated into its absorption and emission parts, the absorption parts are removed, the emission parts are reweighted from $n_{\boldsymbol{Q}\nu} + 1$ to one, and the gap is shifted. Without data resolved by branch the separation is not unique. Sect. 13.5 carries out the transfer and finds that the untransferred 77 K data overestimate the millikelvin dark-photon rate of silicon by more than a factor of two within about 20 meV of the edge.

10.7.3 Axion-like-particle absorption

The axion-like particle couples to the electrons through Eq. (8.10), $(g_{aee}/m_e)[(\nabla\phi)\cdot\boldsymbol{s}_e + (\partial_t\phi)\mathcal{J}_e^{\alpha\alpha}]$ with the spin current traced over its two indices. For a particle of the halo the spin-density term is negligible. Its coefficient carries $|\boldsymbol{p}_\phi|$ instead of m_ϕ. Without spin–orbit coupling the total spin also commutes with the Hamiltonian, so $\boldsymbol{s}_e(\boldsymbol{p}_\phi)$ connects different states only at first order in $\boldsymbol{p}_\phi$. Its amplitude is then of order p_ϕ^2, against m_ϕ times an electron velocity for the spin current. Only the row $\mathcal{J}_e \equiv \mathcal{J}_e^{\alpha\alpha}(0)$ is left, with $g_\mathcal{J}^\phi = g_{aee}m_\phi/m_e$ per unit amplitude, and Eq. (8.19) gives $\Gamma = (g_{aee}^2 m_\phi/2m_e^2)\,S_{\mathcal{J}\mathcal{J}}(0, m_\phi)$.

The spin current is related to the ordinary current by a spin trace. An electron–hole pair $(c\sigma', v\sigma)$ is created by $\mathcal{J}_e$ with the amplitude $\sum_\alpha v_{cv}^\alpha (S^\alpha)_{\sigma'\sigma}$, with $\boldsymbol{v}_{cv} = \langle c|\boldsymbol{p}/m_e|v\rangle$, and by the current along a unit vector $\hat{\boldsymbol{e}}$ with $\hat{\boldsymbol{e}}\cdot\boldsymbol{v}_{cv}\,\delta_{\sigma'\sigma}$. Summed over the spins, the first gives $\sum_{\alpha\beta} v_{cv}^\alpha v_{cv}^{\beta*}\,\mathrm{Tr}(S^\alpha S^\beta) = |\boldsymbol{v}_{cv}|^2/2$, and the second $2|\hat{\boldsymbol{e}}\cdot\boldsymbol{v}_{cv}|^2$, which in a cubic crystal averages

to $2|\boldsymbol{v}_{cv}|^2/3$. For independent electrons without spin–orbit coupling, therefore, $S_{\mathcal{J}\mathcal{J}} = \frac{3}{4}S_{jj}$, three Cartesian directions times the quarter that Sect. 10.5.5 found for every spin row. The transverse current response is the conductivity, $S_{jj}(0,\omega) = 2\omega\sigma_1/e^2$ (Sects. 4.4 and 8.2.3), and

Absorption of an axion-like particle through its electron coupling

$$\Gamma = \frac{3\,m_\phi^2}{4\,m_e^2}\,\frac{g_{aee}^2}{e^2}\,\sigma_1(m_\phi), \qquad \mathcal{R} = \frac{\rho_\chi}{\rho_T\,m_\phi}\,\frac{3\,m_\phi^2\,g_{aee}^2}{4\,m_e^2\,e^2}\,\sigma_1(m_\phi), \tag{10.32}$$

for independent electrons without spin–orbit coupling [27, Eq. (10), 26, Eq. (47)]. There is no in-medium factor: σ_1 enters, not the loss function.

The absence of mixing is the spin trace again. A photon-reducible chain needs a proper correlation of $\mathcal{J}_e$ with the charge density, and a single spin operator traced over an unpolarized pair vanishes, $\operatorname{Tr} S^\alpha = 0$, so that $\tilde{R}_{\mathcal{J}Z} = 0$ (Sect. 8.4.5). The parity of the particle says the same: a pseudoscalar cannot mix with the photon of a medium that has neither handedness nor magnetization [26, Sect. III B, footnote 5]. Compared to the dark photon, then, the axion-like particle is not suppressed where the target absorbs light and not enhanced at the plasmon. At equal couplings the ratio of the two rates is $(3m_\phi^2/4m_e^2e^2)|\varepsilon|^2$, which at 10 eV is 1.6×10^{-8}: the factor m_ϕ^2/m_e^2 is the price of a coupling to spin that a slow electron transmits only through its velocity. Fig. 10.14b shows the rate from the measured σ_1. The empirical-pseudopotential model of Sect. 4.6 gives 0.8 to 1.15 times the measured value between 4 and 18 eV, and half of it at 20 eV, far above the transitions to which its three form factors were fitted.

Eq. (10.32) uses the measured σ_1 for a response that differs from the one measured. The current creates electron–hole pairs in the spin singlet, and the spin current in the triplet. With the interaction switched on, the singlet pairs feel both kernels of the Bethe–Salpeter equation and the triplet pairs only the direct one, Eq. (10.23). The excitonic attraction is common to both, and the measured edge carries it correctly. The local fields, however, which are the rest of the exchange kernel, are not. In the model of Sect. 4.6 they lower the main ε_2 peak of silicon from 41 to 34 and move it up by 60 meV. The measured σ_1 therefore understates the axion-like response near its peak by about a fifth, and by less where the local fields are weak. Spin–orbit coupling is the second caveat. It ties the spin to the orbital motion, the spin trace no longer factorizes, and the factor $\frac{3}{4}$ acquires corrections set by the spin–orbit splitting over the transition energy, $\Delta_{\mathrm{SO}} = 0.044$ eV in silicon against several eV (Sect. 3.3). In germanium, GaAs and heavier targets the two responses must be computed separately.

The photon coupling, $g_{a\gamma\gamma}\phi\,\boldsymbol{E}\cdot\boldsymbol{B}$ of Sect. 1.4, needs a field to act on. In a static laboratory field $\boldsymbol{B}_0$ it reads $g_{a\gamma\gamma}\phi\,\boldsymbol{E}\cdot\boldsymbol{B}_0 = -g_{a\gamma\gamma}\phi\,(\partial_t\boldsymbol{A})\cdot\boldsymbol{B}_0$ in the gauge without scalar potential, and after an integration by parts in time it is $\boldsymbol{J}_a\cdot\boldsymbol{A}$, with a source current along the field,

$$\boldsymbol{J}_a = g_{a\gamma\gamma}\,(\partial_t\phi)\,\boldsymbol{B}_0,$$

which oscillates at m_ϕ with the wavevector $\boldsymbol{p}_\phi$. The medium answers as it answers any current. For a sample much larger than the skin depth $(m_\phi \operatorname{Im}\sqrt{\varepsilon})^{-1}$ its boundaries do not matter [28, footnote 1], and at $\boldsymbol{p}_\phi \to 0$ the curl of the magnetic field drops out of the Ampère–Maxwell law $\nabla\times\boldsymbol{B} = \boldsymbol{J} + \partial_t\boldsymbol{E}$ of Sect. 4.4, with a relative error of order p_ϕ^2/ω^2. The induced current is $\boldsymbol{J}_{\mathrm{ind}} = \sigma_{\mathrm{cond}}\cdot\boldsymbol{E}$, with the conductivity tensor, so that $0 = \boldsymbol{J}_a + \sigma_{\mathrm{cond}}\cdot\boldsymbol{E} - \mathrm{i}\omega\boldsymbol{E} = \boldsymbol{J}_a - \mathrm{i}\omega\,\varepsilon\cdot\boldsymbol{E}$, by Eq. (4.50). With the amplitude ϕ_0 of $\phi = \mathrm{Re}(\phi_0 \mathrm{e}^{-\mathrm{i}m_\phi t})$, $\boldsymbol{J}_a$ has the amplitude $-\mathrm{i}m_\phi g_{a\gamma\gamma}\phi_0\boldsymbol{B}_0$, and the field is

$$\boldsymbol{E} = \frac{1}{\mathrm{i}\omega}\,\varepsilon^{-1}\cdot\boldsymbol{J}_a = -g_{a\gamma\gamma}\,\phi_0\,\varepsilon^{-1}(m_\phi)\cdot\boldsymbol{B}_0.$$

The power that this field dissipates per unit volume, averaged over a period, is $\frac{1}{2}\,\mathrm{Re}(\boldsymbol{J}^*_{\mathrm{ind}}\cdot\boldsymbol{E}) = \frac{\omega}{2}\boldsymbol{E}^\dagger\cdot\varepsilon''\cdot\boldsymbol{E}$, with $\varepsilon'' = (\varepsilon-\varepsilon^\dagger)/2\mathrm{i}$ the absorptive part of the tensor. Since $(\varepsilon^{-1})^\dagger(\varepsilon-\varepsilon^\dagger)\varepsilon^{-1} = (\varepsilon^{-1})^\dagger - \varepsilon^{-1}$, the power is $\frac{\omega}{2}g_{a\gamma\gamma}^2|\phi_0|^2\,\boldsymbol{B}_0\cdot\mathrm{Im}[-\varepsilon^{-1}]\cdot\boldsymbol{B}_0$, where $\mathrm{Im}[-\varepsilon^{-1}]$ denotes the absorptive part of $-\varepsilon^{-1}$. The number density of the field is its energy density $\frac{1}{2}m_\phi^2|\phi_0|^2$ over m_ϕ. Dividing the power by it and by the energy ω of one quantum,

Absorption of an axion-like particle in a static field

$$\Gamma = \frac{g_{a\gamma\gamma}^2 B_0^2}{m_\phi}\,\hat{\boldsymbol{B}}_0\cdot\mathrm{Im}\big[-\varepsilon^{-1}(m_\phi)\big]\cdot\hat{\boldsymbol{B}}_0, \qquad \mathcal{R} = \frac{\rho_\chi}{\rho_T}\,\frac{g_{a\gamma\gamma}^2 B_0^2}{m_\phi^2}\,\hat{\boldsymbol{B}}_0\cdot\mathrm{Im}\big[-\varepsilon^{-1}\big]\cdot\hat{\boldsymbol{B}}_0, \tag{10.33}$$

the inverse dielectric tensor projected on the direction of the field [28, Eqs. (6) and (7)]. In a cubic crystal the projection is the loss function $L(m_\phi)$, and the rate is that of a dark photon with $\zeta = g_{a\gamma\gamma}B_0/m_\phi$.

The axion-like particle in a magnet is therefore a dark photon in disguise, polarized along $\boldsymbol{B}_0$, and Fig. 10.14a gives its rate in silicon with that substitution. A field of 10 T is $B_0 = 1953\,\mathrm{eV}^2$ in the units of Appendix A.2, and with $g_{a\gamma\gamma} = 10^{-10}\,\mathrm{GeV}^{-1}$ the equivalent mixing at 10 eV is 2.0×10^{-17}, or 0.58 events per kilogram-year in silicon. The field also acts on the target, through the Zeeman and orbital energies, but $\mu_\mathrm{B}B_0 = 0.58\,\mathrm{meV}$ at 10 T is negligible against electronic energies, and the derivation never used the symmetry of ε, only its absorptive part. The two couplings of the axion-like particle add without interference: the cross term needs a correlation of the spin current with the charge current, which vanishes without spin–orbit coupling for the same reason as $\tilde{R}_{\mathcal{J}Z}$.

In a crystal of lower symmetry the projection of the inverse is not the inverse of the projection. For a uniaxial crystal with the field at an angle θ to the optic axis, $\hat{\boldsymbol{B}}_0\cdot\varepsilon^{-1}\cdot\hat{\boldsymbol{B}}_0 = \cos^2\theta/\varepsilon_\parallel + \sin^2\theta/\varepsilon_\perp$, while the dielectric function along the field is $\varepsilon_{BB} = \cos^2\theta\,\varepsilon_\parallel + \sin^2\theta\,\varepsilon_\perp$, and

$$\hat{\boldsymbol{B}}_0\cdot\varepsilon^{-1}\cdot\hat{\boldsymbol{B}}_0 - \frac{1}{\varepsilon_{BB}} = \frac{\cos^2\theta\,\sin^2\theta\,(\varepsilon_\parallel-\varepsilon_\perp)^2}{\varepsilon_\parallel\,\varepsilon_\perp\,\varepsilon_{BB}}. \tag{10.34}$$

The difference is second order in the anisotropy and vanishes on the principal axes. Near a longitudinal resonance of one axis it is not small, because $1/\varepsilon_\parallel$ or $1/\varepsilon_\perp$ is large there while ε_{BB} is not near zero. The isotropic formula with the dielectric function along the field then misses the resonance, and Eq. (10.33) keeps it.

Limit Check 10.8. Absorption with in-medium mixing

Dark photon: $\mathcal{R} = (\rho_\chi/\rho_T m_\phi)\,\zeta_\mathrm{eff}^2\sigma_1(m_\phi)$, $\zeta_\mathrm{eff}^2 = \zeta^2 m_\phi^4/|m_\phi^2 - \Pi_\mathrm{em}|^2$ [27, Eqs. (6) and (7)], which for $m_\phi^2 \gg |\Pi_\mathrm{em}|$ reduces to the vacuum mixing, $\Gamma = \zeta^2\sigma_1$ [25, Eq. (20)]. Axion-like particle in a field: $\Gamma = (g_{a\gamma\gamma}B_0)^2\,\mathrm{Im}[-1/\varepsilon(m_\phi)]/m_\phi$ for an isotropic target [28, Eq. (7)].

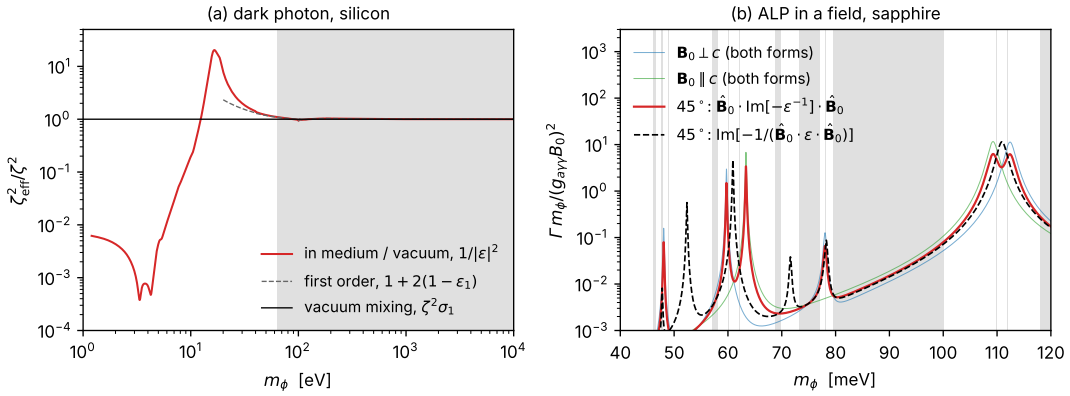


Figure 10.15: (a) Dark photon in silicon: the in-medium rate over the vacuum-mixing rate, $\zeta^2_{\rm eff}/\zeta^2 = 1/|\varepsilon(m_\phi)|^2$ (red), against the vacuum-mixing limit (black) and its first-order correction $1 + 2(1 - \varepsilon_1)$ (dashed, drawn above 20 eV). Measured ε at 300 K below 40 eV [29] and from atomic scattering factors above [30]. Gray: masses where $1/|\varepsilon|^2$ is within 10 % of one, all masses above 63 eV. A halo momentum $|\boldsymbol{p}_\phi| = 10^{-3} m_\phi$ changes the transverse and longitudinal rates by 10^{-6}, invisible here. (b) Axion-like particle in a field at 45° to the optic axis of sapphire: the rate of Eq. (10.33), $\hat{\boldsymbol{B}}_0 \cdot \mathrm{Im}[-\varepsilon^{-1}] \cdot \hat{\boldsymbol{B}}_0$ (red), against the isotropic formula evaluated with the dielectric function along the field, $\mathrm{Im}[-1/\varepsilon_{BB}]$ (dashed), and the two principal axes, where both forms agree (thin). Phonon dielectric functions of sapphire in the damped factorized form $\varepsilon = \varepsilon_\infty \prod_\nu (\omega^2_{\mathrm{L}\nu} - \omega^2 - \mathrm{i}\omega\gamma_{\mathrm{L}\nu})/(\omega^2_{\mathrm{T}\nu} - \omega^2 - \mathrm{i}\omega\gamma_{\mathrm{T}\nu})$ with $\varepsilon_\infty = 3.2$ and the frequencies and 77 K widths of Ref. [32, Eq. (37), Tables I and II] (ordinary ray, longitudinal 112.4, 78.0, 59.7 and 48.1 meV with widths 1.98, 0.55, 0.17 and 0.17 meV, transverse 78.6, 70.5, 54.4 and 47.7 meV with widths 0.47, 0.40, 0.19 and 0.17 meV; extraordinary ray, longitudinal 109.2 and 63.3 meV with widths 1.98 and 0.17 meV, transverse 72.2 and 49.3 meV with widths 0.40 and 0.19 meV), whose widths for the field along the axis are borrowed from the perpendicular modes. Gray: masses where the two forms agree within 10 %. Silicon is cubic, and there the two coincide.

Below 63 eV the medium changes the dark-photon rate of silicon by factors from 4×10^{-4} to 20, and in an anisotropic crystal with the field off its principal axes the isotropic formula misses the longitudinal resonances of each axis by up to a factor of several hundred.

10.7.4 Scalar absorption and the kinetic-energy row

A scalar coupled to $\bar\psi\psi$ acts on the target through Eq. (8.11), $-g_{\phi ee}\phi\,[n_e(\boldsymbol{p}_\phi) - \mathcal{T}_e(\boldsymbol{p}_\phi)/m_e]$: on the number of electrons and, at next order in $1/m_e$, on their kinetic energy. For scattering the density row dominates. For absorption at $|\boldsymbol{p}_\phi| \approx 10^{-3} m_\phi$ the order is reversed, because the density row connects different states only through $\boldsymbol{p}_\phi$, $\langle f|n_e(\boldsymbol{p}_\phi)|0\rangle \simeq \mathrm{i}\boldsymbol{p}_\phi \cdot \langle f|\sum_j \boldsymbol{r}_j|0\rangle$, while the kinetic energy does not commute with the crystal Hamiltonian and connects them already at $\boldsymbol{p}_\phi = 0$ (Sect. 8.1.5). The density row is suppressed by v^2 in the rate and the kinetic row is not.

The in-medium mixing needs a second look. Sect. 8.4.5 found that the scalar mixes with the photon through the proper correlations $\tilde R_{n_e Z}$ and $\tilde R_{\mathcal{T}_e Z}$ of its two rows with the charge. The first is the density response, of order p_ϕ^2. For the second, expand in $\boldsymbol{p}_\phi$: $\tilde R_{\mathcal{T}_e n_e}(\boldsymbol{p}_\phi, \omega) = \boldsymbol{p}_\phi \cdot \boldsymbol{F}(\omega) + O(p_\phi^2)$, where $\boldsymbol{F}$ is a vector built from the crystal alone. It must

be invariant under every operation of the point group, and such a vector exists only in a polar group, one that leaves some direction fixed under every operation. Crystals with a polar group are pyroelectric: they carry a spontaneous electric polarization along that direction. Silicon (O_h), GaAs (T_d) and sapphire (D_{3d}) have none, and in them $\tilde{R}_{\mathcal{T}_e n_e} = O(p_\phi^2)$. The photon-reducible part, $\tilde{R}_{\mathcal{T}_e Z}\, v_{\rm C}(p_\phi)\, \varepsilon^{-1}\, \tilde{R}_{Z\mathcal{T}_e}$, is then of order $p_\phi^4 \cdot e^2/p_\phi^2 = e^2 p_\phi^2$ and vanishes with the density row. In a polar crystal, such as wurtzite GaN or ZnO, $\boldsymbol{F} \neq 0$ and the mixing survives at $\boldsymbol{p}_\phi \to 0$ as a term $(e^2/\omega^2)\, \hat{\boldsymbol{p}}_\phi \cdot \tilde{R}_{\mathcal{T}_e \boldsymbol{j}}\, \varepsilon^{-1}\, \tilde{R}_{\boldsymbol{j}\mathcal{T}_e} \cdot \hat{\boldsymbol{p}}_\phi$, with $n_e = \boldsymbol{p}_\phi \cdot \boldsymbol{j}_e/\omega$ on the energy shell (Eq. (4.29)). Keeping only the kinetic row, with $g^\phi_{\mathcal{T}_e} = g_{\phi ee}/m_e$ in Eq. (8.19),

Scalar absorption

$$\Gamma = -\frac{g_{\phi ee}^2}{m_\phi\, m_e^2}\, \mathrm{Im}\, R_{\mathcal{T}\mathcal{T}}(0, m_\phi) = \frac{g_{\phi ee}^2}{2\, m_\phi\, m_e^2}\, S_{\mathcal{T}\mathcal{T}}(0, m_\phi), \tag{10.35}$$

with $R_{\mathcal{T}\mathcal{T}}$ the response of the kinetic-energy row $\mathcal{T}_e(0) = \sum_j p_j^2/2m_e$ at $T = 0$, for every crystal whose point group has no polar vector, up to corrections of order v^2 from the other rows [26, Eqs. (55) and (56)]. The response is unscreened, and no optical measurement gives it.

In the notation of Mitridate and collaborators the row is $\bar{v}^2 = \mathcal{T}_e/m_e$ and the coupling $d_{\phi ee}$, with $g_{\phi ee} = d_{\phi ee}\sqrt{4\pi}\, m_e/M_{\rm Pl}$ (Sect. 1.4). For independent electrons the response is a joint density of states weighted by the kinetic matrix elements,

$$S^0_{\mathcal{T}\mathcal{T}}(0,\omega) = \frac{4\pi}{\mathcal{V}} \sum_{cv\boldsymbol{k}} \big|\langle c\boldsymbol{k}|\mathcal{T}_e|v\boldsymbol{k}\rangle\big|^2\, \delta(\omega - \varepsilon_{c\boldsymbol{k}} + \varepsilon_{v\boldsymbol{k}}), \tag{10.36}$$

spin summed, with the one-electron element $\langle c|p^2/2m_e|v\rangle = \sum_m \boldsymbol{p}_{cm} \cdot \boldsymbol{p}_{mv}/2m_e$, a sum over all bands, the core included. Since c and v are eigenstates of $p^2/2m_e + V$ with different energies, the element is also $-\langle c|V|v\rangle$: the kinetic row measures the matrix elements of the crystal potential itself. Those are largest where the potential is deep, near the nuclei, and there a pseudo-wavefunction is smooth where the true state oscillates. The rate is therefore sensitive to the core region in a way that the conductivity is not, and a calculation needs the all-electron states or their reconstruction from a projector-augmented-wave basis (Appendix E.16).

The kinetic row also follows a different selection rule. It is even under inversion, while the momentum is odd. At Γ the top valence states of silicon and the lowest conduction states have opposite parity, which is what makes the direct transition at 3.4 eV allowed for light. For the scalar it is forbidden: in the model of Sect. 4.6 the kinetic matrix element between them vanishes at Γ to machine precision, while the momentum element does not, and the scalar spectrum rises from the vertical gap more slowly than ε_2. Below the direct gap the scalar, like the dark photon, is absorbed only with a phonon, and its rate there needs the phonon-assisted kinetic row of Appendix E.14.

The kinetic row has a sum rule of its own. The first-moment identity Eq. (4.28) at $\boldsymbol{q} = 0$ gives $\int_0^\infty \mathrm{d}\omega\, \omega\, S_{\mathcal{T}\mathcal{T}} = (\pi/\mathcal{V})\langle[[\mathcal{T}_e, H], \mathcal{T}_e]\rangle$ at $T = 0$. Take electrons that move independently in a local one-body potential, $H = \sum_j [p_j^2/2m_e + V(\boldsymbol{r}_j)]$. Writing $\mathcal{T}_e = H - \sum_j V(\boldsymbol{r}_j)$ in the outer commutator, the part with H has zero expectation value in an eigenstate, and what remains is $-\langle[[\mathcal{T}_e, V], V]\rangle$. For one electron $[p^2, V] = -\mathrm{i}(\boldsymbol{p} \cdot \nabla V + \nabla V \cdot \boldsymbol{p})$, and a second commutator with V leaves $[[p^2, V], V] = -2|\nabla V|^2$. Per electron, then, $\frac{1}{2}\langle[[\mathcal{T}_e, H], \mathcal{T}_e]\rangle = \langle|\nabla V|^2\rangle/2m_e$, and

$$\int_0^\infty \mathrm{d}\omega\, \omega\, S_{\mathcal{T}\mathcal{T}}(0,\omega) = \frac{\pi\, \boldsymbol{n}_e}{m_e}\, \langle|\nabla V|^2\rangle, \tag{10.37}$$

with the average taken per electron. Compared to the f-sum rule Eq. (4.30), which has

$\pi n_e q^2/m_e$, the squared momentum kick q^2 is replaced by the mean squared force on an electron. For the absorption rate it fixes $\int \mathrm{d}m_\phi\, m_\phi^2 \Gamma = \pi g_{\phi ee}^2 n_e \langle|\nabla V|^2\rangle/2m_e^3$.

The conditions are restrictive: a potential that is local and one-body, and a mean squared force that is finite. A nonlocal pseudopotential keeps the identity in the form $-\langle[[\mathcal{T}_e, V_{\mathrm{NL}}], V_{\mathrm{NL}}]\rangle$, but that is no longer $\langle|\nabla V|^2\rangle$, just as it changes the f-sum rule (Eq. (4.32)). The Coulomb repulsion between the electrons turns $\langle|\nabla V|^2\rangle$ into the mean squared total force on an electron, a two-body and three-body average. The decisive failure is the Coulomb potential of a nucleus, which makes the force diverge: $|\nabla V|^2 = Z^2\alpha^2/r^4$, and an s state, with a finite density at the nucleus, gives $\int r^2 \mathrm{d}r/r^4 = \infty$. The valence states of silicon have s character inside the core, where they oscillate to stay orthogonal to the $1s$ and $2s$ states, so for the real crystal the right-hand side of Eq. (10.37) is infinite. The spectrum of the kinetic row has a tail that decays too slowly for its first moment to converge, and Eq. (10.37) cannot normalize it as the f-sum rule normalizes the density response. What the sum rule does provide is a check of a calculation within its own model, for a local and regular potential such as an empirical pseudopotential, or the valence block of a calculation whose potential is regular.

Limit Check 10.9. Scalar absorption through the kinetic-energy row

$\Gamma = -(g_{\phi ee}^2/m_\phi m_e^2)\,\mathrm{Im}\, R_{\mathcal{T}\mathcal{T}}(0, m_\phi)$, from the kinetic-energy term of the nonrelativistic scalar coupling [26, Eqs. (18) and (56)]; its energy-weighted integral is Eq. (10.37) for a local, regular one-body potential.

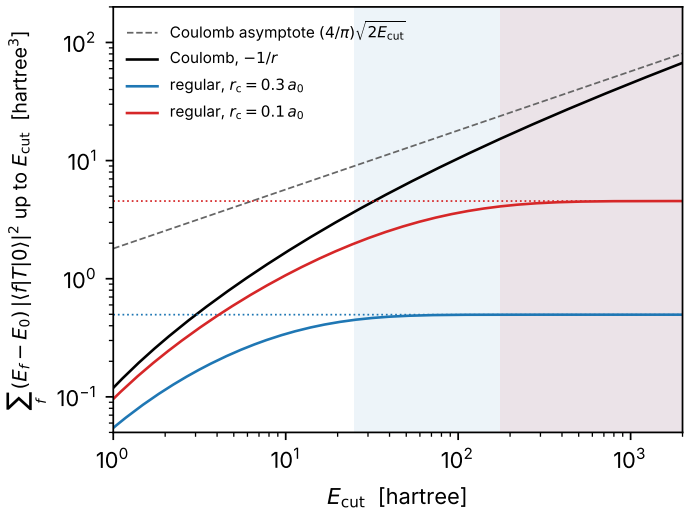


Figure 10.16: Energy-weighted spectrum of the kinetic row, $\sum_f (E_f - E_0)|\langle f|p^2/2m_e|0\rangle|^2$ over final states up to an excitation energy E_{cut}, for one electron in its ground state (atomic units), against E_{cut}. Black: Coulomb potential $-1/r$, hydrogen; dashed, its asymptote $(4/\pi)\sqrt{2E_{\mathrm{cut}}}$, from the plane-wave limit of the continuum states. Blue and red: the regular potentials $-1/(r^2 + r_c^2)^{1/2}$ with $r_c = 0.3$ and $0.1\,a_0$; dotted, their sum rules $\langle|\nabla V|^2\rangle/2$, Eq. (10.37). Shaded: E_{cut} above which each regular spectrum holds 90 % of its sum rule, 25 and 176 hartree. Only s states contribute, since the kinetic energy conserves angular momentum. Radial equation on a uniform grid of $0.004\,\mathrm{a}_0$ in a box of $30\,\mathrm{a}_0$; on the full grid the spectral sum and the double commutator agree to six digits.

For the Coulomb potential of a nucleus the energy-weighted spectrum grows as $E_{\mathrm{cut}}^{1/2}$ without limit, so the sum rule exists only for regular potentials, and its value grows as

the core radius shrinks.

The empirical-pseudopotential model of Sect. 4.6 is local and regular, and it satisfies Eq. (10.37) exactly within its basis of 113 plane waves. On a grid of 16^3 $\boldsymbol{k}$ points the spectral first moment and the double commutator agree to six digits, at $\langle|\nabla V|^2\rangle/2m_e = 1.30 \times 10^3\,\mathrm{eV}^3$ per valence electron. Only 2 % of it lies below 10 eV and 41 % below 40 eV. Even in a model with a regular potential, then, the sum rule is carried by transitions far above the masses that matter, and it constrains the low-energy spectrum only weakly. Fig. 10.14c shows the model's scalar rate in silicon: $9.8 \times 10^{29}\, g_{\phi ee}^2$ per kilogram-year at 10 eV, four orders of magnitude above the axion-like rate at equal coupling. The comparison with the electron's momentum makes the size plausible. The kinetic matrix elements are a few eV, so $\mathcal{T}_e/m_e \sim 10^{-5}$, while the axion-like amplitude carries m_ϕ/m_e times an electron velocity, $\sim 10^{-7}$. The absolute value of the model curve is not a prediction, for the reason given above, and the all-electron response of Appendix E.16 is what fixes it.

The dashed curve of Fig. 10.14c is the density row alone, screened with the measured loss function, $(g_{\phi ee}^2/2m_\phi)(2p_\phi^2/e^2)L$ at $|\boldsymbol{p}_\phi| = 10^{-3}m_\phi$. It is one of the terms of order v^2 in Eq. (10.35). At 5 eV it is 3×10^{-3} of the model's kinetic rate and at 10 eV 2×10^{-2}. Near the plasma frequency, however, the density row is enhanced by the same factor $1/|\varepsilon|^2$ that enhances the dark photon, and between 16 and 18 eV it is as large as the model's kinetic rate. The all-electron calculation of Mitridate and collaborators finds the kinetic term dominant in silicon and germanium [26, Fig. 1 and Sect. IV]; its kinetic matrix elements include the core oscillations that our model lacks. The other terms of order v^2, the cross term between the rows and the mixing through $\tilde{R}_{\mathcal{T}_e Z}$, have the same size and have not been computed. Within a few eV of the plasma frequency the margin by which Eq. (10.35) holds is therefore open, and it is fixed by the same all-electron calculation, extended to order p_ϕ^2.

The kinetic row has no optical counterpart. The scalar reads a response that only a calculation provides, weighted toward the cores of the atoms, and its reach in any target is only as good as that calculation. A boson that acts on the target twice, absorbed in pairs or scattered off an electron, asks the same rows for a response of second order, which is the subject of Sect. 10.8.

10.8 Two-insertion processes

A dark photon of 100 keV that meets an electron in silicon is more likely to scatter off it than to be absorbed. It leaves as an ordinary photon of 91.8 keV, and the electron recoils with the remaining 8.2 keV. This is dark Compton scattering, and above about 55 keV it is the main way a silicon detector registers a dark photon. Its amplitude carries two insertions of a field: one of the dark photon and one of the photon. A complex scalar that couples quadratically to the electrons, through $\phi^\dagger\phi$, is a second example. It cannot be absorbed singly, since the coupling destroys a particle only together with an antiparticle, and the target absorbs the pair, the energy $2m_\phi$ and the momentum of both. Each process asks the crystal for more than the single row of Sect. 10.7. We will write the rate of a process with two insertions from the second-order amplitude of Sect. 2.4, reduce pair absorption to the single-absorption rates of Sect. 10.7 at twice the mass, and derive dark Compton scattering from the seagull term of the electron Hamiltonian. The exact tree-level rate then gives the range of validity of the non-relativistic form, and it corrects a published rate for axion-like particles.

10.8.1 Second-order transition operator

To second order in the interaction $H' = H_{\rm int}$, the Lippmann–Schwinger series Eq. (2.53) gives $T_{fi} = H'_{fi} + \sum_m H'_{fm}H'_{mi}/(E_i - E_m + \mathrm{i}0^+)$, and the rate is $2\pi|T_{fi}|^2\delta(E_f - E_i)$ (Eq. (2.52)). A field can enter T_{fi} twice in two ways. In the first, one term of $H_{\rm int}$ is already quadratic in the field and linear in a target operator, so that both insertions sit at one vertex. The first-order term of T then removes two quanta at once, and the target answers with one of its rows O_a at the total momentum and energy of the pair. The rate needs only the response matrix S_{ab} of Eq. (8.15). In the second way, each insertion comes with its own target operator, and the second-order term of T joins them through the target propagator. With insertions a and b that bring $(\boldsymbol{k}_1, \omega_1)$ and $(\boldsymbol{k}_2, \omega_2)$, the target sees the effective operator

$$T^{(2)}_{ab} = O_b(\boldsymbol{k}_2)\,\frac{1}{E_i + \omega_1 - H_0 + \mathrm{i}0^+}\,O_a(\boldsymbol{k}_1) + O_a(\boldsymbol{k}_1)\,\frac{1}{E_i + \omega_2 - H_0 + \mathrm{i}0^+}\,O_b(\boldsymbol{k}_2), \qquad (10.38)$$

the two time orderings of Eq. (2.54), and the rate is the response of $T^{(2)}_{ab}$, a four-point function of the target operators, at $(\boldsymbol{k}_1 + \boldsymbol{k}_2, \omega_1 + \omega_2)$. When neither denominator can vanish, because ω_1 and ω_2 lie below every excitation that O_a and O_b reach, $T^{(2)}_{ab}$ is to leading order a static polarizability of the target, and the four-point function collapses to a two-point function. A vector field coupled to the electron current brings a contact term of its own, the seagull $e^2\boldsymbol{A}^2/2m_e$ of the electron Hamiltonian. When the frequencies lie far above the binding energies of the electrons, the second-order terms are small against it, and the seagull alone gives the Thomson limit. Dark Compton scattering is this case, and the seagull is where we start in Sect. 10.8.3. A coupling that has both a contact term and two linear ones gives amplitudes of both kinds, and their interference needs a three-point response of the target, linear in a quadratic response function. We come back to it among the open problems.

10.8.2 Pair absorption

Take a complex scalar ϕ of mass m_ϕ, whose particles and antiparticles are equally abundant, each with the number density $\rho_\chi/2m_\phi$. Bauer, Perez-Soler and Shergold write its lowest couplings to electrons, in Hamiltonian form, as [33, Eq. (3)]

$$H_{\rm int} = \int \mathrm{d}^3x \left[m_e\,\phi^\dagger\phi\;\bar\psi(G_{\rm S} + \mathrm{i}G_{\rm P}\gamma^5)\psi + \mathrm{i}(\phi^\dagger \overleftrightarrow{\partial}_\mu \phi)\;\bar\psi\gamma^\mu(G_{\rm V} + G_{\rm A}\gamma^5)\psi \right], \qquad (10.39)$$

with $\phi^\dagger \overleftrightarrow{\partial}_\mu \phi = \phi^\dagger\partial_\mu\phi - (\partial_\mu\phi^\dagger)\phi$ and four real couplings of dimension eV^{-2}, for scalar, pseudoscalar, vector and axial-vector electron bilinears. They are dimension-six operators, of order $\Lambda_{\rm NP}^{-2}$ for a new-physics scale $\Lambda_{\rm NP}$, and we quote rates at $G = G_{\rm F}$, the Fermi constant. To first order in the couplings the interaction Hamiltonian is minus the interaction Lagrangian, derivative terms included. Each operator is quadratic in ϕ and linear in the electrons, so pair absorption is a contact process of the first kind.

The rate. Quantize ϕ in the box. The field operator removes a particle of momentum $\boldsymbol{p}$ with the factor $\mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{x} - \mathrm{i}Et}/\sqrt{2E\mathcal{V}}$ and $\phi^\dagger$ removes an antiparticle of momentum $\bar{\boldsymbol{p}}$ with $\mathrm{e}^{\mathrm{i}\bar{\boldsymbol{p}}\cdot\boldsymbol{x} - \mathrm{i}\bar{E}t}/\sqrt{2\bar{E}\mathcal{V}}$, with E and $\bar{E}$ their energies. Removing both from $\phi^\dagger\phi$ leaves the plane wave $\mathrm{e}^{\mathrm{i}(\boldsymbol{p}+\bar{\boldsymbol{p}})\cdot\boldsymbol{x}}/(\mathcal{V}\sqrt{4E\bar{E}})$. In $\mathrm{i}\phi^\dagger \overleftrightarrow{\partial}_\mu \phi$ the derivatives bring down $-\mathrm{i}p_\mu$ and $-\mathrm{i}\bar{p}_\mu$ with opposite signs, and the same plane wave comes with the factor $(p - \bar{p})_\mu$. Every operator of Eq. (10.39) therefore acts on the target as a row at the total momentum of the pair, $\boldsymbol{q} = \boldsymbol{p} + \bar{\boldsymbol{p}}$, with a coefficient $g_a^{\phi\phi}$ per unit pair amplitude. The coefficients are $m_e G_{\rm S}$ and $m_e G_{\rm P}$ against the bilinears $\bar\psi\psi$ and $\bar\psi\mathrm{i}\gamma^5\psi$, and $G_{\rm V}(p - \bar{p})^\mu$ and $G_{\rm A}(p - \bar{p})^\mu$ against the vector and axial currents. The golden rule, summed over final states, turns $\sum_f 2\pi|\langle f|O_b(\boldsymbol{q})|i\rangle|^2\delta(\omega - E_f + E_i)$ into $\mathcal{V}\,S_{bb}(\boldsymbol{q}, \omega)$, as in Eq. (8.19), and one pair in the box is absorbed at the rate $\sum_{ab} g_a^{\phi\phi *} g_b^{\phi\phi} S_{ab}/(4E\bar{E}\mathcal{V})$. The

box holds $\mathscr{V}^2$ times $(\rho_\chi/2m_\phi)^2$ pairs, distributed over the two velocities with the laboratory distribution f_χ of Sect. 1.2. The number of pairs absorbed per unit time and volume is

$$\frac{\mathit{\Gamma}_{\rm pair}}{\mathscr{V}} = \Big(\frac{\rho_\chi}{2m_\phi}\Big)^2 \int \mathrm{d}^3v\, \mathrm{d}^3\bar v\, f_\chi(\boldsymbol{v}) f_\chi(\bar{\boldsymbol{v}})\, \frac{\sum_{ab} g_a^{\phi\phi *} g_b^{\phi\phi}\, S_{ab}(\boldsymbol{p}+\bar{\boldsymbol{p}},\, E+\bar E)}{4E\bar E}, \tag{10.40}$$

with $\boldsymbol{p} = m_\phi \boldsymbol{v}$, $\bar{\boldsymbol{p}} = m_\phi \bar{\boldsymbol{v}}$ and $E + \bar E = 2m_\phi + K_{\rm pair}$, where $K_{\rm pair} = \frac{1}{2} m_\phi (v^2 + \bar v^2)$ is the kinetic energy of the pair. The event rate per unit target mass is $\mathit{\Gamma}_{\rm pair}/(\mathscr{V}\rho_T)$. This is the golden rule and phase-space measure of Bauer and collaborators [33, Eqs. (5) and (7)] with S_{ab} in place of a single atomic transition. Their truncated Gaussian momentum distribution of each species [33, Eq. (8)], written per degree of freedom, integrates to the number density $\rho_\chi/2m_\phi$ used here, so their rates equal ours at the same density. Their rates assume $\rho_\chi = 0.39\,\mathrm{GeV\,cm^{-3}}$ and scale as ρ_χ^2, so they are larger than at the $0.3\,\mathrm{GeV\,cm^{-3}}$ of Sect. 1.2 by $(0.39/0.3)^2 = 1.69$.

The response is sampled close to a single point. The total momentum is at most $2m_\phi v_{\rm max} = 5.3\times10^{-3}\, m_\phi$, and the pair's kinetic energy has the mean $\bar K_{\rm pair} = m_\phi \langle v^2\rangle = 1.60\times10^{-6}\, m_\phi$ in the laboratory, for the halo of Sect. 1.2 ($\langle v^2\rangle^{1/2} = 379\,\mathrm{km/s}$), with a spread of half that. For an electronic transition at $2m_\phi = 10\,\mathrm{eV}$ the momentum times the lattice constant is $qa < 10^{-4}$. Wherever S_{ab} is smooth on the scale $\bar K_{\rm pair}$ in energy, it can be taken at $(\boldsymbol{0}, 2m_\phi)$, and

$$\frac{\mathit{\Gamma}_{\rm pair}}{\mathscr{V}} = \Big(\frac{\rho_\chi}{2m_\phi}\Big)^2 \frac{\sum_{ab} \langle g_a^{\phi\phi *} g_b^{\phi\phi}\rangle\, S_{ab}(0, 2m_\phi)}{4m_\phi^2}, \tag{10.41}$$

with the average over the two velocities left only on the coefficients. Pair absorption is single absorption at the energy $2m_\phi$, with a tensor of coefficients that may depend on the velocities. Which rows the four couplings reach, and with what coefficients, follows from the non-relativistic reduction of the electron bilinears, which Sects. 1.4 and 8.1.5 have done for linear couplings. The reduction does not care whether the field multiplying a bilinear is ϕ or $\phi^\dagger\phi$.

Scalar. By Eq. (8.11), $\bar\psi\psi$ is $n_e - \mathcal{T}_e/m_e$ at low energy. The density row connects different states only at order q, and the kinetic row survives at $\boldsymbol{q} = 0$, as for a single scalar (Sect. 10.7.4). With $g^{\phi\phi} = m_e G_{\rm S}$ against $\mathcal{T}_e/m_e$, Eq. (10.41) gives $(\rho_\chi/2m_\phi)^2 G_{\rm S}^2 S_{\mathcal{T}\mathcal{T}}(0, 2m_\phi)/4m_\phi^2$. Compare it with Eq. (10.35) for a scalar of mass $2m_\phi$ and coupling $g_{\phi ee}$, absorbed at the rate $g_{\phi ee}^2 S_{\mathcal{T}\mathcal{T}}(0, 2m_\phi)/(4m_\phi m_e^2)$ per particle. The pair rate per unit volume equals the rate of a scalar of mass $2m_\phi$ at the number density $\rho_\chi/2m_\phi$ when

$$g_{\phi ee}^2 \;\to\; g_{\rm eff}^2 = \frac{\rho_\chi}{2m_\phi}\, \frac{m_e^2 G_{\rm S}^2}{m_\phi}. \tag{10.42}$$

Each particle is absorbed as if it were a scalar of twice its mass, coupled with the strength that the density of its partners supplies. The rate per unit volume grows as ρ_χ^2/m_ϕ^4 at fixed response, and this is what makes pair absorption interesting at small masses.

Pseudoscalar. The non-relativistic form of $\bar\psi \mathrm{i}\gamma^5\psi$ times a field is Eq. (1.40), with $g_{aee}\phi$ replaced by $m_e G_{\rm P}\phi^\dagger\phi$. It has two terms. The spin-density term carries the gradient of the field, which on the pair is $\mathrm{i}\boldsymbol{q}$, and is negligible. The spin-current term carries its time derivative, which on the pair is $-\mathrm{i}(E+\bar E) \simeq -2\mathrm{i}m_\phi$, not the momentum. The pseudoscalar pair is therefore absorbed through the axioelectric row $\mathcal{J}_e$ of Sect. 10.7.3, with the coefficient $2m_\phi G_{\rm P}$ in place of $g_{aee} m_\phi/m_e$, and without any velocity suppression. With $S_{\mathcal{J}\mathcal{J}} = \frac{3}{4} S_{jj}$ and $S_{jj}(0,\omega) = 2\omega\sigma_1/e^2$ (Sect. 10.7.3),

$$\frac{\mathit{\Gamma}_{\rm pair}}{\mathscr{V}} = \Big(\frac{\rho_\chi}{2m_\phi}\Big)^2 G_{\rm P}^2\, S_{\mathcal{J}\mathcal{J}}(0, 2m_\phi) = \Big(\frac{\rho_\chi}{2m_\phi}\Big)^2 \frac{3m_\phi\, G_{\rm P}^2}{e^2}\, \sigma_1(2m_\phi), \tag{10.43}$$

which is Eq. (10.32) at the mass $2m_\phi$ with $g_{aee}^2 \to g_{\text{eff}}^2$ of Eq. (10.42), G_{S} replaced by G_{P}. It inherits the caveats of the axioelectric row: the measured σ_1 belongs to singlet pairs and the row creates triplets, and spin–orbit coupling spoils the factor $\frac{3}{4}$.

Vector. The coefficient $(p-\bar{p})^\mu$ is orthogonal to the total four-momentum of the pair, exactly: $(p+\bar{p})\cdot(p-\bar{p}) = E^2-\bar{E}^2-\boldsymbol{p}^2+\bar{\boldsymbol{p}}^2 = m_\phi^2-m_\phi^2 = 0$. A dark photon of four-momentum k couples to the electron current through its polarization ϵ^μ, with $k\cdot\epsilon = 0$. The vector pair is therefore a dark photon of four-momentum $p+\bar{p}$, mass close to $2m_\phi$ and polarization $(p-\bar{p})^\mu$, whose length squared is $-(p-\bar{p})^2 = |\boldsymbol{p}-\bar{\boldsymbol{p}}|^2-(E-\bar{E})^2 \simeq m_\phi^2|\boldsymbol{v}-\bar{\boldsymbol{v}}|^2$, coupled with G_{V} in place of ζe. Since it couples to the electron current, it mixes with the photon of the medium as a dark photon does, and the in-medium chain of Sect. 10.7.1 applies. A dark photon at rest is absorbed at $\zeta^2\omega L(\omega)$, which by Eq. (8.19) is $(\zeta e)^2/2\omega$ times the screened current response along a unit polarization; that response is therefore $2\omega^2 L/e^2$, and

$$\frac{\Gamma_{\text{pair}}}{\mathscr{V}} = \Big(\frac{\rho_\chi}{2m_\phi}\Big)^2 \frac{2G_{\text{V}}^2}{e^2}\, m_\phi^2\, \langle|\boldsymbol{v}-\bar{\boldsymbol{v}}|^2\rangle\, L(2m_\phi), \tag{10.44}$$

with corrections of order v^2 from the difference between longitudinal and transverse polarizations (Eq. (10.30)). The relative velocity is the same in every frame, and for the halo of Sect. 1.2 $\langle|\boldsymbol{v}-\bar{\boldsymbol{v}}|^2\rangle^{1/2} = 402\,\text{km/s}$, close to $\sqrt{3}v_0 = 412\,\text{km/s}$ for a Maxwellian without the escape cut. The derivative coupling brings the factor $\langle|\boldsymbol{v}-\bar{\boldsymbol{v}}|^2\rangle = 1.8\times10^{-6}$.

Axial vector. The space components of $\bar{\psi}\gamma^\mu\gamma^5\psi$ reduce to twice the spin density $\boldsymbol{s}_e$, and the time component to a spin–velocity term like $\mathcal{J}_e$. The time component comes with $E-\bar{E} = O(m_\phi v^2)$. The space components come with $\boldsymbol{p}-\bar{\boldsymbol{p}} = O(m_\phi v)$, but they act through $\boldsymbol{s}_e(\boldsymbol{q})$, and without spin–orbit coupling or magnetic order $\boldsymbol{s}_e(0)$ is the total spin, which commutes with the Hamiltonian and connects no two different states. Both amplitudes are therefore of order v^2, and the rate of order v^4. The spin-density row comes back at $\boldsymbol{q}=0$ only where the target has spin-split levels: the Zeeman sublevels of an atom in a field, which Bauer and collaborators use in hydrogen [33, Eq. (23)], or the magnon of an ordered magnet (Sect. 14.3). Their squared amplitude for the spin flip, G_{A}^2 times the square of the relative velocity transverse to the field over four, is the integrand of Eq. (10.40) with $g^{\phi\phi,\mu} = G_{\text{A}}(p-\bar{p})^\mu$ against $2\boldsymbol{s}_e$ and the normalization $1/4E\bar{E}$.

Fig. 10.17a shows the three unsuppressed rates of silicon at $G = G_{\text{F}}$. They are small. At $m_\phi = 5\,\text{eV}$, where the pair deposits 10 eV, the kinetic row of the pseudopotential model of Sect. 10.7.4 gives 1.6×10^{-12} events per kilogram-year, the pseudoscalar 4.0×10^{-17} and the vector 4.7×10^{-24}. Three events in a kilogram-year would need $G_{\text{S}} = 1.4\times10^6\,G_{\text{F}}$, an interaction stronger than the weak one by a factor of a million, at a scale $\Lambda_{\text{NP}} \approx 0.25\,\text{GeV}$. In the language of Eq. (10.42), $G = G_{\text{F}}$ at 5 eV corresponds to $g_{\text{eff}} = 1.3\times10^{-21}$. The scaling ρ_χ^2/m_ϕ^4 favors much lighter particles, and this is where Bauer and collaborators find pair absorption competitive: spin flips of atoms at µeV energies, whose splitting a magnetic field tunes. A crystal has no electronic transition there. The density row of the scalar, which the kinetic row dwarfs by a factor of 50 at 10 eV, matches it near the plasmon, as for single absorption (Sect. 10.7.4).

The reduction to Eq. (10.41) assumes a smooth response. A target with a sharp line at ΔE resolves the kinetic energy of the pair instead: the rate is spread over $2m_\phi$ between $\Delta E - 4\bar{K}_{\text{pair}}$ and ΔE, and the peak it reaches depends on how the width of the line compares with $\bar{K}_{\text{pair}}$. This is the regime of the atomic transitions of Ref. [33], whose line shape follows the kinetic-energy distribution of the pair. In silicon the absorption continua vary on scales of meV, and Eq. (10.41) holds; only a line narrower than $\bar{K}_{\text{pair}}$, about 1.6 µeV at $m_\phi = 1\,\text{eV}$, would resolve the kinetic energy of the pair. Limit Check 10.10 shows where the reduction fails.

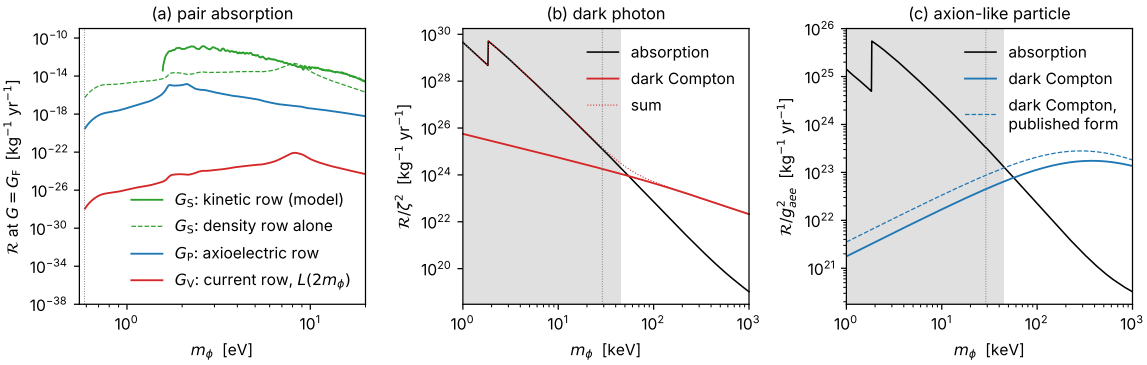


Figure 10.17: Two-insertion rates of bosonic dark matter in silicon per kilogram-year, with $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$, $\rho_T = 2.33\,\mathrm{g\,cm^{-3}}$ and the halo of Sect. 1.2. (a) Pair absorption of a complex scalar of mass m_ϕ at couplings $G = G_\mathrm{F}$, Eq. (10.41). Scalar through the kinetic row in the empirical-pseudopotential model of Sect. 10.7.4, which has its own vertical gap at 3.12 eV and is a model, not a prediction (solid green). Scalar through the density row alone with $\langle q^2\rangle^{1/2} = 2.14 \times 10^{-3}\, m_\phi$ and the measured loss function (dashed). Pseudoscalar, Eq. (10.43), and vector, Eq. (10.44), from the measured optical constants of float-zone silicon at 300 K [29], spliced at 40 eV to atomic scattering factors [30]. Dotted: $2m_\phi$ at the excitonic edge of cold silicon, 1.1728 eV with $E_\mathrm{g}(0) = 1.1692$ eV; the 300 K data are wrong within tens of meV of it (Sect. 10.7.2). (b) Dark photon: absorption, Eq. (10.31) (black; from the optical constants below 29 keV, dotted line, and from the photoelectric cross section of the XCOM tables (the photon cross-section database of the National Institute of Standards and Technology) [34] above, with $\varepsilon = 1$), and dark Compton scattering on all 14 electrons, Eq. (10.48) (red). (c) Axion-like particle: absorption, Eq. (10.32) (black), dark Compton scattering, Eq. (10.51) (solid blue), and the published form, larger by $(m_\phi + 2m_e)/(m_\phi + m_e)$ (dashed). Gray in (b) and (c): $m_\phi < 45$ keV, where the electron recoil ω_C falls below the K-shell binding of 1.84 keV and counting all 14 electrons as free overstates the Compton rate.

Limit Check 10.10. Pair absorption

Pair absorption as single absorption at $\omega = 2m_\phi$ and $\boldsymbol{q} \to 0$, Eq. (10.41), with the constant coefficients $m_e G_\mathrm{S}$, $m_e G_\mathrm{P}$ and the velocity-dependent $G_\mathrm{V,A}(p-\bar{p})^\mu$ [33, Eqs. (3) and (5)–(8)].

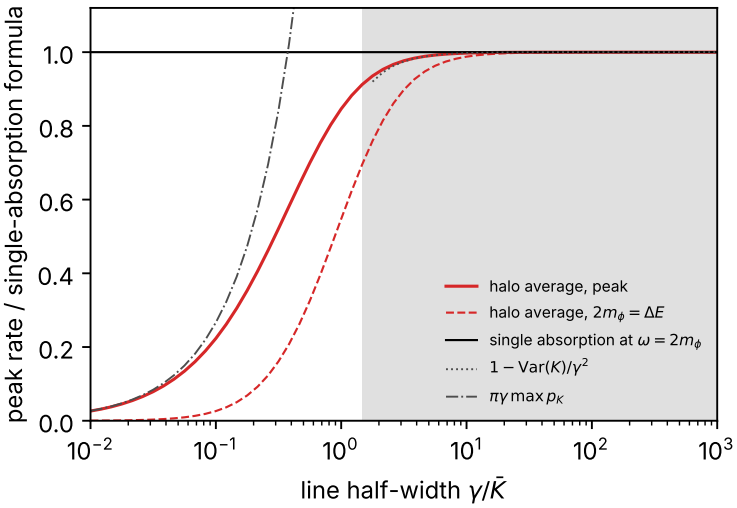


Figure 10.18: Peak absorption rate of a model target with one Lorentzian line of half-width γ, as the halo average of Eq. (10.40) over the pair kinetic energy K_pair (red), divided by the single-absorption formula Eq. (10.41) (black), against $\gamma/\bar{K}_\mathrm{pair}$ with $\bar{K}_\mathrm{pair} = m_\phi\langle v^2\rangle = 1.60\times10^{-6}\,m_\phi$. Solid red: peak over the detuning $\Delta E - 2m_\phi$; dashed: at $2m_\phi = \Delta E$. Dotted: the leading correction for broad lines, $1 - \mathrm{Var}(K_\mathrm{pair})/\gamma^2$ with $\mathrm{Var}(K_\mathrm{pair}) = 0.253\,\bar{K}^2_\mathrm{pair}$; dash-dotted: the narrow-line limit $\pi\gamma \max p_K$, with p_K the distribution of K_pair, which the formula of Ref. [33] for a discrete transition reproduces. Halo of Sect. 1.2, 2×10^6 velocity pairs; the dependence on $\boldsymbol{q}$ is dropped ($qa < 10^{-4}$). Gray: $\gamma \geq 1.5\,\bar{K}_\mathrm{pair}$, peak within 10 % of the formula.

For a line narrower than about $\bar{K}_\mathrm{pair}$, 1.6 µeV at $m_\phi = 1\,\mathrm{eV}$, the deposited energy follows the kinetic-energy distribution of the pair, and the formula overestimates the peak by $\bar{K}_\mathrm{pair}/\gamma$ up to a factor of order one.

The pair-absorption rates need nothing that single absorption does not already compute: the absorption tables of the charge-like rows and the spin responses at $(\boldsymbol{q} \simeq 0, 2m_\phi)$, with the velocity moments of the vector and axial coefficients taken over two copies of the halo. Appendix E.26 specifies them, and the reduction to single absorption at $2m_\phi$ is their check.

10.8.3 Dark Compton scattering

A dark photon couples to the electrons as a photon does, with e replaced by ζe (Sect. 1.4). An electron in the field of both therefore sees the total vector potential $\boldsymbol{A} + \zeta\boldsymbol{A}'$, and its Hamiltonian is $(\boldsymbol{p} + e\boldsymbol{A} + \zeta e\boldsymbol{A}')^2/2m_e$ plus the Pauli term. Expanded, it has the two linear couplings $(e/m_e)\boldsymbol{A}\cdot\boldsymbol{p}$ and $(\zeta e/m_e)\boldsymbol{A}'\cdot\boldsymbol{p}$, symmetrized, and a seagull that contains both fields at one point, $(\zeta e^2/m_e)\,\boldsymbol{A}\cdot\boldsymbol{A}'$. Absorbing a dark photon of polarization $\hat{\boldsymbol{e}}$ and emitting a photon of momentum $\boldsymbol{k}_\gamma$, energy $\omega_\gamma = |\boldsymbol{k}_\gamma|$ and polarization $\hat{\boldsymbol{e}}_\gamma$ is a process with two insertions. The seagull does it at first order, and the linear couplings at second order through Eq. (10.38).

Per unit amplitude of each field, the target sees the transition operator

$$T^{\mathrm{C}} = \frac{\zeta e^2}{m_e}\,(\hat{\boldsymbol{e}}\cdot\hat{\boldsymbol{e}}^*_\gamma)\,n_e(\boldsymbol{q}) + \zeta e^2\Big[\hat{\boldsymbol{e}}^*_\gamma\cdot\boldsymbol{j}_e\,\frac{1}{E_i+m_\phi-H_0}\,\hat{\boldsymbol{e}}\cdot\boldsymbol{j}_e + \hat{\boldsymbol{e}}\cdot\boldsymbol{j}_e\,\frac{1}{E_i-\omega_\gamma-H_0}\,\hat{\boldsymbol{e}}^*_\gamma\cdot\boldsymbol{j}_e\Big], \tag{10.45}$$

with the momentum transfer $\boldsymbol{q} = \boldsymbol{p}_\phi - \boldsymbol{k}_\gamma \simeq -\boldsymbol{k}_\gamma$ and the current rows $\boldsymbol{j}_e$ at the momenta of their own fields. Both fields carry the factor $1/\sqrt{2\omega\mathcal{V}}$ of a quantum of energy ω, as in Eq. (8.19). Summed over the momenta and polarizations of the photon, the golden rule gives the rate per dark photon

$$\Gamma = \int \frac{\mathrm{d}^3k_\gamma}{(2\pi)^3\,2\omega_\gamma}\,\frac{1}{2m_\phi}\sum_{\hat{\boldsymbol{e}}_\gamma} S_{T^{\mathrm{C}}T^{\mathrm{C}}}(\boldsymbol{q},\,m_\phi-\omega_\gamma), \tag{10.46}$$

averaged over the three polarizations $\hat{\boldsymbol{e}}$ of the dark photon. The target takes the energy $\omega = m_\phi - \omega_\gamma$ and the momentum $\boldsymbol{q}$, whose size is close to m_ϕ. A dark photon of 100 keV probes the electrons at $|\boldsymbol{q}| = 92\,\mathrm{keV}$, or $47\,\text{Å}^{-1}$, far beyond every reciprocal-lattice vector and every inverse orbital radius except those of the innermost shells.

The Thomson limit. For a free electron at rest both second-order terms vanish. If the dark photon is absorbed first, its vertex $\hat{\boldsymbol{e}}\cdot\boldsymbol{p}$ acts on an electron at rest, with $\boldsymbol{p}_\phi$ negligible. If the photon is emitted first, its vertex is $\hat{\boldsymbol{e}}^*_\gamma\cdot(\boldsymbol{p}_i+\boldsymbol{p}_m)/2 = -\hat{\boldsymbol{e}}^*_\gamma\cdot\boldsymbol{k}_\gamma/2 = 0$, since the photon is transverse. Only the seagull remains, and $S_{nn}(\boldsymbol{q},\omega) = 2\pi\boldsymbol{n}_e\,\delta(\omega - q^2/2m_e)$ for electrons at rest. The polarization sum $\sum_{\hat{\boldsymbol{e}}_\gamma}|\hat{\boldsymbol{e}}\cdot\hat{\boldsymbol{e}}^*_\gamma|^2 = 1-(\hat{\boldsymbol{e}}\cdot\hat{\boldsymbol{k}}_\gamma)^2$, averaged over $\hat{\boldsymbol{e}}$, is $\frac{2}{3}$. The angular integral of Eq. (10.46) is then 4π, and the energy integral removes the delta function with the Jacobian $1+\omega_\gamma/m_e$:

$$\Gamma = \boldsymbol{n}_e\,\zeta^2\sigma_{\mathrm{Th}}\,\frac{\omega_\gamma}{m_\phi\,(1+\omega_\gamma/m_e)}, \qquad \omega_\gamma + \frac{\omega_\gamma^2}{2m_e} = m_\phi, \qquad \sigma_{\mathrm{Th}} = \frac{e^4}{6\pi m_e^2}, \tag{10.47}$$

with $\sigma_{\mathrm{Th}} = 6.65\times10^{-25}\,\mathrm{cm}^2$ the Thomson cross section. For $m_\phi \ll m_e$ the fraction tends to one, and each electron scatters the dark photon with the cross section $\zeta^2\sigma_{\mathrm{Th}}$ times the speed of light, as if it were a photon at rest in the frame of the electron. The three polarizations of the massive vector make the result isotropic, where Thomson scattering of a photon is not. The electron keeps the recoil energy $\omega = m_\phi - \omega_\gamma$, which for $m_\phi \ll m_e$ is $m_\phi^2/2m_e$.

In a crystal the seagull row is the density at $|\boldsymbol{q}| \simeq m_\phi$, and Eq. (10.46) needs S_{nn} at large momentum. There the dark matter resolves single electrons, and the impulse approximation of Worked example 4.3 applies, as for the core shells of Sect. 10.6. An electron whose binding is small against the recoil scatters as if free, and its deposit is a line at the free recoil, broadened by its momentum in the atom: the Compton profile of its shell. The binding corrections are the second-order terms of Eq. (10.45), which no longer vanish for a bound electron. An electron bound more strongly than the recoil can absorb barely contributes. In silicon the recoil reaches the K-shell binding, 1.84 keV, at $m_\phi = 45\,\mathrm{keV}$, and the L-shell binding, about 100 eV, at 10 keV. Above 45 keV all 14 electrons count, $\boldsymbol{n}_e/\rho_T = 3.00\times10^{23}\,\mathrm{g}^{-1}$. The medium also changes the phase space of the photon through the refractive index $n_r(\omega_\gamma) = 1-\omega_p^2/2\omega_\gamma^2$. It differs from one by 2×10^{-7} at 50 keV with the all-electron $\omega_p = 31.1\,\mathrm{eV}$. The in-medium mixing of the dark photon is, at these masses, the vacuum mixing (Sect. 10.7.1).

The exact tree-level rate. The relativistic amplitude of $V + e \to \gamma + e$ consists of the two Compton diagrams with one photon made massive. Squared, summed over the spins of the electrons and the polarizations of the photon, and averaged over the three polarizations of the dark photon, it is a standard Dirac trace. The term $k^\mu k^\nu/m_\phi^2$ of the massive polarization sum drops out because the electron current is conserved, $k_\mu\mathcal{M}^\mu = 0$. Hochberg, von Krosigk, Kuflik and Yu evaluated the trace and integrated it over the final states [35, Eqs. (7)–(12)].

For a dark photon and an electron at rest the photon energy is fixed by two-body kinematics at $\omega_\gamma = m_\phi(m_\phi + 2m_e)/2(m_\phi + m_e)$, the electron takes

$$\omega_{\rm C} = m_\phi - \omega_\gamma = \frac{m_\phi^2}{2(m_\phi + m_e)},$$

and the rate per dark photon is [35, Eq. (2)]

$$\Gamma = n_e\, \zeta^2\, \frac{e^4}{24\pi m_e^2}\, \frac{(m_\phi + 2m_e)(m_\phi^2 + 2m_e m_\phi + 2m_e^2)}{(m_\phi + m_e)^3}, \tag{10.48}$$

with their kinetic mixing ε written as ζ. The event rate per unit mass is $(\rho_\chi/m_\phi\rho_T)\Gamma$. At threshold the traces give the averaged squared amplitude $|\bar{\mathcal{M}}|^2 = \frac{4}{3}e^4\zeta^2(m_\phi^2 + 2m_e m_\phi + 2m_e^2)/[m_e(m_\phi + m_e)]$, and with the two-body phase space, $\Gamma = n_e|\bar{\mathcal{M}}|^2\omega_\gamma/[16\pi m_\phi m_e(m_\phi + m_e)]$, this is Eq. (10.48) exactly, at every m_ϕ/m_e. Eq. (10.48) is therefore not a non-relativistic approximation: it is the exact tree-level rate on a free electron at rest. For $m_\phi \gg m_e$ it falls to a quarter of the Thomson value, $n_e\zeta^2\sigma_{\rm Th}/4$.

Expanding both rates in $x = m_\phi/m_e$, in units of $n_e\zeta^2\sigma_{\rm Th}$,

$$\text{Eq. (10.48)}:\ 1 - \tfrac{3}{2}x + \tfrac{5}{2}x^2 - \tfrac{15}{4}x^3 + O(x^4), \qquad \text{Eq. (10.47)}:\ 1 - \tfrac{3}{2}x + \tfrac{5}{2}x^2 - \tfrac{35}{8}x^3 + O(x^4). \tag{10.49}$$

The seagull with non-relativistic recoil is correct through second order in x, and the first difference, $\frac{5}{8}x^3$, comes at third order. One eighth, $\frac{1}{8}x^3$, is kinematic: with the exact two-body ω_γ in Eq. (10.47) the cubic coefficient becomes $-\frac{17}{4}$. The remaining $\frac{1}{2}x^3$ is the Dirac amplitude beyond the seagull, the coupling of the photon to the spin and the relativistic structure of the Compton amplitude that the Klein–Nishina formula contains for photons. At 100 keV the seagull is 0.3 % low; it stays within 1 % up to 165 keV and within 10 % up to 517 keV, close to m_e.

Fig. 10.17b compares dark Compton scattering in silicon with absorption. Absorption falls with the photoelectric cross section, roughly as m_ϕ^{-3} per particle above the K edge, while the Compton rate per particle hardly changes below m_e. The two cross at $m_\phi = 55$ keV, and at 100 keV dark Compton scattering is 6.2 times more frequent. In germanium, whose photoelectric cross section is 15 times larger at 100 keV, they cross at 144 keV. The photon that leaves carries most of the energy, 92 keV at $m_\phi = 100$ keV, and travels about 2 cm in silicon before it interacts [34].

Axion-like particles. An axion-like particle couples to the spin current of the electrons through $(\partial_t\phi)$, the second term of Eq. (1.40). Minimal coupling turns the gradient in that term into $\nabla + \mathrm{i}e\boldsymbol{A}$ for an electron of charge $-e$, and the spin current $\psi^\dagger\boldsymbol{\sigma}\cdot\overleftrightarrow{\nabla}\psi$ acquires $2\mathrm{i}e\boldsymbol{A}\cdot\psi^\dagger\boldsymbol{\sigma}\psi$. Multiplied by $\mathrm{i}g_{aee}(\partial_t\phi)/4m_e^2$ this is a seagull for the axion-like particle,

$$H_{\rm sg} = \frac{e\, g_{aee}}{m_e^2}\, (\partial_t\phi)\, \boldsymbol{A}\cdot\boldsymbol{s}_e,$$

which absorbs the particle and emits a photon at one point by flipping, or not flipping, a spin. For a free electron at rest the second-order terms of the photon's orbital coupling vanish as for the dark photon. The time derivative gives m_ϕ. The spin factor $\frac{1}{2}\sum_{\hat{\boldsymbol{e}}_\gamma}\mathrm{Tr}[(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{e}}_\gamma^*)(\boldsymbol{\sigma}\cdot\hat{\boldsymbol{e}}_\gamma)]/4$ is $\frac{1}{2}$, and the integrals that led to Eq. (10.47) give

$$\Gamma = n_e\, \frac{e^2 g_{aee}^2}{8\pi m_e^4}\, \frac{m_\phi\, \omega_\gamma}{1 + \omega_\gamma/m_e} \;\longrightarrow\; n_e\, \frac{e^2 g_{aee}^2 m_\phi^2}{8\pi m_e^4} \quad (m_\phi \ll m_e). \tag{10.50}$$

The Pauli coupling of the photon to the spin, joined to the spin current of the particle, adds second-order terms at relative order x, so Eq. (10.50) is right only at leading order, within

10 % below 49 keV. The exact tree-level rate, from the Lagrangian $g_{aee}\phi\,\bar\psi \mathrm{i}\gamma^5\psi$ and the same two diagrams with the pseudoscalar vertex γ^5 (here the photon polarization sum is the only one), is

$$\Gamma = n_e \,\frac{e^2 g_{aee}^2}{16\pi m_e^2}\,\frac{m_\phi^2\,(m_\phi + 2m_e)}{(m_\phi + m_e)^3}, \tag{10.51}$$

whose expansion in x starts as $n_e(e^2 g_{aee}^2 m_\phi^2/8\pi m_e^4)(1-\frac{5}{2}x+\frac{9}{2}x^2)$, in agreement with Eq. (10.50) at leading order. The squared amplitude behind it agrees, at every mass, with the one printed in Ref. [35, Eq. (15)], in the normalization that turns their vector amplitude into Eq. (10.48). The rate they give [35, Eqs. (3) and (17)] has $(m_\phi + 2m_e)^2/(m_\phi + m_e)^4$ in place of $(m_\phi + 2m_e)/(m_\phi + m_e)^3$, and is therefore larger than Eq. (10.51) by

$$\frac{m_\phi + 2m_e}{m_\phi + m_e}, \tag{10.52}$$

which is 2 for $m_\phi \ll m_e$ and tends to 1 only for $m_\phi \gg m_e$. Its small-mass limit, $4m_\phi^2/m_e^2$ in place of $2m_\phi^2/m_e^2$ in units of $n_e e^2 g_{aee}^2/16\pi m_e^2$, contradicts Eq. (10.50), which follows from the non-relativistic coupling of Sect. 1.4 without any trace. Their differential rate [35, Eq. (16)], integrated over the photon angle at $v \to 0$, gives Eq. (10.51) exactly. The factor appears only in their integrated rate [35, Eqs. (3) and (17)], so the error enters in the step from the differential to the integrated form. Fig. 10.17c shows both forms in silicon: with Eq. (10.51), dark Compton scattering overtakes absorption of an axion-like particle at 57 keV, and with the published form at 46 keV.

Worked example 10.8: The axion-like Compton rate at 100 keV

At $m_\phi = 100$ keV, $x = 0.1957$. The photon leaves with $\omega_\gamma = m_\phi(m_\phi+2m_e)/2(m_\phi+m_e) = 91.82$ keV, and the electron takes $\omega_C = 8.18$ keV, above the K-shell binding of silicon and below that of germanium, 11.1 keV. The factor Eq. (10.52) is $(100+1022)/(100+511) = 1122/611 = 1.836$. Per electron and per unit coupling, Eq. (10.51) gives $\Gamma/(n_e g_{aee}^2) = 1.34\times10^{-25}\,\mathrm{cm}^2$ times the speed of light, and the published form $2.46\times10^{-25}\,\mathrm{cm}^2$. The non-relativistic Eq. (10.50) gives $1.62\times10^{-25}\,\mathrm{cm}^2$, 21 percent high, as its error of relative order x allows.

In silicon, with $n_e/\rho_T = 3.00\times10^{23}\,\mathrm{g}^{-1}$ and the photoelectric cross section $0.0250\,\mathrm{cm}^2/\mathrm{g}$ at 100 keV, the Compton rate of Eq. (10.51) is 5.1 times the absorption rate of Eq. (10.32), and the published form makes it 9.4 times. The total rate of an axion-like particle of 100 keV in silicon is then overstated by $(1+9.4)/(1+5.1) = 1.70$. In germanium absorption still dominates at this mass (the Compton rate is 0.30 of it), and the total is overstated by 1.19. Where dark Compton scattering is the whole signal, a bound on g_{aee} scales as the inverse square root of the rate, and a bound set with the published form is too strong by $\sqrt{1.836} = 1.36$.

Limit Check 10.11. Dark Compton scattering

Vector on a free electron at rest: $\Gamma = n_e\zeta^2(e^4/24\pi m_e^2)(m_\phi+2m_e)(m_\phi^2+2m_e m_\phi+2m_e^2)/(m_\phi+m_e)^3$ [35, Eq. (2)], against the seagull form Eq. (10.47). Axion-like particle: $\Gamma = n_e(e^2 g_{aee}^2/16\pi m_e^2)\,m_\phi^2(m_\phi+2m_e)^2/(m_\phi+m_e)^4$ [35, Eq. (3)], against the tree-level Eq. (10.51).

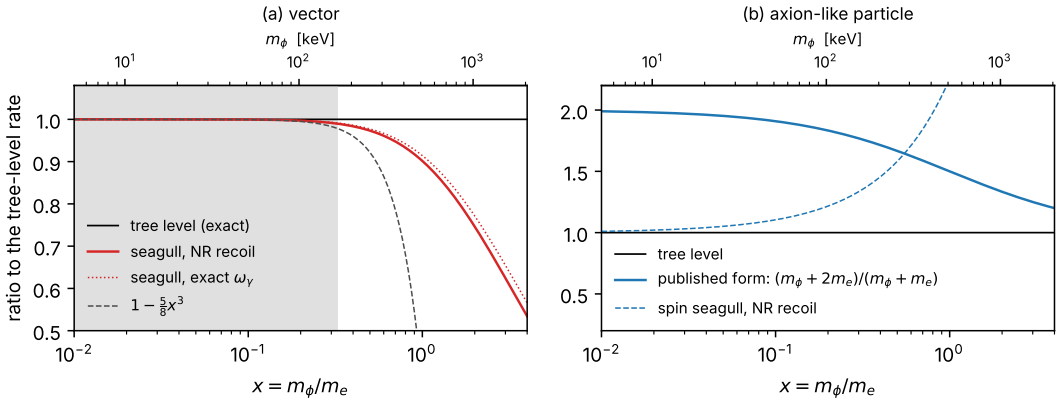


Figure 10.19: Dark Compton scattering on a free electron at rest, as ratios to the exact tree-level rate, against $x = m_\phi/m_e$ (top axis: m_ϕ). (a) Vector: the seagull with non-relativistic recoil, Eq. (10.47) (red), and with the exact two-body photon energy (dotted), against Eq. (10.48), the exact tree-level rate (black). Dashed: $1 - \frac{5}{8}x^3$, the leading correction of Eq. (10.49). Gray: $x < 0.32$ ($m_\phi < 165\,\mathrm{keV}$), seagull within 1 %. (b) Axion-like particle: the published rate over Eq. (10.51), which is the factor $(m_\phi + 2m_e)/(m_\phi + m_e)$ at every x (solid), and the spin seagull Eq. (10.50) (dashed), correct at leading order. Nothing is shaded: the published form differs from the tree-level rate at every mass.

The seagull form of the vector rate drifts from the exact one above about 0.2 MeV, by 1 % at 165 keV and 10 % at 517 keV, as the Dirac structure of the amplitude takes over, while the published axion-like rate is too large by a factor that falls from 2 at small mass to 1 well above m_e; bound electrons with binding near ω_C break both formulas.

10.8.4 Open problems

Two pieces of the two-insertion response are not computed for any crystal. The first is the interference between a contact amplitude and a second-order one, which a dark-matter model with both kinds of coupling produces. For the stopping of a charged particle the same term is the Barkas correction, odd in the charge of the projectile. It is linear in the quadratic response of the target, a three-point function, and no reduction to two-point responses exists. The quadratic response of the electron gas is known and serves as a check. Appendix E.26 specifies it mapped onto the crystal by the local-density approximation; a crystal calculation at second order in time-dependent perturbation theory would go beyond that. The second is the bound-state correction to dark Compton scattering for the shells whose binding is close to ω_C. They are the K shell of silicon between 45 and about 100 keV, the K shell of germanium, whose binding of 11.1 keV the recoil reaches at 118 keV, and the L shells below about 20 keV. There the second-order terms of Eq. (10.45) no longer vanish, and the rate needs the shell Compton profiles or the large-q density response, together with the refractive index at the photon energy, as Appendix E.26 specifies.

10.9 Inelastic and relativistic dark matter

A dark-matter particle of 1 MeV that arrives in an excited state 10 eV above its ground state carries three times more energy in its mass splitting than in its motion, whose kinetic energy

is at most 3.5 eV. A particle accelerated by a cosmic-ray electron can move at nearly the speed of light, and a fermion that is absorbed hands over almost its whole rest energy. Each of these cases changes one ingredient of the master formula of Sect. 8.2 and leaves the others alone. A mass splitting changes the kinematics, through the term $-\Delta_\chi$ in $\omega_{\boldsymbol{q}}$. A relativistic particle changes the dark-matter tensor $\mathfrak{D}_{ab}$, and an outgoing light fermion changes both. The target enters, as before, through the response matrix of the rows of Eq. (8.3), and the silicon tables of Sect. 10.2 provide it. We will derive the rate of each case and find the lines that exothermic dark matter leaves in the spectrum. For a fast flux we will bound the error of the non-relativistic density row, and for boosted and fermionic dark matter we will show where the published formulas differ from the conserved-current result. An excited state can also decay inside the detector without scattering, and we end with the rate of that decay inside the crystal.

10.9.1 Inelastic and exothermic scattering

For inelastic dark matter the potential and $\bar{\sigma}_e$ keep their elastic form (Sect. 1.4). The time component of the current that connects χ_1 to χ_2 reduces to one, as for the elastic current, up to corrections of order Δ_χ/m_χ (Eq. (C.128)), and the recoil term of Eq. (8.13) is smaller by the same fraction. The only change that matters is the term $-\Delta_\chi$ in $\omega_{\boldsymbol{q}}$. The derivation of Sect. 10.2 then goes through step by step. The angular integral of Eq. (A.42) produces η at the minimum velocity of Eq. (1.23), taken in absolute value:

Inelastic electron-recoil rate

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{\rho_\chi\,\bar{\sigma}_e}{8\pi^2\alpha\,\rho_T\,m_\chi\,\mu_{\chi e}^2}\int \mathrm{d}q\; q^3\,\eta(v_{\min})\,F_{\mathrm{DM}}(q)^2\,\bar{L}(q,\omega), \qquad v_{\min} = \left|\frac{\omega+\Delta_\chi}{q} + \frac{q}{2m_\chi}\right|, \tag{10.53}$$

at $T = 0$, with $\bar{L}$ the loss function averaged over the directions of $\boldsymbol{q}$. It is exact under the conditions of Eq. (10.4), and with ε_2 of the independent-particle head in place of $\bar{L}$ it is the unscreened rate of Sect. 10.1.

The absolute value is needed only when $\omega + \Delta_\chi < 0$, which at $T = 0$ means exothermic dark matter depositing less than $|\Delta_\chi|$. Gu, Wu and Zhu write the unscreened form of Eq. (10.53) with the crystal form factor [36, Eqs. (10), (11), (15), (16) and (20)]; in the box below their rate carries the factor q^{-2} of the elastic form-factor rate [1, Eq. (3.13)], which their printed form lacks, and the absolute value in $v_{\min}$, which they handle through a piecewise $q_{\min}$.

An endothermic splitting is paid from the kinetic energy. Since $\omega \le \frac{1}{2}m_\chi v^2 - \Delta_\chi$ (Eq. (1.24)), a pair across the gap needs $m_\chi \ge 2(\Delta_\chi + E_{\mathrm{g}})/v_{\max}^2$: 0.60 MeV for $\Delta_\chi = 1$ eV and 3.2 MeV for 10 eV, with the step-model gap of 1.11 eV that accompanies the tables of Sect. 10.2. An exothermic splitting is paid by the dark matter, and as Sect. 8.2.4 found, it shifts the band of allowed (q,ω) up by $|\Delta_\chi|$, down to $q \to 0$. Fig. 10.20 shows the consequence in silicon at 10 MeV. With a heavy mediator, per kilogram-year at $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$, the rate is 314 events for $\Delta_\chi = 10$ eV, 5.9×10^4 for elastic scattering and 7.5×10^5 for $\Delta_\chi = -10$ eV. With a light mediator the three numbers are 6.2, 9.8×10^3 and 5.2×10^5, and the exothermic spectrum has a sharp peak at $\omega = |\Delta_\chi|$. Splittings of a few eV move the rate by orders of magnitude, in either direction.

An exothermic particle is not limited by the kinetic energy of the halo, and its spectrum is set by the splitting. To see how, complete the square in $\omega_{\boldsymbol{q}}$ with $\Delta_\chi < 0$,

$$\omega_{\boldsymbol{q}} = |\Delta_\chi| + \boldsymbol{q}\cdot\boldsymbol{v} - \frac{q^2}{2m_\chi} = |\Delta_\chi| + \tfrac{1}{2}m_\chi v^2 - \frac{|\boldsymbol{q} - m_\chi\boldsymbol{v}|^2}{2m_\chi}.$$

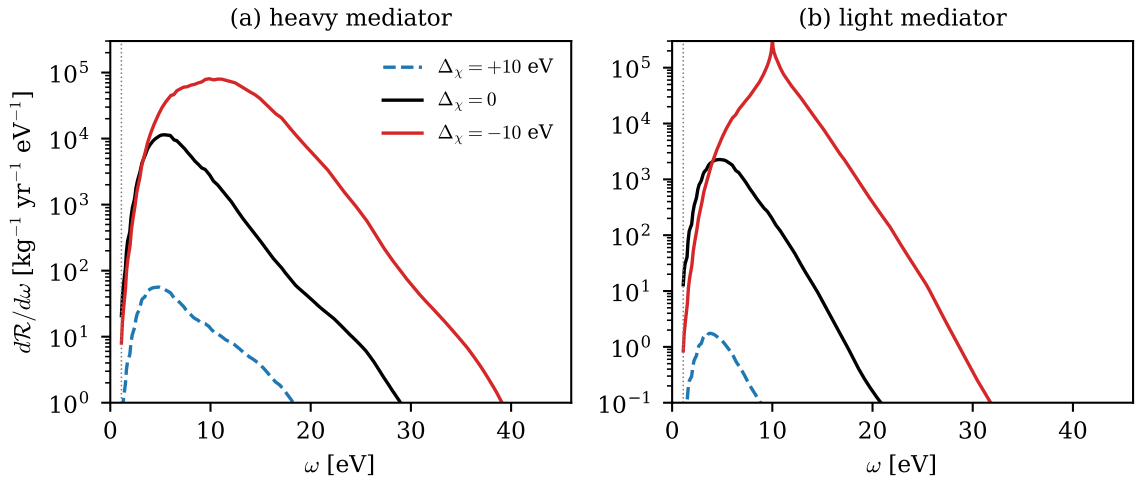


Figure 10.20: Electron-recoil spectrum of silicon for inelastic dark matter of mass $m_\chi = 10\,\mathrm{MeV}$ with splittings $\Delta_\chi = 10\,\mathrm{eV}$ (endothermic, dashed), 0 (elastic) and $-10\,\mathrm{eV}$ (exothermic), from Eq. (10.53). Inputs: the random-phase loss function with local fields of the tables of Sect. 10.2, the standard halo of Sect. 1.2, $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$ and $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm^2}$. Dotted: 1.11 eV, the step-model gap that accompanies the tables, below the low-temperature gap of 1.17 eV to which their scissor shifts the bands. Below $|\boldsymbol{q}| = 493\,\mathrm{eV}$, the first tabulated momentum, the loss function is held at its value there. (a) Heavy mediator, $F_\mathrm{DM} = 1$. (b) Light mediator, $F_\mathrm{DM} = (q_0/q)^2$; the exothermic spectrum peaks at $\omega = |\Delta_\chi|$, where the momentum transfer goes to zero.

At fixed $\omega < |\Delta_\chi|$ the delta function of Eq. (8.16) puts $\boldsymbol{q}$ on a sphere of radius k_* centered at $m_\chi \boldsymbol{v}$, with $k_*^2 = q_\Delta^2 + m_\chi^2 v^2$ and

$$q_\Delta = \sqrt{2m_\chi(|\Delta_\chi| - \omega)}, \qquad v_\Delta = q_\Delta/m_\chi.$$

For any function G of $|\boldsymbol{q}|$, $\int \mathrm{d}^3q\,\delta(\omega - \omega_{\boldsymbol{q}})\,G = m_\chi k_* \int \mathrm{d}^2\hat{\boldsymbol{n}}\,G(|m_\chi \boldsymbol{v} + k_*\hat{\boldsymbol{n}}|)$. On the sphere $q^2 = k_*^2 + m_\chi^2 v^2 + 2k_* m_\chi \boldsymbol{v}\cdot\hat{\boldsymbol{n}}$, and the term linear in $\boldsymbol{v}$ averages to zero because the sphere is centered at $m_\chi \boldsymbol{v}$. The integral is therefore $4\pi m_\chi q_\Delta G(q_\Delta)[1 + O(v^2/v_\Delta^2)]$, exactly $4\pi m_\chi k_* G$ for constant G. The rate per particle no longer depends on its velocity, and the average over the halo is trivial. With $G = |\tilde{V}|^2\bar{S}$, $\bar{S} = (2q^2/e^2)\bar{L}$ and $\rho_\chi/m_\chi\rho_T$ per unit mass,

Exothermic shell

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{\rho_\chi\,\bar{\sigma}_e\,q_\Delta^3\,F_\mathrm{DM}(q_\Delta)^2\,\bar{L}(q_\Delta,\omega)}{4\pi^2\alpha\,\rho_T\,\mu_{\chi e}^2}\left[1 + O(v^2/v_\Delta^2)\right], \qquad \omega < |\Delta_\chi|, \tag{10.54}$$

independent of the velocity distribution as long as v_Δ exceeds the speeds of the halo. The spectrum samples the response along the parabola $\omega = |\Delta_\chi| - q^2/2m_\chi$.

For constant $\bar{L}$ and a heavy mediator the correction is about $2.5\langle v^2\rangle/v_\Delta^2$, with $\langle v^2\rangle^{1/2} = 1.3\times10^{-3}$ in the laboratory frame: at $m_\chi = 0.1\,\mathrm{MeV}$ and $\omega = 3\,\mathrm{eV}$, Eq. (10.53) exceeds Eq. (10.54) by 2.9 %. Since $v_\Delta^2 = 2(|\Delta_\chi| - \omega)/m_\chi$, the shell form needs m_χ well below $2|\Delta_\chi|/v_\mathrm{max}^2$, which is 2.9 MeV for $|\Delta_\chi| = 10\,\mathrm{eV}$; at 10 MeV the halo still matters, and Fig. 10.20 uses Eq. (10.53).

The parabola explains where exothermic spectra peak. On a free electron at rest the

response is a line, $S \propto \delta(\omega - q^2/2m_e)$, and it crosses the parabola once, at

$$\omega_{\rm line} = \frac{\mu_{\chi e}}{m_e}\,|\Delta_\chi|, \qquad q_{\rm line} = \sqrt{2\mu_{\chi e}|\Delta_\chi|},$$

the energy shared between the two recoiling particles. A heavy dark-matter particle hands the whole splitting to the electron, and a light one keeps part of it. In a crystal the electron line exists only where the free-electron picture holds, $q_{\rm line} \gg q_0$, that is $\omega_{\rm line} \gg q_0^2/2m_e = 13.6\,\text{eV}$. For smaller splittings the parabola crosses the region in which the response is that of bound electrons, and the peak sits where $q_\Delta^3 \bar{L}$ is largest. The factor q_Δ^3 falls toward $\omega = |\Delta_\chi|$, while $\bar{L}$ rises from the gap toward the plasma frequency, and the product peaks in between.

A light mediator turns the end of the spectrum into a line. As $\omega \to |\Delta_\chi|$, $q_\Delta \to 0$ and $\bar{S} \to (q^2/2\pi\alpha)L(0,\omega)$, the optical limit, so that with $F_{\rm DM}^2 = q_0^4/q^4$

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} \to \frac{\rho_\chi\,\bar{\sigma}_e\,q_0^4\,L(0,\omega)}{4\pi^2\alpha\,\rho_T\,\mu_{\chi e}^2\,q_\Delta} \propto \left(|\Delta_\chi| - \omega\right)^{-1/2}. \tag{10.55}$$

The divergence is integrable. Within $\frac{1}{2}m_\chi v^2$ of the end point, where v_Δ falls below the halo speeds, the exact kernel softens it to a logarithm, since there the momentum transfer can go to zero for every speed of the halo. Its strength is the optical loss function at $\omega = |\Delta_\chi|$: near the end point the exothermic particle acts like a boson of mass $|\Delta_\chi|$ absorbed at $\boldsymbol{q} \simeq 0$ (Sect. 8.2.4), and like the dark photon of Sect. 10.7 it reads L rather than ε_2.

Worked example 10.9: Exothermic lines in silicon

Take $\Delta_\chi = -10\,\text{eV}$ in silicon and the masses 0.1, 0.3, 1 and 3 MeV. The free-electron lines would sit at $\omega_{\rm line} = 1.64$, 3.70, 6.62 and 8.54 eV, at $q_{\rm line} = 1.29$, 1.94, 2.60 and 2.96 keV. Every one of these momenta lies below $q_0 = 3.73\,\text{keV}$; even an infinitely heavy particle reaches only $q_{\rm line} = (2m_e|\Delta_\chi|)^{1/2} = 3.20\,\text{keV}$. A 10 eV splitting therefore never meets free electrons in silicon.

The silicon spectrum from Eq. (10.53), with the random-phase loss function with local fields and a heavy mediator, peaks at 6.95, 6.19, 5.99 and 7.29 eV for the four masses, with half-maximum edges between about 4 and 8.5 eV (up to 11.8 eV at 3 MeV). At 0.1 MeV the peak lies 5.3 eV above the free-electron line, at 1 MeV 0.6 eV below it: the peak barely moves with the mass because it is set by the loss function, not by the recoil. The momentum transfer at the peak is $q_\Delta = 0.78$, 1.51, 2.83 and 4.03 keV, from $0.21\,q_0$ to $1.08\,q_0$. Per kilogram-year at $\bar{\sigma}_e = 10^{-37}\,\text{cm}^2$ the totals are 2.1×10^5, 3.1×10^5, 8.7×10^5 and 1.5×10^6. The shell form Eq. (10.54) reproduces them to 0.91, 0.81, 0.63 and 0.73. The largest speeds of the halo reach v_Δ at the peak for $m_\chi \gtrsim 1\,\text{MeV}$, with $v_{\rm max}/v_\Delta = 0.34$, 0.53, 0.94 and 1.97. At the lightest mass the residual comes from the tables: binned in 100 eV steps of $|\boldsymbol{q}|$, the loss function varies by about 15 % from bin to bin at these momenta, and the shell form samples a single q.

With a light mediator every mass gives a line at $\omega = 10.0\,\text{eV}$, an integrable cusp whose width is set by the energy resolution of the detector rather than by the halo. The last 0.5 eV below $|\Delta_\chi|$ holds 40, 42, 47 and 60 % of the rate, and the 0.1 eV around $|\Delta_\chi|$ holds 15, 12, 8 and 6.5 %. Its height follows from Eq. (10.55) with $L(0, 10\,\text{eV})$, which is 0.247 measured (Worked example 10.7) and 0.218 in the first bin of the tables. The two mediators therefore leave different signatures for the same splitting: a broad bump near 6 eV whose position hardly depends on the mass, or a line at the splitting itself. Neither sits at $\mu_{\chi e}|\Delta_\chi|/m_e$.

Nothing new is needed from the material: Eq. (10.53) uses the same loss functions as the elastic rate, now on a grid of ω that extends to $|\Delta_\chi| + \frac{1}{2} m_\chi v_{\max}^2$, and the light-mediator line needs the optical limit, which measurements give (Appendices E.6 and E.16).

Limit Check 10.12. Inelastic electron recoil in crystals

For a mass splitting Δ_χ, the unscreened crystal rate per unit target mass is $\mathrm{d}\mathcal{R}/\mathrm{d}\omega = [\rho_\chi \bar{\sigma}_e \alpha m_e^2/(m_\chi M_{\mathrm{cell}} \mu_{\chi e}^2)] \int \mathrm{d}q\, q^{-2} \eta(v_{\min}) F_{\mathrm{DM}}^2 |f_{\mathrm{crys}}(q,\omega)|^2$ with $v_{\min} = |(\omega + \Delta_\chi)/q + q/2m_\chi|$ [36, Eqs. (10), (11), (15), (16) and (20)], with the factor q^{-2} of [1, Eq. (3.13)] restored and the absolute value written out.

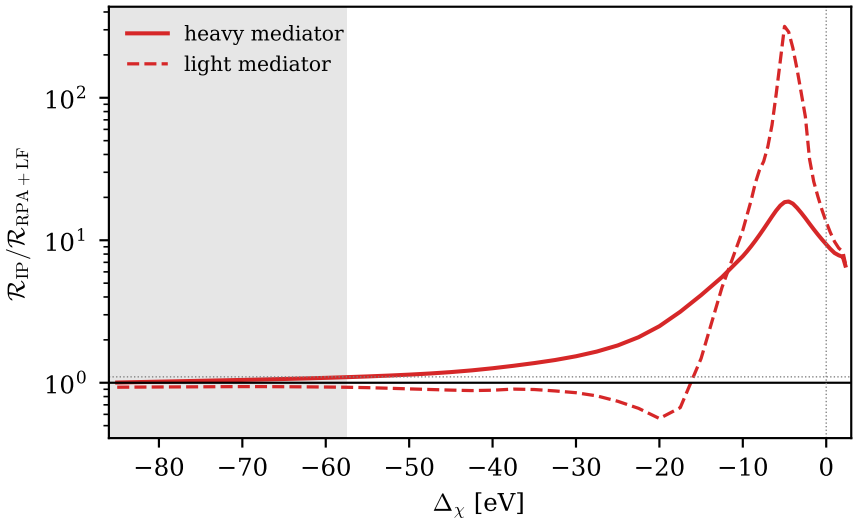


Figure 10.21: Ratio of the unscreened rate to the screened rate of silicon at $m_\chi = 1\,\mathrm{MeV}$, total above 1.11 eV, the step-model gap that accompanies the tables, against the splitting Δ_χ. Unscreened: ε_2 of the independent-particle head of the tables of Sect. 10.2, standing in for the crystal form factor of the box; screened: the random-phase loss function with local fields, Eq. (10.53). Standard halo of Sect. 1.2. Solid: heavy mediator; dashed: light mediator. Gray: $|\Delta_\chi| \geq 57.5\,\mathrm{eV}$, where the heavy-mediator ratio is within 10 % of one. The tables end at 89 eV, which fixes the left edge.

Below a splitting of about 50 eV the unscreened formula overestimates the exothermic rate by up to 18 for a heavy mediator and by up to 300 for a light one, whose line at $|\Delta_\chi|$ is suppressed by $|\varepsilon(|\Delta_\chi|)|^2$, and near the plasmon a light mediator reverses the sign of the correction.

Screening is the leading correction to the box, as in elastic scattering, and its size depends on the splitting. At 1 MeV the elastic rate is overestimated by 9.4 for a heavy mediator. An exothermic splitting of 5 eV raises the overestimate to 18.5, because the rate now collects around 5 eV at small q, where screening is strongest. Larger splittings move the weight up toward and above the plasma frequency, where the loss function and ε_2 approach each other: the ratio is 2.5 at −20 eV, 1.5 at −30 eV and 1.14 at −50 eV. For a light mediator the line at $|\Delta_\chi|$ samples $L(0, |\Delta_\chi|)$ against ε_2, a ratio $|\varepsilon|^2$ that reaches 317 at −5 eV and falls below one near the plasmon, 0.56 at −20 eV, where $|\varepsilon| < 1$ enhances the screened rate. Above 99 eV the $2p$ shell of silicon opens (Sect. 10.6), and the valence tables no longer suffice.

10.9.2 Boosted dark matter

A light halo particle can be accelerated after it leaves the halo's equilibrium. Cosmic-ray electrons and protons of MeV to GeV energies cross the dark-matter halo of the Galaxy, and a small fraction of them scatter on dark-matter particles along the way, which then carry kinetic energies comparable to their mass. The resulting flux at the Earth is the line-of-sight integral of the dark-matter density times the cosmic-ray flux and the scattering cross-section. For electrons, Liang and collaborators write it as $\mathrm{d}\Phi_\chi/\mathrm{d}T_\chi = D_{\mathrm{eff}}(\rho_\chi/m_\chi)\int \mathrm{d}T_e\,(\mathrm{d}\sigma_{\chi e}/\mathrm{d}T_\chi)(\mathrm{d}\Phi_e/\mathrm{d}T_e)$, with T_χ and T_e the kinetic energies and $D_{\mathrm{eff}} = 8.02\,\mathrm{kpc}$ an effective distance for a halo profile integrated to 10 kpc [37, Eq. (1)]. The flux is small, since only one halo particle in many meets a cosmic ray, but it reaches speeds close to that of light and with them energies far above $\frac{1}{2}m_\chi v_{\mathrm{max}}^2$.

The Sun accelerates a second component. Halo particles that fall into it scatter on the electrons of the solar core, at $1.57\times10^7\,\mathrm{K}$ at the center [38, Table 5], whose thermal speed $(3T/m_e)^{1/2}$ is about 0.09. Some leave with speeds of that order, far above the escape speed of the Galaxy but well below that of light. The DAMIC-M collaboration, Dark Matter in CCDs (DAMIC) at Modane, has searched for this solar-reflected component [39]. In both cases the halo average of Eq. (8.16) is replaced by an average over the flux. For a flux that is isotropic in the laboratory, the number density per unit speed is $(\mathrm{d}\Phi_\chi/\mathrm{d}v)/v$, and the non-relativistic rate keeps its form with

$$\frac{\rho_\chi}{m_\chi}\,\eta(v_{\mathrm{min}}) \;\to\; \int_{v_{\mathrm{min}}} \mathrm{d}v\;\frac{1}{v^2}\,\frac{\mathrm{d}\Phi_\chi}{\mathrm{d}v},$$

the replacement of Ref. [39, Eqs. (9)–(11)]. The real fluxes are anisotropic, the solar one pointing away from the Sun, and the angular average of the response that makes η sufficient is then an assumption. For speeds near that of light the non-relativistic rows themselves must be checked, and this is what we do now.

The non-relativistic potential of Sect. 8.1 assumed that the energy deposited is small against the momentum transfer, so that the mediator acts instantaneously. For a relativistic particle ω can approach q, and the mediator must be kept whole. Take a vector mediator of mass m_V coupled with strengths y_χ and y_e to the dark-matter current $J_\chi^\mu = \bar u(p')\gamma^\mu u(p)$ and to the electron current $J_e^\mu = (n_e, \boldsymbol{j}_e)$. We write $\mathcal{Q} = |\boldsymbol{q}|$ for the momentum transfer, to keep it apart from the four-momentum, and $E_\chi = (p^2 + m_\chi^2)^{1/2}$ for the energy, so that $\omega = E_\chi(\boldsymbol{p}) - E_\chi(\boldsymbol{p}-\boldsymbol{q})$ by Eq. (8.12). The amplitude is $y_\chi y_e(J_\chi^0 n_e - \boldsymbol{J}_\chi\cdot\boldsymbol{j}_e)/(\mathcal{Q}^2 - \omega^2 + m_V^2)$. Both currents are conserved. For the target, Eq. (4.29) gives $\hat{\boldsymbol{q}}\cdot\boldsymbol{j}_e \to (\omega/\mathcal{Q})n_e$ between eigenstates; for the dark matter, the Dirac equation gives $\hat{\boldsymbol{q}}\cdot\boldsymbol{J}_\chi = (\omega/\mathcal{Q})J_\chi^0$. Splitting each spatial current into its part along $\hat{\boldsymbol{q}}$ and its transverse part, $\boldsymbol{J}_\chi\cdot\boldsymbol{j}_e = (\omega^2/\mathcal{Q}^2)J_\chi^0 n_e + \boldsymbol{J}_{\chi\mathrm{T}}\cdot\boldsymbol{j}_{e\mathrm{T}}$, and the amplitude becomes

$$y_\chi y_e\,\frac{J_\chi^0\,n_e\,(1-\omega^2/\mathcal{Q}^2) - \boldsymbol{J}_{\chi\mathrm{T}}\cdot\boldsymbol{j}_{e\mathrm{T}}}{\mathcal{Q}^2 - \omega^2 + m_V^2}. \tag{10.56}$$

For a massless mediator the density term is $J_\chi^0 n_e/\mathcal{Q}^2$, the instantaneous Coulomb interaction, and only the transverse term propagates at the speed of light. Retardation enters the density row of a massive mediator as the factor $(1-\omega^2/\mathcal{Q}^2)$, and it comes from keeping the longitudinal current, which continuity ties to the density.

The density row needs the spin average of $|J_\chi^0|^2$. Write $\gamma\cdot p = \gamma^\mu p_\mu$ for the four-momentum $p = (E_\chi, \boldsymbol{p})$. The spinors of Eq. (C.127) obey $\sum_s u(p,s)\bar u(p,s) = \gamma\cdot p + m_\chi$, and the traces needed are $\mathrm{Tr}(\gamma^\mu\gamma^\nu) = 4g^{\mu\nu}$, $\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho})$, and zero for an

odd number of matrices. Then

$$\frac{1}{2}\sum_{s,s'}\left|\bar{u}(p',s')\gamma^0 u(p,s)\right|^2 = \frac{1}{2}\,\mathrm{Tr}\left[(\gamma\cdot p' + m_\chi)\gamma^0(\gamma\cdot p + m_\chi)\gamma^0\right]$$
$$= 2(E_\chi E'_\chi + \boldsymbol{p}\cdot\boldsymbol{p}' + m_\chi^2) = (2E_\chi - \omega)^2 - \mathcal{Q}^2,$$

where the four-matrix trace gives $4(2E'_\chi E_\chi - p'\cdot p)$ with $p'\cdot p = E'_\chi E_\chi - \boldsymbol{p}'\cdot\boldsymbol{p}$, and the last step uses $E'_\chi = E_\chi - \omega$ and $2\boldsymbol{p}\cdot\boldsymbol{p}' = p^2 + p'^2 - \mathcal{Q}^2$. Dividing by the normalization $2E_\chi \cdot 2E'_\chi$ of the two dark-matter states, squaring Eq. (10.56) and writing the couplings through $\bar{\sigma}_e$ as in Eq. (1.34),

Relativistic density row

$$\mathfrak{D}_{nn} = \frac{\pi\bar{\sigma}_e}{\mu_{\chi e}^2}\,\frac{(2E_\chi-\omega)^2 - \mathcal{Q}^2}{4E_\chi(E_\chi-\omega)}\left(1-\frac{\omega^2}{\mathcal{Q}^2}\right)^2\left[\frac{q_0^2+m_V^2}{\mathcal{Q}^2-\omega^2+m_V^2}\right]^2, \tag{10.57}$$

which tends to $\pi\bar{\sigma}_e F_{\mathrm{DM}}^2/\mu_{\chi e}^2$ as $v\to 0$. The rate per particle is Eq. (8.15) with $S_{nn} = (2\mathcal{Q}^2/e^2)\bar{L}$, and per unit target mass $\mathrm{d}\mathcal{R}/\mathrm{d}\omega = \rho_T^{-1}\int \mathrm{d}v\, v^{-1}(\mathrm{d}\Phi_\chi/\mathrm{d}v)\,\mathrm{d}\Gamma/\mathrm{d}\omega$ for an isotropic flux.

This constructs the relativistic tensor that Sect. 8.2.3 left open, for the density row. The transverse row enters through $\boldsymbol{J}_{\chi\mathrm{T}}$, whose square for a particle much heavier than the energy deposited is $4E_\chi^2 v_\mathrm{T}^2$, with $v_\mathrm{T}^2 = v^2 - (\boldsymbol{v}\cdot\hat{\boldsymbol{q}})^2 = v^2 - \omega^2/\mathcal{Q}^2$. For a massless mediator the medium answers a transverse current through its transverse dielectric function ε^T of Sect. 4.4, and the propagator $1/\mathcal{Q}^2$ of the density row becomes $1/(\mathcal{Q}^2 - \omega^2\varepsilon^\mathrm{T})$ in the transverse row. After averaging over the directions of $\boldsymbol{q}$ the density and the transverse current are uncorrelated, and the two rows add. The loss function of the density row is replaced by

$$\bar{L}(\mathcal{Q},\omega) \;\to\; \bar{L}(\mathcal{Q},\omega) + \left(v^2 - \frac{\omega^2}{\mathcal{Q}^2}\right)\mathrm{Im}\left[\frac{-\mathcal{Q}^2}{\omega^2\varepsilon^\mathrm{T}(\mathcal{Q},\omega) - \mathcal{Q}^2}\right], \tag{10.58}$$

the form of Fermi's energy-loss formula for a fast charge [40], which a calculation of the transverse response of silicon writes for dark matter [41, Eq. (2)].

We can bound the retardation factor. For any particle whose energy is a convex function of its momentum, $E_\chi(p) - E_\chi(p') \le (\mathrm{d}E_\chi/\mathrm{d}p)(p-p')$, and $p - p' \le \mathcal{Q}$ by the triangle inequality, so

$$\omega \le v\,\mathcal{Q}, \qquad \left(1 - v_{\chi,\max}^2\right)^2 \le \left(1-\frac{\omega^2}{\mathcal{Q}^2}\right)^2 \le 1, \tag{10.59}$$

with $v_{\chi,\max}$ the largest speed in the flux. A rate computed without the factor is therefore too large by at most $(1 - v_{\chi,\max}^2)^{-2}$. For speeds up to 0.1 this is 2 %, and up to 0.2 it is 9 %. For the solar-reflected flux, with typical speeds of 0.1, the non-relativistic rate with η replaced by the flux integral therefore stands at the level of a few percent. For cosmic-ray-boosted dark matter, with $v\to 1$, the bound says nothing, and the answer depends on where in $(\mathcal{Q},\omega)$ the rate collects.

The published treatment of cosmic-ray-boosted dark matter in silicon and germanium writes the density row as Eq. (10.57) without the factor $(1-\omega^2/\mathcal{Q}^2)^2$, with the mediator form factor evaluated at the four-momentum transfer $\mathcal{Q}^2 - \omega^2$ [37, Eqs. (4)–(6)]. Its supplementary derivation couples $\bar{u}\gamma^0 u$ to the density alone and drops the longitudinal current that continuity ties to it. Both forms reduce to the same non-relativistic limit, so that limit does not test the difference. The ratio of the two rates is the average of $(1-\omega^2/\mathcal{Q}^2)^2$ over the published integrand, which Eq. (10.59) confines to $[(1 - v_{\chi,\max}^2)^2, 1]$. For a heavy mediator the rate

of silicon collects at $\mathcal{Q}$ of keV, where $\omega/\mathcal{Q}$ is tiny: the factor $(1-\omega^2/\mathcal{Q}^2)^2$ changes the published rate by 10^{-4} and the transverse row by at most 10^{-4}, and the heavy-mediator limit is unchanged. For a massless mediator the published form factor $q_0^4/(\mathcal{Q}^2-\omega^2)^2$ grows without limit as $\mathcal{Q}\to\omega$, where the exact longitudinal interaction is the Coulomb $q_0^4/\mathcal{Q}^4$. For a particle of speed v the exact density row of silicon is then 0.997 of the published one at $v=0.1$, 0.94 at 0.5, 0.65 at 0.9 and 0.17 at 0.99 (Limit Check 10.13). Because the boosted flux is itself proportional to $\bar\sigma_e$, a change R of the detection rate moves the limit by $R^{-1/2}$. For the light mediator the shift of $\bar\sigma_e$ is below 1 % for $m_\chi \geq 10\,\mathrm{keV}$. Below about 3 keV it is undetermined: the published row has a pole at $\mathcal{Q}=\omega$, and its weight depends on the treatment of low kinetic energies of the flux, which the source does not state. Depending on that treatment the ratio of the density rows ranges from 0.05 to 0.93 at 1 keV. Appendix E.16 specifies the recast with the exact row.

The transverse row of Eq. (10.58) contains the Cherenkov term of Fermi's formula, standard for fast charged particles in matter and in the photo-absorption-ionization model of energy loss in silicon [40, 42]; what is new here is only its size in a boosted-dark-matter rate in silicon with a massless mediator. Between the gap and the direct gap silicon is nearly transparent, with $\varepsilon_1 = 12.5$ at the gap rising to 44.9 at 3.3 eV in the measured data at 300 K. A particle faster than the light of the medium, $v > \varepsilon_1^{-1/2}$, which is 0.28 at 1.2 eV and 0.15 at 3.3 eV, emits Cherenkov radiation. The condition $\omega^2\varepsilon_1 = \mathcal{Q}^2$ is then met inside the kinematic range $\mathcal{Q} \geq \omega/v$, and the transverse term has a pole there, broadened only by the small ε_2. We take the measured ε in place of ε^{T}, valid at these small $\mathcal{Q}$, and a massless mediator. Below 2 eV the transverse row then exceeds the density row, which is suppressed there by $1/|\varepsilon|^2$, by a factor of 49 at $v=0.3$ and 530 at $v=0.5$. Integrated over the spectrum above the gap, the transverse share is 2.7×10^{-5} at $v=0.1$, 1 % at 0.5 and 5.5 % at 0.99; in the plasmon window of 10 to 20 eV it is 7×10^{-4} at 0.5 and 7×10^{-3} at 0.99. A model of the homogeneous electron gas finds the transverse loss function four to six orders of magnitude below the longitudinal one at the plasmon, and corrections of up to 8 % below 1 eV [41]. The two results differ for two reasons: the gas has no transparent window above a gap, and the rate weights momenta down to ω/v, where the transverse response is largest. The Cherenkov photons are real photons, and they make electron–hole pairs only where the crystal absorbs them, within $1/(2\omega\,\mathrm{Im}\,n_r)$, which is 0.4 mm at 1.2 eV and 12 µm at 1.5 eV in the warm data and longer in a cold crystal (Sect. 10.7.2). Whether they count is a question of the size and temperature of the detector.

Limit Check 10.13. Boosted dark matter: the density row

Relativistic density row of Ref. [37, Eqs. (4)–(6)]: $\Gamma(\boldsymbol{p}) = \int \mathrm{d}^3q/(2\pi)^3\,|V|^2(2\mathcal{Q}^2/e^2)\,\mathrm{Im}[-1/\varepsilon(\mathcal{Q},\omega)]$ with $|V|^2 = \pi\bar\sigma_e[(2E_\chi-\omega)^2 - \mathcal{Q}^2]F_{\mathrm{DM}}^2/[4\mu_{\chi e}^2E_\chi(E_\chi-\omega)]$ and F_{DM} at $\mathcal{Q}^2-\omega^2$; for slow fluxes the non-relativistic rate with $(\rho_\chi/m_\chi)\eta \to \int \mathrm{d}v\,v^{-2}\mathrm{d}\Phi_\chi/\mathrm{d}v$ [39, Eqs. (9)–(11)]; transverse row as in Eq. (10.58) [41, Eq. (2)].

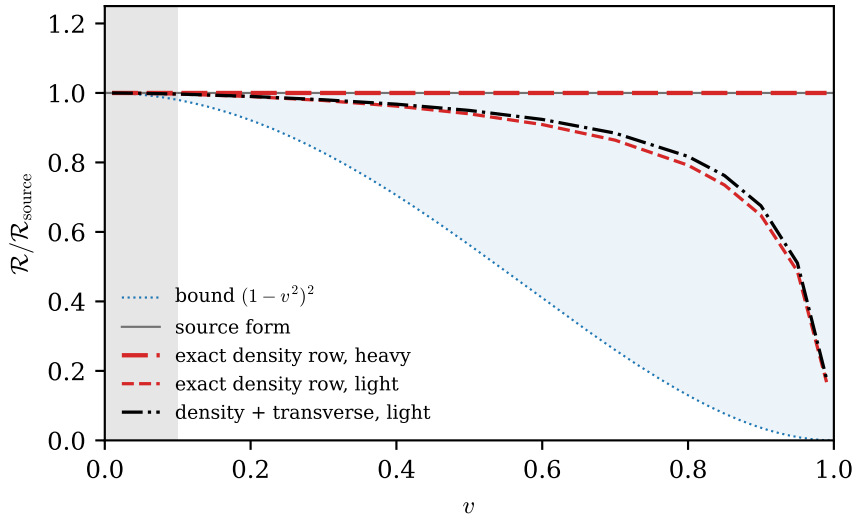


Figure 10.22: Electron-recoil rate of silicon for dark matter of speed v, much heavier than the energy deposited, divided by the rate of the published density row, against v. Red, long dashes: the exact density row Eq. (10.57) with a heavy mediator, which coincides with the published form to 10^{-4}; red, short dashes: the same with a massless mediator; dash-dotted: the massless mediator with the transverse row of Eq. (10.58) added. Blue band: the range $[(1-v^2)^2, 1]$ allowed by Eq. (10.59). Gray: $v \leq 0.1$, where the bound guarantees agreement to 2 %. Rates are integrated over ω from 1.11 eV, the step-model gap that accompanies the tables, to 89 eV and over $\mathcal{Q}$ from ω/v to 22.5 keV. Loss function: random-phase with local fields, from the tables of Sect. 10.2, held at its value at 493 eV below that momentum. Transverse function: the measured $\varepsilon(\omega)$ of silicon at 300 K [29], with its Cherenkov pole integrated exactly; the transverse row counts the Cherenkov photons emitted above the gap as deposited energy, wherever they are absorbed.

Above about $v = 0.3$ the published density row overestimates the rate for a light mediator by an amount that grows without bound as $v \to 1$; the published light-mediator limit moves by less than 1 % for $m_\chi \geq 10$ keV and is undetermined below about 3 keV.

The published row errs on the dark-matter side, not in the target: the response is the loss function in whatever many-body approximation the calculation uses, while $\mathfrak{D}$ omits a factor that the conserved currents fix. The transverse row is the leading correction on the target side, and it is first order in v^2 times the transverse response.

10.9.3 Fermionic absorption on electrons

A fermion χ of 100 keV that is absorbed on an electron and leaves as a massless $\tilde{\nu}$ deposits $\omega = m_\chi - q$ (Eq. (8.14)). On a free electron at rest this is the recoil $q^2/2m_e$, so $q \simeq m_\chi$ and $\omega \simeq m_\chi^2/2m_e = 9.8$ keV, a deposit that no halo particle of that mass could give by scattering. The momentum transfer is set by the rest energy and not by the speed of the dark matter, so that the rate is a single line through the response, independent of the halo. Sect. 8.2.3 showed that the master formula holds with a sum over the momenta of $\tilde{\nu}$ and left the dark-matter tensor to be built here.

Take the operators $\Lambda_F^{-2}(\bar{\tilde{\nu}} P_L \chi)(\bar{\psi}\psi)$ and $\Lambda_F^{-2}(\bar{\tilde{\nu}} \gamma_\mu P_R \chi)(\bar{\psi}\gamma^\mu\psi)$ of Sect. 8.1.5, with the chirality projectors $P_{L,R} = (1 \mp \gamma^5)/2$. Let $p = (m_\chi, \mathbf{0})$ be the four-momentum of χ at rest

and $k = (q, -\boldsymbol{q})$ that of $\tilde{\nu}$, so that $k \cdot p = m_\chi q$. On the target side the scalar bilinear is the density row n_e and the vector bilinear the current $J_e^\mu = (n_e, \boldsymbol{j}_e)$, up to terms of order q/m_e (Eq. (C.128)). On the dark-matter side, averaged over the spin of χ and summed over that of $\tilde{\nu}$, and with the trace identities of Sect. 10.9.2 together with $\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^5) = 0$ and $P_L(\gamma\cdot p + m_\chi)P_R = (\gamma\cdot p)P_R$,

$$\begin{aligned}\frac{1}{2}\,\mathrm{Tr}\big[(\gamma\cdot k)P_L(\gamma\cdot p + m_\chi)P_R\big] &= k\cdot p = m_\chi q,\\ \frac{1}{2}\,\mathrm{Tr}\big[(\gamma\cdot k)\gamma^\mu(\gamma\cdot p)\gamma^\nu P_L\big] &= k^\mu p^\nu + k^\nu p^\mu - g^{\mu\nu}k\cdot p + \mathrm{i}\epsilon^{\mu\nu\alpha\beta}k_\alpha p_\beta,\end{aligned}$$

up to the sign convention of the antisymmetric term. That term pairs with an antisymmetric current response, which a target with inversion symmetry and without magnetic order does not have, and we drop it. The relativistic normalization of the two dark-sector states divides by $2m_\chi \cdot 2q$. For the scalar operator the tensor is the number $\mathfrak{D}_{nn} = 1/4\Lambda_F^4$. For the vector operator, $\mathfrak{D}^{00} = 1/4\Lambda_F^4$, $\mathfrak{D}^{0i} = -q^i/(4q\Lambda_F^4)$ and $\mathfrak{D}^{ij} = \delta^{ij}/4\Lambda_F^4$, contracted with the responses of $J_\mu = (n_e, -\boldsymbol{j}_e)$. Continuity, $\hat{\boldsymbol{q}}\cdot\boldsymbol{j}_e \to (\omega/q)n_e$, turns the longitudinal current into the density, and the density, cross and longitudinal terms add to $(1+\omega/q)^2 S_{nn} = (m_\chi/q)^2 S_{nn}$. The transverse current, two components, keeps the weight one. With $\mathrm{d}^3q/(2\pi)^3 = q^2\mathrm{d}\omega/2\pi^2$ along the line $q = m_\chi - \omega$, and with $\sigma_F = m_\chi^2/4\pi\Lambda_F^4$,

Fermionic absorption on electrons

$$\begin{aligned}\frac{\mathrm{d}\Gamma_S}{\mathrm{d}\omega} &= \frac{\sigma_F}{2\pi}\,\frac{q^2}{m_\chi^2}\,\bar{S}_{nn}(q,\omega),\\ \frac{\mathrm{d}\Gamma_V}{\mathrm{d}\omega} &= \frac{\sigma_F}{2\pi}\Big[\bar{S}_{nn}(q,\omega) + \frac{q^2}{m_\chi^2}\,\mathrm{Tr}\,\bar{S}_{\boldsymbol{j}_\mathrm{T}\boldsymbol{j}_\mathrm{T}}(q,\omega)\Big], \qquad q = m_\chi - \omega,\end{aligned} \tag{10.60}$$

for the scalar and vector operators, at $v \to 0$ and for a massless $\tilde{\nu}$; the event rate per unit mass is $(\rho_\chi/m_\chi\rho_T)\Gamma$, independent of the velocity distribution.

For free electrons at rest, $\bar{S}_{nn} = 2\pi n_e\delta(\omega - q^2/2m_e)$ and the transverse current vanishes, and both rates are $n_e\sigma_F$ up to corrections of relative order m_χ/m_e, from the recoil of the electron. The operators' scale enters as σ_F, which Dror and collaborators call σ_e [43, Eqs. (1), (2) and (10)]; we keep $\bar{\sigma}_e$ for the reference cross-section of scattering.

For an atomic shell the density response is the ionization form factor of Eq. (2.22): with the binding energy E_B and the ejected energy $E_\mathrm{ej} = \omega - E_B$, the response per atom is $\pi|f_\mathrm{ion}(p_\mathrm{ej}, q)|^2/2E_\mathrm{ej}$, both spins counted in $|f_\mathrm{ion}|^2$ as in Ref. [1, Eqs. (A.18)–(A.21)]. The scalar rate of Eq. (10.60) then becomes $\mathrm{d}\langle\sigma v\rangle_S/\mathrm{d}E_\mathrm{ej} = \sigma_F q^2|f_\mathrm{ion}|^2/(4E_\mathrm{ej}m_\chi^2)$, which is Ref. [43, Eq. (10)] term by term. Their general rate [43, Eq. (7)] is the same formula with the dark-matter tensor written through the free-electron squared amplitude, and their ionization form factor [43, Eq. (8)], with $4p_\mathrm{ej}^3$ in place of $2p_\mathrm{ej}^3$, counts the two spins explicitly. The convention matters: a table that counts one spin and a formula that assumes two differ by a factor of two.

The vector operator is where the two treatments part. Ref. [43, Eq. (9)] multiplies $|f_\mathrm{ion}|^2$ by the squared amplitude of absorption on a free electron, evaluated at the bound momentum $q = m_\chi - \omega$. Its ratio to the density term of Eq. (10.60) is

$$\begin{aligned} r &= \frac{q\,[2m_e m_\chi(2m_e + m_\chi) - 2m_e^2 q - q^2(2m_e + m_\chi)]}{2m_e^2 m_\chi^2}\\ &= 1 - \frac{\omega^2}{m_\chi^2} + \frac{\omega(m_\chi - \omega)(2m_\chi - \omega)}{m_e m_\chi^2} + O\Big(\frac{m_\chi^2}{m_e^2}\Big).\end{aligned} \tag{10.61}$$

At free kinematics, $\omega = q^2/2m_e \ll m_\chi$, the ratio is one up to $O(m_\chi^2/m_e^2)$. For a bound electron $\omega \geq E_B$, and when m_χ is a few times E_B the term $-\omega^2/m_\chi^2$ is of order one. The free-electron amplitude assumes that the longitudinal current is $(q/2m_e)n_e$, the free value, while continuity requires $(\omega/q)n_e$, which near the threshold of an inner shell is larger by an order of magnitude or more. The free-vertex form is therefore not current-conserving at bound kinematics, and it underestimates the rate. Within the non-relativistic, local-potential framework of Eq. (10.60) the vector rate is the density term plus the transverse-current term, a sum of squares that can only add, so the current-conserving rate exceeds the free-vertex form by at least the factor $1/r$ wherever $r < 1$. In a hydrogen-like model of the xenon $1s$ shell, with $E_B = 33.3\,\mathrm{keV}$ [43, Table I], $r = 0.59$ at $m_\chi = 56.5\,\mathrm{keV}$, the mass at which Dror and collaborators fit an excess of electron recoils, and 0.89 at 100 keV: the free vertex is low by at least a factor 1.7 at the fitted mass. Only the relativistic corrections to the bound states, of relative size $(Z\alpha)^2 \approx 0.155$ for xenon $1s$ (Appendix C.15), have a sign that is not known.

A light mediator changes the rate in a way that the dark-matter speed does not control. For the scalar operator mediated by a scalar of mass m_{med} with Yukawa couplings y_{ee} and $y_{\chi\tilde{\nu}}$, Λ_F^{-2} becomes $y_{ee}y_{\chi\tilde{\nu}}/(m_{\mathrm{med}}^2 - t)$, with t the squared four-momentum transfer. On the $\tilde{\nu}$ leg, $t = (p-k)^2 = \omega^2 - q^2 = 2m_\chi\omega - m_\chi^2$, fixed by the rest energy, unlike the $-q^2$ of scattering. The rate is then the contact rate times $[m_{\mathrm{med}}^2/(m_{\mathrm{med}}^2 + m_\chi^2 - 2m_\chi\omega)]^2 = 1 - 2(m_\chi^2 - 2m_\chi\omega)/m_{\mathrm{med}}^2 + \ldots$, the correction of relative order $(m_\chi/m_{\mathrm{med}})^2$ that Cox, Dolan and Wood attach to the contact limit [44, Eqs. (1), (2), (14) and (15)]. It also tilts the spectrum, raising the high-ω end.

The two halves of this section meet when $\tilde{\nu}$ is massive. With $m_{\tilde{\nu}} = m_\chi + \Delta_\chi$ and $|\Delta_\chi| \ll m_\chi$, the outgoing fermion is slow, and Eq. (8.13) gives ω exactly as for inelastic scattering, with the recoil term computed with m_χ instead of $m_{\tilde{\nu}}$ and the tensor multiplied by $m_\chi/m_{\tilde{\nu}}$, both corrections of relative order Δ_χ/m_χ (Appendix D.24). Absorption into a nearly degenerate fermion is exothermic or endothermic scattering, and Eq. (10.53) applies.

In practice the responses on the line $\omega = m_\chi - q$ reach far above the valence scale. For m_χ from the gap to 1 MeV the momentum transfer runs into the cores of the atoms. The density row and the transverse current row are then needed from all-electron states, with Dirac–Hartree–Fock orbitals where $q \gtrsim m_e/3$. Appendix E.26 specifies the calculation, which also sizes the term that the free-vertex form omits.

Limit Check 10.14. Fermionic absorption on electrons

Per atomic shell, $\mathrm{d}\langle\sigma v\rangle/\mathrm{d}E_{\mathrm{ej}} = \overline{|\mathcal{M}|^2}\, q\, |f_{\mathrm{ion}}|^2/(64\pi m_\chi m_e^2 E_{\mathrm{ej}})$ with $q = m_\chi - \omega$; scalar operator $\sigma_F q^2 |f_{\mathrm{ion}}|^2/(4E_{\mathrm{ej}}m_\chi^2)$; vector operator $\sigma_F q |f_{\mathrm{ion}}|^2[2m_e m_\chi(2m_e + m_\chi) - 2m_e^2 q - q^2(2m_e + m_\chi)]/(8m_e^2 m_\chi^2 E_{\mathrm{ej}})$ [43, Eqs. (7)–(10) and (30)], with $|f_{\mathrm{ion}}|^2$ counting both spins [1, Eqs. (A.18)–(A.21)].

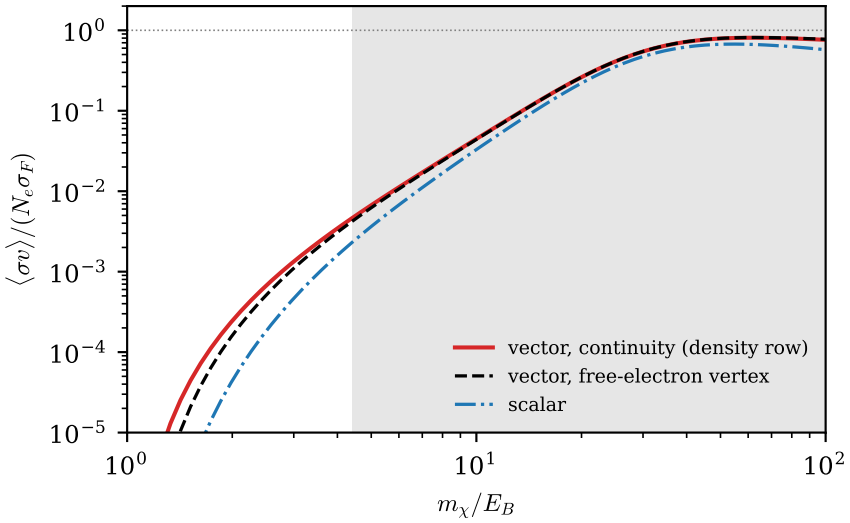


Figure 10.23: Absorption of a fermion on the $1s$ shell of silicon in a hydrogen-like model, per electron and in units of the free-electron value σ_F, against m_χ/E_B. Solid: vector operator, density term of Eq. (10.60) with continuity; dashed: the free-electron vertex of the box; dash-dotted: scalar operator, Eq. (10.60), equal to the box. The transverse-current term of the vector operator is not included. Model: non-relativistic bound–free transitions of a $1s$ shell of two electrons with nuclear charge $Z_{\text{eff}} = 11.6$, chosen to give the measured binding $E_B = 1839\,\text{eV}$ [30]; its continuum oscillator strength is 0.435 at $q \to 0$ and tends to one at large q, as for hydrogen. Gray: $m_\chi \geq 4.4\,E_B$, where the two vector forms agree within 10 %. Dotted: the free-electron value; the left edge of the plot is the threshold $m_\chi = E_B$.

When the mass falls below a few binding energies, the free-electron vertex lies below the current-conserving density term by tens of percent, and the transverse current, not included here, can only raise the current-conserving rate further.

The box treats the atom as independent electrons and takes the vector tensor from free kinematics. In the hydrogen-like model the bound region is also where the rate is small: at $m_\chi = 2E_B$ the momentum transfer is at most a twentieth of the shell's momentum $Z_{\text{eff}}\alpha m_e = 43\,\text{keV}$, and the rate is 2×10^{-4} of the free value. The departure matters wherever a search lives near a shell threshold, as the xenon example shows.

10.9.4 In-medium decay of an excited state

An excited state χ_2 that survives from the early universe, or that a scattering in the Earth produces, can decay to χ_1 inside the detector without touching anything, emitting a photon of energy $|\Delta_\chi|$. A transition magnetic moment μ_M between the two states, $H = -\mu_\text{M}\boldsymbol{\sigma}\cdot\boldsymbol{B}$ with $\boldsymbol{\sigma}$ the Pauli matrices between the spin states of χ_2 and χ_1, gives the vacuum rate

$$\Gamma_\text{vac} = \frac{\mu_\text{M}^2 |\Delta_\chi|^3}{\pi}, \tag{10.62}$$

for $|\Delta_\chi| \ll m_\chi$. This is the electric-dipole rate $\omega^3|\boldsymbol{d}|^2/3\pi$ of Heaviside–Lorentz units, with $|\boldsymbol{d}|^2$ replaced by the spin sum $\mu_\text{M}^2 \sum_f |\langle f|\boldsymbol{\sigma}|i\rangle|^2 = 3\mu_\text{M}^2$. Baryakhtar and collaborators estimate the lifetime for a dipole operator $\bar{\chi}_2 \sigma^{\mu\nu}\chi_1 F_{\mu\nu}$ with coefficient $1/\Lambda_d$ as $\tau^{-1} \sim \pi|\Delta_\chi|^3/\Lambda_d^2$ [45, Eqs. (2.1) and (2.2)]. In the non-relativistic limit $\sigma^{\mu\nu}F_{\mu\nu} \to -2\boldsymbol{\sigma}\cdot\boldsymbol{B}$, so that $\mu_\text{M} \simeq 2/\Lambda_d$,

and Eq. (10.62) gives $4|\Delta_\chi|^3/\pi\Lambda_d^2$, the same scaling. No operator of the target appears in the amplitude, and the master formula does not apply as it stands. The medium enters through the photon.

The decay rate follows from the golden rule with the magnetic field of the medium as the final state. At $T = 0$ the spectral density of the field is twice the imaginary part of its propagator, and in a medium the transverse propagator is $1/(\mathcal{Q}^2 - \omega^2\varepsilon^{\mathrm{T}})$, with $\boldsymbol{B} = \mathrm{i}\boldsymbol{q} \times \boldsymbol{A}$. The spin average gives $\frac{1}{2}\,\mathrm{Tr}(\sigma^i\sigma^j) = \delta^{ij}$, the transverse projection of $\boldsymbol{q} \times \boldsymbol{A}$ gives $2\mathcal{Q}^2$, and with $\omega = |\Delta_\chi|$, neglecting the recoil of χ_1,

$$\Gamma = \frac{2\mu_{\mathrm{M}}^2}{\pi^2} \int_0^\infty \mathrm{d}\mathcal{Q}\; \mathcal{Q}^4\; \mathrm{Im}\, \frac{1}{\mathcal{Q}^2 - \omega^2\varepsilon^{\mathrm{T}}(\mathcal{Q},\omega)}. \tag{10.63}$$

In vacuum $\mathrm{Im}[1/(\mathcal{Q}^2 - \omega^2 - \mathrm{i}0)] = \pi\delta(\mathcal{Q}^2 - \omega^2)$, and Eq. (10.63) returns Eq. (10.62). In a transparent medium, $\varepsilon_2 \to 0$, the delta function sits at $\mathcal{Q} = \omega n_r$ with the refractive index $n_r = \sqrt{\varepsilon}$, and the rate is $n_r^3\Gamma_{\mathrm{vac}}$. The same calculation with the electric field, $\boldsymbol{E} = \mathrm{i}\omega\boldsymbol{A}$, gives $n_r\Gamma_{\mathrm{vac}}$ for an electric dipole, local-field factors aside. The difference has a simple reason: the density of photon states grows as n_r^2 times the group index n_{grp}, the electric field per photon falls as $1/(n_r n_{\mathrm{grp}})$, and the magnetic field per photon is n_r times the electric one.

Where the medium absorbs, Eq. (10.63) changes character. At large $\mathcal{Q}$ the integrand tends to $\omega^2\varepsilon_2^{\mathrm{T}}$, and the integral grows with the momentum up to which $\varepsilon_2^{\mathrm{T}}(\mathcal{Q},\omega)$ stays finite. That large-momentum part is not a photon. It is energy handed directly to the electrons near the particle through a virtual photon, a scattering $\chi_2 + \text{target} \to \chi_1 + \text{excitation}$. Its rate is the master formula with $\Delta_\chi < 0$, the magnetic-dipole operators of Sect. 8.1 and the transverse current row: the exothermic process of Sect. 10.9.1 with a different vertex. The decay thus splits at an interface. The radiative part, a photon of energy $|\Delta_\chi|$ that is absorbed later or escapes, lies outside the response picture and is listed among the processes outside the framework in Sect. 22.3. The non-radiative part lies inside it and needs ε^{T}, or the transverse current response, at finite $\mathcal{Q}$ (Appendix E.16).

Silicon shows the three regimes. Below the gap it is transparent, with $n_r = 3.44$ at 0.5 eV and 3.54 at 1.1 eV, and an excited state with a splitting there decays $n_r^3 = 41$ to 44 times faster inside the crystal than in vacuum, into a photon that the crystal does not absorb. Between 3.2 and 22.5 eV, where $\mathrm{Im}\, n_r > 0.1\,\mathrm{Re}\, n_r$, the decay is non-radiative, and its rate is the master-formula rate. Above the plasma frequency n_r approaches one, from below as $1 - \omega_p^2/2\omega^2$ for a free electron gas and with small excursions at the $2p$ edge, and the vacuum rate holds within 10 % for every splitting above 55 eV.

Limit Check 10.15. In-medium decay of an excited state

Vacuum decay through a transition dipole: $\tau^{-1} \sim \pi|\Delta_\chi|^3/\Lambda_d^2$ for the operator $(1/\Lambda_d)\bar{\chi}_2\sigma^{\mu\nu}\chi_1 F_{\mu\nu}$ [45, Eqs. (2.1) and (2.2)], in the book's form $\Gamma_{\mathrm{vac}} = \mu_{\mathrm{M}}^2|\Delta_\chi|^3/\pi$ (Eq. (10.62)) with $\mu_{\mathrm{M}} \simeq 2/\Lambda_d$.

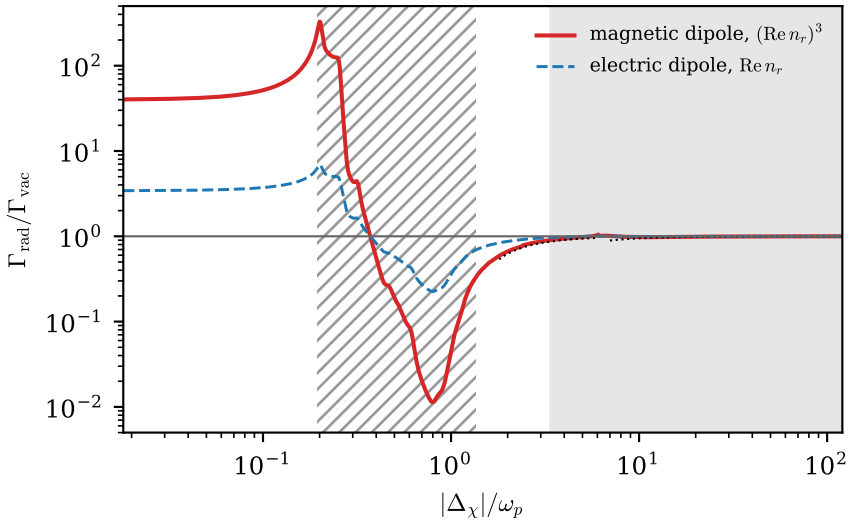


Figure 10.24: Radiative decay rate of an excited state inside silicon over the vacuum rate, against the splitting $|\Delta_\chi|$ in units of the valence plasma frequency $\omega_p^{\mathrm{val}} = 16.6\,\mathrm{eV}$. Red: magnetic dipole, $(\mathrm{Re}\, n_r)^3$; blue dashed: electric dipole, $\mathrm{Re}\, n_r$, local-field factors omitted. $n_r = \sqrt{\varepsilon}$ from the measured dielectric function of silicon at 300 K below 40 eV [29] and from atomic scattering factors above [30]. Dotted: the free-electron-gas asymptote $n_r^3 \simeq 1 - 3\omega_p^2/2\omega^2$, with ω_p^{val} below the $2p$ edge and the all-electron 31.1 eV above. Hatched: $\mathrm{Im}\, n_r > 0.1\,\mathrm{Re}\, n_r$, 3.2 to 22.5 eV, where the radiative factor loses its meaning and the decay is a non-radiative transfer given by Eq. (10.63) at finite $\mathcal{Q}$. Gray: splittings above 55 eV, $3.3\,\omega_p^{\mathrm{val}}$, where $(\mathrm{Re}\, n_r)^3$ is within 10 % of one.

Below about $3\omega_p^{\mathrm{val}}$ the vacuum rate fails inside a solid: it is enhanced by $n_r^3 \approx 40$ in the transparent window of silicon below the gap and replaced by non-radiative transfer to the electrons where the crystal absorbs.

No response of the target enters the vacuum rate; the medium acts only through ε^{T} at $\mathcal{Q} = \omega n_r$, and in a transparent medium the leading correction is the local field at the particle, which the macroscopic ε does not contain.

10.9.5 Open problems

Two published rates differ from the conserved-current results of this section. The free-electron vertex used for fermionic absorption on bound electrons underestimates the rate by at least $1/r$ when the dark-matter mass is a few binding energies, a factor 1.7 for the xenon $1s$ shell at 56.5 keV. The transverse current of the shell along the line $\omega = m_\chi - q$ can only raise the rate further, and only the relativistic corrections of a shell with $(Z\alpha)^2 \approx 0.155$ have an unknown sign. The full rate needs the transverse current response with Dirac–Hartree–Fock states (Appendix E.26), and the effect on the published limits is not known. The density row of cosmic-ray-boosted dark matter in the published treatment lacks the factor $(1 - \omega^2/\mathcal{Q}^2)^2$, which for a massless mediator lowers the rate of a particle at $v = 0.9$ by a third and of one at $v = 0.99$ by a factor of six. The heavy-mediator limit is unchanged. For the light mediator the limit moves by less than 1 % for $m_\chi \geq 10\,\mathrm{keV}$ and is undetermined below about 3 keV, where it depends on the low-energy end of the flux; fixing it needs the published calculation redone with the exact density row and the transverse row, Cherenkov pole included (Appendix E.16).

The target needs no new response for any of them beyond the density and transverse

current rows of Sect. 10.4, evaluated where these processes reach. Each of these rates inherits the uncertainties of the responses it uses, and Sect. 10.10 turns them into error bands.

10.10 Nuisance parameters of the electronic rate

A dark-matter particle of 3 MeV with a heavy mediator and $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$ makes 5.2×10^4 electron–hole pairs per kilogram-year in silicon if the electrons are independent, 2.5×10^4 if they screen in the random-phase approximation, and 1.85×10^4 once local fields are included. All three numbers come from the same published tables, counted from the low-temperature gap of 1.17 eV, and they differ only in the many-body approximation. The last one is the best computed so far, but it is not the rate. The electron–hole attraction raises it, the conduction bands above the edge are uncertain at the level of a tenth of an eV, and two published tabulations of the same independent-particle response disagree by a third at this mass. Sect. 8.8 turned each truncation into a nuisance parameter ϑ_k, with the computed or estimated step to the next order as its template. We give the electronic sector those templates with their sizes in silicon, collect them in one table, and put an error band of a factor 1.48 on the rate at 3 MeV. The open calculations stated in Sects. 10.4 to 10.9 are grouped at the end by the calculation that would close them.

10.10.1 Many-body spread

The hierarchy of approximations of Sect. 4.6 is not an expansion in a small parameter, and its steps do not shrink. At 3 MeV with a heavy mediator, random-phase screening lowers the independent-particle rate by 52 %, and local fields lower what remains by a further 26 %; for a light mediator the steps are 76 % and 18 %. The ratio of the independent-particle rate to the screened rate with local fields is 2.8 for a heavy mediator and 5.1 for a light one, between the values 2.3 and 9 that Sect. 10.2 found at 10 and 1 MeV. A spread of this kind is not an uncertainty of the best approximation. Once an approximation is computed it becomes the central value, and the nuisance that remains is the step after it (Sect. 8.8). For silicon that step is the electron–hole attraction, the excitonic (vertex) correction of the Bethe–Salpeter equation, and no finite-momentum calculation of it exists for silicon (Sect. 10.5).

Its size can be estimated from the model of Fig. 10.11. At $|\boldsymbol{q}| = 0.91\,\mathrm{keV}$ the attraction raises the screened loss function by 27 to 29 % from the onset to 3.5 eV, by 49 % at 4 eV, and by about 30 % above, where part of the rise is an artifact of four conduction bands. A particle of 3 MeV deposits at most $\frac{1}{2} m_\chi v_{\max}^2 = 10.5\,\mathrm{eV}$, and the mean energy of its spectrum is 4.7 eV. We take $\delta_{\mathcal{X}} = +0.28$ over that window, an estimate from a model at one momentum transfer, with the coupling block of Eq. (10.22) adding about ± 0.10 to it. The template has a definite sign: at $\vartheta_{\mathcal{X}} = 1$ the rate rises. By the f-sum identity Eq. (8.76) a many-body template is a tilt. Inside the 10.5 eV window it raises the spectrum nearly uniformly and so acts like a normalization; the weight it moves comes from energies above the window, which the particle cannot reach. The same template, read at 10 MeV, reaches 35 eV and samples part of the compensating loss, so one $\vartheta_{\mathcal{X}}$ changes the rates at 3 and 10 MeV by different fractions. The attraction also adds new support. The bound lines below the pair continuum carry the weight of the free continuum within $8.0\,E_x^b = 0.12\,\mathrm{eV}$ of the edge (Sect. 10.5.2), which in the tables holds 4×10^{-4} of the heavy-mediator rate at 3 MeV and 1.5×10^{-3} of the light-mediator rate. The tables resolve energies to 0.2 eV, so these shares are orders of magnitude, and they matter only for thresholds at the gap.

A second nuisance sits inside the approximation already computed. At momentum transfers of two to three times the shortest reciprocal-lattice vector the local fields lower the per-q rate of the tables by 1.67 at 8 keV and 1.83 at 10 keV (Sect. 10.2.2). That size rests on a

spectrum that holds only 17 % of the f-sum below 89 eV at 10 keV. The model of Fig. 10.2, built from a local pseudopotential with no input in common with the tables but the crystal structure, gives the same suppression. At 5 eV and between 2 and 2.8 q_0 it finds the screened structure factor with local fields at 0.54 to 0.61 of the independent-particle one, with the head screening already within 10 % of one above 2.2 q_0. The tables at the same energy and momenta give 0.40 to 0.48, with the head screening at 0.77 to 0.97 of the independent-particle value. Two calculations that share neither potential nor basis therefore agree that local fields hold the response near half of its unscreened value there, to within a quarter of each other. Both, however, lack the core-region oscillations of the valence states and the high-energy tail of the spectrum, and an all-electron calculation with enough bands is needed to fix the size (Appendices E.2 and E.6). The template is the change of the rate if the suppression above $2q_0 = 7.46$ keV were absent, $\vartheta = 1$ restoring the random-phase value there. It is +0.067 at 3 MeV for a heavy mediator, where only 7.5 % of the rate comes from those momenta, +0.009 for a light one, and +0.24 at 10 MeV, where they hold 29 % of the rate.

A third source is not a truncation of the label but of the numerics. The crystal form factor of Sect. 10.1 and the independent-particle head of the tables of Sect. 10.2 are the same quantity in the same approximation, computed with different codes, pseudopotentials and grids. With the same halo, each from its own onset, they give 1.60×10^5 and 1.36×10^5 events per kilogram-year at 10 MeV for a heavy mediator, a ratio of 1.18, and 1.29 for a light one. At 3 MeV the ratio is 1.34 for a heavy mediator and 1.55 for a light one. The difference sits at small q, where the lighter particle collects its rate, and the f-sum rule locates it. Between 0.3 q_0 and 0.7 q_0 the first moment of the form-factor table is 1.1 to 1.2 times the valence value (Sect. 10.1), and that of the independent-particle head of the tables is 0.93 at every such momentum, partly because those tables end at 89 eV. The scissor shift accounts for about ten percent of the excess of the first table, and the rest has no identified cause. Neither table conserves the f-sum at small q, and the two violations lie on either side of it. The spread is measured with independent particles, but the central value is screened, and screening is strongest exactly at the small q where the tables disagree. There a normalization error of ε_2 is partly screened out: if ε_2 and, by Kramers–Kronig, $\varepsilon_1 - 1$ are too large by $1 + \delta$ where $\varepsilon_1 \gg 1$, then $L \simeq \varepsilon_2/\varepsilon_1^2$ moves by about $1/(1 + \delta)$, opposite in sign and no larger. The independent-particle ratio is therefore an upper bound on the spread of the screened rate, and we carry it as such. Sect. 8.8 named this case: for a truncation that breaks the f-sum the violation measures the error, and the spread between the calculations is the nuisance until a converged calculation passes the test (Appendix E.7).

The Bethe–Salpeter step carries one more pitfall, an evaluation error rather than a truncation. A Bethe–Salpeter calculation that keeps only the resonant poles has the right ε_2 and the wrong ε_1, and it overstates the loss function by 2.12 at 1.39 keV and 1.3 eV, falling to within 15 % of one above 6.7 keV (Eq. (10.21)). Both time orderings remove it. It is listed below because a rate that inherits it is wrong by a factor that no nuisance of the correct calculation covers.

10.10.2 Gap and threshold nuisances

The tables carry a scissor shift that places the conduction bands at the measured gap of silicon at zero temperature, 1.17 eV [3, Sect. III B], the gap a detector at millikelvin temperatures has, with the zero-point renormalization included in the measurement (Sect. 3.4). The rate is counted from that gap, and the threshold mass is $m_{\min} = 2E_{\mathrm{g}}/v_{\max}^2 = 0.333$ MeV. The 1.11 eV that often accompanies these tables in analysis codes is a different number, the gap of the step model of pair creation with 3.6 eV per pair (Sect. 1.5), not the gap of the response. The energy nodes of the tables lie 0.2 eV apart and do not resolve the edge itself.

What remains depends on what is measured. The edge itself is known to a few meV, and a shift of 5 meV changes the 3 MeV rate by 0.4 %. A scissor, however, moves every conduction band by the same amount, and quasiparticle corrections in silicon differ from state to state by about 0.1 eV (Sect. 3.4). The bands above the edge are therefore uncertain at that level even when the edge is not. A shift of the whole spectrum by 0.1 eV bounds this effect from above: it changes the 3 MeV rate by -7.5 and $+8.0$ percent for a heavy mediator, by ∓ 10 percent for a light one, and the 10 MeV heavy-mediator rate by 4 %. The effect grows toward the threshold mass, where the spectrum samples only the first eV above the edge. For a target whose gap has not been measured the edge itself carries the 0.1 eV, and $\delta \ln m_{\min} = \delta \ln E_{\mathrm{g}}$ turns it into 8.5 % in the threshold mass of silicon (Worked example 3.7). Near that mass the halo supply changes by orders of magnitude over a factor of two in m_χ, and the rate there depends on the gap more than on any other entry of the table below. The exciton binding of 14.7 meV is a threshold nuisance of the same kind, which lowers the first line to 1.155 eV (Worked example 10.5), and Table 8.6 carries it.

10.10.3 Nuisance table of the electronic sector

Table 10.2 lists the nuisances of the electronic sector. Each row names the truncation, the type of its template in the classes of Sect. 8.8, its size in silicon where it is known, how the size was obtained, and the section where the corresponding open calculation is stated. A next order computed in a model gives a template whose sign and shape are known and whose value may change once the real calculation is done. An estimate from a small parameter gives the size to a factor of order one. A spread between published calculations at the same order stands in for a template where the error lies outside the truncation label. An inequality gives a range without a central value, and for some items only the omission can be named.

Two of these nuisances change the silicon rate at 3 MeV by more than ten percent: the electron–hole attraction, by about a quarter, and the spread between tabulations of the independent-particle response, by a third for a heavy mediator and by half for a light one. At 10 MeV the local fields at large momentum transfer join them, with a quarter, and the tabulation spread falls to a fifth. The gap stays below ten percent above 3 MeV but grows without limit toward the threshold mass. Outside the density rows the largest items are not nuisances in the sense of a computed step. The cross term of a velocity coupling and the final states of the germanium $3d$ shell are known only between bounds, a factor of ten apart for germanium near 30 MeV. A limit set on such a coupling has no central value yet, and the bound itself is what it can use (Sect. 10.4).

One rule governs how the rows enter a likelihood. A truncation that two rates share acts on both with one ϑ_k (Sect. 8.8). The attraction raises the rate at every mass and for both mediators at once, by different fractions, and a combination of masses or experiments that treated it as independent nuisances would let the data play one against the other. The inelastic, exothermic, boosted and fermionic rates of Sect. 10.9 read the same loss function and transverse current at shifted kinematics, and they carry the same nuisances with the same parameters; they add no rows of their own. Where one rate uses the tables and another the measured optical constants, as the scattering and absorption rates of Sects. 10.2 and 10.7 do, the two share the many-body truncation only through what the measurement contains, and the absorption rate carries the temperature row instead.

Worked example 10.10: The nuisance band on the silicon rate at 3 MeV

Take $m_\chi = 3\,\mathrm{MeV}$, a heavy mediator, $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$ and the standard halo of Sect. 1.2, with $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$. The kinematic window ends at $\omega = 10.5\,\mathrm{eV}$ and $q = 2 m_\chi v_{\max} =$

Table 10.2: Electronic nuisances. Type in the classes of Sect. 8.8: normalization, tilt (in ω or in q), threshold, new support, and class mix (weight moved between the zero-phonon and phonon-dressed sectors without changing the total). Basis of the size: "model" (next order computed in a model), "tables" (evaluated from the published tables), "est." (small-parameter estimate), "spread" (published calculations at the same order), "bound" (only an inequality is known), "named" (no size). Sizes are for silicon unless germanium is named, with rates at 3 MeV for a heavy mediator. The last column gives the section where the open calculation is stated.

Truncation	Type	Size	Basis	Section
Electron–hole attraction ($\mathcal{X}$: RPA + LF $\to$ BSE)	tilt; new support	L up by 27–29 % near the edge at 0.91 keV; rate $\delta_{\mathcal{X}} \approx +0.28$; lines 4×10^{-4} of the rate	model	Sect. 10.5
Coupling block dropped (BSE in Tamm–Dancoff form)	tilt	about 10 % of the proper ε_2	est.	Sect. 10.5
Resonant-only real part (evaluation error)	tilt in q	L $\times 2.12$ at 1.39 keV, 1.3 eV; within 15 % above 6.7 keV	tables	Sect. 10.5
Local fields at $q \gtrsim 2q_0$ (bands, pseudo-states)	tilt in q	per-q factor 1.67 (8 keV), 1.83 (10 keV); rate +0.066 if absent above $2q_0$ (+0.24 at 10 MeV)	spread	Sect. 10.2
Independent-particle tabulation (code, grid, pseudopotential)	normalization at small q	IP rates differ by 1.34 (1.18 at 10 MeV), an upper bound for the screened rate; f-sum 1.1–1.2 against 0.93 at 0.3–0.7 q_0	spread	Sect. 10.1
Quasiparticle gap (rigid scissor)	threshold	edge measured, few meV; bands 0.1 eV: rate $-7.5/+8.0$ %; $m_{\min}$ 8.5 % without a measured gap	spread	Sect. 3.4
Plasmon decay into pairs with phonons ($n_g = 2$)	class mix	branching 10^{-3} to 10^{-2}	est.	Sect. 10.3
Local fields of the transverse current	normalization	-4 to -15 % (model); -10 to -30 % (est.)	model	Sect. 10.4
Density–current cross term of a velocity coupling	normalization	rate at least 16 (IP), about 29 (screened) times the static-electron formula at 10 MeV	bound	Sect. 10.4
Triplet response beyond independent pairs	tilt	weight $\times 2.2$ below 3 eV (model); exchange splitting $\sim 10^{-3}$ of the binding	model	Sect. 10.5
Final states of the germanium 3d shell	normalization	3d/valence 0.07 (plane waves) against 0.69 (Fermi factor) at 30 MeV	bound	Sect. 10.6
Screening of core transitions	normalization	macroscopic 2 % at 12 keV (Ge); local fields not computed	est.	Sect. 10.6
Relativistic corrections (Pauli order)	normalization	about 10^{-3}, at most 2 % (Ge 3d); next order 3×10^{-6}	est.	Sect. 10.6
Spin–orbit mixing at the germanium edge	tilt	up to 10 % in amplitude	est.	Sect. 10.6
Terms of order p_ϕ^2 in scalar absorption	new support near ω_p	density row equal to the kinetic rate at 16–18 eV (model)	named	Sect. 10.7
Room-temperature optical data near the edge (n_T)	threshold	more than $\times 2$ within about 20 meV of the edge	est.	Sect. 13.5
Bound shells in dark Compton scattering	tilt	K shell of Si at 45–100 keV, of Ge up to 118 keV; L shells below 20 keV	named	Sect. 10.8
Axial pair absorption with spin–orbit coupling	new support	of order v^4 without it	named	Sect. 10.8
Average over the directions of $\boldsymbol{q}$	tilt in time	first anisotropic term $\ell = 4$	named	Sect. 11.7

15.9 keV. The central value is the loss-function rate Eq. (10.4) with the random-phase loss function with local fields of the tables of Sect. 10.2, counted from the low-temperature gap of 1.17 eV:

$$\mathcal{R}^{\mathfrak{T}} = 1.85 \times 10^{4}\,\mathrm{kg^{-1}\,yr^{-1}},$$

about 51 events per kilogram and day. Its spectrum has a mean energy of 4.7 eV, and 10 % of it lies below 3 eV.

Four nuisances act on this number with sizes in hand. In the power form Eq. (8.72), $\mathcal{R}(\vartheta) = \mathcal{R}^{\mathfrak{T}} \prod_k (1+\delta_k)^{\vartheta_k}$, each enters through $\ln(1+\delta_k)$:

$$\begin{aligned}
&\text{attraction, } \delta_{\mathcal{X}} = +0.28: && \ln(1+\delta) = +0.247,\\
&\text{tabulation spread (bound), } +0.335: && +0.289,\\
&\text{band energies, } 0.1\,\text{eV}: && -0.078,\\
&\text{local fields above } 2q_0,\ +0.066: && +0.064.
\end{aligned}$$

The four are separate truncations with independent standard-normal ϑ_k. To first order the logarithm of the rate then has the Gaussian width of Eq. (C.123), the four logarithms added in quadrature,

$$\sigma_{\ln \mathcal{R}} = \left(0.247^2 + 0.289^2 + 0.078^2 + 0.064^2\right)^{1/2} = 0.393,$$

and the rate lies between $\mathcal{R}^{\mathfrak{T}}/\mathrm{e}^{0.393}$ and $\mathcal{R}^{\mathfrak{T}}\mathrm{e}^{0.393}$ at one standard deviation:

$$1.25 \times 10^{4}\,\mathrm{kg^{-1}\,yr^{-1}} \le \mathcal{R} \le 2.75 \times 10^{4}\,\mathrm{kg^{-1}\,yr^{-1}},$$

a factor of 1.48 either way. The attraction and the tabulation spread carry 93 % of the variance. Taking the attraction at the upper value of the model, +0.49, widens the factor to 1.65; the coupling block, ± 0.10 on $\delta_{\mathcal{X}}$, changes $\ln(1+\delta_{\mathcal{X}})$ by 0.075.

The band is not symmetric about the best guess. The attraction has a known sign, and $\vartheta_{\mathcal{X}} = 1$ moves the central value to 2.37×10^{4}; the band of the power form describes the uncertainty of where the next approximation lands, not a symmetric spread around the present one. The tabulation spread has no known sign, since the two tables miss the f-sum on opposite sides. For a light mediator the same steps give a central value of 6.0×10^{3} and a factor of 1.67, with the tabulation spread at +0.55 and the gap at −0.10, and the local fields at large q below one percent.

The band covers the truncations that the density response of silicon carries at this mass. It does not cover the halo, which the likelihood of Eq. (8.78) holds fixed, nor the detector, whose nuisances Sect. 21.4 adds. It is also wider than the step between the two best computed approximations, 2.49×10^{4} to 1.85×10^{4}. That step is not an uncertainty, since it has been taken; the band covers the next step and the spread between calculations in the same approximation.

10.10.4 Open problems

A finite-momentum Bethe–Salpeter calculation, an all-electron random-phase calculation with local fields, crystal final states for the core, and second-order calculations of plasmon decay and bound Compton scattering would remove most rows of Table 10.2. A finite-momentum Bethe–Salpeter calculation for silicon and germanium, with the coupling block kept and the bound lines added from computed masses and vertices, fixes the attraction, the coupling block, the triplet channel and the spin–orbit mixing at the germanium edge (Sects. 10.5 and 10.6;

Appendix E.15). An all-electron random-phase calculation with local fields, with enough bands to hold the f-sum at 10 keV and with the current and kinetic rows carried as external rows of the Dyson equation. It fixes the local fields at large q, the screening of core transitions, the transverse-current fields, the in-medium ratio of velocity couplings and the order-p_ϕ^2 terms of scalar absorption (Sects. 10.2, 10.4 and 10.7; Appendices E.6 and E.16). Until these rows are computed, Eq. (10.15) is all a limit on a velocity coupling can use. Crystal conduction states for the germanium $3d$ shell, and a four-component response for the bound on relativistic terms, settle the core rows (Sect. 10.6). A second-order electron–phonon calculation at the plasma frequency, and shell Compton profiles, settle the plasmon branching and the bound-shell correction to dark Compton scattering (Sects. 10.3 and 10.8; Appendix E.26). The spread between tabulations needs no new physics, only one converged calculation that passes the f-sum test at small q.

The quadratic response, two published rates and the self-trapped excitons of NaI are open items that are not nuisances of the rates computed here. The interference between a contact amplitude and a second-order one needs the quadratic response of the crystal, which no reduction to two-point functions supplies (Sect. 10.8.4). The free-electron vertex of fermionic absorption on a bound shell underestimates the rate by at least $1/r$, a factor 1.7 for xenon $1s$ at 56.5 keV, with only the $(Z\alpha)^2$ correction of unknown sign; and the density row of cosmic-ray-boosted dark matter without its retardation factor overstates the rate at relativistic speeds. The boosted limit is unchanged for a heavy mediator and undetermined below about 3 keV for a light one; how far the absorption vertex moves published limits is not known (Sect. 10.9.5). And the self-trapped excitons of NaI, where the Wannier picture fails, are open in every quantity that the rate needs (Sect. 10.5). For silicon the conclusion is simpler. At 3 MeV the rate of a density coupling at clamped nuclei is known to within a factor of 1.5 at one standard deviation, and two items dominate that factor, the attraction and the disagreement between published independent-particle tables. One calculation would remove each.

Summary

An electron–hole pair in silicon costs at least the gap, and the rate of making pairs is a contraction of the master formula with the response of the electrons (Sect. 10.1). For independent electrons the structure factor is a sum over valence–conduction pairs with plane-wave vertices at the unfolded momentum transfer, and its angle average, rescaled, is the crystal form factor on which most published limits rest. Its normalization is where tabulations go wrong, and the f-sum rule tests it. From the published silicon form factor a 10 MeV particle with a heavy mediator and $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$ gives 1.60×10^5 events per kilogram-year, 27 % of them from $q > 2q_0$, where pseudo-wavefunctions lose accuracy. The excess of its f-sum at small q is only partly explained by the scissor shift.

Screening enters through one function, the loss function, and the loss-function rate is exact for a coupling to the total charge and for a coupling to the electron density at clamped nuclei or above the phonon band (Sect. 10.2). In the published silicon tables the 10 MeV rate is 1.36×10^5 for independent particles (two independent-particle tabulations differ by 18 %, Sect. 10.10.1), 8.6×10^4 with random-phase screening and 5.9×10^4 with local fields. The independent-particle rate is too high by 2.3 for a heavy mediator and 3.7 for a light one, and at the large momentum transfers that carry the heavy-mediator rate the local fields, not the head screening, do most of the work. The f-sum and the static step bound the rate of any target, loosely for silicon at MeV masses. A mediator that couples to electrons alone breaks the bound on the static step once the lattice relaxes, through the acoustic peak of the

electrons that follow the nuclei.

The electron gas at the valence density reproduces the silicon rate within a factor 1.5 between 4.2 and 9.5 eV, and fails below the absorption peaks, where it has no gap, and at large energies, where bound electrons carry momenta the gas lacks (Sect. 10.3). Its plasmon is the pole of the inverse dielectric function, but halo dark matter cannot reach it: the slowest plasmon needs a particle faster than $2860\,\mathrm{km\,s^{-1}}$. A plasmon made otherwise decays into pairs, and into pairs with phonons with a branching estimated at 10^{-3} to 10^{-2}.

Couplings to the electron spin and current need responses beyond the density (Sect. 10.4). Continuity, spin rotation and the product of inversion and time reversal reduce them to twelve real functions per momentum and energy in silicon and germanium, in a singlet block, screened, and a triplet block, unscreened. A velocity coupling adds the transverse current to the density, and a formula that keeps only the dark matter's velocity underestimates its rate in silicon at 10 MeV by at least a factor 16 for independent particles, and about 29 with the density row screened.

The electron–hole attraction moves weight toward the gap and adds bound lines below it (Sect. 10.5). In silicon the $1s$ line lies at 1.155 eV, below the low-temperature gap of 1.17 eV. In a model at 0.91 keV the attraction raises the screened loss function by 27 to 29 % near the edge. An evaluation that drops the antiresonant poles overstates the loss function by a factor 2.12 at small momentum transfer, and dropping the coupling block costs about 10 %. The triplet channel has the singlet's lines without its local fields or screening.

Core shells enter at large momentum transfer (Sect. 10.6). The germanium $3d$ shell opens at 11.3 keV and overtakes the valence at 12.5 keV; whether it dominates a rate depends on its final states, and with crystal states it does so above about 30 MeV at a one-pair threshold. A filled, flat shell responds as a sum of independent atoms, without interference with the valence. Relativistic corrections to the response are about 10^{-3} and at most 2 % for the shells halo dark matter reaches, and they shift the f-sum by -0.24 and -0.76 percent in silicon and germanium.

Absorbed bosons sample the response at zero momentum (Sect. 10.7). A dark photon is absorbed with the loss function at its mass, suppressed 761 times at 3 eV in silicon and enhanced 20 times at the plasmon, 16.4 eV. An axion-like particle couples to the spin current, whose response without spin–orbit coupling is three quarters of the current response and is not screened; in a static magnetic field it converts like a dark photon. A scalar couples to the kinetic energy of the electrons, and mixes with the photon only at order p_ϕ^2 in crystals without a polar axis.

Two insertions of the dark field reach rates that one insertion forbids (Sect. 10.8). The absorption of a particle and its antiparticle is single absorption at twice the mass, and the pseudoscalar pair is absorbed through the axioelectric row without velocity suppression. Dark Compton scattering is the main signal of a dark photon in silicon above 55 keV; its seagull form is exact to 1 % below 165 keV, and the published axion-like rate is too large by $(m_\phi + 2m_e)/(m_\phi + m_e)$, 1.836 at 100 keV.

Inelastic, exothermic, boosted and fermionic dark matter use the same responses at other kinematics (Sect. 10.9). An endothermic splitting of 1 eV raises the lightest mass that makes pairs to 0.60 MeV, and an exothermic one turns the spectrum into a halo-independent shell. For boosted particles with a massless mediator the exact density row is 0.65 of the published one at $v = 0.9$, and above $v = \varepsilon_1^{-1/2}$ the transverse row adds Cherenkov emission. An excited state below the gap decays radiatively 41 to 44 times faster inside silicon than in vacuum.

Each approximation of the chapter is a nuisance with a size (Sect. 10.10). At 3 MeV with a heavy mediator the silicon rate at the low-temperature gap is 1.85×10^4 events per kilogram-year, uncertain by a factor of 1.48 at one standard deviation. The electron–hole attraction and the spread between published independent-particle tables dominate; the gap

and the local fields at large q add less than ten percent each. Outside the density rows, the cross term of velocity couplings and the final states of the germanium $3d$ shell are known only between bounds. Throughout the chapter the nuclei were held fixed. Chap. 11 releases them, and the energy a light particle deposits goes into phonons.

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Chapter 11

Lattice excitations

Below about 10 MeV the largest free recoil energy that a nucleon-coupled dark-matter particle gives a silicon nucleus falls to the phonon energies, and the nucleus can no longer be treated as free. At that mass the largest momentum transfer is about 50 keV, the free recoil energy $q^2/2M$ is 50 meV, and that is the energy of the phonons themselves. Lighter particles give the nucleus less than one phonon's worth, and the energy goes into the lattice instead. It goes into a single acoustic or optical phonon at the lowest masses, into a few phonons in between, and, only at the top of the range, into a recoil broadened by the motion of the nucleus in its well. The electronic channels of Chap. 10 need an electron to cross a gap of an eV; these channels need only a phonon of a few meV, and with a meV threshold they reach masses fifty times smaller. What is the rate, across that whole range, and which numbers of a material decide it?

From the harmonic crystal of Chap. 5 and the master formula of Chap. 8, each regime of Fig. 11.1 reduces to a few material numbers: sound speeds, Born effective charges and Debye–Waller exponents, each checked against the fuller result. Throughout, the electrons stay in their ground state and follow the nuclei. The Migdal effect and phonon-assisted electron recoil, which start from this sector, follow in Chaps. 12 and 13.

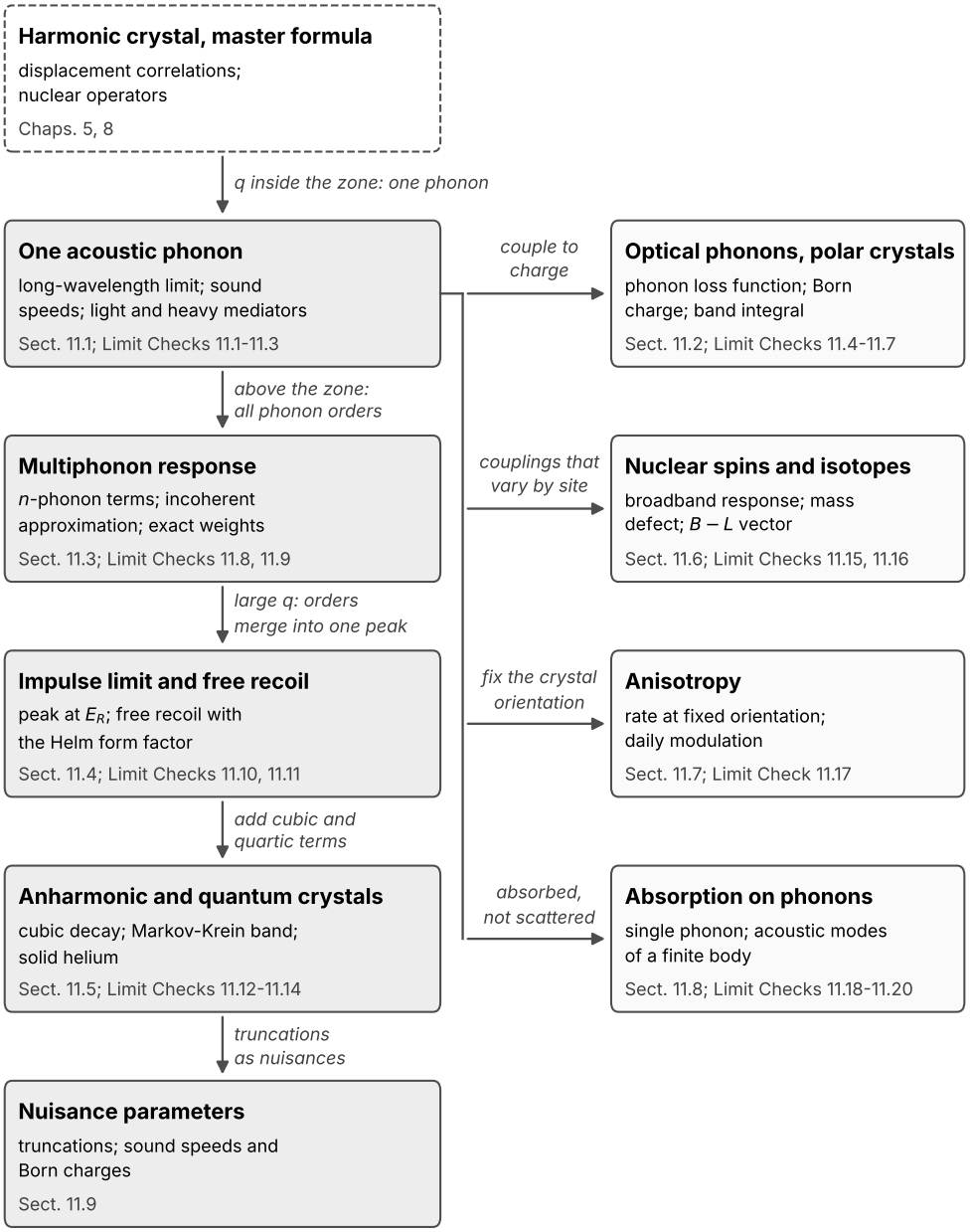


Figure 11.1: The lattice regimes of the nuclear response and their branches. Left, as the momentum transfer grows: one acoustic phonon inside the zone (Sect. 11.1; Limit Checks 11.1 to 11.3); all phonon orders above it (Sect. 11.3; Limit Checks 11.8 and 11.9); the orders merging into one peak at E_R, whose limit is the free nuclear recoil (Sect. 11.4; Limit Checks 11.10 and 11.11); cubic and quartic terms (Sect. 11.5; Limit Checks 11.12 to 11.14); the truncations as nuisances (Sect. 11.9). Right: coupling to charge excites optical phonons of polar crystals (Sect. 11.2; Limit Checks 11.4 to 11.7); couplings that vary from site to site, nuclear spins and isotopes (Sect. 11.6; Limit Checks 11.15 and 11.16); a crystal of fixed orientation and the daily modulation (Sect. 11.7; Limit Check 11.17); and absorption on phonons and acoustic modes (Sect. 11.8; Limit Checks 11.18 to 11.20).

11.1 Acoustic phonons

A dark-matter particle of 100 keV transfers at most $2m_\chi v_{\max} = 0.53\,\text{keV}$ of momentum. Given to a free silicon nucleus, that momentum would deposit $q^2/2M = 5.4\,\mu\text{eV}$, far below any threshold. Given to the crystal, it can create one longitudinal acoustic phonon of energy $c_{\text{LA}}q \approx 16\,\text{meV}$, because the whole crystal takes up the momentum and the energy is set by the sound speed rather than by the mass of a nucleus. This is the kinematic match that makes single phonons the channel of the lightest dark matter that couples to nucleons (Sect. 1.3). At long wavelengths its rate depends on the material only through the sound speeds, and for a light mediator not even through these.

11.1.1 The one-phonon response

Take a dark-matter particle that couples to protons and neutrons with the same potential $\tilde{V}_N(\boldsymbol{q})$, the spin-independent coupling $\mathcal{O}_1$ of Sect. 8.1. By Eq. (8.8), a nucleus of species κ then carries the coupling $A_\kappa \tilde{V}_N(\boldsymbol{q}) F_N(q)$, and the dark matter sees the target through the single operator

$$O_N(\boldsymbol{q}) = \sum_\kappa A_\kappa\, n_\kappa(\boldsymbol{q}),$$

the nucleon number density of the crystal at momentum $\boldsymbol{q}$. The nuclear form factor F_N of Eq. (2.26) is one to better than 10^{-3} at the momentum transfers of this section, which stay below a few keV, and we drop it. The coupling has no electronic row, so the electrons enter only through the forces they transmit between nuclei: the phonons are those of the Born–Oppenheimer surface (Sect. 5.1), and the final states lie in the electron-following sector of Table 8.5. Screening, which acts on rows that couple to charge (Sect. 8.4), does not enter. The dark photon, which couples to charge, is the subject of Sect. 11.2.

The rate per unit target mass is Eq. (8.16) with one row, $\mathfrak{D} = |\tilde{V}_N(\boldsymbol{q})|^2$ and $S = S_A$, where $\omega_{\boldsymbol{q}} = \boldsymbol{q}\cdot\boldsymbol{v} - q^2/2m_\chi$ is the energy the dark matter loses (Eq. (1.22)), not a phonon frequency, and S_A is the response Eq. (4.13) of O_N, the nuclear block $\sum_{\kappa\kappa'} A_\kappa A_{\kappa'} S_{\kappa\kappa'}$. Its one-phonon part follows from the amplitude of Sect. 5.3. To first order in the displacement, a nucleus of species κ emits a phonon $\boldsymbol{Q}\nu$ with the amplitude

$$\langle n_{\boldsymbol{Q}\nu}+1|\, n_\kappa(\boldsymbol{q})\, |n_{\boldsymbol{Q}\nu}\rangle = \mathrm{i}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}\, \frac{\boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*}{\sqrt{2M_\kappa\omega_{\boldsymbol{Q}\nu}}}\, \sqrt{N(n_{\boldsymbol{Q}\nu}+1)}\, \sum_{\boldsymbol{G}} \delta_{\boldsymbol{q},\boldsymbol{Q}+\boldsymbol{G}}.$$

The full exponential multiplies this by $\mathrm{e}^{-W_\kappa(\boldsymbol{q})}$: the generating formula Eq. (5.58) for a pair of nuclei, expanded to first order in the displacement correlation $\mathcal{C}_{\kappa\kappa'}$, is the one-phonon term with the factor $\mathrm{e}^{-W_\kappa - W_{\kappa'}}$ in front. The amplitudes of the nuclei of one cell add with their phases. We square, sum over the final phonon states and divide by the volume $\mathcal{V} = N\Omega$ as in Eq. (4.13). The factor N cancels, and the Kronecker delta leaves in each branch one phonon wavevector $\boldsymbol{Q}$, the reduction of $\boldsymbol{q}$ to the first Brillouin zone:

One-phonon nuclear response

$$S_A^{(1)}(\boldsymbol{q},\omega) = \frac{2\pi}{\Omega}\sum_\nu \frac{\left|\sum_\kappa A_\kappa\, \mathrm{e}^{-W_\kappa(\boldsymbol{q})}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}\, \boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*/\sqrt{M_\kappa}\right|^2}{2\omega_{\boldsymbol{Q}\nu}} \times \Big[(n_{\boldsymbol{Q}\nu}+1)\,\delta(\omega-\omega_{\boldsymbol{Q}\nu}) + n_{\boldsymbol{Q}\nu}\,\delta(\omega+\omega_{\boldsymbol{Q}\nu})\Big], \tag{11.1}$$

with $\boldsymbol{Q} = \boldsymbol{q} - \boldsymbol{G}$ in the first zone, $\boldsymbol{q}$ itself unreduced in the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}$, and the phase convention of Eq. (5.32).

The absorption term follows from the second matrix element of Eq. (5.34) with $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$, which leaves the modulus unchanged. In a detector at 10 mK, $n_{\boldsymbol{Q}\nu}$ vanishes for every phonon above a few tens of μeV (Sect. 5.3), and only emission remains. At $T = 0$ the same amplitude, with the sum over $\boldsymbol{G}$, is that of Campbell-Deem and collaborators [1, Eq. (7)]. On the truncation label of Sect. 8.5, Eq. (11.1) is the point $\mathcal{K}$ = one-phonon, $n_A = 0$, with $n_T = 0$ for the emission term and $n_T = 1$ for the absorption term, and $n_B = 0$ throughout.

Three factors of Eq. (11.1) set the response: the branch weights, the conservation of crystal momentum and the Debye–Waller factor. The weight of a branch is the projection of $\boldsymbol{q}$ on the displacement of each nucleus, divided by the phonon energy, so that soft modes polarized along $\boldsymbol{q}$ dominate. The momentum is conserved up to a reciprocal-lattice vector, so that each $\boldsymbol{q}$ excites one phonon per branch, with the energy fixed by the dispersion: the response is a set of sharp lines in ω at fixed $\boldsymbol{q}$, not a continuum. And the Debye–Waller factor e^{-2W_κ} of Eq. (5.56) measures the competition with the multiphonon terms of Sect. 11.3. In silicon, $2W_\kappa = q^2\langle u_x^2\rangle$ is 3×10^{-3} at $q = 2.25\,\mathrm{keV}$ with the zero-temperature $\langle u_x^2\rangle = 2.45 \times 10^{-3}\,\text{Å}^2$ of Worked example 5.7, so in the first Brillouin zone the factor is one to 3×10^{-3}.

11.1.2 The long-wavelength limit

Most of the rate of light dark matter comes from momentum transfers well inside the first zone. For a crystal with a lattice constant of a few Å, the scale of the zone is

$$q_{\mathrm{BZ}} = \Big(\frac{6\pi^2}{\Omega}\Big)^{1/3}, \tag{11.2}$$

the radius of the sphere with the volume of the Brillouin zone, one wavevector per primitive cell. For silicon, $\Omega = a^3/4$ and $q_{\mathrm{BZ}} = 1.139\,\text{Å}^{-1} = 2.25\,\mathrm{keV}$, between the distances from Γ to L (1.98 keV) and to X (2.28 keV). It should not be confused with the Debye wavevector Q_{D} of Eq. (5.23), which counts one wavevector per atom and is larger by $n_{\mathrm{at}}^{1/3}$.

For $q \ll q_{\mathrm{BZ}}$, Sect. 5.2 removes the reciprocal-lattice vector, the internal motion of the cell and the curvature of the dispersion. The reciprocal-lattice vector is $\boldsymbol{G} = 0$ and $\boldsymbol{Q} = \boldsymbol{q}$. The acoustic polarization vectors move every nucleus of the cell together, $\boldsymbol{e}_{\kappa\nu} = (M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu(\hat{\boldsymbol{q}})$, with $\hat{\boldsymbol{e}}_\nu$ the unit polarizations of the Christoffel problem (Appendix C.18), and the dispersion is linear, $\omega_{\boldsymbol{q}\nu} = c_\nu(\hat{\boldsymbol{q}})\,q$. And the phases $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}$ and the Debye–Waller factors are one up to corrections of order qa and $q^2\langle u_x^2\rangle$. The sum over nuclei in Eq. (11.1) then becomes $q\,(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)\sum_\kappa A_\kappa/\sqrt{M_{\mathrm{cell}}}$, the optical branches drop out, since in them the center of mass of the cell is at rest (Sect. 5.2), and at $T = 0$

$$S_A^{(1)}(\boldsymbol{q},\omega) = \frac{\pi\, q\, A_{\mathrm{cell}}^2}{\Omega\, M_{\mathrm{cell}}} \sum_{\nu=1}^{3} \frac{(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2}{c_\nu(\hat{\boldsymbol{q}})}\, \delta\big(\omega - c_\nu(\hat{\boldsymbol{q}})\, q\big),$$

with $A_{\mathrm{cell}} = \sum_\kappa A_\kappa$ the number of nucleons in a cell. Dividing by $\rho_T = M_{\mathrm{cell}}/\Omega$, and writing the nuclear masses as $M_\kappa = A_\kappa m_u$ with $m_u = 931.494\,\mathrm{MeV}$ the atomic mass unit,

Long-wavelength acoustic structure factor

$$\frac{S_A^{(1)}(\boldsymbol{q},\omega)}{\rho_T} = \frac{\pi q}{m_u^2} \sum_{\nu=1}^{3} \frac{(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2}{c_\nu(\hat{\boldsymbol{q}})}\, \delta\big(\omega - c_\nu(\hat{\boldsymbol{q}})\, q\big), \qquad q \ll q_{\mathrm{BZ}}. \tag{11.3}$$

Per unit mass, the coherent sum over the A_{cell} nucleons of a cell has left nothing of the composition. The material enters through its sound speeds and the polarizations of its acoustic branches. The weights $(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2$ sum to one over the three branches, by the completeness

of the polarizations, and along a symmetry direction the longitudinal branch takes all of it. In a general direction the quasi-transverse branches, whose polarizations are not exactly perpendicular to $\hat{\boldsymbol{q}}$ (Appendix C.18), take a small share. The binding energy of the nuclei and the isotopic mixture make M_κ/A_κ differ from m_u by less than a percent, which is the accuracy of the replacement.

Campbell-Deem and collaborators derived the longitudinal term of Eq. (11.3), with m_p in place of m_u, and noted that the composition drops out of the rate per unit mass [1, Eq. (10), 2, Eqs. (27), (30)]; they apply it up to $q = 2\pi/a$, which the limit check below tests.

Limit Check 11.1. Long-wavelength acoustic structure factor

At momentum transfers small compared with the zone, the one-phonon response per unit mass of a crystal to a coupling to nucleon number is carried by the longitudinal acoustic branch, $S/\rho_T = (\pi q/m_u^2)\,\delta(\omega - c_{\mathrm{LA}}q)/c_{\mathrm{LA}}$, the same for every composition [2, Eq. (30) with Eq. (27), 1, Eq. (10)]. Off the symmetry directions the quasi-transverse branches share the weight, as in Eq. (11.3).

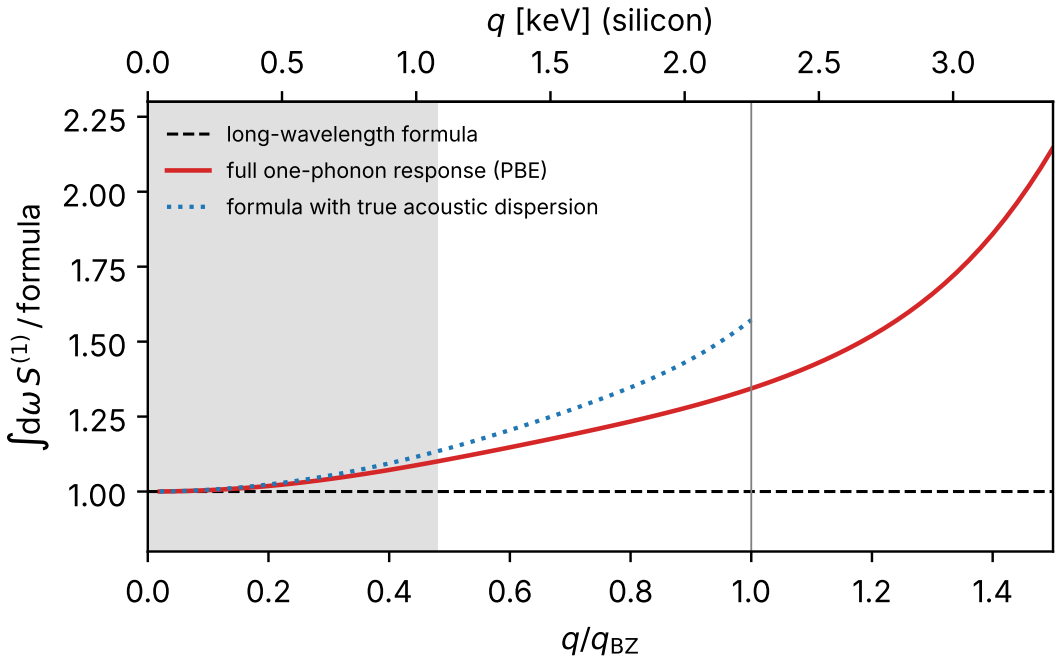


Figure 11.2: The ω-integrated one-phonon response of silicon at $T = 0$, Eq. (11.1) with all six branches, Umklapp terms and the Debye–Waller factor (solid), divided by the long-wavelength form Eq. (11.3) (dashed, one), both averaged over the directions of $\boldsymbol{q}$, against q/q_{BZ}, with $q_{\mathrm{BZ}} = 2.25\,\mathrm{keV}$. Phonons from the PBE force constants of Sect. 5.2, Fourier-interpolated; the formula uses the sound speeds and polarizations of the same force constants, so that the ratio tends to one as $q \to 0$. Dotted: the long-wavelength polarizations with the true acoustic dispersion $\omega_{\boldsymbol{q}\nu}$ in place of $c_\nu q$, the leading correction. Gray: $q < 0.48\,q_{\mathrm{BZ}} = 1.08\,\mathrm{keV}$, where the two agree within 10 %; they agree within 5 % below $0.34\,q_{\mathrm{BZ}}$. At q_{BZ} the formula is 26 % low.

Toward the zone boundary the acoustic branches bend below $c_\nu q$, and since the weight goes as $1/\omega$, the formula underestimates the response; beyond q_{BZ} the phonon carries only the reduced momentum, and the response grows as q^2, the incoherent form of Sect. 11.3.

The departure is driven by the dispersion: the dotted curve, which keeps the long-wavelength polarizations and replaces $c_\nu q$ alone by the true $\omega_{\boldsymbol{q}\nu}$, accounts for most of it below $q_{\mathrm{BZ}}/2$. The Debye–Waller factor is irrelevant here. The rest of the difference comes from the polarizations and the phases $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}$, which change at order $(qa)^2$ and pull the other way. For a rate weighted toward small q, as with a light mediator, the region below $q_{\mathrm{BZ}}/2$ carries nearly everything and the formula is accurate. For a heavy mediator the rate is weighted toward the largest q that kinematics allows, $2m_\chi v_{\max}$, and the response there stays within 10 % of the formula only for masses up to about $0.48\, q_{\mathrm{BZ}}/2v_{\max} = 0.20\,\mathrm{MeV}$ in silicon.

11.1.3 Orientation-averaged rate

A crystal at a fixed orientation sees the dark-matter wind from one direction, and its rate depends on that direction, which is the daily modulation of Sect. 11.7. Averaged over the orientations of the crystal, the response enters through its average over the directions of $\boldsymbol{q}$, and the velocity integral of Eq. (8.16) acts on the delta function alone. It gives $\eta(v_{\min})/2q$ per unit $\mathrm{d}^3q/(2\pi)^3$, with η the mean inverse speed of Eq. (1.28) (Sect. 8.2). For a phonon of energy $\omega = c_\nu q$ and elastic scattering, Eq. (1.23) gives

$$v_{\min} = c_\nu + \frac{q}{2m_\chi},$$

the sound speed plus half the speed the dark matter loses. We write the potential as in Sect. 1.4, $|\tilde{V}_N(q)|^2 = \pi\sigma_p F_{\mathrm{DM}}(q)^2/\mu_{\chi p}^2$, with σ_p the dark-matter–proton cross-section, $\mu_{\chi p}$ the reduced mass and F_{DM} the form factor of Eq. (1.33) with $q_0 = \alpha m_e$. Inserting Eq. (11.3) into Eq. (8.16), integrating ω above a threshold ω_{th} and collecting $4\pi\cdot\pi\cdot\pi/(2(2\pi)^3) = 1/4$,

One-acoustic-phonon rate per unit mass

$$\mathcal{R} = \frac{\rho_\chi\,\sigma_p}{4\,m_\chi\,\mu_{\chi p}^2\,m_u^2}\left\langle \sum_{\nu=1}^{3} \frac{(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2}{c_\nu} \int_{\omega_{\mathrm{th}}/c_\nu}^{q_{\mathrm{up},\nu}} \mathrm{d}q\; q^2\, F_{\mathrm{DM}}(q)^2\, \eta\Big(c_\nu + \frac{q}{2m_\chi}\Big)\right\rangle_{\hat{\boldsymbol{q}}}, \tag{11.4}$$

with c_ν and $\hat{\boldsymbol{e}}_\nu$ taken at $\hat{\boldsymbol{q}}$, the brackets the average over the sphere of directions, and the upper limit $q_{\mathrm{up},\nu} = \min[2m_\chi(v_{\max} - c_\nu),\, q_{\mathrm{BZ}}]$.

The kinematic upper limit is where η vanishes, $v_{\min} = v_{\max}$; the zone cut q_{BZ} is the range of Eq. (11.3), beyond which the multiphonon response of Sect. 11.3 takes over. The rate is nonzero only when the lower limit lies below the upper one, $\omega_{\mathrm{th}}/c_\nu \le 2m_\chi(v_{\max} - c_\nu)$. This is the *acoustic edge* of Sect. 9.2,

$$m_{\min}^{\mathrm{ac}} = \frac{\omega_{\mathrm{th}}}{2c_\nu(v_{\max} - c_\nu)} \simeq \frac{\omega_{\mathrm{th}}}{2c_\nu v_{\max}}, \tag{11.5}$$

set by the fastest direction and branch with longitudinal weight, $c_{\mathrm{LA,max}}$. It lies above the threshold mass $2\omega_{\mathrm{th}}/v_{\max}^2$ of Eq. (1.25) by $v_{\max}/4c_\nu$, about 21 in silicon. The reason is that a phonon of energy ω_{th} must carry the momentum $\omega_{\mathrm{th}}/c_\nu$, far more than the dark matter needs to give up ω_{th} to a free target. For silicon at $\omega_{\mathrm{th}} = 1\,\mathrm{meV}$, $c_{\mathrm{LA,max}} = 9.35\,\mathrm{km\,s^{-1}}$ along [111] and $m_{\min}^{\mathrm{ac}} = 6.1\,\mathrm{keV}$, against 0.28 keV for the threshold mass. Griffin and collaborators give the same edge, $m_{\chi,\min} \sim 20\,\mathrm{keV}\,(\omega_{\min}/\mathrm{meV})(10^{-5}/c_s^{\mathrm{LA}})$ for their halo [3, Eq. (36)].

Fig. 11.3 shows Eq. (11.4) for silicon, germanium and diamond. For a light mediator the curves merge at high mass; for a heavy one they level off at plateaus set by the sound speed.

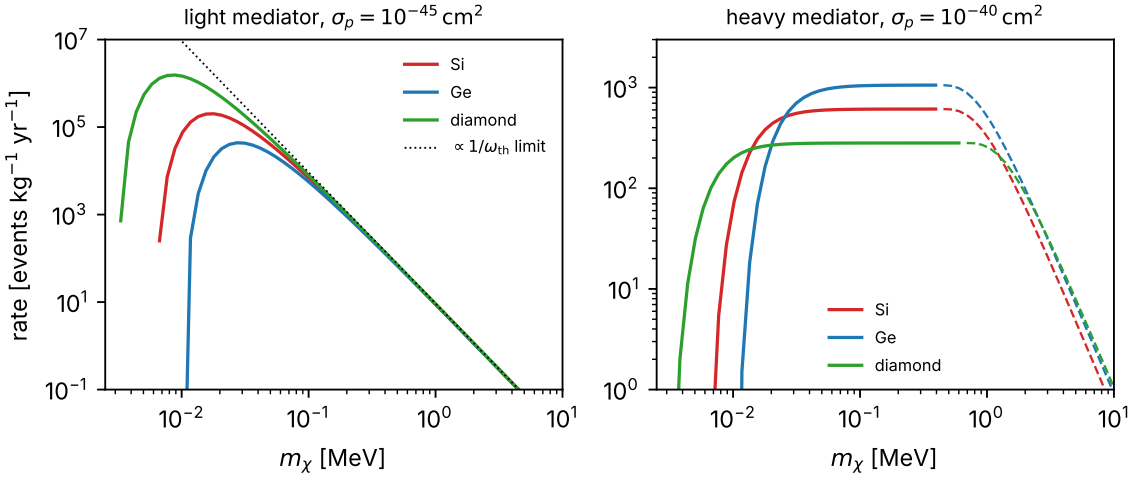


Figure 11.3: Rate of single-acoustic-phonon emission per unit target mass, Eq. (11.4), for silicon, germanium and diamond at a threshold $\omega_{\mathrm{th}} = 1\,\mathrm{meV}$, averaged over the orientation of the crystal, with equal couplings to protons and neutrons, $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$ and the standard halo of Sect. 1.2. Sound speeds and polarizations from the Christoffel problem of the measured room-temperature elastic tensors (Worked example 11.2); momentum transfers $q \leq q_{\mathrm{BZ}}$, Debye–Waller factor one. Left: light mediator, $F_{\mathrm{DM}} = (q_0/q)^2$, $\sigma_p = 10^{-45}\,\mathrm{cm}^2$; dotted, the target-independent limit Eq. (11.6). Right: heavy mediator, $F_{\mathrm{DM}} = 1$, $\sigma_p = 10^{-40}\,\mathrm{cm}^2$; solid up to $q_{\mathrm{BZ}}/2v_{\mathrm{max}}$ (0.42, 0.41 and 0.65 MeV), dashed above, where the fall is an artifact of the cut $q \leq q_{\mathrm{BZ}}$ and the multiphonon and Umklapp terms of Sect. 11.3 take over. The edges are 6.1 (Si), 10.2 (Ge) and 3.1 keV (diamond).

11.1.4 Light mediator

For a light mediator, $F_{\rm DM}^2 = (q_0/q)^4$, the integrand of Eq. (11.4) falls as q^{-2}, and the rate is dominated by the smallest momentum transfer that a phonon above threshold can carry. In the variable $u = q/2m_\chi$, the integral of one branch and direction is

$$\frac{q_0^4}{2m_\chi}\int_{u_-}^{u_+} \mathrm{d}u\,\frac{\eta(c_\nu+u)}{u^2}, \qquad u_- = \frac{\omega_{\rm th}}{2m_\chi c_\nu}, \quad u_+ = v_{\rm max} - c_\nu.$$

Far above the edge, where $u_- \ll u_+$, the integral tends to $\eta(c_\nu)/u_-$, and since $u_- = \omega_{\rm th}/2m_\chi c_\nu$, the q-integral of each branch and direction is $q_0^4 c_\nu \eta(c_\nu)/\omega_{\rm th}$. Its factor c_ν cancels the weight $1/c_\nu$ in the brackets of Eq. (11.4). The shift of η by c_ν is of relative order $c_\nu^2/v_0^2 \sim 10^{-3}$, since the laboratory speed distribution vanishes as v^2 at small speeds and η is flat at the origin. By the completeness $\sum_\nu (\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2 = 1$ the directions and branches then add up to one, and

$$\mathcal{R}_{\rm light} = \frac{\rho_\chi\,\sigma_p\,q_0^4\,\eta(0)}{4\,m_\chi\,\mu_{\chi p}^2\,m_u^2\,\omega_{\rm th}}. \tag{11.6}$$

Nothing of the target is left. The rate per unit mass is the same for every crystal, inversely proportional to the threshold, and it falls as m_χ^{-3} for $m_\chi \ll m_p$. Griffin and collaborators found this behavior in their comparison of targets: the rate is dominated by the lowest detectable momentum transfer, and "the dependence on the target properties drops out" [3, Eq. (35) and the text after it]. The light-mediator curves of Fig. 11.3 merge with Eq. (11.6) at high mass. They separate near the edges, which differ by a factor of three between diamond and germanium.

How far above the edge must a mass be? Write $\eta(c_\nu+u) = \eta(0) - [\eta(0) - \eta(c_\nu+u)]$. The first term integrates to $\eta(0)(1/u_- - 1/u_+)$. In the second, the bracket vanishes as u^2 at small u, so its integral from 0 to u_- is of order u_-, and to first order in u_- the integral runs from 0 to u_+. Defining

$$\kappa_\eta = \frac{u_+}{\eta(0)}\int_0^{u_+}\mathrm{d}u\,\frac{\eta(0)-\eta(u)}{u^2}, \tag{11.7}$$

and using $u_-/u_+ = m_{\rm min}^{\rm ac}/m_\chi$ from Eq. (11.5), the integral is $(\eta(0)/u_-)[1-(1+\kappa_\eta)\,m_{\rm min}^{\rm ac}/m_\chi]$ up to terms of second order in $m_{\rm min}^{\rm ac}/m_\chi$. The rate is therefore

$$\frac{\mathcal{R}}{\mathcal{R}_{\rm light}} = \Big[1 - (1+\kappa_\eta)\frac{m_{\rm min}^{\rm ac}}{m_\chi} + O\Big(\frac{m_{\rm min}^{\rm ac}}{m_\chi}\Big)^2\Big], \qquad \kappa_\eta = 3.14, \tag{11.8}$$

with $\mathcal{R}_{\rm light}$ the limit Eq. (11.6), for $m_\chi \le q_{\rm BZ}/2v_{\rm max}$, where the kinematic limit u_+ applies. The value of κ_η is a property of the halo alone: it is Eq. (11.7) for the standard halo of Sect. 1.2 with $u_+ = v_{\rm max}$, the limit $c_\nu \ll v_{\rm max}$. At the sound speed of silicon, $u_+ = v_{\rm max} - c_\nu$ and $\kappa_\eta = 3.09$. The material enters only through the edge, and the term 1 of $1+\kappa_\eta$ is the loss of the momentum transfers above u_+, the term κ_η the loss of the slow particles that the halo lacks. For a crystal the edge in Eq. (11.8) is the average over directions and branches, weighted like the rate, which is $m_{\rm min}^{\rm ac}$ evaluated at the speed $\langle\sum_\nu(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2/c_\nu\rangle^{-1}$, $8.956\,\mathrm{km\,s^{-1}}$ for silicon (Worked example 11.2), where it is 6.39 keV. Above $q_{\rm BZ}/2v_{\rm max}$ the zone cut replaces u_+, and the bracket tends to $1-\omega_{\rm th}/(c_\nu q_{\rm BZ})$, 0.985 for silicon at 1 meV.

Limit Check 11.2. Light-mediator single-phonon rate

For a light mediator the one-phonon rate per unit target mass is $\mathcal{R} = \rho_\chi\sigma_p q_0^4\eta(0)/(4m_\chi\mu_{\chi p}^2 m_u^2\omega_{\rm th})$, inversely proportional to the threshold and independent of the target [3, Eq. (35)].

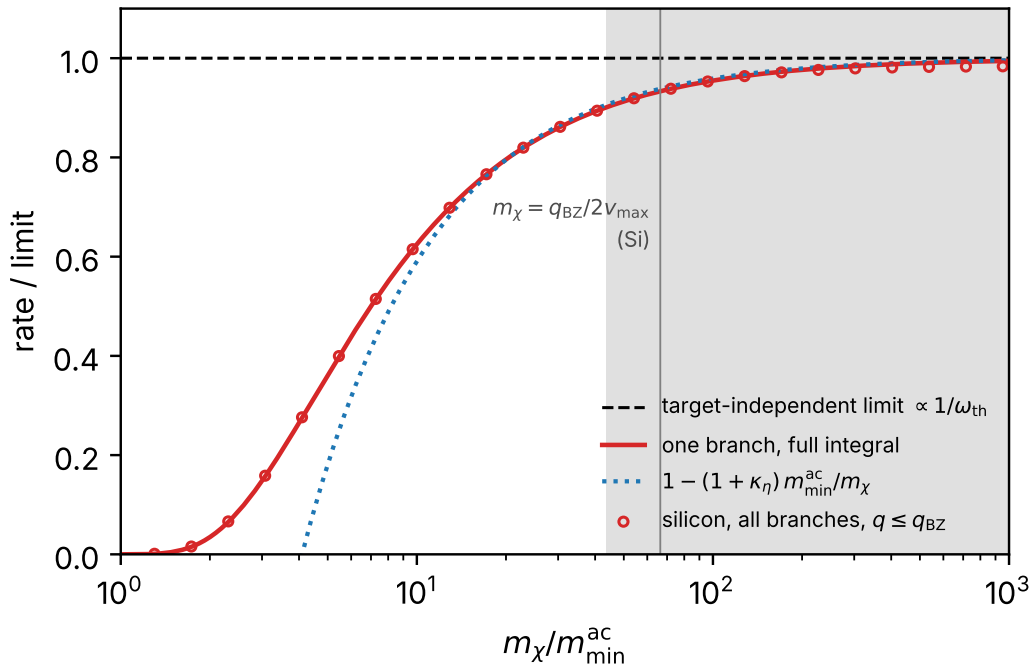


Figure 11.4: The light-mediator rate Eq. (11.4), divided by its target-independent limit Eq. (11.6) (dashed, one), against $m_\chi/m_{\rm min}^{\rm ac}$. Solid: one branch of speed $9.0\,\mathrm{km\,s^{-1}}$ without the zone cut; the curve changes by 3 % at $m_\chi = 10\,m_{\rm min}^{\rm ac}$ between speeds of 3 and $18\,\mathrm{km\,s^{-1}}$. Circles: silicon, all branches and directions, with $q \le q_{\rm BZ}$; on this axis $m_{\rm min}^{\rm ac} = 6.39\,\mathrm{keV}$ at $\omega_{\rm th} = 1\,\mathrm{meV}$, the edge for the polarization-weighted inverse speed $8.956\,\mathrm{km\,s^{-1}}$ of Eq. (11.9) (the fastest longitudinal speed, $9.35\,\mathrm{km\,s^{-1}}$, the convention of Sect. 9.2, gives 6.1 keV); the vertical line marks $q_{\rm BZ}/2v_{\rm max}$, above which the zone cut holds the ratio at 0.985. Dotted: the first-order correction Eq. (11.8) with $\kappa_\eta = 3.09$. Gray: $m_\chi \ge 44\,m_{\rm min}^{\rm ac}$, where the rate is within 10 % of the limit. The first-order form is within 10 % of the full rate for $m_\chi \ge 9\,m_{\rm min}^{\rm ac}$. Standard halo of Sect. 1.2.

Within a few times the edge the momentum window $[\omega_{\rm th}/c_\nu,\, 2m_\chi v_{\rm max}]$ closes and only the fastest particles of the halo contribute, so the rate falls far below the limit and depends on the material through its edge.

Worked example 11.1: The light-mediator correction at three times the edge

Take silicon at $\omega_{\rm th} = 1\,\mathrm{meV}$, where $m_{\rm min}^{\rm ac} = 6.39\,\mathrm{keV}$ for the polarization-weighted inverse speed $8.956\,\mathrm{km\,s^{-1}}$ of Eq. (11.9) (6.1 keV for the fastest longitudinal speed), and $m_\chi = 3\,m_{\rm min}^{\rm ac} = 19.2\,\mathrm{keV}$. The momentum transfers that can create a phonon above threshold run from $\omega_{\rm th}/c = 33\,\mathrm{eV}$ to $2m_\chi(v_{\rm max} - c) = 100\,\mathrm{eV}$, a factor of three. With $\kappa_\eta = 3.09$ at this sound speed, the first-order factor is $1 - 4.09/3 = -0.36$: negative, and therefore meaningless. The full integral of Eq. (11.4), with all branches and directions, gives $\mathcal{R}/\mathcal{R}_{\rm light} = 0.149$. At three times the edge the rate is a seventh of the target-independent value. The expansion behind Eq. (11.8) has failed there, because the speeds that contribute, $v_{\rm min}$ between $c + u_-$ and $v_{\rm max}$, all lie in the tail of the halo above $270\,\mathrm{km\,s^{-1}}$, where η is far from flat. At ten times the edge, 63.9 keV, the full ratio is 0.625 and the first-order form 0.591; at 44 times, 0.28 MeV, the full ratio is 0.902 and the first-order form 0.907.

The target-independent statement holds within 10 % only from about 0.28 MeV in silicon, and for any two materials only above 44 times the higher of their edges.

11.1.5 Heavy mediator

For a heavy mediator, $F_{\mathrm{DM}} = 1$, the integrand of Eq. (11.4) grows as q^2, and the rate is dominated by the largest momentum transfer, $2m_\chi(v_{\max} - c_\nu)$. With $u = q/2m_\chi$ the integral of one branch is $8m_\chi^3 \int_{u_-}^{u_+} \mathrm{d}u\, u^2 \eta(c_\nu + u)$. Well above the edge and with $c_\nu \ll v_{\max}$, it tends to $8m_\chi^3 I_0$, with $I_0 = \int_0^{v_{\max}} \mathrm{d}u\, u^2 \eta(u) = 5.32 \times 10^{-7}$ for the standard halo. For $m_\chi \ll m_p$ the reduced mass is $\mu_{\chi p} \approx m_\chi$, and the factors of m_χ cancel:

$$\mathcal{R} = \frac{2\rho_\chi\, \sigma_p\, I_0}{m_u^2} \Big\langle \sum_\nu \frac{(\hat{\boldsymbol{q}} \cdot \hat{\boldsymbol{e}}_\nu)^2}{c_\nu} \Big\rangle_{\hat{\boldsymbol{q}}}. \tag{11.9}$$

The rate per unit mass is independent of the dark-matter mass and inversely proportional to a polarization-weighted mean speed: soft crystals win. The plateaus of Fig. 11.3 nearly stand in the ratio of the inverse speeds, $8.956/5.321 = 1.68$ for germanium against silicon and $8.956/18.16 = 0.49$ for diamond. The computed rates at 0.3 MeV stand in the ratios 1.73 and 0.46. The difference is the shift of η by c_ν, which matters at the large u that a heavy mediator favors, where η is not flat, and lowers the rate of a stiff crystal by several percent per $10\,\mathrm{km\,s^{-1}}$. Griffin and collaborators found the same ordering, $R \propto \sigma_n/c_s$, between 0.1 and 1 MeV [3, Eq. (37)].

The plateau is reached only well above the edge and ends near 0.42 MeV. Near the edge the lower cut removes part of I_0, and the rate approaches Eq. (11.9) only slowly, because η falls steeply toward $v_{\max}$ and the integrand is weighted toward small u. In silicon the rate reaches 90 % of the plateau value (654 events per kilogram and year at $\sigma_p = 10^{-40}\,\mathrm{cm}^2$) only at 45 keV, seven times the edge. However, it never exceeds 94 % of it, and diamond never exceeds 87 %, because Eq. (11.9) drops the shift of η by c_ν. Above $q_{\mathrm{BZ}}/2v_{\max}$, 0.42 MeV for silicon, the momentum transfers leave the region where Eq. (11.3) holds, and by Fig. 11.2 the response at the largest momentum transfers is already off by more than 10 % from about 0.2 MeV. Above that, Umklapp processes and multiphonons take over (Sect. 11.3). Whether Eq. (11.9) ranks two materials correctly, and over which window of masses, is a question of figures of merit; Sect. 17.5 answers it, and names the inverse of the bracket the effective sound speed of the channel.

11.1.6 Sound speeds from the elastic tensor

The rates above need the sound speeds in every direction, and with them three averages: the largest longitudinal speed for the edge, the polarization-weighted inverse speed of Eq. (11.9), and, in much of the literature, a single sphere-averaged longitudinal speed. All three follow from the elastic tensor C_{ijkl} and the density through the Christoffel equation Eq. (C.156): in each direction $\hat{\boldsymbol{q}}$ the eigenvalues of $\mathsf{M}_{ik} = C_{ijkl}\hat{q}_j\hat{q}_l$ are $\rho_T c_\nu(\hat{\boldsymbol{q}})^2$, and the eigenvectors are the polarizations $\hat{\boldsymbol{e}}_\nu$ of Eq. (11.3). The *sphere-averaged longitudinal speed* is

$$\bar{c}_{\mathrm{LA}} = \langle c_{\mathrm{LA}}(\hat{\boldsymbol{q}}) \rangle = \frac{1}{4\pi} \int \mathrm{d}^2\hat{\boldsymbol{q}}\; c_{\mathrm{LA}}(\hat{\boldsymbol{q}}), \tag{11.10}$$

with the quasi-longitudinal branch, the largest eigenvalue, in every direction. Griffin, Inzani, Trickle, Zhang and Zurek define the sound speed of their target tables in this way, as the average of ω_{LA}/q over a sphere of radius 10 eV about Γ, computed from density-functional

phonons [3, App. A 3]. Here it is computed from a measured tensor, and it is not the Debye-averaged speed $\bar{c}$ of Eq. (5.25), which weights the three branches by c_ν^{-3} and is dominated by the slow transverse waves.

In practice the elastic constants are either measured or computed by density-functional theory, from finite strains with relaxed internal coordinates or from perturbation theory with strain as the perturbation; reading them off the slopes of interpolated acoustic branches is much less reliable (Appendix E.4). Some codes and databases store the shear rows of the Voigt matrix in the order xy, yz, zx instead of yz, xz, xy (Appendix C.18). For a cubic crystal the error is invisible, since $C_{44} = C_{55} = C_{66}$, but for a lower symmetry it exchanges the shear constants and produces wrong speeds along the low-symmetry directions. The tensor must therefore be in Voigt order before it enters Eq. (11.10), and the check of Appendix E.4, a speed from the tensor against the slope of the computed dispersion along one direction, catches the error.

Worked example 11.2: The sphere-averaged longitudinal speed of silicon

The room-temperature constants of silicon are $C_{11} = 165.6$, $C_{12} = 63.9$ and $C_{44} = 79.5$ GPa, with $\rho_T = 2.33\,\mathrm{g\,cm^{-3}}$ (Appendix C.18). Along the three symmetry directions Eq. (C.157) gives $\rho_T c_{\mathrm{LA}}^2 = 165.6$, 194.3 and 203.8 GPa, and $c_{\mathrm{LA}} = 8.43$, 9.13 and $9.35\,\mathrm{km\,s^{-1}}$ along [100], [110] and [111]. The speed varies by 11 % over the sphere, and the average needs all directions. A closed form exists to first order in the anisotropy $\Delta_C = C_{11} - C_{12} - 2C_{44}$, which is -57.3 GPa for silicon and zero for an isotropic medium. By Appendix C.18 the longitudinal eigenvalue is $\rho_T c_{\mathrm{LA}}^2 = C_{11} - 2\Delta_C s_Q$ to first order, with $s_Q = \hat{q}_x^2\hat{q}_y^2 + \hat{q}_y^2\hat{q}_z^2 + \hat{q}_z^2\hat{q}_x^2$, and since $\langle \hat{q}_x^4 \rangle = \frac{1}{5}$ and $\langle \hat{q}_x^2\hat{q}_y^2 \rangle = \frac{1}{15}$, $\langle s_Q \rangle = \frac{1}{5}$. The averaged eigenvalue is $C_{11} - \frac{2}{5}\Delta_C = 188.5$ GPa, and its square root over ρ_T is $8.995\,\mathrm{km\,s^{-1}}$. It coincides with the longitudinal speed $[(B + \frac{4}{3}G_{\mathrm{V}})/\rho_T]^{1/2}$ of an isotropic medium with the Voigt-averaged shear modulus $G_{\mathrm{V}} = (C_{11} - C_{12} + 3C_{44})/5$, as it must, since both are the first-order average.

The exact average is a quadrature. On a product grid of Gauss–Legendre nodes in $\cos\theta$ and equally spaced nodes in ϕ (Appendix C.18), with the 3×3 Christoffel matrix diagonalized at each node, $\bar{c}_{\mathrm{LA}} = 9.019$, 9.017 and $9.017\,\mathrm{km\,s^{-1}}$ on grids of 72, 288 and 1152 directions. The converged value,

$$\bar{c}_{\mathrm{LA}} = 9.017\,\mathrm{km\,s^{-1}},$$

lies 0.2 % above the first-order value, between the extremes 8.43 and $9.35\,\mathrm{km\,s^{-1}}$. The other averages differ from it by less than a percent: $1/\langle 1/c_{\mathrm{LA}} \rangle = 9.011\,\mathrm{km\,s^{-1}}$, and the inverse of the rate coefficient of Eq. (11.9), $\langle \sum_\nu (\hat{\boldsymbol{q}} \cdot \hat{\boldsymbol{e}}_\nu)^2 / c_\nu \rangle^{-1}$, is $8.956\,\mathrm{km\,s^{-1}}$. The same procedure yields $\bar{c}_{\mathrm{LA}} = 5.37\,\mathrm{km\,s^{-1}}$ for germanium and $18.20\,\mathrm{km\,s^{-1}}$ for diamond, from the tensors 131, 49, 68 GPa [4] and 1081, 125, 579 GPa [5], as tabulated in Ref. [6, Table III], with the densities 5.33 and $3.52\,\mathrm{g\,cm^{-3}}$ that follow from the lattice constants 5.658 and 3.567 Å.

For comparison, the target table of Ref. [3, Table I] lists $c_s^{\mathrm{LA}} = 2.84 \times 10^{-5}$ for silicon, $8.51\,\mathrm{km\,s^{-1}}$, 5.6 % below the value from the measured tensor; for germanium and diamond its values are 10 % and 1.5 % below. The force constants of Sect. 5.2, from the same kind of calculation at the measured lattice constant, give $8.83\,\mathrm{km\,s^{-1}}$ for silicon. Which speeds of that table are consistent with measured elasticity, and what that does to rankings of targets, is examined in Sect. 17.5.

The anisotropy of a cubic crystal is measured by the Zener ratio $A_{\mathrm{Z}} = 2C_{44}/(C_{11} - C_{12})$,

one for an isotropic medium, 1.56 for silicon, 1.66 for germanium and 1.21 for diamond. How much does it matter that a rate is written with one speed? The heavy-mediator rate Eq. (11.9) weights each direction by $\sum_\nu (\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2/c_\nu$, and the single-speed form replaces that by $1/\bar{c}_{\mathrm{LA}}$. The curvature of $1/c$ and the longitudinal weight of the transverse branches separate the two, both at second order in Δ_C. For the curvature, Jensen's inequality gives $\langle 1/c_{\mathrm{LA}}\rangle \geq 1/\langle c_{\mathrm{LA}}\rangle$, with a gap set by the variance of c_{LA} over the sphere, which is first order in Δ_C. The transverse weight arises because the polarizations are not exactly longitudinal. The Christoffel matrix is $\mathsf{M} = C_{44}\mathbb{1} + (C_{12}+C_{44})\hat{\boldsymbol{q}}\hat{\boldsymbol{q}} + \Delta_C\,\mathrm{diag}(\hat{q}_i^2)$, an isotropic part plus the anisotropy. To first order in Δ_C the longitudinal eigenvector acquires a transverse admixture, so the transverse branches carry the weight $\sum_{\mathrm{TA}}(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2 = \Delta_C^2\, p_Q/(C_{12}+C_{44})^2$, with $p_Q = \sum_i \hat{q}_i^6 - (\sum_i \hat{q}_i^4)^2$, and since they are slower, each unit of that weight counts more. The limit check below compares the two forms.

Limit Check 11.3. Sphere-averaged longitudinal speed

A crystal enters the one-phonon nuclear rate through one sound speed, the longitudinal speed averaged over the sphere of directions, $\bar{c}_{\mathrm{LA}} = \langle c_{\mathrm{LA}}\rangle$ [3, App. A 3], from the Christoffel problem of the elastic tensor in Voigt order.

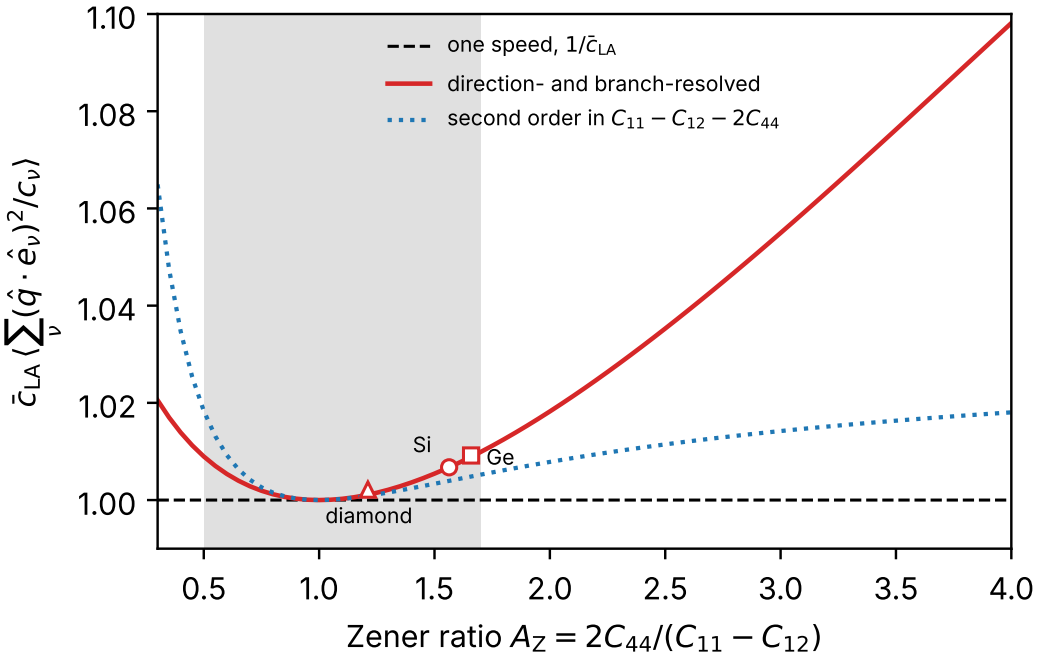


Figure 11.5: Direction- and branch-resolved coefficient of the heavy-mediator rate, $\langle\sum_\nu(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2/c_\nu\rangle$ of Eq. (11.9), in units of the single-speed form $1/\bar{c}_{\mathrm{LA}}$ (dashed, one), against the Zener ratio, for cubic tensors with the C_{11}, C_{12} and density of silicon and C_{44} varied (solid). Dotted: the expansion to second order in $\Delta_C = C_{11}-C_{12}-2C_{44}$, with the eigenvalue $C_{11}-2\Delta_C s_Q + \Delta_C^2 p_Q/(C_{12}+C_{44})$ and the transverse weight given in the text; it converges slowly, since $\Delta_C/(C_{12}+C_{44}) = -0.40$ for silicon. Points: silicon (1.0067), germanium (1.0092) and diamond (1.0021) with their own tensors. Gray: $0.50 \leq A_Z \leq 1.70$, where the single speed is within 1 %. Sphere averages on a 48×96 Gauss–Legendre product grid.

In strongly anisotropic crystals the slow quasi-transverse branches take up longitudinal

weight, and the single speed underestimates the rate by 10 % at $A_Z = 4$. No average serves the kinematic edge, which is set by the fastest direction.

For the three running crystals the single speed is good to a percent, and the limit check shows why: the Jensen gap is below 0.1 %, and almost all of the difference is the longitudinal weight of the transverse branches. The edge is a different matter. It is set by $c_{\mathrm{LA,max}}$, which is 3.7 % above $\bar{c}_{\mathrm{LA}}$ in silicon, and an edge computed with $\bar{c}_{\mathrm{LA}}$ is high by that amount.

A tabulated speed can be tested without the full tensor. The largest eigenvalue of the Christoffel matrix is at least its expectation value in the unit vector $\hat{\boldsymbol{q}}$. For cubic symmetry that expectation value is never below the bulk modulus $B = (C_{11} + 2C_{12})/3$ of Eq. (C.155), which gives Eq. (C.158):

$$c_{\mathrm{LA}}(\hat{\boldsymbol{q}}) \geq \sqrt{B/\rho_T} \qquad \text{(cubic crystals, every direction)}. \tag{11.11}$$

The bound is 6.48 km s^{-1} for silicon, 3.79 km s^{-1} for germanium and 11.23 km s^{-1} for diamond, 23, 24 and 36 % below their slowest longitudinal speeds. It is proved for cubic crystals only; for lower symmetries the uniform compression and the uniform strain no longer coincide (Appendix C.18), and the bound must be replaced by the Christoffel eigenvalue itself. A measured bulk modulus thus sets a floor that any cubic sound speed in a table must clear, and Sect. 17.5 applies it.

The same tensor gives the Debye temperature of Eq. (5.23), which detector studies quote as a measure of stiffness, although it is not an input to the rates above. In the convention of Anderson the speeds are those of an isotropic polycrystal, $v_L = [(B + \frac{4}{3}G)/\rho_T]^{1/2}$ and $v_T = (G/\rho_T)^{1/2}$, and the wavevector sphere counts every atom [7]. The shear modulus G is the Hill average, the mean of the Voigt value G_{V} and the Reuss value $G_{\mathrm{R}} = 5(C_{11} - C_{12})C_{44}/[4C_{44} + 3(C_{11} - C_{12})]$:

Elastic Debye temperature

$$\Theta_{\mathrm{D}} = v_m\Big(\frac{6\pi^2 n_{\mathrm{at}}}{\Omega}\Big)^{1/3}, \qquad \frac{3}{v_m^3} = \frac{1}{v_L^3} + \frac{2}{v_T^3}, \tag{11.12}$$

with n_{at} the number of atoms in the primitive cell of volume Ω, so that $(6\pi^2 n_{\mathrm{at}}/\Omega)^{1/3}$ is the Debye wavevector Q_{D} of Eq. (5.23).

For silicon, $G_{\mathrm{V}} = 68.0$ and $G_{\mathrm{R}} = 64.9$ GPa, $v_L = 8.95$, $v_T = 5.34$ and $v_m = 5.91$ km s^{-1} with the Hill average, and $\Theta_{\mathrm{D}} = 648$ K; the Voigt and Reuss bounds give 655 and 641 K. The exact angular average of Worked example 5.3, which uses $\bar{c}$ of Eq. (5.25) in place of v_m, gives 645 K. Germanium gives 375 and 373 K, diamond 2248 and 2246 K; the values quoted in detector studies, 645, 374 and 2220 K, agree with the Hill values within 1.2 % [8, Table I, 9, Table I]. The convention matters more than the averaging: counting one wavevector per primitive cell instead of per atom lowers Θ_{D} by $n_{\mathrm{at}}^{1/3}$, to 512 K for silicon (Sect. 5.2).

A coupling to charge, the dark photon, also excites the optical branches of a polar crystal, through the Born effective charges of Sect. 5.5; Sect. 11.2 derives that rate. And above q_{BZ} the single phonon gives way to the multiphonon response, which Sect. 11.3 builds from the same displacement correlations.

11.2 Optical phonons in polar crystals

A dark-matter particle of 100 keV that scatters in GaAs through a light dark photon most often leaves behind one longitudinal optical phonon of 36.35 meV. The dark photon does not

couple to the nucleons. It couples to the charge, and when the gallium and arsenic sublattices move against each other every cell carries the dipole of their Born charges (Sect. 5.5), whose field the electrons screen. In the acoustic waves of Sect. 11.1 the cell moves as a whole and stays neutral, and at long wavelength a coupling to charge has no vertex there; in the optical branch of GaAs its vertex is $Z^*/\varepsilon_\infty = 0.19$ per cell at every small momentum transfer (Worked example 8.6). Silicon and germanium, whose Born charges vanish, have no such channel. A polar target enters the rate through one band integral of its loss function, and the crystals with the largest band integrals are not the ones that reach the lightest dark matter.

11.2.1 The phonon loss function

A dark photon, heavy or light, couples to the charge-number density n_Z, and so does any particle with a small electric charge (Worked example 8.1). For this row Sect. 8.4 gave an exact answer: $S_{ZZ} = (2q^2/e^2)\, L/(1 - \mathrm{e}^{-\beta\omega})$, with $L = \mathrm{Im}[-(\varepsilon^{-1})_{00}]$ the loss function of the total dielectric matrix of electrons and nuclei. Below the band gap the electrons add a constant to ε, and every structure of L there comes from the ions. We need ε at the frequencies of the optical phonons and at momentum transfers small compared with the zone, where the long-range Coulomb line is the macroscopic field and the rank-one form Eq. (8.36) holds.

The dielectric function along $\hat{\boldsymbol{q}}$ follows from the energy Eq. (5.47). The electrons follow adiabatically and contribute $\varepsilon_\infty(\hat{\boldsymbol{q}}) \equiv \hat{\boldsymbol{q}} \cdot \varepsilon_\infty \cdot \hat{\boldsymbol{q}}$. The ions respond as the oscillators of the analytic zone-center problem, the eigenmodes of $D^{\mathrm{an}}_{\kappa\kappa'} = C^{\mathrm{an}}_{\kappa\kappa'}/(M_\kappa M_{\kappa'})^{1/2}$ with real polarization vectors and frequencies $\omega_{\mathrm{T}\nu}$, each driven by the field through its mode charge (Appendix D.14). Collecting the $3n$ components of the n sublattices into one vector,

$$\varepsilon(\hat{\boldsymbol{q}},\omega) = \varepsilon_\infty(\hat{\boldsymbol{q}}) + \frac{e^2}{\Omega}\, \boldsymbol{p}^{\mathrm{T}} \big(D^{\mathrm{an}} - \omega^2\big)^{-1} \boldsymbol{p}, \qquad p_{\kappa\alpha}(\hat{\boldsymbol{q}}) = \frac{(Z^*_\kappa \cdot \hat{\boldsymbol{q}})^\alpha}{\sqrt{M_\kappa}}, \tag{11.13}$$

which is Eq. (5.53) for any number of atoms and any direction. Expanded in the eigenmodes of D^{an}, the second term is a sum of poles at the transverse frequencies, each weighted by the square of the mode charge along $\hat{\boldsymbol{q}}$.

The loss function needs $1/\varepsilon$, and the structure of Eq. (11.13) makes the inversion exact. The ionic term is a matrix element of the inverse of $A = D^{\mathrm{an}} - \omega^2$ between one vector, and the Sherman–Morrison identity, $(A + \boldsymbol{b}\boldsymbol{b}^{\mathrm{T}})^{-1} = A^{-1} - A^{-1}\boldsymbol{b}\boldsymbol{b}^{\mathrm{T}}A^{-1}/(1 + \boldsymbol{b}^{\mathrm{T}}A^{-1}\boldsymbol{b})$, gives $\boldsymbol{b}^{\mathrm{T}}(A + \boldsymbol{b}\boldsymbol{b}^{\mathrm{T}})^{-1}\boldsymbol{b} = \boldsymbol{b}^{\mathrm{T}}A^{-1}\boldsymbol{b}/(1 + \boldsymbol{b}^{\mathrm{T}}A^{-1}\boldsymbol{b})$. With $\boldsymbol{b} = e\boldsymbol{p}/[\Omega\varepsilon_\infty(\hat{\boldsymbol{q}})]^{1/2}$, Eq. (11.13) reads $\varepsilon = \varepsilon_\infty(\hat{\boldsymbol{q}})(1 + \boldsymbol{b}^{\mathrm{T}}A^{-1}\boldsymbol{b})$, and therefore

$$\frac{1}{\varepsilon(\hat{\boldsymbol{q}},\omega)} = \frac{1}{\varepsilon_\infty(\hat{\boldsymbol{q}})}\Big[1 - \boldsymbol{b}^{\mathrm{T}}\big(D^{\mathrm{an}} + \boldsymbol{b}\boldsymbol{b}^{\mathrm{T}} - \omega^2\big)^{-1}\boldsymbol{b}\Big]. \tag{11.14}$$

The matrix $D^{\mathrm{an}} + \boldsymbol{b}\boldsymbol{b}^{\mathrm{T}}$ is the dynamical matrix Eq. (5.51): $\boldsymbol{b}\boldsymbol{b}^{\mathrm{T}}$ is exactly its nonanalytic term. Inverting ε has traded the poles of ε, at the transverse frequencies, for the eigenfrequencies of the full dynamical matrix, which include the longitudinal ones. Let ω_ν and $\boldsymbol{e}_\nu$ be its eigenpairs for the direction $\hat{\boldsymbol{q}}$. Then $\boldsymbol{b}^{\mathrm{T}}(D^{\mathrm{dyn}} - \omega^2)^{-1}\boldsymbol{b} = \sum_\nu (\boldsymbol{b}\cdot\boldsymbol{e}_\nu)^2/(\omega_\nu^2 - \omega^2)$, and with $\omega \to \omega + \mathrm{i}0^+$ each term has the imaginary part $\pi(\boldsymbol{b}\cdot\boldsymbol{e}_\nu)^2\,\delta(\omega - \omega_\nu)/2\omega_\nu$ for $\omega > 0$. The loss function in the phonon band is a set of lines,

$$L(\hat{\boldsymbol{q}},\omega) = \frac{\pi e^2}{2\Omega\,\varepsilon_\infty(\hat{\boldsymbol{q}})^2} \sum_\nu \frac{(\boldsymbol{p}\cdot\boldsymbol{e}_\nu)^2}{\omega_\nu}\, \delta(\omega - \omega_\nu).$$

Nothing in these steps needs $\boldsymbol{q} = 0$ except the vertex. At a wavevector inside the zone, D^{an} becomes the dynamical matrix at $\boldsymbol{Q}$ with the macroscopic field removed, each component of $\boldsymbol{p}$ takes the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}$ of its site, and $D^{\mathrm{an}} + \boldsymbol{b}\boldsymbol{b}^{\mathrm{T}}$ is the full dynamical matrix at $\boldsymbol{Q}$. Multiplying by $2q^2/e^2$:

Born-charge one-phonon structure factor

$$S^{(1)}_{ZZ}(\boldsymbol{q},\omega) = \frac{2\pi}{\Omega}\sum_{\nu}\frac{\left|\sum_{\kappa} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}\, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^* \cdot Z^*_\kappa \cdot \boldsymbol{q}/\sqrt{M_\kappa}\right|^2}{2\omega_{\boldsymbol{Q}\nu}\left(\hat{\boldsymbol{q}}\cdot\varepsilon_\infty\cdot\hat{\boldsymbol{q}}\right)^2}\,\delta(\omega-\omega_{\boldsymbol{Q}\nu}), \tag{11.15}$$

at $T = 0$, for q small compared with the zone and ω far below the gap, with the phonons of the full dynamical matrix, nonanalytic term included, and the phase convention of Eq. (11.1).

The structure is that of the nucleon response Eq. (11.1), with the nucleon number A_κ replaced by the Born charge projected on $\boldsymbol{q}$ and the whole vertex divided by $\varepsilon_\infty(\hat{\boldsymbol{q}})$. Each branch contributes the square of its dipole along $\boldsymbol{q}$, screened by the electrons, divided by its energy. At the zone center the acoustic branches drop out by the neutrality Eq. (5.50), and in a cubic crystal so do the transverse optical ones, whose dipole is perpendicular to $\boldsymbol{q}$. What remains is the longitudinal mode, at ω_L. The proper response of the ions has its poles at ω_T, but the transverse pole of ε is a zero of $1/\varepsilon$, and a coupling to charge excites the longitudinal phonon, as in absorption (Sect. 5.5).

Trickle and collaborators reach the same result with the ions in the vertex: they couple to $\boldsymbol{q}\cdot Z^*_\kappa$ and screen with the electrons alone [10, Eq. (105)]. The loss function of the total ε and the Born-charge vertex screened by ε_∞ are two resummations of the series Eq. (8.35), one with the ions inside the dielectric function and one with them in the vertex, and Eq. (11.14) is the step between them. The vertex is also familiar from Sect. 6.2. It is the Fröhlich vertex Eq. (D.77) with the electron replaced by the dark particle: the density vertex ρ_{mn} becomes one, and the charge of the electron becomes the coupling of the dark matter. Griffin and collaborators write the dark-photon matrix element in this form, up to the phase convention of the eigenvectors [11, Eq. (19)].

Every input of Eq. (11.15) is an output of density-functional perturbation theory: phonons at $\boldsymbol{Q}$ with the nonanalytic term, Born charges and ε_∞ (Appendix E.4). The rate carries $(Z^*/\varepsilon_\infty)^2$, and the semilocal ε_∞ of GaAs, 26 % too high (Sect. 5.5), would make it 37 % too low. We use instead the computed phonons and Born charge with the measured room-temperature $\varepsilon_\infty = 10.88$ and $\varepsilon_0 = 12.9$ of GaAs [12], or the measured splitting itself (Sect. 11.2.4).

Eq. (11.15) holds when the momentum transfer is small compared with the zone, the energy far below the gap, and the lattice harmonic and cold. At small momentum transfer the charge that moves with a nucleus is its Born charge; corrections to this vertex start at relative order qa. Far below the gap the electrons follow the ions adiabatically and the vertex may be computed at zero frequency; by the argument of Sect. 8.4 the error is of order $(\omega_\mathrm{L}/E_\mathrm{g})^2$, 6×10^{-4} in GaAs. For the cold harmonic lattice, the Debye–Waller factor is $1-8\times10^{-8}$ at the smallest momentum transfer, $q = \omega_\mathrm{L}/v_\mathrm{max} = 13.7\,\mathrm{eV}$, with the root-mean-square displacement of Table 5.1. At 10 mK, $\omega_\mathrm{L}/T = 4.2\times10^4$ and no optical phonon is thermally occupied. The anharmonic width of the longitudinal phonon, a fraction of a meV, spreads the line without moving its weight. The small-q vertex is the condition that matters in practice, and the limit check tests it with the phonons of Fig. 5.5.

Limit Check 11.4. Born-charge single-phonon rate

A particle that couples to charge creates one phonon with the structure factor Eq. (11.15): Born charges, screened by ε_∞, acting on the phonons of the full dynamical matrix [10, Eqs. (92), (105), (106), 11, Eqs. (19), (23), 13, Eq. (61)]. For one longitudinal branch this is the form of Knapen and collaborators [14, Eqs. (4)–(7)].

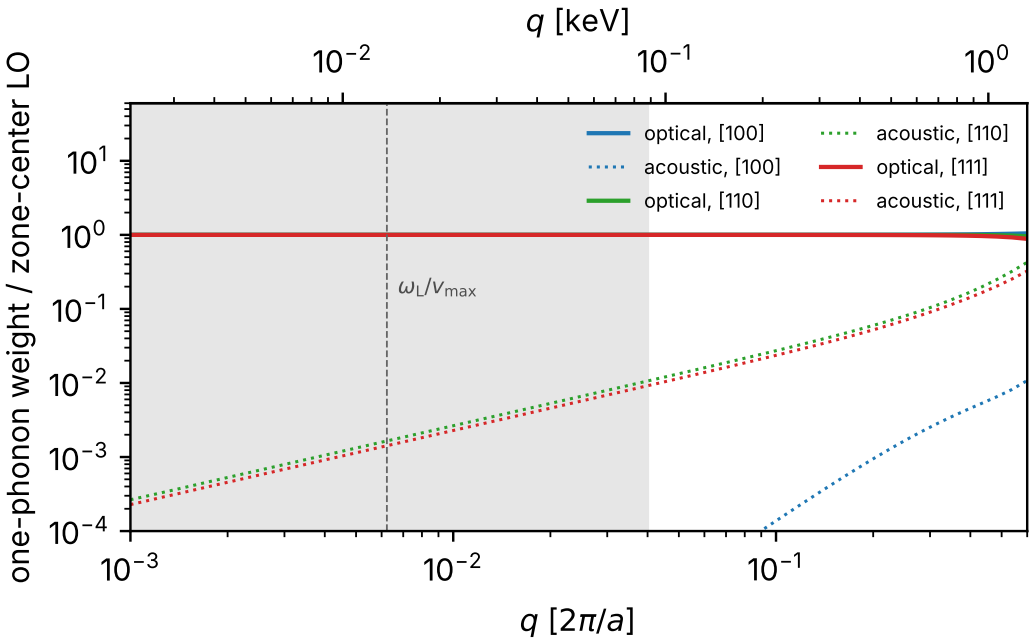


Figure 11.6: One-phonon weight of GaAs in Eq. (11.15), $\sum_\nu |\sum_\kappa e^{i\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa} \boldsymbol{e}^*_{\kappa\nu} \cdot Z^*_\kappa \cdot \hat{\boldsymbol{q}}/\sqrt{M_\kappa}|^2/2\omega_\nu$, divided by its zone-center value $Z^{*2}/2\mu_{\mathrm{red}}\omega_{\mathrm{L}}$, against q along [100], [110] and [111]. Solid: the three optical branches; dotted: the three acoustic branches. Phonons at the actual $\boldsymbol{q}$ from the PBE force constants of Fig. 5.5; Born charges held at their zone-center value. The ratios do not depend on Z^* or ε_∞. Gray: $q < 0.04 \times 2\pi/a = 88\,\mathrm{eV}$, where the optical weight is within 10^{-4} of the formula and the acoustic weight is at most about 1 %. Dashed line: $q = \omega_{\mathrm{L}}/v_{\max}$, the smallest momentum transfer that reaches the longitudinal phonon.

At q comparable with $2\pi/a$ the Born charge no longer describes the charge that moves with a nucleus, and the acoustic branches acquire a piezoelectric charge that the Born charges fix only in part.

The optical weight barely moves. It stays within 10^{-4} of its zone-center value up to 88 eV and within 0.4 % up to 450 eV, where the longitudinal branch has dispersed by 0.3 meV. The acoustic branches are the leading correction, and they grow linearly in q: their weight reaches 1 % of the optical one at 90 eV along [111] and 5 % at 450 eV, while along [100] it vanishes by symmetry. This is the piezoelectric charge of Sect. 6.2. With rigid Born charges an acoustic wave polarizes the cell only through the relative displacement of the sublattices that its strain induces; the dynamical quadrupoles of the displaced atoms, the next term beyond the Born charge in the expansion of the charge that moves with an atom in powers of q, add a contribution of the same order (Appendix E.5), which the limit check does not include. The dotted curves show the size of the correction, not its value.

The same eigenvectors change the nucleon vertex of Worked example 8.6 far more. That vertex vanishes as $q \to 0$, apart from the small differences of A_κ/M_κ, and its value at finite q comes entirely from the order-qa structure of the longitudinal mode. Zone-center eigenvectors with site phases do not capture it: the phases depend on which atom is assigned to the cell, and the eigenvectors at $\boldsymbol{q}$ compensate them. What survives is physical. In zincblende a shear strain displaces the two sublattices against each other, the internal strain behind

the piezoelectric charge above, and by the same coupling an optical wave drags the center of mass along wherever a longitudinal wave contains shear. Along [111] it does, and with the eigenvectors at $\boldsymbol{q}$, $|a_N| = 0.24$ at 13.7 eV and 1.74 at 100 eV along the bond, the values of Worked example 8.6; zone-center eigenvectors with site phases give 0.61 and 4.5, a factor 2.6 too large. Along [100] it does not, at first order in q, and $|a_N| = 2.1 \times 10^{-3}$ at 13.7 eV, close to its value at $q \to 0$, where the zone-center estimate gives 0.35. A coupling to the mass density alone gives the same pattern. The charge vertex, finite at $q = 0$, is insensitive to the correction.

11.2.2 The polar single-phonon rate

The dark matter enters through one potential. A dark photon couples to $-\tilde{V} n_Z$, with the same $\tilde{V}$ on electrons and nuclei, and we normalize it as for electron scattering, $|\tilde{V}|^2 = \pi \bar{\sigma}_e F_{\rm DM}^2 / \mu_{\chi e}^2$ (Eq. (1.34)), with $F_{\rm DM} = (q_0/q)^2$ for a light dark photon. We average the structure factor over the directions of $\boldsymbol{q}$, so that the halo enters through $\eta(v_{\rm min})$ (Eq. (A.42)); the direction dependence that a crystal keeps is the subject of Sect. 11.7. With $S_{ZZ} = (2q^2/e^2)\bar{L}$ at $T = 0$, the rate Eq. (A.44) becomes, for the light mediator,

Polar single-phonon rate

$$\mathcal{R} = \frac{\rho_\chi \, \bar{\sigma}_e \, q_0^4}{2\pi \, m_\chi \, \mu_{\chi e}^2 \, e^2 \, \rho_T} \int_{\rm band} {\rm d}\omega \; \bar{L}(\omega) \, K_\eta(\omega, m_\chi), \qquad K_\eta(\omega, m_\chi) = \int \frac{{\rm d}q}{q} \; \eta(v_{\rm min}(q,\omega)), \tag{11.16}$$

with $\bar{L}$ the loss function at small q, averaged over directions, and the integral over the phonon band.

All powers of q cancel but one. A light mediator weights every logarithmic interval of momentum transfer equally, and the phonon fixes the energy deposited. The rate is then the loss function summed over the band, each line weighted by a kinematic factor that depends only on its energy and the dark-matter mass. Since $v_{\rm min}(q,\omega) \geq (2\omega/m_\chi)^{1/2}$ at every q, the factor vanishes below the mass at which the fastest particle can deposit ω at all, $2\omega/v_{\rm max}^2$ (Eq. (1.25)), which is 10.35 keV for the longitudinal phonon of GaAs. For a single speed v, $\eta = \Theta(v - v_{\rm min})/v$ and $K_\eta = \ln(q_+/q_-)/v$, with $q_\pm = m_\chi v[1 \pm (1 - 2\omega/m_\chi v^2)^{1/2}]$ the ends of the kinematic window.

For a heavy mediator, $F_{\rm DM} = 1$, the same steps replace K_η by $q_0^{-4} \int {\rm d}q \, q^3 \eta$, and the rate is weighted toward the largest momentum transfer, $2 m_\chi v_{\rm max}$. It leaves the region of Fig. 11.6, where the acoustic share stays below a few percent, at about $0.2 \times 2\pi/a$, or $m_\chi \approx 83$ keV in GaAs. The optical phonons alone keep the one-branch form to about 0.1 % at 99 keV; the vertex is what fails first.

Fig. 11.7a shows the result for GaAs. The light-mediator rate rises from the edge to about 840 events per kilogram and year near 29 keV for $\bar{\sigma}_e = 10^{-40}\,{\rm cm}^2$, and falls at larger masses as $\mu_{\chi e}^{-2} m_\chi^{-1}$, with a logarithmic rise of K_η on top. The full calculation, with every optical branch at the actual $\boldsymbol{q}$ and averaged over directions, agrees with the one-branch form of the next subsection to 3×10^{-4} below 300 keV. At 1 MeV it is 4 % lower, because part of the logarithmic integral then comes from momentum transfers beyond the zone. The acoustic branches, with rigid Born charges and a 1 meV threshold, add at most 3 %, near the edge, and about 1 % at 100 keV.

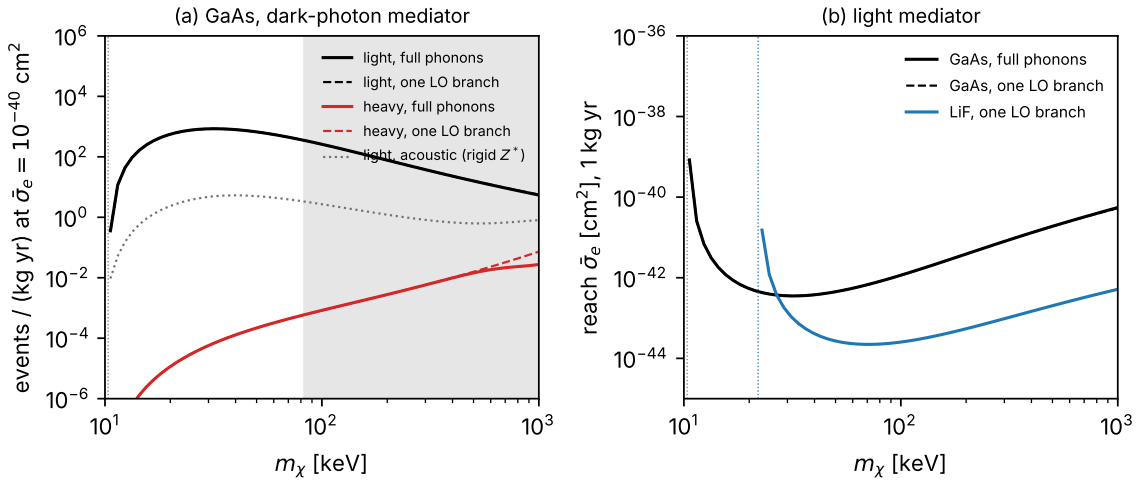


Figure 11.7: Single-phonon scattering of dark matter through a dark photon in polar crystals, for the standard halo of Sect. 1.2 at $T = 0$. (a) Rate of GaAs per kilogram and year at $\bar{\sigma}_e = 10^{-40}\,\mathrm{cm}^2$, light ($F_{\mathrm{DM}} = (q_0/q)^2$) and heavy ($F_{\mathrm{DM}} = 1$) mediator. Solid: Eq. (11.15) with the three optical branches of the PBE phonons of Fig. 5.5, at the actual $\boldsymbol{q}$ (Umklapp included), averaged over 146 directions, in Eq. (A.44). Dashed: one longitudinal branch at the zone center, Eq. (11.17) in Eq. (11.16). Both use $Z^* = 2.11$, the measured $\varepsilon_\infty = 10.88$, $\omega_{\mathrm{L}} = 36.34\,\mathrm{meV}$ (DFPT) and $\rho_T = 5.361\,\mathrm{g\,cm}^{-3}$; for the light mediator the two curves coincide to 3×10^{-4} below 300 keV. Dotted: the acoustic branches with the same rigid Born charges and a 1 meV threshold, an estimate of the piezoelectric correction (Fig. 11.6). Gray: $2m_\chi v_{\mathrm{max}} > 0.2 \times 2\pi/a$, where the Born-charge vertex is no longer reliable for the heavy mediator. (b) Light-mediator reach, three events in one kilogram-year without background: GaAs as in (a), and LiF with one longitudinal branch and the computed $Z^* = 1.05$, $\varepsilon_\infty = 2.02$ and $\omega_{\mathrm{L}} = 77.2\,\mathrm{meV}$ of Ref. [3, Table I]. Vertical lines: the edges $2\omega_{\mathrm{L}}/v_{\mathrm{max}}^2 = 10.35$ and 21.98 keV.

11.2.3 The band integral

GaAs has one polar branch, and the loss function has one line. By Eq. (5.53), $1/\varepsilon = \varepsilon_\infty^{-1}(\omega_\mathrm{T}^2 - \omega^2)/(\omega_\mathrm{L}^2 - \omega^2)$, whose pole at ω_L leads to $L = (\pi\omega_\mathrm{L}/2)\varepsilon_\infty^{-1}(1 - \omega_\mathrm{T}^2/\omega_\mathrm{L}^2)\,\delta(\omega - \omega_\mathrm{L})$. By the Lyddane–Sachs–Teller relation the bracket is $1 - \varepsilon_\infty/\varepsilon_0$, and

$$S_{ZZ}(\boldsymbol{q},\omega) = \frac{\pi q^2 \omega_\mathrm{L}}{e^2}\Big(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0}\Big)\,\delta(\omega - \omega_\mathrm{L}) = \frac{\pi q^2 Z^{*2}}{\Omega\,\mu_\mathrm{red}\,\varepsilon_\infty^2\,\omega_\mathrm{L}}\,\delta(\omega - \omega_\mathrm{L}). \tag{11.17}$$

The second form is Eq. (11.15) with the longitudinal eigenvector of Appendix D.14, $\boldsymbol{e}_1 = (M_2/M_\mathrm{cell})^{1/2}\hat{\boldsymbol{q}}$ and $\boldsymbol{e}_2 = -(M_1/M_\mathrm{cell})^{1/2}\hat{\boldsymbol{q}}$, for which $\sum_\kappa \boldsymbol{e}_\kappa \cdot Z_\kappa^* \cdot \boldsymbol{q}/\sqrt{M_\kappa} = Z^* q/\sqrt{\mu_\mathrm{red}}$; the two forms are equal by Eqs. (5.52) and (5.54). In the language of amplitudes, a particle of charge e' has $\tilde{V} = ee'/q^2$, and $|\tilde{V}|^2 S_{ZZ} = 2\pi\delta(\omega - \omega_\mathrm{L})\,e'^2(\omega_\mathrm{L}/2)(1/\varepsilon_\infty - 1/\varepsilon_0)/q^2$. This is the square of the matrix element of Griffin and collaborators [11, Eqs. (21), (22)], whose two forms, with the Born charge and with the dielectric constants, are the two sides of Eq. (11.17). They reach the form with Z^* by approximating the mean of $1/\sqrt{M_\kappa}$ by $1/\sqrt{\mu_\mathrm{red}}$, which needs nearly equal masses; the exact eigenvectors give $1/\sqrt{\mu_\mathrm{red}}$ for any masses. For one dark-matter speed, Eq. (11.16) without the halo average gives the scattering rate of one dark-matter particle of speed v in the crystal (of dimension energy in natural units),

$$\Gamma(v) = \frac{\bar{\sigma}_e\, q_0^4\, \omega_\mathrm{L}}{4\mu_{\chi e}^2 e^2\, v}\Big(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0}\Big)\ln\frac{1 + (1 - 2\omega_\mathrm{L}/m_\chi v^2)^{1/2}}{1 - (1 - 2\omega_\mathrm{L}/m_\chi v^2)^{1/2}}\,\Theta(m_\chi v^2 - 2\omega_\mathrm{L}), \tag{11.18}$$

which is their closed form [11, Eq. (24)] with $e'^2 = \pi\bar{\sigma}_e q_0^4/(\mu_{\chi e}^2 e^2)$, the relation between e' and $\bar{\sigma}_e$ of Knapen and collaborators [14, Eq. (8)].

With several polar branches, how is the weight of L shared among them? A sum rule answers part of the question exactly. The inverse-moment relation Eq. (4.48), a consequence of causality alone, reads $(2/\pi)\int \mathrm{d}\omega\, L/\omega = 1 - 1/\varepsilon_0$ for the total dielectric function and $1 - 1/\varepsilon_\infty$ with the nuclei clamped. Since the gap separates the two bands, the difference belongs to the phonons:

$$\frac{2}{\pi}\int_\mathrm{band} \mathrm{d}\omega\,\frac{L(\omega)}{\omega} = \frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0}. \tag{11.19}$$

At small q the band is a set of lines, $L = \sum_\nu (\pi\omega_{\mathrm{L}\nu}/2)\,w_\nu^\mathrm{pol}\,\delta(\omega - \omega_{\mathrm{L}\nu})$, with the weights $w_\nu^\mathrm{pol} = e^2(\boldsymbol{p}\cdot\boldsymbol{e}_\nu)^2/(\Omega\varepsilon_\infty^2\omega_{\mathrm{L}\nu}^2)$ read off from the line formula above, and Eq. (11.19) says that $\sum_\nu w_\nu^\mathrm{pol} = 1/\varepsilon_\infty - 1/\varepsilon_0$. The ionic part of the inverse static screening is shared out among the longitudinal branches. The rate needs the zeroth moment instead, the *band integral*

$$\begin{gathered}\mathcal{I}_\mathrm{pol} = \int_\mathrm{band} \mathrm{d}\omega\; L(\omega) = \frac{\pi}{2}\sum_\nu \omega_{\mathrm{L}\nu}\, w_\nu^\mathrm{pol}, \\ \frac{\pi}{2}\Big(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0}\Big)\omega_\mathrm{L,min} \le \mathcal{I}_\mathrm{pol} \le \frac{\pi}{2}\Big(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0}\Big)\omega_\mathrm{T,max}\Big(\frac{\varepsilon_0}{\varepsilon_\infty}\Big)^{1/2},\end{gathered} \tag{11.20}$$

bracketed by the sum rule. The upper end uses $\omega_\mathrm{L,max} \le \omega_\mathrm{T,max}(\varepsilon_0/\varepsilon_\infty)^{1/2}$, which holds because every ratio $\omega_{\mathrm{L}\nu}/\omega_{\mathrm{T}\nu}$ exceeds one and their squares multiply to $\varepsilon_0/\varepsilon_\infty$ (Eq. (D.76)). It needs nothing but the transverse frequencies and the two dielectric constants.

In a crystal of lower symmetry these statements hold along each principal axis separately. Along a principal axis the lossless ε of Eq. (11.13) rises monotonically between consecutive poles, so that each transverse pole is followed by one longitudinal zero, $\omega_\mathrm{T1} < \omega_\mathrm{L1} < \omega_\mathrm{T2} < \ldots$ (Appendix D.14), and $\varepsilon = \varepsilon_\infty \prod_\nu (\omega_{\mathrm{L}\nu}^2 - \omega^2)/(\omega_{\mathrm{T}\nu}^2 - \omega^2)$. The weights are then the residues of $-1/\varepsilon$ at its zeros, and the ε_0 and ε_∞ of the sum rule must belong to the same axis and the same model. For a direction off the principal axes the zeros of $\hat{\boldsymbol{q}}\cdot\varepsilon\cdot\hat{\boldsymbol{q}}$ move with the direction, and the bracket applies axis by axis.

For a diatomic crystal, Eq. (11.17) gives $\mathcal{I}_{\rm pol} = (\pi/2)e^2Z^{*2}/(\Omega\mu_{\rm red}\varepsilon_\infty^2\omega_{\rm L})$, and per unit mass, with $\rho_T = M_{\rm cell}/\Omega$,

$$\frac{\mathcal{I}_{\rm pol}}{\rho_T} = \frac{\pi}{2}\,\frac{e^2Z^{*2}}{M_{\rm cell}\,\mu_{\rm red}\,\varepsilon_\infty^2\,\omega_{\rm L}}, \tag{11.21}$$

in which the cell volume has dropped out. In general Eq. (11.16) is a sum over branches, $\sum_\nu(\pi\omega_{{\rm L}\nu}/2)(w_\nu^{\rm pol}/\rho_T)K_\eta(\omega_{{\rm L}\nu}, m_\chi)$. When K_η changes little across the band, it is $\mathcal{I}_{\rm pol}/\rho_T$ times K_η at the weighted mean frequency $\bar\omega_{\rm pol} = \sum_\nu \omega_{{\rm L}\nu}w_\nu^{\rm pol}/\sum_\nu w_\nu^{\rm pol}$, and the material enters through $\mathcal{I}_{\rm pol}/\rho_T$ alone. Sapphire, with four polar branches on its ordinary axis, tests this reading.

Limit Check 11.5. Polar matrix element and band integral

For one polar branch a particle of charge e' has $|\mathcal{M}|^2 = e'^2(\omega_{\rm L}/2)(1/\varepsilon_\infty - 1/\varepsilon_0)/q^2$, and its rate is Eq. (11.18) [11, Eqs. (21), (22), (24)]; with several branches the light-mediator rate per unit mass is proportional to $\mathcal{I}_{\rm pol}/\rho_T$.

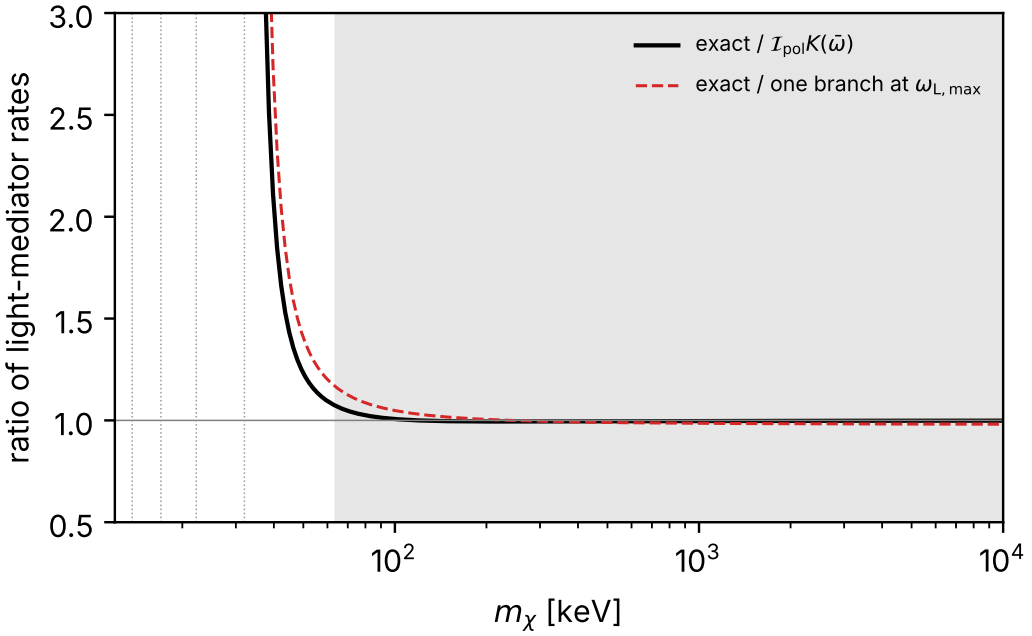


Figure 11.8: Light-mediator rate of the ordinary axis of sapphire, the exact sum over its four longitudinal branches, divided by the band-integral form $\mathcal{I}_{\rm pol}K_\eta(\bar\omega_{\rm pol}, m_\chi)$ (solid) and by the one-branch form with the highest longitudinal frequency and the same $1/\varepsilon_\infty - 1/\varepsilon_0$ (dashed), against m_χ. Lossless factorized ε from the measured frequencies of Ref. [11, Table II] (ordinary ray: longitudinal 112.4, 78.0, 59.7, 48.1 meV; transverse 78.6, 70.5, 54.4, 47.7 meV) and $\varepsilon_\infty = 3.2$ [11, Table I]; weights from the residues of $-1/\varepsilon$, $\bar\omega_{\rm pol} = 109.9$ meV; standard halo of Sect. 1.2. Dotted lines: the four edges $2\omega_{{\rm L}\nu}/v_{\rm max}^2$, from 13.7 to 32.0 keV. Gray: m_χ above twice the highest edge, 64 keV, where the band-integral form is within 7 %.

Within about twice the highest edge the branches open one by one, the lowest first, and the rate follows the individual longitudinal frequencies rather than their weighted sum.

On the ordinary axis of sapphire one branch takes almost all the weight. The longitudinal phonon at 112.4 meV carries 95.2 % of $1/\varepsilon_\infty - 1/\varepsilon_0 = 0.210$, and $\mathcal{I}_{\rm pol} = 36.3$ meV, about

70 % of the way up the bracket Eq. (11.20), [15.9, 45.4] meV. With the highest longitudinal frequency itself, 112.4 meV, in place of its bound $\omega_{\mathrm{T,max}}(\varepsilon_0/\varepsilon_\infty)^{1/2}$, the upper end tightens to 37.1 meV. With $\rho_T = 3.98\,\mathrm{g\,cm^{-3}}$ [11, Table I], $\mathcal{I}_{\mathrm{pol}}/\rho_T = 9.12\,\mathrm{meV\,cm^3\,g^{-1}}$. Above 64 keV the band-integral form is good to 7 %. At 40 keV, however, the true rate is 2.1 times the band-integral form, because the branches between 48 and 78 meV are already well open while the dominant one has barely opened. The one-branch form with the highest frequency does slightly worse at every mass, since it ignores the lower branches altogether. Damping adds a correction of order $\gamma_{\mathrm{L}}/\omega_{\mathrm{L}}$, the width of the line over its frequency. For the dominant phonon of sapphire, $\gamma_{\mathrm{L}} = 16\,\mathrm{cm^{-1}}$ at 77 K [11, Table II], or 1.8 %; the Lorentzian line adds weight in its tails.

The static dielectric constant is the quantity most often tabulated, and the one-branch relation between it and the Born charge, Eq. (5.53) at $\omega = 0$,

$$\varepsilon_0 - \varepsilon_\infty = \frac{e^2 Z^{*2}}{\Omega\,\mu_{\mathrm{red}}\,\omega_{\mathrm{T}}^2}, \tag{11.22}$$

is the one most often used to estimate a polar response. With several branches the right side becomes a sum over the transverse poles, $(e^2/\Omega)\sum_\nu (\boldsymbol{z}_\nu \cdot \hat{\boldsymbol{e}})^2/\omega_{\mathrm{T}\nu}^2$, with the mode charges $\boldsymbol{z}_\nu$ of Sect. 5.5 (Appendix D.14). The rate, however, weights the longitudinal zeros of ε, through w_ν^{pol}, and the two weightings need not agree.

Limit Check 11.6. Static dielectric constant of one polar branch

For one transverse optical branch the lattice adds $\varepsilon_0 - \varepsilon_\infty = e^2 Z^{*2}/(\Omega\mu_{\mathrm{red}}\omega_{\mathrm{T}}^2)$ to the static dielectric constant [15, Chap. II, 11, Eq. (B14) at $\omega = 0$].

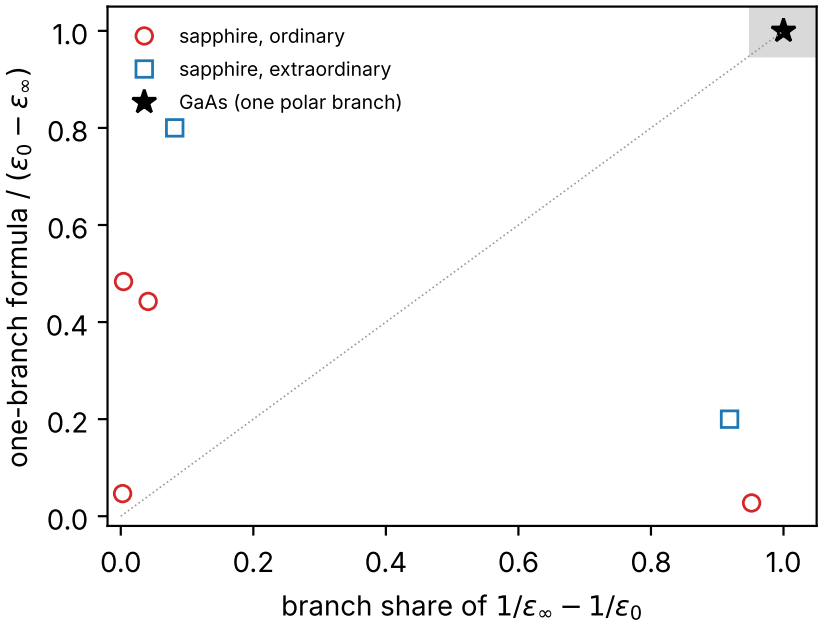


Figure 11.9: Each polar branch of sapphire treated as the only one. Vertical axis: Eq. (11.22) with that branch alone, as a fraction of the full $\varepsilon_0 - \varepsilon_\infty$, which is the sum over all transverse poles (6.58 on the ordinary axis, 8.90 on the extraordinary one). Horizontal axis: the share of the same branch in $1/\varepsilon_\infty - 1/\varepsilon_0$, the weight it carries in the rate. Lossless factorized ε from the frequencies of Ref. [11, Table II], on the ordinary axis those listed in Fig. 11.8 and on the extraordinary axis longitudinal 109.2

and 63.3 meV and transverse 72.2 and 49.3 meV, with $\varepsilon_\infty = 3.2$ [11, Table I]; branches paired in order of frequency. GaAs, with one polar branch, sits at (1, 1). Gray: both shares above 0.95.

With several polar branches the static constant is shared among the transverse poles while the loss function sits at the longitudinal zeros, and no single branch carries both.

The highest branch of sapphire's ordinary axis is the extreme case. It carries 95 % of the weight in the rate and 2.7 % of $\varepsilon_0 - \varepsilon_\infty$, whose bulk sits at the transverse poles of 70.5 and 54.4 meV. The two weights measure different things. A transverse pole contributes its own oscillator strength, and the highest one of sapphire, at 78.6 meV, lies within 0.5 meV of the zero below it and has almost none. The highest longitudinal zero is a collective mode: the macroscopic field of all the dipoles together pushes it up to 112.4 meV, and like a plasmon it collects nearly all of the loss weight. In GaAs both weights belong to the one branch, and Eq. (11.22) and Eq. (11.17) contain the same information. A ranking of polar targets that reads the rate off $\varepsilon_0 - \varepsilon_\infty$, or off a mean optical frequency, fails exactly in the crystals with many branches. This is why $\mathcal{I}_{\mathrm{pol}}$ is built from $1/\varepsilon_\infty - 1/\varepsilon_0$ and the longitudinal frequencies (Sect. 17.5).

11.2.4 The Born charge from the splitting

For a diatomic crystal the rate needs one combination of material numbers, Z^*/ε_∞, the screened dipole per unit displacement. Solving Eq. (5.52) for it,

$$\frac{Z^*}{\varepsilon_\infty} = \Big[\frac{\Omega\,\mu_{\mathrm{red}}\,(\omega_{\mathrm{L}}^2 - \omega_{\mathrm{T}}^2)}{e^2\,\varepsilon_\infty}\Big]^{1/2}, \tag{11.23}$$

which needs two phonon frequencies, the cell, the masses and ε_∞, but not the static constant. For GaAs, Worked example 5.6 has already extracted $Z^* = 2.11$ from the measured splitting, and by Eq. (11.23) $Z^*/\varepsilon_\infty = 0.194$. The error budget follows from the logarithmic derivative,

$$\frac{\delta(Z^*/\varepsilon_\infty)}{Z^*/\varepsilon_\infty} = -\frac{1}{2}\frac{\delta\varepsilon_\infty}{\varepsilon_\infty} + \frac{\omega_{\mathrm{L}}\,\delta\omega_{\mathrm{L}} - \omega_{\mathrm{T}}\,\delta\omega_{\mathrm{T}}}{\omega_{\mathrm{L}}^2 - \omega_{\mathrm{T}}^2}.$$

A common calibration error of the two frequencies cancels to first order. Independent errors are amplified by $\omega^2/(\omega_{\mathrm{L}}^2 - \omega_{\mathrm{T}}^2) \approx 6.9$, so that 0.1 % on each frequency costs 0.9 %, and together with 1 % on ε_∞ the result is good to 1.0 %. The dielectric constant should be the one at the temperature of the detector, since it changes at the percent level between room temperature and liquid-helium temperature [11].

Limit Check 11.7. Born charge over ε_∞ from the LO–TO splitting

The screened Born charge of a diatomic crystal follows from its splitting, $Z^*/\varepsilon_\infty = [\Omega\mu_{\mathrm{red}}(\omega_{\mathrm{L}}^2 - \omega_{\mathrm{T}}^2)/(e^2\varepsilon_\infty)]^{1/2}$, the relation of Born and Huang [15, Chap. II, 11, Eqs. (B11), (B15)], with the Lyddane–Sachs–Teller relation [16].

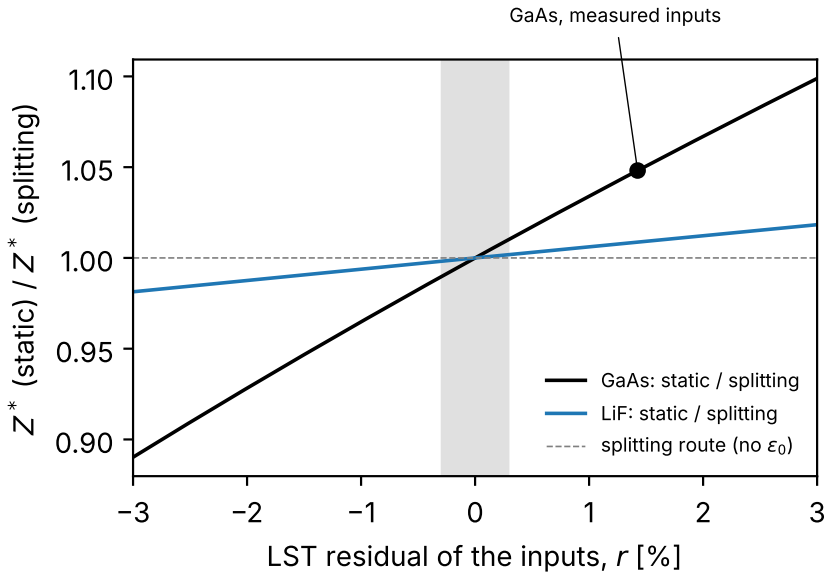


Figure 11.10: Born charge from the static route, $Z^{*2} = \Omega\mu_{\mathrm{red}}\omega_{\mathrm{T}}^2(\varepsilon_0 - \varepsilon_\infty)/e^2$, divided by the one from the splitting, Eq. (11.23) times ε_∞ (dashed, one), against the residual $r = (\varepsilon_0/\varepsilon_\infty)/(\omega_{\mathrm{L}}/\omega_{\mathrm{T}})^2 - 1$ by which the four inputs miss the Lyddane–Sachs–Teller relation. The ratio is $[((1+r)\omega_{\mathrm{L}}^2 - \omega_{\mathrm{T}}^2)/(\omega_{\mathrm{L}}^2 - \omega_{\mathrm{T}}^2)]^{1/2}$. Black: GaAs, $\omega_{\mathrm{L}}/\omega_{\mathrm{T}} = 36.35/33.62$ (Worked example 5.6); blue: LiF, 77.2/33.5 [3, Table I]. Point: GaAs with its measured inputs, $r = +1.4\,\%$, static route 2.21 against 2.11. Gray: $|r| < 0.29\,\%$, where the two routes agree within 1 % for GaAs.

When the inputs do not satisfy the Lyddane–Sachs–Teller relation, the static route amplifies their residual by $\omega_{\mathrm{L}}^2/2(\omega_{\mathrm{L}}^2 - \omega_{\mathrm{T}}^2)$, 3.5 for GaAs, and the splitting is the route that survives.

The amplification is a property of weakly polar crystals. In GaAs the splitting is 8 % of the frequency, and the static route turns a 1.4 % inconsistency of four measurements into a 5 % error in Z^* and 10 % in the rate. In LiF the longitudinal frequency is 2.3 times the transverse one, the factor drops to 0.6, and both routes work.

11.2.5 Rate and reach of polar targets

Worked example 11.3: The band integral of GaAs and LiF

GaAs has one polar branch, and Eq. (11.20) has one term. We take the measured numbers of Worked example 5.6 and its cell, $a = 5.638\,\text{Å}$, the static-lattice constant of Table E.1, the measured value with the thermal and zero-point expansion removed, which is also the cell of the computed phonons; then $\rho_T = M_{\mathrm{cell}}/\Omega = 144.645\,\mathrm{u}/44.80\,\text{Å}^3 = 5.361\,\mathrm{g\,cm^{-3}}$. The room-temperature cell, $a = 5.653\,\text{Å}$, is 0.8 % larger and would lower ρ_T by as much.
Through the dielectric constants. $1/\varepsilon_\infty - 1/\varepsilon_0 = 1/10.88 - 1/12.9 = 0.01439$, so that $\mathcal{I}_{\mathrm{pol}} = (\pi/2)(36.35\,\mathrm{meV})(0.01439) = 0.822\,\mathrm{meV}$ and $\mathcal{I}_{\mathrm{pol}}/\rho_T = 0.153\,\mathrm{meV\,cm^3\,g^{-1}}$.
Through the splitting. By the Lyddane–Sachs–Teller relation $1/\varepsilon_\infty - 1/\varepsilon_0 = (1 - \omega_{\mathrm{T}}^2/\omega_{\mathrm{L}}^2)/\varepsilon_\infty$, and $\mathcal{I}_{\mathrm{pol}} = (\pi/2)(\omega_{\mathrm{L}}^2 - \omega_{\mathrm{T}}^2)/(\varepsilon_\infty\omega_{\mathrm{L}}) = 0.759\,\mathrm{meV}$, or $0.1415\,\mathrm{meV\,cm^3\,g^{-1}}$ per unit mass. The two routes differ by 8 %: the 1.4 % residual of the inputs, amplified by

$\partial \ln \mathcal{I}_{\rm pol}/\partial \ln \varepsilon_0 = \varepsilon_\infty/(\varepsilon_0 - \varepsilon_\infty) = 5.4$. By Fig. 11.10 the splitting is the input to trust.
From a table of computed values. The target table of Griffin and collaborators lists $Z^* = 2.27$, $\varepsilon_\infty = 10.9$ and optical energies from 31.8 to 34.9 meV for GaAs [3, Table I]. In Eq. (11.21), with $M_{\rm cell} = 144.645\,\mathrm{u}$ and $\mu_{\rm red} = 36.11\,\mathrm{u}$, they give $0.170\,\mathrm{meV\,cm^3\,g^{-1}}$, 20 % above the splitting route, mostly because their Born charge is 8 % above the measured one.
LiF. The same table gives $Z^* = 1.05$, $\varepsilon_\infty = 2.02$ and optical energies from 33.5 to 77.2 meV. With $M_{\rm cell} = 25.94\,\mathrm{u}$ and $\mu_{\rm red} = 5.084\,\mathrm{u}$, Eq. (11.21) yields $\mathcal{I}_{\rm pol}/\rho_T = 19.0\,\mathrm{meV\,cm^3\,g^{-1}}$, 112 times the GaAs value of the same table. The factors are: the weaker electronic screening gains $(10.9/2.02)^2 = 29$ and the lighter ions $M_{\rm cell}\mu_{\rm red}$ a factor of 40, while the smaller Born charge gives back $(1.05/2.27)^2 = 0.21$ and the higher frequency $34.9/77.2 = 0.45$.
Edges. The lightest masses that reach the longitudinal phonon are $2\omega_{\rm L}/v_{\rm max}^2 = 10.35\,\mathrm{keV}$ for GaAs and 21.98 keV for LiF.

At large mass the kinematic weights of the two crystals approach each other: at 1 MeV, K_η of LiF is 0.77 times that of GaAs. LiF then wins by a factor of 103, the ratio 134 of its band integral to the splitting value of GaAs, times 0.77. Toward lower masses the ratio falls, to 46 at 100 keV, 17 at 50 keV and 2.6 at 30 keV; at 25 keV GaAs is ahead, and below 22 keV LiF has no rate at all. Fig. 11.7b shows the consequence for the reach: the two curves cross at 26.7 keV, and below it GaAs reaches what LiF cannot.

The order of polar targets by rate is reversed in their reach. The reversal is general. Write the diatomic band integral with the splitting, $\mathcal{I}_{\rm pol}/\rho_T = (\pi/2)(\omega_{\rm L}/\varepsilon_\infty\rho_T)(1 - \omega_{\rm T}^2/\omega_{\rm L}^2)$. At a given screening, density and ratio of the two frequencies, the band integral is proportional to $\omega_{\rm L}$, and so is the edge $2\omega_{\rm L}/v_{\rm max}^2$. The light ions that make $\mathcal{I}_{\rm pol}$ large are the ones that make the phonon hard, through the reduced mass in $\omega_{\rm T}^2 = C_0/\mu_{\rm red}$ and in the splitting Eq. (5.52). A crystal of heavy, soft ions, such as CsI with optical modes between 6.9 and 10.0 meV [3, Table I], reaches down to 2.85 keV, but with a far smaller rate above that. Which polar crystal is best is a question of the mass window, and Sect. 17.5 poses it as a figure of merit. At momentum transfers comparable with the zone, both the Born-charge vertex and the one-phonon picture give way, and the energy is shared among several phonons, which is the response of Sect. 11.3.

11.3 Multiphonon response

Between $q_{\rm BZ} = 2.25\,\mathrm{keV}$, where the long-wavelength phonons of Sect. 11.1 stop describing the response, and about 90 to 140 keV, where the free recoil of Sect. 11.4 takes over (by the two onset criteria compared there), a silicon nucleus that takes up the momentum $\boldsymbol{q}$ leaves behind zero, one, two or a dozen phonons. Dark matter between about 0.4 and 26 MeV transfers its largest momenta, $2m_\chi v_{\rm max}$, inside this window. Sect. 5.6 computed the response of one nucleus as if the others were not there: the phonon number is Poisson distributed with mean $2W_\kappa(\boldsymbol{q})$, and the spectrum of each term follows from the density of states. A crystal adds the pairs of different nuclei, whose interference produced the conservation of crystal momentum in the one-phonon response of Sect. 11.1. How much of that interference survives when two or more phonons share the momentum, and when may a calculation drop it? Dropping the pairs, the incoherent approximation of the literature, is exact on average far above the zone for a lattice without a basis, and in silicon its phonon-number weights are good to 10 % above about 6.6 keV.

11.3.1 The response at all phonon orders

We keep the coupling of Sect. 11.1, a potential $\tilde{V}_N(\boldsymbol{q})$ that sees each nucleus of species κ with a weight c_κ, so that the dark matter couples to the operator $\sum_\kappa c_\kappa n_\kappa(\boldsymbol{q})$. For equal couplings to protons and neutrons $c_\kappa = A_\kappa$, the nucleon number. Nothing below uses that value: the results hold for any coupling that multiplies the density of each species by a number, the coefficient $c_1^{(\kappa)}$ of Sect. 8.4, and in particular for the rigid-ion charge Z_κ. We keep writing S_A for the response, as in Sect. 11.1, and Eq. (8.16) with one row turns it into a rate. The nuclei sit at $\boldsymbol{X}_{\boldsymbol{R}\kappa} = \boldsymbol{R} + \boldsymbol{\tau}_\kappa + \boldsymbol{u}_{\boldsymbol{R}\kappa}$, and $n_\kappa(\boldsymbol{q}) = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_{\boldsymbol{R}\kappa}}$. The correlation form of Eq. (4.13) then contains, for every pair of nuclei, the average $\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_{\boldsymbol{R}\kappa}(t)} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_{\boldsymbol{R}'\kappa'}(0)} \rangle$. The static positions give a phase, and the displacements give the generating formula of a pair, Eq. (D.83). The average depends only on the separation $\Delta\boldsymbol{R} = \boldsymbol{R} - \boldsymbol{R}'$ of the cells. The sum over $\boldsymbol{R}'$ then gives a factor N, and with the volume $\mathscr{V} = N\Omega$ we find

Harmonic nuclear response at all phonon orders

$$S_A(\boldsymbol{q},\omega) = \frac{1}{\Omega} \sum_{\Delta\boldsymbol{R},\kappa,\kappa'} c_\kappa c_{\kappa'}\, \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\Delta\boldsymbol{R}_{\kappa\kappa'}}\, \mathrm{e}^{-W_\kappa(\boldsymbol{q}) - W_{\kappa'}(\boldsymbol{q})} \int_{-\infty}^{\infty} \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\, \mathrm{e}^{\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa'}(\Delta\boldsymbol{R},t)\cdot\boldsymbol{q}}, \quad (11.24)$$

with $\Delta\boldsymbol{R}_{\kappa\kappa'} = \Delta\boldsymbol{R} + \boldsymbol{\tau}_\kappa - \boldsymbol{\tau}_{\kappa'}$ the vector from nucleus κ' in the cell at the origin to nucleus κ in the cell $\Delta\boldsymbol{R}$, the Debye–Waller exponents of Eq. (5.56) and the displacement correlation of Eq. (5.57). It holds for a harmonic crystal in thermal equilibrium at any temperature (Appendix D.31).

This is the classical starting point of neutron scattering from crystals [17, App. N], written for a coupling that differs from species to species. Every quantity in it was computed in Sect. 5.6 from the force constants. What is new is the sum over pairs. Its terms with $\Delta\boldsymbol{R}_{\kappa\kappa'} = 0$, one nucleus scattering at both times, form the *self part*, which is Eq. (5.59) weighted by c_κ^2 and summed over the basis. All other terms form the *distinct part*. Each carries the phase $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\Delta\boldsymbol{R}_{\kappa\kappa'}}$, and its size is set by how strongly the displacements of two nuclei a distance $|\Delta\boldsymbol{R}_{\kappa\kappa'}|$ apart are correlated.

Expanding the last exponential in powers of $\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa'}\cdot\boldsymbol{q}$ sorts the response by the number n of phonons, $S_A = \sum_n S_A^{(n)}$, as in Sect. 5.6. The term $n = 0$ is time independent. Its sum over $\Delta\boldsymbol{R}$ vanishes unless $\boldsymbol{q}$ is a reciprocal-lattice vector, and it is the Bragg peak, $\propto \delta(\omega)\sum_{\boldsymbol{G}} \delta^3(\boldsymbol{q} - \boldsymbol{G})$. It deposits no energy and occupies a set of zero measure in $\boldsymbol{q}$, so it never contributes to a rate. For $n \geq 1$ each factor $\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa'}\cdot\boldsymbol{q}$ is a sum over phonons $\boldsymbol{Q}\nu$ with the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\Delta\boldsymbol{R}}$, and the sum over $\Delta\boldsymbol{R}$ against $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\Delta\boldsymbol{R}}$ forces the phonon wavevectors to add up to $\boldsymbol{q}$ up to a reciprocal-lattice vector. Each phonon i is created ($s_i = +1$) or absorbed ($s_i = -1$), and after the time integral (Appendix D.31)

The n-phonon term

$$S_A^{(n)}(\boldsymbol{q},\omega) = \frac{2\pi}{\Omega}\,\frac{1}{n!} \sum_{\{\nu_i,s_i\}} \Big\langle \prod_{i=1}^{n} \frac{n_{\boldsymbol{Q}_i\nu_i} + \frac{1}{2}(1+s_i)}{2\omega_{\boldsymbol{Q}_i\nu_i}}\, \delta\Big(\omega - \sum_i s_i\omega_{\boldsymbol{Q}_i\nu_i}\Big) \times \Big| \sum_\kappa c_\kappa\, \mathrm{e}^{-W_\kappa}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}\, \frac{\prod_i \boldsymbol{q}\cdot\boldsymbol{e}^{s_i}_{\kappa\nu_i}(\boldsymbol{Q}_i)}{M_\kappa^{n/2}} \Big|^2 \Big\rangle, \quad (11.25)$$

with $\boldsymbol{e}^+ = \boldsymbol{e}^*$ for a created and $\boldsymbol{e}^- = \boldsymbol{e}$ for an absorbed phonon, $\boldsymbol{q}$ unreduced in the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}$, and the brackets the average $N^{-(n-1)}\sum$ over $\boldsymbol{Q}_1, \ldots, \boldsymbol{Q}_{n-1}$ in the zone,

with $\boldsymbol{Q}_n$ fixed by $\sum_i s_i \boldsymbol{Q}_i = \boldsymbol{q} - \boldsymbol{G}$.

At $n = 1$ the brackets are empty, $\boldsymbol{Q}_1 = \boldsymbol{q} - \boldsymbol{G}$ for emission, and Eq. (11.25) is the one-phonon response Eq. (11.1). At $n = 2$ and $T = 0$ the amplitude inside the modulus is the contact term of Campbell-Deem and collaborators [1, Eq. (12)]. For every n the crystal momentum is conserved, and the response at fixed $\boldsymbol{q}$ is a convolution over the zone of n branches. It is the full harmonic answer, and it is also an expensive one: at $n = 2$ the sum runs over one wavevector of the zone for each $\boldsymbol{q}$, at $n = 3$ over two.

On the truncation label of Sect. 8.5, Eq. (11.24) sits at $n_A = 0$ and $n_B = 0$, with every thermal order kept through the factors $n_{\boldsymbol{Q}\nu}$, and with $\mathcal{K}$ = multiphonon, all orders. In a detector at 10 mK only the terms with every $s_i = +1$ survive, at $n_T = 0$.

In neutron scattering, "coherent" and "incoherent" mean something else: they refer to the random variation of the coupling from site to site, between isotopes or nuclear spin states, and the incoherent part of the response is proportional to its variance [2, Eq. (11)]. A nucleon-number coupling varies in the same way, since natural silicon is 92.223 % ^{28}Si, 4.685 % ^{29}Si and 3.092 % ^{30}Si [18]. The variance of A_κ is 0.159, 2.0×10^{-4} of $\langle A_\kappa \rangle^2 = 790.1$, and the isotopic incoherent scattering is negligible here; the isotopes act instead through their masses (Sect. 11.6). In this section, as in the dark-matter literature, "incoherent" means the approximation that keeps only the self part.

The quantities in Eqs. (11.24) and (11.25) come from one harmonic calculation per material: DFPT force constants, interpolated to a fine grid of wavevectors, give the frequencies and polarization vectors, from which the exponents W_κ, the correlations $\mathcal{C}_{\kappa\kappa'}$ and the zone sums follow at negligible cost (Sect. 5.4 and Appendix E.4). Appendix E.17 specifies the tables of the multiphonon response in q, $\hat{\boldsymbol{q}}$ and ω that a rate code reads, for silicon, germanium and GaAs at 10 mK, 1 K and 77 K, together with the coherence corrections derived below.

11.3.2 The incoherent approximation

At momentum transfers far above the zone, the phase $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\Delta\boldsymbol{R}_{\kappa\kappa'}}$ of a distinct term turns through many radians when $\boldsymbol{q}$ changes by a fraction of q_{BZ}. The rest of the term, a polynomial in $\boldsymbol{q}$ times the Debye–Waller factors, barely changes. Any rate integrates over $\boldsymbol{q}$, and the distinct terms then average away. What is left is the self part, which treats each nucleus as an independent oscillator in the potential well of its neighbors:

$$S_A(\boldsymbol{q},\omega) \approx S_A^{\mathrm{inc}}(\boldsymbol{q},\omega) = \frac{2\pi}{\Omega} \sum_\kappa c_\kappa^2 \, \mathrm{e}^{-2W_\kappa(\boldsymbol{q})} \sum_{n=0}^{\infty} \frac{[2W_\kappa(\boldsymbol{q})]^n}{n!} \, P_n^\kappa(\hat{\boldsymbol{q}},\omega). \tag{11.26}$$

Here $P_1^\kappa(\hat{\boldsymbol{q}},\omega)$ is the single-phonon distribution of Sect. 5.6 for nucleus κ, built from the density of states projected on the direction of $\boldsymbol{q}$, $N^{-1}\sum_{\boldsymbol{Q}\nu} |\hat{\boldsymbol{q}}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2 \delta(\omega - \omega_{\boldsymbol{Q}\nu})$, times $[n_{\mathrm{B}}(\omega)+1]/\omega$ and normalized to one; P_n^κ is its n-fold convolution. On a site of cubic symmetry the projected density of states is a third of the atom-projected one, $g_{\mathrm{ph},\kappa}(\omega)/3$, independent of $\hat{\boldsymbol{q}}$, and Eq. (11.26) is Eq. (5.59) summed over the basis. Campbell-Deem, Knapen, Lin and Villarama compute every phonon order above the zone in this way, by dropping the terms between different atoms, which at $T = 0$ gives Eq. (11.26) [2, Eqs. (15), (36)]. For one nucleus bound in an isotropic oscillator of frequency ω_0, $P_n = \delta(\omega - n\omega_0)$ and $2W_\kappa = q^2/2M_\kappa\omega_0$, and the weights are the Poisson law that Kahn, Krnjaic and Mandava wrote for dark-matter scattering, $(q/q_{\mathrm{osc}})^{2n}\mathrm{e}^{-q^2/q_{\mathrm{osc}}^2}/n!$ with $q_{\mathrm{osc}}^2 = 2M_\kappa\omega_0$ [19, Eq. (10)].

The incoherent approximation keeps the self part exactly, so it inherits two exact properties from Sect. 5.6: the phonon number is Poisson distributed, and the first moment of the response, $\int \mathrm{d}\omega\, \omega S_A$, is $(2\pi/\Omega)\sum_\kappa c_\kappa^2 E_R$ with $E_R = q^2/2M_\kappa$. The distinct part adds weight but no

energy. Every distinct term has zero first moment in ω, at every order n, because its time derivative at $t = 0$ contains $\partial_t \mathcal{C}_{\kappa\kappa'}(\Delta\boldsymbol{R}, 0)$, which the completeness of the polarization vectors makes proportional to $\delta_{\kappa\kappa'}\delta_{\Delta\boldsymbol{R},\boldsymbol{0}}$ (Appendix D.31). The first moment of the order n is therefore that of its self part, and for one species

$$\frac{\int \mathrm{d}\omega\, \omega\, S_A^{(n)}(\boldsymbol{q},\omega)}{\int \mathrm{d}\omega\, \omega\, S_A(\boldsymbol{q},\omega)} = \mathrm{e}^{-2W_\kappa}\, \frac{(2W_\kappa)^{n-1}}{(n-1)!}, \tag{11.27}$$

at every $\boldsymbol{q}$ and temperature, coherent terms included. A calculation that stops at order N misses the fraction $\sum_{n\geq N} p_n^{\mathrm{ph}}$ of the recoil energy, the Poisson tail of the weights Eq. (5.60); with several species the fractions are averaged with the weights c_κ^2/M_κ. At 50 keV in silicon, where $2W_\kappa = 1.57$, the orders above two carry 47 % of the energy deposited, and a two-phonon calculation misses all of it.

The tail matters most for a threshold above the phonon band. Since P_n^κ vanishes above $n\,\omega_{\max}$, with $\omega_{\max}$ the highest phonon energy, an energy deposit above E needs at least $\lceil E/\omega_{\max}\rceil$ phonons, and in the incoherent approximation at $T = 0$ its probability is at most the Poisson tail $\sum_{n\geq\lceil E/\omega_{\max}\rceil} p_n^{\mathrm{ph}}$. A free nucleus deposits exactly E_R, nothing above it.

Worked example 11.4: The Poisson tail above three Debye energies

Take silicon at $T = 0$ and a threshold $E = 3\omega_{\mathrm{D}} = 166.8$ meV, with $\omega_{\mathrm{D}} = 55.6$ meV from Worked example 5.7. The computed density of states of Sect. 5.2 ends at $\omega_{\max} = 63.7$ meV (the zone-center optical phonon measured by neutron scattering at room temperature is 64.1 meV, Sect. 5.2), so two phonons reach at most 127 meV and the deposit above E needs three or more. Of the three-phonon spectrum P_3, 1.4 % lies above E; of P_4, P_5 and P_6, 11, 35 and 62 %. With $2W_\kappa = q^2\langle u_x^2\rangle$ and $\langle u_x^2\rangle = 2.45 \times 10^{-3}$ Å^2, the probability of a deposit above E is $\sum_n p_n^{\mathrm{ph}} \int_E^\infty P_n$:

$q = 10$ keV. $E_R = 1.91$ meV and $2W_\kappa = 0.063$. The probability is 6.1×10^{-7}, almost all of it from three phonons; the Poisson bound $\sum_{n\geq 3} p_n^{\mathrm{ph}} = 4.0 \times 10^{-5}$ is 65 times larger, because a three-phonon deposit rarely reaches the top of its band.

$q = 30$ keV. $E_R = 17.2$ meV and $2W_\kappa = 0.566$. The probability is 6.2×10^{-4}, from three, four and five phonons in the proportion 0.38 : 0.43 : 0.16; the bound is 2.0×10^{-2}.

$q = 50$ keV. $E_R = 47.8$ meV, a third of the threshold, and $2W_\kappa = 1.57$. The probability is 1.7×10^{-2}, against the bound 0.21. Four and five phonons contribute most.

A free silicon nucleus deposits 166.8 meV only for $q \geq (2M_\kappa E)^{1/2} = 93.4$ keV. A particle much lighter than the nucleus transfers at most $2m_\chi v_{\max}$, so the free-recoil picture needs $m_\chi \geq 17.6$ MeV for this threshold. Dark matter of 10 MeV reaches $q = 53$ keV, and already at 50 keV the bound lattice deposits more than $3\omega_{\mathrm{D}}$ in 1.7 % of the scatterings. The tail also refuses a shortcut. An oscillator at $\langle\omega^{-1}\rangle^{-1} = 30.4$ meV has the same $2W_\kappa$ as the crystal at $T = 0$ (Worked example 5.7) and the same Poisson weights. It needs six phonons to pass E, however, and gives 8×10^{-11}, 2.8×10^{-5} and 5.6×10^{-3} at the three momentum transfers, too small by factors of 7400, 22 and 3. The number of phonons is set by $\langle\omega^{-1}\rangle$; the energy they can carry is set by the hardest phonons of the spectrum.

When does the incoherent approximation hold? Appendix D.31 answers this by averaging $\boldsymbol{q}$ over a small cell of reciprocal space, the size of the zone, around a given $\boldsymbol{q}$. The phase of a distinct term averages to $j_{\mathrm{BZ}}(\Delta\boldsymbol{R}_{\kappa\kappa'}) = (\Omega/(2\pi)^3) \int_{\mathrm{BZ}} \mathrm{d}^3p\, \mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\Delta\boldsymbol{R}_{\kappa\kappa'}}$, which vanishes for every lattice vector except zero, while the slowly varying factors change across the cell by a relative amount of order $n\, q_{\mathrm{BZ}}/q$. For a lattice with one atom per cell every distinct term joins two nuclei of the same sublattice, and the incoherent response is the average of the exact one over a zone-sized cell, up to corrections of relative order nq_{BZ}/q. With a basis the pairs

on different sublattices are not separated by lattice vectors, and they survive the average: in silicon each of the four bonds keeps a weight $j_{\mathrm{BZ}} \approx 0.44$ in a spherical zone. These terms decay because the correlation of two displacements falls with distance, and the average over the directions of $\boldsymbol{q}$, as in the rate of a crystal of random orientation, reduces each of them further by a spherical Bessel function of $q|\Delta\boldsymbol{R}_{\kappa\kappa'}|$. Hence the condition $q \gg \max(n, 2W_\kappa)\, q_{\mathrm{BZ}}$, and for a crystal with a basis the size of the correction has to be computed.

Limit Check 11.8. Multiphonon incoherent approximation

Above the zone, the nuclear response is the sum of independent atomic responses, each a Poisson series in the phonon number, $S \approx (2\pi/\Omega) \sum_\kappa c_\kappa^2 \mathrm{e}^{-2W_\kappa} \sum_n [(2W_\kappa)^n/n!] P_n^\kappa(\omega)$, with P_n^κ the n-fold convolution of the single-phonon distribution [2, Eqs. (15), (36), 19, Eq. (10)].

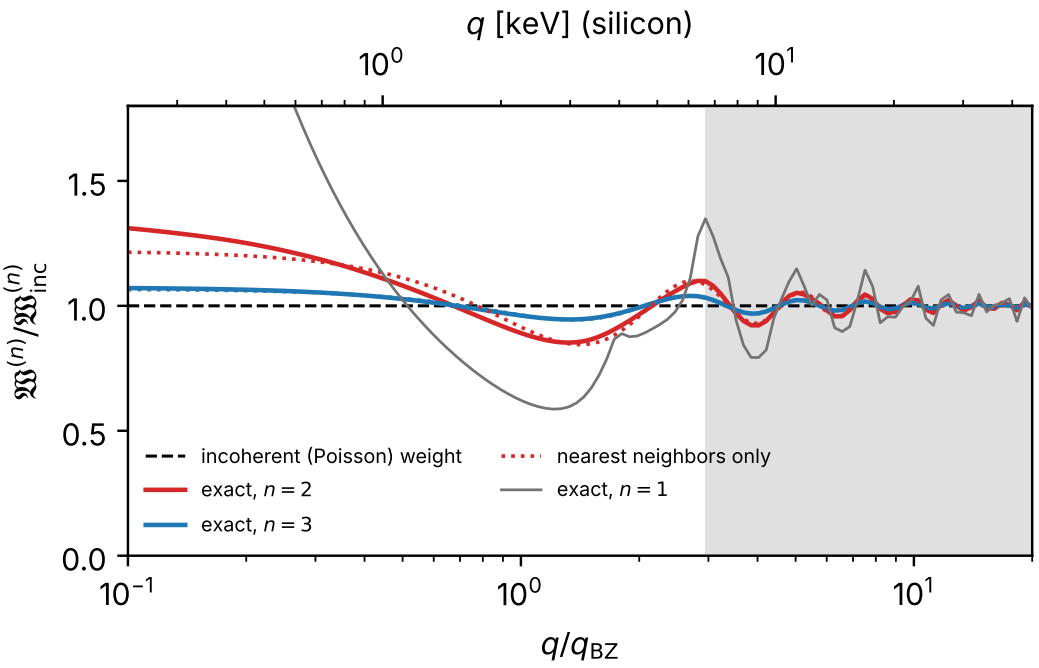


Figure 11.11: The exact ω-integrated n-phonon weight of silicon at $T = 0$, Eq. (11.29), divided by its incoherent value, the Poisson weight of Eq. (11.26) (dashed, one), averaged over the directions of $\boldsymbol{q}$, against q/q_{BZ}, with $q_{\mathrm{BZ}} = 2.25\,\mathrm{keV}$. For one species the couplings and the Debye–Waller factors cancel in the ratio, which is $1 + \delta_n^{\mathrm{coh}}(q)$ for any coupling of the form $\sum_\kappa c_\kappa n_\kappa$. Displacement correlations from the PBE force constants of Sect. 5.2 on a 48^3 grid, pair sum to 40 Å, direction average exact. Solid: two (red) and three (blue) phonons. Dotted: the nearest-neighbor pairs alone, the leading correction. Thin gray: one phonon, which must be summed in reciprocal space. Gray band: $q \geq 2.95\, q_{\mathrm{BZ}} = 6.6\,\mathrm{keV}$, where the two- and three-phonon weights stay within 10 % of the incoherent ones. The three-phonon weight does so at every q, the one-phonon weight only above $7.9\, q_{\mathrm{BZ}}$.

Below about three times the zone momentum the two-phonon weight differs from the Poisson weight by up to 35 %, and pointwise in ω the incoherent spectrum misses the momentum conservation of the phonon pair even far above the zone.

At small q the phases are irrelevant, and every pair adds with the square of its correlation.

The coherent weight then exceeds the incoherent one by 35 %, mostly from the four bonds, along which the longitudinal displacements of neighbors are correlated with a coefficient of 0.44 (a dynamical number, distinct from the geometric weight $j_{\mathrm{BZ}} \approx 0.44$ above, to which it is close by coincidence). Near q_{BZ} the bond phases turn the sign of this correction, and from about $3\,q_{\mathrm{BZ}}$ the remainder oscillates with the period of the bond length and decays: $\delta_2^{\mathrm{coh}} = -0.079$, $+0.014$, -0.018 and -0.025 at 2, 5, 10 and 20 keV. The nearest-neighbor pairs alone account for 0.21 of the 0.35 at small q, and above about $2\,q_{\mathrm{BZ}}$ they follow the full curve within 1 % of the weight. The three-phonon correction is smaller everywhere, since it carries the cube of a correlation coefficient below one: -0.027, $+0.016$, -0.006 and -0.012 at the same momentum transfers. The one-phonon term is the exception. Its pair sum does not converge absolutely, it is sharp in ω at fixed $\boldsymbol{q}$, and its ω-integrated weight approaches the incoherent value slowly, within 10 % above about 18 keV; near each reciprocal-lattice vector the soft acoustic phonons enhance it. Above q_{BZ}, where Eq. (11.3) no longer holds, the one-phonon term is computed from Eq. (11.1) with all branches and Umklapp terms, and its weight grows on average as $q^2\mathrm{e}^{-2W_\kappa}$, the incoherent form, up to these corrections.

Fig. 11.12 shows the three parts of the silicon response at $q = 10$ keV, the momentum transfer at which Sect. 8.5 placed the end of the one-phonon regime. One phonon carries a weight of 0.058 per nucleus, two phonons 1.8×10^{-3} and three or more 4.0×10^{-5}, close to the Poisson weights $2W_\kappa\mathrm{e}^{-2W_\kappa} = 0.059$, $(2W_\kappa)^2\mathrm{e}^{-2W_\kappa}/2 = 1.9 \times 10^{-3}$ and the rest. The exact one-phonon spectrum, averaged over directions, follows the projected density of states. The exact two-phonon spectrum follows the self-convolution of the single-phonon distribution within 17 % in every bin of 1 meV between 22 and 120 meV, where the spectrum exceeds a tenth of its maximum; toward the edges of the band the ratio falls to about one half. Each part fills the band of its phonon number: one phonon up to 64 meV, two up to 127 meV, and the tail beyond.

11.3.3 Exact phonon-order weights

Integrated over ω, the time integral of Eq. (11.24) returns the integrand at $t = 0$, and the n-phonon term becomes a sum over pairs of a static quantity. At $t = 0$ the two terms of Eq. (5.57) combine into $2n_{\boldsymbol{Q}\nu} + 1$, and the displacement correlation becomes the static correlation tensor

$$B_{\kappa\kappa'}(\Delta\boldsymbol{R}) = \mathcal{C}_{\kappa\kappa'}(\Delta\boldsymbol{R}, 0) = \frac{1}{N}\sum_{\boldsymbol{Q}\nu}(2n_{\boldsymbol{Q}\nu} + 1)\,\frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\,\boldsymbol{e}_{\kappa'\nu}(\boldsymbol{Q})^\dagger\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\Delta\boldsymbol{R}}}{2\omega_{\boldsymbol{Q}\nu}\sqrt{M_\kappa M_{\kappa'}}}, \tag{11.28}$$

whose on-site value $B_{\kappa\kappa}(\boldsymbol{0})$ is the displacement tensor B_κ of Eq. (5.35). Off site it measures how the displacements of two nuclei move together: $\langle u^\alpha_{\Delta\boldsymbol{R}\kappa} u^\beta_{\boldsymbol{0}\kappa'}\rangle$.

Exact n-phonon weight

The ω-integrated n-phonon response per cell is

$$\mathfrak{W}^{(n)}(\boldsymbol{q}) = \Omega\int\frac{\mathrm{d}\omega}{2\pi}\,S_A^{(n)}(\boldsymbol{q},\omega) = \frac{1}{n!}\sum_{\Delta\boldsymbol{R},\kappa,\kappa'} c_\kappa c_{\kappa'}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\Delta\boldsymbol{R}_{\kappa\kappa'}}\,\mathrm{e}^{-W_\kappa - W_{\kappa'}}\,\left[\boldsymbol{q}\cdot B_{\kappa\kappa'}(\Delta\boldsymbol{R})\cdot\boldsymbol{q}\right]^n. \tag{11.29}$$

Its self terms are the incoherent weight $\mathfrak{W}^{(n)}_{\mathrm{inc}} = \sum_\kappa c_\kappa^2\,\mathrm{e}^{-2W_\kappa}(2W_\kappa)^n/n!$, and $\delta_n^{\mathrm{coh}}(\boldsymbol{q}) = \mathfrak{W}^{(n)}/\mathfrak{W}^{(n)}_{\mathrm{inc}} - 1$ is the exact correction to the incoherent approximation for any energy window that contains the whole n-phonon band.

This is Eq. (E.95), and a rate code applies it as a factor $1 + \delta_n^{\mathrm{coh}}$ per phonon order. The sum converges for $n \geq 2$ at $T = 0$. The acoustic modes near Γ dominate the correlation at

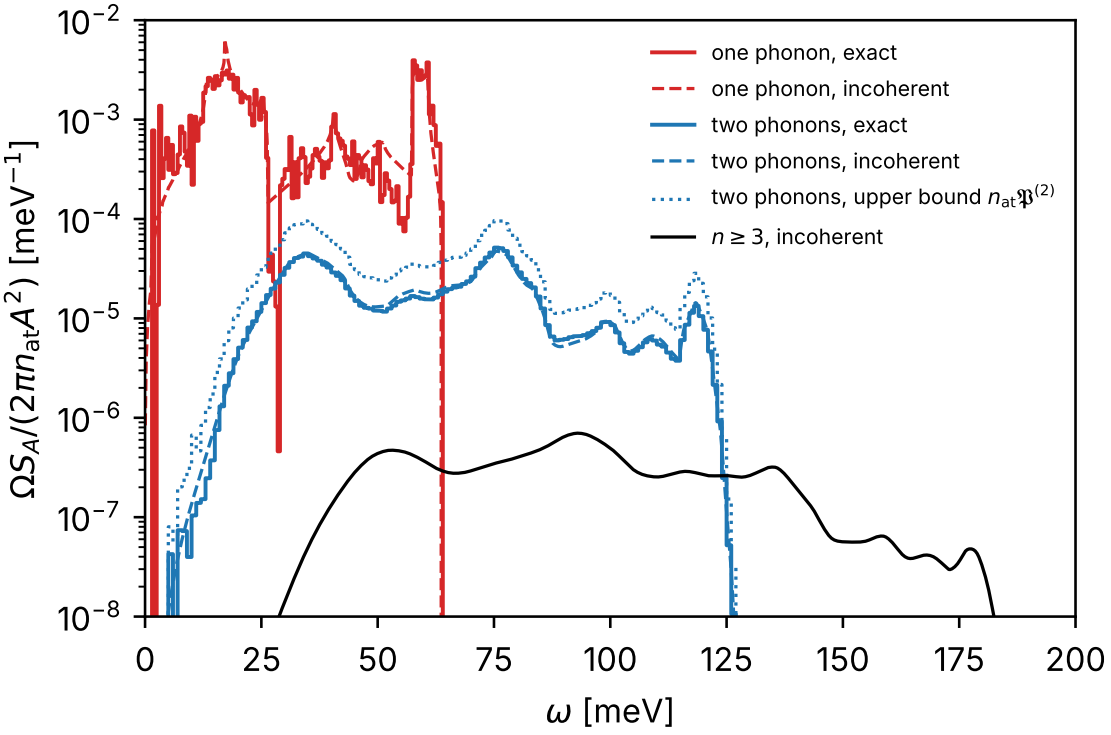


Figure 11.12: Nuclear response of silicon at $|\boldsymbol{q}| = 10\,\mathrm{keV}$ and $T = 0$, per nucleus, $\Omega S_A/(2\pi n_{\mathrm{at}} A^2)$, averaged over the directions of $\boldsymbol{q}$, split by the number of phonons; the elastic weight $\mathrm{e}^{-2W_\kappa} = 0.939$ at $\omega = 0$ is not shown. Red: one phonon, exact (Eq. (11.1), all six branches, histogram of 0.5-meV bins) and incoherent (dashed). Blue: two phonons, exact (Eq. (11.25), 1-meV bins), incoherent (dashed) and the upper end $n_{\mathrm{at}}\mathfrak{P}^{(2)}$ of the certified band Eq. (11.30) (dotted). Black: three or more phonons, incoherent. Phonons from the PBE force constants of Sect. 5.2; incoherent parts from its density of states with $\langle u_x^2 \rangle = 2.45 \times 10^{-3}\,\text{Å}^2$. The exact two-phonon response is averaged over 240 wavevectors with $|\boldsymbol{q}|$ within 1 % of 10 keV on a 24^3 grid commensurate with the zone, each rescaled to 10 keV by $(10\,\mathrm{keV}/|\boldsymbol{q}|)^4$. Weights: one phonon 0.058 (incoherent 0.059), two phonons 1.84×10^{-3} (1.86×10^{-3}), three or more 4.0×10^{-5}.

large distances, and since their weight $1/\omega = 1/c_\nu Q$ has the Fourier transform of $1/Q$ in three dimensions, $B_{\kappa\kappa'}(\Delta\boldsymbol{R})$ falls as $|\Delta\boldsymbol{R}|^{-2}$ (Appendix D.31). Its square falls as $|\Delta\boldsymbol{R}|^{-4}$, faster than the number of pairs grows, and the sum converges absolutely. For $n = 1$ it does not, and the one-phonon term is summed in reciprocal space, where it is Eq. (11.1). At $T > 0$ the factor $2n_{\boldsymbol{Q}\nu} + 1 \approx 2T/\omega$ makes the acoustic weight $1/Q^2$, $B_{\kappa\kappa'}$ falls only as $T/|\Delta\boldsymbol{R}|$, and the two-phonon sum diverges; the last subsection resums it.

In silicon at $T = 0$ the correlation is short ranged in practice. With the force constants of Sect. 5.2 on a 48^3 grid of wavevectors, the on-site value is $\langle u_x^2 \rangle = 2.45 \times 10^{-3}\,\text{Å}^2$, as in Worked example 5.7. Projected on the bond, the longitudinal displacements of two neighbors are correlated with a coefficient of 0.44. The sum of Eq. (11.29) to 40 Å takes seconds for any $\boldsymbol{q}$, and changes by less than 2×10^{-3} beyond 30 Å. It produced the curves of Fig. 11.11. Campbell-Deem and collaborators checked the incoherent two-phonon response against their long-wavelength calculation below the zone and concluded that there it should be taken as an order-of-magnitude estimate [2, Sect. III C]. The exact weight is more favorable: in silicon at $T = 0$ the ω-integrated two-phonon weight is within 35 % of the incoherent one at every q.

The weight leaves open how the remainder is distributed in ω. A pointwise statement follows from the structure of Eq. (11.25), in which the coherent sum over the basis appears inside a squared modulus. By the Cauchy–Schwarz inequality, $|\sum_\kappa a_\kappa|^2 \le n_{\text{at}} \sum_\kappa |a_\kappa|^2$ for the n_{at} nuclei of a cell, term by term in the sum over phonons. The squared amplitudes of the single nuclei, summed with everything else in Eq. (11.25) unchanged, define the *momentum-resolved self density* $\mathfrak{P}^{(n)}(\boldsymbol{q},\omega)$: the response of independent nuclei whose n phonons must still add up to the crystal momentum $\boldsymbol{q}$. For silicon, $n_{\text{at}} = 2$, and with a_1, a_2 the amplitudes of the two nuclei of a cell for one set of phonons, $|a_1 + a_2|^2 = 2(|a_1|^2 + |a_2|^2) - |a_1 - a_2|^2 \le 2(|a_1|^2 + |a_2|^2)$, with equality when the two nuclei move in phase. The momentum constraint is the Kronecker delta $\delta_{\sum_i s_i \boldsymbol{Q}_i,\, \boldsymbol{q}-\boldsymbol{G}}$. Shift $\boldsymbol{q}$ by $\boldsymbol{p}$ in this delta alone, with the vertex factors left at $\boldsymbol{q}$, and average $\boldsymbol{p}$ over a zone-sized cell. The delta becomes $1/N$, the last phonon wavevector is no longer fixed, and the n phonons become independent; the sum over them is the n-fold convolution P_n^κ of the incoherent approximation.

Certified band

At every $\boldsymbol{q}$, ω and n,

$$\begin{gathered} 0 \le S_A^{(n)}(\boldsymbol{q},\omega) \le n_{\text{at}}\, \mathfrak{P}^{(n)}(\boldsymbol{q},\omega), \\ \frac{\Omega}{(2\pi)^3} \int_{\text{BZ}} \mathrm{d}^3 p\; \mathfrak{P}^{(n)}(\boldsymbol{q}+\boldsymbol{p},\omega)\Big|_{\text{vertex at } \boldsymbol{q}} = S_A^{\text{inc},(n)}(\boldsymbol{q},\omega), \end{gathered} \tag{11.30}$$

where the average holds the factors $\boldsymbol{q}\cdot\boldsymbol{e}$ and e^{-W_κ} at $\boldsymbol{q}$ and moves only the momentum constraint, and $S_A^{\text{inc},(n)}$ is the term of order n of Eq. (11.26) (Appendix D.31).

For a lattice with one atom per cell the two sides of the inequality coincide, $S_A^{(n)} = \mathfrak{P}^{(n)}$, and the only error of the incoherent approximation is the momentum constraint, which the zone average removes. The constraint matters at the edges of each band: the lowest two-phonon energies need two soft acoustic phonons, both near Γ, and hence $\boldsymbol{q}$ near a reciprocal-lattice vector. At other $\boldsymbol{q}$ the exact spectrum vanishes at those energies, where the self density does not, and the lower end of the band is reached. For a crystal with a basis the band is wider by the factor n_{at}. In silicon at 10 keV its upper end reaches 2.7 times the incoherent two-phonon spectrum (Fig. 11.12), so it is not a certificate at the level of 10 %. The band bounds the error, but it is too loose to vouch for the incoherent spectrum point by point below a few q_{BZ}; for that we need Eq. (11.25) itself. This is why Appendix E.17 computes the coherent

two-phonon response pointwise up to 10 keV and stores δ_n^{coh} at every q.

11.3.4 Thermal diffuse scattering

In a warm crystal the scattered intensity of X-rays and neutrons does not drop to zero between the Bragg peaks. Thermal phonons fill the space around each peak with a diffuse cloud, which crystallographers subtract when they measure structure factors. Its first-order part is the one-phonon response at $\boldsymbol{q} = \boldsymbol{G} + \boldsymbol{k}$ with a soft acoustic phonon of wavevector $\boldsymbol{k}$. By Eq. (11.1) its weight carries $(2n_{\boldsymbol{k}\nu} + 1)/\omega_{\boldsymbol{k}\nu} \approx 2T/(c_\nu k)^2$, and it grows as $1/|\boldsymbol{q} - \boldsymbol{G}|^2$ toward the peak. The second-order part, two soft phonons whose wavevectors add up to $\boldsymbol{k}$, is milder, and it is exactly the term whose real-space sum diverged above.

In reciprocal space nothing diverges. Summing Eq. (11.25) at $n = 2$ over the creation and absorption of each phonon and integrating over ω turns the thermal factors into $(2n_{\boldsymbol{Q}_1\nu_1} + 1)(2n_{\boldsymbol{Q}_2\nu_2} + 1)$, and $\mathfrak{W}^{(2)}$ becomes an average over one wavevector $\boldsymbol{Q}_1$ of the zone, with $\boldsymbol{Q}_2 = \boldsymbol{k} - \boldsymbol{Q}_1$. Each factor $(2n+1)/2\omega \approx T/\omega^2$ is integrable around $\boldsymbol{Q} = 0$ in three dimensions, and the integral is finite unless both phonons can be soft at once, which happens as $\boldsymbol{k} \to 0$. There the acoustic polarizations move every nucleus of the cell together, $\boldsymbol{e}_{\kappa\nu} = (M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$, the amplitude factorizes into $(\boldsymbol{q}\cdot\hat{\boldsymbol{e}}_1)(\boldsymbol{q}\cdot\hat{\boldsymbol{e}}_2)/M_{\mathrm{cell}}$ times the structure factor of the coupling at $\boldsymbol{G}$, and the completeness of the three polarizations bounds the branch sum by $q^4/(c_{\min}^4 Q_1^2 Q_2^2)$. The remaining integral is $\int \mathrm{d}^3Q_1/(Q_1^2|\boldsymbol{k} - \boldsymbol{Q}_1|^2) = \pi^3/k$ (Appendix D.31). The thermal part of the weight therefore obeys

$$\mathfrak{W}^{(2),\mathrm{th}}(\boldsymbol{q}) \le \Big|\sum_\kappa c_\kappa\, \mathrm{e}^{-W_\kappa}\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_\kappa}\Big|^2 \frac{\Omega\, T^2 q^4}{16\, M_{\mathrm{cell}}^2\, c_{\min}^4\, |\boldsymbol{q} - \boldsymbol{G}|} + (\text{terms regular at } \boldsymbol{q} = \boldsymbol{G}), \quad (11.31)$$

with $c_{\min}$ the lowest acoustic speed, $4.67\,\mathrm{km\,s^{-1}}$ in silicon for the transverse wave along [110] polarized along $[1\bar{1}0]$. The singularity is integrable, and it is weighted by the Bragg structure factor: it vanishes near a reflection that the basis extinguishes, such as (200) in silicon. In the classical limit the singular part is exactly proportional to $1/|\boldsymbol{q} - \boldsymbol{G}|$. Its coefficient lies below the bound and depends on the direction of $\boldsymbol{q} - \boldsymbol{G}$ relative to $\boldsymbol{q}$; for an isotropic medium with the speeds of silicon and $\boldsymbol{q} - \boldsymbol{G}$ parallel to $\boldsymbol{q}$ it is 0.46 of the bound. This $1/|\boldsymbol{q} - \boldsymbol{G}|$ law, against the $1/|\boldsymbol{q} - \boldsymbol{G}|^2$ of the first order, is the classical result for second-order thermal diffuse scattering [20]. On the truncation label it is $\mathcal{K}$ = multiphonon (two phonons, coherent) at $n_T = 2$, since the singular part is quadratic in the occupation numbers; the cross terms linear in $n_{\boldsymbol{Q}\nu}$, at $n_T = 1$, are at most logarithmically singular.

Limit Check 11.9. Second-order thermal diffuse scattering

Near a reciprocal-lattice vector $\boldsymbol{G}$ of a warm crystal, two soft acoustic phonons give the ω-integrated response a singular part that falls as $1/|\boldsymbol{q} - \boldsymbol{G}|$, weighted by the structure factor at $\boldsymbol{G}$ and by T^2 [20]; its bound is Eq. (11.31).

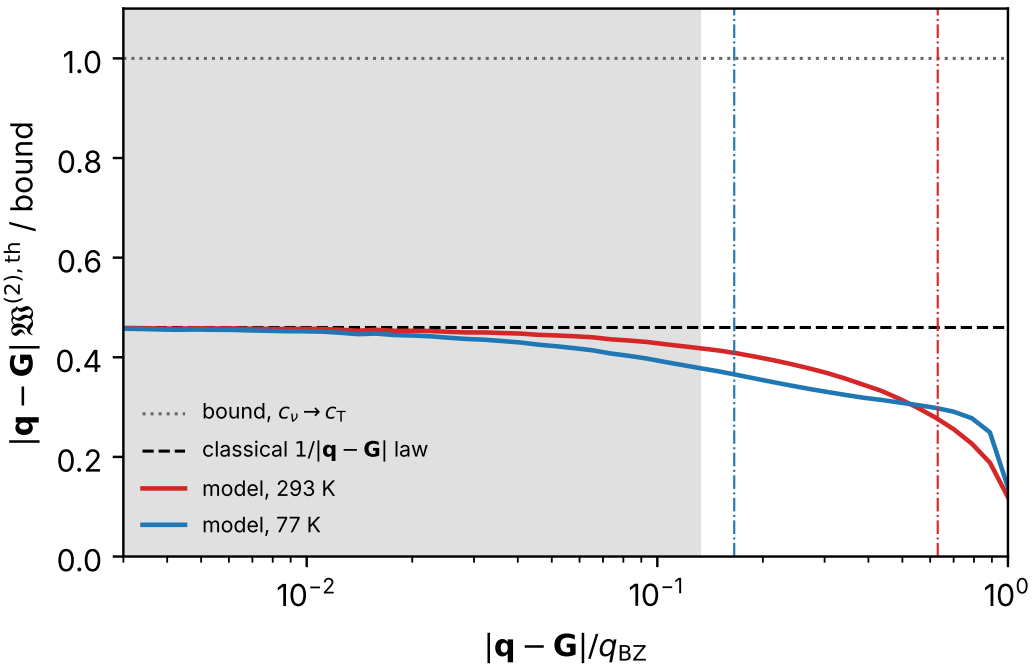


Figure 11.13: Thermal two-phonon weight times $|\boldsymbol{q}-\boldsymbol{G}|$, in units of the bound Eq. (11.31) with c_min replaced by the transverse speed (dotted, one), for $\boldsymbol{G} = \boldsymbol{G}_{111}$ of silicon and $\boldsymbol{q}-\boldsymbol{G}$ parallel to $\boldsymbol{q}$, against $|\boldsymbol{q}-\boldsymbol{G}|/q_\mathrm{BZ}$. Model: isotropic elastic medium with the Hill-averaged speeds 8.95 and 5.34 km s^{-1} of Sect. 11.1, exactly longitudinal and transverse polarizations, wavevectors cut at $|\boldsymbol{Q}| < q_\mathrm{BZ}$, and the full thermal factors $\coth(\omega_1/2T)\coth(\omega_2/2T) - 1$. Dashed: the classical $1/|\boldsymbol{q}-\boldsymbol{G}|$ law, 0.46 of the bound. Solid: the model at 293 K (red) and 77 K (blue). Dash-dotted: the boundary $|\boldsymbol{q}-\boldsymbol{G}| = T/c_\mathrm{T}$ of the classical soft pair at each temperature. Gray: within 10 % of the classical law at 293 K, $|\boldsymbol{q}-\boldsymbol{G}| \le 0.13\,q_\mathrm{BZ}$; at 77 K the same holds below $0.065\,q_\mathrm{BZ}$. At 10 mK the boundary lies at $2\times10^{-5}\,q_\mathrm{BZ}$.

The law needs both phonons to be classical, $c_\nu|\boldsymbol{q}-\boldsymbol{G}| \ll T$. Farther from the peak the thermal occupations fall off and the weight drops below the law, and at the zone scale the dispersion and the regular two-phonon terms take over.

In a dark-matter detector this term is negligible. At 10 mK the classical region ends at $|\boldsymbol{q}-\boldsymbol{G}| = T/c_\mathrm{T} = 48$ meV, with the Hill transverse speed c_T of Sect. 11.1, and the coefficient carries T^2, 10^{-9} of its room-temperature value. It matters for the X-ray and neutron data, taken warm, that calibrate the lattice response, and for any rate quoted at 77 K or above. At room temperature, a tenth of the zone momentum away from the (111) peak of silicon, the singular thermal weight is as large as the whole incoherent two-phonon weight there, 1.07 times it in the model of the limit check. A calculation at $T > 0$ evaluates the two-phonon coherence in reciprocal space and never as the real-space sum of Eq. (11.29).

11.3.5 The charge vertex at large momentum

For a coupling to charge, the dark photon of Sect. 11.2, the electrons are part of the vertex, and c_κ is no longer a number. Every result of this section assumed that it is; for a coupling to nucleons this is exact, since the nucleons move with their nucleus. At momentum transfers

small compared with the zone they follow the nuclei adiabatically, and their effect is the Born effective charge and the screening of Sect. 11.2. Far above the zone the question is how much of the electron cloud a displaced atom carries along, at the momentum transfer q itself. A rigid neutral atom contributes $Z_\kappa - f_\kappa(q)$, with f_κ the form factor of its electron cloud (Sect. 8.6), and what the rigid atom misses is the deformation of the electron density as the atom moves. The deformation obeys one exact constraint. Summed over the atoms of a cell, for a displacement that is the same in every cell, it is the change of the electron density under a uniform translation of the crystal; at the Fourier component $\boldsymbol{G}$ this is $\mathrm{i}G_\alpha$ times the density component itself, since with $n_e(\boldsymbol{q}) = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}$ a translation by $\boldsymbol{u}$ multiplies that component by $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{u}}$. The core shells follow their nucleus closely, polarized by the small residual field at the nucleus, and Appendix E.17 computes their share separately. The valence share is the open problem. How fast the deformation of the bonding charge decays between about 3 and 20 keV is not known: a calculation with pseudopotentials gives densities that are unphysical inside the core radius, above about 10 keV in silicon, and a free-atom model has no bonds. Appendix E.17 specifies the all-electron calculation, displaced atoms in a large cell, with the translation sum rule as its test. Until it is done, the multiphonon response of a charge coupling far above the zone rests on the rigid-atom vertex as an assumption, while that of a nucleon coupling does not.

At 10 keV the one-, two- and many-phonon parts of the harmonic response stand in the ratio of the Poisson weights, each spread over its band by the density of states. From a few zone momenta on, the interference of different nuclei corrects this incoherent picture by a few percent in the weights. As q grows further, $2W_\kappa$ and the recoil parameter x_R of Sect. 8.5 grow as q^2, the Poisson series moves to large phonon numbers, and the spectrum concentrates around E_R with a relative width $x_R^{-1/2}$. Sect. 11.4 follows it into the impulse regime, where the free nuclear recoil of the standard rate appears with a width set by the motion of the nucleus in its well.

11.4 Impulse limit and free nuclear recoil

A dark-matter particle of 20 MeV can transfer up to $2m_\chi v_{\max} = 106$ keV to a silicon nucleus. At $q = 100$ keV the free recoil energy is $E_R = 191$ meV, three times the highest phonon energy, and the Poisson series of Sect. 11.3 has a mean of $2W_\kappa = 6.3$ phonons. Their sum is no longer a set of phonon bands. It is a single peak centered at E_R, 88 meV wide, and as q grows it narrows toward the delta function $\delta(\omega - E_R)$ of the free recoil that Sects. 1.3 and 2.2 assumed. The nucleus recoils almost as if it were free, and what is left of the crystal is the motion of the nucleus in its well at the moment of the kick, which Doppler-broadens the line by an amount set by the kinetic energy of the nuclei; the standard nuclear-recoil rate with the Helm form factor is its limit, reached in silicon above 90 to 140 keV depending on the criterion.

11.4.1 The impulse approximation

At $q = 100$ keV, 44 times the zone momentum, the distinct part of the response is negligible (Sect. 11.3), and each nucleus responds on its own. Its self correlation has an exact form that needs no phonons. The kick $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}$ shifts the momentum of the nucleus by $\boldsymbol{q}$, and since every interaction in the crystal depends on the positions of the nuclei and not on their momenta, the kick changes only the kinetic energy $\boldsymbol{p}^2/2M_\kappa$. The Hamiltonian seen after the kick is the old one plus

$$D = E_R + \frac{\boldsymbol{q}\cdot\boldsymbol{p}}{M_\kappa},$$

the energy that the same kick would give a free particle of momentum $\boldsymbol{p}$. Its mean and variance follow exactly (Appendix D.32). The mean energy deposited is $\langle D\rangle = E_R$, the sum rule Eq. (5.61). The variance is $\langle D^2\rangle - E_R^2 = q^2\langle p_{\hat{q}}^2\rangle/M_\kappa^2$, with $p_{\hat{q}} = \hat{\boldsymbol{q}}\cdot\boldsymbol{p}$, at any temperature and for any potential, anharmonic ones included. Defining the kinetic energy of the nucleus along the direction of the kick, $E_\kappa^{\mathrm{kin}}(\hat{\boldsymbol{q}}) = \langle p_{\hat{q}}^2\rangle/2M_\kappa$, the width of the response is

$$\sigma_\omega = \big[4E_R\, E_\kappa^{\mathrm{kin}}(\hat{\boldsymbol{q}})\big]^{1/2}. \tag{11.32}$$

If the nucleus were free, D would commute with the Hamiltonian, and each momentum $\boldsymbol{p}$ would absorb exactly the energy D. For a bound nucleus the commutator is the force of the neighbors, and the *impulse approximation* drops it: the nucleus is treated as free for the duration of the kick, with the momentum distribution $\mathfrak{f}_\kappa(\boldsymbol{p})$ that it has in the crystal, normalized to one.

Impulse approximation of the nuclear response

$$S_A(\boldsymbol{q},\omega) \simeq \frac{2\pi}{\Omega}\sum_\kappa c_\kappa^2 \int \mathrm{d}^3p\; \mathfrak{f}_\kappa(\boldsymbol{p})\; \delta\Big(\omega - \frac{q^2}{2M_\kappa} - \frac{\boldsymbol{q}\cdot\boldsymbol{p}}{M_\kappa}\Big). \tag{11.33}$$

It keeps the exact mean E_R and the exact width Eq. (11.32). Its leading error is the third cumulant, at $T=0$ equal to $c_3 = (q^2/2M_\kappa^2)\langle\partial_{\hat{q}}^2 V\rangle$ with V the Born–Oppenheimer surface, whose share of the shape falls as $1/q$ (Appendix D.32).

The term $\boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa$ is a Doppler shift: a nucleus moving toward the kick absorbs more energy, one moving away absorbs less. The same picture underlies the impulse approximation of the electrons in Worked example 4.3, where the Fermi momentum takes the place of the zero-point momentum, and neutron Compton scattering, which uses it to measure the momentum distributions of light nuclei in solids and liquids [21]. The approximation is good when the kick is short compared with a vibration period. The time integral of the response is cut off at $t \sim 1/\sigma_\omega$, and the force of the neighbors changes the momentum during that time by a fraction $\sim \langle\omega\rangle/\sigma_\omega$ of its spread. At $T=0$ this fraction is $x_R^{-1/2}$, with the recoil parameter $x_R = E_R/\langle\omega\rangle$ of Sect. 8.5.

In a harmonic crystal $\mathfrak{f}_\kappa$ is a Gaussian at any temperature, since the momentum of each mode is Gaussian in a thermal state of an oscillator. Its width follows from the displacement correlation Eq. (5.57). On a site of cubic symmetry, with the projected density of states $g_{\mathrm{ph},\kappa}/3$ of Sect. 11.3,

$$E_\kappa^{\mathrm{kin}} = \frac{1}{4}\int \mathrm{d}\omega\; \frac{g_{\mathrm{ph},\kappa}(\omega)}{3}\,\omega\coth\frac{\omega}{2T},$$

which at $T=0$ is $\langle\omega\rangle/4$. Half the zero-point energy of each mode is kinetic, so a nucleus carries the kinetic energy $3\langle\omega\rangle/4$, a third of it along each direction. The impulse response of one species is then the Gaussian

$$S_A(\boldsymbol{q},\omega) \simeq \frac{2\pi c_\kappa^2}{\Omega}\,\frac{1}{(2\pi\sigma_\omega^2)^{1/2}}\exp\Big[-\frac{(\omega-E_R)^2}{2\sigma_\omega^2}\Big], \qquad \sigma_\omega^2 = E_R\Big\langle\omega\coth\frac{\omega}{2T}\Big\rangle, \tag{11.34}$$

which at $T=0$ has $\sigma_\omega/E_R = x_R^{-1/2}$, the relative width that Sect. 8.5 found from the cumulants of the Poisson series. Campbell-Deem, Knapen, Lin and Villarama obtain it as the saddle point of the time integral at $t=0$ [2, Eqs. (37)–(39)]. For a nucleus in an isotropic oscillator of frequency ω_0, the spread $q(\omega_0/2M_\kappa)^{1/2}$ of Kahn, Krnjaic and Mandava [19, Eq. (4)] is the same width with $\langle\omega\rangle = \omega_0$. The harmonic crystal gives every cumulant (Appendix D.32): $c_k = E_R\langle\omega^{k-1}\rangle$ for odd k, at any temperature, and $c_k = E_R\langle\omega^{k-1}\coth(\omega/2T)\rangle$ for even k.

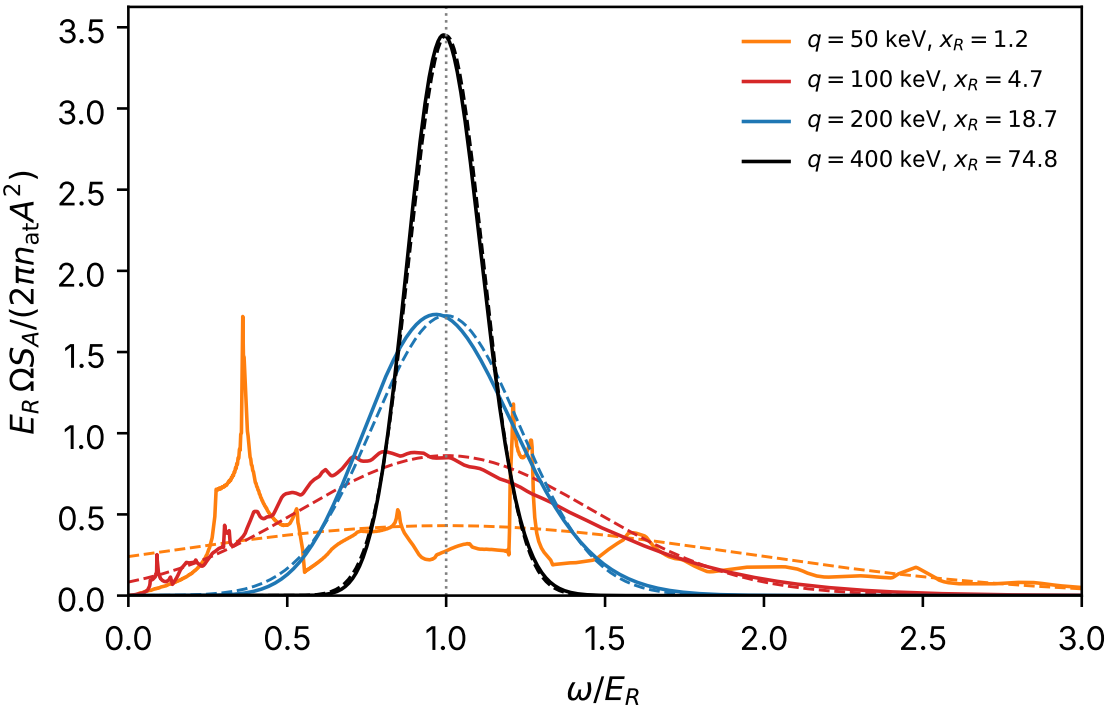


Figure 11.14: Nuclear response of silicon at $T = 0$ per nucleus, $\Omega S_A/(2\pi n_{\mathrm{at}} A^2)$, times E_R, against ω/E_R, at $q = 50$, 100, 200 and 400 keV ($x_R = 1.2$, 4.7, 18.7 and 74.8). Solid: the incoherent response of Eq. (11.26) summed over all phonon orders, from the density of states of the PBE force constants of Sect. 5.2. It is evaluated by Fourier transform of the generating formula Eq. (5.58). The zero-phonon line, of weight $\mathrm{e}^{-2W_\kappa} = 0.21$ at 50 keV and below 2×10^{-3} at the larger q, is not shown. Dashed: the impulse Gaussian Eq. (11.34), mean E_R and relative width $x_R^{-1/2}$. Dotted: the free recoil, $\omega = E_R$. The spectrum narrows around E_R and approaches the Gaussian as q grows; its maximum lies at 0.36, 0.81, 0.97 and 0.99 E_R.

The skewness, $c_3/\sigma_\omega^3 = (\langle\omega^2\rangle/\langle\omega\rangle^2)\, x_R^{-1/2}$ at $T = 0$, is the leading correction to Eq. (11.34). It moves the maximum of the spectrum below E_R by about half the skewness times σ_ω, while the mean stays at E_R.

In silicon the density of states of Sect. 5.2 gives $\langle\omega\rangle = 40.9$ meV, so $E_\kappa^{\mathrm{kin}} = 10.2$ meV at $T = 0$, and $\langle\omega^2\rangle/\langle\omega\rangle^2 = 1.19$. The zero-point motion broadens the line as much as a classical gas at 237 K would, the temperature at which $T/2$ equals E_κ^{kin}. Heating the crystal changes this little until the thermal energy reaches the phonon energies: E_κ^{kin} rises by 2 % at 77 K and by 56 % at 300 K, where the width is 1.25 times its zero-temperature value. At the 10 mK of a detector the zero-temperature values hold.

Fig. 11.14 follows the silicon response from the multiphonon regime into the impulse regime. At $q = 50$ keV, $x_R = 1.2$, the spectrum still shows the phonon bands: its maximum lies at $0.36\,E_R = 17$ meV, on the transverse-acoustic peak of the single-phonon distribution, and a fifth of the weight is in the zero-phonon line. At 100 keV the bands have merged into a peak whose maximum lies at $0.81\,E_R$. The Gaussian Eq. (11.34) has the right height and width there, and it misses by up to a fifth of the peak height, from the skewness and from the remaining structure of the phonon orders. At 200 and 400 keV the deviations fall to 7 and 3 % of the peak, roughly as $x_R^{-1/2}$, and the maximum approaches E_R.

The width needs one number per species, the kinetic energy of the nucleus along $\hat{\boldsymbol{q}}$. The

harmonic value comes from the same phonon calculation as the Debye–Waller exponent (Sect. 5.4 and Appendix E.4), as the first moment of the projected density of states instead of the inverse moment. Beyond the harmonic approximation the nuclear momentum distribution, its kinetic energy and the curvature that sets c_3 are computed by path-integral molecular dynamics, which treats the nuclei as quantum particles on the full Born–Oppenheimer surface; Appendix E.17 specifies that calculation for silicon, germanium and GaAs. The anharmonic change of $E_\kappa^{\rm kin}$ is the subject of Sect. 11.5.

Worked example 11.5: The impulse width at 100 keV in silicon

Take a silicon nucleus, $M = 26.162\,\text{GeV}$, at $T = 0$ and $q = 100\,\text{keV}$. The free recoil energy is $E_R = q^2/2M = 191.1\,\text{meV}$, and with $\langle\omega\rangle = 40.9\,\text{meV}$ the recoil parameter is $x_R = 4.67$. The kinetic energy along $\hat{\boldsymbol{q}}$ is $\langle\omega\rangle/4 = 10.23\,\text{meV}$, and Eq. (11.32) gives

$$\sigma_\omega = (4 \times 191.1\,\text{meV} \times 10.23\,\text{meV})^{1/2} = 88.4\,\text{meV}, \qquad \frac{\sigma_\omega}{E_R} = 0.463 = x_R^{-1/2}.$$

The Doppler reading gives the same number. The nucleus moves along $\hat{\boldsymbol{q}}$ with the root-mean-square speed $(2E_\kappa^{\rm kin}/M)^{1/2} = 265\,\text{m}\,\text{s}^{-1}$, while the kick gives it $q/M = 1.15\,\text{km}\,\text{s}^{-1}$. The relative spread of D is twice the ratio of the two speeds, 0.463.
The Gaussian has a full width at half maximum of 208 meV, larger than E_R itself. The full response of Fig. 11.14 at this q has a width at half maximum of 207 meV. The shape is skewed. The skewness is $1.19 \times 4.67^{-1/2} = 0.55$, which places the maximum about $0.55\,\sigma_\omega/2 = 24\,\text{meV}$ below E_R; the full response peaks at 154 meV, 37 meV below, because the phonon orders still leave ripples on the peak. At 300 K the width grows to 110 meV.
A germanium nucleus is 2.6 times heavier and its phonons are softer. With $\langle\omega\rangle \approx 23.6\,\text{meV}$, an estimate from its Debye temperature of 373 K (below), the same q gives $E_R = 73.9\,\text{meV}$, $x_R = 3.1$ and $\sigma_\omega/E_R = 0.57$: at equal momentum transfer the germanium response is further from free recoil than the silicon one.

On the truncation label of Sect. 8.5, Eq. (11.33) is $\mathcal{K}$ = impulse with $n_B = 0$. It holds at any anharmonic order n_A when $\mathfrak{f}_\kappa$ is the true momentum distribution, and the Gaussian Eq. (11.34) is its value at $n_A = 0$. Temperature enters only through $E_\kappa^{\rm kin}$, with all orders n_T kept by the factor $\coth(\omega/2T)$.

11.4.2 Free recoil and the standard rate

As x_R grows, the relative width $x_R^{-1/2}$ goes to zero and the Gaussian becomes $\delta(\omega - E_R)$ [2, Eq. (42)]. For a coupling to nucleons the vertex of a nucleus is $A_\kappa F_N(q)$, with the elastic form factor F_N of Eq. (2.26). The nucleons move with their nucleus, and the momenta of the impulse regime, up to a few MeV for sub-GeV dark matter, are far too small to excite it internally. Inserting $S_A = 2\pi \boldsymbol{n}_N A^2 F_N(q)^2 \delta(\omega - q^2/2M)$, with $\boldsymbol{n}_N$ the number density of nuclei, into Eq. (8.16) with one row, the response no longer depends on the direction of $\boldsymbol{q}$, the velocity integral gives $\eta(v_{\rm min})$, and the minimum velocity Eq. (1.23) at $\omega = q^2/2M$ is $q/2M + q/2m_\chi = q/2\mu_{\chi N}$ (Appendix D.32). For a heavy mediator,

Free nuclear recoil as the impulse limit

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}E_R} = \frac{\rho_\chi\,\sigma_N}{2m_\chi\,\mu_{\chi N}^2}\,F_N(q)^2\,\eta(v_{\rm min}), \qquad q = (2ME_R)^{1/2}, \qquad v_{\rm min} = \frac{q}{2\mu_{\chi N}}, \tag{11.35}$$

the spin-independent recoil spectrum of Sect. 2.2, with σ_N of Eq. (2.27). It is the limit $x_R \to \infty$ of the crystal response.

This is the standard rate of nuclear-recoil searches [22, 23, Eq. (104)], and with F_N the Helm form Eq. (2.28) and the Lewin–Smith parameters of Sect. 2.2 it is the rate that experiments quote. Kahn and Lin reach the same spectrum from an Einstein oscillator, whose Poisson distribution of phonon numbers concentrates at E_R/ω_0 for large q [24, Sect. IV A, Eqs. (90)–(92)]. On the truncation label the free recoil is $\mathcal{K}$ = impulse at zero width, $n_B = 0$; its mean deposit is exact at every anharmonic order and temperature, and only the width depends on them.

The derivation says what the formula needs: a recoil parameter large enough that the response is a narrow peak, and a rate whose weight in ω, the factor $\eta(v_{\min})$ together with the threshold, does not vary across the width σ_ω. Limit Check 11.10 tests both in silicon, by computing the heavy-mediator rate from the full multiphonon response and dividing it by Eq. (11.35). For a particle of 1 GeV, whose kinematic endpoint lies far above the energies shown, the two agree within 10 % for $x_R \geq 2.3$, or $E_R \geq 93\,\mathrm{meV}$ and $q \geq 70\,\mathrm{keV}$, and within 2 % for $x_R \geq 4.3$. The free form works where the spectrum is not yet narrow, as Campbell-Deem and collaborators observed [2, Sect. III D], because a smooth weight averages over the peak and the mean of the peak is exactly E_R. Below $x_R \approx 2$ the deposit comes in whole phonons, and the ratio swings by factors of two with the bands of the phonon spectrum.

A lighter particle has its endpoint $2\mu_{\chi N}^2 v_{\max}^2/M$ close to the energies that matter, and there η falls steeply across the peak. For 50 MeV the endpoint is at 1.34 eV, $x_R = 32.7$. Between $x_R = 3$ and 8 the free form overestimates the crystal rate by 5 to 21 %, because the width spreads each momentum transfer over a range of energies across which η falls steeply. From $x_R = 14$ upward, however, $4.9\,\sigma_\omega$ below the endpoint, the crystal rate exceeds the free one by more than 10 % and keeps rising. Above the endpoint the free recoil gives nothing, while the Doppler tail of the bound nucleus still deposits energy. This is the impulse-regime counterpart of the Poisson tail of Worked example 11.4, and the reason Kahn, Krnjaic and Mandava found that a bound nucleus extends the reach of a detector below the masses that elastic kinematics allows [19, text after Eq. (4)]. The impulse Gaussian Eq. (11.34), used in the same rate integral, follows the full crystal result within 5 % from $x_R \approx 3.6$ at 1 GeV and 3.8 at 50 MeV, up to three quarters of the endpoint, and it captures the endpoint tail too. Where the free recoil fails at large x_R, the impulse approximation with its width is the correction.

Limit Check 11.10. Free nuclear recoil with the Helm form factor

For $x_R \gg 1$ the nuclear response becomes $(2\pi/\Omega)\sum_\kappa c_\kappa^2 \delta(\omega - q^2/2M_\kappa)$ [2, Eqs. (39), (42)], and the rate is the free recoil Eq. (11.35) with $F_N = F_{\mathrm{Helm}}$ and the Lewin–Smith parameters [22, 23, Eq. (104)].

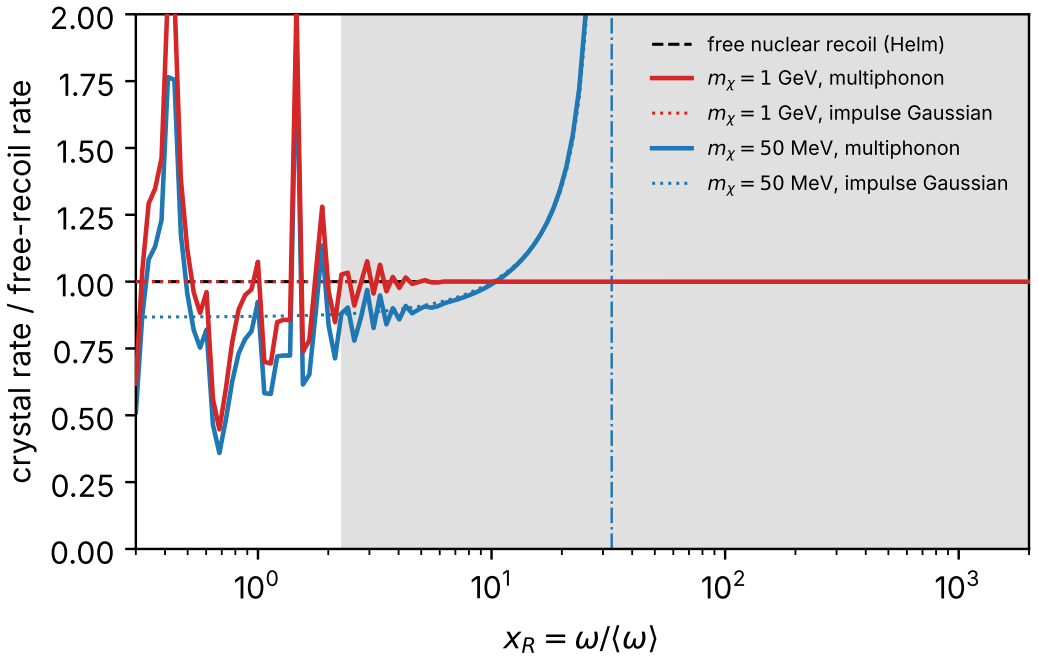


Figure 11.15: Heavy-mediator recoil rate of silicon per unit mass from the crystal response, divided by the free-recoil rate Eq. (11.35) at $E_R = \omega$, against $x_R = \omega/\langle\omega\rangle$ with $\langle\omega\rangle = 40.9\,\text{meV}$. Solid: incoherent multiphonon response at $T = 0$, all orders, from the PBE density of states of Sect. 5.2, integrated over q with $\eta(v_{\min})$ of the halo of Sect. 1.2 and the Helm form factor, for $m_\chi = 1\,\text{GeV}$ (red) and 50 MeV (blue). Dotted: the same with the impulse Gaussian Eq. (11.34), the leading correction. Dash-dotted: the free-recoil endpoint at 50 MeV. Gray band: $x_R \geq 2.3$, where the 1 GeV rate stays within 10 % of the free one. The lighter particle departs from it within about five widths σ_ω of its endpoint.

Below $x_R \approx 2$ the energy arrives in whole phonons and the rate follows the phonon bands; near the kinematic endpoint the Doppler width of the bound nucleus feeds energies that a free nucleus cannot reach.

Where does the impulse regime begin? The onset depends on the criterion. The relative width $x_R^{-1/2}$ falls to one half at $x_R = 4$, where the impulse Gaussian reproduces the rate of the limit check within 5 %. This is $q = 2(2M\langle\omega\rangle)^{1/2}$, the boundary above which Campbell-Deem and collaborators replace the multiphonon sum by the Gaussian [2, text after Eq. (39)]. In silicon it lies at $q = 92.5\,\text{keV}$, $E_R = 164\,\text{meV}$. The stricter criterion of Sect. 8.5, a width below a third of E_R, needs $x_R \geq 9$, at 139 keV and 0.37 eV. For germanium a rough $\langle\omega\rangle$ is enough, and its Debye temperature of 373 K supplies one (Table 5.1): the Debye mean $\frac{3}{4}\omega_\text{D} = 24.1\,\text{meV}$, scaled by the ratio 0.981 of the computed to the Debye mean in silicon, gives $\langle\omega\rangle \approx 23.6\,\text{meV}$. The onsets are then $q = 113\,\text{keV}$ and $E_R = 95\,\text{meV}$ at $x_R = 4$, and 170 keV and 0.21 eV at $x_R = 9$. The heavier nucleus needs a larger momentum transfer but a smaller recoil energy. These onsets are not the 93.4 keV of Worked example 11.4, which is the momentum at which a free nucleus would first deposit three Debye energies. A particle much lighter than the nucleus reaches q only if $2m_\chi v_{\max} \geq q$, so the impulse regime is available in silicon from $m_\chi = 17.5\,\text{MeV}$ and in germanium from 21.3 MeV ($x_R = 4$), or from 26.2 and 32.0 MeV ($x_R = 9$). Even then, as the limit check shows, the free form needs the energies of

interest to lie several widths below the endpoint.

The impulse approximation leaves the crystal at the moment of the kick. It ignores the motion of the nucleus after the kick, and it leaves out anharmonicity, electronic excitation and the displacement of the atom. The kinetic energy and curvature of an anharmonic crystal shift σ_ω and c_3 (Sect. 11.5), but never the mean. The electrons follow the nucleus at $n_B = 0$; the probability that the kick leaves them excited is the Migdal effect of Chap. 12. With the electrons included, the third moment of the full nuclear response contains the electron density at the nucleus and is dominated by those excitations (Sect. 12.5); the curvature in c_3 above is that of the Born–Oppenheimer surface alone. And a recoil above the displacement threshold of the lattice ejects the atom from its site, which changes how the energy is shared among phonons, defects and ionization (Chap. 19) but not the response Eq. (11.33), since the momentum and energy are transferred before the atom moves.

11.4.3 Fermionic absorption on nuclei

A fermion χ of 10 MeV that is absorbed on a silicon nucleus and leaves as a massless $\tilde{\nu}$ transfers the momentum $q = m_\chi - \omega \simeq 10\,\text{MeV}$, and a free nucleus recoils with $E_R = m_\chi^2/2M = 1.91\,\text{keV}$. Scattering at the same mass deposits at most $2\mu_{\chi N}^2 v_{\max}^2/M = 54\,\text{meV}$. Absorption reaches recoil energies $1/4v_{\max}^2$ times larger, and because the momentum transfer is set by the rest energy, the deposit is a line, whatever the velocity distribution [25]. Sect. 10.9 built the rate for the operators $\Lambda_F^{-2}(\bar{\tilde{\nu}} P_L \chi)(\bar{\psi}\psi)$ and $\Lambda_F^{-2}(\bar{\tilde{\nu}}\gamma_\mu P_R \chi)(\bar{\psi}\gamma^\mu\psi)$, Eq. (10.60). For nucleons in place of electrons only the response changes. The vector operator gives $(\sigma_F/2\pi)\bar{S}_{nn}(q,\omega)$, with $\sigma_F = m_\chi^2/4\pi\Lambda_F^4$ and $\bar{S}_{nn}$ the direction-averaged response of the nucleon density, because continuity, exact for the nucleon current, turns the longitudinal current into the density and multiplies it by $(m_\chi/q)^2$. The transverse current of a nucleus is smaller by a factor of order E_κ^{kin}/M or q^2/M^2 and is dropped.

For a nucleus at rest, $\bar{S}_{nn} = 2\pi \boldsymbol{n}_N A^2 F_N(q)^2 \delta(\omega - q^2/2M)$. Along the line $q = m_\chi - \omega$ the argument of the delta function has the slope $1 + q/M$ (Appendix D.32), and

Fermionic absorption on nuclei

$$\Gamma_V = \frac{\sigma_F\, \boldsymbol{n}_N A^2 F_N(q_{\text{abs}})^2}{1 + q_{\text{abs}}/M} = \sigma_F\, \boldsymbol{n}_N A^2 F_N^2 \Big(1 - \frac{m_\chi}{M} + \dots\Big), \qquad m_\chi = q_{\text{abs}} + \frac{q_{\text{abs}}^2}{2M}, \tag{11.36}$$

per dark-matter particle, in a target of free nuclei, at a recoil energy $E_R = q_{\text{abs}}^2/2M \simeq m_\chi^2/2M$; the event rate per unit mass is $(\rho_\chi/m_\chi\rho_T)\Gamma_V$. In a crystal, in the incoherent approximation and for m_χ not within a relative v of the length of a reciprocal-lattice vector, the factor $1 - \mathrm{e}^{-2W_\kappa(q_{\text{abs}})}$ multiplies the rate.

Since $\sigma_F \propto m_\chi^2$, the rate carries the factor $m_\chi^2(1 - m_\chi/M)$. The scalar operator weights the density by q^2/m_χ^2 in addition and gives $m_\chi^2(1 - 2m_\chi/M)$. The difference is the longitudinal current: the density row alone has twice the recoil correction, and continuity halves it. Dror, Elor and McGehee write the rate per target nucleus as $(\rho_\chi/m_\chi)\sigma_F A^2 F_N^2$ for the vector operator, with the Helm form factor at $q = m_\chi$ [25, Eqs. (1) and (4)], which is Eq. (11.36) at $M \to \infty$. In silicon the recoil factor reaches 10 % only at $m_\chi = 3\,\text{GeV}$, where q_{abs} lies far beyond the first zero of the Helm form factor, at 256 MeV, and elastic absorption is negligible; on a proton target it is about 10 % at 100 MeV.

The crystal changes the rate at the other end of the mass range. In the response of Sect. 11.3 the zero-phonon term conserves crystal momentum and deposits no energy, so it absorbs a fermion only if $|\boldsymbol{q}| = m_\chi$ equals the length of a reciprocal-lattice vector, to within

the relative spread v of the dark-matter momenta. At any other mass the lattice must take up the momentum with at least one phonon, and in the incoherent approximation the weight of those terms is $1 - \mathrm{e}^{-2W_\kappa}$, evaluated at $q = q_{\mathrm{abs}}$. In silicon, with $2W_\kappa = q^2\langle u_x^2\rangle$, this is 0.061 at $m_\chi = 10\,\mathrm{keV}$, 0.43 at 30 keV and 0.90 at 60.5 keV. The free-nucleus formula therefore overestimates the absorption rate by a factor of 16 at 10 keV, and it holds to 10 % only above 61 keV, the boundary of Limit Check 11.11. Below about 100 keV the deposit is not a line at $m_\chi^2/2M$ either. It is the multiphonon spectrum of Sect. 11.3 at $q = m_\chi$: mostly one phonon below about 20 keV, a few phonons up to the impulse onset near 90 keV, and the Gaussian Eq. (11.34) above it. The weights of the lowest orders carry the coherence corrections of Sect. 11.3, below 10 % above about 6.6 keV for two and three phonons and above 18 keV for one.

Above the onset the crystal sets the width of the line. The impulse Gaussian has the relative width $x_R^{-1/2} = (2M\langle\omega\rangle)^{1/2}/q_{\mathrm{abs}} = 46.3\,\mathrm{keV}/m_\chi$ in silicon: 4.6 % at 1 MeV, 0.46 % at 10 MeV. The motion of the dark matter shifts the recoil energy by $-2v\cos\theta$ in relative terms, with θ the angle between its velocity and the outgoing $\tilde{\nu}$. For the halo of Sect. 1.2, whose root-mean-square speed in the laboratory is $379\,\mathrm{km\,s^{-1}}$, the root-mean-square shift is $2(\langle v^2\rangle/3)^{1/2} = 1.5\times10^{-3}$, the relative width $\sim 10^{-3}$ that Dror, Elor and McGehee attribute to corrections to the leading kinematics [25]. The two widths are equal at $m_\chi = 32\,\mathrm{MeV}$, and below that mass the line in a crystal is broader than the halo makes it, by the zero-point motion of the nuclei.

Limit Check 11.11. Fermionic absorption on nuclei

A fermion absorbed on free nuclei gives the rate $(\rho_\chi/m_\chi)\sigma_F A^2 F_N^2$ per nucleus, with $\sigma_F = m_\chi^2/4\pi\Lambda_F^4$, at the recoil energy $m_\chi^2/2M$ [25, Eqs. (1), (3) and (4)].

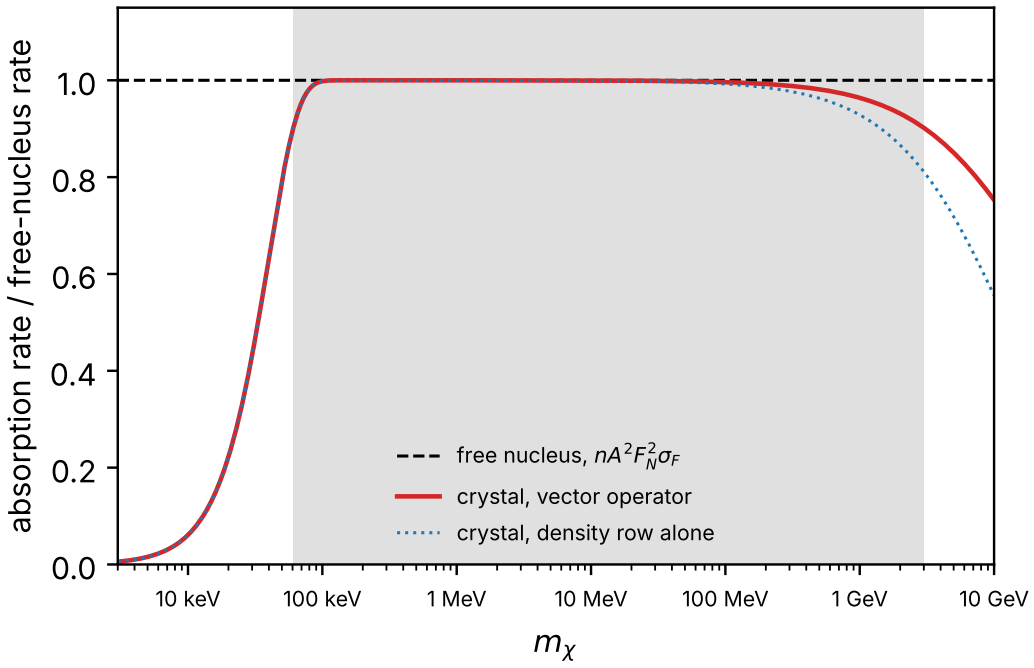


Figure 11.16: Absorption rate of a fermion on silicon nuclei, divided by the free-nucleus rate of the limit check, against m_χ, with the nuclear form factor divided out of both. Solid: the vector operator in the crystal, $[1-\mathrm{e}^{-2W_\kappa(q_{\mathrm{abs}})}]/(1+q_{\mathrm{abs}}/M)$ from Eq. (11.36),

incoherent approximation at $T = 0$ with $\langle u_x^2 \rangle = 2.45 \times 10^{-3}\,\text{Å}^2$ from the PBE density of states of Sect. 5.2. Dotted: the density row alone, as for the scalar operator, with the recoil factor $(1 - m_\chi/M)^2$ to leading order instead of $1 - m_\chi/M$. Gray band: $61\,\text{keV} \le m_\chi \le 3.0\,\text{GeV}$, where the vector rate is within 10 % of the free one.

Below about 60 keV in silicon the absorption must create phonons and the free-nucleus rate overcounts it by $e^{-2W_\kappa}/(1 - e^{-2W_\kappa})$; above a few GeV the recoil factor $1 - m_\chi/M$ and the form factor at q_abs instead of m_χ matter.

On the truncation label the limit check sits at $n_B = 0$ and $n_T = 0$, with $\mathcal{K}$ = multiphonon (incoherent, all orders) below the impulse onset and $\mathcal{K}$ = impulse above it.

From about 90 keV of momentum transfer in silicon, then, a nucleon coupling sees free nuclei. The rate is that of the elastic recoil of Sect. 1.3 with the Helm form factor. The spectrum is a peak at E_R with the width $(4E_R E_\kappa^\text{kin})^{1/2}$, which a harmonic crystal fixes through $\langle\omega\rangle/4$. Only the width and the shape around the mean depend on the crystal. They are also what the anharmonic terms of the potential change first, through the kinetic energy and the curvature of the Born–Oppenheimer surface, and Sect. 11.5 turns to them.

11.5 Anharmonic and quantum-crystal corrections

At its zero-point amplitude the cubic term of the silicon energy surface is 5 % of the harmonic one ($\eta_\text{anh} = 0.049$, Sect. 5.7), and every formula of Sects. 11.3 and 11.4 dropped it. How much does that change a rate? For the impulse peak, by a part in a thousand. For a deposit of several phonons at a momentum transfer of a few keV, by factors of several, because an anharmonic lattice can make many phonons out of one kick, which a harmonic lattice can only do with many kicks. In solid helium the zero-point motion of an atom is a quarter of the distance to its neighbor, and some harmonic frequencies are imaginary. And in a crystal with a soft mode or a light atom no expansion is available at all, and only exact moments of the response, which anharmonicity cannot change, bound it.

11.5.1 Exact moments of the self response

Above a few zone momenta each nucleus responds on its own (Sect. 11.3), and its self response is controlled by the kick operator $D = E_R + \boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa$ of Sect. 11.4: the Hamiltonian after the kick is $H + D$, exactly, whatever the potential, provided it depends on the positions of the nuclei and not on their momenta (Eq. (D.188)). Normalize the response to a distribution in ω, $\mathrm{d}P = S\,\mathrm{d}\omega/\int S\,\mathrm{d}\omega$, with moments $\mu_k = \int \omega^k\,\mathrm{d}P$ and central moments $\bar\mu_k$, as in Appendix C.10. At $T = 0$, writing $A = H - E_0$, which annihilates the ground state, and $\delta D = D - E_R = (q/M_\kappa)p_{\hat q}$, the central moments are $\bar\mu_k = \langle (A + \delta D)^k \rangle$. Every term with A at either end vanishes. The zeroth moment is $\mu_0 = 1$: the weight of the self response is the norm of $e^{i\boldsymbol{q}\cdot\boldsymbol{X}}|0\rangle$. The first is $\mu_1 = E_R$, the sum rule Eq. (5.61). The second and third were derived in Appendix D.32. The fourth needs one more step: of the four terms of $\langle \delta D (A + \delta D)^2 \delta D\rangle$, the term $\langle \delta D\, A^2\, \delta D\rangle$ is the squared norm of $A\,\delta D|0\rangle = [H, \delta D]|0\rangle$, and $[H, p_{\hat q}] = i\,\partial_{\hat q} V$ is the force along $\hat{\boldsymbol{q}}$. The two mixed terms $\langle \delta D\, A\, \delta D^2\rangle$ and its complex conjugate equal $i(q/M_\kappa)^3\langle p_{\hat q}(p_{\hat q}\partial_{\hat q}V + \partial_{\hat q}V\, p_{\hat q})\rangle$, by $[H, p_{\hat q}^2] = i(p_{\hat q}\partial_{\hat q}V + \partial_{\hat q}V\, p_{\hat q})$; for a real ground-state wavefunction the average is real, so the term is imaginary and cancels against its conjugate. Collecting,

Exact moments of the self response

$$\mu_0 = 1, \qquad \mu_1 = E_R, \qquad \bar\mu_2 = 4E_R E_\kappa^{\mathrm{kin}},$$
$$\bar\mu_3 = \frac{q^2}{2M_\kappa^2}\langle \partial_{\hat q}^2 V\rangle, \quad \bar\mu_4 = \frac{q^2}{M_\kappa^2}\langle(\partial_{\hat q} V)^2\rangle + \frac{q^4}{M_\kappa^4}\langle p_{\hat q}^4\rangle, \tag{11.37}$$

for any potential V of the nuclei, with $E_\kappa^{\mathrm{kin}} = \langle p_{\hat q}^2\rangle/2M_\kappa$ the kinetic energy along $\hat{\boldsymbol{q}}$ of Eq. (11.32). The averages are over the ground state at $T = 0$; all five moments hold at any temperature with thermal averages (Appendix D.32).

Each moment is an equal-time average in the ground state: the kinetic energy, the mean curvature of the energy surface, the mean-square force and the fourth moment of the momentum distribution. For a harmonic crystal they reproduce the cumulants of Eq. (D.193); for one oscillator of frequency ω_0, for instance, the mean-square force is $M_\kappa^2\omega_0^4\langle u_{\hat q}^2\rangle = M_\kappa\omega_0^3/2$, so the first term of $\bar\mu_4$ is the fourth cumulant $E_R\omega_0^3$, and the second is $3\bar\mu_2^2$. In an anharmonic crystal the same averages come from a path-integral simulation of the nuclei, which needs no dynamics (Appendix E.19). They are the self part of the moments $\mathscr{M}^k$ of Eq. (8.22) for one species, divided by $\mathscr{M}^0$.

Write its relative correction to the harmonic self response as $\delta_{\mathrm{anh}}(\boldsymbol{q},\omega) = S_A/S_A^{\mathrm{harm}} - 1$. Then the weight $\int S_A^{\mathrm{harm}}\delta_{\mathrm{anh}}\,\mathrm{d}\omega$ and the first moment $\int \omega\, S_A^{\mathrm{harm}}\delta_{\mathrm{anh}}\,\mathrm{d}\omega$ vanish exactly, and the second moment $\int \omega^2 S_A^{\mathrm{harm}}\delta_{\mathrm{anh}}\,\mathrm{d}\omega$ is fixed by the change of the kinetic energy, $4E_R(E_\kappa^{\mathrm{kin}} - E_\kappa^{\mathrm{kin,harm}})$ times the weight. These three linear constraints test any calculation of the anharmonic response. Time reversal adds a selection rule. For a target invariant under time reversal the eigenfunctions are real, $|\langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|0\rangle|^2 = |\langle f|\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|0\rangle|^2$, and the response is even in $\boldsymbol{q}$. A term of first order in $\Phi^{(3)}$ carries three displacements from the cubic term of Eq. (5.68), so it needs an odd number of displacements from the kick to pair them all, and it is odd in $\boldsymbol{q}$. It therefore vanishes. The first anharmonic correction to the rate is of second order in $\Phi^{(3)}$ and first order in $\Phi^{(4)}$, the level $n_A = 2$ of Sect. 8.5, of relative size η_{anh}^2 by Eq. (5.73).

11.5.2 Cubic phonon decay

In the impulse regime the first correction is small. The impulse response Eq. (11.33) depends on the potential only through the momentum distribution $\mathfrak{f}_\kappa$, and its width through E_κ^{kin}. Lin, Shen, Sholapurkar and Villarama model the potential of one nucleus, with its neighbors held fixed, by a one-dimensional anharmonic oscillator [26, Eq. (26)],

$$V(x) = \tfrac{1}{2}M_\kappa\omega_0^2x^2 + \lambda_3\,\omega_0\Big(\frac{x}{\ell_0}\Big)^3 + \lambda_4\,\omega_0\Big(\frac{x}{\ell_0}\Big)^4, \qquad \ell_0 = (2M_\kappa\omega_0)^{-1/2}, \tag{11.38}$$

with ℓ_0 the zero-point amplitude of the oscillator. In this potential the cubic energy at $x = \ell_0$ is $4\lambda_3$ times the harmonic one, and $4\lambda_3$ is the anharmonic parameter of Eq. (5.72) measured on the site coordinate instead of the bond. Fitting an empirical silicon potential along the bond and perpendicular to it, they find $\lambda_3 = 0.006$ to 0.01 and $\lambda_4 = (2\text{ to }3)\times10^{-4}$ [26, Sect. IV], so $4\lambda_3 = 0.024$ to 0.04, of the size of the bond value 0.049 of Sect. 5.7. To second order in λ_3 and first in λ_4 the kinetic energy becomes $E^{\mathrm{kin}} = (\omega_0/4)(1 - 44\lambda_3^2 + 12\lambda_4)$ [26, Eq. (65)], which an exact diagonalization of Eq. (11.38) confirms to 3×10^{-5}. Over the silicon ranges of λ_3 and λ_4 the change lies within $\pm0.2\,\%$, the extremes $(\lambda_3,\lambda_4) = (0.01, 2\times10^{-4})$ and $(0.006, 3\times10^{-4})$. The impulse width $\sigma_\omega \propto (E^{\mathrm{kin}})^{1/2}$ then moves by at most a part in a thousand, 0.09 meV of the 88 meV of Worked example 11.5. Above the impulse onset the harmonic crystal is exact to a part in a thousand.

At smaller q the correction is of a different kind, because anharmonicity changes how many kicks a given number of phonons needs. Expand the self response of Eq. (11.24) in

powers of q. At $\omega \neq 0$ the constant and the averages of single displacements contribute only at $\omega = 0$, and the first term is the correlation of two displacements,

$$S_A(\boldsymbol{q},\omega) = \frac{1}{\Omega}\sum_\kappa c_\kappa^2\, q_\alpha q_\beta \int \mathrm{d}t\; \mathrm{e}^{\mathrm{i}\omega t}\langle u_\kappa^\alpha(t)\, u_\kappa^\beta(0)\rangle + O(q^4),$$

exact for any anharmonicity. Insert the displacement in phonon coordinates, $\boldsymbol{u}_{\boldsymbol{R}\kappa} = \sum_{\boldsymbol{Q}\nu}(2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{-1/2}\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}B_{\boldsymbol{Q}\nu}$ (Sect. 5.3). The correlation of $B_{\boldsymbol{Q}\nu}$ is the spectral function of the dressed propagator $\mathsf{D}_\nu(\boldsymbol{Q},\omega) = 2\omega_{\boldsymbol{Q}\nu}/[\omega^2 - \omega_{\boldsymbol{Q}\nu}^2 - \Pi_\nu(\boldsymbol{Q},\omega)]$ of Eq. (6.29): at $T = 0$ and $\omega > 0$, $\int \mathrm{d}t\,\mathrm{e}^{\mathrm{i}\omega t}\langle B_{\boldsymbol{Q}\nu}(t)B_{-\boldsymbol{Q}\nu}\rangle = -2\,\mathrm{Im}\,\mathsf{D}_\nu(\boldsymbol{Q},\omega)$, which is $2\pi\delta(\omega-\omega_{\boldsymbol{Q}\nu})$ for the free phonon. Here Π_ν is the anharmonic self-energy: the bubble of two cubic vertices of Eq. (5.69), whose imaginary part on shell is $-\omega_{\boldsymbol{Q}\nu}\gamma_{\boldsymbol{Q}\nu}^{\mathrm{anh}}$ (Eqs. (5.70) and (6.30)), and the loop of one quartic vertex, which is real. Off shell, $-\,\mathrm{Im}\,\Pi_\nu(\boldsymbol{Q},\omega) = \omega_{\boldsymbol{Q}\nu}\gamma_{\boldsymbol{Q}\nu}^{\mathrm{anh}}(\omega)$, with $\gamma_{\boldsymbol{Q}\nu}^{\mathrm{anh}}(\omega)$ the width of Eq. (5.70) with the energy ω in place of $\omega_{\boldsymbol{Q}\nu}$ in the delta function of the decay: the rate at which a phonon of momentum $\boldsymbol{Q}$, given the energy ω, decays into two. With the branches kept separate, as in Appendix D.17, the harmonic term gives back the one-phonon response Eq. (11.1), a delta function on each branch. Away from the poles, above the one-phonon band, the first order in Π_ν is $-\,\mathrm{Im}\,\mathsf{D}_\nu = 2\omega_{\boldsymbol{Q}\nu}(-\,\mathrm{Im}\,\Pi_\nu)/(\omega^2-\omega_{\boldsymbol{Q}\nu}^2)^2$, and

First anharmonic correction to the nuclear response

$$S_A^{\mathrm{anh}}(\boldsymbol{q},\omega) = \frac{2\pi}{\Omega}\sum_\kappa c_\kappa^2\,\mathrm{e}^{-2W_\kappa}\,\frac{1}{N}\sum_{\boldsymbol{Q}\nu}\frac{|\boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2}{\pi M_\kappa}\,\frac{\omega_{\boldsymbol{Q}\nu}\,\gamma_{\boldsymbol{Q}\nu}^{\mathrm{anh}}(\omega)}{(\omega^2-\omega_{\boldsymbol{Q}\nu}^2)^2}, \tag{11.39}$$

at $T = 0$, for ω outside the one-phonon band and q above a few q_{BZ}, where the self part is the response. It is of order q^2 and of second order in $\Phi^{(3)}$; the Debye–Waller factor, which the higher orders in q supply, is written to leading order in η_{anh}.

The physical picture is that of Fig. 6.6 with the electron bubble replaced by a pair of phonons: the kick creates one phonon off its energy shell, and it decays into two real ones. Near $\omega = \omega_{\boldsymbol{Q}\nu}$ the same expression, resummed, is the Lorentzian tail of a phonon line of full width $\gamma_{\boldsymbol{Q}\nu}^{\mathrm{anh}}$. Campbell-Deem and collaborators wrote the amplitude of this process for long-wavelength acoustic phonons as the sum of two time orderings, the decay of a virtual phonon and the creation of three from the vacuum [1, Eq. (13)]; the two denominators $1/(\omega-\omega_{\boldsymbol{Q}\nu})$ and $1/(\omega+\omega_{\boldsymbol{Q}\nu})$ combine into the $2\omega_{\boldsymbol{Q}\nu}/(\omega^2-\omega_{\boldsymbol{Q}\nu}^2)$ of D_ν.

How large is Eq. (11.39) against the harmonic two-phonon response? The harmonic term needs two kicks and carries q^4; the anharmonic term needs one kick and the factor $\gamma^{\mathrm{anh}}/\omega \sim \eta_{\mathrm{anh}}^2$. Their ratio is $\sim \eta_{\mathrm{anh}}^2\, 2M_\kappa\omega/q^2$, large at small q. The model Eq. (11.38) makes the counting exact. To first order in λ_3, with $x/\ell_0 = a + a^\dagger$ and the harmonic states $|n\rangle$ in units of ω_0, the ground state and the second level become

$$|\tilde 0\rangle = |0\rangle - \lambda_3\Big(3|1\rangle + \sqrt{\tfrac{2}{3}}\,|3\rangle\Big), \qquad |\tilde 2\rangle = |2\rangle + 6\sqrt{2}\,\lambda_3|1\rangle + \ldots,$$

where the dots stand for $|3\rangle$ and $|5\rangle$, which the kick does not connect to $|\tilde 0\rangle$ at first order in q. The coefficient $6\sqrt{2}$ is $\langle 1|(a+a^\dagger)^3|2\rangle$ divided by the level spacing. With $k = q\ell_0$ the kick is $\mathrm{e}^{\mathrm{i}k(a+a^\dagger)} = 1 + \mathrm{i}k(a+a^\dagger) - \frac{1}{2}k^2(a+a^\dagger)^2 + \ldots$. The constant term vanishes between orthogonal states. The linear term gives $\langle\tilde 2|a+a^\dagger|\tilde 0\rangle = -\lambda_3(3\sqrt{2}+\sqrt{2}) + 6\sqrt{2}\,\lambda_3 = 2\sqrt{2}\,\lambda_3$. The quadratic term gives the harmonic $-k^2/\sqrt{2}$. One amplitude is imaginary and the other real; they do not interfere, and the weight of the second level is

$$|\langle\tilde 2|\mathrm{e}^{\mathrm{i}qx}|\tilde 0\rangle|^2 = \mathrm{e}^{-k^2}\Big(\frac{k^4}{2} + 8\lambda_3^2k^2 + \ldots\Big), \qquad \frac{\text{anharmonic}}{\text{harmonic}} = \frac{16\lambda_3^2}{k^2}, \tag{11.40}$$

which the exact solution confirms. The anharmonic term wins below $k = 4\lambda_3$, that is, below $q_\times = 4\lambda_3(2M_\kappa\omega_0)^{1/2}$: the crossover sits where the momentum transfer, in units of the inverse zero-point amplitude, equals the anharmonic parameter. For silicon, with $\omega_0 = \langle\omega^{-1}\rangle^{-1} = 30.4\,\mathrm{meV}$ as Lin and collaborators choose it, $(2M_\kappa\omega_0)^{1/2} = 39.9\,\mathrm{keV}$ and $q_\times = 1.6\,\mathrm{keV}$ for $\lambda_3 = 0.01$, just below $q_\mathrm{BZ} = 2.25\,\mathrm{keV}$. At q_BZ the two-phonon weight is raised by 50 % for $\lambda_3 = 0.01$ and by 18 % for 0.006.

More phonons need more kicks, and the anharmonic gain grows with their number. Three phonons need three kicks in the harmonic crystal, $k^6/6$, but only two with one cubic vertex. That two-kick amplitude alone gives $24\lambda_3^2k^4$; interference between the one-kick amplitude of order λ_3^2k and the harmonic three-kick amplitude $k^3/\sqrt{6}$ subtracts $6\lambda_3^2k^4$, and the exact solution confirms the net $18\lambda_3^2k^4$. The ratio is then $108\lambda_3^2/k^2$ and the crossover lies at $k = 10.4\lambda_3$, 4.2 keV for $\lambda_3 = 0.01$. In general an n-phonon weight acquires terms $\lambda_3^jk^{2i}$ with $i < n$, and the smaller i the higher the power j [26, Eqs. (39), (45)–(48)]. Limit Check 11.12 shows the result: for $\lambda_3 = 0.01$ the harmonic two-, three- and four-phonon weights hold within 10 % only above 5, 12 and 20 keV, while at q_BZ the four-phonon weight is 27 times the harmonic one. Which rates does this affect? A threshold of ω_th needs at least $\omega_\mathrm{th}/\omega_\mathrm{max}$ phonons, and a particle of mass m_χ reaches at most $q = 2m_\chi v_\mathrm{max}$. High thresholds and masses of a few MeV are therefore where the harmonic multiphonon rate fails. Lin and collaborators find that the anharmonic rate exceeds the harmonic one by up to two orders of magnitude for masses of a few MeV and thresholds of a few phonon energies, in the tail of the spectrum where the rate is small. Above 10 MeV the two agree within their accuracy [26, Sect. IV A]. In their model the anharmonic rate is never below the harmonic one. The reason is that the term of lowest order in q is a square, $|\langle f|\mathrm{i}qx|\tilde{0}\rangle|^2$, which the harmonic crystal lacks for $n \geq 2$. This is a statement about the leading term, not a theorem about the full response.

Limit Check 11.12. Anharmonic corrections to multiphonon weights

For a nucleus in the potential Eq. (11.38), the n-phonon weight carries, besides the harmonic $\mathrm{e}^{-k^2}k^{2n}/n!$, terms $\lambda_3^jk^{2i}$ with $i < n$, which dominate for small $k = q\ell_0$; the kinetic energy changes as $1 - 44\lambda_3^2 + 12\lambda_4$ [26, Eqs. (39), (45)–(48), (65)].

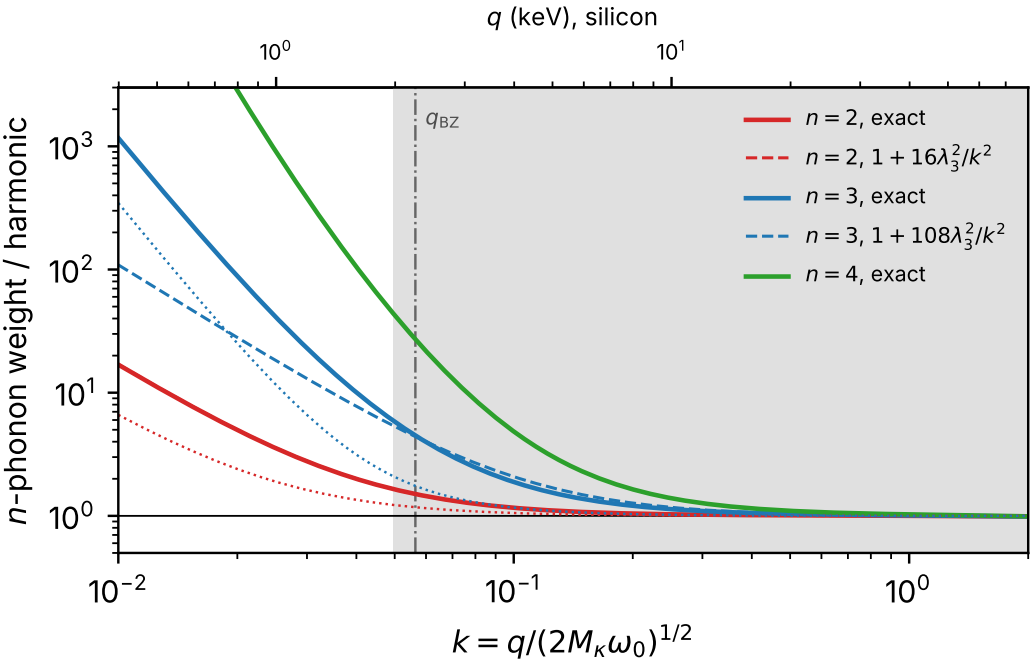


Figure 11.17: Weight of the n-th level of the single-site model Eq. (11.38), divided by the harmonic Poisson weight $\mathrm{e}^{-k^2}k^{2n}/n!$, against $k = q/(2M_\kappa\omega_0)^{1/2}$, for $n = 2$ (red), 3 (blue) and 4 (green). Model with the silicon values $\omega_0 = 30.4\,\mathrm{meV}$, $M_\kappa = 26.16\,\mathrm{GeV}$, $\lambda_3 = 0.01$, $\lambda_4 = 2.5 \times 10^{-4}$ (solid), and $\lambda_3 = 0.006$, $\lambda_4 = 2 \times 10^{-4}$ (dotted), solved exactly in 200 oscillator states. Dashed: the leading anharmonic terms, $1 + 16\lambda_3^2/k^2$ for $n = 2$ (Eq. (11.40)) and $1 + 108\lambda_3^2/k^2$ for $n = 3$. Dash-dotted: $q_{\mathrm{BZ}} = 2.25\,\mathrm{keV}$. Gray band: $k \geq 0.05$, where both leading terms are within 10 % of the exact weights; below it the three-phonon weight is dominated by a term of order $\lambda_3^4 k^2$.

The model fixes the neighbors, so it overstates the effect below q_{BZ}, where translation invariance makes every multiphonon term of a mass-proportional coupling of order q^4; at large k the weights return to the harmonic ones.

The single-site model breaks translation invariance: a nucleus pulled against fixed neighbors feels a force even when the whole crystal moves. In the real crystal the soft acoustic limit Eq. (8.31) is exact, anharmonic terms included, and for a coupling proportional to mass every response at fixed $\omega \neq 0$ vanishes as q^4 for $q \ll q_{\mathrm{BZ}}$. The term of order q^2 in Eq. (11.39) survives because, above q_{BZ}, the self part no longer cancels against the distinct part. Below q_{BZ} the anharmonic two-phonon response has the same q^4 as the harmonic one, and Campbell-Deem and collaborators found it to dominate the harmonic term in GaAs near the on-shell longitudinal-acoustic resonance, where that resonance is kinematically accessible [1, Fig. 2]. The window in which the anharmonic q^2 term dominates is therefore $q_{\mathrm{BZ}} \lesssim q \lesssim q_\times(n)$, open for three and more phonons in silicon and germanium.

On the truncation label of Sect. 8.5, Eq. (11.39) is the first correction along the anharmonic axis, $n_A = 2$, with $\mathcal{K}$ = multiphonon, $n_B = 0$ and $n_T = 0$; at finite temperature the factor $1 + n_B(\omega)$ multiplies it and the thermal factors of Eq. (5.70) enter γ^{anh}. The self-energy needs the cubic force constants of the whole crystal on a grid of $\boldsymbol{Q}$, which density-functional perturbation theory or finite displacements in a supercell provide (Sect. 5.4 and Appendix E.8). Appendix E.19 specifies the calculation of δ_{anh} for silicon, germanium and GaAs, checked

against the three linear constraints above, with the kinetic energy of the struck nucleus taken from the path-integral simulation of Appendix E.17. No such calculation exists yet; the single-site model is the only estimate.

11.5.3 The Markov–Krein band

Perturbation theory in $\Phi^{(3)}$ fails for soft modes, light atoms and quantum crystals, but the moments Eq. (11.37) do not need it. They are static averages, computable for any anharmonicity, and Appendix C.10 shows how much five moments say about a distribution. Every distribution with the moments $\mu_0, \dots, \mu_4$ has its cumulative $F(\xi) = \int_{-\infty}^{\xi} \mathrm{d}P$ inside the Markov–Krein band Eq. (C.93). At each ξ the band is spanned by the canonical representation through ξ: three point masses w_i at y_i, one of them at ξ, that reproduce the five moments, $\sum_i w_i y_i^k = \mu_k$ for $k = 0, \dots, 4$. These are five equations for the two free positions and the three weights. The weight below ξ and the weight up to and including ξ bound $F(\xi)$ from below and above. The width of the band is the mass at ξ, the largest that any distribution with these moments can put there: the Christoffel function $\varrho_{\mathrm{MK}}(\xi)$ of Eq. (C.91). For a Gaussian of unit variance at its mean, $\xi = 0$, the solution is $y = 0, \pm\sqrt{3}$ with $w = \frac{2}{3}, \frac{1}{6}, \frac{1}{6}$: indeed $\sum w = 1$, $\sum w y^2 = 2 \cdot \frac{1}{6} \cdot 3 = 1$ and $\sum w y^4 = 2 \cdot \frac{1}{6} \cdot 9 = 3$, the odd moments vanish by symmetry, and $F(0)$ lies between $\frac{1}{6}$ and $\frac{5}{6}$. At general ξ the Gaussian width is $2/(3 + \xi^4)$ in units of the standard deviation (Eq. (C.94)).

Fig. 11.18 draws the band around the harmonic response of silicon at $q = 100\,\mathrm{keV}$, the momentum of Worked example 11.5, and at 50 keV. The moments are those of the cumulants $c_k = E_R \langle \omega^{k-1} \rangle$ of Eq. (D.193), which the numerical spectrum reproduces to 10^{-6} at 100 keV. The response lies inside the band everywhere, as it must. Near the peak the band is wide: at $\omega = E_R$ it spans 0.19 to 0.86 at 100 keV, a width of 0.67, close to the Gaussian 2/3, and at 50 keV, where the zero-phonon line puts a step of 0.21 at $\omega = 0$ and the phonon bands show, it spans 0.21 to 0.89. In the tails it narrows as $(\xi - \mu_1)^{-4}$, with the coefficient $\bar{\mu}_4 - \bar{\mu}_2^2 - \bar{\mu}_3^2/\bar{\mu}_2$ of Eq. (C.95).

At $T = 0$ the response vanishes for $\omega < 0$, a constraint that the band of Eq. (C.93) does not use. It remains valid, and where every node of the canonical representation lies at $\omega \geq 0$ that representation is itself an admissible response, which makes the band sharp on the half line too. This holds at $\omega = E_R$ for $q = 100\,\mathrm{keV}$, where the nodes sit at 60, 191 and 371 meV. It fails in two windows, 0.65 to $0.85\,E_R$ and 1.61 to $1.82\,E_R$, where one outer node of the three-point solution moves out to infinity and comes back from the other side, below $\omega = 0$; at 50 keV the windows are 0.46 to $1.20\,E_R$ and 2.6 to $3.4\,E_R$. There the half-line point-mass bound Eq. (C.96) is smaller than ϱ_{MK}, by up to 6.8 % at 100 keV and 24.5 % at 50 keV, so the sharp half-line band is narrower at least by that much. We use the band Eq. (C.93) throughout, which is conservative and has explicit extremal distributions.

Worked example 11.6: Markov–Krein band of silicon at the recoil energy

Take silicon at $q = 100\,\mathrm{keV}$ and $T = 0$, with $E_R = 191.1\,\mathrm{meV}$ and $\langle \omega \rangle = 40.9\,\mathrm{meV}$ as in Worked example 11.5. The cumulants $c_k = E_R \langle \omega^{k-1} \rangle$ give the standard deviation $\sigma_\omega = 88.4\,\mathrm{meV}$, the skewness $c_3/\sigma_\omega^3 = 0.553$ and the standardized fourth moment $3 + c_4/c_2^2 = 3 + (\langle \omega^3 \rangle / \langle \omega \rangle^3)/x_R = 3.333$, against 0 and 3 for a Gaussian. For the Gaussian the canonical representation through the mean is the one worked out above, at 0 and $\pm\sqrt{3}$ standard deviations. For silicon the same construction, Eq. (C.89) with these moments, puts the nodes at 59.8, 191.1 and 371.3 meV with weights 0.191, 0.670 and 0.139. The skewness pushes the outer nodes upward and moves weight to the lower one, but leaves

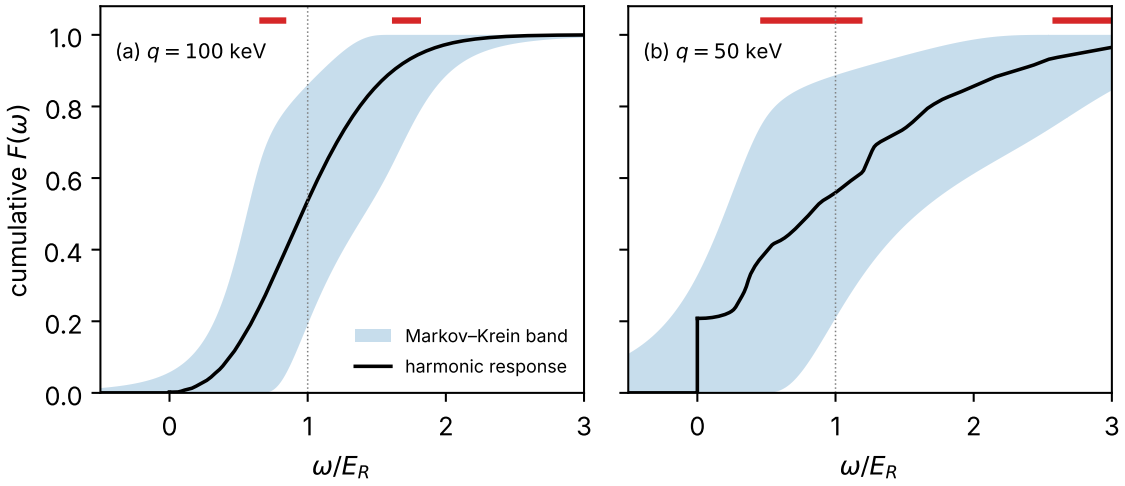


Figure 11.18: Cumulative nuclear response $F(\omega)$ of silicon at $T = 0$ (black), the fraction of the weight of the self response at energies up to ω, against ω/E_R, with the Markov–Krein band of its first five moments (blue). (a) $q = 100\,\mathrm{keV}$, $E_R = 191.1\,\mathrm{meV}$, $x_R = 4.67$; (b) $q = 50\,\mathrm{keV}$, $E_R = 47.8\,\mathrm{meV}$, $x_R = 1.17$, where the zero-phonon weight $\mathrm{e}^{-2W_\kappa} = 0.21$ is a step at $\omega = 0$. Response: harmonic, incoherent, all phonon orders, from the PBE density of states of Sect. 5.2; moments from $c_k = E_R\langle\omega^{k-1}\rangle$. Red bars: values of ω at which a node of the canonical representation lies below $\omega = 0$, so that the band drawn is wider than the sharp band of distributions confined to $\omega \geq 0$; elsewhere it is sharp for them too. Dotted: $\omega = E_R$.

the central weight at $\varrho_{\mathrm{MK}}(E_R) = 0.670$, almost the Gaussian 2/3. So

$$0.191 \leq F(E_R) \leq 0.861,$$

while the harmonic response has $F(E_R) = 0.537$. A detector with its threshold at E_R sees between 14 and 81 % of the events of this momentum transfer, whatever the dynamics, if only the five moments are right.

Energy bins narrow the band little. Between $E_R - \sigma_\omega$ and $E_R + \sigma_\omega$ the band allows any weight from 0.07 to 1.00 (the harmonic value is 0.68); between $E_R \pm 2\sigma_\omega$ it allows 0.78 to 1.00 (0.96). In the tail it is worse in relative terms: above $E_R + 3\sigma_\omega = 456\,\mathrm{meV}$ the harmonic response has the weight 5.6×10^{-3}, and the band allows anything from 0 to 3.4×10^{-2}, six times more.

Compare these widths with what anharmonicity does to the moments of silicon. The kinetic energy changes by at most ±0.2 %, which moves σ_ω by at most 0.09 meV and the cumulative distribution by at most 2.5×10^{-4}, nearly three thousand times less than the width of the band at the peak. For silicon the perturbative correction of the previous subsection is the tool; the band is for targets where no correction can be computed.

The band is absolute and not relative. Near the peak it is wide; in the tail its width falls as a power, while every true threshold tail falls faster, as a Gaussian in the impulse regime or exponentially in the Poisson tail of Sect. 11.3. A rate above a threshold far in the tail is smaller than the band width, and five moments leave it undetermined. Limit Check 11.13 makes this quantitative. In silicon at 100 keV the band fixes the weight above a threshold to 10 % only when the threshold lies more than $1.84\,\sigma_\omega$ below the mean, at $\omega \leq 28\,\mathrm{meV}$, where

nearly all of the response lies above threshold anyway. At three standard deviations above the mean the band is six times the weight, and at five standard deviations 63 times.

Limit Check 11.13. The Markov–Krein band

Every distribution with the moments $\mu_0, \dots, \mu_4$ has $\sum_{y_i<\xi} w_i \le F(\xi) \le \sum_{y_i\le\xi} w_i$, of width $\varrho_{\mathrm{MK}}(\xi)$, with (y_i, w_i) the canonical representation through ξ; far from the mean $\varrho_{\mathrm{MK}} \simeq (\bar\mu_4 - \bar\mu_2^2 - \bar\mu_3^2/\bar\mu_2)/(\xi - \mu_1)^4$ [27, 28].

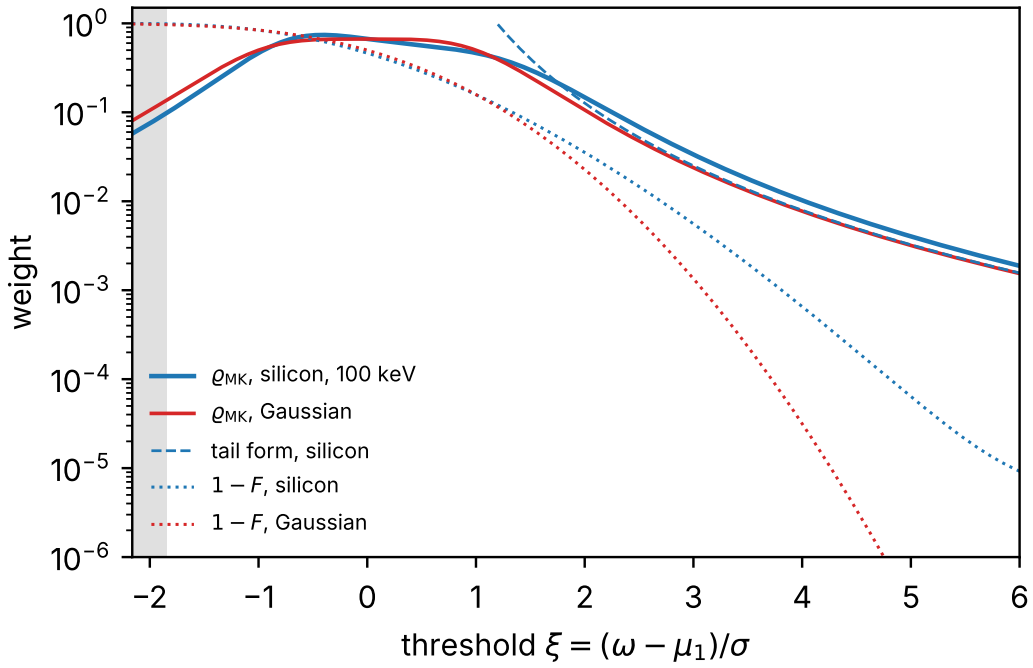


Figure 11.19: Width ϱ_{MK} of the band (solid) and weight $1 - F$ above a threshold (dotted), against the threshold ξ in units of the standard deviation from the mean, for the harmonic silicon response at $q = 100\,\mathrm{keV}$ (blue; skewness 0.553, standardized fourth moment 3.333) and for a Gaussian (red), the two model distributions whose five moments define their bands. Dashed: the tail form Eq. (C.95) for silicon, the leading term far from the mean, with the numerator 2.03 (2 for the Gaussian); the exact width exceeds it by 34 % at $\xi = 3$ and 20 % at $\xi = 6$, a correction of relative order $1/\xi$ from the skewness. The silicon curves start at $\omega = 0$. Gray band: thresholds at which the band fixes the silicon weight above threshold to 10 %, $\varrho_{\mathrm{MK}} \le 0.1\,(1 - F)$, that is $\xi \le -1.84$.

Above about two standard deviations below the mean the absolute width exceeds a tenth of the weight above threshold, and in the tail it exceeds the weight itself: the band bounds the bulk of a response, never the ratio of two tails.

On the truncation label the band is exact in n_A and in n_T when the moments are, and it makes no assumption about $\mathcal{K}$. Its use is twofold. For a strongly anharmonic target at a detector temperature it is the only rigorous statement about the response, since the moments come from a static simulation and every real-time method has a problem at millikelvin (below). And for any approximate dynamics it is an acceptance test: a computed response outside the band of its exact moments is wrong. Appendix E.19 specifies both, the moments by path-integral molecular dynamics with a machine-learned potential and the band per $\boldsymbol{q}$, for $KTaO_3$, $SrTiO_3$, LiH and solid helium.

11.5.4 Solid helium

Helium solidifies only under a pressure of a few MPa even at the lowest temperatures [29], because the zero-point energy of so light an atom in so shallow a well, 10.8 K deep for a pair [30], competes with the binding itself. Such a solid is a *quantum crystal*: the zero-point amplitude of an atom is a sizable fraction of the distance to its neighbors, about a quarter near melting, and the harmonic expansion fails at its lowest order. The Lindemann ratio $r_\mathrm{L} = \langle u^2 \rangle^{1/2} / d_\mathrm{nn}$, the root-mean-square displacement over the nearest-neighbor distance d_nn, measures it. At the static lattice positions an atom sits far out on the attractive tail of its neighbors' potentials, where the curvature is negative, and the static force constants give imaginary frequencies. The self-consistent harmonic approximation of Eq. (5.75) was devised for this case: averaged over the Gaussian spread of the atom, the curvature includes the steep repulsion that the atom samples on its excursions, and it is positive. Compression reduces r_L. Diffusion Monte Carlo calculations with a pair potential fitted to dispersion-corrected density-functional theory find $r_\mathrm{L} = 0.12$ at 2 GPa and 0.10 from 14 GPa up to 150 GPa [31, Table I]. At such pressures the static lattice is stable, and the quasi-harmonic approximation of Sect. 5.7 becomes a reasonable starting point.

Ashour and Griffin proposed solid helium under pressure as a target for light dark matter that couples to nucleons [29]. At low pressure its phonon energies are below 3 meV and its sound speed is of order 10^{-6}, too low for single-phonon reach. Between 1 and 40 GPa the sound speed rises more than tenfold and the phonon energies nearly a hundredfold, and at 40 GPa the single-phonon reach of hcp helium rivals that of diamond. Their rate [29, Eq. (2)] is Eq. (8.16) with one row and the coupling written through a dark-matter–nucleon cross-section. Their response [29, Eqs. (4) and (5)] is Eq. (11.1) with $c_\kappa = A_\kappa F_N(q)$ and the Debye–Waller exponent of Sect. 5.6, all from the phonons of the static Hessian at the compressed volume, computed with the PBE functional and a dispersion correction. They leave out the low pressures, where the density functional fails and the Born–Oppenheimer phonons miss the zero-point motion.

When is the quasi-harmonic response good enough, and what does the self-consistent one add? Limit Check 11.14 answers in a cell model, in which one atom moves in the field of its twelve neighbors held fixed on an fcc lattice and the Schrödinger equation is solved exactly. The pair potential is a Morse form with the well of the helium potential, so that only one shape parameter is chosen (the curvature at the minimum of a 12-6 potential with the same well). The model gives $r_\mathrm{L} = 0.23$ at $21\,\mathrm{cm^3\,mol^{-1}}$, near melting, where its first excitation lies at 2.2 meV, and 0.117 and 0.094 at the volumes of the Monte Carlo values 0.12 and 0.10 above. The static curvature turns negative for molar volumes above $17\,\mathrm{cm^3\,mol^{-1}}$, $r_\mathrm{L} \geq 0.20$. Below that the quasi-harmonic weight of a single phonon, $q^2/2M\omega$ per direction, overshoots the exact one, because the static curvature is too soft; the error falls below 10 % only for $r_\mathrm{L} \leq 0.115$, a molar volume of $6.5\,\mathrm{cm^3\,mol^{-1}}$, about the 2 GPa of the Monte Carlo calculation. The self-consistent harmonic weight stays within 10 % up to $r_\mathrm{L} = 0.227$ and is 10 % low at $21\,\mathrm{cm^3\,mol^{-1}}$. The kinetic energy that sets the impulse width behaves alike: at $r_\mathrm{L} = 0.117$ the quasi-harmonic value is 9 % low and the self-consistent one within 0.1 %.

Limit Check 11.14. Solid helium: quasi-harmonic and self-consistent phonons

The single-phonon rate of compressed helium is Eq. (11.1) with $c_\kappa = A_\kappa F_N$, phonons and Debye–Waller exponent from the static Hessian at the compressed volume [29, Eqs. (2), (4), (5)]; the self-consistent correction replaces the Hessian by $\Phi^{(2)}_\mathrm{aux}$ of Eq. (5.75).

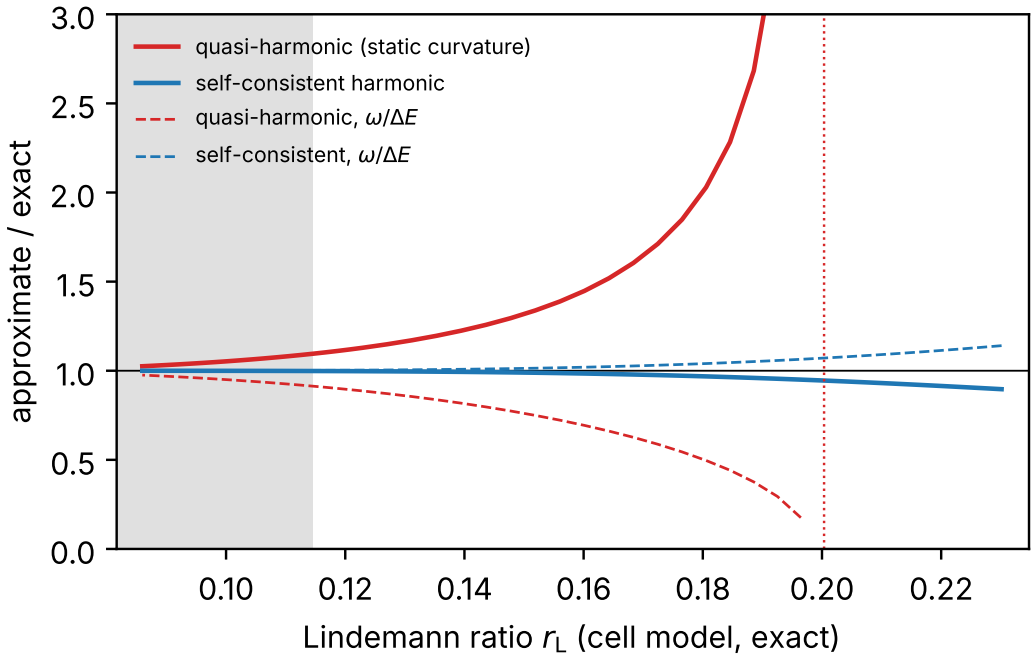


Figure 11.20: Cell model of solid ^{4}He: one atom in the spherically averaged field of 12 fixed fcc neighbors, Morse pair potential with the well depth 10.8 K and position 2.967 Å of the Aziz potential and a curvature equal to that of a 12-6 potential with the same well. Solid: one-phonon weight per unit q^2, $1/2M\omega$, from the static curvature (red, quasi-harmonic) and from the self-consistent curvature (blue), divided by the exact dipole weight of the lowest $l = 1$ level, against the exact Lindemann ratio r_L, for molar volumes from 3 to 21 $\mathrm{cm^3\,mol^{-1}}$. Dashed: the corresponding phonon energies over the exact excitation energy ΔE. Dotted: $r_L = 0.20$, above which the static curvature is negative. Gray band: $r_L \leq 0.115$, where the quasi-harmonic weight is within 10 %; the self-consistent weight is within 10 % up to 0.227 and is 0.896 of the exact weight at $r_L = 0.230$, the largest molar volume.

Above a Lindemann ratio of about 0.12 the static Hessian overstates the one-phonon weight and the Debye–Waller exponent, and above 0.20 it has no real frequencies; the self-consistent phonons hold to near melting, where correlations at short range remain.

The model fixes the neighbors and uses a Morse wall, so its numbers are a guide, not a prediction for helium; it lands within 0.003 and 0.006 of the two Monte Carlo ratios. The quasi-harmonic response of Ashour and Griffin is safe in most of their pressure range and needs the self-consistent force constants at its low end. And at the lowest pressures, where a helium detector would be easiest to build, neither is exact. The self-consistent weight is 10 % low near melting even in the model, and in the real crystal the Gaussian lets an atom overlap its neighbors, which a real atom avoids by correlated motion. There the response needs the exact moments and the band of Limit Check 11.13. On the truncation label the self-consistent response sums the even anharmonic orders into $\Phi^{(2)}_{\mathrm{aux}}$ at all n_A. The cubic terms enter the physical frequencies through a bubble (Appendix E.8); in the cubic cell of the model the centroid sits on the site and that bubble vanishes. Appendix E.19 specifies the self-consistent force constants of compressed helium from 1 to 40 GPa and path-integral moments near melting.

The same reasoning sorts the other targets in which anharmonicity is not small. A soft mode, such as the one that drives the structural transition of $SrTiO_3$, has a harmonic frequency that is small or imaginary, and its self-consistent frequency depends on temperature at order one. A light atom, such as hydrogen in LiH, has a zero-point amplitude larger by $(M_{\mathrm{Si}}/M_{\mathrm{H}})^{1/4} \approx 2.3$ than a silicon atom in the same well, since $\ell_{\mathrm{zp}} \propto M^{-1/4}$ (Sect. 5.7), and η_{anh} grows with it.

Open problem: the response of strongly anharmonic targets at detector temperatures. For soft-mode and light-atom crystals, and for solid helium near melting, no expansion parameter is small, and the nuclear response at 10 mK has not been computed. The exact moments are within reach: they are static averages, and a path-integral simulation gives them at any temperature and anharmonicity. The dynamics is not. Ring-polymer and centroid molecular dynamics return Kubo-transformed correlations. Their conversion to S multiplies the error by ω/T, so they hold only for $T \gtrsim \omega/10$; the time-dependent self-consistent harmonic method excites at most two phonons at a time and cannot follow a multiphonon response at $q \gg q_{\mathrm{BZ}}$ (Appendix E.19). At the operating temperature of a cryogenic detector the response of these targets is therefore bounded by the band of their exact moments and by nothing sharper. What would close the gap is a real-time method for quantum nuclei that is exact at low temperature, or a measurement: neutron Compton scattering measures the momentum distribution, and with it the impulse width, of light nuclei directly [21].

The nuclei carry one more property that the lattice sector has so far ignored, their spin and their mass number, which Sect. 11.6 takes up.

11.6 Nuclear-spin and isotope couplings

One silicon nucleus in 21 is ^{29}Si and carries a nuclear spin, and in a crystal without a magnetic field the spins point in random directions. The three stable isotopes also differ in their number of neutrons per unit mass, by 6.7 % from ^{28}Si to ^{30}Si. The couplings of Sects. 11.1 and 11.2 are the same on every site of a sublattice, and they conserve crystal momentum. A coupling to the nuclear spin, or to the neutron number of a nucleus at fixed mass, changes at random from site to site. The lattice is still perfect, but to the dark matter the crystal is disordered, crystal momentum is no longer conserved, and a probe that brings almost no momentum, such as an absorbed boson, can create a phonon of any wavevector in the zone. The response then follows the phonon density of states, a broad band, instead of the single line of the zone-center optical phonon. Unpolarized nuclear spins respond in this way, and so do the isotopes to a vector boson coupled to baryon minus lepton number, which in silicon is absorbed almost entirely through them.

11.6.1 Broadband response of random couplings

Label the nuclei of a crystal by I, with masses M_I and displacements $\boldsymbol{u}_I$ from their equilibrium positions. In a harmonic crystal the displacements are sums over normal modes ν of the actual configuration, with polarization vectors $\boldsymbol{e}_{I\nu}$ normalized over all sites and the mode frequencies ω_ν, $\boldsymbol{u}_I = \sum_\nu \boldsymbol{e}_{I\nu}(b_\nu + b_\nu^\dagger)/\sqrt{2M_I\omega_\nu}$. In a perfect crystal ν is the pair $\boldsymbol{Q}\nu$ of Eq. (5.32); when the masses vary from site to site, the modes carry no wavevector and ν is just a label. Consider a target operator whose one-phonon matrix element is linear in the displacements,

$$\langle 1_\nu |\, O \,| 0 \rangle = \sum_I \frac{\boldsymbol{v}_I \cdot \boldsymbol{e}_{I\nu}^*}{\sqrt{2M_I\omega_\nu}},$$

with a vertex $\boldsymbol{v}_I$ for each site. The density row at small $\boldsymbol{q}$ has this form, with $\boldsymbol{v}_I = \mathrm{i}c_I\boldsymbol{q}$ (Sect. 11.1). So has every row built from the nuclear velocities $\boldsymbol{P}_I/M_I$. In a harmonic crystal

the momentum is M_I times the time derivative of the displacement, and the derivative of a matrix element brings down $\mathrm{i}\omega$, with ω the energy of the final state: $\langle f|\boldsymbol{P}_I|0\rangle = \mathrm{i}M_I\omega\langle f|\boldsymbol{u}_I|0\rangle$.

Part of every vertex does nothing. Write $\boldsymbol{v}_I = M_I\bar{\boldsymbol{v}} + \delta\boldsymbol{v}_I$ with $\bar{\boldsymbol{v}} = \sum_I \boldsymbol{v}_I / \sum_I M_I$. The uniform translation of the crystal is a zero mode with $\boldsymbol{e}_{I0} \propto \sqrt{M_I}$, every vibration is orthogonal to it, $\sum_I \sqrt{M_I}\,\boldsymbol{e}_{I\nu} = 0$, and the mass-proportional part drops out of every one-phonon element. Mitridate and collaborators noted this coupling-to-mass cancellation for the zone-center optical phonons of a perfect crystal [32, Eq. (44)]. It holds for any configuration of masses, not only at the zone center of a perfect crystal: a coupling proportional to the mass of each nucleus couples to the total momentum, which the forces of the lattice conserve. Only the deviations $\delta\boldsymbol{v}_I$ create phonons.

Suppose now that the $\delta\boldsymbol{v}_I$ are random: zero on average, independent from site to site and isotropic, $\langle \delta v_I^\alpha\, \delta v_J^{\beta *}\rangle = \delta_{IJ}\delta^{\alpha\beta}\langle|\delta\boldsymbol{v}_I|^2\rangle/3$. Averaging the squared matrix element over the disorder removes every term with $I \neq J$, together with the phases that enforced crystal momentum. Each site then contributes through its own vibrations, measured by the local density of states

$$g_I(\omega) = \sum_\nu |\boldsymbol{e}_{I\nu}|^2\, \delta(\omega - \omega_\nu),$$

normalized to three by the completeness of the polarization vectors; in a crystal of one isotope it is the atom-projected density of states $g_{\mathrm{ph},\kappa}$ of Sect. 5.3. Summing over the modes with Eq. (4.13) gives

Broadband one-phonon response of a random coupling

$$\langle S_{OO}(\boldsymbol{q}\to 0,\omega)\rangle = \frac{2\pi}{\mathscr{V}} \sum_I \langle|\delta\boldsymbol{v}_I|^2\rangle\, \frac{g_I(\omega)}{6M_I\,\omega}\,\big[n_{\mathrm{B}}(\omega)+1\big], \tag{11.41}$$

for $\omega > 0$, with the Bose factor of Sect. 5.3 and the average over the disorder in the brackets. It is exact for a harmonic crystal when the random vertex is independent of the masses and force constants, with g_I the local density of states at site I of the actual crystal.

Unlike the single zone-center line that a coherent coupling excites (Sect. 11.2), the response is a band: every phonon of energy ω contributes, whatever its wavevector, since the random vertex supplies the missing crystal momentum. And it does not depend on how the vertices of neighboring sites are related, only on the variance at each site. On the truncation label of Sect. 8.5 it is the point $\mathcal{K}$ = one-phonon, $n_A = 0$, $n_B = 0$, with $n_T = 0$ at 10 mK; the coherent sum of Eq. (11.1) is replaced by its incoherent counterpart, and this time without approximation.

11.6.2 Unpolarized nuclear spins

A coupling linear in the nucleon spin acts on a nucleus of isotope i through its spin $\boldsymbol{J}_I^{\mathrm{nuc}}$, with the coefficient of Eq. (8.8). It enters the target through the spin rows of Eq. (8.3), the spin density $\boldsymbol{s}_\kappa(\boldsymbol{q}) = \sum_I \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}\boldsymbol{J}_I^{\mathrm{nuc}}$ and the spin current $\mathcal{J}_\kappa^{\alpha\beta}$. In zero field the spin states of a nucleus are split only by its quadrupole and dipolar interactions, by at most about 0.1 µeV, against $T = 0.86$ µeV at 10 mK. A spin $\frac{1}{2}$ such as ^{29}Si has no quadrupole splitting at all, and its states are equally occupied. A nucleus with a quadrupole moment keeps $\langle\boldsymbol{J}^{\mathrm{nuc}}\rangle = 0$ but can be slightly aligned at millikelvin temperatures; we do not pursue this correction. The spins also barely act on the lattice: the forces between the atoms are electronic, and the hyperfine coupling that ties a nuclear spin to anything else is far weaker. The state of the target is then a product of an equal mixture of spin states for each nucleus and the phonon

state, and

$$\langle J_I^a J_{I'}^b \rangle = \delta_{II'}\, \delta^{ab}\, \frac{J_{\mathrm{nuc},i}(J_{\mathrm{nuc},i}+1)}{3},$$

with $J_{\mathrm{nuc},i}$ the spin of isotope i. Two different nuclei have uncorrelated spins, so in the correlation form of Eq. (4.13) every pair of distinct nuclei drops out of the spin-density response, at every phonon order and every temperature. What remains is the self part of Sect. 11.3, one nucleus at both times, for the nuclei that carry a spin:

Spin-density response of unpolarized nuclei

$$\begin{aligned} S_{s^a s^b}(\boldsymbol{q},\omega) &= \delta^{ab} \sum_i \frac{J_{\mathrm{nuc},i}(J_{\mathrm{nuc},i}+1)}{3}\, S_i^{\mathrm{self}}(\boldsymbol{q},\omega), \\ S_i^{\mathrm{self}}(\boldsymbol{q},\omega) &= \frac{1}{\mathscr{V}} \int_{-\infty}^{\infty} \mathrm{d}t\; \mathrm{e}^{\mathrm{i}\omega t} \sum_{I\in i} \big\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I(t)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I(0)} \big\rangle, \end{aligned} \tag{11.42}$$

the sum running over the isotopes with a spin. The self response is Eq. (11.26) for these nuclei with $c_\kappa^2 \to 1$; its one-phonon term at $T = 0$ on a cubic site is $(2\pi/\mathscr{V}) N_i\, \mathrm{e}^{-2W_i}\, (q^2/2M_i\omega)\, g_i(\omega)/3$, with N_i the number of nuclei of isotope i.

The incoherent approximation of Sect. 11.3, there valid only far above the zone, is exact here at every momentum transfer. The spin-dependent operator $\mathcal{O}_4 = \boldsymbol{S}_\chi \cdot \boldsymbol{S}_N$ has the dark-matter tensor $\mathfrak{D}_{s^a s^b} = |c_4^{(i)}|^2 \delta^{ab} |\boldsymbol{S}_\chi|(|\boldsymbol{S}_\chi|+1)/3$ of Eq. (8.7), and the contraction in the rate Eq. (8.16) is $\sum_i |c_4^{(i)}|^2\, |\boldsymbol{S}_\chi|(|\boldsymbol{S}_\chi|+1)\, J_{\mathrm{nuc},i}(J_{\mathrm{nuc},i}+1)\, S_i^{\mathrm{self}}/3$. In silicon only ^{29}Si contributes, with $J_{\mathrm{nuc}} = \frac{1}{2}$, an abundance of 4.685 %, and $c_4^{(i)} = 0.312\, c_4^{(n)}$ for a coupling to neutron spins alone (Sect. 8.1.3).

The loss of crystal momentum changes the kinematics. A coherent one-phonon response at $q \ll q_{\mathrm{BZ}}$ offers only acoustic phonons of energy $c_\nu q$, and the reach stops at the acoustic edge Eq. (11.5). The self response at the same q spans the whole band, and any energy up to the kinetic energy of the dark matter can be deposited. A spin-dependent coupling of unpolarized nuclei therefore reaches down to the kinematic limit $2\omega_{\mathrm{th}}/v_{\mathrm{max}}^2$ of a free target, Eq. (1.25), 0.28 keV at a 1 meV threshold, 21 times below the acoustic edge of silicon. The weight, however, is small: the self response grows as q^2, and at the smallest momentum transfers it is tiny.

Absorption needs the spin current. An axion-like particle couples to nucleons as to electrons in Sect. 1.4, $\mathcal{L} \supset g_{aNN}\, \phi\, \bar\psi_N \mathrm{i}\gamma^5 \psi_N$ for $N = p, n$, and the reduction that gave Eq. (8.10) gives for each nucleon $(g_{aNN}/m_N)[(\nabla\phi)\cdot\boldsymbol{S}_N + (\partial_t\phi)\, \boldsymbol{v}_N \cdot \boldsymbol{S}_N]$, with $\boldsymbol{v}_N$ the nucleon velocity. Inside a nucleus the velocity of each nucleon is that of the nucleus, $\boldsymbol{P}_I/M_i$, plus an internal part that averages to zero, and the Wigner–Eckart theorem replaces the weighted sum of nucleon spins by $g_{ai}\boldsymbol{J}_I^{\mathrm{nuc}}$ with

$$g_{ai} = \frac{g_{app}\langle S_p\rangle + g_{ann}\langle S_n\rangle}{J_{\mathrm{nuc},i}},$$

as in Eq. (8.8). The interaction is then $m_N^{-1} \sum_i g_{ai}[(\nabla\phi)\cdot\boldsymbol{s}_i + (\partial_t\phi)\, \mathcal{J}_i^{\alpha\alpha}]$ at $\boldsymbol{q} = \boldsymbol{p}_\phi$. The gradient term rotates a nuclear spin and leaves the lattice as it was. Its final states lie at the spin splittings, below a μeV, and never at $\omega = m_\phi$; it would excite a phonon only through the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{p}_\phi\cdot\boldsymbol{u}_I}$, a correction of order $p_\phi^2\langle u_x^2\rangle \approx 2\times10^{-18}$ at 50 meV. The time-derivative term moves the nucleus. At $\boldsymbol{p}_\phi \to 0$ it is exactly linear in the nuclear momenta, $\mathcal{J}_i^{\alpha\alpha} = \sum_{I\in i} \boldsymbol{P}_I \cdot \boldsymbol{J}_I^{\mathrm{nuc}}/M_i$, so in a harmonic crystal it creates exactly one phonon, with the vertex $\delta\boldsymbol{v}_I = \mathrm{i}\omega\, (g_{ai} m_\phi/m_N)\, \boldsymbol{J}_I^{\mathrm{nuc}}$ in Eq. (11.41), whose variance over the spin states is $\omega^2 (g_{ai} m_\phi/m_N)^2 J_{\mathrm{nuc},i}(J_{\mathrm{nuc},i}+1)$. The absorption formula Eq. (8.19) with $\omega_\phi = m_\phi$ and the

event rate per unit mass $(\rho_\chi/m_\phi\rho_T)\Gamma_{\rm abs}$ then give, at $T=0$,

$$\mathcal{R} = \frac{\pi\rho_\chi m_\phi}{6\,m_N^2 M_{\rm cell}} \sum_\kappa \sum_i x_{\kappa i}\, g_{ai}^2\, \frac{J_{{\rm nuc},i}(J_{{\rm nuc},i}+1)}{M_i}\, g_i(m_\phi), \tag{11.43}$$

with $x_{\kappa i}$ the abundance of isotope i on the sites κ of the cell and $M_{\rm cell}$ the mass of the cell. The rate is broadband: from the lowest acoustic phonons to the top of the optical band, every axion mass is absorbed at the rate set by the density of states at $\omega = m_\phi$. Per unit mass, light nuclei with a large spin weight are favored through the factor $1/M_i M_{\rm cell}$. In silicon the spin weight is $x_{29}(0.312)^2 \cdot \frac{3}{4} = 3.4\times10^{-3}$ in units of g_{ann}^2, and the proton coupling enters with a coefficient ten times smaller, through $\langle S_p\rangle = 0.016$. Eq. (11.43) is the rate of Bloch, Knapen, Madden and Marocco [33, Eq. (31)], who couple the axion to nucleons through $g_N \partial_\mu\phi\, \bar\psi_N\gamma^\mu\gamma^5\psi_N$. On shell this is our coupling with $g_{aNN} = 2m_N g_N$, and their $g_p\lambda_j$, ξ_j and $D_j(\omega)$ are our $g_{ai}/2m_N$, the weight $\lambda_j^2 J(J+1)/M$ averaged over isotopes, and $g_i/3$.

The spin sits on ^{29}Si, however, which is 3.2 % heavier than the mean silicon atom, and its local density of states g_{29} is not the $g_{{\rm ph},\kappa}$ of the crystal; the literature uses the host density of states for every isotope.

11.6.3 An isotopic mass defect

How does a single heavier atom vibrate in a crystal of lighter ones? Its neighbors push it with the same springs, and only its inertia differs. The displacement response of a harmonic crystal, the retarded Green function $\mathsf{G}(\omega) = (\Phi - \omega^2\mathsf{M} - {\rm i}0)^{-1}$ with Φ the matrix of force constants of Sect. 5.2 and M the diagonal matrix of masses, answers this. In a crystal of mean mass $\bar M$ the element on one site, for one Cartesian direction, is

$$\mathsf{G}_{\rm h}(\omega) = \frac{1}{\bar M}\int_0^\infty {\rm d}\omega'\, \frac{g_{{\rm ph},\kappa}(\omega')/3}{\omega'^2 - \omega^2 - {\rm i}0}, \qquad {\rm Im}\,\mathsf{G}_{\rm h}(\omega) = \frac{\pi\, g_{{\rm ph},\kappa}(\omega)}{6\bar M\omega},$$

the spectral sum over the modes weighted by the squared polarization on the site; on a cubic site it is a multiple of the identity. Now give the atom on site I the mass $M_i = \bar M(1-\delta_i^M)$, with the mass defect δ_i^M positive for a lighter atom. The change in $\Phi - \omega^2\mathsf{M}$ is the single entry $\delta_i^M \bar M\omega^2$ on that site, and since it acts on one site only, the Dyson equation $\mathsf{G} = \mathsf{G}_{\rm h} - \mathsf{G}_{\rm h}\,\delta_i^M\bar M\omega^2\,\mathsf{G}$ closes on the element at I. The local density of states follows from ${\rm Im}\,\mathsf{G}$ as for the host, with M_i for $\bar M$:

Local vibrations of one mass defect

$$\mathsf{G}_i(\omega) = \frac{\mathsf{G}_{\rm h}(\omega)}{1 + \delta_i^M\,\bar M\omega^2\,\mathsf{G}_{\rm h}(\omega)}, \qquad \frac{g_i(\omega)}{g_{{\rm ph},\kappa}(\omega)} = (1-\delta_i^M)\,\frac{{\rm Im}\,\mathsf{G}_i(\omega)}{{\rm Im}\,\mathsf{G}_{\rm h}(\omega)}. \tag{11.44}$$

It is exact for one defect in a harmonic host, with the force constants unchanged.

This is the classical result of Lifshitz [34], worked out for an isotope in a cubic crystal by Dawber and Elliott [35]. The real part of $\mathsf{G}_{\rm h}$ follows from the density of states by a Kramers–Kronig integral (Sect. 4.3), and nothing else is needed: one harmonic calculation of the pure crystal gives the vibrations of every isotope in it. To first order in the defect,

$$\frac{g_i(\omega)}{g_{{\rm ph},\kappa}(\omega)} = 1 + \delta_i^M\, h_{\rm iso}(\omega) + O\big((\delta_i^M)^2\big), \qquad h_{\rm iso}(\omega) = -1 - 2\bar M\omega^2\,{\rm Re}\,\mathsf{G}_{\rm h}(\omega). \tag{11.45}$$

At low frequency $h_{\rm iso}\to -1$: a heavier atom moves with the long waves like its neighbors and carries more weight in proportion to its mass. Higher in the band ${\rm Re}\,\mathsf{G}_{\rm h}$ decides. Below a sharp peak of the density of states it is positive, $h_{\rm iso} < 0$, and a heavier atom gains weight;

just above the peak it is large and negative, and the heavier atom loses weight. A heavy atom thus shifts weight down from the top of each band. A light atom with δ_i^M large enough does more: where $1 + \delta_i^M \bar{M}\omega^2 \operatorname{Re} \mathsf{G}_\mathrm{h} = 0$ above the band, G_i has a pole, and the atom vibrates in a local mode that the host cannot carry away. The sum rule $\int g_i \, \mathrm{d}\omega = 3$ holds in every case, with the local mode included.

For silicon we take the density of states of the PBE force constants of Sect. 5.2, which ends at $\omega_\mathrm{max} = 63.7\,\mathrm{meV}$, and the natural mixture of masses: 92.223 % ^{28}Si, 4.685 % ^{29}Si and 3.092 % ^{30}Si [18], with atomic masses 27.977, 28.976 and 29.974 u [36]. The mean mass is $\bar{M} = 28.0855\,\mathrm{u}$, and the defects are $\delta_i^M = +0.0039$, -0.0317 and -0.0672. A local mode requires $\delta_i^M > 0.22$, an atom 22 % lighter than its neighbors, so no isotope of silicon has one.

Limit Check 11.15 compares its local density of states with the host one that Eq. (11.43) assumes in the literature.

Limit Check 11.15. Incoherent spin absorption

Unpolarized nuclear spins absorb an axion-like particle broadband, at the rate $\mathcal{R}$ of Eq. (11.43) with the host density of states $g_{\mathrm{ph},\kappa}$ in place of g_i for every isotope [33, Eqs. (25)–(31)].

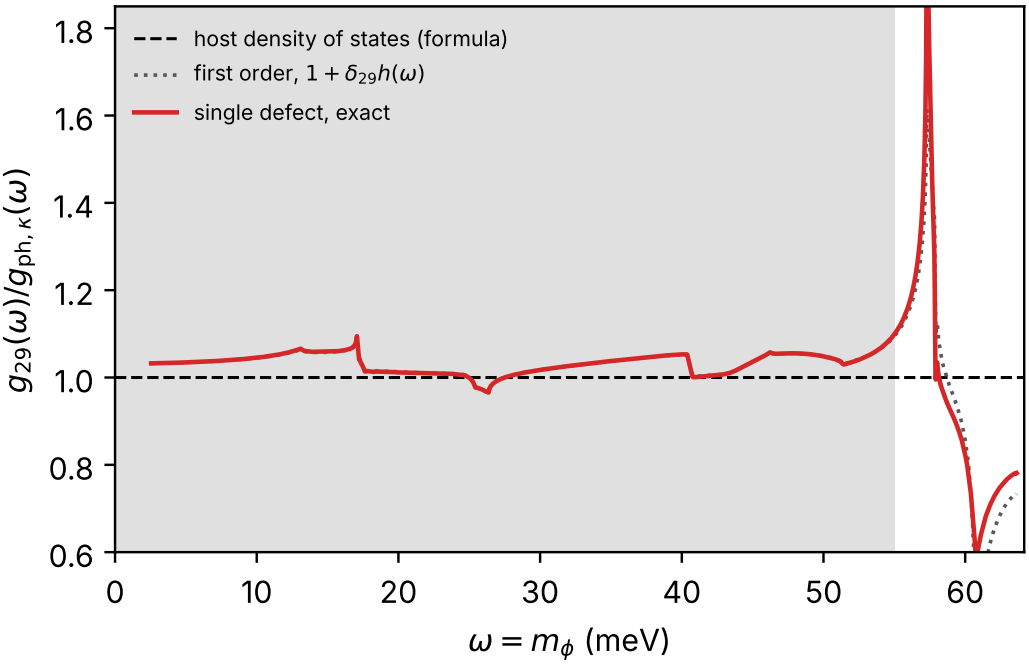


Figure 11.21: Rate of Eq. (11.43) for natural silicon, where the spin sits on ^{29}Si, with the exact local density of states of one ^{29}Si atom from Eq. (11.44) (red), divided by the same rate with the host density of states (dashed, the formula), against the axion mass. The ratio $g_{29}/g_{\mathrm{ph},\kappa}$ does not depend on the couplings. Dotted: the first order in the mass defect, $1 + \delta_{29} h_\mathrm{iso}(\omega)$ with $\delta_{29} = -0.0317$, Eq. (11.45). PBE density of states of Sect. 5.2, $T = 0$. Gray: within 10 % of the formula, $m_\phi < 55.0\,\mathrm{meV} = 0.865\,\omega_\mathrm{max}$.

Near the top of the optical band the heavier isotope moves weight from the band edge to just below it, and the host density of states errs by up to a factor of 2.

Below 55 meV the heavier spin carrier changes the rate by -3.5 to $+10$ %, mostly by 1 to 5 % upward, and the first order of Eq. (11.45) reproduces the exact ratio to 0.5 %. At the

sharp lower edge of the optical band the ratio rises to about 2, with a peak of 2.05 at 57.3 meV in the PBE spectrum, and between 60 and 63 meV it falls to 0.6, with a minimum of 0.59 near 60.7 meV. The band integral is unchanged, since both densities of states are normalized to three, so the weight is moved and not lost. This is the correlation between the vertex and the mass: the vertex sits on an atom whose mass, and therefore whose vibrations, differ from the average. Bloch and collaborators assume that the isotopes leave the phonon spectrum unchanged, which holds for the crystal as a whole when one isotope dominates [33, footnote 5]. The local spectrum at a minority isotope still differs, and in silicon the host density of states is good to 10 % over 86 % of the band.

Eq. (11.44) itself treats one defect in a perfect host. How far does it carry into a crystal that contains many?

Limit Check 11.16. Lifshitz mass-defect Green function

One atom of mass $\bar{M}(1-\delta^M)$ in a harmonic host: $\mathsf{G}_i = \mathsf{G}_\mathrm{h}/(1+\delta^M \bar{M}\omega^2 \mathsf{G}_\mathrm{h})$ on its site [34, 35].

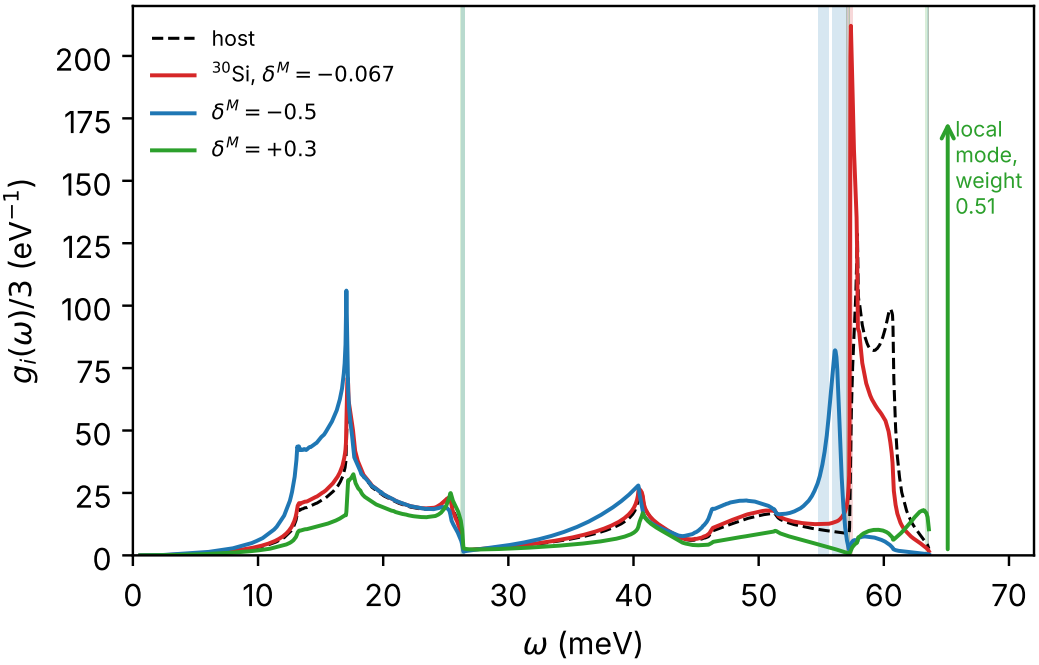


Figure 11.22: Local density of states per direction, $g_i(\omega)/3$, of one mass defect in silicon from Eq. (11.44), with the PBE density of states of Sect. 5.2 (dashed, the host). Red: ^{30}Si, $\delta^M = -0.067$. Blue: a heavy defect, $\delta^M = -0.5$. Green: a light defect, $\delta^M = +0.3$, which puts 51 % of its weight into a local mode at 65.1 meV, above the band (arrow). The integral of each curve, local mode included, is one. Shaded in the color of each curve: frequencies where the leading pair term exceeds 10 %. The pair term is the effect of a second defect anywhere in the crystal on the first, for the natural isotope mixture around ^{30}Si and for a concentration of 1 % of the strong defects; pair term to first order in the scattering of the second defect, from the host Green function.

The single-defect form fails where defects interact: within a meV of the sharp edges of the host density of states in natural silicon, and over windows of a few meV around them for defects of tens of percent in mass at concentrations of a percent.

The pair term in the caption needs no new calculation. A second defect j at site $\boldsymbol{r}$ changes the host function on site I to $\mathsf{G}_\mathrm{h}+\mathsf{G}_{I\boldsymbol{r}}t_j\mathsf{G}_{\boldsymbol{r}I}$, with $t_j=-\delta_j^M\bar{M}\omega^2/(1+\delta_j^M\bar{M}\omega^2\mathsf{G}_\mathrm{h})$ its scattering strength, and the sum over all $\boldsymbol{r}\neq I$ of $\mathsf{G}_{I\boldsymbol{r}}\mathsf{G}_{\boldsymbol{r}I}$ is $(\mathsf{G}^2)_{II}-\mathsf{G}_\mathrm{h}^2=\partial\mathsf{G}_\mathrm{h}/\partial(\bar{M}\omega^2)-\mathsf{G}_\mathrm{h}^2$, since G is the inverse of $\Phi-\bar{M}\omega^2$. For the natural mixture referenced to $\bar{M}$ the first order in the second defect cancels, $\sum_j x_j\delta_j^M=0$, and what remains is of order $g_\mathrm{md}=\sum_j x_j(\delta_j^M)^2=2.0\times10^{-4}$, the mass variance of Appendix E.12. It is the self-energy behind the isotope scattering of phonons: its imaginary part gives the width $\frac{\pi}{2}g_\mathrm{md}\,\omega^2 g_{\mathrm{ph},\kappa}(\omega)/3$, which at low frequency is the Rayleigh rate of Eq. (E.81), and at 50 meV is 2.5×10^{-4} of ω. For ^{30}Si in natural silicon the pair term stays below 2.2 % up to 55 meV and passes 10 % only within a meV of the sharp edge of the optical band at 57 meV. In silicon the isotopes can be treated one at a time.

11.6.4 Absorption of a baryon-minus-lepton vector

A vector boson can couple to baryon number minus lepton number, $B-L$, with strength g_{B-L}, the same for protons and neutrons, $+g_{B-L}$, and for electrons, $-g_{B-L}$ (Sect. 1.4). A neutral atom of isotope i, with A_i nucleons and Z electrons, carries the charge $A_i-Z=N_i$, its neutron number. Light enough, the vector can be the dark matter, and it is then absorbed like a dark photon. At rest it couples to the spatial current of its charges (Sect. 8.1.5),

$$g_{B-L}\Big[\sum_\kappa A_\kappa\boldsymbol{j}_\kappa-\boldsymbol{j}_e\Big]=g_{B-L}\sum_\kappa N_\kappa\boldsymbol{j}_\kappa+g_{B-L}\Big[\sum_\kappa Z_\kappa\boldsymbol{j}_\kappa-\boldsymbol{j}_e\Big].$$

The bracket $\sum_\kappa Z_\kappa\boldsymbol{j}_\kappa-\boldsymbol{j}_e$ is the electric current in units of e, and that part of the coupling is a dark photon with $\zeta=g_{B-L}/e$, and its rate is the dark-photon rate with in-medium mixing, $\zeta^2\sigma_1/|\varepsilon|^2$ (Sect. 10.7):

$$\Gamma_\mathrm{EM}=\Big(\frac{g_{B-L}}{e}\Big)^2\omega\,\mathrm{Im}\Big[-\frac{1}{\varepsilon(\omega)}\Big], \tag{11.46}$$

at $\omega=m_\phi$, with $\sigma_1=\omega\varepsilon_2$. In silicon this part creates no single phonon. Its one-phonon vertex is the Born charge, and the Born charges of silicon vanish (Sect. 5.5). Below the electronic gap it is absorbed only by pairs of phonons, through the dipole that a displacement induces on top of another, which is the weak lattice absorption of silicon in the far infrared.

The neutron-number current $g_{B-L}\sum_\kappa N_\kappa\boldsymbol{j}_\kappa$ couples each nucleus to its neutron number. In a crystal of one isotope the vertex is $\mathrm{i}\omega g_{B-L}N_\kappa\hat{\boldsymbol{\epsilon}}$, with $\hat{\boldsymbol{\epsilon}}$ the polarization of the vector. It is proportional to M_κ on both sites of silicon, and the coupling-to-mass cancellation removes it: a monatomic crystal of one isotope does not absorb a $B-L$ vector into one phonon at all. Natural silicon does, because the neutron number per unit mass,

$$\mathfrak{b}_i=\frac{N_i}{M_i},$$

is 0.5004, 0.5177 and 0.5338 u^{-1} for the three isotopes. The mass-proportional part of the vertex is set by the mass-weighted mean $\bar{\mathfrak{b}}=\sum_i x_iN_i/\sum_i x_iM_i=0.502\,35\,\mathrm{u}^{-1}$, with x_i the abundances, and the remainder $\delta\boldsymbol{v}_I=\mathrm{i}\omega g_{B-L}M_i(\mathfrak{b}_i-\bar{\mathfrak{b}})\hat{\boldsymbol{\epsilon}}$ has zero mean and is independent from site to site. Its direction is fixed, but on a cubic site every direction gives the same projection of the local density of states. Eq. (11.41) with the vertex in S_{OO} and Eq. (8.19) give, per absorbed quantum at $T=0$ and with Ω_at the volume per atom,

Isotope-disorder absorption of a $B-L$ vector

$$\begin{aligned}\Gamma_{\rm iso}(\omega) &= \frac{\pi g_{B-L}^2}{2\Omega_{\rm at}} \sum_i x_i M_i (\mathfrak{b}_i - \bar{\mathfrak{b}})^2 \, \frac{g_i(\omega)}{3} \\ &= \frac{\pi g_{B-L}^2 \bar{M} \sigma_{\mathfrak{b}}^2}{2\Omega_{\rm at}} \, \frac{g_{{\rm ph},\kappa}(\omega)}{3} \Big[1 + \langle \delta^M \rangle_{\mathfrak{b}} \, h_{\rm iso}(\omega) + O(g_{\rm md}) \Big],\end{aligned} \tag{11.47}$$

at $\omega = m_\phi$ inside the one-phonon band and zero above it, with $\sigma_{\mathfrak{b}}^2 = \sum_i x_i (M_i/\bar{M})(\mathfrak{b}_i - \bar{\mathfrak{b}})^2$ the mass-weighted variance of $\mathfrak{b}$ and $\langle \delta^M \rangle_{\mathfrak{b}}$ the mass defect averaged with the weights $x_i M_i (\mathfrak{b}_i - \bar{\mathfrak{b}})^2$. In natural silicon $\sigma_{\mathfrak{b}}^2/\bar{\mathfrak{b}}^2 = 1.88 \times 10^{-4}$ and $\langle \delta^M \rangle_{\mathfrak{b}} = -0.054$.

The two parts of the coupling reach different final states, one phonon or two, and they add without interference. The isotope phonons do not mix with the photon either, since the Born charge of every silicon atom vanishes whatever its mass. The vertex of an isotope sits on the atom whose mass differs. The isotope-resolved form of Eq. (11.47) takes this into account through the local density of states of each isotope, Eq. (11.44), and the terms between different sites, which vanish to first order in the mass defects, are the pair terms of Limit Check 11.16. A vector coupled to baryon number alone would see almost nothing. Its charge per mass, A_i/M_i, differs between the isotopes only through their binding energies, and the variance is 9×10^{-11} of the squared mean, two million times below that of $B-L$.

How large is the isotope channel against the electromagnetic one? The answer depends on the temperature, through the two-phonon absorption. Consider a pair process in which the field is absorbed and two phonons of energies ω_1 and ω_2 are created, $\omega = \omega_1 + \omega_2$. Its rate is proportional to $(1+n_1)(1+n_2)$ and that of the reverse process, stimulated emission, to $n_1 n_2$, with the Bose occupations $n_{1,2}$, so the net absorption carries $1 + n_1 + n_2 \geq 1$. In a difference process a phonon ω_2 is absorbed and a phonon $\omega_1 = \omega + \omega_2$ created, with the net weight $n_2(1+n_1) - n_1(1+n_2) = n_2 - n_1 \geq 0$, which vanishes at $T = 0$. Suppose the second-order dipole moments and the phonon frequencies do not depend on temperature. Then every term of the absorption coefficient falls as the crystal cools, and

$$\alpha(\omega; T \to 0) \leq \alpha(\omega; 293\,{\rm K}). \tag{11.48}$$

The frequencies do shift, by about 1 % between room temperature and zero, and the lines narrow, so the inequality holds for the absorption integrated over a window wider than the shift and the room-temperature width; where the room-temperature spectrum is flat over that window, it holds point by point. Below the top of the one-phonon band the far-infrared absorption of silicon at room temperature is dominated by difference processes [37], which freeze out completely, so the bound is far from saturated at millikelvin temperatures.

Worked example 11.7: The isotope channel against the electromagnetic one at 50 meV

Take natural silicon and $m_\phi = 50\,{\rm meV}$, inside the one-phonon band. With $\bar{\mathfrak{b}} = 0.502\,35\,{\rm u}^{-1}$ and $\sigma_{\mathfrak{b}}^2 = 1.879 \times 10^{-4}\,\bar{\mathfrak{b}}^2 = 4.742 \times 10^{-5}\,{\rm u}^{-2}$, the product $\bar{M}\sigma_{\mathfrak{b}}^2 = 1.332 \times 10^{-3}\,{\rm u}^{-1}$ is $1.430 \times 10^{-12}\,{\rm eV}^{-1}$ with $1\,{\rm u} = 931.494\,{\rm MeV}$. The PBE density of states of Sect. 5.2 gives $g_{{\rm ph},\kappa}/3 = 15.9\,{\rm eV}^{-1}$ at 50 meV, and the volume per atom is $\Omega_{\rm at} = a^3/8 = 20.02\,{\rm Å}^3 = 2.606 \times 10^{-9}\,{\rm eV}^{-3}$. Eq. (11.47) with the host density of states gives

$$\frac{\Gamma_{\rm iso}}{g_{B-L}^2} = \frac{\pi \times 1.430 \times 10^{-12}\,{\rm eV}^{-1} \times 15.9\,{\rm eV}^{-1}}{2 \times 2.606 \times 10^{-9}\,{\rm eV}^{-3}} = 1.37 \times 10^{-2}\,{\rm eV}.$$

The correction for the heavier carriers of the vertex is $\langle \delta^M \rangle_{\mathfrak{b}} h_{\rm iso} = (-0.054)(-1.55) =$

+8.3 % to first order, with $h_{\mathrm{iso}}(50\,\mathrm{meV}) = -1.55$ from the same density of states; the exact single-defect densities give +8.0 %, and $\Gamma_{\mathrm{iso}}/g_{B-L}^2 = 1.48 \times 10^{-2}\,\mathrm{eV}$.
For the electromagnetic part we need $\mathrm{Im}(-1/\varepsilon)$ at 50 meV ($403\,\mathrm{cm}^{-1}$). The measured absorption coefficient of silicon at 293 K is $\alpha = 0.55\,\mathrm{cm}^{-1}$ there [37]; we read it off a published figure [38, Fig. 3], and the reading is uncertain by about 10 %. With $\varepsilon_2 = \alpha n_r/\omega$, $n_r = 3.42$ and $\omega = 2534\,\mathrm{cm}^{-1}$ as an inverse length (2π times the wavenumber), $\varepsilon_2 = 7.43 \times 10^{-4}$, and with $\varepsilon_1 = 11.7$, $\mathrm{Im}(-1/\varepsilon) = \varepsilon_2/(\varepsilon_1^2 + \varepsilon_2^2) = 5.42 \times 10^{-6}$. Eq. (11.46) with $e^2 = 4\pi\alpha = 0.0917$ gives $\Gamma_{\mathrm{EM}}/g_{B-L}^2 = 0.050\,\mathrm{eV} \times 5.42 \times 10^{-6}/0.0917 = 2.96 \times 10^{-6}\,\mathrm{eV}$. By Eq. (11.48) this is an upper bound at millikelvin temperatures, and

$$\frac{\Gamma_{\mathrm{iso}}}{\Gamma_{\mathrm{EM}}} \geq \frac{1.48 \times 10^{-2}\,\mathrm{eV}}{2.96 \times 10^{-6}\,\mathrm{eV}} = 5.0 \times 10^{3} \qquad (50\,\mathrm{meV},\ T \to 0).$$

At 50 meV, in a valley of the density of states between the longitudinal acoustic and optical features, the PBE spectrum, softer by about 1 % than the room-temperature neutron data of Sect. 5.2, shifts $g_{\mathrm{ph},\kappa}$ by a comparable amount. Neither this nor the reading of α changes the conclusion: in a cold silicon crystal a $B - L$ vector of 50 meV is absorbed by the isotopes thousands of times more often than by the electromagnetic channel.

Fig. 11.23 shows the two channels across the band. The isotope channel follows the density of states, from the acoustic peaks near 17 meV to the optical peak near 58 meV, and vanishes at the top of the band. Against the room-temperature electromagnetic bound it wins by factors from 2×10^2, at the top of the band where the density of states falls to zero, to 1.4×10^4. Above the band only the electromagnetic part remains, and at the two-phonon peak near 76 meV it is 17 times stronger than at 50 meV. The size of the isotope channel is set by the isotopic variance: a crystal enriched in ^{28}Si suppresses it in proportion to the depletion of the minority isotopes, while the electromagnetic part is unchanged, which makes enrichment a control of the signal.

The inputs of these rates come from one harmonic calculation of the pure crystal. Its density of states and Eq. (11.44) give the local density of states of every isotope, and the force constants are those of Sect. 5.2 and Appendix E.4. The exact rate of a disordered crystal includes every isotope and the pair terms. It comes from diagonalizing the Hessian of a supercell of a few thousand atoms with the isotopes drawn at random, and Appendix E.17 specifies that calculation; the single-defect correction of Eq. (11.47) is its check. The spin expectation values $\langle S_p \rangle$ and $\langle S_n \rangle$ of each odd isotope, which set g_{ai} and $c_4^{(i)}$, come from nuclear shell-model calculations, and Appendix E.27 specifies them with their spread.

One input is missing. The electromagnetic channel of Eq. (11.46) needs $\varepsilon(\omega)$ of silicon inside the one-phonon band at the temperature of the detector, and we know of no measurement of the two-phonon absorption of silicon below 65 meV at low temperature. The room-temperature data give only the bound Eq. (11.48), dominated by difference processes that are absent in a cold crystal. The zero-temperature absorption requires the second-order dipole moments, the derivatives of the Born charges with respect to the displacements, which a first-principles calculation can supply (Appendix E.17); until it or a cold measurement exists, the margin of 5×10^3 at 50 meV is a lower limit.

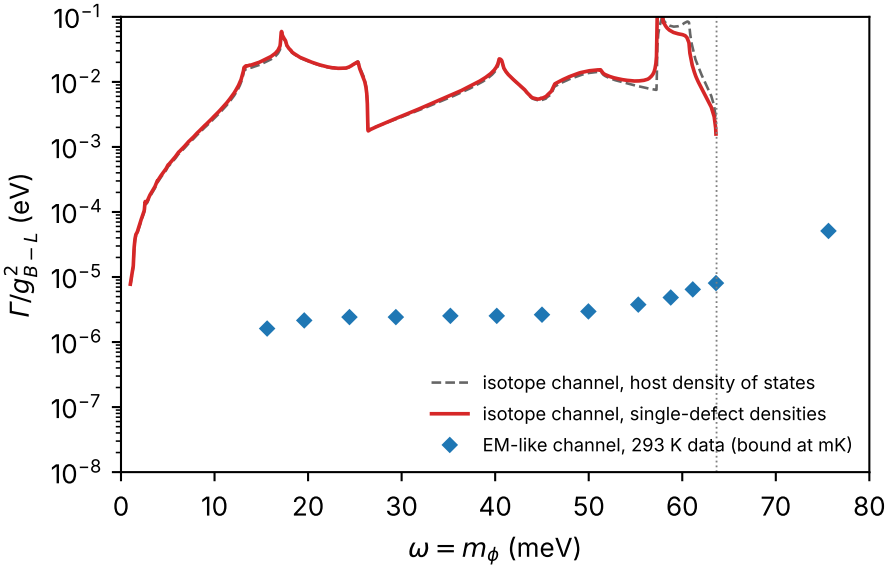


Figure 11.23: Absorption rate per quantum of a $B - L$ vector in natural silicon, over the squared coupling, against its mass. Red: the isotope-disorder channel of Eq. (11.47) with the exact single-defect densities of states of the three isotopes, Eq. (11.44); dashed: the same with the host density of states for all. PBE phonons of Sect. 5.2, $T = 0$; the dotted line marks the top of the PBE band, 63.7 meV (the measured band top is 64.1 meV). Blue diamonds: the electromagnetic channel Eq. (11.46), from the measured absorption coefficient at 293 K of Ikezawa and Ishigame [37], read from Fig. 3 of Deinzer and Strauch [38] with an uncertainty of about $0.05\,\mathrm{cm}^{-1}$ plus 10 %, converted with $n_r = 3.42$ and $\varepsilon_1 = 11.7$. At millikelvin temperatures the diamonds are upper bounds, Eq. (11.48). The diamond at 76 meV is the two-phonon peak, above the one-phonon band.

11.7 Anisotropy and daily modulation

At 45°N the dark-matter wind arrives from almost straight overhead when Cygnus crosses the meridian, and from close to the horizon twelve sidereal hours later (Fig. 1.3). A crystal fixed to the laboratory floor is therefore probed along different crystal directions in the course of each sidereal day. An isotropic target such as a liquid has the same rate in every orientation; a crystal responds differently along different crystal axes, so its rate follows the sidereal day. The size of the effect differs widely between crystals: below 1 % in silicon, for electron recoils and for phonons alike, and 6 % for multiphonon events in sapphire above a threshold of 300 meV. Two factors set it, the angular structure of the crystal response and the sharpness with which the halo singles out one direction of $\boldsymbol{q}$, and the two can be computed separately, for the electronic responses of Chap. 10 and the nuclear responses of this chapter alike.

11.7.1 Rate at fixed orientation

Take a coupling through one density, the electron density of Sect. 10.1 or the nuclear operator of Sect. 11.1, with a potential $\tilde{V}(q)$ that depends only on $|\boldsymbol{q}|$. The rate per unit target mass is Eq. (8.16) with one row, $\mathfrak{D} = |\tilde{V}(q)|^2$, and the direction of $\boldsymbol{q}$ enters both the structure factor $S(\boldsymbol{q},\omega)$ and the velocity kernel $\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\,\delta(\omega-\omega_{\boldsymbol{q}})$. For elastic scattering, $\omega_{\boldsymbol{q}} = \boldsymbol{q}\cdot\boldsymbol{v} - q^2/2m_\chi$ (Eq. (A.14)), and the delta function becomes $q^{-1}\delta(\hat{\boldsymbol{q}}\cdot\boldsymbol{v} - v_{\min})$ with $v_{\min} = \omega/q + q/2m_\chi$ (Eq. (1.23)). The laboratory distribution is the galactic one shifted by the Earth's velocity, $f_\chi(\boldsymbol{v}) = f_{\mathrm{gal}}(\boldsymbol{v}+\boldsymbol{v}_{\mathrm{E}})$ (Eq. (1.8)). Substituting $\boldsymbol{u} = \boldsymbol{v} + \boldsymbol{v}_{\mathrm{E}}$ turns the velocity integral into

$$\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\,\delta(\omega-\omega_{\boldsymbol{q}}) = \frac{1}{q}\int \mathrm{d}^3u\, f_{\mathrm{gal}}(\boldsymbol{u})\,\delta(\hat{\boldsymbol{q}}\cdot\boldsymbol{u} - v_{\min} - v_{\mathrm{E}}\,\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{v}}_{\mathrm{E}}) = \frac{1}{q}\,\hat{f}_{\mathrm{gal}}(v_{\min}+v_{\mathrm{E}}x),$$

with $x = \hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{v}}_{\mathrm{E}}$ the cosine of the angle between the momentum transfer and the Earth's velocity. The function $\hat{f}_{\mathrm{gal}}(s)$ is the integral of the isotropic galactic distribution over the plane $\hat{\boldsymbol{q}}\cdot\boldsymbol{u} = s$, the Radon transform of f_{gal}. For the truncated Maxwellian of Eq. (1.3), the two Gaussian integrals across the plane run over the disc $u_\perp^2 < v_{\mathrm{esc}}^2 - s^2$, and

$$\hat{f}_{\mathrm{gal}}(s) = \frac{\pi v_0^2}{N_{\mathrm{esc}}}\left[\mathrm{e}^{-s^2/v_0^2} - \mathrm{e}^{-v_{\mathrm{esc}}^2/v_0^2}\right]\Theta(v_{\mathrm{esc}}-|s|),$$

with N_{esc} of Eq. (1.4). The halo enters only through this function of one variable, and the crystal only through $S(\boldsymbol{q},\omega)$ at fixed $|\boldsymbol{q}|$, a function on the sphere of directions.

Both factors are now expanded on the sphere. The halo factor is a function of x alone and has a Legendre series, $\hat{f}_{\mathrm{gal}}(v_{\min}+v_{\mathrm{E}}x) = \sum_\ell \frac{1}{2}(2\ell+1)\,\mathcal{H}_\ell\,P_\ell(x)$, with coefficients

$$\mathcal{H}_\ell(v_{\min}) = \int_{-1}^{1} \mathrm{d}x\; P_\ell(x)\,\hat{f}_{\mathrm{gal}}(v_{\min}+v_{\mathrm{E}}x), \tag{11.49}$$

the multipoles of the halo. The crystal factor is expanded in real spherical harmonics $Y_{\ell m}$, the real and imaginary parts of those of Appendix C.3, taken in crystal axes, with components $S_{\ell m}(|\boldsymbol{q}|,\omega) = \int \mathrm{d}^2\hat{\boldsymbol{q}}\,Y_{\ell m}(\hat{\boldsymbol{q}})\,S(\boldsymbol{q},\omega)$. The addition theorem Eq. (C.21) holds for the real harmonics too. It gives $\int \mathrm{d}^2\hat{\boldsymbol{q}}\,P_\ell(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{v}}_{\mathrm{E}})\,Y_{\ell' m}(\hat{\boldsymbol{q}}) = [4\pi/(2\ell+1)]\,\delta_{\ell\ell'}\,Y_{\ell m}(\hat{\boldsymbol{v}}_{\mathrm{E}})$, and the angular integral pairs each multipole of the response with the halo multipole of the same order:

Rate of a crystal at fixed orientation

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega}(t) = \frac{\rho_\chi}{m_\chi\,\rho_T}\int \frac{q\,\mathrm{d}q}{4\pi^2}\,|\tilde{V}(q)|^2 \sum_{\ell m}\mathcal{H}_\ell\big(v_{\min}(q,\omega)\big)\,S_{\ell m}(q,\omega)\,Y_{\ell m}\big(\hat{\boldsymbol{v}}_{\mathrm{E}}(t)\big), \tag{11.50}$$

with $\hat{\boldsymbol{v}}_{\mathrm{E}}(t)$ the direction of the Earth's velocity in crystal axes, which turns with the sidereal day. For a cubic crystal the sum over m at each order is one cubic harmonic Y_ℓ^{c} of Eq. (C.25), $\sum_m S_{\ell m} Y_{\ell m}(\hat{\boldsymbol{v}}_{\mathrm{E}}) = S_\ell\, Y_\ell^{\mathrm{c}}(\hat{\boldsymbol{v}}_{\mathrm{E}})$ with $S_\ell(|\boldsymbol{q}|,\omega) = \int \mathrm{d}^2\hat{\boldsymbol{q}}\, Y_\ell^{\mathrm{c}}(\hat{\boldsymbol{q}})\, S(\boldsymbol{q},\omega)$, for $\ell \leq 10$.

The monopole term reproduces the isotropic rate. Since $\int_{-1}^{1} \mathrm{d}x\, \hat{f}_{\mathrm{gal}}(v_{\mathrm{min}} + v_{\mathrm{E}}x)/2q$ is the average of the velocity integral over the directions of $\boldsymbol{q}$, which Eq. (A.42) identified with $\eta/2q$, the zeroth halo multipole is the mean inverse speed, $\mathcal{H}_0(v_{\mathrm{min}}) = \eta(v_{\mathrm{min}})$. With $S_{00}Y_{00} = \bar{S}$, the term $\ell = 0$ of Eq. (11.50) is the isotropic rate of Eq. (A.42). The other terms carry all of the time dependence, through $Y_{\ell m}(\hat{\boldsymbol{v}}_{\mathrm{E}}(t))$, and each is weighted by the product of a property of the crystal, $S_{\ell m}$, and a property of the halo, $\mathcal{H}_\ell$. The same separation underlies the harmonic method of Al-Adulrazzaq and collaborators [39, Eqs. (6), (10), (17)]. Stratman and Lin integrate the same rate directly over the directions of $\boldsymbol{q}$ [40, Eqs. (28), (29), (31)]. For a coupling with several rows, $|\tilde{V}|^2 S$ becomes the contraction of Eq. (8.16); as long as $\mathfrak{D}_{ab}$ depends on $|\boldsymbol{q}|$ alone, each S_{ab} is expanded in the same way. A coupling that depends on the dark-matter velocity puts powers of $\boldsymbol{v}$ under the velocity integral, and its halo factor is then no longer a function of x alone.

How quickly the halo multipoles fall with ℓ decides how much of a crystal's anisotropy reaches the rate. With the standard halo of Sect. 1.2, the ratio $\mathcal{H}_2/\mathcal{H}_0$ is 0.13, 0.37, 0.51 and 0.70 at $v_{\mathrm{min}} = 300$, 500, 600 and 700 km s^{-1}, while $\mathcal{H}_4/\mathcal{H}_0$ is only -0.016, 0.017, 0.069 and 0.27. The reason is the shape of the halo factor. As a function of x it is a Gaussian of width $v_0/v_{\mathrm{E}} = 0.95$, centered at $x = -v_{\mathrm{min}}/v_{\mathrm{E}}$: the momentum transfer prefers to point along the wind, against $\boldsymbol{v}_{\mathrm{E}}$, but only loosely, and a broad function has small high multipoles. Near the kinematic edge, $v_{\mathrm{min}} \to v_{\mathrm{esc}} + v_{\mathrm{E}}$, only a small cap of directions around $-\hat{\boldsymbol{v}}_{\mathrm{E}}$ is allowed, and every multipole approaches $\mathcal{H}_0$ in magnitude. At small v_{min} the preference reverses: a particle that needs only a small velocity component along $\boldsymbol{q}$ is best served by a $\boldsymbol{q}$ nearly perpendicular to its velocity, and $\mathcal{H}_2/\mathcal{H}_0 = -0.13$ at $v_{\mathrm{min}} = 0$. The quadrupole changes sign at $v_{\mathrm{min}} = 196$ km s^{-1} and the hexadecapole at 138 and 435 km s^{-1}, so the phase of a modulation can flip with the mass or the energy deposited, as the annual modulation does at 211 km s^{-1} (Sect. 1.2). The halo therefore filters the anisotropy of a crystal: the quadrupole passes at the level of tens of percent, the hexadecapole at a few percent, and the higher orders only for the fastest particles.

We measure the result by the daily-modulation amplitude

$$A_{\mathrm{day}} = \frac{\mathcal{R}_{\mathrm{max}} - \mathcal{R}_{\mathrm{min}}}{\mathcal{R}_{\mathrm{max}} + \mathcal{R}_{\mathrm{min}}}, \tag{11.51}$$

the half-range of the rate over one sidereal day divided by its mean, for a rate integrated above a threshold. It depends on the crystal, on its mounting and on the latitude, which together fix the path of $\hat{\boldsymbol{v}}_{\mathrm{E}}(t)$ in crystal axes (Eq. (1.17)), and on the dark-matter mass and the threshold, which fix the values of v_{min} the rate samples.

11.7.2 Selection rules

The first restriction is evenness. With inversion symmetry, or with time-reversal symmetry, $S(\boldsymbol{q},\omega) = S(-\boldsymbol{q},\omega)$ (Appendix A.4), and every component with odd ℓ vanishes. This holds in silicon, germanium and GaAs, with or without spin–orbit coupling, at every $|\boldsymbol{q}|$, every ω and every temperature, and for the loss function as well as for S. The odd halo multipoles are large, $\mathcal{H}_1/\mathcal{H}_0 = -0.54$ at 300 km s^{-1}, but no even response can use them.

A response with odd parts needs both symmetries broken, as in a magnetically ordered crystal without a center of inversion. For a density its odd part is then small at small momentum

transfer. For $f \neq i$ the matrix element starts as $\langle f|n_e(\boldsymbol{q})|i\rangle = \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{d}_{fi} - \frac{1}{2}q_\alpha q_\beta Q_{fi}^{\alpha\beta} + O(q^3)$, with $\boldsymbol{d}_{fi}$ and $Q_{fi}^{\alpha\beta}$ the matrix elements of $\sum_j \boldsymbol{r}_j$ and $\sum_j r_j^\alpha r_j^\beta$; the term of order zero vanishes by orthogonality. The difference between $\boldsymbol{q}$ and $-\boldsymbol{q}$ is the interference of the two leading terms,

$$|\langle f|n_e(\boldsymbol{q})|i\rangle|^2 - |\langle f|n_e(-\boldsymbol{q})|i\rangle|^2 = 2\,\mathrm{Im}\big[(\boldsymbol{q}\cdot\boldsymbol{d}_{fi})\,q_\alpha q_\beta Q_{fi}^{\alpha\beta *}\big] + O(q^5),$$

which is of third order in q. Inversion forbids it, because $\boldsymbol{d}_{fi}$ connects states of opposite parity and Q_{fi} states of the same parity; time reversal forbids it, because it makes both matrix elements real in a suitable basis. Where it survives, it multiplies the odd halo multipoles, and the rate acquires a part that is odd in the direction of the wind: opposite wind directions in crystal axes give it opposite signs. Such nonreciprocal targets are treated in Sect. 14.5. A chiral crystal that keeps time reversal has an even S, and its two enantiomers modulate identically.

The second restriction is the point group. A function of direction invariant under the cubic group has no component with $\ell = 2$ (Appendix C.3), so in silicon, germanium and GaAs the anisotropy of every response, electronic or nuclear, starts at $\ell = 4$ with Y_4^{c}, followed by Y_6^{c} and Y_8^{c}. A uniaxial crystal, such as trigonal sapphire or a hexagonal polytype of SiC, has one invariant direction, the c axis $\hat{\boldsymbol{c}}$, and allows $\ell = 2$ with $m = 0$ about it. To this order the rate is $\mathcal{R}_0 + \mathcal{R}_2 P_2(\hat{\boldsymbol{c}}\cdot\hat{\boldsymbol{v}}_{\mathrm{E}})$, the constant plus a constant times $\cos^2\theta_{\mathrm{lab}}$ that Sect. 1.2 wrote down by symmetry alone, and Eq. (11.50) now supplies the two constants. At $\ell = 4$ a trigonal crystal adds terms with $|m| = 3$, and a hexagonal one first breaks the symmetry about $\hat{\boldsymbol{c}}$ at $\ell = 6$. Combined with the halo filter, the counting already explains the main pattern: for the same anisotropy of the response, a uniaxial crystal modulates through $\mathcal{H}_2$, a cubic one through $\mathcal{H}_4$, roughly ten times less except near the kinematic edge.

11.7.3 Electronic anisotropy

How anisotropic is the loss function of silicon? The covalent bonds point along the four $\langle 111\rangle$ directions, the conduction-band minima lie along $\langle 100\rangle$, and an electron–hole pair of a few eV made with a momentum transfer of a few keV feels both. Fig. 11.25(a) shows the ratio L_4/L_0 of the cubic-harmonic components of the loss function, in the empirical pseudopotential model of Sect. 3.3 that Limit Check 10.1 used, with the random-phase head of the dielectric matrix and no local fields. Each component is fitted to the loss function computed along ten directions of the irreducible wedge of the cube. The k grid has the full cubic symmetry, so the components with $\ell = 2$ and odd ℓ vanish identically and cannot leak in as artifacts; refining it from 12^3 to 16^3 points changes L_4/L_0 by at most 0.01. At $q = q_0 = 3.73\,\mathrm{keV}$ the ratio is 0.18 at 3 eV, 0.05 at 5 eV and below 0.01 above 8 eV. The hexadecapole is not dominant: L_6/L_0 is of the same size, and near the gap the components beyond $\ell = 8$ carry an rms deviation of 10 % of $\bar{L}$ or more, a sign of structure on small angular scales. Away from the gap the response is closer to that of a uniform gas, which has no preferred direction. GaAs, with the same lattice and a different basis, has components of the same size.

The halo multipoles now decide how much of this survives in the rate. Limit Check 11.17 follows it at 5 eV for a 10 MeV particle. Wherever $v_{\min}$ is well below the kinematic edge, $\mathcal{H}_4/\mathcal{H}_0$ is a few percent and the anisotropic part of the rate at each q stays below a few tenths of a percent, carried by the term $\ell = 4$ alone. Only near the edge, where the halo admits a narrow cone of directions, does the rate depend strongly on the direction of the wind, by tens of percent, and there the components $\ell \geq 6$ enter as well. Integrated over the spectrum of a 2 MeV particle, the components combine into a daily modulation of about 1 % (Worked example 11.8).

These numbers are those of a model, and the model has two known gaps. The empirical pseudopotential reproduces the bands, not the wavefunctions at large momentum, and the

calculation stops at $q = 2.8\,q_0$. Local fields add the crystal's own reciprocal vectors to the response at $q + \boldsymbol{G}$ and are known to change the direction-averaged rate by a factor of 1.3 to 1.5 at 10 MeV (Sect. 10.2); their effect on L_4 has not been computed. The cubic-harmonic components of the silicon loss function from a first-principles calculation with local fields, on a grid of directions fine enough for the structure near the gap, are therefore an open quantity. So, with them, is the electronic daily modulation of silicon beyond a model estimate. The calculation is specified in Appendix E.16: the full-$\boldsymbol{q}$ loss function on Lebedev grids of increasing order, projected on Y_ℓ^{c} for $\ell = 4$, 6 and 8, with the components $\ell = 1$, 2 and 3 as built-in tests that must vanish.

Evenness also settles a point in the literature. Al-Adulrazzaq and collaborators compute the electronic daily modulation of silicon and GaAs at 10 MeV from the full loss function and find about one percent. They attribute the nearly sinusoidal shape of the signal at 5 eV, with one period per sidereal day, to the dipole component of the loss function [39, Sect. III B]. The dipole component vanishes identically in both crystals, as do all odd components. A period of one sidereal day does not need $\ell = 1$: when the crystal axes are tilted with respect to the Earth's axis, $Y_4^{\mathrm{c}}(\hat{\boldsymbol{v}}_{\mathrm{E}}(t))$ contains every harmonic of the sidereal frequency up to the fourth, and which one dominates depends on the mounting. Their Migdal rate in the soft limit, in which the electronic and nuclear momenta separate, depends on the loss function only through its components with $\ell = 0$ and 2 at each $|\boldsymbol{k}|$ [39, Eqs. (37)–(41)]. In a cubic crystal the component $\ell = 2$ vanishes at every $|\boldsymbol{k}|$, so the soft-limit Migdal rate of silicon, germanium and GaAs has no daily modulation at all. Their computed modulation of the Migdal rate is much smaller than that of the electron rate, as it must be, and any residue is numerical. Beyond the soft limit the Migdal response is the subject of Chap. 12.

11.7.4 Nuclear anisotropy

The nuclear response carries its direction dependence in two places. In the incoherent approximation Eq. (11.26), each nucleus enters through its Debye–Waller exponent $2W_\kappa(\boldsymbol{q}) = \boldsymbol{q} \cdot B_\kappa \cdot \boldsymbol{q}$ (Eq. (5.56)) and through the spectrum $P_1^\kappa(\hat{\boldsymbol{q}}, \omega)$, which is built from the density of states projected on $\hat{\boldsymbol{q}}$. Both are quadratic forms in $\hat{\boldsymbol{q}}$. The projected density of states is $\hat{\boldsymbol{q}} \cdot D_\kappa(\omega) \cdot \hat{\boldsymbol{q}}$, with the tensor

$$D_\kappa^{\alpha\beta}(\omega) = \frac{1}{N} \sum_{\boldsymbol{Q}\nu} \mathrm{Re}\big[e_{\kappa\nu}^{\alpha}(\boldsymbol{Q})\, e_{\kappa\nu}^{\beta}(\boldsymbol{Q})^*\big]\, \delta(\omega - \omega_{\boldsymbol{Q}\nu}),$$

whose trace is the atom-projected density of states $g_{\mathrm{ph},\kappa}(\omega)$ and whose integral against $\coth(\omega/2T)/(2M_\kappa\omega)$ is B_κ. Stratman and Lin write the incoherent response with this tensor and compute its daily modulation in sapphire and SiC [40, Eqs. (18)–(24)]. At a site of cubic symmetry every second-rank tensor is a multiple of the unit tensor (Appendix C.3). Then $B_\kappa = \langle u_x^2 \rangle \mathbb{1}$ and $D_\kappa = (g_{\mathrm{ph},\kappa}/3)\mathbb{1}$, and the direction of $\boldsymbol{q}$ drops out of Eq. (11.26) term by term:

Isotropy of the incoherent response of cubic crystals

If every nucleus sits at a site of cubic symmetry, the incoherent nuclear response Eq. (11.26) is independent of the direction of $\boldsymbol{q}$ at every phonon order and every temperature. The direction dependence of the harmonic nuclear response of silicon, germanium, diamond and GaAs is entirely in its distinct part, the pair terms of Eq. (11.24).

The temperature enters only through the Bose factors, which multiply each phonon by a scalar, so the statement holds at any T. The published tensors of Stratman and Lin show it

to numerical precision: for silicon the three diagonal elements of B_κ agree to 2×10^{-4} and the off-diagonal ones are below $10^{-7}\,\text{Å}^2$, and the two sites of GaAs agree to 8×10^{-4}. Their finding that the multiphonon rates of silicon and GaAs do not modulate [40, Sect. IV] is therefore exact within their approximation, not a numerical accident.

The anisotropy of the cubic nuclear response is entirely in the coherent terms, and Sect. 11.3 computed them. At long wavelength the one-phonon weight per direction is $\sum_\nu(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2/c_\nu(\hat{\boldsymbol{q}})$ of Eq. (11.3). With the measured elastic constants of Sect. 11.1 its component $\ell=4$ is $S_4/S_0=0.028$, and $\ell=6$ is below 10^{-3}. The PBE force constants of Sect. 5.2 give $S_4/S_0=0.025$ for $q\le0.3\,q_{\text{BZ}}$, and the dispersion of the branches raises $|S_6/S_0|$ to 0.026 at $0.7\,q_{\text{BZ}}$. Beyond the zone the one-phonon weight is strongly anisotropic. Near each reciprocal-lattice vector the acoustic phonons at $\boldsymbol{q}-\boldsymbol{G}$ are soft and their weight $1/\omega$ is large. The vectors $\boldsymbol{G}$ of a shell point in definite directions, and the weight is concentrated along them: just beyond the shell $|\boldsymbol{G}_{111}|=3.95\,\text{keV}=1.76\,q_{\text{BZ}}$, at $q=1.79\,q_{\text{BZ}}$, $S_4/S_0=-0.36$, and S_6/S_0 reaches -0.49 at $2.9\,q_{\text{BZ}}$; beyond $5\,q_{\text{BZ}}$, $|S_4/S_0|$ still reaches 0.2 and $|S_6/S_0|$ 0.4. The two-phonon weight, a sum over pairs of the square of a correlation, is less anisotropic, with $|S_4/S_0|$ and $|S_6/S_0|$ of about 0.1 at most. From three phonons on the incoherent part dominates, and the coherent corrections, and with them the anisotropy, are below 3 % (Sect. 11.3). These components come from the same force constants and the same pair correlations as the direction-averaged weights of Sect. 11.3, and Appendix E.17 specifies their tabulation for silicon, germanium and GaAs, with the incoherent components $\ell\ge4$ as a test that must vanish to numerical noise.

A cubic crystal therefore modulates through its nuclear response by less than 1 %, and appreciably just at masses whose largest momentum transfer, $2m_\chi v_{\max}$, ends close to a shell of reciprocal-lattice vectors, where the halo multipoles are large and one shell dominates. In silicon that happens near 1 and 2 MeV (Worked example 11.8). At other masses the shells inside the kinematic range contribute with alternating signs and largely cancel, and above a few MeV the incoherent part dominates the rate: the amplitude is 0.04 % at 3 and 4 MeV and 10^{-4} at 5 and 10 MeV.

A crystal whose nuclei sit at sites of lower symmetry has an anisotropic incoherent response already. To first order in the anisotropy of B_κ, write $\boldsymbol{q}\cdot B_\kappa\cdot\boldsymbol{q}=q^2[\bar b_\kappa+\frac{2}{3}\Delta b_\kappa P_2(\hat{\boldsymbol{c}}\cdot\hat{\boldsymbol{q}})]$ for a uniaxial site, with $\bar b_\kappa=\frac{1}{3}\operatorname{Tr}B_\kappa$ and $\Delta b_\kappa=B_\kappa^{cc}-B_\kappa^{aa}$ the difference between the elements along and across the axis. The Poisson weight $p_n^{\text{ph}}=\mathrm{e}^{-2W_\kappa}(2W_\kappa)^n/n!$ of n phonons (Eq. (5.60)) changes by $\delta(2W_\kappa)(n/2W_\kappa-1)$ in its logarithm, so that

$$\frac{\delta p_n^{\text{ph}}}{p_n^{\text{ph}}}=(n-2\bar W_\kappa)\,\frac{2\Delta b_\kappa}{3\bar b_\kappa}\,P_2(\hat{\boldsymbol{c}}\cdot\hat{\boldsymbol{q}}),\tag{11.52}$$

with $2\bar W_\kappa=q^2\bar b_\kappa$. Orders below the mean phonon number lose weight along the axis of larger displacement and orders above it gain. A threshold well inside the multiphonon tail needs n far above $2\bar W_\kappa$. The rate above it therefore inherits the anisotropy of B_κ multiplied by the phonon number, while the projected spectrum $P_1^\kappa(\hat{\boldsymbol{q}},\omega)$ adds an anisotropy of its own whenever the threshold cuts through the phonon band. For sapphire the published tensors give, weighted with A_κ^2 over the four Al and six O sites of the cell, $B^{cc}/B^{aa}-1=+0.059$; for 2H-SiC only +0.008, because the larger displacement along the axis of the Si sites is partly offset by the C sites.

Limit Check 11.17. Daily modulation from the anisotropic response

The rate of a crystal at fixed orientation pairs each multipole $S_{\ell m}$ of the response with the halo multipole $\mathcal{H}_\ell$ of the same order, $\mathrm{d}\mathcal{R}/\mathrm{d}\omega(t)=(\rho_\chi/m_\chi\rho_T)\int q\,\mathrm{d}q\,(4\pi^2)^{-1}|\tilde V|^2\sum_{\ell m}\mathcal{H}_\ell S_{\ell m}Y_{\ell m}(\hat{\boldsymbol{v}}_{\text{E}}(t))$, Eq. (11.50), for the loss func-

tion [39, Eqs. (10), (25)] and for the incoherent nuclear response with direction-dependent Debye–Waller exponents and projected densities of states [40, Eqs. (19), (24)]. Its first anisotropic term is $\ell = 4$ in a cubic crystal and $\ell = 2$ in a uniaxial one.

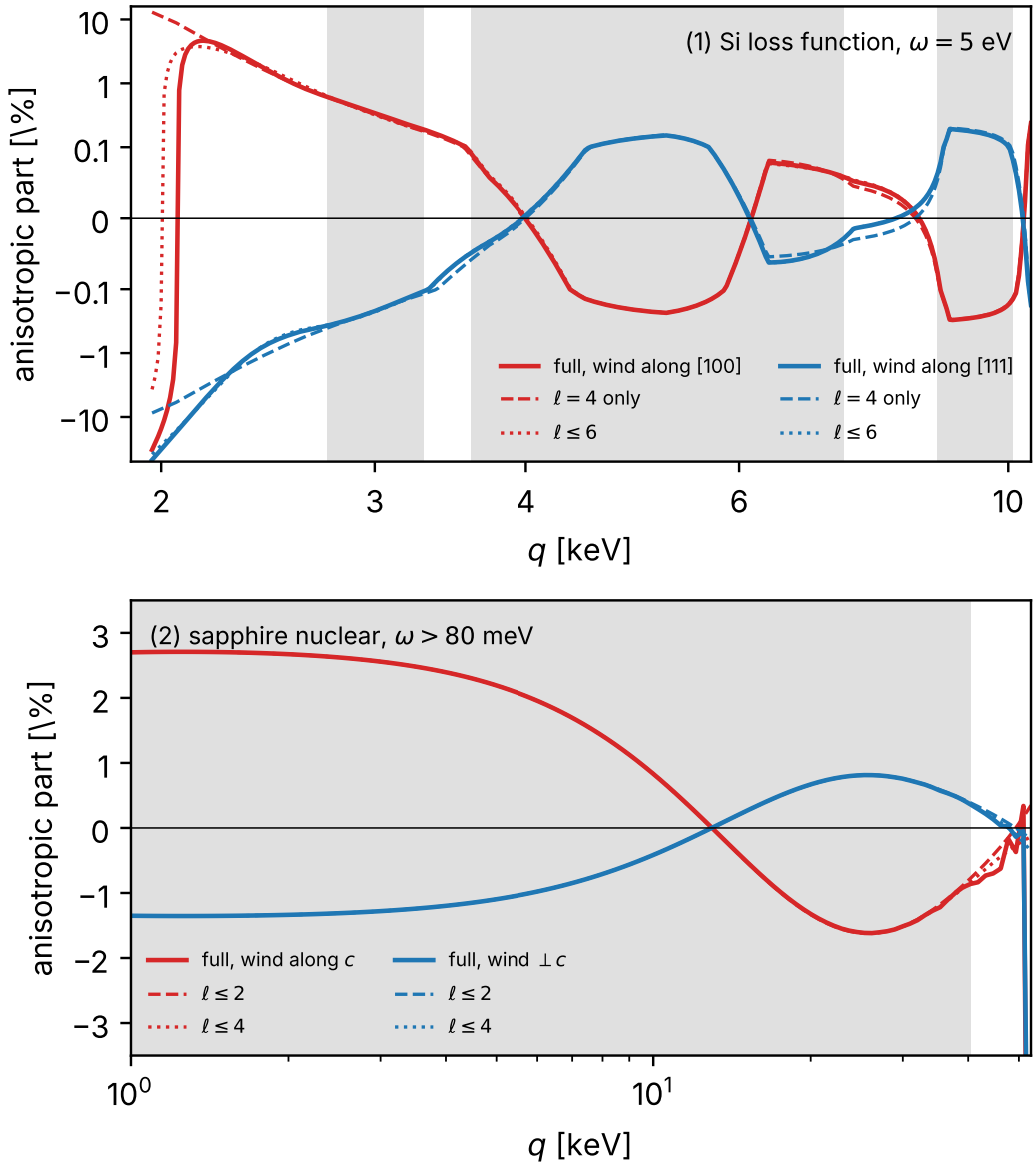


Figure 11.24: Anisotropic part of the rate at fixed momentum transfer for a wind along a fixed crystal axis, $\sum_{\ell\geq 1} \mathcal{H}_\ell \sum_m S_{\ell m} Y_{\ell m}(\hat{\boldsymbol{n}})/(\mathcal{H}_0 S_{00} Y_{00})$ from Eq. (11.50), against q, for $m_\chi = 10\,\mathrm{MeV}$, a heavy mediator and the standard halo of Sect. 1.2. (1) Silicon loss function at $\omega = 5\,\mathrm{eV}$ in the empirical pseudopotential model (random-phase head, no local fields), wind along [100] (red) and [111] (blue), from the kinematic edge to the end of the model at $2.8\,q_0$; solid: all fitted components to $\ell = 8$; dashed: $\ell = 4$ alone, the formula; dotted: $\ell \leq 6$, the leading correction (symmetric logarithmic axis, linear below 0.1 %). (2) Sapphire, incoherent nuclear response with direction-dependent Debye–Waller exponents and projected densities of states from the site tensors of Stratman and Lin, integrated above 80 meV, wind along the c axis (red) and perpendicular to

it (blue); solid: direct integral over the directions of $\boldsymbol{q}$; dashed: $\ell \leq 2$, the formula; dotted: $\ell \leq 4$. Gray bands: q where the truncation at $\ell \leq 4$ is within 10 % of the full anisotropic part (or within 10^{-4} of the rate where that part is below 10^{-3}): in panel (1) from 2.7 keV up, except near 3.4, 8 and 10 keV, where $\mathcal{H}_4$ or L_4 passes close to zero and $\ell = 6$ takes over; in panel (2) up to 40 keV, against an edge at 53 keV.

Near the kinematic edge, where the halo admits only a narrow cone of directions, every multipole contributes, and a rate computed from the first anisotropic term alone misses the anisotropy there.

11.7.5 Daily-modulation amplitudes

The time dependence comes from the path of $\hat{\boldsymbol{v}}_{\mathrm{E}}(t)$ in crystal axes. In equatorial axes the wind direction stays on a cone of half-angle $\theta_{\mathrm{E}} = 42.5°$ around the Earth's rotation axis, and Eq. (1.17) gives its components along the local east, north and vertical. A crystal mounted with its axes along these three directions sees $\hat{\boldsymbol{v}}_{\mathrm{E}}(t)$ with exactly these components, and any other mounting is a fixed rotation of them. The amplitude of a term ℓ is then the range of $\sum_m S_{\ell m} Y_{\ell m}$ along the path. For a uniaxial crystal with its c axis vertical at 45°N, the angle to the wind runs from 2.5° to 87.5°, and P_2 sweeps from 0.997 to −0.497, nearly its full range. For a cubic crystal with [001] vertical at the same latitude, Y_4^{c} runs from −0.39 to 0.64, close to its full range of −0.43 at [111] to 0.65 at [100]. The rate oscillates mainly at twice the sidereal frequency, with a maximum when the wind arrives from overhead, along [001]. A cubic crystal mounted with [001] along the Earth's axis instead sees the wind circle at a fixed angle of 42.5° from [001]. Then Y_4^{c} varies through its azimuthal part alone, $\cos 4\varphi$, which gives four maxima per sidereal day and an amplitude six to ten times smaller.

Fig. 11.25(b) shows the uniaxial case. Sapphire's rate depends on the direction of the wind almost only through the term $\ell = 2$, since the terms with $\ell \geq 4$ contribute less than 5×10^{-4} of the rate, and its amplitude depends on which phonons the threshold admits. Above 1 meV, where the acoustic and low optical phonons carry most of the one-phonon weight, the one-phonon rate along the axis and across it differ by 0.6 %, and the rate of all phonon numbers by 1.0 %, since these modes move all ten atoms of the cell almost isotropically. Above 80 meV the highest optical modes alone remain, and they are polarized mostly in the basal plane: the one-phonon rate along the axis is 7 % below the rate across it. Above 300 meV the rate comes from three or more phonons, and Eq. (11.52) multiplies the 6 % anisotropy of the displacements by the phonon number: the difference grows to 12 %. Mounted with its axis vertical at 45°N, a sapphire crystal then modulates with $A_{\mathrm{day}} = 0.48\,\%$ for all phonons above 1 meV, 3.5 % for one phonon above 80 meV and 6.4 % above 300 meV, at 10 MeV. At 30 MeV the amplitudes for the two higher thresholds fall to 1.2 and 0.6 %. 2H-SiC, whose displacements are nearly isotropic, modulates less: 0.08, 0.61 and 3.5 % for the same three cases at 10 MeV. These are estimates within the incoherent approximation, which holds for q above a few q_{BZ}, here 1.75 keV for the ten-atom cell of sapphire, and therefore for masses of a few MeV and above. They agree in pattern with the multiphonon calculation of Stratman and Lin, who find up to 11 % in sapphire at 10 MeV above 300 meV, somewhat less in SiC and none in silicon and GaAs [40, Sect. IV and App. C]. They use a different halo ($v_0 = 220$, $v_{\mathrm{esc}} = 500$ and $v_{\mathrm{E}} = 240\,\mathrm{km\,s^{-1}}$) and put the crystal axis along the wind at the start of the day. At lower masses the momentum transfers stay inside the zone, and the coherent single-phonon response decides the rate. The full phonon calculation of Griffin and collaborators finds daily modulation in sapphire of up to 30 % at masses near 50 keV, for a threshold that leaves only the optical branches. With the acoustic branches open it finds much less, because the sound speeds of sapphire are nearly isotropic [11, Sects. III C and IV C]. The same holds for the

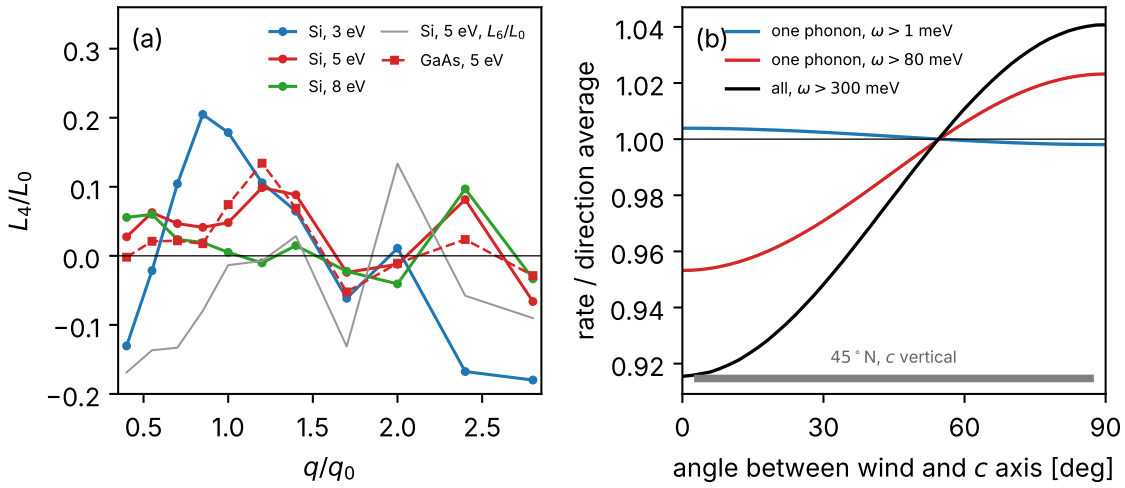


Figure 11.25: Anisotropy of a cubic and of a uniaxial crystal. (a) Hexadecapole of the silicon loss function, the ratio L_4/L_0 of its components on the cubic harmonics Y_4^c and Y_0^c, against q/q_0, $q_0 = 3.73\,\mathrm{keV}$, at $\omega = 3$, 5 and 8 eV (bins of 1 eV); thin gray: L_6/L_0 at 5 eV; dashed: GaAs at 5 eV. Empirical pseudopotential model of Sect. 3.3, head of the random-phase dielectric matrix, no local fields, ten directions of the irreducible wedge of the cube on a k grid with the full cubic symmetry, components $\ell = 0$, 4, 6 and 8 fitted by least squares. (b) Rate of a sapphire crystal for a wind at a fixed angle to its c axis, divided by the rate averaged over all wind directions, at $m_\chi = 10\,\mathrm{MeV}$ with a heavy mediator coupled to nucleon number: one phonon above 1 meV (blue) and above 80 meV (red), and all phonon numbers above 300 meV (black). Incoherent nuclear response at $T = 0$ with the direction-dependent Debye–Waller exponents and projected densities of states of the published site tensors of Stratman and Lin [40], whose A_κ^2-weighted anisotropy is $B^{cc}/B^{aa} - 1 = +0.059$. Bar: the range of angles swept in one sidereal day by a c axis along the vertical at 45°N.

hexagonal polytypes of SiC, with the largest modulation in 2H [9, Sects. IV B and V].

In silicon both the electron and the nucleon rates modulate, and with the same phase.

Worked example 11.8: Daily modulation of a silicon crystal at 45°N

A silicon crystal sits at latitude 45°N with [001] along the local vertical, [100] pointing east and [010] north. Dark matter of mass 2 MeV scatters through a heavy mediator, either on electrons, through the loss function, or on nucleons, through the nuclear response. At this mass the momentum transfers reach $2m_\chi v_{\max} = 10.6\,\mathrm{keV} = 2.84\,q_0$; the electronic model ends at $2.8\,q_0$, and the sliver above it, at the kinematic edge where the rate vanishes, is dropped. The energy deposited on electrons stays below $m_\chi v_{\max}^2/2 = 7.0\,\mathrm{eV}$.

Geometry. By Eq. (1.17), the wind direction in crystal axes is

$$\hat{\boldsymbol{v}}_\mathrm{E} = \big(-\cos\delta_\mathrm{E}\sin h_\mathrm{E},\ \cos\ell_\mathrm{lat}\sin\delta_\mathrm{E} - \sin\ell_\mathrm{lat}\cos\delta_\mathrm{E}\cos h_\mathrm{E},\ \sin\ell_\mathrm{lat}\sin\delta_\mathrm{E} + \cos\ell_\mathrm{lat}\cos\delta_\mathrm{E}\cos h_\mathrm{E}\big),$$

with $\delta_\mathrm{E} = 47.5°$ and the hour angle h_E advancing at the sidereal rate. Along this path Y_4^c of Eq. (C.25) runs from -0.394 to 0.640, its maximum at $h_\mathrm{E} = 0$, when the wind arrives nearly along [001]. Let $\mathcal{R}_\ell$ be the term ℓ of Eq. (11.50), integrated over q and ω without its factor Y_ℓ^c, and $\bar{\mathcal{R}}$ the rate averaged over orientations. If the hexadecapole dominates, $\mathcal{R}(t)/\bar{\mathcal{R}} = 1 + \sqrt{4\pi}\,(\mathcal{R}_4/\mathcal{R}_0)\,Y_4^\mathrm{c}(\hat{\boldsymbol{v}}_\mathrm{E}(t))$, and

$$A_\mathrm{day} = \sqrt{4\pi}\,\frac{0.640 + 0.394}{2}\,\frac{\mathcal{R}_4}{\mathcal{R}_0} = 1.83\,\frac{\mathcal{R}_4}{\mathcal{R}_0}.$$

Electronic part. The integrals $\mathcal{R}_\ell = \sum_\omega \int \mathrm{d}q\, q^3 \mathcal{H}_\ell(v_\mathrm{min}) L_\ell(q,\omega)$ over the spectrum, with the components of Fig. 11.25(a) in bins of 1 eV from 1.5 to 7.5 eV, give $\mathcal{R}_4/\mathcal{R}_0 = 4.9\times10^{-3}$, while $\mathcal{R}_6/\mathcal{R}_0 = -3\times10^{-5}$ and $\mathcal{R}_8/\mathcal{R}_0 = -7\times10^{-6}$ are negligible. The components beyond $\ell = 8$, which the ten directions do not resolve, enter with $\mathcal{H}_{10}$; estimated from the residual of the fit, they can shift the rate by about 10^{-3}. The hexadecapole of the response, a few percent, passes because 47 % of the rate of so light a particle comes from $v_\mathrm{min} > 600\,\mathrm{km\,s^{-1}}$, where the halo is sharp: weighted with the rate, $\mathcal{H}_4/\mathcal{H}_0$ is 0.09. The amplitude is $A_\mathrm{day} = 1.83\times4.9\times10^{-3} = 0.9\,\%$, uncertain by about 0.1 percentage point from those higher components. The rate oscillates twice per sidereal day, with its maximum when Cygnus crosses the meridian.

Lattice part. For the nucleon coupling, $\mathcal{R}_\ell = \sum_n \int \mathrm{d}q\, q\, \mathcal{H}_\ell(q/2m_\chi)\,\mathrm{e}^{-2W}(2W)^n/n!\, s_{n,\ell}(q)$, with $2W = q^2\langle u_x^2\rangle$, $\langle u_x^2\rangle = 2.45\times10^{-3}\,\text{Å}^2$, and $s_{n,\ell}$ the cubic components of the exact one- and two-phonon weights of Sect. 11.3, divided by their incoherent values; three and more phonons are treated as isotropic, their coherent corrections being below 3 %. In v_min the deposit of n phonons enters as $n\times30.4\,\mathrm{meV}/q$, the mean energy of the single-phonon distribution, which matters only at the smallest q, where the rate is small. The momentum transfers cross the shells $\boldsymbol{G}_{111}$ and $\boldsymbol{G}_{220}$, where the one-phonon weight is most anisotropic, and $\mathcal{R}_4/\mathcal{R}_0 = 3.2\times10^{-3}$, so $A_\mathrm{day} = 0.58\,\%$, in phase with the electronic part since both $\mathcal{R}_4$ are positive. A detector that records both channels therefore sees an amplitude between 0.6 and 0.9 %, the mean of the two weighted by their rates.

Mounting. Over 3000 random orientations of the crystal at this latitude the largest amplitudes are 0.93 and 0.60 %, so the vertical [001] is close to the best choice. With [001] along the Earth's axis instead, the wind circles at a fixed angle to [001], only the term $\cos4\varphi$ of Y_4^c varies, and the amplitudes drop to 0.15 and 0.06 %, with four maxima per sidereal day.

GaAs. The same mounting and mass give $\mathcal{R}_4/\mathcal{R}_0 = 3.8 \times 10^{-3}$ and $A_{\text{day}} = 0.7\,\%$ for the electronic part, in the same model and with the same uncertainty. For the nuclear part, with the one-phonon weights of the force constants of Sect. 5.5 and $\langle u_x^2 \rangle \approx 1.9 \times 10^{-3}\,\text{Å}^2$, they give $\mathcal{R}_4/\mathcal{R}_0 = 1.6 \times 10^{-3}$ and $A_{\text{day}} = 0.29\,\%$. Both parts are in phase with those of silicon, and the combined amplitude of a GaAs detector lies between 0.3 and 0.7 %.

At 1 MeV, where $2m_\chi v_{\max} = 5.3\,\text{keV}$ lies just beyond the shell $\boldsymbol{G}_{111}$, the lattice amplitude of silicon is 0.77 %, with the opposite phase, because S_4 is negative there. (Near $\boldsymbol{G}_{200}$ the two sublattices cancel, as for the Bragg peak, and the soft phonons add nothing.) At 0.7 and 1.5 MeV the amplitude is only 0.13 and 0.18 %. Each of these amplitudes is a product of an anisotropy of the response and the halo multipole of the same order. Cubic crystals, whose first term is $\ell = 4$, stay below about 1 % in both sectors. Uniaxial crystals reach several percent, more for thresholds that select phonons polarized along one axis or the multiphonon tail, and at the lowest masses the coherent single-phonon response of a polar uniaxial crystal reaches tens of percent. Only near the kinematic edge does the halo pass every multipole, and a measurement there needs the full direction-resolved response.

11.8 Absorption on phonons and acoustic modes

A dark photon of mass 36.35 meV that is absorbed in GaAs deposits its whole energy in one longitudinal optical phonon, and its momentum, $m_\phi v \approx 46\,\mu\text{eV}$, is a part in 10^8 of the zone. Absorption samples the crystal at a single point, $\boldsymbol{q} = \boldsymbol{p}_\phi$ and $\omega = m_\phi$ (Sect. 8.2.3), and below the electronic gap the only excitations at that point are the optical phonons at the zone center. For bosons of a neV or a μeV the picture changes again. Their de Broglie wavelength $2\pi/m_\phi v$, a kilometer at 1 μeV, is larger than any detector, and the field is a classical wave that oscillates in phase over the whole body. What absorbs it is a normal mode of the body itself, at MHz in a millimeter crystal and at kHz in a ton-scale bar. A scalar coupled to electrons sees the electrons that a longitudinal phonon carries with each nucleus, and in GaAs, where the two sublattices carry nearly equal numbers per unit mass, its rate is a near-cancellation that the rigid-ion picture overstates by a factor of 7 to 9.

11.8.1 Single-phonon absorption

The absorption rate per quantum is the master formula Eq. (8.19) at one point. Collect the couplings of the field into one row, $O_\phi = \sum_a g_a^\phi O_a(\boldsymbol{p}_\phi)$; then $\Gamma_{\text{abs}} = S_{O_\phi O_\phi}(\boldsymbol{p}_\phi, \omega_\phi)/2\omega_\phi$, and in a cold insulator below the gap S is the one-phonon structure factor. The kinematics leaves no choice of branch. An acoustic phonon of momentum p_ϕ has the energy $c_\nu p_\phi = c_\nu v\, m_\phi$, and absorbing the boson would need $c_\nu v = 1$, while sound speeds are $\sim 10^{-5}$ and $v \sim 10^{-3}$ in units of the speed of light. An optical phonon at $\boldsymbol{Q} = \boldsymbol{p}_\phi$ is the zone-center phonon to a part in 10^{15}, since the dispersion starts at order $(p_\phi a)^2$. The momentum transfer is far below the 0.05 eV at which the finite-q structure of the longitudinal mode becomes visible (Worked example 8.6), and the zone-center vertex is exact here. The finite-q corrections of Sect. 11.2, the piezoelectric charge of the acoustic branches and the dynamical quadrupoles, play no role.

For a row built from densities the one-phonon structure factor has the form of Eq. (11.1) and Eq. (11.15): each sublattice contributes its displacement projected on $\boldsymbol{p}_\phi$, times a coupling per site. With $\omega_\phi \simeq m_\phi$ and $T = 0$,

$$\Gamma_{\text{abs}} = \frac{\pi}{2\Omega\, m_\phi} \sum_\nu \frac{\left| \sum_\kappa \boldsymbol{p}_\phi \cdot \mathcal{G}_\kappa^\phi \cdot \boldsymbol{e}_{\kappa\nu}(0)^* / \sqrt{M_\kappa} \right|^2}{\omega_\nu}\, \delta(m_\phi - \omega_\nu), \tag{11.53}$$

with ω_ν and $\boldsymbol{e}_{\kappa\nu}(0)$ the zone-center phonons of the full dynamical matrix for the direction $\hat{\boldsymbol{p}}_\phi$, nonanalytic term included. The tensor $\mathcal{G}_\kappa^\phi$ is the coupling of the field to the motion of site κ, and finding it is the whole problem. The naive answer is the coefficient of each species times the number of particles that move with the nucleus. For the nucleons that number is fixed. For the electrons it is not, because how many electrons follow a nucleus depends on the macroscopic field that the motion itself creates.

The Born charge answers half of the question. By its definition in Sect. 5.5, Z_κ^* is the charge that moves with nucleus κ at zero macroscopic field: the nuclear charge Z_κ minus the electrons that follow. At zero field the electrons that follow thus number $Z_\kappa \mathbb{1} - Z_\kappa^*$, and this is the vertex of the electron density in the proper response. A longitudinal phonon, however, carries a macroscopic field, and the electrons respond to it as well. The screened response Eq. (8.36) adds that field, and Worked example 8.6 has done the algebra for a nucleon row. The electron row differs in one respect: it also correlates with the charge through the electrons alone, at fixed nuclei.

Let us do the algebra once for any rows. Take frequencies near the optical phonons and neglect local fields. The screened response is then the rank-one form Eq. (8.36), $R_{ab} = \tilde{R}_{ab} + v_\mathrm{C}\tilde{R}_{aZ}\tilde{R}_{Zb}/\varepsilon$ with $\varepsilon = 1 - v_\mathrm{C}\tilde{R}_{ZZ}$, built from the proper responses $\tilde{R}$. The proper response of rows a and b then has a phonon part and an electronic part,

$$\tilde{R}_{ab} = \frac{q^2}{\Omega}\,\boldsymbol{p}_a^\mathrm{T}(\omega^2 - D^\mathrm{an})^{-1}\boldsymbol{p}_b + \chi_{ab},$$

with $\boldsymbol{p}_a$ the $3n$-vector of the proper site couplings of row a divided by $\sqrt{M_\kappa}$ and projected on $\hat{\boldsymbol{q}}$, as in Eq. (11.13), and χ_{ab} the real response of the electrons at clamped nuclei. For the charge row, $\boldsymbol{p}_Z = \boldsymbol{p}$ is built from the Born charges and $v_\mathrm{C}\chi_{ZZ} = 1 - \varepsilon_\infty$. Write $x_a = (q^2/\Omega)\,\boldsymbol{p}_a^\mathrm{T}(\omega^2 - D^\mathrm{an})^{-1}\boldsymbol{p}$ and $x = x_Z$, so that $\varepsilon = \varepsilon_\infty - v_\mathrm{C}x$. The full phonon propagator $(\omega^2 - D^\mathrm{an} - \boldsymbol{b}\boldsymbol{b}^\mathrm{T})^{-1}$, with $\boldsymbol{b}\boldsymbol{b}^\mathrm{T}$ the nonanalytic term of Eq. (11.14), follows from the Sherman–Morrison identity, and between rows it gives $x_a x_b$-type terms:

$$\frac{q^2}{\Omega}\,\boldsymbol{p}_a^\mathrm{T}(\omega^2 - D^\mathrm{an} - \boldsymbol{b}\boldsymbol{b}^\mathrm{T})^{-1}\boldsymbol{p}_b = \frac{q^2}{\Omega}\,\boldsymbol{p}_a^\mathrm{T}(\omega^2 - D^\mathrm{an})^{-1}\boldsymbol{p}_b + \frac{v_\mathrm{C}\,x_a x_b}{\varepsilon_\infty - v_\mathrm{C}x}.$$

Insert $\tilde{R}_{aZ} = x_a + \chi_{aZ}$ into Eq. (8.36) and expand the product $(x_a + \chi_{aZ})(x_b + \chi_{bZ})$. The $x_a x_b$ term is the one just written. The terms linear in χ and the $\chi\chi$ term are reproduced by shifting each vertex by $\lambda_a\boldsymbol{p}$ with $\lambda_a = v_\mathrm{C}\chi_{aZ}/\varepsilon_\infty$, since $(q^2/\Omega)\boldsymbol{p}_a^\mathrm{T}(\omega^2 - D^\mathrm{an} - \boldsymbol{b}\boldsymbol{b}^\mathrm{T})^{-1}\boldsymbol{p} = \varepsilon_\infty x_a/\varepsilon$ by the same identity; what remains of the $\chi\chi$ term is a real constant. The result is exact within the rank-one form:

$$R_{ab} = \frac{q^2}{\Omega}\,\boldsymbol{p}_a'^\mathrm{T}(\omega^2 - D^\mathrm{an} - \boldsymbol{b}\boldsymbol{b}^\mathrm{T})^{-1}\boldsymbol{p}_b' + \chi_{ab} + \frac{v_\mathrm{C}\chi_{aZ}\chi_{Zb}}{\varepsilon_\infty}, \qquad \boldsymbol{p}_a' = \boldsymbol{p}_a + \frac{v_\mathrm{C}\chi_{aZ}}{\varepsilon_\infty}\,\boldsymbol{p}. \quad (11.54)$$

The transverse poles are gone, the phonons are those of the full dynamical matrix, and each row keeps its own vertex, shifted along the Born-charge vector by its electronic correlation with the charge. The constant is real and does not absorb. For the electron density, $n_Z = -n_e + \sum_\kappa Z_\kappa n_\kappa$ gives $\chi_{eZ} = -\chi_{ee} = -\chi_{ZZ}$, so that $v_\mathrm{C}\chi_{eZ} = \varepsilon_\infty - 1$, and the shift adds $(1 - 1/\varepsilon_\infty)Z_\kappa^*$ to the proper vertex $Z_\kappa \mathbb{1} - Z_\kappa^*$.

Site coupling of a density row at the longitudinal pole

A field that couples to $\sum_\kappa g_{n_\kappa}^\phi n_\kappa + g_{n_e}^\phi n_e$ creates the zone-center phonons of Eq. (11.53)

with the site couplings

$$\mathcal{G}^{\phi}_{\kappa} = g^{\phi}_{n_\kappa}\mathbb{1} + g^{\phi}_{n_e}\Big(Z_\kappa\mathbb{1} - \frac{Z^*_\kappa}{\varepsilon_\infty(\hat{\boldsymbol{p}}_\phi)}\Big), \tag{11.55}$$

for ω far below the gap, without local fields, and in the harmonic lattice. A longitudinal phonon carries $Z_\kappa - Z^*_\kappa/\varepsilon_\infty$ electrons with nucleus κ.

The total charge, the nucleon number and the electron density each test Eq. (11.55). For the total charge, $g^{\phi}_{n_e} = -\lambda$ and $g^{\phi}_{n_\kappa} = \lambda Z_\kappa$, and the nuclear charge cancels to leave $\lambda Z^*_\kappa/\varepsilon_\infty$, the screened Born charge of Eq. (11.15). For the nucleons, $g^{\phi}_{n_e} = 0$ and the vertex is the bare coefficient, unscreened, as Worked example 8.6 found. For the electrons alone the vertex is new, and its physical reading is short. At zero field the electrons that follow nucleus κ number $Z_\kappa - Z^*_\kappa$. In the longitudinal wave the macroscopic field is $-\boldsymbol{P}/\varepsilon_\infty$, the electrons polarize in it, and they move another $Z^*_\kappa(1 - 1/\varepsilon_\infty)$ along with the nucleus. Altogether the nucleus drags $Z_\kappa - Z^*_\kappa/\varepsilon_\infty$ electrons, its own charge minus its screened Born charge. Rigid ions of charge Q_κ, whose electron number is $Z_\kappa - Q_\kappa$, give the same structure with Q_κ in place of Z^*_κ.

Optical phonons do not move the center of mass, $\sum_\kappa \sqrt{M_\kappa}\,\boldsymbol{e}_{\kappa\nu}(0) = 0$, and by the coupling-to-mass cancellation of Sect. 11.6.1 the sum in Eq. (11.53) is unchanged when $\mathcal{G}^{\phi}_{\kappa} \to \mathcal{G}^{\phi}_{\kappa} - \lambda M_\kappa\mathbb{1}$ for any λ: a coupling proportional to the mass excites no optical phonon at all [32, Eq. (44)]. What Eq. (11.53) measures is how far the site couplings deviate from proportionality to the masses. For two atoms in the cell, with the longitudinal eigenvector of Sect. 11.2, $\boldsymbol{e}_1 = (M_2/M_{\text{cell}})^{1/2}\hat{\boldsymbol{p}}_\phi$ and $\boldsymbol{e}_2 = -(M_1/M_{\text{cell}})^{1/2}\hat{\boldsymbol{p}}_\phi$, the sum collapses to $p_\phi\,\mu_{\text{red}}^{1/2}\,d$ with the mass-weighted difference

$$d = \frac{\mathcal{G}^{\phi}_1}{M_1} - \frac{\mathcal{G}^{\phi}_2}{M_2}, \tag{11.56}$$

in a cubic crystal, where the tensors are scalars. Only the longitudinal mode couples, since $\boldsymbol{p}_\phi \cdot \boldsymbol{e}$ vanishes for the transverse ones.

The rate takes a compact form through the loss function. By Eq. (11.17) the longitudinal line of L has the weight $\pi e^2 Z^{*2}/(2\Omega\mu_{\text{red}}\varepsilon_\infty^2\omega_{\text{L}})$. Dividing Eq. (11.53) by it, averaging $p_\phi^2 = m_\phi^2 v^2$ over the halo and multiplying by $\rho_\chi/m_\phi\rho_T$,

$$\mathcal{R} = \frac{\rho_\chi}{\rho_T}\,\frac{\langle v^2\rangle}{e^2}\Big(\frac{\varepsilon_\infty\,\mu_{\text{red}}\,d}{Z^*}\Big)^2 L(m_\phi). \tag{11.57}$$

The form is more than a convenience. In the one-branch crystal, with the damping carried by the phonon propagator, Eq. (11.54) makes the full response of any density row proportional to $L(\omega)$ at every frequency, not only at the pole. The line shape is that of the loss function, peaked at ω_{L}, with no feature at ω_{T}. For a coupling proportional to the charge the bracket is one, and Eq. (11.57) is the photon-like rate of Mitridate and collaborators [32, Eq. (42)]. Compared with the dark photon below, it carries $\langle v^2\rangle$, since a scalar couples to a density, whose matrix elements vanish as $p_\phi \to 0$. For the standard halo of Sect. 1.2, $\langle v^2\rangle^{1/2} = 379\,\text{km/s}$. A photon-like scalar then reaches the dark-photon sensitivity at $g = \zeta e/\langle v^2\rangle^{1/2} = 240\,\zeta$; with $q = \omega v_0$ and $v_0 = 230\,\text{km/s}$ [32, Eq. (43)], the factor is 395.

For a scalar that couples to nucleon number the bracket is $\varepsilon_\infty\mu_{\text{red}}d_N/Z^*$, with d_N the difference of nucleon number per mass between the two species, 2.6×10^{-5} per u in GaAs (Worked example 8.6). The bracket is 4.8×10^{-3}, and the rate is 2×10^{-5} of the photon-like one. Masses are nearly proportional to nucleon numbers, and the coupling-to-mass cancellation is nearly complete. In a crystal of mixed isotopes the cancellation fails site by site, the translation symmetry that selects the zone center is broken, and the absorption spreads over the whole phonon band (Sect. 11.6).

A dark photon does not couple to a density. At rest it couples to the current, with any of three polarizations, and Sect. 8.4.5 found its rate exactly, $\Gamma_{\rm net} = \zeta^2 m_\phi L(m_\phi)$, the loss function again, now without the factor v^2. Why a transverse field excites the longitudinal phonon is the physics of the phonon polariton (Sect. 5.5). The photon of the crystal and its transverse optical phonon mix into two branches near $Q \sim \varepsilon_0^{1/2}\omega_{\rm T}$, about 0.12 eV in GaAs. The upper branch starts at $Q = 0$ at the zero of ε, which is $\omega_{\rm L}$; the lower one starts at zero frequency. The dark photon sits at $Q = p_\phi$, some 2600 times below the crossover, where the upper branch is flat at $\omega_{\rm L}$ and the distinction between transverse and longitudinal has disappeared: at $Q \to 0$ a cubic crystal has $\varepsilon^{\rm T} = \varepsilon$ (Eq. (4.52)). With the line of Eq. (11.20), the rate integrated over the mass is

$$\int \mathrm{d}m_\phi\, \mathcal{R} = \frac{\rho_\chi}{\rho_T}\, \zeta^2\, \mathcal{I}_{\rm pol}, \tag{11.58}$$

the band integral of Sect. 11.2, independent of the widths of the lines. Dark-photon absorption is single-phonon scattering through a dark photon, read at $q \to 0$.

Limit Check 11.18. Single-phonon absorption

A boson absorbed on optical phonons has the rate $\Gamma_{\rm abs} = (\pi/2\Omega m_\phi) \sum_\nu |\sum_\kappa \boldsymbol{p}_\phi \cdot \mathcal{G}^\phi_\kappa \cdot \boldsymbol{e}^*_{\kappa\nu}(0)/\sqrt{M_\kappa}|^2 \delta(m_\phi - \omega_\nu)/\omega_\nu$, Eq. (11.53) [32, Eqs. (2), (7), (14), (16)]. For a dark photon it is $\mathcal{R} = (\rho_\chi/\rho_T)\zeta^2\, \mathrm{Im}[-1/\varepsilon(m_\phi)]$ [14, Eqs. (2), (3)], and for scalars Eq. (11.57), with the rigid-ion site couplings of Ref. [32, Eqs. (40)–(44)] replaced by Eq. (11.55).

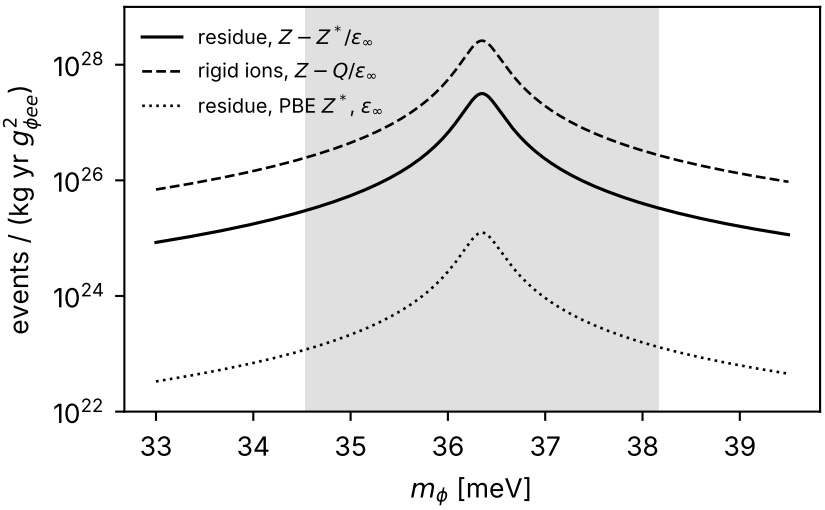


Figure 11.26: Absorption of a scalar coupled to electrons in GaAs, events per kilogram-year and per $g^2_{\phi ee}$, against m_ϕ, from Eq. (11.57). Solid: site couplings $Z_\kappa - Z^*_\kappa/\varepsilon_\infty$ with $Z^* = 2.11$ and $\varepsilon_\infty = 10.88$ (Worked example 5.6); a change of Z^*/ε_∞ by $\pm 1\,\%$ moves it by $\pm 9\,\%$, less than the width of the line on this scale. Dashed: the rigid-ion couplings $Z_\kappa - Q_\kappa/\varepsilon_\infty$ with $Q = \pm 3$, the structure of Ref. [32, Eqs. (40), (41)], 8.2 times higher. Dotted: site couplings $Z_\kappa - Z^*_\kappa/\varepsilon_\infty$ with the PBE values $Z^* = 2.094$ and $\varepsilon_\infty = 13.70$ of Sect. 5.5, 250 times lower. Loss function of one damped branch with $\omega_{\rm T} = 33.62$, $\omega_{\rm L} = 36.35$ meV and widths set to 1 % of each frequency [32, Sect. III A]; standard halo of Sect. 1.2, $\langle v^2\rangle^{1/2} = 379$ km/s, $\rho_T = 5.361\,\mathrm{g\,cm^{-3}}$. Gray: $|m_\phi - \omega_{\rm L}| < 5\gamma_{\rm L}$, where a line of constant width describes the phonon.

Where the mass-weighted difference d is small against each of its terms, the corrections

that Eq. (11.55) drops are no longer small against the rate, and the formula fixes only its order of magnitude.

11.8.2 The near-cancellation in GaAs

A scalar that couples to electrons through $g_{\phi ee}$ acts on the row $-g_{\phi ee} n_e$ (Eq. (8.11)), so that $g^{\phi}_{n_e} = -g_{\phi ee}$ and $g^{\phi}_{n_\kappa} = 0$, and by Eq. (11.55) each site contributes $Z_\kappa - Z^*_\kappa/\varepsilon_\infty$ electrons. GaAs is the obvious target, polar and well measured, and it turns out to be close to the worst case.

Worked example 11.9: The electron-coupled scalar in GaAs

We take the inputs of Sect. 11.2: $Z^* = 2.11$ from the splitting and the measured $\varepsilon_\infty = 10.88$, so that $Z^*/\varepsilon_\infty = 0.1939$, with $Z^*_{\mathrm{Ga}} = +Z^*$ and $Z^*_{\mathrm{As}} = -Z^*$; atomic numbers 31 and 33; mean atomic masses 69.723 u and 74.9216 u; $\mu_{\mathrm{red}} = 36.11$ u.
Site couplings. Gallium carries $31 - 0.1939 = 30.806$ electrons and arsenic $33 + 0.1939 = 33.194$, and each site couples with $-g_{\phi ee}$ times its count.
Mass-weighted difference. Split Eq. (11.56) into its two parts. The nuclear charges per unit mass are $Z_{\mathrm{Ga}}/M_{\mathrm{Ga}} = 0.44462$ and $Z_{\mathrm{As}}/M_{\mathrm{As}} = 0.44046$ per u, and their difference is $+4.16 \times 10^{-3}$. The Born charges contribute $-(Z^*/\varepsilon_\infty)(1/M_{\mathrm{Ga}} + 1/M_{\mathrm{As}}) = -0.1939 \times 0.02769 = -5.37 \times 10^{-3}$. The sum is $d = -1.21 \times 10^{-3}$ per u, times $-g_{\phi ee}$. The two terms are 3.4 and 4.4 times larger than their sum.
Rate. The bracket of Eq. (11.57) is $\varepsilon_\infty \mu_{\mathrm{red}} d/Z^* = 10.88 \times 36.11 \times 1.214 \times 10^{-3}/2.11 = 0.226$, and its square is 0.051. An electron-coupled scalar is absorbed in GaAs at 5 % of the rate of a photon-like scalar of the same coupling.
Sensitivity. Since the Born-charge term is 4.4 times $|d|$, a fractional change of Z^*/ε_∞ changes d 4.4 times as much and the rate 8.8 times as much. The splitting route of Sect. 11.2.4 fixes Z^*/ε_∞ to 1.0 %, and these inputs limit the rate to 9 %. Exact cancellation would need $Z^*/\varepsilon_\infty = 0.1501$. The values $Z^* = 2.16$ and $\varepsilon_\infty = 10.9$, also found in the literature, would give $d = -1.33 \times 10^{-3}$, a rate 20 % higher.
Rigid ions. With the formal charges $Q = \pm 3$ in place of Z^* [32, Eqs. (40), (41)], gallium carries $31 - 3/10.88 = 30.724$ electrons and arsenic 33.276, $d = -3.48 \times 10^{-3}$ per u, and the rate is 8.2 times larger. For Z^* between 2.09 and 2.16 the factor runs from 9.0 to 6.7. The rigid ion changes the electrons each site drags by 0.08 out of 31, but the rate measures the small difference between two sites.

The worked example is the reason for the caution in Limit Check 11.18. Mitridate and collaborators state the rigid-ion assumption and give the Born-charge fix for the photon coupling [32, footnote 2]. Their GaAs projection, however, takes the electron numbers 28 and 36 for the two sites [32, text after Eq. (44)]. These are the zero-field counts $Z_\kappa - Q_\kappa$ of rigid ions with $Q = \pm 3$. In the longitudinal wave the macroscopic field moves another $Q_\kappa(1 - 1/\varepsilon_\infty)$ electrons with each nucleus, as for Eq. (11.55). The counts that enter d are therefore $Z_\kappa - Q_\kappa/\varepsilon_\infty = 30.724$ and 33.276, those of Worked example 11.9; inserted literally into Eq. (11.56), 28 and 36 would give $d = -0.079$ per u and a rate thousands of times larger. With the screened rigid-ion counts the rate is high by the factor 7 to 9 of Worked example 11.9, and the reach in $g_{\phi ee}$ by about 3. At the line center, with widths of 1 %, three events in a kilogram-year need $g_{\phi ee} = 3.1 \times 10^{-14}$, against 1.1×10^{-14} with rigid ions. The qualitative conclusion survives, and is strengthened: III–V crystals, whose neighbors in the periodic table have nearly equal charge per mass, suppress the absorption of an electron-coupled scalar.

The kinetic energy of the core electrons, the isotopic masses and the frequency dependence of the electronic response are left out of Eq. (11.55); each is small in absolute terms but not against d. The scalar density of an electron is its number density times $1 - v^2/2$ (Eq. (8.11)), and the core electrons that ride with a nucleus move fast. Their kinetic-energy row enters through the same factor $\mathrm{i}\boldsymbol{p}_\phi \cdot \boldsymbol{u}_\kappa$ as their density, and it reduces the electrons each site carries by $\langle \mathcal{T} \rangle_\kappa / m_e$, the kinetic energy of those electrons over the electron mass. By the virial theorem this is the binding energy of the atom over m_e, tens of keV over 511 keV, of order 0.1 for gallium and arsenic. As a parametric estimate, its mass-weighted difference between the two sites is $\Delta d \approx +1.2 \times 10^{-4}$ per u, a tenth of d and of the opposite sign, so that it lowers $|d|$ by about 10 % and the rate by about 19 %. No calculation has included it yet, and Sect. 11.9 carries it as a nuisance. The masses enter at the same level: 0.01 u on the mean mass of gallium moves d by 5 %, so an isotopically enriched crystal must use its own masses. The frequency dependence of the electronic response, of relative order $(\omega_\mathrm{L}/E_\mathrm{g})^2 = 6 \times 10^{-4}$ (Sect. 11.2), is amplified by 4.4 to 0.3 % of d and is negligible.

In practice the inputs come from Appendix E.4. Density-functional perturbation theory delivers Z^* robustly, to a percent, and ε_∞ poorly. With the PBE values of Sect. 5.5, $Z^* = 2.094$ and $\varepsilon_\infty = 13.70$, the ratio $Z^*/\varepsilon_\infty = 0.153$ lies within 2 % of exact cancellation, $d = -7.6 \times 10^{-5}$ per u, and the rate falls by a factor of 250 (the dotted curve of Limit Check 11.18). The test of the computed values against the measured splitting, Eq. (E.25), matters more here than any convergence parameter, and the measured splitting with Eq. (11.23) is the safer input; Appendix E.17 specifies the calculation, the masses and the phonon widths, which three-phonon perturbation theory supplies (Sect. 5.7).

11.8.3 Dark photons in GaAs and sapphire

For a dark photon the vertex is the screened Born charge, there is no cancellation, and the rate is the measured or computed loss function, $\mathcal{R} = (\rho_\chi/\rho_T)\zeta^2 L(m_\phi)$. Two numbers characterize each line: its area, which is fixed by the band integral Eq. (11.58), and its width, which is not. For GaAs, $\mathcal{I}_\mathrm{pol} = 0.759\,\mathrm{meV}$ from the splitting (Worked example 11.3), and a Lorentzian of full width γ_L peaks at $L = 2\mathcal{I}_\mathrm{pol}/\pi\gamma_\mathrm{L}$. A width of 1 % of the frequency leads to $L = 1.33$ at the center, ten times narrower gives 13.3, and the area stays at 0.76 meV. The reach on the line center scales as $\gamma_\mathrm{L}^{1/2}$, while the reach averaged over a scan of masses depends only on the area.

Fig. 11.27 shows both crystals. GaAs has one line, and three events in a kilogram-year reach $\zeta = 2.9 \times 10^{-17}$ at its center. Sapphire has six polar branches between the two polarizations. The strongest is the collective top branch of the ordinary ray, at 112.4 meV, which carries 95 % of the ordinary weight (Sect. 11.2.3). With its measured 77 K width of 2.0 meV it reaches $\zeta = 1.0 \times 10^{-17}$, and the lines at 59.7 and 63.3 meV reach 2×10^{-17}. Per unit mass the band integral of sapphire is 64 times that of GaAs, $35.9\,\mathrm{meV}/3.98\,\mathrm{g\,cm^{-3}}$ against $0.759\,\mathrm{meV}/5.361\,\mathrm{g\,cm^{-3}}$; here 35.9 meV is the polarization average $\frac{2}{3}$ ordinary $+\frac{1}{3}$ extraordinary of the absorbed field, while the ordinary ray alone gives the 36.3 meV of Sect. 11.2.3. The same per-mass band integral sets the rate of scattering through a light dark photon (Sect. 11.2.3). The widths are the weak input. Those of sapphire were measured at 77 K, and at millikelvin temperatures the anharmonic part, the thermal part of Eq. (5.70), shrinks while the spontaneous decay stays, so the lines should narrow by a factor of order one [11]. Between the lines the absorption is multiphonon, which the measured ε contains and the factorized model does not, and it fills the gaps of Fig. 11.27b [11, 14].

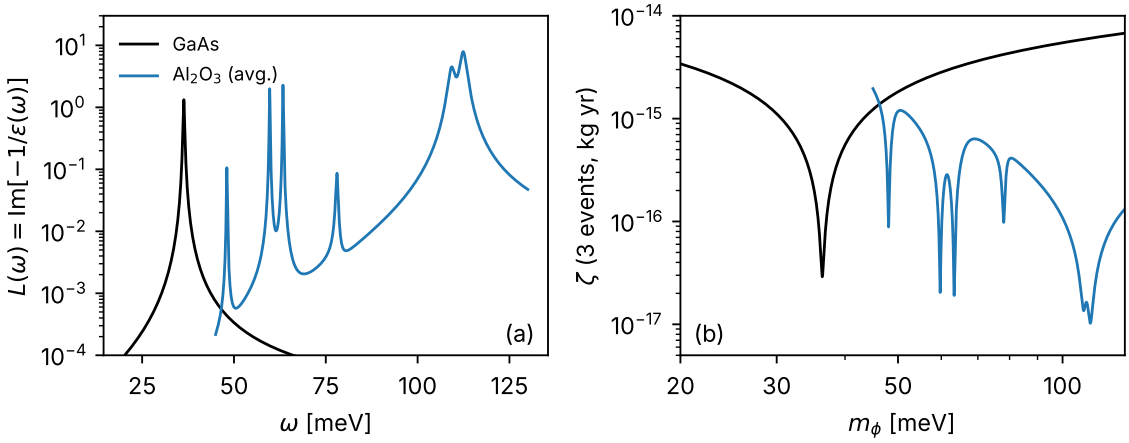


Figure 11.27: Absorption of dark-photon dark matter on single optical phonons. (a) Loss function $L = \mathrm{Im}[-1/\varepsilon(\omega)]$ at $q \to 0$ from the damped factorized dielectric function $\varepsilon_\infty \prod_\nu (\omega_{\mathrm{L}\nu}^2 - \omega^2 - \mathrm{i}\omega\gamma_{\mathrm{L}\nu})/(\omega_{\mathrm{T}\nu}^2 - \omega^2 - \mathrm{i}\omega\gamma_{\mathrm{T}\nu})$. GaAs (black): $\omega_{\mathrm{T}} = 33.62$, $\omega_{\mathrm{L}} = 36.35\,\mathrm{meV}$, $\varepsilon_\infty = 10.88$, so that the line carries the splitting weight $(1 - \omega_{\mathrm{T}}^2/\omega_{\mathrm{L}}^2)/\varepsilon_\infty = 0.0133$ (Sect. 11.2), with an assumed width of 1 % of each frequency. Sapphire (blue): frequencies and 77 K widths of Ref. [11, Table II], ordinary ray, longitudinal 112.4, 78.0, 59.7 and 48.1 meV with widths 1.98, 0.55, 0.17 and 0.17 meV, transverse 78.6, 70.5, 54.4 and 47.7 meV with widths 0.47, 0.40, 0.19 and 0.17 meV; extraordinary ray, longitudinal 109.2 and 63.3 meV with widths 1.98 and 0.17 meV, transverse 72.2 and 49.3 meV with widths 0.40 and 0.19 meV, the extraordinary widths taken from the ordinary ray as there, and $\varepsilon_\infty = 3.2$; averaged over polarizations as $\frac{2}{3}L_{\mathrm{o}} + \frac{1}{3}L_{\mathrm{e}}$ [11, Eq. (40)]; drawn above 45 meV, since below the lowest transverse line the model gives $|L| < 10^{-4}$ of the wrong sign. (b) Kinetic mixing ζ for three events in one kilogram-year without background, $\mathcal{R} = (\rho_\chi/\rho_T)\zeta^2 L(m_\phi)$, with $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$ and $\rho_T = 5.361$ and $3.98\,\mathrm{g\,cm^{-3}}$. Single-phonon lines only: the multiphonon absorption between them and the electronic absorption above the gap are not included.

11.8.4 Acoustic modes of a finite body

The lowest thickness mode of a quartz plate 1 mm thick rings at about 3 MHz, an energy of 12 neV. A dark photon of that mass has a de Broglie wavelength of order 100 km and stays coherent for a fraction of a second, $t_{\rm coh} = 1/m_\phi v_0^2$, which is 1.04 ms at 1 μeV and scales as $1/m_\phi$. Even the wavelength of light at its frequency, $2\pi/m_\phi$, is 100 m. Over the plate and over many periods the field is one classical wave, $\phi \propto \cos(m_\phi t)$, uniform in space. Crystal momentum no longer selects anything: a body small against the wavelength has no translation symmetry at that scale, and the role of $\boldsymbol{p}_\phi$ passes to the shape of the normal modes of the body. The acoustic modes, which could not absorb in an infinite crystal, now can, because a finite body has acoustic modes at nonzero frequency at zero wavevector of the field.

The field inside the body. A dark photon at rest acts on charges as an electric field $\zeta\boldsymbol{E}'$, with $|\boldsymbol{E}'| = m_\phi|A'| = (2\rho_\chi)^{1/2}$ since $\rho_\chi = m_\phi^2|A'|^2/2$. In an infinite medium the induced polarization screens it to $\zeta\boldsymbol{E}'/\varepsilon$, the in-medium mixing of Eq. (8.39). A body of size $L_{\rm s} \ll 2\pi/m_\phi$ is in the quasi-static regime, and its surface charges set the field instead. Take an ellipsoid with depolarization factor $N_{\rm dp}$ along the field. The induced polarization $(\varepsilon_0 - 1)\boldsymbol{E}_{\rm in}$ produces the depolarizing field $-N_{\rm dp}(\varepsilon_0 - 1)\boldsymbol{E}_{\rm in}$, so that $\boldsymbol{E}_{\rm in} = \zeta\boldsymbol{E}' - N_{\rm dp}(\varepsilon_0 - 1)\boldsymbol{E}_{\rm in}$, or

$$\boldsymbol{E}_{\rm in} = \frac{\zeta\boldsymbol{E}'}{1 + N_{\rm dp}(\varepsilon_0 - 1)} = f_{\rm sh}\,\frac{\zeta\boldsymbol{E}'}{\varepsilon_0}, \qquad f_{\rm sh} = 1,\ \ \frac{3\varepsilon_0}{\varepsilon_0 + 2},\ \ \varepsilon_0, \tag{11.59}$$

for a field normal to a slab ($N_{\rm dp} = 1$), in a sphere ($N_{\rm dp} = 1/3$) and parallel to a slab ($N_{\rm dp} = 0$). The static constant ε_0 appears because MHz frequencies lie far below the optical phonons. The slab with the field along its normal coincides with the bulk, and a field along a thin plate is not screened at all. A conducting shield of size $R_{\rm s}$ around the experiment suppresses the field further by $\min\{1, (m_\phi R_{\rm s})^2\}$ [41, Eq. (5)].

The coupling. In a piezoelectric crystal a strain $S_{\beta\gamma}$ polarizes the cell, $P_\alpha = \sum_{\beta\gamma} e^{\rm pz}_{\alpha\beta\gamma} S_{\beta\gamma}$ (Sect. 6.2), and the field couples through $\delta H = -\int \mathrm{d}^3x\, \boldsymbol{E}_{\rm in}\cdot\boldsymbol{P}$. We take a plate cut so that its axis z carries the dominant element, $e^{\rm pz} \equiv e^{\rm pz}_{zzz}$, and a field component $E_{\rm in}\cos\theta$ along z, with θ the angle between the polarization of the dark photon and the axis. Then $\delta H = -E_{\rm in}\cos\theta\,\cos(m_\phi t)\, e^{\rm pz}\int \mathrm{d}^3x\,\partial_z u_z$.

The modes. The displacement of the body is $\boldsymbol{u}(\boldsymbol{x}) = \sum_n \boldsymbol{U}_n(\boldsymbol{x})(b_n + b_n^\dagger)$, with the mode functions normalized as $\int \mathrm{d}^3x\,\rho_T\,\boldsymbol{U}_n\cdot\boldsymbol{U}_{n'} = \delta_{nn'}/2\omega_n$, the finite-body form of Eq. (5.32). A plano-convex plate of central thickness L_0 traps its thickness modes near the axis. Across it they have a Gaussian profile of radius r_n [41, Eqs. (2), (3)],

$$\boldsymbol{U}_n = \frac{\hat{\boldsymbol{z}}}{\sqrt{M_n\omega_n}}\,\cos\frac{n\pi z}{L_0}\,\exp\Big(-\frac{x^2+y^2}{2r_n^2}\Big), \qquad M_n = \rho_T\pi r_n^2 L_0, \qquad \omega_n = \frac{n\pi c_{\rm L}}{L_0},$$

which satisfies the normalization, since the cosine squared averages to one half and the Gaussian squared integrates to πr_n^2; M_n is the mass of the mode and $c_{\rm L}$ the longitudinal sound speed of the plate. The polarization integral is

$$\int \mathrm{d}^3x\,\partial_z U_{n,z} = \frac{\cos n\pi - 1}{\sqrt{M_n\omega_n}}\int \mathrm{d}x\,\mathrm{d}y\,\mathrm{e}^{-(x^2+y^2)/2r_n^2} = -\frac{4\pi r_n^2}{\sqrt{M_n\omega_n}},$$

for odd n, and zero for even n, whose two faces move together and leave no net polarization.

The power. Write $\cos(m_\phi t) = (\mathrm{e}^{-\mathrm{i}m_\phi t} + \mathrm{e}^{\mathrm{i}m_\phi t})/2$ and keep the part $\delta H_0\,\mathrm{e}^{-\mathrm{i}m_\phi t}$ that absorbs. With $E_0 = f_{\rm sh}\zeta(2\rho_\chi)^{1/2}/\varepsilon_0$ the amplitude of $E_{\rm in}$ and $\pi r_n^2 = M_n/\rho_T L_0$,

$$|\langle n|\delta H_0|0\rangle|^2 = \Big(\frac{E_0\cos\theta\, e^{\rm pz}}{2}\Big)^2\frac{16\pi^2 r_n^4}{M_n\omega_n} = \frac{8 f_{\rm sh}^2\zeta^2\rho_\chi (e^{\rm pz})^2 M_n\cos^2\theta}{\varepsilon_0^2\rho_T^2 L_0^2\,\omega_n}.$$

The golden rule gives the absorbed power $m_\phi \cdot 2\pi|\langle n|\delta H_0|0\rangle|^2\delta(m_\phi-\omega_n)$. A mode of full width γ_n replaces the delta function by the line $(2\omega_n^2\gamma_n/\pi)/[(m_\phi^2-\omega_n^2)^2+\omega_n^2\gamma_n^2]$, which near resonance is the normalized Lorentzian, and

$$P_n = 32\, f_{\rm sh}^2\zeta^2\,\frac{\rho_\chi}{\rho_T^2}\,\frac{(e^{\rm pz})^2}{\varepsilon_0^2}\,\frac{M_n}{L_0^2}\,\frac{m_\phi\,\omega_n\,\gamma_n\cos^2\theta}{(m_\phi^2-\omega_n^2)^2+\omega_n^2\gamma_n^2}. \tag{11.60}$$

The prefactor $32 = 2\pi\times 8\times 2/\pi$ collects the golden rule, the squared matrix element and the line. On resonance P_n is proportional to $1/\gamma_n = \mathfrak{Q}/\omega_n$, with $\mathfrak{Q}$ the quality factor of the line. The power grows with the mass of the mode and falls as $1/L_0^2$, since a thinner plate turns the same displacement into more strain.

The field's own width. The halo field is not monochromatic. Its energies are spread over $\omega = m_\phi(1+v^2/2)$, with the distribution $F_\phi(\omega)\,{\rm d}\omega = f_{\rm sp}(v)\,{\rm d}v$ inherited from the speed distribution $f_{\rm sp}$. A record long compared with $\min(t_{\rm coh}, 1/\gamma_n)$ absorbs the mean power of the spread field (Sect. 8.2.3). That is Eq. (11.60) averaged over F_ϕ, with $\gamma_n = \omega_n/\mathfrak{Q}_{\rm p}$ the width of the mode alone. On resonance this is Eq. (11.60) with $\mathfrak{Q}$ replaced by

$$\mathfrak{Q}_{\rm eff} = \omega_n \int {\rm d}\omega\, F_\phi(\omega)\,\frac{\omega\,\omega_n\gamma_n}{(\omega^2-\omega_n^2)^2+\omega_n^2\gamma_n^2}. \tag{11.61}$$

A broad mode, $\mathfrak{Q}_{\rm p}$ small, sees the field as a single line and $\mathfrak{Q}_{\rm eff} = \mathfrak{Q}_{\rm p}$. A narrow mode sees the field's spread instead: its line integrates to $\pi/2$ times a delta function, and

$$\mathfrak{Q}_{\rm eff} \to \mathfrak{Q}_{\rm DM} = \frac{\pi}{2}\,\omega_n F_\phi(\omega_n) = \frac{\pi}{2}\,\Big|\frac{{\rm d}\eta}{{\rm d}v}\Big|, \qquad v = \Big(\frac{2(\omega_n-m_\phi)}{m_\phi}\Big)^{1/2}, \tag{11.62}$$

where the second form uses $F_\phi = f_{\rm sp}/(m_\phi v)$ and $\eta(v_{\rm min}) = \int_{v_{\rm min}} {\rm d}v\, f_{\rm sp}(v)/v$ (Eq. (1.28)). For the standard halo of Sect. 1.2 it peaks at $\mathfrak{Q}_{\rm DM} = 1.34\times 10^6$, for a mode tuned to $v = 256\,{\rm km/s}$, that is $\omega_n = m_\phi(1+3.6\times 10^{-7})$; through η it follows the annual modulation of the Earth's velocity. Trickle writes the signal with $\min\{\mathfrak{Q}_{\rm p}, \mathfrak{Q}_{\rm DM}\}$ and estimates $\mathfrak{Q}_{\rm DM}\sim 10^6$ [41, text after Eq. (7)], which Eq. (11.62) makes precise.

Limit Check 11.19. Bulk acoustic resonator

A plano-convex piezoelectric resonator absorbs from a dark-photon field the power Eq. (11.60), $P_n = 32\zeta^2(\rho_\chi/\rho_T^2)[(e^{\rm pz})^2/\varepsilon_0^2](M_n/L_0^2)\cos^2\theta\,\mathfrak{Q}_{\rm s}/m_\phi$ on resonance, with $\mathfrak{Q}_{\rm s} = \min\{\mathfrak{Q}_{\rm p}, \mathfrak{Q}_{\rm DM}\}$ [41, Eqs. (5), (7), (8)], for $f_{\rm sh} = 1$.

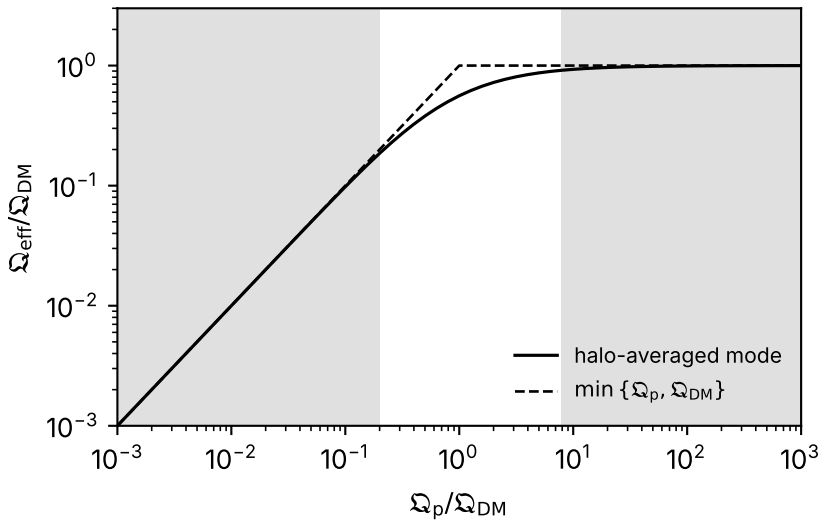


Figure 11.28: Effective quality factor of a resonator mode in the halo field, Eq. (11.61), in units of $\mathfrak{Q}_{\rm DM} = 1.34 \times 10^6$, against the quality factor of the mode, $\mathfrak{Q}_{\rm p}/\mathfrak{Q}_{\rm DM}$, with the mode tuned to the frequency that maximizes it. Solid: Eq. (11.61) with F_ϕ from the speed distribution of the standard halo of Sect. 1.2; dashed: $\min\{\mathfrak{Q}_{\rm p}, \mathfrak{Q}_{\rm DM}\}$. Gray: $\mathfrak{Q}_{\rm p}/\mathfrak{Q}_{\rm DM} < 0.2$ and > 7.9, where the two agree within 10 %. At $\mathfrak{Q}_{\rm p} = \mathfrak{Q}_{\rm DM}$ the prescription is 1.8 times too high.

Where the widths of the mode and of the field are comparable, the signal is their convolution and neither width alone sets it; for records shorter than the coherence time the mean power is not a rate at all.

The inputs that Eq. (11.60) needs are material and geometric. The elastic and piezoelectric tensors come from density-functional perturbation theory (Appendix E.4), and the shapes and frequencies of the modes of the actual body from finite elements. Add the shape factor $f_{\rm sh}$ of Eq. (11.59) and the measured quality factor of the mode, which for acoustic resonators of this kind lies between 10^6 and 10^{10} [41]. Appendix E.26 specifies them. A mode with $\mathfrak{Q}_{\rm p}$ above 10^7 is in the narrow regime, where the signal is set by the field, not by the resonator. Its ring-down time, $\mathfrak{Q}_{\rm p}/\omega_n$, then exceeds the coherence time of the field, and the mean power of Eq. (11.61) is reached only after many coherence times. How the signal builds up over shorter records, and how it fluctuates, is the coherent regime of Sect. 14.6.

11.8.5 Resonant-mass detectors

At masses of a peV to a neV the scalar of Sect. 11.8.2 does something else. Its coupling $g_{\phi ee}\phi\bar\psi\psi$ adds $-g_{\phi ee}\phi$ to the electron mass, and with $\phi = \phi_0\cos(m_\phi t)$, $\phi_0 = (2\rho_\chi)^{1/2}/m_\phi$, the electron mass oscillates, $\delta m_e/m_e = -g_{\phi ee}\phi/m_e$. A scalar coupled to $F_{\mu\nu}F^{\mu\nu}$ makes the fine-structure constant oscillate in the same way. Arvanitaki, Dimopoulos and Van Tilburg write both with dimensionless couplings normalized to the strength of gravity [42, Eqs. (1), (3), (4)]. Such a field changes the size of every atom. The Coulomb Hamiltonian of electrons and clamped nuclei, Eq. (2.48) with $M_\kappa \to \infty$, contains α and m_e only through the Bohr radius $a_0 = 1/(\alpha m_e)$, which sets every length, and $\alpha^2 m_e$, which sets every energy. The lattice

constant is a fixed number of Bohr radii, and

$$h_{\mathrm{mod}} \equiv \frac{\delta a}{a} = -\frac{\delta\alpha}{\alpha} - \frac{\delta m_e}{m_e}. \tag{11.63}$$

The scaling fails only through physics that the Coulomb Hamiltonian leaves out. Relativistic corrections are of relative order $(Z\alpha)^2$ for the inner electrons. They add to $\mathrm{d}\ln a/\mathrm{d}\ln\alpha$ a term that fully relativistic calculations put at -1.97×10^{-2} for copper and -2.2×10^{-4} for silicon; the finite nuclear mass adds a dependence on the proton-to-electron mass ratio, -1.12×10^{-3} per unit fractional change for copper and -0.86×10^{-3} for silicon [43, Table II]. For a light crystal Eq. (11.63) holds to about 10^{-3}.

A resonant-mass detector reads out such a change. It is a massive elastic body, a cylindrical bar or a sphere, of low internal loss, whose lowest mode is watched by a transducer at its own frequency: a mechanical resonator, read at its mode frequency. The detectors were built for gravitational waves, and the cryogenic ton-scale aluminum bars of the last generation ring near 900 Hz [42], an energy of 3.7 peV.

Why should an oscillating size excite a mode at all? A static change of a excites nothing. The body simply takes its new equilibrium shape, with every material point $\boldsymbol{\xi}$ moved to $(1+h_{\mathrm{mod}})\boldsymbol{\xi}$, and there is no wave. Write the displacement in the laboratory as $\boldsymbol{u}_{\mathrm{lab}} = \boldsymbol{u} + h_{\mathrm{mod}}(t)\,\boldsymbol{\xi}$, with $\boldsymbol{u}$ the displacement from the instantaneous equilibrium. The elastic stress responds to the strain relative to that equilibrium, $\sigma = C:(\nabla\boldsymbol{u}_{\mathrm{lab}} - h_{\mathrm{mod}}\mathbb{1}) = C:\nabla\boldsymbol{u}$, for a uniform h_{mod} in a cubic or isotropic body, with C the elastic tensor. Newton's law holds in the laboratory, $\rho_T\ddot{\boldsymbol{u}}_{\mathrm{lab}} = \nabla\cdot\sigma$, and

$$\rho_T\,\ddot{\boldsymbol{u}} = \nabla\cdot(C:\nabla\boldsymbol{u}) - \rho_T\,\ddot{h}_{\mathrm{mod}}\,\boldsymbol{\xi}. \tag{11.64}$$

The scalar acts as a body force $-\rho_T\ddot{h}_{\mathrm{mod}}\boldsymbol{\xi}$, the inertial force of the moving equilibrium. It is proportional to $\omega^2 h_{\mathrm{mod}}$: the stiffness of the body never enters, and a slow modulation is followed adiabatically. Projected on a mode $\boldsymbol{u}_n$ of the body, the force is $F_n = -\ddot{h}_{\mathrm{mod}}\int \mathrm{d}^3\xi\,\rho_T\,\boldsymbol{\xi}\cdot\boldsymbol{u}_n$. For a sphere of radius R_{sph} and mass M_{sph} only the radial modes $\boldsymbol{u}_n = \hat{\boldsymbol{r}}u_n(r)$ couple, and $F_n = -\ddot{h}_{\mathrm{mod}}M_{\mathrm{sph}}R_{\mathrm{sph}}J_n$ with $J_n = 3R_{\mathrm{sph}}^{-4}\int_0^{R_{\mathrm{sph}}}\mathrm{d}r\,r^3u_n(r)$, which Arvanitaki and collaborators obtain by treating the modulus as a scalar gravitational wave with the curvature $\delta_{ij}\ddot{h}$ [42, Eqs. (7), (8)].

A thin bar of length L_{b} and sound speed c_{b}, free at both ends, shows the whole response in closed form. Along the bar, Eq. (11.64) with $h_{\mathrm{mod}}\mathrm{e}^{-\mathrm{i}\omega t}$ has the solution $u_{\mathrm{lab}} = h_{\mathrm{mod}}\sin(k\xi)/[k\cos(kL_{\mathrm{b}}/2)]$, $k=\omega/c_{\mathrm{b}}$, which satisfies the free-end condition $\partial_\xi u_{\mathrm{lab}} = h_{\mathrm{mod}}$ at $\xi = \pm L_{\mathrm{b}}/2$. Measured from the instantaneous equilibrium, the end moves by

$$u(L_{\mathrm{b}}/2) = \frac{h_{\mathrm{mod}}L_{\mathrm{b}}}{2}\Big(\frac{\tan x}{x} - 1\Big), \qquad x = \frac{\omega L_{\mathrm{b}}}{2c_{\mathrm{b}}} = \frac{\pi}{2}\,\frac{\omega}{\omega_1}, \tag{11.65}$$

with $\omega_1 = \pi c_{\mathrm{b}}/L_{\mathrm{b}}$ the lowest mode that couples. Below it, $\tan x/x - 1 \simeq x^2/3$, and the response is $(\pi^2/24)\,h_{\mathrm{mod}}L_{\mathrm{b}}(\omega/\omega_1)^2$, the ω^2 of the drive. At each odd mode it resonates. Far above, the end stays where it was in the laboratory while the equilibrium moves, and the relative displacement tends to $-h_{\mathrm{mod}}L_{\mathrm{b}}/2$. The lowest mode alone, $u_1 = \sin(\pi\xi/L_{\mathrm{b}})$ with mass $M_{\mathrm{b}}/2$ and force $2\omega^2 h_{\mathrm{mod}}M_{\mathrm{b}}L_{\mathrm{b}}/\pi^2$, gives $(4/\pi^2)h_{\mathrm{mod}}L_{\mathrm{b}}(\omega/\omega_1)^2/(1-\omega^2/\omega_1^2)$; its low-frequency coefficient, $4/\pi^2 = 0.405$, falls short of $\pi^2/24 = 0.411$ by the higher odd modes, whose shares $4/\pi^2 n^4$ sum to the difference.

Limit Check 11.20. Resonant-mass drive of a scalar

A scalar that changes α or m_e strains a body by $h_{\rm mod} = -\delta\alpha/\alpha - \delta m_e/m_e$ and drives its modes with the force $F_n = -\ddot{h}_{\rm mod} M_{\rm sph} R_{\rm sph} J_n$, proportional to ω^2 [42, Eqs. (6)–(8)].

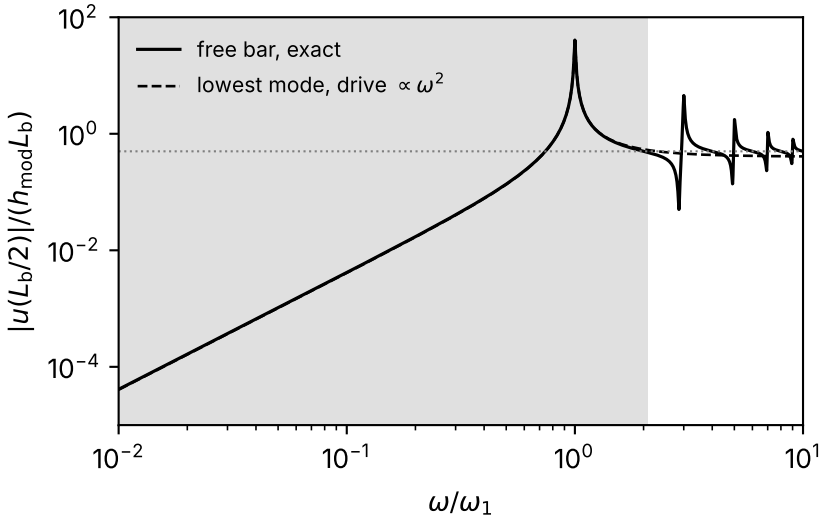


Figure 11.29: Displacement of the end of a free bar from its instantaneous equilibrium, in units of $h_{\rm mod} L_{\rm b}$, against ω/ω_1. Solid: the exact response Eq. (11.65); dashed: the lowest mode alone with the ω^2 drive. Every mode has the width $\omega_n/\mathfrak{Q}$ with $\mathfrak{Q} = 100$ for visibility (resonant-mass detectors reach 10^6 to 10^7). Dotted: the high-frequency limit 1/2. Gray: $\omega < 2.1\,\omega_1$, where the lowest mode is within 10 % of the exact response. The field is uniform over the bar while $m_\phi v L_{\rm b} \ll 1$. With $v = 10^{-3}$ and $c_{\rm b} = 5\,{\rm km/s}$ this holds up to $\omega \approx 2 \times 10^7 \omega_1$, far off scale.

Above the lowest mode the higher modes and their antiresonances take over, and the response approaches the rigid limit, in which the body no longer follows its equilibrium.

Between the optical-phonon absorption at tens of meV and the response of a whole bar at a peV, the scalar field of Sect. 11.8.2 is classical, and over the time it stays coherent a resonator responds to its amplitude rather than to a rate. That regime, in which the golden rule gives way to the coherent drive, is the subject of Sect. 14.6.

11.9 Nuisance parameters of the lattice rates

A dark-matter particle of 30 MeV with a heavy mediator and $\sigma_p = 10^{-40}\,{\rm cm}^2$ makes 5.26×10^3 events per kilogram-year in silicon above a threshold of 100 meV. The number assumes a harmonic crystal, nuclei that scatter independently and the phonons of the PBE force constants of Sect. 5.2. The free nuclear recoil of the standard rate gives 4.91×10^3 for the same threshold, and above 400 meV it gives a sixteenth of the crystal rate. That difference is not an uncertainty. The crystal response is the better calculation, and the free recoil is its limit at large x_R (Sect. 11.4). The uncertainty is how far the crystal number moves when the truncations of this chapter are lifted: the dropped pairs of nuclei, the harmonic potential, the computed phonon frequencies. Sect. 8.8 turned each truncation into a nuisance parameter ϑ_k with the next order as its template. For the silicon rate at 30 MeV the band is a quarter of a percent, set by the phonon frequencies, against a factor of 1.48 for the electronic rate at 3 MeV (Sect. 10.10).

11.9.1 Truncations of the nuclear response

The incoherent approximation is the first truncation of the multiphonon response, and its next order is computed. Eq. (11.29) gives the exact ω-integrated weight of each phonon order as a static sum over pairs of nuclei, and $\delta_n^{\mathrm{coh}}(\boldsymbol{q})$ is the template. In silicon the two- and three-phonon weights stay within 10 % of the incoherent ones above 6.6 keV, the one-phonon weight only above $7.9\,q_{\mathrm{BZ}}$, and the two-phonon weight is within 35 % at every q (Limit Check 11.8). A rate code multiplies each order by $1+\delta_n^{\mathrm{coh}}$, so this is the first of the three cases of Sect. 8.8: the central value moves, and the nuisance that remains is the step after it. That step is the distribution of the coherent weight in ω inside each band, which Eq. (11.30) bounds at every point but does not fix. Its bound is loose: at 10 keV the upper end reaches 2.7 times the incoherent two-phonon spectrum. The residual is a tilt within each band, and it matters for a threshold inside a band at momentum transfers below a few q_{BZ}, where Appendix E.17 computes the coherent response pointwise. Far above the zone the template dies quickly with the phonon number, because each order carries one more power of the correlation between neighbors, 0.44 along a bond. At 30 MeV it raises the rate above 100 meV by 2.6×10^{-4}.

Anharmonicity is the second truncation, and by time reversal its first nonvanishing order is $n_A=2$ (Sect. 11.5). Above the impulse onset the zeroth and first moments of the recoil spectrum are exact on any energy surface, and the template is the constrained tilt Eq. (8.77), which keeps them. The second moment carries the kinetic energy of the nucleus along $\boldsymbol{q}$, which the cubic and quartic terms change by at most ±0.2 % in single-site models with the silicon anharmonic parameters. The third carries the curvature of the Born–Oppenheimer surface, which changes at relative order $\eta_{\mathrm{anh}}^2=2.4\times10^{-3}$. The first is computed in a model, the second only estimated from the small parameter, and both are small. Between q_{BZ} and about 20 keV the situation reverses. There an anharmonic crystal makes two phonons from one kick, and the same model raises the two-phonon weight by 50 % at q_{BZ} and the four-phonon weight 27 times; the harmonic two-, three- and four-phonon weights hold within 10 % only above 5, 12 and 20 keV. These sizes come from one oscillator with a fitted three-phonon coefficient, so they fix the order of magnitude and the shape of the template, not its value. A calculation of the cubic and quartic force constants of silicon with the bubble self-energy fixes the value (Appendix E.19). The same order shifts the optical phonon by $\eta_{\mathrm{anh}}^2\omega_{\mathrm{opt}}=0.15$ meV, a threshold nuisance (Table 8.6).

For soft-mode and light-atom crystals, and for solid helium near melting, neither a computed next order nor a small parameter exists. The moments of the response remain exact static averages, and the Markov–Krein band of Eq. (C.93) is all that can be said. Its width is 2/3 at the mean of a Gaussian, and at the recoil energy of silicon at 100 keV it spans 0.19 to 0.86 of the cumulative weight (Limit Check 11.13). This is the third case of Sect. 8.8: a range without a central value. A likelihood uses the edges of the band, as it uses the bound on a velocity coupling in Sect. 10.10, and no width is invented for it. Solid helium sits between the cases: the quasi-harmonic weight is within 10 % of the exact one for Lindemann ratios up to 0.115, and the self-consistent one up to 0.227 (Limit Check 11.14).

The phonons themselves are an input, not a truncation of the label, and their error plays the part of the spread between tabulations in Sect. 10.10. Of the eight measured frequencies of silicon at Γ, X and L, the PBE force constants of Sect. 5.2 miss by 2.6 % on average and the zone-center optical mode by 0.6 %. They give a sphere-averaged longitudinal speed of $8.83\,\mathrm{km\,s^{-1}}$ against $9.017\,\mathrm{km\,s^{-1}}$ from the measured elastic tensor (Worked example 11.2). We take as template a uniform rescaling of all phonon frequencies by 2.6 %. It moves the Poisson weights through $2W_\kappa\propto\langle\omega^{-1}\rangle$ and the impulse width through $\langle\omega\rangle^{1/2}$ together, and since the deviations of single modes have both signs, it is an upper template.

Temperature enters at two orders, and both templates are computed and vanish in a

detector. The phonon-absorption branch ($n_T = 1$) carries e^{-21550} at 10 mK (Sect. 8.5). The second-order thermal diffuse scattering near a reciprocal-lattice vector ($n_T = 2$) has a coefficient proportional to T^2, 10^{-9} of its room-temperature value at 10 mK. At 293 K, a tenth of q_{BZ} away from the (111) peak, it equals the whole incoherent two-phonon weight (Limit Check 11.9). It matters for the warm X-ray and neutron data that calibrate the lattice response, not for the rate.

The charge vertex above the zone and the free recoil are not nuisances of this kind. For a coupling to charge, the vertex far above the zone depends on how much of the valence charge a displaced atom carries, which is unknown between about 3 and 20 keV (Sect. 11.3); a nucleon coupling is unaffected. The row can only be named. And the free nuclear recoil, used in place of the crystal response, is an evaluation error, as the resonant-only Bethe–Salpeter real part is in Table 10.2. It holds within 10 % for $x_R \geq 2.3$ at 1 GeV; at 50 MeV it is high by 5 to 21 % between $x_R = 3$ and 8 and misses the Doppler tail of the crystal above its endpoint (Limit Check 11.10). The free recoil and the rigid-ion vertex of Sect. 11.8 are not uncertainties of a correct calculation but errors of a simpler one, by up to factors of 16 and 8.2. No nuisance of the crystal calculation covers this difference, and a published limit that rests on the free recoil below the impulse onset inherits it in full.

11.9.2 Sound speeds and Born charges

The single-phonon rates of Sects. 11.1 and 11.2 carry few truncations once the full one-phonon response is used, and their uncertainty sits in a handful of material numbers. For a nucleon coupling and a heavy mediator, the plateau Eq. (11.9) is proportional to $\langle \sum_\nu (\hat{\boldsymbol{q}} \cdot \hat{\boldsymbol{e}}_\nu)^2 / c_\nu \rangle$, and the acoustic edge Eq. (11.5) to $1/c_{\mathrm{LA,max}}$. A measured elastic tensor fixes both through the Christoffel problem. A single averaged speed costs 0.67 % in silicon (Fig. 11.5), and the edge computed with $\bar{c}_{\mathrm{LA}}$ in place of $c_{\mathrm{LA,max}}$ is 3.7 % too high. The spread between sources of the speeds is larger: a target table built from density-functional phonons lists speeds 5.6 % below the measured tensor for silicon and 10 % below for germanium (Worked example 11.2), and the plateau scales inversely with them. The tensors are room-temperature values, and the change on cooling has not been sized. For a light mediator above 44 times the edge, nothing of the material enters at the 10 % level; closer to the edge the material enters only through the edge. The acoustic rate is therefore known to the precision of the elastic tensor whenever a measured one is used, and to a few percent when computed speeds are used instead.

A polar rate scales as $(Z^*/\varepsilon_\infty)^2$, through the band integral Eq. (11.19) and the one-branch form Eq. (11.17). From the measured splitting, Eq. (11.23) gives Z^*/ε_∞ of GaAs to 1.0 %, 2 % in the rate. A semilocal ε_∞ fails at this level (Table 11.1), and the static route through $\varepsilon_0 - \varepsilon_\infty$ amplifies the inconsistency of four measured inputs by 3.5, from 1.4 % to 10 % in the rate (Fig. 11.10). The dielectric constant also changes at the percent level between room and liquid-helium temperature. The leading truncation of the polar vertex is the charge carried by acoustic waves, which is new support in the rate. With rigid Born charges it adds at most 3 %, near the edge, and about 1 % at 100 keV at a 1 meV threshold, and the dynamical quadrupoles add a term of the same order that has not been computed (Appendix E.5). The polar rate is therefore dominated by its dielectric input, and a first-principles ε_∞ is not good enough.

The same ratio decides absorption, with a power that depends on the coupling. Dark-photon absorption integrates to $\zeta^2 \mathcal{I}_{\mathrm{pol}}/\rho_T$ (Eq. (11.58)) and inherits the polar row unchanged. The absorption of an electron-coupled scalar in GaAs is a near-cancellation between the electrons per unit mass of the two sublattices and their Born charges (Worked example 11.9). A change of 1 % in Z^*/ε_∞ moves its rate by 9 %, and with the PBE inputs of Sect. 5.5 the rate falls by a factor of 250. Three small terms enter at the level of the site difference d itself. The kinetic energy of the core electrons reduces the electrons each site carries by about 0.1; as

a parametric estimate from the virial theorem it lowers $|d|$ by a tenth and the rate by about 19 % (Sect. 11.8.2). A shift of 0.01 u in the mean mass of gallium moves d by 5 %, and the frequency dependence of the electronic response by 0.3 %. The rigid-ion vertex, which takes the electron numbers of the free atoms, is an evaluation error that raises the rate 8.2 times. The phonon widths, an assumed 1 % of each frequency, set the peak height in proportion to $1/\gamma_{\mathrm{L}}$ and the reach at the line center as $\gamma_{\mathrm{L}}^{1/2}$, but not the integral over a scan of masses. For a resonator in the coherent field, the prescription $\min\{\mathfrak{Q}_{\mathrm{p}}, \mathfrak{Q}_{\mathrm{DM}}\}$ is 1.8 times too high where the two quality factors are equal (Limit Check 11.19).

The broadband channels of Sect. 11.6 carry their own inputs. The isotope correlation of the exact single-defect densities of states, +8.0 % at 50 meV, is a computed next order and moves the central value. The electromagnetic channel against which the isotope channel is compared rests on a room-temperature absorption coefficient read from a published figure, to about 10 % (Worked example 11.7), and on the missing cold measurement; the ratio $\Gamma_{\mathrm{iso}}/\Gamma_{\mathrm{EM}} \geq 5.0 \times 10^3$ is a bound. The spin channels need the spin expectation values of protons and neutrons in each odd isotope, from nuclear shell-model calculations whose spread Appendix E.27 specifies. The daily modulation of Sect. 11.7 is a ratio of rates, and most normalization nuisances cancel in it; what does not cancel is the angular structure. Its electronic part in silicon, 0.9 % at 2 MeV, comes from a model without local fields, which change the averaged rate by a factor of 1.3 to 1.5 and do something not yet computed to its $\ell = 4$ component. The sapphire and SiC amplitudes are incoherent-approximation estimates, valid above a few MeV; at the lowest masses the coherent single-phonon modulation of sapphire reaches 30 % in the published calculation.

11.9.3 Nuisance table of the lattice sector

Table 11.1 lists the nuisances of the lattice sector in the format of Table 10.2. A next order computed with the force constants of Sect. 5.2 or with measured inputs is marked “computed”, and its central value has already moved.

A truncation that two rates share carries one parameter in both (Sect. 8.8). The phonon frequencies of one set of force constants enter the one-phonon, multiphonon and impulse rates of that crystal together, at every mass and threshold, and a fit that gave each its own ϑ would let the data trade them. The measured elastic tensor of Sect. 11.1 is a different input, and an acoustic rate built from it shares no parameter with the multiphonon rate built from computed phonons. The ratio Z^*/ε_∞ of a polar crystal is one input of its dark-photon scattering rate, its dark-photon absorption and its electron-scalar absorption, and one ϑ moves all three, by different factors. The halo stays fixed, as in Eq. (8.78), and the detector nuisances belong to Sect. 21.4.

Worked example 11.10: The nuisance band on the silicon multiphonon rate at 30 MeV

Take $m_\chi = 30\,\mathrm{MeV}$, a heavy mediator, equal couplings to protons and neutrons with $\sigma_p = 10^{-40}\,\mathrm{cm}^2$, and the standard halo of Sect. 1.2 with $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$. The threshold $\omega_{\mathrm{th}} = 100\,\mathrm{meV}$ and the cross-section σ_p are illustrative choices; the threshold lies above the one-phonon band, so that every event counted contains at least two phonons. The largest momentum transfer is $2m_\chi v_{\max} = 159\,\mathrm{keV}$, where $E_R = 0.48\,\mathrm{eV}$ and $x_R = 11.8$. The central value is Eq. (8.16) with one row and the incoherent harmonic response of all orders at $T = 0$, from the PBE density of states of Sect. 5.2, the Helm form factor, and the velocity integral done for an orientation-averaged crystal:

$$\mathcal{R}^{\mathfrak{T}} = 5.26 \times 10^3\,\mathrm{kg^{-1}\,yr^{-1}},$$

Table 11.1: Lattice nuisances. Type in the classes of Sect. 8.8: normalization, tilt (in ω, in q, in the dark-matter mass or in time), threshold and new support. Basis of the size: "computed" (next order computed, central value moved), "model" (next order computed in a model), "est." (small-parameter or parametric estimate), "spread" (published or computed inputs against measured ones), "bound" (only an inequality is known), "named" (no size). Sizes are for silicon unless another crystal is named; Worked example 11.10 is the silicon rate at 30 MeV above 100 meV, and d the site difference of Worked example 11.9. The last column gives the section where the item is treated. Items without a row: the optical-phonon shift of 0.15 meV (Table 8.6), the quantum-crystal and germanium-spectrum models (Sects. 11.4 and 11.5), the absorption line widths and resonator prescription (Sect. 11.8), and the nuclear spin inputs (Appendix E.27).

Truncation or input	Type	Size	Basis	Section
Pairs of nuclei dropped (incoherent approximation)	normalization per order	within 10 % above 6.6 keV ($n = 2, 3$); Worked example 11.10: $+2.6 \times 10^{-4}$	computed	Sect. 11.3
Coherent weight inside a band	tilt	up to 2.7× incoherent at 10 keV	bound	Sect. 11.3
Phonon frequencies (PBE)	tilt	2.6 % mean, uniform rescale (upper estimate); Worked example 11.10: +0.24 % (+3.1 % above 400 meV)	spread	Sect. 5.2
Anharmonic recoil width, skewness ($n_A = 2$)	tilt (constrained)	kinetic energy within ±0.2 %; third cumulant $\sim \eta_{\text{anh}}^2$	model; est.	Sect. 11.5
Anharmonic few-phonon weights ($n_A = 2$)	tilt in q	$n = 2$: +50 %, $n = 4$: 27× at q_{BZ}; harmonic to 10 % above 5, 12, 20 keV	model	Sect. 11.5
Strong anharmonicity (soft modes, light atoms, helium)	any	Markov–Krein band; Si, 100 keV: 0.19 to 0.86 at E_R	bound	Sect. 11.5
Thermal diffuse scattering ($n_T = 2$)	new support	$\propto T^2$; 10^{-9} of room temperature at 10 mK	model	Sect. 11.3
Charge vertex above the zone	tilt in q	unknown at 3–20 keV; nucleon coupling unaffected	named	Sect. 11.3
Free recoil for the crystal (evaluation error)	tilt	10 % from $x_R = 2.3$ (1 GeV); Worked example 11.10: −6.8 %, ×0.062 above 400 meV	computed	Sect. 11.4
Sound speeds	normalization; threshold	tables −5.6 % (Si), −10 % (Ge); single speed 0.67 %; edge +3.7 % with $\bar{c}_{\text{LA}}$	spread	Sect. 11.1
Z^*/ε_∞, polar rates	normalization	splitting route: rate 2 %; semilocal ε_∞: −37 %; static route: +10 %	spread	Sect. 11.2
Charge of acoustic waves	new support	at most 3 % near the edge, 1 % at 100 keV (GaAs); quadrupoles not computed	est.	Sect. 11.2
Z^*/ε_∞, GaAs scalar absorption	normalization	1 % gives 9 %; PBE inputs: /250	spread	Sect. 11.8
Core kinetic energy; mean masses (GaAs scalar)	normalization	$\Delta d \approx -0.1\, d$, rate −19 % (parametric); 0.01 u: 5 % of d	est.	Sect. 11.8
Rigid-ion vertex (evaluation error)	normalization	GaAs scalar rate ×8.2	computed	Sect. 11.8
Single-isotope density of states	normalization	+8.0 % at 50 meV	computed	Sect. 11.6
Cold in-band absorption of silicon	normalization	$\Gamma_{\text{iso}}/\Gamma_{\text{EM}} \geq 5.0 \times 10^3$; warm datum 10 %	bound	Sect. 11.6
Electronic cubic harmonics	tilt in time	Si $A_{\text{day}} = 0.9$ % without local fields	model	Sect. 11.7

with $\mathcal{R}^{\mathfrak{T}}$ the rate at the central values of the truncation labels (Sect. 10.10), or 14 events per kilogram and day. Half of it comes from momentum transfers below 80 keV, where $x_R = 3.0$, and 80 % from between 54 and 109 keV, x_R from 1.4 to 5.6: the rate sits across the impulse onset.

The computed coherent step multiplies each phonon order by $1 + \delta_n^{\mathrm{coh}}(q)$ from Eq. (11.29). Between 30 and 159 keV, $|\delta_2^{\mathrm{coh}}|$ stays below 0.021 and $|\delta_8^{\mathrm{coh}}|$ below 1.5×10^{-4}, and the rate rises by 2.6×10^{-4}, which leaves 5.26×10^3 unchanged at three digits. Three nuisances remain, each entering the power form Eq. (8.72) through $\ln(1 + \delta_k)$:

phonon frequencies, $+2.6\,\%$:	$\ln(1+\delta) = +2.4 \times 10^{-3}$,
anharmonic kinetic energy, $-0.14\,\%$ ($\lambda_3 = 0.01$,	-8×10^{-5},
$\lambda_4 = 2.5 \times 10^{-4}$) :	
anharmonic third cumulant (est.) :	$+5 \times 10^{-5}$.

The kinetic-energy template is a dilation of each spectrum about E_R by $(1+\delta E^{\mathrm{kin}}/E^{\mathrm{kin}})^{1/2}$, which keeps the area and the mean and changes the second moment by the computed amount; the rate changes by 0.12 times the relative change of the width, and the extreme $-0.2\,\%$ of the range of Sect. 11.5 would give -1.2×10^{-4}. For the third cumulant, the whole difference between the incoherent spectrum and a Gaussian of the same mean and variance changes the rate by 1.9 %, and the anharmonic change of the cumulants is of relative order η_{anh}^2. The anharmonic few-phonon weights barely enter: below 20 keV, where they are large, E_R is under 8 meV, and those momentum transfers carry 3×10^{-4} of the rate. With independent standard-normal ϑ_k the widths add in quadrature, Eq. (C.123),

$$\sigma_{\ln \mathcal{R}} = \left[(2.4 \times 10^{-3})^2 + (8 \times 10^{-5})^2 + (5 \times 10^{-5})^2\right]^{1/2} = 2.4 \times 10^{-3},$$

and the rate lies between 5.25×10^3 and 5.27×10^3 per kilogram-year at one standard deviation, a factor of 1.0024 either way. The phonon frequencies carry all of it.

The free recoil gives 4.91×10^3, 6.8 % lower, 29 band widths away. Raising the threshold to 400 meV, near the endpoint 0.48 eV, changes the picture. The crystal rate falls to $110\,\mathrm{kg^{-1}\,yr^{-1}}$, half of it from momentum transfers below 118 keV, $x_R = 6.5$. The free recoil gives 6.8, a sixteenth of it, because above 400 meV most events come from the Doppler tail of the recoil spectrum, which the free nucleus does not have. The rate now samples the width of that spectrum, the dilation coefficient grows from 0.12 to 2.2, and the frequency template becomes +3.1 %; the band is a factor of 1.031. The anharmonic kinetic energy gives -1.5×10^{-3}, still a twentieth of it.

Without perturbation theory the same rate would be far less certain. At 80.4 keV, the median momentum transfer of the rate above 100 meV, the five exact moments of the response allow any weight above threshold between 0.19 and 0.95 by the Markov–Krein band, against the harmonic 0.58. For silicon the next order is small and computed, and the band is a few percent at most. For a soft-mode crystal the moment band would be the answer.

11.9.4 Open problems

An all-electron calculation of the density response of displaced atoms in a large cell, with the translation sum rule as its test, fixes the charge vertex above the zone. A coupling to charge needs that vertex for its multiphonon rates far above q_{BZ} (Sect. 11.3; Appendix E.17). A

real-time method for quantum nuclei that is exact at low temperature, or a measurement of the momentum distribution by neutron Compton scattering, replaces the moment band of soft-mode and light-atom crystals and of solid helium near melting by a response (Sect. 11.5; Appendix E.19). The zero-temperature two-phonon absorption of silicon in the one-phonon band, from the derivatives of the Born charges with respect to the displacements or from a cold measurement, turns the lower bound on the isotope channel into a ratio (Sect. 11.6; Appendix E.17). And a first-principles calculation of the cubic-harmonic components of the silicon loss function with local fields fixes the electronic part of its daily modulation (Sect. 11.7; Appendix E.16).

Estimated rows need calculations rather than new physics. The cubic and quartic force constants of silicon, with the bubble self-energy and a path-integral kinetic energy as the anchor, give the anharmonic template whose shape the single-site model only suggests (Appendix E.19). The dynamical quadrupoles of GaAs size the acoustic charge of the polar rate (Appendix E.5). And the kinetic energy of the core electrons of gallium and arsenic, a property of the free atoms, settles the core row of the scalar absorption (Sect. 11.8). For silicon and a nucleon coupling the conclusion is simpler than for the electrons. At 30 MeV the lattice rate is known to a quarter of a percent above 100 meV and to 3 % near the endpoint. The largest error a published rate can carry on the lattice side is the use of the free recoil below the impulse onset.

Summary

A nucleon-coupled particle too light to give a nucleus a free recoil above threshold excites phonons, because the whole crystal takes up the momentum (Sect. 11.1). A particle of 100 keV transfers at most 0.53 keV, and that momentum creates a longitudinal acoustic phonon of about 16 meV. At momentum transfers well inside the zone the one-phonon response per unit mass depends on the material only through the sound speeds and polarizations of the acoustic branches, and in silicon it stays within 10 % of the long-wavelength form up to $0.48\,q_{\mathrm{BZ}}$. The acoustic edge, set by the fastest longitudinal sound speed, 6.1 keV for silicon at a 1 meV threshold, lies 21 times above the free-target threshold mass. For a light mediator the rate per unit mass is the same in every crystal, but only from 44 times the edge of the polarization-weighted speed (6.39 keV in silicon), and at three times that edge it is a seventh of that value. For a heavy mediator it reaches a plateau proportional to the polarization-weighted inverse sound speed, 654 events per kilogram and year in silicon at $\sigma_p = 10^{-40}\,\mathrm{cm}^2$. The sphere-averaged longitudinal speed of silicon is $9.017\,\mathrm{km\,s^{-1}}$ from its measured elastic tensor.

A dark photon couples to charge and, in a polar crystal, excites the longitudinal optical phonon through the screened Born charge (Sect. 11.2). The rate is the loss function inside the phonon band, and a crystal enters through one band integral, a longitudinal frequency and a share of $1/\varepsilon_\infty - 1/\varepsilon_0$ per branch. For GaAs $Z^*/\varepsilon_\infty = 0.194$ from the measured splitting, and the splitting route avoids the amplification of inconsistent static inputs. LiF has a band integral 112 times that of GaAs and a rate 103 times larger at 1 MeV, yet GaAs reaches lower masses, below the crossing at 26.7 keV: light ions raise the band integral and the edge together.

Above the zone a nucleus leaves behind several phonons, Poisson distributed in the self part of the response (Sect. 11.3). The incoherent approximation that keeps only that part gives the two- and three-phonon weights of silicon within 10 % from about 6.6 keV. The exact ω-integrated weight, a static sum over pairs, corrects it at every momentum transfer, and a band bounds the remainder pointwise, loosely. The pairs add weight but no energy, so the first moment fixes the share of every phonon order, and at 50 keV the orders above two carry 47 %

of the energy. Second-order thermal diffuse scattering is singular near each reciprocal-lattice vector in a warm crystal and negligible at 10 mK. The charge vertex far above the zone is an open problem for couplings to charge, not for nucleon couplings.

At large momentum transfer the phonon orders merge into one peak at E_R (Sect. 11.4). At 100 keV in silicon it lies at 191 meV and is 88 meV wide, a Doppler width set by the kinetic energy of the nucleus in its well; the zero-point motion broadens it as a classical gas at 237 K would. The free nuclear recoil with the Helm form factor is the limit of this peak, and it holds for a heavy particle within 10 % from $x_R = 2.3$; the impulse picture sets in between 90 and 140 keV in silicon, and between 113 and 170 keV in germanium, depending on the criterion. Fermionic absorption on nuclei carries the factor $m_\chi^2(1 - m_\chi/M)$, and in a crystal the free-nucleus formula overstates it 16 times at 10 keV, because the zero-phonon weight cannot be used.

Anharmonicity enters the rate first at second order, by time reversal (Sect. 11.5). In silicon it changes the impulse width by less than a part in a thousand. Between q_{BZ} and about 20 keV it changes the few-phonon weights by factors that grow with the number of phonons, up to 27 for four phonons at q_{BZ} in a single-site model. The exact moments of the response hold for any potential and bound its cumulative weight through the Markov–Krein band, 2/3 wide at the mean of a Gaussian. Solid helium, with a zero-point amplitude a quarter of the neighbor distance, needs self-consistent phonons, and strongly anharmonic targets at detector temperatures have only that band.

Couplings that change from site to site break crystal-momentum conservation and give a broadband response that follows the density of states (Sect. 11.6). One silicon nucleus in 21 carries a spin, and unpolarized spins make the incoherent approximation exact for spin couplings. The isotopes differ by 6.7 % in neutrons per unit mass, and a baryon-minus-lepton vector is absorbed in silicon through them at least 5.0×10^3 times faster than through the electromagnetic channel at 50 meV. The vibration of a single isotope follows from the Lifshitz Green function, with a local mode only for an atom more than 22 % lighter than its neighbors.

A crystal at fixed orientation sees the wind along changing crystal directions, and its rate follows the sidereal day (Sect. 11.7). In a cubic crystal the first anisotropic term has $\ell = 4$, and odd terms need both inversion and time reversal broken. A silicon crystal at 45°N modulates by 0.9 % in its electronic rate and 0.58 % in its lattice rate at 2 MeV, in phase; a GaAs detector between 0.3 and 0.7 %, and sapphire by 6 % for multiphonon events above 300 meV.

Absorption samples the crystal at zero momentum (Sect. 11.8). A scalar coupled to electrons sees the electrons that a longitudinal phonon carries with each nucleus. In GaAs the two sublattices carry nearly equal numbers per unit mass, and the rate is a near-cancellation, overstated 7 to 9 times by rigid ions and moved by 9 % by a percent in Z^*/ε_∞. At neV to µeV masses the field is coherent over the body, and the acoustic modes of a resonator absorb it; the effective quality factor of the halo is 1.34×10^6. A scalar that changes the electron mass or the fine-structure constant drives a resonant mass as ω^2, since a static change of size excites nothing.

Each truncation of the chapter is a nuisance with a size (Sect. 11.9). At 30 MeV with a heavy mediator the silicon rate above 100 meV is 5.26×10^3 events per kilogram-year, uncertain by a quarter of a percent, almost all of it from the computed phonon frequencies. The free recoil gives 4.91×10^3, and above 400 meV a sixteenth of the crystal rate. The polar rates are set by Z^*/ε_∞, which a computed ε_∞ misses by enough to move a rate by 37 %. Throughout the chapter the electrons stayed in their ground state and followed the nuclei. When a nucleus is kicked they do not follow at once, and Chap. 12 computes how often a nuclear recoil also excites them.

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Chapter 12

The Migdal effect

A dark-matter particle of 100 MeV gives a silicon nucleus at most 5.3 eV. That recoil would have to be seen through the phonons it makes, since a nucleus this slow produces hardly any ionization (Chap. 19). When a nucleus is kicked, however, its electrons do not follow at once, and a small fraction of nuclear recoils leaves the atom ionized or the crystal with an electron–hole pair. That fraction is the Migdal effect. It matters because the electrons are far easier to see than the recoil: a pair needs only the gap, and the charge detectors of Sect. 1.5 count it one electron at a time. A coupling to nucleons can therefore be searched for through electrons, at masses well below those where the recoil alone is visible. The question is how large the effect is in a crystal rather than in an isolated atom, and how to compute it without counting the same energy twice, once in the lattice response of Chap. 11 and again in the electronic response of Chap. 10. In the decomposition of Sect. 8.6 it is the first mixed sector, the nuclear density reaching the electrons; Chap. 13 treats the second and Sect. 17.4 their interference. We derive every form of the effect from one response that resolves the energy given to the lattice and to the electrons; the published formulas are its truncations (Fig. 12.1).

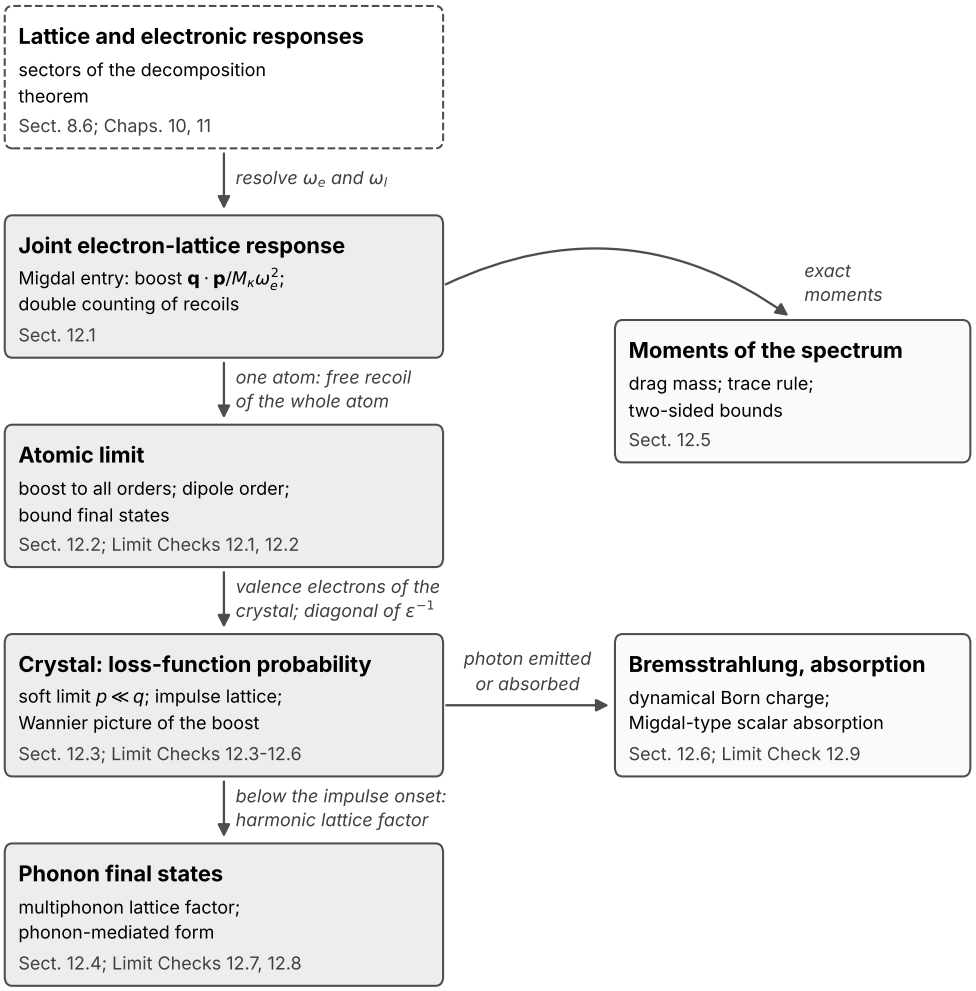


Figure 12.1: The Migdal entry of the joint electron–lattice response and its truncations. Resolving the energy given to the electrons, ω_e, and to the lattice, ω_l, in the sectors of Sect. 8.6 gives the joint response, whose Migdal entry is a boost $\boldsymbol{q} \cdot \boldsymbol{p}/M_\kappa\omega_e^2$ (Sect. 12.1). One atom with a free recoil gives the atomic limit (Sect. 12.2; Limit Checks 12.1 and 12.2); the valence electrons of the crystal with the diagonal of ε^{-1} give the loss-function probability (Sect. 12.3; Limit Checks 12.3 to 12.6); below the impulse onset the harmonic lattice factor gives the phonon final states (Sect. 12.4; Limit Checks 12.7 and 12.8). Branches: the exact moments of the spectrum (Sect. 12.5), and the emission or absorption of a photon or scalar (Sect. 12.6; Limit Check 12.9).

12.1 The joint electron–lattice response

At a momentum transfer of 50 keV a silicon nucleus receives the recoil energy $E_R = q^2/2M_\kappa = 47.8\,\mathrm{meV}$. The lattice takes it as a few phonons: the Debye–Waller exponent is $2W_\kappa = 1.57$, and in 21 % of the kicks the lattice is left as it was (Sect. 11.3). In about two kicks in a million, in the model of Fig. 12.2 below, the electrons do not follow and a pair is made as well, with a few eV. Such an event deposits about 6 eV, of which 48 meV sit in the lattice and the rest in the electrons, and a detector treats the two parts differently. The electronic energy makes charge at once; the lattice energy lies below the gap and reaches only a phonon sensor (Chaps. 18 and 20). The response $S_{ab}(\boldsymbol{q},\omega)$ of the master formula Eq. (8.16) cannot keep the two apart, because it adds all final states of the same total energy ω. We need a response with two energy arguments.

We will define it, place the sectors of Sect. 8.6 in its plane, and compute its Migdal entry at the leading nonadiabatic order, where it becomes the response of the electrons to the field of a nucleus that suddenly moves. Every published form of the Migdal effect is a truncation of that entry. The same object then fixes how Migdal and lattice rates are added without counting a kick twice, and what changes when the target has no gap.

12.1.1 The joint response

Every prompt final state f of Sect. 8.7.1 has a definite electronic content: none in the electron-following sector, and otherwise an electron–hole pair, an exciton or a core hole, with or without phonons. We give it an *electronic energy* E_f^{el}, zero in $\mathcal{P}_{\mathrm{ad}}$ and otherwise the zero-phonon energy of its electronic excitation; for a pair this is the quasiparticle energy $\tilde{E}_m$ of Eq. (8.59). The rest of its energy, $E_f - E_f^{\mathrm{el}}$, belongs to the lattice: phonons and a recoil. Sorting the final-state-resolved response Eq. (8.64) by the two energies defines the *joint response*

$$S_{ab}(\boldsymbol{q};\omega_l,\omega_e) = \frac{2\pi}{\mathcal{V}} \sum_{i,f} p_i \,\langle i|O_a(\boldsymbol{q})^\dagger|f\rangle\langle f|O_b(\boldsymbol{q})|i\rangle\, \delta\big(\omega_e - E_f^{\mathrm{el}} + E_i^{\mathrm{el}}\big)\, \delta\big(\omega_l - (E_f - E_f^{\mathrm{el}}) + (E_i - E_i^{\mathrm{el}})\big), \tag{12.1}$$

with ω_e the energy given to the electrons and ω_l the energy given to the lattice. At $T \ll E_{\mathrm{g}}$ every initial state has $E_i^{\mathrm{el}} = 0$. Integrating over ω_e at fixed total energy removes one delta function and returns the response of Sect. 4.2,

$$\int \mathrm{d}\omega_e \; S_{ab}(\boldsymbol{q};\omega-\omega_e,\omega_e) = S_{ab}(\boldsymbol{q},\omega). \tag{12.2}$$

The properties of the resolved response carry over (Appendix D.33). Where the electronic energy is a good label, each point of the plane is a sum of outer products with positive weights, and the matrix $S_{ab}(\boldsymbol{q};\omega_l,\omega_e)$ is positive semidefinite at every (ω_l,ω_e). The rate into a region of the plane is Eq. (8.16) with the joint response Eq. (12.1) in place of $S_{ab}(\boldsymbol{q},\omega)$ and $\omega = \omega_l + \omega_e$ in the delta function, a double spectrum $\mathrm{d}^2\mathcal{R}/\mathrm{d}\omega_l\,\mathrm{d}\omega_e$. It is this double spectrum, not $\mathrm{d}\mathcal{R}/\mathrm{d}\omega$, that the cascade of Chaps. 18 to 20 turns into charge and phonon signals.

The electronic energy is a label, and it is defined where Sect. 8.6 defines labels. Inside the off-shell window it is unique to relative order θ_{ep}. Near a resonant zero-phonon pair the Dyson convention Eq. (8.59) decides: its delta part sits at $(\omega_l,\omega_e) = (0,\tilde{E}_m)$, and its finite-part and principal-value parts at the pair energy and $\omega_l = \pm\omega_{\boldsymbol{Q}\nu}$ of the phonon made with it. We take the zero-phonon energy, and not the vertical energy at the equilibrium positions, because the two differ by the energy the lattice gains when it relaxes around the excitation, and that energy ends up in phonons. For a pair spread over many cells the difference is negligible, since a delocalized pair displaces no nucleus (Sect. 8.6); for a localized excitation it is not, as Worked example 12.1 shows.

At $T = 0$ the three sectors of Eq. (8.48) fall on three parts of the plane,

$$S_{ab}(\boldsymbol{q};\omega_l,\omega_e) = \underbrace{\delta(\omega_e)\, S^{\mathrm{ad}}_{ab}(\boldsymbol{q},\omega_l)}_{\text{electron-following}} + \underbrace{\delta(\omega_l)\, S^{0}_{ab}(\boldsymbol{q},\omega_e)}_{\text{zero-phonon electronic}} + \underbrace{S^{+}_{ab}(\boldsymbol{q};\omega_l,\omega_e)}_{\text{phonon-dressed electronic}} \quad , \tag{12.3}$$

with S^{ad}_{ab} and S^{0}_{ab} the sector responses Eq. (8.49), each a function of one energy. The electron-following sector is the line $\omega_e = 0$, the zero-phonon electronic sector the line $\omega_l = 0$, and the phonon-dressed sector fills the quadrant $\omega_l, \omega_e > 0$. In a warm target phonons can also be absorbed, and ω_l takes either sign.

Each row of the master formula occupies its own parts of the plane. The electron density puts the phonons excited through the electrons that follow the nuclei (Eq. (8.51)) on the line $\omega_e = 0$, the clamped-nuclei response of Chap. 10 on the line $\omega_l = 0$, and the phonon sideband of Chap. 13 at $\omega_l = \pm\omega_{\boldsymbol{Q}\nu}$ for one phonon. The nuclear density puts the lattice response of Chap. 11, single phonons, multiphonons and free recoil, on the line $\omega_e = 0$. Its weight off that line is the Migdal effect, which the rest of this chapter computes. The interference $S_{e\kappa}$ of the two densities is the Born-charge interference of Sect. 8.4 on the line $\omega_e = 0$. Off that line it is the interference of a Migdal amplitude with an electron-recoil amplitude into the same final state, a zero-phonon pair at $\omega_l = 0$ or a pair with a phonon at $\omega_l = \omega_{\boldsymbol{Q}\nu}$. Positivity bounds it at every point of the plane, $|S_{e\kappa}|^2 \le S_{ee} S_{\kappa\kappa}$ (Eq. (4.20)), and Sects. 12.5 and 17.4 compute it.

12.1.2 The boosted electronic response

The nuclear density reaches the electronic sectors only through the nonadiabatic coupling. Eq. (8.53) gives the amplitude at its first order: a kick $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$ excites the electrons to Ψ_j with the factor $-\mathrm{i}\,\boldsymbol{q}\cdot\boldsymbol{d}^I_{j0}/(M_\kappa \Delta E_j)$, times the kick of the nuclear wavefunction. To turn this into a response we write the derivative coupling as the matrix element of an operator whose response we know. By Eq. (5.11), $\boldsymbol{d}^I_{j0} = -\boldsymbol{F}^I_{j0}/\Delta E_j$, with $\boldsymbol{F}^I_{j0}$ the matrix element of the force that nucleus I exerts on the electrons. Let the nucleus act on the electrons through a potential v_κ that moves rigidly with it, the bare Coulomb potential $-Z_\kappa e^2/(4\pi r)$ or a pseudopotential (Sect. 6.1). With the Fourier convention Eq. (A.6), the projected force is a sum of density components,

$$\boldsymbol{q}\cdot\nabla_{\boldsymbol{X}_I} V_{eN} = -\mathrm{i}\int \frac{\mathrm{d}^3p}{(2\pi)^3}\, (\boldsymbol{q}\cdot\boldsymbol{p})\, v_\kappa(\boldsymbol{p})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{X}_I}\, n_e(\boldsymbol{p}). \tag{12.4}$$

The nucleus is, for its electrons, an external potential with Fourier components $v_\kappa(\boldsymbol{p})$, and moving it by $\delta\boldsymbol{X}$ changes each component by $-\mathrm{i}\boldsymbol{p}\cdot\delta\boldsymbol{X}$ times itself. Here $\boldsymbol{p}$ is the momentum given to the electrons. Inserting Eq. (12.4) in Eq. (8.53) (Appendix D.33.2),

$$\big\langle \Psi^{\mathrm{nuc}\prime}\Psi_j \big| n_\kappa(\boldsymbol{q}) \big| \Psi^{\mathrm{nuc}}\Psi_0 \big\rangle = \sum_{I\in\kappa} \int \frac{\mathrm{d}^3p}{(2\pi)^3}\, \frac{\boldsymbol{q}\cdot\boldsymbol{p}}{M_\kappa \Delta E_j^2}\, v_\kappa(\boldsymbol{p})\, \langle \Psi_j | n_e(\boldsymbol{p}) | \Psi_0 \rangle\, \big\langle \Psi^{\mathrm{nuc}\prime} \big| \mathrm{e}^{\mathrm{i}(\boldsymbol{q}-\boldsymbol{p})\cdot\boldsymbol{X}_I} \big| \Psi^{\mathrm{nuc}} \big\rangle. \tag{12.5}$$

Read from right to left, the nucleus takes the kick $\boldsymbol{q}$, passes the momentum $\boldsymbol{p}$ to the electrons through its own field and keeps $\boldsymbol{q}-\boldsymbol{p}$; the electrons are excited by the density component $n_e(\boldsymbol{p})$, as by any external potential. The factor $\boldsymbol{q}\cdot\boldsymbol{p}/(M_\kappa\Delta E_j) = \boldsymbol{v}_N\cdot\boldsymbol{p}/\Delta E_j$ is the sudden amplitude Eq. (6.6), and the remaining $v_\kappa/\Delta E_j$ turns the force into a derivative coupling.

Eq. (12.5) takes the transition density $\langle\Psi_j|n_e(\boldsymbol{p})|\Psi_0\rangle$ at the equilibrium positions, the Condon approximation of Sect. 5.6, and the vibrational final states of the ground surface, so that ΔE_j is the vertical excitation energy; for a delocalized pair it is also the zero-phonon energy. Freezing the transition density in the frame of the crystal makes the lattice recoil with $\boldsymbol{q}-\boldsymbol{p}$, as in the crystal formulas of Knapen, Kozaczuk and Lin [1, Eq. (3)] and of Berghaus and collaborators [2, Eq. (3.8)]. For an isolated atom, whose electrons move with the nucleus,

the same steps give $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}$: the whole atom recoils (Appendix D.33.5). Valence electrons in a crystal lie between the two cases, and which frame their transition densities follow, that of the crystal or that of the struck ion, is an open question. The difference is a Herzberg–Teller term, the dependence of the transition density on the displacement in Eq. (8.52). It changes the momentum the lattice keeps, not the electronic spectrum: at $q = 5\,\mathrm{keV}$ it changes the fraction of Migdal events that emit phonons by a factor 2.4 (Sect. 12.4).

Squaring Eq. (12.5), summing over nuclei and final states, and using the conservation of crystal momentum by the electronic transition (Appendix D.33.3), we obtain the entry of the joint response that holds the Migdal effect.

The Migdal entry of the joint response

At Born–Oppenheimer order $n_B = 2$, with Condon transition densities and lattice final states on the ground surface, the nuclear density of species κ has, for $\omega_e > 0$,

$$S_{\kappa\kappa}(\boldsymbol{q};\omega_l,\omega_e) = \frac{1}{2\pi}\int_{\mathrm{BZ}}\frac{\mathrm{d}^3k}{(2\pi)^3}\sum_{\boldsymbol{G}\boldsymbol{G}'}\underbrace{\frac{[\boldsymbol{q}\cdot(\boldsymbol{k}+\boldsymbol{G})]\,[\boldsymbol{q}\cdot(\boldsymbol{k}+\boldsymbol{G}')]}{M_\kappa^2\,\omega_e^4}}_{\text{boost}}\underbrace{v_\kappa(\boldsymbol{k}+\boldsymbol{G})^*\,v_\kappa(\boldsymbol{k}+\boldsymbol{G}')}_{\text{vertex}} \times\underbrace{S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k},\omega_e)}_{\text{electrons}}\underbrace{S^{\mathrm{ad}}_{\kappa;\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}-\boldsymbol{k},\omega_l)}_{\text{lattice}}, \tag{12.6}$$

where $S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k},\omega) = (2\pi/\mathcal{V})\sum_{j\neq 0}\langle\Psi_0|n_e(\boldsymbol{k}+\boldsymbol{G})^\dagger|\Psi_j\rangle\langle\Psi_j|n_e(\boldsymbol{k}+\boldsymbol{G}')|\Psi_0\rangle\,\delta(\omega-\Delta E_j)$ is the density response matrix of the electrons at clamped nuclei and $S^{\mathrm{ad}}_{\kappa;\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{Q},\omega)$ the electron-following response Eq. (8.49) between $n_\kappa(\boldsymbol{Q}-\boldsymbol{G})$ and $n_\kappa(\boldsymbol{Q}-\boldsymbol{G}')$. Its zero-phonon part, at $\omega_l = 0$, lies in $\mathcal{P}_0$, and the rest in $\mathcal{P}_+$.

The electronic factor is the full response of the interacting electrons, with screening and local fields inside it, and the vertex is the bare potential of the nucleus. The moving nucleus is an external charge to the electrons, exactly as the dark matter is in the loss-function rate of Sect. 10.2. If the core electrons are frozen into a pseudopotential, v_κ is that pseudopotential and S^{ee} the response of the valence electrons, which is the setting of Sect. 12.3. The lattice factor is the response of Chap. 11, evaluated at the momentum $\boldsymbol{q}-\boldsymbol{k}-\boldsymbol{G}$ that the nucleus keeps.

The size of the effect follows from the moments of S^{ee}. The first electronic moment of the Migdal probability is fixed by the drag mass, $\int \mathrm{d}\omega_e\,\omega_e\,\mathrm{d}P_{\mathrm{Mig}}/\mathrm{d}\omega_e = E_R\,\delta M_\kappa/M_\kappa$ for an isotropic drag mass (Eq. (8.56)). With a typical electronic energy $\langle\omega_e\rangle$, therefore,

$$P_{\mathrm{Mig}} \approx \frac{E_R}{\langle\omega_e\rangle}\,\frac{\delta M_\kappa}{M_\kappa},$$

the recoil energy in units of the electronic one, times the share of the mass that the electrons carry. For silicon at $q = 50\,\mathrm{keV}$, with the atomic estimate $\delta M_\kappa/M_\kappa = 2.7\times10^{-4}$ of Sect. 8.6 and $\langle\omega_e\rangle \approx 6\,\mathrm{eV}$, this is 2×10^{-6}. The atomic count is the sum of the drag masses over all nuclei, while a kick probes the element of the struck nucleus alone, which in silicon is about twice as large (Sect. 12.5.2); the agreement with the model below is therefore partly fortuitous. Since E_R grows as q^2, so does P_{Mig}, and the Migdal effect matters where the recoil itself is invisible: it trades the factor P_{Mig} for an energy $\langle\omega_e\rangle/E_R$ times larger, 120 times at 50 keV in silicon.

Fig. 12.2 shows the plane for silicon at $q = 50\,\mathrm{keV}$, in a model that keeps the valence electrons, the diagonal $\boldsymbol{G} = \boldsymbol{G}'$ and the incoherent lattice factor (Appendix D.33.4). The line

$\omega_e = 0$ carries the lattice spectrum of Sect. 11.3, with the zero-phonon weight $e^{-2W_\kappa} = 0.21$ and the mean E_R. The Migdal weight, $P_{\text{Mig}} = 1.86 \times 10^{-6}$ in this valence-only model, or 3.9×10^{-5} per eV of recoil energy, sits at ω_e above the gap: half of it below 4.4 eV, 89 % below 10 eV, with a mean of 5.9 eV. Along ω_l the Migdal weight repeats the lattice spectrum. Its zero-phonon part, the line $\omega_l = 0$, holds 21 % of it, and the rest has the peaks of the phonon density of states. The mean lattice energy of a Migdal event is 48.4 meV, above E_R by $\langle p^2\rangle/2M_\kappa$, the recoil of the momentum the electrons take, with $\langle p^2\rangle^{1/2} = 5.6$ keV for the point ion of the model (6.1 keV with the ion charge $Z_{\text{ion}}(p)$ of Sect. 12.3). The full silicon value, with the core-resolving ion charge $Z_{\text{ion}}(p)$, is 6.07×10^{-5} per eV of recoil energy (Sect. 12.3). The map is close to a product of the lattice spectrum and the electronic spectrum. That product is the soft limit $p \ll q$, which Sect. 12.3 makes precise; here p/q is about 0.1.

The leading entry neglects the nuclear motion in the energy denominators, the Herzberg–Teller term and the displacement of the excited surface. The energy denominators of the boost neglect the nuclear motion against ω_e, at relative order $v_N p/\omega_e$ and $(\omega_l/\omega_e)^2$; at the momenta of Fig. 12.2 these are 2×10^{-3} and 10^{-4}. The exact denominators are Eq. (12.24) for a free nucleus and Eq. (12.30) for a bound one. The Condon approximation drops the change of the transition density with the displacements, the Herzberg–Teller term above. The ground-surface final states drop the displacement of the excited surface, which moves energy from ω_e to ω_l without changing their sum, and which is negligible for a delocalized pair. Worked example 12.1 shows the last two at work where they are not small.

The dielectric matrix of the electrons, with local fields, on a grid of $(\boldsymbol{k}, \omega_e)$ comes from Appendix E.6, and the lattice response on a grid of momenta from Appendix E.17. Appendix E.18 specifies their convolution for silicon and germanium, all-electron at the shallow cores, with the checks that the moments of Sect. 12.5 impose.

Worked example 12.1: The joint response of a displaced oscillator

We take the model of Worked example 8.8: the bond coordinate x of silicon, with $\omega_0 = 64.1$ meV and zero-point length $\ell_0 = 0.048$ Å, and an electron with two levels split by $\Delta_{\text{tl}} = 1.17$ eV. The diagonal coupling $g_{\text{d}} = -\sqrt{2}\,\omega_0$ displaces the excited surface, with $S_{\text{HR}} = 2$, and the off-diagonal coupling $g_{\text{n}} = 20$ meV is the source of the nonadiabatic coupling. The kick is e^{iqx} with $q\ell_0 = 1$, that is $q = 41.1$ keV, whose recoil energy on the coordinate is $E_R = \omega_0$. We diagonalize the model exactly and label each eigenstate by the Born–Oppenheimer product state it overlaps most, as in Worked example 8.8. A state on the excited surface with vibrational level m gets $\omega_e = 1.0463$ eV, the zero-phonon energy of the upper level, and $\omega_l = m \times 65.5$ meV, the quanta of the excited surface. Both are read off the exact diagonalization, in which g_{n} mixes the two levels; the displaced harmonic surface alone gives $\Delta_{\text{tl}} - S_{\text{HR}}\omega_0 = 1.0418$ eV and ω_0, and the ground surface has the quanta 63.4 meV.

The exact plane. The line $\omega_e = 0$ carries the kick on the ground surface, with weight $1 - P_{\text{Mig}}$. The Migdal weight $P_{\text{Mig}} = 3.94 \times 10^{-6}$ of Worked example 8.8 lies on the line $\omega_e = 1.0463$ eV as a comb in ω_l, with the fractions 0.097, 0.227, 0.265, 0.205, 0.119 and 0.055 for $m = 0$ to 5. The lattice receives on average 0.153 eV in a Migdal event, more than twice the recoil energy.

The leading entry with ground-surface final states. This is Eq. (12.6): every Migdal event sits at the vertical energy $\omega_e = \Delta_{\text{tl}} = 1.17$ eV, and the lattice factor is the kick of the ground surface, Poisson distributed with mean $(q\ell_0)^2 = 1$. The fractions are 0.368, 0.368, 0.184, 0.061, and the mean lattice energy is $E_R = 64.1$ meV. The total energy per event, $\Delta_{\text{tl}} + E_R = 1.234$ eV, is close to the exact 1.199 eV, but the split between electrons and lattice is wrong.

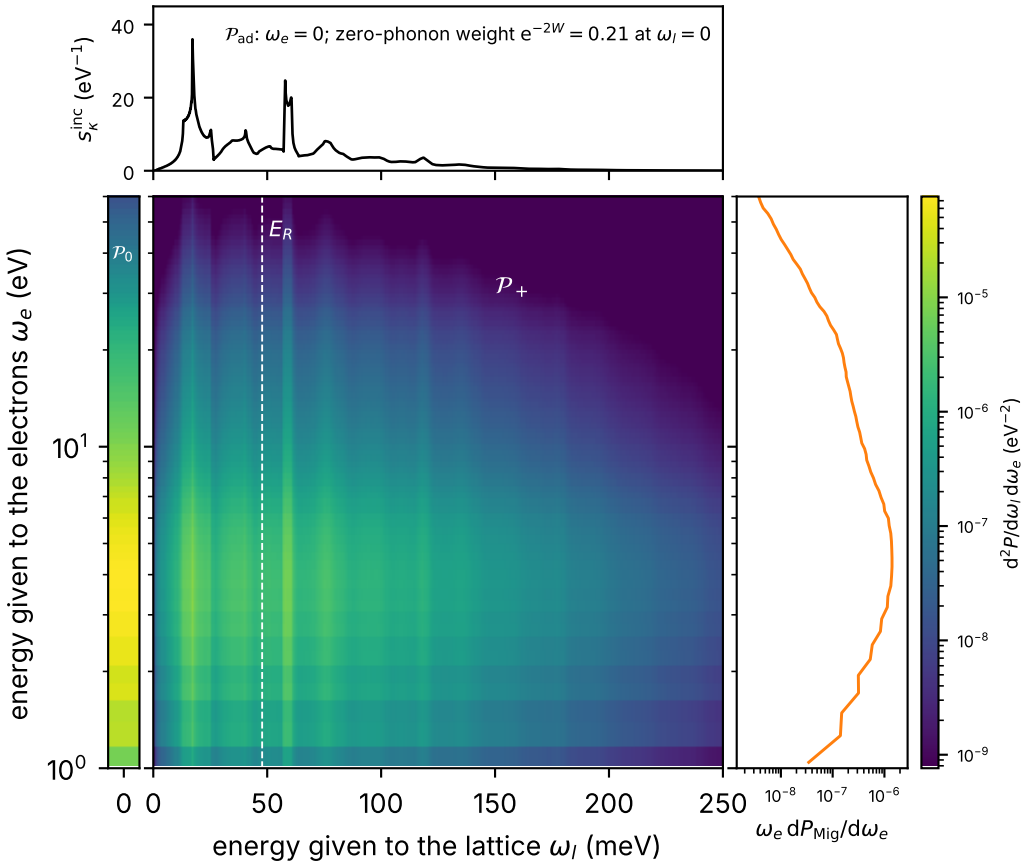


Figure 12.2: The joint response of silicon at $q = 50\,\mathrm{keV}$ in the plane of the energy ω_l given to the lattice and the energy ω_e given to the electrons, per nucleus and normalized as probabilities, at $T = 0$. Top: the electron-following line $\omega_e = 0$, the incoherent lattice spectrum $s_\kappa^{\rm inc}$ of one nucleus without its zero-phonon part $e^{-2W} = 0.21$ at $\omega_l = 0$ ($E_R = 47.8\,\mathrm{meV}$, $2W = 1.57$, phonon density of states from density-functional perturbation theory). Main panel: the Migdal probability density $d^2P/d\omega_l\, d\omega_e$ in the phonon-dressed sector $\mathcal{P}_+$; the dashed line marks E_R. Left strip: the zero-phonon Migdal line $\omega_l = 0$ (sector $\mathcal{P}_0$), drawn as a bin of $1\,\mathrm{meV}$ in ω_l on the same color scale. Right: the marginal $\omega_e\, dP_{\rm Mig}/d\omega_e$. Model: Eq. (12.6) with valence electrons only, a point pseudo-ion of charge 4 ($v_\kappa = -4e^2/p^2$, no core form factor) and the diagonal $\boldsymbol{G} = \boldsymbol{G}'$. The electronic factor is the random-phase loss function of silicon with local fields of Knapen, Kozaczuk and Lin [3] (threshold $1.11\,\mathrm{eV}$, the step-model gap that accompanies the tables, $p \le 22.5\,\mathrm{keV}$, $\omega_e \le 89\,\mathrm{eV}$, averaged over directions); the transition densities are Condon in the frame of the crystal (lattice factor at $|\boldsymbol{q} - \boldsymbol{p}|$), and the lattice response is incoherent. In this model $P_{\rm Mig} = 1.86 \times 10^{-6}$, of which 21 % is zero-phonon.

Excited-surface final states. Projecting the kicked state on the vibrational states of the excited surface moves the relaxation energy where it belongs. For surfaces of equal frequency the overlap $\langle m_{\mathrm{up}}|\mathrm{e}^{\mathrm{i}qx}|0_{\mathrm{lo}}\rangle$ is that of a coherent state whose amplitude adds the imaginary kick $\mathrm{i}q\ell_0$ to the real displacement $-S_{\mathrm{HR}}^{1/2}$. The weights are then Poisson with mean $(q\ell_0)^2 + S_{\mathrm{HR}} = 3$, and the mean lattice energy is $3\omega_0 = 0.192\,\mathrm{eV}$, against E_R before. On the computed surfaces, with their unequal quanta, the same projection with the derivative coupling evaluated at $x = 0$, the Condon approximation, gives the mean lattice energy 0.188 eV and the zero-phonon fraction 0.056 instead of $\mathrm{e}^{-3} = 0.050$. The relaxation energy $S_{\mathrm{HR}}\omega_0 = 0.128\,\mathrm{eV}$ has moved from ω_e to ω_l, and the total $1.0463\,\mathrm{eV} + 0.188\,\mathrm{eV} = 1.234\,\mathrm{eV}$ is unchanged. The probability is 3.52×10^{-6}, the boost formula of Worked example 8.8, and its zero-phonon fraction 0.056 falls short of the exact 0.097.
Beyond Condon. The derivative coupling and the gap both vary along x: the gap is $\Delta_{\mathrm{tl}} + g_{\mathrm{d}}x/\ell_0$, smaller on the side of the displaced minimum. Keeping $q\,d_{10}(x)/[(M/2)\Delta E(x)]$ inside the vibrational matrix element, as a function of x, gives $P_{\mathrm{Mig}} = 4.00 \times 10^{-6}$, the fractions 0.104, 0.236, 0.267, 0.201, 0.113 and 0.051, and the mean lattice energy 0.148 eV, all within a few percent of the exact values. The excess of 12 % that Worked example 8.8 traced to the displaced surface is this Herzberg–Teller term, and it acts most on the zero-phonon line.
The joint response resolves what the total sums over: two forms of the leading entry with the same total energy disagree by 0.12 eV, about twice the recoil energy, on how much of it the electrons receive. The electronic energy must be the zero-phonon energy for the lattice to receive the relaxation it actually makes. And a strongly displaced excited surface, as for an exciton bound to a defect, needs the non-Condon term. For an electron–hole pair spread over N cells, S_{HR} is of order $1/N$, and the ground-surface form Eq. (12.6) is the right one.

12.1.3 Published Migdal formulas as truncations

The literature computes the Migdal effect with four formulas, each derived in its own framework. All four are truncations of Eq. (12.6): each keeps the boost and makes a choice for the vertex, the electronic factor and the lattice factor (Table 12.1). The atomic formula of Ibe and collaborators takes one isolated atom, all its electrons and a free recoil of the whole atom; to all orders in v_N it is Migdal's boost Eq. (6.8), of which Eq. (12.6) is the first order. The crystal formula of Knapen, Kozaczuk and Lin keeps the diagonal $\boldsymbol{G} = \boldsymbol{G}'$, the impulse limit of the lattice and the soft limit $p \ll q$. In that limit the lattice factor integrates to one, and the electronic factor becomes the loss function with the coefficient 4α. Berghaus and collaborators keep the harmonic lattice factor at $\boldsymbol{q} - \boldsymbol{k} - \boldsymbol{G}$, which reaches momenta below the impulse onset; their formula keeps the off-diagonal loss-function elements, and their evaluation drops them. Liang and collaborators write the inertia of the nucleus as a virtual phonon between the kick and the electron–phonon vertex, which replaces $1/\omega_e^2$ in the boost by a phonon propagator and agrees with it to relative order $(\omega_{\mathrm{ph}}/\omega_e)^2$, the bound-nucleus correction of Sect. 6.1. Appendix D.33 carries out each reduction.

Local fields enter the electronic factor twice. On the diagonal of ε^{-1} they lower the silicon probability by a factor 1.35 against a scalar dielectric function (Limit Check 12.3). Through the off-diagonal elements $\boldsymbol{G} \neq \boldsymbol{G}'$ they couple different momenta. No table in this book contains that part, and the published estimates disagree; in an empirical-pseudopotential random-phase model of silicon it raises the rate by a factor of about 1.4 on the largest computed set and about 1.2 when extrapolated, flat in m_χ; the factor still falls as reciprocal-lattice vectors are added,

Table 12.1: The published Migdal formulas as truncations of the Migdal entry Eq. (12.6) of the joint response. Each keeps the boost $\boldsymbol{q}\cdot\boldsymbol{p}/(M_\kappa\omega_e^2)$ at Born–Oppenheimer order $n_B = 2$; the columns give the choices for the vertex and electronic factor, for the lattice factor, and the section and derivation that treat the form.

Form	Vertex and electrons	Lattice	Treated in
Atomic [4, Eqs. (55)–(58)]	one atom, all electrons, bare nucleus; the full boost to all orders	free recoil of the atom, argument $\boldsymbol{q}$	Sect. 12.2, Appendix D.33.5
Crystal, loss function [1, Eqs. (3) and (6)]	valence electrons, ion charge Z_{ion}, diagonal $\boldsymbol{G} = \boldsymbol{G}'$: loss function	impulse limit; soft limit $p \ll q$	Sect. 12.3, Appendix D.33.4
Crystal, phonon final states [2, Eq. (3.8)]	valence electrons, full loss-function matrix; diagonal in the evaluation	harmonic multiphonon response at $\boldsymbol{q} - \boldsymbol{k} - \boldsymbol{G}$	Sect. 12.4, Appendix D.33.6
Phonon-mediated [5, Eqs. (3.1)–(3.4)]	valence electrons, bare vertex screened afterwards	virtual phonon in place of $1/\omega_e^2$; incoherent multiphonon response	Sect. 12.4, Appendix D.33.6

so 1.2 is not a lower bound, and the full calculation is still open (Sect. 12.3.2). The same matrix changes the electron-recoil rate of silicon by a factor of 1.5 at 10 MeV (Sect. 10.2.2). The atomic formula, applied to the valence electrons of a crystal, puts the wrong electrons in the electronic factor. The valence electrons of silicon are not those of a silicon atom, and their gap is 1.17 eV instead of the atomic binding of several eV, a difference that ω_e^{-4} magnifies. For the core shells, whose electrons do move with their nucleus, the atomic form is the right truncation (Sect. 12.2).

12.1.4 Double counting of recoils

The joint response holds the lattice and the Migdal parts of one nuclear response, and its moments fix how they add. Take the incoherent response, one nucleus at a time. The total weight of $n_\kappa(\boldsymbol{q})$ is $2\pi\boldsymbol{n}_\kappa$, the density of nuclei times 2π, whatever the final states; integrated over the plane, Eq. (12.1) has that weight. The Migdal entry takes $2\pi\boldsymbol{n}_\kappa P_{\text{Mig}}(q)$ of it, and the line $\omega_e = 0$ keeps

$$\int \mathrm{d}\omega_l \; S^{\text{ad}}_{\kappa\kappa}(\boldsymbol{q},\omega_l) = 2\pi\boldsymbol{n}_\kappa\big[1 - P_{\text{Mig}}(q)\big]. \tag{12.7}$$

The first moment of the line, by Eq. (8.56), is that of a nucleus of mass $M_\kappa + \delta M_\kappa$. The lattice responses of Chap. 11 are normalized to unit weight and use the bare mass. Added to a Migdal rate, they count the kicks that excited the electrons twice, once as pure recoils on the line $\omega_e = 0$ and once as Migdal events. The double count is a fraction P_{Mig} of the kicks, but in each of them the lattice energy is not small: Fig. 12.2 shows that a Migdal event carries the whole recoil, 48.4 meV on average against $E_R = 47.8$ meV, and the electronic energy on top of it. The consistent sum takes the lattice response with the weight $1 - P_{\text{Mig}}$ and the mass $M_\kappa + \delta M_\kappa$, the first prescription of Sect. 8.6, and the joint response does this by construction. In silicon both corrections are below 10^{-3} for $E_R \lesssim 16$ eV, which covers the phonon and multiphonon momenta of Chap. 11, and the rate of a phonon detector does not see them there; at 500 eV the weight $1 - P_{\text{Mig}}$ is a 3 % correction (Sect. 12.5.3). The counting of Migdal and lattice events, however, is exact, and Sect. 12.5 completes it with the higher moments.

The same entry fixes the lattice factor of the Migdal rate itself. A Migdal event leaves the lattice in the state that the kick $\boldsymbol{q} - \boldsymbol{p}$ makes, so the lattice factor of a Migdal rate below the impulse onset is the multiphonon response of Sect. 11.3, not a free recoil. Combining a free-recoil Migdal formula with a multiphonon lattice rate is not a double count, but it mixes two truncations of one response, and a rate built that way is labeled by neither. On the scale of Sect. 8.5 the two pieces of a rate should carry the same lattice regime $\mathcal{K}$.

Measured nuclear-recoil signals obey the same rule from the other side. A calibration with neutrons measures the ionization of nuclear recoils, and the Migdal electrons of those recoils are part of what it measures. A nuclear-recoil rate converted to charge with a measured yield already contains the Migdal ionization, and a computed Migdal rate added to it counts that ionization twice. The yield and its Migdal part are separated in Sect. 19.2.

12.1.5 Targets without a gap

The construction used the gap three times: to give each final state an electronic energy, to make the energy denominators of the boost large compared with the lattice energies, and to keep the low-energy end of S^{ee} empty. In a metal, a superconductor or a semimetal all three fail.

The labels survive in the form that Sect. 8.6 gives them for gapless targets. The sector projectors $\mathcal{P}^{\mathrm{qp}}_{n,m}$ count quasiparticles and phonons after a dressing transformation, and ω_e becomes the energy of the quasiparticles created. It is defined to within their width, and the joint response is meaningful on bins of the plane that satisfy the bin bound Eq. (8.63).

The leading entry does not survive unchanged. In a metal the loss function grows linearly from $\omega_e = 0$, since electron–hole pairs at the Fermi surface cost nothing, and with the factor ω_e^{-4} the Migdal probability Eq. (12.6) integrates as $\int \mathrm{d}\omega_e\, \omega_e^{-3}$ and diverges at small ω_e. The divergence marks the failure of the sudden expansion, not a large effect. Once ω_e falls to the phonon energies, the electrons follow the nucleus instead of being left behind. The boost denominator ω_e^2 must then be replaced by the full one. For a bound nucleus that is the phonon propagator $\omega_e^2 - \omega_{\mathrm{ph}}^2$ of the phonon-mediated form, whose pole at $\omega_e = \omega_{\mathrm{ph}}$ is the ordinary electron–phonon coupling of Chap. 6. The part of the plane with $\omega_e \lesssim \omega_{\mathrm{D}}$ then belongs to the electron–phonon response of the metal, computed with Fan–Migdal and Eliashberg methods (Sects. 6.3 and 6.4), and only $\omega_e \gg \omega_{\mathrm{D}}$ is a Migdal process in the sense of this chapter. The electronic factor also changes character: S^{ee} contains the intraband response of the conduction electrons, which screens the moving ion at small ω_e, so that a slowly moving ion in a metal carries no charge. The full response handles this, and a formula that screens the vertex statically does not. A superconductor reverses the hierarchy: its pair-breaking edge $2\Delta_{\mathrm{BCS}}$ lies far below the phonon energies, the electrons near the edge are slower than the lattice, and a kicked nucleus breaks pairs through the phonons it emits (Chaps. 15 and 20).

The next sections take the truncations of Table 12.1 one at a time, starting with the isolated atom, where the boost can be done to all orders.

12.2 The atomic limit

A germanium atom that recoils with 1 keV loses an electron to the continuum in one recoil in 580 according to the calculation of Ibe and collaborators, and in one recoil in 610 according to that of Cox and collaborators; in one recoil in 230 its electrons are left excited or ionized. Most of these events come from two shells, the $3d$ shell, bound by about 30 eV and smaller than an angstrom, and the valence shell. A crystal of germanium keeps the first nearly as it is in the atom and rebuilds the second into bands. In the atom the Migdal entry Eq. (12.6) of the joint response can be evaluated to all orders in the momentum given to the electrons,

and every number can be checked against a sum rule. With these results we can give the probability, shell by shell, for xenon, germanium and silicon atoms at any recoil energy.

12.2.1 The boost of a free atom

Take one atom at rest in its ground state, with a nucleus of mass M at $\boldsymbol{X}$ and N_e electrons at $\boldsymbol{r}_e$. Dark matter couples to the nucleus through $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}$, to first order in the coupling. The Coulomb energy depends only on the positions of the electrons relative to the nucleus, $\boldsymbol{\rho}_e = \boldsymbol{r}_e - \boldsymbol{X}$. We trade $\boldsymbol{X}$ for the center of mass $\boldsymbol{R}_{\rm cm} = (M\boldsymbol{X} + m_e \sum_e \boldsymbol{r}_e)/m_A$ and keep the $\boldsymbol{\rho}_e$, with

$$m_A = M + N_e m_e, \tag{12.8}$$

the mass of the atom up to the binding energy of its electrons, less than a part in 10^6. The chain rule gives $\nabla_{\boldsymbol{X}} = (M/m_A)\nabla_{\boldsymbol{R}_{\rm cm}} - \sum_e \nabla_{\boldsymbol{\rho}_e}$ and $\nabla_{\boldsymbol{r}_e} = (m_e/m_A)\nabla_{\boldsymbol{R}_{\rm cm}} + \nabla_{\boldsymbol{\rho}_e}$. In the kinetic energy $-\nabla^2_{\boldsymbol{X}}/2M - \sum_e \nabla^2_{\boldsymbol{r}_e}/2m_e$ the cross terms between $\nabla_{\boldsymbol{R}_{\rm cm}}$ and $\nabla_{\boldsymbol{\rho}_e}$ cancel, and the terms in $\nabla^2_{\boldsymbol{R}_{\rm cm}}$ add up to $1/2m_A$. The Hamiltonian separates exactly,

$$H = -\frac{\nabla^2_{\boldsymbol{R}_{\rm cm}}}{2m_A} + H_{\rm rel}, \qquad H_{\rm rel} = -\sum_e \frac{\nabla^2_{\boldsymbol{\rho}_e}}{2m_e} - \frac{1}{2M}\Big(\sum_e \nabla_{\boldsymbol{\rho}_e}\Big)^2 + V_{eN} + V_{ee}. \tag{12.9}$$

The internal Hamiltonian is the clamped-nucleus Hamiltonian of Sect. 5.1 plus one term, the kinetic energy $(\sum_e \boldsymbol{p}_e)^2/2M$ of a nucleus that balances the momentum of its electrons. Its diagonal part turns m_e into the reduced mass, and its off-diagonal part, $-M^{-1}\sum_{e<e'} \nabla_{\boldsymbol{\rho}_e}\cdot\nabla_{\boldsymbol{\rho}_{e'}}$, is the mass-polarization term of atomic physics. Both change energies and wavefunctions at relative order m_e/M.

The kick separates just as exactly. Since $\boldsymbol{X} = \boldsymbol{R}_{\rm cm} - (m_e/m_A)\sum_e \boldsymbol{\rho}_e$,

$$\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}} = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_{\rm cm}}\ \exp\Big(-\mathrm{i}\boldsymbol{q}_e\cdot\sum_e \boldsymbol{\rho}_e\Big), \qquad \boldsymbol{q}_e = \frac{m_e}{m_A}\,\boldsymbol{q} = m_e \boldsymbol{v}_A, \tag{12.10}$$

with $\boldsymbol{v}_A = \boldsymbol{q}/m_A$ the velocity of the recoiling atom. The first factor gives the whole atom the momentum $\boldsymbol{q}$ and the kinetic energy $q^2/2m_A$. The second acts on the internal state alone, as a momentum $-\boldsymbol{q}_e$ given to every electron. For hydrogen this is the separation of Sect. 6.1. With N_e electrons the center of mass still separates exactly; what the other electrons change is the internal Hamiltonian, through the mass-polarization term, and the boost mass, m_A in place of the proton mass. The joint response Eq. (12.1) of a gas of n_T such atoms per unit volume, incoherent between atoms, is therefore

Migdal effect of a free atom

$$S_{\kappa\kappa}(\boldsymbol{q};\omega_l,\omega_e) = 2\pi n_T\,\delta\Big(\omega_l - \frac{q^2}{2m_A}\Big)\sum_f P_f(\boldsymbol{q})\,\delta(\omega_e - E_f + E_0),$$
$$P_f(\boldsymbol{q}) = \Big|\Big\langle \Psi_f\Big|\exp\Big(-\mathrm{i}\boldsymbol{q}_e\cdot\sum_e \boldsymbol{\rho}_e\Big)\Big|\Psi_0\Big\rangle\Big|^2, \tag{12.11}$$

with Ψ_f and E_f the eigenstates and energies of $H_{\rm rel}$ and $\boldsymbol{q}_e = m_e\boldsymbol{q}/m_A$. It holds to first order in the dark-matter coupling and to all orders in q, with no assumption about time.

The term $f = 0$ is the line $\omega_e = 0$, the electron-following sector of Eq. (12.3), with weight $P_0 = 1 - P_{\rm Mig}$. The rest is the Migdal entry. All of it sits on the line $\omega_l = q^2/2m_A$: a free atom

has no phonons, and the lattice energy of every event is the recoil of the atom. Eq. (12.11) is the electron-cloud transition factor of Ibe and collaborators, who derive it in the same way and write the argument as the final velocity of the atom times m_e [4, Eqs. (55), (57) and (73)]. Their phase $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}_e\cdot\boldsymbol{x}}$ in the electron coordinates and the opposite sign of Kahlhoefer and Su [6, Eq. (1)] leave $|\cdot|^2$ unchanged.

Sect. 6.1 reached the same structure from the other side, as the sudden boost Eq. (6.8) with the velocity $v_N = q/M$ of the bare nucleus, and Appendix D.33.5 reached it as the atomic truncation of Eq. (12.6). Both carry M, and both are right to the order they claim. The exact separation fixes the next order: the boost mass is m_A, and at fixed q the electrons receive $m_e q/m_A$, smaller than $m_e q/M$ by the factor $1 - N_e m_e/m_A$. The electrons that follow the nucleus add their mass to it. This is the drag mass of Eq. (8.56), $\delta M = N_e m_e$ for an atom, and $M + \delta M = m_A$. The probability changes by $2N_e m_e/m_A$, 5.5×10^{-4} for silicon and 4.8×10^{-4} for germanium, far below any other uncertainty of the calculation, but the counting of Migdal and lattice events in Sect. 12.5 needs it, and in a crystal the same role falls to $M_\kappa + \delta M_\kappa$ with the tensor drag mass of Sect. 8.6. What the separation leaves to be solved is the internal problem. The states Ψ_f are those of H_{rel}, and the atomic calculations used below solve the clamped-nucleus problem instead, an error of relative order m_e/M in each P_f.

The rate of a free-atom target follows from Eq. (8.16) with the joint response Eq. (12.1). For a spin-independent coupling the nuclear vertex is $AF_N(q)$, the delta function of ω_l fixes $q = (2m_A E_R)^{1/2}$, and energy conservation needs $\omega_{\boldsymbol{q}} = E_R + \omega_e$. With the minimum velocity Eq. (1.23) at $\omega = E_R + \omega_e$, the free-recoil rate Eq. (11.35) becomes

$$\frac{\mathrm{d}^2\mathcal{R}}{\mathrm{d}E_R\,\mathrm{d}\omega_e} = \frac{\rho_\chi\,\sigma_N}{2m_\chi\,\mu_{\chi N}^2}\,F_N(q)^2\,\eta\Big(\frac{q}{2\mu_{\chi N}} + \frac{\omega_e}{q}\Big)\,\frac{\mathrm{d}P}{\mathrm{d}\omega_e}(q), \tag{12.12}$$

a rate per target atom, with $\mu_{\chi N}$ the reduced mass of the dark matter and the atom and $\mathrm{d}P/\mathrm{d}\omega_e = \sum_{f\neq 0} P_f\,\delta(\omega_e - E_f + E_0)$. This is the rate of Ibe and collaborators [4, Eqs. (91) and (92)]. The electronic energy raises $v_{\min}$ by ω_e/q, and at fixed ω_e the recoil energy can no longer be arbitrarily small, since $v_{\min}$ grows without bound as $q \to 0$. The total deposit is at most the kinetic energy $\frac{1}{2}m_\chi v_{\max}^2$ of the dark matter, with the largest laboratory speed $v_{\max} = 795\,\mathrm{km\,s^{-1}}$ of the halo of Sect. 1.2, 35 eV at 10 MeV and 3.5 keV at 1 GeV, and this, more than the probability, decides which shells a light particle can ionize.

12.2.2 Shell probabilities at dipole order

The electrons of an atom receive little momentum. A recoil energy E_R gives them $q_e = m_e(2E_R/m_A)^{1/2}$, 88 eV for germanium at $E_R = 1\,\mathrm{keV}$, and the multipole parameter of Sect. 8.5 is $\xi_D = q_e r_{\mathrm{orb}} = 0.07$ for its valence shell, of mean radius $2.9\,a_0$, and 0.02 for the $3d$ shell. The boost can be expanded. With $D_{\hat{q}} = \hat{\boldsymbol{q}}\cdot\sum_e \boldsymbol{\rho}_e$ the dipole operator of the electrons along the kick, the amplitude of an excited state is $-\mathrm{i}q_e\langle\Psi_f|D_{\hat{q}}|\Psi_0\rangle$ plus terms of order q_e^2, and

$$P_f = q_e^2\,\big|\langle\Psi_f|D_{\hat{q}}|\Psi_0\rangle\big|^2, \qquad f \neq 0, \tag{12.13}$$

at multipole order $n_D = 2$. Its corrections, of relative order ξ_D^2, come from the quadrupole and monopole terms of the boost and from the dipole term at second order. For hydrogen Eq. (12.13) sums to $q_e^2\langle z^2\rangle = q_e^2 a_0^2$, the small-$v_N$ form of Eq. (6.9).

Two sums over final states are fixed without solving for any of them. Closure over $f \neq 0$ gives the total probability, and the double commutator of $D_{\hat{q}}$ with H_{rel} gives its first moment,

$$P_{\mathrm{Mig}} = q_e^2(\langle D_{\hat{q}}^2\rangle - \langle D_{\hat{q}}\rangle^2), \qquad \int \mathrm{d}\omega_e\,\omega_e\,\frac{\mathrm{d}P}{\mathrm{d}\omega_e} = \frac{N_e m_e}{M}\,E_R, \tag{12.14}$$

with expectation values in the ground state. The total holds at dipole order; the first moment is exact to all orders in q_e. It is the f-sum rule Eq. (4.35) of the internal Hamiltonian, whose kinetic term Eq. (12.9) gives the double commutator $N_e/m_e + N_e^2/M = N_e m_A/(m_e M)$, multiplied by $q_e^2/2$. At $E_R = 1\,\mathrm{keV}$ it is 0.27 eV for silicon, 0.24 eV for germanium and 0.23 eV for xenon: the mean energy the electrons take per recoil, carried by the small fraction of events that excite them. It is also the drag-mass term of Eq. (8.56), $\boldsymbol{q}\cdot\delta M\cdot\boldsymbol{q}/2M^2$ with $\delta M = N_e m_e$, up to a relative m_e/M. Any table of atomic Migdal probabilities must reproduce it, and the core shells, whose large excitation energies weight the sum, dominate it.

In practice the atom is described by one Slater determinant of orbitals, from a Hartree–Fock or Dirac–Hartree–Fock calculation, and Eq. (12.13) becomes a sum of one-electron transitions. The dipole operator is a sum of one-electron operators, so at this order it moves one electron, from an occupied orbital a of a shell $n\ell$ to an empty orbital b. A transition into an occupied orbital is blocked by the Pauli principle. Summed over the orbitals of the shell and the degenerate final ones, the probability per unit energy of an ejected electron is

$$\frac{\mathrm{d}P_{n\ell}}{\mathrm{d}E_{\mathrm{ej}}} = q_e^2 \sum_{a\in n\ell}\sum_{b} \left|\langle b|\hat{\boldsymbol{q}}\cdot\boldsymbol{\rho}|a\rangle\right|^2 \delta(E_{\mathrm{ej}} - \epsilon_b), \tag{12.15}$$

with ϵ_b the energy of a continuum orbital normalized per unit energy and $\omega_e = E_B + E_{\mathrm{ej}}$, with E_B the binding energy of the shell. Bound empty orbitals b give the excitation probabilities of the shell. Only $\ell' = \ell \pm 1$ is reached, and for a closed shell the result does not depend on the direction of $\boldsymbol{q}$. This is the leading-order formula of Ibe and collaborators [4, Eqs. (86)–(90)], and since every term carries q_e^2, a table at one q_e gives every other. The orbitals are atomic Hartree–Fock orbitals and continuum states in the field of the ion, computed as for the ionization form factors of dark-matter–electron scattering (Appendix E.7); Eq. (12.15) is the dipole limit of the ionization form factor Eq. (2.22), with q_e in place of q.

Limit Check 12.1. Atomic Migdal effect at dipole order

For a free atom recoiling with q, each shell is ionized with $\mathrm{d}P_{n\ell}/\mathrm{d}E_{\mathrm{ej}} = q_e^2 \sum_{a,b} |\langle b|\hat{\boldsymbol{q}}\cdot\boldsymbol{\rho}|a\rangle|^2 \delta(E_{\mathrm{ej}} - \epsilon_b)$, $q_e = m_e q/m_A$, and the rate is Eq. (12.12) [4, Eqs. (73), (86)–(92)].

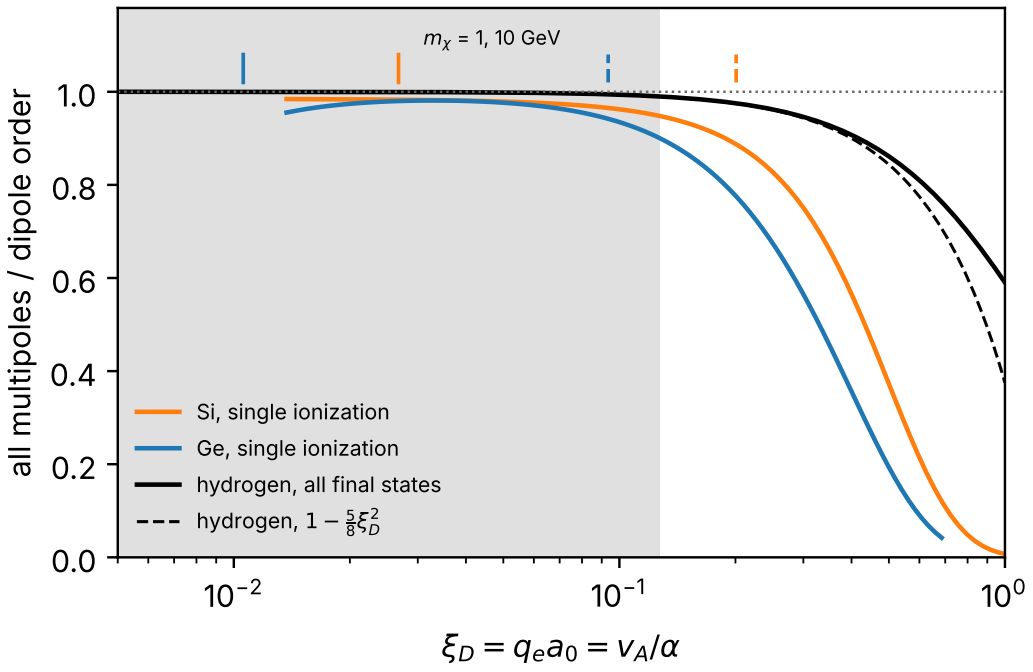


Figure 12.3: The Migdal probability with the full boost $\exp(-i\boldsymbol{q}_e \cdot \sum_e \boldsymbol{\rho}_e)$, all multipoles, divided by its dipole-order value, against $\xi_D = q_e a_0 = v_A/\alpha$. Black: hydrogen, all final states, $[1 - F_{1s}^2]/\xi_D^2$ from Eq. (6.9), exact. Dashed: its leading correction, $1 - \frac{5}{8}\xi_D^2$. Orange and blue: single ionization of silicon and germanium atoms, Dirac–Hartree–Fock tables with multipoles up to $L = 10$ and at dipole order [7]; a velocity-independent part of the tabulated full probabilities, 1.6×10^{-4} (Si) and 1.3×10^{-4} (Ge), which no transition between orthogonal states can have, is subtracted. Ticks: the largest recoil velocity for $m_\chi = 1\,\mathrm{GeV}$ (solid) and 10 GeV (dashed), $v_A = 2\mu_{\chi N} v_{\max}/m_A$. Gray: $\xi_D \leq 0.127$, where both atoms stay within 10 % of the dipole result; silicon does so up to 0.188.

Beyond $\xi_D \approx 0.13$ in germanium, reached by dark matter heavier than 14 GeV (silicon: 0.19, 9 GeV), an electron that is ionized can no longer leave the rest of the outer shell unexcited, and the single-ionization probability falls below the dipole value much earlier than in hydrogen.

On the truncation label of Eq. (8.44) Limit Check 12.1 sits at $n_B = 2$, $n_D = 2$, $\mathfrak{M}$ = sudden and $\mathcal{K}$ = impulse. The dipole formula fails first in the outer shell, as Cox and collaborators showed. The exclusive probability of one ionization multiplies the transition of one electron by the probability that every other electron stays in its orbital. That product feels the bound–bound transitions of the outer shell at order ξ_D^2, with r_{orb} the outer-shell radius [7, Sect. II C]. Below $\xi_D = 0.1$ all three curves of the box lie within 7 % of one. The offset of about 2 % of the two tabulated atoms at the smallest ξ_D, where the full and the dipole tables must agree, measures the numerical precision of the tables. For a particle of 1 GeV, the endpoints are $\xi_D = 0.027$ in silicon and 0.011 in germanium, and Eq. (12.13) is as good as the atomic structure behind it.

Fig. 12.4 shows the Migdal ionization spectra of silicon and germanium atoms at $E_R = 1\,\mathrm{keV}$ against the electronic energy transfer ω_e, the energy a detector records. The valence shells, bound by a few eV, dominate below a few tens of eV and fall as a power of ω_e. The core shells open at their binding energies in ω_e and, because a tightly bound orbital couples to fast electrons, keep their spectrum flat up to energies of the order of their binding before they fall. The silicon $2p$ shell and the germanium $3d$ shell carry the spectrum from about 50 eV to about a keV, and deeper shells beyond. Integrated over ω_e, the ionization probability at 1 keV is 2.09×10^{-3} for silicon, of which the core shells $1s$, $2s$ and $2p$ give 7.9×10^{-4}, and 1.63×10^{-3} for germanium in the Dirac–Hartree–Fock tables of Cox and collaborators. Xenon, from the tables of Ibe and collaborators, gives 1.98×10^{-3}. The three numbers are close because at fixed E_R the factor $q_e^2 = 2m_e^2 E_R/m_A$ falls with the mass of the atom while the dipole sum $\langle D_q^2 \rangle$ grows with the number of outer electrons. Two independent calculations exist for germanium, the relativistic local-potential one of Ibe and collaborators and the Dirac–Hartree–Fock one of Cox and collaborators. They agree within 0.5 % on the $3d$ and $3p$ shells and within 13 % on the valence $4p$ shell, and differ by 6 % on the total, 1.73×10^{-3} against 1.63×10^{-3}, the size of the uncertainty that the atomic structure adds.

Worked example 12.2: The Migdal probability of a germanium atom at 1 keV

A germanium atom, $m_A = 72.630\,\mathrm{u} = 67.654\,\mathrm{GeV}$, recoils with $E_R = 1\,\mathrm{keV}$. Dark matter must be at least 2.27 GeV to give it this energy at $v_{\max}$. The momentum transfer is $q = (2m_A E_R)^{1/2} = 11.63\,\mathrm{MeV}$, the atom moves at $v_A = q/m_A = 1.72 \times 10^{-4}$, or $51.5\,\mathrm{km\,s^{-1}}$, and each electron receives $q_e = m_e v_A = 87.9\,\mathrm{eV}$, so that $q_e a_0 = v_A/\alpha = 0.024$. The tables of Ibe and collaborators are given at $q_e = m_e \times 10^{-3} = 511\,\mathrm{eV}$ [4, Table II],

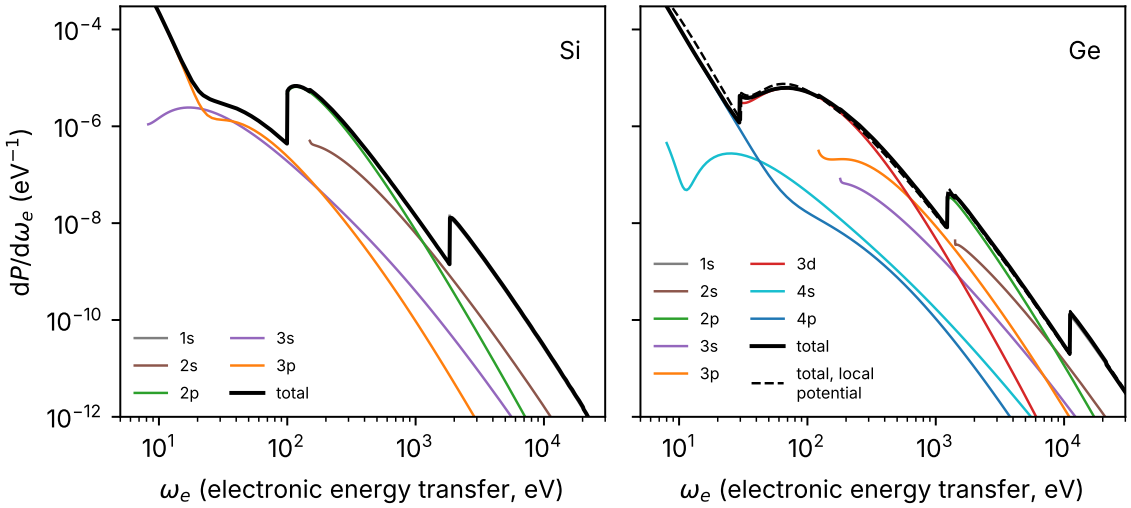


Figure 12.4: Migdal ionization spectra of free silicon (left) and germanium (right) atoms at the recoil energy $E_R = 1\,\mathrm{keV}$, per unit ω_e (electronic energy transfer), by initial shell (colors) and in total (black). Each shell is placed at $\omega_e = E_B + E_{\mathrm{ej}}$, its binding energy plus the kinetic energy of the ejected electron: measured binding energies of the core shells (Sect. 3.5) and, for the valence shells, the measured first ionization energy of the atom, 8.15 eV (Si) and 7.90 eV (Ge), exact for the outer p shell and a lower limit for the valence s shell. Dipole-order Dirac–Hartree–Fock tables of Cox and collaborators [7], with the spin–orbit partners of each shell added, rescaled from $q_e = m_e \times 10^{-3}$ to $q_e = m_e(2E_R/m_A)^{1/2}$, 141 eV for silicon and 88 eV for germanium. Dashed: total for germanium from the local-potential calculation of Ibe and collaborators [4], same rescaling. Integrated probabilities: 2.09×10^{-3} (Si), 1.63×10^{-3} (Ge, Dirac–Hartree–Fock) and 1.73×10^{-3} (Ge, local potential). At other recoil energies every curve scales as E_R.

and at dipole order we multiply them by $(87.9/511)^2 = 0.0296$.
The ionization probabilities are then 8.2×10^{-4} from the $3d$ shell and 7.8×10^{-4} from the $4p$ shell. The $3p$, $2p$ and $4s$ shells add 7.1×10^{-5}, 2.1×10^{-5} and 1.8×10^{-5}, and the deeper shells less. The total is 1.73×10^{-3}: in this calculation one recoil in 580 ejects an electron. The shells with $n \leq 3$ give 9.3×10^{-4} of it. The table also lists transitions to bound orbitals. They add up to 8.75×10^{-2} at the reference q_e, or 2.59×10^{-3} at 1 keV, more than the ionization, and the single transition $4s \to 4p$ gives 1.18×10^{-3} of it. Every transition together has the probability 4.3×10^{-3}, one recoil in 230.
The sum rule Eq. (12.14) checks the table. At the reference q_e the exact first moment is $q_e^2 N_e/2m_e = 32 \times (511\,\text{eV})^2/(2 \times 511\,\text{keV}) = 8.18\,\text{eV}$. The ionization spectra, weighted with $\omega_e = E_B + E_{\text{ej}}$ and the binding energies of the table, give 7.28 eV, and the bound transitions, weighted with the differences of the tabulated level energies, 0.63 eV. The tabulated transitions hold 96.8 % of the moment; the rest lies in what the table omits, levels above $n' = 6$ and electrons faster than 70 keV, and in the relativistic corrections to the nonrelativistic sum rule. At 1 keV the moment is $(N_e m_e/M)E_R = 0.242\,\text{eV}$, bound transitions included. Its ionization part, $7.28\,\text{eV} \times 0.0296 = 0.215\,\text{eV}$, carried by one event in 580, gives an ionizing event 124 eV on average, and a $3d$ ionization 135 eV.
Inside a germanium crystal the $3d$ shell keeps this probability. Its binding is 29.4 eV below the valence maximum (Sect. 10.6) where the table uses a computed 35 eV, and the spectrum shifts with it. The valence part, 8.0×10^{-4} of ionization and most of the bound transitions, does not carry over: in the crystal the $4s$ and $4p$ orbitals form the valence and conduction bands, $4s \to 4p$ is a transition across the gap, and Sect. 12.3 computes it from the loss function.

12.2.3 Core shells inside a crystal

Eq. (12.11) assumes three things that a crystal can violate: that the atom recoils freely, that it is isolated, and that the kick is over before the electrons respond.

Free recoil. A nucleus bound in a lattice recoils like a free one once the recoil parameter $x_R = E_R/\langle\omega\rangle$ is large, above the impulse onset $x_R = 4$ of Sect. 11.4: $q = 92.5$ keV and $E_R = 164$ meV in silicon, 113 keV and 95 meV in germanium. Above it the line $\omega_l = q^2/2m_A$ of Eq. (12.11) becomes the impulse peak of Eq. (11.33), of relative width $x_R^{-1/2}$, 5×10^{-3} at 1 keV in germanium. The electronic factor is untouched. After the kick the neighbors pull the nucleus back on the time $1/\omega_{\text{ph}}$, and the model of Sect. 6.1 found that this multiplies the excitation probability by $[1 - (\omega_{\text{ph}}/\Delta E)^2]^{-2}$. For the germanium $3d$ shell, with the optical phonon of 37.6 meV (Table 5.1), the factor exceeds one by 3×10^{-6}, and for silicon $2p$ by 8×10^{-7}. Below the onset the lattice keeps the recoil in phonons, and the lattice factor is the multiphonon response of Sect. 12.4.

Suddenness. The kick lasts about as long as the dark matter takes to cross the nucleus, 6×10^{-21} s for germanium at $v_{\max}$ (Sect. 6.1). Against the $3d$ excitation energy this is $\Delta E\,\tau_{\text{acc}} = 3 \times 10^{-4}$, and for silicon $2p$ 7×10^{-4}, deep in the sudden region of Fig. 6.2. The free atom needs no such condition, since Eq. (12.11) is exact for it; in a crystal the condition is what lets the nucleus be treated as kicked first and decelerated afterward.

An isolated atom. The electrons must move with their own nucleus and not with the crystal. A core orbital does. The silicon $2p$ shell has a mean radius of 0.28 Å and the germanium $3d$ shell 0.39 Å, against bonds of 2.35 Å and 2.45 Å. Their binding energies, 99.5 eV (the spin–orbit centroid) and 29.4 eV below the valence maximum, lie far below a valence band about 12 eV wide. Such a shell forms a flat band whose orbitals are those of the atom, which first-principles

calculations confirm for exactly these two shells (Appendix E.7). The valence orbitals, of mean radius 1.5 Å, reach the neighbors, and their transitions become the band-to-band transitions of Chap. 10. Their Migdal rate is the crystal formula of Sect. 12.3, with the gap in place of the atomic binding. The atomic valence numbers above do not carry over: the crystal changes the valence channel by more than an order of magnitude (Sect. 12.3.3).

The impulse width spreads the lattice energy by $x_R^{-1/2}$ without changing the probability, and the deceleration changes the probability by $(\omega_{\rm ph}/\Delta E)^2$. The mass that sets q_e is $M_\kappa+\delta M_\kappa$, with the core electrons in δM_κ exactly and the valence part a tensor (Sect. 8.6), an effect of order 10^{-4}. The largest correction is not in these three conditions but in the final state. Near its threshold the ejected core electron enters the conduction bands of the crystal and not the atomic continuum, and Sect. 10.6 found that the choice of final states changes the germanium $3d$ contribution to an electron-scattering rate by an order of magnitude near its opening. The atomic tables are reliable for core electrons ejected with tens of eV or more; within a few eV of the edge they are an estimate. With these provisos the core shells of a crystal take their Migdal probability from the atom, and Sect. 12.3 adds them to the valence rate.

12.2.4 Excitation without ionization

A kick can also leave an electron in an empty bound orbital. Kahlhoefer and Su proposed to detect this in superfluid helium, where an excited atom binds to a neighbor and the excimer emits an ultraviolet photon [6]. The first excited level of helium lies at 20.6 eV, so a recoil far too weak to excite a neighbor by collision can still excite its own electrons. The probability is the sum of Eq. (12.11) over the bound final states, and at dipole order, with the oscillator strengths Eq. (4.36),

$$P_{\rm exc} = q_e^2 \sum_{f\,\text{bound}} \left|\langle \Psi_f | D_{\hat q} | \Psi_0 \rangle\right|^2 = \frac{q_e^2}{2m_e} \sum_{f\,\text{bound}} \frac{f_{f0}^{\rm osc}}{E_f - E_0}. \tag{12.16}$$

The rate is Eq. (12.12) with $\mathrm{d}P/\mathrm{d}\omega_e$ replaced by a sum of lines at the excitation energies, which is the form of Kahlhoefer and Su [6, Eq. (6)]. Only levels reached by the dipole operator appear: from the $1s^2$ ground state of helium, the $1snp\,{}^1P$ levels. With the measured oscillator strengths of these lines [8], 0.276 for $2\,{}^1P$ at 21.22 eV and 0.073 for $3\,{}^1P$ at 23.09 eV, Eq. (12.16) gives $P_{\rm exc} = 0.177\,(v_A/\alpha)^2$ for the first line, $0.220\,(v_A/\alpha)^2$ with the second, and $0.263\,(v_A/\alpha)^2$ with all lines up to the series limit at 24.59 eV. A helium atom recoiling with 10 eV, which dark matter of 52 MeV can produce, has $v_A/\alpha = 0.010$ and is excited with probability 2.7×10^{-5}.

Any finite list of lines underestimates $P_{\rm exc}$, since every term of Eq. (12.16) is positive; the first two helium lines hold 84 % of it. A sharper statement compares the bound part with the ionization. Every bound level lies below the ionization energy I and every continuum state above it. Weighting the oscillator strengths with $1/(E_f - E_0)$ therefore favors the bound part. Let $F_{\rm exc}$ and $F_{\rm ion}$ be the oscillator strength in the bound lines and in the continuum, and $P_{\rm ion} = P_{\rm Mig} - P_{\rm exc}$ the ionization probability. Then $P_{\rm exc} \geq q_e^2 F_{\rm exc}/2m_e I$ and $P_{\rm ion} \leq q_e^2 F_{\rm ion}/2m_e I$, and

$$\frac{P_{\rm exc}}{P_{\rm ion}} \geq \frac{F_{\rm exc}}{F_{\rm ion}}. \tag{12.17}$$

An estimate of the bound share from the oscillator strength alone, the share of the Thomas–Reiche–Kuhn sum Eq. (4.37) in the lines, is a lower bound. For hydrogen the lines hold 0.565 of the oscillator strength (Sect. 4.3) but 0.717 of the dipole probability, of which the $2p$ level gives 0.555 and the levels with $n \leq 3$ give 0.644.

Limit Check 12.2. Migdal excitation of bound final states

A free atom recoiling with q is left in the bound level f with $P_f = |\langle \Psi_f | \exp(-i\boldsymbol{q}_e \cdot \sum_e \boldsymbol{\rho}_e) | \Psi_0 \rangle|^2$, and the rate is the free-recoil rate times $\sum_f P_f \, \delta(\omega_e - E_f + E_0)$ over a list of levels [6, Eqs. (1) and (6)]; at dipole order this is Eq. (12.16).

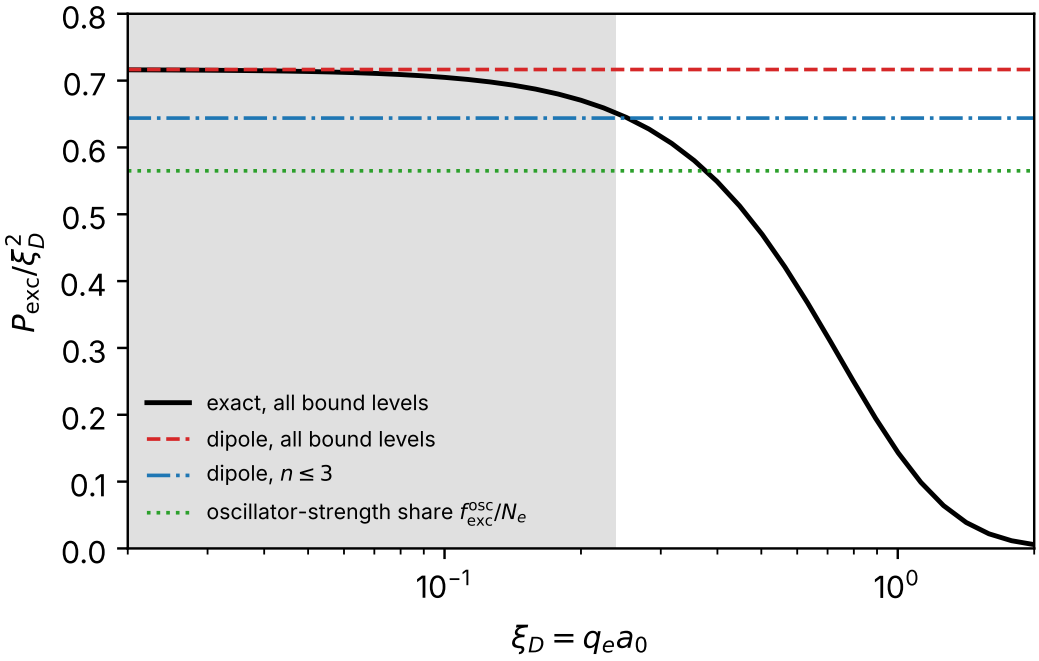


Figure 12.5: Probability that a kick leaves a hydrogen atom in a bound excited level, divided by $\xi_D^2 = (q_e a_0)^2$, against ξ_D. Solid: exact, the full boost and every bound level, from radial integrals with the spherical Bessel functions $j_\ell(q_e r)$ for $n \le 40$ and $\ell \le 12$ and an n^{-3} tail. Dashed: dipole order, all np levels, 0.717. Dash-dotted: dipole order, levels with $n \le 3$ only, 0.644. Dotted: the bound share of the oscillator strength times the total dipole probability, 0.565, the lower bound Eq. (12.17). Gray: $\xi_D \le 0.24$, where the dipole sum over all levels is within 10 % of the exact result.

Beyond $\xi_D \approx 0.24$ the boost carries the electron more and more into the continuum, and the dipole sum overstates the bound probability, which falls to a fifth of it at $\xi_D = 1$.

Limit Check 12.2 sits at $n_B = 2$, $n_D = 2$, $\mathfrak{M}$ = sudden and $\mathcal{K}$ = impulse, as Limit Check 12.1 does. For dark matter lighter than 1 GeV in helium, ξ_D stays below 0.16, where the dipole sum for hydrogen is within 4 % of the exact one; the finite list of lines, 16 % short in helium, sets the larger error of the published estimate.

In germanium the bound transitions outnumber the ionizations. The table of Worked example 12.2 lists 8.75×10^{-2} in bound transitions against 5.86×10^{-2} in ionization at the reference q_e, mostly $4s \to 4p$ and $4p \to 4d$, $5s$, and since the table stops at $n' = 6$ the true bound share is larger still. In a crystal these valence transitions do not exist as such. The bound final states of a crystal are excitons, whose lines below the pair continuum carry as much weight as the free pairs within $8E_x^b$ of the edge would (Sect. 10.5). The Migdal entry Eq. (12.6) weights each final state with ω_e^{-4}, which favors the lines below the edge for the same reason as Eq. (12.17). The exciton share of the Migdal spectrum is therefore at least the share of the lines in the electronic response near the edge. In silicon, with $E_x^b = 14.7$ meV, and in germanium, whose indirect exciton is bound by about 5 meV [9], that share is negligible.

In wide-gap crystals with binding energies of an eV and more it is not, and the lines are a separate, optically detectable signal (Appendix E.15).

In a crystal the core shells inherit the atomic probability, q_e^2 times their dipole sum, while the valence electrons do not: for them the crystal changes the probability by more than an order of magnitude, 47 times the atomic valence value in silicon, and Sect. 12.3 replaces the atom by the loss function of the crystal.

12.3 The loss-function Migdal probability

A dark-matter particle of 300 MeV can give a silicon nucleus a momentum of 1.6 MeV and a recoil energy of 47 eV. The momentum is seven hundred times the size of the Brillouin zone, the recoil energy a thousand times that of a phonon, and the nucleus leaves its site as a free particle (Sect. 11.4). The electrons it sets in motion take a momentum of a few keV, and half of the time they receive less than 4.6 eV, a few times the gap. In this regime the Migdal entry Eq. (12.6) of the joint response simplifies twice. The lattice factor collapses to a free recoil, and the electronic factor becomes the loss function $L(\boldsymbol{p},\omega_e)$ that Sect. 10.2 used for the electron-recoil rate. The probability that a kick leaves an electron–hole pair is then fixed by one material function, an integral of L over momenta weighted by ω_e^{-4}, and it comes out thirty times larger in crystalline silicon than in a silicon atom. The same function sets the rate per unit target mass, which Sect. 17.5 uses to compare materials. In silicon the formula needs two conditions. Beyond the first zone it needs the dielectric matrix with local fields. And the nucleus must move slowly against the electrons: at recoil energies above about 160 eV, reached by dark matter heavier than about 0.5 GeV, its motion during the electronic transition changes the probability by more than 10 %. At low mass, where the electrons take a momentum comparable with the kick, the formula holds unchanged for $q \gtrsim 2q_{\mathrm{BZ}}$ (Sect. 12.4).

12.3.1 Migdal probability through the loss function

Keeping the diagonal $\boldsymbol{G}=\boldsymbol{G}'$ of the electronic factor, a free recoil and the soft limit $p \ll q$ takes Eq. (12.6) to the loss function. On the diagonal the integral over the zone and the sum over $\boldsymbol{G}$ then combine into one integral over the unreduced momentum $\boldsymbol{p}=\boldsymbol{k}+\boldsymbol{G}$ given to the electrons. By Eq. (4.45), with the diagonal inverse dielectric matrix Eq. (E.40), the diagonal element of the density response at $\boldsymbol{p}$ is the loss function at that momentum:

$$S^{ee}_{\boldsymbol{G}\boldsymbol{G}}(\boldsymbol{k},\omega_e) = \frac{2p^2}{e^2}\, L(\boldsymbol{p},\omega_e), \qquad \omega_e > 0,\ T \ll E_{\mathrm{g}}. \tag{12.18}$$

The local fields on the diagonal stay inside L; the dropped elements $\boldsymbol{G} \neq \boldsymbol{G}'$ carry the part of the local fields that couples different momenta, and we return to both below. The free recoil treats each nucleus as if it were alone and in the impulse regime, so that the lattice factor is the free recoil of Sect. 11.4 at the momentum $\boldsymbol{q}-\boldsymbol{p}$ the nucleus keeps, $S^{\mathrm{ad}}_{\kappa}(\boldsymbol{q}-\boldsymbol{p},\omega_l) = 2\pi \boldsymbol{n}_\kappa\, \delta(\omega_l - |\boldsymbol{q}-\boldsymbol{p}|^2/2M_\kappa)$. The *soft limit* $p \ll q$ then says that the electrons take a small part of the momentum, the lattice factor becomes $2\pi \boldsymbol{n}_\kappa\, \delta(\omega_l - E_R)$ with $E_R = q^2/2M_\kappa$, and it no longer depends on $\boldsymbol{p}$. The Migdal entry then factorizes into the free recoil and a probability per kick,

$$S_{\kappa\kappa}(\boldsymbol{q};\omega_l,\omega_e) = 2\pi \boldsymbol{n}_\kappa\, \delta(\omega_l - E_R)\, \frac{\mathrm{d}P_{\mathrm{Mig}}}{\mathrm{d}\omega_e}. \tag{12.19}$$

The vertex is the potential of the moving ion. In silicon the ten electrons of the $1s$, $2s$ and $2p$ shells are bound by 100 eV or more and follow their nucleus; for the valence electrons

whose response is L, nucleus and core act together as one ion. Its potential is

$$v_\kappa(\boldsymbol{p}) = -\frac{Z_{\rm ion}(p)\,e^2}{p^2}, \qquad Z_{\rm ion}(p) = Z_\kappa - f_\kappa^{\rm core}(p), \tag{12.20}$$

with $f_\kappa^{\rm core}(p)$ the form factor of the core electrons, the Fourier transform of their density. This defines the *valence ionization factor* $Z_{\rm ion}(p)$. At small p it is the ionic charge, 4 for silicon; at large p the probe resolves the core and $Z_{\rm ion}$ rises toward the nuclear charge 14. With the core orbitals of the free atom from Sect. 3.5, $Z_{\rm ion}$ of silicon exceeds 4 by 5 % at 2.4 keV and by 10 % at 3.4 keV, and reaches 4.84 at 5 keV and 6.78 at 10 keV. The split between core and valence is a convention, and $Z_{\rm ion}$ must count exactly the electrons that L leaves out.

Inserting Eqs. (12.18) and (12.20) and the lattice factor in Eq. (12.6), the factors combine as

$$\frac{1}{2\pi}\,\frac{(\boldsymbol{q}\cdot\boldsymbol{p})^2}{M_\kappa^2\omega_e^4}\,\frac{Z_{\rm ion}^2 e^4}{p^4}\,\frac{2p^2}{e^2}\,L\;2\pi\boldsymbol{n}_\kappa = 2\pi\boldsymbol{n}_\kappa\,\frac{e^2}{\pi}\,\frac{Z_{\rm ion}^2(\boldsymbol{v}_N\cdot\boldsymbol{p})^2}{\omega_e^4\,p^2}\,L,$$

with $\boldsymbol{v}_N = \boldsymbol{q}/M_\kappa$ the recoil velocity of Sect. 6.1. Since $e^2/\pi = 4\alpha$, comparison with Eq. (12.19) gives the result.

The crystal Migdal probability in the soft limit

For a kick of momentum $\boldsymbol{q}$ in the impulse regime, with $p \ll q$, the probability per unit electronic energy that a nucleus of species κ leaves the valence electrons excited is

$$\frac{{\rm d}P_{\rm Mig}}{{\rm d}\omega_e} = 4\alpha\int\frac{{\rm d}^3p}{(2\pi)^3}\,Z_{\rm ion}(p)^2\,\frac{(\boldsymbol{v}_N\cdot\boldsymbol{p})^2}{\omega_e^4\,p^2}\,L(\boldsymbol{p},\omega_e) = \frac{4\alpha v_N^2}{3\,\omega_e^4}\,L_{\rm Mig}(\omega_e), \tag{12.21}$$

where the second form holds for a direction-averaged loss function and defines the Migdal-weighted loss function

$$L_{\rm Mig}(\omega_e) = \int\frac{{\rm d}^3p}{(2\pi)^3}\,Z_{\rm ion}(p)^2\,\bar{L}(p,\omega_e). \tag{12.22}$$

This is the formula of Knapen, Kozaczuk and Lin [1, Eq. (6)], here derived as a truncation of the joint response (Appendix D.33.4). Read as a product, it is the probability of a dipole boost, v_N^2, times the response of the electrons to the field of a charge $Z_{\rm ion}$ that has suddenly started to move. It is the same response that the dark matter probes in electron recoil, so the screening and the local fields of Sect. 10.2 enter here unchanged, now through an external charge that is the ion instead of the dark-matter particle. The factor ω_e^{-4} is the square of the boost denominator ω_e^2 of Eq. (12.5). With $v_N^2 = 2E_R/M_\kappa$ the probability grows linearly with the recoil energy, and the material enters only through $L_{\rm Mig}$.

For silicon we evaluate Eq. (12.21) with the loss function of Knapen, Kozaczuk and Lin with local fields [3], the table of Sect. 10.2, counted from 1.11 eV, the step-model gap that accompanies it, rather than from its scissor gap, the physical 1.17 eV; the table stops at 89 eV and 22.5 keV. At $E_R = 100$ eV, reached at $q = 2.29$ MeV with $v_N = 8.74\times10^{-5}$ (26 km/s), the probability is $P_{\rm Mig} = 6.1\times10^{-3}$, or 6.1×10^{-5} per eV of recoil energy. Half of it lies below $\omega_e = 4.6$ eV and 89 % below 10 eV; the mean is 6.0 eV. The momenta are larger than the zone: the median p is 5.7 keV, two and a half times $q_{\rm BZ}$ of Eq. (11.2), and 97 % of the probability comes from $p > q_{\rm BZ}$. The plasmon, which lives at $p \lesssim 2.5$ keV, carries 4 %: the factor ω_e^{-4} favors the single-pair continuum near the gap over the plasmon at 17 eV, as Knapen and collaborators found [1, App. C]. With a point ion of charge 4 instead of $Z_{\rm ion}(p)$ the probability falls to 3.9×10^{-3}, lower by a factor 1.57. Knapen and collaborators report 1.6 with the

tabulated form factor of the Si^{4+} ion, and the point-ion value, 3.9×10^{-5} per eV of recoil, is the one of the model of Fig. 12.2.

Worked example 12.3: The coefficient 4α from the f-sum

The coefficient of Eq. (12.21) can be checked without the diagrams of Eq. (12.6), from a sum rule. Take one Fourier component $\boldsymbol{p}$ of the force of the ion, $F_{\boldsymbol{p}} = \boldsymbol{v}_N \cdot \nabla_{\boldsymbol{X}} V_{eN}$ restricted to that component. By Eq. (12.4) it is $-\mathrm{i}(\boldsymbol{v}_N \cdot \boldsymbol{p})\, v_\kappa(\boldsymbol{p})\, n_e(\boldsymbol{p})/\mathcal{V}$ up to a phase, and the Migdal amplitude to a final state j is $\langle \Psi_j | F_{\boldsymbol{p}} | \Psi_0 \rangle / \Delta E_j^2$. The fifth moment of the probability weighs each final state with ΔE_j^5 and therefore equals the energy-weighted sum $\sum_j \Delta E_j |\langle \Psi_j | F_{\boldsymbol{p}} | \Psi_0 \rangle|^2$. For the density operator this sum is the f-sum rule Eq. (4.30), $\sum_j \Delta E_j |\langle \Psi_j | n_e(\boldsymbol{p}) | \Psi_0 \rangle|^2 = N_e p^2 / 2m_e$ at $T = 0$, which holds for the interacting electrons exactly. Per unit volume the fifth moment is therefore

$$(\boldsymbol{v}_N \cdot \boldsymbol{p})^2 \, |v_\kappa(\boldsymbol{p})|^2 \, \frac{n_e \, p^2}{2m_e} = (\boldsymbol{v}_N \cdot \boldsymbol{p})^2 \, \frac{Z_{\mathrm{ion}}^2 e^4}{p^4} \, \frac{n_e \, p^2}{2m_e}.$$

The loss-function form with an unknown coefficient c in place of 4α gives instead, with the f-sum of the loss function $\int_0^\infty \mathrm{d}\omega \, \omega L = (\pi/2)\omega_p^2$ (Eq. (10.5)) and $\omega_p^2 = e^2 n_e / m_e$,

$$c \, Z_{\mathrm{ion}}^2 \, \frac{(\boldsymbol{v}_N \cdot \boldsymbol{p})^2}{p^2} \int_0^\infty \mathrm{d}\omega \, \omega L = c \, Z_{\mathrm{ion}}^2 \, \frac{(\boldsymbol{v}_N \cdot \boldsymbol{p})^2}{p^2} \, \frac{\pi}{2} \, \frac{e^2 n_e}{m_e}.$$

The two agree if $c = e^2/\pi = 4\alpha$, at every $\boldsymbol{p}$ and for any electron density. The coefficient is the one that makes the Migdal spectrum obey the f-sum of the electrons.
For silicon the check also measures the table. At the valence density, $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$ and $\omega_p^{\mathrm{val}} = 16.6\,\mathrm{eV}$, the table holds 0.94, 0.93, 0.92 and 0.90 of the f-sum at $p = 1$, 2, 3 and 5 keV, so the fifth moment of the computed spectrum is within 10 % of its exact value there. At 7.5 keV it holds 0.62 and at 10 keV only 0.17, because the free-electron peak $p^2/2m_e$ moves past the table's last energy of 89 eV. The probability is far less sensitive than the fifth moment, since ω_e^{-4} weights the low energies the table does contain. The momenta above 10 keV give 2.3 % of P_{Mig}, and an upper bound on the energies beyond the table follows from the same sum rule, $\int_{\omega_{\mathrm{top}}}^\infty \mathrm{d}\omega \, L/\omega^4 \le (\pi/2)\omega_p^2/\omega_{\mathrm{top}}^5$, which at $\omega_{\mathrm{top}} = 89\,\mathrm{eV}$, summed over the momenta of the table, is 1.6 % of the probability below it.

The material input of Eq. (12.21) is the loss function of the valence electrons at momenta up to about 20 keV, where the Migdal weight ends, and energies from the gap to about 50 eV. Appendix E.6 describes its calculation in the random-phase approximation with local fields and its convergence in bands and reciprocal vectors; at these momenta the diagonal elements far from the head are needed, the requirement that Sect. 10.2 met for electron recoil. An all-electron calculation of L at $p \le 30\,\mathrm{keV}$ removes the need for Z_{ion} and puts the core shells in the same table. Appendix E.18 specifies a single code path that convolves this table with the nuclear response of Appendix E.17 and reproduces each truncation of Table 12.1 from one set of inputs, and Appendix E.15 the Migdal lines of excitons below the gap, which the random-phase loss function does not contain.

12.3.2 Validity of the soft limit

Knapen and collaborators state Eq. (12.21) with the scalar dielectric function $\varepsilon(\boldsymbol{p}, \omega)$ and note that it holds only for $\boldsymbol{p}$ in the first zone [1, text after Eq. (6)]. Inside the zone the scalar function is the macroscopic one, and the loss function is the head of the inverse matrix.

Outside it, a scalar function of $|\boldsymbol{p}|$ has no unique meaning, and the derivation above says which element to take: the diagonal element of the inverse at the reciprocal vector that contains $\boldsymbol{p}$, Eq. (E.40). The distinction matters because the Migdal weight sits outside the zone. Limit Check 12.3 compares, momentum by momentum, the loss function that a scalar random-phase ε gives with the diagonal element of the inverse with local fields. They agree within 10 % up to $p = 1.3\,q_{\mathrm{BZ}} = 2.9\,\mathrm{keV}$, which holds 8 % of the probability. Beyond, the scalar function overestimates the weight by 24 % at $2q_{\mathrm{BZ}}$ and 35 % at $3q_{\mathrm{BZ}}$, where the probability peaks, and the total by a factor 1.35. The mechanism is the one of Sect. 10.2: at $|\boldsymbol{p}| \approx 2$–$3\,|\boldsymbol{G}_{111}|$ the charge induced in the cell has a macroscopic component that screens the field of the ion.

Limit Check 12.3. Crystal Migdal effect through the loss function

$\mathrm{d}P/\mathrm{d}\omega = 4\alpha \int \mathrm{d}^3p/(2\pi)^3\, Z_{\mathrm{ion}}(p)^2 |\boldsymbol{v}_N \cdot \boldsymbol{p}|^2\, \mathrm{Im}[-1/\varepsilon(\boldsymbol{p},\omega)]/(\omega^4 p^2)$ with a scalar dielectric function, for $\boldsymbol{p}$ in the first zone [1, Eq. (6)].

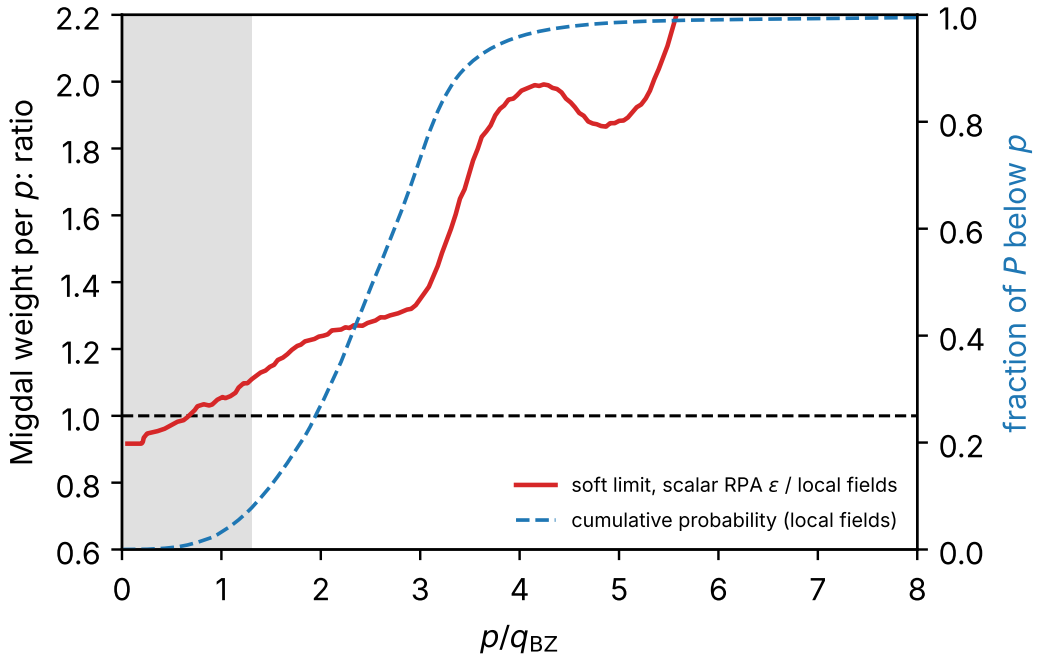


Figure 12.6: Migdal weight of silicon per unit momentum p given to the electrons, $p^2 Z_{\mathrm{ion}}(p)^2 \int_{E_g} \mathrm{d}\omega\, L(p,\omega)/\omega^4$, from the loss function of a scalar random-phase dielectric function at $|\boldsymbol{p}|$ (no local fields), divided by the same with the diagonal element of the inverse dielectric matrix with local fields, Eq. (E.40), against p/q_{BZ} ($q_{\mathrm{BZ}} = 2.25\,\mathrm{keV}$). Both from the direction-averaged tables of Knapen, Kozaczuk and Lin [3] (threshold 1.11 eV, the step-model gap that accompanies them, $\omega \leq 89\,\mathrm{eV}$), smoothed over 0.7 keV in p; $Z_{\mathrm{ion}}(p)$ of Eq. (12.20). The elements $\boldsymbol{G} \neq \boldsymbol{G}'$ are in neither curve. Dashed, right axis: the fraction of P_{Mig} carried by momenta below p. Gray band: $p \leq 1.3\,q_{\mathrm{BZ}}$, where the two agree within 10 %; it holds 8 % of the probability.

Beyond the first zone the scalar form has to be replaced by the diagonal of the inverse dielectric matrix, and the local fields lower the silicon probability by a factor 1.35; the elements $\boldsymbol{G} \neq \boldsymbol{G}'$ are the next correction.

On the truncation label Limit Check 12.3 sits at $n_B = 2$, the order of a Migdal rate (Sect. 8.6), with $\mathfrak{M}$ = sudden, $\mathcal{K}$ = impulse in the soft limit, and $\mathcal{X}$ = random-phase with local fields on

the diagonal.

The elements $\boldsymbol{G} \neq \boldsymbol{G}'$ keep the full matrix in Eq. (12.6). For a direction-averaged target the boost factor averages to $\langle[\hat{\boldsymbol{q}}\cdot(\boldsymbol{k}+\boldsymbol{G})][\hat{\boldsymbol{q}}\cdot(\boldsymbol{k}+\boldsymbol{G}')]\rangle = (\boldsymbol{k}+\boldsymbol{G})\cdot(\boldsymbol{k}+\boldsymbol{G}')/3$. At the random-phase level the response matrix is $S^{ee}_{\boldsymbol{G}\boldsymbol{G}'} = (2|\boldsymbol{k}+\boldsymbol{G}||\boldsymbol{k}+\boldsymbol{G}'|/e^2)\,\mathrm{Im}[-(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}\boldsymbol{G}'}]$, with the symmetrized matrix $\tilde{\varepsilon}_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\mathrm{C}}^{1/2}(\boldsymbol{k}+\boldsymbol{G})\,R^0_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k},\omega)\,v_{\mathrm{C}}^{1/2}(\boldsymbol{k}+\boldsymbol{G}')$, which has the same inverse diagonal as ε. In a crystal with time reversal and inversion it is symmetric, $\tilde{\varepsilon}_{\boldsymbol{G}\boldsymbol{G}'} = \tilde{\varepsilon}_{\boldsymbol{G}'\boldsymbol{G}}$, but not Hermitian at frequencies where it absorbs; $\mathrm{Im}[-(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}\boldsymbol{G}'}]$ denotes the anti-Hermitian part $[(\tilde{\varepsilon}^{-1})^\dagger - \tilde{\varepsilon}^{-1}]_{\boldsymbol{G}\boldsymbol{G}'}/2\mathrm{i}$, which for a symmetric matrix is the imaginary part element by element. With a constant ion charge, L_{Mig} of Eq. (12.22) becomes

$$
\begin{aligned}
L_{\mathrm{Mig}}(\omega) &= \frac{Z_{\mathrm{ion}}^2}{\Omega}\,\mathcal{F}_{\mathrm{Mig}}(\omega),\\
\mathcal{F}_{\mathrm{Mig}}(\omega) &= \sum_{\boldsymbol{G}\boldsymbol{G}'}\int_{\mathrm{BZ}}\frac{\Omega\,\mathrm{d}^3k}{(2\pi)^3}\,\frac{(\boldsymbol{k}+\boldsymbol{G})\cdot(\boldsymbol{k}+\boldsymbol{G}')}{|\boldsymbol{k}+\boldsymbol{G}|\,|\boldsymbol{k}+\boldsymbol{G}'|}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{G}-\boldsymbol{G}')\cdot\boldsymbol{\tau}_\kappa}\,\mathrm{Im}\big[-(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k},\omega)\big],
\end{aligned}
\tag{12.23}
$$

where $\boldsymbol{\tau}_\kappa$ is the position of the struck nucleus in the cell, averaged over the equivalent sites of its species. The indices are in the order of $S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}$ in Eq. (12.6), with $n_e(\boldsymbol{k}+\boldsymbol{G})^\dagger$ on the left of $R^0_{\boldsymbol{G}\boldsymbol{G}'}$; the transposed order reverses the sign of the phase, which leaves silicon unchanged, since its two sites are images under inversion, but not GaAs. The phase belongs to the struck nucleus: the off-diagonal elements depend on where the origin of ε is placed, and the phase is one only when the origin sits on that atom. The diagonal terms carry no phase and give back Eq. (12.22) at any origin. Liang and collaborators write this dimensionless loss spectrum without the phase and do not state an origin [10, Eqs. (12), (13)]; their form is Eq. (12.23) with the origin on the struck atom. Their rate [10, Eq. (11)] is the soft limit with a free recoil, Eq. (12.25) below. The off-diagonal terms need the full matrix, which the tables of Sect. 10.2 do not contain. Knapen and collaborators bound their effect on the rates of silicon and germanium by a factor of order one [1], while Berghaus and collaborators call the off-diagonal loss-function elements small [2]. An empirical-pseudopotential random-phase model of silicon, with the phase of Eq. (12.23), puts the full matrix above the diagonal form on the total, by a factor of 1.38 with 8^3 points $\boldsymbol{k}$ and $|\boldsymbol{G}|^2 \leq 19\,(2\pi/a)^2$ and of about 1.2 when extrapolated in both. The two numbers are not a bracket: the factor still falls as reciprocal-lattice vectors are added, and the fall is not slowing, so 1.2 is not a lower bound. The off-diagonal elements raise the rate at every truncation computed, and the factor is flat in m_χ. The model carries only about two thirds of the diagonal local-field effect of Limit Check 12.3, so its off-diagonal size is a guide, not a bound in either direction. Table 12.3 carries that model size; the full local-field calculation that would replace it stays open (Appendix E.18).

The free recoil is the second approximation. Below the impulse onset of Sect. 11.4, $x_R = 4$ at $q = 92.5\,\mathrm{keV}$ in silicon, the lattice takes the recoil as single phonons and multiphonons, not as a free particle, and Knapen and collaborators bracketed their rates by moving the cut on E_R from $4\langle\omega\rangle$ to $9\langle\omega\rangle$ [1, App. C]. In the soft limit, however, the lattice factor drops out of the probability. It enters the rate only through the energy balance $\omega = \omega_e + \omega_l$ in $v_{\min}$, and the lattice energy has the mean E_R in every regime (Eq. (5.61)) and is small against ω_e: 0.16 eV at the onset, against a few eV. Replacing the free recoil by the incoherent response of a bound silicon nucleus at all phonon orders, including its zero-phonon line, changes the rate by less than 10^{-5} at every mass from 3 to 110 MeV (Limit Check 12.4). The impulse onset is therefore not where the Migdal formula fails.

Dropping $\boldsymbol{p}$ from the lattice factor, the soft limit as defined above, costs nothing for the electronic spectrum, because the lattice factor has unit weight at every $\boldsymbol{q}-\boldsymbol{p}$ (Sect. 12.4). What remains is the boost amplitude, and it expands in $v_N p/\omega_e$, not in p/q. For a free nucleus the kick and the Coulomb exchange with the electrons come in two time orderings [10,

Eq. (1)]. Kicked first, the nucleus moves with $\boldsymbol{q}$, passes $\boldsymbol{p}$ to the electrons and leaves with $\boldsymbol{q}-\boldsymbol{p}$; the energy denominator is $\omega_e - b + a$, with $b = \boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa$ and $a = p^2/2M_\kappa$. With the exchange first, the nucleus recoils with $-\boldsymbol{p}$ before the kick, and the denominator is $\omega_e + a$. The boost factor $(\boldsymbol{v}_N\cdot\boldsymbol{p})/\omega_e^2$ becomes

$$\frac{1}{\omega_e - b + a} - \frac{1}{\omega_e + a} = \frac{b}{(\omega_e - b + a)(\omega_e + a)} = \frac{b}{\omega_e^2}\Big(1 + \frac{b-2a}{\omega_e} + \cdots\Big), \tag{12.24}$$

the free-nucleus case of Eq. (12.30). Its numerator is $\boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa$ at every p: the velocity in the boost is that of the kicked nucleus, whatever momentum the electrons take. Averaged over the angle c between $\boldsymbol{q}$ and $\boldsymbol{p}$, the square gives

$$\Big\langle \frac{b^2}{(\omega_e - b + a)^2(\omega_e + a)^2}\Big\rangle = \frac{v_N^2 p^2}{3\omega_e^4}\Big[1 - \frac{2p^2}{M_\kappa\omega_e} + \frac{9}{5}\,\frac{v_N^2 p^2}{\omega_e^2} + \cdots\Big],$$

the free-nucleus part of Eq. (12.33). The terms odd in c drop out; in a crystal they drop out because the loss function is even under $\boldsymbol{p}\to-\boldsymbol{p}$ in any target with time reversal. There is no term in p/q, so the soft limit does not fail when the electrons take a momentum comparable with the kick. The recoil term $2p^2/M_\kappa\omega_e$ is 2×10^{-3} at the median p near the gap. The Doppler term $\frac{9}{5}v_N^2p^2/\omega_e^2$, the shift $\boldsymbol{v}_N\cdot\boldsymbol{p}$ of the electronic energy by the moving nucleus at second order, grows with the recoil energy. Weighted with the silicon spectrum it is 0.04 at $E_R = 100\,\mathrm{eV}$, 0.13 at 300 eV and 0.42 at 1 keV. It is only the leading term. Near the gap $v_N p/\omega_e$ is already 0.41 at the median p and $E_R = 100\,\mathrm{eV}$, and the higher orders of Eq. (12.24) add as much again. With the exact bracket the soft-limit probability is low by the factor 0.93 at 100 eV, 0.79 at 300 eV and 0.62 at 500 eV, where the leading term alone gives 0.96, 0.90 and 0.85. The soft limit holds within 10 % up to $E_R \approx 160\,\mathrm{eV}$, the spectrum-weighted counterpart of the bound $v_N p/\omega_e \le 0.27$ found near the gap at fixed p in Limit Check 12.7. Toward 1 keV the expansion breaks down. There $v_N p$ at the median momentum is 1.6 eV, above the gap, and the denominator $\omega_e - b + a$ vanishes inside the range of the integral: the kicked nucleus, already moving, ionizes the electrons as a slow ion does. That is an on-shell two-step process, the electronic stopping of the recoiling ion, not a Migdal transition, and the second-order amplitude does not describe it. The momenta where it occurs hold 0.9 % of the soft-limit probability at 500 eV and 3.7 % at 1 keV.

Limit Check 12.4. Crystal Migdal rate with free recoil

$\mathcal{R} = (\rho_\chi/m_\chi)\,[2A^2\sigma_p\,Z_{\mathrm{ion}}^2\,\alpha N_T/(3\mu_{\chi n}^2\Omega M^2)]\int \mathrm{d}^3v\,(f/v)\int p_N^3\,\mathrm{d}p_N\int \mathrm{d}\omega\,\omega^{-4}\,\Theta(v - v_{\min})\,\mathcal{F}_{\mathrm{Mig}}(\omega)$, $v_{\min} = p_N/2\mu_{\chi N} + \omega/p_N$, with N_T the number of nuclei per unit target mass, Ω the cell volume, p_N the momentum transfer, the soft-limit amplitude and $\mathcal{F}_{\mathrm{Mig}}$ of Eq. (12.23) [10, Eqs. (1), (7), (11), (12)].

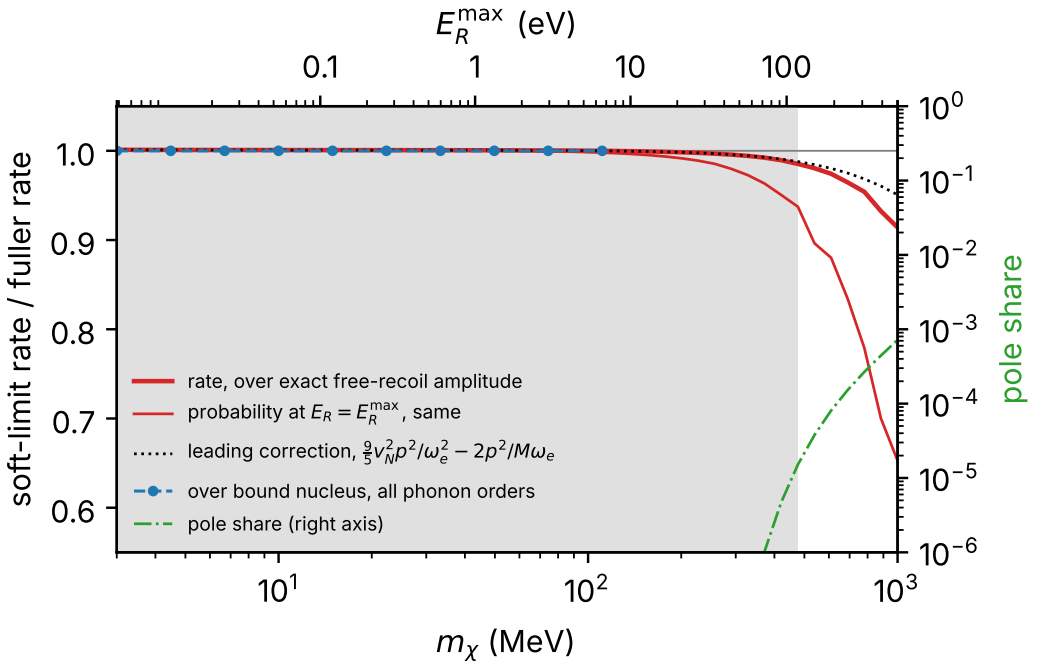


Figure 12.7: Total Migdal rate of silicon ($\omega_e \geq 1.11\,\mathrm{eV}$, heavy mediator) in the soft limit with a free recoil, divided by two fuller results, against the dark-matter mass; top axis, the largest recoil energy $E_R^{\max} = 2\mu_{\chi N}^2 v_{\max}^2/M$. Thick solid: over the rate with the exact free-recoil amplitude Eq. (12.24). Thin solid: the same ratio for the probability at $E_R = E_R^{\max}$, the end of the recoil spectrum. Dotted: the leading correction $\frac{9}{5}v_N^2 p^2/\omega_e^2 - 2p^2/M_\kappa\omega_e$ alone. Dashed with points: over the soft-limit rate with the incoherent response of a bound nucleus at all phonon orders and $T = 0$ (density of states of Sect. 5.2); it differs from one by less than 10^{-5}. Dash-dotted, right axis: the share of the soft-limit rate at momenta where the exact amplitude has a pole, $v_N p \geq \omega_e + p^2/2M_\kappa$, which both rates of the ratio leave out. Every curve uses the same diagonal loss function with local fields and $Z_{\mathrm{ion}}(p)$ in place of $\mathcal{F}_{\mathrm{Mig}}$, and the halo of Sect. 1.2. Gray band: $m_\chi \lesssim 0.5\,\mathrm{GeV}$, where the probability at the end of the recoil spectrum is within 10 % of the exact one.

Above about 0.5 GeV the end of the recoil spectrum passes $E_R \approx 160\,\mathrm{eV}$, where the motion of the nucleus during the electronic transition changes the probability by more than 10 %, and near 1 keV the moving nucleus ionizes the electrons by itself; the binding of the nucleus changes the soft-limit rate by less than 10^{-5} at every mass from 3 to 110 MeV.

Limit Check 12.4 sits at $n_B = 2$ and $\mathfrak{M} =$ sudden, with $\mathcal{K} =$ impulse and $\mathcal{X} =$ random-phase with local fields; the dashed curve shows that, within the soft limit, $\mathcal{K}$ can be changed to the multiphonon regime without changing the rate. From 3 to 150 MeV the soft-limit rate is within 1.1×10^{-3} of the exact free-recoil rate, and at 1 GeV, where the rate integrates over the whole recoil spectrum, it is low by the factor 0.91.

Below the onset the free-recoil amplitude is itself only a model, since the nucleus that keeps $\boldsymbol{q} - \boldsymbol{p}$ is bound, and Sect. 12.4 replaces it by the lattice factor at that momentum. The bound amplitude, too, stays within a few parts in 10^3 of the soft limit (Eq. (12.33)). At low mass, therefore, neither the soft limit nor the lattice limits the formula. The electronic factor

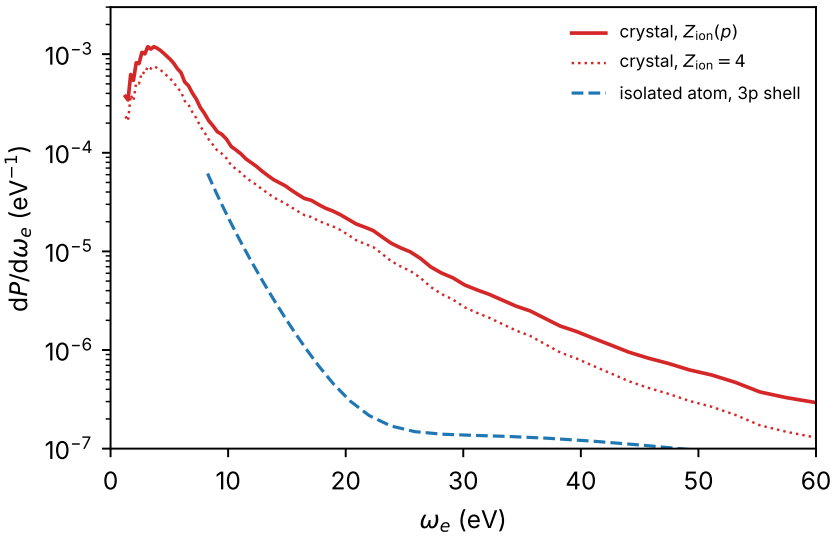


Figure 12.8: Migdal ionization spectra of silicon at the recoil energy $E_R = 100\,\text{eV}$ ($v_N = 8.74 \times 10^{-5}$), against the energy ω_e given to the electrons. Solid: the crystal, Eq. (12.21), with the loss function of Knapen, Kozaczuk and Lin with local fields (diagonal, direction-averaged, threshold 1.11 eV, the step-model gap that accompanies the tables) and $Z_{\text{ion}}(p)$ of Eq. (12.20); $P_{\text{Mig}} = 6.1 \times 10^{-3}$. Dotted: the same with a point ion, $Z_{\text{ion}} = 4$; 3.9×10^{-3}. Dashed: the $3p$ shell of an isolated atom at dipole order, from the Dirac–Hartree–Fock tables of Cox, Dolan, McCabe and Quiney [7] rescaled by v_N^2, with ω_e the kinetic energy of the electron plus the measured ionization energy of the atom, 8.15 eV; 1.2×10^{-4}. The atomic $3s$ (1.0×10^{-5}) and the core shells (8×10^{-5}, nearly all $2p$ above 100 eV) are not drawn.

does: a loss function without local fields, or the scalar function outside the zone, puts the probability 35 % too high (Limit Check 12.3).

The core shells stay frozen in Z_{ion}. Their own excitation is a separate Migdal process of atomic type, with the boost of each core shell (Sect. 12.2). The $2p$ shell of an isolated silicon atom gives 7.1×10^{-5} at $E_R = 100\,\text{eV}$, 1.2 % of the valence probability, at ω_e above 100 eV. And the formula needs no restriction to $m_\chi \ll M_\kappa$: the reduced mass $\mu_{\chi N}$ in v_{min} keeps the recoil of a heavy particle, and only the per-mass form below drops it.

12.3.3 Crystal and atomic spectra of silicon

The crystal formula and the atomic formula of Sect. 12.2 answer the same question for the same nucleus, and they differ by more than an order of magnitude. Fig. 12.8 shows both at $E_R = 100\,\text{eV}$. The crystal probability is 6.1×10^{-3}. The isolated silicon atom, with all its shells and the Dirac–Hartree–Fock tables of Cox and collaborators [7], gives 2.1×10^{-4}, of which 1.3×10^{-4} come from the $3s$ and $3p$ valence shells. The crystal is 29 times the whole atom and 47 times its valence shells.

The factor can be read off the figure. The threshold is lower: the crystal spectrum starts at the gap, 1.1 eV, the atomic one at the ionization energy of 8.15 eV, and with the weight ω_e^{-4} the region between the two holds 84 % of the crystal probability. Above 8 eV the crystal is still larger, six times at 10 eV and sixty times at 20 eV. There the crystal spectrum acts as a loss function, which concentrates the oscillator strength of the valence electrons in the plasmon and the broad interband peak around it, between 10 and 25 eV; the atom has only the tail of its $3p$ continuum there. The vertex differs too. The atom is boosted with all its electrons,

and the boost probes each orbital at the momentum $q_e = m_e v_N$, 45 eV at $E_R = 100\,\mathrm{eV}$, far below the inverse size of any shell. The crystal formula probes the valence electrons at the momenta p of the ion's field, mostly 3 to 9 keV, where $Z_{\rm ion}(p)$ adds the factor 1.6 of the point-ion comparison.

None of the three differences makes the atom a lower bound on the crystal in general. A crystal with a gap larger than the ionization energy of its atoms would not show the first difference, and one with valence electrons as localized as atomic ones would not show the second. The Wannier picture below shows how the two forms connect when the valence bands are flat.

12.3.4 The Migdal rate per unit mass

The rate follows from the free-recoil spectrum Eq. (11.35) by weighting each recoil with its Migdal probability. A recoil of momentum q that also gives the electrons ω_e needs the minimum speed Eq. (1.23) at $\omega = \omega_e + q^2/2M_\kappa$, which is $v_{\rm min} = \omega_e/q + q/2\mu_{\chi N}$. With $\mathrm{d}E_R = q\,\mathrm{d}q/M_\kappa$ and Eq. (12.21), the rate per unit target mass of a crystal of one species, for a heavy mediator, is

$$\begin{aligned}\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega_e} &= \frac{\rho_\chi\,\sigma_N}{2m_\chi\,\mu_{\chi N}^2 M_\kappa}\int \mathrm{d}q\; q\,\eta\Big(\frac{\omega_e}{q}+\frac{q}{2\mu_{\chi N}}\Big)\frac{\mathrm{d}P_{\rm Mig}}{\mathrm{d}\omega_e}\\ &= \frac{2\alpha\,\rho_\chi\,\sigma_N}{3m_\chi\,\mu_{\chi N}^2 M_\kappa^3}\,\frac{L_{\rm Mig}(\omega_e)}{\omega_e^4}\int \mathrm{d}q\; q^3\,\eta\Big(\frac{\omega_e}{q}+\frac{q}{2\mu_{\chi N}}\Big).\end{aligned} \tag{12.25}$$

For a compound each species adds its term with $1/M_\kappa$ replaced by its number of nuclei per unit mass. It is the formula of Limit Check 12.4 with $\mathcal{F}_{\rm Mig}$ replaced by its diagonal.

When ω_e is small against the kinetic energy of the dark matter, it can be dropped from $v_{\rm min}$, and the momentum integral becomes a moment of the halo: with $u = q/2\mu_{\chi N}$, $\int \mathrm{d}q\, q^3\eta(q/2\mu_{\chi N}) = 16\mu_{\chi N}^4\int_0^\infty \mathrm{d}u\, u^3\eta(u) = 4\mu_{\chi N}^4\langle v^3\rangle$, since $\int_0^\infty \mathrm{d}u\, u^3\eta(u) = \int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\, v^3/4$. For the halo of Sect. 1.2, $\langle v^3\rangle^{1/3} = 401\,\mathrm{km/s}$. With $\sigma_N = A^2\sigma_p\mu_{\chi N}^2/\mu_{\chi n}^2$ and $A_\kappa/M_\kappa = 1/m_u$, a compound of N_κ atoms of each species per formula unit, of mass number $A_{\rm cell} = \sum_\kappa N_\kappa A_\kappa$, gives

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega_e} = \underbrace{\frac{8\alpha\,\rho_\chi\,\sigma_p\,\langle v^3\rangle}{3\,m_u^3}\,\frac{\mu_{\chi N}^4}{m_\chi\,\mu_{\chi n}^2}}_{\text{halo and coupling}}\;\underbrace{\frac{1}{\omega_e^4}\sum_\kappa \frac{N_\kappa\,L_{{\rm Mig},\kappa}(\omega_e)}{A_{\rm cell}}}_{\text{material}}, \tag{12.26}$$

where $L_{{\rm Mig},\kappa}$ is Eq. (12.22) with the ion factor of species κ, and $\mu_{\chi N}$ is taken equal for all species, as it is for $m_\chi \ll M_\kappa$. This is the factorized rate of Knapen and collaborators [1, Eqs. (1), (6) and (A36)] in per-mass form. It separates the dark matter from the target. The mass dependence $\mu_{\chi N}^4/(m_\chi\mu_{\chi n}^2)$ grows as m_χ below the nucleon mass and as m_χ^3 between the nucleon and the nuclear mass. This factor has the local exponents 1.01 between 5 and 10 MeV, 1.24 between 100 and 200 MeV, and 2.07 between 3 and 6 GeV, where $\mu_{\chi N}$ already falls below m_χ. The material enters as $L_{\rm Mig}$ per nucleon: at fixed $L_{\rm Mig}$ the nuclear mass drops out, so that light nuclei win only if their ions keep a comparable $L_{\rm Mig}$, and the comparison is a comparison of loss functions weighted by $Z_{\rm ion}^2$ and ω_e^{-4}.

The comparison carries a convention: $Z_{\rm ion}$ must count the electrons that are not in the loss function. For nickel the small-p ion charge is 2, 10 or 18 according as L holds the $4s$ electrons, the $3d$ and $4s$, or also the $3s$ and $3p$, which changes its term by a factor 25 to 81. And the weight ω_e^{-4} makes $L_{\rm Mig}/\omega_e^4$ dominated by the region just above the gap: for a loss function that starts at $E_{\rm g}$, the f-sum bounds $\int \mathrm{d}\omega\, L/\omega^4$ by $(\pi/2)\omega_p^2/E_{\rm g}^5$ at each p, so that a small gap is worth more than a large oscillator strength. Sect. 17.5 uses Eq. (12.26) as the Migdal figure of merit.

Limit Check 12.5. Per-mass Migdal rate with the valence ionization factor

$\mathrm{d}\mathcal{R}/\mathrm{d}\omega \propto \mu_{\chi N}^4 m_\chi^{-1} \mu_{\chi n}^{-2} \omega^{-4} \sum_\kappa N_\kappa L_{\mathrm{Mig},\kappa}(\omega)/A_{\mathrm{cell}}$, the factorized soft-limit rate with free-recoil kinematics in which ω is dropped from energy conservation [1, Eqs. (1), (6)].

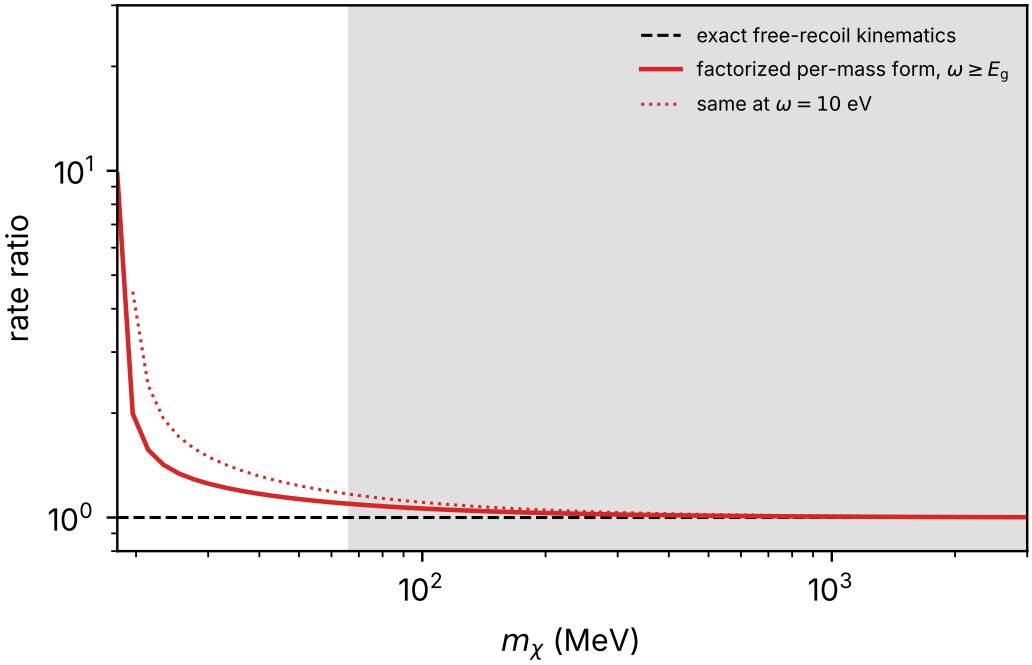


Figure 12.9: Migdal rate of silicon from the per-mass form Eq. (12.26), divided by the rate Eq. (12.25) with the exact free-recoil kinematics, against the dark-matter mass. Solid: total rate for $\omega_e \geq 1.11\,\mathrm{eV}$; dotted: the spectrum at $\omega_e = 10\,\mathrm{eV}$. Both rates: heavy mediator, $\sigma_p = 10^{-38}\,\mathrm{cm}^2$, halo of Sect. 1.2, the loss function with local fields and $Z_{\mathrm{ion}}(p)$, and momentum transfers above the impulse onset $q \geq 92.5\,\mathrm{keV}$. Gray band: $m_\chi \geq 66\,\mathrm{MeV}$, where the total is within 10 %.

Below about 60 MeV the electronic energy is no longer small against the kinetic energy of the dark matter, the per-mass form overestimates the rate, by a factor 1.25 at 30 MeV and 1.85 at 20 MeV, and the exact v_{min} of Eq. (12.25) must be kept.

Limit Check 12.5 sits at the point of Limit Check 12.3 on the truncation label.

For orientation, at $\sigma_p = 10^{-38}\,\mathrm{cm}^2$ the exact rate Eq. (12.25) above the onset is 1.9 events per kg and year at 30 MeV, 30 at 100 MeV and 970 at 1 GeV. These are rates of primary pairs of a few eV, with the spectrum of Fig. 12.8. The number of electrons a detector counts follows from the cascade of Chap. 18. Knapen and collaborators use a local density of $0.4\,\mathrm{GeV/cm}^3$ and an escape speed of 500 km/s, which scale their rates by 4/3 at high mass and lower them near the kinematic threshold.

12.3.5 The Wannier picture of the boost

Before the loss-function form, Liang and collaborators extended the atomic boost to a crystal in a different way [11]. They wrote the valence states in Wannier functions centered on the atoms (Sect. 3.6) and applied Migdal's boost $\mathrm{e}^{\mathrm{i}\boldsymbol{q}_e\cdot\boldsymbol{r}}$, $\boldsymbol{q}_e = (m_e/M_\kappa)\boldsymbol{q}$, only to the Wannier functions of the struck atom, leaving those of every other atom in place [11, Eqs. (2.7), (2.8)]. The picture is physical: the electrons attached to the moving nucleus are left behind, the

others are not disturbed. It agrees with Eq. (12.5) under one hypothesis about the force of the moving nucleus, and the same hypothesis fixes its form.

For one atom the two agree to all orders. For a single nucleus the force on the electrons is the negative of the force on the nucleus, $\nabla_{\boldsymbol{X}} V_{eN} = -\sum_i \nabla_i V_{eN}$, and with a local Hamiltonian the total force on the electrons is a double commutator, $\sum_i \nabla_i V_{eN} = -\mathrm{i}[H, \sum_i \boldsymbol{p}_i] = m_e[H, [H, \sum_i \boldsymbol{r}_i]]$, since the electron–electron forces cancel in the sum. Its matrix element is $m_e \Delta E_j^2 \langle \Psi_j | \sum_i \boldsymbol{r}_i | \Psi_0 \rangle$, and the amplitude Eq. (8.53) becomes $(m_e \boldsymbol{q}/M_\kappa) \cdot \langle \Psi_j | \sum_i \boldsymbol{r}_i | \Psi_0 \rangle$ up to a phase common to all final states, the first order of the boost Eq. (6.8) (Sect. 12.2). In a crystal the force of one nucleus is not the total force on the electrons. The Wannier picture replaces it by the total force restricted to the electrons in the Wannier functions of the struck atom. This is rigid Wannier following. Let $\hat{P}_I^{\mathrm{W}}$ be the projector of Sect. 3.6 on the occupied Wannier functions centered on nucleus I, and $\boldsymbol{F}_e = \sum_i \nabla_i V_{eN}$, the negative of the total force on the electrons. Then

$$\nabla_{\boldsymbol{X}_I} V_{eN} \;\to\; -\boldsymbol{F}_e \, \hat{P}_I^{\mathrm{W}}. \tag{12.27}$$

For independent electrons in a local one-body Hamiltonian h with eigenstates of energy ε_m, $\boldsymbol{F}_e = m_e[h, [h, \boldsymbol{r}]]$ and $\langle c | \boldsymbol{F}_e | m \rangle = m_e (\varepsilon_c - \varepsilon_m)^2 \boldsymbol{r}_{cm}$. Inserting a complete set between $\boldsymbol{F}_e$ and the projector, whose range is the occupied manifold, gives the amplitude of a transition from a valence state v to a conduction state c, with $\omega_{cv} = \varepsilon_c - \varepsilon_v$,

$$A_{cv}^{\mathrm{W}} = \frac{m_e \boldsymbol{q}}{M_\kappa} \cdot \sum_{v'} \frac{(\varepsilon_c - \varepsilon_{v'})^2}{\omega_{cv}^2} \, \boldsymbol{r}_{cv'} \, (\hat{P}_I^{\mathrm{W}})_{v'v}, \tag{12.28}$$

exactly, where v' runs over the occupied manifold. If the manifold is flat, every $\varepsilon_{v'}$ equals ε_v, the weights are one, and the amplitude is $(m_e \boldsymbol{q}/M_\kappa) \cdot \langle c | \boldsymbol{r} \, \hat{P}_I^{\mathrm{W}} | v \rangle$: the first order of the boost of the struck atom's Wannier functions, the picture of Liang and collaborators. A manifold of width W_{v} has $|\varepsilon_{v'} - \varepsilon_v| \le W_{\mathrm{v}}$, so each weight lies between $(1 - W_{\mathrm{v}}/\omega_{cv})^2$ and $(1 + W_{\mathrm{v}}/\omega_{cv})^2$, and

$$\Big| A_{cv}^{\mathrm{W}} - \frac{m_e \boldsymbol{q}}{M_\kappa} \cdot \langle c | \boldsymbol{r} \, \hat{P}_I^{\mathrm{W}} | v \rangle \Big| \le \Big(\frac{2 W_{\mathrm{v}}}{\omega_{cv}} + \frac{W_{\mathrm{v}}^2}{\omega_{cv}^2} \Big) \frac{m_e q}{M_\kappa} \sum_{v'} |\hat{\boldsymbol{q}} \cdot \boldsymbol{r}_{cv'}| \, |(\hat{P}_I^{\mathrm{W}})_{v'v}|. \tag{12.29}$$

The Wannier picture is exact for flat manifolds, the core shells and the isolated atom, and its error for a dispersive valence band is bounded by the ratio of the band width to the excitation energy. For silicon, with occupied bands about 12 eV wide, the bracket is 1.6 at $\omega_{cv} = 20$ eV, 0.54 at 50 eV and 0.25 at 100 eV: near the gap, where the Migdal probability lives, the bound does not constrain the valence amplitude at all.

The order of $\boldsymbol{F}_e$ and $\hat{P}_I^{\mathrm{W}}$ in Eq. (12.27) is not a matter of taste; the acoustic sum rule fixes it. When every nucleus is moved by the same $\delta \boldsymbol{X}$, the electrons see the whole crystal translated, and the gradients with respect to all nuclei add up to the gradient with respect to all electrons, $\sum_I \nabla_{\boldsymbol{X}_I} V_{eN} = -\boldsymbol{F}_e$; this is the electronic side of Eq. (5.18). The Wannier projectors of all sites add up to the projector on the occupied manifold, $\sum_I \hat{P}_I^{\mathrm{W}} = \hat{P}_{\mathrm{occ}}$, and $\hat{P}_{\mathrm{occ}} |v\rangle = |v\rangle$. With the projector on the side of the initial state, as in Eq. (12.27), the sum over sites of Eq. (12.28) is therefore $(m_e \boldsymbol{q}/M_\kappa) \cdot \boldsymbol{r}_{cv}$, the boost of the whole crystal, which a uniform kick must produce. The Hermitian ordering $\frac{1}{2}\{\boldsymbol{F}_e, \hat{P}_I^{\mathrm{W}}\}$ gives exactly half of it for every interband transition, because the term with the projector on the side of the final state vanishes, $\langle c | \hat{P}_I^{\mathrm{W}} = 0$ for c outside the occupied manifold, and it violates the sum rule. In the chain model of Limit Check 12.6 the site sum of Eq. (12.28) reproduces $\boldsymbol{r}_{cv}$ to 2×10^{-15} and the Hermitian ordering gives one half to 3×10^{-16}.

Limit Check 12.6. The Wannier picture of the boost

$A_{cv} = (m_e \boldsymbol{q}/M_\kappa) \cdot \langle c|\boldsymbol{r}\, \hat{P}_I^{\mathrm{W}}|v\rangle$ at first order in q_e: the Migdal boost applied to the occupied Wannier functions of the struck atom only [11, Eqs. (2.7), (2.8)].

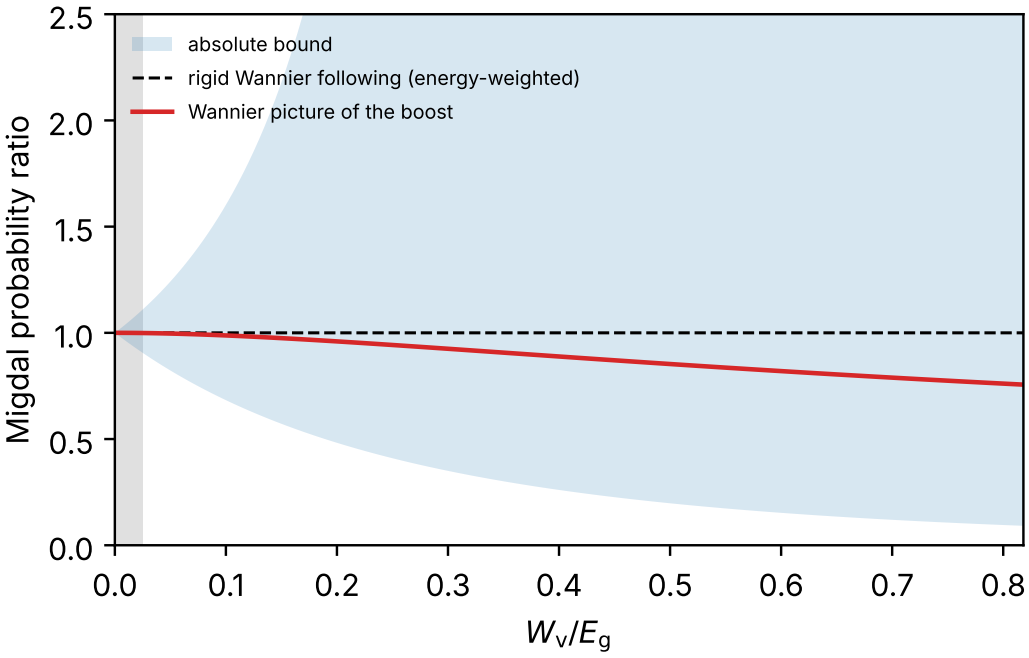


Figure 12.10: Total Migdal probability from the Wannier picture of the boost, divided by that of rigid Wannier following Eq. (12.28), against the width W_v of the valence band in units of the gap, in a model: an open chain of 81 sites with one valence and one conduction orbital per site, nearest-neighbor hoppings giving both bands the width W_v, an intrasite dipole of 0.3 lattice constants, no hopping between the two orbitals (so that the valence Wannier functions are the site orbitals), the valence band filled and the central site struck. The force on the electrons is defined by $\boldsymbol{F}_e = m_e[h,[h,\boldsymbol{r}]]$. Blue: the range allowed by the absolute bound Eq. (12.29) at $\omega_{cv} = E_\mathrm{g}$. Gray band: $W_\mathrm{v} \leq 0.025\, E_\mathrm{g}$, where the bound is within 10 %.

The bound guarantees 10 % only for valence bands narrower than a fortieth of the gap; in the model the actual ratio stays within 10 % up to $W_\mathrm{v} \approx 0.38\, E_\mathrm{g}$, and falls to 0.76 at $0.8\, E_\mathrm{g}$.

On the truncation label Limit Check 12.6 sits at $n_B = 2$ and $\mathfrak{M} =$ sudden, at the first order in q_e.

The model shows the sign of the error: the Wannier picture underestimates. Near the gap the initial state lies at the top of the valence band, every other valence state lies below it, and every weight $(\varepsilon_c - \varepsilon_{v'})^2/\omega_{cv}^2$ exceeds one; setting the weights to one loses the excess. In diamond the loss-function calculation of Liang and collaborators exceeded their Wannier calculation, most near the gap [10, Fig. 4], the direction the weights predict, although the two calculations also differ in their screening. The comparison is between two pictures of the same rigid following, however, and rigid following is itself a hypothesis. The true vertex is the change $\nabla_{\boldsymbol{X}_I} V^{\mathrm{scf}}$ of the self-consistent potential when one atom moves, the quantity of density-functional perturbation theory (Sect. 5.4), and for valence electrons shared between

neighbors Eq. (12.27) is not a controlled approximation to it. Its correction is the amplitude $(\boldsymbol{q}/M_\kappa)\cdot\langle c|\nabla_{\boldsymbol{X}_I}V^{\mathrm{scf}}+\boldsymbol{F}_e\hat{P}_I^{\mathrm{W}}|v\rangle/\omega_{cv}^2$.

Open problem. The ratio of the Migdal probability with the self-consistent vertex of one moving atom to the probability with rigid Wannier following has not been computed for any semiconductor. It decides whether the Wannier picture, which is exact for the core shells, can be used for valence electrons at all, and it is the cleanest test of the frozen-ion vertex $Z_{\mathrm{ion}}(p)$ of Eq. (12.20), which makes the same assumption in momentum space. Appendix E.18 specifies the calculation for silicon, germanium and diamond. It needs the Wannier functions of the occupied bands and the derivatives of the self-consistent potential in the same gauge, evaluates both amplitudes side by side at each momentum, and uses the acoustic sum rule and the drag-mass sum rule of Sect. 12.5 as checks.

12.4 Phonon final states at small momentum

A dark-matter particle of 2 MeV can transfer a momentum of at most $q=2m_\chi v_{\max}=10.6\,\mathrm{keV}$. At $q=10\,\mathrm{keV}$ a silicon nucleus would recoil with $E_R=1.9\,\mathrm{meV}$, less than the energy of most phonons, and the Debye–Waller exponent is $2W_\kappa=0.063$: in 94 % of the kicks the lattice is left exactly as it was (Sect. 11.3). Yet the particle carries enough kinetic energy, $\frac{1}{2}m_\chi v_{\max}^2=7\,\mathrm{eV}$, to make an electron–hole pair. If the electrons do not follow the kicked nucleus, a pair is made while the lattice stays in its ground state, and the whole deposit is electronic. The free recoil of Sect. 12.3 is the wrong lattice factor here; the impulse regime of Sect. 11.4 begins only at 92.5 keV, that is at $m_\chi=17.5\,\mathrm{MeV}$. Below it the lattice factor of the Migdal entry is the multiphonon response of Sect. 11.3 (Sect. 12.1.4).

Does the leading entry Eq. (12.6), derived with a nucleus that is free during the electronic time, hold for a bound nucleus, and with the momentum p taken by the electrons as large as the kick itself? It does, to a few parts in 10^3, for the incoherent lattice factor, which is the coherent zero-phonon process of the whole lattice averaged over directions for $q\gtrsim 2q_{\mathrm{BZ}}$ (for $q\ll q_{\mathrm{BZ}}$ the coherent process obeys Eq. (8.32) instead, as the key result below states). The real question at small q is how a Migdal event shares its energy between electrons and lattice when the lattice can take zero, one or a few phonons.

12.4.1 The amplitude of a bound nucleus

Write the Migdal process as two perturbations acting in turn: the kick $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$ from the dark matter, and the potential of nucleus I acting on the electrons. By Appendix D.33.2, that potential is a sum of Fourier components $v_\kappa(\boldsymbol{p})\,\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{X}_I}\,n_e(\boldsymbol{p})$, and the electronic factor $v_\kappa(\boldsymbol{p})\langle\Psi_j|n_e(\boldsymbol{p})|\Psi_0\rangle$ is common to every term below. What depends on the lattice is the nuclear factor. Let H_N be the Hamiltonian of the nuclei on the ground surface, with eigenstates $|\lambda\rangle$ of energy E_λ, the initial one $|0\rangle$. The two orderings of Eq. (2.54) give, for a final lattice state f and electronic energy ω_e,

$$\mathcal{A}_f(\boldsymbol{q},\boldsymbol{p})=\sum_\lambda\left[\frac{\langle f|\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{X}_I}|\lambda\rangle\langle\lambda|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}|0\rangle}{\omega_e+E_f-E_\lambda}-\frac{\langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}|\lambda\rangle\langle\lambda|\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{X}_I}|0\rangle}{\omega_e+E_\lambda-E_0}\right]. \tag{12.30}$$

In the first ordering the kick acts first and the nucleus then passes $\boldsymbol{p}$ to the electrons; the denominator is the energy of the final state minus that of the intermediate one. In the second the electrons are excited first, by the nucleus at rest, and the intermediate state carries ω_e in addition to the lattice energy. This is the amplitude of Berghaus and collaborators [2, Eq. (3.1)], with their lattice states as the $|\lambda\rangle$.

The electronic energy is large against the lattice energies that the matrix elements allow, and the denominators can be expanded in powers of $1/\omega_e$. Call $A = \mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{X}_I}$ and $B = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$. Since $\langle f|H_N = E_f\langle f|$ and $H_N|0\rangle = E_0|0\rangle$, each power of $(E_f - E_\lambda)$ in the first term turns into a commutator with H_N acting on A from the left, and each power of $(E_\lambda - E_0)$ in the second into the same commutator acting on A from the right. Order by order,

$$\mathcal{A}_f = \frac{\langle f|[A,B]|0\rangle}{\omega_e} + \frac{\langle f|[[A,H_N],B]|0\rangle}{\omega_e^2} + \frac{\langle f|[[[A,H_N],H_N],B]|0\rangle}{\omega_e^3} + \cdots . \tag{12.31}$$

The first term vanishes, since A and B are functions of the same position. For the second, $H_N = \sum_I \boldsymbol{P}_I^2/2M_\kappa + U(\{\boldsymbol{X}\})$ with any interatomic potential U. The potential commutes with A; the kinetic energy does not, because A shifts the momentum of nucleus I by $-\boldsymbol{p}$, $A\boldsymbol{P}_I A^\dagger = \boldsymbol{P}_I + \boldsymbol{p}$. Hence

$$[A,H_N] = \Big(\frac{\boldsymbol{p}\cdot\boldsymbol{P}_I}{M_\kappa} + \frac{p^2}{2M_\kappa}\Big)A, \qquad [[A,H_N],B] = \frac{\boldsymbol{p}}{M_\kappa}\cdot[\boldsymbol{P}_I,B]\,A = \frac{\boldsymbol{q}\cdot\boldsymbol{p}}{M_\kappa}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{q}-\boldsymbol{p})\cdot\boldsymbol{X}_I},$$

and the leading nuclear factor is

$$\mathcal{A}_f^{(2)} = \frac{\boldsymbol{q}\cdot\boldsymbol{p}}{M_\kappa\,\omega_e^2}\,\langle f|\mathrm{e}^{\mathrm{i}(\boldsymbol{q}-\boldsymbol{p})\cdot\boldsymbol{X}_I}|0\rangle. \tag{12.32}$$

This is the nuclear factor of Eq. (12.5), now obtained without any assumption about the binding of the nucleus. The interatomic potential has dropped out, as Berghaus and collaborators showed with their effective Hamiltonian $\nabla_I H_{\chi L}\cdot\nabla_I H_{eL}/(m_N\omega^2)$ [2, Eq. (3.6)]. Esposito and Rocchi obtained the same matrix element from the first nonadiabatic correction to the Born–Oppenheimer product state [12, Eq. (21)], and showed that in an infinite crystal, whose center of mass does not move, the adiabatic part of the amplitude vanishes. Two routes therefore give one result: the boost of Sect. 8.6 and the second-order amplitude with the lattice integrated out.

The lattice enters at the next order, and the expansion shows what controls it. With $X = \boldsymbol{p}\cdot\boldsymbol{P}_I/M_\kappa + p^2/2M_\kappa$, the same steps give $[[[A,H_N],H_N],B] = (\boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa)(2X - \boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa)\,BA$, plus a term $-\mathrm{i}(\boldsymbol{p}\cdot\nabla_I U/M_\kappa)A$ that commutes with B and drops out. The potential first appears at order ω_e^{-4}, through $[\boldsymbol{p}\cdot\nabla_I U, \boldsymbol{P}_I]$, the force constants. Summing $|\mathcal{A}_f|^2$ over all final states, averaging over the direction of $\boldsymbol{p}$ with the weight $(\boldsymbol{q}\cdot\boldsymbol{p})^2$ of the leading term, and collecting the terms of the first two orders,

$$\frac{\sum_f |\mathcal{A}_f|^2}{\sum_f |\mathcal{A}_f^{(2)}|^2} = 1 - \frac{2p^2}{M_\kappa\,\omega_e} + \frac{2\langle\omega_{\mathrm{ph}}^2\rangle}{\omega_e^2} + \frac{9}{5}\,\frac{v_N^2 p^2}{\omega_e^2} + \cdots , \tag{12.33}$$

with $v_N = q/M_\kappa$ and $\langle\omega_{\mathrm{ph}}^2\rangle$ the mean square phonon frequency seen by the nucleus. The first correction is the recoil energy of the momentum given to the electrons, $p^2/2M_\kappa$, over ω_e: the expectation value of $2X - \boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa$ in the kicked state is $\boldsymbol{p}\cdot(\boldsymbol{q}-\boldsymbol{p})/M_\kappa$, whose part odd in $\boldsymbol{p}$ averages away. The second is the binding of the nucleus, the factor $[1-(\omega_{\mathrm{ph}}/\omega_e)^2]^{-2}$ that Sect. 6.1 found for a nucleus kicked in a harmonic well, at its first order. The third is the Doppler shift of the electronic energy by the moving nucleus, $\boldsymbol{v}_N\cdot\boldsymbol{p}$, at second order: for a free nucleus, Eq. (12.30) is $b/[(\omega_e - b + a)(\omega_e + a)]$ with $b = \boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa$ and $a = p^2/2M_\kappa$, whose expansion $(b/\omega_e^2)(1 + b/\omega_e + b^2/\omega_e^2 + \cdots)$ squared and averaged with $\langle c^4\rangle/\langle c^2\rangle = 3/5$ gives the coefficient $9/5$.

None of the three is a ratio p/q, so the leading factor Eq. (12.32) holds when the electrons take as much momentum as the nucleus receives, or more: the velocity in the boost is that of the kicked nucleus, $\boldsymbol{q}/M_\kappa$, as for the free nucleus of Eq. (12.24). For silicon at $p = 5.7\,\mathrm{keV}$,

the median momentum of the Migdal weight (Sect. 12.3), and $\omega_e = 1.2\,\mathrm{eV}$, near the gap, the three terms of Eq. (12.33) are -2.1×10^{-3}, $+2.8 \times 10^{-3}$ (with the silicon phonon spectrum) and at most 5.1×10^{-4} for every q up to the impulse onset. The leading factor is exact to a few parts in 10^3 throughout the region of this section. Limit Check 12.7 below compares it with the full sum.

12.4.2 Migdal rate with phonon final states

Squaring Eq. (12.32), summing over the nuclei and keeping the diagonal $\boldsymbol{G} = \boldsymbol{G}'$ of the electronic response, as in Appendix D.33.4, gives the Migdal entry with the lattice factor of one nucleus at the momentum $\boldsymbol{q} - \boldsymbol{p}$ that it keeps. Below the impulse onset that factor is the harmonic self response of Eq. (11.26), a zero-phonon line of weight e^{-2W_κ} and the n-phonon spectra P_n^κ with the Poisson weights Eq. (5.60).

The Migdal probability with phonon final states

At $n_B = 2$, for the valence electrons with the frozen-core vertex $v_\kappa = -Z_{\mathrm{ion}}(p)e^2/p^2$ of Eq. (12.20) and the diagonal loss function, a kick $\boldsymbol{q}$ of a nucleus of species κ at a cubic site of a harmonic crystal at $T = 0$, with the incoherent lattice factor, leaves electronic energy ω_e and lattice energy ω_l with the probability density

$$\frac{\mathrm{d}^2 P_{\mathrm{Mig}}}{\mathrm{d}\omega_l\,\mathrm{d}\omega_e} = 4\alpha \int \frac{\mathrm{d}^3 p}{(2\pi)^3}\, Z_{\mathrm{ion}}(p)^2\, \frac{(\boldsymbol{v}_N \cdot \boldsymbol{p})^2}{\omega_e^4\, p^2}\, L(\boldsymbol{p}, \omega_e)\, \mathrm{e}^{-2W_\kappa(\boldsymbol{q}-\boldsymbol{p})} \times \Bigg[\underbrace{\delta(\omega_l)}_{\text{zero-phonon, } \mathcal{P}_0} + \underbrace{\sum_{n\geq 1} \frac{[2W_\kappa(\boldsymbol{q}-\boldsymbol{p})]^n}{n!}\, P_n^\kappa(\omega_l)}_{\text{phonon-emitting, } \mathcal{P}_+} \Bigg]. \tag{12.34}$$

Integrated over ω_l, the bracket times e^{-2W_κ} gives one for every $\boldsymbol{p}$, and $\mathrm{d}P_{\mathrm{Mig}}/\mathrm{d}\omega_e$ is Eq. (12.21) at every q in this incoherent lattice form. The coherent zero-phonon process of the whole lattice has the same electronic spectrum, averaged over directions, for $q \gtrsim 2q_{\mathrm{BZ}}$. For $q \ll q_{\mathrm{BZ}}$ Eq. (8.32) fixes it at $(m_e/M_\kappa)^2(2q^2/e^2)L(\boldsymbol{q}, \omega_e)/(2\pi \boldsymbol{n}_\kappa)$, with L the head of $-\varepsilon^{-1}$: in silicon 1/14.9 of Eq. (12.21), with median $\omega_e = 17.4\,\mathrm{eV}$ at the plasmon against 4.5 eV. Between the two the coherent response is not computed (Table 12.3).

This is the rate formula of Berghaus and collaborators [2, Eq. (3.8)], with their partial structure factor in the incoherent harmonic form [2, Eqs. (4.1), (4.5)] and their effective charge $Z(k)$ as Z_{ion}. Their full structure factor $S(\boldsymbol{q} - \boldsymbol{k}_e - \boldsymbol{K}, E)$ is the electron-following response of Eq. (12.6) at the momentum the nucleus keeps.

The sum rule in the key result, which Berghaus and collaborators pointed out, says that the electronic spectrum does not depend on the lattice dynamics. It needs no soft limit, since the lattice factor has unit weight at every $\boldsymbol{q} - \boldsymbol{p}$, and no impulse approximation. The lattice enters the rate only through energy conservation. In the double rate the lattice factor replaces the free recoil of Eq. (12.25). The weight $\mathrm{d}E_R = q\,\mathrm{d}q/M_\kappa$ is unchanged, and the minimum speed becomes $v_{\mathrm{min}} = (\omega_e + \omega_l)/q + q/2m_\chi$, so that for a heavy mediator

$$\frac{\mathrm{d}^2 \mathcal{R}}{\mathrm{d}\omega_l\,\mathrm{d}\omega_e} = \frac{\rho_\chi\, \sigma_N}{2 m_\chi\, \mu_{\chi N}^2 M_\kappa} \int \mathrm{d}q\; q\, \eta\Big(\frac{\omega_e + \omega_l}{q} + \frac{q}{2m_\chi}\Big) \frac{\mathrm{d}^2 P_{\mathrm{Mig}}}{\mathrm{d}\omega_l\,\mathrm{d}\omega_e}. \tag{12.35}$$

Since ω_l is a few meV against an ω_e of a few eV, $\mathrm{d}\mathcal{R}/\mathrm{d}\omega_e$ hardly depends on how the lattice takes its share: replacing the free recoil by the bound response leaves the silicon rate unchanged

(Limit Check 12.4). What the lattice factor decides is the second axis of the joint response, the energy that a phonon sensor sees together with the charge. Berghaus, Essig and McDuffie extended Eq. (12.35) to the nonrelativistic effective operators of the dark-matter–nucleon interaction [13]. Each operator multiplies the same electronic and lattice factors by its own nuclear response, and the lattice factor is again read at the momentum $\boldsymbol{q}-\boldsymbol{p}$ that the nucleus keeps.

Limit Check 12.7. Migdal effect with phonon final states

$\mathrm{d}^2P/\mathrm{d}\omega_l\,\mathrm{d}\omega_e = 4\alpha\int \mathrm{d}^3p/(2\pi)^3\, Z_{\mathrm{ion}}(p)^2(\boldsymbol{v}_N\cdot\boldsymbol{p})^2 L(\boldsymbol{p},\omega_e)\, s_\kappa^{\mathrm{inc}}(\boldsymbol{q}-\boldsymbol{p},\omega_l)/(\omega_e^4 p^2)$, from the effective interaction $\nabla H_{\chi L}\cdot\nabla H_{eL}/(M_\kappa\omega_e^2)$ [2, Eqs. (3.6), (3.8)].

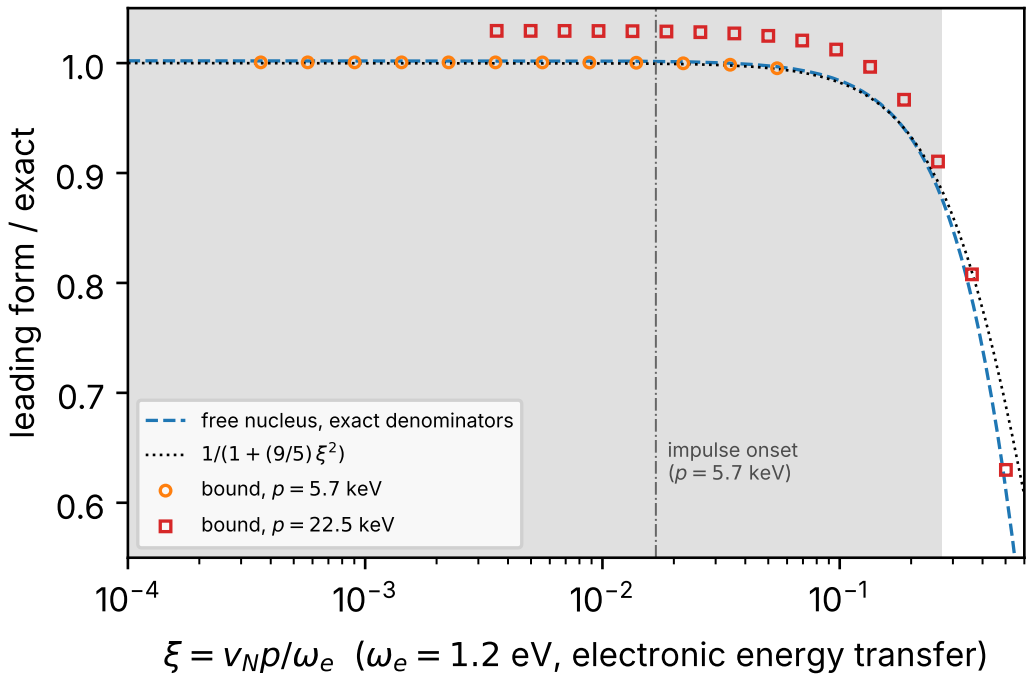


Figure 12.11: Migdal probability summed over all lattice final states, from the leading factor Eq. (12.32), divided by the same from the two time orderings Eq. (12.30) with exact energy denominators, against $v_N p/\omega_e$ at the electronic energy transfer $\omega_e = 1.2\,\mathrm{eV}$. Model: a silicon nucleus in an isotropic harmonic well of frequency $\omega_0 = \langle\omega^{-1}\rangle^{-1} = 30.4\,\mathrm{meV}$, which has the Debye–Waller factor of the full phonon spectrum; direction of $\boldsymbol{p}$ averaged. Circles, $p = 5.7\,\mathrm{keV}$ for $q = 2$ to $300\,\mathrm{keV}$; squares, $p = 22.5\,\mathrm{keV}$ for q up to $700\,\mathrm{keV}$, whose offset of 0.03 at small $v_N p/\omega_e$ is the recoil term $2p^2/M_\kappa\omega_e$. Dashed: free nucleus with exact denominators. Dotted: the leading correction $1/(1+\frac{9}{5}v_N^2p^2/\omega_e^2)$. Dash-dotted: the impulse onset at $p = 5.7\,\mathrm{keV}$. Gray: within 10 %, $v_N p/\omega_e \le 0.27$.

The form fails when the Doppler shift $\boldsymbol{v}_N\cdot\boldsymbol{p}$ approaches ω_e, at $q \approx 1.5\,\mathrm{MeV}$ for $p = 5.7\,\mathrm{keV}$ near the gap, far above the small-momentum region, where it holds to 10^{-3}.

On the truncation label Limit Check 12.7 sits at $n_B = 2$, with $\mathfrak{M}$ = sudden, $\mathcal{K}$ = multiphonon at all orders (harmonic, incoherent), and $\mathcal{X}$ = random-phase with local fields on the diagonal. The diagonal is the approximation that costs most: the plain dielectric form is 35 % high in total, the error concentrated at $p \gtrsim q_{\mathrm{BZ}}$ (Limit Check 12.3), and the off-diagonal elements

$\boldsymbol{G} \neq \boldsymbol{G}'$, a factor of about 1.4 as computed and 1.2 extrapolated, not a lower bound, in a model of silicon (Sect. 12.3.2), still await the full calculation (Appendix E.18).

12.4.3 Zero-phonon, few-phonon and impulse regimes

Fig. 12.12 evaluates Eq. (12.34) for silicon and sorts the Migdal events by the number of phonons they leave. The zero-phonon weight of the kick, $\mathrm{e}^{-2W_\kappa(q)}$, falls to one half at $q = 33.2\,\mathrm{keV}$, reached from $m_\chi = 6.3\,\mathrm{MeV}$. The impulse onset $x_R = 4$ lies at $q = 92.5\,\mathrm{keV}$, from $m_\chi = 17.5\,\mathrm{MeV}$ (Sect. 11.4). Below the first, most Migdal events are zero-phonon: at $q = 10\,\mathrm{keV}$, 92 % of them leave the lattice in its ground state, 7.6 % emit one phonon and 0.5 % two. Between the two boundaries a Migdal event emits a few phonons; at 50 keV the fractions for zero, one, two and three or more are 0.21, 0.32, 0.25 and 0.22. Above the onset the lattice factor is the impulse Gaussian, and the soft-limit treatment of Sect. 12.3 holds.

At small q the electrons' momentum controls the lattice. The momentum p that the electrons take has $\langle p^2\rangle^{1/2} = 6.1\,\mathrm{keV}$ in the weight of Eq. (12.34), and the nucleus keeps $\boldsymbol{q} - \boldsymbol{p}$, with $\langle|\boldsymbol{q} - \boldsymbol{p}|^2\rangle = q^2 + \langle p^2\rangle$. At 5 keV this is 2.5 times q^2. The mean lattice energy of a Migdal event, $\langle|\boldsymbol{q} - \boldsymbol{p}|^2\rangle/2M_\kappa$ by the first moment of Sect. 11.3, is then 1.20 meV, 2.5 times the recoil energy, and at 2 keV it is ten times E_R. The phonon-emitting fraction follows $2W_\kappa(\boldsymbol{q} - \boldsymbol{p})$, not $2W_\kappa(\boldsymbol{q})$: 3.8 % at 5 keV, against $1 - \mathrm{e}^{-2W(q)} = 1.6\,\%$. Above 30 keV the difference fades, and at 50 keV the mean lattice energy, 48.5 meV, exceeds E_R by 1.5 %.

In the exact harmonic lattice factor, its part at $\omega_l = 0$ is the product of two ground-state expectation values of n_κ, which vanish unless $\boldsymbol{q} - \boldsymbol{p}$ is a reciprocal-lattice vector. A zero-phonon Migdal event is a Bragg process: the crystal takes a reciprocal-lattice vector $\boldsymbol{G}$ as a whole, with weight $\mathrm{e}^{-2W_\kappa(\boldsymbol{G})}$ and the structure factor of the cell, and the electrons take exactly $\boldsymbol{p} = \boldsymbol{q} - \boldsymbol{G}$. The incoherent form in Eq. (12.34) replaces this sum over discrete momenta by an integral over $\boldsymbol{p}$, which is its average over the zone (Sect. 11.3). The two agree when the electronic weight varies slowly over q_BZ. For silicon, averaging over the direction of $\boldsymbol{q}$ and keeping the diamond structure factor $|1 + \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}}|^2$, the Bragg sum is 1.08, 1.04 and 1.03 times the incoherent integral at $q = 1.5$, 3 and 5 keV, and within 1 % of it from 10 keV. This check keeps the diagonal $\boldsymbol{G} = \boldsymbol{G}'$ and averages over directions; the cross-site terms $\boldsymbol{G} \neq \boldsymbol{G}'$ of the coherent sum, which carry the screening of Eq. (8.32) at $q \ll q_\mathrm{BZ}$, are not tested by it. The electronic weight is spread over several zones, median $p = 2.5\,q_\mathrm{BZ}$, and the coherent sum samples it densely enough. For a fixed direction of $\boldsymbol{q}$ in a crystal at rest the sum depends on the orientation, which a direction-averaged rate does not see.

Germanium follows the same pattern at shifted boundaries. It has no first-principles phonon spectrum in this book (Table 5.1). An estimate from its Debye temperature of 373 K, scaled by the ratio 0.82 of the computed to the Debye value of $\langle\omega^{-1}\rangle^{-1}$ in silicon, gives $\langle\omega^{-1}\rangle^{-1} \approx 17.6\,\mathrm{meV}$, and $\mathrm{e}^{-2W(q)} = \frac{1}{2}$ at $q \approx 41\,\mathrm{keV}$, from $m_\chi \approx 7.7\,\mathrm{MeV}$. The impulse onset is at 113 keV, from 21.3 MeV (Sect. 11.4). The heavier nucleus makes the recoil term of Eq. (12.33) smaller by the mass ratio 2.6. The binding term is about the same as in silicon, 3×10^{-3} at the germanium gap of 0.67 eV with the phonon energies scaled by the ratio of the Debye temperatures.

The loss function with local fields on a grid of $(\boldsymbol{p}, \omega_e)$ comes from Appendix E.6, the lattice response of silicon, germanium and gallium arsenide at all phonon orders with its coherent part from Appendix E.17, and their convolution from Appendix E.18.

12.4.4 The phonon-mediated form

Liang and collaborators describe the same process with the phonons as the intermediate states [5]. The kick creates on-shell phonons through the multiphonon vertex; one further

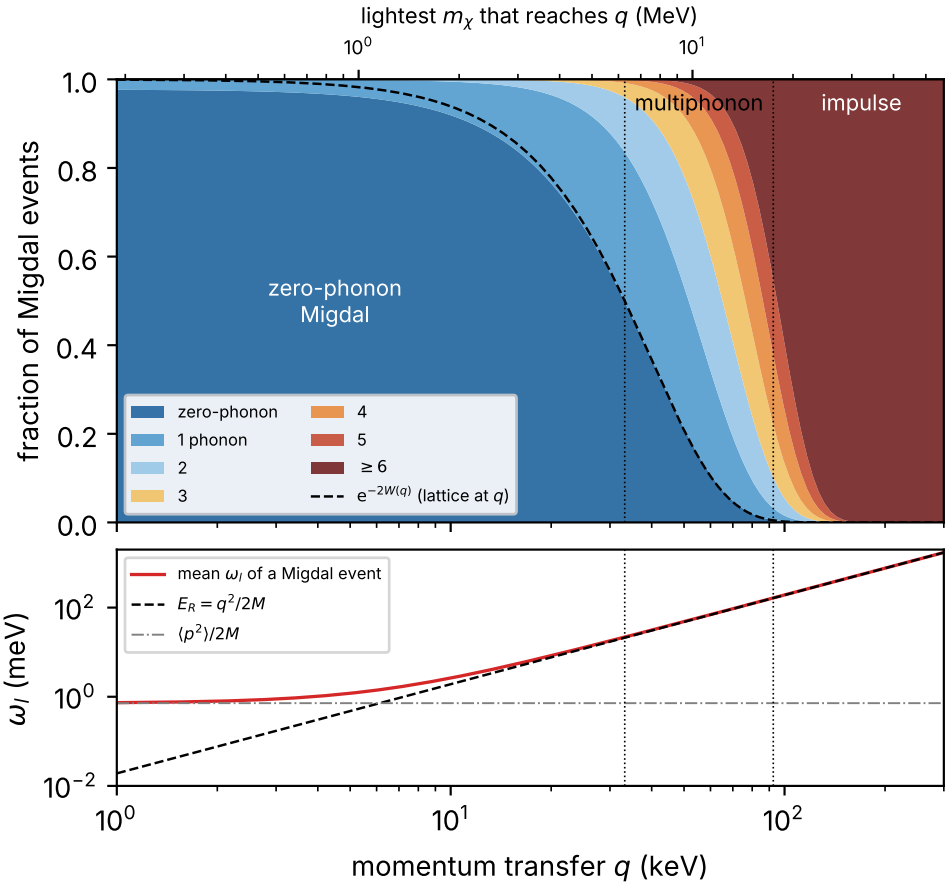


Figure 12.12: The three regimes of the Migdal effect in silicon against the momentum transfer q, at $T = 0$. Top: fraction of Migdal events that leave the lattice with zero, one, two, ... phonons, from Eq. (12.34), with the lattice factor at the momentum $\boldsymbol{q} - \boldsymbol{p}$ that the nucleus keeps, averaged over the momentum $\boldsymbol{p}$ given to the electrons with the weight $(\boldsymbol{q}\cdot\boldsymbol{p})^2 Z_{\rm ion}(p)^2 p^{-2} \int_{E_{\rm g}} {\rm d}\omega_e\, L(p,\omega_e)/\omega_e^4$. Dashed: the zero-phonon weight $e^{-2W(q)}$ of the kick alone. Bottom: mean energy ω_l given to the lattice in a Migdal event, against the recoil energy E_R and the recoil energy $\langle p^2\rangle/2M = 0.72\,\text{meV}$ of the electrons' momentum. Dotted lines: $e^{-2W(q)} = \frac{1}{2}$ at 33.2 keV and the impulse onset at 92.5 keV. Top axis: the lightest dark matter that reaches q, $q/2v_{\rm max}$. Model: diagonal loss function with local fields of Knapen, Kozaczuk and Lin [3] (threshold 1.11 eV, the step-model gap that accompanies the tables, $p \le 22.5\,\text{keV}$), $Z_{\rm ion}(p)$ of Eq. (12.20), incoherent harmonic lattice factor with Poisson weights in $2W(Q) = Q^2\langle u_x^2\rangle$ from the PBE phonon spectrum of Sect. 5.2.

phonon, virtual, carries momentum from the kicked nucleus to the bare electron–phonon vertex and excites a pair of energy $\varepsilon_i - \varepsilon_j$ [5, Eq. (3.1)]. The virtual phonon brings the propagator $1/[(\varepsilon_i - \varepsilon_j)^2 - \omega_{\boldsymbol{Q}\nu}^2]$, which is the bound-nucleus form of the boost denominator, Eq. (D.197). For gaps large against the phonon energies they reduce it to $1/(\varepsilon_i - \varepsilon_j)^2$ [5, Eq. (3.2)], and with the completeness of the polarization vectors the amplitude becomes the leading one, with the lattice factor at $\boldsymbol{q}$ instead of $\boldsymbol{q} - \boldsymbol{p}$.

This form is exact at first order in $\boldsymbol{p}$ for a harmonic lattice. Write the displacement of nucleus I in normal modes, $\boldsymbol{u}_I = \sum_\nu (\boldsymbol{\ell}_\nu a_\nu + \boldsymbol{\ell}_\nu^* a_\nu^\dagger)$, with $|\boldsymbol{\ell}_\nu|^2 = 1/(2NM_\kappa\omega_\nu)$ and ν running over the wavevectors and branches. The kick $B = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_I}$, up to a phase, displaces every mode, $a_\nu B = B(a_\nu + \beta_\nu)$ with $\beta_\nu = \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\ell}_\nu^*$. At first order in $\boldsymbol{p}$, $A = 1 - \mathrm{i}\boldsymbol{p}\cdot\boldsymbol{u}_I$; the 1 drops out of Eq. (12.30) as the first term of Eq. (12.31) does. The operator $\boldsymbol{p}\cdot\boldsymbol{u}_I$ changes one occupation by one, so the intermediate state differs from f (first ordering) or from 0 (second ordering) by one quantum, and the denominators are $\omega_e \mp \omega_\nu$ and $\omega_e + \omega_\nu$. Moving a_ν and $a_\nu^\dagger$ through B with the displacement rule, $a_\nu B = B(a_\nu + \beta_\nu)$ and $B a_\nu^\dagger = (a_\nu^\dagger - \beta_\nu^*)B$, the terms in $\langle f|a_\nu^\dagger B|0\rangle$ cancel between the two orderings, and what is left is proportional to $\langle f|B|0\rangle$:

$$\mathcal{A}_f = -\mathrm{i}\sum_\nu \Big[\frac{(\boldsymbol{p}\cdot\boldsymbol{\ell}_\nu)\beta_\nu}{\omega_e - \omega_\nu} + \frac{(\boldsymbol{p}\cdot\boldsymbol{\ell}_\nu^*)\beta_\nu^*}{\omega_e + \omega_\nu}\Big]\langle f|B|0\rangle = \frac{1}{N}\sum_\nu \frac{(\boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu})(\boldsymbol{p}\cdot\boldsymbol{e}_{\kappa\nu}^*)}{M_\kappa(\omega_e^2 - \omega_\nu^2)}\,\langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}|0\rangle, \quad (12.36)$$

for real polarization vectors $\boldsymbol{e}_{\kappa\nu}$, with $\boldsymbol{\ell}_\nu = \boldsymbol{e}_{\kappa\nu}/(2NM_\kappa\omega_\nu)^{1/2}$. The propagator is exact for every final state f, at all orders in q: the multiphonon blob and the virtual phonon factorize. At a cubic site the polarization sum is isotropic, $N^{-1}\sum_\nu e_{\kappa\nu}^\alpha e_{\kappa\nu}^{\beta*}\,\delta(\omega - \omega_\nu) = \delta_{\alpha\beta}\,g_{\mathrm{ph},\kappa}(\omega)/3$, and the propagator becomes $(\boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa)\int \mathrm{d}\omega\, g_{\mathrm{ph},\kappa}(\omega)/[3(\omega_e^2 - \omega^2)]$.

The phonon-mediated Migdal probability

At first order in $\boldsymbol{p}$, for a harmonic crystal at $T = 0$ and a cubic site, the leading probability Eq. (12.34) becomes

$$\frac{\mathrm{d}^2 P_{\mathrm{Mig}}}{\mathrm{d}\omega_l\,\mathrm{d}\omega_e} = 4\alpha \int \frac{\mathrm{d}^3 p}{(2\pi)^3}\, Z_{\mathrm{ion}}(p)^2\, \frac{(\boldsymbol{v}_N\cdot\boldsymbol{p})^2}{p^2} \left[\int \mathrm{d}\omega\, \frac{g_{\mathrm{ph},\kappa}(\omega)/3}{\omega_e^2 - \omega^2}\right]^2 L(\boldsymbol{p},\omega_e)\,\mathrm{e}^{-2W_\kappa(\boldsymbol{q})} \times \Bigg[\underbrace{\delta(\omega_l)}_{\text{zero-phonon}} + \underbrace{\sum_{n\geq 1} \frac{[2W_\kappa(\boldsymbol{q})]^n}{n!}\, P_n^\kappa(\omega_l)}_{\text{phonon-emitting}}\Bigg], \quad (12.37)$$

with the lattice factor at the kick $\boldsymbol{q}$. For $\omega_e \gg \omega_{\mathrm{ph}}$ the squared bracket is $\omega_e^{-4}[1 + 2\langle\omega_{\mathrm{ph}}^2\rangle/\omega_e^2 + \cdots]$, with $\langle\omega_{\mathrm{ph}}^2\rangle$ the mean of ω^2 over $g_{\mathrm{ph},\kappa}/3$.

With the propagator reduced to ω_e^{-2} this is the rate of Liang and collaborators [5, Eqs. (3.3), (3.4)], whose $T_0(E) = \delta(E)$ is the zero-phonon term and whose T_n with $n \geq 1$ are the phonon-emitting terms. Their factor $\mathrm{Im}[-\tilde{\varepsilon}^{-1}_{\boldsymbol{G}\boldsymbol{G}'}]$ keeps the elements $\boldsymbol{G} \neq \boldsymbol{G}'$ that Eq. (12.37) drops, and their Z_{ion} is the constant valence charge.

The zero-phonon term dominates wherever the phonon-mediated form is needed. At $q = 5\,\mathrm{keV}$ it holds 96 % of the Migdal events, and a rate that kept only the phonon-emitting terms would be low by a factor 26. The first numerical evaluation of the phonon-mediated rate omitted T_0 and found rates far below the free-recoil ones at small mass; with T_0 restored the two agree at all masses [5, Note added], as the sum rule under Eq. (12.34) requires.

The two published forms are complementary truncations of Eq. (12.30). The leading form Eq. (12.34) is exact in $\boldsymbol{p}$ and keeps the first order in $1/\omega_e$; the phonon-mediated form

Eq. (12.37) is exact in $\omega_{\mathrm{ph}}/\omega_e$ and keeps the first order in $\boldsymbol{p}$. Their difference at order p^2 is the lattice factor at $\boldsymbol{q}-\boldsymbol{p}$ against $\boldsymbol{q}$. For the electronic spectrum it is immaterial, since both lattice factors have unit weight. For the lattice spectrum it is not: at 5 keV the phonon-emitting fraction is 3.8 % with the lattice factor at $\boldsymbol{q}-\boldsymbol{p}$ and 1.6 % at $\boldsymbol{q}$, because the electrons' momentum is larger than the kick. In the frame of the crystal, where the transition density is frozen and the nucleus feels the recoil of $\boldsymbol{p}$, the all-orders form is the correct one. The exact sum over a harmonic site gives the zero-phonon fraction 0.967 at $q = 5$ keV and $p = 5.7$ keV, against 0.984 at $\boldsymbol{q}$ alone. The frame itself is the open question of Sect. 12.1.2.

Limit Check 12.8. Phonon-mediated Migdal effect

$\mathrm{d}^2P/\mathrm{d}\omega_l\,\mathrm{d}\omega_e$ as Limit Check 12.7 with the propagator $(\boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa)\sum_\nu |\boldsymbol{e}_\nu|^2/[N(\omega_e^2-\omega_\nu^2)]$ reduced to $\boldsymbol{q}\cdot\boldsymbol{p}/(M_\kappa\omega_e^2)$, the lattice factor at $\boldsymbol{q}$ and its zero-phonon term included [5, Eqs. (3.1)–(3.4)].

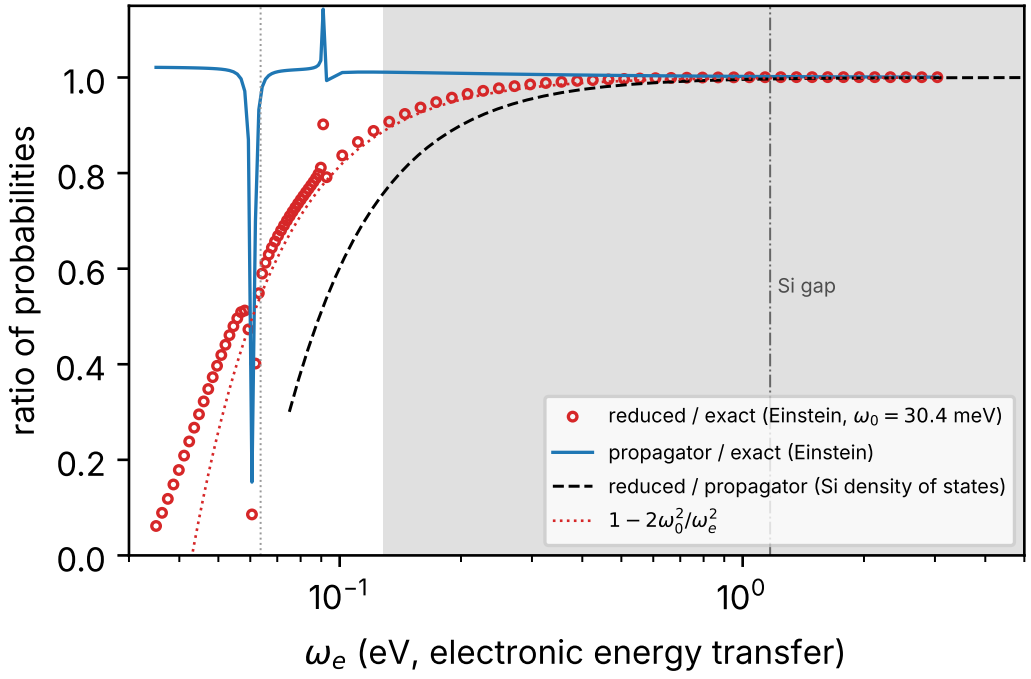


Figure 12.13: Migdal probability summed over lattice final states against the electronic energy transfer ω_e, at $q = 10$ keV and $p = 5.7$ keV, direction of $\boldsymbol{p}$ averaged. Circles: the reduced form over the exact second-order amplitude Eq. (12.30) of a silicon nucleus in an isotropic harmonic well of frequency $\omega_0 = 30.4$ meV. Solid: the unreduced propagator form Eq. (12.36) over the same exact result; its narrow feature at $\omega_e = 2\omega_0$ is the two-phonon intermediate state, of relative weight $p^2/(2M_\kappa\omega_0) = 0.02$, which the first-order form lacks. Dotted: the leading correction $1 - 2\omega_0^2/\omega_e^2$. Dashed: reduced over propagator form with the PBE phonon spectrum of silicon (PBE band top 63.7 meV, dotted vertical line). Gray: reduced form within 10 % of the exact one, $\omega_e \geq 4.2\,\omega_0$. Dash-dotted: the silicon gap.

The reduced form fails where ω_e approaches the phonon band, below 0.20 eV with the silicon spectrum, where the electrons no longer see a sudden kick and the pole at $\omega_e = \omega_\nu$ turns the process into the absorption of a real phonon.

On the truncation label Limit Check 12.8 sits at $n_B = 2$, with $\mathfrak{M}$ = sudden and the bound

denominators $\omega_e^2 - \omega_\nu^2$, $\mathcal{K}$ = multiphonon at all orders with the lattice factor at $\boldsymbol{q}$, and $\mathcal{X}$ = random-phase with local fields. Its leading correction is one step of n_B, from 2 to 4: the phonon energies in the electronic denominators, of relative size $2\langle\omega_{\rm ph}^2\rangle/\omega_e^2$, as Sect. 8.5 counts it.

Silicon and germanium do not need the propagator. At the silicon gap of 1.17 eV the reduced form is low by 2.9×10^{-3}, at the germanium gap by about 3×10^{-3}, and averaged over the silicon spectrum by 3×10^{-4}. It is needed where the gap is comparable with the phonon energies: the narrow-gap and Dirac materials of Chap. 16, whose gaps of tens of meV were the motivation of the phonon-mediated description, and the metals of Sect. 12.1.5. There the region $\omega_e \lesssim 3\omega_{\rm ph}$ is not a Migdal process with a small correction but the electron–phonon response of the target.

Worked example 12.4: The phonon-mediated term at $q = 5\,\mathrm{keV}$

A silicon nucleus kicked with $q = 5\,\mathrm{keV}$ recoils with $E_R = q^2/2M_\kappa = 0.478\,\mathrm{meV}$ and velocity $v_N = 1.91 \times 10^{-7}$, 57 m/s. The momentum is reached by dark matter of $m_\chi \geq q/2v_{\max} = 0.94\,\mathrm{MeV}$; at 1 MeV the largest energy transfer at this q, $qv_{\max} - q^2/2m_\chi$, is 0.75 eV, below the gap, and at 2 MeV it is 7.0 eV. The Debye–Waller exponent is $2W_\kappa(q) = 0.0157$. The momentum is 2.2 times $q_{\rm BZ}$, below the $2.95\,q_{\rm BZ}$ from which Sect. 11.3 shows the incoherent multiphonon terms to hold and the 18 keV from which the incoherent one-phonon term holds.

The probability. By the sum rule of Eq. (12.34), the electronic spectrum is Eq. (12.21) with the recoil energy E_R. At $q = 2.2\,q_{\rm BZ}$ this is the direction-averaged coherent process, within the $\boldsymbol{G} \neq \boldsymbol{G}'$ row of Table 12.3. With $Z_{\rm ion}(p)$, local fields and $\omega_e \geq 1.11\,\mathrm{eV}$, Sect. 12.3 gives 6.07×10^{-5} per eV of recoil energy, so $P_{\rm Mig} = 2.90 \times 10^{-8}$. One kick in 3.4×10^7 makes a pair.

The propagator. With the PBE phonon spectrum, $\langle\omega_{\rm ph}^2\rangle^{1/2} = 44.7\,\mathrm{meV}$. The squared bracket of Eq. (12.37) times ω_e^4 is 1.0029 at 1.17 eV, the leading correction $2\langle\omega_{\rm ph}^2\rangle/\omega_e^2 = 2.9 \times 10^{-3}$, and 1.0003 averaged over the electronic spectrum, half of which lies above 4.6 eV. The virtual phonon is far off shell, and the bound-nucleus correction is negligible.

The zero-phonon term. With the lattice factor at $\boldsymbol{q} - \boldsymbol{p}$, 96.2 % of the Migdal events leave no phonon, $P = 2.79 \times 10^{-8}$: the crystal takes the momentum as a whole, and all of the deposit, a few eV, is electronic. With the lattice factor at $\boldsymbol{q}$, as in Eq. (12.37), the share is 98.4 %.

The phonon-emitting terms. The rest, 1.10×10^{-9}, emits one phonon in 96 % of the cases. Its energy follows the one-phonon spectrum of the incoherent response, $g_{{\rm ph},\kappa}(\omega)/\omega$, with mean $\langle\omega^{-1}\rangle^{-1} = 30.4\,\mathrm{meV}$ and median 21.7 meV: a phonon-emitting Migdal event gives the lattice 64 times the recoil energy. The energy comes from the dark matter, not from the recoil: kinematics bounds only the sum $\omega_e + \omega_l$, by 7.0 eV at 2 MeV, and a phonon costs a small part of it. Averaged over all Migdal events, the lattice receives 1.20 meV, 2.5 times E_R, almost all in the rare events with a phonon.

A check. Take a nucleus in a harmonic well of frequency 30.4 meV, with $p = 5.7\,\mathrm{keV}$ and $\omega_e = 1.2\,\mathrm{eV}$. The exact amplitude Eq. (12.30), summed over final states, is 0.99934 times the leading form Eq. (12.32) and 0.9980 times the propagator form Eq. (12.36). Its zero-phonon fraction is 0.967, against 0.984 for the lattice factor at $\boldsymbol{q}$ alone.

At small q the Migdal signal is a pure charge signal: 96 % of it has no phonon. The few phonon-emitting events carry a full phonon, not a share of E_R, and their number is set by the electrons' momentum rather than by the kick. And the virtual-phonon propagator changes the probability by less than a part in a thousand.

The frame of the transition density, the lattice factor at small $|\boldsymbol{q}-\boldsymbol{p}|$ and the off-diagonal electronic response remain open at small momentum. In the frame of the crystal the lattice keeps $\boldsymbol{q}-\boldsymbol{p}$; if the valence electrons moved rigidly with the nucleus, it would keep $\boldsymbol{q}$. The difference is the Herzberg–Teller term of Sect. 12.1.2, and at 5 keV it is the factor 2.4 between the phonon-emitting fractions above; the electronic spectrum does not depend on it. Below $|\boldsymbol{q}-\boldsymbol{p}| \approx 6.6\,\mathrm{keV}$ the incoherent multiphonon terms are not certified (Limit Check 11.8), while the zero-phonon line, the dominant term, agrees with its diagonal, direction-averaged Bragg sum within 8 %. The elements $\boldsymbol{G} \neq \boldsymbol{G}'$ of the electronic response are common to all crystal forms (Appendix E.18). The frame and the lattice factor change the lattice spectrum of Migdal events, not their number, and Table 12.3 carries all three.

12.5 Moments of the Migdal spectrum

A silicon nucleus kicked to a recoil energy of 500 eV leaves its valence electrons excited in three events out of a hundred and gives them on average 0.18 eV per kick, 0.28 eV with the core shells. The second number needs no excited state of the crystal. It is the recoil energy times the ratio of the drag mass, the mass of the electrons that a slowly moving nucleus carries with it, to the mass of the nucleus. Every moment of the Migdal spectrum is fixed in the same way by the force that the electrons exert on one nucleus. The moments give the size of the effect without a spectrum, test a computed spectrum, bound the Migdal probability from both sides, and fix how a Migdal rate is added to a lattice rate.

12.5.1 Inverse moments of the force spectrum

The force of Eq. (12.4) has a spectrum. With $|0\rangle$ the ground state of the clamped electrons, $|a\rangle$ their excited states and $\Delta_a = E_a - E_0$, we define the spectral measure of its component along the kick and its inverse moments,

$$\mathrm{d}\sigma_\kappa(\Delta) = \sum_{a\neq 0} \left|\hat{\boldsymbol{q}} \cdot \langle a|\nabla_{\boldsymbol{R}_\kappa} V_{eN}|0\rangle\right|^2 \delta(\Delta-\Delta_a), \qquad \mathcal{I}_k = \int \Delta^{-k}\,\mathrm{d}\sigma_\kappa(\Delta), \tag{12.38}$$

for one nucleus $\boldsymbol{R}_\kappa$ of species κ. The leading-order Migdal probability Eq. (8.55), with $\boldsymbol{d}_{a0} = -\boldsymbol{F}_{a0}/\Delta_a$ from Eq. (5.11), resolved in the electronic energy, is

$$\frac{\mathrm{d}P_{\mathrm{Mig}}}{\mathrm{d}\omega_e} = \frac{q^2}{M_\kappa^2\,\omega_e^4}\,\frac{\mathrm{d}\sigma_\kappa}{\mathrm{d}\Delta}\bigg|_{\Delta=\omega_e}. \tag{12.39}$$

It is the electronic marginal of the Migdal entry Eq. (12.6), divided by $2\pi n_\kappa$, once the lattice factor has been integrated over ω_l with its unit weight (Appendix D.34.1). The Migdal spectrum is the force spectrum divided by ω_e^4, and each of its moments is one inverse moment of the force spectrum.

Moments of the Migdal spectrum

At leading order in the boost, for a nucleus of species κ kicked with momentum $\boldsymbol{q}$ and $E_R = q^2/2M_\kappa$,

$$\int \mathrm{d}\omega_e\,\omega_e^k\,\frac{\mathrm{d}P_{\mathrm{Mig}}}{\mathrm{d}\omega_e} = \frac{q^2}{M_\kappa^2}\,\mathcal{I}_{4-k}:$$

$$\underbrace{P_{\mathrm{Mig}} = \frac{q^2\mathcal{I}_4}{M_\kappa^2}}_{k=0}, \quad \underbrace{E_1^{\mathrm{Mig}} = E_R\,\frac{\delta M_\kappa}{M_\kappa}}_{k=1,\ \delta M_\kappa = 2\mathcal{I}_3}, \quad \underbrace{E_2^{\mathrm{Mig}} = \frac{q^2\mathcal{I}_2}{M_\kappa^2}}_{k=2}, \quad \underbrace{E_3^{\mathrm{Mig}} = \frac{q^2\mathcal{I}_1}{M_\kappa^2}}_{k=3}, \tag{12.40}$$

with E_k^{Mig} the kth moment. The first moment is the Born–Huang drag mass of Eq. (8.56) along $\hat{\boldsymbol{q}}$, times E_R/M_κ.

Every moment is proportional to E_R, so each defines a material constant per unit recoil energy. Each also has a meaning in the ground state of the clamped electrons. The first is the drag mass of Born and Huang [14, App. VIII]: $\delta M_\kappa = 2\mathcal{I}_3$ is Eq. (D.166) projected on $\hat{\boldsymbol{q}}$, the identity that Sect. 12.1.2 used to estimate P_{Mig}. It is also the curvature in frequency of the electronic force constant. A nucleus displaced at frequency ω below the gap feels the force constant $\Phi^{\mathrm{el}}_{\kappa\kappa}(\omega) = \langle \partial^2_{\hat{q}} V_{eN}\rangle - 2\int \mathrm{d}\sigma_\kappa(\Delta)\,\Delta/(\Delta^2-\omega^2)$, whose expansion in ω^2 is $\Phi^{\mathrm{el}}_{\kappa\kappa}(0) - \omega^2\delta M_\kappa + O(\omega^4)$ (Appendix D.34.2). The equation of motion $[\Phi(\omega) - \omega^2 M_\kappa]u = 0$ then moves with the mass $M_\kappa + \delta M_\kappa$: the electrons that follow a slow nucleus add their inertia to it. The second moment is the diagonal Born–Huang correction of Appendix D.28, $E_{\mathrm{BH}} = \mathcal{I}_2/2M_\kappa$ along $\hat{\boldsymbol{q}}$, so that $E_2^{\mathrm{Mig}} = 4E_R E_{\mathrm{BH}}$. The third is the second-order energy of the electrons in the field of a displaced nucleus, and Sect. 12.5.5 turns it into the electron density at the nucleus.

The force spectrum of the valence electrons follows from the loss function. Comparing Eq. (12.39) with the soft-limit probability Eq. (12.21) gives, for the diagonal $\boldsymbol{G} = \boldsymbol{G}'$ and the frozen-core vertex of Eq. (12.20),

$$\begin{aligned} \frac{\mathrm{d}\sigma_\kappa}{\mathrm{d}\omega} &= \frac{e^2}{\pi}\int \frac{\mathrm{d}^3p}{(2\pi)^3}\, Z_{\mathrm{ion}}(p)^2\, \frac{(\hat{\boldsymbol{q}}\cdot\boldsymbol{p})^2}{p^2}\, L(\boldsymbol{p},\omega), \\ \delta M_\kappa^{\mathrm{val}} &= \frac{8\alpha}{3}\int \frac{\mathrm{d}^3p}{(2\pi)^3}\, Z_{\mathrm{ion}}(p)^2 \int_0^\infty \frac{\mathrm{d}\omega}{\omega^3}\, \bar{L}(p,\omega), \end{aligned} \tag{12.41}$$

the second form for a direction-averaged loss function, with $e^2/\pi = 4\alpha$. The drag mass of the valence electrons is the ω^{-3} moment of the loss function, weighted with the square of the ion charge that the electrons see at each momentum. It is a property of the clamped crystal, like the static dielectric constant, which fixes the ω^{-1} moment of the same loss function at $p \to 0$ (Sect. 4.4).

The inverse moments come from linear-response calculations without any excited state. The drag mass is the ω^2 coefficient of the on-site force constant, which the frequency-dependent Sternheimer equation of Appendix E.4 returns at a few imaginary frequencies; two static Sternheimer solves per occupied state give $\mathcal{I}_2$, $\mathcal{I}_3$ and $\mathcal{I}_4$ at the level of independent pairs. Appendix E.18 specifies both for silicon, germanium and gallium arsenide, and Appendix E.17 the contact densities of Sect. 12.5.5.

12.5.2 On-site drag mass and trace rule

Two drag masses must be kept apart. Displace every nucleus of a crystal by the same $\boldsymbol{u}$, slowly. The electrons see a rigidly translated potential and follow it exactly, so the whole crystal moves with the mass of nuclei and electrons together. Let δM_{IJ} be the drag-mass tensor between nuclei I and J, defined as in Eq. (D.166) with the forces of the two nuclei. The translation then gives the *trace rule*

$$\sum_J \delta M_{IJ} = Z_\kappa m_e\, \mathbf{1} \quad \text{(one species)}, \qquad \sum_{\kappa\kappa'} \delta M_{\kappa\kappa'}(\boldsymbol{Q}=0) = N_e m_e\, \mathbf{1} \quad \text{per cell}, \tag{12.42}$$

which follows from the f-sum rule of the electrons in velocity form, because the summed force $\sum_J \nabla_J V_{eN}$ is the time derivative of the total electronic momentum (Appendix D.34.3). It holds for any insulator and any interaction among the electrons. For a free atom there is only one nucleus, the sum has one term, and it is the Thomas–Reiche–Kuhn value $N_e m_e$ of

Eq. (8.56). The estimate $14\,m_e$ that Sect. 8.6 took for silicon is this sum: exact, but not what a kick measures.

A dark-matter particle kicks one nucleus. The Migdal moments Eq. (12.40) contain the force of nucleus I alone, and δM_κ in them is the on-site element δM_{II}. Written in phonon wavevectors, δM_{II} is the average over the Brillouin zone of $\delta M_{\kappa\kappa}(\boldsymbol{Q})$, while Eq. (12.42) fixes only the value at $\boldsymbol{Q} = 0$. The two differ by the off-site coupling,

$$\delta M_{II} = Z_\kappa m_e - \sum_{J \neq I} \delta M_{IJ}, \tag{12.43}$$

which no sum rule fixes, in size or in sign. Positivity of the tensor at each $\boldsymbol{Q}$ gives $\delta M_{II} \geq 0$ and nothing more.

Silicon shows what the two elements are. Its cores move rigidly with the nucleus and contribute their own count to every element. The atomic Migdal tables of Cox and collaborators [7], rescaled to dipole order as in Sect. 12.2, confirm this. The first moments of the $1s$, $2s$ and $2p$ ionization spectra, with the measured binding energies, add up to $10.1\,m_e$, of which $7.2\,m_e$ is $2p$. The valence electrons are where the crystal acts. Evaluated on the loss function of Knapen, Kozaczuk and Lin with local fields, the table of Sect. 12.3, Eq. (12.41) gives $\delta M^{\text{val}} = 18.6\,m_e$, and with the cores the on-site drag mass is $28.7\,m_e$, $\delta M_{II}/M = 5.60 \times 10^{-4}$, twice the trace value. In the free atom the four valence electrons carry $4\,m_e$ in all, of which their ionization spectra carry $1.2\,m_e$ and the bound transitions the rest.

Where the valence value comes from decides how far to trust it. In silicon 96 % of the integral comes from $p > q_{\text{BZ}}$ and 61 % from $p > 5\,\text{keV}$, where the ion charge has risen from 4 to 4.84 and the probe resolves the core region. In energy, 67 % lies below 10 eV, near the gap, where the weight ω^{-3} is largest. The table stops at 89 eV; the valence f-sum held by the table bounds the missing part at 24 %. Momenta above 10 keV carry 2.8 % of the integral, so those beyond the table's 22.5 keV are estimated to add less. The approximations matter more than the truncation. A point ion of charge 4 gives $11.8\,m_e$, and the loss function without local fields $24.0\,m_e$. Both are uncontrolled at $p \gtrsim 5\,\text{keV}$. The frozen-core vertex assumes that the valence electrons inside the core region feel the bare nucleus screened by a rigid core. The diagonal $\boldsymbol{G} = \boldsymbol{G}'$ drops the off-diagonal elements of the dielectric matrix, which are sized only in a model and only for the rate (Sect. 12.3.2); the local fields on the diagonal already move the valence drag mass from 24.0 to $18.6\,m_e$.

The excess over the trace value has no sign fixed by anything above. A model shows that it can go either way. In a periodic chain of Gaussian wells, two independent electrons per well and one filled band (Appendix D.34.3), the trace rule Eq. (12.42) holds to 10^{-4} for every depth of the wells. The on-site element equals $2\,m_e$ while the wells are deep and isolated, and falls to $1.29\,m_e$ at a gap of 8.1 eV and to $0.92\,m_e$ at 5.4 eV as they overlap: with short-range wells the neighbors take part of the drag, and $\sum_{J\neq I} \delta M_{IJ} > 0$. The silicon table gives the opposite sign, $\sum_{J\neq I} \delta M_{IJ} = -14.7\,m_e$, from the long-range Coulomb vertex at large p that the chain lacks. No calculation of $\delta M_{\kappa\kappa}(\boldsymbol{Q})$ exists that could decide between the two pictures for silicon. We therefore take the on-site valence drag mass of silicon as $18.6\,m_e$, with the band 12 to $24\,m_e$ from the vertex and the local fields, and carry the band as a nuisance of the chapter (Table 12.3). For germanium the same tables give $39.5\,m_e$, with the band 11.5 to $52\,m_e$: the frozen $3d$ shell is shallow, bound by only 29.4 eV and extended to 0.4 Å, and $Z_{\text{ion}}(p)$ climbs from 4 already at a few keV. The cores of germanium carry $26.8\,m_e$ in the atomic tables without the $1s$, close to the $26\,m_e$ of a rigid core without the $1s$. Appendix E.18 specifies the $\boldsymbol{Q}$-resolved calculation that closes the question, with Eq. (12.42) as its acceptance test.

The difference between the two elements also tells which mass the lattice responses of Chap. 11 should carry. An incoherent kick probes δM_{II}. A coherent phonon of wavevector $\boldsymbol{Q}$

probes $\delta M_{\kappa\kappa'}(\boldsymbol{Q})$, and a long-wavelength acoustic phonon, which translates the whole cell, carries exactly the full mass of the atoms, $M_\kappa + Z_\kappa m_e$. The prescription of Sect. 8.6 and Sect. 12.1.4, mass $M_\kappa + \delta M_\kappa$ in the electron-following response, uses the on-site element for the impulse and multiphonon responses of an incoherent kick.

12.5.3 Bounds on the Migdal probability

Two moments bound the probability from both sides. On the support of the force spectrum every energy exceeds the threshold E_{th} of the electrons, the gap for the valence and the binding energy for a core shell, so $\mathcal{I}_4 \le \mathcal{I}_3/E_{\mathrm{th}}$. The Cauchy–Schwarz inequality for the measure $\mathrm{d}\sigma_\kappa$, applied to the functions Δ^{-1} and Δ^{-2}, gives $\mathcal{I}_3^2 \le \mathcal{I}_2\mathcal{I}_4$. With Eq. (12.40) the two become a statement about the measured quantities.

Bounds on the Migdal probability

For a nucleus kicked with recoil energy E_R, with the Migdal spectrum split into groups of final states s (valence and core shells) with thresholds E_s and drag masses δM_s, $\sum_s \delta M_s = \delta M_\kappa$,

$$\frac{(E_1^{\mathrm{Mig}})^2}{E_2^{\mathrm{Mig}}} \;\le\; P_{\mathrm{Mig}} \;\le\; \sum_s \frac{E_R}{E_s}\,\frac{\delta M_s}{M_\kappa} \;\le\; \frac{E_R}{E_{\mathrm{g}}}\,\frac{\delta M_\kappa}{M_\kappa}. \tag{12.44}$$

The upper bound needs only the drag mass and the threshold, the two numbers a static calculation gives most reliably. It is reached when the whole force spectrum sits at the threshold, and the shell-resolved form is much tighter than the last one because the core shells, which hold a large part of δM_κ, have thresholds a hundred times the gap. The lower bound is reached when the spectrum is a single line. The ratio of the two bounds measures the width of the spectrum in energy.

On the silicon table the valence probability is 19 % of its upper bound and 1.74 times its lower bound; for germanium the numbers are 15 % and 1.92. The mean energy $E_1^{\mathrm{Mig}}/P_{\mathrm{Mig}}$ of the valence spectrum is 6.0 eV in silicon and 5.1 eV in germanium, five to seven times the gap, which is why the upper bound is loose. For the $2p$ shell of the silicon atom the probability is 50 % of its bound, since a core spectrum begins with a sharp edge. The bounds carry the leading-order caveat of Eq. (12.40). At large recoil the motion of the nucleus during the transition, the free-recoil bracket Eq. (12.24), raises the probability by the factor 1.6 at 500 eV (Sect. 12.3.2); the bounds apply to the leading-order spectrum.

The probability, small per event, sets the counting of Migdal and lattice events. On the silicon table the valence probability reaches 10^{-3} at $E_R = 16.5$ eV and grows linearly from there. Above a few hundred electronvolts it is a few percent, and the lattice response that a Migdal rate is added to must then carry the weight $1 - P_{\mathrm{Mig}}$, as Eq. (12.7) requires. The mass correction is smaller. It moves the mean of the impulse response by $-E_R\,\delta M_{II}/M_\kappa$, a few parts in 10^4 of E_R, while the impulse width $E_R x_R^{-1/2}$ of Sect. 11.4 is a percent of E_R at these energies.

Worked example 12.5: The silicon drag mass and the bound on the Migdal probability at 500 eV

A silicon nucleus, $M = 26.16$ GeV, recoils with $E_R = 500$ eV when it receives $q = (2ME_R)^{1/2} = 5.11$ MeV; its velocity is $v_N = 1.96 \times 10^{-4}$. The drag masses of Sect. 12.5.2 are $10.1\,m_e$ for the cores and $18.6\,m_e$ for the valence, so $\delta M_{II} = 28.7\,m_e$ and $\delta M_{II}/M =$

5.60×10^{-4}, against 2.73×10^{-4} for the trace value $14\,m_e$. The first moment is $E_1^{\mathrm{Mig}} = E_R\,\delta M_{II}/M = 0.280\,\mathrm{eV}$: 0.099 eV from the cores and 0.181 eV from the valence.
The bounds follow shell by shell. The valence, with the threshold 1.11 eV of the table, has $P \le 0.181/1.11 = 0.163$; its second moment, $E_2^{\mathrm{Mig}} = 1.89\,\mathrm{eV}^2$, gives $P \ge 0.181^2/1.89 = 0.0174$. The computed spectrum gives $P = 0.0303$, inside the window. The $1s$, $2s$ and $2p$ shells, with thresholds of 1839, 150 and 99.5 eV, give $P \le 8.0 \times 10^{-4}$; the atomic tables give 4.0×10^{-4}. The total probability is $P_{\mathrm{Mig}} = 0.031$, below the shell-resolved bound 0.164 and far below the single-threshold bound $E_R\,\delta M_{II}/(M E_{\mathrm{g}}) = 0.24$ with the physical gap of 1.17 eV. Using the trace value $14\,m_e$ in that last bound gives 0.12. It happens to lie above the true probability here, but it is not a bound, because the kick probes the on-site element.
A lattice response for silicon at $E_R = 500\,\mathrm{eV}$, added to the Migdal rate, carries the weight $1 - P_{\mathrm{Mig}} = 0.969$, a correction of 3.1 %. It also carries the mass $M(1 + 5.60 \times 10^{-4})$, which moves its mean by 0.28 eV, a sixteenth of the impulse width $E_R(E_R/\langle\omega\rangle)^{-1/2} = 4.5\,\mathrm{eV}$ with $\langle\omega\rangle = 40.9\,\mathrm{meV}$ (Sect. 11.4).

12.5.4 The cross first moment

A dark-matter particle that couples to both electrons and nucleons reaches the Migdal final states through two amplitudes, the boost of a kicked nucleus and the direct recoil of an electron, and they interfere. The interference $S_{e\kappa}$ has a vanishing total first moment, Eq. (4.34), and Eq. (8.57) split it into equal and opposite parts on and off the line $\omega_e = 0$. For a free atom the split is exact to all orders. Use the center-of-mass separation of Sect. 12.2. In the elastic final states the atom recoils as a whole with the energy $q^2/2m_A$ and its electrons stay in the ground state; these states carry the first moment $\boldsymbol{\imath}_\kappa(q^2/m_A)F_{\mathrm{cr}}(q)$, and the excited states carry the negative of it (Appendix D.34.4).

Electronic part of the cross first moment

For a free atom of mass m_A, exactly,

$$\mathscr{M}^1_{e\kappa}\big|_{\omega_e>0} = -\boldsymbol{\imath}_\kappa \frac{q^2}{m_A} F_{\mathrm{cr}}(q),$$
$$F_{\mathrm{cr}}(q) = \mathrm{Re}\Big[\langle 0|\sum_e \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\rho}_e}\mathrm{e}^{-\mathrm{i}\boldsymbol{q}_e\cdot\sum_{e'}\boldsymbol{\rho}_{e'}}|0\rangle^*\,\langle 0|\mathrm{e}^{-\mathrm{i}\boldsymbol{q}_e\cdot\sum_{e'}\boldsymbol{\rho}_{e'}}|0\rangle\Big], \tag{12.45}$$

with $\boldsymbol{\rho}_e$ and $\boldsymbol{q}_e = m_e\boldsymbol{q}/m_A$ as in Eq. (12.10). In a crystal, at leading order in κ_{BO}, $m_A \to M_\kappa$ and F_{cr} is defined by the electron-following form of Eq. (8.57), $\mathscr{M}^1_{e\kappa}|_{\mathrm{ad}} = \boldsymbol{\imath}_\kappa q^2 F_{\mathrm{cr}}(\boldsymbol{q})^*/M_\kappa$.

For an atom, $F_{\mathrm{cr}}(q) = f(q)[1 + O(q_e^2\langle\rho^2\rangle, m_e/m_A)]$, the atomic form factor. In a crystal F_{cr} counts the electrons that move with nucleus κ, weighted by the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ at the momentum of the probe. At $q \to 0$ it is the charge that follows the nucleus at zero macroscopic field, $\hat{\boldsymbol{q}}\cdot(Z_\kappa\mathbf{1} - Z^*_\kappa)\cdot\hat{\boldsymbol{q}}$, which is 14 for silicon and, with the Born charge $Z^* = 2.11$ of Sect. 5.5, 28.9 for gallium and 35.1 for arsenic in GaAs. As q grows past the inverse size of each shell, that shell drops out of F_{cr}, and at $q \gg 1/a_{1s}$ only the nucleus is left to recoil with no electrons attached.

The negative part lies in the Migdal cells of $S_{e\kappa}$. Its size is close to the largest that the positivity Eq. (4.20) of the response matrix allows. For the free atom at small q, the Migdal first moment is $\boldsymbol{\imath}_\kappa q^2 N_e m_e/m_A^2$, and that of the electronic sectors of S_{ee} is $\boldsymbol{\imath}_\kappa q^2 N_e/m_e$.

Their geometric mean, $n_\kappa q^2 N_e/m_A$, is reached by Eq. (12.45) when $F_{\rm cr} \to N_e$, because at small q the boost and the direct recoil excite the same dipole, with amplitudes in the fixed ratio $-m_e/m_A$ of Eq. (8.32). Any model of a mixed coupling that keeps the Born-charge interference on the line $\omega_e = 0$ must keep this negative Migdal interference, of equal size, off it.

12.5.5 Third moment and contact density

The third moment uses only the ground state. The inverse moment $\mathcal{I}_1 = \sum_a |\langle a|\partial_{\hat{q}} V_{eN}|0\rangle|^2/\Delta_a$ is half the second-order energy of the electrons when nucleus κ is displaced along $\hat{\boldsymbol{q}}$. Second-order perturbation theory writes the curvature of the Born–Oppenheimer surface as $\partial_{\hat{q}}^2 E_0 = \langle \partial_{\hat{q}}^2 V\rangle - 2\mathcal{I}_1$, with V the electron–nucleus and nucleus–nucleus potentials. For point nuclei the second derivative of the Coulomb potential is a contact term: $\nabla^2_{\boldsymbol{R}} V_{eN} = Z_\kappa e^2 \sum_i \delta^3(\boldsymbol{r}_i - \boldsymbol{R}_\kappa)$, and $\nabla^2_{\boldsymbol{R}} V_{NN} = 0$ at the nucleus. On a cubic site a third of the trace lies along $\hat{\boldsymbol{q}}$, and with the curvature c_3 that Sect. 11.4 gives to the third cumulant of the impulse response we obtain the following result (Appendix D.34.5).

Third moment of the Migdal spectrum

For point nuclei on cubic sites, at leading order in the boost and averaged over the zero-point motion,

$$E_3^{\rm Mig} = \underbrace{\frac{q^2}{2M_\kappa^2}\,\frac{Z_\kappa e^2 n_e^{(\kappa)}}{3}}_{\text{contact}} - \underbrace{c_3}_{\text{lattice}}, \qquad c_3 = \frac{q^2}{2M_\kappa^2}\,\langle \partial_{\hat{q}}^2 E_0\rangle, \tag{12.46}$$

with $n_e^{(\kappa)}$ the electron density at the nucleus of Eq. (8.24). The full third potential moment of the nuclear response, $c_3 + E_3^{\rm Mig}$, is the contact term alone.

The lattice curvature is negligible: in silicon $M\langle\omega^2\rangle$ with $\langle\omega^2\rangle^{1/2} = 44.7\,\text{meV}$ from the phonon density of states of Sect. 11.4 is 1.3×10^{-6} of the contact term. The split that Sect. 11.4 made, c_3 for the electron-following response and the rest for the electrons, is therefore a split of the third moment into one part in a million and the rest. With the contact density of Worked example 8.4, $n_e^{(\kappa)} = 1765\, a_0^{-3}$, the third moment of silicon is $1497\,\text{eV}^2$ per eV of recoil energy. For germanium an all-electron Hartree–Fock calculation of the free atom, done in the same way, gives $n_e^{(\kappa)} = 22\,475\, a_0^{-3}$, converged to 0.2 % in the tight basis functions, and $16\,849\,\text{eV}^2$ per eV of recoil. The valence electrons hold 0.43 % of the contact density in silicon and 0.085 % in germanium. The third moment is a property of the $1s$ and $2s$ shells, whose spectra reach keV, and a valence table sees almost none of it: the silicon table, cut at 89 eV, holds 5×10^{-5} of it.

The same contact density appears in the mixed third moment of Sect. 8.3. At $q \to 0$ on a cubic site Eq. (8.25) reads $\mathcal{M}^3_{e\kappa} = -2n_\kappa(M_\kappa/m_e)(E_3^{\rm Mig} + c_3)[1 + 2n_e/n_e^{(\kappa)}]$: the interference of electrons and nuclei measures the Migdal third moment, up to the uniform-density term $2n_e/n_e^{(\kappa)} = 1.2 \times 10^{-4}$ in silicon. At finite q the $e\kappa$ face carries the bracket of Eq. (8.26), which climbs from $n_e^{(\kappa)}/3$ toward $n_e^{(\kappa)}$, while the $\kappa\kappa$ face keeps Eq. (12.46) at every q. The two faces agree only through that bracket.

Relativity changes the third moment more than anything else. The Dirac Hamiltonian depends on the nuclear position only through V_{eN}, so the derivation of Eq. (12.46) holds for relativistic electrons unchanged. What changes is the density at the nucleus. The Foldy–Wouthuysen reduction of Sect. 10.6, recapped in Appendix C.15, identifies it: the Darwin

term Eq. (C.133) is a contact interaction proportional to $\nabla^2 V$, and every operator that probes $\nabla^2 V$ carries relative corrections of order $(Z\alpha)^2$ from the core s shells. The expansion in $1/m_e$ cannot give their size, because the Dirac $1s$ orbital behaves as $r^{\gamma_\mathrm{D}-1}$ near a point nucleus, with $\gamma_\mathrm{D} = [1-(Z\alpha)^2]^{1/2}$, and its density diverges at $r = 0$. A nucleus of finite size cuts the divergence off, and its charge distribution ρ_nuc replaces the delta function: $\nabla^2_{\boldsymbol{R}} V_{eN} = Z_\kappa e^2 \rho_\mathrm{nuc}(\boldsymbol{r} - \boldsymbol{R}_\kappa)$, so that $\boldsymbol{n}_e^{(\kappa)}$ becomes the density averaged over the nucleus. For a $1s$ orbital of charge Z and a uniform nucleus of radius R_nuc, the ratio of that average to the nonrelativistic contact density is (Appendix D.34.6)

$$\mathcal{R}_\mathrm{rel} = \frac{6}{(2\gamma_\mathrm{D}+1)\,\Gamma(2\gamma_\mathrm{D}+1)} \Big(\frac{2ZR_\mathrm{nuc}}{a_0}\Big)^{2\gamma_\mathrm{D}-2} = 1 + (Z\alpha)^2 \ln\frac{a_0}{2ZR_\mathrm{nuc}} + c_\mathrm{rel}(Z\alpha)^2 + \cdots , \tag{12.47}$$

with the Euler gamma function Γ and c_rel a coefficient of order one without the logarithm. The logarithm is the reason the $1/m_e$ series fails: the correction is of order $(Z\alpha)^2$ times the logarithm of the ratio of the atomic to the nuclear size. With the nuclear radii of Sect. 2.2, 3.47 fm for silicon and 4.77 fm for germanium, $\mathcal{R}_\mathrm{rel}$ is 1.082 for silicon and 1.425 for germanium, and 1.397 and 1.454 for gallium and arsenic. The logarithm alone gives 0.066 of the excess 0.082 in silicon and 0.281 of 0.425 in germanium. We apply the $1s$ factor to every s shell, since all of them behave alike inside the nucleus, and a change of the nuclear radius by 10 % moves $\mathcal{R}_\mathrm{rel}$ by 0.10 % in silicon and 0.53 % in germanium. The remainder without the logarithm, 0.017 in silicon and 0.144 in germanium, depends on the model of the orbital inside the nucleus and is uncertain at order $(Z\alpha)^2$, about 5 % in germanium. The $p_{1/2}$ shells add a contact density of relative order $(Z\alpha)^2$ that the $1s$ factor omits, since their small components are s-like at the origin. The relativistic third moments are $1620\,\mathrm{eV}^2$ per eV of recoil for silicon and $24\,000\,\mathrm{eV}^2$ for germanium, higher by 8 % and 42 % than the nonrelativistic values. Third moments from nonrelativistic all-electron densities are therefore about 30 % low for germanium and gallium arsenide.

Table 12.2 collects the moments of both crystals.

Open problem. The finite-q factor of the electron density at the nucleus, the bracket of Eq. (8.26) that the mixed third moment carries, is known only for spherical atomic densities. For a crystal the valence anisotropy and the relativistic shape of the factor are missing. The valence density adds an anisotropy of tetrahedral symmetry, estimated at 1.3×10^{-3} of the factor in silicon. Relativity changes the factor in two places that do not cancel. Its normalization, the contact density, rises by the $\mathcal{R}_\mathrm{rel}$ of Eq. (12.47). Its quadrupolar part, set by the density at distances $1/q$ from the nucleus, changes by only a few percent; applying the relativistic change of one to the nonrelativistic value of the other is an error of the size of $\mathcal{R}_\mathrm{rel} - 1$. Until both are computed, the mixed third moment at finite q can check the Migdal third moment only to that accuracy in germanium and gallium arsenide, and the relativistic factor itself carries the non-logarithmic remainder and the $p_{1/2}$ contact density, each of relative order $(Z\alpha)^2$. Appendix E.17 specifies the all-electron and finite-nucleus Dirac calculations that remove both gaps.

Table 12.2: Migdal moments of silicon and germanium per nucleus, at leading order in the boost. Drag masses in units of m_e; the other moments per eV of recoil energy E_R. Trace: the rule Eq. (12.42). Cores: first moments of the free-atom ionization spectra at dipole order (Dirac–Hartree–Fock tables of Cox, Dolan, McCabe and Quiney [7], measured binding energies; germanium without $1s$). Valence: Eqs. (12.21) and (12.41) on the loss functions of Knapen, Kozaczuk and Lin, computed with the grid-based projector-augmented wave (GPAW) code, with local fields (diagonal, direction-averaged; thresholds 1.11 eV for Si, the step-model gap that accompanies the table, and 0.744 eV for Ge), with $Z_{\mathrm{ion}}(p)$ from the free-atom core; in brackets the range from a point ion $Z_{\mathrm{ion}} = 4$ to the loss function without local fields. Third moment: Eq. (12.46) with the Hartree–Fock contact densities of the free atoms, nonrelativistic and with the factor Eq. (12.47).

Quantity	Si	Ge
Trace $\sum_J \delta M_{IJ}$	14	32
On-site drag mass, cores	10.1	26.8
On-site drag mass, valence	18.6 [11.8–24.0]	39.5 [11.5–52.1]
Valence P_{Mig}/E_R (eV^{-1})	6.07×10^{-5}	5.89×10^{-5}
Valence mean ω_e (eV)	5.97	5.07
Valence E_2^{Mig}/E_R (eV)	3.77×10^{-3}	2.91×10^{-3}
Contact density $n_e^{(\kappa)}$ (a_0^{-3}), nonrelativistic	1765	22 475
E_3^{Mig}/E_R (eV^2), nonrelativistic	1497	16 849
Relativistic factor $\mathcal{R}_{\mathrm{rel}}$	1.082	1.425
E_3^{Mig}/E_R (eV^2), relativistic	1620	24 008

12.6 Bremsstrahlung and Migdal-type absorption

A nucleus that is suddenly set moving is a charge that radiates. Besides leaving an electron–hole pair behind, in 6.1×10^{-3} of the kicks for silicon at a recoil energy of 100 eV (Sect. 12.3), it emits photons, the bremsstrahlung of the recoil. Kouvaris and Pradler pointed out that such photons, unlike the recoil, are easy to see, and used them to carry xenon searches to lower masses [15]. A second process uses the same nonadiabatic structure with the roles of energy and momentum exchanged. A scalar coupled to nucleons and absorbed by a nucleus gives it almost no momentum but its whole rest energy, and the electrons, which do not follow the nucleus exactly, take that energy [16]. Both effects were worked out for isolated atoms. We will compute both in a crystal. For bremsstrahlung the crystal replaces the charge of the nucleus by a dynamical Born charge, which in silicon and germanium is fixed exactly by the measured dielectric function; for absorption it replaces the atomic photoabsorption cross-section by the loss function.

12.6.1 Photon emission in a medium

The photon is a new final-state particle in this book, and Appendix C.16 sets up what we need: the transverse radiation field as a set of oscillators, its minimal coupling to charges, and the golden-rule rate of emitting one photon. A charge e_{c} at rest that is set moving with velocity $\boldsymbol{v} \ll 1$ in a time short compared with $1/\omega$ radiates the photon number spectrum Eq. (C.140), flat in energy. And inside a transparent medium of refractive index $n_r(\omega)$ the same current radiates n_r times more, Eq. (C.141): the density of photon modes grows as $n_r^2 n_{\mathrm{grp}}$, the field of one photon shrinks as $(n_r n_{\mathrm{grp}})^{-1/2}$, and the group index n_{grp} cancels. Together,

$$\frac{\mathrm{d}N_\gamma}{\mathrm{d}\omega_\gamma} = n_r(\omega_\gamma)\,\frac{2\alpha}{3\pi}\,\frac{e_{\mathrm{c}}^2}{e^2}\,\frac{v^2}{\omega_\gamma}, \tag{12.48}$$

with ω_γ the energy of the photon. The spectrum holds for photon wavelengths long compared with the emitter, the dipole limit, and for a medium that does not absorb at ω_γ, so that the photon is a propagating mode.

A nucleus in an atom does not radiate with its bare charge. Its electrons are pulled along, and the photon sees the dipole of the whole atom. Kouvaris and Pradler computed this in second-order perturbation theory, kick and emission in both orders, and found the bare spectrum, $e_\mathrm{c} = Z_\kappa e$ in Eq. (12.48) [15, Eq. (3)], with Z_κ^2 replaced by $|f(\omega_\gamma)|^2$ [15, Eq. (8)]. Their f is the atomic scattering factor, and in our units it is fixed by the dipole polarizability $\alpha_\mathrm{pol}(\omega)$ of the atom,

$$Z^*_\mathrm{at}(\omega) \equiv f(\omega) = -\frac{m_e\omega^2}{e^2}\,\alpha_\mathrm{pol}(\omega), \tag{12.49}$$

the free-atom limit of the dynamical Born charge of Sect. 12.6.2. A model of electrons bound to the nucleus by springs shows what it means. If the nucleus moves by u at frequency ω, an electron with natural frequency ω_n follows with the amplitude $\omega_n^2 u/(\omega_n^2 - \omega^2)$, and the atom acquires the dipole $-eu\sum_n f_n\omega^2/(\omega_n^2-\omega^2)$, with f_n the oscillator strengths, which sum to Z_κ (Sect. 4.3). At high frequency the electrons stay behind and the dipole is $Z_\kappa eu$, the bare charge. At low frequency they follow, the neutral atom carries no dipole, and Z^*_at falls as ω^2. For silicon, with the static polarizability of 37.3 atomic units [17], that is $4\pi \times 37.3\,a_0^3 = 69.5\,\text{Å}^3$ in Heaviside–Lorentz units, $Z^*_\mathrm{at} = -0.050$ at 1 eV: a photon of an electronvolt sees one twentieth of an electron charge.

12.6.2 Crystal nuclear-recoil bremsstrahlung

In a crystal the charge that moves with a displaced nucleus is the Born effective charge of Sect. 5.5. The Born charge is defined for slow displacements, with the electrons following adiabatically. A photon of an electronvolt oscillates faster than the lattice but not much slower than the electrons. What it sees is the *dynamical Born charge* $Z^*_\kappa(\omega)$: the dipole created per unit displacement of one nucleus of species κ oscillating at frequency ω, all other nuclei at rest, at zero macroscopic field, with the electrons responding at ω. By the Maxwell relation Eq. (5.49), taken at frequency ω, it is also the force on the nucleus per unit field. At $\omega \to 0$ it is the Born charge Z^*_κ, at ω far above every electronic frequency it is Z_κ, and for a free atom it is Z^*_at of Eq. (12.49). The nucleus couples to the macroscopic field of a photon, $\boldsymbol{\mathcal{E}}_\gamma$, through the term of Eq. (5.55) at that frequency,

$$H_\gamma = -e\sum_I \boldsymbol{u}_I \cdot Z^*_\kappa(\omega_\gamma) \cdot \boldsymbol{\mathcal{E}}_\gamma. \tag{12.50}$$

The local fields at the nucleus are inside Z^*_κ, and $\boldsymbol{\mathcal{E}}_\gamma$ is the field of the medium's photon mode of Appendix C.16.

The amplitude. Dark matter kicks nucleus I with $K = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$, and the photon is emitted by $D = -e\boldsymbol{u}_I \cdot Z^*_\kappa \cdot \langle 1_\gamma|\boldsymbol{\mathcal{E}}_\gamma|0\rangle$, which acts on the nucleus only. The process has two time orders. With H the Hamiltonian of the crystal, $|i\rangle$ and $|f\rangle$ its initial and final states and $|n\rangle$ the intermediate ones, the second-order amplitude is

$$A_{fi} = \sum_n \Big[\frac{\langle f|D|n\rangle\langle n|K|i\rangle}{\omega_\gamma - (E_n - E_f)} - \frac{\langle f|K|n\rangle\langle n|D|i\rangle}{\omega_\gamma + (E_n - E_i)}\Big].$$

When ω_γ is large compared with the energy differences in the denominators, we expand in their ratio. At zeroth order the two terms give $\langle f|[D,K]|i\rangle/\omega_\gamma$, which vanishes, since $\boldsymbol{u}_I$ and $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$ commute. At first order, closure turns the energy differences into commutators with H, and

$$A_{fi} = -\frac{1}{\omega_\gamma^2}\,\langle f|\,[[H,D],K]\,|i\rangle.$$

Whatever potential holds the nucleus, it is a function of the positions, so $[H, \boldsymbol{u}_I] = -\mathrm{i}\boldsymbol{P}_I/M_\kappa$, and $[\boldsymbol{P}_I, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}] = \boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$. The double commutator is the kick times $\mathrm{i}e\,(\boldsymbol{q}/M_\kappa)\cdot Z^*_\kappa \cdot \langle 1_\gamma|\boldsymbol{\mathcal{E}}_\gamma|0\rangle$, and the amplitude factorizes into the matrix element of the kick and the emission amplitude of a charge eZ^*_κ that jumps to the velocity $\boldsymbol{v}_N = \boldsymbol{q}/M_\kappa$. This is the classical result of Appendix C.16 for a sudden kick, and the lattice potential has dropped out. The neglected terms are of relative order $(E_n - E_i)/\omega_\gamma$. The displacement $\boldsymbol{u}_I$ changes the number of phonons by one, so the energies in question are phonon energies, and the expansion holds for $\omega_\gamma \gg \omega_{\mathrm{ph}}$: the nucleus moves freely during one period of the photon.

Each kick of nucleus κ thus comes with the photon spectrum Eq. (12.48) for the charge $eZ^*_\kappa(\omega_\gamma)$ and the velocity $\boldsymbol{v}_N$. Inserting the kick into Eq. (8.16) with the joint response of Sect. 12.1.1, and the photon energy as a third energy deposited, gives the rate.

Crystal nuclear-recoil bremsstrahlung

For photon energies $\omega_{\mathrm{ph}} \ll \omega_\gamma$ below the absorption edge of the crystal, and momentum transfers above the zone, where the kicks of different nuclei add incoherently,

$$\frac{\mathrm{d}^2\mathcal{R}}{\mathrm{d}\omega_l\,\mathrm{d}\omega_\gamma} = \frac{1}{\rho_T}\,\frac{\rho_\chi}{m_\chi}\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v}) \int \frac{\mathrm{d}^3q}{(2\pi)^3}\sum_\kappa \underbrace{\mathfrak{D}_{\kappa\kappa}(\boldsymbol{q},\boldsymbol{v})\,S_{\kappa\kappa}(\boldsymbol{q},\omega_l)}_{\text{kick}} \times \underbrace{\frac{2\alpha\, n_r(\omega_\gamma)}{3\pi\,\omega_\gamma}\,\frac{|Z^*_\kappa(\omega_\gamma)\cdot\boldsymbol{q}|^2}{M_\kappa^2}}_{\text{photons per kick}}\,\delta(\omega_l+\omega_\gamma-\omega_{\boldsymbol{q}}), \tag{12.51}$$

with $S_{\kappa\kappa}(\boldsymbol{q},\omega_l)$ the lattice response of Chap. 11 in whatever regime it is, single phonons, multiphonons or free recoil. At a cubic site $|Z^*_\kappa\cdot\boldsymbol{q}|^2 = Z^{*2}_\kappa q^2$.

For a gas of atoms, $n_r = 1$, $Z^*_\kappa \to Z^*_{\mathrm{at}}$ and a free recoil, Eq. (12.51) is the rate of Kouvaris and Pradler [15, Eq. (8)]; with the bare charge Z_κ in place of Z^*_κ it is their bare-nucleus rate [15, Eq. (3)]. Note that their derivation kept only the elastic atomic channel, the atom left in its ground state, and the crystal formula does the same: the electrons appear only through $Z^*_\kappa(\omega_\gamma)$.

The photon must be soft, its momentum $n_r\omega_\gamma$ small compared with q, which for photons of an eV and momentum transfers of keV holds by three orders of magnitude. The wavelength must be long compared with the cell, which makes Z^*_κ the zone-center quantity. The impulse approximation, $E_R \gg \omega_{\mathrm{ph}}$, is not needed, since the lattice potential dropped out of the amplitude. Below the zone, however, the kicks of neighboring nuclei interfere, and the photon sees the dipole of a whole cell. For a coupling to nucleon number the cell is pushed as a unit. Its dipole is then accelerated only through the small difference of nucleon number per mass between its atoms, the cancellation that Worked example 8.6 found for optical phonons. Above the absorption edge the photon is absorbed before it leaves the crystal. The same amplitude then makes an electron–hole pair. That piece is the exchange of a transverse photon between the moving nucleus and the electrons, a correction to the Coulomb amplitude of the Migdal effect (Sect. 12.1) suppressed by powers of the velocities of the charges. The photon pole is bremsstrahlung, and the continuum is Migdal ionization; nothing is counted twice.

The dynamical Born charge of silicon and germanium. What remains is $Z^*_\kappa(\omega)$, and one exact sum rule fixes much of it. Let every nucleus of the crystal move together, by $\boldsymbol{u}(t) = \boldsymbol{u}\,\mathrm{e}^{-\mathrm{i}\omega t}$. In the frame that moves with the nuclei, the electrons see the clamped crystal plus the inertial force $-m_e\ddot{\boldsymbol{u}} = m_e\omega^2\boldsymbol{u}$ on each electron, a uniform force indistinguishable from a uniform field $\boldsymbol{\mathcal{E}}_{\mathrm{in}} = -m_e\omega^2\boldsymbol{u}/e$. At zero macroscopic field the electrons respond to $\boldsymbol{\mathcal{E}}_{\mathrm{in}}$

as to an applied field, with the polarization $[\varepsilon(\omega)-1]\boldsymbol{\mathcal{E}}_{\rm in}$, local fields included, and $\varepsilon(\omega)$ the electronic dielectric function at clamped nuclei. A rigid translation of the neutral cell adds no dipole of its own, so this polarization is the whole dipole per cell, $\Omega\boldsymbol{P} = e\sum_\kappa Z^*_\kappa(\omega)\cdot\boldsymbol{u}$. Hence

$$\sum_\kappa Z^*_\kappa(\omega) = -\frac{\Omega\, m_e\,\omega^2}{e^2}\,\big[\varepsilon(\omega)-1\big], \tag{12.52}$$

with Ω the cell volume. At $\omega\to 0$ the right-hand side vanishes, the neutrality of the Born charges Eq. (5.50). At ω above every electronic frequency, $\varepsilon-1\to-\omega_p^2/\omega^2$ with the plasma frequency of all N_e electrons of the cell, $\omega_p^2 = e^2N_e/(\Omega m_e)$, and the sum tends to $N_e=\sum_\kappa Z_\kappa$, the bare charges. For a dilute gas, $\varepsilon-1$ is the density of atoms times $\alpha_{\rm pol}$, and Eq. (12.52) becomes Eq. (12.49). The two atoms of silicon or germanium are exchanged by the inversion through the bond center, so their dynamical Born charges are equal at every frequency, and

$$Z^*(\omega) = -\frac{\Omega\, m_e\,\omega^2}{2e^2}\,\big[\varepsilon(\omega)-1\big] = -\frac{N_{\rm v}}{2}\,\big[\varepsilon(\omega)-1\big]\frac{\omega^2}{(\omega_p^{\rm val})^2} \qquad \text{(Si, Ge)}, \tag{12.53}$$

with $N_{\rm v}=8$ valence electrons per cell and the valence plasma frequency $\omega_p^{\rm val}$ of Sect. 10.2. No calculation of the phonon response is needed: the measured refractive index gives the dynamical Born charge.

Below the gap ε changes slowly, and $Z^*(\omega)\simeq -C_\kappa\omega^2$ with $C_\kappa = \Omega m_e(\varepsilon_\infty-1)/2e^2$, the coefficient of the small-ω expansion that Appendix E.17 specifies. For silicon, with $\Omega = 40.05\,\text{Å}^3$ and $\varepsilon_\infty = 11.7$, $C_\kappa = 0.155\,\text{eV}^{-2}$; with the measured dielectric function at 1 eV, $\varepsilon = 12.37$ [18], $Z^*(1\,\text{eV}) = -0.165$. The photons per kick then grow as ω_γ^3, and the rate is suppressed by ω_γ^4 against that of a nucleus with a fixed charge. The suppression is that of the free atom, but the crystal is a stronger emitter. The polarizability per atom of the solid, $\Omega(\varepsilon-1)/2 = 228\,\text{Å}^3$ at 1 eV, is 3.3 times that of the free atom, $69.5\,\text{Å}^3$. The emission goes as the square of the charge, $3.3^2 = 10.7$, and the photon gains $n_r = 3.5$ on top. Per photon the silicon crystal emits 33 to 38 times more than its atoms between 0.3 and 1 eV, and germanium, with $\varepsilon_\infty = 16.0$ [19], 86 to 96 times more between 0.3 and 0.65 eV. A free-atom estimate is therefore not an upper bound for the crystal.

Polar crystals. In GaAs the Born charge is $Z^* = 2.11$ at zero frequency (Worked example 5.6), and nothing suppresses the emission at small ω_γ. The sum rule Eq. (12.52) fixes the sum of the two dispersions, $Z^*_{\rm Ga}(\omega) - Z^* + Z^*_{\rm As}(\omega) + Z^*$, from the electronic dielectric function, but not their difference. That difference is a mixed response to a field and a displacement at frequency ω, which frequency-dependent density-functional perturbation theory computes with shifted Sternheimer equations at $\pm\omega$ and a scissor to the measured gap, checked against Eq. (12.52) and the static Born charge (Appendix E.17); the same appendix bounds each dispersion by $|Z^*_\kappa(\omega) - Z^*_\kappa(0)| \le 0.9\,(\omega/1\,\text{eV})^2$. We split the sum equally between the two sites and carry the bound as a band. The rate depends on the difference only through the cross term with Z^*, and below 0.3 eV the band is a few percent wide.

Fig. 12.14 shows the result. In GaAs the photons per kick are almost flat in $\ln\omega_\gamma$, as for a bare charge; half of them lie below 0.36 eV. In silicon and germanium the spectrum climbs as ω_γ^3 up to the absorption edge. The absolute numbers are small. A recoil of 100 eV emits 4.0×10^{-13} photons between 0.1 and 1.1 eV in silicon and 1.7×10^{-10} between 0.1 and 1.27 eV in GaAs, against the Migdal probability of 6.1×10^{-3} in silicon at the same recoil. At 100 MeV the GaAs rate in panel (b) integrates to 9×10^{-7} events per kilogram-year, and the silicon rate to 2×10^{-9}, while the Migdal rate of silicon with the same coupling is 30 (Sect. 12.3.4). The photons of GaAs lie below the gap, where the Migdal effect is closed, but they come with a Migdal signal above the gap that, judged by silicon, is larger by more than seven orders of magnitude. Bremsstrahlung in these crystals is therefore a check of the theory, not a search

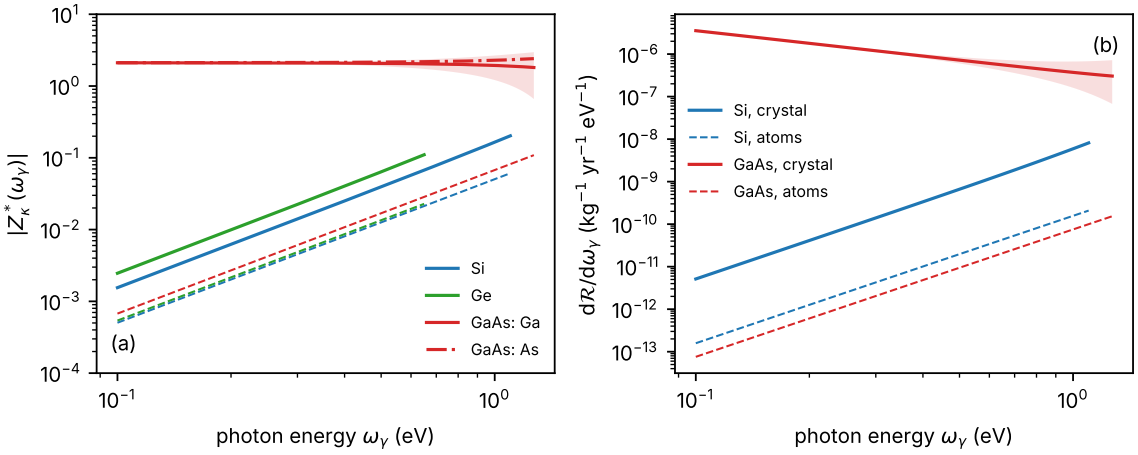


Figure 12.14: Nuclear-recoil bremsstrahlung in silicon, germanium and GaAs below their absorption edges. (a) The dynamical Born charge $|Z^*_\kappa(\omega_\gamma)|$ against the photon energy (solid; dash-dotted for the arsenic site), from the sum rule Eq. (12.52) with measured refractive indices: silicon at 300 K [18] (0.1–1.1 eV), germanium at 293 K [19] (0.1–0.65 eV) and GaAs at 295 K [20, 21] (0.1–1.27 eV, the range of the dispersion formula; its phonon term removed from the electronic ε). GaAs: $Z^* = 2.11$ at zero frequency and the dispersion of the sum split equally between the sites; shaded, the range allowed by the bound $|Z^*_\kappa(\omega) - Z^*_\kappa(0)| \leq 0.9\,(\omega/1\,\mathrm{eV})^2$. Dashed: the free atoms, Eq. (12.49) with the static polarizabilities 37.3 (Si), 40 (Ge) and 50 (Ga) atomic units [17]. (b) Rate per unit photon energy for $m_\chi = 100\,\mathrm{MeV}$, a heavy mediator coupled to nucleon number, $\sigma_p = 10^{-38}\,\mathrm{cm}^2$, the halo of Sect. 1.2, Eq. (12.51) with a free recoil. Solid: crystals; dashed: the same nuclei with the free-atom charges and $n_r = 1$; shaded: the GaAs band of panel (a).

channel. At photon energies of hundreds of eV and above, where Kouvaris and Pradler found the reach of their xenon search, $\varepsilon \to 1$, the crystal emits as its atoms, and Eq. (12.52) returns the bare charges as the electrons fall behind.

Worked example 12.6: The GaAs enhancement over the atomic rate

How much more does a GaAs crystal radiate than a gas of the same atoms? We work at $\omega_\gamma = 1\,\mathrm{eV}$ and then over the transparent band.

The dynamical Born charges. The cell volume is $\Omega = a^3/4 = 44.80\,\text{Å}^3$ with the static-lattice constant $a = 5.638\,\text{Å}$ of Table E.1, the cell of the computed phonons and Born charge of Sect. 5.5, and $\Omega m_e/e^2 = 3.249 \times 10^{-2}\,\mathrm{eV}^{-2}$. The dispersion formula of Skauli and collaborators [20], as tabulated in Ref. [21], gives $n_r = 3.4185$ at 1 eV, and without its phonon term the electronic dielectric function $\varepsilon = 11.688$. By Eq. (12.52) at $\omega = 1\,\mathrm{eV}$ the sum of the charges is $-(3.249 \times 10^{-2})(10.688) = -0.347$. Split equally, $Z^*_{\mathrm{Ga}}(1\,\mathrm{eV}) = 2.11 - 0.174 = 1.936$ and $Z^*_{\mathrm{As}}(1\,\mathrm{eV}) = -2.11 - 0.174 = -2.284$, so that the sum of their squares is 8.965. For any other split the sum of squares is $2(2.11 + b)^2 + 2(0.174)^2$, with b half the difference between the two dispersions; the band of Fig. 12.14 carries the values of b that the bound allows.

The free atoms. With $\alpha_{\mathrm{pol}} = 50$ and 30 atomic units [17], that is $4\pi \times 50\,a_0^3 = 93.1\,\text{Å}^3$ and $55.9\,\text{Å}^3$ in Heaviside–Lorentz units, Eq. (12.49) gives $Z^*_{\mathrm{at}} = -0.0675$ for gallium and -0.0405 for arsenic at 1 eV, and the sum of squares is 6.20×10^{-3}.

Per photon. The two nuclei have nearly the same nucleon number per mass and nearly the same q^2/M_κ^2 at a given recoil, so the ratio of Eq. (12.51) to its atomic form is the ratio of $n_r \sum_\kappa Z^{*2}_\kappa$ to $\sum_\kappa Z^{*2}_{\mathrm{at},\kappa}$: $3.4185 \times 8.965/6.20 \times 10^{-3} = 4.94 \times 10^3$ at 1 eV. The crystal emits five thousand times more. Of that, n_r contributes 3.4 and the Born charge the rest.

Over the band. The ratio grows toward small ω_γ as ω_γ^{-4}, since the atomic charges fall as ω_γ^2 and the crystal charges do not. Integrated with the weight $1/\omega_\gamma$ of Eq. (12.51) from 0.1 to 1.27 eV, $\int n_r \sum_\kappa Z^{*2}_\kappa \,\mathrm{d}\omega_\gamma/\omega_\gamma = 75.4$ and $\int \sum_\kappa Z^{*2}_{\mathrm{at},\kappa}\,\mathrm{d}\omega_\gamma/\omega_\gamma = 4.03 \times 10^{-3}$, a ratio of 1.87×10^4. The full rate at $m_\chi = 100\,\mathrm{MeV}$ gives 1.88×10^4, the same to the precision shown, because the kinematics hardly depends on photon energies so far below the kinetic energy of the dark matter.

Per recoil. A gallium nucleus with $E_R = 100\,\mathrm{eV}$ has $v_N^2 = 2E_R/M = 3.08 \times 10^{-9}$, and with $2\alpha/3\pi = 1.549 \times 10^{-3}$ it emits 1.7×10^{-10} photons in the band. The enhancement over the atom is large, four orders of magnitude, but it multiplies a small number. Its uncertainty is the band of Fig. 12.14: the rate at 100 MeV lies between $-19\,\%$ and $25\,\%$ of its central value.

12.6.3 Migdal-type absorption of a scalar

A scalar of mass 1 keV from the halo, absorbed in silicon, deposits 1 keV with a momentum of only $k = m_\phi v \approx 1\,\mathrm{eV}$. If it couples to nucleons, the nucleus takes that momentum, and its recoil energy, $k^2/2M_\kappa$, is $10^{-11}\,\mathrm{eV}$. The energy must go somewhere else, and it goes to the electrons. Dzuba, Flambaum and Samsonov computed this for atoms [16]. In the language of Sect. 12.2 it is a Migdal effect at the momentum k instead of q. The exact center-of-mass separation Eq. (12.10) turns the kick $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{X}}$ into a kick of the whole atom and the boost $\exp(-\mathrm{i}\boldsymbol{k}_e \cdot \sum_e \boldsymbol{\rho}_e)$ of the electrons, with $\boldsymbol{k}_e = m_e\boldsymbol{k}/m_A$. At the dipole order of Eq. (12.13) the amplitude to excite the electrons to Ψ_f is $-\mathrm{i}k_e\langle\Psi_f|D_{\hat{k}}|\Psi_0\rangle$. What differs from scattering is the energy. The field supplies $\omega = m_\phi$, and the whole of it goes to the electrons, $\omega_e = m_\phi$, while the lattice receives nothing to the precision of interest.

The coupling. A scalar coupled to protons and neutrons, $\mathcal{L} \supset \phi\,(g_{\phi pp}\,\bar{p}p + g_{\phi nn}\,\bar{n}n)$, couples to each nucleus coherently, as in Eq. (8.8), with the nuclear form factor equal to one at these momenta. The interaction is $H_\phi = -\phi \sum_\kappa c_\kappa n_\kappa(\boldsymbol{p}_\phi)$, the nuclear density row of Sect. 8.1.5 with

$$c_\kappa = Z_\kappa\, g_{\phi pp} + (A_\kappa - Z_\kappa)\, g_{\phi nn}, \qquad g^\phi_{n_\kappa} = -c_\kappa,$$

the per-species weight c_κ of Sect. 11.3, which here carries the couplings themselves rather than their ratio to a reference coupling. For equal couplings, $c_\kappa = A_\kappa g_{\phi nn}$.

The acceleration form. For an atom the center-of-mass separation is exact; in a crystal there is no atom to separate, and we need another route to the same amplitude. The equations of motion give it. Whatever the interactions, the potential energy V of the crystal depends on the positions alone, so $[H, \boldsymbol{X}_I] = -\mathrm{i}\boldsymbol{P}_I/M_\kappa$ and $[H, \boldsymbol{P}_I] = \mathrm{i}\nabla_{\boldsymbol{X}_I} V$, and the double commutator is $[H,[H,\boldsymbol{X}_I]] = \nabla_{\boldsymbol{X}_I} V/M_\kappa$. Between eigenstates with $\omega_{fi} = E_f - E_i \neq 0$ this reads $\omega_{fi}^2 \langle f|\boldsymbol{X}_I|i\rangle = \langle f|\nabla_{\boldsymbol{X}_I} V|i\rangle/M_\kappa$. Expanding the kick to first order in $\boldsymbol{k}$,

$$\Big\langle f\Big|\sum_I c_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{X}_I}\Big|i\Big\rangle = \frac{\mathrm{i}}{\omega_{fi}^2}\sum_I \frac{c_\kappa}{M_\kappa}\,\boldsymbol{k}\cdot\langle f|\nabla_{\boldsymbol{X}_I} V|i\rangle + O(k^2), \tag{12.54}$$

the acceleration form of the amplitude. It is exact, at every order of the Born–Oppenheimer expansion. Its matrix element is minus the total force on nucleus I, from the electrons and the other nuclei, between the initial and final states. Between electronic states this is the derivative coupling of Eq. (5.11), and Eq. (12.54) is the boost amplitude Eq. (8.53) with q replaced by k and the excitation energy by ω_{fi}. In scattering the boost is the first term of an expansion in the velocity of the nucleus. Here the momentum is so small that only the dipole term matters, and the equations of motion give it exactly.

Couplings proportional to mass. Let $c_\kappa = \tilde{c} M_\kappa$ for every species, with one constant $\tilde{c}$. The potential depends only on differences of positions, so the gradients with respect to all particles sum to zero, $\sum_I \nabla_{\boldsymbol{X}_I} V = -\sum_j \nabla_{\boldsymbol{r}_j} V$, the sum over electrons: the total force on the nuclei is minus the total force on the electrons. The same double commutator for the electrons, with m_e in place of M_κ, turns the right-hand side of Eq. (12.54) into a dipole matrix element of the electrons, and

$$\Big\langle f\Big|\tilde{c}\sum_I M_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{X}_I}\Big|i\Big\rangle = -\tilde{c}\, m_e\,\langle f|n_e(\boldsymbol{k})|i\rangle\,[1 + O(kr)], \tag{12.55}$$

with r the size of the states involved. The physical reading is the equivalence principle. A field that couples to the mass of every particle cannot, at zero momentum, distinguish them: it pushes the center of mass and nothing else, and its matrix elements between different states start at order k^2. A coupling to the mass of the nuclei is the coupling to the total mass minus that to the electrons, and only the second term excites the crystal. It is the electronic counterpart of the cancellation by which a coupling proportional to mass excites no optical phonon (Eq. (11.53)).

The scalar therefore acts on the electrons as a density coupling of strength $\tilde{c}m_e$. The absorption formula Eq. (8.19) with $S_{ee} = (2k^2/e^2)L$, Eq. (4.45), the electron-density form that holds above the phonon band, gives the rate.

Migdal-type absorption in a crystal

For a scalar whose couplings to the nuclei are proportional to their masses, $c_\kappa = \tilde{c}M_\kappa$,

absorbed with momentum $\boldsymbol{p}_\phi$ and energy $\omega_\phi \simeq m_\phi$ above the phonon band,

$$\Gamma_{\rm abs} = \frac{\tilde{c}^2 m_e^2\, p_\phi^2}{e^2\, m_\phi}\, L(\boldsymbol{p}_\phi \to 0, m_\phi), \qquad \mathcal{R} = \frac{\rho_\chi}{\rho_T}\, \frac{\tilde{c}^2 m_e^2 \langle v^2\rangle}{e^2}\, L(m_\phi), \tag{12.56}$$

with $L = \mathrm{Im}[-1/\varepsilon]$ the loss function at zero momentum and $\langle v^2\rangle$ the mean squared speed of the halo in the laboratory.

For the halo of Sect. 1.2, $\langle v^2\rangle^{1/2} = 379\,\mathrm{km\,s^{-1}}$. The spectrum is that of dark-photon absorption, whose rate is $(\rho_\chi/\rho_T)\zeta^2 L(m_\phi)$ (Sect. 10.7), with ζ^2 replaced by $\tilde{c}^2 m_e^2\langle v^2\rangle/e^2$. Everything Sect. 10.7 found about screening carries over: the scalar is absorbed with the loss function, not with the absorptive part ε_2, suppressed near the absorption peaks of silicon, where $|\varepsilon|^2$ is large, and enhanced at the plasmon. In silicon and germanium, with one species, every coupling to nucleons is proportional to the mass up to the isotopes, so Eq. (12.56) is the whole rate.

The atomic result is the dilute limit. For a gas, $|\varepsilon| \to 1$ and $L \to \varepsilon_2$, and the photoabsorption cross-section σ_γ of one atom fixes $\omega\varepsilon_2 = n_{\rm at}\sigma_\gamma$, with $n_{\rm at}$ the density of atoms. Then $\Gamma_{\rm abs} = n_{\rm at} v\, \sigma_\phi$ with $\sigma_\phi = \tilde{c}^2 m_e^2 (p_\phi/m_\phi)\sigma_\gamma/e^2$, which for $\tilde{c} = g_{\phi nn}/m_p$ is the relation of Dzuba, Flambaum and Samsonov between scalar absorption and photoionization [16, Eqs. (14), (17)]. They cautioned that rates in solids could differ from the atomic estimate, and Eq. (12.56) says by how much. Applied to a crystal with the measured photoabsorption, which in a medium is $n_{\rm at}\sigma_\gamma = \omega\varepsilon_2/n_r$, the atomic formula uses ε_2/n_r where the crystal has $\varepsilon_2/|\varepsilon|^2$. The ratio, $n_r/|\varepsilon|^2$, is the medium factor. Far above the absorption edges, $\varepsilon \simeq 1 - \omega_p^2/\omega^2$ with ω_p the plasma frequency of the electrons bound below ω, so that $n_r \simeq 1 - \omega_p^2/2\omega^2$ and $|\varepsilon|^2 \simeq 1 - 2\omega_p^2/\omega^2$, and

$$\frac{n_r}{|\varepsilon|^2} = 1 + \frac{3}{2}\,\frac{\omega_p^2}{\omega^2} + O\Big(\frac{\omega_p^4}{\omega^4},\ \varepsilon_2^2\Big). \tag{12.57}$$

For silicon at 1 keV, with the twelve electrons per atom bound by less than 1 keV and $\omega_p = 28.8\,\mathrm{eV}$, this is $1 + 1.2\times10^{-3}$. Near the plasmon and the absorption peaks it is far from one.

Limit Check 12.9. Migdal-type absorption of a scalar

$\sigma_\phi = g_{\phi nn}^2 (m_e^2/m_p^2)(p_\phi/m_\phi)\,\sigma_\gamma/4\pi\alpha$ per atom, with σ_γ the photoionization cross-section at $\omega = m_\phi$ [16, Eqs. (14), (17)].

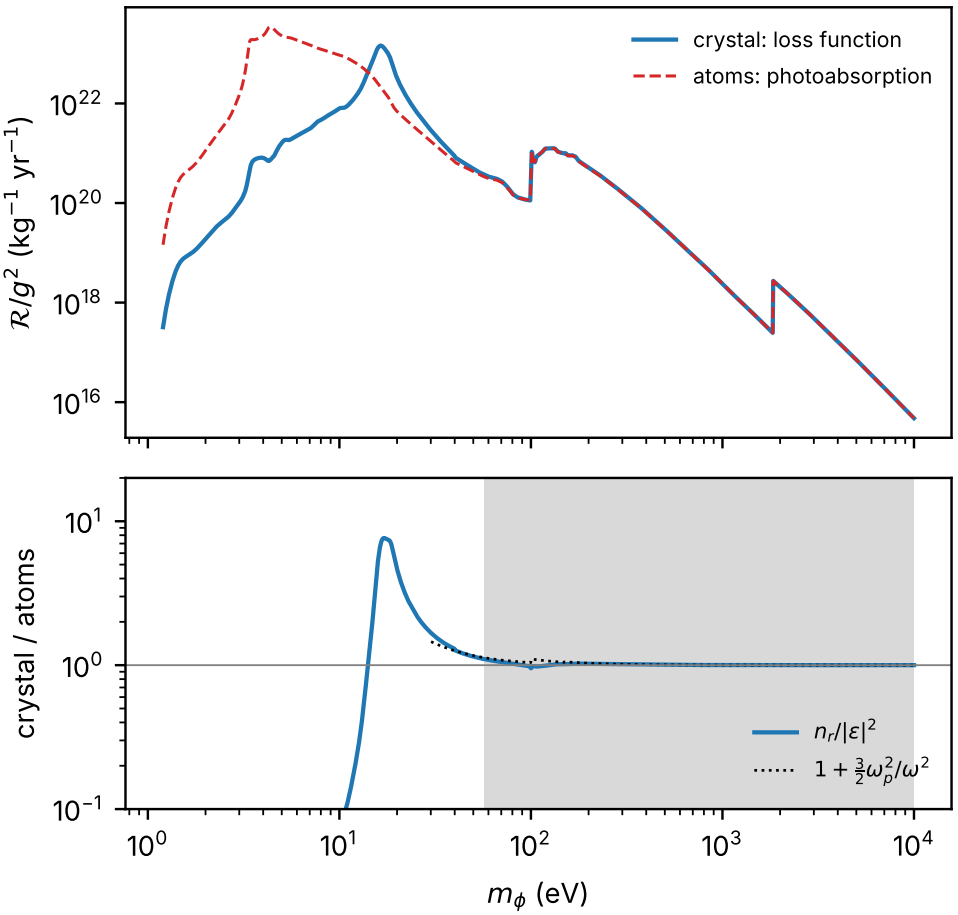


Figure 12.15: Absorption of a scalar coupled equally to protons and neutrons ($c_\kappa = gA_\kappa$) in silicon against its mass m_ϕ. Top: rate per unit mass divided by g^2, from the loss function, Eq. (12.56) (solid), and from the atomic formula with the measured photoabsorption of the crystal, $n_{\rm at}\sigma_\gamma = m_\phi\varepsilon_2/n_r$ (dashed); halo of Sect. 1.2. Bottom: their ratio $n_r/|\varepsilon|^2$, with the expansion Eq. (12.57) (dotted, ω_p from the electrons bound below m_ϕ, binding energies of the L and K shells as in Sect. 10.6). Dielectric function of silicon at 300 K [18] below 40 eV and from the atomic scattering factors [22] above, as in Sect. 10.7. Gray band: $m_\phi \geq 57$ eV, where the atomic formula is within 10 %.

Below about 60 eV the atomic formula misses the screening of the crystal, and is too high by about 400 near the absorption peak at 4.2 eV and too low by 7.7 near the plasmon, at 17 eV.

On the truncation label Limit Check 12.9 sits at dipole order, $n_D = 2$ in the rate, with no Born–Oppenheimer truncation, since Eqs. (12.54) and (12.55) are exact; $\mathcal{X}$ is the measured loss function, and the lattice plays no role at these energies. Its next corrections are the multipoles of order kr, which at 10 keV and a $1s$ radius of a_0/Z are below 10^{-3}, and, outside the shaded band, the full loss function in place of the atomic formula.

Couplings not proportional to mass. If c_κ/M_κ differs between species, write it as $\tilde{c} + \delta\tilde{c}_\kappa$, with $\tilde{c}$ a mean. The mean part is Eq. (12.56). The remainder keeps the forces in Eq. (12.54), $\mathrm{i}\omega^{-2}\sum_I \delta\tilde{c}_\kappa\, \boldsymbol{k}\cdot\langle f|\nabla_{\boldsymbol{X}_I} V|i\rangle$, and these no longer sum to anything simple. For transitions of the electrons the force on a nucleus is the density component of Eq. (12.4), the vertex of

the Migdal entry. The sum over the sites of one species, weighted by $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{X}_I}$, then forces the momentum given to the electrons to be $\boldsymbol{k}+\boldsymbol{G}$. The remainder is therefore a sum over reciprocal-lattice vectors weighted by the basis structure factor of $\delta\tilde{c}_\kappa Z_\kappa$, and it samples the off-diagonal elements $S^{ee}_{\boldsymbol{0G}}$ and $S^{ee}_{\boldsymbol{GG'}}$ of the electronic response at vanishing reduced momentum (Eq. (E.100)). These are nonadiabatic matrix elements in the sense of Sect. 12.1: the response of the electrons to the force of one sublattice, which no optical measurement provides. For the core shells, which belong to one nucleus, the atomic dipole formula with $\delta\tilde{c}_\kappa$ in place of $\tilde{c}$ applies shell by shell. For the valence electrons Appendix E.18 specifies the calculation, with $S^{ee}_{\boldsymbol{GG'}}$ to $|\boldsymbol{G}| = 6(2\pi/a)$ and the test that the (200) channel of the diamond structure vanishes for equal $\delta\tilde{c}_\kappa$. How large the remainder is depends on the coupling. For nucleon number in GaAs the two species differ in A_κ/M_κ by 2.6×10^{-5} per u (Worked example 8.6), and the remainder is negligible. For a coupling to protons alone, Z_κ/M_κ is 0.4446 per u for gallium and 0.4405 for arsenic, and the isotopes of natural silicon differ by up to 7 %; the isotopic part is a random remainder, incoherent between sites, like the isotope scattering of Sect. 11.6.

A scalar may also couple to electrons. Its amplitude adds $\tilde{c}m_e$ to the density coefficient $g^\phi_{n_e} = -g_{\phi ee}$ of Eq. (8.11), and the kinetic row of the electron coupling, which dominates its absorption (Sect. 10.7), interferes with the density row through their cross response. The nucleon coupling adds no new row.

12.6.4 Nuisance table of the Migdal sector

Table 12.3 collects the truncations and inputs of the chapter, each with its size and the section that treats it, in the classes of Sect. 8.8.

For the silicon Migdal probability above a few MeV the table has one large computed row and one large row sized only in a model. The local fields on the diagonal lower the total by a factor 1.35, and the off-diagonal part raises it again, in an empirical-pseudopotential model, by a factor of about 1.4 as computed and 1.2 extrapolated; the factor still falls with the reciprocal-lattice set, so the lower end is not a bound, and the full calculation is open. The table truncation, by contrast, costs a few percent, and the ion charge is fixed by the frozen core. The per-mass form is not an uncertainty of a correct calculation but the error of a simpler one, and its size says where the simpler one may be used. The exact energy denominators move the leading entry by a few parts in 10^3 below the impulse onset; at recoil energies of hundreds of eV the motion of the nucleus raises the probability by up to the factor 1.6 at 500 eV. The headline silicon numbers of the chapter use the weight of the valence ionization factor $Z_{\mathrm{ion}}(p)$, for which the momentum taken by the electrons has $\langle p^2\rangle^{1/2} = 6.1\,\mathrm{keV}$ (Sect. 12.4). The 5.6 keV of Fig. 12.2 belongs to the point-ion model of Sect. 12.1. The counting of Migdal and lattice events depends on the on-site drag mass, which is known only within a factor of two. And the frame of the transition density changes how a Migdal event shares its energy between electrons and lattice without changing how many events there are, which matters to a detector that reads both charge and phonons.

12.6.5 Open problems

The dynamical Born charges of a polar crystal are fixed by Eq. (12.52) only in their sum; the difference between the sites needs the frequency-dependent calculation of Appendix E.17, and until it is done the GaAs bremsstrahlung rate carries the band of Fig. 12.14. And the absorption of a scalar whose couplings are not proportional to the masses needs the off-diagonal response $S^{ee}_{\boldsymbol{GG'}}$ at vanishing reduced momentum, which Appendix E.18 specifies. The open items of the earlier sections are rows of Table 12.3. Above all of them stands a measurement. No crystal Migdal signal has yet been observed. A tagged-recoil measurement,

Table 12.3: Nuisances of the Migdal sector. Type in the classes of Sect. 8.8: normalization, tilt (in ω_e, in q or in the dark-matter mass), threshold and lattice spectrum (the share of a Migdal event that goes to phonons, at fixed electronic energy transfer). Basis of the size: "computed" (next order computed), "model" (next order computed in a model), "est." (parametric estimate), "bound" (only an inequality is known), "named" (no size). Sizes are for silicon unless another crystal is named. The last column gives the section where the item is treated.

Truncation or input	Type	Size	Basis	Section
Local fields on the diagonal of ε^{-1} (scalar ε against $\varepsilon^{-1}_{\boldsymbol{GG}}$)	normalization; tilt in ω_e	factor 1.35 on the total, concentrated at $p \gtrsim q_{\mathrm{BZ}}$ (Limit Check 12.3)	computed	Sect. 12.3
Off-diagonal elements $\boldsymbol{G} \neq \boldsymbol{G}'$ of ε^{-1}	normalization	raises the rate: full over diagonal 1.38 at 8^3 points $\boldsymbol{k}$ and $\lvert\boldsymbol{G}\rvert^2 \leq 19\,(2\pi/a)^2$, about 1.2 extrapolated, in an empirical-pseudopotential random-phase model of Si that carries about 2/3 of the diagonal local-field effect; still falling with $\boldsymbol{G}$, so the lower end is not a bound; flat in m_χ; the full local-field calculation is open; incoherent, zone-averaged form, not applicable at $q \ll q_{\mathrm{BZ}}$	model	Sects. 12.1 and 12.3
Loss-function table ($\omega_e \leq 89\,\mathrm{eV}$, $p \leq 22.5\,\mathrm{keV}$)	normalization	P_{Mig}: $\leq 1.6\,\%$ above 89 eV; $p > 10\,\mathrm{keV}$ holds 2.3 %; drag mass: missing part $\leq 24\,\%$	bound	Sects. 12.3 and 12.5
Ion charge $Z_{\mathrm{ion}}(p)$ against a point ion	normalization	P_{Mig} ×1.57; drag mass 18.6 against $11.8\,m_e$	model	Sects. 12.3 and 12.5
Valence on-site drag mass	normalization of the added lattice rate	Si $18.6\,m_e$ (12 to $24\,m_e$), Ge $39.5\,m_e$ (11.5 to $52\,m_e$); trace value $14\,m_e$ is not the on-site element	bound	Sect. 12.5
Per-mass form of the rate	tilt in m_χ	within 10 % above 66 MeV	computed	Sect. 12.3
Frame of the transition density (lattice at $\boldsymbol{q}-\boldsymbol{p}$ or $\boldsymbol{q}$; Herzberg–Teller)	lattice spectrum	phonon-emitting fraction ×2.4 at 5 keV; +12 % in the displaced-oscillator model	model	Sects. 12.1 and 12.4
Incoherent lattice factor below the zone	lattice spectrum	zero-phonon line within 8 % of the Bragg sum; multiphonon terms uncertified below 6.6 keV	computed; named	Sect. 12.4
Coherent zero-phonon response at $q < 2q_{\mathrm{BZ}}$ (cross-site terms of the lattice factor)	normalization; tilt in ω_e	limit $q \ll q_{\mathrm{BZ}}$ (Eq. (8.32)): 1/14.9 of the diagonal form, median ω_e 17.4 eV against 4.5 eV; within 15 % only for $q \leq 0.75\,\mathrm{keV}$ (model), which carries $\leq 0.2\,\%$ of the rate at every mass; 0.75 keV to $2q_{\mathrm{BZ}}$ unsized, nearly all of the rate up to 1 MeV, 33 % at 2 and 7 % at 3 MeV	computed (limit); model; unsized	Sect. 12.4
Exact energy denominators (recoil of $\boldsymbol{p}$, binding, Doppler; Eqs. (12.24) and (12.33))	tilt in q	near the gap -2.1×10^{-3}, $+2.8\times10^{-3}$ and at most 5.1×10^{-4} below the impulse onset; free recoil: probability within 10 % to $E_R \approx 160\,\mathrm{eV}$, factor 1.6 at 500 eV, rate within 10 % to 1 GeV	computed	Sects. 12.3 and 12.4
Germanium lattice inputs (Debye-scaled)	lattice spectrum	$\langle\omega^{-1}\rangle^{-1} \approx 17.6\,\mathrm{meV}$ from $\Theta_{\mathrm{D}} = 373\,\mathrm{K}$; regime boundaries near 41 and 113 keV; no first-principles phonon spectrum	est.	Sect. 12.4
Dipole order of the atomic boost (cores)	tilt in q	within 10 % to $\xi_D = 0.188$ (Si), 0.127 (Ge)	computed	Sect. 12.2
Bound final states of the atom	normalization	lower bound only	bound	Sect. 12.2
Wannier picture of the boost	normalization	absolute bound 1.6, 0.54, 0.25 at 20, 50, 100 eV; non-rigid ratio open	bound	Sect. 12.3
Relativistic contact density (third moment)	normalization of the check	+8 % (Si), +42 % (Ge); non-logarithmic Dirac remainder and $p_{1/2}$ contact density uncontrolled, each of relative order $(Z\alpha)^2$ (5 % in Ge)	computed; est.	Sect. 12.5
Site difference of $Z^*_\kappa(\omega)$, GaAs bremsstrahlung	normalization	−19 to +25 % at 100 MeV	bound	Sect. 12.6
Atomic formula for	tilt in m_χ	within 10 % above 57 eV; about 400×	computed	Sect. 12.6

in which the energy and direction of each nuclear recoil are fixed independently, for instance by the kinematics of a neutron scattered at a known angle, and the electrons it releases are counted, would test the whole chain at once in silicon at the recoil energies of Sect. 12.5.

Summary

A kicked nucleus leaves energy in two places, the lattice and the electrons, and the joint response resolves the two (Sect. 12.1). It is the response of Sect. 8.7 sorted by the electronic energy transfer ω_e and the lattice energy ω_l; its marginal is the ordinary response, and the sectors of Sect. 8.6 fall on the two axes and the quadrant between them. At the leading nonadiabatic order its Migdal entry is the response of the electrons to the field of a suddenly moving nucleus: a boost $\boldsymbol{q} \cdot \boldsymbol{p}/M_\kappa\omega_e^2$, the bare vertex of the nucleus, the density response of the clamped electrons and the lattice response at the momentum the nucleus keeps. The four published Migdal formulas are truncations of that entry. Its first moment is fixed by the drag mass, and the estimate $P_{\mathrm{Mig}} \approx (E_R/\langle\omega_e\rangle)(\delta M_\kappa/M_\kappa)$ follows. Added to a lattice rate, a Migdal rate requires the lattice response with the weight $1 - P_{\mathrm{Mig}}$ and the dragged mass, and a measured nuclear-recoil yield already contains the Migdal ionization. In a metal the sudden expansion fails at small ω_e.

The isolated atom is the truncation in which the boost can be done to all orders (Sect. 12.2). The center of mass separates exactly, and the electrons receive the momentum $m_e\boldsymbol{q}/m_A$, with m_A the mass of the whole atom. At dipole order the ionization probability at a recoil energy of 1 keV is 2.09×10^{-3} for silicon and 1.63×10^{-3} or 1.73×10^{-3} for germanium in two independent calculations, and the dipole form holds within 10 % up to $\xi_D = 0.188$ and 0.127. Bound final states without ionization add to it, and the oscillator strengths give only a lower bound on their share. In a crystal the atomic form is the right one for the core shells, not for the valence.

At large momentum transfer the crystal Migdal probability is a weighted moment of the loss function, with the coefficient 4α fixed by the f-sum rule and the valence ionization factor $Z_{\mathrm{ion}}(p)$ in the vertex (Sect. 12.3). In silicon it is 6.1×10^{-5} per eV of recoil energy, 29 times the atomic value. A scalar dielectric function overestimates it by about 35 %, an error concentrated at $p \gtrsim q_{\mathrm{BZ}}$; the off-diagonal elements of the dielectric matrix raise it in a model by a factor of about 1.4 as computed and 1.2 extrapolated, the lower end not a bound, and their full calculation is open. The soft limit has no correction in p/q and holds at low mass; it fails only when the nucleus moves fast during the transition, above about 160 eV of recoil. The factorized rate per unit mass holds within 10 % above 66 MeV. The Wannier picture of the boost is exact for flat bands and bounded only in absolute terms for the valence band of silicon.

Below the impulse onset the lattice factor of the Migdal entry is the multiphonon response at the momentum the nucleus keeps (Sect. 12.4). The electronic spectrum does not depend on it, since the lattice factor has unit weight. What it decides is how a Migdal event shares its energy: at 10 keV in silicon 92 % of the events leave the lattice in its ground state, and most events stay zero-phonon until the zero-phonon weight of the kick itself falls to one half, at 33.2 keV. The phonon-mediated form is the same process with the bound-nucleus propagator, exact in the phonon energies at first order in $\boldsymbol{p}$; for silicon and germanium its correction is a few parts in a thousand at the gap, and it matters only where gaps approach phonon energies.

The moments of the Migdal spectrum are inverse moments of the force that the electrons exert on one nucleus (Sect. 12.5). The first is the on-site drag mass, which differs from the trace value of $14\,m_e$ for silicon that a uniform translation fixes; the loss function gives $18.6\,m_e$ for the valence electrons alone, uncertain by a factor of two. Two moments bound the

probability from both sides. The cross first moment fixes the Migdal interference of a mixed coupling, and the third moment is the electron density at the nucleus, which relativity raises by 8 % in silicon and 42 % in germanium.

A recoiling nucleus also radiates, and an absorbed scalar coupled to nucleons gives its energy to the electrons through the same boost (Sect. 12.6). The crystal emits with the dynamical Born charge, which in silicon and germanium follows exactly from the measured dielectric function and suppresses the emission as the fourth power of the photon energy. Silicon still emits 33 to 38 times more than its atoms, and GaAs, with its static Born charge, 1.87×10^4 times more over its transparent band. The rates are seven orders below the Migdal rate, so bremsstrahlung is a check, not a channel. A scalar whose couplings to the nuclei are proportional to their masses is absorbed with the loss function, by the equivalence principle, and the atomic formula holds in silicon only above 57 eV. The chapter's nuisances are collected in Table 12.3. Throughout, the electronic final states were reached by kicking a nucleus; Chap. 13 turns to the other mixed sector, the electrons kicked directly while a phonon carries the missing momentum.

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Chapter 13

Phonon-assisted electron recoil

The lowest electron–hole pair of silicon has a crystal momentum of 1.94 keV: the hole sits at the center of the zone and the electron in one of the six conduction valleys near X (Sect. 3.3). Dark matter lighter than about 0.4 MeV cannot supply that momentum together with the 1.17 eV of the gap, because at the momentum transfer where it deposits the most energy the pairs cost more. It can still create a pair, however, if it creates a phonon at the same time, and the phonon carries the missing crystal momentum for an energy of at most 64 meV. Light does the same between the indirect and the direct gap, and the absorption edge of silicon is a measurement of that process (Sect. 6.5). The question of this chapter is how large the phonon-assisted rate is, and how to compute it without double-counting the zero-phonon rate of Chap. 10. For silicon detectors the answer decides over which masses this is the only electron-recoil channel, and which edges of that window are still uncertain. In the decomposition of Sect. 8.6 its final states form the phonon-dressed electronic sector, the second mixed sector after the Migdal effect of Chap. 12, and its optical limit ties it to the absorption data of Sect. 6.5 (Fig. 13.1).

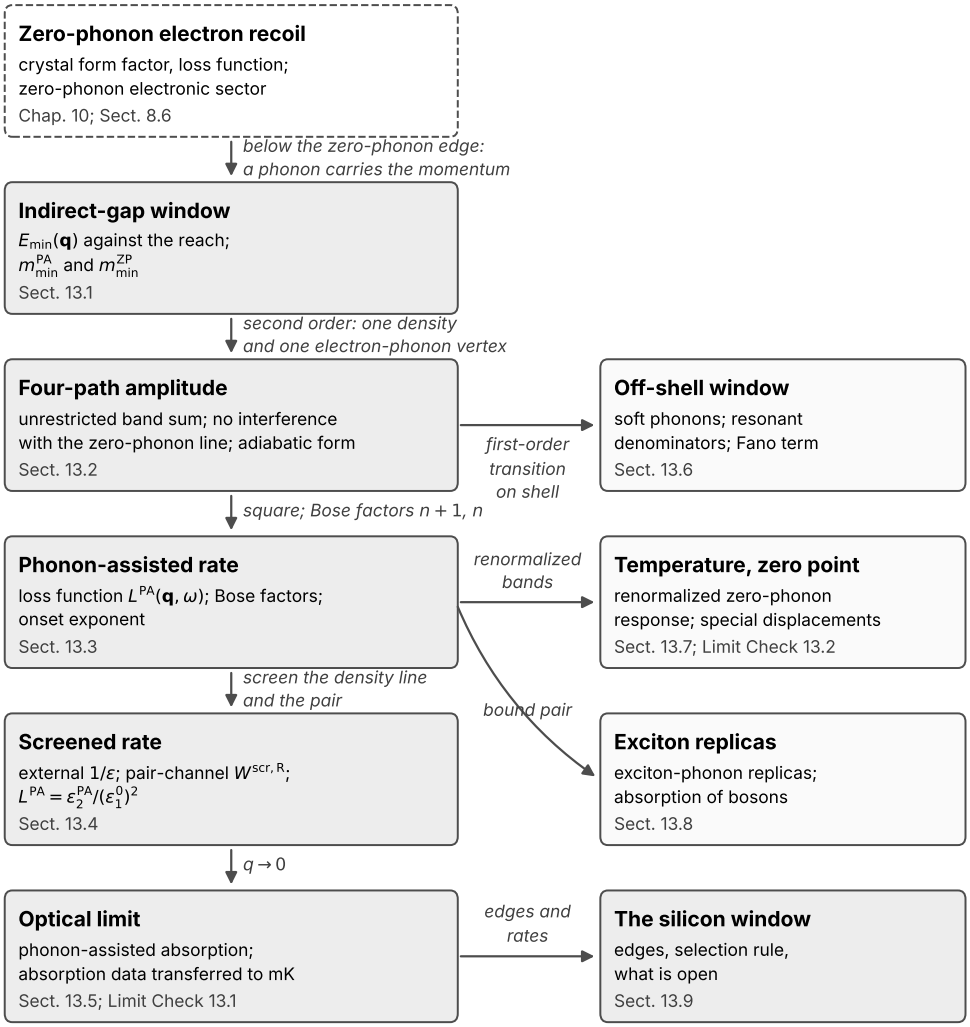


Figure 13.1: Phonon-assisted electron recoil, from the kinematics of the window to the silicon window itself. Below the zero-phonon edge of the electronic response of Chap. 10, a phonon must carry the missing crystal momentum (Sect. 13.1). The amplitude is of second order, with one density vertex and one electron–phonon vertex on four paths (Sect. 13.2); squared and summed over final states it gives the phonon-assisted loss function and rate (Sect. 13.3), which screening dresses on the density line and on the pair (Sect. 13.4); at $q \to 0$ it becomes phonon-assisted optical absorption (Sect. 13.5; Limit Check 13.1). The figure also places the off-shell window outside which the second-order rate is not separately defined (Sect. 13.6), the renormalized bands at finite temperature and zero point (Sect. 13.7; Limit Check 13.2), and the bound pair with its phonon replicas and the absorption of bosonic dark matter (Sect. 13.8). The edges and rates meet in the silicon window (Sect. 13.9).

13.1 The indirect-gap window

A dark-matter particle of 0.35 MeV can deposit up to $\frac{1}{2}m_\chi v_{\max}^2 = 1.23\,\mathrm{eV}$, more than the 1.17 eV gap of silicon, but only at the momentum transfer $q = m_\chi v_{\max} = 0.93\,\mathrm{keV}$. That is less than half of the 1.94 keV that separates the valence-band maximum from the conduction valleys, and at $q = 0.93\,\mathrm{keV}$ the cheapest electron–hole pair along the valley axis costs about 1.9 eV (Fig. 3.7). On its own the particle cannot create a pair. With a phonon of about 15 meV that takes up the remaining 1.0 keV of crystal momentum, it can. We will make this precise: derive the lightest mass that creates a pair without a phonon, $m_{\min}^{\mathrm{ZP}}$, and the lightest that creates one with a phonon, $m_{\min}^{\mathrm{PA}}$, evaluate both in silicon and germanium, and show that a direct-gap crystal such as GaAs has a window as well. Between the two masses lies the window in which the phonon-assisted process of Sect. 9.1 is the only way dark matter scattering on electrons creates an electron–hole pair.

13.1.1 The zero-phonon threshold mass

Two scales decide the kinematics. The largest momentum transfer of a particle of mass m_χ is $2m_\chi v_{\max}$, and it reaches the valley momentum $k_{\mathrm{c}} = 1.94\,\mathrm{keV}$ of Eq. (3.47) only for $m_\chi \geq k_{\mathrm{c}}/2v_{\max} = 0.366\,\mathrm{MeV}$; at that mass it does so by backscattering, with no energy to spare. The energy it can deposit at fixed q is $qv_{\max} - q^2/2m_\chi$ (Eq. (1.26)), largest at $q = m_\chi v_{\max}$. The two requirements conflict: depositing the gap favors $q \approx m_\chi v_{\max}$, while reaching the valley needs $q \approx k_{\mathrm{c}}$, and for light particles $m_\chi v_{\max}$ is well below k_{c}. Worked example 3.5 put numbers on the two extremes. The floor $2E_{\mathrm{g}}/v_{\max}^2 = 0.333\,\mathrm{MeV}$ of Eq. (1.25) assumes the gap can be absorbed at $q = 0.88\,\mathrm{keV}$, which it cannot; depositing the gap at exactly $q = k_{\mathrm{c}}$ needs 0.474 MeV. The zero-phonon threshold lies between them.

To locate it, we turn the kinematic condition Eq. (3.59) into a condition on the mass. At fixed $\boldsymbol{q}$, a pair of energy E is reachable when $qv_{\max} - q^2/2m_\chi \geq E$. For $qv_{\max} > E$ this reads

$$m_\chi \geq \frac{q^2}{2\,(qv_{\max} - E)}.$$

The right-hand side grows with E at fixed q, so the cheapest pair at each $\boldsymbol{q}$, the minimum pair energy $E_{\min}(\boldsymbol{q})$ of Eq. (3.57), decides. Minimizing then over $\boldsymbol{q}$, and writing out the minimum over the hole's crystal momentum $\boldsymbol{k}$ that $E_{\min}$ contains, gives the zero-phonon threshold mass

$$m_{\min}^{\mathrm{ZP}} = \min_{\boldsymbol{q}} \frac{q^2}{2\,[qv_{\max} - E_{\min}(\boldsymbol{q})]} = \min_{\boldsymbol{q},\boldsymbol{k},v,c} \frac{q^2}{2\,[qv_{\max} - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}]}, \tag{13.1}$$

with the minimum over the region where the denominator is positive. The second form is one minimization over six continuous variables, three for $\boldsymbol{q}$ and three for $\boldsymbol{k}$, and it shows that fixing the hole at the valence-band maximum, or restricting $\boldsymbol{q}$ to a few directions, can only raise the result. Both are common simplifications, and both give upper values.

Near a valley the minimization can be done by hand. Along the axis of a silicon valley, the pair energy is the parabola Eq. (3.58), $E_{\min} \simeq E_{\mathrm{g}} + (q - k_{\mathrm{c}})^2/2M_{\mathrm{eh}}$, where $M_{\mathrm{eh}} = m_\parallel^* + m_{\mathrm{hh}}$ is the mass with which the pair as a whole recoils, the sum of the longitudinal electron mass and the heavy-hole mass in the same direction. With $m_\parallel^* = 0.916\,m_e$ and $m_{\mathrm{hh}} = 0.277\,m_e$ along [100] (Table 3.1), $M_{\mathrm{eh}} = 1.19\,m_e$. Inserted in Eq. (13.1), the parabola gives $m_{\min}^{\mathrm{ZP}} = 0.425\,\mathrm{MeV}$ at $q = 1.46\,\mathrm{keV}$. At that point the particle falls 0.48 keV short of the valley, and the pair absorbs the shortfall as kinetic energy, $(0.48\,\mathrm{keV})^2/2M_{\mathrm{eh}} = 0.19\,\mathrm{eV}$, for a total of 1.36 eV. Lighter particles would have to transfer still less momentum, and the kinetic energy of the pair grows quadratically with the shortfall while the energy available grows only linearly with m_χ. The first-principles bands of Fig. 3.7, used along the same axis, give 0.420 MeV at

$q = 1.43\,\mathrm{keV}$, in agreement within the coarse grid of crystal momenta behind them. Along [110] and [111], which pass the valleys at a distance, the same evaluation gives 0.548 and 0.615 MeV.

The pair need not move along the valley axis, and the hole need not sit at the valence-band maximum; the heavy-hole surfaces of silicon are strongly warped (Eq. (3.56)), and a hole that moves into a flat direction lowers the kinetic energy of the pair. The full minimization Eq. (13.1) therefore lands below 0.42 MeV, off the axis, and Sect. 13.9 gives its value and the reason why the first zero-phonon states it reaches are weakly coupled. For the kinematics of this section it is enough that the bracket of Worked example 3.5, 0.333 to 0.474 MeV, has narrowed to about 0.40 to 0.42 MeV. The bound pair does not change this picture. The $1s$ exciton lies $E_x^b = 14.7\,\mathrm{meV}$ below the free pair (Sect. 4.9). Since $\partial m_\chi/\partial E = m_\chi/(qv_{\max} - E)$ at fixed q, it lowers the threshold mass by $m_\chi E_x^b/(qv_{\max} - E) = 0.0025\,\mathrm{MeV}$ at the parabolic optimum, about 0.6 %.

13.1.2 The phonon-assisted onset

A phonon changes the bookkeeping of crystal momentum. The dark matter transfers $\boldsymbol{q}$; the electron ends at $\boldsymbol{k}_e$ in a conduction band, the hole at $\boldsymbol{k}_h$ in a valence band, and a phonon of branch ν and wavevector $\boldsymbol{Q}$ is emitted. Conservation of crystal momentum fixes $\boldsymbol{Q} = \boldsymbol{k}_e - \boldsymbol{k}_h - \boldsymbol{q}$ up to a reciprocal-lattice vector, and the energy deposited is $\varepsilon_{c\boldsymbol{k}_e} - \varepsilon_{v\boldsymbol{k}_h} + \omega_{\boldsymbol{Q}\nu}$. The phonon takes whatever crystal momentum the pair needs, and since no phonon of silicon has more than 64 meV (Sect. 5.2), that momentum costs at most a few percent of the gap. The cheapest final state puts the electron at the bottom of a valley and the hole at the valence-band maximum. At momentum transfer $\boldsymbol{q}$ it costs

$$E_{\min}^{\mathrm{PA}}(\boldsymbol{q}) = E_{\mathrm{g}} + \min_{\nu} \omega_{\boldsymbol{k}_{\mathrm{c}}-\boldsymbol{q},\,\nu}, \tag{13.2}$$

with $\boldsymbol{k}_{\mathrm{c}}$ the valley, among the six and their images under reciprocal-lattice vectors, that makes the phonon cheapest. Compared to the zero-phonon edge, this threshold hardly depends on $\boldsymbol{q}$. It falls to E_{g} only at $\boldsymbol{q} = \boldsymbol{k}_{\mathrm{c}}$, where the acoustic phonon needed has zero energy, and it never rises above $E_{\mathrm{g}} + 64\,\mathrm{meV}$. Moving the electron and the hole off the band extrema could lower the phonon energy but adds kinetic energy to the pair. With the dispersion of Sect. 5.2, the best such trade lowers the threshold by less than 0.02 meV, and we drop it.

Eq. (13.2) assumes that the phonon is emitted. At temperature T a phonon can also be absorbed, which costs $E_{\mathrm{g}} - \omega_{\boldsymbol{Q}\nu}$ but comes with the occupation $n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$ in place of $n_{\mathrm{B}} + 1$ (Sect. 5.3), and E_{g} itself shifts with T (Sect. 13.7). For the transverse acoustic phonon of about 15 meV that sets the onset below, n_{B} is 0.12 at 77 K, 2×10^{-19} at 4 K and 3×10^{-75} at 1 K. The absorption sideband, which would open at 0.329 MeV, is absent in a detector at millikelvin temperatures. There only emission counts, and E_{g} is the gap extrapolated to zero temperature.

The onset mass follows from Eq. (13.2) as $m_{\min}^{\mathrm{ZP}}$ followed from $E_{\min}$:

$$m_{\min}^{\mathrm{PA}} = \min_{\boldsymbol{q}} \frac{q^2}{2\,[qv_{\max} - E_{\min}^{\mathrm{PA}}(\boldsymbol{q})]}. \tag{13.3}$$

For a threshold E independent of q, the minimum sits at $q = 2E/v_{\max}$ and equals the floor $2E/v_{\max}^2$ of Eq. (1.25). Here E varies with q, but slowly. Write $E(q)$ for $E_{\min}^{\mathrm{PA}}$ along a line of $\boldsymbol{q}$ and $E' = \mathrm{d}E/\mathrm{d}q$. Setting the derivative of $q^2/[2(qv_{\max} - E)]$ to zero gives $qv_{\max} - 2E + qE' = 0$. The optimum is therefore $q^* = 2E/(v_{\max} + E')$, and inserting it back,

$$m_{\min}^{\mathrm{PA}} = \frac{2\,[E_{\mathrm{g}} + \omega_\nu(\boldsymbol{Q}^*)]}{v_{\max}^2 - u_\nu^2}, \qquad \boldsymbol{Q}^* = \boldsymbol{k}_{\mathrm{c}} - \boldsymbol{q}^*, \quad q^* \simeq \frac{2\,[E_{\mathrm{g}} + \omega_\nu(\boldsymbol{Q}^*)]}{v_{\max}}, \tag{13.4}$$

where $u_\nu = |E'|$ is the group velocity of the phonon along $\boldsymbol{q}$ at $\boldsymbol{Q}^*$ and the branch ν is the one that gives the smallest result. The slope of the dispersion enters only squared, because q^* is a stationary point: moving q slightly changes the phonon energy to first order but the mass only to second. Phonon group velocities are a few km/s, below the speeds of sound, against $v_{\max} = 794.5\,\mathrm{km\,s^{-1}}$, and $u_\nu^2/v_{\max}^2 \sim 10^{-5}$. To that accuracy the onset is the kinematic floor with the gap raised by one phonon energy, evaluated at the phonon wavevector that the dark matter leaves for the phonon. Eq. (13.4) is implicit, since $\boldsymbol{Q}^*$ depends on q^*, but one or two iterations starting from $\omega = 0$ settle it.

Window kinematics

In an indirect-gap crystal at zero temperature, scattering dark matter creates an electron–hole pair only above the phonon-assisted onset $m_{\min}^{\mathrm{PA}}$ of Eq. (13.4), and without a phonon only above the zero-phonon threshold $m_{\min}^{\mathrm{ZP}}$ of Eq. (13.1). For

$$m_{\min}^{\mathrm{PA}} \le m_\chi < m_{\min}^{\mathrm{ZP}},$$

the phonon-assisted process is the only electron-recoil channel. Bound pairs lower each edge by replacing E_g with $E_\mathrm{g} - E_x^b$ in the corresponding threshold.

Which phonon wins is a question of energy at $\boldsymbol{Q}^*$, not of coupling strength. In silicon the transverse acoustic (TA) branch is the lowest everywhere away from the zone center, and Worked example 13.1 finds $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*) = 14.8\,\mathrm{meV}$ at $|\boldsymbol{Q}^*| = 0.46\,(2\pi/a)$, with $\boldsymbol{q}^*$ along the valley axis. Tilting $\boldsymbol{q}$ away from the axis raises the TA energy at $\boldsymbol{Q}^*$, by 0.7 meV at a tilt of 0.3 rad in a {001} plane, so the onset is on the axis. The longitudinal acoustic and transverse optical branches open later, at 0.341 and 0.350 MeV. Compared to light, dark matter needs a smaller phonon. A photon carries no momentum on the scale of the zone, its phonon must supply all of $\boldsymbol{k}_\mathrm{c}$, and the TA phonon that sets the optical edge has 18.27 meV [1] (Sect. 6.5). A particle at the onset supplies $q^* = 0.90\,\mathrm{keV}$ itself, about half of k_c, and leaves the phonon a shorter wavevector on the flat part of the TA branch.

Fig. 13.2 puts the two thresholds side by side. In panel (a) the zero-phonon edge $E_{\min}(\boldsymbol{q})$ dips to E_g only near k_c, while the phonon-assisted threshold Eq. (13.2) runs almost flat just above E_g at every q. The parabola of a 0.3 MeV particle stays below both. That of a 0.4 MeV particle crosses the phonon-assisted threshold but not, along these directions, the zero-phonon edge; a 0.5 MeV particle reaches both. Panel (b) shows the same comparison as a threshold mass at each q, whose minima are Eqs. (13.1) and (13.3): the phonon-assisted curve has a broad minimum near 0.9 keV, the zero-phonon curve a narrow one near 1.4 keV.

Worked example 13.1: The phonon-assisted onset of silicon

We evaluate Eq. (13.4) with the measured gap extrapolated to zero temperature, $E_\mathrm{g} = 1.17\,\mathrm{eV}$, the valley at $k_\mathrm{c} = 0.85\,(2\pi/a) = 1940\,\mathrm{eV}$ (Worked example 3.5), $v_{\max}^2 = 7.023 \times 10^{-6}$, and the TA branch of the force constants of Sect. 5.2, with $\boldsymbol{q}$ along the valley axis.
First pass, no phonon energy. $q^* = 2E_\mathrm{g}/v_{\max} = 883\,\mathrm{eV}$, which leaves the phonon $Q^* = k_\mathrm{c} - q^* = 1057\,\mathrm{eV} = 0.463\,(2\pi/a)$. The TA energy there is 14.87 meV.
Second pass. With $E = E_\mathrm{g} + \omega_{\mathrm{TA}}(\boldsymbol{Q}^*) = 1.1849\,\mathrm{eV}$, $q^* = 2E/v_{\max} = 894\,\mathrm{eV}$ and $Q^* = 0.458\,(2\pi/a) = 0.53\,\mathrm{Å}^{-1}$, where $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*) = 14.79\,\mathrm{meV}$. A third pass changes nothing at this precision. The slope of the TA branch at Q^* is $u_{\mathrm{TA}} = 2.1\,\mathrm{km\,s^{-1}}$, and

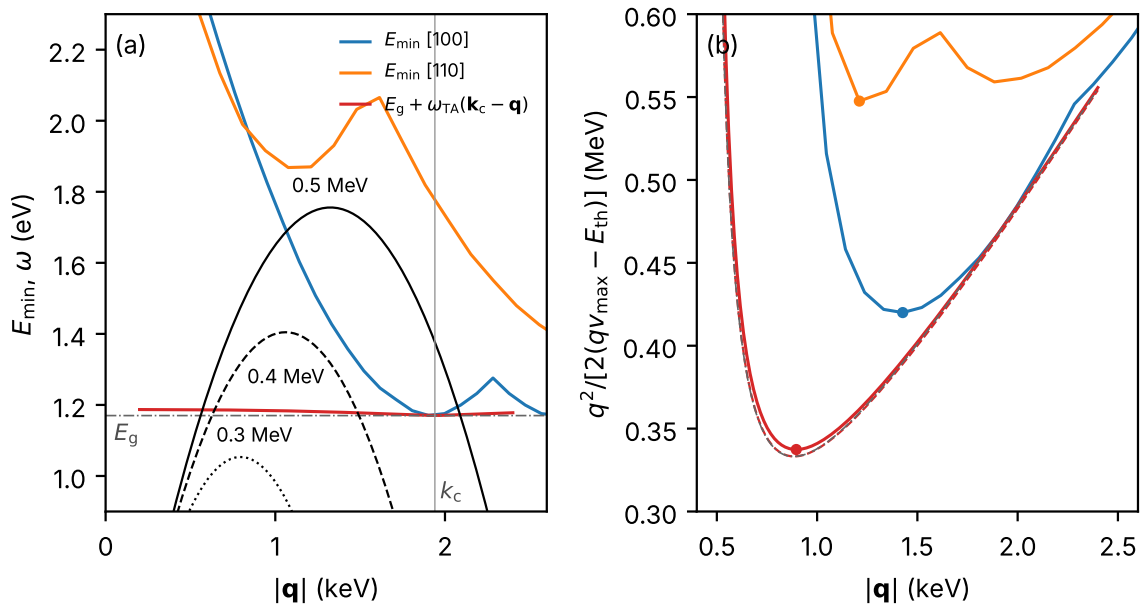


Figure 13.2: Kinematics of the indirect-gap window of silicon. (a) Minimum pair energy $E_{\min}(\boldsymbol{q})$ (Eq. (3.57)) for $\boldsymbol{q}$ along [100] and [110], and the phonon-assisted threshold $E_g + \omega_{TA}(\boldsymbol{k}_c - \boldsymbol{q})$ (Eq. (13.2)) for $\boldsymbol{q}$ along [100], with the electron at the valley minimum $\boldsymbol{k}_c = 0.85\,(2\pi/a)(1,0,0)$ and the hole at the valence-band maximum. Black: the largest energy deposited, $qv_{\max} - q^2/2m_\chi$, for $m_\chi = 0.3$ (dotted), 0.4 (dashed) and 0.5 MeV (solid), with $v_{\max} = 2.650 \times 10^{-3}$. Dash-dotted: $E_g = 1.17$ eV. (b) The lightest mass that reaches each threshold at momentum transfer q, $q^2/[2(qv_{\max} - E_{th})]$; dots mark the minima over q. Red solid: phonon-assisted, free pair, minimum 0.3374 MeV at $q = 0.90$ keV; red dashed: bound pair, E_g lowered by $E_x^b = 14.7$ meV, minimum 0.3332 MeV; grey dash-dotted: the floor $2E_g/v_{\max}^2$ at fixed threshold E_g. Blue and orange: zero-phonon, with minima 0.420 MeV along [100] and 0.548 MeV along [110]; these are upper values for the zero-phonon threshold, since the joint minimum over all directions of $\boldsymbol{q}$ and all hole momenta lies lower. The [111] direction, whose $E_{\min}$ stays above 2.13 eV, is off scale (minimum 0.615 MeV). Bands: PBE calculation of Fig. 3.7 at $a = 5.431$ Å on a grid of 48^3 crystal momenta, without spin–orbit coupling, with the conduction bands shifted up by 0.613 eV so that the indirect gap equals the measured 1.17 eV. Phonons: PBE force constants of Sect. 5.2 at the same lattice constant. Their TA branch is 8 % below the neutron data at X, and scaling it up to the data raises the phonon-assisted minimum to 0.3378 MeV.

$u_{\mathrm{TA}}^2/v_{\max}^2 = 7 \times 10^{-6}$ is negligible. The onset is

$$m_{\min}^{\mathrm{PA}} = \frac{2 \times (1.17\,\mathrm{eV} + 0.014\,79\,\mathrm{eV})}{7.023 \times 10^{-6}} = 0.3374\,\mathrm{MeV},$$

1.3 % above the floor $2E_{\mathrm{g}}/v_{\max}^2 = 0.3332\,\mathrm{MeV}$. The direct minimization of Eq. (13.3) on a grid of q gives the same value, at $q = 0.90\,\mathrm{keV}$.
Sensitivity. An error $\delta\omega$ in the phonon energy moves the onset by $2\,\delta\omega/v_{\max}^2 = 2.8 \times 10^{-4}\,\mathrm{MeV}$ per meV. The TA branch of this calculation is 8 % soft at X; scaling the whole branch up to the neutron value there gives $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*) = 16.1\,\mathrm{meV}$ and an onset of 0.3378 MeV. Had the phonon to supply all of $\boldsymbol{k}_{\mathrm{c}}$, as for light, the 18.27 meV of the optical edge would put the onset at 0.3384 MeV.
Bound pair. The 1*s* exciton replaces E_{g} by $E_{\mathrm{g}} - E_x^b = 1.1553\,\mathrm{eV}$. The same passes give $q^* = 2E/v_{\max} = 0.883\,\mathrm{keV}$, where $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*) = 14.87\,\mathrm{meV}$, and the onset becomes $2 \times 1.1702\,\mathrm{eV}/v_{\max}^2 = 0.3332\,\mathrm{MeV}$; the direct minimization puts the optimum at 0.885 keV with the same onset. The phonon energy and the binding energy nearly cancel, and the bound-pair onset falls on the floor $2E_{\mathrm{g}}/v_{\max}^2$ that a crystal with a direct gap of 1.17 eV would have. The size of this replica line, as against the free-pair continuum above it, is a question for Sect. 13.8.

13.1.3 Germanium and a direct-gap crystal

Germanium has its valleys at the L points, at $|\boldsymbol{k}_{\mathrm{c}}| = |\Gamma\mathrm{L}| = 1.90\,\mathrm{keV}$, almost the momentum of the silicon valleys, but a gap of only 0.744 eV (Sect. 3.3). Its floor $2E_{\mathrm{g}}/v_{\max}^2$ is 0.212 MeV, at $q = 0.56\,\mathrm{keV}$, and the phonon must supply the remaining 1.34 keV, about 0.7 of the way from Γ to L. Along that line the transverse acoustic branch of germanium rises monotonically to 1.9 THz, 7.9 meV, at L [2, 3, Table IV], so $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*) \leq 7.9\,\mathrm{meV}$, and Eq. (13.4) places the phonon-assisted onset between 0.2119 and 0.2141 MeV. The bracket needs no band structure and no computed phonon, only the gap, the position of L and the measured TA branch.

The zero-phonon side of germanium is harder. Worked example 3.6 bracketed it between 0.212 and 0.420 MeV. On the L axis, with the parabola of Eq. (3.58) and $M_{\mathrm{eh}} = m_{\parallel}^* + m_{\mathrm{hh}} = (1.59 + 0.469)\,m_e = 2.06\,m_e$ along [111] (Table 3.1), Eq. (13.1) gives 0.32 MeV at $q = 1.10\,\mathrm{keV}$. The heavy longitudinal mass of the L valleys makes the shortfall cheap, and the zero-phonon edge sits much closer to the onset, in relative terms, than in silicon. That value orients, but it does not bound the edge. The minimum need not lie on the axis, and the heavy hole of germanium is warped and strongly mixed by spin–orbit coupling ($\Delta_{\mathrm{SO}} = 0.296\,\mathrm{eV}$). The plain density-functional bands, moreover, put the Γ state below L (Appendix E.1), so no calculation with those bands can locate the edge. The joint minimization Eq. (13.1) on quasiparticle bands with spin–orbit coupling is an open calculation, and Sect. 13.9 states what is known.

The route through the direct gap at Γ plays no part: as Worked example 3.6 showed, its light pair opens only at a few MeV.

The last remark already contains the answer for a direct-gap crystal. Take GaAs, with both band edges at Γ and a gap of 1.52 eV. Its floor is $2E_{\mathrm{g}}/v_{\max}^2 = 0.433\,\mathrm{MeV}$, and below that mass dark matter scattering creates no pair in GaAs at all, with or without a phonon. At the masses of the silicon and germanium windows a direct-gap crystal of this gap has no electron-recoil channel. Above the floor, however, the same competition returns. A pair at rest at Γ costs exactly E_{g}, but the particle deposits E_{g} only near $q^* = 2E_{\mathrm{g}}/v_{\max} = 1.15\,\mathrm{keV}$, and a zero-phonon pair that carries this momentum has kinetic energy $q^{*2}/2M_{\mathrm{eh}}$. The electron of GaAs is light, $0.067\,m_e$, and with a heavy hole along [100] or [111] the pair mass is $M_{\mathrm{eh}} = 0.42$

or $0.96\,m_e$, a kinetic energy of 3.1 or 1.3 eV, comparable to the gap itself. For a pair of mass M_{eh} at Γ, the reachable condition is $qv_{\max} - q^2/2m_\chi - q^2/2M_{\mathrm{eh}} \geq E_{\mathrm{g}}$. Its left-hand side is largest at $q = \mu v_{\max}$, with $\mu = m_\chi M_{\mathrm{eh}}/(m_\chi + M_{\mathrm{eh}})$, where it equals $\frac{1}{2}\mu v_{\max}^2$. Writing $m_{\mathrm{g}} = 2E_{\mathrm{g}}/v_{\max}^2$ for the floor, the zero-phonon threshold of a direct gap is therefore

$$m_{\mathrm{min}}^{\mathrm{ZP}} = \frac{m_{\mathrm{g}} M_{\mathrm{eh}}}{M_{\mathrm{eh}} - m_{\mathrm{g}}} \qquad (M_{\mathrm{eh}} > m_{\mathrm{g}}), \tag{13.5}$$

and there is none when $M_{\mathrm{eh}} \leq m_{\mathrm{g}}$. For GaAs, $m_{\mathrm{g}} = 0.847\,m_e$: the pair with the [100] heavy hole never opens, and the one with the [111] heavy hole opens at 3.7 MeV, both in the parabolic approximation. A phonon removes the problem as it does in silicon: it carries $\boldsymbol{Q} \approx -\boldsymbol{q}^*$, about half the way to the zone boundary, and leaves the pair at rest. The transverse acoustic branch, the cheapest, stays below the 10.5 meV that the computed dispersion gives at X (Table 5.1), an upper value since the computed branch lies above the measured one there, and Eq. (13.4) gives an onset of 0.433 to 0.436 MeV. The first zero-phonon states that a particle near this mass can reach are those of the L valleys, 1.815 eV above the valence-band maximum at $|\Gamma\mathrm{L}| = 1.90$ keV (Sect. 3.3). Their floor is $2 \times 1.815\,\mathrm{eV}/v_{\max}^2 = 0.517$ MeV, and depositing 1.815 eV at exactly $|\Gamma\mathrm{L}|$ needs 0.560 MeV, so their edge lies between the two. GaAs therefore has a phonon-assisted window too, from 0.43 MeV to above 0.5 MeV, but it begins above the masses at which the silicon window has already closed.

The window is thus not a peculiarity of indirect gaps. It exists in any crystal whose pairs cannot absorb the momentum $q^* \approx 2E_{\mathrm{g}}/v_{\max}$ for less than a phonon energy, that is, whenever $|\boldsymbol{q}^* - \boldsymbol{k}_{\mathrm{c}}|^2/2M_{\mathrm{eh}}$ exceeds the phonon energies at the corresponding $\boldsymbol{Q}^*$. For band masses of order m_e and $q^* \sim 1$ keV the kinetic energy is of order an eV, a hundred times a phonon energy, and the condition is met in silicon, germanium and GaAs alike. What the indirect gap of silicon and germanium adds is position. Their windows start at their own floors, 0.333 and 0.212 MeV, where no direct-gap crystal of comparable gap has any electron recoil, and their optical absorption edges, being phonon-assisted, measure the same kind of amplitude in the limit $q \to 0$ (Sect. 13.5).

13.1.4 Onset precision and the zero-phonon edge

Kinematics fixes the onset, and an upper value for the zero-phonon threshold along any chosen direction. It does not fix where the window ends for a detector, however: a zero-phonon state that is kinematically open may be coupled so weakly that the phonon-assisted rate stays larger above $m_{\mathrm{min}}^{\mathrm{ZP}}$, and the end of the window is then a ratio of two rates, the phonon-assisted rate of Sect. 13.3 against the zero-phonon vertices of Sect. 13.9.

In practice, the quantities this section uses are cheap to compute. The minimum pair energy over the joint space of $\boldsymbol{q}$ and $\boldsymbol{k}$ needs bands interpolated over the zone, with a scissor or a quasiparticle correction to the gap (Sect. 3.4 and Appendix E.1). The onset needs the phonon energy at one wavevector, $\boldsymbol{Q}^*$, from the density-functional perturbation theory of Sect. 5.4. Appendix E.14 specifies these checks for silicon and germanium, including the direct evaluation of $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$ at more than one lattice constant that the softness of the TA branch calls for. The rate itself is a different matter, and it begins with the amplitude of Sect. 13.2.

13.2 The phonon-assisted amplitude

At the onset of Worked example 13.1 a particle of 0.3374 MeV transfers $q^* = 0.90$ keV along a valley axis of silicon and deposits 1.185 eV. The final state has a hole at the valence-band maximum, an electron at the bottom of a conduction valley, 1.94 keV away in crystal momentum, and a transverse acoustic phonon of 14.8 meV that carries the remaining 1.0 keV.

Neither coupling reaches it alone. The density $n_e(\boldsymbol{q})$ creates pairs of crystal momentum $\boldsymbol{q}$, and at $q = 0.90\,\mathrm{keV}$ the cheapest of them costs about 1.9 eV (Fig. 13.2). The electron–phonon coupling creates a phonon and can change the crystal momentum of an electron by any amount, but it cannot lift an electron across a gap of 1.17 eV with the tens of meV that a phonon brings. The amplitude is therefore of second order, with one density vertex and one electron–phonon vertex. Light reaches the same final states (Sect. 6.5), but its phonon must supply all of $\boldsymbol{k}_\mathrm{c}$. Kahn and Lin name this process among the excitations of an indirect-gap crystal near its gap [4, Sect. VI].

We will see that the four paths of this amplitude combine on the energy shell into a one-electron band sum, do not interfere with the zero-phonon line of Chap. 10, and reduce adiabatically to a frozen-phonon derivative, which is what a first-principles calculation evaluates.

13.2.1 Final states and the second-order amplitude

The target is the crystal at the base point of Sect. 8.5 with one coupling added: independent electrons in Bloch states $\psi_{n\boldsymbol{k}}$ with energies $\varepsilon_{n\boldsymbol{k}}$ (Kohn–Sham energies with the conduction bands raised by the scissor shift, as in Sect. 10.1), harmonic phonons $(\boldsymbol{Q}, \nu)$ with energies $\omega_{\boldsymbol{Q}\nu}$, and the electron–phonon Hamiltonian H_ep of Eq. (A.25), linear in the displacements, with the screened vertex $g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q})$ of Eq. (A.26). The dark matter couples to the electron density, the row of Eq. (8.16) with $\mathfrak{D}_{ee} = |\tilde{V}(\boldsymbol{q})|^2$, and everything about the target sits in the matrix elements of $n_e(\boldsymbol{q})$ of Eq. (3.37) that enter the structure factor Eq. (4.13). In the zero-phonon response of Sect. 10.1 that matrix element is the density vertex $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q})$ of Eq. (3.34). Here it is replaced by a second-order amplitude.

The initial state $|i\rangle$ has the valence bands filled, the conduction bands empty and $n_{\boldsymbol{Q}\nu}$ phonons in each mode. A one-phonon final state has an electron in $(c, \boldsymbol{k}')$, a hole in $(v, \boldsymbol{k})$ and the phonon occupations of $|i\rangle$ changed in one mode,

$$|f\rangle = c^\dagger_{c\boldsymbol{k}'} c_{v\boldsymbol{k}}\,|i'\rangle, \qquad \boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{q} + \boldsymbol{Q}, \qquad E_f - E_i = \varepsilon_{c\boldsymbol{k}'} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}, \tag{13.6}$$

where $|i'\rangle$ has one phonon more in $(-\boldsymbol{Q}, \nu)$ (upper sign, emission) or one fewer in $(\boldsymbol{Q}, \nu)$ (lower sign, absorption), as in Eq. (A.28), and $\boldsymbol{k}'$ is folded back into the zone. The momentum transfer $\boldsymbol{q}$ is not folded: the density vertex keeps it whole and carries the Umklapp vector in its plane wave (Eq. (3.34)). Conservation of crystal momentum leaves the phonon whatever the pair needs beyond $\boldsymbol{q}$, $\boldsymbol{Q} = \boldsymbol{k}' - \boldsymbol{k} - \boldsymbol{q}$ modulo a reciprocal-lattice vector, as in Sect. 13.1. Both vertices are diagonal in spin, so the electron and the hole have the same spin, and the spin sum enters only the rate (Sect. 13.3). With spin–orbit coupling the bands are spinors, and the formulas below hold with the band indices running over spinor bands.

The second-order amplitude follows from Eq. (2.51) as for light (Sect. 6.5 and Appendix D.18). The dark matter gives up the energy ω and the momentum $\boldsymbol{q}$ at its own vertex, and its kinetic energy cancels from every denominator. In the ordering in which the density vertex acts first, the intermediate state $|m\rangle$ already holds the energy ω; in the ordering in which the phonon acts first, it does not:

$$\langle f|n_e^{(2)}(\boldsymbol{q})|i\rangle = \sum_m \left[\frac{\langle f|H_\mathrm{ep}|m\rangle\langle m|n_e(\boldsymbol{q})|i\rangle}{E_i + \omega - E_m} + \frac{\langle f|n_e(\boldsymbol{q})|m\rangle\langle m|H_\mathrm{ep}|i\rangle}{E_i - E_m} \right], \qquad \omega = E_f - E_i. \tag{13.7}$$

This matrix element takes the place of $\langle f|n_e(\boldsymbol{q})|i\rangle$ in Eq. (4.13). The condition $\omega = E_f - E_i$ is the energy delta function of the structure factor, and the amplitude is needed only there, on the energy shell. The sum runs over many-electron states with their phonons, and it is finite as long as no E_m equals $E_i + \omega$ or E_i. The second never happens in an insulator. The

first would mean that the density alone creates a real pair at $(\boldsymbol{q},\omega)$, a zero-phonon final state at the same energy, and we will see below that it cannot happen while ω lies below $E_{\min}(\boldsymbol{q})$.

13.2.2 Four paths

Both vertices are one-body operators, $n_e(\boldsymbol{q}) = \sum \rho_{mn} c^\dagger c$ and $H_{\text{ep}} = N^{-1/2} \sum g_{mn\nu} c^\dagger c\, B_{\boldsymbol{Q}\nu}$, and each moves one electron. Acting first on the filled valence bands, a vertex can only move an electron from a filled state into an empty one, so every intermediate state $|m\rangle$ holds exactly one electron–hole pair. (A term that moves an electron back into the state it came from needs a momentum transfer equal to a reciprocal-lattice vector, $\boldsymbol{q} = \boldsymbol{G}$ or $\boldsymbol{Q} = 0$. It has measure zero in the integrals over $\boldsymbol{q}$ and $\boldsymbol{Q}$, and Appendix D.35 shows that even there it adds nothing new.) The second vertex must turn that intermediate pair into the final pair $(c\boldsymbol{k}'; v\boldsymbol{k})$, again by moving one electron. It has two ways to do so. It can move the intermediate electron into $(c,\boldsymbol{k}')$, if the intermediate hole is already the final one, $(v,\boldsymbol{k})$; or it can move the electron of $(v,\boldsymbol{k})$ into the intermediate hole, if the intermediate electron is already the final one, $(c,\boldsymbol{k}')$. With two orderings of the vertices this makes four paths:

$$\begin{aligned}
&\text{A:}\quad \frac{g_{cp\nu}(\boldsymbol{k}+\boldsymbol{q},\boldsymbol{Q})\,\rho_{pv}(\boldsymbol{k},\boldsymbol{q})}{\omega - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}}, \qquad
\text{B:}\quad \frac{\rho_{cp}(\boldsymbol{k}+\boldsymbol{Q},\boldsymbol{q})\,g_{pv\nu}(\boldsymbol{k},\boldsymbol{Q})}{\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} \mp \omega_{\boldsymbol{Q}\nu}}, \qquad p \text{ empty},\\
&\text{C:}\quad -\frac{\rho_{cp}(\boldsymbol{k}+\boldsymbol{Q},\boldsymbol{q})\,g_{pv\nu}(\boldsymbol{k},\boldsymbol{Q})}{\omega - \varepsilon_{c\boldsymbol{k}'} + \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}}}, \qquad
\text{D:}\quad -\frac{g_{cp\nu}(\boldsymbol{k}+\boldsymbol{q},\boldsymbol{Q})\,\rho_{pv}(\boldsymbol{k},\boldsymbol{q})}{\varepsilon_{p,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{c\boldsymbol{k}'} \mp \omega_{\boldsymbol{Q}\nu}}, \qquad p \text{ filled}.
\end{aligned} \tag{13.8}$$

Each term is the contribution of one intermediate band p, with the common factor $N^{-1/2}(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})^{1/2}$ of the phonon vertex taken out. Fig. 13.3 draws them as time-ordered diagrams.

In paths A and B an electron propagates between the vertices. In A the density creates a pair, $(p,\boldsymbol{k}+\boldsymbol{q})$ with $(v,\boldsymbol{k})$, and the electron then scatters into $(c,\boldsymbol{k}')$ by emitting or absorbing the phonon; this is the virtual zero-phonon pair of Sect. 10.1, carried to the final state by the phonon. In B the phonon vertex creates a virtual pair at crystal momentum $\boldsymbol{Q}$, $(p,\boldsymbol{k}+\boldsymbol{Q})$ with $(v,\boldsymbol{k})$, and the density then scatters the electron by $\boldsymbol{q}$. The intermediate band p must be empty in both, since an electron cannot enter a filled state. In paths C and D a hole propagates. In C the density creates the final electron from a filled state $(p,\boldsymbol{k}+\boldsymbol{Q})$, and the phonon vertex then moves an electron from $(v,\boldsymbol{k})$ into the hole it left. In D the phonon vertex creates the final electron from a filled state $(p,\boldsymbol{k}+\boldsymbol{q})$, and the density then fills that hole from $(v,\boldsymbol{k})$. The energies in the denominators are those of the intermediate pairs, with ω added in the density-first orderings A and C and the phonon energy added in the phonon-first orderings B and D, where the phonon has already been emitted or absorbed.

The minus signs of C and D come from the order of the fermion operators. In path A the operators act as $c_c^\dagger c_p\, c_p^\dagger c_v = c_c^\dagger (1-n_p) c_v$, which is $c_c^\dagger c_v$ on a state with p empty. In path C they act as $c_p^\dagger c_v\, c_c^\dagger c_p$, and three exchanges of neighboring operators bring them into the order $-c_c^\dagger\,(c_p^\dagger c_p)\,c_v = -c_c^\dagger\, n_p\, c_v$, which is $-c_c^\dagger c_v$ on a state with p filled (Appendix D.35). The same reordering gives -1 for D. The four paths are those of phonon-assisted absorption (Appendix D.18), with the velocity vertex of light replaced by the density vertex; summed over every band they give the two-term amplitude of phonon-assisted absorption [5, Eq. (200), 6, Eqs. (2) and (3)]. The replacement changes the crystal momenta at which the intermediate states sit: the photon-first intermediate state of Eq. (6.41) sits at the crystal momentum of the hole, and the one of path A at $\boldsymbol{k}+\boldsymbol{q}$.

13.2.3 The unrestricted band sum

On the energy shell the hole paths change form. Insert $\omega = \varepsilon_{c\boldsymbol{k}'} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$ of Eq. (13.6) into their denominators. The denominator of C becomes $\varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$, and with its

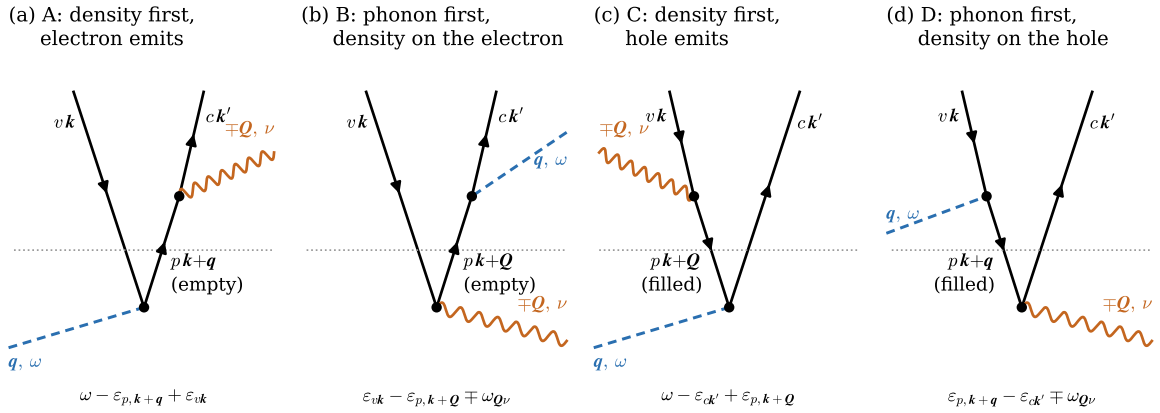


Figure 13.3: The four paths of the phonon-assisted electron-recoil amplitude, Eq. (13.8), as time-ordered diagrams. Time runs upward. Lines with upward arrows are electrons, lines with downward arrows are holes; the dashed line is the dark-matter density vertex, which delivers momentum $\boldsymbol{q}$ and energy ω; the wavy line is the phonon, emitted with $-\boldsymbol{Q}$ or absorbed with $+\boldsymbol{Q}$. The dotted line cuts the intermediate state, whose energy denominator is printed below each panel. Every path ends in the same final state, an electron in $(c, \boldsymbol{k}')$ with $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{q} + \boldsymbol{Q}$ and a hole in $(v, \boldsymbol{k})$. (a) Path A: the density creates a pair with the electron in an empty band p at $\boldsymbol{k} + \boldsymbol{q}$, and the electron then emits or absorbs the phonon. (b) Path B: the phonon vertex creates a pair with the electron in an empty band p at $\boldsymbol{k} + \boldsymbol{Q}$, and the density then moves the electron. (c) Path C: the density creates the final electron and a hole in a filled band p at $\boldsymbol{k} + \boldsymbol{Q}$, and the hole then couples to the phonon. (d) Path D: the phonon vertex creates the final electron and a hole in a filled band p at $\boldsymbol{k} + \boldsymbol{q}$, and the density then moves the hole. Schematic; no line length or angle carries a number.

minus sign, path C becomes the expression of path B for a filled band p. The denominator of D becomes $\varepsilon_{p,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}} - \omega$, and path D becomes the expression of path A for a filled band. A single electron could not take paths A and B through a filled band: Pauli blocking forbids them. In the many-electron amplitude they return, with exactly their one-electron value, as paths C and D. They return as exchange partners. In path C the electron that ends in $(c, \boldsymbol{k}')$ comes from the filled band p, and the electron from $(v, \boldsymbol{k})$ takes its place; exchanging the two identical electrons turns this into path B with the electron from v passing through p. The fermion sign of the exchange and the sign of the hole denominator cancel, and the four paths combine into the amplitude of one electron alone in the bands:

Phonon-assisted amplitude

On the energy shell, the second-order amplitude Eq. (13.7) into the final state Eq. (13.6) is

$$\begin{aligned} \langle f|n_e^{(2)}(\boldsymbol{q})|i\rangle &= \frac{1}{\sqrt{N}}\big(n_{\boldsymbol{Q}\nu} + \tfrac{1}{2} \pm \tfrac{1}{2}\big)^{1/2} M^{\pm}_{cv\nu}(\boldsymbol{k}, \boldsymbol{q}, \boldsymbol{Q}), \\ M^{\pm}_{cv\nu}(\boldsymbol{k}, \boldsymbol{q}, \boldsymbol{Q}) &= \sum_p \left[\frac{g_{cp\nu}(\boldsymbol{k}+\boldsymbol{q}, \boldsymbol{Q})\, \rho_{pv}(\boldsymbol{k}, \boldsymbol{q})}{\omega - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}} + \frac{\rho_{cp}(\boldsymbol{k}+\boldsymbol{Q}, \boldsymbol{q})\, g_{pv\nu}(\boldsymbol{k}, \boldsymbol{Q})}{\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} \mp \omega_{\boldsymbol{Q}\nu}} \right], \end{aligned} \tag{13.9}$$

with $\omega = \varepsilon_{c\boldsymbol{k}'} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$, the upper sign for emission, and the sum over every band p, filled or empty, including $p = v$ and $p = c$.

The amplitude $M^{\pm}$ is dimensionless, like ρ, and of order one in N; the factor $N^{-1/2}$ is that of Eq. (A.25), and it is removed by the independent sum over $\boldsymbol{Q}$ in the rate (Sect. 13.3). In operator form, $M^{\pm}$ is the matrix element between $(v, \boldsymbol{k})$ and $(c, \boldsymbol{k}')$ of the two vertices joined by the resolvent of the one-electron Hamiltonian, with no projector on empty states. The general statement, for any two one-body vertices acting on an independent-particle ground state, is proved in Appendix D.35; Eq. (6.41) is its instance for light.

The band-sum form Eq. (13.9), the band-sum theorem of Appendix D.35, holds under three conditions. It holds on the energy shell only. Off the shell the denominators of C and D differ from those of the blocked electron paths by $\omega - (E_f - E_i)$. A quantity that needs the amplitude off the shell, such as the real part of a phonon-assisted response, has to keep the four paths of Eq. (13.8) as they stand. It holds for independent electrons, where the intermediate and final states are single Slater determinants. With the electron–hole attraction the final pair is an exciton, and the paths are rebuilt from exciton states in Sect. 13.8. And it needs every intermediate state off the shell, which Eq. (13.11) below guarantees inside the window.

Keeping only empty intermediate bands, the natural truncation when the intermediate states are thought of as virtual electrons, gets the amplitude wrong by a factor of order one: Worked example 13.2 finds a factor of 2.4 in the rate of a model, or 1.6 with the other phase convention of its phonon vertex. The intraband terms belong to the sum as well. In path A with $p = v$, the density moves the valence electron within its band and the phonon then lifts it across the gap. In path B with $p = c$, the phonon creates the pair and the density moves the electron within the conduction band. These two terms carry the largest density vertices at small q, since ρ_{vv} and ρ_{cc} tend to one, and they cancel at $\boldsymbol{q} = 0$. There $\rho_{pv}(\boldsymbol{k}, 0) = \delta_{pv}$, path A reduces to $g_{cv\nu}(\boldsymbol{k}, \boldsymbol{Q})/\omega$ and path B to $g_{cv\nu}(\boldsymbol{k}, \boldsymbol{Q})/(\varepsilon_{v\boldsymbol{k}} - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} \mp \omega_{\boldsymbol{Q}\nu}) = -g_{cv\nu}(\boldsymbol{k}, \boldsymbol{Q})/\omega$, and

$$M^{\pm}_{cv\nu}(\boldsymbol{k}, 0, \boldsymbol{Q}) = 0. \tag{13.10}$$

This is exact, because $n_e(0)$ is the number of electrons, which commutes with the Hamiltonian

and causes no transition at any order. The amplitude therefore starts at first order in $\boldsymbol{q}$, as the zero-phonon vertex ρ_{cv} does (Eq. (3.43)), and the limit $\boldsymbol{q} \to 0$ that connects it to phonon-assisted optical absorption is taken in Sect. 13.5. A calculation that truncates the band sum, or treats the intraband terms with a different vertex than the interband ones, loses Eq. (13.10) and with it the small-q behavior of the rate.

Worked example 13.2: Exchange partners in a two-band chain

The smallest system in which all four paths appear has two bands and one phonon mode. Take a chain of $N = 6$ cells with two orbitals per cell, A at $x = 0$ and B at $x = a/2$, on-site energies $+0.4\,\mathrm{eV}$ and $-0.4\,\mathrm{eV}$, and hoppings $1.6\,\mathrm{eV}$ within the cell and $1.0\,\mathrm{eV}$ between cells. The electrons are spinless, since the spin only doubles a rate. The density vertex is $\mathrm{e}^{\mathrm{i}qx}$ on the orbital positions. One dispersionless phonon of energy $\omega_{\mathrm{ph}} = 60\,\mathrm{meV}$ shifts the two on-site energies in opposite directions. Its vertex is $0.2\,\mathrm{eV}$ times the matrix element of $\sum_R \mathrm{e}^{\mathrm{i}QR}(n_{R\mathrm{A}} - n_{R\mathrm{B}})$ between the two Bloch states it connects, with the phase of the displacement wave taken at the origin R of each cell. Crystal momenta are multiples of $2\pi/6a$, shifted by a twist of the boundary condition of $0.3\,\mathrm{rad}$ around the ring, $0.05\,\mathrm{rad}$ per crystal momentum, which lifts the degeneracy of k and $-k$. The lower band is filled and the upper is empty, with the smallest gap near the zone boundary, $1.448\,\mathrm{eV}$ with the twist and $1.442\,\mathrm{eV}$ without it. All parameters are illustrative.

Put the final pair at the gap, (v, k) and (c, k), with $q = 2 \times 2\pi/6a$ and a phonon emitted with $Q = -q$. The energy deposited on the shell is $\omega = 1.448\,\mathrm{eV} + \omega_{\mathrm{ph}} = 1.508\,\mathrm{eV}$, while the cheapest zero-phonon pair at this q costs $E_{\mathrm{min}}(q) = 2.912\,\mathrm{eV}$: the zero-phonon line is closed by $1.40\,\mathrm{eV}$. The four terms of Eq. (13.8) are:

path	intermediate pair	denominator (eV)	term
A	electron $(c, k+q)$, hole (v, k)	-1.552	$+0.0176 - 0.0246\,\mathrm{i}$
B	electron $(c, k+Q)$, hole (v, k)	-3.060	$+0.0439 + 0.0051\,\mathrm{i}$
C	electron (c, k), hole $(v, k+Q)$	-1.492	$+0.0062 - 0.0379\,\mathrm{i}$
D	electron (c, k), hole $(v, k+q)$	-3.120	$-0.0313 - 0.0354\,\mathrm{i}$

In the one-electron sum Eq. (13.9), the blocked forms are path B with $p = v$, with denominator $\varepsilon_{vk} - \varepsilon_{v,k+Q} - \omega_{\mathrm{ph}} = +1.492\,\mathrm{eV}$, and path A with $p = v$, with denominator $+3.120\,\mathrm{eV}$. Their values are $+0.0062 - 0.0379\,\mathrm{i}$ and $-0.0313 - 0.0354\,\mathrm{i}$, equal to C and D to all printed digits: the hole denominators have the opposite sign, and the fermion sign restores it. The intraband density vertex in path A with $p = v$, $|\rho_{vv}|^2 = 0.642$, is larger than the interband one, $|\rho_{cv}|^2 = 0.358$, and the hole paths are not a small correction: $|\mathrm{C} + \mathrm{D}| = 0.077$, against $|\mathrm{A} + \mathrm{B}| = 0.065$ and $|M| = 0.100$.

As a check of the signs and of the counting, we computed Eq. (13.7) by brute force, in the space of all 924 Slater determinants of six electrons in twelve orbitals, with every intermediate state. Its total, $+0.0364 - 0.0927\,\mathrm{i}$, agrees with the unrestricted sum Eq. (13.9) to 1.5×10^{-16}, and no intermediate state other than the four of the table contributes. The sum over the empty band alone, paths A and B, gives $+0.0615 - 0.0195\,\mathrm{i}$, with $|M|^2$ at 0.42 of the correct value: a factor of 2.4 in the rate. With the phase of the displacement wave taken at the orbital positions, $\mathrm{e}^{\mathrm{i}Qx}$ with $Q = -q$, the coupling itself changes: then $|M| = 0.107$, and the empty band alone is off by a factor of 1.6. The size of the error depends on the model; that it is of order one does not.

Move ω off the shell by $0.10\,\mathrm{eV}$. Path A changes as its formula says and B not at all, but C becomes $+0.0066 - 0.0406\,\mathrm{i}$ against $+0.0062 - 0.0379\,\mathrm{i}$ for the blocked form. The many-body total now differs from the unrestricted one-electron sum by 3.7%. The exchange partners are equal only where energy is conserved.

13.2.4 No interference with the zero-phonon line

A dark-matter particle that deposits ω at $\boldsymbol{q}$ could, in principle, produce the same pair with and without a phonon, and the two amplitudes might interfere. At second order in g they do not. The rate is a sum over final states of the squared total amplitude into each. The zero-phonon final states have the phonon occupations of $|i\rangle$; their amplitude is ρ_{cv} at zeroth order in g, with corrections of order g^2 from a phonon emitted and reabsorbed. The final states of Eq. (13.6) have one occupation changed by one; their amplitude starts at first order. A cross term needs two amplitudes into the same final state, and states that differ in the occupation of one mode are orthogonal. Squaring therefore gives, through order g^2, the zero-phonon rate with its own corrections, $|\rho_{cv}|^2$ plus twice the real part of ρ_{cv}^* times the order-g^2 amplitude, and the phonon-assisted rate $|M^{\pm}|^2$, with no term that mixes them. A thermal initial state changes nothing, since its density matrix is diagonal in the phonon occupations and the change of occupation between initial and final state remains a good label.

The order-g^2 corrections to the zero-phonon amplitude, the Fan–Migdal and Allen–Heine self-energies of Sect. 6.3 and the virtual-phonon vertex, therefore belong to the zero-phonon line. They shift and reweight it (Sects. 8.6 and 13.7), and they are never added to the phonon-assisted rate. Inside the window, $\omega < E_{\min}(\boldsymbol{q})$, no zero-phonon final state exists at the energy ω, and the question does not arise at any order. Above $E_{\min}(\boldsymbol{q})$ both channels are open, and the statement still holds at this order. The Fano term of Sect. 8.6 is a different interference: inside the one-phonon final states, between the part of path A in which the zero-phonon pair is real and the paths in which it is not. It belongs to the phonon-dressed sector and, at the level of one-body self-energies, vanishes in silicon and germanium. To order g^2 the response is the sum of the zero-phonon response of Chap. 10 and the phonon-assisted one built from Eq. (13.9), and neither counts the other.

13.2.5 The adiabatic (Herzberg–Teller) form

The denominators of Eq. (13.9) decide both whether the amplitude is finite and how it can be computed. Each is the energy by which an intermediate pair misses the energy shell, and on the shell each is the energy of a pair minus a known amount. For p empty in the first term, the pair $(p, \boldsymbol{k}+\boldsymbol{q};\, v, \boldsymbol{k})$ has crystal momentum $\boldsymbol{q}$, and the denominator is ω minus its energy. For p filled in the second term, path C, it is the energy of the pair $(c, \boldsymbol{k}';\, p, \boldsymbol{k}+\boldsymbol{Q})$, again of crystal momentum $\boldsymbol{q}$, minus ω. Both are at least $E_{\min}(\boldsymbol{q}) - \omega$ in size, with $E_{\min}$ of Eq. (3.57). In the two remaining cases the intermediate pair has crystal momentum $\boldsymbol{Q}$ and costs at least E_{g}, and the denominator differs from its energy by $\omega_{\boldsymbol{Q}\nu}$. The smallest off-shell distance of Eq. (8.40) is therefore bounded below on the shell:

$$\delta_{\mathrm{off}} \geq \min\bigl(E_{\min}(\boldsymbol{q}) - \omega,\; E_{\mathrm{g}} - \omega_{\boldsymbol{Q}\nu}\bigr). \tag{13.11}$$

Below $E_{\min}(\boldsymbol{q})$ every denominator is negative or positive by a finite margin, and the amplitude needs no width, no $\mathrm{i}0^+$ and no subtraction. The only denominators that can approach zero are those of the virtual zero-phonon pairs at $\boldsymbol{q}$, the electron of path A and the hole of path C. As ω approaches $E_{\min}(\boldsymbol{q})$ they become real pairs that emit a phonon after they are created, a cascade, and the second-order amplitude stops being a separate quantity. Sect. 13.6 turns this into the window inside which Eq. (13.9) holds. In silicon the bound for pairs at $\boldsymbol{Q}$ is at least $1.17\,\mathrm{eV} - 64\,\mathrm{meV} = 1.11\,\mathrm{eV}$, and the tighter bound is the margin below the zero-phonon edge. At the onset of Worked example 13.1, $\omega \approx 1.19\,\mathrm{eV}$ against $E_{\min} \approx 1.9\,\mathrm{eV}$, it is about $0.7\,\mathrm{eV}$.

The phonon energy enters the amplitude twice: in the energy balance, where it sets the threshold, and in the denominators, where it is small against δ_{off}. Dropping it from the

denominators, with $\omega = \varepsilon_{c\boldsymbol{k}'} - \varepsilon_{v\boldsymbol{k}}$ in the first term, gives the adiabatic amplitude

$$M^{\mathrm{ad}}_{cv\nu}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q}) = \sum_p \left[\frac{g_{cp\nu}(\boldsymbol{k}+\boldsymbol{q},\boldsymbol{Q})\,\rho_{pv}(\boldsymbol{k},\boldsymbol{q})}{\varepsilon_{c\boldsymbol{k}'} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{q}}} + \frac{\rho_{cp}(\boldsymbol{k}+\boldsymbol{Q},\boldsymbol{q})\,g_{pv\nu}(\boldsymbol{k},\boldsymbol{Q})}{\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}}} \right], \tag{13.12}$$

the same for emission and absorption, with the states degenerate with $(c,\boldsymbol{k}')$ or $(v,\boldsymbol{k})$ left out of the respective term. It is a derivative of the zero-phonon vertex. Replace the phonon operator $B_{\boldsymbol{Q}\nu}$ in H_{ep} by a number b, with its conjugate for $B_{-\boldsymbol{Q}\nu}$. The electrons then feel the static potential of a frozen displacement wave of the mode, Eq. (A.21) with $B \to b$, which couples $\psi_{n\boldsymbol{k}}$ to the states at $\boldsymbol{k} \pm \boldsymbol{Q}$ with the matrix elements $N^{-1/2} b\, g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})$ and their conjugates. By first-order perturbation theory, the valence state $\psi_{v\boldsymbol{k}}$ acquires the admixture $N^{-1/2} b \sum_p \psi_{p,\boldsymbol{k}+\boldsymbol{Q}}\, g_{pv\nu}(\boldsymbol{k},\boldsymbol{Q})/(\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}})$, which is Eq. (E.32), and the conduction state $\psi_{c\boldsymbol{k}'}$ acquires states at $\boldsymbol{k}' - \boldsymbol{Q} = \boldsymbol{k} + \boldsymbol{q}$ through the conjugate coupling. The transition density $\rho_{cv}(\boldsymbol{k},\boldsymbol{q}) = \langle \psi_{c\boldsymbol{k}'} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \psi_{v\boldsymbol{k}} \rangle$, zero in the perfect lattice because $\boldsymbol{k}' \neq \boldsymbol{k} + \boldsymbol{q}$, becomes nonzero at first order in b. The change of the conduction state gives the first term of Eq. (13.12) and the change of the valence state the second, and

$$\frac{\partial \rho_{cv}}{\partial b} = \frac{1}{\sqrt{N}}\, M^{\mathrm{ad}}_{cv\nu}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q}), \qquad \langle f | n^{(2)}_e(\boldsymbol{q}) | i \rangle \simeq \langle n' | B_{\boldsymbol{Q}\nu} | n \rangle \, \frac{\partial \rho_{cv}}{\partial b}, \tag{13.13}$$

where $\langle n' | B_{\boldsymbol{Q}\nu} | n \rangle = (n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})^{1/2}$ is the phonon matrix element. This is the Herzberg–Teller term of Eq. (8.52): the transition density of the frozen crystal, linear in the displacement, taken between the vibrational states. It is also the derivative Eq. (E.31) that Appendix E.4 computes. In this form the unrestricted band sum needs no theorem. First-order perturbation theory of one orbital sums over all other orbitals, filled or empty, and the filled ones are exactly the hole paths C and D.

The adiabatic form drops the phonon energy in the denominators. Expanding each denominator to first order in $\omega_{\boldsymbol{Q}\nu}$,

$$M^{\pm}_{cv\nu} - M^{\mathrm{ad}}_{cv\nu} = \mp\, \omega_{\boldsymbol{Q}\nu} \sum_p \left[\frac{g_{cp\nu}\,\rho_{pv}}{(\varepsilon_{c\boldsymbol{k}'} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{q}})^2} - \frac{\rho_{cp}\,g_{pv\nu}}{(\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}})^2} \right] + O\!\left(\frac{\omega^2_{\boldsymbol{Q}\nu}}{\delta^2_{\mathrm{off}}} \right) M^{\mathrm{ad}}, \tag{13.14}$$

with the arguments of Eq. (13.12). Each term is smaller than its adiabatic counterpart by $\omega_{\boldsymbol{Q}\nu}$ over its own denominator, so the relative error is of order $\omega_{\boldsymbol{Q}\nu}/\delta_{\mathrm{off}}$ as long as the sum is not much smaller than its largest term (Appendix D.35). The correction has opposite signs for emission and absorption, and the adiabatic form, which cannot tell the two apart, misses their difference, of the same relative order.

In silicon, with the phonon energies at the conduction valley (Sect. 6.5) and the interband bound of Eq. (13.11), the estimate $\omega_{\boldsymbol{Q}\nu}/\delta_{\mathrm{off}}$ is 1.6% for the transverse acoustic phonon of 18.27 meV and 5.2% for the transverse optical phonon of 57.73 meV. The margin below the zero-phonon edge is the tighter condition. For the error to stay below 10%, $E_{\mathrm{min}}(\boldsymbol{q}) - \omega$ must exceed 0.18 eV for the transverse acoustic and 0.58 eV for the transverse optical phonon. Deep in the window the adiabatic form is accurate; near its upper edge it fails, on the same denominators that make the amplitude itself resonant. The two-band chain of Worked example 13.2 confirms the pattern. There the frozen-displacement derivative of the transition density, computed by finite differences, agrees with Eq. (13.12) to 5×10^{-10}, and the full amplitude with its phonon of 60 meV differs from the adiabatic one by 1.9% at a zero-phonon margin of 1.40 eV and by 6.2% at 0.62 eV, below $\omega_{\boldsymbol{Q}\nu}/\delta_{\mathrm{off}}$ in both cases.

The same estimate disposes of the crossings of two valence bands at $\boldsymbol{k}$ and $\boldsymbol{k} + \boldsymbol{Q}$, where the adiabatic denominator of the second term vanishes (Appendix E.4). By Eq. (13.11) such a crossing makes path C resonant, so it lies outside the window.

The adiabatic form is what a first-principles calculation evaluates, either as a finite difference of the transition density in a supercell that holds the displaced lattice or as the derivative Eq. (E.31) from density-functional perturbation theory. The second needs more than a phonon calculation provides: besides the part of the change of each valence state in the empty bands, it needs the part in the filled bands, which carries the hole path C, and the change of the conduction state, which carries paths A and D (Eqs. (E.32) to (E.34)). The sum-over-states form Eq. (13.9) avoids the derivative and keeps the phonon energy in the denominators. The f-sum rule of the density vertices tests that the band sum is complete. Appendix E.5 sets out the route, its controls and its operation count, and Appendix E.14 specifies the calculation for silicon. Squared, summed over final states and weighted by the phonon occupations, Eq. (13.9) gives the phonon-assisted rate of Sect. 13.3.

13.3 The phonon-assisted rate

A dark-matter particle of 0.35 MeV that transfers 1.0 keV of momentum to silicon can deposit up to 1.22 eV, while at that momentum transfer the cheapest zero-phonon pair along the valley axis costs about 1.77 eV (Eq. (3.57), with the bands of Fig. 3.7). Every event it causes therefore leaves a phonon behind, a transverse acoustic phonon of 14 meV that carries the missing 0.94 keV. The square of the amplitude of Sect. 13.2, summed over final states, enters the rate exactly as the zero-phonon structure factor of Sect. 10.1 does; we will write it as a structure factor, as a loss function that enters Eq. (10.4), and near its threshold as a definite power law.

13.3.1 The phonon-assisted structure factor

The structure factor counts the transitions that the density operator drives from the thermal state, Eq. (2.67) with the transition sum Eq. (2.65). Inside the off-shell window, $\omega \leq E_{\min}(\boldsymbol{q}) - \delta_{\mathrm{w}}$, with δ_{w} the margin of Sect. 8.6, 0.3 eV in silicon (Sect. 13.6), no zero-phonon pair exists at the energy deposited, and every final state contains a phonon, up to a virtual admixture of relative weight θ_{ep} (Sect. 8.6). The final states are those of Eq. (13.6): a hole in $(v, \boldsymbol{k})$, an electron in $(c, \boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q})$, and one more phonon in $(-\boldsymbol{Q}, \nu)$ or one fewer in $(\boldsymbol{Q}, \nu)$. The matrix element of the density into each of them is Eq. (13.9): the factor $N^{-1/2}$ of the electron–phonon Hamiltonian, the phonon factor $(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})^{1/2}$ of Eq. (A.28), and the dimensionless four-path amplitude $M^{\pm}_{cv\nu}(\boldsymbol{k}, \boldsymbol{q}, \boldsymbol{Q})$, summed over every band.

To sum over the final states, we label them by $\boldsymbol{k}$ and $\boldsymbol{Q}$, each running over the N points of the zone, by the bands c and v, the branch ν, the sign, and the spin. Neither vertex flips the spin, so the spin gives a factor 2. The initial phonon state is thermal, and the average of the squared phonon matrix element is $n_{\boldsymbol{Q}\nu} + 1$ for emission and $n_{\boldsymbol{Q}\nu}$ for absorption, with $n_{\boldsymbol{Q}\nu} = n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$. The electrons start in the ground state: the Fermi factor $f_v(1 - f_c)$ of each transition differs from one by terms of order $\mathrm{e}^{-E_{\mathrm{g}}/2T}$. The energy delta function sets $\omega = \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$. Collecting the factors, $(2\pi/\mathscr{V}) \times 2 \times N^{-1} \sum_{\boldsymbol{k}\boldsymbol{Q}} = (4\pi N/\mathscr{V}) \langle\!\langle \cdots \rangle\!\rangle$ with the double zone average of Eq. (A.11), and $\mathscr{V} = N\Omega$ (Appendix D.35 gives the thermal averages):

Phonon-assisted structure factor

$$S^{\mathrm{PA}}(\boldsymbol{q}, \omega) = \frac{4\pi}{\Omega} \Big\langle\!\Big\langle \sum_{\nu} \sum_{cv} \sum_{\pm} \big(n_{\boldsymbol{Q}\nu} + \tfrac{1}{2} \pm \tfrac{1}{2}\big) \, \big|M^{\pm}_{cv\nu}(\boldsymbol{k}, \boldsymbol{q}, \boldsymbol{Q})\big|^2 \, \delta\big(\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu} - \omega\big) \Big\rangle\!\Big\rangle, \tag{13.15}$$

for $T \ll E_\mathrm{g}$ and ω inside the off-shell window. The phonon-assisted rate is Eq. (8.16) with $\mathfrak{D}_{ee} = \pi\bar{\sigma}_e F_\mathrm{DM}(q)^2/\mu_{\chi e}^2$ (Eq. (1.34)) and $S = S^\mathrm{PA}$.

The number of cells drops out, as it must. The double sum over $\boldsymbol{k}$ and $\boldsymbol{Q}$ contributes N^2, the vertex N^{-1}, and the volume $\mathscr{V} = N\Omega$ the last power, so that S^PA is intensive like the structure factor of a single pair. The crystal momentum of the final electron is $\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}$ reduced to the zone, while $\boldsymbol{q}$ itself stays unreduced inside ρ, as in Sect. 10.1.

The two signs are the two thresholds of Sect. 6.5. Emission costs the phonon energy and comes with $n_{\boldsymbol{Q}\nu}+1$; absorption gains it and comes with $n_{\boldsymbol{Q}\nu}$, and opens at energies below the gap. Their weights stand in the ratio $\mathrm{e}^{-\omega_{\boldsymbol{Q}\nu}/T}$ of Eq. (6.45). At detector temperatures the transverse acoustic phonon of 15 meV is unoccupied ($n_\mathrm{B} < 10^{-74}$ below 1 K, Sect. 13.1). Only emission survives, the lower sign drops out, and Eq. (13.15) vanishes below the onset $E^\mathrm{PA}_\mathrm{min}(\boldsymbol{q})$ of Eq. (13.2).

Eq. (13.15) holds where every denominator of $M^\pm$ is bounded away from zero, which is what the window condition guarantees: a denominator vanishes only where a first-order transition at the same $(\boldsymbol{q},\omega)$ is on shell (Sect. 13.2). In the expansion of Sect. 8.5 the rate is quadratic in the electron–phonon vertex, $n_g = 2$, and the next order costs one power of the off-shell parameter θ_ep, 0.010 to 0.016 in silicon at the absorption edge (Eq. (8.40)). The amplitude is unscreened, so the many-body approximation is $\mathscr{X} = \mathrm{IP}$; Sect. 13.4 screens both the density vertex and the pair. Beyond the window the intermediate states can be real, and the resonant paths need the widths and the finite part of Sect. 13.6.

13.3.2 The phonon-assisted loss function

The fluctuation–dissipation relation Eq. (4.45) defines a loss function for any piece of the structure factor. For the phonon-assisted piece we write

Phonon-assisted loss function

$$\begin{aligned} L^\mathrm{PA}(\boldsymbol{q},\omega) &= \frac{e^2}{2q^2}\left(1-\mathrm{e}^{-\beta\omega}\right) S^\mathrm{PA}(\boldsymbol{q},\omega) \\ &= \frac{8\pi^2\alpha}{\Omega\, q^2}\left\langle\!\!\left\langle \sum_{\nu cv\pm} w_\pm \left|M^\pm_{cv\nu}\right|^2 \delta\left(\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu} - \omega\right)\right\rangle\!\!\right\rangle, \end{aligned} \tag{13.16}$$

with $w_\pm = (1-\mathrm{e}^{-\beta\omega})(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})$. The rate is Eq. (10.4) with $\bar{L}$ replaced by $\bar{L}^\mathrm{PA}$, its average over the directions of $\boldsymbol{q}$.

The second form uses $e^2 = 4\pi\alpha$. The factor $1-\mathrm{e}^{-\beta\omega}$ appears once, in L^PA, and Eq. (10.4) divides it out again, so the rate carries only the phonon factors $n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2}$ of Eq. (13.15). Each weight $w_\pm$ is the rate of a transition minus that of its inverse, as for any dissipative response (Appendix D.35). At a detector's temperature, with $\beta\omega \sim 10^4$ or more, both factors are one, and L^PA is $(e^2/2q^2)S^\mathrm{PA}$ with the emission term alone.

Since the amplitude is unscreened, L^PA is a phonon-assisted imaginary dielectric function at finite momentum, $\varepsilon_2^\mathrm{PA}(\boldsymbol{q},\omega)$, in the same way that the independent-particle structure factor of Sect. 10.1 is $(2q^2/e^2)\,\varepsilon_2^\mathrm{IP}$ (Sect. 10.2). As $\boldsymbol{q}\to 0$ the density vertex reduces to the velocity vertex on shell (Eq. (3.43)), and $\varepsilon_2^\mathrm{PA}(\boldsymbol{q},\omega)$ tends to the phonon-assisted absorption $\varepsilon_2^\mathrm{PA}(\omega)$ of Eq. (6.42); Sect. 13.5 proves this and uses the measured absorption edge as the test of a computed L^PA. And with the other electrons polarizing the medium, a loss function is the imaginary part of $-1/\varepsilon$, not ε_2; Sect. 13.4 shows that the phonon-assisted part then becomes $\varepsilon_2^\mathrm{PA}/(\varepsilon_1^0)^2$ plus the screening of the pair.

The prescriptions of Sect. 8.6 for adding rates apply to L^{PA} directly. A measured loss function or optical conductivity contains both electronic sectors, the zero-phonon line and the phonon sideband, and a computed L^{PA} is never added to it. Below the direct gap the measured optical absorption of silicon is the phonon-assisted loss function at $\boldsymbol{q} \to 0$ and nothing else, at the temperature of the measurement, which is why Sect. 13.5 has to transfer it to millikelvin before it can stand in for L^{PA}.

A first-principles L^{PA} needs the bands, the density vertices and the electron–phonon vertices at three crystal momenta, $\boldsymbol{k}$, $\boldsymbol{k}+\boldsymbol{q}$ and $\boldsymbol{k}+\boldsymbol{Q}$, on fine and independent grids of $\boldsymbol{k}$ and $\boldsymbol{Q}$, as for Eq. (6.42). The route through Wannier interpolation and its controls are in Appendix E.5, and the calculation for silicon, germanium and GaAs, binned in $(|\boldsymbol{q}|, \hat{\boldsymbol{q}}, \omega)$, is specified in Appendix E.14.

13.3.3 The zero-phonon limit

Applied at first order, the same sums over final states give back the structure factor of Sect. 10.1. Without a phonon the matrix element of the density between the ground state and a pair is $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q})$, the final states are labeled by $\boldsymbol{k}$ alone, and the sum is $(2\pi/\mathcal{V}) \times 2 \times \sum_{\boldsymbol{k}}$. With $N^{-1}\sum_{\boldsymbol{k}} = \Omega \int_{\mathrm{BZ}} \mathrm{d}^3k/(2\pi)^3$ this is $4\pi \int_{\mathrm{BZ}} \mathrm{d}^3k/(2\pi)^3 \, |\rho_{cv}|^2 \, \delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}})$, which is Eq. (10.1) term by term. The two structure factors share the $2\pi/\mathcal{V}$ of the golden rule, the spin factor 2 and the cancellation of the volume. The phonon-assisted one has a second zone average, over the phonon wavevector, and the factor N^{-1} of the electron–phonon vertex pays for it. Both therefore enter the same rate formula with the same prefactor. In particular, a phonon-assisted table in the normalization of the crystal form factor, $q^3\Omega\bar{S}^{\mathrm{PA}}/(4\pi\alpha m_e^2)$ as in Eq. (10.3), enters the rate of Limit Check 10.1 by addition to $|f_{\mathrm{crys}}|^2$.

The zero-phonon structure factor is the $n_g = 0$ term of the expansion in the electron–phonon vertex, and S^{PA} is its $n_g = 2$ term. As $g \to 0$, $M^{\pm}$ vanishes linearly, S^{PA} quadratically, the phonon-dressed sector empties, and the rate is the crystal form-factor rate of Limit Check 10.1. At finite coupling the two terms live in different places. Inside the window S^0 is zero, since no pair exists below $E_{\mathrm{min}}(\boldsymbol{q})$, and S^{PA} is the whole electronic response at this order. Above $E_{\mathrm{min}}(\boldsymbol{q})$ both are present. Their final states differ in the number of phonons and are orthogonal, so at order g^2 the two rates add with no interference term (Sect. 13.2).

Adding them is still not free of double counting, because the two sectors share one sum rule. The electron–phonon Hamiltonian is a function of the electron positions and commutes with the density, so the f-sum of the electron density, Eq. (4.28), holds at every order in g; the two electronic sectors together carry it up to a part in 10^3 (Sect. 8.6). The first moment of S^{PA} is positive, and the zero-phonon sector loses the same amount at the same order, through the self-energies and vertex corrections of its pairs. A zero-phonon rate computed with the bare bands and $|\rho_{cv}|^2$, plus S^{PA}, overshoots the f-sum by the first moment of S^{PA}, a relative excess of order θ_{ep}. The renormalized zero-phonon response that removes it is that of Sect. 13.7. Inside the window none of this matters, because there the zero-phonon line has no weight to lose.

13.3.4 The onset exponent

Near its threshold the phonon-assisted rate is a power of the energy above it, and the power is fixed by counting the free momenta of the final state. Take a fixed $\boldsymbol{q}$, $T \to 0$, and an energy deposited $x = \omega - E^{\mathrm{PA}}_{\mathrm{min}}(\boldsymbol{q})$ just above the onset of Eq. (13.2). The hole lies near the valence-band maximum and the electron near the bottom of the valley $\boldsymbol{k}_{\mathrm{c}}$, at small momenta $\boldsymbol{\kappa}_{\mathrm{h}}$ and $\boldsymbol{\kappa}_{\mathrm{e}}$ from the extrema, and the phonon wavevector is $\boldsymbol{Q}^* + \boldsymbol{\kappa}_{\mathrm{e}} - \boldsymbol{\kappa}_{\mathrm{h}}$ with $\boldsymbol{Q}^* = \boldsymbol{k}_{\mathrm{c}} - \boldsymbol{q}$.

The final-state energy above the onset is

$$\frac{\kappa_{\mathrm{e}}^2}{2m_{\mathrm{d}}^*}+\frac{\kappa_{\mathrm{h}}^2}{2m_{\mathrm{h}}}+\boldsymbol{u}_\nu\cdot(\boldsymbol{\kappa}_{\mathrm{e}}-\boldsymbol{\kappa}_{\mathrm{h}})+O(\kappa^2\,\partial^2\omega_\nu),$$

with $\boldsymbol{u}_\nu$ the group velocity of the phonon at $\boldsymbol{Q}^*$ and the valley written with its density-of-states mass, as in Sect. 6.5. The linear term tilts the minimum: completing the square lowers it by $\frac{1}{2}M_{\mathrm{eh}}u_\nu^2$, with $M_{\mathrm{eh}}=1.19\,m_e$ the pair mass of Sect. 13.1. This is the shift that Sect. 13.1 found below 0.02 meV at the onset mass; at $|\boldsymbol{q}|=1.0\,\mathrm{keV}$ it is 0.02 meV for the transverse acoustic branch and 0.18 meV for the longitudinal one. The curvature of the phonon branch is a percent of $1/M_{\mathrm{eh}}$ and changes the masses by as much. What remains is two parabolic pockets that are independent, because the phonon takes up any momentum mismatch. Their joint density of states is Eq. (6.43). With $M^{\pm}$ constant across the pockets, at its value M_ν at the extrema, Eq. (13.15) becomes, per valley, branch and valence band,

$$S^{\mathrm{PA}}(\boldsymbol{q},\omega)\simeq\frac{\Omega\,(m_{\mathrm{d}}^*m_{\mathrm{h}})^{3/2}}{4\pi^2}\,|M_\nu|^2\,x^2\,\Theta(x). \tag{13.17}$$

At fixed $\boldsymbol{q}$ the rate rises as the square of the energy above its onset, the Hall–Bardeen–Blatt law Eq. (6.44) at finite momentum transfer.

The rate of a detector that does not resolve the direction of $\boldsymbol{q}$ uses $\bar{S}^{\mathrm{PA}}$, and the average adds powers. The onset depends on the direction through $\boldsymbol{Q}^*$. Let θ_q be the angle between $\boldsymbol{q}$ and the valley axis, take $q<k_{\mathrm{c}}$ and expand for small θ_q. The phonon wavevector lengthens, $|\boldsymbol{Q}^*|^2=(k_{\mathrm{c}}-q)^2+2k_{\mathrm{c}}q(1-\cos\theta_q)\simeq(k_{\mathrm{c}}-q)^2+k_{\mathrm{c}}q\,\theta_q^2$, and it tilts away from the axis by $\theta_Q\simeq q\theta_q/(k_{\mathrm{c}}-q)$. Writing the phonon energy near the axis as a function of $|\boldsymbol{Q}|$ and θ_Q, the onset is

$$E_{\mathrm{min}}^{\mathrm{PA}}(\theta_q)\simeq E_{\mathrm{g}}+\omega_\nu(k_{\mathrm{c}}-q)+\varkappa_\nu\theta_q^2,\qquad \varkappa_\nu=\frac{\omega_\nu'\,k_{\mathrm{c}}q}{2(k_{\mathrm{c}}-q)}+\frac{\partial^2_{\theta_Q}\omega_\nu}{2}\,\frac{q^2}{(k_{\mathrm{c}}-q)^2}, \tag{13.18}$$

where ω_ν' is the slope of the branch along the axis and $\partial^2_{\theta_Q}\omega_\nu$ its curvature in the tilt angle at fixed $|\boldsymbol{Q}|$, both at $|\boldsymbol{Q}|=k_{\mathrm{c}}-q$. The first term comes from the longer phonon wavevector and is positive for a branch that rises away from Γ; the second is the anisotropy of the branch. Where $\varkappa_\nu>0$ the valley axis is an isolated minimum of the onset, and only a cone of half-angle $(x/\varkappa_\nu)^{1/2}$ around it contributes at energy x above the onset. Integrating Eq. (13.17) over that cone, as Worked example 13.3 does, gives one more power of x.

Both powers are instances of one rule. If the final-state energy has an isolated quadratic minimum in D free variables, the volume below x grows as $x^{D/2}$ and its density as $x^{D/2-1}$; if $|M|^2$ vanishes at the minimum as the square of the distance from it, one more power enters. Two free pockets give $D=6$ and x^2, the two angles of $\hat{\boldsymbol{q}}$ make $D=8$ and x^3. A bound exciton with a phonon has only its center-of-mass momentum free, $D=3$, and its replica rises as $x^{1/2}$ at fixed $\boldsymbol{q}$ and as $x^{3/2}$ after the average (Sect. 13.8). The power of x in $\bar{S}^{\mathrm{PA}}$, and so in $\bar{L}^{\mathrm{PA}}$ and in the spectrum at fixed $|\boldsymbol{q}|$, is the *onset exponent* p_{on}.

Symmetry decides whether M_ν vanishes at the minimum. In silicon, with $\boldsymbol{q}$ on a valley axis, the vectors $\boldsymbol{q}$, $\boldsymbol{Q}^*$ and $\boldsymbol{k}_{\mathrm{c}}$ all lie on the [100] line, and the operations that leave this line fixed form the group C_{4v}. The electron at the bottom of the valley belongs to its identity representation Δ_1, and the threefold valence maximum Γ_{25}' splits on the line into $\Delta_2'\oplus\Delta_5$ [7, Chap. 2]. The density vertex with no reciprocal-lattice vector added to $\boldsymbol{q}$ is invariant (Δ_1): an operation with a fractional translation $\boldsymbol{\tau}$ multiplies it by $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}}$, and this phase cancels against those of the two Bloch states, as Worked example 13.9 shows for the mirror. A longitudinal acoustic displacement lies along the axis and is in phase on the two atoms of the cell. Every operation that fixes the axis maps the axis onto itself and the two atoms

onto each other or themselves, so it leaves this displacement unchanged, and it is Δ_1 as well; the longitudinal optical one is Δ_2', and the transverse pairs are Δ_5. An amplitude is a number and is nonzero only if the product of the representations of electron, vertex, phonon and hole contains Δ_1 (Eq. (C.143)). For the longitudinal acoustic phonon the product is $\Delta_1 \otimes \Delta_1 \otimes \Delta_1 \otimes (\Delta_2' \oplus \Delta_5) = \Delta_2' \oplus \Delta_5$, which does not. The transverse branches and the longitudinal optical one are allowed, because $\Delta_5 \otimes \Delta_5$ and $\Delta_2' \otimes \Delta_2'$ contain Δ_1.

The forbidden branch rises one power later. Off the axis its amplitude grows linearly with the distance from it, $|M|^2$ grows quadratically, and the longitudinal acoustic onset is a kink with $p_{\mathrm{on}} = 4$ on top of the transverse acoustic edge with $p_{\mathrm{on}} = 3$. Spin–orbit coupling mixes the valence representations and lifts the zero, and leaves a part with $p_{\mathrm{on}} = 3$. We estimate its relative weight from the spin–orbit lift of the zero-phonon vertex on the same axis, $|\rho_{cv}|^2 = 3.6 \times 10^{-6}$ against an allowed 0.150 (Sect. 13.9.2), as below 10^{-4}. Beyond $|\boldsymbol{q}| = 2\pi/a = 2.28\,\mathrm{keV}$, halfway between the valley at 0.85 X and the image of the opposite valley at 1.15 X, the nearest pocket is that image, reached with the reciprocal-lattice vector $(2\pi/a)(2,0,0)$ inside the vertex. There the vertex picks up the glide phase of Eq. (C.149) and transforms as Δ_2', the longitudinal optical branch becomes the forbidden one, and the longitudinal acoustic branch is allowed. Its minimum, however, lies off the axis, where the vertex does not vanish, so every branch of that class has $p_{\mathrm{on}} = 3$.

Whether the minimum of each branch lies on the axis depends on the sign of $\varkappa_\nu$. With the silicon dispersion of Sect. 5.2, the transverse acoustic onset has its minimum on the axis for $|\boldsymbol{q}| < 1.34\,\mathrm{keV}$; above that, the anisotropy term of Eq. (13.18) wins and the minima move off the axis into the {001} planes. The longitudinal acoustic branch rises steeply and keeps its minimum on the axis. The optical branches fall away from Γ, the first term of $\varkappa_\nu$ is negative, and their minima lie far off the axis: at $|\boldsymbol{q}| = 1.0\,\mathrm{keV}$ the lowest optical onset over all directions is 44 meV above E_{g}, against 60 meV on the axis. None of this changes an exponent. Off the axis, cubic symmetry leaves isolated minima at which the vertex is generically nonzero, and $p_{\mathrm{on}} = 3$; a minimum spread over a ring around the axis would give 5/2, but the anisotropy of a cubic crystal breaks the ring. In germanium the valleys lie at L on the threefold axes, the group of the line is C_{3v}, the valence maximum splits into $\Lambda_1 \oplus \Lambda_3$, the electron is Λ_1 and the phonons are Λ_1 (longitudinal) and Λ_3 (transverse). Every branch is allowed, and $p_{\mathrm{on}} = 3$ throughout. The two transverse acoustic branches of germanium are degenerate on the [111] axis and split linearly away from it, so their minimum is never on the axis.

Onset exponent

Inside the off-shell window, at zero temperature and fixed $|\boldsymbol{q}|$, the direction-averaged phonon-assisted loss function rises above its onset as $x^{p_{\mathrm{on}}}$ with

$$\begin{aligned} &p_{\mathrm{on}} = 3 \quad \text{(Si: transverse acoustic edge, optical branches; Ge: all branches)},\\ &p_{\mathrm{on}} = 4 \quad \text{(Si: longitudinal acoustic kink)}, \end{aligned} \tag{13.19}$$

two at a fixed direction of $\boldsymbol{q}$, and 3/2 for an exciton replica.

Fig. 13.4 shows the exponents in a model of silicon at $|\boldsymbol{q}| = 1.0\,\mathrm{keV}$, with the onsets taken from the computed dispersion over the whole sphere of directions. Panel (b) gives their range. The averaged transverse acoustic component follows x^3 within 10 % for $x < 1.3\,\mathrm{meV}$, and its local exponent has drifted only to 3.3 at 10 meV; the longitudinal acoustic one follows x^4 for $x < 2.4\,\mathrm{meV}$. The average also costs strength. At 0.1 meV above the onset the averaged transverse acoustic component of one valley is 1.2×10^{-3} of its value with $\boldsymbol{q}$ on that valley's axis, because only a narrow cone of directions is open.

The exponents also need the onset itself inside the window. At the onset the pair sits at

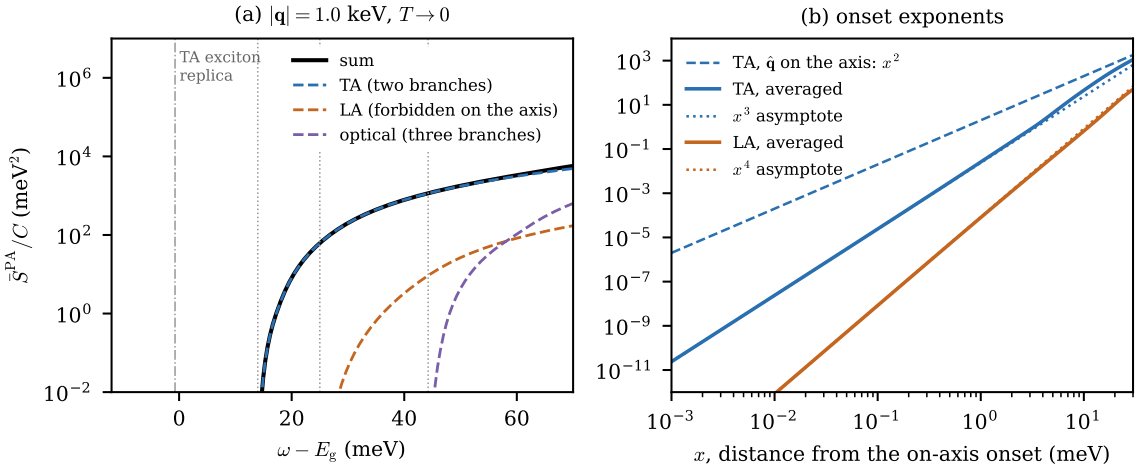


Figure 13.4: The phonon-assisted structure factor of silicon near its threshold, averaged over the directions of $\boldsymbol{q}$ at $|\boldsymbol{q}| = 1.0\,\mathrm{keV}$, at zero temperature (emission only), in units of a common coefficient C. Model: one valley at $\boldsymbol{k}_{\mathrm{c}} = 0.85\,(2\pi/a)$, parabolic pockets and the fixed-$\boldsymbol{q}$ form Eq. (13.17), with the onset $E_{\mathrm{g}} + \omega_\nu(\boldsymbol{k}_{\mathrm{c}} - \boldsymbol{q})$ computed in every direction from the PBE force constants of Sect. 5.2, whose transverse acoustic energy at X is 17.06 meV against 18.57 meV measured. Every branch is given the same $|M_\nu|^2 = C$, except the longitudinal acoustic one, which is given $C \sin^2\theta_Q$ because its amplitude vanishes on the valley axis; the relative heights are therefore illustrative. The tilt of the pair minimum (at most 0.18 meV) is neglected. (a) Components against $\omega - E_{\mathrm{g}}$: the transverse acoustic edge at 13.98 meV, the longitudinal acoustic kink at 25.00 meV, and the optical onset at 44.1 meV, whose minimum lies far off the axis (dotted lines). The dash-dotted line marks the exciton replica of the transverse acoustic edge, $E_x^b = 14.7$ meV lower, whose strength is not modeled. (b) Each acoustic component against the distance x from its own onset. With $\boldsymbol{q}$ held on the axis the transverse acoustic component grows as x^2 (dashed). Averaged over directions it grows as x^3 (solid), and the longitudinal acoustic one as x^4; dotted lines are the asymptotes of Worked example 13.3. The window condition holds over the whole plot: at this $|\boldsymbol{q}|$ the zero-phonon pair continuum starts 0.60 eV above E_{g} (PBE bands of Sect. 3.3 with the measured gap).

the band extrema and the phonon takes all of $\boldsymbol{Q}^*$, while the zero-phonon edge at the same $\boldsymbol{q}$ lies $|\boldsymbol{Q}^*|^2/2M_{\mathrm{eh}}$ above E_{g} (Eq. (3.58)). The window condition $E_{\mathrm{g}} + \omega_\nu(\boldsymbol{Q}^*) \leq E_{\mathrm{min}}(\boldsymbol{q}) - \delta_{\mathrm{w}}$ then reads $|\boldsymbol{Q}^*|^2 - 2M_{\mathrm{eh}}c\,|\boldsymbol{Q}^*| - 2M_{\mathrm{eh}}\delta_{\mathrm{w}} \geq 0$ for a branch $\omega_\nu = c|\boldsymbol{Q}|$, that is,

$$|\boldsymbol{Q}^*| \geq M_{\mathrm{eh}}c + \left(M_{\mathrm{eh}}^2c^2 + 2M_{\mathrm{eh}}\delta_{\mathrm{w}}\right)^{1/2}.$$

With a margin $\delta_{\mathrm{w}} = 0.3\,\mathrm{eV}$, five times the top of the phonon band (Sect. 13.6 fixes the margin), and the phase velocity of the transverse acoustic branch there, $5.0\,\mathrm{km\,s^{-1}}$, the bound is $0.31\,\mathrm{\AA}^{-1}$, or $0.615\,\mathrm{keV}$. In silicon the exponents of Eq. (13.19) therefore hold for $|\boldsymbol{q}| \leq k_{\mathrm{c}} - 0.615\,\mathrm{keV} = 1.33\,\mathrm{keV}$ for parabolic valleys, and for $|\boldsymbol{q}| \leq 1.27\,\mathrm{keV}$ with the computed bands (Sect. 13.6); over this whole range the transverse acoustic minimum lies on the axis. Closer to the valley the onset falls into the band at the edge of the window, where the phonon-assisted rate is not separately defined (Sect. 13.6), and no exponent of it alone exists.

Worked example 13.3: The onset exponent from the joint density of states

Take silicon at $|\boldsymbol{q}| = 1.0\,\mathrm{keV}$, near the valley axis [100], with $k_{\mathrm{c}} = 0.85\,(2\pi/a) = 1.94\,\mathrm{keV}$. On the axis the phonon carries $k_{\mathrm{c}} - q = 0.94\,\mathrm{keV}$, or $0.48\,\mathrm{\AA}^{-1} = 0.41\,\mathrm{X}$, where the computed dispersion of Sect. 5.2 gives $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*) = 13.98\,\mathrm{meV}$ and $\omega_{\mathrm{LA}}(\boldsymbol{Q}^*) = 25.00\,\mathrm{meV}$. The onset $E_{\mathrm{g}} + \omega_{\mathrm{TA}}(\boldsymbol{Q}^*) = 1.184\,\mathrm{eV}$ is reached by dark matter heavier than $0.341\,\mathrm{MeV}$, since $qv_{\mathrm{max}} - q^2/2m_\chi$ must exceed it with $qv_{\mathrm{max}} = 2.650\,\mathrm{eV}$. The zero-phonon edge at this $\boldsymbol{q}$ lies $0.60\,\mathrm{eV}$ above E_{g} on the computed bands of Fig. 3.7 (the parabola Eq. (3.58) gives $0.73\,\mathrm{eV}$), so the window extends to $0.30\,\mathrm{eV}$ above E_{g}, well beyond the energies considered here.

Fixed direction. With $\boldsymbol{q}$ on the axis, Eq. (13.17) gives $S^{\mathrm{PA}} = Cx^2$ for each transverse acoustic branch, $C = \Omega(m_{\mathrm{d}}^* m_{\mathrm{h}})^{3/2}|M_{\mathrm{TA}}|^2/4\pi^2$, with $m_{\mathrm{d}}^* = 0.322\,m_e$. This is the x^2 of Eq. (6.43): the two pockets give six free momenta.

The curvature of the onset. The transverse acoustic slope at $0.48\,\mathrm{\AA}^{-1}$ is $\omega'_{\mathrm{TA}} = 16.9\,\mathrm{meV\,\AA}$, and the first term of Eq. (13.18) is

$$\frac{\omega'_{\mathrm{TA}}\,k_{\mathrm{c}}q}{2(k_{\mathrm{c}} - q)} = 16.9\,\mathrm{meV\,\AA} \times \frac{0.983 \times 0.507\,\mathrm{\AA}^{-2}}{2 \times 0.477\,\mathrm{\AA}^{-1}} = 8.85\,\mathrm{meV}.$$

The anisotropy of the branch lowers this. Evaluated on a cone $\theta_q = 0.02\,\mathrm{rad}$ around the axis, the lower transverse acoustic branch has $\varkappa$ between 2.3 and 8.5 meV depending on the azimuth ϕ, the upper one between 18 and 24 meV, and the longitudinal acoustic one 16.8 meV in every direction. All are positive: the axis is a minimum.

The average. For an onset $\varkappa\theta_q^2$ above its minimum, the average over the sphere of directions, $\bar{S} = (4\pi)^{-1}\int \mathrm{d}\phi\,\sin\theta_q\,\mathrm{d}\theta_q\,S$, is dominated by small θ_q, where $\sin\theta_q \simeq \theta_q$. With $u = \varkappa\theta_q^2$,

$$\frac{1}{4\pi}\int_0^{2\pi}\mathrm{d}\phi\int\theta_q\,\mathrm{d}\theta_q\,C(x - \varkappa\theta_q^2)^2\Theta = \frac{C}{4\pi}\int_0^{2\pi}\mathrm{d}\phi\,\frac{1}{2\varkappa}\int_0^x \mathrm{d}u\,(x-u)^2 = \frac{C\,x^3}{12}\left\langle\frac{1}{\varkappa}\right\rangle_\phi,$$

with $\langle 1/\varkappa\rangle_\phi$ the average over the azimuth: $0.240\,\mathrm{meV^{-1}}$ and $0.047\,\mathrm{meV^{-1}}$ for the two transverse branches. The exponent has risen from 2 to 3, and $p_{\mathrm{on}} = 3$.

The forbidden branch. For the longitudinal acoustic phonon $|M|^2 = C\sin^2\theta_Q \simeq C\,[q/(k_{\mathrm{c}} - q)]^2\theta_q^2$, and the extra $\theta_q^2 = u/\varkappa$ in the integrand gives $\int_0^x u(x-u)^2\mathrm{d}u = x^4/12$, so

$$\bar{S}_{\mathrm{LA}} = \frac{C\,x^4}{48}\left(\frac{q}{k_{\mathrm{c}} - q}\right)^2\left\langle\frac{1}{\varkappa^2}\right\rangle_\phi,$$

and $p_{\mathrm{on}} = 4$. A numerical average over the whole sphere, with the computed onset in every direction, reproduces both asymptotes to 0.2 % at $x = 10^{-3}\,\mathrm{meV}$ (Fig. 13.4).

The cost of the average. Per valley, the ratio of the averaged to the on-axis transverse acoustic structure factor is $(x/24)(\langle 1/\varkappa\rangle_1 + \langle 1/\varkappa\rangle_2)$ for the two branches, 1.2×10^{-3} at $x = 0.1\,\mathrm{meV}$. Only a cone of solid angle $\pi x/\varkappa$ around the valley axis is open, a fraction $x/4\varkappa$ of the sphere.

At fixed $|\boldsymbol{q}|$ the halo enters the rate through $q^3\eta(v_{\min})F_{\mathrm{DM}}^2$ in Eq. (10.4), which is smooth across an onset as long as $v_{\min} < v_{\max}$. The spectrum contributed by each $|\boldsymbol{q}|$ therefore rises with the same exponent as $\bar{L}^{\mathrm{PA}}$. The integral over $|\boldsymbol{q}|$ smooths it further, because the onset moves with $|\boldsymbol{q}|$, and near the onset mass only a corner of the (q,ω) plane is reachable at all. How these factors combine into the dependence of the rate on the dark-matter mass at the edge of the silicon window, and how large the rate is there, is the subject of Sect. 13.9.

13.4 Screening of the phonon-assisted rate

The loss function of Sect. 13.3 uses the bare density vertex. At a momentum transfer of 1 keV and an energy of 1.2 eV, inside the phonon-assisted window, however, the dielectric function of silicon is about 6 in the random-phase approximation (Worked example 13.4). The potential that the dark matter exerts on a valence electron is therefore six times weaker than the bare one, and the bare loss function is too large by a factor of 30 to 50. Screening enters the amplitude of Sect. 13.2 at three places, among which Sect. 8.4 sorted the Coulomb lines: the dark-matter vertex at $(\boldsymbol{q},\omega)$, the electron–phonon vertex at $(\boldsymbol{Q},\omega_{\boldsymbol{Q}\nu})$, and the final pair at $(\boldsymbol{q}+\boldsymbol{Q},\omega\mp\omega_{\boldsymbol{Q}\nu})$. With all three in the amplitude, the loss function keeps one macroscopic factor $1/\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega)^2$ and nothing else outside it; we size each piece in silicon, and Appendix D.36 gives the steps at the random-phase level.

13.4.1 The screened density and phonon vertices

The electron–phonon vertex is already screened. Density-functional perturbation theory returns $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})$ as the matrix element of the self-consistent potential change ∂V_{KS}, which is the bare potential of the displaced ions dressed by every Coulomb chain at $\boldsymbol{Q}+\boldsymbol{G}$, the operator $(1-f_{\mathrm{Hxc}}R^0)^{-1}$ of Eq. (6.10). These are the vertex lines of Sect. 8.4.4. The calculation evaluates them at zero frequency, although they carry the phonon frequency; below the gap the error is of relative order $(\omega_{\boldsymbol{Q}\nu}/E_{\mathrm{g}})^2$, about 3×10^{-3} in silicon (Sect. 8.4.4). Dividing the computed g once more by $\varepsilon(\boldsymbol{Q})$ would count these lines twice, and a bare ionic potential in its place would overstate the long-range part of the coupling by about ε_∞.

The density vertex of Eq. (13.9) is bare. The dark matter applies a potential at $(\boldsymbol{q},\omega)$, and the electrons that it polarizes add their Hartree and exchange–correlation potentials at every $\boldsymbol{q}+\boldsymbol{G}$. An electron feels the total, the external potential times the column $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q},\omega)$ of the inverse dielectric matrix of a test electron (Sect. 8.7). These are the external lines, and in the amplitude they replace the density vertex by

$$\rho^{\mathrm{scr}}_{mn}(\boldsymbol{k},\boldsymbol{q}) = \sum_{\boldsymbol{G}} (\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q},\omega)\,\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G}), \tag{13.20}$$

the vertex of the total potential, as in the screened pair weight Eq. (8.68). The sum over $\boldsymbol{G}$ carries the local fields. Without local fields, where the dielectric matrix is diagonal, and without an exchange–correlation kernel, ρ^{scr} is $\rho/\varepsilon_{\boldsymbol{00}}(\boldsymbol{q},\omega)$, the head screening of Sect. 10.2.

Exactly one factor of $1/\varepsilon$ survives once the local fields are kept. To see this, remove from the Coulomb kernel its macroscopic term, $v_{\mathrm{C}}(q)$ at $\boldsymbol{G}=\boldsymbol{G}'=0$, and let $\bar{R}$ be the response without it (Sect. 4.4). The full response is then $\bar{R}$ plus chains of $\bar{R}$ joined by the

macroscopic line, and each element $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}$ factors into $(\varepsilon^{-1})_{\boldsymbol{00}} = 1/\varepsilon_{\mathrm{mac}}$ times a local-field column (Appendix D.36). In the random-phase approximation,

$$\begin{aligned}\rho^{\mathrm{scr}}_{mn}(\boldsymbol{k},\boldsymbol{q}) &= \frac{\rho^{\mathrm{LF}}_{mn}(\boldsymbol{k},\boldsymbol{q})}{\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega)},\\ \rho^{\mathrm{LF}}_{mn}(\boldsymbol{k},\boldsymbol{q}) &= \rho_{mn}(\boldsymbol{k},\boldsymbol{q}) + \sum_{\boldsymbol{G}\neq 0} v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\,\bar{R}_{\boldsymbol{G}0}(\boldsymbol{q},\omega)\,\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G}),\end{aligned} \tag{13.21}$$

exactly, with $\varepsilon_{\mathrm{mac}}$ the macroscopic dielectric function Eq. (4.44). An exchange–correlation kernel adds its own elements, its head included, to the local-field vertex ρ^{LF} and leaves the factorization intact. Physically, ρ^{LF} is the bare vertex corrected by the microscopic potentials that the macroscopic field induces inside each cell, and $1/\varepsilon_{\mathrm{mac}}$ is the reduction of the macroscopic field itself. Without local fields $\rho^{\mathrm{LF}} = \rho$ and $\varepsilon_{\mathrm{mac}} = \varepsilon_{\boldsymbol{00}}$. The rest of the section shows that every other screening effect is linear in ρ^{LF}, so that this one factor multiplies the whole amplitude: one external $1/\varepsilon(\boldsymbol{q},\omega)$ remains, and it is the macroscopic one.

Inside the window this factor is real. No zero-phonon pair exists at $(\boldsymbol{q},\omega)$, so the independent-particle response at that momentum and energy has no absorptive part, and $\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega)$ is real up to the phonon-assisted terms of order g^2 that this section computes. We write its zero-phonon value $\varepsilon_1^0(\boldsymbol{q},\omega)$, the real part of $\varepsilon_{\mathrm{mac}}$ at $n_g = 0$; its imaginary part ε_2^0 vanishes in the window. The response $R(\boldsymbol{q},\omega)$ is then Hermitian, so the retarded column $\delta_{\boldsymbol{G}0} + (f_{\mathrm{Hxc}}R)_{\boldsymbol{G}0}$ and the advanced one $\delta_{\boldsymbol{G}0} + (f_{\mathrm{Hxc}}R^\dagger)_{\boldsymbol{G}0}$ coincide, and the question of Sect. 8.7, which of the retarded and advanced weights labels a final pair, does not arise for the external lines. It arises at the pair.

13.4.2 The pair channel

Once the phonon has taken the momentum $\boldsymbol{Q}$, the dark matter and the phonon together have disturbed the electrons at a third wavevector, $\boldsymbol{q}+\boldsymbol{Q}$, and at a third frequency,

$$\omega_f = \omega \mp \omega_{\boldsymbol{Q}\nu},$$

the upper sign for emission. These are the crystal momentum and energy of the final pair itself, $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} = \omega_f$ on shell. The disturbance is a density of second order, bilinear in the two vertices, and its Hartree and exchange–correlation potential acts back on the electrons at $(\boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G},\omega_f)$. That potential can create the final pair directly. Summed over the chains that the medium inserts, it is the pair channel of Sect. 8.4.4: the Coulomb lines on the segment of the cycle between the two dark-matter vertices. The four-path amplitude of Sect. 13.2 leaves it out, because it treats the electrons as independent once the two vertices are given.

To find it we follow the density matrix ϱ of the Kohn–Sham electrons to second order in time-dependent mean-field theory (Appendix D.36). The dark matter acts through the screened vertex, $\mathsf{x} = \rho^{\mathrm{scr}}$ at frequency ω, and the phonon through $\mathsf{y} = g$ at frequency $\mp\omega_{\boldsymbol{Q}\nu}$, with the factor $N^{-1/2}(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})^{1/2}$ of Eq. (13.9) taken out. Each field alone produces a first-order density matrix, $\varrho^{\mathsf{x}}_{ab} = \mathsf{x}_{ab}(f_b - f_a)/(\omega - \varepsilon_a + \varepsilon_b + \mathrm{i}0^+)$ and its analog ϱ^{y}, with f_a the occupation of the one-electron state a. At second order, each field acts on the first-order response to the other, which gives the source

$$\mathfrak{s}^{(2)}_{ab} = \sum_m \big[\mathsf{x}_{am}\varrho^{\mathsf{y}}_{mb} - \varrho^{\mathsf{y}}_{am}\mathsf{x}_{mb} + \mathsf{y}_{am}\varrho^{\mathsf{x}}_{mb} - \varrho^{\mathsf{x}}_{am}\mathsf{y}_{mb}\big],$$

a matrix over every pair of states, filled or empty. Its element between the final conduction state and the final valence state is, on shell, the four-path amplitude itself. For an empty state m the source picks up the electron paths of Eq. (13.8); for a filled m the occupation

differences produce the hole paths with their minus sign, and the energy shell turns them into the Pauli-blocked electron paths. So $\mathfrak{s}^{(2)}_{cv} = M^{\pm}[\rho^{\mathrm{scr}}]$, Eq. (13.9) with the screened vertex and the unrestricted band sum: the band-sum theorem of Appendix D.35 follows from the density matrix without Fock space.

The source drives a second-order density of the Kohn–Sham electrons at $\boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G}$,

$$\delta n^{(2)}_{\boldsymbol{G}}(\boldsymbol{q}+\boldsymbol{Q}) = \frac{2}{\mathscr{V}} \sum_{\boldsymbol{k}} \sum_{ab} \frac{\rho_{ab}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G})^{*}\, \mathfrak{s}^{(2)}_{ab}(\boldsymbol{k})}{\omega_f + \varepsilon_{b\boldsymbol{k}} - \varepsilon_{a,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}} + \mathrm{i}0^{+}}, \tag{13.22}$$

the sum running over all pairs of states, including the rearrangements among filled and among empty bands. Its potential is screened by the same chains as any potential at that wavevector and frequency: the electrons respond to it with the bubble $R^0(\boldsymbol{q}+\boldsymbol{Q}, \omega_f)$, their response adds potential, and the series sums to $\tilde{\varepsilon}^{-1} f_{\mathrm{Hxc}}\, \delta n^{(2)}$. In the random-phase approximation $\tilde{\varepsilon}^{-1} f_{\mathrm{Hxc}} = \varepsilon^{-1} v_{\mathrm{C}}$ is the retarded screened interaction $W^{\mathrm{scr,R}}$ of Eq. (4.99). The equation of motion of the pair element $\varrho^{(2)}_{cv}$ is that of a first-order coherence driven at ω_f by $\mathfrak{s}^{(2)}$ plus this potential, and its numerator is the amplitude for creating the pair. Finally, every term is linear in x, so the factor $1/\varepsilon_{\mathrm{mac}}(\boldsymbol{q}, \omega)$ of Eq. (13.21) comes out in front.

Screened phonon-assisted amplitude

In the random-phase approximation and at order g in the amplitude, the screened matrix element of the density into the final state Eq. (13.6) is $N^{-1/2}(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})^{1/2}\, \tilde{M}^{\pm}_{cv\nu}(\boldsymbol{k}, \boldsymbol{q}, \boldsymbol{Q})/\varepsilon_{\mathrm{mac}}(\boldsymbol{q}, \omega)$. The amplitude without the macroscopic factor is

$$\begin{aligned} \tilde{M}^{\pm}_{cv\nu}(\boldsymbol{k}, \boldsymbol{q}, \boldsymbol{Q}) = M^{\pm}_{cv\nu}[\rho^{\mathrm{LF}}] + \sum_{\boldsymbol{G}\boldsymbol{G}'} \rho_{cv}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G}) \\ \times W^{\mathrm{scr,R}}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}+\boldsymbol{Q}, \omega_f)\, \delta n^{(2)}_{\boldsymbol{G}'}(\boldsymbol{q}+\boldsymbol{Q}), \end{aligned} \tag{13.23}$$

where $M^{\pm}[\rho^{\mathrm{LF}}]$ is Eq. (13.9) with the local-field vertex Eq. (13.21), $\delta n^{(2)}$ is Eq. (13.22) built with $\mathsf{x} = \rho^{\mathrm{LF}}$, and the retarded $W^{\mathrm{scr,R}}$ carries the index of the final pair first. With an exchange–correlation kernel, $W^{\mathrm{scr,R}}$ becomes $\tilde{\varepsilon}^{-1} f_{\mathrm{Hxc}}$, and an adiabatic local kernel adds the source of Eq. (13.25).

The four-path term is the amplitude of Sect. 13.2 with the local-field vertex. The pair-channel term is added to it, not multiplied: it does not rescale the four-path term by $1/\varepsilon(\boldsymbol{q}+\boldsymbol{Q}, \omega_f)$, and no $1/|\varepsilon|^4$ appears anywhere. The rate needs Eq. (13.23) only on shell, where $\mathfrak{s}^{(2)}_{cv}$ equals the four-path amplitude.

Fig. 13.5 shows the three kinds of screening as diagrams. With the external chains at $(\boldsymbol{q}, \omega)$, the static chains inside g and the pair chain at $(\boldsymbol{q}+\boldsymbol{Q}, \omega_f)$, every random-phase diagram with one phonon in the final state is counted once (Sect. 8.4.4). The classification is by the position of a line on the diagram, so it holds where the momentum labels coincide.

The pair line is complex. The final pair is on shell, so pairs exist at $(\boldsymbol{q}+\boldsymbol{Q}, \omega_f)$, and the response there has an absorptive part. With an absorptive interaction the share of the total rate that each final pair receives depends on whether the retarded or the advanced interaction is used, while the total does not. The pair that the cascade receives is the out-state of the evolution, and it takes the retarded $W^{\mathrm{scr,R}}$, as the external weight did in Sect. 8.7. For the external lines the choice did not arise, since they are not absorptive in the window.

The momentum of the pair channel is pinned. At the threshold the hole sits near the valence maximum at Γ and the electron near the bottom of a valley at $\boldsymbol{k}_{\mathrm{c}}$. The pair momentum $\boldsymbol{q}+\boldsymbol{Q}$ therefore equals $\boldsymbol{k}_{\mathrm{c}}$ up to a reciprocal-lattice vector, whatever $\boldsymbol{q}$ is, exactly at the onset and within a few tenths of an inverse Å at the top of the window. In silicon

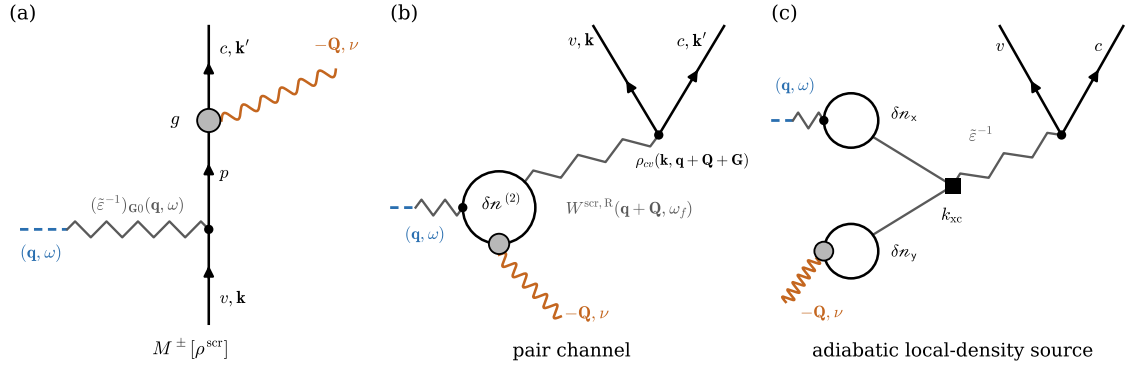


Figure 13.5: Screening of the phonon-assisted amplitude, schematic. Time runs upward; solid lines are electrons and holes, dashed blue lines the dark-matter potential, wavy lines the phonon, and zigzag lines Coulomb chains summed to all orders. (a) The four-path term $M^{\pm}[\rho^{\mathrm{scr}}]$: the dark-matter potential reaches the electron through the external chains at $(\boldsymbol{q}+\boldsymbol{G},\omega)$, the column $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q},\omega)$, and the phonon through the vertex g of density-functional perturbation theory (gray disk), whose chains at $(\boldsymbol{Q}+\boldsymbol{G},\approx 0)$ are inside it. One of the four paths is drawn. (b) The pair channel: the two vertices act on a loop of Kohn–Sham electrons and produce the second-order density $\delta n^{(2)}$ at $(\boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G}',\omega_f)$, whose potential, screened by $W^{\mathrm{scr,R}}(\boldsymbol{q}+\boldsymbol{Q},\omega_f)$, creates the final pair through $\rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G})$. (c) The extra source of an adiabatic local-density kernel: the first-order densities induced by the dark matter and by the phonon meet at one point through k_{xc}, the third derivative of the exchange–correlation energy, and the potential they produce is screened at the pair. In the random-phase approximation panel (c) is absent.

$|\boldsymbol{k}_\mathrm{c}| = 0.85\,(2\pi/a) = 1.94\,\mathrm{keV}$, and ω_f lies within a few tenths of an eV of the gap. One screened interaction, at the six valley momenta and a narrow band of energies, serves the whole window; it is the same one that screens phonon-assisted optical absorption, where $\boldsymbol{q} = 0$ and the pair momentum is $\boldsymbol{Q}$ (Sect. 13.5). Only $\delta n^{(2)}$ depends on $\boldsymbol{q}$.

Two limits can be solved, and they show what the pair channel does. In the first, the applied-potential limit, the source acts on every pair at $\boldsymbol{q}+\boldsymbol{Q}$ as an applied potential would, $\mathfrak{s}^{(2)}_{ab}$ proportional to $\rho_{ab}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{Q})(f_b - f_a)$. Then $\delta n^{(2)}$ is the bubble times that potential, and the two terms of Eq. (13.23) combine into the source screened as an external potential at the pair's own momentum and energy, $\tilde{M}^{\pm} = M^{\pm}[\rho^{\mathrm{LF}}]/\varepsilon(\boldsymbol{q}+\boldsymbol{Q}, \omega_f)$ without local fields (Appendix D.36). In this limit the pair channel reduces the rate by $1/\varepsilon_1(\boldsymbol{k}_\mathrm{c}, \omega_f)^2$, the screening of the final pair's own momentum.

In the second limit the phonon is a rigid translation of the crystal, $\boldsymbol{Q} = 0$, and the vertex is the commutator of the electron momentum with the Kohn–Sham Hamiltonian (Eq. (6.12)). In the frame that moves with the nuclei the dark-matter potential only acquires a phase factor, and the displaced density is screened at the energy of the final pair, $1/\varepsilon(\boldsymbol{q}, \omega_f)$. The four-path term alone gives $1/\varepsilon(\boldsymbol{q}, \omega)$, screened at the energy deposited, and the pair channel supplies exactly the difference, $1/\varepsilon(\boldsymbol{q}, \omega_f) - 1/\varepsilon(\boldsymbol{q}, \omega)$ (Appendix D.36). In the strict limit an acoustic $\omega_{\boldsymbol{Q}\nu}$ vanishes, $\omega_f = \omega$, and the pair term vanishes with it. Evaluated with the transverse acoustic energy of the window instead, the two energies change ε_1 of silicon by 1×10^{-3} and the rate by 2×10^{-3} (Worked example 13.4). The limit therefore tests the sign and the normalization of Eq. (13.23), not its size. It requires a zero-phonon pair at $(\boldsymbol{q}, \omega_f)$ and therefore lies outside the window. Inside it, the phonons that carry the missing momentum have wavevectors near the zone boundary, where the displacement is not a translation of anything, and the pair channel is neither of these limits.

13.4.3 The screened phonon-assisted loss function

The rate follows from the screened amplitude as it followed from the bare one in Sect. 13.3: the golden rule into the bare pairs, summed over $\boldsymbol{k}$, $\boldsymbol{Q}$, bands, branches and spin. That this sum is the whole absorptive part of the one-phonon response, with nothing counted twice, is the optical theorem of the mean-field evolution (Appendix D.36). The diagrams with both dark-matter vertices on one loop give $|M^{\pm}[\rho^{\mathrm{LF}}]|^2$ when cut at the pair. Those with a pair segment give the cross terms and the square of the pair channel, because the absorptive part of the screened interaction is $W^{\mathrm{scr,R}\dagger}$ times that of the bubble times $W^{\mathrm{scr,R}}$, an identity of the same algebra as Eq. (8.67). Together they make $|\tilde{M}^{\pm}|^2$ pair by pair. The amplitude therefore enters Eq. (13.16) with $M^{\pm} \to \tilde{M}^{\pm}/\varepsilon_{\mathrm{mac}}(\boldsymbol{q}, \omega)$.

The macroscopic factor is real up to order g^2 in the window. Write $\varepsilon_{\mathrm{mac}} = \varepsilon_1^0 + \delta\varepsilon_1 + \mathrm{i}\varepsilon_2$, where $\delta\varepsilon_1$ and ε_2 are the phonon-assisted parts, both of order g^2 because $\varepsilon_2^0 = 0$. Then $|\varepsilon_{\mathrm{mac}}|^{-2} = (\varepsilon_1^0)^{-2}[1 + O(g^2)]$, and since $|\tilde{M}^{\pm}|^2$ is already of order g^2, every correction is of order g^4, the next order of the expansion in Sect. 8.5.

Screened phonon-assisted loss function

Inside the off-shell window, where $\varepsilon_2^0(\boldsymbol{q}, \omega) = 0$,

$$
\begin{aligned}
L^{\mathrm{PA}}(\boldsymbol{q}, \omega) &= \frac{\varepsilon_2^{\mathrm{PA}}(\boldsymbol{q}, \omega)}{\varepsilon_1^0(\boldsymbol{q}, \omega)^2} + O(g^4), \\
\varepsilon_2^{\mathrm{PA}}(\boldsymbol{q}, \omega) &= \frac{8\pi^2\alpha}{\Omega\, q^2} \Big\langle\!\Big\langle \sum_{\nu c v \pm} w_{\pm} \left|\tilde{M}^{\pm}_{cv\nu}\right|^2 \delta\big(\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu} - \omega\big) \Big\rangle\!\Big\rangle,
\end{aligned}
\tag{13.24}
$$

with $\tilde{M}^{\pm}$ of Eq. (13.23), the weights $w_{\pm}$ of Eq. (13.16), and ε_1^0 the macroscopic zero-

phonon dielectric function with local fields. The rate is Eq. (10.4) with $\bar{L}$ replaced by the direction average $\bar{L}^{\mathrm{PA}}$.

From here on L^{PA} denotes this screened loss function; the function of Eq. (13.16) is its independent-particle value. The definition of $\varepsilon_2^{\mathrm{PA}}$ in Eq. (13.24) identifies it as the imaginary part, at order g^2, of the macroscopic dielectric function, since $L = \mathrm{Im}(-1/\varepsilon_{\mathrm{mac}}) = \varepsilon_2/|\varepsilon_{\mathrm{mac}}|^2$ must agree with the golden-rule sum term by term. Eq. (13.24) is therefore the familiar $L = \varepsilon_2/|\varepsilon|^2$ of Sect. 4.4, evaluated where the zero-phonon ε_2 vanishes and the phonon-assisted one is small. In that place the squared imaginary part in the denominator is of order g^4, so no condition $\varepsilon_2 \ll \varepsilon_1$ is needed: it holds by construction.

Eq. (13.24) has two simpler forms. One drops the pair channel and the local fields: then $\tilde{M}^{\pm} = M^{\pm}$, $\varepsilon_2^{\mathrm{PA}}$ is the independent-particle loss function of Sect. 13.3, $\varepsilon_1^0 = \varepsilon_{1,\mathbf{00}}$, and the screened loss function is the unscreened one divided by $\varepsilon_{1,\mathbf{00}}(\boldsymbol{q},\omega)^2$. This is the form that a calculation with the bare density vertex and a scalar dielectric function would use. Each of the two omissions is a definite set of diagrams, and neither is a choice of convention. The local fields change both ε_1^0 and the vertex, and the pair channel changes the amplitude by its own term in Eq. (13.23). The other is the optical limit: as $\boldsymbol{q} \to 0$, ε_1^0 becomes the optical $\varepsilon_1(\omega)$ and $\varepsilon_2^{\mathrm{PA}}$ the phonon-assisted absorption of Sect. 6.5, now with the pair channel inside it (Sect. 13.5). A measured absorption edge contains the screening of the amplitude, the local-field wings and the pair channel, at $\boldsymbol{q} = 0$; the external factor is not in ε_2, and a rate normalized to the edge calibrates it only through the value of ε_1^0 at $\boldsymbol{q} \to 0$. Because the pair momentum is pinned at $\boldsymbol{k}_{\mathrm{c}}$, the measured edge tests the same screened interaction that acts at finite $\boldsymbol{q}$. It does not test the second-order density $\delta n^{(2)}$ at finite $\boldsymbol{q}$, which is where the two differ.

Outside the window Eq. (13.24) fails twice. Where zero-phonon pairs exist at $(\boldsymbol{q},\omega)$, ε_2^0 is of order one and enters the denominator, and the external lines become absorptive, so that their retarded and advanced weights differ pair by pair. The one-phonon sideband is then no longer a separate quantity, and its treatment by the Dyson convention and the Fano term of Sect. 8.6 is the subject of Sect. 13.6.

The prescriptions of Sect. 13.3 carry over. A computed L^{PA} is never added to a measured loss function, and a phonon-assisted table in the normalization of the crystal form factor adds to $|f_{\mathrm{crys}}|^2$ only if both are screened at the same level. A rate that uses Eq. (13.24) without the pair channel sits at $\mathcal{X} = \mathrm{RPA}$ with local fields for the external lines and $\mathcal{X} = \mathrm{IP}$ for the pair, and it should be reported that way.

13.4.4 The adiabatic local-density source

The random-phase approximation is linear in the induced density: the Hartree potential of a density δn is $v_{\mathrm{C}}\,\delta n$, whatever its order. The exchange–correlation potential is not. The adiabatic local-density approximation of Sect. 4.6 takes v_{xc} as a function of the density at each point, and expanding it to second order,

$$v_{\mathrm{xc}}(n_{\mathrm{gs}}+\delta n) = v_{\mathrm{xc}}(n_{\mathrm{gs}}) + f_{\mathrm{xc}}\,\delta n + \tfrac{1}{2} k_{\mathrm{xc}}\,\delta n^2 + \cdots , \qquad k_{\mathrm{xc}}(\boldsymbol{r}) = \left.\frac{\mathrm{d}^3\big[n\,\varepsilon_{\mathrm{xc}}(n)\big]}{\mathrm{d}n^3}\right|_{n_{\mathrm{gs}}(\boldsymbol{r})} , \tag{13.25}$$

with f_{xc} the kernel of Eq. (4.71) and k_{xc} the next derivative. The density to first order is the sum of two pieces, δn_{x} induced by the dark matter at $(\boldsymbol{q},\omega)$ and δn_{y} induced by the phonon at $(\boldsymbol{Q}, \mp\omega_{\boldsymbol{Q}\nu})$, the density response that density-functional perturbation theory computes along with g. The square $\frac{1}{2}(\delta n_{\mathrm{x}} + \delta n_{\mathrm{y}})^2$ contains the cross term $\delta n_{\mathrm{x}}\delta n_{\mathrm{y}}$ with coefficient one. In real space it is a product, so its Fourier components lie at $\boldsymbol{q} + \boldsymbol{Q} + \boldsymbol{G}$, and it oscillates at

ω_f. It is a potential at the pair's momentum and energy that no Coulomb chain produces: the second-order source of the kernel itself. The electrons screen it as any applied potential, and it adds

$$\sum_{\boldsymbol{G}} \rho_{cv}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G}) \left[\tilde{\varepsilon}^{-1}(\boldsymbol{q}+\boldsymbol{Q}, \omega_f)\, (k_{\mathrm{xc}}\, \delta n_{\mathrm{x}} \delta n_{\mathrm{y}})\right]_{\boldsymbol{G}}$$

to $\tilde{M}^{\pm}$ (Appendix D.36), with δn_{x} taken without its macroscopic factor, as ρ^{LF} is. The term is the reason Eq. (13.23) is stated at the random-phase level. A calculation that takes g from density-functional perturbation theory in the local-density approximation, and the screening from the adiabatic kernel, has f_{xc} in every linear chain and is consistent only if it keeps this source too.

Its size follows from exchange alone. For an exchange energy density $n\varepsilon_{\mathrm{x}} \propto -n^{4/3}$, the third derivative is $k_{\mathrm{x}} = -\frac{2}{3} f_{\mathrm{x}}/n$, and the gas value $f_{\mathrm{x}} = -e^2/4k_{\mathrm{F}}^2$ of Sect. 4.6 compares with the Coulomb kernel at the pair momentum as $(|\boldsymbol{q}+\boldsymbol{Q}|/2k_{\mathrm{F}})^2$. Against the Hartree part of the pair channel, the new term is then

$$\frac{2}{3}\Big(\frac{|\boldsymbol{k}_{\mathrm{c}}|}{2k_{\mathrm{F}}}\Big)^2 \frac{|\delta n_{\mathrm{x}} \delta n_{\mathrm{y}}|}{n\, |\delta n^{(2)}|},$$

where the last ratio, a product of two first-order densities over the density times a second-order one, is of order one in the electron gas. The valence electrons of silicon, 32 in a cubic cell of 5.431 Å, have $n = 0.200\,\text{Å}^{-3}$ and $k_{\mathrm{F}} = 1.81\,\text{Å}^{-1}$, and with $|\boldsymbol{k}_{\mathrm{c}}| = 0.983\,\text{Å}^{-1}$ the prefactor is $\frac{2}{3} \times 0.074 = 0.05$. The adiabatic source is therefore an estimated twentieth of the pair channel, times a coefficient of order one that is not known. It matters only as much as the pair channel does.

13.4.5 Screening factors in silicon

The external factor is the largest and the best known, and Worked example 13.4 sizes it at $|\boldsymbol{q}| = 1\,\text{keV}$. It changes by less than 5 % across the window at fixed $|\boldsymbol{q}|$, since ε_1 rises slowly toward the direct gap. It does change with $|\boldsymbol{q}|$, however, since the valence electrons screen less at shorter wavelengths: without local fields the static value falls from 10.2 at $\boldsymbol{q} \to 0$ to 6.1–6.6 at 1 keV and to 3.2–3.7 near 2 keV in the same data. A rate integrated over $|\boldsymbol{q}|$ therefore needs ε_1^0 at every $|\boldsymbol{q}|$, not one number.

Two ratios measure what the simpler forms leave out. The local-field ratio $1 + \delta_{\mathrm{LF}}$ is L^{PA} with local fields but without the pair channel, divided by L^{PA} with the head alone. The pair-channel ratio $1 + \delta_{\mathrm{pair}}$ is L^{PA} with local fields and the pair channel, divided by L^{PA} with local fields alone. The full screened loss function is then the head-screened one times $(1 + \delta_{\mathrm{LF}})(1 + \delta_{\mathrm{pair}})$. The local-field ratio has two parts. The macroscopic factor changes from $1/\varepsilon^2_{1,\boldsymbol{00}}$ to $1/\varepsilon^2_{\mathrm{mac}}$, an increase of 25 to 31 % at 1 keV in the data of Worked example 13.4. The vertex changes from ρ to ρ^{LF}, whose wing terms $\boldsymbol{G} \neq 0$ the published dielectric functions do not contain. They need the column $\bar{R}_{\boldsymbol{G}0}(\boldsymbol{q}, \omega)$ and the vertices $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{G})$ at the same $(\boldsymbol{q}, \omega)$; a bound from the zero-phonon vertices alone, Eq. (E.92), costs about ten core-hours. The macroscopic factor and the column $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q}, \omega)$ are standard outputs of a random-phase calculation with local fields, specified for the phonon-assisted grid in Appendix E.14.

The size of the pair channel in silicon is open, and the two solvable limits of Eq. (13.23) bracket it only loosely. If the source acted like an applied potential at the pair momentum, the rate would fall by $1/\varepsilon_1(\boldsymbol{k}_{\mathrm{c}}, \omega_f)^2$. The same data have four mesh points between 1.87 and 2.03 keV. At 1.2 eV their ε_1 is 3.3 to 3.8 without local fields and 2.9 to 3.5 with them, so that factor is 0.07 to 0.12; the imaginary part there is about 10^{-3}, since the final pairs sit at their own edge. If the source were the displacement of a rigid crystal, the pair channel would change the rate by 2×10^{-3} at the phonon energy of the window. Neither limit is a bound on

the real case, and the sign of δ_{pair} is not derived: the source of a zone-boundary phonon is partly a displacement of the bond charge and partly a change of its shape, and the two limits stand for these.

Where the number comes from is known. One screened interaction at the six valley momenta, $W^{\mathrm{scr,R}}(\boldsymbol{k}_{\mathrm{c}}, \omega_f)$, serves every $\boldsymbol{q}$. The second-order density $\delta n^{(2)}$ is a mixed second derivative of the density, with respect to the dark-matter potential and to a phonon displacement, of the kind that density-functional perturbation theory computes for Raman tensors. The route that contains every class at once is a supercell displaced along a phonon pattern, with local fields in the supercell basis: then $\boldsymbol{Q}$ is a reciprocal vector of the supercell, and the pair channel is a local-field effect of the supercell. Three runs of that calculation give δ_{LF} and δ_{pair} separately (Appendix E.16).

The remaining classes are small. The static evaluation of g errs by 3×10^{-3}, and the ionic part of the dielectric function, which a silicon crystal has only through its quadrupoles, enters the external and pair chains below 10^{-4} (Sect. 8.4.4). The frequency at which the external lines are evaluated, ω rather than ω_f, matters at the level of 2×10^{-3} (Worked example 13.4). The adiabatic local-density source is a twentieth of the pair channel times a coefficient of order one.

Worked example 13.4: The external screening factor in silicon at 1 keV

Take $|\boldsymbol{q}| = 1.0\,\mathrm{keV}$ and $\omega = 1.20\,\mathrm{eV}$, just above the transverse acoustic onset at 1.184 eV and well inside the window, which extends to 1.47 eV at this momentum (Worked example 13.3). The random-phase dielectric function of silicon with and without local fields has been published on a mesh of $|\boldsymbol{q}|$ and ω [8, Sect. III B], with the data files of Ref. [9]. Each mesh momentum is the length of one vector $\boldsymbol{q} + \boldsymbol{G}$, and so one direction. Two mesh points lie within 0.1 keV of 1 keV, at 0.944 and 0.986 keV, and we quote the range over the two rather than a fit. Interpolating linearly in ω between the mesh energies 1.04 and 1.26 eV,

$$\varepsilon_{1,\mathbf{00}} = 6.31\text{–}6.93, \qquad \varepsilon_{\mathrm{mac}} = 5.52\text{–}6.20,$$

and $\varepsilon_2 = 0$ at both, as it must be below the zero-phonon edge.

The factor. The screened loss function is the independent-particle one divided by ε_1^2, so the factor is

$$\frac{1}{\varepsilon_{1,\mathbf{00}}^2} = 0.021\text{–}0.025, \qquad \frac{1}{\varepsilon_{\mathrm{mac}}^2} = 0.026\text{–}0.033.$$

The unscreened L^{PA} of Sect. 13.3 overstates the screened one by a factor of 40 to 48 with the head alone and 30 to 38 with the macroscopic function. The local fields of the macroscopic function raise the factor by $(\varepsilon_{1,\mathbf{00}}/\varepsilon_{\mathrm{mac}})^2 = 1.25$ and 1.31 at the two points.

Against the optical constant. The measured $\varepsilon_\infty = 11.7$ (Sect. 4.4) would give $1/\varepsilon_\infty^2 = 0.0073$, too small by a factor of 2.9 to 3.4 against the head and 3.6 to 4.5 against the macroscopic function. Screening at 1 keV is weaker than at $\boldsymbol{q} \to 0$, because the momentum transfer resolves the bonds that do the screening. The same data give 10.2 at $\boldsymbol{q} \to 0$ without local fields and 9.1 with them, 13 and 22 % below the measured value. With bands scissored to the measured gap, that deficit is mostly the electron–hole attraction, the vertex correction that the random-phase approximation leaves out, and the factor at 1 keV carries an error of the same kind, which no measurement at this momentum checks.

Across the window. Between the onset, 1.184 eV, and the window edge, 1.47 eV, the factor changes by 4.2 and 4.8 % at the two points with the head, and by 3.7 and 4.4 % with the macroscopic function.

The energy of the final pair. The logarithmic slope $\mathrm{d}\ln\varepsilon_1/\mathrm{d}\ln\omega$ is 0.07 to 0.08 here. Replacing ω by $\omega_f = \omega - \omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$, with $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*) = 14.0\,\mathrm{meV}$ at this momentum,

changes ε_1 by $0.08 \times 0.014/1.2 = 1 \times 10^{-3}$ and the factor by 2×10^{-3}, the size that the rigid-translation form of the pair channel takes at this phonon energy.

The pair channel is the open piece. Whether the pair-channel ratio is the same at finite $\boldsymbol{q}$ as at $\boldsymbol{q} = 0$ decides how much of it a measured absorption edge calibrates (Sect. 13.5). In both solvable limits it is: the applied-potential factor $1/\varepsilon_1(\boldsymbol{k}_c, \omega_f)^2$ is the same at every $\boldsymbol{q}$, since the pair momentum is pinned, and changes by less than 4×10^{-4} between the phonon energies of the window and of the optical edge, while the rigid-translation term stays below 2×10^{-3} at both momenta. That holds at exact pinning, at the onset; above it the pair momenta spread by about 0.2 keV around $\boldsymbol{k}_c$, where ε_1 changes by several tenths, and the factor cancels only as far as the distributions of final pairs at finite $\boldsymbol{q}$ and at $\boldsymbol{q} = 0$ match, an unsized residual. Between the limits it is not derived, because $\delta\boldsymbol{n}^{(2)}$ depends on $\boldsymbol{q}$ through the phonon at $\boldsymbol{k}_c - \boldsymbol{q}$. A rate normalized to the measured edge therefore needs, beyond the form factor of its vertex, a screening correction relative to that calibration: the external factor of its model dielectric function against the true ε_1^0, the local-field wings at finite $\boldsymbol{q}$ against those at $\boldsymbol{q} = 0$, and the pair ratio at finite $\boldsymbol{q}$ against its optical value. Sect. 13.9 sizes the first two for the silicon estimate, 0.3 to 2.6 together, and leaves the third as an unsized nuisance. Lines inside a loop, the attraction between the electron and the hole of the final pair, lie outside Eq. (13.23); they bind the pair into excitons below the edge and enhance it above, and they are the subject of Sect. 13.8.

13.5 The optical limit

A photon of 1.24 eV absorbed in silicon brings a momentum of 1.24 eV, five parts in 10^4 of the reciprocal-lattice scale $2\pi/a = 2.28$ keV, so its phonon supplies all of the 1.94 keV between hole and electron, while dark matter near the onset mass supplies 0.9 keV or more itself (Sect. 13.1). Both processes have the same final states, and the independent-particle loss function $L_{\mathrm{IP}}^{\mathrm{PA}}(\boldsymbol{q}, \omega)$ of Eq. (13.16) describes both, at different momentum transfers. Its limit $\boldsymbol{q} \to 0$ is the phonon-assisted absorption $\varepsilon_2^{\mathrm{PA}}(\omega)$ of Eq. (6.42). The measured absorption edge then serves twice. It tests a computed L^{PA} at one end of its range of $\boldsymbol{q}$, and it enters the rate of dark-photon absorption directly, once it has been transferred from the temperature of the measurement to that of a detector.

13.5.1 The small-momentum limit

The limit cannot be taken term by term in the amplitude $M_{cv\nu}^{\pm}(\boldsymbol{k}, \boldsymbol{q}, \boldsymbol{Q})$ of Eq. (13.9). Its intraband terms carry density vertices that tend to one and cancel at $\boldsymbol{q} = 0$ (Eq. (13.10)), so the leading term in q is a difference of large terms, each with a different expansion of the bands and vertices. The continuity equation avoids the expansion. For the full Hamiltonian of electrons and nuclei, Eq. (4.29) gives $[H, n_e(\boldsymbol{q})] = \boldsymbol{q} \cdot \boldsymbol{j}_e(\boldsymbol{q})$, exactly, and between any two eigenstates of H

$$(E_f - E_i)\, \langle f | n_e(\boldsymbol{q}) | i \rangle = \langle f | \boldsymbol{q} \cdot \boldsymbol{j}_e(\boldsymbol{q}) | i \rangle .$$

The second-order amplitude Eq. (13.7) is a matrix element of this kind. To first order in H_{ep} the eigenstates of H are $|\tilde{i}\rangle = |i\rangle + \sum_m |m\rangle\langle m|H_{\mathrm{ep}}|i\rangle/(E_i - E_m)$ and $\langle\tilde{f}| = \langle f| + \sum_m \langle f|H_{\mathrm{ep}}|m\rangle\langle m|/(E_f - E_m)$, with unchanged energies, because H_{ep} changes the number of phonons by one and has no diagonal elements. With $E_f = E_i + \omega$, the part of $\langle\tilde{f}|n_e(\boldsymbol{q})|\tilde{i}\rangle$ linear in H_{ep} is Eq. (13.7) term by term. The same construction applied to $\boldsymbol{q} \cdot \boldsymbol{j}_e(\boldsymbol{q})$ gives its

second-order amplitude, and the identity holds order by order on the energy shell:

$$\omega \, \langle f | n_e^{(2)}(\boldsymbol{q}) | i \rangle = \langle f | \big(\boldsymbol{q} \cdot \boldsymbol{j}_e(\boldsymbol{q}) \big)^{(2)} | i \rangle. \tag{13.26}$$

The intermediate states must be off the shell for the expansion to exist, and the off-shell window of Sect. 13.2 guarantees it.

The current needs no cancellation. At $\boldsymbol{q} = 0$ it is the total velocity, $\boldsymbol{j}_e(0) = \sum_j \boldsymbol{p}_j / m_e = \hat{\boldsymbol{V}}$ of Eq. (6.38), which conserves $\boldsymbol{k}$, and its second-order amplitude into the final state of Eq. (13.6) is the one that Sect. 6.5 computed for light, with the photon vertex $(e\mathcal{E}/2\omega)\, \hat{\boldsymbol{e}}_\gamma \cdot \hat{\boldsymbol{V}}$ replaced by $\boldsymbol{q} \cdot \hat{\boldsymbol{V}}$. The same four paths combine into one sum over every band. Write $M^{\pm}_{cv\nu}(\boldsymbol{k}, \boldsymbol{Q}; \hat{\boldsymbol{e}}_\gamma)$ for the optical amplitude Eq. (6.41) with its polarization shown. Then, with $\boldsymbol{j}_e(\boldsymbol{q}) = \hat{\boldsymbol{V}} + O(q)$,

Optical limit of the phonon-assisted amplitude

$$M^{\pm}_{cv\nu}(\boldsymbol{k}, \boldsymbol{q}, \boldsymbol{Q}) = \frac{q}{\omega} \, M^{\pm}_{cv\nu}(\boldsymbol{k}, \boldsymbol{Q}; \hat{\boldsymbol{q}}) + O(q^2), \tag{13.27}$$

on the energy shell inside the off-shell window, with the final electron at $\boldsymbol{k} + \boldsymbol{Q}$ in the limit.

At $\boldsymbol{q} = 0$ the right side vanishes, which is Eq. (13.10): the identity contains the cancellation of the intraband terms and needs no expansion of them. It also gives the amplitude its first power of q, as Eq. (3.43) does for the zero-phonon vertex, with the transition energy ω in the denominator. For a single pair, $\rho_{cv} \to \boldsymbol{q} \cdot \boldsymbol{v}_{cv} / \omega$ on the shell; for a pair with a phonon, the velocity vertex is replaced by the whole second-order optical amplitude.

Eq. (4.29) holds because the potential energy depends on positions alone. A nonlocal pseudopotential breaks this, and $[H, n_e(\boldsymbol{q})]$ acquires the commutator of the nonlocal part. The velocity then contains the term $\mathrm{i}[V_{\mathrm{NL}}, \boldsymbol{r}]$ of Eq. (3.40), which first-principles optical calculations include (Appendix E.7). The electron–phonon vertex of a nonlocal potential, the derivative of V_{NL} with respect to a displacement, contributes a commutator of the same kind. It is a contact term, in which one vertex creates the phonon and couples to the field at once, and Eq. (13.26) holds only if it is kept on the side of the current.

Inserted into the loss function Eq. (13.16), Eq. (13.27) multiplies the optical $|M^{\pm}|^2$ by q^2/ω^2, and the factor q^2 cancels the $1/q^2$ of the Coulomb line. The prefactor becomes $8\pi^2\alpha/(\Omega\omega^2) = g_{\mathrm{deg}} \pi e^2 / (\omega^2 \Omega)$ with $g_{\mathrm{deg}} = 2$ and $e^2 = 4\pi\alpha$. The weights $w_\pm = (1 - \mathrm{e}^{-\beta\omega})(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})$ become $n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2}$, since $\beta\omega > 40$ at room temperature for ω near the gap. The energy delta function loses $\boldsymbol{q}$ from the electron's crystal momentum. What remains is Eq. (6.42) with the polarization $\hat{\boldsymbol{e}}_\gamma = \hat{\boldsymbol{q}}$:

Optical limit of the phonon-assisted loss function

$$\lim_{\boldsymbol{q} \to 0} L^{\mathrm{PA}}_{\mathrm{IP}}(\boldsymbol{q}, \omega) = \varepsilon_2^{\mathrm{PA}}(\omega)\big|_{\hat{\boldsymbol{e}}_\gamma = \hat{\boldsymbol{q}}}, \tag{13.28}$$

for $T \ll E_{\mathrm{g}}$ and ω inside the off-shell window at $\boldsymbol{q} \to 0$, that is, below the direct gap. In a crystal with a center of inversion $L^{\mathrm{PA}}_{\mathrm{IP}}(\boldsymbol{q}, \omega) = L^{\mathrm{PA}}_{\mathrm{IP}}(-\boldsymbol{q}, \omega)$, and the corrections are of relative order q^2.

The optical result and its edge are derived in Sect. 6.5 and Appendix D.18. Near threshold, Eq. (13.28) reduces to the Hall–Bardeen–Blatt edge Eq. (6.44) [10], the $q \to 0$ form of the fixed-$\boldsymbol{q}$ law Eq. (13.17). As $\boldsymbol{q} \to 0$ the phonon wavevector $\boldsymbol{Q}^* = \boldsymbol{k}_{\mathrm{c}} - \boldsymbol{q}$ tends to $\boldsymbol{k}_{\mathrm{c}}$, the onset $E_{\mathrm{g}} + \omega_\nu(\boldsymbol{Q}^*)$ tends to the optical threshold $E_{\mathrm{g}} + \omega_\nu$, and $|M_\nu|^2/q^2$ to the optical $|M_\nu|^2/\omega^2$. The full formula is the one that first-principles calculations of the absorption of silicon evaluate [6, Eq. (1)].

13.5.2 Longitudinal and transverse absorption

Eq. (13.28) selects the polarization along $\boldsymbol{q}$, the longitudinal one. A photon is transverse, and the measured absorption is $\varepsilon_2^{\mathrm{PA}}$ for $\hat{\boldsymbol{e}}_\gamma \perp \boldsymbol{q}$. Both are projections of one tensor. The amplitude is linear in the polarization, $M^{\pm}(\boldsymbol{k},\boldsymbol{Q};\hat{\boldsymbol{e}}) = \hat{e}^{\alpha} M^{\pm\alpha}$, and Eq. (6.42) is $\hat{e}^{\alpha}\hat{e}^{\beta}\varepsilon_2^{\mathrm{PA},\alpha\beta}(\omega)$, with the tensor built from $M^{\pm\alpha}M^{\pm\beta *}$ under the same double zone average. That average runs over every $\boldsymbol{k}$ and $\boldsymbol{Q}$ of the zone, every band and every branch, and the thermal state of the phonons is invariant under the point group. The tensor therefore commutes with every operation of the cubic group, and in silicon it is a multiple of the unit matrix (Appendix C.3). The phonon-assisted part obeys Eq. (4.52) by itself:

$$\varepsilon_2^{\mathrm{PA,T}}(\omega) = \varepsilon_2^{\mathrm{PA,L}}(\omega) = \varepsilon_2^{\mathrm{PA}}(\omega) \qquad (\boldsymbol{q} \to 0).$$

The equality holds for the summed tensor, not transition by transition: a single valley and branch contributes an anisotropic tensor, and the six valleys together restore the cubic symmetry.

The measured absorption between the indirect and the direct gap is therefore the longitudinal $\varepsilon_2^{\mathrm{PA}}$ at $\boldsymbol{q} \to 0$ and at the temperature of the measurement. With the other electrons polarizing the medium, the longitudinal loss function is $\varepsilon_2^{\mathrm{PA}}/(\varepsilon_1^0)^2$ (Eq. (13.24)). Note that Eq. (13.28) holds for the screened amplitude $\tilde{M}^{\pm}$ as well, pair channel included, since the continuity argument uses only the energy shell; the measured $\varepsilon_2^{\mathrm{PA}}$ contains every screening effect of the amplitude at $\boldsymbol{q} \to 0$, though not the external factor. At the edge of silicon $\varepsilon_1^0 = n_r^2 = 12.8$ at 1.24 eV [11], and ε_2 stays below 10^{-2} within 0.2 eV of the gap, so this is the edge part of $\varepsilon_2/|\varepsilon|^2$, the loss function that sets the dark-photon rate Eq. (10.31). A dark photon of mass m_ϕ in the halo carries $|\boldsymbol{p}_\phi| \approx 10^{-3} m_\phi$, about 1 meV at the edge, and sits at $|\boldsymbol{p}_\phi|/(2\pi/a) \approx 5 \times 10^{-7}$. Its longitudinal and transverse modes both absorb with the measured $\varepsilon_2^{\mathrm{PA}}$, and the corrections of order q^2 are negligible. Its rate at the edge, with the phonon-assisted absorption of bosonic dark matter in general, is the subject of Sect. 13.8.

13.5.3 The measured edge as a test

Eq. (13.28) turns every measured absorption spectrum of silicon below 3.4 eV into a test of a computed L^{PA} at one end of its domain. The test runs forward: the calculation is done at the temperature of the measurement, with the occupations $n_{\boldsymbol{Q}\nu}(T)$ and the gap $E_{\mathrm{g}}(T)$ of that temperature, and compared with the data. First-principles calculations of the silicon edge at 249 K to 415 K agree with the measured spectra at every temperature, once the computed spectra are shifted rigidly by 0.15 eV to 0.23 eV to correct the computed gap [6]. Limit Check 13.1 shows how far the simplest form of the limit, the edge of Eq. (6.44) with constant matrix elements and parabolic bands, follows the measured absorption at room temperature.

Limit Check 13.1. Phonon-assisted absorption as the optical limit

$\varepsilon_2^{\mathrm{PA}}(\omega) = (8\pi^2\alpha/\Omega\omega^2)\langle\langle\sum(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})|M^{\pm}_{cv\nu}|^2\delta(\cdots)\rangle\rangle$, which is $n_r\alpha_{\mathrm{abs}}/\omega$ with α_{abs} of [6, Eq. (1)], with the edge $\alpha_{\mathrm{abs}} \simeq \sum_\nu A_\nu[n_\nu(\omega - E_{\mathrm{g}} + \omega_\nu)^2 + (n_\nu + 1)(\omega - E_{\mathrm{g}} - \omega_\nu)^2]$ [10].

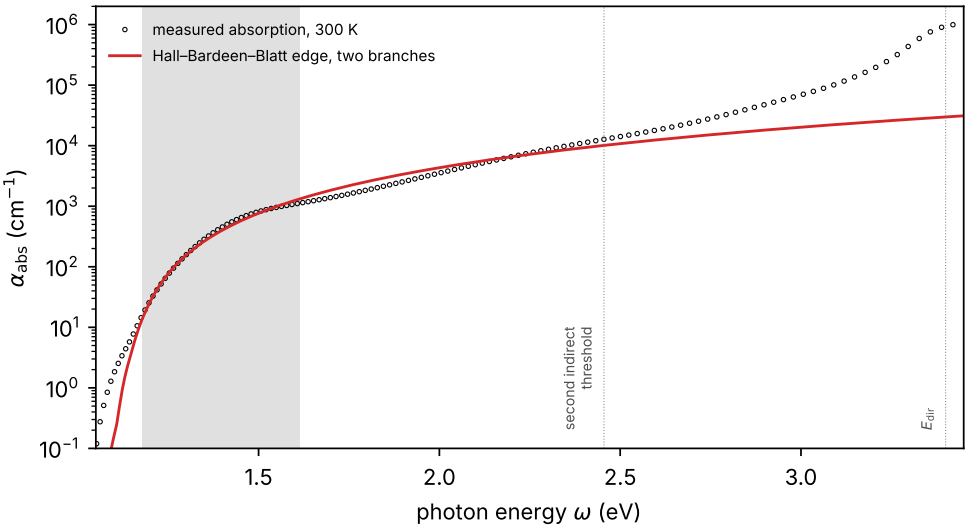


Figure 13.6: Absorption coefficient of silicon at 300 K against the photon energy. Circles: measured, float-zone silicon [11]. Red: the Hall–Bardeen–Blatt edge with free pairs, parabolic bands and constant matrix elements. It has the transverse acoustic and transverse optical branches at $\omega_{\mathrm{TA}} = 18.27\,\mathrm{meV}$ and $\omega_{\mathrm{TO}} = 57.73\,\mathrm{meV}$, the branch weights $A_{\mathrm{TA}} : A_{\mathrm{TO}} = 5.5 : 4.0$ of a fit to the measured edge [12, 13], the gap $E_{\mathrm{g}}(300\,\mathrm{K}) = 1.123\,\mathrm{eV}$ (Sect. 6.3), and one overall scale fitted to the data between 1.18 and 1.30 eV. Gray: the formula lies within 15 % of the data, from 55 to 490 meV above the gap. Dotted: the second indirect threshold, at 2.45 eV at 300 K [13], and the direct gap, $E_{\mathrm{dir}} = 3.4\,\mathrm{eV}$.

Closer to the gap the data exceed the edge formula, by a factor of 1.4 at 36 meV and of 2 at 17 meV, through the excitonic thresholds E_x^b below the free-pair ones and the thermal broadening of the edge; above about 0.5 eV the matrix elements and bands vary across the zone, and above 2.5 eV new thresholds open.

Over half an electronvolt the free-pair power two and the two branches with the measured phonon energies hold with a single scale. The constant matrix element is an approximation of the edge formula, not of Eq. (13.28): the full $\varepsilon_2^{\mathrm{PA}}$ keeps $M^{\pm}$ at every $\boldsymbol{k}$ and $\boldsymbol{Q}$, and only a calculation of it can follow the data above 0.5 eV. Near the gap the excess is the electron–hole attraction, which the independent-particle L^{PA} leaves out. The bound pair lies $E_x^b = 14.7\,\mathrm{meV}$ below the gap at the crystal momentum of the valley, every threshold has an excitonic companion that much lower (Sect. 6.5), and a calculation that is to match the edge within a few tens of meV must include it (Sect. 13.8).

13.5.4 Temperature transfer of absorption data

The measured edge is also an input. A dark photon absorbed between the indirect and the direct gap of a detector at 10 mK sees the phonon-assisted ε_2 of cold silicon. Continuous spectra of the edge are measured at 300 K or at 77 K; the one measurement below 1 K covers a few discrete photon energies from 1.21 eV up [13]. Projections for silicon detectors have used the 77 K data near the gap [14]. Eq. (6.42) contains the temperature in two places, the occupations and the gap, and between them they change the edge by large factors (Sect. 10.7.2).

Take one branch ν and write $\alpha_{\mathrm{abs},\nu}(\omega; T)$ for its part of the absorption. It consists of an

emission component with weight $n_\nu + 1$ and an absorption component with weight n_ν, both functions of the pair energy above the gap, $\omega - E_g(T) \mp \omega_\nu$. Assume that the spectrum of the pairs, the matrix elements and the phonon energy do not change with temperature, and that the gap moves rigidly by $\Delta E_g(T) = E_g(T) - E_g(0) < 0$. At $T \to 0$ only emission survives, and that spectrum is $\alpha^0_{\mathrm{abs},\nu}(\omega)$. At temperature T,

$$\alpha_{\mathrm{abs},\nu}(\omega;T) = (n_\nu + 1)\,\alpha^0_{\mathrm{abs},\nu}(\omega - \Delta E_g) + n_\nu\,\alpha^0_{\mathrm{abs},\nu}(\omega - \Delta E_g + 2\omega_\nu), \tag{13.29}$$

which is Eq. (6.44) with a general spectrum in place of the square. The emission term is the cold spectrum moved down by the gap shift; the absorption term is the same spectrum moved down by a further $2\omega_\nu$.

Eq. (13.29) can be solved for the cold spectrum. Evaluate it at $\omega + \Delta E_g$ and divide by $n_\nu + 1$:

$$\alpha^0_{\mathrm{abs},\nu}(\omega) = \frac{\alpha_{\mathrm{abs},\nu}(\omega + \Delta E_g;T)}{n_\nu + 1} - \frac{n_\nu}{n_\nu + 1}\,\alpha^0_{\mathrm{abs},\nu}(\omega + 2\omega_\nu).$$

The unknown reappears on the right at an energy higher by $2\omega_\nu$, with the factor $n_\nu/(n_\nu+1) = \mathrm{e}^{-\omega_\nu/T}$ of detailed balance (Eq. (6.45)). Substituting the equation into itself and using $1/(n_\nu + 1) = 1 - \mathrm{e}^{-\omega_\nu/T}$,

Inverse temperature transfer

$$\alpha^0_{\mathrm{abs},\nu}(\omega) = (1 - \mathrm{e}^{-\omega_\nu/T}) \sum_{k=0}^{\infty} \left(-\mathrm{e}^{-\omega_\nu/T}\right)^k \alpha_{\mathrm{abs},\nu}(\omega + \Delta E_g(T) + 2k\,\omega_\nu;\, T), \tag{13.30}$$

exact for the spectrum of one branch measured at the temperature T, under the assumptions of Eq. (13.29). The cold absorption is $\sum_\nu \alpha^0_{\mathrm{abs},\nu}$, and the dark-photon rate Eq. (10.31) scales with it.

The series converges geometrically at any temperature, with ratio $\mathrm{e}^{-\omega_\nu/T}$: 0.064 for the transverse acoustic branch at 77 K, 1.7×10^{-4} for the transverse optical one, and 0.49 and 0.11 at 300 K. It needs the data up to a few times $2\omega_\nu$ above ω, and it is the unique solution that vanishes below the cold threshold. Applied to the two-branch edge formula at 77 K or 300 K, sixty terms return its zero-temperature form to 10^{-15}. Each of the assumptions is a property of the spectrum, not of the inversion, and we size them below.

The inversion needs the spectrum of each branch separately, and a measured spectrum is their sum. One spectrum cannot fix two unknown functions. The triangular structure of Eq. (13.29) does allow a solution that starts at the lowest threshold and works upward, but the term with the highest argument at each energy is the absorption replica of the transverse optical branch. Each step of that solution divides by its occupation, $n_{\mathrm{TO}} = 1.7 \times 10^{-4}$ at 77 K, and multiplies every error in the data by $\mathrm{e}^{\omega_{\mathrm{TO}}/T} = 6 \times 10^3$. At 300 K the factor is 9.3. Without spectra resolved by branch, the transfer is therefore ill-posed: the data at one temperature leave the decomposition, and with it the cold spectrum, undetermined. The branch weights must come from elsewhere, from low-temperature spectra in which each threshold is resolved, as in the edge spectroscopy of Sect. 6.5, or from a calculation of $\varepsilon_2^{\mathrm{PA}}$ branch by branch (Appendix E.14). The cold spectrum inherits their error. From 77 K data the inheritance is mild, because only the transverse acoustic branch is occupied: with its weight changed by 25 % either way, the transferred spectrum changes by less than 1 % at 15 and 50 meV above the cold threshold. From 300 K data it changes by −29 and +18 percent at 15 meV.

Fig. 13.7 shows the transfer for silicon. The warm edge starts 57.7 meV below the 300 K gap of 1.123 eV, with the absorption replica of the transverse optical branch, and the cold

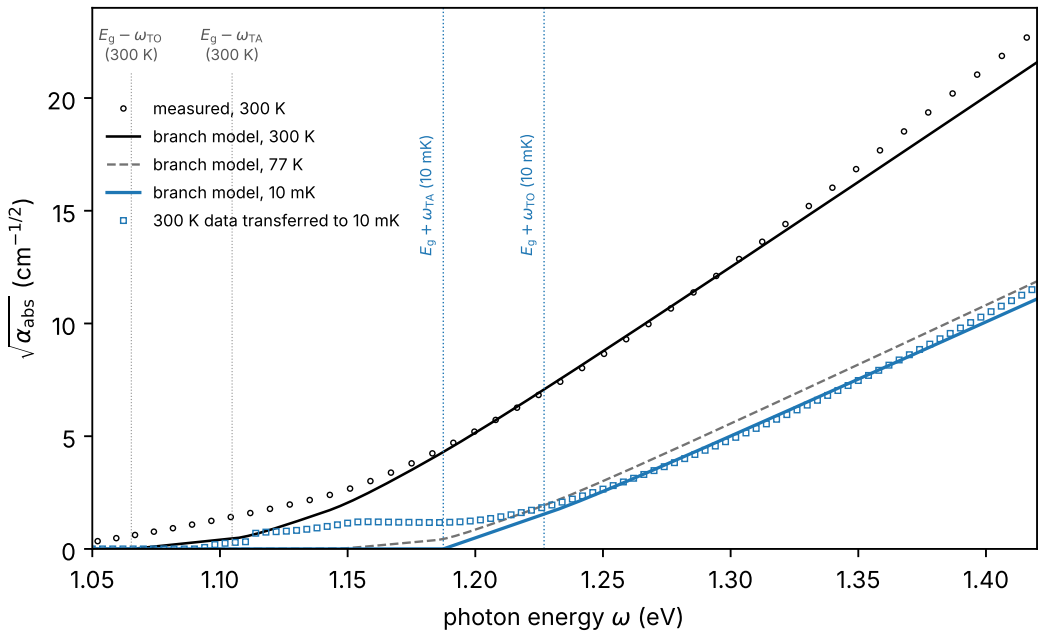


Figure 13.7: The absorption edge of silicon, as $\sqrt{\alpha_{\mathrm{abs}}}$ against the photon energy, at 300 K and transferred to 10 mK. Circles: measured at 300 K [11]. Lines: a two-branch model of the edge. It has free pairs with the Hall–Bardeen–Blatt form of Eq. (6.44), $\omega_{\mathrm{TA}} = 18.27$ meV and $\omega_{\mathrm{TO}} = 57.73$ meV, branch weights 5.5 : 4.0 [12, 13]. The gap is the measured $E_{\mathrm{g}}(T)$ of Sect. 6.3, 1.1692, 1.1652 and 1.1230 eV at 0, 77 and 300 K, with one scale fitted to the 300 K data between 1.18 and 1.30 eV. It is drawn at 300 K (black), 77 K (gray, dashed) and 10 mK (blue). Squares: the 300 K data transferred to 10 mK with Eq. (13.30), each energy split between the branches in the proportion of the model. Dotted: the thresholds $E_{\mathrm{g}} - \omega_\nu$ of phonon absorption at 300 K and the emission thresholds $E_{\mathrm{g}} + \omega_\nu$ of cold silicon. Excitonic thresholds, 14.7 meV below the free-pair ones, are not modeled.

one at $E_g(0) + \omega_{TA} = 1.1875\,\text{eV}$, 0.12 eV higher; at 1.236 eV the measured 300 K absorption, $58\,\text{cm}^{-1}$, is 16 times the cold one. Transferred, the data follow the cold model above 1.25 eV to within 10 %, the accuracy with which the model follows the warm data. Below the cold threshold they keep about $1.4\,\text{cm}^{-1}$, where cold silicon absorbs nothing outside the excitonic thresholds. That residue is the part of the 300 K data that is not of the form Eq. (13.29) assumes, the excitonic companions and the thermal broadening of the warm edge, and the inversion carries it to the cold side unchanged in size. Near the cold threshold the transfer of room-temperature data is no better than this residue.

The assumption of a temperature-independent spectrum fails at a level that measurements show. Stanford and co-workers measured the absorption of silicon at 0.5, 5, 77 and 295 K and fitted the two-branch form to it with an effective gap of 1.134 eV. The fit needs a branch strength that falls with temperature as $\mathrm{e}^{-c_1 T}$ with $c_1 = 1.7 \times 10^{-3}\,\text{K}^{-1}$ [13, Eqs. (6) and (7)]: a factor 0.61 at 295 K and 0.88 at 77 K, relative to the lowest temperature, within their own law for the gap, which differs slightly from the one used here. The occupations and a rigid gap do not carry all of the temperature dependence. The bands are renormalized by the lattice vibrations, not only shifted (Sect. 6.3), and the vertices change with them (Sect. 13.7). Transferred with Eq. (13.30), 295 K data therefore underestimate the cold spectrum by about 40 % on this evidence, and 77 K data by about 12 %, with the opposite sign to the errors of the untransferred data.

Untransferred 77 K data

Used without transfer, the 77 K absorption of silicon overestimates the millikelvin dark-photon rate by the factors

$$2.3\text{–}2.6, \quad 1.7\text{–}1.9, \quad 1.3\text{–}1.5, \quad 1.1\text{–}1.3 \qquad \text{at} \qquad x = 15,\ 25,\ 50,\ 100\,\text{meV},$$

where x is the energy above the cold threshold $E_g(0)+\omega_{TA}$, and by more than a factor of two for x below 18 to 23 meV. The upper values hold for the transverse acoustic branch alone and for the two-branch edge to two digits; the lower ones include the measured fall of the branch strength between the lowest temperature and 77 K. Untransferred 300 K data overestimate the two-branch edge by 38, 16 and 7 at $x = 25$, 50 and 100 meV.

The two causes add. The 77 K gap lies 3.97 meV below the cold one, so that at the same photon energy the warm emission edge has run $x + 3.97\,\text{meV}$. And the transverse acoustic phonon is still occupied, $n_{TA} = 0.068$, which adds stimulated emission and an absorption replica that starts $2\omega_{TA}$ lower. The gap shift dominates close to the threshold, where $(1 + |\Delta E_g|/x)^2$ is large, and the occupation farther away. The factors belong to the free-pair edge, whose power two is the steepest at the threshold. An excitonic edge rises more slowly and its gap-shift factor is smaller. The lower values of the box follow from the branch strength at 77 K, lower than the cold one by the factor 0.88 above, which moves the factor of two from $x = 23$ to 18 meV. These are the corrections that the silicon dark-photon rate of Sect. 10.7.2 needs near its edge, and they apply to every absorption rate of Chap. 10 that uses measured σ_1 below the direct gap.

Worked example 13.5: The transverse acoustic correction at 50 meV

Take the transverse acoustic component of the silicon edge at $x = 50\,\text{meV}$ above the cold threshold, $\omega = E_g(0)+\omega_{TA}+x = 1.2375\,\text{eV}$, with $E_g(0) = 1.1692\,\text{eV}$ and $\omega_{TA} = 18.27\,\text{meV}$, and its 77 K spectrum, with $E_g(77\,\text{K}) = 1.1652\,\text{eV}$ ($\Delta E_g = -3.97\,\text{meV}$) and $n_{TA} = 0.0681$. In the edge form the cold component is $A\,x^2$.

Forward. At 77 K the emission component has run $\omega - E_{\mathrm{g}}(77\,\mathrm{K}) - \omega_{\mathrm{TA}} = 53.97\,\mathrm{meV}$, and the absorption component $\omega - E_{\mathrm{g}}(77\,\mathrm{K}) + \omega_{\mathrm{TA}} = 90.50\,\mathrm{meV}$. Relative to the cold value $A\,(50\,\mathrm{meV})^2$,

$$\underbrace{(n_{\mathrm{TA}}+1)}_{1.0681}\underbrace{\Big(\frac{53.97}{50}\Big)^2}_{1.1651} + \underbrace{n_{\mathrm{TA}}}_{0.0681}\underbrace{\Big(\frac{90.50}{50}\Big)^2}_{3.276} = 1.2443 + 0.2230 = 1.467.$$

The gap shift alone gives 17 %, stimulated emission multiplies it by 1.068, and the absorption replica adds 0.22: the 77 K spectrum is 1.47 times the cold one at this energy. *Inverse.* Eq. (13.30) with $\mathrm{e}^{-\omega_{\mathrm{TA}}/T} = 0.0637$ reads the 77 K spectrum at $\omega + \Delta E_{\mathrm{g}} + 2k\omega_{\mathrm{TA}} =$ 1.2335, 1.2700, 1.3066, 1.3431, 1.3796 eV. In units of the cold value, the terms are

$$+1.19087, \quad -0.21547, \quad +0.02724, \quad -0.00289, \quad +0.00028,$$

and the partial sums 1.19087, 0.97540, 1.00264, 0.99975, 1.00002. The first term alone, the gap-shifted spectrum divided by $n_{\mathrm{TA}} + 1$, is 19 % high, because it still contains the absorption replica; the second removes it, and each further term corrects the previous one by a factor $\mathrm{e}^{-\omega_{\mathrm{TA}}/T}$ and a ratio of squares. Five terms recover the cold value to 2×10^{-5}.

13.5.5 Finite-momentum corrections

A dark-matter particle that scatters never reaches the optical limit. To deposit ω it must transfer at least $q = \omega/v_{\max}$, 0.45 keV at the silicon edge with $v_{\max} = 2.65 \times 10^{-3}$, and in the phonon-assisted window it transfers 0.9 keV to 1.3 keV (Sects. 13.1 and 13.3), 0.4 to 0.6 of $2\pi/a$. At such $\boldsymbol{q}$ the corrections of Eq. (13.28) are not small, for three reasons outside the reach of optical data. The intermediate states of path A sit at $\boldsymbol{k} + \boldsymbol{q}$, and their energy moves by an amount of order $q^2/2m_e$, 0.98 eV at 1 keV, against a distance from the energy shell of $E_{\mathrm{dir}} - \omega \approx 2.2\,\mathrm{eV}$ at the edge. The density vertices $\rho_{pv}(\boldsymbol{k}, \boldsymbol{q})$ depart from their linear form Eq. (3.43) on the scale of $2\pi/a = 2.28\,\mathrm{keV}$, and beyond that a reciprocal-lattice vector inside the vertex reaches valleys that no photon reaches, as the class $(2\pi/a)(2,0,0)$ of Sect. 13.3 does, with its own selection rules. And the phonon itself is a different one: at $\boldsymbol{q} \to 0$ it carries $\boldsymbol{k}_{\mathrm{c}}$, where the transverse acoustic branch is at 18.27 meV, while at $|\boldsymbol{q}| = 1.0\,\mathrm{keV}$ along the valley axis it carries $\boldsymbol{k}_{\mathrm{c}} - \boldsymbol{q}$, at 0.41 X, and the computed branch is at 13.98 meV (Worked example 13.3). The optical edge measures the electron–phonon vertices only at $\boldsymbol{Q} \approx \boldsymbol{k}_{\mathrm{c}}$.

The optical limit therefore tests the parts of a calculation that do not depend on $\boldsymbol{q}$: the bands at the extrema, the gap, the electron–phonon vertices near $\boldsymbol{k}_{\mathrm{c}}$, the unrestricted sum over intermediate bands with its intraband terms, and the normalization of Eq. (13.16). It does not test the finite-$\boldsymbol{q}$ density vertex, the Umklapp classes, the local fields or the vertices at $\boldsymbol{k}_{\mathrm{c}} - \boldsymbol{q}$, and these carry the dark-matter scattering rate. A computed $L^{\mathrm{PA}}(\boldsymbol{q}, \omega)$ that reproduces the measured edge at the temperature of the measurement has passed a necessary check, and the calculation specified for silicon, germanium and GaAs in Appendix E.14 includes it. The finite-$\boldsymbol{q}$ structure has to be controlled by the sum rules and convergence tests of Appendix E.5. Every result so far has also assumed the off-shell window, whose boundary Sect. 13.6 fixes.

13.6 The off-shell window

At $|\boldsymbol{q}| = 1.0\,\mathrm{keV}$ along the valley axis the cheapest zero-phonon pair of silicon costs 1.77 eV, and the phonon-assisted structure factor of Sect. 13.3 was written for energies below it. Two

kinds of denominators in the amplitude Eq. (13.9) can become small. Those of path A with an empty band and of path C with a filled band are the distances of virtual zero-phonon pairs from the energy shell, and they vanish as ω reaches $E_{\min}(\boldsymbol{q})$. The intraband terms, path A with $p = c$ and path B with $p = v$, have denominators of order $|\boldsymbol{Q}|$ and vanish as the phonon becomes a soft acoustic one, while the interband vertices of that phonon grow as $|\boldsymbol{Q}|^{-1/2}$. The soft acoustic phonon produces no divergence: translation invariance fixes its amplitude, and inside the window it is never on the energy shell. The vanishing zero-phonon detunings, however, set the boundary of the region in which the phonon-assisted rate exists as a quantity of its own; we derive that boundary, size the margin δ_{w} of Sects. 8.6 and 13.3 at 0.3 eV, and show how the rate is defined beyond it.

13.6.1 The soft acoustic limit

A long acoustic wave moves every atom of a region by nearly the same amount, and a uniform displacement of the whole crystal is no excitation at all. The electron–phonon vertex reflects this only in part. By Eq. (6.12) its intraband elements vanish as $\boldsymbol{Q} \to 0$, but its interband elements tend to $\mathrm{i}\ell_{\boldsymbol{Q}\nu}(\varepsilon_m - \varepsilon_n)\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{p}_{mn}$, the rigid translation of the electronic states, and with $\ell_{\boldsymbol{Q}\nu} = (2M_{\mathrm{cell}}c_\nu|\boldsymbol{Q}|)^{-1/2}$ they grow without bound. In Eq. (13.9) these large interband terms meet intraband terms of the form $0/0$, a vertex of order $\ell|\boldsymbol{Q}|$ over a denominator of order $|\boldsymbol{Q}|$. Neither kind can be estimated alone. Translation invariance, which Sect. 8.3 wrote as a Ward identity for matrix elements, decides their sum.

The identity we need is Eq. (D.206). Change the vertex by $\delta g_{mn} = (\varepsilon_{m,\boldsymbol{\kappa}+\boldsymbol{Q}} - \varepsilon_{n\boldsymbol{\kappa}} \pm \omega_{\boldsymbol{Q}\nu})\mathrm{t}_{mn}$, the matrix element of a commutator with the unperturbed Hamiltonian, for any one-electron operator $\hat{\mathrm{t}}_{\boldsymbol{Q}}$. On the energy shell every term of Eq. (13.9) then has numerator equal to $\pm$ its denominator, band by band, and the sum over all bands collapses to $\langle c, \boldsymbol{k}'|[\hat{\mathrm{t}}_{\boldsymbol{Q}}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}]|v, \boldsymbol{k}\rangle$ (Appendix D.37). The amplitude is unchanged if this contact term is added back. The useful choice of $\hat{\mathrm{t}}_{\boldsymbol{Q}}$ carries each electron along with the local displacement of the wave, Eq. (D.207). With it the vertex loses its rigid translation, keeps a strain and a velocity coupling, and the contact term becomes a recoil. The vertex that remains, the comoving vertex g^{cm} of Eq. (D.209), vanishes at $\boldsymbol{Q} = 0$ for every pair of bands, by Eq. (6.12). Its first order in $\boldsymbol{Q}$ is the coupling to a uniform strain, the deformation-potential tensor D^n of Eq. (6.13), plus a coupling to the velocity of the moving lattice. And the contact term is $\mathrm{i}\ell_{\boldsymbol{Q}\nu}(\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{q})\,\rho_{cv}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{Q})$, because the density at wavevector $\boldsymbol{q}$ carries momentum $\boldsymbol{q}$. With these, the amplitude has a finite limit.

Soft acoustic limit of the phonon-assisted amplitude

For an acoustic branch ν with $\omega_{\boldsymbol{Q}\nu} = c_\nu|\boldsymbol{Q}|$ and polarization $\hat{\boldsymbol{e}}_\nu$, nondegenerate bands c and v, and $\hat{\boldsymbol{Q}} = \boldsymbol{Q}/|\boldsymbol{Q}|$, the amplitude Eq. (13.9) tends on the energy shell to

$$M^{\pm}_{cv\nu}(\boldsymbol{k}, \boldsymbol{q}, \boldsymbol{Q}) \to \mathrm{i}\,\ell_{\boldsymbol{Q}\nu}\,\rho_{cv}(\boldsymbol{k}, \boldsymbol{q}) \Bigg[\underbrace{\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{q}}_{\text{recoil}} + \underbrace{\frac{\hat{\boldsymbol{Q}} \cdot D^c \cdot \hat{\boldsymbol{e}}_\nu \mp c_\nu\, \hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{p}_{cc}}{\hat{\boldsymbol{Q}} \cdot \boldsymbol{v}^{\mathrm{g}}_c \pm c_\nu} - \frac{\hat{\boldsymbol{Q}} \cdot D^v \cdot \hat{\boldsymbol{e}}_\nu \mp c_\nu\, \hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{p}_{vv}}{\hat{\boldsymbol{Q}} \cdot \boldsymbol{v}^{\mathrm{g}}_v \pm c_\nu}}_{\text{deformation and velocity}} \Bigg], \tag{13.31}$$

with D^c, $\boldsymbol{v}^{\mathrm{g}}_c$, $\boldsymbol{p}_{cc}$ taken at $\boldsymbol{k}+\boldsymbol{q}$ and D^v, $\boldsymbol{v}^{\mathrm{g}}_v$, $\boldsymbol{p}_{vv}$ at $\boldsymbol{k}$, and the upper sign for emission. The leading term is of order $|\boldsymbol{Q}|^{-1/2}$, and the relative correction is of order $|\boldsymbol{Q}|$, for $|\boldsymbol{Q}|$ small compared with the scale over which the group velocities change; at a band extremum of mass m^* this is $|\boldsymbol{Q}| \ll 2m^*c_\nu$.

The limit is derived in Appendix D.37. Each term has a picture. The recoil term is the first order in the displacement of $e^{i\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{u})}$: the dark matter kicks an electron that rides on the lattice, and the lattice under it recoils. The deformation terms are the soft phonon emitted by the final electron after the kick, or by the hole before it, through the strain it produces; their denominators are the detunings of the zero-phonon pair at $\boldsymbol{q}$, which shrink with $|\boldsymbol{Q}|$. The velocity terms are smaller than the deformation terms by $c_\nu|\boldsymbol{p}_{nn}|/|D^n|$, a few times 10^{-3} per band in Worked example 13.6 (and 10^{-5} there once the electron and hole terms cancel), but they are what turns the laboratory-frame intraband vertex into the comoving one. That vertex is $i\ell|\boldsymbol{Q}|\,\hat{\boldsymbol{Q}}\cdot(D^n+\boldsymbol{v}^{\mathrm{g}}_n\otimes\boldsymbol{p}_{nn})\cdot\hat{\boldsymbol{e}}_\nu$ (Eq. (D.211)). It reduces to Eq. (6.13) at a band extremum, and away from one it carries a Galilean term that a frozen-strain calculation does not produce. Projected on $\hat{\boldsymbol{Q}}$ and $\hat{\boldsymbol{e}}_\nu$, divided by its detuning it splits as $(D+v^{\mathrm{g}}p)/(v^{\mathrm{g}}\pm c_\nu)=(D\mp c_\nu p)/(v^{\mathrm{g}}\pm c_\nu)+p$, and the last term cancels against the bare Galilean terms $-p_{cc}+p_{vv}$ of the rigid translation, which leaves the numerators of Eq. (13.31) (Appendix D.37).

The soft acoustic limit of Sect. 8.3 is a different statement. There the probe itself is proportional to mass and its amplitude vanishes as q^2 at fixed energy. Here the probe is the electron density, its momentum $\boldsymbol{q}$ is finite, and the soft object is the phonon. Translation invariance does not make the amplitude vanish. It makes it finite, of order $\ell_{\boldsymbol{Q}\nu}$ rather than of order $\ell_{\boldsymbol{Q}\nu}/|\boldsymbol{Q}|$ as the separate intraband terms would suggest, and it fixes its value through D^n, the band velocities and $\boldsymbol{q}$. The squared amplitude then behaves as $1/|\boldsymbol{Q}|$, and with the phase space $Q^2\mathrm{d}Q$ the emission of soft phonons is integrable, as $Q\,\mathrm{d}Q$ at $T=0$ and as $\mathrm{d}Q$ at finite temperature.

In silicon the deformation terms dominate. With $D\approx 9\,\mathrm{eV}$ along the valley axis (Sect. 6.2) and $c_\nu=8.43\,\mathrm{km\,s^{-1}}$, D/c_ν is 300 to 350 keV, against momentum transfers of 1 to 2 keV near the window, so at the band extrema the recoil term is a correction of a percent or less. It is not negligible everywhere. For a nearly free electron the strain shift at fixed reduced wavevector, D^n, and the Galilean term $\boldsymbol{v}^{\mathrm{g}}_n\otimes\boldsymbol{p}_{nn}$ nearly cancel, since a weak electron–ion potential gives a weak electron–phonon vertex. The recoil term is then of the same size as the rest of the bracket. Worked example 13.6 shows such a case.

The limit tests a computed amplitude, gives the soft emission of a real pair outside the window, and shows that inside the window no soft phonon occurs. For the test, Eq. (13.9) is evaluated as a function, with the energy-shell condition imposed but no delta function, and it must approach Eq. (13.31) with a deviation linear in $|\boldsymbol{Q}|$ for both signs, several directions and decreasing $|\boldsymbol{Q}|$. A failure of the test exposes a common shortcut. Codes set the vertices of the three acoustic modes at $\boldsymbol{Q}=0$ to zero (Sect. 6.2), and an interpolation that carries this into the neighborhood of $\boldsymbol{Q}=0$ removes the rigid-translation part of the interband vertices. That part supplies the recoil term and the bare Galilean terms, $i\ell[\hat{\boldsymbol{e}}_\nu\cdot(\boldsymbol{q}-\boldsymbol{p}_{cc}+\boldsymbol{p}_{vv})]\rho_{cv}$ (Appendix D.37), and in Worked example 13.6 its loss changes $|M|^2$ by a factor of ten. The second use is outside the window, where the zero-phonon pair at $\boldsymbol{q}$ is real and Eq. (13.31) is its amplitude to emit a soft phonon. The denominators vanish on the cone $\hat{\boldsymbol{Q}}\cdot\boldsymbol{v}^{\mathrm{g}}_c=\mp c_\nu$, where the electron outruns the sound wave. This is the Cherenkov emission of Eq. (6.28), a real decay of the pair, and it is regularized by the width like the other resonances below.

Inside the off-shell window, the most important of the three uses, the soft limit is never reached. A final state at $\omega\le E_{\min}(\boldsymbol{q})-\delta_{\mathrm{w}}$ whose electron sits at $\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}$ needs $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}}-\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}}\ge\delta_{\mathrm{w}}$, since the pair at $\boldsymbol{q}$ costs at least $E_{\min}(\boldsymbol{q})$. The electron must therefore fall by δ_{w} over the distance $|\boldsymbol{Q}|$ in crystal momentum, which bounds the phonon wavevector from below by $\delta_{\mathrm{w}}/(v^{\mathrm{g}}_{\max}+c_\nu)$ (Eq. (D.213)). The steepest conduction band of silicon on the symmetry lines of Fig. 3.7 has a slope of 16.7 eV Å, or $2.5\times10^6\,\mathrm{m\,s^{-1}}$, and at $\delta_{\mathrm{w}}=0.3\,\mathrm{eV}$ the bound is $0.018\,\mathrm{Å}^{-1}$. The soft limit needs $|\boldsymbol{Q}|\ll 2m^*c_\nu$, with $m^*c_\nu=1.0\times10^{-3}$ and

$6.7 \times 10^{-3}\,\text{Å}^{-1}$ for transverse emission across the valley and longitudinal emission along it. Inside the window the intraband denominators are zero-phonon detunings of at least δ_{w}, the amplitude has no infrared feature at any temperature, and no subtraction is needed. At the onset itself the phonon is far harder, $|\boldsymbol{Q}^*| \geq 0.31\,\text{Å}^{-1}$ (Sect. 13.3). A computation inside the window can therefore give the acoustic modes at $\boldsymbol{Q} = 0$ zero weight, as long as it keeps the interband vertices near it.

Worked example 13.6: The soft-phonon amplitude in a one-dimensional crystal

A test of Eq. (13.31) needs a vertex computed in the laboratory frame, without the comoving construction, since a model whose orbitals ride on the atoms has the identity built in. Take a chain of rigid ions, lattice constant $a = 3.0\,\text{Å}$, mass 28.09 u, each ion a Gaussian well $-V_0 \mathrm{e}^{-x^2/2\sigma^2}$ with $V_0 = 2.5\,\text{eV}$ and $\sigma = 0.5\,\text{Å}$, and one spinless electron per cell, solved in 41 plane waves. The lowest band is filled and the next empty, with a gap of 1.215 eV at the zone boundary. One longitudinal acoustic branch has $c_\nu = 5.0\,\text{km}\,\text{s}^{-1}$. The vertex is the matrix element of the potential change of ions displaced by $\ell_{\boldsymbol{Q}\nu} \mathrm{e}^{\mathrm{i}QX_j}$, the density vertex is $\mathrm{e}^{\mathrm{i}qx}$, and Eq. (13.9) is summed over 40 bands. All parameters are illustrative.

Put the hole at $k = 0.80\,\pi/a$ and the transfer at $q = 0.30 \times 2\pi/a = 0.628\,\text{Å}^{-1}$, away from the band extrema. The pair at q costs 5.68 eV and $|\rho_{cv}| = 0.965$. The valence state has $v_v^{\text{g}} = 5.91\,\text{eV Å}$, 180 times the sound speed, and $p_{vv} = 0.776\,\text{Å}^{-1}$; the conduction state $v_c^{\text{g}} = 11.05\,\text{eV Å}$ and $p_{cc} = 1.450\,\text{Å}^{-1}$. Straining the crystal at fixed reduced wavevector gives $D^v = -4.563\,\text{eV}$ and $D^c = -15.237\,\text{eV}$, nearly the $-2\times$ kinetic energy of free electrons. The laboratory intraband slopes, read off the computed vertex at $|Q| = 10^{-5} \times 2\pi/a$, are 0.021 eV and 0.774 eV, equal to $D^n + v_n^{\text{g}} p_{nn}$ to 10^{-4} eV: the strain and Galilean terms cancel to a few percent.

The steps of Appendix D.37 check one by one. At $Q = 0.10 \times 2\pi/a$, far from the limit, the computed amplitude equals the comoving amplitude plus the contact term to a relative 3×10^{-14}, with a contact term 7.5 times the total, and at $Q = -0.25 \times 2\pi/a$ for absorption to 4×10^{-16}. The comoving vertex is of order $\ell|Q|$ for every band pair, about $220\,\ell|Q|$ eV at most, while the laboratory interband vertex grows from 0.44 to 4.4 eV as $|Q|$ falls from 10^{-2} to $10^{-4} \times 2\pi/a$. For emission with $Q > 0$ the bracket of Eq. (13.31) is

term	value (Å^{-1})
recoil, q	$+0.6283$
deformation	-0.6076
velocity	-1.1×10^{-5}
total	$+0.0207$

and the full amplitude deviates from the limit by 0.20, 0.020 and 0.0020 at $|Q| = 10^{-2}$, 10^{-3} and $10^{-4} \times 2\pi/a$: linearly, as the relative correction of order $|\boldsymbol{Q}|$ requires. The same holds for absorption and for $Q < 0$. Recoil and deformation nearly cancel here, and with the interband vertices set to zero the recoil is lost and $|M|^2$ comes out 10.4 times too large.

At the band extrema the balance changes. With the hole at $k = \pi/a$ and $q = 2\pi/a$, an Umklapp transfer that puts the electron at the bottom of the conduction band, both velocities vanish, $D^v = -7.42\,\text{eV}$ and $D^c = -7.26\,\text{eV}$, and the bracket is $q \pm (D^c - D^v)/c_\nu = 2.09 \pm 4.91\,\text{Å}^{-1}$, so that the soft amplitude differs by a factor of 2.5 between the two directions of $\boldsymbol{Q}$. The limit is reached only for $|Q| \ll 2m^* c_\nu$, here $1.2 \times 10^{-3}\,\text{Å}^{-1}$ with $m^* = 0.135\,m_e$, and since each deformation term is about $220\,\text{Å}^{-1}$ and they cancel to

$4.9\,\text{Å}^{-1}$, the deviation falls below 1 % only at $|Q| = 10^{-8} \times 2\pi/a$. Setting the interband vertices to zero now removes the recoil term: for emission with $Q > 0$ it halves $|M|^2$, and for the opposite direction it multiplies $|M|^2$ by 2.9. In silicon, where $(D^c - D^v)/c_\nu$ is a hundred times q, this effect is at the percent level.

13.6.2 The off-shell window

The denominators of Eq. (13.9) were bounded on the energy shell in Eq. (13.11). Those of path A with an empty band p and of path C with a filled band are the energies by which a pair at crystal momentum $\boldsymbol{q}$, $(p, \boldsymbol{k}+\boldsymbol{q};\, v, \boldsymbol{k})$ or $(c, \boldsymbol{k}';\, p, \boldsymbol{k}+\boldsymbol{Q})$, misses ω, and they are at least $E_{\min}(\boldsymbol{q}) - \omega$ in size. The other two kinds belong to pairs at crystal momentum $\boldsymbol{Q}$ and are at least $E_{\mathrm{g}} - \omega_{\boldsymbol{Q}\nu}$, which never vanishes. A denominator of Eq. (13.9) therefore vanishes exactly where a first-order transition at the same $(\boldsymbol{q}, \omega)$, the creation of a zero-phonon pair by the density alone, is on shell. Below the zero-phonon edge no such transition exists. Above it, at fixed $(\boldsymbol{q}, \omega)$, the resonance condition is met on a surface of crystal momenta, and the sum over final states crosses that surface. There the squared amplitude has a double pole $1/x^2$ in the detuning x, which is not integrable across a simple zero. It has no principal value either, and Sect. 2.4 showed what it needs: the width of the intermediate state, after which the rate splits into a cascade and a finite part. The second-order rate is then no longer a quantity of its own. Its value depends on the width, and its share of the response depends on a convention.

The edge itself has to be the true one. The bands in the denominators are shifted and broadened by the electron–phonon self-energy (Eq. (6.21)), and the pair edge moves with them, by the zero-point renormalization of the gap at low temperature and by more as the temperature rises (Sect. 13.7). A bound pair lies below the free pairs by its binding energy (Sect. 13.8). We write $E^{\mathrm{ren}}_{\min}(\boldsymbol{q}, T)$ for the lowest energy of a zero-phonon final state at crystal momentum $\boldsymbol{q}$, with renormalized bands and the bound pair included, and use the same renormalized energies in the denominators. A bare edge with renormalized denominators, or the reverse, misplaces the boundary by the renormalization itself. In the silicon numbers of this chapter the conduction bands are shifted so that the indirect gap equals the measured one at low temperature, which places the renormalized edge correctly at the band extrema and treats the renormalization as rigid in $\boldsymbol{k}$.

Window theorem

The phonon-assisted structure factor Eq. (13.15), built from the amplitude Eq. (13.9), is defined at $(\boldsymbol{q}, \omega)$ without widths, without $\mathrm{i}0^+$, without subtraction and without counterterms if and only if every first-order transition at the same $(\boldsymbol{q}, \omega)$ is off shell, $\omega < E^{\mathrm{ren}}_{\min}(\boldsymbol{q}, T)$, with renormalized energies used in the edge and in the denominators alike. It is quoted as a channel of its own in the pointwise window

$$\omega \le E^{\mathrm{ren}}_{\min}(\boldsymbol{q}, T) - \delta_{\mathrm{w}}, \qquad \delta_{\mathrm{w}} \gg \omega_{\max},\ \hat{\Gamma},\ T, \tag{13.32}$$

applied at each $\boldsymbol{q}$ separately, with $\omega_{\max}$ the top of the phonon band and $\hat{\Gamma}$ the largest width of the zero-phonon pairs between the edge and δ_{w} above it. In silicon $\delta_{\mathrm{w}} = 0.3\,\mathrm{eV}$.

The first half is the statement about the denominators above. The second half is about accuracy, and it needs the resummed amplitude of Sect. 8.6. There the virtual pair m enters through its propagator $\mathsf{G}_m(\omega) = 1/[\omega - E_m - \Sigma_m(\omega)]$, Eq. (8.58), and the second-order amplitude keeps only $1/(\omega - E_m)$. The difference is of relative order $\Sigma_m(\omega)/(\omega - E_m)$, and the margin must make it small. With renormalized energies the real part of Σ_m at the pair's

own energy is absorbed into E_m, and three pieces remain. The self-energy changes between E_m and ω, on the scale of the phonon energies, because the thresholds for emitting each phonon sit $\omega_{\boldsymbol{Q}\nu}$ above the edges of the continuum (Sect. 6.3); this requires $\delta_{\mathrm{w}} \gg \omega_{\max}$. The imaginary part, the width, requires $\delta_{\mathrm{w}} \gg \hat{\Gamma}$. And thermal phonons broaden the edge by about T. The same ratio $\omega_{\boldsymbol{Q}\nu}/\delta_{\mathrm{off}}$ sets the error of the adiabatic form (Eq. (13.14)), and the bound pair adds the condition $\delta_{\mathrm{w}} \gg E_x^{\mathrm{b}}$ unless it is already part of the edge.

The parameter θ_{ep} of Eq. (8.40) looks like the controlling one, and taken at face value it would forbid a margin of 0.3 eV. Evaluated at $\delta_{\mathrm{off}} = 0.3\,\mathrm{eV}$ with the largest vertex of silicon, the optical vertex of the holes, about 0.25 eV, it is $(0.25/0.3)^2 \approx 0.7$; the $\theta_{\mathrm{ep}} = 1$ at 0.25 eV of Sect. 8.5 is a bound of this kind, not a size. But θ_{ep} prices one more pair of vertices on intermediate states that all lie δ_{off} from the shell. At the window edge the extra vertices act on the virtual pair, whose self-energy sums $|g|^2$ over one-phonon states spread across an energy range E_{w} of an electronvolt or more, the width of the continuum in the sense of Sect. 2.4. Its size is then $|\Sigma_m| \approx \langle |g|^2 \rangle / E_{\mathrm{w}}$ rather than $\langle |g|^2 \rangle / \delta_{\mathrm{w}}$, and the relative correction $|\Sigma_m|/\delta_{\mathrm{w}}$ is $\theta_{\mathrm{ep}}\,\delta_{\mathrm{w}}/E_{\mathrm{w}} \approx 0.7 \times 0.3/1 \approx 0.2$ at 0.3 eV. Measured on the gap itself this is the zero-point renormalization over the margin, which is what the resummation actually adds. The parameter θ_{ep} bounds the correction near the edge; it does not size it.

The silicon numbers fix the margin (Fig. 13.8(b)). The top of the phonon band, the zone-center optical phonon of 64.1 meV, is 0.21 of 0.3 eV, and the transverse acoustic phonon of the onset, 14.8 meV, is 0.05. The zero-point renormalization of the gap is 56 meV as computed at fixed volume and 71 to 73 meV for the electron–phonon part implied by experiment (Sect. 6.3), 0.19 to 0.24 of the margin. It is absorbed into the renormalized edge, and only its variation across the margin remains. The exciton binding energy, 14.7 meV, is 0.05. At a detector's temperature T is zero for this purpose; at 300 K it would be 0.09.

For the widths we estimate the decay of the pair at the zero-phonon edge by phonon emission, with deformation-potential rates and the constants computed by Li, Graziosi and Neophytou [15, Table III and Eq. (10)]. The electron scatters between valleys by the g-type longitudinal optical and the f-type longitudinal acoustic and transverse optical phonons, the hole emits optical phonons, and both emit acoustic phonons as in Eq. (6.28). The pair at $E_{\min}(\boldsymbol{q})$ shares its kinetic energy in proportion to the masses along the valley axis, 0.77 to the electron. Between $|\boldsymbol{q}| = 0.4$ and 1.4 keV this gives widths of 3 to 21 meV at the edge, 11 meV at 1.0 keV, and 15 meV at 1.0 keV and 0.3 eV above the edge, so that $\hat{\Gamma}/\delta_{\mathrm{w}}$ stays below 0.1. Every ratio is then below about 0.2. A margin of 0.1 eV would put the phonon ratio at 0.64 and the renormalization ratio at 0.56 to 0.73, and one of 0.5 eV would bring them to 0.13 and 0.11 to 0.15 at the price of the window itself: along the valley axis it would end at $|\boldsymbol{q}| = 1.07$ keV instead of 1.27 keV. The margin is therefore $\delta_{\mathrm{w}} = 0.3\,\mathrm{eV}$, and with it the bound $|\boldsymbol{Q}^*| \geq 0.31\,\text{Å}^{-1}$ of Sect. 13.3 holds. Inside the window the second-order rate then carries a relative correction of order 0.2 at the edge, falling as $1/(E_{\min} - \omega)$ toward the onset.

Fig. 13.8(a) shows the window in the plane of $|\boldsymbol{q}|$ and ω. It is a region of that plane, not an interval of ω: at each $\boldsymbol{q}$ it runs from the onset $E_{\min}^{\mathrm{PA}}(\boldsymbol{q})$ of Eq. (13.2) up to $E_{\min}^{\mathrm{ren}}(\boldsymbol{q}, T) - \delta_{\mathrm{w}}$, and it is empty where the two meet. At $|\boldsymbol{q}| = 1.0$ keV it spans 1.184 to 1.467 eV. Along the valley axis it closes at $|\boldsymbol{q}| = 1.27$ keV with the computed bands, 1.33 keV with the parabolic valley of Sect. 13.3, and it reopens on the far side of the Umklapp image of the opposite valley, beyond 3.30 keV. Dark matter just above the onset mass reaches only the bottom of the window. At 0.42 MeV, where the zero-phonon line opens along this axis (Sect. 13.1), the largest deposits already lie in the margin.

Between $E_{\min}^{\mathrm{ren}} - \delta_{\mathrm{w}}$ and $E_{\min}^{\mathrm{ren}}$ the amplitude is still finite, but its corrections grow as $1/(E_{\min}^{\mathrm{ren}} - \omega)$ and reach order one at the edge. Above the edge, within a few widths of it, no test function is smooth on the scale of the detuning, and the Dyson split of Sect. 8.6 has no meaning. The margin and this shell form the edge band. There we quote only totals,

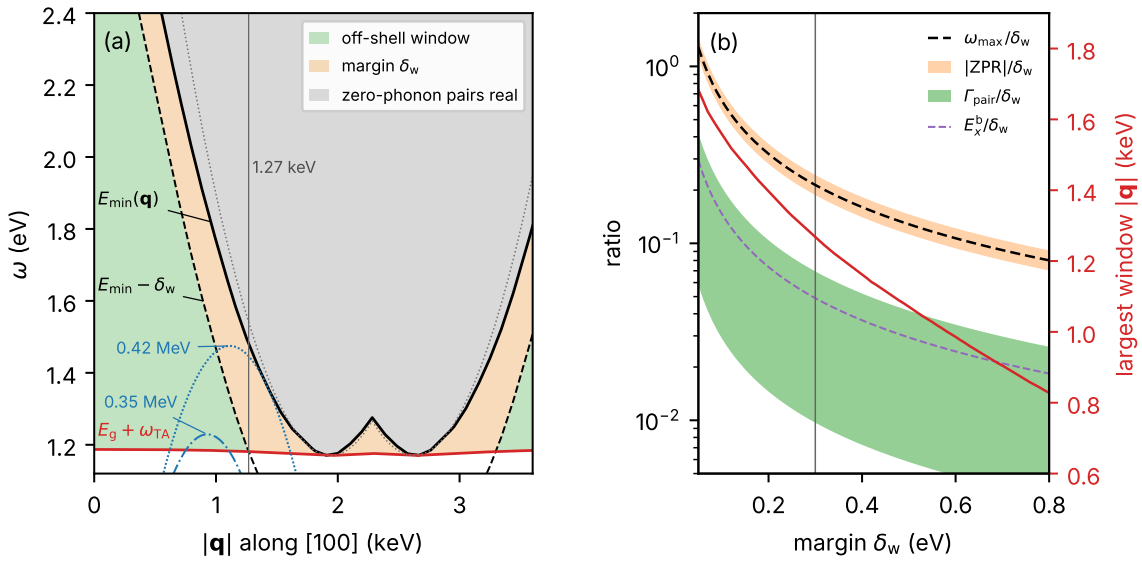


Figure 13.8: The off-shell window of silicon. (a) The plane of momentum transfer $|\boldsymbol{q}|$ along [100], the valley axis, and energy deposited ω. Black: the zero-phonon edge $E_{\min}(\boldsymbol{q})$ from the PBE bands of Fig. 3.7 on a 48^3 grid, with the conduction bands shifted to the measured gap of 1.17 eV (dotted: the parabolic valley, pair mass $1.19\,m_e$); dashed: $E_{\min} - \delta_w$ with $\delta_w = 0.3$ eV; red: the phonon-assisted onset $E_g + \omega_{TA}(\boldsymbol{Q}^*)$ of Eq. (13.2), with the PBE force constants of Sect. 5.2 and $\boldsymbol{Q}^*$ measured to the nearest valley, the Umklapp image of the opposite valley at 2.63 keV included. Green: the off-shell window, where the phonon-assisted rate is a channel of its own; orange: the margin, where it is finite but its corrections grow; gray: zero-phonon pairs are real and the Dyson convention of Sect. 8.6 applies. Blue: the largest energy deposited by dark matter of 0.35 and 0.42 MeV. The window ends at $|\boldsymbol{q}| = 1.27$ keV (1.33 keV with the parabolic valley) and reopens beyond 3.30 keV. (b) The ratios that control the window, against the margin δ_w: top of the phonon band (64.1 meV), zero-point renormalization of the gap (56 to 73 meV), width of the pair at the zero-phonon edge for $0.4 \le |\boldsymbol{q}| \le 1.4$ keV (3 to 21 meV, a deformation-potential estimate with computed constants [15]), and exciton binding energy (14.7 meV). Red, right axis: the largest $|\boldsymbol{q}|$ below the valley at which the window is not empty. The vertical line marks $\delta_w = 0.3$ eV.

computed from the unexpanded amplitude Eq. (8.58), and no phonon-assisted rate of its own; in particular the onset exponents of Eq. (13.19) hold only where the onset lies inside the window, as Sect. 13.3 found for $|\boldsymbol{q}| \leq 1.33\,\mathrm{keV}$ with parabolic valleys and $|\boldsymbol{q}| \leq 1.27\,\mathrm{keV}$ with the computed bands.

In a calculation the window is applied point by point. For each bin of $(|\boldsymbol{q}|, \hat{\boldsymbol{q}})$ the edge is the minimum of the renormalized pair energies on the full grid of $\boldsymbol{k}$, not along a symmetry line, and the bins in ω stop at that edge minus δ_{w} (Appendix E.14). The resonant denominators, path A with p empty and path C with p filled, are written as zero-phonon detunings, $\omega - E_{pv}(\boldsymbol{k}, \boldsymbol{q})$ and $E_{cp}(\boldsymbol{k}+\boldsymbol{Q}, \boldsymbol{q}) - \omega$, so that their smallness inside the window is visible and the energies in them are the ones that define the edge. Any smearing of the energy delta function must be small compared with δ_{w}, or it leaks the zero-phonon continuum into the window.

13.6.3 Resonant denominators beyond the window

Above $E^{\mathrm{ren}}_{\mathrm{min}}(\boldsymbol{q}, T)$ a resonant intermediate state is a real pair. In path A it is the pair $m = (p, \boldsymbol{k}+\boldsymbol{q};\, v, \boldsymbol{k})$ with an electron in an empty band, which then emits or absorbs the phonon. In path C it is $m = (c, \boldsymbol{k}';\, p, \boldsymbol{k}+\boldsymbol{Q})$ with a hole in a filled band, which relaxes into $(v, \boldsymbol{k})$ by the phonon. Near such a state we split the amplitude into the resonant term and the rest. The resonant denominator receives the width of the pair, Eq. (2.57):

$$M^{\pm}_{cv\nu} = \frac{g_{fm}\,\rho_m}{\omega - \tilde{E}_m + \mathrm{i}\Gamma_m/2} + g_{fm}\,\mathcal{A}^{\mathrm{off}}_f, \tag{13.33}$$

with $\rho_m = \rho_{pv}(\boldsymbol{k}, \boldsymbol{q})$ and $g_{fm} = g_{cpv}(\boldsymbol{k}+\boldsymbol{q}, \boldsymbol{Q})$ for path A, $\rho_m = \rho_{cp}(\boldsymbol{k}+\boldsymbol{Q}, \boldsymbol{q})$ and $g_{fm} = g_{pv\nu}(\boldsymbol{k}, \boldsymbol{Q})$ for path C, $\tilde{E}_m$ and Γ_m the renormalized energy and the width of the pair, and $g_{fm}\mathcal{A}^{\mathrm{off}}_f$ the sum of all other terms of Eq. (13.9); the minus sign of path C in Eq. (13.8) is absorbed into $\rho_m g_{fm}$. A path C term has to be taken from the four-path form Eq. (13.8), with its many-body denominator, before the band-sum theorem is used: the width belongs to the propagator of the real pair m, and the theorem holds only on the energy shell (Sect. 13.2). This is the amplitude Eq. (8.58) of the level-and-band model with Υ_m dropped, a correction of order g^2.

Squared, Eq. (13.33) contains the Lorentzian $|g_{fm}\rho_m|^2/[(\omega-\tilde{E}_m)^2+\Gamma_m^2/4]$, which Eq. (2.60) splits into $(2\pi/\Gamma_m)\delta(\omega - \tilde{E}_m)$, the finite part $\mathrm{fp}(\omega - \tilde{E}_m)^{-2}$ and a correction of relative order $\Gamma_m^2/\delta_{\mathrm{F}}^2$. Summed over the final states of the phonon, the delta term is $|\rho_m|^2\delta(\omega - \tilde{E}_m)$ times $\sum_f 2\pi|g_{fm}|^2\delta(E_f - \tilde{E}_m)/\Gamma_m$, the fraction of the decays of m that go into phonon emission. It is the zero-phonon pair, created as a real state, followed by its first decay: the cascade of Sect. 2.4. The Dyson convention Eq. (8.59) assigns it to the zero-phonon sector with its residue, and assigns the finite part, the principal-value cross term with $\mathcal{A}^{\mathrm{off}}_f$ and $|\mathcal{A}^{\mathrm{off}}_f|^2$ to the phonon-dressed one. Outside the window, then, the phonon-assisted structure factor is S^+ of Eq. (8.59) with the substitutions above, summed over the resonances of both paths, and the zero-phonon one is S^0 with renormalized pairs. The two add up to the response to order g^4 and $\Gamma_m^2/\delta_{\mathrm{F}}^2$, against any test function smooth on the scale of the detuning δ_{F} (Sect. 8.6).

A computed phonon-assisted table beyond the window differs from one inside it in three ways: it need not be positive, it must not be mixed with another convention, and it is defined only near narrow lines. The finite part is negative against a narrow test function, so S^+ is not a positive spectrum near $\tilde{E}_m$, and a computed phonon-assisted table may contain negative bins near the edge; only $S^0 + S^+$ is positive. A Lorentzian zero-phonon line added to the resummed square of Eq. (13.33) counts the cascade twice, an error of order one. Added to the finite part it counts the off-shell weight twice, an error of order g^2 (Sect. 8.6). The rate of Sect. 13.3 continued past the edge with a small numerical broadening in place of the width produces a delta term weighted by the inverse broadening, a cascade with an arbitrary weight.

The split is sharp only while $\Gamma_m \ll \delta_{\mathrm{F}}$, with δ_{F} the distance of $\tilde{E}_m$ to the nearest structure of the one-phonon continuum: the zero-phonon edge itself, a threshold for emitting a further phonon, or the onset of a new valley or band. In silicon, with the widths of 3 to 21 meV estimated above and detunings of a few tenths of an eV away from such structures, $\Gamma_m/\delta_{\mathrm{F}}$ is 0.01 to 0.1, and by Eq. (8.61) the zero-phonon weight is fixed to that order. Within a few widths of each structure it is not, and only totals are quoted, as at the edge of the window.

13.6.4 The Fano term near a resonance

When a real pair m decays, the phonon it emits first interferes with the same phonon emitted directly by the off-shell paths, and Eq. (8.62) shifts the branching of the first emission by the Fano coefficients $c_\nu^{\mathrm{F}} = \Gamma_m^{\mathrm{tot}}\,\mathrm{Im}(\rho_m^* \mathcal{A}_\nu^{\mathrm{off}})/|\rho_m|^2$. Their size near a resonance follows from Eq. (13.33). With ϕ_ν the phase of $\rho_m^* \mathcal{A}_\nu^{\mathrm{off}}$,

$$|c_\nu^{\mathrm{F}}| = \frac{\Gamma_m^{\mathrm{tot}}}{\delta_{\mathrm{F}}}\,\frac{|\mathcal{A}_\nu^{\mathrm{off}}|\,\delta_{\mathrm{F}}}{|\rho_m|}\,|\sin\phi_\nu|. \tag{13.34}$$

The middle factor compares the off-shell amplitude with the resonant one at the distance δ_{F} from the line. Both are second-order amplitudes with denominators of that size, and the factor is of order one. The Fano coefficients are therefore of first order in $\Gamma_m/\delta_{\mathrm{F}}$, the same order as the ambiguity of the Dyson split and as the branching of the cascade itself, and they need the phase ϕ_ν in addition.

That phase is what symmetry controls. In a crystal invariant under inversion combined with time reversal, with an even number of electrons, $(\mathsf{PT})^2 = +1$ on the many-electron states, every amplitude can be made real, $\sin\phi_\nu = 0$, and the Fano term vanishes (Sect. 8.6). Silicon and germanium satisfy the hypothesis at the level of one-body self-energies, with spin–orbit coupling as well: the single-electron Kramers pairs cannot all be chosen real, but the many-electron states with an even number of electrons can. If the attraction of electron and hole splits a degenerate zero-phonon manifold in a basis that is not invariant, only the sum of the coefficients over the branches vanishes. GaAs has no center of inversion, and its antisymmetric crystal potential gives ϕ_ν a nonzero value. We estimate $|c_\nu^{\mathrm{F}}| \lesssim 0.3\,\Gamma_m/\delta_{\mathrm{F}}$ to first order, with the middle factor of Eq. (13.34) set to one and $|\sin\phi_\nu| \le 0.3$ assumed for the antisymmetric part of the GaAs potential, and with Fröhlich widths of a few meV just above the conduction minimum and 20 to 60 meV higher up, $\Gamma_m/\delta_{\mathrm{F}}$ is 0.01 to 0.2 away from thresholds (Appendix E.14). The Fano coefficients then move a few percent at most of the first emissions from one branch to another. Since their deviations from the mean average to zero over the branches, they change no total and no number of zero-phonon events, only which phonon the cascade starts with, and the distribution of the number of pairs by at most $\max_\nu |c_\nu^{\mathrm{F}}|$. Near each threshold δ_{F} falls toward Γ_m, Eq. (13.34) reaches order one and the expansion ends; these shells are excluded and quoted as totals. Inside the window no zero-phonon pair is real, and the Fano term does not exist.

Where in $(\boldsymbol{q}, \omega)$ the Fano term of GaAs can be quoted is decided by a map of $\Gamma_m/\delta_{\mathrm{F}}$. It needs the Fan–Migdal widths of the pairs on fine grids and the non-resonant remainder of the same amplitude on the final shell, and Appendix E.14 specifies the calculation, with silicon as the control that must return zero. With the soft phonon, the window and its edge settled, the remaining question about the zero-phonon line is how temperature and zero-point motion move it, which is the subject of Sect. 13.7.

13.7 Temperature and zero-point renormalization

Between room temperature and 10 mK the indirect gap of silicon opens by 46 meV (Worked example 6.3), and the phonon-assisted onset of Worked example 13.1 moves from 0.324 to 0.337 MeV. At 10 mK the zero-point motion still holds the gap below its clamped-nucleus value, by 56 meV as computed at fixed volume and by 71 to 73 meV for the electron–phonon part implied by experiment (Sect. 6.3), and that renormalization is already part of the measured 1.17 eV that every threshold of this chapter uses. It enters a detector calculation in three places. The zero-phonon response that is added to S^{PA} must carry it without counting anything twice, a point Sect. 13.3 left open. The special displacements of Appendix E.8, the cheapest route to renormalized spectra, compute it together with a phonon sideband that cannot be added to the phonon-assisted rate. And the part that adiabatic methods miss must be small enough not to move the window. We end with the temperature dependence of the window edges, which is what decides whether data taken at 77 or 300 K describe a detector at 10 mK.

13.7.1 The renormalized zero-phonon response

At order g^2 a zero-phonon pair is dressed in three ways: its energy is shifted, its weight is reduced, and its transition density acquires the virtual-phonon correction. All three are in the Dyson convention of Sect. 8.6, which assigns to the zero-phonon sector the delta function $z_{\mathrm{qp}}(m)|\tilde{\rho}_m|^2\delta(\omega-\tilde{E}_m)$ of each pair m (Eq. (8.59)). For a pair $(c,\boldsymbol{k}+\boldsymbol{q};\,v,\boldsymbol{k})$ the energy is the difference of the renormalized band energies Eq. (6.21), and the residue of the pair is, to order g^2, the product of the residues of electron and hole, since the slope of the pair self-energy is the sum of the slopes of Σ^{FM} for the two states (Eq. (4.83)). The zero-phonon structure factor at temperature T is then

$$S^0(\boldsymbol{q},\omega;T)=4\pi\sum_{v,c}\int_{\mathrm{BZ}}\frac{\mathrm{d}^3k}{(2\pi)^3}\,z_{\mathrm{qp}}(c,\boldsymbol{k}+\boldsymbol{q})\,z_{\mathrm{qp}}(v,\boldsymbol{k})\,\bigl|\tilde{\rho}_{cv}(\boldsymbol{k},\boldsymbol{q})\bigr|^2\,\delta\bigl(\omega-\varepsilon^{\mathrm{ren}}_{c,\boldsymbol{k}+\boldsymbol{q}}(T)+\varepsilon^{\mathrm{ren}}_{v\boldsymbol{k}}(T)\bigr),\tag{13.35}$$

with $z_{\mathrm{qp}}(n,\boldsymbol{k})=1+\partial_\omega\,\mathrm{Re}\,\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\omega)$ at $\omega=\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}}$, and $|\tilde{\rho}_{cv}|^2=|\rho_{cv}|^2+2\,\mathrm{Re}(\rho^*_{cv}\,\delta\rho_{cv})$, where $\delta\rho_{cv}$ collects the mixing of bands by the off-diagonal self-energy and the virtual-phonon vertex Υ_m of Eq. (8.58). With $z_{\mathrm{qp}}=1$, $\delta\rho=0$ and bare energies it is Eq. (10.1). Temperature enters only through the phonon occupations inside the self-energies, $n_{\boldsymbol{Q}\nu}+1-f$ and $n_{\boldsymbol{Q}\nu}+f$ in Σ^{FM} and $2n_{\boldsymbol{Q}\nu}+1$ in Σ^{AH} (Eqs. (6.19) and (6.20)); the band occupations are zero or one to $\mathrm{e}^{-E_{\mathrm{g}}/2T}$. At 1 K the occupation of every phonon above 10 meV is below 10^{-50}, and Eq. (13.35) is the zero-point response.

The shift of the pair energies is the largest of the three corrections, and most of it is a rigid shift of the gap: in silicon the valence maximum rises by 35 meV and the conduction minimum drops by 22 meV (Worked example E.7). A calculation that scissors its bands to the measured low-temperature gap has this rigid part already, and adding a computed renormalization to it counts the renormalization twice (Appendix E.8). What the scissor does not supply is the variation of the shift across the zone, which Sect. 13.9 needs for the zero-phonon edge and which we bound in the subsection after next. The residues and the vertex correction change the weight of each line by an amount of relative order θ_{ep} (Eq. (8.40)), since each is one more pair of vertices on an intermediate state off the shell; in silicon near the gap $\theta_{\mathrm{ep}}=0.010$ to 0.016 (Sect. 8.5), so the weights move by one or two percent.

The weight that the zero-phonon lines lose reappears in the sideband. With all-electron (local) potentials, the electron–phonon Hamiltonian at first and second order in the displacements is a sum of one-body potentials that commute with the density. The double commutator in Eq. (4.28) is therefore that of the kinetic energy alone, $\mathscr{M}^1_{ee}(\boldsymbol{q})=n_e q^2/m_e$, at every order in g.

To order g^2 the electronic response with a pair in the final state is $S^0(\boldsymbol{q},\omega;T)+S^{\mathrm{PA}}(\boldsymbol{q},\omega;T)$. The density also responds without creating a pair, through the electrons dragged by the ions into a one-phonon state, and that sector has a first moment of order g^2 too, smaller than $n_e q^2/2m_e$ by about $Z_v m_e/M_{\mathrm{cell}} = 8\times10^{-5}$ in silicon. Up to that term the first moment of S^0+S^{PA} equals that of the clamped-nucleus response, and

$$\int_0^\infty \mathrm{d}\omega\;\omega\,[S^0(\boldsymbol{q},\omega;T)-S^0(\boldsymbol{q},\omega)] = -\int_0^\infty \mathrm{d}\omega\;\omega\,S^{\mathrm{PA}}(\boldsymbol{q},\omega;T)+O(g^4), \tag{13.36}$$

where $S^0(\boldsymbol{q},\omega)$ without the temperature argument is Eq. (10.1), and the factor 2π and the sum over $\pm\boldsymbol{q}$ of Eq. (4.28) cancel on the two sides. The zero-phonon line gives up, in energy-weighted weight, exactly what the phonon sideband gains. The Allen–Heine term contributes nothing to this transfer: it is a static local potential, $\frac{1}{2}\sum_{ab}\langle u_a u_b\rangle\,\partial_a\partial_b V$, and a static local potential leaves the double commutator unchanged. The transfer is carried by the Fan–Migdal term and the vertex correction, the processes in which a phonon is emitted and reabsorbed, since S^{PA} is those processes with the phonon left behind. Eq. (13.36) is the precise form of the warning of Sect. 13.3: the bare S^0 plus S^{PA} overshoots the f-sum by the first moment of S^{PA}, and Eq. (13.35) is what removes the excess. Inside the off-shell window none of this matters, since Eq. (13.35) has no weight there; above the zero-phonon edge the bookkeeping is a percent effect on the zero-phonon rate.

13.7.2 Special displacements as a quadrature

The special displacements of Appendix E.8 compute renormalized spectra without vertices, from one supercell with displaced nuclei. Applied to the density response, the Williams–Lax average [16, 17, Eqs. (1) and (3)] is the clamped-nucleus structure factor $S^{\mathrm{cl}}(\boldsymbol{q},\omega;\{x\})$ of the supercell, averaged over the Gaussian distribution of its normal coordinates,

$$S^{\mathrm{WL}}(\boldsymbol{q},\omega;T) = \prod_j \int \mathrm{d}x_j\,\frac{\mathrm{e}^{-x_j^2/2\sigma_j^2}}{\sqrt{2\pi}\,\sigma_j}\;S^{\mathrm{cl}}(\boldsymbol{q},\omega;\{x\}), \qquad \sigma_j^2=\frac{2n_j+1}{2\omega_j}, \tag{13.37}$$

which is Eq. (E.56) with S in place of ε_2. Here $j=1,2,\ldots$ numbers the modes of the supercell, each a pair $(\boldsymbol{Q}\nu)$ of a commensurate wavevector and a branch, in order of increasing frequency, and x_j are the mass-weighted coordinates $x_{\boldsymbol{Q}\nu}$ with widths $\sigma_{\boldsymbol{Q}\nu}$. The average is adiabatic by construction: each configuration is frozen while the probe acts, and the nuclei enter only through the distribution of their positions.

One configuration, with every coordinate at $x_j=\varsigma_j\sigma_j$ and signs $\varsigma_j=(-1)^{j-1}$ that alternate along the frequency-ordered list, reproduces Eq. (13.37) as the supercell grows. Expand any smooth property $O(\{x\})$ of the frozen supercell, a band energy or S^{cl} at fixed $(\boldsymbol{q},\omega)$, about the equilibrium positions. The Gaussian average keeps only even powers of each coordinate,

$$\langle O\rangle_T = O(0)+\frac{1}{2}\sum_j \sigma_j^2\,\frac{\partial^2 O}{\partial x_j^2}+O(\sigma^4),$$

while the single configuration gives

$$O(\{\varsigma\sigma\}) = O(0)+\sum_j \varsigma_j\sigma_j\frac{\partial O}{\partial x_j}+\frac{1}{2}\sum_j\sigma_j^2\frac{\partial^2 O}{\partial x_j^2}+\frac{1}{2}\sum_{j\neq k}\varsigma_j\varsigma_k\,\sigma_j\sigma_k\frac{\partial^2 O}{\partial x_j\partial x_k}+O(\sigma^3).$$

The diagonal second-order terms agree, because $\varsigma_j^2=1$. The linear terms and the cross terms must vanish. Take two neighbors in the list, j and $j+1$. As the supercell grows, the commensurate grid becomes dense, the frequency spacing $\omega_{j+1}-\omega_j$ goes to zero wherever the

phonon density of states is nonzero, and neighbors in the list become effectively degenerate, with the same σ. Their couplings to the electrons are not equal mode by mode, since neighbors in the list can come from distant wavevectors, but averaged over many neighbors they vary smoothly with frequency. The linear terms, which enter with alternating signs, then cancel on average, and the cross terms with any third mode k cancel the same way, since $\partial_j \partial_k O$ and $\partial_{j+1} \partial_k O$ come with opposite signs [17, Eqs. (6)–(9)]. Gaps in the density of states only split the list into ranges in which the argument applies separately. What remains is the diagonal sum, which is $\langle O \rangle_T$ to order σ^2. The odd orders are removed exactly, at any supercell size, by averaging the configuration with its mirror image $x_j \to -x_j$ [17, Eq. (12)]. For the higher even orders the reason is statistical: the sum $\sum_j x_j^2/\sigma_j^2$ over the Gaussian distribution has mean N_m and width $(2N_\mathrm{m})^{1/2}$ for N_m modes. For large supercells, as a result, almost all of the weight of Eq. (13.37) lies on configurations with $|x_j| \approx \sigma_j$, and the special displacement is one of them [17, Sect. III A]. The single configuration is therefore not a random sample of the distribution but a deterministic quadrature of it, whose error falls as the grid of phonon wavevectors becomes dense.

Limit Check 13.2. Zero-point renormalization by special displacements

$\langle \varepsilon_{n\boldsymbol{k}} \rangle_T = \varepsilon_{n\boldsymbol{k}}(\{\boldsymbol{u}^\mathrm{ZG}\})$ for a large supercell, with $\boldsymbol{u}^\mathrm{ZG}$ of Eq. (E.57), and to second order the adiabatic Allen–Heine–Cardona shift Eq. (6.22) [18, 17, Eqs. (5), (10) and (20)].

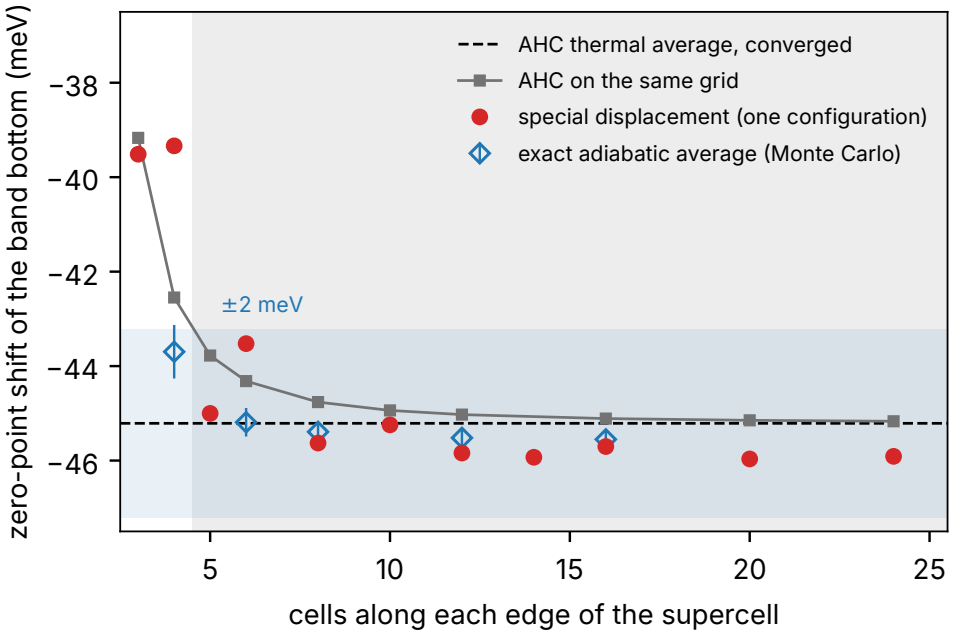


Figure 13.9: Zero-point shift of the band bottom in a model crystal against the size of the supercell. The model: a simple cubic lattice, $a = 2.70\,\text{Å}$, one atom of 28.09 u per cell, central springs of 4.0 and $1.5\,\text{eV Å}^{-2}$ to nearest and next-nearest neighbors (phonons up to 65 meV), and one s band with nearest-neighbor hopping $t(d) = -1\,\text{eV} \times (a/d)^6$ (band-bottom mass $0.52\,m_e$), at $T = 0$. Red: the special displacement, one configuration with every mode at its root-mean-square amplitude and signs alternating in order of frequency, wavevectors equal to their own time-reversal partners left out. Gray squares: the adiabatic Allen–Heine–Cardona average on the grid of phonon wavevectors of the same supercell; dashed: its limit for a dense grid, −45.2 meV, with a band of ±2 meV.

Blue diamonds: the exact adiabatic average of the same supercell, to all orders in the displacements, by Monte Carlo with antithetic pairs (statistical errors shown). Gray: supercells in which the special displacement is within 2 meV of the limit, from $5 \times 5 \times 5$ cells (125 atoms).

In smaller supercells the coarse grid of phonon wavevectors and the uncancelled cross terms between modes dominate, and at any size the special displacement misses the nonadiabatic part of the renormalization and places the phonon-assisted sideband at the wrong energy.

The model of Fig. 13.9 isolates the quadrature. Its band-bottom shift is -45.2 meV in the second-order average, -34.5 meV from the Allen–Heine term and -10.7 meV from the Fan term, comparable to the conduction-minimum shift of silicon. The second-order average on the grid of the supercell converges from above, and is within 2 meV of its limit from $5 \times 5 \times 5$ cells on. The special displacement scatters around it by up to 3 meV in the smallest supercells, where the pairing of neighbors in the frequency list is poor, and settles from $5 \times 5 \times 5$ cells on. The exact average lies 0.4 meV below the second-order one in the larger supercells: the terms of fourth order in the displacements are one percent of the shift, and the special displacement follows the exact average rather than its second-order part, as the argument above requires. For silicon the same convergence has been computed directly: a $4 \times 4 \times 4$ supercell is within ten percent of the converged gap renormalization, and an $8 \times 8 \times 8$ supercell of 1024 atoms gives -57 meV (Worked example E.7). In practice the calculation needs the phonons of the primitive cell on the commensurate grid, one density-functional calculation of the displaced supercell, and an unfolding of its bands onto the primitive zone. The convergence is checked by bracketing, with and without the self-partner wavevectors, and Appendix E.8 gives the procedure and its costs.

What the average contains follows from the same expansion applied to S^{cl}. For a zero-phonon pair f of the clamped crystal the energy $E_f(\{x\})$ appears inside a delta function, and the second-order term shifts it by $\frac{1}{2}\sum_j \sigma_j^2\, \partial_j^2 E_f$, the frozen-phonon form of Eq. (6.22): the zero-phonon line comes out at the adiabatic Allen–Heine–Cardona energies, with a Gaussian width from the first-order terms [17, Eqs. (19)–(21)]. A pair whose crystal momentum differs from $\boldsymbol{q}$ by a phonon wavevector has no transition density in the clamped crystal, and acquires one, linear in x_j, in the displaced supercell. Its squared density averages to $\sigma_j^2 |\partial_j \rho_f|^2$, and $\partial_j \rho_f$ is the adiabatic amplitude M^{ad} of Eq. (13.12) by Eq. (13.13). To second order in the displacements,

$$S^{\mathrm{WL}}(\boldsymbol{q},\omega;T) = S^{0}_{\mathrm{ad}}(\boldsymbol{q},\omega;T) + \underbrace{\frac{4\pi}{\Omega}\Big\langle\Big\langle \sum_{\nu cv} (2n_{\boldsymbol{Q}\nu}+1)\,\big|M^{\mathrm{ad}}_{cv\nu}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q})\big|^2\,\delta\big(\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} - \omega\big)\Big\rangle\Big\rangle}_{S^{\mathrm{PA}}_{\mathrm{ad}}(\boldsymbol{q},\omega;T)} + O(\sigma^4), \tag{13.38}$$

with S^{0}_{ad} the zero-phonon response at the adiabatic renormalized energies and the double average over $\boldsymbol{k}$ and $\boldsymbol{Q}$ as in Eq. (13.15). The second term, $S^{\mathrm{PA}}_{\mathrm{ad}}$, is the adiabatic limit of S^{PA}, the Hall–Bardeen–Blatt sideband [10] with the phonon energy dropped from the delta function [17, Eq. (24)]. The Williams–Lax average, and with it a special-displacement calculation, therefore contains both electronic sectors at once. Adding S^{PA} to it counts the sideband twice. A calculation that wants the cheap renormalized zero-phonon line and the exact sideband must subtract the adiabatic sideband computed with the same vertices, $S^{\mathrm{WL}} + S^{\mathrm{PA}} - S^{\mathrm{PA}}_{\mathrm{ad}}$, or use Eq. (13.35) with Eq. (13.15) from the start.

The sideband of Eq. (13.38) has the right total weight and the wrong position. Compared

with Eq. (13.15) in the adiabatic amplitude,

$$S^{\mathrm{PA}} - S^{\mathrm{PA}}_{\mathrm{ad}} = \frac{4\pi}{\Omega}\Big\langle\Big\langle \sum_{\nu cv} |M^{\mathrm{ad}}|^2 \big[(n+1)\,\delta(E-\omega+\omega_{\boldsymbol{Q}\nu}) + n\,\delta(E-\omega-\omega_{\boldsymbol{Q}\nu}) - (2n+1)\,\delta(E-\omega)\big]\Big\rangle\Big\rangle + O\Big(\frac{\omega_{\boldsymbol{Q}\nu}}{\delta_{\mathrm{off}}}\Big) S^{\mathrm{PA}}, \tag{13.39}$$

with $E = \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}}$, $n = n_{\boldsymbol{Q}\nu}$, and the last term the amplitude error of Eq. (13.14). The bracket integrates to zero over ω, so the weights agree; its first moment is $+\sum |M^{\mathrm{ad}}|^2 \omega_{\boldsymbol{Q}\nu}$, independent of temperature. At $T \gg \omega_{\boldsymbol{Q}\nu}$, or at energies far above the threshold, the three delta functions merge and the adiabatic sideband is exact. At 10 mK it is not. There $n = 0$, the whole sideband sits at E instead of $E + \omega_{\boldsymbol{Q}\nu}$, and the phonon-assisted onset falls by one phonon energy: from 0.3374 MeV to the floor $2E_{\mathrm{g}}/v^2_{\max} = 0.3332$ MeV (Worked example 13.1). Above the onset the adiabatic spectrum is that of the exact one shifted down by ω_ν. With the onset exponent $p_{\mathrm{on}} = 3$ of Eq. (13.19) the ratio at an energy x above the true onset is $(1+\omega_\nu/x)^3$, which falls below 1.1 only for $x > 31\,\omega_\nu$, about 0.46 eV for the transverse acoustic phonon of the onset. At $|\boldsymbol{q}| = 1$ keV the whole off-shell window is 0.28 eV wide along the valley axis and narrower off it (Sect. 13.6), so nowhere inside it does the adiabatic sideband describe the rate to ten percent. Special displacements give the renormalized zero-phonon line and the spectrum far from the edges; the phonon-assisted rate in the window needs the explicit phonon energies of Eq. (13.15).

13.7.3 The nonadiabatic correction to the renormalization

The adiabatic form Eq. (6.22) drops the phonon energy from the Fan–Migdal denominators. Restoring it at $T = 0$, with $\omega = \varepsilon_{n\boldsymbol{k}}$ in Eq. (6.19), an empty intermediate state enters only through phonon emission and an occupied one only through the absorption term with $f = 1$. The difference between the two forms is

$$\mathrm{ZPR}^{\mathrm{na}}_{n\boldsymbol{k}} - \mathrm{ZPR}^{\mathrm{ad}}_{n\boldsymbol{k}} = \frac{1}{N}\sum_{\boldsymbol{Q}\nu m} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2 \left[\frac{1}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} - s_m\omega_{\boldsymbol{Q}\nu}} - \frac{1}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}}\right], \tag{13.40}$$

with $s_m = +1$ for m empty and $s_m = -1$ for m occupied; the Allen–Heine term has no denominator and drops out. For an intermediate band far from $\varepsilon_{n\boldsymbol{k}}$ the bracket is smaller than the adiabatic term by $\omega_{\boldsymbol{Q}\nu}/|\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}|$, a few percent for silicon. The terms that matter are those in which the intermediate state is degenerate with $(n,\boldsymbol{k})$ at $\boldsymbol{Q} = 0$: the band itself and, at the valence maximum of silicon, its partners. Take the valence maximum as a parabolic band of mass m^*, so that $\varepsilon_{n\boldsymbol{k}} - \varepsilon_{n,\boldsymbol{k}+\boldsymbol{Q}} = Q^2/2m^*$, coupled by an optical phonon of energy ω_{O} with a vertex g_{O} that tends to a constant as $Q \to 0$. With $N^{-1}\sum_{\boldsymbol{Q}} = \Omega \int \mathrm{d}^3Q/(2\pi)^3$ the bracket integrates in closed form,

$$\frac{\Omega|g_{\mathrm{O}}|^2}{2\pi^2}\int_0^\infty \mathrm{d}Q\,Q^2\left[\frac{1}{Q^2/2m^* + \omega_{\mathrm{O}}} - \frac{2m^*}{Q^2}\right] = -\frac{\Omega|g_{\mathrm{O}}|^2}{2\pi^2}\,2m^*\int_0^\infty \frac{2m^*\omega_{\mathrm{O}}\,\mathrm{d}Q}{Q^2 + 2m^*\omega_{\mathrm{O}}} = -\frac{\Omega|g_{\mathrm{O}}|^2(2m^*)^{3/2}\sqrt{\omega_{\mathrm{O}}}}{4\pi}. \tag{13.41}$$

The correction lowers the valence maximum, and with it reduces the closing of the gap. It is not of order ω_{O} but of order $\sqrt{\omega_{\mathrm{O}}}$: the adiabatic integrand $2m^*/Q^2$ is integrable in three dimensions, but its weight sits at $Q \lesssim (2m^*\omega_{\mathrm{O}})^{1/2}$, exactly where the phonon energy is not small against the band dispersion. For silicon we take the optical vertex of the holes, $|g_{\mathrm{O}}| \approx 0.25$ eV (Sect. 13.6), $\omega_{\mathrm{O}} = 64.1$ meV, and $\Omega = a^3/4$. Three bands meet at the valence maximum without spin–orbit coupling, with the spherical masses $m_e/(\gamma_1^{\mathrm{Lu}} \mp 2\bar\gamma^{\mathrm{Lu}}) = 0.44$

and $0.16\,m_e$ of the heavy and light holes (Sect. 3.3) and $m_e/\gamma_1^{\mathrm{Lu}} = 0.23\,m_e$ for the third. Eq. (13.41) gives 2.0, 0.4 and 0.8 meV, together 3.2 meV.

The full calculation finds more. Poncé and co-workers evaluated both forms of the Allen–Heine–Cardona theory for silicon in the local-density approximation on the same grids, and their indirect-gap renormalization is −64.3 meV in the adiabatic form and −56.2 meV in the nonadiabatic one [19, Table VI], the 8 meV of Sect. 6.3. Their valence maximum rises by 40.9 meV in the adiabatic form and 34.8 meV in the nonadiabatic one [19, Sect. VI A 2], so 6.1 meV of the difference sits at the valence maximum, about twice the estimate, which rests on a single vertex for all three bands. The remaining 1.9 meV belong to the conduction minimum, which drops by 23.3 meV in the adiabatic form and 21.4 meV in the nonadiabatic one. Symmetry forbids an intravalley optical vertex there (Sect. 6.2), and the degenerate intermediate states are the minima of the other five valleys, reached by intervalley phonons, for which the same mechanism applies. The nonadiabatic correction to the gap renormalization of silicon, defined by Eq. (13.40) as the difference between the two forms of one calculation with the same vertices and grids, is therefore 8 meV, of which Eq. (13.41) accounts for the order of magnitude and the sign, and it is precisely what special displacements miss, since they are adiabatic (Limit Check 13.2). Differences between methods, such as the 9 meV between the nonadiabatic value −56 meV and an adiabatic supercell value −65 meV compiled by Miglio and co-workers [20, Table S2], mix the form of the theory with the method and its inputs, and do not measure this correction.

For the window the correction does not matter, because the window is computed from the measured gap, and a measured gap contains the whole renormalization, adiabatic or not. It would enter in two ways. A calculation that builds the gap from a clamped-nucleus value, from *GW* or a hybrid functional, plus a computed renormalization inherits the choice of form. Then 8 meV on the gap moves the phonon-assisted onset by $0.283\,\mathrm{MeV\,eV^{-1}} \times 8\,\mathrm{meV} = 0.0023\,\mathrm{MeV}$ and the zero-phonon edge by 0.0015 MeV, with the coefficients of Eq. (13.42) below. A calculation that scissors its bands to the measured gap misses only the variation of the renormalization between the band edges and the pair states that the zero-phonon edge reaches, about 0.2 eV above the gap (Sect. 13.9). Between the conduction minimum and the conduction state at Γ, 3.4 eV above the valence maximum, the nonadiabatic renormalization varies by 14 meV, from −21.4 to −7.4 meV [19, Table VI and Sect. VI A 2]. A variation of at most 10 meV over the reached states is therefore a generous bound, and it moves the zero-phonon edge by at most 0.002 MeV. Both numbers are below the spread of the computed edges themselves, 0.405 to 0.408 MeV (Sect. 13.9).

13.7.4 Temperature dependence of the window

At detector temperatures the window does not move. The measured gap $E_{\mathrm{g}}(T) = E_{\mathrm{g}}(0) - aT^2/(T+b)$ of Worked example 6.3, with $a = 4.9 \times 10^{-4}\,\mathrm{eV\,K^{-1}}$ and $b = 655\,\mathrm{K}$, closes by 0.07 meV at 10 K and by less than 1 µeV at 1 K, and the physical shift is smaller still, since the renormalization changes with the occupations $n_{\boldsymbol{Q}\nu}$, which vanish exponentially. For the transverse acoustic phonon of the onset, $n_{\mathrm{B}}(14.8\,\mathrm{meV})$ is 3×10^{-8} at 10 K and 2×10^{-4} at 20 K, so the emission factor $n+1$ is one and the absorption sideband of Sect. 13.1 is absent below about 20 K. The margin δ_{w} of the window is measured from the renormalized edge at each temperature, and at these temperatures nothing in it depends on T. A detector between 10 mK and a few kelvin sees the zero-temperature window: the onset at 0.3374 MeV, its bound-pair replica at 0.3332 MeV, and the zero-phonon edge at 0.405 to 0.408 MeV with its 1*s* line at 0.402 to 0.405 MeV.

At higher temperatures the gap closes, and the edges follow it. A uniform shift $\Delta E_{\mathrm{g}}(T) = E_{\mathrm{g}}(T) - E_{\mathrm{g}}(0)$ of all pair energies moves each threshold mass by the derivative of Eq. (13.1)

or Eq. (13.3) with respect to the threshold energy, taken at the optimal momentum transfer, where the shift of the optimum itself does not contribute to first order. For a threshold E reached at q this derivative is $q^2/[2(qv_{\max} - E)^2]$, and

$$\frac{\partial m_{\min}^{\mathrm{PA}}}{\partial E_{\mathrm{g}}} = \frac{2}{(v_{\max} + u_\nu)^2} = 0.283\,\mathrm{MeV\,eV^{-1}}, \qquad \frac{\partial m_{\min}^{\mathrm{ZP}}}{\partial E_{\mathrm{g}}} = \frac{2\,(m_{\min}^{\mathrm{ZP}})^2}{q_{\mathrm{ZP}}^2} = 0.187\,\mathrm{MeV\,eV^{-1}}. \tag{13.42}$$

For the onset we have used $q^* = 2E/(v_{\max} - u_\nu)$ from Sect. 13.1, with the phonon energy falling as q grows, $E' = -u_\nu$; the group velocity enters to first order here, although it entered the onset itself only squared, and lowers the coefficient by 0.5 % from $2/v_{\max}^2 = 0.285\,\mathrm{MeV\,eV^{-1}}$. For the zero-phonon edge $q_{\mathrm{ZP}} = 1.33\,\mathrm{keV}$ is the momentum transfer of the joint minimum, re-minimized on bands with spin–orbit coupling at the measured lattice constant (Sect. 13.9). The zero-phonon edge moves less than the onset, because it is reached at a larger momentum transfer, where the particle has more energy to spare, and the window widens slightly as it moves down. Two effects come on top of the rigid shift. Absorption sidebands open below the onset, with the weights $n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$, and the phonon energies themselves shift, but a change of one percent in $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$ would move the onset by only 4×10^{-5} MeV. At 300 K the gap falls by $0.26\,\mathrm{meV\,K^{-1}}$, and the onset by 7×10^{-5} MeV per kelvin. Fig. 13.10 shows the gap and the edges from 10 mK to 300 K.

Worked example 13.7: The window shift between 10 mK and 300 K

Gap. At 10 mK, $\Delta E_{\mathrm{g}} = 0$ to 10^{-10} eV. At 300 K, $\Delta E_{\mathrm{g}} = -4.9 \times 10^{-4}\,\mathrm{eV\,K^{-1}} \times (300\,\mathrm{K})^2/955\,\mathrm{K} = -46.2\,\mathrm{meV}$; with a and b varied by their errors, -42.5 to -50.2 meV.
Onset. With $E = E_{\mathrm{g}} + \omega_{\mathrm{TA}}$ lowered by 46.2 meV, the optimum q^* falls from 0.90 to 0.86 keV, the phonon wavevector grows from 0.46 to $0.47\,(2\pi/a)$, and the TA energy there rises from 14.78 to 15.01 meV. Then

$$m_{\min}^{\mathrm{PA}}(300\,\mathrm{K}) = \frac{2 \times (1.17\,\mathrm{eV} - 0.0462\,\mathrm{eV} + 0.015\,01\,\mathrm{eV})}{7.023 \times 10^{-6}} = 0.3243\,\mathrm{MeV},$$

0.0131 MeV below the 0.3374 MeV at 10 mK; Eq. (13.42) gives $0.283\,\mathrm{MeV\,eV^{-1}} \times 46.2\,\mathrm{meV} = 0.0131\,\mathrm{MeV}$ directly. The bound-pair replica moves from 0.3332 to 0.3201 MeV.
Zero-phonon edge. At $q_{\mathrm{ZP}} = 1.33$ keV the edge moves by $0.187\,\mathrm{MeV\,eV^{-1}} \times 46.2\,\mathrm{meV} = 0.0086$ MeV; evaluated exactly at that momentum transfer, 0.405 to 0.408 MeV becomes 0.3966 to 0.3995 MeV, the best value moves from 0.407 to 0.3985 MeV, and the 1s line from 0.402–0.405 to 0.394–0.397 MeV.
Width and sidebands. The window, from the onset to the best zero-phonon edge, spans 0.0696 MeV at 10 mK and 0.0742 MeV at 300 K. At 300 K the absorption sidebands are open: with a TA phonon of 15.2 meV absorbed, occupation 1.25, from 0.3157 MeV; with an optical phonon of 61.0 meV, occupation 0.10, from 0.3027 MeV. At 77 K the shifts are -0.0011 (onset) and -0.0007 MeV (zero-phonon edge), and the TA absorption sideband, occupation 0.12, starts at 0.3278 MeV.
Consequence. A room-temperature calculation of the window, or room-temperature absorption data used for it, places both edges about 0.01 MeV too low and adds a thermal tail that reaches 0.035 MeV below the cold onset with one phonon absorbed, and further with several. The shift is thirty times the 4×10^{-4} MeV by which the softness of the computed TA branch moves the onset (Worked example 13.1), and it has to be removed before a warm number describes a detector at 10 mK.

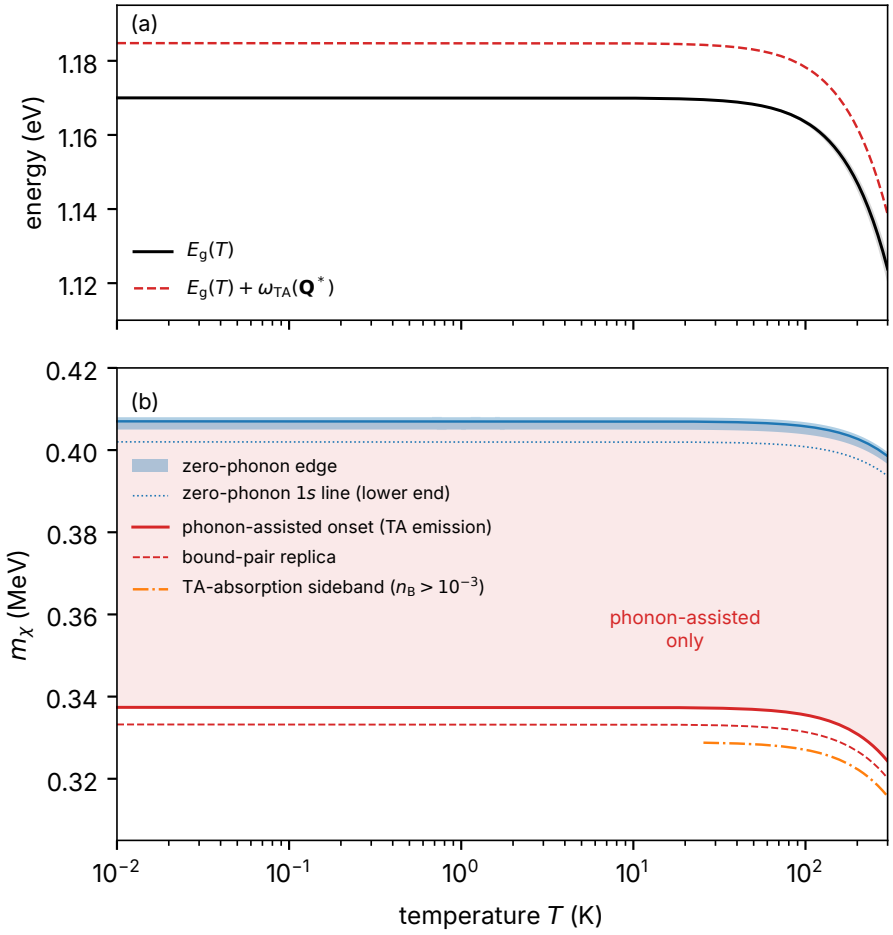


Figure 13.10: The silicon window against temperature, on a logarithmic temperature axis from 10 mK to 300 K. (a) The indirect gap $E_g(T) = 1.17\,\text{eV} + \Delta E_g(T)$, with $\Delta E_g(T)$ from the fit $E_g(0) - aT^2/(T+b)$, $a = 4.9 \times 10^{-4}\,\text{eV K}^{-1}$, $b = 655\,\text{K}$, to photoluminescence data [21, 22, Eq. (11)] (gray: a and b varied by their quoted errors, $0.2 \times 10^{-4}\,\text{eV K}^{-1}$ and 40 K); dashed: the threshold of the phonon-assisted onset, $E_g(T) + \omega_{TA}(\boldsymbol{Q}^*)$. (b) Window edges in dark-matter mass. Red: the phonon-assisted onset, transverse acoustic emission, from Eq. (13.3) with the PBE phonons of Sect. 5.2 and $\boldsymbol{q}$ along the valley axis; red dashed: its bound-pair replica, with $E_g - E_x^b$, $E_x^b = 14.7\,\text{meV}$. Orange dash-dotted: the sideband with a transverse acoustic phonon absorbed, drawn where its occupation exceeds 10^{-3}. Blue band: the zero-phonon edge, 0.405 to 0.408 MeV at low temperature (Sect. 13.9), moved with the gap at the momentum transfer of the joint minimum, 1.33 keV (bands with spin–orbit coupling, Sect. 13.9); blue dotted: the lower end of its 1s line. Shaded red: masses at which phonon-assisted recoil is the only electron-recoil channel. Every shift is a rigid shift of the pair energies with the measured gap.

13.8 Exciton replicas and bosonic absorption

Cold silicon begins to absorb light 14.7 meV below the free-pair threshold $E_\mathrm{g}(0) + \omega_\mathrm{TA} = 1.1875\,\mathrm{eV}$ of Sect. 13.5. The electron and the hole that a photon creates together with a transverse acoustic phonon can stay bound, by $E_x^b = 14.7\,\mathrm{meV}$ (Sects. 4.9 and 6.5). With the measured gap $E_\mathrm{g}(0) = 1.1692\,\mathrm{eV}$ of Sect. 6.3 the bound threshold is $E_\mathrm{g}(0) + \omega_\mathrm{TA} - E_x^b = 1.1728\,\mathrm{eV}$. Every phonon-assisted final state of this chapter has such a bound companion. In the rate of dark-matter scattering the companions are replicas of the exciton lines, one per phonon branch, which open below the free-pair onset of Sect. 13.1. For a boson that is absorbed they matter more: between the indirect gap and the direct gap at 3.4 eV absorption needs a phonon, and at 1.2 eV the bound pairs carry most of it. We rebuild the amplitude of Sect. 13.2 from exciton states, state when the rebuilt amplitude is complete, reduce it near the edge to a closed form, and work out the absorption of dark photons, axion-like particles and scalars at the silicon edge.

13.8.1 Exciton–phonon coupling and the replica amplitude

The Bethe–Salpeter equation of Sect. 4.9 at total crystal momentum $\boldsymbol{P}$ has the eigenstates

$$|\lambda(\boldsymbol{P})\rangle = \sum_{cv\boldsymbol{k}} A^{\lambda}_{cv\boldsymbol{k}}(\boldsymbol{P})\, c^\dagger_{c,\boldsymbol{k}+\boldsymbol{P}} c_{v\boldsymbol{k}}\,|0\rangle,$$

with the hole at $\boldsymbol{k}$, the electron at $\boldsymbol{k}+\boldsymbol{P}$, and the energies $E_\lambda(\boldsymbol{P})$. They include the bound lines and the continuum, in which each free pair is dressed by the attraction. The density creates them from the ground state with $\langle\lambda|n_e(\boldsymbol{q})|0\rangle = \sqrt{g_\mathrm{deg}}\,\rho_{\lambda 0}(\boldsymbol{q})$, the exciton form factor Eq. (10.16). The electron–phonon Hamiltonian Eq. (A.25) scatters the electron or the hole of an exciton, and it creates an exciton from the ground state. With the factor $N^{-1/2}$ and the phonon matrix element taken out, as for $g_{mn\nu}$, the matrix elements between the one-pair states above are

$$\begin{aligned} g^x_{\lambda\lambda'\nu}(\boldsymbol{q},\boldsymbol{Q}) &= \sum_{cv\boldsymbol{k}} A^{\lambda*}_{cv\boldsymbol{k}}(\boldsymbol{q}+\boldsymbol{Q})\Big[\sum_{c'} g_{cc'\nu}(\boldsymbol{k}+\boldsymbol{q},\boldsymbol{Q})\, A^{\lambda'}_{c'v\boldsymbol{k}}(\boldsymbol{q}) - \sum_{v'} A^{\lambda'}_{cv',\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{q})\, g_{v'v\nu}(\boldsymbol{k},\boldsymbol{Q})\Big], \\ g^x_{\lambda 0\nu}(\boldsymbol{Q}) &= \sum_{cv\boldsymbol{k}} A^{\lambda*}_{cv\boldsymbol{k}}(\boldsymbol{Q})\, g_{cv\nu}(\boldsymbol{k},\boldsymbol{Q}), \end{aligned} \tag{13.43}$$

for an exciton λ' at $\boldsymbol{q}$ scattered into λ at $\boldsymbol{q}+\boldsymbol{Q}$, and for an exciton created at $\boldsymbol{Q}$. The first is the exciton–phonon vertex of Antonius and Louie [23, Eq. (49)]: the phonon vertex of the electron, or of the hole, weighted by the envelopes of the two excitons. The hole term carries the minus sign that the reordering of fermion operators gave the hole paths in Sect. 13.2. The density vertex between two excitons, from λ' at $\boldsymbol{Q}$ to λ at $\boldsymbol{q}+\boldsymbol{Q}$, has the same structure,

$$\rho_{\lambda\lambda'}(\boldsymbol{q}) = \sum_{cv\boldsymbol{k}} A^{\lambda*}_{cv\boldsymbol{k}}(\boldsymbol{q}+\boldsymbol{Q})\Big[\sum_{c'} \rho_{cc'}(\boldsymbol{k}+\boldsymbol{Q},\boldsymbol{q})\, A^{\lambda'}_{c'v\boldsymbol{k}}(\boldsymbol{Q}) - \sum_{v'} A^{\lambda'}_{cv',\boldsymbol{k}+\boldsymbol{q}}(\boldsymbol{Q})\, \rho_{v'v}(\boldsymbol{k},\boldsymbol{q})\Big].$$

The second-order amplitude Eq. (13.7) is now evaluated with exciton states. The final state holds the exciton λ at $\boldsymbol{q}+\boldsymbol{Q}$ and one phonon more in $(-\boldsymbol{Q},\nu)$ or one fewer in $(\boldsymbol{Q},\nu)$. With the density first, the intermediate exciton λ' sits at $\boldsymbol{q}$ and the state holds the energy ω. With the phonon first, it sits at $\boldsymbol{Q}$ and the phonon has already been emitted or absorbed:

Exciton–phonon replica amplitude

$$\langle f|n_e^{(2)}(\boldsymbol{q})|i\rangle = \sqrt{\frac{g_{\mathrm{deg}}}{N}}\,(n_{\boldsymbol{Q}\nu}+\tfrac{1}{2}\pm\tfrac{1}{2})^{1/2} M^{\pm}_{\lambda\nu}(\boldsymbol{q},\boldsymbol{Q}),$$
$$M^{\pm}_{\lambda\nu}(\boldsymbol{q},\boldsymbol{Q}) = \sum_{\lambda'}\left[\frac{g^x_{\lambda\lambda'\nu}(\boldsymbol{q},\boldsymbol{Q})\,\rho_{\lambda'0}(\boldsymbol{q})}{\omega - E_{\lambda'}(\boldsymbol{q})} + \frac{\rho_{\lambda\lambda'}(\boldsymbol{q})\,g^x_{\lambda'0\nu}(\boldsymbol{Q})}{-E_{\lambda'}(\boldsymbol{Q})\mp\omega_{\boldsymbol{Q}\nu}}\right], \tag{13.44}$$

on the energy shell $\omega = E_\lambda(\boldsymbol{q}+\boldsymbol{Q})\pm\omega_{\boldsymbol{Q}\nu}$, with λ' running over every exciton state, bound or not, of every pair of bands. The phonon-assisted structure factor with excitonic final states is

$$S^{\mathrm{PA}}(\boldsymbol{q},\omega) = \frac{2\pi g_{\mathrm{deg}}}{N\mathscr{V}}\sum_{\lambda\boldsymbol{Q}\nu\pm}(n_{\boldsymbol{Q}\nu}+\tfrac{1}{2}\pm\tfrac{1}{2})\,\bigl|M^{\pm}_{\lambda\nu}(\boldsymbol{q},\boldsymbol{Q})\bigr|^2\,\delta(E_\lambda(\boldsymbol{q}+\boldsymbol{Q})\pm\omega_{\boldsymbol{Q}\nu}-\omega), \tag{13.45}$$

which replaces Eq. (13.15) near the edge; it does not add to it.

The spin enters once, through the singlet exciton created by the density or by the phonon; the scattering vertices conserve it. For a bound exciton $M^{\pm}_{\lambda\nu}$ grows as $\mathscr{V}^{1/2}$, as $\rho_{\lambda0}$ does (Sect. 10.5), and S^{PA} is intensive.

Without the attraction, Eq. (13.44) is the amplitude of Sect. 13.2. The exciton states are then single pairs, $A^{\lambda}_{cv\boldsymbol{k}} = 1$ for one pair and zero otherwise, and the exciton sums become band sums. The electron term of the first part is then path A of Eq. (13.8) and its hole term path C; in the second part the electron term is path B and the hole term path D, each with its sign and its denominator. Eq. (13.44) is the four-path amplitude written in exciton states, and on the energy shell the band-sum theorem of Sect. 13.2 turns it into Eq. (13.9). The sum over final pairs $(c,v,\boldsymbol{k})$ then carries the factor $(2\pi/\mathscr{V})\times 2\times N^{-1}\sum_{\boldsymbol{kQ}}$, and Eq. (13.45) becomes Eq. (13.15). At $\boldsymbol{q}=0$ the form factor $\rho_{\lambda'0}$ vanishes, and in $\rho_{\lambda\lambda'}(0)$ the electron and hole terms are both $\delta_{\lambda\lambda'}$ and cancel. The amplitude vanishes, as Eq. (13.10) requires of any phonon-assisted density amplitude.

Screening enters as in Sect. 13.4. The density vertex carries the external factor $1/\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega)$, and the exchange interaction of the final pair at its momentum contains the pair channel. What Eq. (13.44) adds is the direct kernel, the electron–hole attraction of Eq. (4.104), in both the intermediate and the final states. That is the excitonic part of the ladder that Sect. 13.4 left out.

The vertices Eq. (13.43) need the electron–phonon vertices and the exciton envelopes in one gauge: a Bloch state enters g and A^{λ} with the same phase, and a phase changed in one but not the other changes $|g^x|$. In practice the vertices come from Wannier interpolation in the gauge of the interpolated bands, and the Bethe–Salpeter equation is solved on the same states. It is needed at the momenta of the silicon pockets, with about 10^3 points of $\boldsymbol{k}$ per pocket to resolve an exciton of radius $a_x = 4.2\,\mathrm{nm}$. Appendix E.15 specifies that calculation. For silicon and germanium it is open, and a converged replica spectrum from it is the open problem of this section.

13.8.2 Completeness and the two-pair test

The sums over λ' in Eq. (13.44) run over every one-pair state at $\boldsymbol{q}$ and at $\boldsymbol{Q}$, including the pairs whose hole lies deep in the valence band, which the hole terms of g^x and $\rho_{\lambda\lambda'}$ reach. Dropping those hole terms is the exciton version of truncating the band sum to empty bands, which cost a factor of 2.4 in the model of Worked example 13.2, with the phase of its displacement wave at the cell origin. A Bethe–Salpeter basis of a few bands on each side of the gap describes the final exciton well, but not the intermediate ones, which reach every band. The f-sum test

of the density vertices (Appendix E.5) applies to the intermediate sums here as it does to Eq. (13.9).

The space of one pair must also be the whole space of intermediate states. For independent electrons every intermediate state of Eq. (13.7) holds exactly one pair, because each vertex moves one electron (Sect. 13.2). With the electron–hole interaction this is no longer exact. The Coulomb interaction contains terms that create a pair while they scatter an electron or a hole, so an exact eigenstate with one pair carries an admixture of states with two pairs, and the exact ground state an admixture of virtual pairs. Eq. (13.44) keeps the Tamm–Dancoff space of one pair for both the intermediate and the final excitons. It is complete if the states with two pairs add nothing beyond the one-pair terms. Without interaction they do not exist, and the formula reduces to the four paths exactly. With the interaction it is unproven, and the completeness of the replica expansion reduces to the size of the two-pair sector. Its parametric estimate is the ratio of the binding energy to the gap, $E_x^b/E_\mathrm{g} \approx 1\,\%$ in silicon, an estimate and not a bound.

A model with an exact two-pair sector would size this correction: a two-band chain of 8 to 12 cells with a contact attraction between electron and hole and one phonon mode, diagonalized with its states of zero, one and two pairs and compared with Eq. (13.44) built from its exact one-pair excitons. Appendix E.15 specifies it.

13.8.3 Replicas near the silicon edge

Near the edge the final exciton lies within a few E_x^b of the band extrema, while every intermediate exciton is far from the energy shell. By Eq. (13.11) the distance is at least 0.7 eV at the onset of Worked example 13.1, and 1.11 eV for absorption between the indirect and the direct gap. This separation of scales leads to a closed form. The vertices $g^x_{\lambda\lambda'\nu}$ and $\rho_{\lambda\lambda'}$ are linear in the final envelope $A^{\lambda*}(\boldsymbol{q}+\boldsymbol{Q})$, so that

$$M^{\pm}_{\lambda\nu}(\boldsymbol{q},\boldsymbol{Q}) = \sum_{\boldsymbol{k}} A^{\lambda*}_{cv\boldsymbol{k}}(\boldsymbol{q}+\boldsymbol{Q})\, M^{x\pm}_{cv\nu}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q}),$$

where $M^{x\pm}$ is the amplitude into the free final pair $(c, \boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q};\, v, \boldsymbol{k})$ with exciton intermediate states. The final pair enters $M^{x\pm}$ also through the energy ω in the denominators, which moves by at most E_x^b between a bound and a free final pair, a relative change $E_x^b/\delta_\mathrm{off}$ of about 2 % at the onset and less in absorption. With free intermediate pairs, $M^{x\pm}$ is the $M^{\pm}$ of Eq. (13.9). The attraction among the intermediate pairs changes it, as it changes the optical spectrum of silicon near its main peak (Sect. 4.9). It does so smoothly across the zone, and changes the size of $M^{x\pm}$ at the extrema, not its variation over the small region of $\boldsymbol{k}$ that the envelope covers. With $M^{x\pm}$ held at its value M^x_ν at the extrema and $\sum_{\boldsymbol{k}} A^{\lambda*}_{\boldsymbol{k}} = \mathcal{V}^{1/2} F^*_\lambda(0)$ (Sect. 4.9),

$$M^{\pm}_{\lambda\nu}(\boldsymbol{q},\boldsymbol{Q}) \simeq \mathcal{V}^{1/2}\, F^*_\lambda(0)\, M^x_\nu, \tag{13.46}$$

the Elliott form Eq. (4.111) with the phonon-assisted amplitude in place of the density vertex [24]. Only s states contribute, and the shape of the edge is fixed by the final pair alone.

The final excitons are labeled by their relative motion, a bound level n or a continuum state, and by their total momentum $\boldsymbol{P} = \boldsymbol{q}+\boldsymbol{Q}$, with energy $E_\mathrm{g} + E^\mathrm{rel} + (\boldsymbol{P}-\boldsymbol{k}_\mathrm{c})^2/2M_\mathrm{eh}$ near a valley by Eq. (4.108), where M_eh is the translational mass of the pair (Sect. 13.1), taken isotropic. The sum over $\boldsymbol{Q}$ in Eq. (13.45) is a sum over $\boldsymbol{P}$. Insert Eq. (13.46) and take $T \to 0$, so that only emission survives. The factors combine as $(2\pi g_\mathrm{deg}/N\mathcal{V})\,\mathcal{V}\sum_{\boldsymbol{P}} = (4\pi/N)\, N\Omega \int \mathrm{d}^3\kappa/(2\pi)^3 = 4\pi\Omega \int \mathrm{d}^3\kappa/(2\pi)^3$, with $\boldsymbol{\kappa} = \boldsymbol{P} - \boldsymbol{k}_\mathrm{c}$. The sum of $|F_\lambda(0)|^2$ over the relative states at fixed $E^\mathrm{rel} = E$ is the bracket of Eq. (10.18), which we write $\mathcal{W}_x(E)$: lines $\delta(E + E_x^b/n^2)/(\pi a_x^3 n^3)$ below zero and the continuum $(2\mu_x)^{3/2}E^{1/2}F_\mathrm{Fermi}/4\pi^2$ above, with

the Fermi factor Eq. (2.25) at $\xi_{\rm C} = (E^b_x/E)^{1/2}$. The integral over $\boldsymbol{\kappa}$ is the density of states of the center-of-mass motion, which the phonon leaves free. Per valley, branch and hole band,

$$S^{\rm PA}(\boldsymbol{q},\omega) \simeq 4\pi\Omega\,|M^x_\nu|^2 \int {\rm d}E\; \mathcal{W}_x(E)\,\frac{(2M_{\rm eh})^{3/2}(x-E)^{1/2}}{4\pi^2}, \qquad x = \omega - E^{\rm PA}_{\rm min}(\boldsymbol{q}),$$

with $E^{\rm PA}_{\rm min}(\boldsymbol{q})$ the free-pair threshold of Eq. (13.2). Without the attraction, $\mathcal{W}_x$ is the free continuum alone, $\int_0^x {\rm d}E\, E^{1/2}(x-E)^{1/2} = \pi x^2/8$, and the result is $\Omega(\mu_x M_{\rm eh})^{3/2}|M_\nu|^2 x^2/4\pi^2$. That is Eq. (13.17), because $\mu_x M_{\rm eh}$ is the product of the electron and hole masses for parabolic bands. Divided by this free-pair edge, the masses cancel. With $(2\mu_x)^{3/2} = 1/[a_x^3 (E^b_x)^{3/2}]$ from Eq. (4.107), each line weight $1/(\pi a_x^3 n^3)$ is $4\pi (E^b_x)^{3/2}/n^3$ in units of the continuum coefficient. Normalizing by $\pi x^2/8$ then gives the edge in closed form.

Replica edge in the Wannier–Mott limit

Near the threshold of one branch, at fixed $\boldsymbol{q}$ and $T \to 0$, a phonon-assisted spectrum whose free-pair form is $A_\nu x^2$ becomes, with excitonic final states,

$$A_\nu \Big[32\,(E^b_x)^{3/2} \sum_{n\geq 1} \frac{(x + E^b_x/n^2)^{1/2}}{n^3} + \frac{8}{\pi}\int_0^x {\rm d}E\; E^{1/2}(x-E)^{1/2}\,\frac{2\pi\xi_{\rm C}}{1-{\rm e}^{-2\pi\xi_{\rm C}}}\Big], \quad (13.47)$$

with $\xi_{\rm C} = (E^b_x/E)^{1/2}$ and each line term counted for $x > -E^b_x/n^2$. The coefficient A_ν is the one of Eq. (13.17) for dark-matter scattering and of Eq. (6.44) for light.

Each bound level becomes a replica edge that opens at $E^{\rm PA}_{\rm min}(\boldsymbol{q}) - E^b_x/n^2$ and rises as a square root: a line of the relative motion, spread by the center-of-mass motion. The continuum starts from zero at the free-pair threshold, but as $(32/3)(E^b_x)^{1/2}x^{3/2}$ instead of x^2, since the Fermi factor makes its relative-motion weight finite at the edge. These are the exponents that Sect. 13.3 anticipated: $\frac{1}{2}$ for a replica and $\frac{3}{2}$ for the excitonic continuum at fixed $\boldsymbol{q}$, and one power more each after the average over the directions of $\boldsymbol{q}$ (Eq. (13.19)). The strength follows from the ratio of the $1s$ term to the free one at equal distance x from their own thresholds, $32(E^b_x/x)^{3/2}$: 32 at $x = E^b_x$ and 5.1 at 50 meV. The weight of the relative motion is redistributed as in Sect. 10.5: the lines together hold what the free pairs hold within $8.0\,E^b_x$ of the edge, and the Fermi factor keeps the continuum above its free value far beyond.

Fig. 13.11a shows Eq. (13.47) for cold silicon. The $1s$ replica of the transverse acoustic branch opens at 1.1728 eV and the $2s$ replica at 1.1838 eV, both below the free-pair threshold at 1.1875 eV; the transverse optical replica opens at 1.2122 eV. With the same $|M^x_\nu|^2$ the excitonic edge of the two branches exceeds the free-pair one by a factor of 81 at 1.200 eV, 16 at 50 meV above the free threshold and 7.3 at 100 meV, and at 1.200 eV the $1s$ replica holds three quarters of it.

The measured spectra test the shape. Fitted in the same interval of the 300 K data, the excitonic form needs a scale 0.154 times that of the free-pair form, since the Fermi factor already supplies part of the absorption there. Neither form follows the data over the whole range (Fig. 13.11b). The free-pair form falls short within 40 meV of the gap, by the factor 2.0 at 17 meV found in Sect. 13.5. The hydrogenic form overshoots there by up to 1.8 and falls short by 1.5 to 2.5 between 1.3 and 1.6 eV, where the parabolic bands and the hydrogenic continuum of Sect. 10.5 no longer hold. The edge region is where the excitonic form belongs, and there the measurements at low temperature are the test. At 1.21 eV, the lowest energy measured at 0.5 K, the fit to the measurements, evaluated at that temperature, is $6.4\,{\rm cm}^{-1}$, the excitonic form 4.0 and the free-pair form $0.77\,{\rm cm}^{-1}$. With the decrease of the branch strength between the lowest temperature and 295 K found in Sect. 13.5, a factor 0.61, the

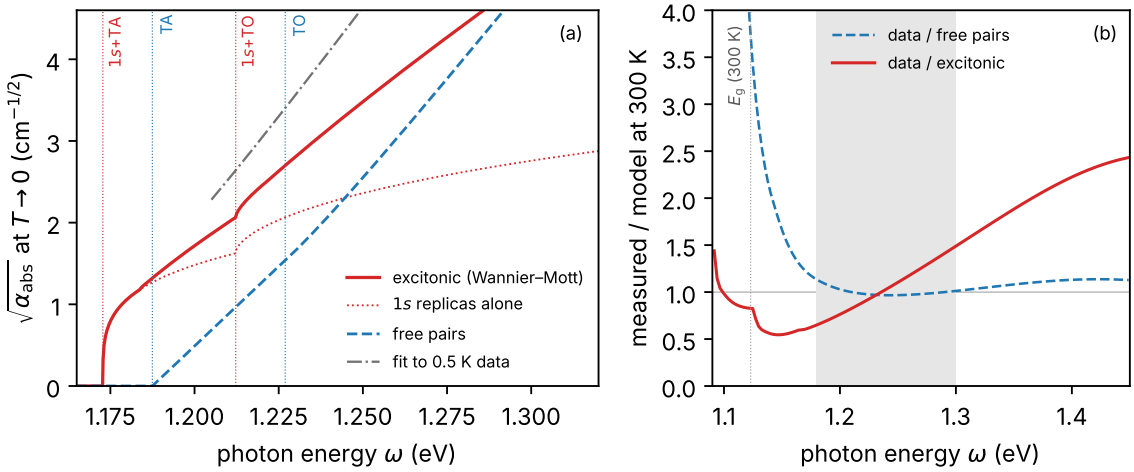


Figure 13.11: Phonon-assisted absorption edge of silicon with excitonic final states. (a) $\sqrt{\alpha_{\mathrm{abs}}}$ at $T \to 0$ against the photon energy. Red: Eq. (13.47) for the transverse acoustic and transverse optical branches, $\omega_{\mathrm{TA}} = 18.27\,\mathrm{meV}$ and $\omega_{\mathrm{TO}} = 57.73\,\mathrm{meV}$, with weights 5.5 : 4.0 [12, 13], $E_{\mathrm{g}}(0) = 1.1692\,\mathrm{eV}$ and $E_x^b = 14.7\,\mathrm{meV}$ for both hole bands; dotted, the two $1s$ replicas alone. Blue, dashed: the free-pair edge of Sect. 13.5. Each model has one scale, fitted as in Sect. 13.5 to the measured 300 K absorption between 1.18 and 1.30 eV [11]. Gray, dash-dotted: the free-pair form fitted to measurements between 0.5 and 295 K, with an effective gap of 1.134 eV, evaluated at 0.5 K and drawn above 1.205 eV, the lowest photon energy measured [13, Eqs. (4)–(7)]. Vertical lines: the $1s$ replica thresholds $E_{\mathrm{g}} - E_x^b + \omega_\nu$ (red) and the free-pair thresholds $E_{\mathrm{g}} + \omega_\nu$ (blue). (b) Measured absorption at 300 K divided by each model at 300 K, both branches with their phonon occupations; gray band, the fitted interval. Effective-mass models with constant matrix elements, not calculations.

excitonic value becomes 6.6 and the free-pair one 1.3. The fit itself needs an effective gap of 1.134 eV, 35 meV below the true one: a free-pair form can follow an excitonic edge only by moving its threshold down.

For dark-matter scattering the same construction applies at each $\boldsymbol{q}$, with A_ν the coefficient of Eq. (13.17) and the thresholds $E^{\mathrm{PA}}_{\mathrm{min}}(\boldsymbol{q}) - E^b_x/n^2$. The binding energy is that of the pair at the momentum $\boldsymbol{q} + \boldsymbol{Q}^*$, near $\boldsymbol{k}_{\mathrm{c}}$, where the hole sits near the valence maximum and the measured 14.7 meV applies; its variation along the dispersion of the pair is Eq. (E.90). The lightest mass at which the $1s$ replica opens follows from Eq. (13.4) with E_{g} replaced by $E_{\mathrm{g}} - E^b_x$, as Sect. 13.1 found, and its value and status belong to Sect. 13.9.

13.8.4 Phonon-assisted absorption of bosonic dark matter

Absorption of a boson probes the response at $\boldsymbol{q} = \boldsymbol{p}_\phi$, about $10^{-3} m_\phi$, and $\omega = m_\phi$ (Eq. (8.19)). In silicon between the excitonic edge at 1.1728 eV and the direct gap $E_{\mathrm{dir}} = 3.4$ eV no vertical pair exists. Every first-order transition at $\boldsymbol{q} \simeq 0$ is then off the energy shell, and the window of Eq. (13.32), with the vertical gap as the zero-phonon edge, extends to $E_{\mathrm{dir}} - \delta_{\mathrm{w}} = 3.1$ eV. Inside it every response row is phonon-assisted. Its structure factor is Eq. (13.15) at $\boldsymbol{q} = 0$ with the vertices of the two rows in place of ρ, or Eq. (13.45) near the edge. Each of the three rates of Sect. 10.7 therefore holds there with its response replaced by the phonon-assisted one. The bosons differ in whether that response is measured.

The dark photon has no phonon-assisted amplitude of its own at first order in g. It couples to the electromagnetic current of electrons and nuclei with ζ times the charge of the photon, and every second-order amplitude into a pair and one phonon reduces to that of light or vanishes. The electron current with one electron–phonon vertex is the amplitude of light, Eq. (6.41) with the velocity vertex, and its square summed over final states is the measured $\varepsilon_2^{\mathrm{PA}}$. The longitudinal polarization couples to the charge density at $\boldsymbol{q} = \boldsymbol{p}_\phi$, and the phonon-assisted density amplitude there is, by continuity, (q/ω) times the velocity amplitude (Eq. (13.27)). It is the same amplitude written another way, and adding the density form of Sect. 13.2 to the current form would count the transition twice. The nuclear current, finally, changes a phonon occupation at $\boldsymbol{Q} \simeq \boldsymbol{p}_\phi$ without moving an electron, while the electron–phonon vertex moves one electron and changes one occupation. Together they either leave two occupations changed, a final state with two phonons, or remove the phonon the current created and leave a vertical pair, which needs $\omega \geq E_{\mathrm{dir}}$; in silicon the Born charges vanish in any case. The rate is Eq. (10.31) with $L = \varepsilon_2^{\mathrm{PA}}/(\varepsilon_1^0)^2$ at the temperature of the detector (Sects. 13.4 and 13.5). At the edge $\varepsilon_2^{\mathrm{PA}}$ is of order 10^{-4}, so the phonon-assisted part of the photon self-energy does nothing to the mixing, and $\zeta_{\mathrm{eff}} = \zeta/\varepsilon_1^0$. What the dark photon needs is the measured edge itself, brought to millikelvin temperatures and resolved into its excitonic structure, as Worked example 13.8 shows.

The axion-like particle couples through the spin current $\mathcal{J}_e$ (Sect. 10.7.3). The electron–phonon vertex is diagonal in spin. The spin structure of the phonon-assisted amplitude is therefore that of the one spin operator in $\mathcal{J}_e$, and the spin trace of Sect. 10.7.3 goes through unchanged: $S^{\mathrm{PA}}_{\mathcal{J}\mathcal{J}} = \frac{3}{4} S^{\mathrm{PA}}_{jj}$ and $\Gamma = (3m_\phi^2/4m_e^2)(g_{aee}^2/e^2)\,\sigma_1^{\mathrm{PA}}$, with $\sigma_1^{\mathrm{PA}} = m_\phi \varepsilon_2^{\mathrm{PA}}$, for independent electrons without spin–orbit coupling. At equal couplings it is $(3m_\phi^2/4m_e^2e^2)|\varepsilon|^2 = 7.3 \times 10^{-9}$ of the dark-photon rate at 1.2 eV. The caveats of Sect. 10.7.3 take a specific form at the edge. The spin current creates triplet pairs, which feel the direct kernel and not the exchange one (Eq. (10.23)). The replica lines are therefore the same as in the measured singlet edge, up to the exchange splitting, which is about 2×10^{-3} of the binding (Sect. 10.5). The pair channel of Sect. 13.4, a Coulomb screening at the pair momentum, is absent from the triplet response, and its size is not known for the singlet either.

The scalar couples through two rows, the density and the kinetic energy (Sect. 10.7.4). The

phonon-assisted density amplitude vanishes at $\boldsymbol{q} = 0$ by Eq. (13.10), so the density row enters at order p_ϕ and its rate at order v^2. The kinetic row, however, does not vanish. Its phonon-assisted amplitude $M^{\pm}_{\mathcal{T},cv\nu}(\boldsymbol{k},\boldsymbol{Q})$ is Eq. (13.9) at $\boldsymbol{q} = 0$ with the vertical kinetic elements $(\mathcal{T}_e)_{mn}(\boldsymbol{k}) = \langle m\boldsymbol{k}|p^2/2m_e|n\boldsymbol{k}\rangle$ in place of ρ_{mn}, and its intraband terms no longer cancel. Path A with $p = v$ and path B with $p = c$ add to $g_{cv\nu}(\boldsymbol{k},\boldsymbol{Q})[(\mathcal{T}_e)_{vv}(\boldsymbol{k}) - (\mathcal{T}_e)_{cc}(\boldsymbol{k}+\boldsymbol{Q})]/\omega$, the difference of the kinetic energies of the two band states, where the density has $\rho_{vv} = \rho_{cc} = 1$. The whole amplitude has a compact form. In the operator form of Eq. (13.9) (Appendix D.35), $\langle c|y\,(\varepsilon_v + \omega - h)^{-1}x + x\,(\varepsilon_v \mp \omega_{\boldsymbol{Q}\nu} - h)^{-1}y|v\rangle$ with $x = p^2/2m_e$ and y the phonon vertex, write $x = h - V$, with h the one-electron Hamiltonian and V everything in it but the kinetic energy. The part h acts on $|v\rangle$ in the first term and on $\langle c|$ in the second, where it returns ε_v and ε_c, and the resolvents return $1/\omega$ and $-1/\omega$ on the shell. With $\varepsilon_c - \varepsilon_v = \omega \mp \omega_{\boldsymbol{Q}\nu}$,

$$M^{\pm}_{\mathcal{T},cv\nu}(\boldsymbol{k},\boldsymbol{Q}) = -\Big(1 \mp \frac{\omega_{\boldsymbol{Q}\nu}}{\omega}\Big)\, g_{cv\nu}(\boldsymbol{k},\boldsymbol{Q}) - M^{\pm}_{V,cv\nu}(\boldsymbol{k},\boldsymbol{Q}), \tag{13.48}$$

where $M^{\pm}_V$ is Eq. (13.9) at $\boldsymbol{q} = 0$ with the matrix elements of V in place of ρ. The first term is the electron–phonon vertex itself. The second is the phonon-assisted amplitude of the crystal potential, the counterpart of $\langle c|p^2/2m_e|v\rangle = -\langle c|V|v\rangle$ of Sect. 10.7.4, and like it weighted toward the cores. Both are of order g in energy units, and nothing cancels them. Eq. (10.35) with the phonon-assisted response is

$$\begin{aligned} \Gamma &= \frac{g^2_{\phi ee}}{2\,m_\phi m_e^2}\, S^{\mathrm{PA}}_{\mathcal{T}\mathcal{T}}(0, m_\phi), \\ S^{\mathrm{PA}}_{\mathcal{T}\mathcal{T}}(0,\omega) &= \frac{4\pi}{\Omega}\Big\langle\!\Big\langle \sum_{\nu cv\pm} (n_{\boldsymbol{Q}\nu} + \tfrac{1}{2} \pm \tfrac{1}{2})\,|M^{\pm}_{\mathcal{T},cv\nu}|^2\,\delta(\varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu} - \omega)\Big\rangle\!\Big\rangle, \end{aligned} \tag{13.49}$$

unscreened, for a point group without a polar vector, inside the window; near the edge the final pairs are the excitons of Eq. (13.45) with the kinetic row in place of the density. No measurement provides this response, and it is not proportional to $\varepsilon_2^{\mathrm{PA}}$, since Eq. (13.48) has no velocity vertex in it. The comparison with the axion-like particle in Sect. 10.7.4 carries over. The kinetic amplitude is of the order of the vertex g, while the axion-like amplitude carries m_ϕ/m_e times an electron velocity. At equal couplings the scalar is therefore absorbed far more strongly, by a factor that only a calculation fixes. Mitridate and collaborators computed the phonon-assisted kinetic absorption for the intraband transitions of a superconducting metal [25, Sect. V, Eq. (72)]; their silicon rate, built from vertical transitions, has below about 3 eV only the tails of the smearing, as they note [25, Sect. IV]. The interband channel of Eqs. (13.48) and (13.49) at the indirect edge of an insulator needs, for its evaluation, the phonon-assisted kinetic row of Appendix E.14, with all-electron kinetic matrix elements (Appendix E.16).

Worked example 13.8: A dark photon of 1.2 eV in cold silicon

At $m_\phi = 1.200\,\mathrm{eV}$ a dark photon sits 12.5 meV above the free-pair threshold $E_{\mathrm{g}}(0) + \omega_{\mathrm{TA}} = 1.1875\,\mathrm{eV}$ and 27.2 meV above the $1s$ replica, below both transverse optical thresholds. Only the transverse acoustic branch absorbs, by emission.

Cold absorption. Stanford and co-workers fitted their data at 0.5 to 295 K with the free-pair form $\alpha_{\mathrm{abs}} = \sum_\nu C_\nu c_0(\omega - E_{\mathrm{g}1} - \omega_\nu)^2$ at low temperature, with the branch weights $C_{\mathrm{TA}} = 5.5$ and $C_{\mathrm{TO}} = 4.0$, the strength $c_0 = 325\,\mathrm{eV}^{-2}\,\mathrm{cm}^{-1}$ and an effective gap $E_{\mathrm{g}1} = 1.134\,\mathrm{eV}$ [13, Eqs. (4)–(7)]. Evaluated at 0.5 K, it gives $\alpha_{\mathrm{abs}} = 5.5 \times 325\,\mathrm{eV}^{-2}\,\mathrm{cm}^{-1} \times (0.0477\,\mathrm{eV})^2 = 4.07\,\mathrm{cm}^{-1}$ from its transverse acoustic term and $0.09\,\mathrm{cm}^{-1}$ from the transverse optical one, which its lowered gap opens: $4.16\,\mathrm{cm}^{-1}$, 6 meV below their lowest photon energy. The

20 K data of Macfarlane and co-workers [1] give $0.21\,\mathrm{cm}^{-1}$ at this energy, twenty times less; near the edge the cold datasets differ by that much (Sect. 13.9.4). The excitonic edge of Fig. 13.11 reaches $2.95\,\mathrm{cm}^{-1}$ (4.8 with the strength factor 0.61), the free-pair edge transferred to millikelvin temperatures in Sect. 13.5 only $0.24\,\mathrm{cm}^{-1}$, and the 300 K data used without transfer $27.3\,\mathrm{cm}^{-1}$.
Loss function. With the fit value $\alpha_{\mathrm{abs}} = 4.16\,\mathrm{cm}^{-1}$, $n_r = 3.565$ from the 300 K data and $\hbar c/\omega = 1.644 \times 10^{-5}\,\mathrm{cm}$,

$$\varepsilon_2^{\mathrm{PA}} = n_r\,\alpha_{\mathrm{abs}}\,\frac{\hbar c}{\omega} = 2.44 \times 10^{-4}, \qquad L = \frac{\varepsilon_2^{\mathrm{PA}}}{(\varepsilon_1^0)^2} = \frac{2.44 \times 10^{-4}}{12.71^2} = 1.51 \times 10^{-6}.$$

The medium suppresses the rate by $|\varepsilon|^2 = 161.5$ against the vacuum mixing.
Rate. With $\rho_\chi = 0.3\,\mathrm{GeV\,cm}^{-3}$ and $\rho_T = 2.33\,\mathrm{g\,cm}^{-3}$, $(\rho_\chi/\rho_T)/\hbar = 6.17 \times 10^{33}$ per kilogram-year, as in Worked example 10.7, and Eq. (10.31) gives

$$\mathcal{R} = 6.17 \times 10^{33} \times 1.51 \times 10^{-6}\,\zeta^2 = 9.3 \times 10^{27}\,\zeta^2 \text{ per kilogram-year.}$$

Three events in a kilogram-year need $\zeta = 1.8 \times 10^{-14}$. The untransferred 300 K data would claim a rate 6.6 times larger and $\zeta = 7.0 \times 10^{-15}$; the transferred free-pair edge a rate 17 times smaller and $\zeta = 7.5 \times 10^{-14}$, and the 20 K data a rate 20 times smaller. At 1.2 eV the correction that matters is the bound pair: in the excitonic edge three quarters of the absorption at this energy is the $1s$ replica, which neither the warm data nor the free-pair transfer represents.

13.9 The silicon window

A silicon detector that counts single electrons sees dark matter of 0.36 MeV only through the phonon-assisted rate, and dark matter of 0.6 MeV almost only through zero-phonon pairs. Kinematics fixed the onset of the first at 0.3332 and 0.3374 MeV and put upper values on the second along chosen directions (Sect. 13.1). Here the zero-phonon side is evaluated with bands over the whole zone: the first zero-phonon edge lies at 0.405 to 0.408 MeV, off the valley axis, and its state is weakly coupled because a glide of the diamond structure forbids its vertex. These edges and their order rest on two independent calculations. Where the phonon-assisted rate stops leading, how large it is, and the windows of germanium and GaAs are known only as estimates or not at all.

13.9.1 The zero-phonon edges

The joint minimum Eq. (13.1) over the momentum transfer and the hole's crystal momentum lies off the valley axis. The search uses the PBE bands of Sect. 3.4, interpolated in Wannier functions, with the conduction bands shifted rigidly to the measured gap $E_{\mathrm{g}}(0) = 1.170\,\mathrm{eV}$. Of 21 starting points, 18 converge on one point up to the cubic symmetry, and a local re-minimization on the plane-wave bands at the measured lattice constant refines it. The optimum has $|\boldsymbol{q}| = 1.33\,\mathrm{keV}$, tilted by 0.18 rad from the [100] axis within a {001} plane. The hole sits at $\boldsymbol{k}_v = (0.126, -0.107, 0)\,(2\pi/a)$, 62 meV below the valence-band maximum, along a direction in which the heavy-hole band is flat (its PBE mass along [110] is about $2.5\,m_e$). The electron sits on the valley axis, about $0.16\,(2\pi/a)$ short of the valley minimum. The pair costs 1.355 eV, and the threshold mass is

$$m_{\mathrm{min}}^{\mathrm{ZP}} = 0.407\,\mathrm{MeV},$$

with spin–orbit coupling, and 0.404 MeV without, within a band of 0.405 to 0.408 MeV that carries the lattice constant and the interpolation. The $1s$ exciton lowers it by $m_\chi E_x^b/(q v_{\max} - E) = 0.0028\,\mathrm{MeV}$ (Sect. 13.1), to 0.402–0.405 MeV.

The joint minimum leaves the axis because, at $|\boldsymbol{q}| = 1.2$ to 1.48 keV, the [100] axis is a local maximum of $E_{\min}(\boldsymbol{q})$ over the directions of $\boldsymbol{q}$, not a minimum. The minima sit 0.13 to 0.22 rad away in the {001} planes, 14 to 107 meV below the on-axis value, and are shallower in the {011} planes. Around the optimum the pair energy rises as $\beta_\theta \vartheta^2 + \beta_z \varphi^2$, with $\beta_\theta = 2.47\,\mathrm{eV\,rad^{-2}}$ for the tilt within the plane and $\beta_z = 3.70\,\mathrm{eV\,rad^{-2}}$ out of it. The warped heavy hole of Eq. (3.56) is the cause: moving the hole into its flat direction costs less than the pair kinetic energy it saves. On the axis the same bands have a local optimum at 0.415 MeV, which the coarser grid of Sect. 13.1 placed at 0.420 MeV and the parabola at 0.425 MeV. The joint minimum is therefore 0.013 to 0.018 MeV below the upper values of Sect. 13.1.

Spin–orbit coupling raises the edge by 0.003 MeV, fifteen times more than on the axis. At the optimum the hole lies 62 meV below the valence maximum, deeper than the split-off energy ($\Delta_{\mathrm{SO}} = 48.6$ meV in PBE, 44 meV measured, Table 3.1), and it does not share the rise by $\Delta_{\mathrm{SO}}/3$ of the Γ_8 maximum to which the gap is referred. With spin–orbit coupling it sits 77 meV below that maximum, and the pair costs 15.5 meV more. On the axis, where the hole is near Γ, the shift is under 1 meV.

Three further edges complete the zero-phonon side. On the axis the two heavy holes are degenerate. A tilt within a {001} plane splits them into one band that is odd under the mirror of that plane and one that is even, and they open at different masses. The first edge belongs to the odd band. The even band reaches the same pair energy only on the axis, and its edge lies at 0.415 to 0.420 MeV, the range set by the lattice constant and spin–orbit coupling. Off the axis its pair energy rises as $\beta_2 \vartheta^2$, with β_2 from $4.5\,\mathrm{eV\,rad^{-2}}$ in the {001} planes to $1.1\,\mathrm{eV\,rad^{-2}}$ in the {011} planes. The third edge is that of the Umklapp class $\boldsymbol{G}' = (\bar{2}\,0\,0)\,2\pi/a$, in which the unreduced crystal momentum of the electron lies beyond the X face of the zone, so that the reciprocal-lattice vector sits inside the density vertex (Sect. 3.1). Restricted to this class, Eq. (13.1) has its minimum on the X face, at $|\boldsymbol{q}| = 1.83$ keV with the hole at $|\boldsymbol{k}_v| = 0.26\,(2\pi/a)$: the line opens at 0.50 MeV, spin–orbit coupling included. The last edge of the class without Umklapp on the [100] axis is far above all of these. Along [110] and [111], 0.548 and 0.615 MeV (Sect. 13.1), the zero-phonon channel is already open through the other edges.

The edges fall in a fixed order:

$$\underbrace{0.3332 < 0.3374}_{\text{phonon-assisted onset}} < \underbrace{0.402\text{–}0.405 < 0.405\text{–}0.408}_{\text{first zero-phonon edge}} < \underbrace{0.415\text{–}0.420}_{\text{second heavy hole}} < \underbrace{\approx 0.50}_{(\bar{2}\,0\,0)\text{ line}} \ \text{MeV}.$$

The order does not depend on the last digits. The phonon-assisted onset moves by 2.8×10^{-4} MeV per meV of phonon energy (Worked example 13.1), and even a transverse acoustic phonon at its largest energy, 18.6 meV at X, would put it at 0.339 MeV, 0.07 MeV below the first zero-phonon edge. The zero-phonon edges are separated by more than their own spreads. Between the onset and the first edge no zero-phonon pair exists at any momentum transfer, and the phonon-assisted rate is the only electron-recoil channel; above the first edge the question becomes one of rates.

These are thresholds, not rates, and they are cheap. The joint minimization needs interpolated bands over the zone with a gap correction, a check of the interpolated pair energy against a direct plane-wave calculation at the optimum (it differs by 11 meV), and a local re-minimization with spin–orbit coupling (Appendix E.1). Appendix E.14 specifies the bins on which the zero-phonon rate is to be computed next to the phonon-assisted one: the cones around the mirror-plane optima, the cone of the second heavy hole and the X-face edge.

13.9.2 The glide-parity selection rule

On the valley axis, without Umklapp, the zero-phonon vertex between the bottom of the valley and the valence maximum vanishes. The electron is invariant under the operations that fix the axis, and the holes have no invariant part (Sect. 10.5.3; in the language of Sect. 13.3, $\Delta_1 \otimes \Delta_1 \otimes (\Delta_2' \oplus \Delta_5)$ contains no Δ_1). Off the axis these operations are gone, and only one survives for a pair that stays in a $\{001\}$ plane through the axis: the mirror of that plane. In diamond it carries the fractional translation $\boldsymbol{\tau}_2$ and is a glide (Appendix C.17). Bloch states with crystal momentum in the plane are its eigenstates, with a glide parity $\eta = \pm 1$ (Eq. (C.149)). The density vertex with $\boldsymbol{q}$ in the plane connects a valence state of parity η_v to the conduction state of reduced parity η_c only if

$$\eta_c \, \eta_v \, \mathrm{e}^{-\mathrm{i}\boldsymbol{G}'\cdot\boldsymbol{w}/2} = 1, \tag{13.50}$$

where $\boldsymbol{G}'$ is the Umklapp vector of the unreduced final momentum and $\boldsymbol{w}$ the lattice vector that the glide gives when applied twice. This is Eq. (C.150) applied to the mirror m_z. The rule decides which zero-phonon line is strong at each of the edges.

Worked example 13.9: The selection rule for the first zero-phonon line

Take the valley on the x axis and the mirror m_z, $(x, y, z) \to (x, y, -z)$, whose plane $z = 0$ contains the axis and the optimum of Sect. 13.9.1. In diamond it occurs as $g = \{\mathsf{m}_z|\boldsymbol{\tau}_2\}$ with $\boldsymbol{\tau}_2 = (a/4)(1, 1, 1)$, as in Eq. (C.147) with x and z exchanged. Applied twice it gives the translation $\boldsymbol{w} = \boldsymbol{\tau}_2 + \mathsf{m}_z\boldsymbol{\tau}_2 = (a/2)(1, 1, 0)$, a lattice vector.

The phase of the operator. Conjugating the vertex gives $g\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}g^{-1} = \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_2}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ for $\boldsymbol{q}$ in the plane. Against the phases $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{w}/2}$ of the two eigenvalues it leaves $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{w}/2-\boldsymbol{\tau}_2)}$, and $\boldsymbol{w}/2 - \boldsymbol{\tau}_2 = (0, 0, -a/4)$ is normal to the plane, so this phase is one. With unreduced labels the vertex therefore needs $\eta_c = \eta_v$.

Reduced labels. Reducing the final momentum by $\boldsymbol{G}'$ multiplies the conduction eigenvalue by $\mathrm{e}^{-\mathrm{i}\boldsymbol{G}'\cdot\boldsymbol{w}/2}$. For $\boldsymbol{G}' = 0$ the factor is one and Eq. (13.50) requires equal parities. For $\boldsymbol{G}' = (\bar{2}\,0\,0)\,2\pi/a$, $\boldsymbol{G}'\cdot\boldsymbol{w}/2 = -(4\pi/a)(a/4) = -\pi$, the factor is -1, and the parities must be opposite. Every vector of the (200) class flips the rule; vectors with $h + k \equiv 0 \pmod 4$ in the plane keep it.

On the axis. The heavy-hole pair Δ_5 splits under the mirror into one even and one odd state, the light hole Δ_2' is odd and the electron Δ_1 even, all relative to the electron's convention. The glide rule alone would allow the even heavy hole at $\boldsymbol{G}' = 0$; the full group of the axis forbids it, as above, and the vertex vanishes on the axis for every hole. In the class $(\bar{2}\,0\,0)$ the rule allows the odd states, and the full group of the axis leaves exactly one of them, the light hole, which carries the vertex there (multiplicity one, $\Delta_1 \otimes \Delta_2' \otimes \Delta_2' \ni \Delta_1$).

At the optimum. Off the axis in the plane the two heavy holes are no longer degenerate, and which of them is on top is not fixed by symmetry. The calculation finds the odd one on top at the optimum. With $\boldsymbol{G}' = 0$, Eq. (13.50) then forbids its vertex everywhere in the plane: computed, $|\rho_{cv}|^2 = 3 \times 10^{-14}$, zero to numerical precision. The even partner, 0.39 eV deeper at the same $\boldsymbol{q}$, has $|\rho_{cv}|^2 = 0.064$. Leaving the plane breaks the glide, and the vertex grows linearly with the distance from it, with gradients $|\partial_{q_z}\rho_{cv}| = 1.57\,\text{Å}$ (the pair relaxing by its center of mass) and $|\partial_{k_z}\rho_{cv}| = 0.32\,\text{Å}$. The first zero-phonon line is second class in the sense of Sect. 10.5.3: its vertex vanishes on a surface through the edge and its strength grows from zero.

The other edges. The second heavy hole is the even band, allowed by Eq. (13.50) as soon as the pair leaves the axis; its vertex grows linearly in the tilt with $|\partial_{q_\perp}\rho_{cv}| = 2.21\,\text{Å}$.

In the class $(\bar{2}\,0\,0)$ the rule inverts and the odd top hole is allowed. At the X-face edge $|\rho_{cv}|^2 = 0.0095$ for the lowest conduction band, 0.028 for the two lowest (the second lies 50 meV higher there), and 0.024 to 0.036 at 0.02 to 0.10 $(2\pi/a)$ inside the class.

The vertex of the $(\bar{2}\,0\,0)$ class is not small, although the (200) Fourier component of every lattice-periodic function of diamond vanishes (Eq. (C.148)). At the extremum of the class, $|\boldsymbol{q}| = 2.64\,\mathrm{keV}$, the light hole carries $\sum_v |\rho_{cv}|^2 = 0.150$, of the size of an ordinary allowed vertex: the four lowest conduction bands together, averaged over directions, give 0.45 at $|\boldsymbol{q}| = 2.28\,\mathrm{keV}$, a number not comparable one-to-one with the single-band 0.150. The extinction removes one Fourier component of the potential, but the vertex is a product of Bloch functions with all their components, and it is not suppressed. A nearly-free-electron estimate, in which the (200) component would carry the transition alone, gives the opposite answer.

Spin–orbit coupling mixes the two glide parities, since the spin also transforms under the mirror, and lifts each zero. On the axis the lift, summed over the four states of the Γ_8 quartet, is $|\rho_{cv}|^2 = 3.6 \times 10^{-6}$, an amplitude of 0.5 % of the allowed $(\bar{2}\,0\,0)$ vertex; in the mirror plane it is 1.3×10^{-4}, 4.5 % in amplitude of the allowed partner. Neither moves an edge or a rate in what follows.

13.9.3 The rate at the onset

Just above $m_{\mathrm{min}}^{\mathrm{PA}}$ only a small corner of the (q, ω) plane is reachable, and three factors vanish together there: the loss function above its threshold, the halo near v_{max}, and the range of momentum transfers. At fixed $|\boldsymbol{q}|$ write $E_{\mathrm{th}}(q)$ for the onset of $\bar{L}^{\mathrm{PA}}$ and $x = \omega - E_{\mathrm{th}}(q)$, so that $\bar{L}^{\mathrm{PA}} \simeq C x^{p_{\mathrm{on}}}$ with the onset exponent of Eq. (13.19). With $v_{\mathrm{min}} = \omega/q + q/2m_\chi$ (Eq. (1.23)), the distance of v_{min} from v_{max} is

$$v_{\mathrm{max}} - v_{\mathrm{min}} = \frac{x_{\mathrm{max}}(q) - x}{q}, \qquad x_{\mathrm{max}}(q) = q v_{\mathrm{max}} - \frac{q^2}{2m_\chi} - E_{\mathrm{th}}(q).$$

The function $x_{\mathrm{max}}(q)$ is the largest energy above threshold that the mass reaches at q. Its maximum over q is zero at the onset mass $m_0 = m_{\mathrm{min}}^{\mathrm{PA}}$ and lies at $q^* = m_0 v_{\mathrm{max}}$ to the accuracy of Eq. (13.4). Since $\partial x_{\mathrm{max}}/\partial m_\chi = q^2/2m_\chi^2$, a mass $m_\chi = m_0 + \delta m$ reaches

$$x_{\mathrm{max}}(q) \simeq x_{\mathrm{max}}^* - \frac{(q - q^*)^2}{2m_0}, \qquad x_{\mathrm{max}}^* = \tfrac{1}{2} v_{\mathrm{max}}^2\, \delta m,$$

where the curvature of E_{th}, of order $10^{-8}\,\mathrm{eV}^{-1}$ for the transverse acoustic branch, is dropped against $1/m_0 = 3 \times 10^{-6}\,\mathrm{eV}^{-1}$. For the standard halo of Table 1.1 the halo function is quadratic near v_{max}, $\eta(v_{\mathrm{min}}) \simeq A_\eta (v_{\mathrm{max}} - v_{\mathrm{min}})^2$ (Eq. (1.29)), and the factors $q^3 F_{\mathrm{DM}}^2$ of Eq. (10.4) are smooth. The energy integral at fixed q is then

$$\int_0^{x_{\mathrm{max}}} \mathrm{d}x\, x^{p_{\mathrm{on}}} (x_{\mathrm{max}} - x)^2 = \frac{2\, x_{\mathrm{max}}^{p_{\mathrm{on}}+3}}{(p_{\mathrm{on}}+1)(p_{\mathrm{on}}+2)(p_{\mathrm{on}}+3)},$$

and the integral over q of $x_{\mathrm{max}}^{p_{\mathrm{on}}+3}$ across the interval where $x_{\mathrm{max}} > 0$, of width $2(2m_0 x_{\mathrm{max}}^*)^{1/2}$, gives $(x_{\mathrm{max}}^*)^{p_{\mathrm{on}}+3} (2m_0 x_{\mathrm{max}}^*)^{1/2} B(\tfrac{1}{2}, p_{\mathrm{on}} + 4)$, with B the beta function. Collecting the powers of x_{max}^*:

Mass dependence at the phonon-assisted onset

$$\mathcal{R}(m_\chi) \propto (m_\chi - m_{\text{min}}^{\text{PA}})^{p_{\text{on}}+7/2}, \tag{13.51}$$

for $\frac{1}{2}v_{\text{max}}^2(m_\chi - m_{\text{min}}^{\text{PA}})$ inside the energy range in which $\bar{L}^{\text{PA}}$ follows $x^{p_{\text{on}}}$; the power is 13/2 at the free-pair transverse acoustic edge of silicon with independent pairs, 6 with the excitonic continuum and 5 at the exciton replica, for either mediator, with the standard halo of Table 1.1, whose $\eta(v_{\text{min}})$ vanishes quadratically at v_{max}. A halo with another behavior at the escape speed changes the 7/2.

The loss function gives $p_{\text{on}} + 1$ powers, the halo two and the width in q one half. A numerical integration of Eq. (10.4) with the computed transverse acoustic onset on the axis, the halo function of Sect. 1.2 and a loss function $Cx^{p_{\text{on}}}$ reproduces the two powers, and their local value stays within 0.06 of them up to 0.01 MeV above each onset (Fig. 13.12(b)). The model loss function is the idealization; the computed one of Fig. 13.4 follows x^3 within 10 % only for $x < 1.3$ meV, which here means $\delta m < 4 \times 10^{-4}$ MeV, since $x_{\text{max}}^* = 3.5$ meV per 0.001 MeV. Its local exponent drifts to 3.3 at 10 meV, so the power of the mass stays between 6.5 and 7 to about 0.003 MeV above the onset. With excitonic final states the free-pair continuum starts as $x^{3/2}$ at fixed $\boldsymbol{q}$ instead of x^2, because the Coulomb attraction keeps its weight finite at the edge (Eq. (13.47)); averaged over directions it has $p_{\text{on}} = 5/2$, and its power of the mass is 6.

The steep power turns any error in the onset into a large error in the rate. An error ϵ in $m_{\text{min}}^{\text{PA}}$ multiplies the rate at δm above it by $(1 - \epsilon/\delta m)^{p_{\text{on}}+7/2}$. The softness of the computed transverse acoustic branch moves the onset by 0.0004 MeV (Worked example 13.1), and at 0.005 MeV above the onset this changes the rate by a factor 0.56 to 0.58. Within a few thousandths of an MeV of the onset only the exponent is a meaningful prediction, and between the replica onset at 0.3332 MeV and the free-pair onset at 0.3374 MeV the whole rate is the replica's, whose strength belongs to Sect. 13.8.

13.9.4 Rates across the window

No first-principles phonon-assisted rate of silicon exists. What exists is an estimate, built to be checked bin by bin against the calculation of Appendix E.14. It takes a measured cold absorption of silicon as the optical limit $\varepsilon_2^{\text{PA}}(\omega)$ of Eq. (13.28), where it contains the excitons, the local fields and the pair channel at $\boldsymbol{q} \to 0$. The primary input is the 0.5 K fit of Stanford and co-workers (Worked example 13.8), 21.5 cm^{-1} at 1.25 eV, or $\varepsilon_2 = 1.2 \times 10^{-3}$. The 20 K data of Macfarlane and co-workers [1] give 10.4 cm^{-1} there, or $\varepsilon_2 = 5.9 \times 10^{-4}$, and twenty times less than the fit at 1.20 eV, below the lowest photon energy that the fit rests on; above 1.37 eV the two agree within 5 %. For the fall-off at finite q it multiplies $\varepsilon_2^{\text{PA}}$ by a form factor of the zero-phonon vertex, $\sum_{cv} |\rho_{cv}(\Gamma, \boldsymbol{q})|^2/q^2$ over its value at $q \to 0$, computed from the PBE wavefunctions over the four lowest conduction bands and averaged over the cubic directions. The form factor falls to 0.12 at $|\boldsymbol{q}| = 1.0$ keV. It then divides by $\varepsilon_1(q, \omega)^2$ from the model dielectric function of Cappellini and co-workers [26], pinned to the measured $\varepsilon_1(\omega)$ at $q \to 0$, and keeps only the off-shell window of Sect. 13.6.

The weakest step is the form factor. The phonon-assisted amplitude is linear in the density vertex. Its intraband terms, however, which cancel at $q = 0$ (Eq. (13.10)), decorrelate at finite q and add amplitude that the interband form factor does not see. The form factor is more likely low than high, and the dipole extrapolation, five to twenty times larger at the momentum transfers reached near 0.40 MeV, is its ceiling. The estimate carries a factor three either way from the form factor, a factor 2.8 from the screening correction of the next paragraph, the spread of the two cold datasets, a factor 2.8 at 0.38 MeV falling to 1.4 at 0.43 MeV, and a factor 1.5 from the shift of the threshold with $\boldsymbol{Q}^*$. Combined, these give a

factor 6.5 at 0.38 to 0.43 MeV, which we round up to a factor of ten to cover the correlations among the inputs. At 0.36 MeV the two datasets alone differ by a factor of nine, and the combined uncertainty is a factor of fifteen.

The screening correction is the part of that budget that Sect. 13.4 sizes only in part. The estimate is normalized to the measured edge, whose ε_2 contains the local-field wings and the pair channel at $q \to 0$; the external factor is calibrated there only because the model ε_1 is pinned to the measured one. What the estimate needs at finite q is therefore the correction relative to that calibration, the product of three ratios (Sect. 13.4). The external factor of the model dielectric function against the true ε_1^0 is 0.60 to 1.45 over the window, with the true value taken between the random-phase $\varepsilon_{\mathrm{mac}}$ of Worked example 13.4 and that value raised by the 29 % by which the random-phase approximation falls below the measured ε_1 at $q \to 0$; the macroscopic local fields, +25 to +31 % at 1 keV, are inside this factor. The local-field wings of the vertex, taken no larger at each momentum than the macroscopic local-field change of $1/\varepsilon_1^2$ there, with either sign and independently at q and at $q \to 0$, give 0.55 to 1.77. Together the correction is 0.3 to 2.6, an estimate. The pair ratio is one in both solvable limits, to 4×10^{-3}, at exact pinning of the pair momentum; the spread of pair momenta above the onset, about 0.2 keV, leaves a residual, and between the limits the ratio is not derived. Both remain an unsized nuisance of the estimate, which the supercell calculation of Appendix E.16 would size.

Take a background-free exposure of 1 kg yr that counts every event with at least one electron. With these inputs, three events correspond to a reference cross section of $4 \times 10^{-32}\,\mathrm{cm}^2$ at 0.40 MeV for a heavy mediator and $2 \times 10^{-34}\,\mathrm{cm}^2$ for a light one, uncertain by a factor of ten. At 0.36 MeV the reach is the band between the two cold datasets, 3×10^{-30} to $3 \times 10^{-29}\,\mathrm{cm}^2$ for a heavy mediator and 1.4×10^{-32} to $1.3 \times 10^{-31}\,\mathrm{cm}^2$ for a light one. There 55 % of the rate with the 0.5 K fit comes from below 1.206 eV, the lowest photon energy of its data, where the fit is an extrapolation with a threshold 21 meV below the cold edge, since its gap was floated to 1.134 eV; the 20 K curve below 1.212 eV is a linear ramp of the estimate from 1.160 eV, not data. The fall by two orders of magnitude over 0.04 MeV is kinematic: near the onset the halo supplies only particles close to v_{max}, and Eq. (13.51) is its limiting form. Cross sections below $10^{-30}\,\mathrm{cm}^2$ are reached in the top 0.04 MeV of the phonon-assisted-only band, 0.03 MeV with the 20 K data. Part of the rate falls in the margin below the zero-phonon edge, where only totals are defined (Sect. 13.6): 12 % at 0.40 MeV and 22 % at 0.42 MeV.

Above the first zero-phonon edge the question is the ratio of the two rates. Both channels are screened by the same external factor at the same (q, ω), so the ratio is that of the proper ε_2. On the zero-phonon side the estimate puts a Wannier–Mott continuum (Sect. 10.5) at each of the 24 equivalent minima of the first edge: six valleys, two mirror planes each, two signs of the tilt. Each has the pair energy $E_0 + \beta_\theta \vartheta^2 + \beta_z \varphi^2$ and the vertex grown from the computed gradients of Worked example 13.9. Neighboring cones touch 49 meV above the edge, and above that each direction is assigned to its lowest cone. Averaged over directions at fixed $|\boldsymbol{q}|$, the first heavy hole gives a zero-phonon to phonon-assisted ratio of 0.020, 0.16 and 0.34 at 14.7, 50 and 100 meV above the edge, and the part lifted by spin–orbit coupling adds at most 0.01. These ratios use the 20 K data on the phonon-assisted side; the 0.5 K fit changes them by at most 5 %, since the two datasets agree at these energies. The second heavy hole adds its own continuum from its edge on, with its vertex assigned entirely to that band, an upper estimate.

To turn this into a statement about masses, each channel is evaluated at the largest energy above the first edge that the mass reaches: 0.015, 0.066, 0.115 and 0.161 eV at 0.41, 0.42, 0.43 and 0.44 MeV. This is again an upper estimate of the integrated ratio. Fig. 13.12(a) shows the result. The ratio is 0.02 at 0.41 MeV, 0.24 to 0.56 at 0.42 MeV, 1.2 to 2.1 at 0.43 MeV and 2.6 to 3.9 at 0.44 MeV, the ranges spanning the edge of the second heavy hole between 0.420 and 0.415 MeV. Above 0.43 MeV the second heavy hole carries most of the zero-phonon

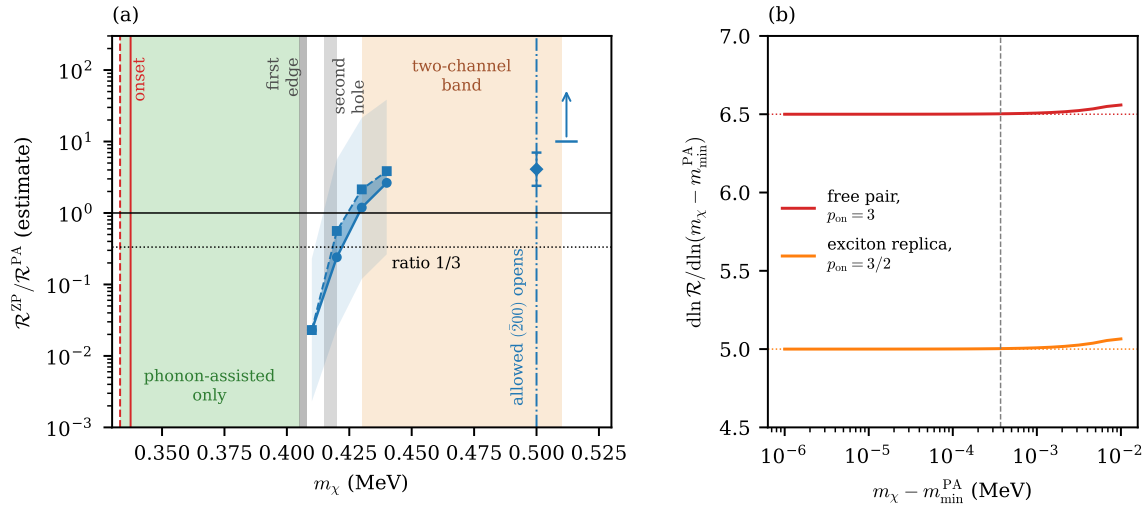


Figure 13.12: Rates across the silicon window. (a) Estimated ratio of the zero-phonon to the phonon-assisted electron-recoil rate against the dark-matter mass. Green band: phonon-assisted only, from the onsets (red: free pair 0.3374 MeV, solid; exciton replica 0.3332 MeV, dashed) to the first zero-phonon edge; gray bands: the first zero-phonon edge (0.405–0.408 MeV) and the edge of the second heavy hole (0.415–0.420 MeV). Blue points: the ratio with the second-hole edge at 0.420 MeV (circles) and 0.415 MeV (squares), each channel at the largest energy above the first edge that the mass reaches; dark band between them, light band the factor of ten of the phonon-assisted side. Diamond: the allowed $(\bar{2}\,0\,0)$ line at its opening, 0.50 MeV (dash-dotted), with one or two conduction bands reached; bar and arrow: ratio of 10 or more from 0.51 MeV. Black lines: ratio one and one third. Orange: the two-channel band. Estimates: phonon-assisted side from the measured cold absorption (20 K data [1]; the 0.5 K fit of Worked example 13.8 changes the ratios by at most 5 %) and a computed vertex form factor; zero-phonon side a Wannier–Mott continuum with computed vertices; between 0.44 and 0.50 MeV no ratio has been evaluated. (b) Local power of the model rate, $\mathrm{d}\ln\mathcal{R}/\mathrm{d}\ln(m_\chi - m_{\mathrm{min}}^{\mathrm{PA}})$, for a loss function $Cx^{p_{\mathrm{on}}}$ above the computed transverse acoustic onset on the valley axis (PBE force constants of Sect. 5.2), the standard halo (Table 1.1) and a heavy mediator, with the asymptotes $p_{\mathrm{on}} + 7/2$ of Eq. (13.51) (dotted). Dashed: the mass at which the largest energy above threshold reaches 1.3 meV, the end of the x^3 range of Fig. 13.4.

side. The two channels are equal at 0.424 to 0.428 MeV. If the phonon-assisted rate sits at the low end of its band, a third of the estimate, they are equal at 0.418 to 0.421 MeV. At the opening of the $(\bar{2}\,0\,0)$ line, 0.50 MeV, the allowed line alone gives a ratio of 2.4 to 7.0 just above its edge, the range set by whether one or two conduction bands are reached. At $0.05\,(2\pi/a)$ inside the class, reached from 0.508 MeV, it gives 10 to 11, just outside the factor of ten of the phonon-assisted side.

The phonon-assisted rate leads the electron recoil of silicon at least to about 0.42 MeV and centrally to about 0.43 MeV. From there to about 0.51 MeV lies a two-channel band, in which the rates are comparable and neither can be quoted alone with confidence. Above about 0.51 MeV the allowed zero-phonon line leads. The biases point both ways. The isotropic exciton mass underestimates the zero-phonon density of states by a factor 1.3 to 1.6, which makes the ratios too low. The form factor, more likely low than high, makes them too high; the second-hole vertex and the evaluation at the largest energy make the zero-phonon side an upper estimate. A rate computed from the amplitude on the same bins as the zero-phonon rate would settle all of it.

13.9.5 Germanium and GaAs

The germanium window has a known lower end and an unknown upper end. Its onset, 0.2119 to 0.2141 MeV, rests only on the gap, the position of L and the measured transverse acoustic branch (Sect. 13.1); the computed PBE branch is 13.5 % low at X and 11.6 % low at L, so its $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*) = 6.70\,\mathrm{meV}$ on a 6^3 grid is too soft by about as much. It places the onset at 0.213 80 MeV for a free pair and 0.212 64 MeV for the bound $1s$ exciton, with an estimated binding of 4.0 meV. Both lie inside the bracket, and scaling the branch to the neutron data moves them by at most 0.0003 MeV, so the softness does not reach the onset (Appendix E.14). The zero-phonon edge is not known. PBE puts the conduction state at Γ below the L valley, so no scissor-corrected PBE calculation can locate it (Appendix E.1), and the joint minimization on quasiparticle bands with spin–orbit coupling is open. The value 0.32 MeV of Sect. 13.1, from the parabola on the L axis, assumes a parabolic pair with the hole at Γ, and it brackets the edge in neither direction. Only the joint minimum restricted to the axis, evaluated on the true bands, would be an upper value. The silicon result does not settle whether the germanium optimum leaves the axis. The [111] valley axis of germanium is already its heavy-hole direction, and the silicon optimum came from a hole moving into a flat direction away from the valley axis. What does carry over is that spin–orbit coupling must be in the bands, and that the hole must be free in the minimization. The selection rule differs as well. The mirror planes that contain a [111] axis are the diagonal planes of the {110} family, which belong to $\bar{4}3m$ and carry no fractional translation (Appendix C.17); there is no glide phase, and on the L axis the vertex at the extrema does not vanish (Sect. 10.5.3). Whether the first zero-phonon state of germanium is weakly coupled has to be checked at its optimum. Until then germanium has a phonon-assisted onset, the onset exponent 3 for every branch (Sect. 13.3), and no window.

GaAs is the control: a direct gap with its own phonon-assisted band. Below its floor of 0.433 MeV it has no electron recoil at all. Above, the pair at Γ cannot absorb the momentum $q^* \approx 1.15\,\mathrm{keV}$ at which the gap is deposited, and a phonon must take it, from an onset of 0.433 to 0.436 MeV (Sect. 13.1). The zero-phonon channel first opens through the L valleys at 1.815 eV, between their floor, 0.517 MeV, and 0.560 MeV, where 1.815 eV is deposited at exactly |ΓL|. Both ends of that bracket are rigorous for the given L energy; only the position of the edge between them needs bands. The phonon-assisted band of GaAs therefore runs from 0.43 MeV to between 0.52 and 0.56 MeV, in the parabolic description of the Γ pair. The top scales as $2E_{\mathrm{L}}/v_{\mathrm{max}}^2$, and for a detector the low-temperature L energy applies. That GaAs reproduces this band, and that its phonon-assisted rate is a small correction wherever a zero-phonon line is open, is the control that Appendix E.14 specifies.

13.9.6 The silicon window in one table

Table 13.1 collects the window in two blocks. The first holds what two independent calculations or a symmetry argument with a numerical check establish: the zero-phonon edges, the lower end of the second heavy hole, the opening of the allowed line, the selection rule and the order of the edges. The second holds what rests on one calculation or an estimate, or is open: the onset, the leads, the two-channel band, the screening correction, the rate and the reach, germanium and GaAs.

13.9.7 Nuisance table of the phonon-assisted sector

Table 13.2 collects the truncations and inputs of the chapter, each with its size and the section that treats it, in the classes of Sect. 8.8.

For the rate in the window, three rows dominate: the finite-q form factor of the estimate, a factor three either way and probably low; the screening correction relative to the optical

Table 13.1: The silicon window, with germanium and GaAs. Masses in MeV. Status: "two independent calculations", "symmetry, confirmed numerically", "derivation", "one calculation", "measured data", "estimate" (a model in which the phonon-assisted rate is known to a factor of ten either way, apart from the unsized pair channel), "open". Bands at the measured lattice constant with spin–orbit coupling, gap $E_g(0) = 1.170\,\mathrm{eV}$, halo of Table 1.1.

Quantity	Value	Status	Section
Established			
First zero-phonon edge, continuum	0.405–0.408 (best 0.407); off the axis, {001} mirror plane, tilt 0.18 rad	two independent calculations	Sect. 13.9.1
First zero-phonon edge, 1s line	0.402–0.405	two independent calculations	Sect. 13.9.1
Second (mirror-even) heavy hole	from 0.415–0.420	0.415: two independent calculations; 0.420: estimate	Sect. 13.9.1
Allowed $(\bar{2}\,0\,0)$ line opens	≈ 0.50 (X face); vertex at the class extremum $\sum_v \lvert\rho_{cv}\rvert^2 = 0.150$	two independent calculations	Sects. 13.9.1 and 13.9.2
Selection rule	first line second class by the glide; $(\bar{2}\,0\,0)$ class allowed; spin–orbit lift 3.6×10^{-6} (axis), 1.3×10^{-4} (plane)	symmetry, confirmed numerically	Sect. 13.9.2
Order of the edges	onset < first edge < second hole < allowed line	two independent calculations	Sect. 13.9.1
Estimated or open			
Phonon-assisted onset	0.3332 (exciton replica), 0.3374 (free pair)	one calculation	Sect. 13.1
Phonon-assisted rate leads	centrally to ≈ 0.43 (0.424–0.428), at least to ≈ 0.42 (0.418–0.421)	estimate	Sect. 13.9.4
Two-channel band	0.43–0.51	estimate	Sect. 13.9.4
Zero-phonon rate leads	above ≈ 0.51 (ratio 2.4–7.0 at 0.50, ≥ 10 pointwise from 0.51)	estimate	Sect. 13.9.4
Screening correction relative to the optical calibration	0.3–2.6 on L^{PA} (external factor and local-field wings); pair channel beyond the calibration not sized	estimate	Sects. 13.4 and 13.9.4
Rate and reach	$\bar{\sigma}_e \approx 3 \times 10^{-30}$–$3 \times 10^{-29}$ (0.36; the 0.5 K fit extrapolated below 1.206 eV carries half of it), $4 \times 10^{-32}\,\mathrm{cm}^2$ (0.40), heavy mediator, 1 kg yr; power $p_{\mathrm{on}} + 7/2$ at the onset	estimate (factor 10 at 0.40; 15 at 0.36); power: derivation	Sects. 13.9.3 and 13.9.4
Germanium	onset 0.2119–0.2141; zero-phonon edge not known	onset from measured data; edge open	Sects. 13.1 and 13.9.5
GaAs	phonon-assisted band 0.43 to 0.52–0.56	estimate (parabolic Γ pair)	Sects. 13.1 and 13.9.5

Table 13.2: Nuisances of the phonon-assisted sector. Type in the classes of Sect. 8.8: normalization, tilt (in ω, in q or in the dark-matter mass), threshold and new support. Basis of the size: "computed" (next order computed), "model" (next order computed in a model), "est." (parametric estimate), "bound" (only an inequality or two limits are known), "named" (no size). Sizes are for silicon unless another crystal is named. The last column gives the section where the item is treated.

Truncation or input	Type	Size	Basis	Section
Transverse acoustic energy at $\boldsymbol{Q}^*$ (PBE branch 8 % soft at X)	threshold	onset +0.0004 MeV with the branch scaled to the neutron data; 2.8×10^{-4} MeV per meV	computed	Sect. 13.1
Exciton binding	threshold	phonon-assisted onset 0.3374 → 0.3332 MeV; zero-phonon edge −0.0028 MeV	computed	Sects. 13.1 and 13.9
Fixed directions of $\boldsymbol{q}$, hole at the valence maximum	threshold	zero-phonon edge 0.405–0.408 against 0.420–0.425 MeV on the axis	computed	Sects. 13.1 and 13.9
Spin–orbit coupling in the bands	threshold	first zero-phonon edge +0.003 MeV; on the axis < 0.0002 MeV	computed	Sect. 13.9
Band sum cut to empty bands	normalization	$\lvert M\rvert^2$ off by a factor 1.6 to 2.4 in a two-band chain (two phase conventions of its phonon vertex)	model	Sect. 13.2
Adiabatic (Herzberg–Teller) amplitude	normalization	1.6 % (TA), 5.2 % (TO) at the edge; 10 % when $E_{\mathrm{min}}(\boldsymbol{q}) - \omega$ falls to 0.18 eV (TA)	est.	Sect. 13.2
Next order in the electron–phonon vertex ($n_g = 4$)	normalization	$\theta_{\mathrm{ep}} = 0.010$–$0.016$ at the absorption edge; ≈ 0.2 at the window edge	est.; bound	Sects. 13.3 and 13.6
Tilt of the pair minimum, branch curvature	threshold	0.02 meV (TA), 0.18 meV (LA) at 1 keV; masses 1 %	computed	Sect. 13.3
External screening $1/(\varepsilon_1^0)^2$	normalization	1/40–1/48 (head), 1/30–1/38 (macroscopic) at 1 keV; 3.7–4.8 % across the window	computed	Sect. 13.4
Screening relative to the optical calibration: external model factor, local-field wings, pair channel	normalization	0.3–2.6 on L^{PA} for the external factor and the wings; pair ratio one in both solvable limits at exact pinning; residual from the 0.2 keV spread of pair momenta and the range between the limits not sized	est.; named	Sects. 13.4 and 13.9
Adiabatic local-density source	normalization	about 5 % of the pair term	est.	Sect. 13.4
Parabolic pockets, constant vertex (edge form)	tilt in ω	within 15 % of the 300 K data from 55 to 490 meV above the gap; factor 2 at 17 meV	computed	Sect. 13.5
Temperature of absorption data	tilt in ω	untransferred 77 K data ×2.3–2.6, 1.7–1.9, 1.3–1.5, 1.1–1.3 at 15, 25, 50, 100 meV; strength factor 0.88 (77 K), 0.61 (295 K); branch weights ±25 %: < 1 % from 77 K	computed	Sect. 13.5
Cold absorption data: 0.5 K fit against 20 K data	normalization; tilt in ω	×20 at 1.20 eV, ×2.1 at 1.25 eV, within 5 % above 1.37 eV; rate ×9 at 0.36 MeV, ×1.8 at 0.40 MeV; 55 % of the 0.36 MeV rate from the fit below its lowest data point	computed	Sects. 13.8 and 13.9
Margin of the off-shell window, pair widths	tilt in q	controlling ratios ≤ 0.21 at $\delta_{\mathrm{w}} = 0.3$ eV; widths 3–21 meV; window to $\lvert\boldsymbol{q}\rvert = 1.33$ keV (parabolic), 1.27 keV (computed bands)	est.; model	Sect. 13.6
Fano term	tilt	zero in Si and Ge; GaAs $\lesssim 0.3\Gamma/\delta_{\mathrm{F}}$, a few percent of first emissions moved	est.	Sect. 13.6
Nonadiabatic zero-point renormalization	threshold	8 meV on the gap; onset 0.0023 MeV if the gap is built from a computed renormalization; with the measured gap only its variation across the zone, zero-phonon edge ≤ 0.002 MeV	computed; bound	Sect. 13.7
Exciton–phonon replicas beyond the two-pair sector	new support	no size	named	Sect. 13.8
Singlet against triplet pairs in axion-like absorption	normalization	no pair channel in the triplet response, size not known; replica lines shifted by the exchange splitting, about 2×10^{-3} of E_x^b	named (open)	Sect. 13.8
Finite-q form factor of the rate estimate	normalization; tilt in m_χ	factor 3; combined factor 6.5 at 0.38–0.43 MeV, quoted as 10	est.	Sect. 13.9
Completeness of the plane-wave band sum (f-sum)	normalization	0.81 → 0.89 of $q^2/2m_e$ with 16 → 100 bands; nonlocal and truncation parts not separated	computed	Sect. 13.9
Zero-phonon side of the ratio (isotropic exciton mass, second-hole vertex)	normalization	density of states low by 1.3–1.6; second-hole term an upper estimate	est.	Sect. 13.9
Spin–orbit lift of symmetry zeros	new support	zero-phonon ratio ≤ 0.01; longitudinal acoustic kink $< 10^{-4}$ on the axis	computed; est.	Sects. 13.3 and 13.9

calibration, whose pair-channel part no calculation sizes; and, near the onset, the spread of the cold data. Every threshold row is small in mass, a few thousandths of an MeV at most, and the order of the edges survives all of them together. What a threshold error costs, however, is not small near the onset, where Eq. (13.51) raises it to the power $p_{\mathrm{on}} + 7/2$.

13.9.8 Open problems

The onset rests on one set of force constants, soft on exactly the branch that sets it, and a direct calculation of $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$ at two lattice constants and with a second pseudopotential would verify it (Appendix E.14). The end of the phonon-assisted lead, the width of the two-channel band and the mass from which the zero-phonon line leads follow from the ratio of the two rates computed on common bins, with the full amplitude on the phonon-assisted side and the mirror-plane cones, the second heavy hole and the X-face edge on the zero-phonon side (Appendix E.14). The same calculation gives the rate and the reach with their controls: the optical limit, the symmetry zeros, the soft acoustic limit and the completeness of the band sum, whose plane-wave f-sum with a norm-conserving pseudopotential reaches 0.81 of $q^2/2m_e$ with 16 bands and 0.89 with 100 at $|\boldsymbol{q}| = 0.1$ to $0.5\,\text{Å}^{-1}$, the bands left out and the nonlocal term not yet separated. The pair channel beyond the optical calibration needs the supercell calculation of Appendix E.16, with local fields in the supercell basis. Germanium needs quasiparticle bands with spin–orbit coupling for its zero-phonon edge and phonons that pass the neutron data on the transverse acoustic branch, and GaAs, the control of the whole chain, its band calculation and its rate with the polar vertex (Appendix E.14).

One measurement would test the chain directly: the cold absorption edge of silicon within 25 meV of its threshold, which no measurement at millikelvin temperatures covers. There the untransferred 77 K data are off by more than a factor of two (Sect. 13.5), and the two cold datasets of Sect. 13.9.4 differ by a factor of twenty at 1.20 eV.

Summary

Dark matter lighter than about 0.4 MeV cannot create an electron–hole pair in silicon on its own: where it deposits the gap it transfers about 0.9 keV, and a pair with that momentum costs more than the gap. A phonon of about 15 meV takes the missing crystal momentum (Sect. 13.1). The lightest mass that creates a pair without a phonon is a joint minimum over the momentum transfer and the hole's crystal momentum. The lightest that creates one with a phonon is the kinematic floor with the gap raised by one phonon energy, at the wavevector the dark matter leaves for the phonon: 0.3374 MeV for the free pair, 0.3332 MeV for the exciton replica. The onset is insensitive to the band structure and rests on one phonon energy. Germanium has its onset at 0.212 to 0.214 MeV from the measured transverse acoustic branch alone, and GaAs, though direct, has a phonon-assisted band of its own above 0.433 MeV, because its light pair cannot absorb the momentum either.

The amplitude is of second order, one density vertex and one electron–phonon vertex on four paths (Sect. 13.2). The exclusion principle turns the paths through filled states into exchange partners of the blocked ones, and on shell the amplitude becomes a one-electron sum over all bands, filled ones included; cutting it to empty bands is wrong by a factor of order one. The intraband terms cancel at $\boldsymbol{q} = 0$, so the amplitude starts at first order in q. It does not interfere with the zero-phonon line, whose final states have one phonon fewer. Its adiabatic form is the derivative of the zero-phonon transition density with respect to a frozen phonon, the Herzberg–Teller term, accurate to a few percent at the absorption edge.

Squared and summed, the amplitude gives the phonon-assisted structure factor and loss function, which enter the rate exactly as the zero-phonon ones do (Sect. 13.3). Inside the

off-shell window they are the whole electronic response; above the zero-phonon edge they add to it, and the f-sum rule they share forbids adding a bare zero-phonon rate to them without the renormalization of Sect. 13.7. At its threshold the direction-averaged loss function rises as the cube of the energy above it, as the fourth power at the longitudinal acoustic kink, which a selection rule on the valley axis forbids, and as the power 3/2 for an exciton replica.

Screening acts on the density vertex and on the final pair (Sect. 13.4). The external factor is one inverse macroscopic dielectric function for the whole amplitude, 1/30 to 1/38 in the loss function at 1 keV in silicon, far from the optical $1/\varepsilon_\infty^2$. The pair channel screens at the pinned momentum of the final pair, the same for every $\boldsymbol{q}$; its size and sign are open, bracketed only by two solvable limits a factor of ten apart, and in both limits a measured optical edge calibrates it at every $\boldsymbol{q}$.

As $\boldsymbol{q} \to 0$ the loss function becomes phonon-assisted optical absorption, by the continuity equation applied to the second-order amplitude (Sect. 13.5). The measured edge of silicon is its test: the Hall–Bardeen–Blatt form, with one fitted scale, lies within 15 % of the 300 K data over 55 to 490 meV above the gap. Absorption data must be transferred to the temperature of a detector branch by branch. The inverse transfer is exact for branch-resolved spectra and ill-posed without them, and untransferred 77 K data overstate the cold edge by more than a factor of two within 18 to 23 meV of it.

Two kinds of small denominator threaten the amplitude (Sect. 13.6). Soft acoustic phonons give finite amplitudes once the recoil, deformation and velocity terms of a Ward identity are kept, and inside the window the soft limit is never reached. Zero-phonon detunings fix the window. The phonon-assisted rate is a channel of its own only where every first-order transition is off shell by a margin, 0.3 eV in silicon. The margin confines it to $|\boldsymbol{q}| \leq 1.33$ keV on the valley axis in the parabolic description and 1.27 keV with the computed bands. Beyond the window only totals are defined, with the Dyson convention, and the Fano term vanishes in silicon and germanium.

At finite temperature and zero point the bands are renormalized, and the special-displacement method evaluates the thermal average as a deterministic quadrature (Sect. 13.7). Between room temperature and 10 mK the gap opens by 46 meV and the onset moves from 0.324 to 0.337 MeV; below a few kelvin the window does not move. The nonadiabatic part of the zero-point renormalization, 8 meV on the silicon gap, is already in the measured gap, and only its variation across the zone, at most 0.002 MeV on the zero-phonon edge, can move the window. Excitons with phonons add replica edges below the free-pair edges, each rising as a square root at fixed $\boldsymbol{q}$, with a completeness condition that reduces to the two-pair sector; at the optical edge of cold silicon they carry most of the absorption. The same vertices give the phonon-assisted absorption of dark photons, axion-like particles and scalars at the indirect edge; the dark photon has no new pair-plus-phonon amplitude at first order in its coupling (Sect. 13.8).

In silicon the first zero-phonon edge lies at 0.405 to 0.408 MeV, off the valley axis in a mirror plane, with its 1s line at 0.402 to 0.405 MeV (Sect. 13.9). The glide of the diamond structure forbids its vertex, so the first zero-phonon line is weak; the second heavy hole opens from 0.415 to 0.420 MeV, and an allowed Umklapp line of full size opens at 0.50 MeV, where the same glide inverts the rule. These edges, the rule and their order are established. Between the onset and the first edge the phonon-assisted rate is the only electron-recoil channel. Its rate rises from the onset as the power $p_{\mathrm{on}} + 7/2$ of the distance in mass, so that near the onset only the exponent is predicted. By an estimate normalized to the measured cold edge, with rates uncertain by a factor of ten apart from the pair channel's dependence on q, it leads to about 0.43 MeV, the two channels share 0.43 to 0.51 MeV, and the zero-phonon line leads above. The nuisances of the chapter are collected in Table 13.2. Germanium and GaAs need quasiparticle band calculations. Throughout, the dark matter coupled to the electron density,

which does not act on spin; Chap. 14 turns to couplings that reach the spins of a magnet.

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Chapter 14

Magnetic excitations

Every electron spin in silicon is paired in a bond, and flipping one costs the 1.17 eV gap, so a dark-matter particle that couples to the spin of the electron rather than to its charge can deposit energy in silicon only above that gap. In NiO the spins of the nickel ions are ordered, and a disturbance of that order, a magnon, costs anywhere from a few meV to 118 meV (Sect. 7.1). The same particle can emit magnons there as a slow neutron does, and a bosonic field with a spin coupling can be absorbed by the uniform precession of a ferromagnet, at a frequency that an applied magnetic field tunes. This chapter asks which magnetic targets respond to which couplings, how far measured neutron data can stand in for computed responses, which crystals respond differently to $\boldsymbol{q}$ and $-\boldsymbol{q}$ because they break inversion or time reversal, and when an ultralight field must be treated as a coherent classical wave rather than as a flux of particles. It builds on the spin waves of Sect. 7.1 and the spin rows of Sect. 8.1, and Fig. 14.1 maps its sections; superconductors, the other targets with meV gaps, follow in the next chapter.

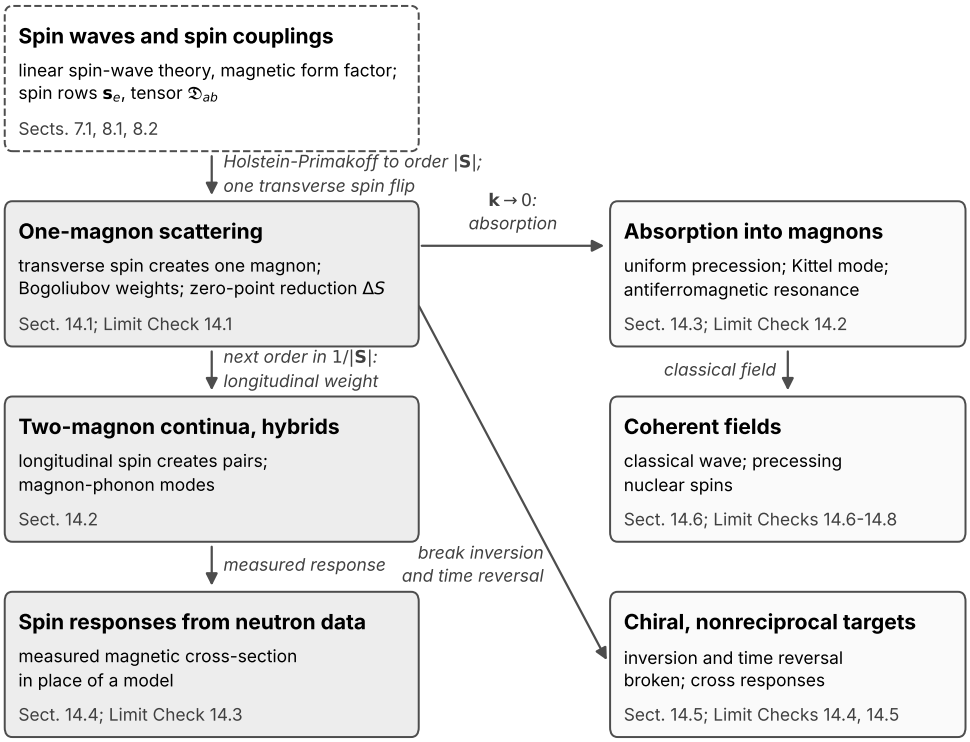


Figure 14.1: Magnetic excitations, from the spin waves and spin couplings of earlier chapters to the targets of this one. A transverse spin coupling creates one magnon at leading order in $1/|\boldsymbol{S}|$, with the Bogoliubov weights of an antiferromagnet and a zero-point reduction of the ordered moment (Sect. 14.1; Limit Check 14.1). At the next order the longitudinal spin creates magnon pairs, and magnons mix with phonons (Sect. 14.2). The measured magnetic neutron cross-section can replace the model response (Sect. 14.4; Limit Check 14.3). At vanishing momentum an absorbed boson drives the uniform precession (Sect. 14.3; Limit Check 14.2); a field whose modes are highly occupied acts as a coherent classical wave, which also drives precessing nuclear spins (Sect. 14.6; Limit Checks 14.6 to 14.8). Targets that break both inversion and time reversal add chiral and nonreciprocal responses (Sect. 14.5; Limit Checks 14.4 and 14.5).

14.1 Magnon scattering

A dark-matter particle of 1 MeV that couples to the spin of the electron can transfer up to 5.3 keV of momentum, twice the distance from Γ to the ordering wavevector of NiO. It can therefore create a magnon anywhere in the zone, and Sect. 7.1 showed that the weight of that magnon depends strongly on where it lands: an antiferromagnet suppresses magnons near Γ and enhances them near the ordering wavevector. Here we turn those weights into a rate, from the lightest mass that can emit a magnon above a few meV to masses where the magnetic form factor cuts it off, and we estimate the first correction in $1/|\boldsymbol{S}|$.

14.1.1 One-magnon response of an antiferromagnet

Every coupling to the electron spin in Table 8.1, from the contact interaction $\mathcal{O}_4 = \boldsymbol{S}_\chi \cdot \boldsymbol{S}_e$ to the velocity-dependent $\mathcal{O}_{12}$–$\mathcal{O}_{15}$, reaches the electrons through the spin row $\boldsymbol{s}_e(\boldsymbol{q})$ of Eq. (8.3), together with the spin current for some of them. In a magnetic insulator the spin density lives on the open shells of the ions, and Eq. (7.14) replaces $\boldsymbol{s}_e(\boldsymbol{q})$ by $F^{\mathrm{mag}}(q)\sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\boldsymbol{S}_j$, a sum over the ionic spins weighted by the magnetic form factor. We take ions whose orbital moment is quenched, as for Ni^{2+}; ions with an orbital moment add an orbital row (Appendix C.19, Sect. 14.3). At linear spin-wave order the transverse components $S^{\pm}$ are linear in the magnon operators and create one magnon, while S^z is quadratic and creates two. The density row creates no magnon at all with the nuclei clamped, so the density–spin blocks of S_{ab} vanish on one-magnon final states.

For the ferromagnet the one-magnon response is Eq. (7.15). For the antiferromagnet we need the matrix elements of $S^{\pm}(\boldsymbol{q})$ at any $\boldsymbol{q}$, not only near Γ and near the ordering wavevector. Appendix D.38 computes them from the operators of Appendix D.19. A momentum transfer outside the zone of one sublattice is reduced by a reciprocal-lattice vector $\boldsymbol{G}$ of that sublattice, and the sites of sublattice B, displaced from those of A by $\boldsymbol{\delta}_0$, pick up the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\delta}_0} = \pm 1$. That phase decides whether the two sublattices add in phase or against each other. Since $J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{q})$ changes sign under a reciprocal vector of the sublattice that maps A onto B, the phase is carried automatically if the coefficients $A_{\boldsymbol{k}}$ and $B_{\boldsymbol{k}}$ of Eq. (7.9) are evaluated at the full momentum transfer rather than at its reduced value. The two degenerate magnons, one created by S^- and one by S^+, precess in opposite senses and have equal weight. Inserted into Eq. (4.13), their circular polarizations add to a linear one:

One-magnon spin response of a two-sublattice antiferromagnet

$$S_{ab}(\boldsymbol{q},\omega) = \frac{\pi|\boldsymbol{S}|}{\Omega_{\mathrm{ion}}}\,|F^{\mathrm{mag}}(q)|^2\, w^{\mathrm{M}}_{\boldsymbol{q}}\,(\delta_{ab} - \hat{z}_a\hat{z}_b)\,\delta(\omega - \omega^{\mathrm{M}}_{\boldsymbol{q}}), \qquad w^{\mathrm{M}}_{\boldsymbol{q}} = \frac{|\boldsymbol{S}|\,(A_{\boldsymbol{q}} - B_{\boldsymbol{q}})}{\omega^{\mathrm{M}}_{\boldsymbol{q}}}, \tag{14.1}$$

for $T = 0$, $\omega > 0$ and moments along $\pm\hat{\boldsymbol{z}}$, with Ω_{ion} the volume per magnetic ion (for NiO the volume of the chemical primitive cell), $\omega^{\mathrm{M}}_{\boldsymbol{q}}$ of Eq. (7.11), and $A_{\boldsymbol{q}}$, $B_{\boldsymbol{q}}$ of Eq. (7.9) at the momentum transfer $\boldsymbol{q}$ itself.

Compared to the ferromagnet of Eq. (7.15), two things change. The antisymmetric part $\mathrm{i}\epsilon_{abc}\hat{z}_c$ is gone, because the antiferromagnet offers both senses of precession; it absorbs a transverse probe of either circular polarization. The weight is no longer one: $w^{\mathrm{M}}_{\boldsymbol{q}}$ is the factor $(u^{\mathrm{M}} \mp v^{\mathrm{M}})^2$ of Eq. (7.16), now written so that it holds at every $\boldsymbol{q}$ (Eq. (D.214)). The matrix in brackets has eigenvalues 1, 1 and 0, so Eq. (14.1) is positive semidefinite as long as $w^{\mathrm{M}}_{\boldsymbol{q}} \geq 0$, which is the stability condition $A_{\boldsymbol{q}} \geq |B_{\boldsymbol{q}}|$ of Sect. 7.1.

The weight has a simple form near the two points where the magnon energy vanishes.

Write $(\omega^{\mathrm{M}}_{\boldsymbol{q}})^2 = |\boldsymbol{S}|^2(A_{\boldsymbol{q}} - B_{\boldsymbol{q}})(A_{\boldsymbol{q}} + B_{\boldsymbol{q}})$. Near Γ the sum $A + B$ tends to $2J^{\mathrm{ex}}_{\mathrm{AB}}(0)$, so

$$w^{\mathrm{M}}_{\boldsymbol{q}} \simeq \frac{\omega^{\mathrm{M}}_{\boldsymbol{q}}}{2|\boldsymbol{S}|J^{\mathrm{ex}}_{\mathrm{AB}}(0)} \quad (\boldsymbol{q} \to 0), \qquad w^{\mathrm{M}}_{\boldsymbol{q}} \simeq \frac{2|\boldsymbol{S}|J^{\mathrm{ex}}_{\mathrm{AB}}(0)}{\omega^{\mathrm{M}}_{\boldsymbol{q}}} \quad (\boldsymbol{q} \to \boldsymbol{Q}_{\mathrm{AF}}),$$

where near the ordering wavevector $\boldsymbol{Q}_{\mathrm{AF}}$, $(\pi/a)(1,1,1)$ in NiO, $B_{\boldsymbol{q}} \to -J^{\mathrm{ex}}_{\mathrm{AB}}(0)$ and the difference $A - B$ takes the large value instead. The scale is the exchange energy $2|\boldsymbol{S}|J^{\mathrm{ex}}_{\mathrm{AB}}(0) = 12|\boldsymbol{S}|(J_1 + J_2) = 211.7\,\mathrm{meV}$ for NiO. A magnon of 5 meV near Γ carries $w^{\mathrm{M}} = 0.024$, a magnon of the same energy near $\boldsymbol{Q}_{\mathrm{AF}}$ carries 42, and a ferromagnet would give 1 to both. Physically, a long-wavelength transverse probe tries to tilt the two sublattices the same way, against the full exchange field. A probe that alternates in sign from one (111) plane to the next tilts them in opposite directions, which rotates the Néel order as a whole and costs almost nothing. Evaluated within $0.01\,\text{Å}^{-1}$ of either point, Eq. (14.1) agrees with these forms to 0.02 %.

The exchange constants that fix $A_{\boldsymbol{q}}$, $B_{\boldsymbol{q}}$ and $\omega^{\mathrm{M}}_{\boldsymbol{q}}$ come, for NiO, from the neutron fit of Eq. (7.5). For a magnet without such data they are computed in DFT $+U$ by the energy-mapping, spin-spiral or Green's-function routes of Appendix E.9, together with the single-ion anisotropies that open the gaps. Appendix E.20.4 specifies these calculations for yttrium iron garnet (YIG), NiO, MnO, Cr_2O_3 and $NiPS_3$. The spread of the computed J_2 of NiO, 11 to 27 meV (Sect. 7.1), scales ω^{M} but leaves $w^{\mathrm{M}}_{\boldsymbol{q}}$ unchanged, since the weight depends only on ratios of exchange constants. A ferrimagnet such as YIG, with 20 magnetic ions per primitive cell, needs the multi-sublattice Bogoliubov transformation in place of Eq. (14.1).

14.1.2 Rate formula and mass edge

The rate follows from Eq. (8.16) with S_{ab} of Eq. (14.1) in the spin block. Since the response is proportional to $\delta_{ab} - \hat{z}_a\hat{z}_b$, the contraction picks out the part of the dark-matter tensor transverse to the moments,

$$\sum_{a,b} \mathfrak{D}_{ab} S_{ab}(\boldsymbol{q},\omega) = \frac{\pi|\boldsymbol{S}|}{\Omega_{\mathrm{ion}}} |F^{\mathrm{mag}}(q)|^2 \, w^{\mathrm{M}}_{\boldsymbol{q}} \left[\mathrm{Tr}\,\mathfrak{D} - \hat{\boldsymbol{z}}\cdot\mathfrak{D}\cdot\hat{\boldsymbol{z}}\right] \delta(\omega - \omega^{\mathrm{M}}_{\boldsymbol{q}}), \tag{14.2}$$

with $\mathfrak{D}_{ab}$ of Eq. (8.7) for the spin rows. For $\mathcal{O}_4$, the spin-dependent contact interaction, $\mathfrak{D}_{ab} = |c_4^{(e)}|^2\delta_{ab}/4$ for dark matter of spin $\frac{1}{2}$, the bracket is $|c_4^{(e)}|^2/2$, and the rate does not depend on the direction of the moments. For $\mathcal{O}_6$, whose tensor is proportional to q_aq_b, the bracket is proportional to $1 - (\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{z}})^2$. Only the component of the momentum transfer perpendicular to the moments then creates a magnon, and the rate of a fixed crystal changes as the Earth turns the wind relative to the moments.

We present rates for $\mathcal{O}_4$ with a heavy mediator, $c_4^{(e)}$ independent of q. The reference cross-section is that of the same operator on a free electron, $\bar{\sigma}_e = \mu^2_{\chi e}\langle|\tilde{V}|^2\rangle/\pi$ as in Sect. 1.4, with the average over both spins. Since $\langle|\boldsymbol{S}_\chi\cdot\boldsymbol{S}_e|^2\rangle = \sum_{a,b}\mathrm{Tr}(S^a_\chi S^b_\chi)\,\mathrm{Tr}(S^a_e S^b_e)/4 = 3/16$,

$$\bar{\sigma}_e = \frac{3\mu^2_{\chi e}|c_4^{(e)}|^2}{16\pi}, \tag{14.3}$$

the definition of Trickle, Zhang and Zurek [1, Eq. (11)] for this operator. The response depends on the direction of $\boldsymbol{q}$ through $w^{\mathrm{M}}_{\boldsymbol{q}}$ and $\omega^{\mathrm{M}}_{\boldsymbol{q}}$, and the full velocity kernel of Appendix A.8 gives the daily modulation of a fixed crystal. Averaged over the orientations of the crystal relative to the wind, the kernel is $\eta(v_{\mathrm{min}})/2q$ per $\mathrm{d}^3q/(2\pi)^3$ (Eq. (A.42)). The rate per unit target mass above a threshold ω_{th} is then

$$\mathcal{R}(\omega_{\mathrm{th}}) = \frac{2\rho_\chi\bar{\sigma}_e|\boldsymbol{S}|}{3\,m_\chi\rho_T\,\mu^2_{\chi e}\,\Omega_{\mathrm{ion}}} \int_0^\infty \mathrm{d}q\, q \left\langle |F^{\mathrm{mag}}(q)|^2\, w^{\mathrm{M}}_{\boldsymbol{q}}\, \eta(v_{\mathrm{min}}(q,\omega^{\mathrm{M}}_{\boldsymbol{q}}))\, \Theta(\omega^{\mathrm{M}}_{\boldsymbol{q}} - \omega_{\mathrm{th}})\right\rangle_{\hat{\boldsymbol{q}}}, \tag{14.4}$$

with $\langle\cdots\rangle_{\hat{\boldsymbol{q}}}$ the average over the directions of $\boldsymbol{q}$ at fixed q and $v_{\min} = \omega/q + q/2m_\chi$ from Eq. (1.23). The numerical factor collects $\pi|c_4^{(e)}|^2/2$ from Eq. (14.2), $16\pi\bar{\sigma}_e/3\mu_{\chi e}^2$ from Eq. (14.3), and $1/4\pi^2$ from the angular integral. At $F^{\rm mag} = 1$, Eq. (14.4) is the orientation average of the one-magnon rate of Trickle, Zhang and Zurek [1, Eqs. (5)–(10)], which their later work extends to all the operators of Table 8.1 [2, Eqs. (34) and (38)]. For one sublattice, $w^{\rm M} = 1$ and the bracket of the ferromagnet replaces that of Eq. (14.2); the result is their ferromagnetic rate [1, Eq. (12)] with the spin density $|\boldsymbol{S}|/\Omega_{\rm ion}$.

The lightest mass that reaches a threshold follows from the linear dispersion near the gapless points, $\omega^{\rm M} \simeq c_{\rm M}|\boldsymbol{q} - \boldsymbol{k}_0|$ with $\boldsymbol{k}_0$ either 0 or $\boldsymbol{Q}_{\rm AF}$ and $c_{\rm M}$ the magnon velocity. The momentum transfer from light dark matter is small, so only the region near Γ counts. A magnon of momentum q can be emitted if $c_{\rm M}q \le qv - q^2/2m_\chi$, that is, for $q \le 2m_\chi(v - c_{\rm M})$, and it lies above threshold for $q \ge \omega_{\rm th}/c_{\rm M}$. Both hold at the fastest speed when

$$m_\chi \ge \frac{\omega_{\rm th}}{2c_{\rm M}(v_{\max} - c_{\rm M})}, \tag{14.5}$$

the magnon analog of the acoustic edge of Sect. 11.1. The speed $c_{\rm M} = 1.39 \times 10^{-4}$ of Worked example 7.2 is that of J_2 alone. With J_1 added, the model of Fig. 14.3 has speeds from 1.29×10^{-4} to 1.36×10^{-4} depending on direction, and the fastest puts the edge for $\omega_{\rm th} = 5\,\text{meV}$ at 7.3 keV. Near the edge, however, the weight is $w^{\rm M} \simeq \omega_{\rm th}/2|\boldsymbol{S}|J^{\rm ex}_{\rm AB}(0) = 0.024$, and the antiferromagnet starts about forty times below a ferromagnet with the same magnon velocity.

14.1.3 Zero-point spin reduction

Linear spin-wave theory is the leading order of an expansion in $1/|\boldsymbol{S}|$, and for Ni^{2+}, with $|\boldsymbol{S}| = 1$, the expansion parameter $1/(2|\boldsymbol{S}|)$ is one half. Before we trust Eq. (14.4) for NiO we need the size of the first correction, and the cleanest handle is a sum rule. Each ionic spin has a fixed length, $\boldsymbol{S}_i^2 = |\boldsymbol{S}|(|\boldsymbol{S}| + 1)$, and the equal-time correlations of one site split that total into three parts. The static moment $\langle S_i^z\rangle^2$ scatters only elastically, into Bragg peaks. The transverse fluctuations $W_\perp = \langle S_i^x S_i^x + S_i^y S_i^y\rangle$ are, at linear order, all one-magnon weight. The longitudinal fluctuations $W_\parallel = \langle (S_i^z)^2\rangle - \langle S_i^z\rangle^2$ create magnon pairs. Each weight is the integral of the corresponding block of S_{ab} over ω and over the zone. Together they fix how the spectral weight that a spin coupling can reach is divided between the one-magnon line, the two-magnon continuum and the elastic peaks, which deposit no energy.

In the Néel state the transverse fluctuations would be $|\boldsymbol{S}|$ and the longitudinal ones zero. The true ground state is not the Néel state. The pair term of Eq. (7.9) mixes in states with a deviation on each sublattice, and each mode leaves the zero-point deviation $(v_{\boldsymbol{k}}^{\rm M})^2$ on every site (Sect. 7.1). Appendix D.38 evaluates the three weights in this ground state, the longitudinal one by Wick's theorem:

Spin weights at linear spin-wave order

$$\Delta S = \frac{1}{N_{\rm s}}\sum_{\boldsymbol{k}} (v_{\boldsymbol{k}}^{\rm M})^2, \qquad W_\perp = |\boldsymbol{S}|\,(1 + 2\Delta S), \qquad W_\parallel = \Delta S\,(1 + \Delta S),$$
$$\frac{W_\parallel}{W_\perp} = \frac{\Delta S\,(1 + \Delta S)}{|\boldsymbol{S}|\,(1 + 2\Delta S)}, \tag{14.6}$$

with the sum over the zone of one sublattice and the ordered moment $\langle S_i^z\rangle = |\boldsymbol{S}| - \Delta S$.

The zero-point spin reduction ΔS depends on the lattice and on ratios of exchange constants, not on $|\boldsymbol{S}|$, so the ratio falls as $1/|\boldsymbol{S}|$ with a coefficient fixed by the lattice. For a ferromagnet $v_{\boldsymbol{k}}^{\rm M} = 0$, and both ΔS and $W_\parallel$ vanish: the fully polarized state is exact, and a longitudinal

probe finds nothing to excite. For an antiferromagnet the zone average of the weight $w_{\boldsymbol{q}}^{\mathrm{M}}$ of Eq. (14.1) is $1+2\Delta S$, which is the same statement as $W_\perp$, and gives a check of any numerical evaluation of the one-magnon response.

The three weights violate the sum rule. Added up, $W_\perp + W_\parallel + \langle S_i^z\rangle^2$ exceeds $|\boldsymbol{S}|(|\boldsymbol{S}|+1)$ by $\Delta S(1+2\Delta S)$, because linear spin-wave theory replaces the square roots of the Holstein–Primakoff transformation Eq. (7.7) by their value at zero deviation. Holding the ordered moment and $W_\parallel$ at the values of Eq. (14.6), the sum rule requires the transverse weight $(|\boldsymbol{S}|-\Delta S)(1+2\Delta S)$, and Appendix D.38 shows that the full square roots, evaluated in the same ground state, give exactly this value. At this order, with $\langle S_i^z\rangle$ and $W_\parallel$ held at their linear spin-wave values, the one-magnon weight of linear spin-wave theory is too large by the fraction $\Delta S/|\boldsymbol{S}|$. This is an estimate of the leading correction, not a bound. The corrected transverse operator also creates three magnons, which lowers the one-magnon part further; however, the change of the ground state at the same order, which moves $\langle S_i^z\rangle$ and $W_\parallel$ as well, is not part of the estimate. A second consequence is less obvious. The one-magnon and two-magnon weights together, $(|\boldsymbol{S}|-\Delta S)(1+2\Delta S)+\Delta S(1+\Delta S) = |\boldsymbol{S}|(1+2\Delta S)-\Delta S^2$, agree with the linear spin-wave one-magnon weight up to ΔS^2. The first correction does not remove weight; it moves the fraction $\Delta S/|\boldsymbol{S}|$ of it from the one-magnon line into the two-magnon continuum, which lies at other energies for the same momentum transfer (Sect. 14.2).

Worked example 14.1: The zero-point spin reduction of NiO

For the type-II order of NiO with the exchange constants Eq. (7.5), $(v_{\boldsymbol{k}}^{\mathrm{M}})^2 = \frac{1}{2}[A_{\boldsymbol{k}}(A_{\boldsymbol{k}}^2 - B_{\boldsymbol{k}}^2)^{-1/2} - 1]$ with $A_{\boldsymbol{k}}$ and $B_{\boldsymbol{k}}$ of Eq. (7.9). We average it over a 100^3 grid of the cube of side $4\pi/a$, the period of the reciprocal lattice of the nickel ions along each cube axis, which holds every wavevector of the zone twice, and find

$$\Delta S = 0.0784.$$

The grid avoids the gapless points at Γ and $\boldsymbol{Q}_{\mathrm{AF}}$, where $(v^{\mathrm{M}})^2$ diverges as the inverse distance; this singularity is integrable. The sum changes by 1.1×10^{-4} from 60^3 to 80^3 points and by 5×10^{-5} from 80^3 to 100^3, as the inverse square of the grid size, and extrapolates to 0.0785. With $J_1 = 0$ it is 0.0783. The reason is that the next-nearest neighbors at $\pm a\hat{\boldsymbol{x}}$, $\pm a\hat{\boldsymbol{y}}$, $\pm a\hat{\boldsymbol{z}}$ link the fcc lattice into four interpenetrating simple-cubic lattices of side a, each a Néel antiferromagnet. Their zero-point reduction is Anderson's 0.078 for the simple-cubic antiferromagnet [3], and the weak J_1 barely changes it.
With $|\boldsymbol{S}| = 1$, the ordered moment is $|\boldsymbol{S}| - \Delta S = 0.9216$ of its classical value, and Eq. (14.6) gives

$$W_\perp = 1.157, \qquad W_\parallel = 0.0846, \qquad \frac{W_\parallel}{W_\perp} = \frac{0.0784\times1.0784}{1\times1.1569} = 0.0731.$$

The sum rule requires $W_\perp = 0.9216\times1.1569 = 1.066$: at this order the one-magnon weight of linear spin-wave theory, and with it the rate Eq. (14.4), is high by about $\Delta S/|\boldsymbol{S}| = 7.8\,\%$. The two-magnon continuum holds 7.3 % of the linear spin-wave one-magnon weight, and the total, one-magnon plus two-magnon, is 0.995 of it. Note that $\Delta S/|\boldsymbol{S}| = 0.078$ is six times smaller than the expansion parameter $1/(2|\boldsymbol{S}|) = 0.5$. In three dimensions the zero-point motion of the spin waves is small, and the expansion in $1/|\boldsymbol{S}|$ converges faster than its parameter suggests.

Limit Check 14.1 sets the literature rate against these weights as $|\boldsymbol{S}|$ varies at the fixed lattice and exchange ratios of NiO, where $\Delta S = 0.078$ throughout.

Limit Check 14.1. One-magnon rate at linear spin-wave order

$\Gamma(\boldsymbol{v}) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} \sum_{a,b} \mathfrak{D}_{ab} S_{ab}(\boldsymbol{q}, \omega_{\boldsymbol{q}})$ with the one-magnon response Eq. (14.1) at $F^{\mathrm{mag}} = 1$, spectral weight $W_\perp = |\boldsymbol{S}|(1 + 2\Delta S)$ per ion [1, Eqs. (5)–(12), 2, Eq. (38)].

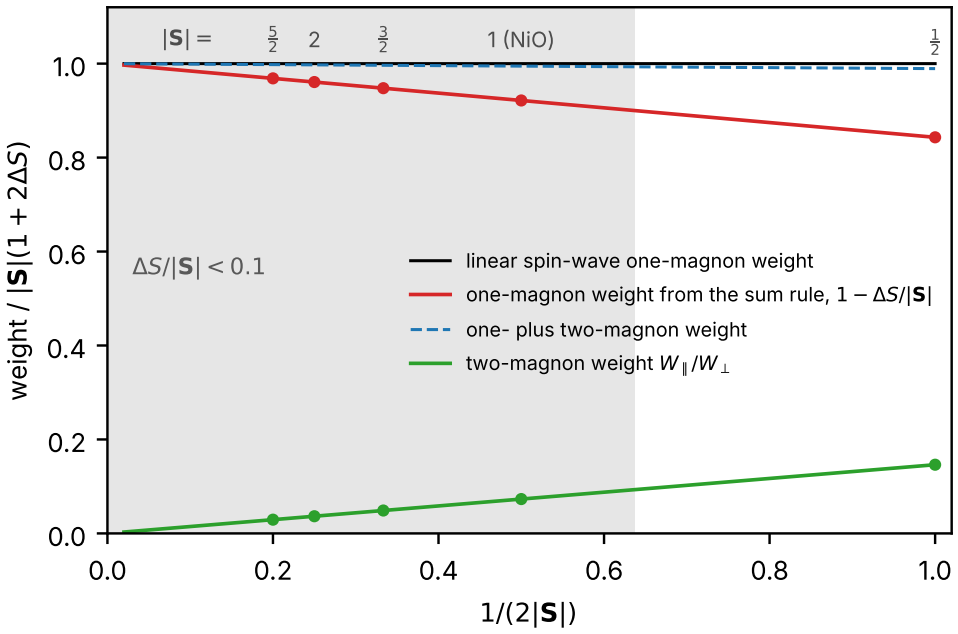


Figure 14.2: Spectral weights of the spin of one magnetic ion against the expansion parameter $1/(2|\boldsymbol{S}|)$, in units of the one-magnon weight of linear spin-wave theory, $|\boldsymbol{S}|(1+2\Delta S)$, which the literature rate contains (black line). Red: the one-magnon weight that the sum rule $\boldsymbol{S}^2 = |\boldsymbol{S}|(|\boldsymbol{S}|+1)$ requires, $1-\Delta S/|\boldsymbol{S}|$. Green: the two-magnon weight $W_\parallel/W_\perp$ of Eq. (14.6). Dashed: one- plus two-magnon weight, $1 - \Delta S^2/[|\boldsymbol{S}|(1 + 2\Delta S)]$. Lattice and exchange ratios of NiO (type-II order, $J_1/J_2 = -0.072$), zero-point reduction $\Delta S = 0.078$ from a 100^3 grid; points at $|\boldsymbol{S}| = \frac{1}{2}$, 1 (NiO), $\frac{3}{2}$, 2 and $\frac{5}{2}$. Gray: $\Delta S/|\boldsymbol{S}| < 0.1$, which holds for $|\boldsymbol{S}| \geq 0.78$.

For $|\boldsymbol{S}| = \frac{1}{2}$ the one-magnon line is overcounted by about 16 % at this order and the two-magnon weight reaches 15 %, and the magnon interactions of the next order, which shift and broaden the magnons, are no longer small.

For Mn^{2+} and Fe^{3+}, with $|\boldsymbol{S}| = \frac{5}{2}$, the same lattice gives a defect of 3.1 % and a two-magnon weight of 2.9 %. The rates of this section use the exchange fitted to the measured magnons, so the shift of the magnon energies by the quartic terms of the Holstein–Primakoff expansion is already absorbed into J_1 and J_2 (Worked example 7.2). The division of weight in Fig. 14.2 is not.

14.1.4 The one-magnon rate of NiO

The inputs for NiO are those of Sect. 7.1: the exchange constants Eq. (7.5), $|\boldsymbol{S}| = 1$, and $a = 4.17\,\text{Å}$, so that the volume per nickel ion is $\Omega_{\mathrm{ion}} = a^3/4 = 18.13\,\text{Å}^3$ and the density, from four formula units per cubic cell, is $\rho_T = 6.84\,\mathrm{g\,cm^{-3}}$. The form factor of Ni^{2+}, whose orbital moment is quenched, is the spin-only $F^{\mathrm{mag}} = \langle j_0(q)\rangle$ of Eq. (C.166). We take $\langle j_0\rangle$ from the tabulated fit $A\mathrm{e}^{-\alpha s^2} + B\mathrm{e}^{-\beta s^2} + C\mathrm{e}^{-\gamma s^2} + D$ with $s = q/4\pi$, in Å^{-1}, and the coefficients $(A, \alpha, B, \beta, C, \gamma, D) = (0.0163, 35.88, 0.3916, 13.22, 0.6052, 4.339, -0.0133)$ [4,

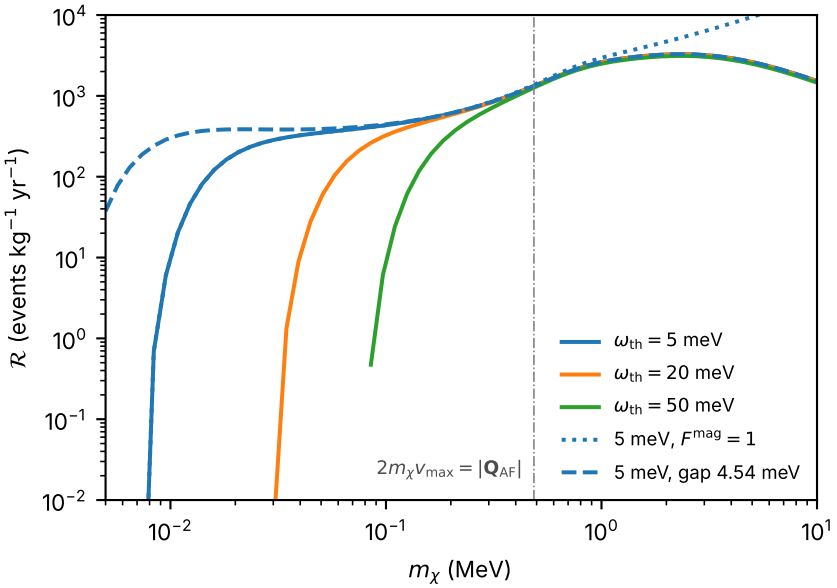


Figure 14.3: One-magnon rate of NiO per kilogram-year for dark matter with the spin-dependent contact interaction $\mathcal{O}_4 = \boldsymbol{S}_\chi \cdot \boldsymbol{S}_e$ (heavy mediator) at the reference cross-section $\bar{\sigma}_e = 10^{-40}\,\mathrm{cm}^2$ of Eq. (14.3), against the dark-matter mass, from Eq. (14.4): $T = 0$, rate averaged over the orientations of the crystal relative to the dark-matter wind, standard halo of Sect. 1.2. Solid: energy thresholds of 5, 20 and 50 meV. Model: Heisenberg exchange $J_1 = -1.37\,\mathrm{meV}$, $J_2 = 19.01\,\mathrm{meV}$ from neutron scattering, $|\boldsymbol{S}| = 1$, $a = 4.17\,\text{Å}$, undistorted fcc lattice without anisotropy (gapless), linear spin-wave theory, magnetic form factor of Ni^{2+}. Dotted: 5 meV threshold without the form factor, $F^{\rm mag} = 1$. Dashed: 5 meV threshold with an easy-axis single-ion anisotropy added to $A_{\boldsymbol{k}}$ and chosen to put the zone-center gap at 4.54 meV, the zone-center magnon of NiO seen in far-infrared absorption. Vertical line: the mass at which the largest momentum transfer, $2m_\chi v_{\rm max}$, reaches the ordering wavevector, 0.49 MeV.

Sect. 4.4.5, Table 4.4.5.2]. It gives $|F^{\rm mag}|^2 = 0.84$ at $|\boldsymbol{Q}_{\rm AF}| = 2.6\,\mathrm{keV}$ and 0.49 at 5.3 keV, the largest momentum transfer of a 1 MeV particle.

Two limits fix the shape of the rate before any integral is done. At the light end, Eq. (14.5) sets the edge, and above it the weight near Γ grows in proportion to the magnon energy. At the heavy end, the momentum transfers sweep many zones and the magnon energies, at most 118 meV, are negligible against qv. Then $v_{\rm min} \simeq q/2m_\chi$, the direction average of $w^{\rm M}_{\boldsymbol{q}}$ over a shell that crosses many zones tends to its zone average $1 + 2\Delta S$, and $\int \mathrm{d}q\, q\, \eta(q/2m_\chi) = 4m_\chi^2 \int \mathrm{d}u\, u\, \eta(u) = 2m_\chi^2 \langle v \rangle$, with $\langle v \rangle$ the mean speed of the dark matter in the laboratory. Eq. (14.4) at $F^{\rm mag} = 1$ becomes

$$\mathcal{R} \simeq \frac{\rho_\chi}{m_\chi} \frac{\langle v \rangle}{\rho_T \Omega_{\rm ion}} \bar{\sigma}_e \Big(\frac{m_\chi}{\mu_{\chi e}}\Big)^2 \frac{|\boldsymbol{S}|(1 + 2\Delta S)}{3/4}. \tag{14.7}$$

This is the flux of dark matter times the number of ions per unit mass times a cross-section per ion. In other words, at large q the magnet looks like a gas of free spins, each with the transverse weight $W_\perp$ in place of the electron's $\frac{3}{4}$, and with the reduced mass $\mu_{\chi e}$ replaced by m_χ for a heavy target. With $\langle v \rangle = 354\,\mathrm{km/s}$ for the halo of Sect. 1.2, the integral Eq. (14.4) at $F^{\rm mag} = 1$ is 0.96 of Eq. (14.7) at 3 MeV and 0.989 at 10 MeV.

Fig. 14.3 shows the full integral. At a threshold of 5 meV the rate rises from the edge, at

7.4 keV on the mass grid of the figure, to 10 events per kilogram-year at 10 keV, 280 at 30 keV and 430 at 100 keV, for $\bar{\sigma}_e = 10^{-40}\,\mathrm{cm}^2$. At 100 keV all of it comes from momentum transfers within half the distance to $\boldsymbol{Q}_{\mathrm{AF}}$, where the weight is suppressed. Once the momentum transfer reaches the ordering wavevector the enhanced magnons there take over, and at 1 MeV, where the rate is 2.6×10^3, only 19 % of it comes from the region near Γ. The rate peaks at 3.2×10^3 near 3 MeV and then falls, because the form factor removes the momentum transfers beyond the inverse radius of the $3d$ shell. It costs a factor 0.89 at 1 MeV, and a factor 11.4 against $F^{\mathrm{mag}} = 1$ at 10 MeV, where the rate is 1.5×10^3. The threshold matters only for the lightest masses. Above 0.3 MeV the rates at 5, 20 and 50 meV differ by less than 20 %, since most of the rate then comes from magnons above 50 meV. Below, the higher thresholds push the edge up to 30 keV and 85 keV. The second is later than Eq. (14.5) gives for 50 meV, because the dispersion bends over below its maximum.

Near its gap the rate depends on what the Heisenberg model leaves out. Magnetic anisotropy, which Sect. 7.1 neglected, lifts the Goldstone modes at Γ and $\boldsymbol{Q}_{\mathrm{AF}}$ to energies below 5 meV. The undistorted J_1–J_2 model also has a third, accidental zero at $\mathrm{X} = (2\pi/a)(1,0,0)$, where $A_{\boldsymbol{k}} = B_{\boldsymbol{k}} = 6J_2 - 2J_1$; the weight vanishes there, and the rhombohedral distortion of the real crystal lifts it. Far-infrared absorption places a zone-center magnon of NiO at 4.54 meV, the value used by Hutchings and Samuelsen [5] and quoted by dos Santos and collaborators [6, Sect. III]. We add an easy-axis term K to $A_{\boldsymbol{k}}$, which keeps the form of Eq. (7.9), and choose $K = 0.0973\,\mathrm{meV}$ so that $|\boldsymbol{S}|[(A_0 + K)^2 - A_0^2]^{1/2} = 4.54\,\mathrm{meV}$. With a 5 meV threshold the rate then grows by a factor of 29 at 10 keV and 1.35 at 30 keV, and changes by less than 3 % above 100 keV. The gap puts magnons just above threshold at vanishing momentum, where the kinematics is easiest, and their weight $K^{1/2}/(2A_0)^{1/2}$ is set by the anisotropy. For dark matter lighter than about 50 keV, the rate of NiO near its gap is set by its anisotropy. The real anisotropy is biaxial: it keeps the moments in the (111) planes (Fig. 7.1) and gaps the two branches differently, so that only one of them reaches 4.54 meV while the in-plane branch lies near 0.6 meV. The one-constant model gaps both at 4.54 meV, and the factor 29 is an upper value. The anisotropy enters through the uniform modes of Sect. 14.3 and the anisotropy constants of Appendix E.20.4. Temperature adds thermal magnons with occupation $\mathrm{e}^{-\omega^{\mathrm{M}}/T}$; at the gap of 4.54 meV the exponent is -5×10^3 at 10 mK and -53 at 1 K, and the zero-temperature response holds at any cryogenic operating point.

14.2 Two-magnon continua and magnon–phonon modes

In NiO, linear spin-wave theory assigns to the component of each nickel spin along the ordered moment a fluctuation weight equal to 7 % of the one-magnon weight (Worked example 14.1), carried entirely by magnon pairs, which that component creates (Sect. 7.1). Each pair shares the momentum transfer between its two members, so at a small momentum transfer the pair can take up any energy from the one-magnon energy to twice the top of the magnon band, 236 meV. The pairs thus decouple the energy deposited from the momentum transfer, which matters for the lightest dark matter. In the effective theory of gapless antiferromagnetic magnons, the two-magnon channel extends the projected reach of NiO, MnO and $\mathrm{Cr_2O_3}$ down to masses of about 1 keV [7]. The lattice adds a second correction. Through spin–orbit coupling a strain changes the magnetic anisotropy energy, and a magnon and a phonon of the same wavevector and energy mix into hybrid modes. Together with the one-magnon weight defect of Sect. 14.1, the pairs, their final-state interaction and the hybrids are the corrections in $1/|\boldsymbol{S}|$ and in the magnetoelastic coupling to the one-magnon rate, and for NiO we size each.

14.2.1 Two-magnon amplitude and response

Take the two-sublattice antiferromagnet of Appendix D.19, with the moments along $\pm\hat{\boldsymbol{z}}$, and the longitudinal ionic spin at wavevector $\boldsymbol{q}$, $S^z(\boldsymbol{q}) = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j} S^z_j$. On sublattice A, $S^z_i = |\boldsymbol{S}| - d^\dagger_{\mathrm{A}i} d_{\mathrm{A}i}$, and on B, $S^z_j = -|\boldsymbol{S}| + d^\dagger_{\mathrm{B}j} d_{\mathrm{B}j}$, exactly, by Eq. (7.7). The constants give the magnetic Bragg peaks, at $\omega = 0$, and deposit no energy. The rest is bilinear in the bosons. With the Fourier convention of Appendix D.19 on each sublattice, the lattice sums give

$$S^z(\boldsymbol{q}) = -\sum_{\boldsymbol{k}} d^\dagger_{\mathrm{A}\boldsymbol{k}}\, d_{\mathrm{A},\boldsymbol{k}-\boldsymbol{q}} + \sum_{\boldsymbol{k}} d^\dagger_{\mathrm{B}\boldsymbol{k}}\, d_{\mathrm{B},\boldsymbol{k}+\boldsymbol{q}} + (\text{Bragg}),$$

with $\boldsymbol{k}$ over the N_s wavevectors of the zone of one sublattice. Now insert the inverse Bogoliubov transformation Eq. (D.115), $d_{\mathrm{A}\boldsymbol{k}} = u^{\mathrm{M}}_{\boldsymbol{k}} a_{\boldsymbol{k}1} - v^{\mathrm{M}}_{\boldsymbol{k}} a^\dagger_{\boldsymbol{k}2}$ and $d_{\mathrm{B}\boldsymbol{k}} = u^{\mathrm{M}}_{\boldsymbol{k}} a_{\boldsymbol{k}2} - v^{\mathrm{M}}_{\boldsymbol{k}} a^\dagger_{\boldsymbol{k}1}$. At $T = 0$ only the terms with two creation operators survive acting on the magnon vacuum $|0\rangle$. The A term contributes $u^{\mathrm{M}}_{\boldsymbol{k}} v^{\mathrm{M}}_{\boldsymbol{k}-\boldsymbol{q}}\, a^\dagger_{\boldsymbol{k}1} a^\dagger_{\boldsymbol{k}-\boldsymbol{q},2}$, and the B term, after the relabeling $\boldsymbol{k}+\boldsymbol{q} \to \boldsymbol{k}$, contributes $-v^{\mathrm{M}}_{\boldsymbol{k}} u^{\mathrm{M}}_{\boldsymbol{k}-\boldsymbol{q}}$ times the same pair. Hence

$$S^z(\boldsymbol{q})|0\rangle = \sum_{\boldsymbol{k}} \mathcal{A}^{2\mathrm{m}}_{\boldsymbol{k}}(\boldsymbol{q})\, a^\dagger_{\boldsymbol{k}1} a^\dagger_{\boldsymbol{k}-\boldsymbol{q},2}|0\rangle, \qquad \mathcal{A}^{2\mathrm{m}}_{\boldsymbol{k}}(\boldsymbol{q}) = u^{\mathrm{M}}_{\boldsymbol{k}} v^{\mathrm{M}}_{\boldsymbol{k}-\boldsymbol{q}} - v^{\mathrm{M}}_{\boldsymbol{k}} u^{\mathrm{M}}_{\boldsymbol{k}-\boldsymbol{q}}, \tag{14.8}$$

apart from the Bragg term. The first magnon carries the crystal momentum $\boldsymbol{k}$ and lowers the total S^z by one; the second carries $\boldsymbol{q}-\boldsymbol{k}$, since $a_{\boldsymbol{k}2}$ destroys $-\boldsymbol{k}$ in the convention of Appendix D.19, and raises S^z by one. The pair has the momentum $\boldsymbol{q}$ and no net spin, as a longitudinal operator requires. The wavevector $\boldsymbol{k}-\boldsymbol{q}$ may leave the zone of one sublattice. We then compute u^{M} and v^{M} at that wavevector from $A_{\boldsymbol{k}}$ and $B_{\boldsymbol{k}} = J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{k})$ with its sign. Since $J^{\mathrm{ex}}_{\mathrm{AB}}$ changes by at most a sign under a reciprocal vector of the sublattice, v^{M} does too, and $|\mathcal{A}^{2\mathrm{m}}_{\boldsymbol{k}}|^2$ does not depend on which representative is used.

The amplitude Eq. (14.8) vanishes for a ferromagnet and at $\boldsymbol{q} = 0$, and it is largest near the ordering wavevector. A ferromagnet has $v^{\mathrm{M}} \equiv 0$ and no amplitude, and at $T = 0$ no inelastic longitudinal response at all, beyond linear order as well (Sect. 14.1). In an antiferromagnet the amplitude vanishes at $\boldsymbol{q} = 0$, where $u_{\boldsymbol{k}} v_{\boldsymbol{k}} - v_{\boldsymbol{k}} u_{\boldsymbol{k}} = 0$, and at every reciprocal vector of the lattice of magnetic ions, where $S^z(\boldsymbol{q})$ is the total S^z, which commutes with H_H and creates nothing. For small $\boldsymbol{q}$, with $u^{\mathrm{M}} = \cosh\theta_{\boldsymbol{k}}$ and $v^{\mathrm{M}} = \sinh\theta_{\boldsymbol{k}}$, the amplitude is $\mathcal{A}^{2\mathrm{m}}_{\boldsymbol{k}} = \sinh(\theta_{\boldsymbol{k}-\boldsymbol{q}} - \theta_{\boldsymbol{k}}) \simeq -\boldsymbol{q}\cdot\nabla_{\boldsymbol{k}}\theta_{\boldsymbol{k}}$, and the weight grows as q^2. Near the ordering wavevector $v^{\mathrm{M}}_{\boldsymbol{k}-\boldsymbol{q}}$ has the opposite sign to $v^{\mathrm{M}}_{\boldsymbol{k}}$, the two terms add, and the amplitude approaches $2u^{\mathrm{M}}_{\boldsymbol{k}} v^{\mathrm{M}}_{\boldsymbol{k}} = \sinh 2\theta_{\boldsymbol{k}}$, its largest value. Physically, $S^z(\boldsymbol{q})$ near the ordering wavevector measures the fluctuation of the staggered moment, which the zero-point motion of the spin waves makes large.

The response follows as in Sect. 7.1. The electron spin density along the moments is $s_{e,z}(\boldsymbol{q}) = F^{\mathrm{mag}}(q) S^z(\boldsymbol{q})$ by Eq. (7.14). The pair states of Eq. (14.8) are orthonormal, and each appears once in the sum over $\boldsymbol{k}$. With $N = 2N_\mathrm{s}$ magnetic ions, Ω_{ion} the volume per ion and $\mathcal{V} = N\Omega_{\mathrm{ion}}$, Eq. (4.13) gives:

Two-magnon spin response of a collinear antiferromagnet

$$S^{(2\mathrm{m})}_{zz}(\boldsymbol{q},\omega) = \frac{2\pi}{\Omega_{\mathrm{ion}}} |F^{\mathrm{mag}}(q)|^2 \frac{1}{2N_\mathrm{s}} \sum_{\boldsymbol{k}} \left|\mathcal{A}^{2\mathrm{m}}_{\boldsymbol{k}}(\boldsymbol{q})\right|^2 \delta(\omega - \omega^{\mathrm{M}}_{\boldsymbol{k}} - \omega^{\mathrm{M}}_{\boldsymbol{k}-\boldsymbol{q}}), \tag{14.9}$$

for $T = 0$ and $\omega > 0$, with z along the moments and $\mathcal{A}^{2\mathrm{m}}$ of Eq. (14.8). Its weight and that of the one-magnon line at the same $\boldsymbol{q}$ are in the ratio $w_\parallel(\boldsymbol{q})/w_\perp(\boldsymbol{q})$ of the weights

per ion

$$w_{\parallel}(\boldsymbol{q}) = \frac{1}{2N_{\mathrm{s}}} \sum_{\boldsymbol{k}} \left|\mathcal{A}^{2\mathrm{m}}_{\boldsymbol{k}}(\boldsymbol{q})\right|^2, \qquad w_{\perp}(\boldsymbol{q}) = |\boldsymbol{S}|\, w^{\mathrm{M}}_{\boldsymbol{q}}, \tag{14.10}$$

with $w^{\mathrm{M}}_{\boldsymbol{q}}$ the one-magnon weight of Eq. (14.1).

Here $w_{\perp}$ is the ω-integral of $S_{xx}+S_{yy}$ in units of $(2\pi/\Omega_{\mathrm{ion}})|F^{\mathrm{mag}}|^2$. The one-magnon weight is of order $|\boldsymbol{S}|$ and the two-magnon weight of order $|\boldsymbol{S}|^0$, since the Bogoliubov coefficients do not depend on $|\boldsymbol{S}|$: the pairs are a correction of relative order $1/|\boldsymbol{S}|$. Eq. (14.9) is the longitudinal structure factor that Huberman and collaborators compared with the two-magnon continuum of the square-lattice antiferromagnet Rb_2MnF_4 measured by neutron scattering [8, Eq. (5)]. A dark-matter model enters through $\mathfrak{D}_{zz} S^{(2\mathrm{m})}_{zz}$ in Eq. (8.16), in the frame of the moments. For a coupling isotropic in the electron spin, $\mathfrak{D}_{ab} \propto \delta_{ab}$, the two-magnon and one-magnon parts of $\sum_a \mathfrak{D}_{aa} S_{aa}$ at fixed $\boldsymbol{q}$ are in the ratio $w_{\parallel}/w_{\perp}$, whatever the direction of the moments in each domain.

The zone averages of the two weights are the local weights of Sect. 14.1. Averaged over $\boldsymbol{q}$, $v^{\mathrm{M}}_{\boldsymbol{k}-\boldsymbol{q}}$ takes both signs equally often, the cross term of $|\mathcal{A}^{2\mathrm{m}}|^2$ drops out, and with $\langle (u^{\mathrm{M}})^2\rangle = 1+\Delta S$ and $\langle (v^{\mathrm{M}})^2\rangle = \Delta S$, where ΔS is the zero-point spin reduction of Sect. 14.1, the average of $w_{\parallel}$ is $\frac{1}{2}\cdot 2(1+\Delta S)\Delta S = W_{\parallel}$, while the average of $w_{\perp}$ is $|\boldsymbol{S}|(1+2\Delta S) = W_{\perp}$ (Sect. 14.1). Eq. (14.10) thus resolves the ratio $W_{\parallel}/W_{\perp}$ of Sect. 14.1 in momentum transfer, which is what a rate needs, since light dark matter samples only small q.

Fig. 14.4 shows the two weights of NiO in linear spin-wave theory with the exchange constants of Eq. (7.5). At small q along [111] the one-magnon weight rises as $w_{\perp} \simeq 0.29\, qa$, and the two-magnon weight as $w_{\parallel} \simeq 0.015\,(qa)^2$, so the ratio vanishes linearly. At $\boldsymbol{q} = (\frac{1}{4},\frac{1}{4},\frac{1}{4})2\pi/a$, a momentum transfer of 1.29 keV, $w_{\perp} = 1.00$ and $w_{\parallel} = 0.081$. Toward the ordering wavevector $w_{\perp}$ diverges while $w_{\parallel}$ stays finite, near 0.25, and the ratio falls again, averaging over the zone to the $W_{\parallel}/W_{\perp}$ of Worked example 14.1.

14.2.2 Kinematic boundaries of the continuum

At fixed $\boldsymbol{q}$, Eq. (14.9) has weight only between the lower edge $\omega_-(\boldsymbol{q}) = \min_{\boldsymbol{k}}(\omega^{\mathrm{M}}_{\boldsymbol{k}} + \omega^{\mathrm{M}}_{\boldsymbol{k}-\boldsymbol{q}})$ and the upper edge $\omega_+(\boldsymbol{q}) = \max_{\boldsymbol{k}}(\omega^{\mathrm{M}}_{\boldsymbol{k}} + \omega^{\mathrm{M}}_{\boldsymbol{k}-\boldsymbol{q}}) \le 2\max_{\boldsymbol{k}} \omega^{\mathrm{M}}_{\boldsymbol{k}}$. Putting one magnon at a zero of the dispersion, $\boldsymbol{k}=0$, shows that $\omega_- \le \omega^{\mathrm{M}}_{\boldsymbol{q}}$; for a dispersion that rises linearly and then bends over, as in NiO, $\omega^{\mathrm{M}}_{\boldsymbol{k}} + \omega^{\mathrm{M}}_{\boldsymbol{k}-\boldsymbol{q}} \ge \omega^{\mathrm{M}}_{\boldsymbol{q}}$, and the lower edge is the one-magnon branch itself. The weight at the lower edge is small, because $|\mathcal{A}^{2\mathrm{m}}|^2$ diverges only as $1/k$ near $\boldsymbol{k}=0$ while the phase space closes as k^2. In NiO the continuum at $(\frac{1}{4},\frac{1}{4},\frac{1}{4})2\pi/a$ runs from the one-magnon energy, 105.8 meV, to 236 meV, and at the ordering wavevector it runs from zero, where both magnons sit at zeros of the dispersion, to 236 meV (Fig. 14.4b). The anisotropy that Eq. (7.6) leaves out gaps the magnons by less than 5 meV in NiO; a gap ω_{gap} in both branches moves the lower edge to at least $2\omega_{\mathrm{gap}}$ and leaves the rest of the continuum in place. At $T>0$ the operator S^z also scatters a thermal magnon into another, with weight proportional to the occupation, which vanishes for temperatures far below the gap.

The continuum changes what dark matter can deposit. Energy conservation, Eq. (1.22), allows $\omega \le qv - q^2/2m_\chi$, and the one-magnon channel opens at Eq. (14.5). A pair, however, can take any energy in its continuum, which at small q reaches far above $c_{\mathrm{M}}q$, so the only limit is Eq. (1.24), $\omega \le \frac{1}{2}m_\chi v^2$, reached at $q = m_\chi v$. For a threshold E_{th} the two-magnon channel opens at the threshold mass Eq. (1.25), $2E_{\mathrm{th}}/v^2_{\mathrm{max}}$. The one-magnon channel opens

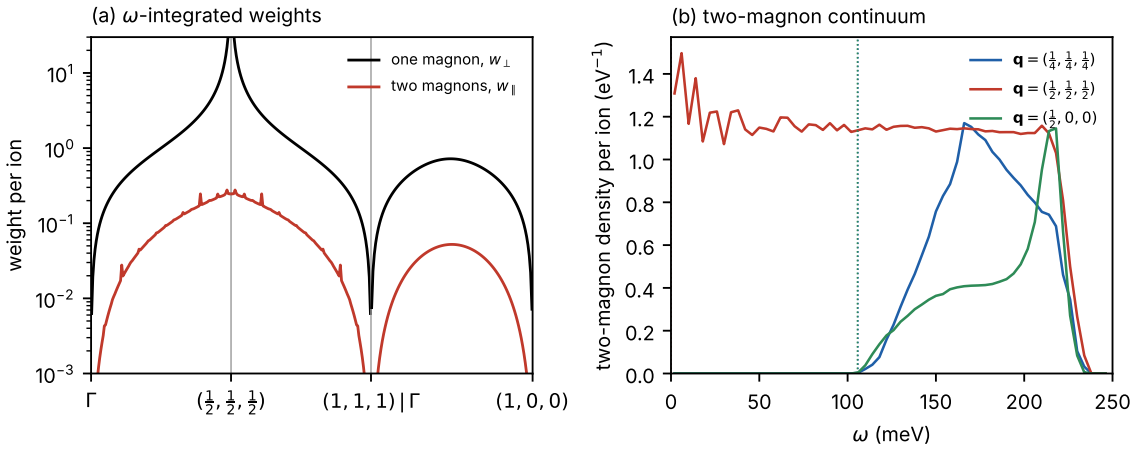


Figure 14.4: One- and two-magnon weights of NiO in linear spin-wave theory: Heisenberg model with $J_1 = -1.37\,\mathrm{meV}$ and $J_2 = 19.01\,\mathrm{meV}$ (antiferromagnetic $J^{\mathrm{ex}} > 0$, pairs counted once), $|\boldsymbol{S}| = 1$, type-II order on the undistorted fcc lattice, no anisotropy, $T = 0$, magnetic form factor set to one. (a) Weights per ion, integrated over energy, of the transverse one-magnon response, $w_\perp$ (both branches), and of the longitudinal two-magnon response, $w_\parallel$. The path runs along [111] from Γ through the ordering wavevector $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ to $(1, 1, 1)$, which is equivalent to Γ, and from Γ to $(1, 0, 0)$; wavevectors in units of $2\pi/a$, $a = 4.17\,\text{Å}$. Both vanish at Γ and $(1, 1, 1)$, where the total longitudinal spin is conserved; $w_\perp$ diverges at the ordering wavevector and vanishes at $(1, 0, 0)$, a soft point of the undistorted model, where $w_\parallel$ is small and would vanish for $J_1 = 0$. (b) Two-magnon density per ion, $(\Omega_{\mathrm{ion}}/2\pi) S_{zz}^{(2\mathrm{m})}/|F^{\mathrm{mag}}|^2$, against energy at three wavevectors (4-meV bins, 3×10^6 sampled wavevectors). The dotted lines mark the one-magnon energy at the same wavevector, zero at the ordering wavevector.

at $E_{\mathrm{th}}/[2c_{\mathrm{M}}(v_{\max}-c_{\mathrm{M}})]$, larger by the factor

$$\frac{v_{\max}^2}{4c_{\mathrm{M}}(v_{\max}-c_{\mathrm{M}})} \simeq \frac{v_{\max}}{4c_{\mathrm{M}}} \qquad (c_{\mathrm{M}} \ll v_{\max}). \tag{14.11}$$

For NiO, with the fastest magnon speed of the J_1–J_2 model, $c_{\mathrm{M}} = 1.36 \times 10^{-4}$ (Sect. 14.1), the factor is 5.1: at $E_{\mathrm{th}} = 1\,\mathrm{meV}$ one magnon needs $m_\chi \geq 1.46\,\mathrm{keV}$ and two magnons $m_\chi \geq 0.285\,\mathrm{keV}$. NiO gains little because its magnons are fast, $c_{\mathrm{M}} \approx v_{\max}/19$; a magnet with slower magnons gains in proportion to $v_{\max}/c_{\mathrm{M}}$. Near the two-magnon edge, however, the momentum transfer $q = m_\chi v_{\max}$ is below an eV, $(qa)^2 \sim 10^{-6}$, and the q^2 law of $w_\parallel$ makes the rate tiny there; the linear law of $w_\perp$ suppresses the one-magnon rate near its own edge less. The reach in mass thus lies well above the kinematic edge, more so for the pair.

14.2.3 Final-state interactions and bound pairs

The two magnons of Eq. (14.8) do not move independently. Linear spin-wave theory dropped the terms of fourth order in the bosons, and they act between the members of the pair. For an antiparallel bond the longitudinal part $S_i^z S_j^z = (|\boldsymbol{S}| - n_{\mathrm{A}i})(-|\boldsymbol{S}| + n_{\mathrm{B}j})$ contains $-n_{\mathrm{A}i}n_{\mathrm{B}j}$: two deviations on neighboring sites lower the energy of the bond by J_{ij}^{ex}, an attraction. The next term of the square root in Eq. (7.7), $S_i^+ \simeq (2|\boldsymbol{S}|)^{1/2}[1 - n_{\mathrm{A}i}/4|\boldsymbol{S}|]d_{\mathrm{A}i}$, adds quartic terms of the same size. The quadratic Hamiltonian is of order $J^{\mathrm{ex}}|\boldsymbol{S}|$ and the quartic one of order J^{ex}, so the interaction is smaller than the energies of the pair by a factor of order $1/|\boldsymbol{S}|$. In the ferromagnetic chain of Worked example 7.1 the attraction is visible directly: on a ring of ten spins $\frac{1}{2}$ the lowest level of total spin $N/2-2$ lies at $0.328|J^{\mathrm{ex}}|$, below the $0.382|J^{\mathrm{ex}}|$ of the cheapest pair of free magnons.

At leading order in $1/|\boldsymbol{S}|$ the interaction acts in the pair channel only. Let $U_{\boldsymbol{q}}^{\mathrm{mm}}(\boldsymbol{k},\boldsymbol{k}')$ be the matrix element of the quartic Hamiltonian between the pair states $a_{\boldsymbol{k}1}^\dagger a_{\boldsymbol{k}-\boldsymbol{q},2}^\dagger|0\rangle$ and $a_{\boldsymbol{k}'1}^\dagger a_{\boldsymbol{k}'-\boldsymbol{q},2}^\dagger|0\rangle$ of total momentum $\boldsymbol{q}$, and $\mathsf{G}_{\boldsymbol{q}}^{2\mathrm{m}}(\omega)$ the free pair propagator, a diagonal matrix in $\boldsymbol{k}$ with elements $1/(\omega - \omega_{\boldsymbol{k}}^{\mathrm{M}} - \omega_{\boldsymbol{k}-\boldsymbol{q}}^{\mathrm{M}} + \mathrm{i}0^+)$. Since $-\operatorname{Im}\mathsf{G}_{\boldsymbol{q}}^{2\mathrm{m}} = \pi\delta(\omega - \omega_{\boldsymbol{k}}^{\mathrm{M}} - \omega_{\boldsymbol{k}-\boldsymbol{q}}^{\mathrm{M}})$, Eq. (14.9) is $-(2/\mathscr{V})|F^{\mathrm{mag}}|^2 \operatorname{Im}[\mathcal{A}^\dagger \mathsf{G}_{\boldsymbol{q}}^{2\mathrm{m}} \mathcal{A}]$, with $\mathcal{A}$ the vector of amplitudes Eq. (14.8). Repeated scattering of the pair sums the ladder series $\mathsf{G} + \mathsf{G}U\mathsf{G} + \ldots$:

$$S_{zz}^{(2\mathrm{m})}(\boldsymbol{q},\omega) = -\frac{2}{\mathscr{V}}\,|F^{\mathrm{mag}}(q)|^2\,\operatorname{Im}\Big[\mathcal{A}^\dagger\,\mathsf{G}_{\boldsymbol{q}}^{2\mathrm{m}}(\omega)\big(1 - U_{\boldsymbol{q}}^{\mathrm{mm}}\mathsf{G}_{\boldsymbol{q}}^{2\mathrm{m}}(\omega)\big)^{-1}\mathcal{A}\Big]. \tag{14.12}$$

For collinear order the Hamiltonian has no cubic terms, because every term of $\boldsymbol{S}_i \cdot \boldsymbol{S}_j$ is even in the bosons. No process of the same order then turns the pair into one magnon or three. Noncollinear order adds a cubic vertex, and with it the decay of one magnon into two. A *bound pair* is a pole of Eq. (14.12) below the lower edge, at the energy $\omega_{\mathrm{b}}(\boldsymbol{q}) < \omega_-(\boldsymbol{q})$ where $\det[1 - U_{\boldsymbol{q}}^{\mathrm{mm}}\mathsf{G}_{\boldsymbol{q}}^{2\mathrm{m}}(\omega_{\mathrm{b}})] = 0$; it appears in S_{zz} as a line with the residue of the pole projected on $\mathcal{A}$.

The interaction cannot change the energy-integrated two-magnon weight. Since $\mathsf{G}_{\boldsymbol{q}}^{2\mathrm{m}}(1 - U_{\boldsymbol{q}}^{\mathrm{mm}}\mathsf{G}_{\boldsymbol{q}}^{2\mathrm{m}})^{-1} = (\omega - H_{\boldsymbol{q}}^{2\mathrm{m}} + \mathrm{i}0^+)^{-1}$, with $H_{\boldsymbol{q}}^{2\mathrm{m}}$ the free pair energies plus $U_{\boldsymbol{q}}^{\mathrm{mm}}$, a Hermitian matrix, and the spectral function of a resolvent integrates to the unit matrix, $\int \mathrm{d}\omega\,(-1/\pi)\operatorname{Im}(\omega - H_{\boldsymbol{q}}^{2\mathrm{m}} + \mathrm{i}0^+)^{-1} = 1$, the integral of Eq. (14.12) over ω is $(2\pi/\mathscr{V})|F^{\mathrm{mag}}|^2\mathcal{A}^\dagger\mathcal{A}$ whatever U^{mm} is. The final-state interaction moves the weight $w_\parallel(\boldsymbol{q})$ within the continuum, and into a bound pair if there is one; only the corrections to the ground state, at the next order, change $w_\parallel$ itself.

Whether a bound pair exists depends on the strength of the attraction compared with the width of the continuum. In three dimensions the density of pair states vanishes at a generic band edge, so a weak attraction does not bind. In a cubic Heisenberg ferromagnet Wortis

found that the total momentum of the pair decides how many bound states exist, and that a sufficiently strong longitudinal (Ising) anisotropy puts bound states below all continuum energies [9]. Single-ion anisotropy binds in a different way. With $H_{\mathrm{an}} = -D_{\mathrm{an}} \sum_j (S_j^z)^2$ and an easy axis, $D_{\mathrm{an}} > 0$, two deviations on one site cost $4D_{\mathrm{an}}(|\boldsymbol{S}| - 1)$ and two on different sites $2D_{\mathrm{an}}(2|\boldsymbol{S}| - 1)$, so the pair on one site is lower by $2D_{\mathrm{an}}$. This on-site attraction exists only for $|\boldsymbol{S}| \geq 1$, when a site can hold two deviations, and above a critical anisotropy the resulting single-ion bound state exists throughout the zone [10]. A dark-matter particle sees these states only if the spin density creates them. In a ferromagnet without anisotropy the pair carries $\Delta S^z = -2$, while one spin operator changes S^z by at most one, so the bound pair is invisible at $T = 0$. Easy-plane or biaxial anisotropy breaks the conservation of the total S^z and makes $v^{\mathrm{M}} \neq 0$. The pair then has an amplitude of the order of the ratio of anisotropy to exchange. In an antiferromagnet the longitudinal pair is created directly by Eq. (14.8), and a bound pair would be a line of S_{zz} below the continuum.

For NiO the interaction is not small, $1/(2|\boldsymbol{S}|) = 0.5$. The vertex $U_{\boldsymbol{q}}^{\mathrm{mm}}$ follows from the spin Hamiltonian symbolically, but the solution of Eq. (14.12) for NiO or YIG, and the existence, energy and weight of a bound pair below the NiO continuum, are open. Appendix E.20 specifies the calculation: one linear solve per $(\boldsymbol{q}, \omega)$ on a grid of 30^3 to 40^3 wavevectors, with the inputs of Appendix E.20.4.

14.2.4 Magnon–phonon hybrids

A strain of the crystal changes the exchange constants, and it changes the magnetic anisotropy energy. Only the second couples one magnon to one phonon in a collinear magnet. Expanding J_{ij}^{ex} to first order in the displacements gives $\sum (\partial J_{ij}^{\mathrm{ex}}/\partial \boldsymbol{u}) \cdot \boldsymbol{u}\ \boldsymbol{S}_i \cdot \boldsymbol{S}_j$, and $\boldsymbol{S}_i \cdot \boldsymbol{S}_j$ expanded about a collinear state has no term linear in the bosons, as the bond expansions of Sect. 7.1 show. Exchange striction therefore produces a phonon together with two magnons, a vertex of third order that couples phonons to the continuum of Eq. (14.9) and damps them, but no mixing at quadratic order. A term linear in the transverse spin needs an energy that depends on the angle between the magnetization and the strain axes, which spin–orbit coupling and the dipolar interaction provide. For a cubic crystal with magnetization direction $\boldsymbol{m} = \boldsymbol{M}/M_{\mathrm{s}}$, saturation magnetization M_{s} and strain $u_{\alpha\beta} = \frac{1}{2}(\partial_\alpha u_\beta + \partial_\beta u_\alpha)$, the magnetoelastic energy density is [11]

$$f_{\mathrm{me}} = B_1 \sum_\alpha m_\alpha^2 u_{\alpha\alpha} + B_2 \sum_{\alpha \neq \beta} m_\alpha m_\beta u_{\alpha\beta}, \tag{14.13}$$

with two magnetoelastic constants B_1 and B_2. With $\boldsymbol{m}$ along $\hat{\boldsymbol{z}}$, the terms of first order in the transverse components are $2B_2(m_x u_{xz} + m_y u_{yz})$: a precessing magnetization exerts an oscillating shear stress.

Take a ferromagnet with one spin of length $|\boldsymbol{S}|$ per cubic cell of side a and a magnon $a_{\boldsymbol{k}}$ of energy $\omega_{\mathrm{m}}(\boldsymbol{k})$. For a wavevector along the magnetization, the shear strains u_{xz} and u_{yz} belong to the two transverse acoustic phonons $b_{\boldsymbol{k}x}$, $b_{\boldsymbol{k}y}$, of energy $\omega_{\mathrm{p}}(\boldsymbol{k}) = c_{\mathrm{TA}} k$. Per cell, $m_x = S^x/|\boldsymbol{S}|$ with $S^x \simeq (|\boldsymbol{S}|/2)^{1/2}(d + d^\dagger)$ and $S^y \simeq -\mathrm{i}(|\boldsymbol{S}|/2)^{1/2}(d - d^\dagger)$ by Eq. (7.7), so the coupling is $(2/|\boldsymbol{S}|)^{1/2} B_{\mathrm{me}}[(d + d^\dagger)u_{xz} - \mathrm{i}(d - d^\dagger)u_{yz}]$ with $B_{\mathrm{me}} = B_2 a^3$. The displacement is Eq. (5.32) with one mass M_{cell} per cell, and $u_{\alpha z} = \frac{1}{2}\partial_z u_\alpha$ brings down $\mathrm{i}k/2$. Collecting the terms that convert a phonon into a magnon,

$$H_{\mathrm{mp}} = \sum_{\boldsymbol{k}} (\mathrm{i} g_{\boldsymbol{k}}^{\mathrm{mp}}\, a_{\boldsymbol{k}}^\dagger b_{\boldsymbol{k}+} + \mathrm{h.c.}) + \ldots, \qquad g_{\boldsymbol{k}}^{\mathrm{mp}} = \frac{B_{\mathrm{me}}\, k}{\left[2|\boldsymbol{S}| M_{\mathrm{cell}}\, \omega_{\mathrm{p}}(\boldsymbol{k})\right]^{1/2}}, \tag{14.14}$$

where $b_{\boldsymbol{k}+} = (b_{\boldsymbol{k}x} + \mathrm{i} b_{\boldsymbol{k}y})/\sqrt{2}$ is the circular phonon that rotates with the precession, and the dots are the terms that create or destroy a magnon and a phonon together. This is the

vertex of Ref. [12, Eqs. (3.23) and (3.25)] for a wavevector along the magnetization. The other circular phonon does not couple: the magnet selects one sense of rotation, as in Eq. (7.15). Since $g^{\mathrm{mp}}_{\boldsymbol{k}} \propto k^{1/2}$, the coupling vanishes for uniform modes.

Near a crossing $\omega_{\mathrm{m}}(\boldsymbol{k}_\times) = \omega_{\mathrm{p}}(\boldsymbol{k}_\times)$ the two modes form a two-level problem. Neglecting the terms in the dots, which shift the energies only at relative order $(g^{\mathrm{mp}}/\omega)^2$, the energies of the *magnon–phonon hybrids* are

$$\omega_\pm = \tfrac{1}{2}(\omega_{\mathrm{m}}+\omega_{\mathrm{p}}) \pm \big[\tfrac{1}{4}(\omega_{\mathrm{m}}-\omega_{\mathrm{p}})^2 + |g^{\mathrm{mp}}_{\boldsymbol{k}}|^2\big]^{1/2}, \qquad |X^{\mathrm{m}}_\pm|^2 = \frac{1}{2}\Big[1 \pm \frac{\omega_{\mathrm{m}}-\omega_{\mathrm{p}}}{\big[(\omega_{\mathrm{m}}-\omega_{\mathrm{p}})^2 + 4|g^{\mathrm{mp}}_{\boldsymbol{k}}|^2\big]^{1/2}}\Big], \tag{14.15}$$

with the magnon content $|X^{\mathrm{m}}_\pm|^2$ and the phonon content $|X^{\mathrm{p}}_\pm|^2 = 1 - |X^{\mathrm{m}}_\pm|^2$. At the crossing the two branches are split by the hybridization gap $2|g^{\mathrm{mp}}_{\boldsymbol{k}_\times}|$ and share both characters equally; at a detuning $\delta_{\mathrm{mp}} = \omega_{\mathrm{m}} - \omega_{\mathrm{p}}$ large compared with g^{mp}, the mostly-magnon branch keeps a phonon content $|g^{\mathrm{mp}}|^2/\delta^2_{\mathrm{mp}}$. In a general magnet the magnons of all sublattices and all phonon branches enter one quadratic boson form. Colpa's diagonalization [13], the method of Appendix C.7 for many modes, with the metric $\mathrm{diag}(1,-1)$ on the annihilation and creation parts, gives the hybrid energies ω_h and contents X^{m}_h, X^{p}_h.

A hybrid carries both the spin response and the lattice response. Its state is $X^{\mathrm{m}*}_h|\mathrm{m}\rangle + X^{\mathrm{p}*}_h|\mathrm{p}\rangle$ at leading order. The spin density acts on the magnon part and the density n_κ of a nuclear species on the phonon part. The one-magnon line of $S_{s_e s_e}$ and the one-phonon line of $S_{n_\kappa n_\kappa}$ at the same $\boldsymbol{q}$ are therefore each split into the two hybrid lines, with weights $|X^{\mathrm{m}}_h|^2$ and $|X^{\mathrm{p}}_h|^2$. An off-diagonal element appears,

$$S_{s_{e,a} n_\kappa}(\boldsymbol{q},\omega) = \frac{2\pi}{\mathscr{V}} \sum_h X^{\mathrm{m}}_h X^{\mathrm{p}*}_h \, \langle \mathrm{m}|s_{e,a}(\boldsymbol{q})|0\rangle^* \langle \mathrm{p}|n_\kappa(\boldsymbol{q})|0\rangle \, \delta(\omega-\omega_h). \tag{14.16}$$

Each hybrid line is a rank-one 2×2 matrix in the spin and nuclear rows, so it saturates the interference bound Eq. (8.21). This is the off-diagonal element of Sect. 8.2.5 that a magnet supplies. A dark-matter model with both a spin coupling and a nucleon coupling interferes with itself at the hybrid lines, with a cross term of either sign as large as twice the geometric mean of the two single-coupling rates.

Worked example 14.2: The hybridization gap of a model ferromagnet

Model the long-wavelength magnons of yttrium iron garnet (YIG) as a ferromagnet with one effective spin per cubic cell, $a = 12.375\,\text{Å}$ [14], of length $|\boldsymbol{S}| = M_{\mathrm{s}}a^3/2\mu_{\mathrm{B}} = 14.23$ for $4\pi M_{\mathrm{s}} = 1750\,\mathrm{G}$, with μ_{B} the Bohr magneton. With the wavevector along the magnetization the magnetostatic term drops out, and $\omega_{\mathrm{m}}(k) = \omega_{\mathrm{Lar}} + D_{\mathrm{sw}}k^2$, with the exchange stiffness $D_{\mathrm{sw}} = 2\mu_{\mathrm{B}} \times 5.17 \times 10^{-9}\,\mathrm{Oe\,cm^2} = 5.985\,\mathrm{meV\,nm^2}$ and the Larmor energy $\omega_{\mathrm{Lar}} = 2\mu_{\mathrm{B}}B = 11.58\,\mu\mathrm{eV}$ in an internal field $B = 0.1\,\mathrm{T}$, chosen for illustration. The transverse sound speed is $c_{\mathrm{TA}} = 3.843\,\mathrm{km/s} = 1.282 \times 10^{-5}$, the mass density $5.17\,\mathrm{g\,cm^{-3}}$, so $M_{\mathrm{cell}} = 5.50\,\mathrm{TeV}$, and $B_2 = 6.96 \times 10^6\,\mathrm{erg\,cm^{-3}}$, so $B_{\mathrm{me}} = B_2 a^3 = 8.23\,\mathrm{meV}$ [12, Eqs. (2.13), (3.7), (3.15) and (3.16)].

The crossing condition $\omega_{\mathrm{Lar}} + D_{\mathrm{sw}}k^2 = c_{\mathrm{TA}}k$ has two roots. The lower one, at $k_\times = 4.63 \times 10^{-3}\,\mathrm{nm^{-1}}$ (a wavelength of $1.36\,\mu\mathrm{m}$), lies just above the uniform mode, at $\omega_\times = 11.71\,\mu\mathrm{eV}$ (2.83 GHz). There Eq. (14.14) gives $g^{\mathrm{mp}} = 0.176\,\mu\mathrm{eV}$ and a gap $2g^{\mathrm{mp}} = 0.351\,\mu\mathrm{eV}$, or 85 MHz, 3.0 % of $\omega_\times$. Since the two dispersions cross with slopes that differ by $c_{\mathrm{TA}} - 2D_{\mathrm{sw}}k_\times$, the modes are mixed, with $|X^{\mathrm{p}}|^2 \geq 0.146$ for $|\delta_{\mathrm{mp}}| \leq 2g^{\mathrm{mp}}$, only for wavevectors within 3.1 % of $k_\times$; at $|\delta_{\mathrm{mp}}| = 10g^{\mathrm{mp}}$ the phonon content is down to 0.010. The upper root, at $k = 0.418\,\mathrm{nm^{-1}}$ and $\omega = 1.06\,\mathrm{meV}$, has a gap of $3.34\,\mu\mathrm{eV}$, 0.32 % of its energy, at $ka = 0.52$, where the continuum model is at the edge of its validity. The full quadratic

Hamiltonian of the modes $\pm\boldsymbol{k}$ at the lower crossing adds the terms that create a magnon and a phonon together and keeps both transverse phonons. Its exact diagonalization changes the gap by a relative 3.5×10^{-5}, of the order of $(g^{\mathrm{mp}}/\omega_\times)^2 = 2.3 \times 10^{-4}$, and leaves the counter-rotating phonon uncoupled at ω_{p}.

Acoustic hybrids are out of reach of absorption. A bosonic field of mass m_ϕ is absorbed at the momentum $p = m_\phi v \approx 10^{-3} m_\phi$, while an acoustic crossing sits at $k_\times = \omega_\times / c_{\mathrm{TA}}$, so that $p/k_\times = v c_{\mathrm{TA}}$, about 10^{-8}. In the model of Worked example 14.2 a field resonant with the uniform magnon, $m_\phi = \omega_{\mathrm{Lar}}$, has $p = 1.16 \times 10^{-8}\,\mathrm{eV}$; the phonon at that momentum has almost no energy, the detuning is ω_{Lar}, and Eq. (14.14) gives a phonon content of 3×10^{-12}. Absorption thus sees the uniform magnon of Sect. 14.3 and the zone-center optical phonons of Sect. 11.8 as separate modes. A hybrid dominates the absorption only where a zone-center optical phonon lies within a few g^{mp} of a uniform magnon, and where symmetry allows a coupling at $\boldsymbol{k} = 0$ through the anisotropy that the optical displacement modulates. Antiferromagnetic resonances in the terahertz range can approach low optical phonons, and an applied field tunes the magnon through them. For $|m_\phi - \omega_h| \lesssim g^{\mathrm{mp}}$, a field that couples to the lattice, such as a dark photon through the Born charges of a polar magnet, is then absorbed at the magnon frequency with up to half of its phonon strength. A field that couples to the spin is absorbed at the phonon frequency with up to half of its magnon strength. The hybrid takes over whenever the borrowed coupling, weighted by $|X_h^{\mathrm{p}}|^2$ or $|X_h^{\mathrm{m}}|^2$, exceeds the direct one; away from the window the borrowed fraction falls as $(g^{\mathrm{mp}}/\delta_{\mathrm{mp}})^2$.

14.2.5 The $1/|\boldsymbol{S}|$ corrections as nuisances

In the language of Sect. 8.8, the continuum adds new support, the one-magnon weight defect is a normalization, and the hybrids tilt the spectrum. The two-magnon continuum, computed in linear spin-wave theory by Eq. (14.9), belongs in the central value, and what a rate of light dark matter needs is its momentum dependence: along [111] the ratio $w_\parallel / w_\perp$ peaks at 0.082 near 1.4 keV and stays below 0.05 under 0.5 keV. Its own correction is the final-state interaction of Eq. (14.12), of relative order $1/(2|\boldsymbol{S}|) = 0.5$. By the sum rule of the ladder it moves weight within the continuum, and into a bound pair below the lower edge if NiO has one, so that on the full spin response it is an estimate of about $0.5 \times 0.073 \approx 0.04$, moved in energy rather than added. The one-magnon defect of Sect. 14.1, 0.078 for NiO at every $\boldsymbol{q}$, moves a limit set by the one-magnon channel by about that fraction, with the sign that weakens it. Magnon–phonon mixing is confined to $|\omega_{\mathrm{m}} - \omega_{\mathrm{p}}| \lesssim 2|g^{\mathrm{mp}}|$ around each crossing, where it moves weight between the spin and nuclear rows without changing the summed spin weight of the two hybrid lines; it is 3 % in energy at the gigahertz crossing of Worked example 14.2, and unsized for NiO, whose magnetoelastic constants are not among the measured inputs of this book. Thermal magnons carry $\mathrm{e}^{-\omega_{\mathrm{gap}}/T}$ and vanish at the temperatures of a cryogenic detector. The magnetoelastic constants B_1 and B_2 of a target without ultrasonic data follow from the strain derivative of the magnetic anisotropy energy, computed with spin–orbit coupling in DFT $+U$ at a few strained cells (Appendix E.20).

14.3 Absorption into magnons

An axion-like particle of mass 10 µeV has a Compton wavelength $2\pi/m_\phi$ of 12 cm and, at the speed v_E of the Earth through the halo (Sect. 1.2), a de Broglie wavelength $2\pi/m_\phi v_E$ of 148 m. Across a crystal of centimeter size its gradient is uniform, and through the spin row of Eq. (8.10) it acts on every electron spin like one oscillating magnetic field. It brings the momentum

$|\boldsymbol{p}_\phi| \approx 10^{-8}\,\mathrm{eV}$ and the energy m_ϕ, and it can be absorbed only by an excitation at vanishing wavevector with energy m_ϕ. In an ordered magnet that excitation is the uniform precession of the spins. An applied field $\boldsymbol{B}_0$ tunes it at 116 µeV per tesla, which places absorption in ferromagnets and ferrimagnets between a few µeV and about a meV; an antiferromagnet has its uniform mode near 1 meV without any field, 1.1 meV in MnF_2. In both kinds of magnet the rate is a single line, whose weight follows from a conservation law in the first case and from two energies of the spin Hamiltonian in the second. Only the $\boldsymbol{k} = 0$ mode enters.

14.3.1 The spin row at vanishing momentum

With the spin row of Eq. (8.10), the axion-like particle has the coefficient $g^{\phi}_{s_e,a} = \mathrm{i} g_{aee}\, p_{\phi,a}/m_e$ per unit amplitude, from $\nabla\phi = \mathrm{i}\boldsymbol{p}_\phi\phi$ for the plane wave. Eq. (8.19) then gives

$$\Gamma_{\mathrm{abs}} = \frac{g_{aee}^2}{2\omega_\phi m_e^2} \sum_{a,b} p_{\phi,a}\, p_{\phi,b}\, S_{ab}(\boldsymbol{p}_\phi, \omega_\phi), \tag{14.17}$$

with a, b the Cartesian components of $\boldsymbol{s}_e$, as in Eq. (7.15). Only the part of S_{ab} symmetric in a and b survives the contraction with $p_{\phi,a}p_{\phi,b}$.

The other rows of the particle create no magnons. The spin current $\mathcal{J}_e^{\alpha\alpha}$ of Eq. (8.10), which carries absorption across an electronic gap (Sect. 10.7), multiplies the polar momentum of each electron by its axial spin and is therefore odd under inversion. Magnons are states of the ground configuration of the magnetic ions, and every multiplet of a configuration has the parity of its orbitals, and $\mathcal{J}_e^{\alpha\alpha}$ therefore has no matrix element between them. It creates charge excitations above the gap of the insulator instead, and reaches magnons only through a crystal field that breaks inversion at the ion together with spin–orbit coupling, which we neglect. The rows of a scalar, n_e and $\mathcal{T}_e$ of Eq. (8.11), act on the orbital coordinates alone; without spin–orbit coupling they commute with every spin operator, and the scalar has no magnon channel at leading order.

A dark photon reaches the spins through its magnetic field. The electron moment $-(e/m_e)\boldsymbol{s}_e$, with $g_e = 2$, couples to ζ times the field $\boldsymbol{B}' = \nabla \times \boldsymbol{A}' = \mathrm{i}\boldsymbol{p}_\phi \times \hat{\boldsymbol{e}}\, A'$ of a dark photon of polarization $\hat{\boldsymbol{e}}$, and the interaction $\zeta(e/m_e)\boldsymbol{B}'\cdot\boldsymbol{s}_e$ is again the spin row [15, Eqs. (55), (59)]. Its coefficient is $g^{\phi}_{s_e,a} = \mathrm{i}\zeta e\,(\boldsymbol{p}_\phi \times \hat{\boldsymbol{e}})_a/m_e$: everything below holds for the dark photon with $g_{aee}\boldsymbol{p}_\phi$ replaced by $\zeta e\,\boldsymbol{p}_\phi \times \hat{\boldsymbol{e}}$. Note that $|\boldsymbol{B}'| \sim v|\boldsymbol{E}'|$: the magnetic coupling is suppressed by the dark-matter speed relative to the electric one, which is why it is subleading wherever electric dipoles exist. We write the vacuum ζ. In-medium mixing (Sect. 8.4.5) replaces it by ζ_{eff}, which here depends on the dielectric constant of the magnet and on the conducting surfaces around it, because the wavelength $2\pi/m_\phi$ exceeds the sample. The electric field of the dark photon couples to electric dipoles, and the magnons of a centrosymmetric magnet carry none; magnets in which they do are the multiferroics of Sect. 14.5.

The momentum is small on the scale of the lattice: for YIG, with $a = 12.375\,\text{Å}$, $|\boldsymbol{p}_\phi|a = 5 \times 10^{-11}$ at $m_\phi = 10\,\mu\mathrm{eV}$. To that accuracy $S_{ab}(\boldsymbol{p}_\phi, \omega) = S_{ab}(0, \omega)$, the magnetic form factor of Eq. (7.14) is $F^{\mathrm{mag}}(0) = 1$, and the spin row becomes

$$\boldsymbol{s}_e(0) = \sum_j \lambda_{S,j}\, \boldsymbol{S}_j, \tag{14.18}$$

where $\boldsymbol{S}_j$ is the angular momentum that labels the low-energy states of ion j and $\lambda_{S,j}$ is the fraction of it carried by electron spin. For an ion without orbital moment, such as Fe^{3+} and Mn^{2+}, $\lambda_S = 1$ and Eq. (14.18) is Eq. (7.14) at $\boldsymbol{q} = 0$.

Table 14.1: Ground multiplets of trivalent rare-earth ions from Hund's rules: total spin S, orbital angular momentum L, total angular momentum J, Landé factor g_J, and the spin fraction $\lambda_S = g_J - 1$ to which an axion-like particle couples (Eq. (14.19)). The last column gives $\lambda_S/(g_J/2)$, the spin coupling relative to the moment that a magnetic field, a neutron and a dark photon see.

Ion	S	L	J	g_J	λ_S	$\lambda_S/(g_J/2)$
Ce^{3+} $(4f^1)$	$\frac{1}{2}$	3	$\frac{5}{2}$	$\frac{6}{7}$	$-\frac{1}{7}$	-0.33
Sm^{3+} $(4f^5)$	$\frac{5}{2}$	5	$\frac{5}{2}$	$\frac{2}{7}$	$-\frac{5}{7}$	-5.0
Gd^{3+} $(4f^7)$	$\frac{7}{2}$	0	$\frac{7}{2}$	2	1	1.0
Dy^{3+} $(4f^9)$	$\frac{5}{2}$	5	$\frac{15}{2}$	$\frac{4}{3}$	$\frac{1}{3}$	0.50
Er^{3+} $(4f^{11})$	$\frac{3}{2}$	6	$\frac{15}{2}$	$\frac{6}{5}$	$\frac{1}{5}$	0.33

14.3.2 The Landé factor

In an ion with an open shell, the spin–orbit coupling Eq. (3.53) of each electron, summed over the shell, splits each term of total orbital angular momentum $\boldsymbol{L}$ and total spin $\boldsymbol{S}$ into multiplets of total angular momentum $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$. Within one multiplet every vector operator is proportional to $\boldsymbol{J}$, by the Wigner–Eckart theorem used in Sect. 7.1, and the constant follows by projecting onto $\boldsymbol{J}$: $\boldsymbol{S} \to [\langle \boldsymbol{S}\cdot\boldsymbol{J}\rangle/J(J+1)]\,\boldsymbol{J}$, with $\boldsymbol{S}\cdot\boldsymbol{J} = \frac{1}{2}(\boldsymbol{J}^2 + \boldsymbol{S}^2 - \boldsymbol{L}^2)$. The electron spin of the ion is therefore

$$\boldsymbol{S} = \lambda_S\,\boldsymbol{J}, \qquad \lambda_S = \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} = g_J - 1, \tag{14.19}$$

where the *Landé factor* g_J is defined by the magnetic moment of the multiplet, $\mu_B(\boldsymbol{L} + 2\boldsymbol{S}) = \mu_B(\boldsymbol{J} + \boldsymbol{S}) = g_J\mu_B\boldsymbol{J}$, so that $g_J = 1 + \lambda_S$ [16, Eqs. (30), (34)]. The axion-like particle couples to $\lambda_S\boldsymbol{J}$ and the dark photon, through $\boldsymbol{B}'$, to the moment $(g_J/2)\boldsymbol{J}$ in units of $2\mu_B$, the same combination that a magnetic field and a neutron see (Appendix C.19). Table 14.1 lists the ground multiplets of trivalent rare-earth ions, which follow from Hund's rules. Their $4f$ shells sit inside the filled $5s$ and $5p$ shells, so that spin–orbit coupling exceeds the crystal field and J is a good quantum number. For Ce^{3+} and Sm^{3+} the spin points against $\boldsymbol{J}$, and in Sm^{3+} the spin coupling exceeds the magnetic one by a factor of five in amplitude. Sm^{3+} is also the most fragile entry: its $J = \frac{7}{2}$ multiplet lies close above the ground multiplet, and since $g_J = \frac{2}{7}$ is small, a small admixture changes $\lambda_S/(g_J/2)$ strongly. The crystal field splits the multiplet, but since the projection holds for every state within it, Eq. (14.19) survives the splitting; it fails as soon as the crystal field or exchange admixes a different J multiplet.

The $3d$ ions of most magnetic insulators behave differently. There the crystal field, of energy $\Delta_{\rm CF}$ of order an eV, exceeds the spin–orbit coupling, and it leaves an orbital ground state $|0\rangle$ that is nondegenerate and can be chosen real. Then $\langle 0|\boldsymbol{L}|0\rangle = 0$, the orbital moment is quenched, the low-energy states are the $2S+1$ states of the spin, and the electron spin is $\boldsymbol{S}$ itself: $\lambda_S = 1$. Spin–orbit coupling within the term, $\xi_{\rm ion}\boldsymbol{L}\cdot\boldsymbol{S}$, admixes excited orbital states $|n\rangle$ at first order, $|0\rangle \to |0\rangle - \sum_n |n\rangle\langle n|\xi_{\rm ion}\boldsymbol{L}\cdot\boldsymbol{S}|0\rangle/(E_n - E_0)$. The orbital moment in the admixed state is $\langle L_a\rangle = -2\xi_{\rm ion}\sum_b \Lambda_{ab}S_b$, with the real symmetric tensor $\Lambda_{ab} = \sum_n \mathrm{Re}\,\langle 0|L_a|n\rangle\langle n|L_b|0\rangle/(E_n - E_0)$, so the moment $\boldsymbol{L} + 2\boldsymbol{S}$ acquires the spectroscopic g tensor of the ion, $g_{{\rm ion},ab} = 2\delta_{ab} - 2\xi_{\rm ion}\Lambda_{ab}$. The shift of $g_{\rm ion}$ from 2 is of first order in $\xi_{\rm ion}/\Delta_{\rm CF}$, while the spin changes only at second order, through the normalization of the admixed state. Setting $\lambda_S = g_{\rm ion} - 1$ for such an ion would therefore err at first order: for an ion with $g_{\rm ion} = 2.2$ it gives $\lambda_S = 1.2$ instead of $1 - O(\xi_{\rm ion}^2/\Delta_{\rm CF}^2)$, and a rate too large by 1.44. The rule $\lambda_S = g_J - 1$ belongs to J multiplets only. The spectroscopic $g_{\rm ion}$ of a quenched ion converts a

measured magnetic response into the spin response, the conversion that Sect. 14.4 applies to neutron data.

14.3.3 The Larmor line

Take a magnet with rotation-invariant exchange, such as the Heisenberg model Eq. (7.6), in a uniform field, and give all ions the same λ_S and the same spectroscopic factor g_{ion}. We take $\hat{\boldsymbol{z}}$ along the net spin, which the field selects opposite to itself because the electron moment is opposite to its spin, so that the Zeeman energy is $-\omega_{\text{Lar}} S^z_{\text{tot}}$ with the Larmor frequency $\omega_{\text{Lar}} = g_{\text{ion}}\mu_B|\boldsymbol{B}_0|$, 115.8 µeV or 28.0 GHz per tesla for $g_{\text{ion}} = 2$. The total spin $\boldsymbol{S}_{\text{tot}} = \sum_j \boldsymbol{S}_j$ commutes with the exchange, and the Zeeman term leads to $[H, S^-_{\text{tot}}] = \omega_{\text{Lar}} S^-_{\text{tot}}$. Acting on any eigenstate, S^-_{tot} therefore produces an eigenstate exactly ω_{Lar} higher, however strong and complicated the exchange, and S^+_{tot} one exactly ω_{Lar} lower [17, Eqs. (2.4), (2.5)]. By Eq. (14.18), $\boldsymbol{s}_e(0) = \lambda_S \boldsymbol{S}_{\text{tot}}$, and all of $S_{ab}(0,\omega)$ sits at $\omega = \pm\omega_{\text{Lar}}$ (and at $\omega = 0$ for S^z_{tot}, which is conserved).

The weight follows from the spin algebra. In Eq. (4.13), the line at $+\omega_{\text{Lar}}$ comes from the S^-_{tot} parts of $s_{e,x} = \lambda_S(S^+_{\text{tot}} + S^-_{\text{tot}})/2$ and $s_{e,y} = \lambda_S(S^+_{\text{tot}} - S^-_{\text{tot}})/2\mathrm{i}$, and its strength is the thermal average $\langle S^+_{\text{tot}} S^-_{\text{tot}}\rangle$. The commutator gives $\langle S^+_{\text{tot}} S^-_{\text{tot}}\rangle - \langle S^-_{\text{tot}} S^+_{\text{tot}}\rangle = 2\langle S^z_{\text{tot}}\rangle$, and detailed balance for a transition of energy ω_{Lar} gives $\langle S^-_{\text{tot}} S^+_{\text{tot}}\rangle = \mathrm{e}^{-\beta\omega_{\text{Lar}}}\langle S^+_{\text{tot}} S^-_{\text{tot}}\rangle$. With $\mathfrak{s}_{\text{net}} = \langle S^z_{\text{tot}}\rangle/N$ the net spin per unit cell of volume Ω, N cells and $\mathscr{V} = N\Omega$:

The Larmor line

$$S_{ab}(0,\omega) = \frac{\pi\lambda_S^2\,\mathfrak{s}_{\text{net}}}{\Omega}\,\frac{\delta_{ab} - \hat{z}_a\hat{z}_b + \mathrm{i}\epsilon_{abc}\hat{z}_c}{1-\mathrm{e}^{-\beta\omega_{\text{Lar}}}}\,\delta(\omega-\omega_{\text{Lar}}), \tag{14.20}$$

for $\omega > 0$, and in Eq. (14.17)

$$\Gamma_{\text{abs}} = \frac{\pi g_{aee}^2\lambda_S^2\,\mathfrak{s}_{\text{net}}\,p_{\phi\perp}^2}{2m_e^2\,\omega_\phi\,\Omega}\,\frac{\delta(\omega_\phi-\omega_{\text{Lar}})}{1-\mathrm{e}^{-\beta\omega_{\text{Lar}}}}, \tag{14.21}$$

with $p^2_{\phi\perp} = |\boldsymbol{p}_\phi|^2 - (\boldsymbol{p}_\phi\cdot\hat{\boldsymbol{z}})^2$, for exchange invariant under spin rotations and uniform λ_S and g_{ion}, at any temperature.

At $T = 0$ and with $\mathfrak{s}_{\text{net}} \to |\boldsymbol{S}|$, Eq. (14.20) is the one-magnon response Eq. (7.15) of the ferromagnet at $\boldsymbol{q}_r = 0$, now shown to hold beyond linear spin-wave theory. It holds for a ferrimagnet, whose net spin is the difference of its sublattices, and for a paramagnet, whose $\mathfrak{s}_{\text{net}}$ is the magnetization induced by the field. The counted rate carries $1/(1-\mathrm{e}^{-\beta\omega_{\text{Lar}}})$; the net rate Γ_{net} of Eq. (8.19) carries $\mathfrak{s}_{\text{net}}$ alone. With one sublattice, Eq. (14.21) is the rate of Mitridate and collaborators for a single magnon branch [16, Eqs. (27), (35)], and with $g_{aee}/m_e = 1/f$ and $\lambda_S = 1$ it is the golden-rule rate $(\pi/2)|V|^2\delta(m_\phi - \omega_{\text{Lar}})$ of the two-level problem of Chigusa, Moroi and Nakayama [15, Eqs. (25)–(27)].

The Larmor line is also a selection rule. YIG has 20 magnetic ions in its primitive cell and 20 magnon branches; 19 of them are gapped at $\boldsymbol{k} = 0$, and Eq. (14.20) leaves them no weight. Mitridate and collaborators found this numerically and explained it by the classical torque: a uniform field $\boldsymbol{f}$ that precesses every spin alike changes no exchange energy, $\mathrm{d}(\boldsymbol{S}_j\cdot\boldsymbol{S}_{j'})/\mathrm{d}t \propto (\boldsymbol{f}_j - \boldsymbol{f}_{j'})\cdot(\boldsymbol{S}_j\times\boldsymbol{S}_{j'})$ [16, Eq. (28)]. The commutator states the same thing exactly. It also shows what lifts the rule. If $\lambda_{S,j}$ differs between sublattices, $\boldsymbol{s}_e(0)$ is no longer proportional to $\boldsymbol{S}_{\text{tot}}$, and the gapped branches acquire weight in proportion to $(\lambda_{S,j} - \lambda_{S,j'})^2$; anisotropic exchange or single-ion anisotropy breaks the commutation of $\boldsymbol{S}_{\text{tot}}$ with H and moves the uniform line away from ω_{Lar}.

Eq. (14.21) has a delta function, and a real line has a width. The width of the field itself is $\sim m_\phi v_0^2$, a fraction 6×10^{-7} of m_ϕ, set by the coherence time $t_{\rm coh}=1/m_\phi v_0^2$ of Sect. 8.2.3, 0.10 ms at 10 μeV. The uniform mode of a good magnet is broader. For the uniform mode of YIG (the Kittel mode of Sect. 14.3.4), Mitridate and collaborators assume a full width $\Gamma_{\rm M}=10^{-5}\omega$, a free parameter that they set to the width of that mode [16, Sect. V B]; it is an assumed value, not a millikelvin measurement. The delta function then becomes a Lorentzian of full width $\Gamma_{\rm M}$, whose peak is $2/\pi\Gamma_{\rm M}$. Averaging $p_{\phi\perp}^2=m_\phi^2 v_\perp^2$ over the halo and multiplying by $\rho_\chi/m_\phi\rho_T$, the event rate per unit target mass on resonance at $T\to0$ is

$$\mathcal{R}_{\rm peak}=\frac{\rho_\chi\, g_{aee}^2\,\langle v_\perp^2\rangle\,\lambda_S^2\,\mathfrak{s}_{\rm net}}{m_e^2\,\rho_T\Omega\,\Gamma_{\rm M}},\tag{14.22}$$

where $\rho_T\Omega$ is the mass of the unit cell. The boson mass has dropped out except through $\Gamma_{\rm M}$. For the standard halo, $\langle v_\perp^2\rangle=v_0^2+|\boldsymbol{v}_E|^2\sin^2\theta_E$, with θ_E the angle between $\hat{\boldsymbol{z}}$ and the velocity of the Earth through the halo, and the rate is modulated daily by up to a factor 0.47 as the laboratory rotates (Sect. 1.2). The golden-rule mean needs an exposure long compared with the shorter of $t_{\rm coh}$ and $1/\Gamma_{\rm M}$, and a single record matches it only beyond $t_{\rm coh}$ (Sect. 14.6.1); for shorter exposures the field drives the mode coherently, which Sect. 14.6 treats. The exchange dispersion of the mode, $\omega=\omega_{\rm Lar}+k^2/2M_{\rm sw}$ with $M_{\rm sw}=1/2D_{\rm sw}=3.25$ MeV for the stiffness of YIG in Worked example 14.2 (Chigusa and collaborators quote about 3.5 MeV [15, Eqs. (19), (20)]), shifts the line by 10^{-18} of m_ϕ at $k=m_\phi v_E$. The same line also absorbs the energy that exothermic dark matter releases when its splitting $|\Delta_\chi|$ matches $\omega_{\rm Lar}$, a channel common to phonons, magnons and plasmons that the comparison of channels later in the book treats.

14.3.4 The Kittel mode and demagnetizing fields

A magnetized body produces its own field, and the Larmor line assumes there is none. A uniformly magnetized ellipsoid has a uniform internal field: in Heaviside–Lorentz units with $\mu_0=1$ it is $\boldsymbol{B}_0-\mathsf{N}\boldsymbol{M}$, where the demagnetizing tensor N is diagonal along the axes of the ellipsoid with elements $N_x^{\rm dm}+N_y^{\rm dm}+N_z^{\rm dm}=1$. A sphere has $N_i^{\rm dm}=\frac13$, a thin film $N^{\rm dm}=1$ along its normal and 0 in its plane. The field adds the magnetostatic energy $\frac12\mathcal{V}\sum_i N_i^{\rm dm}M_i^2$, and in the magnet of Sect. 14.3.3 the magnetization is $\boldsymbol{M}=2\mu_B\boldsymbol{S}_{\rm tot}/\mathcal{V}$ for $g_{\rm ion}=2$. Let $\boldsymbol{B}_0$ and the net spin lie along the z axis of the ellipsoid. To second order in the transverse components, with $S_{\rm tot}^z=S_{\rm tot}-(S_{{\rm tot},x}^2+S_{{\rm tot},y}^2)/2S_{\rm tot}$, the energy of the uniform mode is

$$H_{\rm u}=\frac{1}{2S_{\rm tot}}\Big[\omega_{{\rm Kit},x}\,S_{{\rm tot},x}^2+\omega_{{\rm Kit},y}\,S_{{\rm tot},y}^2\Big],\qquad \omega_{{\rm Kit},i}=\omega_{\rm Lar}+\big(N_i^{\rm dm}-N_z^{\rm dm}\big)\,\omega_M,\tag{14.23}$$

where $\omega_M=2\mu_B M_{\rm s}$ is the saturation magnetization $M_{\rm s}=|\boldsymbol{M}|$ in units of frequency ($2\mu_B\mu_0 M_{\rm s}$ in SI units) and $S_{\rm tot}=N\mathfrak{s}_{\rm net}$. With the Holstein–Primakoff form of Eq. (7.7) for the total spin, $S_{{\rm tot},x}=S_{\rm tot}^{1/2}\hat{x}$ and $S_{{\rm tot},y}=S_{\rm tot}^{1/2}\hat{p}$, where $\hat{x}=(a_0+a_0^\dagger)/\sqrt2$ and $\hat{p}=-{\rm i}(a_0-a_0^\dagger)/\sqrt2$ obey $[\hat{x},\hat{p}]={\rm i}$. Eq. (14.23) is then an oscillator $\frac12(\omega_{{\rm Kit},x}\hat{x}^2+\omega_{{\rm Kit},y}\hat{p}^2)$, whose frequency is the Kittel formula [18]

$$\omega_{\rm Kit}=\big(\omega_{{\rm Kit},x}\,\omega_{{\rm Kit},y}\big)^{1/2}.\tag{14.24}$$

In its ground state $\langle\hat{x}^2\rangle=\frac12(\omega_{{\rm Kit},y}/\omega_{{\rm Kit},x})^{1/2}$, so the transverse weights of Eq. (14.20) become elliptical, $|\langle1|S_{{\rm tot},x}|0\rangle|^2=\frac12 S_{\rm tot}(\omega_{{\rm Kit},y}/\omega_{{\rm Kit},x})^{1/2}$ and $|\langle1|S_{{\rm tot},y}|0\rangle|^2=\frac12 S_{\rm tot}(\omega_{{\rm Kit},x}/\omega_{{\rm Kit},y})^{1/2}$. In Eq. (14.21) this replaces $\delta(\omega_\phi-\omega_{\rm Lar})$ by $\delta(\omega_\phi-\omega_{\rm Kit})$ and $p_{\phi\perp}^2$ by

$$p_{\phi,x}^2\Big(\frac{\omega_{{\rm Kit},y}}{\omega_{{\rm Kit},x}}\Big)^{1/2}+p_{\phi,y}^2\Big(\frac{\omega_{{\rm Kit},x}}{\omega_{{\rm Kit},y}}\Big)^{1/2}.\tag{14.25}$$

Averaged over transverse directions of $\boldsymbol{p}_\phi$, the weight grows by $(\omega_{\mathrm{Kit},x}+\omega_{\mathrm{Kit},y})/2\omega_{\mathrm{Kit}} \geq 1$. The sum of the two transverse weights is not conserved by the shape, and the line moves; the Larmor theorem fails because the magnetostatic energy is not invariant under a rotation of $\boldsymbol{S}_{\mathrm{tot}}$.

For a sphere every N_i^{dm} is equal, and $\omega_{\mathrm{Kit}} = \omega_{\mathrm{Lar}}$ with circular weights: the Larmor line survives. A thin film with $\boldsymbol{B}_0$ in its plane has $\omega_{\mathrm{Kit}} = [\omega_{\mathrm{Lar}}(\omega_{\mathrm{Lar}}+\omega_M)]^{1/2}$, and with $\boldsymbol{B}_0$ along its normal $\omega_{\mathrm{Kit}} = \omega_{\mathrm{Lar}} - \omega_M$. All of this assumes a body saturated along $\boldsymbol{B}_0$, which needs an internal field $|\boldsymbol{B}_0| - N_z^{\mathrm{dm}} M_{\mathrm{s}} > 0$; below it the sample breaks into domains and Eq. (14.23) does not apply. A uniaxial anisotropy field B_A along $\boldsymbol{B}_0$ adds $g_{\mathrm{ion}}\mu_B B_A$ to both $\omega_{\mathrm{Kit},i}$, moving the line without changing its weights.

Beyond the uniform field of an ellipsoid, the dipolar interaction couples the precession at different points of the sample and produces the magnetostatic modes of finite wavevector. The dark-matter field is uniform over the sample, since $1/m_\phi v \gg$ its size, and in an ellipsoid it drives the Kittel mode alone; in a body of any other shape the internal field is nonuniform and the weight spreads over several magnetostatic modes. The limits $\boldsymbol{k} \to 0$ and sample size $\to \infty$ do not commute: the mode at $\boldsymbol{k} \to 0$ in an infinite crystal has no demagnetizing field along $\boldsymbol{k}$, while the uniform mode of any finite body has the field of its shape. Absorption needs the second. Limit Check 14.2a compares the Larmor line with the film, and Appendix E.20.4 specifies the linear spin-wave calculation with shape and dipolar sums for real samples.

Worked example 14.3: The Kittel-mode rate of YIG for a 10 μeV axion-like particle

YIG, $Y_3Fe_5O_{12}$, has a body-centered cubic lattice with $a = 12.375\,\text{Å}$ [14] and a primitive cell of volume $\Omega = a^3/2 = 947.6\,\text{Å}^3$ that holds four formula units, of mass $\rho_T\Omega = 2951.8\,\mathrm{u}$. Its 20 Fe^{3+} ions, each with $|\boldsymbol{S}| = \frac{5}{2}$ and no orbital moment, order as a ferrimagnet, 12 against 8 [16, Sect. III], so that $\mathfrak{s}_{\mathrm{net}} = 4 \times \frac{5}{2} = 10$ and $\lambda_S = 1$. The saturation magnetization is $M_{\mathrm{s}} = 20\mu_B/\Omega$, $\mu_0 M_{\mathrm{s}} = 0.246\,\mathrm{T}$, or $37.8\,\mathrm{emu/g}$, close to the measured $38\,\mathrm{emu/g}$ at 0.1 K [15]; then $\omega_M = 28.5\,\mu\mathrm{eV}$. (The model of Worked example 14.2 used the room-temperature $4\pi M_{\mathrm{s}} = 1750\,\mathrm{G}$; at low temperature it is 2460 G.)

Field. Since in a sphere the line sits at ω_{Lar}, $m_\phi = 10\,\mu\mathrm{eV}$ needs $|\boldsymbol{B}_0| = m_\phi/2\mu_B = 86.4\,\mathrm{mT}$. The sphere saturates for $|\boldsymbol{B}_0| > \mu_0 M_{\mathrm{s}}/3 = 82.0\,\mathrm{mT}$, that is for lines above $9.49\,\mu\mathrm{eV}$; at $10\,\mu\mathrm{eV}$ the internal field is only $4.4\,\mathrm{mT}$. At 10 mK, $\beta m_\phi = 11.6$ and the thermal factor of Eq. (14.21) is $1 + 9 \times 10^{-6}$.

Rate. The assumed width is $\Gamma_{\mathrm{M}} = 10^{-5} m_\phi = 1.0 \times 10^{-10}\,\mathrm{eV}$, a full width of 24 kHz, with $1/\Gamma_{\mathrm{M}} = 6.6\,\mu\mathrm{s}$. With $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$, $\hat{\boldsymbol{z}}$ perpendicular to $\boldsymbol{v}_E$, so that $\langle v_\perp^2 \rangle = v_0^2 + v_E^2 = 1.33 \times 10^{-6}$, and $g_{aee} = 10^{-13}$, Eq. (14.22) gives

$$\mathcal{R}_{\mathrm{peak}} = 3.63\,\mathrm{s^{-1}\,kg^{-1}} = 1.15 \times 10^8\,\mathrm{kg^{-1}\,yr^{-1}}.$$

A sphere 1 cm across, of 2.7 g, would absorb 35 particles an hour while its line sits on m_ϕ, within 24 kHz. With $\hat{\boldsymbol{z}}$ along $\boldsymbol{v}_E$ the rate falls to 0.47 of this value. (It is the coherent rate of [15, Eq. (38)] at a coherence time $4/\Gamma_{\mathrm{M}}$, without their cavity factor $\frac{1}{2}$.)

Shape. A thin film with the field in its plane reaches $10\,\mu\mathrm{eV}$ at $\omega_{\mathrm{Lar}} = 3.16\,\mu\mathrm{eV}$, $|\boldsymbol{B}_0| = 27.3\,\mathrm{mT}$, and it is saturated at any field. By Eq. (14.25) the component of $\boldsymbol{p}_\phi$ in the plane of the film counts 3.16 times and the normal one 0.316 times; averaged, the rate is 1.74 times that of the sphere.

14.3.5 Antiferromagnetic resonance

An antiferromagnet has no net spin, $\mathfrak{s}_{\rm net} = 0$ in zero field, and the Larmor line of Eq. (14.20) is empty. With isotropic exchange the uniform mode is the Goldstone mode at $\omega = 0$, and Eq. (7.16) assigns it a weight that vanishes linearly in k. Anisotropy changes both. Take two sublattices, A along $+\hat{\boldsymbol{z}}$ and B along $-\hat{\boldsymbol{z}}$, with an easy axis along $\hat{\boldsymbol{z}}$. The uniform mode then depends on two energies, written as frequencies. The exchange frequency $\omega_E = |\boldsymbol{S}|J^{\rm ex}_{\rm AB}(0)$ is the cost of one deviation in the exchange field of the other sublattice, and the anisotropy frequency ω_A is the cost of one deviation away from the easy axis. In fields, $\omega_E = g_{\rm ion}\mu_B B_E$ and $\omega_A = g_{\rm ion}\mu_B B_A$. At $\boldsymbol{k} = 0$ the bosonic Hamiltonian Eq. (7.9) has $|\boldsymbol{S}|A_0 = \omega_E + \omega_A$ and $|\boldsymbol{S}|B_0 = \omega_E$, since the intra-sublattice terms cancel at $\boldsymbol{k} = 0$ and the anisotropy adds only to the diagonal. Eq. (7.11) then gives the two degenerate modes at

$$\omega_{\rm AFMR} = \big[(\omega_E + \omega_A)^2 - \omega_E^2\big]^{1/2} = \big[\omega_A(2\omega_E + \omega_A)\big]^{1/2}, \tag{14.26}$$

the frequency of *antiferromagnetic resonance*, the geometric mean of the anisotropy frequency and twice the exchange frequency. Its large size is the exchange enhancement already seen in the linear dispersion of Sect. 7.1: a uniform tilt of one sublattice drags the other through the pair term, and the precession works against the full exchange field.

The weight is Eq. (7.16) at $\boldsymbol{k} = 0$. In the uniform transverse spin $S^x_{\rm tot} = (|\boldsymbol{S}|N_{\rm s}/2)^{1/2}(d_{\rm A0} + d^\dagger_{\rm A0} + d_{\rm B0} + d^\dagger_{\rm B0})$, the inverse of Eq. (7.10) gives $d_{\rm A0} + d^\dagger_{\rm B0} = (u_0^{\rm M} - v_0^{\rm M})(a_{01} + a^\dagger_{02})$, and each mode is created with the amplitude $(|\boldsymbol{S}|N_{\rm s}/2)^{1/2}(u_0^{\rm M} - v_0^{\rm M})$. Summed over the two modes and divided by the number $N_{\rm s}$ of antiparallel pairs, with $(u_0^{\rm M} - v_0^{\rm M})^2 = (A_0 - B_0)/(A_0^2 - B_0^2)^{1/2} = \omega_A/\omega_{\rm AFMR}$:

Uniform weight of a two-sublattice antiferromagnet

$$\frac{1}{N_{\rm s}}\sum_{\nu=1,2}\big|\langle\nu|S^x_{\rm tot}|0\rangle\big|^2 = |\boldsymbol{S}|\,\frac{\omega_{\rm AFMR}}{2\omega_E + \omega_A} \;\simeq\; |\boldsymbol{S}|\Big(\frac{\omega_A}{2\omega_E}\Big)^{1/2} \quad (\omega_A \ll \omega_E), \tag{14.27}$$

at linear spin-wave order and $T = 0$, against $|\boldsymbol{S}|$ for a ferromagnetic pair. In Eq. (14.21) the antiferromagnet at $\boldsymbol{B}_0 = 0$ enters with $\mathfrak{s}_{\rm net} \to 2|\boldsymbol{S}|\omega_{\rm AFMR}/(2\omega_E + \omega_A)$ per pair cell, $\delta(\omega_\phi - \omega_{\rm Lar}) \to \delta(\omega_\phi - \omega_{\rm AFMR})$ and $\mathrm{i}\epsilon_{abc}\hat{z}_c \to 0$.

The same factor $\omega_{\rm AFMR}/(2\omega_E + \omega_A)$ that raises the frequency suppresses the weight, so the product of the two is the anisotropy frequency ω_A: an antiferromagnet trades rate for reach. The two modes carry opposite circular polarizations, one lowering and one raising $S^z_{\rm tot}$ (Sect. 7.1), and their contributions to $\mathrm{i}\epsilon_{abc}$ cancel. A field along the easy axis adds $\mp g_{\rm ion}\mu_B|\boldsymbol{B}_0|$ to the diagonal energies of the two modes. Since it couples to the conserved $S^z_{\rm tot}$, it commutes with the Bogoliubov transformation, and the line splits into $\omega_{\rm AFMR} \pm g_{\rm ion}\mu_B|\boldsymbol{B}_0|$ with both weights unchanged, up to the spin-flop field $B_{\rm sf} = [B_A(2B_E - B_A)]^{1/2}$, where the lower mode has fallen to 0.01 meV in MnF_2 and the sublattices turn perpendicular to the field. Shape enters only through the small net moment of the precession, and it shifts $\omega_{\rm AFMR}$ by a fraction of order ω_M/ω_E, 1.6×10^{-2} in MnF_2 with ω_M formed from the sublattice magnetization, $\mu_0 M = 0.74\,\mathrm{T}$. A ferrimagnet such as YIG has modes of both kinds. Besides its Larmor line, its 19 gapped branches at $\boldsymbol{k} = 0$ are exchange modes in which the sublattices precess against each other. With uniform λ_S and isotropic exchange their weight is the zero found in Sect. 14.3.3.

The exchange and anisotropy frequencies are measured by neutron scattering and magnetic resonance, or computed by DFT $+U$ with the energy-mapping methods of Appendix E.9; Appendix E.20.4 specifies them, with the $g_{\rm ion}$ tensors, for the magnets of this chapter. MnF_2 is the standard two-sublattice antiferromagnet: Mn^{2+} ions with $|\boldsymbol{S}| = \frac{5}{2}$ and no orbital moment,

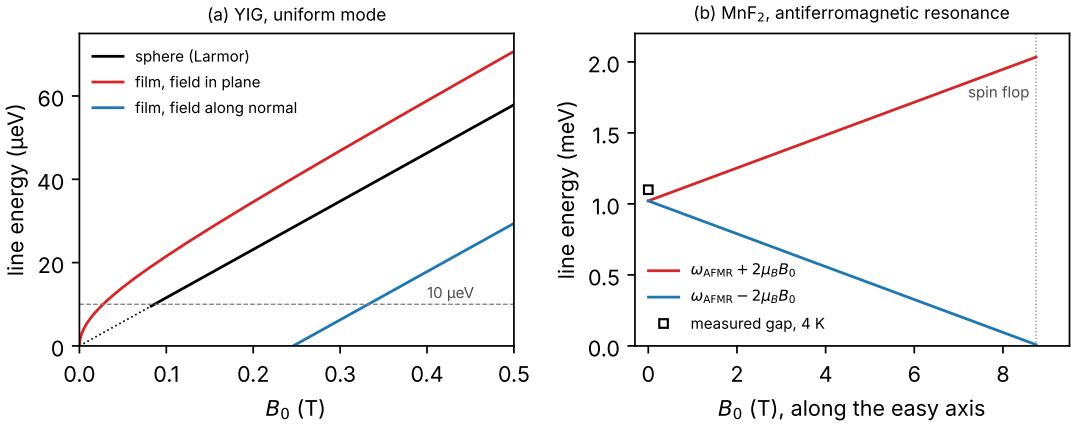


Figure 14.5: Absorption lines of YIG and MnF_2 against the applied field B_0. (a) The uniform mode of YIG ($g_{\text{ion}} = 2$, $\mu_0 M_s = 0.246\,\text{T}$). A sphere absorbs at the Larmor frequency $2\mu_B B_0$ once it is saturated, $B_0 > \mu_0 M_s/3$ (dotted below). A thin film absorbs at $[\omega_{\text{Lar}}(\omega_{\text{Lar}} + \omega_M)]^{1/2}$ with the field in its plane and at $\omega_{\text{Lar}} - \omega_M$ with the field along its normal, $\omega_M = 28.5\,\mu\text{eV}$. The dashed line marks the $10\,\mu\text{eV}$ of the worked example. (b) The two antiferromagnetic-resonance lines of MnF_2, $\omega_{\text{AFMR}} \pm 2\mu_B B_0$ with $\omega_{\text{AFMR}} = 1.021\,\text{meV}$ from $B_E = 47.05\,\text{T}$ and $B_A = 0.82\,\text{T}$, for a field along the easy axis up to the spin flop at $8.75\,\text{T}$, both from the fitted fields; each line carries half of the weight $|\mathbf{S}|\omega_{\text{AFMR}}/(2\omega_E + \omega_A) = 0.232$ per pair. The square is the measured zone-center gap at $4\,\text{K}$, $1.1\,\text{meV}$, which goes with a measured spin flop near $9.4\,\text{T}$.

$\lambda_S = 1$, one antiparallel pair per rutile cell of mass $185.9\,\text{u}$, and an easy axis along c set mainly by the dipolar interaction [19]. Neutron scattering finds a zone-center gap of $1.1\,\text{meV}$ at $4\,\text{K}$ [19], and the spin flop occurs near $9.4\,\text{T}$ [20, Fig. 1]. Since $\omega_{\text{AFMR}} \simeq g_{\text{ion}}\mu_B B_{\text{sf}}$ for $\omega_A \ll \omega_E$, the two measurements agree with each other: $1.1\,\text{meV}$ corresponds to $9.5\,\text{T}$. The exchange and anisotropy fields fitted by Vaidya and collaborators, $B_E = 47.05\,\text{T}$ and $B_A = 0.82\,\text{T}$ [20], give $\omega_E = 5.447\,\text{meV}$ and $\omega_A = 0.0949\,\text{meV}$, and Eq. (14.26) gives $\omega_{\text{AFMR}} = 1.021\,\text{meV}$ (247 GHz) with $B_{\text{sf}} = 8.75\,\text{T}$, both 7 % below the measured pair. Keeping ω_E and fitting ω_A to the measured gap instead gives $\omega_A = 0.110\,\text{meV}$ and a spin flop at $9.4\,\text{T}$. With the fitted fields the weight factor is 0.0929, within half a percent of $(\omega_A/2\omega_E)^{1/2} = 0.0933$, and a pair absorbs with 0.232 against 2.5 for a ferromagnetic pair; with ω_A fitted to the gap the factor is 0.100, or 0.250 per pair. Per unit mass this is 0.74 (0.79) of the weight of YIG. With the same assumed ratio $\Gamma_{\text{M}}/\omega = 10^{-5}$, the peak rate Eq. (14.22) at the resonance is 7.2×10^{-3} of that of YIG at $10\,\mu\text{eV}$, $2.6 \times 10^{-2}\,\text{s}^{-1}\,\text{kg}^{-1}$ for $g_{aee} = 10^{-13}$, for either set of values: the rate goes as $|\boldsymbol{S}|\omega_A/\omega_{\text{AFMR}}^2$, because Γ_{M} grows with the frequency. A field along c sweeps the lower line from $1.1\,\text{meV}$ nearly to zero at the spin flop (Fig. 14.5), and one crystal covers every mass below $1\,\text{meV}$ with a weight that stays the same along the sweep.

Limit Check 14.2. Absorption into magnons

(a) The Larmor statement: at $\boldsymbol{k} \to 0$ all transverse spin weight of a magnet with isotropic exchange and uniform λ_S and g_{ion} lies in one line at ω_{Lar} with weight $2\langle S^z_{\text{tot}}\rangle$ [17, Eqs. (2.4), (2.5), 15, Eqs. (25)–(27), 16, Eq. (27)]; for a J multiplet the axion-like particle couples with $\lambda_S = g_J - 1$. (b) The uniform weight $|\boldsymbol{S}|\omega_{\text{AFMR}}/(2\omega_E + \omega_A)$ per antiparallel pair, Eq. (14.27).

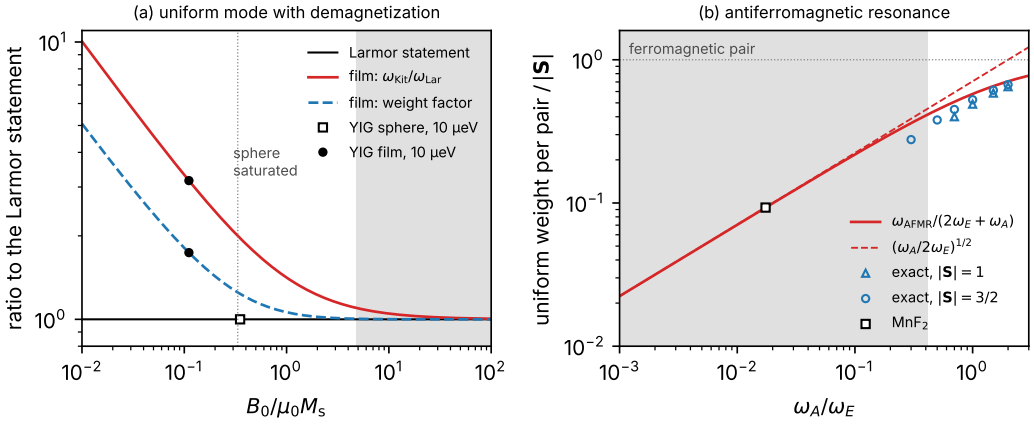


Figure 14.6: (a) Uniform mode of a thin film with the field in its plane against the applied field over the shape-anisotropy field, $B_0/\mu_0 M_s$: line position $\omega_{Kit}/\omega_{Lar} = (1+\mu_0 M_s/B_0)^{1/2}$ (solid) and weight averaged over transverse directions of $\boldsymbol{p}_\phi$ (dashed), both relative to the Larmor statement (black, 1). A saturated sphere follows the Larmor statement exactly; the markers place YIG at 10 μeV. Gray: both ratios within 10 % of 1, $B_0 \geq 4.8\,\mu_0 M_s$. For a J multiplet both curves keep their shape, the rate scales by $(g_J-1)^2$ and the line moves to $g_J\mu_B B_0$. (b) Uniform weight per antiparallel pair over $|\mathbf{S}|$ against ω_A/ω_E: the linear spin-wave result (solid) and its exchange-suppressed form $(\omega_A/2\omega_E)^{1/2}$ (dashed). Symbols: the exact total weight $\langle(S^x_{tot})^2\rangle$ of the eight-site easy-axis (XXZ) cluster of the text for $|\mathbf{S}| = 1$ and $\frac{3}{2}$, where that cluster is ordered (the splitting of its two lowest states below 1 % of ω_{AFMR}). Gray: the square-root form within 10 %, $\omega_A \leq 0.42\,\omega_E$. The square marks MnF_2 with the fitted fields. The isotropic Heisenberg magnet, with zero uniform weight, is the left end.

Outside the gray regions the line of a non-spherical sample leaves ω_{Lar} and its weight becomes elliptical, the antiferromagnetic weight departs from the square-root form, and in the eight-site cluster the zero-point fluctuations remove 5 to 23 % of the linear spin-wave weight for $|\mathbf{S}| = \frac{3}{2}$ (ω_A/ω_E from 2 to 0.3).

Linear spin-wave theory itself adds no correction to the Larmor line and one to the antiferromagnetic weight. Eq. (14.20) is exact for any spin and temperature, and its only inputs are λ_S, g_{ion} and the net magnetization. The antiferromagnetic weight is of order $|\boldsymbol{S}|^0$ in the expansion of Sect. 14.1, and an exactly solvable cluster shows its zero-point correction directly. Put spins $|\boldsymbol{S}|$ on the eight corners of a cube, each coupled to its three neighbors on the other sublattice by $H = J\sum_{\langle ij\rangle}(S^x_i S^x_j + S^y_i S^y_j + \Delta_{XXZ} S^z_i S^z_j)$ with $J > 0$ and $\Delta_{XXZ} \geq 1$, twelve bonds and open boundaries. Linear spin-wave theory at $\boldsymbol{k} = 0$ maps it onto Eq. (14.26) with $\omega_E = 3J|\boldsymbol{S}|$ and $\omega_A = 3J|\boldsymbol{S}|(\Delta_{XXZ}-1)$, so that Eq. (14.27) becomes $|\boldsymbol{S}|[(\Delta_{XXZ}-1)/(\Delta_{XXZ}+1)]^{1/2}$. The exact weight is the equal-time correlation $\langle 0|(S^x_{tot})^2|0\rangle$ in the ground state of the sector $S^z_{tot} = 0$, divided by the four antiparallel pairs; it contains every final state that S^x_{tot} reaches. At $\omega_A = \omega_E$ it is 0.85 and 0.91 of Eq. (14.27) for $|\boldsymbol{S}| = 1$ and $\frac{3}{2}$ (Fig. 14.6b). The cluster cannot reach the small anisotropies of real antiferromagnets, $\omega_A/\omega_E = 0.017$ in MnF_2, because below $\omega_A \approx 0.3\,\omega_E$ for $|\boldsymbol{S}| = \frac{3}{2}$, and about $0.6\,\omega_E$ for $|\boldsymbol{S}| = 1$, its two lowest states mix and it no longer orders. For Mn^{2+} the expansion parameter $1/2|\boldsymbol{S}|$ is 0.2, which sets the order of the correction at small ω_A/ω_E but not its size.

14.4 Spin responses from neutron data

The full magnon spectrum of YIG, all twenty branches up to about 110 meV, has been measured by inelastic neutron scattering with neutrons of 120 meV [21, 22]. Such a neutron carries 15.0 keV of momentum, the largest momentum of a dark-matter particle of 5.7 MeV, and its spin couples to the same spin density $\boldsymbol{s}_e(\boldsymbol{q})$ as a dark-matter particle with a spin-dependent coupling to electrons (Appendix C.19). The measured cross-section can therefore replace the model response of Sect. 14.1, with every magnon branch, every multi-magnon process and the magnetic form factor already included. Berlin, Millar, Trickle and Zhou did exactly this for YIG [22]. A neutron, however, couples only to the part of the spin density perpendicular to the momentum transfer, and how much of the dark-matter response that part fixes depends on the coupling and on how much model input is added for the magnet.

14.4.1 Conversion of neutron spectra

Neutron spectra are reported per formula unit of the crystal. With N_{fu} formula units in a cell of volume Ω, the measured magnetic structure factor $S^{\mathrm{neu}}(\boldsymbol{q},\omega)$ is defined by $\mathrm{d}^2\sigma/\mathrm{d}\Omega\,\mathrm{d}E' = (p'_n/p_n)\,S^{\mathrm{neu}}$ for one formula unit, with $\mathrm{d}\Omega$ here the solid angle of the outgoing neutron. Dividing Eq. (C.165) by the number of formula units in the sample, $\mathscr{V}N_{\mathrm{fu}}/\Omega$, and solving for the response,

$$\mathrm{Tr}\big[P_T\,S(\boldsymbol{q},\omega)\big] = \sum_{a,b}(P_T)_{ab}\,S_{ab}(\boldsymbol{q},\omega) = \frac{2\pi N_{\mathrm{fu}}}{(\gamma_n r_{\mathrm{cl}})^2\,\Omega}\,S^{\mathrm{neu}}(\boldsymbol{q},\omega), \tag{14.28}$$

with $(P_T)_{ab} = \delta_{ab} - \hat{q}_a\hat{q}_b$ and $(\gamma_n r_{\mathrm{cl}})^2 = 0.2906\,\mathrm{b}$. This is the relation of Berlin et al. [22, Eq. (26)], whose susceptibility is $\chi''_{ab} = 2(1-\mathrm{e}^{-\beta\omega})S_{ab}$ [22, Eq. (B4)]: their spin operator is $\boldsymbol{\sigma}$ where ours is $\boldsymbol{\sigma}/2$, a factor 4, and their response carries $\pi/\mathscr{V}$ where S_{ab} carries $2\pi/\mathscr{V}$, a factor $\frac{1}{2}$. The thermal vibration of the ions enters through the Debye–Waller factor, which the dark matter sees in the same way, so it stays inside S_{ab} (Appendix C.19).

Two corrections turn Eq. (14.28) into the response that a detector needs. The first is temperature. The spectrum is measured at a temperature T_{d}, below 10 K for the YIG data [22, App. B 2], while a detector runs at millikelvin. By the fluctuation–dissipation theorem Eq. (4.18), $(1-\mathrm{e}^{-\beta\omega})S_{ab}$ is the dissipative response, and for free magnons this response does not depend on temperature, as for the thermal oscillator of Sect. 4.2. Hence

$$S_{ab}(\boldsymbol{q},\omega;T\to 0) = \big(1-\mathrm{e}^{-\omega/T_{\mathrm{d}}}\big)\,S_{ab}(\boldsymbol{q},\omega;T_{\mathrm{d}}), \qquad \omega>0,$$

which removes the thermal population of the measured spectrum. The residual is set by the magnon interactions, which make the response follow the thermal reduction of the ordered moment; it is small for T_{d} well below the ordering temperature.

The second correction is the orbital moment. A neutron couples to the magnetization, spin and orbital, while spin-coupled dark matter couples to the spin alone. For a quenched $3d$ ion of spin $\boldsymbol{S}$, Sect. 14.3 showed that spin–orbit coupling induces an orbital moment $\boldsymbol{L} = (g_{\mathrm{ion}}-2)\boldsymbol{S}$ at first order, with g_{ion} the spectroscopic g factor of the ion (Sect. 14.3.2), taken isotropic here, while the spin itself changes only at second order, $\lambda_S = 1 - O(\xi^2_{\mathrm{ion}}/\Delta^2_{\mathrm{CF}})$. In the dipole approximation of Appendix C.19 the spin part of the moment carries the radial integral $\langle j_0\rangle$ and the orbital part $\langle j_0\rangle + \langle j_2\rangle$. The moment $2\boldsymbol{S}\langle j_0\rangle + \boldsymbol{L}(\langle j_0\rangle + \langle j_2\rangle)$ is then $g_{\mathrm{ion}}\boldsymbol{S}\,F_g^{\mathrm{mag}}(q)$, and the neutron couples to

$$\boldsymbol{s}_e(\boldsymbol{q}) \to \frac{g_{\mathrm{ion}}}{2}\,F_g^{\mathrm{mag}}(q)\sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_j}\,\boldsymbol{S}_j, \qquad F_g^{\mathrm{mag}}(q) = \langle j_0(q)\rangle + \Big(1-\frac{2}{g}\Big)\langle j_2(q)\rangle,$$

while the dark matter couples to $\lambda_S \langle j_0(q)\rangle \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_j}\boldsymbol{S}_j$. The two operators differ by a number at each q, so their responses differ by its square:

Data-driven spin response

$$\mathrm{Tr}\big[P_T\, S(\boldsymbol{q},\omega)\big] = C_{\mathrm{sn}}(q)\,\frac{2\pi N_{\mathrm{fu}}}{(\gamma_n r_{\mathrm{cl}})^2\,\Omega}\,S^{\mathrm{neu}}(\boldsymbol{q},\omega), \qquad C_{\mathrm{sn}}(q) = \left[\frac{\lambda_S\,\langle j_0(q)\rangle}{(g_{\mathrm{ion}}/2)\,F_g^{\mathrm{mag}}(q)}\right]^2, \tag{14.29}$$

for one species of quenched magnetic ion, with S_{ab} the response of the electron spin density and S^{neu} measured per formula unit, its thermal population removed. At $q \to 0$, $C_{\mathrm{sn}} = [\lambda_S/(g_{\mathrm{ion}}/2)]^2$.

For an anisotropic g_{ion} tensor, $g_{\mathrm{ion}}/2$ becomes the matrix $g_{\mathrm{ion},ab}/2$ acting on both indices of S_{ab}. For a J multiplet, g_{ion} is replaced by g_J, λ_S by $g_J - 1$ (Eq. (14.19)) and F_g^{mag} by the dipole form factor Eq. (C.167); at $q \to 0$, C_{sn} is the square of the last column of Table 14.1. It is 1 for Gd^{3+}, $\frac{1}{9}$ for Er^{3+} and 25 for Sm^{3+}, whose spin points against its moment.

For Ni^{2+} in NiO the literature value of the Landé factor, as quoted by Kritzell et al., is 2.18 [23]; we round it to $g_{\mathrm{ion}} \simeq 2.2$. With $\lambda_S = 1$,

$$C_{\mathrm{sn}}(0) = \left(\frac{2}{2.2}\right)^2 = 0.83 :$$

the spin response of NiO is 17 % below what the neutron cross-section suggests if read as pure spin. Setting $\lambda_S = g_{\mathrm{ion}} - 1$ instead, the rule that holds only for J multiplets, would give $(1.2/1.1)^2 = 1.19$ and an error of the opposite sign. The q dependence enters through $\langle j_2\rangle$, which vanishes at $q = 0$ and grows with q. With the tabulated radial integrals of Ni^{2+} [4, Sect. 4.4.5], the same fit form as for $\langle j_0\rangle$ in Sect. 14.1 multiplied by s^2 with coefficients $(1.708, 11.016, 1.2147, 4.103, 0.315, 1.533, 0.0018)$, the ratio $\langle j_2\rangle/\langle j_0\rangle$ is 0.035 at $|\boldsymbol{Q}_{\mathrm{AF}}| = 2.6\,\mathrm{keV}$ and 0.15 at 5.3 keV. $C_{\mathrm{sn}}(q)/C_{\mathrm{sn}}(0)$ is then 0.994 and 0.973, and it stays within 10 % of one up to 10 keV. The value of g_{ion} is the larger uncertainty. Magnetic x-ray diffraction on NiO, which separates the spin and orbital moments, finds $L/S = 0.34$ [24], as quoted by Kwon and Min, whose DFT $+U$ with spin–orbit coupling gives 0.36 [25]. Read in the quenched picture as $g_{\mathrm{ion}} = 2 + L/S = 2.34$, this lowers $C_{\mathrm{sn}}(0)$ to 0.73, 12 % below the value at $g_{\mathrm{ion}} = 2.2$. The orbital moment of Ni^{2+} is thus a nuisance on the normalization of the data-driven rate, at the level of 10 %. The g_{ion} tensor and the orbital moments are computed in DFT $+U$ with spin–orbit coupling as part of the magnet inputs of Appendix E.20.4, and the radial integrals from Dirac–Fock or Wannier-projected shells (Appendix E.20).

Limit Check 14.3 shows how far the spin-only reading of the data can be trusted as the orbital admixture grows.

Limit Check 14.3. Data-driven spin response

$\mathrm{Tr}[P_T S(\boldsymbol{q},\omega)] = 2\pi N_{\mathrm{fu}} S^{\mathrm{neu}}(\boldsymbol{q},\omega)/[(\gamma_n r_{\mathrm{cl}})^2\Omega]$, for a neutron coupled to the electron spin alone [22, Eqs. (8), (11), (26)].

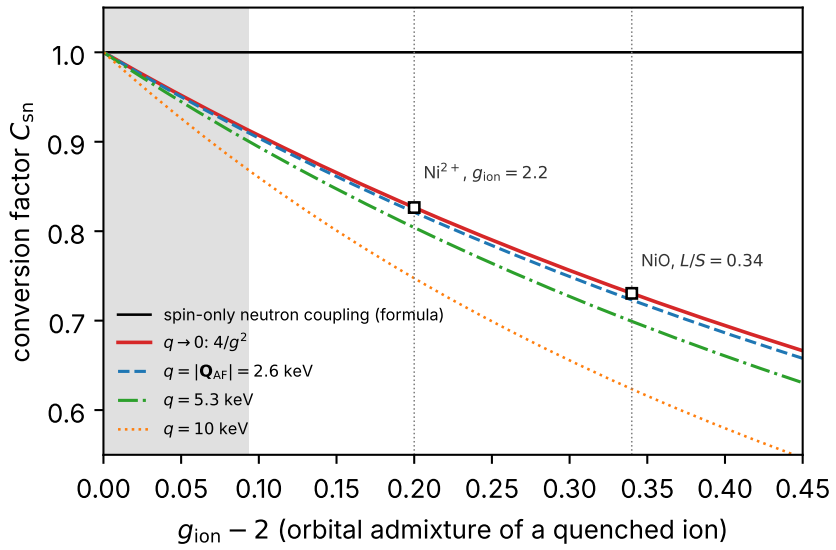


Figure 14.7: Ratio of the spin response to the response inferred from neutron data by the spin-only formula, $C_{sn}(q)$ of Eq. (14.29), against the shift $g_{ion} - 2$ of the spectroscopic g_{ion} factor of a quenched ion, with $\lambda_S = 1$ and the dipole form factor $\langle j_0 \rangle + (1 - 2/g_{ion})\langle j_2 \rangle$. Black: the formula, ratio 1. Red: $q \to 0$, $4/g^2$. Dashed, dash-dotted and dotted: $q = 2.6\,\text{keV}$ (the ordering wavevector of NiO), 5.3 keV (largest momentum transfer at $m_\chi = 1\,\text{MeV}$) and 10 keV, with the radial integrals of Ni^{2+}. Squares: $g_{ion} = 2.2$ (octahedral Ni^{2+}) and $g_{ion} = 2.34$ (NiO with $L/S = 0.34$ from magnetic x-ray diffraction). Gray: $g_{ion} - 2 \leq 0.094$, where the formula holds within 10 % for $q \leq 5.3\,\text{keV}$.

Outside the gray band the orbital moment shifts the neutron response from the spin response at first order in $g_{ion} - 2$, and at large q the orbital form factor $\langle j_2 \rangle$ adds to the shift.

How much of the dark-matter kinematic region the data reach depends on the instrument. The YIG spectrum used by Berlin et al. was taken at 120 meV on a time-of-flight spectrometer whose detectors cover 3 to 60 degrees from the beam, with an energy resolution of 5 meV at zero energy transfer [22, Sect. III B and App. D]. The neutron's energy is small next to that of a 5.7 MeV dark-matter particle, 20 eV, but it suffices for the whole magnon band. Data below 5 meV are dominated by elastic scattering and are discarded. The crystal was rotated about one axis only, so at large q only a narrow band of directions $\hat{\boldsymbol{q}}$ is covered. Berlin et al. fill the gaps in two steps. They average the measured S^{neu} over the directions covered at fixed $|\boldsymbol{q}|$ and ω, which is justified well outside the first zone, since a small change in the direction of $\boldsymbol{q}$ moves its reduced part across the whole zone. They then extend the averaged response at constant value from the smallest measured q down to $q_{1\text{BZ}} = 987\,\text{eV}$, which is $2\pi/a$ for their $a = 12.56\,\text{Å}$; with $a = 12.375\,\text{Å}$, $2\pi/a = 1002\,\text{eV}$ is the distance from Γ to the zone corner H of the bcc lattice. The largest sphere inside the zone has the radius $\sqrt{2}\pi/a = 708\,\text{eV}$, so below a mass of 134 keV, where $2m_\chi v_{\text{max}}$ reaches it, every momentum transfer lies inside the first zone, where there are few data and no extrapolation; the data-driven reach falls rapidly below about 200 keV. Near 1 MeV the data-driven rate is in close agreement with the Heisenberg-model rate fitted to the same spectrum [22, Fig. 3]. At larger masses the data contain the form factor of Fe^{3+}, which the model omits and which suppresses the rate by an order of magnitude for $q \gtrsim 10\,\text{keV}$. They also contain phonons, excited by the nuclear scattering of the neutron with a relative weight that grows as $q^2/(M\omega)$, about $(q/30\,\text{keV})^2$ for nuclei of 50 GeV and

$\omega \sim 20\,\mathrm{meV}$ [22, App. C]. Spin-flip scattering with the polarization along $\hat{\boldsymbol{q}}$ contains all the magnetic scattering and no coherent nuclear scattering (Appendix C.19), and it removes the phonons without a model, up to the two thirds of the nuclear-spin-incoherent scattering that it keeps, which is small for Y, Fe and O. A modelled phonon part subtracted from unpolarized data must leave a response that is still positive semidefinite (Eq. (4.20)).

14.4.2 Transverse projection of the neutron cross-section

Which couplings Eq. (14.28) calibrates depends on the dark-matter tensor $\mathfrak{D}_{ab}$ of Eq. (8.7) for the spin row. For the operators of Table 8.1 with dark matter of spin $\frac{1}{2}$ there are three structures. The contact interaction $\mathcal{O}_4$ gives $\mathfrak{D}_{ab} = |c_4^{(e)}|^2\delta_{ab}/4$. The pseudoscalar couplings give $\hat{q}_a\hat{q}_b$: $\mathcal{O}_{10}$ gives $|c_{10}^{(e)}|^2 q_a q_b/m_e^2$ and $\mathcal{O}_6$ gives $|c_6^{(e)}|^2 q^2 q_a q_b/4m_e^4$. Operator $\mathcal{O}_9 = \mathrm{i}\boldsymbol{S}_\chi\cdot(\boldsymbol{S}_e\times\boldsymbol{q}/m_e)$, and the spin–spin part $(\boldsymbol{S}_\chi\times\boldsymbol{q})\cdot(\boldsymbol{S}_e\times\boldsymbol{q})$ of magnetic-dipole and anapole dark matter, give the transverse projector: for $\mathcal{O}_9$, $\mathrm{Tr}_\chi[(\boldsymbol{q}\times\boldsymbol{S}_\chi)_a(\boldsymbol{q}\times\boldsymbol{S}_\chi)_b]/2 = q^2(P_T)_{ab}/4$, so $\mathfrak{D}_{ab} = |c_9^{(e)}|^2 q^2 (P_T)_{ab}/4m_e^2$. Since $\delta_{ab} = (P_T)_{ab} + \hat{q}_a\hat{q}_b$, the three contractions are

$$\sum_{a,b}\delta_{ab}S_{ab} = \mathrm{Tr}\big[P_T S\big] + \hat{\boldsymbol{q}}\cdot S\cdot\hat{\boldsymbol{q}}, \qquad \sum_{a,b}(P_T)_{ab}S_{ab} = \mathrm{Tr}\big[P_T S\big], \qquad \sum_{a,b}\hat{q}_a\hat{q}_b S_{ab} = \hat{\boldsymbol{q}}\cdot S\cdot\hat{\boldsymbol{q}}. \tag{14.30}$$

The neutron data fix the transverse couplings completely: the rates of $\mathcal{O}_9$, magnetic-dipole and anapole dark matter follow from Eq. (8.16) with S^{neu} and nothing else. For $\mathcal{O}_4$, however, positivity gives only a lower bound. The longitudinal element $\hat{\boldsymbol{q}}\cdot S\cdot\hat{\boldsymbol{q}}$ is a diagonal element of a positive semidefinite matrix and cannot be negative, and the transverse part alone is a lower bound on the rate of the contact interaction. For $\mathcal{O}_6$ and $\mathcal{O}_{10}$ the data at one $\boldsymbol{q}$ say nothing. Tensors built from the dark-matter velocity, as for $\mathcal{O}_7$ and $\mathcal{O}_{12}$–$\mathcal{O}_{15}$, also need the mixed elements between the direction of $\boldsymbol{q}$ and the two directions perpendicular to it.

Absorption needs exactly the missing element. The spin row of an axion-like particle is proportional to its momentum, and the absorption rate Eq. (14.17) contracts S_{ab} with $p_{\phi,a}p_{\phi,b}$: it is $p_\phi^2\,\hat{\boldsymbol{p}}_\phi\cdot S\cdot\hat{\boldsymbol{p}}_\phi$, a longitudinal element [22, Eqs. (15), (17)]. Its momentum, $p_\phi \sim 10^{-3}m_\phi$, is also far below any momentum transfer a neutron spectrometer resolves. At $q\to 0$ the full tensor is fixed instead by the magnetic permeability measured in resonance or terahertz experiments [22, footnote 3], and in the model of Sect. 14.3 by the Larmor and Kittel lines.

That the longitudinal element is missing at each q, whatever the beam, follows from the form of the amplitude. Fig. 14.8 shows the geometry.

Neutron scattering and the longitudinal spin response

At a given $\boldsymbol{q}$, magnetic neutron scattering measures only the block of $S_{ab}(\boldsymbol{q},\omega)$ transverse to $\hat{\boldsymbol{q}}$: unpolarized beams its trace $\mathrm{Tr}[P_T S]$, polarization analysis its elements. Without a model of the magnet (of the spin density within the cell, or the spin-isotropy hypothesis of Sect. 14.4.3), the longitudinal element $\hat{\boldsymbol{q}}\cdot S\cdot\hat{\boldsymbol{q}}$ and the mixed elements between $\hat{\boldsymbol{q}}$ and the transverse plane are measured by no neutron experiment, polarized or not, and no symmetry of the crystal relates them to the transverse block at the same $\boldsymbol{q}$. Within the localized-moment model of Eq. (14.29), transverse data at several wavevectors $\boldsymbol{q}+\boldsymbol{G}$ determine the full tensor.

The first part is the perpendicular-projection rule of magnetic neutron scattering (Appendix C.19). Every amplitude in Eqs. (C.164) and (C.168) contains $\boldsymbol{s}_e$ only through $\boldsymbol{s}_\perp = P_T\boldsymbol{s}_e$, because a spin density modulated along $\boldsymbol{q}$ produces no magnetic field at that wavevector (Eq. (C.163)). In the frame of Appendix C.19, with x along $\hat{\boldsymbol{q}}$, polarization analysis separates S_{yy}, S_{zz} and the chiral part of S_{yz}, and full polarimetry reaches its symmetric part as well [26]. That is at most four of the nine real numbers of the Hermitian matrix S_{ab}.

For the second part, let h be an element of the magnetic point group, acting as a rotation matrix R on positions and on the axial spin. It maps the response at $\boldsymbol{q}$ onto the response at $\mathsf{R}\boldsymbol{q}$, $S(\mathsf{R}\boldsymbol{q}) = \mathsf{R}\,S(\boldsymbol{q})\,\mathsf{R}^{\mathrm{T}}$, possibly transposed if h contains time reversal; the transpose does not change the symmetric contractions. Since also $P_T(\mathsf{R}\hat{\boldsymbol{q}}) = \mathsf{R}\,P_T(\hat{\boldsymbol{q}})\,\mathsf{R}^{\mathrm{T}}$,

$$\mathrm{Tr}\big[P_T(\mathsf{R}\hat{\boldsymbol{q}})\,S(\mathsf{R}\boldsymbol{q})\big] = \mathrm{Tr}\big[\mathsf{R}P_T(\hat{\boldsymbol{q}})\mathsf{R}^{\mathrm{T}}\mathsf{R}S(\boldsymbol{q})\mathsf{R}^{\mathrm{T}}\big] = \mathrm{Tr}\big[P_T(\hat{\boldsymbol{q}})\,S(\boldsymbol{q})\big],$$

and a measurement at a symmetry-equivalent wavevector repeats the same equation. The elements that leave $\hat{\boldsymbol{q}}$ fixed map the direction of $\boldsymbol{q}$ onto itself and the transverse plane onto itself, so they constrain the transverse block and the longitudinal element separately and never relate one to the other. A wavevector $\boldsymbol{q}+\boldsymbol{G}$ that points elsewhere belongs to a different operator, $\boldsymbol{s}_e(\boldsymbol{q}+\boldsymbol{G})$, which the exact response does not tie to $\boldsymbol{s}_e(\boldsymbol{q})$.

The localized-moment form that Eq. (14.29) already uses supplies that tie. For one magnetic ion per primitive cell, as for NiO in its chemical cell, $\boldsymbol{s}_e(\boldsymbol{k}+\boldsymbol{G}) = F^{\mathrm{mag}}(|\boldsymbol{k}+\boldsymbol{G}|)\sum_j \mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{R}_j}\boldsymbol{S}_j$ with $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{R}_j} = 1$, so that

$$S_{ab}(\boldsymbol{k}+\boldsymbol{G},\omega) = |F^{\mathrm{mag}}(|\boldsymbol{k}+\boldsymbol{G}|)|^2\,\mathrm{e}^{-2W(\boldsymbol{k}+\boldsymbol{G})}\,S^{\mathrm{lat}}_{ab}(\boldsymbol{k},\omega), \tag{14.31}$$

for every reciprocal vector $\boldsymbol{G}$, with S^{lat} the response of the lattice spins and the Debye–Waller factor e^{-2W} taken at $\boldsymbol{k}+\boldsymbol{G}$, beside the form factor. Each $\boldsymbol{G}$ gives the transverse trace $\mathrm{Tr}\,S^{\mathrm{lat}} - \hat{\boldsymbol{n}}_{\boldsymbol{G}}\cdot S^{\mathrm{lat}}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{G}}$ along its own direction $\hat{\boldsymbol{n}}_{\boldsymbol{G}} = (\boldsymbol{k}+\boldsymbol{G})/|\boldsymbol{k}+\boldsymbol{G}|$, a linear equation for the six components of the symmetric part of $S^{\mathrm{lat}}(\boldsymbol{k},\omega)$, and six generic directions fix all six. For NiO the four shortest vectors of type (111) and two of type (200) already give a system of full rank. The longitudinal element at the momentum transfer of the dark matter then follows from Eq. (14.31), and as $\boldsymbol{k}\to 0$ so does the tensor that absorption needs, resolution permitting; neutron diffraction fixes the directions of ordered moments in the same way. The route rests on the dipole-approximation form factor, which gives every spin component the same radial shape, and on a known magnetic basis; with several magnetic ions per cell, as in YIG, the phases $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}}$ of the basis enter and more wavevectors are needed. The rates of $\mathcal{O}_6$ and $\mathcal{O}_{10}$ and the absorption of axion-like particles are therefore not measurable without a model, but they follow from the transverse data within the localized-moment model under these assumptions.

14.4.3 The longitudinal response under isotropy

Spin-rotation symmetry about the moments supplies the missing element at one $\boldsymbol{q}$, without a model of the spin density but at the price of a hypothesis on the Hamiltonian: spin–orbit coupling and the dipolar interaction are neglected. In a collinear magnet with moments along $\pm\hat{\boldsymbol{m}}$, and with spin–orbit coupling neglected, the Hamiltonian is invariant under rotations of all spins about $\hat{\boldsymbol{m}}$, independently of the lattice; the Heisenberg model of Sect. 7.1 is the example. A rotation of the spin indices alone then leaves $S_{ab}(\boldsymbol{q},\omega)$ unchanged at every $\boldsymbol{q}$, and the only tensors with that property are built from δ_{ab}, $\hat{m}_a\hat{m}_b$ and $\epsilon_{abc}\hat{m}_c$:

$$S_{ab}(\boldsymbol{q},\omega) = S_{\parallel}\,\hat{m}_a\hat{m}_b + S_{\perp}\,(\delta_{ab} - \hat{m}_a\hat{m}_b) + \mathrm{i}S_{\mathrm{c}}\,\epsilon_{abc}\hat{m}_c,$$

with three functions of $\boldsymbol{q}$ and ω. The one-magnon response Eq. (14.1) has this form with $S_{\parallel} = 0$, and the two-magnon response Eq. (14.9) is all $S_{\parallel}$. The antisymmetric part drops out of every symmetric contraction. With θ_m the angle between $\hat{\boldsymbol{q}}$ and $\hat{\boldsymbol{m}}$,

$$\mathrm{Tr}\big[P_T S\big] = S_{\parallel}\sin^2\theta_m + S_{\perp}(1+\cos^2\theta_m), \qquad \hat{\boldsymbol{q}}\cdot S\cdot\hat{\boldsymbol{q}} = S_{\parallel}\cos^2\theta_m + S_{\perp}\sin^2\theta_m. \tag{14.32}$$

Polarization analysis now suffices at a single $\boldsymbol{q}$. With $\hat{\boldsymbol{m}}$ in the scattering plane at an angle $\theta_m \neq 0, \frac{\pi}{2}$ to $\hat{\boldsymbol{q}}$, the vertical element is $S_{zz} = S_{\perp}$ and the in-plane element $S_{yy} =$

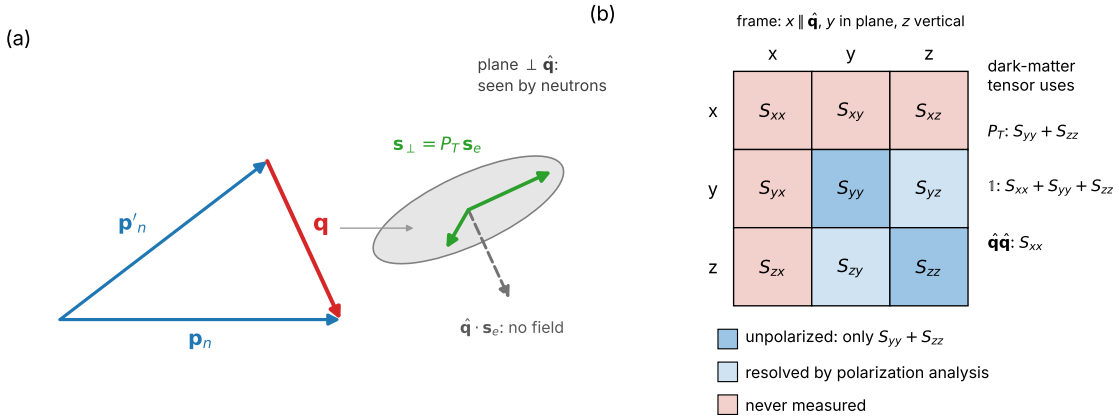


Figure 14.8: The neutron geometry and the transverse projector. (a) A neutron of momentum $\mathbf{p}_n$ leaves with $\mathbf{p}'_n$ and gives the target $\boldsymbol{q} = \mathbf{p}_n - \mathbf{p}'_n$. Of the spin density at wavevector $\boldsymbol{q}$, only the part in the plane perpendicular to $\hat{\boldsymbol{q}}$, $\boldsymbol{s}_\perp = P_T \boldsymbol{s}_e$, produces a magnetic field and scatters the neutron; the component along $\hat{\boldsymbol{q}}$ produces none. (b) The spin response S_{ab} in the frame with x along $\hat{\boldsymbol{q}}$, y in the scattering plane and z vertical. Unpolarized neutrons measure the sum $S_{yy} + S_{zz}$; polarization analysis resolves the transverse block; the elements in the first row and column are not measured at this $\boldsymbol{q}$. Right-hand columns of (b): the elements that the dark-matter tensors proportional to P_T (for example $\mathcal{O}_9$), to the unit matrix ($\mathcal{O}_4$) and to $\hat{q}_a \hat{q}_b$ ($\mathcal{O}_6$, $\mathcal{O}_{10}$) contract. Schematic.

$S_\parallel \sin^2\theta_m + S_\perp \cos^2\theta_m$, which equals $\hat{\boldsymbol{q}} \cdot S \cdot \hat{\boldsymbol{q}}$ with cos and sin exchanged and fixes $S_\parallel$. Unpolarized data give one equation for two unknowns at each $\boldsymbol{q}$, and a second measurement is needed at another angle with the same $S_\parallel$ and $S_\perp$. That is the content of the *isotropy hypothesis*: in the window used, $S_\parallel$ and $S_\perp$ depend on $\boldsymbol{q}$ only through $|\boldsymbol{q}|$, or do not change when the moments are turned at fixed $\boldsymbol{q}$. The second form is the cleaner one to realize. A soft magnet can be turned by an applied field with $\boldsymbol{q}$ held fixed in the crystal; a YIG sphere saturates above 82 mT (Worked example 14.3). In the spin-rotation-invariant model the two settings then measure exactly the same $S_\parallel$ and $S_\perp$, and only spin–orbit coupling and the magnetic dipolar interaction break the hypothesis. The dipolar term, $\omega_M \sin^2\theta_{\boldsymbol{q}}$ in the spin-wave energy with $\omega_M = 28.5\,\mu\text{eV}$ for YIG (Worked example 14.3), is negligible above about 0.1 meV, far below the 5 meV where the data start. Given $P_i = \text{Tr}[P_T S]$ at two angles θ_1 and θ_2, with $c_i = \cos^2\theta_i$, the solution of the two equations Eq. (14.32) is

$$S_\parallel = \frac{P_1(1+c_2) - P_2(1+c_1)}{2(c_2 - c_1)}, \qquad S_\perp = \frac{P_2(1-c_1) - P_1(1-c_2)}{2(c_2 - c_1)}, \tag{14.33}$$

and the longitudinal element follows from Eq. (14.32) for any direction of $\boldsymbol{q}$ relative to $\hat{\boldsymbol{m}}$.

Two limits recover the shortcuts of Berlin et al. A paramagnet in zero field, or a sample of many randomly oriented domains, is invariant under all rotations, so $S_\parallel = S_\perp$ and $\hat{\boldsymbol{q}} \cdot S \cdot \hat{\boldsymbol{q}} = \frac{1}{2}\,\text{Tr}[P_T S]$ [22, Eq. (35)]. An ordered magnet described at linear spin-wave order has $S_\parallel = 0$ at one-magnon energies, and one angle fixes $S_\perp = \text{Tr}[P_T S]/(1 + \cos^2\theta_m)$ [22, Eqs. (36), (37)]. The second shortcut sets the two-magnon continuum to zero. In NiO that continuum holds 7.3 % of the one-magnon weight (Worked example 14.1), entirely in $S_\parallel$, and where it is the only response, at energies outside the one-magnon line, the shortcut misses all of the longitudinal element. The two-angle inversion keeps it.

Worked example 14.4: The two-angle inversion for an isotropic paramagnet

Take a paramagnet in zero field, so that $S_\parallel = S_\perp = S_0$ at the $(|\boldsymbol{q}|, \omega)$ of interest, and measure $\mathrm{Tr}[P_T S]$ with $\boldsymbol{q}$ perpendicular and then parallel to a fixed crystal axis $\hat{\boldsymbol{m}}$: $c_1 = 0$, $c_2 = 1$. Both measurements give $2S_0$, and Eq. (14.33) reduces to $S_\parallel = P_1 - P_2/2$ and $S_\perp = P_2/2$, which returns S_0 for both. The hypothesis is tested by the data themselves: $P_2 - P_1 = (c_2 - c_1)(S_\perp - S_\parallel)$ vanishes.

Errors decide whether the inversion is worth doing. Give each measurement a statistical error of 5 %, $0.1S_0$, independent between the two. Then $S_\perp$ carries $0.050S_0$, while $S_\parallel$ carries $(\sqrt{5}/2) \times 0.1S_0 = 0.112S_0$, more than twice the relative error of the data. The longitudinal element needed for $\mathcal{O}_6$ along $\hat{\boldsymbol{m}}$ is $S_\parallel$ itself, with the same 11 %; at $\cos^2\theta = \frac{1}{3}$ it is $P_1/3 + P_2/6$, with 3.7 %. The isotropy test, $P_2 - P_1 = 0 \pm 0.141S_0$, limits $|S_\perp - S_\parallel|$ to about 14 % of S_0 at one standard deviation.

A smaller rotation costs more. With $c_2 = \sin^2 30^\circ = 0.25$ instead, the denominator $2(c_2 - c_1)$ drops from 2 to 0.5, $S_\parallel = 2.5P_1 - 2P_2$ and its error grows to $0.32S_0$, 32 %. The one-angle shortcut $\frac{1}{2}\,\mathrm{Tr}[P_T S]$ has the 5 % of the data, but it is right only if the paramagnet is in fact isotropic, which the two-angle measurement is there to check.

A spin-wave model fitted to the transverse data supplies the longitudinal element as well, with the fit uncertainty as a normalization nuisance (Sect. 8.8); a longitudinal response measured at dark-matter momenta, without such a model or the localized-moment form, does not exist.

14.5 Chiral and nonreciprocal targets

Density-functional perturbation theory gives α-quartz the gyration $g_{33} = 0.125\,\mathrm{bohr}$ along its optic axis, and with it a rotation of the plane of polarization by 21.6° per millimeter for light of 2.105 eV [27, Table I, Eq. (4)]; the mirror-image crystal turns the plane the other way. The rotation is a response odd in the momentum of the light, and at a photon momentum of 2.1 eV it is small, $q\,g_{33} = 7 \times 10^{-5}$. Dark matter that deposits the same energy brings at least $1/v_{\max} = 377$ times more momentum, so a response odd in $\boldsymbol{q}$ may matter more for dark matter than for light. Sect. 11.7 showed that the density response stays even in $\boldsymbol{q}$ unless inversion and time reversal are both broken; in a crystal, magnetic order breaks time reversal. Here we find which other responses become odd when only one of the two is broken, in which crystals, and which couplings of the dark matter see them. Three detectors are built on broken symmetry: phonons that carry angular momentum, molecular magnets held in a metastable state, and multiferroics, in which electric and magnetic order coexist.

14.5.1 Reciprocity of the response matrix

Time reversal relates the response at $\boldsymbol{q}$ to that at $-\boldsymbol{q}$ for every pair of rows. Each row of Eq. (8.3) goes over into itself at $-\boldsymbol{q}$ under the antiunitary time-reversal operator T, up to a sign σ_a^{T}: $\mathsf{T} O_a(\boldsymbol{q}) \mathsf{T}^{-1} = \sigma_a^{\mathsf{T}} O_a(-\boldsymbol{q})$, since T conjugates $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ and reverses momenta and spins. The density and the kinetic-energy density have $\sigma^{\mathsf{T}} = +1$, the current and the spin density -1, and the spin current $+1$, as the product of two reversed vectors. For an antiunitary operator, $\langle f|A|i\rangle^* = \langle \mathsf{T}f|\mathsf{T}A\mathsf{T}^{-1}|\mathsf{T}i\rangle$. We apply it to both matrix elements in Eq. (4.13); the time-reversed states carry the same energies and thermal weights, and $S_{ab}(\boldsymbol{q},\omega)^* = \sigma_a^{\mathsf{T}}\sigma_b^{\mathsf{T}} S_{ab}(-\boldsymbol{q},\omega)$,

Table 14.2: Parts of the response odd in $\boldsymbol{q}$, or odd under inversion of the crystal, and the symmetries they need, from Eq. (14.34). "Gyrotropic" point groups are those with $N_{\mathrm{gyr}} > 0$ in Eq. (C.146).

Quantity	Needs	Treated in
Density $S(\boldsymbol{q},\omega) - S(-\boldsymbol{q},\omega)$	inversion and time reversal broken	Sect. 11.7
Antisymmetric current part $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}$	inversion broken, gyrotropic group	Sect. 14.5.2, Worked example 14.5
Symmetric current part, odd in $\boldsymbol{q}$	inversion and time reversal broken	Sect. 14.5.1
Phonon angular momentum $\boldsymbol{L}^{\mathrm{ph}}$	inversion or time reversal broken	Sect. 14.5.3
Static spin–spin-current block $R_{s^z\mathcal{J}}(0,\omega\to 0)$	inversion and time reversal broken	Sect. 14.5.5

and with $S^*_{ab} = S_{ba}$,

$$\begin{aligned} S_{ab}(\boldsymbol{q},\omega) &= \sigma^{\mathsf{T}}_a \sigma^{\mathsf{T}}_b \, S_{ba}(-\boldsymbol{q},\omega) \quad \text{(time reversal)},\\ S_{ab}(\boldsymbol{q},\omega) &= \sigma^{\mathsf{P}}_a \sigma^{\mathsf{P}}_b \, S_{ab}(-\boldsymbol{q},\omega) \quad \text{(inversion)}. \end{aligned} \tag{14.34}$$

The second relation follows in the same way from the unitary inversion P, with $\sigma^{\mathsf{P}} = +1$ for the density, the kinetic-energy density and the spin density and -1 for the current and the spin current. For the density block alone both relations give $S(\boldsymbol{q}) = S(-\boldsymbol{q})$ (Appendix A.4).

The current block is the first that differs. Hermiticity splits it into a real symmetric part and i times a real antisymmetric part, and an antisymmetric 3×3 matrix is a vector:

$$S_{j^\alpha j^\beta}(\boldsymbol{q},\omega) = S^{\mathrm{R}}_{\alpha\beta}(\boldsymbol{q},\omega) + \mathrm{i}\,\epsilon_{\alpha\beta\gamma}\,\mathcal{G}^{\gamma}_{\mathrm{OA}}(\boldsymbol{q},\omega). \tag{14.35}$$

With $\sigma^{\mathsf{T}}_j = -1$ twice, time reversal gives $S_{j^\alpha j^\beta}(\boldsymbol{q}) = S_{j^\alpha j^\beta}(-\boldsymbol{q})^*$: the symmetric part is even in $\boldsymbol{q}$ and $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}$ is odd. Since inversion makes both even, $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}$ needs a crystal without inversion and survives time reversal, while an odd symmetric part needs both symmetries broken. A crystal without inversion, mirror planes or rotation–reflections is chiral, and exists in two forms related by a mirror, its enantiomers; $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}$ changes sign between them. Physically, $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}$ measures whether the electrons circulate one way more easily than the other along $\boldsymbol{q}$, which only a crystal without inversion allows. For rows of opposite time-reversal sign, such as the spin density and the spin current, Eq. (14.34) makes the symmetric part of the block odd in $\boldsymbol{q}$, so its value at $\boldsymbol{q} = 0$, which a static field would probe, needs inversion broken as well. Table 14.2 collects the cases that this section uses.

14.5.2 Finite-momentum optical activity

The vector $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}$ is the finite-momentum generalization of natural optical activity. To see it, write the antisymmetric part of the proper current response as $\mathrm{i}\epsilon_{\alpha\beta\gamma}\mathcal{G}^{\gamma}_{\mathrm{R}}$, with $\boldsymbol{\mathcal{G}}_{\mathrm{R}}$ complex. The fluctuation–dissipation theorem Eq. (4.18), $(1-\mathrm{e}^{-\beta\omega})S_{ab} = \mathrm{i}(R_{ab} - R^*_{ba})$, turns it into $\boldsymbol{\mathcal{G}}_{\mathrm{OA}} = -2\,\mathrm{Im}\,\boldsymbol{\mathcal{G}}_{\mathrm{R}}/(1-\mathrm{e}^{-\beta\omega})$, and the dielectric tensor of Sect. 4.4, $\varepsilon^{\alpha\beta} = \delta^{\alpha\beta} - (e^2/\omega^2)[(\boldsymbol{n}_e/m_e)\delta^{\alpha\beta} + \tilde{R}^{\alpha\beta}_{jj}]$, acquires the antisymmetric term $-\mathrm{i}(e^2/\omega^2)\epsilon_{\alpha\beta\gamma}\mathcal{G}^{\gamma}_{\mathrm{R}}$. Optical activity is defined by the term linear in $\boldsymbol{q}$ of that tensor, $\varepsilon^{\alpha\beta}(\boldsymbol{q},\omega) = \varepsilon^{\alpha\beta}(\omega) + \mathrm{i}\epsilon_{\alpha\beta m} g_{ml}(\omega) q_l + O(q^2)$, with the gyration tensor g_{ml} of Appendix C.17 in the normalization of the dielectric tensor [27, Eqs. (2)–(4)]. Comparing the two antisymmetric terms,

$$\mathcal{G}^{\gamma}_{\mathrm{OA}}(\boldsymbol{q},\omega) = \frac{2\omega^2}{e^2}\,\frac{\mathrm{Im}\, g_{\gamma l}(\omega)\, q_l}{1-\mathrm{e}^{-\beta\omega}} + O(q^3), \tag{14.36}$$

with no term of order q^2, since $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}$ is odd. Only the absorptive part of the gyration enters the structure factor. Its reactive part, $\mathrm{Re}\, g$, rotates the polarization, and the two are related by the Kramers–Kronig relations Eq. (4.24), which hold for $g(\omega)$ because it is a coefficient of a causal response.

The observable is circular dichroism, the difference between the absorption of the two helicities. A transverse field of polarization $\hat{\boldsymbol{e}}_\pm = (\hat{\boldsymbol{e}}_1 \pm \mathrm{i}\hat{\boldsymbol{e}}_2)/\sqrt{2}$, with $\hat{\boldsymbol{e}}_1 \times \hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{q}}$, couples to the row $\hat{\boldsymbol{e}}_\pm \cdot \boldsymbol{j}_e$, and Eq. (8.19) needs $\sum_{\alpha\beta} \hat{e}^*_{\pm\alpha} \hat{e}_{\pm\beta} S_{j^\alpha j^\beta}$. The symmetric part yields the transverse response S_{T}, and the antisymmetric part gives $\mathrm{i}(\hat{\boldsymbol{e}}^*_\pm \times \hat{\boldsymbol{e}}_\pm)\cdot \boldsymbol{\mathcal{G}}_{\mathrm{OA}}$ with $\hat{\boldsymbol{e}}^*_\pm \times \hat{\boldsymbol{e}}_\pm = \pm\mathrm{i}\hat{\boldsymbol{q}}$:

$$S_\pm(\boldsymbol{q},\omega) = S_{\mathrm{T}}(\boldsymbol{q},\omega) \mp \hat{\boldsymbol{q}} \cdot \boldsymbol{\mathcal{G}}_{\mathrm{OA}}(\boldsymbol{q},\omega), \qquad \frac{S_- - S_+}{S_- + S_+} = \frac{\hat{\boldsymbol{q}} \cdot \boldsymbol{\mathcal{G}}_{\mathrm{OA}}}{S_{\mathrm{T}}} = \frac{q\ \mathrm{Im}\, g_{\hat{q}\hat{q}}}{\varepsilon_2^{\mathrm{T}}}, \tag{14.37}$$

with $g_{\hat{q}\hat{q}} = \hat{\boldsymbol{q}} \cdot g \cdot \hat{\boldsymbol{q}}$; the last form uses Eq. (14.36) and $\varepsilon_2^{\mathrm{T}} = (e^2/2\omega^2)(1 - \mathrm{e}^{-\beta\omega}) S_{\mathrm{T}}$ at the order kept. Reversing $\boldsymbol{q}$ at fixed helicity exchanges $\hat{\boldsymbol{e}}_1$ and $\hat{\boldsymbol{e}}_2$ up to sign and flips $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}$, so the dissymmetry depends on the helicity, not on the direction of travel: natural optical activity is reciprocal. For light, $q = n_r\omega$ with n_r the refractive index, and the helicity components of ε are $\varepsilon^{\mathrm{T}} \mp g_{\hat{q}\hat{q}} q$. Then $n_\pm = n_r \mp g_{\hat{q}\hat{q}}\omega/2$, and per unit length the plane of linear polarization turns by

$$\rho(\omega) = \frac{\omega}{2}\, \mathrm{Re}(n_- - n_+) = \frac{\omega^2}{2}\, \mathrm{Re}\, g_{\hat{q}\hat{q}}(\omega), \tag{14.38}$$

the rotatory power of Zabalo and Stengel [27, Eq. (4)], while the absorption coefficients differ by $2\omega^2\, \mathrm{Im}\, g_{\hat{q}\hat{q}}$.

Fig. 14.9 sketches what changes for dark matter. At optical momenta the dissymmetry is of order $q|g| \sim 10^{-4}$. Dark matter at energy ω brings $q \geq \omega/v_{\max}$, and at $q = 1\,\mathrm{keV}$ the quartz value of g_{33} gives $q\, g_{33} = 0.034$. That number sets a scale only, since Eq. (14.36) holds for q small against the inverse lattice spacing, while dark-matter momenta of keV reach the zone boundary, and above the gap of quartz the absorptive $\mathrm{Im}\, g$, not the static g_{33}, is the relevant input. The full $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}(\boldsymbol{q},\omega)$ at keV momenta is not known for any crystal. Appendix E.20 specifies the calculation for tellurium or α-quartz: the current response at $+\boldsymbol{q}$ and $-\boldsymbol{q}$ with spin–orbit coupling and the nonlocal part of the current vertex, checked against the density-functional optical activity at small q and against zero in silicon.

Dark matter sees $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}$ only through a coupling tensor with an antisymmetric current block. In Table 8.1 the current row of $\mathcal{O}_5$ has the potential $\tilde{V}_{j^\delta} = -(\mathrm{i}/m_e) c_5 \epsilon_{\mu\gamma\delta} S^\mu_\chi q^\gamma$, and that of $\mathcal{O}_8$ has $\tilde{V}_{j^\delta} = -c_8 S^\delta_\chi$. Each alone leads to a real symmetric $\mathfrak{D}_{jj}$, $\frac{1}{4}|c_8|^2\delta_{\alpha\beta}$ and $(|c_5|^2/4m_e^2)(q^2\delta_{\alpha\beta} - q_\alpha q_\beta)$. Their interference, with $\mathrm{Tr}_\chi(S^\mu_\chi S^\nu_\chi)/2 = \delta_{\mu\nu}/4$ in Eq. (8.7), gives

$$\mathfrak{D}_{j^\alpha j^\beta} \supset -\frac{\mathrm{i}}{2m_e}\, \mathrm{Re}(c_5^* c_8)\, \epsilon_{\alpha\beta\gamma} q^\gamma,$$

purely imaginary and antisymmetric, and its contraction with Eq. (14.35) is $(1/m_e)\, \mathrm{Re}(c_5^* c_8)\, \boldsymbol{q} \cdot \boldsymbol{\mathcal{G}}_{\mathrm{OA}}$. Since $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}$ changes sign between enantiomers, the difference of their rates isolates this term. The density rows of $\mathcal{O}_5$ and $\mathcal{O}_8$ meet the current rows through the block S_{nj}. Inversion maps one enantiomer onto the other and, by Eq. (14.34) with $\sigma^{\mathsf{P}}_n \sigma^{\mathsf{P}}_j = -1$, gives $S^{\mathrm{L}}_{nj}(\boldsymbol{q}) = -S^{\mathrm{R}}_{nj}(-\boldsymbol{q})$, so the enantiomer-odd part $S^{\mathrm{R}}_{nj}(\boldsymbol{q}) - S^{\mathrm{L}}_{nj}(\boldsymbol{q})$ is a vector even in $\boldsymbol{q}$. Averaged over crystal orientations at fixed $\boldsymbol{q}$ it must point along $\hat{\boldsymbol{q}}$, which an even function cannot do, so it averages to zero; it also enters contracted with $\boldsymbol{q} \times \bar{\boldsymbol{v}}_\chi$, orthogonal to $\hat{\boldsymbol{q}}$. Inserting the interference term in Eq. (8.16) and averaging over orientations with the kernel $\eta(v_{\min})/2q$ of Eq. (A.42),

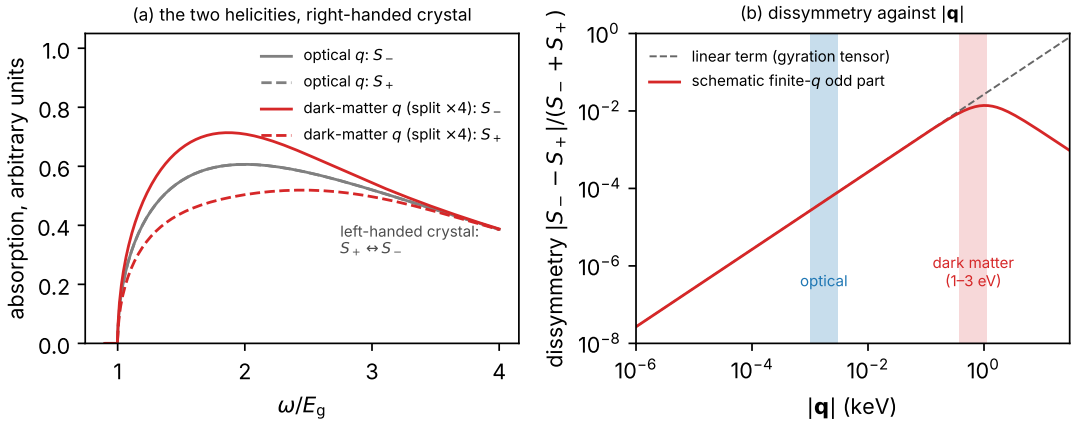


Figure 14.9: Circular dichroism of a chiral crystal such as tellurium. Schematic: all curves are model curves, not computed for tellurium. (a) Absorption of the two helicities, $S_\pm = S_\mathrm{T} \mp \hat{\boldsymbol{q}}\cdot\boldsymbol{\mathcal{G}}_\mathrm{OA}$ (Eq. (14.37)), for a model band edge at E_g in the right-handed form, at an optical momentum (gray, split invisible) and at a dark-matter momentum (red, split exaggerated four times). The left-handed form exchanges S_+ and S_-. (b) The dissymmetry against $|\boldsymbol{q}|$: the linear term $q|g|$ with $|g| = 0.1\,\mathrm{bohr}$, the order of the computed gyration of trigonal crystals [27, Table I] (dashed), and a schematic turnover at the inverse of a 6 Å helix pitch (red). Blue band: photons of 1 to 3 eV; red band: the smallest dark-matter momenta at the same energies, ω/v_max. The odd part beyond the linear term is not known.

Enantiomer difference of the rate

$$\frac{\mathrm{d}\mathcal{R}^\mathrm{R}}{\mathrm{d}\omega} - \frac{\mathrm{d}\mathcal{R}^\mathrm{L}}{\mathrm{d}\omega} = \frac{\rho_\chi}{m_\chi\,\rho_T}\int \frac{q\,\mathrm{d}q}{2\pi^2}\,\eta(v_\mathrm{min})\,\frac{q}{m_e}\,\mathrm{Re}\big[c_5^*(q)\,c_8(q)\big]\,\bar{\mathcal{G}}_\mathrm{OA}(q,\omega), \tag{14.39}$$

with $\bar{\mathcal{G}}_\mathrm{OA}(q,\omega)$ the average of $\hat{\boldsymbol{q}}\cdot\boldsymbol{\mathcal{G}}_\mathrm{OA}$ over the directions of $\boldsymbol{q}$ in the right-handed form.

Operator $\mathcal{O}_8 = \boldsymbol{S}_\chi\cdot\boldsymbol{v}^\perp$ is odd under inversion and $\mathcal{O}_5$ is even. Only dark matter whose couplings violate parity therefore has different rates in the two enantiomers, for example a particle with both a magnetic dipole moment, which produces $\mathcal{O}_5$, and an anapole moment, which produces $\mathcal{O}_8$ (Sect. 8.1). The crystal supplies the other parity-odd factor. Because $\boldsymbol{q}\cdot\boldsymbol{\mathcal{G}}_\mathrm{OA}$ is even in $\boldsymbol{q}$, the difference is not a wind asymmetry: at fixed orientation it contains the multipoles $\ell = 0$ and $\ell = 2$ of Eq. (11.50) and modulates with the even halo multipoles. Relative to the transverse-current part of the rate of $\mathcal{O}_8$ alone, $\frac{1}{2}|c_8|^2 S_\mathrm{T}$, the interference term is the dissymmetry of Eq. (14.37) at dark-matter momenta times $2|c_5|q/(m_e|c_8|)$.

Worked example 14.5: Point-group test for odd responses in α-quartz

α-Quartz, like tellurium, has the point group 32 (D_3): the identity, two rotations by $\pm 2\pi/3$ about the optic axis $\hat{\boldsymbol{z}}$, and three rotations by π about twofold axes in the basal plane. All six are proper, $\det \mathsf{R} = 1$, and the class values of $1 + 2\cos\theta_\mathsf{R}$ are 3, 0 and -1 (Appendix C.17).

Scalar responses. An odd part of a scalar such as $S(\boldsymbol{q},\omega)$ starts with the harmonics $\ell = 1$, which transform as a polar vector. By Eqs. (C.144) and (C.145) their invariants number $(3 + 2\cdot 0 + 3\cdot(-1))/6 = 0$. The harmonics $\ell = 3$, with characters $\sin(\frac{7}{2}\theta)/\sin(\frac{1}{2}\theta) = 7$, 1, -1, have $(7 + 2 - 3)/6 = 1$ invariant. Quartz keeps time reversal, however, and by

Eq. (14.34) its density response is even anyway; a magnetic crystal with the same point group could have a scalar odd part starting at $\ell = 3$, never at $\ell = 1$, because the group is not polar.
The current response. The gyration tensor transforms with $(\det \mathsf{R})(1 + 2\cos\theta_{\mathsf{R}})^2 = 9,\ 0,\ 1$ on the three classes, and Eq. (C.146) gives $N_{\mathrm{gyr}} = (9 + 0 + 3)/6 = 2$. Projecting all nine components with $|\mathcal{P}|^{-1}\sum_{\mathsf{R}}(\det \mathsf{R})\,\mathsf{R} \otimes \mathsf{R}$ leaves the two diagonal tensors $\mathrm{diag}(1, 1, 0)$ and $\mathrm{diag}(0, 0, 1)$, so $g = \mathrm{diag}(g_{11}, g_{11}, g_{33})$. The density-functional values are $g_{11} = -0.071\,\mathrm{bohr}$ and $g_{33} = 0.125\,\mathrm{bohr}$ [27, Table I].
Consequences. Along the optic axis, $g_{\hat{q}\hat{q}} = g_{33}$ and Eq. (14.38) gives $\rho = 21.6\,^\circ\,\mathrm{mm}^{-1}$ at 2.105 eV. Averaged over directions, $g_{\hat{q}\hat{q}}$ becomes $(2g_{11} + g_{33})/3 = -0.0057\,\mathrm{bohr}$, only -0.045 of g_{33}, because the two components have opposite signs. If the same cancellation held for $\mathrm{Im}\, g$ above the gap, the orientation-averaged enantiomer difference Eq. (14.39) of quartz would be twenty times smaller than the difference for $\boldsymbol{q}$ along the optic axis. A crystal held with its axis along the wind would gain that factor back through the multipole $\ell = 2$. GaAs, with $N_{\mathrm{gyr}} = 0$ (Worked example C.1), has no $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}$ at linear order although it lacks inversion, and no enantiomers.

14.5.3 Chiral phonons

In GaAs, which lacks inversion, the atoms in a phonon of a general wavevector move on ellipses, and a phonon of the lowest band carries on average 0.12 units of angular momentum. In the normalization of Eq. (5.32), the angular momentum of mode $\boldsymbol{Q}\nu$ in units of $\hbar$ is

$$\boldsymbol{L}^{\mathrm{ph}}_{\boldsymbol{Q}\nu} = \sum_{\kappa} \boldsymbol{L}^{\mathrm{ph}}_{\kappa,\boldsymbol{Q}\nu}, \qquad \boldsymbol{L}^{\mathrm{ph}}_{\kappa,\boldsymbol{Q}\nu} = -\mathrm{i}\,\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^* \times \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}), \tag{14.40}$$

the definition of Romao and collaborators and of Matas, Krizek and Romao [28, Eq. (6), 29, Eq. (11)]. Each term is real, and $|\boldsymbol{L}^{\mathrm{ph}}_{\boldsymbol{Q}\nu}| \le \sum_\kappa |\boldsymbol{e}_{\kappa\nu}|^2 = 1$, with equality for circular motion. Time reversal gives $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$ for a nondegenerate mode, so $\boldsymbol{L}^{\mathrm{ph}}_{-\boldsymbol{Q}\nu} = -\boldsymbol{L}^{\mathrm{ph}}_{\boldsymbol{Q}\nu}$. Inversion maps the axial vector $\boldsymbol{L}^{\mathrm{ph}}$ at $\boldsymbol{Q}$ onto itself at $-\boldsymbol{Q}$, and with both symmetries it vanishes: with the force constants of Sect. 5.2 silicon yields $|\boldsymbol{L}^{\mathrm{ph}}| < 2 \times 10^{-7}$ at every point sampled, numerical noise. In a magnetically ordered crystal any phonon can carry angular momentum.

The angular momentum labels the final state; the coupling is the density coupling of Eq. (11.1), which creates mode $\boldsymbol{Q}\nu$ with the amplitude given there, and multiplying each term of that response by $\delta^3(\boldsymbol{L} - \boldsymbol{L}^{\mathrm{ph}}_{\boldsymbol{Q}\nu})$ gives the final-state-resolved one-phonon response $S^{(1)}_A(\boldsymbol{q}, \omega; \boldsymbol{L})$, whose integral over $\boldsymbol{L}$ is $S^{(1)}_A(\boldsymbol{q}, \omega)$. With $S^{(1)}_A$ even in $\boldsymbol{q}$ and $\boldsymbol{L}^{\mathrm{ph}}$ odd, the mean angular momentum of the created phonons survives the integration over $\boldsymbol{q}$ only through the part of the halo kernel odd in $\hat{\boldsymbol{q}} \cdot \hat{\boldsymbol{v}}_{\mathrm{E}}$, the odd multipoles of Eq. (11.49). To first order in $\hat{\boldsymbol{v}}_{\mathrm{E}}$ it is $\langle L_i \rangle = G_{ij}\hat{v}_{\mathrm{E},j}$, and the crystal tensor G_{ij} joins an axial index to a polar one, like the gyration tensor. Its invariants are counted by Eq. (C.146): in GaAs there are none, so the phonons that dark matter creates there carry no net angular momentum along the wind at this order, while α-quartz and tellurium allow two constants (Worked example 14.5).

Chiral phonons are detected through their magnetic moment. A nucleus that carries its Born charge eZ^*_κ on a closed orbit has the moment $(eZ^*_\kappa/2M_\kappa)\,\boldsymbol{u} \times \boldsymbol{P}$. Summed over the nuclei, this operator is bilinear in the phonon operators. It has no matrix element between the ground state and one phonon, and it measures a property of the occupied mode. In a

state with one phonon in mode $\boldsymbol{Q}\nu$,

$$\boldsymbol{\mu}^{\mathrm{ph}}_{\boldsymbol{Q}\nu} = \sum_{\kappa} \frac{eZ^*_{\kappa}}{2M_{\kappa}}\, \boldsymbol{L}^{\mathrm{ph}}_{\kappa,\boldsymbol{Q}\nu}, \tag{14.41}$$

which is the moment of Refs. [28, Eq. (7), 29, Eq. (12)] for isotropic Born charges. With the PBE Born charges of Sect. 5.5, $Z^* = \pm 2.09$, GaAs reaches at most $0.015\,\mu_N$ in its lowest band, and no mode exceeds $Z^*_{\mathrm{Ga}} m_p / M_{\mathrm{Ga}} = 0.030\,\mu_N$, which a mode with $|\boldsymbol{L}^{\mathrm{ph}}| = 1$ on the gallium sublattice alone would have; μ_N is the nuclear magneton, 1/1836 of μ_B. Phonon moments of order μ_B have been reported in some $4f$ paramagnets and topological semimetals [28, Sect. II]. They are larger than Eq. (14.41) by four to five orders of magnitude and come from the coupling of the lattice to the orbital and spin moments of the electrons, which the Born charge does not contain; computing them needs the derivatives of the crystal-field states with respect to the atomic displacements, separately for the spin and the orbital moment (Appendix E.19). A coupling to the ionic current, such as that of a particle with a magnetic dipole moment, creates single phonons through the transverse part of that current, at first order in q, and not through $\boldsymbol{\mu}^{\mathrm{ph}}$.

The detector that Romao and collaborators propose reads out phonons of the lowest band, the only band whose phonons live for microseconds or longer [29, Sect. III], after the phonons created by the dark matter have decayed into it. A phonon Hall effect, the transverse deflection of a heat current carried by chiral phonons, collects them at the surfaces, where magnetometers sense their moments [28, Sect. IV]. The rate is the one-phonon rate [29, Eqs. (3), (4)], Eq. (11.1) in Eq. (8.16), times a detection probability that Matas and collaborators take as the average of $|\boldsymbol{L}^{\mathrm{ph}}|$ over the lowest band, with each event ending as one phonon placed at random in that band. They quote 0.07 for GaAs, 0.22 for $LiNbO_3$ and 0.16 for the framework CuCl(pyrimidine) [29, Sect. III], and Romao and collaborators 0.15 for InF_3(4,4′-bipyridine) [28, Sect. VII]. Two steps carry the estimate: the created phonons sample the band as the average assumes, and their chirality survives the decay. Limit Check 14.4a tests the first in GaAs. The second is not protected, since L^{ph}_z is not conserved. For $H = \frac{1}{2}\boldsymbol{p}\cdot\boldsymbol{p} + \frac{1}{2}\boldsymbol{u}\cdot D\boldsymbol{u}$ in mass-weighted coordinates and $L_z = \boldsymbol{u}\cdot J_z\boldsymbol{p}$, with J_z the generator of rotations about $\hat{\boldsymbol{z}}$ acting on the displacement of each site, $[H, L_z] = \frac{\mathrm{i}}{2}\,\boldsymbol{u}\cdot[J_z, D]\,\boldsymbol{u}$, which vanishes only if the force constants are isotropic about every site. A crystal conserves only the pseudo-angular momentum of its discrete rotations, the eigenvalue of a rotation by $2\pi/n$ defined modulo n, and the anharmonic decay changes $\boldsymbol{L}^{\mathrm{ph}}$. The probability that a created phonon ends as a detectable chiral phonon is the solution of a linear system over the three-phonon decay rates, $\boldsymbol{\mathcal{N}}^{\mathrm{ch}} = (\mathbb{1} - \mathsf{M})^{-1}\boldsymbol{b}$, which Appendix E.19 sets up for InF_3(4,4′-bipyridine) with the angular momentum recomputed for each daughter.

14.5.4 Single-molecule magnets

The twelve manganese ions of a Mn_{12}-acetate molecule lock into a total spin $|\boldsymbol{S}| = 10$, held along the molecular axis by an axial anisotropy. Electron spin resonance measures $D = 0.454\,\mathrm{cm}^{-1} = 0.0563\,\mathrm{meV}$ and a quartic term $B^0_4 = -2.0\times10^{-5}\,\mathrm{cm}^{-1}$ [30, Fig. 2], so that in a field along the axis the molecule has

$$H_{\mathrm{mol}} = -D\,S_z^2 + B^0_4\,O^0_4 - \omega_{\mathrm{Lar}} S_z, \tag{14.42}$$

with the axial term $-DS_z^2$ of Alves and collaborators [31, Eq. (3)], the quartic operator $O^0_4 = 35S_z^4 - [30|\boldsymbol{S}|(|\boldsymbol{S}|+1) - 25]S_z^2 + \mathrm{const}$ of Edwards and collaborators [30], and the Zeeman term of Sect. 14.3.3. The levels $m = -|\boldsymbol{S}|, \ldots, |\boldsymbol{S}|$ form two wells separated by a barrier; at 1 T the top lies at $m = -1$, 4.6 meV (53 K) above $m = -|\boldsymbol{S}|$. A crystal cooled in a field

and then given the reversed field keeps its molecules in the upper well, at $m = -|\boldsymbol{S}|$, for the relaxation time $\tau_\mathrm{R} \approx \tau_0 \exp[(U - \frac{1}{2}\Delta E_\mathrm{Zee})/T]$ of thermal activation over the barrier [31, Eq. (4)]. The molecules then store the Zeeman energy ΔE_Zee, and a local deposit of heat can release it in a front that propagates through the crystal, a magnetic avalanche, which is the readout proposed by Alves and collaborators.

The molecules sit far apart in their crystal, and we treat them as independent, with number density n_mol, populations p_m, and the spin row $\boldsymbol{s}_e(0) = \lambda_S \sum \boldsymbol{S}$ of Eq. (14.18), $\lambda_S = 1$ for manganese ions with quenched orbital moments. With $s_{e,x} = \lambda_S (S^+ + S^-)/2$ and $|\langle m \pm 1|S^\pm|m\rangle|^2 = |\boldsymbol{S}|(|\boldsymbol{S}|+1) - m(m\pm 1)$, Eq. (4.13) gives

$$S_{xx}(0,\omega) = \frac{\pi n_\mathrm{mol} \lambda_S^2}{2} \sum_m p_m \Big\{ \big[|\boldsymbol{S}|(|\boldsymbol{S}|+1) - m(m+1)\big] \delta(\omega - \omega_{m,m+1}) + \big[|\boldsymbol{S}|(|\boldsymbol{S}|+1) - m(m-1)\big] \delta(\omega - \omega_{m,m-1}) \Big\}, \tag{14.43}$$

with $\omega_{m,m'} = E_{m'} - E_m$ the level differences of Eq. (14.42), and $S_{yy} = S_{xx}$. Without the quartic term, $\omega_{m,m+1} = -D(2m+1) - \omega_\mathrm{Lar}$. A molecule in the lower well, $m = |\boldsymbol{S}|$, absorbs through S^- at $D(2|\boldsymbol{S}| - 1) + \omega_\mathrm{Lar}$. A molecule held at $m = -|\boldsymbol{S}|$ absorbs through S^+ at

$$\omega_\mathrm{meta} = D\,(2|\boldsymbol{S}| - 1) - \omega_\mathrm{Lar}, \tag{14.44}$$

with the weight $2|\boldsymbol{S}|$: the metastable line, which the field moves down by 0.116 meV per tesla while it moves the line of the lower well up. For Mn_{12}-acetate, $D(2|\boldsymbol{S}| - 1) = 1.07$ meV; the quartic term raises the zero-field line to 1.21 meV, and at 1 T the metastable line sits at 1.098 meV, 12.7 K. The next line of the upper well, from $m = -9$, lies 0.18 meV lower. As for the Larmor line, the spin algebra fixes the total: summed over all lines of a well, the net absorptive weight $\sum_m (p_m - p_{m+1})[|\boldsymbol{S}|(|\boldsymbol{S}|+1) - m(m+1)]$ is $2|\langle S_z\rangle|$, however the population is spread.

An axion-like particle reaches the line through Eq. (14.17), with p_ϕ^2 replaced by its component perpendicular to the molecular axis, and a dark photon through its magnetic field, as in Sect. 14.3. The electric field of a dark photon couples instead to the electronic and vibrational dipoles of the molecules. For it, Alves and collaborators take the rate from the measured absorption coefficient $\alpha(\omega)$ of the crystal, with the in-medium mixing set to its vacuum value: in our notation $\mathcal{R} = (\rho_\chi / m_\phi \rho_T)\, \zeta^2 \alpha(m_\phi)$ [31, Eqs. (9), (10), (17), (18)]. The absorption rate of Sect. 8.2.3 with the mixing of Eq. (8.39) is $\zeta^2 \sigma_1 / |\varepsilon|^2$, and $\alpha = \sigma_1 / n_r$, so the rate is that expression times $n_r / |\varepsilon|^2$. An axion-like particle in a static field is a dark photon with $\zeta = g_{a\gamma\gamma} B_0 / m_\phi$ (Eq. (10.33)), and its rate is quadratic in $g_{a\gamma\gamma} B_0$, as the reach of Ref. [31, Eq. (13)] assumes. The avalanche needs a deposit of about 10^{-2} eV in molecules of radius 2 to 3 nm [31, Eq. (6), Fig. 6]; an absorbed boson deposits its whole energy $\omega_\phi \simeq m_\phi$, and for any nonrelativistic speed the trigger is a condition on m_ϕ alone. Their boosted fraction, which scales the scattering rate, therefore does not apply to absorption: for $m_\phi \geq 10^{-2}$ eV halo absorption triggers with the full halo density, below it halo absorption cannot trigger, and a boosted population would need its own energy spectrum. The avalanche is a thermal response of the detector, treated with the readout models later in the book.

Limit Check 14.4 compares the two literature descriptions with the fuller final-state-resolved responses. In GaAs the average of $|\boldsymbol{L}^\mathrm{ph}|$ over the lowest band is 0.124 with the force constants of Sect. 5.5, against the 0.07 quoted from another dynamical matrix. The grid, the masses and the weighting cannot produce a factor 2. A Gamma-centred or shifted 10^3 grid gives 0.126 or 0.118, equal or swapped masses 0.115 or 0.112, and energy or inverse-energy weighting 0.135 or 0.117. Scaling the long-range dipole part of the force constants between zero and 1.5 times its value moves the result between 0.175 and 0.107, within ±40 %. The

average of a single Cartesian component, $\langle|L_x^{\rm ph}|\rangle$, is 0.062, close to 0.07, which is a possible origin of the difference. Weighted with the one-phonon rate, the lowest-band phonons that dark matter creates directly approach 0.112, within 10 % of the average, for $m_\chi \geq 2.2\,\mathrm{MeV}$, where they carry 30 % of the one-phonon rate. Below 0.8 MeV the lowest band takes less than 1 % of the rate, and all created phonons together carry less than 0.03 below 0.2 MeV, because the long-wavelength acoustic phonons that a light particle can create move their atoms almost on straight lines. For light dark matter the detected chirality is thus set by the decay, which Appendix E.19 specifies. For the molecular magnet the literature picture, every molecule in $m = -|\boldsymbol{S}|$, gives the full line weight $2|\boldsymbol{S}|$. Thermal population of the upper well lowers the weight of the line at $\omega_{\rm meta}$ by stimulated emission and moves it to the neighboring lines. The line keeps 90 % of $2|\boldsymbol{S}|$ for $\omega_{\rm meta}/T \geq 3.0$, that is below 4.2 K for Mn_{12}-acetate at 1 T.

Limit Check 14.4. Chiral phonons and molecular magnets

(a) Probability of a detectable chiral phonon taken as the average of $|\boldsymbol{L}^{\rm ph}|$, Eq. (14.40), over the lowest band, applied to the one-phonon rate [28, Eqs. (6), (7), 29, Eqs. (3), (4), (11), (12)]. (b) Metastable line of a molecular magnet held in $m = -|\boldsymbol{S}|$, at $\omega_{\rm meta}$ of Eq. (14.44) with weight $2|\boldsymbol{S}|$ [31, Eqs. (3), (4)].

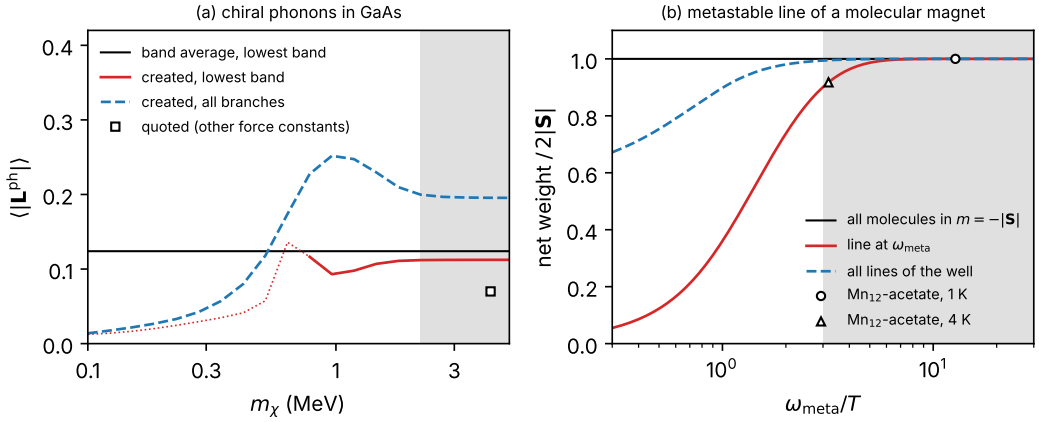


Figure 14.10: (a) GaAs, from the PBE force constants of Sect. 5.5. Black: the average of $|\boldsymbol{L}^{\rm ph}|$ over the lowest band, 0.124. Red: the same average weighted with the one-phonon rate of a heavy mediator coupled to nucleon number (Eq. (11.1), orientation-averaged halo kernel, $T = 0$), for the lowest-band phonons that the dark matter creates directly, against m_χ; dotted where the lowest band takes less than 1 % of the rate. Blue dashed: the same for all created phonons. Square: 0.07, the average quoted for GaAs from another dynamical matrix [29, Sect. III]. Gray: red within 10 % of black, $m_\chi \geq 2.2\,\mathrm{MeV}$. (b) Mn_{12}-acetate ($|\mathbf{S}| = 10$, $D = 0.454\,\mathrm{cm}^{-1}$, $B_4^0 = -2.0 \times 10^{-5}\,\mathrm{cm}^{-1}$, 1 T, $g_{\rm ion} = 2$, $\omega_{\rm meta} = 1.098\,\mathrm{meV}$): net absorptive weight over $2|\mathbf{S}|$ against $\omega_{\rm meta}/T$, for thermal populations inside the upper well. Red: the line at $\omega_{\rm meta}$; blue dashed: all lines of the well, $2|\langle S_z\rangle|/2|\mathbf{S}|$. Gray: the line within 10 % of $2|\mathbf{S}|$, $\omega_{\rm meta}/T \geq 3.0$. Markers: 1 K and 4 K.

Outside the gray regions the band average misstates the chirality of the phonons that a light particle creates, and a warm molecular crystal spreads the metastable weight over the lines of its well.

14.5.5 Magnetoelectric response of multiferroics

Hexagonal $Lu_{1-x}Sc_xFeO_3$ with $x \approx 0.4$ becomes ferroelectric at about 1200 K. Below its magnetic ordering temperature $T_{CM} \approx 160\,\mathrm{K}$ the canted spins of its iron ions add a net magnetization parallel to the polarization. For hexagonal $LuFeO_3$ ($x = 0$), density-functional calculations give a polarization of $10.5\,\mu\mathrm{C\,cm^{-2}}$ along the c axis and a net moment of $0.02\,\mu_B$ per iron [32, Sect. IV A]. A crystal with both orders breaks inversion and time reversal and keeps their product. Røising and collaborators propose that an axion field shifts the magnetization of such a multiferroic through its coupling to the electron, $\delta E = -\mathrm{i} g_{aee}(\partial_t\phi/2m_e^2)\langle\boldsymbol{\sigma}\cdot\nabla\rangle$ [32, Eq. (5)]. With $\boldsymbol{p} = -\mathrm{i}\nabla$ and $\boldsymbol{\sigma} = 2\boldsymbol{S}$, the spin current of Eq. (8.3) at $\boldsymbol{q} = 0$ is $\mathcal{J}_e^{\alpha\alpha}(0) = \sum_j \boldsymbol{p}_j\cdot\boldsymbol{S}_j/m_e = -\mathrm{i}\sum_j \boldsymbol{\sigma}_j\cdot\nabla_j/2m_e$, so the coupling is $(g_{aee}/m_e)(\partial_t\phi)\,\mathcal{J}_e^{\alpha\alpha}(0)$, the second term of Eq. (8.10) at $\boldsymbol{p}_\phi \to 0$. By the Kubo formula of Sect. 4.2, the spin density then responds as

$$\frac{\delta\langle s_e^z(0)\rangle(\omega)}{\mathscr{V}} = \frac{g_{aee}}{m_e}\,R_{s^z\mathcal{J}}(0,\omega)\,(\partial_t\phi)(\omega) = -\mathrm{i}\,\frac{g_{aee}m_\phi}{m_e}\,R_{s^z\mathcal{J}}(0,m_\phi)\,\phi(\omega), \tag{14.45}$$

with $R_{s^z\mathcal{J}}$ the retarded response of s_e^z to $\mathcal{J}_e^{\alpha\alpha}$ and the magnetization $-2\mu_B$ times the left-hand side. The two rows have opposite signs under both symmetries, $\sigma_s^{\mathsf{T}}\sigma_{\mathcal{J}}^{\mathsf{T}} = \sigma_s^{\mathsf{P}}\sigma_{\mathcal{J}}^{\mathsf{P}} = -1$, so by Eq. (14.34) inversion removes the block at $\boldsymbol{q} = 0$ and time reversal removes its symmetric part, the part that survives as $\omega \to 0$. The adiabatic response that Røising and collaborators consider thus needs both symmetries broken, as in $Lu_{1-x}Sc_xFeO_3$ below T_{CM}.

The Lehmann form of Sect. 4.2, $R_{ab} = \mathscr{V}^{-1}\sum_{i,n}(p_i - p_n)\langle i|O_a^\dagger|n\rangle\langle n|O_b|i\rangle/(\omega - \omega_{ni} + \mathrm{i}0^+)$, decides what the response is made of. Far below T_{CM} the states n reached at $\boldsymbol{q} = 0$ from the ordered state are its uniform magnon modes, in linear spin-wave theory as in Sect. 14.3. Each mode ν with both matrix elements nonzero contributes a pair of poles at $\pm\omega_\nu$, which combine for real matrix elements into a term proportional to $2\omega_\nu/(\omega^2 - \omega_\nu^2)$. Two conditions restrict the modes. If the Hamiltonian conserves the total spin component S_{tot}^z and $s_e^z(0) = \lambda_S S_{\mathrm{tot}}^z$, then $s_e^z(0)$ has no time dependence and $R_{s^z\mathcal{J}}(0,\omega)$ vanishes at every $\omega \neq 0$. The response needs a spin–orbit coupling, a Dzyaloshinskii–Moriya interaction or a single-ion anisotropy that does not commute with S_{tot}^z, such as the one that cants the spins of hexagonal $Lu_{1-x}Sc_xFeO_3$. And $\mathcal{J}_e^{\alpha\alpha}$ is odd under inversion, and it has matrix elements within the ground configuration of an ion only through a crystal field without inversion together with spin–orbit coupling (Sect. 14.3). The response is thus at least of second order in spin–orbit coupling: one order for the vertex, and one or two for the breaking of S_{tot}^z, by a Dzyaloshinskii–Moriya term or a single-ion term. One mode ω_{soft} with a damping γ_{soft} added by hand is the single damped oscillator of Ref. [32, Eq. (10)],

$$\Big|\frac{\delta M}{\theta_0}\Big| = \frac{|\chi_\theta\alpha_{\mathrm{me}}M_0/2\alpha_M - cP_0/2m_P^2|}{\left[(\omega_a^2 - \omega_{\mathrm{soft}}^2)^2 + \gamma_{\mathrm{soft}}^2\omega_a^2\right]^{1/2}}, \tag{14.46}$$

in which θ_0 is the amplitude of $\theta = \phi\, g_{a\gamma\gamma}$, the axion field times its photon coupling (their axion enters the free energy through a term $\chi_\theta\theta\boldsymbol{P}\cdot\boldsymbol{M}$, which does not separate the electron coupling of Eq. (14.45)), M_0 and P_0 are the static magnetization and polarization, α_{me} is the magnetoelectric coefficient of the term $\alpha_{\mathrm{me}}\boldsymbol{M}\cdot\boldsymbol{P}$, α_M the coefficient of $|\partial_t\boldsymbol{M}|^2$, m_P the stiffness of the polarization fluctuations, $\omega_a = m_\phi$ the frequency of the field, and $c = 1$. It is one mode of the Lehmann sum, written for the longitudinal magnetization. Its static limit, proportional to $\omega_{\mathrm{soft}}^{-2}$, is the isothermal susceptibility only if the magnon populations relax faster than the axion oscillates.

The temperature dependence is where the two descriptions part. In the Ginzburg–Landau free energy the quadratic coefficient is linear in $T - T_{CM}$. Below T_{CM} the longitudinal stiffness is then proportional to $T_{CM} - T$, and the static response grows as $(T_{CM} - T)^{-1}$. Their analysis adds a temperature-dependent magnetoelectric coefficient and a critical slowing down

$\tau \propto |T - T_{\mathrm{CM}}|^{-\nu z}$, with ν the correlation-length exponent and z the dynamic exponent [32, Eq. (9), Sect. III D]. That expansion holds near T_{CM}, where linear spin-wave theory does not. Limit Check 14.5 shows the two regimes in a model ferromagnet with the spin $\frac{5}{2}$ of Fe^{3+}. The Landau form agrees within 10 % with the Weiss mean field of the same magnet only for $T \geq 0.94\,T_{\mathrm{CM}}$, inside the critical region where mean field itself fails. At $T \leq 0.37\,T_{\mathrm{CM}}$, where thermal magnons reduce the moment by less than 10 %, the controlled result is that of spin waves over the anisotropy gap. The Landau form exceeds it by 1.4 at $0.3\,T_{\mathrm{CM}}$ and by 4.5 at $0.1\,T_{\mathrm{CM}}$ for a gap of $0.02\,T_{\mathrm{CM}}$. Near T_{CM} dynamic scaling predicts the structure $R_{s^z\mathcal{J}}(0,\omega;T) = \chi_\times(T)\,\Phi(\omega\tau(T))$, and Eq. (14.46) is the case of a single relaxing mode; the amplitude $\chi_\times$ and the scaling function Φ of a real material are not known. The exchange, Dzyaloshinskii–Moriya and anisotropy constants of the spin model for $T \ll T_{\mathrm{CM}}$ come from the DFT $+U$ energy mappings of Appendix E.20.4, with spin–orbit coupling included for the vertex of $\mathcal{J}_e^{\alpha\alpha}$.

Limit Check 14.5. Multiferroic cross response

Static magnetoelectric response of a multiferroic proportional to $\omega_{\mathrm{soft}}^{-2} \propto (T_{\mathrm{CM}} - T)^{-1}$, the adiabatic limit of the single damped oscillator Eq. (14.46) [32, Eqs. (5), (10)].

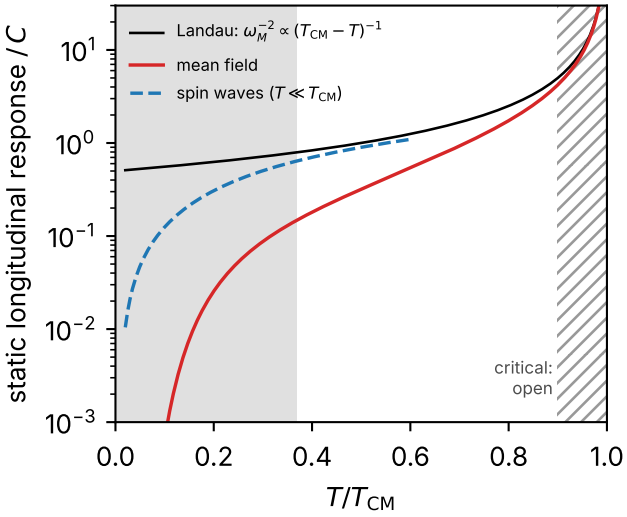


Figure 14.11: Static longitudinal response of a model ferromagnet with spin $|\mathbf{S}| = \frac{5}{2}$, in units of the Curie constant $C = |\mathbf{S}|(|\mathbf{S}|+1)/3$, against T/T_{CM}. Black: the Landau form $C/2(T_{\mathrm{CM}} - T)$, to which the static limit of Eq. (14.46) is proportional. Red: the Weiss mean field of the same magnet. Blue dashed: thermal spin waves on a simple-cubic lattice with nearest-neighbor exchange fixed by the mean-field relation $T_{\mathrm{CM}} = zJ|\mathbf{S}|(|\mathbf{S}|+1)/3$ with $z = 6$, so that the temperature axis is in mean-field units (the Curie point of this model lies below T_{CM}), and an easy-axis gap of $0.02\,T_{\mathrm{CM}}$, shown for $T \leq 0.6\,T_{\mathrm{CM}}$. Gray: thermal reduction of the moment below 10 %, $T \leq 0.37\,T_{\mathrm{CM}}$, where spin waves are the controlled result. Hatched: $T > 0.9\,T_{\mathrm{CM}}$, critical fluctuations, open. The Landau form is within 10 % of mean field only for $T \geq 0.94\,T_{\mathrm{CM}}$.

Away from T_{CM} the single soft mode of the Landau expansion does not exist, and the response is that of the uniform magnons and their thermal population; near T_{CM} it is not known.

14.6 Coherent fields and precessing nuclear spins

An axion-like field of mass 1 neV oscillates at 241.8 kHz, and its phase wanders on a time of about a second. A ^{207}Pb nucleus in a field of 26.8 mT precesses at the same frequency. Over a record shorter than a second the field acts on the nuclear spins as one classical wave with a fixed amplitude and phase, the spins tip together, and their transverse magnetization grows in proportion to the time until the spin coherence time T_2 caps it, after 17 ms in the ferroelectric PMN-PT. The amplitude of the field then changes only on the scale of a second. Whether the golden-rule rate describes such a record depends on the shorter of the two times, and how the sensitivity of a search grows with the length of the record depends on the statistics of the random amplitude.

14.6.1 The coherent classical-field regime

A light boson of mass m_ϕ has about $\rho_\chi/(m_\phi^4 v_0^3)$ quanta in a volume $(1/m_\phi v_0)^3$, 4.6×10^{39} at 1 neV, and below a few eV it is a classical field (Sect. 8.2.3). It is a sum of modes with random phases, one for each velocity in the halo, and by the central limit theorem a Gaussian random field [33, Eqs. (3)–(10)]. A Gaussian field with zero mean is fixed by its two-point function. For a free nonrelativistic field with the velocity distribution $f_\chi(\boldsymbol{v})$ of Sect. 1.2, each mode oscillates at $\omega_{\boldsymbol{v}} = m_\phi(1+v^2/2)$, and the average over the random phases is

$$\langle\phi(\boldsymbol{x},t)\,\phi(\boldsymbol{x}',t')\rangle = \frac{\rho_\chi}{m_\phi^2}\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\cos\big[\omega_{\boldsymbol{v}}(t-t') - m_\phi\boldsymbol{v}\cdot(\boldsymbol{x}-\boldsymbol{x}')\big],$$

normalized so that $\langle\phi^2\rangle = \rho_\chi/m_\phi^2$, the mean square of a wave $\phi_0\cos(m_\phi t)$ with $\frac12 m_\phi^2\phi_0^2 = \rho_\chi$. With the transforms of Appendix A.3 in space and time, the power spectrum of the field is

$$\mathcal{S}_\phi(\boldsymbol{k},\omega) = \frac{\pi\rho_\chi}{m_\phi^2}\,(2\pi)^3\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\Big[\delta^3(\boldsymbol{k}+m_\phi\boldsymbol{v})\,\delta(\omega-\omega_{\boldsymbol{v}}) + \delta^3(\boldsymbol{k}-m_\phi\boldsymbol{v})\,\delta(\omega+\omega_{\boldsymbol{v}})\Big]. \quad (14.47)$$

The positive-frequency weight sits at $\boldsymbol{k} = -m_\phi\boldsymbol{v}$ in the sign convention of Eqs. (A.6) and (A.7), while the target receives the momentum $\boldsymbol{p}_\phi = m_\phi\boldsymbol{v}$, as in Eq. (8.19); the sign matters only for targets whose response is odd in $\boldsymbol{q}$ (Sect. 14.5). Integrated over $\boldsymbol{k}$, the positive-frequency part is $(\pi\rho_\chi/m_\phi^2)F_\phi(\omega)$, with

$$F_\phi(\omega)\,\mathrm{d}\omega = F_\chi(v)\,\mathrm{d}v, \qquad \omega - m_\phi = \tfrac12 m_\phi v^2, \quad (14.48)$$

the energy distribution of the field of Sect. 11.8, normalized to one and inherited from the laboratory speed distribution F_χ of Sect. 1.2. The line is one-sided: it starts at m_ϕ and ends at $m_\phi(1+v_{\max}^2/2)$, 3.5×10^{-6} above it.

The width of the line sets the coherence time. We define it, as Sect. 8.2.3 estimated it, by

$$t_{\mathrm{coh}} = \frac{1}{m_\phi v_0^2} = 1.04\,\mathrm{s}\times\frac{1\,\mathrm{neV}}{m_\phi}, \quad (14.49)$$

with $v_0 = 238$ km/s. Some papers include a factor 2π and call $2\pi/(m_\phi v^2)$ the coherence time [34, 35], 6.6 s at 1 neV. For the halo of Sect. 1.2 seen from the Earth, F_ϕ peaks at $0.58/t_{\mathrm{coh}}$ above m_ϕ, its equivalent width $\Delta\omega_{\mathrm{eq}} = [\int \mathrm{d}\omega\, F_\phi^2]^{-1}$ is $2.67/t_{\mathrm{coh}}$, and the correlation of the complex amplitude of the field with itself, $|\int \mathrm{d}\omega\, F_\phi(\omega)\,\mathrm{e}^{-\mathrm{i}\omega t}|$, falls to one half at $t = 1.47\,t_{\mathrm{coh}}$. Over the coherence length $1/m_\phi v_0$, 249 km at 1 neV, the field is uniform, and every laboratory sample sees the same amplitude.

We need two things from linear response: the signal that a readout sees, and when its average turns into the golden-rule rate. A coupling $\sum_b g_b^\phi O_b$ of the absorption rows of

Sect. 8.1.5 drives the expectation value of any readout operator $O_{\rm rd}$ through the Kubo formula Eq. (4.3), with the retarded response $R_{{\rm rd},b}(\boldsymbol{p}_\phi, t)$. The readout can be the operator that the field couples to, as for the magnetization of the spins below, or another, as for the voltage across a piezoelectric resonator (Sect. 11.8). Either way the signal is linear in the amplitude of the field and in the coupling. Its square averaged over the Gaussian ensemble is quadratic in both and contains $\mathcal{S}_\phi$. The energy that the target absorbs is the case $O_{\rm rd} = \sum_b g_b^\phi O_b$, and its average reproduces the golden rule, but only after a long enough record. Take the coupling on from $t' = 0$ to t. The amplitude of a transition $i \to f$ driven by one mode of frequency ω contains $\int_0^t {\rm d}t'\, {\rm e}^{{\rm i}(E_f - E_i - \omega)t'}$, whose square is $2\pi t\, \mathcal{K}_t(\omega - E_f + E_i)$ with the Fejér kernel

$$\mathcal{K}_t(x) = \frac{t}{2\pi} \frac{\sin^2(xt/2)}{(xt/2)^2}, \qquad \int {\rm d}x\, \mathcal{K}_t(x) = 1. \tag{14.50}$$

The Gaussian average removes the cross terms between modes, as in Appendix D.25, and the mean number of quanta absorbed per unit volume in the record is $t\,(\rho_\chi/m_\phi)$ times the rate Eq. (8.19) averaged over the line F_ϕ, with the $\delta(\omega - E_f + E_i)$ in S_{ab} replaced by $\mathcal{K}_t$. As $t \to \infty$ the kernel becomes the delta function and the mean is the golden-rule rate times t. The replacement matters only where both the field line and the target line are narrow compared with $1/t$. Since the kernel acts on their convolution, whose width is at least the larger of the two widths, the golden-rule mean holds for

$$t \gg \min\!\left(t_{\rm coh},\, 1/\Gamma_{\rm tg}\right).$$

Here $\Gamma_{\rm tg}$ is the width of the target's line: $1/T_2$ for nuclear spins, the full width $\Gamma_{\rm M}$ of a magnon line (Sect. 14.3), γ_n for a resonator mode (Sect. 11.8). For a record shorter than both, the energy absorbed grows as t^2. That is the regime of a nuclear spin driven by a field of neV mass over a record shorter than its T_2, the coherent regime that Sect. 8.2.3 named.

The golden-rule mean is not yet what one record measures. A single realization of the field matches the ensemble only when the record spans many coherence times, so that it samples many independent amplitudes. For a detector that counts absorbed quanta, the intensity $|\Phi|^2$ of the complex amplitude Φ of the field is exponentially distributed within one coherence time, and a record of $N_{\rm coh}$ independent coherence times gives ${\rm Var}\, N = \bar{N} + \bar{N}^2/N_{\rm coh}$, the bunching of chaotic light. The Poisson statistics of Eq. (1.51) hold when $N_{\rm coh} \gg \bar{N}$, which counting searches satisfy by a wide margin: the YIG search of Worked example 14.3 at 10 μeV sees 35 events per hour in 3.4×10^7 coherence times. A readout of the mean field, such as the magnetization below, is a different matter, since its signal is proportional to the amplitude of the field rather than to its intensity.

14.6.2 Bloch equations and nuclear magnetic resonance

A nucleus with spin $\boldsymbol{J}^{\rm nuc}$ and magnetic moment $\mu_{\rm nuc}$ in a static field $\boldsymbol{B}_0 = B_0 \hat{\boldsymbol{z}}$ has the Zeeman energy $-\omega_{\rm Lar} J_z^{\rm nuc}$, with the Larmor frequency $\omega_{\rm Lar} = \mu_{\rm nuc} B_0 / J_{\rm nuc}$; for $\mu_{\rm nuc} > 0$ the lower levels have the spin along the field. A Hamiltonian $-\boldsymbol{b} \cdot \boldsymbol{J}^{\rm nuc}$, with $\boldsymbol{b}$ a vector of energy, makes the spin precess, ${\rm d}\boldsymbol{J}^{\rm nuc}/{\rm d}t = \boldsymbol{J}^{\rm nuc} \times \boldsymbol{b}$; the Zeeman term has $\boldsymbol{b} = \omega_{\rm Lar} \hat{\boldsymbol{z}}$. For ^{207}Pb, with $J_{\rm nuc} = \frac{1}{2}$ and $\mu_{\rm nuc} = 0.5926$ nuclear magnetons, $\omega_{\rm Lar}/2\pi = 9.03\,{\rm MHz}$ per tesla [36, Eq. (S11)].

A sample of $n_{\rm nuc}$ such nuclei per unit volume carries the magnetization $\boldsymbol{M}$, equal to $n_{\rm nuc}\mu_{\rm nuc}/J_{\rm nuc}$ times $\langle \boldsymbol{J}^{\rm nuc} \rangle$, and a spin $\frac{1}{2}$ in equilibrium has the polarization $p_{\rm pol} = 2\langle J_z^{\rm nuc} \rangle = \tanh(\omega_{\rm Lar}/2T)$ and $M_0 = n_{\rm nuc}\mu_{\rm nuc} p_{\rm pol}$. For ^{207}Pb at 4.4 T and 4.2 K, $p_{\rm pol} = 2.3 \times 10^{-4}$, and with $n_{\rm nuc} = 3.4 \times 10^{27}\,{\rm m}^{-3}$ in PMN-PT, a relaxor ferroelectric of lead magnesium niobate and lead titanate, $\mu_0 M_0 = 2.9\,{\rm nT}$, the equilibrium value of Aybas and collaborators [36, Eqs. (S12), (S13)]; the magnetization they measured after preparing the sample was 1.9 nT [36]. Searches

therefore polarize the spins in a large field, or optically, and then work at a lower field while the polarization relaxes; the sample is out of equilibrium, with a polarization fixed by its preparation.

The averaged spins obey the same precession, with two relaxation times added phenomenologically by Bloch [37]:

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{M} \times \boldsymbol{b}(t) - \frac{M_x \hat{\boldsymbol{x}} + M_y \hat{\boldsymbol{y}}}{T_2} - \frac{M_z - M_0}{T_1}\, \hat{\boldsymbol{z}}, \tag{14.51}$$

with $\boldsymbol{b}(t) = \omega_{\mathrm{Lar}} \hat{\boldsymbol{z}} + \boldsymbol{b}_\phi(t)$ including any drive. The longitudinal time T_1 is set by the exchange of energy with the lattice and reaches hours in insulators at cryogenic temperatures [34]. The transverse time T_2 is set by the loss of phase between spins, mainly through their magnetic dipole interactions, and is of order a millisecond in a solid; pulse sequences that average the dipolar coupling extend it by orders of magnitude. For ^{207}Pb in PMN-PT, $T_2 = 16.7\,\mathrm{ms}$. Spins at inequivalent sites, or with different chemical shifts, precess at different frequencies, and this inhomogeneous broadening spreads the resonance of the whole sample over a width Γ_{inh}, $2\pi \times 78\,\mathrm{kHz}$ in PMN-PT [36, Eqs. (S24), (S25)]. In the response formalism, the spins form an absorber with a Lorentzian line. With the transverse decay e^{-t/T_2} inserted in Eq. (4.3), the operator J_-^{nuc}, which raises the Zeeman energy by ω_{Lar}, has the retarded response

$$R_{J_- J_-}(\omega) = \frac{2\langle J^{\mathrm{nuc}}_{z,\mathrm{tot}} \rangle / \mathscr{V}}{\omega - \omega_{\mathrm{Lar}} + \mathrm{i}/T_2},$$

from $\langle [J_+, J_-] \rangle = 2\langle J_z \rangle$ for the total spin of the sample. Its imaginary part is negative, an absorber in the sense of Sect. 4.2, only if $\langle J_z^{\mathrm{nuc}} \rangle > 0$, that is, if the lower Zeeman levels are the more populated.

14.6.3 The nuclear-spin precession signal

Two couplings of an axion-like field reach the nuclear spins. The first is the gradient coupling of Sect. 11.6.2: each nucleus of isotope i feels $(g_{ai}/m_N)(\nabla\phi) \cdot \boldsymbol{J}^{\mathrm{nuc}}$, the nuclear spin row of Eq. (8.3) with the coefficient $g^\phi_{s_i,a} = \mathrm{i} g_{ai} p_{\phi,a}/m_N$. Sect. 11.6.2 found that this term does not reach the phonons, since its final states sit at the spin splittings; an applied field moves those splittings up to m_ϕ. For a wave $\phi_0 \cos(m_\phi t - \boldsymbol{p}_\phi \cdot \boldsymbol{x})$ the coupling is $-\boldsymbol{b}_\phi \cdot \boldsymbol{J}^{\mathrm{nuc}}$ with

$$\boldsymbol{b}_\phi(t) = -\frac{g_{ai}}{m_N} \sqrt{2\rho_\chi}\, \boldsymbol{v}\, \sin(m_\phi t),$$

at the sample, a pseudo-magnetic field along the velocity of the dark matter. Aybas and collaborators write the same coupling with $g_{aNN} = g_{ai}/m_N$ [36, Eqs. (S49)–(S53)]. The second coupling is the oscillating electric dipole moment that an axion induces in nucleons, $d_n = g_d \phi$. In a polar crystal the electrons screen the electric field at the nucleus, and what survives, through the Schiff moment (the part of the nuclear electric-dipole distribution that the screening leaves), is an energy shift $\pm\Delta\varepsilon(t)$ of the two levels of a spin $\frac{1}{2}$ quantized along an effective field $\boldsymbol{E}^*$ fixed by the crystal [34]. That is again a spin coupling, with $\boldsymbol{b}_\phi = 2\Delta\varepsilon(t)\hat{\boldsymbol{E}}^*$. For ^{207}Pb in PMN-PT, $\Delta\varepsilon = d_n E^*$ with $E^* = 340\,\mathrm{kV/cm}$, uncertain by about 50 % [36, Eqs. (S44), (S45)]. The scalar couplings of Sect. 8.1.5 act on orbital coordinates and leave the nuclear spins alone, as for the electron spins of Sect. 14.3.

Only the component of $\boldsymbol{b}_\phi$ transverse to $\boldsymbol{B}_0$ tips the spins. A linear drive $b_\perp \cos(m_\phi t)\hat{\boldsymbol{x}}$ is the sum of two circular components of amplitude $b_\perp/2$, and near resonance only the one that rotates with the precession acts. In the frame that rotates at m_ϕ, and dropping the counter-rotating component, the spins see a static field of strength

$$\Omega_{\mathrm{Rabi}} = \tfrac{1}{2} b_\perp,$$

with the detuning $\delta\omega_{\rm Lar} = \omega_{\rm Lar} - m_\phi$. For the gradient coupling $\Omega_{\rm Rabi} = g_{ai}\sqrt{2\rho_\chi}\,v_\perp/2m_N$, and for the dipole coupling of a spin $\frac{1}{2}$ with $\boldsymbol{E}^* \perp \boldsymbol{B}_0$, $\Omega_{\rm Rabi} = \Delta\varepsilon$, the shift of each level. Write $M_+ = M_x + \mathrm{i}M_y$. In Eq. (14.51) the precession about $\boldsymbol{B}_0$ gives $\mathrm{d}M_+/\mathrm{d}t = -\mathrm{i}\omega_{\rm Lar}M_+$, and the drive adds $\mathrm{i}b_\perp\cos(m_\phi t)M_z$. As long as the spins are barely tipped, $M_z = M_0$. For $\tilde{M}_+ = \mathrm{e}^{\mathrm{i}m_\phi t}M_+$, keeping the co-rotating half of the drive,

$$\frac{\mathrm{d}\tilde{M}_+}{\mathrm{d}t} = -\Big(\mathrm{i}\delta\omega_{\rm Lar} + \frac{1}{T_2}\Big)\tilde{M}_+ + \mathrm{i}\,\Omega_{\rm Rabi}M_0,$$

a linear equation with a constant source. With the spins along the field at $t = 0$, $\tilde{M}_+(0) = 0$, its solution is the magnetization that a search records.

Transverse magnetization of a driven nuclear-spin ensemble

$$\begin{aligned}
\tilde{M}_+(t) &= \mathrm{i}\,\Omega_{\rm Rabi}M_0\,\frac{1-\mathrm{e}^{-(\mathrm{i}\delta\omega_{\rm Lar}+1/T_2)t}}{\mathrm{i}\delta\omega_{\rm Lar}+1/T_2},\\
|M_\perp| &= \Omega_{\rm Rabi}M_0\,\frac{2|\sin(\delta\omega_{\rm Lar}t/2)|}{|\delta\omega_{\rm Lar}|} \quad (t \ll T_2),\\
|M_\perp| &= \frac{\Omega_{\rm Rabi}M_0T_2}{(1+\delta\omega_{\rm Lar}^2T_2^2)^{1/2}} \quad (t \gg T_2),
\end{aligned} \tag{14.52}$$

for a field of fixed amplitude and phase, with $\Omega_{\rm Rabi}t \ll 1$ and $\Omega_{\rm Rabi}^2T_1T_2 \ll 1$. The magnetization precesses at m_ϕ in the laboratory, about $\boldsymbol{B}_0$.

On resonance the magnetization grows linearly, $|M_\perp| = \Omega_{\rm Rabi}M_0t$, until T_2 caps it at $\Omega_{\rm Rabi}M_0T_2$. This is the transient of [34, Eq. (1)], whose envelope $\sin(\delta\omega_{\rm Lar}t)/\delta\omega_{\rm Lar}$ is half as wide as the solution of the Bloch equation, and the steady state $M_\perp = u_{\rm sp}M_0\Omega_{\rm Rabi}T_2$ of [36, Eq. (1)]. The spectral factor $u_{\rm sp}$ averages the Lorentzian of Eq. (14.52) over the inhomogeneous line, $u_{\rm sp} \simeq \pi/(\Gamma_{\rm inh}T_2) = 3.8\times10^{-4}$ for ^{207}Pb in PMN-PT [36, Eqs. (S22)–(S25)]: only the spins within about $1/T_2$ of m_ϕ respond, a fraction $u_{\rm sp}$ of the sample. The steady state is also the Kubo response of the magnetization, $\langle J_+\rangle$ driven through $R_{J_-J_-}$ of Sect. 14.6.2. The imaginary part of the same response, inserted in Eq. (8.19), is the rate at which the spins absorb quanta.

The drive is the Gaussian field of Eq. (14.47), whose complex amplitude in the rotating frame wanders on the scale $t_{\rm coh}$. Let $\Omega_{\rm rms}$ be the Rabi frequency of a fixed wave with the mean energy density, $\phi_0 = \sqrt{2\rho_\chi}/m_\phi$. For the gradient coupling, whose direction is random as well, we replace $v_\perp^2$ by its average $\langle v_\perp^2\rangle$; this is an approximation, since $v_\perp^2$ is correlated with the speed and hence with ω, and the exact form weights the line by the mean of $v_\perp^2$ at each speed. The equation for $\tilde{M}_+$ is linear, so each mode of the field contributes its own solution of the form Eq. (14.52), at its own detuning, and the Gaussian average adds their squares:

$$\big\langle|M_\perp(t)|^2\big\rangle = \Omega_{\rm rms}^2M_0^2\int\mathrm{d}\omega\,F_\phi(\omega)\,\frac{\big|1-\mathrm{e}^{-[\mathrm{i}(\omega_{\rm Lar}-\omega)+1/T_2]t}\big|^2}{(\omega_{\rm Lar}-\omega)^2+1/T_2^2}. \tag{14.53}$$

For $t \ll t_{\rm coh}$ all modes are in phase over the record, and the magnetization has the form Eq. (14.52) with a random amplitude: $|M_\perp|$ is Rayleigh distributed, and its square is exponential with the mean $\Omega_{\rm rms}^2M_0^2t^2$ on resonance. For $t_{\rm coh} \ll t \ll T_2$ the phase of the drive wanders and the magnetization random-walks, $\langle|M_\perp|^2\rangle = 2\pi F_\phi(\omega_{\rm Lar})\,\Omega_{\rm rms}^2M_0^2\,t$. For $t \gg T_2$ it saturates, at $\pi F_\phi(\omega_{\rm Lar})\,\Omega_{\rm rms}^2M_0^2\,T_2$ if $T_2 \gg t_{\rm coh}$, the balance between the golden-rule rate of Eq. (8.19) and the decay $1/T_2$. If instead $T_2 \ll t_{\rm coh}$, the spin line is broad compared with the field line, and the fixed-amplitude steady state with a Rayleigh amplitude holds at every time. Limit Check 14.6 shows both cases.

Limit Check 14.6. Transverse magnetization of a driven nuclear-spin ensemble

$|M_\perp| = \Omega_{\rm Rabi} M_0 t$ for $t \ll T_2$ and $\Omega_{\rm Rabi} M_0 T_2$ for $t \gg T_2$, on resonance, for a drive of fixed amplitude [34, Eq. (1), 36, Eq. (1)].

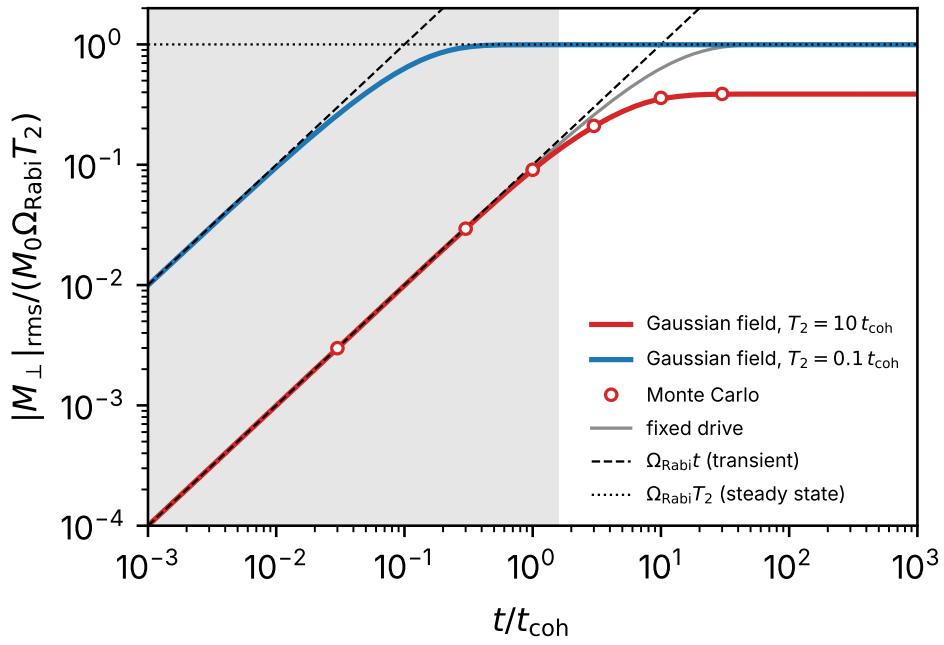


Figure 14.12: Root-mean-square transverse magnetization of a nuclear-spin ensemble driven on resonance by an axion-like field, in units of $\Omega_{\rm Rabi} M_0 T_2$, against the time over the coherence time $t_{\rm coh} = 1/m_\phi v_0^2$. Gray: the linear solution Eq. (14.52) for a drive of fixed amplitude, $1 - \mathrm{e}^{-t/T_2}$, with its two literature limits, the transient $\Omega_{\rm Rabi}t$ (dashed) and the steady state $\Omega_{\rm Rabi}T_2$ (dotted). Colored: the same equation driven by the Gaussian field of the standard halo of Sect. 1.2 seen from the Earth, Eq. (14.53), with the Larmor frequency at the peak of the field line and $\Omega_{\rm Rabi}$ the value for the mean energy density; $T_2 = 10\,t_{\rm coh}$ (red) and $0.1\,t_{\rm coh}$ (blue). Circles: average over 1500 Monte Carlo realizations of the field for $T_2 = 10\,t_{\rm coh}$, each a sum of 2000 plane waves with random phases. Gray band: $t \le 1.6\,t_{\rm coh}$, where the field-averaged magnetization is within 10 % of the fixed-drive formula for $T_2 = 10\,t_{\rm coh}$. For $T_2 = 0.1\,t_{\rm coh}$ the two agree within 0.6 % at all times.

For $T_2 \gg t_{\rm coh}$ and $t \gtrsim t_{\rm coh}$ the magnetization stops growing linearly and saturates at 0.39 of the fixed-drive value for $T_2 = 10\,t_{\rm coh}$; for longer T_2 the ratio falls as $(T_2/t_{\rm coh})^{-1/2}$.

The linear solution assumes that the drive does not deplete M_z. At resonance and for $t \ll T_2$ the exact solution of Eq. (14.51) is $M_\perp = M_0 \sin(\Omega_{\rm Rabi} t)$, below the linear one by the fraction $(\Omega_{\rm Rabi}t)^2/6$. In the steady state the z component of Eq. (14.51) in the rotating frame gives $M_z = M_0/(1 + \Omega_{\rm Rabi}^2 T_1 T_2)$, and since T_1 is long, saturation sets in at drives with $\Omega_{\rm Rabi}T_2$ still small. Aybas and collaborators correct their limit by 30 % for it at their detection threshold, $\Omega_{\rm Rabi} = 0.23\,\mathrm{rad/s}$ [36, Sect. III]. The signal competes with noise from the spins themselves and from the magnetometer. The spin noise follows from the fluctuation–dissipation theorem Eq. (4.18) at the spin temperature $T_{\rm s}$ of a prepolarized sample, $p_{\rm pol} = \tanh(\omega_{\rm Lar}/2T_{\rm s})$. Since $\mathrm{Im}\, R_{J_- J_-} \propto p_{\rm pol}$ while the symmetrized thermal factor is $\coth(\omega_{\rm Lar}/2T_{\rm s}) = 1/p_{\rm pol}$, the spin noise does not depend on the polarization [34, Eq. (A1)], and prepolarization raises the

signal only.

The material inputs are the density and moment of the spin-carrying isotope, the polarization a preparation reaches, T_1 and T_2, the inhomogeneous line and, for the dipole coupling, E^*; Appendix E.26 specifies their calculation for candidate targets.

14.6.4 The exponential-bin likelihood

A search records a readout for a time t_{int}, the magnetometer signal of the precessing spins or the voltage of a resonator. It is demodulated at a carrier near m_ϕ into a complex signal $X(t)$ that holds both quadratures, and frequencies are measured from the carrier. The signal part is linear in the complex amplitude Φ of the field, $X_{\mathrm{sig}}(t) = \int \mathrm{d}t'\, \mathcal{G}_{\mathrm{rd}}(t-t')\, \Phi(t')$ with a response kernel $\mathcal{G}_{\mathrm{rd}}$, and is thus Gaussian as well, with the power spectrum $\mathcal{S}_{\mathrm{sig}}(\omega) = |\mathcal{G}_{\mathrm{rd}}(\omega)|^2 \mathcal{S}_\phi(\omega)$ up to the normalization of Φ. For the spins, $|\mathcal{G}_{\mathrm{rd}}|^2$ is the Lorentzian of Eq. (14.52); for a target much broader than the field line it is a constant, and $\mathcal{S}_{\mathrm{sig}}$ has the shape F_ϕ. The noise, from the spins, the magnetometer or the amplifier, is Gaussian with the spectrum $\mathcal{S}_{\mathrm{n}}(\omega)$. The data are reduced to the periodogram at the frequencies $\omega_k = 2\pi k/t_{\mathrm{int}}$,

$$\mathsf{P}_k = \frac{1}{t_{\mathrm{int}}} \Big| \int_0^{t_{\mathrm{int}}} \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega_k t} X(t) \Big|^2 .$$

Its mean is the spectrum convolved with the Fejér kernel Eq. (14.50), $\langle \mathsf{P}_k \rangle = \int \mathrm{d}\omega\, \mathcal{K}_{t_{\mathrm{int}}}(\omega_k - \omega)[\mathcal{S}_{\mathrm{sig}}(\omega) + \mathcal{S}_{\mathrm{n}}(\omega)]$, which tends to $\mathcal{S}_{\mathrm{sig}}(\omega_k) + \mathcal{S}_{\mathrm{n}}(\omega_k)$ for a long record. The Fourier integral of a Gaussian process is a complex Gaussian variable, with independent real and imaginary parts of equal variance, and the sum of signal and noise is one such variable. Its squared modulus is exponentially distributed:

$$p(\mathsf{P}_k) = \frac{1}{\bar{\mathsf{P}}_k}\, \mathrm{e}^{-\mathsf{P}_k/\bar{\mathsf{P}}_k}, \qquad \bar{\mathsf{P}}_k = \mathsf{s}_k + \mathsf{b}_k, \tag{14.54}$$

with s_k and b_k the mean signal and noise powers of bin k. This is the distribution of Foster, Rodd and Safdi, with the mean $\mathsf{s}_k \propto F_\chi(v)/v \propto F_\phi(\omega_k)$ for a long record [33, Eq. (24)]. Signal and noise combine in the amplitudes, not in the powers; the sum of two exponential variables would not be exponential. For a record shorter than t_{coh} the whole line falls into one bin, and Eq. (14.54) states that the power in that bin, the square of the Rayleigh amplitude of the field, fluctuates by 100 % from one realization to the next. For a longer record the line spreads over about $t_{\mathrm{int}}\Delta\omega_{\mathrm{eq}}/2\pi$ independent bins, each with its own exponential fluctuation. Fig. 14.13 shows single realizations, and Limit Check 14.7 the approach of the mean to its long-record form.

Limit Check 14.7. Gaussian-field signal and its power spectrum

Exponential periodogram bins with mean $\mathsf{s}(\omega) \propto F_\phi(\omega)$, the line shape of Eq. (14.48), for a field built from Rayleigh-distributed amplitudes with random phases [33, Eqs. (8), (10), (24)]; for an absorber, the golden-rule mean of Eq. (8.19) averaged over the line.

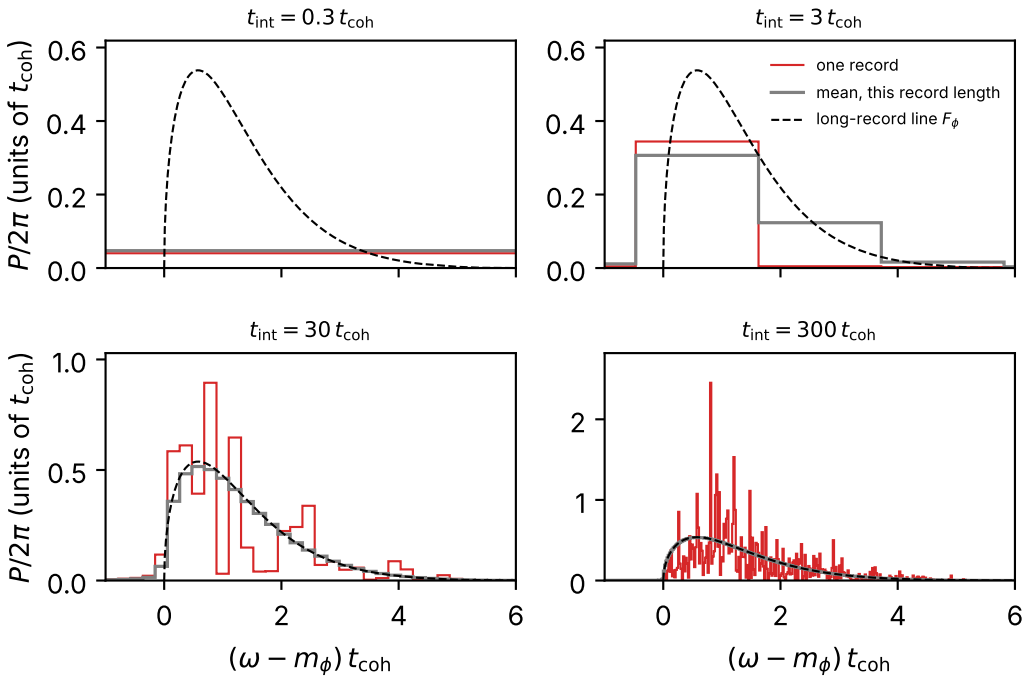


Figure 14.13: Periodogram of an axion-like field of unit mean power for records of $t_{\rm int} = 0.3$, 3, 30 and $300\,t_{\rm coh}$, against the energy above the rest energy in units of $1/t_{\rm coh} = m_\phi v_0^2$. Red: one Monte Carlo realization of the field, a sum of 3000 plane waves with random phases and with velocities from the standard halo of Sect. 1.2 seen from the Earth ($v_0 = 238\,{\rm km/s}$, $v_{\rm esc} = 544\,{\rm km/s}$, $v_E = 250.5\,{\rm km/s}$). Bins of width $2\pi/t_{\rm int}$, one centered on the peak of the line. Gray: the mean periodogram for that record length, the line shape convolved with the Fejér kernel Eq. (14.50). Dashed: the line shape F_ϕ of Eq. (14.48), the mean for a long record. Below $t_{\rm coh}$ one bin holds the whole line, with a height proportional to $t_{\rm int}$. Above it the bins resolve the line and scatter about it with the exponential distribution Eq. (14.54).

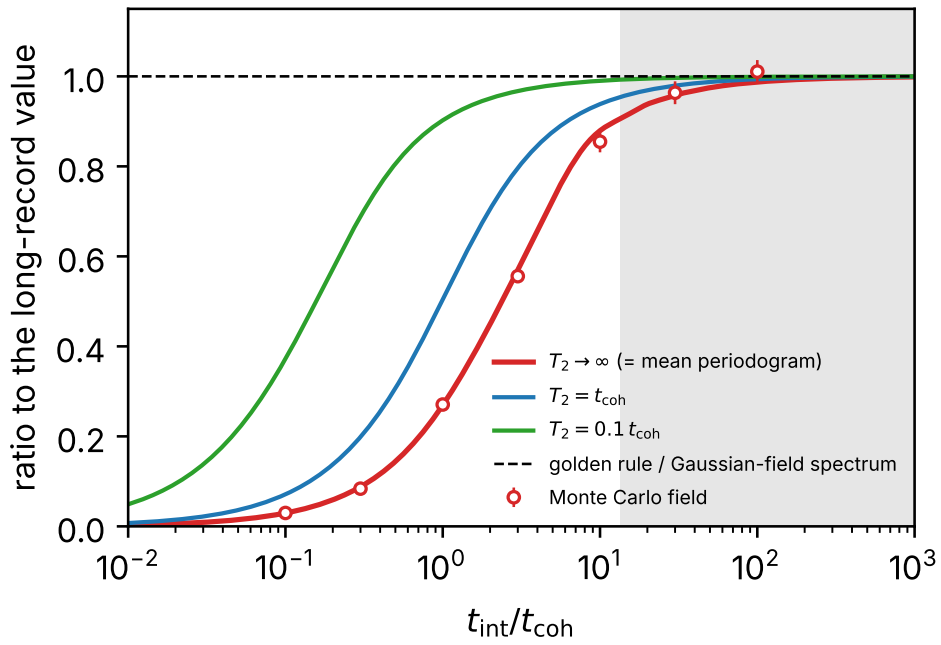


Figure 14.14: Mean over the Gaussian field, for a record of length $t_{\rm int}$, divided by its long-record value, against $t_{\rm int}/t_{\rm coh}$. Red: the mean periodogram at the peak of the line, $\int \mathrm{d}\omega\, F_\phi(\omega)\mathcal{K}_{t_{\rm int}}(\omega_{\rm pk}-\omega)/F_\phi(\omega_{\rm pk})$, which is also the energy absorbed by spins with $T_2 \to \infty$ at the peak frequency divided by the golden-rule value; circles: 1500 Monte Carlo realizations, with the scatter of their mean. Blue and green: energy absorbed by linearly driven spins with $T_2 = t_{\rm coh}$ and $0.1\,t_{\rm coh}$, divided by the golden-rule rate times $t_{\rm int}$. Standard halo of Sect. 1.2. Gray band: the red curve within 10 % of the long-record value, $t_{\rm int} \geq 13\,t_{\rm coh}$; for $T_2 = t_{\rm coh}$ and $0.1\,t_{\rm coh}$ the same accuracy needs $6.5\,t_{\rm coh}$ and $t_{\rm coh}$. The variance of the Monte Carlo bins equals the square of their mean within the sampling error, as the exponential law requires.

Below $t_{\rm coh}$ and T_2 the mean grows as $t_{\rm int}^2$ rather than $t_{\rm int}$, with the ratio $t_{\rm int}/2\pi F_\phi(\omega_{\rm pk})$, and above them it approaches the golden rule as $1 - 1.25\,t_{\rm coh}/t_{\rm int}$, the leakage of the Fejér kernel.

The bins of a stationary process are independent as long as the leakage of the Fejér kernel between neighbors is negligible. The likelihood of the periodogram is then the product of the distributions Eq. (14.54),

$$-2\ln \mathsf{L} = 2\sum_k \Big[\ln \bar{\mathsf{P}}_k + \frac{\mathsf{P}_k}{\bar{\mathsf{P}}_k}\Big], \tag{14.55}$$

the exponential-bin likelihood of Foster, Rodd and Safdi [33, Eq. (29)], in which the signal enters through s_k and the coupling through its square, $\mathsf{s}_k \propto g^2$. Limits and discoveries follow from the likelihood ratio t of Eq. (C.120) and its thresholds Eq. (C.121) (Appendix C.14). How the sensitivity grows with $t_{\rm int}$ follows from the expected ratio for data without signal, $\mathsf{P}_k = \mathsf{b}_k$. Let $x_k = \mathsf{s}_k/\mathsf{b}_k$. Then $\mathsf{t} = 2\sum_k[\ln(1+x_k) - x_k/(1+x_k)]$, and for a weak signal, expanding to second order,

$$\mathsf{t} \simeq \sum_k x_k^2 = \frac{t_{\rm int}}{2\pi}\int \mathrm{d}\omega\, \frac{\mathcal{S}_{\rm sig}(\omega)^2}{\mathcal{S}_{\rm n}(\omega)^2},$$

where the sum over bins of width $2\pi/t_{\rm int}$ became an integral. Let $\sigma_{\rm sig}^2$ be the mean square of the signal, proportional to g^2, and take white noise. For a record shorter than $t_{\rm coh}$ one bin holds the signal, $\mathsf{s} = \sigma_{\rm sig}^2 t_{\rm int}$ and $\mathsf{b} = \mathcal{S}_{\rm n}$, so a fixed threshold on t means a fixed x, and

$$\sigma_{\rm sig} \propto \mathcal{S}_{\rm n}^{1/2}\, t_{\rm int}^{-1/2} \quad (t_{\rm int} \ll t_{\rm coh}), \qquad \sigma_{\rm sig} \propto \mathcal{S}_{\rm n}^{1/2}\, (t_{\rm int} t_{\rm coh})^{-1/4} \quad (t_{\rm int} \gg t_{\rm coh}). \tag{14.56}$$

For a long record the integral gives $\mathsf{t} = 2\pi t_{\rm int}\sigma_{\rm sig}^4/(\Delta\omega_{\rm eq}\mathcal{S}_{\rm n}^2)$ with $\Delta\omega_{\rm eq} = 2.67/t_{\rm coh}$, which is the second law. Inside one coherence time the signal adds in amplitude and the noise in power, the familiar $t^{-1/2}$ of a coherent sinusoid. Beyond it the signal power stops growing in each bin, and only the number of bins over which the line is averaged grows, as $t_{\rm int}$, which halves the exponent. These are the scalings of Budker and collaborators [34, Eqs. (A7), (A8)]; the long-record form, with the coefficient of the weak-signal ratio, is the expected limit of Foster, Rodd and Safdi [33, Eqs. (54), (63), (69)]. For the spins, $\sigma_{\rm sig} = |M_\perp|$ is proportional to g, and the coupling at the threshold scales as $\sigma_{\rm sig}$.

The amplitude of the field within one coherence time is a random draw, and an analysis that ignores this sets too strong a limit. A search shorter than $t_{\rm coh}$ that takes the amplitude fixed at its mean value $\sqrt{2\rho_\chi}/m_\phi$ excludes every coupling whose signal would have exceeded the noise in that one realization; however, the field may have been weak during the record, with probability $1 - \mathrm{e}^{-a}$ for an intensity below a times its mean. Centers and collaborators found that limits with a fixed amplitude are too strong by a factor of 3 in a Bayesian treatment and 2.7 in a frequentist one, up to 10 with other priors, and 8.4 for gradient couplings [35, Eq. (7)]. The exponential-bin likelihood already contains the random amplitude, so the correction is made once: with the likelihood Eq. (14.55), or with a fixed-amplitude likelihood followed by the correction factor, never both.

The size of the correction can be checked with an exact Neyman construction. Replace the line of the standard halo, convolved with the Fejér kernel, by $N_{\rm eff}$ equal bins, with $N_{\rm eff} = 1/\sum_k p_k^2$ the participation number of the bin fractions p_k ($N_{\rm eff} \to 1$ below $t_{\rm coh}$ and $t_{\rm int}\Delta\omega_{\rm eq}/2\pi$ above). The bins have the width $2\pi/t_{\rm int}$, with one bin centered on the peak of the line. The statistic is the summed periodogram Q of those bins, with noise b per bin and total signal $\mathsf{s} = \sigma_{\rm sig}^2 t_{\rm int}$ at every record length. With the Rayleigh amplitude of Eq. (14.54), Q follows a Gamma distribution of shape $N_{\rm eff}$ and scale $\mathsf{b} + \mathsf{s}/N_{\rm eff}$; with a fixed amplitude and random phase, $2\mathsf{Q}/\mathsf{b}$ follows a noncentral χ^2 distribution with $2N_{\rm eff}$ degrees of freedom and noncentrality $2\mathsf{s}/\mathsf{b}$. The 95 % upper limit $\mathsf{s}_{\rm up}$ solves $P(\mathsf{Q} \le \mathsf{Q}_{\rm obs}\,|\,\mathsf{s}_{\rm up}) = 0.05$, with $\mathsf{Q}_{\rm obs}$ the median of the Gamma distribution of shape $N_{\rm eff}$ and scale b for data without signal. For the median of data without signal and one bin, $\mathsf{s}_{\rm up}/\mathsf{b} = 12.5$ with the Rayleigh amplitude and 3.17 with a fixed one, so that the fixed amplitude sets a coupling limit lower by $(12.5/3.17)^{1/2} = 1.99$. For data at the detection threshold, $\mathsf{Q}_{\rm obs} = \mathsf{b}\ln 20$, the same construction gives 57.4 and 7.72 and a factor 2.73, which reproduces the frequentist 2.7 of Centers and collaborators: it is the same random amplitude, counted with a different statistic. Limit Check 14.8 shows both treatments against the record length.

Limit Check 14.8. Sensitivity scalings of the exponential-bin likelihood

Threshold amplitude $\propto \mathcal{S}_{\rm n}^{1/2} t_{\rm int}^{-1/2}$ for $t_{\rm int} < t_{\rm coh}$ and $\propto \mathcal{S}_{\rm n}^{1/2}(t_{\rm int}t_{\rm coh})^{-1/4}$ for $t_{\rm int} > t_{\rm coh}$ [34, Eqs. (A7), (A8), 33, Eqs. (54), (63), (69)].

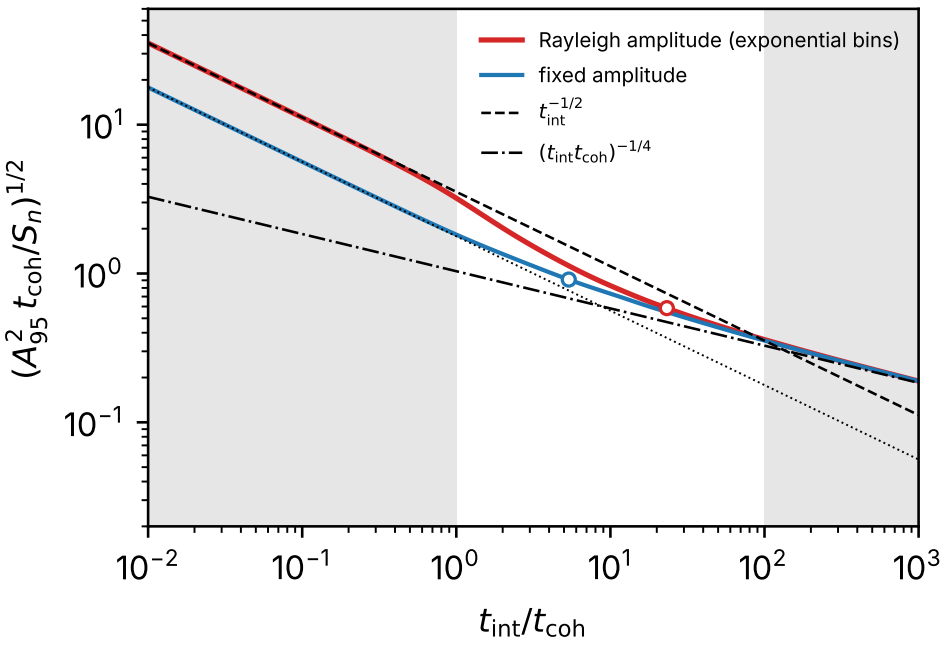


Figure 14.15: Signal amplitude at the median expected 95 % upper limit, $(\sigma_{\rm sig}^2 t_{\rm coh}/\mathcal{S}_{\rm n})^{1/2}$, proportional to the coupling, against the record length over the coherence time, for white noise of spectrum $\mathcal{S}_{\rm n}$. Red: the signal as a Gaussian variable with a Rayleigh amplitude, the exponential-bin likelihood Eq. (14.55). Blue: the signal power in each bin taken as fixed, with a random phase. Both curves follow from the Neyman construction of the text, with $N_{\rm eff}$ equal bins for the line of the standard halo of Sect. 1.2. Dashed: $t_{\rm int}^{-1/2}$ for the Rayleigh amplitude (dotted: fixed amplitude); dash-dotted: $(t_{\rm int}t_{\rm coh})^{-1/4}$, common to both. Circles: the crossover, where the local slope is $-3/8$, at $23\,t_{\rm coh}$ (Rayleigh) and $5.4\,t_{\rm coh}$ (fixed). Gray: the Rayleigh curve within 10 % of the $t_{\rm int}^{-1/2}$ law ($t_{\rm int} \le t_{\rm coh}$) or of the $(t_{\rm int}t_{\rm coh})^{-1/4}$ law ($t_{\rm int} \ge 100\,t_{\rm coh}$).

Neither power law holds between $t_{\rm coh}$ and $100\,t_{\rm coh}$, where the Rayleigh curve falls as steeply as $t_{\rm int}^{-0.67}$, faster than either law, because the record begins to average over independent amplitudes.

The fixed amplitude sets a limit lower by 1.99 for records below $t_{\rm coh}$, by 1.76 at $t_{\rm coh}$ and by 1.5 % at $100\,t_{\rm coh}$. The two power laws, extended, meet at $136\,t_{\rm coh}$ for the Rayleigh amplitude and at $8.7\,t_{\rm coh}$ for the fixed one, so the crossover lies between a few and about a hundred coherence times and its coefficient is not universal.

Worked example 14.6: The scaling crossover for a 1 neV axion-like field

An axion-like field of $m_\phi = 1\,\text{neV}$ oscillates at $m_\phi/2\pi = 241.8\,\text{kHz}$. By Eq. (14.49), $t_{\rm coh} = 1.044\,\text{s}$, and its line has the equivalent width $\Delta\omega_{\rm eq}/2\pi = 2.67/(2\pi t_{\rm coh}) = 0.41\,\text{Hz}$, 1.7×10^{-6} of the frequency. Over the coherence length, $1/m_\phi v_0 = 249\,\text{km}$, the field is uniform.

A ^{207}Pb sample resonates at this frequency in $B_0 = 241.8\,\text{kHz}/(9.034\,\text{MHz}\,\text{T}^{-1}) = 26.8\,\text{mT}$. Its $T_2 = 16.7\,\text{ms}$ is $0.016\,t_{\rm coh}$, so the spin line, of half width $1/T_2 = 60\,\text{s}^{-1}$, is 63 times wider than the field line, $m_\phi v_0^2 = 0.96\,\text{s}^{-1}$. The spins are in the case $T_2 \ll t_{\rm coh}$ of Limit

Check 14.6: the magnetization reaches the steady state of Eq. (14.52) after a few T_2, about 0.05 s, and its amplitude then follows the Rayleigh amplitude of the field, which changes on the scale of a second.
The sensitivity depends on how long a record at one frequency can be. With the construction of Limit Check 14.8, the threshold coupling falls by a factor 3.46 from $t_{\rm int} = 0.1\,{\rm s}$ to 1 s, more than the $\sqrt{10} = 3.16$ of the coherent law, since the record starts to sample a second amplitude. The local slope reaches $-3/8$ at $23\,t_{\rm coh} = 24\,{\rm s}$. A record of one hour spans 3447 coherence times and a day 82 730. Going from an hour to a day improves the limit by 2.24, close to $24^{1/4} = 2.21$ and far from the $24^{1/2} = 4.9$ that a coherent signal would give. At 0.1 s the fixed-amplitude analysis would claim a limit 1.99 times lower; at one hour the two agree to better than 0.1 %.

A search that scans the field over a decade of frequency in a fixed total time spends less time at each frequency as the line narrows toward low masses. The crossover then decides whether to dwell longer at one frequency or to scan more. At $m_\phi = 1\,\mu{\rm eV}$, $t_{\rm coh} = 1.04\,{\rm ms}$, and every practical record lies deep in the $(t_{\rm int} t_{\rm coh})^{-1/4}$ regime; at 1 peV, $t_{\rm coh} = 17\,{\rm min}$, and the random amplitude must be in the likelihood.

Summary

A dark-matter particle that couples to the electron spin creates one magnon through the transverse spin, at leading order in $1/|\boldsymbol{S}|$ (Sect. 14.1). In a two-sublattice antiferromagnet the one-magnon response carries the weight $w^{\rm M}_{\boldsymbol{q}} = |\boldsymbol{S}|(A_{\boldsymbol{q}} - B_{\boldsymbol{q}})/\omega^{\rm M}_{\boldsymbol{q}}$, which suppresses magnons near the zone center and enhances them near the ordering wavevector: 0.024 against 42 for magnons of 5 meV in NiO. Inserted in the master rate, the response gives the one-magnon rate of Trickle, Zhang and Zurek. For NiO at $\bar\sigma_e = 10^{-40}\,{\rm cm}^2$ it opens near 7.3 keV for a 5 meV threshold and peaks at 3.2×10^3 events per kilogram-year near 3 MeV. Near its edge the rate depends on the magnetic anisotropy, which the Heisenberg model leaves out. The spin-length sum rule estimates that linear spin-wave theory overstates the one-magnon weight by $\Delta S/|\boldsymbol{S}|$ at the next order, 7.8 % for NiO, where the zero-point reduction $\Delta S = 0.078$ is that of four interleaved simple-cubic antiferromagnets.

The longitudinal spin creates magnon pairs (Sect. 14.2). Their weight vanishes in a ferromagnet and at $\boldsymbol{q} = 0$, grows as q^2, and averages to 0.073 of the one-magnon weight in NiO, the part of the $\Delta S/|\boldsymbol{S}| = 0.078$ moved out of the one-magnon line that arrives in the continuum (the two differ by ΔS^2). Because a pair shares its momentum, it reaches the energy $\frac{1}{2} m_\chi v^2$ at any mass, and the two-magnon channel opens at masses lower by $v^2_{\rm max}/4c_{\rm M}(v_{\rm max} - c_{\rm M})$, a factor 5.1 for NiO. Final-state interactions move the pair weight within the continuum without changing its integral; whether NiO has a bound pair is open. A magnon mixes with a phonon at linear order only through the dependence of the anisotropy energy on strain, which spin–orbit coupling and the dipolar interaction provide; the strain dependence of the exchange couples a phonon only to magnon pairs. The hybrids carry an off-diagonal spin–nuclear response that saturates the interference bound of Sect. 8.3.

A boson absorbed at $\boldsymbol{k} \to 0$ drives the uniform precession (Sect. 14.3). When the exchange is invariant under spin rotations, the uniform transverse spin response is one line at the Larmor frequency, with weight $2\langle S^z_{\rm tot}\rangle$, at any spin and temperature. The spin couples through λ_S, the fraction of the ionic angular momentum carried by the spin, which equals $g_J - 1$ only within a J multiplet and is one for quenched $3d$ ions. The shape of the sample moves the line to the Kittel frequency and reweights its polarizations. For YIG at 10 µeV and $g_{aee} = 10^{-13}$, with an assumed line width of 10^{-5} of the frequency, the peak rate is $3.6\,{\rm s}^{-1}\,{\rm kg}^{-1}$. An antiferromagnet

absorbs at its resonance with the exchange-suppressed weight $|\boldsymbol{S}|\omega_{\mathrm{AFMR}}/(2\omega_E+\omega_A)$ per pair of sublattices, about $0.1\,|\boldsymbol{S}|$ in MnF_2, whose resonance lies at the measured 1.1 meV.

Measured neutron spectra can replace a computed spin response, within limits (Sect. 14.4). The magnetic cross-section at a given momentum transfer contains only the part of the spin response transverse to it, so without a model of the spin density no neutron measurement, polarized or not, gives the longitudinal element $\hat{\boldsymbol{q}}\cdot S\cdot\hat{\boldsymbol{q}}$. Within the localized-moment model, transverse data at several wavevectors $\boldsymbol{q}+\boldsymbol{G}$ fix the full tensor, and an isotropy hypothesis recovers it from two scattering angles. For quenched ions the neutron sees the moment with the spectroscopic g, the dark matter the spin, and the factor $[\lambda_S/(g_{\mathrm{ion}}/2)]^2$ converts one into the other, 0.83 for Ni^{2+} at $g_{\mathrm{ion}}=2.2$ and 0.73 at 2.34.

Targets without inversion or time-reversal symmetry add responses that a collinear magnet lacks (Sect. 14.5). The current response of a chiral crystal acquires a part odd in $\boldsymbol{q}$, the finite-momentum optical activity, and the point group decides whether it exists; in α-quartz the orientation-averaged gyration is only -0.045 of its value along the optic axis. Phonons of the lowest band of GaAs carry on average 0.12 units of angular momentum. Single-molecule magnets offer metastable lines, such as the line of Mn_{12}-acetate at 1.098 meV in 1 T, which keeps 90 % of its weight below 4.2 K. A multiferroic below its ordering temperature has a magnetoelectric cross response; near that temperature the response is open.

An ultralight field is a classical Gaussian wave, coherent for $t_{\mathrm{coh}}=1/m_\phi v_0^2$, 1.04 s at 1 neV (Sect. 14.6). Its average reproduces the golden-rule rate for records longer than the shorter of t_{coh} and the inverse width of the target's line, while a single record matches the ensemble only beyond t_{coh}. A polarized nuclear-spin ensemble driven by the gradient or the dipole coupling of an axion-like field obeys the Bloch equations. Its transverse magnetization grows as $\Omega_{\mathrm{Rabi}}M_0 t$ and saturates at $\Omega_{\mathrm{Rabi}}M_0T_2$, and a Gaussian drive turns the linear growth into a random walk once the record exceeds t_{coh}. Each periodogram bin of signal plus noise is exponentially distributed. The resulting likelihood gives sensitivities that scale as $t_{\mathrm{int}}^{-1/2}$ below the coherence time and as $(t_{\mathrm{int}}t_{\mathrm{coh}})^{-1/4}$ above it, and it already contains the random amplitude, which a fixed-amplitude analysis misses by a factor 1.99 for the median limit and 2.73 at the detection threshold.

Table 14.3 collects the truncations and inputs of the chapter with their sizes, and Table 14.4 the inputs of its targets. For magnon scattering in NiO two rows of nearly equal size act in opposite directions: the one-magnon line is high by about 7.8 %, and the two-magnon continuum adds 7.3 % at other energies, so that the total spin weight is right to half a percent while its distribution in energy is not. Near the lightest masses the anisotropy row dominates every other by more than an order of magnitude, and it is a model, not a calculation. For absorption into the uniform mode, the shape of the sample and the width of the line set the rate, and neither depends on the spin-wave expansion. In the coherent regime the largest effects are statistical: the length of the record against t_{coh} and T_2, and whether the likelihood contains the random amplitude.

Table 14.3: Nuisances of the magnetic sector and of the coherent regime. Type in the classes of Sect. 8.8: normalization, tilt (in ω or in the dark-matter mass), threshold and new support. Basis of the size: "computed" (next order or exact result computed), "model" (computed in a model), "est." (parametric estimate), "input spread" (spread or limitation of a measured or computed input), "unsized" (no size known). Sizes are for NiO, YIG, MnF_2, GaAs, Mn_{12}-acetate and ^{207}Pb in PMN-PT as named. The last column gives the section where the item is treated.

Truncation or input	Type	Size	Basis	Section
One-magnon weight of linear spin-wave theory	normalization	high by about $\Delta S/\lvert\boldsymbol{S}\rvert$ at the next order: 7.8 % (NiO), 3.1 % for $\lvert\boldsymbol{S}\rvert = \frac{5}{2}$ on the same lattice, 16 % for $\lvert\boldsymbol{S}\rvert = \frac{1}{2}$	est.	Sect. 14.1
Two-magnon continuum	new support	0.073 of the one-magnon weight averaged over the zone (NiO); $w_\parallel/w_\perp < 0.05$ below 0.5 keV, peak 0.082 near 1.4 keV	computed	Sect. 14.2
Final-state interaction of the pair; bound pairs	tilt; new support	about 0.04 of the spin response, moved in energy at fixed integrated weight (NiO); a bound pair below the continuum not known	est.; unsized	Sect. 14.2
Magnetic anisotropy near the gap	threshold; tilt in m_χ	NiO at a 5 meV threshold, zone-center gap 4.54 meV in a one-constant model: rate ×29 at 10 keV (an upper value, since the in-plane branch lies lower), ×1.35 at 30 keV, < 3 % above 100 keV	model	Sect. 14.1
Exchange constants from DFT $+U$ instead of neutron data	threshold	computed J_2 of NiO 11 to 27 meV against 19.01: magnon energies ×0.58 to 1.42, weights unchanged	input spread	Sect. 14.1
Magnon–phonon mixing	tilt	hybridization gap 3.0 % of the crossing energy at 2.83 GHz (YIG model, 0.1 T); NiO without measured magnetoelastic constants	model; unsized	Sect. 14.2
Thermal magnons	new support	$e^{-\omega_{gap}/T}$, exponent -5×10^3 at 10 mK for the 4.54 meV gap of NiO	computed	Sects. 14.1 and 14.2
Sample shape and demagnetizing fields	threshold; normalization	YIG film: line and weight within 10 % of the Larmor line for $B_0 \geq 4.8\,\mu_0 M_s$; weight ×1.74 at 10 µeV; saturated sphere exact	computed	Sect. 14.3
Zero-point correction to the antiferromagnetic-resonance weight	normalization	5 to 23 % for $\lvert\boldsymbol{S}\rvert = \frac{3}{2}$ at $\omega_A/\omega_E = 2$ to 0.3 (eight-site cluster); MnF_2 ($\lvert\boldsymbol{S}\rvert = \frac{5}{2}$, $\omega_A/\omega_E = 0.017$) not sized	model; unsized	Sect. 14.3
Antiferromagnetic resonance of MnF_2	threshold; normalization	measured 1.1 meV and spin flop 9.4 T; fitted fields 7 % lower in both; weight 8 % higher with ω_A fitted to the gap; peak rate unchanged	input spread	Sect. 14.3
Width Γ_M of the uniform magnon line	normalization	peak rate $\propto 1/\Gamma_M$; $10^{-5}\omega$ for YIG is an assumed value, not a millikelvin measurement	unsized	Sect. 14.3
Spin fraction λ_S of quenched $3d$ ions	normalization	$1 - O(\xi_{ion}^2/\Delta_{CF}^2)$; the rule $\lambda_S = g_{ion} - 1$ would give ×1.44 at $g_{ion} = 2.2$	est.	Sect. 14.3
In-medium mixing of an absorbed dark photon	normalization	ζ_{eff} set by the dielectric constant and the enclosure	unsized	Sect. 14.3
Conversion of neutron data for quenched ions	normalization	$[\lambda_S/(g_{ion}/2)]^2 = 0.83$ for Ni^{2+} at $g_{ion} = 2.2$, 0.73 at 2.34 ($L/S = 0.34$), 12 %; q dependence 0.973 at 5.3 keV	computed; input spread	Sect. 14.4
Longitudinal spin response from neutron data	new support	not measurable at one $\boldsymbol{q}$ without a model of the spin density; fixed by transverse data at several $\boldsymbol{q} + \boldsymbol{G}$ in the localized-moment model, or by an isotropy hypothesis	unsized	Sect. 14.4
Chiral-phonon detection probability as a band average	normalization	GaAs: 0.124 (this book) against 0.07 (Matas et al.; a single Cartesian component gives 0.062, a possible origin); rate-weighted value within 10 % of the band average only for $m_\chi \geq 2.2$ MeV; lowest band below 1 % of the rate under 0.8 MeV	computed; input spread	Sect. 14.5
Chiral frameworks: three-phonon branching with angular momentum, giant-moment amplitudes	normalization	no size	unsized	Sect. 14.5
Thermal population of the upper well of a molecular magnet	normalization	Mn_{12}-acetate at 1 T: line keeps 90 % of $2\lvert\boldsymbol{S}\rvert$ for $\omega_{meta}/T \geq 3.0$, $T \leq 4.2$ K	computed	Sect. 14.5
Multiferroic cross response near T_{CM}	normalization; tilt	derived for $T \ll T_{CM}$ only; no size near T_{CM}	unsized	Sect. 14.5
Record length against t_{coh} and $1/\Gamma$	normalization	mean within 10 % of the golden rule for $t_{rec} \geq 13\,t_{coh}$ ($T_2 \to \infty$), $6.5\,t_{coh}$	computed	Sect. 14.6

Table 14.4: Inputs of the magnetic targets of Chap. 14. NiO, YIG and MnF_2 enter through linear spin-wave theory or the Larmor theorem; rare-earth ions through their J multiplets; ^{207}Pb in the ferroelectric PMN-PT through the Bloch equations. Values and sources as in the sections listed in the last column; the corrections beyond each model are in Table 14.3.

Target	Moments	Model constants	Modes and widths	Section
NiO	Ni^{2+}, $\lvert\boldsymbol{S}\rvert = 1$, orbital moment quenched, $\lambda_S = 1$; $\Omega_{\text{ion}} = 18.13\,\text{Å}^3$, $\rho_T = 6.84\,\text{g}\,\text{cm}^{-3}$	$J_1 = -1.37$, $J_2 = 19.01\,\text{meV}$ (neutron fit); type-II order, $a = 4.17\,\text{Å}$	zone-center magnon 4.54 meV (far infrared); band top 118 meV; $c_{\text{M}} = 1.29$ to 1.36×10^{-4}	Sects. 14.1 and 14.2
YIG	ferrimagnet, net spin $\mathfrak{s}_{\text{net}} = 10$ per cell of $947.6\,\text{Å}^3$, $\lambda_S = 1$	$\mu_0 M_{\text{s}} = 0.246\,\text{T}$; spin stiffness $5.985\,\text{meV}\,\text{nm}^2$; $B_2 = 6.96 \times 10^6\,\text{erg}\,\text{cm}^{-3}$	uniform mode at 115.8 μeV per tesla (sphere); $\Gamma_{\text{M}} = 10^{-5}\omega$ assumed	Sects. 14.2 and 14.3
MnF_2	Mn^{2+}, $\lvert\boldsymbol{S}\rvert = \frac{5}{2}$, $\lambda_S = 1$	$B_E = 47.05\,\text{T}$, $B_A = 0.82\,\text{T}$ (fit)	antiferromagnetic resonance 1.1 meV and spin flop 9.4 T (measured); 1.021 meV and 8.75 T (fitted fields)	Sect. 14.3
Rare-earth ions	J multiplets, $\lambda_S = g_J - 1$ from $-5/7$ (Sm^{3+}) to 1 (Gd^{3+})	Hund's rules	crystal-field levels within the multiplet	Sect. 14.3
^{207}Pb in PMN-PT	$J_{\text{nuc}} = \frac{1}{2}$, $\mu_{\text{nuc}} = 0.5926\,\mu_N$; $n_{\text{nuc}} = 3.4 \times 10^{27}\,\text{m}^{-3}$	$\omega_{\text{Lar}}/2\pi = 9.03\,\text{MHz}$ per tesla; $E^* = 340\,\text{kV/cm}$	$T_2 = 16.7\,\text{ms}$; $\Gamma_{\text{inh}}/2\pi = 78\,\text{kHz}$; $p_{\text{pol}} = 2.3 \times 10^{-4}$ at 4.4 T, 4.2 K	Sect. 14.6

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Chapter 15

Superconductors

The gap of a superconductor is more than three thousand times smaller than that of silicon. In aluminum $\Delta_{\rm BCS} = 0.18\,\text{meV}$, so breaking a pair costs 0.36 meV, and a dark-matter particle of 0.10 keV can do it (Worked example 7.3). Below a third of an MeV no halo particle can cross the gap of silicon (Sect. 1.3), yet a superconductor still responds to a coupling to the electron density, and the BCS theory gives that response in closed form. Whether the rate of breaking pairs can be computed as reliably as the semiconductor rates of Chap. 10 is the question of this chapter.

The response of independent quasiparticles, however, violates charge conservation, and the correction that restores it is largest at the small momentum transfers that light dark matter favors (Sect. 7.2). At the edge $\omega = 2\Delta_{\rm BCS}$ the coherence factors make the absorptive part jump, and the screening that follows from that jump cannot be expanded. And the sensors are thin films, often amorphous and strongly disordered, whereas the BCS model describes a clean bulk crystal. As a result the loss functions in use for aluminum differ by up to a factor of three near the edge, although far above it they agree to a few percent (Fig. 15.1).

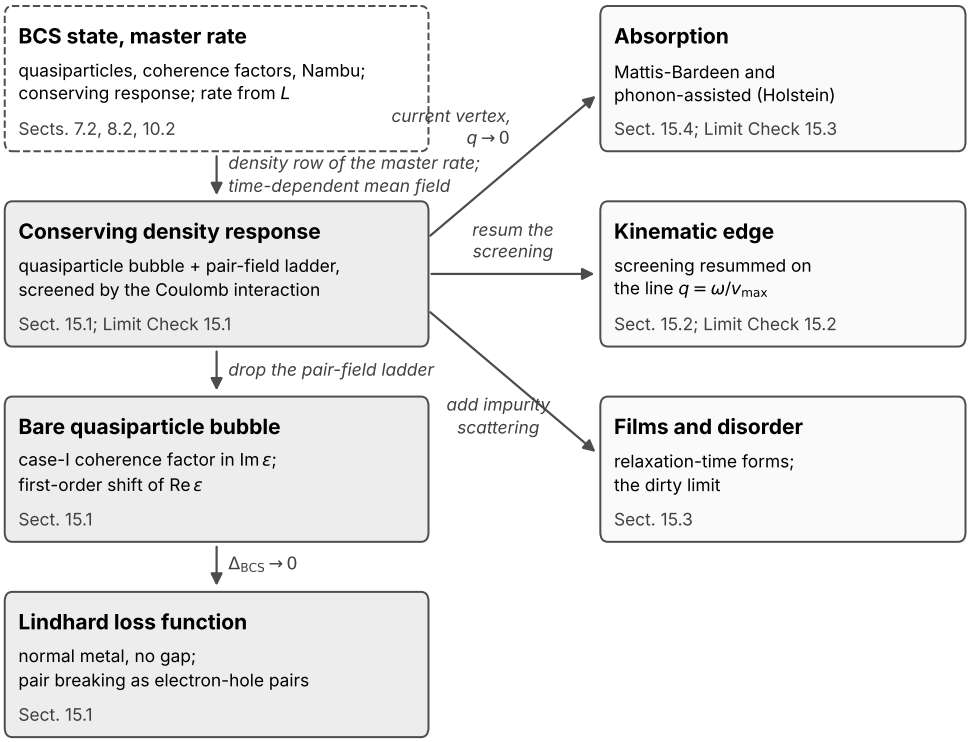


Figure 15.1: Superconductors, from the BCS state of Sect. 7.2 and the loss-function rate of Sects. 8.2 and 10.2 to the targets of this chapter. The density response that conserves charge adds the pair-field ladder to the quasiparticle bubble and is screened by the Coulomb interaction (Sect. 15.1; Limit Check 15.1). Dropping the ladder leaves the bare quasiparticle bubble, whose absorptive part carries the case-I coherence factor; letting the gap close leaves the Lindhard function of the normal metal (Sect. 15.1). On the kinematic edge $q = \omega/v_{\max}$ the screening is resummed (Sect. 15.2; Limit Check 15.2). Impurity scattering and the film geometry change the response of real sensors (Sect. 15.3). At vanishing momentum the current and the spin current give the absorption of dark photons and axion-like particles, with and without a phonon (Sect. 15.4; Limit Check 15.3).

15.1 Pair breaking by dark matter

A dark-matter particle of 1 keV that scatters in aluminum deposits at most 3.5 meV, about ten times the pair-breaking threshold, and transfers at most 5.3 eV of momentum. To deposit ω it must transfer at least $q = \omega/v_{\max}$, with $v_{\max} = 2.650 \times 10^{-3}$ the largest speed of a halo particle in the laboratory (Eq. (1.9)). The line $q = \omega/v_{\max}$ in the (q, ω) plane, the *kinematic edge*, is a different object from the *pair-breaking edge* $\omega = 2\Delta_{\mathrm{BCS}}$, the threshold in energy. On the kinematic edge the ratio of the gap to the Doppler width of the Fermi surface, $\Delta_{\mathrm{BCS}}/qv_{\mathrm{F}} = (v_{\max}/v_{\mathrm{F}})(\Delta_{\mathrm{BCS}}/\omega)$, takes its largest value, 0.20 at the threshold, and this is the ratio that Sect. 7.2 found to control the correction that restores charge conservation. Three forms of the loss function are in use at these momenta: the Lindhard function of the normal metal, the same function multiplied by the case-I coherence factor of Eq. (7.30), and the bare quasiparticle bubble screened in the random-phase approximation. We will build the loss function from the conserving response of Eq. (7.34), reduce it step by step to each of the three, and see where each step fails.

15.1.1 The loss-function rate of a superconductor

A mediator that couples to the electron number acts through the density row of the master formula, with $\mathfrak{D}_{ee} = |\tilde{V}(\boldsymbol{q})|^2$ and $S = (2q^2/e^2)\, L/(1 - \mathrm{e}^{-\beta\omega})$ (Sect. 10.2). At 10 mK the threshold has $\beta\omega = 418$, so the thermal factor is one, and the spectrum is Eq. (10.4) with $\bar{L}$ replaced by the loss function of the superconductor, which is isotropic in the model used here. For a dark photon, which couples to the total charge, this is exact. For a mediator that couples to electrons alone it holds with the nuclei clamped, and the lattice then enters only through the pairing interaction already contained in Δ_{BCS}. The ions' own response appears in the loss function through the phonon pole at $\omega = c_{\mathrm{LA}}q$, and above that pole it is a correction of relative order $(c_{\mathrm{LA}}q/\omega)^2$. The longitudinal sound speed along [100] is $c_{\mathrm{LA}} = (C_{11}/\rho)^{1/2} = 6.47\,\mathrm{km\,s^{-1}}$, from the elastic constant $C_{11} = 114.3\,\mathrm{GPa}$ extrapolated to zero temperature by Kamm and Alers, as quoted by Malica and Dal Corso [1, Table 2], and the low-temperature density $\rho = 2.73\,\mathrm{g\,cm^{-3}}$ (the room-temperature $\rho_T = 2.70\,\mathrm{g\,cm^{-3}}$ enters the rates). The correction is then $(c_{\mathrm{LA}}/v_{\max})^2 = 6.6 \times 10^{-5}$ on the kinematic edge. For $m_\chi < 3.2\,\mathrm{keV}$ even the largest momentum transfer, $2m_\chi v_{\max}$, excites phonons only below $2\Delta_{\mathrm{BCS}}$.

The dielectric function is the proper density response screened by the Coulomb interaction, $\varepsilon = 1 - v_{\mathrm{C}}(q)\tilde{R}$ (Sect. 4.4). In the normal metal the random-phase approximation puts the independent-particle bubble in place of $\tilde{R}$, which gives the Lindhard function of Eq. (4.59). In the superconductor, however, the proper response must itself conserve charge, and Sect. 7.2 showed that the bare quasiparticle bubble does not. The response that does is the time-dependent mean field in the three Nambu channels $(\boldsymbol{\eta}, \mathsf{s}_1, \mathsf{s}_2)$ of Eq. (7.32), the pair-field ladder of Eq. (7.34) [2]. In field-theory language the ladder is the vertex correction that restores the Ward identity of charge. With the Coulomb interaction added in the density channel, this is the generalized random-phase approximation:

Conserving loss function of a superconductor

$$\varepsilon^{\mathrm{cons}}(q,\omega) = 1 - v_{\mathrm{C}}(q)\, R^{\mathrm{cons}}(q,\omega), \qquad L^{\mathrm{cons}}(q,\omega) = \mathrm{Im}\Big[-\frac{1}{\varepsilon^{\mathrm{cons}}(q,\omega)}\Big], \tag{15.1}$$

with $R^{\mathrm{cons}} = R_{\boldsymbol{\eta}\boldsymbol{\eta}}$ the density response of Eq. (7.34), which contains the pair-field ladder and no Coulomb interaction.

The Coulomb interaction is a function of the local density and commutes with it, so the

screened response obeys the continuity equation and the f-sum rule whenever $R^{\rm cons}$ does (Sect. 7.2). The core electrons of aluminum add a constant of order one to ε, negligible against $k_{\rm TF}^2/q^2 \geq 5.8\times10^5$ for $q \leq 5.3\,\text{eV}$. Here $k_{\rm TF} = 4.05\,\text{keV}$ belongs to the free-electron gas of aluminum that the whole book uses: three electrons per atom in the fcc cell of side $a = 4.05\,\text{Å}$, so $n_e = 1.81\times10^{23}\,\text{cm}^{-3}$, $k_{\rm F} = 1.75\,\text{Å}^{-1}$, the bare electron mass, $v_{\rm F} = k_{\rm F}/m_e = 6.75\times10^{-3}$ and $\omega_p = 15.78\,\text{eV}$. The amplitude channel s_1 decouples from the density in a metal whose density of states is symmetric about the Fermi level (Sect. 7.2); the asymmetry over the energy range of the pairs is of relative order $\Delta_{\rm BCS}/E_{\rm F} = 1.5\times10^{-5}$, and we drop it.

15.1.2 The conserving response in closed form

For aluminum we set the temperature to zero, take the density of states $\mathcal{N}_{\rm F}$ constant over the energies of the pairs, and take the metal clean. We also linearize the band across the shell, $\xi_{\boldsymbol{k}+\boldsymbol{q}} = \xi_{\boldsymbol{k}} + v_{\rm F} q\mu$ with μ the cosine of the angle between $\boldsymbol{k}$ and $\boldsymbol{q}$. The dropped term $q^2/2m_e$ is below $3\times10^{-5}\,\text{eV}$ for $q \leq 5.3\,\text{eV}$, whereas $v_{\rm F} q \geq 2.5\,\omega$. With these simplifications every bubble of Eq. (7.34) reduces to one function. At $T = 0$ the density and pair operators either create a pair of quasiparticles, with energy $E + E'$ for $E = E_{\boldsymbol{k}}$ and $E' = E_{\boldsymbol{k}+\boldsymbol{q}}$, or destroy one, and the bubbles are

$$R^0_{AB}(q,\omega) = \frac{1}{\mathcal{V}}\sum_{\boldsymbol{k}}\Big[\frac{M_A^{\rm N*}M_B^{\rm N}}{\omega - E - E' + \mathrm{i}0^+} - \frac{M_A^{\rm N}M_B^{\rm N*}}{\omega + E + E' + \mathrm{i}0^+}\Big], \tag{15.2}$$

with $M_A^{\rm N}$ the amplitude with which the Nambu-channel operator O_A (superscript N for Nambu) creates the pair. For the density it is $M_{\boldsymbol{\eta}}^{\rm N} = uv' + vu'$ of Eq. (7.28). The phase operator is $O_{\mathsf{s}_2} = -\mathrm{i}(B_{\boldsymbol{q}}^\dagger - B_{-\boldsymbol{q}})$, with $B_{\boldsymbol{q}}^\dagger$ the pair creation operator of Appendix D.21. It therefore creates the same pair through $B_{\boldsymbol{q}}^\dagger$, with the amplitude $M_{\mathsf{s}_2}^{\rm N} = -\mathrm{i}(uu' + vv')$; this phase reproduces $R^0_{\mathsf{s}_2\boldsymbol{\eta}}$ of Sect. 7.2 at $q = 0$, and the sign of the ladder below does not depend on it. With Eq. (7.23),

$$|M_{\boldsymbol{\eta}}^{\rm N}|^2 = \frac{EE' - \xi\xi' + \Delta_{\rm BCS}^2}{2EE'}, \qquad |M_{\mathsf{s}_2}^{\rm N}|^2 = \frac{EE' + \xi\xi' + \Delta_{\rm BCS}^2}{2EE'},$$
$$M_{\boldsymbol{\eta}}^{\rm N*}M_{\mathsf{s}_2}^{\rm N} = -\frac{\mathrm{i}\Delta_{\rm BCS}(E+E')}{2EE'},$$

where the mixed product follows from $(uv' + vu')(uu' + vv') = u'v'(u^2+v^2) + uv(u'^2 + v'^2) = \Delta_{\rm BCS}/2E' + \Delta_{\rm BCS}/2E$.

The linear band turns these into functions of $E + E'$. Since $E^2 = \xi^2 + \Delta_{\rm BCS}^2$, the product $(EE' + \xi\xi' + \Delta_{\rm BCS}^2)(EE' - \xi\xi' - \Delta_{\rm BCS}^2) = E^2E'^2 - (\xi\xi' + \Delta_{\rm BCS}^2)^2$ equals $\Delta_{\rm BCS}^2(\xi' - \xi)^2$, and $EE' + \xi\xi' + \Delta_{\rm BCS}^2$ is half of $(E+E')^2 - (\xi' - \xi)^2$. For the density amplitude we write $EE' - \xi\xi' + \Delta_{\rm BCS}^2 = (EE' - \xi\xi' - \Delta_{\rm BCS}^2) + 2\Delta_{\rm BCS}^2$ and use the product for the first bracket. With $\xi' - \xi = v_{\rm F} q\mu$,

$$EE' + \xi\xi' + \Delta_{\rm BCS}^2 = \tfrac{1}{2}\big[(E+E')^2 - v_{\rm F}^2 q^2\mu^2\big], \qquad |M_{\boldsymbol{\eta}}^{\rm N}|^2 = \frac{\Delta_{\rm BCS}^2(E+E')^2}{EE'\big[(E+E')^2 - v_{\rm F}^2q^2\mu^2\big]}. \tag{15.3}$$

The quantity $\omega_\mu^2 = \omega^2 - v_{\rm F}^2q^2\mu^2$ is the energy of the pair in the frame that moves with the Doppler shift, and a pair at angle μ can be created only if $\omega_\mu \geq 2\Delta_{\rm BCS}$, the counterpart of the threshold Eq. (7.27). What is left is the $q = 0$ pair sum $F(\omega)$ of Sect. 7.2, now at each angle. A hyperbolic substitution evaluates it (Appendix D.39):

$$F_\mu(\omega) = \frac{\mathcal{N}_{\rm F}}{2}\int \mathrm{d}\xi\,\frac{E+E'}{EE'\,[\omega^2 - (E+E')^2]} = -\frac{\mathcal{N}_{\rm F}}{2\Delta_{\rm BCS}^2}\,\frac{\arcsin\sqrt{\mathfrak{p}_\mu}}{\sqrt{\mathfrak{p}_\mu(1-\mathfrak{p}_\mu)}}, \qquad \mathfrak{p}_\mu = \frac{\omega_\mu^2}{4\Delta_{\rm BCS}^2}, \tag{15.4}$$

for $0 < \mathfrak{p}_\mu < 1$, with $F_0 = F$, and continued to the other ranges of $\mathfrak{p}_\mu$ in Eq. (D.218). Only at $\mathfrak{p}_\mu > 1$, where pairs can be created, is F_μ complex.

The density bubble follows from Eqs. (15.2) and (15.3) with $\mathcal{V}^{-1}\sum_{\boldsymbol{k}} \to \mathcal{N}_\mathrm{F}\int \mathrm{d}\xi \int_{-1}^{1} \mathrm{d}\mu/2$. A partial-fraction split of its integrand into an F_μ term and an elementary term, and the gap equation, give (Appendix D.39), with $\langle X\rangle_\mu = \int_0^1 \mathrm{d}\mu\, X$,

$$\begin{aligned} R^0_{\boldsymbol{\eta\eta}} &= \Big\langle \frac{2\mathcal{N}_\mathrm{F} v_\mathrm{F}^2 q^2\mu^2 + 4\Delta_\mathrm{BCS}^2\omega^2 F_\mu}{\omega_\mu^2}\Big\rangle_\mu , \\ R^0_{\mathbf{s}_2\boldsymbol{\eta}} = -R^0_{\boldsymbol{\eta}\mathbf{s}_2} = 2\mathrm{i}\Delta_\mathrm{BCS}\,\omega\,\langle F_\mu\rangle_\mu , &\qquad 1 + \tfrac{1}{2}V_\mathrm{BCS}R^0_{\mathbf{s}_2\mathbf{s}_2} = \tfrac{1}{2}V_\mathrm{BCS}\langle \omega_\mu^2 F_\mu\rangle_\mu . \end{aligned} \tag{15.5}$$

At $q = 0$ these are the bubbles of Sect. 7.2. The last expression is the denominator of the phase channel in Eq. (7.34), and the coupling V_BCS drops out of the result:

Conserving density response of a clean superconductor

$$R^\mathrm{cons}(q,\omega) = \underbrace{\Big\langle \frac{2\mathcal{N}_\mathrm{F} v_\mathrm{F}^2 q^2\mu^2 + 4\Delta_\mathrm{BCS}^2\omega^2 F_\mu(\omega)}{\omega^2 - v_\mathrm{F}^2 q^2\mu^2}\Big\rangle_\mu}_{\text{bare bubble } R^0_{\boldsymbol{\eta\eta}}} \underbrace{- \frac{4\Delta_\mathrm{BCS}^2\omega^2\,\langle F_\mu(\omega)\rangle_\mu^2}{\langle(\omega^2 - v_\mathrm{F}^2 q^2\mu^2)F_\mu(\omega)\rangle_\mu}}_{\text{pair-field ladder}}, \tag{15.6}$$

with F_μ of Eq. (15.4), for $T = 0$, constant $\mathcal{N}_\mathrm{F}$, a linear band and no impurities.

At $q = 0$ the ladder cancels the bubble for every ω, which is Eq. (7.35). At small q the cancellation is incomplete, and $R^\mathrm{cons} \to 2\mathcal{N}_\mathrm{F} v_\mathrm{F}^2 q^2/3\omega^2 = n_e q^2/m_e\omega^2$, the free-particle tail whose coefficient is the f-sum rule; at $v_\mathrm{F} q = 0.01\,\Delta_\mathrm{BCS}$ the two agree to one part in 10^5, while the bare bubble stays at its $q = 0$ value. At $\omega \to 0$ the bubble gives $-2\mathcal{N}_\mathrm{F}$, the Thomas–Fermi value, and the ladder vanishes: pairing leaves the static screening of the metal unchanged. As $\Delta_\mathrm{BCS} \to 0$ at fixed ω/qv_F, the response approaches the Lindhard function at $q \ll k_\mathrm{F}$, to 6×10^{-5} at $v_\mathrm{F} q = 10^3\,\Delta_\mathrm{BCS}$ and 8×10^{-7} at $10^4\,\Delta_\mathrm{BCS}$.

Below $2\Delta_\mathrm{BCS}$ no pairs can be created, F_μ is real, and L^cons vanishes except where $\varepsilon^\mathrm{cons}$ does. The denominator of the ladder vanishes at one frequency $\Omega_\mathrm{AB}(q) < 2\Delta_\mathrm{BCS}$: the Anderson–Bogoliubov mode of Sect. 7.2, the Goldstone mode of the broken phase symmetry, now at finite q. There R^cons has a pole, so $\varepsilon^\mathrm{cons}$ has one too and the loss function has a zero, not a peak. The Coulomb interaction moves the weight of the mode to the plasmon, as Sect. 7.2 anticipated (the Anderson–Higgs mechanism), and a sum rule proves that no loss weight is left below $2\Delta_\mathrm{BCS}$ (Appendix D.39). Through the Kramers–Kronig relation the pole still shifts $\mathrm{Re}\,\varepsilon^\mathrm{cons}$ above $2\Delta_\mathrm{BCS}$. On the kinematic edge the pole term alone exceeds the whole shift relative to the Lindhard function at $\omega = 2.2\,\Delta_\mathrm{BCS}$, where the rest of the ladder acts the other way. It is 58 % of the shift at $2.5\,\Delta_\mathrm{BCS}$, 16 % at $3\Delta_\mathrm{BCS}$, and nothing at $5\Delta_\mathrm{BCS}$, where the edge momentum has pushed the mode into the pair continuum. The pole explains the shift only for $y \lesssim 2.3$.

Appendix E.20 sets out the calculation of the conserving density response of aluminum on the full grid of momentum transfers and energies, with impurity scattering. The clean limit needs only $\mathcal{N}_\mathrm{F}$, v_F and Δ_BCS, and no first-principles calculation.

15.1.3 The case-I coherence factor

Dropping the ladder leaves the bare bubble $R^0_{\boldsymbol{\eta\eta}}$, screened by the Coulomb interaction, $\varepsilon^\mathrm{bare} = 1 - v_\mathrm{C} R^0_{\boldsymbol{\eta\eta}}$. Gan and collaborators wrote this dielectric function [3, Eq. (8)] and the rate built on it [3, Eq. (9)]. Its absorptive part comes from the imaginary part of F_μ alone,

$\operatorname{Im} F_\mu = -\pi\mathcal{N}_\mathrm{F}/[4\Delta_\mathrm{BCS}^2\sqrt{\mathfrak{p}_\mu(\mathfrak{p}_\mu-1)}]$ for $\omega_\mu > 2\Delta_\mathrm{BCS}$. Changing the variable of the angular average from μ to ω_μ, $\mathrm{d}\mu = -\omega_\mu\,\mathrm{d}\omega_\mu/[v_\mathrm{F}q(\omega^2-\omega_\mu^2)^{1/2}]$, gives

$$\operatorname{Im} R^0_{\eta\eta} = -\frac{4\pi\mathcal{N}_\mathrm{F}\Delta_\mathrm{BCS}^2\omega^2}{v_\mathrm{F}q}\int_{2\Delta_\mathrm{BCS}}^{\omega}\frac{\mathrm{d}\omega'}{\omega'^2\sqrt{\omega'^2-4\Delta_\mathrm{BCS}^2}\,\sqrt{\omega^2-\omega'^2}},$$

provided $v_\mathrm{F}q \geq (\omega^2-4\Delta_\mathrm{BCS}^2)^{1/2}$, so that the whole interval of ω_μ lies inside $0\leq\mu\leq 1$. The normal metal has $\operatorname{Im} R^0 = -\pi\mathcal{N}_\mathrm{F}\omega/v_\mathrm{F}q$ below $\omega = qv_\mathrm{F}$ (Sect. 4.5). The ratio of the two depends on $y=\omega/\Delta_\mathrm{BCS}$ alone:

Absorptive part of the bare quasiparticle bubble

$$\operatorname{Im}\varepsilon^\mathrm{bare}(q,\omega) = \mathfrak{g}_\mathrm{I}\Big(\frac{\omega}{\Delta_\mathrm{BCS}}\Big)\operatorname{Im}\varepsilon_\mathrm{Lin}(q,\omega) \qquad (v_\mathrm{F}q \geq (\omega^2-4\Delta_\mathrm{BCS}^2)^{1/2}),$$
$$\mathfrak{g}_\mathrm{I}(y) = 4y\int_2^y \frac{\mathrm{d}u}{u^2\sqrt{u^2-4}\sqrt{y^2-u^2}} = \Big(1+\frac{2}{y}\Big)\mathrm{E}(k) - \frac{4\,\mathrm{K}(k)}{y+2}, \qquad k=\frac{y-2}{y+2}, \tag{15.7}$$

with K and E the complete elliptic integrals of the first and second kind of modulus k, for $y>2$; $\mathfrak{g}_\mathrm{I}=0$ below.

The function $\mathfrak{g}_\mathrm{I}$ is the case-I ratio $S_\mathrm{I}/S_\mathrm{n}$ of Eq. (7.30), obtained there by counting pairs in the energies of the two quasiparticles and here by integrating over the angle; the two integrals agree with the elliptic form to 10^{-12}. It jumps to $\pi/2$ at $y=2$, is 1.322 at $y=3$ and 1.051 at $y=10$, and Table 15.1 lists it at the other energies we use. Allen's theory of phonon-assisted absorption in metals contains the elliptic form [4], as quoted by Hochberg, Lin and Zurek [5, Eq. (17)]. The corresponding case-II form is that of Mattis and Bardeen [6]. Under the simplifications above Eq. (15.7) is exact: the absorptive part of the bare bubble carries no correction in powers of $\Delta_\mathrm{BCS}/qv_\mathrm{F}$. Its condition holds for every momentum transfer of halo dark matter, since $v_\mathrm{F}q \geq v_\mathrm{F}\omega/v_\mathrm{max} = 2.5\,\omega$.

The pair-field ladder changes the absorptive part. Its relative size is the vertex fraction $\mathfrak{v} = \operatorname{Im}(R^\mathrm{cons}-R^0_{\eta\eta})/\operatorname{Im} R^0_{\eta\eta}$, which follows from Eq. (15.6) by quadrature over μ (Appendix D.39). On the kinematic edge it is largest near the threshold, +0.44 at $y=2.2$, and falls to +0.004 at $y=10$ (Table 15.1). It does not fall monotonically with q, however: at $y=5$ it is +0.052 on the edge and still +0.044 at four times the edge momentum, and at $y=10$ it is nearly three times larger there than on the edge.

At the threshold itself the ladder replaces the jump of Eq. (7.30). The bare absorptive part jumps to $\pi/2$ times the normal value at $y=2$; in the conserving response it starts from zero and rises steeply. At the edge momentum of $y=2.05$, $\operatorname{Im}\varepsilon^\mathrm{cons}/\operatorname{Im}\varepsilon_\mathrm{Lin}$ is 0.14 at $y-2=10^{-6}$, 0.64 at 10^{-3} and 1.30 at 10^{-2}, and it passes the bare value $\pi/2$ before $y-2=0.03$, that is within 5 μeV of the threshold. For a rate integrated over energy, the conserving response therefore keeps a step at $2\Delta_\mathrm{BCS}$, smoothed over a few microelectronvolts. Whether the loss function keeps it depends on the real part of ε, which we take up next.

At $q = 0.3\,\mathrm{eV}$, $v_\mathrm{F}q = 11\,\Delta_\mathrm{BCS}$, every energy that halo dark matter can deposit lies within $4.4\,\Delta_\mathrm{BCS}$ of zero, and the bare bubble and the conserving response bracket the Lindhard function from below and from above (Fig. 15.2). At $q = 1\,\mathrm{eV}$, $v_\mathrm{F}q = 38\,\Delta_\mathrm{BCS}$, the differences are confined to the first few Δ_BCS above the edge.

15.1.4 First-order BCS screening correction

Since the absorptive parts of $\varepsilon^\mathrm{bare}$ and ε_Lin differ only by the factor $\mathfrak{g}_\mathrm{I}$, their difference is $\delta\operatorname{Im}\varepsilon = \operatorname{Im}\varepsilon_\mathrm{Lin}\,[\mathfrak{g}_\mathrm{I}(y)\Theta(y-2)-1]$: pairing removes the absorption below $2\Delta_\mathrm{BCS}$ and piles

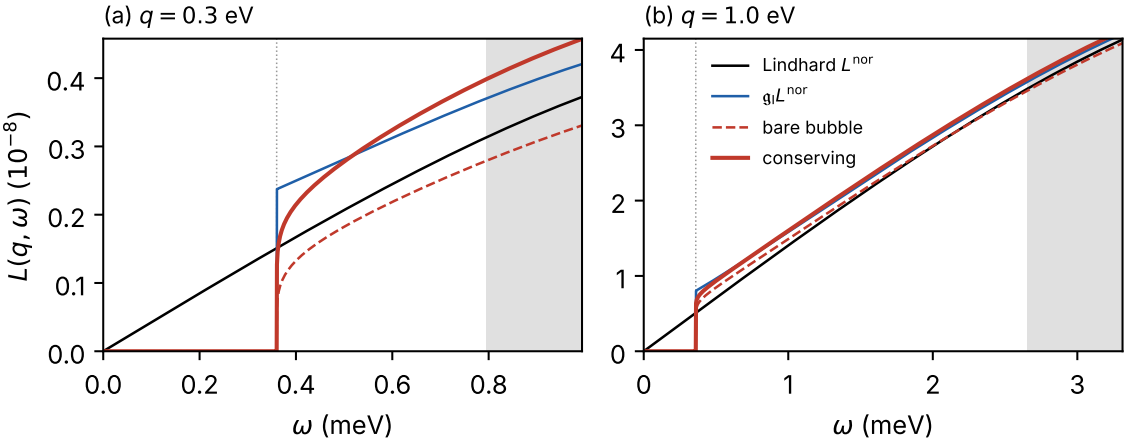


Figure 15.2: Loss function of aluminum near the pair-breaking edge at the momentum transfers (a) $q = 0.3\,\mathrm{eV}$ and (b) $q = 1\,\mathrm{eV}$, against the energy ω, for the bulk gap $\Delta_{\rm BCS} = 0.18\,\mathrm{meV}$ (dotted line at $2\Delta_{\rm BCS}$). Thick red: conserving response, Eqs. (15.1) and (15.6); dashed red: bare quasiparticle bubble; blue: the case-I factor times the normal loss function, $\mathfrak{g}_{\rm I} L^{\rm nor}$; black: Lindhard loss function of the normal metal, $L^{\rm nor}$. Halo dark matter reaches the energies left of the gray band, $\omega \le q v_{\rm max}$. All curves are clean, $T = 0$, with a constant density of states, $v_{\rm F} = 6.75 \times 10^{-3}$ and $k_{\rm TF} = 4.05\,\mathrm{keV}$. At $0.3\,\mathrm{eV}$ and $\omega \ge 2.2\,\Delta_{\rm BCS}$ the bare bubble lies 11 to 22 % below the Lindhard function and the conserving one 27 to 34 % above it. At $1\,\mathrm{eV}$ all four agree within 5 % at the largest reachable energy.

it up above. The half-range Kramers–Kronig relation Eq. (4.25), applied to $\varepsilon - 1 = -v_{\rm C}\tilde{R}$, turns this into a difference of real parts. Below $q v_{\rm F}$ the Lindhard function has $\mathrm{Im}\,\varepsilon_{\rm Lin} = (\pi/2)(k_{\rm TF}^2/q^2)(\omega/q v_{\rm F})$, from $\mathrm{Im}\,R^0 = -\pi \mathcal{N}_{\rm F}\omega/v_{\rm F} q$ and $k_{\rm TF}^2 = 2e^2\mathcal{N}_{\rm F}$. With $\omega' = \Delta_{\rm BCS} y'$,

First BCS correction to the screening

$$\begin{aligned} \mathrm{Re}\,\varepsilon^{\rm bare} - \mathrm{Re}\,\varepsilon_{\rm Lin} &= \frac{2}{\pi}\,\mathrm{PV}\int_0^\infty \mathrm{d}\omega'\,\frac{\omega'\,\delta\,\mathrm{Im}\,\varepsilon(\omega')}{\omega'^2 - \omega^2} = \frac{k_{\rm TF}^2}{q^2}\,\frac{\Delta_{\rm BCS}}{q v_{\rm F}}\,\mathfrak{s}_{\rm BCS}(y), \\ \mathfrak{s}_{\rm BCS}(y) &= \mathrm{PV}\int_0^\infty \mathrm{d}y'\,\frac{y'^2\,[\mathfrak{g}_{\rm I}(y')\Theta(y'-2) - 1]}{y'^2 - y^2}, \\ L^{\rm bare} &\simeq \mathfrak{g}_{\rm I}(y)\,L^{\rm nor}\Big[1 - 2\,\frac{\Delta_{\rm BCS}}{q v_{\rm F}}\,\mathfrak{s}_{\rm BCS}(y)\Big], \end{aligned} \tag{15.8}$$

to first order in $\Delta_{\rm BCS}/q v_{\rm F}$, with $L^{\rm nor} = \mathrm{Im}[-1/\varepsilon_{\rm Lin}]$, for $\omega \ll q v_{\rm F}$, where $\mathrm{Re}\,\varepsilon_{\rm Lin} \simeq k_{\rm TF}^2/q^2 \gg \mathrm{Im}\,\varepsilon_{\rm Lin}$.

The integral uses $\mathrm{Im}\,\varepsilon_{\rm Lin} \propto \omega'$ up to infinity, whereas the true function stops at $\omega' \simeq q v_{\rm F}$. The integrand of $\mathfrak{s}_{\rm BCS}$ falls as $\ln y'/y'^2$, since $\mathfrak{g}_{\rm I} - 1 \simeq (2\ln 2y - 1)/y^2$ for $y \gg 1$ (0.0505 against 0.0499 at $y = 10$). The region $y' > q v_{\rm F}/\Delta_{\rm BCS}$ therefore changes $\mathfrak{s}_{\rm BCS}$ by a term of relative order $(\Delta_{\rm BCS}/q v_{\rm F})\ln(q v_{\rm F}/\Delta_{\rm BCS})$, which is of second order in the real part. As a check, the shift of the real part of the bare bubble, evaluated from Eq. (15.6) without the ladder at $y = 5$ and multiplied by $q v_{\rm F}/\Delta_{\rm BCS}$, is 1.0531 at $v_{\rm F} q = 100\,\Delta_{\rm BCS}$, against $\mathfrak{s}_{\rm BCS}(5) = 1.0529$.

At $y = 0$, $\mathfrak{s}_{\rm BCS}(0) = \int_0^\infty \mathrm{d}y'\,[\mathfrak{g}_{\rm I}\Theta(y'-2) - 1] = 0$: the absorption that pairing piles up above the edge is exactly the absorption it removes below, weighted by $1/\omega$, and the static screening is unchanged, as found above for the full response. The function $\mathfrak{s}_{\rm BCS}$ is positive at

every y we have evaluated, from 0.1 to 50, so the superconductor's $\mathrm{Re}\,\varepsilon$ exceeds the normal one and the correction lowers the loss function. It diverges logarithmically at $y = 2$, from the jump of $\mathfrak{g}_\mathrm{I}$, is 2.03 at $y = 3$ and 0.50 at $y = 10$ (Table 15.1), and approaches $\pi^2/2y$ beyond (0.248 at $y = 20$).

The logarithm drives the bare loss function to zero at the threshold. At the edge momentum of $y = 2.05$, $L^\mathrm{bare}/L^\mathrm{nor}$ is 0.03 at $y - 2 = 10^{-9}$ and still only 0.30 at $y - 2 = 0.03$. The conserving loss function also vanishes at $2\Delta_\mathrm{BCS}$, because its absorptive part does, but it is back to $0.97\,L^\mathrm{nor}$ by $y - 2 = 0.03$. The jump of Eq. (7.30) thus survives in L^cons as a step a few microelectronvolts wide and is screened away in L^bare.

For halo dark matter the ratio $\Delta_\mathrm{BCS}/qv_\mathrm{F}$ is at most $(v_\mathrm{max}/v_\mathrm{F})/y$, reached on the kinematic edge. Being of first order, the correction is then at most $2(v_\mathrm{max}/v_\mathrm{F})\,\mathfrak{s}_\mathrm{BCS}(y)/y$ of the loss function, and it falls as $1/q$ away from the edge. For aluminum, with $v_\mathrm{max}/v_\mathrm{F} = 0.392$, this maximum is 3.9 % at $y = 10$ and 17 % at $y = 5$. At $y = 3$ it is 53 % and at $y = 2.2$ it exceeds one, so near the threshold the expansion fails (Sect. 15.2). The full bare bubble confirms the first-order term where it is small: on the edge of $y = 10$ its loss function is 0.935 of $\mathfrak{g}_\mathrm{I}L^\mathrm{nor}$ against $1 - 0.039 = 0.961$, and at four times the edge momentum of $y = 5$ it is 0.953 against 0.959.

The pair-field ladder shifts the real part as well, and near the edge its shift is comparable to the first-order term and of the opposite sign. The first-order coefficient returns only at large momentum transfers: at $y = 5$ the shift of the conserving real part times $qv_\mathrm{F}/\Delta_\mathrm{BCS}$ is 0.954 at $v_\mathrm{F}q = 10^2\,\Delta_\mathrm{BCS}$, 1.048 at $10^3\,\Delta_\mathrm{BCS}$ and 1.053 at $10^4\,\Delta_\mathrm{BCS}$. Eq. (15.8) is therefore the leading term of both responses as $\Delta_\mathrm{BCS}/qv_\mathrm{F} \to 0$, but at the momenta where it is large enough to matter it describes the bare bubble only. Over all momentum transfers that halo dark matter reaches, the clean conserving loss function lies between 0.980 and 1.085 times $\mathfrak{g}_\mathrm{I}L^\mathrm{nor}$ for $y \geq 3$, whereas the bare bubble falls to 0.53 of the conserving one. Sect. 15.2 gives the values on the edge, and Sect. 15.3.3 the change with the mean free path.

15.1.5 Truncations of the conserving response

The loss functions used for superconducting targets are truncations of Eq. (15.1). Dropping the pair-field ladder keeps the bare bubble, with its absorptive part Eq. (15.7) and its own real part [3, Eq. (8)]. Keeping the case-I absorptive part but screening it with the normal metal gives $\mathrm{Im}\,\varepsilon^\mathrm{bare}/|\varepsilon_\mathrm{Lin}|^2 = \mathfrak{g}_\mathrm{I}L^\mathrm{nor}$, the approximation that Gan and collaborators compare with their full result [3, Eq. (10)] and that Hochberg and collaborators use for directional detection [7, Eq. (2) and text]. Closing the gap, $\Delta_\mathrm{BCS} \to 0$ at fixed ω, leaves the Lindhard loss function [8, Eqs. (1), (S.13)], whose unscreened structure factor below qv_F is the free-gas $S_\mathrm{n} = m_e^2\omega/\pi q$ of Sect. 7.2 [9, Eq. (3.4)]. The truncations are controlled by different parameters. Dropping the ladder is controlled by $\Delta_\mathrm{BCS}/qv_\mathrm{F}$, and replacing the real part by the normal one by the same ratio times $\mathfrak{s}_\mathrm{BCS}(y)$. Closing the gap is controlled by y alone, through $\mathfrak{g}_\mathrm{I}(y) - 1$. On the kinematic edge all three controlling parameters are functions of y, and that is where each truncation is worst.

Limit Check 15.1. Loss function of a superconductor near the pair-breaking edge

Lindhard loss function $L^\mathrm{nor} = \mathrm{Im}[-1/\varepsilon_\mathrm{Lin}]$ in the rate Eq. (10.4), with $\mathrm{Im}\,\varepsilon_\mathrm{Lin} = (\pi/2)(k_\mathrm{TF}^2/q^2)(\omega/qv_\mathrm{F})$ below qv_F [8, Eqs. (1), (S.13), 9, Eq. (3.4)]; with the case-I factor, $\mathfrak{g}_\mathrm{I}L^\mathrm{nor}$ [3, Eq. (10), 7, Eq. (2) and text]; bare bubble [3, Eqs. (8), (9)].

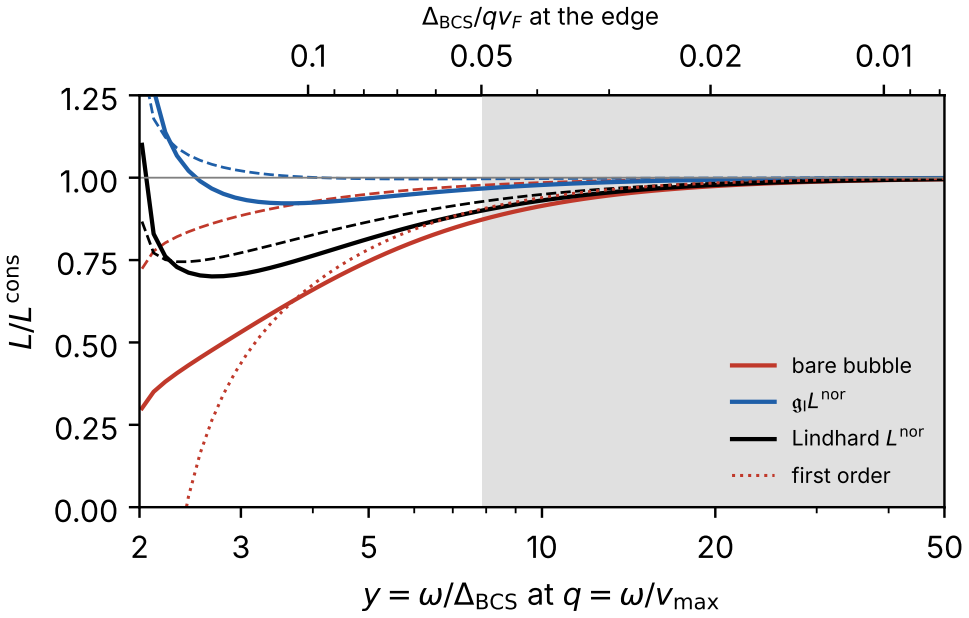


Figure 15.3: Loss function of aluminum on the kinematic edge $q = \omega/v_{\max}$ in the three truncations, divided by the conserving loss function Eq. (15.1), against $y = \omega/\Delta_{\mathrm{BCS}}$ (top axis: $\Delta_{\mathrm{BCS}}/qv_{\mathrm{F}} = 0.392/y$ there). Red: bare quasiparticle bubble; dotted red: its first-order form Eq. (15.8); blue: $\mathfrak{g}_{\mathrm{I}}L^{\mathrm{nor}}$; black: Lindhard L^{nor}. Dashed: the same at four times the edge momentum. Clean, $T = 0$, constant density of states, $v_{\mathrm{F}} = 6.75 \times 10^{-3}$. "Within 10 %" means that the smaller of the two loss functions is at least 0.9 of the larger. Gray: $y \geq 7.9$, where the Lindhard function is within 10 % of the conserving result on the edge and at all larger momenta. The product $\mathfrak{g}_{\mathrm{I}}L^{\mathrm{nor}}$ is within 10 % already for $y \geq 2.25$.

Below $y \approx 3$ on the edge the Lindhard function falls about 30 % below the conserving loss function, and the clean conserving result itself assumes that impurities do not scatter the electrons within the size of a pair (Sect. 15.3.1).

Fig. 15.3 orders the truncations. The Lindhard function, which has no gap, is within 10 % of the conserving result for $y \geq 7.9$ on the edge, where $\Delta_{\mathrm{BCS}}/qv_{\mathrm{F}} \leq 0.050$, and for $y \geq 6.3$ at four times the edge momentum. At $y = 5$ it is 0.815 of the conserving value on the edge, and in the conserving response the case-I factor $\mathfrak{g}_{\mathrm{I}}(5) = 1.15$ accounts for most of that. The bare bubble is the worst of the three below $y \approx 5$, since its first-order screening correction is large there and the ladder that would cancel it is missing. The product $\mathfrak{g}_{\mathrm{I}}L^{\mathrm{nor}}$, which keeps the absorptive part of the pairs and the screening of the normal metal, is within 9 % of the conserving result for $y \geq 3$ at every momentum transfer that halo dark matter reaches. Gan and collaborators found that the bare and the normal $|\varepsilon|^2$ agree for energy transfers above $5\Delta_{\mathrm{BCS}}$ [3, Sect. V]. On the kinematic edge of $y = 5$ the clean model gives $|\varepsilon^{\mathrm{bare}}|^2/|\varepsilon_{\mathrm{Lin}}|^2 = 1.26$, against 1.17 from the first-order term of Eq. (15.8) alone.

All of these are clean-limit statements. The structure that separates the curves near the edge, the steep rise of the conserving absorption over a few microelectronvolts and the logarithm of $\mathfrak{s}_{\mathrm{BCS}}$, is narrower than Δ_{BCS}. Impurity scattering, at a rate given by v_{F} over the mean free path, smooths it once that rate is no longer small compared with the width. For $y \geq 3$ the curves rest on the smooth parts of $\mathfrak{g}_{\mathrm{I}}$ and $\mathfrak{s}_{\mathrm{BCS}}$, which vary on the scale of Δ_{BCS}

itself.

15.1.6 The aluminum gap and the rate

The gap enters the loss function through $y = \omega/\Delta_{\mathrm{BCS}}$, in $\mathfrak{g}_{\mathrm{I}}$, in $\mathfrak{s}_{\mathrm{BCS}}$ and in the ladder, and not only through the threshold. For bulk aluminum it is measured, $\Delta_{\mathrm{BCS}} = 0.18\,\mathrm{meV}$ [10, Table III], and it agrees within one percent with the weak-coupling value $1.764\,T_c = 0.179\,\mathrm{meV}$ for the measured $T_c = 1.18\,\mathrm{K}$ (Worked example 7.3). Gan and collaborators use 0.17 meV [3, Sect. IV]. Hochberg and collaborators take $\Delta = 0.3\,\mathrm{meV}$ for aluminum, which they define as half of the superconducting gap [7, p. 2]. That is the film convention $2\Delta_{\mathrm{BCS}} = 0.6\,\mathrm{meV}$ of Sect. 1.5, 1.67 times the bulk value. It describes thin films, whose critical temperatures exceed the bulk value and in ultraclean films approach twice it [11]. A rate for a bulk crystal uses 0.18 meV. A rate for a film uses the gap measured on that film, and Sect. 15.3 adds what the film changes in the response itself.

Worked example 15.1: The aluminum rate at 1 keV with the bulk and the film gap

Take $m_\chi = 1\,\mathrm{keV}$, a light mediator, $F_{\mathrm{DM}} = (\alpha m_e/q)^2$, the reference cross section $\bar{\sigma}_e = 10^{-40}\,\mathrm{cm}^2$, the standard halo of Sect. 1.2, and aluminum with $\rho_T = 2.70\,\mathrm{g\,cm^{-3}}$, $v_{\mathrm{F}} = 6.75 \times 10^{-3}$ and $k_{\mathrm{TF}} = 4.05\,\mathrm{keV}$. The particle deposits at most $\frac{1}{2} m_\chi v_{\mathrm{max}}^2 = 3.51\,\mathrm{meV}$, which is $19.5\,\Delta_{\mathrm{BCS}}$ for the bulk gap and $11.7\,\Delta_{\mathrm{BCS}}$ for the film gap, and transfers at most $2 m_\chi v_{\mathrm{max}} = 5.30\,\mathrm{eV}$, so that $q/k_{\mathrm{TF}} \leq 1.3 \times 10^{-3}$ and $v_{\mathrm{F}} q \leq 199\,\Delta_{\mathrm{BCS}}$ (bulk). Every point lies in the regime of Eq. (15.6).

Eq. (10.4) with the conserving loss function, integrated numerically over $2\Delta_{\mathrm{BCS}} \leq \omega \leq 3.51\,\mathrm{meV}$ and over the momenta with $v_{\mathrm{min}} \leq v_{\mathrm{max}}$ (Gauss–Legendre sums, in ω graded toward $2\Delta_{\mathrm{BCS}}$ and in q between the two roots of $v_{\mathrm{min}}(q,\omega) = v_{\mathrm{max}}$), gives

$$\mathcal{R} = 3.26\,\mathrm{kg^{-1}\,yr^{-1}}\ (\Delta_{\mathrm{BCS}} = 0.18\,\mathrm{meV}), \qquad \mathcal{R} = 2.65\,\mathrm{kg^{-1}\,yr^{-1}}\ (\Delta_{\mathrm{BCS}} = 0.30\,\mathrm{meV}).$$

The bulk gap raises the rate by a factor of 1.23. The lower threshold alone would do more: with the Lindhard function cut at $2\Delta_{\mathrm{BCS}}$, which has no coherence factor, the factor is 1.30. The difference comes from the edge. With the film gap the window $2\Delta_{\mathrm{BCS}} < \omega < 3\Delta_{\mathrm{BCS}}$, where $\mathfrak{g}_{\mathrm{I}}$ is largest, carries 37 % of the conserving rate; with the bulk gap it carries 20 %. For a heavy mediator, $F_{\mathrm{DM}} = 1$, the rate at the same cross section is $3.8 \times 10^{-13}\,\mathrm{kg^{-1}\,yr^{-1}}$, and the bulk gap raises it by 1.13.

The choice of loss function matters less than the choice of gap. With the bulk gap and the light mediator, the bare bubble gives 0.947 of the conserving rate, $\mathfrak{g}_{\mathrm{I}} L^{\mathrm{nor}}$ gives 1.010, and the Lindhard function cut at the threshold gives 0.850. The normal metal with no gap at all gives 1.024: at this mass the pile-up of pair states above $2\Delta_{\mathrm{BCS}}$ almost exactly replaces the rate lost below it. With the film gap the same ratios are 0.903, 1.027, 0.806 and 1.26. These are clean-limit values, and the part of the rate below $3\Delta_{\mathrm{BCS}}$ is the part that impurity scattering can change (Sect. 15.3).

15.2 Screening at the kinematic edge

A particle that deposits $\omega = 2.2\,\Delta_{\mathrm{BCS}}$ in aluminum with the smallest momentum transfer it can, $q = \omega/v_{\mathrm{max}}$, sees a first-order screening correction of 1.43 in Eq. (15.8), so the first-order loss function is negative there. Which particles scatter that close to the kinematic edge? A

particle at the threshold mass $m_{\min} = 0.10\,\mathrm{keV}$ of Worked example 7.3 must move at $v_{\max}$ and give up its whole kinetic energy, which takes the momentum transfer $q = m_{\min} v_{\max} = 2\omega/v_{\max}$, twice the edge value. Heavier particles can deposit the same energy with less momentum: from $v_{\min}(q,\omega) = v_{\max}$ the smallest momentum transfer at $\omega = 2\Delta_{\mathrm{BCS}}$ is 1.17 times the edge value for $2m_{\min}$ and only 1.03 times it at 1 keV. For every particle a few times heavier than $m_{\min}$, the scatters just above $2\Delta_{\mathrm{BCS}}$ therefore lie close to the kinematic edge, and there we need the loss function without expanding in the screening correction. We will write it in resummed form for the bare quasiparticle bubble and for the conserving response of Sect. 15.1, compare both with the normal-state screening $\mathfrak{g}_{\mathrm{I}} L^{\mathrm{nor}}$, and use the comparison to fix the nuisances of a rate calculation near the edge.

15.2.1 The resummed loss function

At $q \ll k_{\mathrm{F}}$ the Lindhard function Eq. (4.56) depends on ω and q only through the reduced frequency $\mathfrak{w} = \omega/qv_{\mathrm{F}}$, the common limit of the arguments $\omega/qv_{\mathrm{F}} \pm q/2k_{\mathrm{F}}$ as $q/2k_{\mathrm{F}} \to 0$. To first order in q/k_{F} the bracket of Eq. (4.56) is the derivative $\partial_{\mathfrak{w}}[(1-\mathfrak{w}^2)\ell(\mathfrak{w})] = 2 - 2\mathfrak{w}\,\ell(\mathfrak{w})$, with $\ell(\mathfrak{w}) = \ln[(\mathfrak{w}+1)/(\mathfrak{w}-1)]$ and $\ell' = 2/(1-\mathfrak{w}^2)$. With $\varepsilon_{\mathrm{Lin}} = 1 - v_{\mathrm{C}} R^0$ and $k_{\mathrm{TF}}^2 = 2e^2\mathcal{N}_{\mathrm{F}}$, this gives $R^0 = -2\mathcal{N}_{\mathrm{F}}\mathfrak{f}_{\mathrm{L}}(\mathfrak{w})$ and, on the real axis,

$$\varepsilon_{\mathrm{Lin}}(q,\omega) = 1 + \frac{k_{\mathrm{TF}}^2}{q^2}\Big[\mathrm{Re}\,\mathfrak{f}_{\mathrm{L}}(\mathfrak{w}) + \mathrm{i}\,\frac{\pi\mathfrak{w}}{2}\Big], \qquad \mathfrak{f}_{\mathrm{L}}(\mathfrak{w}) = 1 - \frac{\mathfrak{w}}{2}\,\ell(\mathfrak{w}),$$
$$\mathrm{Re}\,\mathfrak{f}_{\mathrm{L}} = 1 - \frac{\mathfrak{w}}{2}\ln\frac{1+\mathfrak{w}}{1-\mathfrak{w}} \qquad (\mathfrak{w} < 1). \tag{15.9}$$

The function $\mathfrak{f}_{\mathrm{L}}$ is analytic in the upper half plane of $\mathfrak{w}$; at real $\mathfrak{w} < 1$ the retarded branch of ℓ gives the imaginary part $\pi\mathfrak{w}/2$. It has $\mathfrak{f}_{\mathrm{L}}(0) = 1$, the static limit, and $\mathfrak{f}_{\mathrm{L}} \to -1/3\mathfrak{w}^2$ for $|\mathfrak{w}| \gg 1$, the free-particle tail of Eq. (4.58). The kinematics of Sect. 1.3 require $q \geq \omega/v_{\max}$, so $\mathfrak{w} \leq v_{\max}/v_{\mathrm{F}}$, with equality on the edge. In aluminum $v_{\max}/v_{\mathrm{F}} = 0.392$, and on the edge $\mathrm{Re}\,\mathfrak{f}_{\mathrm{L}} = 0.837$ at every ω. The real part of the normal dielectric function there is 16 % below the Thomas–Fermi value k_{TF}^2/q^2, and the ratio of imaginary to real part,

$$\mathfrak{r}_\varepsilon = \frac{\mathrm{Im}\,\varepsilon_{\mathrm{Lin}}}{\mathrm{Re}\,\varepsilon_{\mathrm{Lin}}} = \frac{\pi\mathfrak{w}}{2\,\mathrm{Re}\,\mathfrak{f}_{\mathrm{L}}(\mathfrak{w})} = 0.736,$$

is not small. The first-order form Eq. (15.8) assumes $\omega \ll qv_{\mathrm{F}}$, where $\mathfrak{f}_{\mathrm{L}} \to 1$ and $\mathfrak{r}_\varepsilon \to 0$; on the edge neither holds, even before the gap enters. With the "1" of ε dropped, which changes nothing at these momenta (Sect. 15.1.1), every loss-function ratio below depends on $y = \omega/\Delta_{\mathrm{BCS}}$ and $qv_{\mathrm{F}}/\Delta_{\mathrm{BCS}}$ alone, and on the edge on y and $v_{\max}/v_{\mathrm{F}}$. The aluminum numbers therefore apply to any clean weak-coupling superconductor with the same velocity ratio.

Sect. 15.1 gives the two ingredients of the bare bubble. Its absorptive part is $\mathfrak{g}_{\mathrm{I}}(y)\,\mathrm{Im}\,\varepsilon_{\mathrm{Lin}}$, exactly, by Eq. (15.7); its real part exceeds $\mathrm{Re}\,\varepsilon_{\mathrm{Lin}}$ by $(k_{\mathrm{TF}}^2/q^2)(\Delta_{\mathrm{BCS}}/qv_{\mathrm{F}})\mathfrak{s}_{\mathrm{BCS}}(y)$, by Kramers–Kronig, Eq. (15.8). Neither statement needs the shift to be small. Inserted in $L = \mathrm{Im}\,\varepsilon/|\varepsilon|^2$ without expansion, they give the loss function exactly within the model (Eq. (D.219)):

Resummed loss function on the kinematic edge

$$L^{\mathrm{bare}} = \mathfrak{g}_{\mathrm{I}}(y)\,L^{\mathrm{nor}}\,\frac{1+\mathfrak{r}_\varepsilon^2}{(1+\delta_{\mathfrak{s}})^2 + \mathfrak{g}_{\mathrm{I}}^2(y)\,\mathfrak{r}_\varepsilon^2}, \qquad \delta_{\mathfrak{s}} = \frac{\Delta_{\mathrm{BCS}}}{qv_{\mathrm{F}}}\,\frac{\mathfrak{s}_{\mathrm{BCS}}(y)}{\mathrm{Re}\,\mathfrak{f}_{\mathrm{L}}(\mathfrak{w})}, \tag{15.10}$$

for a clean superconductor at $T = 0$ with constant density of states, $q \ll k_{\mathrm{F}}$ and $\omega < qv_{\mathrm{F}}$.

Table 15.1: Loss function of aluminum on the kinematic edge $q = \omega/v_{\max}$ ($\omega/qv_{\rm F} = v_{\max}/v_{\rm F} = 0.392$), divided by $\mathfrak{g}_{\rm I}L^{\rm nor}$, the value with normal-state screening, against $y = \omega/\Delta_{\rm BCS}$. Columns: case-I ratio $\mathfrak{g}_{\rm I}$ of Eq. (15.7); Kramers–Kronig shift $\mathfrak{s}_{\rm BCS}$ of Eq. (15.8); first-order term $2(\Delta_{\rm BCS}/qv_{\rm F})\mathfrak{s}_{\rm BCS}$; bare bubble in the resummed form Eq. (15.10), with the Kramers–Kronig integral taken to infinity and cut at $\omega' = qv_{\rm F}$; vertex fraction $\mathfrak{v}$ of Sect. 15.1.3; conserving response Eq. (15.6), clean, and with Born scattering by impurities at the mean free paths $\ell_e = 1\,\mu$m and 30 nm (Sect. 15.3.3), each divided by $\mathfrak{g}_{\rm I}$ times the normal state at the same scattering rate. All at $T = 0$, constant density of states, $q \ll k_{\rm F}$.

				bare, resummed			conserving		
y	$\mathfrak{g}_{\rm I}$	$\mathfrak{s}_{\rm BCS}$	first order	to ∞	cut	$\mathfrak{v}$	clean	1 µm	30 nm
2.05	1.552	6.01	2.30	0.222	0.267	0.22	0.681	0.365	0.045
2.2	1.500	4.01	1.43	0.331	0.401	0.44	0.867	0.600	0.187
2.5	1.418	2.82	0.89	0.450	0.534	0.35	0.999	0.854	0.569
3	1.322	2.03	0.53	0.575	0.660	0.21	1.069	1.021	1.008
5	1.151	1.05	0.17	0.801	0.856	0.052	1.067	1.057	1.091
10	1.051	0.50	0.04	0.937	0.957	0.004	1.023	1.020	1.021

The gap and the case-I factor both lower the loss function below $\mathfrak{g}_{\rm I}L^{\rm nor}$. The gap stiffens the real part of ε, by the fraction $\delta_{\mathfrak{s}}$. And the same case-I factor that raises $\mathrm{Im}\,\varepsilon$ in the numerator also raises $|\varepsilon|$ in the denominator, through $\mathfrak{g}_{\rm I}^2\mathfrak{r}_\varepsilon^2$, which is not small with $\mathfrak{r}_\varepsilon = 0.736$ even at $\Delta_{\rm BCS}/qv_{\rm F} \to 0$. For $\delta_{\mathfrak{s}} \ll 1$ and $\omega \ll qv_{\rm F}$, Eq. (15.10) returns Eq. (15.8); unlike the first-order form it is positive for every $y > 2$. As $y \to 2$, $\mathfrak{s}_{\rm BCS}$ grows as $(\pi/2)\ln[1/(y-2)]$ (Appendix D.39), and $L^{\rm bare}$ falls to zero as the inverse square of that logarithm.

The Kramers–Kronig integral of Eq. (15.8) runs over all ω', whereas the Lindhard absorption it is built on stops at $\omega' \simeq qv_{\rm F}$. Near the edge the difference is no longer of second order, and two prescriptions bracket it: the integral as written, and the integral cut at $\omega' = qv_{\rm F}$, which on the edge is $y' = y\,v_{\rm F}/v_{\max}$. The cut lowers the shift from 4.01 to 2.92 at $y = 2.2$ and from 2.03 to 1.15 at $y = 3$. The bare bubble itself, its real part computed directly from Eq. (15.6) without the ladder, lies within 1.5 % of the uncut prescription at every y of Table 15.1.

On the edge of aluminum, Eq. (15.10) gives $L^{\rm bare}/\mathfrak{g}_{\rm I}L^{\rm nor} = 0.33$ at $y = 2.2$ with the integral to infinity and 0.40 with the cut; Table 15.1 gives both prescriptions at the other energies. The first-order term reaches 0.3, a size at which its square is no longer negligible, at $y = 3.81$ with the uncut shift and at $y = 3.00$ with the cut one. The resummation turns the case-I enhancement into its opposite: a bubble that raises $\mathrm{Im}\,\varepsilon$ by 50 % at $y = 2.2$ ends up with a loss function a third of $\mathfrak{g}_{\rm I}L^{\rm nor}$, and half of $L^{\rm nor}$.

15.2.2 The conserving response in resummed form

Eq. (15.10) is a statement about any response whose absorptive part is a multiple $\mathfrak{g}$ of the normal one and whose real part is the normal one times $1 + \delta$ (Eq. (D.219)). The conserving response of Eq. (15.6) enters with

$$\mathfrak{g} = \mathfrak{g}_{\rm I}(1 + \mathfrak{v}), \qquad \delta = \frac{\mathrm{Re}(\varepsilon^{\rm cons} - \varepsilon_{\rm Lin})}{\mathrm{Re}\,\varepsilon_{\rm Lin}}, \tag{15.11}$$

with $\mathfrak{v}$ the vertex fraction of Sect. 15.1.3; both follow from Eq. (15.6) by one quadrature over μ. This separates what the pair-field ladder does to the two parts of ε, and on the edge it does more to the real part. At $y = 3$ the ladder raises the absorptive part from 1.322 to $1.322 \times 1.214 = 1.605$ times normal, while it moves the real part from 1.33 times normal, the

bare value, to 0.60 times normal. The modulus of ε ends close to the normal one, and the loss function close to $\mathfrak{g}_\mathrm{I} L^\mathrm{nor}$: $L^\mathrm{cons} = 1.07\,\mathfrak{g}_\mathrm{I} L^\mathrm{nor} = 1.41\,L^\mathrm{nor}$. Nearer the threshold the shift overshoots, and at $y = 2.2$ the real part of $\varepsilon^\mathrm{cons}$ is negative, $1 + \delta = -0.18$: the condensate responds inductively there. The superconductor nonetheless screens about as strongly as the normal metal, with $|\varepsilon^\mathrm{cons}|/|\varepsilon_\mathrm{Lin}| = 1.29$ at $y = 2.2$, 1.07 at 3 and 0.99 at 5. Along the whole edge the conserving value lies within 10 % of $\mathfrak{g}_\mathrm{I} L^\mathrm{nor}$ for $y \geq 2.25$ and above it from $y = 2.5$ on, by at most 8.4 % at $y = 3.6$; $L^\mathrm{cons}/L^\mathrm{nor}$ peaks at 1.43 at $y = 2.70$. These are clean values; at $y \leq 2.5$ they hold only for mean free paths $\mathfrak{l}_e \gtrsim 10\,\mu\mathrm{m}$ (Sect. 15.3.3).

Near the threshold, $y \lesssim 2.3$, the ladder is large because the Anderson–Bogoliubov pole of Eq. (15.6) lies just below the gap, at 1.927, 1.966 and $1.992\,\Delta_\mathrm{BCS}$ for the edge momenta of $y = 2.2$, 2.5 and 3 (Sect. 15.1.2). At $y \geq 3$ the shift comes from the pair continuum of the ladder instead. Gan and collaborators identified this phase-mode term and estimated it as of relative order $(\Delta_\mathrm{BCS}/v_\mathrm{F} q)^2$ [3, Sect. V]. The relative size $|R^\mathrm{cons} - R^0_{\eta\eta}|/|R^0_{\eta\eta}|$ is $27(\Delta_\mathrm{BCS}/q v_\mathrm{F})^2$ on the edge at $y = 3$, and its coefficient falls only slowly with q, to 18 at $q v_\mathrm{F} = 100\,\Delta_\mathrm{BCS}$ and 11 at $10^3\,\Delta_\mathrm{BCS}$. The power of their estimate is right, but its coefficient is an order of magnitude larger than one; this is why the bare and conserving loss functions differ so much on the edge.

Limit Check 15.2. Resummed loss function on the kinematic edge

$L = \mathrm{Im}\,\varepsilon^\mathrm{bare}/|\varepsilon^\mathrm{bare}|^2$ with the bare BCS dielectric function [3, Eqs. (8), (9)], in the resummed form Eq. (15.10).

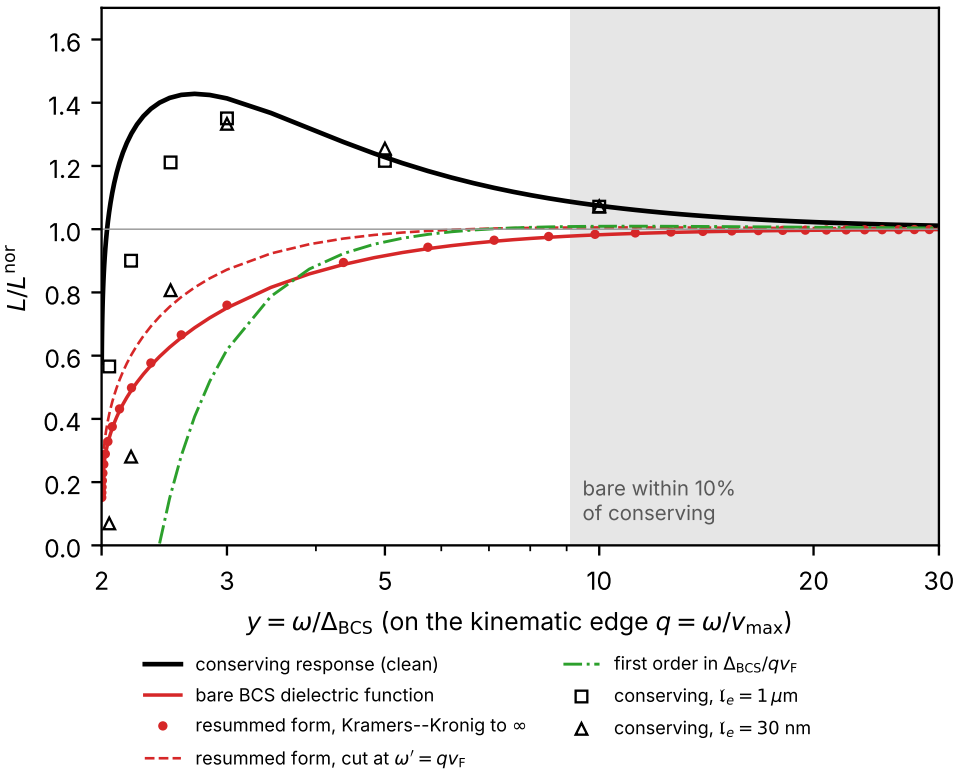


Figure 15.4: Loss function of aluminum on the kinematic edge $q = \omega/v_\mathrm{max}$, divided by the normal-state Lindhard value, against $y = \omega/\Delta_\mathrm{BCS}$ ($v_\mathrm{max}/v_\mathrm{F} = 0.392$, $T = 0$, constant density of states, $q \ll k_\mathrm{F}$). Black: conserving response Eq. (15.6), clean.

Red: bare BCS dielectric function, evaluated directly (solid), and the resummed form Eq. (15.10) with the Kramers–Kronig integral taken to infinity (dots) and cut at $\omega' = qv_F$ (dashed). Green dash-dotted: first order in Δ_{BCS}/qv_F, Eq. (15.8), where positive. Open symbols: conserving response with Born scattering by impurities, mean free path 1 µm (squares) and 30 nm (triangles). Gray: the bare loss function at least 0.9 of the conserving one, $y \geq 9.1$. The normal-state screening $\mathfrak{g}_I L^{nor}$ is in Fig. 15.3.

Below $y = 9.1$ on the edge the Kramers–Kronig shift of the bare bubble, which carries its violation of charge conservation, overstates $|\varepsilon|$: at $y = 2.2$ the bare loss function is 0.38 of the conserving one, and a rate there needs Eq. (15.11) or $\mathfrak{g}_I L^{nor}$.

Worked example 15.2: The edge suppression of aluminum at $y = 3$

Take $\omega = 3\Delta_{BCS} = 0.54\,\text{meV}$ with the bulk gap $\Delta_{BCS} = 0.18\,\text{meV}$. The smallest momentum transfer is $q = \omega/v_{max} = 0.54 \times 10^{-3}\,\text{eV}/2.650 \times 10^{-3} = 0.204\,\text{eV}$, so $1/q = 0.97\,\mu\text{m}$. With $v_F = 6.75 \times 10^{-3}$, $qv_F = 1.38\,\text{meV} = 7.65\,\Delta_{BCS}$, so $\mathfrak{w} = 0.392$ and $\Delta_{BCS}/qv_F = 0.131$.

Normal metal. Eq. (15.9) gives $\text{Re}\,\mathfrak{f}_L = 1 - 0.196\ln(1.392/0.608) = 0.837$ and $\mathfrak{r}_\varepsilon = (\pi/2)(0.392)/0.837 = 0.736$.

Bare bubble. The absorptive part rises by $\mathfrak{g}_I(3) = 1.322$. With $\mathfrak{s}_{BCS}(3) = 2.034$, $\delta_\mathfrak{s} = 0.131 \times 2.034/0.837 = 0.318$. In Eq. (15.10) the numerator is $1 + 0.736^2 = 1.542$ and the denominator $1.318^2 + 1.322^2 \times 0.736^2 = 1.736 + 0.947 = 2.684$, so $L^{bare} = 0.575\,\mathfrak{g}_I L^{nor} = 0.760\,L^{nor}$. The first-order form gives $1 - 2 \times 0.131 \times 2.034 = 0.468$ for the same ratio, 19 % lower. With the Kramers–Kronig integral cut at qv_F the shift falls to 1.15 and the ratio rises to 0.660.

Conserving response. Eq. (15.11) has $\mathfrak{g} = 1.322 \times 1.214 = 1.605$ and $1 + \delta = 0.596$. Then $|\varepsilon^{cons}|/|\varepsilon_{Lin}| = [(0.596^2 + 1.605^2 \times 0.736^2)/(1 + 0.736^2)]^{1/2} = 1.065$, and $L^{cons} = 1.605\,L^{nor}/1.065^2 = 1.414\,L^{nor} = 1.069\,\mathfrak{g}_I L^{nor}$. In a 30 nm film the ratio is 1.008 (Table 15.1).

The bare bubble predicts a loss function a quarter below the normal metal at this energy and momentum. The conserving response predicts 41 % above it, almost all of it the case-I factor, and the mean free path changes that by less than 6 %: at $y = 3$ the edge value is known to 6 %.

15.2.3 Corrections and nuisances

The resummed forms inherit the hypotheses of Eq. (15.6), and Sects. 15.1.1 and 15.1.2 bound the terms they drop: the band curvature, the "1" and the core electrons in ε, and the particle–hole asymmetry that couples the amplitude channel, each below 10^{-4} at these momenta. Thermal quasiparticles add the scattering channel of Sect. 7.2 with weight $e^{-\Delta_{BCS}/T}$, e^{-209} at 10 mK. Within the model every correction to the absorptive part of order $(\Delta_{BCS}/qv_F)^2$ is in the ladder, since the bare absorptive part has none, and Eq. (15.6) contains the ladder to all orders. Outside the model lie disorder (Sect. 15.3.3) and the frequency dependence of the gap in strong coupling, which is small in aluminum, where $2\Delta_{BCS}/T_c = 3.54$ against the weak-coupling 3.53 (Sect. 7.2), and is not sized here. So is the film geometry, once the length $1/q$, about a micrometer on the edge, exceeds the film thickness (Sect. 15.3.4).

The superconducting part of a rate carries three nuisances of the kinds of Sect. 8.8, which the chapter summary collects. The first is the gap, a threshold nuisance Eq. (8.74): $\delta E = 2\,\delta\Delta_{BCS}$ moves the edge, and at fixed ω it moves y. Its size is the uncertainty of the gap

of the sensor in use, the bulk 0.18 meV or the measured film gap of Sect. 15.1.6. The second is the relaxation rate $\gamma_{\rm rel}$ of the film, a tilt that acts only near the edge. At $y \geq 3$ it moves the loss function by less than 6 % relative to $\mathfrak{g}_{\rm I} L^{\rm nor}$, and at $y = 2.2$ by a factor of 4.6 between the clean limit and $\mathfrak{l}_e = 30\,\text{nm}$ (Sect. 15.3.3). The third is the edge remainder, the difference between the loss function a rate calculation uses and the conserving one. With $\mathfrak{g}_{\rm I} L^{\rm nor}$ as the central value, the remainder at $y \geq 3$ has no fixed sign. In the clean limit it runs from $-2.0\,\%$ to $+8.5\,\%$ over all momentum transfers that halo dark matter reaches, the lower end at about four times the edge momentum of $y = 3$. With impurities it reaches $+9.1\,\%$ on the edge, and off the edge it has not been computed. It is therefore a two-sided normalization nuisance, $-2\,\%$ to $+9\,\%$, with the standard-normal constraint of Eq. (8.72). The spread between the two Kramers–Kronig prescriptions of the bare bubble cannot serve as its template, since the conserving value lies outside that spread at every y of Table 15.1.

Near the threshold the conserving loss function depends strongly on the mean free path, and a rate calculation excludes those energies from its signal region. The criterion we adopt is that the edge value of $L^{\rm cons}/\mathfrak{g}_{\rm I} L^{\rm nor}$ vary by at most 10 % over the mean free paths from 30 nm to 100 µm. It holds at $y = 3$, where the values agree to 6 %, and fails at $y = 2.5$, where they span a factor of 1.75. The boundary lies between the two, and the energies $y < 3$ are excluded. In that window the remainder is not bounded at all for a film: for $\mathfrak{l}_e \lesssim 300\,\text{nm}$ the conserving value at $y = 2.2$ lies at or below even the cut bare value, so no central value taken from the bare bubble is conservative. In a crystal with $\mathfrak{l}_e \geq 10\,\mu\text{m}$ the clean column of Table 15.1 holds to 4 % down to $y = 2.2$, and the excluded energies can be restored with it. The deviations from $\mathfrak{g}_{\rm I} L^{\rm nor}$ are largest on the edge, so the exclusion covers the momenta above it as well.

15.3 Films and disorder

An electron in a clean aluminum crystal crosses a Cooper pair, 2.4 µm across (Sect. 7.2), without scattering, and the loss functions of Sects. 15.1 and 15.2 assume exactly that. The sensors that count broken pairs, however, are films. In a 50 nm aluminum film an electron reaches a surface after at most about 50 nm. In the amorphous tungsten silicide of the QROCODILE dark-matter search, whose sensors are superconducting microwires patterned from a layer 2 nm thick [12], it scatters after a few tenths of a nanometer, about once per interatomic distance. At the momentum transfers of keV-mass dark matter these collisions raise the normal-state loss function of the aluminum film about twentyfold. They also change the loss function at the pair-breaking edge, and a film thinner than $1/q$ screens like a sheet rather than like a bulk metal. We will compute these effects for aluminum with a relaxation rate, see why a relaxation rate cannot describe amorphous WSi, and end with the measured film properties that a rate calculation needs.

15.3.1 Mean free path and Anderson's theorem

The rate $\gamma_{\rm rel}$ at which collisions relax the momentum of an electron (Sect. 4.5) sets two lengths. An electron at the Fermi surface travels the *mean free path*

$$\mathfrak{l}_e = \frac{v_{\rm F}}{\gamma_{\rm rel}}, \qquad D_{\rm diff} = \frac{v_{\rm F}\mathfrak{l}_e}{3}, \tag{15.12}$$

between collisions, and over many collisions it performs a random walk with diffusion constant $D_{\rm diff}$. The second relation follows from free flights of exponentially distributed length, which have $\langle s^2 \rangle = 2\mathfrak{l}_e^2$, one per time $1/\gamma_{\rm rel}$ in a random direction, so that $\langle r^2 \rangle = 2\mathfrak{l}_e^2 \gamma_{\rm rel} t = 6 D_{\rm diff} t$. The dc resistivity measures $\mathfrak{l}_e$. The Drude conductivity of Sect. 4.5 is $n_e e^2/(m_e \gamma_{\rm rel})$ in SI

Table 15.2: Disorder parameters of two superconducting sensor materials. Aluminum film: free-electron gas with $k_\mathrm{F} = 1.75\,\text{Å}^{-1}$ and $v_\mathrm{F} = 6.75 \times 10^{-3}$, mean free path taken equal to the thickness of 50 nm, coherence length $v_\mathrm{F}/\pi\Delta_\mathrm{BCS}$ with the bulk gap 0.18 meV. Amorphous W_xSi_{1-x}: 5 nm films with $x = 0.75$–0.85 [14, Table I]; the range of ℓ_e comes from combining the measured diffusion constant (lower end) or the measured resistivity (upper end) with the free-electron gas at $E_\mathrm{F} = 7\,\text{eV}$; the coherence length is the measured Ginzburg–Landau value. The pair-breaking edge is $q = 2\Delta_\mathrm{BCS}/v_\mathrm{max}$, the smallest momentum transfer at which a halo particle can deposit $2\Delta_\mathrm{BCS}$; WSi has $\Delta_\mathrm{BCS} = 0.263\,\text{meV}$.

	Al film, 50 nm	amorphous WSi
mean free path ℓ_e	50 nm	0.13–0.41 nm
relaxation rate γ_rel	27 meV	2.5–7.7 eV
$k_\mathrm{F}\ell_e$	874	1.8–5.5
ℓ_e/ξ_coh	0.021	0.02–0.06
$q\ell_e$ at the pair-breaking edge	0.034	$(1.4\text{–}4.1)\times10^{-4}$
$q\ell_e$ at $q = 1\,\text{keV}$	253	0.7–2.1

units, and with $n_e = k_\mathrm{F}^3/3\pi^2$ and $v_\mathrm{F} = \hbar k_\mathrm{F}/m_e$ the resistivity ρ_el obeys

$$\rho_\mathrm{el}\,\ell_e = \frac{3\pi^2\hbar}{e^2 k_\mathrm{F}^2}, \tag{15.13}$$

a constant of the metal. For the free-electron gas of aluminum (Sect. 15.1.1) it is $3.98 \times 10^{-16}\,\Omega\,\text{m}^2$, and a film with $\ell_e = 50\,\text{nm}$ has a residual resistivity of $0.80\,\mu\Omega\,\text{cm}$ and $\gamma_\mathrm{rel} = 26.7\,\text{meV}$. We take ℓ_e equal to the thickness, $d_\mathrm{film} = 50\,\text{nm}$, as for a film whose surfaces scatter electrons diffusely; a measured residual resistivity replaces this choice through Eq. (15.13).

Which theory applies depends on three ratios: $q\ell_e$, ℓ_e/ξ_coh and $k_\mathrm{F}\ell_e$. The first, $q\ell_e$, is the distance between collisions in units of the length $1/q$ of the perturbation. For $q\ell_e \gg 1$ the electrons are *ballistic* on the scale $1/q$, and the response is that of the clean gas; for $q\ell_e \ll 1$ they are *diffusive*, and a density disturbance spreads by diffusion. The second compares ℓ_e with the coherence length $\xi_\mathrm{coh} = v_\mathrm{F}/\pi\Delta_\mathrm{BCS}$, the size of a pair (Sect. 7.2). A superconductor with $\ell_e \gg \xi_\mathrm{coh}$ is *clean*, and one with $\ell_e \ll \xi_\mathrm{coh}$ is in the *dirty limit*: each pair is spread over many impurities. For aluminum $\xi_\mathrm{coh} = 2.4\,\mu\text{m}$ with the free-electron v_F, and the coherence length tabulated for aluminum is $1.6\,\mu\text{m}$ [13, Ch. 10, Table 5]; the clean results of Sects. 15.1 and 15.2 need a mean free path long compared with either, which only bulk high-purity crystals have. The third ratio, $k_\mathrm{F}\ell_e$, is the mean free path in units of the electron wavelength. Disorder is weak for $k_\mathrm{F}\ell_e \gg 1$, where an electron still has a well-defined momentum between collisions, and strong near the Ioffe–Regel limit $k_\mathrm{F}\ell_e \approx 1$, where it does not. Table 15.2 lists the three numbers for the aluminum film and for amorphous WSi. The aluminum film is weakly disordered, dirty, and diffusive at the momenta that keV-mass dark matter transfers; WSi is strongly disordered.

In the dirty limit the plane waves of Sect. 7.2 are no longer eigenstates, and the BCS construction has to be redone. Anderson [15] noticed that nonmagnetic impurities do not break time reversal. Every exact one-electron eigenstate ψ_n of the disordered metal has a time-reversed partner ψ_n^* at the same energy. He paired ψ_n with ψ_n^*, in place of $\boldsymbol{k}$ with $-\boldsymbol{k}$, and took the attraction constant between the exact states within ω_D of the Fermi level. The reduced Hamiltonian of Eq. (7.20) then leads to the gap equation of Eq. (7.26) with the density of states of the exact states at the Fermi level in place of $\mathcal{N}_\mathrm{F}$. Weak disorder changes that density of states only at relative order $1/k_\mathrm{F}\ell_e$, and nonmagnetic disorder leaves Δ_BCS and T_c at their clean values. This is *Anderson's theorem*. Magnetic impurities break time

reversal, the partner states no longer pair, and the gap falls. For a weakly disordered film the pair-breaking threshold is still $2\Delta_{\rm BCS}$, with the gap of the film, measured through its T_c or by tunneling (Sect. 15.1.6). The response, on the other hand, does change with disorder, because the matrix elements of $e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$ between exact states are not those between plane waves.

15.3.2 Relaxation-time loss functions

Collisions enter the loss function through the Mermin dielectric function of Eq. (4.62), which relaxes the electrons toward local equilibrium, conserves their number and is causal (Appendix D.9). At the momentum transfers of light dark matter $q \ll k_{\rm F}$, and the Lindhard bubble at a complex frequency z in the upper half plane is Eq. (15.9) continued off the real axis,

$$R^0(q,z) = -2\mathcal{N}_{\rm F}\, \mathfrak{f}_{\rm L}(\mathfrak{w}), \qquad \mathfrak{w} = \frac{z}{q v_{\rm F}}, \tag{15.14}$$

where $\ell(\mathfrak{w}) = \ln[(\mathfrak{w}+1)/(\mathfrak{w}-1)]$ in $\mathfrak{f}_{\rm L}$ is the principal logarithm of the complex argument, the retarded branch of Eq. (C.75). With $\varepsilon_{\rm Lin} - 1 = (k_{\rm TF}^2/q^2)\,\mathfrak{f}_{\rm L}$ from Eq. (4.53), the factors $i\gamma_{\rm rel}/\omega$ of Eq. (4.62) combine into

$$\varepsilon_{\rm Mer}(q,\omega) = 1 + \frac{k_{\rm TF}^2}{q^2}\,\frac{(\omega + i\gamma_{\rm rel})\,\mathfrak{f}_{\rm L}(\mathfrak{w}_\gamma)}{\omega + i\gamma_{\rm rel}\,\mathfrak{f}_{\rm L}(\mathfrak{w}_\gamma)}, \qquad \mathfrak{w}_\gamma = \frac{\omega + i\gamma_{\rm rel}}{q v_{\rm F}}. \tag{15.15}$$

The size of $|\mathfrak{w}_\gamma|$ is the size of $1/q\mathfrak{l}_e$. For dark matter $\omega \le q v_{\rm max} < q v_{\rm F}$, so $|\omega + i\gamma_{\rm rel}|/q v_{\rm F}$ lies between $\gamma_{\rm rel}/q v_{\rm F} = 1/q\mathfrak{l}_e$ and $1/q\mathfrak{l}_e + v_{\rm max}/v_{\rm F}$.

In the diffusive regime, $q\mathfrak{l}_e \ll 1$, the tail $\mathfrak{f}_{\rm L} \simeq -q^2 v_{\rm F}^2/3(\omega + i\gamma_{\rm rel})^2$ applies. Since then also $\omega \ll \gamma_{\rm rel}$, the numerator of the fraction in Eq. (15.15) is $iq^2 D_{\rm diff}$ and the denominator $\omega + iq^2 D_{\rm diff}$, with $D_{\rm diff}$ of Eq. (15.12). The dielectric function becomes

$$\varepsilon_{\rm Mer} \simeq 1 + \frac{k_{\rm TF}^2 D_{\rm diff}}{D_{\rm diff} q^2 - i\omega} \qquad (q\mathfrak{l}_e \ll 1), \tag{15.16}$$

the diffusion pole: a density disturbance of wavevector q decays at the rate $D_{\rm diff} q^2$. For $\omega \ll D_{\rm diff} q^2$ it reduces to the static screening $k_{\rm TF}^2/q^2$, and for $\omega \gg D_{\rm diff} q^2$ to $ik_{\rm TF}^2 D_{\rm diff}/\omega$, the Drude form $i\sigma_{\rm cond}/\omega$ of Eq. (4.50) with the dc conductivity $k_{\rm TF}^2 D_{\rm diff} = \omega_p^2/\gamma_{\rm rel}$, by Eq. (4.53). The loss function is the same in both regimes. With $|\varepsilon| \gg 1$, $\mathrm{Im}\,\varepsilon_{\rm Mer} = k_{\rm TF}^2 D_{\rm diff}\omega/(D_{\rm diff}^2 q^4 + \omega^2)$ and $|\varepsilon_{\rm Mer}|^2 = k_{\rm TF}^4 D_{\rm diff}^2/(D_{\rm diff}^2 q^4 + \omega^2)$, so $L = \omega/k_{\rm TF}^2 D_{\rm diff} = \omega\gamma_{\rm rel}/\omega_p^2$ at every $\omega/D_{\rm diff} q^2$: the Drude tail of Sect. 4.5, independent of q.

In the ballistic regime, $q\mathfrak{l}_e \gg 1$, the small argument $|\mathfrak{w}_\gamma| \ll 1$ applies, and on the retarded branch $\ell(\mathfrak{w}) \simeq -i\pi + 2\mathfrak{w}$, so $\mathfrak{f}_{\rm L} \simeq 1 + i\pi\mathfrak{w}_\gamma/2$. The fraction in Eq. (15.15) is then $1 + i\pi\omega/2q v_{\rm F}$, with $\gamma_{\rm rel}$ gone at this order, and $L = \pi\omega q/(2k_{\rm TF}^2 v_{\rm F})$. This is the low-energy corner of the clean Lindhard loss function of Sect. 4.5, written with Eq. (4.53).

Loss function of a metal with collisions

For $q \ll k_{\rm F}$, $q \ll k_{\rm TF}$ and $\omega \le q v_{\rm max}$, the Mermin loss function has the limits

$$L_{\rm Mer}(q,\omega) \simeq \begin{cases} \omega\gamma_{\rm rel}/\omega_p^2 = \omega/\sigma_{\rm dc}, & q\mathfrak{l}_e \ll 1, \\ \pi\omega q/(2k_{\rm TF}^2 v_{\rm F}), & q\mathfrak{l}_e \gg 1, \end{cases} \tag{15.17}$$

with $\sigma_{\rm dc} = \omega_p^2/\gamma_{\rm rel}$ the dc conductivity. The ballistic form is $\pi q\mathfrak{l}_e/6$ times the diffusive one. In SI units the diffusive form is $L = \varepsilon_0\omega\rho_{\rm el}$, fixed by the measured resistivity alone.

Collisions therefore raise the loss function at small q, by the factor $6/\pi q\mathfrak{l}_e$. In a clean metal a slow potential of long wavelength is absorbed only by the electrons that move almost

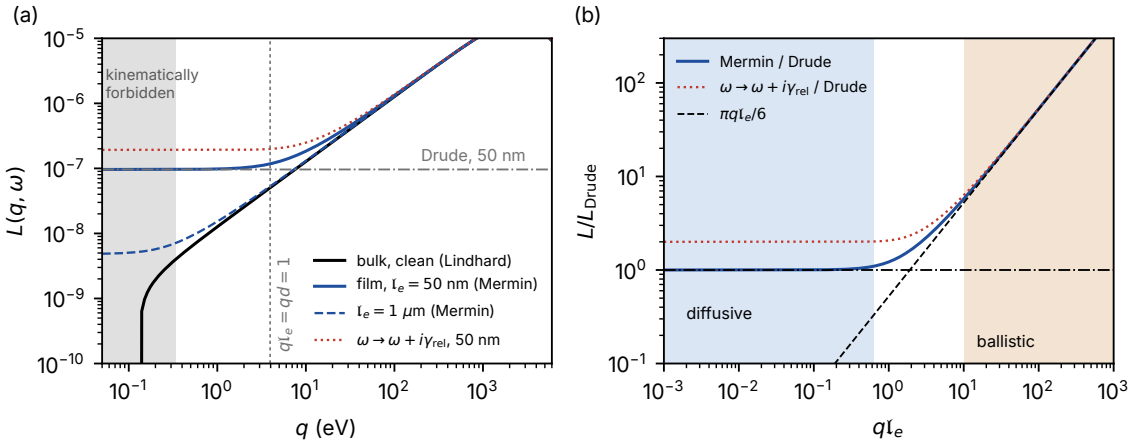


Figure 15.5: Normal-state loss function of aluminum in a film and in the bulk. (a) $L(q,\omega)$ at $\omega = 0.90\,\text{meV}$ ($5\Delta_{\text{BCS}}$ with the bulk gap) against the momentum transfer q, for a free-electron gas with $k_{\text{F}} = 1.75\,\text{Å}^{-1}$ and $v_{\text{F}} = 6.75 \times 10^{-3}$. Curves: clean Lindhard function (black), Mermin function with mean free path $\ell_e = 50\,\text{nm}$ (blue, $\gamma_{\text{rel}} = 26.7\,\text{meV}$) and $\ell_e = 1\,\mu\text{m}$ (blue dashed), Lindhard function at the complex frequency $\omega + \text{i}\gamma_{\text{rel}}$ with $\ell_e = 50\,\text{nm}$ (red dotted), and the Drude tail $\omega\gamma_{\text{rel}}/\omega_p^2$ for 50 nm (gray). Gray band: $q < \omega/v_{\text{max}}$, out of reach of halo dark matter. Dotted vertical line: $q\ell_e = 1$, which for this film is also $qd_{\text{film}} = 1$. (b) Ratio of the Mermin loss function (blue) and of the complex-frequency Lindhard loss function (red dotted) to the Drude tail, against $q\ell_e$, for $q \ll k_{\text{F}}$ (Eq. (15.15)) and $\omega/\gamma_{\text{rel}} = 0.034$; dashed, the ballistic asymptote $\pi q\ell_e/6$. Shaded: within 10 % of the diffusive limit (left) and of the ballistic limit (right), that is a ratio to the limit of at most 1.1.

perpendicular to $\boldsymbol{q}$, which keep pace with it (Landau damping), and their share is ω/qv_{F}. Collisions let every electron at the Fermi surface absorb, and the absorbed power is the Joule heat of the induced current, set by the resistivity. Fig. 15.5b shows the ratio $L_{\text{Mer}}/L_{\text{Drude}}$ against $q\ell_e$. It is within 10 % of one for $q\ell_e \leq 0.64$ and within 10 % of $\pi q\ell_e/6$ only from $q\ell_e = 9.8$; at $q\ell_e = 1.5$ it is 1.43, almost twice the ballistic asymptote. Neither form holds in between, and there Eq. (15.15) has to be evaluated.

Fig. 15.5a shows the consequence for aluminum. At the smallest momentum transfer that can deposit 0.90 meV, $q = \omega/v_{\text{max}} = 0.34\,\text{eV}$, the 50 nm film has a loss function 24 times that of the clean crystal. The asymptotic $6/\pi q\ell_e$ is 22; the clean value lies below its own small-ω form there, since $\omega/qv_{\text{F}} = 0.39$. The enhancement persists up to $q \approx 1/\ell_e = 3.9\,\text{eV}$, which is the largest momentum transfer of dark matter lighter than 0.75 keV, and above that the two curves merge. A film with $\ell_e = 1\,\mu\text{m}$ lies between. Collisions in a film thus change the loss function for keV-mass dark matter, whereas at the momentum transfers of MeV-mass dark matter the film electrons are ballistic.

The two relaxation-time forms that appear in projections fail in opposite directions. The Drude function $1 - \omega_p^2/\omega(\omega + \text{i}\gamma_{\text{rel}})$, used at finite q, has no diffusion pole and screens a static charge perfectly at every q, which is wrong for $\omega < D_{\text{diff}}q^2$. Its loss function is nonetheless correct for $q\ell_e \ll 1$, as shown above, and wrong by the factor $\pi q\ell_e/6$ in the ballistic regime. The Lindhard function at the complex frequency $\omega + \text{i}\gamma_{\text{rel}}$ is the substitution that Sect. 4.5 showed not to conserve electron number. In the diffusive regime it has $\mathfrak{f}_{\text{L}}(\mathfrak{w}_\gamma) \simeq -q^2v_{\text{F}}^2/3(\omega + \text{i}\gamma_{\text{rel}})^2$, so $\varepsilon - 1 \simeq (\omega_p^2/\gamma_{\text{rel}}^2)(1 + 2\text{i}\omega/\gamma_{\text{rel}})$. This is the finite static dielectric constant of an insulator, with a loss function $2\omega\gamma_{\text{rel}}/\omega_p^2$, twice the Drude tail, at every $q\ell_e \ll 1$ (Fig. 15.5b), not only at $q = 0$. The QROCODILE analysis gives the Lindhard function of WSi a plasmon width of

$0.1E_\mathrm{F}$ [12, supplement]. If that width is added as $\omega \to \omega + \mathrm{i}\gamma_\mathrm{rel}$, as in Ref. [8, Eq. (S.13)], it is this substitution. At the momentum transfers of a keV that the analysis needs, the electrons of WSi are in any case neither diffusive nor ballistic (Sect. 15.3.5).

The plasmon width and the relaxation rate are different numbers. A Drude form fitted to the measured aluminum plasmon has $\omega_p = 14.9\,\mathrm{eV}$ and a width of $0.863\,\mathrm{eV}$ [8, Fig. S1], 32 times the relaxation rate of the 50 nm film. The plasmon decays into electron–hole pairs by interband transitions and by collisions alike, while the tail at meV energies, by Eq. (15.17), is set by the dc resistivity, which measures momentum relaxation at the Fermi surface only. A tail extrapolated from the plasmon width overstates the loss function of the 50 nm film at meV energies by that factor of 32.

A causal response alone is not enough: every calculation of the superconducting ε, which at $T = 0$ is real below $2\Delta_\mathrm{BCS}$ and has its branch cut on the pair continuum $|\omega| \geq 2\Delta_\mathrm{BCS}$, must avoid three errors. First, the logarithms of the Lindhard function and of the BCS bubbles must be taken on the retarded branch Eq. (C.75). The other branch, which the limit of a vanishing width does not exclude by itself, suppresses the computed rate of an aluminum detector at low mass by up to seven orders of magnitude [8]. Second, a time-ordered bubble coincides with the retarded one at $T = 0$ only for $\omega > 0$, whereas a Kramers–Kronig integral and a sum rule need $\omega < 0$ as well, and a calculation at $T > 0$ needs the thermal factors of the retarded form. Third, a relaxation rate must enter through a construction that keeps the response conserving and dissipative, as the Mermin form does (Appendix D.9). The Lindhard function at $\omega + \mathrm{i}\gamma_\mathrm{rel}$ is analytic but not conserving, and a local-field factor with an imaginary part, fitted to the aluminum plasmon, makes the loss function negative at small ω [8, Fig. S4], which a passive medium cannot do, since L is proportional to the structure factor, a sum of positive transition rates (Sect. 4.4).

15.3.3 The superconducting film

On the kinematic edge of $y = \omega/\Delta_\mathrm{BCS} = 2.2$ the momentum transfer is $q = 2.2\,\Delta_\mathrm{BCS}/v_\mathrm{max} = 0.15\,\mathrm{eV}$, whatever v_F is, so an aluminum film with $\ell_e = 30\,\mathrm{nm}$ has $q\ell_e = 0.023$ there. Its electrons are diffusive on the scale of the probe, and the clean response of Sect. 15.1 does not apply. The conserving response with Born scattering by nonmagnetic impurities at $T = 0$ replaces it. In Nambu space the impurity potential couples through $\boldsymbol{\eta}$, the Nambu matrix of the density, so the Born self-energy is $\boldsymbol{\eta}\,\mathsf{G}\,\boldsymbol{\eta}$ averaged over the Fermi surface. The same impurity ladder dresses the density vertex and the two pair vertices; with both, the response stays conserving (Appendix E.20). Its normal state is the Mermin function Eq. (4.62) at the same relaxation rate $\gamma_\mathrm{rel} = v_\mathrm{F}/\ell_e$. Table 15.1 lists its edge values for 1 µm and 30 nm, divided by $\mathfrak{g}_\mathrm{I}$ times that normal state, and we use the same normalization for every number in this subsection.

At $y \geq 3$ the ratio hardly moves: it stays between 1.008 and 1.091 for every mean free path from 100 µm down to 30 nm. At $y \leq 2.5$, however, it moves a great deal. At $y = 2.2$ it falls from 0.87 in the clean limit to 0.60 at $\ell_e = 1\,\mu\mathrm{m}$, 0.38 at 300 nm and 0.19 at 30 nm, and at $y = 2.5$ from 1.00 to 0.57. The clean values hold to 4 % down to $y = 2.2$ only for $\ell_e \geq 10\,\mu\mathrm{m}$. Below about $\ell_e = 300\,\mathrm{nm}$ the conserving value at $y = 2.2$ falls below even the cut bare value of Table 15.1, 0.40. In a film the suppression near the threshold therefore returns, now through the inductive response of the condensate and not through the bare bubble's shift.

The local limit of short mean free path shows why. When $q\ell_e \ll 1$ and $D_\mathrm{diff}q^2 \ll \Delta_\mathrm{BCS}$, neither an electron nor a pair resolves the spatial variation of the potential, and the film responds locally. The induced density then follows from the induced current by the continuity equation, longitudinal and transverse responses coincide (Eq. (4.52)), and $\varepsilon = 1 + \mathrm{i}\sigma_\mathrm{cond}(\omega)/\omega$ by Eq. (4.50), in the superconducting state as in the normal one. The conductivity of a dirty

superconductor is the Mattis–Bardeen conductivity, derived in Sect. 15.4.1 and Appendix D.40. At $T = 0$ its real part σ_1/σ_n is the case-II pair-creation integral of Eq. (7.30), which is zero below $2\Delta_{\mathrm{BCS}}$ and rises continuously from zero above it. Its imaginary part σ_2 is the Kramers–Kronig partner of σ_1, and it contains the weight $(\pi^2/2)\Delta_{\mathrm{BCS}}\sigma_n$ missing from σ_1 below $2\Delta_{\mathrm{BCS}}$, which the condensate carries at $\omega = 0$. Since $|\varepsilon| \gg 1$, $-1/\varepsilon = \mathrm{i}\omega/\sigma_{\mathrm{cond}}$, and

$$L = \frac{\omega\,\sigma_1}{\sigma_1^2 + \sigma_2^2}, \qquad \frac{L_{\mathrm{s}}}{L_{\mathrm{n}}} = \frac{\sigma_n\,\sigma_1}{\sigma_1^2 + \sigma_2^2} \qquad (q\ell_e \ll 1,\ D_{\mathrm{diff}}q^2 \ll \Delta_{\mathrm{BCS}}), \tag{15.18}$$

with $L_{\mathrm{n}} = \omega/\sigma_n$ the Drude tail of Eq. (15.17) and L_{s} the superconducting loss function. Near the threshold the ratio is small for two reasons. By continuity the density couples through the current, which is odd under time reversal, so the coherence factor is that of case II, which vanishes at $2\Delta_{\mathrm{BCS}}$, rather than that of case I with its step of $\pi/2$. And σ_2 exceeds σ_1 there, 0.757 against 0.139 times σ_n at $y = 2.2$: the supercurrent of the condensate short-circuits the slow potential. Divided by $\mathfrak{g}_{\mathrm{I}}$, the normalization of Table 15.1, Eq. (15.18) gives 0.157 at $y = 2.2$, 0.985 at $y = 3$, 1.094 at $y = 5$ and 1.023 at $y = 10$, against 0.187, 1.008, 1.091 and 1.021 in the 30 nm column: Eq. (15.18) is the end point of the table, and it needs only the measured conductivity of the film. At 30 nm and $y = 2.2$ the conserving value still lies 20 % above the local limit, although $D_{\mathrm{diff}}q^2 = 0.042\,\Delta_{\mathrm{BCS}}$ is already small there; the origin of that excess has not been established.

15.3.4 Film thickness

A film thicker than $1/q$ responds as the bulk. Its surfaces spread the momentum along the normal over a range $\sim 1/d_{\mathrm{film}}$, a relative correction of order $1/qd_{\mathrm{film}}$, and they carry surface plasmons near $\omega_p/\sqrt{2} \approx 11\,\mathrm{eV}$, far above the energies that dark matter deposits at these momenta. A film thinner than $1/q$ responds as a sheet. The external potential is then uniform across the thickness, and the film responds to its in-plane wavevector $\boldsymbol{q}_\parallel$ with the induced density $d_{\mathrm{film}}R(q_\parallel, \omega)\phi$ per area. This sheet of charge produces the potential $e^2/[(\varepsilon_a + \varepsilon_b)q_\parallel]\,\mathrm{e}^{-q_\parallel|z|}$ per unit density, with ε_a and ε_b the dielectric constants of the media on the two sides. The two-dimensional Fourier transform of $e^2/4\pi r$ is $e^2/2q_\parallel$, and the image charges of the two half spaces replace the 2 by $\varepsilon_a + \varepsilon_b$. In the bulk the same density produces e^2/q^2. Therefore

$$\varepsilon_{\mathrm{sh}}(q_\parallel, \omega) - 1 = \frac{q_\parallel d_{\mathrm{film}}}{\varepsilon_a + \varepsilon_b}\,[\varepsilon(q_\parallel, \omega) - 1] \qquad (qd_{\mathrm{film}} \ll 1), \tag{15.19}$$

and the screening factor $1/|\varepsilon|^2$ in the structure factor $-2\,\mathrm{Im}\,R/|\varepsilon|^2$ becomes $1/|\varepsilon_{\mathrm{sh}}|^2$, larger by $[(\varepsilon_a + \varepsilon_b)/q_\parallel d_{\mathrm{film}}]^2$ while both dielectric functions are large. For a slab in vacuum this is the leading term in $q_\parallel d_{\mathrm{film}}$ of the exact slab response of Lasenby and Prabhu [16, Eq. (19)], whose next term is of relative order $q_\parallel d_{\mathrm{film}}$. A thin metal film screens a long-wavelength potential far less than the bulk, because the field of its induced charge spreads outside it. The factor depends on the direction of $\boldsymbol{q}$ through $q_\parallel$, which makes the rate of a thin film directional. For this reason the QROCODILE analysis solves the electrostatic problem of a single 2 nm WSi layer in vacuum with a Drude dielectric function; it finds the anisotropy strong at $q \sim 1\,\mathrm{nm}^{-1}$ and already significant at $q \sim 1\,\mathrm{keV}$ for energies above 4 eV [12, supplement]. For the 50 nm aluminum film $qd_{\mathrm{film}} = 1$ at $q = 3.9\,\mathrm{eV}$, the same momentum at which $q\ell_e = 1$; for dark matter lighter than 0.75 keV the film is a diffusive sheet at every momentum transfer. Stacks of sheets and the full slab need the response of a layered medium, which we do not treat here.

Worked example 15.3: The Mermin loss function of a 50 nm aluminum film

Take the free-electron gas of aluminum with $k_\mathrm{F} = 3.45\,\mathrm{keV}$ and $v_\mathrm{F} = 6.75 \times 10^{-3}$, for which $k_\mathrm{TF} = 4.05\,\mathrm{keV}$ and $\omega_p = 15.78\,\mathrm{eV}$ by Eq. (4.53), and a mean free path $\mathfrak{l}_e = d_\mathrm{film} = 50\,\mathrm{nm} = 0.253\,\mathrm{eV}^{-1}$. Then $\gamma_\mathrm{rel} = v_\mathrm{F}/\mathfrak{l}_e = 26.7\,\mathrm{meV}$ and $D_\mathrm{diff} = 5.70 \times 10^{-4}\,\mathrm{eV}^{-1}$ ($337\,\mathrm{cm^2\,s^{-1}}$). Let a dark-matter particle deposit $\omega = 5\Delta_\mathrm{BCS} = 0.90\,\mathrm{meV}$ with the bulk gap, at the smallest momentum transfer that allows it, $q = \omega/v_\mathrm{max} = 0.340\,\mathrm{eV}$. Then $q\mathfrak{l}_e = 0.086$: the film is diffusive.

Mermin function. With $qv_\mathrm{F} = 2.29\,\mathrm{meV}$, $\mathfrak{w}_\gamma = 0.392 + 11.62\,\mathrm{i}$, and Eq. (15.14) gives $\mathfrak{f}_\mathrm{L} = (2.449 + 0.165\,\mathrm{i}) \times 10^{-3}$, within 0.5 % of the tail $-1/3\mathfrak{w}_\gamma^2$. Eq. (15.15) yields $\varepsilon_\mathrm{Mer} = (0.407 + 10.35\,\mathrm{i}) \times 10^6$, and the diffusion pole Eq. (15.16), with $D_\mathrm{diff}q^2 = 0.066\,\mathrm{meV}$, is $(0.755 + 10.33\,\mathrm{i}) \times 10^6$: the imaginary parts agree, and the real parts differ at the order $\omega/\gamma_\mathrm{rel}$ that the pole drops. The loss function is $L_\mathrm{Mer} = 9.65 \times 10^{-8}$, 0.2 % above the Drude tail $\omega\gamma_\mathrm{rel}/\omega_p^2 = 9.63 \times 10^{-8}$. The same number is $\varepsilon_0\omega\rho_\mathrm{el}$ with the residual resistivity $0.80\,\mu\Omega\,\mathrm{cm}$ of Eq. (15.13).

Comparisons. The clean Lindhard function at the same point is $\varepsilon_\mathrm{Lin} = (1.19 + 0.88\,\mathrm{i}) \times 10^8$, with $L = 4.01 \times 10^{-9}$: the film absorbs 24 times more. The Lindhard function at $\omega + \mathrm{i}\gamma_\mathrm{rel}$ is $\varepsilon = (3.48 + 0.234\,\mathrm{i}) \times 10^5$ and $L = 1.93 \times 10^{-7}$, twice the Mermin value; its imaginary part is 440 times too small and $|\varepsilon|$ 30 times too small, the screening of an insulator.

Superconducting film. The film is dirty ($\mathfrak{l}_e/\xi_\mathrm{coh} = 0.021$), and $D_\mathrm{diff}q^2 = 0.37\,\Delta_\mathrm{BCS}$: Eq. (15.18) applies roughly. At $y = 5$ it gives $L_\mathrm{s}/L_\mathrm{n} = 1.259$ and $L_\mathrm{s} = 1.21 \times 10^{-7}$. In the normalization of Table 15.1 that is 1.094, and clean, intermediate and dirty films lie between 1.057 and 1.094 at this energy, within 4 %; at $y = 2.2$ the clean value is 5.5 times the local limit.

Thickness. Finally $qd_\mathrm{film} = 0.086$: the film is a sheet at this momentum, and in vacuum Eq. (15.19) reduces the screening by $q_\parallel d_\mathrm{film}/2$ for $\boldsymbol{q}$ in the plane, which multiplies the structure factor by $(2/0.086)^2 = 540$ to leading order in $q_\parallel d_\mathrm{film}$; the exact slab response of Lasenby and Prabhu gives 588. The bulk loss function computed here is the input to the sheet response, not the final answer.

15.3.5 Amorphous tungsten silicide

Amorphous $\mathrm{W}_x\mathrm{Si}_{1-x}$ has no lattice, and its transport is measured, not computed. In 5 nm films with $x = 0.75$–0.85 the sheet resistance, the resistance of a square of film, is 326–417 Ω, a resistivity $\rho_\mathrm{el} = R_\square d_\mathrm{film}$ of 163–209 μΩ cm, 205 to 263 times that of the aluminum film of Worked example 15.3 [14, Table I]. The magnetic field that destroys superconductivity rises below T_c at a rate set by how fast the electrons diffuse, and its slope yields the diffusion constant 0.70–0.75 $\mathrm{cm^2\,s^{-1}}$. In a diffusive conductor the current is the diffusion current of the electrons that a field pushes off equilibrium, so the conductivity is $\sigma = e^2(2\mathcal{N}_\mathrm{F})D_\mathrm{diff}$, the Einstein relation. With the measured resistivity and diffusion constant of each film it gives the density of states $2\mathcal{N}_\mathrm{F} = (2.7\text{–}3.3) \times 10^{47}\,\mathrm{m^{-3}\,J^{-1}}$, and at most $3.4 \times 10^{47}\,\mathrm{m^{-3}\,J^{-1}}$ for any combination of the measured values. The table of Zhang and collaborators lists up to $4.1 \times 10^{47}\,\mathrm{m^{-3}\,J^{-1}}$ for $x = 0.85$, which their own resistance and diffusion constant do not reproduce. The Ginzburg–Landau coherence length, the size of a pair measured near T_c, is 7.0–7.4 nm, and T_c is 3.8–4.1 K. The 2 nm $\mathrm{W}_{0.45}\mathrm{Si}_{0.55}$ layer of the QROCODILE detector has $T_c = 1.73\,\mathrm{K}$ [12], which gives $\Delta_\mathrm{BCS} = 1.764\,T_c = 0.263\,\mathrm{meV}$. Gan and collaborators use $\Delta = 0.26\,\mathrm{meV}$ and $E_\mathrm{F} = 7.0\,\mathrm{eV}$ for WSi with a clean BCS response [3].

The free-electron gas at $E_\mathrm{F} = 7\,\mathrm{eV}$ has $k_\mathrm{F} = 2.67\,\mathrm{keV}$, $v_\mathrm{F} = 5.23 \times 10^{-3}$ and $2\mathcal{N}_\mathrm{F} = 1.13 \times 10^{47}\,\mathrm{m^{-3}\,J^{-1}}$. The density of states from the Einstein relation is 2.4–2.9 times larger: the

gas does not describe these electrons, and the mean free path depends on which free-gas number is combined with the data. With the measured D_{diff} and Eq. (15.12), $\ell_e = 3D_{\mathrm{diff}}/v_{\mathrm{F}} = 0.13$–$0.14\,\mathrm{nm}$ and $k_{\mathrm{F}}\ell_e = 1.8$–$1.9$. With the measured resistivity and the free-gas $\omega_p = 10.8\,\mathrm{eV}$, $\gamma_{\mathrm{rel}} = \varepsilon_0\omega_p^2\rho_{\mathrm{el}} = 2.5$–$3.3\,\mathrm{eV}$, $\ell_e = 0.32$–$0.41\,\mathrm{nm}$ and $k_{\mathrm{F}}\ell_e = 4.3$–$5.5$. Either way $k_{\mathrm{F}}\ell_e = 2E_{\mathrm{F}}/\gamma_{\mathrm{rel}}$ is a few: the relaxation rate is comparable with the Fermi energy and the mean free path with the electron wavelength $2\pi/k_{\mathrm{F}} = 0.46\,\mathrm{nm}$. The QROCODILE width $0.1E_{\mathrm{F}} = 0.7\,\mathrm{eV}$ is 3.6 to 11 times smaller than these rates.

At the momentum transfers of MeV-mass dark matter, $q = 1$ to $5\,\mathrm{keV}$, this leaves no small parameter. There $q/k_{\mathrm{F}} = 0.37$–1.9 and $q\ell_e = 0.7$–3.4 with the shorter mean free path, while Eq. (15.14) needs $q \ll k_{\mathrm{F}}$, the limits Eq. (15.17) need $q\ell_e$ far from one, and a relaxation time itself needs $k_{\mathrm{F}}\ell_e \gg 1$, so that an electron has a momentum to relax. The Mermin function, the Drude function and the Lindhard function with a width are then all outside the conditions under which they were derived, and no correction factor to any of them can be quoted. At the pair-breaking edge the situation is the opposite. There $q = 2\Delta_{\mathrm{BCS}}/v_{\mathrm{max}} = 0.20\,\mathrm{eV}$ and $q\ell_e \approx 10^{-4}$, and the diffusion pole Eq. (15.16) holds for any conductor, since it needs only the continuity equation and Ohm's law with the Einstein relation. The normal-state loss function there is $\varepsilon_0\omega\rho_{\mathrm{el}} = (1.2$–$1.5) \times 10^{-5}$ at $\omega = 2\Delta_{\mathrm{BCS}}$, fixed by the measured resistivity.

The superconducting part lies outside mean-field BCS. WSi is deep in the dirty limit, $\ell_e/\xi_{\mathrm{coh}} \approx 0.02$, where Anderson's theorem would keep the clean gap and Eq. (15.18) would give the edge. But the theorem rests on disorder being weak: the density of states of the exact states, and the attraction between partners, change at relative order $1/k_{\mathrm{F}}\ell_e$, which is not small here. In addition, slow diffusion enhances the Coulomb repulsion between electrons, the Altshuler–Aronov effect. The enhanced repulsion competes with the pairing attraction and lowers T_c, the Finkel'stein mechanism [17, Sect. III.2]. BCS theory contains neither, and in strongly disordered films the gap also varies from place to place.

The rate of an amorphous film therefore needs the configurational average of the loss function over the amorphous structures, $\langle \mathrm{Im}[-(\varepsilon^{-1})_{\mathbf{00}}] \rangle$, and the distribution of the local gap. The average differs from the loss function of the averaged ε at the order of the variance of ε. Appendix E.20 specifies the calculation, and a reader can skip its ingredients here. Simulated cooling of a melt with a machine-learned interatomic potential supplies the amorphous structures, and density-functional theory a ground state for each. The conductivity computed from those states (the Kubo–Greenwood formula) is checked against the measured resistivity, and the dielectric matrix is computed in the random-phase approximation with local fields. The self-consistent BCS equations solved in real space (Bogoliubov–de Gennes) on a Wannier tight-binding model of each structure, with the attraction fitted to the measured T_c, give the local gap. The disorder-averaged loss function of amorphous WSi at momentum transfers of a keV, and the distribution of its local gap, are open; the calculation of Appendix E.20 is designed to supply them.

A rate for a superconducting film thus needs measured properties besides the bulk material. The residual resistivity fixes ℓ_e through Eq. (15.13), and with it both the normal-state loss function at $q\ell_e \ll 1$ and the position of the film between the clean and dirty edges. The critical temperature or the tunneling gap of the film fixes Δ_{BCS}, which sets the threshold and the variable y of every ratio above. The thickness, compared with $1/q$, decides between bulk and sheet screening, and the dielectric constants of substrate and cover enter the sheet. And the strength of disorder, $k_{\mathrm{F}}\ell_e$, which follows from the same resistivity, decides whether a mean-field BCS response applies at all.

15.4 Absorption in superconductors and metals

A dark photon of 1 meV that is absorbed in aluminum gives up all of its energy but brings a momentum of only $m_\phi v \approx 1\,\mu\text{eV}$. An electron at the Fermi surface that takes up 1 meV must change its momentum by at least $\omega/v_\text{F} = 0.15\,\text{eV}$, a hundred thousand times more. A clean metal therefore cannot absorb the boson with one electron alone. A third body must take up the momentum: an impurity, which does so elastically, or a phonon, which carries off the momentum and part of the energy. Sect. 10.7 expressed the rates of the dark photon and the axion-like particle through the optical conductivity σ_1 at the boson mass. In a metal below its interband threshold, about 0.5 eV in aluminum [5, p. 4], that conductivity comes from these two processes and from electron–electron collisions, a channel we name but do not size. Within a few gaps of the pair-breaking edge $2\Delta_\text{BCS} = 0.36\,\text{meV}$ the superconducting state changes both processes through the coherence factors of Sect. 7.2: the phonon-assisted process carries the case-I factor and the impurity-assisted one the case-II factor. We will derive both from the structure of the current in the Nambu notation of Sect. 7.2 and recover the formula of Allen that the dark-matter literature uses for the phonon-assisted part. In a clean singlet superconductor without spin–orbit coupling, the spin current, which absorbs the axion-like particle, breaks pairs without a third body.

15.4.1 Impurity-assisted absorption

At $\boldsymbol{q} \to 0$ a dark photon, like a photon, couples to the total velocity $\hat{\boldsymbol{V}}$ of the electrons, Eq. (6.38). In a metal below its interband threshold only the band that crosses the Fermi level contributes, and $\hat{\boldsymbol{V}} = \sum_{\boldsymbol{k}\sigma} \boldsymbol{v}_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}\sigma} c_{\boldsymbol{k}\sigma}$ with the band velocity $\boldsymbol{v}_{\boldsymbol{k}} = \partial\xi_{\boldsymbol{k}}/\partial\boldsymbol{k}$. Time reversal makes $\xi_{-\boldsymbol{k}} = \xi_{\boldsymbol{k}}$, and therefore $\boldsymbol{v}_{-\boldsymbol{k}} = -\boldsymbol{v}_{\boldsymbol{k}}$. Relabeling $\boldsymbol{k} \to -\boldsymbol{k}$ in the spin-down term and writing the result with the Nambu spinors $\Psi_{\boldsymbol{k}}$ of Eq. (7.32) (Appendix D.40),

$$\hat{\boldsymbol{e}} \cdot \hat{\boldsymbol{V}} = \sum_{\boldsymbol{k}} (\hat{\boldsymbol{e}} \cdot \boldsymbol{v}_{\boldsymbol{k}})\, \Psi^\dagger_{\boldsymbol{k}} \Psi_{\boldsymbol{k}} = \sum_{\boldsymbol{k}} (\hat{\boldsymbol{e}} \cdot \boldsymbol{v}_{\boldsymbol{k}})(\gamma^\dagger_{\boldsymbol{k}1}\gamma_{\boldsymbol{k}1} - \gamma^\dagger_{\boldsymbol{k}2}\gamma_{\boldsymbol{k}2}). \tag{15.20}$$

The density is the matrix $\boldsymbol{\eta}$ in Nambu space; the uniform current is the unit matrix. The reason is the sign of the velocity under $\boldsymbol{k} \to -\boldsymbol{k}$, which the density, an even function of $\boldsymbol{k}$, does not have. The unit matrix commutes with $h_{\boldsymbol{k}}$, so the current commutes with H_MF, and in the quasiparticle basis it only counts: $\gamma^\dagger_{\boldsymbol{k}1}$ carries the velocity $\boldsymbol{v}_{\boldsymbol{k}}$ of its electron $(\boldsymbol{k}\uparrow)$ and $\gamma^\dagger_{\boldsymbol{k}2}$ the velocity $\boldsymbol{v}_{-\boldsymbol{k}}$ of $(-\boldsymbol{k}\downarrow)$. It creates no pairs. A clean BCS superconductor therefore absorbs nothing at $\boldsymbol{q} \to 0$ and $\omega > 0$, and the whole intraband weight of the f-sum rule sits at $\omega = 0$, as the supercurrent. The clean normal metal behaves the same way, with a Drude peak of zero width. The weight is $(\pi/2)(\omega_p^\text{intra})^2$, with the intraband plasma frequency

$$(\omega_p^\text{intra})^2 = 2e^2\, \mathcal{N}_\text{F}\, \langle (\hat{\boldsymbol{e}} \cdot \boldsymbol{v})^2 \rangle_\text{FS}, \tag{15.21}$$

where $\langle \dots \rangle_\text{FS}$ is the average over the Fermi surface and the factor 2 counts the spins. For free electrons it is ω_p of Sect. 4.3, 15.78 eV for the three valence electrons of aluminum; the computed bands of aluminum give 12.29 eV [10, Table IV], which we use.

An impurity breaks the conservation of momentum, and with it the commutation. The one-electron eigenstates α of the disordered metal have energies ξ_α and no definite momentum, but time reversal still pairs each with a partner $\bar\alpha$ of the same energy, and for nonmagnetic disorder the paired state is built from $(\alpha\uparrow, \bar\alpha\downarrow)$ with the same gap (Sect. 15.3). The current acquires matrix elements $\langle\alpha|\hat{\boldsymbol{e}} \cdot \hat{\boldsymbol{v}}|\beta\rangle$ between different states, and since it is odd under time reversal it is still the unit matrix in the Nambu space of $(\alpha, \bar\alpha)$. Its pair-creation amplitude then carries $u^\text{B}_\alpha v^\text{B}_\beta - v^\text{B}_\alpha u^\text{B}_\beta$, the case-II factor of Eq. (7.28), which vanishes for two quasiparticles at the gap edge (Appendix D.40). In the dirty limit the relaxation rate $\gamma_\text{rel} = v_\text{F}/\ell_e$ far exceeds

Δ_{BCS} ($\ell_e \ll \xi_{\mathrm{coh}}$, Sect. 15.3), and the squared matrix elements change only over energies of order γ_{rel}. Over the few Δ_{BCS} where the coherence factors matter they are constant, and ξ_α and ξ_β are independent. The ratio to the normal metal is then the case-II integral of Eq. (7.30):

Impurity-assisted absorption (Mattis–Bardeen)

$$\sigma_1^{\mathrm{MB}}(\omega) = \sigma_n(\omega)\, \mathfrak{g}_{\mathrm{II}}\Big(\frac{\omega}{\Delta_{\mathrm{BCS}}}\Big), \qquad \sigma_n(\omega) = \frac{\sigma_{\mathrm{dc}}\, \gamma_{\mathrm{rel}}^2}{\omega^2 + \gamma_{\mathrm{rel}}^2},$$
$$\mathfrak{g}_{\mathrm{II}}(y) = \Big(1 + \frac{2}{y}\Big)\mathrm{E}(k) - \frac{4}{y}\,\mathrm{K}(k), \tag{15.22}$$

for $\omega \geq 2\Delta_{\mathrm{BCS}}$ and $\gamma_{\mathrm{rel}} \gg \Delta_{\mathrm{BCS}}$, with σ_n the Drude conductivity of the normal metal, σ_{dc} its dc value, and K, E the complete elliptic integrals of the modulus $k = (y-2)/(y+2)$ of Eq. (15.7) (Eq. (D.224)).

This is the absorption of light by a superconductor that Mattis and Bardeen found [6], whose zero-temperature ratio Sect. 7.2 met as the case-II pair-creation spectrum. The function $\mathfrak{g}_{\mathrm{II}}$ starts from zero at the edge, is 0.67 at $4\Delta_{\mathrm{BCS}}$ and comes within 10 % of one at $8.1\,\Delta_{\mathrm{BCS}}$; its case-I partner $\mathfrak{g}_{\mathrm{I}}$ of Sect. 15.1 has the same closed form with $4/(y+2)$ in place of $4/y$. No ladder of the pair field corrects Eq. (15.22). In the conserving response Eq. (7.34) the current would couple to the amplitude and phase channels through bubbles that pair one odd factor $\boldsymbol{v}$ with an even pair vertex, and these vanish. The bubble of independent quasiparticles in the exact impurity states is already the conserving response at $\boldsymbol{q} \to 0$, and through the continuity equation it is the long-wavelength limit of the density response of Sect. 15.1.

Impurity-assisted absorption is large in a film. The 50 nm aluminum film of Worked example 15.3, with $\ell_e = 50\,\mathrm{nm}$, has $\gamma_{\mathrm{rel}} = 26.7\,\mathrm{meV}$, 148 gaps, and with the free-electron plasma frequency its dc conductivity is $\sigma_{\mathrm{dc}} = \omega_p^2/\gamma_{\mathrm{rel}} = 9.35 \times 10^3\,\mathrm{eV}$, a residual resistivity of $0.80\,\mu\Omega\,\mathrm{cm}$. At 1 meV the paired film absorbs with $\sigma_1^{\mathrm{MB}} = 7.56 \times 10^3\,\mathrm{eV}$, 1.4×10^6 times the phonon-assisted value derived next. Intrinsic absorption dominates only in a pure sample, as Hochberg, Lin and Zurek assumed [5], and in a film the rate follows the residual resistivity of the sample, not a property of aluminum.

15.4.2 Phonon-assisted absorption

In a pure crystal the third body is a phonon. The absorption is then of second order, with one current vertex and one electron–phonon vertex, as for phonon-assisted absorption across the gap of silicon (Sect. 6.5), and the phonon carries the momentum that the electrons need. At 1 meV in aluminum the phonon has at most $\omega - 2\Delta_{\mathrm{BCS}} = 0.64\,\mathrm{meV}$. With the sound speeds along [100] from the zero-temperature elastic constants, $c_{\mathrm{LA}} = 6.47\,\mathrm{km\,s^{-1}}$ and $c_{\mathrm{TA}} = (C_{44}/\rho)^{1/2} = 3.40\,\mathrm{km\,s^{-1}}$ for $C_{44} = 31.6\,\mathrm{GPa}$ [1, Table 2] and $\rho = 2.73\,\mathrm{g\,cm^{-3}}$, that is a momentum of at most 29.7 or 56.4 eV. This is two hundred times the 0.15 eV that the electrons need, and 61 times less than $k_{\mathrm{F}} = 3.45\,\mathrm{keV}$, so only long-wavelength acoustic phonons take part near the edge. In a superconductor the question is which coherence factor the final pair carries, and the Nambu form Eq. (15.20) answers it exactly.

Take the electron–phonon Hamiltonian Eq. (A.25) restricted to the band at the Fermi level, with the vertex $g_\nu(\boldsymbol{k}, \boldsymbol{Q})$, and the final states with one quasiparticle in $(\boldsymbol{k} + \boldsymbol{Q}\uparrow)$, one in $(-\boldsymbol{k}\downarrow)$ and an emitted phonon $(-\boldsymbol{Q}, \nu)$. The second-order amplitude has the two time orderings of Eq. (13.7), with the current in place of the density. The current is diagonal in the quasiparticle basis, so every intermediate state reached by the current is the state it acted on. The sum over intermediate states collapses, and the amplitude is the matrix element

of the phonon vertex times the change of the current eigenvalue, $J_f - J_i$, divided by $-\omega$ (Eq. (D.223)). The ground state carries no current, and by Eq. (15.20) the final pair carries $\hat{\boldsymbol{e}} \cdot (\boldsymbol{v}_{\boldsymbol{k}+\boldsymbol{Q}} - \boldsymbol{v}_{\boldsymbol{k}})$. The phonon vertex creates the pair through its spin-up and spin-down terms, as the density does in Eq. (7.28). With the phase convention in which time reversal gives $g_\nu(-\boldsymbol{k}-\boldsymbol{Q}, \boldsymbol{Q}) = g_\nu(\boldsymbol{k}, \boldsymbol{Q})$,

Phonon-assisted amplitude in a superconductor

$$T_{fi} = -\frac{\hat{\boldsymbol{e}} \cdot (\boldsymbol{v}_{\boldsymbol{k}+\boldsymbol{Q}} - \boldsymbol{v}_{\boldsymbol{k}})}{\omega} \, \frac{1}{\sqrt{N}} \, g_\nu(\boldsymbol{k}, \boldsymbol{Q}) \, (u^{\mathrm{B}}_{\boldsymbol{k}+\boldsymbol{Q}} v^{\mathrm{B}}_{\boldsymbol{k}} + v^{\mathrm{B}}_{\boldsymbol{k}+\boldsymbol{Q}} u^{\mathrm{B}}_{\boldsymbol{k}}), \tag{15.23}$$

for $|f\rangle = \gamma^\dagger_{\boldsymbol{k}+\boldsymbol{Q},1} \gamma^\dagger_{\boldsymbol{k}2} b^\dagger_{-\boldsymbol{Q}\nu} |\mathrm{BCS}\rangle$ at $T = 0$, with $\omega = E_{\boldsymbol{k}+\boldsymbol{Q}} + E_{\boldsymbol{k}} + \omega_{\boldsymbol{Q}\nu}$ and N the number of unit cells.

No intermediate state needs to be far from the energy shell. The only input is that the current commutes with the mean-field Hamiltonian, and the result holds for any ω and any phonon energy. The current supplies only the velocity difference, the factor that suppresses forward scattering. The pair itself is created by the phonon, through a vertex even under time reversal, and so it carries the case-I factor of the density. Phonon-assisted absorption has the coherence factor of the density, impurity-assisted absorption that of the current: the third body decides. The same steps with the density in place of the current give zero, because its eigenvalue at $\boldsymbol{q} = 0$ is the conserved electron number, as in Eq. (13.10).

The conductivity follows from the absorbed power as in Sect. 6.5,

$$\sigma_1 = \frac{\pi e^2}{\omega \mathscr{V}} \sum_f |T_{fi}|^2 \, \delta(E_f - E_i - \omega).$$

The band velocity and the vertex change across the Fermi surface on scales of eV, whereas $\xi_{\boldsymbol{k}}$ and $\xi_{\boldsymbol{k}+\boldsymbol{Q}}$ range only over ω. The sum over $\boldsymbol{k}$ and $\boldsymbol{Q}$ therefore splits into an average over the Fermi surface and independent integrals over the two energies. The integrals give the pair spectrum $2x \, \mathfrak{g}_{\mathrm{I}}(x/\Delta_{\mathrm{BCS}})$ at the pair energy $x = \omega - \omega_{\boldsymbol{Q}\nu}$, the case-I integral of Eq. (7.30). The average defines the transport Eliashberg function,

$$\alpha^2 F_{\mathrm{tr}}(\omega) = \frac{1}{2 \langle (\hat{\boldsymbol{e}} \cdot \boldsymbol{v})^2 \rangle_{\mathrm{FS}} \mathscr{N}_{\mathrm{F}} \Omega \, N^2} \sum_{\boldsymbol{k}\boldsymbol{Q}\nu} |g_\nu(\boldsymbol{k}, \boldsymbol{Q})|^2 \big[\hat{\boldsymbol{e}} \cdot (\boldsymbol{v}_{\boldsymbol{k}+\boldsymbol{Q}} - \boldsymbol{v}_{\boldsymbol{k}})\big]^2 \delta(\xi_{\boldsymbol{k}}) \, \delta(\xi_{\boldsymbol{k}+\boldsymbol{Q}}) \, \delta(\omega - \omega_{\boldsymbol{Q}\nu}), \tag{15.24}$$

with Ω the volume of the unit cell, so that $\mathscr{V} = N\Omega$. This is the Eliashberg function of Eq. (6.34) with each pair of Fermi-surface points weighted by the change of the velocity. For a spherical Fermi surface in a cubic crystal the weight averages to $\langle [\hat{\boldsymbol{e}} \cdot (\boldsymbol{v}' - \boldsymbol{v})]^2 \rangle = 2(1 - \cos\theta_{\boldsymbol{k}\boldsymbol{k}+\boldsymbol{Q}}) \langle (\hat{\boldsymbol{e}} \cdot \boldsymbol{v})^2 \rangle_{\mathrm{FS}}$, and with the factor $\frac{1}{2}$ of the definition the weight is $1 - \cos\theta_{\boldsymbol{k}\boldsymbol{k}+\boldsymbol{Q}}$, which removes forward scattering. The transport coupling constant is $\lambda_{\mathrm{tr}} = 2 \int \mathrm{d}\omega \, \alpha^2 F_{\mathrm{tr}}(\omega)/\omega$. With Eq. (15.21) (Appendix D.40),

Phonon-assisted (Holstein) absorption

$$\begin{aligned} \sigma_1^{\mathrm{S}}(\omega) &= \frac{(\omega_p^{\mathrm{intra}})^2}{\omega^2} \, \frac{2\pi}{\omega} \int_0^{\omega - 2\Delta_{\mathrm{BCS}}} \mathrm{d}\omega' \, \alpha^2 F_{\mathrm{tr}}(\omega') \, (\omega - \omega') \, \mathfrak{g}_{\mathrm{I}}\Big(\frac{\omega - \omega'}{\Delta_{\mathrm{BCS}}}\Big), \\ \sigma_1^{\mathrm{N}}(\omega) &= \frac{(\omega_p^{\mathrm{intra}})^2}{\omega^2} \, \gamma_{\mathrm{H}}(\omega), \qquad \gamma_{\mathrm{H}}(\omega) = \frac{2\pi}{\omega} \int_0^{\omega} \mathrm{d}\omega' \, (\omega - \omega') \, \alpha^2 F_{\mathrm{tr}}(\omega'), \end{aligned} \tag{15.25}$$

for the superconductor and the normal metal at $T = 0$, ω below the interband threshold and above the impurity rate.

The normal-metal form is Allen's formula for the phonon-assisted optical conductivity [4], as quoted by Hochberg, Lin and Zurek [5, Eqs. (13), (16)]: a Drude conductivity at $\omega \gg \gamma_{\mathrm{H}}$ with the frequency-dependent scattering rate $\gamma_{\mathrm{H}}(\omega)$ of an electron that absorbs ω and emits a phonon. The superconducting form, with its case-I factor and the bound $\omega' \leq \omega - 2\Delta_{\mathrm{BCS}}$, is also Allen's, in the form that Hochberg, Lin and Zurek use [5, Eq. (17)]; the derivation above shows that it is exact under the stated hypotheses, not only for intermediate states far off shell. Two of those hypotheses can be sized. The pair created by the phonon is not dressed by the ladder of Eq. (7.34). That correction is of relative order $(\Delta_{\mathrm{BCS}}/v_{\mathrm{F}}Q)^2$ at the phonon momentum Q (Sect. 7.2); averaged over the integrand of Eq. (15.25) with the ω'^4 law below, it is 4×10^{-4} at $\omega = 2.2\,\Delta_{\mathrm{BCS}}$ (2.6×10^{-4} for the most energetic phonon) and 1.4×10^{-6} at 1 meV. At millikelvin temperatures there are no thermal phonons to absorb, and thermal quasiparticles are suppressed by $\mathrm{e}^{-\Delta_{\mathrm{BCS}}/T}$ (Sect. 7.2).

The dark-matter literature evaluates Eq. (15.25) with a Debye transport function, $\alpha^2 F_{\mathrm{tr}}(\omega') = 2\lambda_{\mathrm{tr}}\omega'^4/\omega_{\mathrm{D}}^4$ below ω_{D}, which carries the coupling λ_{tr}. The fourth power is that of long-wavelength longitudinal phonons: $\alpha^2 F$ grows as ω'^2 (Sect. 6.4), and the transport weight $1 - \cos\theta \propto Q^2$ adds two more powers. The rate is then $\gamma_{\mathrm{H}} = (2\pi/15)\lambda_{\mathrm{tr}}\omega^5/\omega_{\mathrm{D}}^4$ below ω_{D}, and above it $\gamma_{\mathrm{H}} = (4\pi/5)\lambda_{\mathrm{tr}}\omega_{\mathrm{D}}(1 - 5\omega_{\mathrm{D}}/6\omega)$, the result of Ref. [5, Eq. (16)], and the ratio of the two lines of Eq. (15.25) is their Eq. (17) (Appendix D.40). Below ω_{D} that ratio depends only on the power law, not on λ_{tr} or ω_{D}. For aluminum we take $\lambda_{\mathrm{tr}} = 0.39$, the value that Savrasov and Savrasov extract from the measured resistivity with their computed plasma frequency [10, Table IV], against 0.37 from their linear-response calculation, and $\omega_{\mathrm{D}} = 37$ meV [5]. Limit Check 15.3 compares the superconducting rate with the normal-metal formula.

Limit Check 15.3. Phonon-assisted absorption: the normal-metal limit

Phonon-assisted absorption of a superconductor, Eq. (15.25) with $\alpha^2 F_{\mathrm{tr}} = 2\lambda_{\mathrm{tr}}\omega'^4/\omega_{\mathrm{D}}^4$ [5, Eqs. (13), (16), (17)], following Allen [4] as quoted there; for $\omega \gg \Delta_{\mathrm{BCS}}$ it becomes the normal-metal formula $\sigma_1^{\mathrm{N}} = (\omega_p^{\mathrm{intra}})^2\gamma_{\mathrm{H}}/\omega^2$.

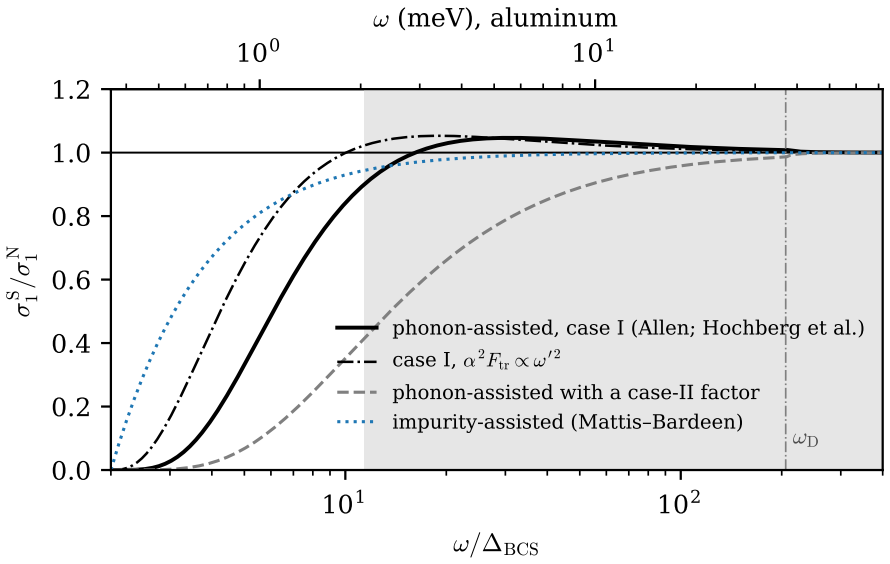


Figure 15.6: Phonon-assisted conductivity of a superconductor over that of the normal metal, $\sigma_1^{\mathrm{S}}/\sigma_1^{\mathrm{N}}$, against $\omega/\Delta_{\mathrm{BCS}}$ (top axis: ω for aluminum, $\Delta_{\mathrm{BCS}} = 0.18$ meV), at $T = 0$. Black: Eq. (15.25) with the case-I factor and the Debye law $\alpha^2 F_{\mathrm{tr}} \propto \omega'^4$, which is the

ratio of Hochberg, Lin and Zurek and, below ω_D (dash-dotted vertical line, 37 meV), independent of λ_{tr} and ω_D; dash-dotted: the same with the softer law $\alpha^2 F_{tr} \propto \omega'^2$, a test of the shape. Thin line at one: the normal-metal formula of Allen. Dashed: the phonon-assisted integral with the case-II factor in place of the case-I factor, for comparison. Dotted: impurity-assisted absorption in the dirty limit, $\mathfrak{g}_{II}(\omega/\Delta_{BCS})$, Eq. (15.22). Gray: the case-I result within 10 % of the normal-metal formula, $\omega \geq 11.4\,\Delta_{BCS}$ (2.05 meV).

Below $11.4\,\Delta_{BCS}$ the normal-metal formula overstates the rate, by a factor of 2.4 at 1 meV and without bound toward the edge, where the superconducting rate falls as $(\omega - 2\Delta_{BCS})^5$; there the result also depends on the shape of $\alpha^2 F_{tr}$ below a few meV (0.42 or 0.73 of the normal value at 1 meV for the two laws).

The gap removes the absorption near the edge for a reason of phase space, not of coherence. The case-I factor is largest at the edge, where $\mathfrak{g}_I = \pi/2$, but a phonon of energy ω' is available only up to $\omega - 2\Delta_{BCS}$. Since $\alpha^2 F_{tr}$ grows as ω'^4, the gap removes the most energetic phonons, which carry most of the weight. Above $16\,\Delta_{BCS}$ the pile-up of the case-I factor wins and the superconductor absorbs slightly more than the normal metal, by at most 4.7 %, at $31\,\Delta_{BCS}$. Had the pair carried the case-II factor of impurity-assisted absorption, the rate would reach 90 % of the normal value only at $54\,\Delta_{BCS}$. Fig. 15.7a shows the two processes on an absolute scale. The phonon-assisted conductivity of the normal metal grows as ω^3 up to ω_D and falls as ω^{-2} above. The impurity-assisted conductivity of the 50 nm film, however, is flat up to γ_{rel} and stays above the phonon-assisted one at every energy of the figure, by a factor of 1.6 at 0.1 eV and 1.3 at 0.3 eV.

Worked example 15.4: The Holstein onset in aluminum

A dark photon of $m_\phi = 1$ meV, $5.56\,\Delta_{BCS}$, is absorbed by pure aluminum with one phonon. The Debye form gives the normal-metal scattering rate $\gamma_H = (2\pi/15)\lambda_{tr}\omega^5/\omega_D^4 = 8.72 \times 10^{-11}$ eV, and with $\omega_p^{intra} = 12.29$ eV the normal-metal conductivity is $\sigma_1^N = (\omega_p^{intra})^2\gamma_H/\omega^2 = 0.0132$ eV. The superconductor has $\sigma_1^S/\sigma_1^N = 0.421$, so $\sigma_1^S = 5.54 \times 10^{-3}$ eV.

Close to the edge the integral of Eq. (15.25) simplifies. For $\omega - 2\Delta_{BCS} \ll \Delta_{BCS}$ the pair energy $\omega - \omega'$ stays near $2\Delta_{BCS}$ and $\mathfrak{g}_I$ near $\pi/2$, and

$$\sigma_1^S \simeq \frac{(\omega_p^{intra})^2}{\omega^2}\,\frac{2\pi}{\omega}\,\frac{2\lambda_{tr}}{\omega_D^4}\,\frac{\pi}{2}\,2\Delta_{BCS}\,\frac{(\omega - 2\Delta_{BCS})^5}{5}.$$

At $\omega = 2.05$ and $2.5\,\Delta_{BCS}$ this is within 0.2 and 2.1 % of the full integral. At $2.2\,\Delta_{BCS}$ the superconductor absorbs with $\sigma_1^S = 4.4 \times 10^{-8}$ eV, against 8.2×10^{-4} eV for the normal metal: in practice the onset is not at $2\Delta_{BCS}$.

The rate needs the loss function, $L = \varepsilon_2/|\varepsilon|^2$ in Eq. (10.31). At 1 meV the condensate dominates ε: in the clean superconductor at $T = 0$ its imaginary conductivity is $\sigma_2 = (\omega_p^{intra})^2/\omega$, and $\varepsilon_1 = 1 - \sigma_2/\omega = -1.51 \times 10^8$. The in-medium factor $1/|\varepsilon|^2$ is 4.4×10^{-17}, and $L = \sigma_1^S\omega^3/(\omega_p^{intra})^4 = 2.43 \times 10^{-16}$. With $\rho_\chi = 0.3$ GeV cm^{-3} and $\rho_T = 2.70$ g cm^{-3}, Eq. (10.31) gives $1.29 \times 10^{18}\,\zeta^2$ events per kilogram-year, and three events need $\zeta = 1.5 \times 10^{-9}$. The normal-metal formula would claim 9.9×10^{-10}. In the 50 nm film of Worked example 15.3 the impurity-assisted conductivity at the same mass is 1.4×10^6 times larger, and the rate with it.

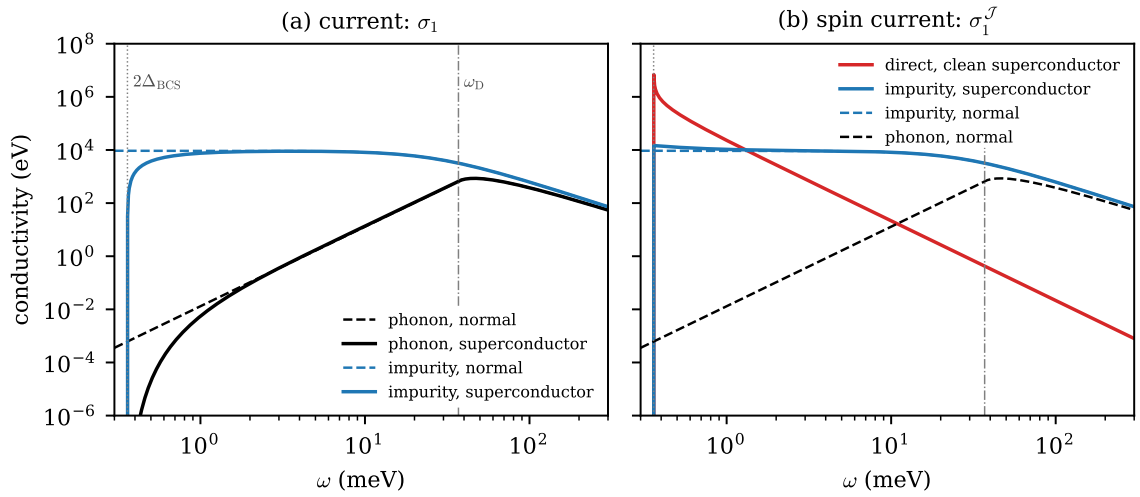


Figure 15.7: Absorption of a uniform field by aluminum at $T = 0$, against the boson energy ω, with $\Delta_{\rm BCS} = 0.18\,\text{meV}$ (dotted vertical line at $2\Delta_{\rm BCS}$) and the computed intraband plasma frequency 12.29 eV. (a) Optical conductivity σ_1, which sets the rates of the dark photon. Black: phonon-assisted, Eq. (15.25), in the normal metal (dashed) and the superconductor (solid), with the Debye transport function, $\lambda_{\rm tr} = 0.39$ and $\omega_{\rm D} = 37\,\text{meV}$ (dash-dotted vertical line). Blue: impurity-assisted, the Drude conductivity of the normal metal (dashed) and the Mattis–Bardeen conductivity of the superconductor (solid), Eq. (15.22), for the 50 nm film of Worked example 15.3 (mean free path 50 nm, $\gamma_{\rm rel} = 26.7\,\text{meV}$, free-electron plasma frequency 15.78 eV in its Drude conductivity). (b) Spin-current conductivity $\sigma_1^{\mathcal{J}}$, which sets the rate of the axion-like particle, normalized to equal σ_1 in the normal metal. Red: direct pair breaking in the clean superconductor, Eq. (15.27) with the quasiparticle factor $1/(1 + \lambda_{\rm ep}) = 0.70$. Blue: impurity-assisted, with the case-I factor in the superconductor (solid) and the Drude form in the normal metal (dashed), same mean free path. Black dashed: the phonon-assisted conductivity of the normal metal, for reference. Interband absorption, above about 0.5 eV, is not included.

15.4.3 Spin-current absorption

The axion-like particle couples through $(\partial_t\phi)\mathcal{J}_e^{\alpha\alpha}$, the spin current of the electrons traced over its two indices, $\mathcal{J}_e = \sum_\alpha \mathcal{J}_e^{\alpha\alpha}$, a scalar like $\boldsymbol{\sigma}\cdot\boldsymbol{v}$ (Sect. 10.7.3). Sect. 10.7.3 reduced its rate to the optical conductivity through the spin trace $S_{\mathcal{J}\mathcal{J}} = \frac{3}{4}S_{jj}$, valid for independent electrons without spin–orbit coupling. In a superconductor we keep the response of the spin current itself and define its conductivity by

$$\sigma_1^{\mathcal{J}}(\omega) = \frac{2e^2}{3\omega}\, S_{\mathcal{J}\mathcal{J}}(0,\omega), \tag{15.26}$$

so that Eq. (10.32) holds with $\sigma_1^{\mathcal{J}}$ in place of σ_1, and $\sigma_1^{\mathcal{J}} = \sigma_1$ in the normal metal. Its component with the spin and the velocity both along $\hat{\boldsymbol{z}}$ is $\mathcal{J}_e^{zz} = \frac{1}{2}\sum_{\boldsymbol{k}} v_{\boldsymbol{k}}^z (c_{\boldsymbol{k}\uparrow}^\dagger c_{\boldsymbol{k}\uparrow} - c_{\boldsymbol{k}\downarrow}^\dagger c_{\boldsymbol{k}\downarrow})$. The relabeling that turned the current into the unit matrix now meets a second sign, that of the spin, and the two cancel: $\mathcal{J}_e^{zz} = \frac{1}{2}\sum_{\boldsymbol{k}} v_{\boldsymbol{k}}^z \Psi_{\boldsymbol{k}}^\dagger \boldsymbol{\eta} \Psi_{\boldsymbol{k}}$, with the matrix of the density. That matrix does not commute with $h_{\boldsymbol{k}}$. In the quasiparticle basis (Eq. (D.221)) the spin current contains the term $\frac{1}{2}v_{\boldsymbol{k}}^z(\Delta_{\mathrm{BCS}}/E_{\boldsymbol{k}})\gamma_{\boldsymbol{k}1}^\dagger\gamma_{\boldsymbol{k}2}^\dagger$, which creates a pair of total momentum zero and energy $2E_{\boldsymbol{k}}$. A uniform charge current moves both electrons of a pair $(\boldsymbol{k}\uparrow, -\boldsymbol{k}\downarrow)$ in the same direction and only boosts the condensate. A spin current drives the spin-up and spin-down electrons in opposite directions, pulls the two members of each pair apart, and breaks pairs.

In a clean superconductor this needs no third body. Without spin–orbit coupling the three components $\mathcal{J}_e^{\alpha\alpha}$ contribute equally and do not interfere, and the golden rule with the amplitude above gives (Appendix D.40)

Spin-current absorption in a clean superconductor

$$\sigma_1^{\mathcal{J}}(\omega) = \frac{2\pi(\omega_p^{\mathrm{intra}})^2\,\Delta_{\mathrm{BCS}}^2}{\omega^2\,(\omega^2 - 4\Delta_{\mathrm{BCS}}^2)^{1/2}}\,\Theta(\omega - 2\Delta_{\mathrm{BCS}}), \tag{15.27}$$

at $T = 0$, without spin–orbit coupling, for a mean free path long compared with the size of a pair; its integral over ω is $(\pi/2)(\omega_p^{\mathrm{intra}})^2$.

In the clean normal metal the spin current commutes with the one-body band Hamiltonian, but not with the interactions. A two-body force enters with the weight of the spin difference, $[p_1^z\sigma_1^z + p_2^z\sigma_2^z, V(\boldsymbol{r}_1 - \boldsymbol{r}_2)] = -\mathrm{i}\,(\partial_z V)(\sigma_1^z - \sigma_2^z)$, which vanishes for parallel spins only. Galilean invariance protects the charge current, whose internal forces cancel in pairs, but not the spin current. Electron–electron collisions (spin Coulomb drag) and normal phonon processes relax it, and its normal-state weight forms a narrow Drude peak rather than a delta function at $\omega = 0$. With a collision rate $\beta\omega^2$ and $\beta = 0.03$ to $0.3\,\mathrm{eV}^{-1}$, an estimate of the Fermi-liquid scale, the normal metal at $T \to 0$ absorbs at 0.72 meV with $\sigma_1^{\mathcal{J}} \approx 5$ to 45 eV, against the phonon-assisted $\sigma_1^{\mathrm{N}} = 4.9 \times 10^{-3}\,\mathrm{eV}$; the superconducting direct term is 1.5×10^3 to 1.5×10^4 times larger still. Pairing keeps the charge-current weight at zero frequency, as the supercurrent, and moves the spin-current weight above $2\Delta_{\mathrm{BCS}}$: 87 % of it lies below $4\Delta_{\mathrm{BCS}}$ and 99.5 % below $20\Delta_{\mathrm{BCS}}$. The total weight, summed over all frequencies, is fixed beyond mean field. For any Hamiltonian that does not depend on spin, the spin current is the time derivative of the spin dipole, so its integrated absorption obeys an exact sum rule: the total weight, interband absorption included, is that of the charge current, $(\pi/2)\omega_p^2$ with the free-electron ω_p; for the conduction band alone, a one-band Hamiltonian with spin-independent interactions, it is $(\pi/2)(\omega_p^{\mathrm{intra}})^2$ exactly. The part near $2\Delta_{\mathrm{BCS}}$ is the coherent quasiparticle weight $(\pi/2)(\omega_p^{\mathrm{intra}})^2(1 + F_1^a/3)/(1 + \lambda_{\mathrm{ep}})$, with the Landau parameter F_1^a of the quasiparticle spin current and the residue $1/(1 + \lambda_{\mathrm{ep}})$ of Eq. (6.36); the remainder lies near ω_{D}, like the Holstein band of the charge current, and at the scale of the Fermi energy. The sum rule

therefore requires Eq. (15.27) near the edge to carry the factor $(1 + F_1^a/3)/(1 + \lambda_{\rm ep})$, which is 0.70 for $F_1^a = 0$ and the measured $\lambda_{\rm ep} = 0.43$ of Sect. 6.4. The bare BCS bubble is conserving in the spin channel: a local s-wave pairing term $\psi_\uparrow^\dagger \psi_\downarrow^\dagger$ commutes with the local spin density, so $H_{\rm MF}$ keeps spin continuity with the bare spin current. A singlet superconductor has zero spin stiffness: a uniform static spin twist is a gauge transformation of any spin-independent Hamiltonian that leaves $\psi_\uparrow \psi_\downarrow$ unchanged, so the singlet carries no spin supercurrent at $T = 0$, as its Pauli susceptibility vanishes (the Knight shift falls to zero), and the weight cannot sit at $\omega = 0$. The lowest spin-one excitations at zero momentum are pairs of quasiparticles at $2\Delta_{\rm BCS}$ or above. In a dirty film the matrix elements of $\mathcal{J}_e$ between impurity states replace the direct term. Since the spin current is even under time reversal, the steps of Eq. (15.22) give $\sigma_n \mathfrak{g}_{\rm I}(\omega/\Delta_{\rm BCS})$, with the factor of the density, which jumps to $\pi/2$ at the edge. That form conserves the normal-state weight exactly for a flat σ_n, since $\int_2^\infty {\rm d}y\, [\mathfrak{g}_{\rm I}(y) - 1] = 2$: the weight above $2\Delta_{\rm BCS}$ replaces the weight below. For the Drude σ_n of the 50 nm film the balance holds to 0.05 %, the residue coming from the Drude shape.

At 1 meV Eq. (15.27) with the factor 0.70 gives $\sigma_1^{\mathcal{J}} = 2.31 \times 10^4$ eV, 4×10^6 times the phonon-assisted $\sigma_1^{\rm S}$ of Worked example 15.4, and it exceeds the phonon-assisted conductivity of the normal metal up to 10.9 meV (Fig. 15.7b). For an axion-like particle of 0.72 meV, $4\Delta_{\rm BCS}$, with $g_{aee} = 10^{-13}$, Eq. (10.32) with $\sigma_1^{\mathcal{J}}$ predicts 0.080 events per kilogram-year in clean aluminum, and three events per kilogram-year need $g_{aee} = 6.1 \times 10^{-13}$ (5.1×10^{-13} for the bare BCS form). Published projections for aluminum put the phonon-assisted $\sigma_1^{\rm S}$ in Eq. (10.32) [5, Eq. (22), 18, Sect. V], which takes the spin-current response equal to the current response, as for independent electrons; the same independent-electron spin trace appears in [18, Eq. (28)] and [19, Sect. IV.2.1]. Recomputed with the phonon-assisted $\sigma_1^{\rm S} = 7.9 \times 10^{-4}$ eV of this section (Debye transport function), that mapping needs 5.6×10^{-9} for the same number of events; this value is our recomputation, not one read from the published curves. The identification fails for paired electrons, because the anomalous, pair-amplitude part of the two responses enters with opposite signs.

Three conditions limit this comparison. First, the clean form needs a mean free path long compared with the size of a pair, $\xi_{\rm coh} \approx 1.6$ µm to 2.4 µm (Sect. 15.3.1), which only bulk high-purity crystals reach. Second, the factor of 10^4 in g_{aee} is a gain over phonon-only projections for clean aluminum alone. For the 50 nm film the published σ_1 mapping, applied to the film's own Mattis–Bardeen σ_1, already gives $g_{aee} = 2.0 \times 10^{-12}$ at 0.72 meV, against 1.5×10^{-12} from the spin channel $\sigma_n \mathfrak{g}_{\rm I}$. Third, the sum rule fixes the total weight and, with the factor $(1 + F_1^a/3)/(1 + \lambda_{\rm ep})$, the coherent weight near the edge, but not the line shape at a given mass. The strong-coupling factor $1/(1 + \lambda_{\rm ep})$ is included in the numbers above and raises the required g_{aee} by 1.20. The quasiparticle interaction dresses the spin current in the form of Leggett, $K_s(\omega) \propto (1 + F)(1 - f)/(1 + Ff)$ with $F = F_1^a/3$, $f(\omega)$ the bare BCS spin-current bubble normalized to $f(0) = 1$, and $\sigma_1^{\mathcal{J}} = -\operatorname{Im} K_s/\omega$ (Appendix D.40). For $|F_1^a/3| \le 0.1$ it changes $\sigma_1^{\mathcal{J}}$ at $4\Delta_{\rm BCS}$ by a factor 0.75 to 1.31 and g_{aee} by -12 to $+15$ %, so that three events need $g_{aee} = 5.4 \times 10^{-13}$ to 7.1×10^{-13}. For $F_1^a < 0$ the denominator vanishes below the edge: a bound spin-one mode with $\ell = 1$ forms, $5.6 \times 10^{-3} \Delta_{\rm BCS}$ below $2\Delta_{\rm BCS}$ at $F_1^a/3 = -0.05$ and $2.1 \times 10^{-2} \Delta_{\rm BCS}$ at -0.1, and takes weight from the continuum. A triplet p-wave interaction acts in the same channel and would dress the vertex further. Spin–orbit coupling, weak in aluminum, mixes the current and the spin current and leaves a fraction of the weight of order the spin-flip rate over $\Delta_{\rm BCS}$ at low frequency; in heavy-element superconductors it is the controlling caveat. The phonon-assisted part of the spin-current response also has a coherence factor near the edge that differs from that of Eq. (15.23), because the spin current creates pairs at intermediate stages. Its size there is open, but it competes with the direct term only above about 10 meV, where it tends to its normal-metal value.

15.4.4 Strong coupling and the transport function

Eq. (15.25) assumes weak coupling: a BCS gap that does not depend on energy, and quasiparticles without the mass enhancement $1+\lambda_{\rm ep}$ of Eq. (6.36). In the Eliashberg theory of Appendix E.9 the electron–phonon interaction is retarded, so the gap $\Delta^{\rm El}(E)$ depends on the energy E of the quasiparticle, and the pair spectrum of the BCS spinors is replaced by spectral functions. Two of them enter the absorption: the density of states $N^{\rm El}(E) = \mathrm{Re}[E/(E^2-\Delta^{\rm El}(E)^2)^{1/2}]$ and the pair amplitude $P^{\rm El}(E) = \mathrm{Re}[\Delta^{\rm El}(E)/(E^2-\Delta^{\rm El}(E)^2)^{1/2}]$, both per unit of the normal density of states. The case-I combination of two quasiparticles of energies E and $E'=\omega-\omega'-E$ becomes $N^{\rm El}(E)N^{\rm El}(E')+P^{\rm El}(E)P^{\rm El}(E')$ in the integrand of $\mathfrak{g}_{\rm I}$. For a constant gap this is $(EE'+\Delta^2_{\rm BCS})/[(E^2-\Delta^2_{\rm BCS})(E'^2-\Delta^2_{\rm BCS})]^{1/2}$, the integrand of Eq. (7.30). The renormalization of the quasiparticles is a separate matter. For the normal metal at $\omega \ll \omega_{\rm D}$ it is known: with the self-energy resummed and the bare current vertex, whose corrections are of order $\omega_{\rm D}/E_{\rm F}$ by Migdal's argument (Sect. 6.4), the conductivity becomes $\sigma_1 \simeq (\omega_p^{\rm intra})^2\gamma_{\rm H}/[\omega^2(1+\lambda_{\rm tr})^2]$ [20, Eq. (3)], with the transport-weighted mass enhancement. In aluminum, with $\lambda_{\rm tr}=0.39$, this factor is 0.52. Whether the same factor multiplies the superconducting form near the edge, where the pair spectrum and the renormalization both depend on energy, is open; Appendix E.20 computes the rate with and without it.

The transport function is computed from first principles with the electron–phonon vertices of density-functional perturbation theory on Wannier-interpolated grids (Appendices E.5 and E.9), with the weight $1-\cos\theta$ in place of one, and Appendix E.20 specifies the conductivity of Eq. (15.25), normal and superconducting, for aluminum and niobium. The model of Fig. 6.7, free electrons scattered by screened empty-core ions, shows what such a calculation must get right. Its transport coupling is $\lambda_{\rm tr}=0.32$, against $\lambda_{\rm ep}=0.44$ for the same model, because the weight $1-\cos\theta$ removes the small-angle scattering, and against 0.37 computed and 0.39 from the resistivity [10, Table IV]. Near the edge, however, it fails completely: below 1 meV it holds 1.25×10^4 times the weight of the ω'^4 law, all of it from umklapp scattering at momentum transfers near $|\boldsymbol{G}_{111}|$ by phonons of small $\boldsymbol{Q}$. A plane wave is not an eigenstate of the crystal near a Bragg plane, and the plane-wave vertex violates the acoustic sum rule of Eq. (6.12), which makes the vertex vanish for $\boldsymbol{Q}\to0$ between states on the Fermi surface. The phonon-assisted rate within a few meV of the edge therefore needs a vertex that obeys the sum rule, and that rate, through the shape of $\alpha^2F_{\rm tr}$ below a few meV, is the least certain input of Limit Check 15.3. Interband absorption above about 0.5 eV is taken from measured optical data, as in Sect. 10.7.

Scalar absorption. The scalar couples to the kinetic-energy row of Sect. 10.7.4, whose Nambu vertex, like the density, is $\boldsymbol{\eta}$, with a weight even in $\boldsymbol{k}$. Its part linear in $\xi_{\boldsymbol{k}}$ couples to the amplitude channel of the pair field (Sect. 7.2), so the scalar response of a superconductor near its edge needs the conserving treatment of Sect. 15.1 extended to that channel, and its size there is open.

Summary

A dark-matter particle that couples to the electron density breaks pairs in a superconductor through the loss function, and the loss function that conserves charge adds the pair-field ladder of Sect. 7.2 to the quasiparticle bubble and screens it with the Coulomb interaction (Sect. 15.1). Its truncations are the three forms in use. The absorptive part of the bare bubble is the Lindhard value times the case-I coherence integral $\mathfrak{g}_{\rm I}(\omega/\Delta_{\rm BCS})$, exactly, with $\mathfrak{g}_{\rm I}=1.500$ at $y=2.2$ falling to 1.051 at $y=10$. The ladder raises that absorption by up to 44 % near the

edge, and the first BCS correction to the screening is of first order in $\Delta_{\mathrm{BCS}}/qv_{\mathrm{F}}$, at most 17 % of the loss function at $y = 5$ and more than one at $y = 2.2$. On the kinematic edge $q = \omega/v_{\max}$ of aluminum the Lindhard function is within 10 % of the conserving loss function for $y \geq 7.9$, and the product $\mathfrak{g}_{\mathrm{I}} L^{\mathrm{nor}}$ within 9 % for $y \geq 3$ at every momentum transfer. The bulk gap of aluminum is the measured 0.18 meV; the 0.3 meV of projections is a film convention. For a 1 keV particle with a light mediator the bulk gap raises the rate by 1.23, more than any choice among the loss functions changes it.

On the kinematic edge the screening must be resummed (Sect. 15.2). There $\omega/qv_{\mathrm{F}} = 0.392$, and the normal dielectric function has an imaginary part 0.736 times its real part. In a clean crystal the conserving loss function, the physical one, is 0.87 times $\mathfrak{g}_{\mathrm{I}} L^{\mathrm{nor}}$ at $y = 2.2$ and 1.07 at $y = 3$ and 5. The bare bubble, resummed in closed form, falls far below it near the edge, to 0.33–0.40 of $\mathfrak{g}_{\mathrm{I}} L^{\mathrm{nor}}$ at $y = 2.2$, because its shift of the real part carries its violation of charge conservation. Impurity scattering changes the conserving value only near the threshold: at $y \geq 3$ the ratio stays between 1.008 and 1.091 for mean free paths from 100 µm to 30 nm, whereas at $y = 2.2$ it falls from 0.87 to 0.19. The superconducting part of a rate therefore carries three nuisances: the gap, the relaxation rate of the film, and the edge remainder, which is two-sided, −2 % to +9 % about $\mathfrak{g}_{\mathrm{I}} L^{\mathrm{nor}}$ for $y \geq 3$. The energies $y < 3$, where the conserving value moves by more than 10 % with the mean free path, are excluded from the signal region. The frequency dependence of the gap in strong coupling is not sized.

Real sensors are films (Sect. 15.3). Nonmagnetic disorder leaves the gap unchanged, by Anderson's theorem, but collisions change the response. At momentum transfers below the inverse mean free path the normal loss function becomes the Drude tail $\omega\gamma_{\mathrm{rel}}/\omega_p^2$, fixed by the residual resistivity, 24 times the clean value for a 50 nm aluminum film at 0.90 meV. At the edge of a dirty film the density couples through the current, with the case-II factor, and the condensate short-circuits a slow potential, so the loss function of the superconducting film starts from zero and is a quarter of the normal one at $y = 2.2$. A film thinner than $1/q$ screens like a sheet, which enhances the structure factor by $[(\varepsilon_a + \varepsilon_b)/q_{\parallel} d_{\mathrm{film}}]^2$ to leading order in $q_{\parallel} d_{\mathrm{film}}$, 540 for the 50 nm film in vacuum at 0.90 meV (588 for the exact slab), and makes the rate directional. Amorphous WSi, with $k_{\mathrm{F}}\ell_e$ of a few, lies outside every relaxation-time form and outside mean-field BCS.

A boson absorbed at $\boldsymbol{q} \to 0$ needs a third body in a metal, and the Nambu structure of the current decides the coherence factor (Sect. 15.4). The uniform current is the unit matrix in Nambu space and commutes with the mean-field Hamiltonian, so a clean superconductor does not absorb it. With impurities it carries the case-II factor of Mattis and Bardeen. With a phonon the second-order amplitude collapses exactly to the velocity difference times the phonon vertex, the pair carries the case-I factor, and the result is Allen's form, which the dark-matter literature evaluates with a Debye transport function. Near the edge phase space, not coherence, removes the phonon-assisted absorption: it reaches 90 % of the normal-metal value at $11.4\,\Delta_{\mathrm{BCS}} = 2.05$ meV and is 0.42 of it at 1 meV, where a dark photon needs $\zeta = 1.5 \times 10^{-9}$ for three events per kilogram-year in pure aluminum. In a 50 nm film impurity-assisted absorption is 1.4×10^6 times larger at that mass. The spin current, which absorbs the axion-like particle, has the Nambu matrix $\boldsymbol{\eta}$ of the density and, in a clean singlet superconductor without spin–orbit coupling, breaks pairs by itself. In the normal metal electron–electron collisions and normal phonon processes relax the spin current, which Galilean invariance does not protect, and an estimate gives $\sigma_1^{\mathcal{J}} \approx 5$ to 45 eV at 0.72 meV. In the superconductor the spin-current weight moves above $2\Delta_{\mathrm{BCS}}$, 87 % of it below $4\Delta_{\mathrm{BCS}}$. The sum rule fixes the total weight; the coherent part near the edge carries the factor $(1 + F_1^a/3)/(1 + \lambda_{\mathrm{ep}})$, 0.70 for $F_1^a = 0$, and the line shape depends on F_1^a through the form of Leggett and on a triplet interaction. At 0.72 meV three events per kilogram-year in a bulk crystal with a mean free path longer than the size of a pair need $g_{aee} = 6.1 \times 10^{-13}$, and 5.4×10^{-13} to 7.1×10^{-13} for $|F_1^a/3| \leq 0.1$,

against 5.6×10^{-9} from our recomputation with the phonon-assisted conductivity. In a 50 nm film the gain largely disappears, 1.5×10^{-12} against 2.0×10^{-12} from the film's Mattis–Bardeen conductivity. The strong-coupling factor $1/(1 + \lambda_{\mathrm{tr}})^2$ of the normal metal is known; whether it carries over to the superconducting phonon-assisted form is open.

Table 15.3 collects the truncations and inputs of the chapter with their sizes, and Table 15.4 the inputs of its targets. For scattering in a bulk crystal the gap and the edge remainder dominate, and both are small above $y = 3$; for a film the mean free path and the thickness enter at least as strongly as the choice of response. For absorption the purity of the sample decides between impurity-assisted and phonon-assisted absorption by six orders of magnitude at 1 meV, and for the axion-like particle in a clean crystal the direct pair breaking by the spin current exceeds the phonon-assisted absorption up to 10.9 meV. How the quasiparticles that dark matter creates are counted is a question of the sensor readout, which this chapter does not treat.

Table 15.3: Nuisances of the superconducting sector. Type in the classes of Sect. 8.8: normalization, tilt (in ω or in the dark-matter mass), threshold and new support. Basis of the size: "computed" (next order or exact result computed), "model" (computed in a model), "est." (parametric estimate), "input spread" (spread or limitation of a measured or computed input), "unsized" (no size known). Sizes are for aluminum unless stated; $y = \omega/\Delta_{\mathrm{BCS}}$, and "edge" is the kinematic edge $q = \omega/v_{\max}$. The last column gives the section where the item is treated.

Truncation or input	Type	Size	Basis	Section
Gap of the sensor	threshold; tilt in y	bulk 0.18 meV against the film convention 0.30 meV: rate $\times 1.23$ at 1 keV (light mediator), $\times 1.13$ (heavy mediator)	input spread	Sects. 15.1 and 15.2
Pair-field ladder dropped (bare bubble)	normalization	absorptive part +44 % at $y = 2.2$ on the edge, +0.4 % at $y = 10$; bare loss function down to 0.53 of the conserving one for $y \geq 3$ and 0.38 at $y = 2.2$; rate 0.947 at 1 keV	computed	Sects. 15.1 and 15.2
Edge remainder (central value $\mathfrak{g}_{\mathrm{I}} L^{\mathrm{nor}}$)	normalization, two-sided	-2 % to +9 % for $y \geq 3$ (clean at all momenta, impurities on the edge); energies $y < 3$ excluded, where the conserving value moves by more than 10 % with the mean free path	computed	Sect. 15.2
Relaxation rate of the film	tilt near the edge	factor 4.6 at $y = 2.2$ between the clean limit and $\mathfrak{l}_e = 30$ nm; below 6 % for $y \geq 3$	computed	Sects. 15.2 and 15.3
Collisions in the normal loss function	normalization	Drude tail $6/\pi q\mathfrak{l}_e$ times the clean value for $q\mathfrak{l}_e \ll 1$: $\times 24$ for the 50 nm film at 0.90 meV; neither limit within 10 % for $0.64 < q\mathfrak{l}_e < 9.8$	computed	Sect. 15.3
Film thinner than $1/q$ (sheet screening)	normalization; tilt in direction	enhancement $[(\varepsilon_a + \varepsilon_b)/q_\parallel d_{\mathrm{film}}]^2$ of the structure factor to leading order in $q_\parallel d_{\mathrm{film}}$, $\times 540$ for $\boldsymbol{q}$ in the plane of the 50 nm film in vacuum at $qd_{\mathrm{film}} = 0.086$ ($\times 588$ for the exact slab); depends on the film model and the media on both sides	model	Sect. 15.3
Strong disorder (amorphous WSi)	normalization	$k_{\mathrm{F}}\mathfrak{l}_e = 1.8$–$5.5$; outside relaxation-time forms and mean-field BCS; no correction factor	unsized	Sect. 15.3
Thermal quasiparticles	new support	$\mathrm{e}^{-\Delta_{\mathrm{BCS}}/T} = \mathrm{e}^{-209}$ at 10 mK	computed	Sect. 15.2
Energy dependence of the gap (strong coupling) in scattering	normalization	size unknown; aluminum is near weak coupling, $2\Delta_{\mathrm{BCS}}/T_c = 3.54$ against 3.53	unsized	Sect. 15.2
Shape of $\alpha^2 F_{\mathrm{tr}}$ below a few meV (phonon-assisted absorption)	normalization; tilt in m_ϕ	$\sigma_1^{\mathrm{S}}/\sigma_1^{\mathrm{N}} = 0.42$ (ω'^4) or 0.73 (ω'^2) at 1 meV; normal-metal formula within 10 % from 11.4 or $7.4\,\Delta_{\mathrm{BCS}}$	model	Sect. 15.4
Strong-coupling renormalization of phonon-assisted absorption	normalization	$\times 0.52$, the factor $1/(1+\lambda_{\mathrm{tr}})^2$ of the normal metal at $\omega \ll \omega_{\mathrm{D}}$; whether it applies to the superconducting form is open	est.	Sect. 15.4
Impurity-assisted absorption	normalization	1.4×10^6 times the phonon-assisted σ_1 at 1 meV for the 50 nm film; follows the residual resistivity	computed	Sect. 15.4
Electron–electron absorption (intrinsic σ_1)	normalization	a third intrinsic channel of σ_1 besides the phonon-assisted one; size unknown	unsized	Sect. 15.4
Spin-current absorption beyond the clean BCS model	normalization; tilt in m_ϕ	integrated weight fixed by the sum rule; coherent weight near the edge times $(1 + F_1^a/3)/(1+\lambda_{\mathrm{ep}})$, with $1/(1+\lambda_{\mathrm{ep}}) = 0.70$ included ($g_{aee} = 6.1 \times 10^{-13}$ at 0.72 meV); Fermi-liquid dressing $(1+F)(1-f)/(1+Ff)$, $F = F_1^a/3$: g_{aee} -12 to +15 % for $\lvert F\rvert \leq 0.1$, and for $F_1^a < 0$ a bound spin-one $\ell = 1$ mode below $2\Delta_{\mathrm{BCS}}$ that takes weight from the continuum; a triplet p-wave interaction; spin–orbit coupling; phonon-assisted part near the edge; clean form only for $\mathfrak{l}_e \gg \xi_{\mathrm{coh}}$	unsized	Sect. 15.4
Scalar absorption near the edge (amplitude channel)	normalization	kinetic row couples to the amplitude channel of the pair field; size unknown	unsized	Sect. 15.4
Interband absorption	new support	above about 0.5 eV; taken from measured optical data	input spread	Sect. 15.4

Table 15.4: Inputs of the superconducting targets of Chap. 15. Values and sources as in the sections listed in the last column; the corrections beyond each model are in Table 15.3.

Target	Pairing	Electrons	Phonons and disorder	Section
Aluminum, bulk crystal	$\Delta_{\mathrm{BCS}} = 0.18\,\mathrm{meV}$ (measured), $T_c = 1.18\,\mathrm{K}$, $2\Delta_{\mathrm{BCS}} = 0.36\,\mathrm{meV}$	free-electron gas, $m^* = m_e$: $k_{\mathrm{F}} = 1.75\,\mathrm{Å}^{-1}$, $v_{\mathrm{F}} = 6.75 \times 10^{-3}$, $k_{\mathrm{TF}} = 4.05\,\mathrm{keV}$, $\omega_p = 15.78\,\mathrm{eV}$; computed intraband $\omega_p^{\mathrm{intra}} = 12.29\,\mathrm{eV}$; $\rho_T = 2.70\,\mathrm{g\,cm^{-3}}$	$c_{\mathrm{LA}} = 6.47\,\mathrm{km\,s^{-1}}$, $c_{\mathrm{TA}} = 3.40\,\mathrm{km\,s^{-1}}$ (along [100], 0 K; isotropic 6.63 and $3.30\,\mathrm{km\,s^{-1}}$, Sect. 20.3); $\lambda_{\mathrm{tr}} = 0.39$ (resistivity), 0.37 (computed); Debye cutoff 37 meV; clean for $\ell_e \gg \xi_{\mathrm{coh}}$, 1.6–2.4 µm	Sects. 15.1, 15.3 and 15.4
Aluminum, 50 nm film	film convention $\Delta_{\mathrm{BCS}} = 0.30\,\mathrm{meV}$ in projections; the measured film gap in a rate	as bulk	$\ell_e = 50\,\mathrm{nm}$, $\gamma_{\mathrm{rel}} = 26.7\,\mathrm{meV}$, $0.80\,\mu\Omega\,\mathrm{cm}$; $\ell_e/\xi_{\mathrm{coh}} = 0.021$	Sects. 15.1 and 15.3
Amorphous W_xSi_{1-x}	$T_c = 1.73\,\mathrm{K}$, $1.764\,T_c = 0.263\,\mathrm{meV}$ (2 nm layer)	density of states from the Einstein relation 2.4–2.9 times the free gas	$\ell_e = 0.13$–$0.41\,\mathrm{nm}$, $\gamma_{\mathrm{rel}} = 2.5$–$7.7\,\mathrm{eV}$, 163–209 µΩ cm (5 nm films)	Sect. 15.3

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Chapter 16

Novel targets

Dark matter must carry at least 0.333 MeV to lift an electron across the gap of silicon (Sect. 1.3), and the bulk crystals of Chaps. 10 to 15 extend the reach downward through phonons, magnons and the pair-breaking edge of a superconductor. Other solids have been proposed for channels that these crystals lack, often with projected reaches that a complete treatment moves by orders of magnitude. The slow linear bands of a Dirac material bring the threshold mass of an electronic excitation below 10 keV (Sect. 7.3). The published reaches, however, leave the remote bands out of the screening of the dark-matter field, and for $ZrTe_5$ they are too strong by a factor of several thousand. The printed rate of quantum dots is too large by 4×10^4 at 10 MeV, and a single-channel model of a nanotube is four times too small. Other targets are limited by their inputs rather than by their formulas. The dark-photon rate on a glass is its measured loss tangent, which changes seventeenfold between preparations, and a color center counts only if a recoil leaves a defect that a readout finds, a yield that no measurement has isolated. In most cases, then, the limit is a material input and not the framework of Chap. 8: a background dielectric constant, a density of tunneling states, the yield of a defect. The tunneling defects of glasses also release heat at millikelvin temperatures, and their kinetics shape the delayed signals of Sect. 19.6 (Fig. 16.1).

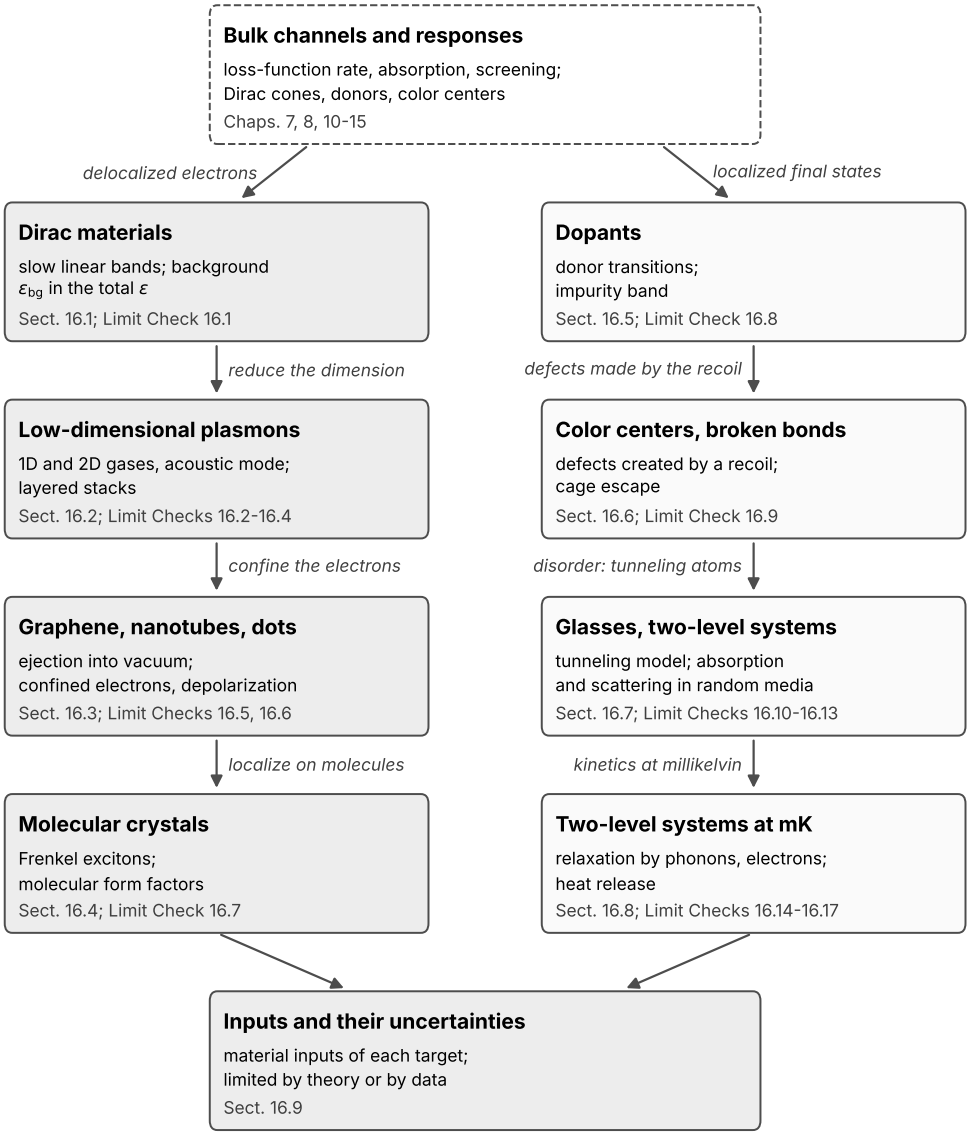


Figure 16.1: Novel targets, from the bulk responses and rates of Chaps. 7 and 8 and Chaps. 10 to 15. Left, delocalized electrons: Dirac materials with the background dielectric constant in the total ε (Sect. 16.1; Limit Check 16.1); electron gases in one and two dimensions and layered stacks (Sect. 16.2; Limit Checks 16.2 to 16.4); graphene, nanotubes and quantum dots (Sect. 16.3; Limit Checks 16.5 and 16.6); molecular crystals and their Frenkel excitons (Sect. 16.4; Limit Check 16.7). Right, localized final states: dopants (Sect. 16.5; Limit Check 16.8), color centers and broken bonds (Sect. 16.6; Limit Check 16.9), the tunneling two-level systems of glasses and random media (Sect. 16.7; Limit Checks 16.10 to 16.13) and their kinetics at millikelvin temperatures (Sect. 16.8; Limit Checks 16.14 to 16.17). Both feed the table of material inputs and their uncertainties (Sect. 16.9).

16.1 Dirac-material rates

In $ZrTe_5$ the remote bands alone have a dielectric constant of about 50, and the Dirac electrons raise it by 9 %, to 55 (Worked example 7.4). A dark photon that couples to an electron–hole pair in the cone reaches it through both: the field is screened by $|\varepsilon|^2 \approx 3000$, not by the factor $|1+\varepsilon_\mathrm{D}/\varepsilon_\mathrm{bg}|^2 = 1.09^2 = 1.2$ that the Dirac electrons contribute alone. The projected reaches of Dirac materials were computed with the second factor [1, 2], and for $ZrTe_5$ the two differ by a factor 5200 to 6400 in the scattering rate (Sect. 16.1.3). We derive the loss function with ε_bg in the total ε, the change of the published reaches, the anisotropic case of $ZrTe_5$, and where one loop fails.

16.1.1 Loss function of a Dirac target

The dielectric function of Eq. (7.42) is a sum, $\varepsilon = \varepsilon_\mathrm{bg} + \varepsilon_\mathrm{D}$, of the background constant and the Dirac part $\varepsilon_\mathrm{D} = -v_\mathrm{C} R^0_\mathrm{D}$. By Eq. (7.44), $\varepsilon_\mathrm{D} = (g_\mathrm{deg}\alpha/6\pi v_\mathrm{D}) B_\mathrm{D}(s)$, with B_D the bracket of that equation: the Coulomb factor e^2/q^2 multiplies the bare pair response, and no ε_bg appears. Since ε_bg is real, the loss function $L = \mathrm{Im}(-1/\varepsilon)$ of Eq. (4.45) has the Dirac pairs in its numerator and the whole crystal in its denominator (Appendix D.41).

Loss function of a Dirac material

$$L(\boldsymbol{q},\omega) = \frac{\mathrm{Im}\,\varepsilon_\mathrm{D}}{|\varepsilon_\mathrm{bg}+\varepsilon_\mathrm{D}|^2} = \frac{g_\mathrm{deg}\alpha}{6 v_\mathrm{D}} \frac{\rho(s)\,\Theta(s-4\Delta_\mathrm{D}^2)}{|\varepsilon_\mathrm{bg}+\varepsilon_\mathrm{D}(\boldsymbol{q},\omega)|^2}, \qquad s = \omega^2 - v_\mathrm{D}^2 q^2, \tag{16.1}$$

with $\rho(s) = (1-4\Delta_\mathrm{D}^2/s)^{1/2}(1+2\Delta_\mathrm{D}^2/s)$ and ε_D from Eq. (7.44). The scattering rate is Eq. (10.4) with this L; the absorption rate of a dark photon is Eq. (10.31) with $L(0, m_\phi)$.

For a dark photon, which couples to the total charge, Eq. (16.1) is exact within the two approximations of Eq. (7.42), a local background and the random-phase Dirac response (Eq. (8.38)). For a mediator that couples to the electron number it holds with the nuclei clamped. Pairs cost at least $2\Delta_\mathrm{D}$, 23.6 meV in $ZrTe_5$. Where the pair energies lie above the optical phonons, ε_bg is the clamped-ion constant of the remote bands. Near a polar phonon it acquires the ionic poles of Sect. 5.5 and depends on ω.

The numerator of Eq. (16.1) does not depend on the background. Deep in the continuum it is $g_\mathrm{deg}\alpha/6v_\mathrm{D}$, which is 6.1 for $v_\mathrm{D} = 4\times10^{-4}$, and all of the dependence on ε_bg sits in the denominator. At weak screening, $|\varepsilon_\mathrm{D}| \ll \varepsilon_\mathrm{bg}$, the loss function tends to the leading form $\mathrm{Im}\,\varepsilon_\mathrm{D}/\varepsilon_\mathrm{bg}^2$, and the rate of a Dirac target falls as $1/(v_\mathrm{D}\varepsilon_\mathrm{bg}^2)$. The factor $|1+\varepsilon_\mathrm{D}/\varepsilon_\mathrm{bg}|^{-2}$ that separates the two forms is the screening by the Dirac electrons themselves, the charge renormalization of quantum electrodynamics with v_D in place of the speed of light. Its size is set by $g_\mathrm{deg}\tilde\alpha$, with $\tilde\alpha = \alpha/(\varepsilon_\mathrm{bg} v_\mathrm{D})$ the effective coupling of Eq. (7.43), times a logarithm: 0.61 at a pair energy of 10 meV in the benchmark of Worked example 16.1, and 0.86 to 0.94 for the rate of $ZrTe_5$.

The inputs are those of Sect. 7.3: the principal velocities, the half-gap, the cutoff and the background tensor. Their values for $ZrTe_5$, and the excitonic calculation that the threshold region needs, are specified in Appendix E.20. A computed dielectric tensor contains the Dirac pairs at the computed gap, and the background must exclude them; for $ZrTe_5$ the Dirac part is 3 to 8 % of each component for the parameters used below (Worked example 16.9), within the spread of the computed tensors.

16.1.2 Background screening in the published rates

The published Dirac rates [1, 2] take the reference cross-section $\bar{\sigma}_e$ in vacuum, sum the pair transitions with a vertex of unit charge, which gives $\operatorname{Im}\varepsilon_{\mathrm{D}}$, and let the medium enter through an in-medium form factor $F_{\mathrm{med}} = 1/\varepsilon_{\mathrm{pub}}$. Here $\varepsilon_{\mathrm{pub}} = 1 + \varepsilon_{\mathrm{D}}/\varepsilon_{\mathrm{bg}}$ (written ε_r there) is the polarization of the Dirac electrons computed with the charge $e/\varepsilon_{\mathrm{bg}}^{1/2}$ [1, Eqs. (2.5), (2.6), (3.7), (3.11)], [2, Eqs. (2.5)–(2.7), (3.3)]. Since $\varepsilon_{\mathrm{pub}} = \varepsilon/\varepsilon_{\mathrm{bg}}$, the background screens the Dirac electrons among themselves but not the field of the dark photon. For absorption the same polarization serves as the photon self-energy, $\Pi = \omega^2(1 - \varepsilon_{\mathrm{pub}})$ [1, Eqs. (4.4), (4.5)], [2, Eqs. (2.25)–(2.27)]. Appendix D.41 derives both constructions and compares them with the total-ε rates.

Published Dirac rates against the total ε

For an isotropic Dirac material, at every $(\boldsymbol{q}, \omega)$ and for any size of ε_{D},

$$L_{\mathrm{pub}} = \frac{\operatorname{Im}\varepsilon_{\mathrm{D}}}{|\varepsilon/\varepsilon_{\mathrm{bg}}|^2} = \varepsilon_{\mathrm{bg}}^2\, L, \qquad \Gamma_{\mathrm{pub}} = \zeta^2 m_\phi\, \frac{\operatorname{Im}\varepsilon_{\mathrm{pub}}}{|\varepsilon_{\mathrm{pub}}|^2} = \varepsilon_{\mathrm{bg}}\, \Gamma, \tag{16.2}$$

so that a background-free reach curve moves by $\bar{\sigma}_e^{\mathrm{lim}} \to \varepsilon_{\mathrm{bg}}^2 \bar{\sigma}_e^{\mathrm{lim}}$ and $\zeta^{\mathrm{lim}} \to \varepsilon_{\mathrm{bg}}^{1/2} \zeta^{\mathrm{lim}}$ at every mass (Eq. (D.233)). For an anisotropic material the factors hold along the principal axes, with $\varepsilon_{\mathrm{bg},ii}$.

The powers differ because the numerators differ. In scattering the numerator is the bare pair spectrum with the vacuum charge, while in absorption $\operatorname{Im}\varepsilon_{\mathrm{pub}} = \operatorname{Im}\varepsilon_{\mathrm{D}}/\varepsilon_{\mathrm{bg}}$ already carries one $1/\varepsilon_{\mathrm{bg}}$, the square of the reduced charge in the loop. The published scattering rate is the exact rate of a mediator that couples to the Dirac electrons and to no other charge, whereas a kinetically mixed dark photon couples to the electromagnetic current of the whole crystal [1, Eq. (2.1)]. The factor survives the rescaling of the cutoff used in the original analysis, in either subtraction scheme (Eq. (D.230)).

How large is the factor in a material where the one-loop dielectric function can be trusted? For an isotropic cone, or along a principal axis with v_i and $\varepsilon_{\mathrm{bg},ii}$ in place of v_{D} and $\varepsilon_{\mathrm{bg}}$, scattering needs $v_{\mathrm{D}} < v_{\mathrm{max}}$ (Sect. 7.3), and the weak-coupling condition $\tilde{\alpha} = \alpha/(\varepsilon_{\mathrm{bg}} v_{\mathrm{D}}) < 1$ then requires $\varepsilon_{\mathrm{bg}} > \alpha/v_{\mathrm{D}} > \alpha/v_{\mathrm{max}} = 2.75$. Wherever both hold, the published scattering rate is too large by at least $(\alpha/v_{\mathrm{max}})^2 = 7.6$, and for a cone slower than v_{max} the published absorption rate, too large by $\varepsilon_{\mathrm{bg}}$, by at least $\alpha/v_{\mathrm{max}} = 2.75$; absorption needs no $v_{\mathrm{D}} < v_{\mathrm{max}}$, and a faster cone has a smaller bound α/v_{D}. The published form is the $\varepsilon_{\mathrm{bg}} \to 1$ limit of Eq. (16.1), and Limit Check 16.1 follows the two rates as the background grows.

The correction also reverses a rule for choosing a target. In the published form the scattering rate grows with $\varepsilon_{\mathrm{bg}}$ toward the unscreened rate of the pairs, and the absorption rate $\propto \varepsilon_{\mathrm{bg}} \operatorname{Im}\varepsilon_{\mathrm{D}}/|\varepsilon_{\mathrm{bg}} + \varepsilon_{\mathrm{D}}|^2$ is largest at $\varepsilon_{\mathrm{bg}} = |\varepsilon_{\mathrm{D}}|$, with $|\varepsilon_{\mathrm{D}}| = 7.6$ at 50 meV in the material of Limit Check 16.1. A maximum of this kind underlies the optimal ratio $\varepsilon_{\mathrm{bg}}/g_{\mathrm{deg}} \sim 10$ found by Hochberg et al. [1, Sect. 4.2]. With the total ε both rates fall monotonically as the background grows at fixed Dirac parameters. The best Dirac target has the smallest background constant that still keeps the coupling weak, $\varepsilon_{\mathrm{bg}}$ near α/v_{D}.

Limit Check 16.1. Background screening of Dirac-material rates

Published scattering and absorption rates of a Dirac material, with the vacuum $\bar{\sigma}_e$ and the in-medium factor $1/\varepsilon_{\mathrm{pub}}$ [1, Eqs. (3.7), (3.11), (4.4), (4.5)], [2, Eqs. (2.7), (2.25)–(2.27), (3.14)]: the $\varepsilon_{\mathrm{bg}} \to 1$ limit of Eq. (16.1) and Eq. (10.31).

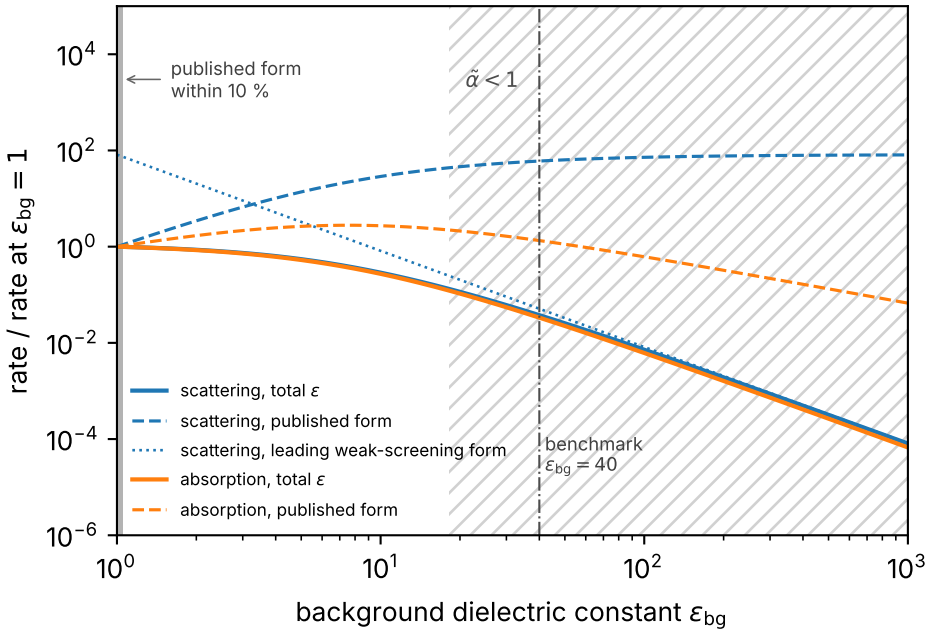


Figure 16.2: Rates of the gapped isotropic Dirac material of Ref. [1] ($v_D = 4 \times 10^{-4}$, $\Delta_D = 2.5\,\mathrm{meV}$, $g_{deg} = 2$, hard cutoff $\Lambda_D = 0.2\,\mathrm{keV}$) against the background dielectric constant at fixed Dirac parameters, each divided by its value at $\varepsilon_{bg} = 1$, where the two forms coincide. Blue: scattering at $m_\chi = 100\,\mathrm{keV}$ through a light mediator, Eq. (10.4), with the total ε (solid), the published form (dashed) and the leading weak-screening form $\mathrm{Im}\,\varepsilon_D/\varepsilon_{bg}^2$ (dotted). Orange: dark-photon absorption at $m_\phi = 50\,\mathrm{meV}$ with the total ε (solid) and the published form (dashed). Gray: $\varepsilon_{bg} \leq 1.049$, where the published scattering rate is within 10 % of the total-ε rate. Hatched: $\tilde{\alpha} < 1$, $\varepsilon_{bg} > \alpha/v_D = 18.2$, where one-loop screening is used. Dash-dotted: the published benchmark $\varepsilon_{bg} = 40$, where the total-ε scattering rate is 0.74 of its leading form. The dashed curves are ε_{bg}^2 and ε_{bg} times the solid ones.

For $\varepsilon_{bg} > 1$ the published form leaves out the screening of the dark-photon field by the remote bands; at large ε_{bg} its scattering rate tends to the unscreened rate of the Dirac pairs, 0.99 of it at $\varepsilon_{bg} = 1000$, while the true rate falls as $1/\varepsilon_{bg}^2$.

Worked example 16.1: The factor ε_{bg}^2 at $\varepsilon_{bg} = 40$

The ideal Dirac material of Ref. [1, Fig. 4] has $v_D = 4 \times 10^{-4}$, $\varepsilon_{bg} = 40$, $g_{deg} = 2$ and a cutoff of 0.2 keV, which we read as a hard cutoff, $v_D\Lambda_D = 80\,\mathrm{meV}$. The coupling is $\tilde{\alpha} = \alpha/(40 \times 4 \times 10^{-4}) = 0.456$, and $g_{deg}\tilde{\alpha} = 0.91$ is the effective coupling quoted there. *One point.* Take the gapless cone at $\boldsymbol{q} \to 0$ and $\omega = 10\,\mathrm{meV}$, so that $\sqrt{s} = 10\,\mathrm{meV}$. Eq. (D.138) gives the bracket $\ln(4v_D^2\Lambda_D^2/s) + i\pi = 5.545 + 3.142\mathrm{i}$, and with $g_{deg}\alpha/(6\pi v_D) = 1.936$, $\varepsilon_D = 10.73 + 6.08\mathrm{i}$. The total is $\varepsilon = 50.73 + 6.08\mathrm{i}$ and $|\varepsilon|^2 = 2611$, so $L = 6.08/2611 = 2.33 \times 10^{-3}$, 0.61 of the leading form $6.08/1600 = 3.80 \times 10^{-3}$. Had the 0.2 keV been a cutoff in the subtraction scheme of the reference, the bracket would gain $\frac{5}{3}$, $\mathrm{Re}\,\varepsilon_D/\varepsilon_{bg}$ would rise from 0.27 to 0.35, and the ratio to the leading form would be 0.54. The published construction divides $\mathrm{Im}\,\varepsilon_D = 6.08$ by $|\varepsilon/\varepsilon_{bg}|^2 = 1.632$ and obtains 3.73,

which is 1600 times L.
Absorption. At $m_\phi = 10\,\mathrm{meV}$ the same ε applies. The published rate is $\zeta^2 m_\phi \,\mathrm{Im}\,\varepsilon_{\mathrm{pub}}/|\varepsilon_{\mathrm{pub}}|^2 = 0.0932\,\zeta^2 m_\phi$, forty times $\zeta^2 m_\phi L$. With the density $10\,\mathrm{g\,cm^{-3}}$ of the reference and $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$, Eq. (10.31) gives $3.35\times10^{30}\,\zeta^2$ events per kg-year, and three events in a kg-year reach $\zeta = 9.5\times10^{-16}$, against 1.5×10^{-16} with the published form, a factor $\sqrt{40} = 6.3$.
Scattering reach. For scattering the reference counts one Dirac valence electron per cell, $n_e = 5\times10^{24}\,\mathrm{kg^{-1}}$ electrons per unit mass in cells of $V_{\mathrm{uc}} = 60\,\mathrm{Å}^3$, which fixes the density $1/(n_e V_{\mathrm{uc}}) = 3.33\,\mathrm{g\,cm^{-3}}$; its absorption estimate takes $10\,\mathrm{g\,cm^{-3}}$ instead, and we keep each with its own rate. For the gapped cone ($\Delta_{\mathrm{D}} = 2.5\,\mathrm{meV}$) at $m_\chi = 100\,\mathrm{keV}$, with the standard halo and three events in a kg-year, Eq. (10.4) with Eq. (16.1) reaches $\bar\sigma_e = 4.4\times10^{-42}\,\mathrm{cm^2}$, and the published form $2.8\times10^{-45}\,\mathrm{cm^2}$, a factor 1600 at this and at every other mass, 3.2 decades. The published curve for this material has its minimum, $3.4\times10^{-46}\,\mathrm{cm^2}$ near 9 keV, read from the reference's figure; corrected, it lies at $5.4\times10^{-43}\,\mathrm{cm^2}$.[a]

[a] The published curve lies about a factor two above the reference's own formula: with the standard halo it reads $5.0\times10^{-45}\,\mathrm{cm^2}$ at 100 keV against the recomputed $2.8\times10^{-45}\,\mathrm{cm^2}$, a factor 1.8. The factor 1600 between the published and the total-ε reach holds either way.

16.1.3 $ZrTe_5$ and other anisotropic targets

The background of an anisotropic crystal enters the longitudinal dielectric function as $\hat{\boldsymbol{q}}\cdot\varepsilon_{\mathrm{bg}}\cdot\hat{\boldsymbol{q}} = \sum_i \varepsilon_{\mathrm{bg},ii}\hat q_i^2$, beside a Dirac part that carries the same weights $\hat q_i^2$ times v_i^2 (Eq. (D.226)). The in-medium factor of Ref. [2, Eq. (3.14)] instead divides each component of the Dirac polarization by its own $\varepsilon_{\mathrm{bg},ii}$ and keeps the vacuum 1 beside them. Along a principal axis the ratio of the two rates is $\varepsilon_{\mathrm{bg},ii}^2$, which for the computed tensor of $ZrTe_5$, $\varepsilon_{\mathrm{bg},ii} = 308.4$, 20.75 and 126.1 [2, Table 2], is 9.5×10^4, 431 and 1.6×10^4. Off the axes it depends on the direction through both sums of Eq. (D.229), and the reach must be recomputed. The background tensor also orders the directions by itself. At equal Dirac numerator, a momentum transfer along y, the axis of the smallest background constant, gives a rate $(308.4/20.75)^2 = 221$ times that along x. The rate of a crystal fixed in the laboratory therefore modulates daily mainly through its background tensor, which neither a harmonic mean of the tensor nor the per-component factor represents.

We evaluate $ZrTe_5$ with the parameter row that the calculation of Worked example 7.4 gives for the measured crystal structure [2, Table 1 and text below it]: the principal velocities 1.1×10^{-3}, 4.5×10^{-4} and 1.0×10^{-3}, the energy cutoff $\tilde\Lambda = \min_i v_i\Lambda_i = 0.191\,\mathrm{eV}$, $g_{\mathrm{deg}} = 2$, and the half-gap $\Delta_{\mathrm{D}} = 11.8\,\mathrm{meV}$, half the computed gap of 23.6 meV. The relaxed cell of Worked example 7.4, the row the reference implements, gives 15.6 meV with its own velocities and cutoff, and an earlier calculation 17.5 meV. The threshold mass is 8.0 keV for the isotropic cone and 9.6, 7.1 and 9.0 keV along the principal axes. We average the loss function over the directions of $\boldsymbol{q}$, which gives the rate averaged over the orientations of the crystal relative to the dark-matter wind, and insert it into Eq. (10.4) with a light mediator, the standard halo and the measured density $6.089\,\mathrm{g\,cm^{-3}}$ [2, Fig. 3]. At $m_\chi = 100\,\mathrm{keV}$ and $\bar\sigma_e = 10^{-38}\,\mathrm{cm^2}$ this yields 108 events per kg-year. Since a background-free reach scales as the inverse rate (Eq. (1.54)), three events in a kg-year reach about $1.5\times10^{-40}\,\mathrm{cm^2}$ near 40 keV (Fig. 16.3a). The published rule, recomputed with the same inputs and the standard halo, gives a rate 5200 to 6400 times larger between 20 keV and 1 MeV. Over the two computed rows (11.8 and 15.6 meV) and the two halos of Fig. 16.3 the factor spans 4800 to 6600, and a halo with

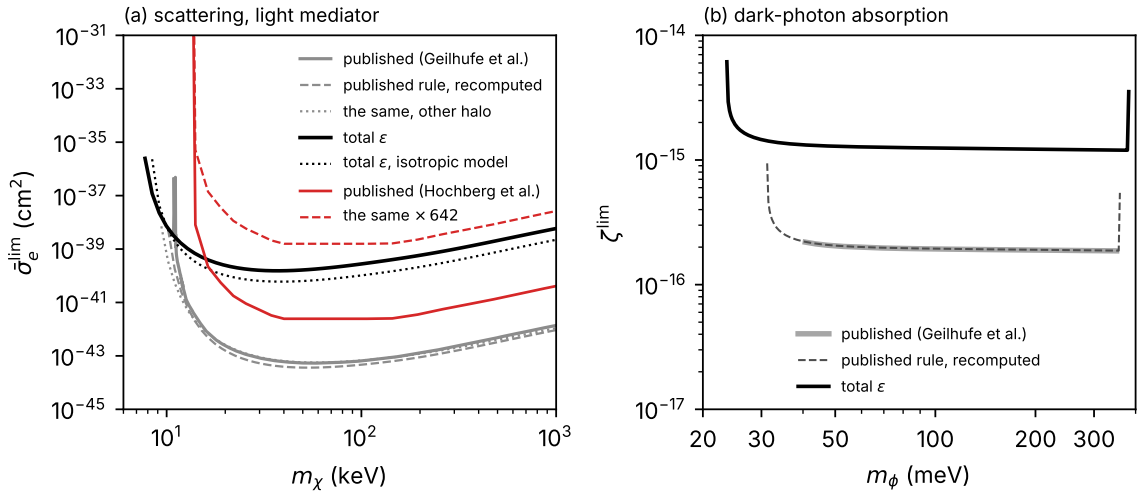


Figure 16.3: $ZrTe_5$ reach as published and with the background in the total dielectric function: three events in one kg-year, no background, $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$. (a) Scattering through a light mediator. Gray solid: the published curve for the computed parameters of Geilhufe et al. [2] (read from their figure). Gray dashed: their in-medium factor, recomputed for their half-gap, 15.6 meV, with the standard halo and the loss function averaged over the directions of $\boldsymbol{q}$; it lies at 0.63 to 0.72 of the published curve above 20 keV, consistent with a difference of halo: the source states none, and with $v_0 = 220$, $v_\mathrm{E} = 232$ and $v_\mathrm{esc} = 600\,\mathrm{km\,s^{-1}}$ (gray dotted) the ratio is 0.8 to 1.1. Black solid: the target with the total ε, Eq. (16.1) with the anisotropic ε_D of Eq. (D.226), measured-structure row ($\Delta_\mathrm{D} = 11.8\,\mathrm{meV}$, $\tilde{\Lambda} = 0.191\,\mathrm{eV}$), density $6.089\,\mathrm{g\,cm^{-3}}$. Black dotted: its isotropic model (geometric-mean velocity, harmonic-mean $\varepsilon_\mathrm{bg} = 50.5$) with the total ε. Red: the published curve of Hochberg et al. [1] for their $ZrTe_5$ parameters, isotropized with $\varepsilon_\mathrm{bg} = 25.34$ (solid), and the same curve times 642 (dashed). (b) Absorption of dark-photon dark matter, each polarization along a principal axis, unpolarized average, density $6.135\,\mathrm{g\,cm^{-3}}$ as in Ref. [2]. Gray: their published curve; dashed: their rule recomputed with $\Delta_\mathrm{D} = 15.6\,\mathrm{meV}$, within 0.5 % of it; black: the total ε with the measured-structure row. Published-rule curves: $\Delta_\mathrm{D} = 15.6\,\mathrm{meV}$, $\tilde{\Lambda} = 0.180\,\mathrm{eV}$. Off-diagonal entries of the tensor ($|\varepsilon_{\mathrm{bg},ij}| \le 1.2$) are dropped.

$v_0 = 220$, $v_\mathrm{E} = 232$ and $v_\mathrm{esc} = 544\,\mathrm{km\,s^{-1}}$ lowers its lower end to 4300. That is more than the square of the harmonic mean of the tensor, $50.5^2 = 2550$, and an isotropic model of the same row, with the geometric-mean velocity and the harmonic-mean background as in Worked example 7.4, overstates the orientation-averaged total-ε rate by a factor 2.3 to 2.7. The curve of Hochberg et al. [1], computed for an isotropic model with $\varepsilon_\mathrm{bg} = 25.34$, the harmonic mean of its own tensor (187.5, 9.8 and 90.9), moves by $25.34^2 = 642$ exactly.

Absorption at rest has no direction of $\boldsymbol{q}$, but the polarization of the dark photon picks a component of the tensor, and each polarization is absorbed with the factor $\varepsilon_{\mathrm{bg},ii}$ of Eq. (16.2). With the total ε the polarization along y, with the smallest background constant, carries 86 % of the unpolarized rate at 100 meV. Recomputed with the published rule, our absorption rate reproduces the published reach of Ref. [2] to 0.5 % over the whole range, which checks the inputs. With the total ε and the measured-structure row, the reach in ζ weakens by a factor of 6.3 to 6.4 between the gap and the cutoff (Fig. 16.3b), to about 1.25×10^{-15} at 100 meV. The factor is the square root of a polarization average and lies between $\varepsilon_{\mathrm{bg},yy}^{1/2} = 4.6$ and $\varepsilon_{\mathrm{bg},xx}^{1/2} = 17.6$. For the cubic Yb_3PbO of the same reference the factors are $42.8^2 = 1832$

and $42.8^{1/2} = 6.5$ identically. Its computed dielectric constant, however, is described there as sensitive to the size of the gap [2, Sect. 4]. It contains the Dirac bands, the background is smaller than 42.8, and the Dirac term is counted twice under either rule. These two factors describe the model, not the material.

16.1.4 Excitonic and strong-coupling corrections

Eq. (16.1) keeps the random-phase response of the Dirac electrons, the first order in $\tilde{\alpha}$ of their polarization, with the expansion parameter $g_{\mathrm{deg}}\tilde{\alpha}\ln[v_{\mathrm{D}}\Lambda_{\mathrm{D}}/\max(\Delta_{\mathrm{D}}, v_{\mathrm{D}}q)]$ of Sect. 7.3. For $ZrTe_5$ that is 1.02 at long wavelength with the measured-structure row (0.93 with the relaxed cell), at the edge of the expansion, and for the benchmark of Worked example 16.1, $g_{\mathrm{deg}}\tilde{\alpha} = 0.91$ times a logarithm of a few. At next order the electron–hole ladder, the self-energy and vertex of the carriers, and the material outside the model all enter.

The ladder of the Bethe–Salpeter equation (Sect. 4.9) is the attraction between the electron and the hole of a pair. Near the edge the pair is nonrelativistic, with reduced mass $\Delta_{\mathrm{D}}/2v_{\mathrm{D}}^2$ and Coulomb coupling $\alpha/\varepsilon_{\mathrm{bg}}$. It binds with the energy $(\Delta_{\mathrm{D}}/2v_{\mathrm{D}}^2)(\alpha/\varepsilon_{\mathrm{bg}})^2/2 = \Delta_{\mathrm{D}}\tilde{\alpha}^2/4$, 0.10 meV for the isotropic model of $ZrTe_5$ with $\tilde{\alpha} = 0.18$, and its continuum carries the Sommerfeld factor $x/(1-\mathrm{e}^{-x})$ with $x = \pi\tilde{\alpha}/b$. Here $b = (1-4\Delta_{\mathrm{D}}^2/s)^{1/2}$ is the speed of each carrier in the rest frame of the pair, in units of v_{D}. In $ZrTe_5$ the factor exceeds 1.5 for $b < 0.66$, that is for pair energies up to 1.33 times the edge, and that region carries 60 % of the rate at 20 keV, 12 % at 100 keV and 3 % at 1 MeV in the isotropic model. Near the threshold mass the reach is therefore uncertain by the Sommerfeld factor, 1.5 or more, until the ladder is solved, which Appendix E.20 specifies. The bound states lie at $\sqrt{s} < 2\Delta_{\mathrm{D}}$ and add absorption lines below the edge.

The self-energy of the carriers renormalizes v_{D} logarithmically in the cutoff, and a vertex correction accompanies it. Each is of relative order $\tilde{\alpha}$, with a coefficient that the one-loop result does not fix. Outside the model, a chemical potential inside a band adds an intraband term, Thomas–Fermi screening with $k_{\mathrm{TF}}^2 = e^2\mathcal{N}_{\mathrm{D}}(E_{\mathrm{F}})/\varepsilon_{\mathrm{bg}}$, which removes the weak screening at $\omega \lesssim v_{\mathrm{D}}k_{\mathrm{F}}$. The background acquires ionic poles near the optical phonons. And the bands bend beyond Λ_{D}, which moves weight between the background and the logarithm of Eq. (7.46).

At $\tilde{\alpha} \gtrsim 1$ none of these is small, and no controlled method is known. The ladder is the controlled correction of first order in $\tilde{\alpha}$ and is uncontrolled beyond it; neither self-consistent *GW* with the Bethe–Salpeter equation nor a functional renormalization group has a small parameter there. The question matters for the slowest cones, which have the most states and the lowest thresholds. BNQ-TTF, with velocities near 2×10^{-4}, has an effective coupling of about 10, and its authors present its one-loop rates as a qualitative estimate [2, footnote 3]. What replaces Eq. (16.1) for such targets is open.

16.2 Low-dimensional plasmons and layered stacks

Rolled into a metallic carbon nanotube, the Dirac electrons of Sect. 16.1 put their whole low-energy response to a density coupling into one line, $\omega = v_{\mathrm{D}}q_{\parallel}/g_c$, with $v_{\mathrm{D}} = 3.02 \times 10^{-3}$ the Dirac velocity of graphene. The Luttinger parameter $g_c \approx 0.2$ (Eq. (16.9)) makes the line 3.4 to 6.9 times faster than any particle of the halo at momenta of 1 to 100 eV. The plasmon of a doped graphene sheet is faster still at the momenta of keV-mass dark matter. In bulk silicon a mode that fast is out of reach: the plasmon of Sect. 10.3 needs a particle at 2860 km s^{-1}. In a tube or a sheet, however, the response depends only on the momentum along the tube or in the plane, and a halo particle can supply the rest of its momentum across it. A mode faster than $v_{\max}$ is then reached by an oblique momentum transfer, and the confined plasmon becomes a final state of its own. Its weight sits in a single line at low

energy, which is what the sum-rule bounds of Sect. 10.2 need to come close to saturation. We will derive the Coulomb kernels of sheets and tubes, the random-phase plasmons in one and two dimensions, the acoustic plasmon of two kinds of carriers, and the dielectric function of a stack of layers, exact at long wavelength, with its leading correction.

16.2.1 Confined electrons and their Coulomb kernels

Consider identical units, wires or sheets, in which electrons move freely in one or two dimensions and are confined across, with no tunneling between units. Write $\boldsymbol{q} = (\boldsymbol{q}_\parallel, \boldsymbol{q}_\perp)$, with $\boldsymbol{q}_\parallel$ along the free directions. Suppose the distribution of the electrons across a unit does not depend on their state along it. The density operator of Eq. (2.38) then factorizes, $n_e(\boldsymbol{q}) = \sum_u \mathrm{e}^{\mathrm{i}\boldsymbol{q}_\perp\cdot\boldsymbol{r}_u} F_\perp(\boldsymbol{q}_\perp)\, n_u(\boldsymbol{q}_\parallel)$. Here $\boldsymbol{r}_u$ locates unit u, n_u is its density operator along the free directions, and the transverse form factor $F_\perp$ is the Fourier transform of the normalized distribution across the unit, with $F_\perp(0) = 1$. For a sheet of uniform thickness d, $F_\perp = \sin(q_z d/2)/(q_z d/2)$; for a tube whose electrons sit on a cylinder of radius r_t, $F_\perp = J_0(q_\perp r_\mathrm{t})$. If the units respond independently, the cross terms between units average to zero at $\omega \neq 0$. The structure factor of Eq. (4.13) then becomes

$$S(\boldsymbol{q},\omega) = \boldsymbol{n}_\mathrm{u}\, |F_\perp(\boldsymbol{q}_\perp)|^2\, S_\mathrm{u}(\boldsymbol{q}_\parallel,\omega), \tag{16.3}$$

with $\boldsymbol{n}_\mathrm{u}$ the number of units per unit transverse area (wires) or length (sheets) and S_u the structure factor of one unit per unit length or area, in the normalization of Sect. 4.2. In Eq. (8.16) the target enters as $S/\rho_T = |F_\perp|^2 S_\mathrm{u}/\rho_\mathrm{u}$, with ρ_u the mass of one unit per unit length or area. The rate per unit mass is then independent of how densely the units are packed, as long as they remain independent. The tube proposal of Zhang and Chen reports its rate per unit tube length [3, Eq. (6)], and Das, Jang and Min write Eq. (16.3) for a sheet of uniform thickness with the areal mass density [4, Eqs. (13), (14)].

Eq. (16.3) changes the kinematics. The structure factor depends on $\boldsymbol{q}_\parallel$ alone, while the dark matter deposits $\omega = \boldsymbol{q}\cdot\boldsymbol{v} - q^2/2m_\chi$ with the full $\boldsymbol{q}$. A mode at $(\boldsymbol{q}_\parallel,\omega)$ is reached by any $\boldsymbol{q}$ with that projection, and the condition $\omega \le q v_\mathrm{max} - q^2/2m_\chi$ only requires $q \gtrsim \omega/v_\mathrm{max}$, which can exceed $q_\parallel$ by any amount. The cost is the form factor, which falls once $q_\perp$ exceeds the inverse thickness. For a tube of radius 0.7 nm, $|J_0|^2$ starts to fall at $q_\perp \approx 1/r_\mathrm{t} = 282\,\mathrm{eV}$, and halo particles reach energies up to $v_\mathrm{max}/r_\mathrm{t} = 0.75\,\mathrm{eV}$ without that suppression. A bulk mode at $(\boldsymbol{q},\omega)$, by contrast, is reached only if $\omega \le q v_\mathrm{max}$ at its own $\boldsymbol{q}$.

The electrons of a unit interact through the Coulomb potential averaged over the transverse distribution. Take a sheet between two half spaces of dielectric constants ε_a and ε_b. Sect. 15.3.4 found the potential of its density wave from the two-dimensional Fourier transform of $e^2/4\pi r$ and the image charges,

$$v_\mathrm{C}^\mathrm{2D}(q_\parallel) = \frac{e^2}{(\varepsilon_a+\varepsilon_b)\,q_\parallel} = \frac{e^2}{2\bar\varepsilon\, q_\parallel}, \qquad \varepsilon_\mathrm{2D} = 1 - v_\mathrm{C}^\mathrm{2D}\tilde R^\mathrm{2D}, \qquad S_\mathrm{2D} = -2\ \mathrm{Im}\,\frac{\tilde R^\mathrm{2D}}{\varepsilon_\mathrm{2D}} \quad (T=0), \tag{16.4}$$

with $\bar\varepsilon = (\varepsilon_a+\varepsilon_b)/2$ and $\tilde R^\mathrm{2D}$ the proper density response of the sheet per unit area, the two-dimensional analogue of Eq. (4.39). The kernel falls as $1/q_\parallel$, not as $1/q^2$: a sheet screens a long-wavelength potential weakly, because the field of its induced charge spreads into the space around it. Give the sheet a finite thickness d, with a uniform distribution across it in a homogeneous medium. Averaging $\mathrm{e}^{-q_\parallel|z-z'|}$ over both positions multiplies the kernel by

$$F_\mathrm{th}(q_\parallel d) = \frac{2}{q_\parallel d} - \frac{2(1-\mathrm{e}^{-q_\parallel d})}{(q_\parallel d)^2} = 1 - \frac{q_\parallel d}{3} + O(q_\parallel^2 d^2). \tag{16.5}$$

For the interlayer distance $d = 3.35\,\text{Å}$ of bilayer graphene the factor is 0.95 at $q_\parallel = 100\,\mathrm{eV}$ and 0.85 at 300 eV, and it reaches 0.9 at 191 eV; Ref. [4, Eq. (17)] keeps the profile in the form

factor but uses the zero-thickness kernel, a correction of this size at the momenta of sub-MeV dark matter.

The bands far from the Fermi level enter the proper response as a static part $\tilde{R}^{\mathrm{2D}}_{\mathrm{bg}}$, which for an insulating background is proportional to $q_\parallel^2$ at small $q_\parallel$. Writing $-\tilde{R}^{\mathrm{2D}}_{\mathrm{bg}} = 2r_0 q_\parallel^2/e^2$ defines the Keldysh length r_0, and

$$\bar{\varepsilon}\,\varepsilon_{\mathrm{2D}}(q_\parallel,\omega) = \bar{\varepsilon} + r_0 q_\parallel - \frac{e^2}{2q_\parallel}\,\tilde{R}^{\mathrm{2D}}_{\mathrm{low}}(q_\parallel,\omega), \tag{16.6}$$

with $\tilde{R}^{\mathrm{2D}}_{\mathrm{low}}$ the response of the bands near the Fermi level. Compared to a bulk material, the remote bands of a sheet do not screen at long wavelength: their contribution $r_0 q_\parallel$ vanishes as $q_\parallel \to 0$, while in three dimensions they give the constant $\varepsilon_{\mathrm{bg}}$ of Sect. 7.3 that Sect. 16.1 places inside the total ε. In a sheet the substrate enters only through $\bar{\varepsilon}$ and the remote bands only through r_0; neither rescales the coupling to the dark matter. A polar substrate makes ε_b depend on frequency, and the sheet plasmon then mixes with its surface phonons.

For a tube the same average over a cylindrical shell of radius r_{t} selects the azimuthally uniform term of the expansion of $1/|\boldsymbol{r}-\boldsymbol{r}'|$ in cylindrical coordinates, $(2/\pi)\int_0^\infty \mathrm{d}k \cos[k(z-z')] I_0(kr_{\mathrm{t}}) K_0(kr_{\mathrm{t}})$, with I_0 and K_0 the modified Bessel functions of order zero [5, Sect. 10.25]. Its Fourier transform along the axis is $2I_0(q_\parallel r_{\mathrm{t}})K_0(q_\parallel r_{\mathrm{t}})$, and with $e^2 = 4\pi\alpha$ (Appendix A.2)

$$v_{\mathrm{C}}^{\mathrm{1D}}(q_\parallel) = \frac{2\alpha}{\varepsilon_{\mathrm{env}}}\, I_0(q_\parallel r_{\mathrm{t}})\, K_0(q_\parallel r_{\mathrm{t}}) \simeq \frac{2\alpha}{\varepsilon_{\mathrm{env}}}\Big[\ln\frac{2}{q_\parallel r_{\mathrm{t}}} - \gamma_{\mathrm{E}}\Big] \quad (q_\parallel r_{\mathrm{t}} \ll 1), \tag{16.7}$$

for a tube in a medium of dielectric constant $\varepsilon_{\mathrm{env}}$, with γ_{E} Euler's constant; the small-argument form follows from $I_0 \to 1$ and $K_0(x) \to \ln(2/x) - \gamma_{\mathrm{E}}$ [5, Eqs. (10.30.1), (10.31.2)]. The logarithmic form falls 10 % below the exact kernel at $q_\parallel r_{\mathrm{t}} = 0.38$, which is 108 eV for $r_{\mathrm{t}} = 0.7$ nm. The logarithm comes from the one-dimensional geometry: the interaction of two lines of charge grows without bound as their wavelength grows.

Appendix E.21 specifies how $\tilde{R}^{\mathrm{2D}}$, r_0 and the corresponding tables for tubes and stacks are computed. The proper response of a monolayer comes from a plane-wave code with the Coulomb interaction cut off between periodic images, or from a Wannier-function bubble. The substrate enters through its measured $\varepsilon_b(\omega)$ with the polar phonons, and the random-phase equations are solved with the kernels above.

16.2.2 Plasmons in one and two dimensions

At frequencies above the pair continuum the proper response of any band takes the form $\tilde{R} \simeq D_{\mathrm{Dr}}\, q_\parallel^2/\omega^2$, with D_{Dr} the Drude weight. The coefficient follows from the intraband part of the Lehmann sum, expanded for $\omega \gg \boldsymbol{q}\cdot\boldsymbol{v}$: $D_{\mathrm{Dr}} = \sum_{\boldsymbol{k}} (-\partial n_{\mathrm{F}}/\partial E)(\hat{\boldsymbol{q}}\cdot\boldsymbol{v}_{\boldsymbol{k}})^2$ per unit area, summed over spin and valleys. For a parabolic band of mass m^* this is $\boldsymbol{n}_{\mathrm{2D}}/m^*$, the areal density over the mass, as the f-sum rule requires. For a Dirac cone, $\boldsymbol{v}_{\boldsymbol{k}} = v_{\mathrm{D}}\hat{\boldsymbol{k}}$ and the Fermi circle has length $2\pi k_{\mathrm{F}}$, so $D_{\mathrm{Dr}} = g_{\mathrm{deg}} E_{\mathrm{F}}/4\pi$, with $g_{\mathrm{deg}} = 4$ for graphene. The zero of $\varepsilon_{\mathrm{2D}}$ in Eq. (16.4) is then the sheet plasmon. For a parabolic band the next term of the high-frequency expansion is that of Eq. (4.58), with the disc average $\langle(\hat{\boldsymbol{q}}\cdot\boldsymbol{v})^2\rangle = v_{\mathrm{F}}^2/4$ in place of the sphere's $v_{\mathrm{F}}^2/5$. Then

$$\omega_{\mathrm{pl}}^2(q_\parallel) = \frac{e^2\, \boldsymbol{n}_{\mathrm{2D}}\, q_\parallel}{(\varepsilon_a + \varepsilon_b)\, m^*} + \frac{3}{4}\, v_{\mathrm{F}}^2 q_\parallel^2 + O(q_\parallel^3). \tag{16.8}$$

Compared to the bulk plasmon, which starts at ω_p, the sheet plasmon is gapless and rises as $\sqrt{q_\parallel}$, because the kernel falls only as $1/q_\parallel$. It stays undamped as long as it lies above the pair continuum, $\omega > q_\parallel v_{\mathrm{F}} + q_\parallel^2/2m^*$. For doped graphene in vacuum, $\omega_{\mathrm{pl}}^2 = 2\alpha E_{\mathrm{F}} q_\parallel$. At

$E_F = 0.2\,\text{eV}$ its phase velocity is 20 times v_{max} at $q_\parallel = 1\,\text{eV}$ and 6.4 times at 10 eV, and the long-wavelength form ends at $q_\parallel = 32\,\text{eV}$, where $\omega = 0.30\,\text{eV}$ and the line meets the interband edge $2E_F - v_D q_\parallel$. Here $v_D = 3ta_{cc}/2 = 3.02 \times 10^{-3}$, from the nearest-neighbor hopping $t = 2.8\,\text{eV}$ and the carbon–carbon distance $a_{cc} = 1.42\,\text{Å}$ [6, text after Eqs. (5) and (7)], the value used for graphene and nanotubes throughout this chapter. Das, Jang and Min quote a hopping of 3.16 eV for their bilayer model [4, Sect. II], and Zhang and Chen take $v = 0.8 \times 10^6\,\text{m s}^{-1} = 2.67 \times 10^{-3}$ for nanotubes [3]. Their rate for bilayer graphene is Eqs. (16.3) and (16.4) at $\varepsilon_a = \varepsilon_b = 1$, with the random-phase response of the four low-energy bands [4, Eqs. (13)–(19)]; on a substrate the 2 in their kernel $e^2/2q$ becomes $\varepsilon_a + \varepsilon_b$. Their gate opens a gap and puts the Fermi level inside it, so their loss function is interband and has no sheet plasmon.

The bands of a metallic nanotube at its Fermi level are linear. The tube has N_{ch} channels at the Fermi level, two for spin times two for the valleys, each with right- and left-moving electrons of velocity $\pm v$; for a nanotube $v = v_D$. For the right movers of one channel, $n_k - n_{k+q} = 1$ on an interval of length q below k_F and the energy denominator is $\omega - vq$; the left movers give the same with $-v$. Their sum per unit length is $(q/2\pi)[1/(\omega - vq) - 1/(\omega + vq)]$, and for N_{ch} channels

$$R^0_{1D}(q_\parallel, \omega) = \frac{N_{ch}\, v\, q_\parallel^2}{\pi\,(\omega^2 - v^2 q_\parallel^2)}.$$

The random-phase response $R = R^0_{1D}/(1 - v_C^{1D} R^0_{1D})$ has its pole where $\omega^2 = v^2 q_\parallel^2[1 + N_{ch} v_C^{1D}/(\pi v)]$, and near it $\text{Im}[1/(\omega^2 - \omega_{pl}^2 + \text{i}0^+)] = -\pi\delta(\omega - \omega_{pl})/2\omega_{pl}$, so that $S = -2\,\text{Im}\,R$ is a single line:

One-dimensional plasmon of linear channels

$$S_{1D}(q_\parallel, \omega) = N_{ch}\, g_c\, |q_\parallel|\, \delta\Big(\omega - \frac{v|q_\parallel|}{g_c}\Big), \qquad g_c(q_\parallel) = \Big[1 + \frac{N_{ch}\, v_C^{1D}(q_\parallel)}{\pi v}\Big]^{-1/2}, \tag{16.9}$$

at $T = 0$ and $\omega > 0$, with v_C^{1D} of Eq. (16.7). It is embedded in three dimensions by Eq. (16.3) with $F_\perp = J_0(q_\perp r_t)$.

For a single channel this is the result of Zhang and Chen [3, Eqs. (3), (4)]. They bosonize a model with forward couplings g_2 and g_4 between right and left movers; with $g_2 = g_4 = v_C^{1D}$ their Luttinger parameter and velocity become g_c and v/g_c of Eq. (16.9). For this model the random-phase result is exact, because the linear band makes every higher-order term in the interaction cancel; that is the content of bosonization. The product of the weight and the velocity, $g_c \cdot v/g_c = v$, does not depend on the interaction. The first moment of the line, $\int \text{d}\omega\, \omega S_{1D} = N_{ch} v q_\parallel^2$, is therefore that of the free channels, the f-sum of the linear bands: the interaction raises the line and lowers its weight in the same proportion.

The charge of a nanotube is carried by all four channels together. The dark matter couples to the total density $n = \sum_i n_i$, and so does the Coulomb interaction. The symmetric combination of the channel densities is the plasmon of Eq. (16.9) with $N_{ch} = 4$. The three other combinations, in which spin or valley densities oscillate against each other, carry no charge, keep $g = 1$, and do not appear in S. The model of Ref. [3] has one right- and one left-moving mode, $N_{ch} = 1$, but its benchmark values $g = 0.2$ to 0.7 are those measured for nanotubes, which belong to the charge mode g_c of the four-channel theory. At fixed g_c its structure factor, and the rate of its Eq. (6), are four times too low. At a fixed interaction instead the comparison is $N_{ch} g_c(N_{ch})$ against $g_c(1)$: with the kernel of a tube of radius 0.7 nm

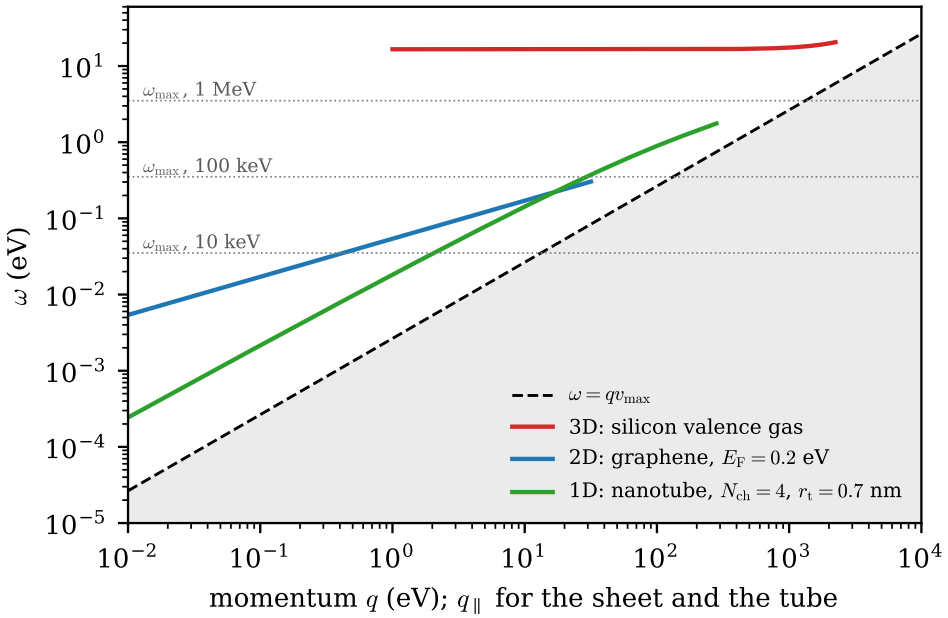


Figure 16.4: Plasmons in three, two and one dimensions against the reach of halo dark matter. Red: the valence-electron gas of silicon, $\omega_{\rm pl}^2 = \omega_p^2 + \frac{3}{5}v_{\rm F}^2 q^2$ with $\omega_p = 16.6\,\text{eV}$, to the quadratic cutoff at 2.24 keV. Blue: doped graphene in vacuum at a Fermi energy of 0.2 eV, $\omega_{\rm pl}^2 = 2\alpha E_{\rm F} q_\parallel$, up to the interband edge at 32 eV. Green: the charge mode of a metallic nanotube, Eq. (16.9) with $N_{\rm ch} = 4$, $r_{\rm t} = 0.7\,\text{nm}$, $\varepsilon_{\rm env} = 1$ and $v_{\rm D} = 3.02 \times 10^{-3}$, to $q_\parallel = 1/r_{\rm t}$. Dashed: $\omega = qv_{\max}$, the largest energy a halo particle deposits with momentum q (gray below it). Dotted: the largest energy $m_\chi v_{\max}^2/2$ for three masses. The bulk mode lies above the dashed line at its own q and is out of reach. The modes of the sheet and the tube depend only on $q_\parallel$, and a particle reaches them with a total momentum $q \geq \omega/v_{\max}$, the rest of it across the sheet or the tube.

in vacuum, $g_c = 0.212$ for four channels and 0.398 for one at $q_\parallel = 10\,\text{eV}$, and the four-channel line is 2.1 times stronger.

The interaction is strong. With $r_{\rm t} = 0.7\,\text{nm}$, $\varepsilon_{\rm env} = 1$ and the graphene velocity, $N_{\rm ch} v_{\rm C}^{\rm 1D}/\pi v_{\rm D}$ is 35 at $q_\parallel = 1\,\text{eV}$ and 8 at 100 eV, and g_c is 0.17, 0.21 and 0.34 at 1, 10 and 100 eV, inside the measured range. The plasmon then moves at 6.9, 5.4 and 3.4 times $v_{\max}$. A bulk mode that fast would be unreachable; the line of a tube is reached by any halo particle that deposits ω with a total momentum $q \gtrsim \omega/v_{\max}$ and a projection $q_\parallel = g_c\omega/v$ on the axis. The rate depends on the angle between the tube and the velocity of the particle, and with it on the time of day, which is the directional signal of Ref. [3]. Fig. 16.4 compares the plasmons of a tube, a sheet and bulk silicon with the reach of the halo.

Eq. (16.9) rests on the linear band. A real band curves on the scale of its Fermi momentum, and the limit check below compares the linear channels with parabolic ones of the same velocity v at $k_{\rm F}$, of mass $m^* = k_{\rm F}/v$. The one-dimensional Lindhard function of one such channel is $(m^*/2\pi q_\parallel)\ln[(\omega^2 - \omega_-^2)/(\omega^2 - \omega_+^2)]$, with $\omega_\pm = q_\parallel v \pm q_\parallel^2/2m^*$ the edges of its pair band, and the zero of the random-phase dielectric function has the closed form $\omega^2 = (\omega_+^2 e^y - \omega_-^2)/(e^y - 1)$, with $y = 2\pi q_\parallel/(N_{\rm ch} v_{\rm C}^{\rm 1D} m^*)$. Expanding it in y, the $q_\parallel^3$ terms cancel and the plasmon frequency is

$$\frac{\omega_{\rm pl}^{\rm par}}{\omega_{\rm pl}} = 1 + \frac{q_\parallel^2}{2k_{\rm F}^2}\, g_c^2 \Big[\frac{1}{4} + \frac{g_c^2}{3(1-g_c^2)}\Big] + O(q_\parallel^4/k_{\rm F}^4).$$

The correction carries g_c^2: the stronger the interaction, the further the plasmon lies above the pairs, and the less the shape of the band matters. In a nanotube the linear band itself ends at the next subbands, whose pairs open at $2v_\mathrm{D}/r_\mathrm{t} = 1.7\,\mathrm{eV}$; below that energy and for $q_\parallel \ll 1/r_\mathrm{t}$ the four channels are the whole low-energy response. In a forest of parallel tubes the Coulomb interaction between tubes adds to v_C^1D once $q_\perp$ is below the inverse spacing, 1 to 2 eV for the spacings of 100 to 200 nm that Ref. [3] considers.

Limit Check 16.2. One-dimensional plasmons of linear channels

$S_\mathrm{1D} = N_\mathrm{ch} g_c |q_\parallel| \delta(\omega - v|q_\parallel|/g_c)$ with $g_c = [1 + N_\mathrm{ch} v_\mathrm{C}^\mathrm{1D}/\pi v]^{-1/2}$; Zhang and Chen give the case $N_\mathrm{ch} = 1$, and a metallic nanotube has $N_\mathrm{ch} = 4$ [3, Eqs. (3)–(6)].

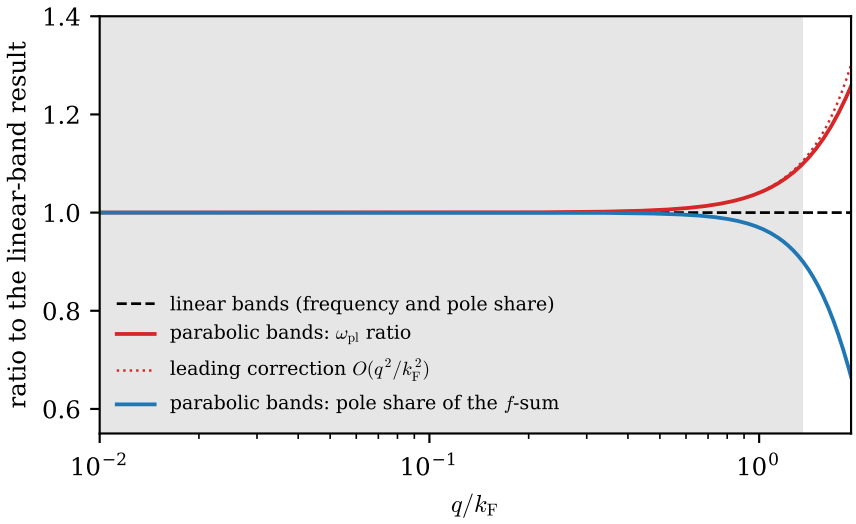


Figure 16.5: Plasmon frequency and pole share of the f-sum for $N_\mathrm{ch} = 4$ parabolic channels, relative to the linear-band result, against $q_\parallel/k_\mathrm{F}$. Both use the kernel of a tube of radius 0.7 nm in vacuum and the velocity $v_\mathrm{D} = 3.02 \times 10^{-3}$ at k_F; the parabolic band, of mass $k_\mathrm{F}/v_\mathrm{D}$ with $k_\mathrm{F} = 1/r_\mathrm{t} = 282\,\mathrm{eV}$, is a model for the curvature that the subbands bring on that scale. Red: frequency ratio; dotted, its leading correction in $q_\parallel^2/k_\mathrm{F}^2$. Blue: share of the f-sum $N_\mathrm{ch} v q_\parallel^2$ in the plasmon line (one for linear bands). Gray: both within 10 % of the linear-band result, $q_\parallel < 1.36\,k_\mathrm{F}$. Along the curve g_c rises from 0.18 to 0.61.

Beyond $q_\parallel \approx 1.4\,k_\mathrm{F}$ the plasmon of curved bands moves up by more than 10 % and loses weight to the pair continuum, which linear bands do not have.

In the random-phase approximation the line of Eq. (16.9) does not decay, and neither does the sheet plasmon above the pair continuum. Such a plasmon is a final state of its own in the sense of Sect. 8.7, unlike the bulk plasmon of silicon, which decays into pairs at once. Its energy leaves the tube or sheet by propagating to the contacts, where Ref. [3] proposes to convert it into quasiparticles of a superconductor, or by damping through disorder, phonons or the substrate. How much of it reaches a sensor is a question for the cascade of Chap. 18, not for the response.

16.2.3 The acoustic plasmon of two components

A material with two kinds of carriers, light electrons of Fermi velocity $v_{\mathrm{F},l}$ and heavy ones of mass m_h and density n_h, has a second collective mode below the plasmon. It is known

as Pines' demon, after its prediction in 1956 [7]. In the window $q v_{\mathrm{F},h} \ll \omega \ll q v_{\mathrm{F},l}$ the light electrons follow the potential statically and screen it, while the heavy ones respond as a free plasma. Take the light response at $q \ll k_{\mathrm{F},l}$ from Eq. (4.56), $R_l^0 = -2\mathcal{N}_{\mathrm{F},l}\, f_3(\nu)$ with $\nu = \omega/q v_{\mathrm{F},l}$ and $f_3(\nu) = 1 - (\nu/2)\ln|(1+\nu)/(1-\nu)| + \mathrm{i}(\pi/2)\nu\,\Theta(1-\nu)$, and the heavy response at high frequency, $n_h q^2/m_h\omega^2$. In three dimensions

$$\varepsilon(q,\omega) = 1 + \frac{k_{\mathrm{TF},l}^2}{q^2}\, f_3(\nu) - \frac{\omega_{p,h}^2}{\omega^2},$$
$$\omega_a(q) = \frac{\omega_{p,h}\, q}{\left(q^2 + k_{\mathrm{TF},l}^2\right)^{1/2}} \simeq v_a q, \qquad v_a = \frac{\omega_{p,h}}{k_{\mathrm{TF},l}} = \Big(\frac{n_h}{2m_h\mathcal{N}_{\mathrm{F},l}}\Big)^{1/2}, \tag{16.10}$$

with $k_{\mathrm{TF},l}^2 = 2e^2\mathcal{N}_{\mathrm{F},l}$ and $\omega_{p,h}^2 = e^2 n_h/m_h$, where ω_a is the zero of ε with $\mathrm{Re}\, f_3 \simeq 1$. The heavy plasma would oscillate at $\omega_{p,h}$; the light electrons move against it and cancel its charge to within a fraction $q^2/k_{\mathrm{TF},l}^2$, and the frequency of the screened oscillation vanishes linearly in q. The mode is consistent if $v_{\mathrm{F},h} \ll v_a \ll v_{\mathrm{F},l}$.

Since its charge is screened, the mode carries little weight in the loss function. Near the zero, $\partial\varepsilon/\partial\omega = 2(1 + k_{\mathrm{TF},l}^2/q^2)/\omega_a$, and the line in $L = \mathrm{Im}(-1/\varepsilon)$ has weight $\pi/(\partial\varepsilon/\partial\omega)$, as in Eq. (4.61). Its first moment, compared with the f-sum $(\pi/2)\omega_{p,h}^2$ of the heavy carriers,

$$\frac{\omega_a \cdot \pi/(\partial\varepsilon/\partial\omega)}{(\pi/2)\,\omega_{p,h}^2} = \Big(\frac{q^2}{q^2 + k_{\mathrm{TF},l}^2}\Big)^2 \quad \text{(3D)}, \qquad \Big(\frac{q_\parallel}{q_\parallel + q_s}\Big)^2 \quad \text{(2D)}, \tag{16.11}$$

where the two-dimensional form follows below. A density coupling acts on light and heavy electrons with the same sign, and it sees the acoustic plasmon only at order $q^4/k_{\mathrm{TF},l}^4$. The mode is neutral at long wavelength, as Hochberg, Novko, Ovadia and Politano note for the demon of Sr_2RuO_4, observed there by electron energy-loss spectroscopy [8]. Their first-principles loss function of that layered metal shows the mode along the stacking axis as a broad peak from 80 to 800 meV, while the plasmon at 1.6 eV dominates in the plane. Their rate is Eq. (4.45) inserted in Eq. (8.16) [8, Eqs. (1), (2)]. Eqs. (16.10) and (16.11) are the two-component random-phase limit of such a loss function, which a calculation of the full crystal contains automatically. Hole-doped diamond, in the same work, has an acoustic plasmon near 200 meV from the oscillation of its valence manifold.

The light electrons also damp the mode, since at $\omega_a < q v_{\mathrm{F},l}$ the acoustic plasmon lies inside their pair continuum. With $\mathrm{Im}\,\varepsilon = (k_{\mathrm{TF},l}^2/q^2)(\pi/2)\nu$ and the slope above, the full width at half maximum, $\Gamma_a = 2\,\mathrm{Im}\,\varepsilon/(\partial\varepsilon/\partial\omega)$, is at $q \ll k_{\mathrm{TF},l}$

$$\frac{\Gamma_a}{\omega_a} = \frac{\pi}{2}\,\frac{v_a}{v_{\mathrm{F},l}} \quad \text{(3D)}, \qquad \frac{\Gamma_a}{\omega_a} = \frac{v_a/v_{\mathrm{F},l}}{\left(1 - v_a^2/v_{\mathrm{F},l}^2\right)^{1/2}} \quad \text{(2D)}. \tag{16.12}$$

The mode is well defined only while $v_a \ll v_{\mathrm{F},l}$, and the condition that made the light electrons static is the same that keeps the mode narrow. In two dimensions every step repeats with the kernel of Eq. (16.4). The static light response gives $v_{\mathrm{C}}^{2\mathrm{D}} \cdot 2\mathcal{N}_{\mathrm{F},l} = q_s/q_\parallel$, with the two-dimensional screening wavevector $q_s = 2e^2\mathcal{N}_{\mathrm{F},l}/(\varepsilon_a + \varepsilon_b)$, and the heavy plasma gives $\omega_{2\mathrm{D},h}^2 = e^2 n_{2\mathrm{D},h} q_\parallel/[(\varepsilon_a + \varepsilon_b)m_h]$. The zero is $\omega_a^2 = \omega_{2\mathrm{D},h}^2/(1 + q_s/q_\parallel) \simeq v_a^2 q_\parallel^2$, with the same $v_a = (n_{2\mathrm{D},h}/2m_h\mathcal{N}_{\mathrm{F},l})^{1/2}$: the charge cancels from the velocity, which depends on neither the dimension nor the substrate. The width differs. At $q_\parallel \ll k_{\mathrm{F},l}$ the two-dimensional Lindhard function below the continuum edge has $\mathrm{Re}\, f_2 = 1$ exactly and $\mathrm{Im}\, f_2 = \nu/(1-\nu^2)^{1/2}$, which gives the second form of Eq. (16.12), without the factor $\pi/2$ and with no shift of the line.

Limit Check 16.3. Acoustic plasmon of two components and its Landau width

$\Gamma_a/\omega_a = (\pi/2)v_a/v_{\mathrm{F},l}$ for the demon of a three-dimensional two-component metal [7], with $v_a = \omega_{p,h}/k_{\mathrm{TF},l}$ from Eq. (16.10) and the loss function entering the rate as in Ref. [8, Eqs. (1), (2)]; $(v_a/v_{\mathrm{F},l})(1 - v_a^2/v_{\mathrm{F},l}^2)^{-1/2}$ for a sheet, from the two-dimensional Lindhard response of the light carriers, Eq. (16.12).

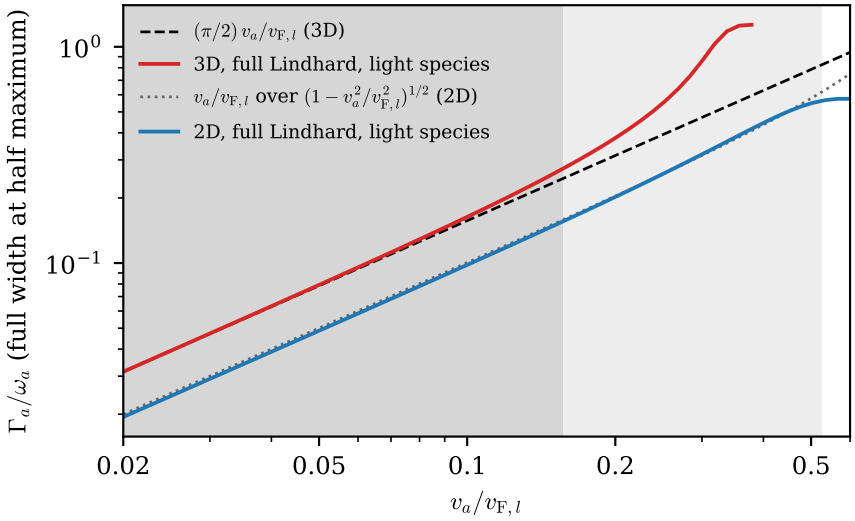


Figure 16.6: Full width at half maximum of the acoustic plasmon over its frequency, against $v_a/v_{\mathrm{F},l}$. Dashed and dotted: Eq. (16.12) in three and two dimensions. Red and blue: the width of the line in $L = \mathrm{Im}(-1/\varepsilon)$ with the full frequency dependence of the light carriers' Lindhard function at $q \ll k_{\mathrm{F},l}$. The heavy carriers are cold, and $q = 0.02\, k_{\mathrm{TF},l}$ (3D) and $q_\parallel = 0.02\, q_s$ (2D); the 3D curve stops at $v_a/v_{\mathrm{F},l} = 0.4$, beyond which the line is too broad and asymmetric for a width to be defined. Dark gray: the 3D form within 10 %, $v_a/v_{\mathrm{F},l} < 0.157$; light gray: the 2D form within 10 %, $v_a/v_{\mathrm{F},l} < 0.53$. At $v_a/v_{\mathrm{F},l} = 0.02$ the 2D width is 3 % below its formula because of the finite $q_\parallel/q_s$.

Beyond the shaded region the light carriers no longer screen statically, the line broadens into their continuum, in three dimensions with a long tail toward high frequency, and the width stops being a small correction to a sharp mode.

Worked example 16.2: The acoustic plasmon of a two-band sheet

Take a sheet suspended in vacuum with two parabolic, spin-degenerate bands of areal density $n_{2\mathrm{D}} = 1.0 \times 10^{13}\,\mathrm{cm}^{-2}$ each, and masses $m_l = 0.2\, m_e$ and $m_h = 4.0\, m_e$, model values chosen to show the mode. Both bands have $k_{\mathrm{F}} = (2\pi n_{2\mathrm{D}})^{1/2} = 156.4\,\mathrm{eV}$ $(0.793\,\mathrm{nm}^{-1})$, so $v_{\mathrm{F},l} = 1.53 \times 10^{-3}$ and $v_{\mathrm{F},h} = 7.65 \times 10^{-5}$. With $2\mathcal{N}_{\mathrm{F},l} = m_l/\pi$, the screening wavevector is $q_s = e^2 m_l/2\pi = 2\alpha m_l = 1.49\,\mathrm{keV}$.

Velocity. $v_a^2 = n_{2\mathrm{D}}\pi/(m_h m_l) = k_{\mathrm{F}}^2/(2m_h m_l)$, so $v_a = 2.42 \times 10^{-4}$, or $72.5\,\mathrm{km\,s}^{-1}$. The mode lies between the continua: $v_a/v_{\mathrm{F},l} = (m_l/2m_h)^{1/2} = 0.158$ and $v_a/v_{\mathrm{F},h} = 3.16$.

Frequency and width at $q_\parallel = 20\,\mathrm{eV}$. Here $q_\parallel/k_{\mathrm{F}} = 0.128$ and $q_\parallel/q_s = 0.0134$. The heavy band alone would oscillate at $\omega_{2\mathrm{D},h} = 41.8\,\mathrm{meV}$; screened, $\omega_a = \omega_{2\mathrm{D},h}(1 + q_s/q_\parallel)^{-1/2} = 4.81\,\mathrm{meV}$. Eq. (16.12) gives $\Gamma_a/\omega_a = 0.160$, a width of 0.77 meV. Both bands together carry the ordinary sheet plasmon at 191.5 meV to leading order (193 meV with the v_{F}^2

term of Eq. (16.8)), far above $q_\parallel v_{\mathrm{F},l} = 30.6\,\mathrm{meV}$.
Weight. By Eq. (16.11) the line carries $(q_\parallel/(q_\parallel + q_s))^2 = 1.75 \times 10^{-4}$ of the heavy band's f-sum. The light band's own pair continuum carries about as much in the lower half of its range, below $\frac{1}{2} q_\parallel v_{\mathrm{F},l} = 15.3\,\mathrm{meV}$: 1.72×10^{-4} of the same sum, spread over all energies up to there.
Check. Now treat both bands with the two-dimensional Lindhard function at $q_\parallel \ll k_\mathrm{F}$, so that the heavy band is no longer assumed cold. The peak of L then sits at 4.97 meV with a width of 0.76 meV ($\Gamma_a/\omega_a = 0.154$), and the first moment within three widths of the peak is 1.76×10^{-4} of the heavy f-sum. The heavy band raises the frequency by 3.5 %, the size of $\frac{3}{4}(v_{\mathrm{F},h}/v_a)^2 = 0.075$ in ω_a^2.
Kinematics. A halo particle deposits 4.81 meV if $m_\chi \geq 2\omega_a/v_{\max}^2 = 1.37\,\mathrm{keV}$, and supplies $q_\parallel = 20\,\mathrm{eV}$ if $2m_\chi v_{\max} \geq 20\,\mathrm{eV}$, or $m_\chi \geq 3.8\,\mathrm{keV}$. The energy needs a total momentum of only $\omega_a/v_{\max} = 1.8\,\mathrm{eV}$, and the in-plane 20 eV decides. The mode, eleven times slower than the fastest halo particles, is reachable by keV-mass dark matter. Its weight, however, is four orders of magnitude below that of the heavy band: a density coupling scatters off this sheet mostly through the pair continuum and the optical plasmon, not through the acoustic one.

16.2.4 Layered stacks

A stack of identical sheets with period a_s in a background ε_b, with no tunneling between them, is the simplest layered target. The sheets interact, and Eq. (16.3) no longer applies. A density wave $\delta n_{2\mathrm{D}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}_\parallel \cdot \boldsymbol{r}_\parallel}$ on sheet n, with the phase $\mathrm{e}^{\mathrm{i}q_z n a_s}$ of the external potential, produces $(e^2/2\varepsilon_b q_\parallel)\mathrm{e}^{-q_\parallel |z - n a_s|}$ at height z, and the sum over sheets at $z = 0$ is a geometric series. Each sheet therefore sees the kernel

$$v_\mathrm{C}^\mathrm{L}(q_\parallel, q_z) = \frac{e^2}{2\varepsilon_b q_\parallel} \sum_n \mathrm{e}^{-q_\parallel |n| a_s + \mathrm{i} q_z n a_s} = \frac{e^2}{2\varepsilon_b q_\parallel} \frac{\sinh q_\parallel a_s}{\cosh q_\parallel a_s - \cos q_z a_s},$$
$$S(\boldsymbol{q}, \omega) = -\frac{2}{a_s}\, \mathrm{Im}\, \frac{\tilde{R}^{2\mathrm{D}}}{1 - v_\mathrm{C}^\mathrm{L} \tilde{R}^{2\mathrm{D}}}, \tag{16.13}$$

at $T = 0$, per unit volume, for sheets thinner than $1/q$, with the background treated as a static dielectric. For $a_s \to \infty$ the kernel becomes that of an isolated sheet, and S is $1/a_s$ times its structure factor. For $q_\parallel a_s, q_z a_s \ll 1$ it becomes $e^2/(\varepsilon_b a_s q^2)$, the bulk kernel acting on the density $n_{2\mathrm{D}}/a_s$. In between, the stack carries a band of layer plasmons. For parabolic sheets, $\omega^2 = v_\mathrm{C}^\mathrm{L} n_{2\mathrm{D}} q_\parallel^2/m^*$ runs at small $q_\parallel$ from the bulk plasma frequency of the density $n_{2\mathrm{D}}/a_s$ at $q_z = 0$, where all sheets oscillate in phase, down to the acoustic mode $\omega^2 = e^2 n_{2\mathrm{D}} a_s q_\parallel^2/4\varepsilon_b m^*$ at $q_z a_s = \pi$, where neighbors oscillate against each other. The dark-matter momentum along the axis is not reduced to a zone, since the sheets have no structure on the scale of $1/q$; S is periodic in q_z with period $2\pi/a_s$ until the profile of a sheet cuts it off.

Layers many atoms thick respond through their own local dielectric functions. Take layers of materials A and B, of thicknesses d_A and $d_B = a_s - d_A$, filling $f = d_A/a_s$, and local dielectric functions $\varepsilon_A(\omega)$, $\varepsilon_B(\omega)$. When the wavelength is long compared with a_s, the field within one period is that of a uniform field across planar boundaries. The tangential field is continuous, so the in-plane displacement is the volume average of ε; the normal displacement is continuous, so the normal field is the volume average of $1/\varepsilon$. The stack is a uniaxial medium

with its optic axis along the stacking direction, and with θ the angle between $\boldsymbol{k}$ and that axis,

$$\varepsilon_\perp = f\varepsilon_A + (1-f)\,\varepsilon_B, \qquad \frac{1}{\varepsilon_\parallel} = \frac{f}{\varepsilon_A} + \frac{1-f}{\varepsilon_B},$$
$$L(\boldsymbol{k},\omega) = \operatorname{Im}\frac{-1}{\varepsilon_\parallel \cos^2\theta + \varepsilon_\perp \sin^2\theta} \qquad (ka_s \to 0). \tag{16.14}$$

Since $\varepsilon_\parallel = \varepsilon_A\varepsilon_B/[(1-f)\varepsilon_A + f\varepsilon_B]$ and $\varepsilon_\perp[(1-f)\varepsilon_A + f\varepsilon_B] = \varepsilon_A\varepsilon_B + f(1-f)(\varepsilon_A-\varepsilon_B)^2$, the inverse $1/(\varepsilon_\parallel\cos^2\theta + \varepsilon_\perp\sin^2\theta)$ equals $[(1-f)\varepsilon_A + f\varepsilon_B]/[\varepsilon_A\varepsilon_B + f(1-f)(\varepsilon_A-\varepsilon_B)^2\sin^2\theta]$. This is identical, at every filling, to the effective longitudinal response of Lasenby and Prabhu [9, Eq. (16)], which they state for $d_A/a_s \ll 1$. The loss function has one line in each direction, at the zero of the denominator. For metal layers in vacuum the line falls toward zero frequency as $\boldsymbol{k}$ approaches the axis: with $\varepsilon_A = \varepsilon_\infty(1-\omega_p^2/\omega^2)$ and small θ and f it sits at $\omega^2 \simeq f\varepsilon_\infty\omega_p^2\theta^2/\varepsilon_B$, the filling fraction times the plasma frequency squared, reduced by the direction [9].

For $\boldsymbol{k}$ in the plane of the layers the response is known in closed form at every ka_s. The source $\mathrm{e}^{\mathrm{i}kz}$ is uniform across the layers, and the potential is $\psi(x)\mathrm{e}^{\mathrm{i}kz}$ with x along the axis. Inside layer j it is the particular solution $1/\varepsilon_j k^2$ plus $C_j\cosh[k(x-x_j)]$, symmetric about the center x_j of the layer. Continuity of ψ and of $\varepsilon\,\partial_x\psi$ at an interface fixes $C_A = (\varepsilon_A-\varepsilon_B)s_B/[k^2\varepsilon_A(\varepsilon_B s_B c_A + \varepsilon_A s_A c_B)]$ and $C_B = -\varepsilon_A s_A C_A/\varepsilon_B s_B$, with $s_j = \sinh(kd_j/2)$ and $c_j = \cosh(kd_j/2)$. The average of $k^2\psi$ over a period is the head of ε^{-1} at the external wavevector:

Stack of two dielectrics, k in the plane

$$\frac{1}{\varepsilon(k,\omega)} = \frac{f}{\varepsilon_A} + \frac{1-f}{\varepsilon_B} - \frac{2}{ka_s}\,\frac{(\varepsilon_A-\varepsilon_B)^2\,s_A s_B}{\varepsilon_A\varepsilon_B\,(\varepsilon_A s_A c_B + \varepsilon_B s_B c_A)}, \tag{16.15}$$

with $L = \operatorname{Im}[-1/\varepsilon]$, exact for local ε_A, ε_B at any ka_s.

At $ka_s \to 0$, $s_j \to kd_j/2$, $c_j \to 1$, and the right side becomes $1/\varepsilon_\perp$, the in-plane limit of Eq. (16.14). Expanding to the next order,

$$\frac{1}{\varepsilon(k,\omega)} = \frac{1}{\varepsilon_\perp} + (ka_s)^2\,\frac{f^2(1-f)^2(\varepsilon_A-\varepsilon_B)^2}{12\,\varepsilon_\parallel\,\varepsilon_\perp^2} + O((ka_s)^4). \tag{16.16}$$

The correction is quadratic in the period over the wavelength and grows with the contrast between the layers. At $ka_s \to \infty$, $s_j/c_j \to 1$ and the response is the volume average of $1/\varepsilon_j$: each layer then screens as its own bulk. One layer alone follows for $d_B \to \infty$ with $\varepsilon_B = 1$. Multiplied by a_s, $\varepsilon^{-1}-1$ becomes the response per unit area, $d_A(1/\varepsilon_A - 1) - 2(\varepsilon_A - 1)^2 s_A/[k\varepsilon_A(\varepsilon_A s_A + c_A)]$, and minus this equals the thin-film response of Lasenby and Prabhu up to the real constant d_A, which does not enter the loss [9, Eq. (19)]: the constant is the vacuum term that their response, normalized to $-d/\varepsilon$ for a bulk slab, does not subtract. Its line sits at $\varepsilon_A = -\coth(kd_A/2)$; for a Drude metal at $kd_A \ll 1$ that is $\omega^2 = \omega_p^2 kd_A/2$, the sheet plasmon of Eq. (16.8) with $n_{2\mathrm{D}} = n_e d_A$ in vacuum. With $|\varepsilon_A| \gg 1$ and $kd_A \ll 1$ the same expression reduces to the sheet screening of Eq. (15.19). For a general direction the same matching, with the Bloch condition $\psi(x+a_s) = \mathrm{e}^{\mathrm{i}k_x a_s}\psi(x)$ across one period, gives ε^{-1} by solving four linear equations; it also approaches Eq. (16.14) at small ka_s.

Limit Check 16.4. Effective medium of a layered stack

For $ka_s \to 0$ a stack of layers with local ε_A, ε_B and filling f has $1/\varepsilon_L = [(1-f)\varepsilon_A + f\varepsilon_B]/[\varepsilon_A\varepsilon_B + f(1-f)(\varepsilon_A-\varepsilon_B)^2\sin^2\theta]$, θ measured from the stacking axis [9, Eq. (16)].

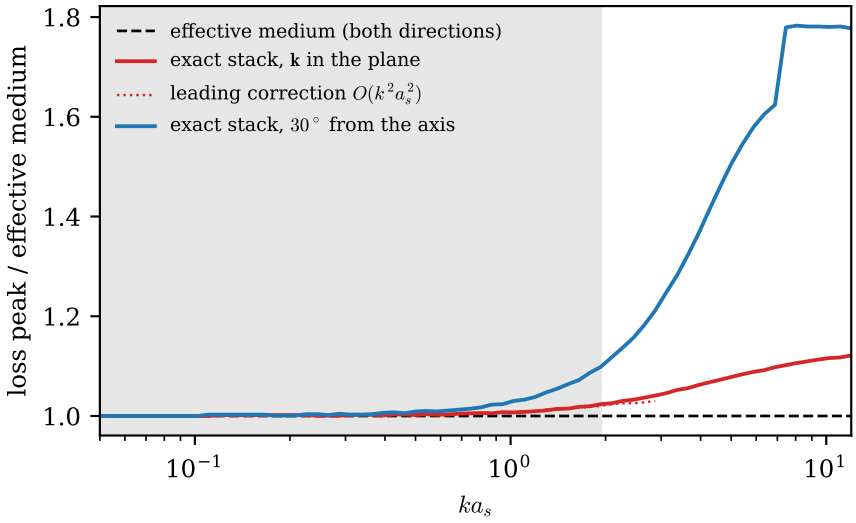


Figure 16.7: Frequency of the loss peak of a model stack relative to the effective medium, against ka_s. Layers of 1 nm with $\varepsilon_A = 10[1 - \omega_p^2/\omega(\omega + \mathrm{i}\gamma)]$, $\omega_p = 0.5\,\mathrm{eV}$, $\gamma = 0.1\,\omega_p$, alternate with 4 nm of vacuum ($a_s = 5\,\mathrm{nm}$, $f = 0.2$), the toy stack of Lasenby and Prabhu. Red: $\boldsymbol{k}$ in the plane, exact (Eq. (16.15)); dotted, the leading correction Eq. (16.16). Blue: $\boldsymbol{k}$ at 30° from the axis, exact solution of the periodic stack. The effective-medium peaks are at 0.4215 and 0.2740 eV. Gray: both within 10 % of the effective medium, $ka_s < 1.94$. At $ka_s = 12$ the peaks are at 0.473 and 0.487 eV, near the surface plasmon of a single layer, $\varepsilon_A = -\varepsilon_B$ (0.477 eV without damping).

Beyond $ka_s \approx 2$ the field varies within a period, the line of an oblique direction moves up by tens of percent toward that of a single layer, and the stack must be treated layer by layer.

A single undamped line in each direction is the case in which the sum-rule bound of Sect. 10.2 can be reached. For $L = A\,\delta(\omega - \omega_{\boldsymbol{k}})$ every moment is $K_n = A\omega_{\boldsymbol{k}}^n$, and the Hölder inequality behind Eq. (10.6) holds as an equality; the bound Eq. (10.7) is then attained when the line sits where $\omega^{-n}\eta$ is largest. A bulk metal places its line at ω_p, far above what light dark matter deposits. A stack moves it down by the filling fraction and the direction, which is why Lasenby and Prabhu proposed it [9]. For the model stack of the limit check, 10 % accuracy at 30° ends at $k = 77\,\mathrm{eV}$, the largest momentum transfer, $2m_\chi v_{\max}$, of a particle of 14.5 keV. Heavier dark matter probes the stack at momenta where the field varies within a period, and needs the full periodic response, with q_z unreduced, that Appendix E.21 specifies, and at $k \gtrsim 1/d_A$ also the nonlocal response $\varepsilon_A(\boldsymbol{k}, \omega)$ of the layer material itself.

16.3 Graphene, nanotubes and quantum dots

An electron of a graphene sheet that receives more than the work function, about 4.3 eV, can leave the sheet without meeting another atom. It then carries into the vacuum the direction and the energy the dark matter gave it, and both can be measured. A bulk crystal turns the same deposit into a cascade of secondary pairs and phonons, and the direction is lost. Carbon nanotubes keep the one-atom thickness and roll the sheet into a cylinder of radius r_t, which quantizes the momentum around the tube and opens a gap in most of them. Quantum dots, nanocrystals a few nanometers across, push confinement to all three directions: a lead sulfide

dot 3.3 nm across has an optical gap of 1.2 eV, against 0.41 eV for the bulk crystal, and it re-emits the energy of a pair as a photon [10]. We will derive the ejection rate of a sheet and the size of its two corrections, the response of a semiconducting tube in the random-phase approximation, and the rate of a dot with its bulk limit and the depolarization factor that its absorption needs.

16.3.1 Electron ejection into vacuum

Take an electron in band i with in-plane crystal momentum $\boldsymbol{k}_\parallel$, bound by $E_B(i,\boldsymbol{k}_\parallel) \geq 0$ below the Fermi level. It leaves with kinetic energy $E_\mathrm{ej} = p_\mathrm{ej}^2/2m_e$ when the dark matter deposits

$$\omega = E_\mathrm{ej} + \Phi_\mathrm{wf} + E_B(i,\boldsymbol{k}_\parallel), \tag{16.17}$$

with Φ_wf the work function, the energy from the Fermi level to the vacuum. For graphene $\Phi_\mathrm{wf} \simeq 4.3\,\mathrm{eV}$, and it can be moved by the choice of substrate [11, text after Eq. (5) and footnote 2]. The largest energy a particle can deposit is its kinetic energy, $m_\chi v_\mathrm{max}^2/2$, so ejection starts at $m_\chi = 2\Phi_\mathrm{wf}/v_\mathrm{max}^2 = 1.22\,\mathrm{MeV}$, at a momentum transfer of at least $\Phi_\mathrm{wf}/v_\mathrm{max} = 1.62\,\mathrm{keV}$. That is already a fifth of the momentum scale of a carbon $2p$ orbital, $\beta = Z_\mathrm{eff}/2a_0 = 7.51\,\mathrm{keV}$ for the hydrogenic orbital with $Z_\mathrm{eff} = 4.03$ that Hochberg et al. fit to the overlap of neighboring orbitals [11, Eq. (2)]. Ejection probes the short-distance structure of the orbitals, as atomic ionization does (Sect. 2.2), and not their tails.

In the independent-particle picture the initial states are the occupied Bloch states $\psi_{i\boldsymbol{k}_\parallel}$ of the π band and the three σ bands, normalized over the sample, and each final state is a hole in the sheet with a free electron. We take the free electron as a plane wave $\mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}}/\mathscr{V}^{1/2}$ of momentum $\boldsymbol{p}$, $|\boldsymbol{p}| = p_\mathrm{ej}$. Its transition form factor Eq. (2.17) is the momentum-space wavefunction of the initial state, $M_\mathrm{PW}(\boldsymbol{p},\boldsymbol{q}) = \tilde{\psi}_{i\boldsymbol{k}_\parallel}(\boldsymbol{p}-\boldsymbol{q})$, and the structure factor of Eq. (4.13) becomes

$$S_\mathrm{ej}(\boldsymbol{q},\omega) = \frac{2\pi}{\mathscr{V}} \sum_{i,\boldsymbol{k}_\parallel,\sigma} \int \frac{\mathrm{d}^3p}{(2\pi)^3}\, |M(\boldsymbol{p},\boldsymbol{q})|^2\, \delta(\omega - E_\mathrm{ej} - \Phi_\mathrm{wf} - E_B(i,\boldsymbol{k}_\parallel)), \tag{16.18}$$

with $M = M_\mathrm{PW}$ for plane waves and σ the spin. Only energy is conserved. Since the bound state is not a momentum eigenstate, it supplies whatever part of $\boldsymbol{q}$ the electron does not leave with, and $\boldsymbol{p}$ and $\boldsymbol{q}$ are independent variables. Inserting Eq. (16.18) and $|\tilde{V}|^2 = \pi\bar{\sigma}_e F_\mathrm{DM}^2/\mu_{\chi e}^2$ (Eq. (1.34)) into Eq. (8.16), with $\rho_T\mathscr{V} = N_\mathrm{cell}M_\mathrm{cell}$, the factors $2\pi\cdot\pi/(2\pi)^3$ combine into $1/4\pi$, and the rate per unit mass is a sum of cross-sections, one for each occupied state:

$$\mathcal{R} = \frac{\rho_\chi}{m_\chi\, N_\mathrm{cell} M_\mathrm{cell}} \sum_{i,\boldsymbol{k}_\parallel,\sigma} \int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\, \langle\sigma v\rangle_{i\boldsymbol{k}_\parallel}(\boldsymbol{v}),$$

$$\langle\sigma v\rangle_{i\boldsymbol{k}_\parallel} = \frac{\bar{\sigma}_e}{\mu_{\chi e}^2} \int \frac{\mathrm{d}^3p}{(2\pi)^3} \int \frac{\mathrm{d}^3q}{4\pi}\, F_\mathrm{DM}(q)^2\, |M(\boldsymbol{p},\boldsymbol{q})|^2\, \delta\Big(E_\mathrm{ej} + \Phi_\mathrm{wf} + E_B + \frac{q^2}{2m_\chi} - \boldsymbol{q}\cdot\boldsymbol{v}\Big). \tag{16.19}$$

For plane waves the cross-section is that of Hochberg et al. [11, Eq. (5)]. They write the sum over states as $2N_\mathrm{C}A_\mathrm{cell}\int \mathrm{d}^2k_\parallel/(2\pi)^2$ per band, with N_C the number of carbon atoms per unit mass and A_cell the area of the unit cell. Since a band of a two-atom cell holds N_C electrons per unit mass, this sum as printed counts the states of each band twice, unless each cluster function carries the weight $\frac{1}{2}$. They take for $\tilde{\psi}$ the tight-binding function of one atom and its three nearest neighbors [11, Eqs. (1), (3), (7)]. A nanotube changes only the sum over states. Going once around the tube must return the Bloch phase, so the momentum around the circumference takes the values n/r_t. Then $A_\mathrm{cell}\int \mathrm{d}^2k_\parallel/(2\pi)^2$ becomes $(A_\mathrm{cell}/2\pi r_\mathrm{t})\sum_n \int \mathrm{d}k_z/2\pi$, with k_z along the axis, which is the rate of Cavoto et al. [12, Eqs. (16)–(18), (23)].

With Bloch states, the in-plane part of $\boldsymbol{p}-\boldsymbol{q}$ is fixed up to a reciprocal-lattice vector by the crystal momentum of the hole. A Bloch state normalized over $N_{\rm cell}$ cells has $|\tilde\psi_{i\boldsymbol{k}_\parallel}(\boldsymbol{k})|^2 \propto \sum_{\boldsymbol{G}} \delta^2(\boldsymbol{k}'_\parallel - \boldsymbol{k}_\parallel - \boldsymbol{G})$, with $\boldsymbol{k}'_\parallel$ the in-plane part of $\boldsymbol{k}$. As a result, each direction of the ejected electron selects one state of each band, with its own binding energy. This is the origin of the diffraction pattern that Hochberg et al. find in the angular distribution of electrons ejected by a cold stream [11, Sect. III]. The halo washes the pattern out but keeps the forward peak: the momentum-space orbital is largest when $\boldsymbol{p}-\boldsymbol{q}$ is small, and the electron leaves close to the direction of $\boldsymbol{q}$, which follows the dark-matter velocity. For the standard halo they find the forward rate about twice the backward rate between 10 MeV and 10 GeV [11, Sect. III], and the daily rotation of a fixed sheet modulates both.

The orbitals and the binding energies are the material input. They come from a density-functional ground state, with maximally localized Wannier functions for the occupied bands and an all-electron orbital for the carbon 1s shell, which a pseudopotential lacks. Appendix E.21 specifies the calculation of the ejection factors of graphene and nanotubes, including the orthogonality correction below.

Orthogonality of the final state. A plane wave is not an eigenstate of the sheet, and it is not orthogonal to the state the electron leaves. As $q \to 0$, $M_{\rm PW}$ tends to $\tilde\psi_{i\boldsymbol{k}_\parallel}(\boldsymbol{p})$, which is not zero, so plane waves predict ejection with no momentum transfer, although $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \to 1$ cannot connect two orthogonal eigenstates. The cure is to orthogonalize the final state to the occupied states, $|\chi_{\boldsymbol{p}}\rangle = |\boldsymbol{p}\rangle - \sum_n |\psi_n\rangle\langle\psi_n|\boldsymbol{p}\rangle$, the orthogonalized plane wave (OPW). Its transition form factor is

$$M_{\rm OPW}(\boldsymbol{p},\boldsymbol{q}) = \tilde\psi_i(\boldsymbol{p}-\boldsymbol{q}) - \sum_{n\in\mathrm{occ}} \tilde\psi_n(\boldsymbol{p})\, f_{i\to n}(\boldsymbol{q}), \tag{16.20}$$

with $f_{i\to n}$ the form factor Eq. (2.17) between occupied states, here written with a single index for $(i,\boldsymbol{k}_\parallel)$. Since $f_{i\to n}(\boldsymbol{q}) = \delta_{in} + O(q)$, the term without momentum transfer cancels and $M_{\rm OPW} = O(q)$. To size the correction we take the hydrogenic $2p_z$ orbital above, $\psi \propto r\mathrm{e}^{-\beta r}\cos\theta$ with $\tilde\psi(\boldsymbol{k}) \propto \beta k_z/(\beta^2+k^2)^3$ [11, Eq. (4)], orthogonalize the plane wave to it alone, and average over the orientation of the sheet. Integrated over the directions of the ejected electron, $|M_{\rm OPW}|^2/|M_{\rm PW}|^2$ runs from 0.46 to 25 at $E_{\rm ej} = 50$ to 100 eV and $q = 5$ to 40 keV. The value depends on q and on the angle between $\boldsymbol{q}$ and the orbital axis, and an orbital of the same form with the more diffuse exponent $1.568/a_0$ of Clementi and Raimondi, used in Appendix E.21, gives 0.4 to 4.3. At 10 eV it reaches 740, but only at $q \gg p_{\rm ej}$, where the plane-wave amplitude has already fallen as $|\boldsymbol{p}-\boldsymbol{q}|^{-5}$ and contributes little. The rate weights each $(\boldsymbol{p},\boldsymbol{q})$ by the halo, so these ratios bound the correction without giving it.

For an orbital averaged over orientations, $|M|^2$ integrated over the directions of $\boldsymbol{p}$ depends only on $p_{\rm ej}$ and q. Call this integral $K(p_{\rm ej},q)$. The velocity integral of Eq. (16.19) then reduces with Eq. (A.42), $\int \mathrm{d}^3q\, g(q) \int \mathrm{d}^3v\, f_\chi\, \delta(\cdots) = 2\pi \int q\,\mathrm{d}q\, \eta(v_{\rm min})\, g(q)$, and $p^2\mathrm{d}p = m_e p_{\rm ej}\,\mathrm{d}E_{\rm ej}$ gives the spectrum per electron

$$\frac{\mathrm{d}\langle\sigma v\rangle}{\mathrm{d}E_{\rm ej}} = \frac{\bar\sigma_e\, m_e\, p_{\rm ej}}{16\pi^3\mu_{\chi e}^2} \int q\,\mathrm{d}q\, \eta\big(v_{\rm min}(q,\omega)\big)\, F_{\rm DM}(q)^2\, K(p_{\rm ej},q), \tag{16.21}$$

with $v_{\rm min} = \omega/q + q/2m_\chi$ and ω from Eq. (16.17). Summed over the electrons of a full atomic shell with plane waves, Eq. (16.21) times $E_{\rm ej}$ is Eq. (2.23) with Eq. (2.24). Fig. 16.8 applies it to the π electrons of graphene, one per atom, with the momentum density of the isolated orbital. Their binding energies are those of the nearest-neighbor band, $E_B = t|1+\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{a}_1}+\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{a}_2}|$ with $\boldsymbol{a}_{1,2}$ the primitive vectors and hopping $t = 2.8$ eV [6, text after Eq. (5)], spread from 0 to 8.4 eV with a mean of 4.41 eV. At $\bar\sigma_e = 10^{-37}\,\mathrm{cm}^2$ and a heavy mediator the π electrons alone

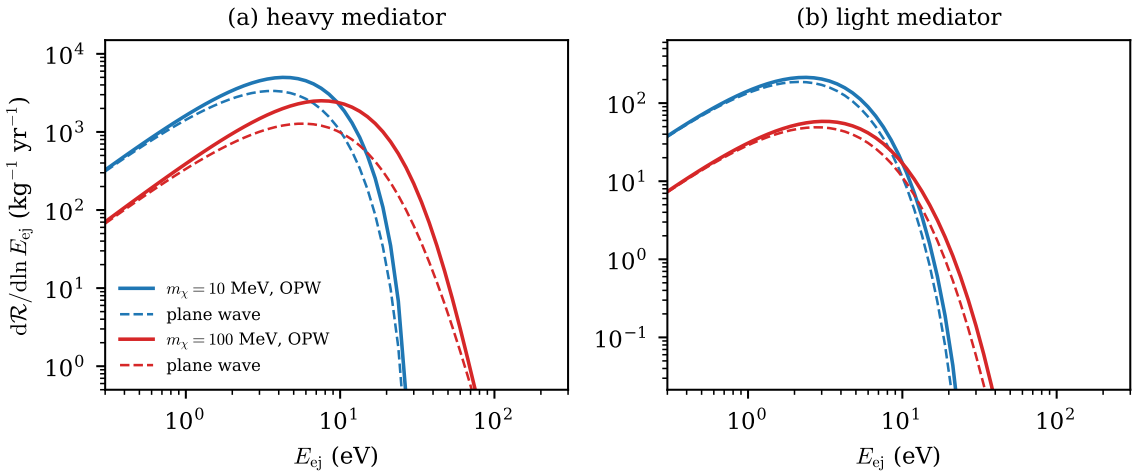


Figure 16.8: Ejection spectra of the π electrons of graphene, $d\mathcal{R}/d\ln E_{ej}$ per kg-year at $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$, against the kinetic energy E_{ej} of the ejected electron, for $m_\chi = 10\,\mathrm{MeV}$ (blue) and 100 MeV (red): (a) heavy mediator, $F_{DM} = 1$; (b) light mediator, $F_{DM} = (\alpha m_e/q)^2$. Dashed: plane-wave final states; solid: final plane wave orthogonalized to the initial orbital, Eq. (16.20). Model: one hydrogenic $2p_z$ orbital per atom with $Z_{eff} = 4.03$, Bloch interference between neighbors dropped, binding energies of the nearest-neighbor π band ($t = 2.8\,\mathrm{eV}$, 0 to 8.4 eV), work function 4.3 eV, sheet orientation averaged, standard halo of Table 1.1. The σ electrons are not included.

give 6.9×10^3 events per kg-year at 10 MeV with plane waves and 1.0×10^4 with orthogonalized ones; at 100 MeV, 2.8×10^3 and 5.2×10^3. The orthogonalization raises the rate, by factors of 1.44 and 1.84, and moves the peak of the spectrum up in energy. For a light mediator the factors are only 1.14 and 1.19. The light mediator weights the smallest momentum transfers, a few keV, where the orthogonality term lowers the squared amplitude (to 0.46–0.74 of the plane-wave value at $q = 5\,\mathrm{keV}$ for $E_{ej} = 50\,\mathrm{eV}$) and so partly cancels the gain at larger q.

Limit Check 16.5. Vacuum ejection with plane-wave final states

The ejection rate of a sheet or a tube with plane-wave final states, $M = \tilde{\psi}_{i\boldsymbol{k}_\parallel}(\boldsymbol{p} - \boldsymbol{q})$ in Eq. (16.19), is the published form [11, Eqs. (5), (7), 12, Eq. (23)].

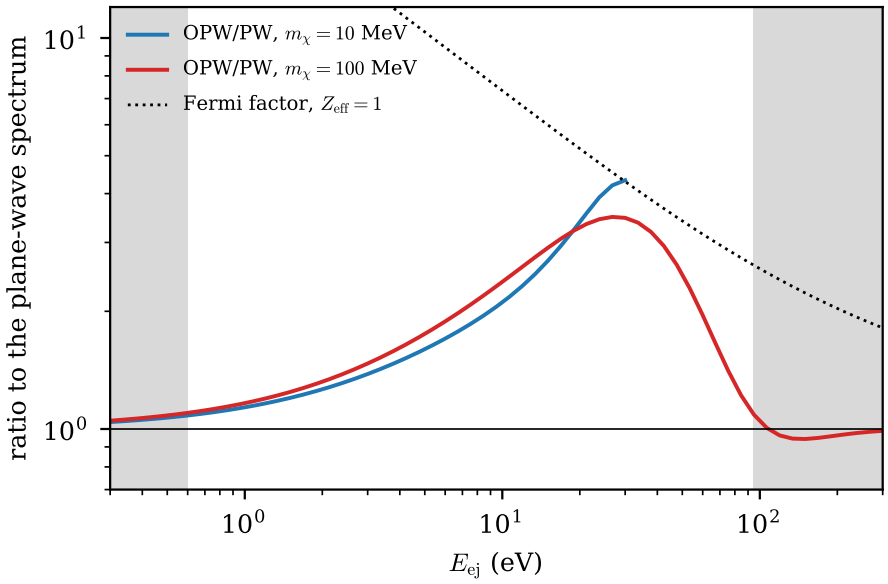


Figure 16.9: Ejection spectrum of graphene π electrons with orthogonalized final states, Eq. (16.20), divided by the plane-wave spectrum, against the kinetic energy E_{ej} of the ejected electron, for $m_\chi = 10\,\text{MeV}$ (blue, which ends at 31 eV) and 100 MeV (red), heavy mediator, in the model of Fig. 16.8. Thin line: the plane-wave form. Dotted: the Fermi factor Eq. (2.25) with $Z_{\text{eff}} = 1$, the largest final-state enhancement an unscreened hole can produce. Gray: the orthogonalized spectrum within 10 % of the plane-wave one, $E_{\text{ej}} < 0.60\,\text{eV}$ and, for 100 MeV, $E_{\text{ej}} > 94\,\text{eV}$.

Between these energies, where most of the rate lies, orthogonality raises the spectrum by factors up to 3.5 at 100 MeV (2.1 at 10 eV and 4.3 at the kinematic end for 10 MeV), and the attraction of the hole can add up to the Fermi factor, 7.3 at 10 eV and 2.6 at 100 eV; neither is fixed until all occupied orbitals and true scattering states are used.

The second correction is the final-state interaction. The ejected electron leaves a hole behind, and an unscreened point charge would enhance the slow part of the spectrum by the Fermi factor Eq. (2.25), 7.3, 3.4 and 2.6 at $E_{\text{ej}} = 10$, 50 and 100 eV. Graphene screens the hole with its own electrons, which pushes the factor toward one, and the ejected electron travels only atomic distances near the nuclei before it is in vacuum; for these two reasons Hochberg et al. leave the factor out [11, text after Eq. (6)]. A semiconducting tube screens less. The two corrections cannot simply be multiplied: an orthogonalized plane wave is not a scattering state of the hole, and multiplying it by a Fermi factor counts part of the same distortion twice. The consistent final state is a scattering state of the sheet's own potential plus the screened hole, which Appendix E.21 specifies.

16.3.2 Semiconducting nanotubes

Below the work function a tube responds through pairs within its own bands. Rolling the sheet quantizes the momentum around the circumference in steps of $1/r_{\text{t}}$, and the Dirac cones of graphene, with $v_{\text{D}} = 3ta_{\text{cc}}/2$ for the carbon–carbon distance $a_{\text{cc}} = 1.42\,\text{Å}$ [6, text after Eqs. (1) and (7)], are cut along lines. A tube is metallic when a line passes through a Dirac point; otherwise the nearest line misses it by a third of the spacing, and the subbands, labeled n, are massive Dirac bands in one dimension with half-gaps $\Delta_n = nv_{\text{D}}/3r_{\text{t}}$ for $n = 1, 2$.

With $t = 2.8\,\mathrm{eV}$, $v_\mathrm{D} = 3.02 \times 10^{-3}$, or $9.1 \times 10^5\,\mathrm{m\,s^{-1}}$, the velocity of Sect. 16.2, and a tube 1 nm across has $2\Delta_1 = 0.80\,\mathrm{eV}$. The edge is reached through momentum across the tube, which the structure factor Eq. (16.3) does not see except through $|F_\perp|^2 = J_0(q_\perp r_\mathrm{t})^2$. At the edge $q_\parallel = 0$ and $\omega = 2\Delta_n$, and since $q \geq \omega/v_\mathrm{max}$ the dark matter needs $q_\perp \geq 2\Delta_n/v_\mathrm{max}$, that is $q_\perp r_\mathrm{t} \geq 2nv_\mathrm{D}/3v_\mathrm{max}$, independent of the radius: 0.76 for the first subband, where $J_0^2 = 0.74$, and 1.52 for the second, where $J_0^2 = 0.25$. For a 1 nm tube the first edge opens at $m_\chi = 4\Delta_1/v_\mathrm{max}^2 = 226\,\mathrm{keV}$.

Each subband pair is $h = v_\mathrm{D} k\sigma_x + \Delta_n\sigma_z$, the one-dimensional form of Eq. (7.38), with k along the axis and $g_\mathrm{deg} = 4$ for spin and valley. The interband response per unit length follows from two symmetries before any integral is done. The model is invariant under Lorentz boosts with v_D in place of the speed of light, and its density is the time component of a conserved current. The response then has the form $q_\parallel^2\,\mathfrak{p}(s)$, with $s = \omega^2 - v_\mathrm{D}^2 q_\parallel^2$ the invariant of Sect. 7.3 and $\mathfrak{p}$ a function of s alone, and it is enough to compute it at small $q_\parallel$. There the pair is made at $\pm k$ with $\omega = 2\varepsilon_k$, $\varepsilon_k = (v_\mathrm{D}^2 k^2 + \Delta_n^2)^{1/2}$. The spinors of the two bands are rotated by the angle θ_k with $\tan\theta_k = v_\mathrm{D} k/\Delta_n$, and the interband overlap is $|\langle +, k+q_\parallel | -, k\rangle|^2 = \frac{1}{4}(\partial_k\theta_k)^2 q_\parallel^2 = q_\parallel^2 v_\mathrm{D}^2\Delta_n^2/4\varepsilon_k^4$. Inserting it in $-\operatorname{Im}\tilde{R} = (g_\mathrm{deg}/2)\int \mathrm{d}k\,|\langle +|-\rangle|^2\,\delta(\omega - \varepsilon_k - \varepsilon_{k+q_\parallel})$, summing the two roots and replacing ω^2 by s,

$$-\operatorname{Im}\tilde{R}_n^\mathrm{1D}(q_\parallel,\omega) = \frac{2g_\mathrm{deg} v_\mathrm{D} q_\parallel^2\Delta_n^2}{s^{3/2}\,(s-4\Delta_n^2)^{1/2}}\,\Theta(s-4\Delta_n^2). \tag{16.22}$$

The inverse square root at $s = 4\Delta_n^2$ is the van Hove singularity of a one-dimensional band edge. The factor multiplying $q_\parallel^2$ integrates to $g_\mathrm{deg} v_\mathrm{D}$ over s, whatever Δ_n; as $\Delta_n \to 0$ it becomes $g_\mathrm{deg} v_\mathrm{D}\delta(s)$, the single line of the metallic tube in Sect. 16.2.

In the random-phase approximation the subbands are screened together by the tube kernel Eq. (16.7), $\varepsilon_\mathrm{1D} = 1 - v_\mathrm{C}^\mathrm{1D}\sum_n \tilde{R}_n^\mathrm{1D}$, with the real part of $\tilde{R}_n^\mathrm{1D}$ from Eq. (16.22) by the Kramers–Kronig relation (Appendix C.5), and $S_\mathrm{1D} = -2\operatorname{Im}(\sum_n \tilde{R}_n^\mathrm{1D}/\varepsilon_\mathrm{1D})$ at $T = 0$ enters Eq. (16.3). At $q_\parallel r_\mathrm{t} \gg 1$ the kernel tends to $\alpha/q_\parallel r_\mathrm{t}$, which is the sheet kernel $2\pi\alpha/q_\parallel$ of Eq. (16.4) (in vacuum) shared over the circumference $2\pi r_\mathrm{t}$: at wavelengths shorter than the radius the tube screens like the flat sheet it was rolled from. Two effects lie outside this response. Transitions between subbands of different n carry angular momentum around the tube and enter through the higher azimuthal harmonics of the transverse form factor. And the electron and the hole of a pair attract each other strongly in one dimension, which binds excitons below each edge and moves the weight of the van Hove singularity into them; that is the Bethe–Salpeter correction of Sect. 4.9 and Sect. 10.5.

The same response sets the absorption of a dark photon by an array of aligned tubes. As $q_\parallel \to 0$, $v_\mathrm{C}^\mathrm{1D}\tilde{R}_n^\mathrm{1D}$ vanishes as $q_\parallel^2\ln(1/q_\parallel r_\mathrm{t})$: an isolated tube does not screen a uniform field along its axis. Take a dilute array of n_u tubes per unit area and a field along the tubes. Eq. (16.3) and $\varepsilon_2 = (e^2/q^2)(-\operatorname{Im} R)$ then give

$$\varepsilon_2^\parallel(\omega) = 8\pi\alpha\, g_\mathrm{deg} v_\mathrm{D}\, n_\mathrm{u} \sum_n \frac{\Delta_n^2}{\omega^3\,(\omega^2 - 4\Delta_n^2)^{1/2}}\,\Theta(\omega - 2\Delta_n), \tag{16.23}$$

and the rate is Eq. (10.31) with $L \simeq \varepsilon_2^\parallel$, since $|\varepsilon| \simeq 1$ for a dilute array; a randomly polarized field couples with one third of it. Dividing by the mass density, which is n_u times the mass of a tube per unit length, removes the packing, as it did for scattering.

16.3.3 Quantum dots

A dot of radius R much larger than the lattice constant has a periodic potential locally and a boundary globally. Its states near the band edges are Bloch functions at the edge, $u_\kappa(\boldsymbol{r})$

for the valence or conduction band, times envelopes $\psi_{nlm} = R_{nl}(r)Y_{lm}$ that vanish at the surface [10, Eqs. (2)–(4)]. The radial parts are $R_{nl} \propto j_l(\chi_{nl} r/R)$, with χ_{nl} the nth zero of the spherical Bessel function: this is the effective-mass picture of Sect. 3.3 in a hard-walled sphere. The levels are $\chi_{nl}^2/2m^*R^2$ for each carrier, and the gap of the dot,

$$\Delta E_{\mathrm{QD}} = E_{\mathrm{g}} + \frac{\pi^2}{2R^2}\Big(\frac{1}{m_e^*} + \frac{1}{m_h^*}\Big), \tag{16.24}$$

exceeds the bulk gap by the confinement energy E_{conf} of the two lowest envelopes [10, Eq. (5)]. The Coulomb attraction of the pair shifts it by an amount $\sim \alpha/\varepsilon R$, small when R is below the exciton radius a_x of Sect. 4.9, about 20 nm in PbS as quoted by Blanco et al. [10]. For the dots 3.3 nm across that they consider, $\Delta E_{\mathrm{QD}} = 1.2\,\mathrm{eV}$ against $E_{\mathrm{g}} = 0.41\,\mathrm{eV}$, so $E_{\mathrm{conf}} = 0.79\,\mathrm{eV}$, and the electron and the hole are confined independently.

Dots suspended in a solvent or a glass at volume fraction $\varphi_{\mathrm{QD}} \ll 1$ respond independently, and each transition of a dot is screened by its host. A pair moves an electron from a valence envelope ψ^h to a conduction envelope ψ^e, and for q small against the inverse lattice constant the matrix element factorizes into the band-edge form factor of one cell times the envelope overlap $\langle \psi^e | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \psi^h \rangle$. The overlaps obey a closure rule: summed over all conduction envelopes, $\sum_e |\langle \psi^e | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \psi^h \rangle|^2 = 1$ for every valence envelope. Confinement therefore redistributes the bulk weight in energy without changing its total. When the levels are dense on the scale of the energy resolution, the sum over pairs becomes the joint density of states of the bulk with the gap moved to ΔE_{QD}, and $S^{\mathrm{QD}} \to \varphi_{\mathrm{QD}}\, S^{\mathrm{bulk}}(\boldsymbol{q}, \omega)$ with $E_{\mathrm{g}} \to \Delta E_{\mathrm{QD}}$, the scissor of Sect. 10.1. Blanco et al. use this limit for deposits of 2.5 to 10 eV, quoting measurements in which the optical response of PbS dots resembles the bulk above about 2.5 eV [10, Sect. III A].

The rate per unit mass of dot material follows from Eq. (A.44). Blanco et al. define the form factor of a dot as $|f_{\mathrm{QD}}(q,\omega)|^2 = \sum_f \sum_{i \in \mathrm{cell}} \delta(\omega - E_f + E_i)\, |\langle f | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | i \rangle|^2$, with the initial states of one cell [10, Eq. (10)], which has the dimension of an inverse energy. The angle-averaged structure factor is then $\bar{S} = (2\pi/\Omega)\langle |f_{\mathrm{QD}}|^2 \rangle$, with Ω the cell volume and the average taken over the directions of $\boldsymbol{q}$. With $\rho_T \Omega = M_{\mathrm{cell}}$,

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\ln\omega} = \frac{\omega\, \rho_\chi\, \bar{\sigma}_e}{2 m_\chi\, M_{\mathrm{cell}}\, \mu_{\chi e}^2} \int q\, \mathrm{d}q\, \eta\big(v_{\min}(q,\omega)\big)\, F_{\mathrm{DM}}(q)^2\, \langle |f_{\mathrm{QD}}(q,\omega)|^2 \rangle. \tag{16.25}$$

In the bulk limit $2\pi\langle |f_{\mathrm{QD}}|^2 \rangle = 4\pi\alpha m_e^2 |f_{\mathrm{crys}}|^2/q^3$ by Eq. (10.3), and Eq. (16.25) becomes Eq. (D.171) times ω with the scissored gap, the form their numerical rates use through the bulk code of Ref. [13]. Their printed rate [10, Eq. (6)] carries $1/8 m_\chi \mu_{\chi e}$ in place of $\omega/2 m_\chi \mu_{\chi e}^2$. With the form factor of their Eq. (10) it exceeds Eq. (16.25) by $\mu_{\chi e}/4\omega$, 4.05×10^4 at $m_\chi = 10\,\mathrm{MeV}$ and $\omega = 3\,\mathrm{eV}$, and with the dimensionless crystal form factor of their Eq. (12) it does not have the dimension of a rate.

Limit Check 16.6. Quantum-dot rate and its bulk limit

The scattering rate of quantum dots is Eq. (16.25) with the bulk crystal form factor and the gap moved to the dot gap, $|f_{\mathrm{crys}}(q, \omega; E_{\mathrm{g}} \to \Delta E_{\mathrm{QD}})|^2$ in Eq. (D.171) [10, Eqs. (6), (10)–(12)].

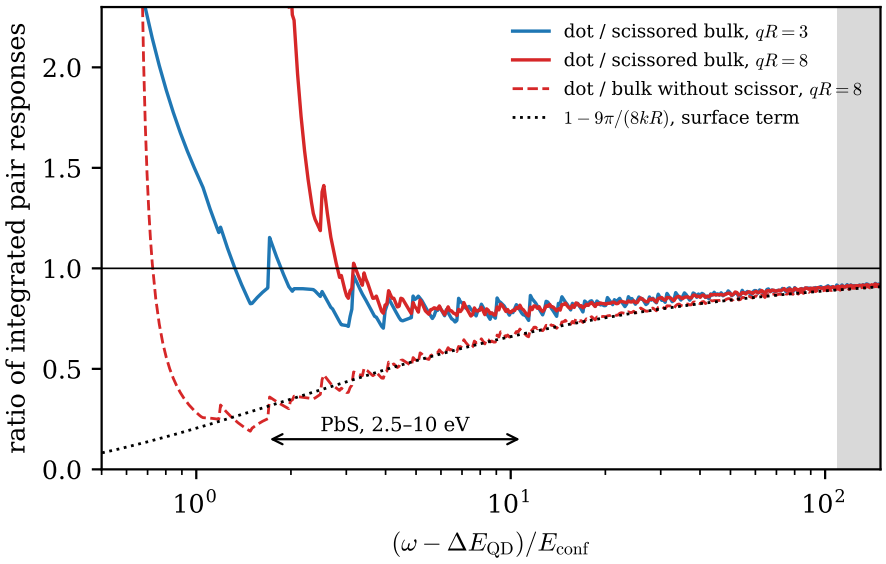


Figure 16.10: Pair response of a hard-walled spherical dot, integrated from its gap up to ω, divided by the same integral of the bulk response with the scissored gap, against $(\omega - \Delta E_{\mathrm{QD}})/E_{\mathrm{conf}}$, for $qR = 3$ (blue) and 8 (red). Model: envelopes of Eq. (16.24), equal electron and hole masses, the band-edge cell factor common to dot and bulk; the direction of $\boldsymbol{q}$ averaged. Dashed: the bulk without the scissor at $qR = 8$. Dotted: $1 - 9\pi/(8kR)$, the surface term of the level count of a sphere at $q \to 0$, with k the envelope momentum. Arrow: the range of deposits of PbS dots 3.3 nm across, 2.5 to 10 eV with $E_{\mathrm{conf}} = 0.79$ eV ($qR \geq 7.5$ there). Gray: both ratios within 10 % of one, above 109 confinement energies ($kR \geq 33$).

Between 3 and 11 confinement energies above the gap, 3.6 to 10 eV for these dots, the hard wall removes states near the surface and the dot carries 0.76 to 1.03 of the scissored bulk response at $qR = 8$ (0.70 to 0.97 at $qR = 3$), a deficit set by kR and not by qR. Nearer the gap the discrete levels dominate the cumulative ratio, which reaches 2.4 to 66 at $qR = 8$.

The deficit comes from the surface. A wavefunction that must vanish at the surface is depleted within about a wavelength of it, and the number of levels of a sphere below momentum k is $Vk^3/6\pi^2 - Sk^2/16\pi$, with V and S its volume and surface. The second term removes the fraction $9\pi/8kR$ of the bulk count, and the scissor, which shifts the bulk by a fixed energy, corrects only at order $1/(kR)^2$. Between 3 and 11 confinement energies above the gap the model lies 15 to 30 % below the scissored bulk over most of the range. A real dot has surface atoms and ligands with states of their own, and its wall is not infinitely high. Compared to such a dot, the hard wall overstates the depletion, and the measured bulk-like optical response above 2.5 eV points to a smaller one.

Absorption and the depolarization factor. A dark photon of mass m_ϕ near the dot gap has a reduced wavelength $1/m_\phi v$ of about 0.2 mm at 1 eV. Over a dot it is a uniform field, and a dot in a host is a dielectric sphere in that field. Inside, the potential is $-E_{\mathrm{in}} r \cos\theta$; outside it is $-E_0 r \cos\theta$ plus a dipole term $\propto \cos\theta/r^2$. Continuity of the potential and of $\varepsilon\, \partial_r \phi$ at $r = R$ gives $E_{\mathrm{in}} = 3\varepsilon_{\mathrm{out}} E_0/(\varepsilon_{\mathrm{in}} + 2\varepsilon_{\mathrm{out}})$, with $\varepsilon_{\mathrm{in}}$ and $\varepsilon_{\mathrm{out}}$ the dielectric functions of the dot and of the host at $\omega = m_\phi$. The polarization of a dilute set of such spheres gives the Maxwell Garnett

dielectric function $\varepsilon_{\rm eff} = \varepsilon_{\rm out}[1 + 3\varphi_{\rm QD} b/(1 - \varphi_{\rm QD} b)]$, $b = (\varepsilon_{\rm in} - \varepsilon_{\rm out})/(\varepsilon_{\rm in} + 2\varepsilon_{\rm out})$. To first order in $\varphi_{\rm QD}$ and for a real $\varepsilon_{\rm out}$, $\mathrm{Im}(-1/\varepsilon_{\rm eff}) = (3\varphi_{\rm QD}/\varepsilon_{\rm out})\,\mathrm{Im}\, b = 9\varphi_{\rm QD}\,\mathrm{Im}\,\varepsilon_{\rm in}/|\varepsilon_{\rm in} + 2\varepsilon_{\rm out}|^2$, the loss function that enters Eq. (10.31). Per unit mass of dot material this replaces the bulk loss function $\mathrm{Im}\,\varepsilon_{\rm in}/|\varepsilon_{\rm in}|^2$ that a rescaling of the measured photon cross-section with the dot's own ε uses [10, Eq. (16)], and the two differ by the depolarization factor

$$\frac{9\,|\varepsilon_{\rm in}|^2}{|\varepsilon_{\rm in} + 2\varepsilon_{\rm out}|^2} > 1 \iff |\varepsilon_{\rm in}|^2 > \frac{\varepsilon_{\rm out}}{2}\,(\mathrm{Re}\,\varepsilon_{\rm in} + \varepsilon_{\rm out}). \tag{16.26}$$

The criterion follows from expanding $|\varepsilon_{\rm in} + 2\varepsilon_{\rm out}|^2 = |\varepsilon_{\rm in}|^2 + 4\varepsilon_{\rm out}\,\mathrm{Re}\,\varepsilon_{\rm in} + 4\varepsilon_{\rm out}^2$. The factor tends to 9 when $|\varepsilon_{\rm in}| \gg \varepsilon_{\rm out}$, equals one when dot and host are the same medium, and falls below one when $|\varepsilon_{\rm in}|$ is of order one; for a real $\varepsilon_{\rm in}$ it exceeds one exactly when $\varepsilon_{\rm in} > \varepsilon_{\rm out}$. The dielectric function used in Ref. [10] is a model of the dot material itself [14], as quoted there: it is $\varepsilon_{\rm in}$, and the factor is missing.

Worked example 16.3: The depolarization factor of a PbS dot in a glass

Take a PbS dot in fused silica. Below the bulk gap, at 3.5 µm (0.354 eV), the dispersion formula of Zemel et al. [15] gives $\varepsilon_{\rm in} = 17.63$, real, and Malitson's fit for silica [16] gives $\varepsilon_{\rm out} = 1.977$, both as tabulated in Ref. [17]. The field inside the dot is $3\varepsilon_{\rm out}/(\varepsilon_{\rm in} + 2\varepsilon_{\rm out}) = 0.275$ of the field in the glass, against $1/\varepsilon_{\rm in} = 0.0567$ in bulk PbS, and the depolarization factor is

$$\frac{9 \times 17.63^2}{(17.63 + 2 \times 1.977)^2} = 6.00.$$

It depends little on the exact value: $\mathrm{Re}\,\varepsilon_{\rm in} = 10$ gives 4.62 and 25 gives 6.71. Wherever $|\varepsilon_{\rm in}|$ is large, as near the absorption edge of a dot, the bulk formula therefore underestimates the absorption rate by a factor of 4.6 to 6.7. At 5 eV silica has $\varepsilon_{\rm out} = 2.276$. If $\varepsilon_{\rm in}$ there were $1 + \mathrm{i}$, an illustrative value near one, the factor would be 0.566, and the bulk formula would overestimate the rate; for $\varepsilon_{\rm in} = 3 + 3\mathrm{i}$ it would be 2.45.

Which of these cases holds over 2.5 to 10 eV, the range of Blanco et al., is open. The size and even the sign of the correction need the dielectric function $\varepsilon_{\rm in}(\omega)$ of real dots of the size used, measured or computed, and no such data enter the published projection. With it, Eq. (16.26) is a one-line evaluation, and Appendix E.21 specifies the envelope and cluster calculations that give the scattering response of the same dots.

16.4 Molecular crystals

Anthracene, $\mathrm{C_{14}H_{10}}$, is three benzene rings fused in a row. Its molecules stack in a monoclinic crystal, two to a cell, and the crystal keeps much of the molecule's spectrum: the lowest singlet absorption lies at 3.1 eV in the crystal against 3.5 eV in the gas [18, Tables I and III]. A particle must carry $2 \times 3.1\,\mathrm{eV}/v_{\rm max}^2 = 0.883\,\mathrm{MeV}$ to make this excitation (Eq. (1.25)), and an organic crystal turns a large fraction of such excitations into scintillation photons that a photodetector can count, at least 63 percent of them in trans-stilbene [19, Sect. IV B]. The electrons that dark matter excites near threshold sit in π orbitals perpendicular to the flat molecule, and all molecules of the crystal point along two directions, so the rate depends on how the dark-matter wind meets the molecular planes and turns with the Earth. The excited electron stays on its molecule, bound to the hole it left: such a localized excitation is a *Frenkel exciton*. (The vacancy–interstitial pair of Sect. 7.4 is an unrelated object of the same name.) We derive the Frenkel exciton from the Bethe–Salpeter equation of Sect. 4.9,

show when the rate reduces to a sum of molecular form factors, and size the corrections and the direction dependence in a model of anthracene.

16.4.1 Molecular-basis Bethe–Salpeter equation

Eq. (4.104) holds in any basis of electron–hole pairs, and for a molecular crystal the natural one is localized. Combine the occupied Bloch states into orbitals $\phi_{v\mathfrak{o}}$ centered on molecule $\mathfrak{o}$, and the empty ones into orbitals $\phi_{c\mathfrak{o}}$, where $\mathfrak{o} = 1, \dots, N_{\mathrm{mol}}$ labels the molecules of the cell at positions $\boldsymbol{r}_{\mathfrak{o}}$ and the cell index is kept implicit. A pair is then an electron in $\phi_{c\mathfrak{o}'}$ and a hole in $\phi_{v\mathfrak{o}}$. If $\mathfrak{o}' = \mathfrak{o}$ the electron stays on the molecule where the hole is; if $\mathfrak{o}' \neq \mathfrak{o}$ the pair is a *charge-transfer* pair, with the electron on a neighbor at a distance d.

The kernels of Eq. (4.105) now take a simple form when the orbitals of different molecules do not overlap. The direct kernel contains the overlap densities $\phi^*_{c\mathfrak{o}'}\phi_{c\mathfrak{o}'''}$ and $\phi^*_{v\mathfrak{o}}\phi_{v\mathfrak{o}''}$, which vanish unless the electron and the hole each stay on their molecule. It therefore acts within one molecule, as the intramolecular attraction, and between the partners of a charge-transfer pair, as their screened attraction at distance d. The exchange kernel contains the transition densities $\phi^*_{c\mathfrak{o}'}\phi_{v\mathfrak{o}}$, which vanish for a charge-transfer pair; on one molecule they are the charge distribution of a molecular transition, and exchange couples such transitions on different molecules through their Coulomb interaction. In the limit of zero overlap the charge-transfer pairs decouple, and the problem separates into two steps.

The first is the molecule. Its own block of H^{exc}, with both kernels, gives the singlet excited states $\mathfrak{x}$ of one molecule in the crystal, with energies $E_{\mathfrak{x}}$ and transition densities $\rho^{(\mathfrak{o})}_{0\mathfrak{x}}(\boldsymbol{r}) = \langle \mathfrak{x}|\hat{n}_e(\boldsymbol{r})|0\rangle$. Their Fourier transforms are the molecular form factors,

$$f^{(\mathfrak{o})}_{0\mathfrak{x}}(\boldsymbol{q}) = \int \mathrm{d}^3 r\; \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}_{\mathfrak{o}})}\, \rho^{(\mathfrak{o})}_{0\mathfrak{x}}(\boldsymbol{r}) = \Big\langle \mathfrak{x}\Big|\sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}_j-\boldsymbol{r}_{\mathfrak{o}})}\Big|0\Big\rangle, \tag{16.27}$$

with the sum over the electrons of the molecule. The singlet factor $\sqrt{g_{\mathrm{deg}}}$ of Sect. 10.5 is inside f, since $\langle\mathfrak{x}|$ is the full many-electron state. Molecules related by a rotation $\mathsf{R}_{\mathfrak{o}}$ have form factors related by $f^{(\mathfrak{o})}_{0\mathfrak{x}}(\boldsymbol{q}) = f_{0\mathfrak{x}}(\mathsf{R}_{\mathfrak{o}}^{-1}\boldsymbol{q})$, and at small q each starts as $\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{d}_{\mathfrak{x}}$, with $\boldsymbol{d}_{\mathfrak{x}} = \langle\mathfrak{x}|\sum_j(\boldsymbol{r}_j - \boldsymbol{r}_{\mathfrak{o}})|0\rangle$ the transition dipole that fixes the oscillator strength Eq. (4.36).

The second step is the crystal. A molecular excitation on $\mathfrak{o}$ can move to $\mathfrak{o}'$ when its transition density and that of the receiving molecule interact, and the Bloch combinations of crystal momentum $\boldsymbol{Q}$ diagonalize the translations. The Frenkel Hamiltonian is

$$\begin{aligned} H^{\mathrm{F}}_{\mathfrak{o}\mathfrak{x},\mathfrak{o}'\mathfrak{x}'}(\boldsymbol{Q}) &= E_{\mathfrak{x}}\,\delta_{\mathfrak{o}\mathfrak{o}'}\delta_{\mathfrak{x}\mathfrak{x}'} + \mathfrak{J}_{\mathfrak{o}\mathfrak{x},\mathfrak{o}'\mathfrak{x}'}(\boldsymbol{Q}), \\ \mathfrak{J}_{\mathfrak{o}\mathfrak{x},\mathfrak{o}'\mathfrak{x}'}(\boldsymbol{Q}) &= {\sum_{\boldsymbol{R}}}'\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} \int \mathrm{d}^3 r\, \mathrm{d}^3 r'\; \rho^{(\mathfrak{o})}_{0\mathfrak{x}}(\boldsymbol{r})^*\, \frac{e^2}{4\pi|\boldsymbol{r}-\boldsymbol{r}'-\boldsymbol{R}|}\, \rho^{(\mathfrak{o}')}_{0\mathfrak{x}'}(\boldsymbol{r}'), \end{aligned} \tag{16.28}$$

where the prime drops the term $\boldsymbol{R} = 0$, $\mathfrak{o}' = \mathfrak{o}$, since a molecule's interaction with itself is already in $E_{\mathfrak{x}}$. The coupling $\mathfrak{J}$ is the exchange kernel $g_{\mathrm{deg}}K^{\mathrm{x}}$ of Eq. (4.105) between molecules, written in real space; as there, its long-wavelength part, the Fourier component at $\boldsymbol{Q}$ itself, is removed and kept in $\varepsilon_{\mathrm{mac}}$. The lattice sum converges only conditionally, and in practice it is done by Ewald's method. The eigenvectors $A^{\lambda}_{\mathfrak{o}\mathfrak{x}}(\boldsymbol{Q})$ and energies $E_\lambda(\boldsymbol{Q})$ of H^{F} are the Frenkel excitons. A molecular level becomes N_{mol} exciton bands. At $\boldsymbol{Q} \to 0$ these are the components into which the line splits in the absorption spectrum of the crystal, the *Davydov splitting*, and in a cell with two inequivalent orientations they absorb light of different polarizations.

The molecular energies and transition densities are the material input. They come from the *GW* and Bethe–Salpeter equations solved for one molecule in a basis of atom-centered Gaussian orbitals, with singlet energies typically within 0.1 eV to 0.3 eV of measurement. The

crystal enters either as the reaction field of a polarizable medium or through a calculation of the whole periodic crystal (Appendix E.6). In anthracene the environment lowers the fundamental gap, the difference between ionization energy and electron affinity, from 6.9 eV in the gas to 3.9–4.2 eV in the crystal [18, Tables I–III], since the polarization of the neighbors stabilizes a separated electron and hole far more than a neutral excitation. A periodic *GW* calculation followed by the Bethe–Salpeter equation places the crystal's lowest singlet at 3.3 eV, against the measured 3.1 eV [18, Table III].

16.4.2 Frenkel-exciton rate

The structure factor follows from Eq. (4.112) with the molecular basis in place of the band pairs. The vertex of the pair $(\mathfrak{o},\mathfrak{x})$ at momentum transfer $\boldsymbol{q}$ is $F_{\mathfrak{o}\mathfrak{x}}(\boldsymbol{q}) = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_{\mathfrak{o}}} f_{0\mathfrak{x}}^{(\mathfrak{o})}(\boldsymbol{q})$, where the phase places the molecule in the cell, and the exciton that a momentum transfer $\boldsymbol{q}$ creates has crystal momentum $\boldsymbol{q}$. Per cell of volume Ω,

$$S^{\mathrm{F}}(\boldsymbol{q},\omega) = \frac{2\pi}{\Omega}\sum_{\lambda} \frac{\left|\sum_{\mathfrak{o}\mathfrak{x}} A^{\lambda}_{\mathfrak{o}\mathfrak{x}}(\boldsymbol{q})^{*}\, F_{\mathfrak{o}\mathfrak{x}}(\boldsymbol{q})\right|^{2}}{|\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega)|^{2}}\, \delta\big(\omega - E_{\lambda}(\boldsymbol{q})\big). \tag{16.29}$$

Each exciton is a coherent superposition of excitations on all molecules, and its weight carries the interference between them through the phases $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_{\mathfrak{o}}}$. A detector does not resolve the individual excitons, however. Integrate Eq. (16.29) over an energy window $\mathcal{B}$ that contains every exciton built from a set of molecular levels and no other, and assume that $\mathfrak{J}$ couples no level of the set to a level outside it and that $\varepsilon_{\mathrm{mac}}$ does not change across the window. Since H^{F} is Hermitian, as in the Tamm–Dancoff form of Sect. 4.9, the eigenvectors restricted to the set are then complete, $\sum_{\lambda\in\mathcal{B}} A^{\lambda}(A^{\lambda})^{\dagger} = 1$, and

$$\int_{\mathcal{B}} \mathrm{d}\omega\; S^{\mathrm{F}}(\boldsymbol{q},\omega) = \frac{2\pi}{\Omega\,|\varepsilon_{\mathrm{mac}}(\boldsymbol{q})|^{2}} \sum_{\mathfrak{o}} \sum_{\mathfrak{x}\in\mathcal{B}} \left|f_{0\mathfrak{x}}^{(\mathfrak{o})}(\boldsymbol{q})\right|^{2}. \tag{16.30}$$

The phases have dropped out. Summed over all its Davydov components and over the whole exciton band, the crystal responds as a set of independent molecules, whatever $\mathfrak{J}$ is. At $q \to 0$, where $|f_{0\mathfrak{x}}|^{2} \to |\boldsymbol{q}\cdot\boldsymbol{d}_{\mathfrak{x}}|^{2}$, Eq. (16.30) becomes the oscillator-strength sum rule of molecular-exciton theory: the Davydov components share the dipole strength of the N_{mol} molecular transitions, $\sum_{\lambda} |\hat{\boldsymbol{e}}\cdot\boldsymbol{d}_{\lambda}|^{2} = \sum_{\mathfrak{o}} |\hat{\boldsymbol{e}}\cdot\boldsymbol{d}_{\mathfrak{x}}^{(\mathfrak{o})}|^{2}$ for every polarization $\hat{\boldsymbol{e}}$. The oscillator strengths themselves add up only to order $\mathfrak{J}/E_{\mathfrak{x}}$, since Eq. (4.36) weights each component with its own energy. Eq. (16.30) is the same sum rule at every momentum transfer, and Worked example 16.4 works it out for anthracene.

Eq. (16.30) needs the window to hold whole manifolds, $\varepsilon_{\mathrm{mac}}$ and η to stay flat across it, and the vertex to be bare. Coupling to a level $\mathfrak{x}'$ outside the window mixes into a state of the window the amplitude $\mathfrak{J}/(E_{\mathfrak{x}} - E_{\mathfrak{x}'})$ of the other level, and its weight changes by

$$\delta W = \frac{2\,\mathrm{Re}\big[F_{\mathfrak{o}\mathfrak{x}}(\boldsymbol{q})^{*}\,\mathfrak{J}_{\mathfrak{o}\mathfrak{x},\mathfrak{o}'\mathfrak{x}'}(\boldsymbol{q})\,F_{\mathfrak{o}'\mathfrak{x}'}(\boldsymbol{q})\big]}{E_{\mathfrak{x}} - E_{\mathfrak{x}'}}, \tag{16.31}$$

which the level outside loses. This intensity borrowing is first order in $\mathfrak{J}/\Delta E$: the cross term of two amplitudes, one of them already small, is as large as the amplitude itself. The second condition is that $\varepsilon_{\mathrm{mac}}$ and the halo factor $\eta(v_{\mathrm{min}})$ stay constant across the window. For a rate the window is set by kinematics: at fixed q, $\eta(v_{\mathrm{min}})$ changes when ω changes by about qv_{0}, which is 1.6 eV at $q = 2\,\mathrm{keV}$, and much faster near threshold, where v_{min} approaches v_{max}. The binned form holds when the exciton bands are narrow on this scale. The third is the bare vertex. The literature uses the form factors of the isolated molecule, without $1/|\varepsilon_{\mathrm{mac}}|^{2}$

and without the local fields that $\mathfrak{J}$ produces at finite $\boldsymbol{q}$; both corrections vanish when the dark matter resolves distances shorter than a molecule, $qa_{\mathrm{mol}} \gg 1$, with a_{mol} the size of the molecule. The static limit of Eq. (4.110) shows why: $\varepsilon_{\mathrm{mac}}(\boldsymbol{q},0) - 1 = (2e^2/\Omega q^2) \sum_{\mathfrak{o}\mathfrak{r}} |f_{0\mathfrak{r}}^{(\mathfrak{o})}|^2 / E_{\mathfrak{r}}$ is finite at $q \to 0$, where it is the polarizability of the molecules, and falls once the form factors pass their maximum near $q \sim 1/a_{\mathrm{mol}}$.

Insert Eq. (16.30) for each molecular level, with $\varepsilon_{\mathrm{mac}} = 1$ and the window shrunk to the line at $E_{\mathfrak{r}}$, and $|\tilde{V}|^2 = \pi \bar{\sigma}_e F_{\mathrm{DM}}^2 / \mu_{\chi e}^2$ (Eq. (1.34)) into Eq. (8.16); integrate over ω; and write the velocity integral at fixed $\boldsymbol{q}$ as $\hat{f}_{\mathrm{gal}}(v_{\mathrm{min}} + v_{\mathrm{E}} \hat{\boldsymbol{q}} \cdot \hat{\boldsymbol{v}}_{\mathrm{E}})/q$, with the Radon transform $\hat{f}_{\mathrm{gal}}$ of Sect. 11.7. The factors $2\pi \cdot \pi/(2\pi)^3$ combine into $1/4\pi$, and with $\rho_T \Omega = M_{\mathrm{cell}}$, the mass of a cell, the rate per unit target mass is

$$\mathcal{R}(\hat{\boldsymbol{v}}_{\mathrm{E}}) = \frac{\rho_\chi \, \bar{\sigma}_e}{m_\chi \, \mu_{\chi e}^2 \, M_{\mathrm{cell}}} \sum_{\mathfrak{o}\mathfrak{r}} \int \frac{\mathrm{d}^3 q}{4\pi q} \, F_{\mathrm{DM}}(q)^2 \, |f_{0\mathfrak{r}}^{(\mathfrak{o})}(\boldsymbol{q})|^2 \, \hat{f}_{\mathrm{gal}}(v_{\mathrm{min}}^{\mathfrak{r}}(q) + v_{\mathrm{E}} \, \hat{\boldsymbol{q}} \cdot \hat{\boldsymbol{v}}_{\mathrm{E}}), \tag{16.32}$$

with $v_{\mathrm{min}}^{\mathfrak{r}} = E_{\mathfrak{r}}/q + q/2m_\chi$ and $\hat{\boldsymbol{v}}_{\mathrm{E}}$ the direction of the Earth's velocity in crystal axes. Averaged over $\hat{\boldsymbol{v}}_{\mathrm{E}}$, the angular integral of $\hat{f}_{\mathrm{gal}}$ gives $2\pi\eta(v_{\mathrm{min}})$ (Eq. (11.49)), and Eq. (16.32) becomes $\rho_\chi \bar{\sigma}_e / (2 m_\chi \mu_{\chi e}^2 M_{\mathrm{cell}}) \sum \int q \, \mathrm{d}q \, \eta \, F_{\mathrm{DM}}^2 \langle |f|^2 \rangle$. This is the rate of Blanco et al. for a liquid of aromatic molecules [20, Eqs. (11), (13), (16)–(19)], with $N_{\mathrm{mol}}/M_{\mathrm{cell}}$ molecules per unit mass and unit detection efficiency. Eq. (16.32) itself is the direction-resolved form they use for the four molecules of trans-stilbene [19, Eqs. (8), (9)]. Their singlet form factor carries the factor $\sqrt{2}$ of Eq. (16.27). The momentum transfer of the earlier work has the opposite sign [20, Eq. (12)], which $|f|^2$ does not see.

16.4.3 Davydov splitting in anthracene

To put numbers on these conditions we use a π-electron model of anthracene, of the kind the literature uses for benzene and trans-stilbene [19, 20]. The carbon positions are those of the measured crystal structure, with $a = 8.553\,\text{Å}$, $b = 6.016\,\text{Å}$, $c = 11.172\,\text{Å}$ and $\beta = 124.6°$ [21], and the second molecule of the cell is the image of the first under the twofold screw axis along b. We use the crystal axes a, b and c', the last perpendicular to a and b. The long axes of both molecules make 31° with c', their short in-plane axes make 27° with b, and the two molecular planes meet at 51.5°, the herringbone packing of the acenes. The π orbitals are combinations of Slater $2p_z$ orbitals with effective charge $Z_{\mathrm{eff}} = 3.15$ on the 14 carbons [20, Eq. (2)], with the nearest-neighbor resonance integral $-0.63/(m_e x^2)$ at the measured bond length x [19, Eq. (A1)]. A self-consistent field of the Pariser–Parr–Pople type refines them; it adds the repulsion between the π electrons to the one-electron (Hückel) model. We use a one-center repulsion of 11.13 eV and the Mataga–Nishimoto form of the two-center repulsion, a standard parametrization that we take as a model choice. The singlets come from all 49 single excitations of the 14 π electrons.

The model places the lowest singlet at 3.85 eV, 0.35 eV above the measured gas value. For the binned form we shift all levels down by 0.75 eV, so that the line sits at the measured crystal absorption, 3.1 eV, which contains the whole shift from gas to crystal. This state, L_a in Platt's notation, is the transition from the highest occupied to the lowest empty orbital, polarized along the short molecular axis, with $|\boldsymbol{d}| = 0.92\,\text{e Å}$ and a direction-averaged oscillator strength of 0.23. The strong long-axis transition, B_b, follows at 4.49 eV with an oscillator strength of 2.46. A third state, L_b, lies only 18 meV above L_a and has no dipole in the model, a consequence of the pairing symmetry of a π-electron model of an alternant hydrocarbon. With zero differential overlap, the approximation behind the Pariser–Parr–Pople model, each transition density is a sum of atomic $2p_z$ densities weighted by transition charges ϱ_μ on the carbons, $\sum_\mu \varrho_\mu = 0$, and $f_{0\mathfrak{r}}(\boldsymbol{q}) = F_{2p}(\boldsymbol{q}) \sum_\mu \varrho_\mu \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_\mu}$, with $\boldsymbol{R}_\mu$ the carbon positions relative

to the molecular center and F_{2p} the form factor of one $2p_z$ density. Keeping the overlap densities of neighboring orbitals raises the direction-averaged $|f|^2$ of L_a by 1.4 to 11 % at $q = 2$ to 4 keV and gives L_b up to 16 % of its weight. The coupling $\mathfrak{J}$ of Eq. (16.28) is evaluated with the transition charges by Ewald summation. The model is a guide to sizes, not a calculation of anthracene: it has no σ electrons, whose polarizability would screen $\mathfrak{J}$, so its couplings are upper estimates.

Worked example 16.4: The Davydov splitting of anthracene and the binned sum

Take the lowest level, L_a, on the two molecules of the cell, in the model above. The twofold screw axis maps the transition dipole $\boldsymbol{d}_1$ of the first molecule onto $\boldsymbol{d}_2 = \mathsf{R}_b \boldsymbol{d}_1$, with R_b the rotation by 180° about b, so that $\boldsymbol{d}_1 + \boldsymbol{d}_2$ points along b and $\boldsymbol{d}_1 - \boldsymbol{d}_2$ lies in the ac plane. At $\boldsymbol{Q} \to 0$ the Ewald sum of Eq. (16.28) gives $\mathfrak{J}_{11} = \mathfrak{J}_{22} = -196.6\,\mathrm{meV}$, the coupling of each molecule to its translated copies, and $\mathfrak{J}_{12} = -15.6\,\mathrm{meV}$ between the two orientations. The 2×2 Frenkel Hamiltonian has the eigenvalues $E_{L_a} + \mathfrak{J}_{11} \mp |\mathfrak{J}_{12}|$. The measured 3.1 eV is the lower of them, so that the shift from gas to crystal is counted once if the molecular level is $E_{L_a} = 3.1 + 0.197 + 0.016 = 3.31\,\mathrm{eV}$:

$$E_- = 3.100\,\mathrm{eV}\ (\parallel b), \qquad E_+ = 3.131\,\mathrm{eV}\ (\perp b),$$

a Davydov splitting of $2|\mathfrak{J}_{12}| = 31\,\mathrm{meV}$, with the lower component polarized along b. Of the measured lowering from the gas, 3.5 to 3.1 eV, the common shift $\mathfrak{J}_{11} = -0.20\,\mathrm{eV}$ then supplies half, and the polarization of the neighbors, which $\mathfrak{J}$ does not contain, lowers the gas value to E_{L_a}. The dipole strengths of the two components are $|\boldsymbol{d}_-|^2 = 1.334\,\mathrm{e}^2\,\text{Å}^2$ and $|\boldsymbol{d}_+|^2 = 0.359\,\mathrm{e}^2\,\text{Å}^2$, and their sum, $1.693\,\mathrm{e}^2\,\text{Å}^2$, equals $2|\boldsymbol{d}_1|^2$ exactly, as the sum rule demands. The direction-averaged oscillator strengths add to 0.460 against $2 \times 0.245 = 0.491$ for two free molecules at E_{L_a}; the ratio, 0.938, is close to 0.941, the mean energy of the two components divided by E_{L_a}, since each oscillator strength carries its own energy.

The same states at finite momentum transfer show what binning does. At $q = 3\,\mathrm{keV}$ along the short axis of the first molecule, the four excitons built from L_a and L_b carry the weights $|\sum A^* F|^2 = 0.0013$ at 3.299 eV, 0.1858 at 3.356 eV and zero in the two L_b states. The two molecules alone carry $|f^{(1)}|^2 = 0.0599$ and $|f^{(2)}|^2 = 0.1272$, since the momentum transfer lies at different angles to them. Coherence concentrates the weight in one exciton, but any window wider than the 58 meV between the two lines returns 0.1871, the incoherent sum. Over the whole zone this band of four states spans 3.04 to 3.40 eV, a width of 0.36 eV.

Limit Check 16.7. Binned Frenkel rate

The Frenkel-exciton rate binned in ω is the incoherent sum of molecular form factors at the molecular line energies, Eq. (16.32) with Eq. (16.30) and $\varepsilon_{\mathrm{mac}} = 1$ [20, Eqs. (11), (13), (16)–(19), 19, Eqs. (8), (9)].

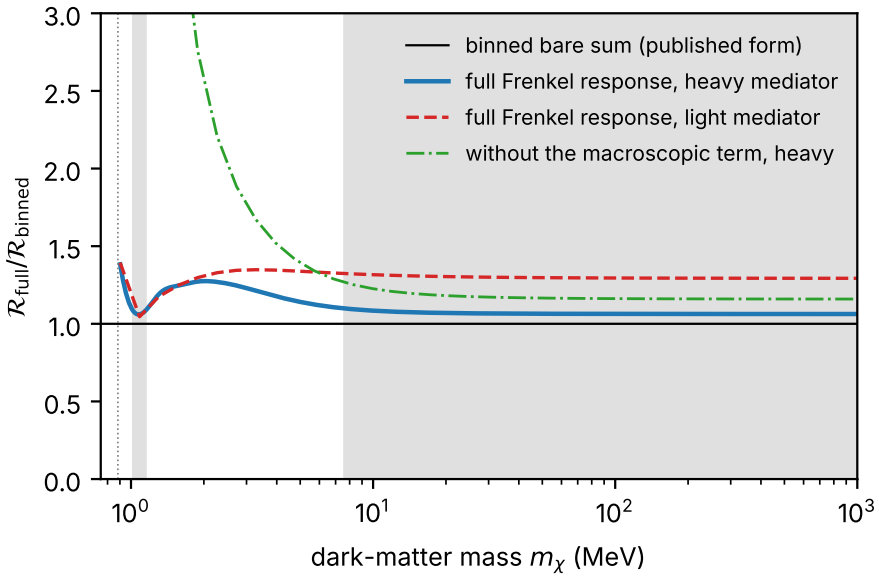


Figure 16.11: Rate of the full Frenkel-exciton response of anthracene divided by the binned rate, against the dark-matter mass, for a heavy (solid) and a light (dashed) mediator, direction-averaged halo. Full response: all 49 π-electron singlets on both molecules, coupled by $\mathfrak{J}$ and by the macroscopic Coulomb term $(e^2/\Omega q^2)FF^\dagger$, so that no factor $1/|\varepsilon_{\mathrm{mac}}|^2$ is needed (Appendix D.12); the molecular levels are shifted by 0.27 eV so that the lowest optically allowed exciton at $\boldsymbol{Q} \to 0$ lies at the measured 3.1 eV of the binned form. Dash-dotted: the full response without the macroscopic term, heavy mediator. Model of Sect. 16.4.3, π electrons only. Gray: the heavy-mediator ratio within 10 % of one. Dotted: the binned threshold, 0.883 MeV.

Just above threshold, between 0.88 and 1.0 MeV, the lowest exciton of the full model lies below the binned line and the ratio reaches 1.38. For a heavy mediator between 1.2 and 7.5 MeV, and for a light mediator at every mass above 1.2 MeV, the mixing of the weak L_a line with the strong B_b transition and the dispersion of the exciton bands change the rate by 10 to 35 %, beyond the reach of the incoherent sum.

Fig. 16.11 separates intensity borrowing far above threshold, band dispersion near it, and the macroscopic term. Above 30 MeV the heavy-mediator rate of the full response exceeds the binned one by 6.3 %, flat in mass, the intensity borrowing of Eq. (16.31). Since $\mathfrak{J}$ is periodic in the crystal momentum, its L_a–B_b element, 0.12 eV at $\boldsymbol{Q} \to 0$ and 0.18 eV at a general point of the zone against an energy difference of 1.4 eV, keeps the binned weight off by several percent at any momentum transfer. A light mediator weights the momentum transfers near the threshold value $\omega/v_{\max} = 1.17$ keV, where $qa_{\mathrm{mol}} \approx 1.6$ for the root-mean-square radius $a_{\mathrm{mol}} = 2.66$ Å of the carbon frame, and there the excess is 29 to 35 % at all masses above 2 MeV. Between 1.2 and 7.5 MeV the heavy-mediator ratio reaches 1.27, because the exciton bands spread the levels over tenths of an eV while the halo factor, near threshold, changes on that scale.

The dash-dotted curve shows that the macroscopic term matters most: without it, the local fields alone would raise the rate ninefold at 1.1 MeV. The π transitions themselves make $\varepsilon_{\mathrm{mac}}(q,0) - 1 = 0.83$ at the threshold momentum and 0.51 at 2 keV in the model, so that the factor $1/|\varepsilon_{\mathrm{mac}}|^2$ of Eq. (16.29) cannot be applied line by line. In the full response the macroscopic term shifts weight up in energy instead of removing it (Sect. 4.9). Since the

model omits the σ electrons, which screen $\mathfrak{J}$, its corrections are probably overstated.

16.4.4 Charge-transfer excitons and other corrections

The zero-overlap limit removes the charge-transfer pairs, and real molecular crystals have some overlap. The simplest description keeps one Frenkel state $|\mathrm{Fr}\rangle$, of energy E_{Fr}, and one charge-transfer state $|\mathrm{CT}\rangle$, of energy E_{CT}, with the electron on a neighbor at distance d. The two are coupled by the transfer integral t_{CT}, the sum of the integrals that move the electron and the hole between the molecules. Both integrals are proportional to the overlap of the orbitals of the two molecules. Diagonalizing the 2×2 problem gives two states $|\pm\rangle$ rotated by the mixing angle θ_{CT},

$$\tan 2\theta_{\mathrm{CT}} = \frac{2t_{\mathrm{CT}}}{E_{\mathrm{CT}} - E_{\mathrm{Fr}}}, \qquad f_{\pm}(\boldsymbol{q}) = \cos\theta_{\mathrm{CT}}\, f_{\mathrm{Fr}}(\boldsymbol{q}) \pm \sin\theta_{\mathrm{CT}}\, f_{\mathrm{CT}}(\boldsymbol{q}), \tag{16.33}$$

with f_{CT} the form factor of the charge-transfer state. Its transition density, the overlap density of an empty orbital on one molecule and an occupied one on the other, integrates to zero for orthogonal orbitals, so $f_{\mathrm{CT}}(0) = 0$; its dipole is of order d times the overlap, and $|f_{\mathrm{CT}}|^2$ enters at order $q^2 d^2$. In the rate the Frenkel line loses the fraction $\sin^2\theta_{\mathrm{CT}}$ of its weight to a new line near E_{CT}, a shift of weight up in energy that a binned rate sees only if the bin ends between the two lines. And f_{CT} carries the phase difference between two molecules, and the charge-transfer contribution is coherent over distances of order d, which no Frenkel model contains. In anthracene the charge-transfer pairs lie near the fundamental gap, 3.9–4.2 eV, some 0.8–1.1 eV above the lowest singlet [18, Tables II and III], and for small mixing $\theta_{\mathrm{CT}}^2 \simeq t_{\mathrm{CT}}^2/(E_{\mathrm{CT}} - E_{\mathrm{Fr}})^2$: a transfer integral of 0.1 eV would move about one percent of the weight. The mixing grows along the acene series, whose larger members have singlets that extend over several molecules [18, Sect. III E].

The nuclei move when a molecule is excited, and each molecular line becomes a Franck–Condon progression of vibronic replicas with Poisson weights set by a Huang–Rhys factor (Eq. (5.63)). In the Condon approximation the progression keeps the total weight of the line, so Eq. (16.30) holds for windows that cover it, while the same overlaps narrow the exciton band of each replica. Above the fundamental gap the crystal has a continuum of separated electrons and holes that the Frenkel model lacks, and Eq. (16.32) omits it, as the published rates do [20]; it raises the rate of masses above $2 \times 4\,\mathrm{eV}/v_{\max}^2 \approx 1.1\,\mathrm{MeV}$. Triplet excitons cost less, 1.9 eV for the lowest triplet of crystalline anthracene [18, Table III], but a spin-independent density coupling cannot create them, and they enter only through the spin couplings of Sect. 10.4. Finally, the π-electron model fails at momentum transfers above about 10 keV, where the σ electrons of the C–C and C–H bonds start to contribute to $|f|^2$. No calculation of the full periodic Bethe–Salpeter response of anthracene at finite $\boldsymbol{q}$ exists to compare with the binned molecular sum and with the zero-overlap model; Appendix E.21 specifies it, with the charge-transfer weight θ_{CT}^2 and the largest change of $|f|^2$ over q as its outputs.

16.4.5 Direction dependence of an anthracene target

A π orbital is odd under reflection through the molecular plane, so the transition density between two π orbitals is even, and with zero differential overlap it is a sum of atomic densities centered in the plane. For $\boldsymbol{q}$ along the normal every phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_\mu}$ is then the same, and $f = F_{2p}\sum_\mu \varrho_\mu$ vanishes. With the overlap densities kept and the measured, slightly puckered geometry, $|f|^2$ of L_a at $q = 2.07\,\mathrm{keV}$ is 1×10^{-7} along the normal, against 0.25 along the short axis. The transition density of L_a also changes sign under reflection through the long axis, so that $|f|^2$ along that axis is only 2×10^{-9}. Near threshold, where only L_a is reached, the

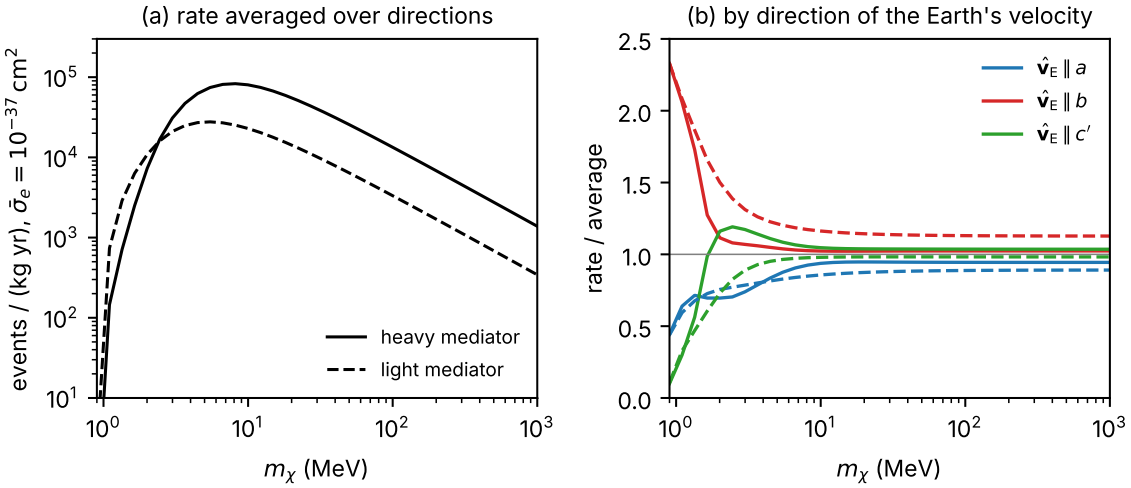


Figure 16.12: Frenkel-exciton rate of an anthracene crystal in the binned form Eq. (16.32), per kg-year at $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$, standard halo of Table 1.1. (a) Rate averaged over the directions of the Earth's velocity $\hat{\boldsymbol{v}}_\mathrm{E}$, heavy (solid) and light (dashed, $F_\mathrm{DM} = (\alpha m_e/q)^2$) mediator. (b) Rate for $\hat{\boldsymbol{v}}_\mathrm{E}$ along the crystal axes a, b and c' (perpendicular to a and b), divided by the average; heavy (solid) and light (dashed) mediator. The rate is even in $\hat{\boldsymbol{v}}_\mathrm{E}$. Model of Sect. 16.4.3: π-electron singlets below 7.5 eV of the Pariser–Parr–Pople model on the measured crystal structure, lowest singlet at 3.1 eV, form factors with zero differential overlap; the continuum above the fundamental gap is not included.

molecule responds to the component of $\boldsymbol{q}$ along its short axis alone. In the crystal both short axes lie within 27° of b, and the halo makes $\boldsymbol{q}$ prefer the direction of the wind when v_min is close to v_max (Sect. 11.7): the rate is largest when the Earth moves along b.

Fig. 16.12 shows the rate of Eq. (16.32) in the model, with the 20 singlets below 7.5 eV, for the Earth's velocity along each crystal axis. At 10 MeV and $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$ the direction-averaged rate is 8.0×10^4 events per kg-year for a heavy mediator and 2.3×10^4 for a light one. The heavy-mediator value is comparable to the independent-particle rate of silicon, 1.36×10^5 (Sect. 10.2). At 1.0 MeV, just above threshold, L_a carries the whole rate, and the largest rate over all directions of the wind, at $\hat{\boldsymbol{v}}_\mathrm{E} \parallel b$, exceeds the smallest, near the long molecular axes, 92 times for a heavy mediator and 69 times for a light one. The contrast falls quickly with the mass: L_a carries 68 % of the rate at 2 MeV and 19 % at 10 MeV, as B_b and the higher levels fill in the directions L_a cannot use. The ratio of the largest to the smallest rate is 2.3 (heavy) and 5.3 (light) at 1.5 MeV, 1.20 and 1.39 at 10 MeV, and 1.17 and 1.29 above 100 MeV.

Whatever the mounting and the latitude, $\hat{\boldsymbol{v}}_\mathrm{E}$ sweeps only part of the sphere in a sidereal day, and the amplitude A_day of Eq. (11.51) cannot exceed $(\mathcal{R}_\mathrm{max} - \mathcal{R}_\mathrm{min})/(\mathcal{R}_\mathrm{max} + \mathcal{R}_\mathrm{min})$ over all directions. In the model this is 0.98 (heavy) and 0.97 (light) at 1.0 MeV, 0.39 and 0.68 at 1.5 MeV, 0.09 and 0.17 at 10 MeV, and 0.08 and 0.13 at 1 GeV. The bound near threshold is a hundred times the daily modulation of silicon, 0.9 % at 2 MeV (Worked example 11.8), because of the crystal class. A cubic crystal has no quadrupole, and its first anisotropic term, the hexadecapole $\ell = 4$, passes the halo only at a few percent; in monoclinic anthracene the quadrupole $\ell = 2$ is allowed and passes at tens of percent of $\mathcal{H}_0$ (Eq. (11.49)). Blanco et al. reach the same conclusion for trans-stilbene, with a modulation near one at threshold and above the ten percent of the annual modulation at large masses [19, Sects. V, VI].

16.5 Dopants

A phosphorus donor in silicon binds its electron by 45.59 meV, and dark matter of 13 keV can set it free (Worked example 7.5). Compared to the host, the donor has sharp bound lines and a ground state built from six valleys. The $1s(A_1) \to 2p_0$ line at 34.11 meV is 4.2 µeV wide in isotopically purified silicon [22, p. 5], and a dark photon whose mass falls on it is absorbed several hundred times more strongly than one in the ionization continuum. Between the valley states, dark matter drives transitions that light of the same energy cannot. The drawback is the number of donor electrons, five parts in 10^9 of the valence electrons at $n_\mathrm{D} = 10^{15}\,\mathrm{cm}^{-3}$. Du et al. proposed doping to $10^{18}\,\mathrm{cm}^{-3}$ to make up for it [23], at which point the donors begin to merge into an impurity band. We will derive the response of the effective-mass donor of Sect. 7.4 to the dark-matter operators, its absorption rate with the host screening counted once, the density at which the isolated-donor picture fails, and the conditions under which a donor excitation is a useful final state.

16.5.1 Response of dilute donors

Donors at $n_\mathrm{D} a^{*3} \ll 1$ act independently, each a hydrogen atom of the crystal (Sect. 7.4). All their transitions lie below the gap of the host, where the host absorbs nothing and its dielectric matrix $\varepsilon_\mathrm{host}$ is real. To first order in n_D the absorptive part of the response then comes from the donors alone, and the host enters through the screening of the vertex. In Eq. (8.35), split the proper response into its host and donor parts. To first order in the donor part, each vertex acquires the host factor $(\varepsilon^{-1}_\mathrm{host})_{\boldsymbol{G0}}$, and the donor structure factor is

$$\begin{aligned} S^\mathrm{D}_{ab}(\boldsymbol{q},\omega) &= 2\pi n_\mathrm{D} \sum_f \langle i|\tilde{O}_a(\boldsymbol{q})^\dagger|f\rangle\langle f|\tilde{O}_b(\boldsymbol{q})|i\rangle\, \delta(\omega - E_f + E_i), \\ \tilde{n}_e(\boldsymbol{q}) &= \sum_{\boldsymbol{G}} \big(\varepsilon^{-1}_\mathrm{host}\big)_{\boldsymbol{G0}}(\boldsymbol{q},\omega)\, n_e(\boldsymbol{q}+\boldsymbol{G}), \end{aligned} \tag{16.34}$$

for $T \ll E^\mathrm{D}_\mathrm{b}$, with $|i\rangle$ the donor ground state and f its excited and ionized states. The dark matter's potential at $\boldsymbol{q}$ is screened by the host into components at every $\boldsymbol{q}+\boldsymbol{G}$, and the donor electron feels all of them. An operator with no proper correlation with the charge (a row of the coupling vector of Sect. 8.4.2), such as the spin density of a host without spin–orbit coupling, enters unscreened, $\tilde{O} = O$. Each donor contributes one electron, and the density does not flip its spin, so there is no factor of two for spin. The scattering rate is Eq. (8.16) with $S = S^\mathrm{D}$ and ρ_T the mass density of the host.

The matrix elements follow from the multivalley donor of Appendix D.23, built on the envelope equation of Kohn and Luttinger [24]. A state s is $\psi_s = \sum_j c^s_j F^s_j(\boldsymbol{r})\, \psi_{c\boldsymbol{k}_j}(\boldsymbol{r})$, with envelopes F^s_j in the six valleys at $\boldsymbol{k}_j$ and valley coefficients c^s_j. The product of two cell-periodic parts is a lattice sum, $u^*_{c\boldsymbol{k}_{j'}} u_{c\boldsymbol{k}_j} = \sum_{\boldsymbol{G}} \mathcal{U}_{j'j}(\boldsymbol{G})\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$, with $\mathcal{U}_{jj}(\boldsymbol{0}) = 1$. Since the envelopes vary slowly over a cell,

$$\langle s'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|s\rangle = \sum_{j,j'} c^{s'*}_{j'} c^s_j \sum_{\boldsymbol{G}} \mathcal{U}_{j'j}(\boldsymbol{G}) \int \mathrm{d}^3r\, F^{s'*}_{j'} F^s_j\, \mathrm{e}^{\mathrm{i}(\boldsymbol{q}+\boldsymbol{k}_j-\boldsymbol{k}_{j'}+\boldsymbol{G})\cdot\boldsymbol{r}}. \tag{16.35}$$

The envelope integral is the Fourier transform of a function a few nanometers across. For the variational envelope of Sect. 7.4 it is $[1 + (r^2_\perp p^2_\perp + r^2_\parallel p^2_\parallel)/4]^{-2}$ at momentum $\boldsymbol{p}$, with $p_\parallel$ along the valley axis, and it is negligible once $p \gg 1/r_\perp = 81\,\mathrm{eV}$. Two kinds of term survive. Intravalley terms, $j = j'$ and $\boldsymbol{G} = \boldsymbol{0}$, carry momentum transfers up to a few hundred eV. Intervalley terms carry $\boldsymbol{q} \simeq \boldsymbol{k}_{j'} - \boldsymbol{k}_j - \boldsymbol{G}$, of order keV.

The same structure fixes the field that the envelope feels. In $\tilde{n}_e$, the component $n_e(\boldsymbol{q}+\boldsymbol{G})$ has intravalley matrix elements $\mathcal{U}_{jj}(-\boldsymbol{G})$ times the envelope integral at $\boldsymbol{q}$. For intravalley

transitions, $\tilde{n}_e = \mathfrak{f}_{\rm D}^{\rm loc}\, n_e/\varepsilon_{\rm host}$, with $\varepsilon_{\rm host} = 1/(\varepsilon^{-1}_{\rm host})_{\mathbf{00}}$ the macroscopic dielectric function of Eq. (4.44) and

$$\mathfrak{f}_{\rm D}^{\rm loc} = 1 + \varepsilon_{\rm host} \sum_{\boldsymbol{G}\neq\mathbf{0}} (\varepsilon^{-1}_{\rm host})_{\boldsymbol{G}\mathbf{0}}\, \mathcal{U}_{jj}(-\boldsymbol{G}). \tag{16.36}$$

The donor electron samples the microscopic field with the weight of the conduction-band density $|u_{c\boldsymbol{k}_j}|^2$ in each cell. Both factors of the sum are small: the off-diagonal elements of $\varepsilon^{-1}_{\rm host}$, which produce the local-field corrections of Sect. 4.6, and the Fourier components of a cell-periodic density at $\boldsymbol{G} \neq \mathbf{0}$. In effective-mass theory the ratio is expected to be close to one, since the donor orbit spans many cells and the Lorentz construction of a local cavity field does not apply to it; it has not been computed. Its value for silicon is an open input, which needs the dielectric matrix of the host and the Bloch functions at the minimum, the inputs of Appendix E.6.

Ionization. For q well below the valley separations, the final state $|\xi\boldsymbol{k}\rangle$, a Coulomb envelope $F_{\boldsymbol{k}}$ in valley ξ, couples to the A_1 ground state with amplitude c_ξ times the envelope overlap. Summed over ξ, with $\sum_\xi |c_\xi|^2 = 1$, the six valleys give the single-valley result $|\int \mathrm{d}^3r\, F^*_{\boldsymbol{k}} F_{1s}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|^2$ [23, Eq. (11)]. For an isotropic hydrogenic envelope of radius $a^* = (2m^*E_{\rm b}^{\rm D})^{-1/2}$ this overlap, summed over the directions of $\boldsymbol{k}$, is the generalized oscillator strength of the hydrogen $1s$ state. With $L = (e^2/2q^2)S$ at $T = 0$ (Eq. (4.45)), the donor part of the loss function is

$$L^{\rm D}(\boldsymbol{q},\omega) = \frac{2^{10}\pi^2\alpha\, m^*\boldsymbol{n}_{\rm D}a^{*4}}{3\,\varepsilon^2_{\rm host}}\, \frac{(3\tilde{q}^2+\tilde{k}^2+1)\exp\!\big[-(2/\tilde{k})\arctan\!\big(2\tilde{k}/(\tilde{q}^2-\tilde{k}^2+1)\big)\big]}{\big[(\tilde{q}+\tilde{k})^2+1\big]^3\big[(\tilde{q}-\tilde{k})^2+1\big]^3\big(1-\mathrm{e}^{-2\pi/\tilde{k}}\big)}, \tag{16.37}$$

with $\tilde{q} = qa^*$, $\tilde{k} = [2m^*(\omega - E_{\rm b}^{\rm D})]^{1/2}a^*$, the arctangent taken in $[0,\pi]$, and $\omega > E_{\rm b}^{\rm D}$. At $q \to 0$ the bracket reduces to the dipole continuum of hydrogen in Worked example 4.5 with $E_B \to E_{\rm b}^{\rm D}$. Du et al. print the bracket with $3\tilde{q}$ in place of $3\tilde{q}^2$ and multiply it by a factor $(E_{\rm eff}/E_0)^2$ [23, Eq. (12)]. The misprint leaves absorption unchanged; with their parameters, a heavy-mediator scattering rate evaluated with the printed bracket differs from the correct one by factors of 1.47, 0.91, 0.59 and 0.56 at 0.02, 0.1, 1 and 10 MeV.

Valley-orbit lines. The $1s$ manifold of phosphorus has its T_2 and E states 11.70 meV and 13.01 meV above the A_1 ground state (Sect. 7.4). At optical momentum the transitions vanish in effective-mass theory, because the valley vectors of Eq. (D.143) are orthogonal and the intervalley terms need keV momenta. Dark matter reaches both. The E states weight the valleys unequally, and the intravalley sum of Eq. (16.35) does not cancel once the envelope integrals differ between valleys. For $\boldsymbol{q}$ along [100], two valleys have $\boldsymbol{q}$ along their axis and four across it. Summed over the two E states,

$$\sum_E |\langle E|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|A_1\rangle|^2 = \frac{2}{9}\Big[\Big(1+\frac{r_\parallel^2 q^2}{4}\Big)^{-2} - \Big(1+\frac{r_\perp^2 q^2}{4}\Big)^{-2}\Big]^2. \tag{16.38}$$

It grows as q^4, as a quadrupole transition does, and peaks at 0.022 for $q = 150\,\mathrm{eV}$, which particles from 29 keV upward can reach. Along [110] the maximum is 0.0032, and along [111], where all six valleys look alike, the transition vanishes, which makes it a directional signal. These numbers assume one effective-mass envelope common to A_1 and E; the central-cell contraction of A_1 changes their size, but not the zeros along [111] and at $q = 0$. The T_2 states pair opposite valleys with opposite signs, and since opposite valleys have identical envelopes their intravalley terms cancel at every q. The transition $A_1 \to T_2$ needs the intervalley terms, at $|\boldsymbol{q}| = 0.685$, 3.60 and 3.88 keV between opposite valleys (the first after an umklapp) and

2.33 and 2.74 keV between perpendicular ones, with the minima at $0.85\,(2\pi/a)$. The lightest particle that delivers 11.70 meV at 0.685 keV has 130 keV, and at 2.33 keV it has 441 keV. The strengths of these lines are set by the coefficients $\mathcal{U}_{j'j}(\boldsymbol{G})$, which need the Bloch functions at the minima (Appendix E.21).

16.5.2 Dopant absorption

A dark photon couples to the total charge, and its absorption rate is Eq. (10.31) with the loss function of the doped crystal. Silicon has no Born charges, and below its gap the host adds only its two-phonon absorption, which at 50 meV gives $\mathrm{Im}(-1/\varepsilon) = 5.4 \times 10^{-6}$ at room temperature and less in a cold crystal (Worked example 11.7). The donor part is the $q \to 0$ limit of $(e^2/2q^2)S^{\mathrm{D}}$ from Eq. (16.34), to first order in n_{D}. With $\tilde{n}_e = n_e/\varepsilon_{\mathrm{host}}$ (taking $f_{\mathrm{D}}^{\mathrm{loc}} = 1$) and $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \simeq 1 + \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}$ between orthogonal states:

Dark-photon absorption by dilute donors

$$L^{\mathrm{D}}(m_\phi) = \frac{\varepsilon_2^{\mathrm{D}}(m_\phi)}{\varepsilon_{\mathrm{host}}^2}, \qquad \mathcal{R} = \zeta^2\,\frac{\rho_\chi}{\rho_T}\,L^{\mathrm{D}}(m_\phi),$$
$$\varepsilon_2^{\mathrm{D}}(\omega) = 4\pi^2\alpha\, n_{\mathrm{D}} \sum_f |\langle f|\hat{\boldsymbol{e}}\cdot\boldsymbol{r}|i\rangle|^2\,\delta(\omega - E_f + E_i), \tag{16.39}$$

for $n_{\mathrm{D}} \ll n_{\mathrm{c}}$ and $|\varepsilon^{\mathrm{D}}| \ll \varepsilon_{\mathrm{host}}$, with $\hat{\boldsymbol{e}}$ the polarization of the dark photon and ε^{D} the donor part of the dielectric function, whose imaginary part is $\varepsilon_2^{\mathrm{D}}$. In a cubic crystal L^{D} does not depend on $\hat{\boldsymbol{e}}$.

The host screening appears once, through the vertex $\tilde{n}_e$, and $\varepsilon_2^{\mathrm{D}}$ is the proper response of the donors, which contains none. The same result follows from writing $\varepsilon = \varepsilon_{\mathrm{host}} + \varepsilon^{\mathrm{D}}$ and expanding $\mathrm{Im}(-1/\varepsilon)$ to first order in ε^{D} with a real $\varepsilon_{\mathrm{host}}$. Per donor, the absorption is that of a hydrogen atom whose matrix elements are those of the envelope, suppressed by $1/\varepsilon_{\mathrm{host}}^2 = 1/130$ at the liquid-helium constant $\varepsilon_{\mathrm{host}} = 11.4$ of Sect. 7.4. In this the dopant differs from the Dirac material of Sect. 16.1, where the background constant sits inside the response of the same electrons.

The A_1 ground state has every valley with weight 1/6, and each valley's p_0 states carry the dipole along its axis and its $p_\pm$ states across it. The cubic average of $|\langle f|\hat{\boldsymbol{e}}\cdot\boldsymbol{r}|i\rangle|^2$ is then $|Z_f|^2/3$ for a p_0 line and $2|X_f|^2/3$ for a $p_\pm$ line, where Z_f and X_f are the dipole elements of one valley along and across its axis, with X_f summed over the two $p_\pm$ partners. We obtain them from the numerical solution of Eq. (7.48) that gives Fig. 7.5, extended to the dipole matrix elements, with the measured line energies. The effective-mass $1s$ envelope, bound by 31.3 meV, is larger than the true ground state, which the central-cell potential deepens to 45.59 meV. To size this, we add a Gaussian well of range 3.2 or 6.3 Å to the s-like channel, tuned to the measured binding. The two ranges give line strengths within 2 % of each other, and for the $2p$ lines a factor of 3.2 to 3.4 below those of the effective-mass envelope (1.9 to 2.7 for the $3p$ lines). For the $1s(A_1) \to 2p_0$ line, $|Z|^2/3 = 13.6\,\text{Å}^2$, an oscillator strength $2m_e\omega|Z|^2/3 = 0.12$, against $43.0\,\text{Å}^2$ and 0.39 with the effective-mass envelope; the $2p_\pm$ line has 0.52 and 1.73. With the central cell the lines carry 25 % of the f-sum of the donor electron, $\sum_f 2m_e(E_f - E_i)|\langle f|\hat{\boldsymbol{e}}\cdot\boldsymbol{r}|i\rangle|^2 = m_e(2/m_\perp^* + 1/m_\parallel^*)/3 = 3.85$ (Eq. (4.37) with band masses), and the continuum the rest; the effective-mass envelope puts 49 % in the lines. Fig. 16.13 shows the resulting spectrum.

The continuum of Du et al. is Eq. (16.37) at $q = 0$ with $E_{\mathrm{b}}^{\mathrm{D}} = 45$ meV, $m^* = 0.3\,m_e$ and $a^* = 23$ atomic units, multiplied by $(E_{\mathrm{eff}}/E_0)^2 = 2.2^2 = 4.84$, a factor they describe as the ratio of the local field at the donor to the average field, chosen to match the measured optical

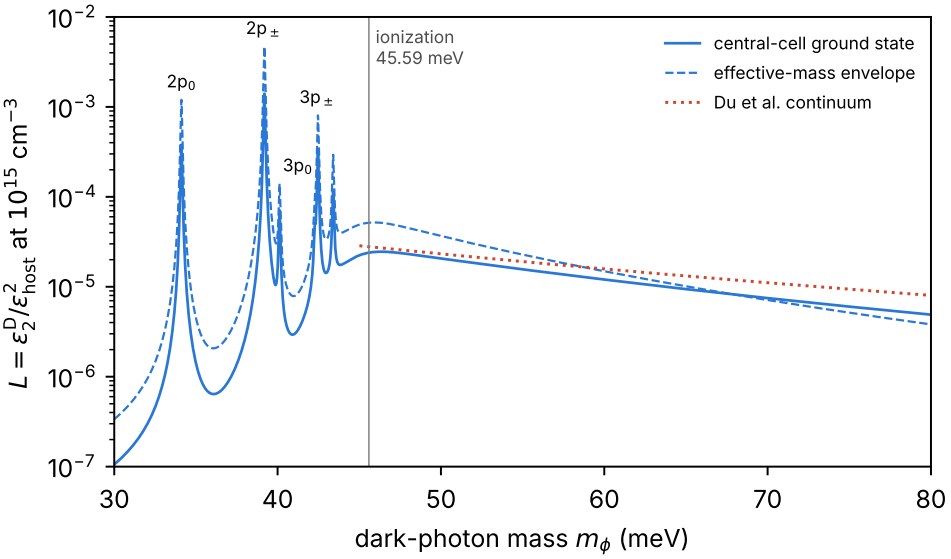


Figure 16.13: Dark-photon loss function of phosphorus donors in silicon, $L^{\mathrm{D}} = \varepsilon_2^{\mathrm{D}}/\varepsilon_{\mathrm{host}}^2$ of Eq. (16.39) at $n_{\mathrm{D}} = 10^{15}\,\mathrm{cm}^{-3}$ and $\varepsilon_{\mathrm{host}} = 11.4$, against the dark-photon mass. Anisotropic effective-mass donor (Eq. (7.48), masses of Table 3.1), solved numerically, with line energies from the measured $1s(A_1)$ at 45.59 meV and the measured $2p_0$, $2p_\pm$, $3p_0$ levels. Solid: ground state with a model central cell tuned to 45.59 meV; dashed: effective-mass envelope. The lines $1s(A_1) \to np_0$, $np_\pm$ ($n = 2$ to 4) are drawn with a display width of 0.1 meV; measured widths are tens of times smaller. The continuum and the higher Rydberg lines are smoothed over 1.5 meV, comparable to the level spacing of the finite box used for the continuum. Dotted: the ionization continuum of Du et al. [23, Eq. (12)] at $q = 0$, with their parameters and their factor $(E_{\mathrm{eff}}/E_0)^2 = 4.84$. It is drawn at the same n_{D} and $\varepsilon_{\mathrm{host}}$.

absorption [23, Sect. VII.2]. In effective-mass theory that ratio is Eq. (16.36), expected close to one but not computed. The factor 4.84 instead compensates the parameters: m^* and a^* together imply a hydrogenic binding $1/2m^*a^{*2} = 85.7\,\mathrm{meV}$, not 45 meV, and the resulting continuum carries 0.30 of the f-sum of an electron of mass m^*, against 0.435 for a consistent hydrogenic donor. After the factor, the published continuum lies 1.13 to 1.48 times above the central-cell model between 50 and 70 meV (Fig. 16.13). Integrated over all energies it carries an oscillator strength of 4.85, more than the 3.85 that the f-sum allows the whole donor, lines included, because it falls more slowly at high energy.

Limit Check 16.8. Dopant absorption

The absorption rate of dark photons on dilute donors is $\mathcal{R} = \zeta^2(\rho_\chi/\rho_T)\,L^{\mathrm{D}}(m_\phi)$, with $L^{\mathrm{D}} = \varepsilon_2^{\mathrm{D}}/\varepsilon_{\mathrm{host}}^2$ the donor loss function of Eq. (16.39) [23, Eqs. (10), (12) and the absorption rate after Eq. (12)].

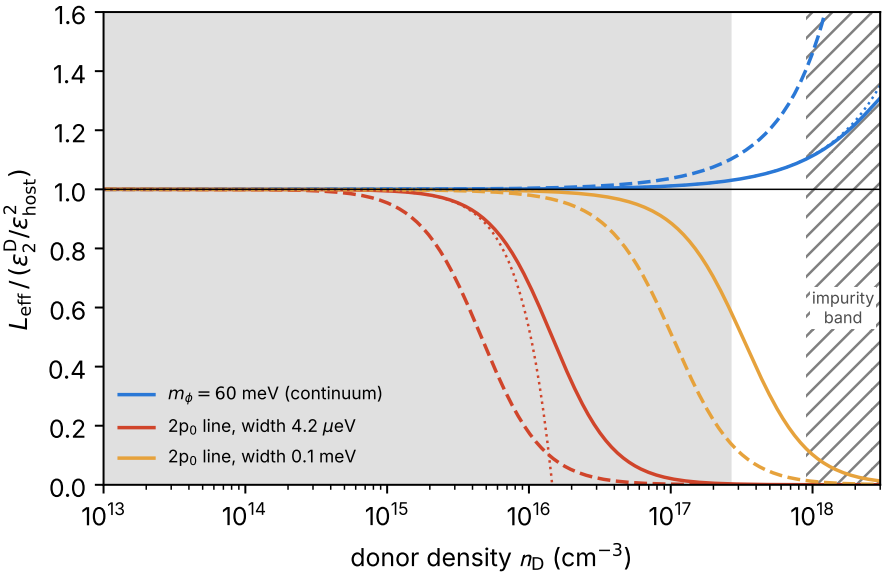


Figure 16.14: Loss function of the effective medium of donors in silicon, $\mathrm{Im}(-1/\varepsilon_{\mathrm{eff}})$ with $\varepsilon_{\mathrm{eff}} = \varepsilon_{\mathrm{host}}(1+2y)/(1-y)$ and $y = \varepsilon^{\mathrm{D}}/3\varepsilon_{\mathrm{host}}$, divided by the published first-order form $\varepsilon_2^{\mathrm{D}}/\varepsilon_{\mathrm{host}}^2$, against the donor density. Donor model of Fig. 16.13, central-cell ground state (solid) and effective-mass envelope (dashed). Blue: $m_\phi = 60\,\mathrm{meV}$, in the continuum. Red and orange: the $1s(A_1) \to 2p_0$ line center, 34.11 meV, for full widths of 4.2 µeV (narrowest measured) and 0.1 meV. Dotted: the expansion $1 - 4\,\mathrm{Re}\,y + 4[3(\mathrm{Re}\,y)^2 - (\mathrm{Im}\,y)^2]$. Gray: the continuum ratio within 10 % of one for both ground states, $n_{\mathrm{D}} < 2.6 \times 10^{17}\,\mathrm{cm}^{-3}$. Hatched: $n_{\mathrm{D}} > 0.3\,n_{\mathrm{c}}$, with $n_{\mathrm{c}} \approx 3 \times 10^{18}\,\mathrm{cm}^{-3}$, where the donors form an impurity band.

Outside the gray region the donors screen the dark photon themselves, and at the narrow line center the first-order form fails already at $n_{\mathrm{D}} = 4.8 \times 10^{15}\,\mathrm{cm}^{-3}$ with the central cell, where $\varepsilon_2^{\mathrm{D}}$ reaches half of $\varepsilon_{\mathrm{host}}$.

The effective medium is the dilute Clausius–Mossotti form, the Maxwell Garnett function of Sect. 16.3 with each donor a polarizable point of polarizability $\alpha_{\mathrm{pol}}(\omega)$ and $\varepsilon^{\mathrm{D}} = n_{\mathrm{D}}\alpha_{\mathrm{pol}}$. Its expansion, $L_{\mathrm{eff}} = (\varepsilon_2^{\mathrm{D}}/\varepsilon_{\mathrm{host}}^2)[1 - 4\,\mathrm{Re}\,y + O(y^2)]$, shows the two ways the first-order form fails. In the continuum above the lines the donors push $\mathrm{Re}\,\varepsilon^{\mathrm{D}}$ negative, lower $|\varepsilon_{\mathrm{eff}}|$ and raise the absorption, by 10 % at $8.8 \times 10^{17}\,\mathrm{cm}^{-3}$ with the central cell and at $2.6 \times 10^{17}\,\mathrm{cm}^{-3}$ with

the effective-mass envelope. At the center of a line ε^{D} is nearly imaginary, and the ratio becomes $1/[1+(4/9)(\varepsilon_2^{\mathrm{D}}/\varepsilon_{\mathrm{host}})^2]$: the absorbed field is screened by the donors as the plasmon screens it in Sect. 10.7. A narrow line saturates at a density proportional to its width.

Worked example 16.5: The $2p_0$ line of Si:P for a dark photon at resonance

Take $n_{\mathrm{D}} = 10^{15}\,\mathrm{cm}^{-3}$ and a dark photon of mass $m_\phi = 34.11\,\mathrm{meV}$, the binding of $1s(A_1)$, 45.59 meV, minus that of $2p_0$, 11.48 meV. With the central-cell line strength $|Z|^2/3 = 13.6\,\text{Å}^2$, the line integral of Eq. (16.39) is

$$\int \varepsilon_2^{\mathrm{D}}\,\mathrm{d}\omega = 4\pi^2\alpha\, n_{\mathrm{D}}\,\frac{|Z|^2}{3} = 7.73\times10^{-6}\,\mathrm{eV}.$$

The narrowest measured full width of the line, in isotopically purified silicon, is $\Gamma_{\mathrm{line}} = 0.034\,\mathrm{cm}^{-1} = 4.22\,\mu\mathrm{eV}$, close to the lifetime width $1/T_1 = 3.2\,\mu\mathrm{eV}$ of the $2p_0$ state, which decays by phonon emission in $T_1 = 205\,\mathrm{ps}$ [22, p. 5]. For a Lorentzian line the peak is $2/\pi\Gamma_{\mathrm{line}}$ times the integral, $\varepsilon_2^{\mathrm{D}} = 1.17$, and

$$L^{\mathrm{D}} = \frac{1.17}{11.4^2} = 8.99\times10^{-3},$$

which the effective medium lowers by half a percent, to 8.94×10^{-3}. The dark photon's own spread in energy, $m_\phi v^2/2 \approx 0.017\,\mu\mathrm{eV}$, lies well inside the line. With $\rho_\chi/\rho_T = 0.3\,\mathrm{GeV\,cm}^{-3}/2.33\,\mathrm{g\,cm}^{-3}$, the rate is

$$\mathcal{R} = \zeta^2\,\frac{\rho_\chi}{\rho_T}\,L^{\mathrm{D}} = 5.5\times10^{31}\,\zeta^2\ \text{per kg-year},$$

and three events per kg-year need $\zeta = 2.3\times10^{-16}$. In the continuum at 50 meV the same crystal has $L^{\mathrm{D}} = 2.1\times10^{-5}$ and needs $\zeta = 4.9\times10^{-15}$, twenty times more. The effective-mass envelope would give $\zeta = 1.3\times10^{-16}$ at the line. The gain holds over a window of masses as wide as the line, $\Gamma_{\mathrm{line}}/m_\phi = 1.2\times10^{-4}$, and the peak, proportional to $n_{\mathrm{D}}/\Gamma_{\mathrm{line}}$, grows with the donor density only while the line keeps its width.

16.5.3 The impurity band

Uncompensated Si:P becomes a metal at low temperature above a donor density $n_{\mathrm{c}} \approx 3\times10^{18}\,\mathrm{cm}^{-3}$, as quoted by Du et al. [23, Sect. II]. This is the *Mott critical density*, the donor density of the metal–insulator transition named in Sect. 7.4. Their projections use $n_{\mathrm{D}} = 10^{18}\,\mathrm{cm}^{-3}$, and their comparison with optical data $1.8\times10^{18}\,\mathrm{cm}^{-3}$, so $n_{\mathrm{D}}/n_{\mathrm{c}} \approx 0.3$ to 0.6. At these densities the donors are not isolated. Donors sit at random positions, and the probability that a given donor has a neighbor within a distance d is $1-\exp(-4\pi n_{\mathrm{D}} d^3/3)$. For d equal to the orbit diameter, $2r_\perp = 4.86\,\mathrm{nm}$, this is 4.8×10^{-4} at $10^{15}\,\mathrm{cm}^{-3}$, but 0.38 at $10^{18}\,\mathrm{cm}^{-3}$ and 0.58 at $1.8\times10^{18}\,\mathrm{cm}^{-3}$. The electron of such a pair hops between the two cores. The hopping splits the $1s$ level of the pair into bonding and antibonding states, shifts its lines, and over a random ensemble spreads the discrete spectrum into a band of states, the *impurity band*. Optical spectra at these densities show it: the lines broaden and the gap below the first line closes [23, Sect. VII.6].

The effective medium of Limit Check 16.8 shows the same failure from the other side. The static polarizability of an isolated donor, from the numerical solution, is $\alpha_{\mathrm{pol}}(0)/\varepsilon_{\mathrm{host}} = 4\pi\times3.8\times10^{-20}\,\mathrm{cm}^3$ with the effective-mass envelope and $4\pi\times1.1\times10^{-20}\,\mathrm{cm}^3$ with the

central cell. At $10^{18}\,\mathrm{cm}^{-3}$ it raises the static dielectric function of the crystal by a factor of 1.15 to 1.57, and at $1.8\times10^{18}\,\mathrm{cm}^{-3}$ by 1.28 to 2.21. The Clausius–Mossotti form diverges at $n_\mathrm{D}\alpha_\mathrm{pol}(0)=3\varepsilon_\mathrm{host}$, at $2.1\times10^{19}\,\mathrm{cm}^{-3}$ with the central cell and at $6.3\times10^{18}\,\mathrm{cm}^{-3}$ with the effective-mass envelope, both above the measured n_c. The transition, however, comes first from pairs and clusters, whose polarizability and overlap exceed those of isolated donors, and the response of isolated donors, in a dilute medium or an effective one, is wrong at order one near the ionization edge for $n_\mathrm{D}/n_\mathrm{c}\approx0.5$.

What replaces it is the disorder average $\langle S^\mathrm{D}(\boldsymbol{q},\omega)\rangle$ of a random ensemble of donors, each with its six-valley orbitals and central cell. The hopping between two donors oscillates with their separation, because the valley phases $\mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{r}}$ interfere, so a single-valley envelope gets it wrong. Appendix E.21 specifies this average as a tight-binding problem on 10^4 to 10^5 random sites with hopping from the overlaps of the multivalley envelopes, checked against density-functional clusters of a few phosphorus atoms for the central cell. The optical data that Du et al. use at $1.8\times10^{18}\,\mathrm{cm}^{-3}$ contain the impurity band at $q=0$; how it reshapes S^D at the momenta of scattering, and the absorption near the edge, is the open problem of this section.

16.5.4 Donors as final states

A donor excitation is useful when the deposit could have made nothing else, when the detector can count it before it is lost, and when the donors are dilute. The first condition is kinematic. Between 13 keV and the 0.333 MeV of the silicon gap (Sect. 1.3), the donor is the only electronic excitation of the crystal that dark matter can reach. Above that it adds a valley structure that the host lacks. The $A_1\to E$ line of Eq. (16.38) makes the rate depend on the direction of $\boldsymbol{q}$ relative to the crystal axes, and the daily rotation of the laboratory turns this into a modulation.

The second condition depends on the final state. An ionized donor leaves an electron in the conduction band, which a field can collect and a charge-coupled device can count one at a time; this is the readout that Du et al. propose [23, Sect. VII.1]. The donors then must stay neutral without the dark matter. Thermal ionization grows as $\mathrm{e}^{-E_\mathrm{b}^\mathrm{D}/T}$, and from the measured capture coefficient of electrons on ionized phosphorus Du et al. estimate that at $10^{18}\,\mathrm{cm}^{-3}$ a temperature of about 5 K keeps it below one electron per kg-year. The bias that collects the electron also drives hopping conduction through the impurity band, which a layer of undoped silicon between the doped volume and the readout blocks, as in the blocked-impurity-band infrared detectors [23, Sect. VII.1]. A bound excitation, such as $2p_0$, has no charge to collect. It returns to the ground state by emitting phonons, within 205 ps for $2p_0$ [22, p. 5], and the energy is then a few phonons of at most 34 meV. In a doped crystal phonons above 11.70 meV can drive the $1s(A_1)\to1s(T_2)$ and $1s(E)$ transitions of other donors, and only those below are expected to travel ballistically to a sensor [23, Sect. VII.6]. A bound excitation therefore needs a sensor of single phonons of about 10 meV, which no present detector provides, while ionization needs a single-electron counter, which exists. Both need the third condition, $n_\mathrm{D}\ll n_\mathrm{c}$, with the bound set by the impurity band of Sect. 16.5.3.

16.6 Color centers and broken bonds

A fluorine atom knocked off its site in lithium fluoride leaves an anion vacancy, and the vacancy binds an electron: an F center, which absorbs light near 5.1 eV [25]. A dark-matter particle of 0.5 GeV can give a fluorine nucleus at most 188 eV (Eq. (1.21)), a few times the tens of eV that a displacement costs. A molecule struck on one of its atoms comes apart in the same way when the kick exceeds its bond energy, 4.75 eV for H_2. Both final states persist,

both sit in a crystal transparent to the light that reads them, and both can be counted singly. Budnik, Cheshnovsky, Slone and Volansky proposed color centers as a target for sub-GeV dark matter [26], and Essig, Mardon, Slone and Volansky the breaking of chemical bonds [27]. We will write both rates as rates into final-state classes (Sect. 8.7) multiplied by the probability that the class ends as a readable defect, recover the two published formulas as limits, and say which of the probabilities have been measured.

16.6.1 The color-center creation rate

A color center forms after the prompt window of Sect. 8.7. Within that window, a femtosecond after the kick, the struck nucleus moves with the momentum it received and its neighbors have not yet responded; the class Λ is a recoil of species κ with energy deposit ω along $\hat{\boldsymbol{q}}$. The defect forms during the collision cascade that follows, within picoseconds, and its fate is decided later still, when the close pair recombines or does not and when the vacancy captures an electron or does not. The rate of creating readable centers is therefore the hand-off formula Eq. (8.69) with the cascade distribution $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ replaced by the probability $Y_{\mathrm{CC}}(\Lambda)$ that a seed of class Λ ends as a center that the readout registers. How a recoil seeds the cascade is made precise in Sect. 18.3, and the energy that defects store in Sect. 19.4. The conditions of Sect. 8.7 apply directly: the yield depends on the direction of the recoil relative to the crystal axes, so $\hat{\boldsymbol{q}}$ must stay inside the class sum, and a rate averaged over directions before the yield is applied is wrong near threshold; Eq. (16.41) below is the orientation-averaged form, exact only for recoil directions isotropic in the crystal frame.

The yield is a product of four probabilities, each fixed at a different stage:

$$Y_{\mathrm{CC}}(E_R, \hat{\boldsymbol{r}}_R) = p_{\mathrm{FP}}(E_R, \hat{\boldsymbol{r}}_R)\, p_{\mathrm{surv}}\, p_q\, f_{\mathrm{opt}}. \tag{16.40}$$

The first, p_{FP}, is the probability that a nucleus set moving with kinetic energy E_R along $\hat{\boldsymbol{r}}_R$ leaves a Frenkel pair once the collision has died out. Its lowest crossing is the displacement threshold $E_{\mathrm{d}}(\hat{\boldsymbol{r}}_R)$ of Eq. (7.54), and molecular dynamics computes it direction by direction (Appendix E.11). For an F center the displaced atom must be an anion, either struck directly or hit by a struck cation. The second, p_{surv}, is the probability that the close pair survives recombination over the time between the event and the readout. At millikelvin temperatures thermal hopping is frozen (Sect. 7.4), and only tunneling recombination and the phonons of later events act. The third, p_q, is the probability that the vacancy ends in the charge state of the readable center: an F center is a vacancy with one electron, while an empty anion vacancy absorbs elsewhere. The fourth, f_{opt}, is the probability that the readout registers a center that is present. None of them enters the response; they are properties of the final state, and the truncation label of Sect. 8.5 applies only to the seed.

The seed is an impulsive recoil. Displacement needs E_R of 10 to 60 eV, three orders of magnitude above the phonon energies, so the recoil parameter $x_R = E_R/\langle\omega\rangle$ is several hundred and the nuclear response is the Gaussian Eq. (11.34) of Sect. 11.4. The momentum transfer, near 1 MeV for a fluorine recoil of 27 eV, is far above the zone scale, where the coherence corrections of the multiphonon response are already below 10 % at 18 keV in silicon and keep falling (Sect. 11.3). Only the self response of each species remains. For a heavy mediator we insert that response and the yield into Eq. (8.69), average over orientations as in Sect. 11.4, and count one nucleus of species κ per cell of mass M_{cell}. The rate per unit target mass is

$$\begin{aligned} \mathcal{R}_{\mathrm{CC}} = \sum_{\kappa} \frac{M_\kappa}{M_{\mathrm{cell}}} \frac{\rho_\chi\, \sigma_{N,\kappa}}{2 m_\chi\, \mu_{\chi\kappa}^2} \int \mathrm{d}E_R \int \mathrm{d}\omega\, \frac{\mathrm{e}^{-(\omega - E_R)^2/2\sigma_\omega^2}}{(2\pi\sigma_\omega^2)^{1/2}} \\ \times\, \eta\Big(\frac{\omega}{q} + \frac{q}{2m_\chi}\Big)\, \big\langle Y_{\mathrm{CC},\kappa}(\omega, \hat{\boldsymbol{r}}_R)\big\rangle_{\hat{\boldsymbol{r}}_R}, \end{aligned} \tag{16.41}$$

with $q = (2M_\kappa E_R)^{1/2}$, $\sigma_\omega^2 = E_R\langle\omega\rangle_\kappa$ at $T = 0$, $\langle\omega\rangle_\kappa$ the mean phonon energy of the site, $\sigma_{N,\kappa}$ the nuclear cross-section of Eq. (2.27) with $F_N = 1$, and $\mu_{\chi\kappa}$ the reduced mass with nucleus κ. In the impulse response the deposit ω equals the change of kinetic energy of the nucleus, so the yield is evaluated at ω; the initial kinetic energy, $3\langle\omega\rangle_\kappa/4$, is a few hundredths of an eV and is dropped. The direction average of Y is exact only when the recoil directions are isotropic in the crystal frame. For a crystal fixed to the rotating Earth they are not, and the full form keeps $\hat{\boldsymbol{r}}_R = \hat{\boldsymbol{q}}$ inside the velocity integral, which is the origin of the daily modulation of the rate [26, Eq. (10)].

Eq. (16.41) reduces to the published rate in two steps. As $\sigma_\omega \to 0$ the Gaussian becomes $\delta(\omega - E_R)$ and the minimum velocity becomes $q/2\mu_{\chi\kappa}$, which is the free recoil Eq. (11.35) per species. Setting then $p_{\mathrm{surv}} = p_q = f_{\mathrm{opt}} = 1$ and $p_{\mathrm{FP}} = \Theta(E_R - E_{\mathrm{d}}(\hat{\boldsymbol{r}}_R))$ gives the rate of Budnik et al. [26, Eqs. (1), (5), (7)]. They write it as an integral over q^2 with the step $\Theta(q - q_{\min}(\hat{\boldsymbol{q}}))$, $q_{\min}^2 = 2M_\kappa E_{\mathrm{d}}$, and with a per-nucleon cross-section and the coherence factor $[f_P Z + f_N(A - Z)]^2$. For equal couplings to protons and neutrons that factor is A^2, and their per-nucleon cross-section, the σ_p of Eq. (2.27), times $A^2/\mu_{\chi p}^2$ is $\sigma_{N,\kappa}/\mu_{\chi\kappa}^2$. For the threshold surface they take $q_{\min}(\hat{\boldsymbol{q}}) = (2M_\kappa \Delta E_B)^{1/2}[1 + a\sin(4\theta)\cos(4\varphi)]$ in crystal axes, with a mean binding energy ΔE_B and $a = 0.4$ [26, Eq. (2) and Table I]. Their formula counts a dislocated nucleus, not a particular defect, and assumes that the vacancy fills with an electron at once.

The Doppler width is the first correction, and it is small. For a fluorine nucleus in LiF the optical phonons lie between 33.5 and 77.2 meV [28, Table I], and we take these two values as the range of $\langle\omega\rangle_{\mathrm{F}}$; at $E_R = 27$ eV the width is 0.95 to 1.44 eV, 3.5 to 5.4 % of the threshold. The zero-point momentum of the nucleus along $\hat{\boldsymbol{q}}$, $(M_{\mathrm{F}}\langle\omega\rangle_{\mathrm{F}}/2)^{1/2}$, is 17 to 26 keV, or 4.6 to 7.0 times αm_e. Budnik et al. estimated the momentum spread of the initial state as αm_e and required it to be small against $q_{\min}$ [26, Eq. (6)]. The nucleus moves several times faster than that estimate, but $q_{\min} = 978$ keV, so the ratio is 0.018 to 0.027 and the classical treatment stands. Limit Check 16.9 shows where the width matters: only where the largest recoil energy is less than 1.22 to 1.36 times the threshold, for site phonon energies of 33.5 to 77.2 meV.

Electronic deposits make centers too. In alkali halides an exciton localizes on the lattice as a self-trapped exciton, which can convert within picoseconds into an F center and an interstitial halogen, the H center, and so store its energy as a Frenkel pair (Appendix E.21). The yield of an electronic class is then $\langle N_{\mathrm{X}}\rangle(\omega)\, p_{\mathrm{FH}}\, p_{\mathrm{surv}}\, f_{\mathrm{opt}}$, with $\langle N_{\mathrm{X}}\rangle$ the number of excitons a deposit ω makes and p_{FH} the probability that a self-trapped exciton becomes a pair rather than decaying. The seed is the electronic response of Chap. 10, and the route opens at the energy of the free exciton, which lies below the band gap of LiF, 14.2 eV [25]. A particle needs $m_\chi \geq 2 \times 14.2\,\mathrm{eV}/v_{\max}^2 = 4.0$ MeV to deposit the gap, an upper bound on the threshold of this route and at least 46 times less than the 0.186 GeV that a fluorine displacement at 27 eV requires. For a search the electronic route is a background as much as a channel, since a γ ray that deposits 100 keV has the energy for thousands of excitons. Araujo et al. find the response of LiF to ^{60}Co γ rays a factor 50 below that to fast neutrons at equal dose [29, Sect. III].

The threshold surface and p_{FP} come from recoil molecular dynamics on machine-learned potentials (Appendix E.11). The levels and optical cross-sections of the centers come from excited-state calculations in supercells, by ΔSCF, the difference of the self-consistent-field (SCF) total energies of two electronic configurations, with hybrid functionals (Appendix E.10). The conversion of a self-trapped exciton and the survival of the close pair at millikelvin need surface hopping and the tunneling paths of Appendix E.22; Appendix E.21 specifies these for LiF, NaF, MgO and LiH.

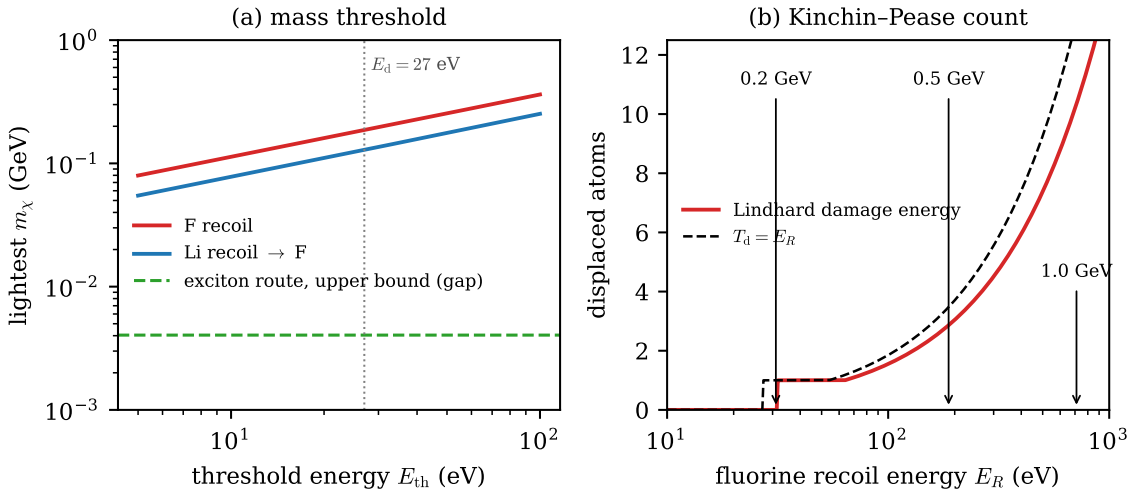


Figure 16.15: F-center creation in LiF. (a) Lightest dark-matter mass that can create an F center, against the threshold energy of a step: a fluorine recoil above the threshold (red), a lithium recoil that passes the threshold energy to a fluorine neighbor in one head-on collision, transfer fraction 0.788 (blue), and the mass that deposits the band gap, 14.2 eV, 4.0 MeV, an upper bound on the threshold of the exciton route (green, dashed); Eq. (1.21) with $v_{max} = 2.650 \times 10^{-3}$. Dotted: $E_d = 27$ eV, the threshold used in binary-collision simulations of LiF. (b) Atoms displaced by a fluorine recoil in the Kinchin–Pease count with $E_d = 27$ eV, with the damage energy of the Lindhard partition for $Z = 9$, $A = 19$ (red; an extrapolation below about 1 keV) and with no electronic loss (black, dashed). Arrows: the largest fluorine recoil energy for $m_\chi = 0.2$, 0.5 and 1.0 GeV, 31, 188 and 711 eV.

16.6.2 F centers in lithium fluoride

LiF is the material in which color-center readout of nuclear recoils has been demonstrated. Its F center has a nondegenerate ground state and a threefold excited state, with absorption measured near 5.1 eV and computed at 4.85 to 4.90 eV, and its luminescence has never been observed [25]. What is read in practice are aggregates of vacancies: the neutral pair F_2^0, emitting at 650 nm, and the charged triple F_3^+, emitting at 525 nm, both excited at 450 nm [29, Sect. I]. They form when single vacancies diffuse together, which takes hours at room temperature: after a γ exposure the two emissions grow with time constants of 9 h (F_2^0) and 6.5 h (F_3^+) [29, Sect. III]. Along the dense track of a fast-neutron recoil, about 1200 fluorine vacancies per micrometer in simulation, the vacancies meet quickly, and Araujo et al. estimate one emitting aggregate for every two or three of them [29, Sect. II].

The displacement threshold of LiF has not been computed direction by direction. Cogswell, Goel and Huber run their binary-collision simulations of LiF with $E_d = 27$ eV and find that half the asymptotic vacancy efficiency is reached at a recoil energy of 80 eV [30, Table I]. Fig. 16.15 turns such thresholds into dark-matter masses. A fluorine recoil above 27 eV needs $m_\chi \geq 0.186$ GeV, and the requirement moves from 0.113 to 0.255 GeV as the threshold moves from 10 to 50 eV. A lithium nucleus is lighter and takes more energy from the same particle, but it makes an F center only by displacing a fluorine neighbor, and one collision passes on at most the fraction $4M_{Li}M_F/(M_{Li} + M_F)^2 = 0.788$ of its energy (for ^{7}Li); the requirement at 27 eV is then 0.129 GeV. Displacing the lithium itself, at 0.114 GeV, leaves a cation vacancy, which is not an F center.

The number of vacancies a recoil makes follows from two classic estimates. The Lindhard

partition splits a recoil of energy E_R into an electronic share $y_L E_R$ and a damage energy $T_\mathrm{d} = (1 - y_L)E_R$, with

$$\begin{aligned} y_L &= \frac{k_L\, g_L(\epsilon_L)}{1 + k_L\, g_L(\epsilon_L)}, \qquad & k_L &= 0.133\, Z^{2/3} A^{-1/2}, \\ g_L(\epsilon_L) &= 3\epsilon_L^{0.15} + 0.7\epsilon_L^{0.6} + \epsilon_L, \qquad & \epsilon_L &= 11.5\, \frac{E_R}{\mathrm{keV}}\, Z^{-7/3}, \end{aligned} \tag{16.42}$$

in the simplified form for a target of one element [31, Eq. (3)]; Sect. 19.3 derives it. The Kinchin–Pease count then gives no vacancy for $T_\mathrm{d} < E_\mathrm{d}$, one for $E_\mathrm{d} \le T_\mathrm{d} < 2E_\mathrm{d}$, and $T_\mathrm{d}/2E_\mathrm{d}$ above, as quoted by Cogswell et al. [30, Sect. II]. Both estimates are rough at the energies of sub-GeV recoils. The Lindhard form already misses the measured ionization of silicon recoils of 100 eV (Appendix E.23), and its use for a fluorine recoil in a compound is a model choice. The Kinchin–Pease count ignores the direction dependence of E_d and recombination. They set the scale, and Worked example 16.6 uses them for that.

Worked example 16.6: The yield of F centers per keV of fluorine recoil

Take a fluorine recoil of 1 keV in LiF. With $Z = 9$ and $A = 19$, Eq. (16.42) gives $k_L = 0.132$, $\epsilon_L = 0.0683$, $g_L = 2.21$ and $y_L = 0.226$: the recoil gives 226 eV to the electrons and leaves a damage energy of 774 eV. With $E_\mathrm{d} = 27$ eV the Kinchin–Pease count is $774/54 = 14.3$ displaced atoms, and 18.5 if no energy went to the electrons. The count does not say how the displacements divide between the two sublattices. All of them on the anion sublattice gives 14.3 fluorine vacancies per keV, an upper value; an equal share gives 7.2.

A fluorine vacancy becomes an F center only with an electron. The electronic share of 226 eV makes at most $226/14.2 = 16$ electron–hole pairs, since each costs at least the gap, so at 1 keV the electrons and the vacancies are comparable in number, before trapping and recombination take their share.

At the energies that sub-GeV dark matter reaches the balance changes. A particle of 0.5 GeV gives at most 188 eV, a damage energy of 155 eV and 2.9 displacements, but at most 2.3 pairs. At 0.3 GeV the largest recoil is 69 eV, about one displacement and at most 0.7 pairs; at 0.2 GeV, 31 eV, which reaches $T_\mathrm{d} = E_\mathrm{d}$ only without electronic loss. Near threshold a recoil therefore makes one vacancy and less than one free electron, and the charge factor p_q in Eq. (16.40) decides how many vacancies become F centers. The per-keV yield of the first paragraph, 7 to 14 fluorine vacancies, is an upper estimate that the four factors of Eq. (16.40) reduce by amounts that no measurement fixes. And since vacancies at millikelvin temperatures do not move, these few centers stay single and must be read one by one.

16.6.3 Bond breaking and cage escape

Take a diatomic molecule of atoms with masses M_1 and M_2 in a molecular solid, and let the dark matter strike atom 1. With the center of mass $\boldsymbol{R}_\mathrm{CM}$ and the bond vector $\boldsymbol{r} = \boldsymbol{R}_1 - \boldsymbol{R}_2$, the struck atom sits at

$$\boldsymbol{R}_1 = \boldsymbol{R}_\mathrm{CM} + \frac{M_2}{M_\mathrm{mol}}\,\boldsymbol{r} = \boldsymbol{R}_\mathrm{CM} + \frac{\mu_{12}}{M_1}\,\boldsymbol{r},$$

with $M_\mathrm{mol} = M_1 + M_2$ and $\mu_{12} = M_1 M_2 / M_\mathrm{mol}$ the reduced mass of the bond. Assume that the forces of the neighbors act on the center of mass only, and depend neither on the length of the bond nor on its orientation. The Hamiltonian then splits into $H_\mathrm{CM} + H_\mathrm{int}$, the operators of the

two parts commute, and so do their Heisenberg evolutions. The self correlation of the struck atom, whose Fourier transform is its response (Eq. (D.188)), factorizes, with $\boldsymbol{q}_{12} = (\mu_{12}/M_1)\boldsymbol{q}$ the kick on the bond vector,

$$\left\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_1(t)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_1} \right\rangle = \left\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_{\mathrm{CM}}(t)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_{\mathrm{CM}}} \right\rangle \left\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}_{12}\cdot\boldsymbol{r}(t)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}_{12}\cdot\boldsymbol{r}} \right\rangle,$$

and a product in time is a convolution in frequency:

Response of an atom in a molecule

$$\begin{aligned} S_{11}(\boldsymbol{q},\omega) &= \int \frac{\mathrm{d}\omega'}{2\pi}\, S_{\mathrm{CM}}(\boldsymbol{q},\omega')\, S_{\mathrm{int}}(\boldsymbol{q}_{12},\omega-\omega'), \\ S_{\mathrm{int}}(\boldsymbol{q}_{12},\omega) &= 2\pi \sum_f \left|\langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}_{12}\cdot\boldsymbol{r}}|0\rangle\right|^2 \delta(\omega - E_f + E_0). \end{aligned} \tag{16.43}$$

S_{CM} is the lattice response of a rigid molecule of mass M_{mol}, and S_{int} the response of the bond to the kick $\boldsymbol{q}_{12}$ on its internal coordinate. The part of S_{int} with final states above the dissociation limit, $E_f - E_0 > E_{\mathrm{bond}}$, is the response for breaking the bond.

The hypothesis holds best for molecules that rotate freely in the solid, with an internal motion that does not feel the cage. It fails for the fragments of a broken bond, which move through the cage and collide with its walls; that failure is what the escape probability below absorbs.

Both factors are impulsive at the momenta that break bonds. A classical bond breaks when the kinetic energy $q_{12}^2/2\mu_{12}$ given to the internal motion exceeds E_{bond}, which requires

$$q \geq q_{\mathrm{min}} = \frac{M_1}{\mu_{12}}\,(2\mu_{12}E_{\mathrm{bond}})^{1/2}, \tag{16.44}$$

the threshold of Essig et al. [27, Eq. (3.3)]. For H_2, with their $E_{\mathrm{bond}} = 4.75\,\mathrm{eV}$, $q_{\mathrm{min}} = 134\,\mathrm{keV}$, and a particle needs $m_\chi \geq 25.9\,\mathrm{MeV}$ to transfer it. The bond vibrates at $\omega_{\mathrm{vib}} = 0.546\,\mathrm{eV}$ [32], so the internal recoil parameter $E_{\mathrm{bond}}/\omega_{\mathrm{vib}}$ is 8.7, and the center-of-mass recoil, a few eV against intermolecular phonons of meV, is deep in the impulse regime of Sect. 11.4. In the impulse approximation each factor shifts the deposit by a Doppler term. The center of mass shifts it by $\boldsymbol{q}\cdot\boldsymbol{P}_{\mathrm{CM}}/M_{\mathrm{mol}}$, which changes the minimum velocity $\omega/q + q/2m_\chi$ by $(\langle\omega\rangle_{\mathrm{CM}}/2M_{\mathrm{mol}})^{1/2}$ per standard deviation, 2×10^{-6} for H_2 with an assumed $\langle\omega\rangle_{\mathrm{CM}} = 10\,\mathrm{meV}$, against $v_{\mathrm{max}} = 2.650\times10^{-3}$. It does not decide whether the bond breaks and plays no further role. The internal factor shifts the energy given to the bond by $\boldsymbol{q}_{12}\cdot\boldsymbol{p}_{12}/\mu_{12}$, with $\boldsymbol{p}_{12}$ the relative momentum in the ground state, and that shift decides the outcome. In the impulse approximation the bond breaks when

$$\frac{q_{12}^2}{2\mu_{12}} + \frac{\boldsymbol{q}_{12}\cdot\boldsymbol{p}_{12}}{\mu_{12}} > E_{\mathrm{bond}}. \tag{16.45}$$

Averaging over the ground state gives the dissociation probability of a kick $\boldsymbol{q}$. Essig et al. take the ground state of a free rotor, a Gaussian in the bond length $|\boldsymbol{r}|$ about its equilibrium value, isotropic in direction [27, Eq. (3.12)]. With the harmonic width $(\mu_{12}\omega_{\mathrm{vib}})^{-1/2}$ the projection of $\boldsymbol{p}_{12}$ on $\hat{\boldsymbol{q}}$ has the root-mean-square value $(\mu_{12}\omega_{\mathrm{vib}}/6)^{1/2}$, 6.53 keV or 1.75 times αm_e for H_2.

The published rates are limits of Eqs. (16.43) and (16.45). With S_{CM} the free recoil of the molecule and $\boldsymbol{p}_{12}\to 0$, the deposit is $q^2/2M_{\mathrm{mol}} + \mu_{12}q^2/2M_1^2 = q^2/2M_1$, the minimum velocity is $q/2\mu_{\chi 1}$, and the rate is the free recoil Eq. (11.35) of atom 1 cut at q_{min}, the classical rate of Essig et al. [27, Eq. (4.3)]. Their quantum rate keeps the spread of the ground state, with plane-wave final states whose internal momentum is set by the energy above the dissociation limit, the improved Born approximation [27, Eqs. (3.11), (3.17), (4.5)]. In that approximation

the bond breaks when $|\boldsymbol{q}_{12} + \boldsymbol{p}_{12}|^2/2\mu_{12} > E_{\rm bond}$. This differs from Eq. (16.45) by the initial kinetic energy of the bond, $\langle p_{12}^2\rangle/2\mu_{12} = \omega_{\rm vib}/4 = 0.136\,\text{eV}$ for H_2: the Born form counts the final kinetic energy of the fragments from zero, the impulse form counts the energy the kick adds. Their estimate of the quantum shift of the mass threshold is $1 - \Delta p/(2\mu_{12}E_{\rm bond})^{1/2}$ with Δp the momentum spread [27, Eq. (3.8)], 0.90 for H_2 with the spread above.

Limit Check 16.9. Color-center creation and bond breaking

The published color-center rate is the free nuclear recoil Eq. (11.35) cut by a step at the displacement threshold, $\Theta(E_R - E_{\rm d})$ [26, Eqs. (1), (2), (5), (7)]. The published classical bond-breaking rate is the free recoil of the struck atom cut at $q_{\rm min}$ of Eq. (16.44) [27, Eqs. (3.3), (4.3)].

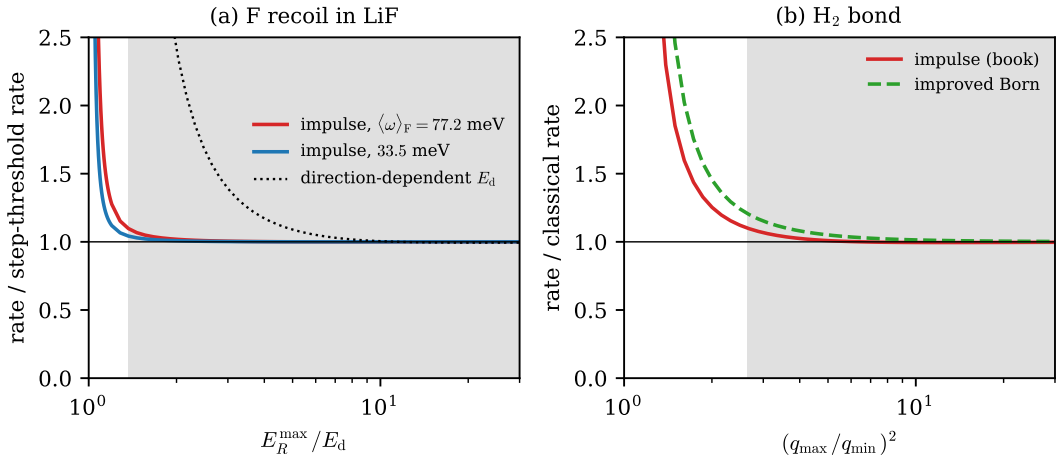


Figure 16.16: Rates from the impulse response of the bound target divided by the published forms, heavy mediator, standard halo, $T = 0$. (a) Fluorine recoils in LiF with $E_{\rm d} = 27\,\text{eV}$, against the ratio of the largest recoil energy to the threshold: Eq. (16.41) with the yield a step at $\omega = E_{\rm d}$, for $\langle\omega\rangle_{\rm F} = 77.2\,\text{meV}$ (red, the top of the LiF optical band, an upper value) and 33.5 meV (blue). Dotted: the direction-dependent threshold $E_{\rm d}(1 + a\sin 4\theta\cos 4\varphi)^2$ with $a = 0.4$, recoil directions isotropic in the crystal, divided by the step at $E_{\rm d}$. (b) H_2 as a free rotor: Eq. (16.43) with Eq. (16.45) over the Gaussian ground state (red) and with the improved Born condition (green, dashed), against $(q_{\rm max}/q_{\rm min})^2$, $q_{\rm max} = 2\mu_{\chi 1}v_{\rm max}$. Gray: the red curve within 10 % of one, $E_R^{\rm max} \geq 1.36\,E_{\rm d}$ ($m_\chi \geq 0.218\,\text{GeV}$) in (a) and $(q_{\rm max}/q_{\rm min})^2 \geq 2.65$ ($m_\chi \geq 42.9\,\text{MeV}$) in (b).

Closer to threshold both rates depend on the zero-point motion of the struck nucleus or bond and stay nonzero below the classical endpoint, and at every mass they still lack the yield factors of Eq. (16.40) and the escape probability of the fragments of a bond, which multiply both curves.

Limit Check 16.9 sizes the two zero-point corrections. For LiF the Doppler width changes the rate by less than 2 % once the largest recoil energy exceeds twice the threshold, and at the rate that the step form gives for $E_R^{\rm max} = 1.1\,E_{\rm d}$ it lowers the mass by 1.4 %. The directional structure of the threshold matters more: with the angular model of Budnik et al. the rate stays more than 10 % away from the step value up to $E_R^{\rm max} = 4.8\,E_{\rm d}$, or $m_\chi = 0.41\,\text{GeV}$, because the threshold surface reaches down to $0.36\,E_{\rm d}$. For H_2 the impulse rate equals the classical rate of $m_\chi = 27.2\,\text{MeV}$ already at 22.8 MeV, a shift of the mass scale by a factor 0.84;

the improved Born rate does so at 20.3 MeV, a factor 0.75. Essig et al. estimate that quantum effects lower the mass threshold by about 10 % [27, Sect. 3, Eq. (3.8)], less than either, since their estimate shifts $q_{\min}$ by one standard deviation of the momentum only.

A broken bond is not yet a signal. In a solid the two fragments start inside the cage of their neighbors, and they recombine unless they escape it, the cage effect of photochemistry. The rate of separated fragments is the rate of dissociating seeds times the escape probability $p_{\mathrm{esc}}(E_{\mathrm{int}}, \hat{\boldsymbol{r}})$, which depends on the energy E_{int} the fragments share above the dissociation limit and on the direction of the bond relative to the cage. Electronic deposits break bonds as well: an excitation into a repulsive state, the antibonding σ^* surface of the bond, dissociates the molecule without any nuclear kick, at the electronic thresholds of the molecular crystals of Sect. 16.4. Neither p_{esc} nor the dissociation yields of these electronic states have been quantified at the momenta and temperatures of a dark-matter search. Appendix E.21 specifies their calculation by molecular dynamics, with the free-molecule rate of Essig et al. as the acceptance test.

16.6.4 Measured yields and single-center readout

Of the inputs to Eq. (16.40), the displacement threshold is the one with measurements, and these fix only its lowest point. Electron irradiation of silicon sees damage set in at 12.9 ± 0.6 eV, where density-functional molecular dynamics puts the minimum of $E_{\mathrm{d}}(\hat{\boldsymbol{r}})$ (Sect. 7.4). The shape of p_{FP} above the minimum is a computed quantity: the defect probability becomes nonzero below the average threshold and approaches one only at two to three times it [30, Sect. II]. In materials where an interstitial recombines with its own vacancy a direction-resolved threshold is not even well defined (Appendix E.11). For ionic crystals the measured responses are those of dense tracks at room temperature: fast neutrons, γ rays of ^{60}Co, and the α and triton tracks of neutron capture on ^{6}Li [29]. The aggregates that LiF forms along such tracks are stable for months in vacuum at room temperature and anneal away at 350 °C [29, Sect. III]. None of these measurements isolates a single recoil of tens to hundreds of eV, and none is made at millikelvin temperatures.

A single-center readout at millikelvin temperatures must work without aggregation. A sub-GeV recoil makes one to three vacancies (Worked example 16.6), and at 10 mK they do not move, so the emitting aggregates read in LiF tracks do not form and the readout must find a single F center. In LiF that center has no observed luminescence [25], so it must be found through its absorption near 5.1 eV, in the ultraviolet, or converted into a state that emits. Budnik et al. estimate the photon budget of an emitting center in a light-guiding rod. A photon flux of $10^{21}\,\mathrm{cm^{-2}\,s^{-1}}$ and an absorption cross-section of $10^{-17}\,\mathrm{cm^2}$ excite it 10^4 times per second; with unit emission efficiency, a collection efficiency of 0.1 and a detector efficiency of 0.5, a center yields 5×10^5 detected photons in 10^3 s [26, footnote 1]. The same estimate puts the detection limit at about one new center among 10^5 existing ones in a rod [26]. The existing centers set the background. Commercial crystals are not certified below a defect concentration of 10^{-9}, and annealing at up to 1000 °C removes much of it [26]; fluorescence measurements of LiF put an upper limit of 10^{-12} to 10^{-10} on the centers present before irradiation [30]. At millikelvin temperatures the readout light must also deposit less heat than the detector tolerates, which none of these estimates addresses.

An existing center can also be a target. The negatively charged nitrogen–vacancy center of diamond needs $m_\chi \geq 0.554$ MeV and a momentum transfer near 1.47 keV to reach its zero-phonon line at 1.945 eV (Sect. 7.4) [33], through a transition density between many-electron states that Appendix E.21 computes in an active space. The rate scales with the number of centers, a small fraction of the atoms of a crystal, which is why centers serve better as readouts than as targets.

Four yields remain open at millikelvin temperatures: the conversion p_{FH} of a self-trapped exciton into a Frenkel pair, the survival p_{surv} of a close pair with the charge factor p_q at recoil energies of tens to hundreds of eV, the efficiency f_{opt} of reading a single center, and the escape p_{esc} of the fragments of a bond with the yields of repulsive electronic states. Each can be measured directly. Neutrons scattered at a tagged angle in LiF at 10 mK give recoils of known energy, and counting single F centers per tagged recoil measures the product $p_{\mathrm{FP}}p_{\mathrm{surv}}p_q f_{\mathrm{opt}}$ as a function of E_R. X-rays of 5.9 keV absorbed at millikelvin temperatures give p_{FH} from the F-center yield per absorbed energy together with the quenching of the exciton luminescence. The photodissociation quantum yield of a molecular crystal at millikelvin temperatures, with the fragments detected by infrared absorption, gives p_{esc}.

16.7 Glasses, random media and two-level systems

A superconducting resonator made on a film of thermally grown SiO_2 loses a fraction 3×10^{-4} of its stored energy per radian at 6 GHz and below 100 mK, when it holds only a few photons [34, Table I]. At higher power or temperature the loss falls. The absorbers are groups of atoms in the glass that can sit in either of two configurations and tunnel between them, typically 10^{20} per eV and cm^3, with splittings that reach down to micro-electronvolts. At these energies a crystal has only its long acoustic phonons. The same systems absorb a dark photon of mass 25 µeV as readily as a microwave photon of that energy, and a dark-matter particle that scatters in the glass can flip one of them. A glass is also the plainest example of a random medium. Its disorder raises two further questions: how its vibrations absorb a dark photon once momentum conservation no longer selects one optical phonon, and when a light particle that scatters coherently off a whole body can be treated in the Born approximation. We derive the response of the two-level systems to a classical strain or field, the dark-photon absorption rate it implies and an estimate of the scattering rate. For the disordered solid we derive its infrared absorption below the Ioffe–Regel frequency and the cross-section above which the Born treatment of a random medium fails. The same tunneling systems release heat at millikelvin temperatures, and their kinetics are the subject of Sect. 16.8.

16.7.1 The standard tunneling model

Let $|\mathrm{L}\rangle$ and $|\mathrm{R}\rangle$ be the ground states of a group of atoms localized in the two wells of a double-well potential. In their span the Hamiltonian is Eq. (C.169) without the bath,

$$H_{\mathrm{TLS}} = \tfrac{1}{2}\Delta_{\mathrm{as}}\,\sigma_z - \tfrac{1}{2}\Delta_0\,\sigma_x, \qquad \Delta_{\mathrm{tl}} = \left(\Delta_{\mathrm{as}}^2 + \Delta_0^2\right)^{1/2}, \tag{16.46}$$

with the Pauli matrices of Appendix C.20 in the localized basis, the asymmetry Δ_{as} between the two wells, the tunneling amplitude Δ_0 and the splitting Δ_{tl} of the eigenstates, the two-level splitting of Sect. 2.5. The tunneling amplitude is exponentially small in the barrier, $\Delta_0 = \omega_{\mathrm{w}}\mathrm{e}^{-\lambda_{\mathrm{tun}}}$, with $\omega_{\mathrm{w}} \sim$ the vibration energy in one well. For a particle of mass M_{tun} crossing a barrier of height V_{b} between wells a distance d_{tun} apart, the tunneling exponent is $\lambda_{\mathrm{tun}} = (d_{\mathrm{tun}}/2)(2M_{\mathrm{tun}}V_{\mathrm{b}})^{1/2}$ [35, Sect. II]. In the eigenbasis the matrix σ_z, which measures in which well the atoms sit, becomes

$$\sigma_z = \frac{\Delta_{\mathrm{as}}}{\Delta_{\mathrm{tl}}}\,\tau_z + \frac{\Delta_0}{\Delta_{\mathrm{tl}}}\,\tau_x, \tag{16.47}$$

with τ_z and τ_x the Pauli matrices of the eigenstates and the sign of τ_x fixed by the phases of the eigenvectors. A perturbation that couples to the position of the atoms therefore drives transitions with the amplitude $\Delta_0/\Delta_{\mathrm{tl}}$: a system with $\Delta_0 \ll \Delta_{\mathrm{as}}$ has its eigenstates localized in the two wells, and the perturbation cannot move it from one to the other.

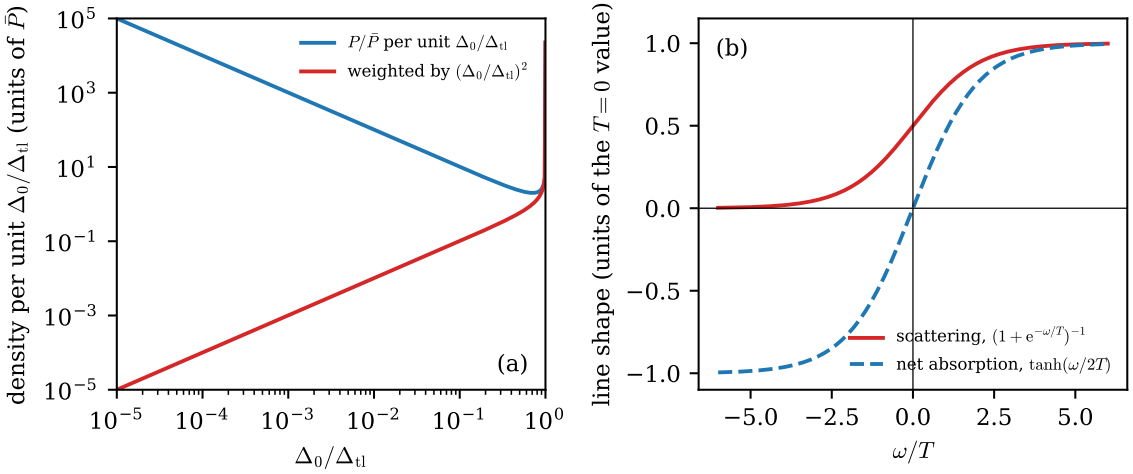


Figure 16.17: The standard tunneling model. (a) Density of two-level systems per unit $u = \Delta_0/\Delta_{tl}$ at fixed splitting, in units of $\bar{P}$: $1/[u(1-u^2)^{1/2}]$ (blue), whose integral grows as $\ln(2/u_{min})$ with the lower cutoff u_{min}. Weighted by the squared transition amplitude u^2 (red), it integrates to one, Eq. (16.49); 87 % of that comes from $u > 0.5$. (b) Line shapes of the ensemble at temperature T, in units of their $T = 0$ values. Red: the scattering structure factor $(1+e^{-\omega/T})^{-1}$ of Eq. (16.57), nonzero for $\omega < 0$ through the thermal population of the upper level; blue, dashed: the net absorption of a classical field, $\tanh(\omega/2T)$, Eq. (16.51).

In a glass every system has its own Δ_{as} and λ_{tun}. The standard tunneling model of Phillips and of Anderson, Halperin and Varma takes both to be independent and uniformly distributed, with density $\bar{P}$ per unit asymmetry, unit exponent and unit volume [36, Eq. (8)]. A system and its mirror image are the same system, so we count $\Delta_{as} \geq 0$ only; this convention defines $\bar{P}$, and with both signs counted every rate below would carry a factor 2. Changing variables from $(\Delta_{as}, \lambda_{tun})$ to (Δ_{tl}, Δ_0), with $d\lambda_{tun} = -d\Delta_0/\Delta_0$ and $\partial\Delta_{as}/\partial\Delta_{tl} = \Delta_{tl}/\Delta_{as}$ at fixed Δ_0, gives

$$P(\Delta_{tl}, \Delta_0) = \frac{\bar{P}\,\Delta_{tl}}{\Delta_0\,(\Delta_{tl}^2 - \Delta_0^2)^{1/2}}, \qquad 0 < \Delta_0 \leq \Delta_{tl}, \tag{16.48}$$

the distribution of Ref. [35, Eq. (12)]. Its integral over Δ_0 diverges logarithmically at small Δ_0, and the number of systems per unit splitting depends on a lower cutoff of the tunneling amplitude that no experiment fixes. A transition probability, however, carries the weight $(\Delta_0/\Delta_{tl})^2$ of Eq. (16.47), and

$$\int_0^{\Delta_{tl}} d\Delta_0\, P(\Delta_{tl}, \Delta_0)\, \frac{\Delta_0^2}{\Delta_{tl}^2} = \bar{P} \int_0^1 \frac{u\,du}{(1-u^2)^{1/2}} = \bar{P}, \tag{16.49}$$

with $u = \Delta_0/\Delta_{tl}$. The systems that dominate the count do not couple, and those that couple are counted by $\bar{P}$ alone, at every splitting. Fig. 16.17(a) shows the two densities.

A strain field and an electric field both shift the energies of the two wells. The atoms move between configurations with different elastic and electric dipoles. To first order the asymmetry becomes $\Delta_{as} + 2\gamma_{TLS}u_d + 2\boldsymbol{p}_{TLS}\cdot\boldsymbol{E}$, with u_d the component of the strain tensor $u_{\alpha\beta}$ along the deformation of the defect, γ_{TLS} its deformation potential and $\boldsymbol{p}_{TLS}$ half the difference of the electric dipoles of the two configurations [36, Eq. (16)]. The perturbation is

$$H_1 = (\gamma_{TLS}u_d + \boldsymbol{p}_{TLS}\cdot\boldsymbol{E})\,\sigma_z \equiv bF\,\sigma_z, \tag{16.50}$$

where F stands for the applied field (u_d or one component of $\boldsymbol{E}$) and b for the coupling to it. By Eq. (16.47) it has a transverse part, which drives transitions at $\omega = \Delta_\mathrm{tl}$, and a longitudinal part, which modulates the splitting.

The transverse part gives the resonant response. The power a field $F\cos\omega t$ deposits per unit volume is $\frac{1}{2}\omega\chi''(\omega)F^2$, with χ'' the absorptive part of the susceptibility of the coupled quantity $b\sigma_z$. For one system the golden rule of Sect. 2.1 gives an absorption rate from the lower level and an equal rate of stimulated emission from the upper one, and the populations of Sect. 2.5 differ by $p_0 - p_1 = \tanh(\Delta_\mathrm{tl}/2T)$. Hence $\chi''(\omega) = \pi b^2(\Delta_0/\Delta_\mathrm{tl})^2 \tanh(\Delta_\mathrm{tl}/2T)\,\delta(\omega - \Delta_\mathrm{tl})$ for $\omega > 0$, where the delta function stands for the Lorentzian of Eq. (2.56) with the width of the system, small against ω and T in an insulating glass at millikelvin temperatures. Summing over the ensemble with Eqs. (16.48) and (16.49),

$$\chi''_\mathrm{res}(\omega) = \pi\bar{P}\,\langle b^2\rangle\,\tanh\frac{\omega}{2T}, \tag{16.51}$$

independent of ω, with $\langle b^2\rangle$ averaged over the orientations of the systems. Its two physical cases are the dielectric loss and the acoustic loss. For the electric field the orientation average is $\langle(\boldsymbol{p}_\mathrm{TLS}\cdot\hat{\boldsymbol{E}})^2\rangle = p_\mathrm{TLS}^2/3$, and in Heaviside–Lorentz units the polarization is $\chi\boldsymbol{E}$, so the imaginary part of the permittivity is χ'' itself. Against the real part ε_r of the glass, the loss tangent is

$$\tan\delta(\omega,T) = \frac{\varepsilon_2}{\varepsilon_r} = \tan\delta_0\,\tanh\frac{\omega}{2T}, \qquad \tan\delta_0 = \frac{\pi\bar{P}\,p_\mathrm{TLS}^2}{3\varepsilon_r}, \tag{16.52}$$

the low-power loss tangent $\tan\delta_0$ at $T \to 0$. For a sound wave of polarization j, longitudinal or transverse, the coupling is the deformation potential γ_j and the elastic modulus $\rho_T c_j^2$ takes the place of ε_r, so the internal friction is $Q^{-1} = \pi C_j \tanh(\omega/2T)$, with the tunneling strength $C_j = \bar{P}\gamma_j^2/\rho_T c_j^2$ [37, Eqs. (5), (6)]. Both are the response of the standard tunneling model to a classical field [35, Eqs. (4), (12), (13), (15)]. In vitreous silica $C = 2.6\times10^{-4}$ on average over the two polarizations [37, Table I], and Golding, Graebner and Schutz find $C_\mathrm{l} = 1.9\times10^{-4}$ and $C_\mathrm{t} = 2.0\times10^{-4}$ at 500 MHz, as quoted by Buchenau et al. [37, Sect. III].

The longitudinal part of Eq. (16.50), $bF(\Delta_\mathrm{as}/\Delta_\mathrm{tl})\tau_z$, shifts the splitting by $2bF\Delta_\mathrm{as}/\Delta_\mathrm{tl}$. In equilibrium the population difference would follow the shifted splitting at once; in fact it relaxes toward it at the rate $\Gamma_1 = 1/\tau_1$ of the system. The static susceptibility of the coupled quantity is $2b^2(\Delta_\mathrm{as}/\Delta_\mathrm{tl})^2\,\partial_{\Delta_\mathrm{tl}}\tanh(\Delta_\mathrm{tl}/2T)$, and a population that relaxes with one time constant turns it into the Debye form $\chi_\mathrm{rel}(\omega) = \chi_\mathrm{rel}(0)/(1 - \mathrm{i}\omega\tau_1)$, whose absorptive part is $\chi_\mathrm{rel}(0)\,\omega\tau_1/(1+\omega^2\tau_1^2)$ [35, Eq. (6)]. In an insulating glass Γ_1 is set by the emission and absorption of one thermal phonon,

$$\Gamma_1 = K_\mathrm{ph}\,\Delta_\mathrm{tl}^3\Big(\frac{\Delta_0}{\Delta_\mathrm{tl}}\Big)^2\coth\frac{\Delta_\mathrm{tl}}{2T}, \qquad K_\mathrm{ph} = \frac{1}{2\pi\rho_T}\Big(\frac{\gamma_\mathrm{l}^2}{c_\mathrm{LA}^5} + \frac{2\gamma_\mathrm{t}^2}{c_\mathrm{TA}^5}\Big), \tag{16.53}$$

which Sect. 16.8 derives [35, Eq. (7)]. Summed over the ensemble and divided by the resonant value at $T = 0$, the full loss is

$$\frac{\tan\delta}{\tan\delta_0} = \tanh\frac{\omega}{2T} + \frac{2}{\pi}\int_0^\infty\frac{\mathrm{d}\Delta_\mathrm{tl}}{2T}\,\mathrm{sech}^2\frac{\Delta_\mathrm{tl}}{2T}\int_0^1\mathrm{d}u\,\frac{(1-u^2)^{1/2}}{u}\,\frac{\omega\tau_1}{1+\omega^2\tau_1^2}, \tag{16.54}$$

with $u = \Delta_0/\Delta_\mathrm{tl}$ and $\tau_1 = 1/\Gamma_1$ from Eq. (16.53); the same ratio holds for $Q^{-1}/\pi C_j$. The relaxation part has two limits. Since $\tau_1^{-1} \propto u^2$, the fastest systems are those with $u = 1$. When they relax fast, $\omega\tau_1(u=1) \ll 1$, the u integral is dominated by the slower systems with $u^2 \sim \omega\tau_1(u=1)$, where $\omega\tau_1 \sim 1$; it equals $\pi/4$, and the relaxation part saturates at $\frac{1}{2}$, the tunneling plateau $Q^{-1} = \pi C/2$ [37, Eq. (10)]. When even the fastest systems relax slowly, $\omega\tau_1(u=1) \gg 1$, the integrand is $1/\omega\tau_1$, the u integral gives $\frac{1}{3}$, and the energy integral

gives $\int_0^\infty y^3\,\mathrm{d}y/\sinh y = \pi^4/8$, so the relaxation part is $(\pi^3/12)K_{\mathrm{ph}}T^3/\omega$. Compared to the resonant part, $\omega/2T$ at $T \gg \omega$, it grows as T^4/ω^2.

Limit Check 16.10. Two-level response to a classical strain or field

The resonant loss of the standard tunneling model, $\tan\delta = \tan\delta_0 \tanh(\omega/2T)$ with $\tan\delta_0 = \pi\bar{P}p_{\mathrm{TLS}}^2/3\varepsilon_r$, and $Q^{-1} = \pi C_j \tanh(\omega/2T)$ for sound [35, Eqs. (4), (12), (13), (15)].

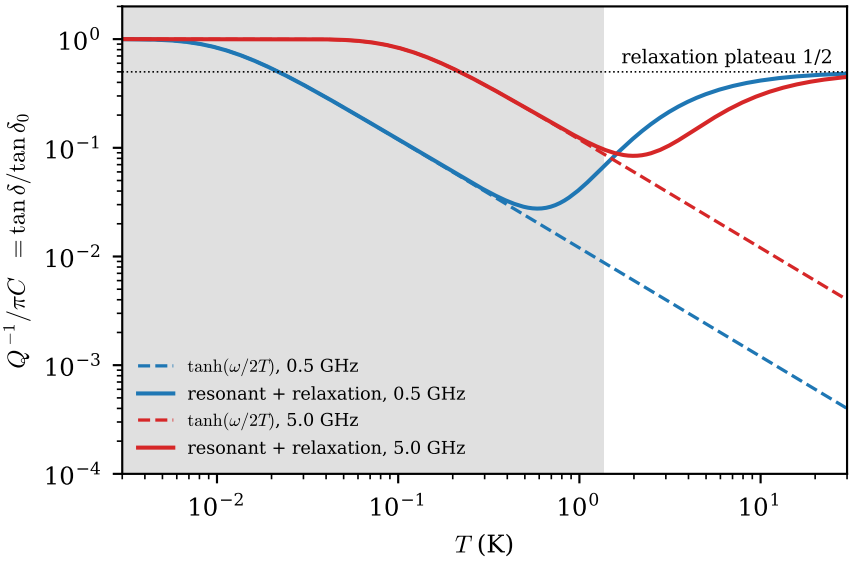


Figure 16.18: Loss of the ensemble in vitreous silica, divided by its value at $T = 0$, against temperature, at 0.5 GHz (blue) and 5 GHz (red). Dashed: $\tanh(\omega/2T)$. Solid: resonant plus relaxation part, Eq. (16.54), with the one-phonon rate Eq. (16.53) for $\rho_T = 2.2\,\mathrm{g\,cm^{-3}}$, $c_{\mathrm{LA}} = 5.8$ and $c_{\mathrm{TA}} = 3.8\,\mathrm{km\,s^{-1}}$ [37, Table I], an assumed $\gamma_{\mathrm{l}} = 1\,\mathrm{eV}$ and $\gamma_{\mathrm{l}}/\gamma_{\mathrm{t}} = 1.6$ [38], so that $K_{\mathrm{ph}} = 4.63 \times 10^4\,\mathrm{eV^{-2}}$. Dotted: the relaxation plateau. Gray: the full loss within 10 % of the tanh law at 5 GHz, $T < 1.36\,\mathrm{K}$ (0.43 K at 0.5 GHz). Above these temperatures the population of slowly relaxing systems lags the field, and the loss rises toward the plateau $\tan\delta_0/2$ instead of falling as $\omega/2T$.

Limit Check 16.10 puts the relaxation correction where the low-temperature estimate puts it: 10 % of the resonant loss at $T = [0.6\,\omega^2/(\pi^3 K_{\mathrm{ph}})]^{1/4}$, which is 1.34 K at 5 GHz against 1.36 K from the full integral. At 10 mK the relaxation part at 5 GHz is 4×10^{-9} of the resonant one. The value of γ_{l} enters only through K_{ph}, and since the crossover temperature scales as $K_{\mathrm{ph}}^{-1/4}$, a deformation potential twice as large moves it down by a factor $2^{1/2}$. The tunneling model has its own limits at these temperatures: above about 2 K, relaxation by several phonons adds to Eq. (16.53) [35, Sect. II]. The second correction is saturation. A field strong enough to drive transitions faster than the systems relax equalizes the two populations and removes the resonant loss; for sound the resonant term falls as $[1 + I/I_{\mathrm{c}}]^{-1/2}$ with the intensity I, and the critical intensity is $\sim 10^{-7}\,\mathrm{W\,cm^{-2}}$ [37, Eq. (8)]. This is why the resonator loss falls at high power and why $\tan\delta_0$ is the low-power value. In practice the tunneling parameters, the distributions of Δ_{as} and λ_{tun}, the deformation potentials and the dipole moments, are computed from ensembles of model glasses, found by searching their energy landscapes for pairs of nearby minima (Appendix E.22). The computed $\bar{P}p_{\mathrm{TLS}}^2$ is then checked against the measured $\tan\delta_0$ of the same preparation.

16.7.2 Dark-photon absorption on two-level systems

A dark photon of mass m_ϕ mixes with the photon through ζ and is absorbed at the rate $\Gamma = \zeta^2 m_\phi L(m_\phi)$, with $L = \mathrm{Im}(-1/\varepsilon)$ the loss function at $\boldsymbol{p}_\phi \to 0$, and the rate per unit target mass is $\mathcal{R} = (\rho_\chi/\rho_T)\zeta^2 L$ (Eq. (10.31)). The momentum $\boldsymbol{p}_\phi \approx m_\phi v$ of a dark photon of 25 μeV is 3×10^{-8} eV for $v = 354\,\mathrm{km\,s^{-1}}$, a wavelength of about 40 m, so the local dielectric function of the glass is all we need. For a glass with $\varepsilon = \varepsilon_r(1 + \mathrm{i}\tan\delta)$ the loss function is $\tan\delta/[\varepsilon_r(1+\tan^2\delta)]$, and with Eq. (16.52),

Dark-photon absorption on two-level systems

$$\mathcal{R} = \frac{\rho_\chi}{\rho_T}\,\zeta^2\,\frac{\tan\delta_0}{\varepsilon_r}\,\tanh\frac{m_\phi}{2T}\,[1 + \mathcal{O}(\tan^2\delta_0)]. \tag{16.55}$$

The rate per unit mass is independent of m_ϕ wherever $m_\phi \gg 2T$, and it is fixed by two measured numbers, the low-power loss tangent and the permittivity.

Note that the factor $\tanh(m_\phi/2T)$ appears once. It is the population difference of the absorbing systems, and the net rate already subtracts the stimulated emission that the dark-matter field drives from the upper level, just as the dielectric loss does; a second thermal factor from the field would count that subtraction twice. The in-medium suppression $1/\varepsilon_r$ is the factor $1/|\varepsilon|^2$ of Eq. (10.31) times the ε_r in $\varepsilon_2 = \varepsilon_r\tan\delta$: the field inside the glass is screened, and the absorbers respond to the screened field. And the dark-matter field drives each system far below saturation. At the coupling that Worked example 16.7 finds for three events per kg-year, the field $\zeta(2\rho_\chi)^{1/2}$ of Sect. 11.8 gives a Rabi frequency below $6 \times 10^{-7}\,\mathrm{s^{-1}}$ for a dipole of 1.1 D, against a relaxation rate of $10^6\,\mathrm{s^{-1}}$ for a system with $\Delta_{\mathrm{tl}} = m_\phi$ at 6 GHz.

What the two-level systems add is a loss that is flat in frequency, from the lowest masses at which m_ϕ exceeds $2T$ up to the energies at which the flat distribution Eq. (16.48) ceases to hold. The loss tangents that fix the rate are measured near 6 GHz (25 μeV) and below 100 mK, and they depend on how the oxide was made. Thermal SiO_2 has $\tan\delta_0 = 3.0$ to 3.3×10^{-4}, oxide deposited from a plasma 2.7 to 2.9×10^{-3}, SiN_x 1.0 to 2.0×10^{-4} and hydrogenated amorphous silicon 2.2 to 2.5×10^{-5} in lumped-element resonators [34, Table I]. A second deposited oxide reached 5×10^{-3} [39]. Such films are a few hundred nanometers thick, and no millikelvin value has been measured for bulk fused silica.

Worked example 16.7: The dark-photon rate on thermal SiO_2

Take thermal SiO_2 with $\tan\delta_0 = 3 \times 10^{-4}$, an assumed static permittivity $\varepsilon_r = 3.9$, and $\rho_T = 2.2\,\mathrm{g\,cm^{-3}}$, at $T = 10$ mK ($T = 0.862$ μeV). A dark photon of 6 GHz, $m_\phi = 24.81$ μeV, has $m_\phi/2T = 14.4$, so $\tanh(m_\phi/2T) = 1$ to eight digits, and the loss function is $L = 3 \times 10^{-4}/3.9 = 7.69 \times 10^{-5}$. The prefactor is $\rho_\chi/\rho_T = (0.3\,\mathrm{GeV\,cm^{-3}})/(2.2\,\mathrm{g\,cm^{-3}}) = 0.136\,\mathrm{GeV/g}$, which divided by $\hbar$ is 2.07×10^{23} per gram per second. Then

$$\mathcal{R} = 5.03 \times 10^{29}\,\zeta^2 \text{ per kg-year},$$

and three events in one kg-year need $\zeta = 2.44 \times 10^{-15}$. The upper end of the measured range, 3.3×10^{-4}, raises the rate by 1.10, and the deposited oxide with 5×10^{-3} by 16.7, which lowers ζ by a factor 0.24. At 100 mK the thermal factor drops to 0.894, which is why a loss tangent quoted at that temperature must be divided by 0.894 to give $\tan\delta_0$. The target mass is the problem. A film of 1 μm on 100 $\mathrm{cm^2}$ weighs 22 mg, and three events in a year then need $\zeta = 5.2 \times 10^{-13}$. Each absorption leaves 24.8 μeV in one two-level system, which decays by emitting a phonon of that energy. The loss tangent also fixes

$\bar{P} p_{\mathrm{TLS}}^2 = 3\varepsilon_r \tan\delta_0/\pi$ and nothing else: with $\bar{P} = 1.2 \times 10^{20}\,\mathrm{eV}^{-1}\,\mathrm{cm}^{-3}$, the value that the polarization-averaged $C = 2.6 \times 10^{-4}$ implies for $\gamma_{\mathrm{l}} = 1\,\mathrm{eV}$, Eq. (16.52) requires $p_{\mathrm{TLS}} = 0.23\,e\,\text{Å} = 1.1\,\mathrm{D}$.

Limit Check 16.11. Dark-photon absorption on two-level systems

$\mathcal{R} = (\rho_\chi/\rho_T)\zeta^2 \tan\delta_0 \tanh(m_\phi/2T)/\varepsilon_r$, the loss function $\mathrm{Im}(-1/\varepsilon) = \varepsilon_2/|\varepsilon|^2$ of Eq. (10.31) with the electric loss of the tunneling model [36, Eq. (20)].

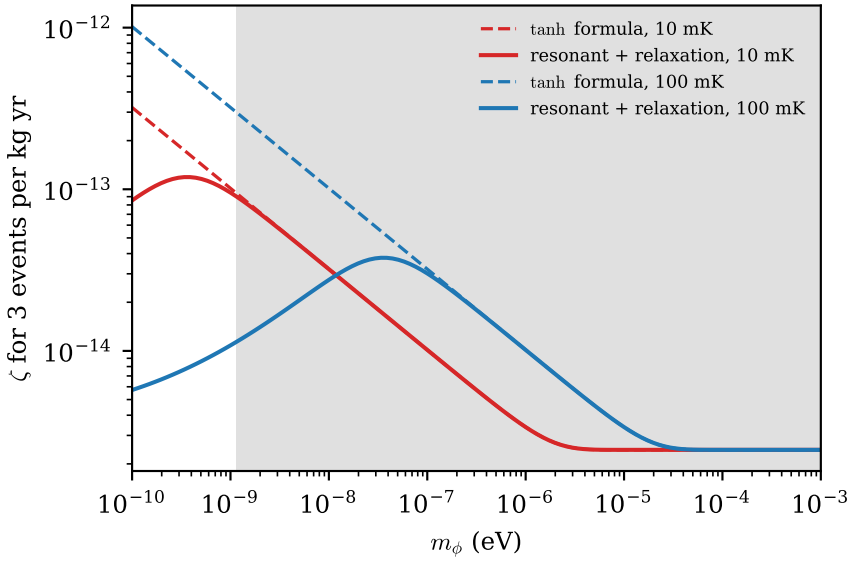


Figure 16.19: Kinetic mixing for three absorptions per kg-year on thermal SiO_2 ($\tan\delta_0 = 3 \times 10^{-4}$, $\varepsilon_r = 3.9$, $\rho_T = 2.2\,\mathrm{g\,cm}^{-3}$), against the dark-photon mass, at 10 mK (red) and 100 mK (blue). Dashed: Eq. (16.55). Solid: $\varepsilon_2/|\varepsilon|^2$ with ε_2 from the resonant plus relaxation loss Eq. (16.54), K_{ph} as in Fig. 16.18. Gray: within 10 % at 10 mK, $m_\phi > 1.15\,\mathrm{neV}$ (0.28 MHz); at 100 mK the bound moves to 115 neV.

Below these masses the relaxation of systems with splittings near T absorbs more than the resonant systems at $\Delta_{\mathrm{tl}} = m_\phi$, whose population difference vanishes as $m_\phi/2T$.

The mass of Limit Check 16.11 scales as T^2: with $\tanh(m_\phi/2T) \approx m_\phi/2T$ and the low-temperature relaxation part $(\pi^3/12)K_{\mathrm{ph}}T^3/m_\phi$, the two are in the ratio $(\pi^3/6)K_{\mathrm{ph}}T^4/m_\phi^2$, which reaches 0.1 at $m_\phi \propto T^2$. Above a millielectronvolt the vibrations of the glass absorb more than its two-level systems (Sect. 16.7.4).

16.7.3 Dark-matter scattering on two-level systems

A dark-matter particle that couples to the electron density sees a two-level system through the change of that density between the two configurations. Projected on $|\mathrm{L}\rangle$ and $|\mathrm{R}\rangle$, the density operator at momentum $\boldsymbol{q}$ is $\frac{1}{2}(n_{\mathrm{LL}} + n_{\mathrm{RR}}) - \frac{1}{2} f_{\mathrm{TLS}}(\boldsymbol{q})\,\sigma_z$, up to terms of relative order $\mathrm{e}^{-\lambda_{\mathrm{tun}}}$, with the vertex

$$f_{\mathrm{TLS}}(\boldsymbol{q}) = \langle \mathrm{R}|n_e(\boldsymbol{q})|\mathrm{R}\rangle - \langle \mathrm{L}|n_e(\boldsymbol{q})|\mathrm{L}\rangle. \tag{16.56}$$

The first term is the static density and scatters elastically; the second flips the system with the amplitude of Eq. (16.47), $|\langle e|n_e(\boldsymbol{q})|g\rangle|^2 = \frac{1}{4}|f_{\mathrm{TLS}}|^2(\Delta_0/\Delta_{\mathrm{tl}})^2$. The structure factor of

Eq. (2.67) sums this over systems, with the weights p_0 and p_1 of Sect. 2.5 for absorption at $\omega = \Delta_{\rm tl}$ and emission at $\omega = -\Delta_{\rm tl}$, and with Eqs. (16.48) and (16.49) the ensemble gives

$$S(\boldsymbol{q},\omega) = \frac{\pi}{2}\,\bar{P}\,\langle|f_{\rm TLS}(\boldsymbol{q})|^2\rangle\,\frac{1}{1+\mathrm{e}^{-\omega/T}}, \tag{16.57}$$

flat in ω inside the band of the two-level systems and obeying detailed balance, Eq. (2.68), as Fig. 16.17(b) shows. The average runs over the systems and their orientations.

The vertex has two parts. The atoms that tunnel carry their electrons from one well to the other, the core term. And the surrounding glass relaxes elastically around each configuration, which moves all its electrons, the host term. For the host term take the defect as an elastic dipole. In configuration R it exerts on the glass a force dipole that differs from that of L by $-2\gamma_{\rm TLS}$ along its deformation, since the energy of a force dipole D in a strain is $-\sum_{\alpha\beta}\mathsf{D}_{\alpha\beta}u_{\alpha\beta}$ and the asymmetry shifts by $2\gamma_{\rm TLS}u_{\rm d}$. In an isotropic continuum the static displacement field of a force dipole has longitudinal Fourier component $\boldsymbol{q}\cdot\tilde{\boldsymbol{u}}(\boldsymbol{q}) = -\mathrm{i}\,\hat{\boldsymbol{q}}\cdot\mathsf{D}\cdot\hat{\boldsymbol{q}}/\rho_T c_{\rm LA}^2$, and since the adiabatic electrons follow the atoms, the electron density changes by $\delta n_e(\boldsymbol{q}) = \mathrm{i}n_e\,\boldsymbol{q}\cdot\tilde{\boldsymbol{u}}$ with n_e the density of all electrons. Hence

$$f_{\rm host} = \frac{2n_e\,\hat{\boldsymbol{q}}\cdot\gamma_{\rm TLS}\cdot\hat{\boldsymbol{q}}}{\rho_T c_{\rm LA}^2}, \qquad \bar{P}\,\langle|f_{\rm host}|^2\rangle = \frac{4n_e^2\,C_{\rm l}}{\rho_T c_{\rm LA}^2}, \tag{16.58}$$

where the orientation average of $(\hat{\boldsymbol{q}}\cdot\gamma_{\rm TLS}\cdot\hat{\boldsymbol{q}})^2$ is $\gamma_{\rm l}^2$, the coupling to longitudinal strain. The deformation potential drops out of the product: the host term of the scattering rate is fixed by the measured tunneling strength $C_{\rm l}$, not by the loss tangent. The form holds in a window of momentum transfers. The momentum transfer must be static, $q \gg \omega/c_{\rm LA}$; for ω near $c_{\rm LA}q$ the elastic response has a pole, and the process becomes the emission of a real phonon that the two-level system absorbs later, which belongs to the phonon channel of Sect. 11.1. The wavelength must also be long against the defect and the atomic spacing, where continuum elasticity holds. No dielectric factor is added: the adiabatic density already contains the screening of the electrons (Appendix E.22).

For the core term the simplest model moves one oxygen ion, with its eight electrons, rigidly by $\boldsymbol{d}$. With the origin at the midpoint, $f_{\rm core} = 8(\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{d}/2} - \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{d}/2}) = 16\mathrm{i}\sin(\boldsymbol{q}\cdot\boldsymbol{d}/2)$, and its orientation average is $\langle|f_{\rm core}|^2\rangle = 128[1-\sin(qd)/qd]$. The core amplitude is imaginary and the static host amplitude is real, so the two add without interference, for every orientation and not only on average; an off-center core would restore a cross term. For a displacement along $\hat{\boldsymbol{q}}$ the two are equal near $q_\times = |f_{\rm host}|/(8d)$, where the host term stops dominating; with the orientation-averaged core, $\langle|f_{\rm core}|^2\rangle^{1/2} \simeq 4.62\,qd$, the crossing moves up by $\sqrt{3}$.

With a heavy mediator, $\mathfrak{D}_{ee} = \pi\bar{\sigma}_e/\mu_{\chi e}^2$ and the structure factor Eq. (16.57) in the isotropic rate Eq. (10.2), the minimum velocity is $q/2m_\chi$ up to ω/q, negligible for $\omega \lesssim 0.1\,\mathrm{meV}$, and a q-independent vertex leaves $\int q\,\mathrm{d}q\,\eta(q/2m_\chi) = 2m_\chi^2\langle v\rangle$. The rate is

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{\rho_\chi}{m_\chi\rho_T}\,\langle v\rangle\,\bar{\sigma}_e\Big(1+\frac{m_\chi}{m_e}\Big)^2\,\frac{\bar{P}\langle|f_{\rm TLS}|^2\rangle}{4\,(1+\mathrm{e}^{-\omega/T})}, \tag{16.59}$$

per unit energy and target mass, a cross-section $\bar{\sigma}_e(1+m_\chi/m_e)^2|f_{\rm TLS}|^2(\Delta_0/\Delta_{\rm tl})^2/4$ per system. The factor $(1+m_\chi/m_e)^2 = m_\chi^2/\mu_{\chi e}^2$ converts the reference cross-section on a free electron into the cross-section on a heavy target. For a q-dependent core vertex the q integral is done with η.

Consider amorphous SiO_2 at $m_\chi = 1\,\mathrm{MeV}$ and $\bar{\sigma}_e = 10^{-38}\,\mathrm{cm}^2$. Thirty electrons per formula unit at $2.2\,\mathrm{g\,cm^{-3}}$ give $n_e = 6.62\times10^{23}\,\mathrm{cm^{-3}}$, and $\rho_T c_{\rm LA}^2 = 7.40\times10^{10}\,\mathrm{Pa} = 4.62\times10^{23}\,\mathrm{eV\,cm^{-3}}$. With the measured longitudinal $C_{\rm l} = 1.9\times10^{-4}$, Eq. (16.58) gives $\bar{P}\langle|f_{\rm host}|^2\rangle = 7.20\times10^{20}\,\mathrm{eV^{-1}\,cm^{-3}}$, and with the halo of Table 1.1, $\langle v\rangle = 354.1\,\mathrm{km\,s^{-1}}$, and

$(1+m_\chi/m_e)^2 = 8.74$, the host term is 6.6×10^{-3} per kg, day and meV. The core term needs $\bar{P}$ itself. For an assumed $\gamma_\mathrm{l} = 1\,\mathrm{eV}$, $\bar{P} = C_\mathrm{l}\rho_T c_\mathrm{LA}^2/\gamma_\mathrm{l}^2 = 8.8\times10^{19}\,\mathrm{eV}^{-1}\,\mathrm{cm}^{-3}$, $|f_\mathrm{host}| = 2.86$, and one oxygen moving by 1 Å adds 1.6×10^{-2}, 2.5 times the host term. The total,

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} \approx 2.3\times10^{-2} \text{ per kg, day and meV} \qquad (m_\chi = 1\,\mathrm{MeV},\ \bar{\sigma}_e = 10^{-38}\,\mathrm{cm}^2),$$

is flat over the band of the two-level systems, and over $0 < \omega < 0.1\,\mathrm{meV}$ it amounts to 2.3×10^{-3} events per kg-day. The polarization-averaged $C = 2.6\times10^{-4}$ would raise both terms by 1.37, to 3.1×10^{-2} per kg, day and meV.

This number is an estimate, uncertain by about a factor 3. The heavy-mediator weight $q\,\eta(q/2m_\chi)$ has its median at $q = 1.73\,\mathrm{keV}$, where $qd = 0.88$ for $d = 1$ Å. The host and core terms cross lower, at $q_\times = 0.71\,\mathrm{keV}$ for a displacement along $\hat{\boldsymbol{q}}$ and 1.2 keV on orientation average. Most of the rate therefore comes from momentum transfers outside the window of Eq. (16.58), near the scale at which the glass stops being a continuum. The core term is a rigid ion, which misses the redistribution of the valence electrons of the moving oxygen, and one geometry stands for many. Appendix E.22 specifies the calculation that replaces both: the density difference Eq. (16.56) from two self-consistent calculations of the same glass cell, averaged over an ensemble of systems. Whatever its value, the rate is small because $\bar{P}$ times the band of energies, about 10^{16} systems per cm^3, is a small number of targets, and each takes at most 0.1 meV. Thin films under a sensor hold far less mass still and matter as absorbers of phonons and as sources of delayed heat (Sect. 16.8), not as targets.

16.7.4 Vibrational absorption of glasses

In a crystal a dark photon is absorbed on phonons only at the zone-center optical frequencies, where its tiny momentum matches a mode (Sect. 11.8). A glass has no zone center. Every vibrational mode can carry a dipole, and the absorption is spread over the whole vibrational spectrum [40]. The rate is still $\mathcal{R} = (\rho_\chi/\rho_T)\zeta^2 L(m_\phi)$, now with the dielectric function of the vibrations of the glass [40, Eq. (2)], and the question is what fixes L at the lowest masses.

Take one configuration of the glass in the harmonic approximation, with modes ν of frequency ω_ν, eigenvectors $\boldsymbol{e}_{I\nu}$ normalized over all N atoms, masses M_I and Born charges Z_I^*. Its dielectric function has the rank-one form of Sect. 11.8, with the Hessian of the glass in place of the dynamical matrix,

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{e^2}{\mathcal{V}}\sum_\nu \frac{|p_\nu|^2}{\omega_\nu^2 - \omega^2 - \mathrm{i}\omega\Gamma_\nu}, \qquad p_\nu = \sum_I \frac{\hat{\boldsymbol{\epsilon}}\cdot Z_I^*\cdot \boldsymbol{e}_{I\nu}}{M_I^{1/2}}, \tag{16.60}$$

for a field along $\hat{\boldsymbol{\epsilon}}$ and mode widths Γ_ν. For narrow modes the imaginary part is $\varepsilon_2(\omega) = (\pi e^2/2\omega\mathcal{V})\sum_\nu |p_\nu|^2\delta(\omega-\omega_\nu)$. With the vibrational density of states $g(\omega)$ normalized to one, $\sum_\nu \delta(\omega-\omega_\nu) = 3Ng(\omega)$, and with the infrared coupling $C^\mathrm{IR}(\omega)$, the mean of $|p_\nu|^2$ over the modes at frequency ω,

$$\varepsilon_2(\omega) = \frac{3\pi e^2 \boldsymbol{n}_\mathrm{at}}{2\omega}\, g(\omega)\, C^\mathrm{IR}(\omega). \tag{16.61}$$

The absorption coefficient $\alpha_\mathrm{abs} = \omega\varepsilon_2/n_r$, with $n_r = \varepsilon_r^{1/2}$ the refractive index, is then a product of the density of states and a coupling, the form in which infrared spectroscopy of glasses is analyzed [41, Eq. (2) and text before Eq. (1)].

Below the Ioffe–Regel frequency ω_IoR, the frequency at which the mean free path of a vibration falls to one wavelength, the modes are still sound waves, scattered weakly by the disorder. For a plane wave of branch ν with sound speed c_ν, wavevector $\boldsymbol{k}$ and polarization $\hat{\boldsymbol{e}}_\nu$, every atom moves with the same velocity, $\boldsymbol{e}_{I\nu} = \hat{\boldsymbol{e}}_\nu \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}_I}(M_I/N\bar{M})^{1/2}$ with $\bar{M}$ the mean

atomic mass. The masses cancel in p_ν, which becomes a Fourier component of the field of Born charges, and averaged over the modes at fixed $\omega = c_\nu k$,

$$C^{\mathrm{IR}}(\omega) = \frac{1}{\bar{M}} \sum_\nu \hat{w}_\nu \, S^Z_\nu(\omega/c_\nu), \qquad S^Z_\nu(k) = \frac{1}{N} \Big\langle \Big| \sum_I \hat{\boldsymbol{\epsilon}} \cdot Z^*_I \cdot \hat{\boldsymbol{e}}_\nu \, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}_I} \Big|^2 \Big\rangle, \tag{16.62}$$

with $\hat{w}_\nu = c_\nu^{-3} / \sum_{\nu'} c_{\nu'}^{-3}$ the fraction of modes at fixed ω in branch ν, as in Eq. (E.101). The behavior of S^Z_ν at small k decides the low-frequency absorption. A crystal has $S^Z_\nu(k) = 0$ for acoustic modes as $k \to 0$, since the Born charges of a cell add to zero. In a glass the charges fluctuate from site to site. If they obey only global neutrality, uncorrelated fluctuations leave $S^Z_\nu(0^+)$ equal to the mean square fluctuation of the projected charge, a constant. If instead they are neutral within every local unit, as within the SiO_4 tetrahedra of silica, the sum is that of dipoles and $S^Z_\nu \propto k^2$. Expanding to second order,

$$C^{\mathrm{IR}}(\omega) = A^{\mathrm{IR}} + B^{\mathrm{IR}}\omega^2, \qquad A^{\mathrm{IR}} = \frac{1}{\bar{M}} \sum_\nu \hat{w}_\nu S^Z_\nu(0^+), \quad B^{\mathrm{IR}} = \frac{1}{\bar{M}} \sum_\nu \frac{\hat{w}_\nu S^{Z\prime\prime}_\nu(0)}{2c_\nu^2}, \tag{16.63}$$

the law Taraskin, Simdyankin, Elliott, Neilson and Lo found in terahertz spectra of glasses [41, Eq. (1)]. Their fit to vitreous silica at room temperature writes $\alpha_{\mathrm{abs}} = (A^{\mathrm{IR}} + B^{\mathrm{IR}}\tilde{\omega}^2) g(\tilde{\omega})$ in spectroscopic units, with α_{abs} and the wavenumber $\tilde{\omega} = \omega/2\pi c$ in cm^{-1} and g normalized per cm^{-1}. It gives $A^{\mathrm{IR}} = 4000\,\mathrm{cm}^{-2}$ and $B^{\mathrm{IR}} = 0.3$, dimensionless, with the Debye density of states of a 330 K Debye temperature [41, Fig. 1]. Harmonic absorption is net absorption and does not depend on temperature, but a room-temperature spectrum also contains anharmonic and relaxational absorption whose size at millikelvin temperatures has not been separated; there the fit is a model value.

Limit Check 16.12. Glass absorption below the Ioffe–Regel frequency

$\alpha_{\mathrm{abs}}(\omega) = (A^{\mathrm{IR}} + B^{\mathrm{IR}}\tilde{\omega}^2)\, g(\tilde{\omega})$ for $\omega < \omega_{\mathrm{IoR}}$ [41, Eq. (1)], inserted in the rate $(\rho_\chi/\rho_T)\zeta^2 \,\mathrm{Im}(-1/\varepsilon)$ [40, Eq. (2)].

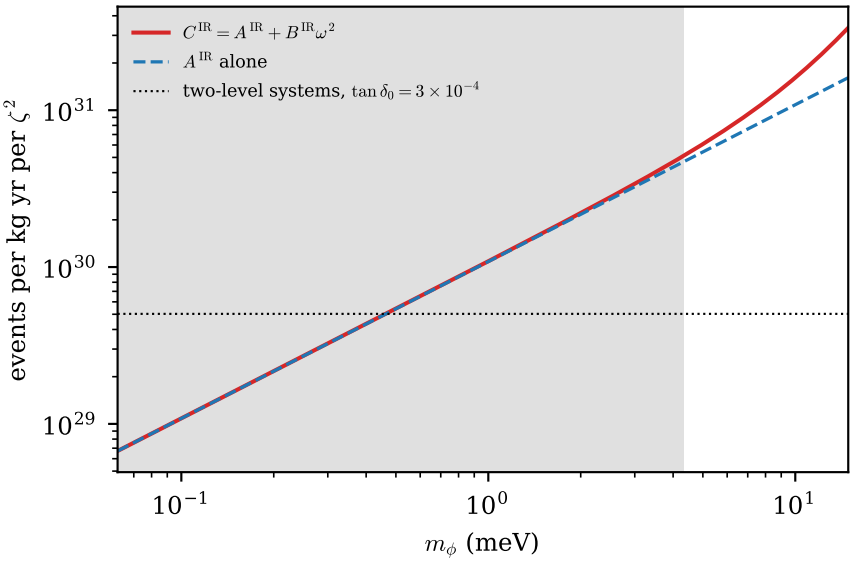


Figure 16.20: Dark-photon absorptions per kg-year per ζ^2 in vitreous silica against the mass, from Eq. (16.61) with $C^{\mathrm{IR}} = A^{\mathrm{IR}} + B^{\mathrm{IR}}\omega^2$ (red) and with the constant term alone (blue, dashed). Inputs: the terahertz fit $A^{\mathrm{IR}} = 4000\,\mathrm{cm}^{-2}$, $B^{\mathrm{IR}} = 0.3$ of Taraskin et

al., a Debye density of states with $\Theta_{\rm D} = 330\,{\rm K}$, $\varepsilon_r = 3.9$ and $\rho_T = 2.2\,{\rm g\,cm^{-3}}$. Dotted: two-level absorption, Eq. (16.55), for thermal SiO_2 at saturation. Gray: below the Ioffe–Regel frequency of silica, $\omega/2\pi c \approx 35\,{\rm cm^{-1}}$ (4.3 meV), where the B term is at most 9 % of the A term.

Above $\omega_{\rm IoR}$ the modes are no longer plane waves, S^Z_ν at one wavevector no longer describes them, and the coupling departs from $A^{\rm IR} + B^{\rm IR}\omega^2$ in a way that depends on the glass.

Limit Check 16.12 shows how the two channels of a glass divide the masses. With $\varepsilon_2 = n_r\alpha_{\rm abs}/(2\pi\tilde\omega)$, the vibrational absorption grows as ω times $C^{\rm IR}$. At 1.24 meV ($\tilde\omega = 10\,{\rm cm^{-1}}$) it gives $\alpha_{\rm abs} = 0.10\,{\rm cm^{-1}}$, $\varepsilon_2 = 3.1\times10^{-3}$ and $1.4\times10^{30}\zeta^2$ absorptions per kg-year. It equals the two-level absorption of thermal SiO_2 at 0.46 meV, and that of the deposited oxide with $\tan\delta_0 = 5\times10^{-3}$ at 6.4 meV, an extrapolation above $\omega_{\rm IoR}$. The constant term carries the low-frequency absorption of silica entirely: the curvature term reaches the constant only at $\tilde\omega = (A^{\rm IR}/B^{\rm IR})^{1/2} = 115\,{\rm cm^{-1}}$, three times above $\omega_{\rm IoR}$. That constant is the incoherent sum of random charges, and a model of the glass that makes every tetrahedron neutral by construction misses it (Appendix E.19). Bloch, Knapen, Li, Madden and Marocco propose amorphous membranes for masses of 50 to 200 meV, above $\omega_{\rm IoR}$, where the mean free path of a mode is shorter than its wavelength and the absorption follows from the measured or computed $\varepsilon(\omega)$ of Eq. (16.60) [40]. The same random couplings give a broadband absorption in a crystal with isotope disorder, which Sect. 11.6 treats.

16.7.5 Random media and the Born limit

A dark-matter particle of 20 eV moving at $v = 10^{-3}$ has a de Broglie wavelength $2\pi/m_\chi v$ of about 60 µm. A momentum transfer $\sim m_\chi v$ cannot resolve atoms, and the particle scatters off the smoothed density $n(\boldsymbol{r})$ of nucleons of a whole body. Liu, Matsumoto, Sheng and Xing pointed out that a random two-phase material scatters such particles far more strongly than a homogeneous one [42]. In the first Born approximation for the body, the amplitudes of all nucleons add with their phases, and for a spin-independent interaction

$$\frac{{\rm d}\sigma_{\rm tot}}{{\rm d}\Omega_q} = \frac{{\rm d}\sigma_{\chi N}}{{\rm d}\Omega_q}\left|\int {\rm d}^3r\, {\rm e}^{{\rm i}\boldsymbol{q}\cdot\boldsymbol{r}}\, n(\boldsymbol{r})\right|^2, \tag{16.64}$$

the coherent sum of Sect. 2.2 with the nucleus replaced by the body [42, Eq. (2)]. Write $n = n_0 + \delta n$ with n_0 the mean nucleon density, and average over the random structure. The cross term vanishes, and the two remaining terms are the homogeneous body, $n_0^2|\int_{\mathscr V}{\rm d}^3r\,{\rm e}^{{\rm i}\boldsymbol{q}\cdot\boldsymbol{r}}|^2$, which falls as $(qR)^{-4}$ for a sphere of radius R once $qR \gg 1$, and the fluctuations, $\int {\rm d}^3r_1{\rm d}^3r_2\,{\rm e}^{{\rm i}\boldsymbol{q}\cdot(\boldsymbol{r}_2-\boldsymbol{r}_1)}\langle\delta n(\boldsymbol{r}_1)\delta n(\boldsymbol{r}_2)\rangle$. For a correlation that decays as $\langle\delta n(\boldsymbol{r}_1)\delta n(\boldsymbol{r}_2)\rangle = \delta n^2 {\rm e}^{-|\boldsymbol{r}_2-\boldsymbol{r}_1|/\xi_{\rm c}}$ over a correlation length $\xi_{\rm c}$, the transform $\int {\rm d}^3r\,{\rm e}^{{\rm i}\boldsymbol{q}\cdot\boldsymbol{r}}{\rm e}^{-r/\xi_{\rm c}} = 8\pi\xi_{\rm c}^3/(1+q^2\xi_{\rm c}^2)^2$ times the volume $\mathscr V = 4\pi R^3/3$ gives, per squared number of nucleons $n_0^2\mathscr V^2$,

$$|F_{\rm fluc}(q)|^2 = \frac{\delta n^2}{n_0^2}\,\frac{6\,\xi_{\rm c}^3}{R^3(1+q^2\xi_{\rm c}^2)^2}, \tag{16.65}$$

for $R \gg \xi_{\rm c}$, the result of Ref. [42, Eqs. (9)–(12)]. The body acts as a collection of incoherent clusters of size $\xi_{\rm c}$, each scattering coherently. At $qR \gg 1$ and $q\xi_{\rm c} \gg 1$ the sphere-averaged homogeneous term is $9/[2(qR)^4]$, and the ratio of the two is $(4/3)(R/\xi_{\rm c})(\delta n/n_0)^2$. For the Ti–6Al–4V alloy of the test masses of the MICROSCOPE satellite test of the equivalence principle, the authors infer $n_0 = 2.69\times10^{24}\,{\rm cm^{-3}}$ and $\delta n = 7.29\times10^{22}\,{\rm cm^{-3}}$ from the phase fractions of its microstructure [42, Eq. (14)]. Then $(\delta n/n_0)^2 = 7\times10^{-4}$, and for that value

the homogeneous term the authors drop is the larger one below a few hundred eV on an equal-mass sphere.

A large coherent cross-section, however, means a strong interaction with the medium, and the Born approximation for a whole body is not the Born approximation for one nucleus. At wavelengths long against the spacing of nuclei, the medium acts on the particle as an optical potential. It is the sum of the pseudopotentials $2\pi b_{\chi N}/\mu_{\chi N}$ of the nucleons, with $b_{\chi N}$ the scattering length, $\sigma_{\chi N} = 4\pi b_{\chi N}^2$, and $\mu_{\chi N} \approx m_\chi$ for a light particle:

$$V_{\rm opt} = \frac{2\pi b_{\chi N} n_0}{\mu_{\chi N}}. \tag{16.66}$$

The plane wave inside the body is undistorted only if this potential is small against the kinetic energy, $|V_{\rm opt}| \ll E_{\rm kin} = m_\chi v^2/2$. Squaring $|b_{\chi N}| \ll m_\chi^2 v^2/4\pi n_0$ gives the condition on the cross-section,

$$\sigma_{\chi N} \ll \sigma_{\rm crit} = \frac{m_\chi^4 v^4}{4\pi n_0^2}, \qquad \frac{|V_{\rm opt}|}{E_{\rm kin}} = \left(\frac{\sigma_{\chi N}}{\sigma_{\rm crit}}\right)^{1/2}. \tag{16.67}$$

Since $\sigma_{\rm crit} \propto v^4$, the condition must hold over the velocities that dominate the observable, not at one speed. The homogeneous term of Eq. (16.64) contains the scattering of the wave by the surface of the body. For a planar face its first-order form can be compared with the exact reflection off a potential step, which shows where the first-order treatment of the medium breaks down. For a wave whose kinetic energy of motion normal to the face is E_n, the wavenumber inside is $k' = k(1 - V_{\rm opt}/E_n)^{1/2}$ and the reflectance is $R_{\rm refl} = |(k - k')/(k + k')|^2$, total for $V_{\rm opt} > E_n$. Expanding k' to first order in $V_{\rm opt}$ gives the Born value $(V_{\rm opt}/4E_n)^2$, the square of the first-order amplitude of the step.

Limit Check 16.13. Random media and the Born limit

First Born approximation for a whole body, $\mathrm{d}\sigma_{\rm tot}/\mathrm{d}\Omega_q = (\mathrm{d}\sigma_{\chi N}/\mathrm{d}\Omega_q)\,|\int \mathrm{d}^3 r\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\, \boldsymbol{n}(\boldsymbol{r})|^2$, with the fluctuation form factor Eq. (16.65) [42, Eqs. (2), (9)–(12), (14)].

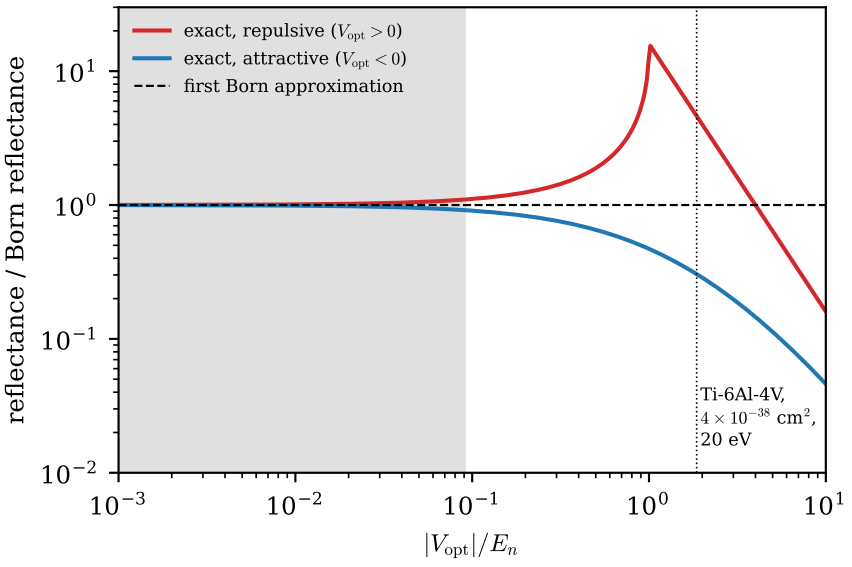


Figure 16.21: Exact reflectance of a planar face of a homogeneous medium with optical potential Eq. (16.66), divided by its first Born approximation $(V_{\rm opt}/4E_n)^2$, against $|V_{\rm opt}|/E_n$, for a repulsive (red) and an attractive (blue) medium. E_n is the kinetic energy of the motion normal to the face. Gray: the ratio within 10 % of one, $|V_{\rm opt}|/E_n < 0.092$.

Dotted: $V_{\mathrm{opt}}/E_{\mathrm{kin}} = 1.86$, the value at $\sigma_{\chi N} = 4 \times 10^{-38}\,\mathrm{cm}^2$, $m_\chi = 20\,\mathrm{eV}$, $v = 10^{-3}$ in Ti–6Al–4V ($n_0 = 2.69 \times 10^{24}\,\mathrm{cm}^{-3}$).

Beyond $|V_{\mathrm{opt}}| \approx E_n$ a repulsive body reflects every particle that reaches it, and the force on it scales with the optical potential of each material, not with the square of its density fluctuations.

The leading correction is linear: the ratio of Limit Check 16.13 is $1 + V_{\mathrm{opt}}/E_n$ for small V_{opt}/E_n, 1 % at $V_{\mathrm{opt}}/E_n = 0.01$, of opposite sign for the two signs of the potential. Below σ_{crit} the uniform part of the medium only shifts the wavenumber inside, $k \to k'$, a phase that drops out of the fluctuation term once the wave is distorted by it, and the fluctuation cross-section keeps corrections of relative order $V_{\mathrm{opt}}/E_{\mathrm{kin}}$. In Ti–6Al–4V at $m_\chi = 20\,\mathrm{eV}$, Eq. (16.67) gives $\sigma_{\mathrm{crit}} = 1.16 \times 10^{-38}\,\mathrm{cm}^2$ at $v = 10^{-3}$, $2.41 \times 10^{-38}\,\mathrm{cm}^2$ at $v = 1.2 \times 10^{-3}$ and $5.7 \times 10^{-37}\,\mathrm{cm}^2$ at the largest laboratory speed $v_{\max} = 2.650 \times 10^{-3}$. Liu et al. quote a reach of $\sigma_{\chi N} \approx 4 \times 10^{-38}\,\mathrm{cm}^2$ near 20 eV from the MICROSCOPE bound on differential acceleration [42, Sect. IV], where $|V_{\mathrm{opt}}|/E_{\mathrm{kin}} = 1.86$ at $v = 10^{-3}$ and 1.29 at $v = 1.2 \times 10^{-3}$, before the projection on the normal of a face lowers E_n further. At that cross-section the first-order treatment of the medium does not apply to the speeds near the peak of the distribution. Whether the true force on the test masses is larger or smaller than the Born estimate depends on the sign of $b_{\chi N}$, on reflection at the faces and on the composition difference of the two masses. Appendix E.19 specifies the calculation by transfer matrices over the faces and the velocity distribution; until it is done, a bound from this channel is a conjecture.

16.7.6 Open problems for glass targets

For a given film the parameters of the standard tunneling model are known only in combination: the loss tangent fixes $\bar{P}p^2_{\mathrm{TLS}}$, the acoustic loss $\bar{P}\gamma_{\mathrm{l}}^2$ and $\bar{P}\gamma_{\mathrm{t}}^2$, and neither fixes $\bar{P}$; the core term, the relaxation corrections and the kinetics of Sect. 16.8 need all three. For the films of real detectors (amorphous Al_2O_3 in junctions, SiO_2 and SiN_x under sensors, amorphous WSi) these have not been measured on the same sample. Beyond the values, the model itself is an assumption: that Δ_{as} and λ_{tun} are independent and flat, at every energy from the lowest masses to the top of the band, and that the systems do not interact. Its successes, the linear specific heat and the logarithmic shift of the sound velocity, test averages, not the distributions, and the ratio $\gamma_{\mathrm{l}}/\gamma_{\mathrm{t}}$, between 1.44 and 1.84 in 18 glasses, is nearly universal without the model giving a reason [38]. One film measured three ways would settle the first question: the dielectric loss, the acoustic loss, and the strain and field dependence of single systems resolved in a resonator, which give γ_{TLS} and p_{TLS} directly [36]. The second needs the distributions computed for a model glass of stated preparation (Appendix E.22).

16.8 Two-level systems at millikelvin

One gram of fused silica cooled to 10 mK is still giving off 46 pW an hour later, and the power falls only as $1/t$ (Worked example 16.8). For a detector held at 10 mK this is a heat load that lasts for weeks. The heat comes from the tunneling two-level systems of Sect. 16.7. Each was in equilibrium at a few kelvin, and each gives up its excess energy when it next relaxes, which for a defect behind a thick barrier takes hours or years. The same relaxation rates decide how fast a two-level system in a sensor film absorbs the energy of a phonon and returns it, and they are the kinetics behind the delayed signals of Sect. 19.6. We compute the relaxation by one phonon and by the conduction electrons of a normal or superconducting metal, its ohmic scaling limit at strong coupling, the crossover from activated hopping to tunneling, and the

heat released by a cooled glass. The defect is the two-configuration system of Eq. (C.169), treated in the spin–boson language of Appendix C.20.

16.8.1 Relaxation by one phonon

A two-level system relaxes when it passes from the upper to the lower of its two eigenstates and gives the splitting to its surroundings. The environment couples to the configuration through $\frac{1}{2}\sigma_z X$, with $X = \sum_k \lambda_k (d_k + d_k^\dagger)$ the bath coordinate of Eq. (C.169). By Eq. (16.47) only the off-diagonal element $\Delta_0/\Delta_{\mathrm{tl}}$ of σ_z in the eigenbasis changes the state, so a symmetric defect couples at full strength and a strongly asymmetric one hardly at all. By the golden rule of Sect. 2.1 the emission of a bath quantum of energy Δ_{tl} has the rate $2\pi \sum_k (\Delta_0 \lambda_k / 2\Delta_{\mathrm{tl}})^2 (n_{\mathrm{B}} + 1)\, \delta(\Delta_{\mathrm{tl}} - \omega_k)$, and absorption has the same rate with n_{B} in place of $n_{\mathrm{B}} + 1$. A population difference relaxes at the sum of the two, which carries $2n_{\mathrm{B}} + 1 = \coth(\Delta_{\mathrm{tl}}/2T)$:

$$\Gamma_1 = \frac{1}{2}\Big(\frac{\Delta_0}{\Delta_{\mathrm{tl}}}\Big)^2 J_{\mathrm{s}}(\Delta_{\mathrm{tl}}) \coth \frac{\Delta_{\mathrm{tl}}}{2T}. \tag{16.68}$$

The bath enters only through its spectral density $J_{\mathrm{s}}(\omega) = \pi \sum_k \lambda_k^2 \delta(\omega - \omega_k)$ of Eq. (C.170) at the splitting. The rate is first order in J_{s}, and it holds as long as $\Gamma_1 \ll \Delta_{\mathrm{tl}}$, so that the eigenstates are defined; its upward and downward parts obey detailed balance (Sect. 2.5).

A long acoustic wave strains the solid, and the strain shifts the energies of the two configurations by different amounts. The coupling is that of Eq. (16.50), and the deformation potential depends on the orientation of the defect relative to the wave. As in Sect. 16.7, γ_{l}^2 and γ_{t}^2 are its mean squares over orientations for a longitudinal wave and for each transverse wave; acoustic experiments on glasses give about 1 eV [36, Sect. IV A]. Summing the strain of every phonon mode of Sect. 5.3 at the defect (Appendix D.42) gives

$$J_{\mathrm{s}}^{\mathrm{ph}}(\omega) = \frac{\omega^3}{\pi \rho_T}\Big(\frac{\gamma_{\mathrm{l}}^2}{c_{\mathrm{LA}}^5} + \frac{2\gamma_{\mathrm{t}}^2}{c_{\mathrm{TA}}^5}\Big) = 2K_{\mathrm{ph}}\, \omega^3, \tag{16.69}$$

with c_{LA} and c_{TA} the sound speeds and K_{ph} the constant of Eq. (16.53). The bath is superohmic, as Appendix C.20 anticipated: the strain of one phonon grows as $\omega^{1/2}$ and the number of modes as ω^2. With Eq. (16.68) it gives the one-phonon rate $\Gamma_1^{\mathrm{ph}} = K_{\mathrm{ph}} \Delta_{\mathrm{tl}}^3 (\Delta_0/\Delta_{\mathrm{tl}})^2 \coth(\Delta_{\mathrm{tl}}/2T)$ of Eq. (16.53). This is Jäckle's rate, in the form of Meißner et al. [35, Eq. (7)] and of Sahling et al. [43, Eqs. (4), (5)], whose prefactor $8\pi^3/\rho h^4$ is $1/2\pi\rho$ with $h = 2\pi$. The fifth power of the sound speed makes the transverse waves dominate. At equal deformation potentials they carry $2(c_{\mathrm{LA}}/c_{\mathrm{TA}})^5 = 16.6$ times the longitudinal rate in vitreous silica, with the sound speeds of Sect. 16.7, and 65 times in aluminum. A thermal defect, with $\Delta_{\mathrm{tl}} \approx T$ and fixed $\Delta_0/\Delta_{\mathrm{tl}}$, relaxes at a rate $\propto T^3$, and since $\Delta_0 \propto \mathrm{e}^{-\lambda_{\mathrm{tun}}}$ the rate spans many decades across the broad distribution of tunneling exponents λ_{tun}. A defect of 5 GHz, $\Delta_{\mathrm{tl}} = 20.7\,\mu\mathrm{eV}$, with $\Delta_0 = 0.3\,\Delta_{\mathrm{tl}}$ and $\gamma_{\mathrm{l}} = \gamma_{\mathrm{t}} = 1\,\mathrm{eV}$, relaxes in aluminum at $2.1 \times 10^5\,\mathrm{s}^{-1}$ below about 50 mK ($2.5 \times 10^5\,\mathrm{s}^{-1}$ at 0.1 K; Fig. 16.22).

16.8.2 Relaxation by conduction electrons

In a metal the defect has a second bath. When it tunnels, the potential that scatters the conduction electrons changes, and the Fermi sea must rearrange. The excitations of the sea are electron–hole pairs, and near the Fermi level their number at energy ω grows as ω while the matrix elements stay constant, so the bath is ohmic, Eq. (C.171), with $J_{\mathrm{s}}^{\mathrm{el}}(\omega) = 2\pi\alpha_{\mathrm{ohm}}\omega$ for ω far below the cutoff. In Eq. (16.68) it gives

$$\Gamma_1^{\mathrm{el}} = \pi\alpha_{\mathrm{ohm}}\Big(\frac{\Delta_0}{\Delta_{\mathrm{tl}}}\Big)^2 \Delta_{\mathrm{tl}} \coth \frac{\Delta_{\mathrm{tl}}}{2T}, \tag{16.70}$$

the rate of Black and coworkers, as quoted by Meißner et al. [35, Eq. (8)] with their coupling $W_{\mathrm{el}}^2 = 4\alpha_{\mathrm{ohm}}$. For a thermal defect it grows only linearly in T, against T^3 for one phonon, and below a kelvin the electrons win by orders of magnitude. For the 5 GHz defect above in normal aluminum, $\Gamma_1^{\mathrm{el}} = 2.5 \times 10^9\,\mathrm{s}^{-1}$ at 10 mK with $\alpha_{\mathrm{ohm}} = 0.28$, against $2.1 \times 10^5\,\mathrm{s}^{-1}$ for one phonon.

The coupling α_{ohm} is fixed by how differently the two configurations scatter electrons at the Fermi level. Let the two configurations act on the electrons as contact potentials $\pm V_{\mathrm{d}}$ over a volume Ω. To second order in V_{d} the electron–hole pairs then form a Gaussian bath, and their spectral density gives (Appendix D.42)

$$\alpha_{\mathrm{ohm}} = 4K_{\mathrm{b}}, \qquad K_{\mathrm{b}} = (\mathcal{N}_{\mathrm{F}} V_{\mathrm{d}} \Omega)^2, \tag{16.71}$$

with $\mathcal{N}_{\mathrm{F}}$ the density of states per spin. This is the coupling K of Maillet et al. [44, Eq. (4)], whose rate $4\pi K \Delta_0^2 \coth(\Delta_{\mathrm{tl}}/2T)/\Delta_{\mathrm{tl}}$ is Eq. (16.70); their fits to the damping of aluminum-coated nanomechanical resonators give $K = 0.07$ and 0.11, that is $\alpha_{\mathrm{ohm}} = 0.28$ and 0.44, with a single-spin density of states. Beyond the Born limit the coupling is Eq. (E.73), a sum over the squared differences $\tilde{\delta}_\nu$ of the scattering phases of the two configurations. For one contact channel per spin the phases are $\mp \arctan(\pi \mathcal{N}_{\mathrm{F}} V_{\mathrm{d}} \Omega)$, and $\alpha_{\mathrm{ohm}} = (4/\pi^2) \arctan^2(\pi K_{\mathrm{b}}^{1/2})$, which saturates at one when the phase difference reaches π. The Born value is 10 % high already at $K_{\mathrm{b}} = 0.017$ (Limit Check 16.14). At the fitted $K_{\mathrm{b}} = 0.07$ the same contact potential gives $\alpha_{\mathrm{ohm}} = 0.19$, not 0.28; read the other way, the measured rate needs a contact potential 32 % stronger than the Born estimate.

A first-principles α_{ohm} needs the two relaxed configurations of the defect, from a nudged-elastic-band search (Appendix E.11). It also needs the scattering matrices of the conduction electrons at the Fermi level in both, from an impurity Green's function or from the overlap of the two many-electron ground states as a function of the supercell size (Appendix E.10). Appendix E.22 specifies the calculation for a crowdion in tungsten, hydrogen in niobium and two-level systems in aluminum, its oxide and amorphous WSi, with the Born limit and the resonator fits as acceptance tests. No such calculation exists yet, and the fitted values are the only numbers in hand.

Below T_c the electrons pair. A defect with $\Delta_{\mathrm{tl}} < 2\Delta_{\mathrm{BCS}}$ cannot break a pair, and it can only scatter the thermal quasiparticles of Sect. 7.2, whose number falls as $\mathrm{e}^{-\Delta_{\mathrm{BCS}}/T}$. A static change of potential is even under time reversal, so the quasiparticle matrix elements carry the case-I coherence factor of the electron density, and the golden-rule integral over the quasiparticle energies is done in Appendix D.42. For Δ_{tl} and T both small compared with Δ_{BCS} it gives

$$\frac{\Gamma_1^{\mathrm{s}}}{\Gamma_1^{\mathrm{el}}} = \frac{2}{1 + \mathrm{e}^{\Delta_{\mathrm{BCS}}/T}} \, \sinh \frac{\Delta_{\mathrm{tl}}}{2T} \, K_1\Big(\frac{\Delta_{\mathrm{tl}}}{2T}\Big), \tag{16.72}$$

with $K_1(u) = \int_0^\infty \mathrm{d}\theta\, \mathrm{e}^{-u\cosh\theta} \cosh\theta$ the modified Bessel function. The last two factors tend to one as $\Delta_{\mathrm{tl}}/T \to 0$, and the remaining factor $2/(1 + \mathrm{e}^{\Delta_{\mathrm{BCS}}/T})$, twice the occupation of the gap edge, is the superconducting form of Meißner et al. [35, Eq. (9)], which they state for $\Delta_{\mathrm{tl}} \ll \Delta_{\mathrm{BCS}}$. The same factor gives the ratio of ultrasonic attenuation in the superconducting and normal states, and for the same reason: both probes scatter quasiparticles through the density. The Bessel factor arises because Δ_{tl} is not small compared with T. Normal electrons accept Δ_{tl} from the whole Fermi sea, while quasiparticles crowd at the gap edge and the energy must be taken up within their thermal spread. The factor is 0.86 for a thermal defect with $\Delta_{\mathrm{tl}} = T$ and 0.71 at $\Delta_{\mathrm{tl}} = 2T$, and the form of the source holds within 10 % only for $\Delta_{\mathrm{tl}} \le 0.7\,T$ (Limit Check 16.14).

Fig. 16.22 puts the three rates together for one defect in aluminum. In the normal state the electrons dominate at every temperature. Below T_c the suppression is so strong that one

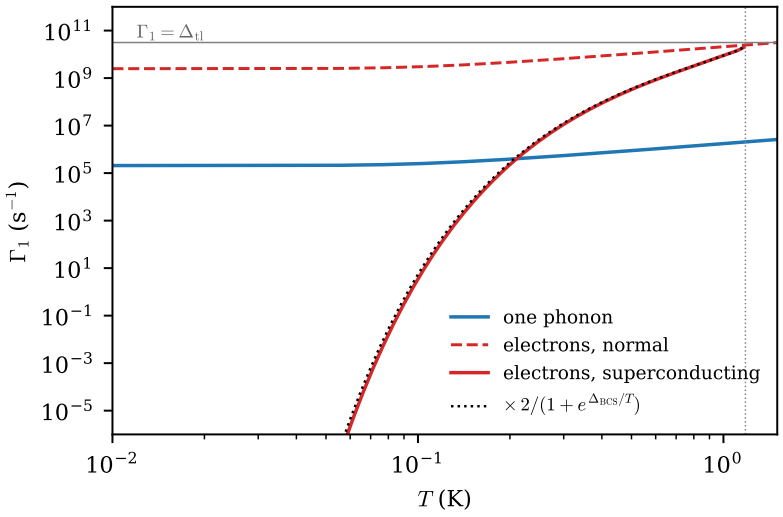


Figure 16.22: Relaxation rates of one two-level system in aluminum against temperature, for $\Delta_{tl} = 20.7\,\mu\text{eV}$ (5 GHz) and $\Delta_0 = 0.3\,\Delta_{tl}$. Blue: one phonon, Eq. (16.53), with the model deformation potentials $\gamma_l = \gamma_t = 1\,\text{eV}$ and the isotropic sound speeds $c_{LA} = 6.63\,\text{km s}^{-1}$, $c_{TA} = 3.30\,\text{km s}^{-1}$ and density $2.70\,\text{g cm}^{-3}$ of aluminum (Sect. 20.3). Red dashed: conduction electrons in the normal state, Eq. (16.70), with $\alpha_{ohm} = 0.28$, continued below $T_c = 1.18\,\text{K}$ (vertical line) as for a film held normal. Red solid: the superconducting state, from the quasiparticle integral of Appendix D.42 with the weak-coupling gap $\Delta_{BCS}(T)$; black dotted: the normal rate times $2/(1+e^{\Delta_{BCS}/T})$. The gray line marks $\Gamma_1 = \Delta_{tl}$, which the normal-state rate reaches at 1.5 K, where the golden rule fails.

phonon takes over below 0.21 K, about $T_c/6$, and at 50 mK the electron rate is $2\times10^{-9}\,\text{s}^{-1}$. A superconducting film at the base temperature of a dilution refrigerator thus leaves its two-level systems coupled to phonons alone, as an insulator would.

Limit Check 16.14. Two-level relaxation by phonons and electrons

The published rates are Jäckle's one-phonon rate and Black's electron rate, Eqs. (16.53) and (16.70), with the superconducting factor $2/(1+e^{\Delta_{BCS}/T})$ [35, Eqs. (7)–(9)], and the Born-limit coupling $\alpha_{ohm} = 4K_b$ [44, Eq. (4)].

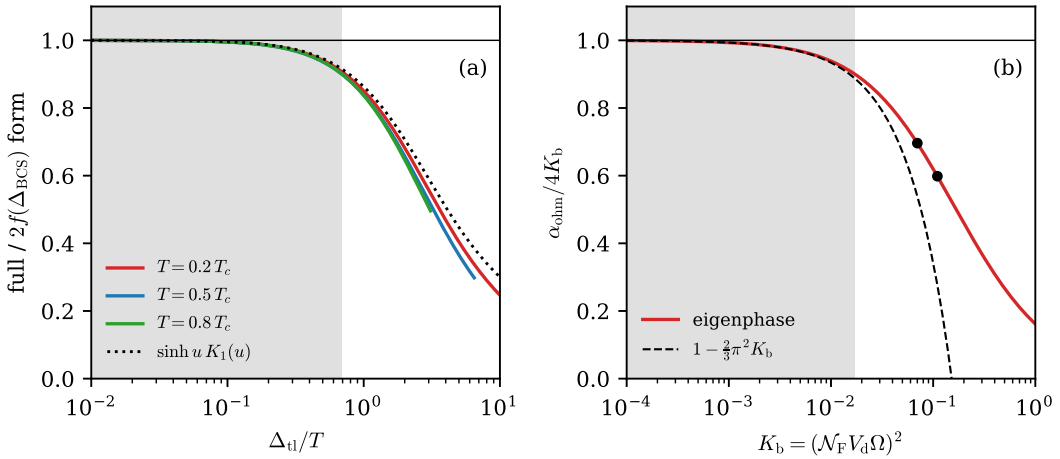


Figure 16.23: (a) Electron relaxation rate of a two-level system in superconducting aluminum, from the quasiparticle integral with the case-I coherence factor and the weak-coupling gap ($T_c = 1.18\,\mathrm{K}$), divided by the normal rate times $2/(1+\mathrm{e}^{\Delta_{\mathrm{BCS}}/T})$, against Δ_{tl}/T at $T = 0.2$, 0.5 and $0.8\,T_c$; each curve ends at $\Delta_{\mathrm{tl}} = 2\Delta_{\mathrm{BCS}}(T)$. Dotted: the closed form $\sinh u\,K_1(u)$, $u = \Delta_{\mathrm{tl}}/2T$, of Eq. (16.72). Gray: within 10 % of one, $\Delta_{\mathrm{tl}} \le 0.69$–$0.73\,T$. (b) Ohmic coupling from the scattering phases of one contact channel per spin, $\alpha_{\mathrm{ohm}} = (4/\pi^2)\arctan^2(\pi K_{\mathrm{b}}^{1/2})$, divided by the Born value $4K_{\mathrm{b}}$, against K_{b}. Dashed: the leading correction $1 - \frac{2}{3}\pi^2 K_{\mathrm{b}}$. Dots: the resonator fits $K_{\mathrm{b}} = 0.07$ and 0.11. Gray: within 10 %, $K_{\mathrm{b}} \le 0.017$.

For $\Delta_{\mathrm{tl}} \gtrsim T$ the quasiparticles cannot take up the splitting as freely as normal electrons, and for $K_{\mathrm{b}} \gtrsim 0.02$ the phase shift of a single channel is no longer small, so that the Born coupling overstates α_{ohm}.

16.8.3 The ohmic scaling limit

The fitted couplings are not small, and Fig. 16.22 shows the consequence: the normal-state rate of Eq. (16.70) reaches the splitting itself at 1.5 K, and for a symmetric defect, $\Delta_0 = \Delta_{\mathrm{tl}}$, it is at least $0.88\,\Delta_{\mathrm{tl}}$ at every temperature, since $\Gamma_1^{\mathrm{el}}/\Delta_{\mathrm{tl}} = \pi\alpha_{\mathrm{ohm}}\coth(\Delta_{\mathrm{tl}}/2T)$ with $\alpha_{\mathrm{ohm}} = 0.28$. A defect damped at a rate comparable to its splitting has no eigenstates to relax between. The electrons then dephase the two configurations faster than the defect tunnels, and the defect hops between $|\mathrm{L}\rangle$ and $|\mathrm{R}\rangle$ at the incoherent rate of Eq. (C.174), second order in Δ_0 and exact in α_{ohm}. Appendix C.20 evaluates it for the ohmic bath, Eq. (C.175), and shows that for $\Delta_0, T \ll \omega_{\mathrm{cut}}$ the cutoff enters only through the renormalized tunneling amplitude Δ_{r} of Eq. (C.176). For a symmetric defect the relaxation rate $\Gamma_{\mathrm{LR}} + \Gamma_{\mathrm{RL}}$ is then

$$\Gamma_{\mathrm{LR}} + \Gamma_{\mathrm{RL}} = \frac{\Gamma(1+\alpha_{\mathrm{ohm}})}{\Gamma(1-\alpha_{\mathrm{ohm}})}\,\frac{\Delta_{\mathrm{r}}^{2(1-\alpha_{\mathrm{ohm}})}}{\alpha_{\mathrm{ohm}}}\,(2\pi T)^{2\alpha_{\mathrm{ohm}}-1}, \qquad \Delta_{\mathrm{r}} \ll T \ll \omega_{\mathrm{cut}}, \tag{16.73}$$

with $\Gamma(z)$ Euler's gamma function; the form follows from Eq. (C.175) at $\Delta_{\mathrm{as}} = 0$ with the duplication and reflection formulas of Γ. The rate depends on Δ_0 and ω_{cut} only through Δ_{r}, which is what makes it a scaling limit. It grows as the temperature falls whenever $\alpha_{\mathrm{ohm}} < \frac{1}{2}$, the anti-Arrhenius law: at $\alpha_{\mathrm{ohm}} = 0.28$ it scales as $T^{-0.44}$, and as $\alpha_{\mathrm{ohm}} \to 0$ as $1/T$. A thermally activated rate would do the opposite. Appendix C.20 traces the growth to the slower dephasing by the electrons at low temperature, over the time $1/T$, which wins against the shrinking dressed amplitude.

Qureshi obtained the same dynamics from the bosonized electron gas [45]. Without electron-assisted tunneling his Laplace-transformed correlation function of σ_z [45, Eqs. (8), (9)] is the noninteracting-blip result of Appendix C.20, and his coupling K is α_{ohm} in the normalization of Eq. (C.171), not K_{b}. For a symmetric defect and $\alpha_{\mathrm{ohm}} \ll 1$ he reduces it to [45, Eq. (10)]

$$\begin{aligned} \hat{C}(z) &\simeq \frac{1}{z + \tilde{\Delta}_0^2/(z+\gamma_0)}, \qquad \gamma_0 = 2\pi\alpha_{\mathrm{ohm}}T, \\ \tilde{\Delta}_0^2 &= \Gamma(1-2\alpha_{\mathrm{ohm}})\cos(\pi\alpha_{\mathrm{ohm}})\,\Delta_0^2\Big(\frac{2\pi T}{\omega_{\mathrm{cut}}}\Big)^{2\alpha_{\mathrm{ohm}}}. \end{aligned} \tag{16.74}$$

Its poles describe both regimes of this section. For $\tilde{\Delta}_0 > \gamma_0/2$ the defect oscillates at a frequency near $\tilde{\Delta}_0$ and its σ_z correlation decays at $\gamma_0/2 = \pi\alpha_{\mathrm{ohm}}T$. For a symmetric defect with $T \gg \Delta_{\mathrm{tl}}$ this is half of Eq. (16.70), the decay of an off-diagonal quantity at $\Gamma_1/2$, so Black's rate is the weakly damped side of Eq. (16.74). For $\tilde{\Delta}_0 < \gamma_0/2$ the motion is overdamped and the slow pole gives the relaxation rate $\tilde{\Delta}_0^2/\gamma_0$. By Eq. (C.176), $\tilde{\Delta}_0^2 = \Delta_{\mathrm{r}}^{2(1-\alpha_{\mathrm{ohm}})}(2\pi T)^{2\alpha_{\mathrm{ohm}}}$,

and $\tilde{\Delta}_0^2/\gamma_0$ is Eq. (16.73) without the factor $\Gamma(1+\alpha_{\rm ohm})/\Gamma(1-\alpha_{\rm ohm}) = 1-2\gamma_{\rm E}\alpha_{\rm ohm}+O(\alpha_{\rm ohm}^2)$, which is 10 % below one at $\alpha_{\rm ohm} = 0.091$ and 0.71 at the fitted 0.28 (Limit Check 16.15). The boundary $\tilde{\Delta}_0 = \gamma_0/2$ is a temperature,

$$T_{\rm coh} = \frac{\Delta_{\rm r}}{2\pi}\Big(\frac{2}{\alpha_{\rm ohm}}\Big)^{1/(1-\alpha_{\rm ohm})}, \tag{16.75}$$

below which a symmetric defect tunnels coherently, with weak damping, and above which it hops. For $\alpha_{\rm ohm} \ll 1$ it is $\Delta_{\rm r}/\pi\alpha_{\rm ohm}$, the temperature at which $\pi\alpha_{\rm ohm}T$ reaches the splitting; at $\alpha_{\rm ohm} = 0.28$ the small-coupling form of Eq. (16.74) puts it at $2.4\,\Delta_{\rm r}$, where the scaling form Eq. (16.73) is itself at the edge of its range. An asymmetric defect, $\Delta_{\rm as} \gg \Delta_0$, is weakly damped as long as Eq. (16.70) stays below $\Delta_{\rm tl}$, and most defects of a glass are of this kind (Sect. 16.7); the scaling limit concerns the nearly symmetric ones and light interstitials in metals.

Limit Check 16.15. The ohmic scaling limit

The published relaxation rate of a symmetric tunneling defect in a metal is $\tilde{\Delta}_0^2/\gamma_0$, from the slow pole of Eq. (16.74), for $\alpha_{\rm ohm} \ll 1$ [45, Eqs. (8)–(10)].

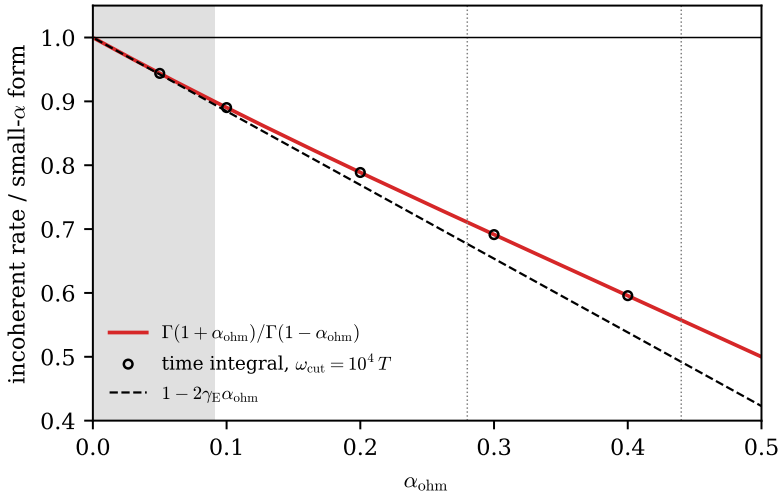


Figure 16.24: Incoherent relaxation rate of a symmetric defect in an ohmic bath, Eqs. (16.73) and (C.174), divided by the small-coupling form $\tilde{\Delta}_0^2/\gamma_0$, against $\alpha_{\rm ohm}$. Red: the scaling limit $\Gamma(1+\alpha_{\rm ohm})/\Gamma(1-\alpha_{\rm ohm})$. Open circles: the time integral of Eq. (C.174) done numerically with the full ohmic correlation exponent at $\omega_{\rm cut} = 10^4\,T$. Dashed: the leading correction $1-2\gamma_{\rm E}\alpha_{\rm ohm}$. Dotted lines: the Born values $\alpha_{\rm ohm} = 0.28$ and 0.44 of the aluminum resonator fits. Gray: within 10 %, $\alpha_{\rm ohm} \le 0.091$.

Outside the gray band the small-coupling form overstates the rate, by 1/0.71 at $\alpha_{\rm ohm} = 0.28$, and both forms fail below $T \sim \Delta_{\rm r}$, where the defect tunnels coherently.

16.8.4 Thermally activated hopping and tunneling

A defect whose barrier $V_{\rm b}$ of Sect. 16.7 is large compared with T can still cross it classically, at the rate $\Gamma = \Gamma_0 e^{-V_{\rm b}/T}$ of harmonic transition-state theory (Appendix E.11), with Γ_0 the attempt rate. In one dimension $\Gamma_0 = \omega_{\rm w}/2\pi$, with $\omega_{\rm w}$ the frequency of the well, and in many dimensions Γ_0 is Vineyard's ratio of the products of normal-mode frequencies at the minimum and at the saddle. This is the transition-state rate of Lawrence [46, Eq. (1)] with a classical,

harmonic reactant partition function and transmission coefficient $\kappa = 1$. Tunneling through the top of the barrier raises the rate. For a barrier that is an inverted parabola of frequency $\omega^\ddagger$ the correction is exact,

$$\kappa_{\mathrm{PB}} = \frac{\omega^\ddagger/2T}{\sin(\omega^\ddagger/2T)} = 1 + \frac{1}{24}\Big(\frac{\omega^\ddagger}{T}\Big)^2 + \frac{7}{5760}\Big(\frac{\omega^\ddagger}{T}\Big)^4 + \dots , \tag{16.76}$$

and its first two terms are Wigner's correction κ_{W} [46, Eqs. (2), (3)]. The parabolic correction diverges at the crossover temperature

$$T_\times = \frac{\omega^\ddagger}{2\pi}, \tag{16.77}$$

below which the particle no longer crosses near the top. It tunnels through the barrier at the energy that maximizes the product of the thermal weight and the transmission, along the instanton, the periodic path in imaginary time of period $1/T$, whose action S_{inst} replaces V_{b}/T in the exponent [46, Eq. (4)]. As $T \to 0$ the instanton becomes the tunneling path between the two minima, its action becomes twice the tunneling exponent λ_{tun} of Sect. 16.7, and the hopping turns into the tunneling between $|\mathrm{L}\rangle$ and $|\mathrm{R}\rangle$ with amplitude $\Delta_0 \propto \mathrm{e}^{-\lambda_{\mathrm{tun}}}$ that the previous sections assumed. For an atom with $\omega^\ddagger = 10\,\mathrm{meV}$, $T_\times = 18.5\,\mathrm{K}$, so at millikelvin temperatures every defect motion is tunneling, and a defect whose λ_{tun} is too large for it to tunnel within the time of an experiment is frozen. With an attempt rate of $10^{13}\,\mathrm{s}^{-1}$ and a year of observation, the bound is $\lambda_{\mathrm{tun}} \le 23.6$ when no energy-gap suppression adds to the exponent (Appendix E.22).

Limit Check 16.16 tests these forms on a barrier whose quantum rate is known exactly, the symmetric Eckart barrier $V(x) = V_{\mathrm{b}}\,\mathrm{sech}^2(x/L)$ with the hydrogen-atom parameters of Lawrence, $V_{\mathrm{b}} = 6\omega^\ddagger/\pi$. The exact rate exceeds the classical one by a factor 6.2 at $T_\times$ and 2.2 at $1.5\,T_\times$, so tunneling matters well above the crossover. Wigner's correction holds within 10 % above $2.1\,T_\times$ and the parabolic form above $1.5\,T_\times$, while the instanton rate stays about 18 % low between 0.25 and $0.5\,T_\times$, an error that the higher terms of its asymptotic expansion in $\hbar$ would reduce [46, Sect. III A].

Activation also fixes which defects of a glass contribute to its heat release. On cooling, defects with large barriers stay in equilibrium by activation until the activated rate falls below the rate at which the cooling changes V_{b}/T, which is $V_{\mathrm{b}}|\dot{T}|/T^2$. Equating the two gives the freezing temperature

$$T_{\mathrm{frz}} = \frac{V_{\mathrm{b}}}{\ln(\Gamma_0 T_{\mathrm{frz}}^2/V_{\mathrm{b}}|\dot{T}|)}, \tag{16.78}$$

the form of Sahling et al. [43, Eq. (8)] with their $\tau_0 = 1/\Gamma_0$. Below T_{frz} the populations of these defects stop following the temperature, and the excess they keep is released later by tunneling.

Limit Check 16.16. Hopping and tunneling through one barrier

The published rates are transition-state theory with Wigner's or the parabolic-barrier transmission, Eq. (16.76), above the crossover temperature Eq. (16.77), and the instanton rate below it [46, Eqs. (1)–(4)].

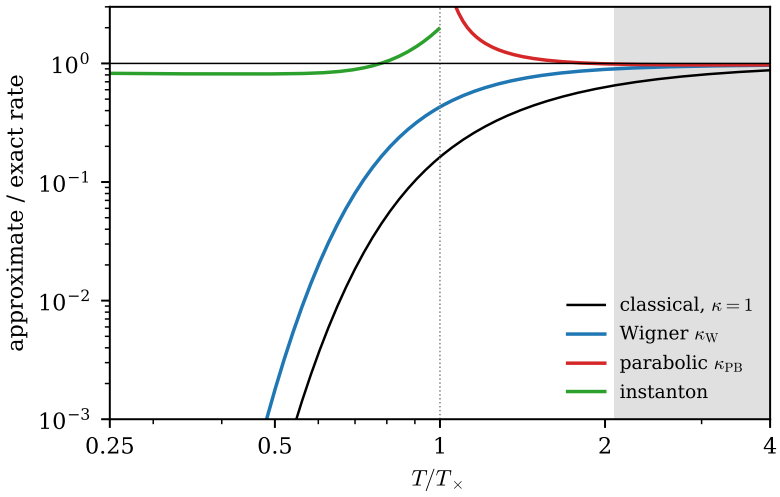


Figure 16.25: Approximate thermal rates through a symmetric Eckart barrier divided by the exact quantum rate, against $T/T_\times$, for the hydrogen-atom parameters of Lawrence ($V_b = 248\,\mathrm{meV}$, $\omega^\ddagger = 130\,\mathrm{meV}$, $T_\times = 240\,\mathrm{K}$). Black: classical transition-state theory. Blue: with Wigner's correction. Red: with the parabolic-barrier correction, divergent at $T_\times$ (dotted line). Green: the instanton rate, for $T < T_\times$. The exact rate integrates the exact transmission probability of the Eckart barrier over the Boltzmann factor. Gray: Wigner's form within 10 %, $T \geq 2.1\,T_\times$.

Near $T_\times$ the expansion in $\omega^\ddagger/T$ diverges and the instanton is not yet formed, so neither form holds there, and below $T_\times$ the transition-state forms underestimate the rate by orders of magnitude.

16.8.5 Heat release of a cooled glass

A glass held at T_{hi} and cooled quickly to T_{lo} keeps its slow defects in the populations of T_{hi}. A defect of splitting Δ_{tl} has the mean energy $-\frac{1}{2}\Delta_{tl}\tanh(\Delta_{tl}/2T)$ in equilibrium, so it holds the excess $\frac{1}{2}\Delta_{tl}[\tanh(\Delta_{tl}/2T_{lo}) - \tanh(\Delta_{tl}/2T_{hi})]$, which it gives to the phonons at its rate Γ_1 of Eq. (16.53), evaluated at T_{lo}. In the standard tunneling model the defects are spread uniformly in the asymmetry Δ_{as} and in the tunneling exponent λ_{tun}, with density $\bar{P}$ per unit energy, per unit λ_{tun} and per unit volume [43, Eq. (1)]. The slow defects have $\Delta_0 \ll \Delta_{as}$, so $\Delta_{tl} \simeq \Delta_{as}$ is distributed uniformly too. Their excess energy, summed over Δ_{tl}, is

$$\int_0^\infty \mathrm{d}\Delta_{tl}\,\frac{\Delta_{tl}}{2}\Big[\tanh\frac{\Delta_{tl}}{2T_{lo}} - \tanh\frac{\Delta_{tl}}{2T_{hi}}\Big] = \frac{\pi^2}{12}(T_{hi}^2 - T_{lo}^2),$$

since $1 - \tanh(x/2T) = 2/(\mathrm{e}^{x/T}+1)$ and $\int_0^\infty \mathrm{d}x\,x/(\mathrm{e}^{x/T}+1) = \pi^2T^2/12$. The rate depends on the tunneling exponent through $\Delta_0^2 \propto \mathrm{e}^{-2\lambda_{tun}}$, so at fixed Δ_{tl} a uniform density in λ_{tun} is a uniform density $\frac{1}{2}$ per unit $\ln\Gamma_1$. A defect relaxes at time t with probability $\Gamma_1\mathrm{e}^{-\Gamma_1 t}\mathrm{d}t$, and summing over λ_{tun} gives $\int \mathrm{d}\lambda_{tun}\,\Gamma_1\mathrm{e}^{-\Gamma_1 t} = \frac{1}{2}\int \mathrm{d}\Gamma_1\,\mathrm{e}^{-\Gamma_1 t} = 1/2t$ whenever the rates extend far on both sides of $1/t$. The power released by a volume $\mathcal{V}$ of glass is therefore

Heat release of a cooled glass

$$\dot{Q}_{TLS}(t) = \frac{\pi^2}{24}\,\bar{P}\,\mathcal{V}\,\frac{T_{hi}^2 - T_{lo}^2}{t}, \qquad \Gamma_{max}^{-1} \ll t \ll \Gamma_{min}^{-1}, \tag{16.79}$$

with $\Gamma_{\max}$ and $\Gamma_{\min}$ the fastest and slowest one-phonon rates at T_{lo} of the defects with $\Delta_{\mathrm{tl}} \sim T_{\mathrm{hi}}$, which hold most of the excess. This is the law of Sahling et al. [43, Eq. (6)]. The $1/t$ comes from the uniform density in λ_{tun}, and the prefactor depends neither on the deformation potentials nor on the sound speeds, which only shift the window in time. Since every decade of time draws on defects in a range of λ_{tun} of width $\frac{1}{2}\ln 10$, the heat released between t_1 and t_2 is the prefactor times $\ln(t_2/t_1)$, and the heat measured at a time t comes from defects with $\Gamma_1 \approx 1/t$. If the glass starts above T_{frz} of Eq. (16.78), the defects that release at time t were last in equilibrium at T_{frz}, and T_{frz} replaces T_{hi} [43, Eq. (7)]; the release then no longer grows with the starting temperature.

Limit Check 16.17. Heat release of a cooled glass

The published heat release after a quench from T_{hi} to T_{lo} is Eq. (16.79) [43, Eqs. (1), (6)–(8)].

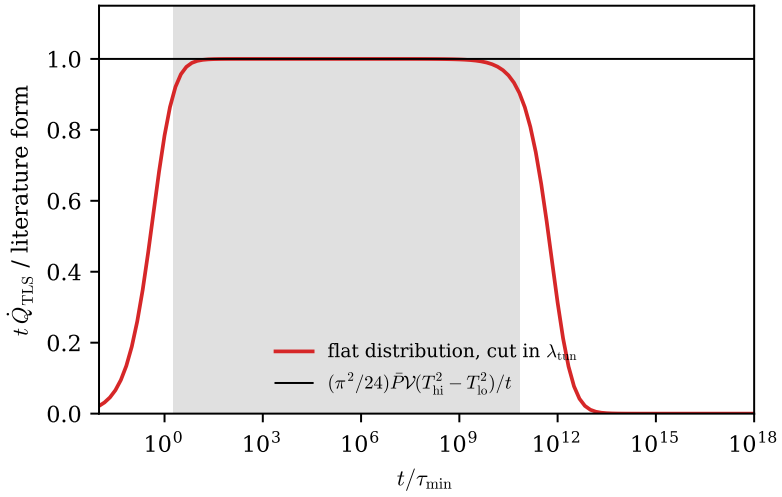


Figure 16.26: Heat release of a model glass times t, divided by the prefactor of Eq. (16.79), against t in units of $\tau_{\min}$, the relaxation time of the fastest defect at $\Delta_{\mathrm{tl}} = T_{\mathrm{hi}}$. Red: the uniform distribution cut at $\lambda_{\min} \le \lambda_{\mathrm{tun}} \le \lambda_{\min} + 14$, with the exact splitting $\Delta_{\mathrm{tl}} = (\Delta_{\mathrm{as}}^2 + \Delta_0^2)^{1/2}$, $\Delta_0 \le 0.1\,T_{\mathrm{lo}}$, the one-phonon rate Eq. (16.53) at T_{lo} and an instantaneous quench with $T_{\mathrm{lo}}/T_{\mathrm{hi}} = 1/480$ (10 mK after 4.8 K). Gray: within 10 % of one, $1.8\,\tau_{\min} \le t \le 7.1 \times 10^{10}\,\tau_{\min}$, 10.6 of the 12.2 decades that the cut spans.

Outside the window the release falls below the $1/t$ law, at early times because the fastest defects have already relaxed and at late times because the distribution of tunneling exponents ends.

Inside its window Eq. (16.79) is exact for the uniform distribution: in Limit Check 16.17 it holds within 2×10^{-4} from 10^2 to $10^8\,\tau_{\min}$. The tests of the law are therefore tests of the distribution. Heat-release measurements on vitreous silica follow $1/t$ and $T_{\mathrm{hi}}^2 - T_{\mathrm{lo}}^2$, but they give a density $\bar{P}$ four times smaller than the specific heat measured over seconds, 2.0×10^{44} against $8.0 \times 10^{44}\,\mathrm{J^{-1}\,m^{-3}}$ [47, Table 1]. Sahling et al. attribute the difference to defects whose barriers are lowered by local stresses frozen in on cooling. These relax within about 100 s, before the heat-release measurement starts, and a small fraction of them, 5 %, freezes at 14.5 K instead of 4.8 K and adds to the release after a quench from higher temperatures [47]. The specific-heat value, $1.28 \times 10^{20}\,\mathrm{eV^{-1}\,cm^{-3}}$, is close to the $1.2 \times 10^{20}\,\mathrm{eV^{-1}\,cm^{-3}}$ that the polarization-averaged acoustic tunneling strength implies for $\gamma_{\mathrm{l}} = 1\,\mathrm{eV}$ (0.88×10^{20} with the

longitudinal C_l; Sect. 16.7).

Worked example 16.8: Heat release of one gram of fused silica at 10 mK

Take 1 g of vitreous silica, $\mathcal{V} = 0.455\,\mathrm{cm}^3$ at the density $\rho_T = 2.2\,\mathrm{g\,cm}^{-3}$ of Sect. 16.7, cooled from room temperature to $T_\mathrm{lo} = 10\,\mathrm{mK}$. The defects that release over hours to weeks freeze at $T_\mathrm{frz} = 4.8\,\mathrm{K}$, and the heat-release density is $\bar{P} = 2.0 \times 10^{44}\,\mathrm{J}^{-1}\,\mathrm{m}^{-3} = 3.20 \times 10^{19}\,\mathrm{eV}^{-1}\,\mathrm{cm}^{-3}$ [47, Table 1]. With T_frz in place of T_hi, and $T_\mathrm{lo}^2/T_\mathrm{frz}^2 = 4 \times 10^{-6}$ negligible, Eq. (16.79) gives

$$\begin{aligned} t\,\dot{Q}_\mathrm{TLS} &= \frac{\pi^2}{24}\,k_\mathrm{B}^2\,\bar{P}\,\mathcal{V}\,T_\mathrm{frz}^2 \\ &= 0.4112 \times (1.381 \times 10^{-23}\,\mathrm{J\,K}^{-1})^2 \times 2.0 \times 10^{44}\,\mathrm{J}^{-1}\,\mathrm{m}^{-3} \\ &\qquad \times 4.55 \times 10^{-7}\,\mathrm{m}^3 \times (4.8\,\mathrm{K})^2 \\ &= 1.64 \times 10^{-7}\,\mathrm{J}. \end{aligned}$$

One hour after the cooling the sample releases $1.64 \times 10^{-7}\,\mathrm{J}/3600\,\mathrm{s} = 46\,\mathrm{pW}$, after a day 1.9 pW, and after a week 0.27 pW. Between the first hour and the end of the first week it gives off $1.64 \times 10^{-7}\,\mathrm{J} \times \ln 168 = 0.84\,\mu\mathrm{J}$, or $5 \times 10^{12}\,\mathrm{eV}$. With the specific-heat density the power would be four times larger. The fraction of defects that freezes at 14.5 K is $0.05 \times 3 = 0.15$ of the normal density, with 3 the ratio of the stress-lowered defects to the normal density, $(8.0 - 2.0)/2.0$. By the square of its freezing temperature it adds another 1.4 times the normal release after a quench from above 14.5 K, with the freezing temperatures of the table of Sahling et al. Their text quotes 5.5 and 14.1 K, which give 0.99, and states that the anomalous part is roughly twice the normal one [47]. A hundred grams of fused silica near a detector at 10 mK thus deposits 4.6 nW an hour after cooling and 0.19 nW a day later, a load and a source of delayed events that Sect. 19.6 takes up.

16.8.6 Interacting two-level systems

Every result of this section treats the defects as independent, yet two defects interact through the strain fields each creates, with a coupling that falls as $1/r^3$. Its typical size relative to the spread of their splittings is the tunneling strength $C = \bar{P}\gamma_\mathrm{l}^2/\rho_T c_\mathrm{LA}^2$, 2.6×10^{-4} in vitreous silica (Sect. 16.7). Thermal flips of neighbors shift the splitting of a given defect in time, which is spectral diffusion; it broadens single lines, and qubit spectroscopy sees it directly [36, Sect. IV]. An ensemble whose spectrum is flat in Δ_tl is unchanged by a normalized shift kernel away from the edges of the band. Spectral diffusion alone therefore leaves the ensemble responses of Sect. 16.7 intact, but it changes which defects relax at a given time and so the window of Eq. (16.79). The phonon emitted by one relaxing defect can trigger the relaxation of another. If each release triggers on average b_br more, a branching process, the mean size of a cascade of releases is $1/(1 - b_\mathrm{br})$, finite for $b_\mathrm{br} < 1$, and the distribution of the cascade size N_casc approaches $N_\mathrm{casc}^{-3/2}$ as $b_\mathrm{br} \to 1$. Such avalanches would turn the steady $1/t$ release into bursts. None of this is quantified for a real glass: b_br needs the density of barriers below the energy of the emitted phonons and the flux of ballistic phonons through the triggering volume, and no calculation or measurement of it exists. Whether the delayed events of a cooled detector are independent relaxations or avalanches of this kind is the open question that Sect. 19.6 inherits.

16.9 Material inputs of the novel targets

Two first-principles calculations of the dielectric tensor of $ZrTe_5$ give 20.75 and 9.8 along its slow axis, and the scattering reach of the material moves by a factor 2.8 between them (Worked example 16.9). Both numbers enter Eq. (16.1) exactly; no approximation of the response is involved, and a measurement of the tensor would settle the question. Most targets of this chapter are limited in the same way. The loss tangent of amorphous SiO_2 changes seventeenfold between preparations. The oscillator strength of a donor line in silicon changes by a factor 3.3 between two models of the donor ground state. And the probability that a single nuclear recoil of a few hundred eV in LiF leaves a readable F center has not been measured. Table 16.1 lists, for each target, the inputs its rate needs, how well they are known, how their uncertainty moves the reach, and where the related open problem is stated. From it we read reach bands and sort the targets into those limited by theory, by a calculation and by data.

16.9.1 Measured, computed and unmeasured inputs

Some inputs of Table 16.1 are measured constants that vary from sample to sample: the loss tangent of a glass, the width of a donor line, the work function of a sheet on its substrate, the Luttinger parameter of a nanotube. The spread between samples is then the uncertainty of the input, and a detector that characterizes its own crystal or film removes it. A donor line, for example, is 4.22 µeV wide in isotopically purified silicon and broadens at the densities where the impurity band forms, and the resonant rate of Worked example 16.5 scales as the inverse width.

Others are computed constants, such as the dielectric tensor and the half-gap of a Dirac material, the Bloch-overlap coefficients $\mathcal{U}_{j'j}(\boldsymbol{G})$ of a donor, or the ohmic coupling α_{ohm} of a tunneling defect. Their uncertainty is the spread between calculations, or the spread of a calculation over its functional and numerical choices (Appendices E.1 to E.5). Note that such a spread is an estimate, not a bound: two calculations need not bracket the true value. For $ZrTe_5$ the two published tensors differ by factors of 0.61, 0.47 and 0.72 along the three axes, and the computed half-gaps span 11.8 to 17.5 meV, beside a measured value of 11.75 meV (Sect. 16.1.3).

A third group are yields and parameters that no measurement isolates. A color center counts only if the recoil displaces an anion, the pair survives, the vacancy holds an electron and the readout finds it. None of the four factors of Eq. (16.40) has been measured for single recoils of tens to hundreds of eV at millikelvin temperatures (Sect. 16.6.4). The loss tangent of a glass fixes the product of the density of two-level systems and the square of their dipole moment, and the acoustic loss the product with the squared deformation potential, but neither fixes the factors (Sect. 16.7.6). The core term of the scattering rate needs them one by one, and the book's estimate rests on an assumed deformation potential of 1 eV and an assumed permittivity of 3.9. For these inputs Sect. 8.8 assigns no width, and a reach is conditional on the value assumed.

16.9.2 Theory-limited and data-limited targets

A target is limited by theory when its response has no controlled approximation at the values its inputs take, so that better inputs would not sharpen the rate. Dirac cones at strong coupling, donors near the Mott density and interacting two-level systems are of this kind. A Dirac cone with $\tilde{\alpha} \gtrsim 1$, such as BNQ-TTF with an effective coupling of about ten, has no small parameter beyond the random-phase response (Sect. 16.1.4). Silicon doped to the densities of the published donor projections, $n_{\mathrm{D}}/n_{\mathrm{c}} \approx 0.3$ to 0.6, is no longer a set of isolated

Table 16.1: Material inputs of the novel targets: values, uncertainty and effect on the reach, what limits the target ("theory": no controlled approximation for the response at the input values; "calculation": a known formalism whose evaluation Appendices E.20 to E.22 specifies; "data": a measurement of the sample or of a yield), and the section where the related open problem is stated. The truncations of the responses are in Table 16.2.

Target	Inputs (values)	Uncertainty and effect on the reach	Limited by	Open problem
Dirac material ($ZrTe_5$)	background tensor $\varepsilon_{\mathrm{bg},ii} = 308.4$, 20.75, 126.1 (computed; earlier 187.5, 9.8, 90.9); velocities, cutoff 0.191 eV and half-gap $\Delta_{\mathrm{D}} = 11.8$ meV of the measured-structure row (computed half-gaps 11.8–17.5 meV; measured 11.75 meV)	between the two tensors $\bar{\sigma}_e^{\mathrm{lim}}$ moves by 2.8 to 2.9 and ζ^{lim} by 1.8 to 2.0 (Worked example 16.9); the Dirac term inside the computed tensor, 3 to 8 %, moves $\bar{\sigma}_e^{\mathrm{lim}}$ by 0.88; the computed half-gaps move the threshold mass along the slow axis from 7.1 to 10.6 keV	data (tensor, gap); calculation (edge ladder)	Sect. 16.1 ($\tilde{\alpha} \gtrsim 1$)
Metallic nanotubes, doped graphene	g_c (measured 0.2–0.7; 0.17–0.34 for $r_{\mathrm{t}} = 0.7$ nm in vacuum); Fermi energy; substrate	line weight $\propto N_{\mathrm{ch}} g_c$; first edge of a semiconducting tube at $q_\perp r_{\mathrm{t}} = 0.76$ ($J_0^2 = 0.74$); Fermi energy and substrate set the sheet plasmon	data	none
Layered stacks	layer $\varepsilon_A(\omega)$, thicknesses, filling	effective medium within 10 % for $ka_s < 1.94$, which in the model stack covers $m_\chi \lesssim 14.5$ keV; above, periodic and nonlocal responses	calculation	none
Vacuum ejection (graphene)	work function $\Phi_{\mathrm{wf}} \simeq 4.3$ eV (substrate-dependent); occupied orbitals and binding energies	threshold mass $2\Phi_{\mathrm{wf}}/v_{\mathrm{max}}^2 = 1.22$ MeV moves with Φ_{wf}; orthogonalized final states move $\bar{\sigma}_e^{\mathrm{lim}}$ by 0.69 (10 MeV) and 0.54 (100 MeV), heavy mediator; hole attraction up to the Fermi factor, 7.3 at 10 eV	calculation (final state); data (Φ_{wf})	none
Quantum dots (PbS in silica)	dot gap 1.2 eV at 3.3 nm; $\varepsilon_{\mathrm{in}}(\omega)$ of the dots; host $\varepsilon_{\mathrm{out}}$	depolarization factor 0.57 to 6.7 for the cases of Worked example 16.3, ζ^{lim} moved by 1.33 to 0.39, sign unknown over 2.5–10 eV; hard wall 0.76–1.03 of the scissored bulk at 3–11 confinement energies	data ($\varepsilon_{\mathrm{in}}$)	Sect. 16.3
Molecular crystals (anthracene)	lowest singlet 3.1 eV (measured); form factors and Frenkel coupling $\mathfrak{J}$ (model); crystal structure (measured)	corrections to the binned rate 6 to 38 % in the π-electron model, $\bar{\sigma}_e^{\mathrm{lim}}$ moved by 0.94 to 0.72; charge-transfer admixture about one percent for $t_{\mathrm{CT}} = 0.1$ eV	calculation (periodic finite-q Bethe–Salpeter)	Sect. 16.4
Dopants (Si:P)	level energies (measured); line strengths (model); line width ≥ 4.22 μeV; donor density against $n_{\mathrm{c}} \approx 3 \times 10^{18}\,\mathrm{cm}^{-3}$; local-field ratio $\mathfrak{f}_{\mathrm{D}}^{\mathrm{loc}}$	strengths of the $2p$ lines differ by 3.2–3.4 between ground-state models, $\zeta^{\mathrm{lim}} = 1.3$ to 2.3×10^{-16} at the $2p_0$ line; resonant rate $\propto n_{\mathrm{D}}/\Gamma_{\mathrm{line}}$; $\mathfrak{f}_{\mathrm{D}}^{\mathrm{loc}}$ close to one, unsized; isolated-donor response wrong at order one for $n_{\mathrm{D}}/n_{\mathrm{c}} \approx 0.5$	data (strengths, width); theory (near n_{c})	Sect. 16.5
Color centers (LiF)	displacement threshold $E_{\mathrm{d}} = 27$ eV (simulation input); site phonon energy 33.5–77.2 meV; yields p_{FP}, p_{surv}, p_q, f_{opt}, p_{FH}	lightest mass 0.113–0.255 GeV for $E_{\mathrm{d}} = 10$–50 eV; a directional threshold moves the rate by more than 10 % up to 0.41 GeV; yields unmeasured, the reach scales as $1/Y_{\mathrm{CC}}$	data	Sect. 16.6
Bond breaking (H_2, molecular crystals)	bond energy 4.75 eV and $\omega_{\mathrm{vib}} = 0.546$ eV (measured); cage escape p_{esc}; yields of repulsive electronic states	mass scale ×0.84 (impulse) to ×0.75 (improved Born); p_{esc} and electronic yields unmeasured	data	Sect. 16.6
Glasses (a-SiO_2)	$\tan\delta_0$; $\varepsilon_r = 3.9$ (assumed); acoustic tunneling strength C_{l} (measured); $\bar{P}$, γ_{l} (1 eV assumed), dipole moment	$\tan\delta_0 = 3 \times 10^{-4}$ (thermal oxide) to 5×10^{-3} (deposited oxide): absorption rate ×16.7, ζ^{lim} ×4.1; scattering host term fixed by C_{l} alone, core term needs $\bar{P}$ and γ_{l}: 2.3×10^{-2} per kg, day and meV at 1 MeV with $C_{\mathrm{l}} = 1.9 \times 10^{-4}$ (3.1×10^{-2} with the polarization-averaged C), uncertain by a factor 3	data	Sect. 16.7
Two-level systems at millikelvin	$\alpha_{\mathrm{ohm}} = 4K_{\mathrm{b}}$ (Born), $K_{\mathrm{b}} = 0.07$, 0.11 fitted (aluminum); $\bar{P}$ of vitreous silica 2.0×10^{44} (heat release) or $8.0 \times 10^{44}\,\mathrm{J}^{-1}\,\mathrm{m}^{-3}$ (specific heat)	heat release fourfold between the two $\bar{P}$; first-principles α_{ohm} needs the scattering phases of both configurations; interactions unsized	data ($\bar{P}$); calculation (α_{ohm}); theory (interactions)	Sect. 16.8

donors, and the response of isolated donors, dilute or in an effective medium, is wrong at order one near the ionization edge (Sect. 16.5.3). And two-level systems that interact through their strain fields undergo spectral diffusion and can trigger avalanches of releases, which the independent-defect kinetics of Sect. 16.8 does not contain.

A second group is limited by a calculation whose formalism is known. The electron–hole ladder near the edge of a Dirac material changes the rate of $ZrTe_5$ by the Sommerfeld factor, 1.5 or more, at 20 keV and by less above 100 keV (Sect. 16.1.4), and Appendix E.20 specifies it. So do the final state of an ejected electron, the periodic Bethe–Salpeter response of a molecular crystal at finite $\boldsymbol{q}$, the excitons of a semiconducting tube and the valley-orbit coefficients of a donor. The response of a layered stack beyond its effective medium and the ohmic coupling α_{ohm} from first principles complete the group, with their calculations specified in Appendices E.10, E.11, E.21 and E.22. In each case the next term has a definite form and is computable, but no computation of it exists for the material.

The remaining targets are limited by data. Their inputs are properties of the sample that a measurement fixes: the dielectric tensor and the gap of $ZrTe_5$, the dielectric function $\varepsilon_{\mathrm{in}}(\omega)$ of real quantum dots, the oscillator strengths of the donor lines, the displacement threshold of LiF as a function of direction, the four yields of a color center, and the parameters of the two-level systems of a given film; Sect. 16.6.4 names the measurements of the yields. For these inputs the response formula is in hand, and the uncertainty of the rate is that of the sample.

16.9.3 Reach bands from input uncertainties

An input x on which the rate depends as a power, $\mathcal{R} \propto x^{-p}$ over the energies the rate samples, is a normalization nuisance in the sense of Sect. 8.8. By Eq. (1.54) it moves a scattering reach as $\bar{\sigma}_e^{\mathrm{lim}} \propto x^p$, so a range $x_1 < x < x_2$ becomes a band of constant width $(x_2/x_1)^p$ in $\bar{\sigma}_e^{\mathrm{lim}}$ wherever p does not change with the mass. An absorption reach moves as $\mathcal{R}^{-1/2}$, and its band is half as wide on a logarithmic scale. An input that sets a threshold acts instead through the low-mass wall, $\delta \ln m_{\mathrm{min}} = \delta \ln E_{\mathrm{th}}$ by Eq. (1.25), and near the wall its band widens without limit (Eq. (8.74)). An input without a size, such as the yield of a color center, gives no band: the reach is drawn for an assumed value and moves as its inverse.

For a Dirac material the power follows from Eq. (16.1). At fixed Dirac parameters,

$$\frac{\partial \ln L}{\partial \ln \varepsilon_{\mathrm{bg}}} = -\frac{2\,\varepsilon_{\mathrm{bg}}\,\mathrm{Re}(\varepsilon_{\mathrm{bg}}+\varepsilon_{\mathrm{D}})}{|\varepsilon_{\mathrm{bg}}+\varepsilon_{\mathrm{D}}|^2} = -2\Big[1-\frac{\mathrm{Re}\,\varepsilon_{\mathrm{D}}}{\varepsilon_{\mathrm{bg}}}+O\Big(\frac{|\varepsilon_{\mathrm{D}}|^2}{\varepsilon_{\mathrm{bg}}^2}\Big)\Big], \tag{16.80}$$

since $\varepsilon_{\mathrm{bg}}$ is real and ε_{D} does not depend on it. The reach therefore moves almost as $\varepsilon_{\mathrm{bg}}^2$, less by twice the fraction of the screening that the Dirac electrons supply. For $ZrTe_5$ that fraction is 3 to 8 % at long wavelength (Worked example 16.9).

Worked example 16.9: The reach band of $ZrTe_5$ from its background tensor

Take the Dirac parameters of Sect. 16.1.3, the measured-structure row with $\Delta_{\mathrm{D}} = 11.8$ meV, and the scattering reach of Fig. 16.3a: light mediator, orientation-averaged loss function, three events in a kg-year. The central tensor is $\varepsilon_{\mathrm{bg},ii} = 308.4$, 20.75 and 126.1 [2, Table 2]. The earlier calculation of Worked example 7.4 gives 187.5, 9.8 and 90.9 [1, Table 1], smaller by factors of 0.61, 0.47 and 0.72. We combine it with the same velocities, gap and cutoff, which isolates the effect of the tensor; the two calculations also differ in their velocities.

Estimate. At weak screening $L \to \mathrm{Im}\,\varepsilon_{\mathrm{D}}/(\hat{\boldsymbol{q}}\cdot\varepsilon_{\mathrm{bg}}\cdot\hat{\boldsymbol{q}})^2$. If $\mathrm{Im}\,\varepsilon_{\mathrm{D}}$ were the same in every direction, the reach would move by the ratio of the direction averages of $(\hat{\boldsymbol{q}}\cdot\varepsilon_{\mathrm{bg}}\cdot\hat{\boldsymbol{q}})^{-2}$, which is 0.327 for the earlier tensor against the central one. Directions within 30° of the slow axis y, where the background is smallest, carry 64 % of that average for the central

tensor, so the band is set mainly by $\varepsilon_{\mathrm{bg},yy}$, $(9.8/20.75)^2 = 0.22$ along the axis itself.
Full calculation. With Eq. (16.1) and the anisotropic ε_{D} of Eq. (D.226), the earlier tensor gives a reach 0.344 to 0.362 times the central one between 20 keV and 1 MeV, and 0.31 just above threshold. At the best mass, 36 keV, the reach lies between $1.53 \times 10^{-40}\,\mathrm{cm}^2$ and $5.5 \times 10^{-41}\,\mathrm{cm}^2$, a band of 2.8 that is nearly constant in mass. The full ratio lies above the estimate because the smaller tensor lets the Dirac electrons screen more: the effective coupling along y rises from 0.144 to 0.305.
Power. Scaling the whole central tensor by 1.1 moves the reach by 1.1^p with $p = 1.86$ to 1.94 between 20 keV and 1 MeV, and $p = 1.85$ at 12 keV, as Eq. (16.80) predicts for $\mathrm{Re}\,\varepsilon_{\mathrm{D}}/\varepsilon_{\mathrm{bg}}$ of a few percent. A tensor uncertain by a common factor of two would give a band of 3.6 to 3.8.
Dirac content. Removing the static Dirac term of Eq. (7.47), 10.0, 1.67 and 8.3 for this row, 3 to 8 % of the components, leaves the background 298.4, 19.08 and 117.8, and moves the reach by 0.881 to 0.885, a correction smaller than the band.
Absorption. For dark-photon absorption the earlier tensor moves ζ^{lim} by 0.49 to 0.55 between the gap and the cutoff, from 1.24×10^{-15} to 6.4×10^{-16} at 100 meV. The polarization along y carries most of the rate, and for it alone $\zeta^{\mathrm{lim}} \propto \varepsilon_{\mathrm{bg},yy}$ at weak screening, which would give 0.47.
Gap. The computed half-gaps of 11.8 to 17.5 meV put the threshold mass along y between 7.1 and 10.6 keV, with 9.4 keV for the relaxed-cell row at 15.6 meV; the measured 11.75 meV gives 7.1 keV. Close to the wall the threshold band exceeds the tensor band, since it diverges there.

Above 100 keV, where the region near the pair edge carries less than a fifth of the rate, the tensor band is the largest sized uncertainty of the $ZrTe_5$ reach.

Summary

A Dirac material screens the field of the dark matter with its whole dielectric function, $\varepsilon = \varepsilon_{\mathrm{bg}} + \varepsilon_{\mathrm{D}}$, and its loss function is $\mathrm{Im}\,\varepsilon_{\mathrm{D}}/|\varepsilon_{\mathrm{bg}} + \varepsilon_{\mathrm{D}}|^2$ (Sect. 16.1). The published rates divide the pair spectrum by $|\varepsilon/\varepsilon_{\mathrm{bg}}|^2$, which screens the Dirac electrons among themselves but not the field of the dark photon. They overstate scattering by $\varepsilon_{\mathrm{bg}}^2$ and absorption by $\varepsilon_{\mathrm{bg}}$ at every mass; wherever the one-loop response holds, these factors are at least $(\alpha/v_{\mathrm{max}})^2 = 7.6$ for scattering and, for cones slower than v_{max}, $\alpha/v_{\mathrm{max}} = 2.75$ for absorption. The rule for choosing a target reverses: the best Dirac target has the smallest background constant that keeps $\tilde{\alpha}$ below one. $ZrTe_5$, with its computed anisotropic tensor and the parameters of its measured structure (half-gap 11.8 meV), has a threshold mass of 7.1 to 9.6 keV depending on the axis and an orientation-averaged rate of 108 events per kg-year at 100 keV and $\bar{\sigma}_e = 10^{-38}\,\mathrm{cm}^2$. Its reach lies near $1.5 \times 10^{-40}\,\mathrm{cm}^2$ at 40 keV, 5200 to 6400 times weaker than with the published rule for the standard halo; its absorption reach, $\zeta \approx 1.25 \times 10^{-15}$, is weaker by 6.3. At this half-gap the one-loop expansion parameter of $ZrTe_5$ is about one. The electron–hole ladder raises the rate near the pair edge by a Sommerfeld factor of 1.5 or more, and cones with $\tilde{\alpha} \gtrsim 1$ have no controlled theory.

Electrons confined to tubes and sheets respond to the momentum along them only, so a halo particle reaches a mode faster than v_{max} through an oblique momentum transfer, at the price of the transverse form factor (Sect. 16.2). A sheet interacts through $e^2/(\varepsilon_a + \varepsilon_b)q_{\parallel}$; its remote bands screen only through the Keldysh term $r_0 q_{\parallel}$, which vanishes at long wavelength, and its substrate only through $\bar{\varepsilon}$. A metallic nanotube puts its density response into one line, $N_{\mathrm{ch}} g_c |q_{\parallel}| \delta(\omega - v|q_{\parallel}|/g_c)$, exact for linear bands, with $g_c = 0.17$ to 0.34 for a tube of 0.7 nm in

vacuum and a plasmon 3.4 to 6.9 times faster than $v_{\max}$. The charge mode involves all four channels, and the single-channel model with the measured g_c is four times low. Two kinds of carriers carry an acoustic plasmon whose velocity is the same in two and three dimensions, but whose weight is screened to $(q^2/(q^2+k_{\mathrm{TF},l}^2))^2$ of the heavy f-sum in three dimensions and $(q_\parallel/(q_\parallel+q_s))^2$ in two, 1.75×10^{-4} in the sheet of Worked example 16.2. A layered stack is a uniaxial medium as $ka_s\to0$, identical at any filling to the response of Lasenby and Prabhu. For $\boldsymbol{k}$ in the plane its response is known in closed form at every ka_s, with a leading correction quadratic in ka_s. A single undamped line in each direction reaches the sum-rule bound of Sect. 10.2.

An electron of graphene that receives more than the work function leaves the sheet with the direction of the momentum transfer, from 1.22 MeV upward (Sect. 16.3). Orthogonalizing the plane-wave final state to the occupied orbitals raises the rate of the π electrons by 1.44 at 10 MeV and 1.84 at 100 MeV for a heavy mediator. The attraction of the hole can add up to the Fermi factor, 7.3 at 10 eV; the two cannot be multiplied, and the consistent final state is a scattering state of the sheet with the screened hole. A semiconducting tube responds through massive Dirac subbands with a one-dimensional van Hove edge, reached at $q_\perp r_{\mathrm{t}}\geq0.76$ for any radius, from 226 keV for a tube 1 nm across. A hard-walled dot carries 0.76 to 1.03 of the scissored bulk response between 3 and 11 confinement energies above its gap, a deficit $9\pi/8kR$ from the levels lost near the surface. The printed dot rate exceeds the correct one by $\mu_{\chi e}/4\omega$, and absorption by a dot in a host carries the depolarization factor $9|\varepsilon_{\mathrm{in}}|^2/|\varepsilon_{\mathrm{in}}+2\varepsilon_{\mathrm{out}}|^2$, 6.00 for PbS in silica below its gap and unknown in size and sign over 2.5 to 10 eV.

The excitons of a molecular crystal follow from the Bethe–Salpeter equation in a molecular basis (Sect. 16.4). Integrated over a window that holds whole manifolds, the crystal responds as a set of independent molecules at every momentum transfer, the Davydov sum rule; coupling to levels outside the window changes the weight at first order in $\mathfrak{J}/\Delta E$. In a π-electron model of anthracene the Davydov splitting of the lowest singlet is 31 meV. For a heavy mediator the full response exceeds the binned rate by 6.3 % above 30 MeV, by up to 27 % between 1.2 and 7.5 MeV and by 38 % just above threshold; for a light mediator it does so by 29 to 35 % above 2 MeV. The macroscopic Coulomb term must sit in the exciton Hamiltonian, not in a factor $1/|\varepsilon_{\mathrm{mac}}|^2$. A monoclinic crystal allows the quadrupole that a cubic crystal forbids, and the daily modulation of anthracene is bounded by 0.98 at 1 MeV and by 0.08 to 0.13 at 1 GeV, against 0.9 % for silicon at 2 MeV.

Dilute donors in silicon add a response to first order in their density, with the host screening the vertex once (Sect. 16.5). Their ionization is the generalized oscillator strength of a hydrogenic $1s$ state. The published bracket carries a misprint that changes heavy-mediator scattering by factors from 1.47 at 20 keV to 0.56 at 10 MeV, and a factor 4.84 that compensates its parameters and gives the continuum an oscillator strength of 4.85, above the 3.85 of the whole donor. The valley-orbit line $1s(A_1)\to1s(E)$ appears at finite momentum through the mass anisotropy and vanishes along [111]. The $2p_0$ line of Si:P at 34.11 meV reaches $\zeta=2.3\times10^{-16}$ at $10^{15}\,\mathrm{cm}^{-3}$, twenty times below the continuum, over a relative mass window of 1.2×10^{-4}. The first-order form fails at the line center from $4.8\times10^{15}\,\mathrm{cm}^{-3}$, where the donors screen the field themselves, and the published densities, 0.3 to 0.6 of the Mott density, lie in the impurity band.

A color center forms after the prompt window: its rate is the recoil rate times the yield $Y_{\mathrm{CC}}=p_{\mathrm{FP}}p_{\mathrm{surv}}p_q f_{\mathrm{opt}}$ (Sect. 16.6). A fluorine recoil in LiF above the simulation threshold of 27 eV needs $m_\chi\geq0.186\,\mathrm{GeV}$; the zero-point motion of the struck nucleus changes the rate by less than 10 % above 1.22 to 1.36 times the threshold, while a direction-dependent threshold matters up to 0.41 GeV. Near threshold a recoil makes one vacancy and less than one free electron. The response of an atom in a molecule is the recoil of the center of mass convolved with the internal response, and the bond breaks when the internal part exceeds the bond

energy. For H_2 quantum effects lower the mass scale by 0.84, or 0.75 in the improved Born approximation. Four yields at millikelvin temperatures are open, and each can be measured directly (Sect. 16.6.4).

Glasses add tunneling two-level systems to the final states (Sect. 16.7). In the standard tunneling model they respond to strain and to the electric field, and a dark photon is absorbed at a rate proportional to $\zeta^2 \tan\delta_0 \tanh(m_\phi/2T)/\varepsilon_r$, the dielectric loss of the glass read at the dark-photon mass. On thermal SiO_2, with $\tan\delta_0 = 3\times10^{-4}$, three events in a kg-year at 6 GHz need $\zeta = 2.44\times10^{-15}$, and a film of 22 mg needs 5.2×10^{-13}. Dark matter of 1 MeV with $\bar\sigma_e = 10^{-38}\,\mathrm{cm}^2$ scatters off the two-level systems of amorphous SiO_2 about 2.3×10^{-3} times per kg-day below 0.1 meV. The estimate is uncertain by a factor of three, and its host term is fixed by the measured longitudinal tunneling strength, not by the loss tangent. Below the Ioffe–Regel frequency the vibrations of silica absorb through the constant term A^{IR} of random Born charges, and overtake the two-level systems of thermal SiO_2 above 0.46 meV. Coherent scattering off a random medium leaves the Born approximation once the optical potential exceeds about a tenth of the normal kinetic energy, and at the reach quoted for a test-mass bound, $4\times10^{-38}\,\mathrm{cm}^2$, it is 1.86 times the kinetic energy.

One gram of fused silica cooled from room temperature to 10 mK still releases 46 pW an hour later, and the power falls as $1/t$ (Sect. 16.8). The heat comes from two-level systems whose populations froze near 4.8 K and which relax by emitting one phonon over hours to weeks. For a uniform distribution of tunneling exponents the release is $(\pi^2/24)\bar P \mathcal{V}(T_{\mathrm{hi}}^2 - T_{\mathrm{lo}}^2)/t$ exactly, but the density $\bar P$ inferred from heat release in vitreous silica is four times smaller than the one from its specific heat. In a metal the conduction electrons relax a defect too, through an ohmic bath with $\alpha_{\mathrm{ohm}} = 4K_{\mathrm{b}}$ in the Born limit. The fits to aluminum-coated resonators give $\alpha_{\mathrm{ohm}} = 0.28$ and 0.44, where the Born value is already more than 10 % high and the small-coupling rate fails; the defect then hops incoherently at the rate of the ohmic scaling limit. A superconductor suppresses the electron rate by $2/(1+\mathrm{e}^{\Delta_{\mathrm{BCS}}/T})$ for $\Delta_{\mathrm{tl}} \ll T$; the quasiparticle golden rule adds a factor 0.86 at $\Delta_{\mathrm{tl}} = T$ and 0.71 at $2T$, so thermal defects relax 14 to 29 % slower. Above the crossover temperature $\omega^\ddagger/2\pi$ a defect crosses its barrier by activation, already enhanced 6.2-fold by tunneling at the crossover for an Eckart barrier, and below it the defect tunnels. Whether the delayed events of a cooled detector are independent relaxations or avalanches of triggered releases is open, and Sect. 19.6 inherits the question.

Table 16.2 collects the truncations of the chapter with their sizes, and Table 16.1 the material inputs. Most targets are limited by data: the background tensor of $ZrTe_5$ moves its reach by a factor 2.8 between two calculations (Worked example 16.9). The loss tangent, the oscillator strengths and the yields move the reaches of glasses, donors and color centers by factors that only a measurement of the sample fixes. Three targets are limited by theory: Dirac cones at strong coupling, donors near the Mott density, and interacting two-level systems. The corrections to published rates are larger still: $\varepsilon_{\mathrm{bg}}^2$ for Dirac materials and $\mu_{\chi e}/4\omega$ for quantum dots exceed the input bands of these targets by three orders of magnitude or more. They are errors of a formula, not uncertainties of an input, and once corrected they leave no band behind.

Table 16.2: Truncations of the responses of the novel targets. Type in the classes of Sect. 8.8: normalization (norm.), tilt, threshold and new support. Basis of the size: "computed" (next order or exact result computed), "model" (computed in a model), "est." (parametric estimate), "unsized" (no size known). The material inputs and their spreads are in Table 16.1. The last column gives the section where the item is treated.

Truncation	Type	Size	Basis	Section
Electron–hole ladder near a Dirac edge	tilt near threshold; new support	Sommerfeld factor above 1.5 below 1.33 times the edge (60, 12, 3 % of the $ZrTe_5$ rate at 20, 100, 1000 keV); bound states 0.10 meV below it	est.	Sect. 16.1
Self-energy and vertex of the Dirac carriers	norm.	relative order $\tilde\alpha = 0.18$ ($ZrTe_5$), coefficient not fixed	est.	Sect. 16.1

Table 16.2, continued

Truncation	Type	Size	Basis	Section
Random-phase response at $\tilde{\alpha} \gtrsim 1$	norm.	no controlled method; BNQ-TTF has an effective coupling near 10	unsized	Sect. 16.1
Isotropic model of an anisotropic cone	norm.; tilt in direction	orientation-averaged $ZrTe_5$ rate overstated by 2.3 to 2.7	computed	Sect. 16.1
Material outside the cone model	tilt	doping (screening at $\omega \lesssim v_D k_F$); ionic poles of ε_{bg}; bands bending beyond Λ_D	unsized	Sect. 16.1
Linear bands of a metallic tube	tilt in $q_\parallel$	plasmon frequency and weight within 10 % for $q_\parallel < 1.36\,k_F$ (model with $k_F = 1/r_t$); subband pairs from $2v_D/r_t = 1.7$ eV	model	Sect. 16.2
Kernels of tubes and sheets	norm.; tilt in $q_\parallel$	logarithmic tube kernel 10 % low at $q_\parallel r_t = 0.38$; sheet thickness $F_{th} = 0.95$ (100 eV), 0.85 (300 eV) for $d = 3.35$ Å	computed	Sect. 16.2
Environment of tubes and sheets	norm.; tilt	Coulomb coupling between tubes below 1–2 eV (spacings 100–200 nm); surface phonons of a polar substrate	unsized	Sect. 16.2
Effective medium of a stack	tilt in direction	within 10 % for $ka_s < 1.94$ in the model stack, leading correction quadratic in ka_s; nonlocal layer response at $k \gtrsim 1/d_A$	model	Sect. 16.2
Plane-wave final state of an ejected electron	norm.; tilt in E_{ej}	orthogonalized final states: rate ×1.44 (10 MeV), ×1.84 (100 MeV) heavy, ×1.14, ×1.19 light; spectrum up to ×3.5; σ electrons left out	model	Sect. 16.3
Attraction of the hole	tilt in E_{ej}	at most the Fermi factor of an unscreened hole, 7.3, 3.4 and 2.6 at 10, 50 and 100 eV; screened value unknown	est.	Sect. 16.3
Excitons and intersubband transitions of a tube	new support	below each subband edge	unsized	Sect. 16.3
Hard-wall envelopes of a dot	norm. near the gap	0.76–1.03 of the scissored bulk at 3–11 confinement energies ($qR = 8$), 2.4–66 nearer the gap; leading term $-9\pi/8kR$	model	Sect. 16.3
Depolarization of a dot in its host (absorption)	norm.	6.00 for PbS in silica below its gap, 4.6–6.7 for $\mathrm{Re}\,\varepsilon_{in} = 10$–$25$, below one for $\lvert\varepsilon_{in}\rvert \sim 1$; unknown at 2.5–10 eV	computed	Sect. 16.3
Binned Frenkel rate	norm.; tilt near threshold	6.3 % above 30 MeV (heavy), up to 27 % at 1.2–7.5 MeV and 38 % at 0.88–1.0 MeV, 29 to 35 % above 2 MeV (light); π-electron model, probably overstated	model	Sect. 16.4
Charge-transfer admixture	norm.; new support	$\theta_{CT}^2 \approx 1$ % for an illustrative $t_{CT} = 0.1$ eV	est.	Sect. 16.4
Frenkel model beyond the π singlets	new support; norm.	ionization continuum above about 1.1 MeV; σ electrons in $\lvert f\rvert^2$ above about 10 keV	unsized	Sect. 16.4
Ground state and local field of a donor	norm.	central cell: $2p$ line strengths 1/3.2–1/3.4 of the effective-mass values; local-field ratio expected close to one, not computed	model	Sect. 16.5
First order in the donor density	norm.	continuum within 10 % below 2.6–8.8×10^{17} cm^{-3}; line center fails from 4.8×10^{15} cm^{-3} (4.2 μeV width)	computed	Sect. 16.5
Isolated donors near the Mott density	norm.; tilt near the edge	order one near the ionization edge for $n_D/n_c \approx 0.5$	unsized	Sect. 16.5
Step threshold for a bound nucleus	threshold	LiF: within 10 % for $E_R^{max} \geq 1.22$–$1.36\,E_d$, 2 % above $2E_d$	computed	Sect. 16.6
Direction dependence of the displacement threshold	threshold	more than 10 % up to $E_R^{max} = 4.8\,E_d$ (0.41 GeV) in an angular model	model	Sect. 16.6
Classical bond breaking	threshold	H_2: within 10 % for $m_\chi \geq 42.9$ MeV; mass scale ×0.84 (impulse), ×0.75 (improved Born)	model	Sect. 16.6
Scattering on two-level systems	norm.	2.3×10^{-2} per kg, day and meV at 1 MeV and $\bar{\sigma}_e = 10^{-38}$ cm^2 in a-SiO_2 with $C_l = 1.9 \times 10^{-4}$ (3.1×10^{-2} with the polarization-averaged C), uncertain by a factor 3; most of the rate from momenta outside the window of the host form; rigid-ion core term	est.	Sect. 16.7
Glass absorption above the Ioffe–Regel frequency	tilt in m_ϕ	below 4.3 meV the B^{IR} term is at most 9 % of the A^{IR} term (Limit Check 16.12); above, the modes are no longer sound waves	computed	Sect. 16.7
Born approximation for a random medium	norm.	within 10 % for $\lvert V_{opt}\rvert/E_n < 0.092$ (Limit Check 16.13); at the published reach 4×10^{-38} cm^2 of a test-mass bound, $\lvert V_{opt}\rvert/E_{kin} = 1.86$, far outside; sign of the true force open	computed	Sect. 16.7
Superconducting factor $2/(1 + e^{\Delta_{BCS}/T})$, valid for $\Delta_{tl} \ll T$	norm.	quasiparticle golden rule adds 0.86 at $\Delta_{tl} = T$, 0.71 at $2T$; within 10 % only for $\Delta_{tl} \leq 0.7\,T$ (Limit Check 16.14)	computed	Sect. 16.8

Table 16.2, continued

Truncation	Type	Size	Basis	Section
Born-limit ohmic coupling	norm.	10 % high from $K_\mathrm{b} = 0.017$; at the fitted $K_\mathrm{b} = 0.07$ one contact channel gives $\alpha_\mathrm{ohm} = 0.19$, not 0.28 (Limit Check 16.14)	model	Sect. 16.8
Small-coupling relaxation in an ohmic bath	norm.	within 10 % of the scaling limit only for $\alpha_\mathrm{ohm} \leq 0.091$; aluminum fits give 0.28 and 0.44 (Limit Check 16.15)	computed	Sect. 16.8
Uniform distribution of tunneling exponents (heat release)	tilt in time	$1/t$ law within 10 % over 10.6 of the 12.2 decades of the model distribution (Limit Check 16.17); $\bar{P}$ from heat release four times below specific heat	model	Sect. 16.8
Wigner and parabolic forms of activated hopping	norm.	Eckart barrier: within 10 % above 2.1 and $1.5\,T_\times$; instanton about 18 % low at 0.25–$0.5\,T_\times$ (Limit Check 16.16)	model	Sect. 16.8
Independent two-level systems	norm.; tilt in time	strain coupling of relative size $\bar{P}\gamma^2/\rho_T c^2 \approx 2$ to 3×10^{-4} (vitreous silica); spectral diffusion moves the window of the heat release; avalanches unquantified	unsized	Sect. 16.8

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Chapter 17

Comparison across channels

Dark matter of 1 MeV that scatters through a heavy mediator coupled to electrons gives a kilogram of silicon 17 events a year at $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$ by creating electron–hole pairs. If the mediator couples to the electrons alone, the same particle gives 7×10^5 events through acoustic phonons, carried by the electrons that move with the nuclei; if it is a dark photon, it gives none that way (Sect. 10.2). Both numbers come from one section; most comparisons an experimentalist needs are spread over seven chapters. The multiphonon rate is in Chap. 11, the Migdal rate in Chap. 12, the phonon-assisted rate in Chap. 13, magnons and superconductors in Chaps. 14 and 15, and the special targets in Chap. 16, each computed with the approximations its own excitation allows. A detector, however, is built for one material and one threshold, and the question it faces runs the other way: for a given coupling, which excitation of which material gives the largest rate and the lowest reachable mass, and why?

Read along the interaction axis, the atlas of Chap. 9, which places each excitation of a solid and each coupling in the section that treats them, answers this question at the level of the response. For each coupling it gives the channel that dominates at each mass and threshold, the few material numbers that set its rate and reach, and how firmly those numbers rank materials once the uncertainty of every input is carried through. Which experiment to build also depends on the energy per quantum, the readout and the backgrounds, which the guide to next targets at the end of the book combines with these rankings (Fig. 17.1).

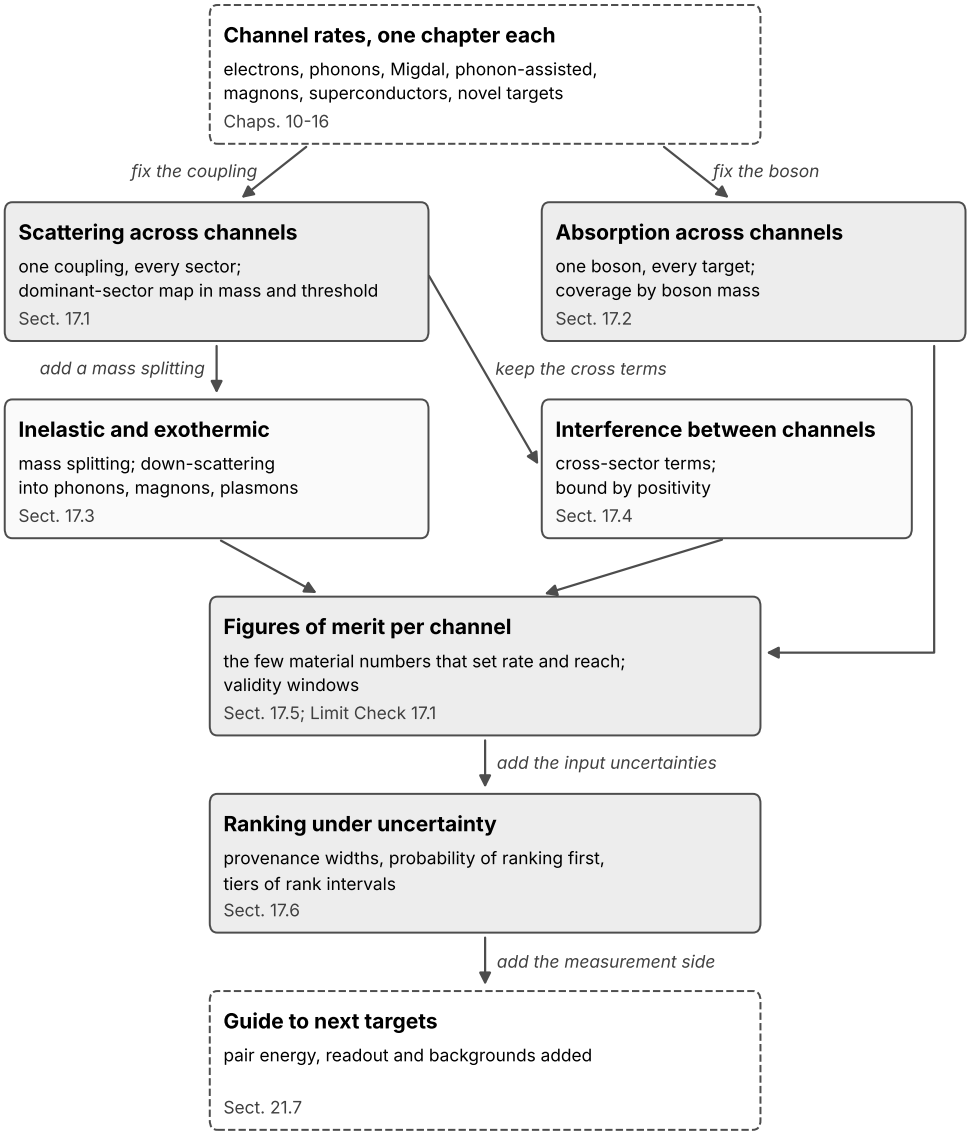


Figure 17.1: Comparison across channels, from the rates of Chaps. 10 to 16. At fixed coupling: scattering in every sector and its dominant-sector map (Sect. 17.1); at fixed boson: absorption on every target (Sect. 17.2). Side branches: dark matter with a mass splitting, including down-scattering into collective modes (Sect. 17.3), and the interference between sectors with its positivity bound (Sect. 17.4). All feed the figures of merit of each channel with their validity windows (Sect. 17.5; Limit Check 17.1), and the ranking under the uncertainty of the inputs (Sect. 17.6), which the guide to next targets at the end of the book combines with the measurement side.

17.1 Scattering across channels

A silicon calorimeter with a threshold of 0.1 eV records 3×10^4 events per kilogram and year from dark matter of 10 MeV that couples to nucleons through a heavy mediator with $\sigma_p = 10^{-38}\,\mathrm{cm}^2$, nearly all of them multiphonon excitations of the lattice. At a threshold of 0.5 eV it records 1.5 events, and every one of them is a Migdal ionization. Between the two thresholds the dominant final state moves from the electron-following sector of the decomposition theorem to the two electronic sectors (Sect. 8.6), and the rate drops by four orders of magnitude. For each coupling, the choice of target and threshold depends on where in mass and threshold these changes fall. We will draw that map for silicon from the scattering rates of Chaps. 10 to 13, explain each boundary by kinematics or by screening, and then follow it into germanium and GaAs.

17.1.1 Kinematic ceilings of the sectors

At fixed coupling the rate is Eq. (8.16) with one coupling matrix $\mathfrak{D}$, and by the decomposition theorem Eq. (8.50) the structure factor is the sum of the electron-following, zero-phonon and phonon-dressed sectors. Each cell of Table 8.5 was computed in its own chapter, with the approximations its excitation allows, and the rates add cell by cell. The bottom row of the table, the interference between electrons and nuclei, vanishes for a coupling to one species alone (Sect. 17.4). We compare two such couplings: a nucleon coupling, with equal couplings to protons and neutrons and the cross-section σ_p per nucleon of Sect. 11.1, and an electron coupling with the reference cross-section $\bar{\sigma}_e$. Both have a heavy mediator. The threshold ω_{th} acts on the total deposited energy, as in a calorimeter. A charge readout records the electronic sectors alone, and how much of a lattice deposit a phonon sensor collects is a question for the chapters on readout.

Which sector dominates follows first from how much energy each can absorb. A single phonon carries at most the top of the phonon band, 64.1 meV in silicon (Sect. 5.2); the PBE density of states used for the maps ends at $\omega_{\max} = 63.7\,\mathrm{meV}$ (Sect. 11.3). A nucleus recoiling freely absorbs at most $E_R^{\max} = 2\mu_{\chi N}^2 v_{\max}^2/M$ (Eq. (1.21)), which is 0.054 eV at 10 MeV and 5.3 eV at 100 MeV. An electronic excitation can absorb up to the whole kinetic energy, $\frac{1}{2} m_\chi v_{\max}^2$ (Eq. (1.24)), since the lattice can take the momentum without taking the energy; this is 3.5 eV at 1 MeV and 35 eV at 10 MeV. For $m_\chi \ll M$ the two ceilings differ by the factor $M/4m_\chi$, 6.5×10^3 at 1 MeV. Below the lattice ceiling the electron-following sector wins whenever the coupling reaches the nuclei, because an electronic excitation costs a small probability per kick. Above it the electronic sectors alone are open. The boundary does not sit exactly at the ceiling, however: a multiphonon spectrum extends above E_R by its Poisson weights, and an impulse peak by its width.

The second factor is screening. A coupling to electrons reaches the lattice through the electrons that move with the nuclei, and whether that charge is seen depends on what else the probe couples to (Sect. 8.4). It reaches the lattice in full or not at all, according to whether it couples to the electrons alone or to the total charge.

17.1.2 Nucleon coupling in silicon

The nucleon-coupling map uses four responses. Below the zone boundary, $q \le q_{\mathrm{BZ}} = 2.25\,\mathrm{keV}$, the one-phonon response is the long-wavelength acoustic form Eq. (11.3), with the sound speeds and polarizations of the measured elastic tensor. Since its linear branches would run above every real phonon near q_{BZ}, we cap its deposits at $\omega_{\max}$. Above q_{BZ} the lattice response is the incoherent harmonic response Eq. (11.26) of the computed phonon density of states of Sect. 5.2, split into its one-phonon term and the terms with two or more phonons, with the

Table 17.1: The multiphonon–Migdal crossover in silicon: nucleon coupling, heavy mediator, $\sigma_p = 10^{-38}\,\mathrm{cm}^2$, threshold on the total deposited energy. Largest momentum transfer and its Debye–Waller exponent; crossover threshold $\omega_{\mathrm{th}}^{\times}$ (lattice rate equal to the Migdal rate) and its ratio to the free-recoil endpoint; Migdal rate at the lowest thresholds (events per kilogram and year) and its zero-phonon share.

m_χ (MeV)	1	3	6.3	10	17.5	30	100
$q_{\max}$ (keV)	5.3	15.9	33.4	53.0	92.7	159	528
$2W(q_{\max})$	0.02	0.16	0.70	1.76	5.40	15.9	175
$\omega_{\mathrm{th}}^{\times}$ (eV)	0.178	0.187	0.242	0.307	0.466	0.826	5.33
$\omega_{\mathrm{th}}^{\times}/E_R^{\max}$	331	38.8	11.3	5.73	2.84	1.71	1.00
Migdal rate	1.9×10^{-3}	0.12	0.68	1.5	3.4	6.9	30
zero-phonon share	0.97	0.94	0.83	0.65	0.33	0.10	0.001

Helm form factor of Sect. 2.2. The n-phonon spectra are built by repeated convolution of the one-phonon distribution, a sum of positive terms, so the tail far above E_R is computed to the precision of the arithmetic. Above $q = 300\,\mathrm{keV}$, where $2W = 56$, the response is the impulse Gaussian Eq. (11.34). The Migdal rate uses the soft-limit probability Eq. (12.21) with the loss function with local fields and $Z_{\mathrm{ion}}(p)$ of Sect. 12.3. Since the sum rule Eq. (12.34) holds at every momentum transfer in its incoherent form (below $2q_{\mathrm{BZ}}$ see the qualifier of the Migdal band below and Table 17.9), the lattice only has to conserve energy, and we let it take its mean share, $q^2/2M$, so that the deposited energy is $\omega_e + q^2/2M$. The share of Migdal events that leave the lattice in its initial state is $\langle \mathrm{e}^{-2W(|\boldsymbol{q}-\boldsymbol{p}|)}\rangle$, averaged over the momentum $\boldsymbol{p}$ the electrons take with the weight of Sect. 12.4. We label a point of the map "one phonon" when the one-phonon rate above threshold exceeds the rate with two or more phonons, and among the rest "free recoil" when the median momentum transfer of the lattice rate lies above the impulse onset, 92.5 keV (Sect. 11.4). The electron rates are counted from 1.11 eV, the gap of the step model of pair creation that accompanies the tables of the loss function in their package; the tables themselves are scissored to the measured 1.17 eV (Sect. 10.10), and the difference lowers the onset of the electron sector from 0.333 to 0.316 MeV. The tables stop at 89 eV; the threshold axis stops at 50 eV.

Fig. 17.2a shows the result. Below the band top the electron-following sector holds the whole map, and inside it the one-phonon term leads up to 20 MeV at a threshold of 1 meV, although the kicks of a 10 MeV particle already reach $2W = 1.8$. The reason is that the incoherent response gives the one-phonon term the Poisson weight $2W\mathrm{e}^{-2W}$, and the sum of all higher orders overtakes it only at $2W = 1.26$; the heavy mediator weights the largest momentum transfers, so the boundary follows the typical kick. Note that the one-phonon contours below 3.4 MeV, where $q_{\max} < 18\,\mathrm{keV}$, carry the coherent correction of Sect. 11.3, larger than 10 % below 18 keV; the boundary near 20 MeV, where the kicks lie far above 18 keV, does not. Multiphonon deposits take over above 20 MeV at the lowest thresholds and, at every mass, at thresholds above the band top. By the median criterion the free recoil takes over above 60 MeV at 1 meV, and above 22 MeV at higher thresholds. At the lowest thresholds the lattice rate exceeds the Migdal rate by 6×10^7 at 1 MeV, 8×10^5 at 10 MeV and 2×10^4 at 100 MeV, roughly the inverse of the Migdal probability per kick averaged over the kicks.

The Migdal effect holds the band between the end of the lattice spectrum and the largest deposit. It opens at $2E_{\mathrm{g}}/v_{\max}^2$, 0.316 MeV with the gap of the tables and 0.333 MeV with the physical 1.17 eV. Up to 1 MeV its rate comes from momentum transfers between about 0.75 keV and $2q_{\mathrm{BZ}}$, where the diagonal soft limit of Sect. 12.4 is not known to equal the coherent response and the coherent response is not computed. Near threshold the factor is likely a large suppression, since the exact small-q spectrum sits at the plasmon, beyond the

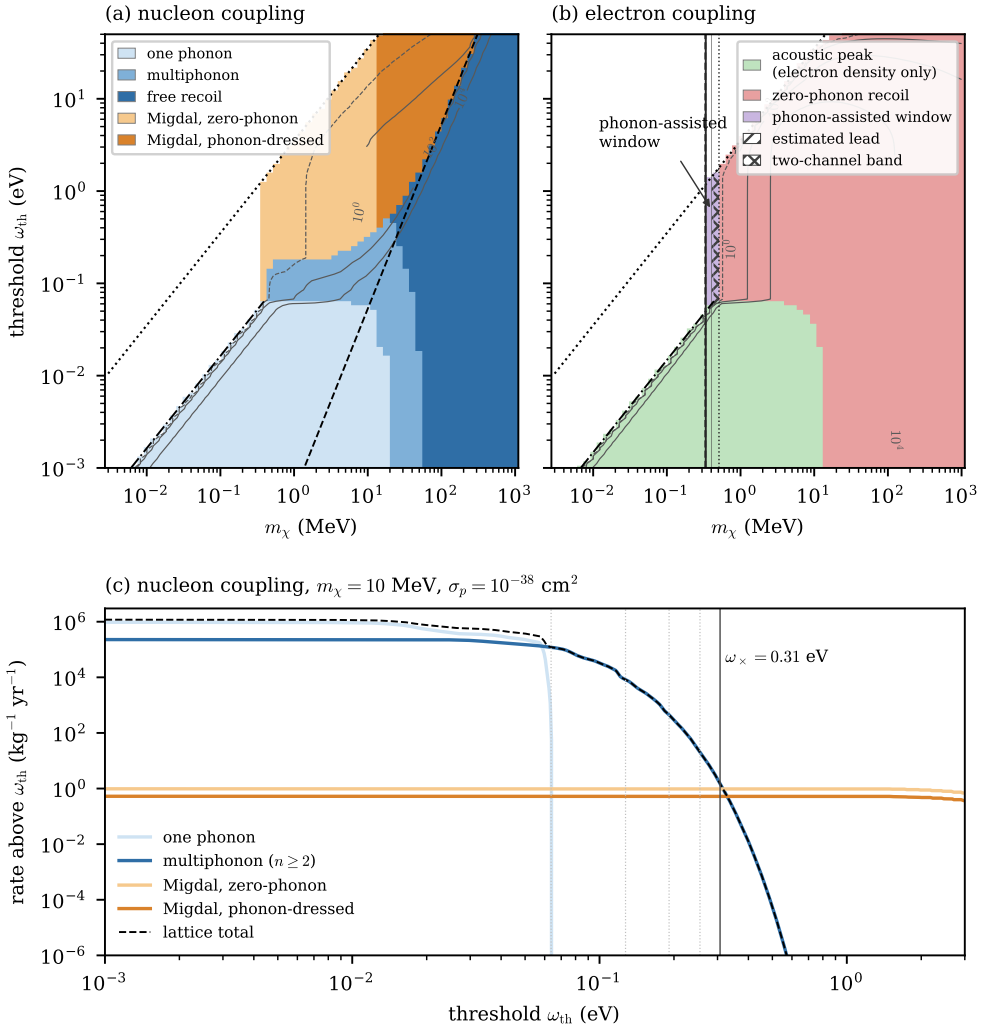


Figure 17.2: Dominant sector of silicon in the plane of dark-matter mass and threshold on the total deposited energy, heavy mediator, standard halo (Table 1.1), $T = 0$. (a) Nucleon coupling: electron-following sector split into one phonon, two or more phonons and free recoil; Migdal effect split by whether more than half of the events leave the lattice in its initial state; below about 2 MeV its rate carries the unsized coherent factor of Table 17.9 (7 % of it at 3 MeV). One-phonon deposits are capped at the top of the PBE band, 63.7 meV. Gray contours: the dominant rate, 10^{-2} to 10^{4} events per kilogram and year at $\sigma_p = 10^{-38}\,\mathrm{cm}^2$. (b) Electron coupling: zero-phonon electron recoil (loss function with local fields, counted from 1.11 eV) against the acoustic peak of an electron-only coupling in the model of Limit Check 10.2 (linear longitudinal branch to $|\boldsymbol{q}| = 2\pi/a$, 28 electrons per cell following the nuclei), which a dark photon does not have. Phonon-assisted window from Table 13.1: onsets 0.3332 (dashed) and 0.3374 MeV (solid) and first zero-phonon edge 0.405–0.408 MeV (dark band), computed; estimated phonon-assisted lead to 0.43 MeV (hatched) and two-channel band 0.43–0.51 MeV (cross-hatched). Contours at $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$. Lines: dotted, the largest deposit $\frac{1}{2}m_\chi v_{\rm max}^2$; dashed, the free-recoil endpoint $E_R^{\rm max}$; dash-dotted, the acoustic edge Eq. (11.5) with $c_{\rm LA,max} = 9.35$ km/s in (a) and the model's 8.43 km/s in (b). White: no channel, or a rate below 10^{-12} events per kilogram and year. (c) Rates above threshold at $m_\chi = 10$ MeV, nucleon coupling; vertical dotted lines at multiples of the band top, solid line at the crossover threshold $\omega_{\rm th}^{\times}$ of Worked example 17.1.

energies open there. The Migdal band at 0.32–1 MeV and the 1 MeV column of Table 17.1, all of whose rate lies in that band (29 % of it below q_{BZ}), carry that unsized factor; at 2 MeV it acts on a third of the rate, and from 3 MeV on on 7 % or less (Table 17.9). Table 17.1 gives the crossover threshold $\omega_{\mathrm{th}}^{\times}$, where the lattice and the Migdal rates are equal, computed on a grid of thresholds spaced by about 0.5 %. Three regimes explain its values. Below 3 MeV the kicks are soft, $2W(q_{\max}) \leq 0.16$, and the crossover stays near 2.8 to 2.9 $\omega_{\max}$. A deposit above $2\omega_{\max}$ needs three phonons, whose weight scales as $(2W)^3$ against $2W$ for the Migdal probability; the three-phonon spectrum falls so steeply at its top that a smaller $2W$ moves the crossover little. Between 3 and 20 MeV the crossover climbs through the Poisson tail of the multiphonon spectrum, as Worked example 17.1 works out at 10 MeV. Above the impulse onset the lattice deposit is a peak at E_R with a Gaussian tail, and the crossover moves onto the free-recoil endpoint. At 300 MeV it lies at 0.98 $E_R^{\max}$, below the endpoint, because $\eta(v_{\min})$ vanishes there faster than the Migdal rate falls. Above 410 MeV the endpoint passes the 89 eV end of the tables, and the Migdal side of the map would need the core shells (Sect. 12.2). Inside the band a charge readout at the single-electron level records the Migdal events with no lattice background from the dark matter itself.

The two Migdal labels split at a mass. A zero-phonon Migdal event needs the lattice to absorb the kick coherently, with probability e^{-2W} at the momentum the nucleus keeps. The rate-weighted share of such events falls with mass (Table 17.1), and it equals the phonon-dressed share at 13.2 MeV. This lies above the 6.3 MeV at which a kick at the largest momentum transfer gives $\mathrm{e}^{-2W} = \frac{1}{2}$ (Sect. 12.4), since the halo puts most kicks below the largest one. At 300 MeV the Migdal rate at the lowest thresholds reaches 130 events per kilogram and year.

Worked example 17.1: The multiphonon–Migdal crossover in silicon at 10 MeV

Kinematics. For $m_\chi = 10$ MeV the largest momentum transfer is $q_{\max} = 2\mu_{\chi N} v_{\max} = 53.0$ keV, and a free recoil there would take $E_R^{\max} = 53.7$ meV. With the mean phonon energy $\langle\omega\rangle = 40.9$ meV the recoil parameter of Sect. 8.5 is $x_R = E_R/\langle\omega\rangle = 1.31$, below the impulse onset $x_R = 4$ (Sect. 11.4). The lattice therefore takes the energy in whole phonons, with Poisson weights of mean $2W = q^2\langle u_x^2\rangle = 1.76$, where $\langle u_x^2\rangle = 2.45 \times 10^{-3}$ Å^2 is the mean square displacement of Sect. 11.3.

Migdal side. The Migdal probability per kick is 6.07×10^{-5} per eV of recoil energy (Sect. 12.3), or 3.26×10^{-6} at $q_{\max}$, all of it at $\omega_e \geq 1.11$ eV.

Lattice side, a bound. A deposit above ω needs at least $\lceil \omega/\omega_{\max} \rceil$ phonons (Sect. 11.3). The Poisson probability of eleven or more phonons at $2W = 1.76$ is 2.6×10^{-6}, below the Migdal probability, and eleven phonons at the band top carry 0.70 eV. The lattice tail at $q_{\max}$ falls below the Migdal probability at some threshold under 0.70 eV.

Lattice side, computed. The incoherent spectrum at $q_{\max}$ is 8.1×10^{-3} per eV at 0.3 eV, where seven phonons give 35 % of it, and 1.24×10^{-4} per eV at 0.4 eV, a fall by a factor 65 per 0.1 eV. The weight above 0.4 eV is then $1.24 \times 10^{-4} \times 0.024\,\mathrm{eV} = 3.0 \times 10^{-6}$, equal to the Migdal probability: per kick, the crossover sits near 0.4 eV. The bound is loose by 0.3 eV because the phonons of a deposit near 0.3 eV carry 43 meV each on average, well below the band top.

Rates. Integrated over momentum transfer and halo, the lattice rate above 0.1, 0.2, 0.3 and 0.5 eV is 3.3×10^4, 283, 2.2 and 5.5×10^{-5} events per kilogram and year at $\sigma_p = 10^{-38}$ cm^2, against a Migdal rate of 1.5 (0.98 zero-phonon, 0.53 phonon-dressed) that does not change below 1.1 eV. The two are equal at $\omega_{\mathrm{th}}^{\times} = 0.307$ eV, or 5.7 $E_R^{\max}$ and 4.8 $\omega_{\max}$. The rate crossover lies below the per-kick value because softer kicks lose more of their tail than of their Migdal probability: the tail scales as a high power of $2W \propto q^2$, the Migdal

probability as q^2. Near $\omega_{\text{th}}^{\times}$ the lattice rate falls by a factor e every 0.020 eV. A factor of two in the tail, the size of the coherent corrections to multiphonon spectra near their band edges (Sect. 11.3), moves $\omega_{\text{th}}^{\times}$ by 0.014 eV.

17.1.3 Electron coupling and screening

An electron coupling reaches the zero-phonon, phonon-dressed and electron-following sectors of silicon. The zero-phonon sector gives the loss-function rate Eq. (10.4), open above the first zero-phonon edge at 0.405 to 0.408 MeV, and we evaluate it with the loss function with local fields. The phonon-dressed sector gives the phonon-assisted rate of Sect. 13.3, open from 0.3332 or 0.3374 MeV; for the window between the two we take Table 13.1, where estimates of the phonon-assisted rate are all that exists. The electron-following sector gives the acoustic peak of Sect. 10.2, carried by the electrons that move with the nuclei. We evaluate it in the model of Limit Check 10.2: a linear longitudinal branch with $c_{\text{LA}} = 8.43\,\text{km/s}$ up to $|\boldsymbol{q}| = 2\pi/a = 2.28\,\text{keV}$, where it reaches 64 meV, and all 28 electrons of the cell following the nuclei.

Fig. 17.2b shows the map. Below 0.3332 MeV the acoustic peak is the one open channel, and it is open up to its top at 64 meV. Between 0.3332 and 0.405 MeV the phonon-assisted channel is the one electronic channel; by the estimates of Sect. 13.9 it leads centrally to 0.43 MeV, the two channels are comparable from 0.43 to 0.51 MeV, and above that the zero-phonon rate leads. Wherever the acoustic peak is open, it dominates the low thresholds: it exceeds the electron-recoil rate by 4.4×10^4 at 1 MeV and by 1.2 at 10 MeV, and the two are equal at 12.3 MeV, above which the electron recoil leads at every threshold. The crossing reflects the model, however. Its peak stops at 2.28 keV, a momentum transfer that dark matter of 0.43 MeV already reaches, and above that mass the peak falls as $1/(m_\chi \mu_{\chi e}^2)$, while the electron recoil keeps gaining phase space at large q up to 10 MeV. Umklapp terms, the transverse branches and multiphonon deposits of the following charge, all absent from the model, would add to the peak. The dispersion and the loss of following charge at finite q would take from it (Sect. 10.2). The mass of 12.3 MeV is a property of the model, not of silicon.

Which of the low-threshold regions exists is decided by screening. For a coupling proportional to the total charge, as for a dark photon, the exactness theorem Eq. (8.38) makes S_{ZZ} proportional to the loss function at every (q, ω) with the nuclei free to move. The loss function of silicon has no acoustic peak at order q^2: the Born charges vanish, so the electrons that follow a displaced nucleus cancel its charge, and the acoustic weight of S_{ZZ} starts at order q^4 (Sect. 8.4). For a dark photon the green region of Fig. 17.2b is empty to the order computed in this book, and silicon below 0.3332 MeV has no channel at all. For a mediator that couples to the electrons alone, on the other hand, nothing cancels the following charge. Its share of the f-sum is just $Z_\kappa m_e/M_\kappa = 2.73 \times 10^{-4}$ (Sect. 8.6), but it lies at phonon energies, where the electronic sectors have no weight, and the rate at 1 MeV grows from 17 to 7×10^5 events per kilogram and year at $\bar{\sigma}_e = 10^{-37}\,\text{cm}^2$. Above the gap the same distinction separates the two couplings by the ratio of the electron-only loss function to the total one, which equals one at clamped nuclei and above the phonon band to the order of the Migdal effect (Sect. 8.4). In silicon, then, a detector that probes the electron coupling below the gap tests whether the mediator couples to the total charge or to the electrons alone. A nucleon coupling is unscreened at phonon energies in silicon for the same reason that the dark photon is screened away: its vertex is not the charge, and the vanishing Born charges remove its one screening term (Sect. 8.4).

Table 17.2: Edges of the dominant-sector map in silicon, germanium and GaAs, heavy mediator, standard halo. Masses in MeV unless marked. Status: computed (one or two calculations), measured data, estimate, or a property of a model, as marked in each cell.

Edge	Si	Ge	GaAs
Acoustic edge, $\omega_{\mathrm{th}} = 1\,\mathrm{meV}$	6.1 keV	10.2 keV	not evaluated
Lattice sector of a dark photon below the gap	none at order q^2 ($Z^* = 0$)	none at order q^2 ($Z^* = 0$)	polar phonon from 10.35 keV
Phonon-assisted onset	0.3332 (replica), 0.3374 (free pair); one calculation	0.2119–0.2141; measured data	0.433–0.436; estimate
Zero-phonon electronic edge	0.405–0.408; two calculations	open	L channel 0.517–0.560 (Γ pair closed)
Phonon-assisted rate leads	to ≈ 0.43; estimate	open	to 0.52–0.56; estimate
Kinematic floor $2E_{\mathrm{g}}/v_{\mathrm{max}}^2$	0.333	0.212	0.433
$\mathrm{e}^{-2W} = \frac{1}{2}$ at q_{max}	6.3	≈ 7.7; estimate	not evaluated
Impulse onset, $x_R = 4$	17.5	21.3; estimate	not evaluated
Multiphonon–Migdal crossover at 10 MeV	0.307 eV (Table 17.1)	not evaluated	not evaluated
Acoustic peak of an electron-only coupling = electron recoil, $\omega_{\mathrm{th}} = 1\,\mathrm{meV}$	12.3; property of the model	not evaluated	not evaluated

17.1.4 Germanium and GaAs

In germanium and GaAs the silicon maps change through the edges of each sector, which Table 17.2 collects. Germanium is heavier and softer. Its acoustic edge at a 1 meV threshold, Eq. (11.5) with $c_{\mathrm{LA,max}} = 5.60\,\mathrm{km/s}$ from the elastic tensor of Sect. 11.1, is 10.2 keV, against 6.1 keV in silicon. Its phonon-assisted onset, 0.2119 to 0.2141 MeV, lies below silicon's because the gap is smaller. Its zero-phonon edge, however, is not known (Sect. 13.9), so the electron-coupling map of germanium has an open boundary where silicon's has a computed one. In the nucleon-coupling map the zero-phonon Migdal kick ends later, with $\mathrm{e}^{-2W} = \frac{1}{2}$ at 7.7 MeV, and the impulse onset lies at 21.3 MeV, both from an estimated mean phonon frequency (Sects. 11.4 and 12.4). Since its phonons are softer, with a Debye temperature of 373 K against 645 K (Sect. 11.1), a deposit of given energy needs more phonons, and the Migdal band should begin at a lower threshold than in silicon at the same mass.

GaAs shows what screening does when the Born charges do not vanish. A dark photon couples to its longitudinal optical phonon through $Z^* = 2.11$ (Sect. 11.2). The polar one-phonon rate opens at $2\omega_{\mathrm{L}}/v_{\mathrm{max}}^2 = 10.35\,\mathrm{keV}$ with $\omega_{\mathrm{L}} = 36.35\,\mathrm{meV}$ and, for a light mediator at $\bar{\sigma}_e = 10^{-40}\,\mathrm{cm}^2$, rises to 840 events per kilogram and year near 29 keV (Fig. 11.7). Below the gap the total-charge map of GaAs is filled by the electron-following sector, where silicon's is empty. The electronic sectors open at 0.433 MeV, the kinematic floor of the direct gap. There a phonon must still take the momentum, because the pair at the zone center cannot absorb the 1.15 keV at which the gap is deposited. The phonon-assisted channel leads from 0.433 to between 0.52 and 0.56 MeV, where the L valleys open the zero-phonon channel (Sect. 13.9). A nucleon coupling, by contrast, barely sees the polar mode. Its vertex on the optical phonon at small q is set by the difference of the mean nucleon number per mass of the two sublattices, 1.001072 per u for gallium and 1.001046 per u for arsenic, a difference of 2.59×10^{-5} per u (Worked example 8.6). At finite q the phase of the two sites adds to it, and the map of a nucleon coupling in GaAs has the structure of silicon's, with acoustic, multiphonon and Migdal regions.

The maps are limited first by the electron-following sector of an electron-only coupling, which is known in silicon only as the long-wavelength model of Sect. 10.2. A calculation

of the density that follows the nuclei at momentum transfers up to several times the zone, with Umklapp and multiphonon terms, would fix the mass at which electron recoil takes over the low thresholds (Appendix E.16). The phonon-assisted side of the window is an estimate known to a factor of ten, and a rate from the amplitude on the same bins as the zero-phonon rate would turn the hatched bands into computed boundaries (Appendix E.14). Germanium lacks its zero-phonon edge and, beyond the Debye estimate of Sect. 5.3, a computed phonon density of states, so neither its electronic window nor its multiphonon–Migdal crossover can be drawn. The crossover in silicon is robust against the shape of the multiphonon tail, but it inherits the factor 1.35 of local fields and the open off-diagonal correction of the Migdal probability (Sect. 12.3), which move it by 0.006 eV at 10 MeV.

17.2 Absorption across channels

A dark photon of 25 µeV is absorbed by the tunneling atoms of an oxide film, one of 36 meV by a longitudinal optical phonon of GaAs, and one of 16 eV by the plasmon of silicon. Three events in a kilogram and year need $\zeta = 2.4 \times 10^{-15}$, 2.9×10^{-17} and 1.0×10^{-17} for the three (Worked example 16.7 and Sects. 10.7.1 and 11.8.3). The numbers come from three chapters, but each is Eq. (8.19) evaluated at one point, $\boldsymbol{q} = \boldsymbol{p}_\phi$ of order $10^{-3} m_\phi$ and $\omega = m_\phi$, with the response of a different target. Unlike scattering, absorption leaves no kinematic freedom: the mass fixes the energy, and a target either has an excitation at that point or does not absorb. Comparing the channels is then a matter of coverage, of which material absorbs at each mass, and by how much the medium screens it. Below a few µeV a second question enters, since the quanta are then softer than the thermal energy of a readout at millikelvin temperatures, and the field must be measured as a wave.

17.2.1 In-medium mixing across targets

Each boson probes one response function at that point. The dark photon measures the loss function $L(m_\phi)$ of the total dielectric function, Eq. (10.31), and every excitation that carries charge enters through it: interband pairs, optical phonons, tunneling atoms, donors, Dirac pairs and quasiparticle pairs alike. The axion-like particle with its electron coupling measures the spin-current response, which for independent electrons without spin–orbit coupling is the optical conductivity σ_1, Eq. (10.32), and which in a superconductor becomes $\sigma_1^{\mathcal{J}}$ of Eq. (15.27); in a magnet it reaches the uniform spin precession instead, Eq. (14.22). Its photon coupling in a static field $\boldsymbol{B}_0$ measures $\hat{\boldsymbol{B}}_0 \cdot \mathrm{Im}[-\varepsilon^{-1}] \cdot \hat{\boldsymbol{B}}_0$, the dark-photon response with $\zeta = g_{a\gamma\gamma} B_0 / m_\phi$ (Eq. (10.33)). The scalar coupled to electrons measures the kinetic-energy row $S_{\mathcal{T}\mathcal{T}}$ in a crystal without a polar vector, Eq. (10.35), and in a polar crystal the electrons that each nucleus carries, Eq. (11.55). Per unit target mass the three rates are proportional to L/ρ_T, to $m_\phi \sigma_1 / \rho_T$ and to $S_{\mathcal{T}\mathcal{T}}/(m_\phi^2 \rho_T)$.

The dark photon, and the axion-like particle in a field, mix with the photon of the medium, and for them the mixing sets a ceiling. At fixed ε_1 the loss function $L = \varepsilon_2/(\varepsilon_1^2 + \varepsilon_2^2)$ is largest at $\varepsilon_2 = |\varepsilon_1|$, where $\partial L / \partial \varepsilon_2$ vanishes, so that

$$L(m_\phi) \leq \frac{1}{2\,|\varepsilon_1(m_\phi)|}. \tag{17.1}$$

A material absorbs a dark photon strongly only where ε_1 is small. Below its gap silicon has $\varepsilon_1 = 11.7$ at room temperature, so $L \leq 0.043$, and the cold edge reaches 1.5×10^{-6} at 1.2 eV (Worked example 13.8). Amorphous SiO_2, with $\varepsilon_r = 3.9$, allows 0.13, and its two-level systems give $L = \tan\delta_0 / \varepsilon_r = 7.7 \times 10^{-5}$ for the loss tangent $\tan\delta_0 = 3 \times 10^{-4}$ of a thermal oxide (Eq. (16.55)). The background constants of $ZrTe_5$, 20.75 to 308 along its three axes, cap the

Dirac pairs at 0.024 for the polarization along the axis with the smallest constant and at 0.0016 along the axis with the largest. At 1 meV the condensate of aluminum has $\varepsilon_1 = -1.5 \times 10^8$, so $L \leq 3.3 \times 10^{-9}$, and the phonon-assisted pairs reach 2.4×10^{-16} (Worked example 15.4). The large values of L sit where ε_1 crosses zero, at the longitudinal modes. At the longitudinal optical phonon of GaAs the width of the line alone limits it, $L = 2\mathcal{I}_{\rm pol}/\pi\gamma_{\rm L} = 1.33$, with $\gamma_{\rm L}$ the width of the longitudinal line (Sect. 11.8.3); it is 7.9 at the top longitudinal line of sapphire, 112.4 meV, and 4.49 at the plasmon of silicon, 16.4 eV. The deepest minima of the dark-photon reach curves are longitudinal modes, with $\zeta \approx 10^{-17}$ for three events in a kilogram and year. The same mixing places the phonon line of a polar crystal at $\omega_{\rm L}$, not at the transverse frequency where ε_2 peaks: at $\omega_{\rm T} = 33.62$ meV GaAs has $L = 0.0054$.

Each target then carries its own version of the screening factor. The two-level systems of a glass are screened once by the static constant, through $1/\varepsilon_r$. Dilute donors are screened once by the host, a factor $1/\varepsilon_{\rm host}^2 = 1/130$ on the absorption of each donor, with the cold constant 11.4 of silicon (Sect. 16.5.2). In a Dirac material the published rates omitted one power of the background constant, the screening by the remote bands. With it included, the reach in ζ weakens by the polarization average of $\varepsilon_{{\rm bg},ii}^{1/2}$, a factor of 6.3 to 6.4 for $ZrTe_5$ (Sect. 16.1). A clean superconductor screens a field of 1 meV by $1/|\varepsilon|^2 = 4.4 \times 10^{-17}$, and it reaches just $\zeta = 1.5 \times 10^{-9}$ with a kilogram and year. A magnet absorbs a dark photon through its magnetic field, $\boldsymbol{B'} = \mathrm{i}\boldsymbol{p}_\phi \times \hat{\boldsymbol{e}}\, A'$, which is smaller than the electric field by the speed v, and the field inside the sample depends on its dielectric constant and on the conductors around it (Sect. 14.3).

The axion-like particle has no mixing, and the ceiling Eq. (17.1) does not apply to it. As a result, the material that screens a dark photon most is among the best absorbers. Clean aluminum absorbs a particle of 0.72 meV through the spin current, and three events in a kilogram and year need $g_{aee} = 6.1 \times 10^{-13}$ (Sect. 15.4), the same order as the best that silicon reaches, 1.4×10^{-13} at 4.3 eV. The silicon optimum lies near the main peak of ε_2, the region of strongest suppression for the dark photon. The two bosons thus favor opposite regions of the same spectrum, the dark photon the zeros of ε_1 and the axion-like particle the maxima of ε_2, and the ratio of their rates at equal couplings is $(3m_\phi^2/4m_e^2e^2)|\varepsilon|^2$ (Sect. 10.7.3). Near that peak the measured σ_1 understates the triplet response by a fifth, and in heavier crystals spin–orbit coupling breaks the factor $\frac{3}{4}$ between the two responses (Sect. 10.7.3). In a static field the photon coupling follows the dark photon instead, and the factor $1/m_\phi$ in $\zeta = g_{a\gamma\gamma}B_0/m_\phi$ favors the light lines. At 10 T the reaches of the GaAs and sapphire phonon lines correspond to $g_{a\gamma\gamma} = 5.4 \times 10^{-13}$ and $5.9 \times 10^{-13}\,\mathrm{GeV}^{-1}$, that of the silicon plasmon to $8.8 \times 10^{-11}\,\mathrm{GeV}^{-1}$.

A scalar coupled to electrons does not mix in a crystal without a polar vector, where the mixing is of order p_ϕ^2 (Sect. 10.7.4). In a polar crystal the medium enters through the vertex: each nucleus carries $Z_\kappa - Z_\kappa^*/\varepsilon_\infty$ electrons, and in GaAs the two sublattices nearly cancel, leaving 5 % of the rate of a photon-like scalar (Worked example 11.9). A scalar coupled to the nuclei in proportion to their masses, $c_\kappa = \tilde{c}M_\kappa$, is absorbed through the Migdal-type process of Eq. (12.56), whose spectrum is again the loss function, with ζ^2 replaced by $\tilde{c}^2m_e^2\langle v^2\rangle/e^2$; its coverage is the dark-photon curve of each target.

17.2.2 Coverage by mass

Fig. 17.3 collects the reaches of the targets of the preceding chapters on one mass axis, from 1 neV to 10 keV, and Table 17.3 reads it decade by decade. Every curve is the rate that its home section derived, with that section's inputs, converted to the coupling that gives three events in a kilogram and year without background. One point is new. The Kittel line of YIG absorbs a dark photon through the substitution $g_{aee}\boldsymbol{p}_\phi \to \zeta e\,\boldsymbol{p}_\phi \times \hat{\boldsymbol{e}}$ of Sect. 14.3. For a random

Table 17.3: Absorbing targets of the preceding chapters by decade of boson mass, with the best reach of each, three events in a kilogram and year without background. Lines are narrow and cover one mass unless tuned; continua cover the whole range given. A dash means no target in the book absorbs that boson at leading order in that decade.

m_ϕ	dark photon, ζ	axion-like, g_{aee}	scalar, $g_{\phi ee}$
1 neV–1 µeV	two-level systems, $\gtrsim 3\times10^{-15}$; acoustic modes of bodies (Sect. 11.8.4); coherent readout	— (nuclear spins couple through g_{aNN}, Sect. 14.6)	resonant masses (Sect. 11.8.5)
1 µeV–0.1 meV	two-level systems, 2.4×10^{-15}; YIG line, 7.6×10^{-17} at 10 µeV (vacuum ζ)	YIG line, 1.6×10^{-17} at 10 µeV	—
0.1 meV to 1 meV	two-level systems to 0.46 meV, then glass vibrations; aluminum from 0.36 meV, 1.5×10^{-9}	YIG line, 1.7×10^{-16} at 1.16 meV; aluminum, 6.1×10^{-13}	—
1 meV to 10 meV	glass vibrations to 4.3 meV, 7.6×10^{-16}	MnF_2 line, 1.9×10^{-16} at 1.1 meV; aluminum, 9.1×10^{-12} at 10 meV	—
10 meV to 100 meV	GaAs line, 2.9×10^{-17}; Si:P $2p_0$, 2.3×10^{-16}; $ZrTe_5$ from 23.6 meV, 1.25×10^{-15}	— (nucleon coupling: Sect. 11.6)	GaAs line, 3.1×10^{-14}
0.1 eV to 1 eV	sapphire, 1.0×10^{-17} at 112 meV; $ZrTe_5$ to 0.38 eV; none from 0.38 eV to 1.17 eV	—	—
1 eV to 10 eV	silicon from 1.1728 eV; 4.4×10^{-17} at 10 eV	silicon, 1.4×10^{-13} at 4.3 eV	silicon (model), 1.1×10^{-15}
10 eV to 100 eV	silicon plasmon, 1.0×10^{-17} at 16.4 eV	silicon, 3–5 $\times10^{-13}$	silicon (model)
0.1 keV to 10 keV	silicon, 1.9×10^{-16} at 100 eV, 2.6×10^{-15} at 1 keV	silicon to 1 keV, 3–5 $\times10^{-13}$	—

polarization the dark-photon rate is the axion-like rate times $e^2\langle|(\boldsymbol{v}\times\hat{\boldsymbol{e}})_\perp|^2\rangle/\langle v_\perp^2\rangle = 0.0451$, with the halo averages taken along the static field, so that $\zeta = g_{aee}/0.212$. At 10 µeV, where the axion-like reach is $g_{aee} = 1.62\times10^{-17}$, this gives the vacuum $\zeta = 7.6\times10^{-17}$.

The dark photon is covered almost everywhere, with two weak stretches. Between 0.38 eV, where the Dirac cone of $ZrTe_5$ ends at twice its cutoff, and 1.17 eV, where cold silicon begins to absorb, no target of the book absorbs at leading order; the sapphire lines and their multiphonon background end near 0.12 eV. A narrow-gap semiconductor or a Dirac material with a wider cone would fill this gap, with the screening ceiling of its background constant. The second weak stretch is the meV decade, where the glass absorbs through its disordered vibrations at $\zeta \approx 10^{-15}$ and the superconductor is screened out.

The axion-like particle with its electron coupling has fewer targets. Magnons cover 10 µeV to a meV, as far as a laboratory field tunes their lines; clean aluminum covers 0.36 meV to 11.5 meV, above which its phonon-assisted absorption takes over; and silicon covers everything above its edge that the tables reach. Between 10 meV and 1.17 eV, however, no target of the preceding chapters covers this coupling. Phonons absorb an axion-like particle through its nucleon coupling alone, on the spin-carrying isotopes of a crystal (Sect. 11.6), and a Dirac material would need its spin-current response computed with its strong spin–orbit coupling, which the factor $\frac{3}{4}$ of Eq. (10.32) does not contain. The scalar has two targets: the GaAs phonon line, suppressed by the near-cancellation of its sublattices, and the kinetic row of silicon above its direct gap, which in Fig. 17.3c is a model with pseudo-wavefunctions. Between the indirect edge and 3.1 eV silicon absorbs a scalar only with a phonon, through the kinetic

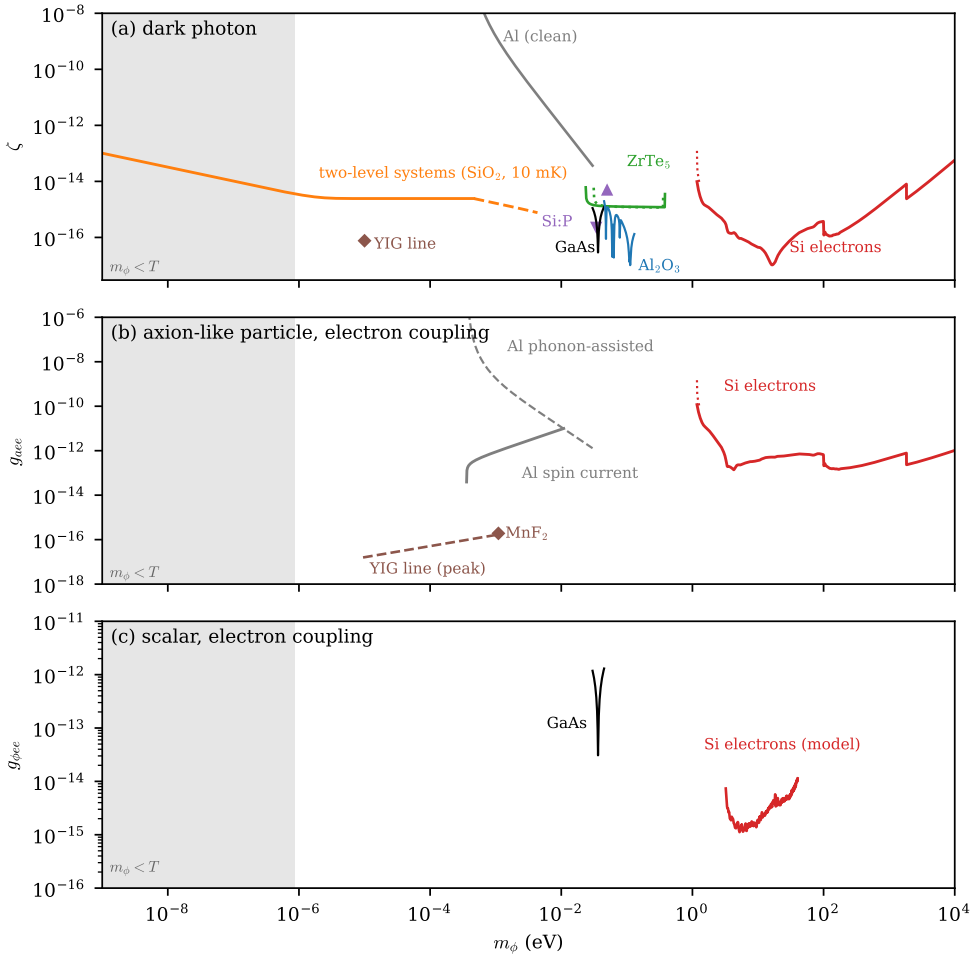


Figure 17.3: Absorption coverage: coupling for three events in one kilogram and year without background, $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$, against the boson mass. (a) Dark photon. Two-level systems of a thermal SiO_2 film at 10 mK, $\tan\delta_0 = 3 \times 10^{-4}$, $\varepsilon_r = 3.9$, Eq. (16.55) (solid, up to 0.46 meV), and the vibrational absorption of the glass up to its Ioffe–Regel frequency, 4.3 meV (dashed; Limit Check 16.12). Clean aluminum, phonon-assisted pair creation with the Debye transport function, Eq. (15.25). Si:P at $10^{15}\,\mathrm{cm^{-3}}$: the $2p_0$ line at 34.11 meV (down triangle) and the continuum at 50 meV (up triangle), Sect. 16.5.2. $ZrTe_5$ with the background in the total ε, polarization average: measured-structure parameters with half-gap 11.8 meV and cutoff $\tilde{\Lambda} = 0.191$ eV (solid), relaxed-cell parameters with 15.6 meV and 0.180 eV (dotted), as in Fig. 16.3. GaAs and sapphire single-phonon lines, widths of Fig. 11.27. Silicon electrons from measured optical constants at 300 K, drawn from the excitonic edge of cold silicon, 1.1728 eV, with the cold excitonic edge of Eq. (13.47) dotted up to 1.3 eV; the room-temperature data overstate the cold rate near the edge (Sect. 13.5). Diamond-shaped point: the Kittel line of a YIG sphere at 10 µeV, vacuum ζ, polarization random. (b) Axion-like particle, electron coupling. YIG Kittel line on resonance as it is tuned from 9.5 µeV to 1.16 meV (10 T), Eq. (14.22), and the MnF_2 resonance at 1.1 meV (diamond-shaped point), both with the assumed width $\Gamma_\mathrm{M} = 10^{-5}\omega$. Clean aluminum: direct pair creation by the spin current, Eq. (15.27) with the quasiparticle factor $1/(1+\lambda_\mathrm{ep}) = 0.70$, up to 10.9 meV (solid), and the phonon-assisted σ_1 that published projections use (dashed). Silicon electrons from the measured σ_1, Eq. (10.32). (c) Scalar, electron coupling: the GaAs phonon line with the site couplings of Eq. (11.55), and the kinetic-energy row of silicon in the empirical-pseudopotential model of Fig. 10.14. Gray: $m_\phi < T$ for $T = 10$ mK, where counting has lost its advantage over a coherent readout (Sect. 17.2.3).

row of Sect. 13.8, which no measurement provides; Appendix E.14 specifies its calculation.

A tuned line and a continuum are compared differently. On resonance the Kittel line of YIG reaches $g_{aee} = 1.6 \times 10^{-17}$ at 10 µeV, but only within its full width $\Gamma_{\rm M} = 10^{-5} m_\phi$. A search that scans the line across one e-fold of mass with uniform exposure spends a fraction $(\pi/2)\Gamma_{\rm M}/m_\phi = 1.6 \times 10^{-5}$ of its time on each mass. Averaged over the scan the reach is 4.1×10^{-15}, still a hundred times better than the continuum of clean aluminum at its best mass. The MnF_2 resonance at 1.1 meV gives 1.9×10^{-16} on resonance and 4.8×10^{-14} averaged. The peak values rest on the assumed ratio $\Gamma_{\rm M}/\omega = 10^{-5}$ (Sect. 14.3). The averaged ones do not depend on it, since the area of the line, not its height, sets what a scan delivers, as for the phonon lines of Eq. (11.58). The fixed lines of GaAs, sapphire and Si:P cover their own masses alone.

Worked example 17.2: The handover from GaAs phonons to silicon electrons

GaAs absorbs a dark photon at its longitudinal optical phonon, 36.35 meV, with all the weight of the line, $\mathcal{I}_{\rm pol} = 0.759\,\text{meV}$, within a few widths, and with multiphonon absorption up to a few phonon energies. Its electrons absorb from its gap, 1.52 eV at low temperature (Sect. 3.3). Silicon absorbs from its excitonic edge. At which mass does silicon take over?

The phonon line does not decide it. The damped factorized model of Fig. 11.27 gives $\varepsilon_2 = 2.5 \times 10^{-4}$ at 1.17 eV, an absorption coefficient of $4.3\,\text{cm}^{-1}$ with the refractive index 3.48 used below. That value is an artifact of the model: with different widths in its numerator and denominator it tends to $\varepsilon_2 = \varepsilon_\infty(\gamma_{\rm L} - \gamma_{\rm T})/\omega$ far above the line, a constant conductivity. A Lorentz oscillator with a single width gives $\varepsilon_2 = 4.8 \times 10^{-7}$ and $0.008\,\text{cm}^{-1}$ at the same energy. A pure crystal transmits there, and what it absorbs depends on its defects.

Equal rates. Per unit mass the rates are equal when $L_{\rm GaAs}/\rho_{\rm GaAs} = L_{\rm Si}/\rho_{\rm Si}$. Below a gap $\varepsilon_2 \ll \varepsilon_1 = n_r^2$, and with $\varepsilon_2 = n_r \alpha_{\rm abs}/\omega$ the loss function is $L = \alpha_{\rm abs}/(n_r^3 \omega)$. With $n_r = 3.48$ for GaAs at 22 °C, from the dispersion formula of Skauli et al. [1] as tabulated in Ref. [2], $n_r = 3.56$ for silicon at 300 K [3], and densities 5.361 and $2.33\,\text{g cm}^{-3}$, GaAs needs $(3.48/3.56)^3 \times 5.361/2.33 = 2.15$ times the absorption coefficient of silicon to match it.

The silicon edge. At millikelvin temperatures silicon absorbs from the 1s replica of the transverse acoustic branch, $E_{\rm g}(0) + \omega_{\rm TA} - E_x^b = 1.172\,77\,\text{eV}$, and Eq. (13.47) rises there as a square root. With the scale of Sect. 13.8,

$$\alpha_{\rm abs}^{\rm Si} = 13.3\,\text{cm}^{-1}\,\left[(m_\phi - 1.172\,77\,\text{eV})/\text{eV}\right]^{1/2},$$

which is $0.13\,\text{cm}^{-1}$ at 0.1 meV above the threshold and $0.42\,\text{cm}^{-1}$ at 1 meV.

The handover. Setting $2.15\,\alpha_{\rm abs}^{\rm Si}$ equal to the absorption of the GaAs crystal places the handover 0.012 meV above the threshold for $0.1\,\text{cm}^{-1}$ and 1.2 meV above it for $1\,\text{cm}^{-1}$. A crystal absorbing $10\,\text{cm}^{-1}$ would move it 40 meV up; at that distance the further replicas of Eq. (13.47) have opened, and the square-root onset alone would have put it at 0.12 eV. For any GaAs crystal transparent to better than $1\,\text{cm}^{-1}$, silicon takes over within a meV of its excitonic edge, at $m_\phi \simeq 1.173\,\text{eV}$, and the square-root onset of the replica makes the handover sharp. A meV above the threshold silicon reaches $\zeta = 5.6 \times 10^{-14}$, two thousand times weaker than the GaAs line at its center.

The worked example uses a model of the cold edge. The cold measurements near the edge differ among themselves by a factor of twenty (Sect. 13.8), which moves the absorption of

silicon but not the place of the handover, since the threshold is fixed by the gap, the phonon and the exciton binding.

17.2.3 Coherent and counting readout

A YIG sphere tuned to 10 µeV counts the quanta it absorbs (Worked example 14.3). A ^{207}Pb sample at 1 neV reads the transverse magnetization that the field drives (Sect. 14.6). Both measure the absorptive part of a response near $\omega = m_\phi$, but the first counts quanta of energy m_ϕ and the second measures the amplitude of a field. Which readout is better at a given mass depends on the thermal occupation $n_{\rm B}(m_\phi) = 1/(\mathrm{e}^{m_\phi/T} - 1)$ of the mode that is read, with T the temperature of that mode.

Take a record of length $t_{\rm int}$ longer than the coherence time. The signal is spread over $N_{\rm coh} \simeq t_{\rm int}\Delta\omega_{\rm eq}/2\pi$ independent frequency bins, with $\Delta\omega_{\rm eq} = 2.67/t_{\rm coh}$ the equivalent width of the line (Sect. 14.6.4), and it delivers $\bar N_{\rm sig}$ absorbed quanta. A counter whose detection band is the line sees, in the same bins, $\bar N_{\rm th} = n_{\rm B}(m_\phi)N_{\rm coh}$ thermal quanta. They are chaotic, and their number fluctuates with the variance $\bar N_{\rm th} + \bar N_{\rm th}^2/N_{\rm coh} = N_{\rm coh} n_{\rm B}(m_\phi)[n_{\rm B}(m_\phi)+1]$ of Sect. 14.6.1. A linear readout, a phase-preserving amplifier or a magnetometer, measures both quadratures of the mode, and its noise is at least $n_{\rm B}(m_\phi) + 1$ quanta per bin: the thermal quanta plus one zero-point quantum, half of it present at the input and half added by the amplifier [4, Eq. (9)]. This per-bin noise has the rms $n_{\rm B}(m_\phi)+1$, and summed in quadrature over the bins it fluctuates by $[n_{\rm B}(m_\phi)+1]N_{\rm coh}^{1/2}$. The significance of the signal is then $\bar N_{\rm sig}/\{N_{\rm coh} n_{\rm B}(m_\phi)[n_{\rm B}(m_\phi)+1]\}^{1/2}$ for the counter and $\bar N_{\rm sig}/\{N_{\rm coh}^{1/2}[n_{\rm B}(m_\phi) + 1]\}$ for the linear readout, whose square is the weak-signal statistic $\mathtt{t}$ of Sect. 14.6.4 with $x_k = \bar N_{\rm sig}/\{N_{\rm coh}[n_{\rm B}(m_\phi) + 1]\}$ in each bin. At equal significance the counter needs fewer signal quanta by the ratio

Counting against linear readout

$$\frac{\bar N_{\rm sig}^{\rm lin}}{\bar N_{\rm sig}^{\rm cnt}} = \Big[\frac{n_{\rm B}(m_\phi) + 1}{n_{\rm B}(m_\phi)}\Big]^{1/2} = \mathrm{e}^{m_\phi/2T}, \tag{17.2}$$

for the same absorber, record and line, as long as the counter sees thermal quanta in the record and its detection band is the line width $\Delta\omega_{\rm eq}$. A counter that sees a band B wider than the line collects more thermal quanta, and the ratio is multiplied by $(B/\Delta\omega_{\rm eq})^{-1/2}$. In the coupling, which enters the signal squared, the ratio is $\mathrm{e}^{m_\phi/4T}$. For $n_{\rm B}(m_\phi) \ll 1$ the Poisson treatment of Lamoreaux and collaborators [4, Eqs. (12)–(14)] is the limit of this result; their cavity counter sees the whole cavity band, which carries the extra factor $(Q_{\rm c}/2\pi Q_a)^{1/2}$ of their Eq. (18), with $Q_{\rm c}$ and Q_a the quality factors of the cavity and of the line.

Three regimes follow. For $m_\phi \ll T$ the ratio tends to one. Each mode holds many thermal quanta, their fluctuation dominates either readout, and a counter gains nothing over a measurement of the field, while it would have to resolve single quanta far below the thermal energy. At 1 neV a mode at 4.2 K holds 3.6×10^5 quanta and one at 10 mK holds 861, and every search there reads the field. For $m_\phi \approx T$ the gain is modest, 1.65 in the signal and 1.28 in the coupling at $m_\phi = T$. For $m_\phi \gg T$ the counter wins exponentially, until the record holds no thermal quantum at all. Requiring zero thermal counts with 95 % probability, $\bar N_{\rm th} < 0.05$, the counter needs three signal quanta (Sect. 1.5) [4, Eqs. (19), (20)], whatever the mass. With $\Delta\omega_{\rm eq} = 2.67\, m_\phi v_0^2$ this happens above 17 µeV for a one-hour record and 20 µeV (4.9 GHz) for one day at 10 mK, and above 0.19 meV and 0.22 meV at 100 mK, that is, at $m_\phi \approx (20\text{–}26)\,T$. The linear readout still needs $\bar N_{\rm sig} = (2.71\, N_{\rm coh})^{1/2}[n_{\rm B}(m_\phi) + 1]$ at the threshold $\mathtt{t} = 2.71$ of Eq. (C.121), the one-sided 95 % level. For instance, at 24.8 µeV, the 6-GHz mass of Worked

example 16.7, a day at 10 mK spreads the line over 8.7×10^8 bins, and the linear readout needs 4.9×10^4 quanta against the counter's three, a factor of 127 in the coupling. At 10 µeV the same record still holds 3200 thermal quanta, and the counter wins by 330 in the signal and 18 in the coupling.

The gray band of Fig. 17.3, $m_\phi < T$ (0.86 µeV, or 210 MHz at 10 mK), is where the advantage of counting has fallen below $e^{1/2}$ and tends to one, and the field is read coherently. Above that mass the counter gains $e^{m_\phi/4T}$ in the coupling, and above $20\,T$ it is limited by the absorbed quanta alone, with a reach that improves as $t_{\rm int}^{-1/2}$ against the $(t_{\rm int}t_{\rm coh})^{-1/4}$ of the field measurement (Eq. (14.56)). Two effects reduce the advantage. A linear readout of one quadrature of a squeezed mode can go below one zero-point quantum [4], which narrows the gap by the degree of squeezing. A real counter, moreover, has dark counts besides the thermal quanta, and they add to its background. The background-free counting that the reach curves of Fig. 17.3 assume holds where both are absent.

17.3 Inelastic scattering into collective modes

A dark-matter particle of 10 keV at the largest laboratory speed $v_{\rm max}$ carries 35 meV of kinetic energy, just short of the 36.35 meV longitudinal optical phonon of GaAs, and elastic scattering cannot create that phonon below 10.35 keV (Sect. 11.2). Let the same particle arrive in an excited state 1 eV above its ground state and leave in the lower one. It then emits the phonon at every speed of the halo, at a rate of 2.6×10^4 events per kilogram and year for a light dark photon at $\bar\sigma_e = 10^{-40}\,{\rm cm}^2$. Sect. 10.9 followed the splitting Δ_χ through the electron-recoil spectrum of silicon. Here we follow it into the collective modes, phonons, magnons and plasmons, which Sect. 8.2.4 showed to be the natural final states of down-scattering: the band of allowed $(\boldsymbol{q},\omega)$ moves up by $|\Delta_\chi|$, and a mode whose energy lies below the splitting is always reached. One formula covers all the modes, and the dispersion of each mode decides where its line sits, how wide it is, and how much of the splitting it takes. We will then compare the electronic and lattice channels of silicon and the endothermic thresholds, $\Delta_\chi > 0$, of every channel.

17.3.1 Down-scattering into a collective mode

Every collective response of Chaps. 10, 11 and 14 is, at $T = 0$, a sum of lines. The one-phonon response Eq. (11.1), the one-magnon response Eq. (14.1) and the plasmon pole of Eq. (4.61) each put a weight on the dispersion surface $\omega = \omega_\nu(\boldsymbol{q})$ of a branch ν. Contracted with the dark-matter tensor, the integrand of Eq. (8.16) becomes

$$\sum_{ab} \mathfrak{D}_{ab}(\boldsymbol{q})\, S_{ab}(\boldsymbol{q},\omega) = \frac{2\pi}{\Omega} \sum_\nu \mathcal{Y}_\nu(\boldsymbol{q})\, \delta\big(\omega - \omega_\nu(\boldsymbol{q})\big), \tag{17.3}$$

which defines the mode weight $\mathcal{Y}_\nu \geq 0$. For the nucleon coupling of Sect. 11.1, in the notation of Eq. (11.1) (nucleon numbers A_κ, polarization vectors $\boldsymbol{e}_{\kappa\nu}$), $\mathcal{Y}_\nu = |\tilde V_N|^2 |\sum_\kappa A_\kappa e^{-W_\kappa} e^{i\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa} \boldsymbol{q}\cdot \boldsymbol{e}^*_{\kappa\nu}/\sqrt{M_\kappa}|^2/2\omega_\nu$. For the spin coupling $\mathcal{O}_4$ in an antiferromagnet, with $\Omega = \Omega_{\rm ion}$, the contraction Eq. (14.2) gives $\mathcal{Y} = \frac14 |\boldsymbol{S}||c_4|^2 |F^{\rm mag}|^2 w^{\rm M}_{\boldsymbol{q}}$, with the coupling c_4, the magnetic form factor $F^{\rm mag}$ and the magnon weight $w^{\rm M}_{\boldsymbol{q}}$ of Sect. 14.1. For the plasmon of an electron gas coupled through the electron density, $S = (2q^2/e^2)L$ and the pole carries the weight $\pi/(\partial\varepsilon_1/\partial\omega)$, so that $\mathcal{Y} = \Omega|\tilde V|^2 q^2/[e^2\,\partial\varepsilon_1/\partial\omega]$.

Down-scattering has $\Delta_\chi < 0$, and energy conservation, $\omega_{\boldsymbol{q}} = \omega_\nu(\boldsymbol{q})$ with $\omega_{\boldsymbol{q}}$ of Eq. (1.22), reads

$$|\Delta_\chi| + \boldsymbol{q}\cdot\boldsymbol{v} - \frac{q^2}{2m_\chi} = \omega_\nu(\boldsymbol{q}).$$

We first set $\boldsymbol{v} = 0$. Along each direction $\hat{\boldsymbol{q}}$ the condition fixes the magnitude of the momentum transfer at the roots $q_\nu(\hat{\boldsymbol{q}})$ of

$$|\Delta_\chi| - \frac{q_\nu^2}{2m_\chi} = \omega_\nu(q_\nu \hat{\boldsymbol{q}}), \tag{17.4}$$

and we write $v_\nu = q_\nu/m_\chi$ for the speed of the outgoing particle relative to the incoming one. For a mode whose energy is small against the splitting, $q_\nu \simeq (2m_\chi|\Delta_\chi|)^{1/2}$; at a fixed deposit ω this is the shell momentum q_Δ of Sect. 10.9. As a function of q along $\hat{\boldsymbol{q}}$, the argument of the delta function has the slope $-(q/m_\chi + u_\nu)$, with $u_\nu = \hat{\boldsymbol{q}} \cdot \nabla_{\boldsymbol{q}}\, \omega_\nu$ the group velocity of the mode along $\boldsymbol{q}$. Writing $\mathrm{d}^3 q = q^2 \mathrm{d}q\, \mathrm{d}^2\hat{\boldsymbol{q}}$, the radial integral picks up $q_\nu^2/(q_\nu/m_\chi + u_\nu)$ at each root. The rate of one particle, $\Gamma = \int \mathrm{d}\omega \int \mathrm{d}^3 q/(2\pi)^3 \sum \mathfrak{D}_{ab} S_{ab}\, \delta(\omega - \omega_{\boldsymbol{q}})$, is then $(2\pi)^{-3}(2\pi/\Omega) \sum_\nu \int \mathrm{d}^2\hat{\boldsymbol{q}}\, q_\nu^2 \mathcal{Y}_\nu/(q_\nu/m_\chi + u_\nu)$, which summed over branches and directions reads

Exothermic rate into a collective mode

$$\Gamma = \frac{m_\chi}{\pi\Omega} \sum_\nu \Big\langle \frac{q_\nu\, \mathcal{Y}_\nu(q_\nu \hat{\boldsymbol{q}})}{1 + u_\nu/v_\nu} \Big\rangle_{\hat{\boldsymbol{q}}} \Big[1 + O(v^2/v_\nu^2)\Big], \qquad \mathcal{R} = \frac{\rho_\chi}{m_\chi \rho_T}\, \Gamma, \tag{17.5}$$

with q_ν the roots of Eq. (17.4), $\langle\cdot\rangle_{\hat{\boldsymbol{q}}}$ the average over directions, and $\mathcal{R}$ the rate per unit target mass. It holds for v_ν above the speeds of the halo and for a weight $\mathcal{Y}_\nu$ that varies little over momenta $m_\chi v_{\max}$.

At finite $\boldsymbol{v}$ the surface of Eq. (17.4) moves by a vector of order $m_\chi \boldsymbol{v}$ and changes its radius at second order. The term linear in $\boldsymbol{v}$ averages to zero over a weight and a dispersion even in $\boldsymbol{q}$, as for the electronic shell of Eq. (10.54), and the correction to the rate is of order v^2/v_ν^2. The deposited energy, by contrast, shifts at first order, and Sect. 17.3.2 shows that this sets the width of the line. The factor $(1 + u_\nu/v_\nu)^{-1}$ is new compared with the electronic shell, where the energy is integrated at fixed ω. It is close to one for phonons and magnons, whose group velocities, 10^{-5} to 10^{-4}, lie far below v_ν, but not for the plasmon, whose dispersion is steep.

The velocity independence has a simple reading in the incoherent limit. For a free scatterer the rate per particle of elastic scattering is $n\sigma v$, and that of exothermic down-scattering is $n\sigma v_\nu$: the final momentum, and with it the phase space, is fixed by the splitting rather than by the incoming speed. The magnons of NiO show it. With the spin coupling $\mathcal{O}_4$, a heavy mediator and $\bar{\sigma}_e = 10^{-40}\,\mathrm{cm}^2$ (the setting of Fig. 14.3), a particle of 1 MeV with $|\Delta_\chi| = 10\,\mathrm{eV}$ has $q_\nu = 4.45\,\mathrm{keV}$, beyond the ordering wavevector at 2.58 keV. The shell then crosses many magnetic zones, and the direction average of the weight is 1.149, close to its zone average 1.156 (in the spin-wave theory of Worked example 14.1, $1 + 2\Delta S = 1.157$, with ΔS the zero-point spin reduction). Eq. (17.5) yields 8210 events per kilogram and year, within 3 % of the full integral over the halo, 8000, against 2620 for elastic scattering at the same mass. Three factors make up the ratio of 3.1. The speed ratio is $v_\nu/\langle v\rangle = 3.77$, with the mean laboratory speed $\langle v\rangle = 354\,\mathrm{km/s}$. The squared magnetic form factor averages to 0.60 on the shell at 4.45 keV, against 0.89 over the elastic rate at 1 MeV (Sect. 14.1), a factor 0.68. The magnon weight, 1.149 on the shell against 1.010 over the elastic rate, adds a factor 1.14. Together these give 2.90, and the remaining 8 % reflects the elastic rate, whose halo weighting at this mass is not exactly proportional to $\langle v\rangle$.

17.3.2 Line positions and widths

Each root of Eq. (17.4) deposits $\omega_\nu(q_\nu\hat{\boldsymbol{q}})$. The splitting is shared between the outgoing dark matter, which keeps $q_\nu^2/2m_\chi$, and the mode, and the dispersion decides the share. A parabolic branch, $\omega_\nu(\boldsymbol{q}) = \omega_\nu(0) + q^2/2m_\nu^*$ with a dispersion mass m_ν^* of either sign, makes this exact for any $\boldsymbol{v}$. Energy conservation becomes $|\Delta_\chi| - \omega_\nu(0) + \boldsymbol{q}\cdot\boldsymbol{v} = q^2/2\mu_\nu$ with $\mu_\nu = m_\chi m_\nu^*/(m_\chi + m_\nu^*)$,

and completing the square,

$$|\boldsymbol{q}-\mu_\nu\boldsymbol{v}|^2=q_\nu^2+\mu_\nu^2v^2,\qquad q_\nu^2=2\mu_\nu\big(|\Delta_\chi|-\omega_\nu(0)\big),$$
$$\omega_\nu(q_\nu)=\omega_\nu(0)+\frac{\mu_\nu}{m_\nu^*}\big(|\Delta_\chi|-\omega_\nu(0)\big). \tag{17.6}$$

The momentum transfers lie on a sphere centered at $\mu_\nu\boldsymbol{v}$, and at $\boldsymbol{v}=0$ the mode takes the fraction $\mu_\nu/m_\nu^*=m_\chi/(m_\chi+m_\nu^*)$ of the energy above its bottom. A free electron at rest is the branch $\omega(0)=0$, $m^*=m_e$, and Eq. (17.6) returns the line $(\mu_{\chi e}/m_e)|\Delta_\chi|$ of Sect. 10.9.1. A flat branch, $|m_\nu^*|\gg m_\chi$, takes little more than $\omega_\nu(0)$ and leaves the rest of the splitting to the dark matter. A steep branch, $m_\nu^*\ll m_\chi$, takes nearly all of it.

The width follows from the same sphere. On it $q^2=q_\nu^2+2\mu_\nu^2v^2+2\mu_\nu\boldsymbol{v}\cdot(\boldsymbol{q}-\mu_\nu\boldsymbol{v})$, and the deposited energy $\omega_\nu(0)+q^2/2m_\nu^*$ differs from the line $\omega_\nu(q_\nu)$ by $(\mu_\nu q_\nu v/m_\nu^*)\cos\theta$ plus a shift of order v^2, with θ the angle between $\boldsymbol{v}$ and $\boldsymbol{q}-\mu_\nu\boldsymbol{v}$. For a weight that is the same in every direction, $\cos\theta$ is uniform on the sphere, and a particle of speed v spreads its energy uniformly over a band of full width $2\mu_\nu q_\nu v/|m_\nu^*|$. Averaging over the halo with the factor $1/v$ of this normalization turns the band into η:

$$\frac{1}{\Gamma_\nu}\frac{\mathrm{d}\Gamma_\nu}{\mathrm{d}\omega}=\frac{|m_\nu^*|}{2\mu_\nu q_\nu}\,\eta\Big(\frac{|m_\nu^*|\,|\omega-\omega_\nu(q_\nu)|}{\mu_\nu q_\nu}\Big), \tag{17.7}$$

to first order in v/v_ν, normalized by $\int_0^\infty\eta(u)\,\mathrm{d}u=1$. The line has the shape of $\eta(v_{\min})$ reflected about its center. It is flat near the center, where every particle contributes, falls to half height where η does, at 278 km/s or 0.35 of $v_{\max}$, and ends at the full width $2\mu_\nu q_\nu v_{\max}/|m_\nu^*|$. For a branch that is not parabolic, the same steps with the dispersion linearized about q_ν replace q_ν/m_ν^* by u_ν and μ_ν by m_χ, as long as $|u_\nu|\ll v_\nu$: the full width is $2|u_\nu|m_\chi v_{\max}$. Intrinsic widths of the mode, from anharmonic decay of a phonon or Landau and interband damping of a plasmon, add by convolution.

Three kinds of mode follow from these formulas. An acoustic branch, $\omega=c_\nu q$, has $q_\nu=m_\chi[(c_\nu^2+2|\Delta_\chi|/m_\chi)^{1/2}-c_\nu]$ and a line at $c_\nu q_\nu$ whose full width relative to its energy is $2v_{\max}/v_\nu$. It is a line as long as v_ν is well above $v_{\max}$ and q_ν still lies inside the zone. Antiferromagnetic magnons near Γ behave the same way. In NiO at 0.1 MeV and $|\Delta_\chi|=1$ eV, $q_\nu=435$ eV, and the magnons emitted at $\boldsymbol{v}=0$ have 55.1 meV with a spread of 0.9 meV over directions. However, $v_{\max}/v_\nu=0.61$, and with the mean group velocity 1.1×10^{-4} the halo spreads them over a full width $2|u_\nu|m_\chi v_{\max}=60$ meV, more than the line energy. Their weight there, 0.28 on average, carries the suppression near Γ of Sect. 14.1, and Eq. (17.5) yields 5130 events per kilogram and year against 5710 from the full integral and 434 for elastic scattering. Once q_ν crosses many zones, the deposited energies sample the whole branch, and the spectrum becomes the weighted density of states: 100 meV on average, with a spread of 15 meV, for NiO at 1 MeV and $|\Delta_\chi|=10$ eV.

The magnons of a ferromagnet start at the Larmor frequency and disperse quadratically near Γ, a parabolic branch with $\omega_\nu(0)=\omega_{\mathrm{Lar}}$ and a positive m_ν^*. As $|\Delta_\chi|$ approaches ω_{Lar}, $q_\nu\to0$ and Eq. (17.5) vanishes in proportion to q_ν for a coupling without q dependence. The uniform precession of Sect. 14.3 is the end point of the exothermic spectrum, where the halo speeds take over from the splitting.

A flat optical branch gives a narrow line at its own energy, and Worked example 17.3 works it out for GaAs.

The plasmon is the steep case. Expanding the gas dispersion of Sect. 4.5, $\omega_{\mathrm{pl}}\simeq\omega_p+\alpha_{\mathrm{pl}}q^2/m_e$, gives $m_{\mathrm{pl}}^*=m_e/2\alpha_{\mathrm{pl}}$. At the valence density of silicon, $E_{\mathrm{F}}=12.47$ eV, $\omega_p=16.61$ eV and $\alpha_{\mathrm{pl}}=\frac{3}{5}E_{\mathrm{F}}/\omega_p=0.451$, so that $m_{\mathrm{pl}}^*=567$ keV, lighter than the dark matter of most electron-recoil searches. A particle of 1 MeV has $\mu_{\mathrm{pl}}=362$ keV and hands the plasmon 0.638

of the splitting above ω_p, a particle of 10 MeV 0.946. At $|\Delta_\chi| = 20\,\mathrm{eV}$ and 1 MeV, Eq. (17.6) puts the line at 18.77 eV and $q_\nu = 1.57\,\mathrm{keV}$, and the exact zero of the Lindhard function at 18.83 eV and 1.53 keV. The steep dispersion that hands the plasmon the splitting also makes the line wide: the full width from the halo is 5.3 eV, more than the measured 3.66 eV width of the silicon loss peak (Sect. 4.5). Since the plasmon exists only below the Landau-damping momentum, the line exists only in a window of splittings. The parabolic dispersion meets the upper edge of the pair continuum, $qv_\mathrm{F} + q^2/2m_e$, at 2.30 keV, and the window closes 7.3 eV above ω_p at 1 MeV.

Worked example 17.3: Exothermic emission of an optical phonon in GaAs

Take a dark photon, $\bar\sigma_e = 10^{-40}\,\mathrm{cm}^2$ as in Sect. 11.2, $m_\chi = 30\,\mathrm{keV}$ and $\Delta_\chi = -1\,\mathrm{eV}$. The splitting lies below the gap of GaAs, 1.52 eV, so no electronic excitation is open, and the dark photon reaches the lattice through the longitudinal optical phonon alone. Its line in the loss function carries $(\pi\omega_\mathrm{L}/2)(1 - \omega_\mathrm{T}^2/\omega_\mathrm{L}^2)/\varepsilon_\infty = 7.59 \times 10^{-4}\,\mathrm{eV}$ with the measured $\omega_\mathrm{T} = 33.62\,\mathrm{meV}$, $\omega_\mathrm{L} = 36.35\,\mathrm{meV}$ and $\varepsilon_\infty = 10.88$ (Eq. (11.17), splitting route). With this line in place of the averaged loss function $\bar L$ of Eq. (10.54), that equation times $(1 + u_\nu/v_\nu)^{-1}$ is Eq. (17.5) for the charge coupling:

$$\mathcal{R} = \frac{\rho_\chi\,\bar\sigma_e\,q_\nu^3\,F_\mathrm{DM}(q_\nu)^2\,\omega_\mathrm{L}}{2e^2\rho_T\,\mu_{\chi e}^2}\,\frac{1}{\varepsilon_\infty}\Big(1 - \frac{\omega_\mathrm{T}^2}{\omega_\mathrm{L}^2}\Big)\frac{1}{1 + u_\nu/v_\nu}.$$

Kinematics. With the computed dispersion of the longitudinal branch of Sect. 5.5, shifted to the measured ω_L at Γ and averaged over [100], [110] and [111], Eq. (17.4) gives $q_\nu = 240.5\,\mathrm{eV}$, 0.109 of $2\pi/a$ and inside the region where the Born-charge vertex holds (Sect. 11.2). Then $v_\nu = 8.0 \times 10^{-3}$ and $v_\mathrm{max}/v_\nu = 0.33$. The estimate $2.5\langle v^2\rangle/v_\nu^2$ of Sect. 10.9 puts the halo correction to the rate at 6 %.
Line. The branch falls with q, $u_\nu = -6.2\times10^{-7}$, and the line sits at 36.279 meV, 0.071 meV below ω_L, between 36.256 and 36.314 meV depending on the direction. Its full width from the halo, $2|u_\nu|m_\chi v_\mathrm{max}$, is 0.099 meV, and its half-maximum width 0.35 of that. Both are below the anharmonic width of the phonon, a fraction of a meV (Sect. 11.2). The dark matter keeps 0.964 eV; the detector sees one optical phonon.
Rate. For a light dark photon, $F_\mathrm{DM} = (q_0/q)^2$, the formula yields 1817 events per kilogram and year, and the full integral over the halo with the absolute value in v_min (Eq. (10.53)) 1810. Elastic scattering at the same mass leads to 842, so the splitting doubles the rate while fixing the energy deposited. For a heavy dark photon the shell form is at 0.0314 and the full integral 0.0334, against 5.7×10^{-5} for elastic scattering, a gain of 585. The two mediators respond oppositely to the splitting, since the light one weights q_ν^{-1} and the heavy one q_ν^3. At $|\Delta_\chi| = 0.5\,\mathrm{eV}$ the full integrals give 2600 for the light dark photon and 0.012 for the heavy one; the shell form yields 2620 and 0.010, off by 12 % for the heavy mediator, which weights the largest momenta, at $v_\mathrm{max}/v_\nu = 0.48$.
Below the elastic edge. At 10 keV, under the edge $2\omega_\mathrm{L}/v_\mathrm{max}^2 = 10.35\,\mathrm{keV}$, elastic scattering is closed. The same splitting gives $q_\nu = 138.8\,\mathrm{eV}$, $v_\mathrm{max}/v_\nu = 0.19$, and 2.6×10^4 events per kilogram and year for the light dark photon, with the shell form within 0.1 % of the full integral.

17.3.3 Electronic and collective channels in silicon

Fig. 17.4 puts the two couplings of silicon side by side at 1 MeV, each at its reference cross-section. Their elastic rates are 976 events per kilogram and year for single phonons and 16.7

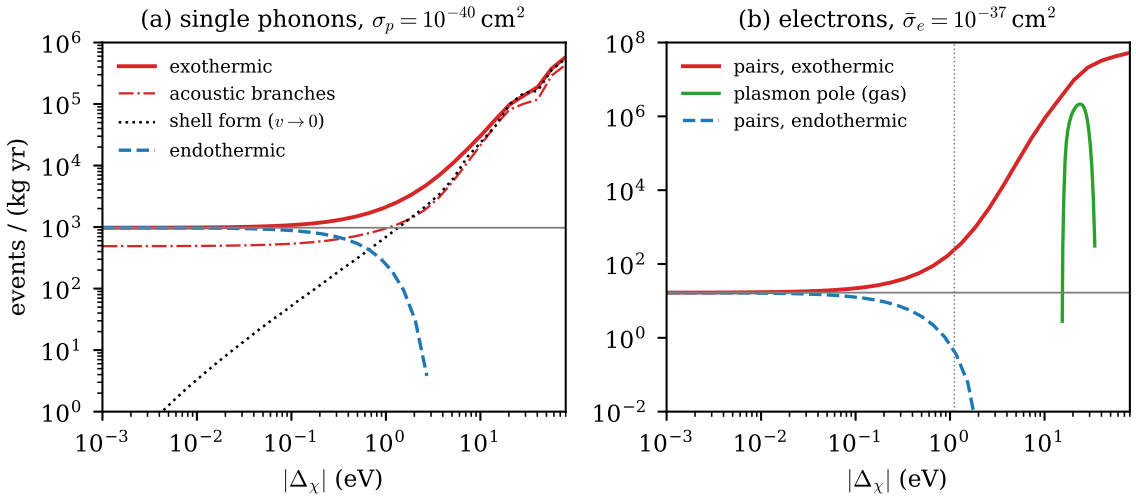


Figure 17.4: Rates of silicon per kilogram and year against the mass splitting for $m_\chi = 1\,\mathrm{MeV}$, heavy mediator, standard halo of Sect. 1.2, $T = 0$. Solid: exothermic ($\Delta_\chi = -|\Delta_\chi|$); dashed: endothermic ($\Delta_\chi = +|\Delta_\chi|$); gray line: elastic scattering. (a) Single phonons for a coupling to the nucleon number, $\sigma_p = 10^{-40}\,\mathrm{cm}^2$: the one-phonon response Eq. (11.1) with all six branches of the computed phonons of Sect. 5.2 at the actual $\boldsymbol{q}$ (Umklapp and Debye–Waller factor included), averaged over the orientations of the crystal, deposits $\omega \geq 1\,\mathrm{meV}$; dash-dotted: the three lowest branches; dotted: the shell form Eq. (17.5). (b) Electrons for a coupling to the electron density, $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$: pairs from Eq. (10.53) with the random-phase loss function with local fields of the tables of Sect. 10.2, above 1.11 eV, the step-model gap of their package (dotted vertical line; the tables are scissored to 1.17 eV, Sect. 10.10), the loss function held at its first tabulated momentum, 493 eV, below it; green: the plasmon pole of an electron gas at the valence density of silicon, Eq. (10.10) with the splitting in v_min, a model of the plasmon part of the electronic rate.

for pairs. A particle of that mass carries up to $\frac{1}{2}m_\chi v_{\max}^2 = 3.51\,\mathrm{eV}$, and a splitting well below this changes little: at $|\Delta_\chi| = 0.12\,\mathrm{eV}$ the two rates are 1.12 and 1.39 times their elastic values. Above a few eV the splitting dominates, and the two channels respond to it differently.

The single-phonon rate rises from 2.0×10^3 at 0.94 eV to 3.2×10^4 at 10.3 eV and 5.8×10^5 at 80 eV. The momentum transfer $q_\nu \simeq (2m_\chi|\Delta_\chi|)^{1/2}$ grows from 1.4 keV to 12.6 keV, from 0.6 to 5.6 q_{BZ}. The one-phonon weight grows as q^2 times the Debye–Waller factor, whose exponent $2W$ is still just 0.10 at 80 eV; two-phonon emission is smaller than one-phonon emission by a factor of order W. The deposited energy does not follow the splitting. Beyond the zone the shell crosses every branch, the spectrum is the weighted density of phonon states, and at 10 eV the mean deposit is 34.5 meV, with 42 % above 40 meV in the optical branches. A detector reading phonons sees the same energies as for elastic scattering, with a rate raised by a factor of 31 at 10 eV. Between 10.3 and 80 eV the shell form lies at 0.78 to 1.10 of the full integral. The residual comes from the condition on the weight in Eq. (17.5), not from the halo speeds, whose correction is $(v_{\max}/v_\nu)^2 \leq 0.34$ there: the one-phonon weight varies on the scale $q_{\mathrm{BZ}} = 2.25\,\mathrm{keV}$, smaller than $m_\chi v_{\max} = 2.65\,\mathrm{keV}$, and a single momentum per direction misses that structure.

The electronic rate rises far more steeply, from 165 at 0.94 eV to 9.6×10^5 at 10.3 eV and 5.3×10^7 at 80 eV, a gain of 3×10^6 over the elastic rate. Elastic scattering at 1 MeV barely crosses the gap, so any energy the splitting adds opens phase space that was nearly closed. Here the deposit does follow the splitting: Sect. 10.9 found the spectrum peaked between the gap and the plasma frequency, where $q_\Delta^3 \bar{L}$ is largest, and an electronic detector reads energies of order $|\Delta_\chi|$. With an electronic coupling the splitting becomes visible energy; with a nucleon coupling it goes into the outgoing particle, and the phonon deposit stays below 64 meV.

The plasmon pole opens inside the electronic channel. For the electron gas at 1 MeV it requires $\omega_{\mathrm{pl}}(q) - q v_{\max} + q^2/2m_\chi \leq |\Delta_\chi|$ at some q below the Landau-damping momentum; the minimum is 15.4 eV, above the 14.7 eV found for a heavy particle in Sect. 10.3, since a particle of 1 MeV keeps $q^2/2m_\chi$. The pole rate then grows to 2.2×10^6 events per kilogram and year at $|\Delta_\chi| = 23.8\,\mathrm{eV}$ and falls beyond, as q_ν approaches the damping momentum and the weight of the line goes to zero. Between 18 and 22 eV it is 0.09 to 0.16 of the electronic rate of the tables. The two are different models, an undamped pole in a gas against a broad loss peak in a crystal; in both, the plasmon is a minority share of exothermic electron recoil even where it is open. In a detector the plasmon decays into pairs (Sect. 10.3.4), and its line, 5.3 eV wide from the halo alone at 20 eV, merges with the pair spectrum around it.

17.3.4 Endothermic thresholds

An endothermic splitting is paid from the kinetic energy, and every channel acquires a lightest mass. For a mode with a gap and a parabolic dispersion, Eq. (17.6) with $\Delta_\chi > 0$ gives $v_{\min} = (\Delta_\chi + \omega_\nu(0))/q + q/2\mu_\nu$. It is smallest at $q = [2\mu_\nu(\Delta_\chi + \omega_\nu(0))]^{1/2}$, and a particle at $v_{\max}$ reaches the mode if

$$\mu_\nu \geq \frac{2\big(\Delta_\chi + \omega_\nu(0)\big)}{v_{\max}^2}. \tag{17.8}$$

For a flat branch $\mu_\nu = m_\chi$, and this is Eq. (1.25) with $\omega_{\mathrm{th}} \to \Delta_\chi + \omega_\nu(0)$; electron–hole pairs follow with $\omega_\nu(0) \to E_{\mathrm{g}}$, as in Sect. 10.9.1. For a gapless linear branch with a threshold ω_{th}, $v_{\min} = \Delta_\chi/q + c_\nu + q/2m_\chi$ must fall below $v_{\max}$ at some $q \geq \omega_{\mathrm{th}}/c_\nu$. Its unconstrained minimum lies at $q = (2m_\chi\Delta_\chi)^{1/2}$, and

$$m_\chi \geq \begin{cases} \dfrac{2\Delta_\chi}{(v_{\max} - c_\nu)^2}, & (2m_\chi\Delta_\chi)^{1/2} \geq \omega_{\mathrm{th}}/c_\nu, \\ \dfrac{\omega_{\mathrm{th}}}{2c_\nu\big(v_{\max} - c_\nu - c_\nu\Delta_\chi/\omega_{\mathrm{th}}\big)}, & \text{otherwise.} \end{cases} \tag{17.9}$$

Table 17.4: Lightest dark-matter mass that reaches each channel for the endothermic splittings $\Delta_\chi = 0$, 10 meV and 1 eV, from Eqs. (17.8) and (17.9) with $v_{\max} = 2.650 \times 10^{-3}$. Gapped channels: the low-temperature gap of silicon, 1.17 eV; the optical phonon of silicon at 64.1 meV (Sect. 5.2); the longitudinal optical phonon of GaAs at 36.35 meV (Worked example 5.6). Linear channels: the largest longitudinal sound speed of silicon, 9.35 km/s, with $\omega_{\text{th}} = 1$ meV (Sect. 11.1); the fastest magnon speed of NiO in the J_1–J_2 model, $c_{\text{M}} = 1.36 \times 10^{-4}$, with $\omega_{\text{th}} = 5$ meV (Sect. 14.1).

Channel	$\Delta_\chi = 0$	10 meV	1 eV
Si, electron–hole pairs	0.333 MeV	0.336 MeV	0.618 MeV
Si, optical phonon	18.3 keV	21.1 keV	303 keV
GaAs, longitudinal optical phonon	10.35 keV	13.2 keV	295 keV
Si, acoustic phonon	6.12 keV	6.95 keV	292 keV
NiO, magnon	7.31 keV	8.20 keV	316 keV
Si, plasmon	closed at every mass		

At $\Delta_\chi = 0$ the momentum threshold always binds, and Eq. (17.9) reduces to the acoustic edge Eq. (11.5) and its magnon analog Eq. (14.5).

Table 17.4 collects the numbers. A splitting of 10 meV moves the lattice and magnon edges by 12 to 28 % and the electronic one by 1 %, because it adds to tens of meV for the collective modes and to an eV for pairs. A splitting of 1 eV erases the differences among the collective channels: all of them then need 0.3 MeV, set by Δ_χ and the halo. The phonon of GaAs or the magnon of NiO then reaches dark matter lighter than the gap of silicon does by a factor of two, against 32 for the GaAs phonon and 46 for the NiO magnon in elastic scattering. For the plasmon, Eq. (17.8) needs $\mu_{\text{pl}} \geq 2\omega_p/v_{\max}^2 = 4.73$ MeV already at $\Delta_\chi = 0$, while μ_{pl} never exceeds $m_{\text{pl}}^* = 567$ keV: the dispersion that hands the plasmon most of an exothermic splitting closes it to endothermic and elastic scattering at every mass. In Fig. 17.4 the same thresholds appear at fixed mass. At 1 MeV the electronic rate closes at $\Delta_\chi = \frac{1}{2} m_\chi v_{\max}^2 - E_{\text{g}} = 2.40$ eV with the gap of the tables, and the single-phonon rate at 3.51 eV less the threshold.

17.4 Interference between channels

A dark photon that deposits 1.2 eV at $q = 1$ keV in silicon can leave an electron–hole pair and a transverse acoustic phonon behind in two ways. It can push the electrons, which then emit the phonon through the electron–phonon coupling: the phonon-assisted amplitude of Sect. 13.2. Or it can push a nucleus, whose electrons lag behind and are excited while the nucleus sheds the phonon: the Migdal amplitude of Sect. 12.1. Both end in the same state, and the rate contains their product as well as their squares. Such products of an electron and a nuclear amplitude fill the bottom row of Table 8.5, the cross response $S_{e\kappa}$ in each of the three sectors. In the electron-following sector the product cancels the phonon rate of a dark photon in silicon completely. In the two electronic sectors, at small momentum transfer, it raises the rate by 5.5 parts in 10^4. We will derive the cross term of the phonon-dressed sector, find when it vanishes, bound it by positivity, and fix it at small q by momentum conservation.

17.4.1 The three cross cells

A mediator that couples to the electrons with the potential $\tilde{V}_e(\boldsymbol{q})$ and to the nuclei of species κ with $\tilde{V}_\kappa(\boldsymbol{q})$ has the cross elements $\mathfrak{D}_{e\kappa} = \tilde{V}_e^* \tilde{V}_\kappa$ of Eq. (8.7), and the rate Eq. (8.16) contains, besides the electron and the nuclear terms, $2\,\text{Re}\sum_\kappa \mathfrak{D}_{e\kappa} S_{e\kappa}$. A dark photon couples to $-\tilde{V} n_Z$

(Worked example 8.1), so that $\tilde{V}_e = \tilde{V}$ and $\tilde{V}_\kappa = -Z_\kappa \tilde{V}$. A coupling to electrons alone has $\tilde{V}_\kappa = 0$, and a coupling to nucleons alone has $\tilde{V}_e = 0$. For these two the cross response never enters the rate, although $S_{e\kappa}$ itself is not zero, which is why Sect. 17.1 could compare the channels cell by cell. For a dark photon the three cells of $S_{e\kappa}$ differ in size by more than three orders of magnitude.

Electron-following cell. In this sector the electron density acts on the nuclei as $\bar{\rho}_e(\boldsymbol{q}; u)$ of Eq. (8.51). At long wavelength the electrons that follow nucleus κ carry the charge $F_\kappa^{\rm fol} = \hat{\boldsymbol{q}} \cdot (Z_\kappa \mathbf{1} - Z_\kappa^*) \cdot \hat{\boldsymbol{q}}$, the $q \to 0$ value of $F_{\rm cr}$ in Eq. (12.45). A phonon then sees the nucleus and its cloud with amplitudes in the ratio $Z_\kappa : F_\kappa^{\rm fol}$, of opposite sign for a dark photon, and the nuclear, electron and cross terms of its rate stand in the ratio $Z_\kappa^2 : (F_\kappa^{\rm fol})^2 : -2Z_\kappa F_\kappa^{\rm fol}$, which add up to Z_κ^{*2}. In silicon $F^{\rm fol} = Z = 14$, and the cross term, -392 against $196 + 196$, cancels the other two exactly: a dark photon makes no phonon in silicon at long wavelength (Sect. 10.2). In a polar crystal the cross term removes all but Z^{*2}, which ε_∞ screens further, and what survives is the optical-phonon rate of Sect. 11.2; Worked example 17.4 gives the numbers for GaAs. This cell is never a correction.

Zero-phonon cell. The electronic cells carry the opposite first moment. By Eq. (8.57) the energy-weighted cross response of the electron-following sector reappears with the opposite sign in the two electronic sectors together. Measured against the f-sum of the electrons, $n_e q^2/m_e$, it is m_e/M_κ, 1.95×10^{-5} in silicon. Momentum conservation distributes it state by state. In a crystal of one element, Eq. (8.32) gives $\sum_\kappa \langle f|n_\kappa(\boldsymbol{q})|0\rangle = -(m_e/M_\kappa)\langle f|n_e(\boldsymbol{q})|0\rangle + O(q^2)$ for every final state at finite energy. For a dark photon the amplitude into any electronic final state is then $\tilde{V}(1 + Z_\kappa m_e/M_\kappa)\langle f|n_e|0\rangle$, and the zero-phonon rate gains the fraction $(1 + Z_\kappa m_e/M_\kappa)^2 - 1 = 5.47 \times 10^{-4}$ in silicon. Of this, the cross cell carries $2Z_\kappa m_e/M_\kappa$ and the zero-phonon Migdal cell $(Z_\kappa m_e/M_\kappa)^2 = 7.5 \times 10^{-8}$. The picture is the inertial argument behind the sum rule Eq. (12.52): the field that pushes the electrons also accelerates the nuclei, and in the frame of the nuclei every electron feels the additional force m_e times their acceleration, $Z_\kappa m_e/M_\kappa$ times the force of the field. This is the ionic correction to the zero-phonon line. A dark photon sees it through the loss function of the total dielectric matrix of electrons and nuclei (Sect. 8.4), so a loss function that includes the nuclei, computed or measured, already contains it.

In a crystal of several elements momentum conservation fixes the mass-weighted sum alone, $\sum_\kappa M_\kappa \langle f|n_\kappa|0\rangle = -m_e\langle f|n_e|0\rangle + O(q^2)$ (Eq. (8.31)). As in Sect. 8.3, write $\tilde{V}_\kappa = \lambda M_\kappa + \delta\tilde{V}_\kappa$. The part in proportion to mass contributes $-\lambda m_e\langle f|n_e|0\rangle$ to the amplitude, and the remainder $\delta\tilde{V}_\kappa$ needs the amplitudes of the separate sublattices, which momentum conservation does not supply. For a dark photon $\lambda = -\bar{z}\tilde{V}$, with $\bar{z} = \sum_\kappa w_\kappa Z_\kappa/M_\kappa$ a mean of Z_κ/M_κ whose weights $w_\kappa = -M_\kappa\langle f|n_\kappa|0\rangle/(m_e\langle f|n_e|0\rangle)$ sum to one. If both sublattices carry positive weights, the cross cell lies between $2m_e$ times the smallest and the largest Z_κ/M_κ of the crystal, with m_e in u.

17.4.2 The Migdal–phonon-assisted cross term

Inside the window the one-phonon states of the phonon-dressed sector are those of Eq. (13.6): an electron in $(c, \boldsymbol{k}')$, a hole in $(v, \boldsymbol{k})$, $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{q} + \boldsymbol{Q}$, and one phonon emitted into $(-\boldsymbol{Q}, \nu)$ or absorbed from $(\boldsymbol{Q}, \nu)$. The electron density reaches them through the phonon-assisted amplitude Eq. (13.9), $N^{-1/2}(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})^{1/2} M^{\pm}_{cv\nu}(\boldsymbol{k}, \boldsymbol{q}, \boldsymbol{Q})$. The nuclear density reaches them through the boost Eq. (12.5), which for a pair of energy $\Delta E_j = \omega_e$ reads

$$\sum_{I \in \kappa} \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \, \frac{\boldsymbol{q} \cdot \boldsymbol{p}}{M_\kappa \omega_e^2} \, v_\kappa(\boldsymbol{p}) \, \langle \Psi_j | n_e(\boldsymbol{p}) | \Psi_0 \rangle \, \langle \Psi^{\rm nuc\prime} | \mathrm{e}^{\mathrm{i}(\boldsymbol{q}-\boldsymbol{p})\cdot \boldsymbol{X}_I} | \Psi^{\rm nuc} \rangle,$$

with v_κ the potential of the nucleus, or its pseudopotential. The nucleus takes the kick $\boldsymbol{q}$, passes $\boldsymbol{p}$ to the electrons through its own field and keeps $\boldsymbol{K} = \boldsymbol{q} - \boldsymbol{p}$. Three steps turn this into the phonon-dressed amplitude.

First, the lattice factor is taken at its one-phonon term. With $\boldsymbol{X}_I = \boldsymbol{R} + \boldsymbol{\tau}_\kappa + \boldsymbol{u}_{\boldsymbol{R}\kappa}$ and the displacement Eq. (A.21), the part of $\mathrm{e}^{\mathrm{i}\boldsymbol{K}\cdot\boldsymbol{X}_I}$ linear in $\boldsymbol{u}$ emits $(-\boldsymbol{Q}, \nu)$ with the matrix element

$$\mathrm{i}\,\frac{\boldsymbol{K}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})}{(2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{1/2}}\,(n_{\boldsymbol{Q}\nu}+1)^{1/2}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{K}+\boldsymbol{Q})\cdot\boldsymbol{R}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{K}\cdot\boldsymbol{\tau}_\kappa}\,\mathrm{e}^{-W_\kappa(\boldsymbol{K})},$$

with the Debye–Waller factor of Eq. (A.29); absorption of $(\boldsymbol{Q}, \nu)$ gives the same with $n_{\boldsymbol{Q}\nu}^{1/2}$. Second, the sum over the N cells $\boldsymbol{R}$ gives N times a Kronecker delta that keeps only $\boldsymbol{K} = -\boldsymbol{Q} - \boldsymbol{G}$, so that the electrons receive $\boldsymbol{p}_{\boldsymbol{G}} = \boldsymbol{q} + \boldsymbol{Q} + \boldsymbol{G}$. Their matrix element $\langle\Psi_j|n_e(\boldsymbol{p}_{\boldsymbol{G}})|\Psi_0\rangle$ is the density vertex $\rho_{cv}(\boldsymbol{k}, \boldsymbol{p}_{\boldsymbol{G}})$ of Eq. (3.34), whose crystal momentum $\boldsymbol{k}+\boldsymbol{p}_{\boldsymbol{G}}$ is $\boldsymbol{k}'$ up to $\boldsymbol{G}$, as required. Third, the integral over $\boldsymbol{p}$ becomes a sum, $\int \mathrm{d}^3p/(2\pi)^3 \to \mathscr{V}^{-1}\sum_{\boldsymbol{p}}$, which the delta reduces to the sum over $\boldsymbol{G}$, and the factor $N/\mathscr{V} = 1/\Omega$ remains. Collecting the factors, the nuclear amplitude of species κ is $N^{-1/2}(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})^{1/2} M^{\mathrm{Mig},\pm}_{\kappa,cv\nu}$, with the dimensionless amplitude

$$\begin{aligned} M^{\mathrm{Mig},\pm}_{\kappa,cv\nu}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q}) = {} & \frac{\mathrm{i}}{\Omega M_\kappa \omega_e^2\,(2M_\kappa\omega_{\boldsymbol{Q}\nu})^{1/2}} \sum_{\boldsymbol{G}} (\boldsymbol{q}\cdot\boldsymbol{p}_{\boldsymbol{G}})\,[(\boldsymbol{q}-\boldsymbol{p}_{\boldsymbol{G}})\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})] \\ & \times v_\kappa(\boldsymbol{p}_{\boldsymbol{G}})\,\rho_{cv}(\boldsymbol{k},\boldsymbol{p}_{\boldsymbol{G}})\,\mathrm{e}^{\mathrm{i}(\boldsymbol{q}-\boldsymbol{p}_{\boldsymbol{G}})\cdot\boldsymbol{\tau}_\kappa - W_\kappa(\boldsymbol{q}-\boldsymbol{p}_{\boldsymbol{G}})}, \end{aligned} \tag{17.10}$$

where $\omega_e = \varepsilon_{c\boldsymbol{k}'} - \varepsilon_{v\boldsymbol{k}}$ is the pair energy. Read from left to right, the nucleus is boosted to the velocity $\boldsymbol{q}/M_\kappa$, its field gives the electrons $\boldsymbol{p}_{\boldsymbol{G}}$, and the momentum it keeps, $-\boldsymbol{Q} - \boldsymbol{G}$, displaces it along the phonon polarization. The factor i of M^{Mig} matches the $-\mathrm{i}$ that the vertex g of Eq. (A.26) carries in Fourier form, so the ratio $M^{\mathrm{Mig}}/M^{\pm}$ is real. Both amplitudes carry the same phonon factor $N^{-1/2}(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})^{1/2}$, and the cross response follows as S^{PA} did in Sect. 13.3, with one amplitude in place of the other.

Migdal–phonon-assisted cross response

In the off-shell window, at first Born–Oppenheimer order and first order in the electron–phonon vertex, the interference of the electron and the nuclear density of species κ into one-phonon pair states is

$$\begin{aligned} S^{+}_{e\kappa}(\boldsymbol{q};\omega_l,\omega_e) = \frac{4\pi}{\Omega}\Big\langle\!\Big\langle \sum_{cv\nu\pm} (n_{\boldsymbol{Q}\nu} + \tfrac{1}{2} \pm \tfrac{1}{2})\, M^{\pm}_{cv\nu}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q})^*\, M^{\mathrm{Mig},\pm}_{\kappa,cv\nu}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q}) \\ \times \delta\big(\omega_e - \varepsilon_{c\boldsymbol{k}'} + \varepsilon_{v\boldsymbol{k}}\big)\,\delta\big(\omega_l \mp \omega_{\boldsymbol{Q}\nu}\big)\Big\rangle\!\Big\rangle, \end{aligned} \tag{17.11}$$

with the zone averages over $\boldsymbol{k}$ and $\boldsymbol{Q}$ and the spin factor of Eq. (13.15). It enters the rate as $2\,\mathrm{Re}\sum_\kappa \mathfrak{D}_{e\kappa} S^{+}_{e\kappa}$, in the joint response of Eq. (12.1).

Each factor of the product depends on the phases chosen for the Bloch states and the polarization vectors, and the product does not, since every such phase enters once in each factor. The two amplitudes must, however, be built from one set of bands, phonons and vertices, the rule of Sect. 8.6 for adding sectors. The cross term is linear in the derivative coupling, which is why it sits at first Born–Oppenheimer order, half the order of a Migdal rate.

Two approximations of Eq. (17.10) come from the boost. It keeps the transition density at the equilibrium positions, in the frame of the crystal. If the valence transition densities follow the struck ion instead, the lattice keeps $\boldsymbol{q}$ and not $\boldsymbol{q}-\boldsymbol{p}_{\boldsymbol{G}}$, and the factor $(\boldsymbol{q}-\boldsymbol{p}_{\boldsymbol{G}})\cdot\boldsymbol{e}_{\kappa\nu}$ changes

with it (Sect. 12.1). The frame changes the fraction of Migdal events that emit phonons by a factor 2.4 at 5 keV (Sect. 12.4), and it enters the cross term directly. The second approximation is the sum over $\boldsymbol{G}$, which must be kept whole: dropping its off-diagonal products in $|M^{\mathrm{Mig}}|^2$ is the diagonal approximation of Sect. 12.3, whose error is open (Appendix E.18). Momentum conservation fixes the exact sum at small q, and any approximate amplitude can be tested against it.

The labels ω_e and ω_l are those of Sect. 8.6. Inside the window ω_e is unique to relative order θ_{ep}, the off-shell parameter of Sect. 8.5, and Eq. (17.11) is the whole cross response of the phonon-dressed sector at this order. Near a resonant pair the Dyson convention Eq. (8.59) assigns the delta part to the zero-phonon cell and the finite parts to the phonon-dressed one, for the cross response as for S^{PA}, and only the sum of the two electronic cells is free of convention.

17.4.3 Vanishing of the cross term

The cross term is absent from the rate in four cases. First, a coupling to one species alone has $\mathfrak{D}_{e\kappa} = 0$.

The second is a mismatch of phonon number. A zero-phonon Migdal event leaves the phonon occupations unchanged, and a phonon-assisted pair has one phonon more or less. The final states are orthogonal, and the two amplitudes do not interfere, by the argument of Sect. 13.2 for the zero-phonon line. Migdal events with two or more phonons interfere with phonon-assisted amplitudes of second order in the electron–phonon vertex alone. At the order of Eq. (17.11) the one-phonon term of the Migdal lattice factor is the one that takes part, and at $q = 5\,\mathrm{keV}$ it is 96 % of the phonon-emitting Migdal events (Sect. 12.4).

The third is the spin. Both vertices of Eq. (17.11) are diagonal in spin, and both amplitudes make singlet pairs. A coupling through the electron spin makes triplet pairs, which the nuclear density does not make in a crystal without spin–orbit coupling and magnetic order, and its cross term with the nuclear density vanishes by the selection rule of Sect. 10.4.

The fourth is the limit $\boldsymbol{q} \to 0$. The phonon-assisted amplitude vanishes there, Eq. (13.10), because $n_e(0)$ is the electron number. The Migdal amplitude vanishes with its factor $\boldsymbol{q} \cdot \boldsymbol{p}_{\boldsymbol{G}}$, because a nucleus that is not kicked is not boosted. Both vanish linearly, so their ratio has a limit.

17.4.4 Positivity bound and the small-momentum limit

The joint response is positive semidefinite at every point (ω_l, ω_e) where the labels exist (Sect. 12.1), and in particular on the one-phonon line. Here and below S^{PA} stands for the phonon-assisted response resolved in both labels, $S^{\mathrm{PA}}(\boldsymbol{q}; \omega_l, \omega_e)$, like $S^{+}_{e\kappa}$; its integral over ω_l is the response of Sect. 13.3. The Cauchy–Schwarz inequality Eq. (4.20) then gives $|S^{+}_{e\kappa}|^2 \le S^{\mathrm{PA}} S^{+}_{\kappa\kappa}$ pointwise, with $S^{+}_{\kappa\kappa}$ the one-phonon Migdal response of the same final states. With Eq. (8.21), taking the electron row as one group and the nuclear rows as the other,

$$\left| 2\,\mathrm{Re} \sum_{\kappa} \mathfrak{D}_{e\kappa} S^{+}_{e\kappa} \right| \le 2 \Big(\mathfrak{D}_{ee} S^{\mathrm{PA}} \sum_{\kappa\kappa'} \mathfrak{D}_{\kappa\kappa'} S^{+}_{\kappa\kappa'} \Big)^{1/2}, \tag{17.12}$$

pointwise, in every bin and for the integrated rate. The restriction to one sector needs care. The phonon-dressed sector is positive by itself inside the window, but outside it only on bins wider than the detuning, because the Dyson convention puts finite parts into it that are negative against a narrow bin (Sect. 8.6). Outside the window the bound is applied to the two electronic sectors together.

At small q the bound is reached. Apply Eq. (8.32) to a one-phonon pair state: for a crystal of one element and to first order in q, the exact summed nuclear amplitude is $-(m_e/M_\kappa)$ times the electron amplitude, with the same ratio for every final state. Written in the notation of Eq. (17.10), which stands here for the exact amplitude and not for its crystal-frame approximation, this is the following result.

Phonon-dressed interference at small momentum transfer

For a crystal of one element, nuclear mass M_κ, and every one-phonon pair state,

$$\sum_\kappa M^{\mathrm{Mig},\pm}_{\kappa,cv\nu}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q}) = -\frac{m_e}{M_\kappa}\, M^{\pm}_{cv\nu}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q})\,[1+O(q/q_{\mathrm{BZ}})], \tag{17.13}$$

so that the summed cross response is $-(m_e/M_\kappa)S^{\mathrm{PA}}$, the summed one-phonon Migdal response is $(m_e/M_\kappa)^2 S^{\mathrm{PA}}$, and Eq. (17.12) holds as an equality. A coupling $\tilde{V}_\kappa = \tilde{V}_{\mathrm{nuc}}$ on every nucleus gives the phonon-dressed rate of Eq. (8.16) with

$$\mathfrak{D}_{ee}S^{\mathrm{PA}} \;\to\; \left|\tilde{V}_e - \frac{m_e}{M_\kappa}\tilde{V}_{\mathrm{nuc}}\right|^2 S^{\mathrm{PA}}. \tag{17.14}$$

Below the zone, then, the one-phonon Migdal effect is the phonon-assisted electron recoil seen through the boost: the same spectrum, with the weight $(m_e/M_\kappa)^2 = 3.8\times10^{-10}$ in silicon. The response matrix of the two rows has rank one, which is the condition of Sect. 8.3 for the bound to be reached, and the sign of the cross term follows from the coupling alone. For a dark photon the bracket of Eq. (17.14) is $1 + Z_\kappa m_e/M_\kappa$, and the phonon-assisted rate gains 5.47×10^{-4} in silicon, the same fraction as the zero-phonon line. For a coupling in proportion to mass, $\tilde{V}_e : \tilde{V}_\kappa = m_e : M_\kappa$, the bracket vanishes. The cross term is then not absent but cancels the other two, and the rate sits on the lower edge of Eq. (8.21); the electronic final states are reached at the next order in q alone, as the soft acoustic limit Eq. (8.31) requires. For $\tilde{V}_{\mathrm{nuc}} = -(M_\kappa/m_e)\tilde{V}_e$ the rate sits on the upper edge, four times the electron term. In a crystal of several elements, Eq. (17.13) holds for the mass-weighted sum, and the remainder $\delta\tilde{V}_\kappa$ of Sect. 17.4.1 is bounded by Eq. (17.12) alone.

The remainder of Eq. (17.13) is one power of q higher. It comes from the momentum flux of Eq. (8.31) and carries no small coefficient: its scale is the momentum over which the pair amplitudes change in the zone. Scattering that deposits the gap needs $q \geq E_{\mathrm{g}}/v_{\max}$, which is 0.19 of $2\pi/a$ in silicon and 0.26 in GaAs. The trouble is that the silicon window lies at 0.39 to 0.57 of $2\pi/a$, where the phonon-assisted amplitude is already off its optical limit by a factor of order one (Sect. 13.5), with a dipole extrapolation five to twenty times larger in rate (Sect. 13.9.4). There the relation sets the scale of the cross term, not its value, and a value needs Eq. (17.10) with the same bands and phonons as the phonon-assisted rate. The relation is exact in the limit that absorption reaches, $q = m_\phi v \approx 10^{-3}\omega$. There the same identity reduces the absorption of a scalar coupled to nuclei in proportion to mass to the electron loss function, the result of Sect. 12.6 at dipole order, and splits a coupling of any other form into that part and the remainder of Appendix E.18.

Fig. 17.5 places the cross term in the plane of q and ω. It exists wherever a pair with one phonon is kinematically allowed, above the phonon-assisted threshold, but it changes an observable rate by more than a correction to a correction in the window alone. There the phonon-dressed sector is the whole electronic rate of silicon between 0.3374 and 0.405 MeV (Table 13.1), and of GaAs from 0.433 MeV to the opening of the L valleys at 0.52 to 0.56 MeV (Sect. 13.1).

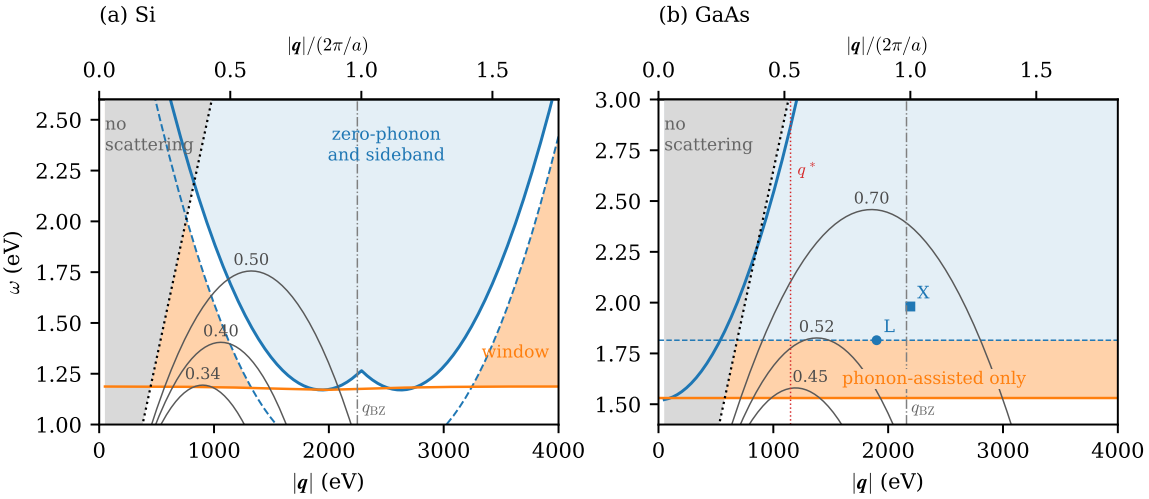


Figure 17.5: Where the Migdal–phonon-assisted interference enters, in the plane of momentum transfer and energy deposited by scattering. (a) Silicon. Thick blue: zero-phonon edge of the parabolic valleys along [100], $E_{\min} = E_g + (q-K)^2/2M_{eh}$ with $E_g = 1.17\,\text{eV}$, $M_{eh} = 1.193\,m_e$, $K = k_c = 0.85\,(2\pi/a)$ and its images under $(2,0,0)\,2\pi/a$, a model that lies 0.13 eV above the computed bands at 1 keV. Dashed blue: the same lowered by the off-shell margin of the window, $\delta_w = 0.3\,\text{eV}$ (Sect. 8.6). Orange line: the phonon-assisted threshold $E_g + \omega_{TA}(k_c - q)$. Orange: the window, where the phonon-dressed sector is the only electronic final state and the cross term of Eq. (17.11) is a fraction of the whole electronic rate. White strip: the edge shell within δ_w of the zero-phonon edge, where only the sum of the two electronic sectors is defined. Light blue: zero-phonon pairs, with the phonon-dressed sector and its cross term a correction. (b) GaAs: edge of the Γ pair with the [111] heavy hole ($E_g = 1.52\,\text{eV}$, $M_{eh} = 0.960\,m_e$), the L and X valleys at 1.815 and 1.981 eV (points at $|\Gamma L|$ and $|\Gamma X|$), and the phonon-assisted threshold with the upper value 10.5 meV of the transverse acoustic branch. Orange: below both the Γ edge and the L energy, phonon-assisted pairs only; the L valleys reach the line $\omega = 1.815\,\text{eV}$ near $|\Gamma L|$ only. Red dotted: $q^* = 1.15\,\text{keV}$ of the GaAs window. Both panels: the largest deposit $qv_{\max} - q^2/2m_\chi$ for the masses marked in MeV (gray), the envelope $qv_{\max}$ (dotted; gray region unreachable by scattering), and q_{BZ}, the radius of the sphere with the volume of the zone (dash-dotted). Top axes: q in units of $2\pi/a$; Eq. (17.13) holds for $q \ll q_{BZ}$.

Worked example 17.4: The cross terms of a heavy dark photon in GaAs

Take a dark photon with a heavy mediator, $F_{\rm DM} = 1$, so that $\tilde{V}_e = \tilde{V}$ and $\tilde{V}_\kappa = -Z_\kappa \tilde{V}$ with $\tilde{V}$ independent of q, scattering in GaAs at $T = 0$. Below 0.433 MeV the particle makes no pair at all. Up to the opening of the L valleys at 0.52 to 0.56 MeV it makes pairs only with a phonon, at momentum transfers near $q^* = 1.15$ keV, 0.52 of $2\pi/a = 2.193$ keV (Fig. 17.5b).

Electron-following cell. With $|Z^*| = 2.11$ (Sect. 5.5), a gallium nucleus carries $Z = 31$ and its cloud $F^{\rm fol} = 28.89$ at long wavelength; an arsenic nucleus $Z = 33$ and $F^{\rm fol} = 35.11$. The separate terms are $961 + 834.6$ and $1089 + 1232.7$, the cross terms -1791.2 and -2317.3, and each total is $Z^{*2} = 4.452$: the cross term removes 99.75 % and 99.81 % of the two separate terms. The optical phonons that the dark photon makes are those of Sect. 11.2. Without the cross cell their rate would be $(Z^2 + (F^{\rm fol})^2)/Z^{*2} = 403$ times (gallium) and 521 times (arsenic) larger, and the bare nuclear charges alone would give 216 and 245 times more, before the screening by ε_∞ that both cases share.

Zero-phonon cell. Above the opening of the L valleys the zero-phonon rate gains $2m_e\bar{z}$ from the ionic correction. The two values of Z/M, 0.4446 per u for gallium and 0.4405 per u for arsenic, differ by 0.9 %. With $m_e = 5.486 \times 10^{-4}$ u the cross cell is 4.88×10^{-4} for weight on gallium alone and 4.83×10^{-4} for weight on arsenic alone, and lies between the two for any positive weights. The zero-phonon Migdal cell is $(m_e\bar{z})^2 = 5.9 \times 10^{-8}$.

Phonon-dressed cell. In the window the whole electronic rate is phonon-assisted, and at leading order in q, Eq. (17.14) with the mass-weighted mean gives it the same fraction, 4.8 to 4.9×10^{-4}, and the one-phonon Migdal rate 5.9×10^{-8} of it. Since $\mathfrak{D}$ has no q dependence for a heavy mediator, the fraction is the same at every momentum transfer at this order, and the integrated rate carries it unchanged. At q^*, however, $q/(2\pi/a) = 0.52$, and the remainder of Eq. (17.13) is not small. The cross term is of order 5×10^{-4} of the phonon-assisted rate, with an uncertainty of order one in that fraction.

Screening. The dark photon couples to the total charge, and both rows share the screening $1/\varepsilon$ of the total-charge row (Sect. 8.4), which cancels from every fraction above.

Verdict. The phonon-assisted rate of the silicon window is an estimate known to a factor of ten (Table 13.1), and the GaAs window is known through its edges (Sect. 13.9.5). The two electronic cross terms lie more than three orders of magnitude below such uncertainties. For a dark photon in GaAs the electron-following cell is the one that matters, and it decides the phonon rate.

At small q the nuclear amplitude in the electronic sectors is $(m_e/M_\kappa)\tilde{V}_{\rm nuc}/\tilde{V}_e$ times the electron amplitude. The cross term competes with the electron term when the coupling to a nucleus exceeds that to an electron by $M_\kappa/m_e = 5.1 \times 10^4$ in silicon, or, per nucleon, by $m_{\rm u}/m_e = 1823$. A scalar mixed with the Higgs boson, whose couplings follow the masses up to the nucleon matrix element, comes close to this; a $B - L$ vector, with couplings of equal size per nucleon and per electron, is far from it. For such a mediator a Migdal rate and an electron-recoil rate computed separately and added miss the true rate by the full cross term.

17.5 Figures of merit per channel

The heavy-mediator plateau of single-phonon emission is 1101 events per kilogram and year in germanium and 654 in silicon at $\sigma_p = 10^{-40}\,{\rm cm}^2$ (Sect. 11.1). The ratio, 1.68, is the ratio of two sound speeds, and once the elastic tensors are known no phonon calculation is needed to obtain it. A number of this kind, a property of the material alone that the rate or the reach of one channel is proportional to, is what a comparison of targets needs. Such numbers exist

for most channels of Chaps. 10 to 12, but each holds over a window of masses, and outside it the ratio of two rates is not the ratio of the two numbers. We will derive one such number per channel, with its window and inputs.

17.5.1 Rate and reach figures of merit

A figure of merit is either a reach number or a rate number. A *reach figure of merit* is the lightest mass a channel can see, set by kinematics and one material energy or speed: the gap floor $2E_{\rm g}/v_{\rm max}^2$ of electronic excitations (Eq. (1.25)), the acoustic edge Eq. (11.5), the polar edge $2\omega_{\rm L}/v_{\rm max}^2$. These are exact statements about where a rate starts, and they rank materials without a window. A *rate figure of merit* F_A of material A is a number that makes the rate per unit target mass factorize at fixed dark-matter parameters, $\mathcal{R}_A(m_\chi) = G(m_\chi)\, F_A$, with G the same for every material. No real rate factorizes exactly, and the useful statement is about pairs. For two materials A and B we write

$$\frac{\mathcal{R}_A(m_\chi)}{\mathcal{R}_B(m_\chi)} = \frac{F_A}{F_B}\,[1+\delta_{AB}(m_\chi)], \qquad W_{AB} = \{m_\chi : |\delta_{AB}(m_\chi)| < 0.1\}, \tag{17.15}$$

and call the contiguous range W_{AB} the *pairwise window* of the figure of merit, the masses at which the ratio of the two figures of merit gives the ratio of the rates to 10 %. The tolerance is a convention, chosen to match the accuracy of most inputs. A window belongs to a pair, not to a material, because the corrections δ of two materials partly cancel when their properties are close and add when they are not. Several materials have a common window only if every pair among them has one, and Sect. 17.5.3 shows that this fails for sound speeds. Table 17.5 collects the figures of merit of every channel with their windows and the provenance of their inputs.

17.5.2 Gap floor and valence f-sum

An electronic excitation needs at least $E_{\rm g}$, and the dark matter can deposit at most $\frac{1}{2}m_\chi v_{\rm max}^2$ (Eq. (1.24)). The floor $2E_{\rm g}/v_{\rm max}^2$ is 0.333 MeV for silicon with its physical gap of 1.17 eV, 0.212 MeV for germanium (0.744 eV) and 0.433 MeV for GaAs (1.52 eV); with the step-model gap of 1.11 eV that accompanies the silicon tables (Sect. 10.10), it is 0.316 MeV. Near the floor the rate is set by the spectrum within a few phonon energies of the edge. The gap is a measured input, and it must be the gap at the operating temperature of the detector, since the floor moves by the same fraction as the gap.

A rate figure of merit is harder. The electron-recoil rate weights the loss function over the window $\omega \le q v_{\rm max}$ by the halo factor. Worked example 10.2 showed that silicon puts a tiny part of its f-sum where a 1 MeV particle can reach it: the computed rate is 4×10^6 times below the all-electron ceiling. Since the weight of the loss function inside the window varies between materials by orders of magnitude, no single number ranks electron-recoil rates; the loss function in the window is the figure of merit itself. What a single number can give is the ceiling. For it to be useful, the ceiling must count the electrons that absorb in the window and no others, and that needs a partial f-sum.

Split the dielectric function of an insulator with clamped nuclei into the contributions of the valence electrons and of the core,

$$\varepsilon(\omega) = \varepsilon_{\rm val}(\omega) + \varepsilon_{\rm core}(\omega) - 1,$$

each of them one plus the polarization of its group, at $q \to 0$. Three conditions make the valence part separable. (i) The two groups add, so that each obeys its own f-sum, $\int_0^\infty {\rm d}\omega\, \omega\, {\rm Im}\,\varepsilon_{\rm val} = (\pi/2)(\omega_p^{\rm val})^2$ with $(\omega_p^{\rm val})^2 = e^2 n_{\rm val}/m_e$; this neglects the transfer of oscillator

Table 17.5: Figures of merit per channel, with the masses at which they rank two materials and the provenance of their inputs. Rates per unit target mass; $\omega_{\rm th} = 1\,\text{meV}$ for the phonon channels; standard halo of Sect. 1.2. "Pairwise" refers to the window Eq. (17.15). The absorption figures of merit are exact at each mass, since absorption fixes both q and ω.

Channel	Figure of merit	Where it ranks	Inputs	Silicon
Electronic, reach	$2E_{\rm g}/v_{\rm max}^2$	exact	$E_{\rm g}(T)$, measured	0.333 MeV
Electronic, rate	$L(q,\omega)$ in the window; ceiling $\propto Z_{\rm val}/A_{\rm cell}$	no scalar ranks rates; ceiling with $\omega_p^{\rm val}$ for $qv_{\rm max} < \omega_{\rm core}$ (Eq. (17.16))	L computed; $n_{\rm val}$, $\omega_p^{\rm val}$ measured	0.142
Acoustic, reach	$m_{\rm min}^{\rm ac}$ at $c_{\rm LA,max}$, Eq. (11.5)	exact	C_{ij}, ρ_T	6.1 keV
Acoustic, heavy mediator	$1/c_{\rm NR1}$, Eq. (17.17)	pairwise: from about 4 edges of the softer crystal to the smaller $q_{\rm BZ}/2v_{\rm max}$; empty if speeds differ by $\gtrsim 13.5\,\text{km/s}$	C_{ij} measured or computed	8.956 km/s
Acoustic, light mediator	none (target independent)	equal rates above 44 times the higher edge	edge only	–
Polar, rate	$\mathcal{I}_{\rm pol}/\rho_T$ with $\bar{\omega}_{\rm pol}$	with $K_\eta(\bar{\omega}_{\rm pol}, m_\chi)$ at each mass; $\mathcal{I}_{\rm pol}/\rho_T$ alone only where the K_η agree (LiF/GaAs above 52 MeV)	ε_∞, ε_0, $\omega_{{\rm T}\nu}$, $\omega_{{\rm L}\nu}$ measured, or Z^* computed	–
Polar, reach	$2\omega_{{\rm L}\nu}/v_{\rm max}^2$, lowest weighted branch	exact	$\omega_{{\rm L}\nu}$, measured	–
Migdal	$F_{\rm Mig}$; pole limit Eq. (17.19)	impulse regime (Si above 17.5 MeV); ranking open	$L(p,\omega)$, $Z_{\rm ion}(p)$ computed	pole: 0.570, 16.6 eV
Absorption	$L(0,m_\phi)/\rho_T$, $m_\phi\sigma_1/\rho_T$, scalar response per mass (Sect. 17.2)	exact at each mass	optical data or computed	Sect. 17.2

strength between shells that the Pauli principle causes (Worked example 4.7). (ii) The valence absorption ends below the first core edge $\omega_{\rm core}$, and the core does not absorb below it. (iii) Over the valence spectrum the core part is real and equal to its static value, $\varepsilon_{\rm core}(\omega) \simeq \varepsilon_{\rm core}(0)$, which holds to order $\omega^2/\omega_{\rm core}^2$. By the Kramers–Kronig relation at $\omega = 0$ (Eq. (C.41)),

$$\varepsilon_{\rm core}(0) - 1 = \frac{2}{\pi} \int_{\omega_{\rm core}}^{\infty} \mathrm{d}\omega \, \frac{\mathrm{Im}\, \varepsilon_{\rm core}(\omega)}{\omega}.$$

Below $\omega_{\rm core}$ we then have $\varepsilon = \varepsilon_{\rm core}(0) + \varepsilon_{\rm val} - 1$. Consider the model function $\varepsilon_{\rm m}(\omega) = \varepsilon_{\rm core}(0) + \varepsilon_{\rm val}(\omega) - 1$ at all frequencies, which equals ε below the core edge to order $\omega^2/\omega_{\rm core}^2$. Above the valence spectrum $\varepsilon_{\rm val} - 1 \to -(\omega_p^{\rm val})^2/\omega^2$, so the function $g(\omega) = \varepsilon_{\rm m}^{-1}(\omega) - 1/\varepsilon_{\rm core}(0)$ falls as $(\omega_p^{\rm val})^2/[\varepsilon_{\rm core}(0)^2 \omega^2]$. It is analytic in the upper half plane, being the inverse of a causal response with a constant subtracted, and the argument that turns the large-frequency tail of ε^{-1} into the f-sum of the loss function (Eq. (4.46), Appendix C.5) applies to it. For the true ε the same tail is an intermediate asymptote, valid between the valence spectrum and the core edge, since at $\omega \to \infty$ the true g tends to $1 - 1/\varepsilon_{\rm core}(0)$. Since $\mathrm{Im}[-g] = L$ below $\omega_{\rm core}$, the loss function collected up to a cutoff ω_c between the valence spectrum and the core edge is

Partial (valence) f-sum of the loss function

$$\int_0^{\omega_c} \mathrm{d}\omega \, \omega \, L(\omega) = \frac{\pi}{2} \, \frac{(\omega_p^{\rm val})^2}{\varepsilon_{\rm core}(0)^2} \simeq \frac{\pi}{2} \, (\omega_p^{\rm val})^2, \tag{17.16}$$

with $\omega_p^{\rm val}$ the plasma frequency of the valence electrons alone.

The core screens the valence plasmon and lowers its weight by $\varepsilon_{\rm core}(0)^{-2}$. The absorption $\mathrm{Im}\, \varepsilon$ collected over the same range carries no such factor, because ε adds where ε^{-1} does not. For silicon, $n_{\rm val} = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$ gives $\omega_p^{\rm val} = 16.6\,\mathrm{eV}$ (Worked example 4.10), the measured plasmon energy, $16.6 \pm 0.1\,\mathrm{eV}$ by electron energy loss [5, Table I]. The core term follows from the photoabsorption above the L edge at 99.8 eV: $\varepsilon_{\rm core}(0) - 1 = 0.0154$, so the core lowers the valence weight of L by the factor 0.970. A valence-only calculation, such as the tables of Sect. 10.2, has no core and satisfies Eq. (17.16) with $\varepsilon_{\rm core} = 1$; this is why 16.6 eV is the right normalization of those tables (Worked example 12.3).

Limit Check 17.1. Partial f-sum of the valence electrons

Below the first core edge the valence electrons carry the f-sum of the loss function $\int_0^{\omega_c} \omega L \, \mathrm{d}\omega = (\pi/2)(\omega_p^{\rm val})^2$, with $\omega_p^{\rm val} = 16.6\,\mathrm{eV}$ for silicon, the measured plasmon energy [5, Table I].

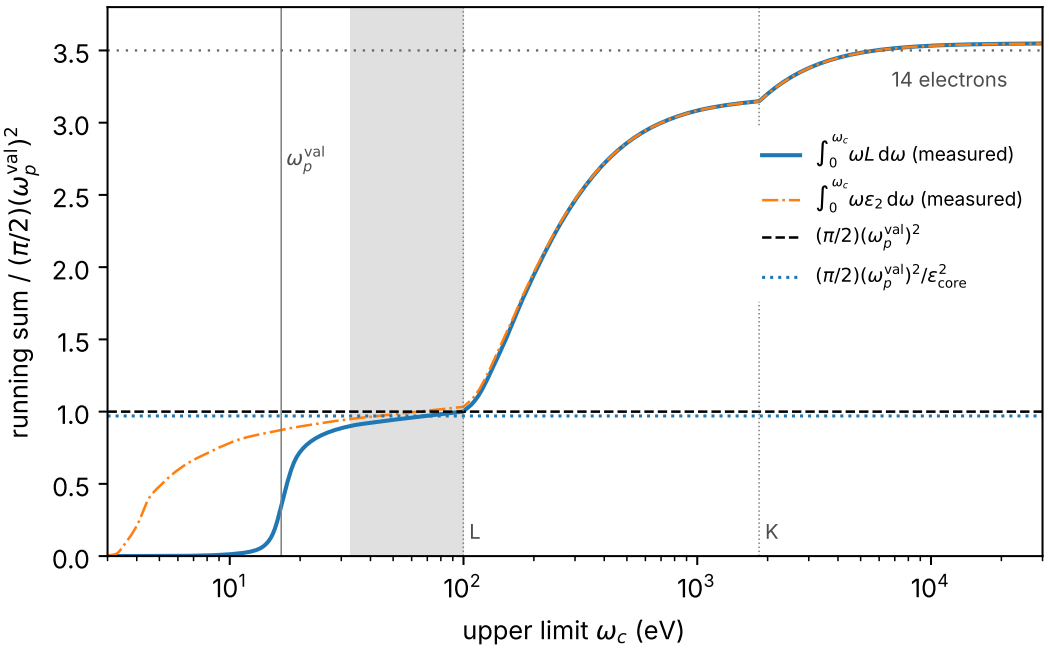


Figure 17.6: Running f-sums of silicon at $q \to 0$, $\int_0^{\omega_c} \omega L \, \mathrm{d}\omega$ (solid) and $\int_0^{\omega_c} \omega \varepsilon_2 \, \mathrm{d}\omega$ (dash-dotted), in units of $(\pi/2)(\omega_p^{\mathrm{val}})^2 = 433\,\mathrm{eV}^2$, against the upper limit ω_c. Data: the measured optical constants of Franta and coworkers at 300 K below 35 eV [3] and the photoabsorption tables of Henke and coworkers above, with $\varepsilon = 1 - (\omega_a^2/\omega^2)(f_1 - \mathrm{i}f_2)$, $\omega_a^2 = e^2 n_{\mathrm{at}}/m_e$ [6]; moving the join to 30 or 40 eV changes the L-sum below the L edge by 0.8 %. Dashed: Eq. (17.16) with $\varepsilon_{\mathrm{core}} = 1$; dotted: with the core term, 0.970. Vertical lines: ω_p^{val} and the L and K edges at 99.8 and 1839 eV; the dotted horizontal line at 3.5 is the sum of all 14 electrons per atom. Gray: ω_c from 32.7 eV to the L edge, where the L-sum lies within 10 % of the formula; just below the edge it is 1.000.

Below about $2\omega_p^{\mathrm{val}}$ the plasmon is not yet collected, and above the L edge the core electrons add their own sum, which rises toward 3.5 times the valence value.

Just below the edge the measured L-sum is 1.000 of $(\pi/2)(\omega_p^{\mathrm{val}})^2$ and the ε_2-sum 1.032. The two corrections to Eq. (17.16) are of similar size and opposite sign. The core screening lowers the L-sum by 3 %, and the valence region holds 3 % more absorption than four electrons, the strength moved up from the blocked core transitions. The data fix each of them to 1.4 %, the amount by which the two sets together overshoot the all-electron sum of 14. The approach to the plateau is slow, because the plasmon at 16.4 eV is broad: the L-sum reaches 0.9 at 32.7 eV.

The partial sum turns the valence plasma frequency into a usable ceiling. Worked example 10.2 found that ω_p^{val} in place of the all-electron 31.1 eV does not give a bound, since the core electrons hold the rest of the f-sum. When the window lies entirely below the core edge, however, the energy integral of the bound runs over the valence spectrum alone, and by Eq. (17.16) its weight is $(\pi/2)(\omega_p^{\mathrm{val}})^2$ to the few-percent accuracy of the two corrections; at finite q the condition is $q v_{\mathrm{max}} < \omega_{\mathrm{core}}$. This is not a theorem, since the corrections need not stay small at finite q, but it lowers the ceiling of silicon by $(31.1/16.6)^2 = 3.5$ for every mass whose largest deposit, $\frac{1}{2} m_\chi v_{\mathrm{max}}^2$, lies below the L edge, $m_\chi < 28.4\,\mathrm{MeV}$. Per unit target mass the ceiling is proportional to $(\omega_p^{\mathrm{val}})^2/\rho_T = e^2 Z_{\mathrm{val}}/(m_e m_u A_{\mathrm{cell}})$, with Z_{val} valence electrons in a cell of A_{cell} mass units: the number of valence electrons per nucleon, 0.142 for silicon, 0.055

for germanium, 0.333 for diamond and 0.308 for LiF. This orders the ceilings, not the rates, which differ from them by the factor 10^6 of Worked example 10.2.

17.5.3 Acoustic edge and effective sound speed

The one-phonon rate of Eq. (11.4) has three figures of merit, one for each limit of the mediator and one for the edge. The reach figure of merit is the acoustic edge Eq. (11.5), evaluated at the fastest longitudinal speed $c_{\mathrm{LA,max}}$: 6.1 keV for silicon, 10.2 keV for germanium and 3.1 keV for diamond at $\omega_{\mathrm{th}} = 1\,\mathrm{meV}$. It is exact kinematics and ranks without a window. A stiff crystal reaches lower masses.

For a light mediator there is no rate figure of merit. The rate per unit mass tends to the target-independent value Eq. (11.6), and the material enters through the correction factor $1 - (1+\kappa_\eta)\, m_{\mathrm{min}}^{\mathrm{ac}}/m_\chi$ alone, with the constant κ_η of Eq. (11.8). Two materials have equal rates within 10 % above 44 times the higher of their two edges (Limit Check 11.2), and below that the edge, a reach number, decides.

For a heavy mediator, Eq. (11.9) gives a plateau inversely proportional to the polarization-weighted mean of the inverse sound speeds. Its inverse is an effective sound speed of the one-phonon nuclear rate,

$$c_{\mathrm{NR1}} = \Big\langle \sum_\nu \frac{(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2}{c_\nu(\hat{\boldsymbol{q}})} \Big\rangle_{\hat{\boldsymbol{q}}}^{-1}, \qquad \mathcal{R} = \frac{2\rho_\chi\,\sigma_p\,I_0}{m_u^2\,c_{\mathrm{NR1}}}, \tag{17.17}$$

so that the figure of merit is $1/c_{\mathrm{NR1}}$. It is 8.956 km/s for silicon, 5.321 km/s for germanium and 18.16 km/s for diamond, within one percent of the sphere-averaged longitudinal speeds (Limit Check 11.3) for these weakly anisotropic crystals. The speeds and polarizations come from the Christoffel problem of the elastic tensor (Eq. (C.156)). The tensor is measured from the speed of ultrasound or from light scattered by thermal sound waves, or computed by density-functional theory from finite strains or from perturbation theory with strain as the perturbation (Appendix E.4); Appendix E.17 specifies the elastic constants needed for the candidate targets. The two figures of merit pull in opposite directions: a soft lattice gives a large heavy-mediator rate and a high edge.

Eq. (17.17) drops two things, and each sets one end of the window. At low mass the lower cut $u_- = \omega_{\mathrm{th}}/(2m_\chi c_\nu)$ of the integral $\int u^2\eta(c_\nu+u)\,\mathrm{d}u$ removes the slowest part of the halo integral. Since $u_- = (v_{\mathrm{max}} - c_\nu)\, m_{\mathrm{min}}^{\mathrm{ac}}/m_\chi$, it is a function of the mass in units of the edge, and it closes the window from below at a few times the edge of the softer crystal. At high mass the momentum transfers reach the zone scale, Eq. (11.3) stops holding, and the window ends at $m_{\mathrm{hi}} = q_{\mathrm{BZ}}/2v_{\mathrm{max}}$ of the crystal with the smaller zone, 0.42 MeV for silicon (Sect. 11.1). Between the two ends the window would be complete if the plateau were exact. It is not, and the reason is the shift of the halo integral by the sound speed.

Far above the edge, the integral of one branch and direction is $8m_\chi^3\int_0^{v_{\mathrm{max}}-c_\nu}\mathrm{d}u\,u^2\eta(c_\nu+u)$. With $v = c_\nu + u$ it becomes $8m_\chi^3\int_{c_\nu}^{v_{\mathrm{max}}}\mathrm{d}v\,(v-c_\nu)^2\eta(v)$, and expanding the square,

$$\int_{c_\nu}^{v_{\mathrm{max}}}\mathrm{d}v\,(v-c_\nu)^2\eta(v) = I_0 - 2c_\nu I_1 + c_\nu^2 + O(c_\nu^3), \qquad I_1 = \int_0^{v_{\mathrm{max}}}\mathrm{d}v\,v\,\eta(v),$$

where I_0 is the plateau integral of Sect. 11.1. The coefficient of c_ν^2 is $\int_0^\infty \eta\,\mathrm{d}v = 1$, the normalization of the velocity distribution, since η is its integral of f_χ/v above v_{min}. The integral from 0 to c_ν is of third order. Each branch and direction carries the weight $(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2/c_\nu$, so the term $-2c_\nu I_1$ contributes $-2I_1\langle\sum_\nu(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2\rangle = -2I_1$ by completeness, independent of the crystal. The rate in units of the plateau is

$$\frac{\mathcal{R}\,c_{\mathrm{NR1}}\,m_u^2}{2\rho_\chi\sigma_p I_0} = 1 - \frac{2I_1}{I_0}\,c_{\mathrm{NR1}} + O(c^2), \qquad \frac{2I_1}{I_0} = \frac{5.88}{v_{\mathrm{max}}} = \frac{1}{135\,\mathrm{km/s}}, \tag{17.18}$$

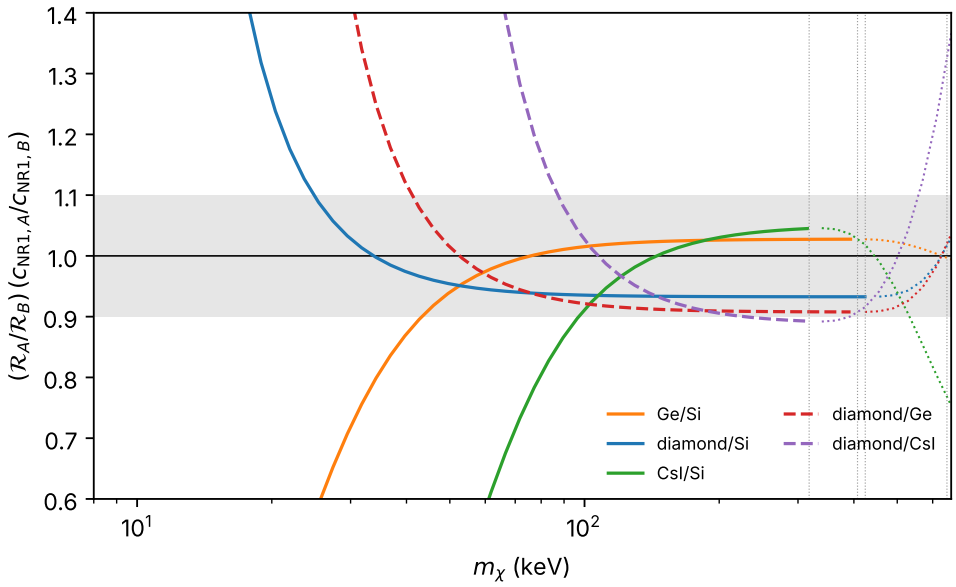


Figure 17.7: Heavy-mediator one-phonon rates of two crystals divided by the ratio of their figures of merit, $(\mathcal{R}_A/\mathcal{R}_B)(c_{\mathrm{NR1},A}/c_{\mathrm{NR1},B})$, against the dark-matter mass, from Eq. (11.4) with all branches and directions, $q \le q_{\mathrm{BZ}}$, $\omega_{\mathrm{th}} = 1\,\mathrm{meV}$, the orientation average and the standard halo of Sect. 1.2. Elastic tensors: measured room-temperature values for Si, Ge and diamond (Worked example 11.2); CsI at 4.2 K [7] with the room-temperature density $4.529\,\mathrm{g\,cm^{-3}}$. Gray: the 10 % band of the pairwise window Eq. (17.15). Each curve is solid up to the smaller $q_{\mathrm{BZ}}/2v_{\max}$ of its pair (vertical dotted lines: 0.32, 0.41, 0.42 and 0.65 MeV for CsI, Ge, Si and diamond) and dotted above. At high mass the curves settle at $1-(c_{\mathrm{NR1},A}-c_{\mathrm{NR1},B})/(135\,\mathrm{km/s})$, Eq. (17.18).

for the standard halo of Sect. 1.2, with η evaluated at $v_{\min} = c_\nu + q/2m_\chi$ as in Eq. (11.4). For silicon the bracket is 0.934 at first order and 0.935 at second, against the largest computed ratio 0.936; for diamond 0.866 and 0.872, against 0.874. This is why the plateau of Sect. 11.1 is never reached: the halo is not flat at the large speeds that a heavy mediator favors.

The shift is linear in the sound speed and does not cancel between materials. For two crystals in the middle of their window,

$$\frac{\mathcal{R}_A\, c_{\mathrm{NR1},A}}{\mathcal{R}_B\, c_{\mathrm{NR1},B}} \simeq 1 - \frac{c_{\mathrm{NR1},A} - c_{\mathrm{NR1},B}}{135\,\mathrm{km/s}}.$$

Roughly, the inverse speeds rank two crystals to 10 % as long as their effective speeds differ by less than 13.5 km/s, which diamond and germanium, 12.8 km/s apart, barely meet. The deviations of several crystals from a common reference add linearly in their speeds, and two crystals that both agree with silicon to 10 % can disagree with each other by up to twice that. Fig. 17.7 shows the ratio for five pairs. Germanium against silicon enters the band at 43 keV and settles at 1.027; diamond against silicon enters at 25 keV and settles at 0.933; CsI, a soft cubic crystal discussed below, enters at 97 keV and reaches 1.045 at its zone cap. Diamond against CsI, however, is inside the band from 87 to 227 keV alone and leaves it below the cap, where both still agree with silicon. A window "against silicon" does not give the ranking of the materials among themselves, and every pair needs its own.

Worked example 17.5: The pairwise window of diamond against silicon

Diamond has $c_{\mathrm{NR1}} = 18.16\,\mathrm{km/s}$ and silicon 8.956 km/s, so the figure of merit predicts a heavy-mediator rate ratio of $8.956/18.16 = 0.493$: the plateaus of Eq. (17.17) are 322.6 and 654.0 events per kilogram and year at $\sigma_p = 10^{-40}\,\mathrm{cm}^2$ and $\omega_{\mathrm{th}} = 1\,\mathrm{meV}$. The edges at the effective speeds are 3.19 keV for diamond and 6.39 keV for silicon, and the zone caps $q_{\mathrm{BZ}}/2v_{\max}$ are 0.65 and 0.42 MeV, so the window can extend at most to 0.42 MeV.
The full rates Eq. (11.4) give the correction $1+\delta$ of Eq. (17.15), the rate ratio divided by 0.493:

m_χ (keV)	10	20	30	45	100	300	420
$\mathcal{R}_{\mathrm{Si}}$	66.0	432.5	546.7	590.6	609.8	612.0	612.2
$\mathcal{R}_{\mathrm{C}}$	199.3	268.0	277.3	280.2	281.4	281.6	281.7
$1+\delta$	6.12	1.256	1.028	0.962	0.935	0.933	0.933

with the rates in events per kilogram and year. At 10 keV silicon is less than two edges above its threshold and diamond leads by a factor of three, against the predicted half. The lower cut explains it: at 20 keV it removes the halo below $u_- = 251\,\mathrm{km/s}$, $0.32\,v_{\max}$, for silicon, but only below 124 km/s for diamond, whose phonons of 1 meV carry half the momentum. The ratio crosses 1.1 at 24.8 keV, 3.9 times the edge of silicon, and the window opens there.
Above 100 keV the correction no longer depends on the mass. It is the shift Eq. (17.18): to first order $1-(18.16-8.956)/135 = 0.932$, and with the exact single-speed integrals $\int_c^{v_{\max}}(v-c)^2\eta\,\mathrm{d}v/I_0$, 0.8724 for diamond and 0.9354 for silicon, the ratio is 0.9327, the computed value to four digits. The window is

$$W_{\mathrm{C,Si}} = 25\,\mathrm{keV}\ \text{to}\ 0.42\,\mathrm{MeV},$$

inside which $1+\delta$ runs from 1.088 down to 0.933. The figure of merit ranks the pair correctly above 14 keV, 2.2 times the edge of silicon, where the two rates cross; below that the cut of the softer crystal reverses the order. Inside $W_{\mathrm{C,Si}}$ it also gives the ratio to 10 %, but even there it is biased by 7 % against the stiffer crystal. Against germanium, 12.8 km/s softer, diamond settles at 0.908, at the edge of the band.

Tabulated sound speeds feed most published comparisons of targets, and they can be tested against measured elasticity without any phonon calculation. For a cubic crystal the floor Eq. (11.11), $c_{\mathrm{LA}} \geq \sqrt{B/\rho_T}$, needs the bulk modulus alone; for any crystal the Christoffel problem of a measured tensor gives the range of longitudinal speeds over all directions. The target table of Griffin and collaborators lists one longitudinal speed per material, a sphere average of ω_{LA}/q from density-functional phonons [8, Table I and App. A 3]. Table 17.6 tests five of its entries against measured tensors, and all five fail.

The nitrides and ZnO are listed above every longitudinal speed of their measured tensors; the older handbook constants of the nitrides give the ratios 1.59 and 1.52 instead of 1.55. CsI is listed below them: its room-temperature bulk modulus, 13.5 GPa [13, Table I], puts the floor at 1.726 km/s, 25 % above the tabulated value, and the tabulated speed is 42 % below the slowest direction of the measured tensor. $CaWO_4$ is listed 8.4 % below its sphere average, 4.65 km/s, and below its slowest direction. For silicon the same table is 5.6 % below the measured average (Worked example 11.2). The errors matter for the figures of merit. With its measured tensor CsI has $c_{\mathrm{NR1}} = 2.40\,\mathrm{km/s}$ and a heavy-mediator figure of merit 3.73 times that of silicon, against 6.49 from the tabulated speed; its acoustic edge, from $c_{\mathrm{LA,max}} = 2.46\,\mathrm{km/s}$, is 23.1 keV. $CaWO_4$ has $c_{\mathrm{NR1}} = 4.58\,\mathrm{km/s}$, 1.96 times silicon against

Table 17.6: Tabulated longitudinal sound speeds [8, Table I] against measured elastic constants (GPa). Hexagonal crystals: C_{11}, C_{12}, C_{13}, C_{33}, C_{44}, the recommended values of Vurgaftman and Meyer for the nitrides as quoted in [9, Table I], and the Brillouin values for ZnO as quoted in [10, Table III]. CsI (cubic, 4.2 K): C_{11}, C_{12}, C_{44} [7, abstract]. $CaWO_4$ (tetragonal, point group $4/m$, which allows $C_{16} = -C_{26}$): ultrasonic data of Farley and Saunders [11] as quoted in [12, p. 16]. Densities from the measured lattice constants (CsI at room temperature). Range: the longitudinal speeds of the Christoffel problem over all directions.

Crystal	Measured constants	ρ_T (g cm^{-3})	Range (km/s)	Tabulated (km/s)	Tabulated/range
GaN	390, 145, 106, 398, 105	6.09	7.64–8.08	12.50	1.55 of max
AlN	396, 137, 108, 373, 116	3.26	10.52–11.02	17.09	1.55 of max
ZnO	206, 118, 118, 211, 44	5.67	6.03–6.10	12.53	2.05 of max
CsI	27.37, 7.93, 8.25	4.529	2.37–2.46	1.379	0.58 of min
$CaWO_4$	$C_{11} = 145.9$, $C_{33} = 127.4$, $C_{12} = 62.6$, $C_{13} = 39.2$, $C_{16} = -19.2$, $C_{44} = 33.5$, $C_{66} = 38.7$	6.115	4.36–5.17	4.257	0.98 of min

2.10, and an edge of 11.0 keV.

17.5.4 Polar band integral and polar edge

For a light dark photon the single-phonon rate of a polar crystal is the band integral $\mathcal{I}_{\rm pol}$ of the loss function, each line weighted by the kinematic factor $K_\eta(\omega, m_\chi)$ of Eq. (11.16). Sect. 11.2 takes K_η at the weighted mean frequency $\bar\omega_{\rm pol}$, which makes $\mathcal{I}_{\rm pol}/\rho_T$ of Worked example 11.3 the natural figure of merit. It factorizes only if K_η is the same for both crystals, and it is not, because K_η depends on the frequency. For a single speed $K_\eta = \ln(q_+/q_-)/v$ (Sect. 11.2.2), and far above the edge $\ln(q_+/q_-) \simeq \ln(2m_\chi v^2/\omega)$, so the ratio of the weights of two crystals approaches one only logarithmically:

$$\frac{K_\eta(\omega_A, m_\chi)}{K_\eta(\omega_B, m_\chi)} \simeq 1 - \frac{\ln(\omega_A/\omega_B)}{\ln(2m_\chi v^2/\omega_B)}, \qquad v \simeq v_0.$$

The halo average puts the effective speed in the logarithm near v_0. For LiF against GaAs, longitudinal energies of 77.2 and 36.35 meV, this gives 0.79 at 1 MeV and 0.87 at 10 MeV, against the full weights 0.77 and 0.87, and it reaches 0.9 at 54 MeV against the full 52 MeV; at 100 keV the full ratio is 0.34. Sapphire, with $\bar\omega_{\rm pol} = 109.9$ meV, reaches the band at 1.7 GeV. The pairwise window of $\mathcal{I}_{\rm pol}/\rho_T$ alone is thus empty below tens of MeV for crystals whose optical energies differ by a factor of two, which covers most pairs of interest. The figure of merit of the polar channel is a pair of numbers, $\mathcal{I}_{\rm pol}/\rho_T$ and $\bar\omega_{\rm pol}$, and the rate ratio at a given mass is $(\mathcal{I}_{{\rm pol},A}/\rho_{T,A})K_\eta(\bar\omega_{{\rm pol},A}, m_\chi)$ over the same for B. Its inputs, and their fragility on the dielectric route, are those of Worked example 11.3 and Limit Check 11.7; a crystal known through ε_∞, ε_0 and $\omega_{\rm T}$ alone is ranked by its interval.

The reach figure of merit is the polar edge, $2\omega_{{\rm L}\nu}/v_{\rm max}^2$ for the lowest longitudinal branch with weight: 10.35 keV for GaAs, 21.98 keV for LiF, and between 13.7 and 32.0 keV for the four branches of sapphire. Rate and reach rank in opposite orders, since a high optical frequency raises the band integral, at fixed screening, density and ratio $\omega_{\rm L}/\omega_{\rm T}$, and raises the edge by the same factor; the reach curves of LiF and GaAs cross at 26.7 keV (Sect. 11.2).

17.5.5 Migdal plasmon-pole factor

In the per-mass Migdal rate Eq. (12.26) the material enters as $\omega_e^{-4}\sum_\kappa N_\kappa L_{{\rm Mig},\kappa}/A_{\rm cell}$, with N_κ atoms of species κ in a cell of $A_{\rm cell}$ mass units and $L_{{\rm Mig},\kappa}$ the loss function seen by its ion.

Above a threshold the figure of merit is its integral,

$$F_{\mathrm{Mig}} = \sum_{\kappa} \frac{N_\kappa}{A_{\mathrm{cell}}} \int_{\omega_{\mathrm{th}}} \mathrm{d}\omega \, \frac{L_{\mathrm{Mig},\kappa}(\omega)}{\omega^4},$$

valid where Eq. (12.26) is: in the impulse regime, above 17.5 MeV for silicon (Sect. 11.4), with the per-mass form within 10 % of the exact free-recoil rate above 66 MeV (Limit Check 12.5), for $m_\chi \ll M_\kappa$ and for deposits small against the kinetic energy of the dark matter. The floor is the electronic one, $2E_{\mathrm{g}}/v_{\mathrm{max}}^2$. F_{Mig} needs the loss function at the large momenta of the ion's field, mostly 3 to 9 keV in silicon (Sect. 12.3), and no tabulated quantity gives it. Two limits show what it depends on.

Suppose first that at every p the valence loss function is a single plasmon pole carrying the partial f-sum Eq. (17.16), $L(p,\omega) = (\pi/2)[(\omega_p^{\mathrm{val}})^2/\omega_{\mathrm{pl}}(p)]\,\delta(\omega - \omega_{\mathrm{pl}}(p))$. Then $\int \mathrm{d}\omega\, L/\omega^4 = (\pi/2)(\omega_p^{\mathrm{val}})^2/\omega_{\mathrm{pl}}(p)^5$, which tends to $(\pi/2)(\omega_p^{\mathrm{val}})^{-3}$ at small p, and with the small-p ion charges

$$F_{\mathrm{Mig}} \propto \sum_{\kappa} \frac{N_\kappa Z_{\mathrm{ion},\kappa}^2}{A_{\mathrm{cell}}} \, \frac{\pi}{2} \, (\omega_p^{\mathrm{val}})^{-3} \qquad \text{(plasmon pole)}. \tag{17.19}$$

The nuclear factor $\sum_\kappa N_\kappa Z_{\mathrm{ion},\kappa}^2/A_{\mathrm{cell}}$ favors light nuclei: 0.570 for silicon ($Z_{\mathrm{ion}} = 4$) and 1.928 for LiF ($Z_{\mathrm{ion}} = 1$ and 7 with the $1s$ shells in the cores), a factor 3.38. The pole factor gives it back. LiF has eight valence electrons per formula unit in a cubic cell of 4.020 Å holding four of them [8, Table III], so $\omega_p^{\mathrm{val}} = 26.1$ eV, and the pole factors stand in the ratio 3.87. In the pole limit LiF has 0.87 of the Migdal figure of merit of silicon. Note, however, that the valence count of LiF includes the shallow fluorine $2s$ shell, and that the lithium $1s$ edge lies not far above the valence spectrum, so condition (ii) of the partial f-sum holds there only marginally. Germanium, with $\omega_p^{\mathrm{val}} = 15.6$ eV and $Z_{\mathrm{ion}} = 4$ with the $3d$ shell in the core, has 0.47 of it. Its $3d$ shell is shallow and polarizable, so $\varepsilon_{\mathrm{core}}(0) - 1$ is far larger than the 0.0154 of silicon, and this factor carries an unsized error from the screening by the core and its hybridization with the valence bands.

Eq. (17.19) ignores the dispersion of the pole. The Migdal weight sits at p of several keV, not at $p \to 0$, and with a dispersing pole the integral over p depends on the material through $\omega_{\mathrm{pl}}(p)$. For a pole $\omega_{\mathrm{pl}}^2 = \omega_p^2 + (3/5)v_{\mathrm{F}}^2 p^2$ integrated over all p, $\int p^2 \mathrm{d}p\, \omega_p^2/\omega_{\mathrm{pl}}^5 \propto v_{\mathrm{F}}^{-3} \propto (\omega_p^{\mathrm{val}})^{-2}$; for $\omega_{\mathrm{pl}}^2 = \omega_p^2 + (p^2/2m_e)^2$ it scales as $(\omega_p^{\mathrm{val}})^{-3/2}$. LiF then has 1.37 to 1.72 of the figure of merit of silicon instead of 0.87, and germanium 0.42 to 0.44. The pole limit is uncertain by a factor of two in its scaling between materials, and even the order of LiF and silicon depends on the dispersion.

The opposite limit puts the weight at the gap. For any loss function that starts at E_{g}, the partial f-sum bounds $\int_{E_{\mathrm{g}}} \mathrm{d}\omega\, L/\omega^4$ by $(\pi/2)(\omega_p^{\mathrm{val}})^2/E_{\mathrm{g}}^5$ at each p, the bound Sect. 12.3 states with the full plasma frequency. For silicon the bound exceeds the pole value by $(\omega_p^{\mathrm{val}}/E_{\mathrm{g}})^5 = 5.7 \times 10^5$, and between silicon and LiF ($E_{\mathrm{g}} = 14.2$ eV) it changes the order: the bound on the energy integral of silicon is 1.1×10^5 times that of LiF, and 3.2×10^4 times with the nuclear factor. Silicon is close to the gap end. Its computed spectrum puts 4 % of the Migdal probability in the plasmon and half of it below 4.6 eV (Sect. 12.3), so the weight just above the gap, not the plasmon, sets its figure of merit. The pole scaling Eq. (17.19) is a statement about crystals whose loss function is dominated by the plasmon at the momenta of the ion's field. Where real crystals sit between the two limits can be decided only with the computed $L(p,\omega)$ near the gap at momenta of several keV, with the valence split of Z_{ion} and L made consistently. The ranking of Migdal targets that would follow is an open problem, which the guide to next targets at the end of the book takes up.

Table 17.7: Provenance widths: the Gaussian width σ_j^{prov} of $\ln x_j$, or the prior, assigned to each kind of input of a figure of merit. The values are conventions. The widths for computed sound speeds and for room-temperature gaps are tested against the book's crystals in Sect. 17.6.1; the prior on X_{pole} is a device (Sect. 17.6.4).

Kind of input	Example	Width
Measured, spread stated	sapphire $\mathcal{I}_{\mathrm{pol}}$ through $\varepsilon_\infty = 3.1$–3.3; GaAs $\mathcal{I}_{\mathrm{pol}}$, half the spread of its dielectric and splitting routes	0.03; 0.04
Measured, no error stated	gap at low temperature; speed from a measured tensor; frequency of a measured phonon	0.01; 0.02; 0.005
Measured at room temperature, used at low temperature	gap of a crystal without low-temperature data	room-temperature value plus a shift uniform on 0.03–0.10 eV
Computed, materials database	sound speed from a computed elastic tensor; density of a computed cell	0.086; 0.030
Computed, density-functional perturbation theory	Born charge Z^*, ε_∞; optical-phonon energy	0.07; 0.05
Estimated	interband enhancement X_{pole} of the Migdal pole factor	log-uniform on [1, 3]

17.6 Ranking under uncertainty

Paratellurite, TeO_2, has the softest lattice of the crystals in our pool: the inverse of its heavy-mediator coefficient, the effective sound speed c_{NR1} of Sect. 17.5, is 3.75 km/s, against 4.58 km/s for $CaWO_4$ and 8.96 km/s for silicon. Its single-phonon rate per unit mass should then be 2.39 times silicon's, and 1.22 times that of $CaWO_4$. The two speeds have different origins, however. The one of $CaWO_4$ follows from an ultrasonic measurement of its elastic tensor; the one of TeO_2 from a tensor computed with a density functional, whose errors are of order 10 % and are not stated with the number. Whether TeO_2 beats $CaWO_4$ turns on those errors. We will give each input a width set by how it was obtained and read off, for each channel, the probability of ranking first, the interval of ranks and the tier.

17.6.1 Provenance widths

Every input of a figure of merit is one of three kinds. A *measured* input comes from an experiment on the crystal: the low-temperature gap of silicon, 1.17 eV (Table 3.1), or the elastic constants of germanium, 131, 49 and 68 GPa (Sect. 11.1). A *computed* input comes from a first-principles calculation, such as the elastic tensors of a materials database or the Born charges of a density-functional perturbation calculation. An *estimated* input is a parametric guess or a prior, such as the shift of a gap from room temperature to the millikelvin temperatures of a detector. We give each input a lognormal distribution, $\ln x_j$ Gaussian about $\ln x_{j0}$ with width σ_j^{prov}, or a stated uniform prior. The provenance width σ_j^{prov} is a convention, set by the kind of input rather than worked out number by number; Table 17.7 lists the values we use.

Two widths carry most of the rankings, and the book's own crystals test both. For computed sound speeds we take the elastic tensors of the JARVIS (Joint Automated Repository for Various Integrated Simulations) database, obtained with one exchange–correlation functional, OptB88vdW, for every crystal [14]. Silicon, germanium and diamond have measured tensors (Sect. 11.1), so the database can be compared with experiment on the same quantity. Solving the Christoffel problem for both tensors on a grid of 48×96 directions gives database values

of c_{NR1} below the measured ones by 2.9 % for silicon, 10.2 % for germanium and 1.0 % for diamond. The root mean square of the logarithmic differences is 0.065, inside the width 0.086 we assign. All three differences have the same sign, however, with a mean of -0.048: the errors of one method are not independent from crystal to crystal, a point we return to in Sect. 17.6.4. For gaps, the shift from room temperature to low temperature is 0.05, 0.08 and 0.10 eV for silicon, germanium and GaAs (Table 3.1); a crystal whose gap is known at room temperature alone gets a shift drawn uniformly between 0.03 and 0.10 eV. Neither width is a bound: a crystal whose computed tensor errs by 25 %, or whose gap moves by 0.2 eV on cooling, is not excluded by the convention.

The halo enters every edge through the same power of $v_{\max}$, so a common factor on it multiplies all edges alike and changes no order in any draw. We keep $v_{\max} = 2.650 \times 10^{-3}$ fixed.

The crystals we rank are silicon, germanium, GaAs and diamond, and seven insulators that cryogenic experiments use or propose: sapphire, LiF, NaF, $CaWO_4$, TeO_2, Li_2MoO_4 and γ-$LiAlO_2$. Each probability below is taken over this pool. The gaps of silicon, germanium and GaAs are the low-temperature values 1.17, 0.744 and 1.52 eV of Table 3.1, and that of diamond, 5.47 eV, is a low-temperature value as well; sapphire (8.8 eV), $CaWO_4$ (5.2 eV), NaF (11.5 eV) and LiF (14.2 eV) enter with room-temperature gaps [8, Table I]. The tensors of silicon, germanium and diamond are the measured ones of Sect. 11.1, and $CaWO_4$ enters with the measured tensor of Table 17.6, $c_{\mathrm{NR1}} = 4.58$ km/s and $c_{\mathrm{LA,max}} = 5.17$ km/s. GaAs, sapphire, TeO_2, γ-$LiAlO_2$, LiF and NaF enter with database tensors and the densities of their computed cells; Li_2MoO_4 has no tensor in the database and is not ranked in the two nuclear channels. The polar inputs of GaAs and sapphire are those of Sect. 11.2: $\mathcal{I}_{\mathrm{pol}}/\rho_T = 0.1415$ for GaAs (the splitting route of Worked example 11.3) and 9.12 for the ordinary axis of sapphire, both from measured frequencies. For LiF and NaF, Eq. (11.21) with the computed $Z^* = 1.05$ and 0.98, $\varepsilon_\infty = 2.02$ and 1.78, and highest optical energies 77.2 and 49.9 meV [8, Table I] gives 19.0 and 9.95 meV cm^3 g^{-1}.

17.6.2 Monte Carlo rank statistics

With every input a distribution, the figure of merit of each material is sampled by Monte Carlo as in Appendix C.14: we draw $N_{\mathrm{MC}} = 2 \times 10^4$ sets of inputs, evaluate the figure of merit of every material on each set, and record the order in every draw. Three quantities follow.

Ranking under uncertainty

For each material A of a pool and each draw k of the inputs, let $R_A^{(k)}$ be the rank of A in that draw. Then

$$P_A(\text{first}) = \frac{1}{N_{\mathrm{MC}}} \sum_{k=1}^{N_{\mathrm{MC}}} \mathbb{1}[R_A^{(k)} = 1], \qquad \text{rank interval} = [R_A^{16}, R_A^{84}], \tag{17.20}$$

where R_A^{16} and R_A^{84} are the 16th and 84th percentiles of $R_A^{(k)}$. The tier of A is its height $h(A)$, Eq. (C.117), in the interval order Eq. (C.116) of the 68% intervals of the figure of merit. All three are conditional on the provenance widths of Table 17.7.

The probabilities sum to one over the pool in every channel, and the sampling error of each is at most $[p(1-p)/N_{\mathrm{MC}}]^{1/2} \le 0.0035$. For a pair of lognormal figures of merit the probability that one beats the other has the closed form Eq. (C.124), which checks the sampling. For instance, TeO_2 against GaAs in the heavy-mediator channel compares two computed speeds with $\Delta = \ln(c_{\mathrm{NR1}}^{\mathrm{GaAs}}/c_{\mathrm{NR1}}^{\mathrm{TeO_2}}) = 0.242$ and $\sigma^{\mathrm{prov}} = 0.086$ each; the formula gives

$\Phi(0.242/0.122) = 0.977$, with Φ the standard normal distribution function, and the Monte Carlo 0.976. For $CaWO_4$ against GaAs, a measured and a computed speed, the two give 0.689 and 0.688.

The probability of ranking first rewards a lead that the widths cannot overturn, not the size of the lead. TeO_2 leads the heavy-mediator channel with $P(\text{first}) = 0.968$, and its 68% interval of $c_{\text{NR1}}(\text{Si})/c_{\text{NR1}}$ is $[2.19, 2.60]$; $CaWO_4$, measured, has the narrow interval $[1.92, 2.00]$ and $P(\text{first}) = 0.010$. With the computed-speed width doubled to 0.172, TeO_2 falls to $P(\text{first}) = 0.781$ and GaAs, also computed, rises from 0.022 to 0.135, while $CaWO_4$ gains just from 0.010 to 0.073. Neither value is the probability that TeO_2 is the better target; each is the probability under one assignment of widths.

The rank interval says where a material stands when it does not lead. Silicon, in the same channel, has the rank interval 8–8: it is eighth in at least 68% of the draws, ahead of sapphire and diamond alone, whose lattices are stiffer. $CaWO_4$ and GaAs share the interval 2–3. A rank interval that spans several places marks a material whose position the inputs do not fix; it says nothing about which neighbor it trades places with.

Fig. 17.8 shows the electron-recoil floor $2E_g/v_{\max}^2$, the lightest mass at which dark matter can cross the gap (Eq. (1.25)), over the 24 crystals of the target table, a wider list than our pool. The lead salts and InSb, whose gaps lie below 0.3 eV, hold the four lowest floors, with PbSe at 0.0484 MeV and $P(\text{first}) = 1.000$. Germanium is fifth at 0.212 MeV, silicon seventh at 0.333 MeV. Gaps measured to a percent separate almost every pair, and the 24 crystals fall into 21 tiers. The pairs that do not separate are the instructive ones. Germanium beats GaSb with probability 0.973, but only because GaSb's gap, 0.720 eV, is a room-temperature value and is shifted upward: unshifted, its floor of 0.205 MeV would lie below germanium's. With the shift drawn from 0 to 0.15 eV instead, the probability falls to 0.837. The order of these two crystals is a statement about the temperature dependence of a gap that has not been measured at the detector's temperature, and the widths make this visible. The upward shift is the convention of the tetrahedral semiconductors. The gaps of the lead salts, by contrast, shrink on cooling; their listed gaps, however, are low-temperature values, which the convention leaves unshifted.

Within our pool the electron-recoil floor ranks without ambiguity: germanium, silicon, GaAs, $CaWO_4$, diamond, sapphire, NaF and LiF, each in its own tier and with a rank interval of one place, from 0.212 MeV to 4.06 MeV. TeO_2, Li_2MoO_4 and γ-$LiAlO_2$ are not ranked, because no measured gap enters the tables we use.

17.6.3 Tiers

A probability of ranking first is a property of one material against the whole pool. A grouping of all materials into classes that the uncertainties can tell apart needs more: two materials should share a class when their intervals overlap, and be in different classes otherwise. Overlap is not transitive, so no grouping into tied classes can satisfy both conditions in general (Appendix C.13). The interval order resolves the conflict. Material A is resolved above B when the whole 68% interval of A lies on the better side of the whole interval of B, Eq. (C.116), and the tier of a material is its height in this order, one plus the length of the longest chain of materials resolved above it, Eq. (C.117). Each tier is an antichain, so its members are tied in the exact sense that none is resolved from any other. Every member of tier $k > 1$ is resolved below some member of tier $k - 1$.

A tier asserts only that each member is resolved below some member of the tier above. A wide interval can therefore rise. The nuclear edge $m_{\text{min}}^{\text{ac}}$ at $\omega_{\text{th}} = 1$ meV, Eq. (11.5) with $c_{\text{LA,max}}$, puts GaAs, with a computed tensor and the interval 10.3–12.3 keV, in tier 6. It puts $CaWO_4$, measured at 10.8–11.2 keV, in tier 7, although the median edge of $CaWO_4$,

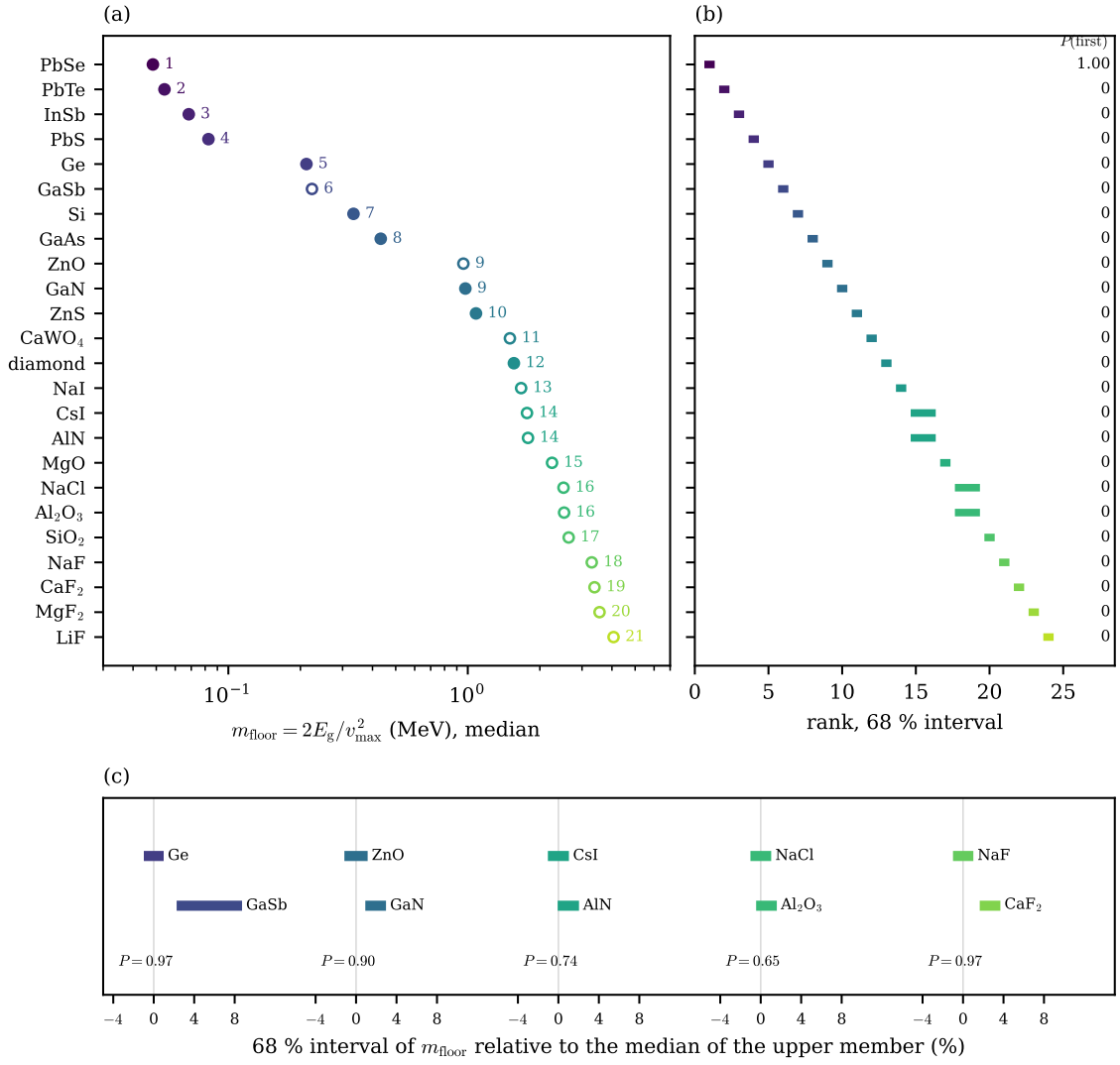


Figure 17.8: Rank intervals for the electron-recoil floor $2E_g/v_{max}^2$, $v_{max} = 2.650 \times 10^{-3}$, of the 24 crystals in the target table of Griffin and collaborators [8, Table I], with the low-temperature gaps of Table 3.1 for silicon, germanium and GaAs. (a) Median floor; filled symbols: gap measured at low temperature, open symbols: room-temperature gap plus an upward shift drawn uniformly on 0.03–0.10 eV; gap width 0.01 in the logarithm for all. Numbers and colors: tier (height in the interval order of the 68% intervals). (b) Rank, 16th to 84th percentile over 2×10^4 draws, and the probability of ranking first. (c) The 68% intervals of five close pairs, relative to the median of the member with the lower floor, with the probability P that it has the lower floor. Pairs in one tier overlap; pairs in different tiers do not, whatever P.

11.0 keV, is the lower. Nothing is resolved between GaAs and germanium, whose measured edge is 10.0–10.4 keV, while germanium is resolved above $CaWO_4$. Tiers never invent an order that the intervals do not contain, and they discard the information about medians that $P(\text{first})$ keeps; the two are read together. Fig. 17.8c shows both at once. ZnO and GaN share a tier although ZnO has the lower floor with probability 0.90, and NaF and CaF_2 are in different tiers with probability 0.97 for NaF: the tier boundary falls where the intervals stop overlapping, not at a fixed probability.

Worked example 17.6: Tiers of five intervals

Take the five crystals that come next to one another in the heavy-mediator channel, with the 68% intervals of $c_{\mathrm{NR1}}(\mathrm{Si})/c_{\mathrm{NR1}}$ from the Monte Carlo at independent errors: TeO_2 $[2.191, 2.598]$, GaAs $[1.721, 2.039]$, $CaWO_4$ $[1.918, 1.996]$, germanium $[1.649, 1.717]$ and NaF $[1.378, 1.634]$. Larger is better, so the intervals need no reorientation.

Order by upper end. The upper ends are 2.598, 2.039, 1.996, 1.717 and 1.634, so the crystals are processed in the order TeO_2, GaAs, $CaWO_4$, germanium, NaF. When a crystal is reached, every crystal that can be resolved above it has already been processed, since its lower end, and hence its upper end, exceeds the current upper end.

Heights. TeO_2: nothing processed, height 1. GaAs: TeO_2's lower end 2.191 exceeds GaAs's upper end 2.039, so TeO_2 is resolved above it, and $h(\mathrm{GaAs}) = 1 + 1 = 2$. $CaWO_4$: TeO_2 ($2.191 > 1.996$) is above it, GaAs ($1.721 < 1.996$) is not; $h = 2$. Germanium: TeO_2 (2.191), GaAs (1.721) and $CaWO_4$ (1.918) all have lower ends above 1.717, and the highest of them has height 2, so $h(\mathrm{Ge}) = 3$. NaF: all four lower ends exceed 1.634, the largest height among them is 3, and $h(\mathrm{NaF}) = 4$.

Tiers. $\{TeO_2\}$, $\{\mathrm{GaAs}, CaWO_4\}$, $\{\mathrm{Ge}\}$, $\{\mathrm{NaF}\}$. GaAs and $CaWO_4$ share tier 2 because their intervals overlap, although $CaWO_4$'s lies inside GaAs's. Among the five, $P(\text{first})$ is 0.968 for TeO_2, 0.022 for GaAs and 0.010 for $CaWO_4$, and the rank intervals are 1–1, 2–3, 2–3, 4–4 and 5–5.

What the tiers rest on. Germanium is in tier 3 because GaAs's lower end, 1.721, exceeds germanium's upper end, 1.717, by 0.004. The tier boundary is as sharp as the percentiles that define it, and a change of the confidence level or of one width can flip it, while the probability that GaAs beats germanium, 0.887, varies smoothly with the widths. Both are conditional on the convention that GaAs's computed speed has the width 0.086 and germanium's measured one 0.02.

17.6.4 Correlated method errors

The three database speeds that we could check against experiment were all low. If a method errs in the same direction for every crystal, the ratio of two of its results is more accurate than either result, and treating their errors as independent understates how well it ranks them. We model the shared part with one parameter, the method correlation r_{mat}. Each input of a computed kind draws its Gaussian variable as

$$z_j = r_{\mathrm{mat}}^{1/2}\, z_{\mathrm{method}} + (1 - r_{\mathrm{mat}})^{1/2}\, z'_j, \tag{17.21}$$

with z_{method} common to every crystal computed with the same method and z'_j its own, both standard normal. The marginal of every input is unchanged, so are the medians and the 68% intervals, and so are the tiers built from them, which are blind to correlation. Joint statements are what move. The methods that share a variable are the database speeds, the densities of computed cells, each quantity of the perturbation calculations, the room-to-low-temperature

shift of the gaps and the pole prior X_{pole}. Inputs measured independently never share one. By Eq. (C.124) the variance of $\ln(x_A/x_B)$ is $\sigma_A^2 + \sigma_B^2 - 2r_{\mathrm{mat}}\sigma_A\sigma_B$, which vanishes at $r_{\mathrm{mat}} = 1$ for two inputs of one method and equal widths.

Correlation sharpens comparisons inside a method. TeO_2 beats GaAs, both from database tensors, with probability 0.976 at $r_{\mathrm{mat}} = 0$, 0.997 at 0.5 and 1.000 at 1. Its probability of ranking first rises from 0.968 to 0.985 and 0.987, since its strongest challenger in the draws, GaAs, now moves with it. LiF beats NaF in the polar rate, both through computed Born charges, with probability 0.987 at independent errors and 1.000 at full correlation.

Correlation also resolves ties that the data do not resolve. The Migdal figure of merit is known only between two limits, the plasmon pole and a loss function concentrated at the gap, which differ by 5.7×10^5 for silicon (Sect. 17.5.5). We rank with the pole form Eq. (17.19) times a factor X_{pole} for the interband weight below the plasma frequency, drawn log-uniformly between 1 and 3, with the median $\sqrt{3} = 1.73$. This prior is a device, not a description of the uncertainty. For silicon, the one crystal with a computed spectrum, the plasmon carries 4 % of the Migdal probability (Sect. 12.3), so its own X_{pole} is about 25, far outside the prior; and X_{pole} depends on the gap, which differs from crystal to crystal (Sect. 17.5.5). The medians of the Migdal row, and the probabilities built on them, are therefore artifacts of the prior. A wider prior would only widen the intervals; a gap-dependent one would favor small-gap crystals (Sect. 17.5.5), and so would the dispersion of the pole for LiF (0.87 against 1.37 to 1.72). The order is therefore open, not merely unresolved. At $r_{\mathrm{mat}} = 0$ six crystals share tier 1 and no crystal ranks first with probability above 0.315 (NaF; then silicon 0.290, LiF 0.147). At $r_{\mathrm{mat}} = 1$ the prior cancels from every ratio, the ranking falls back on the values at $X_{\mathrm{pole}} = 1$, and NaF ranks first with probability 1.000, although its value exceeds silicon's by 2.4 %. That resolution is only as good as the assumption that the interband weight raises the integral by the same factor in every crystal, which the factor's dependence on the gap rules out. We read rankings at $r_{\mathrm{mat}} = 0$ and report the correlated variant beside them: where the two disagree, the ranking rests on an assumption about the method, not on the inputs.

No correlation helps when one material is measured and the other computed. Germanium has a measured tensor and GaAs a database one, and GaAs has the larger figure of merit with probability 0.887. The database speeds of silicon, germanium and diamond are low by 4.9 % on average; moved onto GaAs, that offset lowers its figure of merit from 1.87 to 1.78 and the probability to 0.744. The same offset on TeO_2 lowers the probability that it beats the measured $CaWO_4$ from 0.987 to 0.955. The comparison of a measured with a computed value carries the offset of the method in full, and only a measurement of the second crystal removes it.

17.6.5 Rankings of the scattering channels

Table 17.8 collects the rankings of our eleven crystals in the scattering channels for which Sect. 17.5 defines a figure of merit. Each row holds in the window of its figure alone. The electron-recoil floor and the two edges are kinematic and hold at every mass; the heavy-mediator figure holds pairwise, in the window W_{AB} of two crystals; the polar band integral ranks two crystals by itself where their kinematic weights K_η agree, LiF against GaAs above 52 MeV (Sect. 17.5.4).

Silicon, the running example of the book, leads no row and sits in the middle of every row in which it is ranked. The two pairs of rows that run in opposite directions, as Sect. 17.5 found, are extreme here. Diamond is first in the nuclear edge and last in the heavy-mediator rate, and GaAs, first in polar reach, is last in polar rate, 134 times below LiF and 64 times below sapphire.

Diamond's nuclear edge rests on its measured tensor, and it leads sapphire, the runner-up

Table 17.8: Response-level rankings of the scattering channels for silicon, germanium, GaAs, diamond, sapphire, LiF, NaF, $CaWO_4$, TeO_2, Li_2MoO_4 and γ-$LiAlO_2$. Figures of merit of Sect. 17.5; inputs and provenance widths of Sect. 17.6.1 and Table 17.7; 2×10^4 draws. Leader: median and 68% interval. P(first) at independent method errors, with the value at $r_{\mathrm{mat}} = 1$ in parentheses. Following tiers separated by vertical bars, members of one tier in braces, medians given. Edges at $\omega_{\mathrm{th}} = 1\,\mathrm{meV}$; $v_{\mathrm{max}} = 2.650 \times 10^{-3}$. Migdal row: medians relative to silicon at $X_{\mathrm{pole}} = 1$, so they include the prior median $\sqrt{3} = 1.73$ of X_{pole} (Sect. 17.6.4), and they carry no ordering information.

Channel, figure of merit	Leader (tier 1)	P(first)	Following tiers	Not ranked
Electron-recoil floor $2E_{\mathrm{g}}/v_{\mathrm{max}}^2$ (MeV), lower	Ge, 0.212 [0.210, 0.214]	1.000 (1.000)	Si 0.333 | GaAs 0.433 | $CaWO_4$ 1.50 | diamond 1.56 | sapphire 2.52 | NaF 3.29 | LiF 4.06	TeO_2, Li_2MoO_4, γ-$LiAlO_2$: no measured gap
Heavy-mediator single phonon $c_{\mathrm{NR1}}(\mathrm{Si})/c_{\mathrm{NR1}}$, higher	TeO_2, 2.39 [2.19, 2.60]; computed tensor	0.968 (0.987)	{$CaWO_4$ 1.96, GaAs 1.87} | Ge 1.68 | NaF 1.50 | {LiF 1.21, γ-$LiAlO_2$ 1.11} | Si 1.00 | sapphire 0.83 | diamond 0.49	Li_2MoO_4: no elastic tensor
Nuclear edge $m_{\mathrm{min}}^{\mathrm{ac}}$ with $c_{\mathrm{LA,max}}$ (keV), lower	diamond, 3.11 [3.05, 3.18]; measured tensor	1.000 (1.000)	sapphire 5.16 | Si 6.12 | {γ-$LiAlO_2$ 6.80, LiF 7.39} | NaF 8.83 | {Ge 10.2, GaAs 11.3} | $CaWO_4$ 11.0 | TeO_2 12.4	Li_2MoO_4
Polar rate $\mathcal{I}_{\mathrm{pol}}/\rho_T$ ($\mathrm{meV\,cm^3\,g^{-1}}$), higher	LiF, 18.9 [15.5, 23.2]; computed Z^*, ε_∞	0.987 (1.000)	{NaF 9.93, sapphire 9.12} | GaAs 0.141	Si, Ge, diamond: nonpolar; $CaWO_4$, TeO_2, Li_2MoO_4, γ-$LiAlO_2$: no consistent mode set
Polar reach $2\omega_{\mathrm{L}}/v_{\mathrm{max}}^2$ (keV), lower	GaAs, 10.35 [10.30, 10.40]; measured ω_{L}	1.000 (1.000)	NaF 14.2 | LiF 22.0 | sapphire 32.0	as polar rate
Migdal pole factor times X_{pole}, higher	no reading: tier 1 {NaF 1.77, Si 1.73, LiF 1.51, Li_2MoO_4 1.47, γ-$LiAlO_2$ 1.46, GaAs 0.84}	largest 0.315, NaF (1.000)	Ge, sapphire, TeO_2, $CaWO_4$, diamond in tier 2	–
Light-mediator single phonon	no ranking: target-independent per mass far above the edge (Eq. (11.6))	–	–	–

on a computed tensor, by a factor 1.66. LiF's polar lead rests on computed Born charge, ε_∞ and phonon energy, and survives because the factor of two to the next tier exceeds the width of those three inputs together.

Crystals outside the pool can lead. CsI, with its tensor measured at 4.2 K, has $c_{\mathrm{NR1}} = 2.40\,\mathrm{km/s}$ (Table 17.6), and 2.35 km/s for a cold cell at most 4.5 % denser than the room-temperature one. Its heavy-mediator figure of merit, 3.7 to 3.8, exceeds TeO_2's by more than half; its drawback is a nuclear edge of 23 keV. The lead salts, as Fig. 17.8 shows, have electron-recoil floors 2.6 to 4.4 times below germanium's. Every probability in Table 17.8 passes to the guide to next targets at the end of the book with the provenance widths of Table 17.7 attached.

Summary

At fixed coupling the rate of a crystal adds sector by sector, and which sector dominates follows from two facts: how much energy each sector can absorb, and whether the coupling is screened (Sect. 17.1). A single phonon carries at most the top of the band, 64.1 meV in silicon; a freely recoiling nucleus at most $E_R^{\max}$, 0.054 eV at 10 MeV; an electronic excitation up to the whole kinetic energy, 35 eV at 10 MeV. For a nucleon coupling with a heavy mediator the electron-following sector holds every threshold below the band top, with one phonon leading up to 20 MeV at 1 meV, and the Migdal effect holds the band between the end of the lattice spectrum and the largest deposit. The multiphonon–Migdal crossover threshold in silicon is 0.178, 0.307 and 5.33 eV at 1, 10 and 100 MeV: pinned near three times the band top for soft kicks, climbing through the Poisson tail of the multiphonon spectrum between 3 and 20 MeV, and sitting on the free-recoil endpoint above the impulse onset. For an electron coupling, screening decides the low thresholds. A dark photon couples to the total charge, its structure factor is the loss function, and silicon, with vanishing Born charges, has no lattice channel for it at order q^2; a mediator coupled to the electrons alone sees the electrons that follow the nuclei and raises the rate at 1 MeV from 17 to 7×10^5 events per kilogram and year at $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$. In GaAs the Born charge $Z^* = 2.11$ reverses the picture: the polar phonon opens to the dark photon at 10.35 keV and is nearly closed to a nucleon coupling.

For absorption each boson measures one response at $q = p_\phi$ and $\omega = m_\phi$ (Sect. 17.2). The dark photon sees the loss function, which screening caps at $L \leq 1/(2|\varepsilon_1|)$, so it is absorbed strongly near the zeros of ε_1: the longitudinal phonon of GaAs, the highest polar branch of sapphire and the plasmon of silicon. The axion-like particle sees the conductivity or the spin current and has no such ceiling; in a clean superconductor its spin current breaks pairs directly. Between 0.38 and 1.17 eV no target of the book covers the dark photon at leading order. A counter of quanta whose band is the line beats a quantum-limited linear readout of the same mode by $\mathrm{e}^{m_\phi/2T}$ in signal, so its advantage vanishes for masses well below the temperature, 0.86 µeV at 10 mK, and above twenty times the temperature the counter is limited by the absorbed quanta alone.

Dark matter with a mass splitting can lose its excitation energy to a collective mode (Sect. 17.3). When the mode's group velocity exceeds the halo speeds and its weight is smooth over $m_\chi v_{\max}$, the rate is independent of the halo to order v^2/v_ν^2, the line sits where the mode's dispersion meets the shell of fixed momentum, and its width has the shape of η. In silicon at 1 MeV a splitting of 10 eV raises the single-phonon rate 31 times over elastic scattering, and the shell form gives 0.78 to 1.10 of the full integral. For GaAs at 30 keV and $\Delta_\chi = -1$ eV, down-scattering into the longitudinal phonon gives 1810 events per kilogram and year against 842 for elastic scattering. An endothermic splitting of 1 eV brings every collective channel to 0.3 MeV, half the pair threshold of silicon, and the plasmon is closed at every mass.

Cross terms between sectors appear for couplings to both electrons and nuclei (Sect. 17.4). The interference between the Migdal and the phonon-assisted amplitudes is bounded by the positivity of the response matrix, Eq. (17.12), and at small momentum transfer the bound is reached: the summed Migdal amplitude of a one-phonon pair state is $-(m_e/M_\kappa)$ times the phonon-assisted one, so the phonon-dressed rate carries the factor $|\tilde{V}_e - (m_e/M_\kappa)\tilde{V}_{\rm nuc}|^2$, Eq. (17.14). For a dark photon this raises the phonon-assisted rate of silicon by 5.47×10^{-4}; for a coupling in proportion to mass the bracket vanishes. In the windows of silicon and GaAs, at a fifth to three fifths of the way to $2\pi/a$, the relation fixes the cross term in order of magnitude.

Each channel then reduces to a few material numbers, each valid in a window (Sect. 17.5): the floor $2E_{\rm g}/v_{\rm max}^2$, the inverse effective sound speed $1/c_{\rm NR1}$ for a heavy mediator, the acoustic edge, the polar band integral $\mathcal{I}_{\rm pol}/\rho_T$ and the polar edge, and the Migdal figure of merit. The reach figures of merit are kinematic and exact; the rate figures of merit hold in windows. No scalar ranks electron-recoil rates, and the partial f-sum of the valence electrons caps them through $\omega_p^{\rm val}$, 16.6 eV in silicon. The Migdal figure of merit lies between a plasmon-pole limit and a limit with the weight at the gap, which differ by 5.7×10^5 in silicon, and the pole limit itself depends on the dispersion of the plasmon, so its ranking is open. A heavy-mediator ranking holds pairwise, in the window W_{AB} of two crystals, because the shift of η by the sound speed breaks a common window; at 0.3 MeV the computed rate ratios of germanium and diamond to silicon are 1.73 and 0.46 against the speed ratios 1.69 and 0.49 (Sect. 17.5).

Rankings by these numbers carry the uncertainty of their inputs (Sect. 17.6). Each input gets a provenance width by how it was obtained, measured, computed or estimated; the widths are conventions, and the database sound speeds of silicon, germanium and diamond, low by 1 to 10 % and all in the same direction, show both the size of one convention and why method errors correlate. Monte Carlo propagation gives the probability that a crystal ranks first and its interval of ranks; heights in the interval order of the 68% intervals give tiers, which never assert an order the intervals do not contain. Among eleven crystals germanium leads the electron-recoil floor, TeO_2 the heavy-mediator single-phonon rate (probability 0.968, on a computed tensor), diamond the nuclear edge, LiF the polar rate and GaAs the polar reach. Silicon leads none. Stiff lattices and light polar ions win one row of each pair and lose the other. The Migdal channel has no reading at independent errors, its largest probability of ranking first being 0.315, and the reading it acquires when the pole prior is shared by all crystals is an artifact of that assumption. The light-mediator single-phonon rate does not rank crystals at all. A shared method error sharpens comparisons inside one method and cannot help a comparison of a measured with a computed value, which keeps the method's offset in full.

Table 17.9 collects the truncations and inputs of the chapter with their sizes. The comparison maps are limited by three inputs that only a calculation or a measurement can supply: the density that follows the nuclei at finite momentum, for the electron-only lattice channel; the phonon-assisted rate from the amplitude, for the silicon window; and the germanium zero-phonon edge and phonon density of states. The rankings are limited by computed inputs where measured ones are missing, above all the tensor of TeO_2 and the loss function $L(p, \omega)$ that the Migdal row needs. How the energy per quantum, the readout and the backgrounds change these rankings is the subject of the guide to next targets at the end of the book.

Table 17.9: Nuisances of the comparison across channels. Type in the classes of Sect. 8.8: normalization, tilt, threshold and new support; "order" marks an item that acts on a ranking rather than on a rate. Basis of the size: "computed" (next order or exact result computed), "model" (computed in a model), "est." (parametric estimate), "input spread" (spread or limitation of a measured or computed input), "convention" (a width assigned by convention), "unsized" (no size known). The last column gives the section where the item is treated.

Truncation or input	Type	Size	Basis	Section
Coherent corrections to the multiphonon tail	threshold	a factor of two in the tail moves the crossover in silicon at 10 MeV by 0.014 eV	model	Sect. 17.1
Local fields and off-diagonal elements of the Migdal probability	normalization	diagonal factor 1.35 moves the crossover at 10 MeV by 0.006 eV; off-diagonal size open (Sect. 12.3)	computed; unsized	Sect. 17.1
Coherent Migdal response at $q < 2q_{\mathrm{BZ}}$	normalization; tilt in ω_e	diagonal form 14.9× high as $q \to 0$ (Eq. (8.32)), quantitative only below about 0.75 keV; 0.75 keV to $2q_{\mathrm{BZ}}$ unsized, nearly all of the silicon Migdal rate up to 1 MeV (29 % of it from $q < q_{\mathrm{BZ}}$ at 1 MeV), 33 % at 2 MeV, 7 % at 3 MeV (Table 12.3)	computed (limit); unsized	Sect. 17.1
Electron-following sector of an electron-only coupling (long-wavelength model)	normalization; tilt in m_χ	equality with electron recoil at 12.3 MeV is a property of the model; Umklapp, transverse and multiphonon terms add, dispersion and loss of following charge subtract	unsized	Sect. 17.1
Phonon-assisted side of the silicon window	normalization	estimate known to a factor of ten; lead to about 0.43 MeV, two-channel band 0.43–0.51 MeV	est.	Sect. 17.1
Germanium zero-phonon edge and phonon density of states	threshold	not known; the germanium electronic window and crossover are not drawn	unsized	Sect. 17.1
Half-gap and cutoff of $ZrTe_5$	threshold	measured-structure (11.8 meV) and relaxed-cell (15.6 meV) parameters move the absorption onset from 23.6 to 31.2 meV; the reach at 100 meV changes by 2.5 %	input spread	Sect. 17.2
Counting against linear readout	normalization	a counter band B wider than the line multiplies the advantage by $(B/\Delta\omega_{\mathrm{eq}})^{-1/2}$; squeezed readout and dark counts reduce it	unsized	Sect. 17.2
Exothermic shell form	normalization	halo-independent to $O(v^2/v_\nu^2)$; silicon phonons 0.78–1.10 of the full integral at 1 MeV, the weight varying on the zone scale	computed	Sect. 17.3
Electron-gas plasmon against crystal tables	normalization	plasmon-pole rate 0.09–0.16 of the tables' electronic rate at 18–22 eV (two models)	model	Sect. 17.3
Migdal–phonon-assisted cross term	normalization	bounded by positivity, Eq. (17.12); exact at small q (Eq. (17.14): $+5.47 \times 10^{-4}$ for a dark photon in silicon); in the windows, at 0.2–0.6 of $2\pi/a$, order of magnitude only	computed (limit); est.	Sect. 17.4
Shift of η by the sound speed (heavy mediator)	tilt in m_χ; order	rate ratios differ from the speed ratios by up to 7 % at 0.3 MeV (germanium, diamond against silicon); rankings hold pairwise in W_{AB}	computed	Sect. 17.5
Interband weight of the Migdal pole factor	normalization; order	pole and gap limits differ by 5.7×10^5 in silicon; the dispersion of the pole changes LiF over silicon from 0.87 to 1.4–1.7; the ranking uses the pole form with a prior X_{pole} log-uniform on [1, 3], a device (silicon's own value is about 25), and gives no reading at independent errors	est.	Sects. 17.5 and 17.6
Provenance widths	order	database speeds 0.086 (three checked crystals: root mean square 0.065); gap shift 0.03–0.10 eV; doubling the speed width lowers TeO_2's probability of ranking first from 0.968 to 0.781	convention	Sect. 17.6
Offset of the computed elastic tensors	order	database speeds low by 4.9 % on average (silicon, germanium, diamond); GaAs against germanium 0.887 → 0.744 with the offset	input spread	Sect. 17.6
Method correlation r_{mat}	order	sharpens one-method comparisons (TeO_2 against GaAs 0.976 at 0, 1.000 at 1); resolves the Migdal tie artificially (NaF 0.315 → 1.000)	computed	Sect. 17.6
Room-temperature gaps used at low temperature	threshold; order	germanium against GaSb 0.973 with the shift 0.03–0.10 eV, 0.837 with 0–0.15 eV; reversed without shift	est.	Sect. 17.6

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Part III

Measurement

Chapter 18

Ionization yield

A deposit of 10 eV in silicon has enough energy for eight electron–hole pairs across the 1.17 eV gap of silicon at millikelvin temperatures, yet the cascade model of Sect. 18.1 leaves two or three in nearly every event. The measured mean energy per pair is three times the gap (Sect. 18.4), and the rest of the energy ends in phonons. Between the deposit and the count lies a cascade of hot carriers that ionize, emit phonons and slow down, all within about a picosecond. The mean energy per pair $\varepsilon_{\mathrm{eh}}$ and the Fano factor F_{eh} of the pair number set the threshold of every electron-recoil search in a semiconductor, since at the few-pair level one ionization more or less moves an event from one bin to the next. For a long time both have been fitted to calibration data rather than computed: the cascade model used for silicon has one constant, tuned to the measured $\varepsilon_{\mathrm{eh}}$ (Appendix E.13).

This chapter asks how far the two numbers can be derived from the same rates as the rest of the book. The hot carrier is itself a projectile of the master formula. It scatters off the electrons and the lattice of the target as a dark-matter particle with a Coulomb coupling would, which Sect. 4.8 already found for the quasiparticle lifetime of an electron gas. The cascade also inherits the split of energy between the electron and the hole that the dark-matter interaction made. A pair made at a momentum transfer of a few keV shares its energy between electron and hole differently from a pair made by a photon (Sect. 8.7), and at a few pairs that difference can be a whole ionization. With this chapter the book turns from the response of the target to the measurement: the chapters that follow take the deposit to the quanta a sensor counts, then to a limit and to a guide to the next targets (Fig. 18.1).

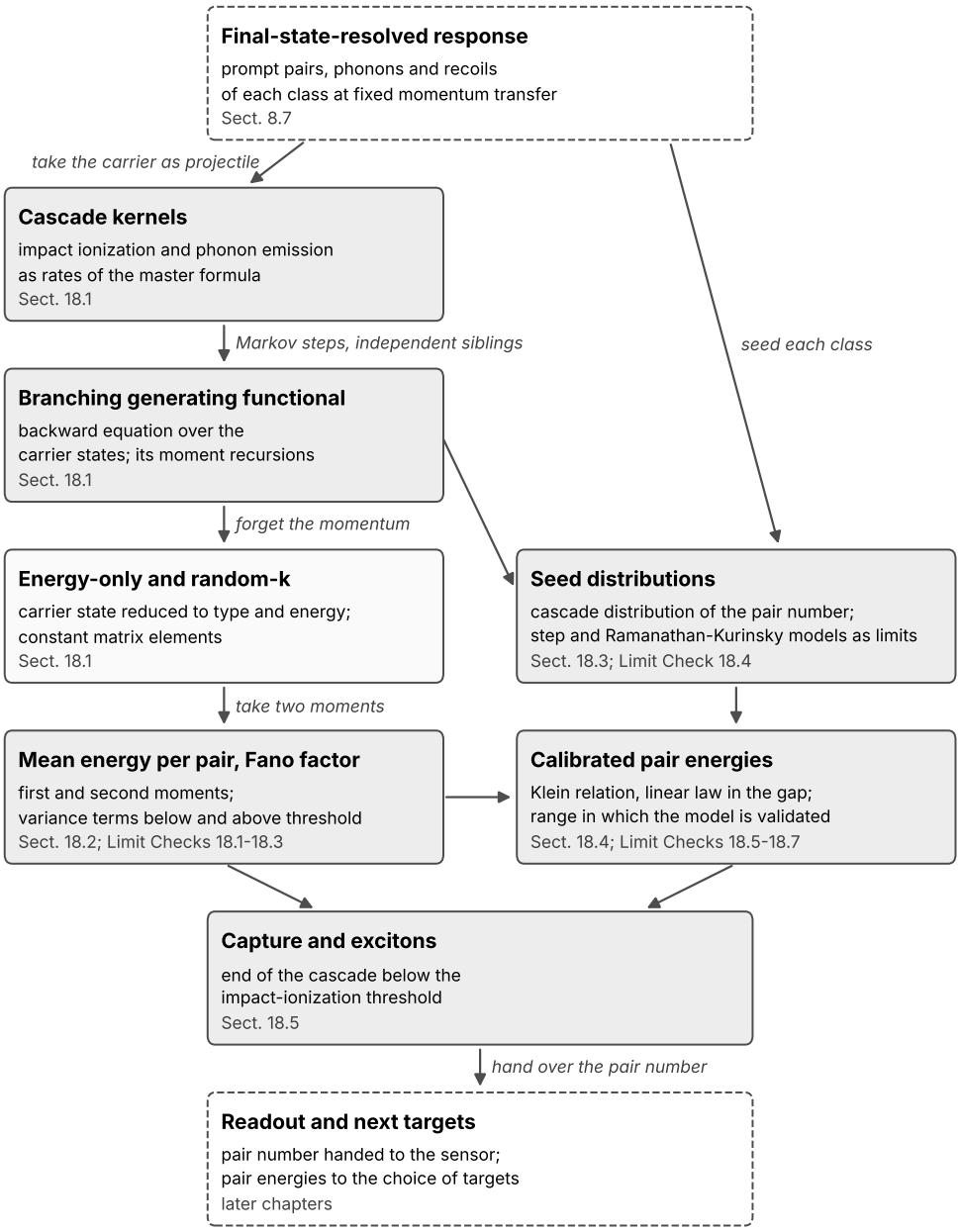


Figure 18.1: Ionization yield, from the final-state-resolved response of Sect. 8.7 to the pair number. Main line: impact ionization and phonon emission as rates of the master formula with the carrier as projectile, the branching generating functional of the cascade and its reduction to carrier energies alone (Sect. 18.1), then the mean energy per pair and the Fano factor as its first two moments (Sect. 18.2; Limit Checks 18.1 to 18.3). Side branches: the seeding of the cascade by each final-state class, with the step model and the Ramanathan–Kurinsky model as its limits (Sect. 18.3; Limit Check 18.4), and the calibrated pair energies of silicon, germanium, GaAs and the wide-gap crystals with the range in which the cascade model is validated (Sect. 18.4; Limit Checks 18.5 to 18.7). Both feed the end of the cascade below the impact-ionization threshold, capture and excitons (Sect. 18.5), which hands the pair number to the readout and the pair energies to the choice of new targets.

18.1 The carrier cascade

In the cascade model of silicon built in this section, an electron with 50 eV of kinetic energy makes about 13 new electron–hole pairs before every carrier has fallen below the threshold for impact ionization, and the remaining 35 eV goes to the lattice. Fig. 18.2 shows one such cascade. Each step in it is a collision of a hot carrier with the target: either it lifts a valence electron across the gap, or it emits a phonon. Both are scattering events of a projectile off the electrons or the nuclei of the crystal, and we will see that each is a rate of the master formula Eq. (8.15) with the carrier in the place of the dark-matter particle. The cascade is then a branching process whose kernels are those rates, and its statistics follow from one generating functional that obeys a recursion in energy. Two reductions make the recursion solvable: the energy-only reduction drops the momentum of each carrier, and the random-k reduction also replaces the matrix elements by constants. Both must keep the seed, the electron and hole that the dark matter left behind, because at a few pairs how they share the energy changes the count.

18.1.1 Impact ionization as a master-formula rate

A carrier in the Bloch state $n\boldsymbol{k}$, an electron in a conduction band or a hole in a valence band, interacts with every other charge of the crystal through the Coulomb potential. It is charged, so it couples to the total charge density n_Z of electrons and nuclei, as a dark photon does (Sect. 8.4). In a collision it gives a momentum $\boldsymbol{p}$ to the target and ends in a state n', $\boldsymbol{k}-\boldsymbol{p}$; we write $\boldsymbol{p}$ for the momentum transfer of the carrier and keep $\boldsymbol{q}$ for the dark-matter momentum transfer that made the seed. Because the carrier is a Bloch wave and not a plane wave, the momenta $\boldsymbol{p}+\boldsymbol{G}$ for all reciprocal-lattice vectors $\boldsymbol{G}$ lead to the same final carrier state, and their amplitudes add. With $\boldsymbol{p}$ in the first zone, the amplitude to reach n', $\boldsymbol{k}-\boldsymbol{p}$ and the target state $|f\rangle$ is

$$\frac{1}{\mathcal{V}}\sum_{\boldsymbol{G}} v_{\mathrm{C}}(|\boldsymbol{p}+\boldsymbol{G}|)\,\rho_{n'n}(\boldsymbol{k},-\boldsymbol{p}-\boldsymbol{G})\,\langle f|n_Z(\boldsymbol{p}+\boldsymbol{G})|i\rangle,$$

with the overlap $\rho_{n'n}$ of Eq. (3.34) in the place of the plane-wave overlap of a free projectile. This is the interaction Eq. (8.6) with one coupling row for each $\boldsymbol{G}$, the potential $\tilde{V}_{\boldsymbol{G}} = v_{\mathrm{C}}(|\boldsymbol{p}+\boldsymbol{G}|)\,\rho_{n'n}(\boldsymbol{k},-\boldsymbol{p}-\boldsymbol{G})$, and the target operators $n_Z(\boldsymbol{p}+\boldsymbol{G})$. The master formula Eq. (8.15) then yields the rate directly, with two changes. The sum over final projectile states runs over the bands n' and the first zone instead of over all momenta. And the final carrier state must be empty, which adds the factor $1-f_{n',\boldsymbol{k}-\boldsymbol{p}}$. The response matrix that appears is $S_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p},\omega)$ of the total charge, at the energy $\omega=\varepsilon_{n\boldsymbol{k}}-\varepsilon_{n',\boldsymbol{k}-\boldsymbol{p}}$ that the carrier loses.

The fluctuation–dissipation theorem turns this response into the screened interaction. With R the retarded density response of Sect. 8.7, at $\omega>0$ it reads $S=-2\,(1+n_{\mathrm{B}}(\omega))\,(R-R^{\dagger})/2\mathrm{i}$ as a matrix in $\boldsymbol{G}$, the matrix form of the relation used in Sect. 8.7, with $1/(1-\mathrm{e}^{-\beta\omega})=1+n_{\mathrm{B}}(\omega)$. Sandwiched between Coulomb kernels, $v_{\mathrm{C}}Rv_{\mathrm{C}}=W^{\mathrm{scr}}-v_{\mathrm{C}}$ by Eq. (4.99), and the bare v_{C} is real and diagonal and drops out of the anti-Hermitian part. With the dielectric matrix of electrons and nuclei together in W^{scr}:

Impact-ionization kernel

$$\Gamma_{\mathrm{ii}}(n\boldsymbol{k}) = -2\sum_{n'}\int_{\mathrm{BZ}}\frac{\mathrm{d}^3p}{(2\pi)^3}\sum_{\boldsymbol{G}\boldsymbol{G}'}\rho^{*}_{\boldsymbol{G}}\Big[\frac{W^{\mathrm{scr}}-W^{\mathrm{scr}\dagger}}{2\mathrm{i}}\Big]_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p},\omega)\;\rho_{\boldsymbol{G}'}\;\big[1-f_{n',\boldsymbol{k}-\boldsymbol{p}}\big]\big[1+n_{\mathrm{B}}(\omega)\big], \tag{18.1}$$

with $\rho_{\boldsymbol{G}}=\rho_{n'n}(\boldsymbol{k},-\boldsymbol{p}-\boldsymbol{G})$ and $\omega=\varepsilon_{n\boldsymbol{k}}-\varepsilon_{n',\boldsymbol{k}-\boldsymbol{p}}$. Without the off-diagonal elements

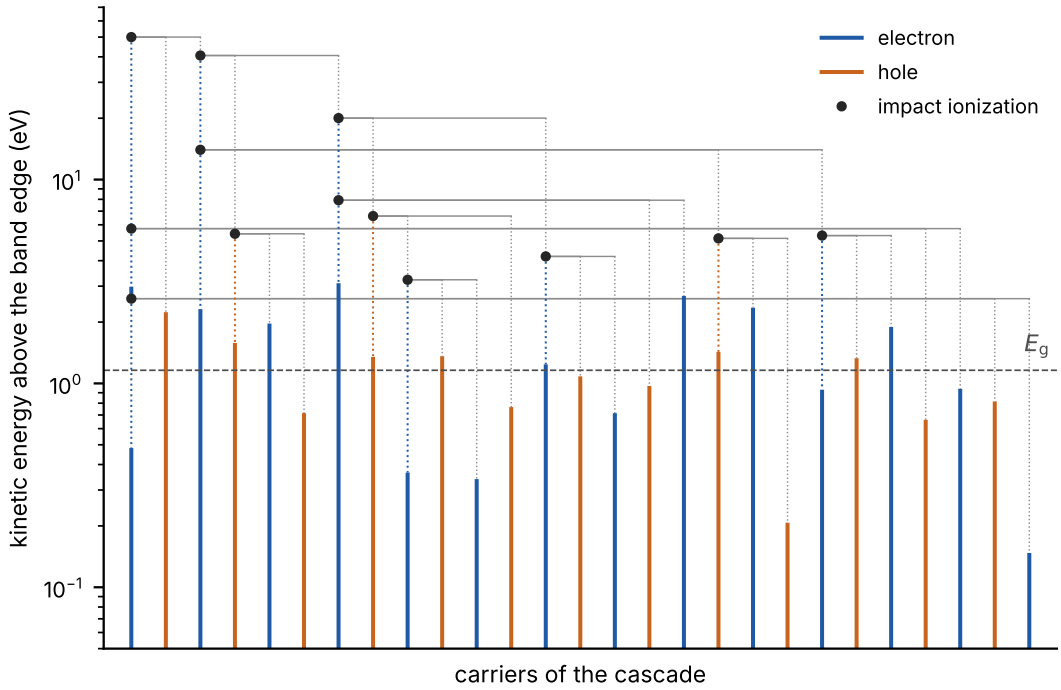


Figure 18.2: One sampled cascade started by an electron with 50 eV of kinetic energy in silicon, in the energy-only random-k model of Sect. 18.1.4: free-particle densities of states, constant matrix elements, impact ionization above $E_g = 1.160\,\text{eV}$ (silicon at 123 K; threshold ratio $\theta_{ii} = 1$), one optical phonon of 63 meV, the rate ratio Eq. (E.87) of the Alig–Bloom model with $A_{AB} = 5.703\,\text{eV}^3$ (calibrated in Sect. 18.4), and holes confined to the 12 eV valence band. Each column is one carrier (blue electrons, orange holes), drawn from the kinetic energy at which it was created down to the band edge, on a logarithmic axis. Solid segments are runs of phonon emission; a dot marks an impact ionization, after which the carrier continues at the lower energy reached along the dotted segment, and the grey lines lead to the two carriers it created. This cascade has 13 ionizations, 27 carriers and 176 phonon emissions, all by carriers below 6 eV, and leaves 34.9 eV as heat. On average the model makes 12.99 new pairs, 3.85 eV per pair at this energy, with a variance of 0.119 times the mean (generating function of Sect. 18.3).

$\boldsymbol{G} \neq \boldsymbol{G}'$ the sum over $\boldsymbol{G}$ and the integral over the zone combine into one integral over all $\boldsymbol{p}$,

$$\Gamma_{\mathrm{ii}}(n\boldsymbol{k}) \simeq \sum_{n'} \int \frac{\mathrm{d}^3 p}{(2\pi)^3}\, 2v_{\mathrm{C}}(p)\, \bigl|\rho_{n'n}(\boldsymbol{k}, -\boldsymbol{p})\bigr|^2\, L(\boldsymbol{p}, \omega)\, \bigl[1 - f_{n', \boldsymbol{k}-\boldsymbol{p}}\bigr]\bigl[1 + n_{\mathrm{B}}(\omega)\bigr], \quad (18.2)$$

with L the loss function Eq. (4.45) of the total dielectric matrix at the unreduced momentum $\boldsymbol{p}$.

For a hole the occupations are those of holes, $1 - f \to f$ in the valence bands. For a free-electron gas, $\rho = 1$ and Eq. (18.2) is the out-scattering part of the lifetime Eq. (4.102): the step from Sect. 4.8 to here is the step from plane waves to Bloch waves and from the electron gas to a crystal with nuclei. The two terms of Eq. (18.1) with $\boldsymbol{G} \neq \boldsymbol{G}'$ are local fields, and their anti-Hermitian combination is the one the master formula delivers. It is real for any W^{scr}, since $(W^{\mathrm{scr}} - W^{\mathrm{scr}\dagger})/2\mathrm{i}$ is a Hermitian matrix and $\rho^\dagger A \rho$ is real for every Hermitian A. The element-wise imaginary part of $W^{\mathrm{scr}}_{\boldsymbol{G}\boldsymbol{G}'}$ agrees with it only when $W^{\mathrm{scr}}_{\boldsymbol{G}\boldsymbol{G}'} = W^{\mathrm{scr}}_{\boldsymbol{G}'\boldsymbol{G}}$, which holds in silicon and germanium with the origin at an inversion center but not in GaAs, the same distinction that separated the retarded and advanced pair weights of Sect. 8.7.

The kernel is the rate at which the carrier scatters out of $n\boldsymbol{k}$, not its quasiparticle width, and the difference has two parts. By the cutting rule of Sect. 4.8, $-2\,\mathrm{Im}\,\Sigma^{\mathrm{R}} = \Gamma_{\mathrm{out}} + \Gamma_{\mathrm{in}}$, and the in-scattering term carries the product $f\, n_{\mathrm{B}}$ of an occupied final state and a thermal excitation of energy $\omega \geq E_{\mathrm{g}}$, which vanishes for a hot carrier at $T \ll E_{\mathrm{g}}$. The width also carries the quasiparticle residue, $\Gamma_{\mathrm{qp}} = -2 z_{\mathrm{qp}}\, \mathrm{Im}\,\Sigma^{\mathrm{R}}$ by Eq. (4.83), with $z_{\mathrm{qp}} \approx 0.8$ for the hot carriers of semiconductors [1, Sect. II]. Eq. (18.1) is first order in W^{scr} and sets $z_{\mathrm{qp}} = 1$. The factor z_{qp} is a nearly uniform 20 percent on every impact-ionization rate, and Appendix E.13 shows how a calibrated model absorbs it. The exchange of the carrier with the electron it excites is a vertex correction, absent from Eq. (18.1) as it was from Eq. (4.102); it enters only beyond first order in W^{scr}, and Sect. 18.5 lists it among the open problems.

The plasmon is inside the kernel. In silicon ε_2 is still 0.224 at the plasmon, the zero of ε_1 (Worked example 4.10), so the weight of the plasmon peak is that of electron–hole pairs. A carrier that "emits a plasmon" and the plasmon that "decays into a pair" are one term of Eq. (18.1), and a cascade built on it must not add a separate plasmon channel. The plasmon becomes a particle of the cascade only where its pole is undamped and carries no pair weight, as in aluminum below the critical momentum q_{c} of Sect. 4.5; it then needs a decay kernel of its own.

The loss function vanishes for $\omega < E_{\mathrm{g}}$ at $T \ll E_{\mathrm{g}}$, so a carrier must give up at least the gap. Energy conservation alone would let it ionize as soon as its kinetic energy x above the band edge exceeds E_{g}, but the three final carriers must also carry its crystal momentum. Take parabolic bands with an electron mass m_{e}^* and a hole mass m_{h}^*. At the threshold energy, the lowest at which an electron can make a pair, all three products move with a common velocity $\boldsymbol{u}$, so that none of the energy is spent on motion in their center-of-mass frame. Momentum conservation gives $m_{\mathrm{e}}^* \boldsymbol{v}_0 = (2m_{\mathrm{e}}^* + m_{\mathrm{h}}^*)\boldsymbol{u}$ for an initial velocity $\boldsymbol{v}_0$, energy conservation $\frac{1}{2} m_{\mathrm{e}}^* v_0^2 = E_{\mathrm{g}} + \frac{1}{2}(2m_{\mathrm{e}}^* + m_{\mathrm{h}}^*)u^2$, and eliminating $\boldsymbol{u}$,

$$x_{\mathrm{th}} = \theta_{\mathrm{ii}} E_{\mathrm{g}}, \qquad \theta_{\mathrm{ii}} = \frac{2m_{\mathrm{e}}^* + m_{\mathrm{h}}^*}{m_{\mathrm{e}}^* + m_{\mathrm{h}}^*}, \quad (18.3)$$

with θ_{ii} the impact-ionization threshold ratio. Equal masses give $\theta_{\mathrm{ii}} = 3/2$; for a hole the two masses exchange places. In a real band structure the threshold is soft rather than sharp, because a crystal conserves momentum only up to $\boldsymbol{G}$ and the bands are not parabolic. Just

above threshold few final states satisfy both conservation laws, and the computed rate scatters by large factors from one $\boldsymbol{k}$ to the next [1, Fig. 2].

18.1.2 The phonon-emission kernel

The carrier also couples to the displacements of the nuclei, through the vertex $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})$ of Eq. (6.11). Take the phonon coordinate of branch ν at $\boldsymbol{Q}$ as the target operator and g as the potential. The response of a harmonic phonon coordinate is $2\pi[(n_{\boldsymbol{Q}\nu}+1)\delta(\omega-\omega_{\boldsymbol{Q}\nu})+n_{\boldsymbol{Q}\nu}\delta(\omega+\omega_{\boldsymbol{Q}\nu})]$, and the master formula with the Pauli factor of the final carrier gives the out-scattering part of Eq. (6.27), the phonon kernel $\Gamma_{\mathrm{ph}}(n\boldsymbol{k})$ of Eq. (E.86). At detector temperatures only the emission term survives. The vertex is screened statically, as density-functional perturbation theory computes it, and the corrections are of higher order in $\omega_{\boldsymbol{Q}\nu}/E_{\mathrm{g}}$ (Sect. 6.2).

In a polar crystal the two kernels overlap. The total dielectric matrix in W^{scr} contains the ionic response, and at ω near the longitudinal optical frequency ω_L its loss function has the Fröhlich line: Eq. (18.1) already contains the emission of a longitudinal optical phonon through the macroscopic field. The long-range vertex $g^L = \mathrm{i}C_{\mathrm{F}}\rho_{mn}/Q$ of Eq. (6.15) is the same Coulomb line, seen from the phonon side. The phonon kernel of a polar crystal must therefore use the short-range vertex alone $g^S = g - g^L$, the split that Sect. 6.2 introduced for Wannier interpolation. The subtraction removes exactly the term that Eq. (18.1) contains only if the two calculations share the Born charges, ε_∞ and ω_L, the multipoles, the long-range cutoff, and the model of the ionic response itself. The Born charges, ε_∞ and ω_L come from one calculation. If g^L carries quadrupoles as well as dipoles, so must the ionic part of the dielectric matrix; otherwise only the dipole g^L is subtracted. When g^L is cut off over $\boldsymbol{G}$ by a Gaussian, the ionic head of W^{scr} at $\boldsymbol{p}=\boldsymbol{Q}+\boldsymbol{G}$ carries the same cutoff, or the Gaussian tail is counted twice. And the ionic part of W^{scr} must be restricted to the same model as g^L, its macroscopic multipole head with that cutoff: a full ionic dielectric matrix over all $\boldsymbol{G}$, as density-functional perturbation theory gives it, also contains the short-range coupling, which g^S would then count a second time. In silicon, with $Z^*=0$, there is no Fröhlich line, and the dipole term vanishes; the last condition still applies to the short-range ionic response if W^{scr} includes it.

The first phonon emission in a cascade has a special place. A pair of the zero-phonon sector $\mathcal{P}_0$ leaves the dark-matter interaction dressed, with the weight $z_{\mathrm{qp}}(m)$ of Sect. 8.6, and its first emission is the first step of the cascade, made by Γ_{ph}. A pair of the sector $\mathcal{P}_+$ arrives with its phonon already emitted, and that phonon's energy goes to heat once; the cascade does not emit it again (Sect. 8.7).

Cascade kernels

Every kernel of the hot-carrier cascade is a rate of the master formula Eq. (8.15) with a projectile from inside the target. Impact ionization has one coupling row per reciprocal-lattice vector, the potential $v_{\mathrm{C}}(|\boldsymbol{p}+\boldsymbol{G}|)\,\rho_{n'n}(\boldsymbol{k},-\boldsymbol{p}-\boldsymbol{G})$ and the response of the total charge, Eq. (18.1); phonon emission has the potential $g^S_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})$ and the one-phonon response. Both carry the Pauli factor of the final carrier state. The kernels hold at first order in W^{scr} and g, with $z_{\mathrm{qp}}=1$ and the carrier distinguishable from the electrons it excites.

The kernels for capture on an impurity and for the decay of a core hole by Auger emission have the same structure, with the impurity potential or the Coulomb interaction between bound and free states as the vertex; Sect. 18.5 treats capture.

18.1.3 The cascade as a branching process

A cascade particle is in a state x: its type (electron or hole; a plasmon, a core hole or a trapped carrier where these occur), its band and its crystal momentum. In a collision of channel c it turns into particles in the states $x'_1, \dots, x'_m$ at the rate density $\mathscr{K}_c(x \to x'_1 \dots x'_m)$, read off the kernels above. Impact ionization turns one carrier into three: the carrier itself, at lower energy, and the electron and hole it created. Phonon emission turns one carrier into one and passes $\omega_{\boldsymbol{Q}\nu}$ to the lattice. The total rate out of x is $\Gamma_x = \sum_c \int \mathscr{K}_c$. Each collision leaves an additive deposit $\delta\boldsymbol{N}_c$, a vector with one entry per quantity we count: one more pair for an ionization, $\omega_{\boldsymbol{Q}\nu}$ more heat for an emission. A carrier whose kinetic energy is below $\theta_{\mathrm{ii}} E_{\mathrm{g}}$ can no longer ionize. It relaxes to its band edge, and its kinetic energy is its last deposit of heat; capture and exciton formation, which change this last step, are the subject of Sect. 18.5.

The cascade is a branching process if collisions are Markovian, carriers evolve independently, and every line terminates; each of these can fail. The first is the Markov property: a collision is short compared with the time between collisions, and no coherence survives from one collision to the next. This is the incoherent limit of Appendix E.13, valid when the total rate of a carrier is small compared with its energy above the band edge, and least reliable within a width of the threshold. The second is that the carriers of one cascade evolve independently, with no Pauli blocking and no Coulomb interaction among them. Pauli blocking needs final-state occupations near one, in the states the hot carriers scatter into, a few eV above the band edges. With the conduction masses of Table 3.1, six valleys give a density-of-states mass of $1.06\, m_e$, and parabolic bands hold 7.5×10^{21} states per cm^3 per eV at 1 eV above the edge. If the pairs of a deposit share a sphere of radius 10 nm, an assumption of this estimate, a 1 keV deposit makes 267 pairs at 6.4×10^{19} per cm^3, which fills less than 10^{-2} of the states within an eV. Blocking is negligible. The Coulomb interaction is not, at the same density: with the exciton radius $a_x = 4.2$ nm of silicon (Sect. 4.9), the pair density times a_x^3 is 4.7 at 1 keV and 0.47 at 100 eV. As a result, the hypothesis is marginal for the deposits of tens of eV that light dark matter makes, and for keV deposits it is an assumption whose error has not been estimated. The third hypothesis is termination: every particle reaches a terminal state with probability one. It holds because every collision lowers the energy of the colliding carrier and its products by at least the smallest phonon energy or the gap, at temperatures where absorption of phonons is negligible.

Under these hypotheses the statistics of the whole cascade follow from one function. Let $\boldsymbol{N}$ be the total deposit of the cascade started by a particle in state x, with entries such as the pair number N_{eh} and the heat E_{heat}, and let $\boldsymbol{\lambda}$ be a vector of conjugate variables of the same length. The *branching generating functional* $\mathfrak{G}_x(\boldsymbol{\lambda}) = \langle \mathrm{e}^{\boldsymbol{\lambda}\cdot\boldsymbol{N}} \,|\, x \rangle$ obeys

Backward equation of the cascade

$$\Gamma_x \, \mathfrak{G}_x(\boldsymbol{\lambda}) = \sum_c \int \mathrm{d}x'_1 \cdots \mathrm{d}x'_m \, \mathscr{K}_c(x \to x'_1 \dots x'_m) \, \mathrm{e}^{\boldsymbol{\lambda}\cdot\delta\boldsymbol{N}_c} \prod_{j=1}^{m} \mathfrak{G}_{x'_j}(\boldsymbol{\lambda}), \tag{18.4}$$

with $\mathfrak{G}_x = \mathrm{e}^{\boldsymbol{\lambda}\cdot\boldsymbol{N}_{\mathrm{term}}(x)}$ on the terminal states, whose deposit $\boldsymbol{N}_{\mathrm{term}}(x)$ is fixed. The physical solution is the minimal one, obtained by iterating from the terminal states.

Appendix D.43 derives it by first-step analysis: condition on the first collision, and the products start independent copies of the same process. It is the type-and-deposit functional Eq. (C.102) of a branching process, with the branching ratios $\int \mathscr{K}_c / \Gamma_x$ in place of the probabilities p_c, written in the exponential form that suits deposits that are not integers. Time does not appear. The ratios of rates enter, not the rates, since the cascade at a given energy depends on which collision comes first, not on when; the absolute rates matter for the Markov hypothesis

and for the timing of the signal. The minimality condition rules out the spurious fixed points that Appendix C.11 met in the toy branching law $1-p+ps^2$, and here it holds automatically, because every channel lowers the energy. The moments obey linear equations, Eqs. (C.103) and (C.104) with the same substitution, and every term on their right refers to products of lower energy than x, so a single sweep upward in energy solves them.

Each collision conserves energy, and that ties the entries of $\boldsymbol{N}$ together. Without capture, excitons and escaping photons, a deposit ω that starts with one pair ends, in every event, as

$$\omega = N_{\mathrm{eh}} E_{\mathrm{g}} + E_{\mathrm{heat}}, \tag{18.5}$$

since each pair holds E_{g} once its carriers sit at the band edges and everything else has gone to phonons. For a carrier injected alone, with kinetic energy x, the same holds with $\omega \to x$ and N_{eh} counting the new pairs. Since this holds on every path, the joint functional of pairs and heat is fixed by that of the pairs alone, $\mathfrak{G}(\lambda_N, \lambda_H) = \mathrm{e}^{\lambda_H \omega}\, \mathfrak{G}(\lambda_N - \lambda_H E_{\mathrm{g}}, 0)$, and every statement about the fluctuations of N_{eh} is a statement about those of the heat. Sect. 18.2 builds $\varepsilon_{\mathrm{eh}}$ and F_{eh} on this identity.

Worked example 18.1: The generating function of a two-step toy cascade

Measure energies in units of E_{g}, take the threshold $\theta_{\mathrm{ii}} = 3/2$, and let the only phonon carry $1/2$, far more than a real phonon, so that the cascade has at most two steps. A carrier at $x = 2$ ionizes with probability p_a; the three carriers then share $x - 1 = 1$, all are below threshold, and the cascade ends. Otherwise, with probability $1 - p_a$, it emits a phonon and sits at $x = 3/2$. There it ionizes with probability p_b, the products sharing $1/2$, or emits a second phonon and stops at $x = 1$, below threshold. Let $s = \mathrm{e}^{\lambda_N}$ count new pairs. All products of the ionization at $3/2$ are terminal, with $\mathfrak{G} = 1$, so Eq. (18.4) read from the bottom up gives $\mathfrak{G}_{3/2}(s) = p_b s + 1 - p_b$, and then

$$\mathfrak{G}_2(s) = p_a\, s + (1 - p_a)\, \mathfrak{G}_{3/2}(s) = p_a\, s + (1 - p_a)(p_b\, s + 1 - p_b).$$

The heat is $2 - N_{\mathrm{eh}}$ on every path: an ionization at 2 leaves heat 1, at $3/2$ heat $1/2+1/2$, and two emissions heat $1/2+1/2+1$. Multiplying each path by $\mathrm{e}^{\lambda_H E_{\mathrm{heat}}}$ gives $\mathrm{e}^{2\lambda_H}\, \mathfrak{G}_2(s\, \mathrm{e}^{-\lambda_H})$, the energy identity of the text.

Now seed the cascade with one pair whose electron and hole both start at $x = 2$ and follow the same rules. The seed pair counts once, and the two carriers are independent, so

$$\mathfrak{G}_\Lambda(s) = s\, \mathfrak{G}_2(s)^2.$$

With $p_a = 3/5$ and $p_b = 2/5$, one carrier makes $\mathfrak{G}_2'(1) = 19/25$ new pairs on average. Expanding $\mathfrak{G}_\Lambda$ in powers of s yields the probabilities of 1, 2 and 3 pairs, $36/625$, $228/625$ and $361/625$. The mean is $63/25 = 2.52$ and the variance $228/625 = 0.365$, so the deposit $\omega = 1 + 2 + 2 = 5$ costs $5/2.52 = 1.98$ gap energies per pair and the variance is 0.145 of the mean. The product form is what independence buys: the generating function of the pair is the product of those of its carriers, and a cascade of any depth is built the same way, one level at a time from the bottom.

18.1.4 Energy-only and random-k reductions

The backward equation Eq. (18.4) runs over every band and crystal momentum of every carrier, which is more than a calculation can tabulate. Its first reduction removes the momentum. Suppose that momentum-randomizing collisions, the scattering by intervalley

and acoustic phonons, are much faster than the collisions that change the energy, $\Gamma_k \gg \Gamma_E$ with $\Gamma_E = \Gamma_{\text{ii}} + \Gamma_{\text{ph}}$ restricted to optical emission. These phonons carry a few tens of meV against the eV of a hot carrier, so they turn its momentum around while barely changing its energy. Between two energy-changing collisions a carrier then spreads uniformly over its surface of constant energy, and after its first collision its future depends only on its type t and kinetic energy x: $\mathfrak{G}_{n\boldsymbol{k}} \to \mathfrak{G}_t(x)$. The kernels become iso-energy averages,

$$\bar{\mathscr{K}}_t(x \to \dots) = \frac{\sum_{n\boldsymbol{k}} \delta(x - \varepsilon_{n\boldsymbol{k}})\, \mathscr{K}(n\boldsymbol{k} \to \dots)}{\sum_{n\boldsymbol{k}} \delta(x - \varepsilon_{n\boldsymbol{k}})}, \tag{18.6}$$

with $\varepsilon_{n\boldsymbol{k}}$ measured from the band edge of type t, and Eq. (18.4) becomes a recursion in one variable per type. The same reduction follows from a second hypothesis, which does not need fast randomization. It is that the products of each ionization are spread over their energy surfaces independently of the momentum of the parent, as they nearly are when the momentum transfer $\boldsymbol{p}$ in Eq. (18.1) ranges over many reciprocal-lattice vectors.

Neither hypothesis is safe in silicon. Above 3 to 5 eV, impact ionization is as fast as phonon emission, and in the model of Fig. 18.2 the two are equally likely at 5.9 eV; there Γ_E is no longer small against Γ_k, at exactly the energies where the number of ionizations is decided. The second hypothesis has not been proved. We expect an error of roughly Γ_E/Γ_k per generation. This is an estimate, not a bound; Appendix E.24 sets out the calculation that would give one. What the reduction does fix is where the momentum memory sits. The first generation, the seed carriers and the first products they make, keeps the crystal momenta it was born with, and through them the dark-matter momentum transfer $\boldsymbol{q}$ that made the seed. Every later generation sees energies alone.

A full-band calculation of the cascade in silicon, with the kernels of Sects. 18.1.1 and 18.1.2 on empirical pseudopotential bands, the deformation-potential phonons of Lundstrom's textbook [2], as quoted by Li et al. [3, Table III] (who credit the hole acoustic value also to Jacoboni and Reggiani [4]), and no fitted constant (Sect. 18.3.3), shows how the reductions compare with a calculation that keeps band structure and momentum. The exchange factor of Eq. (18.1), the ratio of the full Kane rate to its direct part, comes out 0.66 to 0.84 for electrons and 0.51 to 0.68 for holes above 3 eV, inside the range of 0.5 to 1 that Kotani and van Schilfgaarde give [1, Eq. (5)]. The pair energy is decided by the competition of impact ionization and phonon emission between about 2.5 and 4.5 eV: tripling the ionization rate below 4.5 eV lowers ε_{eh} from 4.15 to 3.73 eV, while doubling it above 5 eV leaves it unchanged, since almost every carrier there ionizes either way. Near threshold phonon emission decides the count: at 4 eV fewer than 1 percent of the events have two or more pairs, while without phonon emission nearly all would. And the momentum memory is negligible within the model: the full-band cascade and the same cascade with every carrier redrawn on its energy surface after each collision give nearly the same distributions. That comparison is internal to the deformation-potential model, in which every optical or intervalley emission redraws the crystal momentum anyway; it tests only the memory of the first collision.

The second reduction replaces the matrix elements by constants and drops momentum conservation, the random-k approximation. Take the golden-rule form of impact ionization, a sum over the two final electrons and the hole with a squared screened matrix element, as written by Kane and quoted by Kotani and van Schilfgaarde [1, Eq. (5)]. With a constant matrix element and with every final momentum allowed, the sum is the three-body density of final states. Let $\rho_t(x)$ be the density of states of a carrier of type t at kinetic energy x and $\bar{t}$ the other type. A carrier of type t at x ionizes at a rate proportional to

$$\mathfrak{P}_t(y) = \int_0^\infty \mathrm{d}x_1\, \mathrm{d}x_2\, \mathrm{d}x_3\; \rho_t(x_1)\, \rho_t(x_2)\, \rho_{\bar{t}}(x_3)\; \delta(y - x_1 - x_2 - x_3), \qquad y = x - E_{\text{g}}, \tag{18.7}$$

and its three products have the joint density $\rho_t(x_1)\rho_t(x_2)\rho_{\bar{t}}(x_3)/\mathfrak{P}_t(y)$ on the plane $x_1 + x_2 + x_3 = y$. The phonon rate is proportional to the density of states after one emission, $\rho_t(x - \omega_{\mathrm{AB}})$, for a single optical phonon of energy ω_{AB}. The ratio of the two rates is

$$\frac{\Gamma_{\mathrm{ph}}(x)}{\Gamma_{\mathrm{ii}}(x)} = A_{\mathrm{AB}}\,\frac{\rho_t(x - \omega_{\mathrm{AB}})}{\mathfrak{P}_t(x - E_{\mathrm{g}})}, \tag{18.8}$$

with one constant A_{AB} that collects the two squared matrix elements. The threshold is now $\theta_{\mathrm{ii}} = 1$, since momentum no longer constrains the products. For free-particle densities of states, $\rho_t(x) = x^{1/2}$, the integral is the Dirichlet integral $\int \prod_j x_j^{a-1}\,\delta(y - \sum_j x_j) = \Gamma(a)^3 y^{3a-1}/\Gamma(3a)$ at $a = 3/2$. Since $\Gamma(3/2) = \sqrt{\pi}/2$ and $\Gamma(9/2) = 105\sqrt{\pi}/16$,

$$\mathfrak{P}(y) = \frac{\Gamma(3/2)^3}{\Gamma(9/2)}\,y^{7/2} = \frac{2\pi}{105}\,y^{7/2},$$

and Eq. (18.8) becomes the ratio Eq. (E.87) of the model of Alig, Bloom and Struck, the Alig–Bloom model, as Ramanathan and Kurinsky implement it [5, Eq. (8)]. With ρ_t in $\mathrm{eV}^{1/2}$ and $\mathfrak{P}$ in $\mathrm{eV}^{7/2}$, A_{AB} carries eV^3. Ramanathan and Kurinsky keep each hole below the 12 eV width of the silicon valence band by capping its energy and re-splitting the excess among the other products [5, Sect. III A]. We use a variant that cuts the hole density of states at the band width, so that a split with a hole above it never occurs, and fix A_{AB} by the measured pair energy of silicon at 123 K, where $E_{\mathrm{g}} = 1.160\,\mathrm{eV}$; Sect. 18.4 derives the value, $A_{\mathrm{AB}} = 5.703\,\mathrm{eV}^3$.

Fig. 18.2 is drawn with this model, its products sampled from the free-particle density, the scaled Dirichlet distribution with all three parameters 3/2, with a hole drawn above 12 eV rejected. The rate ratio falls steeply with energy, and the cascade is decided by the competition at a few eV. Every phonon of the cascade in the figure is emitted by a carrier below 6 eV; above that energy the carriers ionize.

18.1.5 Seeding the cascade

A dark-matter interaction does not inject one carrier: it leaves a final state of a class Λ (Sect. 8.7), typically an electron at $(E_{\mathrm{e}}, \boldsymbol{k}_{\mathrm{e}})$ and a hole at $(E_{\mathrm{h}}, \boldsymbol{k}_{\mathrm{h}})$, plus phonons or a core hole in other sectors. By the independence hypothesis the generating functional of the cascade it starts is the product over its particles,

$$\mathfrak{G}_\Lambda(\boldsymbol{\lambda}) = \mathrm{e}^{\lambda_N}\,\mathfrak{G}_{\mathrm{e}}(E_{\mathrm{e}}, \boldsymbol{k}_{\mathrm{e}}; \boldsymbol{\lambda})\,\mathfrak{G}_{\mathrm{h}}(E_{\mathrm{h}}, \boldsymbol{k}_{\mathrm{h}}; \boldsymbol{\lambda}), \tag{18.9}$$

for a pair, with the seed pair counted by e^{λ_N}, as in Worked example 18.1. The mean pair number of the Ramanathan–Kurinsky model, one plus the mean yield of each carrier averaged over the split of the energy [5, Eq. (5)], is the first moment of Eq. (18.9) at $\boldsymbol{\lambda} = 0$. Sect. 18.3 builds $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ from it.

Three properties of Eq. (18.9) make the operator structure of the final state matter. First, at a fixed deposit ω the product depends on how $\omega - E_{\mathrm{g}}$ is split between the electron and the hole, and in the first generation on their crystal momenta. Second, $\mathfrak{G}_t$ is not linear in the carrier energy, so a cascade started from the average split is not the average of the cascades started from the actual splits. In the model of Fig. 18.2, a 10 eV deposit whose 8.84 eV above the gap is split equally yields 2.985 pairs on average and two pairs in 4.5 % of events. A uniform distribution of splits, whose mean is the equal split, yields 2.691 pairs, and two pairs in 36 % of events. Giving everything to the electron yields 3.003, with two pairs in 14 %. At a few pairs the distribution of the split, not only its mean, sets the count. Third, that distribution depends on the operator that made the pair and on $\boldsymbol{q}$: at 10 eV in silicon the

hole carries about 0.9 eV more for a dark-matter kick at q_0 than for an absorbed photon (Worked example 8.9). The cascade has to be seeded class by class from the resolved response $S_{ab}(\boldsymbol{q}, \omega; \Lambda)$, with the coupling tensor contracted inside the sum over classes, as the hand-off Eq. (8.69) requires. A seed distribution fitted to photon data at the same ω replaces the splits of one operator and one momentum transfer by those of another.

The kernels are computed from first principles with the methods of Appendix E.13: impact ionization from the *GW* screened interaction on quasiparticle bands, with local fields, and phonon emission from vertices of density-functional perturbation theory interpolated with Wannier functions. Appendix E.24 specifies the full cascade for silicon, germanium and GaAs, with the first generation resolved in crystal momentum and the later ones reduced to energies, and the calculation of the momentum memory between collisions.

18.2 Mean energy per pair and the Fano factor

A 5.9 keV X-ray absorbed in silicon at 123 K leaves on average 1574 electron–hole pairs. If each pair were an independent event, the count would scatter by $\sqrt{1574} = 39.7$ from one X-ray to the next. The measured spread is 13.7, a Fano factor $F_{\mathrm{eh}} = 0.119$ at a mean energy per pair $\varepsilon_{\mathrm{eh}} = 3.749\,\mathrm{eV}$ (Table 18.1). These two numbers fix where the bins of one, two and three pairs sit on the axis of deposited energy, and how sharp they are. The cascade of Sect. 18.1 produces both. A pair that costs more heat leaves less energy for the pairs that follow, and this constraint, not the randomness of each collision, makes the count sub-Poissonian. We will derive $\varepsilon_{\mathrm{eh}}$ and F_{eh} as the first two moments of the cascade and split F_{eh} into three terms that the energy-only cascade computes exactly. An exactly solvable limit, Rényi's random parking, checks the construction, and the final-state estimate of Subashiev and Luryi falls below the full value in every case.

18.2.1 Moments of the heat

Every event of the cascade obeys the energy identity Eq. (18.5), $\omega = N_{\mathrm{eh}} E_{\mathrm{g}} + E_{\mathrm{heat}}$, as long as no carrier is captured, no exciton forms and no photon leaves the crystal before the pairs are counted; Sect. 18.5 removes each of these conditions. At fixed ω the pair number and the heat are therefore one random variable, not two. Taking the mean and the variance of the identity at fixed ω, with $\bar{n} = \langle N_{\mathrm{eh}} \rangle$ the mean pair number,

$$\varepsilon_{\mathrm{eh}} = \frac{\omega}{\bar{n}} = E_{\mathrm{g}} + \frac{\langle E_{\mathrm{heat}} \rangle}{\bar{n}}, \qquad F_{\mathrm{eh}} = \frac{\mathrm{Var}\, N_{\mathrm{eh}}}{\bar{n}} = \frac{\mathrm{Var}\, E_{\mathrm{heat}}}{E_{\mathrm{g}}^2\, \bar{n}}. \tag{18.10}$$

Both ratios depend on ω for small deposits. The symbols $\varepsilon_{\mathrm{eh}}$ and F_{eh} denote their limits at large ω, where $\bar{n}$ and $\mathrm{Var}\, N_{\mathrm{eh}}$ grow in proportion to ω (Sect. 18.2.5); Appendix D.43 derives these relations from the generating functional, Eq. (D.242). Note that F_{eh} has nothing to do with the Fano coefficients c_ν^{F} of Eq. (8.62), which describe the interference of a discrete state with a continuum; the two carry the name of the same physicist.

The cascade makes its heat in two ways. Let $E_<$ be the kinetic energy that each carrier holds when it falls below the threshold $\theta_{\mathrm{ii}} E_{\mathrm{g}}$ for good, summed over carriers. That carrier then relaxes to its band edge by emitting phonons, but the amount is fixed at the moment it becomes terminal. Let $E_>$ be the energy of the phonons emitted while carriers are still above threshold. With $E_{\mathrm{heat}} = E_< + E_>$, expanding the variance in Eq. (18.10) gives

Fano factor as cascade moments

$$F_{\mathrm{eh}} = \mathcal{V}_< + \mathcal{V}_> + 2\mathcal{V}_{<>}, \qquad \mathcal{V}_< = \frac{\operatorname{Var} E_<}{E_{\mathrm{g}}^2\,\bar{n}}, \quad \mathcal{V}_> = \frac{\operatorname{Var} E_>}{E_{\mathrm{g}}^2\,\bar{n}}, \quad \mathcal{V}_{<>} = \frac{\operatorname{Cov}(E_<, E_>)}{E_{\mathrm{g}}^2\,\bar{n}}, \tag{18.11}$$

together with $\varepsilon_{\mathrm{eh}} = \omega/\bar{n}$, in the limit of large ω. Each term is a variance or covariance of an additive deposit of the cascade, computed by the moment recursions of Appendix D.43, where the decomposition is Eq. (D.243).

The decomposition itself is an identity. Its content is that every term can be computed. A deposit is additive if each collision adds a fixed amount to it: one unit of N_{eh} per ionization, ω_{AB} to $E_>$ per phonon emitted above threshold, and x to $E_<$ when a carrier at kinetic energy x becomes terminal. For such a deposit the first and second moments obey Eqs. (D.240) and (D.241), the energy-only form of the recursions Eqs. (C.103) and (C.104). We condition on the first collision of a carrier, and the three products of an ionization start independent subtrees whose moments are already known at lower energy. The second moment $M_t(x)$ of a deposit then collects four terms after an ionization, written out in Eq. (D.241): the square of the deposit of the collision, twice its product with the mean deposit of the three subtrees, the sum of their second moments, and twice the sum of the products of their means taken in pairs. Each term is a convolution of known moments with the densities of states of the products, divided by $\mathfrak{P}_t$. After a phonon emission the carrier simply continues at $x - \omega_{\mathrm{AB}}$. The variances in Eq. (18.11) are $M_t - m_t^2$ for the deposits $E_<$ and $E_>$, and the covariance follows from the second moment of $E_< - E_>$, a route independent of the two variances.

The model is the random-k cascade of Sect. 18.1.4. Both carriers have free-particle densities of states $\rho_t(x) \propto x^{1/2}$, and the hole density of states stops at the valence band width B_{h}. The threshold ratio is $\theta_{\mathrm{ii}} = 1$, and one optical phonon of energy ω_{AB} competes with ionization through the rate ratio Eq. (18.8). Measured in units of the gap, the cascade then depends on the crystal through two numbers, $\lambda_{\mathrm{AB}} = A_{\mathrm{AB}}/E_{\mathrm{g}}^3$ and $w_{\mathrm{AB}} = \omega_{\mathrm{AB}}/E_{\mathrm{g}}$. The first sets the strength of phonon emission relative to impact ionization: one gap above threshold, at $x = 2E_{\mathrm{g}}$, the rate ratio Eq. (18.8) is $(105/2\pi)(2 - w_{\mathrm{AB}})^{1/2}\lambda_{\mathrm{AB}} = 23.3\,\lambda_{\mathrm{AB}}$. The second is the energy lost in one phonon step. For silicon at 123 K, $E_{\mathrm{g}} = 1.160\,\mathrm{eV}$ [5, Eq. (11)], $\omega_{\mathrm{AB}} = 63\,\mathrm{meV}$ and $B_{\mathrm{h}} = 12\,\mathrm{eV}$ (Appendix E.13). With the one constant that Sect. 18.4 fixes by the measured pair energy, $A_{\mathrm{AB}} = 5.703\,\mathrm{eV}^3$, silicon has $\lambda_{\mathrm{AB}} = 3.654$ and $w_{\mathrm{AB}} = 0.0543$. The seed is an electron with all of $\omega - E_{\mathrm{g}}$, its hole at the band edge. The slopes of $\bar{n}$ and $\operatorname{Var} N_{\mathrm{eh}}$ in ω, fitted for seed energies between $20E_{\mathrm{g}}$ and $40E_{\mathrm{g}}$, give $\varepsilon_{\mathrm{eh}} = 3.232\,E_{\mathrm{g}} = 3.749\,\mathrm{eV}$ and $F_{\mathrm{eh}} = 0.114$. In practice the recursions are solved on grids of step $E_{\mathrm{g}}/74$ and $E_{\mathrm{g}}/148$ and extrapolated to zero step (Appendix D.43). A first-principles cascade would replace both kernels by those of Sect. 18.1; Appendix E.24 sets it up so that the three terms of Eq. (18.11) come out separately.

For a single deposit the slopes are only the end point. The approach to them is set by the constant terms of the mean and the variance, $\bar{n} = \omega/\varepsilon_{\mathrm{eh}} + b_N + o(1)$ and $\operatorname{Var} N_{\mathrm{eh}} = F_{\mathrm{eh}}\omega/\varepsilon_{\mathrm{eh}} + c_N + o(1)$, whose existence Sect. 18.2.5 discusses. The constant b_N depends on the seed (Sect. 18.3.4), and for the one-sided seed of silicon $c_N = -0.02$. The pair energy at finite ω lies below its limit by the fraction $b_N\varepsilon_{\mathrm{eh}}/\omega$.

Limit Check 18.1. Pair energy and Fano factor of a finite deposit

The energy-dependent pair energy and Fano factor $\varepsilon(\omega) = \omega/\bar{n}(\omega)$ and $F(\omega) = \operatorname{Var} N_{\mathrm{eh}}(\omega)/\bar{n}(\omega)$ tend to constants at high energy [5, Eqs. (2)–(3)]; the constants

are the cascade moments $\varepsilon_{\mathrm{eh}}$ and F_{eh} of Eq. (18.11).

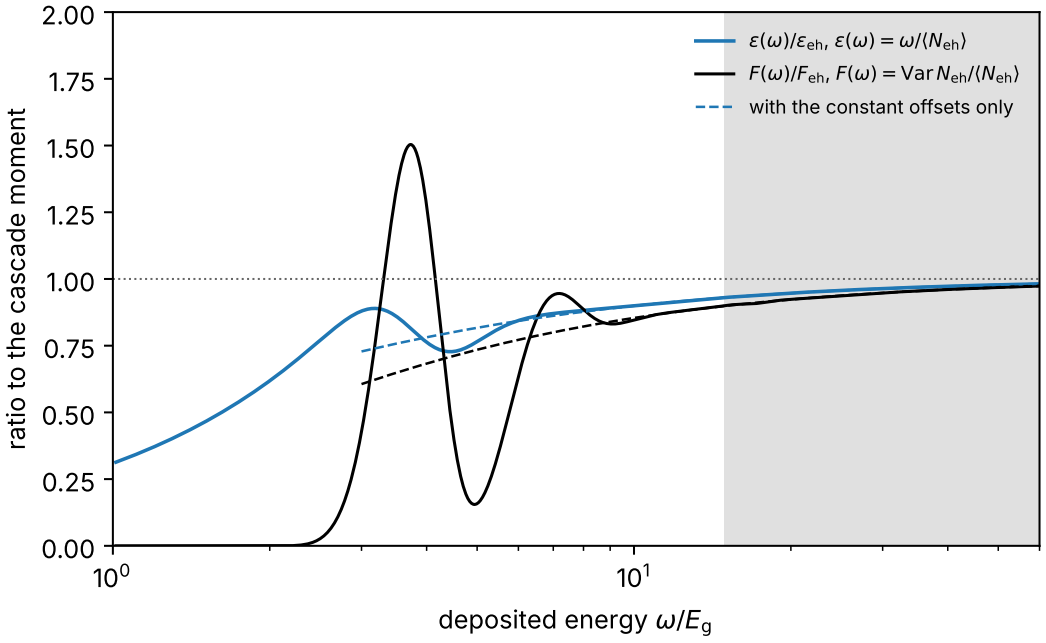


Figure 18.3: Pair energy (blue) and Fano factor (black) of a deposit ω in silicon, in units of their high-energy limits $\varepsilon_{\mathrm{eh}} = 3.232\,E_{\mathrm{g}}$ and $F_{\mathrm{eh}} = 0.114$, against ω/E_{g}. Energy-only random-k cascade with $E_{\mathrm{g}} = 1.160\,\mathrm{eV}$, $\omega_{\mathrm{AB}} = 63\,\mathrm{meV}$, $A_{\mathrm{AB}} = 5.703\,\mathrm{eV}^3$, $\theta_{\mathrm{ii}} = 1$, hole band width 12 eV; seed electron with all of $\omega - E_{\mathrm{g}}$, hole at the band edge; moment recursions on a grid of step $E_{\mathrm{g}}/74$, the ratios taken on that grid. Dashed: the limits corrected by the constant terms only, $\varepsilon(\omega)/\varepsilon_{\mathrm{eh}} = (1 + b_N\varepsilon_{\mathrm{eh}}/\omega)^{-1}$ and $F(\omega)/F_{\mathrm{eh}} = (1 + c_N\varepsilon_{\mathrm{eh}}/F_{\mathrm{eh}}\omega)(1 + b_N\varepsilon_{\mathrm{eh}}/\omega)^{-1}$, with the constant terms of this seed fitted between 30 and $60E_{\mathrm{g}}$ (b_N of Sect. 18.3.4, $c_N = -0.02$; the slopes are fitted between 20 and $40E_{\mathrm{g}}$). Gray: $\omega \geq 14.9\,E_{\mathrm{g}} = 17.3\,\mathrm{eV}$, where both ratios lie within 10 % of 1.

Below about $15E_{\mathrm{g}}$ the yield rises in steps of one pair, and $F(\omega)$ swings between 0 and $1.5\,F_{\mathrm{eh}}$, so that a search at a few pairs needs the full distribution of the pair number (Sect. 18.3) rather than two moments.

At $\omega \leq 2E_{\mathrm{g}}$ the seed electron cannot ionize and $F(\omega) = 0$: one pair, always. The first ionizations then make the variance rise and fall as each new pair becomes possible. Above $10E_{\mathrm{g}}$ the constant terms describe the approach to within 1 %: at $\omega = 10E_{\mathrm{g}}$ the computed ratios differ from the dashed curves by 0.07 % in ε and 0.9 % in F.

18.2.2 The three variance terms in silicon

In silicon the three terms of Eq. (18.11) are $\mathcal{V}_< = 0.092$, $\mathcal{V}_> = 0.290$ and $2\mathcal{V}_{<>} = -0.268$, and they add to $F_{\mathrm{eh}} = 0.114$. The largest term is not the residual energy below threshold but the phonon energy above it. The mean phonon energy emitted above threshold is 0.92 eV per pair, about 15 phonons of 63 meV. If those phonons were independent emissions, their number would be Poisson, $\mathrm{Var}\,E_> = \omega_{\mathrm{AB}}\langle E_>\rangle$, and $\mathcal{V}_> = w_{\mathrm{AB}}\langle E_>\rangle/(E_{\mathrm{g}}\bar{n}) = 0.043$. The computed term is 6.7 times larger. The reason is the steep energy dependence of Eq. (18.8). Within a few phonon steps of threshold the probability that the next collision ionizes is below 10^{-4}; it

is 0.004 at 2 eV and reaches a few percent only at 2.5 to 3 eV (Appendix E.13). As a result, a carrier in this range emits a run of phonons whose length has a long, nearly geometric tail. Two carriers that start at the same energy can end their runs many phonons apart.

Most of this variance is canceled by the covariance, which removes 92 % of $\mathcal{V}_>$. Energy conservation along each carrier line ties the two heats together. A carrier that emits more phonons just above threshold either falls below threshold with a smaller residual, or ionizes later, at a lower energy, into smaller products. What a long run of phonons adds to $E_>$, it takes from $E_<$. The net Fano factor is therefore a small difference of two large terms, which has a practical consequence for any calculation: an error of 1 % in $\mathcal{V}_>$ alone moves F_{eh} by 2.5 %. The covariance is the correlation between the two kinds of heat along one line of descent, not between sibling carriers.

Fig. 18.4 follows the three terms as phonon emission grows stronger, at the phonon step of silicon. Without phonons, $E_> = 0$ and $F_{\text{eh}} = \mathcal{V}_<$ is pure partition noise: the scatter of the residual energies with which the carriers end, 0.081 for parabolic bands at $\theta_{\text{ii}} = 1$. At $\lambda_{\text{AB}} = 0.003$ the phonon term is still small, $\mathcal{V}_> = 0.009$, and the covariance is positive, $+0.010$. From $\lambda_{\text{AB}} \approx 0.1$ on, both large terms grow, until at $\lambda_{\text{AB}} = 30$ they reach $\mathcal{V}_> = 0.55$ and $2\mathcal{V}_{<>} = -0.61$. Across these four decades F_{eh} itself rises only from 0.085 to 0.132, and for λ_{AB} between 1 and 10 it stays between 0.105 and 0.122. In this model a Fano factor near 0.11 is generic for a semiconductor. A full-band cascade of silicon with no fitted constant gives the same 0.114 (Sect. 18.4.3). The measured value tests the shape of the rate ratio within a few phonon energies of threshold, where the two large terms are made, more than it tests the constant A_{AB} (Sect. 18.4).

18.2.3 The random-parking limit

Rényi's random-parking problem is the one limit of the cascade with an exact solution. Take a constant electron density of states, a valence band so narrow that every hole is born at the band edge, $\theta_{\text{ii}} = 1$, and no phonons. An electron of kinetic energy $x \geq E_{\text{g}}$ then ionizes into a hole at rest and two electrons that share $x - E_{\text{g}}$ with uniform probability, since the joint density of states of Eq. (18.7) is flat on the line $x_1 + x_2 = x - E_{\text{g}}$. In units of E_{g} this is the random parking problem of Rényi [6]. A car of unit length parks at a uniformly random place in a free gap of length x and leaves two gaps, which are filled in the same way until every gap is shorter than a car. The jamming density ρ_∞, the number of cars per unit length at the end, is the number of pairs per unit energy, so that $\varepsilon_{\text{eh}} = E_{\text{g}}/\rho_\infty$. Subashiev and Luryi proposed this limit as a test of Fano-factor models, since both moments are known exactly [6, Sect. I]. Worked example 18.2 derives them from the recursions of Sect. 18.2.1.

Worked example 18.2: The random-parking Fano factor

Measure energies in units of E_{g}. Let $M(x)$ be the mean number of new pairs made by an electron of kinetic energy x. The hole of each ionization is terminal with zero residual, and the two electrons share $x - 1$ uniformly, so Eq. (C.103) with one pair per ionization reads $M(x) = 0$ for $x < 1$ and

$$M(x) = 1 + \frac{2}{x-1} \int_0^{x-1} M(y)\,\mathrm{d}y, \qquad x \geq 1.$$

Put $x = z + 1$ and multiply by z: $zM(z+1) = z + 2\int_0^z M(y)\,\mathrm{d}y$. Take the Laplace transform $m(s) = \int_0^\infty \mathrm{e}^{-sx} M(x)\,\mathrm{d}x$. Since M vanishes below 1, the transform of $M(z+1)$

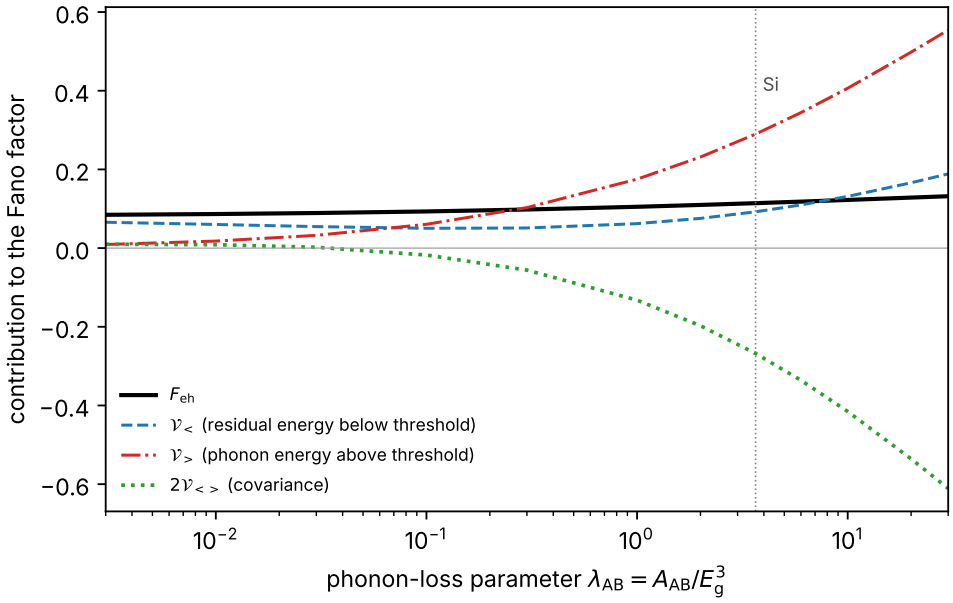


Figure 18.4: Fano factor F_{eh} (solid) and its three terms, Eq. (18.11), against the phonon-loss parameter $\lambda_{\mathrm{AB}} = A_{\mathrm{AB}}/E_{\mathrm{g}}^3$ at the phonon step of silicon, $w_{\mathrm{AB}} = \omega_{\mathrm{AB}}/E_{\mathrm{g}} = 0.0543$. Dashed: $\mathcal{V}_<$, from the residual energies of the carriers when they fall below threshold; dash-dotted: $\mathcal{V}_>$, from the phonon energy emitted above threshold; dotted: $2\mathcal{V}_{<>}$, their covariance. Energy-only random-k cascade with parabolic densities of states, $\theta_{\mathrm{ii}} = 1$, hole band width $12\,\mathrm{eV}/E_{\mathrm{g}}$; asymptotic slopes for seed energies between $X/2$ and X, $X = 40E_{\mathrm{g}}$ ($60E_{\mathrm{g}}$ for $\lambda_{\mathrm{AB}} > 5$); moment recursions on grids of step $E_{\mathrm{g}}/74$ and $E_{\mathrm{g}}/148$, extrapolated to zero step. The vertical line marks silicon, $\lambda_{\mathrm{AB}} = 3.654$, where the terms are $0.092 + 0.290 - 0.268 = 0.114$.

is $\mathrm{e}^{s}m(s)$, and multiplication by z becomes $-\mathrm{d}/\mathrm{d}s$. With $\phi(s)=\mathrm{e}^{s}m(s)$,

$$\phi'(s)+\frac{2\mathrm{e}^{-s}}{s}\,\phi(s)=-\frac{1}{s^{2}},\qquad \phi(s)=\int_{s}^{\infty}\frac{\mathrm{d}t}{t^{2}}\,\exp\bigl[2E_{1}(s)-2E_{1}(t)\bigr],$$

the solution that vanishes as $s\to\infty$, with $E_{1}(s)=\int_{s}^{\infty}\mathrm{e}^{-u}u^{-1}\mathrm{d}u$ the exponential integral. For every $s>0$, exactly, $E_{1}(s)=-\gamma_{\mathrm{E}}-\ln s+\mathrm{Ein}(s)$, with Euler's constant γ_{E} and $\mathrm{Ein}(s)=\int_{0}^{s}(1-\mathrm{e}^{-u})u^{-1}\mathrm{d}u\to 0$. Hence $\mathrm{e}^{2E_{1}(s)}\to\mathrm{e}^{-2\gamma_{\mathrm{E}}}/s^{2}$ as $s\to 0$, and inserting the same identity for $E_{1}(t)$ under the integral,

$$m(s)\to\frac{\rho_{\infty}}{s^{2}},\qquad \rho_{\infty}=\int_{0}^{\infty}\exp\bigl[-2\,\mathrm{Ein}(t)\bigr]\,\mathrm{d}t=0.74759792.$$

A double pole ρ_{∞}/s^{2} is the transform of $\rho_{\infty}x$, so $M(x)\to\rho_{\infty}x$: the pair energy is $\varepsilon_{\mathrm{eh}}=E_{\mathrm{g}}/\rho_{\infty}=1.3376\,E_{\mathrm{g}}$. The second moment obeys Eq. (C.104) with the same kernel. Solved on grids of step 0.01 and 0.005 and extrapolated linearly in the step, it yields $F_{\mathrm{eh}}=0.051040$ and $1/\varepsilon_{\mathrm{eh}}=0.747599$, against the closed form $F_{\mathrm{park}}=0.0510387$ [6, Eq. (44)].

The renewal estimate of Sect. 18.2.4 needs only the final state. At jamming, the number of gaps of size g per unit length is [6, Eq. (19)]

$$G(g)=2\int_{0}^{\infty}t\,\exp\bigl[-gt-2\,\mathrm{Ein}(t)\bigr]\,\mathrm{d}t,\qquad 0\le g\le 1.$$

Each gap is the residual energy of one terminal electron, and there is one per pair. Integrating, $\int G\,\mathrm{d}g=\rho_{\infty}$ (one gap per car) and $\int(1+g)G\,\mathrm{d}g=1$ (cars and gaps fill the line), both to 30 digits. Each pair costs $1+g$, with $\langle g\rangle=0.337617=1/\rho_{\infty}-1$ and $\mathrm{Var}\,g=0.079437$, so that Eq. (18.12) gives

$$F_{\mathrm{ren}}=\frac{\mathrm{Var}\,g}{(1+\langle g\rangle)^{2}}=0.0443977=0.870\,F_{\mathrm{park}}.$$

The final-state statistics are exact here, and still the estimate misses 13 % of the Fano factor. (The values printed in Ref. [6], $\rho_{\infty}=0.7475987$ and 0.0439766, carry misprints in the seventh and third digits.)

Silicon is far from this limit in two ways. Its densities of states rise as $x^{1/2}$, and phonons compete with ionization. The density of states alone, at $\lambda_{\mathrm{AB}}\to 0$, raises $\varepsilon_{\mathrm{eh}}$ from $1.338\,E_{\mathrm{g}}$ to $1.585\,E_{\mathrm{g}}$ and F_{eh} from 0.051 to 0.081. Limit Check 18.2 keeps the parking kernel and adds the phonons.

Limit Check 18.2. Random-parking limit

For a constant electron density of states, holes born at the band edge, $\theta_{\mathrm{ii}}=1$ and no phonons, the cascade is Rényi's random parking, with $E_{\mathrm{g}}/\varepsilon_{\mathrm{eh}}=\rho_{\infty}=\int_{0}^{\infty}\mathrm{e}^{-2\,\mathrm{Ein}(t)}\mathrm{d}t=0.74759792$ [6, Eq. (23)] and $F_{\mathrm{eh}}=F_{\mathrm{park}}=0.0510387$ [6, Eq. (44)].

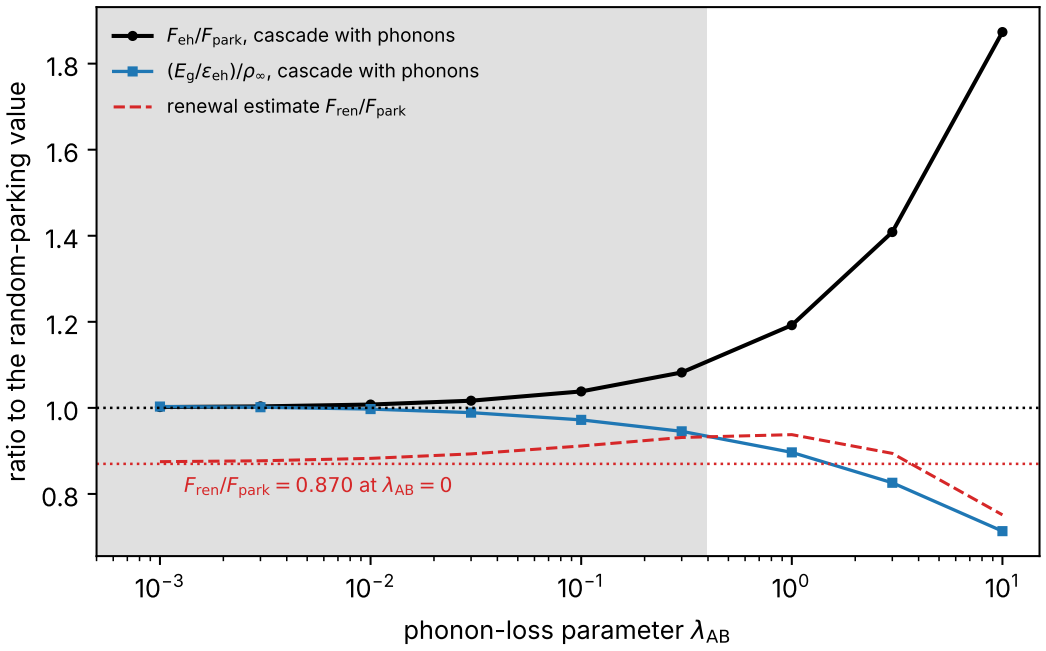


Figure 18.5: The parking kernel with phonons: F_{eh}/F_{park} (black) and $(E_g/\varepsilon_{eh})/\rho_\infty$ (blue) against λ_{AB}, with one phonon step $w_{AB} = 0.0543$ (silicon) and the phonon-to-ionization ratio $\lambda_{AB}/(x/E_g - 1)$ of Eq. (18.8) for a constant density of states, where $\mathfrak{P}(y) = y$ and A_{AB} carries eV, so that here $\lambda_{AB} = A_{AB}/E_g$. Dashed red: the renewal estimate F_{ren} of Eq. (18.13); dotted red: its parking value 0.870. Moment recursions on a grid of step $E_g/74$, which places the values at $\lambda_{AB} \to 0$ 0.4 % (ρ) and 0.1 % (F) above the exact ones. Gray: $\lambda_{AB} < 0.40$, where both ratios are within 10 % of 1 (bisection in $\ln \lambda_{AB}$).

For $\lambda_{AB} \gtrsim 0.4$ the carriers lose their last energy above threshold to phonons, which lowers the pair density and raises F_{eh}, the trend that carries the parking values toward those of silicon.

Relative to the parking values computed on the same grid, the phonons lower E_g/ε_{eh} and raise F_{eh} by $0.8\,\lambda_{AB}$ for $\lambda_{AB} \le 0.003$, and by 3.2 % and 3.7 % at $\lambda_{AB} = 0.1$. At $\lambda_{AB} = 3$ the parking kernel leads to $F_{eh} = 0.072$, 1.41 times F_{park}; the rest of the way to the 0.114 of silicon is the shape of the densities of states.

18.2.4 The renewal estimate

A shortcut to F_{eh} uses only the statistics of the final state. Suppose that each pair costs the gap plus a heat c_i, that the costs are independent and identically distributed with mean $\varepsilon_{eh} - E_g$ and variance σ^2_{pair}, and that pairs are made until ω is spent. This is the single renewal chain of Appendix C.11 with one loss per pair, and Eq. (C.109) gives $\mathrm{Var}\, N_{eh} \simeq \sigma^2_{pair}\omega/\varepsilon^3_{eh}$, that is,

$$F_{ren} = \frac{\sigma^2_{pair}}{\varepsilon^2_{eh}}, \tag{18.12}$$

the uncorrelated final-state estimate of Subashiev and Luryi [6, Eq. (7)]. The denominator is ε_{eh}, not E_g. With N_{eh} held fixed, the heat would scatter by $\bar{n}\sigma^2_{pair}$, and Eq. (18.10) would give σ^2_{pair}/E^2_g. At fixed ω, however, a pair that costs more heat leaves less energy for the pairs

after it, and this constraint multiplies the variance by $(E_g/\varepsilon_{eh})^2$. The difference is large. Take Shockley's uniform population, the assumption behind Klein's relation in Sect. 18.4: parabolic bands with the threshold at $\theta_{ii} = 3/2$, and each terminal carrier spread uniformly in $\boldsymbol{k}$ below threshold. Its residual energy then has the density $x^{1/2}$ on $[0, \frac{3}{2}E_g]$, with mean $\frac{9}{10}E_g$ and variance $\frac{27}{175}E_g^2$. Two carriers per pair give $\sigma^2_{pair} = \frac{54}{175}E_g^2$ and $\varepsilon_{eh} = E_g + 2 \times \frac{9}{10}E_g = \frac{14}{5}E_g$. Held at fixed N_{eh}, this population has $F = 54/175 = 0.31$, 2.6 times the measured 0.119; at fixed ω, however, it has $(54/175)(5/14)^2 = 27/686 = 0.039$, three times below it.

In a cascade the costs are built from computed statistics. Each pair ends with one terminal electron and one terminal hole, whose residual energies have the variances σ_e^2 and σ_h^2, and the phonon heat above threshold is taken as Poisson. Then

$$F_{ren} = \frac{\sigma_e^2 + \sigma_h^2 + \omega_{AB}\langle E_> \rangle/\bar{n}}{\varepsilon_{eh}^2}, \tag{18.13}$$

which keeps the one-carrier statistics of the final state exactly and drops every correlation. In random parking it is the gap-law value 0.0443977 of Worked example 18.2. Limit Check 18.3 compares it with the full F_{eh}.

Limit Check 18.3. Renewal estimate of the Fano factor

For independent, identically distributed costs per pair with variance σ^2_{pair}, the Fano factor is $F_{ren} = \sigma^2_{pair}/\varepsilon^2_{eh}$ [6, Eq. (7)], built for the cascade as Eq. (18.13).

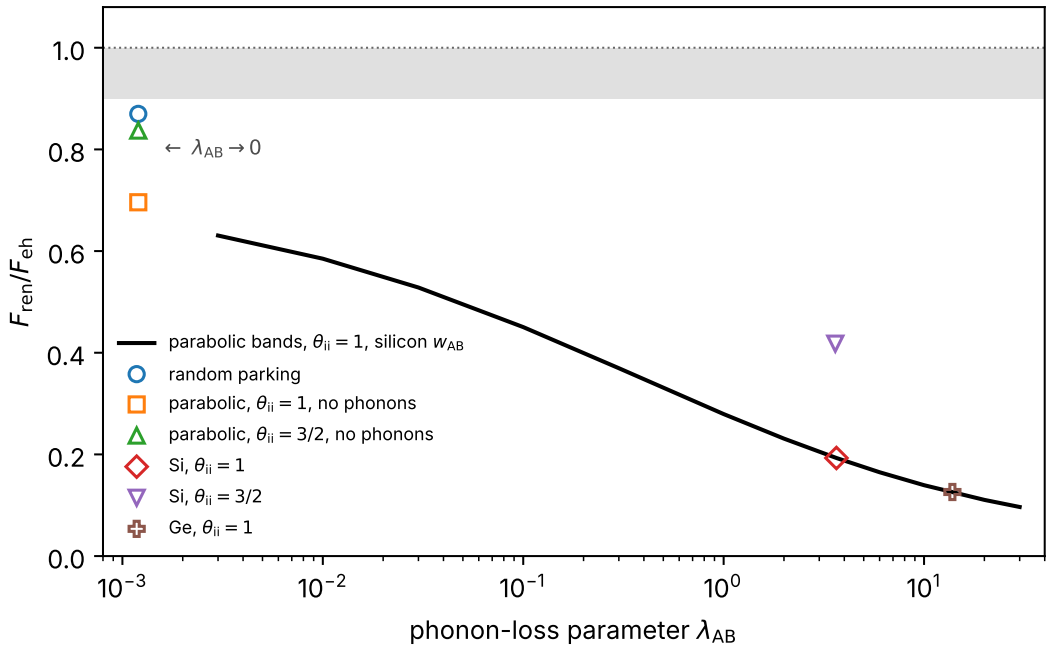


Figure 18.6: Ratio of the renewal estimate Eq. (18.13) to the full Fano factor Eq. (18.11) against λ_{AB}. Line: parabolic densities of states, $\theta_{ii} = 1$, $w_{AB} = 0.0543$, hole band width $12\,\text{eV}/E_g$ (the kernel of Fig. 18.4). Markers at $\lambda_{AB} \to 0$, extrapolated from grid steps 0.01 and $0.005\,E_g$: random parking (0.870), parabolic bands without phonons at $\theta_{ii} = 1$ (0.696) and 3/2 (0.837). Further markers: silicon at $\theta_{ii} = 1$ (0.193) and at $\theta_{ii} = 3/2$ with A_{AB} recalibrated to the pair energy, $5.629\,\text{eV}^3$ (0.418); germanium with $E_g = 0.744\,\text{eV}$, $\omega_{AB} = 37.6\,\text{meV}$, the silicon A_{AB} and the hole density of states cut at the germanium valence-band width, 12.6 eV [7, Table II] (0.126). Markers and line are continuum limits of the moment recursions. The gray band, within 10 % of the full value, contains no

computed case.

The estimate drops the correlations of the cascade, all of which raise the variance here: it lies below F_{eh} in every computed case, by 13 % in random parking, by a factor 5 in silicon and by a factor 8 in germanium.

The ratio falls from 0.70 to 0.10 as λ_{AB} grows, because the phonon heat above threshold is far from Poisson (Sect. 18.2.2), while Eq. (18.13) counts it as Poisson and has no covariance term at all. Without phonons the gap is smaller but still there: $F_{\mathrm{eh}} - F_{\mathrm{ren}} = 0.025$ at $\theta_{\mathrm{ii}} = 1$ and 0.013 at $\theta_{\mathrm{ii}} = 3/2$, with exact one-carrier statistics. This difference measures every departure from independent, identical costs together: correlations between the products of one ionization, between the electron and the hole of one pair, and between distant carriers, and the differences between the cost laws of different pairs. It does not isolate any one of them. In random parking Subashiev and Luryi find the correlation correction positive at intermediate filling and at jamming, and they note that correlations between the two halves of a split are not the only ones [6, Sect. IV]. A Fano factor fitted with a final-state model therefore needs a broader residual distribution, or a larger phonon variance, than the cascade actually has.

18.2.5 The linear mean yield

The definitions Eq. (18.10) assume that $\bar{n}$ grows in proportion to ω. For a single chain of independent losses, Appendix C.11 proved this with the key renewal theorem, Eq. (C.108). The cascade is not a single chain, since an ionization splits the energy among three carriers, and no loss law is common to all its steps. The leading term still follows from renewal theory if we follow a unit of energy instead of a carrier. Give each ionization of a carrier at kinetic energy x the weight 1, spread evenly over the x units of energy that the carrier holds. The number of ionizations in a cascade started at kinetic energy E is then the integral of the collected weights over all units. Taking the mean,

$$\bar{n}_{\mathrm{sub}}(E) = E \left\langle \sum_j \frac{1}{x_j} \right\rangle_u , \tag{18.14}$$

where u is a unit chosen uniformly in $[0, E]$, the sum runs over the ionizations along the chain of carriers that hold u, x_j is the kinetic energy of the holder at the j-th of them, and $\langle \cdots \rangle_u$ is the mean over u and over the cascade. A holder that emits phonons first ionizes below the energy at which it received the unit; $x_1 = E$ only if the first collision of the seed carrier is an ionization. The carriers that hold one unit form a chain. At an ionization the unit passes to one of the three products with probability equal to that product's share of the energy, or it leaves the chain into the gap of the new pair, with probability E_{g}/x. At a phonon emission it leaves with probability ω_{AB}/x. Appendix D.44 shows that Eq. (18.14) is exact for any branching law that conserves energy.

At high energy the shares no longer depend on x. In the random-k model the products divide $x - E_{\mathrm{g}}$ in proportions whose law tends to a fixed law, and the probability of leaving into the gap falls as E_{g}/x. Without a hole cutoff the shares tend to the Dirichlet law with parameters 3/2, and the ratio Eq. (18.8) of phonon emission to ionization falls as $x^{1/2}/x^{7/2} = x^{-3}$. With the hole density of states cut at B_{h}, the model solved here, $\mathfrak{P}_{\mathrm{e}}(y) \propto y^2$ for $y \gg B_{\mathrm{h}}$, the ratio falls as $x^{-3/2}$, the share of the hole tends to zero, and the limit law is the Beta law with parameters 3/2, 3/2 for the two electrons; the argument below is unchanged. The logarithm $\ln x_j$ thus performs a random walk with steps $\ln(x_{j+1}/x_j) \leq 0$ whose law becomes independent of x. By the renewal theorem in the variable $\ln x$, the distribution with which the walk first arrives in the region within a few gaps of threshold converges as $E \to \infty$.

The sum in Eq. (18.14) is dominated by that region, since the high-energy terms are small, $1/x_j$ each. Its mean converges. With the seed pair counted,

Linear mean yield

$$\bar{n}(\omega) = \frac{\omega}{\varepsilon_{\mathrm{eh}}} + o(\omega), \qquad \frac{1}{\varepsilon_{\mathrm{eh}}} = \lim_{E\to\infty} \Big\langle \sum_j \frac{1}{x_j} \Big\rangle_u , \tag{18.15}$$

for any energy-conserving cascade whose share law becomes independent of the energy at high energy, approaches that limit fast enough, is not concentrated on a lattice in $\ln x$, and is regular near threshold. Appendix D.44 states the four conditions and gives the proof.

The pair energy is the inverse of the mean inverse energy collected along the history of one unit. The share law must be spread out in $\ln x$. A cascade that always splits in fixed proportions would make the walk lattice, and $\bar{n}/\omega$ would oscillate in $\ln \omega$ forever instead of converging.

Two statements that the definitions also use are not part of Eq. (18.15). The first is the constant term b_N, which depends on the seed (Sect. 18.3.4). In the one-phonon model every energy loss is a whole combination of E_{g} and ω_{AB}, and as for the deterministic chain of Appendix C.11 the constant then carries a periodic remainder that never dies out. We can therefore fit the slopes over a range of seed energies, $[X/2, X]$, which averages the remainder. The second is the linear growth of the variance, $\mathrm{Var}\, N_{\mathrm{eh}} = F_{\mathrm{eh}}\omega/\varepsilon_{\mathrm{eh}} + O(1)$, which the computed slopes obey but which needs the same second-order estimates as b_N.

18.2.6 Constant term and the renewal bound

The constant term of the yield and the status of the renewal estimate as a lower bound are open. The constant term and the variance of the yield at second order in renewal theory have not been derived for the branching cascade. The expansions $\bar{n} = \omega/\varepsilon_{\mathrm{eh}} + b_N + o(1)$ and $\mathrm{Var}\, N_{\mathrm{eh}} = F_{\mathrm{eh}}\omega/\varepsilon_{\mathrm{eh}} + O(1)$ need second-order estimates for the walk of Eq. (18.14) and the rate at which the cascade forgets its seed, and neither is in hand. The constant matters at a few pairs: at 10 eV in silicon, $\omega/\varepsilon_{\mathrm{eh}} = 2.67$, and the constant term of the one-sided seed adds 13 % to it. The second open question is whether the renewal estimate Eq. (18.13) is a lower bound on F_{eh} for every cascade of this class. It is in every case computed here and in random parking, but there is no proof. A proof would make the final-state statistics a guaranteed floor for the Fano factor, and a counterexample would show which correlations can lower the variance.

18.3 Seed distributions and the step model

At $\omega = 5$ eV in silicon the step model of Eq. (1.43) puts every event in the two-pair bin. The cascade of Sect. 18.1, with the constant calibrated in Sect. 18.4, puts two pairs in 17 % of the events started by a dark-matter kick at $q_0 = \alpha m_e$ and one pair in the rest. A search with a two-pair threshold therefore records about one in six of the 5 eV events that the step model promises. The difference comes from two places: how the 3.84 eV above the gap is shared between the electron and the hole, and how often each carrier emits a phonon instead of ionizing at a few eV. Both enter through the hand-off Eq. (8.70). Once a final-state class has been turned into the energies of its two carriers, the generating function of Sect. 18.1 yields the whole pair-number distribution. The step model and the Ramanathan–Kurinsky model are limits of that distribution in which the seed is replaced by a fixed rule, and we will see that they fail in the few-pair bins, where the rule matters most.

18.3.1 The seed distribution

A final-state class Λ of the zero-phonon pair sector puts an electron at $E_{\mathrm{e}}(\Lambda)$ above the conduction minimum and a hole at $E_{\mathrm{h}}(\Lambda)$ below the valence maximum, with $E_{\mathrm{e}}+E_{\mathrm{h}}=\omega-E_{\mathrm{g}}$. In the energy-only reduction of Sect. 18.1.4 the cascade of the pair depends on these two energies; the crystal momenta $\boldsymbol{k}+\boldsymbol{q}$ and $\boldsymbol{k}$ matter only for the first collisions. We keep the energies and drop the momenta. The price is the first-generation memory that Sect. 18.1.4 locates, and Appendix E.24 specifies the calculation that keeps it. The input from the dark-matter interaction is then a distribution of carrier energies. The *seed distribution* at momentum transfer $\boldsymbol{q}$, energy ω and velocity $\boldsymbol{v}$ is

$$P_{\mathrm{seed}}(E_{\mathrm{e}},E_{\mathrm{h}}|\boldsymbol{q},\omega,\boldsymbol{v})=\frac{\sum_{\Lambda}\sum_{a,b}\mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})\,S_{ab}(\boldsymbol{q},\omega;\Lambda)\,\delta\big(E_{\mathrm{e}}-E_{\mathrm{e}}(\Lambda)\big)\,\delta\big(E_{\mathrm{h}}-E_{\mathrm{h}}(\Lambda)\big)}{\sum_{a,b}\mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})\,S_{ab}(\boldsymbol{q},\omega)}, \tag{18.16}$$

the probability density that the interaction leaves the electron at E_{e} and the hole at E_{h}. The classes add up to $S_{ab}(\boldsymbol{q},\omega)$ by Eq. (8.64), so P_{seed} integrates to one when every class at $(\boldsymbol{q},\omega)$ is a pair, and to the pair share otherwise; the other sectors seed their own cascades. A pair made together with a phonon of energy $\omega_{\boldsymbol{Q}}$ has $E_{\mathrm{e}}+E_{\mathrm{h}}=\omega-\omega_{\boldsymbol{Q}}-E_{\mathrm{g}}$, and the phonon energy goes to heat once (Sect. 8.7). With one coupling row the seed distribution is independent of $\boldsymbol{v}$ and of the dark-matter model, as the class weights of Sect. 8.7.3 are; with several rows it depends on both. In the zero-phonon sector energy conservation leaves one variable, the electron's share $f_{\mathrm{e}}=E_{\mathrm{e}}/(\omega-E_{\mathrm{g}})$ of the energy above the gap.

The generating function of Eq. (18.9) turns the seed into a pair number. Keep only the pair-number variable, $s=\mathrm{e}^{\lambda_N}$, and let $\mathfrak{G}_t(x;s)$ be the generating function of the number of ionizations in the subtree of a carrier of type t at kinetic energy x, the energy-only solution of Eq. (18.4). The seed pair is counted by the factor s, and the two subtrees are independent, so the cascade distribution of a class is the coefficient

$$P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)=[s^{N_{\mathrm{eh}}}]\;s\,\mathfrak{G}_{\mathrm{e}}\big(E_{\mathrm{e}}(\Lambda);s\big)\,\mathfrak{G}_{\mathrm{h}}\big(E_{\mathrm{h}}(\Lambda);s\big), \tag{18.17}$$

where $[s^n]$ takes the coefficient of s^n. It depends on Λ only through the two energies. Writing it as $\int \mathrm{d}E_{\mathrm{e}}\,\mathrm{d}E_{\mathrm{h}}\,\delta(E_{\mathrm{e}}-E_{\mathrm{e}}(\Lambda))\,\delta(E_{\mathrm{h}}-E_{\mathrm{h}}(\Lambda))\,P_{\mathrm{casc}}(N_{\mathrm{eh}}|E_{\mathrm{e}},E_{\mathrm{h}})$ inside the class sum of Eq. (8.70) and exchanging the sum with the integral, the sum becomes the numerator of Eq. (18.16):

$$P(N_{\mathrm{eh}}|\boldsymbol{q},\omega,\boldsymbol{v})=\int \mathrm{d}E_{\mathrm{e}}\,\mathrm{d}E_{\mathrm{h}}\;P_{\mathrm{seed}}(E_{\mathrm{e}},E_{\mathrm{h}}|\boldsymbol{q},\omega,\boldsymbol{v})\;P_{\mathrm{casc}}(N_{\mathrm{eh}}|E_{\mathrm{e}},E_{\mathrm{h}}). \tag{18.18}$$

Its generating function is $s\int P_{\mathrm{seed}}\,\mathfrak{G}_{\mathrm{e}}\mathfrak{G}_{\mathrm{h}}$, a power series with nonnegative coefficients that converges on $|s|\le 1$. Evaluated at N_z roots of unity, the discrete Fourier sum Eq. (E.111) returns $\sum_j P(N_{\mathrm{eh}}+jN_z)$, the exact probability once N_z exceeds every pair number with appreciable weight.

The recursion alone guarantees three things: the distribution is normalized, it is exactly one pair below $(1+\theta_{\mathrm{ii}})E_{\mathrm{g}}$, and its variance splits into a cascade part and a seed part. First, each $\mathfrak{G}_t$ is a probability generating function, because the recursion of Eq. (18.4) maps products and mixtures of generating functions to generating functions; Eq. (18.18) is then normalized and nonnegative. Second, below $\theta_{\mathrm{ii}}E_{\mathrm{g}}$ no carrier ionizes, $\mathfrak{G}_t=1$, and $P_{\mathrm{casc}}=\delta_{N_{\mathrm{eh}}1}$ exactly. The first ionization becomes possible at $\omega=(1+\theta_{\mathrm{ii}})E_{\mathrm{g}}$, for a seed that gives all the energy to one carrier, which is 2.32 eV in silicon with $\theta_{\mathrm{ii}}=1$. Third, the moments separate the cascade from the seed. Let $m_t(x)$ and $v_t(x)$ be the mean and variance of the number of ionizations in one subtree, and let $\langle\cdot\rangle_{\mathrm{seed}}$ average over P_{seed}. The law of total variance gives

$$\langle N_{\mathrm{eh}}\rangle=1+\big\langle m_{\mathrm{e}}(E_{\mathrm{e}})+m_{\mathrm{h}}(E_{\mathrm{h}})\big\rangle_{\mathrm{seed}},\qquad \mathrm{Var}\,N_{\mathrm{eh}}=\big\langle v_{\mathrm{e}}+v_{\mathrm{h}}\big\rangle_{\mathrm{seed}}+\mathrm{Var}_{\mathrm{seed}}\big(m_{\mathrm{e}}+m_{\mathrm{h}}\big). \tag{18.19}$$

The mean is the expression of Sect. 18.1.5. The second term of the variance is the spread of the subtree means over the seed. It vanishes when $m_e + m_h$ takes one value on the whole support of the seed, for a single split and in the forced region, where both means are zero.

P_{seed} is the per-pair weight of Eq. (8.66) or of the screened form Eq. (8.68), binned in the two carrier energies, which is how a library of computed responses stores it (Appendix E.7). The functions $\mathfrak{G}_t$ on an energy grid are the cascade tables of Appendix E.24. The seeded distribution then costs one integral per momentum transfer, energy and pair of coupling rows.

18.3.2 Pair-number distributions in silicon

Fig. 18.7 shows Eq. (18.18) at $\omega = 10\,\text{eV}$ for six seeds run through one cascade. The cascade is the random-k model of Sect. 18.1.4: free-particle densities of states, $\theta_{\text{ii}} = 1$, one phonon of $\omega_{\text{AB}} = 63\,\text{meV}$, holes confined to the 12 eV valence band, and the rate ratio Eq. (18.8) with $A_{\text{AB}} = 5.703\,\text{eV}^3$, the value that Sect. 18.4 fixes by the measured pair energy at $E_g = 1.160\,\text{eV}$. The dark-matter seed comes from the empirical-pseudopotential bands of Worked example 8.9: the pair weights of the density at $\boldsymbol{q}_0 = \alpha m_e \hat{\boldsymbol{x}}$, without screening. The photon seed is the optical limit of the same bands. The model bands have a gap of 0.82 eV, so each seed is taken at the same excess energy $\omega - E_g$ as in the model. To compare two pair-number distributions we use the total-variation distance $d_{\text{TV}}(P,Q) = \frac{1}{2}\sum_N |P(N) - Q(N)|$, the largest difference that the two assign to any set of pair numbers, for instance to the events above a threshold. Screening hardly moves the shares: at 10 eV the screened out-state seed of Worked example 8.9 and its unscreened counterpart give distributions with $d_{\text{TV}} = 0.010$.

The dark-matter seed yields two pairs with probability 0.414, three with 0.547 and four with 0.039. The mean, 2.626 pairs, is 3.81 eV per pair, 1.6 % above the high-energy value. The photon seed yields 0.400, 0.555 and 0.045, and its mean is 2.645. The step model puts every event in the three-pair bin, which overstates the mean by 14 % and misses the two-pair bin entirely. The three splits of Fig. 18.7, whose means Sect. 18.1.5 quoted, bracket what a fixed rule can do at this energy. The equal split puts three pairs in 92 % of events, the uniform split behaves like the band seeds, and giving everything to one carrier fills the four-pair bin with probability 0.143.

The variance shows where the seed enters. For the dark-matter seed Eq. (18.19) splits $\text{Var}\, N_{\text{eh}} = 0.314$ into 0.283 from the subtrees and 0.031 from the spread of the seed. The equal split has no seed term, and its variance is 0.079, a quarter of the band-seed value. The subtree mean is far from linear in the share. At the equal split the two 4.42 eV carriers make 2.985 pairs in all; at $f_e = \frac{1}{4}$, 2.21 eV on one carrier and 6.63 eV on the other, they make 2.436; with all 8.84 eV on one carrier, 3.003. A carrier at 2.21 eV almost never ionizes, while one at 4.42 eV usually does once, so the count depends on how many carriers end up a little above 4 eV.

Worked example 18.3: The step model against the cascade at 5 eV

Take $\omega = 5\,\text{eV}$ in silicon with the constants of Fig. 18.7. The step model puts $N_{\text{eh}} = 1 + \lfloor (5 - 1.160)/3.749 \rfloor = 1 + \lfloor 1.024 \rfloor = 2$ pairs for every event; the literature values $E_g = 1.11\,\text{eV}$ and $\varepsilon_{\text{eh}} = 3.6\,\text{eV}$ of Sect. 1.5 give 2 as well.

In the cascade, the pair shares $\omega - E_g = 3.84\,\text{eV}$. Consider first a seed that gives all of it to one carrier. Below E_g the carrier cannot ionize; above, the probability that its next collision is an ionization is $p_{\text{ii}}(E) = [1 + \Gamma_{\text{ph}}/\Gamma_{\text{ii}}]^{-1}$ from Eq. (18.8), which is 0.0041 at 2 eV, 0.0184 at 2.5 eV, 0.049 at 3 eV, 0.100 at 3.5 eV and 0.145 at 3.84 eV. As long as the carrier does not ionize, it walks down in steps of ω_{AB}: from 3.84 eV it makes 43 collisions

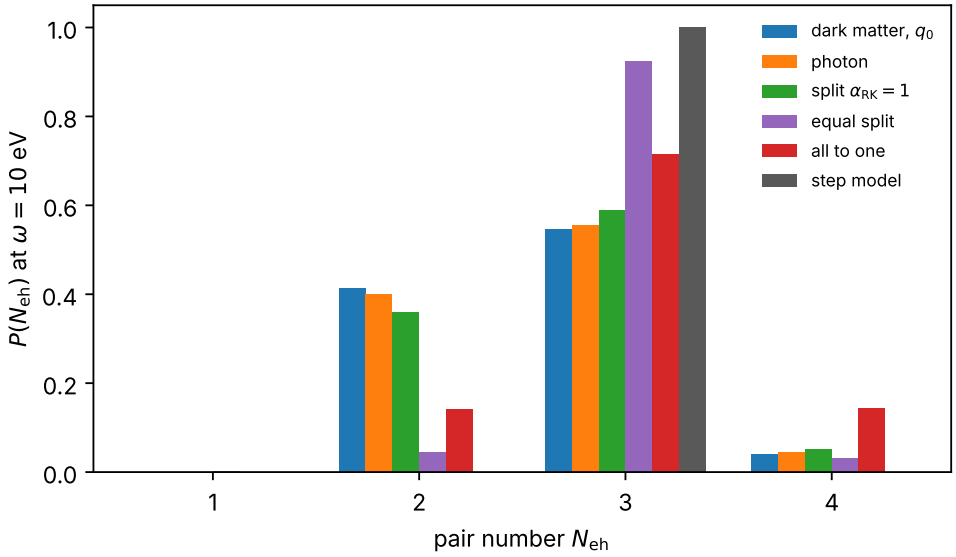


Figure 18.7: Pair-number distribution $P(N_{\mathrm{eh}})$ at $\omega = 10\,\mathrm{eV}$ in silicon, Eq. (18.18), for six seeds in one cascade: the energy-only random-k model of Sect. 18.1.4 with $E_{\mathrm{g}} = 1.160\,\mathrm{eV}$, $\omega_{\mathrm{AB}} = 63\,\mathrm{meV}$, $\theta_{\mathrm{ii}} = 1$, holes below 12 eV and $A_{\mathrm{AB}} = 5.703\,\mathrm{eV}^3$, calibrated to $\varepsilon_{\mathrm{eh}} = 3.749\,\mathrm{eV}$. The seeds are a dark-matter kick at $q_0 = \alpha m_e$ along [100] (independent-particle pair weights of the density, empirical-pseudopotential bands of Worked example 8.9) and an absorbed photon (optical limit of the same bands). The others are the split distributions Eq. (18.20) with $\alpha_{\mathrm{RK}} = 1$, $\alpha_{\mathrm{RK}} \to \infty$ (equal split) and $\alpha_{\mathrm{RK}} \to 0$ (all the energy to one carrier); and the step model Eq. (1.43) with $\varepsilon_{\mathrm{eh}} = 3.749\,\mathrm{eV}$. Band seeds are used at the excess energy $\omega - E_{\mathrm{g}}$ of the model bands, whose gap is 0.82 eV. Means: 2.626 (dark matter), 2.645 (photon), 2.691, 2.985 and 3.003 (the three splits), 3 (step model). $P(1)$ and $P(5)$ are below 0.002 for every seed. Probabilities from the generating function at $N_z = 32$ roots of unity; no distribution in this section has a mean above six.

above E_g. The probability that none of them is an ionization is

$$P_{no} = \prod_{k=0}^{42} [1 - p_{ii}(3.84\,\mathrm{eV} - k\,\omega_{AB})] = 0.176.$$

Replacing the sum of $-\ln(1-p_{ii}) \approx p_{ii}/(1-p_{ii})$ over steps by an integral over energy, $\ln P_{no} \approx -\omega_{AB}^{-1} \int_{E_g}^{3.84\,\mathrm{eV}} \mathrm{d}E\; \Gamma_{ii}/\Gamma_{ph} = -1.735$, which leads to the same 0.176. If the carrier does ionize, the three products share at most $3.84 - 1.16 = 2.68\,\mathrm{eV}$, and a second ionization follows with probability 0.002. This seed therefore leaves one pair with probability 0.176 and two with 0.822. A seed that splits the energy equally leaves two carriers at 1.92 eV, each of which escapes ionization with probability 0.990, and one pair in 98 % of events. The dark-matter seed at q_0 lies in between. At this energy the electron takes 2.08 eV of the 3.84 eV on average, and 26 % of the weight has one carrier with more than three quarters. Averaging $P_{no}(E_e)P_{no}(E_h)$ over the seed, Eq. (18.18) yields

$$P(1) = 0.831, \qquad P(2) = 0.169, \qquad P(3) = 6 \times 10^{-5},$$

a mean of 1.169 pairs and 4.28 eV per pair. The step model puts the whole event in the two-pair bin. Above a two-pair threshold it therefore counts 5.9 times as many 5 eV events as this cascade does; the full-band cascade of Sect. 18.3.3 raises the factor to about ten. The photon seed has $P(2) = 0.190$, 12 % more than the dark-matter seed.

18.3.3 The full-band cascade as reference

Every distribution so far comes from the energy-only cascade with its one constant fitted to the measured pair energy. A cascade with no fitted constant replaces the random-k kernels by those of Sect. 18.1 on full bands, in the standard full-band Monte Carlo form [8]. In the calculation we use as reference, the bands are the empirical pseudopotential bands of Cohen and Bergstresser [9], with the form factors refit to measured transition energies; impact ionization is the golden rule with the diagonal, statically screened Coulomb kernel [10], its direct and exchange terms combined as in Kane's form quoted by Kotani and van Schilfgaarde [1, Eq. (5)]; and phonon emission uses the deformation potentials and phonon energies of Lundstrom's textbook [2], as quoted by Li and coworkers [3, Table III], who credit the hole acoustic value also to Jacoboni and Reggiani [4]; the gap is $E_g = 1.160\,\mathrm{eV}$. The seeds are the pairs a dark-matter particle with a heavy mediator leaves in the same bands, with their crystal momenta. The calculation is varied jointly over the screening (static and dynamic), the phonon couplings (these values against first-principles ones, which weigh electron losses by 0.52 and hole losses by 3.0) and the exchange factor near threshold (0.5 to 1.0), and the ranges below are the envelope of these variants, for pair probabilities with the error of the $\boldsymbol{k}$ grid added.

At 5 eV the full-band cascade puts two or more pairs in a fraction 0.097 of the dark-matter events, with the envelope $[0.030, 0.173]$, against the 0.169 of the energy-only cascade in Worked example 18.3. The step model, which puts every event there, then overcounts the two-pair bin about tenfold, and at least sixfold anywhere in the envelope; the thirtyfold end comes from one extreme corner alone, static screening of the weaker kind with tripled hole couplings and the lowest exchange factor. The energy-only cascade remains the working model of this chapter, because it is calibrated and fast; the full band shows how far its kernels are from those of another model without a fitted constant. Recalibrated to the pair energy of each full-band variant, so that only the shape of the kernels differs, it lies 0.12 to 0.20 in d_{TV} from

the full-band distribution between 6 and 8 eV, where the soft onset of impact ionization near threshold matters most, and the distance grows with the pair energy of the variant. On the same footing the Ramanathan–Kurinsky split with $\alpha_{\mathrm{RK}} = 1$ lies 0.11 to 0.12 away, the equal split 0.45 to 0.50 and the step model 0.79 to 0.83.

18.3.4 Ramanathan–Kurinsky and step limits

The Ramanathan–Kurinsky model is Eq. (18.18) after four replacements [5, Sect. II]. First, the seed distribution is replaced by a distribution of the electron's share that is the same for every process at a given ω and symmetric under the exchange of electron and hole [5, Eq. (6)], in the one-parameter Beta form [5, Eq. (7)]

$$p(f_{\mathrm{e}}|\alpha_{\mathrm{RK}}) = \frac{f_{\mathrm{e}}^{\alpha_{\mathrm{RK}}-1}(1-f_{\mathrm{e}})^{\alpha_{\mathrm{RK}}-1}}{B(\alpha_{\mathrm{RK}},\alpha_{\mathrm{RK}})}, \tag{18.20}$$

with B the Beta function. It interpolates between all the energy on one carrier ($\alpha_{\mathrm{RK}} \to 0$), a uniform share ($\alpha_{\mathrm{RK}} = 1$) and the equal split ($\alpha_{\mathrm{RK}} \to \infty$); their $P(E,\omega)$, normalized to two carriers, is $[p(f)+p(1-f)]/(\omega - E_{\mathrm{g}})$ at $f = E/(\omega - E_{\mathrm{g}})$. The value of α_{RK} at each ω is fitted to the quantum yield of absorbed photons. Second, the energy-only reduction is applied from the first collision on, with one yield function $\langle N(E)\rangle = m_{\mathrm{e}}(E) = m_{\mathrm{h}}(E)$ for electrons and holes. Third, the kernels are those of the random-k model with free-particle densities of states and one phonon, with the rate ratio Eq. (18.8) [5, Eq. (8)], the ionization probability per collision $[1 + \Gamma_{\mathrm{ph}}/\Gamma_{\mathrm{ii}}]^{-1}$ [5, Eq. (10)] and the hole energy capped at the valence band width. Fourth, a separate plasmon channel is added. They see no feature of it in any yield and find that the width of the hole band, not the plasmon, visibly changes the yield; in the book's accounting the channel would count the weight of the ionization channel twice (Sect. 18.1.1).

With these replacements the mean of Eq. (18.18) is their yield formula [5, Eq. (5)],

$$n(\omega) = 1 + \int_0^{\omega - E_{\mathrm{g}}} \mathrm{d}E\; P(E,\omega)\,\langle N(E)\rangle,$$

and the coefficients of Eq. (18.17), averaged over Eq. (18.20), are the pair probabilities $p_n(\omega)$ that they tabulate from a Monte Carlo simulation of this cascade. Above 50 eV they replace p_n by a Gaussian in n centered at $\omega/\varepsilon_{\mathrm{eh}}$ with variance $F_{\mathrm{eh}}n$ [5, Eq. (14)]. This is the renewal limit of Eqs. (C.108) and (C.109) with the constant term b_N of the mean (Sect. 18.2.1) set to zero. That term depends on the seed. In the cascade of Fig. 18.7, with $\omega/\varepsilon_{\mathrm{eh}}$ subtracted at the pair energy of the same calculation, it is 0.008 for the uniform split and 0.35 for a seed that gives all the energy to one carrier. The model is the photon seed in a phenomenological form, and the fit of α_{RK} enters exactly where the seed matters. Between 4 and 7.5 eV the photon data drive α_{RK} from about 1 toward the equal split, and between 8 and 50 eV, where no data exist, it is extrapolated [5, Sect. IV B].

The step model is the further limit in which every pair has a fixed price. Let the first pair cost E_{g} and every further one exactly $\varepsilon_{\mathrm{eh}}$. The energy $\omega - E_{\mathrm{g}}$ left after the first pair is then spent by a renewal chain with the deterministic loss $\varepsilon_{\mathrm{eh}}$, which fits $\lfloor(\omega - E_{\mathrm{g}})/\varepsilon_{\mathrm{eh}}\rfloor$ losses (Appendix C.11). The pair number is

$$N_{\mathrm{eh}}^{\mathrm{step}}(\omega) = 1 + \left\lfloor \frac{\omega - E_{\mathrm{g}}}{\varepsilon_{\mathrm{eh}}} \right\rfloor, \qquad P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda) = \delta_{N_{\mathrm{eh}},N_{\mathrm{eh}}^{\mathrm{step}}(\omega)}, \tag{18.21}$$

which is Eq. (1.43), with a Fano factor of zero. Both the seed and the competition with phonons have been replaced by one price per pair. In the forced region, $\omega < (1+\theta_{\mathrm{ii}})E_{\mathrm{g}}$, the step model is exact. Above it, it is wrong in two ways. The mean is wrong even on average.

Over one period of ω the floor function loses $\frac{1}{2}$ on average (Appendix C.11), so the step model has the mean $(\omega - E_g)/\varepsilon_{eh} + \frac{1}{2} = \omega/\varepsilon_{eh} + 0.19$ in silicon, against $\omega/\varepsilon_{eh} + b_N$ in the cascade, with b_N between 0.008 and 0.35 depending on the seed. That error falls as $1/\langle N_{eh}\rangle$. The distribution never converges. For a single bin n the distance d_{TV} is $1 - P(n)$. The cascade spreads over a width $\sigma_N = (F_{eh}\omega/\varepsilon_{eh})^{1/2}$ that grows with ω: at 20.2 eV the computed distribution of the dark-matter seed has $\sigma_N = 0.79$ and its largest bin holds 0.45, so the step model misplaces more than half of the events there, and the fraction grows with the width.

Limit Check 18.4 compares each phenomenological seed with the dark-matter seed from the gap to 21 eV, the range of the few-pair bins in silicon.

Limit Check 18.4. Step model and Ramanathan–Kurinsky model

The Ramanathan–Kurinsky pair probabilities are the coefficients of Eq. (18.17) averaged over the split distribution Eq. (18.20), with mean $n(\omega) = 1 + \int \mathrm{d}E\, P(E,\omega)\langle N(E)\rangle$ and the rate ratio Eq. (18.8) [5, Eqs. (5)–(10)]. The step model is Eq. (18.21) [11, Eq. (5.1)].

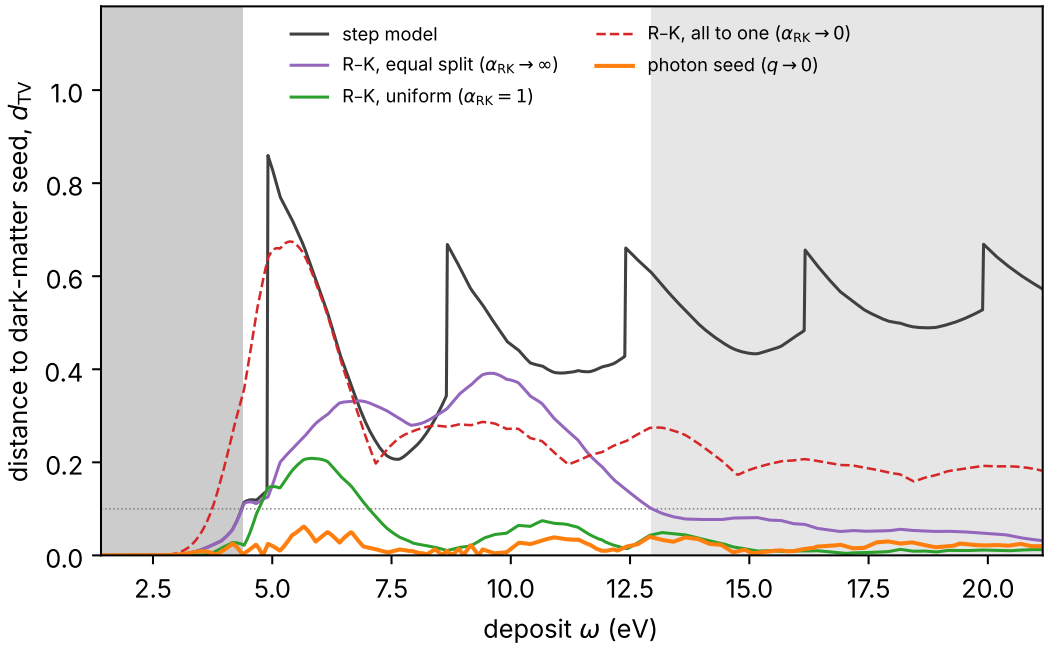


Figure 18.8: Distance d_{TV} between the pair-number distribution of each phenomenological seed and that of the dark-matter seed at $q_0 = \alpha m_e$ along [100], against the deposit ω in silicon. All seeds run through one cascade, the energy-only random-k model of Sect. 18.1.4 with $E_g = 1.160$ eV, $\omega_{AB} = 63$ meV, $\theta_{ii} = 1$, holes below 12 eV and $A_{AB} = 5.703\,\mathrm{eV}^3$. Black: step model, $\varepsilon_{eh} = 3.749$ eV. Purple, green and red dashed: the Ramanathan–Kurinsky (R–K) splits Eq. (18.20) with $\alpha_{RK} \to \infty$, 1 and $\to 0$, with the seeds that put a hole above 12 eV removed. Orange: the photon ($q \to 0$) seed of the same empirical-pseudopotential bands as the dark-matter seed. Band seeds are computed every 0.25 eV of the excess energy $\omega - E_g$ (model gap 0.82 eV) and interpolated linearly; the curves use a 0.02 eV grid. Dark gray: $\omega < 4.37$ eV, where the step model and the splits with $\alpha_{RK} = 1$ and $\to \infty$ are within 0.1 of the dark-matter seed. Light gray: $\omega > 12.95$ eV, where the splits with $\alpha_{RK} = 1$ and $\to \infty$ are. Dotted: $d_{TV} = 0.1$.

Between 4.4 eV and 13 eV the split of the energy decides the count, so it must come from the bands and the operator that made the pair. There the splits with $\alpha_{RK} = 1$

and $\to \infty$ reach distances of 0.21 and 0.39, the one-sided split 0.67, and the step model is off by up to 0.86 at its first step, $\omega = E_\mathrm{g} + \varepsilon_\mathrm{eh} = 4.91\,\mathrm{eV}$.

Below 4.37 eV both models hold because few carriers ionize: the dark-matter seed leaves one pair with probability above 0.9, and so does any split that avoids putting the whole energy on one carrier. The step model leaves this region at 4.37 eV and never returns: at its first step it moves every event into the two-pair bin while the cascade puts 0.14 there, and its distance stays above 0.2 up to 21 eV, above 0.39 beyond 9 eV. The splits with $\alpha_\mathrm{RK} = 1$ and $\alpha_\mathrm{RK} \to \infty$ come within 0.04 of the dark-matter seed at 21 eV. The split with $\alpha_\mathrm{RK} \to 0$ stays 0.18 away even there. At that energy the valence band no longer holds the whole energy, and the seed becomes one-sided, with a mean 0.36 pairs above that of the dark-matter seed. A shift of the mean by a fixed amount gives a distance that decreases only as $\sigma_N^{-1} \propto \omega^{-1/2}$.

The photon seed is the case that matters in practice, because a photon calibration measures it, and it is the error that Sect. 8.7.3 left open. In the energy-only cascade, with both seeds from the same band calculation on one $\boldsymbol{k}$ grid, the photon seed stays within 0.062 of the dark-matter seed everywhere, with the largest distance at 5.67 eV, where the means differ by 4.6 %. At 10 eV the distance is 0.014, and above 15 eV it is at most 0.03. These numbers are internal to that model. They measure only the difference of the energy shares, and they carry the error of the photon seed's $\boldsymbol{k}$ grid: in the full-band calculation of Sect. 18.3.3 the photon seed moves by 0.11 to 0.14 between grids, as much as the difference itself, so the full-band value is unresolved. The valleys and crystal momenta of the first generation, which the dark matter populates differently from a photon, are dropped by the energy-only reduction, and the calculation that keeps them (Appendix E.24) would add their share of the error. For absorption of a dark photon there is no such error: its seed is the photon's (Sect. 8.7.3).

18.3.5 Pair number at the readout

Every distribution so far is the pair number at the end of the prompt cascade, about a picosecond after the deposit, when every carrier has reached its band edge. A detector reads the event microseconds to seconds later, and what it reads depends on more than the number. The induced charge depends on where the pairs are and how far each carrier drifts. Capture costs one gap per captured electron, and excitons change the measured Fano factor (Sect. 18.5). A phonon readout records the heat with its spectrum. The cascade therefore records, besides N_eh, further quantities $\mathfrak{m}$: the positions of the carriers at the end of the prompt cascade, the electron and hole counts separately, the spectrum of the emitted phonons and the energy stored in defects. Let $P_\mathrm{det}(\mathsf{R}|N_\mathrm{eh}, \mathfrak{m})$ be the probability that the readout records R, a charge, a phonon energy or both, at the time it reads. The distribution of the recorded quantity for an event of class Λ is

$$P_\mathrm{obs}(\mathsf{R}|\Lambda) = \sum_{N_\mathrm{eh}} \int \mathrm{d}\mathfrak{m}\; P_\mathrm{casc}(N_\mathrm{eh}, \mathfrak{m}|\Lambda)\; P_\mathrm{det}(\mathsf{R}|N_\mathrm{eh}, \mathfrak{m}), \tag{18.22}$$

and it replaces P_casc in Eq. (8.69) when the rate is written for the recorded quantity. The quantities $\mathfrak{m}$ are read off the same cascade sample as the pair number, so the generating function of Eq. (18.4) carries them as further components of the deposit.

The step model and the Ramanathan–Kurinsky model are statements about one marginal of Eq. (18.22): the pair number at the end of the prompt cascade, with $P_\mathrm{det} = \delta_{\mathsf{R}N_\mathrm{eh}}$. Every later process acts on that marginal and on $\mathfrak{m}$ together. If each pair is collected independently with probability f_det, the recorded number is thinned binomially, its Fano factor is $f_\mathrm{det}F_\mathrm{eh} + 1 - f_\mathrm{det}$ by Eq. (C.100), and $\mathfrak{m}$ drops out. If collection depends on position or on the carrier type, it

does not, and the rate needs the joint law. A model of the prompt marginal, however good, fixes only the first factor of Eq. (18.22). Sect. 18.5 treats the second up to the end of the cascade; the readout itself is the subject of a later chapter.

18.4 Calibrated pair energies

Back at keV deposits, where two moments describe the pair number, a charge-resolving silicon sensor at 123 K measures $\varepsilon_{\mathrm{eh}} = (3.749 \pm 0.001)\,\mathrm{eV}$ and $F_{\mathrm{eh}} = 0.119 \pm 0.002$ for 5.9 keV X-rays [12]. The cascade of Sects. 18.1 and 18.2, reduced to energy alone and given free-particle kernels, has one constant that a measurement must fix. Fixed by the first number, it predicts the second. It predicts both numbers for every other crystal as well, including the crystals ranked in Chap. 17, for most of which neither has been measured. A guide to new targets needs a pair energy for each of them, and it needs to know which of these numbers rest on comparison with data. With its one constant fixed in silicon, the cascade comes within 5 to 6 % of the measured pair energies of germanium and GaAs, explains why Klein's slope 14/5 fails for wide gaps, and reduces to the line $\varepsilon_{\mathrm{eh}} \approx 1.61 E_{\mathrm{g}} + k_2$.

18.4.1 The calibrated cascade

The model is the random-k cascade of Sect. 18.1.4 with free-particle densities of states, threshold ratio $\theta_{\mathrm{ii}} = 1$, one optical phonon of energy ω_{AB} and, in silicon, a hole band cut at $B_{\mathrm{h}} = 12\,\mathrm{eV}$, solved by the moment recursions of Sect. 18.2.1. In units of the gap it depends on a crystal through the two numbers defined there, the phonon strength $\lambda_{\mathrm{AB}} = A_{\mathrm{AB}}/E_{\mathrm{g}}^3$ and the phonon step $w_{\mathrm{AB}} = \omega_{\mathrm{AB}}/E_{\mathrm{g}}$. A cascade with large λ_{AB} loses most of its energy to phonons on the way down and settles more slowly: germanium, with $\lambda_{\mathrm{AB}} > 5$, needs slopes fitted for seed energies between 30 and $60E_{\mathrm{g}}$, against 20 to $40E_{\mathrm{g}}$ for silicon. All values are the continuum limit of the recursions.

The constant is fixed in silicon. With $E_{\mathrm{g}} = 1.160\,\mathrm{eV}$ at 123 K [5, Eq. (11)] and $\omega_{\mathrm{AB}} = 63\,\mathrm{meV}$ (Appendix E.13), the measured pair energy requires

$$A_{\mathrm{AB}} = 5.703\,\mathrm{eV}^3, \tag{18.23}$$

so that $\lambda_{\mathrm{AB}} = 3.65$ and $w_{\mathrm{AB}} = 0.0543$ for silicon. The same constant leads to $F_{\mathrm{eh}} = 0.114$ (Sect. 18.2.2). The value lies 10 % above the 5.2 that Alig and coworkers fitted to the pair energy of silicon [13, Sect. II] and that Ramanathan and Kurinsky use at 300 K [5, Table I]. Everything else in this section uses Eq. (18.23) unchanged: one constant for all crystals. That universality is an assumption. It is the reason the comparison with other crystals tests the model, and the first suspect where the comparison fails.

A first-principles cascade replaces both kernels: the impact-ionization rate by the imaginary part of the *GW* self-energy on quasiparticle bands, and the phonon rate by the golden rule with the computed electron–phonon vertex, with no constant left to fit (Appendix E.13). Appendix E.24 specifies the calculation for silicon, germanium and GaAs and for the wide-gap crystals of Chap. 17. Its results would replace every model number in this section. Two calculations of this kind exist for silicon. Nielsen and Fischetti use density-functional bands and phonons, a loss-function kernel without exchange and a plasmon channel, and find $\varepsilon_{\mathrm{eh}} = 3.10\,\mathrm{eV}$ with a time-dependent density-functional loss function and 3.50 eV with a *GW*/Bethe–Salpeter one [13, Sect. III]; they give the same method for six of the crystals below (the last column of Table 18.2). The full-band cascade of Sect. 18.3.3 uses empirical pseudopotential bands and deformation-potential phonons and gives $\varepsilon_{\mathrm{eh}} = 4.15\,\mathrm{eV}$, with the envelope $[3.95, 4.73]$ eV of its variants and a systematic of at least $\pm 0.4\,\mathrm{eV}$ for the class of

model, the shift between Nielsen and Fischetti's two loss functions. Without a fitted constant, Nielsen and Fischetti lie below the measured 3.749 eV and the full-band cascade above it. The two do not meet even with matched ingredients: with first-principles phonon couplings for both carriers and without exchange, as Nielsen and Fischetti have them, the full-band cascade still gives 4.11 eV, and the remaining 0.6 eV to their 3.50 eV is unexplained. It sits where the pair energy is decided, in the rate of impact ionization between threshold and about 4.5 eV (Sect. 18.1.4). At 100 eV the full-band cascade turns 71.6 % of the deposit into heat, with the envelope [70.2, 75.0] %.

18.4.2 Ramanathan–Kurinsky planes

Ramanathan and Kurinsky sampled the same model event by event, with two additions: an explicit plasmon channel for carriers above the plasmon energy, and a re-splitting of energy whenever a hole would exceed the band width [5, Sect. III A]. They summarized their results at deposits of about 100 eV by planes in the gap and the constant, Eq. (18.24) in Limit Check 18.5, which the moment solution at their parameters reproduces.

Limit Check 18.5. Monte Carlo planes of the random-k cascade

Ramanathan and Kurinsky's least-squares planes through their Monte Carlo, with $\omega_{\mathrm{AB}} = 63\,\mathrm{meV}$ and $B_{\mathrm{h}} = 12\,\mathrm{eV}$ [5, Eqs. (12)–(13)]:

$$\varepsilon_{\mathrm{eh}} = 1.7E_{\mathrm{g}} + 0.084A_{\mathrm{AB}} + 1.3, \qquad F_{\mathrm{eh}} = -0.028E_{\mathrm{g}} + 0.0015A_{\mathrm{AB}} + 0.14, \tag{18.24}$$

energies in eV and A_{AB} in eV^3.

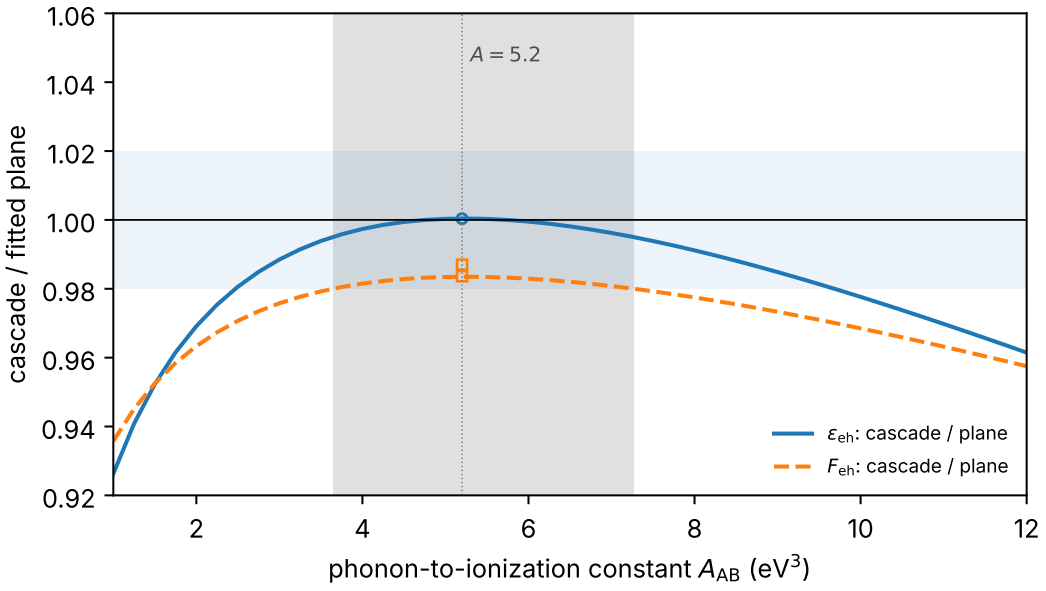


Figure 18.9: The deterministic moment solution of the energy-only random-k cascade divided by the planes Eq. (18.24), against the constant A_{AB} at $E_{\mathrm{g}} = 1.124\,\mathrm{eV}$, the 300 K slice of the planes. Solid: $\varepsilon_{\mathrm{eh}}$; dashed: F_{eh}. Open symbols: $E_{\mathrm{g}} = 1.0$ and 1.2 eV at $A_{\mathrm{AB}} = 5.2$. Light band: ±2 %, the level at which Ramanathan and Kurinsky find F_{eh} constant over their plane [5, Sect. IV A]. Gray: A_{AB} from 3.65 to 7.27, where both ratios lie within 2 % of one. Parameters of Ramanathan and Kurinsky: $\theta_{\mathrm{ii}} = 1$, $\omega_{\mathrm{AB}} = 63\,\mathrm{meV}$, $B_{\mathrm{h}} = 12\,\mathrm{eV}$. The solution has no plasmon channel and no hole re-splitting.

Table 18.1: Mean energy per pair and Fano factor of the energy-only random-k cascade with the single constant Eq. (18.23), against measurements. Germanium gap at 77 K from the fit of Wei, Wang and Mei [14, Eq. (7)]; other gaps from Table 3.1 and Appendix E.13. Measurements: silicon [12]; germanium, the pair energy and the range of five Fano-factor experiments quoted by Mei and coworkers [15, Sect. 4.3]; GaAs, the room-temperature value of Bertuccio and Maiocchi as quoted by Nielsen and Fischetti [13, Fig. 1]. Residual: model over measured, minus one.

	T	E_g (eV)	ε_{eh} (eV)		F_{eh}		Status
			model	measured	model	measured	
Si	123 K	1.160	3.749	3.749 ± 0.001	0.114	0.119 ± 0.002	ε fitted; F predicted, 2.4σ low
	low	1.17	3.766	–	0.114	–	model
Ge	77 K	0.735	2.815	2.96	0.125	0.06–0.13	ε residual $-4.9\,\%$; F inside the spread
	low	0.744	2.831	–	0.124	–	model
GaAs	300 K	1.42	3.964	≈ 4.2	0.104	–	ε residual $-5.6\,\%$
	low	1.52	4.132	–	0.102	–	model

Outside the gray range the planes, being linear fits, miss the curvature of the cascade in A_{AB}, by 6 to 7 % at $A_{AB} = 1$ and 4 % at 12.

At five points around silicon the solution agrees with the planes to 1.2 % in ε_{eh} and 2.4 % in F_{eh}, and its slope $\partial\varepsilon_{eh}/\partial E_g = 1.70$ matches the 1.7 of the plane. Of course, this only shows that two codes solve the same model; it says nothing about whether the model describes silicon, since both share the free-particle bands, the equal matrix elements, the single phonon and the fitted constant.

18.4.3 Silicon, germanium and GaAs

Table 18.1 sets the model against the measurements. Each crystal is computed twice: at the gap of the measurement temperature, for the comparison, and at its low-temperature gap of Table 3.1, for a detector at millikelvin. Temperature enters the model only through $E_g(T)$. Phonon absorption is left out, which costs little: the occupation of the 37.6 meV germanium phonon is 0.0035 at 77 K. The phonon energy ω_{AB} is the measured zone-center optical energy, 37.6 meV for germanium (Table 5.1) and the longitudinal 36.35 meV for GaAs (Sect. 5.5); neither crystal has a hole cutoff.

In silicon the pair energy is fitted, and the Fano factor is the test. The model yields $F_{eh} = 0.114$, 2.4σ below 0.119 ± 0.002, with no uncertainty assigned to the model; the temperature dependence measured by Lowe and Sareen implies 0.118 ± 0.004 at 123 K, as quoted in the same work [12], about 1σ above the model. The comparison is sharper than it looks, because F_{eh} hardly depends on the fitted constant. Raising A_{AB} by 10 % raises ε_{eh} by 1.2 % and F_{eh} by 0.65 %, secant slopes at the silicon gap of $\partial\varepsilon_{eh}/\partial A_{AB} = 0.076\,\text{eV}^{-2}$ and $\partial F_{eh}/\partial A_{AB} = 0.0013\,\text{eV}^{-3}$. The calibration leaves F_{eh} where the structure of the model puts it, and the structural choices move it more than the constant does. Recalibrated with the threshold at $\frac{3}{2}E_g$ instead of E_g, the model gives 0.113; with a density of states rising as x instead of $x^{1/2}$, 0.102. Neither choice reaches 0.119. Nor does the full-band cascade of Sect. 18.3.3, which has no fitted constant: it gives $F_{eh} = 0.114$, the envelope of its variants $[0.101, 0.115]$, the same value as the energy-only model, and no variation within its class moves F_{eh} toward the measurement. Both calculations share what they leave out, the plasmon

channel, excitonic final states and momentum-dependent phonon couplings, so this agreement does not show that the gap to 0.119 is structural. As Sect. 18.2.2 showed, F_{eh} is set within a few phonon energies of threshold, where the free-particle form of Eq. (E.87) is least reliable (Appendix E.13). The model value also agrees with the 0.115 of Ramanathan and Kurinsky [5, Sect. IV A], the same model sampled differently (Limit Check 18.5); the two together count as one prediction, not two.

Germanium tests the pair energy. With the constant fixed in silicon, the model yields 2.81 eV at 77 K, a residual of $-4.9\,\%$ against the measured 2.96 eV [15, Sect. 4.3]. Its larger phonon strength, $\lambda_{\mathrm{AB}} = 14.4$ against 3.65 for silicon, puts $\varepsilon_{\mathrm{eh}}$ at $3.83E_{\mathrm{g}}$, against $3.23E_{\mathrm{g}}$ in silicon. The measured Fano factors of germanium scatter from 0.06 to 0.13 [15, Sect. 4.3], a spread wider than the model's own sensitivity, so the model value 0.125 is consistent with the data but not tested by them. No germanium pair energy has been measured at millikelvin; the value of a bolometric detector rests on extrapolation from 77 K [14], and the model's 2.83 eV at the low-temperature gap is one such extrapolation.

GaAs is the polar contrast. Its pair energy has been measured at room temperature, about 4.2 eV by Bertuccio and Maiocchi as quoted by Nielsen and Fischetti [13, Fig. 1], against the model's 3.96 eV, a residual of $-5.6\,\%$. With $\lambda_{\mathrm{AB}} = 1.99$ and $w_{\mathrm{AB}} = 0.026$, GaAs loses less energy to phonons than silicon, and $\varepsilon_{\mathrm{eh}} = 2.79E_{\mathrm{g}}$. Its optical phonons couple to the carriers through the long-range Fröhlich field, which the constant matrix element of Eq. (E.87) does not distinguish from the short-range deformation-potential coupling of silicon. Its strongly unequal electron and hole masses and its Γ, L and X conduction valleys make a single parabolic random-k kernel especially crude. No Fano factor of GaAs has been measured. Both residuals are negative: with the silicon constant, the model puts too little heat into each pair in germanium and GaAs.

18.4.4 The Klein relation as a truncation

Before cascade calculations, pair energies were estimated from Shockley's energy balance and fitted lines through measurements; Klein's form $\varepsilon_{\mathrm{eh}} = \frac{14}{5}E_{\mathrm{g}} + r_K\omega_K$ is the best known [14, Eq. (3)]. The cascade contains it as a truncation, and the truncation explains why the slope 14/5 is too steep for wide-gap crystals.

Energy conservation along every path of the cascade gives an exact identity. Each pair contributes its gap E_{g}; each carrier, once it falls below threshold for good, leaves its residual kinetic energy as heat; and phonons emitted above threshold take the rest. In the reduced cascade every pair ends with exactly one terminal electron and one terminal hole: the seed's hole starts at the band edge, and each ionization adds one pair and one carrier of each kind. Let $\bar{E}_{\mathrm{res}}$ be the mean residual energy per terminal carrier and r_K the mean number of phonons of energy $\omega_K = \omega_{\mathrm{AB}}$ emitted above threshold per pair. The residual heat $E_<$ and the phonon heat $E_>$ of Sect. 18.2.1 then have means $2\bar{E}_{\mathrm{res}}\bar{n}$ and $r_K\omega_K\bar{n}$, and dividing the energy identity $\omega = N_{\mathrm{eh}}E_{\mathrm{g}} + E_< + E_>$ by $\bar{n}$ at large ω gives

$$\varepsilon_{\mathrm{eh}} = E_{\mathrm{g}} + 2\bar{E}_{\mathrm{res}} + r_K\omega_K. \tag{18.25}$$

This is Shockley's balance, $\varepsilon = 2E_f + E_i + rE_R$ in his notation, with E_f the mean final kinetic energy of a carrier, E_i the threshold energy, E_R the phonon energy and r the number of phonons per pair, and with E_i replaced by the gap in the pair term [13, Eq. (1)]. In silicon the computed terms are $1.160 + 1.666 + 0.923\,\mathrm{eV} = 3.749\,\mathrm{eV}$: the residual energy per carrier is $\bar{E}_{\mathrm{res}} = 0.718E_{\mathrm{g}}$, and each pair carries 14.6 phonons emitted above threshold.

Klein's relation follows from Eq. (18.25) under two assumptions. The first (K1) puts the threshold at $\theta_{\mathrm{ii}} = \frac{3}{2}$, the value Eq. (18.3) for parabolic bands of equal mass with crystal momentum conserved. The second (K2) takes the carriers below threshold to be distributed

uniformly in $\boldsymbol{k}$, Shockley's uniform population [13, Sect. II]. With the parabolic density of states $\rho(x) \propto x^{1/2}$ on $0 \le x \le \theta_{\mathrm{ii}} E_{\mathrm{g}}$, the mean residual is

$$\bar{E}_{\mathrm{res}} = \frac{\int_0^{\theta_{\mathrm{ii}} E_{\mathrm{g}}} x \cdot x^{1/2}\,\mathrm{d}x}{\int_0^{\theta_{\mathrm{ii}} E_{\mathrm{g}}} x^{1/2}\,\mathrm{d}x} = \frac{3}{5}\,\theta_{\mathrm{ii}} E_{\mathrm{g}}, \tag{18.26}$$

which is $0.9E_{\mathrm{g}}$ at $\theta_{\mathrm{ii}} = \frac{3}{2}$. Inserted in Eq. (18.25),

$$\varepsilon_{\mathrm{eh}} = E_{\mathrm{g}} + 2 \times 0.9 E_{\mathrm{g}} + r_K \omega_K = \frac{14}{5} E_{\mathrm{g}} + r_K \omega_K. \tag{18.27}$$

Klein treated $r_K \omega_K$ as a constant fitted to the data, about 0.5 to 0.6 eV as quoted in the recent literature [13, Sect. II], [16, Sect. IV].

The truncation is (K2), and it can be checked where the phonon term vanishes. Without phonons, $\lambda_{\mathrm{AB}} \to 0$, the cascade yields $\varepsilon_{\mathrm{eh}} = 1.585 E_{\mathrm{g}}$ at $\theta_{\mathrm{ii}} = 1$ and $2.056 E_{\mathrm{g}}$ at $\theta_{\mathrm{ii}} = \frac{3}{2}$. The uniform population leads to $2.2 E_{\mathrm{g}}$ and $2.8 E_{\mathrm{g}}$, too high by 39 % and 36 %. The error lies entirely in $\bar{E}_{\mathrm{res}}$: the carriers do not fill the states below threshold uniformly. Each ionization splits the available energy three ways in proportion to the joint density of states, and the products that land below threshold land mostly near the bottom, as in random parking (Sect. 18.2). In the cascade $\bar{E}_{\mathrm{res}} = 0.293 E_{\mathrm{g}}$ and $0.528 E_{\mathrm{g}}$, against $0.6 E_{\mathrm{g}}$ and $0.9 E_{\mathrm{g}}$ from Eq. (18.26). Phonons change this. A carrier that emits phonons near threshold reaches its last ionization with less energy to spare and more often stalls just above threshold, so $\bar{E}_{\mathrm{res}}$ grows with λ_{AB}. Limit Check 18.6 shows the competition.

Limit Check 18.6. Klein relation

Klein's form of the pair energy [14, Eq. (3)], Eq. (18.27): $\varepsilon_{\mathrm{eh}} = \frac{14}{5} E_{\mathrm{g}} + r_K \omega_K$.

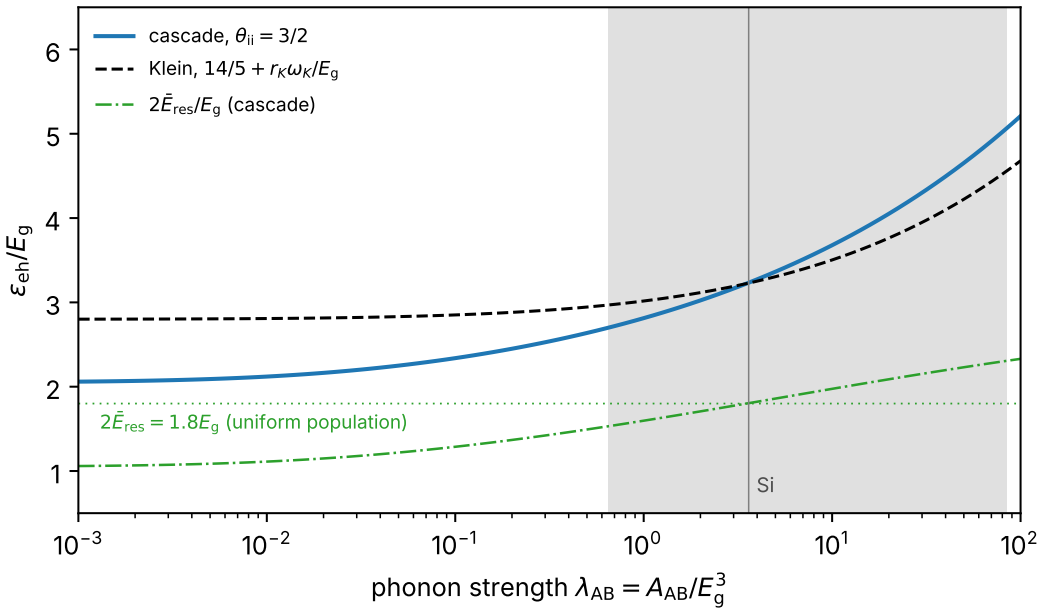


Figure 18.10: Pair energy in units of the gap against the phonon strength $\lambda_{\mathrm{AB}} = A_{\mathrm{AB}}/E_{\mathrm{g}}^3$ (Sect. 18.2.1). Solid: the energy-only random-k cascade with $\theta_{\mathrm{ii}} = \frac{3}{2}$, $w_{\mathrm{AB}} = 0.0543$ (the silicon value) and no hole cutoff. Dashed: Klein's relation with the cascade's own phonon term, $\frac{14}{5} + r_K \omega_K / E_{\mathrm{g}}$, so that the two curves differ only through the uniform population (K2); by Eq. (18.25) their difference is $2(0.9 E_{\mathrm{g}} - \bar{E}_{\mathrm{res}})/E_{\mathrm{g}}$. Dash-dotted:

$2\bar{E}_{\rm res}/E_{\rm g}$ of the cascade; dotted: the uniform-population value 1.8. Gray: $\lambda_{\rm AB}$ from 0.65 to 84, where Klein's relation lies within 10 % of the cascade. Vertical line: this $\theta_{\rm ii} = \frac{3}{2}$ cascade with its constant recalibrated to the silicon pair energy, $A_{\rm AB} = 5.629\,{\rm eV}^3$, $\lambda_{\rm AB} = 3.61$.

Below $\lambda_{\rm AB} \approx 0.65$, the regime of wide gaps and weak phonon loss, the residual energy falls far below the uniform value and Klein's relation overstates $\varepsilon_{\rm eh}$, by up to 36 %.

Klein's relation is exact where $\bar{E}_{\rm res} = 0.9E_{\rm g}$, at $\lambda_{\rm AB} = 3.5$. The $\theta_{\rm ii} = \frac{3}{2}$ cascade recalibrated to silicon sits almost on that point, at $\lambda_{\rm AB} = 3.61$, where Klein's form differs from it by 0.1 %. This is a coincidence of silicon's parameters, not a vindication of (K2): the calibrated model of Table 18.1, with $\theta_{\rm ii} = 1$ and $\bar{E}_{\rm res} = 0.718E_{\rm g}$, lies 11 % below Klein's form with its own phonon term. At small $\lambda_{\rm AB}$ the relation overshoots, by 22 % at $\lambda_{\rm AB} = 0.1$ and 36 % in the limit. Since $\lambda_{\rm AB} = A_{\rm AB}/E_{\rm g}^3$ falls steeply with the gap, wide-gap crystals sit at small $\lambda_{\rm AB}$: diamond, with $E_{\rm g} = 5.47\,{\rm eV}$, has $\lambda_{\rm AB} = 0.035$. The slope 14/5 is therefore too steep at large gaps, which is the known failure of Klein's line for diamond, 15.9 eV against a measured value near 13 eV [17, Sect. III]. At large $\lambda_{\rm AB}$ the comparison reverses: carriers stall above threshold, $\bar{E}_{\rm res}$ exceeds $0.9E_{\rm g}$, and Klein's form undershoots, by 10 % at $\lambda_{\rm AB} = 84$. The fitted constant $r_K\omega_K$ of the original relation absorbs part of either error at the gaps used in the fit, which is why the relation describes narrow- and medium-gap semiconductors and fails for the widest gaps.

18.4.5 The linear law of the pair energy

Measured pair energies have been fitted by straight lines in the gap since the 1950s, $\varepsilon_{\rm eh} = k_1E_{\rm g} + k_2$ [13, Sect. I]. The most recent fit, dominated by wide-gap crystals, is $1.54E_{\rm g} + 1.89\,{\rm eV}$ [13, Sect. II]. The cascade explains why such a line works and why its constant is not universal. Over the fit region $0.003 \le \lambda_{\rm AB} \le 100$, $0.01 \le w_{\rm AB} \le 0.1$, which contains 13 of the 24 crystals of Chap. 17, sixty solutions of the cascade are reproduced by

$$\frac{\varepsilon_{\rm eh}}{E_{\rm g}} = 1.612 + 2.329\,\bigl(\lambda_{\rm AB}\,w_{\rm AB}^{0.9637}\bigr)^{0.2403}, \tag{18.28}$$

to 0.61 % at most and 0.21 % on average (Fig. 18.11). This is an observed fit of the model, not a derivation; its exponents carry no meaning beyond the fit region. Inserting $\lambda_{\rm AB} = A_{\rm AB}/E_{\rm g}^3$ turns it into a nearly linear law in the gap, because the power of $E_{\rm g}$ in the second term, $1 - (3 + 0.9637) \times 0.2403 = 0.048$, is almost zero:

$$\varepsilon_{\rm eh} \approx 1.61\,E_{\rm g} + k_2, \qquad k_2 = 2.329\,\bigl(A_{\rm AB}\,\omega_{\rm AB}^{0.9637}\bigr)^{0.2403}\,E_{\rm g}^{0.048}, \tag{18.29}$$

energies in eV and $A_{\rm AB}$ in ${\rm eV}^3$. The slope is the phonon-free part of the pair energy at the random-k threshold, 1.61 against the exact 1.585 of Sect. 18.4.4; the constant k_2 is the phonon loss. Since $A_{\rm AB}$ is the same for all crystals, k_2 depends on the phonon energy, as $\omega_{\rm AB}^{0.23}$, and only weakly on the gap. For silicon it is 1.88 eV. Over the 24 crystals of Chap. 17 it spans 1.21 to 2.53 eV, and 1.51 to 2.53 eV for those inside the fit region: soft lattices, with small $\omega_{\rm AB}$, lose less to phonons, and stiff ones more.

Limit Check 18.7. Linear law of the pair energy

The observed linear law of the cascade, Eq. (18.29), $\varepsilon_{\rm eh} \approx 1.61E_{\rm g} + k_2$ with $k_2 = 1.2$ to 2.5 eV, agrees in form with the empirical fit $\varepsilon_{\rm eh} = 1.54E_{\rm g} + 1.89\,{\rm eV}$ [13, Sect. II].

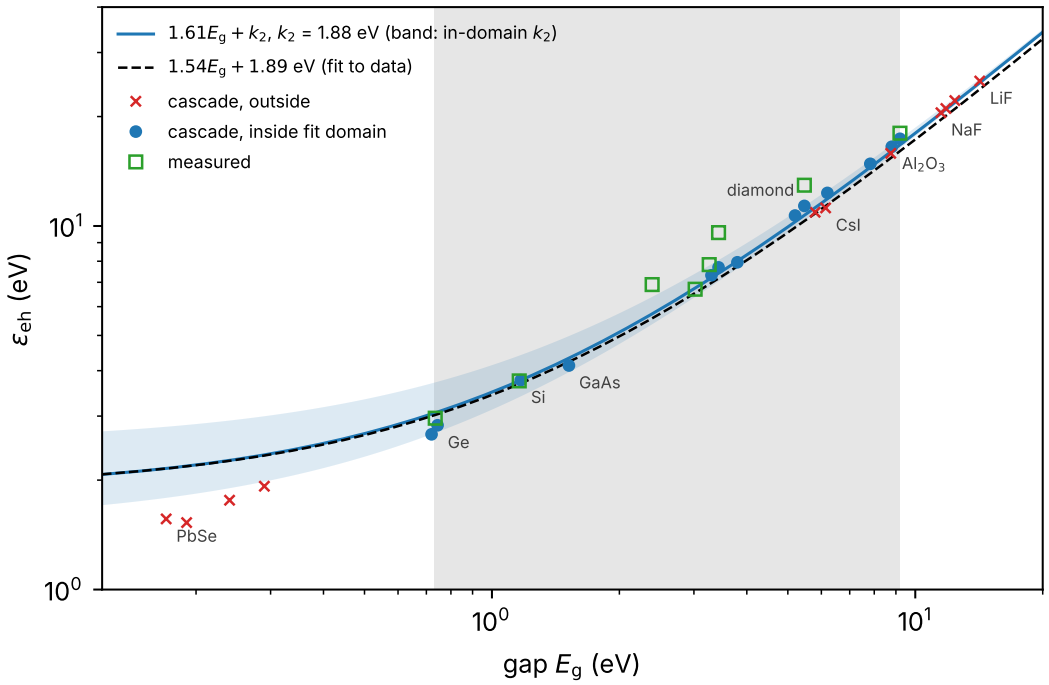


Figure 18.11: Pair energy against the gap. Solid line: Eq. (18.29) with the silicon value $k_2 = 1.88\,\text{eV}$; band: k_2 over the crystals inside the fit region, 1.51 to 2.53 eV. Dashed: the empirical line. Circles: the cascade with the constant Eq. (18.23) for the 24 crystals of Griffin and coworkers [18, Table I] (gap and highest optical phonon energy; silicon, germanium and GaAs at their low-temperature gaps of Table 3.1); filled inside the fit region $0.003 \le \lambda_{AB} \le 100$, $0.01 \le w_{AB} \le 0.1$ of Eq. (18.28), crosses outside it. Squares: measured pair energies of Table 18.2 and silicon. Gray: the gaps of the measured crystals, 0.73 to 9.2 eV. Slopes fitted for seed energies of 12.5 to $25E_g$ (30 to $60E_g$ for $\lambda_{AB} > 5$), a window that changes the silicon value by less than 0.03 % against the 20 to $40E_g$ of Table 18.1.

Outside the fit region, the law with each crystal's own k_2 still follows the cascade to 1.4 % for these 24 crystals, but there it extrapolates the model beyond any comparison with data: the lead salts and InSb have $\lambda_{AB} > 200$, and the halides and fluorides a phonon step below $0.006E_g$.

The two lines have the same form for different reasons. The empirical constant 1.89 lies inside the range of k_2 and close to the value for silicon. The empirical slope 1.54 sits below the model's 1.61, pulled down by diamond and the nitrides, for which the model with the silicon constant gives too little (Sect. 18.4.6). Nielsen and Fischetti note that the measured $\varepsilon_{eh}(E_g)$ is not linear at all [13, Sect. II]; in the model, the nonlinearity sits in k_2, which changes from crystal to crystal with the phonon energy.

Worked example 18.4: Pair energy of GaAs from the linear law

A GaAs detector at millikelvin has $E_g = 1.52\,\text{eV}$ (Table 3.1) and a longitudinal optical

Table 18.2: Pair energies of the cascade with the silicon constant Eq. (18.23) against measurements. Inputs: germanium as in Table 18.1 (77 K); diamond, GaN and α-quartz, gap and highest optical phonon energy of Griffin and coworkers [18, Table I]; the SiC polytypes, gap and longitudinal optical energy of Griffin and coworkers [19, Table I]. Measurements: germanium [15, Sect. 4.3]; diamond, the inverse-variance mean of four measurements (13.1, 13.25, 13.05 and 12.82 eV) compiled by Nielsen and Fischetti [17, Sect. III]; 4H-SiC [20]; 6H-SiC as quoted by Griffin and coworkers [19, Table I]; 3C-SiC and SiO_2 as quoted by Nielsen and Fischetti [13, Table II]; GaN, electron-beam induced current, as quoted by Nielsen and coworkers [16, Sect. IV]. Residual: model over measured, minus one. Last column: the density-functional full-band Monte Carlo of Nielsen and coworkers [13, Table II], [17, Sect. III], [16, Sect. IV].

	E_g (eV)	ω_{AB} (meV)	λ_{AB}	w_{AB}	model (eV)	measured (eV)	residual	full band (eV)
Ge	0.735	37.6	14.4	0.051	2.815	2.96	−4.9 %	–
3C-SiC	2.39	120.5	0.42	0.050	6.121	6.90	−11.3 %	6.6
6H-SiC	3.02	120.0	0.21	0.040	7.167	6.7	+7.0 %	–
4H-SiC	3.26	120.0	0.16	0.037	7.565	7.83 ± 0.02	−3.4 %	7.3
GaN	3.43	88.9	0.14	0.026	7.694	9.59	−19.8 %	8.9
diamond	5.47	161	0.035	0.029	11.357	12.95	−12.3 %	12.87
SiO_2	9.2	149	0.0073	0.016	17.373	18.0	−3.5 %	16.7

phonon at $\omega_{AB} = 36.35$ meV (Sect. 5.5). With Eq. (18.23),

$$\lambda_{AB} = \frac{5.703}{1.52^3} = 1.624, \qquad w_{AB} = \frac{0.03635}{1.52} = 0.02391.$$

The combination in Eq. (18.28) is $\lambda_{AB} w_{AB}^{0.9637} = 0.04448$, and its power 0.2403 is 0.4733. Then

$$\frac{\varepsilon_{eh}}{E_g} = 1.612 + 2.329 \times 0.4733 = 2.714, \qquad \varepsilon_{eh} = 2.714 \times 1.52\,\text{eV} = 4.125\,\text{eV}.$$

In the form Eq. (18.29), $1.61 E_g = 2.45$ eV and $k_2 = 1.68$ eV, smaller than silicon's 1.88 eV because the GaAs phonon carries 0.58 of the silicon phonon energy. The cascade itself yields 4.132 eV (Table 18.1), so the fit costs 0.2 %. The empirical line gives 4.23 eV and Klein's relation with $r_K \omega_K = 0.5$ eV leads to 4.76 eV. At the room-temperature gap of 1.42 eV the linear law gives 3.96 eV, against the measured value of about 4.2 eV [13, Fig. 1]: the residual of the model in Table 18.1, carried through the fit. For a GaAs detector the defensible pair energy is therefore ≈ 4.1 eV with an uncertainty set by the model's residuals, not by the fit.

18.4.6 The validated range

Table 18.2 compares the cascade with the measured pair energies of seven crystals besides silicon, which fixes the constant. The inputs are the gap and the highest optical phonon energy of each crystal, longitudinal for the SiC polytypes.

Over the seven crystals the residuals run from −19.8 % to +7.0 %, with a mean of −6.9 % and a root mean square of 10.5 %. The model is good to 7 % for germanium, 4H- and 6H-SiC and quartz, and it falls short by 11 to 20 % for 3C-SiC, diamond and GaN. Six of the seven residuals are negative, as are those of germanium and GaAs in Table 18.1: with the silicon constant the model tends to give too little heat per pair. The likeliest culprit is the universal

constant, since the light-atom and polar lattices that it misses most have phonon couplings unlike those of silicon, but the comparison does not prove it. The density-functional full-band Monte Carlo does better for diamond, 3C-SiC and GaN and worse for 4H-SiC and quartz, and it has no fitted constant. Its 3.1 eV for silicon, 14 % below the room-temperature value of about 3.6 eV that its authors compare with, and 17 % below the 3.749 eV at 123 K, shows that it is not yet a calibration-free substitute either [13, Sect. III].

The Fano factor is barely tested beyond silicon. The germanium measurements span a factor two, and the only wide-gap values, both for 4H-SiC, disagree with each other: $F_{\mathrm{eh}} = 0.1$ from a linewidth and an upper limit of 0.04 from another, as compiled by Gsponer and coworkers [20, Table 2]. The model gives 0.093. If the upper limit holds, the model fails for 4H-SiC, since 0.04 lies below every Fano factor the cascade produces in this section; a linewidth limit, however, rests on subtracting the noise from the measured width [20, Eq. (3)], and an overestimated noise drives it down. Which of the two applies cannot be decided from the published data.

These comparisons define the *validated range* of the cascade model: the region of phonon strength and phonon step spanned by the crystals with a measured pair energy, $0.007 \leq \lambda_{\mathrm{AB}} \leq 14.4$ and $0.016 \leq w_{\mathrm{AB}} \leq 0.054$, gaps from 0.73 to 9.2 eV. Inside it a model pair energy carries the spread of the residuals, -20 to $+7$ %, with the lower part more likely for light-atom and polar crystals. Of the crystals ranked in Chap. 17, silicon, germanium, GaAs, diamond, quartz, $CaWO_4$, AlN, GaN and ZnO lie inside it (Fig. 18.11). Sapphire and MgO lie just outside, with phonon steps of 0.012 and 0.011, and GaSb at its edge, with $\lambda_{\mathrm{AB}} = 15.3$.

Three groups lie well outside, each for a physical reason. The alkali halides and the fluorides, NaI, CsI, NaCl, LiF, NaF, CaF_2 and MgF_2, have phonon steps of 0.0016 to 0.006 of the gap. A carrier near threshold then emits hundreds of phonons before it ionizes, a regime that no measured crystal samples, and their narrow anion valence bands, which the model treats without a cutoff, confine the holes far more than in any validated crystal. They also bind excitons by an eV or more. A carrier between $E_{\mathrm{g}} - E_x^b$ and the threshold can create a bound pair directly (Appendix E.24); the cascade of Sect. 18.1 has no such channel, and whether a bound exciton counts as a pair depends on the readout (Sect. 18.5). The lead salts and InSb have gaps below 0.3 eV and λ_{AB} from 230 to 1160: their cascades are dominated by phonons, and at gaps this small processes absent from the model, Auger recombination and scattering by free carriers, can compete with impact ionization. For these crystals the model numbers in Fig. 18.11, for example 25.0 eV for LiF, 20.5 eV for NaF, 11.2 eV for CsI and 1.56 eV for PbSe, are estimates without a measured anchor, and any guide to new targets must mark them as such. Neither the pair energy nor the Fano factor of sapphire, $CaWO_4$, LiF or NaF has been measured. The first-principles cascades of Appendix E.24, with Fröhlich vertices for the polar crystals and the Bethe–Salpeter equation for the halides, are the route to replacing these estimates.

18.5 Capture and excitons

A carrier in silicon whose kinetic energy has fallen below the impact-ionization threshold $\theta_{\mathrm{ii}} E_{\mathrm{g}}$ (Sect. 18.1) can no longer add a pair. It reaches the band edge by emitting phonons within about a picosecond, and from then on the number of pairs N_{eh} whose mean and variance Sect. 18.2 computes is fixed. A detector at 50 mK reads the event microseconds to seconds later. In that interval an electron can be captured by an ionized donor, or an electron and a hole can bind into an exciton. The exciton can bind to a neutral impurity, and the pair can recombine, giving its energy to the lattice or to a photon that leaves the crystal. Each of these processes changes what a charge or a heat readout records, and none of their rates has

been measured at millikelvin temperatures in silicon or germanium. Without the rates one can still derive how each process changes the energy balance and the pair number. A captured electron costs exactly one gap; a bound pair costs nothing in pair number and returns its binding energy as heat. Both change the Fano factor that a readout measures in ways that follow from counting alone.

18.5.1 End-of-cascade capture

At the end of the cascade every carrier sits at a band edge, and the deposit is shared by the energy balance Eq. (18.5): E_g for each pair and the heat E_heat for the phonons emitted on the way down. This holds for an electron recoil, with no energy stored in defects and no photon leaving the crystal. Once carriers can be captured, not all of the energy outside the pairs is heat, and we name it separately,

$$E_\mathrm{np} \equiv \omega - N_\mathrm{eh} E_\mathrm{g}, \tag{18.30}$$

the energy not held by pairs, equal to E_heat as long as nothing is captured. At fixed ω the pair number and E_np fluctuate together, $\mathrm{Var}\, N_\mathrm{eh} = \mathrm{Var}\, E_\mathrm{np}/E_\mathrm{g}^2$. From here on N_eh counts the electrons in the conduction band, free or bound to a hole in an exciton. An electron captured on an impurity level no longer counts.

Let an electron be captured on a level E_j below the conduction-band edge E_c. The energy of a charge in the crystal is fixed relative to the Fermi level μ_F of the electrodes, since that is the energy the carrier gives up when it reaches one. A free electron holds $E_c - \mu_F$, a free hole $\mu_F - E_v$, with E_v the valence-band edge, and a pair the sum, E_g. After the capture the electron holds $E_j - \mu_F$, the phonons emitted at capture carry $E_c - E_j$, and the hole, now without a partner, still holds $\mu_F - E_v$. The three shares add to E_g whatever the level, and with N_eh lowered by one, Eq. (18.30) counts all three in E_np; only the share $E_c - E_j$ is heat. A captured hole leaves N_eh unchanged; it changes the charge that the electrodes see, which is a matter of the readout.

Whether the electrons are captured is random. Let electron i be captured with probability f_i, independently of the others once the cascade is given. The number of electrons left, N', then has conditional mean $\sum_i (1-f_i)$ and conditional variance $\sum_i f_i(1-f_i)$, a sum of Bernoulli variances. The law of total variance, $\mathrm{Var}\, N' = \langle \mathrm{Var}(N'|\mathrm{cascade})\rangle + \mathrm{Var}\langle N'|\mathrm{cascade}\rangle$, gives

$$\langle N'\rangle = \Big\langle \sum_i (1-f_i) \Big\rangle, \qquad \mathrm{Var}\, N' = \Big\langle \sum_i f_i(1-f_i) \Big\rangle + \mathrm{Var}\Big[\sum_i (1-f_i)\Big], \tag{18.31}$$

with the averages taken over cascades. If every electron has the same capture probability f_cap, unrelated to the cascade, the sums become $(1-f_\mathrm{cap})N_\mathrm{eh}$ and $f_\mathrm{cap}(1-f_\mathrm{cap})N_\mathrm{eh}$, and the pair number is thinned binomially.

Capture at the end of the cascade

Each electron captured before the readout raises the energy not held by pairs by exactly E_g,

$$E_\mathrm{np} \to E_\mathrm{np} + E_\mathrm{g}, \tag{18.32}$$

of which $E_c - E_j$ goes to phonons at the capture, $E_j - \mu_F$ stays with the trapped electron and $\mu_F - E_v$ with the unpaired hole. Let the capture probability f_cap be equal for all electrons and independent of the cascade. The mean energy per counted pair and the Fano factor of the counted pairs are then

$$\varepsilon'_\mathrm{eh} = \frac{\varepsilon_\mathrm{eh}}{1-f_\mathrm{cap}}, \qquad F'_\mathrm{eh} = (1-f_\mathrm{cap})F_\mathrm{eh} + f_\mathrm{cap}. \tag{18.33}$$

A capture probability that depends on the carrier's state at the end of the cascade requires the general form Eq. (18.31).

The Fano factor of silicon is small, and Eq. (18.33) makes it fragile: a loss raises it by $f_{\text{cap}}(1 - F_{\text{eh}})$, almost one for one. The cascade value 0.114 and the measured 0.119 ± 0.002 (Sect. 18.4) differ by an amount that a loss of $f_{\text{cap}} = 0.54\,\%$ (0.32 to 0.77 %) would produce, and the same loss raises ε_{eh} by 0.55 %, which the calibration of the cascade on the measured pair energy absorbs. The Fano factor alone therefore cannot separate a sub-percent loss of pairs from the truncations of the cascade itself.

The binomial form fails when the capture probability depends on the carrier. The probability of capture depends on the energy x_i that carrier i still has after its last impact ionization. Those terminal energies are correlated with the pair number: Sect. 18.2 found that the residual energies of the terminal carriers anticorrelate with the phonon energy emitted above threshold. With $f_i = f(x_i)$ the variance term $\operatorname{Var}\sum_i (1 - f_i)$ is not, in general, $(1 - \bar{f})^2 \operatorname{Var} N_{\text{eh}}$ with $\bar{f}$ the mean capture probability, and Eq. (18.31) has to be evaluated on the distribution of terminal energies that the cascade produces.

Capture is a collision of the kind that Sect. 18.1 turned into master-formula rates, with the potential of the impurity coupling the free carrier to a bound state and phonons carrying off the energy. Its size follows from a simpler argument. A carrier with kinetic energy E is bound by an attractive center only if it comes within the distance at which the Coulomb energy equals its kinetic energy, $r_C(E) = \alpha/(\varepsilon_0 E)$, and loses energy there. With $\hbar c$ restored, $\alpha = 1.44\,\text{eV nm}$, and with the static constant of silicon at liquid-helium temperature, $\varepsilon_0 = 11.4$ (Sect. 7.4), $\alpha/\varepsilon_0 = 0.126\,\text{eV nm}$, so a carrier of 0.1 eV has $r_C = 1.26\,\text{nm}$ and a capture cross-section of at most $\pi r_C^2 = 5.0 \times 10^{-14}\,\text{cm}^2$; at 0.35 eV, $r_C = 0.36\,\text{nm}$ and $\pi r_C^2 = 4.1 \times 10^{-15}\,\text{cm}^2$. At a density of centers of $10^{12}\,\text{cm}^{-3}$ the mean free path for capture is then at least 20 cm, and it falls to micrometers only near $10^{18}\,\text{cm}^{-3}$. Capture of hot carriers during the cascade is negligible in a pure crystal. The cold carriers that the cascade leaves behind are a different matter. At the thermal energy of 50 mK, $k_B T = 4.3\,\mu\text{eV}$, the Coulomb radius is $\alpha/(\varepsilon_0 k_B T) = 29\,\mu\text{m}$, and a cold carrier finds any ionized center within tens of micrometers. The silicon detectors of CDMS II, the second phase of the Cryogenic Dark Matter Search (CDMS), for example, were operated with their ionized trapping sites neutralized by regular exposure to infrared light-emitting diodes, which left the bulk in a metastable state with low trapping [21, Sect. 2]. Capture on neutral centers and capture during drift belong to the readout, and every one of them adds the same E_{g} to E_{np}.

Worked example 18.5: One captured electron in ten pairs

An electron recoil of $\omega = 37.5\,\text{eV}$ in silicon makes about ten pairs at the measured $\varepsilon_{\text{eh}} = 3.75\,\text{eV}$ of Sect. 1.5. Take an event with exactly ten. With $E_{\text{g}} = 1.17\,\text{eV}$, the energy balance gives $E_{\text{heat}} = E_{\text{np}} = 37.5 - 10 \times 1.17 = 25.80\,\text{eV}$ in phonons. Let one electron be captured by an ionized phosphorus donor, whose ground state lies 45.6 meV below the band edge (Sect. 7.4). The pair number falls to nine and $E_{\text{np}} = 37.5 - 9 \times 1.17 = 26.97\,\text{eV}$, larger by exactly E_{g}. Of that 1.17 eV, 45.6 meV is emitted as phonons when the electron drops onto the donor, and 1.124 eV stays as the potential energy of the trapped electron and of its unpaired hole, divided between them according to where μ_F lies. A charge readout calibrated at 3.75 eV per pair reports $9 \times 3.75 = 33.75\,\text{eV}$; the energy per counted pair of this event is $37.5/9 = 4.17\,\text{eV}$, 11.1 % above ε_{eh}.

Now let each of the electrons be captured with probability $f_{\text{cap}} = 0.1$ instead. The mean number counted is nine, the same as before, and $\varepsilon'_{\text{eh}} = 3.75/0.9 = 4.17\,\text{eV}$ by Eq. (18.33).

With the measured $F_{\rm eh} = 0.119$, however, the Fano factor rises to $F'_{\rm eh} = 0.9 \times 0.119 + 0.1 = 0.207$, and the standard deviation of the counted pairs from $(0.119 \times 10)^{1/2} = 1.09$ to $(0.207 \times 9)^{1/2} = 1.37$. A capture of fixed size shifts the yield; a capture at random also broadens it. Of the variance $0.207 \times 9 = 1.86$, the thinned cascade contributes $0.9^2 \times 1.19 = 0.96$ and the binomial choice of which electrons are captured $0.1 \times 0.9 \times 10 = 0.90$, almost as much.

18.5.2 Excitons and the measured Fano factor

A cold electron and hole left near each other at zero field bind. The binding energy of the silicon exciton, $E_x^b = 14.7\,\text{meV}$ (Sect. 4.9), is 3400 times $k_B T$ at 50 mK, and a pair closer than the 29 µm Coulomb radius found above cannot be pulled apart by thermal motion. Binding is a two-body process that waits for both carriers to cool by emitting acoustic phonons, slow at millikelvin temperatures, so it takes place after the cascade has fixed $N_{\rm eh}$. Since $N_{\rm eh}$ counts bound pairs, an exciton changes neither $N_{\rm eh}$ nor $\varepsilon_{\rm eh}$ nor $F_{\rm eh}$. What it changes is where the energy sits: a bound pair holds $E_{\rm g} - E_x^b$, and the binding energy goes to the lattice.

A heat readout, however, sees more than $E_{\rm np}$. With N_X bound pairs at readout and no capture, the phonon energy is $E_{\rm heat} = E_{\rm np} + N_X E_x^b$. Its mean rises by E_x^b per bound pair, at most $E_x^b/\varepsilon_{\rm eh} = 0.39\,\%$ of ω in silicon. Define the Fano factor inferred from the phonon energy, $F_{\rm heat} \equiv \text{Var}\, E_{\rm heat}/(E_{\rm g}^2 \langle N_{\rm eh}\rangle)$. It differs from $F_{\rm eh} = \text{Var}\, E_{\rm np}/(E_{\rm g}^2\langle N_{\rm eh}\rangle)$ by the variance of N_X and its covariance with $E_{\rm np}$,

$$F_{\rm heat} = F_{\rm eh} + \frac{(E_x^b)^2\,\text{Var}\, N_X + 2E_x^b\,\text{Cov}(E_{\rm np}, N_X)}{E_{\rm g}^2\langle N_{\rm eh}\rangle}. \tag{18.34}$$

When every pair binds, $N_X = N_{\rm eh}$ and $E_{\rm heat} = \omega - N_{\rm eh}(E_{\rm g} - E_x^b)$, so $F_{\rm heat} = F_{\rm eh}(1 - E_x^b/E_{\rm g})^2$: 2.5 % below $F_{\rm eh}$ in silicon, and 0.7 to 1.3 % in germanium, whose indirect exciton is bound by a few meV, 2.5 and 5 meV in absorption [22, 23], against a gap of 0.744 eV.

A charge readout is affected only if the exciton is never split. A bound pair is neutral and drifts as a whole, so it induces no charge on the electrodes. Let each pair independently end as such an exciton with probability $p_{\rm ns}$. The counted charge is then thinned exactly as by capture, and by the algebra of Eq. (18.33) its Fano factor is

$$F_{\rm meas} = (1 - p_{\rm ns})F_{\rm eh} + p_{\rm ns}, \tag{18.35}$$

where $F_{\rm meas}$ is the Fano factor of the measured charge, before the corrections of the readout itself. A heat readout thus sees $F_{\rm heat}$ and a charge readout $F_{\rm meas}$, while $F_{\rm eh}$, which counts bound and free pairs alike, is the cascade value that neither changes. In a biased detector the field can separate the pair before it binds. The force $e\mathcal{E}$ of a field $\mathcal{E}$ exceeds the attraction $\alpha/(\varepsilon_0 r^2)$ beyond the radius

$$r_{\mathcal{E}} = \Big(\frac{\alpha}{\varepsilon_0 e \mathcal{E}}\Big)^{1/2}, \tag{18.36}$$

which in silicon is 0.56 µm at 4 V/cm and 0.11 µm at 100 V/cm. A pair that ends the cascade farther apart than $r_{\mathcal{E}}$ is separated; a closer one can bind, and once bound it stays bound, since the field that ionizes the exciton is of order $E_x^b/(ea_x) = 3.5 \times 10^4\,\text{V/cm}$, with $a_x = 4.2\,\text{nm}$ (Sect. 4.9). Exciton formation thus changes the measured Fano factor through Eq. (18.35) when the readout field is weak enough that $r_{\mathcal{E}}$ exceeds the separation of the pairs at the end of the cascade, a distribution the cascade calculation has to supply (Appendix E.24). When excitons form from carriers of different pairs, as in a dense track, the loss is not independent

from pair to pair, its probability grows with the density, and the variance acquires a part that grows faster than $\langle N_{\mathrm{eh}} \rangle$.

One process acts on the cascade itself. A carrier with energy between $E_{\mathrm{g}} - E_x^b$ and the ionization threshold can create a bound pair directly, which acts for that pair like a gap lowered by E_x^b. With the slope $\partial \varepsilon_{\mathrm{eh}} / \partial E_{\mathrm{g}} = 1.70$ of the calibrated cascade at fixed A_{AB} (Sect. 18.4.3), the pair energy moves by at most $1.70 E_x^b = 25\,\mathrm{meV}$ in silicon, 0.67 %, and 8.6 meV in germanium, 0.30 %, for the larger of its two measured binding energies. In ionic crystals with excitons bound by about 1 eV, the bound pair and not the free pair may be the unit the cascade produces, and the excitons self-trap (Sect. 10.5); their pair energies lie outside the range in which the cascade model was validated (Sect. 18.4).

18.5.3 Rate equations at millikelvin

After the cascade each pair is in one of four kinds of state: a free pair F, a free exciton X, an exciton B_j bound to a neutral impurity of species j, or a pair with a carrier trapped on a center of species j. In the dilute clouds of electron recoils each pair evolves on its own, with its own partner or with a fixed background center. Every transition is then first order in the occupation of the pair's own state. For the occupation probabilities p_F, p_X and $p_{B,j}$ of one pair,

$$\begin{aligned} \frac{\mathrm{d}p_F}{\mathrm{d}t} &= -(\Gamma_{FX} + \Gamma_{FT})\, p_F, \\ \frac{\mathrm{d}p_X}{\mathrm{d}t} &= \Gamma_{FX}\, p_F - \Big(\Gamma_X + \sum_j \Gamma_{XB,j}\Big) p_X, \\ \frac{\mathrm{d}p_{B,j}}{\mathrm{d}t} &= \Gamma_{XB,j}\, p_X - \Gamma_{B,j}\, p_{B,j}. \end{aligned} \tag{18.37}$$

Here Γ_{FX} is the rate at which the pair binds and Γ_{FT} the rate at which one of its carriers is trapped. The free exciton decays at the rate Γ_X, a fraction η_X of it by luminescence, and binds to impurities of density n_j at the rate $\Gamma_{XB,j} = \sigma_{XB,j} v_X n_j$, with cross-section $\sigma_{XB,j}$ and exciton speed v_X. The bound exciton decays at the rate $\Gamma_{B,j}$, a fraction $\eta_{B,j}$ of it by luminescence. In silicon a bound exciton decays mostly by an Auger process: the pair recombines and gives its energy to the extra carrier of the complex, which is ejected and leaves the impurity ionized. Its lifetime at the phosphorus donor is 200 to 300 ns at liquid-helium temperature, as quoted by Lo and coworkers [24]. No reverse steps appear: dissociating an exciton or freeing it from an impurity costs E_x^b or more, thousands of times $k_B T$ at 50 mK.

The trapped carriers follow the occupation n_j^{tr} of each species of center,

$$\frac{\mathrm{d}n_j^{\mathrm{tr}}}{\mathrm{d}t} = \sigma_{\mathrm{cap},j} u\, n_{\mathrm{c}} (n_j - n_j^{\mathrm{tr}}) - \Gamma_j^{\mathrm{em}} n_j^{\mathrm{tr}} - \sigma_j^{\mathrm{rec}} u\, n_{\mathrm{v}} n_j^{\mathrm{tr}}, \tag{18.38}$$

written for electrons, with free-electron and free-hole densities n_{c} and n_{v} and carrier speed u. The center captures with the cross-section $\sigma_{\mathrm{cap},j}$, emits at the rate Γ_j^{em}, and loses its electron to a free hole with the cross-section σ_j^{rec}; holes obey the same equation with the roles exchanged. Thermal emission is frozen. For the phosphorus donor the Boltzmann exponent at 50 mK is $(45.6\,\mathrm{meV})/k_B T = 1.06 \times 10^4$. A field lowers the barrier of a Coulomb center: the potential $-\alpha/(\varepsilon_0 r) - e\mathcal{E}r$ along the field has its maximum at $r_{\mathcal{E}}$ of Eq. (18.36), lower than the zero-field barrier by $2(\alpha e \mathcal{E}/\varepsilon_0)^{1/2}$, the Poole–Frenkel lowering. At 100 V/cm it is 2.25 meV, and the exponent stays at 1.0×10^4. A trapped carrier is released only athermally, by the phonons and hot carriers of later events or by infrared light, so the occupations carry the history of earlier events into later ones, the source of delayed charge (Appendix E.25).

Without reverse steps the matrix of Eq. (18.37) is lower triangular in the order F, X, B_j, and its decay constants γ_k are its diagonal entries, $\Gamma_{FX} + \Gamma_{FT}$, $\Gamma_X + \sum_j \Gamma_{XB,j}$ and $\Gamma_{B,j}$. The

fraction of the gap energy still held by pairs and excitons,

$$r_{\mathrm{gap}}(t) = p_F + p_X + \sum_j p_{B,j} = \sum_k w_k \, \mathrm{e}^{-\gamma_k t}, \qquad \sum_k w_k = 1, \tag{18.39}$$

decays as a sum of exponentials when the γ_k are distinct, up to the binding energies. An exciton bound to a phosphorus donor sits a further 5 meV or so below the free exciton [24], so r_{gap} overstates the held energy by at most $(E_x^b + 5\,\mathrm{meV})/E_{\mathrm{g}} = 1.7\,\%$ in silicon. The weights w_k need not be positive: a transfer from X to B_j without loss delays the decay, and its weight is negative. To see where the gap energy ends up, we integrate the occupations over time, $\int_0^\infty p \, \mathrm{d}t = -M^{-1} p(0)$ for the matrix M of Eq. (18.37) and a pair that starts free. They are $\int p_F = 1/(\Gamma_{FX} + \Gamma_{FT})$, $\int p_X = \Gamma_{FX} \int p_F / (\Gamma_X + \sum_j \Gamma_{XB,j})$ and $\int p_{B,j} = \Gamma_{XB,j} \int p_X / \Gamma_{B,j}$, and multiplying each by its outgoing rates gives

$$b_{\mathrm{trap}} = \frac{\Gamma_{FT}}{\Gamma_{FX} + \Gamma_{FT}}, \qquad b_{\mathrm{light}} = (1 - b_{\mathrm{trap}}) \, \frac{\Gamma_X \eta_X + \sum_j \Gamma_{XB,j} \eta_{B,j}}{\Gamma_X + \sum_j \Gamma_{XB,j}}, \qquad b_{\mathrm{heat}} = 1 - b_{\mathrm{trap}} - b_{\mathrm{light}} \tag{18.40}$$

for the fractions of the gap energy that end trapped, as escaping light and as phonons. The trapped and escaping fractions are lost to a heat readout at any window, while the phonon fraction arrives only as fast as r_{gap} decays.

Fig. 18.12 shows the populations of a pair in silicon at 50 mK for one set of rates. The bound-exciton decay uses the measured Auger lifetime; the other rates are chosen only to show how the solution behaves. The pair binds within a nanosecond, the bound-exciton population peaks at 0.15 near 0.45 µs, half of the gap energy is still held by pairs and excitons at 0.83 µs, when phonons carry 0.445 of it, and 1 % at 4.4 µs. The weights of r_{gap} are +1.28 for the free exciton, −0.33 for the bound one and +0.05 for the free pair. A heat readout that closes its window at 1 µs misses the 42 % of the gap energy still held in this model, plus the 5.6 % already trapped or radiated, and the window dependence enters the heat yield of any phonon readout. In the dense tracks of nuclear recoils the pairs are not independent, and two-body processes take over. Binding of carriers from different pairs moves gap energy from pairs to excitons and leaves r_{gap} unchanged; exciton–exciton annihilation and Auger recombination of free carriers remove it, and they make r_{gap} fall as a power of time at a rate set by the track density. Since that density differs between nuclear and electron recoils of the same energy, so does r_{gap} at a given window.

18.5.4 Unknown rates and unsized truncations

The rates of Eqs. (18.37) and (18.38) are the main unknowns of this section. In silicon and germanium at 10 to 50 mK, no measurement or calculation gives the rate at which a cold pair binds, or the coefficient of binding between carriers of different pairs. The same holds for the capture cross-sections of cold and drifting carriers on charged and neutral centers, as functions of carrier energy, temperature and field, and for the cross-sections for binding of excitons to each impurity species. Also unknown are the decay rate and luminescent fraction of the free exciton, the luminescent fractions of the bound excitons, and the coefficients of exciton–exciton annihilation and of Auger recombination that govern dense tracks. The thresholds at which dense pairs condense into electron–hole droplets, the degenerate electron–hole liquid of germanium and silicon, are open as well at millikelvin temperatures. What is known are the binding energies, 14.7 meV in silicon and a few meV in germanium, and the Auger lifetimes of bound excitons at liquid-helium temperature, a property of the complex in its ground state that should carry over to lower temperatures. The excitons themselves cool by emitting acoustic phonons, slowly at these temperatures, so their temperature need not be that of

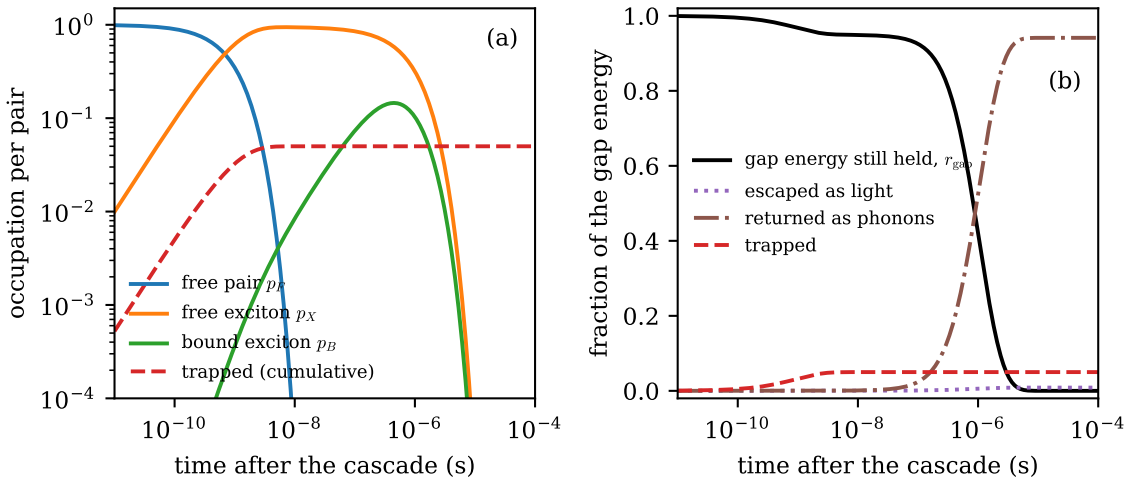


Figure 18.12: End-of-cascade populations of one electron–hole pair in silicon at 50 mK, from the rate equations Eq. (18.37) with one species of neutral impurity. (a) Occupations of the free pair, the free exciton and the bound exciton, and the cumulative trapped fraction, against the time after the cascade. (b) Fraction of the gap energy still held by pairs and excitons, r_{gap} of Eq. (18.39), and the cumulative fractions returned as phonons, escaped as light and trapped. Rates: bound-exciton Auger decay 250 ns, inside the measured 200 to 300 ns of the phosphorus donor [24], without light. Illustrative, since none is measured at 50 mK: binding of the pair 1 ns, trapping $\Gamma_{FT} = 0.053\Gamma_{FX}$, binding of the free exciton to impurities 1 µs, free-exciton decay 10 µs with luminescent fraction 0.1. Asymptotic split Eq. (18.40): 0.050 trapped, 0.009 light, 0.941 phonons.

the lattice, and rates measured at a few kelvin cannot simply be transferred. Appendix E.24 specifies the calculation of the exciton and capture parameters and Appendix E.25 the charge and exciton kinetics of a detector.

The cascade hands over less than these equations need. No material has a computed capture probability $f(x)$ as a function of the terminal energy of a carrier, nor the distribution of terminal energies on which Eq. (18.31) averages it. Nor is the separation of the pairs at the end of the cascade, which decides through Eq. (18.36) whether a field prevents binding. In the ionic crystals, where the cascade quantum may be a bound pair, the exciton binding energies and the binding to impurities are open as well.

Two approximations of the cascade itself remain unsized. A constant factor of 0.8 on the impact-ionization rate, the quasiparticle residue left out of Eq. (18.1), changes $\varepsilon_{\mathrm{eh}}$ by about 3 %, and calibrating on the measured pair energy removes it (Appendix E.24). A factor that varies with the carrier energy between threshold and about 10 eV, as the residue and the carrier–electron exchange may, changes F_{eh}, and its size is not known. For the energy-only reduction, the probability that a carrier keeps its momentum between two events that change its energy is estimated to be below about 0.1 within 1 eV of threshold, where intervalley scattering dominates, and to reach one half above 3 to 5 eV in silicon. A full-band cascade finds the resulting memory negligible, but only within the deformation-potential model of the phonons (Sect. 18.1.4); with first-principles couplings, which do not redraw the crystal momentum at every emission, neither this probability nor the spread over the iso-energy surface that it multiplies has been computed.

Summary

A hot carrier in a semiconductor is a projectile of the master formula (Sect. 18.1). Its impact ionization is the rate Eq. (18.1), with one coupling row per reciprocal-lattice vector and the response of the total charge. The kernel is therefore the screened interaction and, without local fields, the loss function at the momentum the carrier gives up; the plasmon of silicon is inside that kernel and is not a separate channel. Phonon emission is the same formula with the electron–phonon vertex, and in a polar crystal only its short-range part may be added to the screened interaction, which already contains the Fröhlich line. Momentum conservation puts the threshold at $\theta_{\mathrm{ii}} E_{\mathrm{g}}$, with $\theta_{\mathrm{ii}} = 3/2$ for equal masses. The cascade is a branching process whose statistics follow from one generating functional and its backward equation Eq. (18.4), and the energy balance Eq. (18.5) ties every fluctuation of the pair number to one of the heat. Two reductions make it solvable, the energy-only reduction and the random-k model, and silicon violates the hypothesis of the first above 3 to 5 eV. The cascade must be seeded class by class: at 10 eV an equal split of the energy between electron and hole yields 2.985 pairs on average, a uniform distribution of splits with the same mean 2.691.

The mean energy per pair and the Fano factor are the first two moments of that generating functional (Sect. 18.2). At fixed deposit the pair number and the heat are one random variable, so F_{eh} is the variance of the heat per pair in units of E_{g}^2. It splits into the variances of the residual energy below threshold and of the phonon energy above it and their covariance, $0.092 + 0.290 - 0.268 = 0.114$ in silicon. The largest term is the phonon energy above threshold, 6.7 times its Poisson value, and the residual energies of the terminal carriers anticorrelate with it. The random-parking limit reproduces the closed form $F = 0.0510387$, and the renewal estimate built on the final carriers alone falls below the full value in every case computed. Below about $15 E_{\mathrm{g}}$ the yield rises in steps of one pair, so that a search at a few pairs needs the whole distribution.

Seeded from the resolved response, the generating functional yields the cascade distribution

$P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ of each final-state class (Sect. 18.3). At 5 eV in silicon a dark-matter kick at q_0 puts two pairs in 17 % of events, where the step model puts two in every event. The step model and the Ramanathan–Kurinsky model are limits in which the seed is replaced by a fixed rule; they describe the pair number at the end of the cascade and fail in the few-pair bins. In the energy-only cascade a photon calibration seeds the pair number within $d_{\mathrm{TV}} = 0.062$ of a dark-matter kick at q_0; this model-internal number carries the photon seed's $\boldsymbol{k}$-grid error, and the full-band value is unresolved. Between 4.4 and 13 eV, however, the split of the energy decides the count, and the phenomenological models fix that split by rule. A full-band cascade with no fitted constant serves as reference: at 5 eV it puts two or more pairs in a fraction 0.097 [0.030, 0.173] of the dark-matter events, so that the step model overcounts the two-pair bin about tenfold and at least sixfold, and the calibrated energy-only cascade lies 0.12 to 0.20 in d_{TV} from it between 6 and 8 eV.

With its one constant, $A_{\mathrm{AB}} = 5.703\,\mathrm{eV}^3$, fixed by the measured pair energy of silicon, the cascade yields $F_{\mathrm{eh}} = 0.114$ against the measured 0.119 ± 0.002, 2.4σ below, and the structural choices of the model move F_{eh} more than the constant does (Sect. 18.4). The full-band cascade, without a fitted constant, gives $\varepsilon_{\mathrm{eh}} = 4.15\,\mathrm{eV}$ [3.95, 4.73], with a systematic of at least $\pm 0.4\,\mathrm{eV}$ for its class of model, against the measured 3.749 eV, and the same $F_{\mathrm{eh}} = 0.114$ [0.101, 0.115]; a gap of 0.6 eV to the density-functional calculation of Nielsen and Fischetti remains unexplained. The model pair energy of germanium at 77 K lies 4.9 % below the measurement, that of GaAs at room temperature 5.6 %. The Klein relation $\frac{14}{5}E_{\mathrm{g}} + r_K\omega_K$ is the cascade with a uniform population below threshold; it overstates $\varepsilon_{\mathrm{eh}}$ by up to 36 % for wide gaps and weak phonon loss. Over its parameters the cascade reduces to the observed linear law $\varepsilon_{\mathrm{eh}} \approx 1.61E_{\mathrm{g}} + k_2$, with k_2 from 1.2 to 2.5 eV and 1.88 eV for silicon. Against the seven crystals besides silicon with a measured pair energy, quartz included, the residuals of the model run from −19.8 % to +7.0 % (root mean square 10.5 %), which defines its validated range; the halides, fluorides, lead salts and InSb lie outside it, and their pair energies remain estimates.

Below the threshold the pair number can still change (Sect. 18.5). Each electron captured before the readout raises the energy not held by pairs by exactly E_{g}, of which only the depth of the level is heat, and a capture probability f_{cap} shared by all electrons turns $\varepsilon_{\mathrm{eh}}$ into $\varepsilon_{\mathrm{eh}}/(1 - f_{\mathrm{cap}})$ and F_{eh} into $(1 - f_{\mathrm{cap}})F_{\mathrm{eh}} + f_{\mathrm{cap}}$. A loss of 0.54 % would close the gap between the cascade and the measured Fano factor of silicon. Hot carriers are captured rarely in a pure crystal, with a capture length of at least 20 cm at $10^{12}\,\mathrm{cm}^{-3}$ centers, while cold carriers at 50 mK reach any ionized center within tens of micrometers. Excitons change neither the pair number nor its moments. They return their binding energy as heat, which lowers a Fano factor inferred from heat by 2.5 % in silicon when every pair binds. An exciton that is never split thins the measured charge, Eq. (18.35), whenever the field is too weak to separate the pair before it binds. Linear rate equations for free pairs, free and bound excitons and traps give the gap energy still held as a sum of exponentials with weights of either sign, and an exact split of where it ends. Almost none of their rates is known at millikelvin temperatures.

Table 18.3 collects the truncations and inputs of the chapter with their sizes. The pair energies of the wide-gap and polar crystals ranked in Chap. 17 need the full cascade of Appendix E.24, and the end of the cascade needs the rate set of Sect. 18.5.4. The readout of the pairs, with its trapping and collection, comes next.

Table 18.3: Nuisances of the ionization yield. Type in the classes of Sect. 8.8, normalization, tilt, threshold and new support, with "width" for an item that acts on the variance of the pair number rather than on its mean. Basis of the size: "computed" (next order or exact result computed), "model" (computed in a model), "est." (parametric estimate), "input spread" (spread or limitation of a measured or computed input), "unsized" (no size known). The last column gives the section where the item is treated.

Truncation or input	Type	Size	Basis	Section
Independent carriers (Coulomb interaction within one cascade)	normalization; width	pair density times a_x^3 is 0.47 at 100 eV and 4.7 at 1 keV in a 10 nm sphere; error not estimated	unsized	Sect. 18.1
Energy-only reduction	normalization; width	error roughly Γ_E/Γ_k per generation; impact ionization and phonon emission equally likely at 5.9 eV in the random-k model; first-generation memory negligible in a full-band cascade, within the deformation-potential model only	est.; model	Sects. 18.1 and 18.5
Impact ionization at first order in W^{scr}	normalization; width	a constant factor ($z_{\mathrm{qp}} \approx 0.8$) moves $\varepsilon_{\mathrm{eh}}$ by about 3 % and is absorbed by calibration; the energy dependence and the carrier–electron exchange act on F_{eh}	est.; unsized	Sects. 18.1 and 18.5
Step model or average seed instead of the resolved seed	tilt at a few pairs	at 5 eV two pairs in 17 % of events for a kick at q_0, against every event in the step model; at 10 eV, 2.985 pairs from the equal split against 2.691 from uniform splits; a photon seed in place of the dark-matter seed, $d_{\mathrm{TV}} \leq 0.062$ in the energy-only model, unresolved in the full band (photon seed moves 0.11–0.14 between $\boldsymbol{k}$ grids); full band: step model overcounts the two-pair bin at 5 eV about tenfold, at least sixfold	model	Sects. 18.1 and 18.3
Random-k kernels instead of full-band kernels	tilt at a few pairs	at matched pair energy $d_{\mathrm{TV}} = 0.12$–0.20 between 6 and 8 eV; two or more pairs at 5 eV in 0.169 of dark-matter events against 0.097 $[0.030, 0.173]$ in the full band	model	Sect. 18.3
Full-band model class (bands, screening, phonon couplings)	normalization	$\varepsilon_{\mathrm{eh}} = 4.15$ eV $[3.95, 4.73]$ against 3.749 eV measured; class systematic $\geq \pm 0.4$ eV; 0.6 eV unexplained against Nielsen and Fischetti	input spread	Sect. 18.4
Renewal (final-state) estimate of F_{eh}	width	below the full value in every computed case; no general proof	computed; unsized	Sect. 18.2
Constant terms of the mean and variance of the yield	normalization at a few pairs	b_N from 0.008 (uniform split) to 0.35 (one-sided seed), $c_N = -0.02$, in the silicon model; no general derivation	model; unsized	Sect. 18.2
Cascade model against the silicon Fano factor	width	0.114 against 0.119 ± 0.002, 2.4σ; threshold at $\frac{3}{2}E_{\mathrm{g}}$ gives 0.113, a density of states linear in x 0.102	input spread; model	Sect. 18.4
Klein relation (uniform population below threshold)	normalization	0.1 % for the $\theta_{\mathrm{ii}} = \frac{3}{2}$ cascade at silicon's constant; overstates $\varepsilon_{\mathrm{eh}}$ by 22 % at $\lambda_{\mathrm{AB}} = 0.1$ and up to 36 % (Limit Check 18.6)	computed	Sect. 18.4
One phonon constant for all crystals	normalization	residuals (model over measured, minus one) germanium −4.9 %, GaAs −5.6 %; over the seven crystals of the validated range −19.8 % to +7.0 %, root mean square 10.5 %, the largest model error of the chapter; crystals outside the range are estimates	input spread	Sect. 18.4
Capture at the end of the cascade	normalization; width	$\varepsilon_{\mathrm{eh}}/(1 - f_{\mathrm{cap}})$, $(1 - f_{\mathrm{cap}})F_{\mathrm{eh}} + f_{\mathrm{cap}}$; $f_{\mathrm{cap}} = 0.54$ % would close the silicon Fano gap; hot-carrier capture length ≥ 20 cm at 10^{12} cm^{-3}	est.	Sect. 18.5
Capture probability depending on the terminal energy	width	general form Eq. (18.31); $f(x)$ not computed	unsized	Sect. 18.5
Exciton binding in a heat readout	normalization; width	mean heat $+E_x^b$ per bound pair, at most 0.39 % of ω in silicon; $F_{\mathrm{heat}}/F_{\mathrm{eh}} = (1 - E_x^b/E_{\mathrm{g}})^2$ if every pair binds, −2.5 % (Si), −0.7 to −1.3 % (Ge)	computed (limit)	Sect. 18.5
Excitons never split, in a charge readout	width	$(1 - p_{\mathrm{ns}})F_{\mathrm{eh}} + p_{\mathrm{ns}}$; field separation beyond 0.56 µm at 4 V/cm; p_{ns} unknown	unsized	Sect. 18.5
Direct creation of bound pairs by carriers below threshold	normalization	at most 25 meV (0.67 %) in silicon, 8.6 meV (0.30 %) in germanium	est.	Sect. 18.5
Gap energy still held at the end of the window	normalization (heat)	42 % at 1 µs (plus 5.6 % already trapped or radiated), 1 % at 4.4 µs for the illustrative rates of Fig. 18.12; power law	model	Sect. 18.5

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Chapter 19

Low-energy nuclear recoils

A dark-matter particle of 200 MeV gives a silicon nucleus at most 21 eV. The nucleus leaves its site at 12 km/s, reaches its first neighbor within 20 fs, and in less than a picosecond the energy has been shared out. Most of it becomes heat. Along some directions the atom does not return to its site, and a vacancy and an interstitial keep several eV between them for as long as their barriers allow. Above a few hundred eV a part also goes to the electrons and ends as charge. A phonon sensor sees the heat, a charge readout sees the pairs, and neither sees the stored energy until it is released, possibly long after the event. Searches turn these signals into a recoil energy with yields taken from Lindhard's model of ion slowing, which treats the electrons as a gas without a gap and stores no energy in the lattice. That model was built for faster ions, and it is least reliable below a keV, where light dark matter deposits its energy.

The response of Chap. 11 ends when the momentum and the energy have been transferred; it does not ask what the nucleus does next (Sect. 11.4). This chapter takes the nucleus from that moment, through its motion on the Born–Oppenheimer surface and its collisions, to the moment the shares are fixed, and writes one balance law that counts every share once. The electrons that the recoil excites start the hot-carrier cascade of Chap. 18, now seeded by a slow ion instead of a dark-matter vertex. The energy stored in defects links the recoil to the delayed events that limit low-threshold detectors, and so to the readout of the next chapter (Fig. 19.1).

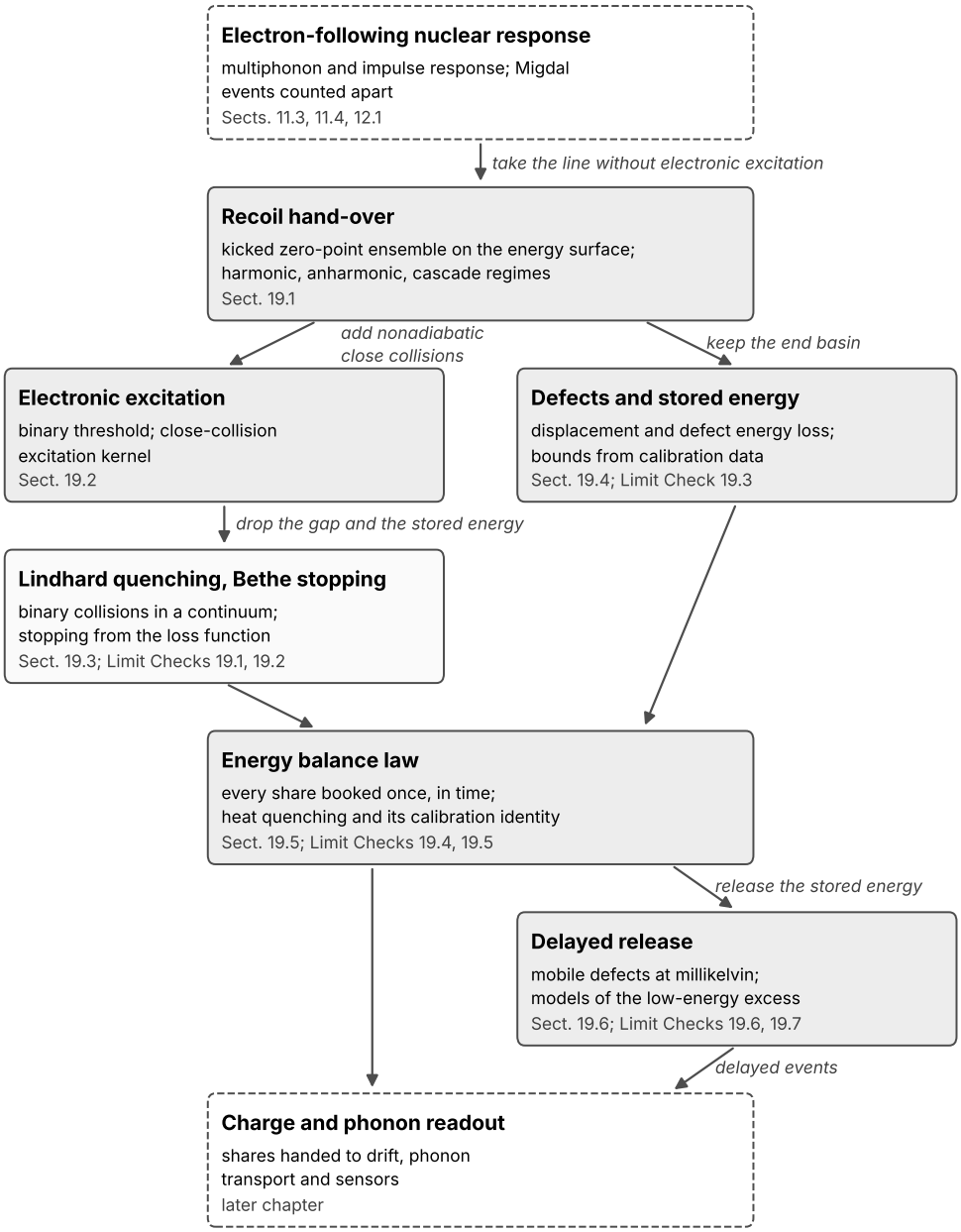


Figure 19.1: Low-energy nuclear recoils, from the electron-following nuclear response (Sects. 11.3 and 11.4; Migdal events counted apart, Sect. 12.1.4) to the shares a detector reads. Main line: the hand-over of the kicked nucleus to dynamics on the Born–Oppenheimer surface, the regimes of that dynamics and the partition of the recoil energy into heat, stored energy and charge (Sect. 19.1), then the energy balance law with heat quenching, its calibration identity and the stored-energy bound (Sect. 19.5; Limit Checks 19.4 and 19.5). Left branch: electronic excitation in close collisions and the binary threshold (Sect. 19.2), and Lindhard quenching and Bethe stopping as its limits (Sect. 19.3; Limit Checks 19.1 and 19.2). Right branch: displacement, the energy stored in defects and the bounds that calibration data set on it (Sect. 19.4; Limit Check 19.3), and the delayed release of that energy by mobile defects (Sect. 19.6; Limit Checks 19.6 and 19.7), which feeds the readout as delayed events.

19.1 Nuclear recoil on the Born–Oppenheimer surface

A silicon nucleus kicked with 12.5 eV along $\langle 111 \rangle$ either swings out and returns to its site, or pushes a neighbor into the open tetrahedral site and leaves a defect behind (Sect. 7.4). The nuclear response of Chap. 11 does not distinguish the two outcomes. It gives the rate of the kick and the energy ω deposited, and both are fixed before the atom has moved (Sect. 11.4). Which outcome occurs is decided afterward, in some 25 fs, by the motion of the struck atom between its neighbors on the Born–Oppenheimer surface E^{ad} of Sect. 5.1. To follow it we will need three things: the state in which the response leaves the crystal, the regimes in which that state can be propagated by classical dynamics, and a definition of the shares into which the energy finally splits. The first is exact, the regime boundaries follow from four dimensionless numbers, and the rest of the chapter computes and bounds the partition.

19.1.1 The hand-over from the nuclear response

The kick either leaves the electrons in their ground state or also excites them. On the plane of the joint response of Sect. 12.1, the line $\omega_e = 0$ holds the kicks after which the electrons stay in their ground state. It carries the weight $1 - P_{\mathrm{Mig}}(q)$ of Eq. (12.7) and the dressed mass $M_\kappa + \delta M_\kappa$; the rest is the Migdal entry, in which the same kick also leaves an electronic energy $\omega_e > 0$ (Sect. 12.1.4). Both kinds of event set the nucleus moving with the same lattice factor. We follow the nucleus here, and a Migdal event hands its electronic energy to the cascade of Chap. 18 in addition. In silicon the weight $1 - P_{\mathrm{Mig}}$ differs from one by less than 10^{-3} for $E_R \lesssim 16\,\mathrm{eV}$ and by 1.9 % at 500 eV with the exact second-order bracket, 3 % in the soft limit (Sects. 12.1.4 and 12.3.2).

At small recoil parameter $x_R = E_R/\langle\omega\rangle$ the final states of the line $\omega_e = 0$ are states of a few phonons, and the response already resolves them by branch and momentum (Eq. (11.25)). Here there is nothing left to follow: the response already lists the phonons, and their energy ω goes as heat straight to the phonon readout (Sect. 8.7). At $T > 0$ some terms of the response absorb a thermal phonon instead of creating one; they carry negative heat, and the net lattice energy, created minus absorbed, is ω.

At large x_R the response no longer lists phonons. In the impulse regime it is a peak of width σ_ω around the free recoil energy $E_R = q^2/2M_\kappa$ (Eq. (11.33)), and the nucleus that received the kick moves far enough to collide with its neighbors. The state it moves in is known exactly. Within the Born–Oppenheimer approximation the kick acts on the nuclear wave function alone, and multiplies it by $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$, with $\boldsymbol{X}_I$ the position of the struck nucleus I of species κ. Write $\{\boldsymbol{X}\}$ and $\{\boldsymbol{p}\}$ for the positions and momenta of all nuclei, and describe the initial state ϱ_0 of the lattice, ground or thermal, by its Wigner function

$$\mathcal{W}_0(\{\boldsymbol{X}\},\{\boldsymbol{p}\}) = \int \frac{\mathrm{d}\{\boldsymbol{y}\}}{(2\pi)^{3N}}\, \langle \{\boldsymbol{X} - \boldsymbol{y}/2\}|\varrho_0|\{\boldsymbol{X} + \boldsymbol{y}/2\}\rangle\, \mathrm{e}^{\mathrm{i}\sum_J \boldsymbol{p}_J\cdot\boldsymbol{y}_J},$$

whose marginals are the distributions of the positions and of the momenta. After the kick the density matrix is $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}\varrho_0\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$. Its matrix element between $\{\boldsymbol{X} - \boldsymbol{y}/2\}$ and $\{\boldsymbol{X} + \boldsymbol{y}/2\}$ is that of ϱ_0 times $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{X}_I - \boldsymbol{y}_I/2)}\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{X}_I + \boldsymbol{y}_I/2)} = \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{y}_I}$. The extra phase shifts the momentum variable of the Fourier integral:

$$\mathcal{W}_+(\{\boldsymbol{X}\},\{\boldsymbol{p}\}) = \mathcal{W}_0(\{\boldsymbol{X}\},\{\boldsymbol{p}\})\big|_{\boldsymbol{p}_I \to \boldsymbol{p}_I - \boldsymbol{q}}. \tag{19.1}$$

The kick leaves every position where it was and moves the momentum of one nucleus by $\boldsymbol{q}$, for any q and any potential.

The energy this state carries follows at once. The Hamiltonian of the nuclei is $H = \sum_J \boldsymbol{p}_J^2/2M_J + E^{\mathrm{ad}}(\{\boldsymbol{X}\})$, a function of the momenta plus a function of the positions, and for

such a Hamiltonian the phase-space function that represents it in the Wigner picture is the classical $H_{\mathrm{cl}}(\{\boldsymbol{X}\},\{\boldsymbol{p}\})$ itself. Every point of phase space therefore gains

$$H_{\mathrm{cl}}(\{\boldsymbol{X}\},\{\boldsymbol{p}\}+\boldsymbol{q}\hat{\boldsymbol{e}}_I)-H_{\mathrm{cl}}(\{\boldsymbol{X}\},\{\boldsymbol{p}\})=E_R+\frac{\boldsymbol{q}\cdot\boldsymbol{p}_I}{M_\kappa}, \tag{19.2}$$

with $\hat{\boldsymbol{e}}_I$ marking the momentum of nucleus I. This is the energy D that the kick gives a free particle of momentum $\boldsymbol{p}_I$ (Sect. 11.4). Now let each point of $\mathcal{W}_+$ move along its classical trajectory on E^{ad}. Since classical motion conserves H_{cl}, each trajectory carries its deposited energy Eq. (19.2) unchanged, and the distribution of the deposit over the ensemble is the distribution of $E_R+\boldsymbol{q}\cdot\boldsymbol{p}_I/M_\kappa$ over the momentum marginal $\mathfrak{f}_\kappa(\boldsymbol{p}_I)$. That is the impulse approximation Eq. (11.33), term by term.

Recoil hand-over

In the impulse regime, the nuclear response hands over the ensemble $\mathcal{W}_+$ of Eq. (19.1): positions and momenta drawn from the Wigner function of the lattice before the kick, ground or thermal, with the momentum of the struck nucleus raised by $\boldsymbol{q}$. Propagated classically on E^{ad}, the ensemble deposits $\omega=E_R+\boldsymbol{q}\cdot\boldsymbol{p}_I/M_\kappa$ per trajectory, which reproduces Eq. (11.33) with its exact mean E_R and its exact width σ_ω of Eq. (11.32). It misses the third cumulant c_3 of Eq. (D.190), the force of the neighbors during the kick, and the quantum corrections to the motion after it.

The two omissions set the validity. The share of c_3 in the shape of the peak falls as $x_R^{-1/2}$ (Sect. 11.4). The quantum corrections to the trajectories are small once the recoil holds many vibrational quanta, which Sect. 19.1.2 makes quantitative: in silicon, above about 0.4 eV. For a harmonic lattice the Wigner function is a positive Gaussian, at the zero-point widths at $T=0$ and with the $\coth(\omega/2T)$ widths of Sect. 11.4 at $T>0$, and it can be sampled point by point. Anharmonic zero-point distributions come from path-integral sampling (Appendix E.17).

The struck atom starts with a spread of kinetic energies. At $T=0$, $\sigma_\omega=(E_R\langle\omega\rangle)^{1/2}$, which for silicon at the lowest displacement threshold, $E_R=12.5\,\mathrm{eV}$ with $\langle\omega\rangle=40.9\,\mathrm{meV}$, is 0.72 eV, or 5.7 % of the threshold. A simulation that starts the other atoms at rest, as for a classical crystal at millikelvin, has no such spread. The direction of the recoil is that of $\boldsymbol{q}$ only on average. Let $E_\kappa^{\mathrm{kin}}=\langle\omega\rangle/4=10.2\,\mathrm{meV}$ be the mean zero-point kinetic energy of the nucleus per Cartesian component (Sect. 11.4). The transverse momentum of the struck nucleus then has the rms value $(2M_\kappa E_\kappa^{\mathrm{kin}})^{1/2}$ per component, against the kick $q=(2M_\kappa E_R)^{1/2}$, so the direction scatters by the rms angle $(E_\kappa^{\mathrm{kin}}/E_R)^{1/2}$ in each transverse plane: 1.6° at 12.5 eV, or 2.3° in polar angle. And the displacement threshold becomes a smooth step. Let $\hat{\boldsymbol{r}}_R=\hat{\boldsymbol{q}}$ be the recoil direction and run every trajectory of the ensemble until its motion has been carried off into the lattice. The basin of the perfect crystal on E^{ad} is the set of configurations that relax to it. The probability $p_{\mathrm{FP}}(E_R,\hat{\boldsymbol{r}}_R)$ that the end state lies outside this basin, in a Frenkel pair (FP) or another defect, rises from 0 to near 1 over a range of E_R, and we take as the displacement threshold its lowest crossing of one half,

$$E_{\mathrm{d}}(\hat{\boldsymbol{r}}_R)=\min\{E_R:\ p_{\mathrm{FP}}(E_R,\hat{\boldsymbol{r}}_R)\geq\tfrac{1}{2}\}. \tag{19.3}$$

When every trajectory of the ensemble is the same, p_{FP} is a step function and Eq. (19.3) is the deterministic threshold Eq. (7.54). The spread σ_ω of the deposit spreads the step over about 0.7 eV in silicon, before the spread of the initial positions adds its own; the measured onset of damage, $12.9\pm0.6\,\mathrm{eV}$ [1], carries an uncertainty of the same size. The step width is a prediction of the response and not a parameter of the dynamics. It matters for light dark matter, whose recoil spectrum falls steeply near its endpoint, so that the few recoils above E_{d} sit on the edge of the step.

19.1.2 Regimes of the recoil dynamics

Whether the handed-over ensemble may be propagated classically, harmonically, without defects and without electronic excitation is decided by four numbers: the quantum ratio $\lambda_{\mathrm{dB}}/\ell_R$, the anharmonic ratio η_R, the displacement threshold E_{d} and the Massey parameter ξ_{M}. Each compares a property of the struck atom at kinetic energy E_R with a scale of the crystal, and Fig. 19.2(b) plots them for silicon.

The quantum and the anharmonic numbers use the amplitude of the struck atom's first excursion. Before its energy has spread, the atom moves in the cage of neighbors that have not yet responded, and for an estimate we treat that cage as an oscillator of the mean phonon frequency. Its turning point lies at

$$\ell_R = \Big(\frac{2E_R}{M_\kappa \langle\omega\rangle^2}\Big)^{1/2},$$

from $E_R = \frac{1}{2} M_\kappa \langle\omega\rangle^2 \ell_R^2$. Classical mechanics holds when the de Broglie wavelength $\lambda_{\mathrm{dB}} = 2\pi/(2M_\kappa E_R)^{1/2}$ is short compared with this amplitude. The ratio is

$$\frac{\lambda_{\mathrm{dB}}}{\ell_R} = \frac{2\pi}{(2M_\kappa E_R)^{1/2}} \Big(\frac{M_\kappa \langle\omega\rangle^2}{2E_R}\Big)^{1/2} = \frac{\pi\langle\omega\rangle}{E_R},$$

independent of the mass: the motion is classical when the recoil holds many quanta $\langle\omega\rangle$. The corrections to classical trajectories are of second order in $\hbar$, and we take their relative size to be $(\lambda_{\mathrm{dB}}/\ell_R)^2$, an estimate. The harmonic approximation holds when ℓ_R is short compared with the length $3\Phi^{(2)}/|\Phi^{(3)}|$ over which the force constants change by their own size. This is the anharmonic parameter η_{anh} of Eq. (5.72) with the recoil amplitude in place of the zero-point length,

$$\eta_R = \frac{|\Phi^{(3)}|\,\ell_R}{3\Phi^{(2)}},$$

and by the counting of Eq. (5.73) the anharmonic corrections to energies and widths are of order η_R^2. For silicon, the bond-coordinate values of Sect. 5.7 give $3\Phi^{(2)}/|\Phi^{(3)}| = 0.98\,\text{Å}$.

The third number is the displacement threshold itself: below $E_{\mathrm{d}}(\hat{\boldsymbol{r}}_R)$ the trajectory relaxes back into the basin of the perfect crystal, above it a defect survives with probability p_{FP}. The fourth decides whether the electrons follow. A nucleus at speed $v_N = (2E_R/M_\kappa)^{1/2}$ changes the potential seen by the electrons over a time ℓ_{int}/v_N, with $\ell_{\mathrm{int}} \approx 1\,\text{Å}$ the range over which a close collision distorts the bonds. The electrons follow adiabatically if this time is long compared with $1/E_{\mathrm{g}}$, the time scale of the lowest excitation. The ratio is the Massey parameter

$$\xi_{\mathrm{M}} = \frac{E_{\mathrm{g}}\,\ell_{\mathrm{int}}}{v_N},$$

and for a perturbation that is smooth on the scale ℓ_{int} the probability of an electronic transition falls as $\mathrm{e}^{-\xi_{\mathrm{M}}}$. The bulk gap overstates the adiabaticity of a close collision, since the local gap of the compressed pair closes as the atoms approach, and Sect. 19.2 treats that channel.

The numbers divide the recoil energies into a quantum, a classical, a displacement and a nonadiabatic regime. Their boundaries follow from the definitions; their values depend on the criterion and on the cage estimate, and we quote them as estimates at the level of 0.1.

Quantum. For $(\pi\langle\omega\rangle/E_R)^2 > 0.1$, that is $E_R < \pi\langle\omega\rangle/\sqrt{0.1}$, the recoil holds too few quanta for classical motion. The response is the whole answer here: its phonons are the final state and the energy is heat.

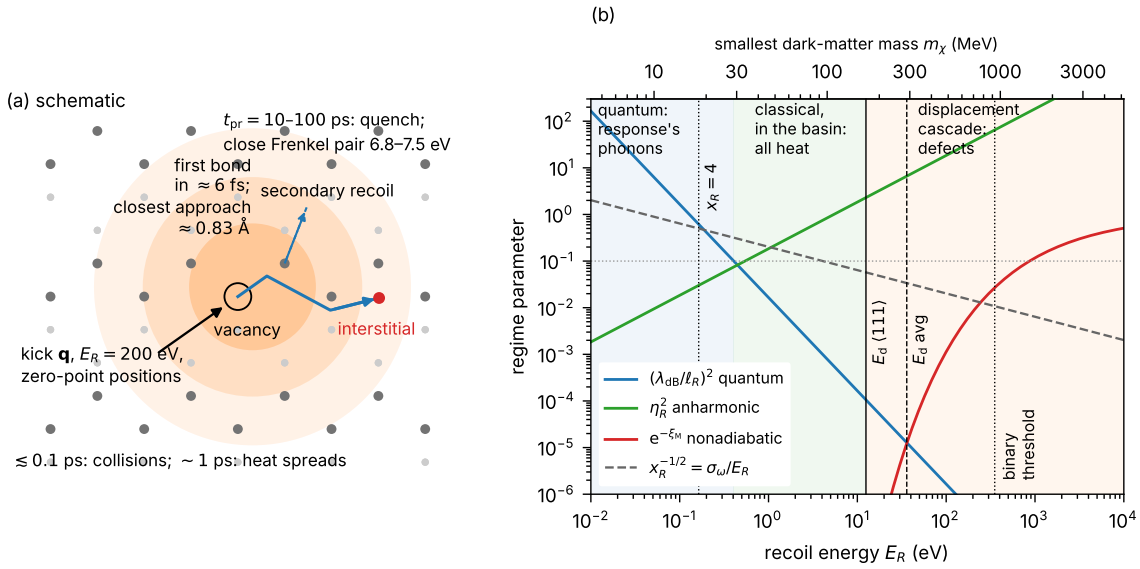


Figure 19.2: A silicon nuclear recoil after the hand-over. (a) Schematic of a 200 eV recoil in a $(1\bar{1}0)$ plane of the diamond lattice (dark dots: atoms in the plane; light dots: the neighboring planes). The kick starts the struck atom from its zero-point position; it crosses its first bond in about 6 fs, with a closest approach of 0.83 Å in a head-on collision, and sets a secondary atom moving. Within about 0.1 ps the collisions are over, and within about a picosecond the heat (shaded rings) spreads through the crystal. At the partition time t_{pr} of Sect. 19.1.4, 10 to 100 ps, the end state is quenched: here a vacancy and an interstitial, which as a close Frenkel pair store 6.8 to 7.5 eV. Paths and end positions are drawn by hand. (b) The regime parameters against the recoil energy E_R at $T = 0$. Curves: the quantum correction $(\lambda_{dB}/\ell_R)^2 = (\pi\langle\omega\rangle/E_R)^2$, the anharmonic parameter η_R^2 with $3\Phi^{(2)}/|\Phi^{(3)}| = 0.98$ Å, the adiabatic suppression $e^{-\xi_M}$ with $E_g = 1.17$ eV and $\ell_{int} = 1$ Å, and the relative width $x_R^{-1/2} = \sigma_\omega/E_R$ of the handed-over deposit, with $\langle\omega\rangle = 40.9$ meV. The dotted horizontal line is the criterion 0.1. Vertical lines: impulse onset $x_R = 4$, displacement thresholds 12.5 eV ($\langle 111\rangle$) and 36 eV (Frenkel pair, direction average), and the binary electron–hole threshold 352 eV of Sect. 19.2. Shading: quantum regime, where the phonons of the response are the final state (blue); classical motion inside the basin of the perfect crystal, where all energy becomes heat (green); displacement cascade (orange). Top axis: the smallest dark-matter mass that reaches E_R at $v_{max} = 2.650 \times 10^{-3}$.

Classical, inside the basin. From there up to E_d the motion is classical and adiabatic, anharmonic above $\eta_R^2 = 0.1$, and the trajectory relaxes back to the perfect crystal. Born–Oppenheimer molecular dynamics is controlled, and all of the energy becomes heat. What the dynamics adds to the response is the spectrum of the phonons the recoil leaves, which anharmonic collisions redistribute.

Displacement cascade. Above E_d some trajectories end in a defect, and the energy splits into heat and stored energy. A recoil many times E_d sets secondary atoms moving above their own thresholds, the collision cascade of Fig. 19.2(a).

Nonadiabatic. Where $\mathrm{e}^{-\xi_\mathrm{M}}$ is no longer negligible, the electrons are excited along the path, and part of the energy becomes charge.

The quantum and harmonic boundaries lie close together: in silicon there is no wide window in which the motion is both harmonic and classical. The hand-over, however, does not need one. It takes place at the instant of the kick, and its only requirement on the later motion is that it be classical.

Worked example 19.1: Regime boundaries for silicon

A silicon nucleus of the natural isotope mixture has $M_\kappa = 28.0855\,\mathrm{u} = 26.16\,\mathrm{GeV}$, as in Sect. 5.3, and $\langle\omega\rangle = 40.9\,\mathrm{meV}$ (Sect. 11.4). The cage stiffness is $M_\kappa\langle\omega\rangle^2 = 26.16\,\mathrm{GeV} \times (0.0409\,\mathrm{eV})^2/(\hbar c)^2 = 11.24\,\mathrm{eV\,Å^{-2}}$, with $\hbar c = 1973.27\,\mathrm{eV\,Å}$.
Classical boundary. $(\pi\langle\omega\rangle/E_R)^2 = 0.1$ gives $E_R = \pi \times 0.0409\,\mathrm{eV}/0.316 = 0.41\,\mathrm{eV}$, or $x_R = 9.9$. At 1 eV, $\lambda_\mathrm{dB} = 0.054\,\mathrm{Å}$ against $\ell_R = 0.42\,\mathrm{Å}$, and the correction estimate is 0.017.
Harmonic boundary. $\eta_R^2 = 0.1$ means $\ell_R = 0.316 \times 0.98\,\mathrm{Å} = 0.31\,\mathrm{Å}$, reached at $E_R = \frac{1}{2} \times 11.24\,\mathrm{eV\,Å^{-2}} \times (0.31\,\mathrm{Å})^2 = 0.54\,\mathrm{eV}$. At 1 eV, $\eta_R^2 = 0.18$.
Displacement. The lowest threshold is 12.5 eV along ⟨111⟩, the direction average for a Frenkel pair 36 eV [2, Table IV]. At 12.5 eV the cage amplitude is $\ell_R = 1.49\,\mathrm{Å}$, more than half the bond length of 2.35 Å: the cage estimate has run out, as it should, since the atom is leaving its cage. The atom moves at $v_N = 9.27\,\mathrm{km/s}$ and crosses one bond in 25 fs, less than half the shortest phonon period of 65 fs ($2\pi/\omega_\mathrm{opt}$ with $\omega_\mathrm{opt} = 64.1\,\mathrm{meV}$, Sect. 5.7), so the collision is over before the lattice has responded. The handed-over deposit has the width $\sigma_\omega = (12.5\,\mathrm{eV} \times 0.0409\,\mathrm{eV})^{1/2} = 0.72\,\mathrm{eV}$, and the energy ends in some $12.5\,\mathrm{eV}/\langle\omega\rangle \approx 300$ phonons if no defect survives.
Adiabaticity. At 20 eV, $v_N = 11.7\,\mathrm{km/s}$ and $\xi_\mathrm{M} = 1.17\,\mathrm{eV} \times 1\,\mathrm{Å}/(\hbar v_N) = 15$, with $\hbar = 6.582 \times 10^{-16}\,\mathrm{eV\,s}$, so $\mathrm{e}^{-\xi_\mathrm{M}} = 3 \times 10^{-7}$. Since $\xi_\mathrm{M} \propto E_R^{-1/2}$, the suppression reaches 0.1 ($\xi_\mathrm{M} = 2.3$) at 0.87 keV, and $\xi_\mathrm{M} = 1$ at 4.6 keV.
Dark-matter masses. By Eq. (1.21) with $v_\mathrm{max} = 2.650 \times 10^{-3}$, the classical boundary needs $m_\chi \geq 28\,\mathrm{MeV}$, the harmonic boundary 32 MeV, the lowest threshold 154 MeV, the average threshold 262 MeV, and $\mathrm{e}^{-\xi_\mathrm{M}} = 0.1$ needs 1.3 GeV.

19.1.3 Acceptance tests for recoil molecular dynamics

In the classical regimes the handed-over ensemble is propagated by molecular dynamics on E^ad. A recoil cascade needs cells of thousands of atoms and thousands of trajectories, far beyond direct density-functional dynamics. In practice the surface is therefore a machine-learned interatomic potential, trained on density-functional energies and forces and joined to a screened Coulomb repulsion at short distance. Appendix E.11 describes the method,

and Appendix E.23 specifies the calculations that this chapter needs. The hand-over and the partition set five requirements on such a calculation: the surface along the recoil path, the initial ensemble, the reference of the stored energy, the harmonic limit and the electronic channel.

The energy surface must be right along the path of the struck atom, not only near equilibrium. The threshold is set by the energy between the neighbors that the atom squeezes past. Potentials that agree on elastic constants and defect energies can still disagree on thresholds by several eV; the Stillinger–Weber potential overshoots the density-functional threshold along $\langle 111 \rangle$ by 4 to 6 eV (Sect. 7.4). The natural target is the width of the step in p_{FP}, about 0.7 eV at the lowest silicon threshold, checked against density-functional dynamics along a few directions. Closer in, the closest approach of a head-on collision is 1.49 Å at 20 eV and 0.83 Å at 200 eV (Table E.4), where the screened Coulomb repulsion takes over; the splice between the two must be complete above the turning point of threshold collisions, so that the dynamics near E_{d} never feels it.

The initial ensemble must be the kicked zero-point ensemble of the hand-over, not a lattice at rest. Classical propagation of a Wigner ensemble is exact for a quadratic Hamiltonian, since the quantum evolution of the Wigner function then reduces to the classical Liouville flow, so the zero-point motion of the parts of the crystal that stay harmonic is carried correctly. Where the collision makes the surface anharmonic, classical dynamics lets the zero-point energy of one mode flow into others and leaves the mode below its quantum ground state, which quantum dynamics forbids. The error is of the order of the zero-point energy of the atoms the collision disturbs, a few tens of meV each, and the definition of the stored energy below is built so that it does not enter.

The stored energy must be measured between minima. The atoms of the initial ensemble sit on average half their zero-point energy above the minimum of E^{ad}, which is $3\langle\omega\rangle/4 = 31$ meV per atom in silicon (the zero-point energy of Sect. 5.3 is 123 meV per cell of two atoms). In a cell of 4096 atoms that is 126 eV, more than ten times the energy of a Frenkel pair. The end state and the perfect crystal are therefore both relaxed to their minima, by a quench at fixed cell without annealing, and the stored energy is the difference Eq. (E.79). The potential's own error on defect energies sets a floor: a potential whose formation energies lie within 7 % of density-functional theory [3, Sect. III E] carries an error of a few tenths of an eV on a defect that stores several eV. One table of these minimum energies, with one potential and cell, must serve the stored energy of Sect. 19.4 and the release paths of Sect. 19.6, or the two disagree by that error.

The harmonic limit provides an acceptance test. With E^{ad} replaced by its harmonic expansion, the classical propagation of $\mathcal{W}_+$ is exact. The mean energy that the kick leaves in a normal mode s is then $|\boldsymbol{q} \cdot \boldsymbol{e}_{\kappa s}|^2/2NM_\kappa$, with $\boldsymbol{e}_{\kappa s}$ the polarization vector of the mode on the struck site and N the number of cells. It is half the square of the mass-weighted momentum the kick gives the mode, since the cross term with the mode's own momentum averages to zero over the ensemble. These shares sum to E_R by the completeness of the polarization vectors, and they are the energy-weighted phonon content of the harmonic response of Sect. 11.3 at the same $\boldsymbol{q}$. A calculation that propagates the kicked ensemble on the true surface must recover this spectrum for recoils small enough to stay harmonic, and the mean and width of Eq. (11.33) for all others.

The electronic channel is missing from Born–Oppenheimer dynamics. A common remedy adds a friction force, minus the electronic stopping power times the direction of motion. Below the binary threshold of Sect. 19.2 that friction fails (Sect. 19.3.3), and at any energy it supplies a mean loss without its distribution. The calculations of Appendix E.23 replace it, where it matters, by excitation events drawn at each close collision from a kernel computed with time-dependent density-functional theory (Sect. 19.2).

19.1.4 Partition of the recoil energy

The shares of the recoil energy change with time, and the partition needs a time. The collisions are over within about 0.1 ps, and the hot carriers that the recoil excites finish their cascade within a picosecond (Chap. 18). On the other side, defects at millikelvin cannot cross barriers of chemical size by thermal activation, since a barrier of 0.1 eV at 10 mK carries the Boltzmann factor $\mathrm{e}^{-1.2\times10^5}$ (Sect. 7.4); they move only by tunneling or with the phonons of later events (Sect. 19.6). The drift of the pairs to the electrodes takes longer still (Sect. 20.1). Any time t_{pr} between 10 and 100 ps after the recoil lies between the two groups of processes, and we fix the partition there.

At t_{pr} the energy ω deposited in the kick has gone three ways. Part has gone to the electrons: the energy ω_e of a Migdal event, if the kick was one, plus whatever the moving nucleus excited along its path. The nuclei keep the rest, as kinetic energy and as potential energy on E^{ad}, and the end state of the quench at t_{pr} splits it into the energy of the minimum the crystal has reached and the vibrations about it.

Partition of the recoil energy

At the time t_{pr},

$$\omega = E_{\mathrm{lat}} + E_{\mathrm{st}} + \omega_e, \qquad E_{\mathrm{st}} = E^{\mathrm{ad}}_{\mathrm{min}}(\mathrm{end}) - E^{\mathrm{ad}}_{\mathrm{min}}(\mathrm{perfect}), \tag{19.4}$$

where ω_e is the energy given to the electrons and E_{st} the stored energy, the difference of the minima of E^{ad} in the basin of the end state and in that of the perfect crystal. The lattice share E_{lat} is the vibrational energy in excess of the zero-point energy of the perfect crystal. After the cascade of Chap. 18 has turned ω_e into N_{eh} pairs,

$$\omega = N_{\mathrm{eh}}E_{\mathrm{g}} + E_{\mathrm{heat}} + E_{\mathrm{st}}, \qquad E_{\mathrm{heat}} = E_{\mathrm{lat}} + (\omega_e - N_{\mathrm{eh}}E_{\mathrm{g}}). \tag{19.5}$$

The energy balance follows from the conservation of the total energy along each trajectory. Before the kick the nuclei hold the minimum of the perfect crystal and its zero-point energy; the kick adds ω, and the electrons take away ω_e. At t_{pr} the nuclei hold the minimum of the end basin and the vibrational energy about it, and subtracting the two gives Eq. (19.4). The second form uses the cascade identity Eq. (18.5) for each electronic excitation, $\omega_e = N_{\mathrm{eh}}E_{\mathrm{g}} + E_{\mathrm{heat}}$ of the cascade, and collects all phonon energy, from the lattice and from the cascade, in one E_{heat}. For an electronic deposit, with $E_{\mathrm{lat}} = E_{\mathrm{st}} = 0$, it is Eq. (18.5) itself. Capture at the end of the cascade moves energy out of the pairs as in Sect. 18.5.

The stored energy is configurational: it compares minima of E^{ad} and contains no vibrational energy. The zero-point energy of the end basin differs from that of the perfect crystal, since a defect softens or stiffens the modes around it, and the difference sits in E_{lat}, on the vibrational side. It is part of the heat but cannot reach a sensor, and Sect. 19.5 counts it separately. The stored energy is also not the displacement threshold, which is a barrier on the path rather than a difference of end points. A close Frenkel pair keeps about half of the lowest silicon threshold, and the rest of the threshold energy becomes heat (Sect. 7.4).

The partition is random. Each trajectory of the handed-over ensemble ends with its own $(E_{\mathrm{lat}}, E_{\mathrm{st}}, \omega_e)$, and in the impulse regime the ensemble is fixed by E_R, the direction $\hat{\boldsymbol{r}}_R$ and the temperature. The cascade turns the electronic excitations into pairs with the distribution P_{casc} of Sect. 18.3, evaluated at the seed energies the recoil produced, which lie near the gap. The result is a joint distribution $P(E_{\mathrm{heat}}, E_{\mathrm{st}}, N_{\mathrm{eh}}|E_R, \hat{\boldsymbol{r}}_R, T)$, which Appendix E.23 specifies for silicon and germanium from 1 eV to 10 keV. Its correlations matter as much as its means, since a recoil that stores energy in a defect leaves less heat and, through the same collision,

may give fewer pairs.

Below E_d the adiabatic suppression makes ω_e negligible, $\mathrm{e}^{-\xi_\mathrm{M}} = 5 \times 10^{-9}$ at 12.5 eV, and every trajectory returns to the perfect crystal, so the recoil deposits all of its energy as heat. In silicon this covers every electron-following recoil from dark matter below about 150 MeV (Worked example 19.1). The charge that these recoils produce therefore comes from the Migdal entry of the response and from the soft tail of electronic excitation of Sect. 19.2, not from the recoil dynamics. After t_pr the stored energy decreases as defects release it, and the pairs recombine or are collected; the balance law of Sect. 19.5 carries this time dependence, with Eq. (19.4) as its starting point.

19.2 Electronic excitation by the recoil

A silicon nucleus recoiling with 100 eV moves at 26 km/s, eighty times slower than the valence electrons at the Fermi velocity. A neutron calibration finds that such recoils leave on average 2.7 electron–hole pairs, an ionization yield of $0.102^{+0.034}_{-0.030}$ [4, Table II]. Yet a binary collision between the nucleus and a valence electron cannot create a single pair until the recoil carries about 350 eV. The charge at 100 eV must therefore come from processes that the binary picture leaves out, and these are the processes that set the charge signal of light dark matter, whose nuclear recoils in silicon stay below 100 eV for masses under 0.44 GeV. The question is which processes those are, and how much of the charge the Migdal effect of Chap. 12 already supplies.

19.2.1 The binary electron–hole threshold

The recoiling ion is a projectile of the master formula, as the hot carrier of Sect. 18.1 was. It couples to the electron density through its screened charge, and the energy it hands to the electrons at momentum $\boldsymbol{k}$ is $\omega = \boldsymbol{k}\cdot\boldsymbol{v}_N - k^2/2M_\kappa \le k v_N$, with $v_N = q/M_\kappa$ the nuclear velocity after the kick. The stopping power Eq. (4.54) contains this constraint as the upper limit $\omega \le k v_N$ of its inner integral; Sect. 19.3 evaluates that integral. Here we can stay with a simpler question: where can the integrand be nonzero at all?

Consider first a free electron gas with the valence density of the crystal and a gap E_g added by hand. Its loss function vanishes outside the particle–hole continuum of Sect. 4.2, which for $k > 2k_\mathrm{F}$ begins at $\omega_-(k) = k^2/2m_e - k v_\mathrm{F}$. An electron–hole pair needs $\omega \ge E_\mathrm{g}$, and the ion supplies at most $\omega = k v_N$. Both conditions hold at some k only if the line $\omega = k v_N$ rises above $\omega_-(k)$ at an energy of at least E_g. The line and the edge cross at $k = 2m_e(v_N + v_\mathrm{F})$, where $\omega = 2m_e v_N(v_N + v_\mathrm{F})$, and this is the largest energy the ion can deposit in one electronic excitation:

$$\omega_\mathrm{max} = 2m_e v_N\,(v_N + v_\mathrm{F}). \tag{19.6}$$

The same number follows from a collision of two particles. In the rest frame of the heavy ion an electron moving against it at v_F approaches with speed $v_\mathrm{F} + v_N$ and is reflected with the same speed, and in the crystal frame it leaves with $v_\mathrm{F} + 2v_N$. Its kinetic energy rises by $\frac{1}{2}m_e[(v_\mathrm{F} + 2v_N)^2 - v_\mathrm{F}^2]$, which is Eq. (19.6). The heavy ion acts as a moving wall: it reverses the electron's velocity in its own frame, and the energy it gives is linear in v_N.

Setting $\omega_\mathrm{max} = E_\mathrm{g}$ gives the threshold speed. Since $v_N \ll v_\mathrm{F}$ at every recoil energy of interest, the term v_N^2 can be dropped.

Binary electron–hole threshold

A slow ion creates an electron–hole pair in a binary collision with a free-electron-like valence band only above the speed $v_{N,\mathrm{th}} = E_\mathrm{g}/(2m_e v_\mathrm{F})$, that is, above the recoil energy

$$E_{R,\mathrm{th}} = \frac{1}{2} M_\kappa v_{N,\mathrm{th}}^2 = \frac{M_\kappa}{m_e}\,\frac{E_\mathrm{g}^2}{16 E_\mathrm{F}}, \tag{19.7}$$

with $E_\mathrm{F} = m_e v_\mathrm{F}^2/2$ the Fermi energy of the valence electrons. It is 352 eV in silicon and 399 eV in germanium.

The threshold is the gap multiplied by two numbers, the mass ratio M_κ/m_e, which is large, and $E_\mathrm{g}/16E_\mathrm{F}$, which is small. In silicon the product puts $E_{R,\mathrm{th}}$ at three hundred times the gap. Keeping the term v_N^2 in Eq. (19.6) and solving the quadratic, $v_N = \frac{1}{2}[(v_\mathrm{F}^2 + 2E_\mathrm{g}/m_e)^{1/2} - v_\mathrm{F}]$, lowers the threshold by 4.4 % to 336 eV in silicon and by 3.1 % to 387 eV in germanium. For germanium we use $E_\mathrm{g} = 0.744\,\mathrm{eV}$ and $a = 5.658\,\text{Å}$ (Sect. 3.3), so that $k_\mathrm{F} = 1.736\,\text{Å}^{-1}$, $E_\mathrm{F} = 11.48\,\mathrm{eV}$ and $M_\kappa/m_e = 132\,396$. The heavier nucleus raises the threshold, and the smaller gap, which enters squared, lowers it again.

Worked example 19.2: The binary threshold of silicon

Silicon has four valence electrons per atom and eight atoms in the cubic cell of side $a = 5.431\,\text{Å}$, so the valence density is $n_e = 32/a^3 = 0.199\,76\,\text{Å}^{-3}$. The free-electron Fermi wavevector is $k_\mathrm{F} = (3\pi^2 n_e)^{1/3} = 1.8085\,\text{Å}^{-1}$, or $k_\mathrm{F} = 1.8085\,\text{Å}^{-1} \times 1973.27\,\mathrm{eV\,Å} = 3.569\,\mathrm{keV}$, as in Fig. 4.3. Then

$$E_\mathrm{F} = \frac{k_\mathrm{F}^2}{2m_e} = \frac{(3.569\,\mathrm{keV})^2}{2 \times 511.0\,\mathrm{keV}} = 12.46\,\mathrm{eV}, \qquad v_\mathrm{F} = \frac{k_\mathrm{F}}{m_e} = 6.984 \times 10^{-3} = 2094\,\mathrm{km/s}.$$

With the low-temperature gap $E_\mathrm{g} = 1.17\,\mathrm{eV}$ the threshold speed is $v_{N,\mathrm{th}} = E_\mathrm{g}/(2m_e v_\mathrm{F}) = 1.17\,\mathrm{eV}/(2 \times 511.0\,\mathrm{keV} \times 6.984 \times 10^{-3}) = 1.639 \times 10^{-4}$, or 49.2 km/s. The silicon nucleus has $M_\kappa = 28.0855 \times 931.494\,\mathrm{MeV} = 26.16\,\mathrm{GeV}$, so $M_\kappa/m_e = 51\,197$, and

$$E_{R,\mathrm{th}} = \frac{M_\kappa}{m_e}\,\frac{E_\mathrm{g}^2}{16E_\mathrm{F}} = 51\,197 \times \frac{(1.17\,\mathrm{eV})^2}{16 \times 12.46\,\mathrm{eV}} = 51\,197 \times 6.87 \times 10^{-3}\,\mathrm{eV} = 352\,\mathrm{eV}.$$

For comparison, Eq. (19.6) at $E_R = 100\,\mathrm{eV}$, where $v_N = 8.74 \times 10^{-5}$, gives $\omega_\mathrm{max} = 0.63\,\mathrm{eV}$: about half the gap, too little for a binary collision at the measured 100 eV point to make a pair. A nucleus of 352 eV has $E_R/E_\mathrm{g} = 300$, and dark matter needs a mass of at least 0.84 GeV to give it that energy (Sect. 19.1).

19.2.2 Softening by the valence momentum tail

Eq. (19.7) assumed that no valence electron moves faster than v_F. In a crystal this is not so. A Bloch state of crystal momentum $\boldsymbol{k}$ has components at every $\boldsymbol{k} + \boldsymbol{G}$, and near each nucleus the valence orbitals oscillate to stay orthogonal to the core; as a result the valence momentum density extends beyond k_F. An electron of momentum p moving against the ion gains up to $2m_e v_N(v_N + p/m_e)$, and it can be promoted across the gap if

$$p \geq p_\mathrm{min}(E_R) = \frac{E_\mathrm{g}}{2v_N} - m_e v_N, \qquad v_N = (2E_R/M_\kappa)^{1/2}. \tag{19.8}$$

In the free gas the fraction of valence electrons above $p_{\min}$ is $1-(p_{\min}/k_{\mathrm{F}})^3$, which vanishes where $p_{\min}=k_{\mathrm{F}}$, at the threshold with v_N^2 kept, 336 eV. In silicon at 100 eV, $p_{\min}=3.37\,\text{Å}^{-1}=1.86\,k_{\mathrm{F}}$. In the language of the loss function, the free-gas support ends at $\omega_-(k)$ and the crystal's $L(\boldsymbol{k},\omega)$ at $k\gtrsim 2k_{\mathrm{F}}$ and $\omega\geq E_{\mathrm{g}}$ is the low-energy tail of the Bethe ridge $\omega\approx k^2/2m_e$, whose width is set by the momentum distribution of the valence electrons. The tail is small but not zero; below $E_{R,\mathrm{th}}$ the linear-response excitation is therefore suppressed, not forbidden.

Fig. 19.3(b) shows how much the tail changes the count. It compares the free-gas fraction with the same fraction computed from the valence momentum density of the empirical pseudopotential model of Sect. 3.3, summed over the four valence bands on a 10^3 grid of the zone. In the model, 23 % of the valence momentum density lies beyond k_{F}, 2.2 % beyond $1.5\,k_{\mathrm{F}}$ and 0.34 % beyond $2k_{\mathrm{F}}$. The fraction of electrons fast enough to be promoted falls smoothly through $E_{R,\mathrm{th}}$ instead of vanishing there: it is 0.27 at 352 eV, where the free gas has 0.07, then 0.047 at 200 eV, 5.6×10^{-3} at 100 eV and 4×10^{-4} at 50 eV. The model's pseudo-orbitals are smooth inside the cores, where the true valence orbitals oscillate, and its plane waves stop at $5.3\,\text{Å}^{-1}$, close to $p_{\min}$ at 50 eV; its tail is therefore smaller than the real one, and the numbers indicate the shape of the softening, not its size. The fraction is also not yet a yield: an electron above $p_{\min}$ is promoted only in a head-on encounter, and the conduction states must accept it.

The zero-point momentum of the struck nucleus spreads its recoil energy by about 2 eV at $E_R=100$ eV (Sect. 19.1), which is negligible against 352 eV. The lattice matters in another way: the ion moves among nuclei rather than through a uniform gas, and it repeatedly passes within an angstrom of one of them.

19.2.3 Close collisions and the excitation kernel

Within a few femtoseconds a recoil of a few hundred eV strikes a neighbor, closer than a bond length, and passes on part of its energy; the neighbor strikes the next atom, and so on (Sect. 19.1). In such an encounter two atoms come within an angstrom of each other, and the electrons around them briefly belong to a quasi-molecule, $\mathrm{Si_2}$ or $\mathrm{Ge_2}$, whose levels depend on the distance between the nuclei. The bulk gap no longer sets the excitation threshold. As the nuclei approach, a filled level and an empty one can cross, or nearly cross, and an electron can be promoted at the crossing.

How likely that is depends on how fast the crossing is passed. Away from crossings the Massey parameter $\xi_{\mathrm{M}}=E_{\mathrm{g}}\ell_{\mathrm{int}}/v_N$ of Sect. 19.1 measures adiabaticity, and nonadiabatic excitation is suppressed roughly as $\mathrm{e}^{-\xi_{\mathrm{M}}}$; in silicon, with $\ell_{\mathrm{int}}=1$ Å, ξ_{M} is 6.8 at $E_R=100$ eV, 3.6 at $E_{R,\mathrm{th}}$ and 2.1 at 1 keV. At an avoided crossing the relevant gap is the local splitting $2h_{\mathrm{LZ}}$ of the two levels, with h_{LZ} the coupling between the crossing diabatic states. The relevant speed is the rate $v|\Delta F_{\mathrm{LZ}}|$ at which their energy difference sweeps through zero, where v is the relative speed of the nuclei and ΔF_{LZ} the difference of the slopes of the two diabatic levels against the internuclear distance. The electrons jump from the lower adiabatic level to the upper one in one passage with the Landau–Zener probability [5, 6]

$$P_{\mathrm{LZ}}=\exp\!\left(-\frac{2\pi h_{\mathrm{LZ}}^2}{v\,|\Delta F_{\mathrm{LZ}}|}\right). \tag{19.9}$$

A collision passes each crossing twice, on the way in and on the way out. If the phase between the two passages is averaged over impact parameters, the electrons end up excited with probability $2P_{\mathrm{LZ}}(1-P_{\mathrm{LZ}})$, one jump and one adiabatic passage in either order. Unlike $\mathrm{e}^{-\xi_{\mathrm{M}}}$, Eq. (19.9) is set by the small splitting at the crossing, not by the bulk gap, and it is why close collisions can excite electrons well below $E_{R,\mathrm{th}}$.

The excitation cannot be computed in linear response in the ion's charge. The Born parameter of the collision, $Z_\kappa\alpha/v_\mathrm{F}$, is 15 for silicon and 35 for germanium. The ion's own electrons screen its charge, but the parameter exceeds one already for a unit charge, and the ion perturbs the electrons near it strongly; and the Massey parameter near 1 keV is of order one, so the electrons neither follow adiabatically nor stay frozen. The natural object is a kernel tabulated from a nonperturbative calculation. It is the distribution

$$\frac{\mathrm{d}P}{\mathrm{d}\omega_e}(b_\mathrm{imp}, E_\mathrm{coll}) \tag{19.10}$$

of the electronic excitation energy ω_e left behind when two Si or two Ge atoms collide in the crystal at impact parameter b_imp and relative energy E_coll, in units of eV^{-1}. Its integral over ω_e is the probability that the collision excites the electrons at all. Along the trajectory of a recoil, every close approach j draws an excitation from the kernel at its own (b_j, E_j); with the trajectories of the partition of Sect. 19.1, this gives the excitation spectrum of the whole recoil, $\mathrm{d}\mathcal{N}_\mathrm{exc}/\mathrm{d}\omega_e = \langle\sum_j(\mathrm{d}P/\mathrm{d}\omega_e)(b_j, E_j)\rangle$. The spectrum has three parts, which a calculation should keep apart: binary excitation above $E_{R,\mathrm{th}}$, promotion at level crossings, and the linear-response excitation through the momentum tail of Sect. 19.2.2.

Each excitation then seeds the carrier cascade of Chap. 18, and the pairs follow from the cascade distribution $P_\mathrm{casc}(N_\mathrm{eh}|\omega_e)$ at that seed, not from $\omega_e/\varepsilon_\mathrm{eh}$. The mean pair number of a recoil is

$$\langle N_\mathrm{eh}\rangle(E_R) = \int \mathrm{d}\omega_e \, \frac{\mathrm{d}\mathcal{N}_\mathrm{exc}}{\mathrm{d}\omega_e} \, \bar{n}(\omega_e), \tag{19.11}$$

with $\bar{n}(\omega_e)$ the mean of P_casc. The difference matters because slow-ion excitations lie near the gap, where the cascade is far from its linear law $\bar{n} = \omega/\varepsilon_\mathrm{eh} + b_N$ (Sect. 18.2). Take the silicon cascade of Chap. 18, with the energy above the gap split uniformly between electron and hole. A seed between E_g and 2.9 eV gives one pair (mean 1.000), where $\omega_e/\varepsilon_\mathrm{eh}$ gives 0.52 at 1.95 eV and 0.78 at 2.9 eV. The two agree to within a few percent only from about 4 eV up (1.05 against 1.06 at 3.96 eV, 1.65 against 1.60 at 6.0 eV). Each near-gap excitation gives a pair with certainty.

The kernel carries the main theory uncertainty of the charge channel, and it comes from exchange and correlation. Real-time time-dependent density-functional theory with adiabatic semilocal functionals places level crossings that involve charge transfer at the wrong distances and makes the gap of a distorted configuration too small; already at equilibrium the semilocal gap of silicon is 0.56 eV against the measured 1.17 eV (Sect. 3.3). Since a too-small gap moves crossings outward and makes promotion too easy, the error has a known sign but no known size. Mean-field dynamics, in which the nuclei move in the average field of the evolving electrons (Ehrenfest dynamics), also yields the mean excitation energy, not its distribution, so the kernel must be built from the projection of the evolved orbitals on the conduction states. The exchange–correlation nuisance of the kernel is then the spread between three treatments that err differently: Ehrenfest dynamics with a hybrid functional; surface hopping with decoherence, in which the nuclei follow one electronic surface at a time and jump between surfaces at random; and Eq. (19.9) with diabatic couplings of the quasi-molecule from multireference calculations, which describe its electrons with several determinants. Appendix E.23 specifies the kernel on 80 points, E_coll from 50 eV to 5 keV and b_imp up to 1.5 Å, with this nuisance and an acceptance test at high energy, where the mean energy lost per unit path must approach the electronic stopping Eq. (4.54).

19.2.4 The ionization yield of silicon recoils

What a charge readout records is the ionization yield, the mean pair number of a nuclear recoil relative to that of an electron recoil of the same energy,

$$Y_q(E_R) = \frac{\langle N_{\mathrm{eh}}\rangle(E_R)\,\varepsilon_{\mathrm{eh}}}{E_R}. \tag{19.12}$$

An electron recoil of $E_R \geq 100\,\mathrm{eV}$ lies on the linear law of Sect. 18.2, where $E_R/\varepsilon_{\mathrm{eh}}$ is its mean pair number to within the constant $b_N \leq 0.35$, about 1 % of the 27 pairs at 100 eV. The literature also writes Y_q as the quenching factor, the electron-equivalent energy over the recoil energy. Lindhard's model, Eq. (16.42), identifies Y_q with the electronic share y_L of the slowing-down; Sect. 19.3 derives it and states where it holds. Eq. (19.11) shows why the identification fails at low energy: the share y_L counts energy, $\bar{n}(\omega_e)$ counts pairs, and near the gap the two are not proportional.

The silicon measurement [4] used a neutron beam of 55.7 keV on a 4 mm silicon crystal read out through its phonons, with the scattered neutron tagged at a fixed angle to fix E_R. Without bias the phonon signal measures E_R; with a bias V_{bias} the drifting pairs add $N_{\mathrm{eh}}eV_{\mathrm{bias}}$, so the total phonon energy is $E_R + N_{\mathrm{eh}}eV_{\mathrm{bias}}$, and the yield is extracted with 3.8 eV per pair [4, Eq. (1)]. This definition takes the phonon energy of the recoil to be the recoil energy, so that no energy is stored in defects; the stored share, a few percent of E_R (Sect. 19.4), lies inside the quoted errors. With the measured $\varepsilon_{\mathrm{eh}} = 3.749\,\mathrm{eV}$ of Sect. 18.4 the yields would be 1.3 % lower.

Fig. 19.3(a) shows the result. Between 0.1 keV and 3.9 keV the yield rises only from 0.10 to 0.24, and the two points below $E_{R,\mathrm{th}}$, at 0.10 and 0.22 keV, agree with each other within their errors: there is no sign of an ionization threshold above 100 eV. The Lindhard yield lies above the data at every energy below a keV, by a factor of 1.5 at 100 eV (0.153 against 0.102) and by 1.4 at 0.46 keV. Germanium, however, shows the opposite deviation. Recoils of 254 eV, below its binary threshold of 399 eV, give (64 ± 8) eV of electron-equivalent energy at 77 K, a quenching factor of (25 ± 3) % against Lindhard's 14.3 % [7, abstract and Table IV].

The measured Fano factor of the nuclear-recoil charge, $F_{\mathrm{NR}} = \mathrm{Var}\,N_{\mathrm{eh}}/\langle N_{\mathrm{eh}}\rangle$, says more. It is $0.9^{+0.7}_{-0.4}$ at 0.10 keV, $0.5^{+0.2}_{-0.1}$ at 0.22 keV and $1.8^{+0.6}_{-0.5}$ at 0.46 keV, and it grows to $8.4^{+2.4}_{-1.9}$ at 3.91 keV [4, Table II], seventy times the electron-recoil value $F_{\mathrm{eh}} = 0.119$. Suppose the electronic energy E_{el} of a recoil were converted at a constant $\varepsilon_{\mathrm{eh}}$. The law of total variance would then give $\mathrm{Var}\,N_{\mathrm{eh}} = F_{\mathrm{eh}}\langle E_{\mathrm{el}}\rangle/\varepsilon_{\mathrm{eh}} + \mathrm{Var}\,E_{\mathrm{el}}/\varepsilon_{\mathrm{eh}}^2$, the cascade fluctuation at fixed E_{el} plus the fluctuation of E_{el} itself, so that

$$F_{\mathrm{NR}} = F_{\mathrm{eh}} + \frac{\mathrm{Var}\,E_{\mathrm{el}}}{\varepsilon_{\mathrm{eh}}\langle E_{\mathrm{el}}\rangle}. \tag{19.13}$$

At 3.91 keV, with $\langle E_{\mathrm{el}}\rangle = Y_q E_R = 923\,\mathrm{eV}$ and 3.8 eV per pair, the measured F_{NR} requires a spread of 170 eV, 18 % of the mean: recoils of the same energy differ in how much of it reaches the electrons, because the sequence of collisions that each one makes is different. At 100 eV the excitations sit near the gap and Eq. (19.13) no longer applies; there F_{NR} near one is what a few independent excitations of about one pair each would give. The Fano factors thus test the distribution of the excitations, not only their mean.

19.2.5 Migdal electrons in the measured yield

The kick that starts the recoil already excites electrons. In the sudden picture of Chap. 12 the nucleus is set in motion and its electrons are left behind for an instant, and with probability P_{Mig} the cloud does not relax back to the ground state. Since a neutron calibration measures

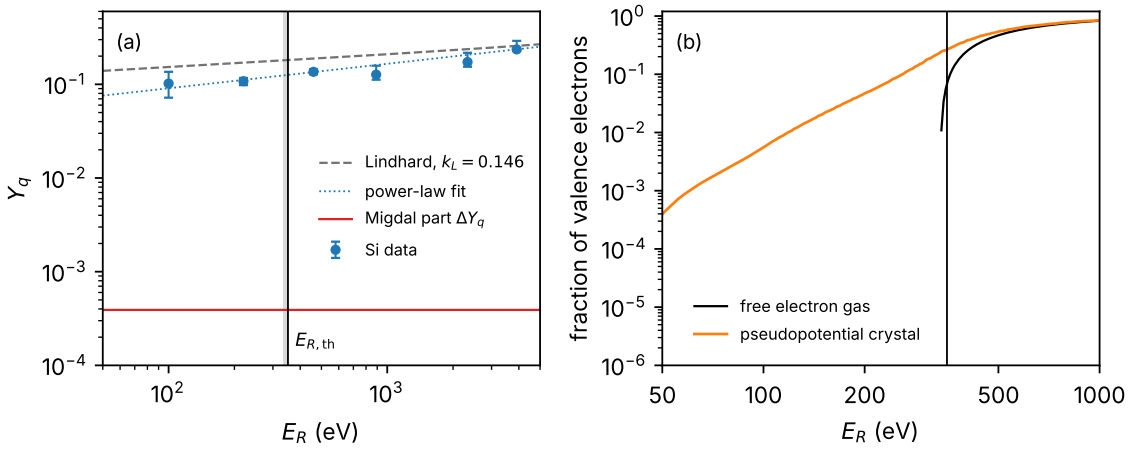


Figure 19.3: The charge channel of silicon recoils. (a) Measured ionization yield Y_q of silicon nuclear recoils (points: neutron scattering, total uncertainties, with 3.8 eV per pair [4, Table II]) against the recoil energy, with the power-law fit of the same work, $0.302(E_R/10\,\mathrm{keV})^{0.261}$ (dotted), and Lindhard's yield Eq. (16.42) with $k_L = 0.146$ (dashed). Vertical line: the binary threshold $E_{R,\mathrm{th}} = 352\,\mathrm{eV}$ of Eq. (19.7); gray band: down to 336 eV, the threshold with the v_N^2 term kept. Red: the part $\Delta Y_q^{\mathrm{Mig}}$ of the yield carried by Migdal electrons, Eq. (19.14), from the soft-limit spectrum of Sect. 12.3 converted to pairs by the silicon cascade of Chap. 18 with a uniform split of energy between electron and hole. (b) Fraction of valence electrons fast enough to be promoted across the gap $E_\mathrm{g} = 1.17\,\mathrm{eV}$ by a head-on binary collision, $p \geq p_{\min}(E_R)$ of Eq. (19.8): free electron gas of the valence density (black), which vanishes at 336 eV, the threshold with the v_N^2 term kept, and the valence momentum density of the empirical pseudopotential model of Sect. 3.3 (four bands, 10^3 points in the zone, plane waves to $5.3\,\mathrm{Å}^{-1}$), whose smooth pseudo-orbitals understate the high-momentum tail; vertical line as in (a).

the charge of real recoils, its yield contains these Migdal electrons, as Sect. 12.1 anticipated. The Migdal part of the yield is the integral Eq. (19.11) taken over the soft-limit spectrum Eq. (12.21):

$$\Delta Y_q^{\mathrm{Mig}} = \frac{\varepsilon_{\mathrm{eh}}}{E_R} \int \mathrm{d}\omega_e \, \frac{\mathrm{d}P_{\mathrm{Mig}}}{\mathrm{d}\omega_e} \, \bar{n}(\omega_e). \tag{19.14}$$

In the soft limit $\mathrm{d}P_{\mathrm{Mig}}/\mathrm{d}\omega_e$ is proportional to $v_N^2 = 2E_R/M_\kappa$, so the factor $1/E_R$ cancels and $\Delta Y_q^{\mathrm{Mig}}$ does not depend on the recoil energy. For silicon, the spectrum of Sect. 12.3 has $P_{\mathrm{Mig}} = 6.07 \times 10^{-3}$ at $E_R = 100\,\mathrm{eV}$ with a mean electronic energy of 5.97 eV. The energy-only cascade of Chap. 18 turns one such excitation into 1.69 pairs on average with a uniform split of the energy above the gap, or 1.92 if the electron takes all of it; the full-band cascade of Sect. 18.3.3 puts fewer events in the two-pair bin, and every Migdal excitation makes at least one pair. With 3.8 eV per pair, the convention of the data,

$$\Delta Y_q^{\mathrm{Mig}} = \frac{3.8\,\mathrm{eV}}{100\,\mathrm{eV}} \times 6.07 \times 10^{-3} \times 1.69 = 3.9 \times 10^{-4},$$

and 4.4×10^{-4} for the one-sided split; one pair per excitation gives the model-free lower edge 2.3×10^{-4}. The upper value is 0.4 % of the measured yield at 100 eV and 0.3 % at 0.46 keV. Both are soft-limit values, and the exact second-order bracket of Sect. 12.3.2 lowers them, multiplying the Migdal probability by 0.93 at 100 eV and by 0.62 at 500 eV. The measured charge of silicon recoils at these energies is therefore almost entirely collisional.

The decomposition fixes how a rate is assembled. A nuclear-recoil rate converted to charge with a measured yield already contains the Migdal ionization of those recoils, and adding a computed Migdal rate on top counts it twice. Conversely, a computed yield must count the Migdal excitation at the kick and the collisional excitation after it, and nothing else. The trajectory calculation therefore starts after the kick, from the nuclear momentum and site distribution that the response hands over (Sect. 19.1), and the lattice factor of the recoils that leave no Migdal electron carries the weight $1 - P_{\mathrm{Mig}}$ of Sect. 12.1, a 0.6 % correction at 100 eV. Sect. 12.3.2 met their boundary from the Migdal side: toward 1 keV the second-order Migdal amplitude acquires a vanishing denominator, because the already moving nucleus ionizes the electrons on shell as a slow ion does, and that part belongs to the collisional channel of this section.

Their balance shifts at low energy. The number of Migdal pairs per recoil falls in proportion to E_R. The collisional channel loses on two counts, fewer close collisions and the adiabatic suppression, whose factor $\mathrm{e}^{-\xi_{\mathrm{M}}}$ falls from 1.1×10^{-3} at 100 eV to 2.6×10^{-7} at 20 eV, against a factor of five for the Migdal probability. If promotion at crossings follows the same trend, an estimate that only the kernel can confirm, the Migdal electrons become the dominant source of charge somewhere between a few tens of eV and 100 eV, the recoil energies of dark matter below about 0.44 GeV. That makes the Migdal rate of Chap. 12, and not an extrapolated yield, the charge signal of the lightest nuclear-recoil searches.

19.2.6 Open problems

No calculation predicts the measured silicon yields and Fano factors below the binary threshold, or the germanium point at 254 eV. The missing input is the close-collision kernel Eq. (19.10) for Si–Si and Ge–Ge collisions, with its exchange–correlation spread. Once it exists, the chain of partition trajectories, kernel and cascade also yields the probability of a recoil without charge, which sets the efficiency of a charge readout at the lowest recoil energies (Appendix E.23).

19.3 Stopping of slow ions

Nearly every nuclear-recoil search in a semiconductor converts measured charge into recoil energy with one formula, Lindhard's electronic share $y_L(E_R)$ of Eq. (16.42). For silicon it gives $y_L = 0.302$ at 10 keV, and the measured yields of Fig. 19.3 lie below it, by the factor 1.5 of Sect. 19.2.4 at 100 eV. The formula is the solution of an integral equation for an ion slowing down by independent binary collisions, with electronic losses taken from the stopping of a uniform electron gas. In the language of Sect. 19.1 it is the partition Eq. (19.4) with $E_{\rm st} = 0$ and ω_e computed from continuum stopping. We will derive Lindhard's equation from that partition and see which of its assumptions fail below a keV. The Bethe formula, derived from the loss function of Sect. 4.4, then gives the opposite, first-order limit: there the solid enters through one mean excitation energy, and the limit sets in only above about twice the Fermi velocity, far above any recoil.

Two objects carry Lindhard's name. The Lindhard function $\varepsilon_{\rm Lin}$ of Sect. 4.5 is the dielectric function of the free electron gas; Lindhard quenching is the electronic share y_L of a slowing-down ion. Their only connection is that the electronic stopping behind y_L was first estimated from an electron gas.

19.3.1 Binary collisions of a slow ion

A silicon recoil of 1 keV moves at $v_N = 2.76 \times 10^{-4}$, a twenty-fifth of the Fermi velocity of the valence electrons (Worked example 19.2). It loses energy in two ways, which Lindhard's model treats as separate: elastic collisions with other nuclei, screened by their electrons, and excitation of the electrons themselves.

The nuclear collisions are those of two atoms interacting through a screened Coulomb potential $(Z_\kappa^2\alpha/r)\,\phi_{\rm TF}(r/a_L)$, with $\phi_{\rm TF}$ the Thomas–Fermi screening function of a neutral atom [8, 9] and $a_L = 0.8853\,a_0/(\sqrt{2}\,Z_\kappa^{1/3})$ Lindhard's screening length for two equal atoms [10, Sect. II A]. For silicon, $a_L = 0.1375$ Å. The natural energy scale is the Coulomb energy of the pair at the distance a_L. For equal masses the center-of-mass frame carries half of the ion's kinetic energy E, and the reduced energy is

$$\epsilon_L = \frac{a_L}{2Z_\kappa^2 e_{\rm G}^2}\,E \equiv c_L E, \tag{19.15}$$

with $e_{\rm G}^2 = \alpha$ the Gaussian charge squared, 14.40 eV Å in conventional units. For silicon $c_L = 0.024\,35\,{\rm keV}^{-1}$, the value $11.5\,Z_\kappa^{-7/3}\,{\rm keV}^{-1}$ of Eq. (16.42): a recoil of 1 keV has $\epsilon_L = 0.024$, and $\epsilon_L = 1$ is reached only at 41 keV.

A collision that deflects the pair by the angle $\theta_{\rm s}$ in the center-of-mass frame gives the struck atom the energy $T_n = E\sin^2(\theta_{\rm s}/2)$. Lindhard's insight was that in the reduced variable $x_L = \epsilon_L \sin(\theta_{\rm s}/2)$ the cross-section of a screened Coulomb potential depends on one function of one variable,

$$\mathrm{d}\sigma_n = \pi a_L^2\,\frac{f_{\rm LSS}(x_L)}{x_L^2}\,\mathrm{d}x_L, \tag{19.16}$$

with $t = x_L^2$ the form of [11, Eq. (2)]. The reason is visible in the small-angle (impulse) approximation, where the deflection is the transverse momentum the potential imparts along a straight path divided by the momentum. With $\beta = b/a_L$ the impact parameter in units of a_L and $I(\beta) = \int_0^\infty \mathrm{d}\zeta\,\phi_{\rm TF}(r)/r$, $r = (\beta^2+\zeta^2)^{1/2}$, the deflection is $\theta_{\rm s} = -\epsilon_L^{-1}\,\mathrm{d}I/\mathrm{d}\beta$, so that $x_L \simeq \epsilon_L\theta_{\rm s}/2 = -\frac{1}{2}\,\mathrm{d}I/\mathrm{d}\beta$ is a function of β alone. Then $\mathrm{d}\sigma_n = 2\pi a_L^2\beta\,\mathrm{d}\beta$ gives $f_{\rm LSS}(x_L) = 2x_L^2\beta\,|\mathrm{d}\beta/\mathrm{d}x_L|$, with no reference to the energy. An unscreened potential, $\phi_{\rm TF} = 1$, has $\mathrm{d}I/\mathrm{d}\beta = -1/\beta$ and gives the Rutherford form $f_{\rm LSS} = 1/2x_L$. The screened function rises as a power of x_L at small x_L, peaks at 0.43 near $x_L = 0.14$ and approaches the Rutherford form

for $x_L \gg 1$. The mean energy lost to nuclei per unit reduced path, $s_n(\epsilon_L) = \epsilon_L^{-1} \int_0^{\epsilon_L} f_{\mathrm{LSS}}\, \mathrm{d}x_L$, peaks at 0.41 near $\epsilon_L = 0.3$.

The electronic losses are a friction. For a projectile much slower than the electrons, the dielectric stopping Eq. (4.54) samples only the low-energy side of the loss function, and the energy loss per unit path is proportional to v_N. Worked example 4.9 showed this for a damped oscillator, and Limit Check 19.2 below shows it for the electron gas. Since $E \propto v_N^2$, the reduced electronic stopping per unit reduced path length $\rho_L = \pi a_L^2 n_{\mathrm{at}} x$, with x the path length and n_{at} the atomic density, is

$$\left.\frac{\mathrm{d}\epsilon_L}{\mathrm{d}\rho_L}\right|_e = k_L\, \epsilon_L^{1/2}, \qquad k_L = 0.133\, Z_\kappa^{2/3} A^{-1/2}, \tag{19.17}$$

with A the atomic mass in atomic units [12, Eq. (4)]. For silicon $k_L = 0.146$. The coefficient comes from Lindhard and Scharff's estimate of the friction of an ion in an electron gas. It contains a factor of about $Z_\kappa^{1/6}$, 1.55 for silicon, inserted by hand to match the stopping data of their time, as Sarkis et al. point out [11, Sect. "Scaling length"]. In physical units Eq. (19.17) assigns a silicon ion in silicon an electronic stopping of 0.88, 2.77 and 8.76 eV/Å at 0.1, 1 and 10 keV.

The friction law is an empirical scaling at these speeds, not a derived limit. The dielectric formula Eq. (4.54) is first order in the ion's charge, and its expansion parameter for a silicon ion among the valence electrons is the Born parameter $Z_\kappa \alpha / v_{\mathrm{F}} = 14.6$ (Sect. 19.2.3). Taken at face value, the electron gas of the valence density gives a proton at the speed of a 1 keV silicon recoil a stopping of 0.32 eV/Å. The first-order stopping of the silicon ion would be $Z_\kappa^2 = 196$ times that, 64 eV/Å, which is 23 times the value of Eq. (19.17). The ion's own electrons screen its charge, and the slow collision is far from linear response. Eq. (19.17) keeps only the scaling with v_N and fits the coefficient.

19.3.2 Lindhard's integral equation

Follow the ion of Sect. 19.1 as a sequence of collisions. Each collision transfers T_n to a target atom, which starts its own slowing-down, and energies $T_{e,i}$ to electrons. Let $T_{\mathrm{d}}(E)$ be the mean energy that an ion starting with kinetic energy E eventually gives to atomic motion, summed over itself and all the atoms it sets moving: the damage energy of Eq. (16.42), as a function of the starting energy. The electronic share is the rest, $E - T_{\mathrm{d}}(E)$. In the partition Eq. (19.4) at $E_{\mathrm{st}} = 0$, T_{d} is the part of E_{lat} delivered by nuclear collisions and $E - T_{\mathrm{d}}$ is ω_e; the phonons that the electrons later emit in their cascade belong to E_{heat} but not to T_{d}.

Two assumptions turn this into an equation. The first is that collisions are binary, with targets at rest and uncorrelated with each other, so that the struck atom starts an independent copy of the same process. The second is that the ion has no memory, so that after a collision it is again an ion of the new energy in a fresh medium. Then the mean damage energy before a collision equals the sum of the mean damage energies of the three objects after it. These are the scattered ion with energy $E - T_n - \sum_i T_{e,i}$, the struck atom with $T_n - U$, where U is the energy spent breaking its binding, and the excited electrons. Averaging over all collisions, which is the integral over the cross-section,

$$\int \mathrm{d}\sigma_{n,e} \left[T_{\mathrm{d}}\Big(E - T_n - \sum_i T_{e,i}\Big) + T_{\mathrm{d}}(T_n - U) - T_{\mathrm{d}}(E) + \sum_i T_{\mathrm{d}}^{(e)}(T_{e,i}) \right] = 0, \tag{19.18}$$

which is Lindhard's basic integral equation [13], as quoted by Sarkis et al. [12, Eq. (3)], with $T_{\mathrm{d}}^{(e)}$ the damage energy an excited electron produces. The equation holds for the mean of any quantity that adds up over the collisions; the damage energy is one such quantity. It is

the first-moment equation of a branching process, the nuclear counterpart of the cascade of Sect. 18.1.

Lindhard reduced Eq. (19.18) with five further approximations, as quoted by Sarkis et al. [12, Sect. II]: (A) excited electrons give no appreciable energy back to atoms, so the last term is dropped; (B) the binding U is set to zero; (C) the energy given to electrons in one collision is small compared to E; (D) nuclear and electronic collisions are separate events; (E) the nuclear transfer T_n is small compared to E where it matters for the electronic term. With (C) the first term is expanded to first order in the electronic transfer,

$$T_{\mathrm{d}}\Big(E - T_n - \sum_i T_{e,i}\Big) \simeq T_{\mathrm{d}}(E - T_n) - T'_{\mathrm{d}}(E) \sum_i T_{e,i},$$

where (E) puts the derivative at E rather than at $E - T_n$. By (D) each collision is either nuclear, with $T_{e,i} = 0$, or electronic, with $T_n = 0$. The electronic collisions then contribute $-T'_{\mathrm{d}}(E) \int \mathrm{d}\sigma_e \sum_i T_{e,i}$, and the integral is the electronic stopping per atom, $(\mathrm{d}E/\mathrm{d}x)_e/n_{\mathrm{at}}$. The nuclear collisions contribute the bracket with $\sum_i T_{e,i} = 0$ and $U = 0$, integrated with Eq. (19.16). In reduced units, $\epsilon_{\mathrm{d}} = c_L T_{\mathrm{d}}$, with Eq. (19.17) for the stopping and $T_n/E = x_L^2/\epsilon_L^2$,

$$k_L\, \epsilon_L^{1/2}\, \epsilon'_{\mathrm{d}}(\epsilon_L) = \int_0^{\epsilon_L} \mathrm{d}x_L\, \frac{f_{\mathrm{LSS}}(x_L)}{x_L^2} \Big[\epsilon_{\mathrm{d}}\Big(\epsilon_L - \frac{x_L^2}{\epsilon_L}\Big) + \epsilon_{\mathrm{d}}\Big(\frac{x_L^2}{\epsilon_L}\Big) - \epsilon_{\mathrm{d}}(\epsilon_L)\Big], \tag{19.19}$$

Lindhard's simplified equation [12, Eq. (5)], with $\epsilon_{\mathrm{d}} \to \epsilon_L$ as $\epsilon_L \to 0$. The left side is the energy the ion would have delivered to atoms but gave to electrons along a unit of reduced path; the right side is what the nuclear collisions along the same path redistribute between the ion and the atoms it strikes. The upper limit $x_L = \epsilon_L$ is the head-on collision, which hands the whole energy to the struck atom.

The small-energy behavior shows where the formula's shape comes from. Continue the screened cross-section below the tabulated range as Lindhard's power law $f_{\mathrm{LSS}} = \lambda_L x_L^{1/3}$ and try $\epsilon_{\mathrm{d}} = \epsilon_L - C\epsilon_L^p$ and keep the first order in k_L. The leading term ϵ_L makes the bracket vanish identically, since the energies of ion and target add up to ϵ_L. With $x_L^2/\epsilon_L = \epsilon_L s$ the right side becomes $\frac{1}{2}\lambda_L C\, \epsilon_L^{p-2/3} J$, with $J = \int_0^1 \mathrm{d}s\, s^{-4/3}[1 - (1-s)^p - s^p]$, and the left side is $k_L \epsilon_L^{1/2}$ to this order. Matching the powers gives $p = 7/6$ and $C = 2k_L/(\lambda_L J)$:

$$y_L = 1 - \frac{\epsilon_{\mathrm{d}}}{\epsilon_L} \simeq \frac{2k_L}{\lambda_L J}\, \epsilon_L^{1/6}, \qquad J = 0.4742. \tag{19.20}$$

With the screened cross-section used below, $\lambda_L = 1.283$ and the coefficient is $3.29\, k_L$. This is the origin of the leading term $3\epsilon_L^{0.15}$ of g_L in Eq. (16.42). With this continuation the electronic share falls as the sixth root of the energy, because a slow ion keeps handing its energy on to slower atoms, each of which loses a fraction to electrons. The law is a property of the assumed $x_L^{1/3}$ continuation, not of the screened Coulomb potential: the Thomas–Fermi screening function falls as r^{-3}, the potential as r^{-4}, and $f_{\mathrm{LSS}} \propto x_L^{1/2}$ at the smallest x_L. With that tail the nuclear and electronic stopping scale alike and y_L tends to a nonzero constant as $\epsilon_L \to 0$.

Eq. (19.19) has no closed solution for a screened potential. Lindhard and coworkers solved it numerically, and the parametrization of Eq. (16.42), $y_L = k_L g_L/(1 + k_L g_L)$ with $g_L = 3\epsilon_L^{0.15} + 0.7\epsilon_L^{0.6} + \epsilon_L$, is Lewin and Smith's fit to their solution [14], as quoted by Sorensen [10, Eqs. (5)–(7)]. We solve the equation again to see how well the fit represents it. The cross-section is built as above, from the Thomas–Fermi function solved numerically ($\phi'_{\mathrm{TF}}(0) = -1.58807$) and the impulse deflection, and continued below $x_L = 2 \times 10^{-3}$, where Lindhard did not tabulate f_{LSS} [10, Sect. II D], by the power law $\lambda_L x_L^{1/3}$. The equation

involves ϵ_{d} only at energies up to ϵ_L, so it can be integrated upward from the asymptotic form Eq. (19.20); the right end of the integral, which involves $\epsilon'_{\mathrm{d}}(\epsilon_L)$ itself, is solved exactly at each step. Halving both grids changes y_L by less than 5×10^{-4}.

Limit Check 19.1. Lindhard quenching

Lindhard's electronic share of a recoil in a one-element target [13], [15, Eq. (3)], Eq. (16.42): $y_L = k_L g_L(\epsilon_L)/[1 + k_L g_L(\epsilon_L)]$, $g_L = 3\epsilon_L^{0.15} + 0.7\epsilon_L^{0.6} + \epsilon_L$, with k_L and ϵ_L of Eqs. (19.15) and (19.17).

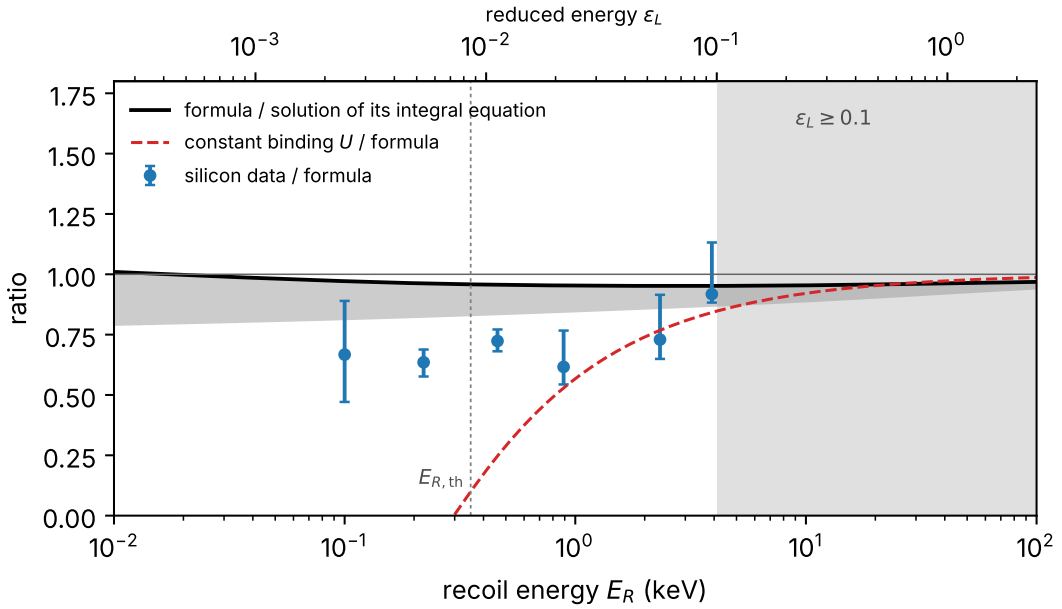


Figure 19.4: Silicon, $k_L = 0.146$, against the recoil energy. Solid: the formula divided by the numerical solution of the binary-collision equation Eq. (19.19) with the Thomas–Fermi cross-section in the impulse approximation, continued below $x_L = 2 \times 10^{-3}$ as $x_L^{1/3}$; dark band: the same with continuation exponents from 1/3 to the local slope 0.363 of the computed cross-section, a sensitivity to that choice. Points: the measured silicon yields [4, Table II] divided by the formula. Dashed: the constant-binding model with $U = 0.15\,\mathrm{keV}$ [12, Eq. (13), Table 3] divided by the formula. Gray: $\epsilon_L \geq 0.1$, $E_R \geq 4.1\,\mathrm{keV}$, where the formula describes the silicon data [12, Sect. II]. Vertical line: the binary threshold $E_{R,\mathrm{th}} = 352\,\mathrm{eV}$.

Below a few keV the measured yields fall 8 to 38 % under the formula, a deficit comparable to the formula's sensitivity to its softest collisions; the stored energy, the gap and the conversion to pairs that the model omits also act there.

With the $x_L^{1/3}$ continuation the solution and the fit agree to 3 % below 100 eV, and the fit lies 4 % to 5 % below the solution between 1 and 20 keV, where y_L is 0.210 against 0.220 at 1 keV. The continuation matters more than the fit. Continuing the computed cross-section at its own local slope, 0.363 at $x_L = 2 \times 10^{-3}$, raises the solution by 13 % at 1 keV and by 10 % at 4 keV; an exponent of 0.30 lowers it by 11 % at 1 keV, and 0.40 raises it by 40 %. The dark band of Fig. 19.4 is therefore a sensitivity to one modeling choice, not an uncertainty. Nothing places the true yield between the two continuations: below the matching point the Thomas–Fermi cross-section steepens toward $x_L^{1/2}$, while binding and the lattice remove the

softest transfers altogether. Below $\epsilon_L = 2 \times 10^{-3}$, about 80 eV, the whole cross-section of a slow secondary lies in the continued region, and the binary model does not determine the yield of a keV recoil there. Those are the soft collisions of the slowest secondaries, at energies of tens of eV, where atoms are bound and the binary model has least claim to hold. The equation propagates their treatment upward because every recoil ends as a population of such secondaries.

19.3.3 Implicit assumptions of Lindhard quenching

Read against the partition of Sect. 19.1, the chain from Eq. (19.18) to y_L makes three assumptions that its list of approximations does not name: no stored energy, electronic friction down to zero speed, and conversion to pairs in proportion. Each fails below a keV in silicon.

The first is that no energy is stored. With $U = 0$ every collision returns its full transfer to atomic motion, and an atom that ends displaced keeps no energy in the lattice: $E_{\mathrm{st}} = 0$ in Eq. (19.4). Restoring U does not repair this. In the constant-binding model of Sarkis et al. [12, Eqs. (9), (13)] the struck atom pays U before it moves, the lower limit of the integral in Eq. (19.19) becomes $(\epsilon_L c_L U)^{1/2}$, and the yield vanishes near $E_R = 2U$. Their fit to the silicon data above 0.68 keV gives $U = 0.15 \pm 0.06$ keV [12, Table 3], and with the fitted correction terms the yield reaches zero at 298 eV (Fig. 19.5); the measured yield at 100 eV is $0.102^{+0.034}_{-0.030}$. The later model of the same authors replaces U by a function $U(E) = U_{\mathrm{FP}} + U_{\mathrm{EDFT}}(E)$: a constant $U_{\mathrm{FP}} = 23.6$ eV, which the authors call the Frenkel-pair creation energy and take from averaged displacement thresholds, plus an energy-dependent excitation of the inner electrons. It adds electronic straggling and a modern stopping model, and it fits 102 points from 0.68 keV to 3 MeV with $\chi^2/\mathrm{dof} = 1870/101$; its threshold lies just above U_{FP} [11, Eqs. (2)–(4), Fig. 3]. The configurational energy of a Frenkel pair in silicon is 6.8 to 9.7 eV (Sect. 7.4), a third of U_{FP}; in the partition that part is stored energy and the rest of a threshold-sized U_{FP} is heat. In both models, however, the U spent by secondary atoms appears as an argument of $T_{\mathrm{d}}(T_n - U)$, is missing from the damage energy, and therefore lands in the electronic share $E - T_{\mathrm{d}}$ by complement. Energy that the partition stores in defects is counted as ionization.

The second is that electronic losses follow the friction law down to zero speed. Eq. (19.17) gives a silicon ion of 100 eV an electronic stopping of 0.88 eV/Å, while a binary collision at that energy cannot create a pair at all (Sect. 19.2.1), and the momentum tail lets through a fraction 5.6×10^{-3} of the valence electrons (Sect. 19.2.2). Below $E_{R,\mathrm{th}}$ the friction assigns losses to final states that are strongly suppressed. The modified equation of Sarkis et al. also keeps a term $-\frac{1}{2} k_L \epsilon_L^{3/2} \epsilon''_{\mathrm{d}}$, obtained from the second-order expansion with the mean nuclear transfer $\epsilon_L/2$ in a collision that also excites electrons [12, Eq. (9), App. B]. Under approximation (D) every collision has $T_n = 0$ or $\sum_i T_{e,i} = 0$, so the cross term $T_n \sum_i T_{e,i}$ that produces it vanishes identically, and the term contradicts the separation used to obtain the stopping.

The third is that the electronic share becomes charge in proportion, $N_{\mathrm{eh}} = y_L E_R / \varepsilon_{\mathrm{eh}}$, with every pair collected and none trapped. Then $Y_q = y_L$. The excitations of a slow ion lie near the gap, where the cascade gives one pair per excitation rather than $\omega_e/\varepsilon_{\mathrm{eh}}$ (Eq. (19.11)), so y_L and Y_q differ even if y_L were exact. Trapping and incomplete collection lower the measured charge further (Sects. 18.5 and 19.5).

The approximations that the list does name are weakest at the same energies. The medium is a random gas of atoms, with no channels and no dependence on direction, and collisions are binary and independent, which fails in the dense cascades of germanium recoils of a few keV. The reduced energy is also convention-dependent. Sarkis et al. define it in their later work with the universal screening length and an effective projectile charge $Z_1 = 12$. This maps the same recoil energy to a reduced energy 10 % higher than Eq. (19.15) [11, Sect. “The nuclear recoil QF”], so that fitted constants cannot be carried from one convention to the other.

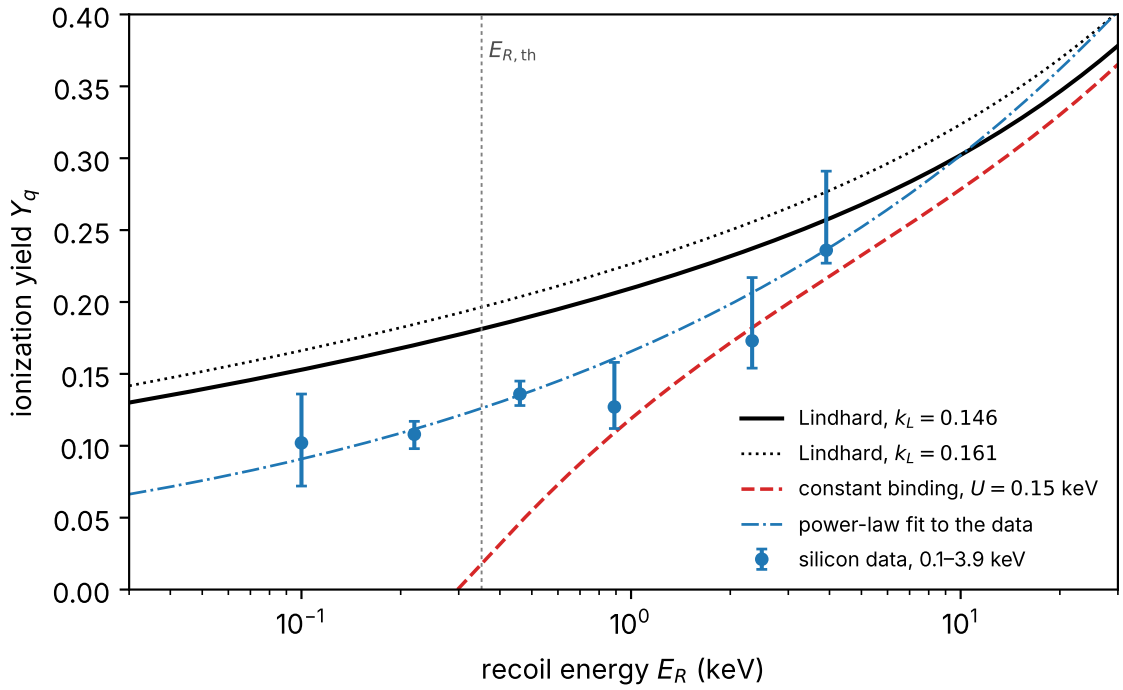


Figure 19.5: Lindhard and modified-Lindhard yields of silicon against the measured ionization yield. Solid: Lindhard's formula Eq. (16.42) with $k_L = 0.146$; dotted: with $k_L = 0.161$, the value of a fit of the numerical constant-binding solution to the silicon data [12, Table 4]. Dashed: the constant-binding model of Sarkis et al. with $U = 0.15\,\mathrm{keV}$, $C_0 = 9.1 \times 10^{-3}$ and $C_1 = 3.33 \times 10^{-5}$ [12, Eq. (13), Table 3], which vanishes at 298 eV. Points: the silicon measurement with total uncertainties [4, Table II]; dash-dotted: its power-law fit $0.302(E_R/10\,\mathrm{keV})^{0.261}$, pinned to the formula at 10 keV. Vertical line: the binary threshold $E_{R,\mathrm{th}} = 352\,\mathrm{eV}$ (Sect. 19.2).

Fig. 19.5 shows the result of these assumptions. Raising k_L moves the whole curve up and away from the sub-keV data. Adding a constant binding brings the curve down to the data near 2 keV only by introducing a cutoff that the data at 0.10 and 0.22 keV contradict. The data themselves follow the power law $E_R^{0.26}$ from 0.1 keV to 3.9 keV, flatter than y_L and with no threshold. Each modified model fixes one assumption and lets its fitted constants absorb the rest; a good fit therefore does not tell us which assumption was wrong. The calculation that would show it is the partition of Sect. 19.1 with stored energy, the excitation kernel of Sect. 19.2 and the cascade conversion, specified in Appendix E.23. Above about 4 keV, where stored energy and the gap are small fractions of the electronic share, Lindhard's formula describes the silicon data and needs no repair.

19.3.4 Bethe stopping from the loss function

At the opposite end of the speed range the stopping of an ion is a derived result. Take the dielectric stopping Eq. (4.54) of a heavy projectile of charge $Z_1 e$ and speed v at $T = 0$, and suppose v is large compared to the speeds of the electrons that hold the weight of the loss function. Split the momentum integral at a value q_0 chosen so that two conditions hold together: $q_0 v$ lies far above the energies where L has its weight, and q_0 is small compared to the inverse size of the atoms, so that $L(q,\omega) \simeq L(0,\omega)$ for $q < q_0$. Both are possible only if v is large.

Below q_0 the loss function is its optical limit, and the condition $\omega \le qv$ becomes $q \ge \omega/v$. Exchanging the order of integration,

$$\int_0^{q_0} \frac{\mathrm{d}q}{q} \int_0^{qv} \mathrm{d}\omega\, \omega L(0,\omega) = \int_0^{\infty} \mathrm{d}\omega\, \omega L(0,\omega)\, \ln\frac{q_0 v}{\omega},$$

where the upper limit of the ω integral moved to infinity because $q_0 v$ lies above the whole spectrum. Above q_0 the line $\omega = qv$ lies above the spectrum at every q until it meets the free-electron recoil $\omega = q^2/2m_e$, at $q = 2m_e v$. Between q_0 and $2m_e v$ the inner integral is therefore the whole f-sum $(\pi/2)\omega_p^2$ of Eq. (4.46), and beyond $2m_e v$ it is zero, since the Bethe ridge has moved out of reach. The second piece is $(\pi/2)\omega_p^2 \ln(2m_e v/q_0)$. Writing the f-sum as $\int_0^\infty \mathrm{d}\omega\, \omega L(0,\omega) = (\pi/2)\omega_p^2$ in the first piece, q_0 cancels between the two, and with $\omega_p^2 = 4\pi\alpha \boldsymbol{n}_e/m_e$ the prefactor $2Z_1^2\alpha/\pi v^2$ of Eq. (4.54) times the f-sum $(\pi/2)\omega_p^2$ equals $4\pi Z_1^2 \alpha^2 \boldsymbol{n}_e/(m_e v^2)$.

Bethe stopping from the loss function

For a heavy projectile of charge $Z_1 e$ and speed v well above the speeds of the electrons that hold the weight of the loss function, in first order in Z_1,

$$-\frac{\mathrm{d}E}{\mathrm{d}x} = \frac{4\pi Z_1^2 \alpha^2 \boldsymbol{n}_e}{m_e v^2}\, \ln\frac{2m_e v^2}{\bar{I}}, \tag{19.21}$$

with the mean excitation energy defined by the logarithmic moment of the optical loss function,

$$\ln \bar{I} = \frac{2}{\pi\omega_p^2} \int_0^{\infty} \mathrm{d}\omega\, \omega\, L(0,\omega)\, \ln\omega, \tag{19.22}$$

where $\boldsymbol{n}_e$ and ω_p count all the electrons of the target.

This is the nonrelativistic limit of the Bethe equation of the Particle Data Group. Its bracket $\frac{1}{2}\ln(2m_e\beta^2\gamma^2 W_{\max}/I^2) - \beta^2$, with $W_{\max} = 2m_e\beta^2\gamma^2$ for a heavy projectile, becomes $\ln(2m_e v^2/I)$ up to terms of order v^4, and its prefactor $Kz^2(Z/A)/\beta^2$ per unit mass is

$4\pi\alpha^2 n_e/(m_e v^2)$ per unit length [16, Eq. (34.5)]. The Particle Data Group describes I as a geometric average of the excitation energies weighted by their oscillator strengths [16, Sect. 34.2.1]; Eq. (19.22) makes the weight precise for a condensed medium. It is the loss function, not ε_2, because the projectile couples to the total potential, and the logarithm sits inside the f-sum weight ωL. The prefactor of Eq. (19.21) depends on the electron density alone, as Sect. 4.4 anticipated; the solid enters only through $\bar{I}$, and only logarithmically.

Worked example 19.3: The mean excitation energy of silicon

Start with the four valence electrons in the single damped oscillator of Worked example 4.9 at $q \to 0$: $L(0,\omega) = \omega_p^2 \Gamma_\mathrm{w} \omega / [(\omega^2 - \bar{E}^2)^2 + \Gamma_\mathrm{w}^2 \omega^2]$ with $\omega_p = 16.6\,\mathrm{eV}$, $\bar{E} = 17.4\,\mathrm{eV}$ and $\Gamma_\mathrm{w} = 3.66\,\mathrm{eV}$. For a narrow line Eq. (19.22) would give $\bar{I} = \bar{E}$, since all the weight sits at one energy. The damping spreads it: the model puts 88 % of its f-sum between 10 and 30 eV, but 7.8 % between 30 and 100 eV and 2.4 % above, in a tail $\omega L \propto \Gamma_\mathrm{w} \omega_p^2/\omega^2$ that the logarithm weights heavily. Numerically,

$$\ln \bar{I}_\mathrm{val} = \frac{2}{\pi \omega_p^2} \int_0^\infty \mathrm{d}\omega\, \omega L(0,\omega) \ln \frac{\omega}{1\,\mathrm{eV}} = 3.012, \qquad \bar{I}_\mathrm{val} = 20.3\,\mathrm{eV},$$

17 % above $\bar{E}$; a damping of 2 eV brings it to 19.0 eV. The inverse moment of Eq. (4.48) fixed $\bar{E}$ from the low-energy side of the spectrum; $\bar{I}$ is set by the high-energy side, and the two mean energies agree only for a narrow line.

The valence electrons are four of fourteen, and the core changes the answer by an order of magnitude. Take the measured loss function of Sect. 10.7, which extends to 30 keV. For all fourteen electrons $\omega_p = 31.05\,\mathrm{eV}$ and $(\pi/2)\omega_p^2 = 1514\,\mathrm{eV}^2$; the tabulated spectrum holds $1537\,\mathrm{eV}^2$, 1.5 % more. Split at 90 eV, below the L edge, and at 1.83 keV, below the K edge, the f-sum weight w_j and the logarithmic mean energy ω_j of each band are

$$\text{valence: } 0.280,\ 19.5\,\mathrm{eV}; \qquad \text{L shell: } 0.607,\ 246\,\mathrm{eV}; \qquad \text{K shell: } 0.113,\ 3.17\,\mathrm{keV},$$

so that, with the spectrum normalized to its own f-sum, $\ln(\bar{I}/1\,\mathrm{eV}) = \sum_j w_j \ln(\omega_j/1\,\mathrm{eV}) = 0.832 + 3.340 + 0.910 = 5.082$ and $\bar{I} = 161\,\mathrm{eV}$. Rescaling the spectrum to the exact f-sum leaves this value unchanged. It lies 7 % below the adopted 173.0 eV of the Particle Data Group [17], and the 1.5 % excess of the tabulated f-sum is a measure of the error of the data. The L shell holds 61 % of the oscillator strength and the valence electrons, which set every dark-matter rate of Chap. 10, 28 %; without the core shells $\bar{I}$ would be the 19.5 eV of the valence band.

The derivation needs two conditions, and both fail for recoils. The split at q_0 requires $2m_e v^2 \gg \bar{I}$, so that the logarithm is large: a proton of 1 MeV has $2m_e v^2 = 2.2\,\mathrm{keV}$, thirteen times the $\bar{I}$ of silicon, while a silicon recoil of 10 keV has 0.78 eV, 4.5×10^{-3} of it, and Eq. (19.21) would give a negative stopping. The first-order treatment requires $Z_1\alpha/v \ll 1$: 0.16 for the proton, 117 for the recoil. Limit Check 19.2 shows the first condition in an electron gas, where both formulas can be evaluated exactly.

Limit Check 19.2. Bethe stopping

The Bethe formula in its nonrelativistic limit [16, Eq. (34.5)], Eq. (19.21): $-\mathrm{d}E/\mathrm{d}x = (4\pi Z_1^2 \alpha^2 n_e/m_e v^2) \ln(2m_e v^2/\bar{I})$, with $\bar{I}$ from Eq. (19.22).

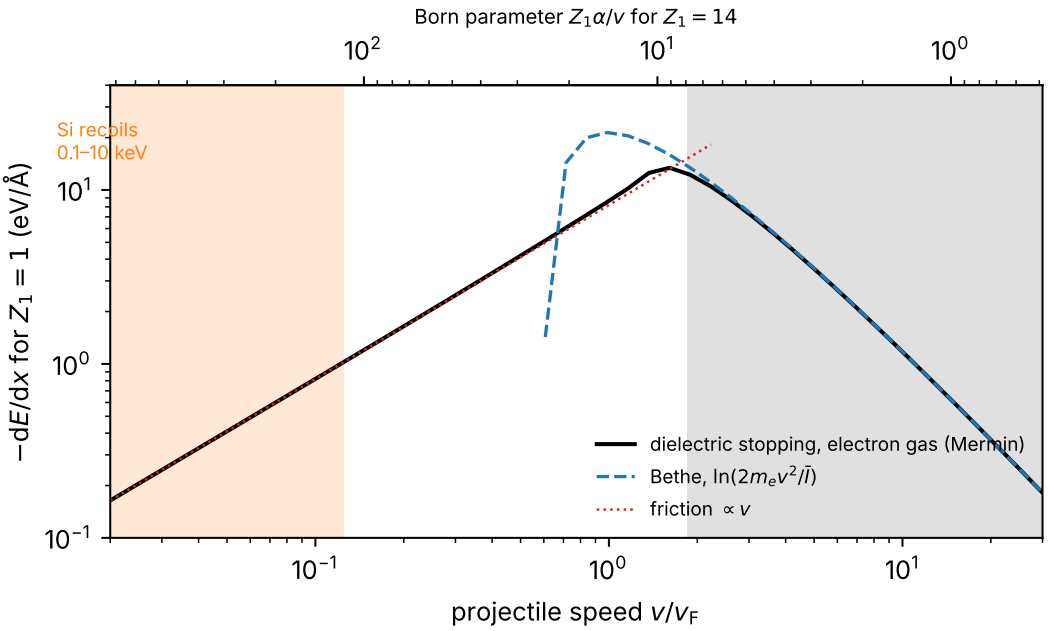


Figure 19.6: Stopping of a heavy projectile with $Z_1 = 1$ in the electron gas of the silicon valence density ($k_F = 1.8085\,\text{Å}^{-1}$, $v_F = 6.98 \times 10^{-3}$, $\omega_p = 16.6\,\text{eV}$), against its speed. Solid: the dielectric stopping Eq. (4.54) with the Mermin dielectric function of Sect. 4.5 and relaxation rate $0.1\,\omega_p$, a modeling choice that keeps the f-sum. Dashed: Eq. (19.21) with the model's own $\bar{I} = 17.9\,\text{eV}$ from Eq. (19.22). Dotted: the friction law, linear in v, fitted at the lowest speed. Gray: $v \geq 1.84\,v_F$, where the Bethe formula lies within 10 % of the dielectric stopping. Orange: the speeds of silicon recoils from 0.1 to 10 keV. Top axis: the Born parameter $Z_1\alpha/v$ of a silicon ion, $Z_1 = 14$.

Below about $2v_F$ the momentum transfers that reach the loss function no longer exhaust its f-sum, and the stopping turns over into the friction regime, where it is set by the low-energy side of L and not by $\bar{I}$.

The approach to the Bethe form is fast once it starts: the dielectric stopping lies within 5 % of Eq. (19.21) above $2.27\,v_F$ and within 1 % above $4.3\,v_F$. The Bethe logarithm vanishes at $v = (\bar{I}/2m_e)^{1/2} = 0.60\,v_F$, below which the formula has no meaning. In the friction regime the same Mermin electron gas gives a stopping of $1173\,\text{eV/Å}$ times v for $Z_1 = 1$, constant to 0.1 % up to $v = 0.1\,v_F$. Silicon recoils, at 0.013 to $0.13\,v_F$, sit deep in that regime, with Born parameters from 117 to over a thousand. The corrections that the Particle Data Group lists for slower projectiles, shell corrections, the Barkas term of order Z_1^3 and the Bloch term, extend the Bethe formula down to about $\beta \approx 0.05$, and between $\beta = 0.01$ and 0.05 no theory is satisfactory [16, Sect. 34.2.6]. Recoils from dark matter move at $\beta < 10^{-3}$. Computing $L(q, \omega)$ to the core energies that set $\bar{I}$, with local fields and the core states of Appendix E.15, is the task of Appendices E.6 and E.16; for slow ions it fixes the first-order stopping of Sect. 19.2.2, not the yield.

A fast heavy ion, by contrast, leaves a track of 10^2 to 10^4 correlated collisions in a lattice already disturbed by the previous ones, and what a track detector reads is a defect configuration, not an excitation spectrum; damage-track and paleo-detector searches need Eqs. (19.17) and (19.21) as input to a transport calculation of that track.

19.4 Defects and stored energy

A ^{206}Pb nucleus that recoils into a germanium detector with 103 keV is reconstructed 6.08 % low, once the calibration of surface events has been corrected [18]. Silicon nuclear recoils of 10 to 100 keV give 4.8 % less phonon signal than electron recoils of the same energy [15]. A Frenkel pair in silicon stores about 7 eV, and a recoil of a few keV makes tens of them, so energy stored in defects can produce deficits of this size. Collection can too, and the two data sets attribute similar deficits to opposite causes. How much of a deficit of 5 to 6 % is stored energy? We will count the defects a recoil leaves, and then read the two calibrations as upper bounds on the stored share.

19.4.1 Stored energy of a displacement

The stored energy of one trajectory is the difference of two minima of the energy surface, Eq. (19.4). The trajectories of the kicked ensemble end in different basins b: the perfect crystal, a bond defect, a close Frenkel pair at one of several separations, or a cluster. With $P_b(E_R, \hat{\boldsymbol{r}}_R)$ the probability that a recoil of energy E_R along $\hat{\boldsymbol{r}}_R$ ends in basin b, and $E_{\mathrm{st},b}$ the energy of that basin's minimum above the perfect crystal, the mean stored energy is

$$\langle E_{\mathrm{st}}\rangle(E_R, \hat{\boldsymbol{r}}_R) = \sum_b P_b(E_R, \hat{\boldsymbol{r}}_R)\, E_{\mathrm{st},b}, \qquad \sum_b P_b = 1. \tag{19.23}$$

Only the end points of the trajectories enter. The barrier crossed on the way sets the probabilities, and the displacement threshold $E_{\mathrm{d}}(\hat{\boldsymbol{r}}_R)$ is the energy at which the probability of leaving the perfect basin first reaches one half (Eq. (19.3)).

Near the threshold a recoil makes at most one defect, and Eq. (19.23) has two terms beyond the perfect crystal: $p_{\mathrm{FP}}E_{\mathrm{FP}}$ for a Frenkel pair of stored energy E_{FP} and the analogous term for a bond defect. In silicon a close Frenkel pair stores 6.8 to 7.5 eV, a vacancy and an interstitial far apart store the sum of their formation energies, 8.1 to 9.7 eV, and a bond defect 2.6 to 3.0 eV [2, Table III] (Sect. 7.4). The step of p_{FP} at E_{d} has the zero-point width of Sect. 19.1. A recoil just above the lowest threshold that leaves a close pair stores more than half of its energy, and the stored fraction $\langle E_{\mathrm{st}}\rangle/E_R$ is largest there and falls as the excess energy goes to heat.

The sharp-threshold limit sets $p_{\mathrm{FP}} = \Theta(E_R - E_{\mathrm{d}}(\hat{\boldsymbol{r}}_R))$, with a direction-dependent threshold from molecular dynamics. Dinmohammadi et al. use it, with a silicon threshold surface from 17.5 to 77.5 eV, as the lower limit of the recoil-energy integral of the rate, under the assumption that the ionization threshold equals the displacement threshold [19, Eq. (8)]. The partition separates the two. Crossing E_{d} moves energy from heat into E_{st}; in a calorimeter it therefore produces a deficit, not a threshold. The ionization threshold belongs to the electrons: the binary threshold and the close-collision kernel of Sect. 19.2.

A defect made at t_{pr} need not survive to the readout. With p_{surv} the probability that a close pair has not recombined by then, as in Sect. 16.6, the stored energy at readout carries the factor p_{surv}. Recombination during the cascade itself, before t_{pr}, is part of P_b.

Eq. (19.23) is computed by molecular dynamics on the energy surface with the quench protocol of Sect. 19.1. Appendix E.23 specifies the resulting tables of $P(E_{\mathrm{heat}}, E_{\mathrm{st}}, N_{\mathrm{eh}})$ for silicon and germanium from 1 eV to 10 keV. Published scans with analytical potentials report the mean energy left in defects against recoil energy up to a few hundred eV [20, Fig. 1]. They take potential-energy differences over runs of 20 ps, without a quench, and their minimum thresholds, 18 eV for silicon and 13 eV for germanium, are properties of the potentials (Appendix E.11).

19.4.2 Displacement count and defect energy loss

Above a few times E_d the recoil displaces several atoms, and the stored energy is the number of surviving defects times their energy. The number follows from the binary-collision picture of Sect. 19.3, with a rule that decides which collisions displace. The rule of Kinchin and Pease, as used in Sect. 16.6, has three cases. A moving atom of energy E that transfers $T < E_\mathrm{d}$ leaves its target in place, and T becomes heat. If $T > E_\mathrm{d}$ and the atom keeps $E - T > E_\mathrm{d}$, both move on and one more atom is displaced. If $T > E_\mathrm{d}$ but $E - T < E_\mathrm{d}$, the atom stops in the site it emptied, a replacement, and the count does not change. Electronic losses are removed, so that the energy of the primary is the damage energy T_d of Sect. 19.3, and no binding energy is spent.

Let $n(E)$ be the mean number of further displacements produced by a moving atom of energy E. An atom with $E < 2E_\mathrm{d}$ cannot leave two moving atoms above E_d, so $n(E) = 0$ there. The first collision transfers T with probability $\mathrm{d}\sigma(E,T)/\sigma$; after it, the three cases give the future counts $n(E-T)$, $1 + n(T) + n(E-T)$ and $n(T)$. Since n vanishes below $2E_\mathrm{d}$, all three are $n(E-T) + n(T) + \Theta(T-E_\mathrm{d})\Theta(E-T-E_\mathrm{d})$. Averaging over the first collision and multiplying by σ,

$$\int \mathrm{d}\sigma(E,T)\Big[n(E)-n(E-T)-n(T)-\Theta(T-E_\mathrm{d})\,\Theta(E-T-E_\mathrm{d})\Big] = 0, \qquad N_\mathrm{FP}(T_\mathrm{d}) = 1+n(T_\mathrm{d}), \tag{19.24}$$

where N_FP counts the primary as the first displaced atom. The total cross-section diverges for a screened potential; the integral is finite, however, since the bracket vanishes as $T \to 0$. Eq. (19.24) is the counting analog of Lindhard's Eq. (19.19): a first-collision balance for an additive quantity of the binary branching process.

For hard spheres the transfer is uniform, $\mathrm{d}\sigma \propto \mathrm{d}T/E$ on $0 < T < E$. Then Eq. (19.24) reads $n(E) = (2/E)\int_0^E n(T)\,\mathrm{d}T + 1 - 2E_\mathrm{d}/E$ for $E \geq 2E_\mathrm{d}$. With $N_\mathrm{FP} = 1 + n$, which equals one on $E_\mathrm{d} \leq T < 2E_\mathrm{d}$, this becomes $N_\mathrm{FP}(E) = (2/E)\int_{E_\mathrm{d}}^E N_\mathrm{FP}(T)\,\mathrm{d}T$. A linear $N_\mathrm{FP} = cE$ above $2E_\mathrm{d}$ turns the right side into $2E_\mathrm{d}/E + cE - 4cE_\mathrm{d}^2/E$, which equals cE only for $c = 1/2E_\mathrm{d}$:

$$N_\mathrm{FP}(T_\mathrm{d}) = \frac{T_\mathrm{d}}{2E_\mathrm{d}} \qquad (T_\mathrm{d} \geq 2E_\mathrm{d};\ \text{hard spheres}), \tag{19.25}$$

the Kinchin–Pease count quoted in Sect. 16.6. Each displacement costs $2E_\mathrm{d}$ of damage energy on average.

A screened potential favors small transfers. With the Thomas–Fermi cross-section of Eq. (19.16), in which $T/E = x_L^2/\epsilon_L^2$, and its continuation $f_\mathrm{LSS} \propto x_L^{1/3}$ below $x_L = 2 \times 10^{-3}$ (Sect. 19.3.2), which covers silicon recoils up to about 80 eV ($\epsilon_L = 8.8 \times 10^{-4}$ at 36 eV), the transfer distribution is $\mathrm{d}\sigma \propto T^{-4/3}\mathrm{d}T$. More of the energy then leaves in collisions too small to displace, and fewer atoms are displaced per unit of damage energy. We solve Eq. (19.24) numerically with the cross-section of Sect. 19.3, by marching upward in energy; the new value of n enters each step linearly, through the small transfers and the near-head-on collisions, and is solved for exactly. The asymptotic count is $N_\mathrm{FP} = \xi_\mathrm{d} T_\mathrm{d}/2E_\mathrm{d}$ with $\xi_\mathrm{d} = 0.70$, within 3×10^{-3} for silicon thresholds of 12.5, 20 and 36 eV, because the power law has no energy scale. At $T_\mathrm{d} = 10E_\mathrm{d}$ the effective factor is still 0.72. The published formula of Norgett, Robinson and Torrens, as quoted by Nordlund et al., keeps the Kinchin–Pease form and multiplies it by 0.8 to account for realistic interatomic potentials [21, Eq. (1)].

Limit Check 19.3. Displacement count

The displacement count of a recoil of damage energy T_d [21, Eq. (1)]: $N_\mathrm{FP} = 0$ for $T_\mathrm{d} < E_\mathrm{d}$, 1 for $E_\mathrm{d} \leq T_\mathrm{d} < 2E_\mathrm{d}/0.8$, and $0.8\,T_\mathrm{d}/2E_\mathrm{d}$ above; the Kinchin–Pease count

Eq. (19.25) without the factor 0.8.

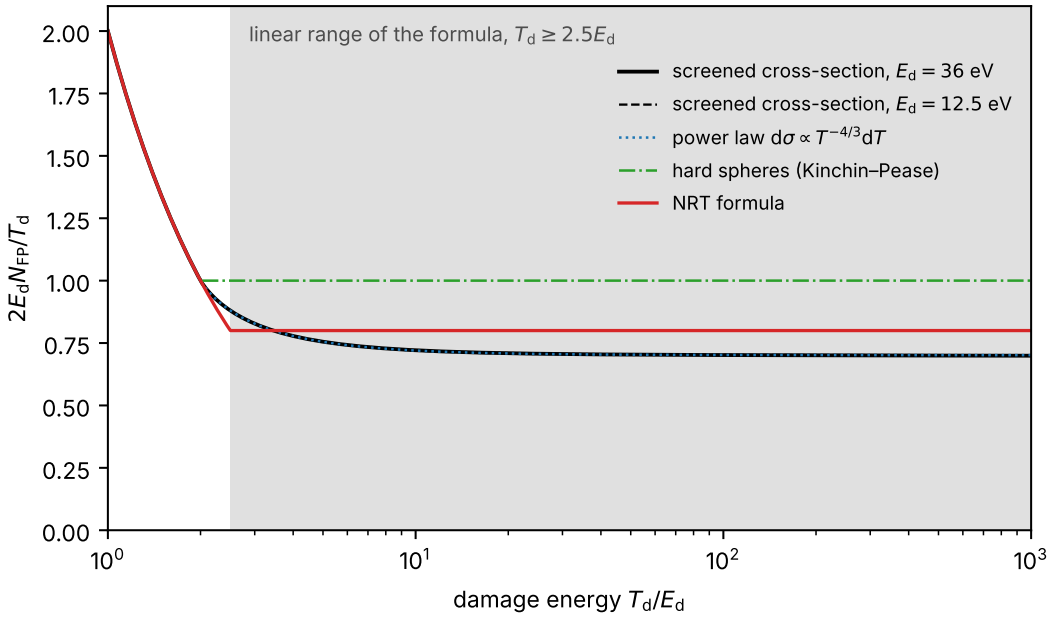


Figure 19.7: Displaced atoms per Kinchin–Pease unit, $2E_dN_{FP}/T_d$, against T_d/E_d, from the counting equation Eq. (19.24) with the Kinchin–Pease rules, no binding energy and no electronic loss. Solid and dashed black: the Thomas–Fermi cross-section of Sect. 19.3 (impulse deflection, continued below $x_L = 2 \times 10^{-3}$ as $x_L^{1/3}$) for silicon with $E_d = 36$ and 12.5 eV; dotted: the power law $T^{-4/3}$; dash-dotted: hard spheres, which reproduce Eq. (19.25) exactly. Red: the formula. Gray: $T_d \geq 2.5E_d$, the formula's linear range. In its linear range the formula lies 11 to 14 % above the screened count for $T_d \geq 10E_d$ and 20 % below the hard-sphere count; below $2.5E_d$ the count is set by the direction dependence of the threshold and by recombination, which no counting rule contains.

The factor in front of the count is therefore a property of the cross-section and of the rules, not a constant of the crystal: 1 for hard spheres, 0.8 in the formula, 0.70 for the screened cross-section that Sect. 19.3 uses for Lindhard quenching. The rules leave out binding and recombination, and both lower the number of defects. A binding energy paid by every displaced atom lowers the count. Recombination inside a dense cascade lowers the number of defects that survive; in metals the surviving number from molecular dynamics is a third to a quarter of the formula's count [21], while silicon and germanium recombine much less [20, Sect. II B].

The defect energy loss follows by multiplying the count by the surviving fraction and by the energy per pair. Writing E_{FP} for the mean stored energy of a surviving displacement,

$$\frac{\langle E_{st} \rangle}{E_R} = \frac{\xi_d}{2} \frac{T_d}{E_R} \frac{p_{surv} E_{FP}}{E_d} \qquad (T_d \gg E_d). \tag{19.26}$$

The stored fraction depends on the recoil energy only through the damage fraction $T_d/E_R = 1 - y_L$, which falls from 0.70 at 10 keV to 0.49 at 100 keV for silicon in Lindhard's model, Eq. (16.42). At fixed $p_{surv}E_{FP}$ it decreases as $1/E_d$: a higher threshold means fewer displacements, each storing the same energy. A measured stored fraction fixes only the combination $\xi_d p_{surv} E_{FP}/E_d$.

Worked example 19.4: Energy stored by a 300 eV silicon recoil

A silicon recoil of 300 eV needs $m_\chi \geq 0.770\,\mathrm{GeV}$ by Eq. (1.21), with $M_\kappa = 26.16\,\mathrm{GeV}$ (natural isotope mixture, as in Sect. 19.1) and $v_{\max} = 2.650 \times 10^{-3}$. Its measured ionization yield, interpolated linearly in $\ln Y_q$ against $\ln E_R$ between 0.108 at 0.22 keV and 0.136 at 0.46 keV [4, Table II], is $Y_q = 0.119$. The electronic share is at most $Y_q E_R = 36\,\mathrm{eV}$, since the near-gap excitations of a slow ion each cost less than $\varepsilon_{\mathrm{eh}}$ per pair (Sect. 19.2), so $T_{\mathrm{d}} \geq 264\,\mathrm{eV}$; we take this value.
With the average Frenkel-pair threshold $E_{\mathrm{d}} = 36\,\mathrm{eV}$, $T_{\mathrm{d}}/E_{\mathrm{d}} = 7.34$. The hard-sphere count Eq. (19.25) gives 3.67 displacements, the formula of Limit Check 19.3 2.94, and the screened count of Eq. (19.24) 2.69. If every displacement survives as a close pair of 6.8 to 7.5 eV, the recoil stores 18.3 to 20.1 eV, 6.1 to 6.7 % of its energy; as separated pairs of 8.1 to 9.7 eV, 22 to 26 eV, or 7.3 to 8.7 %. With the threshold for a defect of either kind, 24 eV, the screened count is 3.95, and bond defects alone would store 10 to 12 eV.
These fractions are comparable to the deficits measured at 10 to 100 keV, and a calorimeter calibrated with electron recoils would read this recoil about 20 eV low.

19.4.3 The measured defect fraction

Both measurements see the stored energy through a phonon deficit. They compare the phonon signal of nuclear recoils with that of electron recoils of the same energy, after removing the Luke phonons that the drifting pairs make in the bias field. The reconstructed energy of a nuclear recoil over its true energy is the phonon-yield ratio Y_{ph}, and its deficit $f_{\mathrm{DF}} = 1 - Y_{\mathrm{ph}}$ is the defect fraction. Sect. 19.5.3 derives Y_{ph} from the balance law, with the reconstruction written out.

The silicon measurement used neutrons from a ^{252}Cf source on CDMS II detectors at 4 V/cm (bias 4 V). A resonance in the elastic cross-section puts a feature near 20 keV in the recoil spectrum, which fixes the energy scale against the simulated spectrum. The fit yields $Y_{\mathrm{ph}} = 0.952^{+0.009}_{-0.007}$ at 95 % confidence over 10 to 100 keV [15, Sects. 2 and 3.2], so $f_{\mathrm{DF}} = 0.048$. The authors attribute it to collection and note that the absorbed fraction may depend on the relative amounts of primary and Luke phonons [15, Sect. 2].

The germanium measurement used ^{206}Pb nuclei that recoil with 103 keV into the surface of SuperCDMS detectors at about 50 mK, from ^{210}Po decays in sources facing two detectors whose faces were held at ±2 V. The measured spectrum near the endpoint is matched to the simulated one by scaling the simulated energies by $1 - f$. Electron recoils from γ rays and β particles at the same surfaces give the scale factor f_{sur} of surface events, and [18, Eqs. (1), (3), (4)]

$$f_{\mathrm{DF}} = 1 - \frac{1 - f}{1 - f_{\mathrm{sur}}}.$$

This is the same ratio, $1 - Y_{\mathrm{ph}}$, with surface electron recoils as the calibrating events. The two detectors give $f_{\mathrm{DF}} = (6.22 \pm 0.11)\,\%$ and $(5.87 \pm 0.13)\,\%$. They differ by 2.06 standard deviations, so the uncertainty of the weighted mean is inflated by that factor, and with a systematic term the result is $(6.08 \pm 0.18)\,\%$ [18, Table II, Sect. V A]. The ^{206}Pb recoils are observed at an ionization yield near 0.3 [18, Sect. II]. Surface events suffer incomplete charge collection [15, Sect. 2.1], so 0.3 is a lower edge of the electronic share, and the damage fraction is $T_{\mathrm{d}}/E_R \leq 0.7$. The authors attribute the deficit to defects. They also note that the two detectors were biased with opposite polarities, so that each collected mainly carriers of one sign [18, Sect. V C]; a polarity-dependent trapping term is missing energy other than stored energy, which differs between the two.

Table 19.1: Upper bounds on the stored energy of nuclear recoils from calibration data, under conditions (a)–(e) of Sect. 19.5.3. Silicon damage fractions $1 - y_C$ from the yield parameterization of the silicon measurement at 10 and 100 keV [15, Eq. (4)]; germanium $T_\mathrm{d}/E_R \leq 0.7$ from the observed surface-event yield near 0.3, so its entry in the last column is the smallest value the data allow. Last column: the bound read through Eq. (19.26) with the factor 0.8 of Limit Check 19.3, and in parentheses with the screened factor 0.70. Sources: silicon [15, Sects. 2 and 3.2]; germanium [18, Table II, Sect. V A].

	Data	f_DF	upper bound (95 % C.L.) on $\langle E_\mathrm{st}\rangle/E_R$	$p_\mathrm{surv}E_\mathrm{FP}/E_\mathrm{d}$
Si	neutron recoils, 10–100 keV, 4 V/cm; $Y_\mathrm{ph} = 0.952^{+0.009}_{-0.007}$ (95 %)	0.048	0.055	0.19 (0.21) at 10 keV, 0.27 (0.31) at 100 keV
Ge	^{206}Pb surface recoils, 103 keV, ±2 V, 50 mK	0.0608 ± 0.0018	0.064	≥ 0.23 (≥ 0.26)

19.4.4 Upper bounds on the stored energy

The defect fraction contains the stored energy once, next to collection terms and other missing energy (Eq. (19.35)). Sect. 19.5.3 shows from the exact Y_ph that the stored fraction is at most the defect fraction, $E_\mathrm{st}/\omega \leq f_\mathrm{DF}$, under five conditions on the collection of primary and Luke phonons, on the other missing energy and on the pair energy the reconstruction adopts, conditions (a)–(e) of Sect. 19.5.3 (Eq. (19.36)). For upper limits at 95 % confidence we take the lower edge of the silicon interval, $Y_\mathrm{ph} \geq 0.945$, and for germanium the one-sided limit $\hat{f} + 1.645\sigma$ of a Gaussian measurement (Appendix C.14), $0.0608 + 1.645 \times 0.0018 = 0.0638$. The bounds are

$$\frac{\langle E_\mathrm{st}\rangle}{E_R} \leq 0.055 \quad \text{(Si, 10–100 keV)}, \qquad \frac{\langle E_\mathrm{st}\rangle}{E_R} \leq 0.064 \quad \text{(Ge, } ^{206}\text{Pb at 103 keV)} \qquad \text{(95 \% C.L.)}.$$

With the measured silicon pair energy in place of the one the silicon reconstruction adopts, the silicon bound under conditions (a)–(d) tightens to 0.049. If the zero-point change of the defects is negative, both bounds widen by about 10^{-3}, to 0.056 and 0.065. Table 19.1 collects the bounds and their inputs. On the branch $\mathfrak{r}_\mathrm{coll} < 1$, where the mean-field transport estimate lies in most configurations (0.81–1.02, Sect. 20.5.3), the silicon bound loosens, to 0.081 at $\mathfrak{r}_\mathrm{coll} = 0.9$ (Limit Check 19.5).

Read through Eq. (19.26), the bounds constrain $p_\mathrm{surv}E_\mathrm{FP}/E_\mathrm{d}$. With the factor 0.8 of Limit Check 19.3, silicon allows at most 0.19 at 10 keV and 0.27 at 100 keV, where the damage fraction is smaller; the screened factor 0.70 raises both by 14 %. For silicon with $E_\mathrm{d} = 36$ eV, full survival predicts a stored fraction of 0.055 to 0.061 at 10 keV for close pairs and 0.066 to 0.079 for separated pairs, against the bound 0.055. The data therefore allow nearly complete survival of close pairs, $p_\mathrm{surv} \leq 0.90$ to 1.00, and require $p_\mathrm{surv} \leq 0.70$ to 0.84 if the pairs separate. They do not yet constrain p_surv at 100 keV. For germanium the entry 0.23 is not conservative. The damage fraction 0.7 is an upper edge, and a smaller one raises the bound, to 0.27 at $T_\mathrm{d}/E_R = 0.6$; the primary is also a ^{206}Pb nucleus in germanium, of unequal mass, outside the self-recoil form of the count.

The bounds hold at the calibration energies. Light dark matter produces silicon recoils below a keV, where the stored fraction is larger in Worked example 19.4 and the damage fraction rises toward one. Carrying the 10 keV constraint on $p_\mathrm{surv}E_\mathrm{FP}/E_\mathrm{d}$ to 300 eV would give a stored fraction of at most 6.6 % with the count of Limit Check 19.3. Since survival in the sparse cascade of a sub-keV recoil need not equal survival in the denser cascades of 10 to 100 keV, this is an extrapolation, not a bound.

19.4.5 The germanium displacement threshold

From the germanium defect fraction and simulations with the binary-collision code TRIM (Transport of Ions in Matter), Agnese et al. extract an average displacement threshold of $E_\mathrm{d} = 19.7^{+0.6}_{-0.5}\,\mathrm{eV}$, since the simulated fraction of energy lost to defects decreases monotonically with the threshold [18, Sect. V B]. Earlier determinations from theory and molecular dynamics span 7 to 30 eV [18, Sect. I]. Read through Eqs. (19.26) and (19.35), the extraction rests on four choices: the energy stored per displacement, the counting rule, the attribution of the measured deficit, and the medium of the simulation. Each changes the number.

The first is the energy stored per displacement. Eq. (19.26) fixes E_d only together with $p_\mathrm{surv}E_\mathrm{FP}$, and the monotonic decrease of the defect fraction with E_d is the factor $1/E_\mathrm{d}$ at fixed stored energy per displacement. With the factor 0.8 and $T_\mathrm{d}/E_R = 0.7$, $E_\mathrm{d} = 19.7\,\mathrm{eV}$ corresponds to $p_\mathrm{surv}E_\mathrm{FP} = 4.3\,\mathrm{eV}$, and every further eV per surviving displacement raises the inferred threshold by 4.6 eV; 6 eV would give 28 eV. The stored energy per displacement in the simulation is not stated [18, Sect. V B]. The second is the counting rule: at fixed $p_\mathrm{surv}E_\mathrm{FP} = 6\,\mathrm{eV}$ the hard-sphere count, the formula and the screened count give 35, 28 and 24 eV. The third is the measured quantity. By Eq. (19.35), f_DF contains the collection terms and the other missing energy, and the two detectors of opposite polarity differ by 0.35 percentage points, a difference that a polarity-dependent trapping term would produce and stored energy would not. Reading all of f_DF as stored energy overstates the stored energy whenever conditions (a)–(c) of Sect. 19.5.3 hold, and with it the number of displacements, and so understates E_d at fixed E_FP. The fourth is the medium. The simulation uses simple potentials with parameters tuned to implantation data, and its systematic uncertainty does not include modeling imperfections [18, Sect. V B]. As a binary-collision code it treats the target as a random medium (Sect. 19.3.3), while the threshold of a crystal depends on the direction and on the path through the lattice (Sect. 7.4).

The value 19.7 eV is therefore conditional on a model of stored energy per displacement, a counting rule and the attribution of the whole deficit to defects. A direct determination would compute the stored energy of Eq. (19.23) for ^{206}Pb recoils in germanium by molecular dynamics on an accurate energy surface, and compare it with f_DF corrected for collection by Eq. (19.36). The threshold would then come out of the same calculation instead of being fitted through a count.

19.5 The energy balance law

A silicon recoil of 1 keV makes about 34 electron–hole pairs, at the measured ionization yield $Y_q = 0.127$ (Sect. 19.2) and $\varepsilon_\mathrm{eh} = 3.75\,\mathrm{eV}$, and their gap energy is 40 eV. The rest of the deposit is phonons and, if the recoil displaced atoms, stored energy. A millisecond later the phonons have been absorbed at the surfaces of the crystal, most pairs have returned their gap energy as phonons and a few are caught on traps, while the defects still hold their energy. A phonon sensor reads the share that reached it within its readout window, and a calibration made with electron recoils converts that reading into an energy. The conversion is correct when electron and nuclear recoils of the same energy deliver the same share. To see when that holds, we write one law that tracks every share at every time. Applied to the silicon calibration of the CDMS II detectors, the law shows that the measured ratio fixes a combination of stored energy, phonon collection and the pair energy of the reconstruction, which a second bias can separate.

19.5.1 Reservoirs of the deposit

At the partition time t_{pr} the deposit ω is split as in Eq. (19.5): $N_{\mathrm{eh}}E_{\mathrm{g}}$ in pairs, E_{st} in defects and E_{heat} in phonons. After t_{pr} the energy keeps moving. Pairs bind into excitons, recombine, are captured on impurities or drift to the electrodes, defects release their energy (Sect. 19.6), and the phonons leave the crystal through its surfaces. Five reservoirs hold the energy at any time t. Write $E_{\mathrm{dep}} \equiv \omega$ for the deposit when it is divided among them.

The carriers hold $E_{\mathrm{pairs}}(t)$. Each energy is counted from the Fermi level μ_F of the electrodes, as in Sect. 18.5: a free electron holds $E_c - \mu_F$, a free hole $\mu_F - E_v$, an exciton $E_{\mathrm{g}} - E_x^b$, an electron trapped on a level E_j holds $E_j - \mu_F$ and a trapped hole $\mu_F - E_j$. To these we add the electrostatic energy of the charge left in the crystal, the Coulomb energy of separated carriers, their images in the electrodes and any space charge. The phonons hold $E_{\mathrm{phon}}(t)$, the vibrational energy above the thermal bath; its value at t_{pr} is the prompt phonon energy $E_{\mathrm{prompt}} \equiv E_{\mathrm{phon}}(t_{\mathrm{pr}})$. The defects hold the stored energy $E_{\mathrm{st}}(t)$, a difference of minima of E^{ad} as in Eq. (19.4). The escaped energy $E_{\mathrm{esc}}(t)$ is carried out of the crystal by quanta that no surface absorbs, in practice luminescence photons. Last, $E_{\mathrm{sens}}(t)$ is the energy absorbed at the surfaces: by the sensor films, of which a part becomes signal, and by holders and uninstrumented faces. A bias source at fixed voltage adds the work $W_{\mathrm{bias}}(t)$ it has done on the drifting charge.

Energy balance law

At every time t after the deposit, and for every event,

$$E_{\mathrm{dep}} + W_{\mathrm{bias}}(t) = \underbrace{E_{\mathrm{pairs}}(t)}_{\text{carriers}} + \underbrace{E_{\mathrm{phon}}(t)}_{\text{phonons}} + \underbrace{E_{\mathrm{st}}(t)}_{\text{defects}} + \underbrace{E_{\mathrm{esc}}(t)}_{\text{escaped}} + \underbrace{E_{\mathrm{sens}}(t)}_{\text{absorbed}}, \tag{19.27}$$

with carrier energies counted from the Fermi level of the electrodes, the stored energy configurational, and the change of zero-point energy of the occupied defect basins inside E_{phon}.

Appendix D.45 proves Eq. (19.27) from the conservation of the total energy of crystal, sensors, radiation and bias source. Apart from the work of the source, the total energy is conserved. Between collisions it equals the sum of the reservoir energies, since the interactions of excited carriers with each other are either long-range Coulomb energy, which E_{pairs} contains, or short-range energy that acts during a collision alone. Every process after t_{pr} is a short-range transition that conserves energy, or a drift whose energy the bias work and the electrostatic energy account for. The law holds for each realization of the random sequence of captures, hops and absorptions, and hence also in the mean. Its content is the completeness of the reservoir list. At t_{pr} and zero field, with no capture, it is the partition: $E_{\mathrm{pairs}} = N_{\mathrm{eh}}E_{\mathrm{g}}$, $E_{\mathrm{prompt}} = E_{\mathrm{heat}}$, $E_{\mathrm{esc}} = E_{\mathrm{sens}} = 0$, and Eq. (19.27) is Eq. (19.5).

The Fermi level of the electrodes is the one reference for the carrier energies that keeps the law closed. A carrier that reaches an electrode gives up its energy relative to the Fermi level of that electrode, as phonons at the contact; the CDMS II analysis names these the recombination phonons [15, Sect. 2]. Counted from another level μ, the carriers still in the crystal hold an energy whose derivative with respect to μ is the number of holes minus the number of electrons among them. That number is zero as long as every carrier still has its partner. Once one carrier of a pair has been collected and the other has not, energies counted from $\mu \neq \mu_F$ change with the arbitrary μ, and the balance no longer closes. A hole left behind when its electron is collected thus stores $\mu_F - E_v$, plus its image energy, and not half the gap.

The stored energy enters once. The displacement threshold appears nowhere in Eq. (19.27).

The stored energy decreases through releases alone, each of which moves its energy to E_{phon}, or to E_{esc} if a photon carries it away. A calibration that subtracts an energy per displaced atom from the heat and also counts the later release as heat counts the same energy twice.

The zero-point energy of the occupied basins sits inside E_{phon}. A defect softens or stiffens the modes of its neighbors and so changes the zero-point energy of the crystal by an amount $\Delta E_{\text{zp}}(t)$, which Sect. 19.1.4 placed in the lattice share. It is the energy of the vibrational ground state, and no sensor can absorb it. Its size follows from the zero-point energy per silicon atom, $3\langle\omega\rangle/2 = 61\,\text{meV}$, half of it the potential energy of Sect. 19.1.3. A point defect that shifts the frequencies of its four neighbors by some tens of percent changes ΔE_{zp} by about 0.07 eV, an estimate, against the several eV a Frenkel pair stores. The sign is not fixed: softened modes make ΔE_{zp} negative, stiffened ones positive. A defect that hops from basin i to basin j releases into the phonons the configurational difference plus the change of zero-point energy between the two basins. Summed along any release path, the zero-point terms telescope like the configurational ones, so the sum rule of Sect. 19.6 holds for both separately. A sensor can draw on $E_{\text{phon}} - \Delta E_{\text{zp}}$ alone.

The Luke effect enters the balance through the bias work. A carrier drifting at constant velocity in the field of a biased detector gains from the field the energy $e\mathcal{E}$ per unit path and gives it to phonons as fast as it gains it: the Neganov–Trofimov–Luke effect of Eq. (1.44), whose emission spectrum Sect. 20.1 treats. Between two planar electrodes at fixed potentials, a carrier that crosses a fraction s of the gap induces the charge es on the electrodes, and the source does the work $eV_{\text{bias}}s$ on it (Ramo's theorem, Appendix D.45). The total work is

$$W_{\text{bias}} = eV_{\text{bias}}N_{\text{ind}}, \tag{19.28}$$

where N_{ind} is the induced charge in units of e, equal to the number of pairs if all are collected and smaller if carriers are trapped on the way. Of this work, W_{bias} minus the change of the electrostatic energy goes to phonons; for pairs born neutral and fully collected the electrostatic term vanishes at the end. For an electron recoil, whose pair number is $\omega/\varepsilon_{\text{eh}}$, the ratio of the Luke phonon energy to the deposit is the Luke gain

$$g_{\text{Luke}} = \frac{eV_{\text{bias}}}{\varepsilon_{\text{eh}}},$$

1.07 at 4 V and 26.7 at 100 V in silicon. Carriers already drift during the cascade, before t_{pr}, over the short distance they travel in that time; that work is part of the same W_{bias} and of the same Luke phonons.

Fig. 19.8 follows the five reservoirs of a 1 keV silicon recoil at zero field. The pair kinetics are those of Fig. 18.12: the gap energy is released within microseconds, except for the 5 % of pairs that the illustrative rates trap and the 0.9 % that leave as light. The phonons are absorbed with a single time constant of 58 µs, the fall time of the pulse of a 10.6 g silicon phonon detector [22, p. 2]. After a millisecond the surfaces have absorbed 943 eV, the traps keep 2.0 eV and the defects their full 55 eV. For a stated detector the evolution of the reservoirs is a transport calculation: the charge kinetics, Luke emission and collection at the surfaces of Appendix E.25, started from the partition tables of Appendix E.23.

19.5.2 Heat quenching

Without bias the phonon sensors measure the heat. Their readout integrates the energy absorbed during a window t_{win}, from microseconds to a second depending on the sensor, and the phonons still in the crystal at the end of the window are absorbed with the same pulse shape whatever made them. Energy that has not become available phonons by t_{win} is missing

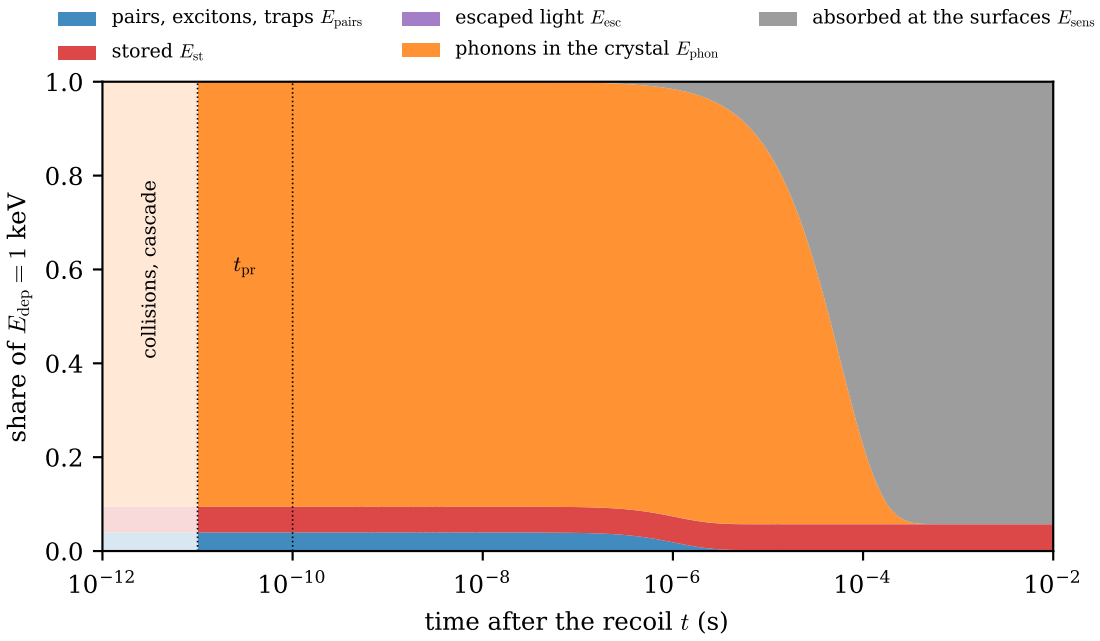


Figure 19.8: Shares of the balance law Eq. (19.27) for a 1 keV silicon recoil at zero field, against the time after the recoil, as fractions of E_{dep}. At the partition time t_{pr} (dotted lines, 10 to 100 ps): $N_{\mathrm{eh}} = 33.9$ pairs from the measured yield $Y_q = 0.127$ at 0.89 keV [4, Table II] with $\varepsilon_{\mathrm{eh}} = 3.75$ eV, holding $N_{\mathrm{eh}} E_{\mathrm{g}} = 39.6$ eV with $E_{\mathrm{g}} = 1.17$ eV; stored energy set to 55 eV, the silicon bound of Table 19.1, for display, and kept frozen; phonons 905 eV. Pair kinetics: the illustrative rates of Fig. 18.12 (one free pair per pair, 5 % trapped, luminescent fraction 0.1 of free-exciton decay, bound-exciton Auger decay 250 ns); trapped carriers stay in E_{pairs}. Phonons absorbed at the surfaces with one exponential time constant, 58 µs, the pulse fall time of a silicon phonon detector [22, p. 2], a modeling choice. Shares at 10 ms: absorbed 0.943, stored 0.055, trapped 0.002, escaped 3×10^{-4}. The collisions and the cascade before t_{pr} are not resolved.

from the signal for good. It is the sum of the energy still held by carriers, the energy still stored, the escaped energy and the zero-point change,

$$E_{\mathrm{miss}}(t_{\mathrm{win}}) = E_{\mathrm{pairs}}(t_{\mathrm{win}}) + E_{\mathrm{st}}(t_{\mathrm{win}}) + E_{\mathrm{esc}}(t_{\mathrm{win}}) + \Delta E_{\mathrm{zp}}(t_{\mathrm{win}}).$$

The sensors turn a fraction ϵ_{abs} of the available phonon energy into signal, $\epsilon_{\mathrm{abs}}(E_{\mathrm{dep}} - E_{\mathrm{miss}})$; the rest goes into E_{sens} as loss on uninstrumented faces and in the films. This fraction is often well below one, but it is the same for every event of a given kind of recoil at a given place. Let $\epsilon^{\mathrm{NR}}_{\mathrm{prim}}$ and $\epsilon^{\mathrm{ER}}_{\mathrm{prim}}$ be its values for the primary phonons of nuclear and electron recoils, the sole phonons at zero field, and define the collection offset $\delta_{\mathrm{coll}} \equiv 1 - \epsilon^{\mathrm{NR}}_{\mathrm{prim}}/\epsilon^{\mathrm{ER}}_{\mathrm{prim}}$. The heat quenching is the ratio of the heat signals of a nuclear and an electron recoil of the same energy,

$$Y_{\mathrm{ph}}(0) = (1 - \delta_{\mathrm{coll}})\,\frac{1 - d_{\mathrm{NR}}}{1 - d_{\mathrm{ER}}}, \qquad d_{\mathrm{NR,ER}} \equiv \frac{E^{\mathrm{NR,ER}}_{\mathrm{miss}}(t_{\mathrm{win}})}{\omega}, \tag{19.29}$$

the zero-field value of the phonon-yield ratio Y_{ph} defined in Sect. 19.5.3. The absolute efficiency has dropped out, and with it the large losses at the surfaces; what remains is the difference between the two kinds of recoil.

Unit heat quenching, $Y_{\mathrm{ph}}(0) = 1$, is the truncation that phonon-only searches use to read a nuclear-recoil energy from a scale calibrated with electron recoils [22, p. 4]. Eq. (19.29) states what it requires: equal collection of the two phonon populations, $\delta_{\mathrm{coll}} = 0$, and equal missing energy, $d_{\mathrm{NR}} = d_{\mathrm{ER}}$. The second condition does not hold by accident. An electron recoil puts the fraction $E_{\mathrm{g}}/\varepsilon_{\mathrm{eh}} = 0.312$ of its energy into pair gap energy, a silicon recoil of 100 eV a fraction Y_q times as large, so the two missing energies are equal when both are zero. The truncation therefore holds when, within the read window, every pair has recombined or been collected with its gap energy returned as phonons, no defect holds energy, and nothing has escaped.

The gap-energy term is the one that depends on the window. Let $f_{\mathrm{ret}}(t)$ be the fraction of the gap energy of the pairs returned as phonons by the time t, and $1 - f_{\mathrm{ret}}(t)$ the rest: held by free pairs and excitons, $r_{\mathrm{gap}}(t)$ of Eq. (18.39), plus what has gone to traps and to light. The nuclear recoil holds Y_q times the pair energy of the electron recoil. At zero field Eq. (19.29) becomes the gap-energy release law

$$Y_{\mathrm{ph}}(0) = (1 - \delta_{\mathrm{coll}})\,\frac{1 - Y_q\,(E_{\mathrm{g}}/\varepsilon_{\mathrm{eh}})(1 - f^{\mathrm{NR}}_{\mathrm{ret}}) - (E_{\mathrm{st}} + \Delta E_{\mathrm{zp}})/\omega}{1 - (E_{\mathrm{g}}/\varepsilon_{\mathrm{eh}})(1 - f^{\mathrm{ER}}_{\mathrm{ret}})}, \tag{19.30}$$

with f_{ret}, E_{st} and ΔE_{zp} taken at t_{win} and the escape of light counted in $1 - f_{\mathrm{ret}}$. Nothing in it is small. If no gap energy has come back, $f_{\mathrm{ret}} = 0$, a silicon recoil of 100 eV with $Y_q = 0.102$ has $Y_{\mathrm{ph}}(0) = 0.968/0.688 = 1.407$. Its heat signal is 41 % larger than that of an electron recoil of the same energy, because the electron recoil holds more of its energy in pairs. The first-order form, $1 + (1 - Y_q)(E_{\mathrm{g}}/\varepsilon_{\mathrm{eh}})(1 - f_{\mathrm{ret}})$, is 1.280 and not usable at this size. With the illustrative rates of Fig. 18.12, a window of 1 µs has $f_{\mathrm{ret}} = 0.525$ and $Y_{\mathrm{ph}}(0) = 1.156$; after 10 µs all free pairs and excitons have decayed, $f_{\mathrm{ret}} = 0.941$, and the 5.9 % that went to traps and to light leave $Y_{\mathrm{ph}}(0) = 1.017$ at any longer window. The trapped fraction never comes back at millikelvin, where emission from traps is frozen (Sect. 18.5.3).

When the kinetics are linear, each pair decays on its own and $f^{\mathrm{NR}}_{\mathrm{ret}} = f^{\mathrm{ER}}_{\mathrm{ret}}$ at every time, a sum of exponentials whose rates are those of Eq. (18.37). The tracks of nuclear recoils are denser than the clouds of electron recoils of the same energy, however, and in a dense track the pairs meet. Binding of carriers from different pairs moves gap energy from free pairs into excitons and changes nothing in $1 - f_{\mathrm{ret}}$. Two-body processes that return gap energy to the lattice do: two excitons that meet annihilate, one of them giving up its energy, and three free carriers recombine by the Auger process. Write $n(t)$ for the density of pairs in the track,

n_0 for its initial value, and take $1 - f_{\rm ret} \simeq n/n_0$ while no other channel acts. Annihilation gives $\mathrm{d}n/\mathrm{d}t = -n^2/(n_0 t_0)$, and Auger recombination $\mathrm{d}n/\mathrm{d}t = -n^3/(2n_0^2 t_0)$, with t_0 the time scale each sets at the initial density. Integrating,

$$1 - f_{\rm ret}(t) = \frac{1}{1 + t/t_{\rm rel}} \quad \text{(annihilation)}, \qquad 1 - f_{\rm ret}(t) = \left(1 + \frac{t}{t_{\rm rel}}\right)^{-1/2} \quad \text{(Auger)}. \tag{19.31}$$

At $t_{\rm win} \gg t_{\rm rel}$ the nuclear recoil then misses $Y_q(E_{\rm g}/\varepsilon_{\rm eh})\, t_{\rm rel}/t_{\rm win}$ or $Y_q(E_{\rm g}/\varepsilon_{\rm eh})(t_{\rm rel}/t_{\rm win})^{1/2}$ of its energy, a power of the window and not an exponential, with $t_{\rm rel}$ set by the density of its own track. The heat quenching of nuclear recoils depends on the window whenever a two-body channel acts within it. Diffusion dilutes a track as it decays, and a compact cloud then stops losing energy by two-body processes and returns to the linear law, so Eq. (19.31) holds while the track stays narrow. The rates that decide which regime a detector is in are not known at millikelvin (Sect. 18.5.4); Appendices E.24 and E.25 specify their calculation.

Limit Check 19.4. Unit heat quenching

The heat signals of nuclear and electron recoils of the same energy are taken to be equal [22, p. 4]: $Y_{\rm ph}(0) = 1$.

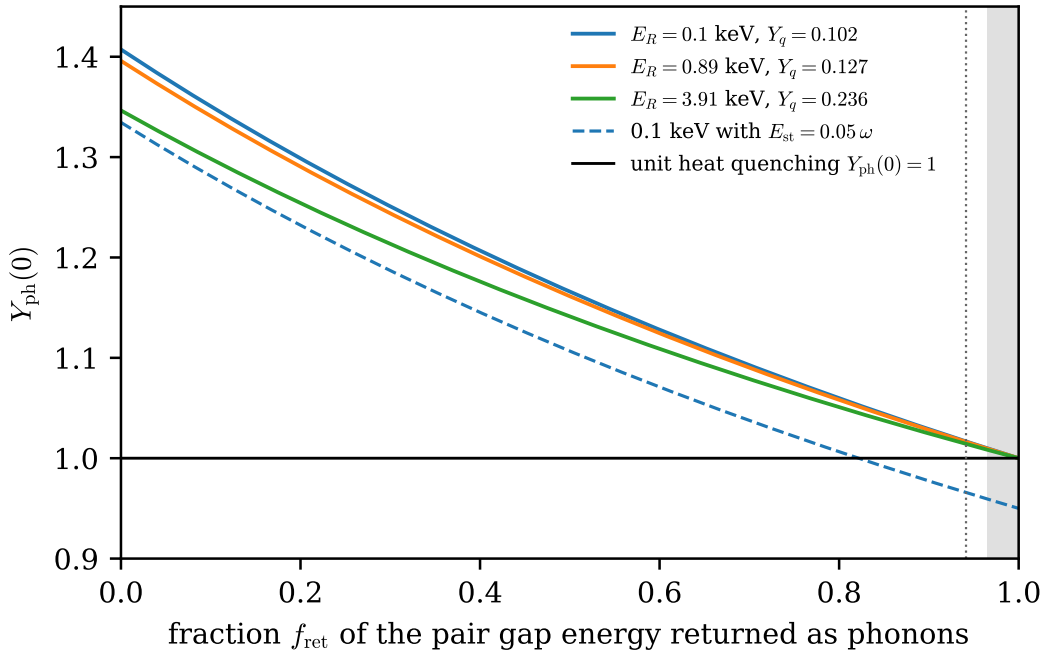


Figure 19.9: Heat quenching of silicon recoils from the release law Eq. (19.30) with $\delta_{\rm coll} = 0$, $\Delta E_{\rm zp} = 0$ and the same returned fraction $f_{\rm ret}$ for both recoils, against $f_{\rm ret}$. Solid: no stored energy, at the measured yields Y_q [4, Table II] of 0.10, 0.89 and 3.91 keV; $E_{\rm g} = 1.17$ eV, $\varepsilon_{\rm eh} = 3.75$ eV. Dashed: 0.1 keV with a stored energy of $0.05\,\omega$. Horizontal line: the truncation. Gray: $f_{\rm ret} \geq 0.965$, where the 0.1 keV curve lies within 1 % of one. Dotted: $f_{\rm ret} = 0.941$, the asymptotic phonon share of the illustrative kinetics of Fig. 18.12; there 5 % of the pairs are trapped.

When part of the gap energy is still held by pairs, excitons or traps at the end of the window, the electron recoil, with four to ten times more energy in pairs, loses more of its heat than the nuclear recoil, and the ratio rises above one, by up to 41 % at 100 eV; stored energy pulls it below.

The truncation demands nearly complete release. Within 1 % at 100 eV requires $1 - f_{\mathrm{ret}} \leq 0.035$, and the bound is nearly the same over the measured range, 0.036 at 0.89 keV and 0.041 at 3.91 keV, since the factor $(1 - Y_q)$ changes little while Y_q stays small. A trapped fraction of a few percent, which no window removes, is already a percent-level violation, of opposite sign to the stored energy. The two effects can cancel in a single measurement, which is one reason a heat measurement alone determines neither.

19.5.3 The phonon-yield ratio

With a bias the phonon signal has three parts: the primary phonons of the recoil, the Luke phonons of the drift and the recombination phonons that carriers emit at the contacts [15, Sect. 2]. The recombination phonons carry gap energy returned within the window, and the balance law counts them with the primaries, in $E_{\mathrm{dep}} - E_{\mathrm{miss}}$; when drift collects the pairs within the window, E_{miss} holds the trapped carriers and their electrostatic energy, and the stored, escaped and zero-point energy. The sensors collect primary and Luke phonons with different efficiencies, because the two populations are born with different spectra and in different places. Let $\epsilon_{\mathrm{prim}}^{\mathrm{ER}}$ and $\epsilon_{\mathrm{prim}}^{\mathrm{NR}} = (1 - \delta_{\mathrm{coll}})\epsilon_{\mathrm{prim}}^{\mathrm{ER}}$ be the efficiencies for the primary phonons of the two recoils and ϵ_{Luke} that for Luke phonons, and define the collection ratio

$$\mathfrak{r}_{\mathrm{coll}} \equiv \frac{\epsilon_{\mathrm{Luke}}}{\epsilon_{\mathrm{prim}}^{\mathrm{ER}}},$$

the efficiency for Luke phonons over that for the primary phonons of an electron recoil. Let the electron recoil be fully collected, so that its bias work is $g_{\mathrm{Luke}}\omega$ by Eq. (19.28), and define the ionized fraction f_{ion} of a nuclear recoil as its bias work over that of an electron recoil of the same energy, $f_{\mathrm{ion}} = N_{\mathrm{ind}}^{\mathrm{NR}}/N_{\mathrm{ind}}^{\mathrm{ER}}$. It equals the ionization yield Y_q of Eq. (19.12) when both recoils are collected alike and is smaller if the carriers of the dense track are trapped more often. Per unit deposit the two signals are then

$$\frac{S_{\mathrm{ER}}}{\epsilon_{\mathrm{prim}}^{\mathrm{ER}}\,\omega} = 1 - d_{\mathrm{ER}} + \mathfrak{r}_{\mathrm{coll}}\, g_{\mathrm{Luke}}, \qquad \frac{S_{\mathrm{NR}}}{\epsilon_{\mathrm{prim}}^{\mathrm{ER}}\,\omega} = (1 - \delta_{\mathrm{coll}})(1 - d_{\mathrm{NR}}) + \mathfrak{r}_{\mathrm{coll}}\, g_{\mathrm{Luke}} f_{\mathrm{ion}}. \quad (19.32)$$

The CDMS II calibration turns these signals into energies in two steps [15, Eq. (1), Sect. 2.1]. The charge channel is calibrated with photon lines, so that an electron recoil of energy ω reads the electron-equivalent energy $E_Q = \omega$ and a nuclear recoil $E_Q = f_{\mathrm{ion}}\omega$. The phonon scale is set with bulk electron recoils, taken to have unit ionization yield, so that an electron recoil reads $E_\phi = \omega(1 + eV_{\mathrm{bias}}/\varepsilon_{\mathrm{cal}})$, and a nuclear recoil is assigned the energy $E_\phi - E_Q eV_{\mathrm{bias}}/\varepsilon_{\mathrm{cal}}$. Note that $\varepsilon_{\mathrm{cal}}$ is the pair energy the reconstruction adopts, 3.82 eV for silicon [15, Sect. 2], not the true $\varepsilon_{\mathrm{eh}}$. It enters both the scale and the subtraction, through the gain $g_{\mathrm{cal}} \equiv eV_{\mathrm{bias}}/\varepsilon_{\mathrm{cal}}$, while the Luke phonons follow the true pair number and g_{Luke}. The phonon-yield ratio Y_{ph} is the reconstructed energy of the nuclear recoil over its true energy, which the CDMS II analysis calls the phonon collection efficiency for nuclear recoils relative to electron recoils, and its deficit is the defect fraction $f_{\mathrm{DF}} \equiv 1 - Y_{\mathrm{ph}}$. From Eq. (19.32),

$$Y_{\mathrm{ph}} = (1 + g_{\mathrm{cal}})\,\frac{(1 - \delta_{\mathrm{coll}})(1 - d_{\mathrm{NR}}) + \mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} f_{\mathrm{ion}}}{1 - d_{\mathrm{ER}} + \mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}}} - f_{\mathrm{ion}} g_{\mathrm{cal}}. \quad (19.33)$$

The absolute efficiency $\epsilon_{\mathrm{prim}}^{\mathrm{ER}}$ has cancelled. With no missing energy and $\mathfrak{r}_{\mathrm{coll}} = 1$ the ratio is $1 + (g_{\mathrm{cal}} - g_{\mathrm{Luke}})(1 - f_{\mathrm{ion}})/(1 + g_{\mathrm{Luke}})$, not one. For silicon at 4 V, with $g_{\mathrm{Luke}} = 1.067$, $g_{\mathrm{cal}} = 1.047$ and $f_{\mathrm{ion}} = 0.324$, the pair energy alone produces a deficit of 0.0064, against the half-width 0.008 of the measured band. At zero bias Eq. (19.33) is the heat quenching Eq. (19.29), and multiplying out its denominator relates the two.

Calibration identity

The phonon-yield ratio at bias V_{bias} and the heat quenching $Y_{\mathrm{ph}}(0)$ of the same recoils, with the same missing energies, are related by

$$Y_{\mathrm{ph}} = \frac{(1+g_{\mathrm{cal}})\,Y_{\mathrm{ph}}(0) + f_{\mathrm{ion}}(\mathfrak{r}' g_{\mathrm{Luke}} - g_{\mathrm{cal}})}{1+\mathfrak{r}' g_{\mathrm{Luke}}}, \qquad \mathfrak{r}' = \frac{\mathfrak{r}_{\mathrm{coll}}}{1-d_{\mathrm{ER}}}. \tag{19.34}$$

When $\mathfrak{r}' g_{\mathrm{Luke}} = g_{\mathrm{cal}}$, the calibrated ratio equals the heat quenching at every bias; for a reconstruction with the true pair energy this is $\mathfrak{r}' = 1$, Luke phonons collected like the available primary phonons of an electron recoil.

The ratio $\mathfrak{r}'$ differs from $\mathfrak{r}_{\mathrm{coll}}$ by the missing energy of the calibrating electron recoils. Flashing, the exposure of the detectors to infrared light that neutralizes ionized traps before data taking [15, Sect. 2], keeps that energy small. Away from $\mathfrak{r}' g_{\mathrm{Luke}} = g_{\mathrm{cal}}$ the two ratios differ even when no energy is missing, since the Luke phonons, a larger share of the electron-recoil signal than of the nuclear one, then carry a different weight in the two.

The expansion of Eq. (19.33) to first order in δ_{coll} and the missing energies, derived in Appendix D.45, displays the separate contributions:

$$\begin{aligned} 1 - Y_{\mathrm{ph}} &= \frac{(\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} - g_{\mathrm{cal}})(1-f_{\mathrm{ion}})}{1+\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}}} + \frac{1+g_{\mathrm{cal}}}{1+\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}}}\left[\delta_{\mathrm{coll}} + d_{\mathrm{NR}} - C_{\mathrm{cal}}\, d_{\mathrm{ER}}\right], \\ C_{\mathrm{cal}} &= \frac{1+\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} f_{\mathrm{ion}}}{1+\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}}}. \end{aligned} \tag{19.35}$$

The collection term is exact, and it is the whole answer when no energy is missing; it contains the pair-energy offset $g_{\mathrm{Luke}} - g_{\mathrm{cal}}$ together with $\mathfrak{r}_{\mathrm{coll}} - 1$. The calibration coefficient C_{cal} measures how much a deficit of the calibrating electron recoils matters. Such a deficit lowers the phonon scale and so raises every reconstructed nuclear-recoil energy. However, it acts through the primary part of the signal alone, and the nuclear recoil has less Luke signal to offset; as a result $C_{\mathrm{cal}} < 1$. For silicon at $\mathfrak{r}_{\mathrm{coll}} = 1$ and $f_{\mathrm{ion}} = 0.324$ it is 0.651 at 4 V and 0.382 at 40 V, and 0.62 to 0.74 at 4 V for the ionized fractions of 10 to 100 keV. Since Eq. (19.33) is linear in δ_{coll} and in d_{NR} separately, the first-order form errs through their product and through d_{ER}: by at most 3.0×10^{-3} when each of the three lies between 0 and 0.05.

Write the missing energy of the nuclear recoil as $d_{\mathrm{NR}} = e_{\mathrm{st}} + x_{\mathrm{NR}}$, with $e_{\mathrm{st}} \equiv E_{\mathrm{st}}/\omega$ the stored fraction, and $x_{\mathrm{ER}} = d_{\mathrm{ER}}$ for the electron recoil, so that x collects trapped, escaped and zero-point energy. Electron recoils store almost no energy. The largest energy an electron of kinetic energy E gives a silicon nucleus is $2E(E+2m_e)/M_\kappa$, which reaches the lowest threshold, 12.5 eV, at $E = 0.14$ MeV; whatever the calibrating electrons do store counts in x_{ER}. Appendix D.45 shows that Eq. (19.33) implies, without approximation,

$$\begin{aligned} (1 - Y_{\mathrm{ph}} - e_{\mathrm{st}})(1 - x_{\mathrm{ER}} + \mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}}) &= (\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} - g_{\mathrm{cal}})(1 - e_{\mathrm{st}} - f_{\mathrm{ion}}) \\ &\quad + (1+g_{\mathrm{cal}})\left[\delta_{\mathrm{coll}}(1-e_{\mathrm{st}}) + (1-\delta_{\mathrm{coll}})x_{\mathrm{NR}}\right] - (1 - e_{\mathrm{st}} + f_{\mathrm{ion}} g_{\mathrm{cal}})\, x_{\mathrm{ER}}. \end{aligned} \tag{19.36}$$

The factor $1 - x_{\mathrm{ER}} + \mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}}$ is positive. The right side is non-negative, and the stored fraction is therefore at most the defect fraction, $E_{\mathrm{st}}/\omega \le 1 - Y_{\mathrm{ph}}$, under five conditions on collection, missing energy and the reconstruction:

(a) $\delta_{\mathrm{coll}} \ge 0$: the primary phonons of nuclear recoils are collected no better than those of electron recoils.

(b) $\mathfrak{r}_{\mathrm{coll}} \ge 1$: Luke phonons are collected no worse than the primaries of electron recoils.

(c) $(1+g_{\mathrm{cal}})(1-\delta_{\mathrm{coll}})x_{\mathrm{NR}} \geq (1-e_{\mathrm{st}}+f_{\mathrm{ion}}g_{\mathrm{cal}})x_{\mathrm{ER}}$: at small deficits, the other missing energy of the nuclear recoil is at least $(1+f_{\mathrm{ion}}g_{\mathrm{cal}})/(1+g_{\mathrm{cal}})$ times that of the electron recoil. It holds, for instance, when the electron recoils lose nothing to traps after flashing and the zero-point change of the defects is not negative.

(d) $e_{\mathrm{st}} \leq 1 - f_{\mathrm{ion}}$, met with a wide margin at stored fractions of a few percent.

(e) $\varepsilon_{\mathrm{eh}} \leq \varepsilon_{\mathrm{cal}}$: the true pair energy is no larger than the one the reconstruction adopts, so that with (b) $\mathfrak{r}_{\mathrm{coll}}g_{\mathrm{Luke}} \geq g_{\mathrm{cal}}$. For germanium it refers to the pair energy of the germanium analysis, whose offset is not evaluated here.

The measured silicon pair energy, 3.75 eV (Sect. 18.4), meets (e) against the 3.82 eV of the CDMS II reconstruction; were $\varepsilon_{\mathrm{eh}}$ above $\varepsilon_{\mathrm{cal}}$, the first term would turn negative and the bound would loosen. At 95 % confidence the silicon datum has $Y_{\mathrm{ph}} \geq 0.945$, so $E_{\mathrm{st}}/\omega \leq 0.055$ under (a)–(e). With the book's $\varepsilon_{\mathrm{eh}}$ the pair-energy term is known, and under (a)–(d) the bound is tighter, 0.049, attained at $\mathfrak{r}_{\mathrm{coll}} = 1$ with no other deficit, since the first term of Eq. (19.36) grows with $\mathfrak{r}_{\mathrm{coll}}$. A negative zero-point change violates (c) and widens the bound by $|\Delta E_{\mathrm{zp}}|/\omega$. With 0.1 eV per displacement and the displacement count of Limit Check 19.3, that is 8×10^{-4} at 10 keV, and the bound for either sign of ΔE_{zp} is 0.056. Sect. 19.4 collects the silicon and germanium bounds in Table 19.1.

Condition (b) is the weak one. Luke phonons are emitted with energies of a few meV at most, softer than the primary phonons. In the films that collect phonons, the part of their spectrum below the pair-breaking energy $2\Delta_{\mathrm{BCS}}$ of aluminum, 0.36 meV (Sect. 1.5), is lost energy, as is the part absorbed at bare surfaces. The sign of $\mathfrak{r}_{\mathrm{coll}} - 1$ is therefore not fixed by the balance law. The mean-field transport estimate of Sect. 20.5.3 puts the collection ratio between 0.81 and 1.02, and between 0.82 and 1.01 at the mean Luke energy of silicon electrons at 1 V/cm, for one or two films of any thickness down to 60 nm with energy conserved; values up to 1.5 arise only for a selective sink that the estimate does not identify in a detector. For a stated detector Appendix E.25 specifies the calculation. On the branch $\mathfrak{r}_{\mathrm{coll}} < 1$, with $\delta_{\mathrm{coll}} = x_{\mathrm{NR}} = x_{\mathrm{ER}} = 0$, Eq. (19.33) solved for the stored energy is

$$\frac{E_{\mathrm{st}}}{\omega} = 1 - Y_{\mathrm{ph}} + \frac{(g_{\mathrm{cal}} - \mathfrak{r}_{\mathrm{coll}}g_{\mathrm{Luke}})(Y_{\mathrm{ph}} - f_{\mathrm{ion}})}{1 + g_{\mathrm{cal}}}, \tag{19.37}$$

so the inferred stored fraction grows linearly as $\mathfrak{r}_{\mathrm{coll}}$ falls. For the central silicon datum it is 0.042 at $\mathfrak{r}_{\mathrm{coll}} = 1$, 0.058 at 0.95 and 0.075 at 0.9, and the 95 % bound reaches 0.081 at 0.9. The bound is void if nothing bounds $\mathfrak{r}_{\mathrm{coll}}$ from below.

Limit Check 19.5. Stored energy from the phonon-yield ratio

The defect fraction read as stored energy [18, Eq. (4)], [15, Eq. (1)]: $E_{\mathrm{st}}/\omega = f_{\mathrm{DF}} = 1 - Y_{\mathrm{ph}}$, the case $\mathfrak{r}_{\mathrm{coll}} = 1$, $\delta_{\mathrm{coll}} = x_{\mathrm{NR}} = x_{\mathrm{ER}} = 0$, $\varepsilon_{\mathrm{cal}} = \varepsilon_{\mathrm{eh}}$ of Eq. (19.33), with $Y_{\mathrm{ph}} = 0.952^{+0.009}_{-0.007}$ (95 % C.L.) measured in silicon at 4 V [15, Sect. 4].

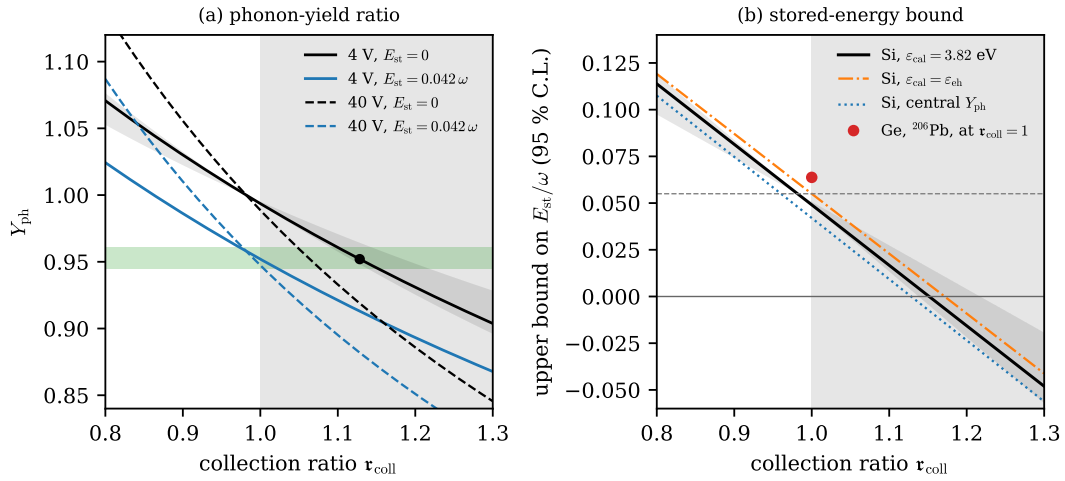


Figure 19.10: Silicon recoils against the collection ratio, from the exact identity Eq. (19.33) with $\delta_{coll} = d_{ER} = 0$, $g_{Luke} = 1.067$ ($\varepsilon_{eh} = 3.75\,\text{eV}$) and $g_{cal} = 1.047$ ($\varepsilon_{cal} = 3.82\,\text{eV}$) at 4 V, ten times both at 40 V, and $f_{ion} = 0.324$, the yield parameterization of the measurement at the 20 keV resonance that dominates the calibration [15, Eq. (4)]. (a) Phonon-yield ratio. Black: no stored energy; blue: stored fraction 0.042; solid at 4 V, dashed at 40 V. Dark band: f_{ion} from 0.269 to 0.497, the same parameterization at 10 and 100 keV. Green: the measured $0.952^{+0.009}_{-0.007}$. Dot: $\mathfrak{r}_{coll} = 1.128$, where collection alone reproduces it. (b) Upper bound on E_{st}/ω at 95 % confidence, Eq. (19.37) at $Y_{ph} = 0.945$, with the band of (a); dotted: the central Y_{ph}; dash-dotted: a reconstruction with the true pair energy; dashed line: the literature reading 0.055. Red point: germanium at $\mathfrak{r}_{coll} = 1$. Gray: $\mathfrak{r}_{coll} \geq 1$, where under (a)–(e) the bound does not exceed f_{DF}.

Below $\mathfrak{r}_{coll} = 1$ the bound exceeds f_{DF}, 0.081 at $\mathfrak{r}_{coll} = 0.9$ for silicon; above $\mathfrak{r}_{coll} = 1.128$ the central silicon datum needs no stored energy, and the datum alone cannot tell the two readings apart (Worked example 19.5).

The collection offset has a part that the balance law fixes. The recombination phonons at the contacts carry the gap energy of the collected pairs, the fraction E_g/ε_{eh} of an electron-recoil deposit and $f_{ion}E_g/\varepsilon_{eh}$ of a nuclear one, and they are born at the surfaces next to the sensors, with their own spectrum. Let ρ_g be their collection efficiency over that of the hard primary phonons, ϵ_{hard}. A recoil whose gap fraction is $\chi_g E_g/\varepsilon_{eh}$, with $\chi_g = 1$ for the electron and $\chi_g = f_{ion}$ for the nuclear recoil, then has the primary efficiency $\epsilon_{hard}[1 + (\rho_g - 1)\chi_g E_g/\varepsilon_{eh}]$. The gap part of the offset is

$$\delta_{coll}^{gap} = \frac{(1 - f_{ion})(E_g/\varepsilon_{eh})(\rho_g - 1)}{1 + (E_g/\varepsilon_{eh})(\rho_g - 1)}. \tag{19.38}$$

In silicon at $f_{ion} = 0.324$, $\rho_g - 1 = 0.21$ gives $\delta_{coll}^{gap} = 0.042$, which alone accounts for the deficit of the datum beyond the pair-energy offset, and a change of 0.02 in ρ_g moves it by 0.0037. Contacts sit on the same faces as the sensors, so a difference of this size is possible; its value needs a transport calculation for the detector (Appendix E.25). The rest of the offset comes from the primary spectra, which differ little once down-conversion has erased their origin (Sect. 20.2); from the depth profiles of the calibration sources, since neutrons and gammas do not interact at the same depths; and from escape at the surfaces, which differs between the two.

19.5.4 Two-bias inversion of the silicon datum

The CDMS II silicon detectors were biased at 4 V across 1 cm, and their neutron calibration gives $Y_{\mathrm{ph}} = 0.952^{+0.009}_{-0.007}$ over 10 to 100 keV [15, Sects. 2, 4]. Worked example 19.5 reads this one number in the two limiting ways.

Worked example 19.5: The degeneracy of the silicon datum between stored energy and collection

The Luke gain is $g_{\mathrm{Luke}} = 4\,\mathrm{V}/3.75\,\mathrm{V} = 1.067$, and the reconstruction uses $g_{\mathrm{cal}} = 4\,\mathrm{V}/3.82\,\mathrm{V} = 1.047$. The calibration rests on the spectral feature that an elastic resonance of silicon puts at recoil energies near 20 keV [15, Sect. 3.2]. There the ionization of nuclear recoils is collected in full and follows the yield parameterization of the same analysis [15, Eq. (4), Sect. 4], $y_C = [1/(a_C E_R) + 1/y_L]^{-1}$ with $a_C = 0.247\,\mathrm{keV}^{-1}$ and y_L Lindhard's yield Eq. (16.42). It gives $f_{\mathrm{ion}} = 0.324$; Lindhard's yield alone, which overestimates the low-energy data that analysis cites, gives 0.346; the same analysis notes that its fit improves if y_C is about 5 % low at 10 to 20 keV, which the Lindhard variant covers. We set $\delta_{\mathrm{coll}} = x_{\mathrm{NR}} = x_{\mathrm{ER}} = 0$.

Stored energy alone. With $\mathfrak{r}_{\mathrm{coll}} = 1$, Eq. (19.37) is

$$e_{\mathrm{st}} = 1 - Y_{\mathrm{ph}} + \frac{(g_{\mathrm{cal}} - g_{\mathrm{Luke}})(Y_{\mathrm{ph}} - f_{\mathrm{ion}})}{1 + g_{\mathrm{cal}}} = 0.048 - 0.006 = 0.042,$$

and 0.049 at the lower edge of the band. The pair-energy offset takes an eighth of the deficit; read with $\varepsilon_{\mathrm{cal}} = \varepsilon_{\mathrm{eh}}$, the datum would give 0.048 and 0.055. Lindhard's f_{ion} changes the stored fraction by 2×10^{-4}.

Collection alone. With $E_{\mathrm{st}} = 0$ the collection term of Eq. (19.35) remains, $1 - Y_{\mathrm{ph}} = (\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} - g_{\mathrm{cal}})(1 - f_{\mathrm{ion}})/(1 + \mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}})$, and solving for the collection ratio,

$$\mathfrak{r}_{\mathrm{coll}} = \frac{(1 - Y_{\mathrm{ph}}) + g_{\mathrm{cal}}(1 - f_{\mathrm{ion}})}{g_{\mathrm{Luke}}(Y_{\mathrm{ph}} - f_{\mathrm{ion}})} = \frac{0.048 + 0.708}{0.670} = 1.128.$$

Luke phonons collected 13 % better than the primaries reproduce the datum with no stored energy. The band of the datum maps onto $1.099 \le \mathfrak{r}_{\mathrm{coll}} \le 1.152$, and Lindhard's f_{ion} gives 1.134. The mean-field transport estimate of Sect. 20.5.3 puts the collection ratio between 0.81 and 1.02 with energy conserved, and only a selective sink that the estimate does not identify in a detector reaches higher values. Collection alone, $\mathfrak{r}_{\mathrm{coll}} = 1.128$, is therefore not reached by the transport estimate, but it is not excluded until computed kernels test real detectors (Appendix E.25); the two-bias measurement of Sect. 19.5.4 decides.

Mixtures. Every point of the line Eq. (19.37) reproduces the datum: half of the stored-energy reading, $e_{\mathrm{st}} = 0.021$, needs $\mathfrak{r}_{\mathrm{coll}} = 1.064$. A gap offset Eq. (19.38) with $\rho_g - 1 = 0.21$ reproduces it with neither.

Energy dependence. At fixed $\mathfrak{r}_{\mathrm{coll}} = 1.128$ the collection term scales with $1 - f_{\mathrm{ion}}$, and f_{ion} grows with the recoil energy: the parameterization gives 0.269 at 10 keV and 0.497 at 100 keV, hence $Y_{\mathrm{ph}} = 0.948$ and 0.964, the second just above the upper edge 0.961 of the band. The same analysis finds, however, that the ionization collected at 4 V/cm falls to about 75 % of the parameterized yield at 100 keV [15, Sect. 4]; with $f_{\mathrm{ion}} = 0.75 \times 0.497 = 0.373$ the prediction is 0.956, inside the band. A stored fraction independent of energy predicts 0.951 to 0.954 over the range. At the present precision the energy dependence does not separate the readings.

A second bias separates the collection ratio from the rest. Multiply Eq. (19.33) by its denominator and divide by $1 - d_{\mathrm{ER}}$: at the bias V_i, with gains $g_{\mathrm{Luke},i}$ and $g_{\mathrm{cal},i}$,

$$(1 + g_{\mathrm{cal},i})\, Y_{\mathrm{ph}}(0) - g_{\mathrm{Luke},i}\,(Y_i - f_{\mathrm{ion}})\,\mathfrak{r}' = Y_i + f_{\mathrm{ion}} g_{\mathrm{cal},i}, \tag{19.39}$$

with Y_i the phonon-yield ratio measured at that bias and $Y_{\mathrm{ph}}(0)$ and $\mathfrak{r}'$ those of Eq. (19.34). The relation is exact and linear in the two unknowns. Two biases give two equations whose determinant is $(Y - f_{\mathrm{ion}})(g_{\mathrm{Luke},1} - g_{\mathrm{Luke},2})$ when the two measured ratios are equal and does not vanish in general, so the pair $(Y_{\mathrm{ph}}(0), \mathfrak{r}')$ follows by solving a 2×2 system (Appendix D.45). For silicon at 4 and 40 V, collection alone ($\mathfrak{r}_{\mathrm{coll}} = 1.128$, no stored energy) predicts $Y_{\mathrm{ph}} = 0.919$ at 40 V, stored energy alone 0.947, since the pair-energy offset grows with the bias, and the half-and-half mixture 0.932. Solving Eq. (19.39) with these values returns the inputs. The difference between the readings, 0.028, is about 3.6 times the half-width of the present band, and an uncertainty of 0.005 on the 40 V ratio moves the inverted $\mathfrak{r}'$ by 0.020 to 0.026. At still higher bias the Luke phonons dominate both signals and Y_{ph} tends to $f_{\mathrm{ion}} + [Y_{\mathrm{ph}}(0) - f_{\mathrm{ion}}]\varepsilon_{\mathrm{eh}}/(\varepsilon_{\mathrm{cal}}\mathfrak{r}')$, which is 0.912 for collection alone.

The inversion holds under hypotheses that are each a missing input. The collection ratio must not change between the two biases, yet the spectrum of Luke phonons depends on the field; the residual is of order $(\mathrm{d}\mathfrak{r}_{\mathrm{coll}}/\mathrm{d}V_{\mathrm{bias}})\,\Delta V_{\mathrm{bias}}$, which the transport calculation of Appendix E.25 provides. The ionized fraction must be measured at both biases with the charge channel, since trapping depends on the field. The offset and the missing energies must also be the same at both biases, which requires both to collect every pair within the window; a zero-bias run does not qualify as the second point, because the gap energy of uncollected pairs then enters through Eq. (19.30). Finally, the reconstruction must use the same $\varepsilon_{\mathrm{cal}}$ at both biases and the true $\varepsilon_{\mathrm{eh}}$ must be known, since Eq. (19.39) carries the two gains separately. A third bias overdetermines Eq. (19.39) and tests the first three hypotheses.

What the inversion cannot do is split $Y_{\mathrm{ph}}(0) = (1 - \delta_{\mathrm{coll}})(1 - d_{\mathrm{NR}})/(1 - d_{\mathrm{ER}})$ into collection offset and stored energy, since both enter it in the same way. Three further measurements or calculations split it. Transport calculations of the efficiencies of each phonon population in the detector that took the data fix the gap and spectral parts of δ_{coll} (Appendix E.25). Neutron and gamma sources with matched depth profiles remove its position part. And two read windows, short and long, separate stored energy released within the window from energy that stays frozen, since the release term of d_{NR} is the one that grows with t_{win}.

19.6 Delayed release and the low-energy excess

A close Frenkel pair in silicon holds 6.8 to 7.5 eV (Sect. 7.4), and at 10 mK it can give them back only by tunneling. For an estimated barrier of 0.1 eV over 1.5 Å the two-parabola tunneling exponent is 27.5, about 14 % above the largest exponent that allows recombination within a year (Sect. 16.8.4). Whether the pair releases its energy during a run or keeps it for good thus turns on one number that is not known. Meanwhile cryogenic detectors with thresholds of tens of eV record an excess of events that rises toward threshold and decays over days after cooldown. A silicon crystal glued to its holder shows more than a hundred times the rate of an identical crystal hung from wire bonds [23]. Energy stored in defects is the obvious suspect. Its release at millikelvin obeys an exact sum rule, which fixes how much energy delayed events can carry and at what energy per event; together with energy conservation it bounds the excess, and it leaves collective release of elastic energy as the candidate.

19.6.1 Defect kinetics at millikelvin

At $t_{\rm pr}$ the recoil leaves the crystal in one of a set of configurations i, each a basin of $E^{\rm ad}$ with the configurational energy $E_i^{\rm conf}$, the minimum of the basin measured from the minimum of the perfect crystal (Eq. (19.4)). The perfect crystal is one of them, with $E^{\rm conf}=0$. Let $p_i(t)$ be the probability that the defect complex is in configuration i at time t, and Γ_{ij} the rate of a hop from i to j. Hops are rare compared with the vibrations inside a basin, so the populations obey the master equation

$$\frac{{\rm d}p_i}{{\rm d}t}=\sum_{j\neq i}\bigl[\Gamma_{ji}\,p_j-\Gamma_{ij}\,p_i\bigr],\qquad p_i(t_{\rm pr})\ \text{from the partition at } t_{\rm pr}. \tag{19.40}$$

The stored energy at time t is the mean configurational energy, $E_{\rm st}(t)=\sum_i p_i(t)E_i^{\rm conf}$, which equals the $E_{\rm st}$ of the partition at $t=t_{\rm pr}$.

The rates at millikelvin are tunneling rates. A barrier of chemical size cannot be crossed thermally, and below the crossover temperature $T_\times$ of Eq. (16.77), tens of kelvin for an atom, every hop goes through the barrier (Sect. 16.8.4). A hop from i to j releases $\varepsilon_{ij}^{\rm hop}=E_i^{\rm conf}-E_j^{\rm conf}$, and its tunneling amplitude is $\Delta_0=\omega_{\rm w}{\rm e}^{-\lambda_{\rm tun}}$ in the notation of Sect. 16.7. The golden rule Eq. (C.174) gives the rate as $\Delta_0^2/4$ times the time integral of the bath correlation at the energy released. Let $A(\varepsilon)$ be the line shape, the probability per unit energy that the bath accepts the energy ε. Then

$$\Gamma_{ij}=\frac{\pi}{2}\,\omega_{\rm w}^2\,{\rm e}^{-2\lambda_{\rm tun}}\,A(\varepsilon_{ij}^{\rm hop}+\Delta E_{{\rm zp},ij}), \tag{19.41}$$

where $\Delta E_{{\rm zp},ij}$ is the decrease of the zero-point energy of the local modes in the hop, so that the argument is the energy the bath actually receives. For the two-level systems of Sect. 16.8 the energy is below the top of the phonon band and one phonon takes it; then $A(\varepsilon)=J_{\rm s}(\varepsilon)\coth(\varepsilon/2T)/\pi\varepsilon^2$, which turns Eq. (19.41) into Eq. (16.68). An eV release is far above the phonon band, so the bath can take it only as many quanta at once, in the local modes k that are displaced between the two minima, with Huang–Rhys factors $S_{{\rm HR},k}$ and the reorganization energy $E_{\rm reorg}=\sum_k S_{{\rm HR},k}\omega_k$ of Eq. (E.106). At $T\ll\omega_k$ the correlation of one displaced mode is the Poisson series of Eq. (5.60), and the size of the rearrangement decides between two regimes.

If the hop is a small rearrangement, $E_{\rm reorg}\ll\varepsilon^{\rm hop}$, the energy must go into $n_q=\varepsilon/\omega_{\rm acc}$ quanta of modes that are barely displaced. For one accepting mode of energy $\omega_{\rm acc}$ and total factor S, the Poisson weight of n_q quanta gives $A(\varepsilon)\,\omega_{\rm acc}={\rm e}^{-S}S^{n_q}/n_q!$, the energy-gap law. For $S=1$, $\omega_{\rm acc}=60\,{\rm meV}$ and $\varepsilon=1\,{\rm eV}$ it is 2.7×10^{-15}, which adds 16.8 to the tunneling exponent. If instead the hop is a large rearrangement, $E_{\rm reorg}\approx\varepsilon^{\rm hop}$, the reaction coordinate itself takes the energy. Then A is a Gaussian of width $(\sum_k S_{{\rm HR},k}\omega_k^2)^{1/2}$ centered at $E_{\rm reorg}$, and there is no gap suppression. The two-state reduction behind Eq. (19.41) is then not controlled either, and the rate is the instanton rate on the full surface (Sect. 16.8.4 and Appendix E.22). Taking the prefactor, of order $\omega_{\rm w}$, as $\Gamma_0=10^{13}\,{\rm s}^{-1}$, a hop happens within the observation time $t_{\rm obs}$ if

$$2\lambda_{\rm tun}+\ln\frac{1}{\omega_{\rm w}A(\varepsilon)}\le\ln(\Gamma_0 t_{\rm obs}), \tag{19.42}$$

and for one year the right side is 47.2. With $\omega_{\rm w}A\approx1$ this is the bound $\lambda_{\rm tun}\le23.6$ of Sect. 16.8.4. With the energy-gap factor above it becomes $\lambda_{\rm tun}\le6.8$, and for a 7 eV release the logarithm of the factor is about 440, so a weakly coupled eV release never happens.

For a close Frenkel pair in silicon the two-parabola exponent $\lambda_{\rm tun}=(d_{\rm tun}/2)(2M_\kappa V_{\rm b})^{1/2}$ with $M_\kappa=26.16\,{\rm GeV}$, $V_{\rm b}=0.1\,{\rm eV}$ and $d_{\rm tun}=1.5\,{\rm \AA}$ is 27.5, and a quartic barrier of the same height and width raises it by 4/3, to 36.7. In the weak-coupling regime the pair is frozen.

In the strong-coupling regime it recombines within a year only if the true exponent lies at least 14 % below the two-parabola value. A split interstitial in aluminum, with an estimated barrier of 0.1 eV over 1.4 Å, has 25.2 and needs 6 %. Aluminum is superconducting at these temperatures, so its conduction electrons do not take the energy (Sect. 16.8.2). The verdict for both defects is therefore conditional, and the condition is one number per path: the tunneling exponent along the actual minimum-energy path, with $E_{\mathrm{reorg}}/\varepsilon^{\mathrm{hop}}$, to about 5 %.

19.6.2 The release-path sum rule

Every hop puts energy into the crystal, and its distribution in energy and time is what a detector sees. For one seed, the expected number of hops per unit time and per unit released energy is

$$\mathcal{R}_{\mathrm{hop}}(E,t) = \sum_{i\neq j} p_i(t)\,\Gamma_{ij}\,\delta(E-\varepsilon_{ij}^{\mathrm{hop}}), \tag{19.43}$$

with the configurational energy in the delta function. Its integral over E and t is the expected number of hops $\langle N_{\mathrm{hop}}\rangle$, not one: a pair may pass through several configurations before it recombines. The energy it releases is fixed by its end points alone.

Release-path sum rule

For populations that obey Eq. (19.40) on a closed set of configurations,

$$\int_{t_{\mathrm{pr}}}^{t} \mathrm{d}t' \int \mathrm{d}E\; E\,\mathcal{R}_{\mathrm{hop}}(E,t') = E_{\mathrm{st}} - E_{\mathrm{st}}(t), \tag{19.44}$$

the stored energy of the partition minus the energy still stored at t.

The proof takes two lines. By Eq. (19.40),

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_i p_i E_i^{\mathrm{conf}} = \sum_{i\neq j}(\Gamma_{ji}p_j - \Gamma_{ij}p_i)E_i^{\mathrm{conf}} = -\sum_{i\neq j}\Gamma_{ij}\,p_i(E_i^{\mathrm{conf}} - E_j^{\mathrm{conf}}) = -\int \mathrm{d}E\; E\,\mathcal{R}_{\mathrm{hop}}(E,t),$$

where the middle step exchanges the labels in the first term. Integrating from t_{pr} to t gives Eq. (19.44). Neither detailed balance nor any model of the rates enters, so the rule holds for every rate law of Sect. 19.6.1 and also for rates that change in time. These include the rates stimulated by a heat pulse, by illumination or by the carriers of a later event.

The rule is exact under five conditions on how a calculation accounts for the energy: every hop counted, one table of configurational energies, photons counted as escaped energy, no probability leaving the listed configurations, and configurational energies throughout. Every hop must be counted, including uphill hops with $\varepsilon^{\mathrm{hop}} < 0$, in which the defect absorbs heat from the lattice. The partition and the release paths must use one table of configurational energies, from one potential and one cell, as Sect. 19.1.3 requires; otherwise the two sides differ by the error of the potential, a few tenths of an eV. A hop that emits a photon still releases its energy, which then appears among the escaping quanta of the balance law of Sect. 19.5 instead of the phonons. No probability may leave the listed configurations, $\sum_i p_i(t) = 1$: the master equation describes one seed, and recombination of an interstitial with the vacancy of another event lies outside it, since it needs the joint state of both. Finally, the energies are configurational. The energy that the bath receives in a hop is $\varepsilon_{ij}^{\mathrm{hop}} + \Delta E_{\mathrm{zp},ij}$, where the zero-point change is below a tenth of an eV, an estimate (Sect. 19.5.1). The zero-point changes add up along any path to the zero-point change of the end basin, which Sects. 19.1.4 and 19.5.1 count in the phonons, not in E_{st}. Read with these conditions, Eq. (19.44) is the

stored-energy term of the balance law Eq. (19.27): what leaves $E_{\mathrm{st}}(t)$ enters the phonons, at the energies and times of $\mathcal{R}_{\mathrm{hop}}$.

Worked example 19.6: The release-path sum rule for one Frenkel pair

Take a close Frenkel pair C in silicon with $E_{\mathrm{C}}^{\mathrm{conf}} = 7.0\,\mathrm{eV}$, which can recombine directly to the perfect crystal 0 or first relax into a bond defect B with $E_{\mathrm{B}}^{\mathrm{conf}} = 2.8\,\mathrm{eV}$, both inside the ranges of Sect. 7.4. Separation to an isolated vacancy and interstitial, which costs 0.6 to 2.9 eV more, is closed at 10 mK. No path of this kind has been computed (Appendix E.22), so we take illustrative strong-coupling exponents in Eq. (19.42), with $\Gamma = \Gamma_0 \mathrm{e}^{-2\lambda_{\mathrm{tun}}}$: $\lambda_{\mathrm{tun}} = 22.5$ for C→B, 24.0 for C→0 and 25.0 for B→0. The rates are $\Gamma_{\mathrm{CB}} = 2.86 \times 10^{-7}$, $\Gamma_{\mathrm{C0}} = 1.43 \times 10^{-8}$ and $\Gamma_{\mathrm{B0}} = 1.93 \times 10^{-9}\,\mathrm{s}^{-1}$. The pair leaves C after 38.5 days on average, through B with probability $\Gamma_{\mathrm{CB}}/(\Gamma_{\mathrm{CB}} + \Gamma_{\mathrm{C0}}) = 0.953$.
The master equation for three states is solved by two exponentials, $p_{\mathrm{C}} = \mathrm{e}^{-\Gamma_{\mathrm{C}} t}$ with $\Gamma_{\mathrm{C}} = \Gamma_{\mathrm{CB}} + \Gamma_{\mathrm{C0}}$, and $p_{\mathrm{B}} = \Gamma_{\mathrm{CB}}(\mathrm{e}^{-\Gamma_{\mathrm{C}} t} - \mathrm{e}^{-\Gamma_{\mathrm{B0}} t})/(\Gamma_{\mathrm{B0}} - \Gamma_{\mathrm{C}})$. After one year $p_{\mathrm{C}} = 10^{-4}$, $p_{\mathrm{B}} = 0.902$ and $p_0 = 0.098$, and the stored energy is $E_{\mathrm{st}}(t) = 0.902 \times 2.8\,\mathrm{eV} = 2.526\,\mathrm{eV}$. The expected numbers of hops are $\Gamma_{ij} \int_0^t p_i\,\mathrm{d}t'$: 0.9525 hops C→B releasing 4.2 eV each, 0.0474 hops C→0 releasing 7.0 eV, and 0.0505 hops B→0 releasing 2.8 eV. Together they release

$$0.9525 \times 4.2\,\mathrm{eV} + 0.0474 \times 7.0\,\mathrm{eV} + 0.0505 \times 2.8\,\mathrm{eV} = 4.474\,\mathrm{eV} = 7.0\,\mathrm{eV} - 2.526\,\mathrm{eV},$$

as Eq. (19.44) requires, in $\langle N_{\mathrm{hop}} \rangle = 1.05$ hops, 4.26 eV per hop. After ten years 5.539 eV has been released in 1.43 hops, and after a hundred years 6.994 eV in 1.95 hops: the number of hops depends on the path, the energy only on the end points. A sensor that absorbs a quarter of the phonon energy of a release, as the sensors of the glued and suspended crystals do [23], which absorb phonons before they thermalize, records the three kinds of hop as events of 1.05, 1.75 and 0.70 eV, each shifted by a quarter of its zero-point change.

19.6.3 The delayed-release spectrum

A detector sees the sum of $\mathcal{R}_{\mathrm{hop}}$ over all seeds, from the recoils of earlier events and from the defects frozen in at cooldown, with each release converted into a recorded energy. A release of energy ε puts $\varepsilon + \Delta E_{\mathrm{zp}}$ into athermal phonons, or into a photon on the radiative branch, and the sensor absorbs a fraction f_{ph} of the phonon energy, which depends on where the release happens and on the phonon spectrum. A release is an event only if $f_{\mathrm{ph}}(\varepsilon + \Delta E_{\mathrm{zp}}) \geq E_{\mathrm{th}}$. Below threshold it is heat, a load on the sensor that adds to its noise but makes no event.

Defects frozen in at cooldown follow the same kinetics, and two families dominate: the two-level systems of amorphous material near the detector and the point defects of the crystal. The two-level systems of the amorphous films and glues near a detector relax by emitting single phonons of energy Δ_{tl} (Sect. 16.8). After a quench from the freezing temperature T_{frz} to T_{lo}, a defect of splitting Δ_{tl} keeps the excess population $\frac{1}{2}[\tanh(\Delta_{\mathrm{tl}}/2T_{\mathrm{lo}}) - \tanh(\Delta_{\mathrm{tl}}/2T_{\mathrm{frz}})]$, and the uniform density in λ_{tun} makes the fraction relaxing per unit time $1/2t$. The steps of Eq. (16.79), kept at fixed splitting, give the number of phonons emitted per unit energy and time by a volume $\mathcal{V}$,

$$\frac{\mathrm{d}\mathcal{R}_{\mathrm{TLS}}}{\mathrm{d}E} = \frac{\bar{P}\mathcal{V}}{4t}\Big[\tanh\frac{E}{2T_{\mathrm{lo}}} - \tanh\frac{E}{2T_{\mathrm{frz}}}\Big], \tag{19.45}$$

whose first moment is Eq. (16.79). The mean quantum is $1.19\,T_{\mathrm{frz}}$, 0.49 meV for $T_{\mathrm{frz}} = 4.8\,\mathrm{K}$:

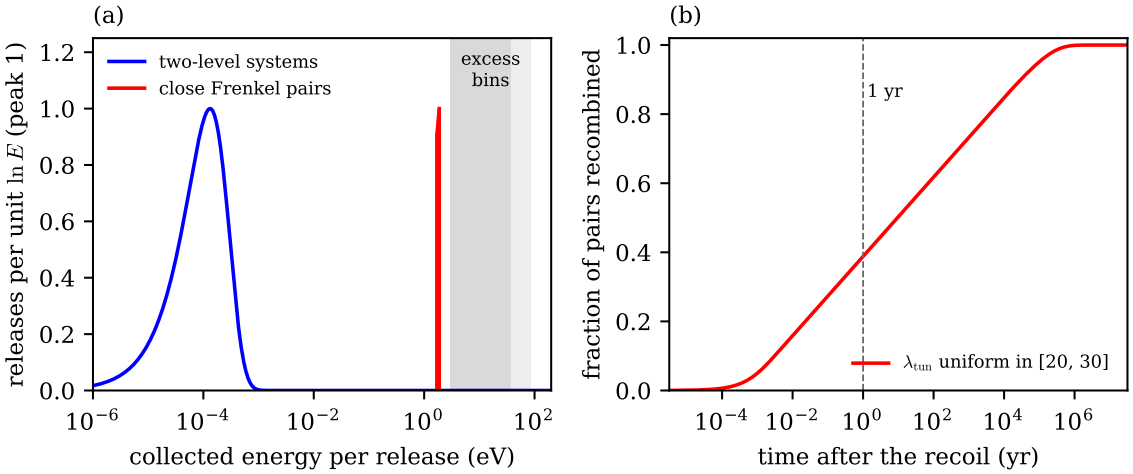


Figure 19.11: Delayed releases from two defect families. (a) Distribution of the energy one release gives a sensor that absorbs a fraction $f_{\mathrm{ph}} = 0.25$ of the phonon energy, per unit logarithm of the energy, each family scaled to a peak of one. Blue: two-level systems frozen at $T_{\mathrm{frz}} = 4.8\,\mathrm{K}$ and relaxing at 10 mK, Eq. (19.45), mean quantum 0.49 meV. Red: recombination of close Frenkel pairs in silicon, with configurational energies spread uniformly between 6.8 and 7.5 eV (Sect. 7.4), zero-point changes neglected. Gray: the 3–38 and 38–85 eV bins of energy absorbed in the sensor in which a glued silicon crystal shows its excess of events [23]. (b) Fraction of a family of close pairs recombined by time t, for strong coupling, $\Gamma = 10^{13}\,\mathrm{s}^{-1}\,\mathrm{e}^{-2\lambda_{\mathrm{tun}}}$, with λ_{tun} spread uniformly between 20 and 30, an illustrative range around the two-parabola estimates. The fraction grows as $\ln t$ (a rate $\propto 1/t$) between 6.5 h and 3.6×10^5 years; 0.39 of the pairs recombine within the first year. In the weak-coupling regime none does.

the release of a glass is all heat, 46 pW per gram of fused silica an hour after cooling (Worked example 16.8), and none of it can be an event.

The second family is the point defects of the crystal itself, close Frenkel pairs and bond defects from earlier recoils, which release eV per hop in the strong-coupling regime and nothing in the weak one. Fig. 19.11 shows both. Their time laws carry the distribution of the paths. A uniform distribution of tunneling exponents leads to $1/t$, as for the glass, and a single path to an exponential. Correlated recombination of a pair that diffuses in three dimensions follows the first-passage tail $t^{-3/2}$, and recombination with defects of other seeds, outside Eq. (19.40), decays as $(1 + t/t_0)^{-2}$. A defect that moves along a row toward its own vacancy, down the gradient of their elastic interaction, arrives after a peaked delay instead. The time law alone, however, does not identify the source, since stress relaxation in a glue also leads to $1/t$ (Sect. 19.6.6).

Fig. 19.11 makes the scale of a single release plain. A close pair gives a sensor that absorbs a quarter of the phonons between 1.70 and 1.88 eV, below the lowest bin of the excess, and a two-level system gives a ten-thousandth of that. Single releases add to the heat load and, through the spectrum, to the noise of a sensor with a threshold of an eV or less; for thresholds of a few eV they are invisible. Amorphous pockets left by keV recoils are the one point-defect source that releases more at once: a pocket that recrystallizes releases of order 0.1 eV per atom, an estimate, so a pocket of 120 atoms gives 3 eV to such a sensor (Appendix E.23).

The inputs of all this are computable from first principles. Barriers and minimum-energy paths come from the nudged elastic band, which relaxes a chain of configurations between

the two minima onto the path, on the density-functional or machine-learned surface of Appendices E.10 and E.11, the Huang–Rhys factors from the Hessians at both minima, and the exponent along the true path from ring-polymer instantons. Appendix E.22 specifies these calculations for close pairs in silicon and germanium and for interstitials in aluminum, with Eq. (19.44) as the acceptance test of the kinetic Monte Carlo, the stochastic sampling of hop sequences that turns the rates into $\mathcal{R}_{\mathrm{hop}}$. Appendix E.23 specifies the pocket distribution, and Appendix E.25 the inputs of the low-energy excess and the test of the two-stage factorization below.

19.6.4 Defectons

A light defect that tunnels between equivalent sites faster than the disorder around it can detune them becomes delocalized. This matters for a detector because a mobile defect reaches its partner, or a surface, on a time set by its band and not by a barrier. If the defect at site $\boldsymbol{r}$ tunnels to each neighbor with amplitude A_{dfn}, the Bloch combinations of the localized states are stationary, and the defect becomes a quasiparticle, a defecton, with a band

$$\varepsilon^{\mathrm{dfn}}(\boldsymbol{k}) = E^{\mathrm{loc}} - A_{\mathrm{dfn}} \sum_{i=x,y,z} \cos k_i a, \tag{19.46}$$

on a simple cubic lattice of spacing a [24, Eqs. (1)–(3)]. The band width is $\Delta_{\mathrm{dfn}} = zA_{\mathrm{dfn}}$, with $z = 6$ neighbors, the mass at the bottom $1/A_{\mathrm{dfn}}a^2$, and the speed at most $A_{\mathrm{dfn}}a$. In solid helium, where the idea was born, a ^{3}He impurity has $\Delta_{\mathrm{dfn}} \sim 10^{-4}\,\mathrm{K}$; the light interstitials of metals, hydrogen in niobium for example, are candidates in the materials of detectors [24]. A narrow band has a weakness. In a hop the defecton cannot change its energy by more than Δ_{dfn}, so a gradient of the potential energy ϕ with $|\nabla\phi|a > \Delta_{\mathrm{dfn}}$ localizes it [24, Eq. (17)]. Around a defect whose elastic interaction falls as $\phi = V_0(a/r)^3$, the gradient criterion reads $3V_0(a/r)^4 > \Delta_{\mathrm{dfn}}$, and the region $r < R_0 = a(3V_0/\Delta_{\mathrm{dfn}})^{1/4}$ is closed to band motion.

In a perfect crystal at $\Delta_{\mathrm{dfn}} \ll T$ the defecton moves through the band ballistically between collisions with phonons, and its diffusion constant falls as T^{-9}. The exponent follows from four facts: the kinematics of a narrow band, the Rayleigh form of two-phonon scattering, the small momentum change per collision, and the band velocity. First, the defecton cannot absorb or emit one phonon: that needs $|\varepsilon^{\mathrm{dfn}}(\boldsymbol{k}+\boldsymbol{Q}) - \varepsilon^{\mathrm{dfn}}(\boldsymbol{k})| = \bar{c}Q$, and the left side is at most $A_{\mathrm{dfn}}aQ$, far below $\bar{c}Q$ for a narrow band. The leading process is therefore two-phonon scattering, $\boldsymbol{Q} \to \boldsymbol{Q}'$ with $\omega' \simeq \omega$. Second, its amplitude is proportional to ω. For a coupling quadratic in the strain, the elastic inhomogeneity of the defect, the amplitude carries $QQ'/(\omega\omega')^{1/2} \propto \omega$. The linear deformation coupling at second order has two time orderings with energy denominators $\pm\omega$ that cancel at leading order, and what remains is the second difference of the band, $\propto A_{\mathrm{dfn}}a^2QQ'$, again $\propto \omega$. The cross-section is therefore $\propto (Qa)^4$, Rayleigh scattering [24, Eq. (49)]. Third, the number of thermal phonons grows as T^3, so the collision rate grows as T^7; but each collision changes the crystal momentum by at most about $T/\bar{c}$, much less than $1/a$ [24, Eq. (54)], and $(1/Qa)^2 \propto T^{-2}$ collisions are needed to randomize it. The transport rate grows as T^9. Fourth, $D = v^2\tau_{\mathrm{tr}}/3$ with $v \sim A_{\mathrm{dfn}}a$ gives $D \propto \Delta_{\mathrm{dfn}}^2 T^{-9}$. Pushkarov computes the rate of growth of the mean squared crystal momentum,

$$\Gamma_{k^2} \equiv \frac{\mathrm{d}\langle(\Delta\boldsymbol{k})^2\rangle}{\mathrm{d}t} = \alpha_{\mathrm{P}}\,\sigma_{\mathrm{P}}\,\frac{\bar{c}}{a^3}\Big(\frac{T}{\theta_p}\Big)^9, \qquad \theta_p = \frac{\bar{c}}{2a}, \qquad \alpha_{\mathrm{P}} = \frac{\pi^5\zeta(9)}{240\,\zeta(8)} = 1.27, \tag{19.47}$$

where $\bar{c} = (c_{\mathrm{LA}}^{-4} + 2c_{\mathrm{TA}}^{-4})^{-1/4}$ is his average sound speed and σ_{P} the strength of the defect's elastic perturbation relative to that of a vacancy [24, Eqs. (48), (58)]. The characteristic temperature is θ_p, about an eighth of the Debye temperature.

These four facts fix the exponent; the prefactor needs the motion in the band. Since each collision changes $\boldsymbol{k}$ by a small step, $\boldsymbol{k}$ diffuses, with $D_1 = \Gamma_{k^2}/6$ per Cartesian component, and detailed balance at temperature T adds the drift $-D_1 \nabla_{\boldsymbol{k}} \varepsilon^{\mathrm{dfn}}/T$. The band Eq. (19.46) is a sum over components, so each component moves on its own circle $\vartheta = k_x a$, with velocity $v_x = A_{\mathrm{dfn}} a \sin\vartheta$. The diffusion constant is the time integral of the velocity correlation, $D = \int_0^\infty \mathrm{d}t\, \langle v_x(0) v_x(t)\rangle$. It equals $(A_{\mathrm{dfn}} a)^2 \langle \sin\vartheta\, \chi\rangle$, where χ solves the backward equation $a^2 D_1(\chi'' - U'\chi') = -\sin\vartheta$ with $U = -(A_{\mathrm{dfn}}/T)\cos\vartheta$, and the average is over e^{-U}. With $y = A_{\mathrm{dfn}}/T$, one integration gives $\mathrm{e}^{y\cos\vartheta}\chi' = \mathrm{e}^{y\cos\vartheta}/(y a^2 D_1) + C$, and periodicity of χ fixes $C = -1/[y a^2 D_1 I_0(y)]$, with I_0 the modified Bessel function. An integration by parts then gives $\langle \sin\vartheta\, \chi\rangle = [1 - I_0(y)^{-2}]/(y^2 a^2 D_1)$, and

Defecton diffusion in a phonon gas

$$D = \frac{6T^2}{\Gamma_{k^2}}\Big[1 - I_0\Big(\frac{A_{\mathrm{dfn}}}{T}\Big)^{-2}\Big] \xrightarrow{T \gg \Delta_{\mathrm{dfn}}} D_9 = \frac{3A_{\mathrm{dfn}}^2}{\Gamma_{k^2}} = \frac{\Delta_{\mathrm{dfn}}^2}{12\Gamma_{k^2}} \propto T^{-9}. \tag{19.48}$$

At $T \ll A_{\mathrm{dfn}}$ the bracket tends to one and $D = 6T^2/\Gamma_{k^2} \propto T^{-7}$, the diffusion of a particle of mass $1/A_{\mathrm{dfn}} a^2$ with the friction that the same phonons exert. These are the two limits of Pushkarov's interpolation [24, Eqs. (61), (64)], whose low-temperature coefficient is half ours because it rests on an estimated collision time.

Limit Check 19.6. Defecton diffusion

The published law is $D_T = (a\bar{c}/3\alpha_{\mathrm{P}}\sigma_{\mathrm{P}})[\varepsilon(p)/\theta_p]^2(\theta_p/T)^9$, for $\Delta_{\mathrm{dfn}} \ll T \ll \theta_p$ [24, Eq. (61)], which is D_9 of Eq. (19.48) with $\varepsilon(p) = \Delta_{\mathrm{dfn}}/4$.

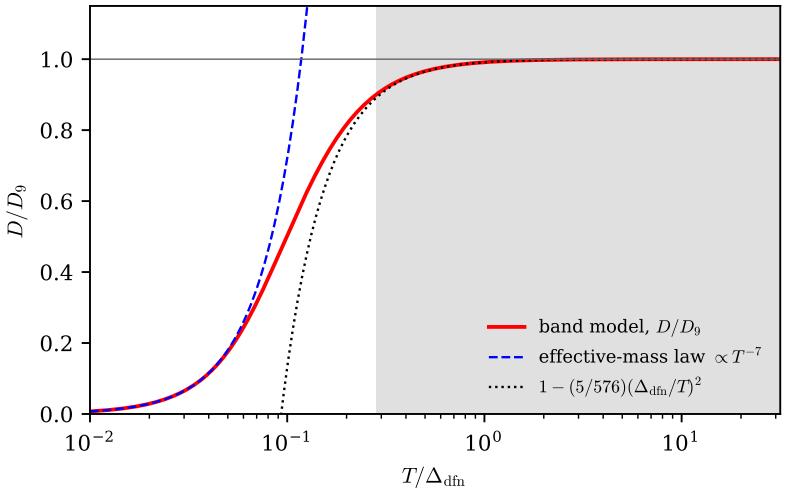


Figure 19.12: Diffusion constant of a defecton in a simple cubic tight-binding band, Eq. (19.48), divided by its T^{-9} limit D_9, against the temperature in units of the band width. The ratio depends on T/Δ_{dfn} alone, since the phonon rate Γ_{k^2} of Eq. (19.47) cancels. Red: the band model, crystal momentum diffusing at a constant rate with the detailed-balance drift. Blue dashed: the effective-mass law $D = 6T^2/\Gamma_{k^2} \propto T^{-7}$. Dotted: the leading correction $1 - \frac{5}{576}(\Delta_{\mathrm{dfn}}/T)^2$. Gray: within 10 % of the T^{-9} law, $T \geq 0.28\,\Delta_{\mathrm{dfn}}$. The upper edge $T \ll \theta_p$ lies outside the plot and depends on the crystal. Below $0.28\,\Delta_{\mathrm{dfn}}$ the band is no longer uniformly populated and the law steepens toward T^{-7}; above θ_p the phonons are no longer long-wave and the Rayleigh law of the

cross-section fails.

The T^{-9} law holds better than its derivation suggests. The ratio D/D_9 expands as $1-(5/16)(A_{\mathrm{dfn}}/T)^2$, so the law is within 10 % for $T \geq 0.28\,\Delta_{\mathrm{dfn}}$ and within 1 % for $T \geq 0.93\,\Delta_{\mathrm{dfn}}$, although the condition stated with it is $T \gg \Delta_{\mathrm{dfn}}$. The relevant energy is the hopping amplitude, a sixth of the band width. At $T = 0.1\,\Delta_{\mathrm{dfn}}$ the local exponent is -7.9. For a detector the consequence runs the other way from intuition: a defecton moves faster the colder the crystal, until strain gradients localize it. In a real crystal the gradients from impurities, dislocations and the defect's own partner set R_0, and a band narrower than the elastic interaction at a few lattice spacings never carries the defect far.

19.6.5 Crowdions

An extra atom in a close-packed row of a body-centered cubic metal spreads over many neighbors as a kink in the row, a crowdion. In tungsten, the metal of many transition-edge sensors, the self-interstitial takes this form along $\langle 111\rangle$ [25]. The Frenkel–Kontorova chain of Appendix C.20 describes it: atoms of mass M_κ at spacing a_{ch}, springs K_{ch}, and the potential $V_{\mathrm{s}}\sin^2(\pi z/a_{\mathrm{ch}})$ of the surrounding crystal, Eq. (C.177). For a slowly varying displacement the chain obeys the sine-Gordon equation, and its kink has the profile Eq. (C.180), $u(z) = (2a_{\mathrm{ch}}/\pi)\arctan \mathrm{e}^{-(z-z_0)/d_{\mathrm{kink}}}$, with width $d_{\mathrm{kink}} = a_{\mathrm{ch}}^2[K_{\mathrm{ch}}/2\pi^2 V_{\mathrm{s}}]^{1/2}$ [25, Eq. (2)]. Its rest energy is $E_{\mathrm{kink}} = 4V_{\mathrm{s}}d_{\mathrm{kink}}/a_{\mathrm{ch}}$, Eq. (C.181) [25, Eq. (3)]. Since the sine-Gordon equation is Lorentz invariant with the chain's sound speed, the crowdion moves as a particle of mass

$$M_{\mathrm{cr}} = \frac{2}{\pi^2}\,\frac{a_{\mathrm{ch}}}{d_{\mathrm{kink}}}\,M_\kappa, \tag{19.49}$$

the kink mass Eq. (C.182), a standard result of the Frenkel–Kontorova model [26]. The continuum kink can sit anywhere; the discrete row makes it cross the Peierls–Nabarro barrier $E_{\mathrm{PN}} \simeq 16\pi^2 V_{\mathrm{s}}(d_{\mathrm{kink}}/a_{\mathrm{ch}})^2\,\mathrm{cosech}(\pi^2 d_{\mathrm{kink}}/a_{\mathrm{ch}})$ of Eq. (C.184) to move by one spacing.

For tungsten, $V_{\mathrm{s}} \approx 1\,\mathrm{eV}$ and $K_{\mathrm{ch}}a_{\mathrm{ch}}^2 = 50$ to $100\,\mathrm{eV}$ are typical values [25]. They give $d_{\mathrm{kink}} = 1.59$ to $2.25\,a_{\mathrm{ch}}$, $E_{\mathrm{kink}} = 6.4$ to $9.0\,\mathrm{eV}$, $M_{\mathrm{cr}} = 0.127$ to $0.090\,M_\kappa$ and a barrier of 1.2×10^{-4} to $3.6\times10^{-7}\,\mathrm{eV}$, with $a_{\mathrm{ch}} = \frac{\sqrt{3}}{2}\times 3.165\,\text{Å} = 2.741\,\text{Å}$. Atomistic calculations put the barrier near a meV instead [25], because the real substrate potential of tungsten is not a single sine. A particle of mass M_{cr} in the periodic potential $\frac{1}{2}E_{\mathrm{PN}}[1-\cos(2\pi z_0/a_{\mathrm{ch}})]$ has a ground band whose width follows from Mathieu's equation. The free-recoil energy $\pi^2/2M_{\mathrm{cr}}a_{\mathrm{ch}}^2$ is 0.12 meV for the narrower kink, equal to its barrier within 3 %, and 0.17 meV for the wider one, and the band is 90 to 166 µeV, 1.0 to 1.9 K. With the atomistic barrier of 1 meV and the same masses it narrows to 13 to 34 µeV, 0.15 to 0.39 K. Either way the band is wider than the temperature of a detector, so a crowdion is a candidate for coherent motion at millikelvin, a defecton of one dimension.

Limit Check 19.7. Crowdion profile and kink mass

The published crowdion is the sine-Gordon kink, $u(z) = (2a_{\mathrm{ch}}/\pi)\arctan \mathrm{e}^{-(z-z_0)/d_{\mathrm{kink}}}$ with $E_{\mathrm{kink}} = (2a_{\mathrm{ch}}/\pi)(2V_{\mathrm{s}}K_{\mathrm{ch}})^{1/2}$ [25, Eqs. (2), (3)], of mass Eq. (19.49) and barrier Eq. (C.184).

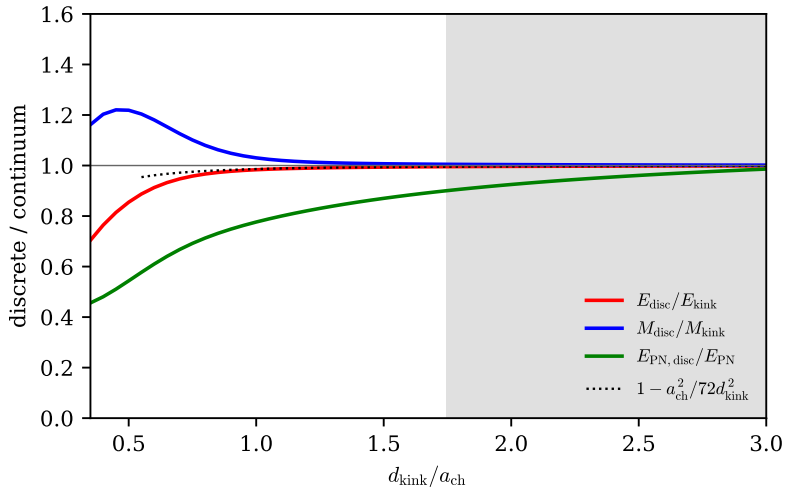


Figure 19.13: Discrete Frenkel–Kontorova crowdion against the continuum kink, as ratios, against the kink width in units of the row spacing. The discrete chain of Eq. (C.177) is relaxed numerically with the kink centered between two atoms (the minimum) and on an atom (the saddle). Red: rest energy, against E_{kink}. Blue: translation mass $M_\kappa \sum_n (u_{n+1} - u_n)^2/a_{\mathrm{ch}}^2$, the lattice form of $M_\kappa \sum_n (\partial u_n/\partial z_0)^2$, against Eq. (19.49). Green: saddle minus minimum, against E_{PN} of Eq. (C.184). Dotted: the leading correction to the energy, $1 - a_{\mathrm{ch}}^2/72d_{\mathrm{kink}}^2$. Gray: all three within 10 %, $d_{\mathrm{kink}} \geq 1.74\,a_{\mathrm{ch}}$, set by the barrier.

Well below one row spacing the kink is no longer a smooth profile: the barrier falls to half the continuum value near $0.4\,a_{\mathrm{ch}}$, and the mass and energy deviate by tens of percent.

The energy and the mass of the kink hold far beyond the barrier. Replacing the spring energy $\frac{1}{2}K_{\mathrm{ch}} \sum (u_{n+1} - u_n)^2$ by its continuum form drops the next term of $4\sin^2(ka_{\mathrm{ch}}/2) = k^2a_{\mathrm{ch}}^2 - k^4a_{\mathrm{ch}}^4/12$, which is $-(K_{\mathrm{ch}}a_{\mathrm{ch}}^3/24) \int \mathrm{d}z\, u''^2$. On the kink, $\int \mathrm{d}z\, u''^2 = 2a_{\mathrm{ch}}^2/3\pi^2 d_{\mathrm{kink}}^3$, and to first order the energy changes by $-K_{\mathrm{ch}}a_{\mathrm{ch}}^5/36\pi^2 d_{\mathrm{kink}}^3 = -E_{\mathrm{kink}}\, a_{\mathrm{ch}}^2/72d_{\mathrm{kink}}^2$, 1.4 % at $d_{\mathrm{kink}} = a_{\mathrm{ch}}$. The discrete energy follows this law within 2×10^{-4} above $1.5\,a_{\mathrm{ch}}$, and the mass exceeds Eq. (19.49) by 3 % at one spacing and by less beyond. The barrier is the weak point, 22 % below Eq. (C.184) at $d_{\mathrm{kink}} = a_{\mathrm{ch}}$ and 7.5 % below at $2\,a_{\mathrm{ch}}$, because it is exponentially small and the relaxation of the discrete chain changes it at the same order. For tungsten the model sits inside the shaded region for the stiffer chain and just below its edge for the softer one, so the mass is reliable and the barrier is not, as the atomistic meV already shows.

The conduction electrons, the phonons and the strain field of the parent vacancy decide whether the band survives in a real metal. Bulk tungsten is a normal metal above about 15 mK (Appendix E.22), so its conduction electrons dress the hopping by a power of the ohmic coupling α_{ohm} and, for $\alpha_{\mathrm{ohm}} < 1/2$, damp the coherent motion above a temperature of order $\Delta_{\mathrm{r}}/\pi\alpha_{\mathrm{ohm}}$, with Δ_{r} the renormalized tunneling amplitude of Sect. 16.8.3. The phonons that distinguish neighboring positions multiply the hopping by $\mathrm{e}^{-S_{\mathrm{pol}}(T)}$ of Eq. (E.107), which has not been estimated for a crowdion. The strain field of the vacancy left behind confines the crowdion within R_0 of it: for an elastic interaction constant of about 1 eV, an assumption, and a band of 13 to 166 µeV, $R_0 = 12$ to 22 spacings. Inside R_0 the crowdion slides down the gradient toward its vacancy, and its recombination time is a drift time, peaked, not the $t^{-3/2}$ tail of a free random walk. Appendix E.22 specifies the calculation that decides:

the Peierls profile, the phonon coupling and the interaction with the parent vacancy, from density-functional theory in a tungsten cell.

19.6.6 The low-energy excess

Cryogenic detectors of many materials and designs record excess events whose spectra rise toward threshold, from tens of eV to a few keV [27]. In the measurement of Anthony-Petersen and collaborators, two silicon crystals of $1 \times 1 \times 0.1\,\mathrm{cm}^3$, with identical sensors made on one wafer, were mounted in two ways: one glued to its copper holder with varnish, the other hung from aluminum wire bonds [23]. In energy absorbed by the sensor, the glued crystal shows a rate between 3 and 38 eV more than two orders of magnitude above the suspended one. The rates in the 3–38 and 38–85 eV bins fall with time constants of 6 to 10 days after cooldown, while the rate above 85 eV, from muons and radioactivity, stays constant. A single exponential does not describe the lowest bin of the glued crystal, with $\chi^2 = 4192$ for 68 degrees of freedom [23, Sect. II B]. In germanium detectors without a charge signal, the shape of the excess stays fixed while its rate falls slowly, the rate jumps after the detectors are warmed above 10 K, and tests with adjustable stress on the crystal and different holders show no significant effect. One leading hypothesis there is the cracking of the epoxy that glues the thermistor to the crystal [27, Sect. 2.1.2].

Energy conservation already bounds what any defect source can supply, and we can use it before choosing a model. A source fed by particle events releases at most the energy those events stored, by Eq. (19.44): over any interval it releases the sum of $E_{\mathrm{st},k}(t_1) - E_{\mathrm{st},k}(t_2)$ over the events k that seeded it. A source of elastic energy relaxing since cooldown releases at most the decrease of its relaxable elastic energy U_{rel}. Point defects grown into the crystal or frozen in at cooldown release at most the decrease of their stored energy $E_{\mathrm{st},0}$, by the same sum rule. The energy recorded in excess events over $[t_1, t_2]$ obeys

$$\int_{t_1}^{t_2} \mathrm{d}t \int \mathrm{d}E\; E\, \frac{\mathrm{d}\mathcal{R}_{\mathrm{LEE}}}{\mathrm{d}E\,\mathrm{d}t} \le f_{\mathrm{ph}} \Big\{ \big[U_{\mathrm{rel}}(t_1) - U_{\mathrm{rel}}(t_2)\big] + \big[E_{\mathrm{st},0}(t_1) - E_{\mathrm{st},0}(t_2)\big] + \sum_k \big[E_{\mathrm{st},k}(t_1) - E_{\mathrm{st},k}(t_2)\big] \Big\}, \quad (19.50)$$

with $f_{\mathrm{ph}} \le 1$ the largest collected fraction. Each event obeys the same bound on its own. Three consequences follow without any model: the event-seeded term is small, a single release must be large, and two crystals that differ in mounting differ in their elastic energy. First, electron recoils below 0.14 MeV cannot displace atoms (Sect. 19.5.3), and a 10 eV electron gives a silicon nucleus at most $4m_eE/M_\kappa = 0.78$ meV. Nuclear recoils store at most 5.5 % of their energy in silicon and 6.4 % in germanium at 95 % confidence (Table 19.1). These bounds hold if Luke phonons are collected no worse than primary phonons and the other conditions of Sect. 19.5.3 are met; on the branch $\mathfrak{r}_{\mathrm{coll}} < 1$ the silicon bound loosens to 0.081 at $\mathfrak{r}_{\mathrm{coll}} = 0.9$. Either way, the stored energy of the recoils a detector has seen is a small fraction of the energy it has already recorded from them. Second, at $f_{\mathrm{ph}} = 0.25$ an event of 3 eV needs 12 eV released at once, and one of 38 eV needs 152 eV, the energy of 22 close Frenkel pairs or of a pocket of about 1500 atoms. Third, two crystals cut from one wafer, with the same exposure to particles and the same grown-in defects, differ in mounting by U_{rel}, provided that mounting changes neither f_{ph} nor the rates of stress-assisted defect reactions. The factor of more than a hundred between them is then a statement about elastic energy, not about defects from radiation or growth. The excess events are therefore collective releases of energy stored at cooldown, and the open question is how they happen.

A collective-release model rests on three hypotheses: a reservoir of relaxable elastic energy, relaxation rates spread uniformly in $\ln \Gamma$, and a fixed law for the energies of single events.

The source is a reservoir U_{rel} of relaxable elastic energy, stored at cooldown in the crystal, the glue and the interfaces of the holder. It relaxes by local events, thermally activated or tunneling, whose rates are spread uniformly in $\ln \Gamma$ between $\Gamma_{\min}$ and $\Gamma_{\max}$, as the barriers of a glass are (Sect. 16.8.5). The energies of single events follow a law $p_{\mathrm{LEE}}(E)$ that does not change in time. The second hypothesis fixes the time law by the argument of Eq. (16.79): the fraction relaxing per unit time is $1/[t \ln(\Gamma_{\max}/\Gamma_{\min})]$ for $\Gamma_{\max}^{-1} \ll t \ll \Gamma_{\min}^{-1}$, so each factor e in time releases $U_{\mathrm{rel}}/\ln(\Gamma_{\max}/\Gamma_{\min})$. A fastest rate of order one per day bends the $1/t$ law at early times into the sum of exponentials that the data prefer. With $\langle E \rangle$ the mean of p_{LEE}, the three together give

$$\frac{\mathrm{d}\mathcal{R}_{\mathrm{LEE}}}{\mathrm{d}E\,\mathrm{d}t} = \frac{f_{\mathrm{ph}}\,U_{\mathrm{rel}}}{\ln(\Gamma_{\max}/\Gamma_{\min})\,\langle E \rangle}\,\frac{p_{\mathrm{LEE}}(E)}{t}. \tag{19.51}$$

The size law comes from triggering. If the phonons of one release trigger on average b_{br} further releases, as in Sect. 16.8.6, and the number triggered is Poisson, the total number N_{casc} in one cascade follows the Borel distribution

$$P(N_{\mathrm{casc}}) = \frac{\mathrm{e}^{-b_{\mathrm{br}} N_{\mathrm{casc}}}\,(b_{\mathrm{br}} N_{\mathrm{casc}})^{N_{\mathrm{casc}}-1}}{N_{\mathrm{casc}}!}, \tag{19.52}$$

with mean $1/(1-b_{\mathrm{br}})$. By Stirling's formula it falls as $N_{\mathrm{casc}}^{-3/2}\mathrm{e}^{-N_{\mathrm{casc}}(b_{\mathrm{br}}-1-\ln b_{\mathrm{br}})}$, a power law $N_{\mathrm{casc}}^{-3/2}$ cut off at $N_{\mathrm{casc}} \sim 2/(1-b_{\mathrm{br}})^2$ near the critical value $b_{\mathrm{br}} = 1$. A spectrum rising toward threshold as a power with an exponential cutoff is what such a cascade of elementary releases produces.

Suppose nothing about p_{LEE} is known except the two moments that Eq. (19.44) fixes for a defect source, the number of releases and the energy released. Then the distribution of largest entropy on $E \geq 0$ is the exponential $\mathrm{e}^{-E/E_{\mathrm{slope}}}/E_{\mathrm{slope}}$, with $E_{\mathrm{slope}} = (E_{\mathrm{st}} - E_{\mathrm{st}}(t))/\langle N_{\mathrm{hop}} \rangle$ the mean release per hop, 4.26 eV in Worked example 19.6. It describes missing information; it does not derive the observed exponential spectra.

The model predicts how the excess scales, not its rate. The rate is proportional to U_{rel}, which for a layer under thermal-mismatch stress σ_{str} is of order $\sigma_{\mathrm{str}}^2 \mathcal{V}/Y_{\mathrm{mod}}$ of the relaxing component, with Y_{mod} its Young's modulus. A glue layer on a clamped crystal dominates, and the factor of a hundred between glued and suspended crystals is a ratio of U_{rel} that a thermo-elastic calculation can test. The clock restarts when the relaxing component is warmed above the temperature at which it creeps, as in the jumps after warming above 10 K. A stress applied to the bulk of a crystal that does not creep adds little U_{rel}, which agrees with the null results under adjustable stress if the source sits in the glue of the thermistor. The shape of the spectrum and its time dependence factorize, so the shape does not change as the rate falls, as observed in germanium. A rate needs two inputs that do not exist: U_{rel} with its spread of rates $[\Gamma_{\min}, \Gamma_{\max}]$ for a given holder and glue, and the branching ratio b_{br}, which fixes the size law. Appendix E.25 specifies both. The first comes from a finite-element calculation of the cooldown stress with measured elastic and expansion data, with the relaxable fraction from low-temperature mechanical data of the glue. The second comes from the catalogue of release paths and the transport of the phonon pulse of one release. A fit of Eq. (19.51) with these inputs to a glued and an unglued crystal would then have no free parameter.

19.6.7 Open problems

The release of close Frenkel pairs in silicon and germanium, and of split interstitials in aluminum, at millikelvin (Sect. 19.6.1) needs one calculation per path: the tunneling exponent along the computed minimum-energy path and $E_{\mathrm{reorg}}/\varepsilon^{\mathrm{hop}}$, both to about 5 % (Appendix E.22).

In ionic crystals the electronic route to stored energy is prompt. A self-trapped exciton can turn into a close pair of an F center and an interstitial within picoseconds. Part of the energy of an electron recoil is then stored before the partition time, and the electron-recoil calibration of such crystals is not free of a heat deficit. Per unit deposit the stored energy is the number of excitons per unit energy times the conversion probability, the survival probability of the close pair and its configurational energy. The conversion probability at millikelvin and the survival of the close pair against tunneling recombination are not known.

Carriers of later events can drive defect reactions without heat. A capture or recombination at a defect puts energy into its local modes, and if the reaction coordinate is coupled linearly with Huang–Rhys factor $S_{\mathrm{HR,rxn}}$ and energy ω_{rxn}, the number of quanta it receives at $T = 0$ is Poisson (Eq. (5.60)). The probability that one capture drives the reaction over a barrier V_{b} is the tail

$$p_{\mathrm{rxn}} = \sum_{n \geq V_{\mathrm{b}}/\omega_{\mathrm{rxn}}} \mathrm{e}^{-S_{\mathrm{HR,rxn}}} \, \frac{S_{\mathrm{HR,rxn}}^{n}}{n!}, \tag{19.53}$$

and the stimulated rate is the capture rate times p_{rxn}. It enters Eq. (19.40) as a time-dependent rate, so Eq. (19.44) still holds. The Huang–Rhys factor, the mode energy and the barrier of real defects, with the change of charge state, are not computed.

The two-stage factorization of Eq. (18.22) lets the detector stage see a cascade only through N_{eh} and the quantities $\mathfrak{m}$. Suppose the detector response is built from one class of seeds, photons, and applied to another, dark matter. Its error is then at most the total-variation distance between the two classes' laws of the full microstate at fixed $(N_{\mathrm{eh}}, \mathfrak{m})$, averaged with the probability of $(N_{\mathrm{eh}}, \mathfrak{m})$, and it vanishes exactly when $(N_{\mathrm{eh}}, \mathfrak{m})$ is a sufficient statistic for the seed class. That distance, for the features left out of $\mathfrak{m}$ such as the shape of the charge cloud beyond its centroid and spread, has not been computed; Appendix E.25 specifies the test.

Summary

The nuclear response of Chap. 11 hands a recoil over to dynamics on the Born–Oppenheimer surface through one exact step: the Wigner function of the lattice with the momentum of the struck nucleus raised by $\boldsymbol{q}$ (Sect. 19.1). Propagated classically, this ensemble reproduces the impulse approximation with its exact mean and width and misses the third cumulant. In silicon the dynamics passes from quantum to classical at 0.41 eV, leaves the harmonic regime near 0.54 eV, displaces atoms from 12.5 to 36 eV depending on direction, and excites electrons nonadiabatically from about 0.9 keV. Below the displacement threshold a recoil deposits all of its energy as heat, which covers every electron-following recoil from dark matter lighter than about 150 MeV. At the partition time, 10 to 100 ps after the kick, the energy splits into a lattice share, a configurational stored energy and the energy given to the electrons, and after the cascade into pairs, heat and stored energy, Eq. (19.5).

The charge channel of a slow recoil has a soft threshold (Sect. 19.2). A binary collision with a valence electron can bridge the gap only above $E_{R,\mathrm{th}} = (M_\kappa/m_e)E_{\mathrm{g}}^2/16E_{\mathrm{F}}$, 352 eV in silicon and 399 eV in germanium, yet the measured silicon yield shows no threshold down to 100 eV. Two channels act below it, the tail of the valence momentum distribution and level crossings in close collisions, and the second is nonperturbative, with a Born parameter of 15. The Migdal part of the measured yield does not depend on the recoil energy in the soft limit and is at most 0.4 % of the silicon yield at 100 eV. A measured yield already contains that Migdal ionization, so adding a computed Migdal rate to a rate converted with it counts the Migdal electrons twice. Lindhard quenching is the limit with no stored energy, continuum stopping to zero speed and proportional conversion to pairs (Sect. 19.3). Its keV yield is

sensitive to how the softest collisions of the slowest secondaries are modeled: continuing the cross-section at its local slope instead of Lindhard's power law raises it by 13 % at 1 keV, and its $\epsilon_L^{1/6}$ law at small energies follows from that power law. The silicon data lie 8 to 38 % below the formula between 0.1 and 3.9 keV. Bethe stopping, with the mean excitation energy from the loss function (161 eV from the measured silicon spectrum, against the adopted 173 eV), holds within 10 % only above $1.84\,v_\mathrm{F}$, while silicon recoils move at 0.013 to $0.13\,v_\mathrm{F}$.

The stored energy of a recoil depends on the end points of its trajectories alone (Sect. 19.4). A close Frenkel pair in silicon keeps 6.8 to 7.5 eV, and the displacement count with a screened cross-section is 0.70 times the Kinchin–Pease value, against the 0.8 of the standard formula. Under conditions on phonon collection, other missing energy and the pair energy of the reconstruction, the measured deficits of nuclear recoils bound the stored share at 95 % confidence by 0.055 in silicon, 0.049 with the measured pair energy, and by 0.064 in germanium. The displacement threshold of 19.7 eV inferred for germanium is conditional on the defect model of the simulation that produced it.

One balance law per event counts every share of the deposit once: pairs counted from the Fermi level of the electrodes, phonons, configurational stored energy, escaped quanta and the energy absorbed at the surfaces (Sect. 19.5). Unit heat quenching requires that, within the read window, every pair has recombined or been collected with its gap energy returned as phonons, and that no energy is stored. With no gap energy returned, a silicon recoil of 100 eV gives 41 % more heat than an electron recoil. With bias, the calibration identity relates the phonon-yield ratio to the heat quenching through the collection ratio of Luke phonons and the pair energy that the reconstruction adopts. The silicon reconstruction uses 3.82 eV per pair against the measured 3.75 eV, which alone lowers the ratio by 0.006. The silicon datum $Y_\mathrm{ph} = 0.952$ is then reproduced either by a stored fraction of 0.042 or, with no stored energy, by a collection ratio of 1.128. The mean-field transport estimate of Sect. 20.5.3 puts the collection ratio between 0.81 and 1.02 with energy conserved; values up to 1.5 need a selective sink that the estimate does not identify in a detector. The transport estimate therefore does not reach a collection ratio of 1.128, but only computed kernels for real detectors can exclude it. A second bias decides between the two: at 40 V the two readings differ by 0.028.

The stored energy is released after the partition time by tunneling (Sect. 19.6). The release-path sum rule Eq. (19.44) makes the energy released by all hops equal to the stored energy minus what is still stored, for any rates, under five conditions on how the energy is counted. Whether close Frenkel pairs in silicon and split interstitials in aluminum release at all at millikelvin is conditional: frozen in the weak-coupling regime, and in the strong-coupling regime only if the true tunneling exponent lies 14 % or 6 % below the two-parabola estimate. A single release gives a sensor that absorbs a quarter of the phonons less than 2 eV, and two-level systems give quanta of half a meV, so neither produces the excess events of tens of eV. A defecton diffuses as T^{-9}, within 10 % already from 0.28 times its band width, and a crowdion is a Frenkel–Kontorova kink with a tenth of the mass of a tungsten atom, whose band in tungsten, 0.01 to 0.2 meV, is wider than the temperature of a detector. Energy conservation bounds the low-energy excess by the stored energy of the events that seeded it and of the grown-in defects, plus the decrease of relaxable elastic energy, and the factor of a hundred between glued and suspended crystals places it in the elastic energy. The collective-release model predicts how the excess scales, with that energy, as $1/t$ and with a fixed spectral shape, but not its rate.

Table 19.2 collects the truncations and inputs of the chapter with their sizes. The charge, phonons and quasiparticles that the balance law leaves at the end of the read window are not yet a signal; the next chapter follows them to the sensors.

Table 19.2: Nuisances of the low-energy nuclear-recoil response. Type in the classes of Sect. 8.8, normalization, tilt, threshold and new support, with "delayed" for an item that moves energy in time rather than changing the prompt yield. Basis of the size: "computed" (next order or exact result computed), "model" (computed in a model), "est." (parametric estimate), "input spread" (spread or limitation of a measured or computed input), "unsized" (no size known). The last column gives the section where the item is treated.

Truncation or input	Type	Size	Basis	Section
Classical propagation of the kicked Wigner ensemble	tilt	misses the third cumulant of the impulse response (share $\propto x_R^{-1/2}$) and quantum effects after the kick; exact for a quadratic surface	computed (structure)	Sect. 19.1
Regime boundaries of the recoil dynamics	threshold	silicon 0.41 and 0.54 eV, 12.5–36 eV, 0.87–4.6 keV, each at a criterion of 0.1 with one-site estimates	est.	Sect. 19.1
Machine-learned surface along the recoil path	threshold; normalization	thresholds must match to the 0.7 eV width of the step; analytical potentials miss by 4–6 eV; formation energies within 7 % give a few tenths of an eV per defect	input spread	Sects. 19.1 and 19.4
Charge below the binary threshold	normalization; threshold	measured silicon yield 0.10 at 100 eV with no threshold; the close-collision kernel is not computed; adiabatic functionals place charge-transfer crossings too early (sign known)	unsized	Sect. 19.2
Migdal part of a measured yield	normalization	2.3–4.4 $\times 10^{-4}$ in Y_q (one pair per excitation to the one-sided split), at most 0.4 % of the silicon yield at 100 eV; counted once	computed	Sect. 19.2
Lindhard quenching below a few keV	normalization	data 8–38 % below the formula at 0.1–3.9 keV; formula within 3–5 % of its own equation	input spread; computed	Sect. 19.3
Soft collisions of the slowest secondaries	normalization	a sensitivity, not an uncertainty: +13 % at 1 keV (+10 % at 4 keV) for the local slope 0.363 against $x_L^{1/3}$; −11 % and +40 % at 1 keV for exponents 0.30 and 0.40	model	Sect. 19.3
Prefactor of the displacement count	normalization	0.70 (screened cross-section) against 0.8 (standard formula) and 1 (hard spheres); formula 11–14 % above the screened count for $T_\mathrm{d} \geq 10E_\mathrm{d}$	computed	Sect. 19.4
Stored share of a nuclear recoil	normalization (heat)	at most 0.055 (silicon; 0.049 under (a)–(d) with the measured pair energy) and 0.064 (germanium) at 95 % confidence under conditions (a)–(e) of Sect. 19.5.3; 0.056 and 0.065 for either sign of the zero-point change; silicon 0.081 at $\mathfrak{r}_\mathrm{coll} = 0.9$; germanium $E_\mathrm{d} = 19.7$ eV conditional on the defect model	input spread	Sects. 19.4 and 19.5
Gap energy unreturned in the read window	normalization (heat)	$Y_\mathrm{ph}(0) = 1.407$ at 100 eV with none returned; within 1 % of one needs at most 3.5–4.1 % unreturned; power laws in time if two-body losses act	computed; model	Sect. 19.5
Collection ratio of Luke phonons	normalization	1.128 alone reproduces the silicon $Y_\mathrm{ph} = 0.952$ (band 1.099–1.152; 1.134 with Lindhard's yield); not reached by the mean-field transport estimate of Sect. 20.5.3 (0.81–1.02 with energy conserved; up to 1.5 only for a selective sink), not excluded until computed detector kernels exist; the two-bias measurement decides; the inferred stored fraction rises from 0.042 at $\mathfrak{r}_\mathrm{coll} = 1$ to 0.058 at 0.95 and 0.075 at 0.9	computed	Sect. 19.5
Pair energy adopted by the reconstruction	normalization	3.82 eV against the measured 3.75 eV: −0.0064 in Y_ph at 4 V, an eighth of the silicon deficit, growing with bias (stored energy alone reads 0.947 at 40 V); the bound holds while $\varepsilon_\mathrm{eh} \leq \varepsilon_\mathrm{cal}$	computed	Sect. 19.5
Collection offset from recombination phonons	normalization	$\rho_g - 1 = 0.21$ alone gives the silicon deficit beyond the pair-energy offset; 0.02 in ρ_g moves it by 0.0037; detector transport not computed	est.; unsized	Sect. 19.5
Collection ratio constant between two biases	normalization	residual $(\mathrm{d}\mathfrak{r}_\mathrm{coll}/\mathrm{d}V_\mathrm{bias})\Delta V_\mathrm{bias}$, not computed; the two readings of the silicon datum differ by 0.028 at 40 V	unsized	Sect. 19.5
Zero-point change of the defect basins	normalization (heat)	below 0.1 eV per point defect, of either sign, in the phonons but not absorbable; a negative change widens the stored-energy	est.	Sects. 19.1, 19.5 and 19.6

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Chapter 20

Charge, phonon and quasiparticle readout

A 100 eV electron recoil in silicon ends, within a picosecond, as about 27 electron–hole pairs that hold 31 eV of gap energy, and some 69 eV of phonons near the point of the event. A nuclear recoil of 1 keV leaves about 34 pairs, close to 0.9 keV of phonons and up to about 55 eV stored in defects (Sect. 19.5). A detector records only the charge induced on its electrodes and the energy its films turn into quasiparticles. The pairs must drift across a centimeter of crystal to the electrodes, and a carrier captured on the way contributes only part of its charge. The phonons must reach a superconducting film on the surface, cross into it and break Cooper pairs, and on the way they decay, scatter off isotopes and bounce off bare surfaces. Each step multiplies the signal by an efficiency and adds a fluctuation of its own.

These efficiencies decide whether the energy balance of Chap. 19 can be read at all. The silicon calibration of Sect. 19.5 measures one combination of stored energy and collection efficiencies, and Luke phonons collected about 13 % better than the primary phonons of an electron recoil explain the datum with no stored energy at all. Whether Luke phonons are in fact collected better, and whether the phonons of a nuclear recoil reach the sensors in a different state from those of an electron recoil, are transport questions. This chapter asks what a charge or a phonon readout measures, given transport in the crystal and across the interfaces, and whether the measurement can still separate nuclear from electron recoils. Transport simulations of whole detectors combine all the steps (Appendix E.12), and how closely they reproduce the relations derived here has not been tested. What the readout measures feeds the exclusion limits of the next chapter and the readout column of its guide to next targets (Fig. 20.1).

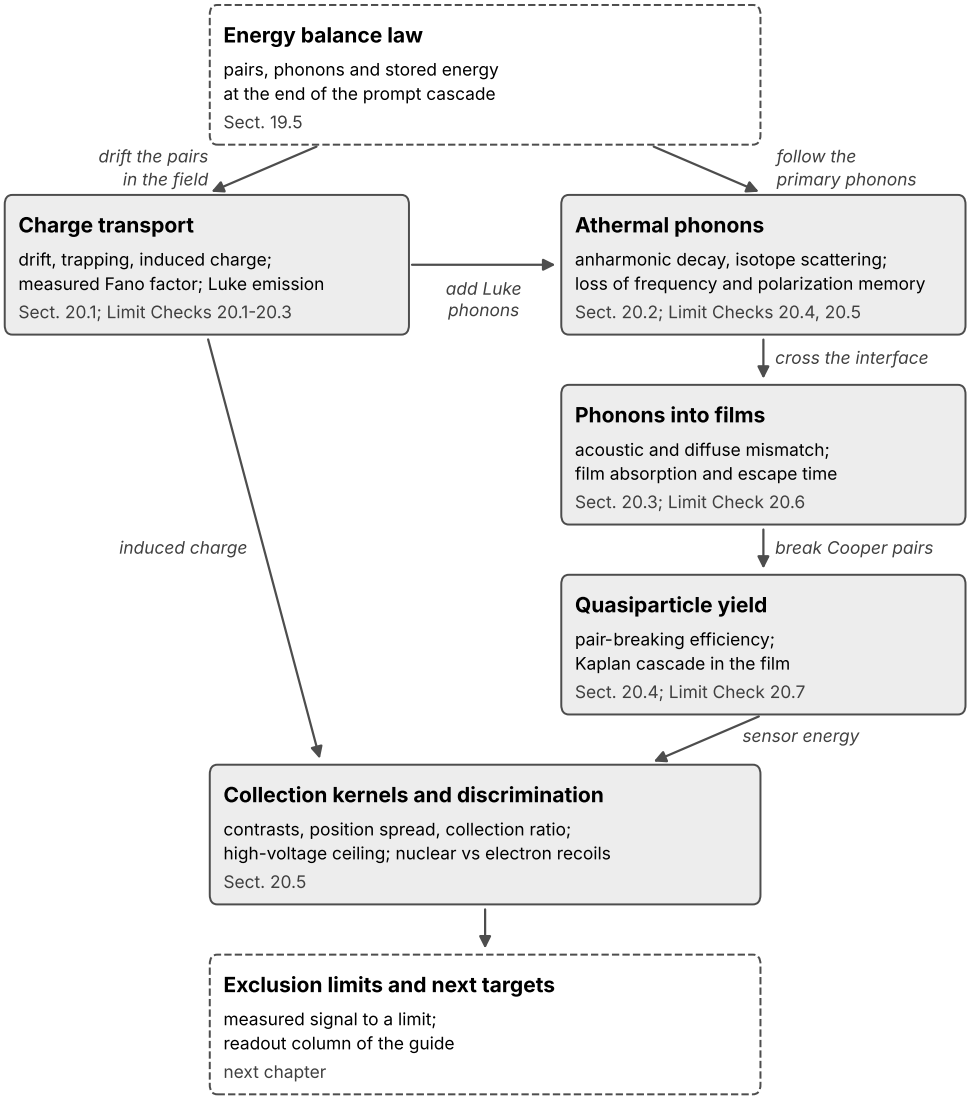


Figure 20.1: Charge, phonon and quasiparticle readout, from the reservoirs of the energy balance law at the end of the prompt cascade (Sect. 19.5) to the signals a detector records. Left: drift, trapping and the charge induced on the electrodes, with the measured Fano factor and the Luke phonons of the drifting carriers (Sect. 20.1; Limit Checks 20.1 to 20.3). Right: the primary phonons and the Luke phonons decay and scatter in the substrate (Sect. 20.2; Limit Checks 20.4 and 20.5), cross into the sensor films (Sect. 20.3; Limit Check 20.6) and break Cooper pairs there (Sect. 20.4; Limit Check 20.7). Both branches meet in the collection kernels, which decide whether phonon or charge readout separates nuclear from electron recoils (Sect. 20.5).

20.1 Charge transport and the measured charge

The charge amplifier of a semiconductor detector integrates the current that flows into an electrode while the carriers move, and a carrier captured halfway across the crystal leaves half a charge in the integral and does half its share of Luke work. As a result, the charge we read is not the pair number N_{eh} of Sect. 18.2 but N_{eh} times a gain that depends on where the event happened and how far each carrier got, and the fluctuations of that gain add to those of the cascade. At millikelvin temperatures the carriers drift in fields of a few V/cm, trapped charge does not escape thermally, and every drifting carrier radiates phonons. We will see that trapping lowers the read charge and the Luke heat by the same factor, but spreads the measured charge in a way that no pair statistics removes.

20.1.1 Induced charge and Ramo's theorem

Consider electrodes k held at potentials V_k around a crystal with static dielectric constant ε_0 (a pure number in the Heaviside–Lorentz units of the book, Sect. 5.5), and a point charge q inside at $\boldsymbol{r}$. The charge on electrode s is the sum of what the bias puts there and what q induces, and in a linear medium the two add. The induced part follows from Green's reciprocity theorem. Compare two configurations of the same conductors. In the first, q sits at $\boldsymbol{r}$ and every electrode is grounded, and electrode k carries the induced charge Q_k^{ind}. In the second, there is no charge inside, electrode s is held at unit potential and the others are grounded; the potential inside is then the *weighting potential* $\varphi_s(\boldsymbol{r})$, the solution of $\nabla\cdot[\varepsilon_0(\boldsymbol{r})\nabla\varphi_s]=0$ with $\varphi_s=1$ on electrode s and 0 on the others. Reciprocity states that the sum of charges times potentials of one configuration, taken with the potentials of the other, is the same either way. The first configuration has charges Q_k^{ind} on the electrodes and q at $\boldsymbol{r}$; in the second the electrode potentials are δ_{ks} and the potential at $\boldsymbol{r}$ is $\varphi_s(\boldsymbol{r})$. The second has charges only on the electrodes, and in the first all of them are grounded. Hence $Q_s^{\mathrm{ind}}+q\varphi_s(\boldsymbol{r})=0$. When q moves from $\boldsymbol{r}_1$ to $\boldsymbol{r}_2$, the external circuit must supply the change, and the charge delivered to electrode s is

$$\Delta Q_s=-q\,\big[\varphi_s(\boldsymbol{r}_2)-\varphi_s(\boldsymbol{r}_1)\big], \tag{20.1}$$

with the current $I_s=-q\,\boldsymbol{v}\cdot\nabla\varphi_s$ in its lead while the carrier moves at velocity $\boldsymbol{v}$. This is the Shockley–Ramo theorem [1, 2], [3, Eq. (2)]. The weighting potential is a property of the electrode geometry alone. It is not the potential the carrier moves in. The applied bias, other carriers and trapped charges change the field and the trajectory, but by linearity they only add their own contributions to Q_s, and the charge induced by q at a given point stays the same. Eq. (20.1) is therefore exact in the presence of space charge.

For two planar electrodes at $z=0$ and $z=d_{\mathrm{det}}$, the weighting potential of the electrode at $z=d_{\mathrm{det}}$ is $\varphi=z/d_{\mathrm{det}}$. Let it collect electrons, so that holes drift toward $z=0$. Let a pair be born at depth z, and let its electron drift a path s_e toward the anode and its hole a path s_h toward the cathode. By Eq. (20.1) the pair delivers the charge $e\,\mathfrak{q}$ to the anode, with

$$\mathfrak{q}=\frac{s_e+s_h}{d_{\mathrm{det}}},\qquad 0\le s_e\le d_{\mathrm{det}}-z,\quad 0\le s_h\le z. \tag{20.2}$$

A pair that is fully collected gives $\mathfrak{q}=1$ wherever it was born, since its two carriers together cross the whole gap. A pair whose electron is captured after a third of the way to the anode, from the midplane, gives $\frac{1}{2}+\frac{1}{6}=\frac{2}{3}$ if its hole arrives.

The same theorem gives the work of the bias. To hold electrode k at V_k while the charge it carries changes by ΔQ_k, its source does the work $V_k\Delta Q_k$. Summing Eq. (20.1) over the electrodes, the work done on a carrier that moves from $\boldsymbol{r}_1$ to $\boldsymbol{r}_2$ is $-q\sum_k V_k[\varphi_k(\boldsymbol{r}_2)-\varphi_k(\boldsymbol{r}_1)]$.

The sum $\varphi_{\rm bias}(\boldsymbol{r}) \equiv \sum_k V_k \varphi_k(\boldsymbol{r})$ is the potential the electrodes produce in the empty crystal, so that

$$W_{\rm bias} = \sum_{\rm carriers} q\,[\varphi_{\rm bias}(\boldsymbol{r}_{\rm start}) - \varphi_{\rm bias}(\boldsymbol{r}_{\rm end})]. \tag{20.3}$$

The bias work is the charge times the drop of the vacuum potential of the electrodes, whatever space charge is present. Between two planes at 0 and $V_{\rm bias}$, $\varphi_{\rm bias} = V_{\rm bias} z/d_{\rm det}$, and each pair contributes $eV_{\rm bias}\mathfrak{q}$: this is Eq. (19.28), $W_{\rm bias} = eV_{\rm bias}N_{\rm ind}$, with $N_{\rm ind} = \sum_i \mathfrak{q}_i$ the charge read on the anode in units of e. Appendix D.45 shows that the Luke heat is $W_{\rm bias}$ minus the change $\Delta E_{\rm es}$ of the electrostatic energy. For a single trapped pair that change is negligible. Two charges 10 µm apart in silicon have the Coulomb energy $\alpha/(\varepsilon_0 r) = 1.3 \times 10^{-5}$ eV, with the liquid-helium value $\varepsilon_0 = 11.4$ (Sect. 7.4), against the bias work $eV_{\rm bias}s/d_{\rm det} = 4$ meV done on a carrier over those 10 µm in a 1 cm crystal at 4 V.

With more than two electrodes the read charge and the bias work weight a trajectory differently. The charge read on electrode s follows φ_s and the work follows $\varphi_{\rm bias}$, and the two are proportional only where $\varphi_s \propto \varphi_{\rm bias}$, as for a pair of planar electrodes. Some detectors place interleaved electrodes on each face to tag events close to a surface. Near them the weighting potentials change over the electrode pitch, and a carrier trapped there adds to the read charge and to the bias work in proportions that differ from its share of the path. The reconstruction $E_R = E_\phi - E_Q eV_{\rm bias}/\varepsilon_{\rm eh}$ of Sect. 19.5 subtracts the Luke heat through the read charge E_Q, and it inherits that difference for carriers trapped near the faces. (CDMS writes E_Q with 3.82 eV per pair [4]; the subtraction does not depend on that choice.)

20.1.2 Trapping and the Hecht relation

A drifting carrier at millikelvin temperatures is lost when an ionized or neutral impurity captures it, with the cross-sections $\sigma_{{\rm cap},j}$ of Sect. 18.5. If capture events are independent, the probability of surviving a further path ds is $1 - {\rm d}s/\lambda_{\rm d}$, and the probability of surviving a path s is $\mathrm{e}^{-s/\lambda_{\rm d}}$. The drift length is $\lambda_{\rm d} = 1/\sum_j n_j^{\rm tr}\sigma_{{\rm cap},j}$ for trap densities $n_j^{\rm tr}$ and cross-sections taken per unit path; with a trapping time $\tau_{\rm tr}$ instead, $\lambda_{\rm d} = v_{\rm dr}\tau_{\rm tr}$ with the drift velocity $v_{\rm dr}$. Electrons and holes have separate drift lengths, $\lambda_{\rm d}^e$ and $\lambda_{\rm d}^h$. The mean of a path that is exponentially distributed but cut off at the distance L to the electrode is $\langle \min(s, L)\rangle = \lambda_{\rm d}(1 - \mathrm{e}^{-L/\lambda_{\rm d}})$.

The classic measurement creates the charge at one face, with an alpha source or a short-range photon, so that one carrier type crosses the whole crystal and the other is collected at once. By Eq. (20.2) the mean collected fraction is then the mean path over $d_{\rm det}$,

$$\frac{Q}{eN_{\rm eh}} = \frac{\lambda_{\rm d}}{d_{\rm det}}\big(1 - \mathrm{e}^{-d_{\rm det}/\lambda_{\rm d}}\big), \tag{20.4}$$

the Hecht relation [5], [3, Eqs. (1)–(4)]. It is linear in $\lambda_{\rm d}$ for $\lambda_{\rm d} \ll d_{\rm det}$, where the carriers stop after a mean path $\lambda_{\rm d}$, and approaches $1 - d_{\rm det}/2\lambda_{\rm d}$ for $\lambda_{\rm d} \gg d_{\rm det}$. Dark-matter events happen anywhere in the bulk, and both carriers then drift. Applying the cut-off mean path to each carrier, a pair at depth z induces on average

$$\bar{\mathfrak{q}}(z) = \frac{\lambda_{\rm d}^e}{d_{\rm det}}\Big(1 - \mathrm{e}^{-(d_{\rm det}-z)/\lambda_{\rm d}^e}\Big) + \frac{\lambda_{\rm d}^h}{d_{\rm det}}\Big(1 - \mathrm{e}^{-z/\lambda_{\rm d}^h}\Big), \tag{20.5}$$

and averaging over a uniform depth,

$$\langle\bar{\mathfrak{q}}\rangle = \sum_{c=e,h} \frac{\lambda_{\rm d}^c}{d_{\rm det}}\Big[1 - \frac{\lambda_{\rm d}^c}{d_{\rm det}}\big(1 - \mathrm{e}^{-d_{\rm det}/\lambda_{\rm d}^c}\big)\Big] \simeq 1 - \frac{d_{\rm det}}{6\lambda_{\rm d}^e} - \frac{d_{\rm det}}{6\lambda_{\rm d}^h}. \tag{20.6}$$

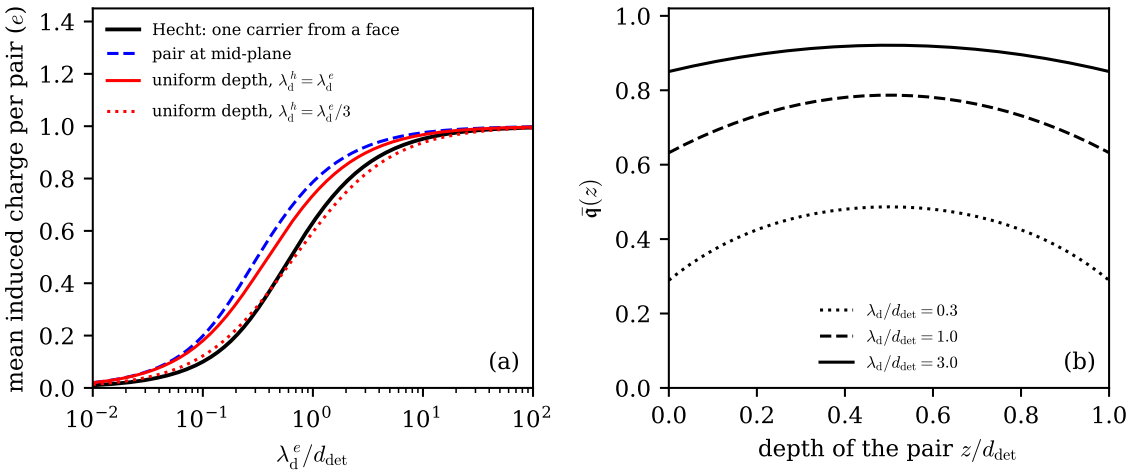


Figure 20.2: Charge collection with exponential trapping in a planar detector of thickness d_{det}. (a) Mean induced charge per pair, in units of e, against the electron drift length: the Hecht relation Eq. (20.4) for one carrier crossing from a face (black), a pair at the midplane with equal drift lengths (blue dashed), and the depth average Eq. (20.6) for uniform events with $\lambda_d^h = \lambda_d^e$ (red) and $\lambda_d^h = \lambda_d^e/3$ (red dotted, an illustrative ratio). (b) Mean induced charge of a pair against its depth, Eq. (20.5) with equal drift lengths 0.3, 1 and 3 d_{det}; the electron-collecting electrode is at $z = d_{det}$.

and at second order in d_{det}/λ_d each carrier adds $+d_{det}^2/24\lambda_d^2$, so the leading correction is accurate to a fraction $d_{det}/4\lambda_d$ of itself. Each carrier of a bulk event loses a third of what the carrier of a face event loses at the same drift length: for $\lambda_d \gg d_{det}$ the loss over a path L is $L^2/2\lambda_d d_{det}$, and the mean square path to an electrode is $d_{det}^2/3$. Fig. 20.2 compares the three cases. At $\lambda_d = d_{det}$ the face event collects 0.632 of its charge, a uniform bulk event 0.736 and an event at the midplane 0.787; at $\lambda_d = 10\,d_{det}$ the face and bulk values are 0.952 and 0.968. The depth dependence, Fig. 20.2(b), is what turns trapping into a spread of the measured charge (Sect. 20.1.4).

Hecht wrote the drift length as $\lambda_d = \mu_{mob}\tau_{tr}V_{bias}/d_{det}$, for a drift velocity proportional to the field, $v_{dr} = \mu_{mob}\mathcal{E}$ with mobility μ_{mob}, and a field-independent trapping time. Measuring Q against the bias and fitting Eq. (20.4) in this form is the standard way to obtain the mobility–lifetime product. Neither assumption enters the derivation above, which needs only exponential survival along the path. At millikelvin temperatures both fail. A carrier accelerates until it moves faster than sound and then sheds its energy into Luke phonons (Sect. 20.1.6); its speed stays within a few times the sound speed, and the largest momentum it reaches grows only as the cube root of the field [6, Eqs. (24)–(25)]. If the capture cross-section per unit path does not depend on the carrier speed, λ_d is independent of the bias and the collected fraction does not change with it. If capture is a rate at fixed trapping time, $\lambda_d = v_{dr}\tau_{tr}$ stops growing once the velocity saturates, and the collected fraction saturates below one.

Limit Check 20.1. Hecht collection

For charge created at one face and drifting in a uniform field [5], [3, Eqs. (1)–(4)]: $Q/eN_{eh} = (\lambda_d/d_{det})(1 - e^{-d_{det}/\lambda_d})$ with $\lambda_d = \mu_{mob}\tau_{tr}V_{bias}/d_{det}$.

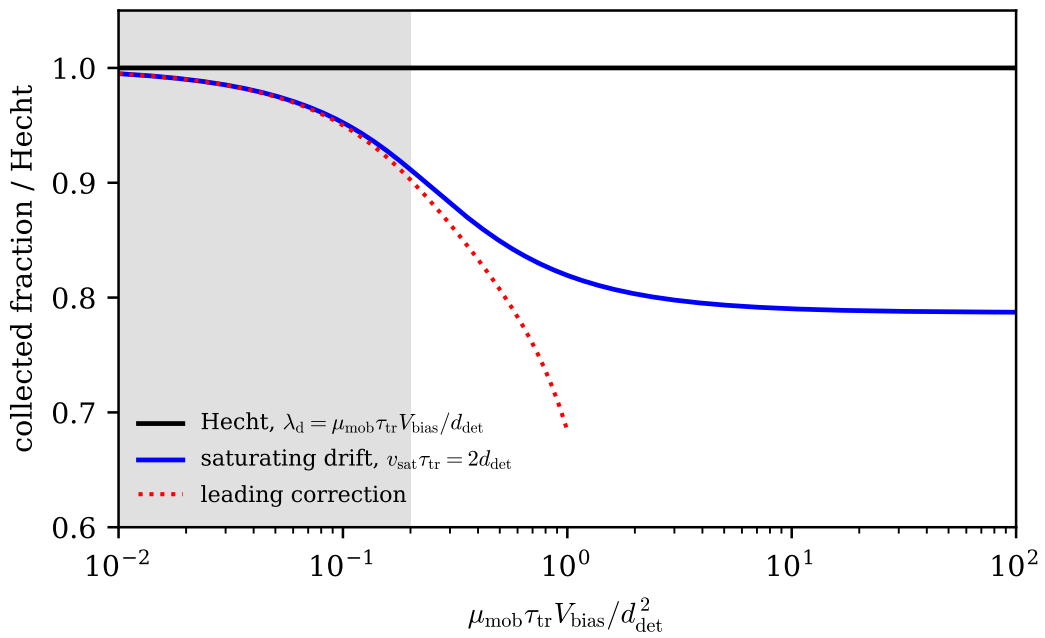


Figure 20.3: Collected fraction with a saturating drift velocity divided by the Hecht relation, against the bias in units of $d_{\mathrm{det}}^2/\mu_{\mathrm{mob}}\tau_{\mathrm{tr}}$. Blue: Eq. (20.4) with $\lambda_{\mathrm{d}} = v_{\mathrm{dr}}\tau_{\mathrm{tr}}$ and the interpolation $v_{\mathrm{dr}} = \mu_{\mathrm{mob}}\mathcal{E}/(1 + \mu_{\mathrm{mob}}\mathcal{E}/v_{\mathrm{sat}})$, at $v_{\mathrm{sat}}\tau_{\mathrm{tr}} = 2\,d_{\mathrm{det}}$ (an illustrative value). Red dotted: the leading correction, $\lambda_{\mathrm{d}} = \mu_{\mathrm{mob}}\tau_{\mathrm{tr}}\mathcal{E}(1 - \mu_{\mathrm{mob}}\mathcal{E}/v_{\mathrm{sat}})$. Gray: $\mu_{\mathrm{mob}}\mathcal{E} \le 0.1\,v_{\mathrm{sat}}$, where the deficit is at most 8.9 %. At high bias the collected fraction saturates at 0.787 of the charge instead of approaching one.

Once the drift velocity no longer follows the field, the collected charge stops rising with the bias, and a fit of the Hecht form returns a mobility–lifetime product that depends on the bias range used.

The deficit of the collected fraction is $\mu_{\mathrm{mob}}\mathcal{E}/v_{\mathrm{sat}}$ to first order, since the collection is linear in λ_{d} while $\lambda_{\mathrm{d}} \ll d_{\mathrm{det}}$. A survival law that is not exponential, as when the trap occupation changes during the drift, changes Eq. (20.4) itself. Drift, Luke emission and trapping together are computed by transport Monte Carlo (Appendix E.12).

20.1.3 Bias work with trapping

Trapping lowers the Luke heat with the read charge. Averaged over a uniform depth, Eq. (20.3) gives $W_{\mathrm{bias}} = eV_{\mathrm{bias}}N_{\mathrm{eh}}\langle\bar{\mathrm{q}}\rangle$, and for an electron recoil that deposits ω the Luke phonon energy is $g_{\mathrm{Luke}}\langle\bar{\mathrm{q}}\rangle\,\omega$, with the Luke gain $g_{\mathrm{Luke}} = eV_{\mathrm{bias}}/\varepsilon_{\mathrm{eh}}$ of Sect. 19.5. The published phonon energy counts every pair as fully collected.

Limit Check 20.2. Bias work and the Luke phonon energy

The phonon energy of n_Q pairs drifted across the bias V_{bias} is $E_\phi = E_R + n_Q eV_{\mathrm{bias}}$, with Luke part $n_Q eV_{\mathrm{bias}}$ [4, Eq. (1)].

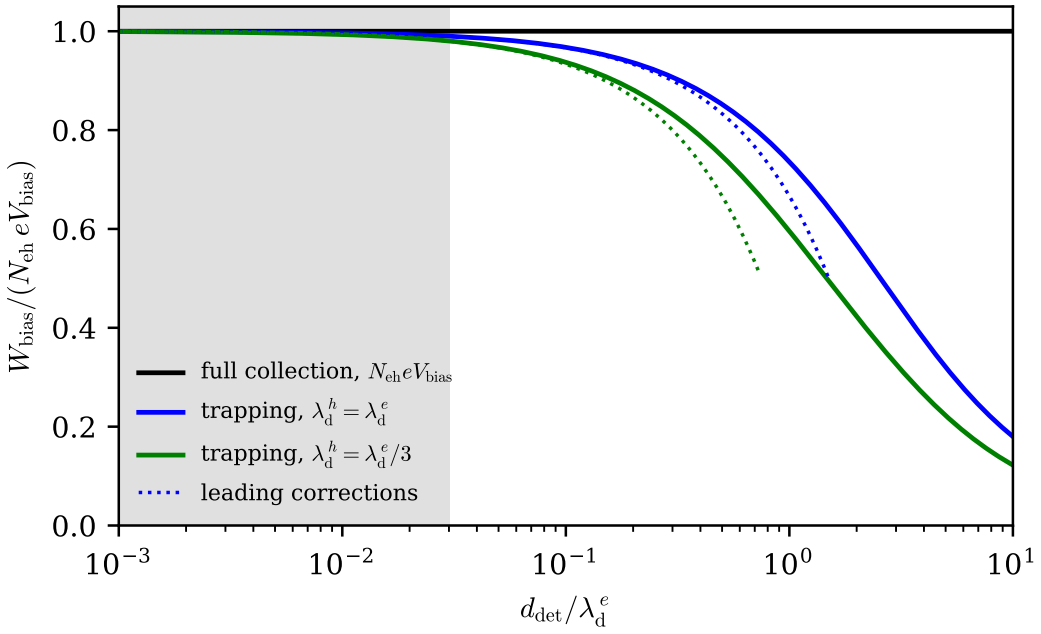


Figure 20.4: Bias work of uniformly distributed events with exponential trapping, Eq. (20.3) with the depth average Eq. (20.6), over the full-collection value $N_{\mathrm{eh}}eV_{\mathrm{bias}}$, against the crystal thickness in units of the electron drift length. Blue: equal drift lengths; green: $\lambda_{\mathrm{d}}^{h} = \lambda_{\mathrm{d}}^{e}/3$ (an illustrative ratio). Dotted: the leading correction $1 - d_{\mathrm{det}}/6\lambda_{\mathrm{d}}^{e} - d_{\mathrm{det}}/6\lambda_{\mathrm{d}}^{h}$. Gray: $d_{\mathrm{det}} \leq 0.030\,\lambda_{\mathrm{d}}$, where the work is within 1 % of full collection for equal drift lengths ($d_{\mathrm{det}} \leq 0.015\,\lambda_{\mathrm{d}}^{e}$ for the unequal pair).

When the drift length is no longer much longer than the crystal, the Luke heat falls below $N_{\mathrm{eh}}eV_{\mathrm{bias}}$ and depends on the depth of the event; it follows the read charge, not the pair number.

Since the read charge of a planar detector carries the same factor $\bar{\mathfrak{q}}$ event by event, the subtraction $E_Q eV_{\mathrm{bias}}/\varepsilon_{\mathrm{eh}}$ removes the Luke heat of partly collected pairs exactly. What trapping changes is the rest of the signal. A trapped carrier keeps its share of the gap energy, which the balance law counts as missing energy (Sect. 19.5), and a nuclear recoil has an ionized fraction f_{ion} below its ionization yield if its dense track traps more often.

20.1.4 Fano factor of the measured charge

If each pair is either collected in full or lost, independently and with the same probability f_{det} everywhere, the read charge is the pair number thinned binomially, and its Fano factor is $f_{\mathrm{det}}F_{\mathrm{eh}} + 1 - f_{\mathrm{det}}$ by Eq. (C.100). Trapping does not work this way. A trapped carrier leaves a partial charge, and the mean collection depends on the depth of the event. Let the event be at $\boldsymbol{r}$, and let the induced charges $\mathfrak{q}_i$ of its pairs, Eq. (20.2), be independent and identically distributed given $\boldsymbol{r}$, with mean $\bar{\mathfrak{q}}(\boldsymbol{r})$ and variance $\sigma_{\mathfrak{q}}^2(\boldsymbol{r})$, and independent of the pair number. This holds when the carriers drift independently after the prompt cascade, which excludes recombination and screening inside a dense track. The read charge is $N_{\mathrm{ind}} = \sum_{i=1}^{N_{\mathrm{eh}}} \mathfrak{q}_i$. At fixed $\boldsymbol{r}$, a sum of a random number of independent terms has mean $\bar{N}_{\mathrm{eh}}\bar{\mathfrak{q}}$ and variance $\bar{N}_{\mathrm{eh}}\sigma_{\mathfrak{q}}^2 + \bar{\mathfrak{q}}^2\,\mathrm{Var}\,N_{\mathrm{eh}}$, with $\mathrm{Var}\,N_{\mathrm{eh}} = F_{\mathrm{eh}}\bar{N}_{\mathrm{eh}}$. Averaging over the position with the law of total

variance, $\mathrm{Var}\, N_{\mathrm{ind}} = \langle \mathrm{Var}(N_{\mathrm{ind}}|\boldsymbol{r})\rangle + \mathrm{Var}_{\boldsymbol{r}}(\bar{N}_{\mathrm{eh}}\bar{\mathfrak{q}})$, and dividing by the mean $\bar{N}_{\mathrm{eh}}\langle\bar{\mathfrak{q}}\rangle$,

$$F_{\mathrm{meas}} = \underbrace{\frac{F_{\mathrm{eh}}\langle\bar{\mathfrak{q}}^2\rangle}{\langle\bar{\mathfrak{q}}\rangle}}_{\text{cascade}} + \underbrace{\frac{\langle\sigma_{\mathfrak{q}}^2\rangle}{\langle\bar{\mathfrak{q}}\rangle}}_{\text{partial charge}} + \underbrace{\bar{N}_{\mathrm{eh}}\frac{\mathrm{Var}_{\boldsymbol{r}}\,\bar{\mathfrak{q}}}{\langle\bar{\mathfrak{q}}\rangle}}_{\text{position}}, \tag{20.7}$$

where $\langle\cdot\rangle$ and $\mathrm{Var}_{\boldsymbol{r}}$ are taken over the distribution of event positions.

The cascade term is the Fano factor of the pairs, scaled by the collection. For all-or-nothing collection with one probability, $\bar{\mathfrak{q}} = f_{\mathrm{det}}$, $\sigma_{\mathfrak{q}}^2 = f_{\mathrm{det}}(1 - f_{\mathrm{det}})$ and the position term vanishes, which returns the binomial result. The unsplit excitons of Eq. (18.35) are the same thinning with $f_{\mathrm{det}} = 1 - p_{\mathrm{ns}}$, and losses that act independently on each pair multiply their f_{det}. The partial-charge term is never larger than in the binomial case. Since $0 \le \mathfrak{q} \le 1$, $\mathfrak{q}^2 \le \mathfrak{q}$, so $\langle\mathfrak{q}^2\rangle \le \bar{\mathfrak{q}}$ and $\sigma_{\mathfrak{q}}^2 \le \bar{\mathfrak{q}}(1 - \bar{\mathfrak{q}})$ at every position. At a fixed position, a loss shared as partial charges by many pairs fluctuates less than the same mean loss concentrated on a few pairs that are lost entirely. The position term is of a different kind. It grows with $\bar{N}_{\mathrm{eh}}$, and its contribution to the relative variance of the read charge, $\mathrm{Var}_{\boldsymbol{r}}\,\bar{\mathfrak{q}}/\langle\bar{\mathfrak{q}}\rangle^2$, does not fall with energy. It is a spread of the gain across the crystal, which no pair statistics removes, and it dominates at large deposits.

The published correction for incomplete collection uses the binomial picture, but writes the Fano term unthinned.

Limit Check 20.3. Fano factor of the measured charge

With charge collection efficiency f_{det} [7, Eqs. (A6)–(A7)]: $F_{\mathrm{meas}} = F_{\mathrm{eh}}/f_{\mathrm{det}} + 1 - f_{\mathrm{det}}$.

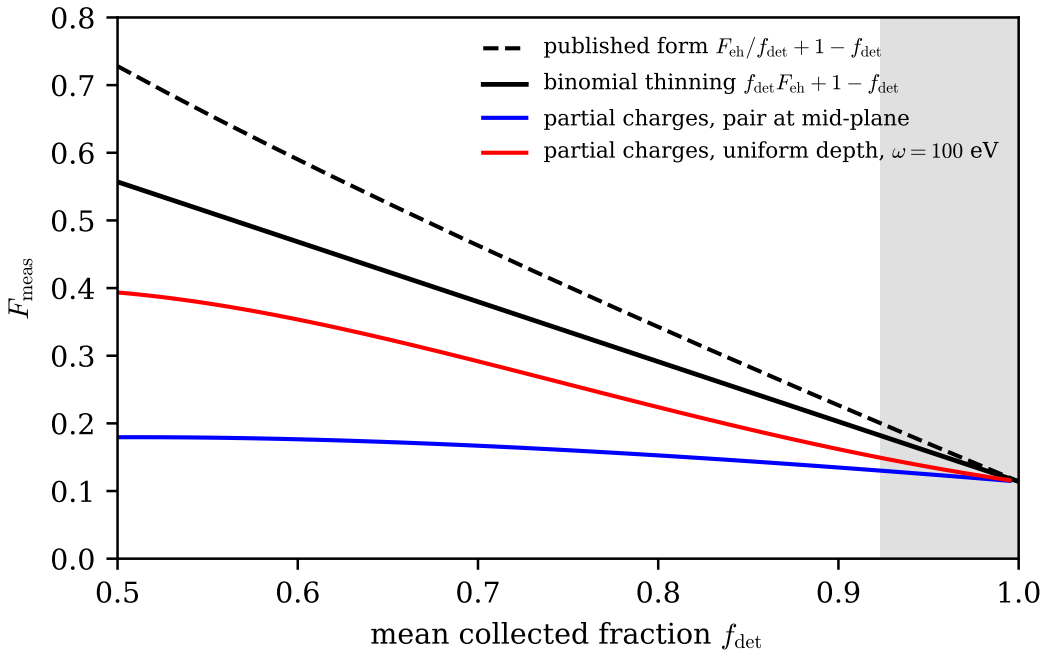


Figure 20.5: Fano factor of the measured charge against the mean collected fraction, for the cascade value $F_{\mathrm{eh}} = 0.114$ (Sect. 18.2). Dashed: the published form. Black: exact binomial thinning, Eq. (C.100). Blue: Eq. (20.7) for pairs at the midplane of a planar detector with equal electron and hole drift lengths, where trapping leaves partial charges and no position spread. Red: the same for events at uniform depth with $\bar{N}_{\mathrm{eh}} = 26.7$ (an electron recoil of 100 eV at $\varepsilon_{\mathrm{eh}} = 3.75$ eV), where the position term enters. The

collected fraction is varied through the common drift length. Gray: $f_{\mathrm{det}} \geq 0.923$, where the published form lies within 10 % of exact thinning.

The published form exceeds exact thinning by $F_{\mathrm{eh}}(1/f_{\mathrm{det}} - f_{\mathrm{det}}) \approx 2F_{\mathrm{eh}}(1 - f_{\mathrm{det}})$, and once trapping leaves partial charges or depends on depth neither form describes the measured charge.

Worked example 20.1: The measured Fano factor at 95 percent collection in silicon

Take the cascade value $F_{\mathrm{eh}} = 0.114$ of Sect. 18.2 and a mean collected fraction of 0.95, and compare three ways of losing the missing 5 %, and the published formula. (Sect. 20.5 uses the measured value 0.119 for electron-recoil signals.)

All-or-nothing loss, uniform. Binomial thinning gives $0.95 \times 0.114 + 0.05 = 0.158$. The published form gives $0.114/0.95 + 0.05 = 0.170$, higher by $0.114\,(1/0.95 - 0.95) = 0.012$.

Trapping, every pair at the midplane. With equal drift lengths, Eq. (20.5) at $z = d_{\mathrm{det}}/2$ is $2(\lambda_{\mathrm{d}}/d_{\mathrm{det}})(1 - \mathrm{e}^{-d_{\mathrm{det}}/2\lambda_{\mathrm{d}}})$, which equals 0.95 at $\lambda_{\mathrm{d}} = 4.83\,d_{\mathrm{det}}$. Each carrier's path is exponential with mean λ_{d}, cut at $d_{\mathrm{det}}/2$, and its second moment is $\langle \min(s, L)^2 \rangle = 2\lambda_{\mathrm{d}}^2[1 - \mathrm{e}^{-L/\lambda_{\mathrm{d}}}(1 + L/\lambda_{\mathrm{d}})]$. The two carriers give $\sigma_{\bar{\mathrm{q}}}^2 = 0.0156$, a third of the binomial $0.95 \times 0.05 = 0.0475$. Eq. (20.7) gives $F_{\mathrm{meas}} = (0.114 \times 0.9025 + 0.0156)/0.95 = 0.125$.

Trapping, uniform depth. Now Eq. (20.6) equals 0.95 at $\lambda_{\mathrm{d}} = 6.41\,d_{\mathrm{det}}$. Integrating Eq. (20.5) and the variance over depth gives $\langle \bar{\mathrm{q}}^2 \rangle = 0.9026$, $\langle \sigma_{\bar{\mathrm{q}}}^2 \rangle = 0.0230$ and $\mathrm{Var}_z\, \bar{\mathrm{q}} = 1.16 \times 10^{-4}$. The cascade and partial-charge terms give 0.1325. The position term is $1.22 \times 10^{-4}\,\bar{N}_{\mathrm{eh}}$: 0.003 for a 100 eV recoil ($\bar{N}_{\mathrm{eh}} = 26.7$), 0.033 at 1 keV and 0.19 at 5.9 keV. It equals 0.01 at 82 pairs, about 0.3 keV.

The same mean collection thus yields a measured Fano factor between 0.125 and 0.158 at 100 eV, depending on how the charge is lost, and nearly three times the cascade value at 5.9 keV if the loss depends on depth. A Fano factor measured at 5.9 keV constrains F_{eh} to 0.01 only if $\mathrm{Var}_z\, \bar{\mathrm{q}}$ is below 6×10^{-6}.

20.1.5 Space charge of trapped carriers

At millikelvin temperatures a trapped carrier is not re-emitted thermally. A donor level 45 meV below the conduction band carries the Boltzmann factor $\mathrm{e}^{-1.04\times10^4}$ at 50 mK. A field lowers the Coulomb barrier of a charged center: the potential energy $-\alpha/(\varepsilon_0 r) - e\mathcal{E}r$ of the carrier along the field has its maximum at the field separation radius $r_{\mathcal{E}} = (\alpha/\varepsilon_0 e\mathcal{E})^{1/2}$ of Eq. (18.36), which lowers the barrier by the Poole–Frenkel amount $2e\mathcal{E}r_{\mathcal{E}}$. In silicon that is 0.45 meV at 4 V/cm and 2.25 meV at 100 V/cm, far below 45 meV. Thermal and field-assisted emission are frozen, and trapped charge is released only by the phonons of later events, by drifting carriers that ionize the center on impact, or by infrared light. Detectors are therefore illuminated at intervals with infrared diodes, which neutralize the ionized trapping sites and keep trapping low; the practice is called flashing [4, Sect. 2]. Between flashes the trapped charge accumulates and changes the field.

In a planar detector at fixed bias the field change follows from Gauss's law and the constraint that the voltage across the crystal does not change. Let $\varrho_{\mathrm{tr}}(z)$ be the net trapped charge per unit volume, averaged over the transverse directions. The field change obeys $\mathrm{d}\Delta\mathcal{E}/\mathrm{d}z = \varrho_{\mathrm{tr}}/\varepsilon_0$, so $\Delta\mathcal{E}(z) = \Delta\mathcal{E}(0) + \varepsilon_0^{-1} \int_0^z \varrho_{\mathrm{tr}}\,\mathrm{d}z'$, and fixed bias requires $\int_0^{d_{\mathrm{det}}} \Delta\mathcal{E}\,\mathrm{d}z = 0$.

Solving for $\Delta\mathcal{E}(0)$ and integrating by parts,

$$\Delta\mathcal{E}(z) = \frac{1}{\varepsilon_0}\left[\int_0^z \varrho_{\mathrm{tr}}(z')\,\mathrm{d}z' - \frac{1}{d_{\mathrm{det}}}\int_0^{d_{\mathrm{det}}} (d_{\mathrm{det}} - z')\,\varrho_{\mathrm{tr}}(z')\,\mathrm{d}z'\right], \tag{20.8}$$

in Heaviside–Lorentz units, with ε_0 the static dielectric constant; in SI units multiply ε_0 by the vacuum permittivity $\varepsilon_0^{\mathrm{SI}}$. A sheet of n_{sh} charges e per unit area at depth z_0 makes the field jump by $en_{\mathrm{sh}}/\varepsilon_0$ across it, with the change $-en_{\mathrm{sh}}(d_{\mathrm{det}} - z_0)/\varepsilon_0 d_{\mathrm{det}}$ below and $+en_{\mathrm{sh}}z_0/\varepsilon_0 d_{\mathrm{det}}$ above, so the largest change lies between one half and the whole jump. For any distribution of one sign with n_{sh} charges per unit area in all, both terms in the bracket of Eq. (20.8) lie between 0 and en_{sh}, so their difference does not exceed en_{sh}. Hence the sheet bound

$$\max_z |\Delta\mathcal{E}(z)| \le \frac{en_{\mathrm{sh}}}{\varepsilon_0}, \tag{20.9}$$

which a sheet approaches as it nears a face. There, however, the whole change sits in the thin slab between the sheet and its electrode, and the field in the rest of the crystal is unchanged: the charge only images onto the electrode. A sheet at depth z_0 changes the field on its far side by $en_{\mathrm{sh}}z_0/\varepsilon_0 d_{\mathrm{det}}$, so a distortion over a finite part of the drift region needs more charge than Eq. (20.9), and a sheet at the midplane, which changes the field by half the jump over the whole crystal, needs twice as much. Charges of both signs give at most $e(n_{\mathrm{sh}}^{+} + n_{\mathrm{sh}}^{-})/\varepsilon_0$, and less where electrons and holes trapped at similar depths cancel.

We measure the distortion by $\xi_{\mathrm{SC}} \equiv \max_z |\Delta\mathcal{E}|/\mathcal{E}$. In silicon, with $\varepsilon_0 = 11.4$, one charge per square centimeter moves the field by $1.59 \times 10^{-7}\,\mathrm{V/cm}$. A distortion $\xi_{\mathrm{SC}} = 1\,\%$ at $1\,\mathrm{V/cm}$ then needs at least $6.3 \times 10^4\,\mathrm{cm}^{-2}$, the limit of a sheet next to a face, which distorts only a vanishing slab; a sheet at the midplane distorts the whole crystal by that amount at $12.6 \times 10^4\,\mathrm{cm}^{-2}$. At the $4\,\mathrm{V/cm}$ of the calibration of Sect. 19.5 both need four times more. Between flashes the trapped density grows as $n_{\mathrm{sh}}(t) \approx R\bar{N}_{\mathrm{eh}}p_{\mathrm{tr}}t$, for R events per unit area and time with $\bar{N}_{\mathrm{eh}}$ pairs each, of which a fraction p_{tr} leaves a trapped charge of the dominant sign. A trapping probability of $12.7\,\%$ is the value used to model a silicon detector that resolves single charges [8]. With it as p_{tr}, $100\,\mathrm{eV}$ events at $1\,\mathrm{Hz}$ on $1\,\mathrm{cm}^2$ reach $1\,\%$ at $1\,\mathrm{V/cm}$ over the whole crystal in 10 hours (the near-face limit in 5), and a calibration at a thousand times that rate within a minute.

None of this changes Eq. (20.1) or Eq. (20.3), which are exact with any space charge. The distortion acts through the trajectories. It changes the drift velocity along the path and, where capture depends on the carrier speed, the drift length; in three dimensions it bends the field lines, and carriers near the sides of the crystal can end on bare surfaces instead of electrodes. The energy stored in the field of the trapped charge is part of the electrostatic term ΔE_{es} that separates the Luke heat from the bias work (Appendix D.45). A detector run therefore carries a history: the trapped populations, the field and the rate of delayed charge released by later events all depend on the exposure since the last flash.

20.1.6 Luke-phonon emission

A carrier that moves faster than the longitudinal sound speed c_{LA} can emit a longitudinal acoustic phonon and conserve both energy and momentum; a slower one cannot. This is the acoustic analogue of Cherenkov radiation, and the phonons emitted are the Luke phonons that carry the bias work. Their rate follows from the golden rule with the deformation-potential vertex of Eq. (6.13). Take a carrier of isotropic mass m, energy $k^2/2m$ and momentum $\boldsymbol{k}$, and an isotropic deformation potential D^n, which couples it to longitudinal phonons only. A phonon of wavevector $\boldsymbol{Q}$ then has the matrix element $|M_Q|^2 = (D^n)^2 Q/(2\rho_T V c_{\mathrm{LA}})$ in a crystal of volume V and mass density ρ_T, the vertex of Sect. 6.2 with the volume normalization. Luke

phonons carry about a meV, more than ten kelvin. At detector temperatures absorption and stimulated emission are therefore negligible, and

$$\Gamma_{\mathrm{L}}(k) = 2\pi \sum_{\boldsymbol{Q}} |M_Q|^2 \, \delta\Big(\frac{k^2}{2m} - \frac{(\boldsymbol{k}-\boldsymbol{Q})^2}{2m} - c_{\mathrm{LA}} Q\Big).$$

With θ the angle between $\boldsymbol{Q}$ and $\boldsymbol{k}$, the argument of the delta function vanishes at $Q_* = 2(k\cos\theta - k_{\mathrm{s}})$, where $k_{\mathrm{s}} = mc_{\mathrm{LA}}$ is the momentum of a carrier moving at the sound speed, and its derivative with respect to Q there has magnitude $Q_*/2m$. A positive root requires $\cos\theta > k_{\mathrm{s}}/k$, and so $k > k_{\mathrm{s}}$: the Cherenkov condition. Writing $\sum_{\boldsymbol{Q}} = V\int \mathrm{d}^3Q/(2\pi)^3$, the azimuth gives 2π, so the prefactor is $2\pi\cdot 2\pi/(2\pi)^3 = 1/2\pi$, and the radial integral $\int Q^2 \mathrm{d}Q\, Q\, \delta(\ldots)$ gives $Q_*^3 \cdot 2m/Q_*$. Hence

$$\Gamma_{\mathrm{L}}(k) = \frac{(D^n)^2 m}{2\pi\rho_T c_{\mathrm{LA}}} \int_{k_{\mathrm{s}}/k}^{1} Q_*^2 \, \mathrm{d}\cos\theta = \frac{(D^n)^2 m}{2\pi\rho_T c_{\mathrm{LA}}} \, \frac{4k^2}{3}\Big(1 - \frac{k_{\mathrm{s}}}{k}\Big)^3.$$

With the scattering length $l_0 = \pi\rho_T/(2m^3(D^n)^2)$, a length that collects the coupling and the carrier mass, this is

$$\Gamma_{\mathrm{L}}(k) = \frac{c_{\mathrm{LA}}}{3l_0}\Big(\frac{k}{k_{\mathrm{s}}}\Big)^2\Big(1 - \frac{k_{\mathrm{s}}}{k}\Big)^3 \qquad (k > k_{\mathrm{s}}), \tag{20.10}$$

and the phonon is emitted at the polar angle θ with the probability density $\propto (\cos\theta - k_{\mathrm{s}}/k)^2 \sin\theta$, up to $\cos\theta = k_{\mathrm{s}}/k$. The angular law printed by Leman [6, Eq. (22)] and for the G4CMP (Geant4 Condensed Matter Physics) toolkit [9, Eq. (9)] integrates to Eq. (20.10), while the total rates printed beside them [6, Eq. (21)], [9, Eq. (8)] are lower by the factor k_{s}/k.

The phonon energy is $c_{\mathrm{LA}}Q_* = 2c_{\mathrm{LA}}k(\cos\theta - k_{\mathrm{s}}/k)$. Averaging over the angular distribution, in which $u = \cos\theta - k_{\mathrm{s}}/k$ is weighted by u^2 on $[0, 1-k_{\mathrm{s}}/k]$ and so has the mean $\frac{3}{4}(1-k_{\mathrm{s}}/k)$,

$$\langle E_{\mathrm{L}}\rangle = \frac{3}{2}\, mc_{\mathrm{LA}}^2\Big(\frac{k}{k_{\mathrm{s}}} - 1\Big), \qquad E_{\mathrm{L}}^{\max} = 2mc_{\mathrm{LA}}^2\Big(\frac{k}{k_{\mathrm{s}}} - 1\Big). \tag{20.11}$$

The energy scale is set by the carrier mass and the sound speed. For silicon holes, with the isotropic mass $0.5\, m_e$, $D^n = 5.0\,\mathrm{eV}$ and $c_{\mathrm{LA}} = 9.0\,\mathrm{km/s}$ of the CDMS transport model [6, Table II], $mc_{\mathrm{LA}}^2 = 0.230\,\mathrm{meV}$, $l_0 = 7.5\,\mu\mathrm{m}$ and $c_{\mathrm{LA}}/3l_0 = 4.0\times10^8\,\mathrm{s}^{-1}$. A hole at twice the sound speed emits at $2.0\times10^8\,\mathrm{s}^{-1}$ phonons of mean energy 0.35 meV and at most 0.46 meV.

How fast a drifting carrier moves follows, roughly, from the balance between the work of the field and the energy it emits. If the carrier moved along the field at its full speed $c_{\mathrm{LA}}k/k_{\mathrm{s}}$, the balance $e\mathcal{E}c_{\mathrm{LA}}k/k_{\mathrm{s}} = \Gamma_{\mathrm{L}}\langle E_{\mathrm{L}}\rangle$ with Eqs. (20.10) and (20.11) would read

$$\frac{(k/k_{\mathrm{s}} - 1)^4}{(k/k_{\mathrm{s}})^2} = \frac{2e\mathcal{E}l_0}{mc_{\mathrm{LA}}^2}.$$

Since the drift velocity is smaller than the speed, this overestimates k/k_{s}. For silicon holes it gives $k/k_{\mathrm{s}} \approx 4.3$ at 1 V/cm and 7.0 at 4 V/cm, with mean Luke phonon energies of 1.1 and 2.1 meV; for electrons, with the conductivity mass $0.26\, m_e$, the effective mass that sets the drift (Appendix E.12), and $D^n = 9.0\,\mathrm{eV}$ ($mc_{\mathrm{LA}}^2 = 0.120\,\mathrm{meV}$), 1.1 and 2.1 meV. A transport Monte Carlo of silicon electrons puts the mean lower, at 0.84 meV at 1 V/cm (Sect. 20.5.2). Luke phonons are thus born at one to a few meV, between the pair-breaking energy $2\Delta_{\mathrm{BCS}} = 0.36\,\mathrm{meV}$ of bulk aluminum (Sect. 1.5) and the optical-phonon energies near 60 meV at which the primary phonons start. Since the emitted energy is weighted by u^3, the fraction of it below an energy E_{c} is $(E_{\mathrm{c}}/E_{\mathrm{L}}^{\max})^4$, and at birth only 0.3 % of the Luke energy of a hole at 1 V/cm lies below $2\Delta_{\mathrm{BCS}}$. Whether the Luke phonons are then absorbed in the sensor films as efficiently as the down-converted primaries is a transport question, and it decides the sign of $\mathfrak{r}_{\mathrm{coll}} - 1$ in the calibration of Sect. 19.5 (Sect. 20.5).

20.1.7 Open problems

The emission rates of trapped charge at millikelvin temperatures are not known, and with them neither the release of trapped charge over a run nor the growth of the space-charge field between flashes. The rate of each trap class is a sum over the phonon flux times a phonon-ionization cross-section, the flux of drifting carriers times an impact-ionization cross-section, and the infrared illumination. A neutral shallow donor needs a phonon above its binding energy, 45 meV or 10.9 THz for phosphorus in silicon, and such phonons decay within about a nanosecond (Sect. 20.2), so they ionize only near the track of a later event. Luke phonons cannot ionize neutral centers but can ionize the over-charged states D^- and A^+, which bind their extra carrier by a few meV and which flashing itself populates. No calculation gives either cross-section for these centers at the carrier and phonon energies that matter. Without them the trapped populations, the field distortion ξ_{SC} and the delayed charge cannot be followed through an exposure history; Appendix E.25 specifies the calculation.

The spectrum that the Luke rate Eq. (20.10) produces is tested only indirectly. The rate keeps longitudinal phonons only, whereas the uniaxial deformation potential Ξ_u couples the electrons of the silicon valleys to transverse phonons with the vertex $\Xi_u \sin\theta \cos\theta$ (Sect. 6.2), and holes couple to them as well; transverse emission starts at the lower transverse sound speed and gives softer phonons. It takes one isotropic deformation potential and a parabolic band, and treats the anisotropic electron valleys by a Herring–Vogt transformation that does not conserve the phonon momentum exactly (Appendix E.12). The measured drift velocity against field, which transport models reproduce in germanium (Appendix E.12), tests the rate of momentum loss along the field, an angular moment of the emission, not the energy spectrum of the Luke phonons on which their collection depends. A rate with both acoustic branches and the valley-resolved deformation potentials, checked against a phonon signal measured at fixed field, would settle it.

A transport simulation of a detector must pass the tests of this section before its collection kernels are used. They are the Hecht relation at a known trap density, the charge sum (charge read on all electrodes plus the trapped charges weighted by their weighting potentials equals the charge created), the bias work Eq. (20.3) and the measured Fano factor Eq. (20.7). A validation of a transport engine against them at the level of a few percent, which the collection ratio of Sect. 19.5 requires, has not been published. Appendix E.25 specifies it.

20.2 Athermal phonons in the substrate

At the partition time of Sect. 19.1, some 100 ps after a 100 eV electron recoil in silicon, about 69 eV sits in phonons of tens of meV, some two thousand of them near the point of the event. A sensor film on the surface absorbs phonons microseconds later, after they have decayed several times and scattered off isotopes hundreds of times. The phonons of a nuclear recoil of the same energy start from a different spectrum. If that difference survived to the sensor, a phonon readout could tell the two recoils apart by what it absorbs, and the phonon collection of a nuclear recoil could differ from that of the electron recoil used to calibrate it (Sect. 19.5). Two rates decide the question: anharmonic decay, which scales as the fifth power of the phonon energy, and isotope scattering, which scales as the fourth. Together they drive every hard spectrum toward one shape that depends on time only through a single energy scale, and they leave the energy born at low phonon energies as the difference that a linear readout can still see.

20.2.1 Prompt phonons at the partition time

The phonons that carry the prompt energy $E_{\mathrm{prompt}} = E_{\mathrm{phon}}(t_{\mathrm{pr}})$ of Sect. 19.5.1 are *athermal*. Their energies E_{ath}, from meV to tens of meV, exceed the thermal energy of 0.86 µeV at 10 mK by four orders of magnitude or more, and their population is far from equilibrium. It is also dilute: about 2300 phonons of 30 meV, even if confined to a sphere of radius 1 µm, share 6×10^{11} modes, an occupation of 4×10^{-9}. They do not interact with each other. Each phonon interacts with the static crystal alone, scattering off the random isotope masses, and decays spontaneously through the cubic anharmonicity; at 10 mK the Bose factors of Eq. (5.70) vanish, leaving spontaneous decay. The transport is linear in the deposit, and the birth spectrum is the initial condition of a linear evolution.

The two recoils make their phonons in different ways. In an electron recoil the hot carriers of Chap. 18 lose most of their energy by emitting optical and intervalley phonons (Sect. 18.1.2). An optical phonon lives 3.7 ps (Worked example 5.9) and decays mostly into a longitudinal acoustic phonon near 42 meV and a transverse one near 22 meV (Sect. 5.7). By t_{pr} the optical phonons are gone, and the population is acoustic and hard. Below the optical energy a carrier can emit only acoustic phonons, and momentum conservation caps their energy at $2c_{\mathrm{LA}}k$ for a carrier of momentum k. For a silicon electron with the density-of-states mass $0.32\,m_e$ of one valley, the cap is 8.7 meV at a kinetic energy of 64 meV and 6.0 meV at 30 meV. Suppose each carrier enters this regime with a kinetic energy spread uniformly below ω_{opt}. Each of the two carriers then has the mean kinetic energy $\omega_{\mathrm{opt}}/2$, so each pair sheds on average ω_{opt} in phonons below about 6 to 9 meV, and the soft fraction of the deposit is

$$f_{\mathrm{s}} \simeq \frac{\omega_{\mathrm{opt}}}{\varepsilon_{\mathrm{eh}}}. \tag{20.12}$$

This estimate gives 1.7 % in silicon, with the measured $\omega_{\mathrm{opt}} = 64.1$ meV (Sect. 5.2) and $\varepsilon_{\mathrm{eh}} = 3.75$ eV (Sect. 18.2), and 1.3 % in germanium, with 37.6 meV (Table 5.1) and 2.96 eV [6, Table I]. The gap energy $N_{\mathrm{eh}}E_{\mathrm{g}}$, 31 % of an electron recoil in silicon, is not part of the prompt population. It returns as phonons later, where and with which spectrum the release channel decides (Sects. 19.5 and 20.5).

In a nuclear recoil below the displacement threshold the whole deposit is heat (Sect. 19.1). For small recoils the phonons that the nuclear response creates are the final state, with the energy-weighted phonon content of Sect. 11.3. Above a few eV the created spectrum comes from the molecular dynamics of Appendix E.23, which has not been tabulated. A recoil that thermalizes locally fills the modes in proportion to their density of states, and its spectrum is hard as well. Both populations are hard, between about 4 and 15 THz in silicon, and they differ in branch content and in shape; only the electron recoil carries the soft fraction f_{s}.

20.2.2 Anharmonic down-conversion

The linewidth Eq. (5.70) at $T = 0$ gives the decay rate of any phonon. For an acoustic phonon of long wavelength it takes a simple form, because both of its ingredients, the vertex and the phase space, scale as powers of the frequency. Translation invariance makes the vertex small. By the sum rule Eq. (5.67), the third-order constants vanish when summed over the positions of any one atom. In Eq. (5.69) the term of zeroth order in each wavevector cancels whenever the polarizations are those of a uniform translation, as they are for acoustic phonons at small Q. The first surviving term carries one power of each wavevector, $Q_1Q_2Q_3$, multiplied by the third-order elastic constants. With the prefactor $(\omega_1\omega_2\omega_3)^{-1/2}$ and the linear dispersion $\omega_i = c_iQ_i$, the vertex becomes

$$V^{(3)}(1,2,3) = (\omega_1\omega_2\omega_3)^{1/2}\, v_3(\hat{\boldsymbol{Q}}_i, \nu_i), \tag{20.13}$$

with v_3 a function of the directions and branches alone. The squared vertex of a decay $\omega \to \omega_1 + \omega_2$ is then $\omega^3 y(1-y)|v_3|^2$, with $y = \omega_1/\omega$. The phase space is the sum over $\boldsymbol{Q}_1$ at fixed $\boldsymbol{Q}_2 = \boldsymbol{Q} - \boldsymbol{Q}_1$. In polar coordinates about $\boldsymbol{Q}$, the angle integral of the energy delta function is $\int \mathrm{d}\cos\theta\, \delta(\omega - c_1 Q_1 - c_2 Q_2) = Q_2/(c_2 Q Q_1)$, since $\mathrm{d}Q_2/\mathrm{d}\cos\theta = -QQ_1/Q_2$. The remaining integral $(2\pi/c_2 Q)\int \mathrm{d}Q_1\, Q_1 Q_2$ scales as ω^2. The angles at which the decay happens are fixed by y and the speeds, so v_3 contributes a function of y only, and the rate scales as $\omega^3 \times \omega^2$:

$$\Gamma_{\mathrm{anh}} = A_{\mathrm{anh}} \Big(\frac{E_{\mathrm{ath}}}{h} \Big)^5 . \tag{20.14}$$

This is Eq. (E.82). The constant A_{anh} depends on the second- and third-order elastic constants and on the branch of the decaying phonon. The derivation needs normal processes ($\boldsymbol{G} = 0$) and frequencies low enough for linear dispersion, the same long-wavelength condition as for the Rayleigh law of isotope scattering below.

Not every branch decays. In the isotropic continuum a transverse phonon cannot decay at all. Its energy $c_{\mathrm{T}} Q$ would have to equal $c_1 Q_1 + c_2 Q_2$, with both speeds at least c_{T}, the lowest speed, and with $Q_1 + Q_2 \geq Q$ by the triangle inequality. The only solutions are collinear decays into two transverse phonons, for which the isotropic vertex vanishes. A longitudinal phonon decays through the channels $\mathrm{L} \to \mathrm{L}' + \mathrm{T}'$ and $\mathrm{L} \to \mathrm{T} + \mathrm{T}$, with the daughter energies distributed as in Appendix E.12. Anisotropy and dispersion open slow decays of transverse phonons in real crystals, which transport codes neglect [9, Sect. 2.2.2]. In the isotropic continuum A_{anh} is a closed form in five elastic constants, the two Lamé constants and three third-order combinations [10, Eqs. (1), (2)]. For silicon, Tamura's calculation gives $A_{\mathrm{anh}} = 7.41 \times 10^{-56}\,\mathrm{s}^4$ [6, Table I], and a recalculation from cryogenic elastic constants gives $1.15 \times 10^{-55}\,\mathrm{s}^4$ [10, Table 3]. For germanium the published values spread over a factor of ten, from $1.61 \times 10^{-55}\,\mathrm{s}^4$ [11] through $6.8 \times 10^{-55}\,\mathrm{s}^4$ in the recalculation to $1.65 \times 10^{-54}\,\mathrm{s}^4$. Two of them are both attributed to Tamura's work yet disagree by a factor 2.6: $6.43 \times 10^{-55}\,\mathrm{s}^4$ [6, Table I] and $1.65 \times 10^{-54}\,\mathrm{s}^4$ [10, Table 3]. No direct measurement exists for either crystal, and we use the slowest germanium value, $1.61 \times 10^{-55}\,\mathrm{s}^4$. A longitudinal phonon at the top of the silicon acoustic band, 15 THz, decays in 18 ps; one at 1 THz lives 14 µs.

The constants come from the elastic tensors. The second-order constants are measured or computed as in Appendix E.4; the third-order ones follow from the same finite-strain calculations taken one order further, or from ultrasonic measurements under stress. Appendix E.12 sets out how the two rates enter a transport Monte Carlo of a whole detector, together with the kinematics of the daughters and the focusing of the group velocities. Appendix E.25 specifies how the created-phonon spectra of the two recoils are carried through it to the sensors.

20.2.3 Isotope scattering

Natural silicon is 92.2 % ^{28}Si, 4.7 % ^{29}Si and 3.1 % ^{30}Si, and a phonon sees the random masses as point scatterers. Write the isotope on site I as a mass defect $\delta_I^M = (\bar{M} - M_I)/\bar{M}$ about the mean mass $\bar{M}$, whose variance over the isotopes is the mass variance $g_{\mathrm{md}} = \sum_i x_i (\delta_i^M)^2$ of Appendix E.12. To lowest order in g_{md}, the self-energy of the mass disorder in Sect. 11.6 gives every phonon of frequency ω the same width,

$$\Gamma_{\mathrm{iso}}(\omega) = \frac{\pi}{6}\, g_{\mathrm{md}}\, \omega^2\, g_{\mathrm{ph},\kappa}(\omega), \tag{20.15}$$

with $g_{\mathrm{ph},\kappa}$ the site-projected density of states normalized to three. The polarization of the initial phonon drops out because on a site of cubic symmetry, as in silicon, the polarization sum over final states is isotropic: cubic symmetry makes the tensor $N^{-1} \sum_{\boldsymbol{Q}'\nu'} e^{\beta}_{\kappa\nu'} e^{\gamma *}_{\kappa\nu'} \delta(\omega - \omega_{\boldsymbol{Q}'\nu'})$ proportional to $\delta_{\beta\gamma}$, and the normalization fixes the factor $g_{\mathrm{ph},\kappa}/3$. At low frequency the

density of states takes its Debye form, $g_{\mathrm{ph},\kappa} = 3\Omega_{\mathrm{at}}\omega^2\langle c^{-3}\rangle/2\pi^2$, with Ω_{at} the volume per atom and $\langle c^{-3}\rangle$ the average of $\frac{1}{3}\sum_\nu c_\nu^{-3}$ over directions. Eq. (20.15) then becomes the Rayleigh law

$$\Gamma_{\mathrm{iso}} = B_{\mathrm{iso}}\Big(\frac{E_{\mathrm{ath}}}{h}\Big)^4, \qquad B_{\mathrm{iso}} = 4\pi^3 g_{\mathrm{md}}\,\Omega_{\mathrm{at}}\langle c^{-3}\rangle, \tag{20.16}$$

which is Eq. (E.81) and Tamura's result [10, Eq. (4)]. Isotope scattering conserves energy. In the isotropic continuum it also forgets the branch, since the final branch is drawn with a weight proportional to its density of states whatever the initial one. Repeated scattering holds the branches at their equilibrium shares, with a longitudinal share f_{LA} of 9.3 % in silicon and 9.8 % in germanium [9, Sect. 2.2.2], set by the density of states of the anisotropic crystal (the isotropic speeds of Appendix C.18 would give 9.75 % in silicon).

Worked example 20.2: The isotope rate of natural silicon at 1 THz

With the AME2020 masses and the natural abundances, $\bar{M} = 28.0855\,\mathrm{u}$, and the three isotopes have the mass defects $\delta^M = +0.39$, -3.17 and -6.72 % [12, 13]. Their contributions to g_{md} are 1.38×10^{-5}, 4.72×10^{-5} and 1.40×10^{-4}, so $g_{\mathrm{md}} = 2.007\times10^{-4}$, and ^{30}Si, the rarest isotope, carries 70 % of it. The volume per atom is $a^3/8 = 20.02\,\text{Å}^3$. Averaging $\frac{1}{3}\sum_\nu c_\nu^{-3}$ over directions with the measured elastic tensor of Appendix C.18 gives $\langle c^{-3}\rangle = 4.90\times10^{-12}\,\mathrm{s^3\,m^{-3}}$. The isotropic code speeds of Appendix E.12, 9.0 and 5.4 km/s, give $4.69\times10^{-12}\,\mathrm{s^3\,m^{-3}}$, 4 % less, because they miss the slow transverse wave along [110]. Then

$$B_{\mathrm{iso}} = 124.0\times(2.007\times10^{-4})\times(2.002\times10^{-29}\,\mathrm{m^3})\times(4.90\times10^{-12}\,\mathrm{s^3\,m^{-3}}) = 2.44\times10^{-42}\,\mathrm{s^3}.$$

This lies inside the range 2.42 to $2.56\times10^{-42}\,\mathrm{s^3}$ that Tamura, Shields, Ramsbey and Wolfe measured across the phonon polarizations [10, Table 4]. At $E_{\mathrm{ath}}/h = 1$ THz, or 4.14 meV, the rate is $\Gamma_{\mathrm{iso}} = 2.44\times10^6\,\mathrm{s^{-1}}$: a lifetime of 0.41 µs, and 3.9×10^{-7} of the angular frequency. With the rms group speed of the mixed branches, 5.85 km/s for the code speeds, the mean free path is 2.4 mm, comparable with the thickness of a detector and some 4×10^5 transverse wavelengths. The full density of states raises the rate at 1 THz by about 5 % above the Rayleigh law (Limit Check 20.4). Germanium has $g_{\mathrm{md}} = 5.87\times10^{-4}$ from the same tables and slower sound, and $B_{\mathrm{iso}} = 3.67\times10^{-41}\,\mathrm{s^3}$ [6, Table I], fifteen times that of silicon.

The Rayleigh law fails early in silicon. Limit Check 20.4 compares it with Eq. (20.15) evaluated with the density of states of the PBE force constants of Sect. 5.2, integrated over the zone on rays from Γ so that small Q is sampled finely. The ratio of the two rates is $g_{\mathrm{ph},\kappa}/g_{\mathrm{Debye}}$, independent of g_{md}. For the comparison the Debye density uses the PBE sound speeds, whose $\langle c^{-3}\rangle$ is 22 % above the measured one because PBE makes the transverse branches soft; the absolute rate is that of Worked example 20.2. The transverse branches bend over well before the zone boundary. The density of states, and with it the rate, exceeds the Rayleigh law by 10 % already at 5.6 meV, or 1.36 THz, a tenth of the Debye energy $\omega_{\mathrm{D}} = 55.6$ meV.

Limit Check 20.4. Rayleigh law of isotope scattering

The published rate is $\Gamma_{\mathrm{iso}} = B_{\mathrm{iso}}(E_{\mathrm{ath}}/h)^4$ with $B_{\mathrm{iso}} = 4\pi^3 g_{\mathrm{md}}\Omega_{\mathrm{at}}/\langle c^3\rangle$, $\langle c^3\rangle$ the polarization-averaged cubed sound speed [10, Eq. (4)]. The book uses the average of the inverse cube, $\langle c^{-3}\rangle$, which is what the Debye density of states contains; the two forms agree when the three speeds are equal.

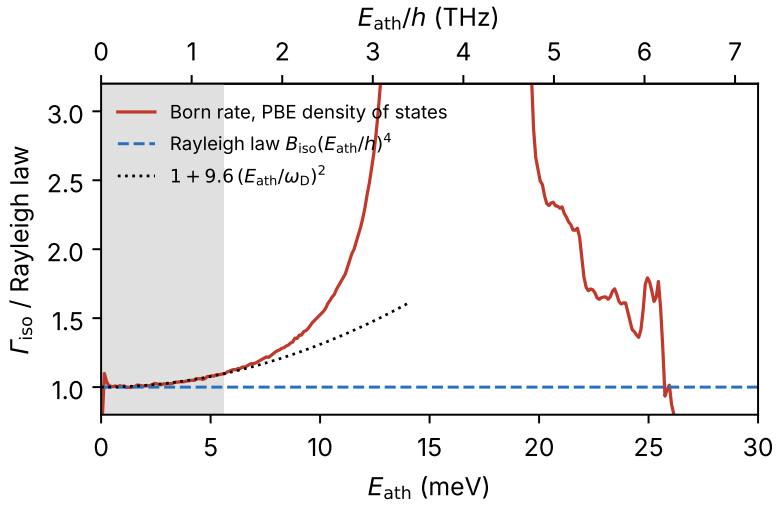


Figure 20.6: Isotope scattering rate of silicon from the Born rate Eq. (20.15) with the full density of states (red), divided by the Rayleigh law Eq. (20.16) (dashed, one), against the phonon energy (lower axis) and frequency (upper axis). The ratio is $g_{\mathrm{ph},\kappa}(\omega)/g_{\mathrm{Debye}}(\omega)$, independent of the mass variance. Density of states from the PBE force constants of Sect. 5.2, integrated in spherical coordinates about Γ to the zone boundary (32 Gauss–Legendre polar points, 64 azimuths, 800 radial steps, 0.1 meV bins, running mean over three bins); the Debye reference uses the PBE sound speeds. Dotted: the leading correction $1 + 9.6\,(E_{\mathrm{ath}}/\omega_{\mathrm{D}})^2$, fitted between 2 and 6 meV, with $\omega_{\mathrm{D}} = 55.6$ meV. Gray: within 10 % of the Rayleigh law, $E_{\mathrm{ath}} \leq 5.6$ meV. The ratio peaks at 8.1 near 17.6 meV, above the plotted range.

Above 1.4 THz the transverse branches of silicon are no longer linear, and the rate follows the density of states rather than the fourth power of the frequency.

20.2.4 The scaling spectrum

The two rates set up a hierarchy. For a longitudinal phonon their ratio is $B_{\mathrm{iso}}/A_{\mathrm{anh}}$ divided by the frequency, and $B_{\mathrm{iso}}/A_{\mathrm{anh}}$ is 33 THz in silicon and 228 THz in germanium with its slowest decay. A phonon scatters off isotopes more often than it decays, by a factor 3.3 at 10 THz in silicon and by more at lower frequency. Between two decays the branches stay close to their equilibrium shares, and a phonon of any branch spends the fraction f_{LA} of its time as a longitudinal phonon, the only branch that decays. The population decays at the mean rate $f_{\mathrm{LA}}\Gamma_{\mathrm{anh}}$, and isotope scattering, which conserves energy, drops out of the evolution of the energy spectrum. Let $\rho_E(E_{\mathrm{ath}}, t)$ be the energy per unit phonon energy, and $P_{\mathrm{d}}(y)$ the energy-weighted distribution of the fraction y that a decay passes to a daughter, counting both daughters: with $p(y)$ the number density of one daughter, $P_{\mathrm{d}}(y) = 2y\,p(y)$, so that $\int_0^1 P_{\mathrm{d}}(y)\,\mathrm{d}y = 1$. Energy flows only downward:

$$\frac{\partial \rho_E(E,t)}{\partial t} = -f_{\mathrm{LA}}\Gamma_{\mathrm{anh}}(E)\,\rho_E(E,t) + \int_E^\infty \frac{\mathrm{d}E'}{E'}\,P_{\mathrm{d}}\Big(\frac{E}{E'}\Big)\,f_{\mathrm{LA}}\Gamma_{\mathrm{anh}}(E')\,\rho_E(E',t). \tag{20.17}$$

In the isotropic continuum P_{d} depends on the energy ratio alone, as the distribution of Appendix E.12 does, since the kinematics of a decay contain no energy scale. The rate

Eq. (20.14) contains one scale, which combines with the time into

$$E_*(t) = h\,(f_{\mathrm{LA}} A_{\mathrm{anh}} t)^{-1/5}, \qquad f_{\mathrm{LA}}\Gamma_{\mathrm{anh}}(E) = \frac{1}{t}\Big(\frac{E}{E_*(t)}\Big)^5. \tag{20.18}$$

A phonon above $E_*(t)$ has had time to decay, and one below it has not. Since E_* is the only energy scale, we look for a solution that depends on the phonon energy only through $y = E/E_*(t)$ and conserves the total energy.

Scaling spectrum of down-conversion

$$\rho_E(E_{\mathrm{ath}}, t) = \frac{E_{\mathrm{prompt}}}{E_*(t)}\,\varphi_*\Big(\frac{E_{\mathrm{ath}}}{E_*(t)}\Big),$$
$$\frac{1}{5}\,\frac{\mathrm{d}}{\mathrm{d}y}\,[y\varphi_*(y)] = -y^5\varphi_*(y) + \int_y^\infty \mathrm{d}y'\, y'^4 P_{\mathrm{d}}\Big(\frac{y}{y'}\Big)\varphi_*(y'), \tag{20.19}$$

with $\int \varphi_*\,\mathrm{d}y = 1$. The spectrum keeps its shape and slides to lower energy as $t^{-1/5}$, and every energy moment scales with $E_*(t)$.

Inserting the form into Eq. (20.17) gives the equation for φ_*. Since $\mathrm{d}E_*/\mathrm{d}t = -E_*/5t$, the time derivative of ρ_E is $(E_{\mathrm{prompt}}/5tE_*)\,\mathrm{d}(y\varphi_*)/\mathrm{d}y$. The loss term is $(E_{\mathrm{prompt}}/tE_*)\,y^5\varphi_*$ by Eq. (20.18), and the gain term becomes the integral after the substitution $E' = E_* y'$. Every term carries the factor E_{prompt}/tE_*, so the time drops out. Integrating over y checks the energy: the left side gives $[y\varphi_*]_0^\infty = 0$, and the two terms on the right cancel after the order of integration is exchanged, since $\int_0^{y'} P_{\mathrm{d}}(y/y')\,\mathrm{d}y = y'$.

Eq. (20.19) says what the spectrum looks like once it has settled, not when it settles. Settling needs a time long compared with the first decay of the hardest phonons, $1/(f_{\mathrm{LA}} A_{\mathrm{anh}} f_{\mathrm{top}}^5)$ for a band top f_{top}. This is 0.19 ns in silicon ($f_{\mathrm{top}} = 15$ THz, $f_{\mathrm{LA}} = 0.093$) and 1.3 ns in germanium ($f_{\mathrm{top}} = 8.64$ THz, $f_{\mathrm{LA}} = 0.098$, $A_{\mathrm{anh}} = 1.61 \times 10^{-55}\,\mathrm{s}^4$). Settling also needs the hierarchy of rates, the fifth-power law, and no absorption or surface scattering in the meantime. In silicon the scale is $E_*(t) = 44.6$, 28.1, 17.7 and 11.2 meV at 1, 10 and 100 ns and 1 µs, and in germanium 37.8, 23.8, 15.0 and 9.5 meV. Since $E_* \propto A_{\mathrm{anh}}^{-1/5}$, the factor of ten among the germanium constants moves E_* by a factor 0.63, and the factor 1.55 between the two silicon values moves it by 0.92.

We have no proof that every birth spectrum approaches φ_*, which is a question about the stability of Eq. (20.19). What we can do is solve Eq. (20.17) for representative births and watch. For the solution we need P_{d}, and we take a symmetric stand-in for the isotropic-continuum distribution: each daughter receives a fraction y drawn from the beta distribution $30y^2(1-y)^2$, which gives $P_{\mathrm{d}}(y) = 60\,y^3(1-y)^2$. On a logarithmic energy grid of step 0.01, Eq. (20.17) becomes a linear system whose exponential we compute exactly. The settled spectrum has its energy-weighted median at $0.66\,E_*$ and 80 % of its energy between $0.36\,E_*$ and $0.97\,E_*$, and its mean is $\langle E_{\mathrm{ath}}\rangle = 0.667\,E_*(t)$. Limit Check 20.5 follows the mean energy of three hard births in silicon: a line at the band top, a spectrum that grows as the cube of the energy up to the band top, and the acoustic pairs of optical decay. The cubic spectrum puts equal energy in every mode of the Debye model, the shape of a locally thermalized recoil. All three reach the scaling law within 10 % after 0.46 ns, a few first-decay times.

Limit Check 20.5. Down-conversion and the scaling spectrum

The published rates are $\Gamma_{\mathrm{anh}} = A\nu^5$ for longitudinal phonons and $\Gamma_{\mathrm{iso}} = B\nu^4$ for all branches [6, Eqs. (6), (8)], with $A = 7.41 \times 10^{-56}\,\mathrm{s}^4$ and $B = 2.43 \times 10^{-42}\,\mathrm{s}^3$ for silicon [6, Table I]. With the mode mixing of Kelsey et al. [9, Sect. 2.2.2] they imply the scaling

spectrum Eq. (20.19), $E_* \propto t^{-1/5}$, and the mean energy $\langle E_{\mathrm{ath}} \rangle = 0.667\, E_*(t)$ for the daughter distribution used here.

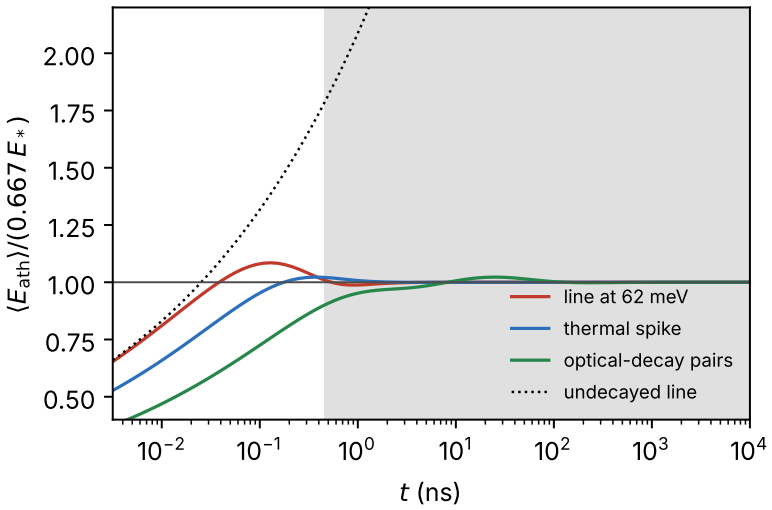


Figure 20.7: Energy-weighted mean phonon energy in silicon divided by the scaling law $0.667\, E_*(t)$ of Eqs. (20.18) and (20.19), against time, for three hard births (models): a line at the band top, 62 meV (red); a spectrum growing as E_{ath}^3 up to 62 meV (blue); and the pairs of longitudinal (42 meV) and transverse (22 meV) phonons from optical decay, Gaussians of 10 % relative width (green). Solution of Eq. (20.17) with $f_{\mathrm{LA}} = 0.093$, $A_{\mathrm{anh}} = 7.41 \times 10^{-56}\,\mathrm{s}^4$, the daughter distribution $P_{\mathrm{d}}(y) = 60\, y^3(1-y)^2$ and no surfaces, on a logarithmic grid of step 0.01 from 0.01 to 18 THz, by the exact matrix exponential. Dotted: the leading correction at early times, the undecayed line, $(f_{\mathrm{LA}} A_{\mathrm{anh}} f_{\mathrm{top}}^5 t)^{1/5}/0.667$. Gray: all three within 10 %, $t \geq 0.46$ ns.

Before the hardest phonons have decayed a few times the mean still depends on the birth spectrum; and where the continuum rates fail, above a few THz in silicon (Limit Check 20.4), the time scale itself is an estimate.

20.2.5 Memory of the birth spectrum

How long does the birth spectrum leave a trace? We measure the trace by the total-variation distance between the normalized energy spectra that two births reach at time t, which is half the integral of their absolute difference. It bounds the difference of the energy that the two deliver into any part of the spectrum. Let $\Delta_{\mathrm{h}}(t)$ be its largest value over pairs of births above an energy E_{mem}. By the scaling form Δ_{h} depends on time only through $E_*(t)/E_{\mathrm{mem}}$, and the solutions of Eq. (20.17) give the threshold: any two births above E_{mem} agree to 10^{-3} once $E_*(t)$ has fallen to between 0.51 and 0.53 E_{mem}. A birth is forgotten when the scale has fallen to about half its energy, and since $E_* \propto t^{-1/5}$ the time this takes grows as the inverse fifth power of the birth energy. Fig. 20.8 shows the consequences for silicon.

The answer depends on how low the hard births reach. Take two births concentrated at the top of the band, the cubic spectrum and a line at 95 % of the band top. They agree to 10^{-3} after 6.8 ns in silicon and 47 ns in germanium with its slowest decay constant (Fig. 20.8(b)), and the fastest germanium constant shortens the latter tenfold. Births spread over the upper factor two of the band take longer, 0.15 µs in silicon and 1.0 µs in germanium. The transverse phonons of optical decay, at 22 meV, are forgotten only when E_* has fallen to about 11 meV,

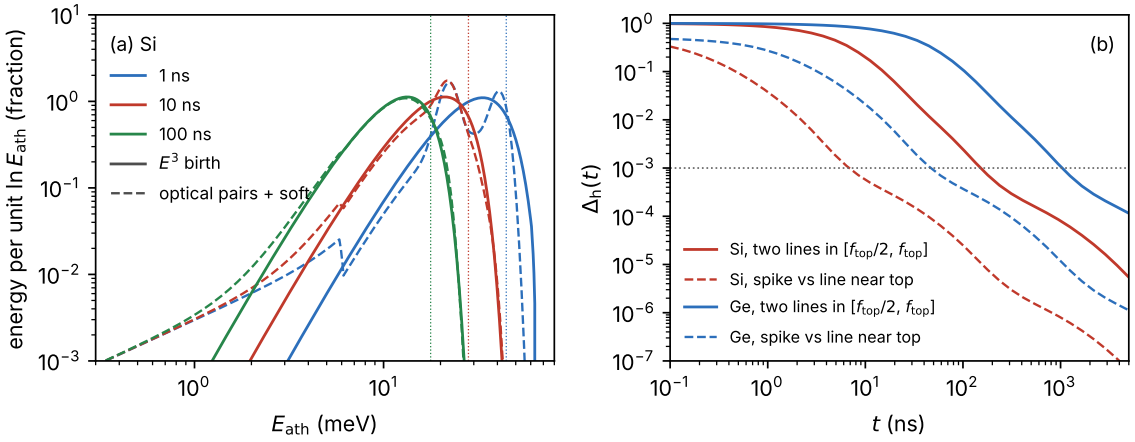


Figure 20.8: Down-conversion of prompt phonons, from Eq. (20.17) in the isotropic continuum with the mean decay rate $f_{\mathrm{LA}} A_{\mathrm{anh}} (E_{\mathrm{ath}}/h)^5$ for every phonon, the daughter distribution $P_{\mathrm{d}}(y) = 60\, y^3 (1-y)^2$ and no surfaces. (a) Energy spectrum of a silicon event, per unit $\ln E_{\mathrm{ath}}$ and normalized to the prompt energy, at 1, 10 and 100 ns (colors), for two births (models). Solid: a spectrum growing as E_{ath}^3 up to the band top, 62 meV, standing for a locally thermalized nuclear recoil. Dashed: the longitudinal (42 meV) and transverse (22 meV) phonons of optical decay, Gaussians of 10 % relative width with the energy shares 42:22 of the two phonons of an optical decay, plus the soft fraction $f_{\mathrm{s}} = 1.7\,\%$ spread uniformly in energy over 0.2 to 6 meV, standing for an electron recoil. Dotted verticals: $E_*(t)$. $A_{\mathrm{anh}} = 7.41 \times 10^{-56}\,\mathrm{s}^4$, $f_{\mathrm{LA}} = 0.093$. (b) Memory of the hard part, $\Delta_{\mathrm{h}}(t)$, for silicon (red) and germanium (blue; $A_{\mathrm{anh}} = 1.61 \times 10^{-55}\,\mathrm{s}^4$, $f_{\mathrm{LA}} = 0.098$, band top 8.64 THz). Solid: the largest total-variation distance between two line births in the upper factor two of the band, $[f_{\mathrm{top}}/2, f_{\mathrm{top}}]$ (six lines). Dashed: one pair concentrated near the top, the E_{ath}^3 spectrum against a Gaussian line at $0.95\, f_{\mathrm{top}}$ of width $0.05\, f_{\mathrm{top}}$. Dotted: 10^{-3}, crossed by the dashed curves at 6.8 ns (Si) and 47 ns (Ge), and by the solid curves at 0.15 µs (Si) and 1.0 µs (Ge).

near 1 µs in silicon. The two model births of Fig. 20.8(a) differ by a total-variation distance of 0.13 at 10 ns, 0.026 at 100 ns, 0.015 at 300 ns and 0.012 at 1 µs. At 1 µs their parts above 6 meV agree to 6×10^{-4}. What remains is the soft fraction: the electron recoil was born with 1.7 % of its energy below 6 meV and the cubic spectrum with 0.01 %, and the decay of the hard parts has only begun to fill that region. All of these times rest on the continuum rates, which in silicon are applied at 5 to 15 THz, above their range (Limit Check 20.4), so they are estimates.

Polarization is forgotten much sooner. Along the scaling trajectory, the number of times a phonon has scattered off isotopes by the time t is

$$N_{\mathrm{mix}}(t) = \int_0^t \mathrm{d}t'\, B_{\mathrm{iso}} \left[E_*(t')/h\right]^4 = 5 B_{\mathrm{iso}} \left(f_{\mathrm{LA}} A_{\mathrm{anh}}\right)^{-4/5} t^{1/5}, \tag{20.20}$$

which is 164 at 1 ns in silicon and 650 at 1 µs; in germanium, with fifteen times the isotope constant, it is 1300 and 5100. Each scattering redraws the branch from the equilibrium shares, so the fraction of phonons that still carry their birth branch is of order $\mathrm{e}^{-N_{\mathrm{mix}}}$.

The phonons also spread out. Take the diffusion constant $v_{\mathrm{ph}}^2/3\Gamma_{\mathrm{iso}}$ of a gas with mean free time $1/\Gamma_{\mathrm{iso}}$ and rms group speed v_{ph}, evaluated at $E_*(t)$. The mean squared distance from the event is then $6 \int \mathrm{d}t'\, v_{\mathrm{ph}}^2/3\Gamma_{\mathrm{iso}} = \frac{10}{9} v_{\mathrm{ph}}^2 (f_{\mathrm{LA}} A_{\mathrm{anh}})^{4/5} t^{9/5}/B_{\mathrm{iso}}$. In silicon, with $v_{\mathrm{ph}} = 5.85\,\mathrm{km/s}$, the rms distance is 8.6 µm at 10 ns, 68 µm at 100 ns and 0.54 mm at 1 µs. In germanium, with $v_{\mathrm{ph}} = 3.52\,\mathrm{km/s}$, it is 1.8 µm, 15 µm and 0.12 mm. This is the quasi-diffuse transport of athermal phonons [9, Sect. 2.2.2]: the phonons random-walk while their mean free path grows with each decay, until it reaches the size of the crystal and they move ballistically. A phonon of 45 meV, the binding energy of a phosphorus donor, belongs to a population that decays within 0.95 ns, by which time it has spread over 1.0 µm. An event closer to a sensor than the rms distance delivers part of its energy before its spectrum has settled, and there the birth is not forgotten. For a silicon detector 4 mm thick with sensors on one face, such as that of Ren et al. [14], that shell is 0.21 % of the volume at 10 ns, 1.7 % at 100 ns and 13.5 % at 1 µs. The memory times above decide how much of the volume counts as bulk.

Sapphire does not fit this picture. Its isotope constant is a hundred times smaller than that of silicon, $B_{\mathrm{iso}} = 2.5 \times 10^{-44}\,\mathrm{s}^3$ calculated and $4 \times 10^{-44}\,\mathrm{s}^3$ measured, while its decay constant lies between 1.88×10^{-55} and $1.27 \times 10^{-54}\,\mathrm{s}^4$. Hernandez et al. trace that spread to the anisotropy and to converting a published rate with the fifth power of a sound speed [10, Tables 3, 4]. Compared to silicon, however, the hierarchy is reversed: mixing outpaces decay only below $B_{\mathrm{iso}}/A_{\mathrm{anh}} = 0.02$ to 0.21 THz. Over the whole primary band a longitudinal phonon decays before it scatters, and the branches are not held at their equilibrium shares. A transverse phonon, which the isotropic continuum does not let decay, scatters $B_{\mathrm{iso}}(E_{\mathrm{ath}}/h)^4 \times 1\,\mu\mathrm{s} = 0.025$, 0.4 and 16 times in a microsecond at 1, 2 and 5 THz. As a result, transverse phonons born below a few THz in sapphire keep their energy and their polarization for microseconds. The single rate $f_{\mathrm{LA}}\Gamma_{\mathrm{anh}}$ of Eq. (20.17) does not exist, and the scaling argument fails at its first step. The fate of these phonons needs the decay rates of transverse phonons in an anisotropic crystal, which the isotropic continuum sets to zero.

20.2.6 Transfer-kernel bound on observable differences

The memory of the spectrum matters through what a readout measures. Take an observable that is linear in the energies delivered to the sensors, with weights between zero and one: the fraction of the energy absorbed by one of two sensors, the fraction that arrives before a given time, or the collection efficiency of a population. Since the transport is linear, a birth spectrum $\hat{\rho}_0(E)$ normalized to unit energy gives $O = \int \mathrm{d}E\, R_O(E)\, \hat{\rho}_0(E)$. Here $R_O(E)$, with $0 \le R_O \le 1$, is the value of the observable for unit energy born at E at the event position,

and Sect. 20.5 builds it from the collection kernels of a detector. Let $\Delta\hat{\rho}_0 = \hat{\rho}_0^{\mathrm{NR}} - \hat{\rho}_0^{\mathrm{ER}}$ be the difference of the birth spectra of the two recoils. Suppose that no sensor absorbs anything before a time t_{mem}, and that births above E_{mem} have been forgotten by then, to the distance $\Delta_{\mathrm{h}}(t_{\mathrm{mem}})$ in energy, branch and position.

Transfer-kernel bound

$$|O^{\mathrm{NR}} - O^{\mathrm{ER}}| \le \int_0^{E_{\mathrm{mem}}} \mathrm{d}E\, |\Delta\hat{\rho}_0(E)| + \Delta_{\mathrm{h}}(t_{\mathrm{mem}}). \tag{20.21}$$

When $\Delta\hat{\rho}_0$ has one sign below E_{mem}, the integral is $\Delta_{\mathrm{soft}} = |\int_0^{E_{\mathrm{mem}}} \Delta\hat{\rho}_0\, \mathrm{d}E|$, the difference of the energy fractions born below E_{mem}.

The response to a birth above E_{mem} is the response to the state it has reached at t_{mem}, and a weight between zero and one integrated against the difference of two normalized distributions is bounded by their total-variation distance. Two births E_1, E_2 above E_{mem} therefore have $|R_O(E_1) - R_O(E_2)| \le \Delta_{\mathrm{h}}$. Both birth spectra have unit energy, so $\int \Delta\hat{\rho}_0\, \mathrm{d}E = 0$ and $O^{\mathrm{NR}} - O^{\mathrm{ER}} = \int (R_O - R_0)\, \Delta\hat{\rho}_0\, \mathrm{d}E$ for any constant R_0. Choose R_0 midway between the largest and the smallest response above E_{mem}. Then $|R_O - R_0| \le \Delta_{\mathrm{h}}/2$ above E_{mem}, where $\int |\Delta\hat{\rho}_0| \le 2$, and $|R_O - R_0| \le 1$ below, which gives the two terms of Eq. (20.21). An observable that is a ratio of delivered energies, such as the share of one of two sensors, is linear to first order in the difference, and the range of its linearized weight replaces the factor one.

Eq. (20.21) makes precise the statement that only the soft part of the birth spectra survives. It rests on two facts, the linearity of the transport and the scaling of Sect. 20.2.5, and it names the one number that the created-phonon spectra of the two recoils must supply, the distance between their parts below E_{mem}. The single-sign condition matters. Two births with the same energy below E_{mem} but different soft shapes, one just above the gap of a sensor film and one just below it, have $\Delta_{\mathrm{soft}} = 0$ and yet give different responses; the bound then needs the full L^1 distance. The branch content of a nuclear recoil above E_{mem} does not enter at all.

For the primary phonons the bound limits the spectral part of the collection offset of Sect. 19.5. With the collection efficiency as the observable, $|\epsilon_{\mathrm{prim}}^{\mathrm{NR}} - \epsilon_{\mathrm{prim}}^{\mathrm{ER}}| \le \int_0^{E_{\mathrm{mem}}} |\Delta\hat{\rho}_0| + \Delta_{\mathrm{h}}$. If E_{mem} lies above the soft band and below every hard birth, and only the electron recoil carries a soft fraction, the integral is $f_{\mathrm{s}} = 1.7\,\%$. The condition on E_{mem} is the weak point. For the model births of Fig. 20.8(a) at 1 µs, the lowest energy above which births agree to 10^{-3}, $E_{\mathrm{mem}} = 21.6\,\mathrm{meV}$, cuts through the transverse daughters of optical decay, and the bound is 0.15, against an actual largest difference of 0.012. At 3 µs, with $E_{\mathrm{mem}} = 17.7\,\mathrm{meV}$, the bound has fallen to 0.026 against 0.009, but the near-face shell of a 4 mm crystal is by then 36 % of the volume. Within the bulk the spectral difference between the two recoils is set by the soft fraction and by how low the hard births reach; near an instrumented face this argument does not bound it.

20.3 Phonons into films

A phonon that has down-converted in silicon to about a terahertz (Sect. 20.2) reaches an aluminum film 60 nm thick on the crystal surface. Silicon and aluminum have almost the same acoustic impedance, and the normal-incidence law of Appendix C.18 lets 99 % of a longitudinal flux through. A fit of measured pulses with a phonon simulation nevertheless returns a silicon-to-aluminum coefficient of 0.3 to 0.55 for such a film [15], and a phonon inside the film that reaches the interface crosses back into the silicon about once in three arrivals. These three numbers measure different things. The transmission of each polarization at every angle, and the reciprocity that ties the two directions together, explain all three, and from

them follow the probability that a phonon entering the film is absorbed there and the time t_l it spends in the film before escaping, which controls the quasiparticle yield of Sect. 20.4.

20.3.1 Acoustic mismatch at oblique incidence

The acoustic mismatch model treats the interface as a welded plane between two elastic continua: displacement and traction are continuous across it, and nothing else happens there [16]. Silicon is cubic, with an anisotropy ratio $2C_{44}/(C_{11}-C_{12}) = 1.56$ (Appendix C.18). Aluminum is nearly isotropic: its elastic constants extrapolated to zero temperature, $C_{11} = 114.3$, $C_{12} = 61.9$ and $C_{44} = 31.6\,\mathrm{GPa}$ [17, Table 2], give the ratio 1.21. We replace each crystal by an isotropic medium with the Voigt averages of its elastic constants, the bulk modulus B of Eq. (C.155) and the shear modulus $(C_{11}-C_{12}+3C_{44})/5$, which is also what transport codes do. Silicon then has the speeds $c_{\mathrm{LA}} = 9.00\,\mathrm{km\,s^{-1}}$ and $c_{\mathrm{T}} = 5.40\,\mathrm{km\,s^{-1}}$ at $\rho_T = 2.33\,\mathrm{g\,cm^{-3}}$, and aluminum $c_{\mathrm{LA}} = 6.63\,\mathrm{km\,s^{-1}}$ and $c_{\mathrm{T}} = 3.30\,\mathrm{km\,s^{-1}}$ at the room-temperature density $2.70\,\mathrm{g\,cm^{-3}}$. The zero-temperature density, $2.73\,\mathrm{g\,cm^{-3}}$, gives 6.59 and $3.28\,\mathrm{km\,s^{-1}}$ and lengthens the escape time below by 1.6 %. A subscript 1 or 2 on a speed or density names the medium. The elastic constants themselves are measured, or computed by density-functional perturbation theory or finite strains (Appendix E.4), and transport simulations take the interface transmissions of this section as tables (Appendix E.12).

Put the interface at $z = 0$, with medium 1 below it. A plane wave of frequency ω that arrives from medium 1 at the angle θ_1 to the normal has the wavevector component $Q_\parallel = \omega \sin\theta_1 / c_{\mathrm{inc}}$ along the interface, where c_{inc} is its speed. Every reflected and transmitted wave it generates has the same ω and the same $Q_\parallel$, which is the acoustic Snell law Eq. (C.162). A wave of speed c in either medium then has the normal component

$$Q_z = \left(\omega^2/c^2 - Q_\parallel^2\right)^{1/2}, \tag{20.22}$$

taken positive. When $Q_\parallel > \omega/c$ the root is imaginary, we take it with a positive imaginary part, and the wave decays away from the interface and carries no energy flux. A wave can have three polarizations: longitudinal (L), transverse in the plane of incidence (SV), and transverse perpendicular to it (SH). The interface is invariant under reflection through the plane of incidence, so SH waves couple only to SH waves, while L and SV waves mix.

For SH waves the displacement u_y and the traction $\sigma_{yz} = \rho_T c_{\mathrm{T}}^2\, \partial_z u_y$ must be continuous. This is the algebra of the normal-incidence derivation in Appendix C.18, with $Q_i \rho_i c_i^2$ there replaced by $\rho_i c_{\mathrm{T},i}^2 Q_{z,i} = \omega Z_{\mathrm{ac},i}^{\mathrm{T}} \cos\theta_i$, where $Z_{\mathrm{ac},i}^{\mathrm{T}} = \rho_i c_{\mathrm{T},i}$ is the transverse impedance. The fraction of the incident energy flux that crosses is

$$t_{\mathrm{SH}}(\theta_1) = \frac{4 Z_{\mathrm{ac},1}^{\mathrm{T}} \cos\theta_1 \,\mathrm{Re}\left(Z_{\mathrm{ac},2}^{\mathrm{T}} \cos\theta_2\right)}{\left|Z_{\mathrm{ac},1}^{\mathrm{T}} \cos\theta_1 + Z_{\mathrm{ac},2}^{\mathrm{T}} \cos\theta_2\right|^2}, \qquad \cos\theta_2 = \left(1 - \frac{c_{\mathrm{T},2}^2}{c_{\mathrm{T},1}^2} \sin^2\theta_1\right)^{1/2}. \tag{20.23}$$

At normal incidence it is Eq. (C.161) with the transverse impedances. Beyond the critical angle $\sin\theta_1 = c_{\mathrm{T},1}/c_{\mathrm{T},2}$, which exists when medium 2 is the faster one, $\cos\theta_2$ is imaginary and $t_{\mathrm{SH}} = 0$: the wave is totally reflected.

For L and SV waves four conditions hold at $z = 0$: continuity of u_x, u_z, $\sigma_{zz} = \rho_T[c_{\mathrm{LA}}^2 \partial_z u_z + (c_{\mathrm{LA}}^2 - 2c_{\mathrm{T}}^2)\partial_x u_x]$ and $\sigma_{xz} = \rho_T c_{\mathrm{T}}^2 (\partial_z u_x + \partial_x u_z)$, the last two being Hooke's law of Appendix C.18 for an isotropic medium. Each wave is $u = a\,\hat{\boldsymbol{e}}\,\mathrm{e}^{\mathrm{i}(Q_\parallel x \pm Q_z z - \omega t)}$ with amplitude a and the sign $+$ for a wave moving toward medium 2 and $-$ for one moving back into medium 1, and unit polarization $\hat{\boldsymbol{e}} = (c/\omega)(Q_\parallel, 0, \pm Q_z)$ for L and $(c/\omega)(\pm Q_z, 0, -Q_\parallel)$ for SV. Collect for each wave the four quantities that must match,

$$\boldsymbol{b} = \Big(\hat{e}_x,\ \hat{e}_z,\ \frac{\rho_T}{\omega}\left[c_{\mathrm{LA}}^2(\pm Q_z)\hat{e}_z + (c_{\mathrm{LA}}^2 - 2c_{\mathrm{T}}^2) Q_\parallel \hat{e}_x\right],\ \frac{\rho_T c_{\mathrm{T}}^2}{\omega}\left[(\pm Q_z)\hat{e}_x + Q_\parallel \hat{e}_z\right]\Big), \tag{20.24}$$

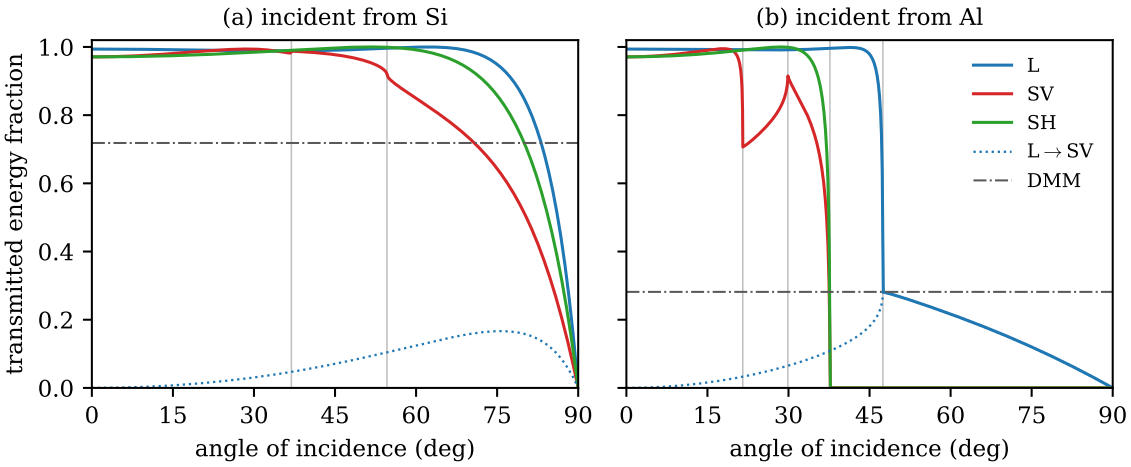


Figure 20.9: Acoustic-mismatch transmission of a welded silicon/aluminum interface against the angle of incidence, for phonons incident (a) from silicon and (b) from aluminum. Solid: fraction of the incident energy flux transmitted into all modes of the other side, for incident longitudinal (L, blue), SV (red) and SH (green) phonons; dotted: the part of the incident longitudinal flux transmitted as SV. Dash-dot: the diffuse-mismatch value $\mathfrak{n}_2/(\mathfrak{n}_1+\mathfrak{n}_2)$, 0.718 in (a) and 0.282 in (b). Gray lines: critical angles, 36.9° and 54.6° in (a); 21.5°, 29.9°, 37.7° and 47.5° in (b). Both media isotropic, with the Voigt averages of the elastic constants: silicon (Appendix C.18) $c_{\mathrm{LA}} = 9.00$, $c_{\mathrm{T}} = 5.40\,\mathrm{km\,s^{-1}}$, $\rho_T = 2.33\,\mathrm{g\,cm^{-3}}$; aluminum, zero-temperature elastic constants at the room-temperature density, 6.63, $3.30\,\mathrm{km\,s^{-1}}$, $2.70\,\mathrm{g\,cm^{-3}}$. The curves do not depend on frequency.

which are the displacement and $\sigma_{zz}/\mathrm{i}\omega$, $\sigma_{xz}/\mathrm{i}\omega$ per unit amplitude. With unit incident amplitude the boundary conditions read $\boldsymbol{b}_{\mathrm{inc}} + a_{\mathrm{rL}}\boldsymbol{b}_{\mathrm{rL}} + a_{\mathrm{rS}}\boldsymbol{b}_{\mathrm{rS}} = a_{\mathrm{tL}}\boldsymbol{b}_{\mathrm{tL}} + a_{\mathrm{tS}}\boldsymbol{b}_{\mathrm{tS}}$, four linear equations for the amplitudes of the reflected (r) and transmitted (t) L and SV waves. A propagating plane wave carries the time-averaged energy flux $\frac{1}{2}\rho_T\omega^2|a|^2c$ along its direction (Appendix C.18), whose normal component is this times $\cos\theta = cQ_z/\omega$. The fraction of the incident flux carried away by the outgoing wave j is therefore

$$t_j = \frac{\rho_j c_j^2\ \mathrm{Re}\,Q_{z,j}\,|a_j|^2}{\rho_1 c_{\mathrm{inc}}^2\,Q_{z,\mathrm{inc}}}, \tag{20.25}$$

and the transmission of an incident L or SV wave is the sum of the two transmitted fractions. We solve the four equations numerically at each angle.

Fig. 20.9 shows the result for silicon and aluminum. From the silicon side, a longitudinal phonon crosses with 0.99 at normal incidence and with more than 0.98 out to 70°, falling below 0.9 only beyond 78°. At oblique incidence part of the transmitted flux is converted to SV, 0.031 at 30° and at most 0.17, near 76°. Transverse phonons cross with 0.97 at normal incidence. The one critical angle that limits transmission from silicon is 54.6°, beyond which an incident SV wave can no longer produce a longitudinal wave in aluminum. From the aluminum side, however, every wave is slower than every wave in silicon, and the critical angles cut transmission off. Transverse phonons beyond $\arcsin(c_{\mathrm{T,Al}}/c_{\mathrm{T,Si}}) = 37.7°$ find no propagating wave in silicon and are totally reflected. Longitudinal phonons lose the longitudinal channel beyond 47.5° and keep only the conversion into silicon transverse waves, which carries 0.28 at that angle and vanishes at grazing incidence.

Frequency drops out of Eqs. (20.23) and (20.25), because $Q_\parallel$ and Q_z both scale with ω. The transmissions are frequency independent as long as the wavelength is long compared with

the lattice spacing, so that the continuum holds, and short compared with the film thickness, so that the film acts as a half-space. In the simulation of the 60 nm films the absorbed phonons peak between 0.7 and 0.9 THz [15, Sect. V]. At 0.8 THz the wavelengths in aluminum are 8.3 nm for longitudinal and 4.1 nm for transverse phonons, so both conditions hold. The longitudinal wavelength equals the thickness at 0.110 THz, close to the pair-breaking threshold $2\Delta_{\mathrm{BCS}} = 0.36\,\mathrm{meV}$, or 0.087 THz, of bulk aluminum (Sect. 7.2); below it the film reflects as a layer, with interference between its two faces.

20.3.2 Angle averages, reciprocity and diffuse mismatch

A sensor film collects phonons arriving from all directions, so what matters is the transmission averaged over an isotropic population. Count the energy flux that such a population sends onto unit area of the interface. A branch has $\mathrm{d}^3Q/(2\pi)^3$ modes in d^3Q, each with occupation $n(\omega)$, energy ω and normal group velocity $\partial\omega/\partial Q_z$. Since $\mathrm{d}Q_z\,\partial\omega/\partial Q_z = \mathrm{d}\omega$, the flux is

$$\int_{Q_z>0} \frac{\mathrm{d}^3Q}{(2\pi)^3}\,\omega\,n(\omega)\,\frac{\partial\omega}{\partial Q_z} = \int \mathrm{d}\omega\,\omega\,n(\omega)\int \frac{\mathrm{d}^2Q_\parallel}{(2\pi)^3}. \tag{20.26}$$

At a given frequency, every patch $\mathrm{d}^2Q_\parallel$ of every branch, in either medium, brings the same flux. In an isotropic medium $\mathrm{d}^2Q_\parallel = 2\pi Q_\parallel \mathrm{d}Q_\parallel = (2\pi\omega^2/c_\nu^2)\sin\theta\cos\theta\,\mathrm{d}\theta$, so the angles enter with the weight $2\sin\theta\cos\theta$ and the branch ν with the weight c_ν^{-2}. The mean transmission of an isotropic population from medium 1 into medium 2 is

$$\bar t_{12} = \frac{1}{\mathfrak{n}_1}\sum_\nu c_{1\nu}^{-2}\int_0^{\pi/2} t_{12,\nu}(\theta)\,2\sin\theta\cos\theta\,\mathrm{d}\theta, \qquad \mathfrak{n}_1 = c_{\mathrm{LA},1}^{-2} + 2c_{\mathrm{T},1}^{-2}, \tag{20.27}$$

where ν runs over L, SV and SH and $t_{12,\nu}(\theta)$ is the transmission of Eq. (20.23) or Eq. (20.25). The interface-weighted density $\mathfrak{n}_1$ is proportional to the rate at which phonons of a given frequency strike unit area from medium 1.

The two directions are not independent. Fix ω and $Q_\parallel$. The interface then scatters N_1 incoming channels from medium 1, the branches of medium 1 that propagate at this $Q_\parallel$, and N_2 from medium 2, into the same number of outgoing ones. Normalize each amplitude so that its squared modulus is the energy flux, as in Eq. (20.25). The matrix S that maps incoming onto outgoing amplitudes is then unitary, because the welded interface neither stores nor dissipates energy. Write it in blocks,

$$\mathsf{S} = \begin{pmatrix} \mathsf{r} & \mathsf{t}' \\ \mathsf{t} & \mathsf{r}' \end{pmatrix},$$

with t the $N_2 \times N_1$ block of transmission from medium 1 into medium 2 and t' the reverse. The upper-left block of $\mathsf{S}^\dagger\mathsf{S} = 1$ gives $\mathsf{r}^\dagger\mathsf{r} + \mathsf{t}^\dagger\mathsf{t} = 1$, and that of $\mathsf{S}\mathsf{S}^\dagger = 1$ gives $\mathsf{r}\mathsf{r}^\dagger + \mathsf{t}'\mathsf{t}'^\dagger = 1$, both in the N_1-dimensional space of medium 1. Taking traces, $\mathrm{Tr}\,\mathsf{t}^\dagger\mathsf{t} = N_1 - \mathrm{Tr}\,\mathsf{r}^\dagger\mathsf{r} = \mathrm{Tr}\,\mathsf{t}'\mathsf{t}'^\dagger$. The left side is the sum of the transmissions of all channels incident from medium 1, the right side that of all channels incident from medium 2. Integrated over $Q_\parallel$ with the weight of Eq. (20.26), this equality becomes a statement about the averages Eq. (20.27).

Reciprocity of interface transmission

For any flat interface that conserves energy, the mean transmissions of isotropic populations in the two directions satisfy

$$\mathfrak{n}_1\,\bar t_{12} = \mathfrak{n}_2\,\bar t_{21}. \tag{20.28}$$

The derivation uses the conservation of $Q_\parallel$ and the conservation of energy, nothing about isotropy or the number of layers. For anisotropic media or a layered interface Eq. (20.28) holds at each frequency, with $\mathfrak{n}_i$ replaced by the area that the branches of medium i cover in the $Q_\parallel$ plane. Evanescent waves are not channels and drop out. In thermal equilibrium the relation is detailed balance: equal temperatures on the two sides give no net flux. It holds, however, for any population, since it is a property of S.

For silicon and aluminum $\mathfrak{n}_{\mathrm{Si}} = 0.0808$ and $\mathfrak{n}_{\mathrm{Al}} = 0.206\,\mathrm{s}^2\,\mathrm{km}^{-2}$, a ratio of 0.392. We find $\bar{t}_{\mathrm{Si}\to\mathrm{Al}} = 0.938$ and $\bar{t}_{\mathrm{Al}\to\mathrm{Si}} = 0.368$. Transmission out of the film is 2.5 times weaker than into it because the slower phonons of aluminum have 2.55 times more modes per unit area of $Q_\parallel$, and most of those modes lie outside the disk that silicon can carry. The transverse phonons beyond 37.7° are the fraction $1 - (c_{\mathrm{T,Al}}/c_{\mathrm{T,Si}})^2 = 0.627$ of the transverse flux and, since transverse phonons carry 0.890 of the flux in aluminum, 0.557 of all phonon flux in the film. They never reach silicon. The impedance mismatch plays almost no part. In the direction into the film the interface hardly distinguishes polarizations: the averages are 0.979 for longitudinal, 0.899 for SV and 0.962 for SH phonons, so longitudinal phonons cross 1.05 times as readily as transverse ones.

The diffuse mismatch model makes the opposite assumption: a phonon arriving at the interface loses its direction and polarization there [16]. A phonon leaving the interface then carries no memory of the side it came from, and the probability of ending in medium 2 is the same for a phonon from medium 1, t_{12}, as for a phonon from medium 2, $1 - t_{21}$. With Eq. (20.28),

$$t_{12}^{\mathrm{DMM}} = \frac{\mathfrak{n}_2}{\mathfrak{n}_1 + \mathfrak{n}_2}, \tag{20.29}$$

which is 0.718 from silicon into aluminum and 0.282 back. The two models serve below as the range of the escape probability.

The normal-incidence impedance law is often taken as the transmission of an interface. Limit Check 20.6 compares it with the angle- and mode-averaged result.

Limit Check 20.6. Impedance law and the averaged transmission

The acoustic-mismatch transmission at normal incidence [16], Eq. (C.161): $t_{12}(0) = 4Z_{\mathrm{ac},1}Z_{\mathrm{ac},2}/(Z_{\mathrm{ac},1} + Z_{\mathrm{ac},2})^2$, the same in both directions.

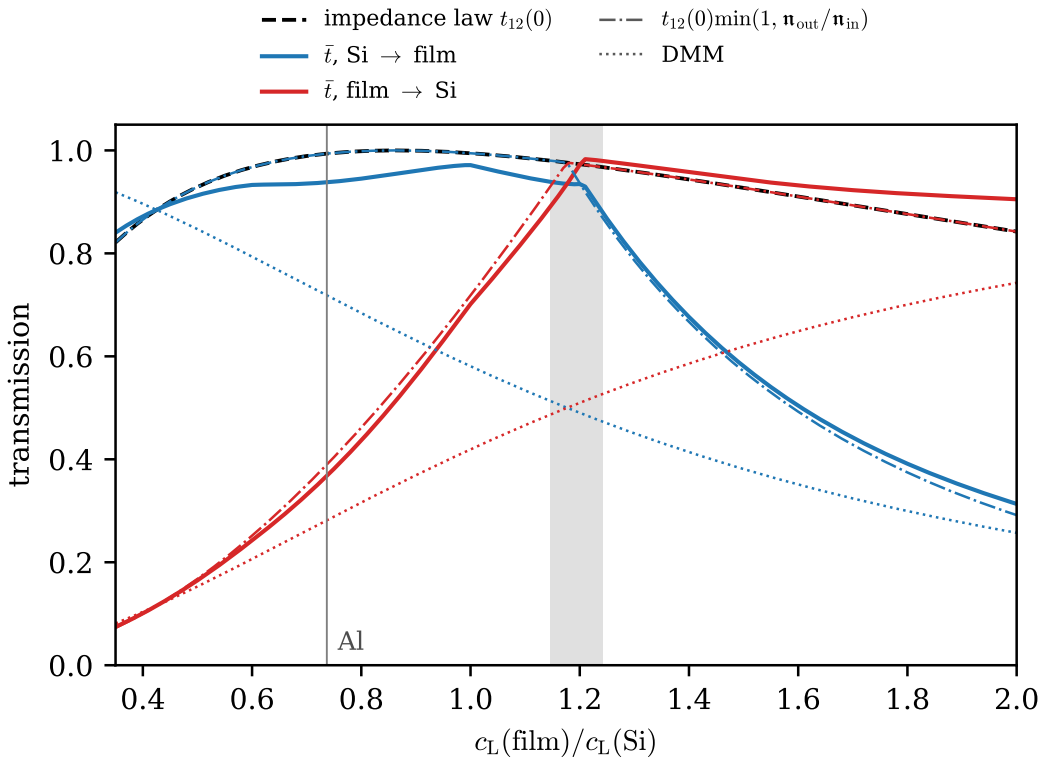


Figure 20.10: Transmission between isotropic silicon and a family of aluminum-like films against the ratio of their longitudinal speeds; the film keeps the density $2.70\,\mathrm{g\,cm^{-3}}$ and the aluminum ratio $c_T/c_{LA} = 0.498$, and $c_{LA}(\text{film})/c_{LA}(\text{Si}) = 0.737$ is aluminum (vertical line). Dashed: the impedance law. Solid: the averages Eq. (20.27) from silicon into the film (blue) and back (red). Dash-dot: the leading correction, reciprocity Eq. (20.28) applied to the impedance law on the slower side, $t_{12}(0)\min(1, \mathfrak{n}_{out}/\mathfrak{n}_{in})$, within 6.9 % of the averages over the whole range. Dotted: the diffuse mismatch Eq. (20.29). Gray: speed ratios 1.146 to 1.242, where the impedance law lies within 10 % of the averages in both directions. In that range $\mathfrak{n}_{film} \approx \mathfrak{n}_{Si}$.

Away from equal interface densities the critical angles of the slower medium's phonons, which the impedance law ignores, remove most of the flux in one direction: for aluminum on silicon the law overstates the escape from the film by a factor 2.70, while reciprocity applied to it, $t_{12}(0)\,\mathfrak{n}_{Si}/\mathfrak{n}_{Al} = 0.390$, comes within 6 % of 0.368.

20.3.3 Film absorption and escape time

Inside a superconducting film a phonon of energy E_{ath} above $2\Delta_{BCS}$ breaks a Cooper pair after a mean free path $\ell_f(E_{ath})$ (Appendix E.12). Take a film of thickness d_f with a free top face and the substrate below. A phonon that moves at the angle θ to the normal travels $2d_f/\cos\theta$ between two arrivals at the interface, where it escapes with probability t_{esc}. If it keeps its angle, each round trip is survived with probability $e^{-y_f/\cos\theta}$ and ends with an escape with probability t_{esc}. Summing the geometric series of round trips, the probability that the phonon is absorbed before escaping is

$$P_{abs}(\theta) = \frac{1 - e^{-y_f/\cos\theta}}{1 - (1 - t_{esc})\,e^{-y_f/\cos\theta}}, \qquad y_f = \frac{2d_f}{\ell_f}. \tag{20.30}$$

A polycrystalline film with a rough top face changes the direction of a phonon at every pass. Each round trip then starts from the flux distribution $2\sin\theta\cos\theta$ of Eq. (20.27), and with $\mu = \cos\theta$ the mean survival over one round trip is

$$\int_0^1 2\mu\, \mathrm{e}^{-y_\mathrm{f}/\mu}\, \mathrm{d}\mu = 2\int_1^\infty \frac{\mathrm{e}^{-y_\mathrm{f} s}}{s^3}\, \mathrm{d}s = 2E_3(y_\mathrm{f}),$$

with $s = 1/\mu$ and E_3 the exponential integral of order three. The escape probability becomes the average $\bar{t}_\mathrm{esc} = \bar{t}_{\mathrm{f}\to\mathrm{sub}}$ of Eq. (20.27), and the absorption probability per entry is

$$P_\mathrm{abs} = \frac{1 - 2E_3(y_\mathrm{f})}{1 - (1 - \bar{t}_\mathrm{esc})\, 2E_3(y_\mathrm{f})}. \tag{20.31}$$

For a thin film, $2E_3(y_\mathrm{f}) = 1 - 2y_\mathrm{f} + O(y_\mathrm{f}^2 \ln y_\mathrm{f})$ and $P_\mathrm{abs} \simeq 4d_\mathrm{f}/(\mathfrak{l}_\mathrm{f}\bar{t}_\mathrm{esc})$: the phonon makes on average $1/\bar{t}_\mathrm{esc}$ round trips of mean length $4d_\mathrm{f}$ before it escapes.

The same counting in time gives the escape time. Let the film hold N_ν phonons of branch ν per unit volume, distributed isotropically. They strike unit area of the interface at the rate $N_\nu c_\nu/4$, of which the fraction $\langle t_\nu\rangle$, the angle integral in Eq. (20.27), escapes. The film holds $d_\mathrm{f}\sum_\nu N_\nu$ phonons per unit area. Let the branches be populated as in equilibrium at a fixed frequency, $N_\nu \propto c_\nu^{-3}$ by Eq. (5.22). The escape rate is then

$$\frac{1}{t_l} = \frac{\sum_\nu N_\nu c_\nu \langle t_\nu\rangle}{4d_\mathrm{f}\sum_\nu N_\nu} = \frac{\mathfrak{n}_\mathrm{f}\,\bar{t}_\mathrm{esc}}{4d_\mathrm{f}\sum_\nu c_\nu^{-3}}.$$

Film escape time

A phonon population that is isotropic and mixed among branches leaves a film of thickness d_f at the rate $1/t_l$, with

$$t_l = \frac{4d_\mathrm{f}}{v_s\,\bar{t}_\mathrm{esc}}, \qquad v_s = \frac{\sum_\nu c_\nu^{-2}}{\sum_\nu c_\nu^{-3}}, \tag{20.32}$$

where v_s is the mean speed of the film's phonons weighted by their density of states and $\bar{t}_\mathrm{esc}$ the mean transmission into the substrate, Eq. (20.27).

Eq. (20.32) assumes that the population is randomized in direction and branch faster than it escapes, that the transmissions are frequency independent, that the top face loses nothing, and that $\bar{t}_\mathrm{esc}$ is the transmission of a clean welded interface, or whatever replaces it. Without the mixing, the 56 % of the flux beyond the transverse critical angle would never leave, and frequency independence needs wavelengths below d_f. With the pair-breaking time $t_\mathrm{pb} = \mathfrak{l}_\mathrm{f}/v_s$, the thin-film limit of Eq. (20.31) reads $P_\mathrm{abs} \simeq t_l/t_\mathrm{pb}$, the first term of the competition $t_l/(t_l + t_\mathrm{pb})$ between escape and pair breaking.

Detector simulations treat the film in a lumped form in which a phonon escapes unabsorbed with probability $\exp(-4d_\mathrm{f}/\mathfrak{l}_\mathrm{f})$ after one round trip [9, Eq. (17)] (Appendix E.12). That is Eq. (20.31) with $\bar{t}_\mathrm{esc} = 1$ and the angle average taken in the exponent. At $\bar{t}_\mathrm{esc} = 1$ it lies above Eq. (20.31) everywhere, by up to 14 %, and within 10 % of it for $d_\mathrm{f}/\mathfrak{l}_\mathrm{f} < 0.074$ and again above 0.55. The larger difference, however, is the escape probability. For aluminum on silicon, with $\bar{t}_\mathrm{esc} = 0.368$, a film in the thin-film limit absorbs $1/\bar{t}_\mathrm{esc} = 2.72$ times more than the lumped form says; at the thickness of Worked example 20.3 the factor is 1.82.

Worked example 20.3: Escape from a 60 nm aluminum film on silicon

The film phonons have $c_{\mathrm{LA}} = 6.63$ and $c_{\mathrm{T}} = 3.30\,\mathrm{km\,s^{-1}}$. The sums in Eq. (20.32) are $\sum_\nu c_\nu^{-2} = \mathfrak{n}_{\mathrm{Al}} = 0.2062\,\mathrm{s^2\,km^{-2}}$ and $\sum_\nu c_\nu^{-3} = 0.0590\,\mathrm{s^3\,km^{-3}}$, so $v_s = 3.50\,\mathrm{km\,s^{-1}}$, close to the transverse speed because transverse modes dominate the density of states.
The escape probability is the average from aluminum into silicon. The acoustic mismatch model gives $\bar{t}_{\mathrm{esc}} = 0.368$, from 0.624 for longitudinal, 0.313 for SV and 0.359 for SH phonons; the diffuse mismatch model gives 0.282. With $d_{\mathrm{f}} = 60\,\mathrm{nm}$,

$$t_l = \frac{4 \times 60\,\mathrm{nm}}{3.50\,\mathrm{km\,s^{-1}} \times 0.368} = 0.187\,\mathrm{ns} \quad \text{(acoustic)}, \qquad 0.244\,\mathrm{ns} \quad \text{(diffuse)},$$

a factor 1.31 between the two models. Near $2\Delta_{\mathrm{BCS}}$ a phonon in aluminum breaks a pair in the time $t_0 \approx 0.26\,\mathrm{ns}$, the characteristic phonon lifetime of Kaplan and coworkers [18], [19, p. 2], which Sect. 20.4 defines as the pair-breaking time at threshold. Hence $t_l/t_0 = 0.72$ (acoustic) or 0.94 (diffuse). A 60 nm film loses its phonons about as fast as they break pairs, which is the regime in which Sect. 20.4 finds the quasiparticle yield sensitive to t_l/t_0. If the mean free path were $\ell_{\mathrm{f}} = v_s t_0 = 0.909\,\mu\mathrm{m}$, then $y_{\mathrm{f}} = 0.132$ and Eq. (20.31) would give $P_{\mathrm{abs}} = 0.422$, against $t_l/(t_l + t_0) = 0.418$; the lumped form gives 0.232, a factor 1.82 less. The room-temperature elastic constants of aluminum, $C_{11} = 106.8$, $C_{12} = 60.7$ and $C_{44} = 28.2\,\mathrm{GPa}$ (Sect. 6.4), give 6.41 and $3.11\,\mathrm{km\,s^{-1}}$, $\bar{t}_{\mathrm{esc}} = 0.327$ (acoustic) and 0.259 (diffuse), and $t_l = 0.223$ and 0.281 ns, or $t_l/t_0 = 0.86$ and 1.08: the zero-temperature constants that apply to a sensor at millikelvin shorten t_l by 16 %.

20.3.4 The fitted silicon-to-aluminum coefficient

Martinez and coworkers illuminated the back of silicon wafers 300 to 380 µm thick, 2 cm square, that carried 60 nm aluminum resonators. They compared the measured phonon pulses with a Monte Carlo simulation using specular reflection, and the silicon-to-aluminum coefficient that reproduced the data lay between 0.30 and 0.55 [15, Sect. V]. Their simulation does not propagate phonons inside the aluminum. A phonon that crosses produces a signal with the pair-breaking efficiency or is removed, and the fitted coefficient's complement includes phonons that enter the film and come back [15, Sect. IV]. The fitted coefficient is therefore the probability that a phonon arriving from silicon is absorbed in the film,

$$T_{\mathrm{fit}} = \bar{t}_{\mathrm{Si}\to\mathrm{Al}}\, P_{\mathrm{abs}}. \tag{20.33}$$

With the acoustic-mismatch value 0.938, the fit gives $P_{\mathrm{abs}} = 0.320$ to 0.586. Inverting Eq. (20.31) with $\bar{t}_{\mathrm{esc}} = 0.368$ gives $\ell_{\mathrm{f}}/d_{\mathrm{f}} = 23.1$ to 8.3, or $\ell_{\mathrm{f}} = 1.39$ to 0.50 µm, and $t_{\mathrm{pb}} = \ell_{\mathrm{f}}/v_s = 0.40$ to 0.14 ns, the same order as t_0. Keeping the angle of each phonon instead, Eq. (20.30) averaged with $t_{\mathrm{esc}} = \bar{t}_{\mathrm{esc}}$ gives 1.29 to 0.46 µm; the diffuse-mismatch pair, 0.718 and 0.282, gives 1.19 to 0.31 µm. The fit thus constrains the pair-breaking length over the thickness, $\ell_{\mathrm{f}}/d_{\mathrm{f}} \approx 5$ to 23 across both models, and says little about the interface. The reason is reciprocity: in the thin-film limit $T_{\mathrm{fit}} \to (\mathfrak{n}_{\mathrm{Al}}/\mathfrak{n}_{\mathrm{Si}})\, 4d_{\mathrm{f}}/\ell_{\mathrm{f}}$ for any interface that conserves energy, by Eqs. (20.28) and (20.31), and the interface model enters only through the finite-thickness corrections. Read as a transmission, the fit would put the escape probability at $(\mathfrak{n}_{\mathrm{Si}}/\mathfrak{n}_{\mathrm{Al}}) \times (0.30 \text{ to } 0.55) = 0.118$ to 0.216 by Eq. (20.28), below both models. The inferred lengths are estimates: they inherit the specular, mode-conserving simulation of the fit and the isotropic averages of this section. An earlier silicon detector model matched its data with a 33 % probability for a phonon to be absorbed at an amorphous-silicon/aluminum interface,

the rest scattering back diffusely [6, Sect. III G]; that coefficient is an absorption probability of the same kind.

20.3.5 Anisotropy and interlayers

The isotropic averages hide the anisotropy of silicon, whose longitudinal speed ranges from 8.43 to 9.35 km s^{-1} and whose transverse speeds from 4.67 to 5.84 km s^{-1} (Eq. (C.157)). Giving the isotropic silicon its [100] speeds, 8.43 and 5.84 km s^{-1}, or its [111] speeds, 9.35 and 5.09 km s^{-1}, moves $\bar{t}_{\mathrm{esc}}$ from 0.368 to 0.327 and 0.405, and t_l for 60 nm from 0.187 to 0.210 and 0.169 ns; $\bar{t}_{\mathrm{Si}\to\mathrm{Al}}$ moves only between 0.927 and 0.943. These are sensitivities, not bounds. The texture of the film matters less, since it changes the longitudinal speed of aluminum by at most 3 % (Appendix C.18), and the Hill average in place of the Voigt average moves t_l by 0.6 %, as one expects for so nearly isotropic a metal. In the anisotropic problem the critical angles depend on the direction in the interface plane, the group velocity is not parallel to $\boldsymbol{Q}$, and the plane waves on each side follow from the Christoffel equation Eq. (C.156); the boundary problem becomes six by six, and Appendix E.25 specifies it. The reciprocity Eq. (20.28) holds there unchanged.

Real interfaces carry a native oxide or a deposited amorphous layer a few nanometers thick. A flat layer that does not dissipate cannot change which waves propagate in silicon, since $Q_{\parallel}$ is conserved through every layer: the critical angles and the 56 % of trapped film flux stay as they are. The layer changes the transmission inside the cone. At normal incidence take a layer of impedance $Z_{\mathrm{ac},3}$, speed c_3 and thickness h. At the quarter-wave frequency $c_3/4h$ the wave crosses it with the phase $\pi/2$, and the layer maps the particle velocity $\dot{u}$ and the force $-\sigma_{zz}$ at its far face onto $\mathrm{i}(-\sigma_{zz})/Z_{\mathrm{ac},3}$ and $\mathrm{i}Z_{\mathrm{ac},3}\dot{u}$ at its near face. A medium of impedance $Z_{\mathrm{ac},2}$ behind it then looks like one of impedance $Z_{\mathrm{ac},3}^2/Z_{\mathrm{ac},2}$, and Eq. (C.161) with this replacement gives the lowest transmission,

$$t_{\mathrm{min}} = \frac{4 Z_{\mathrm{ac},1} Z_{\mathrm{ac},2} Z_{\mathrm{ac},3}^2}{\left(Z_{\mathrm{ac},1} Z_{\mathrm{ac},2} + Z_{\mathrm{ac},3}^2\right)^2},$$

since the transmission oscillates between this value and the bare $t_{12}(0)$ as the frequency varies. Amorphous SiO_2, with $\rho_T = 2.2\,\mathrm{g\,cm^{-3}}$ and $\rho_T c_{\mathrm{LA}}^2 = 7.40 \times 10^{10}\,\mathrm{Pa}$ (Sect. 16.7), has $c_{\mathrm{LA}} = 5.80\,\mathrm{km\,s^{-1}}$ and $Z_{\mathrm{ac}} = 12.8 \times 10^6\,\mathrm{kg\,m^{-2}\,s^{-1}}$. Between silicon and aluminum it gives $t_{\mathrm{min}} = 0.844$, against 0.994 without the layer, at 0.72 THz for a 2 nm layer; a 1 nm layer transmits 0.901 at 0.8 THz. A rough or disordered layer scatters diffusely instead, does not conserve $Q_{\parallel}$, and can release trapped modes; the diffuse mismatch model is the limit of complete randomization. Until the escape probability of a real film is measured, the spread from 0.282 to 0.368, a factor 1.31 in t_l, is the uncertainty to carry.

20.4 Quasiparticle yield in superconductors

A phonon of 0.59 THz, at the soft end of what a silicon crystal delivers to its sensors after down-conversion (Sect. 20.5.2), carries 2.45 meV into an aluminum film, 13.6 times the gap $\Delta_{\mathrm{BCS}} = 0.18\,\mathrm{meV}$ of Sect. 7.2. The sensor counts quasiparticles, and each of them, once it has relaxed to the gap edge, holds Δ_{BCS}. Were all of the phonon's energy spent on breaking pairs, it would make 13.6 quasiparticles. In a film thick enough to keep every phonon it makes about eight, and in a thin film on a substrate fewer than five. The fraction of the phonon energy that ends up in quasiparticles, the quasiparticle or pair-breaking efficiency Y_{qp}, multiplies every phonon collection efficiency ϵ of Sect. 19.5, and it depends on the phonon energy. We will derive it exactly below four gaps and write the cascade of phonons and quasiparticles

that sets it above. Above four gaps it falls with the time t_l that a phonon spends in the film before escaping, which Sect. 20.3 relates to the film thickness and the interface.

20.4.1 Pair breaking below four gaps

Let a phonon of energy $E_{\mathrm{ath}} \geq 2\Delta_{\mathrm{BCS}}$ break a pair in a film at a temperature far below T_c, where thermal quasiparticles are absent (Eq. (7.27)). Within 0.1 to 10 ns [19, p. 1] every quasiparticle it has produced has relaxed to the gap edge, and the rest of its energy has left as phonons that can no longer break pairs. If N_{qp} quasiparticles are left, the yield of the event is

$$Y_{\mathrm{qp}}(E_{\mathrm{ath}}) \equiv \frac{\Delta_{\mathrm{BCS}}\langle N_{\mathrm{qp}}\rangle}{E_{\mathrm{ath}}} = 1 - \frac{\langle E_{\mathrm{sub}} + E_{\mathrm{leak}}\rangle}{E_{\mathrm{ath}}}, \tag{20.34}$$

where E_{sub} is the energy carried off by phonons below $2\Delta_{\mathrm{BCS}}$ and E_{leak} that of phonons above $2\Delta_{\mathrm{BCS}}$ that leave the film before they break a pair. The second form is the energy balance $E_{\mathrm{ath}} = \Delta_{\mathrm{BCS}}N_{\mathrm{qp}} + E_{\mathrm{sub}} + E_{\mathrm{leak}}$, event by event. We count the yield for a phonon that has broken its first pair. Whether an arriving phonon is absorbed at all is the absorption probability of Sect. 20.3, and a sensor efficiency is the product of the two.

The split of the pair energy, the pair-breaking rate and the emission rate of a hot quasiparticle all follow from the golden rule with the coherence factors of Eq. (7.29). The phonon couples to the electron density, so its absorption is case I. Its momentum is E_{ath} divided by the sound speed. Since the longitudinal sound speed of aluminum, 6.63 km/s (Sect. 20.3), is 305 times below $v_{\mathrm{F}} = 6.75 \times 10^{-3}$, or 2.0×10^6 m/s (Sect. 7.2), the condition $v_{\mathrm{F}}q \geq E_{\mathrm{ath}}$ of Eq. (7.30) holds with a wide margin. The energies ξ and ξ' of the two electrons are then independent, the terms odd in them average out, and the creation factor is $\frac{1}{2}(1 + \Delta_{\mathrm{BCS}}^2/EE')$. Each final quasiparticle is weighted with its density of states Eq. (7.25). The probability that one of the two receives the energy E, and the other $E_{\mathrm{ath}} - E$, is then

$$P_{\mathrm{pb}}(E \,|\, E_{\mathrm{ath}}) = \frac{1}{E_{\mathrm{ath}}\mathfrak{g}_{\mathrm{I}}(E_{\mathrm{ath}}/\Delta_{\mathrm{BCS}})} \frac{E(E_{\mathrm{ath}} - E) + \Delta_{\mathrm{BCS}}^2}{\sqrt{E^2 - \Delta_{\mathrm{BCS}}^2}\sqrt{(E_{\mathrm{ath}} - E)^2 - \Delta_{\mathrm{BCS}}^2}}, \tag{20.35}$$

for $\Delta_{\mathrm{BCS}} \leq E \leq E_{\mathrm{ath}} - \Delta_{\mathrm{BCS}}$. The normalization is the case-I ratio $\mathfrak{g}_{\mathrm{I}} = S_{\mathrm{I}}/S_{\mathrm{n}}$ of Eqs. (7.30) and (15.7): the integral of the numerator over the denominator is $E_{\mathrm{ath}}\mathfrak{g}_{\mathrm{I}}$. The split is symmetric about $E_{\mathrm{ath}}/2$ and peaked at both ends, where the density of states diverges. A photon couples through the current, which is case II, and its split has $-\Delta_{\mathrm{BCS}}^2$ in the numerator [19, Eq. (1)].

The same ratio fixes how fast the phonon breaks a pair. In the normal metal a phonon decays into an electron–hole pair at the rate given by Allen's formula Eq. (6.33). The superconductor changes only the phase space of the pair and its coherence factor, which together multiply the normal rate by $\mathfrak{g}_{\mathrm{I}}$. At low energies the Eliashberg function of a metal grows as $\alpha^2F = b_\alpha E_{\mathrm{ath}}^2$, the form behind the factor quadratic in the phonon energy in the spectrum of emitted phonons given by Leman [6, Sect. IV]. With a phonon density of states $\propto E_{\mathrm{ath}}^2$, this α^2F gives a normal-state width linear in E_{ath} (Eq. (6.35)). Measuring the pair-breaking time t_{pb} in units of its value t_0 at threshold, where $\mathfrak{g}_{\mathrm{I}} = \pi/2$, we find

$$\frac{1}{t_{\mathrm{pb}}(E_{\mathrm{ath}})} = \frac{E_{\mathrm{ath}}\,\mathfrak{g}_{\mathrm{I}}(E_{\mathrm{ath}}/\Delta_{\mathrm{BCS}})}{\pi\Delta_{\mathrm{BCS}}\,t_0}, \tag{20.36}$$

the zero-temperature pair-breaking rate. Since $\mathfrak{g}_{\mathrm{I}}$ falls from $\pi/2$ to one, the rate rises linearly with energy at large E_{ath}, the approximate scaling $t_{\mathrm{pb}} \propto 1/E_{\mathrm{ath}}$ of Kaplan et al. [18] as quoted by Guruswamy et al. [19, p. 5], and a phonon of $10\Delta_{\mathrm{BCS}}$ breaks a pair in $0.30\,t_0$. For aluminum, Guruswamy et al. take $t_0 = 0.26$ ns [19, p. 3]. The conserving correction of

Sect. 7.2 plays no role here: it is controlled by $\Delta_{\mathrm{BCS}}/v_{\mathrm{F}}q$, at most 1.6×10^{-3} for a phonon above the threshold.

A quasiparticle of energy E relaxes by emitting a phonon of energy E_{ath} and dropping to $E-E_{\mathrm{ath}}$. This is quasiparticle scattering, with the case-I factor $\frac{1}{2}(1-\Delta_{\mathrm{BCS}}^2/EE')$ after the same average, and the rate per unit phonon energy is

$$\Gamma_{\mathrm{qp,em}}(E_{\mathrm{ath}}\,|\,E)\propto\alpha^2F(E_{\mathrm{ath}})\,\frac{E-E_{\mathrm{ath}}}{\sqrt{(E-E_{\mathrm{ath}})^2-\Delta_{\mathrm{BCS}}^2}}\Big[1-\frac{\Delta_{\mathrm{BCS}}^2}{E(E-E_{\mathrm{ath}})}\Big], \tag{20.37}$$

for $0\le E_{\mathrm{ath}}\le E-\Delta_{\mathrm{BCS}}$, again a rate first given by Kaplan et al. [18]; with $\alpha^2F\propto E_{\mathrm{ath}}^2$ it is the phonon distribution quoted by Leman [6, Sect. IV]. The middle factor is the density of final quasiparticle states, Eq. (7.25) over its normal value. The overall scale, set by the electron–phonon coupling and expressed as a characteristic time of 438 ns for aluminum [19, p. 3], does not affect the yield, because a quasiparticle stays in the film whatever time it takes to relax. At zero temperature it can only lose energy, and it does so until it reaches the gap edge.

Pair-breaking efficiency below four gaps

A phonon of energy E_{ath} that breaks a pair in a superconductor at $T\ll T_c$ leaves exactly two quasiparticles if $E_{\mathrm{ath}}<4\Delta_{\mathrm{BCS}}$, and

$$Y_{\mathrm{qp}}(E_{\mathrm{ath}})=\frac{2\Delta_{\mathrm{BCS}}}{E_{\mathrm{ath}}},\qquad 2\Delta_{\mathrm{BCS}}\le E_{\mathrm{ath}}<4\Delta_{\mathrm{BCS}}, \tag{20.38}$$

whatever the film thickness, the shape of α^2F and the split Eq. (20.35).

The proof is kinematic. A quasiparticle at E can emit a phonon that breaks a further pair only if the phonon has at least $2\Delta_{\mathrm{BCS}}$ and the quasiparticle is left with at least Δ_{BCS}, that is, only if $E\ge3\Delta_{\mathrm{BCS}}$. The hotter of the two quasiparticles from the first pair has at most $E_{\mathrm{ath}}-\Delta_{\mathrm{BCS}}$, since its partner has at least Δ_{BCS}. A third quasiparticle therefore requires $E_{\mathrm{ath}}\ge4\Delta_{\mathrm{BCS}}$. Below that energy each quasiparticle relaxes by emitting phonons of less than $2\Delta_{\mathrm{BCS}}$, which cannot break pairs, and $E_{\mathrm{ath}}-2\Delta_{\mathrm{BCS}}$ leaves as E_{sub}, while $E_{\mathrm{leak}}=0$. Escape cannot compete, because no phonon of the event other than the first is able to break a pair. Guruswamy et al. find the same result in the kinetic equations, a yield that "reduces monotonically and is independent of the phonon loss time" between two and four gaps [19, p. 5]. The argument uses only the threshold Eq. (7.27) and the loss of energy by quasiparticles, so it holds for a photon and for a dark-matter particle that deposits ω in an aluminum target (Sect. 15.1) as well. Each phonon and each photon of the band from $2\Delta_{\mathrm{BCS}}$ to $4\Delta_{\mathrm{BCS}}$, 87 to 174 GHz in aluminum, makes exactly two quasiparticles, and a sensor that counts them has no fluctuation from the film at all.

Worked example 20.4: The yield of a $3\Delta_{\mathrm{BCS}}$ phonon in aluminum

With $\Delta_{\mathrm{BCS}}=0.18$ meV a phonon of $3\Delta_{\mathrm{BCS}}$ has 0.54 meV, or 131 GHz. It lies above the pair-breaking threshold $2\Delta_{\mathrm{BCS}}=0.36$ meV (87.0 GHz). From Eq. (20.36) with $\mathfrak{g}_{\mathrm{I}}(3)=1.322$ (Sect. 15.1), it breaks a pair after $t_{\mathrm{pb}}=\pi t_0/(3\times1.322)=0.792\,t_0=0.206$ ns on average, with $t_0=0.26$ ns. By Eq. (20.35) one quasiparticle receives an energy E between Δ_{BCS} and $2\Delta_{\mathrm{BCS}}$ and the other $3\Delta_{\mathrm{BCS}}-E$, with both ends of that interval favored. The hotter one has at most $2\Delta_{\mathrm{BCS}}$, less than the $3\Delta_{\mathrm{BCS}}$ it would need to emit a pair-breaking phonon. Each quasiparticle relaxes by Eq. (20.37), emitting phonons of at most $E-\Delta_{\mathrm{BCS}}\le\Delta_{\mathrm{BCS}}$, that is below 43.5 GHz. The event ends with $N_{\mathrm{qp}}=2$ every

time, and

$$Y_{\mathrm{qp}}(3\Delta_{\mathrm{BCS}}) = \frac{2}{3}, \qquad E_{\mathrm{sub}} = \Delta_{\mathrm{BCS}} = 0.18\,\mathrm{meV}, \qquad E_{\mathrm{leak}} = 0.$$

The film thickness enters only through the chance that the phonon is absorbed before it escapes, the absorption probability of Sect. 20.3. Once it is absorbed, a film of 30 nm and one of 1 µm give the same two quasiparticles. The 2.45 meV phonon of the opening, at $13.6\,\Delta_{\mathrm{BCS}}$, already has a yield between 0.31 and 0.59 depending on the film (Sect. 20.4.3).

20.4.2 The Kaplan cascade

Above $4\Delta_{\mathrm{BCS}}$ the hot quasiparticles emit phonons that break further pairs, those pairs yield further hot quasiparticles, and the energy comes down in generations, with phonons and quasiparticles alternating. Detector physicists call this the Kaplan cascade, after the rates of Kaplan et al. [18]. Every phonon above $2\Delta_{\mathrm{BCS}}$ now either breaks a pair after t_{pb} or leaves the film after the escape time $t_l = 4d_{\mathrm{f}}/(v_s \bar{t}_{\mathrm{esc}})$ of Sect. 20.3. In the lumped-film model the film is one object: it holds a phonon for an exponentially distributed time of mean t_l, whatever the phonon's energy and position [19, p. 2]. The two processes compete as independent decays, and a phonon breaks a pair with the probability

$$p_{\mathrm{pb}}(E_{\mathrm{ath}}) = \frac{1}{1 + t_{\mathrm{pb}}(E_{\mathrm{ath}})/t_l} = \frac{E_{\mathrm{ath}}\mathfrak{g}_{\mathrm{I}}}{E_{\mathrm{ath}}\mathfrak{g}_{\mathrm{I}} + \pi\Delta_{\mathrm{BCS}}\, t_0/t_l}. \tag{20.39}$$

Hard phonons break pairs before they escape, since t_{pb} falls as $1/E_{\mathrm{ath}}$. The losses, however, concentrate on the phonons a few gaps above threshold, and every cascade passes through them.

In the lumped film a quasiparticle never leaves, so the only times that compete are t_{pb} and t_l, and the yield depends on the film through t_l/t_0 alone. Since the quasiparticles are dilute during the cascade and the temperature is far below T_c, every particle develops independently of the others. The cascade is then a branching process of the kind treated in Sect. 18.1, and its mean obeys linear equations. Let $\bar{N}_{\mathrm{qp}}(E)$ be the mean number of quasiparticles that end at the gap edge from one quasiparticle of energy E, itself included, and $\bar{N}_{\mathrm{ph}}(E_{\mathrm{ath}})$ the same number for one phonon. Averaging over the first step of each,

$$\begin{aligned} \bar{N}_{\mathrm{qp}}(E) &= \int_0^{E-\Delta_{\mathrm{BCS}}} \mathrm{d}E_{\mathrm{ath}}\, w(E_{\mathrm{ath}} \,|\, E)\, [\bar{N}_{\mathrm{qp}}(E - E_{\mathrm{ath}}) + \bar{N}_{\mathrm{ph}}(E_{\mathrm{ath}})], \\ \bar{N}_{\mathrm{ph}}(E_{\mathrm{ath}}) &= p_{\mathrm{pb}}(E_{\mathrm{ath}}) \int \mathrm{d}E\, P_{\mathrm{pb}}(E \,|\, E_{\mathrm{ath}})\, [\bar{N}_{\mathrm{qp}}(E) + \bar{N}_{\mathrm{qp}}(E_{\mathrm{ath}} - E)] \end{aligned}, \tag{20.40}$$

the first for $E \geq 3\Delta_{\mathrm{BCS}}$ and the second for $E_{\mathrm{ath}} \geq 2\Delta_{\mathrm{BCS}}$, with $\bar{N}_{\mathrm{qp}} = 1$ below $3\Delta_{\mathrm{BCS}}$ and $\bar{N}_{\mathrm{ph}} = 0$ below $2\Delta_{\mathrm{BCS}}$. Here $w = \Gamma_{\mathrm{qp,em}}/\int \Gamma_{\mathrm{qp,em}}$ is the normalized spectrum of the emitted phonon from Eq. (20.37). The yield of a phonon that has broken its first pair is $Y_{\mathrm{qp}} = \Delta_{\mathrm{BCS}}\bar{N}_{\mathrm{ph}}(E_{\mathrm{ath}})/[p_{\mathrm{pb}}(E_{\mathrm{ath}})E_{\mathrm{ath}}]$. The equations close on themselves from below: $\bar{N}_{\mathrm{ph}}$ at E_{ath} needs $\bar{N}_{\mathrm{qp}}$ only up to $E_{\mathrm{ath}} - \Delta_{\mathrm{BCS}}$, and $\bar{N}_{\mathrm{qp}}$ at E needs $\bar{N}_{\mathrm{ph}}$ up to $E - \Delta_{\mathrm{BCS}}$. Below $4\Delta_{\mathrm{BCS}}$ they return Eq. (20.38) at once, since $\bar{N}_{\mathrm{qp}} = 1$ for both quasiparticles of the first pair. The only material inputs are the shape of α^2F below E_{ath} and the ratio t_l/t_0; the coupling strength b_α cancels.

We solve Eq. (20.40) on an energy grid of step $0.01\Delta_{\mathrm{BCS}}$ with $\alpha^2F = b_\alpha E_{\mathrm{ath}}^2$. Substituting $E = \Delta_{\mathrm{BCS}} + (E_{\mathrm{ath}} - 2\Delta_{\mathrm{BCS}})\sin^2\theta$ in the split, and a quadratic map onto the edge in the emission integral, removes the inverse-square-root edges of the densities of states, and we

iterate from $\bar{N}_{\mathrm{qp}} = 1$ to convergence. An event-by-event simulation with the same kernels agrees with the solution within 1.2 standard errors, about 0.001. The low-frequency form of α^2F restricts the result to phonon energies well below the transverse peak of the aluminum Eliashberg function at 23 meV (Sect. 6.4), and we stop at $60\Delta_{\mathrm{BCS}} = 10.8\,\mathrm{meV}$. In practice the cascade is run inside detector simulations (Appendix E.12).

In the limit $t_l \to \infty$ nothing escapes. The yield then rises from 0.5 at $4\Delta_{\mathrm{BCS}}$ to a maximum of 0.600 at $5.46\,\Delta_{\mathrm{BCS}}$ and settles at 0.590, between 0.5901 and 0.5905 over the whole range from 10 to $60\Delta_{\mathrm{BCS}}$. This is the infinite-volume value that earlier calculations found for aluminum, niobium and tin, 0.57 to 0.6 [19, p. 2]. About 41 % of a hard phonon's energy is therefore lost even without escape. The loss goes into the sub-gap phonons that every quasiparticle emits on its last steps to the edge, and into the quasiparticles that end between Δ_{BCS} and $3\Delta_{\mathrm{BCS}}$ and have to shed their excess without breaking a pair. The asymptote is independent of energy because the cascade is self-similar while $E \gg \Delta_{\mathrm{BCS}}$. A quasiparticle far above the gap emits phonons with the spectrum E_{ath}^2 on $[0, E]$, which has the mean $3E/4$, and every generation passes energy down in the same proportions until the last few generations near the gap, which set the loss. The split Eq. (20.35) of the first pair matters only near threshold. Without escape, a phonon seed and a photon seed differ by 0.007 at $5\Delta_{\mathrm{BCS}}$ and by less than 0.001 at $10\Delta_{\mathrm{BCS}}$.

20.4.3 Thin films and the Chang–Scalapino equations

In a film the ratio t_l/t_0 decides how much more is lost. Fig. 20.11(a) shows the yield from Eq. (20.40) for t_l/t_0 from 0.5 to 8 and in the infinite-volume limit. For $t_l/t_0 = 4$, a phonon breaks a pair before it escapes with the probability Eq. (20.39) of 0.84 at $3\Delta_{\mathrm{BCS}}$ and 0.93 at $10\Delta_{\mathrm{BCS}}$, and these losses compound over the generations. Above $4\Delta_{\mathrm{BCS}}$ the yield first rises a little, to 0.560 at $5.2\,\Delta_{\mathrm{BCS}}$ for $t_l/t_0 = 4$, and then falls: at $10\Delta_{\mathrm{BCS}}$ it is 0.349, 0.415, 0.477, 0.524 and 0.554 for $t_l/t_0 = 0.5$, 1, 2, 4 and 8, against 0.590 without escape. For $t_l/t_0 = 0.5$ the rise has disappeared, and the yield falls from the start, to 0.238 at $60\Delta_{\mathrm{BCS}}$. The 60 nm film of Worked example 20.3, at $t_l/t_0 = 0.72$ to 0.94, has 0.383 to 0.409 at $10\Delta_{\mathrm{BCS}}$. Two effects compete above four gaps. A hotter phonon can make more pairs, which raises the yield, but its energy passes through more generations of phonons, each of which can escape. The second effect wins at all energies above about $5\Delta_{\mathrm{BCS}}$ for every finite t_l.

Kinetic equations follow the same physics with fewer approximations. Chang and Scalapino wrote coupled Boltzmann equations for the quasiparticle distribution $f(E)$ and the phonon distribution $n(E_{\mathrm{ath}})$ of a superconductor, with the full energy dependence of the rates at finite temperature and the recombination of quasiparticles into $2\Delta_{\mathrm{BCS}}$ phonons [20, 21]. Guruswamy et al. solved them in steady state, with the phonon loss time t_l, for an aluminum film driven by photons of 2 to $10\Delta_{\mathrm{BCS}}$. They used the measured Eliashberg function, $T/T_c = 0.1$ and a weak microwave probe of the kind that reads out a resonator [19, Fig. 4]. The cascade Eq. (20.40) is their equations linearized about the empty superconductor at zero temperature, with α^2F replaced by its low-frequency form, and the difference between the two measures what the linearization and the low-frequency α^2F leave out. Limit Check 20.7 compares them for a photon seed, so that like is compared with like. For $t_l/t_0 \leq 4$ the kinetic equations lie above the cascade by 0.001 to 0.011 at every energy from 5 to $10\Delta_{\mathrm{BCS}}$. For $t_l/t_0 = 8$ the offset is 0.036 to 0.041, nearly constant in energy, and the kinetic equations return 0.59 to 0.61 there, close to the infinite-volume value. The probe itself raises the yield by about 2 % [19, Sect. V]. Counting the excess energy of the quasiparticles above the gap, rather than Δ_{BCS} for each, would raise it by about 4 % [19, Sect. V]; Eq. (20.34) uses the gap-edge count, as they do.

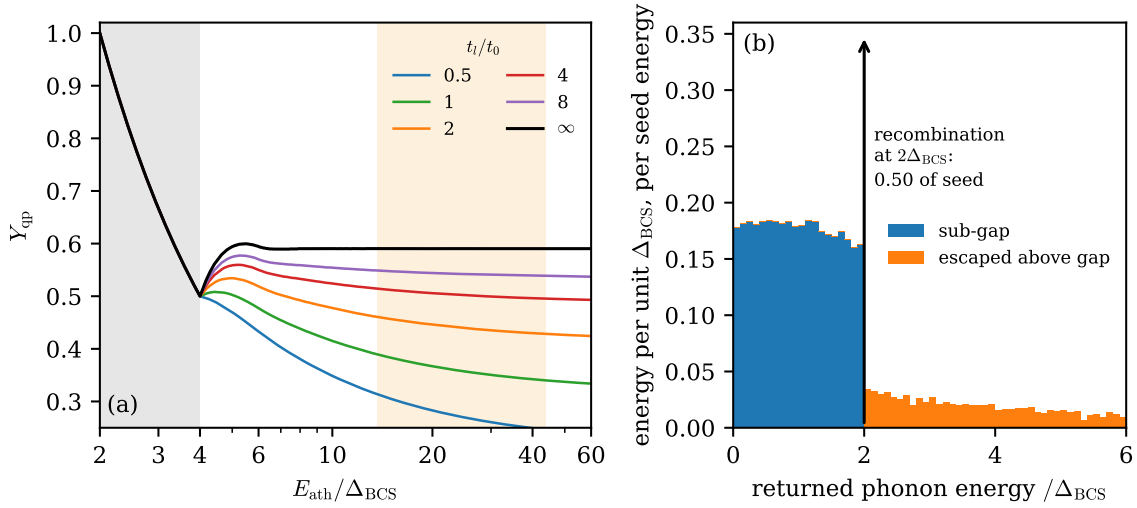


Figure 20.11: Quasiparticle yield of an aluminum film from the Kaplan cascade in a lumped film at zero temperature, Eq. (20.40), with $\alpha^2 F \propto E_{\mathrm{ath}}^2$ and the pair-breaking time Eq. (20.36). (a) Yield $Y_{\mathrm{qp}} = \Delta_{\mathrm{BCS}} \langle N_{\mathrm{qp}} \rangle / E_{\mathrm{ath}}$ of a phonon that has broken its first pair, against its energy in units of the gap, for six ratios of the film escape time to the pair-breaking time at threshold, t_l/t_0. Gray: below $4\Delta_{\mathrm{BCS}}$, where $Y_{\mathrm{qp}} = 2\Delta_{\mathrm{BCS}}/E_{\mathrm{ath}}$ exactly for every film. Tan: 2.45 to 7.88 meV, or 13.6 to 43.8 Δ_{BCS} at $\Delta_{\mathrm{BCS}} = 0.18$ meV (0.59 to 1.91 THz), the band that holds the middle 80 % of the primary-phonon energy absorbed in the mean-field silicon detector of Sect. 20.5.2. (b) Energy spectrum of the phonons that a thick film ($t_l/t_0 = 4$) returns to the substrate when a phonon of $40\Delta_{\mathrm{BCS}}$ cascades in it, per unit energy and per unit seed energy (event simulation, 4000 events). Blue: sub-gap relaxation phonons; orange: phonons above $2\Delta_{\mathrm{BCS}}$ that escape before breaking a pair. Arrow: the share 0.50 held by quasiparticles, which comes back later as recombination phonons of $2\Delta_{\mathrm{BCS}}$.

Limit Check 20.7. Pair-breaking efficiency

A pair-breaking quantum of energy E_{ath} between $2\Delta_{\mathrm{BCS}}$ and $4\Delta_{\mathrm{BCS}}$ yields $Y_{\mathrm{qp}} = 2\Delta_{\mathrm{BCS}}/E_{\mathrm{ath}}$, independent of the film; the infinite-volume yield at high energy is 0.57 to 0.6 [19, pp. 2, 5].

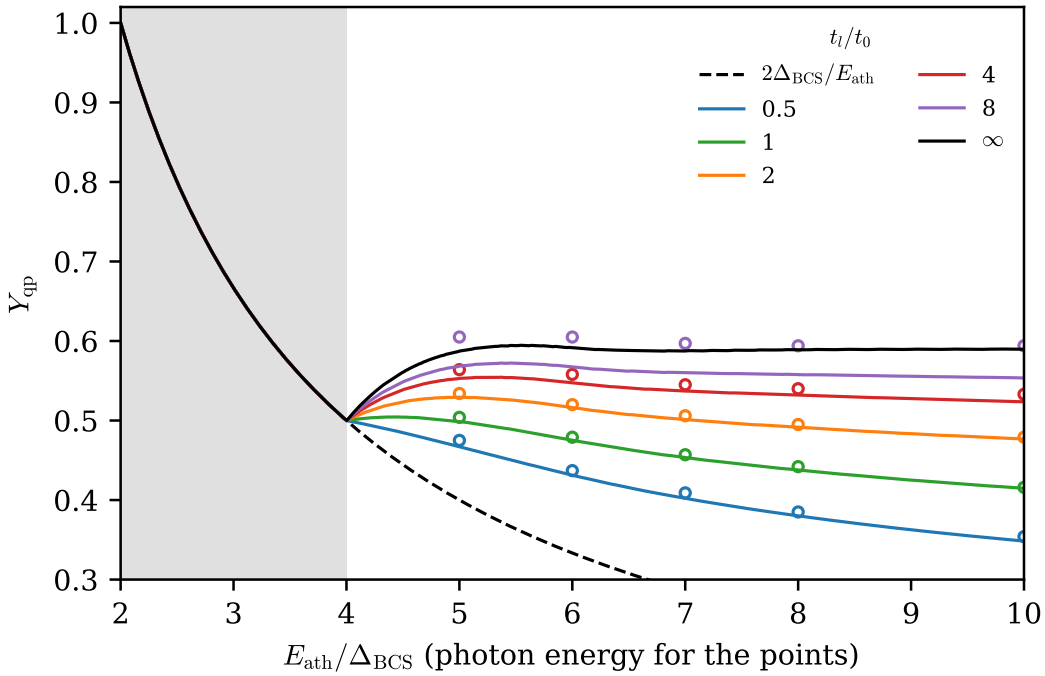


Figure 20.12: Yield of an aluminum film against the energy of the pair-breaking quantum in units of the gap. Dashed: $2\Delta_{\mathrm{BCS}}/E_{\mathrm{ath}}$, Eq. (20.38). Lines: the Kaplan cascade Eq. (20.40) in a lumped film at zero temperature with $\alpha^2F \propto E_{\mathrm{ath}}^2$, for a photon seed (case-II split), at the ratios t_l/t_0 of the escape time to the pair-breaking time at threshold given in the legend. Circles: the coupled kinetic equations of Chang and Scalapino with the measured α^2F of aluminum at $T/T_c = 0.1$ with a microwave probe, read off [19, Fig. 4] at the same t_l/t_0 (colors; digitized to ± 0.005). Gray: $E_{\mathrm{ath}} < 4\Delta_{\mathrm{BCS}}$, where the formula is exact. For $t_l/t_0 \leq 4$ the cascade lies below the circles by at most 0.011; for $t_l/t_0 = 8$, by 0.036 to 0.041.

Above $4\Delta_{\mathrm{BCS}}$ pair-breaking emission sets in, and the yield depends on t_l/t_0 with no closed form.

For detector phonons the energy dependence matters more than any single value. In the mean-field silicon detector of Sect. 20.5.2 the middle 80 % of the absorbed primary energy arrives between 2.45 and 7.88 meV, or 13.6 to 43.8 Δ_{BCS} in aluminum, the tan band of Fig. 20.11(a). Across that band the yield falls by 21.5 % for $t_l/t_0 = 0.5$, by 7.0 % for $t_l/t_0 = 2$ and by 1.9 % for $t_l/t_0 = 8$, and it is flat without escape. Luke phonons, which reach the film softer, are read with the yield near and below four gaps, and primary phonons with the yield of the band. This is the film's contribution to the collection ratio $\mathfrak{r}_{\mathrm{coll}}$ of Sect. 19.5, and Sect. 20.5 combines it with transport in the crystal. For the thin, clean aluminum films of resonator measurements, Guruswamy et al. estimate $t_l/t_0 \approx 0.5$ [19, p. 3]. There the film costs 41 % of the infinite-volume yield at $10\Delta_{\mathrm{BCS}}$, the top of the range they computed, and 47 to 58 % across the band. The 60 nm film of Worked example 20.3 loses 35 to 50 % there, and a thicker film loses less, since $t_l \propto d_{\mathrm{f}}$ (Sect. 20.3).

20.4.4 Recombination phonons from the electrodes

A carrier that reaches an aluminum electrode gives up its energy relative to the Fermi level of the electrode as phonons at the contact (Sect. 19.5). For a Fermi level near midgap this is about half the band gap of the crystal plus the kinetic energy of the carrier. The CDMS II analysis calls these phonons recombination phonons [4, Sect. 2]. In a germanium detector at 3 V they carry 13 % of the phonon signal of an electron recoil, "in a low frequency, ballistic regime that is used to provide a surface-event discriminator" [6, p. 17]. The Kaplan cascade explains why they are soft.

The energy enters the aluminum as a hot electron, far above the Debye energy. It relaxes within the metal by emitting phonons, which break pairs in a fraction of t_0 by Eq. (20.36), and the cascade carries the energy down to the gap before much of it can escape. What leaves the film is of three kinds. Sub-gap phonons, below $2\Delta_{\mathrm{BCS}}$, are emitted by every quasiparticle on its way to the edge. Phonons above $2\Delta_{\mathrm{BCS}}$ escape before they break a pair, mostly within a few gaps of threshold, where t_{pb} is longest. Last, quasiparticles at the edge recombine much later, two at a time, into phonons of $2\Delta_{\mathrm{BCS}}$. In a thick film these may break pairs again and be re-emitted, but every re-emitted phonon still has $2\Delta_{\mathrm{BCS}}$. Fig. 20.11(b) shows the spectrum for a seed of $40\Delta_{\mathrm{BCS}}$ and $t_l/t_0 = 4$. Counting the quasiparticle energy as recombination phonons at $2\Delta_{\mathrm{BCS}}$, 85 % of the returned energy lies at or below $2\Delta_{\mathrm{BCS}}$ and 96 % below $10\Delta_{\mathrm{BCS}}$. For $t_l/t_0 = 8$ the fractions are 92 and 98 %. A seed of $20\Delta_{\mathrm{BCS}}$ changes them by less than 0.02. The hot-electron stage above the seed adds almost nothing to what leaves: by Eq. (20.39), a phonon of $200\Delta_{\mathrm{BCS}}$, near the Debye energy of aluminum, escapes unabsorbed with a probability of 0.4 % at $t_l/t_0 = 4$.

The scale is set by the gap of the electrode, not by the crystal. $2\Delta_{\mathrm{BCS}}$ of aluminum is 0.36 meV or 87 GHz, against the top of the germanium acoustic band, 35.7 meV or 8.64 THz [6, Table I], which sets the scale of its prompt phonons: a factor of 99. In the crystal such a phonon scatters on isotopes at the rate $B_{\mathrm{iso}}(E_{\mathrm{ath}}/h)^4$ of Appendix E.12. With the constants of silicon and germanium [6, Table I] this is $140\,\mathrm{s}^{-1}$ and $2.1 \times 10^3\,\mathrm{s}^{-1}$, mean free paths of 39 m and 1.5 m at the transverse sound speeds, while three-phonon decay of a longitudinal phonon, at $A_{\mathrm{anh}}(E_{\mathrm{ath}}/h)^5$, takes 0.1 to 1 s in germanium for the spread of constants of Sect. 20.2.2, and longer in silicon. A recombination phonon therefore crosses a centimeter crystal ballistically and is absorbed only at surfaces, on a time set by the geometry and the coverage of the films. Its collection efficiency need not equal that of the hard primaries. This is the physical content of the ratio ρ_g of Sect. 19.5, the collection efficiency of the recombination phonons over that of the hard primaries, and it enters Sect. 20.5 as the contrast ϖ_{gap} of release at an electrode.

20.4.5 Open problems

The yield above four gaps rests on the lumped film. A phonon in a real film has a position and a direction, it is absorbed after a path that depends on its energy, and the quasiparticles diffuse. The thick-film offset of 0.04 between the cascade and the kinetic equations is not traced to one of the differences between them: finite temperature, the measured Eliashberg function, recombination and reabsorption of $2\Delta_{\mathrm{BCS}}$ phonons, or the probe. A transport simulation of the film that follows phonons and quasiparticles through its depth, at the temperature and with the α^2F of aluminum, would settle it, and no measurement of Y_{qp} at a fixed phonon energy in a film exists to test either calculation. The lumped film also loses every phonon that escapes; in a detector those phonons return to the crystal and can be absorbed again, which Sect. 20.5.3 shows to cancel the effect of escape on the collection ratio. The yield of the electrode phonons depends on the same film model, and their share that is absorbed in the electrode itself, rather than returned, has not been measured.

20.5 Collection kernels and recoil discrimination

A silicon nucleus recoiling with 100 eV makes 2.7 electron–hole pairs on average, at the measured yield $Y_q = 0.102$ and the 3.8 eV per pair of the measurement [8, Eq. (1), Table II]; an electron recoil of the same energy makes 27. The rest of each deposit becomes phonons, and the phonon sensors see three populations: the primary phonons of the recoil, the Luke phonons of the drifting carriers, and the phonons that return the gap energy of the pairs (Sect. 19.5). Charge readout separates the two recoils by the pair number, and 2.7 against 27 pairs is a separation of many standard deviations. Can a phonon readout do the same, either through the pair number, which sets the size of the Luke and gap-energy populations, or through a difference in the primary phonons themselves? Both routes run through one object, the collection kernel of each sensor. We will define the kernels and compute them for a model detector in the limit where the phonon gas fills the crystal uniformly. From them follows what fixes the sign of the collection ratio $\mathfrak{r}_{\mathrm{coll}} - 1$, which Sect. 19.5 left open, and the ceiling of a readout at high bias.

20.5.1 Collection kernels

Since the transport of Sects. 20.2 to 20.4 is linear in the birth spectrum (Sect. 20.2.1), let a population $X \in \{\mathrm{prim}, \mathrm{Luke}, \mathrm{gap}\}$ be born at the event position $\boldsymbol{r}_{\mathrm{ev}}$ with the spectral energy density $u_X(E_{\mathrm{ath}})$ and the total energy $E_X = \int \mathrm{d}E_{\mathrm{ath}}\, u_X$. Linearity then fixes the energy that sensor s turns into quasiparticles as

$$E_{s,X}(\boldsymbol{r}_{\mathrm{ev}}) = \int \mathrm{d}E_{\mathrm{ath}}\; \mathfrak{C}_s(E_{\mathrm{ath}}; \boldsymbol{r}_{\mathrm{ev}})\, u_X(E_{\mathrm{ath}}), \tag{20.41}$$

where the collection kernel $\mathfrak{C}_s(E_{\mathrm{ath}}; \boldsymbol{r}_{\mathrm{ev}})$ is the fraction of the energy of phonons born with energy E_{ath} at $\boldsymbol{r}_{\mathrm{ev}}$ that ends as quasiparticles in sensor s. It contains everything between birth and signal: down-conversion and isotope scattering in the substrate, transmission into the film and back with probabilities $\bar{t}_{12}$ and $\bar{t}_{\mathrm{esc}}$, pair breaking with the yield Y_{qp}, and loss on bare faces and holders. A phonon below $2\Delta_{\mathrm{BCS}}$ of the film cannot break a pair there and returns to the crystal, and its energy stays in the kernel's bookkeeping until another surface takes it.

The populations enter only through their birth spectra. The kernels belong to the detector; the two recoils differ only in how much energy they put into each population and, possibly, in the spectrum of the primaries. Every readout quantity of Sect. 19.5 and of this section is a linear functional of the kernels. A detector with two sensor channels (two films, or the fast and slow parts of one pulse) has the channel fraction $a_X(\boldsymbol{r}_{\mathrm{ev}}) = E_{2,X}/(E_{1,X} + E_{2,X})$ for each population, where channel 2 is the one that favors soft phonons, and the efficiency $\epsilon_X = (E_{1,X} + E_{2,X})/E_X$.

Readout functionals of the collection kernels

With the channel fractions a_X and efficiencies ϵ_X built from Eq. (20.41),

$$\varpi_O = \big\langle a_{\mathrm{Luke}} - a_{\mathrm{prim}} \big\rangle_{\boldsymbol{r}}, \qquad \varpi_{\mathrm{gap}} = \big\langle a_{\mathrm{gap}} - a_{\mathrm{prim}} \big\rangle_{\boldsymbol{r}},$$
$$\sigma^2_{\mathrm{pos}} = \mathrm{Var}_{\boldsymbol{r}}\big[a_{\mathrm{prim}}(\boldsymbol{r}_{\mathrm{ev}}) - \hat{a}(\hat{\boldsymbol{r}}_{\mathrm{ev}})\big], \qquad \mathfrak{r}_{\mathrm{coll}} = \frac{\epsilon_{\mathrm{Luke}}}{\epsilon^{\mathrm{ER}}_{\mathrm{prim}}}, \tag{20.42}$$

where $\langle\cdot\rangle_{\boldsymbol{r}}$ and $\mathrm{Var}_{\boldsymbol{r}}$ average over the event positions of the calibration, and $\hat{a}(\hat{\boldsymbol{r}}_{\mathrm{ev}})$ is the channel fraction predicted from the position $\hat{\boldsymbol{r}}_{\mathrm{ev}}$ reconstructed from the pulses. The collection contrasts ϖ_O and ϖ_{gap} say how differently a channel sees Luke and gap-energy phonons from primaries; the position spread σ_{pos} is the part of the primary channel

fraction that the position correction leaves.

A detector is therefore characterized for every population and every recoil by two kernels on the plane $(E_{\mathrm{ath}}, \boldsymbol{r}_{\mathrm{ev}})$. The collection ratio of Sect. 19.5 is one of the functionals. It compares two populations of the same detector, so its sign depends only on how the kernels weigh soft Luke phonons against the hard primaries that arrive after down-conversion.

The kernels of a real detector are a transport calculation. The transport Monte Carlo of Appendix E.12 follows phonons launched as monochromatic sources on a grid of energies and positions, which is what turns the tallies into kernels, and Appendix E.25 specifies the grids, the two geometries, the acceptance tests and the pitfalls. Two of the pitfalls decide the answer: Luke birth spectra must come from the carrier transport at each field, and the fate of phonons below $2\Delta_{\mathrm{BCS}}$ must be stated with every result. No kernel of a real detector has been computed, and transport codes are not yet validated against the relations of this chapter (Sect. 20.1.7).

20.5.2 Mean-field kernels

One limit reduces the kernels to a linear equation in energy alone. If phonons cross the crystal many times before a surface absorbs them, the population forgets where it was born, and the kernels lose their dependence on $\boldsymbol{r}_{\mathrm{ev}}$. This holds when the diffusion time across the crystal is short compared with the absorption time, and when the faces scatter diffusely, so that every phonon of energy E_{ath} hits them at the same rate. In a slab of thickness d an isotropic phonon with speed v hits one of the two faces at the rate $v/2d$, since the mean of $|v_z|$ is $v/2$. Let a fraction c_s of the face area carry film s, with transmission $\bar{t}$ into it and pair-breaking threshold $2\Delta_s$, and let a phonon that hits a bare face be lost with probability ℓ_{bare}. The absorption rate per phonon is then

$$\Gamma_{\mathrm{abs}}(E_{\mathrm{ath}}) = \frac{\bar{v}}{2d}\Big[\sum_s c_s\,\bar{t}\,\Theta(E_{\mathrm{ath}} - 2\Delta_s) + \Big(1 - \sum_s c_s\Big)\ell_{\mathrm{bare}}\Big], \tag{20.43}$$

with $\bar{v}$ the number-weighted mean speed of the mixed population and Θ the step function. Down-conversion acts at the mean rate $f_{\mathrm{LA}}\Gamma_{\mathrm{anh}}(E_{\mathrm{ath}})$ of Sect. 20.2.4, since only longitudinal phonons decay and isotope scattering keeps their share at f_{LA}, and a decay at E' passes the energy fraction y to its daughters with the energy-weighted distribution $P_{\mathrm{d}}(y)$ of Eq. (20.17). The number of daughters in $\mathrm{d}E$ is then $P_{\mathrm{d}}(E/E')\,\mathrm{d}E/E$, and the time-integrated number spectrum $\mathcal{N}(E_{\mathrm{ath}})$ of a population with birth number spectrum n_0 obeys

$$[\Gamma_{\mathrm{abs}}(E) + f_{\mathrm{LA}}\Gamma_{\mathrm{anh}}(E)]\mathcal{N}(E) = n_0(E) + \int_E^\infty \frac{\mathrm{d}E'}{E}\,P_{\mathrm{d}}\Big(\frac{E}{E'}\Big) f_{\mathrm{LA}}\Gamma_{\mathrm{anh}}(E')\,\mathcal{N}(E'), \tag{20.44}$$

with $E \equiv E_{\mathrm{ath}}$. Integrating the time-dependent equation for the number spectrum from zero to infinity gives Eq. (20.44), since every phonon is eventually absorbed or lost. The equation is triangular, because daughters are softer than their parent, and it is solved from the top energy down. Sensor s receives $\int \mathrm{d}E\,E\,\Gamma_s(E)\mathcal{N}(E)\,Y_{\mathrm{qp},s}(E)$, where Γ_s is its term in Eq. (20.43). The mean-field kernel is this quantity for a monochromatic source, divided by its energy.

For an estimate we take a silicon slab 4 mm thick, the thickness of a gram-scale detector with an eV-scale threshold [14, Sect. IV]. The isotropic sound speeds $c_{\mathrm{LA}} = 9.00\,\mathrm{km/s}$ and $c_{\mathrm{T}} = 5.40\,\mathrm{km/s}$ of Appendix C.18 give $\bar{v} = 5.75\,\mathrm{km/s}$; we take $f_{\mathrm{LA}} = 0.093$ and $A_{\mathrm{anh}} = 7.41 \times 10^{-56}\,\mathrm{s}^4$, the silicon values of Sect. 20.2, and the stand-in $P_{\mathrm{d}}(y) = 60\,y^3(1-y)^2$. Two films cover 5 % of the face area each: one with the niobium-like threshold $2\Delta_s = 3.0\,\mathrm{meV}$, one of aluminum with 0.36 meV. We take $\bar{t} = 0.9$, close to the acoustic-mismatch average

0.938 for silicon into aluminum (Sect. 20.3), and $\ell_{\rm bare} = 10^{-3}$, the loss per surface bounce measured in CDMS detectors [6, Sect. III G]. The film yield is the Kaplan cascade of Sect. 20.4 in units of each film's gap, run to $120\Delta_s$ and held there above, which affects 0.1 % of the primary energy in aluminum: in a thick film $Y_{\rm qp} = 2\Delta_s/E_{\rm ath}$ below $4\Delta_s$, a maximum of 0.600 at $5.46\,\Delta_s$ and 0.590 above $10\Delta_s$. In a film from which phonons escape, with the ratio t_l/t_0 of Sect. 20.3, $\bar{t}$ is multiplied by the probability $p_{\rm pb}$ of Eq. (20.39) that an entering phonon breaks a pair before it leaves, and the yield is that of Fig. 20.11. A phonon that escapes unbroken re-enters the crystal unchanged, and so do the phonons a film emits after its first pair break, its relaxation phonons and the hot phonons that escape: they return to the crystal with the spectrum the cascade gives them and can be absorbed again, so that energy is conserved. Only returned phonons below $2\Delta_{\rm BCS}$ of aluminum, which no film absorbs, end on the bare faces. The primaries are born at the top of the acoustic band, 62 meV, since their spectrum is forgotten long before they are absorbed (Sect. 20.2).

The primaries come out nearly balanced between the two films. They are absorbed within 15 µs, by then down-converted to a mean energy of 5.0 meV weighted by the absorbed energy (4.0 meV weighted by number), and on first absorption the niobium-like film takes 40.7 % of their energy, aluminum 58.2 %, and the bare faces 1.2 %. Aluminum gets more because 18 % of the absorbed energy arrives below 3 meV, where only aluminum absorbs. The middle 80 % of it arrives between 2.45 and 7.88 meV, the band of Fig. 20.11(a). The relaxation phonons of the niobium-like film lie below its 3 meV threshold but mostly above that of aluminum, and the aluminum film collects them on their return. In thick films the primaries therefore have the channel fraction $a_{\rm prim} = 0.610$ and the efficiency $\epsilon_{\rm prim} = 0.684$; deleting those phonons, as a lumped film does, would give 0.563 and 0.610.

The Luke phonons are softer, and their spectrum is the open input. For a carrier at one speed, Sect. 20.1.6 gives an emitted energy weighted as $E_{\rm ath}^3$ up to $E_{\rm L}^{\rm max} = \frac{4}{3}\langle E_{\rm L}\rangle$, so that the fraction $(x/E_{\rm L}^{\rm max})^4$ of it lies below any x. The balance estimate of Sect. 20.1.6 puts $\langle E_{\rm L}\rangle$ for silicon electrons near 1.1 meV at 1 V/cm, an overestimate by construction; a transport Monte Carlo of the kind described in Appendix E.12 gives 0.84 meV at 1 V/cm, and 0.84 to 4.6 meV for silicon electrons over 1 to 100 V/cm. We scan 0.75 to 4.7 meV, a range that also covers germanium electrons at low field. Real carriers do not drift at one speed: they accelerate in the field, emit, and accelerate again, and transverse phonons, which the one-branch rate leaves out, are emitted softer. Both put more energy at low $E_{\rm ath}$. We therefore use two shapes, labeled by $\langle E_{\rm L}\rangle$: the one-speed spectrum, and a flat number spectrum up to $2\langle E_{\rm L}\rangle$ for a spread of speeds, which puts the fraction $(x/2\langle E_{\rm L}\rangle)^2$ of the energy below x. Which shape and which mean a drifting carrier produces at a given field is still open (Sect. 20.1.7). As a reference, in the strong near-surface fields of germanium detectors the Luke phonons have about 300 GHz for electrons and 700 GHz for holes [11, Sect. 1], 1.2 and 2.9 meV, and in the low bulk field they are softer.

Fig. 20.13(b) shows the contrast this geometry gives. While the whole Luke spectrum lies below the 3 meV threshold of the niobium-like film, only aluminum absorbs Luke phonons, $a_{\rm Luke} = 1$, and $\varpi_O = 1 - a_{\rm prim} = 0.39$. As the mean energy rises, the Luke spectrum reaches above 3 meV and the contrast falls: for the spread of speeds it is 0.12 at 2 meV, 0.045 at 2.9 meV and 0.019 at 4.7 meV. In the mean-field limit the position spread vanishes by construction, so it is the first number a computed kernel adds.

20.5.3 The sign of the collection ratio

The collection ratio differs from one through the part of the Luke spectrum within a few gaps of the aluminum threshold, since the primaries arrive at several meV. Luke energy below $2\Delta_{\rm BCS}$ of aluminum breaks no pairs; it is lost unless a tungsten sensor or another film absorbs

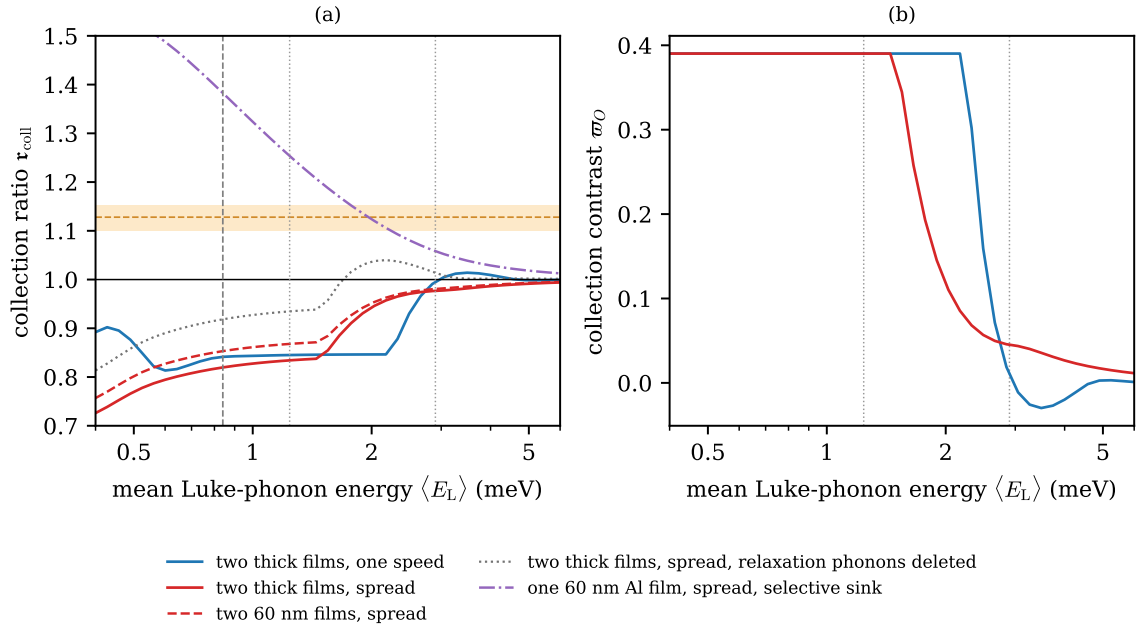


Figure 20.13: Mean-field collection of Luke phonons in a silicon slab 4 mm thick, against the mean Luke-phonon energy. (a) Collection ratio $\mathfrak{r}_{\rm coll}$ of Eq. (20.42). Solid: two thick films ($2\Delta_s = 3.0$ and 0.36 meV, each on 5 % of the face area) with the Kaplan-cascade yield of Sect. 20.4, every phonon a film emits returned to the crystal, for the Luke spectrum of a carrier at one speed (blue, number density $\propto E_{\rm ath}^2$) and for a spread of speeds (red, flat number density up to $2\langle E_{\rm L}\rangle$). Red dashed: two 60 nm films ($t_l/t_0 = 0.72$), spread, phonons returned. Gray dotted: two thick films, spread, with the phonons a film emits after its first pair break deleted. Purple dash-dotted: one 60 nm aluminum film on 10 % of the face area, spread, with the same deletion, a selective sink that the model does not identify in a detector (upper envelope). In all curves phonons below $2\Delta_{\rm BCS} = 0.36$ meV are reflected by the films and eventually lost. Tan band and dashed line: 1.099 to 1.152 and 1.128, the collection ratio that explains the silicon phonon-yield ratio of Sect. 19.5 with no stored energy. Vertical dashed line: 0.84 meV, silicon electrons at 1 V/cm; dotted: 1.24 and 2.9 meV, the near-surface Luke energies of germanium [11]. (b) Collection contrast ϖ_O of the two thick films with phonons returned (channel 2 is the aluminum film), same colors. Model: transmission 0.9 into each film, bare-face loss 10^{-3} per hit, $\bar{v} = 5.75$ km/s, $f_{\rm LA} = 0.093$, $A_{\rm anh} = 7.41 \times 10^{-56}\,{\rm s}^4$, $P_{\rm d}(y) = 60\,y^3(1-y)^2$; primaries born at 62 meV, absorbed in the two thick films with $\epsilon_{\rm prim} = 0.684$, $a_{\rm prim} = 0.610$.

it. Luke energy between $2\Delta_{\rm BCS}$ and $4\Delta_{\rm BCS}$ is converted with $Y_{\rm qp} = 2\Delta_{\rm BCS}/E_{\rm ath}$ in every film, above 0.59 below 0.61 meV and down to 0.5 at $4\Delta_{\rm BCS}$ (Sect. 20.4). Above four gaps a thick film converts with 0.59 to 0.60; a film from which phonons escape converts the primaries, at 14 to 44 $\Delta_{\rm BCS}$, with less than the soft Luke phonons on first absorption, but the phonons it lets out return to the crystal. And a phonon that only one film absorbs bounces longer before it is absorbed, and loses more on the bare faces.

The first two effects act alone in a detector with a single film type. Let $s_{\rm sub}$ be the share of Luke energy born below $2\Delta_{\rm BCS}$. Let sub-gap phonons be absorbed at the same rate as the others and converted with the weight $\kappa_{\rm sub}\bar{Y}_{\rm prim}$, where $\bar{Y}_{\rm prim}$ is the mean yield of the primaries as they are absorbed, and $\kappa_{\rm sub} = 0$ if they are lost and 1 if they are absorbed like hard phonons. Define the near-gap gain

$$s_{\rm ng} = \frac{1}{E_{\rm Luke}} \int_{2\Delta_{\rm BCS}}^{\infty} {\rm d}E_{\rm ath}\; u_{\rm Luke}(E_{\rm ath}) \Big[\frac{Y_{\rm qp}(E_{\rm ath})}{\bar{Y}_{\rm prim}} - 1 \Big].$$

If every phonon above the threshold reaches the film with the same probability per bounce, whatever its energy, the absorbed fraction is common to both populations and cancels in the ratio. The primaries are converted with $\bar{Y}_{\rm prim}$, the Luke phonons with $Y_{\rm qp}(E_{\rm ath})$ above the threshold and with $\kappa_{\rm sub}\bar{Y}_{\rm prim}$ below it, and

$$\mathfrak{r}_{\rm coll} - 1 = s_{\rm ng} - (1 - \kappa_{\rm sub})\, s_{\rm sub} \qquad \text{(one film type)}. \tag{20.45}$$

The mean-field solve of Eq. (20.44) for one thick aluminum film reproduces Eq. (20.45) to 10^{-3}.

The sub-gap share is fixed by the Luke spectrum alone. For the one-speed spectrum it is $s_{\rm sub} = 0.085$ at $\langle E_{\rm L} \rangle = 0.5$ meV, 0.017 at 0.75 meV, 0.011 at 0.84 meV and 5×10^{-3} at 1 meV; for the spread of speeds 0.13, 0.058, 0.046 and 0.032, and still 0.008 at 2 meV. In a thick film $\bar{Y}_{\rm prim} = 0.590$, and the near-gap gain is small: +0.002 for the one-speed spectrum and +0.018 for the spread of speeds at 0.75 meV, since the cascade yield lies above 0.59 between 5 and $8\Delta_{\rm BCS}$ and below it near $4\Delta_{\rm BCS}$. With sub-gap phonons lost, one thick aluminum film then has $\mathfrak{r}_{\rm coll} = 0.985$ and 0.960 at 0.75 meV and 0.994 and 0.968 at 0.84 meV, below one by about $s_{\rm sub}$. If the sensor absorbs the sub-gap phonons, that term vanishes, and $\mathfrak{r}_{\rm coll} = 1.002$ to 1.018 between 0.75 and 1 meV.

Escape does not change this when the phonons a film lets out return to the crystal. A returned phonon reaches a film again and continues the same cascade, so escape followed by return only delays it. For one aluminum film the solve gives the thick-film values within 0.002 at every t_l/t_0 down to the 60 nm of Worked example 20.3, 0.995 and 0.969 at 0.84 meV. This is a numerical result, not an identity: bare-face loss and decay on the return path break it, and it holds at the bare-face loss of 10^{-3}. A sink without signal that takes every phonon, modeled by raising the bare-face loss to 0.3 per hit, lowers $\epsilon_{\rm prim}$ to 0.08 and moves $\mathfrak{r}_{\rm coll}$ of the 60 nm film only to 1.023 and 0.988. A large rise needs a selective sink, one that removes the phonons a film emits after its first pair break and leaves those that have not yet entered a film. The lumped film of Sect. 20.4 makes exactly that assumption. With it, the primaries are read at the yield of the band, about 0.50 at $t_l/t_0 = 4$ and 0.32 at 0.72, while the soft Luke energy keeps $2\Delta_{\rm BCS}/E_{\rm ath}$; the near-gap gain at 0.75 meV becomes +0.13 and +0.11 at $t_l/t_0 = 4$ and +0.61 and +0.49 at 0.72, and one aluminum film at 0.84 meV reaches 1.05 to 1.10 at $t_l/t_0 = 4$, about 300 nm, and 1.30 to 1.53 at 60 nm. These values are an upper envelope of the lumped-film assumption; the estimate does not identify a physical sink in a detector that would realize it.

The two-film geometry adds the third effect. Luke phonons below 3 meV reach only the aluminum film, so they stay 30 µs in the crystal against 15 µs for phonons that both films

absorb, and they lose twice as much on the bare faces, 2.0 % against 1.0 %. With a flat yield of 0.59 above each threshold and sub-gap phonons absorbed by the aluminum sensor, this alone gives $\mathfrak{r}_{\rm coll} = 0.992$, and with sub-gap phonons lost 0.935 at 0.75 meV to 0.998 at 4.7 meV for the spread of speeds. The film cascade with its returned phonons lowers these values further. The primaries reach the niobium-like film in large part, and its relaxation phonons, below 3 meV, return and are collected by the aluminum film, which raises $\epsilon_{\rm prim}$ to 0.684; Luke phonons below 3 meV gain nothing of the kind. Two thick films then give 0.83 and 0.81 at 0.75 meV (one speed and spread), 0.84 and 0.82 at 0.84 meV, and rise to 1.01 near 3.5 meV for one speed, once the Luke spectrum itself reaches the niobium-like threshold. Two 60 nm films give 0.85 to 1.01 over 0.75 to 4.7 meV. Deleting the relaxation phonons of the films instead, as the lumped film does, would give 0.91 to 1.09 for thick films and up to 1.19 at 60 nm; a tenfold bare-face loss, or a sink of 0.3 per hit, lowers all two-film values.

The sign of $\mathfrak{r}_{\rm coll} - 1$ is therefore set by what the detector does with the Luke energy within a few gaps of its film thresholds. With every phonon a film emits returned to the crystal and energy conserved, the mean-field estimate lies between 0.81 and 1.02 over 0.75 to 4.7 meV, one or two films, any film thickness from 60 nm up, sub-gap phonons lost or absorbed, and between 0.82 and 1.01 at the 0.84 meV of silicon electrons at 1 V/cm. A signal-free sink acting on all phonons keeps it below 1.03. The collection ratio 1.128, with its band 1.099 to 1.152, that would explain the CDMS II silicon phonon-yield ratio with no stored energy (Sect. 19.5) therefore lies above the transport estimate; only the selective sink of the lumped film reaches it. Only computed kernels for real detector geometries (Appendix E.25) can exclude collection as the whole explanation, and the two-bias measurement of Sect. 19.5 is what decides between collection and stored energy.

20.5.4 Phonon signatures of the recoil type

A phonon readout separates the two recoils through an observable $O_{\rm obs}$ of each event, such as the channel fraction $E_2/(E_1 + E_2)$ or the share of the pulse in a fast window. At equal total phonon energy, the separation

$$\mathcal{S}_{\rm sep} = \frac{\left|\langle O_{\rm obs}\rangle_{\rm NR} - \langle O_{\rm obs}\rangle_{\rm ER}\right|}{\left[\tfrac{1}{2}(\sigma^2_{\rm NR} + \sigma^2_{\rm ER})\right]^{1/2}}, \tag{20.46}$$

measures the distance between the two distributions in units of their spreads, and we call a route useful where $\mathcal{S}_{\rm sep} \geq 2$. For two Gaussians of equal width cut at the midpoint, $\mathcal{S}_{\rm sep} = 2$ accepts 84 % of one population and rejects 84 % of the other, and a leakage of 10^{-3} needs $\mathcal{S}_{\rm sep} = 6.2$, so "useful" is far from background-free. Let φ_X be the share of population X in the signal. Since the shares sum to one, the channel fraction is $O_{\rm obs} = \sum_X a_X \varphi_X = a_{\rm prim} + \varpi_O\, \varphi_{\rm Luke} + \varpi_{\rm gap}\, \varphi_{\rm gap}$ after the average over positions, and another observable linear in the delivered energies has the same form to first order in the contrasts. There is one route for each population.

Primary phonons. Sect. 20.2 shows that the prompt phonons forget their polarization within a nanosecond, and a hard birth once $E_*(t)$ has fallen to about half its energy. In bulk silicon and germanium this takes 6.8 ns and 47 ns for births near the band top, 0.15 µs and 1.0 µs for births over the upper factor two of the band, and about 1 µs for the 22 meV transverse daughters of optical decay in silicon. These are estimates from the continuum rates and the stand-in daughter distribution of Sect. 20.2. What survives is the soft energy, the share $f_{\rm s} = 1.7\,\%$ of an electron recoil that its carriers shed below the optical phonon energy, Eq. (20.12). A nuclear recoil has Y_q times as many pairs, and its lattice share is born near the optical and zone-boundary energies (Sect. 19.1). If $E_{\rm mem}$ lies above the soft band and

below every hard birth, Eq. (20.21) then gives

$$\left|\langle O_{\rm obs}\rangle_{\rm NR} - \langle O_{\rm obs}\rangle_{\rm ER}\right| \simeq \varpi_{\rm soft}\,\Delta_{\rm soft}, \qquad \Delta_{\rm soft} = (1 - Y_q)\,f_{\rm s}, \tag{20.47}$$

with $\varpi_{\rm soft}$ the contrast of the channel between soft primaries and hard ones. This is an estimate, not a bound: for the model births of Sect. 20.2 at 1 µs the bound itself is 0.15, against an actual difference of 0.012, close to $\Delta_{\rm soft}$. At 100 eV, $\Delta_{\rm soft} = 0.015$. The spread includes the position spread, so $\mathcal{S}_{\rm sep} \le \varpi_{\rm soft}\Delta_{\rm soft}/\sigma_{\rm pos}$, and any $\sigma_{\rm pos}$ above $\Delta_{\rm soft}/2 = 0.008$ excludes a useful separation at every energy, even with a perfect selector, $\varpi_{\rm soft} = 1$. The selector is far from perfect: the soft births have a few meV, and the mean-field estimate of Sect. 20.5.2 puts the energy of the hard primaries at a mean of 5.0 meV when they are absorbed, so no film threshold separates the two bands. The pulse shapes of CDMS detectors depend strongly on how close the event lies to each instrumented face [11, Sect. 2], and a position correction must reach the percent level before this route opens. Phonons born within the memory-loss distance of an instrumented face keep some memory, but they mark a position class, not a recoil type: 13.5 % of the volume of a 4 mm crystal at the 1 µs it takes to forget the transverse daughters of an electron recoil (Sect. 20.2.5). None of this holds for sapphire, where isotope mixing is slower than decay and the memory argument of Sect. 20.2 fails.

Gap energy at zero field. Each pair carries $E_{\rm g}$ until it recombines, so an electron recoil holds $E_{\rm g}/\varepsilon_{\rm eh} = 0.312$ of its energy in pairs, a nuclear recoil Y_q times as much (Sect. 19.5). If this energy comes back later than the primaries, a sensor with two read windows $t_1 < t_2$, with all gap energy returned by t_2, sees the ratio $O_b = E_{\rm sig}(t_1)/E_{\rm sig}(t_2)$. With $f_{\rm ret}(t)$ the returned fraction of Eq. (19.30), no stored energy and equal collection,

$$\Delta O_b = \frac{E_{\rm g}}{\varepsilon_{\rm eh}}\Big\{\big[1 - f_{\rm ret}^{\rm ER}(t_1)\big] - Y_q\big[1 - f_{\rm ret}^{\rm NR}(t_1)\big]\Big\}. \tag{20.48}$$

For equal release laws this is $(1 - Y_q)(E_{\rm g}/\varepsilon_{\rm eh})(1 - f_{\rm ret}) = 0.28\,(1 - f_{\rm ret})$ at 100 eV, twenty times the primary estimate. The route is a timing reading of the pair number, and its size and sign are set by kinetics that are not known at millikelvin. In the dense track of a nuclear recoil, exciton annihilation gives $1 - f_{\rm ret}^{\rm NR} = 1/(1 + t/t_{\rm rel})$, with $t_{\rm rel}$ the time scale set by the initial track density (Eq. (19.31)). Against an exponential release $\mathrm{e}^{-t/t_{\rm rel}}$ of the electron recoil on the same time scale, ΔO_b changes sign at the root of $\mathrm{e}^{-x}(1+x) = Y_q$, which is $t_1 = 3.87\,t_{\rm rel}$ at $Y_q = 0.102$. The site matters as much as the time. Release in the bulk produces hard phonons that lose their memory, and contributes nothing to $\varpi_{\rm gap}$. Release at a surface or an electrode produces soft phonons that travel ballistically, like the recombination phonons that carry 13 % of the phonon signal of germanium detectors at 3 V (Sect. 20.4.4). The zero-field gap-energy channel is therefore unresolved.

Luke phonons at low bias. With a bias the Luke share differs between the recoils because the pair number does. Let all pairs be drifted, with $g = f_{\rm det}\,g_{\rm Luke}$. An electron recoil then has $\varphi_{\rm Luke} = g/(1 + g)$ and a nuclear recoil $Y_q g/(1 + Y_q g)$. The gap shares differ in the same way, so

$$\Delta O_{\rm obs} = \frac{(1 - Y_q)\,(\varpi_O\,g + \varpi_{\rm gap}E_{\rm g}/\varepsilon_{\rm eh})}{(1 + g)(1 + Y_q g)}. \tag{20.49}$$

The Luke term is largest at $g = Y_q^{-1/2}$, where it equals $\varpi_O(1 - Y_q^{1/2})/(1 + Y_q^{1/2})$: at 100 eV, $0.516\,\varpi_O$ at $g = 3.1$, or 13.4 V across a silicon crystal at $f_{\rm det} = 0.873$. Whether the contrast suffices depends on the spreads. We compare the two recoils at the same recoil energy, 100 eV, with the pair numbers of the measured yield in pairs of $\varepsilon_{\rm eh}$, Y_q' of Sect. 20.5.5. The spread

of $O_{\rm obs}$ for each recoil is the position spread, the baseline $\sigma_{\rm ph} = 2.65\,\text{eV}$ of the gram-scale detector [14] divided by that recoil's $E_{\rm ph}$, and the fluctuation of its detected pair number, $\varpi_O(eV_{\rm bias}/E_{\rm ph})(F_{\rm meas}N_{\rm det})^{1/2}$, added in quadrature. The binomial form of the measured-charge Fano factor, $F_{\rm meas} = f_{\rm det}F + 1 - f_{\rm det}$ (Sect. 20.1.4), uses $F_{\rm eh} = 0.119$ for electron recoils, the measured value (the cascade value of Sect. 18.2 is 0.114), the measured $F_{\rm NR} = 0.9$ at 100 eV [8, Table II], and $f_{\rm det} = 0.873$, the complement of the 12.7 % trapping probability in the detector model of the same measurement. With $\varpi_{\rm gap} = 0$, the smallest contrast that reaches $\mathcal{S}_{\rm sep} = 2$ at 100 eV is $\varpi_O = 0.11$, 0.22 and 0.43 for $\sigma_{\rm pos} = 0.02$, 0.05 and 0.10, at biases of 10 to 12 V. The mean-field contrast of about 0.4 for Luke spectra below 1.5 meV is within reach of this requirement, but it is a model, not a measurement. The route reads the pair number with phonons. Its ceiling, with $\varpi_O = 1$ and no other spread, is reached as the bias goes to zero, and it is that of the charge yield itself, 10.8 at 100 eV: the route carries no information beyond the charge yield.

The phonon pulse-shape rejection of SuperCDMS germanium detectors is of this kind. It rejects bulk electron recoils against nuclear recoils at better than 1 in 10^3 at 7 to 20 keV, limited by the statistics of the data [11, Sect. 3], and its information comes from Luke phonons emitted in the strong field near the surfaces.

20.5.5 The high-voltage readout ceiling

At 100 V across a silicon crystal, $g_{\rm Luke} = 26.7$ and the Luke phonons dominate the signal. Detectors run this way read the pair number through the total phonon energy, $E_{\rm ph} = E_{\rm prim} + N_{\rm det}eV_{\rm bias}$ with $N_{\rm det}$ the detected pairs (Sect. 1.5), and an electron recoil of the same $E_{\rm ph}$ is the background to a nuclear recoil. Suppose a second observable reads the Luke share perfectly, with $\varpi_O = 1$, no baseline noise and no position spread. Its separation is the most any phonon observable can extract from the pair number at high bias.

A nuclear recoil of energy E_R makes on average $Y_qE_R/\varepsilon_{\rm src}$ pairs, where $\varepsilon_{\rm src} = 3.8\,\text{eV}$ is the energy per pair with which the measurement converted its charge into the yields [8, Eq. (1)]; write $Y'_q = Y_q\varepsilon_{\rm eh}/\varepsilon_{\rm src}$, so that the pair number is $Y'_qE_R/\varepsilon_{\rm eh}$. Its phonon energy is $E_{\rm ph} = E_R(1 + gY'_q)$, with $g = f_{\rm det}g_{\rm Luke}$ and the primary phonon energy taken equal to the recoil energy, the truncation of Limit Check 19.4 that the measurement uses. The electron recoil with the same $E_{\rm ph}$ has the energy $E_R(1 + gY'_q)/(1 + g)$. Their mean detected pair numbers are

$$\langle N_{\rm det}\rangle_{\rm NR} = f_{\rm det}Y'_q\frac{E_R}{\varepsilon_{\rm eh}}, \qquad \langle N_{\rm det}\rangle_{\rm ER} = f_{\rm det}\frac{E_R}{\varepsilon_{\rm eh}}\,\frac{1 + gY'_q}{1 + g},$$

and their difference is $f_{\rm det}(E_R/\varepsilon_{\rm eh})(1 - Y'_q)/(1 + g)$. The Luke share at fixed $E_{\rm ph}$ is linear in $N_{\rm det}$, so its separation is that of $N_{\rm det}$, with variances $F^X_{\rm eff}\langle N_{\rm det}\rangle_X$.

High-voltage readout ceiling

With an ideal readout of the Luke share at equal total phonon energy, the separation of nuclear from electron recoils is

$$\mathcal{S}_{\rm HV} = \frac{(1 - Y'_q)\,(f_{\rm det}E_R/\varepsilon_{\rm eh})^{1/2}}{\left\{\tfrac{1}{2}(1 + g)\left[F^{\rm NR}_{\rm eff}\,Y'_q(1 + g) + F^{\rm ER}_{\rm eff}(1 + gY'_q)\right]\right\}^{1/2}}, \qquad g = f_{\rm det}\,g_{\rm Luke}, \tag{20.50}$$

with $F_{\rm eff} = f_{\rm det}F + 1 - f_{\rm det}$ when a trapped carrier induces nothing and $F_{\rm eff} = f_{\rm det}F$ when every pair induces the mean charge $f_{\rm det}e$. The cascade and partial-charge terms of Eq. (20.7) lie between these limits.

At zero gain Eq. (20.50) is the separation of an ideal charge-and-heat measurement. As the bias grows it falls as $1/g$, because at equal total phonon energy a larger Luke gain makes the electron recoil smaller, and the two pair numbers converge. In silicon at 220 eV the ceiling is 16.8 at zero bias, 10.7 at 4 V, 3.4 at 30 V, 1.22 at 100 V and 0.43 at 300 V.

Worked example 20.5: The high-voltage ceiling at 220 eV

The measured silicon yield at $E_R = 220\,\text{eV}$ is $Y_q = 0.108^{+0.009}_{-0.010}$, with the pair-number Fano factor $F_{\text{NR}} = 0.5^{+0.2}_{-0.1}$ [8, Table II]. At 100 V, $g_{\text{Luke}} = 100/3.75 = 26.67$, and with $f_{\text{det}} = 0.873$, $g = 23.28$. The yield in pairs of 3.75 eV is $Y'_q = 0.108 \times 3.75/3.8 = 0.1066$, and $E_R/\varepsilon_{\text{eh}} = 58.67$.

The nuclear recoil has $E_{\text{ph}} = 220 \times (1 + 23.28 \times 0.1066) = 765.9\,\text{eV}$, and the electron recoil with the same phonon energy has $765.9/24.28 = 31.54\,\text{eV}$. The mean detected pair numbers are $0.873 \times 0.1066 \times 58.67 = 5.459$ and $0.873 \times 31.54/3.75 = 7.343$, a difference of 1.885 pairs.

With trapped carriers inducing nothing, $F^{\text{NR}}_{\text{eff}} = 0.873 \times 0.5 + 0.127 = 0.5635$ and $F^{\text{ER}}_{\text{eff}} = 0.873 \times 0.119 + 0.127 = 0.2309$. The variances are $0.5635 \times 5.459 = 3.076$ and $0.2309 \times 7.343 = 1.695$, and

$$\mathcal{S}_{\text{HV}} = \frac{1.885}{[\frac{1}{2}(3.076 + 1.695)]^{1/2}} = 1.22.$$

If every pair induces the mean charge instead, the trapping term drops out of both Fano factors and $\mathcal{S}_{\text{HV}} = 1.50$; with full collection, $f_{\text{det}} = 1$, it is 1.32. The most favorable corner of the 1σ bands, $Y_q = 0.098$ and $F_{\text{NR}} = 0.4$, gives 1.36 when the 12.7 % trapped carriers induce nothing ($f_{\text{det}} = 0.873$) and 1.72 in the mean-charge limit; at full collection both limits give 1.52. A useful separation at the central yield would need $F_{\text{NR}} < 0.21$ in the mean-charge case, half the lower edge of the measured band, and no value of F_{NR} reaches it when trapped carriers induce nothing. Even an ideal phonon observable cannot separate the two recoils at 220 eV and 100 V.

Fig. 20.14 extends the example over the measured range. At the central inputs the ceiling stays below 2 at every measured energy up to 890 eV, with its maximum, 1.22, at 220 eV; the Fano factor of nuclear recoils grows to 3.7 at 890 eV and keeps it down. Over the 1σ bands, whose corners treat Y_q and F_{NR} as uncorrelated, the largest value is 1.72, at 220 eV, in the mean-charge limit (1.36 with trapped carriers inducing nothing at $f_{\text{det}} = 0.873$, 1.52 at full collection). With 2σ bands, however, the ceiling exceeds 2 from 100 to 224 eV, up to 2.46 at 100 eV, where the measured yield is least certain; the upper edge rests on the interpolation in $\log E_R$ between the measured rows at 220 and 460 eV. The no-go of a high-voltage phonon readout at $E_R \leq 300\,\text{eV}$ is therefore conditional on four inputs. The yield Y_q and the Fano factor F_{NR} are measured, and not derived below the binary threshold of Sect. 19.2. The collection enters through f_{det} and the spread of the induced charge, which move the ceiling at 220 eV between 1.22 and 1.50. And the result depends on the confidence level at which the bands are taken. It holds at the central values and over the 1σ bands, and fails at 2σ.

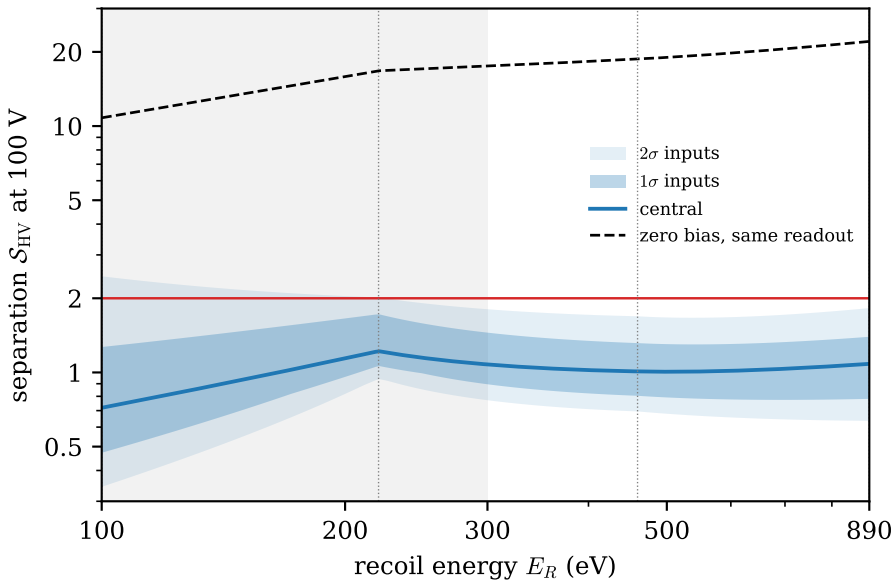


Figure 20.14: High-voltage readout ceiling $\mathcal{S}_{\mathrm{HV}}$ of Eq. (20.50) for silicon recoils at 100 V, against the recoil energy. Line: central measured yield and Fano factor [8, Table II] at 100, 220, 460 and 890 eV (dotted verticals), interpolated linearly in $\log E_R$; detected fraction $f_{\mathrm{det}} = 0.873$, trapped carriers inducing nothing; $F_{\mathrm{eh}} = 0.119$; $\varepsilon_{\mathrm{eh}} = 3.75\,\mathrm{eV}$, yields defined with 3.8 eV per pair. Dark band: the extremes over the corners of the 1σ bands of Y_q and F_{NR}, over $f_{\mathrm{det}} = 0.873$ and 1, and over the two limits of the induced-charge spread; light band: the same with 2σ bands. Dashed: the same ideal readout at zero bias. Red line: $\mathcal{S}_{\mathrm{HV}} = 2$. Gray: the range of the no-go statement, $E_R \leq 300\,\mathrm{eV}$.

Summary

A charge amplifier integrates the charge induced on its electrode (Sect. 20.1). By Ramo's theorem each pair induces e times the fraction of the gap its carriers cross, and the source does the work $eV_{\rm bias}$ times the same fraction, so trapping lowers the read charge and the Luke heat together. With exponential survival over a drift length $\lambda_{\rm d}$, the Hecht relation gives the collected fraction of a face event; at $\lambda_{\rm d} = d_{\rm det}$ a face event collects 0.632 of its charge, a uniform bulk event 0.736 and an event at the midplane 0.787. The Fano factor of the measured charge has three terms, the cascade Fano factor scaled by the collection, the spread of partial charges and the spread of the mean collection over event positions; the last grows with the deposit. At 95 % mean collection in silicon the measured Fano factor of a 100 eV recoil lies between 0.125 and 0.158 depending on how the charge is lost. Trapped charge distorts the field over a finite part of the crystal by a percent at 1 V/cm once some 10^5 charges per cm^2 are trapped, which ordinary event rates reach between flashes. Carriers faster than sound emit Luke phonons by the acoustic Cherenkov process. The mean phonon energy is near 1 to 2 meV in silicon at 1 to 4 V/cm in a single-branch balance estimate, and 0.84 meV for electrons at 1 V/cm in a transport Monte Carlo. The published total rates are low by a factor $k_{\rm s}/k$, and the emission from traps at millikelvin is not known.

The prompt phonons down-convert by anharmonic decay, at a rate $\propto E_{\rm ath}^5$ for longitudinal phonons, and scatter on isotopes at a rate $\propto E_{\rm ath}^4$ (Sect. 20.2). Their spectrum approaches a scaling form in $E_{\rm ath}/E_*(t)$, and a hard birth is forgotten once E_* has fallen to about half its energy. In bulk silicon and germanium this takes 6.8 ns and 47 ns for births near the band top, 0.15 µs and 1.0 µs for births over the upper factor two of the band, and about 1 µs for the transverse daughters of optical decay in silicon. Near an instrumented face the birth is not forgotten, and at 1 µs that region is 13.5 % of a 4 mm crystal. Polarization is forgotten within a nanosecond. These times are continuum estimates, and they do not hold for sapphire, where isotope mixing is slower than decay.

At the interface with a sensor film the averaged transmission is 0.938 from silicon into aluminum and 0.368 back, with the elastic constants of aluminum at zero temperature, tied by the reciprocity $\mathfrak{n}_1\bar{t}_{12} = \mathfrak{n}_2\bar{t}_{21}$ (Sect. 20.3); 56 % of the phonon flux in the film lies beyond the transverse critical angle and escapes only after mixing. A phonon leaves a film of thickness $d_{\rm f}$ in $t_l = 4d_{\rm f}/(v_s\bar{t}_{\rm esc})$, 0.19 ns (acoustic mismatch) or 0.24 ns (diffuse mismatch) for 60 nm of aluminum, comparable to the pair-breaking time. The fitted silicon-to-aluminum coefficient of 0.3 to 0.55 is an absorption probability, and it constrains the ratio of pair-breaking length to film thickness to about 5 to 23 and leaves the escape probability unconstrained.

In the film a phonon between $2\Delta_{\rm BCS}$ and $4\Delta_{\rm BCS}$ breaks one pair and gives the quasiparticle yield $Y_{\rm qp} = 2\Delta_{\rm BCS}/E_{\rm ath}$ (Sect. 20.4). Above $4\Delta_{\rm BCS}$ the Kaplan cascade gives 0.590 without escape, with a maximum of 0.600 at $5.46\,\Delta_{\rm BCS}$, and 0.349 to 0.554 at $10\Delta_{\rm BCS}$ for t_l/t_0 from 0.5 to 8; the yield then falls across the band in which down-converted phonons arrive. Carriers that reach an aluminum electrode return 85 to 92 % of their energy at or below $2\Delta_{\rm BCS}$, phonons that cross the crystal ballistically.

The transport is linear in the birth spectrum, so a detector is described by the collection kernels of its sensors, and every readout quantity is a linear functional of them: the collection contrasts ϖ_O and $\varpi_{\rm gap}$, the position spread and the collection ratio $\mathfrak{r}_{\rm coll}$ (Sect. 20.5). In the mean-field limit the kernels follow from a triangular linear solve. For a 4 mm silicon slab with a niobium-like and an aluminum film, Luke phonons below 3 meV reach only aluminum and $\varpi_O = 0.39$, falling to 0.045 at a mean Luke energy of 2.9 meV. With one film type, $\mathfrak{r}_{\rm coll} - 1 = s_{\rm ng} - (1 - \kappa_{\rm sub})s_{\rm sub}$. The share of Luke energy below $2\Delta_{\rm BCS}$, 0.017 to 0.058 at a mean Luke energy of 0.75 meV depending on the spread of carrier speeds, lowers the ratio if it is lost. The near-gap gain is small in thick films. Phonons that a film lets out return to the

crystal and continue the same cascade, so film escape changes the ratio by less than 0.002, and the relaxation phonons of a niobium-like film, collected by the aluminum film, lower the two-film values. With energy conserved the estimate lies between 0.81 and 1.02 over mean Luke energies of 0.75 to 4.7 meV, and between 0.82 and 1.01 at 0.84 meV; a signal-free sink acting on all phonons keeps it below 1.03. Only a selective sink that deletes the phonons a film emits after its first pair break, as the lumped film assumes, raises one aluminum film to 1.05 to 1.10 near 300 nm and 1.30 to 1.53 at 60 nm, an upper envelope. The value 1.128 that would explain the silicon phonon-yield ratio of Sect. 19.5 without stored energy therefore lies above the transport estimate; only computed kernels for real detectors can exclude it, and the two-bias measurement of Sect. 19.5 decides.

A phonon readout tells the recoil types apart only through their pair numbers. Mode, frequency or angle selection of the prompt phonons cannot separate nuclear from electron recoils in bulk silicon and germanium: only the soft share of an electron recoil, about 1.5 % of its energy at 100 eV, survives the memory loss, and a position spread of the channel fraction above 0.008 hides it. This estimate rests on the memory times above and on a position correction at the percent level. The zero-field gap-energy channel, up to 0.28 of the recoil energy at 100 eV in a two-window reading, is unresolved: its size and sign depend on the release law and site of the gap energy, which are not known at millikelvin. A low-bias readout of the Luke share reads the ionization yield with phonons and needs a contrast ϖ_O of 0.11 to 0.43 for a position spread of 0.02 to 0.10 at 100 eV. At 100 V even an ideal phonon observable stays below a 2σ separation for silicon recoils of 220 eV: 1.22 at the central measured yield and Fano factor. As the Luke gain grows, the pair numbers of the two recoils converge at equal total phonon energy. The no-go at $E_R \leq 300$ eV holds over the 1σ bands, up to 1.72 in the mean-charge limit, 1.36 when trapped carriers induce nothing at $f_{\mathrm{det}} = 0.873$ and 1.52 at full collection, and fails at 2σ. These verdicts are inputs to the readout column of the guide to next targets (Sect. 21.7); all of them assume transport that has not yet been validated against the relations of this chapter.

Table 20.1 collects the truncations and inputs of the chapter with their sizes. The measured charge and phonon signals, with these conditions, are what the exclusion limits of the next chapter start from.

Table 20.1: Nuisances of charge, phonon and quasiparticle readout. Type in the classes of Sect. 8.8, normalization, tilt, threshold and new support, with "delayed" for an item that moves charge or energy in time over an exposure. Basis of the size: "computed" (exact or next-order result computed), "model" (computed in a model), "est." (parametric estimate), "input spread" (spread or limitation of a measured or computed input), "unsized" (no size known). The last column gives the section where the item is treated.

Truncation or input	Type	Size	Basis	Section
Exponential survival of drifting carriers	normalization	collected fraction 0.632 (face), 0.736 (uniform), 0.787 (midplane) at $\lambda_\mathrm{d} = d_\mathrm{det}$; saturating drift velocity caps it below one	computed; model	Sect. 20.1
Fano factor of the measured charge	normalization (width)	0.125–0.158 for a 100 eV recoil at 95 % collection; position term $1.2 \times 10^{-4}\,\bar{N}_\mathrm{eh}$ for depth-dependent trapping	computed	Sect. 20.1
Space charge between flashes	delayed	field change 1 % at 1 V/cm for 6.3–12.6 $\times 10^4$ trapped charges per cm^2, reached in 5–10 h at 1 Hz of 100 eV events	est.	Sect. 20.1
Athermal emission from traps	delayed	phonon- and impact-ionization cross-sections of shallow and over-charged centers not computed	unsized	Sect. 20.1
Luke emission rate and spectrum	normalization (of $\mathfrak{r}_\mathrm{coll}$, ϖ_O)	published total rates low by k_s/k; transverse emission omitted; mean energy 1–2 meV at 1–4 V/cm (single-branch estimate, carrier speed overestimated), 0.84 meV for electrons at 1 V/cm (transport Monte Carlo)	model; unsized	Sects. 20.1 and 20.5
Memory loss of prompt phonons	tilt	6.8 ns (Si), 47 ns (Ge) for births near the band top; 0.15 µs, 1.0 µs over the upper factor two of the band; about 1 µs for the 22 meV transverse daughters in Si; near-face shell 13.5 % of a 4 mm crystal at 1 µs; continuum estimate; fails for sapphire	est.	Sect. 20.2
Interface transmission model	normalization	t_l differs by a factor 1.31 between acoustic and diffuse mismatch; silicon anisotropy moves $\bar{t}_\mathrm{esc}$ from 0.368 to 0.327–0.405; room-temperature aluminum constants lengthen t_l by 20 %; interlayers not included	model	Sect. 20.3
Fitted silicon-to-aluminum coefficient	normalization	read as an absorption probability, $\mathfrak{l}_\mathrm{f}/d_\mathrm{f} \approx 5$–23; inherits the specular simulation of the fit	est.	Sect. 20.3
Quasiparticle yield above $4\Delta_\mathrm{BCS}$	normalization	0.590 without escape; 0.349–0.554 at $10\Delta_\mathrm{BCS}$ for $t_l/t_0 = 0.5$–8; loss across the arrival band 47–58 % at $t_l/t_0 = 0.5$, 35–50 % for 60 nm; lumped film; kinetic equations higher by up to 0.04	model	Sect. 20.4
Soft primary share of an electron recoil	tilt	$f_\mathrm{s} \simeq 0.017$; difference $(1 - Y_q) f_\mathrm{s} = 0.015$ at 100 eV (estimate, not a bound)	est.	Sects. 20.2 and 20.5
Collection ratio of Luke phonons	normalization	mean field, energy conserved (film phonons returned), one or two films, 60 nm to thick: 0.81–1.02 over 0.75–4.7 meV, 0.82–1.01 at 0.84 meV; uniform signal-free sink ≤ 1.03; selective sink of the lumped film (upper envelope) 1.05–1.10 (300 nm), 1.30–1.53 (60 nm); 1.128 not reached; detector kernels not computed	model	Sect. 20.5
Collection contrast of Luke phonons	normalization (low-bias route)	mean field 0.39 for mean Luke energies below about 1.5 meV, 0.12 at 2 meV, 0.045 at 2.9 meV; the route needs 0.11–0.43 for position spreads 0.02–0.10	model	Sect. 20.5
Position spread of the channel fraction	normalization	zero in the mean-field limit by construction; above 0.008 it closes the primary route	unsized	Sect. 20.5
Release law and site of the gap energy at zero field	tilt; delayed	two-window difference up to $0.28(1 - f_\mathrm{ret})$ at 100 eV; sign reverses at $t_1 = 3.87\,t_\mathrm{rel}$ ($Y_q = 0.102$) with dense-track annihilation; release spectra at surfaces and contacts unknown	unsized	Sect. 20.5
Inputs of the high-voltage ceiling	threshold	silicon at 220 eV, 100 V: 1.22 (central), 1.50 (mean-charge collection), 1σ corner 1.72 (mean charge), 1.36 (binomial, $f_\mathrm{det} = 0.873$) or 1.52 (full collection); 2.46 at 100 eV with 2σ bands	input spread	Sect. 20.5

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Chapter 21

Exclusion limits and next targets

A silicon detector that runs for a kilogram-year and records no event above its threshold has measured one number, zero. By Eq. (1.52) it excludes, at 90 percent confidence, every model that predicts more than 2.3 events, and the rest of this book has been about how many events a model predicts. The prediction, however, carries choices that the count does not. The velocity distribution of the halo is known only from models and stellar kinematics, and near the threshold mass only its fastest few percent contribute (Sect. 1.3.5). The response of the crystal is known only at some level of approximation: screening changes the silicon electron-recoil rate at 5 eV by a factor of two (Limit Check 10.1), and the phonon-assisted rate below the zero-phonon edge is known only to a factor of ten (Sect. 13.9). The measured signal is a number of pairs or a phonon energy read through the collection efficiencies of Chap. 20, not a deposited energy. A limit that others can trust states each of these choices, and a limit that others can reuse publishes the responses it was computed from.

The same zero in silicon excludes cross-sections 1.35 times smaller at 3 MeV if local fields are left out of the response, and a halo-independent statement is two to four times weaker than the standard-halo one in the bulk of the halo and orders of magnitude weaker near its largest speed. A second experiment that wants to add its exposure, or a later calculation that changes the response, can use a published limit only if the choices behind it are recorded. And the choice of the next target needs the same rates read across crystals, together with the demonstrations each crystal still lacks. Turning a zero into a limit that can be trusted and reused, and the limits into a choice of target, takes the steps of Fig. 21.1; what remains open is indexed in Chap. 22.

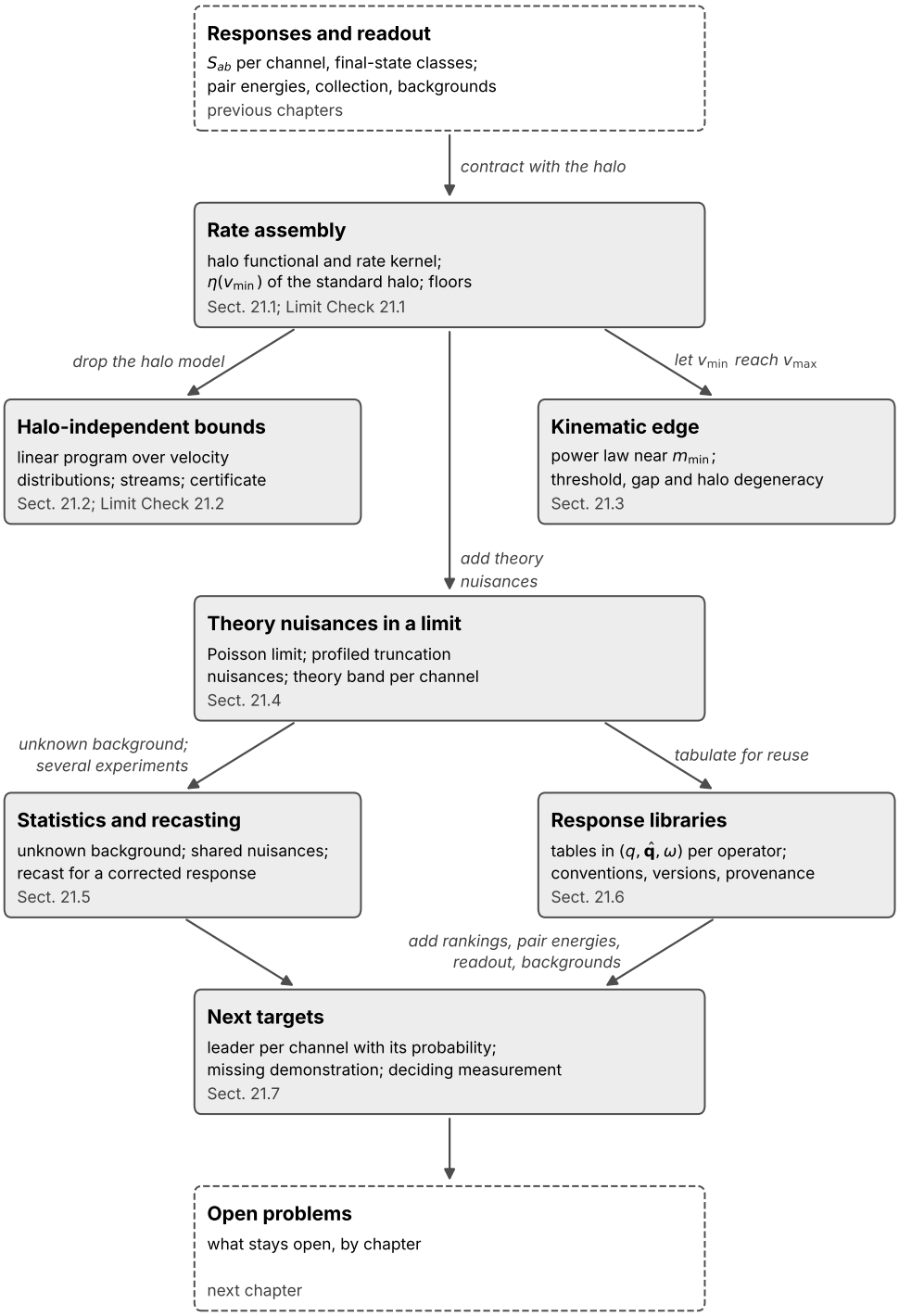


Figure 21.1: From rates to limits and to the guide to next targets. The channel responses and the measured signal of the previous chapters are contracted with the halo into a rate per channel (Sect. 21.1; Limit Check 21.1). Three questions branch from it: what holds for every velocity distribution (Sect. 21.2; Limit Check 21.2), how the limit behaves where the minimum velocity reaches the largest speed in the halo (Sect. 21.3), and how the truncation nuisances of each channel become a band on the limit (Sect. 21.4). The band carries over to unknown backgrounds, combinations and recasts (Sect. 21.5) and fixes what a reusable response library must tabulate (Sect. 21.6). The guide to next targets adds the rankings of the response, the pair energies, the readout and the backgrounds (Sect. 21.7). The last box is the index of open problems of the next chapter.

21.1 Rates as halo functionals

The silicon rate of Worked example 10.1, 1.60×10^5 events per kilogram and year at 10 MeV for $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$, rests on one velocity distribution, the standard halo of Sect. 1.2. Simulated galaxies hold dark matter that this distribution does not describe. In the Via Lactea II simulation, for instance, 22 percent of the dark matter near the Sun's radius is debris stripped from satellites that still survive, and it moves at a galactic speed near $340\,\mathrm{km\,s^{-1}}$ [1]. A limit compared with such a halo has to be recomputed, and the recomputation is cheap because every rate in this book is linear in f_χ. We will write the rate of any channel as one integral of f_χ against a kernel that holds everything else: the coupling, the response, the kinematics and the readout. The standard rate of Essig and collaborators follows from the kernel in a few lines, the threshold masses of Sect. 1.3 follow from where it vanishes, and a halo with a debris flow costs one more integral.

21.1.1 The halo functional of a channel

A detector counts events in bins of some observable, the number of pairs N_eh for a charge readout. The rate in one bin is Eq. (8.69). In it the velocity integral is the outermost one, and everything inside it depends on $\boldsymbol{v}$ but not on the halo. We collect it into one function:

Halo functional of a channel

$$\mathcal{R}_{N_\mathrm{eh}}[f_\chi] = \frac{\rho_\chi}{m_\chi\,\rho_T} \int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\, \Gamma_{N_\mathrm{eh}}(\boldsymbol{v}), \tag{21.1}$$

with the *rate kernel*, the inner integrals of Eq. (8.69) at fixed $\boldsymbol{v}$,

$$\Gamma_{N_\mathrm{eh}}(\boldsymbol{v}) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} \sum_\Lambda \sum_{a,b} \mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})\, S_{ab}(\boldsymbol{q},\omega_{\boldsymbol{q}};\Lambda)\, P_\mathrm{casc}(N_\mathrm{eh}|\Lambda), \tag{21.2}$$

the rate at which one dark-matter particle of velocity $\boldsymbol{v}$ makes events with N_eh pairs.

Summed over N_eh, the kernel is the per-particle rate $\Gamma(\boldsymbol{v})$ of Eq. (8.15). For any other observable, a phonon energy or a bin of total deposited energy, P_casc is replaced by the probability that the event lands in that bin. The halo enters only through f_χ and ρ_χ, the channel only through its kernel, and the rate is a linear functional of f_χ: a halo made of two components gives the sum of their rates, weighted by their shares of the density.

The kernel is never negative, since $\mathfrak{D}$ and S are positive semidefinite matrices (Sect. 8.2) and P_casc is a probability. Adding dark matter at any velocity therefore never lowers a rate, which is the property on which the halo-independent bounds of Sect. 21.2 rest. The kernel vanishes at small speeds. A particle of speed v reaches only the pairs (q,ω) with $v_\mathrm{min}(q,\omega) \le v$ (Sect. 1.3), so $\Gamma(\boldsymbol{v}) = 0$ for v below the onset speed $v_\mathrm{on}(m_\chi)$, the smallest v_min over the support of the response. And for a crystal the kernel depends on the direction of $\boldsymbol{v}$ in crystal axes. The laboratory distribution is $f_\mathrm{gal}(\boldsymbol{v} + \boldsymbol{v}_\mathrm{E}(t))$ (Eq. (1.8)), and the crystal turns with the Earth (Eq. (1.14)). Averaged over an exposure, Eq. (21.1) becomes

$$\bar{\mathcal{R}}_{N_\mathrm{eh}} = \frac{\rho_\chi}{m_\chi\,\rho_T} \int \mathrm{d}^3v_\mathrm{gal}\, f_\mathrm{gal}(\boldsymbol{v}_\mathrm{gal})\, \big\langle \Gamma_{N_\mathrm{eh}}(\boldsymbol{v}_\mathrm{gal} - \boldsymbol{v}_\mathrm{E}(t)) \big\rangle_t, \tag{21.3}$$

where $\langle\cdot\rangle_t$ averages over the live time, with the argument of Γ rotated into crystal axes at each t. The galactic distribution now stands alone, and the exposure-averaged kernel at $\boldsymbol{v}_\mathrm{gal}$ is the rate that a single stream of dark matter at that velocity would produce. Sect. 21.2 builds its bounds on this object.

Absorption fits the same form. The kernel is then $\Gamma_{\rm abs}$ of Eq. (8.19), which depends on $\boldsymbol{v}$ only through the momentum $\boldsymbol{p}_\phi = m_\phi \boldsymbol{v}$ and the energy $\omega_\phi \simeq m_\phi(1 + v^2/2)$. Where the response is smooth on the scale $m_\phi v \approx 10^{-3} m_\phi$, the kernel is constant across the halo and the rate depends on the halo through ρ_χ alone.

21.1.2 The mean inverse speed

For a response averaged over the directions of $\boldsymbol{q}$, as in the crystal form factor of Sect. 10.1, the kernel depends only on the speed. The angular average of the energy delta function is $\Theta(v - v_{\rm min})/(2qv)$ (Sect. 1.3), and for a coupling that does not depend on the dark-matter velocity

$$\Gamma(v) = \frac{1}{v} \int_0^\infty \mathrm{d}\omega \int \frac{q\,\mathrm{d}q}{4\pi^2} \sum_{a,b} \mathfrak{D}_{ab}(q)\, \bar{S}_{ab}(q,\omega)\, \Theta\big(v - v_{\rm min}(q,\omega)\big). \tag{21.4}$$

With $F_\chi(v)$ the speed distribution of Eq. (1.10), the rate is a one-dimensional integral, $\mathcal{R} = (\rho_\chi / m_\chi \rho_T) \int_0^\infty \mathrm{d}v\, F_\chi(v)\, \Gamma(v)$. The product $v\Gamma(v)$ is the response integrated over the part of the (q,ω) plane that a particle of speed v can reach. It is zero below $v_{\rm on}$ and grows with v, because a faster particle reaches every pair that a slower one reaches and more.

The mean inverse speed appears when the order of the integrals is reversed. Since $\eta(u) = \int_u^\infty \mathrm{d}v\, F_\chi(v)/v$ by Eq. (1.28), the speed distribution is $F_\chi(v) = -v\,\mathrm{d}\eta/\mathrm{d}v$. Integrating by parts,

$$\mathcal{R} = \frac{\rho_\chi}{m_\chi\, \rho_T} \int_0^\infty \mathrm{d}u\, \eta(u)\, \frac{\mathrm{d}}{\mathrm{d}u}\big[u\Gamma(u)\big], \tag{21.5}$$

with no boundary terms, because $u\Gamma(u)$ vanishes below $v_{\rm on}$ and η vanishes above $v_{\rm max}$. The derivative of Eq. (21.4) replaces $\Theta(u - v_{\rm min})$ by $\delta(u - v_{\rm min})$: $\mathrm{d}[u\Gamma]/\mathrm{d}u$ is the response of the pairs whose minimum velocity is u, and it is never negative. Inserting it in Eq. (21.5) and doing the u integral sets $u = v_{\rm min}(q,\omega)$ in η, which is the isotropic rate Eq. (A.44) integrated over ω. The kernel form asks how much response a particle of speed v reaches; the form with η asks what fraction of the halo is fast enough for a given pair.

A coupling that depends on the dark-matter velocity adds no freedom to this picture. The operators of Table 8.1 depend on the velocity through $\boldsymbol{v}^\perp$, linearly except for $\mathcal{O}_2$, which is quadratic. So $\mathfrak{D}_{ab}$ is a polynomial in the transverse velocity $\boldsymbol{v}_{\rm T}$, the part of $\boldsymbol{v}$ perpendicular to $\boldsymbol{q}$; at zero mass splitting, $\Delta_\chi = 0$, the longitudinal part is fixed at $v_{\rm min}$ by the delta function, and $v_{\rm T}^2 = v^2 - v_{\rm min}^2$ (Sect. 8.1). For an orientation-averaged response no direction is left in the plane perpendicular to $\boldsymbol{q}$. Odd powers of $\boldsymbol{v}_{\rm T}$ then average to zero, and even powers enter through $v_{\rm T}^2$ alone. The quadratic term has a second speed moment as its halo average:

$$\int \mathrm{d}^3 v\, f_\chi(\boldsymbol{v})\, \frac{v^2 - u^2}{v}\, \Theta(v - u) = 2 \int_u^\infty \mathrm{d}u'\, u' \eta(u') = \eta_2(u). \tag{21.6}$$

To see this, differentiate the left side with respect to u. The boundary term at $v = u$ vanishes, because the integrand is zero there, and what is left is $-2u\,\eta(u)$; both sides vanish for $u \geq v_{\rm max}$. The same step applied to $(v^2 - u^2)^n$ gives $-2nu$ times the moment with $n - 1$, so the quartic term of $\mathcal{O}_2$ is fixed by η_2 and hence by η. The function η_2 of Appendix A.8 is therefore not an independent input. In the kernel form $\Gamma(v)$ contains the extra terms, and Eq. (21.1) holds as written.

Now take the density row, $\mathfrak{D}_{ee} = \pi \bar{\sigma}_e F_{\rm DM}^2 / \mu_{\chi e}^2$ (Eq. (1.34)), and the independent-particle structure factor $\bar{S}^0$ of Sect. 10.1. Eq. (21.5) is then Eq. (10.2), and written with the crystal form factor it is the rate of Limit Check 10.1, divided by the target mass [2, Eqs. (3.13), (3.17)]. In the standard halo η is the closed form of Appendix D.1 [2, App. B]. One step separates that form from what Appendix A.8 prescribes for a rate averaged over an exposure. The

closed form holds at one Earth speed, while over a year $|\boldsymbol{v}_{\rm E}(t)|$ runs from 237.4 to 266.4 km s^{-1} (Sect. 1.2). The exposure average of η is the average of the closed form over those speeds, and the question is when the closed form at $v_{\rm E}^{\rm ref} = 250.5\,{\rm km\,s^{-1}}$, the convention of Appendix A.8 and not the yearly mean of 252.1 km s^{-1}, can stand in for it. Write $|\boldsymbol{v}_{\rm E}(t)| = v_{\rm E}^{\rm ref} + \delta v_{\rm E}(t)$. Over a full year the term of Eq. (1.13) that is linear in the orbital velocity averages to zero, and what is left is $\langle \delta v_{\rm E} \rangle = 1.61\,{\rm km\,s^{-1}}$, almost all of it from the second-order term, and $\langle \delta v_{\rm E}^2 \rangle = 108.3\,{\rm km^2\,s^{-2}}$, an rms of 10.4 km s^{-1}. To second order,

$$\langle \eta(u) \rangle_t - \eta(u)\big|_{v_{\rm E}^{\rm ref}} \simeq \frac{\partial \eta}{\partial v_{\rm E}} \langle \delta v_{\rm E} \rangle + \frac{1}{2} \frac{\partial^2 \eta}{\partial v_{\rm E}^2} \langle \delta v_{\rm E}^2 \rangle . \tag{21.7}$$

In the bulk of the halo both derivatives are small and the correction is a percent. Near the end point $\eta \propto \delta^2$ with $\delta = v_{\rm esc} + v_{\rm E} - u$ (Eq. (1.29)), a convex function of $v_{\rm E}$, and the June excess outweighs the December deficit: the yearly average lies above the value at the mean speed.

Limit Check 21.1. Mean inverse speed of the standard halo

For the truncated Maxwellian of Eq. (1.3) seen at Earth speed $v_{\rm E}$, and $v_{\rm min} < v_{\rm esc} - v_{\rm E}$,

$$\eta(v_{\rm min}) = \frac{\pi v_0^2}{v_{\rm E} N_{\rm esc}} \left\{ \frac{\sqrt{\pi}\, v_0}{2} \left[{\rm erf}\, \frac{v_{\rm min} + v_{\rm E}}{v_0} - {\rm erf}\, \frac{v_{\rm min} - v_{\rm E}}{v_0} \right] - 2 v_{\rm E}\, {\rm e}^{-v_{\rm esc}^2/v_0^2} \right\} ,$$

with the second branch Eq. (D.4) up to $v_{\rm esc} + v_{\rm E}$ [2, Eqs. (B.4) and (B.5)].

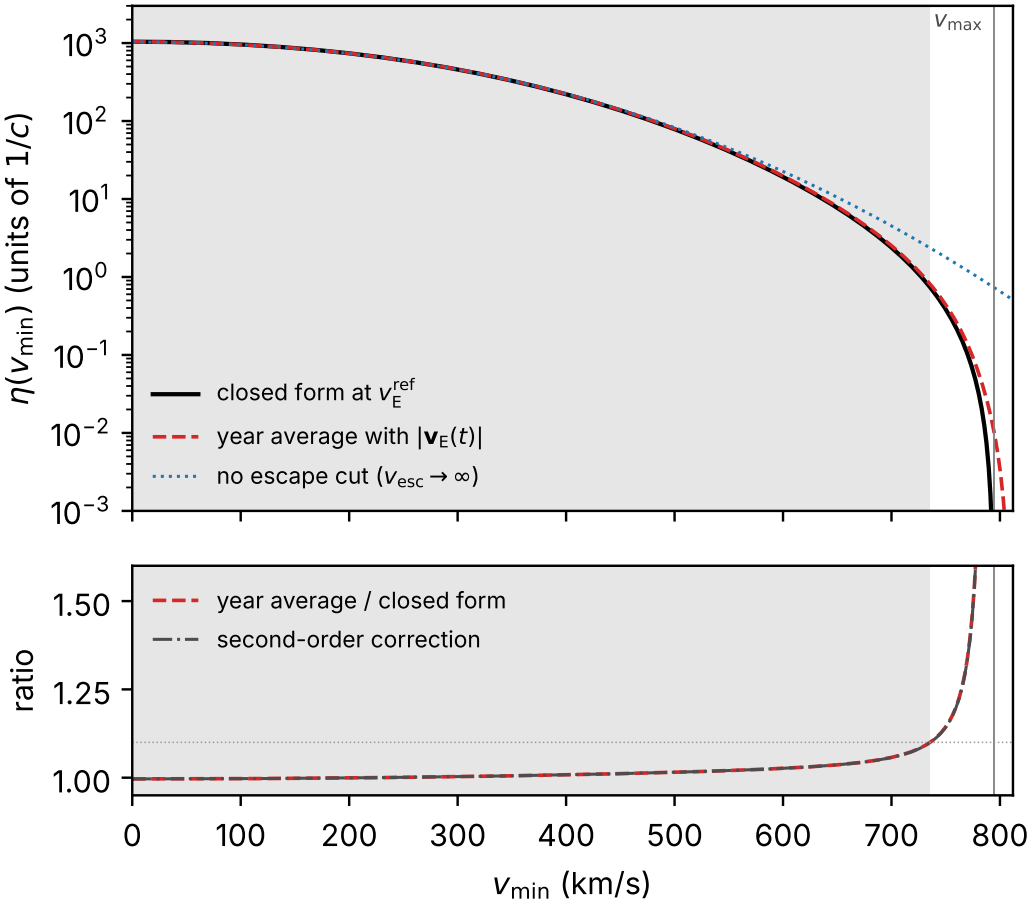


Figure 21.2: Mean inverse speed of the standard halo ($v_0 = 238\,\mathrm{km\,s^{-1}}$, $v_{\mathrm{esc}} = 544\,\mathrm{km\,s^{-1}}$) against v_{min}. Top, in units of $1/c$: the closed form at the reference Earth speed $v_{\mathrm{E}}^{\mathrm{ref}} = 250.5\,\mathrm{km\,s^{-1}}$ (solid). Dashed: the fuller result, the same closed form averaged over a year with the exact Earth speed of Eqs. (1.6) and (1.12) on 7305 equally spaced days. Dotted: the Maxwellian without the escape cut at $v_{\mathrm{E}}^{\mathrm{ref}}$. Bottom: the yearly average over the closed form (dashed) and the second-order correction Eq. (21.7) (dash-dotted). Gray: $v_{\mathrm{min}} \le 735.6\,\mathrm{km\,s^{-1}}$, where the yearly average exceeds the closed form at $v_{\mathrm{E}}^{\mathrm{ref}}$ by less than 10 percent (1 percent up to $430.3\,\mathrm{km\,s^{-1}}$). The second-order correction stays within 1 percent of the yearly average up to $786.3\,\mathrm{km\,s^{-1}}$. Vertical line: $v_{\mathrm{max}} = 794.5\,\mathrm{km\,s^{-1}}$.

Within $59\,\mathrm{km\,s^{-1}}$ of v_{max} the yearly average exceeds the fixed-speed form by more than 10 percent, and above v_{max} it vanishes while the average stays positive up to $810.4\,\mathrm{km\,s^{-1}}$.

At $v_{\mathrm{min}} = 0$ the yearly average is 0.9965 of the closed form: the mean inverse speed itself barely moves. The ratio grows to 1.015 at $500\,\mathrm{km\,s^{-1}}$, 1.026 at 600 and 1.057 at $700\,\mathrm{km\,s^{-1}}$, then turns up steeply, to 1.20 at 760 and 1.78 at $780\,\mathrm{km\,s^{-1}}$. The second-order term follows it within a few parts in a thousand to $780\,\mathrm{km\,s^{-1}}$. The escape cut matters over a much wider range than the annual average does: without it, η would be 1.18 times larger at $600\,\mathrm{km\,s^{-1}}$, 1.90 times at 700 and 6.8 times at $760\,\mathrm{km\,s^{-1}}$. Averaging η over the year raises the silicon rate of Worked example 10.1 by 0.8 percent at 10 MeV and by 4.3 percent at 1 MeV, where the kernel opens near $470\,\mathrm{km\,s^{-1}}$. Closer to the threshold mass the correction grows without bound, and Sect. 21.3 treats that regime.

Fig. 21.3 shows the kernel of the silicon electron-recoil channel for a heavy mediator, built from the tabulated form factor of Fig. 10.3. At 10 MeV it opens at $v_{\mathrm{on}} = 146.9\,\mathrm{km\,s^{-1}}$, which is $(2\omega/m_\chi)^{1/2}$ at the table's first energy, 1.2 eV; the kernel, scaled to a rate per target mass as in Fig. 21.3, is 6.29×10^4 per kilogram and year at $300\,\mathrm{km\,s^{-1}}$, 3.31×10^5 at $500\,\mathrm{km\,s^{-1}}$ and 7.71×10^5 at v_{max}, for $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm^2}$. Integrated against F_χ at $v_{\mathrm{E}}^{\mathrm{ref}}$, it returns $1.599 \times 10^5\,\mathrm{kg^{-1}\,yr^{-1}}$, the rate of Worked example 10.1, and the form Eq. (21.5) returns the same number to five digits. At 1 MeV the kernel opens only near $470\,\mathrm{km\,s^{-1}}$ and the rate, $505\,\mathrm{kg^{-1}\,yr^{-1}}$, comes entirely from the fastest fifth of the halo. At 100 MeV it opens at $46.4\,\mathrm{km\,s^{-1}}$ and changes by a factor of 3.7 between 300 and $700\,\mathrm{km\,s^{-1}}$, against 9.9 at 10 MeV. The heavier the particle, the flatter the kernel over the bulk of the halo, and the less the rate depends on the shape of F_χ.

21.1.3 Threshold masses as floors

Where the kernel vanishes, so does the rate. By Eq. (21.1), a mass gives a nonzero rate only if the halo holds particles faster than $v_{\mathrm{on}}(m_\chi)$. For a response whose onset E_{th} does not depend on q, the smallest v_{min} is $(2E_{\mathrm{th}}/m_\chi)^{1/2}$ (Sect. 1.3), and the condition $v_{\mathrm{on}} < v_{\mathrm{max}}$ is $m_\chi > 2E_{\mathrm{th}}/v_{\mathrm{max}}^2$, the threshold mass of Eq. (1.25). Quoted with $v_{\mathrm{max}} = v_{\mathrm{esc}} + v_{\mathrm{E}}^{\mathrm{ref}}$, that value is a floor, never the edge of a limit: a real onset depends on q, the rate vanishes at the floor, and the floor moves with the Earth speed of the exposure.

The onset of a real response depends on q. Then $v_{\mathrm{on}} = \min_q[E_{\mathrm{th}}(q)/q + q/2m_\chi]$ lies above $(2E_{\mathrm{th}}/m_\chi)^{1/2}$ with E_{th} the global minimum, and the edge lies above the floor. In silicon the floor at the gap, 1.170 eV, is 0.3332 MeV, and at the lowest excitation, the $1s$ exciton at 1.1553 eV, it is 0.3290 MeV. The phonon-assisted rate opens at 0.3374 MeV for a free pair and at 0.3332 MeV for the exciton replica, and the zero-phonon rate only at 0.405 to 0.408 MeV (Sects. 13.1 and 13.9).

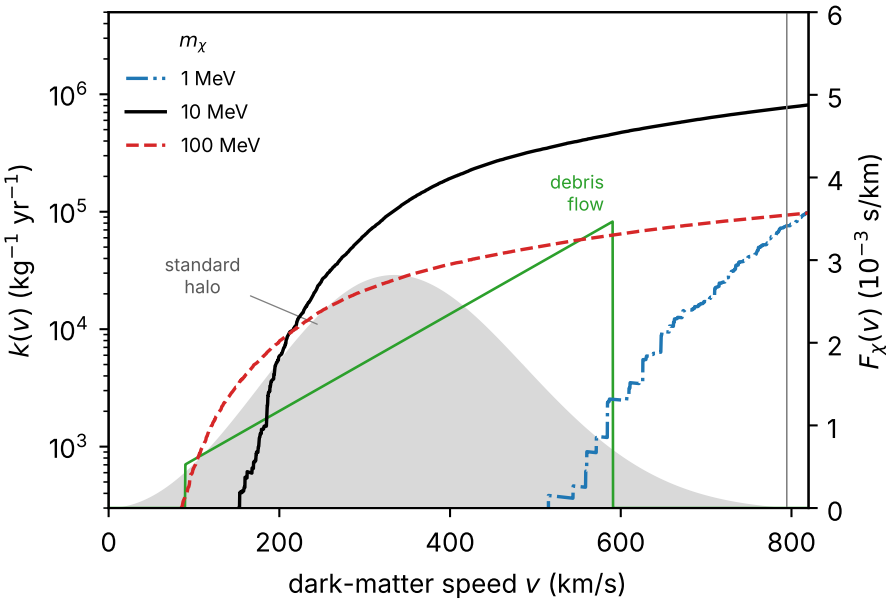


Figure 21.3: Rate kernel of the silicon electron-recoil channel, $\rho_\chi \Gamma(v)/m_\chi \rho_T$ of Eq. (21.4), the rate per unit target mass if all dark matter moved at speed v, for a heavy mediator, $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$ and $\rho_\chi = 0.3\,\mathrm{GeV\,cm}^{-3}$, at $m_\chi = 1$, 10 and 100 MeV (left axis). The response is the unscreened, angle-averaged crystal form factor tabulated by Essig et al. [2] (gap set to 1.11 eV, first energy 1.2 eV; Fig. 10.3); the steps of the 1 MeV curve are the bins of that table at small q. Right axis: laboratory speed distributions at $v_{\mathrm{E}}^{\mathrm{ref}} = 250.5\,\mathrm{km\,s}^{-1}$ of the standard halo (gray, Eq. (1.10)) and of a debris flow of galactic speed $340\,\mathrm{km\,s}^{-1}$ (green, Eq. (21.8)). Vertical line: $v_{\max} = 794.5\,\mathrm{km\,s}^{-1}$. The rate in any halo is the integral of the kernel times its speed distribution.

At the floor the rate is zero. Near $v_{\max}$ the halo supply η vanishes quadratically (Eq. (1.29)), and the reachable part of the response shrinks with it. A finite exposure therefore excludes only masses some distance above the floor, and how far above depends on the exposure, on the onset exponent of the response and on the tail of the halo. Sect. 21.3 derives that distance.

The floor belongs to the halo and to the Earth speed it is computed with. The reference speed fixes $v_{\max} = 794.5\,\mathrm{km\,s^{-1}}$ by convention (Appendix A.8). In June the fastest particles move at $v_{\mathrm{esc}} + |\boldsymbol{v}_{\mathrm{E}}| = 810.4\,\mathrm{km\,s^{-1}}$, and an exposure that includes June has a nonzero rate down to a floor 3.9 percent lower, 0.3202 MeV for the silicon gap; Limit Check 21.1 shows its origin in η. The annual motion is known kinematics, not an uncertainty: a limit uses the rate averaged over its exposure, June included (Sect. 21.3.4), and the floor at the reference speed is the convention for comparing targets. Conversely, a halo component lowers the floor only if it carries particles faster than $v_{\max}$. A debris flow at $340\,\mathrm{km\,s^{-1}}$ reaches at most $590.5\,\mathrm{km\,s^{-1}}$ in the laboratory, and on its own it would give no silicon electron recoil below 0.6031 MeV.

21.1.4 A halo with a debris flow

Kuhlen, Lisanti and Spergel model the debris flow as an isotropic shell in velocity space: every particle has the same galactic speed v_{flow}, in a random direction [1]. In the laboratory a particle of direction $\hat{\boldsymbol{v}}_{\mathrm{gal}}$ has speed v with $v^2 = v_{\mathrm{flow}}^2 + v_{\mathrm{E}}^2 - 2v_{\mathrm{flow}}v_{\mathrm{E}}\cos\theta_{\mathrm{flow}}$, where θ_{flow} is the angle between $\boldsymbol{v}_{\mathrm{gal}}$ and $\boldsymbol{v}_{\mathrm{E}}$. Since $\cos\theta_{\mathrm{flow}}$ is uniform on $[-1,1]$ and $v\,\mathrm{d}v = -v_{\mathrm{flow}}v_{\mathrm{E}}\,\mathrm{d}\cos\theta_{\mathrm{flow}}$, the laboratory speed distribution is [1, Eq. (7)]

$$F_{\mathrm{flow}}(v) = \frac{v}{2v_{\mathrm{flow}}v_{\mathrm{E}}} \quad \text{for } |v - v_{\mathrm{flow}}| < v_{\mathrm{E}}, \tag{21.8}$$

and zero outside, for $v_{\mathrm{flow}} > v_{\mathrm{E}}$. Its mean inverse speed is $\eta_{\mathrm{flow}}(u) = 1/v_{\mathrm{flow}}$ for $u < v_{\mathrm{flow}} - v_{\mathrm{E}}$, falls linearly as $(v_{\mathrm{flow}} + v_{\mathrm{E}} - u)/(2v_{\mathrm{flow}}v_{\mathrm{E}})$ across the window, and vanishes above $v_{\mathrm{flow}} + v_{\mathrm{E}}$ [1, Eq. (8)]. The halo is $f_\chi = (1 - p_{\mathrm{flow}})f_{\mathrm{SHM}} + p_{\mathrm{flow}}f_{\mathrm{flow}}$ at fixed total density, and in Via Lactea II $p_{\mathrm{flow}} = 0.22$ and $v_{\mathrm{flow}} = 340\,\mathrm{km\,s^{-1}}$ [1, Eq. (10) and Fig. 6]. Compared to the standard halo at the reference Earth speed, the flow has fewer slow particles, $\langle 1/v\rangle = 1/v_{\mathrm{flow}}$, or 882 in natural units against 1041, and more fast ones: 55 percent of its particles move faster than $400\,\mathrm{km\,s^{-1}}$, against 36 percent. It has none above $590.5\,\mathrm{km\,s^{-1}}$.

Worked example 21.1: The silicon rate in a halo with a debris flow

Take silicon, $m_\chi = 10\,\mathrm{MeV}$, a heavy mediator and $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$, as in Worked example 10.1, with the kernel of Fig. 21.3 and $v_{\mathrm{E}} = v_{\mathrm{E}}^{\mathrm{ref}}$ for both components. The flow occupies $89.5 < v < 590.5\,\mathrm{km\,s^{-1}}$. Inserting Eq. (21.8) in Eq. (21.1), the factor v of F_{flow} meets the factor $1/v$ of the kernel:

$$\mathcal{R}_{\mathrm{flow}} = \frac{1}{2v_{\mathrm{flow}}v_{\mathrm{E}}}\int_{v_{\mathrm{flow}}-v_{\mathrm{E}}}^{v_{\mathrm{flow}}+v_{\mathrm{E}}} \mathrm{d}v\; v\,\frac{\rho_\chi \Gamma(v)}{m_\chi \rho_T} = \frac{1}{v_{\mathrm{flow}}}\left\langle v\,\frac{\rho_\chi\Gamma(v)}{m_\chi\rho_T}\right\rangle_{\mathrm{window}}.$$

A shell rate is the mean of $v\Gamma(v)$ over its window, divided by its galactic speed. Across the window the kernel rises from zero below $146.9\,\mathrm{km\,s^{-1}}$ through $6.29\times10^4\,\mathrm{kg^{-1}\,yr^{-1}}$ at $300\,\mathrm{km\,s^{-1}}$, 1.92×10^5 at 400 and 3.31×10^5 at $500\,\mathrm{km\,s^{-1}}$, to $4.57\times10^5\,\mathrm{kg^{-1}\,yr^{-1}}$ at $590\,\mathrm{km\,s^{-1}}$. The mean of v times the kernel over the window is $7.313\times10^7\,\mathrm{km\,s^{-1}\,kg^{-1}\,yr^{-1}}$, and

$$\mathcal{R}_{\mathrm{flow}} = \frac{7.313\times10^7\,\mathrm{km\,s^{-1}\,kg^{-1}\,yr^{-1}}}{340\,\mathrm{km\,s^{-1}}} = 2.151\times10^5\,\mathrm{kg^{-1}\,yr^{-1}},$$

the rate if all of the dark matter were in the flow. The form Eq. (21.5) with η_{flow} gives the same number. Per particle the flow is 1.345 times as effective as the standard halo, $\mathcal{R}_{\mathrm{SHM}} = 1.599\times10^5\,\mathrm{kg^{-1}\,yr^{-1}}$, because it puts more particles where the kernel is large.

The mixed halo, by linearity, gives

$$\mathcal{R} = 0.78 \times 1.599 \times 10^5 + 0.22 \times 2.151 \times 10^5 = 1.721 \times 10^5 \,\mathrm{kg^{-1}\,yr^{-1}},$$

7.6 percent above the standard halo, or 471 events per kilogram and day. The spectrum hardly changes shape: its mean moves from 6.01 to 6.06 eV. The flow's own spectrum ends at 19.4 eV, the largest energy a particle at $590.5\,\mathrm{km\,s^{-1}}$ can deposit, against 35 eV for the standard halo, and it peaks at the same 6.5 eV.
The same halo changes the rate differently at other masses. For a light mediator at 10 MeV the mixed rate is 8.2 percent above the standard one, and for a heavy mediator at 100 MeV, 4.9 percent. At 1 MeV, where the kernel opens near $470\,\mathrm{km\,s^{-1}}$, the flow is only 0.39 times as effective as the standard halo, and the mixed rate falls 13 percent below it. A debris flow moves particles from both tails of the speed distribution into its middle. It raises the rate where the kernel opens in the bulk of the halo and lowers it near the threshold mass, which stays at the floor set by the standard component.

21.2 Halo-independent bounds

A dark-matter particle of 2 MeV can make an electron–hole pair in silicon only if it moves faster than $333\,\mathrm{km\,s^{-1}}$ in the laboratory, and a second pair only above $474\,\mathrm{km\,s^{-1}}$, against the largest speed of the standard halo, $v_{\max} = 794.5\,\mathrm{km\,s^{-1}}$. A limit at that mass is therefore a statement about the fast part of the dark matter in the solar neighborhood, the part of the halo that Sect. 1.2 found least certain: changing the escape speed by $25\,\mathrm{km\,s^{-1}}$ changes η at $760\,\mathrm{km\,s^{-1}}$ by a factor of 2.26 or 0.10. A statement that holds for every velocity distribution avoids the question. Because every count is a linear functional of f_χ (Eq. (21.1)), the distributions that a null result allows form a convex set, and the largest value of any other linear quantity over that set is a semi-infinite linear program, with finitely many constraints on a measure over a continuum, the problem of Appendix C.12. We will set up that program for a crystal and solve it in closed form when the response depends only on the speed. We then extend it to the direction-dependent response of a crystal that turns with the Earth, and bound the error of solving it on a grid or with a response that has negative pieces. For silicon at 2 and 10 MeV, dropping the halo model costs a factor of 2.1 to 3.7 in the bulk of the halo and orders of magnitude near $v_{\max}$.

21.2.1 Stream kernels and the linear program

The exposure-averaged rate Eq. (21.3) is an integral over galactic velocities $\boldsymbol{w}$ of the distribution f_{gal} times a kernel, the rate that a single stream at $\boldsymbol{w}$ would produce. Multiplied by the exposure $M_{\mathrm{det}} t_{\mathrm{exp}}$ it is a count. The local density, the cross-section and the mass multiply every rate in the combination $\rho_\chi \bar{\sigma}_e / m_\chi$, which no count can separate from the normalization of f_{gal}; the mass enters the kinematics as well, and the analysis is done at fixed m_χ. We therefore absorb the combination into a nonnegative measure Ξ on the ball $\mathcal{B} = \{|\boldsymbol{w}| \le v_{\mathrm{esc}}\}$ and divide the kernel by $\bar{\sigma}_e$ (Appendix D.46):

$$\mathrm{d}\Xi(\boldsymbol{w}) = \frac{\rho_\chi \bar{\sigma}_e}{m_\chi}\, f_{\mathrm{gal}}(\boldsymbol{w})\, \mathrm{d}^3 w, \qquad \mathsf{k}_i(\boldsymbol{w}) = \frac{M_{\mathrm{det}} t_{\mathrm{exp}}}{\rho_T\, \bar{\sigma}_e} \left\langle \Gamma_i(\boldsymbol{w} - \boldsymbol{v}_{\mathrm{E}}(t)) \right\rangle_t, \tag{21.9}$$

with Γ_i the rate kernel Eq. (21.2) for bin i of the observable. The expected count in bin i is then $\int \mathsf{k}_i \,\mathrm{d}\Xi$. The *stream kernel* $\mathsf{k}_i(\boldsymbol{w})$ is the count that a single stream of unit Ξ-mass at $\boldsymbol{w}$

would leave in bin i; it contains the coupling up to its strength, the response, the final-state weights and the exposure, and nothing about the halo. A measured null result in bin i becomes the constraint $\int \mathsf{k}_i \, \mathrm{d}\Xi \le N_i^{\mathrm{up}}$, with N_i^{up} the upper count of Sect. 1.5 at the chosen confidence.

The quantity to bound is a second linear functional, $\int \mathsf{f} \, \mathrm{d}\Xi$: the mean inverse speed above u_0 in the laboratory, the count of a second experiment, or a modulation amplitude. The mean inverse speed is the object of the standard halo-independent analysis,

$$\tilde{\eta}(u_0) = \int \frac{\Theta(|\boldsymbol{w} - \boldsymbol{v}_{\mathrm{E}}| - u_0)}{|\boldsymbol{w} - \boldsymbol{v}_{\mathrm{E}}|} \, \mathrm{d}\Xi(\boldsymbol{w}) = \frac{\rho_\chi \bar{\sigma}_e}{m_\chi} \, \eta(u_0), \tag{21.10}$$

which is $\tilde{\eta}$ of Appendix A.8 with $\boldsymbol{v}_{\mathrm{E}}$ held fixed. Its value for the standard halo at a given $\bar{\sigma}_e$ is a curve in the plane of u_0 and $\tilde{\eta}$, and a bound on $\tilde{\eta}$ at every u_0 is the halo-independent counterpart of an exclusion curve. The count expected in a second experiment, $\mathsf{f} = \mathsf{k}_j^{\mathrm{B}}$, compares two targets, or one crystal at two orientations, without a halo model. A modulation amplitude, the difference of the kernels of two time bins, is a signed functional, bounded only if every stream on which it is positive is seen by some bin. The program is Eq. (C.111) with these kernels.

By Appendix C.12, any vector $\boldsymbol{\lambda} \ge 0$ with $\sum_i \lambda_i \mathsf{k}_i \ge \mathsf{f}$ on all of $\mathcal{B}$, a *dual certificate*, proves $\int \mathsf{f} \, \mathrm{d}\Xi \le \int \sum_i \lambda_i \mathsf{k}_i \, \mathrm{d}\Xi \le \sum_i \lambda_i N_i^{\mathrm{up}}$ for every allowed halo at once, Eq. (C.112). Checking it needs one function evaluated on the ball, not the optimizer that produced it. And the optimum is attained by a halo of at most n streams for n bins, $\Xi^* = \sum_a \xi_a \delta_{\boldsymbol{w}_a}$ with Ξ-masses ξ_a, placed at the zeros of the dual function $\mathsf{g}_\lambda = \sum_i \lambda_i \mathsf{k}_i - \mathsf{f}$. The extremal halos are therefore cold streams, far from any equilibrium model, and the bound holds for every halo because it holds for them. For nuclear recoils, Feldstein and Kahlhoefer wrote $\tilde{\eta}$ as a sum of steps, each a stream [3, Sect. 3, Eq. (3.1)]. Ibarra and Rappelt solved the program with the normalization of f_χ as an extra equality constraint, which allows one more stream [4, Eqs. (14)–(19)]. Gelmini, Huh and Witte proved from the convex hull of the rates that any likelihood is maximized by a sum of streams [5]. Chen, Gelmini and Takhistov carried the method to electron scattering with the response averaged over directions [6, Sect. 2], and Bernreuther and collaborators wrote the silicon and germanium rates as a matrix acting on steps of $\tilde{\eta}$ [7, Eqs. (2.17)–(2.18)].

All bins of a search constrain one halo, so a bound derived from them is a statement about the event that every bin stays below its N_i^{up}. If each bin is set at confidence $1 - \alpha_i$, that event has probability at least $1 - \sum_i \alpha_i$ by the union bound, and the bounds at all u_0 hold together at that level. The examples below use three pair bins with $N_i^{\mathrm{up}} = \ln 30 = 3.401$ each, the zero-event Poisson limit at 96.7 percent, so that the three bins hold jointly at 90 percent or more.

21.2.2 The bound for a speed-only response

For a response averaged over the directions of $\boldsymbol{q}$, as in the crystal form factor of Sect. 10.1, and with $\boldsymbol{v}_{\mathrm{E}}$ fixed, the stream kernel depends only on the laboratory speed $v = |\boldsymbol{w} - \boldsymbol{v}_{\mathrm{E}}|$. By Eq. (21.4) it is $\mathsf{k}_i(v) = \mathcal{A}_i(v)/v$, with the *step column*

$$\mathcal{A}_i(v) = \frac{M_{\mathrm{det}} t_{\mathrm{exp}}}{\rho_T \, \bar{\sigma}_e} \, v \, \Gamma_i(v) = \frac{M_{\mathrm{det}} t_{\mathrm{exp}}}{\rho_T \, \bar{\sigma}_e} \int \mathrm{d}\omega \int \frac{q \, \mathrm{d}q}{4\pi^2} \, \bar{s}_i(q, \omega) \, \Theta\big(v - v_{\mathrm{min}}(q, \omega)\big), \tag{21.11}$$

where $\bar{s}_i$ is the angle-averaged response contracted with $\mathfrak{D}$ and weighted with the probability $P(i|\Lambda)$ of landing in bin i. The column is the count in bin i for the step $\tilde{\eta}(u) = \Theta(v - u)$: all of $\tilde{\eta}$ sits below the speed v, the halo of a single stream at v. In silicon with the form factor of Limit Check 10.1, the prefactor $M_{\mathrm{det}} t_{\mathrm{exp}} \, \alpha m_e^2 / (M_{\mathrm{cell}} \mu_{\chi e}^2)$, with M_{cell} the mass of the two-atom unit cell, turns the column into a sum over the table, and $\mathcal{A}_i$ comes out in years for $\tilde{\eta}$ in inverse years. The column is the integral of the response over the part of the (q, ω) plane

that a particle of speed v can reach, and it grows with v, since a faster particle reaches every pair that a slower one does. For a positive response it never decreases (Eq. (D.250)).

Monotonicity solves the program. A stream at laboratory speed u_0 saturates the bin that sees it best relative to its bound. No halo does better:

Halo-independent bound for a speed-only response

For a response that depends only on $|\boldsymbol{q}|$, nonnegative bin weights, a fixed $\boldsymbol{v}_{\rm E}$, $0 < u_0 \leq v_{\max}$ and $\max_i \mathcal{A}_i(u_0) > 0$, every velocity distribution allowed by the bins satisfies

$$\tilde{\eta}(u_0) \leq \min_i \frac{N_i^{\rm up}}{\mathcal{A}_i(u_0)}, \tag{21.12}$$

and the bound is attained. The optimal halo is one stream at laboratory speed u_0 in any allowed direction, and the certificate is $\lambda_j = 1/\mathcal{A}_j(u_0)$ on the minimizing bin j, zero on the others.

Appendix D.46 gives the proof: the stream is feasible and reaches the bound, and the certificate holds because $\mathcal{A}_j(v) \geq \mathcal{A}_j(u_0)$ for $v \geq u_0$. With one bin, Eq. (21.12) is the step-function bound of Fox, Liu and Weiner, who placed the limit on a halo of the form $\Theta(v_1 - v_{\min})$ and noted that it corresponds to a stream [8, Eq. (A1)]. With several bins the minimum over bins is the exact optimum, not an estimate, because one dual weight certifies it; the other bins are not saturated by the optimal stream and carry no weight. The closed form fails if a column decreases somewhere or if the kernel and the objective use different speeds. A column that decreases somewhere, which a truncated response with negative pieces can produce, leaves the single-bin value $N_i^{\rm up}/\inf_{v\geq u_0} \mathcal{A}_i(v)$ of Eq. (C.114) and voids the closed form for several bins. And the kernel and the objective must be functions of the same laboratory speed. In galactic coordinates the kernel depends on $|\boldsymbol{w} - \boldsymbol{v}_{\rm E}|$; a bound on the galactic speed integral $\int \Theta(|\boldsymbol{w}| - u_0)|\boldsymbol{w}|^{-1}{\rm d}\Xi$ is a different program, and Eq. (21.12) does not answer it.

Fig. 21.4 applies Eq. (21.12) to a silicon search of one kilogram-year that records no event in the one-, two- and three-pair bins, for a heavy mediator. A deposit ω lands in bin i with the probability $P_{\rm casc}(i|\omega)$ of the cascade of Sect. 18.3, the book's calibrated model with $E_{\rm g} = 1.160\,{\rm eV}$ and $3.749\,{\rm eV}$ per pair, computed for the seed of a dark-matter kick and applied at every q. The probability of the one-pair bin exceeds one half from 1.2 to 6.2 eV, that of the two-pair bin from 6.3 to 9.5 eV and that of the three-pair bin from 9.7 to 13.1 eV, and a second pair is possible from about 2.5 eV. The step model of Sect. 1.5, which puts every deposit between 4.71 and 8.31 eV in the two-pair bin, overcounts that bin about tenfold near 5 eV (Sect. 18.3), and with it the two-pair bin would set the bound at 10 MeV. Each bin gives a curve $N_i^{\rm up}/\mathcal{A}_i(u_0)$ that comes down from infinity at the onset speed of the bin, the smallest $v_{\min}$ among the pairs it can contain, and flattens as the column saturates; the bound is their lower envelope. At 10 MeV the one-pair bin sets it below $532.6\,{\rm km\,s^{-1}}$ and the two-pair bin above. At 2 MeV the one-pair bin sets it at every speed: the two-pair bin opens at $474.3\,{\rm km\,s^{-1}}$ and the three-pair bin at $614.4\,{\rm km\,s^{-1}}$, and at $v_{\max}$ their columns are 2.9 and 250 times smaller. The dashed curves are $\tilde{\eta}$ of the standard halo at the cross-section where it first reaches $N_i^{\rm up}$ in some bin: $\bar{\sigma}_e = 3.72 \times 10^{-42}\,{\rm cm^2}$ at 10 MeV and $1.78 \times 10^{-41}\,{\rm cm^2}$ at 2 MeV, both set by the one-pair bin.

The standard curve lies below the bound at every u_0, as it must. The count of any halo in bin j is $\int_0^\infty \tilde{\eta}(u)\,{\rm d}\mathcal{A}_j(u)$, Eq. (21.5) in these variables, and since $\tilde{\eta}$ never increases and $\mathcal{A}_j$ never decreases, it is at least $\tilde{\eta}(u_0)\mathcal{A}_j(u_0)$. A halo that crossed the bound at some u_0 would overshoot $N_j^{\rm up}$. The ratio of the bound to the standard curve is the cost of dropping the halo model. It is smallest at $379\,{\rm km\,s^{-1}}$ for 10 MeV, a factor 2.12, and at $559\,{\rm km\,s^{-1}}$ for 2 MeV,

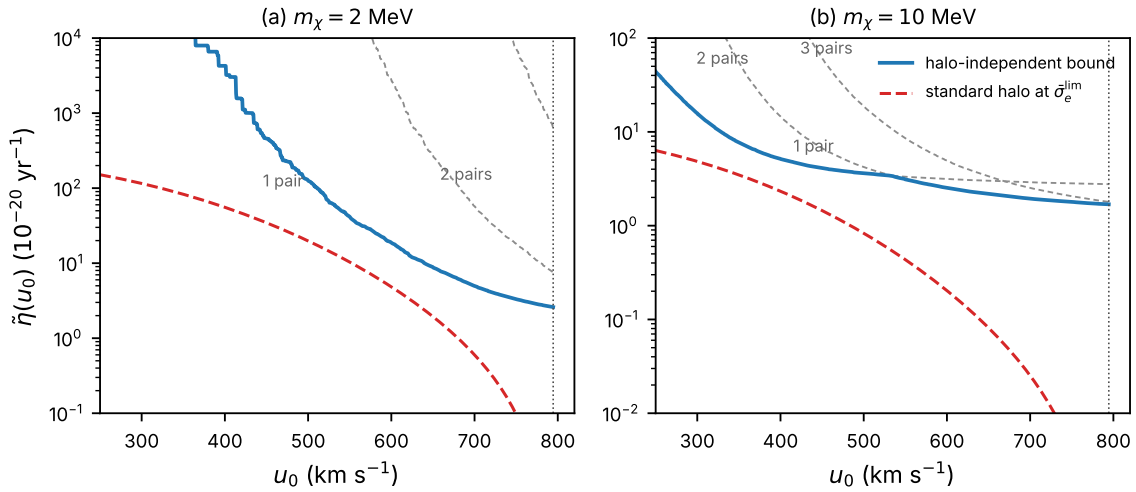


Figure 21.4: Halo-independent bound on $\tilde{\eta}(u_0) = \rho_\chi \bar{\sigma}_e \eta(u_0)/m_\chi$ for silicon, Eq. (21.12) (solid), against $\tilde{\eta}$ of the standard halo at the cross-section of its own limit (dashed), for (a) $m_\chi = 2\,\mathrm{MeV}$ and (b) 10 MeV, heavy mediator. Search: one kilogram-year at the reference Earth speed, no events in the one-, two- and three-pair bins, with the bin probabilities of the energy-only cascade of Sect. 18.3 (3.749 eV per pair), $N_i^{\mathrm{up}} = \ln 30$ in each bin, jointly at least 90 percent confidence. Gray dashed: single-bin bounds $N_i^{\mathrm{up}}/\mathcal{A}_i(u_0)$, labeled by the number of pairs; the solid curve is their lower envelope. Step columns from the angle-averaged crystal form factor of Essig et al. (Fig. 10.3). Standard halo: $v_0 = 238$, $v_{\mathrm{esc}} = 544\,\mathrm{km\,s^{-1}}$, $v_{\mathrm{E}}^{\mathrm{ref}} = 250.5\,\mathrm{km\,s^{-1}}$, at $\bar{\sigma}_e = 1.78 \times 10^{-41}\,\mathrm{cm}^2$ (a) and $3.72 \times 10^{-42}\,\mathrm{cm}^2$ (b). Dotted: $v_{\max} = 794.5\,\mathrm{km\,s^{-1}}$. The steps in (a) are the energy and momentum bins of the table near the one-pair onset.

a factor 3.70. Toward $v_{\max}$ it opens to 12 at $600\,\mathrm{km\,s^{-1}}$ and 77 at $700\,\mathrm{km\,s^{-1}}$ for 10 MeV, because the standard halo has few particles there while a stream can put all its density at one speed. The curves hold for an exposure at the reference Earth speed; over a year the standard curve near $v_{\max}$ follows the year average of Sect. 21.1, 1.06 times higher at $700\,\mathrm{km\,s^{-1}}$. The halo-independent bound is weakest exactly where a limit near its kinematic edge draws its events, the subject of Sect. 21.3.

Worked example 21.2: One stream saturating a bound

Take silicon at $m_\chi = 10\,\mathrm{MeV}$, a heavy mediator, the search of Fig. 21.4 and $u_0 = 500\,\mathrm{km\,s^{-1}}$. The reduced mass is $\mu_{\chi e} = 486.2\,\mathrm{keV}$ (Worked example 10.1), and the prefactor of the column, $M_{\mathrm{det}} t_{\mathrm{exp}}\, \alpha m_e^2/(M_{\mathrm{cell}} \mu_{\chi e}^2)$, is $8.64 \times 10^{22}\,\mathrm{yr}$ for one kilogram-year. Summing the table cells with $v_{\min} \le 500\,\mathrm{km\,s^{-1}}$, each weighted with its bin probability, gives

$$\mathcal{A}_1 = 9.41 \times 10^{19}\,\mathrm{yr}, \qquad \mathcal{A}_2 = 8.07 \times 10^{19}\,\mathrm{yr}, \qquad \mathcal{A}_3 = 1.82 \times 10^{19}\,\mathrm{yr},$$

for the one-, two- and three-pair bins. Divided into $N^{\mathrm{up}} = 3.401$ they give 3.61×10^{-20}, 4.21×10^{-20} and $1.87 \times 10^{-19}\,\mathrm{yr}^{-1}$. The one-pair bin is tightest, and

$$\tilde{\eta}(500\,\mathrm{km\,s^{-1}}) \le \frac{3.401}{9.41 \times 10^{19}\,\mathrm{yr}} = 3.61 \times 10^{-20}\,\mathrm{yr}^{-1}.$$

The halo that reaches it is one stream at a laboratory speed of $500\,\mathrm{km\,s^{-1}}$, 1.668×10^{-3} in natural units, with Ξ-mass $\xi_1 = u_0 N^{\mathrm{up}}/\mathcal{A}_1 = 6.03 \times 10^{-23}\,\mathrm{yr}^{-1}$. It leaves 3.40 events in

the one-pair bin, 2.92 in the two-pair bin and 0.66 in the three-pair bin: only the one-pair bound is saturated. Its galactic speed lies between 249.5 and 750.5 km s^{-1} depending on its direction, and the directions against the Earth's motion are inside the escape speed. The certificate is $\lambda_1 = 1/\mathcal{A}_1 = 1.06 \times 10^{-20}\,\mathrm{yr}^{-1}$ per event, $\lambda_2 = \lambda_3 = 0$, with value $\lambda_1 N^{\mathrm{up}} = 3.61 \times 10^{-20}\,\mathrm{yr}^{-1}$, equal to the stream's. It is valid because $\mathcal{A}_1(v)/\mathcal{A}_1(u_0)$ never drops below 1 between 500 and 794.5 km s^{-1}, where it rises to 1.30; the check needs the column alone, not the program.
The standard halo at its own limit, $\bar{\sigma}_e = 3.72 \times 10^{-42}\,\mathrm{cm}^2$, has $\tilde{\eta}(500\,\mathrm{km\,s}^{-1}) = 8.31 \times 10^{-21}\,\mathrm{yr}^{-1}$, a factor 4.35 below the bound. The factor has two parts. Of the 3.40 standard-halo events in the one-pair bin, 0.86 come from particles faster than 500 km s^{-1}; the other 2.54 are spent on particles that contribute nothing to $\tilde{\eta}(500\,\mathrm{km\,s}^{-1})$, a factor $3.401/0.860 = 3.96$. And the fast particles move above 500 km s^{-1}, where the column is larger, so they produce 1.10 times the count of a stream at exactly 500 km s^{-1} with the same $\tilde{\eta}$. The stream avoids both losses, and $3.96 \times 1.10 = 4.35$.

21.2.3 Directional responses in the laboratory

A crystal resolves directions. Its response depends on $\hat{\boldsymbol{q}}$ in crystal axes, the crystal turns with the Earth, and a stream fixed in the galactic frame sweeps a cone of crystal directions every sidereal day (Sect. 11.7). The stream kernel then depends on the direction of the stream, there is no step column, and the program has to be solved. Take a cubic crystal and a coupling that depends on $|\boldsymbol{q}|$ alone. The bin-weighted response is then $\bar{s}_i[1 + \sum_\ell a_\ell^i Y_\ell^c(\hat{\boldsymbol{q}})]$, with the cubic harmonics of Appendix C.3, $\ell = 4, 6, 8, \ldots$ The energy delta function confines $\hat{\boldsymbol{q}}$ to a circle about the stream direction, at the angle $\arccos(v_{\min}/v)$, and the average of a harmonic over that circle is $P_\ell(v_{\min}/v) Y_\ell^c(\hat{\boldsymbol{v}})$, Eq. (D.248). The kernel of a stream of velocity $\boldsymbol{v}$ in crystal axes is

$$\mathsf{k}_i(\boldsymbol{v}) = \frac{M_{\mathrm{det}} t_{\mathrm{exp}}}{\rho_T\, \bar{\sigma}_e\, v} \int \mathrm{d}\omega \int \frac{q\,\mathrm{d}q}{4\pi^2}\, \bar{s}_i(q,\omega) \Big[1 + \sum_{\ell \geq 4} a_\ell^i(q,\omega)\, P_\ell\Big(\frac{v_{\min}}{v}\Big) Y_\ell^c(\hat{\boldsymbol{v}})\Big] \Theta(v - v_{\min}(q,\omega)), \tag{21.13}$$

before the average over the exposure (Appendix D.46). Pairs with $v_{\min}$ close to v are reached only by momentum transfers along the stream, $P_\ell \to 1$, and carry the full anisotropy of the response. Pairs with $v_{\min} \ll v$ are reached on a wide circle, and their anisotropy is reduced by $P_\ell(v_{\min}/v)$, which for $\ell \geq 4$ is at most 0.375 in magnitude as $v_{\min}/v \to 0$. The anisotropy of the kernel is therefore largest near the onset speed of each bin, where every reachable pair has $v_{\min} \approx v$.

The exposure average turns $Y_\ell^c(\hat{\boldsymbol{v}})$ into its mean over the crystal directions that the stream visits. With $\boldsymbol{v}_{\mathrm{E}}$ fixed, a stream of laboratory speed v and declination δ_{dec} visits, over one sidereal day, the directions of Eq. (1.17) at all hour angles, and the mean depends on δ_{dec}, the latitude and the orientation of the crystal only. Streams with all right ascensions at a given δ_{dec} therefore have the same kernel, and the program may place a stream at whichever right ascension the galactic support allows. That support, $|\boldsymbol{v} + \boldsymbol{v}_{\mathrm{E}}| \leq v_{\mathrm{esc}}$, allows the declination δ_{dec} at speed v for some right ascension if $v^2 + |\boldsymbol{v}_{\mathrm{E}}|^2 - 2v|\boldsymbol{v}_{\mathrm{E}}| \cos(\delta_{\mathrm{dec}} + \delta_{\mathrm{E}}) \leq v_{\mathrm{esc}}^2$, with $\delta_{\mathrm{E}} = 47.5°$ the declination of $\boldsymbol{v}_\odot$ (Sect. 1.2); the cosine is that of the smallest angle between $-\boldsymbol{v}$ and $\boldsymbol{v}_{\mathrm{E}}$ over right ascension. At $v_{\max}$ only $\delta_{\mathrm{dec}} = -\delta_{\mathrm{E}}$ remains, a stream that moves opposite to the Earth and arrives from the direction toward which the Sun moves. The program reduces to two variables, the speed and the declination of the stream.

Limit Check 21.2 solves it for silicon at 2 MeV, where every momentum transfer lies in the range of the model response of Sect. 11.7. The isotropic part of the kernel comes

from the tabulated form factor. The ratios a_ℓ for $\ell = 4, 6, 8$ come from the cubic-harmonic components of the independent-particle structure factor of the empirical-pseudopotential model of Sect. 11.7, applied at the same (q, ω). They are a model, with a gap of 0.82 eV and no local fields, so the plot tests the method rather than measures silicon. The detector sits at 45° north with [001] vertical, [100] east and [010] north, the orientation of Worked example 11.8, and records the search of Fig. 21.4 in its one- and two-pair bins, integrated over sidereal time; the three-pair bin, open only above 614 km s^{-1} at this mass, never binds.

Limit Check 21.2. Speed-only bound against a directional response

For a response averaged over the directions of $\boldsymbol{q}$ and a laboratory-frame stream [8, Eqs. (A1)–(A2)]: $\tilde{\eta}(u_0) \le \min_i N_i^{\mathrm{up}}/\mathcal{A}_i(u_0)$, Eq. (21.12). The formula ignores the anisotropy of the crystal; the plot shows how far the bound of the directional program departs from it.

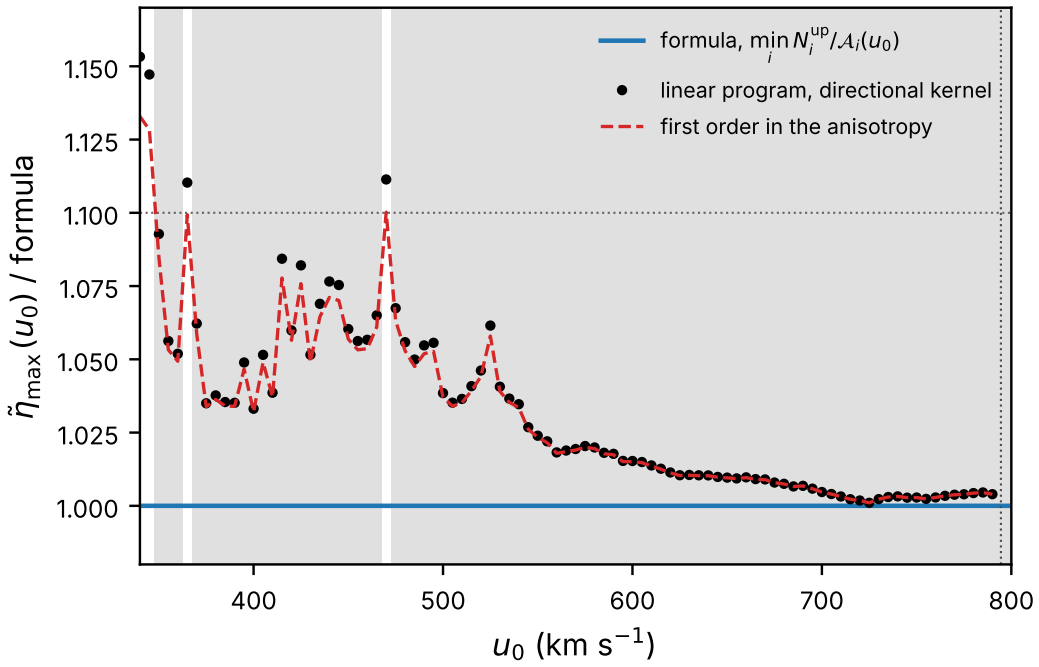


Figure 21.5: Largest $\tilde{\eta}(u_0)$ allowed by a silicon search over all halos, divided by the formula of the box (solid line at 1), against u_0. Dots: the fuller result, the linear program over galactic-frame streams with the direction-dependent kernel Eq. (21.13) of a crystal at 45° north ([001] up, [100] east), integrated over the sidereal day, sampled from 340 km s^{-1} in steps of 5 km s^{-1}. Grid: 2.5 km s^{-1} in speed and 2° in declination, with $|\boldsymbol{v}_{\mathrm{E}}| = 250.5$ km s^{-1} toward declination 47.5° and $v_{\mathrm{esc}} = 544$ km s^{-1}. Dashed: first order in the anisotropy, the formula divided by the smallest daily kernel of the tightest bin among the declinations allowed at u_0. $m_\chi = 2$ MeV, heavy mediator, one kilogram-year with no events in the one- and two-pair bins of the cascade of Sect. 18.3, $N_i^{\mathrm{up}} = \ln 30$. Isotropic response: the crystal form factor of Essig et al.; anisotropy: the $\ell = 4, 6, 8$ ratios of the empirical-pseudopotential model of Sect. 11.7 (a model). Gray: the formula lies within 10 percent of the linear program. Dotted: the 10 percent line and $v_{\max}$.

Within 35 km s^{-1} of the one-pair onset, where every reachable pair sees the full anisotropy

of the response, the formula falls short of the bound by up to 15 percent on the sampled speeds, and only the program gives a valid bound.

The daily average removes most of the anisotropy. In the model the kernel of a stream fixed in crystal axes drops to 0.33 of its angle average in the least favorable direction. The sidereal mean of Y_4^c, however, ranges only over $[-0.16, 0.07]$, against $[-0.43, 0.64]$ for Y_4^c itself. Over the declinations allowed at a given speed the daily kernel of the one-pair bin varies by -3.7 to $+1.8$ percent at $500\,\mathrm{km\,s^{-1}}$ and by ± 1.5 percent at $600\,\mathrm{km\,s^{-1}}$. At every u_0 the program puts one stream at speed u_0, at the allowed declination where the daily kernel of the tightest bin is smallest, $-68°$ at $600\,\mathrm{km\,s^{-1}}$. The bound then exceeds the formula by the corresponding fraction: 1.5 percent at $600\,\mathrm{km\,s^{-1}}$, 0.5 percent at $700\,\mathrm{km\,s^{-1}}$, and no more than 2.0 percent anywhere above $560\,\mathrm{km\,s^{-1}}$. The first-order estimate reproduces the program to 0.04 percent there and to 1.8 percent everywhere, because the optimum stays a single stream on a single bin. Below $560\,\mathrm{km\,s^{-1}}$ the ratio jumps between 1.02 and 1.11 as table cells enter the column one at a time, each with $v_{\min} \approx u_0$ and the full anisotropy, and at $340\,\mathrm{km\,s^{-1}}$, $7\,\mathrm{km\,s^{-1}}$ above the one-pair onset at $332.7\,\mathrm{km\,s^{-1}}$, it reaches 1.15. Closer to the onset the column holds a single table cell and was not sampled. At $790\,\mathrm{km\,s^{-1}}$ the support leaves only declinations between $-56°$ and $-40°$, and the program lies 0.4 percent above the formula. A search binned in sidereal time keeps a kernel per time bin, close to the crystal-frame kernel for short bins, and its bound has to come from the program. The computation, a sweep of the direction-resolved response over streams and sidereal phases followed by a program with one constraint per bin, is specified in Appendix E.27; it needs a response resolved in the direction of $\boldsymbol{q}$, which the libraries of Appendix E.16 and Sect. 21.6 provide.

21.2.4 Grids, certificates and leakage

A program solved on a grid of streams bounds the true optimum from below, since the grid can only restrict the halo, and its dual is a certificate only at the grid points. Between them the dual function can dip below zero. If the dip is at most δ_{cert} on the support of f, the dual still bounds every halo, at a price (Appendix D.46):

$$\int \mathsf{f}\,\mathrm{d}\Xi \le \sum_i \lambda_i N_i^{\mathrm{up}} + \frac{\delta_{\mathrm{cert}}\, N_{\mathrm{tot}}}{\kappa_{\mathrm{v}}}, \tag{21.14}$$

with $N_{\mathrm{tot}} = \sum_i N_i^{\mathrm{up}}$ and κ_{v} the smallest, over the support of f, of the largest bin kernel, the visibility of Eq. (C.113). The second term is the violation times the largest halo mass the bins still allow on the support of f. With $\delta_{\mathrm{cert}} = L_h h$, the Lipschitz constant of the dual function (the largest rate at which it changes per unit velocity) times the covering radius of the grid (the largest distance from a point of the support to the nearest grid point), it is the discretization bracket Eq. (C.115).

For the program of Limit Check 21.2 at $u_0 = 600\,\mathrm{km\,s^{-1}}$, a coarse grid of $10\,\mathrm{km\,s^{-1}}$ and 6° gives $\tilde{\eta}_{\max} = 1.9030 \times 10^{-19}\,\mathrm{yr^{-1}}$ and a fine grid of $1\,\mathrm{km\,s^{-1}}$ and 0.5° gives $1.9049 \times 10^{-19}\,\mathrm{yr^{-1}}$, 0.096 percent more. The coarse dual fails on the fine grid by at most 9.5×10^{-4} of f, so the coarse dual scaled up by that fraction certifies every fine grid point. The Lipschitz bracket, by contrast, is 48 percent wide. The tabulated response is constant on cells of 0.1 eV by $0.02\,q_0$, and the kernel jumps whenever a cell's $v_{\min}$ crosses the speed of the stream, so a finite-difference Lipschitz constant measures the table, not the physics. The same structure gives an exact test. The isotropic part of a tabulated kernel is $\mathcal{A}_i(v)/v$ with $\mathcal{A}_i$ a step function, so the dual function changes its form only at the finitely many cell speeds $v_{\min}$; between them it is smooth. Evaluating the dual at those break speeds, with a Lipschitz bound only for the smooth anisotropic part $P_\ell(v_{\min}/v)$ between them, certifies it on the continuum.

A truncated response can also make a kernel negative, and then the mass bound behind Eq. (21.14) fails for the computed kernel itself. Silicon gives the standard case. Outside the off-shell window, the Dyson convention of Sect. 8.6 splits the electronic response into a zero-phonon part and a phonon-dressed part whose sum is positive and whose parts need not be (Eq. (8.59)). The zero-phonon pair enters the cascade with the energy ω, the phonon-dressed pair with $\omega - \omega_{\mathrm{ph}}$, and where the bin probability changes with energy the two are assigned to different bins. The bin-weighted response is then negative wherever one part is, but only where the bin probability changes within ω_{ph}. Seeding the phonon-dressed pair with the bin probability of its zero-phonon partner gives a kernel that is never negative, and Eq. (21.14) applies to it with the difference of the two kernels added to δ_{cert}, weighted by λ_i. Suppose the number of pairs is stochastically ordered in the seed energy, so that the probability of at least n pairs does not decrease with the energy for any n, as in the cascade of Sect. 18.3. Then each bin probability is the difference of two nondecreasing functions with values between 0 and 1, its total variation is at most 2, and the integral over energy of its change under a shift ω_{ph} is at most $2\omega_{\mathrm{ph}}$, one strip at each side of the bin. Integrated over ω, the misassigned weight is at most $2\omega_{\max}$ times the largest phonon-dressed response (Appendix D.46). Relative to the signal of a bin of width Δ_i, the integral of its probability over energy,

$$\frac{\text{misassigned weight in bin } i}{\text{signal in bin } i} \leq \frac{2\,\omega_{\max}}{\Delta_i}\,\varrho_i^{\mathrm{leak}}, \tag{21.15}$$

with $\varrho_i^{\mathrm{leak}}$ the largest phonon-dressed response over the bin, widened by $\omega_{\max}$ at each side, divided by the average of the total weighted with the bin probability, a number read off the computed response. For silicon, with the highest phonon at $\omega_{\max} = 63.7\,\mathrm{meV}$ (Sect. 11.3) and the two- and three-pair bins of the cascade of Sect. 18.3, whose widths Δ_i are 3.60 and 3.69 eV, close to $\varepsilon_{\mathrm{eh}} = 3.749\,\mathrm{eV}$, the factor $2\omega_{\max}/\Delta_i$ is at most 0.035; the wider one-pair bin, 4.98 eV, gives 0.026. The leakage is a few percent of $\varrho_i^{\mathrm{leak}}$ per bin. In a likelihood it is a multiplicative nuisance on the bin, of the kind Sect. 8.8 defines, with that width; Sect. 21.4 carries it into the band of the limit.

21.3 The kinematic edge

A silicon detector that counts single electrons has no electron-recoil rate below the phonon-assisted onset, 0.3374 MeV for free pairs (Sect. 13.1), and just above it the rate grows as the 13/2 power of the distance in mass (Eq. (13.51)). In the model of Fig. 21.6 the limit on $\bar{\sigma}_e$ is 1.6×10^7 times weaker 0.001 MeV above the onset than at 0.35 MeV, and still 4.6 times weaker 0.010 MeV above it. Every mass up to 0.010 MeV above the onset is tested only by particles within $12\,\mathrm{km\,s^{-1}}$ of $v_{\max}$, a fraction 6.0×10^{-5} of the standard halo. The steep power, the position of the edge and its height therefore carry the halo model as much as the target. We will derive the power for any channel and any truncation of the halo, from the onset of the rate kernel of Sect. 21.1 and the escape tail of the speed distribution. We then work out which combinations of gap, threshold and halo parameters a limit near its edge actually measures.

21.3.1 The rate kernel near its onset

For a response averaged over the directions of $\boldsymbol{q}$, the rate is the speed integral $\mathcal{R} = (\rho_\chi/m_\chi\rho_T)\int \mathrm{d}v\, F_\chi(v)\,\Gamma(v)$ of Sect. 21.1.2. In it F_χ is the speed distribution of Eq. (1.10), and the rate kernel $\Gamma(v)$ of Eq. (21.4) is the rate at which one particle of speed v scatters. The kernel vanishes below the onset speed $v_{\mathrm{on}}(m_\chi)$ (Sect. 21.1.1), and the edge is decided by how it starts there. A particle of mass m_χ and speed v reaches the pair (q,ω) when $v_{\min}(q,\omega) \leq v$, that is when $\omega \leq qv - q^2/2m_\chi$ (Eq. (1.23)). Let $E_{\mathrm{th}}(q)$ be the onset of the direction-averaged

response at fixed $q = |\boldsymbol{q}|$, as in Sect. 13.9.3. We generalize the definition of Sect. 13.3 from the phonon-assisted loss function to any channel: the response rises above the onset as $x^{p_{\rm on}}$, with $x = \omega - E_{\rm th}(q)$. At fixed q the energy integral then runs from $x = 0$ to the largest energy above the onset that the stream reaches,

$$x_{\rm max}(q; v) = qv - \frac{q^2}{2m_\chi} - E_{\rm th}(q),$$

which is the $x_{\rm max}$ of Sect. 13.9.3 with $v_{\rm max}$ replaced by v. The integral contributes $x_{\rm max}^{p_{\rm on}+1}/(p_{\rm on}+1)$ times the smooth factors of the channel: $F_{\rm DM}^2$, the powers of q and the strength of the response.

The onset speed $v_{\rm on}(m_\chi)$ is the speed at which the largest $x_{\rm max}$ reaches zero. Setting $v = v_{\rm max}$ and solving for the mass instead defines the edge mass of the channel,

$$m_{\rm edge} = \min_q \frac{q^2}{2\,[qv_{\rm max} - E_{\rm th}(q)]}, \tag{21.16}$$

which is Eq. (13.3) and Eq. (13.1) written for any onset. It reduces to $m_{\rm min}$ of Eq. (1.25) when $E_{\rm th}$ does not depend on q. Let q^* be the optimal momentum transfer. Since $\partial x_{\rm max}/\partial v = q$, a stream slightly faster than $v_{\rm on}$ reaches $x_{\rm max}(q^*) = q^*(v - v_{\rm on})$ at the optimum, and near it

$$x_{\rm max}(q; v) \simeq q^*(v - v_{\rm on}) - \frac{(q - q^*)^2}{2m_\chi},$$

where, as in Sect. 13.9.3, the curvature of $E_{\rm th}(q)$ is dropped against $1/m_\chi$. That holds at the phonon-assisted onset; at the zero-phonon edge, where the onset rises steeply in q, the curvature changes the prefactor $\Gamma_{\rm on}$ below but not the exponent, which needs only a positive total curvature. The range of q with $x_{\rm max} > 0$ has a width proportional to $(v - v_{\rm on})^{1/2}$, and the integral of $x_{\rm max}^{p_{\rm on}+1}$ over it is $(2m_\chi)^{1/2}[q^*(v - v_{\rm on})]^{p_{\rm on}+3/2}B(\frac{1}{2}, p_{\rm on} + 2)$, with B the Euler beta function. The kernel therefore starts as

$$\Gamma(v) \simeq \Gamma_{\rm on}\,(v - v_{\rm on})^{p_{\rm on}+3/2}, \qquad v \to v_{\rm on}^+, \tag{21.17}$$

with a coefficient $\Gamma_{\rm on}$ that collects the strength of the response at the onset, $F_{\rm DM}^2(q^*)$ and the powers of q^* and m_χ. The power $p_{\rm on} + 3/2$ counts $p_{\rm on} + 1$ from the energy integral and $\frac{1}{2}$ from the width in q.

The $\frac{1}{2}$ becomes 1 when $x_{\rm max}(q)$ has a corner at its maximum. If $x_{\rm max} \simeq x_{\rm max}(q^*) - s_\pm|q - q^*|$ with slopes $s_\pm$ on the two sides, the width in q is linear in $v - v_{\rm on}$ and the power is $p_{\rm on} + 2$. A corner arises when the onset itself has a kink at q^*, or when the range of q ends there. The second is the case of the free nuclear recoil with a detector threshold. Its response is the line $\omega = q^2/2M$, a delta function at fixed q, so the energy integral is a constant, which is the case $p_{\rm on} = -1$. The threshold cuts the range of momentum transfers at $q_{\rm th} = (2ME_{\rm th})^{1/2}$, and near the edge the optimum sits at that cut. The kernel of the free nuclear recoil therefore starts linearly, with power $-1 + 2 = 1$.

21.3.2 The escape tail of the halo

The other factor of the speed integral, $F_\chi(v)$, vanishes at $v_{\rm max}$, and how it vanishes depends on how the galactic distribution goes to zero at the escape speed. The standard halo of Table 1.1 is cut by hand (Eq. (1.3)), so the distribution in the galactic frame stays finite up to $v_{\rm esc}$ and then drops to zero. Equilibrium distribution functions built for the density profiles of

simulated halos go to zero smoothly instead. Lisanti and co-workers write such a family as [9, Eq. (1)]

$$f_{\rm gal}(u) \propto \left[\exp\left(\frac{v_{\rm esc}^2 - u^2}{k_{\rm tail}\, v_0^2}\right) - 1\right]^{k_{\rm tail}} \Theta(v_{\rm esc} - u),$$

which near the escape speed behaves as $(v_{\rm esc}-u)^{k_{\rm tail}}$ [9, Eq. (11)]. The *tail exponent* $k_{\rm tail}$ is zero for the sharp cut of the standard halo and one for the King model, $f_{\rm gal} \propto \mathrm{e}^{-u^2/v_0^2} - \mathrm{e}^{-v_{\rm esc}^2/v_0^2}$ [9, Eq. (12)]. Their fits to the velocity distributions of simulated galactic halos prefer $k_{\rm tail}$ between 1.5 and 3.5 [9, Sect. II, Eq. (10), and Fig. 3]. Near the escape speed, then, $f_{\rm gal} \simeq f_{\rm esc}\, y^{k_{\rm tail}}$ with $y = v_{\rm esc} - u$ and a constant $f_{\rm esc}$; for the standard halo $f_{\rm esc} = \mathrm{e}^{-v_{\rm esc}^2/v_0^2}/N_{\rm esc}$, the height of the normalized distribution at the cut.

The laboratory speeds near $v_{\rm max} = v_{\rm esc} + v_{\rm E}$ come from a small cap of velocity space. The fastest particle has its galactic velocity antiparallel to $\boldsymbol{v}_{\rm E}$, so take the unit vector $\hat{\boldsymbol{e}} = -\boldsymbol{v}_{\rm E}/v_{\rm E}$ and write a laboratory velocity as $\boldsymbol{v} = (v_{\rm max} - s)\hat{\boldsymbol{e}} + \boldsymbol{v}_\perp$, with s and $|\boldsymbol{v}_\perp|$ small. To second order in them the laboratory speed and the galactic speed are

$$v \simeq v_{\rm max} - s + \frac{v_\perp^2}{2v_{\rm max}}, \qquad u = |\boldsymbol{v} + \boldsymbol{v}_{\rm E}| \simeq v_{\rm esc} - s + \frac{v_\perp^2}{2v_{\rm esc}}.$$

A particle with $v \geq v_{\rm max} - \delta$ that is bound to the halo therefore has $0 \leq y \leq \delta - a_\perp v_\perp^2$, with $a_\perp = 1/2v_{\rm esc} - 1/2v_{\rm max} = v_{\rm E}/(2v_{\rm esc}v_{\rm max})$: a lens of depth δ and radius $(\delta/a_\perp)^{1/2}$. At fixed $\boldsymbol{v}_\perp$ the volume element is $\mathrm{d}s = \mathrm{d}y$, and the fraction of particles in the lens is

$$\int \mathrm{d}^2 v_\perp \int_0^{\delta - a_\perp v_\perp^2} \mathrm{d}y\, f_{\rm esc}\, y^{k_{\rm tail}} = \frac{\pi f_{\rm esc}\, \delta^{k_{\rm tail}+2}}{a_\perp (k_{\rm tail}+1)(k_{\rm tail}+2)}.$$

Its derivative with respect to δ is the speed distribution at $v = v_{\rm max} - \delta$:

$$F_\chi(v) \simeq F_{\rm esc}\, (v_{\rm max} - v)^{k_{\rm tail}+1}, \qquad F_{\rm esc} = \frac{2\pi f_{\rm esc}\, v_{\rm esc} v_{\rm max}}{(k_{\rm tail}+1)\, v_{\rm E}}, \tag{21.18}$$

valid for $v_{\rm max} - v \ll v_{\rm E}, v_{\rm esc}$. Dividing by $v \simeq v_{\rm max}$ and integrating once more, $\eta(v_{\rm min}) \propto (v_{\rm max} - v_{\rm min})^{k_{\rm tail}+2}$. For $k_{\rm tail} = 0$ this is the quadratic edge of Eq. (1.29), with its coefficient $A_\eta = \pi v_{\rm esc} \mathrm{e}^{-v_{\rm esc}^2/v_0^2}/(v_{\rm E} N_{\rm esc}) = F_{\rm esc}/2v_{\rm max}$, so the lens reproduces the expansion of the closed form. A numerical integration of η for each member of the family, with v_0, $v_{\rm esc}$ and $v_{\rm E}$ of Table 1.1, gives the power $k_{\rm tail} + 2$ to 0.002 between 0.1 and $1\,\mathrm{km\,s^{-1}}$ below $v_{\rm max}$, and the coefficient to 0.06 % at $0.1\,\mathrm{km\,s^{-1}}$. For the smooth members the power law needs one more condition: the bracket of $f_{\rm gal}$ is linear in y only for $y \ll k_{\rm tail} v_0^2/2v_{\rm esc}$, which is $52\,\mathrm{km\,s^{-1}}$ for the King model.

A smooth tail empties the lens. With the same v_0 and $v_{\rm esc}$, the 6.0×10^{-5} of the standard halo within $12\,\mathrm{km\,s^{-1}}$ of $v_{\rm max}$, which decide every silicon mass up to 0.010 MeV above the onset, become 4.7×10^{-6} in the King model and 3.7×10^{-8} for $k_{\rm tail} = 2.5$.

21.3.3 The edge power law

Both factors of the speed integral now start as powers, one at $v_{\rm on}$ and one at $v_{\rm max}$, and the integral is a beta function. Write $\delta_{\rm on} = v_{\rm max} - v_{\rm on}(m_\chi)$ for the window of speeds that contribute. Inserting Eqs. (21.17) and (21.18),

$$\begin{aligned}\mathcal{R} &\simeq \frac{\rho_\chi}{m_\chi \rho_T}\, F_{\rm esc}\, \Gamma_{\rm on} \int_{v_{\rm on}}^{v_{\rm max}} \mathrm{d}v\, (v_{\rm max} - v)^{k_{\rm tail}+1} (v - v_{\rm on})^{p_{\rm on}+3/2} \\ &= \frac{\rho_\chi}{m_\chi \rho_T}\, F_{\rm esc}\, \Gamma_{\rm on}\, B\big(k_{\rm tail}+2,\, p_{\rm on}+\tfrac{5}{2}\big)\, \delta_{\rm on}^{p_{\rm on}+k_{\rm tail}+7/2}.\end{aligned}$$

The window is linear in the distance from the edge mass. The edge mass of Eq. (21.16), as a function of $v_{\max}$, changes by $\mathrm{d}m_{\rm edge}/\mathrm{d}v_{\max} = -q^{*3}/[2(q^* v_{\max} - E_{\rm th}(q^*))^2] = -2m_{\rm edge}^2/q^*$, since only the explicit dependence counts at the minimum. A mass m_χ just above $m_{\rm edge}$ is therefore at its own edge for the speed $v_{\rm on} = v_{\max} - \delta_{\rm on}$ with

$$\delta_{\rm on} \simeq \frac{q^*}{2m_{\rm edge}^2}\,(m_\chi - m_{\rm edge}).$$

For a q-independent onset, $q^* = m_{\rm min} v_{\max}$ and this is the window $(v_{\max}/2)(m_\chi - m_{\rm min})/m_{\rm min}$ of Sect. 1.3.5; at the phonon-assisted onset of silicon, Eq. (13.4) gives $q^* = m_{\rm min}^{\rm PA}(v_{\max} \pm u_\nu)$, with u_ν the group velocity of the phonon at $\boldsymbol{Q}^*$, equal to $m_{\rm min}^{\rm PA} v_{\max}$ up to $u_\nu/v_{\max} \sim 3\times 10^{-3}$.

Edge power law

Let the direction-averaged response of a channel rise as $x^{p_{\rm on}}$ above its onset, and let the galactic distribution of the halo vanish as $(v_{\rm esc} - u)^{k_{\rm tail}}$ at the escape speed. If the largest energy above the onset, $x_{\max}$, has a smooth maximum in q, the rate near the edge mass $m_{\rm edge}$ of Eq. (21.16) is

$$\mathcal{R}(m_\chi) \propto (m_\chi - m_{\rm edge})^{\,p_{\rm on}+k_{\rm tail}+7/2}. \tag{21.19}$$

If $x_{\max}$ has a corner, the power is $p_{\rm on} + k_{\rm tail} + 4$. The response contributes $p_{\rm on} + 1$, the width in q one half (one at a corner), the halo $k_{\rm tail} + 1$, and the overlap of the two onsets one more. The power holds for either mediator, as long as $F_{\rm DM}$ is smooth at $q^* \neq 0$.

For the standard halo, $k_{\rm tail} = 0$, Eq. (21.19) is Eq. (13.51). For the King model the silicon edge steepens from 13/2 to 15/2, and over the range preferred by simulated halos to between 8 and 10. The kernel carries the target and the speed distribution the halo, and their exponents add because the rate is a convolution of the two near $v_{\max}$. The free nuclear recoil with a sharp threshold has a linear kernel and the power $3 + k_{\rm tail}$. For silicon nuclei and a 10 eV threshold in the standard halo, a direct integration over the recoil energy confirms it to 0.002 up to 1 % above the edge.

The power also sets how close to the edge an exposure reaches, the distance that Sect. 21.1.3 left open. A background-free exposure excludes a mass at which it predicts more than 2.3 events (Eq. (1.52)). Near the edge the predicted number is proportional to $M_{\rm det} t_{\rm exp} \bar\sigma_e (m_\chi - m_{\rm edge})^n$, with the exposure $M_{\rm det} t_{\rm exp}$ and n the power of Eq. (21.19). At fixed $\bar\sigma_e$ the lowest excluded mass therefore lies at a distance from the edge proportional to $(M_{\rm det} t_{\rm exp} \bar\sigma_e)^{-1/n}$. At the free-pair edge of silicon, $n = 13/2$, ten times the exposure brings the excluded mass a factor $10^{-2/13} = 0.70$ closer to the edge; with the King tail the factor is 0.74.

Table 21.1 collects the powers of the silicon electron-recoil edges and the free nuclear recoil. The phonon-assisted entries use the onset exponents of Sects. 13.3 and 13.8. For the zero-phonon edge of Sect. 13.9.1 the exponents follow from the counting rule of Sect. 13.3: an isolated quadratic minimum in D free variables gives $x^{D/2-1}$, and a vertex that vanishes linearly at the minimum adds one power. At fixed $|\boldsymbol{q}|$ a free zero-phonon pair has three free variables, the components of the hole's crystal momentum. The average over the two angles of $\hat{\boldsymbol{q}}$ adds two more, since each of the 24 equivalent optima of the first edge is an isolated minimum with curvatures β_θ and β_z (Sect. 13.9.1). With $D = 5$ the pair would start as $x^{3/2}$. The first zero-phonon line, however, is second class: its vertex vanishes in the mirror plane and grows linearly with the distance from it (Sect. 13.9.2), so $p_{\rm on} = 5/2$. The Coulomb attraction lowers the onset exponent of a continuum at fixed $\boldsymbol{q}$ by one half, from $x^{1/2}$ to a step, as Eq. (13.47) does for the phonon-assisted continuum, and the excitonic zero-phonon continuum has $p_{\rm on} = 2$. The $1s$ line is a delta function at fixed $\boldsymbol{q}$; the two angles turn it into a

Table 21.1: Powers of the rate in the dark-matter mass, $\mathcal{R} \propto (m_\chi - m_{\rm edge})^n$, at the edges of silicon, for the sharp-cut standard halo ($k_{\rm tail} = 0$); a halo with tail exponent $k_{\rm tail}$ adds $k_{\rm tail}$ to every entry (Eq. (21.19)). Masses in MeV, with the halo of Table 1.1; $p_{\rm on}$ is the onset exponent of the direction-averaged response at fixed $|\boldsymbol{q}|$; the fourth column says whether $x_{\rm max}(q)$ has a smooth maximum or a corner at q^* (Eq. (21.19)). The zero-phonon edges lie above the phonon-assisted onset, where they add a kink to the total rate. The nuclear-recoil row is for a 10 eV threshold.

Edge	$m_{\rm edge}$	$p_{\rm on}$	$x_{\rm max}$ at q^*	n
Phonon-assisted, free pair (transverse acoustic)	0.3374	3	smooth	13/2
Phonon-assisted, excitonic continuum	0.3374	5/2	smooth	6
Phonon-assisted, exciton replica	0.3332	3/2	smooth	5
Zero-phonon, first edge, free pair	0.405–0.408	5/2	smooth	6
Zero-phonon, first edge, excitonic continuum	0.405–0.408	2	smooth	11/2
Zero-phonon, first edge, 1s line	0.402–0.405	1	smooth	9/2
Free nuclear recoil, sharp threshold	137	−1	corner	3

step and the vertex zero into x^1. Since the phonon-assisted rate is already open at 0.405 MeV, these powers describe how the zero-phonon rate adds to the total, a kink in the rate rather than the edge of the limit.

Fig. 21.6(b) checks Eq. (21.19) in the model of Sect. 13.9.3: the computed transverse acoustic onset on the valley axis, a loss function Cx^3 and the family of halos above. The local power of the rate is $p_{\rm on} + k_{\rm tail} + 7/2$ to 0.002 up to 3×10^{-4} MeV above the onset for $k_{\rm tail} = 0$, 1 and 2.5. For the standard halo it drifts to 6.54 at 0.005 MeV and 6.61 at 0.02 MeV, as the curvature of the speed distribution enters; a light mediator leaves the power at 6.500 near the onset. The model loss function is the idealization. The computed one follows x^3 only for $x < 1.3$ meV (Fig. 13.4), and since the largest energy above the onset grows as $x^*_{\rm max} = \frac{1}{2} v^2_{\rm max}(m_\chi - m_{\rm edge})$, 3.5 meV per 0.001 MeV, the power is exact only within 4×10^{-4} MeV of the onset. At the zero-phonon edge, $x^*_{\rm max} = (q^{*2}/2m^2_{\rm edge})(m_\chi - m_{\rm edge})$ is 5.3 meV per 0.001 MeV; the 1s power holds while $x^*_{\rm max}$ stays below the exciton binding of 14.7 meV, within 0.0028 MeV of the line, and the continuum powers until neighboring optima merge 49 meV above the edge (Sect. 13.9.4), 0.009 MeV above it. The halo side is never the limitation at these masses: 0.01 MeV above the onset the window is $\delta_{\rm on} = 11.8\,{\rm km\,s^{-1}}$, far inside the $173\,{\rm km\,s^{-1}}$ scale of the correction in Eq. (1.29) and the $52\,{\rm km\,s^{-1}}$ range of the King tail.

21.3.4 Degeneracies at the edge

Near its edge a limit is set by three numbers: the edge mass, the power and the height of the rate. Each is a combination of target and halo parameters, and parameters that enter the same combination cannot be told apart by the limit.

The edge mass depends on the onset $E_{\rm th}(q)$ and on $v_{\rm max} = v_{\rm esc} + v_{\rm E}$, and on nothing else. For a q-independent onset it depends on them only through $E_{\rm th}/v^2_{\rm max}$, the rule $\delta \ln m_{\rm min} = \delta \ln E_{\rm th} - 2\,\delta \ln v_{\rm max}$ of Sect. 1.3.5. The phonon-assisted onset of silicon depends on q, but Eq. (13.4) gives $m^{\rm PA}_{\rm min} = 2E_{\rm on}/(v^2_{\rm max} - u^2_\nu)$ with the onset energy $E_{\rm on} = E_{\rm g} + \omega_\nu(\boldsymbol{Q}^*)$, and $u^2_\nu/v^2_{\rm max} \sim 10^{-5}$. To that accuracy the edge measures the single ratio $E_{\rm on}/v^2_{\rm max}$. A one percent error in the onset energy $E_{\rm on}$, whether it comes from the gap, from the phonon energy that the onset adds or from the energy of a threshold of n pairs, moves the edge exactly as a 0.5 % change of $v_{\rm max}$, $4.0\,{\rm km\,s^{-1}}$, does. Turned around, the escape-speed uncertainty of $\pm 25\,{\rm km\,s^{-1}}$ (Sect. 1.2) moves the edge by −6.0 and +6.6 percent, as an error of −71 or +78 meV in the

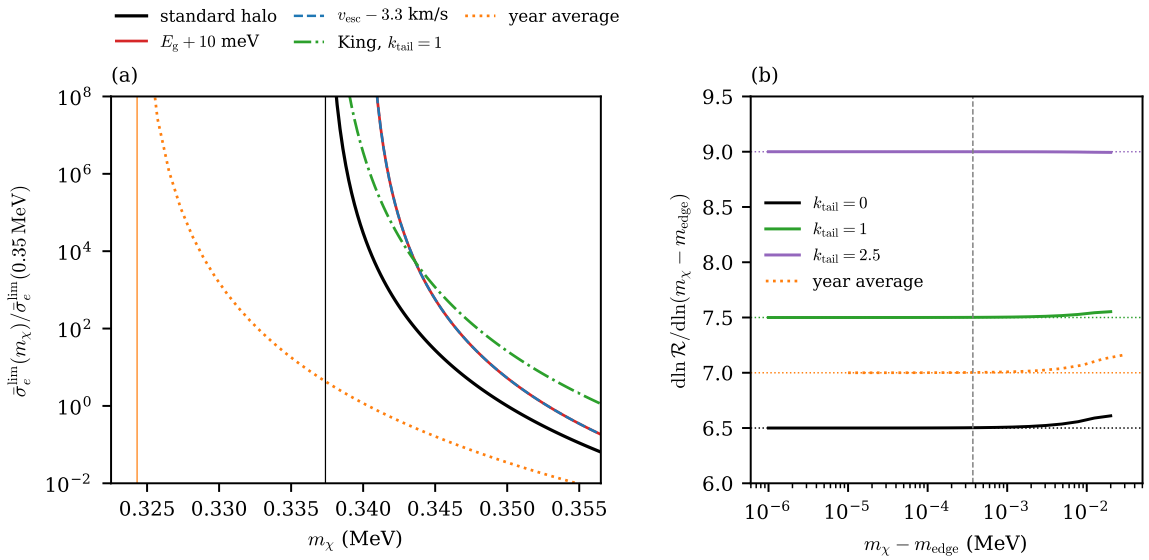


Figure 21.6: The low-mass edge of a silicon electron-recoil limit. Model: direction-averaged loss function Cx^3 above the transverse acoustic onset on the valley axis ($m_{edge} = 0.337\,38$ MeV, PBE force constants of Sect. 5.2, $E_g = 1.17$ eV), heavy mediator, background-free counting at fixed exposure, so that the limit is proportional to $1/\mathcal{R}$. (a) Limit on $\bar{\sigma}_e$ relative to its value at 0.35 MeV. Black: standard halo ($v_0 = 238$, $v_{esc} = 544$, $v_E = 250.5\,km\,s^{-1}$). Red: gap raised by 10 meV. Blue dashed, on top of the red curve: escape speed lowered by $3.3\,km\,s^{-1}$ instead. Green dash-dotted: King model ($k_{tail} = 1$) with the same v_0 and v_{esc}. Orange dotted: an exposure spread uniformly over a year, with v_E between 237.4 and $266.4\,km\,s^{-1}$. Vertical lines: the edge at the reference speed and the June edge, 0.3243 MeV. (b) Local power of the rate, $d\ln\mathcal{R}/d\ln(m_\chi - m_{edge})$, for halos with tail exponents $k_{tail} = 0$, 1 and 2.5 of the family of Lisanti and co-workers [9, Eq. (1)] with the same v_0, v_{esc}, v_E, and for the year average (m_{edge} its June value). Dotted: the powers $p_{on} + k_{tail} + 7/2$ of Eq. (21.19) and $p_{on} + 4$ for the year average. Dashed: the mass at which x^*_{max} reaches 1.3 meV, the end of the x^3 range of the computed loss function.

onset energy would, and the annual cycle of v_E by -3.9 and $+3.4$ percent.

The height of the rate is a product, $F_\mathrm{esc}\Gamma_\mathrm{on}$ in the speed integral above. The reference cross-section and the strength of the response at the onset sit in Γ_on. The height f_esc of the galactic distribution at the cut and the geometric factor $v_\mathrm{esc}v_\mathrm{max}/v_\mathrm{E}$ of the lens sit in F_esc. A null result near the edge bounds only the product. The dispersion v_0 enters the edge only here, through f_esc: for the standard halo $\partial \ln A_\eta/\partial v_0 = 2v_\mathrm{esc}^2/v_0^3 - \partial \ln N_\mathrm{esc}/\partial v_0 = +3.2\,\%$ per $\mathrm{km\,s^{-1}}$, so the uncertainty of $\pm 6\,\mathrm{km\,s^{-1}}$ (Sect. 1.2) scales the rate near the edge by 1.21 and 0.83 at fixed edge mass. The escape speed enters both the edge mass and the height, but in the height its effect nearly cancels. At a fixed edge mass, lowering v_esc raises A_η by $1.76\,\%$ per $\mathrm{km\,s^{-1}}$, since the cut moves into denser parts of the Maxwellian. It also lowers the powers of v_max in the rate. With the q^3 of Eq. (10.4), the energy integral of Sect. 13.9.3 and $q^* = m_\chi v_\mathrm{max}$, the rate at fixed $m_\chi - m_\mathrm{edge}$ and fixed edge mass, with the gap moved together with v_esc, scales as $A_\eta v_\mathrm{max}^{2p_\mathrm{on}+8}$ for a heavy mediator. For $p_\mathrm{on} = 3$ the factor v_max^{14} changes by $1.76\,\%$ per $\mathrm{km\,s^{-1}}$ as well, and the two changes cancel. A light mediator removes four powers of v_max through $F_\mathrm{DM}^2 \propto q^{*-4}$ and leaves $0.5\,\%$ per $\mathrm{km\,s^{-1}}$.

The power is a sum, $p_\mathrm{on} + k_\mathrm{tail}$, and the sum is all that a counting experiment can measure. The two exponents do enter the spectrum separately. At fixed ω the integral over q of $\eta \propto (v_\mathrm{max} - v_\mathrm{min})^{k_\mathrm{tail}+2}$, with $v_\mathrm{max} - v_\mathrm{min} = (x_\mathrm{max}(q) - x)/q$, runs over a range of width $\propto (x_\mathrm{max}^* - x)^{1/2}$ and gives

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} \propto (\omega - E_\mathrm{on})^{p_\mathrm{on}}\,(\omega_\mathrm{top} - \omega)^{k_\mathrm{tail}+5/2}, \qquad \omega_\mathrm{top} = E_\mathrm{on} + x_\mathrm{max}^*,$$

for an onset that is flat near q^*. The response shapes the bottom of the spectrum and the halo its top. The whole spectrum, however, is 3.5 meV wide per 0.001 MeV above the edge, while a silicon detector that counts pairs puts every deposit between E_g and $2E_\mathrm{g}$ into one bin (Sect. 1.5). The bin integrates the spectrum, and with it the two exponents into their sum. A steeper response and a smoother halo give the same limit.

Two handles distinguish the target from the halo. The first is a second edge with a different mixture. At the zero-phonon edge the onset rises steeply with the distance of $\boldsymbol{q}$ from the valley, and the envelope theorem, by which only the explicit dependence of a minimized function on a parameter counts at its minimum (Sect. 21.3.3), applied to Eq. (13.1) leads to $\partial \ln m_\mathrm{min}^\mathrm{ZP}/\partial \ln E_\mathrm{g} = E_\mathrm{g}/(q^* v_\mathrm{max} - E^*)$ and $\partial \ln m_\mathrm{min}^\mathrm{ZP}/\partial \ln v_\mathrm{max} = -q^* v_\mathrm{max}/(q^* v_\mathrm{max} - E^*)$, with E^* the pair energy at the optimum and a rigid shift of the conduction bands. At the joint minimum of Sect. 13.9.1, $|\boldsymbol{q}^*| = 1.33\,\mathrm{keV}$ and $E^* = 1.355\,\mathrm{eV}$, these are 0.54 and -1.62, against 1 and -2 at the phonon-assisted onset. Two edges of one crystal measure the gap and v_max separately, in principle, although the zero-phonon edge appears only as a kink of the rate.

The second handle is time. Over the year v_max moves between 781 and $810\,\mathrm{km\,s^{-1}}$ (Eq. (1.9)) while the gap stays put, and near the edge the rate responds with the full power of Eq. (21.19). An exposure spread over the year has the time-averaged rate. Near the largest Earth speed, on 2 June, the edge mass varies quadratically in time, $m_\mathrm{edge}(t) \simeq m_\mathrm{edge}^\mathrm{June} + \frac{1}{2}\ddot{m}\,(t - t_\mathrm{June})^2$, and the time integral of $(m_\chi - m_\mathrm{edge}(t))^n$ over the interval in which it is positive, of width $\propto (m_\chi - m_\mathrm{edge}^\mathrm{June})^{1/2}$, adds one half to the power. The year-averaged edge sits at the June value, $m_\mathrm{edge}^\mathrm{June} = 0.3243\,\mathrm{MeV}$ for the free-pair onset, $3.9\,\%$ below the edge at the reference speed, with the power $p_\mathrm{on} + k_\mathrm{tail} + 4$ (Fig. 21.6(b)). At 0.010 MeV above the reference edge the averaged rate is 62 times the rate at the reference speed, and at 0.020 MeV still 8.5 times. Near the edge a limit computed at a fixed Earth speed is therefore a statement about that day. A limit uses the exposure average of Eq. (21.3), with its edge at the June value, and the edge at the reference speed is the convention for comparing targets (Sect. 21.1.3).

Worked example 21.3: How a 10 meV gap shift moves the silicon edge

Ten meV is smaller than either energy that separates the silicon onsets, the exciton binding of 14.7 meV and the transverse acoustic phonon at $\boldsymbol{Q}^*$, about 15 meV (Sect. 13.1). That phonon is softer than the 18.27 meV replica of optical absorption, because the dark matter supplies part of the momentum and $\boldsymbol{Q}^*$ is shorter than the valley wavevector. Raise the gap by $\delta E = 10\,\text{meV}$, with the halo of Table 1.1.

Edge. By Eq. (13.4), $\delta m_{\text{edge}} = 2\,\delta E / v_{\text{max}}^2 = 2 \times 0.010\,\text{eV}/(7.02 \times 10^{-6}) = 0.002\,85\,\text{MeV}$. The model of Fig. 21.6 puts the shift at 0.002 83 MeV, 0.84 % of the edge, slightly less because the larger q^* shortens $\boldsymbol{Q}^*$ and lowers the phonon energy that the onset adds.

Limit. At a mass δm above the old edge, Eq. (21.19) multiplies the rate by $(1 - \delta m_{\text{edge}}/\delta m)^{13/2}$. At $\delta m = 0.005$, 0.010 and 0.020 MeV this is 0.0044, 0.115 and 0.371; the model gives 0.0043, 0.114 and 0.368. The limit weakens by factors of 231, 8.8 and 2.7. The power law holds here although the masses lie outside the x^3 range of the computed loss function, because the factor is a ratio of two rates with the same shape.

The halo that mimics it. The same edge follows from a lower v_{max}, by $(v_{\text{max}}/2)\,\delta E/E_{\text{on}}$ with $E_{\text{on}} = \frac{1}{2} m_{\text{edge}} v_{\text{max}}^2 = 1.1848\,\text{eV}$, that is $3.35\,\text{km}\,\text{s}^{-1}$; solving the model for the escape speed gives $3.32\,\text{km}\,\text{s}^{-1}$. Above the common edge the two rates agree to 0.04 % for a heavy mediator, the cancellation of the height derived above, and differ by 1.7 % for a light one. No limit can tell a 10 meV gap error from a $3.3\,\text{km}\,\text{s}^{-1}$ error in the escape speed, less than a seventh of its uncertainty.

The zero-phonon edge. The same gap shift moves $m_{\text{min}}^{\text{ZP}}$ by $m_{\text{min}}^{\text{ZP}}\,\delta E/(q^* v_{\text{max}} - E^*) = 0.4077\,\text{MeV} \times 0.010/2.170 = 0.001\,88\,\text{MeV}$, 0.46 %, while the $3.3\,\text{km}\,\text{s}^{-1}$ halo shift moves it by $1.62 \times 0.42\,\% = 0.68\,\%$, or 0.002 77 MeV. Measured from the phonon-assisted onset, the zero-phonon edge comes 0.000 95 MeV closer under the gap shift and only 0.000 06 MeV closer under the halo shift.

21.4 Theory nuisances in a limit

A silicon detector that records no event above a one-pair threshold in a kilogram-year excludes, at 3 MeV and for a heavy mediator, every cross-section above $1.24 \times 10^{-41}\,\text{cm}^2$, if the rate is the central value of Worked example 10.10. Computed at the random-phase level, without local fields, the same zero excludes everything above $9.23 \times 10^{-42}\,\text{cm}^2$, a limit 1.35 times stronger. The central rate is itself uncertain by a factor of 1.48 at one standard deviation. Which number should the experiment publish, and how wide is the uncertainty on it? We will set the Poisson limit with a known background, put the theory and detector nuisances into its likelihood, and derive how far each nuisance moves the limit. That gives a band of first order in the sizes of the truncations and a profiled limit of second order. We then treat templates of known sign and cells removed from the signal, and read off, channel by channel, which nuisance sets the band.

21.4.1 Poisson limits with a known background

A counting experiment observes N_{obs} events where its background is expected to give N_{bkg}. The classical upper limit at confidence level CL is the signal N^{up} at which data as low as those observed occur with probability $1 - \text{CL}$,

$$\sum_{n=0}^{N_{\text{obs}}} P(n \,|\, N^{\text{up}} + N_{\text{bkg}}) = 1 - \text{CL}, \tag{21.20}$$

with P the Poisson probability Eq. (1.51). Every larger signal would make the observation rarer still, so the set of signals above N^{up} is excluded, and in repeated experiments the true signal lies below N^{up} in a fraction CL of them (Sect. 1.5). For no events the sum has one term, and $N^{\mathrm{up}} = \ln[1/(1-\mathrm{CL})] - N_{\mathrm{bkg}}$, which is $2.303 - N_{\mathrm{bkg}}$ at 90 %. The background lowers the limit even though nothing was seen: with $N_{\mathrm{bkg}} = 1$ the limit is 1.303, and for $N_{\mathrm{bkg}} > 2.303$ the interval is empty. An empty interval is a correct frequentist statement, since such data occur in fewer than 10 % of experiments, but it says nothing about the signal. With one event on a background of one the classical limit is 2.890, and with two on a background of two it is 3.322.

Feldman and Cousins removed the empty intervals by changing the order in which outcomes enter the construction [10, Sect. IV A]. At each trial signal N_{sig} they rank the outcomes n by the likelihood ratio

$$\mathsf{R}_{\mathrm{FC}}(n) = \frac{P(n \mid N_{\mathrm{sig}} + N_{\mathrm{bkg}})}{P\big(n \mid \max(0, n - N_{\mathrm{bkg}}) + N_{\mathrm{bkg}}\big)}, \tag{21.21}$$

the probability of n at the trial signal over its probability at the best signal that is not negative [10, Eq. (4.1)]. Outcomes are accepted in decreasing order of R_{FC} until their summed probability reaches CL, and the interval for N_{obs} is the set of trial signals whose accepted outcomes contain it. The interval is never empty, and it turns from an upper limit into a two-sided interval as N_{obs} grows without the experimenter choosing when. Its 90 % upper ends are 2.44 for no events and no background, 1.61 and 1.08 for no events on a background of one and of three, and 3.36 for one event on a background of one [10, Table IV]. Without background the classical limit is the smaller one, since the ordered construction reserves some probability for two-sided intervals that an upper limit does not need. The book quotes classical upper limits, the N_{90} of Sect. 1.5.

Either number becomes a cross-section limit through Eq. (1.53), with N^{up} in place of N_{90}. A search with several bins could quote the smallest of its per-bin limits, but the joint confidence of such a choice is lower than that of each bin unless the confidence levels are corrected by the union bound of Sect. 21.2.1. The likelihood Eq. (8.78) uses all bins at once, and it is there that the nuisances enter.

21.4.2 Theory and detector nuisances

The likelihood Eq. (8.78) holds the halo fixed and contains the theory nuisances ϑ_k of the response. The detector adds its own. Its efficiency $f_{\mathrm{det},j}$ for events of bin j, its background and its energy scale are calibrated, not known exactly, and each calibration becomes a nuisance $\vartheta_d^{\mathrm{det}}$ with a standard-normal constraint. The expected counts become

$$N_{\mathrm{sig},j} = \bar{\sigma}_e \, M_{\mathrm{det}} t_{\mathrm{exp}} \, f_{\mathrm{det},j}(\vartheta^{\mathrm{det}}) \, r_j(m_\chi; \vartheta), \qquad N_{\mathrm{bkg},j} = N_{\mathrm{bkg},j}(\vartheta^{\mathrm{det}}), \tag{21.22}$$

and L gains the factor $\prod_d \exp[-(\vartheta_d^{\mathrm{det}})^2/2]$. The efficiency is a normalization. The energy scale is a threshold nuisance: through the pair energy $\varepsilon_{\mathrm{eh}}$ it moves the edges of every bin above one pair. The background normalization acts on $N_{\mathrm{bkg},j}$ alone, and it is the one nuisance that the data constrain even when the signal is zero. The split of a pair into the zero-phonon and phonon-dressed sectors adds one more: once both are seeded with the zero-phonon pair distribution (Sect. 21.2.4), their difference is a normalization of at most $0.035 \, \varrho_i^{\mathrm{leak}}$ of a silicon bin (Eq. (21.15)).

The theory nuisances come from the tables that close each channel, Tables 10.2, 11.1, 12.3, 13.2, 14.3, 15.3 and 16.2, and the basis column of each table decides how a row enters. A row sized by a computed or modeled next order, by a small-parameter estimate or by a spread between calculations becomes a ϑ_k with its template in the power form Eq. (8.72). A row known only between bounds has no central value, and the limit is set at the end of the bound

that gives the weaker limit. The density–current cross term of a velocity coupling in silicon, at least 16 times the static-electron value at 10 MeV (Sect. 10.4), and the final states of the germanium $3d$ shell (Sect. 10.6) are of this kind. The tabulation spread of silicon (Sect. 10.10) is a bound of another kind, a bound on the width of a spread rather than on a value: the independent-particle ratio bounds from above how far the screened rate of either table lies from a converged one, and it has no known sign. It enters as a standard-normal nuisance whose width is that bound, a prior that can only overstate the band. A named row without a size gets no width: the limit is reported at the nominal value with the omission stated, by the rule of Sect. 8.8. An evaluation error, finally, such as the free recoil used below the impulse onset (Sect. 11.9) or a Bethe–Salpeter loss function with resonant poles only (Sect. 10.10), is corrected before the limit is set, since no nuisance of the correct calculation covers it.

The halo stays out of this likelihood. Its parameters are common to every experiment, are fixed by convention (Sect. 1.5.5) and are varied in a band of their own, the $\pm 25\,\mathrm{km\,s^{-1}}$ of v_{esc} and $\pm 6\,\mathrm{km\,s^{-1}}$ of v_0 of Sect. 1.2. Near the edge the gap nuisance and the escape speed act through the one combination $E_{\mathrm{on}}/v_{\mathrm{max}}^2$ (Sect. 21.3.4), and there the two are varied together, not as independent entries.

Three numbers describe a limit with nuisances: the nominal limit, with every ϑ at zero; the profiled limit, with every ϑ removed by profiling, Eq. (C.122); and the theory band, the range over which the limit moves when the nuisances range over their constraints. The profiled limit needs a construction, since at the counts of a null search no asymptotic threshold applies (Appendix C.14). The book uses the Neyman construction with the nuisances at their conditional best fit. At each trial cross-section the statistic Eq. (C.122), set to zero where the best fit exceeds the trial value, ranks the possible outcomes. The probability of the outcomes ranked at least as extreme as the observed one is evaluated with every ϑ at $\hat{\hat{\vartheta}}$, its best fit to the observed counts at that cross-section, and the limit is where this probability falls to $1 - \mathrm{CL}$. The constraint terms are held fixed, since a theory nuisance has no auxiliary measurement that a repeated experiment would repeat. The construction is therefore a hybrid of a frequentist count and a fixed constraint; its coverage over the values of ϑ is not guaranteed, and without nuisances it returns the classical limit. Collider searches often quote $\mathrm{CL_s}$ instead, the probability of the outcomes at signal plus background divided by that at background alone [11], which protects a limit from a downward fluctuation of the background; with no background, the case of most limits of this chapter, the two coincide. All three numbers follow from how far the limit moves per unit of each nuisance (Appendix D.47):

Theory band and profiled limit

For data without an excess, the sensitivity of the limit to nuisance k is

$$\beta_k \equiv \frac{\partial \ln \bar{\sigma}_e^{\mathrm{lim}}}{\partial \vartheta_k} = -\sum_j a_j \ln\!\left(1 + \bar{\delta}_{k,j}\right), \tag{21.23}$$

with $\bar{\delta}_{k,j}$ the template of Eq. (8.72) averaged over the signal in bin j and a_j the share with which bin j pulls on the statistic at the nominal limit, its share of the signal when no event is seen (Eq. (D.256)). For independent standard-normal ϑ_k the limit lies, at one standard deviation, in the theory band

$$\bar{\sigma}_e^{\mathrm{lim}}\, \mathrm{e}^{-\sigma_\beta} \le \bar{\sigma}_e^{\mathrm{lim}}(\vartheta) \le \bar{\sigma}_e^{\mathrm{lim}}\, \mathrm{e}^{\sigma_\beta}, \qquad \sigma_\beta^2 = \sum_k \beta_k^2, \tag{21.24}$$

and the profiled limit $\bar{\sigma}_e^{\lim,\mathrm{p}}$ of the construction above lies above the nominal one by

$$\ln \frac{\bar{\sigma}_e^{\lim,\mathrm{p}}}{\bar{\sigma}_e^{\lim}} = \frac{\sigma_\beta^2\, \mathsf{t}'}{2} + O(\sigma_\beta^4), \tag{21.25}$$

where t' is the slope of the statistic t of Appendix C.14 with respect to $\ln \bar{\sigma}_e$ at the nominal limit. For one bin the relation is exact.

For one bin, $\beta_k = -\ln(1+\bar{\delta}_k)$: the band of the limit is the band of the rate turned upside down, and σ_β is the $\sigma_{\ln \mathcal{R}}$ of Worked example 10.10. To this order, any number of nuisances acts as one nuisance of logarithmic width σ_β along the direction of β (Appendix D.47). With no events $\mathsf{t}' = 2N^{\mathrm{up}}$, and the profiled limit lies exactly a factor $\mathrm{e}^{\sigma_\beta^2 N^{\mathrm{up}}}$ above the nominal one: 1.427 for $\sigma_\beta = 0.393$ without background, where $N^{\mathrm{up}} = N_{90}$, and 1.223 on a background of one, where $N^{\mathrm{up}} = 1.303$. A fixed threshold calibrated to the classical limit, the illustration of Eq. (8.79), charges half of this exponent (Appendix D.47). A published limit must say which construction it reports, as it must say whether it quotes a profile or an envelope (Sect. 8.8).

Fig. 21.7 applies this across masses to the silicon electron-recoil limit of a kilogram-year with no event above a one-pair threshold, with the four nuisances of Worked example 10.10 evaluated at each mass. The band is widest at the light end, a factor of 3.4 at 1 MeV for a heavy mediator, where the tabulation spread carries 91 % of the variance; it is 1.48 at 3 MeV and 1.43 at 10 MeV. The tabulation spread, which sits at small q, dominates up to 3.59 MeV and the attraction above, since heavier particles reach larger momentum transfers. Below the dominant nuisance the ranking moves as well: the shift of the band energies passes the attraction below 1.14 MeV, as the spectrum crowds toward the edge, and the local fields above $2q_0$ pass the tabulation spread above 7.04 MeV. For a light mediator, weighted toward small q, the tabulation spread dominates up to 10 MeV. For a heavy mediator the random-phase limit without local fields is stronger than the central one by 1.10 at 1 MeV and 1.46 at 10 MeV. At 10 MeV it lies below the band, because the step it skips, to local fields, has been computed and is no part of the band.

Worked example 21.4: The band of a silicon limit from the random-phase level

Take $m_\chi = 3\,\mathrm{MeV}$, a heavy mediator, 1 kg yr of silicon, a one-pair threshold, no event and no background, so that $N^{\mathrm{up}} = N_{90} = 2.303$ and $\bar{\sigma}_e^{\lim} = N_{90} \times 10^{-37}\,\mathrm{cm}^2/\mathcal{R}$, with $\mathcal{R}$ the rate per kilogram-year at $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$ (Eq. (1.53)). The tables place the conduction bands at the measured gap of 1.17 eV at zero temperature, and the rate is counted from it (Sect. 10.10).

The random-phase level. With the random-phase loss function of the tables, $\mathcal{R} = 2.494 \times 10^4$ and $\bar{\sigma}_e^{\lim} = 9.23 \times 10^{-42}\,\mathrm{cm}^2$.

The computed step. The local fields have been computed, and they move the central value, not the band. They lower the rate to 1.855×10^4, a factor 1.345, and the nominal limit is $1.24 \times 10^{-41}\,\mathrm{cm}^2$.

The band. The four nuisances of Worked example 10.10 have, by Eq. (21.23) with one bin, $\beta_{\mathcal{X}} = -0.247$ (attraction), -0.289 (tabulation spread), $+0.078$ (band energies) and -0.064 (local fields above $2q_0$). Their quadrature sum is $\sigma_\beta = 0.393$, and Eq. (21.24) gives

$$8.38 \times 10^{-42}\,\mathrm{cm}^2 \le \bar{\sigma}_e^{\lim}(\vartheta) \le 1.84 \times 10^{-41}\,\mathrm{cm}^2.$$

The attraction and the tabulation spread carry 93 % of σ_β^2. Taken one at a time to $\vartheta_k = \pm 1$, the four move the limit by factors of 1.28, 1.34, 1.08 and 1.07. The random-phase limit

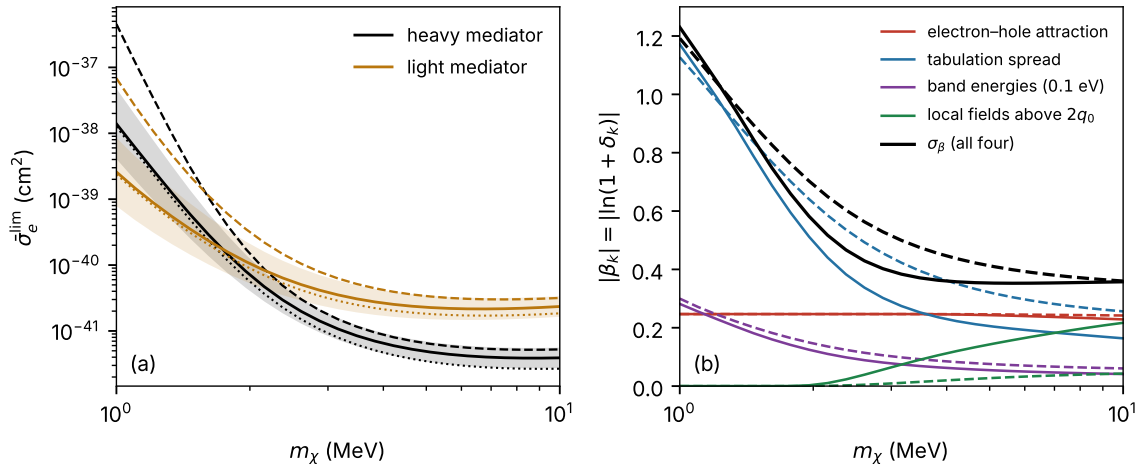


Figure 21.7: A silicon electron-recoil limit with its theory band. Search: 1 kg yr, every event with at least one pair ($\omega \geq 1.17\,\mathrm{eV}$), no event observed, no background, classical 90 % limit $N_{90} = 2.303$; standard halo, $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$. (a) Nominal limit (solid) from the random-phase loss function with local fields of the Knapen–Kozaczuk–Lin tables, whose scissor sets the bands at the measured gap at zero temperature, its band Eq. (21.24) (shaded), the profiled limit (dashed; Neyman construction with the nuisance at its conditional best fit, $\mathrm{e}^{\sigma_\beta^2 N_{90}}$) and the limit at the random-phase level without local fields (dotted), for a heavy (black) and a light mediator (ochre). (b) Sensitivities $|\beta_k|$ of the four nuisances of Worked example 10.10, heavy (solid) and light mediator (dashed). The attraction template is $\delta_\mathcal{X} = +0.28$ for $\omega \leq 10.5\,\mathrm{eV}$ and zero above, which overstates it above 3 MeV, where the compensating weight of the f-sum rule enters the window. The other three are the tabulation spread (ratio of the independent-particle rates of the Essig et al. and Knapen–Kozaczuk–Lin tables), band energies shifted by 0.1 eV, and the random-phase value in place of local fields above $2q_0 = 7.46\,\mathrm{keV}$. Black: their quadrature sum σ_β. Below 1 MeV the first energy bins of one table make the spread unreliable. Above 10 MeV the momentum range of the other, 22.5 keV, truncates the heavy-mediator rate.

lies in the lower part of the band, at 0.74 of the nominal value against the lower edge at 0.675.
The profiled limit. With no events Eq. (21.25) is exact, the factor is $\mathrm{e}^{\sigma_\beta^2 N_{90}} = 1.427$, and $\bar{\sigma}_e^{\mathrm{lim,p}} = 1.77 \times 10^{-41}\,\mathrm{cm}^2$.
The detector. Suppose the efficiency is calibrated to 10 %, a normalization with $\beta = -\ln 1.10 = -0.095$. Then $\sigma_\beta = 0.404$, the band widens from a factor 1.481 to 1.498, and the profiled factor grows from 1.427 to 1.457. At this calibration the detector adds about one percent to the band and two percent to the profiled limit.
A template of known sign. The attraction raises the rate near the gap in every calculation of Sect. 10.5. With $\vartheta_{\mathcal{X}} \geq 0$ and a half-normal constraint the profile sets $\vartheta_{\mathcal{X}} = 0$ at every cross-section (Appendix D.47), and only the three other nuisances, with $\sigma_\beta = 0.306$, are profiled: the factor falls to 1.240 and $\bar{\sigma}_e^{\mathrm{lim,p}} = 1.54 \times 10^{-41}\,\mathrm{cm}^2$. The attraction can then only strengthen the limit, by 1.28 at $\vartheta_{\mathcal{X}} = 1$.
The experiment would therefore quote the nominal limit $1.24 \times 10^{-41}\,\mathrm{cm}^2$, the profiled limit $1.77 \times 10^{-41}\,\mathrm{cm}^2$ (or $1.54 \times 10^{-41}\,\mathrm{cm}^2$ with the one-sided attraction), and the band from $8.4 \times 10^{-42}\,\mathrm{cm}^2$ to $1.84 \times 10^{-41}\,\mathrm{cm}^2$. The random-phase number is none of these.

21.4.3 One-sided nuisances and excluded cells

A template whose sign is known calls for a constraint that respects it, $\vartheta_k \geq 0$ with the weight $\mathrm{e}^{-\vartheta_k^2/2}$ on that side. What it costs depends on the direction (Appendix D.47). If $\vartheta_k > 0$ raises the signal, the profile sets $\vartheta_k = 0$ at every cross-section, the nuisance costs nothing, and the limit at $\vartheta_k = 0$ is the conservative end of its band; the attraction of Worked example 21.4 is of this kind. If $\vartheta_k > 0$ lowers the signal, the profile pulls toward it anyway, since data without an excess prefer less signal, and the one-sided constraint costs as much as a two-sided one. Linear spin-wave theory, for example, overstates the one-magnon weight of NiO by an estimated $\Delta S/|\boldsymbol{S}| = 7.8\,\%$ (Sect. 14.1), and that template, of known sign, lowers the signal. Both statements need the template to have one sign in every bin and the bins to be free of excess, so that every weight a_j is positive.

Some cells of the $(\boldsymbol{q}, \omega)$ plane have no controlled central value, and a limit can be set without them. Removing nonnegative signal from bins without an excess lowers the statistic at every cross-section, so the limit can only rise (Appendix D.47); for one bin it rises by $1/(1-f)$, with f the removed share. Near the pair-breaking edge of aluminum the conserving loss function changes with the mean free path by more than 10 % below $y = \omega/\Delta_{\mathrm{BCS}} = 3$, and those energies are excluded (Sect. 15.2.3). At 1 keV with a light mediator they hold 20 % of the rate for the bulk gap and 37 % for the film gap (Worked example 15.1), and excluding them weakens the limit by 1.25 and 1.59. For a crystal with a mean free path of at least 10 µm the clean loss function holds there, and the cells can be restored. The silicon Migdal rate has a region of this kind as well: the coherent zero-phonon response between 0.75 keV and $2q_{\mathrm{BZ}}$ is not sized (Table 12.3). It carries 33 % of the rate at 2 MeV and 7 % at 3 MeV, so excluding it costs factors of 1.49 and 1.08, and below 1 MeV, where it carries nearly all of the rate, no Migdal limit survives the exclusion. In a bin with an excess the argument fails, and removing signal there can strengthen a limit; such a bin needs the full likelihood.

21.4.4 The dominant nuisance of each channel

Table 21.2 reads the chapter tables along the other axis and gives, for each channel at a reference point, the rows that dominate its band. The dominant rows are missing calculations or missing data. The silicon electron recoil lacks a finite-momentum Bethe–Salpeter calculation

Table 21.2: Dominant theory nuisances of the limit in each channel, read from the nuisance tables of the channel chapters. Sizes are factors or shifts of the rate, and hence, for one bin, of the limit. Basis as in those tables: "computed", "model", "est." (parametric estimate), "spread" (published calculations at the same order), "data" (spread of measured inputs), "bound" (only an inequality is known), "unsized". The last column gives the section, table or figure where the rows are treated.

Channel	Reference point	Dominant nuisances	Basis	Section
Electron recoil, Si, density coupling	1–3.6 MeV, heavy mediator; up to 10 MeV, light	tabulation spread: ×3.2 at 1 MeV, ×1.34 at 3 MeV (heavy), ×1.29 at 10 MeV (light); band energies 0.1 eV: ×0.75 at 1 MeV	spread	Sect. 10.10; 1 MeV: Fig. 21.7
Electron recoil, Si, density coupling	3.6–10 MeV, heavy mediator	electron–hole attraction ×1.28, ×1.26 at 10 MeV (known sign); local fields above $2q_0$ ×1.24 at 10 MeV	model; spread	Sect. 10.10
Electron recoil, velocity couplings	10 MeV	density–current cross term: rate at least 16 (independent particles), about 29 (screened) times the static-electron formula	bound	Sect. 10.4
Core electrons, Ge $3d$	about 30 MeV	crystal final states: $3d$/valence 0.07 against 0.69	bound	Sect. 10.6
Phonon-assisted electron recoil, Si	0.333–0.43 MeV	finite-q form factor, factor 3; screening relative to the optical calibration, pair channel not sized; cold absorption data, rate ×9 at 0.36 MeV	est.; unsized; data	Table 13.2
Migdal effect, Si	up to 3 MeV; above	coherent zero-phonon response from 0.75 keV to $2q_{\mathrm{BZ}}$, 33 % of the rate at 2 MeV; off-diagonal local fields, ×1.2 to 1.4, lower end not bounded	unsized; model	Table 12.3
Nuclear recoil, Si, multiphonon	30 MeV, $\omega_{\mathrm{th}} = 100$ meV	computed phonon frequencies against measured, 0.24 %; free recoil in place of the crystal response (evaluation error) −6.8 %	computed	Sect. 11.9
Polar phonons, GaAs	single phonon	Z^*/ε_∞: 2 % from the measured splitting; a semilocal ε_∞ moves the rate by 37 %	data; computed	Sect. 11.9
Magnons, NiO	near 3 MeV; lightest masses	one-magnon weight high by 7.8 % (known sign) against the two-magnon continuum, +7.3 % at other energies; magnetic anisotropy near the lightest masses	est.; model	Table 14.3
Superconductor, Al	1 keV, light mediator	gap of the sensor, bulk against film: ×1.23; edge remainder −2 to +9 %; $y < 3$ excluded, ×1.25 (bulk) or ×1.59 (film) on the limit	data; computed	Table 15.3
Dirac material, $ZrTe_5$	20–1000 keV, light mediator	background dielectric tensor: reach band ×2.8	data	Worked example 16.9
Pair bins of a charge readout, Si	a few pairs	cascade against the step model: two pairs at 5 eV in 17 % of events against all with the resolved seed of the simplified cascade, about tenfold fewer events in the two-pair bin with the full band	model	Table 18.3

and a converged independent-particle table, the phonon-assisted window its finite-q form factor and pair-channel screening, the Migdal rate its coherent region and off-diagonal local fields, and NiO its anisotropy. The data that are missing are the cold absorption edge of silicon, the background tensor of $ZrTe_5$ and the gap of an aluminum film. Near the edge of every channel the threshold rows take over. A threshold error moves the edge mass by $\delta \ln m_{\mathrm{min}} = \delta \ln E_{\mathrm{th}}$. The rate near the edge rises as the power $p_{\mathrm{on}} + k_{\mathrm{tail}} + 7/2$ of the distance to it (Eq. (21.19)), so a gap shift of 10 meV weakens a silicon limit 0.005 MeV above the edge by a factor of 231 (Worked example 21.3).

For the silicon electron recoil at 3 MeV two calculations would remove 93 % of the variance: a finite-momentum Bethe–Salpeter calculation and an independent-particle table that passes the f-sum test at small q (Sect. 10.10.4). With the band energies and the local fields above $2q_0$ left, the band of Worked example 21.4 would shrink from a factor of 1.48 to 1.11.

21.5 Combining and recasting limits

Two silicon crystals cut from one wafer and read by identical sensors differ by more than two orders of magnitude in their event rate between 3 and 38 eV, because one of them is glued to its holder (Sect. 19.6.6). The lowest pair bins of a charge detector carry a dark rate, charge that leaks across the gap without any deposit (Sect. 1.5.1). No model predicts either rate to the accuracy that a background subtraction needs, so the Poisson limit with a known background of Sect. 21.4 cannot be applied to them as it stands. A limit is also rarely the last use of a measurement. A second experiment wants to add its exposure to the first, and a later calculation of the response, such as the background screening of a Dirac material (Sect. 16.1), changes the expected signal after the limit has been published. We will set limits that hold for any background, combine searches in one likelihood whose theory nuisances are shared, and derive when a published limit can be recast to a new response without the data.

21.5.1 Limits with an unknown background

If part of the background may be unknown, the only expectation that can safely be assumed for it is zero. Counting every observed event as signal then gives the classical limit of Sect. 1.5: the 90 percent upper limit $N_{\mathrm{sig}}^{\mathrm{lim}}$ solves $P(N \le N_{\mathrm{obs}} \,|\, N_{\mathrm{sig}}) = 0.10$ with the Poisson probabilities Eq. (1.51), which gives 2.303, 3.890, 5.322 and 6.681 for zero to three events. The limit holds whatever the background is. The distribution function $P(N \le n \,|\, \mu)$ falls monotonically with the mean, since its derivative with respect to μ is $-P(N = n \,|\, \mu)$, so a signal that leaves a probability below 0.10 for so few events leaves even less once a background $N_{\mathrm{bkg}} \ge 0$ is added to it. The price is that every event, background or not, weakens the limit.

Using the shape of the expected signal gives a stronger limit. At fixed m_χ the spectrum $\mathrm{d}N_{\mathrm{sig}}/\mathrm{d}E$ in the measured variable E is known up to the factor $\bar{\sigma}_e$, and a background of a different shape crowds its events where the signal is small, typically near threshold. Yellin's maximum-gap method uses this [12]. Measure each position by the expected signal count below it, $x(E) = \int^E (\mathrm{d}N_{\mathrm{sig}}/\mathrm{d}E')\,\mathrm{d}E'$. In this variable the signal events of any spectrum are spread uniformly with unit density over $[0, N_{\mathrm{sig}}]$, and the size of a gap between two adjacent events, or between an event and an end of the range, is the signal count expected inside it. For a signal-only experiment the probability that the largest gap is smaller than x depends only on x and N_{sig} [12, Eq. (2)]:

$$C_0(x, N_{\mathrm{sig}}) = \sum_{k=0}^{m} \Big[(kx - N_{\mathrm{sig}})^k - k\,(kx - N_{\mathrm{sig}})^{k-1} \Big] \frac{\mathrm{e}^{-kx}}{k!}, \qquad m = \lfloor N_{\mathrm{sig}}/x \rfloor, \tag{21.26}$$

written in the form that stays finite when N_{sig}/x is an integer; Yellin's printed form has the factor $1 + k/(N_{\mathrm{sig}} - kx)$ in place of the bracket, which is the same function away from those points. The 90 percent limit is the cross-section at which the observed largest gap has $C_0 = 0.90$. At fixed mass the fraction g of the expected signal inside the largest gap does not depend on $\bar{\sigma}_e$, so $N_{\mathrm{sig}}^{\mathrm{lim}}$ solves $C_0(g N_{\mathrm{sig}}, N_{\mathrm{sig}}) = 0.90$. With no events the only gap is the whole range, $g = 1$. Then only $k = 0$ and $k = 1$ contribute, the $k = 1$ term is $-\mathrm{e}^{-N_{\mathrm{sig}}}$, and $C_0 = 1 - \mathrm{e}^{-N_{\mathrm{sig}}}$ recovers $N_{90} = 2.303$ of Eq. (1.52). The method stays valid with any background: added events can only split gaps, never widen them, so the observed largest gap is no larger than the signal alone would have left, and the limit errs on the conservative side.

How much the shape buys depends on where the events fall. One event that leaves 90 percent of the expected signal on one side gives $g = 0.90$ and $N_{\mathrm{sig}}^{\mathrm{lim}} = 2.836$, against 3.890 from counting. The same event in the middle of the signal, $g = 0.5$, gives 7.78, twice the counting limit; the two methods break even at $g = 0.761$. Two events that both lie in the lowest tenth of the expected signal, where a background rising toward threshold would put

them, give $g = 0.90$ and again 2.836, against 5.322 from counting, while two events that cut the signal into thirds give 13.8. The maximum gap gains where the background sits apart from the signal and loses heavily where events are spread through it, and since the method is fixed before the data are seen, an experiment chooses it for the background it expects.

When the events are dense, a gap may be the wrong interval. The optimum-interval method generalizes it [12, Sect. III]. Let $C_n(x, N_{\mathrm{sig}})$ be the probability that, in a signal-only experiment, every interval containing at most n events has an expected count no larger than x. It too depends only on x and N_{sig}; it is computed by Monte Carlo and equals Eq. (21.26) for $n = 0$. We evaluate C_n for the observed events on every interval bounded by events or by the ends of the range, $(N_{\mathrm{obs}} + 1)(N_{\mathrm{obs}} + 2)/2$ intervals, each with its own n and x, and takes the largest value, C_{Max}. The 90 percent limit is the cross-section at which C_{Max} equals the value that 90 percent of signal-only experiments stay below, again from Monte Carlo. That calibration accounts for the interval having been chosen after the data were seen. Both methods need an ordered, continuous variable such as a phonon energy. Integer pair counts leave the events of one bin unordered, and for them the binned analog is a Poisson limit per bin, combined as in Sect. 21.5.3.

Neither method has a likelihood, since the unknown background has none, and so a theory nuisance cannot be profiled in it: Eq. (C.122) needs L. A normalization nuisance scales N_{sig} at fixed shape, and the band it puts on the limit is the envelope, $(1 + \delta_k)^{\pm 1}$, of first order in δ_k, where a likelihood would have charged the second-order cost of Eq. (21.25). A tilt also changes the fractions g, because the gaps are measured in expected counts, and the limit must be recomputed with the tilted spectrum at $\vartheta_k = \pm 1$.

21.5.2 Combining searches with shared nuisances

Two exposures that each set a limit do better together. Their counts are independent, so their Poisson likelihoods multiply (Eq. (C.118)). A combination of experiments $X = \mathrm{A}, \mathrm{B}, \ldots$ is one likelihood, the product of the single-exposure likelihoods Eq. (8.78),

$$\mathsf{L}(\bar{\sigma}_e, \vartheta, \vartheta^{\mathrm{det}}) = \prod_X \mathsf{L}^{(X)}(\bar{\sigma}_e, \vartheta, \vartheta^{\mathrm{det},(X)}). \tag{21.27}$$

Every theory nuisance ϑ_k appears once and acts in every factor, while the detector nuisances $\vartheta^{\mathrm{det},(X)}$, an efficiency, a background normalization or an energy calibration, are one set per experiment with constraints of their own (Sect. 21.4). The theory nuisances are shared because a truncation is a property of the calculation (Sect. 8.8), and every experiment that uses the same computed response inherits the same error. With one ϑ_k per experiment the profile could pull them in opposite directions and fit a difference between the experiments that no calculation can produce. It also pays the constraint once for each, as if the errors were independent, and so understates the cost of the truncation.

Before the product, the signal of every experiment is recomputed from one response at one truncation label, since published signal models from different codes differ by more than many of the truncations do. The tabulation spread of Sect. 21.4.2 enters Eq. (21.27) as one more shared nuisance, not as an error of each experiment. Conventions that differ between publications, the local density, the halo parameters and the pair-creation model that turns energy into pair bins, are choices, not uncertainties, and they are removed by the recomputation, not profiled. Every search on Earth sees the same halo, so if its parameters are varied, one set of them acts on all factors, and near the edge they enter only through the combinations of Sect. 21.3.4. And searches in different materials share the halo and the nuisance of each truncation that is computed the same way for both, with templates δ_k of their own and one ϑ_k (Sect. 8.8). The limit is then set from Eq. (21.27) as from a single likelihood (Sect. 21.4).

Worked example 21.5: Combining two silicon exposures

Take $m_\chi = 10\,\mathrm{MeV}$, a heavy mediator, the standard halo of Sect. 1.2 and the one-pair bin of the cascade of Sect. 18.3, the bin that sets the standard-halo limit at this mass in Sect. 21.2.2. The tabulated silicon response of Sect. 10.1 gives $r = 9.14 \times 10^{41}\,\mathrm{kg^{-1}\,yr^{-1}\,cm^{-2}}$ in this bin, that is 0.914 events per kg-year at $\bar\sigma_e = 10^{-42}\,\mathrm{cm^2}$. Exposure A has $M_{\mathrm{det}} t_{\mathrm{exp}} = 1\,\mathrm{kg\,yr}$, no events and no background. Exposure B has 4 kg yr, three events and a known background of 2.5 events. Both use the same response, which carries one normalization nuisance, $(1+\delta)^\vartheta$ with $\delta = 0.25$, the size of the largest truncations of the silicon rate at 10 MeV in Table 10.2.

Construction. The limit is the Neyman construction of Sect. 21.4.2. For each $\bar\sigma_e$ we sum the Poisson probabilities of all outcomes $(N_\mathrm{A}, N_\mathrm{B})$ up to 60 events each whose statistic Eq. (C.122) is at least the observed one, with the nuisance at its conditional best fit $\hat{\hat\vartheta}$ to the observed counts and the constraint term held fixed. The 90 percent limit is the $\bar\sigma_e$ at which that probability is 0.10.

Without the nuisance. Exposure A alone excludes $2.303/0.914 = 2.52 \times 10^{-42}\,\mathrm{cm^2}$, and B alone $1.14 \times 10^{-42}\,\mathrm{cm^2}$; its three events above a background of 2.5 cost less than its fourfold exposure gains. Together they exclude $0.809 \times 10^{-42}\,\mathrm{cm^2}$, 0.32 of A and 0.71 of B. At the combined limit A expects 0.74 signal events and B 2.96.

One shared nuisance. With Eq. (21.27) and one ϑ the limit is $0.897 \times 10^{-42}\,\mathrm{cm^2}$, a factor 1.109 above the combination without it. The profile pulls the nuisance to $\hat{\hat\vartheta} = -0.46$, toward less signal, in both factors at once.

One nuisance per exposure. Treated as two independent parameters, the nuisances settle at -0.17 (A) and -0.29 (B), and the limit is $0.851 \times 10^{-42}\,\mathrm{cm^2}$, a factor 1.051. Splitting the nuisance halves its cost, and the limit is too strong by 5 %. The worst case within one standard deviation, $\vartheta = -1$, would cost the full factor 1.25.

One exposure. For A alone the shared construction gives a closed form. With no events only the outcome $N_\mathrm{A} = 0$ has a statistic as large as the observed one, so the limit is reached when the profiled expectation $u = N_{\mathrm{nom}}(1+\delta)^{\hat{\hat\vartheta}}$ satisfies $\mathrm{e}^{-u} = 0.10$, $u = N_{90}$. The profile condition of Sect. 8.8, $\hat{\hat\vartheta} = -ku$ with $k = \ln(1+\delta)$, then gives $N_{\mathrm{nom}} = N_{90}\,\mathrm{e}^{k^2 N_{90}}$, a cost of 1.12 for $\delta = 0.25$, which is Eq. (21.25) with $\sigma_\beta = k$ and $\mathrm{t}' = 2N_{90}$.

21.5.3 Recasting a published limit

A published limit $\bar\sigma_e^{\mathrm{lim}}(m_\chi)$ was computed with the rates $r_j^{\mathrm{pub}}(m_\chi)$ per unit cross-section in the bins j of its analysis. A later calculation replaces them by $r_j^{\mathrm{new}}(m_\chi)$, and we write the ratio as the recast factor of bin j,

$$K_j^{\mathrm{rc}}(m_\chi) = \frac{r_j^{\mathrm{pub}}(m_\chi)}{r_j^{\mathrm{new}}(m_\chi)}. \tag{21.28}$$

If $K_j^{\mathrm{rc}} = K^{\mathrm{rc}}$ in every bin, the correction preserves the shape of the expected signal at each mass. Every expected count is then $N_{\mathrm{sig},j} = \bar\sigma_e M_{\mathrm{det}} t_{\mathrm{exp}} r_j^{\mathrm{new}} = (\bar\sigma_e / K^{\mathrm{rc}}) M_{\mathrm{det}} t_{\mathrm{exp}} r_j^{\mathrm{pub}}$. Any procedure that depends on $\bar\sigma_e$ only through the expected counts, which includes a Poisson limit per bin, a profile likelihood over many bins and Yellin's two methods, returns the limit $K^{\mathrm{rc}} \bar\sigma_e^{\mathrm{lim}}$. The recast is then exact without the data.

If K_j^{rc} varies from bin to bin, the correction tilts the expected spectrum, and the recast needs the procedure. One procedure can be recast exactly: a limit defined as the smallest of single-bin limits, $\bar\sigma_e^{\mathrm{lim}} = \min_j N_j^{\mathrm{up}} / (M_{\mathrm{det}} t_{\mathrm{exp}} r_j^{\mathrm{pub}})$, with the upper counts N_j^{up} of the bins

published. Replacing r_j^{pub} by r_j^{new} in that minimum is the new limit. If only the curve is public, the new limit is still bracketed. Each new single-bin limit is K_j^{rc} times the old one, and every old one is at least $\bar{\sigma}_e^{\mathrm{lim}}$, so the new minimum is at least $(\min_j K_j^{\mathrm{rc}})\,\bar{\sigma}_e^{\mathrm{lim}}$. The bin j^* that set the published limit still offers $K_{j^*}^{\mathrm{rc}}\bar{\sigma}_e^{\mathrm{lim}}$, so the new minimum is at most that. A profile likelihood over several bins and the optimum interval have no such form. Their limits depend on how the counts are spread over bins or events and on all the r_j together, and no function of the published curve and the K_j^{rc} reproduces them. Recasting them needs the binned likelihood, or the event list.

Recasting a limit to a new response

With the recast factors Eq. (21.28), a published limit $\bar{\sigma}_e^{\mathrm{lim}}(m_\chi)$ becomes

$$\bar{\sigma}_e^{\mathrm{lim,new}}(m_\chi) = \begin{cases} K^{\mathrm{rc}}(m_\chi)\,\bar{\sigma}_e^{\mathrm{lim}}(m_\chi) & \text{if } K_j^{\mathrm{rc}} = K^{\mathrm{rc}} \text{ for all } j \\ & \text{(any procedure)}, \\ \min_j N_j^{\mathrm{up}} / \left(M_{\mathrm{det}} t_{\mathrm{exp}}\, r_j^{\mathrm{new}}(m_\chi)\right) & \text{if the limit is a minimum over bins} \\ & \text{with } N_j^{\mathrm{up}} \text{ public} \end{cases}, \tag{21.29}$$

and both forms are exact. For a minimum over bins known only through its curve,

$$\min_j K_j^{\mathrm{rc}} \le \frac{\bar{\sigma}_e^{\mathrm{lim,new}}}{\bar{\sigma}_e^{\mathrm{lim}}} \le K_{j^*}^{\mathrm{rc}} \le \max_j K_j^{\mathrm{rc}}, \tag{21.30}$$

with j^* the bin that set the published limit. A correction that varies between bins cannot be recast from the curve of a profile-likelihood or optimum-interval limit.

A minimum over bins carries a cost in confidence. If each of n independent bins is set at 90 percent, the minimum excludes a true signal whenever any one bin does, and it covers with probability 0.9^n: 0.81 for two bins and 0.73 for three. The union bound of Sect. 21.2.1 restores the stated level. A minimum over bins is recastable exactly only with this correction included in the published N_j^{up}.

Halo conventions show both cases of Eq. (21.29). Take a silicon limit at 10 MeV with the search of Sect. 21.2.2: one kg-year, empty one-, two- and three-pair bins of the cascade and $N_j^{\mathrm{up}} = \ln 30$. It was computed with the halo of Essig et al., $v_0 = 230$, $v_{\mathrm{esc}} = 600$, $v_{\mathrm{E}} = 240\,\mathrm{km\,s^{-1}}$ and $\rho_\chi = 0.4\,\mathrm{GeV\,cm^{-3}}$ [2, App. B], and we recast it to the halo of Appendix A.8. The density enters every rate as a factor, $K^{\mathrm{rc}} = 0.4/0.3 = 1.333$ in every bin, and recasts exactly with any procedure. The velocity distribution does not: at fixed density its recast factors are 0.953, 0.903 and 0.866 in the three bins. The old halo, with smaller v_0 and v_{E}, has fewer fast particles in its bulk, and the higher bins need faster particles. With the density the factors are 1.271, 1.204 and 1.154. The published limit, $2.93 \times 10^{-42}\,\mathrm{cm^2}$, was set by the one-pair bin, and the exact recast is $3.72 \times 10^{-42}\,\mathrm{cm^2}$, a factor 1.271. From the curve alone, with the binding bin unknown, Eq. (21.30) gives 1.154 to 1.271. At 2 MeV the factors are 1.156, 1.432 and 4.0 with the density (0.867, 1.074 and 3.0 without it). The higher bins now draw on the tail, which the old halo extends to $v_{\max} = 840\,\mathrm{km\,s^{-1}}$ against 794.5; the three-pair bin opens only above $614\,\mathrm{km\,s^{-1}}$, and the binding one-pair bin gives 1.156.

The same rule decides how far a theory band can be recast. A normalization nuisance has one δ_k in every bin, and its band moves with the central limit. A tilt or a threshold shift has bin-dependent templates, and its band needs the per-bin limits or the likelihood, as the central value of any other tilted correction does. Table 21.3 sorts the corrections met in this book.

Table 21.3: Recast factors Eq. (21.28) of the corrections met in this book, and what a recast from a published curve needs. "Common" means the same factor in every bin at fixed m_χ, recastable exactly with any procedure. "Bin-dependent" means exact only for a single bin or for a minimum over bins with its upper counts public, and bracketed by Eq. (21.30) otherwise.

Correction	Factor	Recast
Local density ρ_χ	common	$\rho_\chi^{\mathrm{pub}}/\rho_\chi^{\mathrm{new}}$, exact
Normalization nuisance (Sect. 8.8)	common	$(1+\delta_k)^{-\vartheta_k}$, exact, band included
Background in the total ε, isotropic Dirac cone (Sect. 16.1)	common	$\varepsilon_{\mathrm{bg}}^2$ for $\bar{\sigma}_e$, $\varepsilon_{\mathrm{bg}}^{1/2}$ for ζ, exact
Halo velocity distribution (v_0, v_{esc}, v_{E})	bin-dependent	higher bins move most; at the edge through $E_{\mathrm{on}}/v_{\mathrm{max}}^2$ (Sect. 21.3)
Screening and local fields, tilt nuisances (Sect. 10.10)	bin-dependent	templates per bin
Gap or threshold shift	bin-dependent	moves the edge, Worked example 21.3
Anisotropic background tensor (Sect. 16.1.3)	depends on $\hat{\boldsymbol{q}}$	ratio of event integrals per mass
Relativistic density row of a boosted flux (Sect. 10.9)	depends on speed	rate ratio R, limit $\times R^{-1/2}$

An exact recast of any procedure needs what its likelihood contains, and a publication that provides it can be reused without approximation: per bin the observed count, the expected background with its covariance, the exposure and efficiency, the response of the detector from deposited energy to bin with its nuisances, and the upper count with its confidence convention; the halo convention and the truncation label of the signal model; and the likelihood itself in machine-readable form.

21.5.4 Dirac and boosted-flux recasts

The published reaches of Dirac materials screen the dark-photon field by the Dirac electrons alone, and the total dielectric function adds the background $\varepsilon_{\mathrm{bg}}$ of the remote bands (Sect. 16.1). For an isotropic cone the published loss function is $\varepsilon_{\mathrm{bg}}^2$ times the true one at every $(\boldsymbol{q},\omega)$, and the published absorption rate $\varepsilon_{\mathrm{bg}}$ times the true one (Eq. (16.2)). The factor is common to every bin and every mass, so by Eq. (21.29) the corrected reach is the published curve times $\varepsilon_{\mathrm{bg}}^2$ in $\bar{\sigma}_e$, and times $\varepsilon_{\mathrm{bg}}^{1/2}$ in the mixing ζ, since the absorption rate grows as ζ^2. Whatever procedure produced the published curve, and whatever its normalization, the recast holds. Nothing else multiplies it: the published $|\varepsilon_{\mathrm{pub}}|^2$ already contains the renormalization of the Dirac charge, $|1+\varepsilon_{\mathrm{D}}/\varepsilon_{\mathrm{bg}}|^{-2}$, with ε_{D} the polarization of the Dirac electrons (Sect. 16.1 and Appendix D.41). Fig. 21.8a recasts the ideal material of Hochberg et al., with $\varepsilon_{\mathrm{bg}} = 40$ [13]. The minimum of its gapped curve, $3.4\times10^{-46}\,\mathrm{cm}^2$ near 9 keV, becomes $5.4\times10^{-43}\,\mathrm{cm}^2$, as in Worked example 16.1, and at 114 keV the gapless curve moves from $5.5\times10^{-45}\,\mathrm{cm}^2$ to $8.8\times10^{-42}\,\mathrm{cm}^2$.

An anisotropic crystal has no common factor. Its published rule divides each component of the Dirac polarization by its own $\varepsilon_{\mathrm{bg},ii}$, the total dielectric function adds the background along $\hat{\boldsymbol{q}}$, and their ratio depends on the direction of $\boldsymbol{q}$ and on (q,ω) (Eq. (D.229)). The published $ZrTe_5$ reach is, however, a single count, three events with no background, and a single bin is the case $n = 1$ of a minimum over bins: the recast is exact once both event integrals are computed at each mass. Their ratio is 5100 to 6400 between 20 keV and 1 MeV (Fig. 21.8b). Toward the threshold it falls, to 2200 at 10 keV and 820 at 7.7 keV, because only momentum transfers near the y axis, with the lowest threshold, 7.1 keV, and the smallest

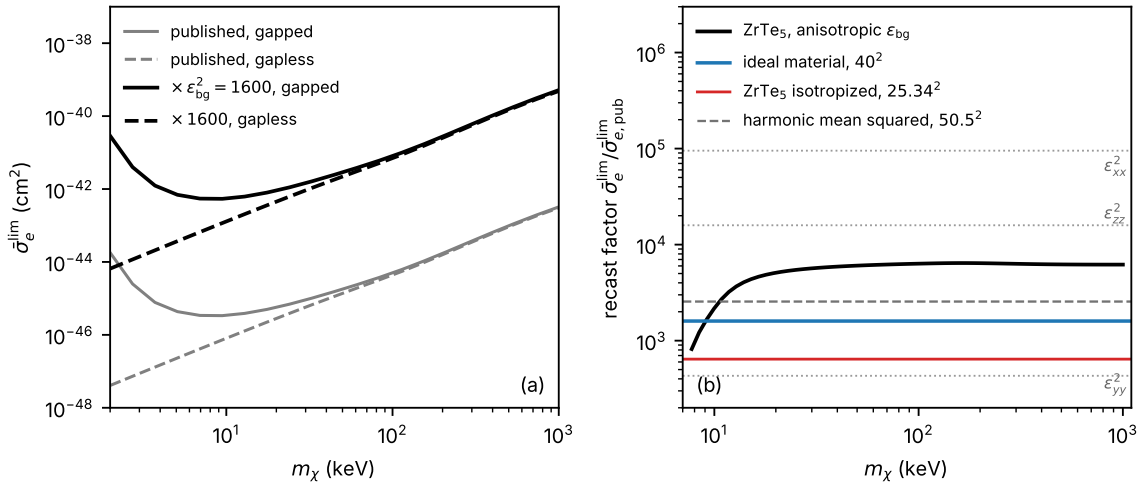


Figure 21.8: Recasting published Dirac-material reaches to the total dielectric function; light mediator, three events in one kg-year, no background. (a) The ideal Dirac material of Hochberg et al. [13] ($v_D = 4 \times 10^{-4}$, $\varepsilon_{bg} = 40$, $g_{deg} = 2$, cutoff 0.2 keV): published curves (gray), gapped ($\Delta_D = 2.5$ meV, solid) and gapless (dashed), read from the reference's figure. Black: the same curves times $\varepsilon_{bg}^2 = 1600$, exact for any procedure because the factor is the same at every $(\boldsymbol{q}, \omega)$. (b) Recast factor, the ratio of the corrected to the published reach, against m_χ. Black: the ratio of the published-rule to the total-ε event integral at each mass for $ZrTe_5$ (Sect. 16.1.3). It uses the anisotropic background tensor $\varepsilon_{bg,ii} = 308.4$, 20.75, 126.1 and the measured-structure row of Geilhufe et al. [14] ($\Delta_D = 11.8$ meV), the book's halo and the rate averaged over orientations of the crystal. Flat lines: $40^2 = 1600$ (blue) and $25.34^2 = 642$ for the isotropized $ZrTe_5$ of Hochberg et al. (red), both exact; the square of the harmonic mean of the tensor, $50.5^2 = 2554$ (gray dashed); the on-axis factors $\varepsilon_{bg,ii}^2 = 431$, 1.6×10^4 and 9.5×10^4 (gray dotted).

background constant, remain kinematically allowed there, and along that axis the factor is $\varepsilon^2_{\mathrm{bg},yy} = 431$. No constant reproduces the curve: the square of the harmonic mean of the tensor, the natural isotropic guess, is low by a factor of two to two and a half above 20 keV and high near threshold. An anisotropic Dirac reach is therefore recast with its rates, not re-plotted.

A boosted flux changes the counting itself. Dark matter accelerated by cosmic rays acquires its speed through the same coupling that the detector measures, so the flux is proportional to $\bar{\sigma}_e$ and the expected count to $\bar{\sigma}_e^2$, as the published analysis of silicon and germanium states [15]. With N^{up} fixed, a change of the detection rate per particle by a ratio R moves the lower edge of the excluded region by $R^{-1/2}$. The exact density row of a relativistic particle carries the factor $(1 - \omega^2/\mathcal{Q}^2)^2$, with $\mathcal{Q}$ the four-momentum transfer, that the published rate omits (Eq. (10.57)). For a heavy mediator it changes the rate of that analysis by 10^{-4}, and the transverse row adds no more than 10^{-4}: the limit is unchanged. For a massless mediator the shift of $\bar{\sigma}_e$ is below one percent for $m_\chi \geq 10\,\mathrm{keV}$. Below about 3 keV it is undetermined. The published density row has a pole at $\mathcal{Q} = \omega$, and its weight depends on the treatment of the lowest kinetic energies of the flux, which the source does not state. Depending on that treatment the ratio of the density rows ranges from 0.05 to 0.93 at 1 keV. Settling it, with the transverse row and its Cherenkov pole included, is the recast that Sect. 10.9.5 leaves open and Appendix E.16 specifies.

21.6 Shared response libraries

At $|\boldsymbol{q}| = \alpha m_e$ and $\omega = 10\,\mathrm{eV}$, the screened structure factor of silicon with local fields is $4.2 \times 10^7\,\mathrm{eV}^2$ in the published random-phase tables of Knapen, Kozaczuk and Lin [16] and $4.9 \times 10^7\,\mathrm{eV}^2$ in the empirical-pseudopotential model of Worked example 8.9. Both calculations sit at the same point of the truncation label, $\mathcal{X} = \mathrm{RPA} + \mathrm{LF}$. They agree on the independent-particle response to 3 percent and differ in the screening, which leaves 0.24 of the unscreened value in the tables and 0.29 in the model (Worked example 21.6). At 3 MeV two tabulations of the same independent-particle response give rates that differ by a third (Sect. 10.10). A limit can therefore be reproduced only from the table it was computed with, and recast for another coupling, halo or readout only if that table holds the rows, directions and final states the new calculation needs. The schema of Table 21.4 lists what a response library must tabulate for this, which conventions each table must state, and which section of Appendix E specifies the calculation that fills it.

21.6.1 Tables of a response library

The axes of a library follow from the rate kernel Eq. (21.2). The velocity of the dark matter enters it in two places: in the coupling matrix $\mathfrak{D}_{ab}(\boldsymbol{q}, \boldsymbol{v})$ and in the energy $\omega_{\boldsymbol{q}}$ fixed by energy conservation. The response $S_{ab}(\boldsymbol{q}, \omega; \Lambda)$ depends on $\boldsymbol{q}$, ω and the final-state class, and $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ on the class alone. A library therefore has no velocity axis and holds nothing about the halo. It tabulates functions of $|\boldsymbol{q}|$, $\hat{\boldsymbol{q}}$ and ω, labeled by a pair of coupling rows (a, b), a class Λ and a pair number. The coupling model enters through $\mathfrak{D}$, built from the operator coefficients of Table 8.1 and contracted by the user as $\sum_{ab} \mathfrak{D}_{ab} S_{ab}$, so the response is stored uncontracted, one function per row pair. Velocity-dependent couplings need no velocity axis either: their kernels are polynomials in the transverse velocity, and after the orientation average every power is fixed by $\eta(v_{\mathrm{min}})$ (Eq. (21.6)).

The electronic response is the largest part. For silicon and germanium the minimal table of Table 10.1 reduces every nonrelativistic operator to 12 real functions per $(\boldsymbol{q}, \omega)$. Six belong to the screened singlet block of the density and the transverse current, and six to the same block

at independent particles, which also fixes the independent-particle triplet block as a quarter of it. Operator $\mathcal{O}_2$ adds four functions in each version, through the kinetic-energy density $\mathcal{T}_e$ that its velocity squared brings in. A crystal without the combined symmetry of inversion and time reversal, such as GaAs, keeps the imaginary parts of the cross terms as well. On the grid of Appendix E.16, 250 values of $|\boldsymbol{q}|$ up to $25\,\alpha m_e$, 30 directions and 1000 energies up to 50 eV, the 12 functions fill 7.5×10^6 nodes, 720 MB in double precision. A screened response at a higher level, a time-dependent density-functional kernel or the Bethe–Salpeter equation, needs the full independent-particle matrix $R^0_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}, \omega)$, a matrix over the reciprocal-lattice vectors of the screening set at every node, too large to store on the same grid. The library keeps it on a coarse subset of the nodes, from which another kernel can be applied without repeating the band calculation.

The direction of $\boldsymbol{q}$ is kept as far as the analyses that use it require. A speed-only kernel, Eq. (21.4), needs the direction average $\bar{S}_{ab}$. The daily modulation (Sect. 11.7) and the directional halo-independent bound of Eq. (21.13) need the dependence on $\hat{\boldsymbol{q}}$ in crystal axes. In a cubic crystal it is carried, away from the edges, by the cubic-harmonic components S_ℓ/S_0 for $\ell = 4$, 6 and 8, since the odd ℓ vanish by time reversal and $\ell = 2$ by cubic symmetry. Near the zero-phonon edge the response changes on a scale of 0.1 rad in direction (Sect. 13.9), and there the library keeps the full grid of directions of Appendix E.16. The lattice tables of Appendix E.17 hold the response per cell, the probability $P_{\mathrm{ph}}(n|\boldsymbol{q}, \omega)$ that n phonons were created and the coherence correction. Their grid, 400 values of q from 0.1 to 300 keV, energies to 1 eV in steps of 0.1 meV and three temperatures, has 1.2×10^7 values per quantity.

The energy grid has to resolve the edges. The power law of the rate near an edge, Eq. (21.19), rests on the onset of the response, and the computed phonon-assisted loss function of silicon follows its x^3 onset only for the first 1.3 meV above the onset, the zero-phonon continuum for 49 meV (Sect. 21.3.3). A step of 0.05 eV, adequate for the bulk of the spectrum, has no node inside the phonon-assisted range, which is 38 times narrower. The library therefore adds a sub-grid with steps of a fraction of a meV within some 50 meV of each onset, and lists for each channel the derived edge entries of Table 21.1: the edge mass m_{edge}, the onset energy and the onset exponent p_{on}.

The final-state tables carry the per-pair weights of Eq. (8.68). A full list is out of reach: the grid of Appendix E.16 has 10^3 points $\boldsymbol{k}$ and 4×56 band pairs in silicon, 2.24×10^5 pairs per node and 1.7×10^{12} numbers, 13 TB for each coupling row. The cascade does not need the list. In the energy-only reduction of Sect. 18.3 it needs the seed distribution $P_{\mathrm{seed}}(E_{\mathrm{e}}, E_{\mathrm{h}}|\boldsymbol{q}, \omega, \boldsymbol{v})$ of Eq. (18.16), which for one coupling row depends on neither the velocity nor the model. In the zero-phonon sector $E_{\mathrm{e}} + E_{\mathrm{h}} = \omega - E_{\mathrm{g}}$, so one variable, the electron share f_e, describes the seed, and 20 bins of it take 1.2 GB per row pair on the full grid. A seed with a phonon adds the phonon energy, since then $E_{\mathrm{e}} + E_{\mathrm{h}} = \omega - \omega_{\boldsymbol{Q}} - E_{\mathrm{g}}$, and a valley label for the electron keeps the momentum memory of Sect. 18.1. The weights are retarded, and the sector split follows one declared convention; both are part of the table. The weights are the retarded ones of Sect. 8.7, which label the late-time state the cascade receives; in silicon and germanium they equal the advanced weights pair by pair, and in GaAs the two differ by 15 percent of the total. The split between the zero-phonon and the phonon-dressed classes follows either the off-shell window or the Dyson convention of Sect. 8.6, and the totals are independent of the choice only when one convention is declared for every table of the library.

The cascade tables hold $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ for each cascade model and parameter set, through the subtree generating functions $\mathfrak{G}_{\mathrm{e}}(x; s)$ and $\mathfrak{G}_{\mathrm{h}}(x; s)$ of Eq. (18.17) on a grid of carrier energies x. Appendix E.24 specifies that grid, a step of about $E_{\mathrm{g}}/50$ up to $20\,E_{\mathrm{g}}$ and the asymptotic Gaussian beyond, with its acceptance tests. The same tables carry the marks $\mathfrak{m}$ of Eq. (18.22) that a readout needs: the phonon spectrum, the separate electron and hole counts, the carrier positions and the stored energy. The library ends there. The detector response

$P_{\mathrm{det}}(\mathsf{R}|N_{\mathrm{eh}},\mathfrak{m})$ of Eq. (18.22) belongs to the experiment, which measures and publishes it with its collection efficiencies (Chap. 20), and the velocity distribution belongs to the analysis, which contracts it with the kernel in Eq. (21.1).

The nuisance tables make the theory band of a limit reproducible. For each truncation k active at a node, the library stores the next-order table $S^{\mathfrak{T}_k^+}$ itself rather than the ratio δ_k of Eq. (8.71). The matrix form Eq. (8.73) is built from $S^{\mathfrak{T}_k^+}$, and a ratio is undefined wherever $S^{\mathfrak{T}}$ vanishes, which is where a new-support template acts (Eq. (8.75)). Beside it stand the type of the template in the classes of Sect. 8.8 (normalization, tilt, threshold or new support), its basis (computed, model, estimate) and a key naming its nuisance parameter ϑ_k. The key is what enforces coherence: one ϑ_k acts on every node, class, target and experiment that shares the approximation (Sect. 8.8), and two tables carrying the same key are varied together. A template known only as an inequality, such as the tabulation spread of Sect. 10.10, a bound on the width of a spread that enters a likelihood with that width (Sect. 21.4.2), is flagged as a bound. A node with no template carries a flag that names the missing calculation. Energies where no central value can be defended, those below $3\Delta_{\mathrm{BCS}}$ at the superconducting edge of an aluminum film (Sect. 15.2), are flagged as excluded. The band of Sect. 21.4 is then built with these omissions named rather than with invented widths. A second calculation at the same truncation label is stored as an alternative table; the spread between the two is the nuisance of the approximations that the label leaves out, as for the two silicon tabulations of Sect. 10.10.

The derived tables are those that analyses would otherwise recompute for every search. The step column $\mathcal{A}_i(v)$ of Eq. (21.11), per bin and cascade model, turns the halo-independent bound of Eq. (21.12) into a minimum over a few numbers, and it follows from the response by cumulative sums over v_{min}. It is also a check: for a nonnegative kernel it never decreases with v (Eq. (D.250)), and a decrease flags either a sign error in a template or the final-state leakage of Eq. (21.15), at most $0.035\,\varrho_i^{\mathrm{leak}}$ per silicon pair bin. The direction-resolved stream kernels of Appendix E.27 complete the set for directional searches.

21.6.2 Conventions declared with each table

A number in a table is usable only with the conventions that turn it into a rate, and each convention that a user has to guess is a possible error of a factor of order one. The momentum $\boldsymbol{q}$ points from the dark matter to the target and ω is the energy given to the target, negative at finite temperature for a de-excitation; detailed balance, $S(\boldsymbol{q},-\omega)=\mathrm{e}^{-\beta\omega}S(\boldsymbol{q},\omega)$, relates the two signs, and the table states the temperature it holds. The temperature is metadata, distinct from the thermal order n_T of the label, which counts powers of the phonon occupations (Sect. 8.5).

Next comes the normalization, the most common source of factors of 2π and 4π. The structure factor of Eq. (4.13) is per volume, in eV^2 in natural units; a table per cell holds ΩS, in eV^{-1}, with $\Omega = 40.05\,\text{Å}^3$ for silicon; a rate per target mass divides by ρ_T in the halo prefactor. Spin is summed. A table of loss functions is turned into structure factors by Eq. (4.45), $S=(2q^2/e^2)\,L$ at zero temperature with $e^2=4\pi\alpha$; in Gaussian units the same symbol e^2 means α and the factor changes by 4π. A dielectric table must say which function it lists: the head ε_{00} of the dielectric matrix, whose loss function $\varepsilon_2/|\varepsilon_{00}|^2$ is the random-phase value without local fields, or the macroscopic function $\varepsilon_{\mathrm{mac}}=1/(\varepsilon^{-1})_{00}$, whose loss function includes them. The two differ by the local-field step, 9 percent at the node of Worked example 21.6. Momenta are unreduced (Appendix A.3); a table that lists $|\boldsymbol{q}+\boldsymbol{G}|$ for the points of its calculation mixes directions at equal length, and an average over bins of $|\boldsymbol{q}|$ is then an average over directions, which the table states.

The row basis and its index order are declared with the matrix. The contraction is

Table 21.4: Schema of a response library for one material. Every table carries the truncation label $\mathfrak{T}$ of Eq. (8.44) with the metadata and conventions of Sect. 21.6.2, a content hash and the list of its parents (Sect. 21.6.3). Energies ω are given to the target, momenta $\boldsymbol{q}$ point from the dark matter to the target and are not reduced to the first zone. The last column names the section of Appendix E that specifies the calculation filling the table.

Table	Axes and labels	Content	Specified in
Material inputs	—	structure and measured lattice constant; the gap of the scissor and the gap used by the cascade, separately; functional, pseudopotentials, cutoffs, grids, number of bands; force constants of the shared harmonic base	Appendices E.1 and E.17
Electronic response	$\lvert\boldsymbol{q}\rvert$, $\hat{\boldsymbol{q}}$, ω; rows (a, b)	minimal table of Table 10.1 (12 real functions in Si and Ge, 20 with $\mathcal{O}_2$), screened and independent-particle, per cell; $R^0_{\boldsymbol{G}\boldsymbol{G}'}$ on a coarse subset of nodes	Appendices E.6, E.7 and E.16
Directional components	$\lvert\boldsymbol{q}\rvert$, ω; ℓ	S_ℓ/S_0 for $\ell = 4$, 6, 8; full direction grid within the edge sub-grid	Appendix E.16
Lattice response	$\lvert\boldsymbol{q}\rvert$, $\hat{\boldsymbol{q}}$, ω; T	response per cell; $P_{\mathrm{ph}}(n\lvert\boldsymbol{q}, \omega)$; branch content; coherence correction	Appendix E.17
Further channels	as the parent table	phonon-assisted loss function; exciton lines and core shells; joint response of the Migdal effect; spin-wave, superconducting and Dirac responses	Appendices E.14, E.15, E.18 and E.20
Final-state classes	$\lvert\boldsymbol{q}\rvert$, ω, rows; f_e, $\omega_{\boldsymbol{Q}}$, valley	seed distribution P_{seed} per class from retarded pair weights; convention of the zero-phonon/phonon-dressed split	Appendices E.6 and E.7
Cascade	carrier energy x; model and parameters	$\mathfrak{G}_{\mathrm{e}}$, $\mathfrak{G}_{\mathrm{h}}$ and $P_{\mathrm{casc}}(N_{\mathrm{eh}}\lvert\Lambda)$; marks $\mathfrak{m}$ for the readout	Appendices E.13 and E.24
Nuisances	as the parent table; key of ϑ_k	next-order table $S^{\mathfrak{T}^+_k}$; type and basis of the template; flags for bounds, missing templates and excluded energies; alternative tables at the same $\mathfrak{T}$	as the parent table
Checks	as the parent table	f-sum fraction; smallest eigenvalue of S_{ab}; Kramers–Kronig residual; detailed balance; symmetry-forbidden parts; cascade normalization, moments and the single-pair limit	as the parent table
Derived	bin i, speed v; channel	step columns $\mathcal{A}_i(v)$; edge mass, onset energy, p_{on}; direction-resolved stream kernels	Appendix E.27

$\sum_{ab} \mathfrak{D}_{ab} S_{ab}$ with the rows of Table 10.1, and the order of the indices of $R^0_{\boldsymbol{G}\boldsymbol{G}'}$ fixes which factor is conjugated. Transposing it conjugates the cross terms, which is immaterial in silicon, where every element is real, but not in GaAs (Sect. 12.3). The final-state tables add the labeling of pairs (retarded) and the convention of the sector split, and a table of Migdal responses states its energy axis, the total electronic energy transfer ω_e (Sect. 12.1).

The truncation label is written out in all eight components of Eq. (8.44), not in the deviation notation that suits running text, together with the reference state (normal, superconducting, spin-wave), the screening set and the split convention. The approximations that the label leaves out, the exchange–correlation functional, the pseudopotentials, the scissor and the grids, are measured by tests and convergence (Appendices E.1 to E.5), and they go into the material inputs with the results of those tests. The scissor gap and the pair-conversion gap need separate entries. The gap to which a scissor shifts the bands and the gap with which a cascade or a step model converts energy into pairs are different numbers in general, and a rate that uses both inherits two thresholds. Near 1 eV the energy nodes of the tables of Worked example 21.6 are 0.22 eV apart, more than the 0.06 eV between the room-temperature and low-temperature gaps of silicon (Sect. 10.10), so the gap of such a table cannot be read off its nodes and has to be stated. The tables of Worked example 21.6 state a scissor to the measured gap at zero temperature, 1.17 eV; the 1.11 eV that accompanies them in their package is the gap of the step model of pair creation (Sect. 10.10). The interpolation rule between nodes is part of the table too: near 10 eV the same loss function changes by 10 percent from one energy node to the next.

21.6.3 Versions and provenance

Every array of a library is identified by a hash of its content. It carries the list of its parents: the structure, the pseudopotentials, the code and its version, the input files, and for a derived table the tables it was computed from. Two arrays with the same name and different content can then not be confused, and a limit can name the exact arrays it used. Versions follow one rule for each kind of change. A change of a truncation label or of a convention changes rates and makes a new major version. Added tables, such as a new row pair, a new material or a new nuisance template, change no existing rate and make a minor version. A numerical refinement, a finer grid or a larger cutoff, is a patch only if no tabulated value moves by more than the convergence tolerance the table declares. A larger move means the old values were outside their stated accuracy, and it makes a major version as well. A published version is never changed and carries a persistent identifier, so a limit cites the library the way it cites a paper.

A version is released with its checks, each stored as a value rather than a verdict. The f-sum rule Eq. (4.30) holds to 2 % below the cutoff of the bands (Appendix E.16), and a spectrum that ends early stores the fraction it holds at each $|\boldsymbol{q}|$. Positivity, Eq. (4.20), is checked through the smallest eigenvalue of every stored matrix S_{ab}, with a tolerance stated by the library, for instance -10^{-6} of its trace; the Kramers–Kronig residual is stored likewise, for instance below 1 %, and the two signs of ω satisfy detailed balance. Symmetry supplies free diagnostics: in silicon and germanium the imaginary parts of the density–current terms and the antisymmetric part of the current block must vanish, and they stay below 10^{-3} of the allowed parts. The cascade tables sum to one and are nonnegative, their derivatives reproduce the moment recursion to 10^{-9}, and below the first impact ionization, at $(1 + \theta_{\mathrm{ii}}) E_{\mathrm{g}} = 2.32\,\mathrm{eV}$ in silicon (Sect. 18.3), they give exactly one pair (Appendix E.24). The step columns never decrease. Where optical, energy-loss or inelastic x-ray measurements exist, the comparison with them is stored in the same way. A table that misses a check by a few percent may still serve an analysis that needs less, and only the stored value lets its user decide.

A limit published with a library names the library and its version, the truncation label of each table it used and the cascade model, and gives per bin the count, the exposure and the detector response. Three things then become recomputations. The limit itself is reproduced from the tables. Its theory band follows from the nuisance tables and their keys (Sect. 21.4). A recast for another coupling, another halo or a corrected response evaluates the kernel Eq. (21.2) again instead of rescaling a published curve, which Sect. 21.5 shows to be exact only under a condition on the bins. The guide to targets of Sect. 21.7 reads its inputs from the same records: the loss function $L(\boldsymbol{k}, \omega)$ of each candidate (Appendix E.16) and its elastic constants (Appendix E.17). The provenance it reports for each figure of merit, measured, computed or estimated, is the provenance stored with those tables.

Worked example 21.6: The silicon loss function at one table node

Knapen, Kozaczuk and Lin computed the random-phase dielectric function of silicon from Kohn–Sham states [16, Sect. III B]. They used the TB09 functional, the meta-generalized-gradient functional of Tran and Blaha (2009) chosen for band gaps, a plane-wave cutoff of 500 eV, an $8 \times 8 \times 8$ grid of $\boldsymbol{k}$ points, 70 bands per spin and a scissor to the measured gap. The tables distributed with their package [17] list ε_1 and ε_2 against ω and $|\boldsymbol{q}+\boldsymbol{G}|$, in one file for the head ε_{00} and in one for $\varepsilon_{\mathrm{mac}}$. We write the entry at the node nearest to $(q_0, 10\,\mathrm{eV})$, with $q_0 = \alpha m_e = 3728.9\,\mathrm{eV}$.

Node. $|\boldsymbol{q}| = 3732.0\,\mathrm{eV} = 1.0008\,q_0$ and $\omega = 9.8585\,\mathrm{eV}$. There the head is $\varepsilon_{00} = 1.838 + 0.591\,\mathrm{i}$ and the macroscopic function $\varepsilon_{\mathrm{mac}} = 1.763 + 0.482\,\mathrm{i}$.

Value. With local fields, $L = \mathrm{Im}(-1/\varepsilon_{\mathrm{mac}}) = 0.1442$. By Eq. (4.45), $2q^2/e^2 = 3.038 \times 10^8\,\mathrm{eV}^2$, so $S = 4.380 \times 10^7\,\mathrm{eV}^2$ per volume and $\Omega S = 0.2283\,\mathrm{eV}^{-1}$ per cell. The same node gives $L = \varepsilon_2/|\varepsilon_{00}|^2 = 0.1585$ without local fields and, from the head's $\varepsilon_2 = 0.591$, the independent-particle $S^{\mathrm{IP}} = 1.796 \times 10^8\,\mathrm{eV}^2$.

Label. $\mathfrak{T} = (n_B = 0,\ n_g = 0,\ n_A = 0,\ \mathcal{X} = \mathrm{RPA} + \mathrm{LF},\ n_T = 0;\ \mathcal{K}, \mathfrak{M}, n_D$ not applicable), with $T = 0$, a normal reference state, external and local-field screening, and no sector split, since a table at $n_g = 0$ has no phonon-dressed part. Off the label: TB09, the cutoff, the grid, the bands and the scissor, as above. The package keeps a separate file of material constants, with $E_{\mathrm{g}} = 1.11\,\mathrm{eV}$ and $\varepsilon_{\mathrm{eh}} = 3.6\,\mathrm{eV}$ for the conversion of energy into pairs by the step model, and an entry of the library would carry that gap beside the gap of the scissor.

Nuisance. The local-field step is computed, so it moves the central value: $\delta_{\mathcal{X}} = 0.1442/0.1585 - 1 = -0.090$ at this node, a tilt by Eq. (8.76). The nuisance that remains is the excitonic correction. No calculation at finite momentum gives it for silicon, so the node carries the flag of a missing template. Sect. 10.10 estimates the step at $+0.28$ over the energies a 3 MeV particle reaches, from a model at 0.91 keV, and a band takes it from there with that basis named.

Checks. At this $|\boldsymbol{q}|$ the table holds 0.916 of the valence f-sum $(\pi/2)\omega_p^2$, $\omega_p = 16.6\,\mathrm{eV}$, below its last energy, 89.2 eV; the missing weight lies above about 70 eV, which the 70 bands do not reach [16, Fig. 1]. The three other nodes within 50 eV of $|\boldsymbol{q}|$, at 3689, 3722 and 3776 eV, give $L = 0.1431$, 0.1402 and 0.1445. The four nodes span 3.0 percent, partly from direction and partly from $|\boldsymbol{q}|$, and the entry states that its direction average is a bin average over such nodes. The next energy node, 10.26 eV, gives $L = 0.1295$, so at 10 eV linear interpolation gives $L = 0.1390$ and $S = 4.22 \times 10^7\,\mathrm{eV}^2$.

Comparison. The empirical-pseudopotential model of Worked example 8.9 gives, at q_0 along [100] and 10 eV, $S^{\mathrm{IP}} = 1.7 \times 10^8\,\mathrm{eV}^2$ against $1.74 \times 10^8\,\mathrm{eV}^2$ here, and $S = 4.89 \times 10^7\,\mathrm{eV}^2$ with local fields against $4.22 \times 10^7\,\mathrm{eV}^2$, a ratio of 0.864. The screening factor S/S^{IP} is 0.29 in the model and 0.242 in the tables, and the local-field step -0.06 against -0.09.

The two calculations carry the same label. What separates them is off the label: a model potential with a gap of 0.82 eV against TB09 with a scissor, and one direction against a mixture. Only the provenance of the table tells a reader which of the two a limit used.

21.7 Next targets

CsI has a heavy-mediator single-phonon rate per unit mass 3.73 times that of silicon, on elastic constants measured at 4.2 K, and 1.56 times that of TeO_2, the leader of the eleven crystals ranked in Sect. 17.6. The lead salts have electron-recoil floors up to 4.4 times below germanium's. Neither family was in that pool: no CsI crystal has been read by a phonon sensor at millikelvin temperatures, and a lead compound carries the ^{210}Pb of its lead, whose β spectrum puts some 700 events per kilogram, day and keV below 1 keV for every becquerel per kilogram. A ranking of responses answers which crystal gives the most rate per unit mass. An experimentalist choosing the target of the next experiment also needs to know what each leader still has to demonstrate, which measurement would change the order, and on what the probabilities rest. We combine the response-level rankings of Sect. 17.6 with the pair energies of Sect. 18.4, the readout verdicts of Sect. 20.5 and the internal backgrounds of the crystals into a guide with one row per channel, Table 21.5. Current capability enters it as a column, never as a filter.

21.7.1 Candidate pools and detector readiness

The *full pool* holds 42 crystals: 23 of the 24 crystals of the target table of Griffin and collaborators [18, Table I], all but InSb, the three SiC polytypes, and the targets of the preceding chapters that have the inputs of at least one figure of merit (TeO_2, Li_2MoO_4, γ-$LiAlO_2$, $SrTiO_3$, $ZrTe_5$, Bi_2Se_3, aluminum, niobium, YIG, NiO, MnO, Cr_2O_3, $NiPS_3$, hexagonal and cubic BN, and polyethylene). Four requirements decide whether a crystal can be the target of a cryogenic experiment now. The crystal operates at millikelvin temperatures after thermal cycling, without a phase transition between 300 K and 10 mK and without absorbing water from the air. At least 1 g of detector-grade crystal can be grown, or an array of instrumented crystals of at least 0.1 g each adds up to a gram. A sensor has been read on its surface at millikelvin temperatures. And its intrinsic rate below 1 keV is at most one event per kilogram, day and keV. A crystal for which all four are documented is a *survivor*. A crystal for which one or more is not documented, though none is shown to be impossible, is a *conditional survivor*. Only positive evidence removes a crystal from the full pool. That evidence exists for two candidates, $KTaO_3$ and InSb, whose natural ^{40}K and ^{115}In give intrinsic rates far above the fourth requirement (Sect. 21.7.4), and both stand outside the 42: InSb is the one crystal of the target table left out, and $KTaO_3$, the strongly anharmonic target of Sect. 11.5, would otherwise enter with the targets of the preceding chapters.

The survivor pool is the eleven crystals of Sect. 17.6: silicon, germanium, GaAs, diamond, sapphire, LiF, NaF, $CaWO_4$, TeO_2, Li_2MoO_4 and γ-$LiAlO_2$. The other 31 are conditional. Membership is a statement about the documented record of a crystal, not about its physics, and it changes with one demonstration. GaAs entered when a 4.3 g crystal was operated at about 10 mK with a thermistor readout [19]. Diamond entered through the array route: two crystals of 0.175 g each, grown by chemical vapor deposition and read by tungsten transition-edge sensors, one with a threshold of 16.8 eV [20]. Six such crystals make a gram, each with its own readout channel and threshold, and their surface-to-volume ratio is $(1/0.175)^{1/3} = 1.79$ times that of a single gram crystal, which raises any surface background in proportion.

The guide reports both pools side by side. The survivor leader of a channel is the best target that the record supports today. The full-pool leader says what a demonstration would gain: if it differs from the survivor leader, the missing demonstration of the full-pool leader is worth as much as its lead. The four requirements form the capability column of Table 21.5, with the status of each for the leaders.

The rankings of the full pool use the method and widths of Sect. 17.6 unchanged: Monte Carlo propagation of every input with its provenance width (Table 17.7), 2×10^4 draws, P(first) and the rank interval of Eq. (17.20), and tiers from the interval order. The eleven survivors enter with the inputs of Sect. 17.6. For the conditional crystals the gaps are those of the target table, a room-temperature gap shifted upward by an amount drawn uniformly between 0.03 and 0.10 eV, and the SiC gaps are room-temperature values of the SiC target table [21, Table I]. Their elastic tensors are computed ones from the same database as in Sect. 17.6 [22], with the width 0.086, except two that are measured: aluminum at zero temperature and CsI at 4.2 K [23], width 0.02. Their polar band integrals come from Eq. (11.21) with computed Born charges, ε_∞ and highest optical energies [18, Table I], or from the measured dielectric constants of the SiC polytypes [21, Table I]; for quartz and MgF_2, whose many polar branches the computed data do not resolve, the band integral is drawn log-uniformly between bounds that differ by factors of 14 and 7.6.

We add one channel that Sect. 17.6 did not rank: the threshold in quanta, measured by the pair energy $\varepsilon_{\mathrm{eh}}$. A lower pair energy turns the same deposit into more pairs and puts the one- and two-pair bins of a charge readout at lower energy (Sect. 18.3). Where a pair energy has been measured, the guide uses the measurement: silicon 3.749 eV at 123 K [24], GaAs about 4.2 eV at room temperature [25, Fig. 1] raised by 1.70×0.10 eV for the shift of its gap on cooling (the slope $\partial\varepsilon_{\mathrm{eh}}/\partial E_{\mathrm{g}}$ of the cascade at fixed constant, Sect. 18.4), diamond and the SiC polytypes as in Fig. 18.11, and germanium as discussed below. For every other crystal the pair energy is the estimate of the cascade of Sect. 18.4 at the tabulated gap and highest optical phonon energy. Inside the validated range of Sect. 18.4.6 we give it a logarithmic width 0.105, the root mean square of the residuals of the seven crystals with measured values; outside it the width is doubled to 0.21, a convention, since no measurement anchors these crystals.

Germanium has no pair energy measured directly at millikelvin temperatures. Its pair energy at 77 K is 2.96 eV, the value Sect. 18.4 tests the cascade against. A detector runs at tens of millikelvin, however, and the one value quoted there, (3.37±0.01) eV, comes from a fit by Wei, Wang and Mei to the ratio of phonon to ionization energy measured in EDELWEISS detectors as a function of bias, $E_{\mathrm{p}}/E_{\mathrm{Q}} = a + eV_{\mathrm{bias}}/\varepsilon_{\mathrm{eh}}$, in which the relative conversion efficiency $a = 0.85$ of primary and Luke phonons is fitted together with $\varepsilon_{\mathrm{eh}}$ [26, Sect. 2.1.4]. The value is therefore model-dependent: it rests on the assumption that one constant a describes the phonon collection at every bias, the quantity whose detector dependence Sect. 20.5 found open. We use 3.37 eV with the width 0.015 of the spread between this fit and their resolution-based value of 3.32 eV. Either value puts germanium ahead of silicon.

Every probability in the guide is conditional on the widths. They are conventions, set by how an input was obtained rather than computed input by input (Sect. 17.6.1). A leader whose lead rests on a computed input loses probability when the convention widens: TeO_2 falls from P(first) = 0.968 to 0.781 when the width of computed sound speeds is doubled. The method correlation r_{mat} of Eq. (17.21) moves the joint probabilities and leaves medians, intervals and tiers alone. The guide reads every row at $r_{\mathrm{mat}} = 0$ and prints the value at $r_{\mathrm{mat}} = 1$ beside it. Where the two differ strongly, as for the Migdal row and for the pair energies of the lead salts, the order rests on an assumption about a method, not on the inputs.

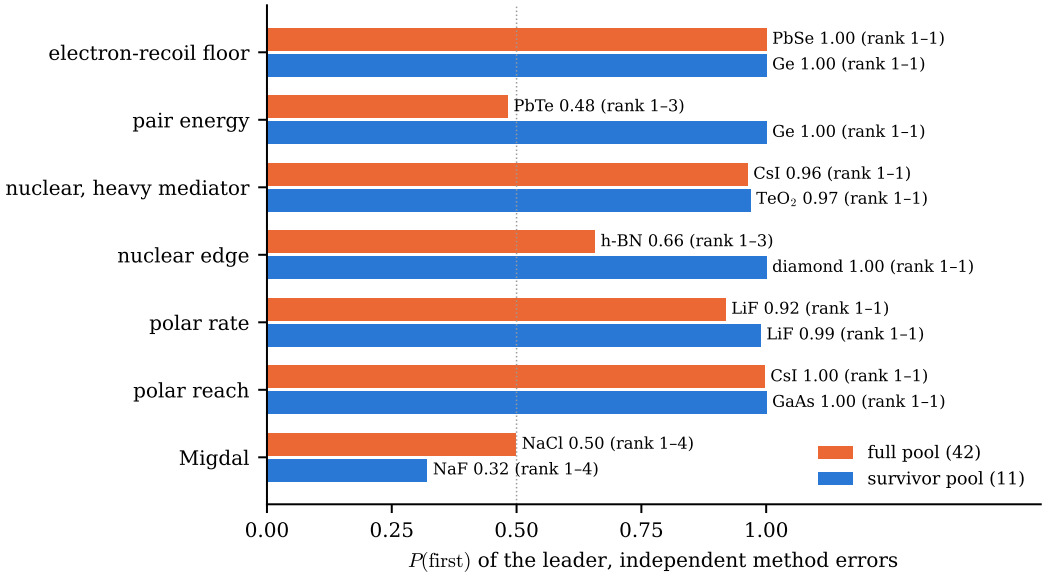


Figure 21.9: Leaders of the scattering channels with their probability of ranking first and their 68% rank interval, for the full pool of 42 crystals (upper bar of each pair) and the pool of eleven survivors (lower bar). Independent method errors, provenance widths of Table 17.7, 2×10^4 Monte Carlo draws. Pair energy: measured values for silicon, germanium, GaAs, diamond and the SiC polytypes; estimates from the cascade of Sect. 18.4 for the rest, with a logarithmic width of 0.105 inside its validated range and 0.21 outside. The dotted line marks $P = 0.5$: below it the leader has less than even odds. The light-mediator single-phonon rate has no row, because it does not depend on the target far above the edge.

21.7.2 The guide to next targets

Table 21.5 is the guide, and Fig. 21.9 shows its leaders with their probabilities of ranking first. Each row holds in the window of its figure of merit (Sect. 17.5): the floor and the edges are kinematic and hold at every mass, the heavy-mediator figure holds pairwise in the window W_{AB} of Eq. (17.15), and the polar band integral ranks crystals by itself only far above their polar edges. Survivor values are those of Table 17.8, except for the pair-energy row, which Sect. 17.6 did not rank.

Table 21.5: The guide to next targets. Leaders of the full pool (42 crystals) and of the survivor pool (eleven crystals, Sect. 21.7.1): median with 68% interval, probability of ranking first at independent method errors with the value at $r_{\mathrm{mat}} = 1$ in parentheses, and 68% rank interval. Every probability is conditional on the provenance widths of Table 17.7 and on the widths of the pair energies stated in Sect. 21.7.1; it is not the probability that a crystal is the better target. Capability: the four requirements of Sect. 21.7.1, operation at millikelvin (mK), gram-scale growth or array (mass), sensor readout (readout) and intrinsic radioactivity (radio), with their status for the leaders: shown (documented in the sources cited in Sect. 21.7.1 and the channel chapters) or not reported (no record found). Edges at $\omega_{\mathrm{th}} = 1\,\mathrm{meV}$; $v_{\mathrm{max}} = 2.650 \times 10^{-3}$. Absorption reaches: coupling for three events in a kilogram and year without background (Table 17.3).

Channel; figure of merit	Full-pool leader	Survivor leader	Capability	Missing demonstration	Deciding measurement or computation	Conditional on
Electron-recoil floor; $2E_{\mathrm{g}}/v_{\mathrm{max}}^2$	PbSe, 0.0484 MeV [0.0479, 0.0489]; 1.000 (1.000); 1–1; PbTe and PbS next	Ge, 0.212 MeV [0.210, 0.214]; 1.000 (1.000); 1–1	Ge: all four shown. Lead salts: mK, mass and readout not reported; radio open (^{210}Pb)	lead salts: a crystal of at least 0.1 g read at mK; ^{210}Pb below 1.38 mBq per kg of compound	^{210}Pb activity of the lead available for growth; low-temperature gap of GaSb	measured low-temperature gaps of the leaders; germanium ahead of GaSb (0.974) only through the shift of GaSb's room-temperature gap
Threshold in quanta; $\varepsilon_{\mathrm{eh}}$	PbTe, 1.52 eV [1.24, 1.87]; 0.482 (1.000); 1–3; one tier with PbSe (0.414) and PbS (0.102)	Ge, 3.37 eV [3.32, 3.42]; 1.000 (1.000); 1–1; then Si 3.75, GaAs 4.37	as above	lead salts: as above; one measured pair energy of a lead salt	pair energy of germanium measured at mK with the phonon collection calibrated separately; first-principles cascades (Appendix E.24)	germanium: the 50 mK fit with a fitted phonon-conversion constant; lead salts and GaSb: cascade estimates outside the validated range (width 0.21)
Heavy-mediator single phonon; $c_{\mathrm{NR1}}(\mathrm{Si})/c_{\mathrm{NR1}}$	CsI, 3.73 [3.65, 3.80]; 0.962 (0.976); 1–1; measured 4.2 K tensor	TeO_2, 2.39 [2.19, 2.60]; 0.968 (0.987); 1–1; computed tensor	CsI: mass and radio shown; mK not reported (slightly hygroscopic); readout not reported. TeO_2: all four, radio with ^{123}Te capture bounded only	CsI: an encapsulated crystal cycled to mK and read by a phonon sensor (Worked example 21.7); TeO_2: none	measured elastic constants of TeO_2; measured ^{123}Te capture rate	pairwise windows (CsI against silicon from 97 keV to 0.32 MeV); computed-speed width 0.086 (doubled: TeO_2 0.781); CsI at the room-temperature density (cold cell: 3.81)
Nuclear edge; $m_{\mathrm{min}}^{\mathrm{ac}}$ with $c_{\mathrm{LA,max}}$	h-BN, 2.95 keV [2.72, 3.21]; 0.656 (0.728); 1–3; one tier with diamond (0.181), polyethylene (0.119), cubic BN (0.043): no resolved leader	diamond, 3.11 keV [3.05, 3.18]; 1.000 (1.000); 1–1; measured tensor	diamond: array route, six crystals of 0.175 g, readout shown. h-BN: mass, mK and readout not reported	diamond: six crystals each with its own threshold and low-energy-excess rate; for every crystal a phonon threshold near 1 meV (shown: 16.8 eV)	measured elastic constants of h-BN, polyethylene and cubic BN	$\omega_{\mathrm{th}} = 1\,\mathrm{meV}$; h-BN and polyethylene reach their edges only for momentum along their stiff directions (computed tensors)
Polar rate; $\mathcal{I}_{\mathrm{pol}}/\rho_T$	LiF, 19.0 [15.5, 23.3]; 0.918 (0.931); 1–1; quartz 0.066 from its wide interval	LiF, 18.9 [15.5, 23.2]; 0.987 (1.000); 1–1	LiF: all four shown	LiF: the ^{6}Li neutron-activation background has not been measured	measured ε_0 and Born charge of LiF at low temperature; mode data of quartz	computed Z^*, ε_∞ and optical energy of LiF; masses far above the polar edges

Channel; figure of merit	Full-pool leader	Survivor leader	Capability	Missing demonstration	Deciding measurement or computation	Conditional on
Polar reach; $2\omega_{\mathrm{L}}/v_{\max}^{2}$	CsI, 2.85 keV [2.71, 2.99]; 0.996 (1.000); 1–1	GaAs, 10.35 keV [10.30, 10.40]; 1.000 (1.000); 1–1; measured ω_{L}	GaAs: all four shown (4.3 g at 10 mK [19]). CsI: as above	GaAs: a threshold below its longitudinal phonon, 36.35 meV; CsI: as above	low-temperature longitudinal optical frequency of CsI	CsI: computed highest optical energy, 10.0 meV
Migdal; pole factor times X_{pole}	no resolved leader: NaCl 0.498 (1.000); 1–4	no resolved leader: NaF 0.315 (1.000); 1–4	–	–	$L(\boldsymbol{k},\omega)$ of every candidate (Appendix E.16)	the pole prior; the dispersion of the pole; the coherent response below $2q_{\mathrm{BZ}}$ (Sect. 12.3)
Light-mediator single phonon	no ranking: the rate per unit mass is within 10 % of the target-independent Eq. (11.6) above $44\,m_{\min}^{\mathrm{ac}}$ (Limit Check 11.2); the target enters through the nuclear edge		–	–	–	–
Dark-photon absorption	per decade (Table 17.3): GaAs line 2.9×10^{-17} at 36.35 meV; sapphire 1.0×10^{-17} at 112 meV; silicon plasmon 1.0×10^{-17} at 16.4 eV; lines cover their own masses only		GaAs, sapphire, silicon: all four shown	a target from 0.38 to 1.17 eV, where none absorbs at leading order	loss function of a narrow-gap or Dirac crystal across that band	measured line widths; screening ceiling $L\le 1/(2\lvert\varepsilon_1\rvert)$
Axion-like absorption, electron coupling	YIG line 1.6×10^{-17} at 10 µeV on resonance, 4.1×10^{-15} averaged over a scan; clean aluminum, direct pair breaking by the spin current, 6.1×10^{-13} at 0.72 meV; silicon 1.4×10^{-13} at 4.3 eV		YIG: mass not reported (millimeter spheres); aluminum: bulk readout not reported	a target from 10 meV to 1.17 eV; aluminum: a bulk crystal with mean free path well above 2.4 µm, read at mK	line width of the magnon resonance; spin-current response of a Dirac crystal with its spin–orbit coupling	YIG peak: assumed width $10^{-5}\omega$; aluminum: clean limit, in films 1.5×10^{-12} against 2.0×10^{-12} from the published mapping (Sect. 15.4.3)
Scalar absorption, electron coupling	GaAs line 3.1×10^{-14}, the only crystal target with a computed rate besides the silicon model		GaAs: all four shown	–	phonon-assisted kinetic row of silicon (Appendix E.14)	near-cancellation of the GaAs sublattice couplings
Magnon scattering, spin coupling	NiO: opens near 7.3 keV at a 5 meV threshold, 3.2×10^{3} events per kg-year near 3 MeV at $\bar{\sigma}_e=10^{-40}\,\mathrm{cm}^2$; no ranking (Sect. 14.1)		NiO: mass and mK not reported	a gram-scale magnetic crystal read at mK	magnetic anisotropy of NiO near the edge	linear spin waves: one-magnon weight high by 7.8 %
Superconductor scattering	aluminum: floor $4\Delta_{\mathrm{BCS}}/v_{\max}^{2}$; energies $y<3$ excluded from the signal region (Sect. 15.2)		aluminum: bulk readout not reported	quasiparticle readout of a bulk absorber	gap and mean free path of the absorber used	bulk gap 0.18 meV against the film convention 0.3 meV (×1.23 at 1 keV, light mediator)

Dirac and other special targets	$ZrTe_5$: threshold 7.1–9.6 keV by axis, reach near $1.5 \times 10^{-40}\,\mathrm{cm}^2$ at 40 keV; glass two-level systems, dark photon 2.44×10^{-15} at 6 GHz; Si:P $2p_0$ line 2.3×10^{-16} at 34.11 meV	$ZrTe_5$: mass, mK and readout not reported	$ZrTe_5$: a gram-scale crystal read at mK	background dielectric tensor of $ZrTe_5$; loss tangent of the sample	$ZrTe_5$: one-loop parameter about one, no controlled theory; background tensor moves the reach by 2.8 (Worked example 16.9)

Germanium leads both electron-recoil rows among the survivors, the floor at 0.212 MeV on its measured low-temperature gap and the threshold in quanta at 3.37 eV per pair, 1.11 times fewer electron volts per pair than silicon. It is the only survivor in the leading tier of two channels, the Migdal row aside. Its lead in quanta holds on the 77 K value as well, which is lower still; what the 50 mK fit leaves open is the number, not the order. In the full pool the lead salts lead both rows. PbSe has the lowest floor, 0.0484 MeV, on a gap measured at low temperature, with $P(\text{first}) = 1.000$. In quanta PbTe, PbSe and PbS share the leading tier, with probabilities 0.482, 0.414 and 0.102. These pair energies are cascade estimates for crystals whose phonon strengths λ_{AB}, the dimensionless ratio of phonon emission to impact ionization in the cascade (Sect. 18.4.6), lie between 230 and 1160, two orders of magnitude outside its validated range, so the tier is real only under the width convention. GaSb, whose estimate lies at the edge of that range, beats germanium's measured value with probability 0.83, so even germanium's place among all crystals rests on an estimate. The full-pool order inside the leading tier moves with the method correlation: at $r_{\text{mat}} = 1$ the three estimates share their error and PbTe ranks first in every draw. Two further steps stand between a lead salt and a target. None has been operated at millikelvin with a phonon or charge readout, and every one carries ^{210}Pb in its bulk (Sect. 21.7.4).

The heavy-mediator rows separate the two pools most clearly. TeO_2 leads the survivors with $P(\text{first}) = 0.968$, on a computed tensor. CsI, a conditional survivor, leads the full pool with $P(\text{first}) = 0.962$ on a measured one, ahead of Bi_2Se_3 and NaI, both computed, with which it is compared at probabilities 0.978 and 0.985. Its lead over TeO_2 is a factor 1.56 in the medians, larger than any width, and it rests on measured constants. What it lacks, however, is capability (Worked example 21.7). In the nuclear edge the stiffest crystals lead: diamond leads the survivors at 3.11 keV on its measured tensor, with $P(\text{first}) = 1.000$, but in the full pool four stiff crystals share the leading tier, h-BN with the largest probability, 0.656, diamond 0.181, polyethylene 0.119 and cubic BN 0.043. No full-pool leader is resolved. h-BN and polyethylene owe their place to one stiff direction, in the hexagonal plane and along the chains, and their tensors are computed. LiF leads the polar rate in both pools, with 0.987 among the survivors and 0.918 in the full pool, where most of the rest goes to quartz through the wide interval of its band integral. The polar reach goes the other way, to soft polar lattices: GaAs among the survivors at 10.35 keV on its measured longitudinal frequency, CsI in the full pool at 2.85 keV on a computed one.

Silicon leads no row, because its gap is wider than germanium's, its lattice is stiffer than TeO_2's and softer than diamond's, and it has no Born charge. Among the survivors it is second in the electron-recoil rows, third in the nuclear edge and eighth in the heavy-mediator rate. What it has in place of a lead is the most complete record of any crystal in this book: responses at several levels of approximation (Chap. 10), a calibrated cascade (Chap. 18) and the readout analysis of Chap. 20.

Two rows have no resolved leader. The Migdal figure of merit has no leader at independent errors: the largest probability of ranking first is 0.315 among the survivors (NaF) and 0.498 in the full pool (NaCl), and both become 1.000 only when the prior X_{pole} is shared by every crystal, an assumption that Sect. 17.6.4 showed fails because the interband weight depends on the gap. What would resolve the row is the loss function $L(\boldsymbol{k}, \omega)$ of each candidate near its gap at momenta of a few keV (Appendix E.16). The light-mediator single-phonon rate does not rank crystals: far above the edge it is the same per unit mass for every target, and the material enters only through the edge that the nuclear-edge row ranks.

By Sect. 20.5, the recoil type of silicon and germanium is read from the charge, not from the phonons. For the guide this means that a leader of an electron-recoil row is read by its charge, through the cascade distribution $P(N_{\text{eh}})$ of Chap. 18, which at 5 eV puts about ten times fewer events in the two-pair bin than the step model (Sect. 18.3.3). A leader of a phonon

row is read by its heat, through a collection efficiency that only the detector's kernels fix. A superconducting absorber is read by quasiparticles, with the yield $2\Delta_{\rm BCS}/E$ below $4\Delta_{\rm BCS}$ and 0.59 above (Sect. 20.4).

The theory band of a limit (Sect. 21.4) and the ranking of targets answer different questions. A nuisance that multiplies the rates of every crystal by the same factor, such as the halo or a shared response approximation, moves every limit and no ranking. A ranking moves only through the inputs that differ between crystals, which is why the provenance widths, and not the bands of Table 21.2, set the probabilities of the guide. Where a band is material-specific, as the background tensor of $ZrTe_5$ or the gap of an aluminum film, the guide lists it in the last column.

21.7.3 Absorption and unranked channels

The absorption rows carry leaders only mass by mass. Each boson measures one response at $q = p_\phi$ and $\omega = m_\phi$ (Sect. 17.2), so two targets compete only where both absorb, and a narrow line covers its own mass alone. The rows of Table 21.5 therefore name the best reach of each mass decade of Table 17.3 and not a probability of ranking first. For the dark photon the leaders are zeros of ε_1, where the screening ceiling $L \leq 1/(2|\varepsilon_1|)$ is highest: the longitudinal phonon of GaAs, the highest polar branch of sapphire and the plasmon of silicon. All three are survivors. The largest gap in coverage is a band with no target: between 0.38 eV, where the Dirac cone of $ZrTe_5$ ends, and 1.17 eV, where cold silicon begins to absorb, no crystal of the book absorbs at leading order. For the axion-like particle with its electron coupling the magnon line of YIG leads from 10 µeV to a meV and clean aluminum from 0.36 to 10.9 meV; neither is a survivor. The aluminum reach rests on direct pair breaking by the spin current (Sect. 15.4.3). It holds for a mean free path well above the size of a pair, 1.6 to 2.4 µm, which only bulk high-purity crystals reach; in a 50 nm film the spin channel and the published mapping give comparable reaches. The scalar has one crystal target with a computed rate, the GaAs phonon line, and the phonon-assisted absorption of silicon below its direct gap is a calculation still to be done (Appendix E.14).

Magnons, superconductors and the special targets of Chap. 16 have no ranking, because no second candidate has the inputs of the same figure of merit. NiO is the magnon target with a computed rate, and its edge is set by the magnetic anisotropy that the Heisenberg model leaves out (Sect. 14.1). Aluminum is the superconducting target; its scattering rate depends more on the gap and mean free path of the absorber actually used than on the choice among loss functions (Sect. 15.2). $ZrTe_5$ is the Dirac target whose computed anisotropic tensor gives a threshold of 7.1 to 9.6 keV by axis, but its one-loop expansion parameter is about one, so its rate has no controlled theory at that half-gap (Sect. 16.1). For each of these rows the guide names what would make a ranking possible: the inputs of a second candidate, and for the leader the measurement of the sample that sets its band.

21.7.4 Internal radioactive backgrounds

Every figure of merit in the guide is a signal figure: it ranks rates, not the backgrounds against which they are seen. A background enters the ranking through a tolerance. Let b be the background rate density in a region of width ΔE. If a search is background limited, the signal must exceed a fixed number of standard deviations of the background count, and its exclusion limit scales as $(b\,\Delta E)^{1/2}/\mathcal{R}$. At the systematics floor, where the background is subtracted only to a fixed fraction, the limit scales as $b\,\Delta E/\mathcal{R}$. Crystal A then keeps its lead over B if $b_A/b_B < (\mathcal{R}_A/\mathcal{R}_B)^2$, or $b_A/b_B < \mathcal{R}_A/\mathcal{R}_B$ at the floor. TeO_2 leads $CaWO_4$ in the heavy-mediator rate by 1.22, so it tolerates 1.48 times $CaWO_4$'s background in a

background-limited search and 1.22 at the floor. LiF tolerates 3.63 times NaF's background in the polar rate, and CsI 1.42 times Bi_2Se_3's. A lead that rests on a rate factor of order one is reversed by a background ratio of order one. The edges are kinematic and do not move with b, but the sensitivity just above them does.

The background that belongs to the crystal itself, and that no shield removes, comes from radioactive nuclei inside it. A β emitter spread uniformly through a crystal much larger than the range of its electrons deposits the whole kinetic energy of the electron, and the antineutrino escapes. The spectrum of deposits is then the β spectrum, normalized to the activity $\mathcal{A}_{\mathrm{act}}$ per unit mass (Appendix E.27). For an allowed transition with endpoint Q_β it is

$$n_\beta(T) \propto p_e E_e (Q_\beta - T)^2 \, \frac{2\pi y_{\mathrm{S}}}{1 - \mathrm{e}^{-2\pi y_{\mathrm{S}}}}, \qquad y_{\mathrm{S}} = \frac{\alpha Z_{\mathrm{d}} E_e}{p_e}, \tag{21.31}$$

with T the kinetic energy of the electron, $E_e = T + m_e$ and p_e its total energy and momentum, Z_{d} the charge of the daughter nucleus, and the last factor the nonrelativistic Fermi function, which accounts for the Coulomb attraction of the daughter. A unique n-th forbidden transition carries in addition the shape factor $\sum_{k=0}^{n} p_e^{2k} q_\nu^{2(n-k)} / [(2k+1)!\,(2n-2k+1)!]$, with $q_\nu = Q_\beta - T$ the momentum of the antineutrino. As $T \to 0$ the factor p_e cancels against the $1/p_e$ of the Fermi function, so the spectrum tends to a constant: the low-energy end of a β spectrum is flat. The rate density below 1 keV is therefore $\mathcal{A}_{\mathrm{act}} f_\beta$ per keV, with f_β the fraction of the spectrum below 1 keV. With $Q_\beta = 18.59$ keV for tritium [27], Eq. (21.31) gives $f_\beta = 0.0892$. Cosmogenic tritium in silicon is produced at sea level at 124 atoms per kilogram and day. It then contributes 0.169 events per kilogram, day and keV below 1 keV after a hundred days and 11.0 at saturation (Appendix E.27). In germanium, at 82 atoms per kilogram and day [28], the two values are 0.112 and 7.31. Cosmogenic activation is reducible: growth, fabrication and storage underground, and in silicon the out-diffusion of hydrogen, lower it. The other cosmogenic nuclides decay by electron capture, and their L and M lines near 1 and 0.1 keV enter the sub-keV region through their L and M fractions (Appendix E.27). Their production rates have been measured for silicon and germanium [28, 29] and not for the other survivors.

A lead compound carries a background that no exposure history changes. Its lead contains ^{210}Pb, half-life 22.2 years, which decays to ^{210}Bi and, five days later, to ^{210}Po; in equilibrium the three activities are equal [27]. In 84 percent of its decays ^{210}Pb feeds the level of ^{210}Bi at 46.5 keV, with a β endpoint of 17.0 keV. The de-excitation of that level, a photon or conversion electrons with their X-rays, is absorbed in the bulk, so these decays deposit at least 46.5 keV and never fall below 1 keV. The other 16 percent go to the ground state, with $Q_\beta = 63.5$ keV. For a daughter as heavy as ^{210}Bi, $Z_{\mathrm{d}} = 83$, the nonrelativistic factor of Eq. (21.31) is crude, and we use instead the traditional relativistic Fermi function of a point charge, with the prefactor 4, evaluated at the nuclear radius $r_A = 1.2 A^{1/3}$ fm [30, Eq. (6)],

$$F_{\mathrm{Fermi}}(Z_{\mathrm{d}}, T) = 4\,(2 p_e r_A)^{2(\gamma_{\mathrm{d}}-1)}\, \mathrm{e}^{\pi y_{\mathrm{S}}}\, \frac{|\Gamma(\gamma_{\mathrm{d}} + \mathrm{i} y_{\mathrm{S}})|^2}{\Gamma(1 + 2\gamma_{\mathrm{d}})^2}, \qquad \gamma_{\mathrm{d}} = \sqrt{1 - Z_{\mathrm{d}}^2 \alpha^2}, \tag{21.32}$$

in place of the last factor of Eq. (21.31), with y_{S} as there. It gives $f_\beta = 0.0445$, against 0.0438 with the nonrelativistic factor. The ^{210}Bi decay to ^{210}Po, $Z_{\mathrm{d}} = 84$ and $Q_\beta = 1161$ keV, adds $f_\beta = 1.26 \times 10^{-3}$ with the allowed shape (9.7×10^{-4} with the nonrelativistic factor), and the α decay of ^{210}Po deposits megaelectronvolts. Per becquerel of ^{210}Pb per kilogram the rate density below 1 keV is

$$\begin{aligned} b_{^{210}\mathrm{Pb}} &= 86\,400\,\mathrm{s\,d^{-1}} \times (0.16 \times 0.0445 + 1.26 \times 10^{-3})\ \mathrm{keV^{-1}} \times 1\,\mathrm{Bq\,kg^{-1}} \\ &= 724 \text{ per kg, day and keV.} \end{aligned} \tag{21.33}$$

Of these, 614.7 come from ^{210}Pb and 109.2, fifteen percent, from ^{210}Bi. A budget of one event per kilogram, day and keV then needs an activity below 1.38 mBq per kilogram of compound:

1.91, 2.23 and 1.60 mBq per kilogram of lead in PbSe, PbTe and PbS. The number is an estimate. The relativistic Fermi function raises it by 5 % over the nonrelativistic 689, mostly through the ^{210}Bi part, and the first-forbidden transition of ^{210}Bi, whose shape departs from the allowed one, moves that part further by an amount not computed here. Whether lead of that purity can be obtained and grown into a crystal decides whether the floor of a lead salt is usable, and it is the deciding measurement of the first two rows of Table 21.5.

Primordial nuclei set the only exclusions of the full pool. Natural potassium carries ^{40}K at 31.7 Bq per gram; 89.3 % of its decays are unique third-forbidden β decays with $Q_\beta = 1311\,\mathrm{keV}$ [27], for which Eq. (21.31) with its shape factor gives $f_\beta = 4.13 \times 10^{-4}$. $KTaO_3$, 14.6 % potassium by mass, holds 4.6 kBq of ^{40}K per kilogram and has a β rate density of 1.5×10^5 per kilogram, day and keV below 1 keV. InSb holds 121 Bq of ^{115}In per kilogram, a fourth-forbidden non-unique emitter with $Q_\beta = 497\,\mathrm{keV}$ whose shape is not known; bracketed between the allowed and the unique fourth-forbidden shapes, its rate density lies between 4.8×10^3 and 3.9×10^4. Both exceed the budget by more than three orders of magnitude, and since only isotopic depletion removes a primordial nucleus, both leave the pool. TeO_2 is the opposite case. Its ^{123}Te is known only to have a half-life above 9.2×10^{16} years [27], so its electron captures are bounded by 693 per kilogram and day, and by the same number per keV if all of them fell in one keV below 1 keV. The bound does not decide whether TeO_2 meets the budget, and its radioactivity entry stays conditional on a measured capture rate.

Gamma lines of nuclei inside the crystal add a Compton continuum. A photon of energy E_γ that scatters once and leaves deposits the electron's recoil energy T, and at $T \to 0$ the Klein–Nishina cross-section per electron tends to

$$\left.\frac{\mathrm{d}\sigma_\mathrm{C}}{\mathrm{d}T}\right|_{T\to 0} = \frac{2\pi r_\mathrm{cl}^2 m_e}{E_\gamma^2}, \tag{21.34}$$

with r_cl the classical electron radius; at 1 keV the full form for the 1461 keV line of ^{40}K differs from it by 2×10^{-4}, and its value there is $1.20 \times 10^{-28}\,\mathrm{cm^2\,keV^{-1}}$. The rate density below 1 keV is the photon emission rate times $n_e \bar{\ell}_\mathrm{esc}\,\mathrm{d}\sigma_\mathrm{C}/\mathrm{d}T$ in the thin-target limit, with n_e the electron density and $\bar{\ell}_\mathrm{esc}$ the mean path from a point inside the crystal to its surface. For a cube of side L the path from a uniformly distributed interior point in an isotropic direction averages $0.448\,L$, not the mean chord $4V/S = 0.667\,L$ of rays crossing the cube from outside. With $n_e = 2 \times 10^{24}\,\mathrm{cm^{-3}}$ and $L = 5\,\mathrm{cm}$ a fraction 5.4×10^{-4} of the 1461 keV photons deposits in each keV below 1 keV. The continuum is flat at low energy like the β spectrum, and for an emitter with a γ line it adds to the β deposits.

Worked example 21.7: What would move CsI into the survivor pool

CsI is a conditional survivor. Two of the four requirements are documented: crystals far above a gram are grown, and neither caesium-133 nor iodine-127 is radioactive, so the intrinsic rate of a pure crystal is zero. Two are not. CsI is slightly hygroscopic, and its operation at millikelvin temperatures after thermal cycling, with a sensor read on it, is not documented. What would its entry change?

The heavy-mediator row. With its elastic constants measured at 4.2 K, the Christoffel problem of Sect. 17.6 gives $c_\mathrm{NR1} = 2.404\,\mathrm{km\,s^{-1}}$ and the figure of merit $8.956/2.404 = 3.73$ times silicon. The width is that of a measured speed, 0.02. Added to the survivors, CsI ranks first in every one of the 2×10^4 draws, and TeO_2, now second, has $P(\mathrm{first}) = 0.000$. The lead is a factor 1.56 in the medians. A colder, denser cell, at most 4.5 % denser than the room-temperature one, lowers the speed by the square root of the density ratio and raises the figure of merit to 3.81. The ranking holds pairwise: against silicon the ratio of the figures of merit gives the ratio of the rates to 10 % from 97 keV up to the zone cap of

CsI at 0.32 MeV (Fig. 17.7).
The polar reach. CsI's highest optical energy, 10.0 meV, a computed value, gives the edge $2\omega_{\mathrm{L}}/v_{\max}^2 = 2.85$ keV, 3.64 times below GaAs's. CsI would lead this row too, with $P(\text{first}) = 1.000$.
The rows it would not change. The softness that wins the heavy-mediator rate loses the nuclear edge: from $c_{\mathrm{LA,max}} = 2.46\,\mathrm{km\,s^{-1}}$ the edge is 23.1 keV, last of the eleven crystals ranked in that row. The floor, 1.77 MeV, would rank sixth, and the pair energy, an estimate of 11.3 eV outside the validated range, fourth to sixth. In the Migdal row CsI would take the largest probability of ranking first, 0.427, without resolving the row.
What decides. CsI's rank in the heavy-mediator row does not hinge on a computed input; its constants are measured, and the cold density moves the figure by 2 %. Two demonstrations decide its entry: an encapsulated crystal that survives cycling to millikelvin temperatures, and a phonon sensor that reads it, deposited on the crystal or on a carrier in thermal contact with it. Once in, its lead must survive its background. It tolerates $1.56^2 = 2.44$ times TeO_2's background rate density in a background-limited search, and the cosmogenic iodine and caesium isotopes, whose production rates have not been measured for CsI, set that rate. The deciding measurement of the row would then move from the elastic constants of TeO_2 to the background of CsI.

21.7.5 Open problems

Three missing inputs limit the guide: the loss functions of the survivors, measured elastic constants, and internal-background rates. The first is the loss function $L(\boldsymbol{k}, \omega)$ of every survivor near its gap at momenta up to a few keV. Without it the Migdal figure of merit is known only between the plasmon-pole and gap-weighted limits, the prior X_{pole} stands in for the interband weight, and the row has no resolved leader in either pool. Appendix E.16 specifies the calculation, random-phase with local fields on a ground state whose gap matches the measured one, tested on the inverse fourth moment against the f-sum bound.

The second is elastic constants. TeO_2, the survivor leader of the heavy-mediator row, rests on a computed tensor; measured constants would remove the method offset that Sect. 17.6.4 found for the computed family, 4.9 % on average for silicon, germanium and diamond, which already lowers the probability that TeO_2 beats the measured $CaWO_4$ from 0.987 to 0.955. The same holds for GaAs, sapphire, LiF, NaF and γ-$LiAlO_2$ among the survivors, and for h-BN, polyethylene and cubic BN, which share the leading tier of the full-pool nuclear edge on computed tensors alone. CsI and diamond enter with measured constants; for CsI the density of the cold cell is missing. Appendix E.17 specifies the computation for the crystals without measurements.

The third is the internal backgrounds, which the guide has only estimated. The sub-keV rates of β emitters use allowed or unique shapes, which are crude for the first-forbidden decay of ^{210}Bi and unknown for ^{115}In; the ^{210}Pb activity of lead that can be grown into a crystal is not known; the ^{123}Te capture rate is bounded only; and cosmogenic production has been measured for silicon and germanium alone. Appendix E.27 specifies the production calculation for the other survivors. Until these numbers exist, the radioactivity column of Table 21.5 holds conditions, not rates, for every crystal but silicon and germanium.

Summary

The rate of any channel is a linear functional of the velocity distribution of the halo, with a rate kernel per bin that the master formula fixes at each velocity and that is nonnegative (Sect. 21.1). For a response averaged over directions the kernel depends on the speed alone, and the rate reduces to an integral of the mean inverse speed $\eta(v_{\min})$; every coupling polynomial in the transverse velocity is fixed by η as well. In the standard halo the closed form of η at the reference Earth speed agrees with the average over a year within 1 % up to $430.3\,\mathrm{km\,s^{-1}}$, and the average exceeds it by less than 10 % up to $735.6\,\mathrm{km\,s^{-1}}$. A threshold mass $2E_{\mathrm{th}}/v_{\max}^2$ is a floor, never an edge: the onset depends on the momentum transfer, the rate vanishes at the floor, and the floor moves with the Earth speed. An exposure that includes June reaches 3.9 % lower for silicon; a limit uses that exposure-averaged rate, while the floor at the reference speed is the convention for comparing targets. A debris flow carrying 22 percent of the local density raises the silicon rate at 10 MeV by 7.6 % and lowers it at 1 MeV by 13 %, both from the kernel alone.

A limit valid for every halo is a semi-infinite linear program in a nonnegative measure (Sect. 21.2). For a direction-averaged response its solution is one stream: $\tilde{\eta}(u_0)$ is bounded by the smallest ratio of upper count to step column over the bins, exactly even with several bins, with a single-bin dual certificate. With the pair bins of the calibrated cascade, dropping the standard halo weakens a silicon limit by at least 2.12 at 10 MeV and 3.70 at 2 MeV. A directional response changes the bound by at most 15 % on the sampled speeds in the model anisotropy of silicon. A discretized program leaks through its grid, 0.096 % at $600\,\mathrm{km\,s^{-1}}$ for the grids used, and the zero-phonon/phonon-dressed split misassigns at most $0.035\,\varrho_i^{\mathrm{leak}}$ of a silicon bin, provided the pair number is stochastically ordered in the seed energy.

Near its low-mass edge a limit follows a power law (Sect. 21.3). The rate rises as $(m_\chi - m_{\mathrm{edge}})^{p_{\mathrm{on}}+k_{\mathrm{tail}}+7/2}$, where p_{on} is the onset exponent of the response and k_{tail} that of the escape tail of the halo, so the free-pair edge of silicon rises with the power 13/2 in the standard halo and 8 to 10 in simulated ones. The edge hides a degeneracy: its position depends on $E_{\mathrm{on}}/v_{\max}^2$ alone, so a gap shift of 10 meV is equivalent to $-3.3\,\mathrm{km\,s^{-1}}$ in escape speed and weakens a limit 0.005 MeV above the edge by a factor of 231. The zero-phonon edge mixes gap and halo differently and breaks the degeneracy.

A count becomes a limit through the Poisson construction with known background, and the truncation nuisances of each channel become a band on it (Sect. 21.4). Each nuisance acts on the limit through its sensitivity β_k, the band is $\mathrm{e}^{\pm\sigma_\beta}$, and profiling in the Neyman construction with the nuisances at their conditional best fit costs $\sigma_\beta^2 \mathsf{t}'/2$ in the logarithm, exactly for one bin: a factor 1.427 for the silicon limit at 3 MeV. For silicon with a heavy mediator the band is a factor of 3.4 at 1 MeV, dominated by the spread of tabulations, and 1.48 at 3 MeV, where two calculations would remove 93 % of its variance. A nuisance of known sign that raises the signal costs nothing, and removing excluded cells never strengthens a limit.

Without a background model every event counts as signal, and the maximum-gap method uses the shape of the signal to do better (Sect. 21.5). A combination of searches is one likelihood with one set of theory nuisances; splitting a shared normalization nuisance per exposure makes the combined limit about 5 % too strong in the example. A published limit can be recast exactly for a correction common to all bins, and for a bin-dependent one only when the limit is the minimum over bins with public upper counts. The background screening of a Dirac material is common for an isotropic background and recasts exactly; for $\mathrm{ZrTe_5}$ it is not, and the recast needs the event integrals.

A response library tabulates functions of the momentum transfer, its direction and the energy, per pair of operator rows, truncation label and final-state class, with no velocity axis

and no halo (Sect. 21.6). Each table declares its conventions, stores its nuisance templates as the next-order tables themselves, and carries a version and a content hash, so that a limit published with the library name can be recomputed: the band and the recast follow from the same records.

The guide to next targets lists, per channel, the leaders of the full pool of 42 crystals and of the eleven survivors, with their probabilities of ranking first, their missing demonstrations and the measurements that would change the order (Sect. 21.7). Among the survivors germanium leads the electron-recoil floor and the threshold in quanta, TeO_2 the heavy-mediator single-phonon rate, diamond the nuclear edge, LiF the polar rate and GaAs the polar reach. The full-pool leaders are conditional survivors: the lead salts in the electron-recoil rows, on estimated pair energies and with bulk ^{210}Pb, which puts 724 events per kilogram, day and keV below 1 keV for every becquerel per kilogram; CsI in the heavy-mediator rate and the polar reach, on measured elastic constants but without a demonstrated millikelvin readout; four stiff crystals sharing the nuclear edge. The Migdal row has no resolved leader and the light-mediator rate no ranking. Every probability is conditional on the provenance widths.

Table 21.6 collects the nuisances that the construction of a limit and of a ranking adds to those of the channels, with their sizes. The halo enters every limit, and near the edge it enters more strongly than any response nuisance; the statistical construction and the treatment of shared nuisances move limits by five to forty-five percent; and the rankings rest on width conventions, estimated pair energies and backgrounds that are conditions rather than rates. What remains open across the book is indexed in Chap. 22.

Table 21.6: Nuisances of limits and rankings. Type in the classes of Sect. 8.8: normalization, tilt, threshold and new support; "order" marks an item that acts on a ranking rather than on a rate. Basis of the size: "computed" (exact or next-order result computed), "model" (computed in a model), "est." (parametric estimate), "convention" (a width or a choice fixed by convention), "unsized" (no size known). Sizes are for silicon unless stated. The last column gives the section where the item is treated.

Truncation or input	Type	Size	Basis	Section
Reference Earth speed in place of the year average	normalization; tilt in m_χ	η within 1 % to 430.3 km s^{-1}, year average higher by less than 10 % to 735.6 km s^{-1}; heavy-mediator rate +0.8 % at 10 MeV, +4.3 % at 1 MeV	computed	Sect. 21.1
Earth speed at the floor	threshold	June Earth speed lowers the floor by 3.9 %; a limit uses the exposure average	computed	Sect. 21.1
Halo substructure	normalization; tilt in m_χ	debris flow with 22 percent of the density: +7.6 % at 10 MeV, −13 % at 1 MeV	model	Sect. 21.1
Standard halo against every halo	normalization	halo-independent bound weaker by at least 2.12 (10 MeV) and 3.70 (2 MeV), cascade pair bins	computed	Sect. 21.2
Directional response in the halo-independent bound	normalization	program over isotropic formula 1.00–1.15 on the sampled speeds (model anisotropy)	model	Sect. 21.2
Discretized program; final-state assignment	normalization	grid leakage 0.096 % at 600 km s^{-1}; misassigned weight at most 0.035 $\varrho_i^{\mathrm{leak}}$ of a bin for a stochastically ordered cascade	computed	Sect. 21.2
Escape tail of the halo	threshold; tilt	edge power 13/2 (standard halo), 15/2 (King), 8–10 (simulated halos); halo fraction within 12 km s^{-1} of $v_{\max}$ from 6.0×10^{-5} to 3.7×10^{-8}	model	Sect. 21.3
Gap or threshold against escape speed at the edge	threshold	10 meV of gap $\equiv -3.3$ km s^{-1}; limit ×231, ×8.8, ×2.7 at 0.005, 0.010, 0.020 MeV above the edge	computed	Sect. 21.3
Year average at the edge	threshold	year-averaged rate over the reference rate 62 at 0.010 MeV and 8.5 at 0.020 MeV above the reference edge	computed	Sect. 21.3

Truncation or input	Type	Size	Basis	Section
Theory band of the channel	normalization	silicon, heavy mediator: factor 3.4 (1 MeV), 1.48 (3 MeV), 1.43 (10 MeV); dominant rows per channel in Table 21.2	computed; model	Sect. 21.4
Statistical construction with a nuisance	normalization	profiled over nominal $e^{\sigma_\beta^2 N^{\mathrm{up}}}$ with no events, 1.427 at $\sigma_\beta = 0.393$ (Neyman construction, nuisance at its conditional best fit); a fixed calibrated threshold charges half the exponent	computed	Sect. 21.4
Excluded cells	normalization	aluminum, $y < 3$ at 1 keV: $\times 1.25$ (bulk), $\times 1.59$ (film); silicon Migdal coherent region: $\times 1.49$ at 2 MeV	computed	Sect. 21.4
Unknown background	normalization	one event: maximum gap 2.836 (event at the edge of the signal region) to 7.78 (event in its middle) against 3.890 for counting	computed	Sect. 21.5
Shared nuisance split per exposure	normalization	combined limit $\times 1.109$ shared against $\times 1.051$ split: about 5 % too strong	computed	Sect. 21.5
Recast of a bin-dependent correction	normalization	bracket between the smallest and largest bin factor, 1.154–1.271 for the halo recast at 10 MeV; a minimum over n bins covers 0.9^n	computed	Sect. 21.5
Pair-bin model of the examples	threshold	cascade bins: the one-pair bin binds at 10 MeV; the step model would overcount the two-pair bin at 5 eV about tenfold against the full-band cascade (Sect. 18.3.3) and make it bind	model	Sects. 21.2 and 21.5
Interpolation between library nodes	normalization	loss function changes 10 % between nodes near 10 eV; four neighboring nodes span 3.0 %; a model at the same truncation label differs by 0.864	computed	Sect. 21.6
Provenance widths	order	doubling the width of computed speeds lowers TeO_2's probability of ranking first from 0.968 to 0.781	convention	Sect. 21.7
Pair energies outside the validated range	order	logarithmic width 0.21 by convention; lead salts share one tier (0.482, 0.414, 0.102), PbTe 1.000 at $r_{\mathrm{mat}} = 1$; GaSb's estimate beats germanium's measured value with probability 0.83	convention	Sect. 21.7
Pair energy of germanium at millikelvin	order	3.37 eV from a fit with a fitted phonon-conversion constant, against 2.96 eV at 77 K; the order against silicon holds for either	est.	Sect. 21.7
^{210}Pb in lead compounds	order	724 per kg, day and keV below 1 keV per Bq/kg (relativistic Fermi function, 5 % above the nonrelativistic value); first-forbidden ^{210}Bi shape not computed for its 15 %; activity of available lead not known	est.	Sect. 21.7
Cosmogenic activation of the survivors	order	tritium in silicon 0.169 (100 days at sea level) to 11.0 (saturation) per kg, day and keV below 1 keV; production not measured except for silicon and germanium	computed; unsized	Sect. 21.7

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Chapter 22

Open problems

A silicon detector sees dark matter with masses between 0.3332 and 0.405 MeV, the first zero-phonon edge, only through the phonon-assisted rate, and that rate is known to a factor of ten (Sect. 13.9). The zero-phonon edge of germanium is not known at all, and no calculation predicts how much charge a silicon recoil of 100 eV releases (Sect. 19.2). Each of these statements stands in the section where its quantity arises, next to the calculation or the measurement that would settle it.

Read one at a time, they do not say where effort pays. A calculation of a few hundred core-hours would turn that silicon estimate into a number with an error bar, while no calculation can sharpen a rate that is as uncertain as the ^{210}Pb content of a crystal. Some statements of the book wait only for a proof and change no number; others are bounds that a single computation would close. At the end of the book we put these items side by side, sorted by the result each would change and by what limits it: theory, calculation or data.

22.1 Open calculations

The phonon-assisted loss function of silicon is specified down to its grids, its Wannier manifold and its tolerances, and its plane-wave stages cost some 350 to 450 core-hours (Appendix E.14). Done, it would turn the estimate of the silicon window into a number with an error bar. Most open calculations of the book are of this kind: specified in Appendix E, and each tied to a definite result. They differ in what they would move, from the reach of the running example to a correction of a few percent on a special target, and in the test that separates a correct result from a converged one. The dark-matter masses quoted below are in MeV; a mass m_χ can deposit at most $m_\chi v_{\max}^2/2$, with $v_{\max} = 2.65 \times 10^{-3}$ the largest dark-matter speed in the laboratory (Sect. 1.2), so 0.333 MeV deposits at most 1.17 eV.

22.1.1 Sum rules and controls

An exact relation holds for every Hamiltonian of a stated class. A result that violates it has an error in its sum over states, in its labeling of final states or in its normalization, whatever it approximates in the interactions. A control compares the result with a measurement that the calculation was not fitted to. The first kind tests completeness and bookkeeping, the second tests the physics, and an open calculation needs both. Exact relations have a limit of their own: an approximation that does not conserve particle number at its vertices violates them by itself, as the bare BCS bubble does (Sect. 7.2), and its error then has to be read from a conserving version of the same response.

The exact relations of the book are the f-sum rule and its shares among the sectors of Sect. 8.6 (Eqs. (4.30), (8.54), (8.56) and (8.57)) and its transfer between zero-phonon lines

and sideband (Eq. (13.36)); positivity of S_{ab} and Kramers–Kronig (Eqs. (4.20) and (4.24)); the translation Ward identity with its soft and boost limits (Eqs. (8.29), (8.31) and (8.32)); the zeros that symmetry imposes; and the conservation laws of the cascade and of detector transport (Eqs. (20.3), (20.4) and (D.242)). Three of them carry an instruction for a calculation. A table that stores S_{ab} with a negative eigenvalue, or a real part that fails the transform, fails before any comparison with data. The acoustic sum rule Eq. (5.18) is imposed on the dynamical matrix, never by setting the electron–phonon vertex to zero at small phonon momentum (Appendix E.14). And a calculation that returns the symmetry zeros at the level of its numerical noise has its phases and labels right: the interference (Fano) term between a zero-phonon state and the phonon-assisted continuum, which vanishes at the level of one-body self-energies in crystals invariant under inversion combined with time reversal (Sect. 8.6), the paths that the glide rule Eq. (13.50) removes on the valley axis of silicon, and the $\ell = 2$ component of any scalar response of a cubic crystal. The controls are measured absorption for a finite-q response in its optical limit, neutron data for computed phonons, the measured pair energy and Fano factor of silicon for a cascade, and the neutron-calibrated charge yields of silicon and germanium. Table 22.1 gives the relation and the control that each calculation must pass.

22.1.2 Priority list of calculations

Table 22.1 lists, each once, the open calculations that would change a printed result, limit or ranking of the book, in the order of the results they change; the other calculation steps appear among the gaps of Table 22.2. A calculation that serves several sections appears in the row of its largest effect, with the others named. The order starts with the silicon electron recoil, the running example and the channel on which most published limits rest.

Table 22.1: Open calculations that would change a printed result, limit or ranking of the book, ordered by the result each would change; the remaining calculation steps are in Table 22.2. Calculation: what is to be computed, with the appendix that specifies it. Result: the number or statement of the book that the calculation replaces, with its present size; masses are dark-matter masses. Check: the exact relation (sum rule, positivity, symmetry zero, conservation law) and the control against data that the result must pass. Section: where the item is stated. The guide is the ranking of crystals by figure of merit in Sect. 21.7, its survivors are the eleven crystals with a documented detector record, and $P(\text{first})$ is the probability that a crystal ranks first in a row of the guide.

Calculation	Result it changes	Check it must pass	Section
Silicon electron recoil			
Phonon-assisted loss function from the four-path amplitude, with the zero-phonon rate on common bins (Appendix E.14)	rate and reach from the onset at 0.333 MeV to the first zero-phonon edge at 0.405 MeV, now an estimate normalized to the measured cold edge; mass up to which the phonon-assisted rate leads (about 0.43 MeV, at least 0.42 MeV); band of comparable rates up to 0.51 MeV	optical limit against the measured 20 K edge; zero rate at $g = 0$; frozen phonon at X; soft acoustic limit Eq. (8.31); symmetry zeros on the valley axis; band-sum f-sum with the nonlocal part separated (Worked example 22.1); transfer rule Eq. (13.36)	Sect. 13.9
Transverse-acoustic frequency $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$ by direct DFPT at two lattice constants and with a second pseudopotential (Appendix E.14)	onset 0.3332 and 0.3374 MeV; 0.3336 and 0.3378 MeV with the branch scaled to neutron data	direct values bracket the interpolated one within 0.3 meV; transverse-acoustic branch against neutron data	Sect. 13.9
Pair channel at finite q in a supercell with local fields (Appendix E.16)	screening correction 0.3 to 2.6 relative to the optical calibration; pair-channel residual unsized	reduces at $q \to 0$ to the optical calibration, which contains the pair channel	Sect. 13.4
Finite-momentum Bethe–Salpeter response of silicon and germanium with the coupling block, bound lines added from computed masses and vertices (Appendix E.15)	attraction, $\times 1.28$ on the rate from 3.6 to 10 MeV (heavy mediator), 39 % of the variance of the one-standard-deviation factor 1.48 at 3 MeV, 93 % together with the converged table of the next row; triplet channel; spin–orbit mixing at the germanium edge	f-sum Eq. (4.30) to a percent; measured optical ε_2 at $q \to 0$; coupling block against its estimate of 10 %	Sects. 10.5 and 10.10

Calculation	Result it changes	Check it must pass	Section
One converged independent-particle table, and all-electron random-phase response with local fields to 10 keV with the current and kinetic rows as external rows (Appendices E.6 and E.16)	tabulation spread, ×3.2 at 1 MeV and ×1.34 at 3 MeV; local fields above $2q_0$, ×1.24 at 10 MeV; velocity couplings, now at least 16 times the static-electron formula	f-sum at small q, where the published tables give 1.1–1.2 against 0.93 at 0.3–0.7 q_0, and at 10 keV; positivity of S_{ab}; Kramers–Kronig	Sects. 10.1, 10.2 and 10.4
Crystal final states of the germanium 3d shell; four-component response (Appendix E.15)	3d share near 30 MeV: 3d/valence between 0.07 and 0.69	relativistic f-sum shift, −0.76 %	Sect. 10.6
Germanium window			
Quasiparticle bands of germanium with spin–orbit coupling; transverse-acoustic phonons on a 6^3 grid (Appendix E.14)	zero-phonon edge, not known: 0.32 MeV in an on-axis parabolic estimate that orients but does not bound it (Sect. 13.1), above the phonon-assisted onset 0.2119–0.2141 MeV; germanium electronic window and crossovers of Sect. 17.1	valley order, L below Γ; both gaps within 0.1 eV of the measured 0.744 and 0.89 eV; transverse mass at L within 0.010 m_e of 0.0815 m_e and mass at Γ within 0.005 m_e of 0.038 m_e; transverse-acoustic branch at X and L against neutron data	Sects. 13.9 and 17.1
Migdal rate and ranking			
Coherent zero-phonon Migdal response of silicon from 0.75 keV to $2q_{\mathrm{BZ}}$, all cross-site terms and local fields (Appendix E.18)	nearly all of the silicon Migdal rate up to 1 MeV, 33 % at 2 and 7 % at 3 MeV, now unsized	boost limit Eq. (8.32) as $q \to 0$ (the diagonal form is 14.9 times too high there); diagonal form above $2q_{\mathrm{BZ}}$; partial sum rule Eq. (8.56)	Sects. 12.4 and 17.1
Off-diagonal $\varepsilon^{-1}_{\boldsymbol{GG'}}$ in the Migdal rate, with an all-electron or PAW ion vertex and a converged $\boldsymbol{G}$ sum (Appendix E.18)	factor 1.2–1.4 from a model whose lower end is not bounded	diagonal local fields, ×1.35 (Limit Check 12.3); plain $1/\varepsilon_{00}$	Sect. 12.3
$\boldsymbol{Q}$-resolved drag mass $\delta M_{\kappa\kappa}(\boldsymbol{Q})$ of crystalline silicon; ratio of the self-consistent to the rigid Wannier amplitude (Appendix E.18)	on-site drag mass δM_{II}, valence part 18.6 m_e bounded to 12–24 m_e (the trace value 14 m_e is exact but is not the on-site element); Wannier picture bounded by 1.6, 0.54 and 0.25 at 20, 50 and 100 eV	trace rule Eq. (12.42), 14 m_e per atom at $\boldsymbol{Q} = 0$; acoustic sum rule; partial sum rule Eq. (8.56)	Sects. 12.3 and 12.5
Loss function $L(\boldsymbol{k}, \omega)$ near the gap, $\lvert\boldsymbol{k}\rvert$ up to a few keV, for every survivor (Appendix E.16)	Migdal row of the guide, without a reading (largest P(first) 0.315 among the survivors, 0.498 in the full pool)	inverse fourth moment against the partial f-sum bound from Eq. (17.16); gap of the ground state equal to the measured one	Sects. 17.5 and 21.7
Charge of a nuclear recoil and the cascade			
Close-collision excitation kernel Eq. (19.10) for Si–Si and Ge–Ge collisions, with its exchange–correlation spread (Appendix E.23)	charge yield below the binary thresholds, 352 and 399 eV; measured silicon yield 0.10 at 100 eV with no threshold, germanium (25 ± 3) % at 254 eV; probability of a recoil without charge	the measured yields and Fano factors of silicon and germanium	Sect. 19.2
Cascade without a fitted constant: impact ionization from the GW self-energy, computed phonon couplings of both carriers, excitonic final states, converged photon seed; then germanium and GaAs with spin–orbit coupling (Appendices E.13 and E.24)	pair energy without a fitted constant, 4.15 eV [3.95, 4.73] with a class systematic of at least ±0.4 eV; two-pair fraction at 5 eV, 0.097 in [0.030, 0.173]; $F_{\mathrm{eh}} = 0.114$; photon against dark-matter seed	measured $\varepsilon_{\mathrm{eh}} = 3.749$ eV and $F_{\mathrm{eh}} = 0.119$ of silicon, 2.96 eV of germanium at 77 K; normalization, moments Eq. (D.242), one pair below $(1 + \theta_{\mathrm{ii}})E_{\mathrm{g}}$	Sects. 18.3 and 18.4
Pair energies of the crystals outside the validated range, and cascade tables P_{casc} for every survivor (Appendices E.13 and E.24)	threshold-in-quanta row of the guide: GaSb ranks ahead of germanium with probability 0.83, and the lead salts are not separated, on estimates with the conventional width 0.21	measured pair energies of the validated range; one pair below $(1 + \theta_{\mathrm{ii}})E_{\mathrm{g}}$ in every table	Sects. 18.4 and 21.7
Binding, trapping and capture rates of pairs at millikelvin in silicon and germanium (Appendix E.24)	capture fraction (0.54 %, 0.32 to 0.77 %, closes the silicon Fano gap); fraction of excitons never split; release law of the gap energy	measured bound-exciton Auger lifetime; the measured silicon Fano factor	Sects. 18.5 and 20.5

Calculation	Result it changes	Check it must pass	Section
Readout and delayed signals			
Collection kernels of a stated detector from validated phonon and charge transport (Appendix E.25)	collection ratio, 0.81–1.02 in the mean-field estimate against the 1.128 that collection alone needs; stored-energy bound	Hecht relation, charge sum, bias work, Fano factor of the drifted charge; energy conservation of film phonons	Sects. 19.5 and 20.5
Emission cross-sections of shallow and over-charged traps; Luke emission into both acoustic branches (Appendices E.12 and E.25)	delayed charge and space-charge field over an exposure; spectrum of Luke phonons	measured drift velocity against field; phonon signal at fixed field	Sect. 20.1
Release paths of close Frenkel pairs: tunneling exponent and $E_{\mathrm{reorg}}/\varepsilon^{\mathrm{hop}}$ to about 5 % (Appendix E.22)	delayed release of stored energy at millikelvin; bound on the low-energy excess	stored-energy sum rule Eq. (19.44)	Sect. 19.6
Lattice, spin, superconducting and novel targets			
Anharmonic template of silicon from cubic and quartic force constants, anchored to a path-integral kinetic energy (Appendix E.19)	few-phonon weights between q_{BZ} and about 20 keV	exact moments with Hankel positivity Eq. (8.23); kinetic energy within ±0.2 %	Sects. 11.5 and 11.9
All-electron density response of displaced atoms above the zone (Appendix E.17)	multiphonon rate of a charge coupling far above q_{BZ}	translation Ward identity Eq. (8.29)	Sects. 11.3 and 11.9
Zero-temperature two-phonon absorption of silicon below 65 meV, from the derivatives of the Born charges (Appendix E.17)	electromagnetic channel of vector absorption in the one-phonon band; the isotope channel now exceeds it at least 5.0×10^3 times at 50 meV	the measured 293 K absorption, reproduced at that temperature; the bound Eq. (11.48)	Sect. 11.6
Cubic-harmonic components of the silicon loss function with local fields (Appendix E.16)	electronic part of the daily modulation	$\ell = 2$ component zero; no first moment in the components with $\ell \geq 4$, since Eq. (4.30) is isotropic	Sect. 11.7
GaAs dynamical quadrupoles (Appendix E.5); kinetic energy of the core electrons of free gallium and arsenic atoms	acoustic charge of the polar rate, at most 3 %; GaAs scalar rate, −19 % in a parametric estimate	small-$\boldsymbol{Q}$ angular dependence of the vertex; virial theorem of the free atom	Sects. 11.2 and 11.8
Magnetic response of NiO beyond linear spin waves, with its anisotropy near the lightest masses (Appendix E.20)	one-magnon weight, high by 7.8 %; rate near the edge	total-spin sum rule (Sect. 14.1)	Sect. 14.1
Two-magnon ladder Eq. (14.12) for NiO and YIG on 30^3 to 40^3 wavevectors (Appendix E.20)	two-magnon continuum, about 0.04 of the NiO spin response moved in energy; existence of a bound pair below the continuum	sum rule of the ladder (weight moved, not added); total-spin sum rule	Sect. 14.2
Current response $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}(\boldsymbol{q}, \omega)$ of tellurium or α-quartz at $\pm\boldsymbol{q}$, with spin–orbit coupling and the nonlocal current vertex (Appendix E.20)	chiral response at keV momenta, now only the scale $q\, g_{33} = 0.034$ at 1 keV in quartz	density-functional optical activity at small q; zero in silicon	Sect. 14.5
Disorder-averaged loss function of amorphous WSi at keV momenta and the distribution of its local gap (Appendix E.20)	rate of an amorphous film, unsized; it differs from the rate of the averaged ε at the order of the variance of ε	computed conductivity against the measured resistivity	Sect. 15.3
Transport function $\alpha^2 F_{\mathrm{tr}}$ of aluminum below a few meV (Appendix E.20)	phonon-assisted absorption within a few meV of $2\Delta_{\mathrm{BCS}}$; the free-electron model holds 1.25×10^4 times too much weight there	acoustic sum rule of the vertex; λ_{tr} against the resistivity value 0.39	Sect. 15.4
Superconducting phonon-assisted absorption with Eliashberg spectral functions, with and without the strong-coupling factor (Appendix E.20)	rate near the edge; the normal-metal factor $(1 + \lambda_{\mathrm{tr}})^{-2} = 0.52$ in aluminum	normal-metal limit at $\omega \ll \omega_{\mathrm{D}}$	Sect. 15.4
Electron–hole ladder of $ZrTe_5$ near its edge (Appendix E.20)	rate below 1.33 times the edge (Sommerfeld factor above 1.5)	f-sum of the loss function Eq. (4.46)	Sect. 16.1

Calculation	Result it changes	Check it must pass	Section
Ejected-electron final states; periodic finite-q Bethe–Salpeter response of a molecular crystal; excitons of a semiconducting tube; valley-orbit coefficients of a donor; layered stack beyond its effective medium; ohmic coupling α_{ohm} of a tunneling defect (Appendices E.10, E.11, E.21 and E.22)	rates of these targets beyond their model forms	Thomas–Reiche–Kuhn sum per donor; partial f-sum per molecule to 2 %; closed-form bubble of an undoped Dirac sheet; Born limit of α_{ohm} and the measured damping of aluminum-covered resonators, $\alpha_{\mathrm{ohm}} = 0.28$ and 0.44	Sect. 16.9
Published limits			
Transverse-current response of bound shells with Dirac–Hartree–Fock states along $\omega = m_\chi - q$ (Appendix E.26)	fermionic absorption on bound electrons, published rate low by at least $1/r$ (1.7 for xenon 1s at 56.5 keV)	continuity Eq. (4.29) of the shell current	Sect. 10.9
Cosmic-ray-boosted limit for a light mediator below about 3 keV, with a stated low-energy treatment of the flux (Appendix E.16)	density-row ratio between 0.05 and 0.93 at 1 keV, depending on that treatment	the heavy-mediator limit, which the exact density row leaves unchanged	Sect. 10.9

No single calculation changes as much as the phonon-assisted one at the head of the table. Up to the first zero-phonon edge it is the whole electron-recoil rate of silicon, and on common bins with the zero-phonon rate it fixes where the zero-phonon line takes over, which Sect. 13.9 places only by estimate. It cannot settle the window alone, however. The rate estimate rests on measured absorption, and the two cold datasets of Sect. 13.9.4 differ by a factor of twenty at 1.20 eV. In Table 21.2 the choice of data moves the rate by a factor of nine at 0.36 MeV, more than the factor three of the finite-q form factor that the calculation replaces. The calculation and a millikelvin measurement of the absorption edge within 25 meV of threshold test each other through the optical limit, and the window needs both.

The direct calculation of $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$ in the second row is the cheapest entry of the table, a few tens of core-hours; it verifies the one computed number in the onset, which moves by 8.5×10^{-5} MeV for a shift of 0.3 meV in that frequency (Appendix E.14). The sum-rule check of the phonon-assisted calculation is the subtle one, since what the calculation adds is a small part of a sum rule that a complete calculation already satisfies, and Worked example 22.1 works it out.

Worked example 22.1: The f-sum check of a phonon-assisted calculation of silicon

Take $|\boldsymbol{q}| = 1\,\mathrm{keV} = 0.507\,\text{Å}^{-1}$ along [100], where the off-shell window of Sect. 13.6 is open and $q^2/2m_e = 0.9785\,\mathrm{eV}$. The phonon-assisted amplitude of Sect. 13.2 sums over intermediate bands m products of a density vertex $\rho_{mv}(\boldsymbol{k}, \boldsymbol{q})$ and an electron–phonon vertex, each pair over an energy denominator. The density vertices of one valence state obey the band-sum form of the f-sum rule, Eq. (E.50): summed over all m, $(\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}})|\rho_{mv}|^2$ gives $\boldsymbol{q} \cdot \boldsymbol{v}_v^{\mathrm{g}}(\boldsymbol{k}) + q^2/2m_e + \delta_{\mathrm{NL}}$. With a norm-conserving PBE pseudopotential the sum at Γ reaches 0.81 of $q^2/2m_e$ with 16 bands and 0.89 with 100, for $|\boldsymbol{q}|$ from 0.1 to $0.5\,\text{Å}^{-1}$ (Sect. 13.9); our test momentum sits at the upper end of that range. The check has three parts: the remainder, the nonlocal term and the transfer rule.

The remainder. The sum over the bands above N is one expectation value, Eq. (E.88), so the truncated sum plus the remainder must equal the right side of Eq. (E.50) to the precision of the eigenvectors. A local model shows what passing looks like. Take the

empirical pseudopotential of Sect. 3.3, which has no nonlocal term ($\delta_{\mathrm{NL}} = 0$), with 531 plane waves, and sum over the four valence bands at Γ. The lowest 8, 16, 30, 60 and 100 bands then give 0.1514, 0.9265, 0.9939, 0.9992 and 0.9996 of $4q^2/2m_e = 3.9139\,\mathrm{eV}$, and the truncated sum plus the remainder equals it to 9×10^{-14}. At $\boldsymbol{k} = (0.10, 0.20, 0.30)\,2\pi/a$ the group-velocity terms lower the target to 3.1135 eV, and the identity closes to 5×10^{-14}. A code that misses the identity by more than its eigenvector precision has an error in its vertices or its Hamiltonian, whatever its physics. Passing it says nothing about convergence in the basis, since the identity is exact in any finite basis.

The nonlocal term. Once the remainder is added, what is left of the shortfall against $q^2/2m_e$ is δ_{NL}, a ground-state expectation value from the same projectors as the Hamiltonian. In the local model 100 bands leave 4×10^{-4} of the sum, so a shortfall of 0.11 at the same N is large on the scale that a smooth local potential sets. The closure says how it divides between bands left out and the nonlocal term. An all-electron calculation, or a PAW calculation whose density vertices are reconstructed inside the augmentation spheres, satisfies the local f-sum of the true electrons and checks the split. The remainder also bounds the part of the amplitude carried by the bands above N. With $\Delta E_m = \varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}} > 0$ there and b_m the electron–phonon vertex divided by its energy denominator, the Cauchy–Schwarz inequality gives

$$\Big| \sum_{m>N} \rho^{*}_{mv}\, b_m \Big|^2 \le \Big[\sum_{m>N} \Delta E_m |\rho_{mv}|^2 \Big] \Big[\sum_{m>N} \frac{|b_m|^2}{\Delta E_m} \Big],$$

whose first factor is the remainder. A small remainder therefore guarantees a small truncation error once the second factor is bounded. The converse does not hold: a large remainder leaves the truncation error unbounded, but small vertices or cancellations can still keep it small.

The transfer rule. The total f-sum cannot see the sideband. The first moment of S^{PA} is of relative order $\theta_{\mathrm{ep}} = 0.010$ to 0.016 of the f-sum (Sects. 8.5 and 13.3), while a converged electronic table holds the f-sum to 2 % (Appendix E.16), 1.25 to 2 times θ_{ep}. A calculation can pass the total test at that tolerance with its sideband missing or doubled. What sees the sideband is Eq. (13.36). The first moment of S^{PA}, integrated over all ω at the test momentum and not only over the window, must equal the first moment that the zero-phonon response loses through its self-energies and vertex corrections, computed from the same vertices. The rule holds up to terms of order g^4 and the share of the electrons that follow the nuclei, $Z_{\mathrm{val}} m_e / M_{\mathrm{cell}} = 7.8 \times 10^{-5}$ of the f-sum with $Z_{\mathrm{val}} = 8$ valence electrons per cell. If the order-g^4 term is estimated as θ_{ep}^2 of the f-sum, the two neglected pieces add to $\theta_{\mathrm{ep}}^2 + Z_{\mathrm{val}} m_e / M_{\mathrm{cell}}$, and relative to the sideband's own weight θ_{ep} the two sides must agree to about 2 %: 1.8 % at $\theta_{\mathrm{ep}} = 0.010$ and 2.1 % at 0.016. A calculation that passes all three parts has a complete band sum, a known nonlocal term and a sideband whose weight is consistent with the renormalization of the lines it accompanies.

Above 1 MeV the zero-phonon rate of silicon is computed, and two calculations would narrow its band. Its spread is dominated by the attraction between electron and hole and by the disagreement between published independent-particle tables (Sect. 10.10). A finite-momentum Bethe–Salpeter calculation removes the first, and one converged table that passes the f-sum at small q removes the second. The all-electron calculation with local fields to 10 keV serves three sections at once: the local fields at large q, the screening of core transitions and the current rows that a velocity coupling needs. Until it exists, a limit on a velocity coupling can use only the bound Eq. (10.15).

The germanium window needs bands before it needs a rate. The phonon-assisted onset follows from the measured transverse-acoustic branch, but the zero-phonon edge depends on the order, the energies and the curvatures of the conduction valleys at L and Γ, which PBE gets wrong. Hybrid functionals are not a safe default either: at the standard exchange fraction a hybrid-functional calculation puts the direct gap of germanium at 0.775 eV, below the indirect gap at 0.823 eV, and no exchange fraction reproduces both measured gaps [1, Fig. 1(a)]; adding spin–orbit coupling does not restore the order (Appendix E.14). A shift of the 4s-like orbitals fitted to both gaps puts them in the right order, but the transverse mass at L then comes out about 24 % heavy and the mass at Γ 39 % heavy. The check in Table 22.1 therefore includes the band masses beside the two gaps, since the edge enters the window through the pair energy at each momentum transfer, and the germanium window stays an estimate until bands pass both. Quasiparticle bands in the *GW* approximation are the next route, under the same check.

The Migdal entries change the light-mass Migdal rate of silicon and the Migdal row of the guide. Up to 1 MeV nearly all of the silicon rate comes from momentum transfers below $2q_{\mathrm{BZ}}$ where neither the boost limit nor the diagonal form is quantitative (Table 12.3). A calculation with the cross-site terms of the lattice factor and local fields would make that part of the rate a number. The off-diagonal elements of ε^{-1} change the rate at all masses (Sect. 12.3.2). The drag mass enters the energy bookkeeping of Eq. (8.56) at a few parts in 10^4; its on-site element is what a kick measures, and the trace rule fixes only its value at zero phonon momentum (Sect. 12.5). With the self-consistent Wannier amplitude the same calculation tests the Wannier picture of the boost, which is exact for core shells, on the valence electrons. The loss function of each survivor near its gap decides whether the Migdal figure of merit ranks crystals at all; between the plasmon-pole and the gap-weighted limits it spans a factor 5.7×10^5 in silicon (Sect. 17.5). A tagged-recoil measurement in silicon would test the whole chain at once (Sect. 12.6.5).

One kernel decides the charge of a low-energy nuclear recoil: the close-collision kernel Eq. (19.10). Since the Born parameter of the collision, $Z_\kappa \alpha / v_{\mathrm{F}}$ with v_{F} the Fermi velocity of the target electrons, is 15 in silicon and 35 in germanium, no linear response describes the close collisions that promote electrons below the binary threshold (Sect. 19.2). The exchange–correlation spread of the non-perturbative methods is then part of the result rather than an error to be converged away. The full-band cascade of Sect. 18.3.3 already has no fitted constant, but its envelope over screening, phonon couplings and exchange factor spans a factor of almost six in the two-pair fraction at 5 eV. Computed impact-ionization rates and phonon couplings remove most of that envelope. The test is the pair energy, which the cascade without a fitted constant misses and which is set by the impact-ionization rate below about 4.5 eV (Sect. 18.4); the Fano factor cannot separate the models, since the full-band and the calibrated cascade give the same value. For crystals outside the validated range of Sect. 18.4.6 the same calculation replaces a conventional width by a computed pair energy, and with it the order of the threshold-in-quanta row of the guide.

The readout entries change the interpretation of measured phonon energies rather than any rate. The collection kernels of a stated detector decide whether collection alone can explain the silicon phonon datum (Sects. 19.5 and 20.5); the two-bias measurement of Sect. 19.5 decides the same question from data.

For the running example the lattice calculations hardly matter: for a nucleon coupling the silicon lattice rate is already known to a quarter of a percent above 100 meV at 30 MeV (Sect. 11.9). They matter for charge couplings, quantum crystals and polar targets, and the spin, superconducting and novel entries each move a special target.

One recast in the last group is settled for the heavy mediator. In the published treatment of cosmic-ray-boosted dark matter in silicon and germanium [2], the density row is evaluated at

the four-momentum transfer of a relativistic particle. With $\mathcal{Q}$ the magnitude of the momentum transfer of that row (Sect. 10.9), the factor $(1-\omega^2/\mathcal{Q}^2)^2$ of the exact density row changes the heavy-mediator rate by 10^{-4}, and the transverse row adds at most 10^{-4}, so that limit is unchanged. Since the boosted flux is itself proportional to $\bar{\sigma}_e$, a change R of the detection rate moves such a limit by $R^{-1/2}$. For a light mediator the shift of $\bar{\sigma}_e$ is below one percent for $m_\chi \geq 10\,\mathrm{keV}$. Below about 3 keV the density row has a pole at $\mathcal{Q}=\omega$ whose weight depends on the low-energy treatment of the boosted flux, which the published calculation does not state; the last row of Table 22.1 is that open part.

22.1.3 Theory, calculation and data limits

A better calculation sharpens a channel only if the channel is limited by calculation. Sect. 16.9.2 sorted the novel targets three ways, and the same sorting applies to every channel of the book. A channel is limited by theory when its response has no controlled approximation at the values its inputs take, so that better inputs or larger grids would not sharpen the rate. It is limited by calculation when the next term has a known form and is computable, but no computation of it exists for the material. It is limited by data when the response formula is in hand and the rate is as uncertain as a property of the sample that a measurement fixes. Fig. 22.1 assigns each channel the class of the item that dominates its band at its reference point, read from the open problems and nuisance tables of its section and from Table 21.2, and marks the classes that contribute.

Of the 22 channels in the figure, 12 are limited first by calculation, 9 by data and 2 by theory; the phonon-assisted window of silicon counts under both calculation and data. The first theory-limited row is that of strongly anharmonic and quantum crystals at detector temperatures, whose response is bounded only by the band of its exact moments (Sect. 11.5). The second joins Dirac crystals at strong coupling and silicon doped near the Mott density, where no small parameter remains (Sects. 16.1 and 16.5). In both, the next step is a method, not a run. Theory also contributes to the recoil charge, where the exchange–correlation spread of the close-collision kernel has a known sign and no known size, and to glass targets, whose two-level systems interact through their strain fields, outside the standard tunneling model of independent systems (Sect. 16.7).

The calculation-limited rows hold the leading electron-recoil and Migdal results of the running example: the silicon rate above 1 MeV, the phonon-assisted window, the germanium edge, the Migdal rate below 3 MeV and the Migdal ranking. The silicon lattice rate belongs here too, since its small band is the difference between computed and measured phonon frequencies. In these channels the framework is complete and the numbers are missing. The data-limited rows are of two kinds. Some are rankings in which computed inputs stand in for measured ones: the elastic tensor of TeO_2 for the heavy-mediator row and the pair energies of crystals outside the validated range for the threshold in quanta (Sect. 21.7). Others are properties of a given sample or detector. The polar rate of GaAs rests on the measured ratio of Born charge to dielectric constant, and the background tensor of $ZrTe_5$ moves its reach by a factor 2.8 (Worked example 16.9). The gap and mean free path of an aluminum film, the parameters of the two-level systems of one film, the collection of a stated detector and the ^{210}Pb and cosmogenic activities of a crystal are of the same kind. For these no calculation in this book substitutes for the measurement.

Four calculations each replace a factor of order ten or an unsized rate in a leading result: the phonon-assisted loss function of silicon with the zero-phonon rate on common bins, the coherent Migdal response of silicon below $2q_{\mathrm{BZ}}$, the loss functions of the survivors near their gaps, and the close-collision kernel. The second holds nearly all of the light-mass Migdal rate, the third gives the Migdal row of the guide a reading, and the fourth decides the charge

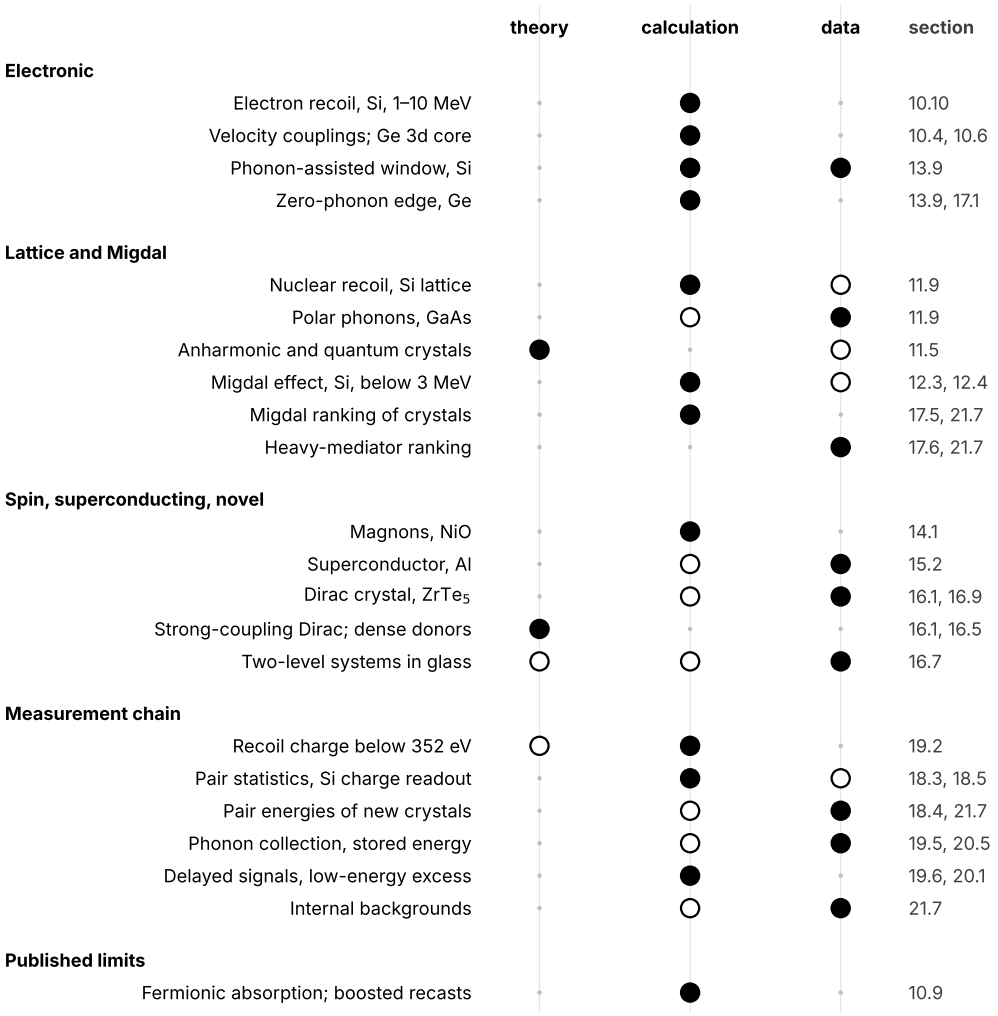


Figure 22.1: Channels of the book by the factor that limits them. Theory: no controlled approximation at the values the inputs take. Calculation: the next term has a known form and is computable, but is not computed for the material. Data: the response is in hand and the rate is as uncertain as a measured property of the sample. Filled circles mark the class of the item that dominates the uncertainty of the channel at its reference point, open circles a class that contributes. The right column gives the section that states the items. The classes are read from the open problems and nuisance tables of each chapter and from Table 21.2.

of every silicon recoil below 352 eV. Germanium bands that pass the valley-order and mass checks would open the zero-phonon window of germanium, the survivor with the lowest electron-recoil threshold mass. On the data side, five measurements move a leading result: the cold absorption edge of silicon near its threshold (Sect. 13.9.8), the pair energy of germanium at millikelvin with its phonon collection calibrated separately, the elastic constants of TeO_2, the ^{210}Pb activity of lead that can be grown into a crystal (Table 21.5), and the two-bias measurement that separates collection from stored energy (Sect. 19.5). A tagged-recoil Migdal measurement in silicon would test the chain from the kick to the count.

22.2 Exact remaining gaps

The decomposition theorem of Sect. 8.6 holds for a finite insulator. For the infinite crystal one estimate is missing, a superadiabatic generator built from exponentially localized terms, and every sector of every rate in this book is added on the strength of the finite-crystal statement with its stated error. At the other end is the silicon rate of a particle with an anapole moment. At 10 MeV it exceeds the static-electron formula at least 16 times, and the local-field screening of the transverse current, which would turn that bound into a number, is open (Sect. 10.4). The first statement needs a proof and changes no number; the second needs a calculation and changes a limit. Both have the same form, a strongest result and one step beyond it. Table 22.2 collects all 74 such items by chapter, with the result each one limits.

22.2.1 Exact remaining gaps and nuisances

Sect. 13.9.8 states the open germanium window in two parts. The phonon-assisted onset lies at 0.212 to 0.214 MeV from the measured transverse acoustic branch alone, and the zero-phonon edge needs quasiparticle bands with spin–orbit coupling. The first part can be used now, and the second can be carried out. Every open item of the book is stated in this form.

Exact remaining gap

An *exact remaining gap* is a pair. Its first part is the strongest statement that the book derives about a quantity or a claim, with the conditions under which it holds: a theorem under stated hypotheses, a bound, an estimate with its factor, or a computed number that awaits an independent check. Its second part is the single missing step, a proof, a calculation or a measurement, that would turn the statement into the full result. Each gap is stated once, in the section where the quantity arises, which gives both parts and, for a calculation, points to its specification in Appendix E.

A gap differs from a nuisance. A nuisance parameter of Sect. 8.8 carries a truncation whose next order is computed or estimated from a small parameter, and it enters a likelihood with a width; the attraction between electron and hole, which raises the silicon electron-recoil rate by $\delta_{\mathcal{X}} \approx +0.28$ over the 3 MeV window, is one (Table 10.2). Sect. 8.8 also names a third case, a truncation with neither a computed next order nor a small parameter, which receives no width. Every such truncation is an exact gap, and so is every claim the book can derive only under a hypothesis it cannot yet remove. The nuisance tables at the end of each chapter carry the first kind of item, and Table 22.2 the second. One item can have both a sized part and an open part. The attraction is such an item: its model size is a nuisance, and the finite-momentum Bethe–Salpeter calculation that would replace the model is a gap. The screening of the phonon-assisted sideband splits the same way.

What a gap limits decides how we can use it, and the table sorts the 74 into five classes. A gap that limits only a proof leaves every number of the book standing. Its missing step

extends a derived statement beyond its conditions, from a finite crystal to an infinite one or from perturbation theory to all orders. Three gaps are of this kind, those of the decomposition theorem. The other 71 involve a number. For 10 of them the book gives a bound or a two-sided bracket, such as the anapole bound above. For 17 it gives an estimate, a model value or a factor that is not a bound, such as the phonon-assisted rate of silicon below the zero-phonon edge (Sect. 13.9). For 39 the number is unsized: the book can say what the result depends on and how to compute or measure it, but assigns it no width. The last 5 are tests. The book computes the number, and the missing step is an independent check of it, a second set of force constants for the phonon-assisted onset or a cold measurement of the silicon absorption edge.

The missing steps are of three kinds. In 56 rows the step is a calculation specified in the section or in Appendix E, in 11 a measurement, and in 7 a proof or a method that does not yet exist, such as a controlled treatment of Dirac targets at $\tilde{\alpha} \gtrsim 1$ (Sect. 16.1) or a real-time method for quantum nuclei at low temperature (Sect. 11.5). Where a section names more than one step, the row gives the decisive one. Sect. 22.1 orders the calculations among these steps by the results they would change and states the check each must pass.

22.2.2 Gaps by chapter

Table 22.2: Exact remaining gaps of the book by chapter, each indexed to the one section that states it. Strongest statement: what that section derives, with its conditions. Missing step: the decisive proof, method, calculation or measurement that would complete it, with the appendix that specifies a calculation. Limits: *proof*, no number of the book changes. *Bounded*, a number is missing and the section gives a bound or a two-sided bracket. *Estimate*, the section gives an estimate that is not a bound. *Unsized*, the section assigns no size. *Test*, the section computes the number and the missing step checks it. The statements are abbreviated; the section named in the first column gives each in full. Survivors: the eleven crystals with a documented detector record (Sect. 21.7). The calculations that would change a printed result, limit or ranking are ordered in Table 22.1.

Gap; section	Strongest statement	Missing step	Limits
Chap. 8. The response framework			
Sector labels outside the off-shell window; Sect. 8.6	Split into zero-phonon and phonon-dressed sectors, and convention independence of the totals, derived in perturbation theory in g with distributional limits; Dyson convention Eq. (8.59)	A limiting-absorption principle for the coupled electron–phonon Hamiltonian at fixed total crystal momentum, uniform in the coupling and the number of cells; it would also sharpen Eq. (8.63) from $(\Gamma\Delta E_{\mathrm{c}})^{1/2}$ to Γ	proof
Uniformity in the number of nuclei; Sect. 8.6	Superadiabatic leak Eq. (8.47) of order $\kappa_{\mathrm{BO}}^{2(n+1)}\alpha^2 m_e$, proved for a finite crystal	A superadiabatic generator built from exponentially localized terms, so that the leak per unit volume stays finite	proof
Dressing of gapless targets; Sect. 8.6	Bin bound Eq. (8.63) for any orthogonal sectors given a dressed vacuum; fails for $U_{\mathrm{qp}} = 1$; connected part of $\mathfrak{h}_{\mathrm{off}}$ estimated at 0.2 eV in aluminum, with no commutator bound uniform in the number of cells	Existence of U_{qp} beyond perturbation theory, with bounds	proof
Chap. 9. Channel atlas			
Symmetry-allowed cells of the channel atlas; Sect. 9.1	Eight cells of Table 9.1 allowed at leading order and computed in no section: a spin coupling to core electrons, Cooper pairs, Migdal transitions, Dirac pairs, phonon-assisted pairs, dopant transitions and Frenkel excitons, and the electron density acting on a freely recoiling atom through its following cloud, whose rigid-atom density $f_\kappa(q)$ is used in the multiphonon vertex (Sect. 11.3)	The eight rates, each from the response of the home section of its row with the vertex of its column	unsized
Chap. 10. Electronic excitations			
Local fields of the transverse current; Sect. 10.4	Static-electron formula low by at least 16 at 10 MeV for independent particles, at least 29 with the density screened (Eq. (10.15)); current screening 4–15 % in a model, 10–30 % estimated, both lowering	Mixed-block screening of Eq. (D.180), which fixes the in-medium ratio r_{med} (Appendix E.16)	bounded
Finite-q Bethe–Salpeter response with coupling block and bound lines; Sect. 10.5	Attraction $\approx +0.28$ over the 3 MeV window from a model, coupling block about 10 % (Eq. (10.22)), resonant-only evaluation off by two (Eq. (10.21)), lines from Eq. (10.18)	Coupled calculation for Si and Ge with the lines added from computed masses and vertices, f-sum to a percent (Appendix E.15)	estimate
Triplet beyond independent particles; Sect. 10.5.5	Same lines and selection rules as the singlet, no local fields, no screening; 2.2 times the independent-particle spectrum below 3 eV in a model	Ladder with the spin-channel exchange–correlation kernel; control: a quarter of the independent density response without interaction (Appendix E.16)	estimate
Self-trapped excitons of NaI; Sect. 10.5.6	Poisson replica series with Huang–Rhys factor S_{HR} for linear coupling, a harmonic lattice and a fixed vertex (Eq. (E.91))	S_{HR}, $\bar{\omega}$ and $\rho_{\lambda 0}(\boldsymbol{q})$ at keV momenta (Appendix E.15)	unsized
Four-component finite-q response; Sect. 10.6.5	At most 2 % for Ge $3d$, central value near 10^{-3}; 10^{-6} beyond two components	A calculation with no-pair projection, the $3d$ binding as a nuisance, checked on the relativistic f-sum shift (Appendix E.15)	bounded
Spin–orbit mixing at the Ge edge; Sect. 10.6.5	Computed and small in Si; up to 10 % in amplitude near the Ge L minimum, estimated	Ge bands with spin–orbit coupling in the measured valley order (Appendix E.15)	estimate

Gap; section	Strongest statement	Missing step	Limits
Interference of contact and second-order amplitudes; Sect. 10.8.4	Linear in the quadratic response, a three-point function with no reduction to two-point responses; known for the electron gas (Barkas term)	Quadratic response of the crystal, in the local-density mapping of Appendix E.26 or at second order in time-dependent perturbation theory	unsized
Bound shells in dark Compton scattering; Sect. 10.8.4	Second-order terms of Eq. (10.45) vanish away from the shells; nonzero for Si K at 45 to about 100 keV, Ge K from 118 keV, L shells below about 20 keV	Compton profiles of the bound shells at these energies, from the large-q density response (Appendix E.26); the refractive index enters as an input	unsized
Fermionic absorption on bound shells; Sect. 10.9.5	Free-electron vertex low by at least $1/r$, 1.7 for xenon $1s$ at 56.5 keV; the transverse current only raises it; sign unknown only at order $(Z\alpha)^2 \approx 0.155$	Transverse current response with Dirac–Hartree–Fock states, and the published limits redone (Appendix E.26)	bounded
Retarded density row of boosted dark matter; Sect. 10.9.5	Heavy mediator: limit unchanged; massless mediator: shift below 1 % for $m_\chi \geq 10$ keV; density-row ratio 0.05 to 0.93 at 1 keV, undetermined below about 3 keV (Sect. 21.5.4)	The published calculation redone with the exact density row and the transverse row, Cherenkov pole included (Appendix E.16)	bounded
Chap. 11. Lattice excitations			
Valence deformation above the zone; Sect. 11.3	Translation sum rule on the cell sum; rigid-atom vertex $Z_\kappa - f_\kappa(q)$; core share computed (Appendix E.17); nucleon couplings unaffected	All-electron density response of displaced atoms in a large cell, 3 to 20 keV, tested on the translation sum rule (Appendix E.17)	unsized
Strongly anharmonic targets; Sect. 11.5	Exact moments by path integrals bound the cumulative weight (Markov–Krein band); Kubo-transformed dynamics only for $T \gtrsim \omega/10$; neutron Compton scattering would measure the momentum distribution	A real-time method for quantum nuclei, exact at low temperature (Appendix E.19)	bounded
Cold in-band dielectric function of Si; Sect. 11.6	$\alpha(\omega; T \to 0) \leq \alpha(\omega; 293\,\mathrm{K})$ (Eq. (11.48)): isotope channel at least 5.0×10^3 times the electromagnetic one at 50 meV; computable from the derivatives of the Born charges (Appendix E.17)	Two-phonon absorption below 65 meV measured at low temperature	bounded
Cubic harmonics of the Si loss function; Sect. 11.7	Pseudopotential-model components to $2.8\,q_0$; local fields change the direction average by 1.3 to 1.5 at 10 MeV	First-principles loss function with local fields on Lebedev grids of increasing order (Appendix E.16)	estimate
Chap. 12. The Migdal effect			
Non-rigid Wannier ratio; Sect. 12.3	Wannier following exact for core shells and flat bands, bounded only in absolute terms for the valence band of Si	Migdal probability with the self-consistent vertex against rigid following, checked on the acoustic and drag-mass sum rules (Appendix E.18)	bounded
Off-diagonal dielectric elements; Sect. 12.3.2	Scalar dielectric function high by about 35 %; $\boldsymbol{G} \neq \boldsymbol{G}'$ elements raise the probability by 1.38 in a model, about 1.2 extrapolated, the lower end not a bound	Converged calculation with the full dielectric matrix (Appendix E.18)	estimate
Contact density at finite q; Sect. 12.5	Known for spherical atoms; valence anisotropy 1.3×10^{-3} in Si; relativistic normalization $\mathcal{R}_\mathrm{rel}$ (Eq. (12.47)) with a quadrupolar part changed by a few percent	All-electron Dirac calculation of the crystal with a finite nucleus (Appendix E.17)	estimate
Site-resolved dynamical Born charges; Sect. 12.6.5	Sum fixed by Eq. (12.52); GaAs bremsstrahlung carries the band of Fig. 12.14	Frequency-dependent charges per site (Appendix E.17)	bounded
Scalar absorption with couplings not proportional to mass; Sect. 12.6.5	Needs $S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}$ at vanishing reduced momentum	That response (Appendix E.18)	unsized
The Migdal chain as a whole; Sect. 12.6.5	Chain from the kick to the electronic spectrum in place (Sects. 12.1 and 12.5); no crystal Migdal signal observed	Tagged-recoil measurement in Si: recoil energy and direction fixed by a scattered neutron, electrons counted	test

Gap; section	Strongest statement	Missing step	Limits
Chap. 13. Phonon-assisted electron recoil			
Exciton–phonon replicas; Sect. 13.8	Replica edges at $E^{\mathrm{PA}}_{\mathrm{min}}(\boldsymbol{q}) - E^{b}_{x}/n^2$, square-root rise at fixed $\boldsymbol{q}$; $1s$ transverse acoustic replica at 1.1728 eV; completeness reduced to the two-pair sector	Bethe–Salpeter solution on Wannier-interpolated states in the gauge of the vertices, about 10^3 points of $\boldsymbol{k}$ per pocket (Appendix E.15)	estimate
Onset on one set of force constants; Sect. 13.9.8	0.3332 MeV (replica), 0.3374 MeV (free pair), set by $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$ and the gap	$\omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$ at two lattice constants and with a second pseudopotential (Appendix E.14)	test
Zero-phonon to phonon-assisted ratio; Sect. 13.9.8	Phonon-assisted lead to ≈ 0.43 MeV (central), at least ≈ 0.42 MeV; two channels 0.43–0.51 MeV; zero-phonon lead above; both rates estimated	Ratio of the two rates on common bins, with the full phonon-assisted amplitude and the complete zero-phonon edge (mirror-plane cones, second heavy hole, X face) (Appendix E.14)	estimate
Rate and reach at first-principles level; Sect. 13.9.8	Estimates normalized to the measured cold edge (Table 13.1)	The same calculation with its controls: optical limit, symmetry zeros, soft acoustic limit, completeness (Appendix E.14)	estimate
Plane-wave f-sum check; Sect. 13.9.8	0.81 of $q^2/2m_e$ with 16 bands, 0.89 with 100, at 0.1–0.5 Å^{-1}	Separation of the omitted bands from the nonlocal term (Appendix E.14)	test
Germanium window and phonons; Sect. 13.9.8	Phonon-assisted onset 0.212–0.214 MeV from the measured transverse acoustic branch; zero-phonon edge not known, so the germanium maps of Sect. 17.1.4 have open boundaries	Quasiparticle bands with spin–orbit coupling in the measured valley order (Appendix E.14)	unsized
Screening beyond the optical calibration; Sect. 13.9.8	External factor and local-field wings 0.3–2.6 relative to the calibration, estimated; pair-channel dependence on q unsized	Supercell calculation with local fields in the supercell basis (Appendix E.16)	estimate
GaAs control of the window; Sect. 13.9.8	Phonon-assisted band 0.43 to 0.52–0.56 MeV ($E_{\mathrm{L}}(0) = 1.815$ eV) in the parabolic description	Band calculation and rate with the polar vertex (Appendix E.14)	estimate
Cold absorption edge of Si; Sect. 13.9.8	Untransferred 77 K data off by more than two near the edge; two cold datasets differ by twenty at 1.20 eV	Absorption within 25 meV of threshold at millikelvin temperatures	test
Chap. 14. Magnetic excitations			
Two-magnon interactions and bound pairs; Sect. 14.2	Ladder Eq. (14.12) with the vertex from the spin Hamiltonian; interaction $1/2\lvert\boldsymbol{S}\rvert = 0.5$ in NiO, not small; existence of a bound pair below the continuum open	Solution of the ladder for NiO on 30^3–40^3 wavevectors (Appendix E.20)	unsized
Longitudinal spin response from neutrons; Sect. 14.4	Transverse block fixes the transverse couplings; for $\mathcal{O}_4$ it is a lower bound; for $\mathcal{O}_6$, $\mathcal{O}_{10}$ one $\boldsymbol{q}$ gives nothing; without the isotropy hypothesis of Sect. 14.4.3, broken only by spin–orbit and dipolar terms, only a model of the spin density	A neutron test of that hypothesis, turning the moments at fixed $\boldsymbol{q}$	bounded
Chirality after decay; Sect. 14.5	GaAs lowest-band $\langle\lvert\boldsymbol{L}^{\mathrm{ph}}\rvert\rangle = 0.124$; below 0.8 MeV that band takes under 1 % of the rate	Survival of chirality through three-phonon decay, $(\mathbb{1} - \mathsf{M})^{-1}\boldsymbol{b}$ (Appendix E.19)	unsized
Large phonon magnetic moments; Sect. 14.5	Born-charge moments at most $0.030\,\mu_N$	Amplitudes of moments of order μ_B from the electronic orbital and spin coupling	unsized
Optical activity at keV momenta; Sect. 14.5	$q\,g_{33} = 0.034$ at 1 keV in quartz, a scale only	Current response at $\pm\boldsymbol{q}$ with spin–orbit coupling and the nonlocal vertex, for tellurium or quartz (Appendix E.20)	unsized
Multiferroic response near ordering; Sect. 14.5	Landau form within 10 % of mean field only for $T \geq 0.94\,T_{\mathrm{CM}}$; spin waves controlled for $T \leq 0.37\,T_{\mathrm{CM}}$ (Limit Check 14.5)	Amplitude $\chi_\times$ and scaling function Φ of the dynamic-scaling form for a real material near T_{CM}	unsized
Chap. 15. Superconductors			

Gap; section	Strongest statement	Missing step	Limits
Amorphous WSi; Sect. 15.3.5	Rate needs the configurational average of the loss function, which differs from that of the averaged ε at the order of its variance	Disorder-averaged loss function at keV momenta and the distribution of the local gap (Appendix E.20)	unsized
Strong-coupling Holstein factor; Sect. 15.4.4	Normal metal at $\omega \ll \omega_{\mathrm{D}}$: factor $(1+\lambda_{\mathrm{tr}})^{-2} = 0.52$ in aluminum	Whether the factor survives in the superconducting form near the edge; rate with and without it (Appendix E.20)	estimate
Phonon-assisted spin current near the edge; Sect. 15.4.3	Coherence factor differs from Eq. (15.23); competes with the direct term only above about 10 meV	Its size near the edge	unsized
Scalar absorption near the edge; Sect. 15.4	The kinetic row couples to the amplitude channel of the pair field	Conserving response of Sect. 15.1 extended to that channel	unsized
Chap. 16. Novel targets			
Dirac targets at strong coupling; Sect. 16.1	Ladder controlled at first order in $\tilde{\alpha}$; BNQ-TTF has $\tilde{\alpha} \approx 10$	A controlled method at $\tilde{\alpha} \gtrsim 1$	unsized
Depolarization of quantum dots; Sect. 16.3	Factor $9\lvert\varepsilon_{\mathrm{in}}\rvert^2/\lvert\varepsilon_{\mathrm{in}}+2\varepsilon_{\mathrm{out}}\rvert^2$ (Eq. (16.26)), above one for real $\varepsilon_{\mathrm{in}} > \varepsilon_{\mathrm{out}}$; also computable (Appendix E.21)	$\varepsilon_{\mathrm{in}}(\omega)$ of real dots over 2.5–10 eV, measured	unsized
Impurity band; Sect. 16.5.3	Isolated-donor response wrong at order one near the ionization edge for $n_{\mathrm{D}}/n_{\mathrm{c}} \approx 0.5$	Disorder average on 10^4–10^5 random sites with multivalley envelopes, checked on clusters (Appendix E.21)	unsized
Local-field ratio of a donor; Sect. 16.5	Expected close to one in effective-mass theory	Dielectric matrix of the host and Bloch functions at the minimum (Appendix E.6)	estimate
Color-center yields at millikelvin; Sect. 16.6	Yield $\langle N_{\mathrm{X}}\rangle p_{\mathrm{FH}} p_{\mathrm{surv}} f_{\mathrm{opt}}$; four factors open; 5.9 keV X-rays and photodissociation at millikelvin would calibrate single factors	Tagged-neutron measurement on LiF at 10 mK, counting centers per recoil	unsized
Tunneling-model parameters of films; Sect. 16.7.6	Loss tangent and acoustic loss fix $\bar{P}p_{\mathrm{TLS}}^2$ and $\bar{P}\gamma^2$, not $\bar{P}$; $\gamma_{\mathrm{l}}/\gamma_{\mathrm{t}}$ 1.44–1.84 in 18 glasses; distributions for a model glass computable (Appendix E.22)	One film measured three ways	unsized
Interacting two-level systems; Sect. 16.8.6	Spectral diffusion leaves a flat ensemble intact; branching gives mean cascade $1/(1-b_{\mathrm{br}})$ and $N_{\mathrm{casc}}^{-3/2}$ as $b_{\mathrm{br}} \to 1$	b_{br} from the barrier density and the ballistic phonon flux	unsized
Chap. 17. Comparison across channels			
Coherent Migdal response at small q; Sect. 17.1	Diagonal soft limit (Sect. 12.4); its unsized factor acts on the whole Si Migdal band at 0.32–1 MeV, on a third of the rate at 2 MeV, on at most 7 % from 3 MeV	Coherent response between about 0.75 keV and $2q_{\mathrm{BZ}}$	unsized
Electron-following density at finite q; Sect. 17.1	Long-wavelength model of Sect. 10.2: acoustic peak and electron recoil equal at 12.3 MeV, a property of the model	Following density to several zone momenta with Umklapp and multiphonon terms (Appendix E.16)	unsized
Chap. 18. Ionization yield			
Constant term and variance; Sect. 18.2.6	$\bar{n} = \omega/\varepsilon_{\mathrm{eh}} + o(\omega)$ (Eq. (18.15)); constants computed per seed	Second-order renewal estimates for the walk of Eq. (18.14) and the forgetting rate of the seed (Worked example 22.2)	unsized
Renewal estimate as a lower value; Sect. 18.2.6	$F_{\mathrm{ren}} \le F_{\mathrm{eh}}$ in every computed case and in random parking	A proof for the class, or a counterexample	unsized
Capture and terminal energies; Sect. 18.5.4	Thinning law Eq. (18.31) given $f(x)$	Computed capture probability $f(x)$, terminal-energy distribution and final pair separations (Appendix E.24)	unsized
Impact ionization beyond first order; Sect. 18.5.4	A constant factor 0.8 moves $\varepsilon_{\mathrm{eh}}$ by about 3 %, removed by calibration	An energy-dependent residue and exchange factor, which changes F_{eh}	unsized

Gap; section	Strongest statement	Missing step	Limits
Memory of crystal momentum; Sect. 18.5.4	Keeping probability below 0.1 within 1 eV of threshold, one half above 3–5 eV; memory negligible in the deformation-potential model	The same with first-principles couplings	estimate
Exciton and trapping rates at millikelvin; Sect. 18.5.4	Binding energies and liquid-helium Auger lifetimes known; luminescent fractions and droplet thresholds at millikelvin also open	Binding, capture and trapping cross-sections of pairs measured at 10–50 mK (Appendices E.24 and E.25)	unsized
Chap. 19. Low-energy nuclear recoils			
Charge yield below the binary threshold; Sect. 19.2.6	No calculation predicts the Si yields and Fano factors there or the Ge point at 254 eV	Close-collision kernel Eq. (19.10) for Si–Si and Ge–Ge with its exchange–correlation spread (Appendix E.23)	unsized
Release of close pairs at millikelvin; Sect. 19.6.7	Energy-gap and instanton pictures both admitted for Si and Ge Frenkel pairs and Al split interstitials	Tunneling exponent along the computed path and $E_{\mathrm{reorg}}/\varepsilon^{\mathrm{hop}}$ to about 5 % (Appendix E.22)	unsized
Stored energy in ionic crystals; Sect. 19.6.7	Electronic route prompt; stored energy per deposit = excitons × conversion × survival × configurational energy of the pair	Conversion and close-pair survival probabilities at millikelvin	unsized
Size law and rate of the excess; Sect. 19.6.6	Borel size law from branching; rate $\propto U_{\mathrm{rel}}$; spectral shape and time dependence factorize; b_{br} as for two-level systems	U_{rel} with its spread of rates for a holder and glue (Appendix E.25)	unsized
Carrier-driven defect reactions; Sect. 19.6.7	Probability per capture Eq. (19.53), a Poisson tail	$S_{\mathrm{HR,rxn}}$, ω_{rxn} and the barrier of real defects with their charge states	unsized
Sufficiency of the marks; Sect. 19.6.7	Error of a photon-built detector response at most the total-variation distance at fixed $(N_{\mathrm{eh}}, \mathfrak{m})$, zero when the marks are sufficient	That distance for the features left out of $\mathfrak{m}$ (Appendix E.25)	unsized
Chap. 20. Charge, phonon and quasiparticle readout			
Trap emission and space charge; Sect. 20.1.7	Emission rate = phonon flux, carrier flux and illumination times cross-sections; 45 meV phonons decay within about 1 ns	Phonon- and impact-ionization cross-sections of neutral and over-charged centers (Appendix E.25)	unsized
Luke spectrum beyond longitudinal phonons; Sect. 20.1.7	Rate Eq. (20.10) with longitudinal phonons, one deformation potential; drift velocities test only the momentum loss	Both acoustic branches with valley-resolved deformation potentials, checked against a phonon signal at fixed field	unsized
Validation of transport; Sect. 20.1.7	Hecht relation, charge sum, bias work Eq. (20.3), Fano factor Eq. (20.7)	An engine passing them to a few percent (Appendix E.25)	test
Thick-film offset of the quasiparticle yield; Sect. 20.4.5	Cascade and kinetic equations differ by 0.04 above four gaps; Y_{qp} measured at fixed phonon energy would test it	Depth-resolved film transport at the measured α^2F	estimate
Return of the gap energy at zero field; Sect. 20.5.4	Two-window contrast $0.28(1 - f_{\mathrm{ret}})$ at 100 eV; sign changes at $t_1 = 3.87\, t_{\mathrm{rel}}$ for bimolecular against exponential release	Release-time law and site at millikelvin, from the rates of Sect. 18.5.4 (Appendix E.25)	unsized
Chap. 21. Exclusion limits and next targets			
Loss functions of the survivors; Sect. 21.7.5	Migdal figure of merit between the plasmon-pole and gap-weighted limits	Random-phase response with local fields at the measured gap, tested on the inverse fourth moment (Appendix E.16)	bounded
Elastic constants; Sect. 21.7.5	Computed tensors offset by 4.9 % on average; TeO_2 beats $CaWO_4$ with probability 0.955 instead of 0.987 ; the other computed-tensor crystals and the cold density of CsI likewise open	Measured elastic constants of TeO_2 (Appendix E.17)	estimate
Internal backgrounds; Sect. 21.7.5	Allowed or unique β shapes with the relativistic Fermi function for ^{210}Pb and ^{210}Bi; first-forbidden ^{210}Bi and ^{115}In shapes not computed; ^{123}Te bounded; cosmogenic rates measured for Si and Ge only (Appendix E.27)	^{210}Pb activity of lead that can be grown into a crystal	estimate

Gap; section	Strongest statement	Missing step	Limits
Chap. 22. Open problems			
Charged-current absorption; Sect. 22.3	Allowed approximation: nuclear strengths times the Fermi function of the daughter (Eq. (21.32)); beyond it the rate is not a dark-matter factor times a target response	Shell-model Fermi and Gamow–Teller strengths per isotope, with quenched axial coupling, for every final state below m_χ	unsized
Charge-density-wave collective modes; Sect. 22.3	Phason as the Goldstone pole and amplitudon of the conserving response on the mean field	That conserving response on a first-principles mean field with the self-consistent distortion	unsized

The three proof gaps concern the sector decomposition on which every rate of the book is built. Chaps. 10 to 13, on electronic, lattice, Migdal and phonon-assisted responses, hold 29 gaps, of which 6 are unsized: the response of a crystal at clamped or harmonic nuclei is a well-posed calculation, and where the book has not carried it out it bounds or estimates the result. Chaps. 18 to 20, on the yield, the low-energy recoil and the readout, hold 17, of which 14 are unsized. What is missing there is mostly a rate at millikelvin temperatures, of capture, trapping, exciton binding, defect release or charge emission, for which neither a measurement nor a calculation exists, and the renewal analysis of the cascade, whose constants the book computes for each model but cannot size for the class.

Several gaps share their missing step. A finite-momentum Bethe–Salpeter calculation for silicon and germanium, with the coupling block and the bound lines, closes the attraction and triplet rows of Sect. 10.5, and with exciton–phonon vertices in the same gauge also the replicas of Sect. 13.8. Quasiparticle bands of germanium with spin–orbit coupling close the spin–orbit row of Sect. 10.6 and the germanium window of Sect. 13.9, and with them the open boundaries of the germanium maps of Sect. 17.1. The branching ratio b_{br} of strain-coupled defects is the missing input both of the avalanche question for two-level systems and of the size law of the low-energy excess (Sects. 16.8.6 and 19.6.6). The millikelvin exciton and trapping rates of Sect. 18.5 also fix the release-time law of the gap energy in Sect. 20.5. The 74 rows therefore stand for fewer independent steps. Table 22.1 lists once each calculation that would change a printed result, limit or ranking; the other calculation steps appear only here.

Eleven gaps need a measurement, and each section specifies it. Two of them test a chain that the book derives from the kick to the count: a tagged-recoil Migdal measurement in silicon, and the cold absorption edge of silicon within 25 meV of its threshold. A third uses the same tagged-neutron method on F centers in LiF at 10 mK, where counting single centers per tagged recoil measures the product of four yields as a function of the recoil energy (Sect. 16.6). The other eight supply inputs that no calculation in the book replaces. On the side of the responses they are the cold two-phonon absorption of silicon in its one-phonon band, a neutron test of the isotropy of the spin response of a magnet, the dielectric function of real quantum dots and three measurements of one amorphous film. On the side of the detector and the crystal they are the millikelvin rates of excitons and traps, the stored-energy yields of ionic crystals, the elastic constants of TeO_2 and the ^{210}Pb activity of lead that can be grown into a crystal.

Worked example 22.2: The constant term of the mean pair number

The constant term of the mean pair number is the first open item of Sect. 18.2.6.
The strongest statement. For any energy-conserving cascade whose share law becomes independent of the energy fast enough at high energy, is not concentrated on a lattice in $\ln x$ (non-arithmetic, in the language of renewal theory) and is regular near threshold, the mean pair number is $\bar{n}(\omega) = \omega/\varepsilon_{\mathrm{eh}} + o(\omega)$, Eq. (18.15), proved in Appendix D.44. In the calibrated silicon cascade $\varepsilon_{\mathrm{eh}} = 3.232\, E_{\mathrm{g}} = 3.232 \times 1.160\,\mathrm{eV} = 3.749\,\mathrm{eV}$.
What the book computes without the gap. The cascade of Sect. 18.3 gives the constant b_N in $\bar{n} = \omega/\varepsilon_{\mathrm{eh}} + b_N$ by direct computation for each seed: 0.35 for a seed that gives all the energy to one carrier, 0.008 for a uniform split of the energy. With the one-sided constants, $b_N = 0.35$ and $c_N = -0.02$ in the variance, the expansion matches the computed pair energy within 0.07 % and the computed Fano factor within 0.9 % at $\omega = 10\, E_{\mathrm{g}}$.
The missing step. Second-order renewal estimates for the walk of $\ln x_j$ in Eq. (18.14), together with the rate at which the cascade forgets its seed. They would show that b_N exists for the whole class, as the average over the periodic remainder that a one-phonon model leaves, and that $\mathrm{Var}\, N_{\mathrm{eh}} = F_{\mathrm{eh}}\omega/\varepsilon_{\mathrm{eh}} + O(1)$.

What it limits. At $\omega = 10\,\mathrm{eV}$ the leading term is $10/3.749 = 2.667$ pairs. The one-sided seed adds 0.35, a mean of 3.017 and 13.1 % of the leading term; the uniform split adds 0.3 %, and the mean $\omega/\varepsilon_{\mathrm{eh}} + 0.19$ of the step model of Sect. 18.3 adds 7.1 %. Each of these numbers comes from a computed cascade, not from the expansion, and none waits for the missing step. The one-sided constant falls below 1 % of the leading term above 131 eV, so the expansion matters only at a few pairs, where the cascade is cheap to compute. What the gap leaves without a size is the constant term of a cascade outside the computed models: whether b_N is a property of every cascade of the class or a fitted constant of each model. The missing step is a proof, and the quantity it limits is unsized.

The same reading of the zero-phonon to phonon-assisted ratio in Sect. 13.9.8 gives a different verdict. There the strongest statement is an estimate, and the missing step, the ratio of the two rates on common bins, moves the mass at which the zero-phonon line of silicon takes over from the phonon-assisted rate.

22.3 Channels outside the framework

A dark-matter fermion of 1 MeV absorbed by a ^{125}Te nucleus in a TeO_2 crystal can turn a neutron into a proton and leave as an electron with 626 keV of kinetic energy. The rate of that process carries a factor of 6.3 that no response of the crystal contains: the attraction that the daughter nucleus, of charge 53, exerts on the outgoing electron. The master formula Eq. (8.15) holds for every process in which the dark-matter side can be traced out before the target, so that a tensor $\mathfrak{D}_{ab}(\boldsymbol{q}, \boldsymbol{v})$ that depends only on the dark-matter side multiplies a response $S_{ab}(\boldsymbol{q}, \omega)$ that depends only on the target. Where that separation fails, the process lies outside the framework of this book.

22.3.1 The scope of the master formula

The derivation of Eq. (8.15) in Appendix D.25 uses three properties of the amplitude besides the assumptions of Sect. 8.2.2. The dark field acts through the rows of Eq. (8.3), one-body operators that conserve the number of particles of every species, once per amplitude (twice for the processes of Sect. 10.8). Whatever leaves on the dark-matter side is a free plane wave. The matrix element is then a dark-matter factor that depends on $\boldsymbol{q}$ and $\boldsymbol{v}$ times $\langle f|O_a(\boldsymbol{q})|i\rangle$, and the sum over final states of the target closes into a two-point function at one momentum transfer. The target starts in a Gibbs state, and the observable is the excitation it is left in, read through a final-state class of Sect. 8.7. A process that violates the first property needs a row that Eq. (8.3) lacks. One that violates the second has no $\mathfrak{D}_{ab}$ at all, because the outgoing particle acts on the target. One that violates the third can have a well-defined rate that no detector reads as an excitation spectrum.

Two of the assumptions of Sect. 8.2.2 also bound the framework. With couplings near 4π and a mediator lighter than about $20A\mu_{\chi T}$, with $\mu_{\chi T}$ the reduced mass of dark matter and nucleus, an attractive potential binds and the first Born approximation fails at order one (Sect. 8.2.2). A target whose density operator has coherences between energy eigenstates, or one driven during the exposure, has no response of the form Eq. (4.13). A last class lies inside the framework: the rate is defined, but no calculation of S_{ab} for that kind of target yet obeys the exact relations of Sect. 8.3. Table 22.3 lists the processes that break one of the three properties, the two that the assumptions exclude, and the one inside the framework that has no conserving response yet.

Table 22.3: Processes outside the framework of the master formula, and one inside it without a computed response. Second column: why the rate is not $\sum_{ab} \mathfrak{D}_{ab} S_{ab}$ of Eq. (8.15), or, for the last row, why no computed response is available. Third column: the calculation or measurement that a prediction needs. Last column: the section that states the process.

Process	Reason	What a prediction needs	Section
Charged-current absorption, $\chi + (A, Z) \to (A, Z{+}1) + e^-$	Species-changing nuclear row; the outgoing electron moves in the field of the final state of the target, so the dark-matter side is an operator on the target	Shell-model Fermi and Gamow–Teller strengths and transition densities per isotope, with quenched axial coupling; Dirac–Coulomb electron waves with finite nuclear size and atomic screening, for the daughter charge	Sect. 22.3.2
Radiative decay of an excited dark state	No target operator in the amplitude; the medium enters through the photon propagator only	Abundance of the excited state at the detector; local field at the decaying particle; absorption of the photon	Sects. 10.9.4 and 22.3.3
Damage tracks of recoiling heavy ions	Born parameter far above one; 10^2 to 10^4 correlated collisions in a lattice damaged by the earlier ones; the readout is a defect configuration	Transport of the ion with the stopping of Eqs. (19.17) and (19.21) as input, and the map from the track to the defects that are read	Sects. 19.3.4 and 22.3.3
Strong dark-matter–target coupling	First Born approximation fails; bound states and resonances	Partial-wave solution of the scattering	Sect. 8.2.2
Target with coherences, or driven during the exposure	No Gibbs or energy-diagonal initial state	Rate for a density operator with coherences between energy eigenstates	Sect. 8.2.2
Charge-density-wave phasons and amplitudons	Inside the framework; the response needs a conserving kernel on the mean field of the distorted crystal	Time-dependent mean-field response with the kernel of the self-consistent distortion; Goldstone and f-sum checks	Sect. 22.3.4

22.3.2 Charged-current absorption

The fermionic absorption of Sect. 8.1.5 has a charged partner. An operator that couples the dark-matter fermion to an electron and a neutron to a proton,

$$\frac{1}{\Lambda_{\mathrm{F}}^2} \left[\bar{p}\, \gamma_\mu (1 + \lambda_A \gamma^5)\, n\right] \left[\bar{e}\, \gamma^\mu P_R\, \chi\right] + \text{h.c.}, \tag{22.1}$$

with p, n, e and χ the proton, neutron, electron and dark-matter fields, Λ_{F} the scale of the operator as for the neutral absorption operators of Sect. 8.1.5, $P_R = (1+\gamma^5)/2$ and the nucleon axial coupling $\lambda_A = 1.27$ in the sign convention of the reference, absorbs χ on a neutron of a nucleus and emits an electron, $\chi + (A, Z) \to (A, Z+1)^{(*)} + e^-$ [3, Eq. (4.1)]; its pure vector form is Eq. (6) of Ref. [4]. For a parent that is stable against β decay the process has a threshold, $m_\chi > m_{\mathrm{th}}^{\mathrm{CC}} = M_{A,Z+1}^{(*)} + m_e - M_{A,Z}$ with nuclear masses, the daughter possibly excited [3, Eq. (4.2)]. Above it the electron takes the kinetic energy $E_{\mathrm{ej}} = m_\chi - m_{\mathrm{th}}^{\mathrm{CC}}$, up to a nuclear recoil of order $m_\chi^2/2M$. The ground state of ^{125}Te has spin $\frac{1}{2}^+$, and the ground state of ^{125}I, $\frac{5}{2}^+$, cannot be reached from it by an allowed transition; the lowest allowed threshold, to an excited state of ^{125}I reached by a Gamow–Teller transition, is 374 keV [3, Table 1].

The nuclear side is a row that Eq. (8.3) does not have. At momentum transfers small against the inverse nuclear radius, the time component of the vector current reduces to the isospin raising operator τ_j^+, which turns nucleon j from a neutron into a proton. The spatial axial current reduces to $\sigma_j^\beta \tau_j^+$, as the density and the spin density of Appendix D.24 do for

the neutral currents. Summed over the nuclei I at positions $\boldsymbol{X}_I$,

$$O_{\mathrm{F}}^{\mathrm{CC}}(\boldsymbol{q}) = \sum_I \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I} \sum_{j\in I} \tau_j^+, \qquad O_{\mathrm{GT},\beta}^{\mathrm{CC}}(\boldsymbol{q}) = \sum_I \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I} \sum_{j\in I} \sigma_j^\beta \tau_j^+, \tag{22.2}$$

the Fermi and Gamow–Teller rows. Every row of Eq. (8.3) conserves the numbers of protons and neutrons; these change the element. Adding these rows, and nothing else, would keep the framework, with a response $S_{ab}^{\mathrm{CC}}(\boldsymbol{q},\omega)$ of the rows in Eq. (22.2).

The electron prevents that. It is charged, and it leaves through the Coulomb field of the daughter nucleus and of the atomic electrons around it. Its final state is a scattering state $\psi_{\boldsymbol{p}_{\mathrm{ej}}}^{(-)}(\boldsymbol{r})$ of that field, not a plane wave, and for a dark-matter particle of momentum $\boldsymbol{p}$ the amplitude is an overlap $\int \mathrm{d}^3r\, \psi_{\boldsymbol{p}_{\mathrm{ej}}}^{(-)*}(\boldsymbol{r})\, \mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}} \langle f|\rho^{\mathrm{CC}}(\boldsymbol{r})|i\rangle$ with the density $\rho^{\mathrm{CC}}(\boldsymbol{r})$ whose Fourier components are the rows in Eq. (22.2). Its square involves the transition density at two points, $\langle i|\rho^{\mathrm{CC}\dagger}(\boldsymbol{r}')|f\rangle\langle f|\rho^{\mathrm{CC}}(\boldsymbol{r})|i\rangle$, weighted by a product of electron waves, so it is not a response at one momentum transfer. The field that distorts the wave, moreover, belongs to the final state $|f\rangle$: the charge $Z+1$ of the daughter and the atomic configuration left behind when the nuclear charge changes. The dark-matter side therefore acts on the target, and no tensor $\mathfrak{D}_{ab}$ exists. The exact rate is the golden rule with the electron in the final state of the combined system, $\sum_f |\langle f; e_{\boldsymbol{p}_{\mathrm{ej}}}^{(-)}|H_{\mathrm{int}}|i;\chi\rangle|^2$. It is the spectral function of an operator that adds an electron to the target and turns a neutron into a proton: a response of the target with one more electron. A condensed-matter reader knows the structure from the x-ray edge, where the core hole left in the final state distorts the outgoing electron.

One limit restores a product. Let the electron wavelength be long against the nuclear radius, $p_{\mathrm{ej}} r_A \ll 1$ with $r_A = 1.2A^{1/3}\,\mathrm{fm}$. The wave can then be evaluated at the nucleus. Its squared amplitude relative to a plane wave is the relativistic Fermi function $F_{\mathrm{Fermi}}(Z_{\mathrm{d}}, E_{\mathrm{ej}})$ of Eq. (21.32), the point-charge function of β decay with its traditional prefactor 4 [5, Eq. (6)], evaluated at the electron's kinetic energy E_{ej} with $Z_{\mathrm{d}} = Z+1$ the charge of the daughter and with the Sommerfeld parameter $y_{\mathrm{S}} = \alpha Z_{\mathrm{d}} E_e / p_{\mathrm{ej}}$ of Eq. (21.31), where $E_e = m_e + E_{\mathrm{ej}}$ is the total energy of the electron. The finite size of the nucleus enters through a further factor L_0 that differs from one by a quantity of order $\alpha Z_{\mathrm{d}} E_e r_A$. In this allowed approximation the rate per isotope is a sum over final nuclear states of $F_{\mathrm{Fermi}}(Z_{\mathrm{d}}, E_{\mathrm{ej}})$ times the electron phase space, proportional to $p_{\mathrm{ej}} E_e$, times the standard allowed combination of β decay [5, Sect. I], the Fermi strength $B_{\mathrm{F}} = |\langle f|\sum_j \tau_j^+|i\rangle|^2$ plus λ_A^2 times the Gamow–Teller strength $B_{\mathrm{GT}} = \sum_\beta |\langle f|\sum_j \sigma_j^\beta \tau_j^+|i\rangle|^2$. The published projections multiply the nucleon-level amplitude by the same kind of function [3, Eq. (4.9)]. Their Eq. (4.10) prints the prefactor $2(1+\gamma_{\mathrm{d}})$, a convention that moves part of L_0 into the function, and $\Gamma(1+\gamma_{\mathrm{d}})^2$ in the denominator where Eq. (21.32) has $\Gamma(1+2\gamma_{\mathrm{d}})^2$. The gamma-function factor alone is therefore larger by $\Gamma(1+2\gamma_{\mathrm{d}})^2/\Gamma(1+\gamma_{\mathrm{d}})^2$, 3.22 at $Z_{\mathrm{d}} = 53$ and 4.0 as $Z_{\mathrm{d}} \to 0$, and the printed function does not reach the nonrelativistic limit.

Worked example 22.3: The Fermi function of ^{125}Te absorption

Take ^{125}Te in $\mathrm{TeO_2}$, the Gamow–Teller threshold of 374 keV and $m_\chi = 1\,\mathrm{MeV}$. The electron leaves with $E_{\mathrm{ej}} = 626\,\mathrm{keV}$ and $p_{\mathrm{ej}} = 1.016\,\mathrm{MeV}$, so $y_{\mathrm{S}} = 0.433$ for the daughter iodine, $Z_{\mathrm{d}} = 53$, and $p_{\mathrm{ej}} r_A = 0.031$ with $r_A = 6.0\,\mathrm{fm}$. The allowed approximation holds. Its corrections in the nuclear operators are of order $(p_{\mathrm{ej}} r_A)^2 \approx 10^{-3}$, and the finite-size Coulomb correction L_0 is of order $\alpha Z_{\mathrm{d}} E_e r_A = 0.013$, about 1 %.

If the master formula applied, the rate would be the contraction of a dark-matter tensor, built from the operator Eq. (22.1) and a plane-wave electron, with the nuclear response of Eq. (22.2) at $q = p_{\mathrm{ej}}$. At these momenta the nuclear strength does not depend on q, so at

fixed matrix element the rate would grow with the electron phase space, as $p_{\rm ej}E_e$. With the distorted wave it is $p_{\rm ej}E_eF_{\rm Fermi}$, and Eq. (21.32) gives $F_{\rm Fermi}(53,626\,{\rm keV}) = 6.34$. The factor belongs to the final state. Evaluated with the charge of the parent, $Z_{\rm d} \to 52$, it is 6.06, 4.3 % lower. A response sums over final states independently of the outgoing particle, and a dark-matter tensor depends on $\boldsymbol{q}$ and $\boldsymbol{v}$ alone, so neither can carry this dependence.
It also changes the threshold behavior. At $E_{\rm ej} = 10\,{\rm keV}$, $F_{\rm Fermi} = 32.5$, and at 1 keV, 102. The product $p_{\rm ej}E_eF_{\rm Fermi}$ tends to a constant, 0.23 of its value at 626 keV for both energies, while $p_{\rm ej}E_e$ alone falls to 0.046 and 0.014 of it. The rate stays finite at the threshold, as the β spectra of Sect. 21.7 do at their low-energy end, where a plane-wave electron would make it vanish as $p_{\rm ej}$.
At low energies the atomic electrons enter as well. The radius of the iodine K shell, $a_0/Z_{\rm d}$, is 998 fm, against $1/p_{\rm ej} = 194\,{\rm fm}$ at 626 keV and 1942 fm at 10 keV. Below about 10 keV the wave is screened by electrons whose configuration, after the sudden change of the nuclear charge, is not stationary, and the distortion then depends on the atomic final state too. The factorized form is an approximation for one final state of the nucleus with the atomic shells as spectators, and it holds because $p_{\rm ej}r_A$ is small and $E_{\rm ej}$ is large; the general rate is a property of the target together with the electron.

The nuclear strengths obey two sum rules. Write $T_\pm = \sum_j \tau_j^\pm$ for the summed raising and lowering operators of isospin, which obey the angular-momentum algebra with $T_z = (Z-N)/2$ in the convention in which a proton has $\tau_z = +1$, so that T_+ turns a neutron into a proton. For the Fermi strength we need the isospin limit, in which the ground state of the parent has total isospin $T_{\rm iso} = (A-2Z)/2$, the smallest value its projection $T_z = -T_{\rm iso}$ allows. It is then the bottom state of its multiplet, and T_-, which would turn one more proton into a neutron, annihilates it. The summed strength is then a ladder norm,

$$\sum_f B_{\rm F} = \langle i|T_-T_+|i\rangle = T_{\rm iso}(T_{\rm iso}+1) - T_{\rm iso}(T_{\rm iso}-1) = 2T_{\rm iso} = A-2Z,$$

and T_+ takes the ground state into a single state, the isobaric analog of the parent. With σ_j^β inserted no such state exists, but the commutator of the summed operators with their conjugates, summed over β, picks up $\sum_\beta \sigma^\beta\sigma^\beta = 3$, and

$$\sum_f B_{\rm F} = A-2Z, \qquad \sum_f B_{\rm GT}(n\to p) - \sum_f B_{\rm GT}(p\to n) = 3(A-2Z), \tag{22.3}$$

the Fermi and Ikeda sum rules, where $A-2Z$ is the neutron excess. The Ikeda rule is an operator identity and holds without isospin symmetry; the Fermi rule in this form, and the concentration of its strength in one state, need the isospin limit. The analog state lies above the parent by the Coulomb energy of the added proton less the neutron–proton mass difference. For ^{125}Te that puts the threshold to reach it at about 13 to 14 MeV: 14.2 MeV for a uniformly charged sphere of radius r_A, and 13.6 MeV once the exchange term of the Coulomb energy of that sphere is subtracted. Below that mass a vector charged current acts only through isospin-breaking admixtures, and the axial current carries the rate through Gamow–Teller transitions. The published projections count the nucleon-level rate once per neutron, $n_j \simeq A-Z = 73$ for ^{125}Te [3, Eq. (4.6)]. The Fermi sum rule allows 21 in total, all of it in the analog state. On the Gamow–Teller side the same count gives $3(A-Z) = 219$ against the Ikeda difference of 63, before quenching. Shell-model Gamow–Teller strengths also exceed measured ones unless the axial coupling in the nucleus is reduced by a factor of

about 0.75, or two-body currents are added [6], which changes the weight λ_A^2 by a factor of about two.

A prediction for a crystal therefore needs three things for each isotope. The first is the nuclear strength: shell-model Fermi and Gamow–Teller strengths for every final state below the dark-matter mass, with the quenched axial coupling or the two-body currents, and their dependence on q through the transition densities, with first-forbidden operators added where $p_{\mathrm{ej}} r_A$ reaches about 0.1 or the allowed matrix element vanishes. The computed strengths must exhaust Eq. (22.3) in a space closed under $\sigma\tau$, which contains every spin–orbit partner of its orbitals. The second is the electron: Dirac–Coulomb waves of the daughter charge, with finite nuclear size and atomic screening, evaluated with the same nuclear radius as the strengths; at 626 keV the parent charge would give a Fermi function 4.3 % too low. The third is the final state that a detector reads: the electron deposits its energy as a fast primary of the kind that Chap. 18 follows, together with the de-excitation of the daughter. Capture of the electron into a bound orbital of the daughter leaves no free particle to carry the energy away, so it occurs only where m_χ matches a transition, within the width of the final atomic state.

22.3.3 Radiative decay and ion tracks

An excited dark state χ_2 that decays to χ_1 and a photon inside the detector has an amplitude with no target operator in it (Sect. 10.9.4). The event rate is then the density of excited states at the detector times their decay rate times the volume, with no flux factor $\rho_\chi v/m_\chi$ and no exposure in target mass, and the medium enters only through the photon density of states at $|\Delta_\chi|$. A prediction needs the abundance of χ_2 at the detector, set by the early universe or by up-scattering in the Earth; the local field at the decaying particle, which the macroscopic ε^{T} of Eq. (10.63) omits; and the absorption length of the photon in the detector, which the optical response gives once the photon exists. The part of the decay that hands energy directly to the electrons through a virtual photon lies inside the framework (Sect. 10.9.4).

A heavy ion leaves a damage track (Sect. 19.3.4). Its rate of energy loss is the Bethe or friction stopping of Eqs. (19.17) and (19.21), a response quantity, but the track is not. A recoiling silicon ion at the speeds that dark matter gives it has a Born parameter $Z_1\alpha/v$, with v the speed of the ion, of 117 to over a thousand, so no single scattering is perturbative. This is the parameter of the moving ion; the Born parameter $Z_\kappa\alpha/v_{\mathrm{F}}$ of Sect. 22.1, set by the electrons it meets, measures the same failure at low speed. A track of 10^2 to 10^4 collisions runs through a lattice that the earlier collisions have already displaced, and the target is not in its Gibbs state for any collision after the first. What a track detector or a paleo-detector reads is the defect configuration that remains, not an excitation spectrum. A prediction needs a transport calculation of the ion through the damaged lattice, with the stopping as input and molecular dynamics for the collisions, and the map from the final configuration to the defects that the readout sees.

22.3.4 Collective modes of a charge-density wave

Below about 260 K the quasi-one-dimensional crystal $(TaSe_4)_2I$ forms a charge-density wave with a single-particle gap $2\Delta_{\mathrm{CDW}} \approx 250$ to 300 meV. It carries a collective mode at 0.23 THz, or 0.951 meV, which Kim and collaborators identify as the massive phason of that wave [7]. The density row of Eq. (8.3) can excite modes that lie 263 to 315 times below the charge gap: with $v_{\max}$ the largest dark-matter speed in the laboratory (Eq. (1.9)), a dark-matter particle heavier than $2\omega/v_{\max}^2 = 271$ eV can deposit 0.951 meV. Nothing in the master formula changes for such a target. What is missing is a response that obeys the exact relations of Sect. 8.3.

In a charge-density wave the electron density is modulated, $n_e(\boldsymbol{r}) = \bar{n}_e + n_1 \cos(\boldsymbol{Q}_{\mathrm{CDW}} \cdot$

$\boldsymbol{r}+\varphi$), with a wave vector $\boldsymbol{Q}_{\mathrm{CDW}}$ that is incommensurate with the lattice in $(TaSe_4)_2I$, and the lattice distorts with it. The order parameter is $\Delta_{\mathrm{CDW}}e^{i\varphi}$. The phason is an oscillation of φ, the amplitudon one of $|\Delta_{\mathrm{CDW}}|$. If the wave is incommensurate, a uniform shift of φ costs no energy, and the phason is a Goldstone mode whose frequency vanishes linearly in its wave vector; impurities and commensurability pin it and open a small gap. A density probe couples to the phason in two places. Near $\boldsymbol{q}=\pm\boldsymbol{Q}_{\mathrm{CDW}}+\boldsymbol{G}$ it meets the modulation, $\delta n_e=-n_1\sin(\boldsymbol{Q}_{\mathrm{CDW}}\cdot\boldsymbol{r}+\varphi)\,\delta\varphi$, and reaching it there takes $m_\chi\geq|\boldsymbol{Q}_{\mathrm{CDW}}|/2v_{\max}$, 189 keV per keV of $|\boldsymbol{Q}_{\mathrm{CDW}}|$. At small $\boldsymbol{q}$ a gradient of φ along the chains moves the condensed electrons with the wave and leaves a charge density proportional to that gradient. That charge feels the long-range Coulomb interaction, which lifts the longitudinal phason, whose wave vector has a component along $\boldsymbol{Q}_{\mathrm{CDW}}$, from zero to the plasma frequency of the condensed electrons. A phason whose wave vector is perpendicular to the chains carries no charge and stays acoustic, or pinned, so the response depends on the direction of $\boldsymbol{q}$. Lee, Rice and Anderson, and Lee and Fukuyama, showed this, as quoted by Kim and collaborators [7]. The condensate plasma frequency is as low as a meV because a moving charge-density wave drags its lattice distortion with it, which gives the condensed electrons a large effective mass [8]. The superconductor of Sect. 15.1 shows the same mechanism, with its phase mode pushed to the plasmon.

The response that captures these modes has the structure of Sect. 15.1. On the mean-field reference with the self-consistent distortion, the proper density response must be the time-dependent mean field in the space of electron states at $\boldsymbol{k}$ and $\boldsymbol{k}+\boldsymbol{Q}_{\mathrm{CDW}}$. Its kernel is the functional derivative of the same self-energy that produced Δ_{CDW}: Hartree–Fock exchange plus the interaction mediated by the phonon at $\boldsymbol{Q}_{\mathrm{CDW}}$. Only then does the broken translation produce the phason pole at zero frequency in the proper response, and only then does the f-sum rule Eq. (4.30) hold. The bare bubble on the reconstructed bands is the counterpart of the nonconserving bubble of Sect. 7.2: it misses the phason and violates the f-sum. The phonon at $\boldsymbol{Q}_{\mathrm{CDW}}$ is dressed by the electronic polarization, and the interaction it mediates may enter either the electronic kernel or the dressed phonon propagator, not both. Screening with $v_{\mathrm{C}}(q)$ as in Eq. (15.1) then gives the loss function, in which the longitudinal phason sits at the condensate plasma frequency. The phason and the amplitudon are composites of electrons and lattice, and assigning them to a sector of Sect. 8.6 is a convention, as for the pair composites of a superconductor.

A calculation for a real crystal needs that kernel on a first-principles mean field. The supercell of the distorted crystal must be relaxed with a functional that produces the distortion, and the electron–phonon vertex at $\boldsymbol{Q}_{\mathrm{CDW}}$ taken from the undistorted cell. The kernel must come from the same functional as the distortion, since a mismatch violates the Ward identity at the order of the mismatch. The checks are the Goldstone limit of the proper response, in which the phason frequency vanishes with its wave vector when $\boldsymbol{Q}_{\mathrm{CDW}}$ is made artificially incommensurate and stays gapped for a commensurate wave; the f-sum rule; and the measured modes, the 0.951 meV phason of $(TaSe_4)_2I$ among them.

Summary

The open problems of the book fall into three kinds. The first are calculations that the framework specifies but that do not exist for the crystals in question, each tied to the result it would change and to an exact relation and a control against data that it must pass (Sect. 22.1, Table 22.1). Sorted by what limits them, 12 of the 22 channels of the book are limited first by calculation, 9 by data and 2 by theory, the phonon-assisted window of silicon counting under both calculation and data. Four calculations each replace a factor of order ten or an unsized rate in a leading result: the phonon-assisted loss function of silicon with the zero-phonon rate

on common bins, the coherent Migdal response of silicon below $2q_{\mathrm{BZ}}$, the loss functions of the survivors near their gaps and the close-collision kernel of a slow recoil. Five measurements do the same on the data side: the cold absorption edge of silicon, the pair energy of germanium at millikelvin, the elastic constants of TeO_2, the ^{210}Pb activity of lead that can be grown into a crystal and the two-bias measurement that separates phonon collection from stored energy. For the heavy mediator the published limit on cosmic-ray-boosted dark matter is unchanged by the exact density row.

An exact remaining gap is the strongest statement the book derives about a quantity, with its conditions, together with the single step that would turn it into the full result (Sect. 22.2). The book contains 74. Three limit only a proof and leave every number standing; of the 71 that involve a number, 10 come with a bound or a bracket, 17 with an estimate, 39 are unsized and 5 are independent checks of a computed number. The missing step is a specified calculation in 56 of them, a measurement in 11 and a proof or method in 7. Several gaps share one missing step, a finite-momentum Bethe–Salpeter calculation for silicon and germanium or quasiparticle bands of germanium with spin–orbit coupling among them, so the 74 rows stand for fewer independent calculations. Unsized gaps dominate both the chapters from magnetic targets to the comparison across channels (16 of 19) and those on the measurement chain (14 of 17), where the rates of capture, trapping and release at millikelvin are neither measured nor computed.

The master formula holds wherever the dark-matter side can be traced out before the target (Sect. 22.3): the dark field acts through species-conserving one-body rows, whatever leaves on the dark side is a free plane wave, and the target starts in a Gibbs state and is read through the excitation it is left in. Charged-current absorption, $\chi + (A, Z) \to (A, Z+1) + e^-$, breaks the first two. Its nuclear rows change the element, and its outgoing electron moves in the Coulomb field of the final state of the target, so the dark-matter side becomes an operator on the target. In the allowed approximation the rate is the relativistic Fermi function of the daughter charge times the phase space $p_{\mathrm{ej}}E_e$ times the Fermi and Gamow–Teller strengths. For ^{125}Te at $m_\chi = 1\,\mathrm{MeV}$ the Fermi function is 6.34, 4.5 % above its value with the charge of the parent, and it keeps the rate finite at threshold. The whole Fermi strength, $A - 2Z$, sits in the isobaric analog state 13 to 14 MeV up, so Gamow–Teller transitions with a quenched axial coupling carry the rate below it, and a count of one conversion per neutron overstates the strength. The radiative decay of an excited dark state has no target operator in its amplitude; the track of a recoiling heavy ion is a transport problem whose stopping alone is a response; strong coupling and targets with coherences fail the assumptions of the formula. The collective modes of a charge-density wave lie inside the framework and need a conserving response whose proper part contains the Goldstone phason.

Most of what the book leaves open is calculation rather than theory. In 20 of its 22 channels the response has a controlled form, the next term is known and its check is stated; what is missing there is the number, and a stated calculation or measurement supplies it.

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Appendix A

Conventions and units

A.1 Natural units

We set $\hbar = c = k_\mathrm{B} = 1$. Energy, mass, momentum, frequency and temperature are then all measured in electronvolts, length and time in eV^{-1}, and speeds are pure numbers, fractions of the speed of light. A dark-matter speed of $250.5\,\mathrm{km\,s^{-1}}$ is 8.356×10^{-4}, and the energy and momentum it carries at mass m_χ are $\frac{1}{2}m_\chi v^2$ and $m_\chi v$ in the same unit. The choice suits a subject in which a particle-physics mass, a crystal momentum, a phonon frequency and a temperature enter one energy-conservation condition. Particle physics uses it routinely, and Ref. [1, Appendix A, Table III] tabulates the astrophysical units it is mixed with; condensed-matter texts often keep $\hbar$ and k_B, and their formulas are converted by restoring the factors below.

A.1.1 Restoring the constants

A formula in natural units is converted to conventional units by inserting the powers of $\hbar$, c and k_B that make every term dimensionally consistent. Four constants suffice; their values follow from the 2022 adjustment of CODATA (the Committee on Data of the International Science Council) [2], in which c, h, e and k_B are exact:

$$\begin{aligned} \hbar c &= 1.973\,270 \times 10^{-5}\,\mathrm{eV\,cm} = 197.3270\,\mathrm{MeV\,fm}, & \hbar &= 6.582\,120 \times 10^{-16}\,\mathrm{eV\,s}, \\ k_\mathrm{B} &= 8.617\,333 \times 10^{-5}\,\mathrm{eV\,K^{-1}}, & c &= 2.997\,925 \times 10^{5}\,\mathrm{km\,s^{-1}}. \end{aligned} \tag{A.1}$$

A length L becomes $L/(\hbar c)$ in eV^{-1}, a time t becomes $t/\hbar$, a temperature T becomes $k_\mathrm{B}T$ and a speed u becomes u/c. A density per unit volume picks up $(\hbar c)^3$, a cross-section $(\hbar c)^{-2}$, and a rate per unit time $\hbar$.

Table A.1 lists the factors. Each entry is a power of one of the products in Eq. (A.1). For example, $1\,\mathrm{cm^{-3}} = (\hbar c)^3/\mathrm{cm^3} = (1.973\,270 \times 10^{-5}\,\mathrm{eV})^3 = 7.6835 \times 10^{-15}\,\mathrm{eV^3}$, and $1\,\mathrm{b} = 10^{-24}\,\mathrm{cm^2}$ divided by $(\hbar c)^2 = 3.8938 \times 10^{-10}\,\mathrm{eV^2\,cm^2}$ gives $2.5682 \times 10^{-15}\,\mathrm{eV^{-2}}$. A mass density in $\mathrm{g\,cm^{-3}}$ first becomes an energy density through $1\,\mathrm{g} = 5.610 \times 10^{23}\,\mathrm{GeV}$ and then picks up $(\hbar c)^3$. Spectroscopic frequencies need care: a frequency ν quoted in THz corresponds to the energy $h\nu = 2\pi\hbar\nu$, while an angular frequency ω in $\mathrm{rad\,s^{-1}}$ corresponds to $\hbar\omega$. Phonon tables usually quote ν, so the optical phonon of silicon at the zone center, at 15.5 THz [3], has $\omega_{Q\nu} = 64.1\,\mathrm{meV}$.

A.1.2 Checking dimensions

In natural units every quantity has a single dimension, a power of energy, and a formula is checked by counting powers of eV. The rate per unit target mass of Eq. (1.47) is the

Table A.1: Conversion factors between conventional units and natural units ($\hbar = c = k_\mathrm{B} = 1$), computed from CODATA 2022 values [2]. Examples in the last column are values used in the book.

Quantity	Conversion	Example
length	$1\,\mathrm{cm} = 5.0677 \times 10^4\,\mathrm{eV}^{-1}$, $1\,\text{Å} = 5.0677 \times 10^{-4}\,\mathrm{eV}^{-1}$	Bohr radius $1/(\alpha m_e) = 0.5292\,\text{Å}$
time	$1\,\mathrm{s} = 1.5193 \times 10^{15}\,\mathrm{eV}^{-1}$	$1\,\mathrm{yr} = 3.156 \times 10^7\,\mathrm{s} = 4.795 \times 10^{22}\,\mathrm{eV}^{-1}$
temperature	$1\,\mathrm{K} = 8.6173 \times 10^{-5}\,\mathrm{eV}$, $1\,\mathrm{eV} = 11\,605\,\mathrm{K}$	$300\,\mathrm{K} = 25.85\,\mathrm{meV}$
speed	$1\,\mathrm{km\,s^{-1}} = 3.3356 \times 10^{-6}$	$v_\mathrm{max} = 794.5\,\mathrm{km\,s^{-1}} = 2.650 \times 10^{-3}$
number density	$1\,\mathrm{cm^{-3}} = 7.6835 \times 10^{-15}\,\mathrm{eV}^3$	$\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}} = 2.31 \times 10^{-6}\,\mathrm{eV}^4$
mass density	$1\,\mathrm{g\,cm^{-3}} = 4.310 \times 10^{18}\,\mathrm{eV}^4$	silicon, $2.33\,\mathrm{g\,cm^{-3}} = 1.004 \times 10^{19}\,\mathrm{eV}^4$
mass	$1\,\mathrm{kg} = 5.610 \times 10^{26}\,\mathrm{GeV}$	–
cross-section	$1\,\mathrm{cm}^2 = 2.5682 \times 10^9\,\mathrm{eV}^{-2}$, $1\,\mathrm{b} = 2.5682 \times 10^{-15}\,\mathrm{eV}^{-2}$	$1\,\mathrm{GeV}^{-2} = 0.3894\,\mathrm{mb} = 3.894 \times 10^{-28}\,\mathrm{cm}^2$
spectroscopy	$1\,\mathrm{THz} = 4.1357\,\mathrm{meV}$ $(h\nu)$, $1\,\mathrm{cm^{-1}} = 0.123\,98\,\mathrm{meV}$	–

standard example: it is dimensionless, as a rate (eV) per unit mass (eV) must be (Sect. 1.5). The dimensions of the main objects are fixed by their definitions: the dynamic structure factor $S(q,\omega)$ has dimension eV^2 (Sect. 4.2), the transform $\tilde{V}(\boldsymbol{q})$ of a potential has eV^{-2}, the density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ and the dielectric function are dimensionless, and $\bar{\sigma}_e$ has eV^{-2}. Final numbers are always converted back: rates to $\mathrm{kg^{-1}\,yr^{-1}}$, cross-sections to cm^2, momenta to keV, and energies to eV or meV. A rate per unit mass in natural units is converted with $1\,\mathrm{kg^{-1}\,yr^{-1}} = 3.718 \times 10^{-59}$, the inverse of the product of 1 kg in eV and 1 yr in eV^{-1}.

The metric signature is $(+,-,-,-)$, with $x^\mu = (t,\boldsymbol{x})$ and $\partial_\mu\partial^\mu = \partial_t^2 - \nabla^2$, so that a free particle obeys $p^\mu p_\mu = E^2 - |\boldsymbol{p}|^2 = m^2$; it is the signature of most particle-physics texts, and Appendix C.9 uses it. Physical values in the text are quoted in conventional units with a thin space, 1.17 eV or $0.3\,\mathrm{GeV\,cm^{-3}}$, and natural-unit values are written without units only for speeds.

A.2 Charge conventions

The elementary charge is $e > 0$ with

$$e^2 = 4\pi\alpha, \qquad \alpha = 1/137.035999, \tag{A.2}$$

the rationalized Heaviside–Lorentz convention of particle physics [2]. The electron carries charge $-e$ and a nucleus of species κ carries $+Z_\kappa e$, and the charge density operator is e times the sum of the nuclear densities weighted by Z_κ minus the electron density. Two electrons at distance r have the interaction energy $e^2/(4\pi r) = \alpha/r$, and Gauss's law states that $\nabla\cdot\boldsymbol{E}$ equals the charge density, without a factor 4π (Appendix C.9). The Coulomb interaction in momentum space, with the Fourier convention of Appendix A.3, is the kernel

$$v_\mathrm{C}(k) \equiv \frac{e^2}{k^2} = \frac{4\pi\alpha}{k^2}, \tag{A.3}$$

the transform of $e^2/(4\pi r)$ by Eq. (C.81) at $m_V = 0$. In a crystal it is a diagonal matrix in reciprocal-lattice vectors, $(v_\mathrm{C})_{\boldsymbol{G}\boldsymbol{G}'} = v_\mathrm{C}(|\boldsymbol{q}+\boldsymbol{G}|)\,\delta_{\boldsymbol{G}\boldsymbol{G}'}$ (Sect. 4.4).

A.2.1 Heaviside–Lorentz units and the loss function

Dark-matter couplings are written in field theory, where Heaviside–Lorentz units are standard: the dark-photon coupling to the electron, $y_e = -\zeta e$, is the electromagnetic one scaled by ζ (Sect. 1.4), and the Yukawa potential of Eq. (1.31) has the same $1/(4\pi r)$ as Coulomb's law. The dielectric response of a solid is expressed in the same units, so the relation between the structure factor and the loss function carries no factors of 4π. The dynamic structure factor and the loss function are related by

$$S(q,\omega) = \frac{2q^2}{e^2}\,\frac{L(q,\omega)}{1-\mathrm{e}^{-\beta\omega}} = \frac{q^2}{2\pi\alpha}\,\frac{L(q,\omega)}{1-\mathrm{e}^{-\beta\omega}}, \tag{A.4}$$

derived in Appendix A.7 [4, Eq. (6), 5, Eq. (S.7)]. In a dark-photon model the factor $1/e^2$ cancels against $y_e^2 = \zeta^2 e^2$, and the rate depends only on ζ^2 and the loss function [5, below Eq. (S.8)].

A.2.2 Converting from Gaussian and SI units

Condensed-matter sources usually use Gaussian units, in which the electron charge e_G obeys $e_\mathrm{G}^2 = \alpha$ (with $\hbar = c = 1$), the Coulomb energy is e_G^2/r and the Coulomb kernel is $4\pi e_\mathrm{G}^2/k^2$. SI sources write $e^2/(4\pi\varepsilon_0^\mathrm{SI} r)$, with $\varepsilon_0^\mathrm{SI}$ the vacuum permittivity; with $\hbar = c = 1$, $e_\mathrm{SI}^2/\varepsilon_0^\mathrm{SI} = 4\pi\alpha$. The three charges are therefore related by

$$e^2 = 4\pi e_\mathrm{G}^2 = \frac{e_\mathrm{SI}^2}{\varepsilon_0^\mathrm{SI}} = 4\pi\alpha. \tag{A.5}$$

A formula is converted by replacing every e^2 of this book with $4\pi e_\mathrm{G}^2$ (to Gaussian) or $e_\mathrm{SI}^2/\varepsilon_0^\mathrm{SI}$ (to SI), and the reverse for the other direction [6, Appendix]. Dimensionless response functions, ε, ε^{-1} and L, have the same value in every system; only the formulas that express them through charges change. The plasma frequency, the dielectric matrix and the optical conductivity convert as follows.

The plasma frequency is

$$\omega_p^2 = \frac{e^2 n_e}{m_e} = \frac{4\pi e_\mathrm{G}^2 n_e}{m_e} = \frac{e_\mathrm{SI}^2 n_e}{\varepsilon_0^\mathrm{SI} m_e},$$

in the three systems; all three give 16.6 eV for the four valence electrons per atom of silicon, $n_e = 2.00\times10^{23}\,\mathrm{cm}^{-3}$ at $a = 5.431$ Å (Sect. 4.3). The dielectric matrix is $\varepsilon_{\boldsymbol{GG'}} = \delta_{\boldsymbol{GG'}} - v_\mathrm{C}(|\boldsymbol{q}+\boldsymbol{G}|)\tilde{R}_{\boldsymbol{GG'}}$ (Appendix A.7); in the Gaussian sources of the local-field literature [7, 8] the same expression appears with $4\pi e_\mathrm{G}^2/|\boldsymbol{q}+\boldsymbol{G}|^2$, and the polarizability there is our $\tilde{R}$ up to the sign and the factor $1/\mathcal{V}$ that each source fixes in its own definition. The optical conductivity enters through $\varepsilon = 1 + \mathrm{i}\sigma_\mathrm{cond}/\omega$, so that $\sigma_1 = \omega\varepsilon_2$; the Gaussian form is $\varepsilon = 1 + 4\pi\mathrm{i}\sigma_\mathrm{cond}/\omega$ and the SI form $\varepsilon = 1 + \mathrm{i}\sigma_\mathrm{cond}/(\varepsilon_0^\mathrm{SI}\omega)$, so a measured conductivity σ_1^SI in $\mathrm{S\,m^{-1}}$ gives $\varepsilon_2 = \sigma_1^\mathrm{SI}/(\varepsilon_0^\mathrm{SI}\omega)$. The prefactor $8\pi^2\alpha/(\Omega\omega^2)$ of the independent-particle ε_2 in the optical limit (Sect. 4.4) reads $8\pi^2 e_\mathrm{G}^2/(\Omega\omega^2)$ in the Gaussian sources, and the two agree under Eq. (A.5).

Fields convert as $\boldsymbol{E} = \boldsymbol{E}_\mathrm{G}/\sqrt{4\pi} = (\varepsilon_0^\mathrm{SI})^{1/2}\boldsymbol{E}_\mathrm{SI}$, so that $e\boldsymbol{E} = e_\mathrm{G}\boldsymbol{E}_\mathrm{G} = e_\mathrm{SI}\boldsymbol{E}_\mathrm{SI}$ and the force on a charge is the same in every system. Magnetic fields convert in the same way, $\boldsymbol{B} = \boldsymbol{B}_\mathrm{G}/\sqrt{4\pi} = (\varepsilon_0^\mathrm{SI})^{1/2}\boldsymbol{B}_\mathrm{SI}$ with $c = 1$. The Bohr magneton is then $\mu_\mathrm{B} = e/(2m_e)$, and the spin coupling of an electron is $g_e\mu_\mathrm{B}\,\boldsymbol{s}\cdot\boldsymbol{B}$, with $\boldsymbol{s}$ its spin and $g_e \simeq 2$; the product of μ_B and the field strength is an energy with the same value in every system, 5.788×10^{-5} eV per tesla.

The symbol ε_0 alone is the static dielectric constant of a crystal, lattice included (Sect. 5.5); the vacuum permittivity appears only as $\varepsilon_0^\mathrm{SI}$, inside quoted SI formulas.

A.3 Fourier transforms and normalization

A function of position is transformed with a plus sign in the exponent,

$$O(\boldsymbol{q}) = \int \mathrm{d}^3 r\; \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\, O(\boldsymbol{r}), \qquad O(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 q}{(2\pi)^3}\; \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\, O(\boldsymbol{q}), \tag{A.6}$$

and a function of time with the same sign,

$$f(\omega) = \int \mathrm{d}t\; \mathrm{e}^{+\mathrm{i}\omega t} f(t), \qquad f(t) = \int \frac{\mathrm{d}\omega}{2\pi}\; \mathrm{e}^{-\mathrm{i}\omega t} f(\omega). \tag{A.7}$$

The factor $(2\pi)^{-1}$ per dimension sits with the inverse transform. Applied to the density of point electrons, $n_e(\boldsymbol{r}) = \sum_i \delta^3(\boldsymbol{r} - \boldsymbol{r}_i)$, Eq. (A.6) gives the density operator

$$n_e(\boldsymbol{q}) = \sum_i \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}, \tag{A.8}$$

and applied to the dark-matter–electron potential it gives $\tilde{V}(\boldsymbol{q}) = \int \mathrm{d}^3 x\, \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}} V(\boldsymbol{x})$, the form used in Eq. (1.32).

A.3.1 Sign of the spatial transform

The sign is fixed by the direction of $\boldsymbol{q}$, which is the momentum the dark matter gives to the target (Appendix A.4). Take dark-matter plane waves $\langle \boldsymbol{r}_\chi | \boldsymbol{p} \rangle = \mathcal{V}^{-1/2} \mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}_\chi}$ in a box of volume $\mathcal{V}$, and a potential $V(\boldsymbol{r}_\chi - \boldsymbol{r}_e)$ between the dark matter and one electron. Shifting the integration variable to $\boldsymbol{x} = \boldsymbol{r}_\chi - \boldsymbol{r}_e$,

$$\langle \boldsymbol{p} - \boldsymbol{q} | V(\boldsymbol{r}_\chi - \boldsymbol{r}_e) | \boldsymbol{p} \rangle = \frac{1}{\mathcal{V}} \int \mathrm{d}^3 r_\chi\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_\chi} V(\boldsymbol{r}_\chi - \boldsymbol{r}_e) = \frac{1}{\mathcal{V}}\, \tilde{V}(\boldsymbol{q})\, \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_e}. \tag{A.9}$$

With Eq. (A.6), the transform of the potential and the operator left on the target both carry $+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}$, and summed over electrons the target operator is $n_e(\boldsymbol{q})$ of Eq. (A.8). With the opposite spatial convention, $\tilde{V}$ would be evaluated at $-\boldsymbol{q}$ and the target operator would be $n_e(-\boldsymbol{q})$. Single-particle form factors follow the same rule: $f_{i\to f}(\boldsymbol{q}) = \langle f | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | i \rangle$ [9, Eq. (3.10)]. Ref. [5, Eq. (S.4)] uses the other spatial convention with the same direction of $\boldsymbol{q}$, and writes the density operator as $\sum_i \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$; Ref. [10, Eq. (4)] keeps $\mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}$ on the target by writing the potential's transform at $-\boldsymbol{q}$. Appendix A.4 gives the condition under which these differences matter.

The time convention fixes where retarded functions are analytic. A response that starts at $t = 0$ and oscillates at a transition energy Ω transforms as $-\mathrm{i}\int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}(\omega-\Omega)t - 0^+ t} = 1/(\omega - \Omega + \mathrm{i}0)$. Its pole lies in the lower half of the complex ω plane, so retarded functions are analytic for $\mathrm{Im}\,\omega > 0$, and their denominators carry $\omega - (E_f - E_i) + \mathrm{i}0$ (Sect. 4.1). A time-dependent perturbation of frequency $\omega > 0$ is $\propto \mathrm{e}^{-\mathrm{i}\omega t}$, the convention of quantum mechanics; engineering texts use $\mathrm{e}^{+\mathrm{i}\omega t}$ (written $\mathrm{e}^{j\omega t}$), and their dielectric functions are the complex conjugates of ours, with $\mathrm{Im}\,\varepsilon < 0$ for absorption. Here $\varepsilon = \varepsilon_1 + \mathrm{i}\varepsilon_2$ with $\varepsilon_2 > 0$ for an absorbing medium at $\omega > 0$.

A.3.2 Box normalization

States are normalized in a box of volume $\mathcal{V}$ with periodic boundary conditions, and the limit $\mathcal{V} \to \infty$ is taken at the end. Allowed wavevectors form a grid of density $\mathcal{V}/(2\pi)^3$, so sums and Kronecker deltas become integrals and delta functions by

$$\sum_{\boldsymbol{q}} \;\longleftrightarrow\; \mathcal{V} \int \frac{\mathrm{d}^3 q}{(2\pi)^3}, \qquad \delta_{\boldsymbol{q}\boldsymbol{q}'} \;\longleftrightarrow\; \frac{(2\pi)^3}{\mathcal{V}}\, \delta^3(\boldsymbol{q} - \boldsymbol{q}'). \tag{A.10}$$

The second follows from the first, since $\sum_{\boldsymbol{q}} \delta_{\boldsymbol{q}\boldsymbol{q}'} = 1$. The box contains N unit cells of volume Ω, so $\mathcal{V} = N\Omega$, and a uniform grid of crystal momenta in the first Brillouin zone has N points (Sect. 3.1). Sums over the zone become

$$\frac{1}{N}\sum_{\boldsymbol{k}\in\mathrm{BZ}} \longleftrightarrow \Omega\int_{\mathrm{BZ}} \frac{\mathrm{d}^3k}{(2\pi)^3}, \qquad \langle\!\langle \cdots \rangle\!\rangle \equiv \frac{1}{N^2}\sum_{\boldsymbol{k},\boldsymbol{Q}}(\cdots) = \Omega^2\int_{\mathrm{BZ}} \frac{\mathrm{d}^3k\,\mathrm{d}^3Q}{(2\pi)^6}\,(\cdots), \tag{A.11}$$

where the double bracket is the zone average over an electron momentum $\boldsymbol{k}$ and a phonon momentum $\boldsymbol{Q}$, used for phonon-assisted processes (Sect. 6.5).

The reason for this bookkeeping is that every intensive quantity must come out independent of N, and the powers of N and $\mathcal{V}$ make that check mechanical. The structure factor of Sect. 4.2 has a prefactor $2\pi/\mathcal{V}$, an independent-particle sum $\sum_{\boldsymbol{k}}$ contributes a factor N, and the density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ is of order one (Appendix A.5). Their product is $2\pi N/\mathcal{V} = 2\pi/\Omega$, independent of the size of the crystal, as an intensive response must be. A factor of N or $\sqrt{N}$ that survives this count signals an error in a normalization. In the same way the rate per unit target mass contains $1/\rho_T$ and no N (Appendix A.8).

A.3.3 Lattice-periodic functions

A function with the periodicity of the lattice, such as the crystal potential or the periodic part of a Bloch state, has components only at reciprocal-lattice vectors. With Eq. (A.6) restricted to one cell,

$$f_{\boldsymbol{G}} = \frac{1}{\Omega}\int_{\Omega} \mathrm{d}^3r\; \mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}} f(\boldsymbol{r}), \qquad f(\boldsymbol{r}) = \sum_{\boldsymbol{G}} f_{\boldsymbol{G}}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}, \tag{A.12}$$

and the transform over the whole crystal is

$$f(\boldsymbol{q}) = \mathcal{V}\sum_{\boldsymbol{G}} f_{\boldsymbol{G}}\,\delta_{\boldsymbol{q},\boldsymbol{G}} \;\longrightarrow\; (2\pi)^3\sum_{\boldsymbol{G}} f_{\boldsymbol{G}}\,\delta^3(\boldsymbol{q}-\boldsymbol{G}),$$

by Eq. (A.10). Plane-wave codes expand in the opposite direction, with coefficients $c_{n\boldsymbol{k}}(\boldsymbol{G})$ of $\mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$ in $u_{n\boldsymbol{k}}(\boldsymbol{r})$ [9, Eq. (3.14)]. A code coefficient $c(\boldsymbol{G})$ is therefore our component at $-\boldsymbol{G}$, and a formula that combines code coefficients with the components of Eq. (A.12) needs $\boldsymbol{G}\to-\boldsymbol{G}$ in one of them. For a real function, $f_{-\boldsymbol{G}} = f^*_{\boldsymbol{G}}$, and the distinction only conjugates; for a Bloch function it does not. Ref. [11, Appendix A] states the transforms between the direct and the reciprocal lattice that the electron–phonon literature uses.

A.3.4 Crystal momenta and reciprocal-lattice matrices

In a crystal, a momentum transfer $\boldsymbol{q}$ is split into a first-zone part and a reciprocal-lattice vector, $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$ with $\boldsymbol{q}_r$ in the first Brillouin zone (Sect. 3.1). Response functions are matrices in reciprocal-lattice vectors, such as $(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}_r,\omega)$: the transform of a function of two positions $\boldsymbol{r}$ and $\boldsymbol{r}'$ with $\mathrm{e}^{+\mathrm{i}(\boldsymbol{q}_r+\boldsymbol{G})\cdot\boldsymbol{r}}$ and $\mathrm{e}^{-\mathrm{i}(\boldsymbol{q}_r+\boldsymbol{G}')\cdot\boldsymbol{r}'}$. The book indexes such a matrix by the unreduced $\boldsymbol{q}$ at $\boldsymbol{G}=0$:

$$(\varepsilon^{-1})_{00}(\boldsymbol{q},\omega) \equiv (\varepsilon^{-1})_{\boldsymbol{G}_q\boldsymbol{G}_q}(\boldsymbol{q}_r,\omega), \tag{A.13}$$

and likewise for every reciprocal-lattice matrix. Thus $(\varepsilon^{-1})_{00}(\boldsymbol{q},\omega)$ at a momentum transfer $\boldsymbol{q}$ outside the first zone is the diagonal element of the inverse dielectric matrix at $\boldsymbol{q}_r$ in the row and column of $\boldsymbol{G}_q$. Both forms appear in the literature, often for the same object; Ref. [4, Eq. (24)] writes the second with an explicit selection $\delta_{\boldsymbol{k},|\boldsymbol{q}+\boldsymbol{G}|}$. The momentum transfer $\boldsymbol{q}$ itself is never folded: it is the momentum the dark matter loses, and the dark-matter kinematics of Sect. 1.3 depend on $|\boldsymbol{q}|$, not on $|\boldsymbol{q}_r|$.

A.4 Kinematic signs

The momentum $\boldsymbol{q}$ is transferred from the dark matter to the target: a dark-matter particle of momentum $\boldsymbol{p}$ leaves with $\boldsymbol{p} - \boldsymbol{q}$. The energy ω is the energy given to the target, so a final target state has $E_f = E_i + \omega$. For a dark-matter particle of laboratory velocity $\boldsymbol{v}$ that changes its mass by $\Delta_\chi = m_{\chi 2} - m_{\chi 1}$ in the collision, energy conservation gives

$$\omega = \omega_{\boldsymbol{q}} \equiv \boldsymbol{q} \cdot \boldsymbol{v} - \frac{q^2}{2m_\chi} - \Delta_\chi, \tag{A.14}$$

as in Eq. (1.22). The minimum velocity needed for a momentum transfer $\boldsymbol{q}$ and an energy deposit ω is $v_{\min}(q,\omega) = |(\omega + \Delta_\chi)/q + q/(2m_\chi)|$ (Eq. (1.23)). The scattering is endothermic for $\Delta_\chi > 0$ and exothermic for $\Delta_\chi < 0$; elastic scattering has $\Delta_\chi = 0$. The component of the dark-matter velocity along $\hat{\boldsymbol{q}}$ that the delta function selects is $(\omega + \Delta_\chi)/q + q/(2m_\chi)$, and the speed integral of every rate depends on $(\boldsymbol{q},\omega)$ only through it and through $\hat{\boldsymbol{q}}$ (Sect. 1.3).

A.4.1 Negative energy transfer

At zero temperature the target starts in its ground state and $\omega \geq 0$. At $T > 0$ it starts in a thermal mixture and can give energy to the dark matter, so $\omega < 0$ is allowed. Such transfers are not added by hand: they are part of the structure factor, which at negative ω is fixed by its value at positive ω through detailed balance, $S(-\boldsymbol{q},-\omega) = \mathrm{e}^{-\beta\omega} S(\boldsymbol{q},\omega)$ (Appendix A.7). A negative ω is therefore suppressed by the Boltzmann factor of the target excitation it de-excites. For electronic excitations across a gap of 1 eV at the 0.1 K or less of a cryogenic detector, $\beta\omega$ exceeds 10^5, and $\omega < 0$ is dropped; for phonons, magnons and other excitations of a few meV in a target at a few kelvin, it is not. The book does not restrict ω to positive values; an expression that holds only for $\omega > 0$ says so.

A.4.2 Direction of the momentum transfer

The direction convention for $\boldsymbol{q}$ differs between papers. Some take $\boldsymbol{q}$ as the momentum gained by the dark matter; Ref. [10, below Eq. (12)] lists papers on each side. Others keep our direction but transform with the opposite sign and write the target vertex as $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ [5, Eq. (S.4)]. Each choice replaces $S(\boldsymbol{q},\omega)$ by $S(-\boldsymbol{q},\omega)$.

The two agree whenever the target responds in the same way at $\boldsymbol{q}$ and $-\boldsymbol{q}$. Inversion symmetry gives this directly, since it maps $n_e(\boldsymbol{q})$ to $n_e(-\boldsymbol{q})$ and leaves the Hamiltonian unchanged. Time-reversal symmetry gives it too: the antiunitary time-reversal operator maps $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ to $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, and the thermal ensemble of a time-reversal-invariant target is mapped to itself. Each term $|\langle f|n_e(\boldsymbol{q})|i\rangle|^2$ of the structure factor then equals $|\langle \tilde{f}|n_e(-\boldsymbol{q})|\tilde{i}\rangle|^2$, with $|\tilde{i}\rangle$ and $|\tilde{f}\rangle$ the time-reversed states, which have the same energies, and the sum over the mapped states is the same sum, so $S(\boldsymbol{q},\omega) = S(-\boldsymbol{q},\omega)$. Silicon, germanium and GaAs are time-reversal invariant, and silicon and germanium also have inversion symmetry. A crystal with neither, for example a magnetically ordered crystal without an inversion center, can have $S(\boldsymbol{q},\omega) \neq S(-\boldsymbol{q},\omega)$, and a daily modulation computed with the wrong sign of $\boldsymbol{q}$ is then wrong. A formula written in the other convention becomes ours under $\boldsymbol{q} \to -\boldsymbol{q}$ in every target quantity.

The momentum transfer $\boldsymbol{q}$ is never reduced to the first Brillouin zone (Appendix A.3). Its first-zone part $\boldsymbol{q}_r$ labels the crystal-momentum change of the target, and its reciprocal-lattice part $\boldsymbol{G}_q$ labels an Umklapp component (Appendix A.3 and Sect. 3.1); both are needed, because $\omega_{\boldsymbol{q}}$ and $v_{\min}$ depend on the full $\boldsymbol{q}$. The phonon wavevector is a different object, written $\boldsymbol{Q}$ and always in the first zone (Appendix A.6).

A.5 Bloch and Wannier conventions

A Bloch state is normalized to one over the whole crystal, and its periodic part to one over a unit cell:

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{N}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\, u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad \int_V \mathrm{d}^3 r\, |\psi_{n\boldsymbol{k}}|^2 = 1, \qquad \int_\Omega \mathrm{d}^3 r\, |u_{n\boldsymbol{k}}|^2 = 1. \tag{A.15}$$

The two normalizations are consistent: $|\psi_{n\boldsymbol{k}}|^2 = |u_{n\boldsymbol{k}}|^2/N$ is periodic, and $\mathscr{V}$ holds N cells. This is the convention of the review of first-principles electron–phonon calculations, Ref. [11, Eq. (29)]. Ref. [9, Eqs. (3.14) and (3.15)] writes the same states as $\mathscr{V}^{-1/2}\sum_{\boldsymbol{G}} u_n(\boldsymbol{k}+\boldsymbol{G})\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}$ with $\sum_{\boldsymbol{G}}|u_n(\boldsymbol{k}+\boldsymbol{G})|^2 = 1$; its plane-wave coefficients are the Fourier components of our $u_{n\boldsymbol{k}}$ times $\Omega^{1/2}$.

A.5.1 Periodic gauge

A Bloch state is fixed only up to a phase, $\psi_{n\boldsymbol{k}} \to \mathrm{e}^{\mathrm{i}\theta_n(\boldsymbol{k})}\psi_{n\boldsymbol{k}}$, and for degenerate or entangled bands up to a unitary mixing [12, Eqs. (6), (7) and (10)]. We fix one part of this freedom everywhere: states at $\boldsymbol{k}$ and at $\boldsymbol{k}+\boldsymbol{G}$ are the same state,

$$\psi_{n,\boldsymbol{k}+\boldsymbol{G}}(\boldsymbol{r}) = \psi_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad \text{equivalently} \qquad u_{n,\boldsymbol{k}+\boldsymbol{G}}(\boldsymbol{r}) = \mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}\, u_{n\boldsymbol{k}}(\boldsymbol{r}). \tag{A.16}$$

This *periodic gauge* lets a crystal momentum outside the first zone be used directly, without bookkeeping of the reciprocal-lattice vector it differs by. It is the gauge in which Wannier functions are built [12, Sect. II A, footnote 1, and Eq. (10)].

A.5.2 The density vertex

The matrix element of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ between Bloch states, the density vertex, is

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) \equiv \langle\psi_{m,\boldsymbol{k}+\boldsymbol{q}}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_{n\boldsymbol{k}}\rangle = \int_\Omega \mathrm{d}^3 r\; u^*_{m,\boldsymbol{k}+\boldsymbol{q}}(\boldsymbol{r})\, u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad \rho_{nn}(\boldsymbol{k},0) = 1. \tag{A.17}$$

The second form follows from Eq. (A.15): the phases $\mathrm{e}^{-\mathrm{i}(\boldsymbol{k}+\boldsymbol{q})\cdot\boldsymbol{r}}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ cancel, the integrand is periodic, and the factor $1/N$ cancels the N cells. When $\boldsymbol{k}+\boldsymbol{q}$ lies outside the first zone it is folded back, $\boldsymbol{k}+\boldsymbol{q} = \boldsymbol{k}'+\boldsymbol{G}$ with $\boldsymbol{k}'$ in the zone, and by Eq. (A.16) the vertex becomes $\int_\Omega u^*_{m\boldsymbol{k}'}\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}u_{n\boldsymbol{k}}$: the Umklapp vector stays in the plane wave. The vertex is of order one and carries no factor of N. Together with the prefactor $2\pi/\mathscr{V}$ of the structure factor and one sum over $\boldsymbol{k}$, this makes S independent of the size of the crystal (Appendix A.3). A vertex defined with an extra factor N, or a missing $1/\sqrt{N}$ in Eq. (A.15), would make S grow with N. In the notation of Ref. [9, Eq. (3.16)], $\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) = f_{[n\boldsymbol{k},m\boldsymbol{k}',\boldsymbol{G}]}$ with $\boldsymbol{q} = \boldsymbol{k}'-\boldsymbol{k}+\boldsymbol{G}$; Ref. [4, Eq. (B2)] defines the same integral over the unit cell, divided by Ω, for u normalized to Ω rather than to one.

At small q and $m \neq n$, the vertex reduces to a velocity matrix element. Expanding $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \simeq 1 + \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}$ and using $\langle m|\boldsymbol{r}|n\rangle = -\mathrm{i}\,\boldsymbol{v}_{mn}/(\varepsilon_m - \varepsilon_n)$, which follows from $\mathrm{i}[H,\boldsymbol{r}] = \boldsymbol{v}$ for $m \neq n$,

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) \xrightarrow{q\to 0} \frac{\boldsymbol{q}\cdot\boldsymbol{v}_{mn}(\boldsymbol{k})}{\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}}, \qquad m \neq n, \tag{A.18}$$

with $\boldsymbol{v}_{mn}(\boldsymbol{k}) = \langle\psi_{m\boldsymbol{k}}|\boldsymbol{v}|\psi_{n\boldsymbol{k}}\rangle$ (Sect. 3.2). The velocity is $\boldsymbol{p}/m_e$ for a local potential; a nonlocal pseudopotential adds $\mathrm{i}[V_{\mathrm{NL}},\boldsymbol{r}]$, where V_{NL} is its nonlocal part, and the momentum matrix elements of a code must then be corrected before they are used as velocities (Appendix E.7).

A.5.3 Gauge consistency

Under a change of gauge $\psi_{n\boldsymbol{k}} \to \mathrm{e}^{\mathrm{i}\theta_n(\boldsymbol{k})}\psi_{n\boldsymbol{k}}$, the vertex and the velocity transform as

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) \to \mathrm{e}^{\mathrm{i}[\theta_n(\boldsymbol{k})-\theta_m(\boldsymbol{k}+\boldsymbol{q})]}\rho_{mn}(\boldsymbol{k},\boldsymbol{q}), \qquad \boldsymbol{v}_{mn}(\boldsymbol{k}) \to \mathrm{e}^{\mathrm{i}[\theta_n(\boldsymbol{k})-\theta_m(\boldsymbol{k})]}\boldsymbol{v}_{mn}(\boldsymbol{k}), \tag{A.19}$$

consistently with Eq. (A.18). A first-order rate contains $|\rho_{mn}|^2$ and does not depend on the gauge. A second-order amplitude does not have this protection: it sums products such as $g_{cm\nu}\rho_{mv}$ over intermediate bands m (Sect. 6.5), and each product is gauge-covariant only if both factors were computed from the same set of states. Every quantity that enters one amplitude, the vertex ρ, the electron–phonon vertex g of Appendix A.6 and any tensor interpolated with Wannier functions, is therefore computed in one gauge, with one set of gauge matrices $U(\boldsymbol{k})$. Codes that diagonalize at each $\boldsymbol{k}$ independently assign random phases, and independent runs for ρ and g do not share them. Projector-augmented-wave and ultrasoft inputs need the augmentation term in ρ at every $\boldsymbol{q}$, or $\rho_{nn}(\boldsymbol{k},0) = 1$ fails (Sect. 3.5).

A.5.4 Wannier functions and band labels

Wannier functions are built from the Bloch states of one gauge [12, Eqs. (10) and (14)],

$$w_n(\boldsymbol{r}-\boldsymbol{R}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}\in\mathrm{BZ}} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} \sum_m U_{mn}(\boldsymbol{k})\, \psi_{m\boldsymbol{k}}(\boldsymbol{r}). \tag{A.20}$$

They are orthonormal over the crystal, and w_n has dimension $\Omega^{-1/2}$ when it is localized within a few cells. The real-space Hamiltonian $H^W(\boldsymbol{R})$ and the interpolated vertices are the matrix elements between them (Sect. 3.6).

With spin–orbit coupling, Bloch states are two-component spinors, $u_{n\boldsymbol{k}}$ carries a spin index, and Eq. (A.17) includes the sum over it; each band then holds one electron per $\boldsymbol{k}$, and the spin factor 2 of the independent-particle formulas is dropped (Appendix A.7). Without spin–orbit coupling the states are spin-degenerate, the band index runs over spatial bands only, and the factor 2 is written explicitly. A degenerate subspace has no preferred basis, so only quantities summed over it, such as $\sum_m |\rho_{mn}|^2$ over the degenerate partners m, are compared between codes.

Band indices are n for a generic band and m for an intermediate one. The letters v and c label occupied (valence) and empty (conduction) bands and appear only as subscripts, $\varepsilon_{v\boldsymbol{k}}$ or ρ_{cv}; bare v is the dark-matter velocity, and c is the speed of light, set to one.

A.6 Phonon conventions

The displacement of nucleus κ in the cell at $\boldsymbol{R}$ is expanded in normal modes as

$$\boldsymbol{u}_{\boldsymbol{R}\kappa} = \sum_{\boldsymbol{Q}\nu} \frac{1}{\sqrt{2NM_\kappa\omega_{\boldsymbol{Q}\nu}}}\, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, B_{\boldsymbol{Q}\nu}, \qquad B_{\boldsymbol{Q}\nu} = b_{\boldsymbol{Q}\nu} + b^\dagger_{-\boldsymbol{Q}\nu}, \tag{A.21}$$

with $\boldsymbol{Q}$ in the first Brillouin zone, ν the branch, M_κ the nuclear mass and $[b_{\boldsymbol{Q}\nu}, b^\dagger_{\boldsymbol{Q}'\nu'}] = \delta_{\boldsymbol{Q}\boldsymbol{Q}'}\delta_{\nu\nu'}$. The polarization vectors are normalized over the basis and obey

$$\sum_\kappa |\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2 = 1, \qquad \boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*, \qquad \omega_{-\boldsymbol{Q}\nu} = \omega_{\boldsymbol{Q}\nu}. \tag{A.22}$$

These are the conventions of Refs. [11, Eqs. (16)–(18)] and [10, Eq. (29)]. The second relation, a choice of phase, makes $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ Hermitian: the terms at $\boldsymbol{Q}$ and $-\boldsymbol{Q}$ are conjugates of each other.

Ref. [11] and most of the phonon literature write the wavevector as $\boldsymbol{q}$ and the operators as $a_{\boldsymbol{q}\nu}$. In this book $\boldsymbol{q}$ is the dark-matter momentum transfer and $a_{\boldsymbol{k}\nu}$ is reserved for magnons (Sect. 7.1), so sources are converted by $\boldsymbol{q} \to \boldsymbol{Q}$ and $a \to b$.

The frequencies and polarization vectors are the eigenvalues and eigenvectors of the dynamical matrix,

$$\sum_{\kappa'\beta} D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q})\, e^{\beta}_{\kappa'\nu}(\boldsymbol{Q}) = \omega^2_{\boldsymbol{Q}\nu}\, e^{\alpha}_{\kappa\nu}(\boldsymbol{Q}), \qquad D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q}) = \frac{1}{\sqrt{M_\kappa M_{\kappa'}}} \sum_{\boldsymbol{R}} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}, \tag{A.23}$$

where $\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})$ is the second derivative of the Born–Oppenheimer energy with respect to the displacements $u^{\alpha}_{\boldsymbol{0}\kappa}$ and $u^{\beta}_{\boldsymbol{R}\kappa'}$, and α,β are Cartesian indices (Sect. 5.2) [11, Eq. (14)]. The sign of the exponent matches Eq. (A.21): a displacement pattern $\propto \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$ is an eigenvector of $D^{\mathrm{dyn}}(\boldsymbol{Q})$, not of $D^{\mathrm{dyn}}(-\boldsymbol{Q})$. A code that uses the opposite sign exchanges $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$ and $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$; by Eq. (A.22) this maps $\boldsymbol{Q}$ to $-\boldsymbol{Q}$ and must be undone before polarization vectors from such a code are combined with vertices from another.

A.6.1 Zero-point length

The scale of the displacement is the zero-point length of a mode,

$$\ell_{\boldsymbol{Q}\nu} \equiv \frac{1}{\sqrt{2M_{\mathrm{cell}}\,\omega_{\boldsymbol{Q}\nu}}}, \tag{A.24}$$

with $M_{\mathrm{cell}} = \sum_\kappa M_\kappa$ and the factor $1/\sqrt{N}$ kept outside. For an acoustic mode at small Q all nuclei move together, $\boldsymbol{e}_{\kappa\nu} = (M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$ with a unit vector $\hat{\boldsymbol{e}}_\nu$, which satisfies Eq. (A.22). The amplitude in Eq. (A.21) is then $(2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{-1/2}(M_\kappa/M_{\mathrm{cell}})^{1/2} = \ell_{\boldsymbol{Q}\nu}/\sqrt{N}$ for every κ, so one mode displaces each nucleus by $\ell_{\boldsymbol{Q}\nu}\hat{\boldsymbol{e}}_\nu/\sqrt{N}$ per unit of $B_{\boldsymbol{Q}\nu}$. Keeping $1/\sqrt{N}$ outside $\ell_{\boldsymbol{Q}\nu}$ matches the vertex normalization below, in which all such factors are explicit. Ref. [11, Eq. (34)] writes the same scale with an arbitrary reference mass M_0 and the ratio $(M_0/M_\kappa)^{1/2}$.

A.6.2 The electron–phonon vertex

The first-order coupling of electrons to phonons has the form

$$H_{\mathrm{ep}} = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}\boldsymbol{Q}} \sum_{mn\nu} g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})\, c^{\dagger}_{m,\boldsymbol{k}+\boldsymbol{Q}} c_{n\boldsymbol{k}}\, B_{\boldsymbol{Q}\nu}. \tag{A.25}$$

The vertex is

$$g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q}) = \frac{1}{\sqrt{2\omega_{\boldsymbol{Q}\nu}}} \sum_{\kappa} \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})}{\sqrt{M_\kappa}} \cdot \langle \psi_{m,\boldsymbol{k}+\boldsymbol{Q}} | \partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}} | \psi_{n\boldsymbol{k}} \rangle. \tag{A.26}$$

Here $\partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}} = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, \partial V_{\mathrm{KS}}/\partial \boldsymbol{u}_{\boldsymbol{R}\kappa}$ is the change of the self-consistent Kohn–Sham potential for a displacement wave of nucleus κ. It is computed with density-functional perturbation theory and therefore screened by the electrons (Sect. 6.2 and Appendix E.4). The matrix element, normalized as in Eq. (A.15), is of order one in N, and so is g; the single factor $1/\sqrt{N}$ of Eq. (A.25) carries the size dependence. This is the normalization of Refs. [11, Eqs. (37) and (38)] and [13, Eq. (2)], and it is what the EPW code outputs, so computed vertices enter the formulas of this book without rescaling. The vertex has dimension of energy.

Hermiticity of H_{ep} fixes the conjugate vertex. The Hermitian conjugate of the term $(\boldsymbol{k},\boldsymbol{Q},m,n)$ is $g^*_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})\, c^{\dagger}_{n\boldsymbol{k}} c_{m,\boldsymbol{k}+\boldsymbol{Q}} B^{\dagger}_{\boldsymbol{Q}\nu}$, and $B^{\dagger}_{\boldsymbol{Q}\nu} = B_{-\boldsymbol{Q}\nu}$. It must equal the term $(\boldsymbol{k}+\boldsymbol{Q},-\boldsymbol{Q},n,m)$ of the sum, so

$$g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})^* = g_{nm\nu}(\boldsymbol{k}+\boldsymbol{Q},-\boldsymbol{Q}), \tag{A.27}$$

which also follows from Eq. (A.26) with $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$ and $(\partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}})^\dagger = \partial_{\kappa,-\boldsymbol{Q}} V_{\mathrm{KS}}$. The vertex is split into a short-range and a long-range part, $g = g^{\mathrm{S}} + g^{\mathrm{L}}$, in polar crystals, where g^{L} is the Fröhlich term (Sect. 6.2). Both carry the gauge of the electronic states, and the rule of Appendix A.5 applies: g and ρ in one amplitude come from one set of states.

A.6.3 Emission and absorption

In Eq. (A.25) the electron gains crystal momentum $\boldsymbol{Q}$. It does so either by absorbing a phonon $(\boldsymbol{Q}, \nu)$, through $b_{\boldsymbol{Q}\nu}$, or by emitting a phonon $(-\boldsymbol{Q}, \nu)$, through $b^\dagger_{-\boldsymbol{Q}\nu}$; both use the same vertex $g(\cdot, \boldsymbol{Q})$. With $n_{\boldsymbol{Q}\nu} = n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$ phonons in the mode, $n_{\mathrm{B}}(x) = (\mathrm{e}^{\beta x} - 1)^{-1}$, the squared matrix elements $|\langle n+1|b^\dagger|n\rangle|^2 = n+1$ and $|\langle n-1|b|n\rangle|^2 = n$ give the thermal weights

$$n_{\boldsymbol{Q}\nu} + \tfrac{1}{2} \pm \tfrac{1}{2}, \tag{A.28}$$

with the upper sign for emission of $-\boldsymbol{Q}$ and the lower for absorption of $+\boldsymbol{Q}$. The energy given to the lattice is $\pm\omega_{\boldsymbol{Q}\nu}$ with the same sign convention. At $T = 0$ only emission survives.

A.6.4 Second order and the Debye–Waller factor

The coupling at second order in the displacements has a vertex quadratic in B, which we call the Allen–Heine vertex g^{AH}; its self-energy is Σ^{AH} (Sect. 6.3). Ref. [11, Eqs. (39) and (40)] calls the same vertex g^{DW}, a "Debye–Waller" vertex. This book reserves "Debye–Waller" for one object only, the reduction of the zero-phonon weight of a momentum transfer $\boldsymbol{q}$ to nucleus κ by the factor $\mathrm{e}^{-2W_\kappa(\boldsymbol{q})}$, with

$$2W_\kappa(\boldsymbol{q}) = \boldsymbol{q} \cdot B_\kappa \cdot \boldsymbol{q}, \qquad B_\kappa = \langle \boldsymbol{u}_\kappa \boldsymbol{u}_\kappa \rangle, \tag{A.29}$$

where B_κ is the thermal displacement tensor, isotropic only at a site of cubic symmetry (Sect. 5.6). The two objects share B_κ and appear next to the same double commutator, that of the first-moment sum rule Eq. (4.28), in different roles. In Sect. 5.6 the factor $\mathrm{e}^{-2W_\kappa(\boldsymbol{q})}$ is the elastic share of the scattering weight, whose mean energy the sum rule fixes at the free recoil energy. In Sect. 6.3 the commutator, taken for a rigid translation, makes Σ^{AH} cancel the Fan–Migdal term for long-wavelength acoustic phonons.

A.7 Response conventions

Response functions are written R, never χ, because χ is the dark-matter particle. For two target operators $O_a(\boldsymbol{q})$ and $O_b(\boldsymbol{q})$, the retarded response and the response matrix are

$$R_{ab}(\boldsymbol{q}, \omega) = -\frac{\mathrm{i}}{\mathscr{V}} \int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} \,\langle [O_a(\boldsymbol{q}, t)^\dagger, O_b(\boldsymbol{q}, 0)] \rangle, \tag{A.30}$$

$$S_{ab}(\boldsymbol{q}, \omega) = \frac{2\pi}{\mathscr{V}} \sum_{i,f} p_i \,\langle i|O_a(\boldsymbol{q})^\dagger|f\rangle\langle f|O_b(\boldsymbol{q})|i\rangle\, \delta(\omega - E_f + E_i), \tag{A.31}$$

with Heisenberg operators $O(t) = \mathrm{e}^{\mathrm{i}Ht} O \mathrm{e}^{-\mathrm{i}Ht}$, thermal averages with Gibbs weights $p_i = \mathrm{e}^{-\beta E_i}/\mathcal{Z}$, and the time convention of Appendix A.3. The dynamic structure factor of the electron density is the diagonal element $S \equiv S_{ee}$ with $O_e = n_e(\boldsymbol{q})$ (Sects. 4.1 and 4.2). Both carry the factor $1/\mathscr{V}$, so both are intensive, and for density operators both have dimension eV^2. Ref. [4, Eqs. (2) and (4)] defines the same objects for the density, with χ for R, $n_{-\boldsymbol{k}}$ for the operator inside S and $\boldsymbol{k}$ for the momentum transfer. The literature's χ, Π and W for response, polarization and loss are listed in Appendix B.2.

Inserting eigenstates of the target into Eq. (A.30) gives its spectral (Lehmann) form,

$$R_{ab}(\boldsymbol{q},\omega) = \frac{1}{\mathcal{V}} \sum_{i,f} p_i \left[\frac{\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle}{\omega - (E_f - E_i) + \mathrm{i}0} - \frac{\langle i|O_b|f\rangle\langle f|O_a^\dagger|i\rangle}{\omega + (E_f - E_i) + \mathrm{i}0} \right], \tag{A.32}$$

with every pole below the real axis, as Appendix A.3 requires. For a diagonal element at $T = 0$ and $\omega > 0$ only the resonant term has an imaginary part, $-\pi\delta(\omega - E_f + E_i)$ times a positive weight, so $\operatorname{Im} R_{aa} < 0$ for $\omega > 0$. The sign is a consequence of the time convention: with $\mathrm{e}^{+\mathrm{i}\omega t}$ in the inverse transform it would be reversed. Analyticity in the upper half plane gives the Kramers–Kronig relations, and they hold for R, for $\varepsilon^{-1} - 1$ and, in a crystal with a gap, for $\varepsilon - 1$ (Sects. 4.1 and 4.4).

A.7.1 Fluctuation–dissipation theorem and detailed balance

Inserting energy eigenstates into Eqs. (A.30) and (A.31) and using $\int_{-\infty}^{\infty} \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} \langle O_a^\dagger(t) O_b\rangle = V S_{ab}(\boldsymbol{q},\omega)$ and $\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} \langle O_b O_a^\dagger(t)\rangle = \mathrm{e}^{-\beta\omega} V S_{ab}(\boldsymbol{q},\omega)$ gives

$$(1 - \mathrm{e}^{-\beta\omega})\, S_{ab}(\boldsymbol{q},\omega) = \mathrm{i}\big[R_{ab}(\boldsymbol{q},\omega) - R_{ba}(\boldsymbol{q},\omega)^*\big], \qquad S_{aa} = -\frac{2\,\operatorname{Im} R_{aa}}{1 - \mathrm{e}^{-\beta\omega}}, \tag{A.33}$$

the fluctuation–dissipation theorem (Sect. 4.2). The diagonal form is Ref. [4, Eq. (3)]. For operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$, such as densities, exchanging i and f in Eq. (A.31) gives detailed balance,

$$S_{ab}(-\boldsymbol{q},-\omega) = \mathrm{e}^{-\beta\omega}\, S_{ba}(\boldsymbol{q},\omega). \tag{A.34}$$

At $T \ll \omega$ the factor $1 - \mathrm{e}^{-\beta\omega}$ is one.

A.7.2 Dielectric function and loss

The longitudinal dielectric matrix of a crystal, with local fields, is built from the Coulomb-proper response $\tilde{R}$. This is the part of the response that is irreducible with respect to one Coulomb line (Sect. 4.4), and

$$\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega) = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\, \tilde{R}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega), \tag{A.35}$$

with $\tilde{R}$ the response of the total charge in units of e and v_{C} from Eq. (A.3). This is the unsymmetrized form; the symmetrized form uses $v_{\mathrm{C}}^{1/2} \tilde{R}\, v_{\mathrm{C}}^{1/2}$, has the same diagonal and differs off the diagonal [4, footnote 3]. The inverse matrix, the macroscopic dielectric function and the loss function are

$$\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega) = \frac{1}{(\varepsilon^{-1})_{00}(\boldsymbol{q},\omega)}, \qquad L(\boldsymbol{q},\omega) = \operatorname{Im}\big[-(\varepsilon^{-1})_{00}(\boldsymbol{q},\omega)\big], \tag{A.36}$$

with the unreduced-$\boldsymbol{q}$ indexing of Eq. (A.13), $\varepsilon = \varepsilon_1 + \mathrm{i}\varepsilon_2$, and $L \geq 0$ for $\omega > 0$. For the electron channel alone, $\varepsilon_e^{-1} = 1 + v_{\mathrm{C}} R_{ee}$; it equals ε^{-1} when the nuclei are clamped. The proper part gives the absorptive part directly, $\varepsilon_2 = -v_{\mathrm{C}} \operatorname{Im} \tilde{R}$.

Combining $\varepsilon_e^{-1} = 1 + v_{\mathrm{C}} R_{ee}$ with Eq. (A.33) gives the relation between the structure factor and the loss function,

$$S(\boldsymbol{q},\omega) = \frac{2q^2}{e^2} \frac{L_e(\boldsymbol{q},\omega)}{1 - \mathrm{e}^{-\beta\omega}}, \qquad \varepsilon_2(\boldsymbol{q},\omega) = \frac{e^2}{2q^2}\,(1 - \mathrm{e}^{-\beta\omega})\, S^{\mathrm{prop}}(\boldsymbol{q},\omega), \tag{A.37}$$

where L_e is the loss of the electron channel and S^{prop} the structure factor built from $\tilde{R}_{ee}$. The first relation is Ref. [4, Eq. (6)], with $e^2 = 4\pi\alpha$. It holds with the total loss L at clamped nuclei, and for a dark photon, which couples to the total charge, it holds exactly with the total ε (Sect. 4.4). The optical conductivity is defined by $\varepsilon = 1 + \mathrm{i}\sigma_{\mathrm{cond}}/\omega$ in the transverse, long-wavelength limit, so that $\sigma_1 = \omega\varepsilon_2$ (Appendix A.2 gives the Gaussian and SI forms).

A.7.3 Spin factor

The structure factor and ε_2 are summed over spin. For a spin-independent vertex without spin–orbit coupling, each band is doubly occupied, and the independent-particle form carries an explicit factor 2:

$$S(\boldsymbol{q},\omega) = \frac{4\pi}{\mathscr{V}} \sum_{v,c,\boldsymbol{k}} |\rho_{cv}(\boldsymbol{k},\boldsymbol{q})|^2 \, \delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}) \qquad (T=0), \tag{A.38}$$

where $4\pi/\mathscr{V}$ is $2 \times 2\pi/\mathscr{V}$ and the sums run over spatial bands. With spin–orbit coupling the factor 2 is dropped and the sums run over spinor bands. Refs. [9, Eq. (A.29)] and [14, Eq. (6)] write the same factor outside their band sums, and Ref. [4, Appendix B, above Eq. (B7)] states that it is implicit in its main-text formulas. A factor of 2 between two published rates for the same material is often this factor.

A.7.4 The first frequency moment

The first frequency moment of the density structure factor is fixed by the f-sum rule. For a target with $S(\boldsymbol{q},\omega) = S(-\boldsymbol{q},\omega)$,

$$\int_{-\infty}^{\infty} \mathrm{d}\omega \; \omega \, S(\boldsymbol{q},\omega) = \frac{\pi \, \boldsymbol{n}_e \, q^2}{m_e}, \tag{A.39}$$

at any temperature, with $\boldsymbol{n}_e$ the mean electron density (Sect. 4.3 and Appendix D.6). The normalization is π over the whole real line. The same rule is also written for the loss function and per particle. Splitting the integral at $\omega = 0$ and using Eq. (A.34) turns the left side into $\int_0^\infty \mathrm{d}\omega \, \omega(1 - \mathrm{e}^{-\beta\omega})S$, and Eq. (A.37) then gives [5, Eq. (S.10)]

$$\int_0^{\infty} \mathrm{d}\omega \; \omega \, L_e(\boldsymbol{q},\omega) = \frac{\pi}{2} \, \omega_p^2, \qquad \omega_p^2 = \frac{e^2 \boldsymbol{n}_e}{m_e}. \tag{A.40}$$

At $T = 0$, where S vanishes for $\omega < 0$, dividing Eq. (A.39) by $2\pi\boldsymbol{n}_e$ gives the per-particle form of many-body texts, $\int_0^\infty \mathrm{d}\omega \, \omega \, S(\boldsymbol{q},\omega)/(2\pi\boldsymbol{n}_e) = q^2/(2m_e)$: the structure factor per electron and without the factor 2π of Eq. (A.31) has first moment $q^2/2m_e$.

A.8 Halo conventions and rate normalization

Every rate in this book is a rate per unit target mass, $\mathcal{R}$ in $\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$. Every halo average uses the standard halo model with the parameters of Ref. [15, Table 1]: $\rho_\chi = 0.3\,\mathrm{GeV\,cm}^{-3}$, $v_0 = 238\,\mathrm{km\,s}^{-1}$, $v_{\mathrm{esc}} = 544\,\mathrm{km\,s}^{-1}$, and the solar velocity $\boldsymbol{v}_\odot = (11.1, 250.2, 7.3)\,\mathrm{km\,s}^{-1}$ in galactic coordinates (Eq. (1.5), Table 1.1). The laboratory distribution is $f_\chi(\boldsymbol{v},t) = N_{\mathrm{esc}}^{-1} \mathrm{e}^{-|\boldsymbol{v}+\boldsymbol{v}_{\mathrm{E}}(t)|^2/v_0^2} \, \Theta(v_{\mathrm{esc}} - |\boldsymbol{v} + \boldsymbol{v}_{\mathrm{E}}(t)|)$ with $\boldsymbol{v}_{\mathrm{E}}(t) = \boldsymbol{v}_\odot + \boldsymbol{v}_\oplus(t)$ (Eq. (1.8)). These are the standard values of the field; the measured ranges are given in Sect. 1.2.

A.8.1 Two Earth speeds

The Earth's speed enters in two roles. Halo integrals, such as $\eta(v_{\min})$ and the daily and annual modulation, use the full time-dependent $\boldsymbol{v}_{\mathrm{E}}(t)$, averaged over the exposure when a time-averaged rate is quoted. Reference scalars, such as kinematic edges, threshold masses and the windows in (q,ω) that a mass can reach, use one fixed speed,

$$v_{\mathrm{E}}^{\mathrm{ref}} \equiv |\boldsymbol{v}_\odot| = 250.5\,\mathrm{km\,s}^{-1}, \qquad v_{\max} \equiv v_{\mathrm{esc}} + v_{\mathrm{E}}^{\mathrm{ref}} = 794.5\,\mathrm{km\,s}^{-1} = 2.650 \times 10^{-3}, \tag{A.41}$$

as in Eq. (1.9). The fixed value keeps every threshold time-independent. Over a year, $v_{\rm esc}+|\boldsymbol{v}_{\rm E}(t)|$ ranges from 781 to 810 km s^{-1} (Sect. 1.2); that spread belongs to the uncertainty of the halo, not to the definition of the edge. The annual mean of $|\boldsymbol{v}_{\rm E}(t)|$ differs from $|\boldsymbol{v}_\odot|$ at second order in $v_\oplus/v_\odot$, and the difference is absorbed in the same way. Papers that quote "$v_{\rm E}$" mean one or the other, and older ones use $v_0 = 220\,{\rm km\,s^{-1}}$ and $v_{\rm E} \approx 232\,{\rm km\,s^{-1}}$ to $244\,{\rm km\,s^{-1}}$ [5, below Eq. (S.9), 14, below Eq. (4)]; to compare with them, first change to their halo parameters.

A.8.2 Speed integrals

The halo enters through the velocity kernel $\int {\rm d}^3v\, f_\chi(\boldsymbol{v})\,\delta(\omega-\omega_{\boldsymbol{q}})$, which depends on $\hat{\boldsymbol{q}}$ and on $v_{\rm min}(q,\omega)$. For a structure factor that depends only on $|\boldsymbol{q}|$, or after averaging S over the directions of $\boldsymbol{q}$, the angular integral over $\hat{\boldsymbol{q}}$ can be done first, and

$$\int \frac{{\rm d}^3q}{(2\pi)^3}\,S(q,\omega)\int {\rm d}^3v\, f_\chi(\boldsymbol{v})\,\delta(\omega-\omega_{\boldsymbol{q}}) = \int \frac{q\,{\rm d}q}{4\pi^2}\,\eta\big(v_{\rm min}(q,\omega)\big)\,S(q,\omega), \tag{A.42}$$

with $\eta(v_{\rm min}) = \langle\Theta(v-v_{\rm min})/v\rangle$ (Eq. (1.28)). Per unit ${\rm d}^3q/(2\pi)^3$ the kernel is $\eta(v_{\rm min})/(2q)$; the factor 1/2 comes from the angular integral $\frac{1}{2}\int_{-1}^{1}{\rm d}\cos\theta\,\delta(\cdots - qv\cos\theta) = \Theta(v-v_{\rm min})/(2qv)$ (Sect. 1.3). The step requires only that the target response be independent of the direction of $\boldsymbol{q}$; the halo itself need not be isotropic, since the integral over $\hat{\boldsymbol{q}}$ at fixed $\boldsymbol{v}$ is the same for every direction of $\boldsymbol{v}$. A second speed moment, $\eta_2(v_{\rm min}) = 2\int_{v_{\rm min}}^{\infty}{\rm d}v'\,v'\eta(v')$, appears for interactions whose strength grows with the dark-matter velocity. Halo-independent analyses absorb the particle physics into $\tilde\eta(v_{\rm min}) = \rho_\chi\bar\sigma_e\eta(v_{\rm min})/m_\chi$ [16, Eq. (2.8)].

For a crystal the structure factor depends on the direction of $\boldsymbol{q}$, and Eq. (A.42) does not apply. The full kernel is then kept. The halo distribution is written in the laboratory frame at the time of the event, the crystal axes are rotated into that frame with the Earth's rotation (Eq. (1.14)), and the integral over $\hat{\boldsymbol{q}}$ is done with $S(\boldsymbol{q},\omega)$ evaluated in crystal axes. The daily modulation of Sect. 1.2 is the result. A time-averaged rate averages this kernel over the exposure; it is not the rate computed with the angle-averaged $\bar S$ unless the exposure samples all orientations of $\boldsymbol{v}_{\rm E}$ relative to the crystal equally, which a fixed detector does not.

A.8.3 Rate per unit target mass

For an interaction through the electron density with transform $\tilde V(\boldsymbol{q})$, a dark-matter particle in the box of Appendix A.3 scatters at the rate $\int {\rm d}^3q\,(2\pi)^{-3}|\tilde V(\boldsymbol{q})|^2 S(\boldsymbol{q},\omega_{\boldsymbol{q}})$. By Eq. (A.9) the matrix element is $\mathcal{V}^{-1}\tilde V\langle f|n_e(\boldsymbol{q})|i\rangle$, the golden rule squares it, the sum over final dark-matter momenta gives $\mathcal{V}\int {\rm d}^3q/(2\pi)^3$, and Eq. (A.31) absorbs the remaining $2\pi/\mathcal{V}$ (Eq. (2.67)). The box holds $\rho_\chi\mathcal{V}/m_\chi$ dark-matter particles and a target mass $\rho_T\mathcal{V}$, so the spectrum per unit target mass is

$$\frac{{\rm d}\mathcal{R}}{{\rm d}\omega} = \frac{\rho_\chi}{m_\chi\,\rho_T}\int {\rm d}^3v\, f_\chi(\boldsymbol{v})\int \frac{{\rm d}^3q}{(2\pi)^3}\,|\tilde V(\boldsymbol{q})|^2\,S(\boldsymbol{q},\omega)\,\delta(\omega-\omega_{\boldsymbol{q}}), \tag{A.43}$$

in which the volume has cancelled. With $|\tilde V|^2 = \pi\bar\sigma_e F_{\rm DM}^2/\mu_{\chi e}^2$ (Eq. (1.34)) and Eq. (A.42),

$$\frac{{\rm d}\mathcal{R}}{{\rm d}\omega} = \frac{\rho_\chi\,\bar\sigma_e}{4\pi\,m_\chi\,\rho_T\,\mu_{\chi e}^2}\int q\,{\rm d}q\;F_{\rm DM}(q)^2\,\eta\big(v_{\rm min}(q,\omega)\big)\,\bar S(q,\omega), \tag{A.44}$$

with $\bar S$ the average of S over the directions of $\boldsymbol{q}$. This is Eq. (1.47), and it agrees with Refs. [4, Eq. (10)] and [5, Eq. (S.9)], the latter after Eq. (A.37). For operators other than the density, $|\tilde V|^2 S$ is replaced by a contraction of a dark-matter-side tensor with the response matrix S_{ab} (Sect. 4.2).

Other papers normalize per target or per unit cell. Ref. [9, Eq. (3.13)] gives the rate of a whole target of $N_{\rm cell}$ cells, and Ref. [14, Eq. (36)] the same with $n_\chi = \rho_\chi/m_\chi$. Dividing by the target mass $N_{\rm cell}M_{\rm cell}$, with $\rho_T = M_{\rm cell}/\Omega$, and comparing with Eq. (A.44) identifies the crystal form factor of Ref. [9] with the angle-averaged structure factor,

$$|f_{\rm crys}(q,\omega)|^2 = \frac{q^3\,\Omega}{4\pi\alpha\, m_e^2}\,\bar{S}(q,\omega) = \frac{q^5\,\Omega}{8\pi^2\alpha^2 m_e^2}\,\varepsilon_2(q,\omega), \tag{A.45}$$

where the second form uses Eq. (A.37) at $T = 0$ without screening and agrees with Ref. [4, Eq. (16)]. The product $\Omega\bar{S}$, of dimension eV^{-1}, is the natural per-cell quantity to tabulate.

A.9 Cross-section conventions

The strength of a dark-matter–electron interaction is quoted as the reference cross-section $\bar{\sigma}_e$ and its momentum dependence as the form factor $F_{\rm DM}(q)$,

$$\bar{\sigma}_e \equiv \frac{\mu_{\chi e}^2}{\pi}\,|\tilde{V}(q_0)|^2, \qquad F_{\rm DM}(q) \equiv \frac{\tilde{V}(q)}{\tilde{V}(q_0)}, \qquad q_0 = \alpha m_e = 3.73\,\mathrm{keV}, \tag{A.46}$$

as in Eq. (1.33). For a mediator of mass m_V, $F_{\rm DM}(q) = (m_V^2 + \alpha^2 m_e^2)/(m_V^2 + q^2)$, which is 1 for a heavy and $(q_0/q)^2$ for a light mediator [4, Eqs. (11) and (12)]. The form factor multiplies the amplitude, so it enters every rate as $F_{\rm DM}^2$. Papers differ in the reference point, the medium and the amplitude normalization of this definition.

The first is the reference point. The momentum $q_0 = \alpha m_e$ is the inverse Bohr radius, typical of bound electrons. A paper that normalizes at another momentum $q_{\rm ref}$ quotes $\sigma_{\rm ref} = \bar{\sigma}_e\, F_{\rm DM}(q_{\rm ref})^2$; for a light mediator this is $\bar{\sigma}_e\,(q_0/q_{\rm ref})^4$, so a limit quoted at $q_{\rm ref} = 2q_0$ is smaller by a factor 16 for the same physics.

The second is the medium. Here $\bar{\sigma}_e$ is defined in vacuum: $\tilde{V}$ is the bare potential between a dark-matter particle and a free electron, and screening, local fields and every other effect of the target are in the response S (Sect. 4.4). The same convention is used in Refs. [4, Eq. (8) and the paragraph after it] and [5, below Eq. (S.8)]. Some papers put the in-medium dielectric factor into the coupling, which is common for dark photons and for Dirac materials with a background dielectric constant. They quote a cross-section that depends on the target; it is converted by removing that factor from the coupling and putting the corresponding screening into the response.

The third is the amplitude normalization. Ref. [9, Eq. (3.8)] defines

$$\bar{\sigma}_e = \frac{\mu_{\chi e}^2\,|\mathcal{M}_{\rm free}(q_0)|^2}{16\pi m_\chi^2 m_e^2},$$

with the relativistically normalized free amplitude $\mathcal{M}_{\rm free}$, averaged over initial and summed over final spins. Comparison with Eq. (A.46) gives the dictionary

$$|\mathcal{M}_{\rm free}(q)|^2 = 16\, m_\chi^2 m_e^2\,|\tilde{V}(q)|^2, \tag{A.47}$$

the factor $4m_\chi m_e$ per amplitude being the ratio of relativistic to nonrelativistic state normalizations. A nonrelativistic effective operator with coefficient c_1 in the relativistic normalization therefore has $\tilde{V} = c_1/(4m_\chi m_e)$ and $\bar{\sigma}_e = \mu_{\chi e}^2|c_1|^2/(16\pi m_\chi^2 m_e^2)$.

The spin average is part of this definition. The amplitude squared in Eq. (A.47) is averaged over the initial spins of the dark matter and the electron and summed over the final spins [9, below Eq. (3.8)]. For a spin-independent interaction this average equals the squared amplitude

of a single spin state, so $\bar{\sigma}_e$ is the same for scalar and fermionic dark matter with the same $\tilde{V}$ [9, below Eq. (2.2)]. The electron spin sum is then not repeated in the target response: there the factor 2 of Appendix A.7 counts occupied spin states, not final spins of a free electron [9, below Eqs. (A.28) and (A.29)].

A.9.1 Differential cross-section

For free elastic scattering in the first Born approximation, $\mathrm{d}\sigma_T/\mathrm{d}\Omega_{\boldsymbol{q}} = \mu_{\chi e}^2|\tilde{V}(\boldsymbol{q})|^2/(4\pi^2)$ (Eq. (2.14)). The momentum transfer at relative speed v is $q^2 = 2\mu_{\chi e}^2 v^2(1-\cos\theta)$, and $\mathrm{d}\Omega_{\boldsymbol{q}} = 2\pi\,\mathrm{d}\cos\theta = \pi\,\mathrm{d}q^2/(\mu_{\chi e}^2 v^2)$; with Eq. (1.34)

$$\frac{\mathrm{d}\sigma_T}{\mathrm{d}q} = \frac{q\,\bar{\sigma}_e}{2\mu_{\chi e}^2 v^2}\,F_{\mathrm{DM}}(q)^2, \qquad 0 \le q \le 2\mu_{\chi e} v, \tag{A.48}$$

as in Eq. (2.15). For $F_{\mathrm{DM}} = 1$ it integrates to $\bar{\sigma}_e$, which is the meaning of the reference cross-section for a heavy mediator. The same normalization holds for nuclei, $\mathrm{d}\sigma_N/\mathrm{d}q = q\sigma_N F_N(q)^2/(2\mu_{\chi N}^2 v^2)$ (Eq. (2.27)). A differential cross-section defined per unit q^2 is $\mathrm{d}\sigma_T/\mathrm{d}q^2 = (2q)^{-1}\mathrm{d}\sigma_T/\mathrm{d}q$; a factor of 2 between two papers often traces to this choice.

A.9.2 Nucleons and nuclei

For a mediator coupled to nucleons, the dark-matter–proton cross-section σ_p is the analogue of $\bar{\sigma}_e$ with $\mu_{\chi p}$, and the spin-independent nuclear cross-section at small q follows from it. It is $\sigma_N = \sigma_p(\mu_{\chi N}/\mu_{\chi p})^2 A^2$ (Eq. (1.37)), with $A^2 \to [Z + (A-Z)y_n/y_p]^2$ for unequal proton and neutron couplings; Ref. [1, Eq. (104)] writes the equivalent rescaling relative to the neutron coupling. For a dark photon, which couples to protons only, $A^2 \to Z^2$. At larger q the nuclear form factor $F_{\mathrm{Helm}}(q)$ multiplies the amplitude (Sect. 2.2). For a dark photon the electron and proton couplings have equal magnitude ζe, so for a heavy dark photon, $m_V \gg q_0$, $\sigma_p/\bar{\sigma}_e = (\mu_{\chi p}/\mu_{\chi e})^2$; this ratio is how electron- and nuclear-scattering limits on the same model are compared.

A.9.3 Kinetic mixing

The dark photon mixes with the photon through the term $+(\zeta/2)F_{\mu\nu}F'^{\mu\nu}$ in the Lagrangian density, with $\mathcal{L} \supset -eA_\mu J^\mu_{\mathrm{em}}$ for the photon (Eq. (C.85)). The sign of the mixing term follows Ref. [1, Eq. (70)], which writes κ for ζ and $+eA_\mu J^\mu_{\mathrm{em}}$ for the photon coupling; Ref. [9, Eq. (2.1)] writes ε and mixes with hypercharge, $(\varepsilon/2\cos\theta_W)F_{Y\mu\nu}F'^{\mu\nu}$, where θ_W is the weak mixing angle. With our signs, the dark photon couples to electrons with $y_e = -\zeta e$ and to protons with $+\zeta e$ (Sect. 1.4). A change of either sign flips the sign of y_e and of $\tilde{V}$, and no rate changes, because every rate contains ζ^2. The symbols used for ζ in the literature are listed in Appendix B.2.

A.9.4 Relativistic dark matter

The form factor of Eq. (A.46) is the nonrelativistic limit, $\omega \ll q$, of the mediator propagator $1/(q^2 - \omega^2 + m_V^2)$ (Appendix C.9). For a light mediator in scattering, $\omega \le q v_{\max}$ and the correction is below 10^{-5}. Dark matter boosted to relativistic speeds is outside this limit: its rates need the full propagator and the coupling to the transverse current, and $F_{\mathrm{DM}}(q)$ alone does not describe them.

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Appendix B

Notation

Every symbol of the book is listed in Appendix B.1 with its meaning, its units and the section of its first use, and every acronym in Appendix B.3 with its expansion. Appendix B.2 gives the symbols that common papers and textbooks use for the same quantities, and the sign and normalization conventions in which they differ from this book.

B.1 List of symbols

Each symbol is listed once, under the chapter or appendix in which it first appears; the last column gives that section. The Units column gives the dimension in natural units ($\hbar = c = k_\mathrm{B} = 1$, Appendix A.1) and, after a semicolon, the conventional unit in which values are quoted. The entry 1 marks a dimensionless quantity, and a dash an object without a dimension of its own, such as an operator, a state or a label.

Symbol	Meaning	Units	Sect.
Chapter 1: Light dark matter			
$\hbar, c, k_\mathrm{B}$	Set to 1	—	1.1
$v_\mathrm{c}(r)$	Circular velocity of a galactic orbit	$\mathrm{km\,s^{-1}}$	1.1
h	Dimensionless Hubble parameter, the Hubble rate today in units of 100 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$	1	1.1
$\Omega_\mathrm{c} h^2$	Cosmological dark-matter density parameter	1	1.1
$\langle \sigma v \rangle$	Thermally averaged annihilation cross-section times relative speed	$\mathrm{cm^3\,s^{-1}}$	1.1
χ	The dark-matter particle	—	1.1
m_χ	Dark-matter mass	eV	1.1
n_χ	Dark-matter number density ρ_χ / m_χ	$\mathrm{eV^3}$; $\mathrm{cm^{-3}}$	1.1
ρ_χ	Local dark-matter mass density, 0.3 $\mathrm{GeV\,cm^{-3}}$	$\mathrm{eV^4}$; $\mathrm{GeV\,cm^{-3}}$	1.1
$f_\chi(v)$	Dark-matter velocity distribution in the laboratory frame	$(\mathrm{velocity})^{-3}$	1.1
v	Dark-matter velocity in the laboratory frame	1; $\mathrm{km\,s^{-1}}$	1.1
q	Momentum transferred from the dark matter to the target	eV	1.1
ω	Energy given to the target (may be negative at $T > 0$)	eV	1.1
v_min	Minimum dark-matter speed $v_\mathrm{min} = \lvert(\omega + \Delta_\chi)/q + q/2m_\chi\rvert$ for a transfer (q, ω)	1; $\mathrm{km\,s^{-1}}$	1.1
$\eta(v_\mathrm{min})$	Mean inverse speed above v_min, $\int \mathrm{d}^3 v f_\chi \Theta(v - v_\mathrm{min})/v$	$(\mathrm{velocity})^{-1}$	1.1
$\mu_{\chi e}$, $\mu_{\chi N}$, $\mu_{\chi p}$	Dark-matter–electron, –nucleus, –proton reduced masses	eV	1.1
$\tilde{V}(q)$	Fourier transform of the dark-matter–electron potential, $\int \mathrm{d}^3 x \mathrm{e}^{+\mathrm{i}q\cdot x} V(x)$	$\mathrm{eV^{-2}}$	1.1

Symbol	Meaning	Units	Sect.
$\bar{\sigma}_e$	Reference dark-matter–electron cross-section, $\mu_{\chi e}^2 \lvert\tilde{V}(q_0)\rvert^2/\pi$, in vacuum; for spin couplings $\mu_{\chi e}^2 \langle\lvert\tilde{V}\rvert^2\rangle/\pi$ with the average over both spins	eV^{-2}; cm^2	1.1
$F_{\mathrm{DM}}(q)$	Dark-matter form factor $(m_V^2 + \alpha^2 m_e^2)/(m_V^2 + q^2)$	1	1.1
ϕ, m_ϕ	Absorbed bosonic dark-matter field and its mass	eV	1.1
$\mathcal{R}$, $\mathrm{d}\mathcal{R}/\mathrm{d}\omega$	Event rate per unit target mass, and its spectrum	$\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	1.1
ρ_T	Target mass density	$\mathrm{g\,cm}^{-3}$	1.1
$S(q,\omega)$	Dynamic structure factor of the electron density	eV^2	1.1
$\rho_{\mathrm{halo}}(r)$	Dark-matter halo mass density at galactocentric radius r (singular isothermal sphere, $\rho_{\mathrm{halo}} \propto r^{-2}$)	eV^4; $\mathrm{GeV\,cm}^{-3}$	1.2
v_0	Circular speed parameter of the standard halo model, 238 km/s	$\mathrm{km\,s}^{-1}$	1.2
v_{esc}	Galactic escape speed, 544 km/s	$\mathrm{km\,s}^{-1}$	1.2
$v_\odot$	Solar velocity in the galactic frame, $\lvert v_\odot\rvert = 250.55$ km/s	$\mathrm{km\,s}^{-1}$	1.2
$v_\oplus(t)$	Earth's orbital velocity relative to the Sun	$\mathrm{km\,s}^{-1}$	1.2
$v_{\mathrm{E}}(t)$	Earth velocity in the galactic frame, $v_\odot + v_\oplus(t)$	$\mathrm{km\,s}^{-1}$	1.2
$v_{\mathrm{E}}^{\mathrm{ref}}$	Reference Earth speed for thresholds and edges: the adopted value of $\lvert v_\odot\rvert$, 250.5 km/s	$\mathrm{km\,s}^{-1}$	1.2
v_{max}	Maximum laboratory speed $v_{\mathrm{esc}} + v_{\mathrm{E}}^{\mathrm{ref}} = 794.5$ km/s	$\mathrm{km\,s}^{-1}$	1.2
N_{esc}	Normalization of the truncated halo distribution	$(\mathrm{velocity})^3$	1.2
Φ_χ	Dark-matter number flux	$\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$	1.2
v_{gal}, $f_{\mathrm{gal}}(v_{\mathrm{gal}})$	Dark-matter velocity in the galactic rest frame; standard-halo distribution in that frame	$\mathrm{km\,s}^{-1}$; $(\mathrm{velocity})^{-3}$	1.2
$F_\chi(v)$	Laboratory speed distribution, $\int \mathrm{d}v F_\chi = 1$	$(\mathrm{velocity})^{-1}$	1.2
ω_{yr}, ω_{sid}	Orbital angular frequency of the Earth, 0.0172 d^{-1}; sidereal angular frequency $2\pi/(23\ \mathrm{h}\ 56\ \mathrm{min}\ 4\ \mathrm{s})$	d^{-1}; s^{-1}	1.2
t_{peak}, t_{E}	Date of largest Earth speed; time at which a laboratory axis and v_{E} share a meridian	d; h	1.2
θ_{E}	Angle between $v_\odot$ and the Earth's rotation axis, 42.5°	rad	1.2
$\hat{n}$	Unit vector along a laboratory-fixed axis	1	1.2
$\theta_{\mathrm{lab}}(t)$, $\theta_{\hat{n}}$	Angle between v_{E} and a laboratory axis $\hat{n}$; polar angle of $\hat{n}$ from the celestial pole	rad	1.2
α_{E}, α_{z}, $\alpha_{\hat{n}}$; δ_{E}; h_{E}	Right ascensions of $v_\odot$, of the zenith and of $\hat{n}$; declination of $v_\odot$; its hour angle	rad	1.2
ℓ_{lat}	Geographic latitude	rad	1.2
u_{esc}	Local escape speed of an attracting body (Sun, Earth), focusing estimate only	$\mathrm{km\,s}^{-1}$	1.2
ϑ, ϑ_{cut}	Angle between v and v_{E} in the derivation of F_χ; its cut-off value	rad	1.2
x_{eq}, y_{eq}, z_{eq}; x_{gal}, y_{gal}, z_{gal}	Equatorial and galactic Cartesian components of a unit vector	1	1.2
$\hat{v}_{\mathrm{east}}$, $\hat{v}_{\mathrm{north}}$, $\hat{v}_{\mathrm{up}}$	Components of the unit vector $\hat{v}_\odot$ along the local east, north and vertical axes of a laboratory	1	1.2
$\Theta(x)$	Heaviside step function	1	1.2
α	Fine-structure constant	1	1.3
m_e	Electron mass	eV	1.3
$\omega_{\boldsymbol{q}}$	Kinematic energy transfer $q \cdot v - q^2/2m_\chi - \Delta_\chi$	eV	1.3
Δ_χ	Dark-matter mass splitting (exothermic for $\Delta_\chi < 0$)	eV	1.3
E_{th}	Threshold energy	eV	1.3
m_{min}	Threshold mass $2E_{\mathrm{th}}/v_{\mathrm{max}}^2$	eV	1.3
q_0	Reference momentum αm_e	eV	1.3
a	Cubic lattice constant; silicon 5.431 Å	Å	1.3
$\boldsymbol{G}_{111}$	Shortest reciprocal-lattice vector of fcc silicon, $\lvert\boldsymbol{G}_{111}\rvert = \sqrt{3}\cdot 2\pi/a = 3.95$ keV $\approx 1.06 q_0$	eV	1.3

Symbol	Meaning	Units	Sect.
θ	Angle between q and v; an integration angle elsewhere, defined where it is used (the scattering angle is θ_s)	rad	1.3
χ_1, χ_2; m_{χ_1}, m_{χ_2}	Lighter and heavier states of inelastic dark matter and their masses, $\Delta_\chi = m_{\chi_2} - m_{\chi_1}$	—; eV	1.3
E_R	Nuclear recoil energy	eV	1.3
E_g	Indirect (fundamental) band gap	eV	1.3
M	Generic nuclear mass in κ_BO	eV	1.3
$\mu_{\chi T}$	Reduced mass of dark matter and a free target of mass M	eV	1.3
G	Reciprocal-lattice vector	eV	1.3
c_ν	Sound speed of branch ν	$\mathrm{km\,s^{-1}}$	1.3
$\mu_{\chi n}$	Dark-matter–nucleon reduced mass (as $\mu_{\chi N}$ with the nucleon)	eV	1.3
e	Elementary charge, $e > 0, e^2 = 4\pi\alpha$; the electron charge is $-e$	1	1.4
m_V	Mediator mass	eV	1.4
y_χ, y_e, y_N	Mediator couplings to dark matter, electrons, nucleons	1	1.4
$V(x)$	Dark-matter–electron potential in position space	eV	1.4
ζ	Dark-photon kinetic mixing	1	1.4
g_{aee}, $g_{a\gamma\gamma}$	Axion-like-particle couplings to electrons and photons	1; $\mathrm{eV^{-1}}$	1.4
$\boldsymbol{p}_\phi$	Momentum of an absorbed bosonic dark-matter particle, $\lvert\boldsymbol{p}_\phi\rvert \approx m_\phi v$	eV	1.4
σ_p, σ_N	Dark-matter–proton and dark-matter–nucleus cross-sections	$\mathrm{cm^2}$	1.4
A, Z	Mass number, atomic number	1	1.4
A'_μ, $F'_{\mu\nu}$	Vector-mediator (dark-photon) field and its field strength	eV; $\mathrm{eV^2}$	1.4
A_μ, $F_{\mu\nu}$, $\tilde{A}_\mu$; $\tilde{F}^{\mu\nu}$	Photon field and field strength; photon after the mixing redefinition; dual field strength $\frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$	eV; $\mathrm{eV^2}$	1.4
ϕ_V	Scalar-mediator field	eV	1.4
J^μ_χ, J^μ_e, J^μ_N, $\boldsymbol{J}_\chi$	Number currents of dark matter, electrons and nucleons; $J^0 = n$	$\mathrm{eV^3}$	1.4
J^μ_em	Electromagnetic current in units of e (an electron counts -1)	$\mathrm{eV^3}$	1.4
n_N, $n_N(r)$	Nucleon number density; of a nucleus, $\int \mathrm{d}^3r\, n_N(r) = A$	$\mathrm{eV^3}$	1.4
y_p, y_n	Mediator couplings to protons and neutrons when they differ from y_N	1	1.4
$g_{\phi ee}$	Scalar-dark-matter–electron Yukawa coupling, $\mathcal{L} \supset g_{\phi ee}\phi\bar{\psi}\psi$	1	1.4
ζ_eff	In-medium effective kinetic mixing for absorption	1	1.4
Π_em	In-medium photon self-energy	$\mathrm{eV^2}$	1.4
$\boldsymbol{\sigma}$	Pauli-matrix vector of an electron	1	1.4
$\mathcal{L}$, $\mathcal{L}_\mathrm{int}$, $\mathcal{L}_{aee}$, $\mathcal{L}_{\phi ee}$	Lagrangian density	$\mathrm{eV^4}$	1.4
$\boldsymbol{E}$, $\boldsymbol{B}$, $\boldsymbol{A}$	Microscopic electric and magnetic fields and vector potential	$\mathrm{eV^2}$	1.4
ψ, $\bar{\psi}$, γ^μ, γ^5	Relativistic (Dirac) electron field and Dirac matrices	$\mathrm{eV^{3/2}}$; 1	1.4
$\boldsymbol{B}_0$	Static laboratory magnetic field	T	1.4
N_eh	Number of electron–hole pairs (ionization yield)	1	1.5
ε_eh	Mean energy per electron–hole pair	eV	1.5
M_det, t_exp	Detector (target) mass and live time; exposure $M_\mathrm{det}t_\mathrm{exp}$	kg; yr	1.5
N_sig, N_obs, N_bkg	Expected signal count, observed count, expected background count	1	1.5
N_{90}	90% confidence upper limit on N_sig (2.303 for $N_\mathrm{obs} = 0$)	1	1.5
$\mathcal{R}_\mathrm{bkg}$	Background rate per unit exposure	$\mathrm{kg^{-1}\,yr^{-1}}$	1.5
$r(m_\chi)$	Rate per unit cross-section, $\mathcal{R} = \bar{\sigma}_e r$	$\mathrm{cm^{-2}\,kg^{-1}\,yr^{-1}}$	1.5
$\bar{\sigma}_e^\mathrm{lim}(m_\chi)$	Cross-section limit	$\mathrm{cm^2}$	1.5
$\langle F_\mathrm{DM}^2\rangle$	Average of F_DM^2 over (q,ω) weighted by η and S	1	1.5

Symbol	Meaning	Units	Sect.
$\mathcal{R}_n$	Rate of events with n electron–hole pairs	$\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	1.5
$\mathcal{R}_{\mathrm{toy}}$	Toy rate of the exclusion-limit worked example	$\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	1.5
V_{bias}	Detector bias voltage	V	1.5
E_{ph}, E_{prim}	Phonon energy measured after Luke amplification, $E_{\mathrm{prim}} + N_{\mathrm{eh}} e V_{\mathrm{bias}}$; energy deposited directly into primary phonons	eV	1.5
A_{mod}	Annual-modulation amplitude	1	1.5
$\lfloor x \rfloor$	Floor function	—	1.5
M_κ, M_{cell}, Z_κ	Nuclear mass, cell mass, nuclear charge number	eV; 1	1.5
Δ_{BCS}	Superconducting gap	meV	1.5
T_{c}	Critical temperature	K	1.5
Chapter 2: Transition rates			
$\Gamma_{i\to f}$, Γ	Transition rate; rate per dark-matter particle	eV	2.1
H', H_{int}	Perturbation; dark-matter–target interaction	eV	2.1
$\mathscr{V}$	Normalization (crystal) volume	eV^{-3}	2.1
$\mathcal{D}_f(E)$	Density of final states	eV^{-1}	2.1
E_i, E_f	Energies of initial and final states	eV	2.1
H_0, H	Unperturbed Hamiltonian; full Hamiltonian $H_0 + H'$	eV	2.1
$c_n(t)$, $c_f(t)$, $c_i(t)$	Expansion amplitudes in the interaction picture	1	2.1
$P_{i\to f}(t)$	Transition probability	1	2.1
ξ	Energy mismatch $E_f - E_i$	eV	2.1
Γ_i	Total rate out of state i, $2\pi\langle\lvert H'\rvert^2\rangle\mathcal{D}_f(E_i)$	eV	2.1
E_{w}	Energy width of a continuum	eV	2.1
$\hat{O}$, O_{fi}	Operator of a harmonic perturbation $H' = \hat{O}\mathrm{e}^{-\mathrm{i}\omega t} + \hat{O}^\dagger \mathrm{e}^{\mathrm{i}\omega t}$; its matrix element $\langle f\lvert\hat{O}\rvert i\rangle$	eV	2.1
O, $O(t)$	Generic operator on the target, unhatted, in formulas valid for any operator; $O(t)$ its Heisenberg form	—	2.1
N_f	Number of final states	1	2.1
τ_{s}	Switching time	eV^{-1}	2.1
ϕ_0	Amplitude of an oscillating bosonic dark-matter field, $\rho_\chi = m_\phi^2\phi_0^2/2$	eV	2.1
σ_T, m_T	Cross-section on, and mass of, one target particle	cm^2; eV	2.1
$\mathcal{M}$	Invariant (relativistic) amplitude; $\lvert\mathcal{M}\rvert^2 = 16 m_\chi^2 m_e^2 \lvert\tilde{V}\rvert^2$ non-relativistically	1	2.1
$\langle p\lvert p'\rangle_{\mathrm{rel}}$, $E_{\boldsymbol{p}}$	Relativistic normalization $2E_{\boldsymbol{p}}(2\pi)^3\delta^3(\boldsymbol{p}-\boldsymbol{p}')$; relativistic energy of momentum $\boldsymbol{p}$	—; eV	2.1
$\mathrm{Si}(z)$	Sine integral	1	2.1
$\hat{T}$	T-matrix operator in $\langle f\lvert\mathrm{i}\hat{T}\rvert i\rangle = \mathrm{i}(2\pi)^4\delta^4\mathcal{M}$	—	2.1
$\boldsymbol{P}_i$, $\boldsymbol{P}_f$	Total initial and final momenta	eV	2.1
$\boldsymbol{p}_\chi$, $\boldsymbol{p}_e$; $\boldsymbol{x}_\chi$, $\boldsymbol{x}_e$	Momenta and positions of the dark matter and the electron in a two-particle state	eV; eV^{-1}	2.1
$\lvert n\rangle$	Generic unperturbed eigenstate	—	2.1
$w(\xi)$, Δ_{w}	Test function in the δ-limit and its energy scale	—; eV	2.1
$\mathrm{d}\Omega_{\boldsymbol{q}}$	Solid-angle element of the scattered momentum	sr	2.2
$f_{i\to f}(q)$	Single-particle transition form factor $\langle f\lvert\mathrm{e}^{\mathrm{i}q\cdot r}\rvert i\rangle$	1	2.2
$F_{1s}(q)$	Hydrogen 1s elastic form factor $16/(4+q^2a_0^2)^2$	1	2.2
a_0	Bohr radius	Å	2.2
$F_{\mathrm{Helm}}(q)$	Helm nuclear form factor	1	2.2
θ_{s}	Scattering angle between initial and final relative momenta	rad	2.2
p, p'; $\boldsymbol{P}_{\mathrm{tot}}$, $\boldsymbol{p}_{\mathrm{rel}}$	Relative momenta before and after a two-body collision; total and relative momenta of a two-body system	eV	2.2
$\boldsymbol{n}_T$	Number density of target atoms or nuclei	eV^3	2.2
ψ_i, ψ_f, ψ_{1s}, ψ_{2p}	Single-particle wavefunctions	$\mathrm{eV}^{3/2}$	2.2

Symbol	Meaning	Units	Sect.
$\tilde{\psi}_i(k)$, $\tilde{\psi}_{nl}(k)$	Momentum-space wavefunction $\int \mathrm{d}^3 r \mathrm{e}^{-\mathrm{i}k\cdot r}\psi_i$; radial momentum wavefunction of shell nl, $\int k^2 dk \lvert\tilde{\psi}_{nl}\rvert^2 = (2\pi)^3$	$\mathrm{eV}^{-3/2}$	2.2
$\tilde{\psi}_{plm}(r)$, $\tilde{\psi}_{p_{\mathrm{ej}}l'm'}(r)$	Outgoing continuum partial wave of an ejected electron of momentum $p(p_{\mathrm{ej}})$, normalized to $(2\pi)^3\delta_{l'l}\delta_{m'm}\delta(p'-p)/p^2$	$\mathrm{eV}^{-3/2}$	2.2
n, l, m	Principal, orbital and magnetic quantum numbers, as subscripts of atomic orbitals only	1	2.2
Y_{lm}	Spherical harmonics	1	2.2
p_{ej}, E_{ej}	Momentum and kinetic energy of an ionized electron	eV	2.2
E_B	Binding energy of an atomic shell, > 0	eV	2.2
$\lvert f_{\mathrm{ion}}(p_{\mathrm{ej}}, q)\rvert^2$	Atomic ionization form factor of a shell	1	2.2
Z_{eff}, $F_{\mathrm{Fermi}}(p)$	Effective charge in the Fermi factor; the Fermi factor	1	2.2
ξ_{C}	Coulomb (Sommerfeld) parameter $Z_{\mathrm{eff}}\alpha m_e/p$	1	2.2
$j_1(x)$	Spherical Bessel function of order one, $(\sin x - x\cos x)/x^2$	1	2.2
λ_{1s}, λ_{2p}	Radial decay constants $2/a_0$ and $3/(2a_0)$ of the hydrogen 1s density and of the $1s \to 2p$ radial integrand	eV	2.2
$V_N(x)$, $\tilde{V}_N(q)$	Dark-matter–nucleon potential and its transform	eV; eV^{-2}	2.2
$F_N(q)$	Nuclear form factor $(1/A)\int \mathrm{d}^3 r \mathrm{e}^{\mathrm{i}q\cdot r}\boldsymbol{n}_N(r)$, $F_N(0) = 1$; F_{Helm} is its model	1	2.2
$\boldsymbol{R}_{\mathrm{cm}}$, $\boldsymbol{r}'_j$	Nuclear center of mass and internal nucleon coordinates	eV^{-1}	2.2
R_{nuc}, s_{skin}	Helm radius and skin thickness	fm	2.2
c_{nuc}, a_{nuc}	Woods–Saxon half-density radius and diffuseness in the Lewin–Smith R_{nuc}	fm	2.2
$q_{\max}$	Largest elastic transfer $2\mu v$	eV	2.2
$c^\dagger$, c	Fermion creation and annihilation operators ($c_{n\boldsymbol{k}}$ for Bloch states)	—	2.3
$\hat{\psi}(r)$	Fermion field operator	$\mathrm{eV}^{3/2}$	2.3
$n_e(q)$	Electron density operator $\sum_i \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$	1	2.3
$v_{\mathrm{C}}(k)$	Coulomb kernel e^2/k^2	eV^{-2}	2.3
γ, γ', $\gamma_1 \ldots \gamma_4$, $\tilde{\gamma}$	Single-particle state labels (spin included; $\tilde{\gamma}$ labels a second basis); orbitals $\psi_\gamma(\boldsymbol{r}, s) = \langle \boldsymbol{r}s\lvert\gamma\rangle$	—	2.3
s, s'	Spin projection $\uparrow$, $\downarrow$	—	2.3
$x = (r, s)$	Position and spin of one particle	—	2.3
n_γ	Occupation number of state γ	1	2.3
$\hat{N}_e$, N_e	Electron number operator $\sum c^\dagger c$ and its value	1	2.3
d_γ, $d_\gamma^\dagger$	Generic boson operators (phonons are $b_{\boldsymbol{Q}\nu}$, magnons $a_{\boldsymbol{k}\nu}$)	—	2.3
E_γ	Single-particle (orbital) energy of state γ; the Hartree–Fock or Kohn–Sham eigenvalue ($\varepsilon_{n\boldsymbol{k}}$ in a crystal)	eV	2.3
S_γ	Number of occupied states before γ in the fixed order	1	2.3
$\hat{O}_1$, $\hat{O}_2$, $\hat{o}$	One- and two-body operators; one-particle operator	—	2.3
$\Psi(x_1, \ldots, x_N)$, $\Psi_{\gamma_1 \ldots \gamma_N}$	N-particle wavefunction of identical particles; the Slater determinant of the orbitals $\gamma_1, \ldots, \gamma_N$	$\mathrm{eV}^{3N/2}$	2.3
o_γ	Eigenvalue of the one-particle operator $\hat{o}$ on the orbital γ (in the basis that diagonalizes $\hat{o}$)	as $\hat{o}$	2.3
$U_1(r)$, $U_2(r)$	One-body potential; pair potential	eV	2.3
$: \cdots :$	Normal ordering	—	2.3
k_{reg}	Infrared regulator of the Coulomb kernel, $v_{\mathrm{C}} \to e^2/(k^2 + k_{\mathrm{reg}}^2)$	eV	2.3
X_I, I	Position of nucleus I; nucleus index ($\kappa(I)$ its species)	Å	2.3
$n_Z(k)$	Charge-number density $-n_e(\boldsymbol{k}) + \sum_\kappa Z_\kappa n_\kappa(\boldsymbol{k})$	1	2.3
H_{ee}, H_{jel}, H_C, H_{Coul}	Electron–electron Coulomb term; jellium Hamiltonian; Coulomb Hamiltonian of electrons and nuclei	eV	2.3
d	Mean electron spacing $n_e^{-1/3}$	Å	2.3
$\lvert\Phi_0\rangle$	A reference Slater determinant	—	2.3

Symbol	Meaning	Units	Sect.
e_{G}, e_{SI}, $\varepsilon_0^{\mathrm{SI}}$	Gaussian elementary charge, $e_{\mathrm{G}}^2 = \alpha$; SI charge and vacuum permittivity	1; —	2.3
r_s	Electron-gas density parameter: Wigner–Seitz radius per electron in units of a_0, $(3/4\pi n_e)^{1/3}/a_0$; 2.0 for silicon's valence density	1	2.3
κ	Nuclear species (basis-site) index	—	2.3
$n_\kappa(q)$	Nuclear density operator of species κ, $\sum_{I\in\kappa} \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_I}$	1	2.3
T_{fi}	Transition (T-matrix) element	eV	2.4
Γ_{w}	Decay width of an intermediate state	eV	2.4
PV, fp	Principal value; Hadamard finite part	—	2.4
$\lvert\psi_{\mathrm{I}}(t)\rangle, H'_{\mathrm{I}}(t), U_{\mathrm{I}}(t,\, t_0)$	Interaction-picture state, perturbation and evolution operator	—; eV; 1	2.4
H'_{mn}	Matrix element $\langle m\lvert H'\rvert n\rangle$	eV	2.4
$\lvert m\rangle, E_m$	Intermediate state and its energy	—; eV	2.4
$c_f^{(2)}(t)$	Second-order amplitude	1	2.4
$\mathrm{i}0^+$	Positive infinitesimal in an energy denominator	eV	2.4
ΔE_i, ΔE_m	Level shift of a state coupled to a continuum, PV $\sum \lvert H'\rvert^2/(E_i - E_f)$ (decorated Δ)	eV	2.4
$\tilde{E}_m$	Shifted energy $E_m + \Delta E_m$	eV	2.4
i_{T}, f_{T}; $E_{i_{\mathrm{T}}}$, $E_{f_{\mathrm{T}}}$	Initial and final target states of a second-order process, in kets and subscripts only; their energies	—; eV	2.4
ω_1, ω_2	Energies of the absorbed and emitted (or two absorbed) quanta	eV	2.4
H_1, H_2; $\hat{O}_a$, $\hat{O}_b$	The two couplings of a second-order process; their target operators	eV	2.4
$w(x)$	Test function of the real denominator variable x	—	2.4
x_{c}	Symmetric cut-off half-width in PV and fp	eV	2.4
δ_1, δ_2, $\mathcal{A}(\omega_1)$	Detunings $\omega_j - (E_m - E_{i_{\mathrm{T}}})$ and two-ordering amplitude	eV; eV^{-1}	2.4
$\Gamma_{m\to f_{\mathrm{T}}}$, $\mathcal{D}_{f_{\mathrm{T}}}$	Rate at which a beam empties m into the band; density of final target states	eV; eV^{-1}	2.4
$\hat{\varrho}$	Density matrix	1	2.5
T, β	Temperature, $\beta = 1/\mathrm{T}$	eV	2.5
p_i	Gibbs weight $\mathrm{e}^{-\beta E_i}/\mathcal{Z}$	1	2.5
$\mathcal{Z}$	Partition function	1	2.5
μ_{ch}	Chemical potential	eV	2.5
$\mathcal{S}_O(\omega)$	Transition sum $\sum_{if} p_i \lvert\langle f\lvert O\rvert i\rangle\rvert^2 \delta(\omega - E_f + E_i)$ of an operator O; identified with S_{ab}	eV^{-1}	2.5
$n_{\mathrm{B}}(x)$, $n_{\mathrm{F}}(x)$	Bose and Fermi functions	1	2.5
n_e	Mean electron number density $N_e/\mathcal{V}$	$\mathrm{Å}^{-3}$	2.5
k_{F}, E_{F}	Fermi wavevector and Fermi energy of a free electron gas	eV	2.5
v_{F}	Fermi velocity of a metal, k_{F}/m_e for the free electron gas	1	2.5
$\mathcal{N}_{\mathrm{F}}$	Density of states per spin and volume at the Fermi level	eV^2	2.5
Δ_{tl}	Splitting of a two-level system	eV	2.5
p_0, p_1, O_{10}	Two-level populations and matrix element $\langle 1\lvert\hat{O}\rvert 0\rangle$	1; eV	2.5
ω_0, λ_{osc}	Oscillator frequency and coupling of the thermal-oscillator worked example	eV	2.5
$U(t)$	Evolution operator	—	2.5
N_e	Number of electrons in the volume $\mathcal{V}$, $n_e = N_e/\mathcal{V}$	1	2.5
Chapter 3: Electrons in crystals			
$\boldsymbol{a}_i$, $\boldsymbol{b}_i$	Primitive vectors of the direct and reciprocal lattices	Å; $\mathrm{Å}^{-1}$	3.1
R	Bravais lattice vector	Å	3.1
Ω	Unit-cell volume	$\mathrm{Å}^3$	3.1
N	Number of unit cells (= number of k points on a uniform grid)	1	3.1

Symbol	Meaning	Units	Sect.
k, k'	Electron crystal momenta in the first zone	eV	3.1
$\boldsymbol{q}_r$, $\boldsymbol{G}_q$	First-zone part and reciprocal vector of $\boldsymbol{q}$, $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$	eV	3.1
Γ, X, L, K	High-symmetry points of the fcc zone; also the point W	—	3.1
$X_{\boldsymbol{R}\kappa}$	Equilibrium nuclear site $\boldsymbol{R} + \tau_\kappa$ of the static lattice	Å	3.1
n_{at}	Number of atoms in the primitive cell (N_{at} counts the atoms of a target or a calculation)	1	3.1
$S^{\mathrm{bas}}(G)$	Basis structure factor $\sum_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\tau_\kappa}$	1	3.1
$T_{\boldsymbol{R}}$	Lattice translation operator acting on all electrons	—	3.1
$\boldsymbol{G}_{hkl}$	Reciprocal-lattice vector $(2\pi/a)(h,k,l)$ of a cubic lattice	eV	3.1
$\tilde{v}_\kappa(G)$, $\tilde{v}(G)$	Transform of the atomic term of species κ in a lattice-periodic sum, $f_{\boldsymbol{G}} = \Omega^{-1}\sum_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\tau_\kappa}\tilde{v}_\kappa(\boldsymbol{G})$ ($\tilde{v}_\kappa = -Z_\kappa v_{\mathrm{C}}$ for a clamped nucleus); its one-species form	as $f_{\boldsymbol{G}}$ Ω	3.1
τ_κ	Position of basis site (atom) κ in the unit cell, bold $\boldsymbol{\tau}_\kappa$	Å	3.1
$\psi_{n\boldsymbol{k}}(r)$, $u_{n\boldsymbol{k}}(r)$	Bloch state normalized over $\mathcal{V}$; periodic part normalized over Ω	$\mathcal{V}^{-1/2}$; $\Omega^{-1/2}$	3.2
n, m	Band indices: generic, intermediate	—	3.2
$\varepsilon_{n\boldsymbol{k}}$	Band energy	eV	3.2
$U_{\boldsymbol{G}}$	Fourier component of the periodic potential	eV	3.2
t_{hop}	Tight-binding hopping amplitude	eV	3.2
$v_n^{\mathrm{g}}(k)$	Group velocity $\nabla_k\varepsilon_{n\boldsymbol{k}}$ ($= v_{nn}(\boldsymbol{k})$)	1	3.2
$\boldsymbol{p}_{mn}(k)$, $v_{mn}(k)$	Momentum and velocity matrix elements between Bloch states; $v_{mn} = \boldsymbol{p}_{mn}/m_e$ for a local potential	eV; 1	3.2
$\rho_{mn}(k,q)$	Density vertex $\langle\psi_{m,\boldsymbol{k}+\boldsymbol{q}}\vert\mathrm{e}^{\mathrm{i}q\cdot\boldsymbol{r}}\vert\psi_{n\boldsymbol{k}}\rangle$, $\rho_{nn}(\boldsymbol{k},0) = 1$; optical limit $q\cdot v_{mn}/(\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}})$	1	3.2
$f_{n\boldsymbol{k}}$	Occupation of state nk	1	3.2
H_{e}	One-electron Hamiltonian $\hat{p}^2/2m_e + U(\boldsymbol{r})$	eV	3.2
$H_{\boldsymbol{k}}$	Bloch Hamiltonian $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}H_{\mathrm{e}}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ acting on $u_{n\boldsymbol{k}}$ (also at shifted arguments, $H_{\boldsymbol{k}+\boldsymbol{q}}$, $H_{\boldsymbol{k}_0+\delta\boldsymbol{k}}$)	eV	3.2
$U(r)$	Effective lattice-periodic one-electron potential	eV	3.2
$\hat{p}$, $\boldsymbol{v}$	Momentum operator $-\mathrm{i}\nabla$; velocity operator $i[H_{\mathrm{e}},\boldsymbol{r}]$	eV; 1	3.2
$\varepsilon^0_{\boldsymbol{k}}$	Free-electron energy $k^2/2m_e$	eV	3.2
ϕ^{at}_j, $\phi^{\mathrm{at}}_{j\boldsymbol{R}}$	Atomic orbital j of tight binding, and its copy in cell $\boldsymbol{R}$	$\text{Å}^{-3/2}$	3.2
$C_{nj}(k)$	LCAO coefficients of band n on the Bloch sum of orbital j	1	3.2
$H^{\mathrm{TB}}(k)$, $S^{\mathrm{ov}}(k)$	Tight-binding Hamiltonian and overlap matrices between Bloch sums	eV; 1	3.2
E_s, E_p	On-site s and p energies of an atom in tight binding (E_p in Sect. 3.2)	eV	3.2
V_{cov}	Covalent energy between two bond hybrids	eV	3.2
$c_{n\boldsymbol{k}}(G)$	Plane-wave coefficients of a Bloch state, of $\mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$ in $\Omega^{1/2}u_{n\boldsymbol{k}}$	1	3.2
$\boldsymbol{G}_{\mathrm{U}}$, N_{FFT}	Umklapp vector of the fold $\boldsymbol{k} + q = \boldsymbol{k}' + \boldsymbol{G}_{\mathrm{U}}$ with $\boldsymbol{k}'$ in the first zone; number of FFT grid points	eV (quoted in Å^{-1}); 1	3.2
v, c	Occupied (valence) and empty (conduction) band labels, only as subscripts	—	3.3
E_{dir}	Minimum direct gap	eV	3.3
$\boldsymbol{k}_{\mathrm{c}}$	Crystal momentum of the conduction-band minimum	eV	3.3
m^*, $m^*_\parallel$, $m^*_\perp$	Effective-mass tensor and its valley components	m_e	3.3
Δ_{SO}	Spin–orbit splitting of the valence-band maximum	eV	3.3
$E_{\mathrm{min}}(q)$	Minimum electron–hole pair energy at fixed q	eV	3.3
δk	Crystal momentum measured from a band extremum k_0, $\delta k = \boldsymbol{k} - k_0$	eV	3.3
E_P, v^x_{cX}	Kane energy $2m_e\vert v^x_{cX}\vert^2$; velocity matrix element between the conduction state and the orbital state $\vert X\rangle$	eV; 1	3.3

Symbol	Meaning	Units	Sect.
$\xi_{\rm SO}$, $\xi_{{\rm SO},\kappa}$, $\xi_{\rm eff}$	Radial spin–orbit constant of an atomic level, of atom κ; its p-weighted crystal average	eV	3.3
$U_{\rm at}(r)$	Central potential energy of an electron in an atom, in the spin–orbit term $H_{\rm so}$	eV	3.3
$m_{\rm hh}$, $m_{\rm lh}$, $m_{\rm h}$	Heavy- and light-hole masses; a generic hole mass	eV; m_e	3.3
$m_{\rm dc}$, $m_{\rm dv}$	Density-of-states masses of the conduction and valence bands	m_e	3.3
$n_{\rm i}$	Intrinsic carrier density	cm^{-3}	3.3
$\gamma_1^{\rm Lu}$, $\gamma_2^{\rm Lu}$, $\gamma_3^{\rm Lu}$, $\bar\gamma^{\rm Lu}$	Luttinger parameters; their spherical average $(2\gamma_2^{\rm Lu}+3\gamma_3^{\rm Lu})/5$	1	3.3
$\boldsymbol{L}$, $\boldsymbol{S}$; j, l	Orbital and spin angular momentum of an electron; total and orbital angular-momentum quantum numbers	1	3.3
$n(r)$	Electron number density (Kohn–Sham density)	Å^{-3}	3.4
$V_{\rm KS}$, $v_{\rm xc}$, $E_{\rm xc}$	Kohn–Sham potential, exchange–correlation potential and energy	eV	3.4
$\Delta_{\rm scis}$	Scissor shift	eV	3.4
E_{N_e}, E_{N_e+1}, E_{N_e-1}	Ground-state energy of $N_e(N_e\pm1)$ electrons; $E^r_{N_e+1}, E^a_{N_e-1}$ are the frozen-orbital removal and addition energies	eV	3.4
$P_{\rm c}(k)$, $H^{\rm s}_{\boldsymbol{k}}$, $v^{\rm s}_{cv}$	Projector $\sum_c \lvert u_{c\boldsymbol{k}}\rangle\langle u_{c\boldsymbol{k}}\rvert$ on the conduction bands; scissor Bloch Hamiltonian $H_{\boldsymbol{k}}+\Delta_{\rm scis}P_{\rm c}$; scissor-rescaled velocity matrix element	1; eV; 1	3.4
$v_{\rm H}(r)$	Hartree potential $\int e^2 n(\boldsymbol{r}')/4\pi\lvert\boldsymbol{r}-\boldsymbol{r}'\rvert$	eV	3.4
$\mathcal{F}[n]$	Universal (Levy–Lieb) functional $\min_{\Psi\to n}\langle H_{\rm kin}+H_{ee}\rangle$	eV	3.4
$H_{\rm kin}$	Kinetic-energy operator $\sum_i p_i^2/2m_e$ (decorated H)	eV	3.4
$K_{\rm s}[n]$	Independent-electron (Kohn–Sham) kinetic energy	eV	3.4
$E_{\rm H}[n]$	Hartree energy	eV	3.4
I, A	Ionization energy, electron affinity	eV	3.4
$E_{\rm g}^{\rm KS}$, $\Delta_{\rm xc}$	Kohn–Sham gap; derivative discontinuity, $E_{\rm g}=E_{\rm g}^{\rm KS}+\Delta_{\rm xc}$	eV	3.4
$\varepsilon_{\rm xc}(n)$, $\varepsilon_{\rm x}(n)$	Exchange–correlation and exchange energy per electron of the uniform gas	eV	3.4
$f_{\rm x}$, $v_{\rm x}$	Exchange-only kernel $-e^2/4k_{\rm F}^2$ and exchange potential $(4/3)\varepsilon_{\rm x}$ of the gas	eV^{-2}; eV	3.4
$E_{\rm qp}(k)$, $\Gamma_{\rm qp}(k)$	Quasiparticle energy and full width	eV	3.4
$\Sigma_{\rm x}$, $\Sigma_{\rm c}$	Exchange and correlation parts of the self-energy Σ; $\Sigma_{\rm x}(x,x')$ is the nonlocal Fock term, $\Sigma_{\rm x}(\boldsymbol{k})=-(e^2k_{\rm F}/2\pi^2)F_{\rm L}(\boldsymbol{k}/k_{\rm F})$ its electron-gas value	eV	3.4
$E_{\rm cut}$	Plane-wave kinetic-energy cutoff	eV	3.5
r_c	Pseudopotential core radius	Å; a_0	3.5
$\tilde p_j$, $\phi_j^{\rm AE}$, $\phi_j^{\rm PS}$	PAW projectors; all-electron and pseudo partial waves	Å$^{-3/2}$	3.5
Ry	Rydberg unit $\alpha^2m_e/2=13.61$ eV	eV	3.5
p_f	Momentum scale of a low-lying final state, $[2m_e(\omega+E_{\rm F})]^{1/2}$	eV	3.5
$u_l(r,\varepsilon)$, $V_l(r)$, P_l	Radial function rR_l at energy ε; pseudopotential of channel l; projector on angular momentum l	—; eV; 1	3.5
ψ^1, $\tilde\psi^1$	One-center all-electron and pseudo partial-wave expansions of a state inside an augmentation sphere	as $\psi_{n\boldsymbol{k}}$	3.5
Φ, ϵ	Normalized two-component state $(1+p^2/8m_e^2)\psi_{\rm A}$ (in Sect. 3.5); energy $E-m_e$ (in Sect. 3.5)	—; eV	3.5
N_{nl}, $\phi_{nl}(p)$	Occupation of the shell nl; its radial momentum-space orbital	1; eV$^{-3/2}$	3.5
$\psi_{\rm A}$, $\psi_{\rm B}$	Large and small two-component parts of a Dirac spinor	eV$^{3/2}$	3.5
$\boldsymbol{\alpha}_{\rm D}$, $\beta_{\rm D}$, $\gamma_{\rm D}$	Dirac matrices; $\gamma_{\rm D}=[1-(Z\alpha)^2]^{1/2}$	1	3.5
$H_{\rm mv}$, $H_{\rm Dar}$, $H_{\rm so}$	Mass–velocity, Darwin and spin–orbit one-body terms	eV	3.5

Symbol	Meaning	Units	Sect.
k_{cut}	Plane-wave cutoff wavevector $(2m_eE_{\mathrm{cut}})^{1/2}$	eV (quoted in bohr^{-1})	3.5
$\tilde{\psi}$, $\tilde{\psi}_{n\boldsymbol{k}}$	Pseudo (smooth) Bloch state of a pseudopotential or PAW calculation	as $\psi_{n\boldsymbol{k}}$	3.5
$\Delta_{\kappa,ij}(q)$	One-center (PAW) correction to the density vertex	1	3.5
E_{core}	Depth of a core shell below the valence maximum	eV	3.5
$p_{1/2}$, $p_{3/2}$, $d_{3/2}$, $d_{5/2}$	Spin–orbit-split p and d shells in spectroscopic labels	—	3.5
$w_n(r-R)$	Wannier function	$\Omega^{-1/2}$	3.6
$U(k)$	Bloch-to-Wannier gauge matrix	1	3.6
$H^W(R)$	Real-space (Wannier) Hamiltonian	eV	3.6
Ω_W	Wannier spread functional	Å^2	3.6
$\psi^W_{n\boldsymbol{k}}$, $u^W_{n\boldsymbol{k}}$	Bloch state and periodic part in the Wannier gauge, $\sum_m U_{mn}(\boldsymbol{k})\psi_{m\boldsymbol{k}}$	as $\psi_{n\boldsymbol{k}}, u_{n\boldsymbol{k}}$	3.6
$\theta_n(k)$	Gauge phase of a Bloch state	1	3.6
$P(k)$, $Q(k)$	Projector $\sum_n \lvert u^W_{n\boldsymbol{k}}\rangle\langle u^W_{n\boldsymbol{k}}\rvert$ on the N_W-dimensional Wannier manifold at $\boldsymbol{k}$, and its complement	1	3.6
P, Q	Projector $\sum_{m\boldsymbol{R}} \lvert w_{m\boldsymbol{R}}\rangle\langle w_{m\boldsymbol{R}}\rvert$ on a group of Wannier functions and its complement	1	3.6
$\phi^{\mathrm{tr}}_n(r)$, $A^{\mathrm{pr}}_{mn}(k)$	Trial orbital of the projection gauge; projection matrix $\langle\psi_{m\boldsymbol{k}}\lvert\phi^{\mathrm{tr}}_n\rangle$	$\Omega^{-1/2}$; 1	3.6
$\bar{r}_n$, $\langle r^2\rangle_n$	Wannier centre $\langle w_{n\boldsymbol{0}}\lvert\boldsymbol{r}\rvert w_{n\boldsymbol{0}}\rangle$; second moment $\langle w_{n\boldsymbol{0}}\lvert r^2\rvert w_{n\boldsymbol{0}}\rangle$	Å; Å^2	3.6
Δk, $\lambda_{\Delta\boldsymbol{k}}$	Vector from a grid point to a neighbour in the finite-difference spread; its weight, $\sum \lambda_{\Delta\boldsymbol{k}}\Delta k_\alpha \Delta k_\beta = \delta_{\alpha\beta}$	eV; eV^{-2}	3.6
$M_{mn}(k,\Delta k)$	Overlap $\langle u^W_{m\boldsymbol{k}}\lvert u^W_{n,k+\Delta k}\rangle_\Omega$	1	3.6
Ω^{I}_W, $\tilde{\Omega}_W$	Gauge-invariant and gauge-dependent parts of the spread, $\Omega_W = \Omega^{\mathrm{I}}_W + \tilde{\Omega}_W$	Å^2	3.6
$H^W(k)$, $\tilde{H}^W(R)$	Wannier-gauge Bloch Hamiltonian $\sum_R \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}H^W(\boldsymbol{R}) = U^\dagger E U$; coarse-grid (aliased) real-space Hamiltonian	eV	3.6
$E(k)$	Diagonal matrix $\mathrm{diag}(\varepsilon_{n\boldsymbol{k}})$ of band energies	eV	3.6
R_{max}	Range of $H^W(\boldsymbol{R})$	Å	3.6
ξ_W	Decay length of a Wannier function, $\lvert w\rvert \sim \mathrm{e}^{-\lvert\boldsymbol{R}\rvert/\xi_W}$	Å	3.6
n_k, n_1, n_2, n_3, n^{irr}_k	Number of points of a $\boldsymbol{k}$ grid, $n_k = n_1n_2n_3$ (points along the three reciprocal primitive vectors); the inequivalent points among them	1	3.6
N_W	Number of Wannier functions	1	3.6
L_{sc}, α_{M}	Supercell side; Madelung constant of the simple-cubic lattice, 2.8373 (in Appendix E.10)	Å; 1	3.6
Chapter 4: Linear response			
$\phi_{\mathrm{ext}}(t)$	External potential coupled to O_b in the Kubo formula	eV	4.1
$R_{ab}(q,\omega)$	Retarded response function of operators O_a, O_b	eV^2 for density operators ($1/\mathcal{V}$ included)	4.1
$O_a(q)$	Target operator coupled to an external field	—	4.1
$H_{\mathrm{tot}}(t)$, H_{ext}	Hamiltonian of the target in an external field, $H + O_b\phi_{\mathrm{ext}}(t)$; the coupling term alone	eV	4.1
$\hat{\varrho}_{\mathrm{I}}(t)$	Interaction-picture density matrix with respect to the full target H	1	4.1
z, z_{p}	Complex frequency $(\mathrm{Re}\,z, \mathrm{Im}\,z)$ of a response or Green's function continued off the real axis; a pole of the continuation in the lower half plane	eV	4.1
A	Generic responding operator of the Kubo derivation	—	4.1
P	Mean power absorbed from a harmonic field, $-(\omega\phi_0^2/2)\mathcal{V}\,\mathrm{Im}\,R_{bb}$	eV^2	4.1

Symbol	Meaning	Units	Sect.
$M_{\rm osc}$	Mass of the oscillator in the classical check	eV	4.1
$\eta_{\rm b}$	Finite broadening of an energy denominator, $\eta_{\rm b} \to 0^+$ in exact forms; also the adiabatic switching rate	eV	4.1
$S_{ab}(q,\omega)$	Response matrix $(2\pi/\mathcal{V})\sum p_i \langle i\vert O_a^\dagger\vert f\rangle\langle f\vert O_b\vert i\rangle \delta(\omega - E_f + E_i)$	eV^2 for density operators	4.2
$\mathcal{G}_{\rm VH}(r,t)$	Van Hove correlation function	$\mathrm{Å}^{-3}$	4.2
$S^{\rm st}(q)$	Static structure factor	1	4.2
$\tilde{V}_b(q)$, $\tilde{V}_e$, $\tilde{V}_{n_e}$	Potential with which the dark matter couples to the target operator $O_b(q)$; $\tilde{V}_e = \tilde{V}_{n_e} = \tilde{V}$ for the density	eV^{-2}	4.2
O_x	Combined operator $\sum_b x_b O_b$ with coefficients x_b in the positivity proof	—	4.2
$S^0(q,\omega)$	Independent-particle structure factor	eV^2	4.2
$\hat{n}_e(r)$	Position-space electron density operator $\sum_j \delta^3(\boldsymbol{r} - \boldsymbol{r}_j)$	$\mathrm{Å}^{-3}$	4.2
$g_{\rm pair}(r)$	Pair distribution function	1	4.2
$\boldsymbol{R}_\ell$, $\boldsymbol{r}_\ell$	Nuclear and electron positions of atom ℓ in the hydrogen-gas example	Å	4.2
$\omega_\pm(q)$	Upper and lower edges of the particle–hole continuum of the free gas, $qv_{\rm F} + q^2/2m_e$ and $\max(0, q^2/2m_e - qv_{\rm F})$	eV	4.2
u	Cosine of the angle between k (or v) and q in angular integrals	1	4.2
$N_{\rm at}$	Number of atoms of a sample or of a (super)cell	1	4.2
$\mathscr{M}^1$	First-moment (double-commutator) matrix	eV^4	4.3
$j_e(q)$	Paramagnetic electron current $\sum \frac{1}{2}\{p/m_e, \mathrm{e}^{\mathrm{i}q\cdot\boldsymbol{r}}\}$	1	4.3
ω_p	Electron plasma frequency, $\omega_p^2 = e^2 n_e/m_e$	eV	4.3
$f_{n0}^{\rm osc}$	Oscillator strength of the transition $0 \to n$	1	4.3
$\bar{K}$	Mean kinetic energy per electron (the relativistic f-sum estimate)	eV	4.3
n_κ, $j_\kappa(q)$	Number density of the nuclei of species κ; their paramagnetic current $\sum_{I\in\kappa} \frac{1}{2}\{P_I/M_\kappa, \mathrm{e}^{\mathrm{i}q\cdot X_I}\}$	$\mathrm{Å}^{-3}$; 1	4.3
$\mathscr{M}^1_{ss'}$, $\mathscr{M}^1_{ee}$, $\mathscr{M}^1_{ZZ}$	Species elements of the first-moment matrix $\mathscr{M}^1$ (electrons e, nuclei κ); its charge-number element	eV^4	4.3
$S_{e\kappa}$, $S_{\kappa e}$	Electron–nucleus elements of the response matrix S_{ab}	eV^2	4.3
$[H,O]_k$	O commuted k times with H	—	4.3
$f_2(\omega)$	Imaginary part of the atomic scattering factor, $(\pi/2)\omega \mathrm{d}f^{\rm osc}/\mathrm{d}\omega$	1	4.3
$\Omega_{\rm arc}$	Radius of the large semicircle of the Kramers–Kronig contour	eV	4.3
$\tilde{R}$	Coulomb-proper (irreducible) response	as R	4.4
$\varepsilon(q,\omega)$, ε_1, ε_2	Longitudinal dielectric function, its real and imaginary parts	1	4.4
$\varepsilon^{\rm T}(q,\omega)$	Transverse dielectric function; $\varepsilon^{\rm T} \to \varepsilon$ as $q \to 0$	1	4.4
$\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}(q,\omega)$	Dielectric matrix with local fields	1	4.4
$\varepsilon_{\rm mac}$	Macroscopic dielectric function $1/(\varepsilon^{-1})_{\mathbf{00}}$	1	4.4
ε_∞	Electronic (static-lattice) dielectric constant	1	4.4
$L(q,\omega)$	Loss function $\mathrm{Im}[-(\varepsilon^{-1})_{\mathbf{00}}]$	1	4.4
$k_{\rm TF}$	Thomas–Fermi wavevector	eV	4.4
$\sigma_{\rm cond}(\omega)$, σ_1, σ_2	Optical conductivity and its real and imaginary parts; $\sigma_1 = \omega\varepsilon_2$	eV	4.4
$-dE/\mathrm{d}x$	Stopping power (energy loss per unit path)	$\mathrm{eV\,Å}^{-1}$	4.4
$\phi_{\rm tot}(q,\omega)$, $\phi_{{\rm tot},\boldsymbol{G}}$, $\phi_{{\rm ext},G}$	Total (external + Hartree) potential energy acting on an electron; components at $q + \boldsymbol{G}$ of the total and external potentials	eV	4.4
$\delta n_e(q,\omega)$, $\delta n_{e,\boldsymbol{G}}$	Amplitude of the induced electron density $\delta\langle n_e(q)^\dagger\rangle/\mathcal{V}$; its component at $q + \boldsymbol{G}$	eV^3	4.4

Symbol	Meaning	Units	Sect.
$R_{\boldsymbol{GG'}}(q,\omega)$, $\tilde{R}_{\boldsymbol{GG'}}$	Density-response matrix in reciprocal-lattice vectors ($O_a = n_e(q+\boldsymbol{G}), O_b = n_e(q+\boldsymbol{G'})$); its proper part	eV^2	4.4
$\bar{R}$, $\bar{R}_{\boldsymbol{00}}$	Response with the $\boldsymbol{G}=0$ Coulomb term removed; $\varepsilon_{\mathrm{mac}} = 1 - v_{\mathrm{C}}(q)\bar{R}_{\boldsymbol{00}}$	eV^2	4.4
R_{ZZ}, S_{ZZ}	Response and structure factor of the charge-number density n_Z	eV^2	4.4
$R^{\alpha\beta}_{jj}$, $\tilde{R}^{\alpha\beta}_{jj}$	Current–current response and its proper part	eV^2	4.4
$\varepsilon^{\alpha\beta}(q,\omega)$, ε^{s}	Dielectric tensor (ε^{T} and ε are its projections); symmetrized dielectric matrix	1	4.4
$\boldsymbol{J}$	Induced current density, $\boldsymbol{J} = \sigma_{\mathrm{cond}}\boldsymbol{E}$	eV^3	4.4
$\varepsilon_{\mathrm{TF}}(q)$	Thomas–Fermi dielectric function $1 + k^2_{\mathrm{TF}}/q^2$	1	4.4
Z_1, M_1	Charge number and mass of a heavy charged projectile in dielectric stopping, $M_1 \to \infty$	1; eV	4.4
$\omega_{\mathrm{res}}(q)$	Resonance frequency of the single-oscillator loss function	eV	4.4
$\bar{E}$	Mean excitation energy of the loss function, $\bar{E}^2 = \int \omega L / \int (L/\omega)$ at $q \to 0$	eV	4.4
R^0	Independent-particle bubble: of the gas at any T, of a crystal, $R^0_{\boldsymbol{GG'}}$, in imaginary frequency $R^0(q, \mathrm{i}\omega^{\mathrm{B}}_j)$	as R	4.5
$\varepsilon_{\mathrm{Lin}}$, $\varepsilon_{\mathrm{Mer}}$	Lindhard and Mermin dielectric functions	1	4.5
$\omega_{\mathrm{pl}}(q)$, q_{c}	Plasmon dispersion; Landau-damping onset	eV	4.5
γ_{rel}	Relaxation rate	eV	4.5
α_{pl}	Plasmon dispersion coefficient, $\omega_{\mathrm{pl}} \simeq \omega_p + \alpha_{\mathrm{pl}} q^2/m_e, \alpha_{\mathrm{pl}} = (3/5)E_{\mathrm{F}}/\omega_p$	1	4.5
$\omega_{p,i}$, γ_i, $\omega_{\mathrm{edge},i}$, $A_i(q)$	Parameters of the i-th term of the Mermin-oscillator fit	eV; eV; eV; 1	4.5
ν, $\nu_\pm$	Reduced frequency z/qv_{F} and $\nu \pm q/2k_{\mathrm{F}}$ of the Lindhard function (ν in Sect. 4.5 and Appendix D.8)	1	4.5
y	$q/2k_{\mathrm{F}}$ in the Lindhard function	1	4.5
$F_{\mathrm{L}}(y)$	Static Lindhard function $\frac{1}{2} + [(1-y^2)/4y]\ln\lvert(1+y)/(1-y)\rvert$	1	4.5
$\ell(w)$	Lindhard logarithm $\ln[(w+1)/(w-1)]$ on the principal branch	1	4.5
R^{Mer}	Mermin (relaxation-time, number-conserving) density response; $\varepsilon_{\mathrm{Mer}} = 1 - v_{\mathrm{C}}R^{\mathrm{Mer}}$	eV^2	4.5
g_{deg}	Degeneracy factor (number of internal states): spin degeneracy 2 of the electron–hole bubble and of the singlet exciton	1	4.6
f_{xc}, f_{Hxc}	Exchange–correlation kernel; Hartree-plus-xc kernel $v_{\mathrm{C}} + f_{\mathrm{xc}}$	eV^{-2}	4.6
$G_{\mathrm{LF}}(q)$	Static local-field factor	1	4.6
$\Delta_{cv\boldsymbol{k}}$	Vertical transition energy $\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}$	eV	4.6
$\varepsilon^{\mathrm{IP}}_2$, $\varepsilon^{\mathrm{IP}}_1$	Independent-particle dielectric function at $q \to 0$	1	4.6
$\boldsymbol{n}_{\mathrm{gs}}(r)$	Ground-state density at which f_{xc} is evaluated	$\mathrm{Å}^{-3}$	4.6
G	Single-particle Green's function	eV^{-1}	4.7
$\mathsf{A}(k,\omega)$	Spectral function	eV^{-1}	4.7
Σ	Electron self-energy	eV	4.7
τ	Imaginary time, $0 \le \tau \le \beta$	eV^{-1}	4.7
ω^{F}_j, ω^{B}_j	Fermionic $(2j+1)\pi T$ and bosonic $2j\pi T$ Matsubara frequencies	eV	4.7
z_{qp}	Quasiparticle residue; $z_{\mathrm{qp}}(n,\boldsymbol{k})$ that of the one-electron state $n\boldsymbol{k}$	1	4.7
H_μ, H_{int}	Grand-canonical Hamiltonian $H - \mu_{\mathrm{ch}}\hat{N}_e$; its interaction part	eV	4.7
G^{R}, G^{A}, G, G_0	Retarded, advanced and time-ordered Green's functions; subscript 0 without interactions; $\mathsf{G}(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{F}}_j), \mathsf{G}(\boldsymbol{k},\tau)$ Matsubara forms	eV^{-1}	4.7

Symbol	Meaning	Units	Sect.
$\mathsf{G}^{\mathrm{R}}_{dd}$, $\mathsf{G}^{\mathrm{R}}_{\boldsymbol{k}d}$, A_{dd}, Σ^{R}_{d}	Green's functions of the level and the mixed level–band function, the spectral function and the self-energy of the level	eV^{-1}; eV^{-1}; eV^{-1}; eV	4.7
$\mathsf{F}^{\mathrm{R}}(k,\omega)$	Retarded correlation of $[c_{\boldsymbol{k}}, H_{\mathrm{int}}]$ with $c^{\dagger}_{\boldsymbol{k}}$ in the equation of motion	1	4.7
$\Sigma^{\mathrm{R}}(k,\omega)$, $\Sigma(k,\mathrm{i}\omega^{\mathrm{F}}_j)$	Retarded and Matsubara self-energies	eV	4.7
T_τ, T	Imaginary-time and real-time ordering symbols	—	4.7
d, $d^{\dagger}$, E_{lev}, E_{w}, $V_{\boldsymbol{k}}$	Level operators (a fermion), level energy, band width and hybridization	—; eV; eV; eV	4.7
$\xi_{\boldsymbol{k}}$, $E_{\boldsymbol{k}}$	Energy of orbital $\boldsymbol{k}$ from the chemical potential, $E-\mu_{\mathrm{ch}}$; Bogoliubov energy $(\xi^2_{\boldsymbol{k}}+\Delta^2_{\mathrm{BCS}})^{1/2}$	eV	4.7
W^{scr}, $W^{\mathrm{scr,R}}$	Screened Coulomb interaction $v_{\mathrm{C}}+v_{\mathrm{C}}Rv_{\mathrm{C}}=\varepsilon^{-1}v_{\mathrm{C}}$, in a crystal the matrix $(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'}v_{\mathrm{C}}(\lvert\boldsymbol{q}+G'\rvert)$; $W^{\mathrm{scr,R}}$ its retarded form; time-ordered when written with a real argument and no further superscript; evaluated at any $(\boldsymbol{q},\omega)$, complex where pairs are open, e.g	eV^{-2}	4.8
ω_{Ein}	Energy of a dispersionless (Einstein) boson; the Einstein energy $\langle\omega^{-1}\rangle^{-1}$ of a phonon spectrum, 30.4 meV for silicon	eV	4.8
$d_{\boldsymbol{Q}}$, $\omega_{\boldsymbol{Q}}$, $\phi_{\boldsymbol{Q}}$, $g(Q)$	Generic boson mode $\boldsymbol{Q}$, its energy, its field $d_{\boldsymbol{Q}}+d^{\dagger}_{-Q}$ and its coupling to electrons, $g(-\boldsymbol{Q})=g(\boldsymbol{Q})^*$	—; eV; —; eV	4.8
$\mathsf{D}_0(Q,\mathrm{i}\omega^{\mathrm{B}}_j)$, $\mathsf{D}_0(Q,\omega)$	Free boson propagator $2\omega_{\boldsymbol{Q}}/((\mathrm{i}\omega^{\mathrm{B}}_j)^2-\omega^2_{\boldsymbol{Q}})$; time-ordered $2\omega_{\boldsymbol{Q}}/(\omega^2-\omega^2_{\boldsymbol{Q}}+\mathrm{i}0^+)$	eV^{-1}	4.8
Γ_{em}, Γ_{out}, Γ_{in}	Golden-rule boson-emission rate; out- and in-scattering rates of a fermion state, $-2\,\mathrm{Im}\,\Sigma^{\mathrm{R}}=\Gamma_{\mathrm{out}}+\Gamma_{\mathrm{in}}$	eV	4.8
E_{c}, $\ell_{\mathrm{b}}(x)$	Conduction-band bottom of the model band and band-integral logarithm $\ln[(x-E_{\mathrm{c}}+\mathrm{i}0^+)/(x-E_{\mathrm{c}}-E_{\mathrm{w}}+\mathrm{i}0^+)]$	eV; 1	4.8
ξ'	ξ_{k-Q}, energy of the intermediate electron in the boson self-energy	eV	4.8
$B(q,\omega)$	Spectral weight of $W^{\mathrm{scr}}-v_{\mathrm{C}}$, $-\,\mathrm{Im}\,W^{\mathrm{scr,R}}/\pi=v_{\mathrm{C}}L/\pi$	eV^{-2}	4.8
E_{cs}	Total energy of the lines crossed by a cut	eV	4.8
$\bar{H}$	Hamiltonian without the Coulomb terms at $\pm\boldsymbol{q}$	eV	4.8
$A^{\lambda}_{cv\boldsymbol{k}}$, E_λ, λ	Exciton envelope, exciton energy, exciton label: eigenpairs of the transition-basis BSE with K^{x} (no $\boldsymbol{G}=0$ term), which depend on $\boldsymbol{q}$	1; eV	4.9
E^b_x, a_x, μ_x	Exciton binding energy, radius, reduced mass of the Wannier–Mott (hydrogenic) exciton	eV; Å; m_e	4.9
$F_\lambda(r)$, F_{ns}	Exciton envelope function in the relative coordinate, $\mathscr{V}^{-1/2}\sum_k A^{\lambda}_{\boldsymbol{k}}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot r}$, $\int\lvert F\rvert^2=1$; hydrogenic envelope	$\mathrm{Å}^{-3/2}$	4.9
K^{x}, K^{d}, H^{exc}	Bethe–Salpeter exchange kernel without its $\boldsymbol{G}=0$ term and direct kernel; resonant (Tamm–Dancoff) BSE Hamiltonian $\Delta+g_{\mathrm{deg}}K^{\mathrm{x}}-K^{\mathrm{d}}$ in the transition basis	eV	4.9
Chapter 5: Lattice dynamics			
$E^{\mathrm{ad}}(u)$	Born–Oppenheimer energy surface $E^{\mathrm{ad}}(u)\equiv E_0(u)$, the ground-state eigenvalue of $H_{\mathrm{el}}(u)$ at fixed nuclei, nuclear repulsion V_{NN} included	eV	5.1
κ_{BO}	Born–Oppenheimer small parameter $(m_e/M)^{1/4}$	1	5.1
F_κ, $F_{\boldsymbol{R}\kappa}$	Zone-center Hellmann–Feynman force on sublattice κ, $-\partial E^{\mathrm{ad}}_{\mathrm{cell}}/\partial u_\kappa$; force on the nucleus κ of cell $\boldsymbol{R}$	$\mathrm{eV\,Å}^{-1}$	5.1
$u_{\boldsymbol{R}\kappa}$, u	Nuclear displacement from equilibrium, $X_I=\boldsymbol{R}+\tau_\kappa+u_{\boldsymbol{R}\kappa}$; the set of all displacements	Å	5.1
$\Phi^{(n)}$	n-th order force constants	$\mathrm{eV\,Å}^{-n}$	5.1
$E^{\mathrm{ad}}_{\mathrm{cell}}$	Energy surface per cell for a uniform displacement of each sublattice	eV	5.1

Symbol	Meaning	Units	Sect.
$H_{\mathrm{el}}(u)$	Electronic Hamiltonian at fixed nuclei, $T_e + V_{ee} + V_{eN} + V_{NN}$	eV	5.1
T_e, V_{ee}, V_{eN}, V_{NN}	Electron kinetic energy; electron–electron, electron–nucleus and nucleus–nucleus interaction energies	eV	5.1
$\Psi_i(r;u)$, $E_i(u)$	Adiabatic electronic states (real) and energies at fixed nuclei, eigenstates of $H_{\mathrm{el}}(u)$; $E_0(u) = E^{\mathrm{ad}}(u)$	—; eV	5.1
$\Psi_i^{\mathrm{nuc}}(u)$, $\tilde{\Psi}_i^{\mathrm{nuc}}(u)$	Nuclear wavefunctions of the adiabatic expansion; coefficients of the expansion in the equilibrium states $\Psi_i(r;0)$	—	5.1
$\mathcal{N}_{ji}$	Nonadiabatic coupling operator between adiabatic states j and i, first- and second-derivative (derivative and scalar) couplings, $\propto 1/M_\kappa$	eV	5.1
ℓ_{zp}	Zero-point length $(M_\kappa\omega)^{-1/2}$ of the bond coordinate (reduced mass $M_\kappa/2$), and the estimate $\kappa_{\mathrm{BO}}a_0$ in the power counting	Å	5.1
ω_{ph}	Generic phonon energy in estimates	meV	5.1
ΔE, F_{j0}	Excitation energy $E_j - E_0$ of the adiabatic states in estimates; matrix element of the force between Ψ_j and Ψ_0 (two state labels)	eV; eV Å^{-1}	5.1
ω_{opt}, $\omega_{\Gamma,\mathrm{opt}}$	Zone-center optical frequency of silicon	meV	5.1
Φ_{opt}, $\Phi^{(3)}_{\mathrm{opt}}$, $\Phi^{(4)}_{\mathrm{opt}}$	Curvature, cubic and quartic coefficients of $E^{\mathrm{ad}}_{\mathrm{cell}}$ along the [111] bond coordinate of silicon	eV Å^{-n}	5.1
$\langle u_x^2\rangle$, $\langle u^2\rangle$	Mean-square displacement of an atom along one Cartesian axis ($= B^{xx}_\kappa$); its trace tr B_κ, $\langle u^2\rangle = 3\langle u_x^2\rangle$ at a cubic site	Å^2	5.1
$D^{\mathrm{dyn}}(Q)$	Dynamical matrix	eV^2	5.2
Q	Phonon wavevector (first zone)	eV	5.2
ν	Phonon branch	—	5.2
$\omega_{\boldsymbol{Q}\nu}$	Phonon frequency	meV	5.2
$e_{\kappa\nu}(Q)$	Phonon polarization vector, $\sum_\kappa \lvert e_{\kappa\nu}\rvert^2 = 1$	1	5.2
$g_{\mathrm{ph}}(\omega)$	Phonon density of states $N^{-1}\sum_{\boldsymbol{Q}\nu}\delta(\omega - \omega_{\boldsymbol{Q}\nu})$, normalized to $3n_{\mathrm{at}}$ modes per cell	eV^{-1}	5.2
Θ_{D}, ω_{D}	Debye temperature and Debye frequency ($\omega_{\mathrm{D}} = \Theta_{\mathrm{D}}$ in natural units)	K; meV	5.2
$\Lambda^{\mathrm{ac},\alpha\beta}(\hat{Q})$, $\Lambda^{\mathrm{ac}}_\nu(\hat{Q})$	Acoustic sound-speed matrix, $\omega^2 = \Lambda^{\mathrm{ac}}Q^2$ at long wavelength, and its eigenvalues c_ν^2	km^2 s^{-2}	5.2
$D_n(Q)$, $T^{\alpha\beta\gamma}$, $\boldsymbol{a}_{\hat{\boldsymbol{e}}}$, $\boldsymbol{a}'_{\hat{\boldsymbol{e}}}$, $\boldsymbol{o}$, ω_{o}	Order-n terms ($n = 0, 1, 2$) of D^{dyn} in powers of Q; first moment $\sum_R \Phi^{(2)}R^\gamma$ of the force constants; acoustic and optical vectors at Γ and an optical frequency at Γ	eV^2; eV Å^{-1}; 1; meV	5.2
a_{c}, K, M_1, M_2, $\omega_{Q\pm}$, ω_0+, ω_{X}	Period, spring constant and masses of the diatomic chain; its two branches, the upper branch at $Q = 0$ and the chain's X-point frequency	Å; eV Å^{-2}; eV; meV	5.2
Q_{D}	Debye wavevector $(6\pi^2 n_{\mathrm{at}}/\Omega)^{1/3}$, one wavevector per atom	Å^{-1}	5.2
$\bar{c}$	Debye-averaged sound speed, $3/\bar{c}^3 = \sum_\nu \langle c_\nu^{-3}\rangle$	km s^{-1}	5.2
$g_{\mathrm{D}}(\omega)$	Debye density of states $9n_{\mathrm{at}}\omega^2/\omega_{\mathrm{D}}^3$ per cell below ω_{D}	eV^{-1}	5.2
μ_{red}	Reduced mass: of the diatomic chain, of a polar mode	eV	5.2
$\gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu}$	Grüneisen parameter $-\partial \ln\omega_{\boldsymbol{Q}\nu}/\partial\ln\Omega$ of mode $Q\nu$	1	5.2
$b_{\boldsymbol{Q}\nu}$, $b^\dagger_{\boldsymbol{Q}\nu}$	Phonon operators	—	5.3
$B_{\boldsymbol{Q}\nu}$	Displacement field $b_{\boldsymbol{Q}\nu} + b^\dagger_{-Q\nu}$	—	5.3
$n_{\boldsymbol{Q}\nu}$	Phonon occupation $n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$	1	5.3
$\ell_{\boldsymbol{Q}\nu}$	Zero-point length $(2M_{\mathrm{cell}}\omega_{\boldsymbol{Q}\nu})^{-1/2}$, with the $1/\sqrt{N}$ kept outside	Å	5.3
B_κ	Displacement tensor $\langle u_\kappa u_\kappa\rangle$	Å^2	5.3
H_{N}	Harmonic nuclear Hamiltonian	eV	5.3

Symbol	Meaning	Units	Sect.
$\boldsymbol{p}_{\boldsymbol{R}\kappa}$	Momentum $-\mathrm{i}\nabla_{\boldsymbol{R}\kappa}$ of the nucleus κ in cell $\boldsymbol{R}$	eV	5.3
$u_{\boldsymbol{Q}\nu}$, $p_{\boldsymbol{Q}\nu}$	Normal coordinate and normal momentum of mode $Q\nu$	$\mathrm{eV}^{-1/2}$; $\mathrm{eV}^{1/2}$	5.3
$g_{\mathrm{ph},\kappa}(\omega)$	Atom-projected phonon density of states $N^{-1}\sum_{\boldsymbol{Q}\nu}\lvert e_{\kappa\nu}(\boldsymbol{Q})\rvert^2\delta(\omega-\omega_{\boldsymbol{Q}\nu})$, normalized to three	eV^{-1}	5.3
$\bar{\omega}$	Typical phonon frequency of the estimate	meV	5.3
$\partial_{\kappa\alpha,\boldsymbol{Q}}$	Derivative with respect to the amplitude of the displacement wave $u^{\alpha}_{\boldsymbol{R}\kappa}\propto \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$, $\sum_R \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\partial/\partial u^{\alpha}_{\boldsymbol{R}\kappa}$	$\mathrm{Å}^{-1}$	5.4
$\Phi^{\mathrm{SC}}_{\kappa\kappa'}(R_s)$, T, R_s	Aliased (supercell) force constants $\sum_T \Phi^{(2)}(R_s+T,0)$ of an N_Q^3 supercell; supercell translation; cell vector within the supercell	$\mathrm{eV\,Å}^{-2}$; Å	5.4
$\hat{P}^{\mathrm{c}}$	Projector $1-\hat{P}^{\mathrm{occ}}$ on the empty Kohn–Sham states	1	5.4
E_{NN}, E_{el}, v_κ	Nucleus–nucleus (Ewald) energy; electronic energy; potential of one nucleus or pseudo-ion of species κ, $v_\kappa(\boldsymbol{r})$ at the origin, its Fourier component $v_\kappa(G)$, $v_\kappa(\boldsymbol{p})$ at the momentum given to the electrons	eV	5.4
$\Delta^{\mathrm{ASR}}_\kappa$, ω_{ASR}	Acoustic-sum-rule violation $\sum_{\boldsymbol{R}\kappa'}\Phi^{(2)}_{\kappa\kappa'}(0,\boldsymbol{R})$ of computed force constants; the Γ acoustic frequency it leaves before the repair	$\mathrm{eV\,Å}^{-2}$; meV	5.4
$D^{\mathrm{dyn},el}$	Electronic part of the dynamical matrix	eV^2	5.4
$\psi^{(l)}$, $E^{(l)}$	Order-l coefficients of an orbital and of the energy in a perturbation of strength λ	—; eV	5.4
$\partial_{\kappa\alpha,Q}V_{\mathrm{KS}}$	First-order change of the Kohn–Sham potential for a displacement wave of sublattice κ along α	$\mathrm{eV\,Å}^{-1}$	5.4
T_d, O_h	Tetrahedral and octahedral point groups (Schoenflies labels)	—	5.4
H_{KS}, V_{ext}	Kohn–Sham Hamiltonian; external (ionic) potential of DFPT	eV	5.4
a_{sh}, $\hat{P}^{\mathrm{occ}}_{\boldsymbol{k}+\boldsymbol{Q}}$	Sternheimer shift (larger than the valence bandwidth); projector on the occupied states at $\boldsymbol{k}+\boldsymbol{Q}$	eV; 1	5.4
N_Q, Q_c	Phonon-grid points per direction; radius of the Γ-cell sphere	1; eV (quoted in $\mathrm{Å}^{-1}$)	5.4
$\mathcal{E}$, $\boldsymbol{P}$	Macroscopic electric field; macroscopic polarization	eV^2; eV^2	5.5
Z^*_κ	Born effective-charge tensor $\partial F_\kappa/\partial\mathcal{E}$ (in units of e)	1	5.5
ε_0	Static dielectric constant $\varepsilon(0,\omega\to 0)$, lattice included: a pure number, never the vacuum permittivity	1	5.5
ω_{L}, ω_{T}	Longitudinal and transverse optical frequencies at $\boldsymbol{Q}\to 0$	meV	5.5
$\boldsymbol{D}$	Electric displacement field $\mathcal{E}+\boldsymbol{P}$	eV^2	5.5
$H_{\mathcal{E}}(t)$	Coupling $-e\sum u\cdot Z^*\cdot\mathcal{E}$ of the nuclei to a uniform macroscopic field	eV	5.5
Z^*_{Ga}, Z^*_{As}, Z^*	Born charges of GaAs by species; their common magnitude in a two-atom cubic crystal, $Z^*_{\mathrm{Ga}}=-Z^*_{\mathrm{As}}=Z^*$	1	5.5
$\boldsymbol{u}_{\mathrm{rel}}$	Relative displacement $\boldsymbol{u}_1-\boldsymbol{u}_2$ of the two sublattices of a diatomic polar crystal	Å	5.5
$C^{\mathrm{an}}_{\kappa\kappa'}$, $C^{\mathrm{na}}_{\kappa\kappa'}(\hat{Q})$, C_0	Analytic (zero-field) and nonanalytic (dipole) zone-center force constants; the single constant of the cubic diatomic crystal	$\mathrm{eV\,Å}^{-2}$	5.5
$\boldsymbol{z}_\nu$	Mode charge vector $\sum_\kappa e_{\kappa\nu}\cdot Z^*_\kappa/\sqrt{M}_\kappa$	$\mathrm{eV}^{-1/2}$	5.5
$E^{\mathrm{ad}}_{\mathcal{E}}(u,\mathcal{E})$	Energy per cell at fixed macroscopic field; E^{ad} at $\mathcal{E}=0$	eV	5.5
$\omega_{\mathrm{T}\nu}$, $\omega_{\mathrm{L}\nu}$, $\omega_{0\nu}$	Transverse and longitudinal zone-center frequencies of polar mode ν (generalized Lyddane–Sachs–Teller relation); frequency of branch ν at Γ without the field ($\omega_{\mathrm{T}},\omega_{\mathrm{L}}$ for a single polar mode)	meV	5.5
$W_\kappa(q)$	Debye–Waller exponent, $2W_\kappa(\boldsymbol{q})=\boldsymbol{q}\cdot B_\kappa\cdot\boldsymbol{q}$	1	5.6
$\mathcal{C}_{\kappa\kappa'}(R,t)$, $\mathcal{C}_{\kappa\kappa'}(t)$	Displacement correlation function $\langle u_{\boldsymbol{R}\kappa}(t)u_{\boldsymbol{0}\kappa'}(0)\rangle$; $\mathcal{C}_{\kappa\kappa'}(t)$ where the cell is clear	$\mathrm{Å}^2$	5.6

Symbol	Meaning	Units	Sect.
S_{HR}	Huang–Rhys factor	1	5.6
$P_1(\omega)$, $P_n(\omega)$	Single-phonon distribution of the self response ($\propto g_{\mathrm{ph}}(\lvert\omega\rvert)[n_{\mathrm{B}}(\omega)+1]/\omega$, unit norm); its n-fold convolution, the n-phonon distribution	eV^{-1}	5.6
p_n^{ph}	Probability that a momentum transfer involves n phonons	1	5.6
H_{g}, H_{e}	Nuclear Hamiltonians on the ground and excited electronic surfaces of a displaced oscillator	eV	5.6
Δx, $A(E)$, w_p	Shift of the equilibrium coordinate between the two surfaces; absorption line shape of the displaced oscillator; weight of the line $E_{\mathrm{ZPL}}+p\omega_0$	Å; eV^{-1}; 1	5.6
$I_p(z)$	Modified Bessel function of the first kind	1	5.6
E_{ZPL}	Zero-phonon-line energy of a color center	eV	5.6
ℓ_0	Zero-point length $(2M\omega_0)^{-1/2}$ of a single oscillator of mass M and frequency ω_0	Å	5.6
$\gamma_{\boldsymbol{Q}\nu}$, $\gamma_{\boldsymbol{Q}\nu}^{\mathrm{anh}}$	Phonon linewidth (full width); its anharmonic part	meV	5.7
η_{anh}	Anharmonic small parameter	1	5.7
$V^{(3)}(1,2,3)$	Three-phonon vertex, an energy; the labels $i=(Q_i\nu_i)$	meV	5.7
H_3	Cubic part of the nuclear Hamiltonian in phonon fields (H_N is the harmonic part)	eV	5.7
$\Delta\omega_{\boldsymbol{Q}\nu}^{\mathrm{anh}}$	Anharmonic frequency shift	meV	5.7
$\gamma_{\Gamma,\mathrm{opt}}$, $\gamma_{\Gamma,\mathrm{opt}}^{\mathrm{anh}}$	Width (FWHM) and anharmonic width of the zone-center optical phonon of silicon ($\gamma_{\boldsymbol{Q}\nu}$ at $\boldsymbol{Q}=\Gamma,\nu=$ opt)	meV	5.7
$\mathcal{F}_{\mathrm{trial}}$, V_{trial}, δX_a	Free energy and potential of the SCHA trial Hamiltonian H_{trial}; displacement $X_a-\bar{X}_a$ from the centroid	eV; eV; Å	5.7
$\varepsilon^{\alpha\beta}$	Homogeneous strain tensor of the quasi-harmonic argument	1	5.7
$\rho_{\mathrm{pair}}(\omega_1)$	Two-phonon density of final states resolved by the energy of one daughter, $N^{-1}\sum\delta(\omega_1-\omega_{\boldsymbol{Q}\nu_1})\delta(\omega_{\mathrm{opt}}-\omega_{\boldsymbol{Q}\nu_1}-\omega_{\boldsymbol{Q}\nu_2})$	eV^{-2}	5.7
S_{ent}	Entropy	1	5.7
$\tilde{\varrho}$, H_{trial}	Trial density matrix of the SCHA; trial harmonic Hamiltonian	1; eV	5.7
X_a, $\bar{X}_a$, $\bar{X}$	Cartesian nuclear coordinates of a supercell; their centroids	Å	5.7
$\Phi_{\mathrm{aux}}^{(2)}$, $\Phi_{\mathrm{aux},ab}^{(2)}$, $\tilde{\Phi}^{(2)}$, $\tilde{\Phi}_{ij}^{(2)}$	SCHA auxiliary force constants; TDEP effective second-order force constants	$\mathrm{eV\,Å}^{-2}$	5.7
Chapter 6: Electron–phonon coupling			
H_{ep}	Electron–phonon Hamiltonian, general bare form	eV	6.1
v_N	Nuclear velocity after a kick, q/M	1	6.1
$\boldsymbol{d}_{ji}^{\boldsymbol{R}\kappa}$	Derivative-coupling vector $\langle\Psi_j\lvert\nabla_{\boldsymbol{R}\kappa}\Psi_i\rangle=\boldsymbol{F}_{ji}/(E_i-E_j)$, antisymmetric, $\boldsymbol{d}_{ii}=0$, a Berry connection	eV (quoted in $\mathrm{Å}^{-1}$)	6.1
ΔE_j, $\boldsymbol{F}_{ji}$	Excitation energy E_j-E_0 of the adiabatic state j; matrix element $\langle\Psi_j\lvert\nabla_{\boldsymbol{R}\kappa}V_{eN}\lvert\Psi_i\rangle$ of the force	eV; $\mathrm{eV\,Å}^{-1}$	6.1
τ_{acc}	Acceleration time of a kicked nucleus	eV^{-1} (quoted in s)	6.1
ω_{e}	Frequency of the bound-electron oscillator of the solvable model	eV	6.1
P_0	Probability that the electrons stay in the ground state after a kick, $\mathrm{e}^{-\lvert\alpha_{\mathrm{d}}\rvert^2}$ in the model, $\lvert F_{1s}\rvert^2$ for hydrogen	1	6.1
M_{H}, μ_{ep}, a_{ep}	Hydrogen-atom mass m_p+m_e; electron–proton reduced mass m_em_p/M_{H}; its Bohr radius $1/(\alpha\mu_{ep})$	eV; eV; Å	6.1
$\mu_{\chi\mathrm{H}}$	Dark-matter–hydrogen-atom reduced mass	eV	6.1

Symbol	Meaning	Units	Sect.
$\boldsymbol{\rho}$, $\boldsymbol{R}_{\mathrm{cm}}$	Relative coordinate $\boldsymbol{r}_e - \boldsymbol{X}$ and center of mass $(m_p\boldsymbol{X} + m_e\boldsymbol{r}_e)/M_{\mathrm{H}}$ of the hydrogen atom; $\boldsymbol{\rho}_e$ and $\boldsymbol{R}_{\mathrm{cm}}$ of a many-electron atom with the mass m_A	Å	6.1
$g_{mn\nu}(k, Q)$	Electron–phonon vertex	eV	6.2
g^{S}, g^{L}	Short- and long-range parts of g	eV	6.2
D^n, $D^n_{\alpha\beta}$, d^n_ν	Acoustic deformation-potential tensor and its Cartesian components; optical deformation potential of band n and zone-center mode ν	eV; eV Å^{-1}	6.2
C_{F}, α_{F}	Fröhlich coupling strength $(g^{\mathrm{L}} = \mathrm{i}C_{\mathrm{F}}\rho_{mn}/Q, C^2_{\mathrm{F}} = (e^2\omega_{\mathrm{L}}/2\Omega)(1/\varepsilon\infty - 1/\varepsilon_0))$; dimensionless Fröhlich constant $\alpha(1/\varepsilon\infty - 1/\varepsilon_0)(m^*/2\omega_{\mathrm{L}})^{1/2}$	eV^2 (natural units); 1	6.2
$\partial_{\kappa\alpha,Q} v_{\mathrm{KS}}$, v_{KS}	Lattice-periodic part of the potential change, $\partial_{\kappa\alpha,Q} V_{\mathrm{KS}} = \mathrm{e}^{\mathrm{i}Q\cdot\boldsymbol{r}}\partial_{\kappa\alpha,Q} v_{\mathrm{KS}}$ (lower-case v marks the periodic part)	eV Å^{-1}	6.2
$\Delta_{\boldsymbol{Q}\nu} v_{\mathrm{KS}}$	Mass-weighted mode potential $\sum_{\kappa\alpha}(M_{\mathrm{cell}}/M_\kappa)^{1/2}\mathrm{e}^\alpha_{\kappa\nu}\partial_{\kappa\alpha,Q} v_{\mathrm{KS}}$; $g_{mn\nu} = \ell_{\boldsymbol{Q}\nu}\langle u_{m,\boldsymbol{k}+\boldsymbol{Q}}\vert\Delta_{\boldsymbol{Q}\nu} v_{\mathrm{KS}}\vert u_{n\boldsymbol{k}}\rangle_\Omega$	eV Å^{-1}	6.2
$S_{\alpha\beta}$, E_1	Strain tensor $\frac{1}{2}(\partial_\alpha u_\beta + \partial_\beta u_\alpha)$ of a long acoustic wave; isotropic deformation potential $\Omega\partial\varepsilon_n/\partial\Omega$ (Bardeen–Shockley)	1; eV	6.2
Ξ_d, Ξ_u, $\hat{k}_{\mathrm{c}}$	Herring–Vogt dilatation and uniaxial deformation potentials of a conduction valley; unit vector along the valley axis	eV; 1	6.2
$e^{\mathrm{pz}}_{\alpha\beta\gamma}$, g^{pz}	Piezoelectric tensor; piezoelectric vertex, $\propto Q^{-1/2}$	eV^2 (quoted in C m^{-2}); eV	6.2
$\boldsymbol{t}$, $\hat{T}_{\boldsymbol{t}}$	Rigid translation of the whole crystal; its translation operator $\mathrm{e}^{-\mathrm{i}\boldsymbol{t}\cdot\hat{\boldsymbol{p}}}$	Å; —	6.2
g^W, $\boldsymbol{R}_e$, $\boldsymbol{R}_p$	Real-space (Wannier-gauge) electron–phonon matrix $g^W_{m'n'\kappa\alpha}(\boldsymbol{R}_e, \boldsymbol{R}_p)$; lattice vectors of the electron hop and of the displaced atom	eV Å^{-1}; Å	6.2
Σ^{FM}, Σ^{AH}, g^{AH}	Fan–Migdal and Allen–Heine self-energies; second-order (Allen–Heine) vertex	eV	6.3
$\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}}$	Phonon-renormalized band energy	eV	6.3
$H^{(2)}_{\mathrm{ep}}$	Second-order electron–phonon Hamiltonian $(1/N)\sum g^{\mathrm{AH}} c^\dagger c B B$	eV	6.3
ZPR	Zero-point renormalization $\varepsilon^{\mathrm{ren}}(T=0) - \varepsilon$ of a state; of a gap, conduction minus valence	meV	6.3
$\Delta\varepsilon_{n\boldsymbol{k}}(T)$	Phonon-induced shift $\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}}(T) - \varepsilon_{n\boldsymbol{k}}$	eV	6.3
$\hat{p}_\nu$	$\hat{e}_\nu \cdot \hat{p}$, momentum along a polarization	eV	6.3
Ξ, $Q_{\max}$	Effective deformation constant of one branch, Ξ^2 averaged over directions; largest emitted wavevector $2(\boldsymbol{k} - m^* c)$	eV; eV	6.3
m^*_{d}	Density-of-states mass $(m^*_\parallel m^{*2}_\perp)^{1/3}$ of one valley, 0.322 m_e in Si	eV (quoted in m_e)	6.3
$\Pi_\nu(Q, \omega)$	Phonon self-energy of branch ν	eV^2	6.4
$\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}$	Electron–phonon part of the phonon linewidth, $-\operatorname{Im}\Pi_\nu(\boldsymbol{Q}, \omega_{\boldsymbol{Q}\nu})/\omega_{\boldsymbol{Q}\nu}$, a full width	meV	6.4
$\alpha^2 F(\omega)$	Eliashberg spectral function	1	6.4
λ_{ep}	Electron–phonon coupling constant $2\int \mathrm{d}\omega\alpha^2 F(\omega)/\omega$	1	6.4
$\mathsf{D}_\nu(Q, \omega)$, D	Dressed phonon propagator $2\omega_{\boldsymbol{Q}\nu}/(\omega^2 - \omega^2_{\boldsymbol{Q}\nu} - \Pi_\nu)$	eV^{-1}	6.4
$\tilde{\omega}_{\boldsymbol{Q}\nu}$, $\omega^{\mathrm{ren}}_{\boldsymbol{Q}\nu}$	Pole $\omega_{\boldsymbol{Q}\nu} + \operatorname{Re}\Pi_\nu/2\omega_{\boldsymbol{Q}\nu}$ of the dressed phonon; nonadiabatic (renormalized) frequency, $\omega_{\boldsymbol{Q}\nu}$ being the adiabatic DFPT one	meV	6.4
$\lambda_{\boldsymbol{Q}\nu}$	Mode coupling $\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}/(2\pi\mathcal{N}_{\mathrm{F}}\Omega\omega^2_{\boldsymbol{Q}\nu})$, λ_{ep} its zone average	1	6.4
$\langle\ldots\rangle_{\mathrm{FS}}$	Fermi-surface average	—	6.4

Symbol	Meaning	Units	Sect.
$w(q)$	Screened empty-core ionic form factor per cell $-3e^2\cos(qr_c)/[\Omega q^2\varepsilon_{\rm Lin}(q,0)]$ of the Al model	eV	6.4
$\omega_{\rm TA}$, $\omega_{\rm LA}$, $\omega_{\rm TO}$, $I_{\rm TA}$, $I_{\rm LA}$	Peak energies and weights of the two-peak α^2F model of Al	meV	6.4
$\varepsilon_2^{\rm PA}$	Phonon-assisted imaginary dielectric function; at finite $\boldsymbol{q}$, $\varepsilon_2^{\rm PA}(\boldsymbol{q},\omega)$ is the phonon-assisted ε_2 whose $\boldsymbol{q}\to 0$ limit is the optical one	1	6.5
$\alpha_{\rm abs}(\omega)$	Optical absorption coefficient; $\alpha_{{\rm abs},\nu}(\omega;T)$ its branch-ν component at temperature T	$\rm cm^{-1}$	6.5
$\hat{e}_\gamma$	Unit polarization vector of a photon	1	6.5
$\hat{\boldsymbol{V}}$, H_A	Total electron velocity operator $\sum_j \boldsymbol{v}_j = \sum \boldsymbol{v}_{mn}c^\dagger c$; coupling $e\boldsymbol{A}\cdot\hat{\boldsymbol{V}}$ of a uniform vector potential to the electrons	1; eV	6.5
$M^{\pm}_{cv\nu}(k,Q)$, $M^{\pm}_{cv\nu}(k,q,Q)$	One-electron phonon-assisted amplitude summed over all intermediate bands p, upper sign emission: with two arguments the optical amplitude of the velocity vertex, with three the dimensionless density amplitude at momentum transfer $\boldsymbol{q}$; $M^{\pm}_{cv\nu}(\boldsymbol{k},\boldsymbol{Q};\hat{e})$ the optical amplitude along the polarization $\hat{e}$, the $\boldsymbol{q}\to 0$ limit of $(\omega/q)M^{\pm}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q})$ along $\hat{e}=\hat{q}$; the argument count is part of the name	1 (a velocity); 1	6.5
$\kappa_{\rm ext}$	Extinction coefficient $\varepsilon_2/2n_r, \varepsilon=(n_r+{\rm i}\kappa_{\rm ext})^2$	1	6.5
ω_ν, n_ν, A_ν, $A_{\rm TA}$, $A_{\rm TO}$, $n_{\rm TA}$, $n_{\rm TO}$	Energy and occupation of the phonon of branch ν at $\boldsymbol{Q}=\boldsymbol{k}_{\rm c}$; Hall–Bardeen–Blatt coefficient of branch ν	meV; 1; $\rm cm^{-1}\,eV^{-2}$	6.5
$\boldsymbol{\kappa}_{\rm e}$, $\boldsymbol{\kappa}_{\rm h}$	Electron and hole crystal momenta measured from the band extrema; their kinetic energies are $E_{\rm e}, E_{\rm h}$	eV	6.5
$n_r(\omega)$	Refractive index of the medium	1	6.5
M_ν	Value of $M^{\pm}_{cv\nu}$ at the band extrema	1	6.5
Chapter 7: Magnets, superconductors, Dirac materials and defects			
$J_{ij}^{\rm ex}$	Exchange constant	meV	7.1
$U_{\rm H}$	On-site (Hubbard) repulsion	eV	7.1
$\boldsymbol{S}_j$	Ionic spin operator of site j	1	7.1
$a_{\boldsymbol{k}\nu}$, $a^\dagger_{\boldsymbol{k}\nu}$	Magnon operators	—	7.1
$u^{\rm M}_{\boldsymbol{k}}$, $v^{\rm M}_{\boldsymbol{k}}$	Magnon Bogoliubov coefficients	1	7.1
$F^{\rm mag}(q)$	Magnetic form factor	1	7.1
$s_e(q)$	Electron spin-density operator $\sum_i {\rm e}^{{\rm i}\boldsymbol{q}\cdot\boldsymbol{r}_i}\sigma_i/2$	1	7.1
$K_{\rm ex}$	Exchange integral of two orthogonal orbitals, the Coulomb self-energy of $\phi_a^*\phi_b$, positive	eV	7.1
$E_{\rm orb}$	Sum of the two one-electron orbital energies in the Dirac identity of the two-electron problem	eV	7.1
$\Delta_{\rm CT}$	Charge-transfer energy of a charge-transfer insulator, $\Delta_{\rm CT}<U_{\rm H}$ in NiO	eV	7.1
d_j, $d_{{\rm A}i}$, $d_{{\rm B}j}$, $d_{{\rm A}\boldsymbol{k}}$, $d_{{\rm B}\boldsymbol{k}}$	Holstein–Primakoff bosons of the ferromagnet; of sublattices A and B, and their transforms at wavevector k; special cases of the generic d_γ	—	7.1
$J^{\rm ex}_{\rm AA}(k)$, $J^{\rm ex}_{\rm AB}(k)$	Intra- and inter-sublattice exchange sums of a two-sublattice magnet	meV	7.1
$A_{\boldsymbol{k}}$, $B_{\boldsymbol{k}}$	Diagonal and pair coefficients of the two-sublattice spin-wave Hamiltonian; $\lvert\boldsymbol{S}\rvert A_{\boldsymbol{k}}$ is the diagonal energy of the boson problem	meV	7.1
$\omega^{\rm M}_{\boldsymbol{k}}$	Magnon energy	meV	7.1
$N_{\rm s}$	Number of sites per sublattice	1	7.1
$E_{\rm cl}$, $E_{\rm gs}$	Classical (fully polarized or Néel) energy of the Heisenberg model; ground-state energy of the linear spin-wave Hamiltonian, $E_{\rm cl}+\sum_k(\omega^{\rm M}_{\boldsymbol{k}}-\lvert\boldsymbol{S}\rvert A_{\boldsymbol{k}})$	eV	7.1

Symbol	Meaning	Units	Sect.
$\theta_{\boldsymbol{k}}$	Bogoliubov angle: of the magnon problem, $\tanh 2\theta_{\boldsymbol{k}} = B_{\boldsymbol{k}}/A_{\boldsymbol{k}}$; of the BCS problem	1	7.1
$\zeta_{\boldsymbol{k}}$	Type-II J_2-only structure factor $(\cos k_x a + \cos k_y a + \cos k_z a)/3$	1	7.1
$S^{\pm}(q)$, $S^{\pm}_j$	Transverse ionic spin at wavevector $\boldsymbol{q}$, $\sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_j} S^{\pm}_j$; raising and lowering components of $\boldsymbol{S}_j$	1	7.1
S^a_{tot}, S^-_{tot}	Total spin components $\sum_j S^a_j (a = x, y, z)$ and total lowering operator	1	7.1
$\hat{O}_{\chi a}(q)$	Dark-matter-side operator of a spin-dependent coupling $-\sum_a \hat{O}_{\chi a}(\boldsymbol{q}) s_{e,a}(\boldsymbol{q})$, as Trickle, Zhang and Zurek write it	—	7.1
$\lvert\boldsymbol{S}\rvert$	Spin length of a magnetic ion, $\boldsymbol{S}^2 = \lvert\boldsymbol{S}\rvert(\lvert\boldsymbol{S}\rvert + 1)$	1	7.1
$\boldsymbol{S}_{\mathrm{tot}}$, $S_{\mathrm{tot},x}$, $S_{\mathrm{tot},y}$, S^z_{tot}, $S^{\pm}_{\mathrm{tot}}$	Total spin $\sum_j \boldsymbol{S}_j$ of a magnet and its components	1	7.1
H_{H}, J_1, J_2, $J^{\mathrm{ex}}(Q)$	Heisenberg Hamiltonian $\sum_{\langle ij\rangle} J^{\mathrm{ex}}_{ij} \boldsymbol{S}_i \cdot \boldsymbol{S}_j$; NiO nearest- and next-nearest-shell constants in that convention; lattice Fourier transform $\sum_R J^{\mathrm{ex}}_{0\boldsymbol{R}} \mathrm{e}^{\mathrm{i}Q\cdot\boldsymbol{R}}$	eV; meV; meV	7.1
$u^{\mathrm{B}}_{\boldsymbol{k}}$, $v^{\mathrm{B}}_{\boldsymbol{k}}$	BCS coherence factors	1	7.2
V_{BCS}	BCS pairing constant, a volume times an energy, acting within the shell $\lvert\xi_{\boldsymbol{k}}\rvert < \omega_{\mathrm{D}}$; $\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}} = \lambda_{\mathrm{ep}}$	eV^{-2} ($\mathrm{eV\,\AA}^3$)	7.2
H_{BCS}, H_{MF}	Reduced BCS Hamiltonian; its mean-field form	eV	7.2
E_{bind}	Binding energy of a Cooper pair above a frozen Fermi sea, $2\omega_{\mathrm{D}} e^{-2/\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}}}$	eV	7.2
$\gamma_{\boldsymbol{k}1}$, $\gamma_{\boldsymbol{k}2}$	Bogoliubov quasiparticle operators	—	7.2
$b_{\boldsymbol{k}}$	Pair amplitude $\langle c_{-k\downarrow} c_{\boldsymbol{k}\uparrow}\rangle$, $(\Delta_{\mathrm{BCS}}/2E_{\boldsymbol{k}}) \tanh(E_{\boldsymbol{k}}/2T)$	1	7.2
$\lvert BCS\rangle$	BCS ground state $\Pi_k (u^{\mathrm{B}}_{\boldsymbol{k}} + v^{\mathrm{B}}_{\boldsymbol{k}} c^{\dagger}_{\boldsymbol{k}\uparrow} c^{\dagger}_{-k\downarrow})\lvert 0\rangle$	—	7.2
γ_{E}	Euler's constant 0.5772	1	7.2
$\mathcal{N}_{\mathrm{qp}}(E)$	BCS quasiparticle density of states per volume and spin orientation	eV^2	7.2
S_{I}, S_{II}, S_{n}	Bare pair-creation spectra of the density (case I) and of s^z_e (case II) and their normal-state value	as S(q, ω)	7.2
$\Psi_{\boldsymbol{k}}$	Nambu spinor $(c_{\boldsymbol{k}\uparrow}, c^{\dagger}_{-k\downarrow})^{\mathsf{T}}$	—	7.2
η, s_1, s_2, s_a, s_3	Nambu matrices: $\eta = \mathrm{diag}(1, -1)$ and the off-diagonal $\mathsf{s}_1, \mathsf{s}_2$; s_a either of them $(a = 1, 2)$; $\mathsf{s}_3 \equiv \eta$ in the trace algebra	1	7.2
$h_{\boldsymbol{k}}$	Nambu Bloch matrix $\xi_{\boldsymbol{k}}\eta - \Delta_{\mathrm{BCS}}\mathsf{s}_1$	eV	7.2
$O_A(q)$	Nambu bilinear $\sum_k \Psi^{\dagger}_{\boldsymbol{k}+\boldsymbol{q}} A \Psi_{\boldsymbol{k}}$ of a 2×2 matrix A; $O_{\boldsymbol{\eta}}(q) = n_e(q)$	1	7.2
K	Channel kernel $\mathrm{diag}(0, -V_{\mathrm{BCS}}/2, -V_{\mathrm{BCS}}/2)$ of the conserving response in the channels $\eta, \mathsf{s}_1, \mathsf{s}_2$; its Coulomb entry $K_{\boldsymbol{\eta}\boldsymbol{\eta}} = v_{\mathrm{C}}(q)$	eV^{-2}	7.2
$F(\omega)$	$q = 0$ pair sum $\mathscr{V}^{-1} \sum'_k [E_{\boldsymbol{k}}(\omega^2 - 4E^2_{\boldsymbol{k}})]^{-1}$	1	7.2
$\mathsf{V}_{\boldsymbol{\eta}}$, $\mathsf{V}_{\boldsymbol{j}}$	Dressed density and current vertices in Nambu form	1; 1	7.2
c_2	Phase-channel coefficient of the dressed density vertex, $\mathsf{V}_{\boldsymbol{\eta}} = \eta + \mathsf{c}_2\mathsf{s}_2$, equal to $-2\mathrm{i}\Delta_{\mathrm{BCS}}/\omega$ at $q = 0$	1	7.2
c_{AB}	Speed of the Anderson–Bogoliubov mode, $v_{\mathrm{F}}/\sqrt{3}$ in a neutral superconductor	1	7.2
$c_{\boldsymbol{k}\sigma}$, $c^{\dagger}_{\boldsymbol{k}\sigma}$, $n_{\boldsymbol{k}\sigma}$, $\psi_\sigma(r)$	Electron operators, occupation and field operator with the spin label $\sigma = \uparrow, \downarrow$	—; 1; $\Omega^{-1/2}$	7.2
v_{D}	Dirac velocity	1	7.3
Δ_{D}	Dirac half-gap	meV	7.3
$\varepsilon_{\mathrm{bg}}$, $\varepsilon_{\mathrm{bg},ii}$	Background (remote-band) dielectric constant; principal components of the background dielectric tensor of an anisotropic material	1	7.3
$\tilde{\alpha}$	Effective Dirac coupling $\alpha/(\varepsilon_{\mathrm{bg}} v_{\mathrm{D}})$	1	7.3
Λ_{D}	Dirac momentum cutoff	eV	7.3
σ^{d}_i, $\boldsymbol{\sigma}^{\mathrm{d}}$	Pauli matrices acting within a Kramers doublet	1	7.3

Symbol	Meaning	Units	Sect.
$\mathbb{1}_2$	2×2 unit matrix	1	7.3
$M(\delta k)$, M_i	Interband velocity block of the four-state $\boldsymbol{k} \cdot p$ Hamiltonian, $M = \sum_i \delta k_i M_i$	1	7.3
v_i	Principal Dirac velocities of an anisotropic cone, the singular values of the interband velocity block (v_D when equal)	1	7.3
$\mathcal{N}_\mathrm{D}(E)$	Density of states per volume of one Dirac conduction band	eV^2	7.3
$\tilde{\Lambda}$, Λ_x, Λ_y, Λ_z	Energy cutoff on $(\sum_i v_i^2 \delta k_i^2)^{1/2}$ of an anisotropic cone; per-axis momentum cutoffs	eV	7.3
R_D^0	Independent-pair (interband) response of the Dirac electrons ($\varepsilon = \varepsilon_\mathrm{bg} - v_\mathrm{C} R_\mathrm{D}^0$)	eV^2	7.3
a^*, E_b^D	Effective Bohr radius $\varepsilon_0 m_e a_0/m^*$; donor binding energy	Å; meV	7.4
$\boldsymbol{n}_\mathrm{D}$	Defect or dopant number density	cm^{-3}	7.4
$E_\mathrm{d}(\hat{r})$	Displacement threshold energy of a recoil in the direction $\hat{r}$	eV	7.4
$r_\perp$, $r_\parallel$, e_ecc	Radii of the Kohn–Luttinger variational envelope across and along the valley axis; eccentricity $(1 - r_\parallel^2/r_\perp^2)^{1/2}$ of the envelope spheroid	Å; 1	7.4
$F_j(r)$, c_j	Donor envelope of valley j; valley coefficients of a multivalley donor state	$\mathrm{Å}^{-3/2}$; 1	7.4
γ_m	Valley mass ratio $m_\perp^*/m_\parallel^*$	1	7.4
Δ_0, Δ_g, Δ_f	Valley-orbit matrix elements of the donor 1s manifold: intravalley, opposite valleys, perpendicular valleys	meV	7.4
E_{A_1}, E_E, E_{T_2}	Valley-orbit levels of the donor 1s manifold, labelled by the irreducible representations of T_d	meV	7.4
$E_\mathrm{g}^\mathrm{ad}(Q)$, $E_\mathrm{e}^\mathrm{ad}(Q)$	Ground- and excited-state Born–Oppenheimer surfaces of a color center along the configuration coordinate	eV	7.4
Q, ΔQ	Mass-weighted configuration coordinate of a color center; separation of the minima of the two surfaces	$\mathrm{eV}^{-1/2}$	7.4
w_ZPL	Zero-phonon fraction $\mathrm{e}^{-S_\mathrm{HR}}$ of a color-center line at $T = 0$	1	7.4
$\boldsymbol{d}$	Transition dipole of a color center	eV^{-1} (e Å)	7.4
Chapter 8: The response framework			
$\mathcal{O}_i$	Non-relativistic effective operators ($\mathcal{O}_2 = (v^\perp)^2$)	—	8.1
$v^\perp$	Transverse velocity of the effective theory	1	8.1
$s_\kappa(q)$, $\boldsymbol{s}_s(\boldsymbol{q})$, $\boldsymbol{j}_s(\boldsymbol{q})$	Nuclear spin density of species κ, $\sum_{I \in \kappa} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I} \boldsymbol{J}_I^\mathrm{nuc}$; spin density and paramagnetic current of species s	1	8.1
$\mathcal{T}_e(q)$, $\mathcal{T}_s(q)$, $\mathcal{T}_\kappa$, $\mathcal{T}$	Kinetic-energy density operator of the electrons, of species s, of nuclei κ (mass M_κ)	eV	8.1
$\mathfrak{D}_{ab}(q, v)$, $\mathfrak{D}$, $\mathfrak{D}_\mathrm{R}$, $\mathfrak{D}_\mathrm{I}$	Dark-matter-side tensor $\mathrm{Tr}_\chi[\tilde{V}_a^\dagger \tilde{V}_b]/(2\lvert\boldsymbol{S}_\chi\rvert + 1)$, Hermitian and positive semidefinite, contracting S_{ab}; the matrix; its real symmetric and imaginary antisymmetric parts	model-dependent	8.1
$\bar{v}_\chi$, $\bar{v}_{\chi,z}$	Mean dark-matter velocity $(\boldsymbol{p} + \boldsymbol{p}')/2m_\chi = \boldsymbol{v} - \boldsymbol{q}/2m_\chi$; its component along the magnetization	1	8.1
$c_i^{(s)}(q)$	Coefficient of $\mathcal{O}_i$ for target particle $s \in \{e, p, n\}$ or nucleus κ, non-relativistic normalization, $c_1^{(e)} = \tilde{V}$	eV^{-2}	8.1
$\boldsymbol{S}_\chi$, $\lvert\boldsymbol{S}_\chi\rvert$	Dark-matter spin operator; its spin quantum number	1	8.1
$\boldsymbol{S}_e$, $\boldsymbol{S}_N$	Spin of a target electron ($\boldsymbol{\sigma}/2$) and of a nucleon in the operator basis	1	8.1
$\mathcal{J}_s^{\alpha\beta}(\boldsymbol{q})$, $\mathcal{J}_e$, $\mathcal{J}$	Spin-current operator of species s, $\sum_j \frac{1}{2}\{\boldsymbol{p}_j^\alpha/m_s, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\} S_j^\beta$	1	8.1
$\boldsymbol{J}_I^\mathrm{nuc}$, J_nuc	Spin operator of nucleus I; its quantum number	1	8.1

Symbol	Meaning	Units	Sect.
$\langle S_p\rangle$, $\langle S_n\rangle$	Proton and neutron spin expectation values in the nuclear state $M = J_{\rm nuc}$	1	8.1
H_{aee}, $H_{\phi ee}$	Interaction Hamiltonians of axion-like and scalar absorption on electrons	eV	8.1
Λ_F	Scale of a fermionic-absorption operator $\Lambda_F^{-2}(\bar\tilde\nu\Gamma\chi)(\bar\psi\Gamma'\psi)$, neutral or charged-current	eV	8.1
$m_{\tilde\nu}$	Mass of the outgoing fermion $\tilde\nu$	eV	8.1
$E_{\rm out}$, $m_{\rm out}$	Energy and mass of the outgoing dark-sector particle ($m_{\rm out} = m_\chi + \Delta_\chi$ for inelastic dark matter; $E_{\rm out} = \lvert\boldsymbol{p} - \boldsymbol{q}\rvert$ for a massless $\tilde\nu$)	eV	8.1
Tr_χ	Trace over the dark-matter spin space	—	8.1
$\boldsymbol{y}_j$	Position of nucleon j relative to the center of its nucleus	fm	8.1
m_N	Nucleon mass (normalizes $\boldsymbol{q}/m_N$ in the nucleon operators)	eV	8.1
$v_{\rm T}$, $v_{e\rm T}$	Dark-matter velocity transverse to $\boldsymbol{q}$ and the electron velocity transverse to $\boldsymbol{q}$	1	8.1
$\tilde\nu$, $\bar\tilde\nu$	Final-state fermion of fermionic absorption; its Dirac conjugate	—	8.1
g_a^ϕ, $g_{n_e}^\phi$, $g_{\mathcal{T}_e}^\phi$, $g_{\mathcal{J}}^\phi$	Coefficient of target row $O_a(\boldsymbol{p}_\phi)$ per unit amplitude of an absorbed bosonic field, derivative and polarization factors included; for a scalar $g_{n_e}^\phi = -g_{\phi ee}, g_{\mathcal{T}_e}^\phi = g_{\phi ee}/m_e$; for an axion-like particle $g_{\mathcal{J}}^\phi = g_{aee}m_\phi/m_e$	model-dependent	8.2
$\Gamma_{\rm abs}$, $\Gamma_{\rm net}$	Absorption rate per quantum of a bosonic field (what a detector counts); the net rate $(1 - \mathrm{e}^{-\beta\omega_\phi})\Gamma_{\rm abs}$ at which the field loses quanta	eV	8.2
ω_ϕ	Energy of one quantum of the absorbed field, $m_\phi(1 + v^2/2) \simeq m_\phi$	eV	8.2
N_ϕ	Number of quanta in one mode of the dark-matter field	1	8.2
$\delta_{\rm Born}$	Relative second-Born correction to a Yukawa cross-section, $\pm\mu_{\chi T}y_\chi y_T/[2\pi(m_V + p_{\rm rel})] = \pm\lvert f(0)\rvert m_V \cdot m_V/(m_V + p_{\rm rel})$, + for attraction	1	8.2
$a_{\rm Born}$	First-Born value of the zero-energy scattering length of the Yukawa potential, of magnitude $\lvert f(0)\rvert = \mu_{\chi T}y_\chi y_T/(2\pi m_V^2)$	eV^{-1}	8.2
y_T	Mediator coupling to the target particle, y_e for an electron, Ay_N for a coherent nucleus	1	8.2
$\hat\varrho_\chi$	Spin density matrix of the dark-matter flux, unit trace; $1/(2\lvert\boldsymbol{S}_\chi\rvert + 1)$ when unpolarized	1	8.2
$t_{\rm coh}$	Coherence time of an ultralight dark-matter field	s	8.2
$\mathscr{M}^k$, $\mathscr{M}^k_{ab}$	k-th moment matrix $\int(\mathrm{d}\omega/2\pi)\omega^k[S_{ab}(q,\omega) + S_{ba}(-q,\omega)]$, a commutator for odd k and an anticommutator for even k	eV^{k+3} (density rows)	8.3
$\boldsymbol{g}(q)$	Momentum-density operator	eV	8.3
Γ_A, Γ_B, Γ_{AB}	Rates of two groups A, B of rows alone and their cross term	eV	8.3
$n_{\rm M}(\boldsymbol{q})$	Mass-density operator $\sum_s m_s n_s(\boldsymbol{q})$	eV	8.3
$\mathsf{T}^{\alpha\beta}(\boldsymbol{q})$	Momentum-flux (stress) operator, $[\mathrm{H},\, g^\alpha(\boldsymbol{q})] = q_\beta\mathsf{T}^{\alpha\beta}$	eV^2	8.3
$\bar n_e^{(\kappa)}(y)$	Spherical average of the conditional electron density $n_e^{(\kappa)}(\boldsymbol{y})$	cm^{-3}	8.3
$G_{\rm min}$	Length of the shortest nonzero reciprocal-lattice vector	keV	8.3
$a_{\rm core}$	Size of the innermost electron shell	Å	8.3
$w_\kappa(\boldsymbol{y})$, $\tilde w_\kappa(\boldsymbol{k})$, $w_{ee}(\boldsymbol{r})$, $\tilde w_{ee}(q)$	Electron–nucleus pair potential $-Z_{\kappa\alpha}/y$ and electron–electron pair potential; their Fourier transforms	eV; eV^{-2}	8.3

Symbol	Meaning	Units	Sect.
$n_e^{(\kappa)}$, $n_e^{(\kappa)}(\boldsymbol{y})$	Electron number density at the site of nucleus κ (contact density, mixed third moment); with an argument, the conditional density at displacement $\boldsymbol{y}$ from a nucleus of species κ, averaged over those nuclei and their motion	cm^{-3}	8.3
$\tilde{R}_{aZ}$, $\tilde{R}_{Zb}$, $\tilde{R}_{ZZ}$, $\tilde{R}_{aZ_{\boldsymbol{G}}}$, $\tilde{R}_{Z_{\boldsymbol{G}}Z_{\boldsymbol{G}'}}$, $\tilde{R}_{eZ}$, $\tilde{R}_{Ze}$, $\tilde{R}_{e\kappa}$, $\tilde{R}_{NZ}$	Proper (channel-irreducible) responses of rows a, b to and from the charge row $n_Z(\boldsymbol{q}+\boldsymbol{G})$; the proper charge–charge block, $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\mathrm{C}}\tilde{R}_{Z_{\boldsymbol{G}}Z_{\boldsymbol{G}'}}$ with nuclei	eV^2	8.4
$\delta n_{Z,\boldsymbol{G}}$, n_Z	Induced charge-number density at $\boldsymbol{q}+\boldsymbol{G}$, $\delta\langle n_Z(\boldsymbol{q}+\boldsymbol{G})^\dagger\rangle/\mathscr{V}$; mean charge-number density	eV^3	8.4
ϕ_b	Amplitude of an external field on target row b	eV	8.4
λ_Z	Proportionality constant of the loss-function theorem, $S_{cc} = \lambda_Z^2(2q^2/e^2)L/(1-\mathrm{e}^{-\beta\omega})$	1	8.4
S_{cc}, O_c	Structure factor and operator of a density coupling $O_c = \sum_s c_1^{(s)} n_s$	eV^2 (per eV^{-4} of coupling); —	8.4
$a_N(\boldsymbol{q})$	Nucleon-number amplitude per cell of the optical mode	1	8.4
Π_{T}, Π_{L}, $\Pi_{\mathrm{T,L}}$	Transverse and longitudinal photon self-energies, $\omega^2(1-\varepsilon^{\mathrm{T}})$ and $(\omega^2-k^2)(1-\varepsilon)$ (the two polarizations of Π_{em})	eV^2	8.4
ω'	Frequency on a vertex segment of a diagram	eV	8.4
$\mathfrak{T}$	Truncation label $(n_B, n_g, n_A, \mathcal{X}, n_T, \mathcal{K}, \mathfrak{M}, n_D)$ of the expansion hierarchy	—	8.5
n_B, n_g, n_A, n_T, n_D	Born–Oppenheimer, electron–phonon, anharmonic, thermal and multipole orders	1	8.5
$\mathcal{X}$, $\mathcal{K}$, $\mathfrak{M}$	Many-body approximation; lattice regime (one-phonon, multiphonon, impulse); Migdal regime (sudden, adiabatic)	—	8.5
θ_{ep}, δ_{off}	Off-shell parameter $\langle\lvert g\rvert^2\rangle/\delta_{\mathrm{off}}^2$; off-shell distance of an intermediate state	1; eV	8.5
x_R	Recoil parameter $q^2/(2M_\kappa\langle\omega\rangle) = E_R/\langle\omega\rangle$, $\langle\omega\rangle$ the arithmetic mean of the per-atom phonon density of states	1	8.5
ξ_D	Multipole parameter (momentum times the size of the excited orbital)	1	8.5
$\langle\omega^k\rangle$	Moment $(1/3)\int \mathrm{d}\omega\omega^k g_{\mathrm{ph}}(\omega)/n_{\mathrm{at}}$ of the per-atom phonon density of states; $\langle\omega\rangle = 40.9$ meV, $\langle\omega^{-1}\rangle^{-1} = 30.4$ meV in Si	meV^k	8.5
c_k	k-th cumulant of the energy deposited in the self response, $c_k = E_R\langle\omega^{k-1}\rangle$ at $T=0$	eV^k	8.5
$\mathfrak{T}_0$	Base point of $\mathfrak{T}$: $n_B=0, n_g=0, n_A=0, \mathcal{X}=\mathrm{IP}$, $n_T=0$; $\mathcal{K}, \mathfrak{M}, n_D$ not applicable	—	8.5
q_e, r_{orb}	Momentum given to the electrons; size of the excited orbital	keV; Å	8.5
$\mathcal{P}_{\mathrm{ad}}$, $\mathcal{P}_0$, $\mathcal{P}_+$	Projectors on the electron-following, zero-phonon electronic and phonon-dressed electronic sectors	1	8.6
$\mathcal{P}_{\mathrm{BO}}$	Born–Oppenheimer projector $\int \mathrm{d}u\lvert\Psi_0(u)\rangle\langle\Psi_0(u)\rvert \otimes \lvert u\rangle\langle u\rvert$	1	8.6
$\mathcal{P}_A$, $S^A_{ab}(\boldsymbol{q},\omega)$	Generic sector projector, $A \in \{\mathrm{ad}, 0, +\}$; the sector response	1; eV^2	8.6
δS_{ab}, S^{bin}	Remainder of the decomposition; a response averaged over a bin of width $\Delta\omega_{\mathrm{bin}}$	eV^2	8.6
$\bar{\rho}_e(\boldsymbol{q};u)$, $\bar{n}_\kappa$	Electron density following the nuclei, $\langle\Psi_0(u)\rvert n_e(\boldsymbol{q})\lvert\Psi_0(u)\rangle$ (an operator on nuclear wavefunctions); the nuclear density in the electron-following sector	1	8.6
$\rho_{j0}(\boldsymbol{q};u)$	Transition density of the frozen crystal between adiabatic states j and 0	1	8.6
$f_\kappa(q)$	Form factor of a rigid electron cloud attached to nucleus κ(Z_κ at $\boldsymbol{q}\to 0$ for a neutral atom)	1	8.6

Symbol	Meaning	Units	Sect.
ρ_m, $\tilde{\rho}_m$	Zero-phonon vertex $\langle m\|O_b\|i\rangle$ of pair state m; with its virtual-phonon correction	as O	8.6
g_{fm}	Electron–phonon matrix element $\langle f\|H_{\mathrm{ep}}\|m\rangle$ between a pair and a one-phonon state	eV	8.6
$\mathcal{A}_f^{\mathrm{off}}$, $\mathcal{A}_\nu^{\mathrm{off}}$	Off-shell amplitude per unit coupling into one-phonon state f; on the shell of decay channel ν	$eV^{-1}\cdot[O]$	8.6
$\mathsf{G}_m(\omega)$, $\Sigma_m(\omega)$, $\Upsilon_m(\omega)$	Green's function, self-energy and vertex correction of a zero-phonon pair m(Υ_m over all decay channels)	eV^{-1}; eV; [O]	8.6
Γ_m, $\Gamma_m^{(\nu)}$, Γ_m^{tot}, $\hat{\Gamma}_m$	Width of pair m, its part from decay channel ν, total width $\sum_\nu \Gamma_m^{(\nu)}$, largest width within δ_{F}; $\hat{\Gamma}$ without index: the largest width of the zero-phonon pairs between the edge and δ_{w} above it	eV	8.6
$\tilde{E}_m$, $z_{\mathrm{qp}}(m)$	Quasiparticle energy and residue of a pair state (z_{qp} widened to pairs)	eV; 1	8.6
δ_{F}	Detuning: distance from $\tilde{E}_m$ to the nearest structure of the one-phonon continuum	eV	8.6
c_ν^{F}, $\bar{c}^{\mathrm{F}}$	Fano coefficients of the first emission, $\Gamma_m^{\mathrm{tot}}\,\mathrm{Im}(\rho_m^*\mathcal{A}_\nu^{\mathrm{off}})/\lvert\rho_m\rvert^2$, ν over all decay channels; their width-weighted mean	1	8.6
w_0, x_{w}	Zero-phonon weight in a bin; half-width of a line window	as S; eV	8.6
$\mathcal{P}_{n,m}^{\mathrm{qp}}$, U_{qp}, $\hat{N}_{\mathrm{qp}}$, $\hat{N}_{\mathrm{ph}}$	Quasiparticle- and phonon-number sector projectors of gapless targets; dressing unitary; quasiparticle and phonon number operators	—	8.6
H_{d}, H_{off}, $\mathfrak{h}_{\mathrm{off}}$	Sector-diagonal and off-diagonal parts of H; norm of the off-diagonal leak out of a state	eV	8.6
C_{ab}^{cross}, $\Delta\omega_{\mathrm{bin}}$, w_{b}, f_Δ, $\mathfrak{c}_{\mathrm{b}}$	Cross term between sectors; bin width; bin shape, its kernel and the bin constant ($1/\pi$ for a Gaussian bin)	eV^2; eV; 1; 1; 1	8.6
ΔE_{c}	Width of the decay continuum in the $\mathfrak{h}_{\mathrm{off}}$ estimate	eV	8.6
g_{d}, g_{n}, $\lvert lo\rangle$, $\lvert up\rangle$	Diagonal and off-diagonal couplings of the two-level electron to the bond coordinate; its lower and upper level	eV; eV; —	8.6
δM_κ	Electron drag mass of species κ	eV	8.6
P_{Mig}	Migdal probability	1	8.6
Λ, P_Λ, $P_{\Lambda'}$	Final-state class and its projector (classes refine the sectors $\mathcal{P}_{\mathrm{ad}}, \mathcal{P}_0, \mathcal{P}_+$; Λ' a second class)	—	8.7
$S(q,\omega;\Lambda)$	Final-state-resolved response	eV^2	8.7
$(\tilde{\varepsilon}^{-1})_{G0}(\boldsymbol{q},\omega)$, $\tilde{\varepsilon}^{-1}$	Inverse dielectric matrix of a test electron, $\delta_{\boldsymbol{G}0}+\sum_{\boldsymbol{G}'}(f_{\mathrm{Hxc}})_{\boldsymbol{G}\boldsymbol{G}'}R_{\boldsymbol{G}'0}$	1	8.7
$\Gamma(v;\Lambda)$	Rate of one dark-matter particle into final-state class Λ	eV	8.7
$S^{\mathrm{IP}}(\boldsymbol{q},\omega)$, $S(\boldsymbol{q},\omega;cv\boldsymbol{k})$	Independent-particle structure factor; screened pair-resolved response (class $\Lambda = cv\boldsymbol{k}$, spin summed)	eV^2	8.7
$\mathcal{R}_{N_{\mathrm{eh}}}$	Rate per unit target mass of events with N_{eh} pairs	$\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	8.7
$P(N_{\mathrm{eh}}\|q,\omega,v)$	Pair-number distribution at fixed momentum transfer, energy and velocity	1	8.7
E_{e}, E_{h}	Kinetic energies of the electron and the hole of a prompt pair, measured from the band edges	eV	8.7
$P_{\mathrm{casc}}(N_{\mathrm{eh}}\|\Lambda)$	Cascade distribution of the pair number for a seed class Λ	1	8.7
ϑ_k, $\vartheta_{\mathcal{X}}$	Nuisance parameter of truncation $k:\vartheta_k=0$ the formula, $\vartheta_k=1$ its next order, standard-normal constraint $\mathrm{e}^{-\vartheta_k^2/2}$; $\vartheta_{\mathcal{X}}$ that of the many-body approximation (θ is an angle, Θ the Heaviside function)	1	8.8
$\mathfrak{T}_k^+$	Label one step further along axis k of $\mathfrak{T}$ (next nonvanishing order, next many-body approximation, next multiphonon term)	—	8.8
ΔS_k, δ_k, $\delta_{\mathcal{X}}$	Next-order difference $S^{\mathfrak{T}_k^+}-S^{\mathfrak{T}}$ and its fractional size, the template of nuisance k; $\delta_{\mathcal{X}}$ that of the many-body approximation	eV^2; 1	8.8

Symbol	Meaning	Units	Sect.
δE_k	Edge shift produced by the next order of truncation k (threshold nuisance)	meV	8.8
δ'_k	Fractional size of the next correction to a new-support piece	1	8.8
$\boldsymbol{r}_j(m_\chi;\vartheta)$	Rate per unit cross-section into detector bin j at nuisances ϑ	$\mathrm{cm}^{-2}\,\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	8.8
$N_{\mathrm{obs},j}$, $N_{\mathrm{sig},j}$, $N_{\mathrm{bkg},j}$	Observed, signal and background counts in bin j	1	8.8
ψ, λ_0, λ_1, $m_n(\vartheta)$	Log-template $\ln(1+\delta_A)$, Lagrange multipliers and moments (in $y=\omega-E_R$) of the constrained nuisance (ψ in Sect. 8.8)	1; 1, eV^{-1}; eV^{n+2}	8.8
t^*, N_{nom}, $N_{\mathrm{nom}}^{\mathrm{lim}}$	Threshold of the zero-count cost, nominal signal and its limit value	1	8.8
L, t, t_{p}	Likelihood; $-2\ln$ of the likelihood ratio, the test statistic; its profiled form	1	8.8
Chapter 9: Channel atlas			
$\circ$	Cell mark: allowed at leading order, computed in no section	—	9.1
ω_{th}	Energy threshold of a detector channel	meV	9.2
$m_{\mathrm{min}}^{\mathrm{ac}}$	Acoustic-phonon edge $\omega_{\mathrm{th}}/[2c_\nu(v_{\mathrm{max}}-c_\nu)]$, with c_ν the sound speed of the branch	keV	9.2
$m_{\mathrm{min}}^{\mathrm{ch}}$	Lightest dark-matter mass that reaches channel ch, $\min_q q^2/2[qv_{\mathrm{max}}-E_{\mathrm{ch}}(q)]$	eV; keV, MeV	9.2
$E_{\mathrm{ch}}(q)$	Lowest energy, at or above ω_{th}, of a final state of channel ch that carries momentum q	eV	9.2
$\bar{q}$	Momentum transfer at which the minimum is reached	eV	9.2
c_{ch}	Speed of a linear branch of channel ch, $E_{\mathrm{ch}}=c_{\mathrm{ch}}q$ or $(c_{\mathrm{ch}}^2q^2+E_0^2)^{1/2}$	1	9.2
E_0	Onset energy of a flat or gapped branch	eV	9.2
M_{Ge}	Atomic mass of germanium, in $(M_{\mathrm{Ge}}/M_{\mathrm{Si}})^{1/2}$, the ratio of the free-recoil edges of germanium and silicon at equal recoil energy	eV	9.2
c_{LA}	Longitudinal-acoustic sound speed	$\mathrm{km\,s}^{-1}$	9.2
c_{M}	Magnon velocity near Γ	1	9.2
Chapter 10: Electronic excitations			
$f_{\mathrm{crys}}(q,\omega)$	Crystal form factor, $\lvert f_{\mathrm{crys}}\rvert^2=q^3\Omega\bar{S}^0/(4\pi\alpha m_e^2)$	1	10.1
$\bar{S}^0(q,\omega)$	Independent-particle structure factor averaged over the directions of q	eV^2	10.1
$\mathcal{G}_{\mathrm{rel}}(\boldsymbol{q})$	Lattice term of the static step of an electron-only coupling with a relaxed lattice, $1-\varepsilon_e^{-1}(\boldsymbol{q},0)-[1-\varepsilon_{\mathrm{cl}}^{-1}(\boldsymbol{q})]\to\Omega_{P,e}^2/(c_{\mathrm{LA}}q)^2$	1	10.2
$\bar{L}(q,\omega)$	Loss function averaged over the directions of q	1	10.2
$K_n(\boldsymbol{q})$	n-th moment $\int_0^\infty \mathrm{d}\omega\omega^n L(\boldsymbol{q},\omega)$ of the loss function, $-1\le n\le 1$	eV^{n+1}	10.2
$\eta_n^{\mathrm{max}}(q)$	Maximum over $\omega\ge\omega_{\mathrm{th}}$ of $\omega^{-n}\eta(v_{\mathrm{min}}(q,\omega))$ (sum-rule bounds on the rate)	as η/ω^n	10.2
$\boldsymbol{U_R}$	Rigid displacement of cell $\boldsymbol{R}$ in a long-wavelength acoustic phonon	Å	10.2
L_e^{ac}	Acoustic (electron-following) one-phonon part of the electron-channel loss function at small q	1	10.2
$\varepsilon_{\mathrm{cl}}(\boldsymbol{q})$, $Z_\kappa^{(e)}(\boldsymbol{q})$	Dielectric function at clamped nuclei; electron-following charge of species κ	1; e	10.2
$\Omega_{P,e}$	Electron-following plasma frequency, $\Omega_{P,e}^2=e^2\boldsymbol{n}_e^2/\rho_T$	eV	10.2
ω_p^{val}	Valence plasma frequency	eV	10.2

Symbol	Meaning	Units	Sect.
$Z_L(q)$	Weight (residue) of the plasmon line in the loss function of the gas, $\pi/(\partial\varepsilon_1/\partial\omega) = (\pi\omega_{\rm pl}/2)Z_L, Z_L = (1+x)/(1+2x) \simeq 1-(3/5)q^2v_{\rm F}^2/\omega_p^2$	1	10.3
$\mathcal{R}_{\rm pl}$	Plasmon production rate per unit target mass	$\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	10.3
$v_{\rm th}$	Slowest dark-matter speed that reaches the plasmon, $\omega_+(q_{\rm c})/q_{\rm c} = v_{\rm F} + q_{\rm c}/2m_e$	1	10.3
$B_{\rm pl}^+(\boldsymbol{q})$, ε_2^+	Phonon branching of a plasmon, $\varepsilon_2^+/\varepsilon_2 = S^+/(S^0+S^+)$ at $\omega_{\rm pl}$ (estimate $10^{-3}-10^{-2}$); the phonon-dressed part of ε_2	1	10.3
$f_1^{\rm L}$, $f_2^{\rm L}$	Real and imaginary parts of the Lindhard bracket of Knapen et al., $f_1^{\rm L} + \mathrm{i}f_2^{\rm L} = -R^0/2\mathcal{N}_{\rm F}$, the braced expression	1	10.3
$r_{\rm IP}$	Independent-particle ratio of two responses (sets the mixed-block band)	1	10.4
$j_{\rm L}$, $j_{{\rm T}i}$, $\boldsymbol{j}_{\rm T}$	Components of $\boldsymbol{j}_e(\boldsymbol{q})$ along $\hat{q}$ and along $\hat{e}_i \perp \hat{q} (i=1,2)$; the transverse current	1	10.4
$\mathcal{J}_e^{{\rm T}i,\beta}$, $\mathcal{J}_{\rm T}$	Spin current with its velocity index along $\hat{e}_i$; the transverse spin current	1	10.4
$\bar{v}_{\rm L}$, $\bar{\boldsymbol{v}}_{\rm T}$	Components of $\bar{v}_\chi$ along and across $\hat{q} (\bar{v}_{\rm L} = (\omega+\Delta_\chi)/q$ on shell)	1	10.4
r, $r_{\rm med}$	Ratio Γ_B/Γ_A of the current-row to the density-row rate of a mixed block; its in-medium value	1	10.4
$\delta_{\rm LF}^j$	Rate-averaged local-field change of the transverse-current rate	1	10.4
$\rho_{\lambda 0}(\boldsymbol{q})$	Exciton form factor $\sum_{cv\boldsymbol{k}} A_{cv\boldsymbol{k}}^{\lambda *}\rho_{cv}(\boldsymbol{k},\boldsymbol{q})$; $\langle\lambda\vert n_e(\boldsymbol{q})\vert 0\rangle = \sqrt{g}_{\rm deg}\rho_{\lambda 0}$	1	10.5
$\varepsilon_1^{\rm res}$, $L^{\rm res}$	Real part of the dielectric function and loss function of a resonant-only response	1	10.5
w_λ, C	Weights and positive constant of $\varepsilon_2 = \pi C\sum_\lambda w_\lambda\delta(\omega - E_\lambda)$	1	10.5
$E_{\rm cp}$	Pole energy of a pair coupled to itself in both Bethe–Salpeter blocks, $E_{\rm cp}^2 = \Delta(\Delta + 2{\rm K})$	eV	10.5
$R^{\rm TD}(0)$	Static response of the retarded Tamm–Dancoff block, beside $R^0(0)$ and $R(0)$ of the free and the coupled pair	eV^{-1}	10.5
$H_{\rm T}^{\rm exc}$	Triplet Bethe–Salpeter Hamiltonian $\Delta - K^{\rm d}$	eV	10.5
$S_{\rm core}^0(\boldsymbol{q},\omega)$	Core-shell part of the independent-particle structure factor	eV^2	10.6
$\rho_{c,\kappa nlm}(\boldsymbol{k},\boldsymbol{q})$	Density vertex between a core Bloch sum and a conduction state	1	10.6
$S_{nl}^{\rm at}(q,\omega)$, $S_{\kappa nl}^{\rm at}$	Response of one atom's shell nl (of species κ); $\int {\rm d}\omega S_{nl}^{\rm at} = N_{nl}$ at large q	eV^{-1}	10.6
$N_{nl}^{\rm acc}(q)$	Electrons per atom of shell nl that dark matter can excite at momentum $q(\omega \le qv_{\rm max})$	1	10.6
$\delta_{\rm mv}$, $\delta_{\rm oo}$	Relative mass–velocity and orbit–orbit shifts of the f-sum $\mathscr{M}_{ee}^1$	1	10.6
$H_{\rm oo}$, T_{ij}, $E_{\rm ee}$	Orbit–orbit (Breit–Pauli) term; its position kernel $(\mathbb{1} + \hat{r}_{ij}\hat{r}_{ij})/\boldsymbol{r}_{ij}$; electron–electron repulsion energy $\alpha\langle\sum_{i<j} 1/\boldsymbol{r}_{ij}\rangle$	eV; eV^{-1}; eV	10.6
q_{3d}	Momentum transfer at which the Ge 3d shell overtakes the valence, keV	keV	10.6
$\Gamma_{\rm T}$, $\Gamma_{\rm L}$	Absorption rates of a transverse and a longitudinal dark-photon polarization at finite $\boldsymbol{p}_\phi$	eV	10.7
$\boldsymbol{J}_a$, $\boldsymbol{J}_{\rm ind}$	Source current $g_{a\gamma\gamma}(\partial_{t\phi})\boldsymbol{B}_0$ of an axion-like particle in a static field; the current it induces in the medium, $\sigma_{\rm cond}\cdot\boldsymbol{E}$	eV^3	10.7

Symbol	Meaning	Units	Sect.
ϕ_0	Complex amplitude of a classical bosonic field, $\phi = \mathrm{Re}(\phi_0 \mathrm{e}^{-\mathrm{i}m_\phi t})$	eV	10.7
ε''	Absorptive part $(\varepsilon - \varepsilon^\dagger)/2\mathrm{i}$ of the dielectric tensor ($\mathrm{Im}[-\varepsilon^{-1}]$ likewise for $-\varepsilon^{-1}$)	1	10.7
$\varepsilon_\parallel$, $\varepsilon_\perp$	Principal dielectric functions of a uniaxial crystal along and across the optic axis	1	10.7
$\gamma_{\mathrm{L}\nu}$, $\gamma_{\mathrm{T}\nu}$	Damping of the longitudinal and transverse mode ν in the factorized dielectric function of a polar crystal	meV	10.7
f_1, f_2	Atomic scattering factors (real, imaginary), $\varepsilon_1 = 1 - 4\pi\alpha n_{\mathrm{at}} f_1/m_e\omega^2, \varepsilon_2 = 4\pi\alpha n_{\mathrm{at}} f_2/m_e\omega^2$	1	10.7
n_{at}	Number density of atoms	eV^3; cm^{-3}	10.7
$R_{\mathcal{T}\mathcal{T}}$, $S_{\mathcal{T}\mathcal{T}}$	Response and structure factor of the kinetic-energy row $\mathcal{T}_e$ (R_{ab}, S_{ab} with $a = b = \mathcal{T}_e$)	as R_{ab}, S_{ab}	10.7
$\boldsymbol{F}(\omega)$	Vector coefficient of the $O(\boldsymbol{p}_\phi)$ term of $\tilde{R}_{\mathcal{T}_e n_e}$	—	10.7
$d_{\phi ee}$	Scalar coupling in the normalization of Mitridate et al., $g_{\phi ee} = d_{\phi ee}\sqrt{(4\pi)} m_e/M_{\mathrm{Pl}}$	1	10.7
x	m_ϕ/m_e, the expansion variable of dark Compton scattering	1	10.8
G_{S}, G_{P}, G_{V}, G_{A}, $G_{\mathrm{V,A}}$	Couplings of the dimension-six pair operators of a complex scalar to $\bar\psi\psi, \bar\psi\mathrm{i}\gamma^5\psi, \bar\psi\gamma^\mu\psi, \bar\psi\gamma^\mu\gamma^5\psi$ (G_{V} in Chap. 10)	eV^{-2}	10.8
G_{F}	Fermi constant, 1.1664×10^{-5} GeV^{-2} (reference value of the pair couplings)	eV^{-2}	10.8
Λ_{NP}	New-physics scale of the pair operators	eV	10.8
$\bar p$, $\bar v$, $\bar E$	Momentum (four-momentum $\bar p^\mu$), velocity and energy of the antiparticle of a complex scalar	eV; 1; eV	10.8
$g_a^{\phi\phi}$	Coefficient of target row a per unit pair amplitude: $m_e G_{\mathrm{S}}, m_e G_{\mathrm{P}}, G_{\mathrm{V,A}}(p - \bar p)^\mu$	model-dependent	10.8
K_{pair}, $\bar K_{\mathrm{pair}}$	Kinetic energy of a pair, $m_\phi(v^2 + \bar v^2)/2$; its halo mean $m_\phi\langle v^2\rangle = 1.60 \times 10^{-6} m_\phi$	eV	10.8
p_K	Distribution of K_{pair}	eV^{-1}	10.8
Γ_{pair}	Pairs absorbed per unit time	eV	10.8
g_{eff}	Effective single-absorption coupling of pair absorption, $g_{\mathrm{eff}}^2 = (\rho_\chi/2m_\phi) m_e^2 G_{\mathrm{S}}^2/m_\phi$	1	10.8
$T^{(2)}_{ab}$, T^{C}	Second-order transition operator of two insertions; its dark-Compton form	—	10.8
ω_γ, $\boldsymbol{k}_\gamma$, $\hat e_\gamma$	Energy, momentum and polarization of the emitted photon in dark Compton scattering	eV; eV; 1	10.8
ω_{C}	Electron recoil energy of dark Compton scattering at threshold, $m_\phi - \omega_\gamma = m_\phi^2/2(m_\phi + m_e)$	eV	10.8
σ_{Th}	Thomson cross section $e^4/6\pi m_e^2 = 6.65 \times 10^{-25}$ cm^2	cm^2	10.8
H_{sg}	Spin seagull of the axion-like coupling, $(e g_{aee}/m_e^2)(\partial_t\phi)\boldsymbol{A}\cdot\boldsymbol{s}_e$	eV	10.8
$\mathcal{Q}$	Momentum-transfer magnitude of the relativistic density row of boosted dark matter	eV	10.9
$v_{\chi,\mathrm{max}}$	Largest dark-matter speed of a boosted flux	1	10.9
q_Δ, v_Δ	Exothermic shell momentum $\sqrt{(2m_\chi(\lvert\Delta_\chi\rvert - \omega))}$ and speed q_Δ/m_χ	eV; 1	10.9
k_*, ω_{line}, q_{line}	Radius of the exothermic shell; energy and momentum of the free-electron line, $(\mu_{\chi e}/m_e)\lvert\Delta_\chi\rvert$ and $(2\mu_{\chi e}\lvert\Delta_\chi\rvert)^{1/2}$	eV	10.9
E_χ	Relativistic dark-matter energy $(p^2 + m_\chi^2)^{1/2}$	eV	10.9
$\boldsymbol{J}_{\chi\mathrm{T}}$, $\boldsymbol{j}_{e\mathrm{T}}$	Components of the dark-matter current $\boldsymbol{J}_\chi$ and of the electron current $\boldsymbol{j}_e$ across $\boldsymbol{q}$	1	10.9
D_{eff}	Effective distance of the cosmic-ray-boosted flux, 8.02 kpc	kpc	10.9
σ_F	Fermionic-absorption cross-section $m_\chi^2/(4\pi\Lambda_F^4)$	cm^2	10.9

Symbol	Meaning	Units	Sect.
Γ_S, Γ_V	Absorption rates of the scalar and the vector fermionic operator	eV	10.9
P_L, P_R, $P_{L,R}$	Chirality projectors $(1 \mp \gamma^5)/2$	1	10.9
$m_{\rm med}$, y_{ee}, $y_{\chi\tilde{\nu}}$	Mass of a scalar mediator of fermionic absorption; its Yukawa couplings to electrons and to the $\chi\tilde{\nu}$ pair	eV; 1	10.9
$\mu_{\rm M}$	Transition magnetic moment between χ_2 and $\chi_1, H = -\mu_{\rm M}\boldsymbol{\sigma}\cdot\boldsymbol{B}$	eV^{-1}	10.9
Λ_d	Dipole scale of the operator $(1/\Lambda_d)\bar{\chi}_2\sigma^{\mu\nu}\chi_1 F_{\mu\nu}, \mu_{\rm M} \simeq 2/\Lambda_d$	eV	10.9
$\Gamma_{\rm vac}$	Vacuum decay rate of $\chi_2 \to \chi_1\gamma, \mu_{\rm M}^2\lvert\Delta_\chi\rvert^3/\pi$	eV	10.9
$\mathcal{R}_{\rm IP}$, $\mathcal{R}_{\rm RPA+LF}$, $\mathcal{R}_{\rm source}$	Rate per unit target mass of the inelastic crystal recoil with the unscreened independent-particle ε_2; the same with the screened random-phase loss function with local fields; rate per unit target mass of boosted dark matter from the density row in its published form	$\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	10.9
$\mathcal{R}^{\mathfrak{T}}$	Rate at the central value of a truncation label (as $S^{\mathfrak{T}}$)	$\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	10.10
$\sigma_{\ln\mathcal{R}}$	Gaussian width of $\ln\mathcal{R}$ from independent nuisances, $(\sum_k \ln^2(1+\delta_k))^{1/2}$	1	10.10
Chapter 11: Lattice excitations			
κ_η	Constant of the light-mediator correction, 3.14 for $c \to 0$ and 3.09 at the sound speed of Si	1	11.1
$\bar{c}_{\rm LA}$	Sphere-averaged longitudinal sound speed (Christoffel average)	$\mathrm{km\,s}^{-1}$	11.1
B	Bulk modulus	GPa	11.1
$q_{\rm BZ}$	Zone-boundary momentum scale of a crystal	keV	11.1
$O_N(q)$	Nucleon-number operator $\sum_\kappa A_\kappa n_\kappa(q)$ of a spin-independent nucleon coupling	1	11.1
$A_{\rm cell}$, m_u	Number of nucleons in a primitive cell; atomic mass unit, 931.494 MeV	1; eV	11.1
$q_{{\rm up},\nu}$, $c_{\rm LA,max}$	Upper momentum limit $\min[2m_\chi(v_{\rm max} - c_\nu), q_{\rm BZ}]$ of branch ν; largest longitudinal sound speed over directions (sets the acoustic edge)	keV; $\mathrm{km\,s}^{-1}$	11.1
$\mathcal{R}_{\rm light}$, I_0	Target-independent light-mediator one-phonon rate; halo integral $\int_0^{v_{\rm max}} u^2\eta(u)du$	$\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$; 1	11.1
Δ_C, $A_{\rm Z}$, s_Q, p_Q	Cubic elastic anisotropy $C_{11} - C_{12} - 2C_{44}$; Zener ratio $2C_{44}/(C_{11} - C_{12})$; direction invariants $\sum \hat{q}_i^2\hat{q}_j^2 (i<j)$ and $\sum \hat{q}_i^6 - (\sum \hat{q}_i^4)^2$	GPa; 1	11.1
v_L, v_T, v_m, $G_{\rm V}$, $G_{\rm R}$	Longitudinal, transverse and Debye-mean speeds of an isotropic polycrystal (Anderson convention); Voigt and Reuss shear moduli ($G_{\rm V}$ in Chap. 11) (v_T in Chap. 11)	$\mathrm{km\,s}^{-1}$; GPa	11.1
$S_A(q,\omega)$, $S_A^{(1)}$, $S_A^{(n)}$, $S_A^{\rm inc}$, $S_A^{{\rm inc},(n)}$, $S_A^{\rm harm}$, $S_A^{\rm anh}$	Response of the nucleon-number operator O_N; its one-phonon and n-phonon terms; the incoherent approximation and its order n; the harmonic self response and its anharmonic part	eV^2 (as S)	11.1
$\mathcal{F}_{\rm pol}$	Polar band integral of the single-phonon rate	$\mathrm{meV\,cm^3\,g^{-1}}$ (per mass)	11.2
$K_\eta(\omega, m_\chi)$	Kinematic factor $\int \mathrm{d}q q^{-1}\eta(v_{\rm min})$ of the light-mediator polar rate	$(\mathrm{km\,s}^{-1})^{-1}$	11.2
$\mathcal{I}_{\rm pol}$	As $\mathcal{F}_{\rm pol}$	meV	11.2
$w_\nu^{\rm pol}$, $\bar{\omega}_{\rm pol}$, $\omega_{{\rm L}\nu}$, $\omega_{\rm L1}$, $\omega_{\rm L,max}$, $\omega_{\rm T,max}$, $\gamma_{\rm L}$, $L_{\rm o}$, L_e	Share of branch ν in $1/\varepsilon\infty - 1/\varepsilon_0$; weighted mean longitudinal frequency; longitudinal frequency of branch ν; largest longitudinal and transverse frequencies; full width of a longitudinal line; loss functions of the ordinary and extraordinary rays (L_e in Chap. 11)	1; meV	11.2
$\varepsilon\infty(\hat{q})$	$\hat{q}\cdot\varepsilon\infty\cdot\hat{q}$, the longitudinal electronic dielectric constant along $\hat{q}$	1	11.2

Symbol	Meaning	Units	Sect.
c_κ	Per-species weight of a density coupling $\sum_\kappa c_\kappa n_\kappa(\boldsymbol{q}), c_1^{(\kappa)}$ in units of $\tilde{V}_N(A_\kappa$ for equal couplings to protons and neutrons)	1	11.3
$\mathfrak{W}^{(n)}$, $\mathfrak{W}^{(n)}_{\mathrm{inc}}$, $\mathfrak{W}^{(2),\mathrm{th}}$	Exact, incoherent and thermal ω-integrated n-phonon weight per cell	1	11.3
$\mathfrak{P}^{(n)}$, δ_n^{coh}, $j_{\mathrm{BZ}}(r)$	Momentum-resolved self density (upper end of the certified band); coherence correction $\mathfrak{W}^{(n)}/\mathfrak{W}^{(n)}_{\mathrm{inc}} - 1$; zone average of a pair phase	as S; 1; 1	11.3
q_{osc}, $c_{\min}$	Oscillator momentum $(2M_\kappa\omega_0)^{1/2}$ of a nucleus in an isotropic well; lowest acoustic speed	keV; $\mathrm{km\,s^{-1}}$	11.3
$\Delta R_{\kappa\kappa'}$, $B_{\kappa\kappa'}(\Delta R)$	Pair vector $\Delta R + \tau_\kappa - \tau_{\kappa'}$ from nucleus κ' in the origin cell to κ in cell ΔR; static displacement correlation $\mathcal{C}_{\kappa\kappa'}(\Delta R, 0), B_{\kappa\kappa}(0) = B_\kappa$	Å; $\mathrm{Å}^2$	11.3
$P_n^\kappa(\hat{q},\omega)$	n-phonon distribution of nucleus κ from the density of states projected on $\hat{q}$	eV^{-1}	11.3
$\omega_{\max}$	Top of the phonon band, 63.7 meV in Si; also the highest phonon frequency of the Allen–Dynes context	eV	11.3
D, δD	Kick operator $E_R + q\cdot\boldsymbol{p}/M_\kappa$ (the energy a kick q gives a free particle of momentum $\boldsymbol{p}$) and its fluctuation $(q/M_\kappa)p_{\hat{q}}$	eV	11.4
$p_{\hat{q}}$, $u_{\hat{q}}$, $E_\kappa^{\mathrm{kin}}(\hat{q})$, σ_ω	Momentum and displacement of a nucleus along $\hat{q}$; its kinetic energy along $\hat{q}, \langle p_{\hat{q}}^2\rangle/2M_\kappa$; impulse (Doppler) width $(4E_R E_\kappa^{\mathrm{kin}})^{1/2}, (E_R\langle\omega\rangle)^{1/2}$ for a harmonic site	eV; Å; eV; eV	11.4
$\mathfrak{f}_\kappa(p)$	Momentum distribution of nucleus κ, normalized to one	eV^{-3}	11.4
q_{abs}, Γ_V	Momentum transfer of fermionic absorption on a nucleus, $m_\chi = q_{\mathrm{abs}} + q_{\mathrm{abs}}^2/2M$; absorption rate of the vector operator per dark-matter particle	eV	11.4
λ_3, λ_4, $q_\times$	Cubic and quartic coefficients of the single-site anharmonic model; crossover momentum $4\lambda_3(2M_\kappa\omega_0)^{1/2} = 4\lambda_3/\ell_0$	1; keV	11.5
r_{L}, d_{nn}, M_{Si}, M_{H}	Lindemann ratio $\langle u^2\rangle^{1/2}/d_{\mathrm{nn}}$; nearest-neighbor distance; atomic masses of silicon and hydrogen	1	11.5
$\delta_{\mathrm{anh}}(q,\omega)$	Relative anharmonic correction $S_A/S_A^{\mathrm{harm}} - 1$ of the self response	1	11.5
Γ_{iso}, Γ_{EM}	Isotope-disorder and electromagnetic absorption rates	eV	11.6
$g_i(\omega)$, δ_i^M, $h_{\mathrm{iso}}(\omega)$	Local density of states at a site of isotope i; its mass defect $1 - M_i/\bar{M}$; first-order isotope response function	eV^{-1}; 1; 1	11.6
G_{h}, G_i, $\mathsf{G}_{I\boldsymbol{r}}$, $\mathsf{G}_{\boldsymbol{r}I}$	Site Green function of the host and of one mass defect; host Green function between sites	eV^{-2}	11.6
$x_{\kappa i}$, $g_B - L$, g_{B-L}, $\mathfrak{b}$, $\mathfrak{b}_i$, $\bar{\mathfrak{b}}$, $\sigma_{\mathfrak{b}}$, Ω_{at}	Abundance of isotope i on sublattice κ; $B-L$ gauge coupling; neutron number per mass of isotope i and its mass-weighted mean and spread; volume per atom	1; 1; u^{-1}; $\mathrm{Å}^3$	11.6
S_{OO}	Response of a generic random-vertex operator O	as S	11.6
$S_i^{\mathrm{self}}(q,\omega)$	Self (single-nucleus) response of the nuclei of isotope i	eV^2	11.6
g_{aNN}, g_{app}, g_{ann}, g_{ai}	ALP–nucleon pseudoscalar couplings, $\mathcal{L} \supset g_{aNN}\phi\bar{\psi}^5_{Ni\gamma}\psi_N$; ALP coupling to the spin of isotope $i, (g_{app}\langle S_p\rangle + g_{ann}\langle S_n\rangle)/J_{\mathrm{nuc}}$	1	11.6
$\langle\delta^M\rangle_{\mathfrak{b}}$	Mass defect averaged with the weights $x_i M_i(\mathfrak{b}_i - \bar{\mathfrak{b}})^2$ (Si -0.054)	1	11.6
$S_\ell(\lvert q\rvert,\omega)$	Cubic-harmonic component of a response	as S	11.7
$\mathcal{H}_\ell(v_{\min})$, $\mathcal{H}_0$, $\mathcal{H}_1$, $\mathcal{H}_2$, $\mathcal{H}_4$, $\mathcal{H}_{10}$	Halo multipole, the Legendre coefficient of the Radon transform of f_{gal} along $v_{\mathrm{E}}, \mathcal{H}_0 = \eta$	$(\mathrm{km\,s^{-1}})^{-1}$	11.7
$\hat{f}_{\mathrm{gal}}(s)$, $Y_{\ell m}$, $Y_{\ell' m}$, Y_ℓ^{c}, $S_{\ell m}$, L_ℓ, P_ℓ	Radon transform of the galactic distribution; real spherical harmonics in crystal axes; components of S and L on them (L_ℓ on Y_ℓ^{c}); Legendre polynomials	$(\mathrm{km\,s^{-1}})^{-2}$; 1	11.7

Symbol	Meaning	Units	Sect.
$A_{\rm day}$, $\mathcal{R}_\ell$, $\bar{\mathcal{R}}$, $\mathcal{R}_{\max}$, $\mathcal{R}_{\min}$	Daily-modulation amplitude; term ℓ of the fixed-orientation rate without its harmonic; orientation-averaged rate; extremes over a sidereal day	1; $\mathrm{kg^{-1}\,yr^{-1}}$	11.7
$\bar{b}_\kappa$, Δb_κ, $\bar{W}_\kappa$, $s_{n,\ell}$	Mean and uniaxial anisotropy of B_κ per q^2; mean Debye–Waller exponent; cubic components of the exact n-phonon weight over the incoherent one	$\mathrm{Å}^2$; 1	11.7
$D_\kappa^{\alpha\beta}(\omega)$	Phonon density-of-states tensor of site κ, $N^{-1}\sum \mathrm{Re}[e^\alpha e^{\beta*}]\delta(\omega-\omega_{\boldsymbol{Q}\nu})$; trace $g_{\mathrm{ph},\kappa}$, $(g_{\mathrm{ph},\kappa}/3)\mathbb{1}$ at a cubic site ($\boldsymbol{D}$ is the displacement field)	$\mathrm{eV^{-1}}$	11.7
$\mathcal{G}_\kappa^\phi$, O_ϕ, d	Site coupling tensor of an absorbed field; its row $\sum_a g_a^\phi O_a$; mass-weighted site difference $\mathcal{G}_1^\phi/M_1-\mathcal{G}_2^\phi/M_2$	1; u^{-1}	11.8
χ_{ab}, χ_{aZ}, χ_{bZ}, χ_{Zb}, χ_{ZZ}, χ_{eZ}, χ_{ee}, $\boldsymbol{p}_a$, $\boldsymbol{p}_Z$, x_a, x_Z, $S_{O_\phi O_\phi}$	Clamped-nuclei electronic response of rows a,b; proper site-coupling vector of row a ($\boldsymbol{p}_Z=\boldsymbol{p}$); phonon part of the proper response	as $\tilde{R}$; 1	11.8
$f_{\rm sh}$, $N_{\rm dp}$, $h_{\rm mod}$	Shape factor of the field in a body; depolarization factor; fractional size change $\delta a/a$	1	11.8
$\mathfrak{Q}$, $\mathfrak{Q}_{\rm p}$, $\mathfrak{Q}_{\rm s}$, $\mathfrak{Q}_{\rm DM}$, $\mathfrak{Q}_{\rm eff}$	Quality factor; of a mode; the signal prescription $\min\{\mathfrak{Q}_{\rm p},\mathfrak{Q}_{\rm DM}\}$; of the halo field; effective	1	11.8
M_n, L_0, r_n, U_n, $U_{n,z}$, $c_{\rm L}$, $F_\phi(\omega)$, $e_{zzz}^{\rm pz}$	Mode mass, central thickness, trapped radius, mode function of a resonator; longitudinal speed of the plate; energy distribution of the halo field; dominant piezoelectric element ($c_{\rm L}$ in Sect. 11.8)	eV; nm; μm; —; $\mathrm{km\,s^{-1}}$; $\mathrm{eV^{-1}}$; $\mathrm{eV^2}$ (quoted in $\mathrm{C\,m^{-2}}$)	11.8
$M_{\rm sph}$, $R_{\rm sph}$, J_n, $L_{\rm b}$, $c_{\rm b}$	Mass and radius of a resonant sphere; overlap integral of mode n; length and sound speed of a bar	eV; m; 1; m; $\mathrm{km\,s^{-1}}$	11.8
ω_n, γ_n, b_n, P_n	Frequency, full width and annihilation operator of mode n of a finite body; power absorbed by mode n from the field	eV; eV; 1; $\mathrm{eV^2}$	11.8
Chapter 12: The Migdal effect			
ω_l, ω_e	Energies given to the lattice and to the electrons in the joint response	eV	12.1
$\boldsymbol{F}_{j0}^I$, $\boldsymbol{F}_{a0}$	Force matrix element $\langle\Psi_j\vert\nabla_{\boldsymbol{X}_I}V_{eN}\vert\Psi_0\rangle$	$\mathrm{eV^2}$	12.1
$m_{\rm up}$	Vibrational level on the excited surface of the displaced oscillator	1	12.1
$S^{\rm ad}_{\kappa;\boldsymbol{G}\boldsymbol{G}'}$	Electron-following response between $n_\kappa(\boldsymbol{Q}-\boldsymbol{G})$ and $n_\kappa(\boldsymbol{Q}-\boldsymbol{G}')$	$\mathrm{eV^{-1}\,Å^{-3}}$	12.1
$S^{\rm ad}_{\kappa;\boldsymbol{G}\boldsymbol{G}}$	Diagonal element of the electron-following response between $n_\kappa(\boldsymbol{Q}-\boldsymbol{G})$ and $n_\kappa(\boldsymbol{Q}-\boldsymbol{G}')$	$\mathrm{eV^{-1}\,Å^{-3}}$	12.1
$S_{ab}(\boldsymbol{q};\omega_l,\omega_e)$	Joint response: the response S_{ab} resolved by the energies given to the lattice and to the electrons, $\int \mathrm{d}\omega_e S_{ab}(\boldsymbol{q};\omega-\omega_e,\omega_e)=S_{ab}(\boldsymbol{q},\omega)$	S_{ab} per eV	12.1
$E_f^{\rm el}$	Electronic energy of a final state: 0 in $\mathcal{P}_{\rm ad}$, the zero-phonon energy of its excitation otherwise ($\tilde{E}_m$ for a pair); $E_f-E_f^{\rm el}$ is the lattice energy	eV	12.1
$\boldsymbol{p}$	Momentum given to the electrons by the moving nucleus	eV	12.1
$S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k},\omega)$	Density response matrix of the clamped electrons between $n_e(\boldsymbol{k}+\boldsymbol{G})$ and $n_e(\boldsymbol{k}+\boldsymbol{G}')$	$\mathrm{eV^2}$; $\mathrm{eV^{-1}\,Å^{-3}}$	12.1
m_A	Atomic boost mass	eV	12.2
$H_{\rm rel}$	Internal Hamiltonian of a free atom after the exact center-of-mass separation, with the mass-polarization term	eV	12.2
$D_{\hat{q}}$, $D_{\hat{k}}$	Dipole operator of the electrons along the kick, $\hat{q}\cdot\sum_e \boldsymbol{\rho}_e$	Å	12.2
$P_{\rm exc}$, $P_{\rm ion}$	Migdal probabilities into bound levels and into the continuum, $P_{\rm ion}=P_{\rm Mig}-P_{\rm exc}$	1	12.2
$F_{\rm exc}$, $F_{\rm ion}$	Oscillator strength in the bound lines and in the continuum	1	12.2

Symbol	Meaning	Units	Sect.
ρ_e, $\rho_{e'}$	Electron coordinate relative to the nucleus, $\boldsymbol{r}_e - \boldsymbol{X}$, of a many-electron atom	Å	12.2
$P_{n\ell}$	Migdal ionization probability of shell $n\ell$	1	12.2
j_ℓ	Spherical Bessel function	1	12.2
$\boldsymbol{v}_A$	Velocity $\boldsymbol{q}/m_A$ of the recoiling atom ($q_e = m_e v_A$)	1	12.2
$P_f(\boldsymbol{q})$	Probability of the internal final state f of an atom after a kick	1	12.2
$Z_{\rm ion}(p)$	Valence ionization factor of the crystal Migdal rate, $Z_\kappa - f_\kappa^{\rm core}(p)$, at the momentum p given to the electrons	1	12.3
$L_{\rm Mig}(\omega)$	Migdal-weighted loss function $\int \mathrm{d}^3p/(2\pi)^3 Z_{\rm ion}(p)^2 \bar{L}(p,\omega)$	eV^3	12.3
$\mathscr{F}_{\rm Mig}(\omega)$	Dimensionless Migdal loss spectrum with the off-diagonal $\boldsymbol{G} \neq \boldsymbol{G}'$ terms and the phase $\mathrm{e}^{\mathrm{i}(\boldsymbol{G}-\boldsymbol{G}')\cdot\tau_\kappa}$ of the struck nucleus (one with the origin of ε on that atom), $L_{\rm Mig} = Z_{\rm ion}^2 \mathscr{F}_{\rm Mig}/\Omega$	1	12.3
b, a	$q\cdot p/M_\kappa$ and $p^2/2M_\kappa$ in the free-recoil bracket	eV	12.3
$\boldsymbol{F}_e$	$\sum_i \nabla_i V_{eN}$, minus the total force on the electrons	eV^2	12.3
$\hat{P}_I^{\rm W}$	Projector on the occupied Wannier functions centered on nucleus I	1	12.3
$A_{cv}^{\rm W}$	Migdal amplitude of rigid Wannier following, transition $v \to c$	1	12.3
$W_{\rm v}$	Width of the occupied valence manifold	eV	12.3
$\hat{P}_{\rm occ}$	Projector on the occupied manifold, $\sum_I \hat{P}_I^{\rm W}$	1	12.3
$\omega_{\rm top}$	Last energy of the loss-function table	eV	12.3
$f_\kappa^{\rm core}(p)$	Form factor of the frozen core electrons of species κ, $Z_{\rm ion}(p) = Z_\kappa - f_\kappa^{\rm core}(p)$	1	12.3
$\tilde{\varepsilon}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k},\omega)$	Symmetrized dielectric matrix $\delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\rm C}^{1/2}(\boldsymbol{k}+\boldsymbol{G}) R^0_{\boldsymbol{G}\boldsymbol{G}'} v_{\rm C}^{1/2}(\boldsymbol{k}+\boldsymbol{G}')$, with the inverse diagonal of ε	1	12.3
N_κ	Number of nuclei of species κ: per formula unit in the per-mass rate, in the crystal in $F_{\rm cr}$	1	12.3
$H_{\chi L}$, H_{eL}, k_e	Dark-matter–lattice and electron–lattice Hamiltonians and electron momentum of Berghaus et al.	eV; eV; eV	12.4
$\mathcal{A}_f(\boldsymbol{q},\boldsymbol{p})$	Nuclear factor of the second-order Migdal amplitude into lattice final state f	eV^{-2}	12.4
$\lvert\lambda\rangle$, E_λ	Eigenstates and energies of H_N, the intermediate lattice states of the Migdal amplitude	—; eV	12.4
$\langle\omega_{\rm ph}^2\rangle$	Mean square phonon frequency over $g_{{\rm ph},\kappa}/3$,^{2} in Si	eV^2	12.4
$F_{\rm cr}(q)$	Electronic factor of the cross first moment $-(q^2/m_A)F_{\rm cr}(q)$	1	12.5
$\mathcal{J}_k$, $\mathrm{d}\sigma_\kappa$	Inverse moments $\int \Delta^{-k} d\sigma_\kappa$ of the force spectrum $\mathrm{d}\sigma_\kappa$ of one nucleus	eV^{4-k}	12.5
$E_k^{\rm Mig}$, $E_{\rm BH}$	k-th moment of the Migdal spectrum; diagonal Born–Huang energy $\mathcal{J}_2/2M_\kappa$	eV^k; eV	12.5
δM_{IJ}	Drag-mass tensor between nuclei I and J (trace rule $\sum_J \delta M_{IJ} = Z_\kappa m_e$)	eV	12.5
$\mathcal{R}_{\rm rel}$, $c_{\rm rel}$	Relativistic finite-nucleus factor of the contact density; its non-logarithmic coefficient of order one	1	12.5
$\rho_{\rm nuc}$	Normalized charge distribution of a nucleus of finite size	Å^{-3}	12.5
$Z^*(\omega)$	Dynamical Born charge	1	12.6
$Z^*_{\rm at}(\omega)$	Free-atom limit of the dynamical Born charge, $-m_e\omega^2\alpha_{\rm pol}(\omega)/e^2$	1	12.6
$\alpha_{\rm pol}(\omega)$	Dipole polarizability of an atom	Å^3	12.6
H_ϕ, $g_{\phi pp}$, $g_{\phi nn}$, $\tilde{c}$, σ_ϕ	Scalar–nucleus interaction; scalar couplings to protons and neutrons; common ratio c_κ/M_κ for mass-proportional couplings; scalar absorption cross-section per atom	eV; 1; eV^{-1}; Å^2	12.6

Symbol	Meaning	Units	Sect.
$\bar{n}$, $\bar{p}$	Dirac adjoints of the neutron and proton fields in $\mathcal{L} \supset \phi(g_{\phi pp}\bar{p}p + g_{\phi nn}\bar{n}n)$	—	12.6
C_κ	Small-ω coefficient of the dynamical Born charge, $Z^*_\kappa(0) - Z^*_\kappa(\omega) = C_\kappa\omega^2 + O(\omega^4)$; Si, Ge: $\Omega m_e(\varepsilon_\infty - 1)/2e^2$	eV^{-2}	12.6
$\mathscr{E}_\gamma$, H_γ	Macroscopic electric field of an emitted photon; its coupling $-e\sum_I \boldsymbol{u}_I \cdot Z^*_\kappa(\omega_\gamma)\cdot\mathscr{E}_\gamma$	—; eV	12.6
$\delta\tilde{c}_\kappa$	Species deviation of the coupling per unit mass, $c_\kappa/M_\kappa = \tilde{c} + \delta\tilde{c}_\kappa$	eV^{-1}	12.6
σ_γ	Photoabsorption cross-section per atom, $n_{\mathrm{at}}\sigma_\gamma = \omega\varepsilon_2/n_r$	$\mathrm{Å}^2$	12.6
N_v	Number of valence electrons per cell, 8 in Si and Ge	1	12.6
Chapter 13: Phonon-assisted electron recoil			
$m^{\mathrm{ZP}}_{\mathrm{min}}$, $m^{\mathrm{PA}}_{\mathrm{min}}$	Zero-phonon threshold mass; phonon-assisted onset mass	MeV	13.1
M_{eh}	Translational mass of an electron–hole pair along a direction, $m^*_\parallel + m_{\mathrm{hh}}$	m_e	13.1
q^*, $\boldsymbol{Q}^*$, u_ν	Optimal momentum transfer at the phonon-assisted onset; phonon wavevector $\boldsymbol{k}_\mathrm{c} - \boldsymbol{q}^*$ it leaves; phonon group velocity along $\boldsymbol{q}$ at $\boldsymbol{Q}^*$	eV; eV; 1	13.1
$E^{\mathrm{PA}}_{\mathrm{min}}(\boldsymbol{q})$	Phonon-assisted threshold $E_\mathrm{g} + \min_\nu \omega_{\boldsymbol{k}_\mathrm{c}-\boldsymbol{q},\nu}$	eV	13.1
$\omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$, $\omega_{\mathrm{LA}}(\boldsymbol{Q}^*)$	Phonon energies at the onset wavevector	eV	13.1
$\boldsymbol{k}_h$, $\varepsilon_{c\boldsymbol{k}_e}$, $\varepsilon_{v\boldsymbol{k}_h}$	Crystal momentum of the final hole; band energies of the final electron and hole	eV	13.1
m_g	Floor $2E_\mathrm{g}/v^2_{\mathrm{max}}$ of a direct gap	eV	13.1
$M^{\mathrm{ad}}_{cv\nu}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q})$	Adiabatic (Herzberg–Teller) phonon-assisted amplitude, $\partial\rho_{cv}/\partial b = N^{-1/2}M^{\mathrm{ad}}$	1	13.2
b	Classical amplitude of a frozen displacement wave replacing $B_{\boldsymbol{Q}\nu}$	1	13.2
$L^{\mathrm{PA}}(q,\omega)$	Phonon-assisted loss function	1	13.3
p_{on}	Onset exponent of the phonon-assisted rate	1	13.3
$S^{\mathrm{PA}}(\boldsymbol{q},\omega)$	Phonon-assisted structure factor	as S	13.3
$L^{\mathrm{PA}}_{\mathrm{IP}}(\boldsymbol{q},\omega)$	Independent-particle value of L^{PA}; L^{PA} itself is the screened loss function	1	13.3
$\varkappa_\nu$, θ_q, θ_Q	Curvature of the onset in the tilt θ_q of $\boldsymbol{q}$ from the valley axis; tilt of $\boldsymbol{Q}^*$	eV; rad; rad	13.3
$\bar{S}_{\mathrm{LA}}$	Direction average of the longitudinal acoustic part of S^{PA}	as S	13.3
$w_\pm$	Thermal weight $(1-\mathrm{e}^{-\beta\omega})(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})$ of the phonon-assisted loss function	1	13.3
ρ^{scr}_{mn}, ρ^{LF}_{mn}	Screened density vertex $\sum_G(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})$; local-field vertex, $\rho^{\mathrm{scr}} = \rho^{\mathrm{LF}}/\varepsilon_{\mathrm{mac}}$	1	13.4
$\varepsilon^0_1(\boldsymbol{q},\omega)$	Real part of $\varepsilon_{\mathrm{mac}}$ at $n_g = 0$	1	13.4
ω_f	Energy of the final pair, $\omega \mp \omega_{\boldsymbol{Q}\nu}$	eV	13.4
$\mathfrak{s}^{(2)}_{ab}$	Second-order source of the Kohn–Sham density matrix	—	13.4
$\tilde{M}^\pm_{cv\nu}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q})$	Screened phonon-assisted amplitude without the macroscopic factor (four-path term with ρ^{LF} plus the pair channel)	1	13.4
$\delta n^{(2)}_{\boldsymbol{G}}(\boldsymbol{q}+\boldsymbol{Q})$	Second-order density of the Kohn–Sham electrons at the pair momentum	$\mathrm{Å}^{-3}$	13.4
k_{xc}, k_x	Third density derivative of the (exchange–)correlation energy density, adiabatic local kernel	—	13.4
δ_{LF}, δ_{pair}	Local-field and pair-channel ratios of L^{PA}, minus one	1	13.4
$\tilde{M}^\pm$	Short form of $\tilde{M}^\pm_{cv\nu}$	1	13.4
n_x, n_y, n_{tot}, $n_{\mathrm{tot},\boldsymbol{G}}$	First-order densities $\delta n_\mathrm{x}, \delta n_\mathrm{y}$ induced by the dark matter and by the phonon; total second-order density and its Fourier component	$\mathrm{Å}^{-3}$	13.4

Symbol	Meaning	Units	Sect.
$\varepsilon_2^0(\boldsymbol{q},\omega)$	Imaginary part of $\varepsilon_{\mathrm{mac}}$ at $n_g=0$; zero inside the phonon-assisted window	1	13.4
$\alpha^0_{\mathrm{abs},\nu}(\omega)$	Cold ($T\to 0$) absorption spectrum of branch ν	cm^{-1}	13.5
$\Delta E_{\mathrm{g}}(T)$	Gap shift $E_{\mathrm{g}}(T)-E_{\mathrm{g}}(0)$	eV	13.5
$\varepsilon_2^{\mathrm{PA},\alpha\beta}$, $\varepsilon_2^{\mathrm{PA,T}}$, $\varepsilon_2^{\mathrm{PA,L}}$	Tensor and transverse/longitudinal projections of $\varepsilon_2^{\mathrm{PA}}$; equal at $\boldsymbol{q}\to 0$ in a cubic crystal	1	13.5
$E^{\mathrm{ren}}_{\mathrm{min}}(q,T)$, δ_{w}	Renormalized minimum pair energy; off-shell margin of the window	eV	13.6
$\hat{\mathfrak{t}}_{\boldsymbol{Q}}$, $\mathfrak{t}_{mn}$	One-electron generator of the Ward identity of the amplitude and its matrix elements	—	13.6
$g^{\mathrm{cm}}_{mn\nu}$	Comoving electron–phonon vertex $g-\delta g$	eV	13.6
$\delta g_{mn}(\boldsymbol{\kappa},\boldsymbol{Q})$	Vertex change $(\varepsilon_{m,\boldsymbol{\kappa}+\boldsymbol{Q}}-\varepsilon_{n\boldsymbol{\kappa}}\pm\omega_{\boldsymbol{Q}\nu})\mathfrak{t}_{mn}$ of the Ward identity, $g^{\mathrm{cm}}=g-\delta g$	eV	13.6
ϕ_ν	Phase of $\rho^*_m\mathcal{A}^{\mathrm{off}}_\nu$ in the Fano coefficient	rad	13.6
S^{WL}, S^{cl}, S^0_{ad}, $S^{\mathrm{PA}}_{\mathrm{ad}}$	Williams–Lax average of the structure factor; clamped-nucleus structure factor of the supercell; its adiabatic zero-phonon and phonon-assisted parts	as S	13.7
x_j, σ_j, N_{m}	Mass-weighted normal coordinates of the supercell (frequency-ordered), their Gaussian widths, number of modes	—; —; 1	13.7
$\mathrm{ZPR}^{\mathrm{na}}$, $\mathrm{ZPR}^{\mathrm{ad}}$, g_{O}, ω_{O}, s_m	Nonadiabatic and adiabatic zero-point renormalization; optical vertex and phonon energy of the closed form; $+1$ for an empty, -1 for an occupied intermediate state	eV; eV; eV; 1	13.7
q_{ZP}	Momentum transfer of the joint zero-phonon minimum	eV	13.7
$S^0(\boldsymbol{q},\omega;T)$	Renormalized zero-phonon structure factor; without the argument T the clamped S^0	eV^2	13.7
$g^x_{\lambda\lambda'\nu}$, $M^{\pm}_{\lambda\nu}$, $M^{x\pm}_{cv\nu}$, M^x_ν	Exciton–phonon vertex; replica amplitude; amplitude into a free final pair with exciton intermediate states; its value at the extrema	eV; 1; 1; 1	13.8
$\mathcal{W}_x(E)$	Relative-motion weight of the Wannier–Mott exciton, lines and continuum	$\mathrm{eV}^{-1}\,\mathrm{Å}^{-3}$	13.8
$M^{\pm}_{\mathcal{T}}$, $M^{\pm}_V$	Phonon-assisted amplitudes of the kinetic and of the potential energy at $\boldsymbol{q}=0$	eV	13.8
C_{TA}, C_{TO}, c_0, E_{g1}	Branch weights, strength and effective gap of the free-pair fit of Stanford et al. to the silicon edge	1; $\mathrm{eV}^{-2}\,\mathrm{cm}^{-1}$; eV	13.8
$g^x_{\lambda 0\nu}$, $g^x_{\lambda' 0\nu}$, $\rho_{\lambda\lambda'}$, $\rho_{\lambda' 0}$, $E_{\lambda'}$	Exciton creation vertex of the phonon; density vertex between two excitons; exciton form factor; energy of an intermediate exciton λ'	eV; 1; 1; eV	13.8
σ_1^{PA}	Phonon-assisted optical conductivity $m_\phi\varepsilon_2^{\mathrm{PA}}$ at the boson mass (ALP absorption)	eV	13.8
$S^{\mathrm{PA}}_{\mathcal{T}\mathcal{T}}(0,\omega)$	Phonon-assisted structure factor of the kinetic-energy row $\mathcal{T}_e$	as $S_{\mathcal{T}\mathcal{T}}$	13.8
β_θ, β_z, β_2	Curvatures of the zero-phonon pair energy about the off-axis optimum (tilt in the plane, out of it; second heavy hole)	$\mathrm{eV\,rad}^{-2}$	13.9
$x_{\mathrm{max}}(q)$, x^*_{max}	Largest energy above threshold that a mass reaches at q; its maximum over q, $\frac{1}{2}v^2_{\mathrm{max}}\delta m$	eV	13.9
A_η	Coefficient of the quadratic edge $\eta(v_{\mathrm{min}})\simeq A_\eta(v_{\mathrm{max}}-v_{\mathrm{min}})^2$	—	13.9
m_z	The mirror $(x,y,z)\to(x,y,-z)$, sans-serif m_z	—	13.9
Chapter 14: Magnetic excitations			
$W_\parallel$, $W_\perp$, ΔS	Longitudinal and transverse spin weights; zero-point spin reduction	1	14.1
Ω_{ion}	Volume per magnetic ion in the two-sublattice formulas	$\mathrm{Å}^3$	14.1
$w^{\mathrm{M}}_{\boldsymbol{q}}$	One-magnon weight of a two-sublattice antiferromagnet at the full momentum transfer, $\lvert\boldsymbol{S}\rvert(A_{\boldsymbol{q}}-B_{\boldsymbol{q}})/\omega^{\mathrm{M}}_{\boldsymbol{q}}$	1	14.1

Symbol	Meaning	Units	Sect.
$\boldsymbol{Q}_{\mathrm{AF}}$	Ordering wavevector of a two-sublattice antiferromagnet, $(\pi/a)(1,1,1)$ in NiO	eV	14.1
$\boldsymbol{\delta}_0$	Translation from sublattice A to sublattice $B, \varepsilon_{\boldsymbol{G}} = \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\delta}_0}$	Å	14.1
$\langle v \rangle$	Mean laboratory dark-matter speed, $2\int u\eta(u)du = 354$ km/s	1	14.1
$w_\parallel(\boldsymbol{q})$, $w_\perp(\boldsymbol{q})$	Energy-integrated two-magnon and one-magnon weights per ion at $\boldsymbol{q}$; zone averages $W_\parallel, W_\perp$	1	14.2
$\mathcal{A}^{\mathrm{2m}}_{\boldsymbol{k}}(\boldsymbol{q})$, $U^{\mathrm{mm}}_{\boldsymbol{q}}$, $\mathsf{G}^{\mathrm{2m}}_{\boldsymbol{q}}$, $H^{\mathrm{2m}}_{\boldsymbol{q}}$, ω_{b}	Two-magnon amplitude $u_{\boldsymbol{k}}v_{\boldsymbol{k}-\boldsymbol{q}} - v_{\boldsymbol{k}}u_{\boldsymbol{k}-\boldsymbol{q}}$; quartic pair vertex; free pair propagator; pair Hamiltonian; bound-pair energy	1; eV; eV^{-1}; eV; eV	14.2
$S^{(2m)}_{zz}(\boldsymbol{q},\omega)$	Longitudinal two-magnon spin response	eV^2	14.2
ω_{gap}	Anisotropy gap of a magnon branch	meV	14.2
$g^{\mathrm{mp}}_{\boldsymbol{k}}$, B_{me}, B_1, B_2	Magnon–phonon coupling; magnetoelastic energy per cell B_2a^3; cubic magnetoelastic constants	eV; eV; $\mathrm{erg\,cm^{-3}}$	14.2
ω_{m}, ω_{p}, δ_{mp}, X^{m}_h, X^{p}_h, ω_h, $k_\times$, $\omega_\times$	Magnon and phonon energies at a crossing; detuning $\omega_{\mathrm{m}} - \omega_{\mathrm{p}}$; magnon and phonon contents and energy of hybrid h; momentum and energy of the crossing (ω_{p} in Sect. 14.2)	eV; eV; eV; 1; eV; eV; eV	14.2
D_{sw}, $b_{\boldsymbol{k}\pm}$, $b_{\boldsymbol{k}x}$, $b_{\boldsymbol{k}y}$	Spin-wave stiffness; circular phonon $(b_x \pm \mathrm{i}b_y)/\sqrt{2}$ and its two linear components	$\mathrm{meV\,nm^2}$; —	14.2
M_{s}, $\boldsymbol{m}$, m_x, m_y	Saturation magnetization; its direction $\boldsymbol{M}/M_{\mathrm{s}}$ and the transverse components of the direction	G or $\mathrm{A\,m^{-1}}$; 1	14.2
$u_{\alpha\beta}$, f_{me}	Strain tensor; magnetoelastic energy density $B_1\sum_\alpha m_\alpha^2 u_{\alpha\alpha} + B_2\sum_{\alpha\neq\beta} m_\alpha m_\beta u_{\alpha\beta}$	1; $\mathrm{erg\,cm^{-3}}$	14.2
D_{an}	Single-ion anisotropy constant, $H_{\mathrm{an}} = -D_{\mathrm{an}}\sum(S^z)^2$	meV	14.2
λ_S, ω_{AFMR}, ω_E, ω_A	Spin fraction of the ionic angular momentum, $\boldsymbol{s}_e(0) = \lambda_S\boldsymbol{J}$ ($g_J - 1$ for a J multiplet; 1 for a quenched ion); antiferromagnetic resonance, exchange and anisotropy frequencies	1; meV	14.3
g_J	Landé factor of a J multiplet	1	14.3
M_{sw}	Exchange mass of the uniform mode, $1/2D_{\mathrm{sw}}$	eV	14.3
g_{ion}, $g_{\mathrm{ion},ab}$	Spectroscopic g factor (tensor) of an ion; $g_{\mathrm{ion}} = 2$ for a pure spin	1	14.3
ω_{Lar}	Larmor frequency $g_{\mathrm{ion}}\mu_B\lvert\boldsymbol{B}_0\rvert$ or $\mu_{\mathrm{nuc}}B_0/J_{\mathrm{nuc}}$	eV	14.3
$\mathfrak{s}_{\mathrm{net}}$	Net spin per unit cell, $\langle S^z_{\mathrm{tot}}\rangle/N$	1	14.3
ω_{Kit}, $\omega_{\mathrm{Kit},i}$, $\omega_{\mathrm{Kit},x}$, $\omega_{\mathrm{Kit},y}$, N^{dm}_i, N^{dm}_x, N^{dm}_y, N^{dm}_z, H_{u}	Kittel frequency $(\omega_{\mathrm{Kit},x}\omega_{\mathrm{Kit},y})^{1/2}$ of a saturated ellipsoid; its two stiffnesses $\omega_{\mathrm{Lar}} + (N^{\mathrm{dm}}_i - N^{\mathrm{dm}}_z)\omega_M$; demagnetizing factors along the axes, sum 1; Hamiltonian of the uniform mode (ω_K is the Klein phonon energy)	eV; eV; 1; eV	14.3
$\mathcal{R}_{\mathrm{peak}}$	Event rate per unit target mass on the peak of a magnon line	$\mathrm{s^{-1}\,kg^{-1}}$	14.3
$v_\perp$, $p_{\phi\perp}$, $p_{\phi,x}$, $p_{\phi,y}$	Dark-matter speed and boson momentum perpendicular to the net-spin axis $\hat{z}$; Cartesian components of $\boldsymbol{p}_\phi$	1; eV	14.3
θ_E	Angle between the net-spin axis $\hat{z}$ and $\boldsymbol{v}_E$	rad	14.3
ξ_{ion}, Δ_{CF}, Λ_{ab}	Spin–orbit constant of an ionic term; crystal-field splitting; orbital second-order tensor $\sum_n \mathrm{Re}\langle 0\lvert L_a\rvert n\rangle\langle n\lvert L_b\rvert 0\rangle/(E_n - E_0)$, $g_{\mathrm{ion},ab} = 2\delta_{ab} - 2\xi_{\mathrm{ion}}\Lambda_{ab}$ for a quenched ion	eV; eV; eV^{-1}	14.3
$\boldsymbol{B}'$	Magnetic field $\nabla\times\boldsymbol{A}'$ of a dark photon	eV^2	14.3
ω_M	Saturation magnetization in frequency units, $2\mu_B\mu_0 M_{\mathrm{s}}$	eV	14.3
Γ_{M}	Full width of the uniform-magnon line	eV	14.3
B_E, B_A, B_{sf}, K	Exchange, anisotropy and spin-flop fields; easy-axis term added to $A_{\boldsymbol{k}}$	T; T; T; meV	14.3
Δ_{XXZ}	Exchange anisotropy of the eight-site XXZ cluster	1	14.3
P_T	Transverse projector of the neutron cross-section	1	14.4

Symbol	Meaning	Units	Sect.
$C_{\mathrm{sn}}(q)$, $F_g^{\mathrm{mag}}(q)$	Spin-to-neutron conversion factor; moment form factor $\langle j_0\rangle + (1-2/g_{\mathrm{ion}})\langle j_2\rangle$	1	14.4
$S_{ab}^{\mathrm{lat}}(\boldsymbol{k},\omega)$	Response of the lattice spins, $S_{ab}(\boldsymbol{k}+\boldsymbol{G}) = \lvert F^{\mathrm{mag}}\rvert^2 S_{ab}^{\mathrm{lat}}(\boldsymbol{k})$ for one magnetic ion per cell	eV^2	14.4
$S_\parallel$, $S_\perp$, S_{c}, θ_m, $\hat{m}$	Parts of a spin-rotation-invariant response along, across and around the moment direction $\hat{m}$($W_\parallel, W_\perp$ are equal-time weights); angle between $\hat{q}$ and $\hat{m}$; moment axis of a collinear magnet	eV^2; rad; 1	14.4
$S^{\mathrm{neu}}(\boldsymbol{q},\omega)$, N_{fu}	Measured magnetic neutron structure factor per formula unit, $\mathrm{d}^2\sigma/\mathrm{d}\Omega dE' = (p_n'/p_n)S^{\mathrm{neu}}$; formula units per cell (N_f is the number of final states)	$\mathrm{b\,eV^{-1}}$; 1	14.4
T_{d}	Temperature at which a neutron spectrum was measured (in Sect. 14.4)	K	14.4
$q_{1\mathrm{BZ}}$	Zone-edge momentum of the YIG data analysis of Berlin et al., 987 eV $= 2\pi/a$ for $a = 12.56$ Å	eV	14.4
h	Element of the magnetic point group	—	14.4
r_{cl}	Classical electron radius $\alpha/m_e = 2.818$ fm	fm	14.4
T_{CM}	Ordering temperature of a multiferroic	K	14.5
$\mathcal{G}_{\mathrm{OA}}$, $\bar{\mathcal{G}}_{\mathrm{OA}}$, $S^{\mathrm{R}}_{\alpha\beta}$	Antisymmetric (optical-activity) part of the current block, $S_{j^\alpha j^\beta} = S^{\mathrm{R}}_{\alpha\beta} + \mathrm{i}\varepsilon_{\alpha\beta\gamma}\mathcal{G}^{\gamma}_{\mathrm{OA}}$, odd in $\boldsymbol{q}$, units of S_{jj}; direction average of $\hat{q}\cdot\mathcal{G}_{\mathrm{OA}}$; symmetric part of the current block	as S_{jj}	14.5
σ_a^{T}, σ_a^{P}	Sign of row a under time reversal and inversion, $\mathsf{T}O_a(\boldsymbol{q})\mathsf{T}^{-1} = \sigma_a^{\mathsf{T}}O_a(-\boldsymbol{q})$	1	14.5
$S_\pm$, $\hat{e}_\pm$	Helicity-resolved transverse response $S_{\mathrm{T}} \mp \hat{q}\cdot\mathcal{G}_{\mathrm{OA}}$; helicity vectors $(\hat{e}_1 \pm \mathrm{i}\hat{e}_2)/\sqrt{2}$	as S; 1	14.5
$\rho(\omega)$, $g_{\hat{q}\hat{q}}$	Rotatory power $\omega^2\,\mathrm{Re}\,g_{\hat{q}\hat{q}}/2$; projection $\hat{q}\cdot g\cdot\hat{q}$ of the gyration tensor g_{ml}	$\mathrm{rad\,m^{-1}}$; m	14.5
$\boldsymbol{L}^{\mathrm{ph}}_{\boldsymbol{Q}\nu}$, $\boldsymbol{L}^{\mathrm{ph}}_{\kappa,\boldsymbol{Q}\nu}$, L^{ph}_x, L^{ph}_z, $\boldsymbol{\mu}^{\mathrm{ph}}_{\boldsymbol{Q}\nu}$	Phonon angular momentum $-\mathrm{i}\sum e^*\times e$, its per-site term and components; its Born-charge magnetic moment $\sum(eZ^*_\kappa/2M_\kappa)\boldsymbol{L}^{\mathrm{ph}}_\kappa$	1; 1; μ_N	14.5
L_z, J_z	Angular momentum $\boldsymbol{u}\cdot J_z\boldsymbol{p}$ of the lattice in mass-weighted coordinates; generator of rotations about $\hat{z}$ acting on each site displacement	1	14.5
$\mathcal{N}^{\mathrm{ch}}$, M	Probabilities that a created phonon ends as a detectable chiral phonon, $(\mathbb{1}-\mathsf{M})^{-1}\boldsymbol{b}$; matrix of three-phonon branching ratios	1	14.5
H_{mol}, S_z, ω_{meta}, D, B_4^0, O_4^0	Spin Hamiltonian of a single-molecule magnet; component of the molecular spin along the easy axis; metastable line $D(2\lvert\boldsymbol{S}\rvert-1)-\omega_{\mathrm{Lar}}$; axial and quartic anisotropy constants; Stevens operator	eV; 1; meV; meV; meV; 1	14.5
$\boldsymbol{n}_{\mathrm{mol}}$, $\omega_{m,m'}$, ΔE_{Zee}, τ_{R}, τ_0, U	Number density of molecules; level difference $E_{m'}-E_m$; Zeeman energy stored in the metastable well; relaxation time over the barrier $\tau_0\exp[(U-\frac{1}{2}\Delta E_{\mathrm{Zee}})/T]$ and its attempt time; barrier height (U in Sect. 14.5)	$\mathrm{m^{-3}}$; eV; eV; s; s; eV	14.5
ω_{soft}, γ_{soft}	Soft-mode frequency and damping of the Ginzburg–Landau oscillator	eV	14.5
χ_θ, α_{me}, α_M, m_P	Coefficients of $\theta\boldsymbol{P}\cdot\boldsymbol{M}$, $\boldsymbol{M}\cdot\boldsymbol{P}$ and $\lvert\partial_t\boldsymbol{M}\rvert^2$, and polarization stiffness, in the free energy of Røising et al. (α_M in Sect. 14.5)	—	14.5
$R_{s^z\mathcal{J}}$	Retarded response of s_e^z to the spin current $\mathcal{J}_e^{\alpha\alpha}$ at $\boldsymbol{q}=0$	eV^2	14.5
$\chi_\times$, Φ, $\tau(T)$	Amplitude and scaling function of the critical cross response near T_{CM}, $\chi_\times(T)\Phi(\omega\tau(T))$; critical relaxation time (Φ in Sect. 14.5)	—; 1; s	14.5
$F_\phi(\omega)$, $\Delta\omega_{\mathrm{eq}}$, Φ	Energy distribution of the field line; its equivalent width $[\int F_\phi^2]^{-1}$, $2.67/t_{\mathrm{coh}}$ for the standard halo; complex amplitude of the field in the frame rotating at m_ϕ (in Sect. 14.6)	$\mathrm{eV^{-1}}$; eV; 1	14.6

Symbol	Meaning	Units	Sect.
$\mathcal{S}_\phi(\boldsymbol{k},\omega)$, $\mathcal{S}_{\rm sig}(\omega)$, $\mathcal{S}_{\rm n}(\omega)$	Power spectrum of the classical dark-matter field; power spectra of the signal and of the noise of a readout	eV^{-2}; (readout unit)^2 × time	14.6
$t_{\rm int}$, $N_{\rm coh}$, $\bar{N}$	Length of a record (integration time); number of independent coherence times in it; mean number of absorption counts, Var $N = \bar{N} + \bar{N}^2/N_{\rm coh}$	s; 1; 1	14.6
$\mathcal{K}_t$	Fejér kernel $(t/2\pi)\,\mathrm{sinc}^2(xt/2)$ of a record of length t, $\int \mathcal{K}_t = 1$; $\mathcal{K}_{t_{\rm int}}$ for the record	eV^{-1}	14.6
$\Gamma_{\rm tg}$	Width of the target's line in the golden-rule condition $t \gg \min(t_{\rm coh}, 1/\Gamma_{\rm tg})$	eV	14.6
$\Omega_{\rm Rabi}$, $\Omega_{\rm rms}$, $\delta\omega_{\rm Lar}$	Rabi frequency (half the transverse drive); its value for the mean energy density; detuning $\omega_{\rm Lar} - m_\phi$	s^{-1}	14.6
$p_{\rm pol}$, M_0, $u_{\rm sp}$, $\Gamma_{\rm inh}$	Polarization of the nuclear spins ($\tanh(\omega_{\rm Lar}/2T)$ in equilibrium); equilibrium magnetization; spectral factor $\approx \pi/(\Gamma_{\rm inh}T_2)$ of Aybas et al.; inhomogeneous width	1; $\mathrm{A\,m}^{-1}$; 1; s^{-1}	14.6
M_x, M_y, M_z, M_+, $\tilde{M}_+$, $M_\perp$	Components of the nuclear magnetization; $M_x + \mathrm{i}M_y$; its rotating-frame value $\mathrm{e}^{\mathrm{i}m_\phi t}M_+$; $\lvert M_+\rvert$	$\mathrm{A\,m}^{-1}$	14.6
$\boldsymbol{b}$, $\boldsymbol{b}_\phi$, $b_\perp$	Vector of energy in $H = -\boldsymbol{b}\cdot\boldsymbol{J}^{\rm nuc}$ (precession vector); its dark-matter part; amplitude of its transverse component	eV	14.6
$T_{\rm s}$	Spin temperature of a prepolarized sample, $\tanh(\omega_{\rm Lar}/2T_{\rm s}) = p_{\rm pol}$	K	14.6
P_k, s_k, b_k, x_k, $\sigma_{\rm sig}$, $N_{\rm eff}$, Q, s, b, $\mathsf{s}_{\rm up}$, $\mathsf{Q}_{\rm obs}$	Periodogram bin; mean signal and noise powers; $\mathsf{s}_k/\mathsf{b}_k$; rms signal; participation number of the line; summed power of the line bins; total signal $\sigma_{\rm sig}^2 t_{\rm int}$ and noise of the line; 95 % upper limit on s; observed value of Q	—	14.6
$\mathcal{G}_{\rm rd}$, $O_{\rm rd}$, $R_{{\rm rd},b}$, $X_{\rm sig}$	Response kernel of a readout, $X_{\rm sig} = \mathcal{G}_{\rm rd} * \Phi$; readout operator; its retarded response to row b; signal part of the readout	—	14.6
E^*, $\boldsymbol{J}^{\rm nuc}$, $J_z^{\rm nuc}$, $J_{z,{\rm tot}}^{\rm nuc}$, $\mu_{\rm nuc}$, $J_{\rm nuc}$, $\boldsymbol{n}_{\rm nuc}$	Effective field of the Schiff moment; nuclear spin, its component along $\boldsymbol{B}_0$ and the sum over the sample; nuclear moment (μ_N is the nuclear magneton) and spin length; density of the isotope	$\mathrm{kV\,cm}^{-1}$; 1; 1; 1; μ_N; 1; m^{-3}	14.6
d_n, g_d, $\Delta\varepsilon$	Nucleon electric dipole moment $g_d\phi$ induced by an axion; EDM coupling; shift d_nE^* of each spin level	e cm; GeV^{-2}; eV	14.6
T_1, T_2	Longitudinal and transverse relaxation times of a nuclear-spin ensemble (Bloch equations)	s	14.6
Chapter 15: Superconductors			
$\mathfrak{s}_{\rm BCS}$, $\mathfrak{g}_{\rm I}$, $L^{\rm nor}$	BCS correction function; case-I coherence integral $\mathfrak{g}_{\rm I}(y) = S_{\rm I}/S_{\rm n}$; normal-state loss function $\mathrm{Im}[-1/\varepsilon_{\rm Lin}]$	1	15.1
$L^{\rm bare}$, $L^{\rm cons}$, y, $\mathfrak{r}_\varepsilon$, $\delta_{\mathfrak{s}}$	Loss functions of the bare quasiparticle bubble and of the conserving response; edge variable $\omega/\Delta_{\rm BCS}$; $\mathrm{Im}\,\varepsilon_{\rm Lin}/\mathrm{Re}\,\varepsilon_{\rm Lin}$ on the kinematic edge; Kramers–Kronig shift of $\mathrm{Re}\,\varepsilon$ of the bare bubble relative to $\mathrm{Re}\,\varepsilon_{\rm Lin}$	1	15.1
$R^{\rm cons}$, $\varepsilon^{\rm cons}$, $\varepsilon^{\rm bare}$	Conserving density response (pair-field ladder, no Coulomb) and the dielectric functions $1 - v_{\rm C}R^{\rm cons}, 1 - v_{\rm C}R^0_{\boldsymbol{\eta\eta}}$	eV^2; 1	15.1
$M_A^{\rm N}$	Amplitude with which the Nambu-channel operator O_A creates a quasiparticle pair	1	15.1
$F_\mu(\omega)$, $\mathfrak{p}_\mu$, $\mathfrak{p}$, ω_μ, s_*	$q = 0$ pair sum F at angle μ; $\omega_\mu^2/4\Delta_{\rm BCS}^2$, and $\mathfrak{p}$ its generic value in the continuation; Doppler-frame pair energy $(\omega^2 - v_{\rm F}^2q^2\mu^2)^{1/2}$; inverse-square-root point $v_{\rm F}q\mu = (\omega^2 - 4\Delta_{\rm BCS}^2)^{1/2}$	1; 1; 1; eV; eV	15.1
$\mathfrak{v}$	Vertex fraction $\mathrm{Im}(R^{\rm cons} - R^0_{\boldsymbol{\eta\eta}})/\mathrm{Im}\,R^0_{\boldsymbol{\eta\eta}}$	1	15.1
$\Omega_{\rm AB}(q)$	Energy of the Anderson–Bogoliubov mode at finite q, below $2\Delta_{\rm BCS}$	eV	15.1
$\mathfrak{w}$, $\mathfrak{w}_\gamma$	Reduced frequency $\omega/qv_{\rm F}$ of the Lindhard function at $q \ll k_{\rm F}$; its complex value $(\omega + \mathrm{i}\gamma_{\rm rel})/qv_{\rm F}$	1	15.2

Symbol	Meaning	Units	Sect.
$\mathfrak{f}_{\rm L}(\mathfrak{w})$	Lindhard bracket at $q \ll k_{\rm F}, 1-(\mathfrak{w}/2)\ell(\mathfrak{w}), R^0 = -2\mathcal{N}_{\rm F}\mathfrak{f}_{\rm L}$	1	15.2
$\mathfrak{g}$, δ	Absorptive factor $\operatorname{Im}\varepsilon/\operatorname{Im}\varepsilon_{\rm Lin}$ and real-part shift $\operatorname{Re}\varepsilon/\operatorname{Re}\varepsilon_{\rm Lin}-1$ of an arbitrary response in the resummation	1	15.2
$\mathfrak{l}_e$	Electron mean free path	nm	15.3
$D_{\rm diff}$, $\xi_{\rm coh}$, $\rho_{\rm el}$, $d_{\rm film}$	Diffusion constant $v_{\rm F}\mathfrak{l}_e/3$; BCS coherence length $v_{\rm F}/\pi\Delta_{\rm BCS}$; residual resistivity; film thickness	eV^{-1}; μm; Ω m; nm	15.3
$\varepsilon_{\rm sh}$, ε_a, ε_b	Sheet dielectric function of a film thinner than 1/q; dielectric constants of the media on its two sides	1	15.3
$L_{\rm s}$, $L_{\rm n}$	Loss functions of the superconducting and normal film at the same scattering rate	1	15.3
$L_{\rm Mer}$, $L_{\rm Drude}$, $L_{\rm Lin}$	Loss functions of the Mermin, Drude and Lindhard dielectric functions	1	15.3
$\omega_p^{\rm intra}$, $\sigma_{\rm dc}$, σ_n	Intraband plasma frequency; dc conductivity $\omega_p^2/\gamma_{\rm rel}$; Drude conductivity of the normal film	eV	15.4
$\mathfrak{g}_{\rm II}$	Case-II coherence integral (Mattis–Bardeen), S_{II}/S_n	1	15.4
$\alpha^2F_{\rm tr}$, $F_{\rm tr}$, $\lambda_{\rm tr}$, $\gamma_{\rm H}$	Transport Eliashberg function; transport coupling $2\int\alpha^2F_{\rm tr}/\omega$; Holstein scattering rate	1; 1; 1; eV	15.4
$\bar\alpha$	Time-reversed partner of the one-electron eigenstate α of a disordered metal	—	15.4
$\sigma_1^{\rm MB}$, $\sigma_1^{\rm S}$, $\sigma_1^{\rm N}$, $\sigma_1^{\mathcal{J}}$	Mattis–Bardeen conductivity; phonon-assisted conductivity of the superconductor and of the normal metal; spin-current conductivity	eV	15.4
J_i, J_f, $V^e_{\alpha\beta}$	Current eigenvalues of the initial and final states; matrix element $\langle\alpha\vert\hat e\cdot\hat v\vert\beta\rangle$ between impurity eigenstates	1	15.4
$N^{\rm El}(E)$, $P^{\rm El}(E)$	Eliashberg density of states and pair amplitude per normal density of states, $\operatorname{Re}[E/(E^2-\Delta^{\rm El2})^{1/2}], \operatorname{Re}[\Delta^{\rm El}/(E^2-\Delta^{\rm El2})^{1/2}]$	1	15.4
Chapter 16: Novel targets			
$\varepsilon_{\rm pub}$	Polarization of the Dirac electrons in the published Dirac rates, $1+\varepsilon_{\rm D}/\varepsilon_{\rm bg}=\varepsilon/\varepsilon_{\rm bg}$	1	16.1
$B_{\rm D}(s)$	Bracket of the one-loop Dirac dielectric function, $\varepsilon_{\rm D}=(g_{\rm deg}\alpha/6\pi v_{\rm D})B_{\rm D}$	1	16.1
$L_{\rm pub}$, $\Gamma_{\rm pub}$	Loss function and absorption rate of the published Dirac construction, $\varepsilon_{\rm bg}^2L$ and $\varepsilon_{\rm bg}\Gamma$	1; eV	16.1
$\varepsilon_{\rm D}$, $\varepsilon_{{\rm D},i}$, $\tilde R_{\rm D}$, $R_{\rm DD}$, $F_{\rm med}$, $\varepsilon_{{\rm bg},xx}$, $\varepsilon_{{\rm bg},yy}$	Dirac part $-v_{\rm C}R^0_{\rm D}$ of the dielectric function and its axis-i component; proper Dirac response; random-phase response of the Dirac electrons alone, $R^0_{\rm D}/\varepsilon_{\rm pub}$; in-medium factor $1/\varepsilon_{\rm pub}$ of the published rates; principal components of the background tensor	1; eV^2; eV^2; 1; 1	16.1
$V_{\rm uc}$, Π	Cell volume of the Dirac benchmark of the published rates (60 $\mathrm{Å}^3$); photon self-energy $\omega^2(1-\varepsilon_{\rm pub})$ of the published absorption	$\mathrm{Å}^3$; eV^2	16.1
$\zeta^{\rm lim}(m_\phi)$	Mixing reach of a target for dark-photon absorption	1	16.1
b, x	Carrier speed $(1-4\Delta_{\rm D}^2/s)^{1/2}$ in the rest frame of a Dirac pair, in units of $v_{\rm D}$; Sommerfeld variable $\pi\tilde\alpha/b$	1	16.1
a_s, $N_{\rm ch}$, v_a	Period of a layered stack; number of conduction channels of a nanotube; phase velocity of the acoustic (demon) plasmon	nm; 1; 1	16.2
s_j, c_j	$\sinh(kd_j/2), \cosh(kd_j/2)$ of layer j of a stack	1	16.2
$S_{\rm u}$, $\rho_{\rm u}$, $\boldsymbol{n}_{\rm u}$, $\boldsymbol{r}_u$, $F_{\rm th}$	Structure factor and mass of one unit (wire, sheet) per unit length or area; units per transverse area or length; position of unit u; thickness factor of a sheet kernel	—	16.2

Symbol	Meaning	Units	Sect.
$\tilde{R}^{D}_{\mathrm{low}}$, $\varepsilon_{\mathrm{2D}}$, $\varepsilon_{\mathrm{1D}}$, $\varepsilon_{\mathrm{env}}$, $\bar{\varepsilon}$, S_{2D}, S_{1D}, R_{1D}, D_{Dr}	Proper response of the bands near the Fermi level of a sheet; sheet and tube dielectric functions; dielectric constant around a tube; mean of the two half-space constants; structure factors in two and one dimension; 1D bubble; Drude weight	—	16.2
n_{2D}, $n_{\mathrm{2D},h}$, n_h, $\omega_{p,h}$, $\omega_{\mathrm{2D},h}$, $v_{\mathrm{F},h}$	Areal density; that of the heavy band; heavy-carrier density; heavy plasma frequency in 3D and 2D; heavy Fermi velocity	cm^{-2}; cm^{-2}; cm^{-3}; eV; eV; 1	16.2
n_u, $\tilde{u}$, $\phi_{c\mathfrak{o}'}$, $\phi_{c\mathfrak{o}'''}$, $\phi_{v\mathfrak{o}''}$, $\rho^{(\mathfrak{o}')}_{0\mathfrak{x}'}$	Density operator of unit u along its free directions; Fourier transform of the static displacement field of a force dipole; primed molecular orbitals and transition densities of further molecules in the kernels	—	16.2
$q_{\parallel}$, $q_{\perp}$, $F_{\perp}(\boldsymbol{q}_{\perp})$	Momentum along the free directions of a sheet or tube and across it; transverse form factor of a confined unit, $F_{\perp}(0)=1$ (sinc for a uniform sheet, $J_0(q_{\perp}r_{\mathrm{t}})$ for a shell)	eV; eV; 1	16.2
$v_{\mathrm{C}}^{\mathrm{2D}}$, $v_{\mathrm{C}}^{\mathrm{1D}}$, $v_{\mathrm{C}}^{\mathrm{L}}$	Coulomb kernels of a sheet, $e^2/[(\varepsilon_a+\varepsilon_b)q_{\parallel}]$; of a cylindrical shell, $(2\alpha/\varepsilon_{\mathrm{env}})I_0K_0$; of a stack of sheets	$\mathrm{eV}^{-1}\times$ length (2D, 1D)	16.2
$\tilde{R}^{\mathrm{2D}}$, $\tilde{R}^{\mathrm{2D}}_{\mathrm{low}}$, $\tilde{R}^{\mathrm{2D}}_{\mathrm{bg}}$, r_0	Proper response of a sheet per area; its low-energy and remote-band parts; Keldysh length, $-\tilde{R}^{\mathrm{2D}}_{\mathrm{bg}}=2r_0q_{\parallel}^2/e^2$	$\mathrm{eV}^{-1}\,\mathrm{length}^{-2}$; nm	16.2
I_0, K_0, K_1	Modified Bessel functions of the first and second kind	1	16.2
g_c	Luttinger charge parameter $[1+N_{\mathrm{ch}}v_{\mathrm{C}}^{\mathrm{1D}}/\pi v]^{-1/2}$ of a one-dimensional metal	1	16.2
m_l, m_h, $v_{\mathrm{F},l}$, $k_{\mathrm{TF},l}$, $\mathcal{N}_{\mathrm{F},l}$, ω_a, Γ_a, q_s	Light and heavy masses of the two-component model; Fermi velocity, Thomas–Fermi wavevector and density of states of the light band; frequency and full width of the acoustic plasmon; two-dimensional screening wavevector $2e^2\mathcal{N}_{\mathrm{F},l}/(\varepsilon_a+\varepsilon_b)$	m_e; 1; eV; $\mathrm{eV}^{-1}\,\mathrm{cm}^{-3}$; eV; eV; eV	16.2
$f_3(\nu)$, $f_2(\nu)$	Lindhard functions at $q\ll k_{\mathrm{F}}$ in three and two dimensions, normalized to minus the density of states, $\nu=\omega/qv_{\mathrm{F}}$	1	16.2
ε_A, ε_B, d_A, d_B, f	Dielectric functions, thicknesses and filling d_A/a_s of the two layers of a stack	1; nm; 1	16.2
$\varepsilon_{\mathrm{in}}$, $\varepsilon_{\mathrm{out}}$	Dielectric constants of a quantum dot and of its host (depolarization factor)	1	16.3
E_r, $\tilde{q}$	The recoil-energy and reduced-momentum variables of the quantum-dot and dopant sources	eV; 1	16.3
r_{t}, Φ_{wf}, N_{C}, M_{PW}, M_{OPW}, E_{QD}, E_{conf}, ϕ_{QD}	Tube radius; work function; carbon atoms per unit mass; plane-wave and orthogonalized transition amplitudes of ejection; dot gap ΔE_{QD}; confinement energy; dot volume fraction	nm; eV; kg^{-1}; —; eV; eV; 1	16.3
f_{QD}, $\varepsilon_{\mathrm{eff}}$, c_{ξ}	Dot form factor of Blanco et al. (inverse energy); Maxwell Garnett dielectric function; valley coefficient of the A_1 ground state in valley ξ	eV^{-1}; 1; 1	16.3
S_{ej}, $K(p_{\mathrm{ej}},q)$	Structure factor of vacuum ejection; solid-angle integral of the orientation-averaged ejection amplitude squared	eV^2; eV^{-3}	16.3
Δ_n, $\tilde{R}^{\mathrm{1D}}_n$, $\varepsilon_2^{\parallel}$	Half-gap $nv_{\mathrm{D}}/3r_{\mathrm{t}}$ of nanotube subband n; proper interband response of subband n per length; imaginary dielectric function of an aligned tube array along the axis	eV; —; 1	16.3
a_{mol}	Size of a molecule (Frenkel-exciton validity $qa_{\mathrm{mol}}\gg 1$)	Å	16.4
$\mathfrak{J}$, $\mathfrak{o}$, $\mathfrak{x}$	Frenkel coupling between molecular transitions; molecule label; molecular excited-state label	eV; —; —	16.4
$A^{\lambda}_{\mathfrak{o}\mathfrak{x}}(Q)$, E_{λ}, N_{mol}	Frenkel eigenvectors and energies; molecules per cell	1; eV; 1	16.4
$F_{\mathfrak{o}\mathfrak{x}}(q)$, $f^{(\mathfrak{o})}_{0\mathfrak{x}}(q)$	Vertex $\mathrm{e}^{\mathrm{i}q\cdot r_{\mathfrak{o}}}f^{(\mathfrak{o})}_{0\mathfrak{x}}$ of a molecular transition; molecular form factor	1	16.4
$\mathcal{B}$, δW	Energy window of the binned Frenkel sum; first-order change of its weight by intensity borrowing	eV; 1	16.4

Symbol	Meaning	Units	Sect.
θ_{CT}, t_{CT}, f_{CT}, E_{Fr}, E_{CT}	Frenkel/charge-transfer mixing angle, transfer integral, charge-transfer form factor, energies of the two states	1; eV; 1; eV	16.4
ϱ_μ, F_{2p}	Transition charge on carbon μ (zero differential overlap); form factor of one $2p_z$ density	1	16.4
$H^{\mathrm{F}}_{\mathfrak{o}\mathfrak{x},\mathfrak{o}'\mathfrak{x}'}$, $\mathfrak{J}_{\mathfrak{o}\mathfrak{x},\mathfrak{o}'\mathfrak{x}'}$, $A_{\mathfrak{o}\mathfrak{x}}$, $F_{\mathfrak{o}\mathfrak{x}}$, $F_{\mathfrak{o}'\mathfrak{x}'}$, $f_{0\mathfrak{x}}$, $f^o_{0\mathfrak{x}}$, $\rho^o_{0\mathfrak{x}}$, $d_{\mathfrak{x}}$, $d^o_{\mathfrak{x}}$, d_λ, $\boldsymbol{r}_{\mathfrak{o}}$, R_o, E_{L_a}, $E_{\mathfrak{x}'}$, f_{Fr}	Frenkel Hamiltonian and coupling in the molecular basis; eigenvector components; vertices; molecular form factors and transition densities; transition dipoles of a molecular state and of an exciton; molecular position and rotation; molecular level of L_a; another molecular level; Frenkel form factor	—	16.4
$\phi_{v\mathfrak{o}}$, $\phi_{c\mathfrak{o}}$	Occupied and empty orbitals centered on molecule $\mathfrak{o}$	$\text{Å}^{-3/2}$	16.4
$S^{\mathrm{F}}(\boldsymbol{q},\omega)$, $\mathsf{R}_{\mathfrak{o}}$, R_b, $\boldsymbol{d}_{\mathfrak{x}}$, $\boldsymbol{d}_1$, $\boldsymbol{d}_2$, $\boldsymbol{d}_\pm$	Frenkel-exciton structure factor; rotation mapping the reference molecule onto molecule $\mathfrak{o}$; molecular transition dipole; those of the two molecules of the cell and of the two Davydov components	eV^{-1}; 1; e Å	16.4
$\boldsymbol{n}_{\mathrm{c}}$	Mott critical density of a dopant	cm^{-3}	16.5
$\varepsilon_{\mathrm{host}}$, ε^{D}, $\varepsilon_2^{\mathrm{D}}$	Macroscopic dielectric function of the host of a dopant; donor part of the dielectric function and its imaginary part	1	16.5
S^{D}_{ab}, L^{D}	Donor structure factor (host screening once); donor loss function $\varepsilon_2^{\mathrm{D}}/\varepsilon^2_{\mathrm{host}}$ (replaces W in the dopant limit check; W is reserved for W_κ)	eV^{-1}; 1	16.5
$\mathfrak{f}^{\mathrm{loc}}_{\mathrm{D}}$, $\mathcal{U}_{j'j}(G)$	Local-field ratio of a donor; Fourier coefficients of $u^*_{c\boldsymbol{k}_{j'}} u_{c\boldsymbol{k}_j}$	1	16.5
y, α_{pol}, Γ_{line}	Clausius–Mossotti variable $\varepsilon^{\mathrm{D}}/3\varepsilon_{\mathrm{host}}$; donor polarizability; full width of a donor line	1; cm^3; eV	16.5
$\tilde{O}$, $\tilde{O}_a$, $\tilde{O}_b$, $\tilde{n}_e$, E_{eff}, L_{eff}	Host-screened operators and density of a dopant; local field of Du et al.; loss function of the donor effective medium	—	16.5
F^s_j, c^s_j, $\tilde{k}$	Envelope and valley coefficient of donor state s in valley j; final-state momentum ka^* of the dopant source	$\text{Å}^{-3/2}$; 1; 1	16.5
Y_{CC}, p_q, f_{opt}, p_{FH}, $\langle N_{\mathrm{X}}\rangle$, p_{esc}	Color-center yield $p_{\mathrm{FP}}p_{\mathrm{surv}}p_q f_{\mathrm{opt}}$; charge-state factor; readout efficiency; F–H conversion probability of a self-trapped exciton; mean number of excitons per deposit; cage-escape probability of bond fragments	1	16.6
M_1, M_2, M_{mol}, μ_{12}, $\boldsymbol{q}_{12}$, $\boldsymbol{p}_{12}$	Masses of the atoms of a diatomic molecule and of the molecule; reduced mass of the bond; kick $(\mu_{12}/M_1)\boldsymbol{q}$ on the bond vector; relative momentum	eV; eV; eV	16.6
E_{bond}, ω_{vib}, E_{int}	Bond (dissociation) energy; vibration quantum of the bond; energy shared by the fragments above the dissociation limit	eV	16.6
S_{CM}, S_{int}, S_{11}	Center-of-mass, internal and struck-atom self responses of a molecule	eV^{-1}	16.6
k_L, ϵ_L, $g_L(\epsilon_L)$, y_L, T_{d}	Lindhard constant, reduced energy, its function, electronic share and damage energy (T_{d} in Sects. 16.6, 19.3 and 19.4)	1; 1; 1; 1; eV	16.6
$\mathcal{R}_{\mathrm{CC}}$, M_{F}, M_{Li}, N_{X}, $\mu_{\chi 1}$, $\mu_{\chi\kappa}$, H_{CM}, $\boldsymbol{R}_{\mathrm{CM}}$, P_{CM}	Color-center rate; fluorine and lithium masses; number of excitons; reduced masses with atom 1 and with nucleus κ; center-of-mass Hamiltonian, position and momentum of a molecule	—	16.6
$\tan\delta_0$	Low-power loss tangent of a glass	1	16.7
ε_r	Relative permittivity of a glass (dark-photon absorption on two-level systems)	1	16.7
$\sigma_{\chi N}$	Dark-matter–nucleus cross-section in the random-medium Born criterion	cm^2	16.7
$\bar{P}$	Density of two-level systems per unit asymmetry $\Delta_{\mathrm{as}} \geq 0$, per unit tunneling exponent λ_{tun} and per unit volume	$\text{eV}^{-1}\,\text{cm}^{-3}$	16.7

Symbol	Meaning	Units	Sect.
C^{IR}, A^{IR}, B^{IR}	Glass infrared coupling $C^{\mathrm{IR}} = A^{\mathrm{IR}} + B^{\mathrm{IR}}\omega^2$	—	16.7
σ_{crit}, n_0	Critical cross-section for Born validity in a random medium; mean nucleon number density (the coherent A is in the density; $b_{\chi N}$ is per nucleon)	cm^2; cm^{-3}	16.7
γ_{TLS}, γ_{l}, γ_{t}, $\boldsymbol{p}_{\mathrm{TLS}}$	Deformation potential of a two-level system ($\delta\Delta_{\mathrm{as}} = 2\gamma_{\mathrm{TLS}} u_{\mathrm{d}}$); its orientation-averaged longitudinal and transverse values; half the dipole difference of the two configurations	eV; eV; e Å	16.7
u_{d}	Component of the strain tensor $u_{\alpha\beta}$ along the deformation of a defect	1	16.7
C, C_j, C_{l}, C_{t}	Tunneling strength $\bar{P}\gamma^2/\rho_T c^2$; for polarization j; longitudinal, transverse	1	16.7
K_{ph}, Γ_1, τ_1	One-phonon relaxation constant; relaxation rate of a two-level population; $\tau_1 = 1/\Gamma_1$ in the Debye response	eV^{-2}; eV; eV^{-1}	16.7
λ_{tun}, ω_{w}, d_{tun}, M_{tun}	Tunneling exponent ($\Delta_0 = \omega_{\mathrm{w}} \mathrm{e}^{-\lambda_{\mathrm{tun}}}$); vibration energy in one well; distance between the wells; tunneling mass	1; eV; Å; eV	16.7
f_{TLS}, f_{host}, f_{core}, $q_\times$	Density difference of the two configurations; its elastic (host) and rigid-core parts; momentum at which they are equal	1; 1; 1; eV	16.7
$S_\nu^Z(k)$, $\hat{w}_\nu$, α_{abs}, ω_{IoR}	Born-charge structure factor of branch ν; fraction of modes in branch ν; absorption coefficient (α is the fine-structure constant); Ioffe–Regel frequency (IR is reserved for the infrared C^{IR})	1; 1; cm^{-1}; eV	16.7
V_{opt}, $b_{\chi N}$, ξ_{c}, E_{kin}, E_n, R_{refl}	Optical potential of a medium; dark-matter–nucleon scattering length; correlation length of density fluctuations; kinetic energy; kinetic energy of motion normal to a face; reflectance (R is the body radius)	eV; eV^{-1}; nm; eV; eV; 1	16.7
V_{b}, V_{d}, Ω	Barrier height of a tunneling defect; half the difference of the contact potentials of the two configurations; contact volume	eV; eV; $\mathrm{Å}^3$	16.7
H_{TLS}, τ_x, τ_z, n_{LL}, χ_{res}, χ_{rel}, $\tilde{\omega}$, $\hat{e}_{\mathrm{t}}$, v_{t}, F_{fluc}, σ_{tot}	Two-level Hamiltonian; Pauli matrices of its eigenstates; projected density in configuration L; resonant and relaxation susceptibilities; wavenumber $\omega/2\pi c$ in cm^{-1}; transverse polarization; transverse sound speed (Sahling's $v_{\mathrm{t}} = c_{\mathrm{TA}}$); fluctuation form factor; cross-section of a whole body	—	16.7
p_ν, $g(\omega)$, δn	Dipole of glass mode ν; vibrational density of states normalized to one; fluctuation $\delta n(\boldsymbol{r})$ of the nucleon density of a random medium	$e/(mass)^{1/2}$; eV^{-1}; cm^{-3}	16.7
α_{ohm}, K_{b}	Ohmic coupling of a tunneling defect to conduction electrons, $\alpha_{\mathrm{ohm}} = (1/2\pi^2)\sum_\sigma \sum_\nu \tilde{\delta}_\nu^2$ with $\mathrm{e}^{2\mathrm{i}\tilde{\delta}_\nu}$ the eigenvalues of $S_{\mathrm{R}} S_L^\dagger$; its Born-limit coupling $K_{\mathrm{b}} = (\mathcal{N}_{\mathrm{F}} V_{\mathrm{d}} \Omega)^2$, with V_{d} half the difference of the contact potentials of the two configurations (at $\pm V_{\mathrm{d}}$), so that $\alpha_{\mathrm{ohm}} = 4K_{\mathrm{b}}$; App	1	16.8
T_{hi}, T_{lo}	Initial and final temperatures of a glass that is cooled (heat-release law)	K	16.8
$\omega^\ddagger$, κ_{PB}, κ_{W}, $T_\times$, S_{inst}, T_{frz}	Barrier frequency; parabolic-barrier and Wigner transmission factors; crossover temperature $\omega^\ddagger/2\pi$; instanton action; freezing temperature on cooling	eV; 1; 1; K; 1; K	16.8
$\dot{Q}_{\mathrm{TLS}}$, b_{br}, N_{casc}, T_{coh}	Heat released by a cooled glass; mean number of releases triggered by one; size of a cascade of releases; coherence temperature of a symmetric defect in an ohmic bath	W; 1; 1; K	16.8
Γ_0	Attempt rate of a thermally activated hop, $\Gamma = \Gamma_0 \mathrm{e}^{-V_{\mathrm{b}}/T}$	s^{-1}	16.8
$J_{\mathrm{s}}^{\mathrm{ph}}$, $J_{\mathrm{s}}^{\mathrm{el}}$	Phonon and electron spectral densities of a two-level system	eV	16.8

Symbol	Meaning	Units	Sect.
$\tilde{\Delta}_0$, W_{el}, Q_{TLS}, $\mathcal{O}$	Dressed tunneling amplitude of Qureshi; electron coupling of Meißner et al.; heat of a cooled glass ($\dot{Q}_{\mathrm{TLS}}$); order symbol	—	16.8
γ_0	Electron damping $2\pi\alpha_{\mathrm{ohm}}T$ of Qureshi	eV	16.8
Chapter 17: Comparison across channels			
$\omega_{\mathrm{th}}^{\times}$	Crossover threshold at which the lattice and Migdal rates above threshold are equal	eV	17.1
σ_p	Per-nucleon cross-section of a coupling equal on protons and neutrons, the σ_p	cm^2	17.1
$\bar{N}_{\mathrm{sig}}$, $\bar{N}_{\mathrm{th}}$, $\bar{N}_{\mathrm{sig}}^{\mathrm{lin}}$, $\bar{N}_{\mathrm{sig}}^{\mathrm{cnt}}$, B, Q_{c}, Q_a	Mean absorbed signal quanta in a record; thermal quanta in the N_{coh} bins; signal quanta needed by a linear and by a counting readout at equal significance; detection band of a counter; quality factors of a cavity and of the line	1; 1; 1; 1; eV; 1	17.2
$\mathcal{Y}_\nu(\boldsymbol{q})$, $\mathcal{Y}$	Mode weight of branch ν : $\sum_{ab}\mathfrak{D}_{ab}S_{ab} = (2\pi/\Omega)\sum_\nu \mathcal{Y}_\nu\delta(\omega-\omega_\nu(\boldsymbol{q}))$; $\mathcal{Y}$ with the label dropped for one branch (magnon, plasmon)	eV	17.3
q_ν, v_ν	Root of $\lvert\Delta_\chi\rvert - q^2/2m_\chi = \omega_\nu(q\hat{\boldsymbol{q}})$ along $\hat{\boldsymbol{q}}$; q_ν/m_χ	eV; 1	17.3
m_ν^*, μ_ν; m_{pl}^*, μ_{pl}	Dispersion mass of a parabolic branch, $\omega_\nu(0)+q^2/2m_\nu^*$, and $m_\chi m_\nu^*/(m_\chi+m_\nu^*)$; the same for the plasmon, $m_{\mathrm{pl}}^* = m_e/2\alpha_{\mathrm{pl}}$	eV	17.3
F_κ^{fol}	Charge of the electrons that follow nucleus κ at long wavelength, $\hat{\boldsymbol{q}}\cdot(Z_\kappa\mathbf{1}-Z_\kappa^*)\cdot\hat{\boldsymbol{q}}$($F_\kappa$ stays the Hellmann–Feynman force)	1	17.4
$M_{\kappa,c\nu\nu}^{\mathrm{Mig},\pm}$, $S_{e\kappa}^+$, $S_{\kappa\kappa'}^+$, $\boldsymbol{p}_{\boldsymbol{G}}$, $\boldsymbol{K}$	Dimensionless Migdal amplitude into a one-phonon pair state; Migdal–phonon-assisted cross response and one-phonon Migdal response (joint, two labels); momentum $\boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G}$ given to the electrons; momentum the lattice keeps	1; eV^{-1}; eV^{-1}; eV; eV	17.4
$\tilde{V}_{\mathrm{nuc}}$, $\bar{z}$	Common row potential of every nucleus; mean $\sum_\kappa w_\kappa Z_\kappa/M_\kappa$ of the sublattices with weights summing to one, for a dark photon $\lambda = -\bar{z}\tilde{V}$ (per u in numbers)	eV^{-2}; u^{-1}	17.4
W_{AB}	Pairwise mass window of two materials A, B: the contiguous masses with $\lvert\delta_{AB}\rvert < 0.1$	MeV	17.5
c_{NR1}	Effective sound speed of the one-phonon nuclear-recoil rate, from the structure-factor coefficient	$\mathrm{km\,s}^{-1}$	17.5
F_A, $G(m_\chi)$, δ_{AB}	Rate figure of merit of material A; its material-independent factor; pairwise correction of a ratio of figures of merit	—; —; 1	17.5
$\varepsilon_{\mathrm{val}}$, $\varepsilon_{\mathrm{core}}$, ε_{m}, ω_{core}, Z_{val}, $\boldsymbol{n}_{\mathrm{val}}$, ω_c	Valence and core parts of ε (each one plus its polarization); model function $\varepsilon_{\mathrm{core}}(0)+\varepsilon_{\mathrm{val}}-1$; first core edge; valence electrons per cell; valence electron density, $(\omega_p^{\mathrm{val}})^2 = e^2\boldsymbol{n}_{\mathrm{val}}/m_e$; upper limit of a partial f-sum	1; 1; 1; eV; 1; cm^{-3}; eV	17.5
I_1, u_-, m_{hi}	First moment $\int v\eta\mathrm{d}v$ of the halo; lower cut $\omega_{\mathrm{th}}/(2m_\chi c_\nu)$ of the one-phonon halo integral; upper end $q_{\mathrm{BZ}}/2v_{\mathrm{max}}$ of the heavy-mediator window	1; 1; eV	17.5
F_{Mig}, X_{pole}	Migdal figure of merit $\sum_\kappa(N_\kappa/A_{\mathrm{cell}})\int L_{\mathrm{Mig},\kappa}\omega^{-4}\mathrm{d}\omega$; interband enhancement of its pole form	eV^{-3}; 1	17.5
$P(first)$, r_{mat}	Probability of ranking first; correlation of method errors between materials	1	17.6
σ_j^{prov}, N_{MC}, $R_A^{(k)}$, R_A^{16}, R_A^{84}, $h(A)$, z_{method}, z_j'	Provenance width of input j; Monte Carlo draws; rank of A in draw k; 16th and 84th percentiles of the ranks (rank interval); height in the interval order (tier); shared and own standard normal variables of a computed input	1	17.6

Symbol	Meaning	Units	Sect.
Chapter 18: Ionization yield			
θ_{ii}	Impact-ionization threshold in units of E_{g}; $(2m_{\mathrm{e}}^{*}+m_{\mathrm{h}}^{*})/(m_{\mathrm{e}}^{*}+m_{\mathrm{h}}^{*})$ for parabolic bands, 1 in the random-$\boldsymbol{k}$ reduction	1	18.1
$\mathscr{K}_c$, $\bar{\mathscr{K}}_t$, $\delta\boldsymbol{N}_c$, $\boldsymbol{N}_{\mathrm{term}}(x)$, Γ_x, Γ_k, Γ_E	Rate density of collision channel c of a cascade particle; its iso-energy average over the states of type t at one kinetic energy; deposit vector of channel c; deposit of a terminal state; total rate out of state x; momentum-randomizing rate and energy-changing rate $\Gamma_{\mathrm{ii}}+\Gamma_{\mathrm{ph}}$ of the energy-only reduction	eV; eV; —; —; eV; eV; eV	18.1
$\mathfrak{G}$, $\mathfrak{G}_x(\boldsymbol{\lambda})$, $\mathfrak{G}_\Lambda$, $\mathfrak{G}_{\mathrm{e}}$, $\mathfrak{G}_{\mathrm{h}}$; λ_N, λ_H	Branching generating functional $\langle \mathrm{e}^{\boldsymbol{\lambda}\cdot\boldsymbol{N}}\vert x\rangle$ of the cascade started in state x; that of a seed class, the product of a prompt pair's factor and the electron's and hole's functionals $\mathfrak{G}_{\mathrm{e}},\mathfrak{G}_{\mathrm{h}}$; conjugate variables of the pair number and the heat ($\boldsymbol{\lambda}=(\lambda_N,\lambda_H,\dots)$)	1	18.1
$\rho_t(x)$, $\rho_{\bar{t}}$; t, $\bar{t}$	Density of states of a carrier of type t at kinetic energy x, and of the other type (random-$\boldsymbol{k}$ kernels; ρ_χ,ρ_T are mass densities, ρ_{mn} the density vertex); carrier type e or h, and the other type	$\mathrm{eV}^{1/2}$ (free form)	18.1
$\boldsymbol{p}$	Momentum a cascade carrier gives to the target in one collision	eV	18.1
x_{th}, N_{sub}, $\bar{n}$, $\bar{n}_{\mathrm{sub}}$	Impact-ionization threshold energy $\theta_{\mathrm{ii}}E_{\mathrm{g}}$; number of ionizations in the subtree of a lone carrier; mean pair number $\langle N_{\mathrm{eh}}\rangle$; mean of N_{sub}	eV; 1; 1; 1	18.1
E_{heat}, $E_<$, $E_>$	Heat of a cascade ($E_{\mathrm{heat}}=E_<+E_>$): kinetic energy of the carriers when they fall below threshold for good, summed; phonon energy emitted above threshold	eV	18.1
$\mathfrak{P}_t(y)$	Three-body density of final states $\rho_t*\rho_t*\rho_{\bar{t}}$ of an ionization with $y=x-E_{\mathrm{g}}$ above the gap; $(2\pi/105)y^{7/2}$ for $\rho\propto x^{1/2}$	$\mathrm{eV}^{7/2}$	18.1
F_{eh}	Fano factor of the electron–hole pair number, Var $N_{\mathrm{eh}}/\langle N_{\mathrm{eh}}\rangle$ at large ω; Si model 0.114 ($A_{\mathrm{AB}}=5.703\ \mathrm{eV}^3$), 2.4σ below the measured 0.119 ± 0.002	1	18.2
ρ_∞	Jamming density of the random-parking (Rényi) limit, 0.74759792	1	18.2
$\mathcal{V}_<$, $\mathcal{V}_>$, $\mathcal{V}_{<>}$	Variance terms of F_{eh}	1	18.2
λ_{AB}, w_{AB}	Phonon strength $A_{\mathrm{AB}}/E_{\mathrm{g}}^3$ of the random-k cascade; phonon step $\omega_{\mathrm{AB}}/E_{\mathrm{g}}$	1	18.2
B_{h}	Hole band width at which the hole density of states is cut	eV	18.2
b_N, c_N	Constant terms of the mean and variance of the pair number, $\bar{n}=\omega/\varepsilon_{\mathrm{eh}}+b_N+o(1)$, Var $N_{\mathrm{eh}}=F_{\mathrm{eh}}\omega/\varepsilon_{\mathrm{eh}}+c_N+o(1)$	1	18.2
$M_t(x)$	Second moment of an additive deposit of the subtree of a carrier of type t at kinetic energy x	as *deposit*^{2}	18.2
σ_{pair}, σ_{e}, σ_{h}, F_{ren}, F_{park}	Standard deviation of the cost per pair in the renewal estimate; standard deviations of the residual energies of terminal electrons and holes; renewal (final-state) Fano estimate; random-parking Fano factor 0.0510387	eV; eV; eV; 1; 1	18.2
$P_{\mathrm{seed}}(E_{\mathrm{e}},E_{\mathrm{h}}\vert\boldsymbol{q},\omega,\boldsymbol{v})$, f_{e}	Seed distribution of the carrier energies handed to the cascade (decorated P); electron share $E_{\mathrm{e}}/(\omega-E_{\mathrm{g}})$ of the energy above the gap (decorated f)	eV^{-2}; 1	18.3
$\mathfrak{G}_t(x;s)$, $m_t(x)$, $v_t(x)$	Pair-number generating function of the subtree of a carrier of type t at kinetic energy x; mean and variance of its number of ionizations	1	18.3
α_{RK}, $N_{\mathrm{eh}}^{\mathrm{step}}$	Beta parameter of the Ramanathan–Kurinsky split distribution; pair number of the step model $1+\lfloor(\omega-E_{\mathrm{g}})/\varepsilon_{eh}\rfloor$	1	18.3

Symbol	Meaning	Units	Sect.
$d_{\mathrm{TV}}(P,Q)$	Total-variation distance $\frac{1}{2}\sum_N \lvert P(N)-Q(N)\rvert$	1	18.3
$\mathfrak{m}$, R, $P_{\mathrm{det}}(\mathsf{R}\lvert N_{\mathrm{eh}},\mathfrak{m})$, $P_{\mathrm{obs}}(\mathsf{R}\lvert\Lambda)$	Further quantities recorded by a cascade besides N_{eh}; readout record; detector response; distribution of the recorded quantity (decorated P)	—; —; 1; 1	18.3
$p_{\mathrm{ii}}(E)$	Probability that the next collision of a carrier at kinetic energy E is an impact ionization, $[1+\Gamma_{\mathrm{ph}}/\Gamma_{\mathrm{ii}}]^{-1}$	1	18.3
k_2; r_K, ω_K	Constant of the observed linear law $\varepsilon_{\mathrm{eh}}\approx 1.61E_{\mathrm{g}}+k_2$; phonon-loss coefficient and phonon energy of the Klein relation $(14/5)E_{\mathrm{g}}+r_K\omega_K$	eV; 1, eV	18.4
k_1	Slope of a straight-line fit $\varepsilon_{\mathrm{eh}}=k_1E_{\mathrm{g}}+k_2$ to measured pair energies	1	18.4
$\bar{E}_{\mathrm{res}}$	Mean residual kinetic energy per terminal carrier of the cascade, Shockley's E_f	eV	18.4
E_{np}, E_j, E_v, μ_F	Energy not held by pairs $\omega-N_{\mathrm{eh}}E_{\mathrm{g}}$; level of an impurity of species j; valence-band edge; Fermi level of the electrodes	eV	18.5
f_{cap}, f_i, N', $\varepsilon'_{\mathrm{eh}}$, F'_{eh}	Capture probability of an electron (common; f_i of electron i); number of electrons left after capture; pair energy and Fano factor of the counted pairs	1; 1; 1; eV; 1	18.5
$r_C(E)$, $r_{\mathcal{E}}$	Coulomb capture radius $\alpha/(\varepsilon_0E)$ of a carrier of kinetic energy E; field separation radius $(\alpha/\varepsilon_0e\mathcal{E})^{1/2}$	nm; μm	18.5
N_X, F_{heat}, p_{ns}	Number of bound pairs (excitons) at readout (in Sect. 18.5); Fano factor inferred from the phonon energy; probability that a pair ends as an exciton that is never split	1	18.5
p_F, p_X, $p_{B,j}$, M	Occupation probabilities of one pair as free pair, free exciton, exciton bound to impurity species j; matrix of the linear rate equations	1; s^{-1}	18.5
Γ_{FX}, Γ_{FT}, Γ_X, $\Gamma_{XB,j}$, $\Gamma_{B,j}$, Γ_j^{em}	Rates of pair binding, carrier trapping, free-exciton decay, binding of a free exciton to species j, bound-exciton decay, emission from a trap	s^{-1}	18.5
η_X, $\eta_{B,j}$, $\sigma_{XB,j}$, v_X, $\sigma_{\mathrm{cap},j}$, σ_j^{rec}, u, n_j^{tr}, n_{c}, n_{v}	Luminescent fractions of free and bound exciton decay; exciton binding cross-section and exciton speed; capture and recombination cross-sections of a center; carrier speed; trapped-carrier density; free-electron and free-hole densities	1; cm^2; $\mathrm{cm\,s}^{-1}$; cm^{-3}	18.5
$r_{\mathrm{gap}}(t)$, w_k, γ_k, b_{trap}, b_{light}, b_{heat}	Fraction of the gap energy still held by pairs and excitons; weights and decay constants of its exponentials; fractions of the gap energy that end trapped, as escaping light and as phonons	1; 1; s^{-1}; 1	18.5
F_{meas}, f_{det}	Fano factor of the measured charge; detected fraction of pairs	1	18.5
Chapter 19: Low-energy nuclear recoils			
E_{st}	Energy stored in defects (configurational)	eV	19.1
t_{pr}, E_{lat}, $\mathcal{W}_0$, $\mathcal{W}_+$, $\hat{e}_I$	Partition time (10–100 ps); lattice share, vibrational energy above the zero-point energy of the perfect crystal; Wigner function of the lattice before and after the kick; unit marker of the momentum of nucleus I	ps; eV; —; 1	19.1
ℓ_R, η_R, λ_{dB}, ℓ_{int}, ξ_{M}	Cage amplitude $(2E_R/M_\kappa\langle\omega\rangle^2)^{1/2}$; recoil anharmonic parameter $\lvert\Phi^{(3)}\rvert\ell_R/3\Phi^{(2)}$; de Broglie wavelength of the struck nucleus; interaction length of a close collision; Massey parameter $E_{\mathrm{g}}\ell_{\mathrm{int}}/v_N$	Å; 1; Å; Å; 1	19.1
H_{cl}, $\boldsymbol{e}_{\kappa s}$, $T_{e,i}$, $n(E)$	Classical (Weyl) function of the nuclear Hamiltonian $\sum p^2/2M_\kappa+E^{\mathrm{ad}}$; polarization vector of normal mode s on species κ; energy given to electron i in one collision; mean number of further displacements made by a moving atom of energy E, $N_{\mathrm{FP}}=1+n(T_{\mathrm{d}})$	eV; 1; eV; 1	19.1

Symbol	Meaning	Units	Sect.
$E_{R,\mathrm{th}}$	Binary electron–hole threshold of a nuclear recoil	eV	19.2
$Y_q(E_R)$	Ionization yield of a nuclear recoil relative to an electron recoil of the same energy	1	19.2
$\omega_{\max}$, $v_{N,\mathrm{th}}$, $p_{\min}$	Largest energy a slow ion gives one electron, $2m_e v_N(v_N + v_\mathrm{F})$ (in Sect. 19.2); threshold speed $E_\mathrm{g}/2m_e v_\mathrm{F}$; smallest valence momentum that can be promoted	eV; 1; eV	19.2
h_LZ, ΔF_LZ, P_LZ, b_imp, E_coll, $\mathcal{N}_\mathrm{exc}$, ΔY_q^Mig	Diabatic coupling at a level crossing; difference of the diabatic slopes; Landau–Zener probability; impact parameter and relative energy of a close collision; number of electronic excitations of a recoil; Migdal part of the ionization yield	eV; eV Å^{-1}; 1; Å; eV; 1; 1	19.2
E_el, F_NR	Electronic energy of a recoil (in Chap. 19); Fano factor of the nuclear-recoil charge	eV; 1	19.2
$\bar{I}$	Mean excitation energy of the Bethe formula, $\ln \bar{I} = (2/\pi\omega_p^2)\int \omega L(0,\omega)\ln\omega \mathrm{d}\omega$	eV	19.3
a_L, c_L, ϕ_TF, T_n, x_L, f_LSS, s_n, ρ_L, λ_L	Lindhard screening length $0.8853a_0/(\sqrt{2}Z_\kappa^{1/3})$; reduced-energy constant ($\epsilon_L = c_L E$) (in Sect. 19.3); Thomas–Fermi screening function; energy transfer to the struck atom; reduced transfer variable $\epsilon_L \sin(\theta_\mathrm{s}/2)$; Lindhard's scattering function; reduced nuclear stopping; reduced path length $\pi a_L^2 n_\mathrm{at} x$; coefficient of the $x_L^{1/3}$ continuation	Å; keV^{-1}; 1; eV; 1; 1; 1; 1; 1	19.3
ϵ_d, $T_\mathrm{d}^{(e)}$, U, U_FP, U_EDFT, W_max	Reduced damage energy $c_L T_\mathrm{d}$; damage energy produced by an excited electron; binding energy spent by a struck atom (Sarkis models) (in Sect. 19.3); its constant part, called Frenkel-pair creation energy by its authors; its inner-shell part; largest energy transfer to a free electron (PDG)	1; eV; eV; eV; eV; eV	19.3
f_DF, p_surv	Defect fraction $1 - Y_\mathrm{ph}$ of the deposit; survival probability of close Frenkel pairs	1	19.4
P_b, $E_{\mathrm{st},b}$, E_FP, N_FP, ξ_d, f_sur	Probability that a recoil ends in basin b, and that basin's minimum above the perfect crystal; mean stored energy of a surviving displacement; number of displaced atoms; displacement-count factor of the screened cross-section; scale factor of surface electron recoils (Ge datum)	1; eV; eV; 1; 1; 1	19.4
Y_ph	Phonon-yield ratio: ER-calibrated reconstructed energy of a nuclear recoil over its true energy; $Y_\mathrm{ph}(0)$, its zero-bias value, is the heat quenching	1	19.4
E_dep, E_pairs, E_prompt, E_esc, E_sens	Deposited energy; energy held by carriers counted from μ_F (free, excitons, trapped, plus the electrostatic energy of uncollected charge); prompt phonon energy $E_\mathrm{phon}(t_\mathrm{pr})$; energy carried out of the crystal by unabsorbed quanta (luminescence); energy absorbed at the surfaces	eV	19.5
$E_\mathrm{phon}(t)$, $\Delta E_\mathrm{zp}(t)$, $W_\mathrm{bias}(t)$, N_ind	Phonon energy above the bath (contains ΔE_zp); zero-point shift of the occupied defect basins; work of the bias source; induced charge in units of $e, W_\mathrm{bias} = eV_\mathrm{bias}N_\mathrm{ind}$	eV; eV; eV; 1	19.5
t_win, E_miss, d_NR, d_ER	Read window of a phonon sensor; energy not available as phonons by t_win, $E_\mathrm{pairs} + E_\mathrm{st} + E_\mathrm{esc} + \Delta E_\mathrm{zp}$; E_miss/ω for nuclear and electron recoils	s; eV; 1	19.5
$\mathfrak{r}_\mathrm{coll}$, δ_coll, $\mathfrak{r}'$	Collection ratio: collection efficiency of Luke phonons over that of the primary phonons of an electron recoil, $\epsilon_\mathrm{Luke}/\epsilon_\mathrm{prim}^\mathrm{ER}$; collection offset $1 - \epsilon_\mathrm{prim}^\mathrm{NR}/\epsilon_\mathrm{prim}^\mathrm{ER}$; $\mathfrak{r}' = \mathfrak{r}_\mathrm{coll}/(1 - d_\mathrm{ER})$	1	19.5

Symbol	Meaning	Units	Sect.
ϵ, $\epsilon^{\rm NR}_{\rm prim}$, $\epsilon^{\rm ER}_{\rm prim}$, $\epsilon_{\rm Luke}$, $\epsilon_{\rm hard}$, ρ_g	Absolute phonon collection efficiencies of all phonons, of the primary phonons of the two recoils, of Luke phonons, of hard primaries; ratio of the efficiencies for gap (recombination) phonons and hard primaries (ϵ in Sect. 19.5)	1	19.5
$C_{\rm cal}$	Calibration coefficient $(1+\mathfrak{r}_{\rm coll}g_{\rm Luke}f_{\rm ion})/(1+\mathfrak{r}_{\rm coll}g_{\rm Luke})$, the weight of the ER deficit	1	19.5
$g_{\rm Luke}$, $g_{\rm cal}$, $\varepsilon_{\rm cal}$	Luke gain $eV_{\rm bias}/\varepsilon_{\rm eh}$; gain $eV_{\rm bias}/\varepsilon_{\rm cal}$ that a reconstruction assumes; pair energy adopted by a reconstruction	1; 1; eV	19.5
$f_{\rm ion}$	Ionized fraction: bias work (induced charge) of a nuclear recoil over that of an electron recoil of the same energy, $N^{\rm NR}_{\rm ind}/N^{\rm ER}_{\rm ind}$	1	19.5
$f_{\rm ret}(t)$, $f^{\rm NR}_{\rm ret}$, $f^{\rm ER}_{\rm ret}$, $x_{\rm NR}$, $x_{\rm ER}$, $e_{\rm st}$, Δ_2	Fraction of the pair gap energy returned as phonons by t; its values for the two recoils; missing energy other than stored, per ω; stored fraction $E_{\rm st}/\omega$; determinant of the two-bias system	1	19.5
$S_{\rm ER}$, $S_{\rm NR}$, E_ϕ, E_Q, $k_{\rm cal}$, χ_g, $\epsilon_{\rm abs}$	Phonon signals of electron and nuclear recoils; calibrated phonon energy and electron-equivalent charge energy of a reconstruction; calibration factor of the phonon scale; gap-fraction multiplier; absolute phonon collection efficiency	1; eV; eV; $\rm eV^{-1}$; 1; 1	19.5
$E_{\rm es}$, $N_{\rm h}$, N_e, $G(r,r')$, $\sigma_{n,e}$	Electrostatic energy of the charges in the crystal with images; numbers of uncollected holes and electrons; Green's function of the Poisson equation vanishing on the electrodes; cross-sections of nuclear and electronic collisions	eV; 1; 1; $\rm Å^{-1}$; $\rm Å^2$	19.5
y_C, a_C	Yield parameterization $[1/(a_C E_R)+1/y_L]^{-1}$ of the CDMS II silicon analysis; its constant 0.247 $\rm keV^{-1}$	1; $\rm keV^{-1}$	19.5
$M_{\rm cr}$	Crowdion kink mass	eV	19.6
$E^{\rm conf}_i$, $\varepsilon^{\rm hop}_{ij}$, $\Delta E_{{\rm zp},ij}$, $\mathcal{R}_{\rm hop}$, $N_{\rm hop}$, $p_i(t)$	Configurational energy of basin i; configurational energy released in a hop $i\to j$; decrease of the zero-point energy in the hop; hop rate per unit time and released energy; number of hops; occupation of configuration i	eV; eV; eV; $\rm eV^{-1}\,s^{-1}$; 1; 1	19.6
$A(\varepsilon)$, $E_{\rm reorg}$, $\omega_{\rm acc}$, n_q, $t_{\rm obs}$	Line shape of the bath; reorganization energy $\sum_k S_{{\rm HR},k}\omega_k$; energy of the accepting mode; number of accepted quanta; observation time	$\rm eV^{-1}$; eV; eV; 1; s	19.6
$A_{\rm dfn}$, $\Delta_{\rm dfn}$, Γ^2_k, $\alpha_{\rm P}$, $\sigma_{\rm P}$, θ_p, $\bar c$, R_0, D_9, $\tau_{\rm tr}$	Defecton hopping amplitude and band width $zA_{\rm dfn}$; rate of growth of the mean squared crystal momentum; Pushkarov's constant 1.27; elastic strength relative to a vacancy; Pushkarov's temperature $\bar c/2a$; his mean sound speed; localization radius $a(3V_0/\Delta_{\rm dfn})^{1/4}$; T^{-9} limit of D; transport time	eV; eV; $\rm Å^{-2}\,s^{-1}$; 1; 1; K; $\rm km\,s^{-1}$; Å; $\rm cm^2\,s^{-1}$; s	19.6
$\mathcal{R}_{\rm TLS}$, $\mathcal{R}_{\rm LEE}$, $p_{\rm LEE}$, $U_{\rm rel}$, $E_{{\rm st},0}$, $f_{\rm ph}$, $E_{\rm slope}$	Phonon emission rate of two-level systems; rate of excess events; law of single-event energies; relaxable elastic energy; stored energy of grown-in or frozen-in defects; collected fraction of a release; mean release per hop	$\rm s^{-1}$; $\rm s^{-1}$; $\rm eV^{-1}$; eV; eV; 1; eV	19.6
$Y_{\rm mod}$, $\sigma_{\rm str}$, $p_{\rm rxn}$, $S_{{\rm HR},{\rm rxn}}$, $\omega_{\rm rxn}$, $S_{\rm pol}$	Young's modulus; thermal-mismatch stress; probability that a capture drives a defect reaction; Huang–Rhys factor and energy of the reaction coordinate; polaron exponent	Pa; Pa; 1; 1; eV; 1	19.6
$\varepsilon^{\rm dfn}(\boldsymbol{k})$, $E^{\rm loc}$, D, D_1, $I_0(y)$, Γ_{ij}, V_0	Defecton band $E^{\rm loc}-A_{\rm dfn}\sum\cos k_i a$; energy of the localized defect; defecton diffusion constant; diffusion rate of the crystal momentum per Cartesian component, $\Gamma^2_k/6$; modified Bessel function of order zero; rate of a hop $i\to j$; elastic interaction constant of a point defect, $\phi=V_0(a/r)^3$	eV; eV; $\rm cm^2\,s^{-1}$; $\rm Å^{-2}\,s^{-1}$; 1; $\rm s^{-1}$; eV	19.6

Symbol	Meaning	Units	Sect.
Chapter 20: Charge, phonon and quasiparticle readout			
λ_{d}	Carrier drift length of the Hecht relation	mm	20.1
d_{det}, s_e, s_h, $\boldsymbol{r}_{\mathrm{start}}$, $\boldsymbol{r}_{\mathrm{end}}$	Detector thickness between planar electrodes; drift paths of the electron and the hole of a pair; start and end points of a carrier trajectory	mm; mm; —	20.1
$\mathfrak{q}$, $\mathfrak{q}_i$, $\bar{\mathfrak{q}}(z)$, $\sigma^2_{\mathfrak{q}}$, $\bar{N}_{\mathrm{eh}}$	Induced charge of a pair in units of e; of pair i; its mean for a pair born at depth z; its variance at fixed position; mean pair number of a deposit	1	20.1
v_{dr}, v_{sat}, μ_{mob}	Drift velocity; carrier mobility	$\mathrm{m\,s^{-1}}$; $\mathrm{m\,s^{-1}}$; $\mathrm{m^2\,V^{-1}\,s^{-1}}$	20.1
n_{sh}, ϱ_{tr}, ξ_{SC}, p_{tr}	Trapped charges per unit area (sheet); net trapped charge per unit volume; relative field distortion $\max\lvert\Delta\mathcal{E}\rvert/\mathcal{E}$; fraction of pairs that leave a trapped charge of the dominant sign	$\mathrm{cm^{-2}}$; $\mathrm{C\,cm^{-3}}$; 1; 1	20.1
E_{ath}, $E_*(t)$	Athermal phonon energy; scaling energy of the down-converting spectrum	meV	20.2
f_{LA}	Equilibrium share of longitudinal phonons under isotope mode mixing	1	20.2
$\rho_E(E_{\mathrm{ath}}, t)$, $P_{\mathrm{d}}(y)$, $\phi_*(y)$	Spectral energy density of the athermal phonons (energy per unit E_{ath}); energy-share distribution of the daughters of one decay, $\int P_{\mathrm{d}}\mathrm{d}y = 1$; scaling function	1; 1; 1	20.2
v_3	Direction- and branch-dependent part of the three-phonon vertex of acoustic phonons, $V^{(3)} = (\omega_1\omega_2\omega_3)^{1/2} v_3$	$\mathrm{eV^{-1/2}}$	20.2
f_{s}	Soft fraction of the prompt phonon energy of an electron recoil, $\approx \omega_{\mathrm{opt}}/\varepsilon_{\mathrm{eh}}$	1	20.2
N_{mix}, v_{ph}, r_{rms}	Number of isotope scatterings along the scaling trajectory; rms group speed of the mixed branches; rms distance of quasi-diffuse phonons from the event	1; $\mathrm{km\,s^{-1}}$; μm	20.2
$\Delta_{\mathrm{h}}(t)$, Δ_{soft}, E_{mem}, t_{mem}	Largest total-variation distance between the evolved spectra of births above E_{mem}; difference of the energy fractions born below E_{mem}; energy above which births are forgotten by t_{mem}; time before which nothing is absorbed	1; 1; meV; ns	20.2
$R_O(E_{\mathrm{ath}})$, $\hat{\rho}_0$, $\Delta\hat{\rho}_0$	Response of a linear observable O to unit energy born at $E_{\mathrm{ath}}, 0 \le R_O \le 1$; birth spectrum normalized to unit energy; NR minus ER birth spectra	1; $\mathrm{meV^{-1}}$; $\mathrm{meV^{-1}}$	20.2
f_{top}, g_{Debye}, $p(y)$	Band-top frequency of the acoustic phonons; Debye density of states; number density of the energy fraction of one daughter of a decay, $P_{\mathrm{d}}(y) = 2yp(y)$	THz; per energy; 1	20.2
$\bar{t}_{12}$, $\bar{t}_{\mathrm{esc}}$	Angle-averaged transmission from medium 1 to 2; film escape probability	1	20.3
t_l, t_0, d_{f}	Film escape time; pair-breaking time of a phonon at $2\Delta_{\mathrm{BCS}}$; film thickness	ns; nm	20.3
$\mathfrak{n}_1$, $\mathfrak{n}_2$	Interface-weighted phonon densities of the two sides in the reciprocity $\mathfrak{n}_1\bar{t}_{12} = \mathfrak{n}_2\bar{t}_{21}$	—	20.3
v_s	Mean phonon group speed in a film	$\mathrm{km\,s^{-1}}$	20.3
P_{abs}	Absorption probability of a phonon in a film over its thickness	1	20.3
Q_z, $Q_{z,i}$, $Q_{z,\mathrm{inc}}$, $Q_{z,j}$, $c_{\mathrm{T},i}$, $c_{\mathrm{T,Si}}$	Wavevector component normal to an interface in medium i, of the incident wave, of outgoing wave j; transverse sound speed of medium i, of silicon	$\mathrm{m^{-1}}$; $\mathrm{km\,s^{-1}}$	20.3
a_{rL}, a_{rS}, a_{tL}, a_{tS}, $\boldsymbol{b}_j$, $\boldsymbol{b}_{\mathrm{rL}}$, $\boldsymbol{b}_{\mathrm{rS}}$, $\boldsymbol{b}_{\mathrm{tL}}$, $\boldsymbol{b}_{\mathrm{tS}}$	Amplitudes of the reflected (r) and transmitted (t)L and SV waves for unit incident amplitude; boundary vector of wave j	1	20.3

Symbol	Meaning	Units	Sect.
$t_{\mathrm{SH}}(\theta)$, t_j, t_{esc}, $\bar{t}_{\mathrm{Si}\to\mathrm{Al}}$, $\bar{t}_{\mathrm{Al}\to\mathrm{Si}}$, $\bar{t}_{\mathrm{f}\to\mathrm{sub}}$, $\mathfrak{n}_{\mathrm{Si}}$, $\mathfrak{n}_{\mathrm{Al}}$, $\mathfrak{n}_{\mathrm{f}}$, $\mathfrak{n}_{\mathrm{out}}$, $\mathfrak{n}_{\mathrm{in}}$	SH transmission at angle θ; flux fraction of outgoing wave j; escape probability at fixed angle; angle averages for the named directions; interface-weighted densities of the named media	1; $\mathrm{s}^2\,\mathrm{km}^{-2}$	20.3
T_{fit}, y_{f}, t_{pb}, $P_{\mathrm{abs}}(\theta)$	Fitted silicon-to-aluminum coefficient read as $\bar{t}_{\mathrm{Si}\to\mathrm{Al}}P_{\mathrm{abs}}$; optical depth $2d_{\mathrm{f}}/\mathfrak{l}_{\mathrm{f}}$ of a film; pair-breaking time $\mathfrak{l}_{\mathrm{f}}/v_s$; absorption probability at fixed angle	1; 1; ns; 1	20.3
$Y_{\mathrm{qp}}(E_{\mathrm{ath}})$	Quasiparticle (pair-breaking) efficiency of a phonon of energy E_{ath}	1	20.4
E_{sub}, E_{leak}, $\bar{N}_{\mathrm{qp}}$, $\bar{N}_{\mathrm{ph}}$, p_{pb}, $P_{\mathrm{pb}}(E\|E_{\mathrm{ath}})$, $\Gamma_{\mathrm{qp,em}}$, b_α	Energy carried off by sub-gap phonons; by phonons above $2\Delta_{\mathrm{BCS}}$ that leave the film unabsorbed; mean number of gap-edge quasiparticles from one quasiparticle, one phonon; probability that a phonon breaks a pair before it escapes; split of the pair energy; emission rate of a hot quasiparticle; coefficient of $\alpha^2F = b_\alpha E^2$	meV; meV; 1; 1; 1; meV^{-1}; $\mathrm{s}^{-1}\,\mathrm{meV}^{-1}$; meV^{-1}	20.4
ϖ_O, ϖ_{gap}, F_{NR}	Collection contrasts of the observable and of the gap-energy channel; Fano factor of nuclear-recoil charge	1	20.5
$\mathfrak{C}_s(E_{\mathrm{ath}};\boldsymbol{r}_{\mathrm{ev}})$, $\boldsymbol{r}_{\mathrm{ev}}$, $u_X(E_{\mathrm{ath}})$, E_X, $E_{s,X}$, $E_{1,X}$, $E_{2,X}$, a_X, ϵ_X, ϕ_X, u_{Luke}, E_{Luke}, a_{prim}, a_{Luke}, a_{gap}, ϵ_{prim}, ϕ_{Luke}, ϕ_{gap}	Collection kernel of sensor s: fraction of the energy of phonons born at E_{ath} at $\boldsymbol{r}_{\mathrm{ev}}$ that ends as quasiparticles in s; event position; spectral energy density of population $X \in \{\mathrm{prim}, \mathrm{Luke}, \mathrm{gap}\}$; its energy; energy that sensor s (channel 1, 2) turns into quasiparticles; channel fraction $E_{2,X}/(E_{1,X}+E_{2,X})$; efficiency; share of X in the signal	1; —; $\mathrm{meV}^{-1}\,\mathrm{eV}$; eV; eV; 1; 1; 1	20.5
$\mathcal{N}(E_{\mathrm{ath}})$, Γ_{abs}, ℓ_{bare}, c_s, $2\Delta_s$, $\bar{v}$	Time-integrated number spectrum of the mean-field solve; absorption rate per phonon (in Sect. 20.5); loss per hit on a bare face; face share of film s (the sound speeds are $c_{\mathrm{LA}}, c_{\mathrm{T}}$); pair-breaking threshold of film s; number-weighted mean phonon speed of the crystal	$\mathrm{meV}^{-1}\,\mathrm{s}$; s^{-1}; 1; 1; meV; $\mathrm{km\,s}^{-1}$	20.5
s_{sub}, s_{ng}, κ_{sub}, $\bar{Y}_{\mathrm{prim}}$	Share of Luke energy born below $2\Delta_{\mathrm{BCS}}$; near-gap gain; weight of sub-gap phonons (0 lost, 1 absorbed); mean yield of the primaries as absorbed (0.590 in a thick aluminum film)	1	20.5
O_{obs}, $\mathcal{S}_{\mathrm{sep}}$, $\mathcal{S}_{\mathrm{HV}}$, σ_{pos}, σ_{NR}, σ_{ER}, σ_{ph}, ϖ_{soft}, Δ_{soft}	Observable of an event (channel fraction or fast-window share); separation; high-voltage ceiling; position spread of the channel fraction; spreads of O_{obs} for the two recoils; phonon baseline; contrast of soft against hard primaries; NR – ER difference of the soft energy fractions $(1-Y_q)f_{\mathrm{s}}$	1; 1; 1; 1; 1; eV; 1; 1	20.5
O_b, $E_{\mathrm{sig}}(t)$, t_1, t_2, t_{rel}	Two-window ratio $E_{\mathrm{sig}}(t_1)/E_{\mathrm{sig}}(t_2)$; signal energy collected by t; read windows; time scale of dense-track annihilation	1; eV; s; s; s	20.5
N_{det}, F_{eff}, F_{eff}^X, $\varepsilon_{\mathrm{src}}$, Y_q'	Detected pairs; effective Fano factor of the detected pair number (binomial $f_{\mathrm{det}}F + 1 - f_{\mathrm{det}}$ or mean-charge $f_{\mathrm{det}}F$); for recoil X; pair energy with which measured nuclear-recoil yields are defined; yield in pairs of $\varepsilon_{\mathrm{eh}}, Y_q\varepsilon_{\mathrm{eh}}/\varepsilon_{\mathrm{src}}$	1; 1; 1; eV; 1	20.5
Chapter 21: Exclusion limits and next targets			
$\Gamma_{N_{\mathrm{eh}}}(\boldsymbol{v})$, $\mathcal{R}_{N_{\mathrm{eh}}}[f_\chi]$, $\bar{\mathcal{R}}_{N_{\mathrm{eh}}}$	Rate kernel of the pair bin N_{eh}, the inner integrals at fixed $\boldsymbol{v}$; halo functional of the channel; its exposure average	eV; $\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	21.1
$v_{\mathrm{on}}(m_\chi)$, F_{flow}, f_{flow}, p_{flow}, v_{flow}, η_{flow}, θ_{flow}, $\mathcal{R}_{\mathrm{flow}}$, $\mathcal{R}_{\mathrm{SHM}}$, f_{SHM}	Onset speed of the kernel (smallest v_{min} over the support); debris-flow speed distribution, its galactic distribution, share of the density, galactic speed, mean inverse speed and lab angle; rates in the flow and in the standard halo; the standard-halo distribution	$\mathrm{km\,s}^{-1}$; $1/(\mathrm{km\,s}^{-1})$; 1; …; $\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$	21.1
$\delta v_{\mathrm{E}}(t)$	$\|\boldsymbol{v}_{\mathrm{E}}(t)\| - v_{\mathrm{E}}^{\mathrm{ref}}$; $\langle\delta v_{\mathrm{E}}\rangle = 1.61\ \mathrm{km\,s}^{-1}$, $\langle\delta v_{\mathrm{E}}^2\rangle = 108.3\ (\mathrm{km\,s}^{-1})^2$ over a year	$\mathrm{km\,s}^{-1}$	21.1

Symbol	Meaning	Units	Sect.
$\mathcal{A}_i(v)$, $\bar{s}_i$, a^i_ℓ, a_ℓ, $a^i_{\ell m}$, f_{u_0}, N_{tot}, L_h, δ_h	Step column of bin i; bin-weighted angle-averaged response; its cubic-harmonic and spherical-harmonic ratios (a_ℓ with the bin understood); objective $\Theta(v-u_0)/v$; $\sum_i N^{\mathrm{up}}_i$; Lipschitz constant of the dual and its grid defect	yr; eV; 1; …; 1	21.2
ξ_a	Ξ-mass of stream a of an extremal halo $\Xi^* = \sum \xi_a \delta_{\boldsymbol{w}_a}$	yr^{-1} (with $c=1$)	21.2
δ_{dec}, δ_{E}	Declination of a laboratory stream; declination of $\boldsymbol{v}_\odot$, 47.5°	rad	21.2
δ_{cert}	Defect of a dual certificate, the largest dip of $\sum \lambda_i \mathsf{k}_i - \mathsf{f}$ below zero on supp f	as f	21.2
$\varrho^{\mathrm{leak}}_i$	Leakage ratio of a final-state split: largest phonon-dressed response over bin i widened by $\omega_{\max}$, over the P_i-weighted average of the total	1	21.2
$\tilde{\eta}_{\max}(u_0)$, N^{up}	Largest $\tilde{\eta}(u_0)$ allowed by a search over all halos; classical upper limit on the signal count at confidence CL	as $\tilde{\eta}$; 1	21.2
Δ_i, $P_i(E)$, s^0, s^+, s^{c}_i, ΔP_i, $\tilde{\mathsf{k}}_i$, l_i	Width $\int P_i \mathrm{d}\omega$ of bin i in electronic energy; bin probability as a function of electronic energy; zero-phonon and phonon-dressed parts of the contracted response; common-seeding response; $P_i(E-\omega_{\mathrm{ph}}) - P_i(E)$; computed (signed) stream kernel; its distance to a nonnegative kernel	eV; 1; as $\bar{s}_i$; …; as k_i	21.2
$\mathsf{k}_i(\boldsymbol{w})$, $\mathsf{f}(\boldsymbol{w})$, g_λ	Stream kernel of bin i; kernel of the bounded functional; dual function $\sum \lambda_i \mathsf{k}_i - \mathsf{f}$	—	21.2
E_{on}, m_{edge}, $m^{\mathrm{June}}_{\mathrm{edge}}$, t_{June}, k_{tail}, f_{esc}, F_{esc}, Γ_{on}, q_{th}	Onset energy $E_{\mathrm{g}} + \omega_\nu(\boldsymbol{Q}^*)$ of the edge; edge mass of a limit; its value at the largest Earth speed (2 June) and that date; tail exponent of the galactic distribution at v_{esc}; its height at the cut; lens prefactor; onset strength of the kernel; nuclear-recoil momentum threshold $(2ME_{\mathrm{th}})^{1/2}$	eV; MeV; …; $1/(\mathrm{km\ s^{-1}})^{k+3}$; …; eV	21.3
δ_{on}, ω_{top}, $a_\perp$, $\boldsymbol{v}_\perp$, $\hat{e}$, $s_\pm$, E^*	Speed window $v_{\max} - v_{\mathrm{on}}(m)$ at the edge; top of the edge spectrum $E_{\mathrm{on}} + x^*_{\max}$; lens geometry near $v_{\max}$; corner slopes; pair energy at the zero-phonon optimum	$\mathrm{km\,s^{-1}}$; eV; …	21.3
$\mathsf{R}_{\mathrm{FC}}(n)$, $\vartheta^{\mathrm{det}}_d$, β_k, $\beta_{\mathcal{X}}$, σ_β, a_j, $\bar{\delta}_{k,j}$, t', t_ϑ, $\bar{\sigma}^{\mathrm{lim}}_e$, $\bar{\sigma}^{\mathrm{lim,p}}_e$	Feldman–Cousins ordering ratio; detector nuisance; sensitivity $\partial \ln \bar{\sigma}^{\mathrm{lim}}_e / \partial \vartheta_k$ of the limit; its attraction entry; quadrature sum; bin weights; bin-averaged template; slope of the statistic in $\ln \bar{\sigma}_e$ at the nominal limit; statistic at fixed ϑ; nominal and profiled limits	1; …; cm^2	21.4
$f_{\mathrm{det},j}$, t^*, ℓ_j	Detection efficiency of bin j; threshold of t; Poisson term $2(N - N_{\mathrm{obs},j} \ln N)$ of bin j	1	21.4
$K^{\mathrm{rc}}_j(m_\chi)$, C_{Max}	Recast factor $\boldsymbol{r}^{\mathrm{pub}}_j / \boldsymbol{r}^{\mathrm{new}}_j$ of bin j; largest C_n over intervals of the optimum-interval method	1	21.5
$C_0(x, N_{\mathrm{sig}})$, C_n, g, $\boldsymbol{r}^{\mathrm{pub}}_j$, $\boldsymbol{r}^{\mathrm{new}}_j$, $\bar{\sigma}^{\mathrm{lim,new}}_e$, $\mathsf{L}^{(X)}$, $\vartheta^{\mathrm{det},}(X)$	Yellin's maximum-gap distribution function; probability that every interval with $\le n$ events has expected count $\le x$; fraction of the expected signal in the largest gap; published and corrected rates per unit cross-section into bin j; recast limit; likelihood of experiment X; its detector nuisances	1; 1; 1; as $\boldsymbol{r}_j$; cm^2; 1; 1	21.5
P_{ph}	Phonon-energy distribution of a phonon-dressed seed in the library	eV^{-1}	21.6
$n_\beta(T)$, f_β, y_{S}, Z_{d}, $\mathcal{A}_{\mathrm{act}}$, $b_{^{210}Pb}$, $F_{\mathrm{Fermi}}(Z_{\mathrm{d}}, T)$, γ_{d}, r_A, σ_{C}, $\bar{\ell}_{\mathrm{esc}}$	β spectrum; fraction below 1 keV; Sommerfeld parameter $\alpha Z_{\mathrm{d}} E_e / p_e$; daughter charge; activity per mass; sub-keV rate density per Bq/kg of ^{210}Pb; relativistic point-nucleus Fermi function; $(1 - Z^2_{\mathrm{d}}\alpha^2)^{1/2}$; nuclear radius $1.2A^{1/3}$ fm; Compton cross-section; mean escape path	keV^{-1}; 1; …; per kg d keV; 1; 1; fm; cm^2; cm	21.7

Symbol	Meaning	Units	Sect.
Q_β, E_e, p_e, q_ν, $\mathrm{d}\sigma_\mathrm{C}/\mathrm{d}T$, b	β endpoint energy; total energy $T+m_e$ and momentum of the β electron; antineutrino momentum $Q_\beta - T$ in a forbidden shape factor; Klein–Nishina cross-section per electron and unit kinetic energy at $T \to 0$; background rate density	keV; keV; keV; keV; $\mathrm{cm}^2\,\mathrm{keV}^{-1}$; $\mathrm{kg}^{-1}\,\mathrm{d}^{-1}\,\mathrm{keV}^{-1}$	21.7
Chapter 22: Open problems			
λ_A, $m_\mathrm{th}^\mathrm{CC}$, $O_\mathrm{F}^\mathrm{CC}(q)$, $O_{\mathrm{GT},\beta}^\mathrm{CC}(q)$, B_F, B_GT, $T_\pm$, T_iso	Nucleon axial coupling 1.27; charged-current threshold mass; Fermi and Gamow–Teller rows of the charged current; Fermi and Gamow–Teller strengths; summed isospin raising and lowering operators; total isospin of the parent (in Sect. 22.3)	1; eV; 1; 1; 1	22.3
Δ_CDW, $\boldsymbol{Q}_\mathrm{CDW}$, n_1, φ	Charge-density-wave gap parameter (single-particle gap $2\Delta_\mathrm{CDW}$); its wave vector; amplitude of the density modulation; its phase (in Sect. 22.3)	eV; eV; eV^3; 1	22.3
$\tau_j^\pm$, τ_z, σ_j^β, T_z	Isospin raising (neutron $\to$ proton) and lowering operators of nucleon $j, T_\pm = \sum_j \tau_j^\pm$; isospin projection with $\tau_z = +1$ for a proton, $T_z = (Z-N)/2$; Pauli matrix β of nucleon j (in Sect. 22.3)	1	22.3
$S_{ab}^\mathrm{CC}(q,\omega)$, $\rho^\mathrm{CC}(r)$, $\psi_{\boldsymbol{p}_\mathrm{ej}}^{(-)}(r)$, $e_{\boldsymbol{p}_\mathrm{ej}}^{(-)}$, $M_{A,Z}$, $M_{A,Z+1}^{(*)}$	Response of the charged-current rows; charged-current transition density, whose Fourier components are those rows; incoming-boundary scattering state of the outgoing electron in the field of the final state, and that electron as a final-state label; nuclear masses of the parent and of the daughter (ground or excited) in $m_\mathrm{th}^\mathrm{CC}$ (in Sect. 22.3)	eV^{-1}; eV^3; $\mathrm{eV}^{3/2}$; —; eV	22.3
p, n, e, χ; $\bar{p}$, $\bar{e}$	Proton, neutron, electron and dark-matter Dirac fields of the charged-current operator and their adjoints (in Sect. 22.3)	$\mathrm{eV}^{3/2}$	22.3
$\bar{n}_e$, n_1	Mean electron density of a charge-density wave, $n_e(\boldsymbol{r}) = \bar{n}_e + n_1 \cos(\boldsymbol{Q}_\mathrm{CDW}\cdot\boldsymbol{r} + \varphi)$ (in Sect. 22.3)	eV^3	22.3
Appendix A: Conventions and units			
μ_B, g_e	Bohr magneton $e/2m_e$; electron g-factor	eV^{-1}; 1	A.2
$\boldsymbol{E}_\mathrm{G}$, $\boldsymbol{B}_\mathrm{G}$; $\boldsymbol{E}_\mathrm{SI}$, $\boldsymbol{B}_\mathrm{SI}$	Electric and magnetic fields in Gaussian and in SI units	as quoted	A.2
$f_{\boldsymbol{G}}$	Fourier component of a lattice-periodic function, $\Omega^{-1}\int_\Omega \mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}} f$	as f	A.3
$\langle\langle\cdot\rangle\rangle$	Zone average $N^{-2}\sum_{kQ}$	—	A.3
$\boldsymbol{r}_\chi$, $\boldsymbol{r}_e$	Positions of the dark matter and of one electron	eV^{-1}	A.3
$\tilde{i}$, $\tilde{f}$	Time-reversed initial and final states, as kets $\lvert\tilde{i}\rangle, \lvert\tilde{f}\rangle$	—	A.4
V_NL	Nonlocal part of a pseudopotential	eV	A.5
$D_{\kappa\kappa'}^{\mathrm{dyn},\alpha\beta}$, $\Phi_{\kappa\kappa'}^{(2),\alpha\beta}$	Cartesian components of the dynamical matrix and of the force constants, written as superscripts	as $D^\mathrm{dyn}, \Phi^{(2)}$	A.6
$\hat{e}_\nu$	Unit polarization vector of an acoustic branch ν at small $\boldsymbol{Q}$	1	A.6
b, $b^\dagger$	Annihilation and creation operators of a single phonon mode, written without labels	—	A.6
L_e	Electron-channel loss function $\mathrm{Im}[-(\varepsilon_e^{-1})_{\mathbf{00}}]$	1	A.7
S^prop	Structure factor of $\tilde{R}_{ee}$	eV^2	A.7
O_e, $\varepsilon_e(q,\omega)$	Electron density operator $n_e(\boldsymbol{q})$ as an operator O_a; electron-channel dielectric function, $\varepsilon_e^{-1} = 1 + v_\mathrm{C} R_{ee}$ (equals ε at clamped nuclei)	—; 1	A.7
$R_{ee}(q,\omega)$, $\tilde{R}_{ee}(q,\omega)$, $S_{ee}(q,\omega)$	Density–density response function, its proper part, and the structure factor of the electron channel ($S \equiv S_{ee}$)	eV^2; eV^2; eV^2	A.7
$\bar{S}(q,\omega)$	S averaged over directions of q	eV^2	A.8
$\eta_2(v_\mathrm{min})$	Second speed moment $2\int_{v_\mathrm{min}}^\infty \mathrm{d}v' v' \eta(v')$	1	A.8
$\tilde{\eta}(v_\mathrm{min})$	$\rho_\chi \bar{\sigma}_e \eta / m_\chi$	—	A.8
N_cell	Number of unit cells of a target of mass M_det, $M_\mathrm{det}/M_\mathrm{cell}$	1	A.8

Symbol	Meaning	Units	Sect.
$\mathcal{M}_{\mathrm{free}}$	Relativistic free dark-matter–electron amplitude	1	A.9
q_{ref}, σ_{ref}	Reference momentum transfer of a cross-section conversion and the cross-section $\bar{\sigma}_e F_{\mathrm{DM}}(q_{\mathrm{ref}})^2$ quoted there	eV; cm^2	A.9
Appendix C: Mathematical tools			
$\mathcal{F}_{\boldsymbol{R}}$, $\mathcal{F}(z)$	Generic function in a Poisson sum and its lattice components; Cauchy integral	—	C.2
σ_{br}	Broadening width of a delta function on a $\boldsymbol{k}$ grid	eV	C.2
$\mathrm{d}\Omega_{\hat{n}}$, $\mathrm{d}\Omega_{\hat{\boldsymbol{Q}}}$, n^α	Solid-angle element of the direction $\hat{n}$; Cartesian components of a unit vector	sr; 1	C.3
$P_l(u)$	Legendre polynomial	1	C.3
$Y_l^{\mathrm{c}}(\hat{n})$	Cubic harmonic of order l	1	C.3
$\delta_\Gamma(x)$	Normalized Lorentzian $(\Gamma_{\mathrm{w}}/2\pi)/(x^2+\Gamma_{\mathrm{w}}^2/4)$	eV^{-1}	C.4
$\varrho_R(\omega)$	Spectral density $-\operatorname{Im} R/\pi$ of a response	as R eV^{-1}	C.5
$\hat{D}(\alpha_{\mathrm{d}})$, α_{d}	Displacement operator $\exp(\alpha_{\mathrm{d}} d^\dagger - \alpha_{\mathrm{d}}^* d)$ and its complex amplitude	1	C.6
$\hat{A}$, $\hat{B}$, $\hat{X}$, $\hat{Y}$	Generic operators in the exponential identities	—	C.6
$\boldsymbol{M}$, $\boldsymbol{A}$, …	Generic matrices, bold upright: blocks $\boldsymbol{A},\boldsymbol{B},\boldsymbol{C},\boldsymbol{D}$ of $\boldsymbol{M}$, the Schur complement $\boldsymbol{M}/\boldsymbol{D}$ and $\boldsymbol{N}=(\boldsymbol{M}/\boldsymbol{D})^{-1}$; also $\boldsymbol{U},\boldsymbol{W},\boldsymbol{K},\boldsymbol{X},\boldsymbol{T},\boldsymbol{h},\boldsymbol{\eta}$	—	C.7
g_0, ξ_0	Pairing amplitude and diagonal energy of a two-mode quadratic Hamiltonian	eV	C.7
E_{b}, E_{f}, θ_{b}, θ_{f}, u^{b}, v^{b}, u^{f}, v^{f}	Boson and fermion Bogoliubov energies, angles and coefficients	eV; 1	C.7
$\tilde{d}_j$, $\tilde{c}_j$	Bogoliubov-transformed boson and fermion operators	—	C.7
$K(s)$, $E(s)$, s, s'	Complete elliptic integrals; modulus and complementary modulus (s in Appendix C.8)	1	C.8
μ_0, μ_1	Coefficients of $\mu_0+\mu_1\cos\phi$ in the elliptic reductions	—	C.8
$\rho_{\mathrm{s}}(x)$	Scalar source density	eV^3	C.9
$G_{\mathrm{Y}}(x)$, $\tilde{G}_{\mathrm{Y}}(q)$	Static Green's function $\mathrm{e}^{-mr}/4\pi r$ of the Klein–Gordon operator and its transform $1/(q^2+m^2)$	eV	C.9
$U(x)$	Interaction energy of two static sources	eV	C.9
$\bar{F}_{\mu\nu}$, $\bar{F}^2$	Field strength $\partial_\mu\tilde{A}_\nu-\partial_\nu\tilde{A}_\mu$ of the photon after the mixing redefinition; its square $\bar{F}_{\mu\nu}\bar{F}^{\mu\nu}$	eV^2; eV^4	C.9
H_{s}, H_{v}	Static field energy of the scalar and vector configurations	eV	C.9
ϱ_{MK}	Markov–Krein band width	1	C.10
μ_k, $\bar{\mu}_k$	k-th moment and central moment of a normalized distribution	as x^k	C.10
$F_P(\xi)$	Cumulative distribution of P (decorated F)	1	C.10
$\boldsymbol{H}$, $\boldsymbol{H}_1$	Hankel matrix μ_{i+j} and shifted Hankel matrix μ_{i+j+1}	—	C.10
$\mathsf{K}(x,\xi)$	Reproducing kernel $\boldsymbol{v}(x)^{\mathsf{T}}\boldsymbol{H}^{-1}\boldsymbol{v}(\xi)$	—	C.10
w_ξ, F_{model}	Weight of the canonical three-point representation at its node ξ, equal to $\varrho_{\mathrm{MK}}(\xi)$; cumulative distribution of a model with the same moments as P	1; 1	C.10
$\mathfrak{G}_N(s)$, $\mathfrak{g}(s)$	Probability generating function of N; offspring generating function	1	C.11
$\bar{m}$	Mean offspring number $\mathfrak{g}'(1)$ of a Galton–Watson process, $\langle N_n\rangle=\bar{m}^n$	1	C.11
P_{ext}	Extinction probability	1	C.11
F_{loss}, $\langle X\rangle$, σ_X	Loss distribution of a renewal chain; its mean and width	1; eV; eV	C.11
E_{res}	Undershoot (residual energy) of a renewal chain	eV	C.11
$\bar{Y}(x)$	Mean $\langle Y\vert x\rangle$ of an additive cascade observable $Y=d_c+Y_1+\ldots+Y_k$ started in state x	as Y	C.11
S_n, $S_{N(E)}$	Partial sum $X_1+\ldots+X_n$ of the losses of a renewal chain, $S_0=0$; sum of the $N(E)$ losses that fit into E	eV	C.11

Symbol	Meaning	Units	Sect.
Ξ, $\mathscr{B}$	Nonnegative velocity measure (in Appendices C.12 and D.46 and Sect. 21.2); its compact domain	—	C.12
λ_i, N_i^{up}, κ_{v}	Dual variables; count bounds; visibility constant	1	C.12
$\mathcal{B}_h$, Φ_h, $\boldsymbol{\lambda}_h$, g_{λ_h}, $\boldsymbol{w}_h$	Grid of directions and speeds with covering radius h; grid primal optimum, a lower bound on the optimum of the full program; grid dual solution; its dual function, nonnegative on the grid only; nearest grid point of $\boldsymbol{w}$	—; as the objective; as λ_i; as g_λ; as $\boldsymbol{w}$	C.12
$\succ$, $h(a)$	Resolved-above relation of an interval order; height	—	C.13
I_a, ℓ_a, u_a	Interval $[\ell_a, u_a]$ of an element of an interval order, oriented so that larger values are better; its lower and upper ends	as the ranked quantity	C.13
$\hat{\vartheta}$, $\hat{\hat{\vartheta}}$	Best-fit nuisance; its value maximizing L at fixed signal	1	C.14
$\hat{N}$, $\sigma_{\hat{N}}$, σ_ϑ	Gaussian estimate of N_{sig} and its width; constraint width of a nuisance parameter	1	C.14
$\Phi(z)$	Standard normal distribution function	1	C.14
z_{CL}	Standard-normal quantile at the confidence level of a one-sided upper limit	1	C.14
E, O, H_{D}, H_{FW}	Even and odd parts of H_{D}; Dirac Hamiltonian in a potential; Foldy–Wouthuysen Hamiltonian to $O(m_e v^4)$(H_{Dar} is the Darwin term)	eV	C.15
$d_{\boldsymbol{k}j}$, $\boldsymbol{\epsilon}_{\boldsymbol{k}j}$	Photon annihilation operator and polarization vector of mode $(\boldsymbol{k}, j)$, $j = 1, 2$	1	C.16
$\mathfrak{a}_{\boldsymbol{k}j}$	Classical amplitude of radiation mode $(\boldsymbol{k}, j)$, normalized so that the mode energy is $\omega_k \lvert\mathfrak{a}_{\boldsymbol{k}j}\rvert^2$; quantization replaces it by $d_{\boldsymbol{k}j}$	1	C.16
e_{c}	Charge of a radiating particle	1	C.16
n_{grp}	Group index $dk/\mathrm{d}\omega$ of a medium	1	C.16
H_{rad}, $\mathrm{d}\Omega_{\hat{\boldsymbol{k}}}$, $\tilde{\boldsymbol{J}}(\boldsymbol{k}, \omega)$	Hamiltonian of the free radiation field, with the photon number states as eigenstates; solid-angle element of the photon direction; space-time Fourier transform of a prescribed classical current $\boldsymbol{J}(\boldsymbol{r}, t)$	eV; 1; eV^{-1}	C.16
$\{\mathsf{R}\lvert\boldsymbol{t}\}$, $\mathcal{P}$	Space-group operation; point group	—	C.17
$c^{(a)}(\mathsf{R})$, c_{V}, c_{A}	Irreducible, polar-vector and axial-vector characters	1	C.17
γ_{ijl}, g_{ml}, N_{gyr}	Linear-$\boldsymbol{q}$ coefficient of the current response; gyration tensor; number of gyration invariants	—	C.17
η	Eigenvalue sign of a glide on Bloch states	1	C.17
$\boldsymbol{u}(\boldsymbol{r}, t)$	Continuum displacement field, long-wavelength limit of $u_{\boldsymbol{R}\kappa}$	Å	C.18
u_{ij}, w_{el}	Strain tensor $\frac{1}{2}(\partial_i u_j + \partial_j u_i)$; elastic energy per unit volume $\frac{1}{2} C_{ijkl} u_{ij} u_{kl}$	1; GPa	C.18
σ_{ij}	Stress tensor $C_{ijkl} u_{kl}$	GPa	C.18
ε_I	Voigt strain vector $(u_{11}, u_{22}, u_{33}, 2u_{23}, 2u_{13}, 2u_{12})$	1	C.18
$\mathsf{M}_{ik}(\hat{Q})$	Christoffel matrix $C_{ijkl} \hat{Q}_j \hat{Q}_l$	GPa	C.18
Z_{ac}	Acoustic impedance ρc of a medium	$\mathrm{kg\,m^{-2}\,s^{-1}}$	C.18
ρ_1, ρ_2	Mass densities of the two media of an interface	$\mathrm{g\,cm^{-3}}$	C.18
r_{12}, t_{12}^{amp}, $t_{12}(\theta_1)$	Displacement reflection and transmission amplitudes at normal incidence; energy transmission from medium 1 to 2 at incidence angle θ_1	1	C.18
m_n, γ_n, μ_N	Neutron mass; neutron magnetic moment in nuclear magnetons, -1.913 (in Appendix C.19); nuclear magneton $e/2m_p$	eV; 1; eV^{-1}	C.19
$\boldsymbol{\sigma}_n$, $\boldsymbol{p}_n$, $\boldsymbol{p}'_n$	Neutron Pauli matrices; incident and outgoing neutron momenta	1; eV	C.19
$\boldsymbol{B}_e$, $\boldsymbol{M}_e$	Magnetic field and spin magnetization of the electrons	eV^2	C.19
$\hat{f}_{\mathrm{mag}}$, $\hat{f}_{\mathrm{nuc}}$, $\boldsymbol{s}_\perp$	Magnetic and nuclear neutron amplitude operators; transverse spin density $P_T s_e(\boldsymbol{q})$	fm; 1	C.19

Symbol	Meaning	Units	Sect.
$\varrho_{\rm s}(r)$, $R_{\rm sh}(r)$, $\langle j_l(q)\rangle$	Normalized unpaired-spin density of an ion; radial function of its shell; radial integrals of j_l	Å^{-3}; $\text{Å}^{-3/2}$; 1	C.19
$\boldsymbol{J}_j$	Total angular momentum of magnetic ion j	1	C.19
σ_x, σ_z	Pseudo-spin Pauli matrices of a two-level system in the localized basis $\lvert L\rangle, \lvert R\rangle$	1	C.20
Δ_0, $\Delta_{\rm as}$, $\Delta_{\rm r}$	Tunneling amplitude; asymmetry; renormalized tunneling amplitude	eV	C.20
d_k, ω_k, λ_k	Bath oscillators, their energies and couplings in the spin–boson model (ω_k in Appendix C.20)	—; eV; eV	C.20
$H_{\rm L}$, $H_{\rm R}$	Spin–boson Hamiltonians in the configurations L and R, each with the oscillator equilibria shifted by $\pm\lambda_k/2\omega_k$	eV	C.20
$J_{\rm s}(\omega)$, $\omega_{\rm cut}$	Bath spectral density $\pi\sum_k \lambda_k^2\delta(\omega-\omega_k)$; its high-energy cutoff	eV	C.20
$\mathcal{Q}(t)$, $C(t)$	Bath correlation exponent and overlap $\mathrm{e}^{-\mathcal{Q}}$	1	C.20
$\Gamma_{\rm LR}$, $\Gamma_{\rm RL}$	Incoherent hopping rates between the configurations	eV	C.20
$\Gamma(z)$	Euler's gamma function	1	C.20
$a_{\rm ch}$, $K_{\rm ch}$, $V_{\rm s}$	Spacing, spring constant and substrate-potential amplitude of a Frenkel–Kontorova chain	Å; eV Å^{-2}; eV	C.20
$H_{\rm FK}$	Frenkel–Kontorova Hamiltonian of a row of atoms in a periodic substrate potential	eV	C.20
$c_{\rm ch}$, $\omega_{\rm s}$, $d_{\rm kink}$	Sound speed of the free chain; single-atom frequency in the substrate; kink width $c_{\rm ch}/\omega_{\rm s}$	$\mathrm{km\,s^{-1}}$; meV; Å	C.20
$E_{\rm kink}$, $M_{\rm kink}$, $E_{\rm PN}$	Kink rest energy; kink mass $E_{\rm kink}/c_{\rm ch}^2$; Peierls–Nabarro barrier	eV	C.20
Appendix D: Long derivations			
$\mathcal{U}_{\gamma\tilde\gamma}$, $\mathcal{U}$	Basis-change matrix $\langle\gamma\lvert\tilde\gamma\rangle$; the matrix itself, unitary, with complex conjugate $\mathcal{U}^*$ and adjoint $\mathcal{U}^\dagger$	1	D.2
$K(t-t')$, $\tilde K(s)$, $R(z)$	Memory kernel; its Laplace transform, $\sum_f \lvert H'_{fi}\rvert^2/(s+\mathrm{i}\xi_f)$ for $\mathrm{Re}\,s>0$; level-shift function	eV^2; eV; eV	D.3
$r_{\rm p}$	Residue of the decaying pole, $1/(1-R'(z_{\rm p}))$	1	D.3
h, $h_{\rm HF}$	One-body Hamiltonian $-\nabla^2/2m_e+U_1$ of a determinant; Hartree–Fock (Fock) operator	eV	D.4
$\varrho(x,x')$	One-body density matrix $\sum_{\rm occ}\psi_\gamma(x)\psi^*_\gamma(x')$, the kernel of $\hat\varrho(x=(\boldsymbol{r},s))$	Å^{-3}	D.4
$\lambda_{\gamma\gamma'}$	Lagrange multipliers of orthonormality	eV	D.4
$\mathcal{M}_{\gamma\gamma'}$	Unitary matrix of occupied-orbital rotations	1	D.4
$E_{\rm HF}$, $E^{\rm HF}_{\boldsymbol{k}}$, $E_{\rm x}$	Hartree–Fock total energy; Hartree–Fock band $k^2/2m_e+\Sigma_{\rm x}(k)$ of the gas; exchange energy ($E_{\rm x}/N_e$ per electron)	eV	D.4
$E_{\rm HOMO}$, $E_{\rm LUMO}$	Highest occupied and lowest unoccupied Kohn–Sham eigenvalues	eV	D.5
v_0, $\bar v$	Head and remainder ($\boldsymbol{G}\neq 0$) of the Coulomb matrix (v_0 in Appendices D.7 and D.36)	eV^{-2}	D.7
$\tilde k$	$k/k_{\rm F}$, radial integration variable	1	D.8
$\mathcal{I}(w)$	Radial Lindhard integral $w+\frac12(1-w^2)\ell(w)$	1	D.8
$\delta\varrho_{\boldsymbol{k}}(t)$, $\delta\varrho^{\rm loc}_{\boldsymbol{k}}$	Coherence $\langle c^\dagger_{\boldsymbol{k}}c_{\boldsymbol{k}+\boldsymbol{q}}\rangle$ and its local-equilibrium value	1	D.9
$D_{\boldsymbol{k}}$, $w_{\boldsymbol{k}}$, $w_{\rm tot}$, ς, Φ_ϱ	$\xi_{\boldsymbol{k}}-\xi_{\boldsymbol{k}+\boldsymbol{q}}$; weight $(n_{\boldsymbol{k}}-n_{\boldsymbol{k}+\boldsymbol{q}})/(\xi_{\boldsymbol{k}+\boldsymbol{q}}-\xi_{\boldsymbol{k}})$ and its sum; $\sum_{\boldsymbol{k}}\delta\varrho_{\boldsymbol{k}}$; the functional $\sum\lvert\delta\varrho_{\boldsymbol{k}}\rvert^2/w_{\boldsymbol{k}}\geq 0$	eV; eV^{-1}; eV^{-1}; 1; eV	D.9
$\phi_{\rm loc}(t)$, $\delta\mu$	Local-equilibrium potential and local chemical-potential shift in Mermin's construction	eV	D.9
R^0_γ, $R^0_{\rm st}$	Shorthands $R^0(\boldsymbol{q},\omega+\mathrm{i}\gamma_{\rm rel})$ and $R^0(\boldsymbol{q},0)$	eV^2	D.9
F	Number of closed fermion loops of a diagram	1	D.11
$U_{\rm I}(\tau,\tau')$, $O_{\rm I}(\tau)$, $H_{\rm int,I}$	Imaginary-time interaction-picture evolution $\mathrm{e}^{\tau H_0}\mathrm{e}^{-(\tau-\tau')H_\mu}\mathrm{e}^{-\tau' H_0}$, operator and interaction	1; —; eV	D.11

Symbol	Meaning	Units	Sect.
$\langle\cdot\rangle_0$, $\mathcal{Z}_0$, $\lvert\Phi_0\rangle$	Free grand-canonical average and partition function; Fermi sea	—; 1; —	D.11
A_j, $\hat{A}_j$	Operators of Wick's theorem; the hat marks an operator removed from the string	—	D.11
$K^{\mathrm{x}}_{\mathrm{full}}$, K^{x}_0	Exchange kernel with all $\boldsymbol{G}$; its $\boldsymbol{G}=0$ term, $K^{\mathrm{x}}_{\mathrm{full}} = K^{\mathrm{x}} + K^{\mathrm{x}}_0$	eV	D.12
$H^{\mathrm{exc}}_{\mathrm{full}}$; $E^{\mathrm{full}}_\lambda$, $A^\lambda_{\mathrm{full}}$	BSE Hamiltonian with $K^{\mathrm{x}}_{\mathrm{full}}$ and its eigenpairs	eV; eV; 1	D.12
$\mathfrak{L}$, $\mathfrak{L}^0$, $\bar{\mathfrak{L}}$	Pair propagator $-\langle T c^\dagger_n c_m(\tau) c^\dagger_{m'} c_{n'}(0)\rangle$; its independent-quasiparticle value; its solution with K^{x} in place of $K^{\mathrm{x}}_{\mathrm{full}}$	eV^{-1}	D.12
$K_{\mathrm{r\bar{r}}}$	Coupling (resonant–antiresonant) block of the BSE kernel	eV	D.12
$v_{ab,cd}$	Coulomb matrix element of four orbitals $\psi_a, \psi_b, \psi_c, \psi_d$	eV	D.12
$V^{\mathrm{L}}(r)$	Potential energy of an electron in the macroscopic field of a longitudinal phonon, $\nabla V^{\mathrm{L}} = e\mathcal{E}$	eV	D.14
$\Pi^{\mathrm{b}}_\nu(Q,\omega)$, Π^{b}	Phonon bubble in energy units, $\Pi_\nu = 2\omega_{\boldsymbol{Q}\nu}\Pi^{\mathrm{b}}_\nu$	eV	D.17
E_{band}	Band-width cutoff of the Fermi-surface average	eV	D.17
$\mathcal{F}(k,k')$, $\rho(u)$, b_U	Interband overlap of the Dirac cone per state; pair spectral function $(1-4\Delta_{\mathrm{D}}^2/u)^{1/2}(1+2\Delta_{\mathrm{D}}^2/u)$ in the invariant u; its square-root variable at the cutoff	1	D.22
s	Lorentz invariant $\omega^2 - v_{\mathrm{D}}^2 q^2$ of the Dirac polarization	eV^2	D.22
N_i, z_i, $\boldsymbol{c}_i$, U_{c}, U_{v}	Reduced velocity blocks $M_i M_1^{-1}$, their scalar and vector parts, and the basis changes in the conduction and valence doublets of the uniqueness argument	1	D.22
$Ry^*_\perp$, $a^*_\perp$	Transverse effective Rydberg and Bohr radius of a donor, with the mass $m^*_\perp$	meV; Å	D.23
$n_{\boldsymbol{k}}$, $\phi_{\boldsymbol{k}}$	Number density of quanta in mode $\boldsymbol{k}$; complex amplitude of mode $\boldsymbol{k}$ of a classical field	eV^3; eV	D.25
$\boldsymbol{P}_e$	Total momentum of the electrons, $i m_e[H_e, \sum_i \boldsymbol{r}_i]$	eV	D.26
H_{ad}, h_{BO}, $\bar{O}_a$, $\bar{O}$	$\mathcal{P}_{\mathrm{ad}} H \mathcal{P}_{\mathrm{ad}}$; the Born–Oppenheimer Hamiltonian of the nuclei it becomes after the rotation U_n; restricted operator $\mathcal{P}_{\mathrm{ad}} O_a \mathcal{P}_{\mathrm{ad}}$	eV; eV; as O	D.28
$\hat{e}_1$, $\hat{e}_2$	Orthonormal unit vectors perpendicular to $\hat{q}$	1	D.30
S^0_{ab}	Independent-particle response matrix of any two rows	as S_{ab}	D.30
$s^{\mathrm{inc}}_\kappa(\boldsymbol{Q},\omega)$	Incoherent spectrum of one nucleus, normalized to one, zero-phonon line included	eV^{-1}	D.33
E^{el}_Λ, H_{el}, H_{lat}	Electronic energy of final-state class Λ; operators that multiply each class by its electronic and lattice energy	eV	D.33
A_e, A_κ, X_e	Internal parts of $n_e(q)$ and $n_\kappa(q)$ after the center-of-mass separation; total electron position $\sum_i \boldsymbol{r}_i$	1; Å	D.34
x, y; X, Y	One-body vertex operators of the dark matter and of the phonon in the second-order amplitude and in the second-order density matrix, with matrix elements $\mathsf{x}_{ab}, \mathsf{y}_{ab}$; $X = \sum_{ab} \mathsf{x}_{ab} c^\dagger_a c_b, Y$ their second-quantized forms	—	D.35
D_{A}, D_{B}	Adiabatic denominators of the density-first and phonon-first terms of $M^{\pm}$	eV	D.35
C_{x}, U_{ext}, $c_{\mathcal{A}}$, $\bar{f}$	Exchange coefficient, $n\varepsilon_{\mathrm{x}} = -C_{\mathrm{x}} n^{4/3}$; applied potential of the mean-field evolution; coefficient $\mathrm{i}\ell\hat{e}\cdot\boldsymbol{q}$ of the rigid translation; kernel without its macroscopic Coulomb term	—; eV; 1; —	D.36
$\varepsilon_{\boldsymbol{G}}$	Sublattice phase $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\delta}_0} = \pm 1$	1	D.38
$W^{\mathrm{sr}}_\perp$	Transverse weight required by the spin sum rule, $(\lvert\boldsymbol{S}\rvert - \Delta S)(1+2\Delta S)$	1	D.38
$\mathfrak{h}(y')$, $\mathfrak{k}(\omega)$, $\tilde{\mathfrak{k}}$, $\delta\mathfrak{g}(y')$, Y_{tail}	Principal-value integrand $y'^2\delta\mathfrak{g}/(y'+y)$; $2\mathcal{N}_{\mathrm{F}}/R$; $\mathfrak{k} - 3\omega^2/(qv_{\mathrm{F}})^2$; $\mathfrak{g}_{\mathrm{I}}\Theta(y'-2)-1$; cutoff of the numerical Kramers–Kronig integral	1	D.39

Symbol	Meaning	Units	Sect.
$\Gamma_{\mathrm{ii},t}(x)$, $\Gamma_{\mathrm{ph},t}(x)$	Iso-energy impact-ionization and phonon-emission rates of a carrier of type t at kinetic energy x in the energy-only reduction	s^{-1}	D.43
$\varkappa$, x_{cut}, $S(u)$, ξ, K_x, K_∞, μ_{ren}	Rate exponent of the approach to the scale-free share law; cut energy of the renewal proof; weight collected on the line of unit u; one step $\ln(x_{j+1}/x_j)$ (in Appendix D.44); its law at energy x; its scale-free limit; minus the mean step of K_∞	1; eV; eV^{-1}; 1; —; —; 1	D.44
Appendix E: Ab initio methods			
N_{PW}	Number of plane waves per state, $\Omega k_{\mathrm{cut}}^3/6\pi^2$	1	E.1
N_{b}, N_{occ}, N_{proj}	Numbers of bands, of occupied bands and of nonlocal projectors of a calculation	1	E.1
σ_{sm}, $\tilde{f}(x)$	Occupation-smearing width; smearing function, $f_{n\boldsymbol{k}} = \tilde{f}((\mu_{\mathrm{ch}} - \varepsilon_{n\boldsymbol{k}})/\sigma_{\mathrm{sm}})$	eV; 1	E.1
F_{sm}, S_{sm}	Smeared free energy $E - \sigma_{\mathrm{sm}} S_{\mathrm{sm}}$; occupation entropy	eV; 1	E.1
$E(\sigma_{\mathrm{sm}} = 0)$	Energy per cell of the unsmeared metal	eV	E.1
β_{mix}, ε_λ	Density-mixing parameter; eigenvalues of the static dielectric matrix ε^{T}	1	E.1
$\boldsymbol{n}_{\mathrm{in}}$, $\boldsymbol{n}_{\mathrm{out}}$	Input and output densities of a self-consistency step	eV^3	E.1
V_{loc}, $D_{\kappa,ij}$	Local radial part and projector coefficients of a separable (Kleinman–Bylander) pseudopotential	eV	E.2
q_{opt}, $E^{\mathrm{res}}(q_{\mathrm{opt}})$	Optimization wavevector of an optimized norm-conserving pseudopotential; residual kinetic energy above it	eV (quoted in bohr^{-1}); eV	E.2
$R_l(r)$, j_l, j_{L}	Radial solution of angular momentum l (logarithmic-derivative test); spherical Bessel functions (j_{L} in Appendices E.2 and E.7)	—; 1	E.2
Δ_{gauge}, $\mathcal{V}_0$, $E_1(\mathcal{V})$, $E_2(\mathcal{V})$	Equation-of-state Δ gauge; equilibrium volume and the two equations of state it compares	meV/atom; $\mathrm{Å}^3$; eV	E.2
$\delta_{\mathrm{NL}}(q)$	Nonlocal-potential correction to the f-sum rule	eV	E.2
$E_{\mathrm{qp}}(nk)$	Quasiparticle energy	eV	E.3
$E_{\mathrm{cut}}^{\varepsilon}$	Dielectric-matrix (screening) cutoff	eV	E.3
$\tilde{\omega}_{\boldsymbol{G}\boldsymbol{G}'}(q)$	Plasmon-pole frequency of the element GG'	eV	E.3
$v^Q P_{mn}$, v_{mn}^{KS}	Velocity matrix elements between quasiparticle and between Kohn–Sham states	1	E.3
$\hat{P}_c^{\mathrm{deg}}$, $\mathcal{W}$, $\hat{P}_{\mathcal{W}}^{\perp}$	Projector on the states degenerate with c; band window of the empty-state Sternheimer equation; projector on its complement	1; —; 1	E.4
$\partial\psi_v^{\pm}$	First-order state changes of the frequency-dependent Sternheimer equation at $\pm\omega$	as $\psi_{n\boldsymbol{k}}$	E.4
$\boldsymbol{P}_{\mathrm{el}}$, $u_{v'\boldsymbol{k}+\Delta\boldsymbol{k}}$	Electronic (Berry-phase) polarization; periodic part at the next point of a Berry-phase string of spacing Δk	$\mathrm{e\,Å}^{-2}$; as $u_{n\boldsymbol{k}}$	E.4
$\Delta B^{\alpha\alpha}$	Γ-cell correction to the displacement tensor B_κ	$\mathrm{Å}^2$	E.4
R_{IFC}	Range of the interatomic force constants in units of the cell size (cost estimate)	1	E.4
C_{ij}, C_{ijkl}	Elastic constants in Voigt and Cartesian form	GPa	E.4
c_{TA}, c_{T}	Transverse-acoustic sound speed; transverse sound speed of an isotropic model	$\mathrm{km\,s}^{-1}$	E.4
N_{fine}	Fine-grid points per direction	1	E.5
k_δ, V_{BZ}, Ω_{BZ}	Radius $(2m^*\delta)^{1/2}$ of the sphere of states within δ of a band edge; Brillouin-zone volume $(2\pi)^3/\Omega$	eV; $\mathrm{Å}^{-3}$	E.5
α_{xc}	Strength of the long-range xc kernel $-\alpha_{\mathrm{xc}}/\lvert\boldsymbol{q}+\boldsymbol{G}\rvert^2$	1	E.6
N_G, N_v, N_c, N_k	Numbers of reciprocal-lattice vectors, valence bands, conduction bands and $\boldsymbol{k}$ points of a response calculation	1	E.6
N_{orb}	Number of orbitals of a molecular GW-BSE calculation	1	E.6

Symbol	Meaning	Units	Sect.
t_f, $\boldsymbol{k}_\mathrm{rot}$	Fractional translation of a symmetry operation; rotated wavevector $\boldsymbol{k}+\boldsymbol{G}$	Å; $\mathrm{Å}^{-1}$	E.7
$\psi^\mathrm{at}_{\kappa nlm}$, $R^\mathrm{at}_{nl}(r)$	Atomic orbital of species κ; its radial function	$\mathrm{Å}^{-3/2}$	E.7
C_{jln}, Z_{jl}, $\tilde{n}_{jl}$	Slater-type expansion coefficients, exponents and integer powers of R^at_{nl}	1	E.7
l'_max	Partial-wave cutoff of the continuum state	1	E.7
$\mathcal{F}(\Omega,T)$, $E_0^\mathrm{ad}(\Omega)$	Free energy per cell; static-lattice energy at cell volume Ω	eV	E.8
α_th	Volume thermal-expansion coefficient	K^{-1}	E.8
N_conf, N_eff, w_j	Configurations of an SCHA population; effective sample number $(\sum_j w_j)^2/\sum_j w_j^2$; reweighting weights	1	E.8
N_t, Δ_fit	MD time steps of a TDEP fit; the force mismatch it minimizes	1; $\mathrm{eV}^2\,\mathrm{Å}^{-2}$	E.8
$\Delta\varepsilon^\mathrm{FM}_{n\boldsymbol{k}}(T)$	Fan–Migdal part of the temperature-dependent band-energy shift	eV	E.8
$x_{\boldsymbol{Q}\nu}$, $\sigma_{\boldsymbol{Q}\nu}$	Mass-weighted normal coordinate of mode $Q\nu$; width of its thermal Gaussian	$\mathrm{eV}^{-1/2}$	E.8
$\boldsymbol{u}^\mathrm{ZG}_{\boldsymbol{R}\kappa}$, $\varsigma_{\boldsymbol{Q}\nu}$, $\mathcal{B}$	ZG (special) displacement of nucleus κ in cell R; its signs ±1; the set of wavevectors it uses	Å; 1; —	E.8
E_U, $n^{I\sigma}_{mm'}$, $P^I_{mm'}$	Hubbard energy correction; occupation matrix of the correlated shell of ion I and spin σ; projector on its orbitals	eV; 1; 1	E.9
R^U_{IJ}, $R^{U,0}_{IJ}$, δV^U_I	Interacting and bare responses of the shell occupations to the potential shift δV^U_I of linear-response U	eV^{-1}; eV^{-1}; eV	E.9
E_FM, E_AFI, E_AFII	Energies per ion of the ferromagnetic and the two antiferromagnetic orders of NiO (energy mapping)	eV	E.9
$E(Q,\theta)$	Energy of a spin spiral of wavevector $\boldsymbol{Q}$ and cone angle θ	eV	E.9
Δ_i	Exchange splitting of site i in the LKAG formula	eV	E.9
J^LKAG_{ij}	Exchange constant of the LKAG formula in the convention of its source (pairs twice, unit vectors), $J^\mathrm{ex}_{ij}=-2J^\mathrm{LKAG}_{ij}/\lvert\boldsymbol{S}\rvert^2$	meV	E.9
$Z^\mathrm{El}(\mathrm{i}\omega^\mathrm{F}_j)$, $\Delta^\mathrm{El}(\mathrm{i}\omega^\mathrm{F}_j)$	Eliashberg renormalization function and gap on the fermionic Matsubara axis	1; eV	E.9
μ^*_C	Coulomb pseudopotential	1	E.9
ω_log	Logarithmic average phonon frequency of the Allen–Dynes formula	eV (quoted in K)	E.9
z_D, z'_D	Defect charge number in units of e; a second charge state	1	E.10
$E^\mathrm{f}_\mathrm{D}(z_\mathrm{D})$, E_D, E_host	Formation energy of a defect D in charge state z_D; total energies of the defective and the host supercells	eV	E.10
n_s, μ^at_s	Number of atoms of species s added; atomic chemical potentials	1; eV	E.10
ε_VBM, E_corr	Kohn–Sham energy of the valence-band maximum; finite-size (image-charge) correction	eV	E.10
ΔQ_k, ω_k	Mass-weighted displacement projected on vibronic mode k; its frequency (in Appendix E.10)	$\mathrm{eV}^{-1/2}$; eV	E.10
V_d	Half the difference of the conduction-electron contact potentials of the two configurations, which sit at $\pm V_\mathrm{d}$	eV	E.10
S_L, S_R, $\tilde{\delta}_\nu$, $\tilde{\delta}$	Scattering matrices of the conduction electrons at the Fermi energy in the two configurations; phase-shift differences, the eigenvalues $\mathrm{e}^{2\mathrm{i}\tilde{\delta}_\nu}$ of $S_R S_L^\dagger$	1	E.10
$E^\mathrm{at}_\kappa(\mathcal{N}_I)$, $\mathcal{N}_I$, $\mathcal{N}_s$	Local atomic energy of an interatomic potential; neighborhood of atom I within the cutoff r_c; representative environments of a GAP fit	eV; —; —	E.11
$V^{(2)}$, α_s, M_rep, K_GAP	GAP pair term; kernel weights; number of representative environments; similarity kernel	eV; eV; 1; 1	E.11
$V_\mathrm{ZBL}(r)$, a_U, Z_1, Z_2	ZBL screened pair potential; screening length $0.8854a_0/(Z_1^{0.23}+Z_2^{0.23})$; atomic numbers of the pair	eV; Å; 1	E.11

Symbol	Meaning	Units	Sect.
$r_{\min}$	Head-on distance of closest approach	Å	E.11
f_{sw}, r_{sw}, b_{sw}, $E^{\mathrm{ad}}_{\mathrm{ML}}$	ZBL switch function, its center and steepness; machine-learned energy surface	1; Å; $\mathrm{Å}^{-1}$; eV	E.11
$\hat{r}_R$	Recoil direction	1	E.11
$p_{\mathrm{FP}}(E_R,\hat{r}_R)$, N_{dir}	Probability that a recoil leaves a Frenkel pair once the collision has died out; number of recoil directions	1	E.11
N_{im}	Number of NEB images	1	E.11
$E^{\mathrm{ad}}_{\mathrm{min}}$	Energy surface at a quenched (inherent) structure	eV	E.11
B_{iso}, g_{md}, $\bar{M}$	Isotope rate constant, $\Gamma_{\mathrm{iso}} = B_{\mathrm{iso}}(E_{\mathrm{ath}}/h)^4$; mass-variance coefficient $\langle(M-\bar{M})^2\rangle/\bar{M}^2$; mean isotope mass	s^3; 1; amu	E.12
Γ_{anh}, A_{anh}	Three-phonon (anharmonic) decay rate $A_{\mathrm{anh}}(E_{\mathrm{ath}}/h)^5$ and its constant	s^{-1}; s^4	E.12
E_{L}, $E_{\mathrm{L}'}$, $\theta_{\mathrm{L}'}$	Energies of the parent and of the longitudinal daughter of $L \to L' + T'$ decay; angle of the daughter	meV; 1	E.12
$\Gamma_{\mathrm{L}}(k)$, k_{s}, l_0	Luke emission rate; sound-speed momentum; scattering length	s^{-1}; eV; m	E.12
m_{c}	Conductivity mass, $3/m_{\mathrm{c}} = 1/m^*_{\parallel} + 2/m^*_{\perp}$	m_e	E.12
$\mathfrak{l}_{\mathrm{f}}(E_{\mathrm{ath}})$	Phonon mean free path in a superconducting film	μm	E.12
$\boldsymbol{r}_{\mathrm{ev}}$, ϕ_s, N_{l}	Event position; Ramo weighting potential of electrode s; number of launched phonons	Å; 1; 1	E.12
$\Gamma_{\mathrm{ii}}(nk)$, $\Gamma_{\mathrm{ph}}(nk)$, Γ_{tot}, Γ_{max}	Impact-ionization and phonon rates of a carrier state; their sum; self-scattering ceiling	s^{-1}	E.13
E_0	Seed energy of a cascade, $\varepsilon_{\mathrm{eh}} = E_0/\langle N_{\mathrm{eh}}\rangle$	eV	E.13
A_{AB}	Alig–Bloom constant, ratio of the squared phonon and Coulomb matrix elements times phase-space factors	eV^3	E.13
ω_{AB}	Energy of the single optical phonon of the Alig–Bloom model, 63 meV	eV	E.13
N_{casc}, k_E	Number of simulated cascades (in Appendix E.13); radius $(2m_eE)^{1/2}$ of the free-electron sphere below energy E	1; eV	E.13
M_{D}, M_{E}	Direct and exchange matrix elements of the screened interaction of impact ionization	eV	E.13
ω_{LO}	Longitudinal optical phonon energy at the zone center in the Fröhlich width estimate of GaAs	meV	E.14
$S_\lambda(\boldsymbol{q},\omega)$	Dynamic structure factor of the self-trapped exciton line λ, its vertical creation dressed by a Poisson series of phonons	eV^{-1}	E.15
$\mathfrak{r}_{\mathrm{scr}}(\boldsymbol{q})$	Screening ratio of the phonon-assisted amplitude, screened over reference, $\lvert\tilde{M}\rvert^2/\lvert M\rvert^2$	1	E.16
v_0	Contact Coulomb kernel of the one-dimensional two-band test model (in Appendix E.16)	model units	E.16
v_χ	Speed of a dark-matter particle in units of c, bounding $\omega \le v_\chi \mathcal{Q}$ in the boosted recast	1	E.16
$t_{\max}$, $S^{(2)}_{\mathrm{coh}}(\boldsymbol{q},\omega)$	Time window of the time-domain multiphonon sum; coherent (cross-site) part of the two-phonon structure factor at fixed $\boldsymbol{q}$ and ω	eV^{-1}; eV^{-1}	E.17
N_{core}, $\boldsymbol{p}_{\mathrm{red}}$, $\mathfrak{r}_W(\boldsymbol{q},\omega_e)$	Number of core electrons per atom frozen in the pseudopotential, which move rigidly with the nucleus; reduced (first-zone) momentum whose Umklapp copies $\boldsymbol{p}_{\mathrm{red}} + \boldsymbol{G}$ cover the unreduced momenta; ratio $\lvert A^{\mathrm{DFPT}}\rvert^2/\lvert A^W\rvert^2$ of the squared Migdal amplitudes with the true and with the Wannier vertex	1; keV; 1	E.18
$P_{\mathrm{wall}}(\hat{\boldsymbol{n}})$	Pressure of the reflected dark matter on a face of outward normal $\hat{\boldsymbol{n}}$ of a test mass	Pa	E.19

Symbol	Meaning	Units	Sect.
L_{th}, $\mathsf{M}_{\lambda\lambda'}$, b_λ, $\mathcal{N}_\lambda^{\mathrm{ch}}$	Threshold on $\lvert L^{\mathrm{ph}}\rvert$ of the long-lived chiral band; mean number of daughters in mode λ' per decay of mode λ; indicator of the terminal chiral set, 1 inside and 0 outside; expected number of detectable chiral phonons from one phonon created in mode λ	1	E.19
$P_{\mathrm{K}}(\boldsymbol{u})$, E_{K}, $H_{\mathrm{CF}}(\boldsymbol{u})$	Projector on the ground Kramers doublet of a lanthanide ion at displacement $\boldsymbol{u}$; energy of that doublet; crystal-field Hamiltonian of the 4f shell	1; eV; eV	E.19
$R_{n\ell}(r)$, j_K, $\langle j_K(q)\rangle$	Radial function of the open $n\ell$ shell of a magnetic ion; spherical Bessel function of even order K; its radial average $\int r^2 \mathrm{d}r R_{n\ell}(r)^2 j_K(qr)$	$\text{Å}^{-3/2}$; 1; 1	E.20
a_{D}, $v_{\mathrm{D},x}$, $v_{\mathrm{D},y}$, $v_{\mathrm{D},z}$, $\varepsilon_{\mathrm{bg},zz}$	Bohr radius of the Dirac exciton; principal Dirac velocities along the crystal axes; zz component of the background dielectric tensor	Å; 1; 1	E.20
N_{src}, N_{corr}	Event counts of the published calculation and of the corrected one, whose ratio rescales a published limit	1	E.20
T_{N}, T_{C}	Néel and Curie temperatures of a magnetic target	K	E.20
$\boldsymbol{k}_\iota$	Wavevector of the conduction minimum of one valley of silicon, $\lvert\boldsymbol{k}_\iota\rvert = k_0$	keV	E.21
$R_{\mathrm{OPW}}(E_{\mathrm{kin}})$	Ratio of the ejection factor with the orthogonalized plane wave to that with the plain plane wave, at final kinetic energy E_{kin}	1	E.21
Ψ_L, Ψ_R	Many-electron ground states of the two configurations L and R of a defect in a plane-wave supercell	—	E.22
$\varepsilon_{\mathrm{hop}}$	Energy released in a defect hop	eV	E.22
$\boldsymbol{n}_{\mathrm{pair}}$, C_{Aug}, r_{tr}, λ_{tr}	Electron–hole pair density of a recoil track; Auger recombination coefficient, rate $C_{\mathrm{Aug}}\boldsymbol{n}^3$ per volume; initial radius of a Gaussian track; its pairs per unit length	cm^{-3}; $\mathrm{cm}^6\,\mathrm{s}^{-1}$; nm; nm^{-1}	E.23
N_z, z_j	Number of roots of unity at which a pair-number generating function is evaluated in the discrete Fourier inversion; the roots $\mathrm{e}^{2\pi i j/N_z}$	1	E.24
$\Gamma_{\mathrm{ph,loss}}$, w_{mem}	Rate of the phonon emissions that change the carrier energy; weight of the remembered momentum distribution at the next energy-changing event	s^{-1}; 1	E.24
$\mathcal{G}_\Lambda(z)$, x_e, x_h	Generating function $\sum_N z^N P_{\mathrm{casc}}(N\lvert\Lambda)$ of the pair number for seed class Λ; energies of the seed electron and hole above their band edges	1; eV	E.24
N_{meas}	Measured pair number of an event in the read window	1	E.25
$s_\ell(E_{\mathrm{ath}})$, D_X, S_{surf}, L_X, s_{hard}, s_{soft}	Phonon spectrum of release channel ℓ, energy fraction per unit E_{ath}; exciton diffusion constant; surface recombination velocity; exciton diffusion length $(D_X/\Gamma_X)^{1/2}$; hard-line and soft (Debye) trial spectra of the surface and electrode channels	meV^{-1}; $\mathrm{cm}^2\,\mathrm{s}^{-1}$; $\mathrm{cm}\,\mathrm{s}^{-1}$; μm; meV^{-1}	E.25
$f_{\mathrm{coll}}^{(p)}$	Collection efficiency of the phonon population p: primary phonons of nuclear and of electron recoils, Luke phonons, gap-energy phonons	1	E.25
λ_{dec}, R_k, t_k, t'_k	Decay constant of a cosmogenic nuclide; production rate, length of exposure k, time from its end to the measurement	s^{-1}; $\mathrm{kg}^{-1}\,\mathrm{d}^{-1}$; s; s	E.27

B.2 Literature notation

The dark-matter literature and the condensed-matter literature use the same letters for different objects. The letter χ is the dark-matter particle in one and a susceptibility in the other, q is the momentum transfer in one and the phonon wavevector in the other, and W is

the loss function, the screened interaction or the Debye–Waller exponent, depending on the paper. This book gives each letter one meaning, so several common symbols are renamed. Table B.2 lists the renamings that a reader moving between the book and the papers will meet. The left column gives what the literature writes, the second column what this book writes, and the last column the section where the book's symbol is defined.

Table B.2: Symbols of the literature and of this book. Where a paper's symbol differs from the book's only by a factor or a sign, the factor is given in the Meaning column.

Literature	Here	Meaning	Section
Response and screening			
χ_{AB}, χ, C^R_{AB}	R_{ab}	retarded response function; the literature usually omits the factor $1/\mathcal{V}$	4.1
Π	$\tilde{R}$	polarization, the Coulomb-proper (irreducible) response	4.4
W, $\mathrm{Im}(-1/\varepsilon)$	L	loss function $\mathrm{Im}[-(\varepsilon^{-1})_{00}]$	4.4
W (as in GW)	W^{scr}	screened Coulomb interaction	4.8
$-2\,\mathrm{Im}\,G^{\mathrm{R}}$	A	spectral function; the book's A integrates to one, the literature's to 2π	4.7
$G(\boldsymbol{r},t)$	$\mathcal{G}_{\mathrm{VH}}(\boldsymbol{r},t)$	Van Hove correlation function	4.2
λ_{TF}	$1/k_{\mathrm{TF}}$	Thomas–Fermi screening length	4.4
κ	$\varepsilon_{\mathrm{bg}}$	background (remote-band) dielectric constant	7.3
$\alpha(\omega)$	$\alpha_{\mathrm{abs}}(\omega)$	optical absorption coefficient	6.5
Dark matter and kinematics			
ε, κ, χ	ζ	kinetic mixing of the dark photon	1.4
g_χ, g_e	y_χ, y_e	mediator couplings to dark matter and electrons	1.4
n_χ	n_χ	dark-matter number density ρ_χ/m_χ	1.1
$\boldsymbol{k}$	$\boldsymbol{q}$	momentum transferred from the dark matter to the target	1.3
$\mathrm{d}\Omega$	$\mathrm{d}\Omega_{\boldsymbol{q}}$	solid-angle element of the scattered momentum	2.2
σ_e (fermionic absorption)	σ_F	absorption cross-section of fermionic dark matter	10.9
Q	N_{eh}	number of electron–hole pairs (ionization yield)	1.5
Nuclei and phonons			
$\boldsymbol{q}$, $\boldsymbol{k}$	$\boldsymbol{Q}$	phonon wavevector in the first zone	5.2
r_n, R_1; s	R_{nuc}; s_{skin}	Helm radius; skin thickness	2.2
c, a	c_{nuc}, a_{nuc}	Woods–Saxon half-density radius and diffuseness	2.2
λ, μ (Lamé)	C_{12}, C_{44}	elastic constants of an isotropic medium	C.18
λ	λ_{ep}	electron–phonon coupling constant	6.4
μ^*	μ^*_{C}	Coulomb pseudopotential	E.9
Z, Δ	Z^{El}, Δ^{El}	Eliashberg renormalization function and gap	E.9
Electrons in crystals			
μ	μ_{ch}	chemical potential	2.5
$N(0)$, $N(E_{\mathrm{F}})$	$\mathcal{N}_{\mathrm{F}}$	density of states per spin and volume at the Fermi level	2.5
v_{F} (Dirac material)	v_{D}	Dirac velocity; v_{F} is the Fermi velocity of the electron gas	7.3

Literature	Here	Meaning	Section
χ_{nl}	$\tilde{\psi}_{nl}$	radial momentum-space wavefunction of an atomic shell	2.2
T_s	K_s	Kohn–Sham (independent-electron) kinetic energy	3.4
V_2	V_cov	covalent energy between two bond hybrids	3.2
Ω; Ω_I; $\Omega_\mathrm{D}+\Omega_\mathrm{OD}$	Ω_W; Ω_W^I; $\tilde{\Omega}_W$	Wannier spread and its gauge-invariant and gauge-dependent parts	3.6
V_cell	Ω	unit-cell volume	3.1
χ, χ^0, α	R^U, $R^{U,0}$, δV^U	interacting and bare occupation responses, and the potential shift, of linear-response Hubbard U	E.9
α, K	α_ohm	ohmic coupling of a tunneling defect to conduction electrons	16.8
e (Gaussian)	e_G	Gaussian elementary charge, $e_\mathrm{G}^2=\alpha$; the book's e has $e^2=4\pi\alpha$	A.2
k (elliptic modulus)	s	modulus of the complete elliptic integrals	C.8

The Fourier sign, the sign of the mixing term, the units of charge and the normalization of the structure factor differ from parts of the literature by a sign or a factor rather than by a letter, and a formula taken from a paper must be converted before it is combined with the book's.

Fourier transforms. A function of position is transformed with $\mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ and a function of time with $\mathrm{e}^{+\mathrm{i}\omega t}$, with the factor $(2\pi)^{-1}$ per dimension on the inverse transform (Eqs. (A.6) and (A.7)). The electron density operator is then $n_e(\boldsymbol{q})=\sum_i \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$, with $\boldsymbol{q}$ the momentum given to the target. Some papers use the opposite spatial sign with the same direction of $\boldsymbol{q}$ and write the density operator as $\sum_i \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$ [1, Eq. (S.4)]; others keep $\mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}$ on the target by evaluating the transform of the potential at $-\boldsymbol{q}$ [2, Eq. (4)]. A perturbation of frequency $\omega>0$ goes as $\mathrm{e}^{-\mathrm{i}\omega t}$, so retarded functions are analytic for $\mathrm{Im}\,\omega>0$ and $\varepsilon_2>0$ in an absorbing medium. Engineering texts use $\mathrm{e}^{+\mathrm{i}\omega t}$, and their dielectric functions are the complex conjugates of ours (Appendix A.3).

Sign of the mixing term. The dark photon mixes through $+(\zeta/2)F_{\mu\nu}F'^{\mu\nu}$, with $\mathcal{L}\supset -eA_\mu J^\mu_\mathrm{em}$ for the photon, so that the dark photon couples to electrons with $y_e=-\zeta e$ (Sect. 1.4 and Appendix A.9). Ref. [3, Eq. (70)] has the same mixing sign but writes κ for ζ and $+eA_\mu J^\mu_\mathrm{em}$; Ref. [4, Eq. (2.1)] writes ε and mixes with hypercharge, $(\varepsilon/2\cos\theta_W)F_{Y\mu\nu}F'^{\mu\nu}$, with θ_W the weak mixing angle. Changing either sign flips the signs of y_e and $\tilde{V}$. No rate changes, because every rate contains ζ^2.

Units of charge. Charges are in Heaviside–Lorentz units, $e^2=4\pi\alpha$, so the Coulomb energy of two electrons is $e^2/(4\pi r)$ and the Coulomb kernel is $v_\mathrm{C}(k)=e^2/k^2$ (Appendix A.2). Condensed-matter sources usually use Gaussian units, with $e_\mathrm{G}^2=\alpha$ and kernel $4\pi e_\mathrm{G}^2/k^2$. A formula is converted by replacing every e^2 of this book with $4\pi e_\mathrm{G}^2$ for Gaussian units, or with $e_\mathrm{SI}^2/\varepsilon_0^\mathrm{SI}$ for SI units. The dimensionless response functions ε, ε^{-1} and L have the same values in all three systems.

Normalization of the structure factor. The dynamic structure factor carries $2\pi/\mathcal{V}$ (Eq. (A.31)). It is intensive, has dimension eV^2 for the density, and obeys the f-sum rule $\int \mathrm{d}\omega\,\omega\,S(\boldsymbol{q},\omega)=\pi n_e q^2/m_e$ over the whole real line (Eq. (A.39)). Many-body texts define the structure factor per electron and without the 2π; at $T=0$ theirs is $S/(2\pi n_e)$, with first moment $q^2/2m_e$ (Appendix A.7). Ref. [5, Eqs. (2) and (4)] uses the book's normalization, with χ for R and $\boldsymbol{k}$ for the momentum transfer. For a spin-degenerate band structure the book's S is summed over spin: its independent-particle form carries $4\pi/\mathcal{V}=2\times 2\pi/\mathcal{V}$ outside a sum over spatial bands (Eq. (A.38)). Refs. [4, Eq. (A.29)] and [6, Eq. (6)] also write the

factor 2 outside their band sums, and Ref. [5, Appendix B] leaves it implicit in its main-text formulas. What differs between sources is whether the quantity called S, or the crystal form factor, includes the 2, and a factor of 2 between two published rates for the same material often comes from this.

B.3 Acronyms

The last column gives the section or appendix in which the acronym first appears.

Acronym	Expansion	First use
ACE	atomic cluster expansion	E.23
AHC	Allen–Heine–Cardona	6.3
ALDA	adiabatic local-density approximation	4.6
ALP	axion-like particle	1.4
ANO	atomic natural orbital	E.16
BCS	Bardeen–Cooper–Schrieffer	1.5
BSE	Bethe–Salpeter equation	8.5
CASSCF	complete-active-space self-consistent field	E.19
CBM	conduction-band minimum	3.3
CCD	charge-coupled device	1.5
CDMS	Cryogenic Dark Matter Search	18.5
CMB	cosmic microwave background	1.1
CNT	carbon nanotube	E.21
CODATA	Committee on Data of the International Science Council	A.1
COHSEX	Coulomb-hole plus screened-exchange	E.3
CPU	central processing unit	E.12
DAMIC	Dark Matter in CCDs	10.9
DFPT	density-functional perturbation theory	5.2
DFT	density-functional theory	7.1
DLMF	Digital Library of Mathematical Functions	C.8
DOS	density of states	F
EPW	Electron–Phonon Wannier	6.2
EXCEED-DM	Extended Calculation of Electronic Excitations for Direct detection of Dark Matter (code)	10.4
FFT	fast Fourier transform	E.1
FIRE	fast inertial relaxation engine	E.23
FP	Frenkel pair	19.1
FT	fast transverse	E.12
G4CMP	Geant4 Condensed Matter Physics	20.1
GGA	generalized-gradient approximation	3.4
GIPAW	gauge-including projector-augmented-wave	E.26
GPAW	grid-based projector-augmented wave	12.5
GPU	graphics processing unit	E.22
GRASP	General-purpose Relativistic Atomic Structure Package	E.20
HOMO	highest occupied molecular orbital	D.5
HSE	hybrid functional of Heyd, Scuseria and Ernzerhof	3.4
HSE06	hybrid functional of Heyd, Scuseria and Ernzerhof, 2006 parametrization	10.6
IP	independent particles	4.6
JARVIS	Joint Automated Repository for Various Integrated Simulations	17.6
KK	Kramers–Kronig	E.20
KKR	Korringa–Kohn–Rostoker	E.22
LA	longitudinal acoustic	5.2
LCAO	linear combination of atomic orbitals	3.2

Acronym	Expansion	First use
LDA	local-density approximation	3.4
LF	local fields	8.5
LO	longitudinal optical	4.9
LUMO	lowest unoccupied molecular orbital	D.5
MC	Monte Carlo	E.23
MD	molecular dynamics	E.22
ML	machine-learned	E.21
MLIP	machine-learned interatomic potential	E.23
NEB	nudged elastic band	E.22
NR	nonrelativistic	F
NV	nitrogen–vacancy	E.21
OPW	orthogonalized plane wave	16.3
PAW	projector-augmented-wave	3.5
PBE	generalized-gradient functional of Perdew, Burke and Ernzerhof	3.3
PBE0	hybrid functional built on the functional of Perdew, Burke and Ernzerhof	E.21
QED	quantum electrodynamics	6.3
RAVE	Radial Velocity Experiment	1.2
RCC	relativistic core correlation	E.16
RPA	random-phase approximation	8.5
SALMON	Scalable Ab initio Light–Matter simulator for Optics and Nanoscience (code)	E.23
SCAN	strongly constrained and appropriately normed	E.17
SCF	self-consistent-field	16.6
SCHA	self-consistent harmonic approximation	5.7
SH	transverse polarization perpendicular to the plane of incidence (shear horizontal)	20.3
SHM	standard halo model	1.2
SOC	spin–orbit coupling	8.4
ST	slow transverse	E.12
SV	transverse polarization in the plane of incidence (shear vertical)	20.3
TA	transverse acoustic	5.2
TB09	meta-generalized-gradient functional of Tran and Blaha (2009)	21.6
TDDFT	time-dependent density-functional theory	8.5
TDEP	temperature-dependent effective potential	E.8
TO	transverse optical	4.9
TRIM	Transport of Ions in Matter	19.4
VBM	valence-band maximum	3.3
WEST	Without Empty STates	E.21
WIMP	weakly interacting massive particle	1.1
X2C	exact two-component (scalar-relativistic Hamiltonian)	E.16
XCOM	photon cross-section database of the National Institute of Standards and Technology	10.8
YIG	yttrium iron garnet	14.1
ZBL	Ziegler, Biersack and Littmark	E.11
ZG	Zacharias–Giustino (special displacement)	E.8
ZPR	zero-point renormalization	6.3

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Appendix C

Mathematical tools

C.1 Fourier transforms and delta functions

Fermi's golden rule rests on one limit: the function $\sin^2(\xi t/2)/\xi^2$ of Eq. (2.3) grows in height as t^2 and shrinks in width as $1/t$, and at long times it acts on smooth functions as $(\pi t/2)\,\delta(\xi)$ (Eq. (2.4)). Sect. 2.1 used the limit and the integral $\int \mathrm{d}u\, \sin^2 u/u^2 = \pi$ behind it; Sect. 2.2 used the rule for a delta function of a function. This section proves them, together with the Fourier identities that the rest of the book takes for granted, in the conventions of Appendix A.3.

C.1.1 Delta functions as limits

A delta function is defined by what it does under an integral. For every test function $w(x)$, a smooth function that falls off fast enough for the integrals below to converge,

$$\int_{-\infty}^{\infty} \mathrm{d}x\; \delta(x)\, w(x) = w(0). \tag{C.1}$$

No ordinary function does this; δ is a distribution, a linear rule that assigns a number to each test function [1, Sect. 1.16]. In physics it always appears as the limit of ordinary functions that become tall and narrow, and the useful question is when and how fast such a limit is reached.

The basic construction takes any integrable $f(u)$ with $\int \mathrm{d}u\, f(u) = 1$ and squeezes it, $f_t(x) = t\, f(tx)$. The area stays one while the width shrinks as $1/t$. With $u = tx$,

$$\int \mathrm{d}x\, f_t(x)\, w(x) = \int \mathrm{d}u\, f(u)\, w(u/t) \;\longrightarrow\; w(0) \int \mathrm{d}u\, f(u) = w(0) \qquad (t \to \infty),$$

because $w(u/t) \to w(0)$ at every fixed u, and $|f(u)\, w(u/t)| \le |f(u)| \max|w|$ lets the limit pass under the integral (dominated convergence). The family f_t therefore tends to δ. Three members recur in this book: the Gaussian $f(u) = \mathrm{e}^{-u^2/2}/\sqrt{2\pi}$, which describes a broadened line on a computational grid (Appendix C.2); the Lorentzian $f(u) = \pi^{-1}/(1+u^2)$, which describes a decaying state (Sect. 2.4 and Appendix C.4); and the function of the golden rule, $f(u) = (2/\pi)\sin^2(u/2)/u^2$.

How fast the limit is reached depends on the tails of f. Expanding $w(u/t)$ in powers of u/t, an even f with a finite second moment $\langle u^2\rangle_f$ gives $w(0) + \frac{1}{2}w''(0)\langle u^2\rangle_f/t^2$, a correction of relative order $1/t^2$. The Gaussian is of this kind. The Lorentzian and the golden-rule function fall off only as $1/u^2$, and their second moments diverge. The expansion then fails: their corrections are of order $1/t$, and they are set by the values of w far from the origin, not by its curvature at the origin. Eq. (C.3) below gives the correction for the golden-rule function and Appendix C.4 the one for the Lorentzian.

C.1.2 The golden-rule limit

The golden-rule limit rests on two integrals, $\int_0^\infty \mathrm{d}u\, \sin u/u = \pi/2$ and $\int \mathrm{d}u\, \sin^2 u/u^2 = \pi$. The Dirichlet integral follows from a convergence factor. Since $\int_0^\infty \mathrm{d}u\, \mathrm{e}^{-su} \sin u = 1/(1+s^2)$ for $s > 0$, and $1/u = \int_0^\infty \mathrm{d}s\, \mathrm{e}^{-su}$ for $u > 0$,

$$\int_0^\infty \mathrm{d}u\, \frac{\sin u}{u} = \int_0^\infty \mathrm{d}s \int_0^\infty \mathrm{d}u\, \mathrm{e}^{-su} \sin u = \int_0^\infty \frac{\mathrm{d}s}{1+s^2} = \frac{\pi}{2}.$$

The exchange of the two integrals is legitimate because the double integral converges absolutely once a factor $\mathrm{e}^{-s_0 u}$ with $s_0 > 0$ is included, and the result is continuous as $s_0 \to 0$. The integrand is even, so $\int_{-\infty}^\infty \mathrm{d}u\, \sin u/u = \pi$. Integrating by parts, with $\sin^2 u/u \to 0$ at both ends and $\mathrm{d}(\sin^2 u)/\mathrm{d}u = \sin 2u$,

$$\int_{-\infty}^\infty \mathrm{d}u\, \frac{\sin^2 u}{u^2} = \Big[-\frac{\sin^2 u}{u}\Big]_{-\infty}^\infty + \int_{-\infty}^\infty \mathrm{d}u\, \frac{\sin 2u}{u} = \pi, \tag{C.2}$$

where the last integral is the Dirichlet integral with $u \to 2u$. This is the integral quoted in Sect. 2.1. The central lobe, $|u| < \pi$, holds 90.3 % of it.

The same number follows from a physical reading. The first-order amplitude of Sect. 2.1 is a time integral. Our function is its squared modulus,

$$\Big|\int_0^t \mathrm{d}s\, \mathrm{e}^{\mathrm{i}\xi s}\Big|^2 = \Big|\frac{\mathrm{e}^{\mathrm{i}\xi t} - 1}{\xi}\Big|^2 = \frac{4\sin^2(\xi t/2)}{\xi^2}.$$

By Parseval's theorem, Eq. (C.8) below in its one-dimensional time form, the integral over ξ of the squared transform of a function is 2π times the integral of its squared modulus over time. The function here is one on $0 < s < t$ and zero elsewhere, so $\int \mathrm{d}\xi\, 4\sin^2(\xi t/2)/\xi^2 = 2\pi t$. The area grows linearly in t because the perturbation has acted for a time t.

Now the limit. Write $\sin^2(\xi t/2) = (1 - \cos \xi t)/2$ and split off $w(0)$:

$$\int \mathrm{d}\xi\, \frac{\sin^2(\xi t/2)}{\xi^2}\, w(\xi) = w(0)\, \frac{\pi t}{2} + \frac{1}{2}\int \mathrm{d}\xi\, \frac{w(\xi) - w(0)}{\xi^2} - \frac{1}{2}\int \mathrm{d}\xi\, \cos(\xi t)\, \frac{w(\xi) - w(0)}{\xi^2}.$$

The first term uses Eq. (C.2) with $u = \xi t/2$. In the other two, $w(\xi) - w(0) = w'(0)\xi + O(\xi^2)$ near the origin. The odd part $w'(0)\xi$ makes the integrand odd and singular as $1/\xi$, and the two sides cancel if the integral is taken symmetrically about $\xi = 0$; the remainder is integrable. The second term is then a constant, the Hadamard finite part of $\int \mathrm{d}\xi\, w(\xi)/\xi^2$ defined in Eq. (2.59). The third term tends to zero as $t \to \infty$: the integral of a rapidly oscillating cosine against an integrable function vanishes in the limit (the Riemann–Lebesgue lemma [1, Eq. (1.8.10)]), and the singular odd part contributes nothing by symmetry. Hence

$$\int \mathrm{d}\xi\, \frac{\sin^2(\xi t/2)}{\xi^2}\, w(\xi) = \frac{\pi t}{2}\, w(0) + \frac{1}{2}\, \mathrm{fp}\!\int \mathrm{d}\xi\, \frac{w(\xi)}{\xi^2} + o(1) \qquad (t \to \infty), \tag{C.3}$$

where $o(1)$ stands for terms that vanish as $t \to \infty$. Dividing by $\pi t/2$ proves Eq. (2.4). For a w that varies on an energy scale δ_w, the finite part is of size $w(0)/\delta_w$, so the relative correction is of order $1/(t\,\delta_w)$, the estimate quoted in Sect. 2.1. A constant w has finite part zero, and then the area $\pi t/2$ is exact at every t.

The time derivative of Eq. (2.3) brings in $\sin(\xi t)/\xi$, which has area π at every $t > 0$ by the Dirichlet integral. It is not absolutely integrable, so the dominated-convergence argument does not apply, but the same splitting does. With $w(\xi) = w(0) + [w(\xi) - w(0)]$, the first part gives $\pi w(0)$. The second gives $\int \mathrm{d}\xi\, \sin(\xi t)\, [w(\xi) - w(0)]/\xi$, whose integrand is a sine times a function that is finite at the origin. It vanishes as $t \to \infty$ by the Riemann–Lebesgue

lemma (at large $|\xi|$, where $w(0)/\xi$ is not integrable, the oscillation of the sine still makes the contribution vanish). Hence

$$\frac{\sin \xi t}{\xi} \;\longrightarrow\; \pi\, \delta(\xi) \qquad (t \to \infty). \tag{C.4}$$

The convergence is weaker than for $\sin^2(\xi t/2)/\xi^2$. The function changes sign, its absolute area grows as $\ln t$, and the limit holds only under an integral against a smooth function. This is why Sect. 2.1 reads the rate off the probability, whose kernel is positive, as well as off its derivative.

C.1.3 Delta functions of functions

Let $g(x)$ be smooth with simple roots x_i, that is $g(x_i) = 0$ and $g'(x_i) \neq 0$. Near each root $g(x) \approx g'(x_i)(x - x_i)$, and a delta function of g is supported only at the roots. Changing the variable to $y = g(x)$ in a small interval around x_i, with $\mathrm{d}x = \mathrm{d}y/|g'(x)|$ (the absolute value because the orientation of the interval is restored after the substitution),

$$\int_{x_i-\epsilon}^{x_i+\epsilon} \mathrm{d}x\, \delta(g(x))\, w(x) = \int \mathrm{d}y\, \delta(y)\, \frac{w(x(y))}{|g'(x(y))|} = \frac{w(x_i)}{|g'(x_i)|}.$$

Summing over the roots,

$$\delta(g(x)) = \sum_i \frac{\delta(x - x_i)}{|g'(x_i)|}. \tag{C.5}$$

Two cases appear in Chap. 2. The scaling rule $\delta(cx) = \delta(x)/|c|$ is the case of one root. Energy conservation in elastic scattering, $\delta(p'^2/2\mu_{\chi e} - p^2/2\mu_{\chi e})$ with $p, p' \geq 0$, has the root $p' = p$ with $g'(p) = p/\mu_{\chi e}$ and gives $(\mu_{\chi e}/p)\, \delta(p' - p)$, as used in Sect. 2.2. A double root, where $g'(x_i) = 0$, makes the right side infinite; it signals that the delta function must be kept inside a larger integral, as happens at the van Hove singularities of a density of states (Appendix C.2).

In three dimensions $\delta^3(\boldsymbol{x}) = \delta(x_1)\delta(x_2)\delta(x_3)$. The analogue of Eq. (C.5) for a scalar function of a vector is an integral over the surface on which it vanishes,

$$\int \mathrm{d}^3k\, \delta(g(\boldsymbol{k}))\, w(\boldsymbol{k}) = \int_{g(\boldsymbol{k})=0} \frac{\mathrm{d}S}{|\nabla g(\boldsymbol{k})|}\, w(\boldsymbol{k}), \tag{C.6}$$

which follows from Eq. (C.5) applied along the direction normal to the surface at each of its points, with $\mathrm{d}S$ the element of area. Appendix C.2 uses it for densities of states.

C.1.4 Fourier identities

The transforms of Eqs. (A.6) and (A.7) are inverse to each other because the plane waves are complete,

$$\int_{-\infty}^{\infty} \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} = 2\pi\, \delta(\omega), \qquad \int \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = (2\pi)^3\, \delta^3(\boldsymbol{q}). \tag{C.7}$$

The first is the golden-rule limit once more. The integral over a finite window, $\int_{-T/2}^{T/2} \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} = 2\sin(\omega T/2)/\omega$, tends to $2\pi\delta(\omega)$ as $T \to \infty$ by Eq. (C.4) with $\xi = \omega$ and $t = T/2$. The second is the product of three copies of the first. Inserting the inverse transform of $O(\boldsymbol{q})$ into the forward transform of Eq. (A.6) and using Eq. (C.7) returns $O(\boldsymbol{q})$, which proves that the two transforms are inverse to each other. The transform of $\delta^3(\boldsymbol{r} - \boldsymbol{r}_0)$ is $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_0}$, which is how the density operator Eq. (A.8) arises from point particles, and the transform of a derivative is $\int \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \nabla O = -\mathrm{i}\boldsymbol{q}\, O(\boldsymbol{q})$ after an integration by parts. Hence $-\nabla^2 \to q^2$, the step that solves the static field equations of Appendix C.9.

Parseval's theorem and the convolution theorem follow from Eq. (C.7). Inserting the inverse transforms of f and g into $\int \mathrm{d}^3r\, f^*(\boldsymbol{r})\, g(\boldsymbol{r})$ and doing the $\boldsymbol{r}$ integral first gives Parseval's theorem,

$$\int \mathrm{d}^3r\, f^*(\boldsymbol{r})\, g(\boldsymbol{r}) = \int \frac{\mathrm{d}^3q}{(2\pi)^3}\, f^*(\boldsymbol{q})\, g(\boldsymbol{q}), \tag{C.8}$$

and in one dimension of time $\int \mathrm{d}t\, f^*(t)g(t) = \int \mathrm{d}\omega\, f^*(\omega)g(\omega)/2\pi$. Let $(f*g)(\boldsymbol{r}) = \int \mathrm{d}^3r'\, f(\boldsymbol{r}-\boldsymbol{r}')\, g(\boldsymbol{r}')$ be the convolution. Writing $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{r}')}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}'}$ and shifting the variable of the outer integral gives the convolution theorem,

$$(f*g)(\boldsymbol{q}) = f(\boldsymbol{q})\, g(\boldsymbol{q}). \tag{C.9}$$

The Helm form factor of Sect. 2.2 is an example: a uniform sphere smeared by a Gaussian is the convolution of the two densities, and its transform is the product of $3j_1(qR_{\mathrm{nuc}})/(qR_{\mathrm{nuc}})$ (Appendix C.3) and the Gaussian factor $\mathrm{e}^{-q^2 s_{\mathrm{skin}}^2/2}$ (Appendix C.6).

The time form of Eq. (C.9) explains the structure of response functions. A causal response is a convolution in time of a response kernel with the applied field, and in frequency it becomes a product. The kernel vanishes for negative times, and its transform $\int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}(\cdots)$ converges only with a factor e^{-0^+t}, a decay slower than any physical time scale. For a kernel that oscillates at a single transition energy $\Delta E = E_f - E_i$, as in Appendix A.3,

$$-\mathrm{i}\int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}(\omega-\Delta E)t-0^+t} = \frac{1}{\omega - \Delta E + \mathrm{i}0^+}, \tag{C.10}$$

which is the origin of the infinitesimal $\mathrm{i}0^+$ in every energy denominator of this book (Appendix A.3 and Sect. 2.4). Appendix C.4 gives its meaning as a distribution and Appendix C.5 the analyticity it implies.

C.2 Box normalization and Brillouin-zone sums

A crystal of 10^{23} cells has 10^{23} allowed crystal momenta per band, and no calculation sums over them one by one. Every rate in this book replaces such a sum by an integral, Eqs. (2.10), (A.10) and (A.11), and every first-principles calculation then replaces the integral by a sum again, over a grid of perhaps 10^3 to 10^6 points. This section says when each replacement is accurate and how large its error is. The tool for both is the Poisson summation formula, which also gives the infinite-crystal form of the lattice sum of Sect. 3.1.

C.2.1 From sums to integrals

In a periodic box of side L the allowed momenta in one dimension are $p = 2\pi n/L$ with integer n. The replacement of Eq. (2.10) is $\sum_n g(2\pi n/L) \to (L/2\pi)\int \mathrm{d}p\, g(p)$. Its exact form follows from the Dirac comb. The function $\sum_n \delta(y-n)$ is periodic in y with period one, and its Fourier coefficients over one period are $\int_{-1/2}^{1/2} \mathrm{d}y\, \delta(y)\, \mathrm{e}^{-2\pi\mathrm{i}my} = 1$ for every integer m, so

$$\sum_{n=-\infty}^{\infty} \delta(y-n) = \sum_{m=-\infty}^{\infty} \mathrm{e}^{2\pi\mathrm{i}my}. \tag{C.11}$$

Multiplying by $\mathcal{F}(y)$ and integrating gives the Poisson summation formula, $\sum_n \mathcal{F}(n) = \sum_m \int \mathrm{d}y\, \mathcal{F}(y)\, \mathrm{e}^{2\pi\mathrm{i}my}$ [1, Eq. (1.8.14)]. With $\mathcal{F}(y) = g(2\pi y/L)$ and $p = 2\pi y/L$,

$$\sum_p g(p) = \frac{L}{2\pi} \sum_{m=-\infty}^{\infty} \tilde{g}(mL), \qquad \tilde{g}(x) \equiv \int \mathrm{d}p\, \mathrm{e}^{\mathrm{i}px}\, g(p). \tag{C.12}$$

The term $m = 0$ is the integral. The others are the Fourier transform of g evaluated at multiples of the box size: in position space they are the contributions of the periodic images of the system, which the boundary conditions put at distances mL. If g varies on a momentum scale Δp, its transform is confined to $|x| \lesssim 1/\Delta p$, and the images do not matter once $L\,\Delta p \gg 1$, which is the condition stated in Sect. 2.1.

How small the remainder is depends on the smoothness of g. A Gaussian of width Δp has $\tilde{g}(L) \propto \mathrm{e}^{-L^2\Delta p^2/2}$, and the error is exponentially small. A function with a jump, such as the occupation of a Fermi sea with its step at k_F, has a transform that falls off only as $1/x$, and the error then decreases only as a power of $1/(L\,\Delta p)$. In three dimensions the comb is a product of three combs, the images sit on the lattice of vectors $L\boldsymbol{n}$, and Eq. (C.12) holds with $L/2\pi \to \mathcal{V}/(2\pi)^3$ and a sum over all image vectors. A delta function in g, such as the energy delta of the golden rule, has no meaning on a discrete set of momenta. It must first be broadened by more than the level spacing, as Sect. 2.1 required of the time window, and the limit $\mathcal{V} \to \infty$ is taken before the broadening is removed.

The density of final states is the main application. With Eq. (C.6), the number of states per unit energy of a band $\varepsilon(\boldsymbol{p})$ is

$$\mathcal{D}_f(E) = \sum_{\boldsymbol{p}} \delta\big(E - \varepsilon(\boldsymbol{p})\big) = \mathcal{V} \int \frac{\mathrm{d}^3 p}{(2\pi)^3}\, \delta\big(E - \varepsilon(\boldsymbol{p})\big) = \frac{\mathcal{V}}{(2\pi)^3} \int_{\varepsilon(\boldsymbol{p})=E} \frac{\mathrm{d}S}{|\nabla_{\boldsymbol{p}}\varepsilon|}, \tag{C.13}$$

per spin state. For a free particle the surface is a sphere of area $4\pi p^2$ with $|\nabla_{\boldsymbol{p}}\varepsilon| = p/m$, which gives Eq. (2.11). In a crystal $\nabla_{\boldsymbol{k}}\varepsilon_{n\boldsymbol{k}}$ vanishes at points of every band, by periodicity; there the integrand of Eq. (C.13) diverges. In three dimensions the divergence is integrable and produces square-root kinks in the density of states, the van Hove singularities; in two dimensions it produces a logarithm (Appendix C.8).

C.2.2 The lattice sum

On the Born–von Karman grid of crystal momenta of a block of N cells, Eq. (3.13), the geometric sums of Sect. 3.1 give the lattice sum

$$\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} = N \sum_{\boldsymbol{G}} \delta_{\boldsymbol{k},\boldsymbol{G}}, \tag{C.14}$$

which is Eq. (3.14), together with the count of N crystal momenta in a zone of volume $(2\pi)^3/\Omega$, one per cell. In an infinite crystal, with $\delta_{\boldsymbol{kG}} \to [(2\pi)^3/\mathcal{V}]\,\delta^3(\boldsymbol{k}-\boldsymbol{G})$ by Eq. (A.10) and $\mathcal{V} = N\Omega$, it becomes $\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} = [(2\pi)^3/\Omega] \sum_{\boldsymbol{G}} \delta^3(\boldsymbol{k}-\boldsymbol{G})$, the three-dimensional form of Eq. (C.11).

A sum over all momenta in the box decomposes as $\boldsymbol{q} = \boldsymbol{k} + \boldsymbol{G}$ with $\boldsymbol{k}$ in the zone,

$$\sum_{\boldsymbol{q}} \mathcal{F}(\boldsymbol{q}) = \sum_{\boldsymbol{k}\in\mathrm{BZ}} \sum_{\boldsymbol{G}} \mathcal{F}(\boldsymbol{k}+\boldsymbol{G}) \;\longrightarrow\; \mathcal{V} \int_{\mathrm{BZ}} \frac{\mathrm{d}^3 k}{(2\pi)^3} \sum_{\boldsymbol{G}} \mathcal{F}(\boldsymbol{k}+\boldsymbol{G}), \tag{C.15}$$

which is how an integral over the unreduced momentum transfer $\boldsymbol{q}$ of the dark matter is matched to quantities indexed by $\boldsymbol{q}_r$ and $\boldsymbol{G}_q$ (Appendix A.3).

C.2.3 Zone averages and grids

The zone average of a function of crystal momentum is $\langle\mathcal{F}\rangle_\mathrm{BZ} = N^{-1}\sum_{\boldsymbol{k}} \mathcal{F}(\boldsymbol{k})$, which by Eq. (A.11) equals $\Omega \int_\mathrm{BZ} \mathrm{d}^3k\, \mathcal{F}(\boldsymbol{k})/(2\pi)^3$. A band energy, a squared matrix element or an occupation is periodic in $\boldsymbol{k}$, $\mathcal{F}(\boldsymbol{k}+\boldsymbol{G}) = \mathcal{F}(\boldsymbol{k})$. It therefore has a Fourier series over the direct lattice,

$$\mathcal{F}(\boldsymbol{k}) = \sum_{\boldsymbol{R}} \mathcal{F}_{\boldsymbol{R}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}, \qquad \langle\mathcal{F}\rangle_\mathrm{BZ} = \mathcal{F}_{\boldsymbol{R}=0},$$

where the zone average picks out $\boldsymbol{R} = 0$ by Eq. (C.14) read with the roles of $\boldsymbol{k}$ and $\boldsymbol{R}$ exchanged. A calculation evaluates $\mathcal{F}$ on a uniform grid of $n_k = n_1 n_2 n_3$ points, the allowed momenta of a fictitious block of n_k cells. By the same identity, the grid average of $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}$ is one when $\boldsymbol{R}$ is a lattice vector of that block, $\boldsymbol{R} = \sum_i l_i n_i \boldsymbol{a}_i$ with integer l_i, and zero otherwise. The grid average therefore equals

$$\frac{1}{n_k} \sum_{\boldsymbol{k}\in\text{grid}} \mathcal{F}(\boldsymbol{k}) = \langle \mathcal{F} \rangle_{\text{BZ}} + \sum_{(l_1,l_2,l_3)\neq 0} \mathcal{F}_{\boldsymbol{R}=\sum_i l_i n_i \boldsymbol{a}_i}. \tag{C.16}$$

The error of a grid is the set of Fourier components of the integrand at distances equal to the size of the fictitious block. It is the crystal version of the image terms of Eq. (C.12), and it is small when $\mathcal{F}_{\boldsymbol{R}}$ has decayed over n_i cells.

In an insulator, the quantities summed over occupied bands are analytic functions of $\boldsymbol{k}$, their components $\mathcal{F}_{\boldsymbol{R}}$ decay exponentially with $|\boldsymbol{R}|$, on the scale of the localization length of the Wannier functions (Sect. 3.6), and the grid average converges exponentially in n_i. In a metal, the occupation jumps at the Fermi surface, $\mathcal{F}_{\boldsymbol{R}}$ decays only as a power of $|\boldsymbol{R}|$, and convergence is slow; a smearing of the step, or an interpolation, is needed. Grids displaced from the origin, introduced by Monkhorst and Pack [2], have the same error structure, Eq. (C.16) with a phase on each term (a sign alternating with l_i for a shift of half a grid step), and need fewer symmetry-distinct points. These choices, smearing and the convergence tests are the subject of Appendix E.1.

Energy delta functions on a grid need one more step. The transition sum $N^{-1}\sum_{\boldsymbol{k}} \delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}})$ on n_k points is a set of spikes, not a function of ω. The spikes merge into a curve only after each is broadened by a width σ_{br} larger than the change of the transition energy between neighboring grid points, about $|\nabla_{\boldsymbol{k}}(\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}})|\,|\boldsymbol{b}_i|/n_i$. A Gaussian broadening is preferred to a Lorentzian of the same width, because its error is of order σ_{br}^2 times the curvature of the spectrum, while the slow tails of a Lorentzian give an error of order σ_{br} that depends on the whole spectrum (Appendices C.1 and C.4). The width must then be reduced together with the grid spacing. The alternative is the tetrahedron method, which interpolates the band energies linearly inside small tetrahedra filling the zone and integrates the delta function exactly within each [3]. Either way, a computed spectrum carries a resolution set by the grid, and features narrower than it, such as the onset of absorption at a gap, need a denser grid or an analytic treatment (Sect. 3.3).

C.2.4 Normalization checks

Counting powers of N and $\mathcal{V}$ catches most normalization errors. Each sum over an independent crystal momentum brings a factor N, or $\mathcal{V}/(2\pi)^3$ times an integral, and each plane wave or Bloch state normalized over the crystal brings $\mathcal{V}^{-1/2}$; an intensive quantity must have the powers of N cancel. Each Kronecker delta in crystal momentum that a matrix element produces removes one sum. In the Bloch basis, the product of two lattice sums, Eq. (C.14) for $\boldsymbol{k}$ and again for $\boldsymbol{k}'$, gives $N^2 \delta_{\boldsymbol{k}\boldsymbol{G}} \delta_{\boldsymbol{k}'\boldsymbol{G}'}$, and the square of one Kronecker delta is the same Kronecker delta. In the continuum, the square of $\delta^3(\boldsymbol{k})$ is $\delta^3(0)\,\delta^3(\boldsymbol{k})$ with $\delta^3(0) = \mathcal{V}/(2\pi)^3$, which is the same statement and the reason why rates are worked out in the box (Sect. 2.1).

C.3 Angular averages and spherical functions

A dark-matter rate integrates over the direction of the momentum transfer, and an atomic form factor integrates over the direction of the electron. Sect. 2.2 needed three such integrals: the average of a plane wave over directions, $4\pi \sin(qr)/(qr)$; its first moment, $4\pi \mathrm{i}\, j_1(qr)$; and

the sum rule $\sum_m |Y_{lm}|^2 = (2l+1)/4\pi$ that makes a closed shell spherical. They are special cases of the expansion of a plane wave in spherical harmonics, which is derived here in full. The section closes with the cubic harmonics, the functions of direction that a cubic crystal such as silicon allows.

C.3.1 Plane waves averaged over directions

We write $\hat{\boldsymbol{n}}$ for a unit vector, $\mathrm{d}\Omega_{\hat{n}}$ for the element of solid angle of its direction, and $\int \mathrm{d}\Omega_{\hat{n}} = 4\pi$. With the polar axis along $\boldsymbol{q}$ and $u = \cos\theta$ the cosine of the angle between $\boldsymbol{q}$ and $\hat{\boldsymbol{n}}$, $\mathrm{d}\Omega_{\hat{n}} = \mathrm{d}\phi\,\mathrm{d}u$ and the azimuth gives 2π. For $\boldsymbol{r} = r\hat{\boldsymbol{n}}$,

$$\int \mathrm{d}\Omega_{\hat{n}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = 2\pi \int_{-1}^{1} \mathrm{d}u\, \mathrm{e}^{\mathrm{i}qru} = 4\pi\, \frac{\sin qr}{qr} = 4\pi\, j_0(qr). \tag{C.17}$$

The first moment follows by differentiating with respect to $a = qr$ under the integral, $\int_{-1}^{1} \mathrm{d}u\, u\, \mathrm{e}^{\mathrm{i}au} = -\mathrm{i}\,\mathrm{d}(2\sin a/a)/\mathrm{d}a = 2\mathrm{i}(\sin a - a\cos a)/a^2$, so

$$2\pi \int_{-1}^{1} \mathrm{d}u\, u\, \mathrm{e}^{\mathrm{i}qru} = 4\pi\mathrm{i}\, j_1(qr), \qquad j_1(z) = \frac{\sin z}{z^2} - \frac{\cos z}{z}. \tag{C.18}$$

These are the spherical Bessel functions of orders zero and one [1, Eq. (10.49.3)]. A spherical density $\boldsymbol{n}(r)$ therefore has the transform $4\pi \int_0^\infty \mathrm{d}r\, r^2 j_0(qr)\, \boldsymbol{n}(r)$, a one-dimensional integral; for a density proportional to $\mathrm{e}^{-\beta r}$ the remaining integrals are Laplace transforms of powers times sin and cos. For instance, $\int_0^\infty \mathrm{d}r\, r\, \mathrm{e}^{-\beta r} \sin qr = 2\beta q/(\beta^2+q^2)^2$ and $\int_0^\infty \mathrm{d}r\, r^2 \mathrm{e}^{-\beta r} \cos qr = 2\beta(\beta^2 - 3q^2)/(\beta^2+q^2)^3$. With Eq. (C.18) they combine into

$$\int_0^\infty \mathrm{d}r\, r^3 \mathrm{e}^{-\beta r} j_1(qr) = \frac{1}{q^2}\, \frac{2\beta q}{(\beta^2+q^2)^2} - \frac{1}{q}\, \frac{2\beta(\beta^2-3q^2)}{(\beta^2+q^2)^3} = \frac{8\beta q}{(\beta^2+q^2)^3},$$

the radial integral of the hydrogen $1s \to 2p$ form factor in Sect. 2.2.

Averages of products of components of $\hat{\boldsymbol{n}}$ follow from symmetry. We write n^α for the Cartesian components, with $\alpha, \beta, \gamma, \delta$ as superscripts. The average $\langle n^\alpha n^\beta \rangle = (4\pi)^{-1} \int \mathrm{d}\Omega_{\hat{n}}\, n^\alpha n^\beta$ is a tensor that does not change under rotations, so it is proportional to $\delta^{\alpha\beta}$; contracting with $\delta^{\alpha\beta}$ and using $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}} = 1$ fixes the constant. In the same way the fourth moment is a combination of the three products of Kronecker deltas, with equal weights by the symmetry in its indices. Contracting two indices reduces it to the second moment:

$$\langle n^\alpha n^\beta \rangle = \tfrac{1}{3}\delta^{\alpha\beta}, \qquad \langle n^\alpha n^\beta n^\gamma n^\delta \rangle = \tfrac{1}{15}\big(\delta^{\alpha\beta}\delta^{\gamma\delta} + \delta^{\alpha\gamma}\delta^{\beta\delta} + \delta^{\alpha\delta}\delta^{\beta\gamma}\big). \tag{C.19}$$

Odd moments vanish, since $\hat{\boldsymbol{n}} \to -\hat{\boldsymbol{n}}$ leaves the measure unchanged. As a result, the direction average of $(\boldsymbol{q}\cdot\boldsymbol{v})^2$ is $q^2v^2/3$, and that of a quadratic form $\boldsymbol{q}\cdot B\cdot\boldsymbol{q}$ is $q^2\,\mathrm{Tr}\,B/3$. In general, $\langle (n^x)^{2a} (n^y)^{2b} (n^z)^{2c} \rangle = (2a-1)!!\,(2b-1)!!\,(2c-1)!!/(2a+2b+2c+1)!!$, with $(-1)!! = 1$; it gives $\langle (n^x)^4 \rangle = 1/5$, in agreement with Eq. (C.19).

C.3.2 Spherical harmonics

The spherical harmonics $Y_{lm}(\hat{\boldsymbol{n}})$, with $l = 0, 1, 2, \ldots$ and $m = -l, \ldots, l$, are the eigenfunctions of the squared angular momentum with eigenvalue $l(l+1)$ and of its z component with eigenvalue m. We use the convention, with the Condon–Shortley phase, of most quantum-mechanics texts [1, Eq. (14.30.1)]: $Y_{00} = (4\pi)^{-1/2}$, $Y_{10} = (3/4\pi)^{1/2}\cos\theta$, $Y_{1,\pm1} = \mp(3/8\pi)^{1/2}\sin\theta\,\mathrm{e}^{\pm\mathrm{i}\phi}$, and $Y_{l,-m} = (-1)^m Y_{lm}^*$. They are orthonormal on the sphere and complete,

$$\int \mathrm{d}\Omega_{\hat{n}}\, Y_{lm}^*(\hat{\boldsymbol{n}})\, Y_{l'm'}(\hat{\boldsymbol{n}}) = \delta_{ll'}\delta_{mm'}, \tag{C.20}$$

so any function of direction expands as $\sum_{lm} c_{lm} Y_{lm}$ [1, Eq. (14.30.8)]. The harmonics of order l are the restrictions to the sphere of the homogeneous polynomials of degree l in n^x, n^y, n^z that solve Laplace's equation, and under inversion $\hat{\boldsymbol{n}} \to -\hat{\boldsymbol{n}}$ they change by $(-1)^l$.

The $m = 0$ harmonics are Legendre polynomials, $Y_{l0} = [(2l+1)/4\pi]^{1/2} P_l(\cos\theta)$, with Rodrigues' formula $P_l(u) = (2^l l!)^{-1}\, \mathrm{d}^l (u^2-1)^l / \mathrm{d}u^l$ and $\int_{-1}^{1} \mathrm{d}u\, P_l P_{l'} = 2\delta_{ll'}/(2l+1)$. The addition theorem connects them to the full set [1, Eq. (14.30.9)],

$$P_l(\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}}') = \frac{4\pi}{2l+1} \sum_{m=-l}^{l} Y_{lm}^*(\hat{\boldsymbol{n}})\, Y_{lm}(\hat{\boldsymbol{n}}'). \tag{C.21}$$

Setting $\hat{\boldsymbol{n}}' = \hat{\boldsymbol{n}}$ and using $P_l(1) = 1$ gives Unsöld's theorem,

$$\sum_{m=-l}^{l} |Y_{lm}(\hat{\boldsymbol{n}})|^2 = \frac{2l+1}{4\pi}, \tag{C.22}$$

independent of direction. A filled shell, with every m occupied equally, therefore has a spherical density whatever the shape of its individual orbitals, which is the statement used in Sect. 2.2.

C.3.3 The plane-wave expansion

The function $\mathrm{e}^{\mathrm{i}zu}$ of $u \in [-1,1]$ expands in Legendre polynomials as $\sum_l c_l(z) P_l(u)$, with $c_l(z) = \frac{1}{2}(2l+1)\int_{-1}^{1} \mathrm{d}u\, P_l(u)\, \mathrm{e}^{\mathrm{i}zu}$ by their orthogonality. We evaluate the integral with Rodrigues' formula. Integrating by parts l times moves the derivatives onto the exponential, and the boundary terms vanish because $(u^2-1)^l$ has zeros of order l at $u = \pm 1$:

$$\int_{-1}^{1} \mathrm{d}u\, P_l(u)\, \mathrm{e}^{\mathrm{i}zu} = \frac{(-\mathrm{i}z)^l}{2^l l!} \int_{-1}^{1} \mathrm{d}u\, (u^2-1)^l \mathrm{e}^{\mathrm{i}zu} = \frac{(\mathrm{i}z)^l}{2^l l!}\, I_l(z), \qquad I_l(z) \equiv \int_{-1}^{1} \mathrm{d}u\, (1-u^2)^l \mathrm{e}^{\mathrm{i}zu}.$$

The integrals I_l obey a recursion. Differentiating under the integral and integrating by parts once, with $u(1-u^2)^l = -\mathrm{d}(1-u^2)^{l+1}/\mathrm{d}u\,/[2(l+1)]$,

$$\frac{\mathrm{d}I_l}{\mathrm{d}z} = \mathrm{i}\int_{-1}^{1} \mathrm{d}u\, u(1-u^2)^l \mathrm{e}^{\mathrm{i}zu} = \frac{\mathrm{i}\cdot\mathrm{i}z}{2(l+1)}\, I_{l+1}(z) = -\frac{z}{2(l+1)}\, I_{l+1}(z).$$

Starting from $I_0 = 2\sin z/z$ and iterating $I_{l+1} = -2(l+1)\, z^{-1}\, \mathrm{d}I_l/\mathrm{d}z$ gives

$$I_l(z) = 2^{l+1} l! \left(-\frac{1}{z}\frac{\mathrm{d}}{\mathrm{d}z}\right)^l \frac{\sin z}{z} = 2^{l+1} l!\, z^{-l} j_l(z),$$

where the second step uses $j_l(z) = z^l(-z^{-1}\mathrm{d}/\mathrm{d}z)^l(\sin z/z)$ [1, Eq. (10.49.14)]. Therefore $\int_{-1}^{1} \mathrm{d}u\, P_l\, \mathrm{e}^{\mathrm{i}zu} = 2\mathrm{i}^l j_l(z)$. Hence $c_l = (2l+1)\,\mathrm{i}^l j_l(z)$, and with $z = qr$, $u = \hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{r}}$ and Eq. (C.21),

$$\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = \sum_{l=0}^{\infty} (2l+1)\,\mathrm{i}^l j_l(qr)\, P_l(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{r}}) = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \mathrm{i}^l j_l(qr)\, Y_{lm}^*(\hat{\boldsymbol{q}})\, Y_{lm}(\hat{\boldsymbol{r}}), \tag{C.23}$$

the Rayleigh expansion [1, Eq. (10.60.7)]. Integrating over the directions of $\hat{\boldsymbol{r}}$ keeps only $l = 0$ and returns Eq. (C.17); multiplying by Y_{10} before integrating keeps only $l = 1$ and returns Eq. (C.18).

The expansion is the multipole expansion of a form factor. A transition between orbitals of angular momenta l_i and l_f, through $\langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle$, picks out the terms with $|l_i - l_f| \le l \le l_i + l_f$, and $l_i + l + l_f$ even by parity. The spherical Bessel functions behave as

$$j_l(z) \simeq \frac{z^l}{(2l+1)!!} \quad (z \ll 1), \qquad j_l(z) \simeq \frac{\sin(z - l\pi/2)}{z} \quad (z \gg l), \tag{C.24}$$

[1, Eqs. (10.52.1) and (10.52.3)]. For an orbital of size $r_{\rm orb}$, the multipole l of the form factor is therefore suppressed by $(qr_{\rm orb})^l$ when $qr_{\rm orb} \ll 1$, and the lowest allowed multipole dominates: the monopole for elastic scattering, the dipole for $s \to p$ transitions. When $qr_{\rm orb} \gtrsim 1$ many multipoles contribute with comparable weight, and the transition sees the electron as a particle rather than as a charge distribution. For an atomic valence orbital, $r_{\rm orb}$ is about an Ångström, and $qr_{\rm orb} = 1$ at $q \simeq 2\,\text{keV}$, which is $\sim q_0/2$ (Sect. 1.3); light dark matter sits in the transition region.

C.3.4 Cubic harmonics

A crystal is not spherical, but a cubic crystal comes close. The point group of silicon and germanium, O_h, contains all the rotations and reflections that map a cube onto itself, and the response of such a crystal at fixed $|\boldsymbol{q}|$ is a function of $\hat{\boldsymbol{q}}$ invariant under them. The point group of gallium arsenide, T_d, has only half of these operations and lacks inversion. A response that is even in $\boldsymbol{q}$, such as $S(\boldsymbol{q},\omega)$ in a target with time-reversal symmetry (Appendix A.4), is invariant under $\hat{\boldsymbol{q}} \to -\hat{\boldsymbol{q}}$ all the same, and T_d together with this inversion is O_h. The functions of direction we need are then those invariant under all permutations of (n^x, n^y, n^z) and all changes of their signs.

Which orders l admit such functions? A homogeneous polynomial invariant under the sign changes contains only even powers of each component, so odd l is absent. At degree two, invariance under permutations leaves $(n^x)^2 + (n^y)^2 + (n^z)^2 = 1$, which is the $l = 0$ harmonic, so $l = 2$ is absent too. At degree four the invariants are $\sum_\alpha (n^\alpha)^4$ and 1, and the combination orthogonal to the constant is new. At degree six there is one more, built from $(n^x n^y n^z)^2$. The first cubic harmonics are therefore

$$Y_0^{\rm c} = \frac{1}{\sqrt{4\pi}}, \qquad Y_4^{\rm c}(\hat{\boldsymbol{n}}) = \frac{5}{8}\sqrt{\frac{21}{\pi}}\Big[(n^x)^4 + (n^y)^4 + (n^z)^4 - \frac{3}{5}\Big], \qquad \ldots, \tag{C.25}$$

normalized as in Eq. (C.20). The constant $-3/5$ makes $Y_4^{\rm c}$ orthogonal to $Y_0^{\rm c}$, since $\langle \sum_\alpha (n^\alpha)^4 \rangle = 3/5$. The prefactor follows from the moment formula above: $\langle (\sum_\alpha (n^\alpha)^4)^2 \rangle = 3\langle (n^x)^8 \rangle + 6\langle (n^x)^4 (n^y)^4 \rangle = 3/9 + 6/105 = 41/105$, the variance is $41/105 - 9/25 = 16/525$, and $4\pi \times (5/8)^2 (21/\pi) \times 16/525 = 1$. In spherical harmonics with the polar axis along a cube edge, $Y_4^{\rm c} = (7/12)^{1/2} Y_{40} + (5/24)^{1/2}(Y_{44} + Y_{4,-4})$, the combination given by Von der Lage and Bethe [4], who introduced these functions for electronic states in cubic crystals. The next invariant orders are $l = 6, 8, 10$ with one function each, and $l = 12$ with two.

Consider a property of a cubic crystal that is a second-rank tensor in $\hat{\boldsymbol{q}}$. Examples are the dielectric tensor at $q \to 0$, the displacement tensor B_κ at a site of cubic symmetry (Appendix A.6) or the effective-mass tensor of a nondegenerate band extremum at Γ. Each is a multiple of the unit tensor, since an anisotropy would need an $l = 2$ component, and none exists. The direction dependence of a cubic response at fixed $|\boldsymbol{q}|$ starts at $l = 4$. In silicon the anisotropy of $S(\boldsymbol{q},\omega)$ is a combination of $Y_4^{\rm c}(\hat{\boldsymbol{q}})$, $Y_6^{\rm c}(\hat{\boldsymbol{q}})$ and higher terms, with coefficients that depend on $|\boldsymbol{q}|$ and ω, and it is small where the momentum transfer is small compared with the reciprocal-lattice vectors. Sect. 11.7 expands rates in these functions to compute the daily modulation of Sect. 1.2 for cubic targets.

C.4 Singular denominators

An energy denominator $1/(E_i - E_m + \mathrm{i}0^+)$ is the transform of a causal time evolution, Eq. (C.10), and it is not an ordinary function at $E_m = E_i$. Sect. 2.4 split it into a principal value and a delta function, Eq. (2.58), by taking the limit of a finite width. It also gave the squared denominator a meaning through the Hadamard finite part, Eq. (2.59), with the

expansion Eq. (2.60) of a Lorentzian in powers of its width. Here we prove the splitting for any smooth test function, collect the rules for manipulating these distributions, and give the correction terms of the Lorentzian limits, which control how a broadened spectrum approaches its zero-width limit.

C.4.1 The Sokhotski–Plemelj formula

Let $w(x)$ be a test function, bounded, integrable and differentiable at $x = 0$; a Lipschitz condition $|w(x) - w(0)| \le c|x|$ near the origin is enough. The principal value is the symmetric limit

$$\mathrm{PV}\int \mathrm{d}x\, \frac{w(x)}{x} = \lim_{x_\mathrm{c}\to 0} \int_{|x|>x_\mathrm{c}} \mathrm{d}x\, \frac{w(x)}{x} = \int \mathrm{d}x\, \frac{w(x) - w(0)\,\Theta(X_\mathrm{c} - |x|)}{x}, \tag{C.26}$$

where the second form holds for any fixed $X_\mathrm{c} > 0$. On the symmetric interval $|x| < X_\mathrm{c}$ the subtracted term integrates to zero because $1/x$ is odd, and the remaining integrand is bounded near the origin by the Lipschitz condition. The second form shows that the principal value is finite.

For a finite width $\Gamma_\mathrm{w} > 0$, the real and imaginary parts of $1/(x + \mathrm{i}\Gamma_\mathrm{w}/2)$ act on w separately. The imaginary part is $-\pi$ times the normalized Lorentzian $(\Gamma_\mathrm{w}/2\pi)/(x^2 + \Gamma_\mathrm{w}^2/4)$, which is the squeezed family $t\,f(tx)$ of Appendix C.1 with $f(u) = \pi^{-1}/(1 + u^2)$ and $t = 2/\Gamma_\mathrm{w}$. Its integral against w tends to $w(0)$ by the dominated-convergence argument given there. The real part is $x/(x^2 + \Gamma_\mathrm{w}^2/4)$. Subtracting $w(0)\,\Theta(X_\mathrm{c} - |x|)$, which integrates to zero against this odd kernel,

$$\int \mathrm{d}x\, \frac{x\,w(x)}{x^2 + \Gamma_\mathrm{w}^2/4} = \int \mathrm{d}x\, \frac{x^2}{x^2 + \Gamma_\mathrm{w}^2/4}\, \frac{w(x) - w(0)\,\Theta(X_\mathrm{c} - |x|)}{x}.$$

The first factor lies between 0 and 1 and tends to 1 at every $x \neq 0$; the second is integrable and independent of Γ_w. By dominated convergence the integral tends to the second form of Eq. (C.26). Together,

$$\lim_{\Gamma_\mathrm{w}\to 0^+} \int \mathrm{d}x\, \frac{w(x)}{x - x_0 \pm \mathrm{i}\Gamma_\mathrm{w}/2} = \mathrm{PV}\int \mathrm{d}x\, \frac{w(x)}{x - x_0} \mp \mathrm{i}\pi\, w(x_0), \tag{C.27}$$

after a shift of the variable by x_0. With the lower sign this is the complex conjugate statement. Written as distributions, $1/(x \pm \mathrm{i}0^+) = \mathrm{PV}(1/x) \mp \mathrm{i}\pi\delta(x)$, the Sokhotski–Plemelj formula, Eq. (2.58). The proof used only that w is integrable and Lipschitz at x_0; a sum over a dense set of final states, as in the level shift of Eq. (2.55), satisfies it once the levels are replaced by a smooth density of states (Appendix C.2).

The formula has an equivalent form for functions of a complex variable. Let

$$\mathcal{F}(z) = \frac{1}{2\pi\mathrm{i}} \int \mathrm{d}x\, \frac{w(x)}{x - z}, \tag{C.28}$$

a Cauchy integral, which is analytic in z above and below the real axis. Approaching a point x_0 of the axis from above, $z = x_0 + \mathrm{i}0^+$, and from below, Eq. (C.27) gives

$$\mathcal{F}(x_0 + \mathrm{i}0^+) - \mathcal{F}(x_0 - \mathrm{i}0^+) = w(x_0), \qquad \mathcal{F}(x_0 + \mathrm{i}0^+) + \mathcal{F}(x_0 - \mathrm{i}0^+) = \frac{1}{\pi\mathrm{i}}\,\mathrm{PV}\int \mathrm{d}x\, \frac{w(x)}{x - x_0}. \tag{C.29}$$

The jump across the axis is the density w, and the average of the two sides is its principal-value transform. Every response function of the book has this structure: its imaginary part on the real axis is the density of transitions, and its real part is a principal-value integral over that density. Appendix C.5 draws the consequences, the Kramers–Kronig relations.

C.4.2 Rules for the distributions

Since a distribution is defined by its action on test functions, identities among them are proved by integrating against w.

The product of $\delta(x)$ with a smooth function $h(x)$ is $h(0)\delta(x)$; in particular $x\,\delta(x) = 0$ and $x\,\mathrm{PV}(1/x) = 1$, since $\mathrm{PV}\int \mathrm{d}x\, x\, w(x)/x = \int \mathrm{d}x\, w(x)$. The difference of the two boundary values is a delta function,

$$\frac{1}{x - \mathrm{i}0^+} - \frac{1}{x + \mathrm{i}0^+} = 2\pi\mathrm{i}\,\delta(x), \tag{C.30}$$

which is Eq. (C.27) with both signs. Derivatives act by moving them onto the test function, $\int \mathrm{d}x\, T'(x)\, w(x) = -\int \mathrm{d}x\, T(x)\, w'(x)$, the definition that makes integration by parts valid for distributions [1, Eq. (1.16.8)]. Hence $\mathrm{d}\ln|x|/\mathrm{d}x = \mathrm{PV}(1/x)$, as follows from $\int \mathrm{d}x\, \ln|x|\, w'(x)$ split at $\pm x_\mathrm{c}$, where the boundary terms $\ln x_\mathrm{c}\,[w(x_\mathrm{c}) - w(-x_\mathrm{c})]$ vanish as $x_\mathrm{c} \to 0$.

The derivative of the principal value is minus the finite part. Splitting at $\pm x_\mathrm{c}$ and integrating by parts,

$$-\int_{|x|>x_\mathrm{c}} \mathrm{d}x\, \frac{w'(x)}{x} = \frac{w(x_\mathrm{c}) + w(-x_\mathrm{c})}{x_\mathrm{c}} - \int_{|x|>x_\mathrm{c}} \mathrm{d}x\, \frac{w(x)}{x^2},$$

and as $x_\mathrm{c} \to 0$ the right side becomes minus the bracket of Eq. (2.59), because $w(x_\mathrm{c}) + w(-x_\mathrm{c}) = 2w(0) + O(x_\mathrm{c}^2)$. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}x}\,\mathrm{PV}\,\frac{1}{x} = -\,\mathrm{fp}\,\frac{1}{x^2}. \tag{C.31}$$

The finite part is thus the derivative of a distribution we already have, and it obeys the usual rules: $x^2\,\mathrm{fp}(1/x^2) = 1$, $x\,\mathrm{fp}(1/x^2) = \mathrm{PV}(1/x)$, and $\mathrm{fp}\int \mathrm{d}x\, w(x)/x^2 = 0$ for constant w, the last one by the explicit form of Eq. (2.59).

The square of a denominator follows from the derivative. Since $1/(x + \mathrm{i}\Gamma_\mathrm{w}/2)^2 = -\mathrm{d}[1/(x + \mathrm{i}\Gamma_\mathrm{w}/2)]/\mathrm{d}x$ at every finite width, and derivatives commute with the limit when both act on a test function, Eqs. (2.58) and (C.31) give

$$\frac{1}{(x + \mathrm{i}0^+)^2} = \mathrm{fp}\,\frac{1}{x^2} + \mathrm{i}\pi\,\delta'(x). \tag{C.32}$$

A double pole on one side of the axis is a well-defined distribution. The modulus squared $|x + \mathrm{i}\Gamma_\mathrm{w}/2|^{-2}$ is a different object, because it is the product of a denominator and its conjugate, whose poles lie on opposite sides of the axis and pinch it as $\Gamma_\mathrm{w} \to 0$. Its limit does not exist; its expansion in Γ_w starts with $2\pi\delta(x)/\Gamma_\mathrm{w}$, Eq. (2.60). Products of distributions are in general undefined, and the squared delta function of a rate always comes from such a pinch.

C.4.3 Corrections to the Lorentzian limit

The normalized Lorentzian of full width Γ_w at half maximum is

$$\delta_\Gamma(x) = \frac{1}{\pi}\,\frac{\Gamma_\mathrm{w}/2}{x^2 + \Gamma_\mathrm{w}^2/4} = -\frac{1}{\pi}\,\mathrm{Im}\,\frac{1}{x + \mathrm{i}\Gamma_\mathrm{w}/2}, \tag{C.33}$$

and Eq. (2.60) states how it approaches the delta function. Multiplied by $\Gamma_\mathrm{w}/2\pi$, it reads

$$\int \mathrm{d}x\, \delta_\Gamma(x)\, w(x) = w(0) + \frac{\Gamma_\mathrm{w}}{2\pi}\,\mathrm{fp}\int \mathrm{d}x\, \frac{w(x)}{x^2} - \frac{\Gamma_\mathrm{w}^2}{8}\, w''(0) + O(\Gamma_\mathrm{w}^3). \tag{C.34}$$

The first correction is linear in the width and involves the whole of w, through the finite part. It is the price of the slow $1/x^2$ tails, which give weight $\Gamma_\mathrm{w}/(\pi|x|)$ to the region beyond any $|x|$. A Gaussian of the same width has no linear term: its first correction is proportional

to $\Gamma_{\mathrm{w}}^2\, w''(0)$, and it depends on w only near the origin (Appendix C.1). A test makes the difference concrete. For $w(x) = 1/(1+x^2)$, the Lorentzian integral is exactly $1/(1+\Gamma_{\mathrm{w}}/2)$, by closing the contour in either half plane. The finite part of $\int \mathrm{d}x\, w(x)/x^2$ is $-\pi$, since $1/[x^2(1+x^2)] = 1/x^2 - 1/(1+x^2)$ and the finite part of $\int \mathrm{d}x/x^2$ vanishes, and $w''(0) = -2$. Eq. (C.34) then reads $1 - \Gamma_{\mathrm{w}}/2 + \Gamma_{\mathrm{w}}^2/4 + \ldots$, the expansion of the exact result. At $\Gamma_{\mathrm{w}} = 0.2$ the linear term is a 10 % correction, and a Gaussian of the same full width at half maximum, whose correction is $\frac{1}{2}w''(0)$ times its variance $(\Gamma_{\mathrm{w}}/2.355)^2$, would make an error of 0.7 %.

The Lorentzian is the line shape of exponential decay. A state that decays as $\mathrm{e}^{-\Gamma_{\mathrm{w}}|t|/2}$ in amplitude has, by Eq. (A.7), the spectrum $\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}xt}\mathrm{e}^{-\Gamma_{\mathrm{w}}|t|/2} = 2\pi\delta_\Gamma(x)$. The convolution theorem, Eq. (C.9) in its time form, then fixes the shape of combined lines. Two independent decay processes multiply in time and convolve in energy, $\mathrm{e}^{-\Gamma_1|t|/2}\mathrm{e}^{-\Gamma_2|t|/2} = \mathrm{e}^{-(\Gamma_1+\Gamma_2)|t|/2}$. The convolution of Lorentzians is a Lorentzian whose width is the sum of the two, $\delta_{\Gamma_1} * \delta_{\Gamma_2} = \delta_{\Gamma_1+\Gamma_2}$; widths of independent channels add, as Sect. 2.4 assumed for the total width of an intermediate state. A Lorentzian convolved with a Gaussian, the Voigt profile, keeps the $1/x^2$ tails of the Lorentzian, so the linear correction of Eq. (C.34) survives any additional Gaussian broadening. A width that depends on energy, $\Gamma_{\mathrm{w}}(x)$, keeps the delta limit of Eq. (C.27) at its value at x_0, provided it varies little over a range Γ_{w} around the pole, and its slope enters the finite-width corrections.

C.4.4 Summary of definitions

We collect the objects of this section and Sect. 2.4. Each acts on a test function $w(x)$ that is smooth near $x = 0$.

$$\mathrm{PV}\int \mathrm{d}x\, \frac{w(x)}{x} = \int_0^\infty \mathrm{d}x\, \frac{w(x) - w(-x)}{x}, \qquad \mathrm{fp}\int \mathrm{d}x\, \frac{w(x)}{x^2} = \int_0^\infty \mathrm{d}x\, \frac{w(x) + w(-x) - 2w(0)}{x^2},$$

$$\frac{1}{x \pm \mathrm{i}0^+} = \mathrm{PV}\,\frac{1}{x} \mp \mathrm{i}\pi\delta(x), \qquad \frac{1}{(x \pm \mathrm{i}0^+)^2} = \mathrm{fp}\,\frac{1}{x^2} \pm \mathrm{i}\pi\delta'(x), \qquad \frac{\mathrm{d}}{\mathrm{d}x}\,\mathrm{PV}\,\frac{1}{x} = -\,\mathrm{fp}\,\frac{1}{x^2},$$

$$\frac{1}{x^2 + \Gamma_{\mathrm{w}}^2/4} = \frac{2\pi}{\Gamma_{\mathrm{w}}}\delta(x) + \mathrm{fp}\,\frac{1}{x^2} - \frac{\pi\Gamma_{\mathrm{w}}}{4}\delta''(x) + O(\Gamma_{\mathrm{w}}^2).$$

The first form of the principal value follows from Eq. (C.26) by folding the negative half line onto the positive one. In each case the integrand on the right is finite at the origin, which is what makes these forms convenient for numerical work: a principal-value integral over a tabulated spectrum is computed by folding, not by cutting out an interval around the pole.

C.5 Complex analysis for response functions

A response cannot precede its cause, and this single fact fixes the real part of every response function once its imaginary part is known at all frequencies. The dielectric function of silicon at zero frequency, for example, follows from its measured absorption spectrum. This section derives the analyticity of a retarded function in the upper half of the complex frequency plane, the Kramers–Kronig relations and the sum rules that follow from them, and the continuation of a function known at the imaginary Matsubara frequencies to real frequencies. Contour integration and the residue theorem are assumed (Sects. 4.1 and 4.7 use the results).

C.5.1 Causality and analyticity

Let $R(t)$ be a retarded response: the change of an observable at time t caused by a unit impulse at $t = 0$, so that $R(t) = 0$ for $t < 0$. The response function of Eq. (A.30) is of this

kind. With the time convention of Appendix A.3, its transform is

$$R(z) = \int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}zt} R(t), \qquad z = \omega + \mathrm{i}y. \tag{C.35}$$

For $y > 0$ the factor $\mathrm{e}^{\mathrm{i}zt} = \mathrm{e}^{\mathrm{i}\omega t}\mathrm{e}^{-yt}$ decays, and the integral converges for any $R(t)$ that grows no faster than a polynomial. Differentiating under the integral with respect to z is then allowed, so $R(z)$ is analytic in the upper half plane. The physical response at real ω is the boundary value $R(\omega + \mathrm{i}0^+)$, which is the 0^+ prescription of Eq. (C.10). For $y < 0$ the integral diverges, and the function on the lower half plane is defined, if at all, by analytic continuation. In the spectral (Lehmann) form Eq. (A.32), each term is $1/(z - \Delta E)$ with real ΔE, analytic everywhere except on the real axis. Inserted with $z = \omega + \mathrm{i}0^+$, every pole lies below the axis of real ω, as Appendix A.3 requires.

The sign convention matters at this point. With $\mathrm{e}^{+\mathrm{i}\omega t}$ in the forward transform, as in this book, retarded functions are analytic in the upper half plane; with $\mathrm{e}^{-\mathrm{i}\omega t}$, as in much of the engineering literature, they are analytic in the lower half, and all imaginary parts change sign.

C.5.2 Spectral representation

The Lehmann form makes the structure of $R(z)$ explicit. By Eq. (C.27), the imaginary part of each term on the real axis is a delta function, and the response is recovered from it by

$$R(z) = \int \mathrm{d}\omega' \, \frac{\varrho_R(\omega')}{z - \omega'}, \qquad \varrho_R(\omega) = -\frac{1}{\pi} \operatorname{Im} R(\omega + \mathrm{i}0^+), \tag{C.36}$$

for a diagonal response, $O_a = O_b$, where R is real on the real axis apart from its delta functions. The spectral density ϱ_R is the density of transitions weighted by their squared matrix elements, positive for $\omega > 0$ and negative for $\omega < 0$ (Appendix A.7). At large $|z|$ the denominator can be expanded, $1/(z - \omega') = z^{-1} + \omega' z^{-2} + \dots$, and

$$R(z) = \frac{1}{z} \int \mathrm{d}\omega' \, \varrho_R(\omega') + \frac{1}{z^2} \int \mathrm{d}\omega' \, \omega' \varrho_R(\omega') + O(z^{-3}). \tag{C.37}$$

For the density response the zeroth moment vanishes, since ϱ_R is odd in ω for $S(\boldsymbol{q}, \omega) = S(-\boldsymbol{q}, \omega)$ (Appendix A.4), and the first moment is fixed by the f-sum rule Eq. (A.39). With $-2 \operatorname{Im} R = (1 - \mathrm{e}^{-\beta\omega}) S$ (Eq. (A.33)) and detailed balance, $\int \mathrm{d}\omega' \, \omega' \varrho_R = (2\pi)^{-1} \cdot 2 \int \mathrm{d}\omega \, \omega \, S = n_e q^2 / m_e$, so that $R(z) \simeq n_e q^2 / (m_e z^2)$: at high frequency every electron responds as a free particle. A response function that falls off at least as $1/|z|$ in the upper half plane is all that the following derivations need.

C.5.3 Kramers–Kronig relations

Take z in the upper half plane and integrate $R(z')/(z' - z)$ over a closed contour made of the real axis and a large semicircle above it. The only singularity inside is the pole at $z' = z$, and the semicircle contributes nothing when R falls off as $1/|z'|$, since its length grows as $|z'|$ and the integrand falls off as $|z'|^{-2}$. Cauchy's formula gives $R(z) = (2\pi\mathrm{i})^{-1} \int \mathrm{d}\omega' \, R(\omega')/(\omega' - z)$, the Cauchy integral Eq. (C.28) of the boundary values. Let $z \to \omega + \mathrm{i}0^+$. By Eq. (C.27), $1/(\omega' - \omega - \mathrm{i}0^+) = \mathrm{PV}[1/(\omega' - \omega)] + \mathrm{i}\pi\delta(\omega' - \omega)$, so

$$R(\omega) = \frac{1}{2\pi\mathrm{i}} \, \mathrm{PV} \int \mathrm{d}\omega' \, \frac{R(\omega')}{\omega' - \omega} + \frac{1}{2} R(\omega),$$

where $R(\omega)$ means the boundary value from above. Moving the last term to the left and separating real and imaginary parts,

$$\operatorname{Re} R(\omega) = \frac{1}{\pi} \, \mathrm{PV} \int_{-\infty}^{\infty} \mathrm{d}\omega' \, \frac{\operatorname{Im} R(\omega')}{\omega' - \omega}, \qquad \operatorname{Im} R(\omega) = -\frac{1}{\pi} \, \mathrm{PV} \int_{-\infty}^{\infty} \mathrm{d}\omega' \, \frac{\operatorname{Re} R(\omega')}{\omega' - \omega}. \tag{C.38}$$

These are the Kramers–Kronig relations [5, 6]. Toll showed that they are equivalent to causality, under mild conditions on the response [7].

Most responses have a definite symmetry under $\omega \to -\omega$. If the response in time is real, as for the response of a Hermitian observable to a real field, then $R(-\omega) = R(\omega)^*$ for real ω: the real part is even and the imaginary part odd. The diagonal density response of a target with $S(\boldsymbol{q},\omega) = S(-\boldsymbol{q},\omega)$ and the dielectric function are of this kind. Folding the negative frequencies onto the positive ones, as in Appendix C.4, gives

$$\operatorname{Re} R(\omega) = \frac{2}{\pi}\,\mathrm{PV}\int_0^\infty \mathrm{d}\omega'\,\frac{\omega'\operatorname{Im} R(\omega')}{\omega'^2-\omega^2}, \qquad \operatorname{Im} R(\omega) = -\frac{2\omega}{\pi}\,\mathrm{PV}\int_0^\infty \mathrm{d}\omega'\,\frac{\operatorname{Re} R(\omega')}{\omega'^2-\omega^2}. \tag{C.39}$$

For the first, the negative half axis contributes

$$\int_0^\infty \mathrm{d}\omega'\,\frac{\operatorname{Im} R(-\omega')}{-\omega'-\omega} = \int_0^\infty \mathrm{d}\omega'\,\frac{\operatorname{Im} R(\omega')}{\omega'+\omega},$$

and $1/(\omega'-\omega)+1/(\omega'+\omega) = 2\omega'/(\omega'^2-\omega^2)$; the second follows in the same way with the even real part.

C.5.4 Direct and inverse dielectric functions

The relations hold for any function that is analytic in the upper half plane and falls off at large $|z|$. Three functions of Sect. 4.4 are candidates, and the conditions differ.

The inverse dielectric function obeys them in the form $\varepsilon^{-1}-1$, because $\varepsilon^{-1}-1 = v_{\mathrm{C}}R_{ee}$ for the electron channel (Appendix A.7) is itself a retarded response: the induced potential caused by an external one. With $L = -\operatorname{Im}\varepsilon^{-1}$ and Eq. (C.39) at $\omega = 0$,

$$1 - \operatorname{Re}\varepsilon^{-1}(\boldsymbol{q},0) = \frac{2}{\pi}\int_0^\infty \mathrm{d}\omega'\,\frac{L(\boldsymbol{q},\omega')}{\omega'}. \tag{C.40}$$

In a metal, static screening is perfect at long wavelength, $\varepsilon^{-1}(\boldsymbol{q}\to 0,0) = 0$, and the loss function then obeys $\int_0^\infty \mathrm{d}\omega\, L/\omega = \pi/2$, the perfect-screening sum rule checked on the electron gas in Sect. 4.5.

The dielectric function itself obeys them in the form $\varepsilon - 1$ only if ε has no zeros in the upper half plane, since $1/\varepsilon$ must be analytic there for $\varepsilon - 1 = 1/\varepsilon^{-1} - 1$ to be. Zeros of ε are self-sustained oscillations, and in the upper half plane they would grow in time; a stable medium has none. In a crystal with a gap ε is also finite at $\omega = 0$, and

$$\varepsilon_1(\omega) - 1 = \frac{2}{\pi}\,\mathrm{PV}\int_0^\infty \mathrm{d}\omega'\,\frac{\omega'\varepsilon_2(\omega')}{\omega'^2-\omega^2}, \tag{C.41}$$

which at $\omega = 0$ gives the static electronic dielectric constant from the absorption spectrum, $\varepsilon_\infty - 1 = (2/\pi)\int_0^\infty \mathrm{d}\omega'\,\varepsilon_2(\omega')/\omega'$ at $q \to 0$ with the lattice clamped. In a metal ε has a pole at $\omega = 0$, from the conductivity, $\varepsilon \simeq \mathrm{i}\sigma_{\mathrm{cond}}(0)/\omega$ by Appendix A.7, and the pole must be subtracted before the relations are applied.

At large frequency every electron responds as if free, $\varepsilon(\omega) \to 1 - \omega_p^2/\omega^2$ by Eq. (C.37) and the f-sum rule. Comparing with the large-ω limit of Eq. (C.41), $\varepsilon_1 - 1 \to -(2/\pi\omega^2)\int_0^\infty \mathrm{d}\omega'\,\omega'\varepsilon_2(\omega')$, gives $\int_0^\infty \mathrm{d}\omega\,\omega\,\varepsilon_2 = (\pi/2)\,\omega_p^2$, the sum rule for ε_2 of Sect. 4.4. The Kramers–Kronig relations and the f-sum rule are two faces of one statement about the analytic function $\varepsilon(z)$.

C.5.5 Matsubara frequencies and analytic continuation

At finite temperature, many-body perturbation theory computes response and Green's functions on the imaginary axis, at the bosonic Matsubara frequencies $\mathrm{i}\omega_j^{\mathrm{B}} = 2\pi\mathrm{i}jT$ or the fermionic ones $\mathrm{i}\omega_j^{\mathrm{F}} = (2j+1)\pi\mathrm{i}T$, with integer j (Sect. 4.7). Frequency sums and analytic continuation connect these values with the real axis.

We start with the sums. The Bose function $n_{\mathrm{B}}(z) = 1/(\mathrm{e}^{\beta z} - 1)$, continued to complex z, has poles at $z = \mathrm{i}\omega_j^{\mathrm{B}}$; near each, $\mathrm{e}^{\beta z} \simeq 1 + \beta(z - \mathrm{i}\omega_j^{\mathrm{B}})$, so the residue is $1/\beta = T$. For a function $g(z)$ that is analytic near the imaginary axis and falls off faster than $1/|z|$, the sum $T\sum_j g(\mathrm{i}\omega_j^{\mathrm{B}})$ is therefore the integral of $n_{\mathrm{B}}(z)g(z)/2\pi\mathrm{i}$ around small circles enclosing the poles of n_{B}. Merging the circles and opening the contour to infinity, where the integrand vanishes, leaves the poles z_{p} of g, traversed in the opposite sense:

$$T\sum_j g(\mathrm{i}\omega_j^{\mathrm{B}}) = -\sum_{z_{\mathrm{p}}} \mathrm{Res}_{z=z_{\mathrm{p}}}\big[n_{\mathrm{B}}(z)\, g(z)\big], \qquad T\sum_j g(\mathrm{i}\omega_j^{\mathrm{F}}) = \sum_{z_{\mathrm{p}}} \mathrm{Res}_{z=z_{\mathrm{p}}}\big[n_{\mathrm{F}}(z)\, g(z)\big]. \tag{C.42}$$

The fermionic form uses $n_{\mathrm{F}}(z) = 1/(\mathrm{e}^{\beta z} + 1)$, whose residues at $\mathrm{i}\omega_j^{\mathrm{F}}$ are $-T$. A one-line example fixes the signs. For an oscillator of energy ω_0, take $g(z) = 1/(\omega_0^2 - z^2)$, so that $g(\mathrm{i}\omega_j^{\mathrm{B}}) = 1/(\omega_0^2 + (\omega_j^{\mathrm{B}})^2) > 0$. The poles of g at $z = \pm\omega_0$ have residues $\mp 1/(2\omega_0)$, and with $n_{\mathrm{B}}(-\omega_0) = -1 - n_{\mathrm{B}}(\omega_0)$,

$$T\sum_j \frac{1}{\omega_0^2 + (\omega_j^{\mathrm{B}})^2} = \frac{n_{\mathrm{B}}(\omega_0)}{2\omega_0} + \frac{1 + n_{\mathrm{B}}(\omega_0)}{2\omega_0} = \frac{2n_{\mathrm{B}}(\omega_0) + 1}{2\omega_0}.$$

Up to the factor $1/M$, this is the thermal mean-square displacement of an oscillator of mass M, whose growth with temperature, by the factor $2n_{\mathrm{B}} + 1$, was found directly in Worked example 2.7. Appendix D.10 uses Eq. (C.42) for the sums that diagrams produce.

Analytic continuation is the other step. Suppose a function is computed at the Matsubara frequencies, $R(\mathrm{i}\omega_j^{\mathrm{B}})$. It is turned into the retarded function by finding the function $R(z)$ that is analytic off the real axis, agrees with the computed values at every $\mathrm{i}\omega_j^{\mathrm{B}}$ with $j > 0$, and falls off at infinity, and then setting $z = \omega + \mathrm{i}0^+$. For an expression in closed form this is a substitution, $\mathrm{i}\omega_j^{\mathrm{B}} \to \omega + \mathrm{i}0^+$, made after every occupation factor has been simplified with $\mathrm{e}^{\mathrm{i}\beta\omega_j^{\mathrm{B}}} = 1$. The order of these steps matters. The function $(\mathrm{e}^{\beta z} - 1)\, h(z)$, for any analytic h, vanishes at every bosonic Matsubara frequency, so infinitely many analytic functions take the same values on the Matsubara points. The condition that $R(z)$ falls off at large $|z|$ removes them all, since $\mathrm{e}^{\beta z}$ grows along the real axis, and the continuation is unique [8]. The spectral representation Eq. (C.36) is the form in which uniqueness is manifest: it defines one function on the whole plane off the real axis, equal to the Matsubara function on the imaginary axis and to the retarded function just above the real axis. Numerical data on the imaginary axis are a harder case. Continuation then amplifies small errors exponentially, and a rational (Padé) fit to the Matsubara values, the usual tool, must be checked against sum rules such as Eq. (A.39) (Appendix E.3).

C.6 Gaussian integrals and oscillator identities

When dark matter scatters off a nucleus in a crystal, the target operator is $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}}$, the exponential of a displacement that is a sum of harmonic-oscillator coordinates. Its thermal average gives the Debye–Waller factor, its matrix elements give the probabilities of creating zero, one or many phonons, and both reduce to Gaussian integrals or to their operator analogues. This section collects the identities that Sects. 5.3, 5.6 and 6.1 use: Gaussian integrals in one

and many variables, the Baker–Campbell–Hausdorff formula in the form needed for linear operators, the displacement operator and coherent states, and the statement $\langle \mathrm{e}^{\hat{A}} \rangle = \mathrm{e}^{\langle \hat{A}^2 \rangle/2}$ for operators $\hat{A}$ linear in the oscillator variables.

C.6.1 Gaussian integrals

For $\mathrm{Re}\,\mu > 0$ and any complex η,

$$\int_{-\infty}^{\infty} \mathrm{d}x\, \mathrm{e}^{-\mu x^2 + \eta x} = \sqrt{\frac{\pi}{\mu}}\, \mathrm{e}^{\eta^2/4\mu}. \tag{C.43}$$

For $\eta = 0$ and real μ, squaring the integral and changing to polar coordinates in the plane gives $\int \mathrm{d}x\, \mathrm{d}y\, \mathrm{e}^{-\mu(x^2+y^2)} = 2\pi \int_0^\infty \mathrm{d}r\, r\, \mathrm{e}^{-\mu r^2} = \pi/\mu$. For $\eta \neq 0$, completing the square, $-\mu x^2 + \eta x = -\mu(x - \eta/2\mu)^2 + \eta^2/4\mu$, and shifting the variable by $\eta/2\mu$ gives the result. For complex η the shift moves the path of integration off the real axis; the integrand is analytic and decays at the ends of the strip in between, so by Cauchy's theorem the path can be moved back. The same argument extends the result to complex μ with $\mathrm{Re}\,\mu > 0$, with the branch of $\sqrt{\mu}$ that is positive for real μ. The limit $\mathrm{Re}\,\mu \to 0^+$ gives the Fresnel integral, $\int \mathrm{d}x\, \mathrm{e}^{\mathrm{i}\mu x^2} = \sqrt{\pi/\mu}\, \mathrm{e}^{\mathrm{i}\pi/4}$ for real $\mu > 0$, which appears in the time integrals of a freely recoiling nucleus (Sect. 11.4).

With $\eta = \mathrm{i}q$ and $\mu = 1/(2s^2)$, the Fourier transform of a normalized Gaussian of width s is $\mathrm{e}^{-q^2 s^2/2}$, in any number of dimensions, as for the skin factor of the Helm form factor (Sect. 2.2). Differentiating Eq. (C.43) at $\eta = 0$ with respect to μ gives the moments of the normalized distribution $\sqrt{\mu/\pi}\, \mathrm{e}^{-\mu x^2}$,

$$\langle x^{2n} \rangle = (2n-1)!!\, \langle x^2 \rangle^n, \qquad \langle x^2 \rangle = \frac{1}{2\mu}, \tag{C.44}$$

with odd moments zero. Reading Eq. (C.43) divided by $\sqrt{\pi/\mu}$ as an average gives the generating function

$$\langle \mathrm{e}^{\eta x} \rangle = \mathrm{e}^{\eta^2 \langle x^2 \rangle/2}, \tag{C.45}$$

which is the classical form of the operator identity of Eq. (C.53) below. Expanding both sides in powers of η reproduces Eq. (C.44).

Let $\mathbf{M}$ be a real symmetric positive-definite $n \times n$ matrix and $\boldsymbol{s}$ a vector. In n variables,

$$\int \mathrm{d}^n x\, \mathrm{e}^{-\frac{1}{2}\boldsymbol{x}\cdot\mathbf{M}\cdot\boldsymbol{x} + \boldsymbol{s}\cdot\boldsymbol{x}} = \frac{(2\pi)^{n/2}}{(\det \mathbf{M})^{1/2}}\, \mathrm{e}^{\frac{1}{2}\boldsymbol{s}\cdot\mathbf{M}^{-1}\cdot\boldsymbol{s}}. \tag{C.46}$$

The proof rotates to the eigenvectors of $\mathbf{M}$, which leaves $\mathrm{d}^n x$ unchanged and turns the integral into a product of n copies of Eq. (C.43) with μ equal to half an eigenvalue. The product of the eigenvalues is $\det \mathbf{M}$, and the exponents combine into $\frac{1}{2}\boldsymbol{s}\cdot\mathbf{M}^{-1}\cdot\boldsymbol{s}$. For the normalized distribution, differentiating twice with respect to $\boldsymbol{s}$ at $\boldsymbol{s} = 0$ gives the covariance $\langle x_i x_j \rangle = (\mathbf{M}^{-1})_{ij}$, and differentiating four times gives

$$\langle x_i x_j x_k x_l \rangle = \langle x_i x_j \rangle \langle x_k x_l \rangle + \langle x_i x_k \rangle \langle x_j x_l \rangle + \langle x_i x_l \rangle \langle x_j x_k \rangle, \tag{C.47}$$

the sum over the three ways of pairing the four variables [9]. Every higher even moment is the sum over all pairings in the same way, because the generating function is the exponential of a quadratic form in $\boldsymbol{s}$. In quantum many-body theory this statement is Wick's theorem (Sect. 4.8).

C.6.2 Operator identities

Products of exponentials of operators need care, because $\mathrm{e}^{\hat X}\mathrm{e}^{\hat Y} \neq \mathrm{e}^{\hat X+\hat Y}$ when $\hat X$ and $\hat Y$ do not commute. We first need the transformation of an operator $\hat Y$ by $\mathrm{e}^{\hat X}$. The function $f(\lambda) = \mathrm{e}^{\lambda\hat X}\hat Y\mathrm{e}^{-\lambda\hat X}$ obeys $f'(\lambda) = \mathrm{e}^{\lambda\hat X}[\hat X,\hat Y]\mathrm{e}^{-\lambda\hat X}$, and repeating the step gives the Taylor series

$$\mathrm{e}^{\hat X}\hat Y\mathrm{e}^{-\hat X} = \hat Y + [\hat X,\hat Y] + \frac{1}{2!}[\hat X,[\hat X,\hat Y]] + \frac{1}{3!}[\hat X,[\hat X,[\hat X,\hat Y]]] + \cdots . \tag{C.48}$$

We also need the Baker–Campbell–Hausdorff formula for the case in which the commutator $[\hat X,\hat Y]$ commutes with both $\hat X$ and $\hat Y$, as it does when $\hat X$ and $\hat Y$ are linear in boson operators and $[\hat X,\hat Y]$ is a number. Let $g(\lambda) = \mathrm{e}^{\lambda\hat X}\mathrm{e}^{\lambda\hat Y}$. Then $g'(\lambda) = (\hat X + \mathrm{e}^{\lambda\hat X}\hat Y\mathrm{e}^{-\lambda\hat X})\,g(\lambda) = (\hat X+\hat Y+\lambda[\hat X,\hat Y])\,g(\lambda)$, where the series Eq. (C.48) stops after the commutator. The operators $\hat X+\hat Y$ and $[\hat X,\hat Y]$ commute, so the solution with $g(0)=1$ is $g(\lambda) = \exp[\lambda(\hat X+\hat Y)+\frac12\lambda^2[\hat X,\hat Y]]$, and at $\lambda = 1$

$$\mathrm{e}^{\hat X}\mathrm{e}^{\hat Y} = \mathrm{e}^{\hat X+\hat Y}\,\mathrm{e}^{[\hat X,\hat Y]/2}, \qquad \mathrm{e}^{\hat X}\mathrm{e}^{\hat Y} = \mathrm{e}^{\hat Y}\mathrm{e}^{\hat X}\,\mathrm{e}^{[\hat X,\hat Y]}. \tag{C.49}$$

The second form follows from the first by exchanging $\hat X$ and $\hat Y$ and dividing. For general $\hat X$ and $\hat Y$ the exponent contains nested commutators to all orders [10]; they are not needed here.

C.6.3 Displacement operators and coherent states

We write the oscillator as in Worked example 2.7, $H = \omega_0(d^\dagger d + \frac12)$ with $[d,d^\dagger]=1$, and its coordinate as $x = \ell_0(d+d^\dagger)$ with the zero-point length $\ell_0 = (2M\omega_0)^{-1/2}$ for a mass M; for a phonon mode it is $\ell_{\boldsymbol{Q}\nu}$ of Eq. (A.24). The displacement operator with a complex amplitude α_d is

$$\hat D(\alpha_\mathrm{d}) = \exp(\alpha_\mathrm{d}d^\dagger - \alpha_\mathrm{d}^* d) = \mathrm{e}^{-|\alpha_\mathrm{d}|^2/2}\,\mathrm{e}^{\alpha_\mathrm{d}d^\dagger}\mathrm{e}^{-\alpha_\mathrm{d}^* d}, \tag{C.50}$$

where the second form is Eq. (C.49) with $\hat X = \alpha_\mathrm{d}d^\dagger$, $\hat Y = -\alpha_\mathrm{d}^* d$ and $[\hat X,\hat Y] = |\alpha_\mathrm{d}|^2$. It is unitary, and by Eq. (C.48) it shifts the ladder operators, $\hat D^\dagger d\,\hat D = d + \alpha_\mathrm{d}$, so it shifts the coordinate by $2\ell_0\,\mathrm{Re}\,\alpha_\mathrm{d}$ and the momentum in proportion to $\mathrm{Im}\,\alpha_\mathrm{d}$. A momentum kick is the special case of a real coordinate: $\mathrm{e}^{\mathrm{i}qx} = \mathrm{e}^{\mathrm{i}q\ell_0(d+d^\dagger)} = \hat D(\mathrm{i}q\ell_0)$, which displaces the momentum by q. This is the operator through which dark matter acts on a bound nucleus.

Acting on the ground state, $\mathrm{e}^{-\alpha_\mathrm{d}^* d}|0\rangle = |0\rangle$, and $\mathrm{e}^{\alpha_\mathrm{d}d^\dagger}|0\rangle = \sum_n \alpha_\mathrm{d}^n (d^\dagger)^n|0\rangle/n!$, which is $\sum_n \alpha_\mathrm{d}^n|n\rangle/\sqrt{n!}$ by $(d^\dagger)^n|0\rangle = \sqrt{n!}\,|n\rangle$ (Sect. 2.3). The result is a coherent state,

$$\hat D(\alpha_\mathrm{d})|0\rangle = \mathrm{e}^{-|\alpha_\mathrm{d}|^2/2}\sum_{n=0}^{\infty}\frac{\alpha_\mathrm{d}^n}{\sqrt{n!}}\,|n\rangle, \qquad |\langle n|\hat D(\alpha_\mathrm{d})|0\rangle|^2 = \mathrm{e}^{-|\alpha_\mathrm{d}|^2}\,\frac{|\alpha_\mathrm{d}|^{2n}}{n!}. \tag{C.51}$$

The number of quanta in a displaced ground state is Poisson distributed with mean $|\alpha_\mathrm{d}|^2$. For a kick, $|\alpha_\mathrm{d}|^2 = q^2\ell_0^2 = q^2/(2M\omega_0)$, the recoil energy $q^2/2M$ in units of the quantum ω_0. The probability of leaving the oscillator in its ground state is $\mathrm{e}^{-q^2\ell_0^2}$, and the mean energy deposited is $\omega_0|\alpha_\mathrm{d}|^2 = q^2/2M$, the free recoil energy, at every q. For a 60 meV mode of a silicon nucleus, $M = 26.16\,\mathrm{GeV}$ (the mean atomic mass of natural silicon, as in Sect. 5.3), $q^2\ell_0^2 = 1$ at $q = (2M\omega_0)^{1/2} = 56\,\mathrm{keV}$: below it a kick mostly creates no quantum or one, and far above it the Poisson distribution is broad and the nucleus recoils almost as if free. The same distribution, with $|\alpha_\mathrm{d}|^2$ the Huang–Rhys factor S_HR, describes an electronic transition that displaces the equilibrium position of an oscillator (Sect. 5.6).

Ordering an exponential so that all annihilation operators stand to the right makes thermal and ground-state averages easy. For a real or complex η, Eq. (C.49) with $\hat X = \eta d^\dagger$, $\hat Y = \eta d$ and $[\hat X,\hat Y] = -\eta^2$ gives

$$\mathrm{e}^{\eta(d+d^\dagger)} = \mathrm{e}^{\eta d^\dagger}\mathrm{e}^{\eta d}\,\mathrm{e}^{\eta^2/2}. \tag{C.52}$$

In the ground state, $\langle 0|e^{\eta d^\dagger}e^{\eta d}|0\rangle = 1$, so $\langle 0|e^{\eta(d+d^\dagger)}|0\rangle = e^{\eta^2/2}$. For a kick, $\eta = iq\ell_0$ and the ground-state average is $e^{-q^2\ell_0^2/2}$, the square root of the ground-state survival probability, as it must be. Since $\langle 0|(d+d^\dagger)^2|0\rangle = 1$, the exponent is $\frac{1}{2}\langle 0|(\eta(d+d^\dagger))^2|0\rangle$: the ground-state average of the exponential is the exponential of half the average of the square, as for the classical Gaussian of Eq. (C.45).

C.6.4 Thermal averages of exponentials

The ground state of an oscillator is a Gaussian in x, and so is its thermal state: the density matrix $e^{-\beta H}/\mathcal{Z}$ is Gaussian in the coordinate and the momentum. The identity of the last paragraph therefore extends to any temperature and to any number of independent oscillators. For an operator $\hat{A}$ that is a linear combination of the d_γ and $d_\gamma^\dagger$ of a set of harmonic modes in thermal equilibrium,

$$\langle e^{\hat{A}}\rangle = e^{\langle \hat{A}^2\rangle/2}, \tag{C.53}$$

which is Bloch's identity, proved in Appendix D.15; Mermin gave a short alternative derivation [11]. In the classical limit $T \gg \omega_0$ it is Eq. (C.45), since the coordinates are then classical Gaussian variables. For one mode and $\hat{A} = iqx$, it gives $\langle e^{iqx}\rangle = e^{-q^2\langle x^2\rangle/2}$ with $\langle x^2\rangle = \ell_0^2(2n_B(\omega_0)+1)$, the zero-point value multiplied by the factor found in Worked example 2.7 and in the Matsubara sum of Appendix C.5. For a crystal, $\hat{A} = i\boldsymbol{q}\cdot\boldsymbol{u}_\kappa$ and the result is $e^{-W_\kappa(\boldsymbol{q})}$ with $2W_\kappa(\boldsymbol{q}) = \boldsymbol{q}\cdot B_\kappa\cdot\boldsymbol{q}$, the Debye–Waller factor of Eq. (A.29) at the level of the amplitude.

Combined with Eq. (C.49), Bloch's identity gives the average of a product. For $\hat{A}$ and $\hat{B}$ both linear, $[\hat{A},\hat{B}]$ is a number, $e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}}e^{[\hat{A},\hat{B}]/2}$, and $\langle e^{\hat{A}+\hat{B}}\rangle = e^{\langle(\hat{A}+\hat{B})^2\rangle/2}$. Expanding $(\hat{A}+\hat{B})^2 = \hat{A}^2+\hat{B}^2+\hat{A}\hat{B}+\hat{B}\hat{A}$ and adding $\frac{1}{2}[\hat{A},\hat{B}] = \frac{1}{2}\langle\hat{A}\hat{B}-\hat{B}\hat{A}\rangle$ to the exponent,

$$\langle e^{\hat{A}}e^{\hat{B}}\rangle = \exp\Big[\tfrac{1}{2}\langle\hat{A}^2\rangle + \tfrac{1}{2}\langle\hat{B}^2\rangle + \langle\hat{A}\hat{B}\rangle\Big]. \tag{C.54}$$

With $\hat{A} = -i\boldsymbol{q}\cdot\boldsymbol{u}_\kappa(t)$ and $\hat{B} = i\boldsymbol{q}\cdot\boldsymbol{u}_\kappa(0)$, which are linear because the Heisenberg operators of a harmonic crystal are $d_\gamma(t) = d_\gamma e^{-i\omega_\gamma t}$, the first two terms give $e^{-2W_\kappa(\boldsymbol{q})}$ and the third gives $e^{\langle(\boldsymbol{q}\cdot\boldsymbol{u}_\kappa(t))(\boldsymbol{q}\cdot\boldsymbol{u}_\kappa(0))\rangle}$. This is the generating formula from which Sect. 5.6 builds the multiphonon response: expanding the last exponential in powers of the displacement correlation counts the phonons created, and the zeroth term is the elastic weight e^{-2W_κ}.

C.7 Matrix tools

The response of a crystal is a matrix: in reciprocal-lattice vectors for the dielectric function, in operators for the response matrix S_{ab}, in bands and crystal momenta for excitons. Positive semidefiniteness, block inversion, Dyson-type equations and Bogoliubov transformations recur in Chaps. 4 and 7. Positive semidefiniteness bounds the interference between two couplings. Block inversion turns the inverse dielectric matrix into a macroscopic dielectric function. Dyson-type equations sum geometric series of interactions. Bogoliubov transformations diagonalize Hamiltonians that create excitations in pairs. We use bold upright capitals, $\mathbf{M}$, $\mathbf{A}$, $\mathbf{B}$, for generic matrices, $\mathbf{1}$ for the unit matrix and $\boldsymbol{x}$ for a column vector, and we write $\boldsymbol{x}^\dagger$ for its conjugate transpose.

C.7.1 Positive semidefinite matrices

A Hermitian matrix $\mathbf{M} = \mathbf{M}^\dagger$ is positive semidefinite if $\boldsymbol{x}^\dagger\mathbf{M}\boldsymbol{x} \geq 0$ for every vector $\boldsymbol{x}$. It is equivalent to each of the following conditions. All eigenvalues are nonnegative: take $\boldsymbol{x}$

to be an eigenvector, and conversely expand any $\boldsymbol{x}$ in the orthonormal eigenvectors. $\mathbf{M}$ is a sum of outer products with nonnegative weights, $\mathbf{M} = \sum_f w_f\, \boldsymbol{c}_f \boldsymbol{c}_f^\dagger$ with $w_f \geq 0$: the eigen-decomposition is such a sum, and conversely $\boldsymbol{x}^\dagger \mathbf{M} \boldsymbol{x} = \sum_f w_f |\boldsymbol{c}_f^\dagger \boldsymbol{x}|^2 \geq 0$. Every principal submatrix, the matrix that keeps a subset of rows and the same columns, has nonnegative determinant. The last condition must include all principal submatrices, not only the leading ones: $\mathrm{diag}(0, -1)$ has leading minors 0 and 0 and is not positive semidefinite.

The diagonal elements are then nonnegative, and the 2×2 principal minors give

$$|M_{ab}|^2 \leq M_{aa} M_{bb}, \tag{C.55}$$

the Cauchy–Schwarz inequality for the matrix. The response matrix of Eq. (A.31) has the outer-product form, with one term for each pair of states (i, f), weight $2\pi p_i \delta(\omega - E_f + E_i)/\mathcal{V} \geq 0$ and vector components $c_b = \langle f|O_b|i\rangle$, since $\langle i|O_a^\dagger|f\rangle = c_a^*$. Hence $S_{ab}(\boldsymbol{q}, \omega)$ is positive semidefinite at every $\boldsymbol{q}$ and ω (Sect. 4.2). For a probe that couples to the operators O_a with amplitudes x_a, the rate is proportional to $\sum_{ab} x_a^* S_{ab} x_b \geq 0$, and Eq. (C.55) bounds the interference term between two channels by the geometric mean of their separate rates. A computed S_{ab} with a negative eigenvalue signals an error.

C.7.2 Block inversion

Partition a square matrix into blocks, with $\mathbf{A}$ and $\mathbf{D}$ square and $\mathbf{D}$ invertible:

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \begin{pmatrix} \mathbf{1} & \mathbf{B}\mathbf{D}^{-1} \\ \mathbf{0} & \mathbf{1} \end{pmatrix} \begin{pmatrix} \mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C} & \mathbf{0} \\ \mathbf{0} & \mathbf{D} \end{pmatrix} \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{D}^{-1}\mathbf{C} & \mathbf{1} \end{pmatrix}.$$

Multiplying the first two factors gives $\left(\begin{smallmatrix} \mathbf{M/D} & \mathbf{B} \\ \mathbf{0} & \mathbf{D} \end{smallmatrix}\right)$ with $\mathbf{M/D}$ the middle upper block, and multiplying by the third adds $\mathbf{B}\mathbf{D}^{-1}\mathbf{C}$ to the upper-left block and puts $\mathbf{C}$ in the lower-left one, which returns $\mathbf{M}$. The middle factor contains the Schur complement of $\mathbf{D}$, $\mathbf{M/D} \equiv \mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}$. The outer factors have unit determinant and inverses obtained by changing the sign of their off-diagonal block. Inverting the three factors in reverse order, and writing $\mathbf{N} \equiv (\mathbf{M/D})^{-1}$,

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{N} & -\mathbf{N}\mathbf{B}\mathbf{D}^{-1} \\ -\mathbf{D}^{-1}\mathbf{C}\mathbf{N} & \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{C}\mathbf{N}\mathbf{B}\mathbf{D}^{-1} \end{pmatrix}, \tag{C.56}$$

and the determinants of the three factors give $\det \mathbf{M} = \det \mathbf{D}\, \det(\mathbf{M/D})$. The upper-left block of the inverse is the inverse of the Schur complement. When $\mathbf{A}$ is a single element, the head of a matrix in reciprocal-lattice vectors with $\boldsymbol{G} = \boldsymbol{G}' = 0$, this reads

$$\frac{1}{(\mathbf{M}^{-1})_{00}} = M_{00} - \sum_{\boldsymbol{G}, \boldsymbol{G}' \neq 0} M_{0\boldsymbol{G}} \left[(\mathbf{M}_{\neq 0})^{-1}\right]_{\boldsymbol{G}\boldsymbol{G}'} M_{\boldsymbol{G}'0}, \tag{C.57}$$

where $\mathbf{M}_{\neq 0}$ is the block with $\boldsymbol{G}, \boldsymbol{G}' \neq 0$. Applied to the dielectric matrix it gives the macroscopic dielectric function $\varepsilon_{\mathrm{mac}} = 1/(\varepsilon^{-1})_{00}$ as the head ε_{00} corrected by the local fields that the off-diagonal elements carry (Sect. 4.4 and Appendix D.7).

The same algebra solves eigenvalue problems in a subspace. Split a Hilbert space into a part P of interest and the rest Q, and write the equation $(E - H)\psi = 0$ in blocks. Eliminating $\psi_Q = (E - H_{QQ})^{-1} H_{QP} \psi_P$ leaves

$$\left[H_{PP} + H_{PQ}(E - H_{QQ})^{-1} H_{QP}\right] \psi_P = E\, \psi_P, \tag{C.58}$$

an effective Hamiltonian in P alone, whose energy dependence carries the coupling to the eliminated states. The bracket is E minus the Schur complement of $E - H_{QQ}$ in $E - H$, and the equation is exact. To second order in the coupling, with H_{QQ} replaced by its diagonal, it is the level shift $\sum_f |H'_{fi}|^2/(E_i - E_f)$ of Sect. 2.4; with $E \to E + \mathrm{i}0^+$ and a continuum in Q it acquires the width. Sect. 8.6 uses it to separate the electronic and the phonon-dressed sectors of a target.

C.7.3 Dyson-type equations

Many response objects obey an equation of the form

$$\mathbf{X} = \mathbf{X}_0 + \mathbf{X}_0\mathbf{K}\mathbf{X}, \tag{C.59}$$

with a known bare object $\mathbf{X}_0$ and a kernel $\mathbf{K}$. Its solution is $\mathbf{X} = (\mathbf{1} - \mathbf{X}_0\mathbf{K})^{-1}\mathbf{X}_0$, as follows by moving $\mathbf{X}_0\mathbf{K}\mathbf{X}$ to the left and inverting. When $\mathbf{X}_0$ and $\mathbf{K}$ are invertible this is $\mathbf{X}^{-1} = \mathbf{X}_0^{-1} - \mathbf{K}$, and multiplying out shows that $\mathbf{X} = \mathbf{X}_0 + \mathbf{X}\mathbf{K}\mathbf{X}_0$ holds too. Iterating Eq. (C.59) gives the geometric series $\mathbf{X} = \mathbf{X}_0 + \mathbf{X}_0\mathbf{K}\mathbf{X}_0 + \mathbf{X}_0\mathbf{K}\mathbf{X}_0\mathbf{K}\mathbf{X}_0 + \cdots$, which converges when every eigenvalue of $\mathbf{X}_0\mathbf{K}$ has modulus below one. The closed form remains valid beyond that, as the analytic continuation of the series, and its poles, where $\det(\mathbf{1} - \mathbf{X}_0\mathbf{K}) = 0$, are the collective modes that no finite order of the series contains.

In Chap. 4 this form appears for the density response, for the time-dependent density-functional response and for the single-particle Green's function. The density response, written through its Coulomb-proper part, $R = \tilde{R} + \tilde{R}v_{\mathrm{C}}R$, has $\mathbf{X}_0 = \tilde{R}$ and $\mathbf{K} = v_{\mathrm{C}}$; for a scalar it gives $R = \tilde{R}/(1 - v_{\mathrm{C}}\tilde{R}) = \tilde{R}/\varepsilon$, whose poles at $\varepsilon = 0$ are the plasmons (Sects. 4.4 and 4.5). The time-dependent density-functional response $R = R^0 + R^0 f_{\mathrm{Hxc}} R$ has the kernel f_{Hxc} (Sect. 4.6). The single-particle Green's function obeys $\mathsf{G} = \mathsf{G}_0 + \mathsf{G}_0\Sigma\mathsf{G}$, so that $\mathsf{G}^{-1} = \mathsf{G}_0^{-1} - \Sigma$ and the self-energy shifts and broadens the poles (Sect. 4.7). In a crystal each is a matrix equation in reciprocal-lattice vectors or bands, and the order of the factors matters.

When the kernel has low rank, a smaller inversion suffices. For $\mathbf{K} = \mathbf{U}\mathbf{W}\mathbf{U}^\dagger$ with $\mathbf{U}$ of size $n \times r$ and $r \ll n$, the Woodbury identity

$$(\mathbf{A} + \mathbf{U}\mathbf{W}\mathbf{U}^\dagger)^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{W}^{-1} + \mathbf{U}^\dagger\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{U}^\dagger\mathbf{A}^{-1}, \tag{C.60}$$

reduces the inverse to one of size r. Multiplying its right side by $\mathbf{A} + \mathbf{U}\mathbf{W}\mathbf{U}^\dagger$, the terms combine into $\mathbf{1} + \mathbf{A}^{-1}\mathbf{U}[\mathbf{W} - (\mathbf{W}^{-1} + \mathbf{U}^\dagger\mathbf{A}^{-1}\mathbf{U})^{-1}(\mathbf{1} + \mathbf{U}^\dagger\mathbf{A}^{-1}\mathbf{U}\mathbf{W})]\mathbf{U}^\dagger$, and the bracket vanishes because $\mathbf{1} + \mathbf{U}^\dagger\mathbf{A}^{-1}\mathbf{U}\mathbf{W} = (\mathbf{W}^{-1} + \mathbf{U}^\dagger\mathbf{A}^{-1}\mathbf{U})\mathbf{W}$. It is Eq. (C.56) in another guise, and it is the efficient route when one element, such as the long-range Coulomb head, is treated apart from the rest.

C.7.4 Bogoliubov transformations

A quadratic Hamiltonian that creates excitations in pairs is diagonalized by mixing creation and annihilation operators. Take two boson modes, d_1 and d_2, as for a magnon at $\boldsymbol{k}$ and at $-\boldsymbol{k}$ (Sect. 7.1), with

$$H = \omega_0(d_1^\dagger d_1 + d_2^\dagger d_2) + g_0(d_1 d_2 + d_2^\dagger d_1^\dagger), \qquad \omega_0 > |g_0|. \tag{C.61}$$

New operators $\tilde{d}_1 = u^{\mathrm{b}} d_1 + v^{\mathrm{b}} d_2^\dagger$ and $\tilde{d}_2 = u^{\mathrm{b}} d_2 + v^{\mathrm{b}} d_1^\dagger$, with real coefficients, are bosons if $[\tilde{d}_1, \tilde{d}_1^\dagger] = (u^{\mathrm{b}})^2 - (v^{\mathrm{b}})^2 = 1$; the other commutators vanish identically. We set $u^{\mathrm{b}} = \cosh\theta_{\mathrm{b}}$ and $v^{\mathrm{b}} = \sinh\theta_{\mathrm{b}}$. The inverse is $d_1 = u^{\mathrm{b}}\tilde{d}_1 - v^{\mathrm{b}}\tilde{d}_2^\dagger$ and the same with $1 \leftrightarrow 2$. Substituting and normal ordering with $[\tilde{d}_j, \tilde{d}_j^\dagger] = 1$, with $\tilde{N} = \tilde{d}_1^\dagger\tilde{d}_1 + \tilde{d}_2^\dagger\tilde{d}_2$ and $\tilde{P} = \tilde{d}_1\tilde{d}_2 + \tilde{d}_2^\dagger\tilde{d}_1^\dagger$,

$$\begin{aligned} d_1^\dagger d_1 + d_2^\dagger d_2 &= \cosh 2\theta_{\mathrm{b}}\,\tilde{N} + 2\sinh^2\theta_{\mathrm{b}} - \sinh 2\theta_{\mathrm{b}}\,\tilde{P}, \\ d_1 d_2 + d_2^\dagger d_1^\dagger &= \cosh 2\theta_{\mathrm{b}}\,\tilde{P} - \sinh 2\theta_{\mathrm{b}}\,(\tilde{N} + 1). \end{aligned}$$

The pair term $\tilde{P}$ drops out of H if $\omega_0 \sinh 2\theta_{\mathrm{b}} = g_0 \cosh 2\theta_{\mathrm{b}}$, that is $\tanh 2\theta_{\mathrm{b}} = g_0/\omega_0$. Then $\cosh 2\theta_{\mathrm{b}} = \omega_0/E_{\mathrm{b}}$ and $\sinh 2\theta_{\mathrm{b}} = g_0/E_{\mathrm{b}}$ with

$$H = E_{\mathrm{b}}(\tilde{d}_1^\dagger\tilde{d}_1 + \tilde{d}_2^\dagger\tilde{d}_2 + 1) - \omega_0, \qquad E_{\mathrm{b}} = (\omega_0^2 - g_0^2)^{1/2}. \tag{C.62}$$

The pairing lowers the mode energy from ω_0 to E_b and the ground-state energy by $\omega_0 - E_\mathrm{b}$. The new ground state, annihilated by $\tilde{d}_1$ and $\tilde{d}_2$, is not empty of the old quanta: $\langle d_1^\dagger d_1 \rangle = (v^\mathrm{b})^2 = \sinh^2\theta_\mathrm{b}$. In an antiferromagnet this is the zero-point reduction of the ordered moment. For $|g_0| \geq \omega_0$ no real θ_b exists, E_b is imaginary and the system is unstable. For $\omega_0 = 1$ and $g_0 = 0.6$, Eq. (C.62) gives $E_\mathrm{b} = 0.8$, a ground state at -0.2 and $\sinh^2\theta_\mathrm{b} = 0.125$.

The matrix form generalizes to many modes. With $\Psi = (d_1, d_2^\dagger)^\mathrm{T}$, Eq. (C.61) is $\Psi^\dagger \mathbf{h} \Psi$ minus a constant, with $\mathbf{h}$ the 2×2 matrix with ω_0 on the diagonal and g_0 off it. A linear transformation $\Psi = \mathbf{T}\tilde{\Psi}$ keeps the commutation relations only if $\mathbf{T}\boldsymbol{\eta}\mathbf{T}^\dagger = \boldsymbol{\eta}$ with $\boldsymbol{\eta} = \mathrm{diag}(1, -1)$, because the second component of Ψ is a creation operator; such a $\mathbf{T}$ is called para-unitary. The excitation energies are the eigenvalues of $\boldsymbol{\eta}\mathbf{h}$, here $\pm(\omega_0^2 - g_0^2)^{1/2}$, which come in pairs of opposite sign. For any number of modes, a para-unitary diagonalization with positive excitation energies exists if $\mathbf{h}$ is positive definite [12], the matrix condition for stability.

Fermions differ in one sign. For $c_1 = c_{\boldsymbol{k}\uparrow}$ and $c_2 = c_{-\boldsymbol{k}\downarrow}$, as in a superconductor (Sect. 7.2), take

$$H = \xi_0(c_1^\dagger c_1 + c_2^\dagger c_2) - g_0(c_1^\dagger c_2^\dagger + c_2 c_1). \tag{C.63}$$

The operators $\tilde{c}_1 = u^\mathrm{f} c_1 - v^\mathrm{f} c_2^\dagger$ and $\tilde{c}_2 = u^\mathrm{f} c_2 + v^\mathrm{f} c_1^\dagger$ are fermions if $\{\tilde{c}_1, \tilde{c}_1^\dagger\} = (u^\mathrm{f})^2 + (v^\mathrm{f})^2 = 1$, and we set $u^\mathrm{f} = \cos\theta_\mathrm{f}$, $v^\mathrm{f} = \sin\theta_\mathrm{f}$. The anticommutator adds the two squares where the boson commutator subtracted them, the transformation is unitary instead of para-unitary, and the hyperbolic functions become trigonometric. The inverse is $c_1 = u^\mathrm{f}\tilde{c}_1 + v^\mathrm{f}\tilde{c}_2^\dagger$, $c_2 = u^\mathrm{f}\tilde{c}_2 - v^\mathrm{f}\tilde{c}_1^\dagger$. Substituting and ordering with $\tilde{c}_j\tilde{c}_j^\dagger = 1 - \tilde{c}_j^\dagger\tilde{c}_j$ and $\tilde{c}_2\tilde{c}_1 = -\tilde{c}_1\tilde{c}_2$, with $\tilde{N} = \tilde{c}_1^\dagger\tilde{c}_1 + \tilde{c}_2^\dagger\tilde{c}_2$ and $\tilde{P} = \tilde{c}_1^\dagger\tilde{c}_2^\dagger + \tilde{c}_2\tilde{c}_1$,

$$\begin{aligned} c_1^\dagger c_1 + c_2^\dagger c_2 &= \cos 2\theta_\mathrm{f}\, \tilde{N} + 2\sin^2\theta_\mathrm{f} + \sin 2\theta_\mathrm{f}\, \tilde{P}, \\ c_1^\dagger c_2^\dagger + c_2 c_1 &= \cos 2\theta_\mathrm{f}\, \tilde{P} - \sin 2\theta_\mathrm{f}\, (\tilde{N} - 1). \end{aligned}$$

For the first line, $c_1^\dagger c_1 = (u^\mathrm{f})^2\tilde{c}_1^\dagger\tilde{c}_1 + u^\mathrm{f}v^\mathrm{f}\tilde{P} + (v^\mathrm{f})^2(1 - \tilde{c}_2^\dagger\tilde{c}_2)$, and $c_2^\dagger c_2 = (u^\mathrm{f})^2\tilde{c}_2^\dagger\tilde{c}_2 + u^\mathrm{f}v^\mathrm{f}\tilde{P} + (v^\mathrm{f})^2(1 - \tilde{c}_1^\dagger\tilde{c}_1)$, where the minus sign in the transformation of c_2 cancels against the sign of an anticommuting reordering. For the second line, $c_1^\dagger c_2^\dagger = (u^\mathrm{f})^2\tilde{c}_1^\dagger\tilde{c}_2^\dagger - (v^\mathrm{f})^2\tilde{c}_2\tilde{c}_1 + u^\mathrm{f}v^\mathrm{f}(1 - \tilde{N})$ and $c_2 c_1 = (u^\mathrm{f})^2\tilde{c}_2\tilde{c}_1 - (v^\mathrm{f})^2\tilde{c}_1^\dagger\tilde{c}_2^\dagger + u^\mathrm{f}v^\mathrm{f}(1 - \tilde{N})$. The pair term drops out of Eq. (C.63) if $\xi_0 \sin 2\theta_\mathrm{f} = g_0 \cos 2\theta_\mathrm{f}$, that is $\tan 2\theta_\mathrm{f} = g_0/\xi_0$. Then $\cos 2\theta_\mathrm{f} = \xi_0/E_\mathrm{f}$, $\sin 2\theta_\mathrm{f} = g_0/E_\mathrm{f}$, the coefficient of $\tilde{N}$ is $\xi_0 \cos 2\theta_\mathrm{f} + g_0 \sin 2\theta_\mathrm{f} = E_\mathrm{f}$, the constant is $\xi_0(1 - \cos 2\theta_\mathrm{f}) - g_0 \sin 2\theta_\mathrm{f} = \xi_0 - E_\mathrm{f}$, and

$$H = E_\mathrm{f}(\tilde{c}_1^\dagger\tilde{c}_1 + \tilde{c}_2^\dagger\tilde{c}_2) + \xi_0 - E_\mathrm{f}, \qquad E_\mathrm{f} = \left(\xi_0^2 + g_0^2\right)^{1/2}. \tag{C.64}$$

The pairing now raises the excitation energy, from $|\xi_0|$ to $E_\mathrm{f} \geq |g_0|$: a gap opens even where $\xi_0 = 0$. With $\xi_0 \to \xi_{\boldsymbol{k}}$ and $g_0 \to \Delta_\mathrm{BCS}$ this is the Bogoliubov energy of Sect. 7.2, and $u^\mathrm{f}, v^\mathrm{f}$ become the coherence factors $u_{\boldsymbol{k}}^\mathrm{B}, v_{\boldsymbol{k}}^\mathrm{B}$; in Sect. 7.1 the boson coefficients become $u_{\boldsymbol{k}}^\mathrm{M}, v_{\boldsymbol{k}}^\mathrm{M}$.

C.8 Special functions

The standard halo model cuts a Gaussian off at the escape speed, and every one of its speed integrals is a Gaussian over a finite interval: an error function. The normalization N_esc of Sect. 1.2 and the three branches of $\eta(v_\mathrm{min})$ in Sect. 1.3 and Appendix D.1 are differences of error functions. This section gives the properties used there and the numbers they produce. It then treats the complete elliptic integrals, which appear when a square root of a linear function of a cosine is averaged over an angle, and the logarithm of the Lindhard function of the electron gas (Sect. 4.5), which needs a branch at real frequencies.

C.8.1 The error function

The error function and its complement are [1, Eqs. (7.2.1) and (7.2.2)]

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x \mathrm{d}y\, \mathrm{e}^{-y^2}, \qquad \operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty \mathrm{d}y\, \mathrm{e}^{-y^2}. \tag{C.65}$$

It is odd, $\operatorname{erf}(-x) = -\operatorname{erf}(x)$, it tends to one as $x \to \infty$ by the Gaussian integral Eq. (C.43), and its derivative is $(2/\sqrt{\pi})\mathrm{e}^{-x^2}$. Rescaling the variable gives the Gaussian over a finite interval,

$$\int_{y_1}^{y_2} \mathrm{d}y\, \mathrm{e}^{-y^2/v_0^2} = \frac{\sqrt{\pi}\, v_0}{2} \Big[\operatorname{erf}\Big(\frac{y_2}{v_0}\Big) - \operatorname{erf}\Big(\frac{y_1}{v_0}\Big) \Big], \tag{C.66}$$

which is Eq. (D.1). Higher even powers reduce to it by parts. Since $y\,\mathrm{e}^{-y^2} = -\frac{1}{2}\mathrm{d}\mathrm{e}^{-y^2}/\mathrm{d}y$,

$$\int_0^x \mathrm{d}y\, y^2 \mathrm{e}^{-y^2} = -\frac{x}{2}\,\mathrm{e}^{-x^2} + \frac{1}{2}\int_0^x \mathrm{d}y\, \mathrm{e}^{-y^2} = \frac{\sqrt{\pi}}{4}\operatorname{erf}(x) - \frac{x}{2}\,\mathrm{e}^{-x^2}, \tag{C.67}$$

which, with $x = v_{\mathrm{esc}}/v_0$ and a factor $4\pi v_0^3$, is the normalization Eq. (1.4). Odd powers need no error function: $\int \mathrm{d}y\, y\, \mathrm{e}^{-y^2} = -\frac{1}{2}\mathrm{e}^{-y^2}$.

A Taylor series and an asymptotic series give numbers without tables. For small and moderate x the Taylor series [1, Eq. (7.6.1)]

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n!\,(2n+1)} = \frac{2}{\sqrt{\pi}} \Big(x - \frac{x^3}{3} + \frac{x^5}{10} - \cdots \Big), \tag{C.68}$$

converges for every x, after the terms have grown for $n \lesssim x^2$. At $x = v_{\mathrm{E}}^{\mathrm{ref}}/v_0 = 1.0525$, the value of Sect. 1.2, the partial sums oscillate about the limit and settle to five digits after nine terms, $\operatorname{erf}(1.0525) = 0.86338$. For large x the complement has the asymptotic expansion [1, Eq. (7.12.1)]

$$\operatorname{erfc}(x) \sim \frac{\mathrm{e}^{-x^2}}{x\sqrt{\pi}} \Big(1 - \frac{1}{2x^2} + \frac{3}{4x^4} - \frac{15}{8x^6} + \cdots \Big), \tag{C.69}$$

obtained by integrating $\mathrm{e}^{-y^2} = -(2y)^{-1}\mathrm{d}\mathrm{e}^{-y^2}/\mathrm{d}y$ by parts repeatedly. The series diverges for every x, since the nth coefficient grows as $(2n-1)!!/2^n$, but its terms first decrease, and truncated at the smallest term it is accurate to about the size of that term. At $x = v_{\mathrm{esc}}/v_0 = 544/238 = 2.2857$ the leading term is $\mathrm{e}^{-5.224}/(2.286\sqrt{\pi}) = 1.329 \times 10^{-3}$, the next two terms bring it to 1.238×10^{-3}, and the exact value is 1.227×10^{-3}, so $\operatorname{erf}(2.286) = 0.99877$. Since $\operatorname{erfc}(x) \simeq \mathrm{e}^{-x^2}/(x\sqrt{\pi})$ up to a relative correction $1/(2x^2)$, the fraction of a Gaussian halo beyond the escape speed is small for $v_{\mathrm{esc}} \gg v_0$. The second term of Eq. (C.67), the Gaussian evaluated at the cut, sets the size of the correction, 1.4 % of the normalization, and together with the erfc it gives the 1.5 % of Sect. 1.2.

C.8.2 Complete elliptic integrals

An average over one angle of the square root of a linear function of its cosine is not elementary. Let s be the modulus, $0 \le s < 1$, and $s' = (1 - s^2)^{1/2}$. The complete elliptic integrals of the first and second kind are [1, Eq. (19.2.8)]

$$\mathrm{K}(s) = \int_0^{\pi/2} \frac{\mathrm{d}\theta}{(1 - s^2 \sin^2\theta)^{1/2}}, \qquad \mathrm{E}(s) = \int_0^{\pi/2} \mathrm{d}\theta\, (1 - s^2 \sin^2\theta)^{1/2}. \tag{C.70}$$

The Digital Library of Mathematical Functions (DLMF) writes k for the modulus, which in this book is a crystal momentum; numerical libraries often take the parameter $m = s^2$ as the

argument, so that $\mathrm{K}(s)$ is computed as a function of s^2. Both integrals equal $\pi/2$ at $s=0$. We expand the integrand in powers of $s^2\sin^2\theta$ and use $\int_0^{\pi/2}\mathrm{d}\theta\,\sin^{2n}\theta=(\pi/2)(2n-1)!!/(2n)!!$. This gives [1, Eqs. (19.5.1) and (19.5.2)]

$$\mathrm{K}(s)=\frac{\pi}{2}\Big(1+\frac{s^2}{4}+\frac{9s^4}{64}+\cdots\Big),\qquad \mathrm{E}(s)=\frac{\pi}{2}\Big(1-\frac{s^2}{4}-\frac{3s^4}{64}-\cdots\Big). \tag{C.71}$$

As $s\to1$, $\mathrm{E}(s)\to1$, while $\mathrm{K}(s)$ diverges logarithmically [1, Eq. (19.12.1)],

$$\mathrm{K}(s)=\ln\frac{4}{s'}+O\Big(s'^2\ln\frac{1}{s'}\Big). \tag{C.72}$$

The divergence comes from the end of the interval, $\theta\to\pi/2$, where the integrand behaves as $1/[s'^2+(\pi/2-\theta)^2]^{1/2}$ and its integral over a range of order one is $\ln(1/s')$ plus a constant. At $s'=0.1$, Eq. (C.72) gives 3.689 against the exact 3.696; at $s'=0.01$ they differ by 1×10^{-4}.

The integrals of the form we need reduce to these. For $\mu_0>|\mu_1|$, the substitution $\phi=2\theta$ turns $\cos\phi$ into $1-2\sin^2\theta$ and $\mu_0+\mu_1\cos\phi$ into $(\mu_0+\mu_1)(1-s^2\sin^2\theta)$ with $s^2=2\mu_1/(\mu_0+\mu_1)$; for $\mu_1<0$ the substitution $\phi=\pi-2\theta$ does the same with $\mu_1\to-\mu_1$. Hence, with $s=[2|\mu_1|/(\mu_0+|\mu_1|)]^{1/2}$,

$$\int_0^\pi\frac{\mathrm{d}\phi}{(\mu_0+\mu_1\cos\phi)^{1/2}}=\frac{2\,\mathrm{K}(s)}{(\mu_0+|\mu_1|)^{1/2}},\qquad \int_0^\pi\mathrm{d}\phi\,(\mu_0+\mu_1\cos\phi)^{1/2}=2(\mu_0+|\mu_1|)^{1/2}\,\mathrm{E}(s). \tag{C.73}$$

Integrals of this kind arise when a quantity depends on the distance between a fixed point and a point moving on a circle. The angle between the Earth's galactic velocity and a laboratory axis, Eq. (1.16), has exactly the form $\cos\theta_{\mathrm{lab}}=\mu_0+\mu_1\cos\phi$ with ϕ the sidereal phase. A daily average of a rate that varies as $|\boldsymbol{v}_{\mathrm{E}}-v\hat{\boldsymbol{n}}|^{\pm1}$ for a fixed speed v along the laboratory axis is an elliptic integral. Logarithmic singularities of the same origin appear in the density of states of a two-dimensional band at a saddle point (Appendix C.2), where K diverges as in Eq. (C.72).

C.8.3 The Lindhard logarithms

The independent-particle response of the electron gas (Sect. 4.5) contains logarithms of ratios such as $(u+1)/(u-1)$, where u is a combination of ω and q in units set by the Fermi momentum k_{F} and the Fermi velocity v_{F} (Sect. 2.5). We need their behavior on the real axis, where they are real, and their imaginary part when the frequency acquires the $+\mathrm{i}0^+$ of a retarded function.

For real u the function $\ln|(u+1)/(u-1)|$ is finite except at $u=\pm1$, where it diverges logarithmically but integrably. The static response contains it in the combination

$$F_{\mathrm{L}}(y)=\frac{1}{2}+\frac{1-y^2}{4y}\ln\Big|\frac{1+y}{1-y}\Big|,\qquad y=\frac{q}{2k_{\mathrm{F}}}, \tag{C.74}$$

whose limits are elementary. For $y\to0$, $\ln[(1+y)/(1-y)]=2(y+y^3/3+y^5/5+\dots)$ gives $F_{\mathrm{L}}=1-y^2/3-y^4/15+O(y^6)$, the long-wavelength limit in which the response tends to the density of states at the Fermi level and Thomas–Fermi screening follows (Sect. 4.4). At $y=1$, the momentum $2k_{\mathrm{F}}$ that spans the Fermi sphere, $F_{\mathrm{L}}=\frac{1}{2}$, but its derivative diverges as $\frac{1}{2}\ln|1-y|$: the logarithm has a kink there, the origin of the Kohn anomaly and of Friedel oscillations. For $y\to\infty$, expanding in $1/y$ gives $F_{\mathrm{L}}=1/(3y^2)+O(y^{-4})$.

At finite frequency the argument becomes complex, and the logarithm needs a branch. Take the principal branch, $\ln z=\ln|z|+\mathrm{i}\arg z$ with $-\pi<\arg z\le\pi$, which is analytic off the negative real axis. Let u acquire a small positive imaginary part, $u\to u+\mathrm{i}0^+$, as the retarded

prescription $\omega \to \omega + \mathrm{i}0^+$ requires when u increases with ω. The ratio $(u+1+\mathrm{i}0^+)/(u-1+\mathrm{i}0^+)$ is real and positive for $|u| > 1$. For $|u| < 1$ it is negative: the numerator is positive and the denominator negative, with a small positive imaginary part, so its inverse has a small negative imaginary part and the ratio lies just below the negative real axis. Hence

$$\ln \frac{u+1+\mathrm{i}0^+}{u-1+\mathrm{i}0^+} = \ln\Big|\frac{u+1}{u-1}\Big| - \mathrm{i}\pi\,\Theta(1-|u|). \tag{C.75}$$

The imaginary part is a step. It is nonzero exactly where the frequency lies inside the band of particle–hole energies that a momentum transfer q can create, and its value $-\pi$ produces the negative imaginary part that a retarded response has for $\omega > 0$ (Appendix A.7). The same function follows from Eq. (C.27): $\ln[(u+1)/(u-1)] = \int_{-1}^{1} \mathrm{d}u'/(u-u')$, and the delta-function term of $1/(u-u'+\mathrm{i}0^+)$ contributes $-\mathrm{i}\pi$ when u lies inside the interval of integration. Equivalently, Im of the logarithm counts the on-shell transitions, and its real part is their principal-value transform, as Eq. (C.29) states for any Cauchy integral.

C.9 Classical fields and static propagators

Sect. 1.4 needs two results from classical field theory: the potential that a massive mediator produces around a static source, and the field redefinition that removes kinetic mixing between two vector fields. Both follow from Lagrangian densities and their Euler–Lagrange equations, without quantization. We use natural units, Heaviside–Lorentz charge units (Appendix A.2) and the metric signature $(+,-,-,-)$, with $x^\mu = (t, \boldsymbol{x})$, $\partial_\mu = (\partial_t, \nabla)$ and $\partial^\mu\partial_\mu = \partial_t^2 - \nabla^2$.

C.9.1 Scalar and vector fields with sources

We start with a real scalar field $\phi_V(x)$ of mass m_V. Coupled to a source density $\rho_\mathrm{s}(x)$, it has the Lagrangian density

$$\mathcal{L}_\mathrm{s} = \tfrac{1}{2}\,\partial_\mu\phi_V\,\partial^\mu\phi_V - \tfrac{1}{2}\,m_V^2\phi_V^2 - \phi_V\,\rho_\mathrm{s}. \tag{C.76}$$

Its Euler–Lagrange equation is $\partial_\mu[\partial\mathcal{L}/\partial(\partial_\mu\phi_V)] = \partial\mathcal{L}/\partial\phi_V$. Here it reads $\partial_\mu\partial^\mu\phi_V = -m_V^2\phi_V - \rho_\mathrm{s}$, the Klein–Gordon equation with a source,

$$(\partial_\mu\partial^\mu + m_V^2)\phi_V = -\rho_\mathrm{s}. \tag{C.77}$$

The vector mediator is a massive (Proca) field $A'_\mu(x)$ with field strength $F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu$. Coupled to a current $J^\mu(x)$, it has

$$\mathcal{L}_\mathrm{v} = -\tfrac{1}{4}\,F'_{\mu\nu}F'^{\mu\nu} + \tfrac{1}{2}\,m_V^2 A'_\mu A'^\mu - A'_\mu J^\mu. \tag{C.78}$$

Here $\partial\mathcal{L}_\mathrm{v}/\partial(\partial_\mu A'_\nu) = -F'^{\mu\nu}$, since $\partial(F'_{\alpha\beta}F'^{\alpha\beta})/\partial(\partial_\mu A'_\nu) = 4F'^{\mu\nu}$ (each of $F'_{\mu\nu}$ and $F'_{\nu\mu} = -F'_{\mu\nu}$ contributes $2F'^{\mu\nu}$), and $\partial\mathcal{L}_\mathrm{v}/\partial A'_\nu = m_V^2 A'^\nu - J^\nu$. The field equation is then

$$\partial_\mu F'^{\mu\nu} + m_V^2 A'^\nu = J^\nu. \tag{C.79}$$

Since $F'^{\mu\nu}$ is antisymmetric, $\partial_\nu\partial_\mu F'^{\mu\nu} = 0$, and the divergence of Eq. (C.79) leaves $m_V^2\,\partial_\nu A'^\nu = \partial_\nu J^\nu$. For a conserved current and $m_V \neq 0$ the field obeys $\partial_\nu A'^\nu = 0$; then $\partial_\mu F'^{\mu\nu} = \partial_\mu\partial^\mu A'^\nu$, and every component satisfies $(\partial_\mu\partial^\mu + m_V^2)A'^\nu = J^\nu$. At $m_V = 0$, Eq. (C.79) is Maxwell's equation in Heaviside–Lorentz units: its $\nu = 0$ component is $\nabla\cdot\boldsymbol{E} = J^0$, with $\boldsymbol{E} = -\nabla A^0 - \partial_t\boldsymbol{A}$.

For non-relativistic point particles the sources are number densities and number currents. A particle at $\boldsymbol{x}_i(t)$ contributes $\delta^3(\boldsymbol{x} - \boldsymbol{x}_i)$ to J^0 and $\dot{\boldsymbol{x}}_i\,\delta^3(\boldsymbol{x} - \boldsymbol{x}_i)$ to $\boldsymbol{J}$; a particle that couples with strength y contributes these terms multiplied by y. For the scalar, ρ_s is the coupling-weighted number density, which is the non-relativistic limit of the scalar density $\bar\psi\psi$ [13, Eq. (7)].

C.9.2 The static Green's function

For time-independent sources ∂_t drops out and $\partial_\mu\partial^\mu \to -\nabla^2$. A static source has $\boldsymbol{J}=0$, so only A'^0 is sourced. Eqs. (C.77) and (C.79) become

$$(-\nabla^2 + m_V^2)\phi_V = -\rho_{\rm s}, \qquad (-\nabla^2 + m_V^2)A'^0 = J^0. \tag{C.80}$$

Both are solved by the Green's function $G_{\rm Y}(\boldsymbol{x})$, defined by $(-\nabla^2 + m_V^2)\,G_{\rm Y}(\boldsymbol{x}) = \delta^3(\boldsymbol{x})$. With the Fourier convention of Appendix A.3, $\tilde G_{\rm Y}(\boldsymbol{q}) = \int \mathrm{d}^3x\, \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}} G_{\rm Y}(\boldsymbol{x})$, the operator $-\nabla^2$ becomes q^2 and the delta function becomes 1:

$$\tilde G_{\rm Y}(\boldsymbol{q}) = \frac{1}{q^2 + m_V^2}. \tag{C.81}$$

The inverse transform is $G_{\rm Y}(\boldsymbol{x}) = \int \mathrm{d}^3q\,(2\pi)^{-3}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}/(q^2{+}m_V^2)$. We take the polar axis along $\boldsymbol{x}$ and write $r = |\boldsymbol{x}|$. The angular integral is $\int \mathrm{d}\Omega\,\mathrm{e}^{-\mathrm{i}qr\cos\theta} = 2\pi\int_{-1}^{1}\mathrm{d}u\,\mathrm{e}^{-\mathrm{i}qru} = 4\pi\sin(qr)/(qr)$, which leaves

$$G_{\rm Y}(\boldsymbol{x}) = \frac{1}{2\pi^2 r}\int_0^\infty \mathrm{d}q\,\frac{q\sin(qr)}{q^2+m_V^2} = \frac{1}{4\pi^2 r}\,\mathrm{Im}\int_{-\infty}^{\infty}\mathrm{d}q\,\frac{q\,\mathrm{e}^{\mathrm{i}qr}}{(q-\mathrm{i}m_V)(q+\mathrm{i}m_V)}.$$

The second form holds because the integrand of the first is even in q, which makes the half line half the full line, and because $\sin(qr) = \mathrm{Im}\,\mathrm{e}^{\mathrm{i}qr}$ for real q. Since for $r>0$ the factor $\mathrm{e}^{\mathrm{i}qr}$ decays in the upper half plane, by Jordan's lemma the integral equals the integral over a contour closed there. The contour encloses the pole at $q = \mathrm{i}m_V$, with residue $\mathrm{i}m_V\,\mathrm{e}^{-m_Vr}/(2\mathrm{i}m_V) = \mathrm{e}^{-m_Vr}/2$. The full-line integral is $2\pi\mathrm{i}\times\mathrm{e}^{-m_Vr}/2 = \mathrm{i}\pi\,\mathrm{e}^{-m_Vr}$, its imaginary part is $\pi\,\mathrm{e}^{-m_Vr}$, and

$$G_{\rm Y}(\boldsymbol{x}) = \frac{\mathrm{e}^{-m_Vr}}{4\pi r}. \tag{C.82}$$

A direct check confirms it: for $r>0$, $\nabla^2 G_{\rm Y} = r^{-1}\,\mathrm{d}^2(rG_{\rm Y})/\mathrm{d}r^2 = m_V^2 G_{\rm Y}$, and $(-\nabla^2{+}m_V^2)G_{\rm Y}$ vanishes away from the origin. Near the origin $G_{\rm Y}\to 1/(4\pi r)$, and $-\nabla^2(1/4\pi r) = \delta^3(\boldsymbol{x})$ supplies the source. Eq. (C.82) is the Yukawa form: the field falls off over the range $1/m_V$ and becomes the Coulomb potential $1/(4\pi r)$ as $m_V\to 0$. As a check of the units, a proton and an electron, with sources $\pm e$, then have the interaction energy $-e^2/(4\pi r) = -\alpha/r$ (Eq. (C.84) below, vector sign), the hydrogen potential in Heaviside–Lorentz units.

By superposition the static fields are $A'^0(\boldsymbol{x}) = \int \mathrm{d}^3x'\,G_{\rm Y}(\boldsymbol{x}-\boldsymbol{x}')J^0(\boldsymbol{x}')$ and $\phi_V(\boldsymbol{x}) = -\int \mathrm{d}^3x'\,G_{\rm Y}(\boldsymbol{x}-\boldsymbol{x}')\rho_{\rm s}(\boldsymbol{x}')$. A point particle with coupling y_χ at the origin produces

$$A'^0(\boldsymbol{x}) = y_\chi\,\frac{\mathrm{e}^{-m_Vr}}{4\pi r}, \qquad \phi_V(\boldsymbol{x}) = -y_\chi\,\frac{\mathrm{e}^{-m_Vr}}{4\pi r}. \tag{C.83}$$

C.9.3 Validity of the static limit

Dark matter that scatters moves, and so does the electron it scatters on; the sources are not strictly static. With the time convention $f(\omega) = \int \mathrm{d}t\,\mathrm{e}^{+\mathrm{i}\omega t}f(t)$ of Appendix A.3, ∂_t^2 becomes $-\omega^2$, and the operator $\partial_\mu\partial^\mu + m_V^2$ of Eq. (C.77) becomes $q^2 - \omega^2 + m_V^2$ for a field component of momentum $\boldsymbol{q}$ and energy ω. The full Green's function in momentum space is then $1/(q^2-\omega^2+m_V^2)$, which reduces to Eq. (C.81) when $\omega^2 \ll q^2 + m_V^2$. In a scattering event the mediator carries the energy ω and momentum $\boldsymbol{q}$ given to the target, and kinematics bounds them by $\omega \le q\,v_{\rm max}$ (Sect. 1.3), with $v_{\rm max} = 2.65\times10^{-3}$. The neglected term is then smaller than $v_{\rm max}^2 = 7\times10^{-6}$ relative to q^2, and the static potential is accurate to better than a part in 10^5 for every purpose of this book.

Absorption, however, has no static limit at all. There the boson is the dark matter itself, and its field obeys the source-free equation $(\partial_\mu\partial^\mu + m_\phi^2)\phi = 0$ with $\omega^2 = |\boldsymbol{p}_\phi|^2 + m_\phi^2$: an oscillating, on-shell field, not a static potential. Sect. 1.4 treats the two processes separately for this reason.

C.9.4 Interaction energy of two sources

Since a static configuration has no time derivatives, its energy is $-\int \mathrm{d}^3x\,\mathcal{L}$, written H_s for the scalar and H_v for the vector. For the scalar, Eq. (C.76) with $\partial_t\phi_V = 0$ reads $\mathcal{L}_\mathrm{s} = -\frac{1}{2}(\nabla\phi_V)^2 - \frac{1}{2}m_V^2\phi_V^2 - \phi_V\rho_\mathrm{s}$. Integrating $(\nabla\phi_V)^2$ by parts and using Eq. (C.80),

$$\int \mathrm{d}^3x\,[\tfrac{1}{2}(\nabla\phi_V)^2 + \tfrac{1}{2}m_V^2\phi_V^2] = \tfrac{1}{2}\int \mathrm{d}^3x\,\phi_V(-\nabla^2 + m_V^2)\phi_V = -\tfrac{1}{2}\int \mathrm{d}^3x\,\phi_V\,\rho_\mathrm{s},$$

so $H_\mathrm{s} = \frac{1}{2}\int \mathrm{d}^3x\,\phi_V\rho_\mathrm{s} = -\frac{1}{2}\int \mathrm{d}^3x\,\mathrm{d}^3x'\,\rho_\mathrm{s}(\boldsymbol{x})\,G_\mathrm{Y}(\boldsymbol{x}-\boldsymbol{x}')\,\rho_\mathrm{s}(\boldsymbol{x}')$. For the vector with only A'^0 nonzero, $F'_{0i} = -\partial_i A'^0$ and $-\frac{1}{4}F'_{\mu\nu}F'^{\mu\nu} = -\frac{1}{2}F'_{0i}F'^{0i} = +\frac{1}{2}(\nabla A'^0)^2$, and $\mathcal{L}_\mathrm{v} = \frac{1}{2}(\nabla A'^0)^2 + \frac{1}{2}m_V^2(A'^0)^2 - A'^0J^0$. The same steps lead to $H_\mathrm{v} = +\frac{1}{2}\int \mathrm{d}^3x\,\mathrm{d}^3x'\,J^0(\boldsymbol{x})\,G_\mathrm{Y}(\boldsymbol{x}-\boldsymbol{x}')\,J^0(\boldsymbol{x}')$. At $m_V = 0$ this is $\frac{1}{2}\int \mathrm{d}^3x\,\boldsymbol{E}^2$, the positive electrostatic energy, which fixes the overall sign.

Now take two point particles, one with coupling y_χ at $\boldsymbol{x}_\chi$ and one with y_e at $\boldsymbol{x}_e$. Each double integral contains two self-energy terms, independent of the separation, and two equal cross terms. The cross terms give the interaction energy

$$U(\boldsymbol{x}_e - \boldsymbol{x}_\chi) = \pm\, y_\chi y_e\, G_\mathrm{Y}(\boldsymbol{x}_e - \boldsymbol{x}_\chi) = \pm\, y_\chi y_e\, \frac{\mathrm{e}^{-m_V|\boldsymbol{x}_e-\boldsymbol{x}_\chi|}}{4\pi\,|\boldsymbol{x}_e-\boldsymbol{x}_\chi|}, \tag{C.84}$$

with the upper sign for the vector and the lower for the scalar. Vector exchange repels sources whose couplings have the same sign, as like electric charges repel; scalar exchange attracts them. The same result follows from treating the second particle as a test particle: its term $-y_eA'^0(\boldsymbol{x}_e)$ or $-y_e\phi_V(\boldsymbol{x}_e)$ in the Lagrangian is a potential energy $+y_eA'^0$ or $+y_e\phi_V$, evaluated with the fields of Eq. (C.83). The Fourier transform of Eq. (C.84) is $\pm\,y_\chi y_e/(q^2+m_V^2)$ by Eq. (C.81). The sign drops out of every rate, which depends on its square.

C.9.5 Kinetic mixing

Let the photon A_μ, with field strength $F_{\mu\nu}$, couple to the electromagnetic current eJ^μ_em, where J^μ_em counts each particle with its charge in units of e (an electron counts -1). Let a second vector A'_μ of mass m_V couple to the dark-matter number current J^μ_χ with strength y_χ. Gauge invariance allows a term that mixes the two field strengths, with a dimensionless coefficient ζ:

$$\mathcal{L} = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} - \tfrac{1}{4}F'_{\mu\nu}F'^{\mu\nu} + \tfrac{\zeta}{2}F_{\mu\nu}F'^{\mu\nu} + \tfrac{1}{2}m_V^2A'_\mu A'^\mu - eA_\mu J^\mu_\mathrm{em} - y_\chi A'_\mu J^\mu_\chi. \tag{C.85}$$

The sign of the mixing term follows Ref. [14, Eq. (70)]; other papers use the opposite sign or write ϵ, κ or χ for ζ (Appendix B.2). Only ζ^2 enters a rate.

The kinetic terms are not diagonal, so A_μ and A'_μ are not independent propagating fields. We shift the photon, $A_\mu = \tilde{A}_\mu + \zeta A'_\mu$, so that $F_{\mu\nu} = \bar{F}_{\mu\nu} + \zeta F'_{\mu\nu}$, with $\bar{F}_{\mu\nu} = \partial_\mu\tilde{A}_\nu - \partial_\nu\tilde{A}_\mu$. The photon kinetic term becomes $-\frac{1}{4}\bar{F}^2 - \frac{\zeta}{2}\bar{F}F' - \frac{\zeta^2}{4}F'^2$, with $\bar{F}F' \equiv \bar{F}_{\mu\nu}F'^{\mu\nu}$. The mixing term becomes $\frac{\zeta}{2}\bar{F}F' + \frac{\zeta^2}{2}F'^2$. The cross terms cancel, and

$$\mathcal{L} = -\tfrac{1}{4}\bar{F}_{\mu\nu}\bar{F}^{\mu\nu} - \tfrac{1}{4}(1-\zeta^2)F'_{\mu\nu}F'^{\mu\nu} + \tfrac{1}{2}m_V^2A'_\mu A'^\mu - e\tilde{A}_\mu J^\mu_\mathrm{em} - \zeta e\,A'_\mu J^\mu_\mathrm{em} - y_\chi A'_\mu J^\mu_\chi, \tag{C.86}$$

which differs from Ref. [14, Eq. (72)] only in the sign convention for the current. Shifting the photon, and not A'_μ, leaves the mass term untouched: $\tilde{A}_\mu$ stays massless and the mass matrix stays diagonal. Rescaling A'_μ by $(1-\zeta^2)^{-1/2}$ restores its canonical normalization and changes m_V and the couplings at order ζ^2, which we drop.

For $m_V = 0$ the alternative shift $A'_\mu \to A'_\mu + \zeta A_\mu$ also removes the mixing; the photon then couples to dark matter with strength ζy_χ, and dark matter appears millicharged [14, Eq. (71)]. For $m_V \neq 0$, however, that shift would mix the mass term, and the photon shift

is the one to use. In the new basis the photon couples only to charged particles, and the massive vector couples to dark matter with strength y_χ and to the electromagnetic current with strength ζe. For electrons, $J^\mu_{\rm em}$ contains $-J^\mu_e$, with J^μ_e the electron number current, which makes the electron coupling in the form $-y_e A'_\mu J^\mu_e$ equal to $y_e = -\zeta e$; protons have $+\zeta e$.

The redefinition is exact in vacuum. In a medium the photon acquires a self-energy from the charges it polarizes, the two kinetic terms are no longer the whole quadratic Lagrangian, and mixing reappears at the level of the in-medium propagators (Sects. 1.4 and 4.4).

C.9.6 From the potential to a cross-section

Once the mediator is eliminated, dark matter and an electron interact through the potential energy $V(\boldsymbol{x}) = U(\boldsymbol{x})$ of Eq. (C.84), a function of their separation. Ordinary quantum mechanics then applies. To first order in $V(\boldsymbol{x})$ (the first Born approximation), the amplitude for scattering with momentum transfer $\boldsymbol{q}$ is $-(\mu_{\chi e}/2\pi)\tilde{V}(\boldsymbol{q})$, with $\tilde{V}(\boldsymbol{q})$ the Fourier transform of $V(\boldsymbol{x})$ and $\mu_{\chi e}$ the reduced mass, and the differential cross-section is its square. If $\tilde{V}$ depends on q, the differential cross-section carries $|\tilde{V}(\boldsymbol{q})|^2$, which is how the form factor of Sect. 1.4 enters. If $\tilde{V}$ does not depend on $\boldsymbol{q}$, the angular integral yields $\sigma = \mu^2_{\chi e}|\tilde{V}|^2/\pi$. Sect. 2.2 derives this from the golden rule, with the normalization used in the rest of the book.

C.10 Moment problems

The first five energy moments of a nuclear response at fixed $\boldsymbol{q}$ are equal-time averages, which path-integral simulations give for any anharmonicity, while the response itself needs real-time dynamics. Sect. 11.5 asks what five moments fix about the distribution of energy deposits. The answer is a band around the cumulative distribution, with a width fixed by the moments alone, wide at the peak and narrow in the tails. We derive the band, show why it is an absolute and not a relative bound, and state what changes when the distribution is supported on $\omega \geq 0$.

C.10.1 The Hamburger and Stieltjes problems

Let P be a probability distribution on the real line, with moments

$$\mu_k = \int x^k \,\mathrm{d}P(x), \qquad k = 0, 1, \ldots, 4, \qquad \mu_0 = 1. \tag{C.87}$$

For the response of Sect. 11.5, x is the energy transfer ω and $\mathrm{d}P = S(\boldsymbol{q},\omega)\,\mathrm{d}\omega / \int S\,\mathrm{d}\omega$. The *Hamburger moment problem* asks which numbers μ_k are the moments of some distribution on the real line, and how many distributions share them. The *Stieltjes problem* asks the same for distributions on the half line $x \geq 0$, the case of a response at zero temperature, which has no weight at $\omega < 0$ (Appendix A.4). We treat the truncated problems, in which only $\mu_0, \ldots, \mu_4$ are given.

The necessary condition comes from positivity. For any real polynomial $p(x) = c_0 + c_1 x + c_2 x^2$, written $p(x) = \mathbf{c}^{\mathsf{T}}\mathbf{v}(x)$ with $\mathbf{v}(x) = (1, x, x^2)^{\mathsf{T}}$,

$$0 \leq \int p(x)^2 \,\mathrm{d}P(x) = \mathbf{c}^{\mathsf{T}}\mathbf{H}\,\mathbf{c}, \qquad \mathbf{H} = \int \mathbf{v}(x)\mathbf{v}(x)^{\mathsf{T}}\mathrm{d}P(x) = \begin{pmatrix} \mu_0 & \mu_1 & \mu_2 \\ \mu_1 & \mu_2 & \mu_3 \\ \mu_2 & \mu_3 & \mu_4 \end{pmatrix}. \tag{C.88}$$

The *Hankel matrix* $\mathbf{H}$, whose entries depend only on the sum of their indices, $H_{ij} = \mu_{i+j}$, is therefore positive semidefinite (Appendix C.7). It is positive definite unless P is concentrated

on the two or fewer roots of some quadratic p, and we assume from now on that P has at least three points of support, so that $\mathbf{H}$ is positive definite and invertible. On the half line, $\int x\, p_1(x)^2 \,\mathrm{d}P \geq 0$ for every linear p_1 adds a second condition: the shifted matrix $\mathbf{H}_1$, with entries μ_{i+j+1} for $i, j = 0, 1$, is positive semidefinite as well. For the truncated problems these conditions are also sufficient, up to boundary cases in which a matrix is singular [15]. If all moments of P are given, they fix P uniquely when they grow slowly enough (Carleman's condition, which a Gaussian satisfies); five moments never do, and infinitely many distributions share them. The question is how far apart these distributions can be.

C.10.2 Christoffel function and canonical representation

The bound rests on one function of a point ξ. Define the kernel

$$\mathsf{K}(x, \xi) = \mathbf{v}(x)^{\mathsf{T}} \mathbf{H}^{-1} \mathbf{v}(\xi), \tag{C.89}$$

a quadratic polynomial in x for fixed ξ. It reproduces every quadratic $q(x) = \mathbf{a}^{\mathsf{T}}\mathbf{v}(x)$ under the integral: by Eq. (C.88),

$$\int q(x)\, \mathsf{K}(x, \xi) \,\mathrm{d}P(x) = \mathbf{a}^{\mathsf{T}} \mathbf{H} \mathbf{H}^{-1} \mathbf{v}(\xi) = q(\xi). \tag{C.90}$$

The *Christoffel function* is

$$\varrho_{\mathrm{MK}}(\xi) = \frac{1}{\mathsf{K}(\xi, \xi)} = \frac{1}{\mathbf{v}(\xi)^{\mathsf{T}} \mathbf{H}^{-1} \mathbf{v}(\xi)}. \tag{C.91}$$

It is the smallest value of $\int p^2 \,\mathrm{d}P$ over quadratics with $p(\xi) = 1$: minimizing $\mathbf{c}^{\mathsf{T}}\mathbf{H}\mathbf{c}$ subject to $\mathbf{c}^{\mathsf{T}}\mathbf{v}(\xi) = 1$ with a Lagrange multiplier gives $\mathbf{c} \propto \mathbf{H}^{-1}\mathbf{v}(\xi)$, and the minimum is Eq. (C.91). Hence no distribution with these moments can put more than $\varrho_{\mathrm{MK}}(\xi)$ at the single point ξ: if it puts w there, then $w = w\, p(\xi)^2 \leq \int p^2 \,\mathrm{d}P$ for the minimizing p, so $w \leq \varrho_{\mathrm{MK}}(\xi)$.

The bound is attained by a three-point distribution. Let $\mathsf{K}(\cdot, \xi)$ have nonzero leading coefficient, which fails only at the at most two roots of a quadratic in ξ. Then the cubic $r(x) = (x - \xi)\, \mathsf{K}(x, \xi)$ has three distinct real roots. To see this, note that $\int r(x) s(x) \,\mathrm{d}P = 0$ for every linear s, by Eq. (C.90) applied to the quadratic $(x - \xi)s(x)$, which vanishes at ξ. A polynomial that is orthogonal to all linear polynomials changes sign at least twice on the support of P: if it changed sign at most once, at x_*, the product with $s(x) = \pm(x - x_*)$ would have one sign and a nonzero integral. A real cubic with two sign changes has three real roots. One root is ξ, and it is simple, since $\mathsf{K}(\xi, \xi) > 0$. The other two are the roots of $\mathsf{K}(\cdot, \xi)$, and they are distinct, because a double root would not change sign. Call the three roots $y_1 < y_2 < y_3$; one of them is ξ.

We now show that the three points carry a quadrature exact for every polynomial f of degree up to four. Divide f by r: $f = r\, s + t$, with s linear and t the quadratic that interpolates f at y_1, y_2, y_3. The term $\int r\, s \,\mathrm{d}P$ vanishes, as shown above. The quadratic t is a combination of the three Lagrange polynomials ℓ_i, with $\ell_i(y_j) = \delta_{ij}$, so that $\int t \,\mathrm{d}P = \sum_i w_i f(y_i)$ with $w_i = \int \ell_i \,\mathrm{d}P$. Therefore

$$\int f(x) \,\mathrm{d}P(x) = \sum_{i=1}^{3} w_i\, f(y_i) \qquad \text{for every } f \text{ of degree} \leq 4. \tag{C.92}$$

The weights are positive: ℓ_i^2 has degree four, and Eq. (C.92) gives $\int \ell_i^2 \,\mathrm{d}P = w_i$. The weight at ξ follows from $f = \mathsf{K}(\cdot, \xi)^2$, which vanishes at the two other nodes. By Eq. (C.90) with $q = \mathsf{K}(\cdot, \xi)$ the left side of Eq. (C.92) is $\mathsf{K}(\xi, \xi)$, and the right side is $w_\xi\, \mathsf{K}(\xi, \xi)^2$. So $w_\xi = \varrho_{\mathrm{MK}}(\xi)$. The three-point distribution with nodes y_i and weights w_i has the same five moments as P (take $f = x^k$) and puts the largest admissible mass at ξ. It is the *canonical representation* through ξ.

C.10.3 The Markov–Krein band

Write $F_P(\xi) = P((-\infty, \xi])$ for the cumulative distribution and $F_P(\xi - 0)$ for its limit from below. The band is

Markov–Krein band

$$\sum_{y_i<\xi} w_i \;\le\; F_P(\xi - 0) \;\le\; F_P(\xi) \;\le\; \sum_{y_i\le\xi} w_i, \tag{C.93}$$

with y_i, w_i the canonical representation through ξ. Every distribution with the moments $\mu_0, \dots, \mu_4$ satisfies it, and the two bounds differ by $\varrho_{\mathrm{MK}}(\xi)$ of Eq. (C.91).

The result goes back to Chebyshev, Markov and Stieltjes, and the general form is in Refs. [15] and [16].

The proof compares the indicator function of $(-\infty, \xi]$ with a polynomial U of degree four that lies above it and touches it at the nodes. If $U(x) \ge \Theta(\xi - x)$ for all x, with $\Theta(0) = 1$, and $U(y_i) = \Theta(\xi - y_i)$ at the three nodes, then by Eq. (C.92)

$$F_P(\xi) = \int \Theta(\xi - x)\, \mathrm{d}P(x) \le \int U\, \mathrm{d}P = \sum_i w_i\, U(y_i) = \sum_{y_i\le\xi} w_i.$$

We construct U for the three positions of ξ among the nodes. If $\xi = y_3$, the largest node, $U = 1$ works and the bound is the trivial $F_P \le 1$. If $\xi = y_1$, the smallest, take

$$U(x) = \frac{(x - y_2)^2 (x - y_3)^2}{(y_1 - y_2)^2 (y_1 - y_3)^2}.$$

It is nonnegative everywhere, equals 1 at y_1 and 0 at y_2, y_3, and for $x < y_1$ both factors $|x - y_2|$ and $|x - y_3|$ exceed their values at y_1, so $U \ge 1$ there. If $\xi = y_2$, let U be the quartic with $U(y_1) = U(y_2) = 1$, $U(y_3) = 0$ and $U'(y_1) = U'(y_3) = 0$: five conditions, which fix a polynomial of degree four. Between y_1 and y_2, where U takes the value 1 twice, Rolle's theorem places a root c of U'. The cubic U' then has the three roots $y_1 < c < y_3$ and no other, $U' = A(x - y_1)(x - c)(x - y_3)$. On (y_2, y_3) the function falls from 1 to 0 with no critical point, so $U' < 0$ there; since $(x - y_1)(x - c)(x - y_3) < 0$ on that interval, $A > 0$. The signs of U' now fix the shape: U decreases towards $U(y_1) = 1$ for $x < y_1$, rises on (y_1, c), falls on (c, y_3) through $U(y_2) = 1$ to $U(y_3) = 0$, and rises for $x > y_3$. So $U \ge 1$ for $x \le y_2$ and $U \ge 0$ beyond, which is what we need. The lower bound in Eq. (C.93) is the upper bound applied to the mirrored distribution of $-x$, whose canonical representation through $-\xi$ is the mirror image of that through ξ: the mass of $[\xi, \infty)$ is at most $\sum_{y_i\ge\xi} w_i$, so $F_P(\xi - 0) \ge \sum_{y_i<\xi} w_i$. If the leading coefficient of $\mathsf{K}(\cdot, \xi)$ vanishes, one node runs off to infinity with vanishing weight; since F_P is monotone and the bounds are continuous in ξ, Eq. (C.93) holds there by continuity.

For a standard Gaussian, $\mu_0, \dots, \mu_4 = 1, 0, 1, 0, 3$. The inverse of $\mathbf{H}$ is

$$\mathbf{H}^{-1} = \begin{pmatrix} 3/2 & 0 & -1/2 \\ 0 & 1 & 0 \\ -1/2 & 0 & 1/2 \end{pmatrix},$$

as multiplication by $\mathbf{H}$ confirms. With $\mathbf{v}(\xi) = (1, \xi, \xi^2)^{\mathsf{T}}$, the quadratic form is $\frac{3}{2} + 2(-\frac{1}{2})\xi^2 + \xi^2 + \frac{1}{2}\xi^4$. The two terms in ξ^2 cancel, $\mathsf{K}(\xi, \xi) = \frac{3}{2} + \frac{1}{2}\xi^4$, and

$$\varrho_{\mathrm{MK}}(\xi) = \frac{2}{3 + \xi^4} \qquad \text{(standard Gaussian)}. \tag{C.94}$$

At the mean, $\varrho_{\mathrm{MK}}(0) = 2/3$: the canonical representation through 0 has nodes 0 and $\pm\sqrt{3}$ with weights $\frac{2}{3}$ and $\frac{1}{6}$, and Eq. (C.93) reads $\frac{1}{6} \le F_P(0) \le \frac{5}{6}$. Five moments say almost nothing about the median.

C.10.4 Tails of the band

Far from the mean, $\mathbf{v}(\xi)$ is dominated by its last entry, ξ^2, and $\mathsf{K}(\xi,\xi) \simeq (\mathbf{H}^{-1})_{22}\,\xi^4$, with indices running over 0, 1, 2. For a distribution shifted to zero mean, with central moments $\bar\mu_2, \bar\mu_3, \bar\mu_4$, the corner element of the inverse is the 2×2 minor over the determinant, $(\mathbf{H}^{-1})_{22} = \bar\mu_2/\det\mathbf{H}$, and $\det\mathbf{H} = \bar\mu_2\bar\mu_4 - \bar\mu_3^2 - \bar\mu_2^3$ by expansion along the first row. Hence

$$\varrho_{\mathrm{MK}}(\xi) \simeq \frac{\bar\mu_4 - \bar\mu_2^2 - \bar\mu_3^2/\bar\mu_2}{(\xi-\mu_1)^4}, \qquad |\xi-\mu_1| \to \infty. \tag{C.95}$$

The numerator is positive because $\mathbf{H}$ is positive definite. For the Gaussian it is $3-1=2$, and Eq. (C.94) indeed behaves as $2/\xi^4$. For a skewed example, the gamma distribution of shape 3, the exact ϱ_{MK} exceeds Eq. (C.95) by 4.1 % at 58 standard deviations above the mean and by 1.3 % at 173; at 58 standard deviations below the mean it falls 3.9 % short of it. The next term is smaller by one power of $\xi-\mu_1$.

The band is narrow in the tails in absolute terms, but it is only a power law. A true threshold tail falls much faster: like a Gaussian for a harmonic multiphonon response, or exponentially. At three standard deviations a Gaussian has $1-F_P = 1.35\times10^{-3}$, while the canonical representation through $\xi=3$ leaves $1-F_P$ anywhere in $[0, 2.38\times10^{-2}]$, eighteen times the true value; at four standard deviations the ratio is 240. The band therefore bounds $|F_P - F_{\mathrm{model}}|$ for any model distribution with the right moments, everywhere, by a number that is small in the tail. It does not bound the ratio $(1-F_P)/(1-F_{\mathrm{model}})$, which is what a rate near threshold needs: when the threshold sits far in the tail, the whole predicted rate is smaller than the band width, and five moments leave it undetermined.

C.10.5 Distributions on the half line

At zero temperature the response has no weight below $\omega=0$, and P is a Stieltjes distribution. Every such distribution is also a Hamburger distribution, so Eq. (C.93) remains valid. It is no longer sharp, because a canonical representation with a node below zero is not admissible, and the positivity of $\mathbf{H}_1$ restricts the distributions further. The bound on the mass at a single point $\xi>0$ shows how. Since $x\ge0$ on the support, a mass w at ξ obeys $w\,\xi\,p_1(\xi)^2 \le \int x\,p_1(x)^2\,\mathrm{d}P$ for every linear p_1. Minimizing over p_1 as before gives

$$w \le \min\Big\{\varrho_{\mathrm{MK}}(\xi),\ \big[\xi\,\mathbf{u}(\xi)^{\mathsf{T}}\mathbf{H}_1^{-1}\mathbf{u}(\xi)\big]^{-1}\Big\}, \qquad \mathbf{u}(\xi)=(1,\xi)^{\mathsf{T}}. \tag{C.96}$$

The second entry diverges as $\xi\to0$ and falls only as ξ^{-3} in the tail, so it is weaker than ϱ_{MK} near the origin and far out. It is tighter only at intermediate ξ, where a node of the canonical representation through ξ lies below zero: for silicon at $q=100\,\mathrm{keV}$ it lies below ϱ_{MK} by at most 6.8 %, near $\xi = 0.74\,E_R$ (Sect. 11.5). The sharp Stieltjes band is built from canonical representations whose nodes all lie in $x\ge0$ [15]. For $\xi<0$ the band collapses to $F_P=0$.

C.11 Generating functions, branching and renewal

A 100 eV electron in silicon ends as about 27 electron–hole pairs, each made by a carrier that was itself made by an earlier one. Sects. 18.1 and 18.2 treat this cascade as a branching process and read the mean energy per pair $\varepsilon_{\mathrm{eh}}$ and the Fano factor F_{eh} off its first two moments. Probability generating functions turn sums of independent counts into products. The branching recursion builds the statistics of a whole tree from those of one step. Renewal theory says why the mean pair number becomes linear in the deposited energy, and what is left over at the end.

C.11.1 Probability generating functions

For a random count $N = 0, 1, 2, \ldots$ with probabilities p_n, the *probability generating function* is

$$\mathfrak{G}_N(s) = \langle s^N \rangle = \sum_{n \geq 0} p_n \, s^n, \qquad 0 \leq s \leq 1. \tag{C.97}$$

It is a power series with nonnegative coefficients, so it is increasing and convex on $[0, 1]$, with $\mathfrak{G}_N(0) = p_0$ and $\mathfrak{G}_N(1) = 1$. Derivatives at $s = 1$ give factorial moments: $\mathfrak{G}'_N(1) = \langle N \rangle$ and $\mathfrak{G}''_N(1) = \langle N(N-1) \rangle$. Hence

$$\operatorname{Var} N = \mathfrak{G}''_N(1) + \mathfrak{G}'_N(1) - \mathfrak{G}'_N(1)^2. \tag{C.98}$$

The Poisson distribution of Eq. (1.51) with mean $\bar{N}$ has $\mathfrak{G}(s) = \sum_n \mathrm{e}^{-\bar{N}} (\bar{N}s)^n / n! = \mathrm{e}^{\bar{N}(s-1)}$. Then $\mathfrak{G}'(1) = \bar{N}$, $\mathfrak{G}''(1) = \bar{N}^2$, and $\operatorname{Var} N = \bar{N}$: the Fano factor $\operatorname{Var} N / \langle N \rangle$ of a Poisson count is 1, the reference against which the value 0.1–0.15 of silicon (Sect. 1.5) is sub-Poissonian.

If N_1 and N_2 are independent, $\langle s^{N_1 + N_2} \rangle = \langle s^{N_1} \rangle \langle s^{N_2} \rangle$: the generating function of a sum is the product. If N items are each kept independently with probability f, the kept number N_k is a sum of N independent Bernoulli counts, each with generating function $1 - f + fs$. At fixed N it has the generating function $(1 - f + fs)^N$, and averaging over N,

$$\mathfrak{G}_{N_\mathrm{k}}(s) = \mathfrak{G}_N(1 - f + fs). \tag{C.99}$$

By the chain rule $\langle N_\mathrm{k} \rangle = f \langle N \rangle$ and $\mathfrak{G}''_{N_\mathrm{k}}(1) = f^2 \mathfrak{G}''_N(1) = f^2 (\operatorname{Var} N - \langle N \rangle + \langle N \rangle^2)$. Inserting both into Eq. (C.98) and dividing by $f \langle N \rangle$, the Fano factor of the kept items is

$$F_\mathrm{k} = f \, F_N + 1 - f, \qquad F_N = \operatorname{Var} N / \langle N \rangle. \tag{C.100}$$

Random loss pulls any Fano factor towards the Poisson value 1. This is the rule by which a detector that collects a fraction f_det of the pairs measures $f_\mathrm{det} F_\mathrm{eh} + 1 - f_\mathrm{det}$ (Sect. 20.1).

C.11.2 The branching recursion and extinction

In the simplest branching process, the Galton–Watson process, each individual of generation n has a random number of offspring, independently of all others and with one distribution, of generating function $\mathfrak{g}(s)$ and mean $\bar{m} = \mathfrak{g}'(1)$. Let N_n be the size of generation n, with $N_0 = 1$, and $\mathfrak{G}_n$ its generating function. Given N_n, the next generation is a sum of N_n independent offspring counts, so $\langle s^{N_{n+1}} \mid N_n \rangle = \mathfrak{g}(s)^{N_n}$, and averaging over N_n,

$$\mathfrak{G}_{n+1}(s) = \mathfrak{G}_n\big(\mathfrak{g}(s)\big) = \mathfrak{g}\big(\mathfrak{G}_n(s)\big), \qquad \mathfrak{G}_0(s) = s. \tag{C.101}$$

The first form is what we derived; it makes $\mathfrak{G}_n$ the n-fold composition of $\mathfrak{g}$ with itself, which gives the second. The second form is a *first-step* statement: condition on the offspring of the ancestor, and each child starts an independent copy of the process with n generations to go. Differentiating at $s = 1$ gives $\langle N_n \rangle = \bar{m}^{\,n}$.

The probability that the line has died out by generation n is $\mathfrak{G}_n(0)$. By Eq. (C.101), $\mathfrak{G}_{n+1}(0) = \mathfrak{g}(\mathfrak{G}_n(0))$. Since $\mathfrak{g}$ is increasing and $\mathfrak{G}_1(0) = \mathfrak{g}(0) \geq 0 = \mathfrak{G}_0(0)$, the sequence increases, and being bounded by 1 it converges to a limit P_ext with $\mathfrak{g}(P_\mathrm{ext}) = P_\mathrm{ext}$, by continuity of $\mathfrak{g}$. The equation $\mathfrak{g}(s) = s$ can have two roots in $[0, 1]$, and the iteration selects the smaller: for any root s_*, $0 \leq s_*$ implies $\mathfrak{G}_n(0) \leq \mathfrak{G}_n(s_*) = s_*$ for all n, so $P_\mathrm{ext} \leq s_*$. Hence

Extinction probability

The extinction probability P_{ext} is the smallest root in $[0,1]$ of $\mathfrak{g}(s)=s$. It is 1 for $\bar{m}\le 1$ (excluding the trivial $\mathfrak{g}(s)=s$) and smaller than 1 for $\bar{m}>1$ [17, Chap. I].

The dependence on $\bar{m}$ follows from convexity. The function $\mathfrak{g}(s)-s$ is convex, vanishes at $s=1$ and has the slope $\bar{m}-1$ there. For $\bar{m}\le 1$ a convex function with nonpositive slope at its zero $s=1$ lies above its tangent there, $(\bar{m}-1)(s-1)\ge 0$ on $[0,1)$. A root $s_0<1$ would therefore need $\bar{m}=1$. Convexity would then hold $\mathfrak{g}(s)-s$ between the zero tangent and the zero chord from s_0 to 1, so it would vanish on $[s_0,1]$. A power series that vanishes on an interval vanishes identically, which is the trivial case. For $\bar{m}>1$ the function is negative just below $s=1$ and equals $p_0\ge 0$ at $s=0$, so a root lies in $[0,1)$. For a concrete case let each individual have no offspring with probability $1-p$ and two with probability p, $\mathfrak{g}(s)=1-p+ps^2$. The roots of $ps^2-s+1-p=0$ are $s=1$ and $s=(1-p)/p$. For $p\le\frac{1}{2}$ the smaller root is 1 and the line dies out with certainty; for $p=0.6$ it survives with probability $1-\frac{2}{3}=\frac{1}{3}$, and iterating Eq. (C.101) from $s=0$ converges to $\frac{2}{3}$, not to the other root.

In a cascade, the fixed-point equation of a branching process has more solutions than the physical one, and the physical solution is the *minimal* one, obtained by iterating from the terminal data. In a carrier cascade every collision lowers the energy of the particles it produces, every branch stops below the impact-ionization threshold, and termination has probability 1; the recursion is then solved from low energy upward and the spurious roots never arise (Sect. 18.1).

C.11.3 Branching with types and deposits

A cascade particle carries more than a generation index: a type (electron or hole) and an energy x. Let a particle in state x undergo its next collision in channel c with probability $p_c(x)$. The collision produces particles in states $x'_1,\dots,x'_k$, drawn from a distribution that depends on x and c, and leaves an additive deposit d_c: a pair, an emitted phonon energy, or a residual kinetic energy at the end. Let Y be the total deposit of the tree started by the particle, and $\mathfrak{G}_x(s)=\langle s^Y\,|\,x\rangle$ for an integer deposit. If the subtrees of the k products evolve independently, the first-step argument that gave Eq. (C.101) gives

$$\mathfrak{G}_x(s)=\sum_c p_c(x)\,\Big\langle s^{d_c}\prod_{j=1}^{k}\mathfrak{G}_{x'_j}(s)\Big\rangle_c, \tag{C.102}$$

with $\langle\cdots\rangle_c$ the average over the products of channel c. Moments follow directly from $Y=d_c+Y_1+\cdots+Y_k$, without differentiating. Write $\bar{Y}(x)=\langle Y\,|\,x\rangle$ and $\overline{Y^2}(x)=\langle Y^2\,|\,x\rangle$. Averaging Y and Y^2 over the first collision, and using $\langle Y_iY_j\rangle=\bar{Y}(x'_i)\bar{Y}(x'_j)$ for $i\ne j$ (independent subtrees),

$$\bar{Y}(x)=\sum_c p_c(x)\Big\langle d_c+\sum_j \bar{Y}(x'_j)\Big\rangle_c, \tag{C.103}$$

$$\overline{Y^2}(x)=\sum_c p_c(x)\Big\langle d_c^2+2d_c\sum_j\bar{Y}(x'_j)+\sum_j\overline{Y^2}(x'_j)+2\sum_{i<j}\bar{Y}(x'_i)\bar{Y}(x'_j)\Big\rangle_c. \tag{C.104}$$

These hold for any additive deposit, integer or not. Every term on the right refers to products of lower energy than x, so a single sweep upward in x solves both equations, starting from the terminal states, where $\bar{Y}$ and $\overline{Y^2}$ are known. The variance is $\overline{Y^2}-\bar{Y}^2$, and covariances of two deposits follow from the second moment of their sum.

C.11.4 Renewal theory

For deposits large compared with the gap, the mean pair number is linear in the deposited energy. Renewal theory explains why, and what the corrections are. Strip the cascade to a single chain: a quantity E, the energy, is reduced by independent losses $X_1, X_2, \dots$ with one distribution $F_{\rm loss}$ on $x > 0$, mean $\langle X\rangle$ and second moment $\langle X^2\rangle$. Let $N(E)$ be the number of losses that fit, the largest n with $S_n = X_1 + \cdots + X_n \le E$, and $\bar N(E) = \langle N(E)\rangle$. Conditioning on the first loss $X_1 = x$, either $x > E$ and $N = 0$, or the chain restarts with $E - x$ and one more count. This gives the *renewal equation*

$$\bar N(E) = F_{\rm loss}(E) + \int_0^E \bar N(E - x)\,\mathrm{d}F_{\rm loss}(x). \tag{C.105}$$

Its solutions are controlled by one theorem. Let $Z(E) = z(E) + \int_0^E Z(E - x)\,\mathrm{d}F_{\rm loss}(x)$, with z nonnegative, nonincreasing and integrable (more generally, directly Riemann integrable).

Key renewal theorem

If $F_{\rm loss}$ is not concentrated on a lattice $\{0, d, 2d, \dots\}$ and has finite mean, then

$$\lim_{E\to\infty} Z(E) = \frac{1}{\langle X\rangle}\int_0^\infty z(y)\,\mathrm{d}y. \tag{C.106}$$

The proof is in Ref. [18, Chap. XI.1].

We first apply it to the energy left over at the end, the *undershoot* $E_{\rm res}(E) = E - S_{N(E)}$, the part of E that no further loss fits into. Fix $y \ge 0$ and let $Z(E) = P\big(E_{\rm res}(E) > y\big)$. If the first loss already exceeds E, the residual is E itself, which exceeds y when $E > y$; otherwise the chain restarts with $E - x$. So $Z = z + Z * F_{\rm loss}$ with $z(E) = \Theta(E - y)\,[1 - F_{\rm loss}(E)]$, nonincreasing beyond y. By Eq. (C.106),

$$\lim_{E\to\infty} P(E_{\rm res} > y) = \frac{1}{\langle X\rangle}\int_y^\infty [1 - F_{\rm loss}(x)]\mathrm{d}x, \qquad \langle E_{\rm res}\rangle \to \frac{\langle X^2\rangle}{2\langle X\rangle}. \tag{C.107}$$

The limiting density of the residual is $[1 - F_{\rm loss}(y)]/\langle X\rangle$, and its mean follows by integrating y against it by parts: $\int_0^\infty y[1 - F_{\rm loss}(y)]\,\mathrm{d}y = \frac{1}{2}\int_0^\infty y^2\,\mathrm{d}F_{\rm loss}(y) = \frac{1}{2}\langle X^2\rangle$. A residual is not a typical loss: long losses cover more of the energy axis, so the gap in which the chain ends is biased towards them.

The mean count follows from energy bookkeeping. With the convention $S_0 = 0$, the loss that first overshoots E is X_{N+1}, and $N + 1$ is a stopping time: whether it has occurred is decided by $X_1, \dots, X_{N+1}$. Wald's identity then gives $\langle S_{N+1}\rangle = \langle X\rangle(\bar N + 1)$ [18]. The overshoot $S_{N+1} - E$ obeys a renewal equation of the same kind: if the first loss exceeds E, the overshoot is $X_1 - E$, which exceeds y with probability $1 - F_{\rm loss}(E + y)$; otherwise the chain restarts. With $z(E) = 1 - F_{\rm loss}(E + y)$, Eq. (C.106) gives the limit $\langle X\rangle^{-1}\int_y^\infty [1 - F_{\rm loss}(x)]\,\mathrm{d}x$, the law of Eq. (C.107), with mean $\langle X^2\rangle/(2\langle X\rangle)$. Hence

$$\bar N(E) = \frac{E}{\langle X\rangle} + \frac{\langle X^2\rangle}{2\langle X\rangle^2} - 1 + o(1). \tag{C.108}$$

The leading term is the linear yield, $\bar N(E)/E \to 1/\langle X\rangle$; in the cascade the mean loss per pair is $\varepsilon_{\rm eh}$, and Sect. 18.2 derives $\bar n(\omega) = \omega/\varepsilon_{\rm eh} + o(\omega)$ in this way. The fluctuations follow from the central limit theorem for S_n. $N(E) < n$ means $S_n > E$, and S_n is approximately Gaussian with mean $n\langle X\rangle$ and variance $n\sigma_X^2$, $\sigma_X^2 = \langle X^2\rangle - \langle X\rangle^2$. Setting $n = E/\langle X\rangle + u\,(E\sigma_X^2/\langle X\rangle^3)^{1/2}$

and keeping the leading order in E, $P(N < n) = P(S_n > E) \to \Phi(u)$, the standard normal distribution function. So $N(E)$ is Gaussian at large E with

$$\operatorname{Var} N(E) \simeq \frac{\sigma_X^2}{\langle X \rangle^3}\, E, \qquad \frac{\operatorname{Var} N}{\bar{N}} \to \frac{\sigma_X^2}{\langle X \rangle^2}. \tag{C.109}$$

The Fano factor of a single renewal chain is the squared relative width of one loss. A deterministic loss gives zero, and an exponential loss, for which $\sigma_X = \langle X \rangle$, gives the Poisson value 1. For losses uniform on $[0, 1]$, $\langle X \rangle = \frac{1}{2}$ and $\langle X^2 \rangle = \frac{1}{3}$. Eqs. (C.107) to (C.109) then give $\langle E_{\mathrm{res}} \rangle = \frac{1}{3}$, $\bar{N} - 2E = -\frac{1}{3}$ and a Fano factor $\frac{1}{3}$. A numerical solution of Eq. (C.105) at $E = 20$ reproduces the offset to 10^{-6}, and a simulation of 2×10^5 chains at $E = 50$ the other two values to 1 %.

Each hypothesis of the key renewal theorem fails somewhere in a cascade. A stationary loss law, independent of E, fails when the losses depend on the carrier energy; the theorem then applies only to the high-energy part of the chain. A non-lattice law fails when only a few loss values occur. For a deterministic loss d, $N(E) = \lfloor E/d \rfloor$ and $\bar{N} - E/d = -\{E/d\}$, the negative fractional part, which oscillates forever between -1 and 0. Its average over a period, $-\frac{1}{2}$, is what Eq. (C.108) predicts with $\langle X^2 \rangle = d^2$, but the limit does not exist. A model whose losses are integer combinations of a gap and one phonon energy is of this kind, and its yield carries a periodic remainder. A finite second moment fails for a fast primary that loses energy in knock-on collisions with a tail $\mathrm{d}F_{\mathrm{loss}}/\mathrm{d}x \propto x^{-2}$ up to $E/2$. Then $\langle X^2 \rangle$ grows in proportion to E, the residual of Eq. (C.107) grows without bound, and the corrections to linearity can be logarithmic in E rather than constant. Sect. 18.2 states which reduction of the cascade satisfies which hypothesis.

C.12 Linear programming and duality

Every rate of the book is linear in the dark-matter velocity distribution: double the number of particles moving at one velocity, and their contribution to every bin doubles. A limit that should hold for every velocity distribution, the problem of Sect. 21.2, is then the largest value of a linear quantity over all nonnegative distributions that keep the predicted counts below the observed bounds. That is a linear program with infinitely many variables. Sect. 21.2 uses three of its properties: any feasible dual solution certifies an upper bound, the optimum is attained by a halo made of a few streams, and a grid solution errs by an amount that a Lipschitz constant bounds.

C.12.1 Finite linear programs

A linear program in standard form maximizes $\mathbf{c}^{\mathsf{T}}\mathbf{x}$ over vectors $\mathbf{x} \in \mathbb{R}^M$ with $\mathbf{x} \geq 0$ and $\mathbf{A}\mathbf{x} \leq \mathbf{b}$, where $\mathbf{A}$ is an $n \times M$ matrix and inequalities between vectors hold componentwise. Its *dual* minimizes $\mathbf{b}^{\mathsf{T}}\boldsymbol{\lambda}$ over $\boldsymbol{\lambda} \in \mathbb{R}^n$ with $\boldsymbol{\lambda} \geq 0$ and $\mathbf{A}^{\mathsf{T}}\boldsymbol{\lambda} \geq \mathbf{c}$. For any feasible pair,

$$\mathbf{c}^{\mathsf{T}}\mathbf{x} \leq (\mathbf{A}^{\mathsf{T}}\boldsymbol{\lambda})^{\mathsf{T}}\mathbf{x} = \boldsymbol{\lambda}^{\mathsf{T}}(\mathbf{A}\mathbf{x}) \leq \boldsymbol{\lambda}^{\mathsf{T}}\mathbf{b}. \tag{C.110}$$

The first inequality uses $\mathbf{x} \geq 0$ and $\mathbf{A}^{\mathsf{T}}\boldsymbol{\lambda} \geq \mathbf{c}$, the second $\boldsymbol{\lambda} \geq 0$ and $\mathbf{A}\mathbf{x} \leq \mathbf{b}$. This is *weak duality*: every dual-feasible $\boldsymbol{\lambda}$ bounds the primal maximum from above, whether or not anyone has found the maximum. *Strong duality*, the equality of the two optima, holds for finite programs whenever both are feasible.

The optimum of a finite program is attained at a point with at most n nonzero components, one per constraint. Suppose an optimal $\mathbf{x}$ has more than n positive components. The corresponding columns of $\mathbf{A}$, more than n vectors in $\mathbb{R}^n$, are linearly dependent, so there

is a nonzero $\mathbf{d}$, supported on those components, with $\mathbf{A}\mathbf{d} = 0$. Then $\mathbf{x} \pm t\mathbf{d}$ satisfies every constraint for small $t > 0$, and since $\mathbf{x}$ is optimal, $\mathbf{c}^{\mathsf{T}}\mathbf{d} = 0$. Increasing t until the first component of $\mathbf{x} + t\mathbf{d}$ or $\mathbf{x} - t\mathbf{d}$ reaches zero yields another optimal point with one positive component fewer. Repeating the step leaves at most n.

C.12.2 Semi-infinite programs

Now let the variable be a nonnegative measure Ξ on a compact set $\mathcal{B}$, for Sect. 21.2 the velocities $\boldsymbol{w}$ with $|\boldsymbol{w}| \le v_{\rm esc}$ in the galactic frame. The number of constraints stays finite. Bin i of an experiment predicts $\int \mathsf{k}_i \, \mathrm{d}\Xi$ events, with a continuous kernel $\mathsf{k}_i(\boldsymbol{w}) \ge 0$, the count from a single stream of unit density at $\boldsymbol{w}$; the observations allow at most $N_i^{\rm up} > 0$. The quantity to bound is $\int \mathsf{f} \, \mathrm{d}\Xi$ with a continuous $\mathsf{f} \ge 0$: a halo integral above some speed, or the rate in a second experiment. The program and its dual are

$$\begin{aligned} \Phi_{\max} &= \sup_{\Xi \ge 0} \Big\{ \int_{\mathcal{B}} \mathsf{f} \, \mathrm{d}\Xi \;:\; \int_{\mathcal{B}} \mathsf{k}_i \, \mathrm{d}\Xi \le N_i^{\rm up},\ i = 1, \ldots, n \Big\}, \\ &\min_{\boldsymbol{\lambda} \ge 0} \Big\{ \sum_i \lambda_i N_i^{\rm up} \;:\; \mathsf{g}_\lambda(\boldsymbol{w}) \ge 0 \ \ \forall \boldsymbol{w} \in \mathcal{B} \Big\}, \end{aligned} \tag{C.111}$$

with $\mathsf{g}_\lambda(\boldsymbol{w}) = \sum_i \lambda_i \mathsf{k}_i(\boldsymbol{w}) - \mathsf{f}(\boldsymbol{w})$. The primal has infinitely many variables, the values of Ξ everywhere on $\mathcal{B}$; the dual has infinitely many constraints, one per point. Such programs are called *semi-infinite*.

Weak duality carries over in one line. For feasible Ξ and $\boldsymbol{\lambda}$,

$$\int \mathsf{f} \, \mathrm{d}\Xi \le \int \sum_i \lambda_i \mathsf{k}_i \, \mathrm{d}\Xi = \sum_i \lambda_i \int \mathsf{k}_i \, \mathrm{d}\Xi \le \sum_i \lambda_i N_i^{\rm up}, \tag{C.112}$$

by $\mathsf{g}_\lambda \ge 0$ and $\Xi \ge 0$, then by $\lambda_i \ge 0$ and the primal constraints. A vector $\boldsymbol{\lambda}$ with $\mathsf{g}_\lambda \ge 0$ on all of $\mathcal{B}$ is a *dual certificate*: it proves $\Phi_{\max} \le \sum_i \lambda_i N_i^{\rm up}$ for every halo at once. Checking a certificate requires only the evaluation of one function on $\mathcal{B}$, which is why a halo-independent bound can be verified independently of the optimizer that produced it.

Strong duality needs hypotheses. Let $\mathcal{B}$ be compact, k_i and f continuous, and let every point where $\mathsf{f} > 0$ be seen by some bin, $\kappa_{\rm v} \equiv \min_{\boldsymbol{w} \in \operatorname{supp} \mathsf{f}} \max_i \mathsf{k}_i(\boldsymbol{w}) > 0$. Mass outside $\operatorname{supp} \mathsf{f}$ does not raise the objective and, since $\mathsf{k}_i \ge 0$, only uses up the bounds, so it can be removed. On $\operatorname{supp} \mathsf{f}$, $\sum_i \mathsf{k}_i \ge \kappa_{\rm v}$, and the constraints give

$$\kappa_{\rm v} \, \Xi(\operatorname{supp} \mathsf{f}) \le \int \sum_i \mathsf{k}_i \, \mathrm{d}\Xi \le \sum_i N_i^{\rm up} \equiv N_{\rm tot}, \tag{C.113}$$

so the relevant halos have bounded total mass. The feasible set is then compact in the weak sense in which $\int \mathsf{f} \, \mathrm{d}\Xi$ is continuous, and $\Xi = 0$ satisfies every constraint strictly (the Slater condition). Under these conditions the two optima of Eq. (C.111) are equal and both are attained [19].

C.12.3 Extreme points and streams

The optimal halo is a sum of at most n streams, $\Xi^* = \sum_{a=1}^{n'} \xi_a \, \delta_{\boldsymbol{w}_a}$ with $n' \le n$ and $\xi_a > 0$. On a finite grid of velocities this is the statement proved above for finite programs, with one variable per grid point. On the continuum, Rogosinski's theorem applied to the $n + 1$ functionals $\mathsf{k}_1, \ldots, \mathsf{k}_n, \mathsf{f}$ replaces any optimal Ξ by a measure with the same counts and objective on at most $n + 1$ points [20], and the finite argument above, applied to those points, removes one more. Where the streams sit follows from equality in Eq. (C.112), which strong

duality enforces at the optimum. The first inequality is an equality only if Ξ^* puts no mass where $\mathsf{g}_\lambda > 0$: the streams sit at zeros of g_λ, which, since $\mathsf{g}_\lambda \geq 0$, are its minima. The second is an equality only if $\lambda_i = 0$ for every bin whose bound is not saturated. These are the *complementary-slackness* conditions, and they make the optimum checkable: the streams touch the dual function from below, and only saturated bins carry weight.

With one bin the problem is solved in closed form. Then $\mathsf{g}_\lambda \geq 0$ means $\lambda_1 \geq \mathsf{f}(\boldsymbol{w})/\mathsf{k}_1(\boldsymbol{w})$ wherever $\mathsf{f} > 0$, the smallest certificate is $\lambda_1 = \sup_{\boldsymbol{w}} \mathsf{f}/\mathsf{k}_1$, and the dual value is $N_1^{\mathrm{up}} \sup_{\boldsymbol{w}} \mathsf{f}/\mathsf{k}_1$. A single stream at the maximizer $\boldsymbol{w}_*$, with $\xi_1 = N_1^{\mathrm{up}}/\mathsf{k}_1(\boldsymbol{w}_*)$, saturates the bin and achieves the same value. Hence

$$\Phi_{\max} = N_1^{\mathrm{up}} \sup_{\boldsymbol{w} \in \mathcal{B}} \frac{\mathsf{f}(\boldsymbol{w})}{\mathsf{k}_1(\boldsymbol{w})} \qquad (n = 1). \tag{C.114}$$

The bound is set by the velocity at which the experiment is least sensitive relative to the quantity being bounded; every other halo does better. For a target whose response depends only on speed, k_1 depends only on $|\boldsymbol{w}|$, and Eq. (C.114) becomes the familiar statement that a halo-independent bound is set by the single speed shell that the bin sees worst (Sect. 21.2).

C.12.4 Discretization error

In practice the program is solved on a grid $\mathcal{B}_h \subset \mathcal{B}$ of directions and speeds, with covering radius h: every $\boldsymbol{w} \in \mathcal{B}$ lies within h of a grid point. The grid makes the error two-sided. Restricting Ξ to the grid shrinks the primal feasible set, so the grid primal optimum Φ_h obeys $\Phi_h \leq \Phi_{\max}$. The grid dual solution $\boldsymbol{\lambda}_h$ satisfies $\mathsf{g}_{\lambda_h} \geq 0$ only on $\mathcal{B}_h$, and is no certificate on $\mathcal{B}$. Its violation between grid points is bounded by the Lipschitz constant L_h of g_{λ_h}: for any $\boldsymbol{w}$ and its nearest grid point $\boldsymbol{w}_h$, $\mathsf{g}_{\lambda_h}(\boldsymbol{w}) \geq \mathsf{g}_{\lambda_h}(\boldsymbol{w}_h) - L_h|\boldsymbol{w} - \boldsymbol{w}_h| \geq -L_h h$. Repeating Eq. (C.112) with $\mathsf{f} \leq \sum_i \lambda_i \mathsf{k}_i + L_h h$ on $\operatorname{supp}\mathsf{f}$, and bounding the mass there with Eq. (C.113),

$$\Phi_h \;\leq\; \Phi_{\max} \;\leq\; \sum_i \lambda_{h,i} N_i^{\mathrm{up}} + \frac{L_h\, h\, N_{\mathrm{tot}}}{\kappa_{\mathrm{v}}}. \tag{C.115}$$

The gap between the two sides is the discretization error, and it is computable: L_h follows from the derivatives of the kernels and κ_{v} from their values. It falls linearly with h, and faster in practice, because the optimal streams sit at minima of g_λ, where the function is flat. Refining the grid only near the few streams of the grid optimum, and checking $\mathsf{g}_\lambda \geq 0$ on the refined grid, is the standard way to close it. A bound reported from a grid computation is conservative only if it uses the right side of Eq. (C.115).

The positivity of the kernels was used twice, to discard mass outside $\operatorname{supp}\mathsf{f}$ and in Eq. (C.113). If a truncated response makes some k_i negative at some velocities, Eq. (C.112) still holds for every feasible pair, but the mass bound fails, and with it the control of Eq. (C.115); Sect. 21.2 bounds that leakage separately.

C.13 Interval orders

Two materials whose figures of merit have overlapping 68% intervals cannot be put in order, and a third may be resolved from one of them but not from the other. Sect. 17.6 groups materials into tiers that respect exactly what the intervals resolve, no more. The mathematics is that of an interval order, a partial order built from intervals, and of the heights of its elements.

C.13.1 Partial orders from intervals

A *strict partial order* $\succ$ on a finite set is a relation that is irreflexive (never $a \succ a$) and transitive ($a \succ b$ and $b \succ c$ imply $a \succ c$). Together these imply asymmetry: $a \succ b$ and $b \succ a$

would give $a \succ a$. Two elements with neither $a \succ b$ nor $b \succ a$ are *incomparable*. A partial order differs from a ranking in that incomparability is allowed and is not transitive.

Let each element a carry an interval $I_a = [\ell_a, u_a]$, oriented so that larger values are better; for a figure of merit where smaller is better, use its negative logarithm. Define

$$a \succ b \quad \Longleftrightarrow \quad \ell_a > u_b, \tag{C.116}$$

that is, a is resolved above b when the whole interval of a lies above the whole interval of b. The relation is irreflexive, since $\ell_a \le u_a$, and transitive: $\ell_a > u_b \ge \ell_b > u_c$ gives $\ell_a > u_c$. It is therefore a strict partial order, called an *interval order*. Two elements are incomparable exactly when their intervals overlap.

Incomparability is not transitive. With $I_a = [3, 5]$, $I_b = [2, 4]$ and $I_c = [1, 2.5]$, a overlaps b and b overlaps c, but $\ell_a = 3 > 2.5 = u_c$, so $a \succ c$. A ranking that declares a tied with b and b tied with c must then either tie a with c, which discards a resolved difference, or break one of the ties, which asserts a difference the intervals do not show.

Interval orders have one property that general partial orders lack: if $a \succ b$ and $c \succ d$, then $a \succ d$ or $c \succ b$. To see this, suppose $a \not\succ d$, so $\ell_a \le u_d$. Then $\ell_c > u_d \ge \ell_a > u_b$, and $c \succ b$. Conversely, every finite partial order with this property can be represented by intervals as in Eq. (C.116) [21]. When all intervals have the same length the order is called a semiorder; intervals of different widths, which arise when inputs of different provenance carry different uncertainties, give the general case.

C.13.2 Heights as tiers

A *chain* above a is a sequence $b_1 \succ b_2 \succ \cdots \succ b_k \succ a$. The *height* of a is

$$h(a) = 1 + (\text{length } k \text{ of the longest chain above } a), \tag{C.117}$$

so that $h(a) = 1$ when nothing is resolved above a. The *tier* k is the set of elements of height k.

Each tier is an *antichain*: no two of its members are comparable. If $a \succ b$, any chain above a extended by a is a chain above b, one element longer, so $h(b) \ge h(a) + 1$ and a, b lie in different tiers. Members of one tier are therefore unresolved from each other, and calling them tied within the tier is exact. Every member of tier $k > 1$ is resolved below at least one member of tier $k - 1$. A longest chain above a has $k - 1$ elements, and its last element b has height exactly $k - 1$. The chain above b formed by the first $k - 2$ elements shows $h(b) \ge k - 1$, while a longer chain above b would extend through b to a chain above a of more than $k - 1$ elements.

For interval orders the heights are computed by sorting. Process the elements in decreasing order of u_a. When a is reached, every b with $b \succ a$ has $\ell_b > u_a$, so $u_b \ge \ell_b > u_a$ and b has already been processed; then $h(a) = 1 + \max\{h(b) : \ell_b > u_a\}$, with the maximum over an empty set equal to zero. The cost is at most quadratic in the number of elements.

C.13.3 Interpretation of tiers

Take four materials with intervals $I_A = [8, 10]$, $I_B = [5, 9]$, $I_C = [6, 7]$ and $I_D = [1, 9.5]$. Nothing lies above A, B or D, since no lower end exceeds their upper ends ($\ell_A = 8 < 9 \le u_B, u_D$, and $\ell_B, \ell_C, \ell_D < 9$), so all three have height 1. Only A is resolved above C ($8 > 7$), so C has height 2. The tiers are $\{A, B, D\}$ and $\{C\}$. Material D, whose interval is the widest, sits in the first tier although its midpoint, 5.25, is below that of C, 6.5: nothing is resolved above it. C sits in the second tier although B, which shares the first tier, is not resolved above it.

A tier does not assert that every member of a higher tier beats every member of a lower one: B does not beat C. Nor does it assert that a member of a higher tier is likely to be better than one of a lower tier, since a wide interval rises to the first tier because it is wide. Tiers also depend on the confidence level of the intervals and on the widths assigned to the inputs; at a higher level, the intervals widen and the tiers merge. Tiers never invent an order that the intervals do not contain, but they discard what the intervals say about medians; Sect. 17.6 therefore reports them together with the probability of ranking first, which keeps that information.

C.14 Likelihoods, intervals and profiling

A search that sees no events with no expected background excludes, at 90 % confidence, every signal expectation above $N_{90} = 2.303$ (Eq. (1.52)). Real searches add backgrounds, several bins and uncertain inputs, and Sects. 8.8, 21.4 and 21.5 handle them with likelihoods: the Poisson and Gaussian likelihoods, the likelihood ratio and the intervals it defines, nuisance parameters and their profiling, and, in Sect. 17.6, Monte Carlo propagation of input widths. The meaning of a confidence level, a statement about the procedure in repeated experiments, is that of Sect. 1.5.

C.14.1 Poisson and Gaussian likelihoods

The *likelihood* L is the probability of the observed data, read as a function of the parameters. A counting experiment that observes N_{obs} events, with expected signal N_{sig} and background N_{bkg}, has by Eq. (1.51)

$$\mathsf{L}(N_{\text{sig}}) = \frac{(N_{\text{sig}} + N_{\text{bkg}})^{N_{\text{obs}}}}{N_{\text{obs}}!}\, \mathrm{e}^{-(N_{\text{sig}}+N_{\text{bkg}})}. \tag{C.118}$$

Independent bins multiply, so a spectrum in bins j has the product of factors of this form, one per bin, with expectations $N_{\text{sig},j} + N_{\text{bkg},j}$ [22, Eq. (6)]. Only ratios of likelihoods enter what follows, and factors independent of the parameters, such as $1/N_{\text{obs}}!$, drop out.

When a parameter is measured with a Gaussian error, for example an estimate $\hat{N}$ of N_{sig} with standard deviation $\sigma_{\hat{N}}$ from many events, the likelihood is Gaussian,

$$\mathsf{L}(N_{\text{sig}}) \propto \exp\Big[-\frac{(\hat{N} - N_{\text{sig}})^2}{2\sigma_{\hat{N}}^2}\Big]. \tag{C.119}$$

The Poisson likelihood takes this form at large counts. Expanding $\ln \mathsf{L}$ of Eq. (C.118) to second order about its maximum, at $N_{\text{sig}} + N_{\text{bkg}} = N_{\text{obs}}$, gives $\ln \mathsf{L} = \text{const} - (N_{\text{sig}} + N_{\text{bkg}} - N_{\text{obs}})^2/(2N_{\text{obs}})$, a Gaussian of width $N_{\text{obs}}^{1/2}$. The cubic term of the expansion is smaller by a factor of order $N_{\text{obs}}^{-1/2}$ at one standard deviation from the maximum, so the Gaussian form needs tens of events, not a few.

C.14.2 The likelihood ratio and intervals

The *likelihood ratio* compares a hypothesized value with the best one,

$$\mathsf{t}(N_{\text{sig}}) = -2 \ln \frac{\mathsf{L}(N_{\text{sig}})}{\mathsf{L}(\hat{N}_{\text{sig}})} \geq 0, \tag{C.120}$$

where $\hat{N}_{\text{sig}}$ maximizes L [22, Eq. (8)]. Large t means poor agreement. For the Gaussian likelihood Eq. (C.119), $\hat{N}_{\text{sig}} = \hat{N}$ and $\mathsf{t} = (\hat{N} - N_{\text{sig}})^2/\sigma_{\hat{N}}^2$ exactly. If the data are repeated,

$\hat{N}$ scatters as a Gaussian about the true value, and t at the true value is the square of a standard normal variable, a χ^2 variable with one degree of freedom. The set of $N_{\rm sig}$ with $\mathsf{t}(N_{\rm sig}) \le z^2$, the interval $\hat{N} \pm z\sigma_{\hat{N}}$, then covers the true value with probability $2\Phi(z) - 1$, with Φ the standard normal distribution function. For a likelihood that is not Gaussian, Wilks' theorem states that t at the true value approaches the same χ^2 distribution as the sample grows [23], [22, Sect. 3.1], and the same rule applies. The thresholds are

$$\mathsf{t} \le 1.00,\ 2.71,\ 3.84 \qquad \text{for } 68.27\,\%,\ 90\,\%,\ 95\,\% \text{ (two-sided)}, \tag{C.121}$$

the squares of $z = 1$, 1.645 and 1.960. A 1σ interval is the 68.27 % interval, written "68 %" throughout the book.

An upper limit is one-sided. Data with $\hat{N}_{\rm sig}$ above the tested value are not counted as evidence against it, so the statistic is set to zero there [22, Eq. (14)], and all of the rejected probability lies on one side. For a Gaussian likelihood the upper limit at confidence level CL is $\hat{N} + z_{\rm CL}\sigma_{\hat{N}}$ with $\Phi(z_{\rm CL}) = {\rm CL}$: $z = 1.282$ at 90 % and $z = 1.645$ at 95 %, which correspond to $\mathsf{t} = 1.64$ and 2.71. The same number, 2.71, thus marks a two-sided 90 % interval and a one-sided 95 % limit.

The asymptotic rule fails at small counts, the regime of dark-matter limits. With $N_{\rm obs} = 0$ and $N_{\rm bkg} = 0$, Eq. (C.118) gives $\mathsf{L} = \mathrm{e}^{-N_{\rm sig}}$, maximal at $\hat{N}_{\rm sig} = 0$, and $\mathsf{t} = 2N_{\rm sig}$. The one-sided 90 % threshold 1.64 would give $N_{\rm sig} < 0.82$, while the exact construction of Sect. 1.5 gives 2.303. Limits from a few events must therefore be built from the exact Poisson probabilities, by the classical construction of Sect. 1.5 or by its generalization that orders outcomes by the likelihood ratio and yields one- or two-sided intervals as the data demand [24]; Sect. 21.4 does so.

C.14.3 Nuisance parameters and the profile likelihood

The predicted counts depend on quantities that are not of interest but are not known exactly: a background normalization, an efficiency, or the size of a truncation in the theory (Sect. 8.8). Each becomes a *nuisance parameter* ϑ in the likelihood, usually with a constraint term that encodes what is known about it from a calibration or from a computed next order. The *profile likelihood* replaces each nuisance by its best value at fixed $N_{\rm sig}$,

$$\mathsf{t}_{\rm p}(N_{\rm sig}) = -2\ln \frac{\mathsf{L}\big(N_{\rm sig}, \hat{\hat{\vartheta}}(N_{\rm sig})\big)}{\mathsf{L}(\hat{N}_{\rm sig}, \hat{\vartheta})}, \tag{C.122}$$

where $\hat{\hat{\vartheta}}(N_{\rm sig})$ maximizes L at fixed $N_{\rm sig}$ and the denominator is the overall maximum [22, Eq. (7)]. Wilks' theorem holds for $\mathsf{t}_{\rm p}$ as well, with one degree of freedom per parameter of interest, so Eq. (C.121) still applies.

A Gaussian example shows what profiling does. Let the estimate be shifted by an unknown offset, so that $\hat{N}$ is Gaussian with mean $N_{\rm sig} + \vartheta$ and width $\sigma_{\hat{N}}$, and let a calibration constrain the offset by a Gaussian of mean 0 and width σ_ϑ. Then

$$-2\ln \mathsf{L}(N_{\rm sig}, \vartheta) = \frac{(\hat{N} - N_{\rm sig} - \vartheta)^2}{\sigma_{\hat{N}}^2} + \frac{\vartheta^2}{\sigma_\vartheta^2} + \text{const.}$$

Write $r = \hat{N} - N_{\rm sig}$. Setting the derivative with respect to ϑ to zero, $-(r - \vartheta)/\sigma_{\hat{N}}^2 + \vartheta/\sigma_\vartheta^2 = 0$, gives $\hat{\hat{\vartheta}} = r\sigma_\vartheta^2/(\sigma_{\hat{N}}^2 + \sigma_\vartheta^2)$. Inserting it, the two terms become $r^2\sigma_{\hat{N}}^2/(\sigma_{\hat{N}}^2 + \sigma_\vartheta^2)^2$ and $r^2\sigma_\vartheta^2/(\sigma_{\hat{N}}^2 + \sigma_\vartheta^2)^2$, and their sum is $r^2/(\sigma_{\hat{N}}^2 + \sigma_\vartheta^2)$. The overall minimum is zero, at $N_{\rm sig} = \hat{N}$ and $\vartheta = 0$. Hence

$$\mathsf{t}_{\rm p}(N_{\rm sig}) = \frac{(\hat{N} - N_{\rm sig})^2}{\sigma_{\hat{N}}^2 + \sigma_\vartheta^2}. \tag{C.123}$$

The profiled interval is that of a Gaussian of width $(\sigma_{\hat N}^2+\sigma_\vartheta^2)^{1/2}$: statistical and systematic widths add in quadrature. A normalization nuisance, $N_{\mathrm{sig}}\to N_{\mathrm{sig}}(1+\vartheta)$, gives the same result for the relative widths to first order in ϑ. Profiling broadens the likelihood, and the broadening is the information about N_{sig} lost to the nuisance. A nuisance that the data constrain better than the calibration does is pulled towards the value the data prefer, which a fixed-width envelope would miss; Sect. 21.4 uses this to set limit bands.

C.14.4 Monte Carlo propagation

When a derived quantity depends on many inputs through a formula that is not linear, its distribution is obtained by sampling. Each input x_j is given a distribution, for example lognormal, with $\ln x_j$ Gaussian of mean $\ln x_j^{(0)}$ and width σ_j, a width set by how the input was obtained (Sect. 17.6). We draw N_{MC} sets of inputs, evaluate the output y on each, and read off percentiles. The 16th and 84th percentiles bound the central 68 % interval, because $\Phi(-1)=0.159$; for a Gaussian output they are the mean minus and plus one standard deviation, and for any output they define an interval of the same coverage that need not be symmetric.

Comparisons follow from the same draws. The probability that material A outperforms B is the fraction of draws in which $y_A>y_B$. For lognormal outputs with log-widths σ_A, σ_B and correlation r between their errors, $\ln y_A-\ln y_B$ is Gaussian with mean $\Delta=\ln y_A^{(0)}-\ln y_B^{(0)}$ and variance $\sigma_A^2+\sigma_B^2-2r\sigma_A\sigma_B$, and

$$P(A\succ B)=\Phi\Big(\frac{\Delta}{(\sigma_A^2+\sigma_B^2-2r\sigma_A\sigma_B)^{1/2}}\Big). \tag{C.124}$$

With $\Delta=0.3$, $\sigma_A=0.2$, $\sigma_B=0.25$ and $r=0$, $P(A\succ B)=\Phi(0.94)=0.83$. A positive correlation, as arises when both materials are computed with the same method and share its error, shrinks the variance of the difference and sharpens the comparison; a ratio of two such outputs is more certain than either. The sampling error of an estimated probability p is $[p(1-p)/N_{\mathrm{MC}}]^{1/2}$, at most 0.0035 for $N_{\mathrm{MC}}=2\times10^4$. All such probabilities are conditional on the input distributions: they propagate the assigned widths faithfully, but they cannot be more reliable than the widths themselves.

C.15 Dirac equation and Foldy–Wouthuysen reduction

A 1s electron of germanium moves at $Z\alpha=0.23$ of the speed of light, and its relativistic corrections are a few percent of its binding energy. Sect. 10.6 bounds what such corrections do to the response at large q, Sect. 12.5 needs the electron density at a nucleus with its relativistic correction, and Sect. 10.9 writes the spinor algebra of a fermion absorbed on an electron. All three start from the Dirac equation and its expansion in $1/m_e$. We use natural units, Heaviside–Lorentz charges with the electron charge $-e$ (Appendix A.2) and the metric $(+,-,-,-)$ of Appendix C.9.

C.15.1 The Dirac equation in a potential

The free Dirac field ψ, a four-component spinor, obeys $(\mathrm{i}\gamma^\mu\partial_\mu-m_e)\psi=0$, with the Dirac matrices γ^μ satisfying $\gamma^\mu\gamma^\nu+\gamma^\nu\gamma^\mu=2g^{\mu\nu}$. We work in the Dirac representation,

$$\gamma^0=\beta_{\mathrm D}=\begin{pmatrix}1&0\\0&-1\end{pmatrix},\qquad \gamma^i=\beta_{\mathrm D}\alpha_{\mathrm D}^i,\qquad \alpha_{\mathrm D}^i=\begin{pmatrix}0&\sigma^i\\\sigma^i&0\end{pmatrix},\qquad \gamma^5=\mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3=\begin{pmatrix}0&1\\1&0\end{pmatrix}, \tag{C.125}$$

in 2×2 blocks, with σ^i the Pauli matrices. The subscript D keeps $\boldsymbol{\alpha}_\mathrm{D}$ and β_D apart from the fine-structure constant and the inverse temperature. The matrices obey $(\alpha^i_\mathrm{D})^2=\beta^2_\mathrm{D}=1$, and any two of $\alpha^1_\mathrm{D},\alpha^2_\mathrm{D},\alpha^3_\mathrm{D},\beta_\mathrm{D}$ anticommute.

The electron couples to the electromagnetic potential through $\mathcal{L}\supset -eA_\mu J^\mu_\mathrm{em}$ with $J^\mu_\mathrm{em}=-\bar\psi\gamma^\mu\psi$ for the electron (Eq. (C.85)), $\bar\psi=\psi^\dagger\gamma^0$. The Lagrangian density $\bar\psi(\mathrm{i}\gamma^\mu\partial_\mu-m_e)\psi+eA_\mu\bar\psi\gamma^\mu\psi$ gives the field equation $[\mathrm{i}\gamma^\mu(\partial_\mu-\mathrm{i}eA_\mu)-m_e]\psi=0$. Multiplying by γ^0 and solving for the time derivative, with $\boldsymbol{p}=-\mathrm{i}\nabla$, $A_\mu=(A^0,-\boldsymbol{A})$ and $\gamma^0\gamma^i=\alpha^i_\mathrm{D}$,

$$\mathrm{i}\partial_t\psi=H_\mathrm{D}\psi,\qquad H_\mathrm{D}=\boldsymbol{\alpha}_\mathrm{D}\cdot\boldsymbol{\pi}+\beta_\mathrm{D}m_e+V,\qquad \boldsymbol{\pi}=\boldsymbol{p}+e\boldsymbol{A},\quad V=-eA^0. \tag{C.126}$$

For an electron in the field of a nucleus of charge Ze, $A^0=Ze/(4\pi r)$ and $V=-Z\alpha/r$, with $e^2=4\pi\alpha$. A particle of charge $+e$ has $\boldsymbol{\pi}=\boldsymbol{p}-e\boldsymbol{A}$ and $V=+eA^0$.

The free equation has plane-wave solutions $\psi=u\,\mathrm{e}^{-\mathrm{i}Et+\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{x}}$. Writing $u=(u_\mathrm{A},u_\mathrm{B})$ in upper and lower two-component blocks, Eq. (C.126) with $\boldsymbol{A}=0$, $V=0$ reads $(E-m_e)u_\mathrm{A}=\boldsymbol{\sigma}\cdot\boldsymbol{p}\,u_\mathrm{B}$ and $(E+m_e)u_\mathrm{B}=\boldsymbol{\sigma}\cdot\boldsymbol{p}\,u_\mathrm{A}$. With $(\boldsymbol{\sigma}\cdot\boldsymbol{p})^2=p^2$, nonzero solutions need $E^2=p^2+m_e^2$. For $E=+(p^2+m_e^2)^{1/2}$ and a two-component spinor ξ_s of spin s,

$$u(\boldsymbol{p},s)=\sqrt{E+m_e}\begin{pmatrix}\xi_s\\ \dfrac{\boldsymbol{\sigma}\cdot\boldsymbol{p}}{E+m_e}\,\xi_s\end{pmatrix},\qquad u^\dagger u=2E,\qquad \bar u u=2m_e. \tag{C.127}$$

The lower, or small, component is smaller than the upper by $p/(E+m_e)\simeq v/2$. The solutions with $E=-(p^2+m_e^2)^{1/2}$ describe positrons in field theory; in a bound-state calculation they are the states that a no-pair projection removes. The relativistic 1s state of a point nucleus behaves as $r^{\gamma_\mathrm{D}-1}$ near the origin, with $\gamma_\mathrm{D}=[1-(Z\alpha)^2]^{1/2}$ [25], so its density diverges at $r=0$, weakly: $\gamma_\mathrm{D}-1=-0.005$ for silicon and -0.028 for germanium. A nucleus of finite size removes the divergence.

C.15.2 Bilinears and their non-relativistic limits

Couplings of dark matter to electrons are written with four bilinears, as in Sect. 1.4: the scalar $\bar\psi\psi$, the vector $\bar\psi\gamma^\mu\psi$, the axial vector $\bar\psi\gamma^\mu\gamma^5\psi$ and the pseudoscalar $\bar\psi\mathrm{i}\gamma^5\psi$. Their non-relativistic forms follow from $\psi=(\psi_\mathrm{A},\psi_\mathrm{B})$ with the small component $\psi_\mathrm{B}\simeq(\boldsymbol{\sigma}\cdot\boldsymbol{p}/2m_e)\psi_\mathrm{A}$, the low-energy limit of Eq. (C.127). With the blocks of Eq. (C.125), $\gamma^0\gamma^0=1$, $\gamma^0\gamma^i=\alpha^i_\mathrm{D}$, $\gamma^0\gamma^i\gamma^5=\mathrm{diag}(\sigma^i,\sigma^i)$ and $\gamma^0\mathrm{i}\gamma^5=\mathrm{i}\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$, so

$$\begin{aligned}
\bar\psi\psi&=\psi^\dagger_\mathrm{A}\psi_\mathrm{A}-\psi^\dagger_\mathrm{B}\psi_\mathrm{B}\simeq\psi^\dagger_\mathrm{A}\psi_\mathrm{A},&\qquad \bar\psi\gamma^0\psi&=\psi^\dagger_\mathrm{A}\psi_\mathrm{A}+\psi^\dagger_\mathrm{B}\psi_\mathrm{B}\simeq\psi^\dagger_\mathrm{A}\psi_\mathrm{A},\\
\bar\psi\gamma^i\gamma^5\psi&=\psi^\dagger_\mathrm{A}\sigma^i\psi_\mathrm{A}+\psi^\dagger_\mathrm{B}\sigma^i\psi_\mathrm{B}\simeq\psi^\dagger_\mathrm{A}\sigma^i\psi_\mathrm{A},&\qquad \bar\psi\mathrm{i}\gamma^5\psi&=\mathrm{i}(\psi^\dagger_\mathrm{A}\psi_\mathrm{B}-\psi^\dagger_\mathrm{B}\psi_\mathrm{A})\simeq\frac{1}{2m_e}\nabla\cdot(\psi^\dagger_\mathrm{A}\boldsymbol{\sigma}\psi_\mathrm{A}).
\end{aligned} \tag{C.128}$$

The last form uses $\boldsymbol{\sigma}\cdot\boldsymbol{p}=-\mathrm{i}\boldsymbol{\sigma}\cdot\nabla$: $\mathrm{i}[\psi^\dagger_\mathrm{A}(-\mathrm{i}\boldsymbol{\sigma}\cdot\nabla\psi_\mathrm{A})-(-\mathrm{i}\boldsymbol{\sigma}\cdot\nabla\psi_\mathrm{A})^\dagger\psi_\mathrm{A}]/2m_e=[\psi^\dagger_\mathrm{A}\boldsymbol{\sigma}\cdot\nabla\psi_\mathrm{A}+(\nabla\psi^\dagger_\mathrm{A})\cdot\boldsymbol{\sigma}\psi_\mathrm{A}]/2m_e$. The scalar and the time component of the vector both reduce to the number density, the limit used in Sect. 1.4 and Appendix C.9; they differ at order v^2, where the scalar acquires the kinetic-energy term of Sect. 1.4. The axial vector reduces to the spin density. The pseudoscalar is suppressed by $1/m_e$ and becomes the divergence of the spin density, so a coupling $g\phi\bar\psi\mathrm{i}\gamma^5\psi$ becomes, after integration by parts, $-(g/2m_e)(\nabla\phi)\cdot\psi^\dagger_\mathrm{A}\boldsymbol{\sigma}\psi_\mathrm{A}$, the first term of the axion-like coupling in Sect. 1.4. The spatial vector $\bar\psi\gamma^i\psi=\psi^\dagger_\mathrm{A}\sigma^i\psi_\mathrm{B}+\psi^\dagger_\mathrm{B}\sigma^i\psi_\mathrm{A}$ reduces, by $\sigma^i\sigma^j=\delta^{ij}+\mathrm{i}\epsilon^{ijk}\sigma^k$, to the convection current $(2\mathrm{i}m_e)^{-1}(\psi^\dagger_\mathrm{A}\nabla\psi_\mathrm{A}-\nabla\psi^\dagger_\mathrm{A}\,\psi_\mathrm{A})$ plus the spin current $(2m_e)^{-1}\nabla\times(\psi^\dagger_\mathrm{A}\boldsymbol{\sigma}\psi_\mathrm{A})$, the Gordon decomposition.

C.15.3 The Foldy–Wouthuysen transformation

The large and small components are coupled by $\boldsymbol{\alpha}_{\rm D}\cdot\boldsymbol{\pi}$, the only part of $H_{\rm D}$ that mixes them. Foldy and Wouthuysen removed it order by order in $1/m_e$ by unitary transformations [26]. Call an operator *even* if it commutes with $\beta_{\rm D}$ (block diagonal) and *odd* if it anticommutes (block off-diagonal). We take static fields and $\boldsymbol{A}=0$, the case Sects. 10.6 and 12.5 need, so that

$$H_{\rm D}=\beta_{\rm D}m_e+\mathsf{E}+\mathsf{O},\qquad \mathsf{E}=V,\qquad \mathsf{O}=\boldsymbol{\alpha}_{\rm D}\cdot\boldsymbol{p}. \tag{C.129}$$

The rules are $\beta_{\rm D}\mathsf{O}=-\mathsf{O}\beta_{\rm D}$, $\beta_{\rm D}\mathsf{E}=\mathsf{E}\beta_{\rm D}$, and O^2, a product of two odd operators, is even. Let $X=\beta_{\rm D}\mathsf{O}/(2m_e)$, which is anti-Hermitian, so e^X is unitary, and transform $H'=\mathrm{e}^X H_{\rm D}\mathrm{e}^{-X}=H_{\rm D}+[X,H_{\rm D}]+\frac{1}{2!}[X,[X,H_{\rm D}]]+\frac{1}{3!}[X,[X,[X,H_{\rm D}]]]+\ldots$ (Appendix C.6). The commutators of $\beta_{\rm D}\mathsf{O}$ with the parts of $H_{\rm D}$ follow from these rules:

$$[\beta_{\rm D}\mathsf{O},\beta_{\rm D}]=\beta_{\rm D}\mathsf{O}\beta_{\rm D}-\mathsf{O}=-2\mathsf{O},\qquad [\beta_{\rm D}\mathsf{O},\mathsf{O}]=\beta_{\rm D}\mathsf{O}^2-\mathsf{O}\beta_{\rm D}\mathsf{O}=2\beta_{\rm D}\mathsf{O}^2,$$
$$[\beta_{\rm D}\mathsf{O},\mathsf{E}]=\beta_{\rm D}[\mathsf{O},\mathsf{E}].$$

In each, $\beta_{\rm D}$ was moved to the left with its anticommutation or commutation rule. The first-order term is then

$$[X,H_{\rm D}]=-\mathsf{O}+\frac{\beta_{\rm D}\mathsf{O}^2}{m_e}+\frac{\beta_{\rm D}[\mathsf{O},\mathsf{E}]}{2m_e}.$$

Its first term cancels the odd operator of $H_{\rm D}$; this fixes the choice of X. The commutator of X with each of the three new terms follows from the same rules, using that O^2 is even and $[\mathsf{O},\mathsf{E}]$ is odd: $[X,-\mathsf{O}]=-\beta_{\rm D}\mathsf{O}^2/m_e$, $[X,\beta_{\rm D}\mathsf{O}^2/m_e]=(\beta_{\rm D}\mathsf{O}\beta_{\rm D}\mathsf{O}^2-\beta_{\rm D}\mathsf{O}^2\beta_{\rm D}\mathsf{O})/(2m_e^2)=-\mathsf{O}^3/m_e^2$, and $[X,\beta_{\rm D}[\mathsf{O},\mathsf{E}]/2m_e]=-[\mathsf{O},[\mathsf{O},\mathsf{E}]]/(4m_e^2)$. The same rules give the next orders, $[X,-\beta_{\rm D}\mathsf{O}^2/m_e]=\mathsf{O}^3/m_e^2$, $[X,-\mathsf{O}^3/m_e^2]=-(\beta_{\rm D}\mathsf{O}^4-\mathsf{O}^3\beta_{\rm D}\mathsf{O})/(2m_e^3)=-\beta_{\rm D}\mathsf{O}^4/m_e^3$ and $[X,\mathsf{O}^3/m_e^2]=\beta_{\rm D}\mathsf{O}^4/m_e^3$, so that, up to terms of higher order in $1/m_e$, $\frac{1}{3!}[X,[X,[X,H_{\rm D}]]]=\mathsf{O}^3/(6m_e^2)-\beta_{\rm D}\mathsf{O}^4/(6m_e^3)$ and $\frac{1}{4!}[X,[X,[X,[X,H_{\rm D}]]]]=\beta_{\rm D}\mathsf{O}^4/(24m_e^3)$. Collecting,

$$H'=\beta_{\rm D}\Big(m_e+\frac{\mathsf{O}^2}{2m_e}-\frac{\mathsf{O}^4}{8m_e^3}\Big)+\mathsf{E}-\frac{[\mathsf{O},[\mathsf{O},\mathsf{E}]]}{8m_e^2}+\mathsf{O}',\qquad \mathsf{O}'=\frac{\beta_{\rm D}[\mathsf{O},\mathsf{E}]}{2m_e}-\frac{\mathsf{O}^3}{3m_e^2}. \tag{C.130}$$

The coefficients are $1-\frac12=\frac12$ for $\beta_{\rm D}\mathsf{O}^2/m_e$, $-\frac16+\frac{1}{24}=-\frac18$ for $\beta_{\rm D}\mathsf{O}^4/m_e^3$, and $-\frac12+\frac16=-\frac13$ for O^3/m_e^2.

The power counting is in the electron velocity v: $\mathsf{O}\sim p\sim m_e v$ and $\mathsf{E}\sim m_e v^2$. The even terms of Eq. (C.130) are of order m_e, $m_e v^2$ and $m_e v^4$, while the remaining odd operator O' is of order $m_e v^3$. A second transformation with $X'=\beta_{\rm D}\mathsf{O}'/(2m_e)$ removes O' in the same way. By the first-order formula it adds $\beta_{\rm D}\mathsf{O}'^2/(2m_e)$, of order $m_e v^6$, to the even part, and leaves an odd remainder $\beta_{\rm D}[\mathsf{O}',\mathsf{E}]/(2m_e)$ of order $m_e v^5$, which a third transformation removes with even changes of still higher order. To order $m_e v^4$, the even part of Eq. (C.130) is the final Hamiltonian, and it acts on the upper components alone.

C.15.4 Mass–velocity, Darwin and spin–orbit terms

The three operators of Eq. (C.130) are evaluated with the identity $\alpha_{\rm D}^i\alpha_{\rm D}^j=\delta^{ij}+\mathrm{i}\epsilon^{ijk}\Sigma^k$, where $\boldsymbol{\Sigma}=\mathrm{diag}(\boldsymbol{\sigma},\boldsymbol{\sigma})$ is twice the spin. It gives $\mathsf{O}^2=p^2$ and $\mathsf{O}^4=p^4$. For the double commutator, let $\boldsymbol{W}=\nabla V$. Then $[\mathsf{O},V]=\alpha_{\rm D}^j[p^j,V]=-\mathrm{i}\alpha_{\rm D}^j W^j$, and

$$[\mathsf{O},[\mathsf{O},V]]=-\mathrm{i}\sum_{ij}(\alpha_{\rm D}^i\alpha_{\rm D}^j\,p^iW^j-\alpha_{\rm D}^j\alpha_{\rm D}^i\,W^jp^i).$$

Moving p^i to the right, $p^iW^j=W^jp^i-\mathrm{i}\,\partial_iW^j$, the bracket becomes $(\alpha_{\rm D}^i\alpha_{\rm D}^j-\alpha_{\rm D}^j\alpha_{\rm D}^i)W^jp^i-\mathrm{i}\alpha_{\rm D}^i\alpha_{\rm D}^j\partial_iW^j$. The antisymmetric combination is $2\mathrm{i}\epsilon^{ijk}\Sigma^k$, and the sum $\sum_{ij}\alpha_{\rm D}^i\alpha_{\rm D}^j\partial_iW^j=$

$\nabla^2 V + \mathrm{i}\boldsymbol{\Sigma}\cdot(\nabla\times\nabla V) = \nabla^2 V$. With $\epsilon^{ijk}W^j p^i = -(\boldsymbol{W}\times\boldsymbol{p})^k$,

$$[\mathsf{O},[\mathsf{O},V]] = -\nabla^2 V - 2\boldsymbol{\Sigma}\cdot(\nabla V\times\boldsymbol{p}). \tag{C.131}$$

On the upper components, with $\boldsymbol{\Sigma}\to\boldsymbol{\sigma}=2\boldsymbol{S}$, the Hamiltonian to order $m_e v^4$ is therefore

$$H_{\mathrm{FW}} = m_e + \frac{p^2}{2m_e} + V - \frac{p^4}{8m_e^3} + \frac{\nabla^2 V}{8m_e^2} + \frac{1}{4m_e^2}\,\boldsymbol{\sigma}\cdot(\nabla V\times\boldsymbol{p}). \tag{C.132}$$

The fourth term is the *mass–velocity* correction, the next term of $(p^2+m_e^2)^{1/2}$. The fifth is the *Darwin* term, a smearing of the potential over the Compton wavelength $1/m_e$. The last is the *spin–orbit* coupling; for a central potential, $\nabla V = (\boldsymbol{r}/r)\,\mathrm{d}V/\mathrm{d}r$ and $\boldsymbol{r}\times\boldsymbol{p}=\boldsymbol{L}$, so it equals $(2m_e^2)^{-1}(r^{-1}\mathrm{d}V/\mathrm{d}r)\,\boldsymbol{S}\cdot\boldsymbol{L}$. Eq. (C.132) is the result of Refs. [25, 26] at $\boldsymbol{A}=0$.

C.15.5 The contact density at a nucleus

For the Coulomb potential $V=-Z\alpha/r$, the identity $\nabla^2(1/r) = -4\pi\delta^3(\boldsymbol{r})$ gives $\nabla^2 V = 4\pi Z\alpha\,\delta^3(\boldsymbol{r})$. The Darwin term becomes a contact interaction,

$$H_{\mathrm{Dar}} = \frac{\pi Z\alpha}{2m_e^2}\,\delta^3(\boldsymbol{r}), \qquad \langle H_{\mathrm{Dar}}\rangle = \frac{\pi Z\alpha}{2m_e^2}\,|\psi(0)|^2. \tag{C.133}$$

Its expectation value is fixed by the *contact density* $|\psi(0)|^2$, the electron density at the nucleus, which only s states have. For a hydrogenic ns state of nuclear charge Z, $|\psi_{ns}(0)|^2 = (Z\alpha m_e)^3/(\pi n^3)$, the $n=1$ value being $1/(\pi a_0^3)$ at $Z=1$ (Sect. 2.2). The 1s Darwin shift is then $m_e(Z\alpha)^4/2$, a fraction $(Z\alpha)^2$ of the hydrogenic binding energy $m_e(Z\alpha)^2/2$: 0.72 meV, or 5.3×10^{-5} of the binding, for hydrogen; 28 eV, or 0.010, for $Z=14$; and 0.76 keV, or 0.055, for $Z=32$. The relative size $(Z\alpha)^2$ is that of each relativistic term. The mass–velocity term, $-\frac{5}{8}m_e(Z\alpha)^4$ for 1s, nearly cancels the Darwin term, and the net 1s shift is about a quarter of $(Z\alpha)^2$ of the binding energy, 0.26 % for silicon and 1.46 % for germanium in all-electron calculations of the atoms (Worked example 3.9). Every operator of the response that probes the electron density near the nucleus, such as the third energy moment of Sect. 12.5, picks up corrections of this relative size, $(Z\alpha)^2$, from the core shells.

The density vertex itself changes under the transformation. Since $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ is even, its transform is $\mathrm{e}^X\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\mathrm{e}^{-X} = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} + \beta_{\mathrm{D}}[\mathsf{O},\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}]/(2m_e) - [\mathsf{O},[\mathsf{O},\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}]]/(8m_e^2) + \ldots$, by the same algebra with V replaced by $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$. The odd term connects positive- and negative-energy states and contributes to positive-energy matrix elements only at higher order. Eq. (C.131) with $\nabla\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = \mathrm{i}\boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ and $\nabla^2\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = -q^2\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ gives, on the upper components,

$$\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \;\to\; \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\Big[1 - \frac{q^2}{8m_e^2} + \frac{\mathrm{i}}{4m_e^2}\,\boldsymbol{\sigma}\cdot(\boldsymbol{q}\times\boldsymbol{p})\Big]. \tag{C.134}$$

The scalar correction is $q^2/(8m_e^2) = 4.8\times10^{-5}$ at $q=10\,\mathrm{keV}$. The spin-dependent term is of order qp/m_e^2, larger for core electrons, whose momenta reach $Z\alpha m_e$; Sect. 10.6 bounds both for silicon and germanium.

C.16 The quantized radiation field

The 2p state of hydrogen decays to 1s in 1.60 ns by emitting one photon, and a silicon nucleus kicked by dark matter radiates in the same way, more weakly. Sect. 12.6 computes such emission rates in a crystal. They need the electromagnetic field as a set of quantized oscillators, the coupling of a charge to it, the one-photon emission rate with its dipole limit, and the change of all three in a transparent medium of refractive index $n_r(\omega)$. We use Heaviside–Lorentz units with $e^2 = 4\pi\alpha$ (Appendix A.2) and the box normalization of Appendix C.2, with volume $\mathcal{V}$.

C.16.1 The field as oscillators

In the Coulomb gauge, $\nabla\cdot\boldsymbol{A}=0$ and A^0 is the instantaneous Coulomb potential of the charges, already contained in the Coulomb interaction of Sect. 2.3. The radiation field is the transverse vector potential. In the periodic box it is a sum over wavevectors $\boldsymbol{k}$ and two polarization vectors $\boldsymbol{\epsilon}_{\boldsymbol{k}j}$, $j=1,2$, orthonormal and perpendicular to $\boldsymbol{k}$, which makes $\nabla\cdot\boldsymbol{A}=0$ term by term. Without sources each term obeys the wave equation with $\omega_k=k$:

$$\boldsymbol{A}(\boldsymbol{r},t)=\sum_{\boldsymbol{k}j}\left[c_{\boldsymbol{k}j}\,\boldsymbol{\epsilon}_{\boldsymbol{k}j}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}-\mathrm{i}\omega_k t}+\text{c.c.}\right].$$

The fields are $\boldsymbol{E}=-\partial_t\boldsymbol{A}$ and $\boldsymbol{B}=\nabla\times\boldsymbol{A}$, so each mode contributes $\mathrm{i}\omega_k c_{\boldsymbol{k}j}\boldsymbol{\epsilon}_{\boldsymbol{k}j}$ to $\boldsymbol{E}$ and $\mathrm{i}\boldsymbol{k}\times\boldsymbol{\epsilon}_{\boldsymbol{k}j}c_{\boldsymbol{k}j}$ to $\boldsymbol{B}$, times the plane wave. The energy $\frac{1}{2}\int\mathrm{d}^3r\,(\boldsymbol{E}^2+\boldsymbol{B}^2)$ is evaluated with $\int\mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}-\boldsymbol{k}')\cdot\boldsymbol{r}}=\mathcal{V}\delta_{\boldsymbol{k}\boldsymbol{k}'}$. Terms that pair $\boldsymbol{k}$ with $-\boldsymbol{k}$ oscillate as $\mathrm{e}^{\mp2\mathrm{i}\omega_k t}$ and cancel between $\boldsymbol{E}^2$ and $\boldsymbol{B}^2$, because $(\boldsymbol{k}\times\boldsymbol{\epsilon})\cdot(\boldsymbol{k}\times\boldsymbol{\epsilon}')=k^2\,\boldsymbol{\epsilon}\cdot\boldsymbol{\epsilon}'$ for transverse vectors and $k^2=\omega_k^2$. What remains is $\sum_{\boldsymbol{k}j}2\mathcal{V}\omega_k^2|c_{\boldsymbol{k}j}|^2$. A harmonic oscillator of frequency ω_k whose classical amplitude $\mathfrak{a}_{\boldsymbol{k}j}$ is normalized so that its energy is $\omega_k|\mathfrak{a}_{\boldsymbol{k}j}|^2$ matches this with $c_{\boldsymbol{k}j}=\mathfrak{a}_{\boldsymbol{k}j}/(2\mathcal{V}\omega_k)^{1/2}$. Quantization replaces $\mathfrak{a}_{\boldsymbol{k}j}$ by a boson annihilation operator $d_{\boldsymbol{k}j}$ with the commutators of Eq. (2.35):

$$\boldsymbol{A}(\boldsymbol{r})=\sum_{\boldsymbol{k}j}\frac{1}{(2\mathcal{V}\omega_k)^{1/2}}\left[d_{\boldsymbol{k}j}\,\boldsymbol{\epsilon}_{\boldsymbol{k}j}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}+d^\dagger_{\boldsymbol{k}j}\,\boldsymbol{\epsilon}^*_{\boldsymbol{k}j}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\right],\qquad H_{\mathrm{rad}}=\sum_{\boldsymbol{k}j}\omega_k(d^\dagger_{\boldsymbol{k}j}d_{\boldsymbol{k}j}+\tfrac{1}{2}).\quad\text{(C.135)}$$

The eigenstates of H_{rad} are the photon number states $|\{n_{\boldsymbol{k}j}\}\rangle$, and by Eq. (2.34) the operator $d^\dagger_{\boldsymbol{k}j}$ adds a photon with amplitude $(n_{\boldsymbol{k}j}+1)^{1/2}$. Emission into a mode that already holds n photons is therefore $n+1$ times faster than into an empty one; for $\omega\gg T$ the thermal occupation $n_{\mathrm{B}}(\omega)$ is negligible and only spontaneous emission, the 1, remains. Complex polarization vectors describe circular polarization, and the formulas below hold for them.

C.16.2 Minimal coupling

A non-relativistic particle of mass m and charge e_{c}, with $e_{\mathrm{c}}=-e$ for an electron and $e_{\mathrm{c}}=Z_\kappa e$ for a bare nucleus, couples to $\boldsymbol{A}$ by *minimal coupling*, $\boldsymbol{p}\to\boldsymbol{p}-e_{\mathrm{c}}\boldsymbol{A}$. For the electron this is $\boldsymbol{p}+e\boldsymbol{A}$, the non-relativistic limit of Eq. (C.126). Expanding the kinetic energy,

$$\frac{(\boldsymbol{p}-e_{\mathrm{c}}\boldsymbol{A})^2}{2m}=\frac{p^2}{2m}-\frac{e_{\mathrm{c}}}{m}\,\boldsymbol{A}\cdot\boldsymbol{p}+\frac{e_{\mathrm{c}}^2}{2m}\,\boldsymbol{A}^2,\qquad\text{(C.136)}$$

where $\boldsymbol{p}\cdot\boldsymbol{A}=\boldsymbol{A}\cdot\boldsymbol{p}-\mathrm{i}\nabla\cdot\boldsymbol{A}=\boldsymbol{A}\cdot\boldsymbol{p}$ in the Coulomb gauge. The term linear in $\boldsymbol{A}$ creates or absorbs one photon; the quadratic term acts at first order only in two-photon processes, such as scattering, and drops out of one-photon emission. For several particles the interaction is the sum over particles, and for a prescribed classical current $\boldsymbol{J}(\boldsymbol{r},t)$ it is $H'=-\int\mathrm{d}^3r\,\boldsymbol{J}\cdot\boldsymbol{A}$, the form taken by the linear term when the particle velocity $\boldsymbol{p}/m$ is a given function of time.

C.16.3 The one-photon emission rate

Let the matter make a transition $|i\rangle\to|f\rangle$ with energy release $\omega=E_i-E_f>0$ and emit one photon into the empty field. Only the $d^\dagger$ part of Eq. (C.135) contributes, and the matrix element of the linear term of Eq. (C.136) is

$$\langle f;1_{\boldsymbol{k}j}|H'|i;0\rangle=-\frac{e_{\mathrm{c}}}{m\,(2\mathcal{V}\omega_k)^{1/2}}\,\boldsymbol{\epsilon}^*_{\boldsymbol{k}j}\cdot\boldsymbol{M}_{fi},\qquad\boldsymbol{M}_{fi}=\langle f|\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\,\boldsymbol{p}|i\rangle.$$

The golden rule Eq. (2.5), summed over the photon modes with $\sum_{\boldsymbol{k}} \to \mathcal{V}\int \mathrm{d}^3k/(2\pi)^3 = \mathcal{V}\int \omega_k^2 \mathrm{d}\omega_k\,\mathrm{d}\Omega_{\hat{\boldsymbol{k}}}/(2\pi)^3$, gives the rate per solid angle of emission

$$\frac{\mathrm{d}\Gamma}{\mathrm{d}\Omega_{\hat{\boldsymbol{k}}}} = \frac{e_\mathrm{c}^2\,\omega}{8\pi^2 m^2}\sum_j |\boldsymbol{\epsilon}_{\boldsymbol{k}j}^* \cdot \boldsymbol{M}_{fi}|^2, \qquad |\boldsymbol{k}| = \omega. \tag{C.137}$$

The prefactor $\omega/(8\pi^2)$ collects 2π from the golden rule, $\omega^2/(2\pi)^3$ from the density of modes after the energy delta function is integrated, and $1/(2\omega)$ from the field normalization.

Dipole limit. When the emitting system is small compared with the photon wavelength, $kr \ll 1$, the factor $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ is replaced by 1. A photon of 10 eV has $k = 5.1\times10^{-3}\,\text{Å}^{-1}$, so across an atom $kr \approx 0.005$. If the unperturbed Hamiltonian is $H_0 = p^2/2m + U(\boldsymbol{r})$, then $[H_0, \boldsymbol{r}] = -\mathrm{i}\boldsymbol{p}/m$, and $\langle f|\boldsymbol{p}|i\rangle = \mathrm{i}m(E_f - E_i)\boldsymbol{r}_{fi} = -\mathrm{i}m\omega\,\boldsymbol{r}_{fi}$, with $\boldsymbol{r}_{fi} = \langle f|\boldsymbol{r}|i\rangle$. The sum over the two polarizations is $\sum_j |\boldsymbol{\epsilon}_j^* \cdot \boldsymbol{d}|^2 = |\boldsymbol{d}|^2 - |\hat{\boldsymbol{k}}\cdot\boldsymbol{d}|^2$ for any vector $\boldsymbol{d}$, by completeness of $\{\boldsymbol{\epsilon}_1, \boldsymbol{\epsilon}_2, \hat{\boldsymbol{k}}\}$, and its integral over directions is $4\pi|\boldsymbol{d}|^2(1-\frac{1}{3})$ (Appendix C.3). The total rate is

$$\Gamma_{i\to f} = \frac{e_\mathrm{c}^2\,\omega^3}{3\pi}\,|\boldsymbol{r}_{fi}|^2 = \frac{4}{3}\,\alpha\,\omega^3\,|\boldsymbol{r}_{fi}|^2 \qquad (|e_\mathrm{c}| = e). \tag{C.138}$$

For hydrogen 2p $\to$ 1s, $\omega = \frac{3}{4}\times 13.606\,\text{eV}$ and $|\boldsymbol{r}_{fi}|^2 = (2^{15}/3^{10})\,a_0^2$ for each m substate; the lifetime $1/\Gamma$ is 1.60 ns, the measured value. A transition with $\boldsymbol{r}_{fi} = 0$ proceeds through the next terms of the expansion of $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$, magnetic-dipole and electric-quadrupole, smaller by $(kr)^2$.

Emission by a classical current. For a prescribed current, first-order perturbation theory gives the amplitude of one photon in mode $\boldsymbol{k}j$ as $c_{\boldsymbol{k}j} = -\mathrm{i}\int \mathrm{d}t\,\langle 1_{\boldsymbol{k}j}|H'(t)|0\rangle = \mathrm{i}(2\mathcal{V}\omega_k)^{-1/2}\,\boldsymbol{\epsilon}_{\boldsymbol{k}j}^* \cdot \tilde{\boldsymbol{J}}(\boldsymbol{k},\omega_k)$, with $\tilde{\boldsymbol{J}}(\boldsymbol{k},\omega) = \int \mathrm{d}t\,\mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}\omega t - \mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\,\boldsymbol{J}(\boldsymbol{r},t)$ in the conventions of Appendix A.3. Summing $|c_{\boldsymbol{k}j}|^2$ over modes as above, the number of photons emitted per unit frequency and solid angle is

$$\frac{\mathrm{d}N_\gamma}{\mathrm{d}\omega\,\mathrm{d}\Omega_{\hat{\boldsymbol{k}}}} = \frac{\omega}{16\pi^3}\sum_j |\boldsymbol{\epsilon}_{\boldsymbol{k}j}^* \cdot \tilde{\boldsymbol{J}}(\boldsymbol{k},\omega)|^2. \tag{C.139}$$

A charge at rest that is suddenly set moving with velocity $\boldsymbol{v}$ at $t = 0$ carries $\boldsymbol{J} = e_\mathrm{c}\boldsymbol{v}\,\delta^3(\boldsymbol{r} - \boldsymbol{v}t)\,\Theta(t)$, so $\tilde{\boldsymbol{J}} = e_\mathrm{c}\boldsymbol{v}\int_0^\infty \mathrm{d}t\,\mathrm{e}^{\mathrm{i}(\omega - \boldsymbol{k}\cdot\boldsymbol{v})t} = \mathrm{i}e_\mathrm{c}\boldsymbol{v}/(\omega - \boldsymbol{k}\cdot\boldsymbol{v} + \mathrm{i}0^+) \simeq \mathrm{i}e_\mathrm{c}\boldsymbol{v}/\omega$ for $v \ll 1$. With the polarization sum and angular integral of the dipole case,

$$\frac{\mathrm{d}N_\gamma}{\mathrm{d}\omega} = \frac{e_\mathrm{c}^2\,v^2}{6\pi^2\,\omega} = \frac{2\alpha}{3\pi}\,\frac{v^2}{\omega} \qquad (|e_\mathrm{c}| = e). \tag{C.140}$$

The energy radiated per unit frequency, $\omega\,\mathrm{d}N_\gamma/\mathrm{d}\omega$, is flat. The kick may be treated as sudden when it lasts less than $1/\omega$. For a nucleus bound in a crystal the restoring forces and the response of the electrons modify Eq. (C.140) at the frequencies of the lattice and of the electrons, which Sect. 12.6 includes through the dynamical Born charge.

C.16.4 A photon mode in a medium

Inside a transparent medium with relative permittivity $\varepsilon_r(\omega) = n_r(\omega)^2$ and no magnetic response, the photon is a mode of the macroscopic field. The dispersion relation becomes $k = n_r(\omega)\,\omega$, and the number of modes per unit frequency grows: $\mathcal{V}\,k^2\,\mathrm{d}k/(2\pi)^3 = \mathcal{V}\,n_r^2\,n_\mathrm{grp}\,\omega^2\mathrm{d}\omega/(2\pi)^3$, with $n_\mathrm{grp} = \mathrm{d}k/\mathrm{d}\omega = n_r + \omega\,\mathrm{d}n_r/\mathrm{d}\omega$ the group index. The energy stored in a mode of given amplitude grows as well. The time-averaged energy of a monochromatic field in a transparent dispersive medium is $\frac{1}{4}[\mathrm{d}(\omega\varepsilon_r)/\mathrm{d}\omega\,|\boldsymbol{E}_0|^2 + |\boldsymbol{B}_0|^2]$ per unit volume [27]. With $|\boldsymbol{E}_0| = \omega|\boldsymbol{A}_0|$, $|\boldsymbol{B}_0| = k|\boldsymbol{A}_0| = n_r\omega|\boldsymbol{A}_0|$ and $\mathrm{d}(\omega\varepsilon_r)/\mathrm{d}\omega = n_r^2 + 2n_r\omega\,\mathrm{d}n_r/\mathrm{d}\omega$, the bracket

is $\omega^2|\boldsymbol{A}_0|^2 \cdot 2n_r n_{\rm grp}$, larger than in vacuum by the factor $n_r n_{\rm grp}$. Repeating the quantization, the normalization $(2\mathcal{V}\omega)^{-1/2}$ of Eq. (C.135) becomes $(2\mathcal{V}\omega\, n_r n_{\rm grp})^{-1/2}$; for a medium without dispersion, $n_{\rm grp} = n_r$, this is the quantization of Ref. [28].

In a rate, the squared normalization contributes $1/(n_r n_{\rm grp})$ and the density of modes $n_r^2 n_{\rm grp}$. The group index cancels, and

$$\Gamma^{\rm medium} = n_r(\omega)\,\Gamma^{\rm vac}, \qquad \frac{\mathrm{d}N_\gamma^{\rm medium}}{\mathrm{d}\omega} = n_r(\omega)\,\frac{\mathrm{d}N_\gamma^{\rm vac}}{\mathrm{d}\omega}, \tag{C.141}$$

for the same matrix element or current. The medium must be transparent at ω, so that ε_r is real and the photon is a propagating mode; near an absorption line the photon is not a well-defined final state. And the emitter must couple to the macroscopic field: the local field at an atom differs from the macroscopic one by a local-field factor, which Eq. (C.141) sets to 1. In the dipole factor $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ the wavevector becomes $n_r\omega$, which leaves the dipole limit intact for n_r of a few. Sect. 12.6 applies Eq. (C.141) to nuclear-recoil bremsstrahlung in silicon, germanium and GaAs.

C.17 Point and space groups

Silicon is mapped onto itself by 48 rotations and reflections, half of them combined with a shift by a quarter of the cube diagonal. The second half is why its (200) X-ray reflection vanishes for any charge distribution in the cell, and why some of its electronic transitions are forbidden at the lowest order. Sect. 13.9 uses the shifted operations, glides, to classify the zero-phonon transitions of silicon, and Sect. 14.5 uses point-group characters to decide which crystals allow a response odd in $\boldsymbol{q}$. Both start from the crystal structures of Sect. 3.1.

C.17.1 Symmetry operations

A symmetry operation of a crystal is a map $\boldsymbol{r} \to \mathsf{R}\boldsymbol{r} + \boldsymbol{t}$, written $\{\mathsf{R}|\boldsymbol{t}\}$, with R an orthogonal 3×3 matrix and $\boldsymbol{t}$ a translation, that maps the crystal onto itself. Applying $\{\mathsf{R}'|\boldsymbol{t}'\}$ and then $\{\mathsf{R}|\boldsymbol{t}\}$ gives $\boldsymbol{r} \to \mathsf{R}(\mathsf{R}'\boldsymbol{r} + \boldsymbol{t}') + \boldsymbol{t}$, so

$$\{\mathsf{R}|\boldsymbol{t}\}\{\mathsf{R}'|\boldsymbol{t}'\} = \{\mathsf{R}\mathsf{R}'\,|\,\mathsf{R}\boldsymbol{t}' + \boldsymbol{t}\}, \qquad \{\mathsf{R}|\boldsymbol{t}\}^{-1} = \{\mathsf{R}^{-1}\,|\,-\mathsf{R}^{-1}\boldsymbol{t}\}. \tag{C.142}$$

The operations form the *space group*. It contains the lattice translations $\{1|\boldsymbol{R}\}$ of Eq. (3.1), and each R that occurs comes with a translation $\boldsymbol{t}$ fixed up to a lattice vector. The set of the R is the *point group* $\mathcal{P}$, of order $|\mathcal{P}|$. An orthogonal matrix has $\det\mathsf{R} = \pm 1$. With $\det\mathsf{R} = +1$ it is a proper rotation by an angle θ_R about an axis; with $\det\mathsf{R} = -1$ it is improper, the product of the inversion -1 and the proper rotation $-\mathsf{R}$, and we define θ_R as the angle of $-\mathsf{R}$. A mirror reflection, for example, is the inversion times a rotation by π about the normal of the mirror.

If the origin can be chosen so that every R of the point group occurs with $\boldsymbol{t}$ a lattice vector, the space group is *symmorphic*. Otherwise it is *nonsymmorphic*, and some operations carry a translation that no change of origin removes: a *glide*, a reflection followed by a translation parallel to the mirror, or a *screw*, a rotation followed by a translation along its axis. Applied twice, a glide gives a pure translation, which must be a lattice vector. Crystals have 32 point groups, named in two systems. Silicon and germanium have $m\bar{3}m$ (O_h), the full symmetry of the cube. GaAs has $\bar{4}3m$ (T_d), that of the regular tetrahedron. The 24 proper rotations of the cube form 432 (O), and α-quartz and tellurium have 32 (D_3).

C.17.2 Representations and characters

A *representation* of $\mathcal{P}$ assigns to each R a matrix $\mathsf{D}(\mathsf{R})$ with $\mathsf{D}(\mathsf{R})\mathsf{D}(\mathsf{R}') = \mathsf{D}(\mathsf{R}\mathsf{R}')$. Vectors, tensors and functions of direction each carry one: the matrices describe how their components mix under the operations. A representation is *irreducible* if no basis change splits all its matrices into smaller blocks at once. Its *character* $c(\mathsf{R}) = \operatorname{tr}\mathsf{D}(\mathsf{R})$ is unchanged by basis changes and takes one value on each class of conjugate elements, $\mathsf{S}\mathsf{R}\mathsf{S}^{-1}$ with $\mathsf{S} \in \mathcal{P}$. The irreducible characters $c^{(a)}$ are orthonormal,

$$\frac{1}{|\mathcal{P}|}\sum_{\mathsf{R}\in\mathcal{P}} c^{(a)}(\mathsf{R})^*\, c^{(b)}(\mathsf{R}) = \delta_{ab}, \tag{C.143}$$

so a representation with character c contains the irreducible representation a exactly $n_a = |\mathcal{P}|^{-1}\sum_{\mathsf{R}} c^{(a)}(\mathsf{R})^* c(\mathsf{R})$ times [29, Chap. 3].

The case $a =$ trivial representation, $c^{(a)} = 1$, counts invariants, and it can be proved in two lines. Let $\mathsf{Q} = |\mathcal{P}|^{-1}\sum_{\mathsf{R}}\mathsf{D}(\mathsf{R})$. Since multiplying all elements of a group by one element only reorders them, $\mathsf{D}(\mathsf{S})\mathsf{Q} = \mathsf{Q}$ for every S; hence $\mathsf{Q}^2 = \mathsf{Q}$, and every vector in the range of Q is invariant. Conversely, an invariant vector $\mathbf{x}$ has $\mathsf{Q}\mathbf{x} = \mathbf{x}$. So Q is the projector onto the invariant subspace. Its trace is the dimension of that subspace:

$$n_{\mathrm{inv}} = \frac{1}{|\mathcal{P}|}\sum_{\mathsf{R}\in\mathcal{P}} c(\mathsf{R}). \tag{C.144}$$

For the point group 32, the six elements fall into three classes: the identity 1, the two rotations C_3 by $\pm 2\pi/3$ about the threefold axis, and the three rotations C_2' by π about the twofold axes perpendicular to it. There are three irreducible representations, labeled A_1, A_2 and E in the standard notation, with characters $(1,1,1)$, $(1,1,-1)$ and $(2,-1,0)$ on the classes $(1, C_3, C_2')$. With the class sizes 1, 2 and 3, Eq. (C.143) checks: for the two-dimensional representation with itself, $(4 + 2\cdot 1 + 0)/6 = 1$; for it with A_2, $(2 - 2 + 0)/6 = 0$.

C.17.3 Polar and axial tensors

A *polar* vector, such as $\boldsymbol{q}$, a position or a current, transforms with R itself. Its character is $c_{\mathrm{V}}(\mathsf{R}) = \operatorname{tr}\mathsf{R}$. A proper rotation by θ has the eigenvalues 1 and $\mathrm{e}^{\pm\mathrm{i}\theta}$, so $\operatorname{tr}\mathsf{R} = 1 + 2\cos\theta$; an improper one has the negatives. An *axial* vector, such as an angular momentum, a spin or a magnetic field, is a cross product of two polar vectors and does not change sign under the inversion. It transforms with $(\det\mathsf{R})\mathsf{R}$. Hence

$$c_{\mathrm{V}}(\mathsf{R}) = (\det\mathsf{R})\,(1 + 2\cos\theta_{\mathsf{R}}), \qquad c_{\mathrm{A}}(\mathsf{R}) = 1 + 2\cos\theta_{\mathsf{R}}. \tag{C.145}$$

A tensor transforms as the product of its indices, and its character is the product of theirs. A rank-three polar tensor has c_{V}^3; a second-rank tensor with one polar and one axial index has $c_{\mathrm{V}}c_{\mathrm{A}} = (\det\mathsf{R})(1 + 2\cos\theta_{\mathsf{R}})^2$. For functions of direction, the $2l + 1$ spherical harmonics of order l (Appendix C.3) have the character $\sin[(l + \tfrac{1}{2})\theta]/\sin(\theta/2)$ for a proper rotation, multiplied by $(-1)^l$ for an improper one. A crystal property allowed by symmetry is an invariant tensor, and Eq. (C.144) counts how many independent ones exist.

C.17.4 Responses odd in q

Inversion maps $\boldsymbol{q}$ to $-\boldsymbol{q}$, so a centrosymmetric crystal, one whose point group contains -1, has no response odd in $\boldsymbol{q}$ of any kind. In the sum Eq. (C.144) for an odd-rank polar tensor, the elements R and $-\mathsf{R}$ then contribute with opposite signs and cancel. For a scalar response, such as the structure factor $S(\boldsymbol{q},\omega)$, time reversal removes the odd part as well (Appendix A.4): a

part odd in $\boldsymbol{q}$ needs a crystal with neither inversion nor time-reversal symmetry, a magnetically ordered one.

A tensor response is different. Write the current response at small q as $\chi_{ij}(\boldsymbol{q},\omega) = \chi_{ij}^{(0)}(\omega) + \mathrm{i}\gamma_{ijl}(\omega)\, q_l + O(q^2)$. The coefficient γ_{ijl} is a rank-three polar tensor. In a crystal with time-reversal symmetry, the argument of Appendix A.4, applied to a product of two currents, gives the reciprocity relation $\chi_{ij}(\boldsymbol{q},\omega) = \chi_{ji}(-\boldsymbol{q},\omega)$. The sign that each current takes under time reversal cancels in the product, and the antiunitary conjugation exchanges the order of the two operators. The term linear in $\boldsymbol{q}$ then survives only if γ_{ijl} is antisymmetric in i, j. An antisymmetric pair of indices is equivalent to one axial index, $\gamma_{ijl} = \epsilon_{ijm} g_{ml}$, and the *gyration tensor* g_{ml} has one axial and one polar index. It describes natural optical activity and circular dichroism. By Eqs. (C.144) and (C.145), the number of its independent components in point group $\mathcal{P}$ is

Optical activity allowed by a point group

$$N_{\mathrm{gyr}} = \frac{1}{|\mathcal{P}|} \sum_{\mathsf{R}\in\mathcal{P}} (\det \mathsf{R})\,(1 + 2\cos\theta_{\mathsf{R}})^2. \tag{C.146}$$

Worked example C.1: Gyration invariants of GaAs and of a chiral cubic crystal

The group $\bar{4}3m$ of GaAs has 24 elements in five classes: the identity; eight C_3 ($\theta = 2\pi/3$); three C_2 ($\theta = \pi$); six S_4, improper, with $-S_4$ a rotation by $\pi/2$; and six mirrors σ_d, improper, with $-\sigma_d$ a rotation by π. The values of $(\det \mathsf{R})(1 + 2\cos\theta)^2$ are 9, 0, 1, -1 and -1. With the class sizes, Eq. (C.146) gives $N_{\mathrm{gyr}} = (9 + 0 + 3 - 6 - 6)/24 = 0$. GaAs has no optical activity, although it lacks inversion. It does have one invariant rank-three polar tensor: c_{V}^3 takes the values 27, 0, -1, -1 and 1 on the five classes, and $(27 + 0 - 3 - 6 + 6)/24 = 1$. That invariant is the totally symmetric tensor $|\epsilon_{ijl}|$, nonzero for i, j, l all different. It is the one behind the piezoelectricity of GaAs and the cubic harmonic xyz at $l = 3$. Reciprocity forbids it in the term linear in $\boldsymbol{q}$ of the current response, because it is symmetric in i, j. The group 432 has the 24 proper rotations of the cube, with values 9, 0, 1 on the identity, the eight C_3 and the three C_2, and 1 on the six C_4 ($\theta = \pi/2$) and the six further C_2'. Then $N_{\mathrm{gyr}} = (9 + 0 + 3 + 6 + 6)/24 = 1$: a cubic crystal without mirrors or inversion has isotropic optical activity, $g_{ml} \propto \delta_{ml}$. A direct enumeration of the matrices of each group reproduces all three counts, together with $N_{\mathrm{gyr}} = 0$ for $m\bar{3}m$.

C.17.5 The diamond space group

The diamond structure of silicon and germanium has the space group $Fd\bar{3}m$ (number 227 in the international tables; O_h^7). Its point group is $m\bar{3}m$. With the origin on the atom at $\boldsymbol{\tau}_1 = 0$ of Eq. (3.5), the 24 operations of $\bar{4}3m$, the site symmetry of each atom, occur with $\boldsymbol{t} = 0$. The other 24 occur with $\boldsymbol{t} = \boldsymbol{\tau}_2 = (a/4)(1, 1, 1)$, and they exchange the two atoms of the basis. The inversion $\{-1|\boldsymbol{\tau}_2\}$, for example, is the inversion through the bond center at $\boldsymbol{\tau}_2/2$. No choice of origin removes all these translations, and the group is nonsymmorphic. Zincblende GaAs, with different atoms on the two sites, keeps only the first 24 operations; its space group $F\bar{4}3m$ is symmorphic.

Take the mirror m_x, $(x, y, z) \to (-x, y, z)$, which is not in $\bar{4}3m$. It occurs as

$$g = \{\mathsf{m}_x|\boldsymbol{\tau}_2\}: \quad (x, y, z) \to (-x + \tfrac{a}{4},\ y + \tfrac{a}{4},\ z + \tfrac{a}{4}). \tag{C.147}$$

Points with $x = a/8$ keep their x coordinate, so this is a reflection in the plane $x = a/8$ followed by the translation $(0, a/4, a/4)$ within it: a glide, of the kind called a diamond glide. It maps the atom at 0 to $\boldsymbol{\tau}_2$ and the atom at $\boldsymbol{\tau}_2$ to $(0, a/2, a/2) = \boldsymbol{a}_1$, a lattice point. Applied twice, by Eq. (C.142), it gives $\{1|\boldsymbol{w}\}$ with $\boldsymbol{w} = \boldsymbol{\tau}_2 + \mathsf{m}_x\boldsymbol{\tau}_2 = (a/2)(0,1,1) = \boldsymbol{a}_1$, a lattice translation, as required.

Extinctions. Let $f(\boldsymbol{r})$ be any function invariant under g, $f(\mathsf{m}_x\boldsymbol{r} + \boldsymbol{\tau}_2) = f(\boldsymbol{r})$, and lattice periodic, with components $f_{\boldsymbol{G}}$ (Eq. (A.12)). Substituting $\boldsymbol{r} = \mathsf{m}_x\boldsymbol{r}' + \boldsymbol{\tau}_2$ in $f_{\boldsymbol{G}} = \Omega^{-1}\int_\Omega \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}} f(\boldsymbol{r})$ gives $f_{\boldsymbol{G}} = \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_2} f_{\mathsf{m}_x\boldsymbol{G}}$. For $\boldsymbol{G}$ in the mirror plane, $\boldsymbol{G} = (2\pi/a)(0,k,l)$ with k, l even, $\mathsf{m}_x\boldsymbol{G} = \boldsymbol{G}$ and

$$f_{\boldsymbol{G}} = \mathrm{e}^{\mathrm{i}\pi(k+l)/2} f_{\boldsymbol{G}}, \qquad \text{so} \qquad f_{\boldsymbol{G}} = 0 \quad \text{for } k + l \equiv 2 \pmod 4. \tag{C.148}$$

The components (020) and (002) vanish, and by the glides with m_y and m_z so does (200). They vanish for every function with the symmetry of the crystal: the full crystal potential, the charge density with its bonds, any lattice-periodic response. This is stronger than the extinction of Worked example 3.2, which came from the sum of identical spherical atoms. The (222) component, whose vector lies in no glide plane of the form m_x, m_y, m_z, vanishes only for spherical atoms, and the bond charge makes it nonzero.

Glide parity of Bloch states. Let g act on wavefunctions by $(g\psi)(\boldsymbol{r}) = \psi(g^{-1}\boldsymbol{r})$, with $g^{-1}\boldsymbol{r} = \mathsf{m}_x(\boldsymbol{r} - \boldsymbol{\tau}_2)$. If ψ is a Bloch state of crystal momentum $\boldsymbol{k}$, then $(g\psi)(\boldsymbol{r} + \boldsymbol{R}) = \mathrm{e}^{\mathrm{i}(\mathsf{m}_x\boldsymbol{k})\cdot\boldsymbol{R}}(g\psi)(\boldsymbol{r})$, so $g\psi$ has crystal momentum $\mathsf{m}_x\boldsymbol{k}$. For $\boldsymbol{k}$ in the mirror plane, $k_x = 0$, the operator g maps the Bloch states at $\boldsymbol{k}$ among themselves and commutes with the Hamiltonian, so the states can be chosen as its eigenstates. Since $g^2 = \{1|\boldsymbol{w}\}$ acts as $\psi(\boldsymbol{r} - \boldsymbol{w}) = \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{w}}\psi(\boldsymbol{r})$, the eigenvalues are

$$g\,\psi = \eta\,\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{w}/2}\,\psi, \qquad \eta = \pm 1, \tag{C.149}$$

and η is the *glide parity*. It depends on the label: the same state labeled by $\boldsymbol{k} - \boldsymbol{G}$, with $\boldsymbol{G}$ in the plane, has $\eta\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{w}/2} = \eta\,\mathrm{e}^{-\mathrm{i}\pi(k+l)/2}$.

Consider now the vertex $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ of Sect. 3.2 with $\boldsymbol{q}$ in the mirror plane. Conjugation by g gives $g\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}g^{-1} = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot g^{-1}\boldsymbol{r}} = \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_2}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, using $\mathsf{m}_x\boldsymbol{q} = \boldsymbol{q}$. For eigenstates $|i\rangle$ at $\boldsymbol{k}$ and $|f\rangle$ labeled by the unreduced momentum $\boldsymbol{k} + \boldsymbol{q}$, unitarity of g gives

$$\langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle = \langle gf|g\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}g^{-1}|gi\rangle = \eta_f\eta_i\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{w}/2-\boldsymbol{\tau}_2)}\,\langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle.$$

The vector $\boldsymbol{w}/2 - \boldsymbol{\tau}_2 = (a/4)(-1,0,0)$ is normal to the plane, so the phase is 1 for $\boldsymbol{q}$ in the plane, and the matrix element vanishes unless $\eta_f = \eta_i$. For the reduced momentum $\boldsymbol{k} + \boldsymbol{q} - \boldsymbol{G}$ of the final state, the rule reads

$$\eta_f^{\mathrm{red}} = \eta_i\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{w}/2}, \tag{C.150}$$

so a momentum transfer whose Umklapp vector belongs to the (200) class flips the glide parity, and one with $k + l \equiv 0 \pmod 4$ preserves it. At $\boldsymbol{k} = 0$ and $\boldsymbol{q} = \boldsymbol{G} = (2\pi/a)(0,2,0)$, so that the reduced final momentum $\boldsymbol{k} + \boldsymbol{q} - \boldsymbol{G}$ is again 0, Eq. (C.150) requires a state to have opposite parity to itself, and the diagonal matrix element vanishes: Eq. (C.148) again. Sect. 13.9 applies the rule to the band edges of silicon. A further consequence of the nonsymmorphic operations is that every band of diamond is at least twofold degenerate at the zone-boundary points X, without spin–orbit coupling, a degeneracy that no symmorphic group with the same point group would force [29].

C.18 Continuum elasticity

A longitudinal sound wave in silicon travels at $8.43\,\mathrm{km\,s^{-1}}$ along a cube edge and at $9.35\,\mathrm{km\,s^{-1}}$ along a cube diagonal. Both numbers follow from three elastic constants and the density,

and so do the speeds of the two transverse waves in every direction. Sect. 11.1 needs the longitudinal speed averaged over directions and a lower bound on it from the bulk modulus. Sect. 17.5 needs a polarization-weighted average of the inverse speeds, and Sect. 20.3 the acoustic impedances that decide how much of a phonon flux crosses from a crystal into a metal film. All of them are long-wavelength properties of the acoustic branches of Sect. 5.2: for Q well inside the Brillouin zone, $\omega_{\boldsymbol{Q}\nu} = c_\nu(\hat{\boldsymbol{Q}})\, Q$, and the crystal responds like an elastic continuum.

C.18.1 Strain, stress and the elastic tensor

At long wavelength all atoms of a cell move together in an acoustic wave, and the displacements $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ of Eq. (A.21) become a smooth field $\boldsymbol{u}(\boldsymbol{r}, t)$. A uniform $\boldsymbol{u}$ is a translation and costs no energy. Neither does a rigid rotation, whose gradient $\partial_i u_j$ is antisymmetric. The energy therefore depends on the symmetric part of the gradient, the *strain*

$$u_{ij} = \tfrac{1}{2}(\partial_i u_j + \partial_j u_i), \tag{C.151}$$

which measures changes of length: a segment $\mathrm{d}\boldsymbol{r}$ changes its squared length by $2u_{ij}\,\mathrm{d}r_i\,\mathrm{d}r_j$ to first order. To second order in the strain the energy per unit volume is

$$w_{\mathrm{el}} = \tfrac{1}{2}\, C_{ijkl}\, u_{ij}\, u_{kl}, \tag{C.152}$$

with summation over repeated Cartesian indices. Since u_{ij} is symmetric, only the parts of C_{ijkl} symmetric in ij and in kl contribute, and we take $C_{ijkl} = C_{jikl} = C_{ijlk}$. Since C_{ijkl} is a second derivative of w_{el}, also $C_{ijkl} = C_{klij}$. These symmetries leave 21 independent constants. The *stress* is the derivative $\sigma_{ij} = \partial w_{\mathrm{el}}/\partial u_{ij} = C_{ijkl}u_{kl}$ (Hooke's law); it is the force per unit area, in direction i, that the material on the far side of a surface with normal along j exerts across it. The net force on a volume element is $\partial_j \sigma_{ij}$ per unit volume, and Newton's law for a medium of mass density ρ_T is

$$\rho_T\, \ddot{u}_i = \partial_j \sigma_{ij} = C_{ijkl}\, \partial_j \partial_l u_k, \tag{C.153}$$

where the second form uses $C_{ijkl}u_{kl} = C_{ijkl}\partial_l u_k$, by the symmetry in kl [30].

Tables and codes store C_{ijkl} as a symmetric 6×6 matrix C_{IJ} in *Voigt notation*. Each symmetric pair of indices becomes one index, $11 \to 1$, $22 \to 2$, $33 \to 3$, $23 \to 4$, $13 \to 5$, $12 \to 6$. The strain components are collected as $\epsilon_I = (u_{11}, u_{22}, u_{33}, 2u_{23}, 2u_{13}, 2u_{12})$. The factor 2 on the shear components is needed because u_{23} and u_{32} both occur in the double sum of Eq. (C.152); with it, $w_{\mathrm{el}} = \frac{1}{2} C_{IJ}\epsilon_I\epsilon_J$ and the stress components σ_I, without factors of 2, are $\sigma_I = C_{IJ}\epsilon_J$. The ordering of the shear rows is a common source of error in practice, discussed in Appendix E.4, where the computation of the constants by density-functional perturbation theory or finite strains is described.

A cubic crystal has three constants: $C_{11} = C_{xxxx}$, $C_{12} = C_{xxyy}$ and $C_{44} = C_{yzyz}$, with all others related to them by the cube's symmetry or zero. The Voigt matrix C_{IJ} then has three distinct eigenvalues: $C_{11} + 2C_{12}$ for a uniform dilation, $C_{11} - C_{12}$ (twice) for the two volume-conserving stretches along the cube axes, and C_{44} (three times) for the shears. A crystal is stable when every strain costs energy, which requires

$$C_{11} > |C_{12}|, \qquad C_{11} + 2C_{12} > 0, \qquad C_{44} > 0, \tag{C.154}$$

the Born stability conditions of a cubic crystal. For silicon at room temperature, $C_{11} = 165.6$, $C_{12} = 63.9$ and $C_{44} = 79.5\,\mathrm{GPa}$, with $\rho_T = 2.33\,\mathrm{g\,cm^{-3}}$ [31, Table I].

The *bulk modulus* B is the response to a uniform compression. A strain $u_{ij} = \frac{1}{3}\delta\,\delta_{ij}$ changes the volume by the fraction $\delta = \Delta\mathcal{V}/\mathcal{V}$, and in a cubic crystal it produces the stress $\sigma_{ij} = \frac{1}{3}(C_{11} + 2C_{12})\,\delta\,\delta_{ij}$, a pressure $P = -\frac{1}{3}\sigma_{ii}$. Hence

$$B = -\mathcal{V}\frac{\mathrm{d}P}{\mathrm{d}\mathcal{V}} = \tfrac{1}{3}(C_{11} + 2C_{12}), \tag{C.155}$$

which is 97.8 GPa for silicon. In a crystal of lower symmetry a uniform strain and a uniform pressure do not correspond, and the two give different averages, $C_{iijj}/9$ (Voigt) and the inverse of the compliance sum (Reuss); for cubic crystals they coincide with Eq. (C.155).

An isotropic medium, such as a polycrystal with random grain orientations or a glass, has $C_{11} = C_{12} + 2C_{44}$ and only two independent constants. The literature writes them as the Lamé constants, $\lambda = C_{12}$ and the shear modulus $\mu = C_{44}$, and then $B = C_{11} - \frac{4}{3}C_{44}$. Phonon-transport simulations often give silicon such a tensor, with $C_{12} = 52.4$ and $C_{44} = 68.0$ GPa [31, Table I], so that $C_{11} = 188.4$ GPa and the speeds in every direction are 9.0 and 5.4 km s^{-1}, the values used in Appendix E.12. This is a parameter set of the codes, not a property of silicon: the measured tensor violates the isotropy relation by a factor $2C_{44}/(C_{11} - C_{12}) = 1.56$.

C.18.2 The Christoffel problem

A plane wave $\boldsymbol{u} = \hat{\boldsymbol{e}}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{Q}\cdot\boldsymbol{r} - \omega t)}$ solves Eq. (C.153) if

$$\rho_T\, c^2\, \hat{e}_i = \mathsf{M}_{ik}(\hat{\boldsymbol{Q}})\,\hat{e}_k, \qquad \mathsf{M}_{ik}(\hat{\boldsymbol{Q}}) = C_{ijkl}\,\hat{Q}_j\,\hat{Q}_l, \qquad \omega = cQ. \tag{C.156}$$

This is the Christoffel equation, and M is the Christoffel matrix [30]. By the symmetry of C_{ijkl} in kl, $C_{ijkl}\hat{Q}_j\hat{Q}_l\hat{e}_k = C_{ijkl}\hat{Q}_j\hat{Q}_k\hat{e}_l$. Since $C_{ijkl} = C_{klij}$, M is real and symmetric, and since $\hat{e}_i\mathsf{M}_{ik}\hat{e}_k = C_{ijkl}(\hat{e}_i\hat{Q}_j)(\hat{e}_k\hat{Q}_l)$ is the elastic energy of the strain $\frac{1}{2}(\hat{e}_i\hat{Q}_j + \hat{e}_j\hat{Q}_i)$, it is positive definite in a stable crystal. Each direction $\hat{\boldsymbol{Q}}$ therefore has three positive eigenvalues $\rho_T c_\nu(\hat{\boldsymbol{Q}})^2$ with orthonormal polarizations $\hat{\boldsymbol{e}}_\nu(\hat{\boldsymbol{Q}})$. These are the sound speeds c_ν of Sect. 5.2 and the unit polarizations $\hat{\boldsymbol{e}}_\nu$ of Appendix A.6: Eq. (C.156) is the acoustic block of the dynamical matrix of Eq. (A.23) at small Q, to order Q^2, with $\rho_T = M_{\mathrm{cell}}/\Omega$. In general no polarization is exactly parallel or perpendicular to $\hat{\boldsymbol{Q}}$. The fastest branch, the quasi-longitudinal one, is the branch this book calls LA, with speed $c_{\mathrm{LA}}(\hat{\boldsymbol{Q}})$; the other two are quasi-transverse. The group velocity $\partial\omega/\partial\boldsymbol{Q}$ of each branch is not parallel to $\hat{\boldsymbol{Q}}$ either, which is the origin of the phonon focusing of Appendix E.12.

Along the three symmetry directions of a cubic crystal the polarizations are fixed by symmetry and the eigenvalues are

$$\begin{aligned}
&[100]: && \rho_T c_{\mathrm{LA}}^2 = C_{11}, && \rho_T c_{\mathrm{TA}}^2 = C_{44}\ \text{(twice)},\\
&[110]: && \rho_T c_{\mathrm{LA}}^2 = \tfrac{1}{2}(C_{11} + C_{12} + 2C_{44}), && \rho_T c_{\mathrm{TA}}^2 = C_{44}\ (\hat{\boldsymbol{e}} \parallel [001]),\\
& && && \rho_T c_{\mathrm{TA}}^2 = \tfrac{1}{2}(C_{11} - C_{12})\ (\hat{\boldsymbol{e}} \parallel [1\bar{1}0]),\\
&[111]: && \rho_T c_{\mathrm{LA}}^2 = \tfrac{1}{3}(C_{11} + 2C_{12} + 4C_{44}), && \rho_T c_{\mathrm{TA}}^2 = \tfrac{1}{3}(C_{11} - C_{12} + C_{44})\ \text{(twice)}.
\end{aligned} \tag{C.157}$$

Each follows from M_{ik} written out for that $\hat{\boldsymbol{Q}}$. Along [100], for example, $\mathsf{M}_{ik} = C_{i1k1}$, which is $\mathrm{diag}(C_{11}, C_{44}, C_{44})$. For silicon the longitudinal speeds are 8.43, 9.13 and 9.35 km s^{-1} along [100], [110] and [111], and the transverse speeds are 5.84 (twice), 5.84 and 4.67, and 5.09 (twice), in km s^{-1}. The simplest approximation keeps the measured constants but ignores the anisotropy. This *isotropic* [100] *model* gives every direction the [100] speeds, $\rho_T c_{\mathrm{LA}}^2 = C_{11}$ and $\rho_T c_{\mathrm{TA}}^2 = C_{44}$, which for silicon are 8.43 and 5.84 km s^{-1}. Its Debye speed follows from Eq. (5.25) with direction-independent speeds, $3/\bar{c}^3 = 1/c_{\mathrm{LA}}^3 + 2/c_{\mathrm{TA}}^3$, which gives

$\bar{c} = 6.35\,\mathrm{km\,s^{-1}}$, and its Debye temperature 696 K, 8 % above the 645 K of the full angular average (Worked example 5.3), because it misses the slow transverse wave along [110].

The longitudinal speed in any direction is bounded from below by the bulk modulus. The largest eigenvalue of a symmetric matrix is at least its expectation value in any unit vector, and with the trial vector $\hat{\boldsymbol{Q}}$ itself, $\rho_T c_{\mathrm{LA}}^2 \geq \hat{Q}_i \mathsf{M}_{ik} \hat{Q}_k = C_{ijkl} \hat{Q}_i \hat{Q}_j \hat{Q}_k \hat{Q}_l$, the Rayleigh quotient. For cubic symmetry the quartic form collects C_{11} from the four equal indices. For each pair of axes it collects $2C_{12}$ from $xxyy$ and $yyxx$, and $4C_{44}$ from $xyxy$, $xyyx$, $yxxy$ and $yxyx$:

$$C_{ijkl} \hat{Q}_i \hat{Q}_j \hat{Q}_k \hat{Q}_l = C_{11} \sum_i \hat{Q}_i^4 + (2C_{12} + 4C_{44})\, s_Q = C_{11} - 2(C_{11} - C_{12} - 2C_{44})\, s_Q,$$

with $s_Q = \hat{Q}_x^2 \hat{Q}_y^2 + \hat{Q}_y^2 \hat{Q}_z^2 + \hat{Q}_z^2 \hat{Q}_x^2$, where the second form uses $\sum_i \hat{Q}_i^4 = 1 - 2s_Q$. Since s_Q ranges from 0 along [100] to $\frac{1}{3}$ along [111], and the form is linear in s_Q, it lies between its values in those two directions. They exceed B by $\frac{2}{3}(C_{11} - C_{12})$ and $\frac{4}{3}C_{44}$, both positive by Eq. (C.154).

Bulk-modulus bound on the longitudinal speed

In a cubic crystal that satisfies Eq. (C.154), in every direction,

$$c_{\mathrm{LA}}(\hat{\boldsymbol{Q}}) \geq \sqrt{B/\rho_T}. \tag{C.158}$$

For silicon the bound is $6.48\,\mathrm{km\,s^{-1}}$, 23 % below the slowest longitudinal speed. Physically, a longitudinal wave compresses the medium as a uniform compression does, plus a shear, and the shear only adds stiffness. A tabulated sound speed that violates Eq. (C.158) for a measured B is wrong, which is one of the tests applied to tabulated speeds in Sect. 17.5.

C.18.3 Sphere averages

Rates that sum over directions of $\boldsymbol{Q}$ contain averages of a function $f(\hat{\boldsymbol{Q}})$ over the unit sphere,

$$\langle f \rangle = \frac{1}{4\pi} \int \mathrm{d}\Omega_{\hat{\boldsymbol{Q}}}\, f(\hat{\boldsymbol{Q}}) = \frac{1}{4\pi} \int_{-1}^{1} \mathrm{d}(\cos\theta) \int_0^{2\pi} \mathrm{d}\phi\, f(\theta, \phi). \tag{C.159}$$

Which function is averaged depends on the rate. Sect. 11.1 uses $\langle c_{\mathrm{LA}} \rangle$, written $\bar{c}_{\mathrm{LA}}$, and Sect. 17.5 uses $\langle \sum_\nu (\hat{\boldsymbol{Q}} \cdot \hat{\boldsymbol{e}}_\nu)^2 / c_\nu \rangle$, in which each branch is weighted by the square of its longitudinal component; Debye-type counts of states use $\langle c_\nu^{-3} \rangle$. For an isotropic medium the averages reduce to single values. For a crystal they differ from one another, and they are computed numerically. At each node of a quadrature grid we diagonalize the 3×3 matrix $\mathsf{M}(\hat{\boldsymbol{Q}})$ and evaluates f. Gauss–Legendre nodes in $\cos\theta$ and equally spaced nodes in ϕ, where the trapezoidal rule is exact for trigonometric polynomials, converge quickly for smooth f. The longitudinal speed is smooth, since the largest eigenvalue never meets the others. The transverse speeds touch along [100] and [111], where they are degenerate, so a function of one transverse branch alone has kinks there; a sum over both transverse branches, as in the polarization-weighted average, is smooth. For a cubic crystal the 48 operations of the point group (Appendix C.17) map the sphere onto itself and leave every eigenvalue unchanged, so the average over $\frac{1}{48}$ of the sphere, bounded by the [100], [110] and [111] directions, equals the full average. Sect. 11.1 evaluates $\bar{c}_{\mathrm{LA}}$ for silicon in a worked example.

C.18.4 Acoustic impedance and reflection

When a sound wave meets the interface between two media, part of its energy is reflected. Take two media, labeled 1 for $z < 0$ and 2 for $z > 0$, with densities ρ_1, ρ_2, and a longitudinal

wave of frequency ω incident along z, at normal incidence. In medium 1 the displacement is $u_z = A\,[\mathrm{e}^{\mathrm{i}(Q_1 z-\omega t)} + r_{12}\,\mathrm{e}^{\mathrm{i}(-Q_1 z-\omega t)}]$ and in medium 2 it is $u_z = A\,t_{12}^{\mathrm{amp}}\,\mathrm{e}^{\mathrm{i}(Q_2 z-\omega t)}$, with $Q_i = \omega/c_i$ and c_i the longitudinal speed of medium i along z. A welded interface transmits displacement and force, so u_z and $\sigma_{zz} = \rho_i c_i^2\,\partial_z u_z$ are continuous at $z = 0$:

$$1 + r_{12} = t_{12}^{\mathrm{amp}}, \qquad \mathrm{i}Q_1\rho_1 c_1^2\,(1 - r_{12}) = \mathrm{i}Q_2\rho_2 c_2^2\,t_{12}^{\mathrm{amp}}.$$

Since $Q_i\rho_i c_i^2 = \omega\rho_i c_i$, the second condition contains each medium only through its *acoustic impedance*

$$Z_{\mathrm{ac},i} = \rho_i\,c_i. \tag{C.160}$$

The two conditions give $r_{12} = (Z_{\mathrm{ac},1} - Z_{\mathrm{ac},2})/(Z_{\mathrm{ac},1} + Z_{\mathrm{ac},2})$ and $t_{12}^{\mathrm{amp}} = 2Z_{\mathrm{ac},1}/(Z_{\mathrm{ac},1} + Z_{\mathrm{ac},2})$. The energy flux of a wave is the power per unit area $-\sigma_{zz}\dot{u}_z$; for a real wave $u_z = A\cos(Qz-\omega t)$ its time average is $\frac{1}{2}Z_{\mathrm{ac}}\,\omega^2 A^2$. The fraction of the incident flux that crosses the interface is therefore $Z_{\mathrm{ac},2}|t_{12}^{\mathrm{amp}}|^2/Z_{\mathrm{ac},1}$:

$$t_{12}(0) = \frac{4Z_{\mathrm{ac},1}Z_{\mathrm{ac},2}}{(Z_{\mathrm{ac},1} + Z_{\mathrm{ac},2})^2} = 1 - r_{12}^2. \tag{C.161}$$

The argument 0 is the angle of incidence. The result is symmetric in the two media, $t_{12}(0) = t_{21}(0)$, and equals one when the impedances match, whatever the densities and speeds separately are [32].

Silicon and aluminum are almost matched. Along [100], silicon has $Z_{\mathrm{ac}} = 19.6 \times 10^6\,\mathrm{kg\,m^{-2}\,s^{-1}}$. Aluminum, with $\rho = 2.70\,\mathrm{g\,cm^{-3}}$ and the elastic constants extrapolated to zero temperature, $C_{11} = 114.3$, $C_{12} = 61.9$ and $C_{44} = 31.6\,\mathrm{GPa}$ [33, Table 2], has $c_{\mathrm{LA}} = 6.51\,\mathrm{km\,s^{-1}}$ along [100] and $Z_{\mathrm{ac}} = 17.6 \times 10^6\,\mathrm{kg\,m^{-2}\,s^{-1}}$. Aluminum is nearly isotropic, with $2C_{44}/(C_{11} - C_{12}) = 1.21$. The texture of a film therefore changes its longitudinal speed by at most 3 % above the [100] value, and the isotropic average is $6.63\,\mathrm{km\,s^{-1}}$ (Sect. 20.3). Eq. (C.161) then gives $t_{12}(0) = 0.997$: at normal incidence only a third of a percent of a longitudinal flux is reflected.

At oblique incidence the wavevector component along the interface is the same for every wave that the incident one generates, since the boundary conditions must hold at every point of the plane and at every time. For incidence at angle θ_1 on a wave of speed c_1, a reflected or transmitted wave of speed c' leaves at the angle θ' with

$$\frac{\sin\theta_1}{c_1} = \frac{\sin\theta'}{c'}, \tag{C.162}$$

the acoustic Snell law. In general an incident longitudinal wave generates reflected and transmitted waves of both longitudinal and transverse polarization (mode conversion), with amplitudes fixed by the continuity of all three displacement components and of the three stresses σ_{iz}. When $c' > c_1$ the wave c' exists only for $\sin\theta_1 < c_1/c'$; beyond that critical angle it is evanescent and carries no flux. Since all sound speeds of aluminum lie below the longitudinal speed of silicon, the critical angle matters for phonons leaving the film: a longitudinal phonon from aluminum along [100] reaches longitudinal states in silicon only within $\arcsin(6.51/8.43) = 51^\circ$ of the normal. Sect. 20.3 solves the oblique problem and averages it over angles.

C.19 Magnetic neutron scattering

A neutron carries a magnetic moment of -1.913 nuclear magnetons, and in a magnetic crystal it scatters off the unpaired electron spins about as strongly as off the nuclei. Its spin couples

to the same spin density $\boldsymbol{s}_e(\boldsymbol{q})$ as a dark-matter particle with a spin-dependent coupling to electrons, which is why Sect. 14.4 can turn measured neutron spectra into dark-matter spin responses. The translation needs the cross-section written with the response matrix S_{ab} of Sect. 4.2, the projection that removes the component of the spin density along the momentum transfer, and the magnetic form factor of an ion. Polarized beams then separate the two components that remain.

C.19.1 The magnetic cross-section

A neutron of mass m_n enters with momentum $\boldsymbol{p}_n$ and leaves with $\boldsymbol{p}'_n$. As for dark matter, $\boldsymbol{q} = \boldsymbol{p}_n - \boldsymbol{p}'_n$ is the momentum given to the target and $\omega = (p_n^2 - p_n'^2)/2m_n$ the energy. The neutron moment is $\boldsymbol{\mu}_n = \gamma_n \mu_N \boldsymbol{\sigma}_n$, with $\gamma_n = -1.913$, the nuclear magneton $\mu_N = e/2m_p$ and the Pauli matrices $\boldsymbol{\sigma}_n$ of the neutron spin [34, Chap. 7]. Its interaction with the magnetic field of the electrons is $-\boldsymbol{\mu}_n \cdot \boldsymbol{B}_e(\boldsymbol{r}_n)$.

The spins of the electrons, with $g_e = 2$, form the magnetization $\boldsymbol{M}_e(\boldsymbol{r}) = -2\mu_B\, \boldsymbol{s}_e(\boldsymbol{r})$, where $\boldsymbol{s}_e(\boldsymbol{r})$ is the spin density whose Fourier transform is $\boldsymbol{s}_e(\boldsymbol{q}) = \sum_i \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}\boldsymbol{\sigma}_i/2$ (Sect. 7.1). Without free currents, magnetostatics in Heaviside–Lorentz units reads $\nabla\cdot\boldsymbol{B}_e = 0$ and $\nabla\times\boldsymbol{B}_e = \nabla\times\boldsymbol{M}_e$. In Fourier space, at $\boldsymbol{q} \neq 0$, these are $\boldsymbol{q}\cdot\boldsymbol{B}_e(\boldsymbol{q}) = 0$ and $\boldsymbol{q}\times[\boldsymbol{B}_e(\boldsymbol{q}) - \boldsymbol{M}_e(\boldsymbol{q})] = 0$. The second makes $\boldsymbol{B}_e - \boldsymbol{M}_e$ parallel to $\boldsymbol{q}$, and the first then fixes it:

$$B_{e,a}(\boldsymbol{q}) = \sum_b (P_T)_{ab}\, M_{e,b}(\boldsymbol{q}), \qquad (P_T)_{ab} = \delta_{ab} - \hat{q}_a\hat{q}_b. \tag{C.163}$$

Only the component of the magnetization perpendicular to $\boldsymbol{q}$ produces a field at wavevector $\boldsymbol{q}$. A magnetization that is modulated along its own direction, $\boldsymbol{M}_e \parallel \boldsymbol{q}$, is a pattern of magnetic poles whose demagnetizing field cancels it exactly, and a neutron passing through sees nothing.

To first order in the interaction (the Born approximation, Sect. 2.2), the amplitude for scattering a neutron is $-(m_n/2\pi)$ times the Fourier transform of its potential, here the operator $-\boldsymbol{\mu}_n\cdot\boldsymbol{B}_e(\boldsymbol{q}) = 2\gamma_n\mu_N\mu_B\,\boldsymbol{\sigma}_n\cdot P_T\boldsymbol{s}_e(\boldsymbol{q})$, with P_T acting on the vector index. With $m_n = m_p$, which holds to 0.14 %, the constants combine as $(m_n/2\pi)\cdot 2\gamma_n(e/2m_p)(e/2m_e) = \gamma_n\, e^2/(4\pi m_e)$. This is $\gamma_n r_{\mathrm{cl}}$, with the classical electron radius $r_{\mathrm{cl}} = e^2/(4\pi m_e) = \alpha/m_e = 2.818\,\mathrm{fm}$. The magnetic amplitude operator is therefore

$$\hat{f}_{\mathrm{mag}} = -\gamma_n r_{\mathrm{cl}}\, \boldsymbol{\sigma}_n\cdot\boldsymbol{s}_\perp(\boldsymbol{q}), \qquad \boldsymbol{s}_\perp(\boldsymbol{q}) = P_T\, \boldsymbol{s}_e(\boldsymbol{q}). \tag{C.164}$$

The double-differential cross-section is the flux factor p'_n/p_n times $|\hat{f}_{\mathrm{mag}}|^2$, summed over target final states f with the energy delta function and averaged over the thermal initial states i, with the weights p_i of Eq. (2.62). For an unpolarized beam the average over the neutron spin is $\frac{1}{2}\,\mathrm{Tr}(\sigma_{n,a}\sigma_{n,b}) = \delta_{ab}$. Then $\sum_a s^\dagger_{\perp,a} s_{\perp,a} = \sum_{ab} s^\dagger_{e,a}(P_T)_{ab}s_{e,b}$, since P_T is real, symmetric and equal to its square. Comparing with the definition of S_{ab} in Sect. 4.2, with the operators $O_a = s_{e,a}(\boldsymbol{q})$, we obtain

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}\Omega\,\mathrm{d}E'} = (\gamma_n r_{\mathrm{cl}})^2\,\frac{p'_n}{p_n}\,\frac{\mathcal{V}}{2\pi}\sum_{a,b}(P_T)_{ab}\,S_{ab}(\boldsymbol{q},\omega), \tag{C.165}$$

for the whole sample of volume $\mathcal{V}$; the factor $\mathcal{V}/2\pi$ undoes the normalization of S_{ab}, which is intensive. The scale is $(\gamma_n r_{\mathrm{cl}})^2 = 0.2906\,\mathrm{b}$, to be compared with squared nuclear scattering lengths of 0.1 to 1 b, which is the sense in which the two kinds of scattering are equally strong. This is the standard result of magnetic neutron scattering [34, Chap. 7], [35]. The sum $\sum_{ab}(P_T)_{ab}S_{ab}$ is the combination that Berlin et al. extract from neutron data [36, Eq. (26)]. Their normalization is per formula unit, their spin operator is $\boldsymbol{\sigma}$ in place of $\boldsymbol{\sigma}/2$, and their susceptibility $\chi''_{ab} = 2(1-\mathrm{e}^{-\beta\omega})S_{ab}$ carries $\pi/\mathcal{V}$ where S_{ab} carries $2\pi/\mathcal{V}$ [36, Eq. (B4)]. Since the spin density sits on the ions, which vibrate, the sum over ions carries the Debye–Waller factor $\mathrm{e}^{-2W_\kappa(\boldsymbol{q})}$ of Sect. 5.6, which we absorb into S_{ab}.

C.19.2 Ions and the dipole form factor

In the magnetic insulators of Sect. 14.1 the unpaired electrons sit in an open shell of each magnetic ion j at $\boldsymbol{R}_j$. If the shell has a spin $\boldsymbol{S}_j$ and no orbital moment, as in a 3d ion whose orbital moment is quenched by the crystal field, its spin density is $\boldsymbol{S}_j$ times a normalized density $\varrho_{\rm s}(\boldsymbol{r}-\boldsymbol{R}_j)$ of the unpaired electrons. Then

$$\boldsymbol{s}_e(\boldsymbol{q}) = F^{\rm mag}(q)\sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_j}\,\boldsymbol{S}_j, \qquad F^{\rm mag}(q) = \int \mathrm{d}^3 r\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,\varrho_{\rm s}(\boldsymbol{r}) = \langle j_0(q)\rangle, \tag{C.166}$$

with $F^{\rm mag}(0)=1$. For a spherical shell with radial function $R_{\rm sh}(r)$, the plane-wave expansion Eq. (C.23) keeps only $l=0$, and $\langle j_l(q)\rangle = \int_0^\infty \mathrm{d}r\, r^2 R_{\rm sh}(r)^2 j_l(qr)$ are the radial integrals that tables give for each ion. The form factor falls off when q exceeds the inverse radius of the shell, at a few Å^{-1} for a 3d shell, so magnetic scattering is strongest in the first few Brillouin zones.

If the shell has an orbital moment, as in most rare-earth ions, the orbital currents add to $\boldsymbol{M}_e$. For a J multiplet in the dipole approximation, $qr \lesssim 1$ across the shell, the spin part of the magnetization carries $\langle j_0\rangle$ and the orbital part $\langle j_0\rangle + \langle j_2\rangle$ [34, Chap. 7]. Within the multiplet $\boldsymbol{S} = (g_J-1)\boldsymbol{J}$ and $\boldsymbol{L} = (2-g_J)\boldsymbol{J}$, with g_J the Landé factor of Sect. 14.3 and $\boldsymbol{J}_j$ the total angular momentum of ion j. The moment $2\boldsymbol{S}\langle j_0\rangle + \boldsymbol{L}(\langle j_0\rangle + \langle j_2\rangle)$ then equals $g_J[\langle j_0\rangle + (2/g_J - 1)\langle j_2\rangle]\boldsymbol{J}$, and the neutron couples to

$$\boldsymbol{s}_e(\boldsymbol{q}) \to \frac{g_J}{2}\,F^{\rm mag}(q)\sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_j}\,\boldsymbol{J}_j, \qquad F^{\rm mag}(q) = \langle j_0(q)\rangle + \Big(\frac{2}{g_J}-1\Big)\langle j_2(q)\rangle, \tag{C.167}$$

the dipole form factor. For $g_J = 2$ it reduces to Eq. (C.166). The neutron sees the moment $(g_J/2)\boldsymbol{J} = \boldsymbol{S} + \boldsymbol{L}/2$, while a probe that couples to spin alone sees $\boldsymbol{S} = (g_J-1)\boldsymbol{J}$. The ratio of the two is the conversion factor that Sect. 14.4 applies to neutron data.

C.19.3 Polarization analysis

A polarized beam resolves $\boldsymbol{s}_\perp$ into components. We add the nuclear amplitude, an operator $\hat{f}_{\rm nuc}$ on the target that does not act on the neutron spin, so that the full amplitude is $\hat{f}_{\rm nuc} - \gamma_n r_{\rm cl}\,\boldsymbol{\sigma}_n\cdot\boldsymbol{s}_\perp$. We take the neutron spin quantized along the polarization direction $\hat{\boldsymbol{P}}$, with the two other axes labeled 1 and 2. The Pauli matrices then give the four spin matrix elements

$$\langle \pm|\hat{f}|\pm\rangle = \hat{f}_{\rm nuc} \mp \gamma_n r_{\rm cl}\, s_{\perp,\hat{P}}, \qquad \langle \mp|\hat{f}|\pm\rangle = -\gamma_n r_{\rm cl}\,(s_{\perp,1} \pm \mathrm{i}\, s_{\perp,2}). \tag{C.168}$$

Scattering without spin flip sees the nuclear amplitude and the component of $\boldsymbol{s}_\perp$ along $\hat{\boldsymbol{P}}$. Scattering with spin flip sees only the magnetic components perpendicular to $\hat{\boldsymbol{P}}$, and the difference between the two spin-flip channels measures the chiral combination $\mathrm{i}(s^\dagger_{\perp,1}s_{\perp,2} - s^\dagger_{\perp,2}s_{\perp,1})$ [37, 38].

The standard frame has x along $\hat{\boldsymbol{q}}$, y perpendicular to it in the scattering plane and z vertical. Since $\boldsymbol{s}_\perp$ has no x component, the spin-flip cross-sections, averaged over the two incident spin states, are proportional to $S_{yy}+S_{zz}$ for $\hat{\boldsymbol{P}} \parallel x$, to S_{zz} for $\hat{\boldsymbol{P}} \parallel y$, and to S_{yy} for $\hat{\boldsymbol{P}} \parallel z$. The first channel contains all the magnetic scattering and no coherent nuclear scattering, and the other two separate the two transverse components [38]. The longitudinal element $S_{xx} = \sum_{ab}\hat{q}_a\hat{q}_b S_{ab}$ appears in none of the channels, polarized or not, because every amplitude in Eq. (C.168) contains $\boldsymbol{s}_e$ only through $\boldsymbol{s}_\perp$. A dark-matter response that needs S_{xx}, such as the contact spin coupling or the absorption of an axion-like particle, therefore cannot be taken from neutron data alone; Sect. 14.4 bounds it by positivity and fixes it under spin isotropy.

C.20 Dissipative two-level systems and kinks

A hydrogen atom trapped in a metal, or a group of atoms in a glass, can sit in either of two nearby minima and tunnel between them. In a metal the tunneling is dressed by the conduction electrons, which must rearrange every time the atom moves, and the result is a hopping rate that, for weak coupling ($\alpha_{\rm ohm} < \frac{1}{2}$ in Eq. (C.175)), grows as the temperature falls. A second kind of mobile defect is an extra atom squeezed into a close-packed row of a crystal: it spreads over many neighbors and glides along the row over a barrier of micro-electronvolts. Sect. 16.8 uses the first, through the spin–boson model, and Sect. 19.6 the second, through the Frenkel–Kontorova chain. Below we derive the hopping rate of a tunneling defect in a metal, and the shape, mass and migration barrier of the extra atom.

C.20.1 The spin–boson model

Let $|\mathrm{L}\rangle$ and $|\mathrm{R}\rangle$ be the two localized configurations, with Pauli matrices $\sigma_z = |\mathrm{L}\rangle\langle\mathrm{L}| - |\mathrm{R}\rangle\langle\mathrm{R}|$ and $\sigma_x = |\mathrm{L}\rangle\langle\mathrm{R}| + |\mathrm{R}\rangle\langle\mathrm{L}|$. Their energies differ by the asymmetry $\Delta_{\rm as}$, and the overlap of the two localized wavefunctions through the barrier gives the tunneling amplitude Δ_0. The environment is a set of harmonic modes d_k of energies ω_k, each displaced in opposite directions by the two configurations:

$$H = \tfrac{1}{2}\Delta_{\rm as}\,\sigma_z - \tfrac{1}{2}\Delta_0\,\sigma_x + \sum_k \omega_k\, d_k^\dagger d_k + \tfrac{1}{2}\,\sigma_z \sum_k \lambda_k (d_k + d_k^\dagger). \tag{C.169}$$

Without the modes, the eigenstates are split by $\Delta_{\rm tl} = (\Delta_{\rm as}^2 + \Delta_0^2)^{1/2}$, the two-level splitting of Sect. 2.5. A bath of oscillators is less special than it looks: any environment whose response to the defect is linear has Gaussian fluctuations, and these are reproduced by a suitable set of oscillators [39]. Everything the bath does then enters through its spectral density

$$J_{\rm s}(\omega) = \pi \sum_k \lambda_k^2\, \delta(\omega - \omega_k). \tag{C.170}$$

Its power law at low ω fixes the long-time decay of the bath correlation, Eq. (C.172) below. For acoustic phonons coupled through the strain, the coupling of a mode grows as $\lambda_k^2 \propto \omega_k$, since the strain amplitude of one phonon is $\propto Q(\rho_T \mathcal{V} \omega_{\boldsymbol{Q}\nu})^{-1/2}$, and the density of modes grows as ω^2; then $J_{\rm s} \propto \omega^3$ (superohmic). For conduction electrons, the bath excitations are electron–hole pairs. If the two configurations scatter electrons differently near the Fermi level, the matrix elements are roughly constant. The number of pairs of energy ω grows as ω, and $J_{\rm s}$ is linear (ohmic),

$$J_{\rm s}(\omega) = 2\pi\,\alpha_{\rm ohm}\,\omega\,\mathrm{e}^{-\omega/\omega_{\rm cut}}, \tag{C.171}$$

with a dimensionless coupling $\alpha_{\rm ohm}$ and a cutoff $\omega_{\rm cut}$ of the order of the Fermi energy. Sect. 16.8 computes $\alpha_{\rm ohm}$ from the scattering phase shifts. Ref. [39] writes α for $\alpha_{\rm ohm}$ and Ref. [40] writes K; the normalization of Eq. (C.171) is theirs.

C.20.2 The bath correlation function

In configuration L the term $+\frac{1}{2}\lambda_k(d_k + d_k^\dagger)$ shifts the equilibrium of mode k, and in R the shift is reversed. Completing the square, $H_{\rm L} = \sum_k \omega_k (d_k^\dagger + \lambda_k/2\omega_k)(d_k + \lambda_k/2\omega_k) + \text{const}$, and $H_{\rm R}$ is the same with $\lambda_k \to -\lambda_k$. The two equilibria differ by $\delta_k = \lambda_k/\omega_k$ in the ladder operators. With $b_k = d_k + \lambda_k/2\omega_k$ and the displacement operator $\hat{D}_k = \hat{D}(\delta_k)$ of Eq. (C.50) in b_k, which shifts b_k by δ_k, the two Hamiltonians are related by $H_{\rm R} = \hat{D} H_{\rm L} \hat{D}^\dagger$, with $\hat{D} = \prod_k \hat{D}_k$, up to constants that cancel.

The quantity a transition needs is the correlation $C(t) = \langle \mathrm{e}^{\mathrm{i}H_\mathrm{L}t}\mathrm{e}^{-\mathrm{i}H_\mathrm{R}t}\rangle_\mathrm{L}$, the overlap at time t between the bath evolving with the defect in L and in R, averaged over the equilibrium of L. Since $\mathrm{e}^{-\mathrm{i}H_\mathrm{R}t} = \hat{D}\mathrm{e}^{-\mathrm{i}H_\mathrm{L}t}\hat{D}^\dagger$, $C(t) = \langle \hat{D}(t)\hat{D}^\dagger\rangle_\mathrm{L}$, where $\hat{D}(t) = \mathrm{e}^{\mathrm{i}H_\mathrm{L}t}\hat{D}\mathrm{e}^{-\mathrm{i}H_\mathrm{L}t}$ has $b_k \to b_k\mathrm{e}^{-\mathrm{i}\omega_k t}$. For one mode, write $\hat{D}(t) = \mathrm{e}^{\hat{X}}$ and $\hat{D}^\dagger = \mathrm{e}^{\hat{Y}}$ with $\hat{X} = \delta(b^\dagger\mathrm{e}^{\mathrm{i}\omega t} - b\,\mathrm{e}^{-\mathrm{i}\omega t})$ and $\hat{Y} = -\delta(b^\dagger - b)$. Their commutator is the number $[\hat{X},\hat{Y}] = -2\mathrm{i}\delta^2\sin\omega t$, so Eq. (C.49) gives $\mathrm{e}^{\hat{X}}\mathrm{e}^{\hat{Y}} = \mathrm{e}^{\hat{X}+\hat{Y}}\mathrm{e}^{-\mathrm{i}\delta^2\sin\omega t}$. The operator $\hat{X}+\hat{Y} = \delta(b^\dagger u - b\,u^*)$, with $u = \mathrm{e}^{\mathrm{i}\omega t} - 1$, is linear, and Bloch's identity Eq. (C.53) gives $\langle \mathrm{e}^{\hat{X}+\hat{Y}}\rangle = \exp[-\tfrac{1}{2}\delta^2|u|^2(2n_\mathrm{B}(\omega)+1)]$, with $|u|^2 = 2(1-\cos\omega t)$ and $2n_\mathrm{B}+1 = \coth(\omega/2T)$. Multiplying over modes and using Eq. (C.170),

$$C(t) = \mathrm{e}^{-\mathcal{Q}(t)}, \qquad \mathcal{Q}(t) = \frac{1}{\pi}\int_0^\infty \mathrm{d}\omega\,\frac{J_\mathrm{s}(\omega)}{\omega^2}\Big[\coth\frac{\omega}{2T}\,(1-\cos\omega t) + \mathrm{i}\sin\omega t\Big]. \tag{C.172}$$

At $T = 0$ and for one mode, $C(t) = \exp[-\delta^2(1-\mathrm{e}^{-\mathrm{i}\omega t})] = \mathrm{e}^{-\delta^2}\sum_n(\delta^2)^n\mathrm{e}^{-\mathrm{i}n\omega t}/n!$: the Poisson distribution of the number of bath quanta emitted when the defect jumps, as in the Franck–Condon problem of Sect. 5.6.

For the ohmic bath at $T \ll \omega_\mathrm{cut}$ and $|t| \gg 1/\omega_\mathrm{cut}$ the integral has a closed form. The imaginary part is $2\alpha_\mathrm{ohm}\int_0^\infty \mathrm{d}\omega\,\mathrm{e}^{-\omega/\omega_\mathrm{cut}}\sin(\omega t)/\omega = 2\alpha_\mathrm{ohm}\arctan(\omega_\mathrm{cut}t) \to \pi\alpha_\mathrm{ohm}\operatorname{sgn} t$. For the real part we write $\coth(\omega/2T) = 1 + 2n_\mathrm{B}(\omega)$ and use one integral twice,

$$\int_0^\infty \frac{\mathrm{d}\omega}{\omega}\,\mathrm{e}^{-\omega/\Omega}\,(1-\cos\omega t) = \tfrac{1}{2}\ln(1+\Omega^2t^2),$$

which follows by differentiating with respect to t, since $\int_0^\infty \mathrm{d}\omega\,\mathrm{e}^{-\omega/\Omega}\sin\omega t = \Omega^2 t/(1+\Omega^2t^2)$. With $\Omega = \omega_\mathrm{cut}$ it gives the zero-temperature part, $\alpha_\mathrm{ohm}\ln(1+\omega_\mathrm{cut}^2t^2) \simeq 2\alpha_\mathrm{ohm}\ln(\omega_\mathrm{cut}|t|)$. The thermal part, where the cutoff can be dropped, has $n_\mathrm{B}(\omega) = \sum_{m\geq1}\mathrm{e}^{-m\omega/T}$, and each term is the same integral with $\Omega = T/m$. Summing, $4\alpha_\mathrm{ohm}\cdot\tfrac{1}{2}\sum_m\ln(1+T^2t^2/m^2) = 2\alpha_\mathrm{ohm}\ln[\sinh(\pi T|t|)/(\pi T|t|)]$, by the product $\sinh x/x = \prod_{m\geq1}(1+x^2/\pi^2m^2)$. Together,

$$\mathcal{Q}(t) = 2\alpha_\mathrm{ohm}\ln\Big[\frac{\omega_\mathrm{cut}}{\pi T}\sinh(\pi T|t|)\Big] + \mathrm{i}\pi\alpha_\mathrm{ohm}\operatorname{sgn} t. \tag{C.173}$$

The real part grows without bound, linearly in $|t|$ at times beyond $1/T$, so the overlap decays and the bath forgets which configuration it was prepared in. At $T = 0$ it decays only as a power, $|C| = (\omega_\mathrm{cut}|t|)^{-2\alpha_\mathrm{ohm}}$. This slow decay is the orthogonality catastrophe of a Fermi sea [39]: the static overlap $\exp[-\tfrac{1}{2}\sum_k\delta_k^2]$ of the two bath ground states vanishes, since $\sum_k\delta_k^2 = \pi^{-1}\int\mathrm{d}\omega\,J_\mathrm{s}/\omega^2$ diverges logarithmically at small ω.

C.20.3 Incoherent tunneling rate

When the bath relaxes faster than the defect tunnels, the defect hops between L and R at a rate that is second order in Δ_0. The perturbation $O = -\tfrac{1}{2}\Delta_0|\mathrm{R}\rangle\langle\mathrm{L}| + \mathrm{h.c.}$ moves the defect from L to R, and the thermal golden rule in the correlation form Eq. (2.66), at zero energy given to the external probe, gives $\Gamma_\mathrm{LR} = \int\mathrm{d}t\,\langle O^\dagger(t)O(0)\rangle$. Its evolution under H without the tunneling term contributes $\mathrm{e}^{\mathrm{i}(E_\mathrm{L}-E_\mathrm{R})t}$, with $E_\mathrm{L} - E_\mathrm{R} = \Delta_\mathrm{as}$, times $C(t)$:

$$\Gamma_\mathrm{LR} = \frac{\Delta_0^2}{4}\int_{-\infty}^{\infty}\mathrm{d}t\,\mathrm{e}^{\mathrm{i}\Delta_\mathrm{as}t}\,\mathrm{e}^{-\mathcal{Q}(t)}. \tag{C.174}$$

Expanding $C(t)$ in exponentials, as in the one-mode example, shows the content: each term is an energy-conserving jump in which the defect gives Δ_as to the bath. Ref. [39] obtains the same rate from the path integral and calls the approximation the noninteracting-blip approximation.

For the ohmic bath, Eq. (C.173) makes the integrand of Eq. (C.174)

$$\mathrm{e}^{\mathrm{i}\Delta_{\mathrm{as}}t}\,\mathrm{e}^{-\mathcal{Q}(t)} = \mathrm{e}^{\mathrm{i}\Delta_{\mathrm{as}}t}\Big[\frac{\pi T}{\omega_{\mathrm{cut}}\,\mathrm{i}\sinh\big(\pi T(t-\mathrm{i}0)\big)}\Big]^{2\alpha_{\mathrm{ohm}}}.$$

For $t>0$ the factor $\mathrm{i}^{-2\alpha_{\mathrm{ohm}}}$ gives the phase $\mathrm{e}^{-\mathrm{i}\pi\alpha_{\mathrm{ohm}}}$, and for $t<0$, where $\sinh(\pi T(t-\mathrm{i}0)) = |\sinh|\,\mathrm{e}^{-\mathrm{i}\pi}$, the phase is $\mathrm{e}^{+\mathrm{i}\pi\alpha_{\mathrm{ohm}}}$. The integrand is analytic in the strip $-1/T<\operatorname{Im}t<0$, and we shift the contour to $t=x-\mathrm{i}/2T$. There $\mathrm{i}\sinh(\pi Tx-\mathrm{i}\pi/2)=\cosh(\pi Tx)$ and $\mathrm{e}^{\mathrm{i}\Delta_{\mathrm{as}}t}=\mathrm{e}^{\mathrm{i}\Delta_{\mathrm{as}}x}\mathrm{e}^{\Delta_{\mathrm{as}}/2T}$. The remaining Fourier transform follows from $y=(1+\mathrm{e}^{-2\xi})^{-1}$, which turns $\operatorname{sech}^{2\alpha}\xi\,\mathrm{e}^{\mathrm{i}\nu\xi}\,\mathrm{d}\xi$ into $2^{2\alpha-1}y^{\alpha+\mathrm{i}\nu/2-1}(1-y)^{\alpha-\mathrm{i}\nu/2-1}\mathrm{d}y$, a Beta-function integral:

$$\int_{-\infty}^{\infty}\mathrm{d}\xi\,\frac{\mathrm{e}^{\mathrm{i}\nu\xi}}{\cosh^{2\alpha}\xi}=2^{2\alpha-1}\,\frac{|\Gamma(\alpha+\mathrm{i}\nu/2)|^2}{\Gamma(2\alpha)},$$

with $\Gamma(z)$ Euler's gamma function. With $\xi=\pi Tx$ and $\nu=\Delta_{\mathrm{as}}/\pi T$,

Incoherent rate in an ohmic bath

$$\Gamma_{\mathrm{LR}}=\frac{\Delta_0^2}{4\omega_{\mathrm{cut}}}\Big(\frac{2\pi T}{\omega_{\mathrm{cut}}}\Big)^{2\alpha_{\mathrm{ohm}}-1}\mathrm{e}^{\Delta_{\mathrm{as}}/2T}\,\frac{|\Gamma(\alpha_{\mathrm{ohm}}+\mathrm{i}\Delta_{\mathrm{as}}/2\pi T)|^2}{\Gamma(2\alpha_{\mathrm{ohm}})},\qquad \Gamma_{\mathrm{RL}}=\mathrm{e}^{-\Delta_{\mathrm{as}}/T}\,\Gamma_{\mathrm{LR}}. \tag{C.175}$$

The reverse rate follows from $\Delta_{\mathrm{as}}\to-\Delta_{\mathrm{as}}$, and since $|\Gamma|^2$ is even in Δ_{as} the two satisfy detailed balance. For a symmetric defect, by the duplication formula of Γ, the rate is $\Gamma_{\mathrm{LR}}=(\Delta_0^2/4\omega_{\mathrm{cut}})\,\sqrt{\pi}\,\Gamma(\alpha_{\mathrm{ohm}})/\Gamma(\alpha_{\mathrm{ohm}}+\frac12)\,(\pi T/\omega_{\mathrm{cut}})^{2\alpha_{\mathrm{ohm}}-1}$. It scales as $T^{2\alpha_{\mathrm{ohm}}-1}$, and for $\alpha_{\mathrm{ohm}}<\frac12$ it grows as the temperature falls. Cooling shrinks the dressed amplitude and lengthens the dephasing time. As T falls, fewer slow modes are thermally excited, the tunneling amplitude dressed by the bath shrinks as $\Delta_0(\pi T/\omega_{\mathrm{cut}})^{\alpha_{\mathrm{ohm}}}$, and its square falls as $T^{2\alpha_{\mathrm{ohm}}}$. At the same time the bath dephases the two configurations more slowly, over the time $1/T$ of Eq. (C.173), and the rate is roughly the squared amplitude times that time. At weak coupling the longer time wins. For $\alpha_{\mathrm{ohm}}\ll1$ the relaxation rate $\Gamma_{\mathrm{LR}}+\Gamma_{\mathrm{RL}}$ of a symmetric defect reduces to $\Delta_0^2/(2\pi\alpha_{\mathrm{ohm}}T)$. This is, to leading order in α_{ohm}, the overdamped limit $\tilde\Delta_0^2/\gamma_0$ of the Lorentzian of [40, Eq. (10)], with the dephasing rate $\gamma_0=2\pi\alpha_{\mathrm{ohm}}T$ and the dressed amplitude $\tilde\Delta_0^2=\Gamma(1-2\alpha_{\mathrm{ohm}})\cos(\pi\alpha_{\mathrm{ohm}})\,\Delta_0^2(2\pi T/\omega_{\mathrm{cut}})^{2\alpha_{\mathrm{ohm}}}$ (Sect. 16.8).

The rate depends on Δ_0 and ω_{cut} only through the combination $\Delta_0^2\,\omega_{\mathrm{cut}}^{-2\alpha_{\mathrm{ohm}}}$. Defining, for $\alpha_{\mathrm{ohm}}<1$, the renormalized tunneling amplitude Δ_{r} [39] by

$$\Delta_{\mathrm{r}}^{2(1-\alpha_{\mathrm{ohm}})}=\Gamma(1-2\alpha_{\mathrm{ohm}})\cos(\pi\alpha_{\mathrm{ohm}})\;\Delta_0^2\,\omega_{\mathrm{cut}}^{-2\alpha_{\mathrm{ohm}}}, \tag{C.176}$$

the cutoff drops out of every low-energy quantity: for $\Delta_0,T\ll\omega_{\mathrm{cut}}$ at fixed α_{ohm}, the dynamics depends on Δ_{r}, α_{ohm}, Δ_{as} and T alone. This is the scaling limit. The form of Δ_{r} can be understood by an adiabatic argument. Bath modes much faster than the tunneling follow the defect and dress the tunneling amplitude with their overlap, $\exp[-\frac12\sum_{\omega_k>p\Delta_{\mathrm{r}}}\delta_k^2]=(p\Delta_{\mathrm{r}}/\omega_{\mathrm{cut}})^{\alpha_{\mathrm{ohm}}}$ for an ohmic bath, with p a number of order one. Requiring the dressed amplitude to be Δ_{r} itself gives $\Delta_{\mathrm{r}}=\Delta_0(p\Delta_{\mathrm{r}}/\omega_{\mathrm{cut}})^{\alpha_{\mathrm{ohm}}}$, and therefore $\Delta_{\mathrm{r}}\propto\Delta_0(\Delta_0/\omega_{\mathrm{cut}})^{\alpha_{\mathrm{ohm}}/(1-\alpha_{\mathrm{ohm}})}$, with the prefactor that Eq. (C.176) makes exact. Below $T\sim\Delta_{\mathrm{r}}$ the assumption of a fast bath fails, and for $\alpha_{\mathrm{ohm}}<\frac12$ a symmetric defect then oscillates coherently, with damping, at a frequency of order Δ_{r}. Eq. (C.175) therefore holds for $\Delta_{\mathrm{r}}\ll T\ll\omega_{\mathrm{cut}}$, and the growth of the rate on cooling stops near $T\sim\Delta_{\mathrm{r}}$. For $\alpha_{\mathrm{ohm}}\geq1$ the tunneling amplitude renormalizes to zero and the defect stays localized at $T=0$ [39].

C.20.4 The Frenkel–Kontorova chain

The second model is a row of atoms of mass M_κ, at positions $z_n = na_{\rm ch} + u_n$, joined by springs of constant $K_{\rm ch}$. The row rests in the periodic potential $V_{\rm s}\sin^2(\pi z/a_{\rm ch})$ of the surrounding crystal, whose period equals the chain spacing $a_{\rm ch}$ [41]:

$$H_{\rm FK} = \sum_n \Big[\frac{p_n^2}{2M_\kappa} + \frac{K_{\rm ch}}{2}\,(u_{n+1}-u_n)^2 + V_{\rm s}\sin^2\frac{\pi u_n}{a_{\rm ch}}\Big]. \tag{C.177}$$

An extra atom in the row, a crowdion, corresponds to u_n going from $a_{\rm ch}$ at $n\to-\infty$ to 0 at $n\to+\infty$: the atoms on the left are each pushed one period, so that the row holds one atom more than the potential has minima [42]. The static equations, $\partial H_{\rm FK}/\partial u_n = 0$, are

$$K_{\rm ch}\,(u_{n+1} - 2u_n + u_{n-1}) = \frac{\pi V_{\rm s}}{a_{\rm ch}}\sin\frac{2\pi u_n}{a_{\rm ch}}, \tag{C.178}$$

and the discrete profile is found by minimizing $H_{\rm FK}$ numerically with those boundary values.

When u_n varies slowly from atom to atom, $u_n \to u(z)$ with $u_{n+1} - 2u_n + u_{n-1} \to a_{\rm ch}^2\,\partial_z^2 u$, and the phase $\varphi = 2\pi u/a_{\rm ch}$ obeys the sine-Gordon equation

$$\partial_t^2\varphi - c_{\rm ch}^2\,\partial_z^2\varphi + \omega_{\rm s}^2\sin\varphi = 0, \qquad c_{\rm ch} = a_{\rm ch}\sqrt{K_{\rm ch}/M_\kappa}, \qquad \omega_{\rm s}^2 = \frac{2\pi^2 V_{\rm s}}{M_\kappa a_{\rm ch}^2}, \tag{C.179}$$

where $c_{\rm ch}$ is the sound speed of the free chain and $\omega_{\rm s}$ the frequency of an atom oscillating alone in the potential. A static solution has the first integral $\frac12 d_{\rm kink}^2(\partial_z\varphi)^2 = 1-\cos\varphi$, with the kink width $d_{\rm kink} = c_{\rm ch}/\omega_{\rm s} = a_{\rm ch}^2[K_{\rm ch}/(2\pi^2 V_{\rm s})]^{1/2}$. Since $1-\cos\varphi = 2\sin^2(\varphi/2)$, it becomes $d_{\rm kink}\,\partial_z\varphi = \mp 2\sin(\varphi/2)$, and integrating,

$$u(z) = \frac{2a_{\rm ch}}{\pi}\arctan \mathrm{e}^{-(z-z_0)/d_{\rm kink}}, \tag{C.180}$$

which falls from $a_{\rm ch}$ to 0 over a few $d_{\rm kink}$ around the center z_0, as in [42, Eq. (2)]. The first integral states that the spring energy and the substrate energy are equal at every point. Along the kink $\sin^2(\pi u/a_{\rm ch}) = \sin^2(2\arctan \mathrm{e}^{-s}) = \operatorname{sech}^2 s$, with $s = (z-z_0)/d_{\rm kink}$, and with $\sum_n \to \int \mathrm{d}z/a_{\rm ch}$ the rest energy is

$$E_{\rm kink} = \frac{2V_{\rm s}}{a_{\rm ch}}\int \mathrm{d}z\,\operatorname{sech}^2\frac{z-z_0}{d_{\rm kink}} = \frac{4V_{\rm s}d_{\rm kink}}{a_{\rm ch}} = \frac{2a_{\rm ch}}{\pi}\sqrt{2V_{\rm s}K_{\rm ch}}, \tag{C.181}$$

which is [42, Eq. (3)], with $\beta = K_{\rm ch}$ and $V_0 = V_{\rm s}$ there. The sine-Gordon equation is Lorentz invariant with $c_{\rm ch}$ in place of the speed of light. A kink moving at speed v is therefore the static one with $z - z_0 \to (z - z_0 - vt)/(1-v^2/c_{\rm ch}^2)^{1/2}$. Its energy is $E_{\rm kink}(1-v^2/c_{\rm ch}^2)^{-1/2} \simeq E_{\rm kink} + \frac12 M_{\rm kink}v^2$, with the kink mass

$$M_{\rm kink} = \frac{E_{\rm kink}}{c_{\rm ch}^2} = \frac{2}{\pi^2}\,\frac{a_{\rm ch}}{d_{\rm kink}}\,M_\kappa. \tag{C.182}$$

A wide kink is light: for $d_{\rm kink} = 3a_{\rm ch}$ it weighs 7 % of an atom, because moving it by one spacing moves each of many atoms by a small fraction of a spacing.

The continuum kink has the same energy wherever its center sits. The discrete chain does not, and the difference is the Peierls–Nabarro barrier that a kink must cross to move by one spacing. We keep the continuum profile and evaluate the energy on the atoms [42, Eq. (4)], using the equality of the two energies to write $E(z_0) \simeq 2V_{\rm s}\sum_n \operatorname{sech}^2[(na_{\rm ch} - z_0)/d_{\rm kink}]$. The Poisson summation formula of Eq. (C.12), here $\sum_n f(na_{\rm ch}) = a_{\rm ch}^{-1}\sum_m \tilde f(2\pi m/a_{\rm ch})$ with

$\tilde f(k) = \int \mathrm{d}z\, \mathrm{e}^{-\mathrm{i}kz} f(z)$, and the transform $\int \mathrm{d}z\, \mathrm{e}^{-\mathrm{i}kz} \operatorname{sech}^2(z/d_{\mathrm{kink}}) = \pi k d_{\mathrm{kink}}^2 / \sinh(\pi k d_{\mathrm{kink}}/2)$ give

$$E(z_0) = E_{\mathrm{kink}} + \frac{8\pi^2 V_{\mathrm{s}} d_{\mathrm{kink}}^2}{a_{\mathrm{ch}}^2} \sum_{m\geq 1} m \cos\frac{2\pi m z_0}{a_{\mathrm{ch}}} \operatorname{cosech}\frac{\pi^2 m d_{\mathrm{kink}}}{a_{\mathrm{ch}}}, \tag{C.183}$$

where the $m = 0$ term reproduces Eq. (C.181). The $m = 1$ term dominates, and the barrier, the difference between the largest and smallest $E(z_0)$, is

$$E_{\mathrm{PN}} \simeq \frac{16\pi^2 V_{\mathrm{s}} d_{\mathrm{kink}}^2}{a_{\mathrm{ch}}^2} \operatorname{cosech}\frac{\pi^2 d_{\mathrm{kink}}}{a_{\mathrm{ch}}} \simeq 32\pi^2 \frac{d_{\mathrm{kink}}^2}{a_{\mathrm{ch}}^2} V_{\mathrm{s}}\, \mathrm{e}^{-\pi^2 d_{\mathrm{kink}}/a_{\mathrm{ch}}}. \tag{C.184}$$

The exponential makes wide kinks nearly free: for $d_{\mathrm{kink}} = 2a_{\mathrm{ch}}$ the barrier is $3 \times 10^{-6} V_{\mathrm{s}}$, a few µeV for a substrate potential of an electronvolt. The series of [42, Eq. (5)] is smaller than Eq. (C.183) by a factor 4, and its barrier [42, Eq. (6)] smaller than Eq. (C.184) by a factor 2; a direct lattice sum of $E(z_0)$ confirms the forms given here. The minimum sits with the kink centered between two atoms, $z_0 = a_{\mathrm{ch}}/2$, and the maximum with an atom at its center. Eq. (C.184) is the leading term for wide kinks. Relaxing the discrete chain of Eq. (C.178) at both centers gives a barrier 22 % lower at $d_{\mathrm{kink}} = a_{\mathrm{ch}}$ and 17 % lower at $d_{\mathrm{kink}} = 1.23 a_{\mathrm{ch}}$, and the gap closes as the kink widens. Since the continuum profile and its mass rest on $d_{\mathrm{kink}} \gg a_{\mathrm{ch}}$ as well, the ratio $d_{\mathrm{kink}}/a_{\mathrm{ch}}$ is the parameter that controls the whole continuum description, and it is the one Sect. 19.6 uses to judge it.

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Appendix D

Long derivations

D.1 Standard-halo integrals

This section evaluates the averages of the standard halo model used in Sects. 1.2 and 1.3: the normalization $N_{\rm esc}$, the laboratory speed distribution $F_\chi(v)$, the mean inverse speed $\langle 1/v\rangle$, the mean speed $\langle v\rangle$ and the function $\eta(v_{\rm min})$ for every $v_{\rm min}$. The galactic-frame distribution is $f_{\rm gal}(\boldsymbol{v}_{\rm gal}) = N_{\rm esc}^{-1}\mathrm{e}^{-v_{\rm gal}^2/v_0^2}\Theta(v_{\rm esc}-v_{\rm gal})$, and the laboratory distribution is $f_\chi(\boldsymbol{v}) = f_{\rm gal}(\boldsymbol{v}+\boldsymbol{v}_{\rm E})$ (Eqs. (1.3) and (1.8)). Throughout, $v_{\rm E} = |\boldsymbol{v}_{\rm E}|$ at the time of interest, and we assume $0 < v_{\rm E} < v_{\rm esc}$; the values of Appendix A.8 give $v_{\rm E}/v_{\rm esc} \le 0.49$. Only two integrals are needed:

$$\int_{y_1}^{y_2} \mathrm{e}^{-y^2/v_0^2}\,\mathrm{d}y = \frac{\sqrt{\pi}\,v_0}{2}\left[\mathrm{erf}\left(\frac{y_2}{v_0}\right) - \mathrm{erf}\left(\frac{y_1}{v_0}\right)\right], \tag{D.1}$$

$$\int_{y_1}^{y_2} y\,\mathrm{e}^{-y^2/v_0^2}\,\mathrm{d}y = \frac{v_0^2}{2}\left[\mathrm{e}^{-y_1^2/v_0^2} - \mathrm{e}^{-y_2^2/v_0^2}\right]. \tag{D.2}$$

Eq. (D.1) is the definition of the error function (Appendix C.8); Eq. (D.2) is elementary, since $y\,\mathrm{e}^{-y^2/v_0^2}$ is the derivative of $-\frac{1}{2}v_0^2\mathrm{e}^{-y^2/v_0^2}$.

Normalization

The integrand of $\int \mathrm{d}^3v_{\rm gal}\,\mathrm{e}^{-v_{\rm gal}^2/v_0^2}\Theta(v_{\rm esc}-v_{\rm gal})$ depends only on $y = v_{\rm gal}$; in spherical coordinates the solid angle contributes 4π and $N_{\rm esc} = 4\pi\int_0^{v_{\rm esc}} y^2\mathrm{e}^{-y^2/v_0^2}\mathrm{d}y$. Integrating by parts with $y\,\mathrm{e}^{-y^2/v_0^2}\mathrm{d}y = \mathrm{d}(-\frac{1}{2}v_0^2\mathrm{e}^{-y^2/v_0^2})$,

$$\begin{aligned}\int_0^{v_{\rm esc}} y^2\mathrm{e}^{-y^2/v_0^2}\mathrm{d}y &= \left[-\tfrac{1}{2}v_0^2\,y\,\mathrm{e}^{-y^2/v_0^2}\right]_0^{v_{\rm esc}} + \tfrac{1}{2}v_0^2\int_0^{v_{\rm esc}}\mathrm{e}^{-y^2/v_0^2}\mathrm{d}y \\ &= -\tfrac{1}{2}v_0^2 v_{\rm esc}\mathrm{e}^{-v_{\rm esc}^2/v_0^2} + \frac{\sqrt{\pi}\,v_0^3}{4}\,\mathrm{erf}\left(\frac{v_{\rm esc}}{v_0}\right).\end{aligned}$$

Multiplying by 4π gives Eq. (1.4). As $v_{\rm esc}\to\infty$ it tends to $\pi^{3/2}v_0^3$, the full Gaussian integral.

Speed distribution

The derivation of Eq. (1.10) in Sect. 1.2 uses one fact that we check here: the cut removes directions only for $v > v_{\rm esc} - v_{\rm E}$. The galactic speed $|\boldsymbol{v}+\boldsymbol{v}_{\rm E}|$ ranges over $[\,|v-v_{\rm E}|, v+v_{\rm E}]$ as the direction of $\boldsymbol{v}$ varies. All directions survive if $v + v_{\rm E} < v_{\rm esc}$, none if $|v-v_{\rm E}| > v_{\rm esc}$, that is (since $v_{\rm E} < v_{\rm esc}$) if $v > v_{\rm esc} + v_{\rm E}$; in between, the surviving directions are $-1 \le \cos\vartheta < \cos\vartheta_{\rm cut}$ with $v^2 + v_{\rm E}^2 + 2vv_{\rm E}\cos\vartheta_{\rm cut} = v_{\rm esc}^2$. The angular integral is

$$\int_{-1}^{\cos\vartheta_{\rm cut}} \mathrm{e}^{-2vv_{\rm E}\cos\vartheta/v_0^2}\,\mathrm{d}(\cos\vartheta) = \frac{v_0^2}{2vv_{\rm E}}\left[\mathrm{e}^{2vv_{\rm E}/v_0^2} - \mathrm{e}^{-2vv_{\rm E}\cos\vartheta_{\rm cut}/v_0^2}\right].$$

Multiplying by $2\pi v^2 N_{\rm esc}^{-1} \mathrm{e}^{-(v^2+v_{\rm E}^2)/v_0^2}$ and completing the squares gives both branches of Eq. (1.10), with prefactor $\pi v_0^2 v/(v_{\rm E} N_{\rm esc})$. We write it as $F_\chi(v) = K_{\rm sp}\, v\,[\mathrm{e}^{-(v-v_{\rm E})^2/v_0^2} - e_{\rm cut}(v)]$, with $K_{\rm sp} = \pi v_0^2/(v_{\rm E} N_{\rm esc})$ and $e_{\rm cut}(v) = \mathrm{e}^{-(v+v_{\rm E})^2/v_0^2}$ below $v_{\rm esc} - v_{\rm E}$, $e_{\rm cut}(v) = \mathrm{e}^{-v_{\rm esc}^2/v_0^2}$ above it.

The function $\eta(v_{\rm min})$

Sect. 1.3 defines $\eta(v_{\rm min}) = \langle \Theta(v - v_{\rm min})/v \rangle = \int_{v_{\rm min}}^{\infty} \mathrm{d}v\, F_\chi(v)/v$. The factor v in F_χ cancels the $1/v$, so for $v_{\rm min} \geq 0$

$$\eta(v_{\rm min}) = K_{\rm sp} \left[\int_{v_{\rm min}}^{v_{\rm esc}+v_{\rm E}} \mathrm{e}^{-(v-v_{\rm E})^2/v_0^2} \mathrm{d}v - \int_{v_{\rm min}}^{v_{\rm esc}+v_{\rm E}} e_{\rm cut}(v)\, \mathrm{d}v \right],$$

where the upper limits are those of F_χ. There are three branches.

(i) $0 \leq v_{\rm min} < v_{\rm esc} - v_{\rm E}$. With $y = v - v_{\rm E}$ the first integral is over $v_{\rm min} - v_{\rm E} < y < v_{\rm esc}$. The second has two pieces: $\mathrm{e}^{-(v+v_{\rm E})^2/v_0^2}$ from $v_{\rm min}$ to $v_{\rm esc} - v_{\rm E}$, which with $y = v + v_{\rm E}$ is over $v_{\rm min} + v_{\rm E} < y < v_{\rm esc}$, and the constant $\mathrm{e}^{-v_{\rm esc}^2/v_0^2}$ over an interval of length $2v_{\rm E}$. The Gaussian pieces combine by Eq. (D.1), the $\mathrm{erf}(v_{\rm esc}/v_0)$ terms cancel, and

$$\eta(v_{\rm min}) = \frac{\pi v_0^2}{v_{\rm E} N_{\rm esc}} \left\{ \frac{\sqrt{\pi}\, v_0}{2} \left[\mathrm{erf}\left(\frac{v_{\rm min} + v_{\rm E}}{v_0} \right) - \mathrm{erf}\left(\frac{v_{\rm min} - v_{\rm E}}{v_0} \right) \right] - 2 v_{\rm E}\, \mathrm{e}^{-v_{\rm esc}^2/v_0^2} \right\}. \tag{D.3}$$

(ii) $v_{\rm esc} - v_{\rm E} \leq v_{\rm min} < v_{\rm esc} + v_{\rm E}$. Only the constant part of $e_{\rm cut}$ remains, over the length $v_{\rm esc} + v_{\rm E} - v_{\rm min}$, and

$$\eta(v_{\rm min}) = \frac{\pi v_0^2}{v_{\rm E} N_{\rm esc}} \left\{ \frac{\sqrt{\pi}\, v_0}{2} \left[\mathrm{erf}\left(\frac{v_{\rm esc}}{v_0} \right) - \mathrm{erf}\left(\frac{v_{\rm min} - v_{\rm E}}{v_0} \right) \right] - (v_{\rm esc} + v_{\rm E} - v_{\rm min})\, \mathrm{e}^{-v_{\rm esc}^2/v_0^2} \right\}. \tag{D.4}$$

(iii) $v_{\rm min} \geq v_{\rm esc} + v_{\rm E}$. No particle is fast enough: $\eta(v_{\rm min}) = 0$.

By Eq. (1.23), $v_{\rm min} \geq 0$, so η is never needed at a negative argument.

Checks. At $v_{\rm min} = v_{\rm esc} - v_{\rm E}$ both Eqs. (D.3) and (D.4) give $(\sqrt{\pi} v_0/2)[\mathrm{erf}(v_{\rm esc}/v_0) - \mathrm{erf}((v_{\rm esc} - 2v_{\rm E})/v_0)] - 2v_{\rm E}\mathrm{e}^{-v_{\rm esc}^2/v_0^2}$ inside the braces, so η is continuous. At $v_{\rm min} = v_{\rm esc} + v_{\rm E}$ Eq. (D.4) vanishes, which matches branch (iii). Differentiating either branch with $\mathrm{d}\,\mathrm{erf}(x)/\mathrm{d}x = (2/\sqrt{\pi})\mathrm{e}^{-x^2}$ gives $\mathrm{d}\eta/\mathrm{d}v_{\rm min} = -F_\chi(v_{\rm min})/v_{\rm min}$, as the definition requires. Without the cut ($v_{\rm esc} \to \infty$, $N_{\rm esc} \to \pi^{3/2} v_0^3$) branch (i) becomes $[\mathrm{erf}((v_{\rm min}+v_{\rm E})/v_0) - \mathrm{erf}((v_{\rm min}-v_{\rm E})/v_0)]/(2v_{\rm E})$, the mean inverse speed of a shifted Maxwellian above $v_{\rm min}$. The same branches, in different notation, are given in Ref. [1, Eqs. (B.4) and (B.5)].

Mean inverse speed

At $v_{\rm min} = 0$ the two error functions in Eq. (D.3) are $\mathrm{erf}(v_{\rm E}/v_0)$ and $\mathrm{erf}(-v_{\rm E}/v_0) = -\,\mathrm{erf}(v_{\rm E}/v_0)$, so the bracket is $2\,\mathrm{erf}(v_{\rm E}/v_0)$ and

$$\left\langle \frac{1}{v} \right\rangle = \eta(0) = \frac{\pi v_0^2}{v_{\rm E} N_{\rm esc}} \left[\sqrt{\pi}\, v_0\, \mathrm{erf}\left(\frac{v_{\rm E}}{v_0} \right) - 2 v_{\rm E}\, \mathrm{e}^{-v_{\rm esc}^2/v_0^2} \right],$$

which is Eq. (1.11).

Mean speed

The mean speed is $\langle v \rangle = \int_0^{v_{\rm esc}+v_{\rm E}} v\, F_\chi(v)\, \mathrm{d}v = K_{\rm sp}(I_1 - I_2 - I_3)$. The three integrals are

$$I_1 = \int_0^{v_{\rm esc}+v_{\rm E}} v^2 \mathrm{e}^{-(v-v_{\rm E})^2/v_0^2} \mathrm{d}v, \qquad I_2 = \int_0^{v_{\rm esc}-v_{\rm E}} v^2 \mathrm{e}^{-(v+v_{\rm E})^2/v_0^2} \mathrm{d}v,$$

$$I_3 = \mathrm{e}^{-v_{\rm esc}^2/v_0^2} \int_{v_{\rm esc}-v_{\rm E}}^{v_{\rm esc}+v_{\rm E}} v^2\, \mathrm{d}v.$$

We substitute $y = v - v_{\rm E}$ in I_1 and $y = v + v_{\rm E}$ in I_2:

$$I_1 - I_2 = \int_{-v_{\rm E}}^{v_{\rm esc}} (y + v_{\rm E})^2 {\rm e}^{-y^2/v_0^2} {\rm d}y - \int_{v_{\rm E}}^{v_{\rm esc}} (y - v_{\rm E})^2 {\rm e}^{-y^2/v_0^2} {\rm d}y.$$

On the common interval $v_{\rm E} < y < v_{\rm esc}$ the integrands differ by $(y + v_{\rm E})^2 - (y - v_{\rm E})^2 = 4v_{\rm E}y$; on $-v_{\rm E} < y < v_{\rm E}$ only the first integral contributes, and its term $2v_{\rm E}y$ is odd and drops out. Hence

$$I_1 - I_2 = 4v_{\rm E} \int_{v_{\rm E}}^{v_{\rm esc}} y\, {\rm e}^{-y^2/v_0^2} {\rm d}y + 2 \int_0^{v_{\rm E}} y^2 {\rm e}^{-y^2/v_0^2} {\rm d}y + 2v_{\rm E}^2 \int_0^{v_{\rm E}} {\rm e}^{-y^2/v_0^2} {\rm d}y.$$

The first integral is $\frac{1}{2}v_0^2({\rm e}^{-v_{\rm E}^2/v_0^2} - {\rm e}^{-v_{\rm esc}^2/v_0^2})$ by Eq. (D.2). The second, by the integration by parts used for $N_{\rm esc}$ with $v_{\rm esc}$ replaced by $v_{\rm E}$, is $(\sqrt{\pi}v_0^3/4)\,{\rm erf}(v_{\rm E}/v_0) - \frac{1}{2}v_0^2 v_{\rm E}{\rm e}^{-v_{\rm E}^2/v_0^2}$. The third is $(\sqrt{\pi}v_0/2)\,{\rm erf}(v_{\rm E}/v_0)$. Collecting,

$$I_1 - I_2 = v_{\rm E}v_0^2\, {\rm e}^{-v_{\rm E}^2/v_0^2} - 2v_{\rm E}v_0^2\, {\rm e}^{-v_{\rm esc}^2/v_0^2} + \sqrt{\pi}\, v_0 \left(\frac{v_0^2}{2} + v_{\rm E}^2\right) {\rm erf}\!\left(\frac{v_{\rm E}}{v_0}\right).$$

The last integral is elementary: $I_3 = {\rm e}^{-v_{\rm esc}^2/v_0^2}[(v_{\rm esc}+v_{\rm E})^3 - (v_{\rm esc}-v_{\rm E})^3]/3 = {\rm e}^{-v_{\rm esc}^2/v_0^2}\, 2v_{\rm E}(v_{\rm esc}^2 + v_{\rm E}^2/3)$. Combining the two terms proportional to ${\rm e}^{-v_{\rm esc}^2/v_0^2}$ into $-2v_{\rm E}{\rm e}^{-v_{\rm esc}^2/v_0^2}(v_0^2 + v_{\rm esc}^2 + v_{\rm E}^2/3)$ gives Eq. (1.19).

Check. Without the cut, $K_{\rm sp} = 1/(\sqrt{\pi}\, v_0 v_{\rm E})$ and

$$\langle v \rangle = \frac{v_0}{\sqrt{\pi}} {\rm e}^{-v_{\rm E}^2/v_0^2} + \left(v_{\rm E} + \frac{v_0^2}{2v_{\rm E}}\right) {\rm erf}\!\left(\frac{v_{\rm E}}{v_0}\right),$$

which tends to $2v_0/\sqrt{\pi}$, the mean speed of the Maxwellian, as $v_{\rm E} \to 0$ (expand ${\rm erf}\, x = (2/\sqrt{\pi})(x - x^3/3 + \dots)$), and to $v_{\rm E}$ for $v_{\rm E} \gg v_0$, as it must. For the parameters of Appendix A.8, a direct numerical quadrature of $\int {\rm d}^3v\, f_\chi$ with the weights 1, $1/v$ and v reproduces 1, Eq. (1.11) and Eq. (1.19) to better than 10^{-6} in relative terms.

D.2 One- and two-body operators

This section proves the second-quantized forms of one- and two-body operators used in Sect. 2.3, Eqs. (2.36) and (2.42), and the rule for changing the single-particle basis, Eq. (2.40). The treatment follows Refs. [2, Chap. 1] and [3, Chap. 1]. We work with fermions first and state the changes for bosons at the end.

States

Let $\{|\gamma\rangle\}$ be an orthonormal single-particle basis, spin included. The antisymmetrized N-particle state with orbitals $\gamma_1, \dots, \gamma_N$ is

$$|\gamma_1 \dots \gamma_N\rangle = \frac{1}{\sqrt{N!}} \sum_P (-1)^P\, |\gamma_{P1}\rangle \otimes |\gamma_{P2}\rangle \otimes \cdots \otimes |\gamma_{PN}\rangle, \tag{D.5}$$

where the kth factor of each product is the state of particle k; its wave function is the Slater determinant of Eq. (2.29). Exchanging two labels in the list changes the sign of the state, and two equal labels give zero. If all labels differ, the state has unit norm, since the $N!$ products are orthonormal. Sect. 2.3 identifies this state with $c_{\gamma_1}^\dagger \cdots c_{\gamma_N}^\dagger |0\rangle$, which defines $c_\gamma^\dagger$ through Eq. (2.30). A first-quantized operator that treats all particles alike commutes with every permutation of the particles, so it maps antisymmetric states to antisymmetric states, and it is enough to know its action on the states of Eq. (D.5).

Replacing one orbital

The proofs rest on one lemma. Let the labels $\gamma_1, \ldots, \gamma_N$ be distinct and let γ be any label. Then

$$c^\dagger_\gamma c_{\gamma_j} \, |\gamma_1 \ldots \gamma_j \ldots \gamma_N\rangle = |\gamma_1 \ldots \gamma \ldots \gamma_N\rangle, \tag{D.6}$$

where on the right γ stands in place j and the other labels are unchanged. To prove it, write the state as $c^\dagger_{\gamma_1} \cdots c^\dagger_{\gamma_N}|0\rangle$. By Eq. (2.32), c_{γ_j} anticommutes with each $c^\dagger_{\gamma_k}$, $k \neq j$, because the labels differ. Moving it to the right past the $j-1$ operators in front of $c^\dagger_{\gamma_j}$ gives a sign $(-1)^{j-1}$. There $c_{\gamma_j} c^\dagger_{\gamma_j} = 1 - c^\dagger_{\gamma_j} c_{\gamma_j}$, and the second term gives zero, because c_{γ_j} can then be moved through the remaining creation operators to the vacuum, which it annihilates. So c_{γ_j} deletes $c^\dagger_{\gamma_j}$ from the string at the cost $(-1)^{j-1}$. Next, $c^\dagger_\gamma$ stands at the front of a string of $N-1$ creation operators; moving it back to place j passes $j-1$ of them, each pass costing a sign by $\{c^\dagger_\gamma, c^\dagger_{\gamma_k}\} = 0$. The two signs cancel. If γ equals one of the other labels γ_k, $k \neq j$, both sides of Eq. (D.6) vanish: the left because $(c^\dagger_{\gamma_k})^2 = 0$, the right because a label is repeated. If $\gamma = \gamma_j$, both sides return the original state. A bilinear $c^\dagger_\gamma c_{\gamma'}$ with γ' not in the list gives zero, since $c_{\gamma'}$ moves through all creation operators and annihilates the vacuum.

One-body operators

Let $\hat{O}_1 = \sum_{i=1}^N \hat{o}_i$, with $\hat{o}_i$ the one-particle operator $\hat{o}$ acting on the ith factor of each tensor product. Acting on one product in Eq. (D.5), $\hat{o}_i$ changes only the factor $|\gamma_{Pi}\rangle$. After the sums over i and over permutations, each orbital γ_j of the list is replaced in turn by $\hat{o}|\gamma_j\rangle$:

$$\hat{O}_1|\gamma_1 \ldots \gamma_N\rangle = \sum_{j=1}^N |\gamma_1 \ldots (\hat{o}\gamma_j) \ldots \gamma_N\rangle = \sum_{j=1}^N \sum_\gamma \langle\gamma|\hat{o}|\gamma_j\rangle \, |\gamma_1 \ldots \gamma \ldots \gamma_N\rangle,$$

where $|\gamma_1 \ldots (\hat{o}\gamma_j) \ldots \gamma_N\rangle$ is Eq. (D.5) with $|\gamma_j\rangle$ replaced by $\hat{o}|\gamma_j\rangle$, and the second step expands $\hat{o}|\gamma_j\rangle$ in the basis, using that Eq. (D.5) is linear in each orbital. By Eq. (D.6) each term on the right is $c^\dagger_\gamma c_{\gamma_j}$ acting on the original state. The bilinear $c^\dagger_\gamma c_{\gamma'}$ gives zero for every γ' not in the list. So the sum over j extends to a sum over all labels γ':

$$\hat{O}_1|\gamma_1 \ldots \gamma_N\rangle = \sum_{\gamma\gamma'} \langle\gamma|\hat{o}|\gamma'\rangle \, c^\dagger_\gamma c_{\gamma'} \, |\gamma_1 \ldots \gamma_N\rangle.$$

The two operators agree on a basis of every N-particle space, so they are equal on Fock space. This is Eq. (2.36). The right-hand side does not refer to N: the same operator acts on states of any particle number.

Two-body operators

Let $\hat{O}_2 = \frac{1}{2}\sum_{i\neq j} \hat{u}_{ij}$, where $\hat{u}_{ij}$ is a two-particle operator acting on the factors i and j, symmetric under the exchange of the two particles. Expand it in products of single-particle operators,

$$\hat{u} = \sum_{\gamma_1\gamma_2\gamma_3\gamma_4} \langle\gamma_1\gamma_2|\hat{u}|\gamma_3\gamma_4\rangle \; (|\gamma_1\rangle\langle\gamma_3|) \otimes (|\gamma_2\rangle\langle\gamma_4|),$$

with $\langle\gamma_1\gamma_2|\hat{u}|\gamma_3\gamma_4\rangle = (\langle\gamma_1| \otimes \langle\gamma_2|)\, \hat{u}\, (|\gamma_3\rangle \otimes |\gamma_4\rangle)$ the matrix element between product states, the first factor belonging to the first particle. For a local potential $U_2(\boldsymbol{r} - \boldsymbol{r}')$ it is the integral given in the caption of Table 2.1. For fixed labels, write $\hat{x}_i = (|\gamma_1\rangle\langle\gamma_3|)_i$ and $\hat{y}_j = (|\gamma_2\rangle\langle\gamma_4|)_j$ for these single-particle operators acting on particles i and j. The sum over pairs with $i \neq j$ is the full double sum minus the diagonal,

$$\sum_{i\neq j} \hat{x}_i \hat{y}_j = \Big(\sum_i \hat{x}_i\Big)\Big(\sum_j \hat{y}_j\Big) - \sum_i (\hat{x}\hat{y})_i,$$

and each term on the right is a one-body operator or a product of two. By the one-body result, $\sum_i \hat{x}_i = c^\dagger_{\gamma_1} c_{\gamma_3}$ and $\sum_j \hat{y}_j = c^\dagger_{\gamma_2} c_{\gamma_4}$, while $\hat{x}\hat{y} = |\gamma_1\rangle\langle\gamma_3|\gamma_2\rangle\langle\gamma_4| = \delta_{\gamma_3\gamma_2}|\gamma_1\rangle\langle\gamma_4|$ gives $\sum_i (\hat{x}\hat{y})_i = \delta_{\gamma_2\gamma_3} c^\dagger_{\gamma_1} c_{\gamma_4}$. With Eq. (2.32),

$$c^\dagger_{\gamma_1} c_{\gamma_3} c^\dagger_{\gamma_2} c_{\gamma_4} - \delta_{\gamma_2\gamma_3} c^\dagger_{\gamma_1} c_{\gamma_4} = -c^\dagger_{\gamma_1} c^\dagger_{\gamma_2} c_{\gamma_3} c_{\gamma_4} = c^\dagger_{\gamma_1} c^\dagger_{\gamma_2} c_{\gamma_4} c_{\gamma_3}.$$

Multiplying by the matrix element, summing over the four labels and including the factor $\frac{1}{2}$ gives

$$\hat{O}_2 = \frac{1}{2} \sum_{\gamma_1\gamma_2\gamma_3\gamma_4} \langle\gamma_1\gamma_2|\hat{u}|\gamma_3\gamma_4\rangle \, c^\dagger_{\gamma_1} c^\dagger_{\gamma_2} c_{\gamma_4} c_{\gamma_3}, \tag{D.7}$$

which is Eq. (2.42). The proof uses no property of $\hat{u}$ other than its expansion, so it holds for spin-dependent interactions as well. The subtraction of the diagonal is what turns the product of two bilinears into a normal-ordered product; it removes the interaction of each particle with itself, as in the field-operator argument of Sect. 2.3.

Change of basis

Let $\{|\tilde{\gamma}\rangle\}$ be a second orthonormal basis, with the unitary matrix $\mathcal{U}_{\gamma\tilde{\gamma}} = \langle\gamma|\tilde{\gamma}\rangle$. The antisymmetrized state Eq. (D.5) is linear in each orbital, so substituting $|\tilde{\gamma}_k\rangle = \sum_\gamma \mathcal{U}_{\gamma\tilde{\gamma}_k}|\gamma\rangle$ for every k gives

$$|\tilde{\gamma}_1 \ldots \tilde{\gamma}_N\rangle = \sum_{\gamma'_1 \cdots \gamma'_N} \mathcal{U}_{\gamma'_1\tilde{\gamma}_1} \cdots \mathcal{U}_{\gamma'_N\tilde{\gamma}_N} \, |\gamma'_1 \ldots \gamma'_N\rangle.$$

Written with creation operators, this is $c^\dagger_{\tilde{\gamma}_1} \cdots c^\dagger_{\tilde{\gamma}_N}|0\rangle$ with $c^\dagger_{\tilde{\gamma}} = \sum_\gamma \mathcal{U}_{\gamma\tilde{\gamma}} c^\dagger_\gamma$, the rule of Eq. (2.40); its adjoint gives $c_{\tilde{\gamma}}$. Collect the terms in which the labels γ'_k are a permutation of a fixed ordered set $\{\gamma_1 < \cdots < \gamma_N\}$, and reorder each to the standard order at the sign of the permutation. This shows that the coefficient of $|\gamma_1 \ldots \gamma_N\rangle$ is the determinant of the $N \times N$ submatrix $\mathcal{U}_{\gamma_j\tilde{\gamma}_k}$. A single determinant in one basis is thus a sum of determinants in the other, with minors of $\mathcal{U}$ as coefficients; the two-orbital bond of Sect. 2.3 is the smallest example.

The anticommutators are preserved, $\{c_{\tilde{\gamma}}, c^\dagger_{\tilde{\gamma}'}\} = \sum_{\gamma\gamma'} \mathcal{U}^*_{\gamma\tilde{\gamma}} \mathcal{U}_{\gamma'\tilde{\gamma}'} \delta_{\gamma\gamma'} = (\mathcal{U}^\dagger\mathcal{U})_{\tilde{\gamma}\tilde{\gamma}'} = \delta_{\tilde{\gamma}\tilde{\gamma}'}$. The other two vanish, because the new operators are linear combinations of the old annihilation (or creation) operators alone, which anticommute among themselves. Inserting $c^\dagger_\gamma = \sum_{\tilde{\gamma}} \mathcal{U}^*_{\gamma\tilde{\gamma}} c^\dagger_{\tilde{\gamma}}$ and $c_{\gamma'} = \sum_{\tilde{\gamma}'} \mathcal{U}_{\gamma'\tilde{\gamma}'} c_{\tilde{\gamma}'}$ into Eq. (2.36),

$$\sum_{\gamma\gamma'} \langle\gamma|\hat{o}|\gamma'\rangle c^\dagger_\gamma c_{\gamma'} = \sum_{\tilde{\gamma}\tilde{\gamma}'} \Big[\sum_{\gamma\gamma'} \langle\tilde{\gamma}|\gamma\rangle\langle\gamma|\hat{o}|\gamma'\rangle\langle\gamma'|\tilde{\gamma}'\rangle \Big] c^\dagger_{\tilde{\gamma}} c_{\tilde{\gamma}'} = \sum_{\tilde{\gamma}\tilde{\gamma}'} \langle\tilde{\gamma}|\hat{o}|\tilde{\gamma}'\rangle \, c^\dagger_{\tilde{\gamma}} c_{\tilde{\gamma}'},$$

by completeness of the old basis, used twice. The same substitution in Eq. (D.7) transforms each of the four labels with its own factor of $\mathcal{U}$ or $\mathcal{U}^*$, and completeness turns the matrix element into $\langle\tilde{\gamma}_1\tilde{\gamma}_2|\hat{u}|\tilde{\gamma}_3\tilde{\gamma}_4\rangle$. Both forms are therefore the same in every orthonormal basis. For a continuous basis, such as position, the sums become integrals and the Kronecker deltas Dirac deltas; with $\langle \boldsymbol{r}s|\gamma\rangle = \psi_\gamma(\boldsymbol{r}, s)$ this gives the field operator of Eq. (2.37) and the field-operator form Eq. (2.41).

Bosons

For bosons, Eq. (D.5) has all signs positive and a repeated label is allowed; the state with occupation numbers n_γ then has norm $\prod_\gamma n_\gamma!$, which is why the normalized states carry the factor $(\prod_\gamma n_\gamma!)^{-1/2}$ and $d^\dagger_\gamma$ acts with $\sqrt{n_\gamma + 1}$ (Eq. (2.34)). The lemma Eq. (D.6) holds for the unnormalized states $d^\dagger_{\gamma_1} \cdots d^\dagger_{\gamma_N}|0\rangle$ without any sign, with γ_j replaced in one of its occurrences and a sum over the occurrences when the label is repeated, which matches the

sum over particles in $\hat{O}_1$. The one-body proof then goes through unchanged with $c \to d$. In the two-body proof the commutator $[d_{\gamma_3}, d^\dagger_{\gamma_2}] = \delta_{\gamma_2\gamma_3}$ replaces the anticommutator, and $d^\dagger_{\gamma_1} d_{\gamma_3} d^\dagger_{\gamma_2} d_{\gamma_4} - \delta_{\gamma_2\gamma_3} d^\dagger_{\gamma_1} d_{\gamma_4} = d^\dagger_{\gamma_1} d^\dagger_{\gamma_2} d_{\gamma_3} d_{\gamma_4}$. Annihilation operators commute, so the order of the last two is immaterial and Eq. (D.7) holds with $c \to d$. The change of basis preserves the commutators by the same unitarity argument, and the coefficients of the new states are permanents instead of determinants.

D.3 Wigner–Weisskopf decay

This section derives the exponential decay of a discrete state coupled to a continuum, Eq. (2.55), the Lorentzian distribution of the final states, Eq. (2.56), and the replacement of an infinitesimal by a width in a second-order denominator, Eq. (2.57). The original treatment is that of Weisskopf and Wigner [4]; we follow the resolvent method of Ref. [5, Chap. III], and use the notation of Sects. 2.1 and 2.4.

Exact amplitude equation

A discrete state $|i\rangle$ of energy E_i is coupled by a constant perturbation H' to a set of states $|f\rangle$ of energies E_f, which become a continuum with density $\mathcal{D}_f(E)$ as the volume grows. We assume $H'_{ii} = 0$ and $H'_{ff'} = 0$: the perturbation only connects i with the continuum. A nonzero H'_{ii} is a first-order shift and can be put into E_i; couplings within the continuum are left out of this model. With $\xi_f = E_f - E_i$, the amplitude equations Eq. (2.1) are

$$\mathrm{i}\,\dot{c}_i(t) = \sum_f H'_{if}\,\mathrm{e}^{-\mathrm{i}\xi_f t}\,c_f(t), \qquad \mathrm{i}\,\dot{c}_f(t) = H'_{fi}\,\mathrm{e}^{\mathrm{i}\xi_f t}\,c_i(t), \tag{D.8}$$

with $c_i(0) = 1$ and $c_f(0) = 0$. The second equation integrates to $c_f(t) = -\mathrm{i}H'_{fi}\int_0^t \mathrm{d}t'\,\mathrm{e}^{\mathrm{i}\xi_f t'}c_i(t')$. Inserting this into the first gives a closed equation for c_i,

$$\begin{aligned} \dot{c}_i(t) &= -\int_0^t \mathrm{d}t'\,K(t-t')\,c_i(t'), \\ K(t-t') &= \sum_f |H'_{fi}|^2\,\mathrm{e}^{-\mathrm{i}\xi_f(t-t')} = \int \mathrm{d}E\,\mathcal{D}_f(E)\,\overline{|H'_{fi}|^2}(E)\,\mathrm{e}^{-\mathrm{i}(E-E_i)(t-t')}. \end{aligned} \tag{D.9}$$

No approximation has been made. The kernel K is the Fourier transform of the product $\mathcal{D}_f\overline{|H'_{fi}|^2}$, which varies on the energy scale E_w of Sect. 2.1. It therefore decays on the time scale $1/E_\mathrm{w}$: the continuum has a memory of that length. For the flat continuum of Worked example 2.1, with $\mathcal{D}_f\overline{|H'_{fi}|^2} = \Gamma_i/2\pi$ on $|E - E_i| < E_\mathrm{w}/2$ and zero outside, $K(t-t') = (\Gamma_i/\pi)\sin[E_\mathrm{w}(t-t')/2]/(t-t')$, which starts at $\Gamma_i E_\mathrm{w}/2\pi$ and oscillates away within a few $1/E_\mathrm{w}$.

The Markov approximation

At times short compared with $1/E_\mathrm{w}$, c_i has barely changed and $\dot{c}_i \approx -t\,K(0)$, so $|c_i|^2 \approx 1 - K(0)t^2$: the decay starts quadratically, which is the short-time behavior of Worked example 2.1. At later times, the integral in Eq. (D.9) only receives contributions from $t - t' \lesssim 1/E_\mathrm{w}$. Suppose that c_i changes little over that interval, as it does when its decay rate is small compared with E_w. We can then replace $c_i(t')$ by $c_i(t)$ and extend the lower limit of the integral to $-\infty$:

$$\dot{c}_i(t) = -c_i(t)\int_{-\infty}^t \mathrm{d}t'\,K(t-t').$$

The remaining integral is a sum of terms $\int_{-\infty}^{t} \mathrm{d}t'\, \mathrm{e}^{-\mathrm{i}\xi_f(t-t')}$, which converge with a factor $\mathrm{e}^{-0^+(t-t')}$ and give $-\mathrm{i}/(\xi_f - \mathrm{i}0^+) = \mathrm{i}/(E_i - E_f + \mathrm{i}0^+)$. With the Sokhotski–Plemelj formula Eq. (2.58),

$$\begin{aligned} \int_{-\infty}^{t} \mathrm{d}t'\, K(t-t') &= \mathrm{i} \sum_f \frac{|H'_{fi}|^2}{E_i - E_f + \mathrm{i}0^+} \\ &= \mathrm{i}\ \mathrm{PV} \sum_f \frac{|H'_{fi}|^2}{E_i - E_f} + \pi \sum_f |H'_{fi}|^2\, \delta(E_f - E_i) = \mathrm{i}\Delta E_i + \frac{\Gamma_i}{2}, \end{aligned} \tag{D.10}$$

where Γ_i is the golden-rule rate Eq. (2.6) and ΔE_i the level shift of Eq. (2.55). The equation $\dot{c}_i = -(\Gamma_i/2 + \mathrm{i}\Delta E_i)c_i$ integrates to Eq. (2.55). The approximation requires $\Gamma_i \ll E_\mathrm{w}$, the condition under which the golden-rule window of Eq. (2.7) exists, and holds for $t \gg 1/E_\mathrm{w}$. The memory of the continuum is then short compared with the lifetime, and the decay does not remember its own past: the probability to decay in the next instant is the same at every time, which is the content of an exponential law.

The resolvent and the corrections

The size of the corrections follows from an exact solution of Eq. (D.9) by Laplace transform. With $\tilde{c}_i(s) = \int_0^\infty \mathrm{d}t\, \mathrm{e}^{-st} c_i(t)$ for $\mathrm{Re}\, s > 0$, the convolution becomes a product, $s\tilde{c}_i - 1 = -\tilde{K}(s)\tilde{c}_i$, with $\tilde{K}(s) = \sum_f |H'_{fi}|^2/(s + \mathrm{i}\xi_f)$. With an energy variable $z = \mathrm{i}s$, measured from E_i, which lies in the upper half plane,

$$\tilde{c}_i = \frac{\mathrm{i}}{z - R(z)}, \qquad R(z) = \sum_f \frac{|H'_{fi}|^2}{z - \xi_f}, \tag{D.11}$$

and $c_i(t) = (2\pi)^{-1} \int \mathrm{d}z\, \mathrm{e}^{-\mathrm{i}zt}\, \mathrm{i}/(z - R(z))$ along a line just above the real axis [5, Chap. III]. The function $R(z)$ is analytic off the real axis; just above the axis, by Eq. (2.58), $R(x + \mathrm{i}0^+) = \mathrm{PV} \sum_f |H'_{fi}|^2/(x - \xi_f) - \mathrm{i}\pi \sum_f |H'_{fi}|^2 \delta(x - \xi_f)$, which at $x = 0$ is $\Delta E_i - \mathrm{i}\Gamma_i/2$. Continued below the axis through the continuum, $z - R(z)$ has a zero near $z_\mathrm{p} = \Delta E_i - \mathrm{i}\Gamma_i/2$, as long as R changes little over a range Γ_i around $z = 0$. Closing the contour in the lower half plane picks up this pole, which gives

$$c_i(t) = r_\mathrm{p}\, \mathrm{e}^{-\mathrm{i}z_\mathrm{p}t} + c_i^\mathrm{cut}(t), \qquad r_\mathrm{p} = \frac{1}{1 - R'(z_\mathrm{p})},$$

plus the contribution c_i^cut of the branch points at the edges of the continuum. Since R' is of order Γ_i/E_w, the residue r_p of the pole differs from one by a relative amount of that order, and so does the decay rate, which becomes $\Gamma_i/(1 - R')$ when R is expanded to first order about $z = 0$. For the flat continuum, $R(z) = (\Gamma_i/2\pi) \ln[(z + E_\mathrm{w}/2)/(z - E_\mathrm{w}/2)]$; just above $z = 0$ this is $-\mathrm{i}\Gamma_i/2$, so the shift vanishes, and $R'(0) = 2\Gamma_i/(\pi E_\mathrm{w})$. In Worked example 2.1, $\Gamma_i/E_\mathrm{w} = 0.01$ and $R'(0) = 0.0064$. The survival probability is then $r_\mathrm{p}^2 \mathrm{e}^{-\Gamma_i t/(1-R')}$, which at $t = 1/\Gamma_i$ exceeds e^{-1} by 0.0023, 0.4 % of the depletion: this is the deviation from $1 - \mathrm{e}^{-\Gamma_i t}$ that the exact solution of that model shows once the short-time transient has died out. The branch-point contribution falls off as a power of $1/(E_\mathrm{w} t)$ with a prefactor of order Γ_i/E_w. It overtakes the exponential only after many lifetimes, when the survival probability is already exponentially small, and it plays no role in any rate in this book.

The line shape

With Eq. (2.55), the amplitude of a final state is

$$c_f(t) = -\mathrm{i}H'_{fi}\int_0^t \mathrm{d}t'\,\mathrm{e}^{\mathrm{i}(\xi_f-\Delta E_i)t'-\Gamma_i t'/2} \xrightarrow{t\to\infty} \frac{H'_{fi}}{E_f-E_i-\Delta E_i+\mathrm{i}\Gamma_i/2},$$

whose square is Eq. (2.56). Summed over the continuum, with $\mathcal{D}_f\overline{|H'_{fi}|^2}$ constant over the width Γ_i, it gives $\mathcal{D}_f\overline{|H'_{fi}|^2}\cdot 2\pi/\Gamma_i = 1$, since $\Gamma_i = 2\pi\mathcal{D}_f\overline{|H'_{fi}|^2}$. The probability lost by i is found in the continuum, distributed over a Lorentzian of full width Γ_i at half maximum, which is the natural line shape of the emitted radiation when the continuum is that of the photon modes [4].

A decaying intermediate state

Finally, let a state $|i\rangle$ be coupled weakly, by a constant H_1, to a state $|m\rangle$ that decays into a continuum through a second coupling H_2, with total rate Γ_w and shift ΔE_m from Eq. (D.10) applied to m. Let H_1 act from the remote past, and keep it to first order, so that $c_i = 1$. The amplitude of m then obeys

$$\dot{c}_m(t) = -\mathrm{i}\,(H_1)_{mi}\,\mathrm{e}^{\mathrm{i}(E_m-E_i)t} - \Big(\frac{\Gamma_\mathrm{w}}{2}+\mathrm{i}\Delta E_m\Big)c_m(t),$$

in which the decay term is the Markov form of the coupling to the continuum, valid under the same conditions as before. Its solution with $c_m(-\infty)=0$ is

$$c_m(t) = -\mathrm{i}\,(H_1)_{mi}\int_{-\infty}^t \mathrm{d}t_2\,\mathrm{e}^{\mathrm{i}(E_m-E_i)t_2}\,\mathrm{e}^{-(\Gamma_\mathrm{w}/2+\mathrm{i}\Delta E_m)(t-t_2)} = \frac{(H_1)_{mi}\,\mathrm{e}^{\mathrm{i}(E_m-E_i)t}}{E_i-\tilde{E}_m+\mathrm{i}\Gamma_\mathrm{w}/2}, \qquad \text{(D.12)}$$

with $\tilde{E}_m = E_m+\Delta E_m$. The free propagation $\mathrm{e}^{-\mathrm{i}E_m(t-t_2)}$ of Eq. (2.51) has become a decaying one, and the integral converges without any switching factor. The amplitude of a final state f reached from m by H_2 is, at first order in H_2,

$$c_f(t) = -\mathrm{i}\int_{-\infty}^t \mathrm{d}t_1\,(H_2)_{fm}\,\mathrm{e}^{\mathrm{i}(E_f-E_m)t_1}\,c_m(t_1) = -\mathrm{i}\,\frac{(H_2)_{fm}(H_1)_{mi}}{E_i-\tilde{E}_m+\mathrm{i}\Gamma_\mathrm{w}/2}\int_{-\infty}^t \mathrm{d}t_1\,\mathrm{e}^{\mathrm{i}(E_f-E_i)t_1},$$

which has the form of the second-order amplitude of Sect. 2.4 with the replacement Eq. (2.57), and leads to Eq. (2.52) for the rate. The derivation needs a weak drive, $|(H_1)_{mi}| \ll |E_i-\tilde{E}_m+\mathrm{i}\Gamma_\mathrm{w}/2|$, so that the population $|c_m|^2$ stays small and i is not depleted; on resonance this reads $|(H_1)_{mi}| \ll \Gamma_\mathrm{w}/2$. It also evaluates Γ_w and ΔE_m at the energy of m. Strictly, the resolvent gives them at the energy E_i at which m is passed through. The two agree when the decay continuum of m varies little between E_m and E_i, which always holds near resonance, where the width matters.

D.4 Hartree–Fock equations and Koopmans' theorem

The best single Slater determinant for the electrons of silicon is found by minimizing the energy over the orbitals it is built from. The minimum condition is a set of one-electron equations, the Hartree–Fock equations, in which each electron moves in the electrostatic potential of all the others and feels in addition a nonlocal exchange term from the antisymmetry of the determinant [6]. Their eigenvalues have a precise meaning: with the orbitals held fixed, they are the energies to remove an electron from an occupied orbital or to add one to an empty orbital, which is Koopmans' theorem [7]. This section derives both results for Sect. 3.4 and evaluates them for the electron gas, where they take closed form and show what a single determinant leaves out. The derivation follows Ref. [8, Chap. 2].

The energy of a determinant

Take N_e electrons with clamped nuclei. The Hamiltonian is a one-body part, $h = -\nabla^2/2m_e + U_1(\boldsymbol{r})$, with U_1 the attraction of the nuclei, summed over the electrons, plus the Coulomb pair potential $U_2(\boldsymbol{r}-\boldsymbol{r}') = e^2/(4\pi|\boldsymbol{r}-\boldsymbol{r}'|)$ summed over pairs. Let $|\Phi_0\rangle$ be the determinant Eq. (2.29) of N_e orthonormal orbitals $\psi_\gamma(x)$, with $x = (\boldsymbol{r}, s)$ and $\int \mathrm{d}x = \sum_s \int \mathrm{d}^3r$. Its energy was worked out in Sect. 2.3: the one-body operator contributes $\sum_\gamma \langle\gamma|h|\gamma\rangle$, and the two-body operator contributes the direct term minus the exchange term, summed over pairs of occupied orbitals. In position space,

$$E[\Phi_0] = \sum_\gamma \int \mathrm{d}x\, \psi_\gamma^*(x)\, h\, \psi_\gamma(x) + \frac{1}{2}\sum_{\gamma\gamma'} \int \mathrm{d}x\, \mathrm{d}x'\, U_2(\boldsymbol{r}-\boldsymbol{r}') \Big[|\psi_\gamma(x)|^2 |\psi_{\gamma'}(x')|^2 - \psi_\gamma^*(x)\psi_{\gamma'}^*(x')\psi_{\gamma'}(x)\psi_\gamma(x') \Big], \quad \text{(D.13)}$$

where both sums run over the occupied orbitals. The terms with $\gamma = \gamma'$ cancel inside the bracket, so no electron interacts with itself.

The density matrix $\varrho(x,x') = \sum_\gamma \psi_\gamma(x)\psi_\gamma^*(x')$ has the electron density on its diagonal, $n(\boldsymbol{r}) = \sum_s \varrho(x,x)$, and the direct term is the classical electrostatic energy of that density. The exchange term is $-\frac{1}{2}\int \mathrm{d}x\, \mathrm{d}x'\, U_2\, |\varrho(x,x')|^2$, so ϱ carries all the dependence of the energy on the occupied orbitals. It is unchanged when the occupied orbitals are mixed among themselves by a unitary matrix $\mathcal{M}$, $\psi_\gamma \to \sum_{\gamma'} \psi_{\gamma'} \mathcal{M}_{\gamma'\gamma}$, because $\mathcal{M}\mathcal{M}^\dagger = 1$ leaves ϱ unchanged. The determinant itself changes only by the phase $\det \mathcal{M}$. The energy is therefore a function of the occupied subspace, not of the individual orbitals, and we will use this freedom below.

The variational equations

By the variational principle, $E[\Phi_0] \geq E_0$ for every determinant, where E_0 is the exact ground-state energy. The Hartree–Fock determinant is the one that makes $E[\Phi_0]$ stationary, and its lowest stationary point is the Hartree–Fock ground state. The orbitals must stay orthonormal, and we impose this with a Hermitian matrix of Lagrange multipliers $\lambda_{\gamma'\gamma}$, making stationary

$$E[\Phi_0] - \sum_{\gamma\gamma'} \lambda_{\gamma'\gamma} \Big(\int \mathrm{d}x\, \psi_\gamma^*(x)\psi_{\gamma'}(x) - \delta_{\gamma\gamma'} \Big).$$

A complex function and its conjugate can be varied independently. The derivative with respect to $\psi_\gamma^*(x)$ of the one-body term is $h\psi_\gamma(x)$. In the double sum, ψ_γ^* appears once in the slot γ and once in the slot γ', and since U_2 is symmetric the two contributions are equal and cancel the factor $\frac{1}{2}$. The direct term gives $\int \mathrm{d}x'\, U_2(\boldsymbol{r}-\boldsymbol{r}')\, n(\boldsymbol{r}')\, \psi_\gamma(x)$, and the exchange term gives $-\int \mathrm{d}x'\, U_2(\boldsymbol{r}-\boldsymbol{r}')\, \varrho(x,x')\, \psi_\gamma(x')$. Setting the total derivative to zero,

$$h_{\mathrm{HF}}\, \psi_\gamma = \sum_{\gamma'} \lambda_{\gamma'\gamma}\, \psi_{\gamma'}, \qquad h_{\mathrm{HF}}\, \psi(x) = \Big[-\frac{\nabla^2}{2m_e} + U_1(\boldsymbol{r}) + v_{\mathrm{H}}(\boldsymbol{r}) \Big] \psi(x) + \int \mathrm{d}x'\, \Sigma_{\mathrm{x}}(x,x')\, \psi(x'),$$

with the Hartree potential and the exchange (Fock) term

$$v_{\mathrm{H}}(\boldsymbol{r}) = \int \mathrm{d}^3r'\, \frac{e^2\, n(\boldsymbol{r}')}{4\pi|\boldsymbol{r}-\boldsymbol{r}'|}, \qquad \Sigma_{\mathrm{x}}(x,x') = -\frac{e^2\, \varrho(x,x')}{4\pi|\boldsymbol{r}-\boldsymbol{r}'|}.$$

The operator h_{HF} is Hermitian, since $\Sigma_{\mathrm{x}}(x',x)^* = \Sigma_{\mathrm{x}}(x,x')$. Projecting the equation onto $\psi_{\gamma''}$ gives $\lambda_{\gamma''\gamma} = \langle\gamma''|h_{\mathrm{HF}}|\gamma\rangle$, a Hermitian matrix. A unitary rotation of the occupied orbitals

leaves $h_{\rm HF}$ unchanged, because it depends on them only through $\boldsymbol{n}$ and ϱ, and we can choose the rotation that diagonalizes λ. In that basis the result takes its canonical form,

$$h_{\rm HF}\,\psi_\gamma(x) = E_\gamma\,\psi_\gamma(x), \qquad \boldsymbol{n}(\boldsymbol{r}) = \sum_s \sum_{\gamma\,\mathrm{occ}} |\psi_\gamma(\boldsymbol{r},s)|^2, \quad \varrho(x,x') = \sum_{\gamma\,\mathrm{occ}} \psi_\gamma(x)\psi^*_\gamma(x'). \tag{D.14}$$

The equations are nonlinear, since $h_{\rm HF}$ is built from its own occupied eigenfunctions, and they are solved by iteration until the orbitals that go in are the orbitals that come out. The occupied set is the N_e eigenfunctions of lowest E_γ; the other eigenfunctions of the same operator are the empty, or virtual, orbitals.

The Hartree potential treats the electrons as a continuous charge, and on its own it would let each electron repel itself. The exchange term removes exactly that. Acting on an occupied orbital ψ_γ, the term $\gamma' = \gamma$ of $\Sigma_{\rm x}$ is $-\int \mathrm{d}x'\, U_2 |\psi_\gamma(x')|^2 \psi_\gamma(x)$, which cancels the self-repulsion contained in $v_{\rm H}$. More generally, $-|\varrho(x,x')|^2/\varrho(x,x)$ is a hole in the density of the other electrons of the same spin around an electron at x, the exchange hole. Since $\int \mathrm{d}x''\, \varrho(x,x'')\varrho(x'',x') = \varrho(x,x')$ for orthonormal orbitals, it integrates over x' to exactly one missing electron. The Fock term $\Sigma_{\rm x}$ is the first-order self-energy of Sect. 4.7 in the Coulomb interaction; Appendix D.11 obtains it from a diagram.

Orbital energies and Koopmans' theorem

Taking the expectation value of Eq. (D.14) in ψ_γ leads to the orbital energy,

$$E_\gamma = \langle\gamma|h|\gamma\rangle + \sum_{\gamma'\,\mathrm{occ}} \Big[\langle\gamma\gamma'|U_2|\gamma\gamma'\rangle - \langle\gamma\gamma'|U_2|\gamma'\gamma\rangle\Big],$$

the kinetic and nuclear energy of the orbital plus its direct and exchange interaction with every occupied orbital. Summing E_γ over the occupied orbitals counts every pair twice, and comparison with Eq. (D.13) gives

$$E_{\rm HF} = \sum_{\gamma\,\mathrm{occ}} E_\gamma - \frac{1}{2} \sum_{\gamma\gamma'\,\mathrm{occ}} \Big[\langle\gamma\gamma'|U_2|\gamma\gamma'\rangle - \langle\gamma\gamma'|U_2|\gamma'\gamma\rangle\Big]. \tag{D.15}$$

The total energy is not the sum of the orbital energies, and the difference is the double counting of the interaction.

Now remove the electron in an occupied orbital a and keep all the other orbitals as they are. The new state is a determinant of $N_e - 1$ orbitals, and its energy is Eq. (D.13) without every term that contains a. These are the one-body term $\langle a|h|a\rangle$ and, in the double sum, the terms with $\gamma = a$ or $\gamma' = a$, which are equal in pairs and together give the full sum over γ' of the interaction of a with the others. That is E_a. Adding an electron in an empty orbital r adds the same kind of terms, with the sum over the N_e occupied orbitals. With the orbitals frozen, therefore,

$$E^{(a)}_{N_e-1} - E_{N_e} = -E_a, \qquad E^{(r)}_{N_e+1} - E_{N_e} = E_r. \tag{D.16}$$

The occupied eigenvalues are minus the removal energies and the empty ones are the addition energies. This is Koopmans' theorem. Its hypothesis is the frozen orbitals. In a real removal the other orbitals relax around the hole, and the new determinant of lowest energy is not the one obtained by deleting a column.

In a crystal the relaxation drops out. The orbitals are Bloch states $\psi_{n\boldsymbol{k}}$, each spread over the N cells of the crystal, and removing one electron changes $\boldsymbol{n}$ and ϱ in each cell by an amount of relative order $1/N$. The orbitals respond at first order in that change. The energy, however, is stationary with respect to the orbitals, so the energy gained by relaxing them

is of second order, $1/N^2$ per orbital, and summed over the orbitals of one band it vanishes as $1/N$. Hartree–Fock band energies are therefore the removal and addition energies of the Hartree–Fock description of the crystal, and their difference across the gap is its fundamental gap. What the theorem cannot supply is correlation. When an electron is added, the other electrons move away from it and their polarization lowers its energy; when one is removed, they move toward the hole and raise the cost. A single determinant has no such response, and the Hartree–Fock gap of a semiconductor is far too large. The screening that is missing is what the GW self-energy of Sect. 4.8 adds.

The electron gas

In jellium, Eq. (2.47), plane waves solve Eq. (D.14) for any occupied set of plane waves, since ϱ is then invariant under translations, and the filled Fermi sphere is the natural set. The density is then uniform, its Hartree potential is cancelled by the background (Worked example 2.5), and $\varrho(x,x') = \delta_{ss'}\mathscr{V}^{-1}\sum_{p<k_\mathrm{F}} \mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot(\boldsymbol{r}-\boldsymbol{r}')}$ depends only on $\boldsymbol{r}-\boldsymbol{r}'$. Acting on a plane wave $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$, the exchange term multiplies it by a number. With U_2 written through its transform, $\mathscr{V}^{-1}\sum_{\boldsymbol{k}'} v_\mathrm{C}(k')\,\mathrm{e}^{-\mathrm{i}\boldsymbol{k}'\cdot\boldsymbol{r}}$, and the sum over $\boldsymbol{p}$ turned into an integral,

$$\Sigma_\mathrm{x}(k) = -\frac{1}{\mathscr{V}}\sum_{p<k_\mathrm{F}} v_\mathrm{C}(|\boldsymbol{k}-\boldsymbol{p}|) = -\int_{p<k_\mathrm{F}} \frac{\mathrm{d}^3p}{(2\pi)^3}\,\frac{e^2}{k^2+p^2-2kpu},$$

with u the cosine of the angle between $\boldsymbol{k}$ and $\boldsymbol{p}$. The sum runs over one spin, since ϱ is diagonal in spin: only electrons of the same spin exchange. The angular integral is $\int_{-1}^{1}\mathrm{d}u\,(k^2+p^2-2kpu)^{-1} = (kp)^{-1}\ln|(k+p)/(k-p)|$, which leaves

$$\Sigma_\mathrm{x}(k) = -\frac{e^2}{4\pi^2 k}\int_0^{k_\mathrm{F}} \mathrm{d}p\; p\,\ln\Big|\frac{k+p}{k-p}\Big| = -\frac{e^2k_\mathrm{F}^2}{4\pi^2 k}\int_0^1 \mathrm{d}t\; t\,\ln\Big|\frac{x+t}{x-t}\Big|, \qquad x=\frac{k}{k_\mathrm{F}}.$$

The integrand has the antiderivative $\frac{1}{2}(t^2-x^2)\ln|(x+t)/(x-t)| + xt$, as differentiation shows: the derivative of the logarithm, $2x/(x^2-t^2)$, times $\frac{1}{2}(t^2-x^2)$ is $-x$, which cancels the derivative of xt. Between $t=0$ and 1 it gives $x+\frac{1}{2}(1-x^2)\ln|(1+x)/(1-x)| = 2xF_\mathrm{L}(x)$, with the function F_L of Eq. (C.74) evaluated at x. Hence

$$\Sigma_\mathrm{x}(k) = -\frac{e^2k_\mathrm{F}}{2\pi^2}\,F_\mathrm{L}\!\left(\frac{k}{k_\mathrm{F}}\right), \qquad E^\mathrm{HF}_{\boldsymbol{k}} = \frac{k^2}{2m_e} + \Sigma_\mathrm{x}(k). \tag{D.17}$$

The exchange energy is most negative at the bottom of the band, $\Sigma_\mathrm{x}(0) = -e^2k_\mathrm{F}/2\pi^2$, since $F_\mathrm{L}(0)=1$, and it falls to half that value at the Fermi surface, since $F_\mathrm{L}(1)=\frac{1}{2}$.

The width of the occupied band and the velocity at the Fermi surface show what Hartree–Fock gets wrong in a metal. The band is widened, from E_F to $E_\mathrm{F} + e^2k_\mathrm{F}/4\pi^2 = E_\mathrm{F} + \alpha k_\mathrm{F}/\pi$. At the valence density of silicon, with $k_\mathrm{F} = 3.57\,\mathrm{keV}$ and $E_\mathrm{F} = 12.5\,\mathrm{eV}$ (Sect. 2.5), the widening is 8.29 eV and the band is 20.8 eV wide. And the slope of F_L diverges logarithmically at $x=1$ (Appendix C.8), so the group velocity $\partial E^\mathrm{HF}_{\boldsymbol{k}}/\partial k$ is infinite at the Fermi surface and the density of states there vanishes. Both come from the long range of the bare exchange, $v_\mathrm{C} \propto 1/k^2$, which a sharp Fermi surface turns into a singular $\Sigma_\mathrm{x}(k)$. In the real metal the other electrons screen the exchange, the logarithm disappears, and the velocity at the Fermi surface stays close to k_F/m_e. The screened exchange is the GW self-energy of Sect. 4.8.

The exchange energy of the gas follows from Eq. (D.15). The kinetic part averages to $\frac{3}{5}E_\mathrm{F}$ per electron. Since Σ_x is the interaction of one electron with all the others, the interaction energy is half its average over the occupied states: $E_\mathrm{x}/N_e = \frac{1}{2}\langle\Sigma_\mathrm{x}\rangle$. The average of $F_\mathrm{L}(x)$ over the Fermi sphere is $\int_0^1 \mathrm{d}x\,3x^2F_\mathrm{L}(x)$. Its first part is $\frac{1}{2}$. The second is $\frac{3}{4}\int_0^1 \mathrm{d}x\,(x-x^3)\ln[(1+$

$x)/(1-x)]$; integrating by parts with $-(1-x^2)^2/4$ as the antiderivative of $x-x^3$, which vanishes at $x=1$, turns it into $\frac{3}{4}\int_0^1 \mathrm{d}x\,\frac{1}{2}(1-x^2)=\frac{1}{4}$. The average is $\frac{3}{4}$, and

$$\frac{E_\mathrm{x}}{N_e} = -\frac{3e^2 k_\mathrm{F}}{16\pi^2} = -\frac{3\alpha k_\mathrm{F}}{4\pi} = -\frac{0.916}{r_s}\,\mathrm{Ry}, \tag{D.18}$$

with $\mathrm{Ry} = \alpha^2 m_e/2 = 13.61\,\mathrm{eV}$ and $k_\mathrm{F} = (9\pi/4)^{1/3}/(r_s a_0)$, which follows from $\boldsymbol{n}_e = 3/(4\pi r_s^3 a_0^3)$ and Eq. (2.71). At the silicon valence density, $r_s = 2.0$ and $E_\mathrm{x}/N_e = -6.22\,\mathrm{eV}$, comparable in size to the kinetic energy per electron, $\frac{3}{5}E_\mathrm{F} = 7.5\,\mathrm{eV}$. Exchange is a large effect in a valence electron gas. The local-density approximation of Appendix D.5 is built on Eq. (D.18).

D.5 Hohenberg–Kohn theorems and the Kohn–Sham construction

The ground-state density of the electrons in silicon, a function of three variables, fixes the potential of the nuclei and with it the whole Hamiltonian, every excited state included. This is the first theorem of Hohenberg and Kohn [9]. Their second theorem turns the ground-state energy into the minimum of a functional of the density, and Kohn and Sham [10] made that minimum computable with one-electron equations, the Kohn–Sham equations that every band structure in this book comes from (Sect. 3.4). This section proves the two theorems, derives the equations and the total energy, and states what the eigenvalues mean. It uses the constrained-search formulation of Levy [11], which needs fewer assumptions than the original proof, and follows Ref. [12, Chaps. 6 and 7].

The density determines the potential

Take N_e electrons in an external potential $U_1(\boldsymbol{r})$, the attraction of the clamped nuclei, with

$$H = H_\mathrm{kin} + H_{ee} + \sum_{i=1}^{N_e} U_1(\boldsymbol{r}_i), \qquad H_\mathrm{kin} = \sum_i \frac{\boldsymbol{p}_i^2}{2m_e},$$

and H_{ee} the Coulomb repulsion of Eq. (2.44). Only the last term depends on the material, and it enters every expectation value through the density alone: $\langle\Psi|\sum_i U_1(\boldsymbol{r}_i)|\Psi\rangle = \int \mathrm{d}^3r\, U_1(\boldsymbol{r})\,\boldsymbol{n}(\boldsymbol{r})$, with $\boldsymbol{n}(\boldsymbol{r})$ the electron density of Ψ.

Suppose two potentials U_1 and U_1', which differ by more than a constant, had nondegenerate ground states Ψ and Ψ' with the same density $\boldsymbol{n}$. The two states are different. If they were equal, subtracting the two Schrödinger equations would give $\sum_i[U_1(\boldsymbol{r}_i) - U_1'(\boldsymbol{r}_i)]\Psi = (E_0 - E_0')\Psi$, so that the sum would be constant wherever $\Psi \neq 0$, which for a ground state is almost everywhere, and the potentials would differ by a constant. Since the ground state of H is nondegenerate, Ψ' has a strictly higher energy in H:

$$E_0 < \langle\Psi'|H|\Psi'\rangle = \langle\Psi'|H'|\Psi'\rangle + \int \mathrm{d}^3r\,[U_1(\boldsymbol{r}) - U_1'(\boldsymbol{r})]\,\boldsymbol{n}(\boldsymbol{r}) = E_0' + \int \mathrm{d}^3r\,[U_1 - U_1']\,\boldsymbol{n}.$$

The same argument with the primed and unprimed quantities exchanged gives $E_0' < E_0 + \int \mathrm{d}^3r\,[U_1' - U_1]\boldsymbol{n}$. Adding the two inequalities, the integrals cancel and $E_0 + E_0' < E_0' + E_0$, a contradiction. Hence the ground-state density determines U_1 up to a constant [9]. It also fixes $N_e = \int \mathrm{d}^3r\,\boldsymbol{n}$, so it fixes H, and through H every property of the system. The proof says nothing about how to find any of them from $\boldsymbol{n}$.

The variational principle

For a density n define

$$\mathcal{F}[n] = \min_{\Psi \to n} \langle \Psi | H_{\text{kin}} + H_{ee} | \Psi \rangle, \tag{D.19}$$

where the minimum is over all normalized antisymmetric states of N_e electrons that have the density n [11]. The functional $\mathcal{F}$ is universal: it does not depend on the nuclei, and the same $\mathcal{F}$ serves silicon, germanium and an isolated atom. Every reasonable density, nonnegative, integrating to N_e and with a finite kinetic energy, comes from some determinant, so $\mathcal{F}$ is defined on all of them. For a given potential let

$$E_U[n] = \mathcal{F}[n] + \int \mathrm{d}^3 r \, U_1(\boldsymbol{r}) \, n(\boldsymbol{r}).$$

For any n, the state Ψ_n that realizes the minimum in Eq. (D.19) has $\langle \Psi_n | H | \Psi_n \rangle = E_U[n]$, and by the ordinary variational principle this is at least E_0. The minimum over all states, on the other hand, can be done in two steps, first over the states with a given density and then over the densities:

$$E_0 = \min_{\Psi} \langle \Psi | H | \Psi \rangle = \min_{n} \Big[\min_{\Psi \to n} \langle \Psi | H_{\text{kin}} + H_{ee} | \Psi \rangle + \int \mathrm{d}^3 r \, U_1 n \Big] = \min_{n} E_U[n].$$

The ground-state energy is the minimum of E_U over densities, and the minimum is reached at the ground-state density. This is the second theorem of Hohenberg and Kohn, here without the assumption of a nondegenerate ground state, and without the requirement that every trial density be the ground-state density of some potential. Where $\mathcal{F}$ is differentiable, the minimum under the constraint $\int n = N_e$ obeys $\delta \mathcal{F} / \delta n(\boldsymbol{r}) + U_1(\boldsymbol{r}) = \mu_{\text{ch}}$, with the chemical potential as the Lagrange multiplier.

The Kohn–Sham equations

Nobody knows $\mathcal{F}$, and the kinetic energy is its hardest part. Kohn and Sham computed that part exactly for a system of independent electrons with the same density. Define the independent-electron kinetic energy, written T_{s} in the literature, by the same constrained search over determinants Φ alone,

$$K_{\text{s}}[n] = \min_{\Phi \to n} \langle \Phi | H_{\text{kin}} | \Phi \rangle,$$

and the Hartree energy $E_{\text{H}}[n] = \frac{1}{2} \int \mathrm{d}^3 r \, \mathrm{d}^3 r' \, e^2 n(\boldsymbol{r}) n(\boldsymbol{r}') / (4\pi |\boldsymbol{r} - \boldsymbol{r}'|)$, the classical electrostatic energy of the density. The exchange–correlation energy is everything else:

$$\mathcal{F}[n] = K_{\text{s}}[n] + E_{\text{H}}[n] + E_{\text{xc}}[n]. \tag{D.20}$$

It holds the exchange energy, which removes the self-repulsion contained in E_{H} as in Appendix D.4, the correlation energy, and the part of the true kinetic energy that independent electrons miss. It is the only part that has to be approximated.

Write the minimizing determinant through its orbitals ψ_γ, with occupations n_γ equal to one for the N_e occupied orbitals and zero otherwise, and the density $n(\boldsymbol{r}) = \sum_\gamma n_\gamma \sum_s |\psi_\gamma(\boldsymbol{r}, s)|^2$. Then

$$E_U = \sum_\gamma n_\gamma \int \mathrm{d}x \, \psi_\gamma^*(x) \Big(-\frac{\nabla^2}{2m_e} \Big) \psi_\gamma(x) + \int \mathrm{d}^3 r \, U_1 n + E_{\text{H}}[n] + E_{\text{xc}}[n],$$

which depends on the orbitals explicitly through the kinetic term and implicitly through n. We vary $\psi_\gamma^*(x)$, with Lagrange multipliers for orthonormality chosen diagonal, as in

Appendix D.4. The chain rule gives $\delta E_{\rm H}/\delta\psi^*_\gamma(x) = n_\gamma v_{\rm H}(\boldsymbol{r})\psi_\gamma(x)$, with the Hartree potential $v_{\rm H} = \delta E_{\rm H}/\delta n$ of Appendix D.4, and similarly for $E_{\rm xc}$. Dividing by n_γ,

$$\begin{aligned} \Big[-\frac{\nabla^2}{2m_e} + V_{\rm KS}(\boldsymbol{r})\Big]\psi_\gamma(x) &= E_\gamma\,\psi_\gamma(x), \\ V_{\rm KS}(\boldsymbol{r}) = U_1(\boldsymbol{r}) + v_{\rm H}(\boldsymbol{r}) + v_{\rm xc}(\boldsymbol{r}), \qquad v_{\rm xc}(\boldsymbol{r}) &= \frac{\delta E_{\rm xc}}{\delta n(\boldsymbol{r})}, \end{aligned} \tag{D.21}$$

with the density built from the N_e orbitals of lowest E_γ. These are the Kohn–Sham equations [10]. They have the form of the Hartree–Fock equations Eq. (D.14), but the potential $V_{\rm KS}$ is local, a multiplicative function of $\boldsymbol{r}$, where the Fock term was an integral operator. And if $E_{\rm xc}$ were exact, the density they produce would be the exact ground-state density of the interacting electrons. Filling the lowest orbitals assumes that the minimizing density is the ground-state density of independent electrons in some local potential, the noninteracting v-representability of the density; for the crystals of this book it is taken for granted. In a crystal, γ becomes the band and crystal momentum $n\boldsymbol{k}$, E_γ the band energy $\varepsilon_{n\boldsymbol{k}}$ and n_γ the occupation $f_{n\boldsymbol{k}}$, and the density is Eq. (E.1). The equations are nonlinear, since $V_{\rm KS}$ depends on the density, and they are iterated to self-consistency as Appendix E.1 describes.

Total energy and the eigenvalues

Multiplying Eq. (D.21) by $n_\gamma\psi^*_\gamma$, integrating and summing gives $\sum_\gamma n_\gamma E_\gamma = K_{\rm s} + \int {\rm d}^3r\, V_{\rm KS}\boldsymbol{n}$, and $\int v_{\rm H}\boldsymbol{n} = 2E_{\rm H}$. The ground-state energy is therefore

$$E_0 = \sum_\gamma n_\gamma E_\gamma - E_{\rm H}[\boldsymbol{n}] - \int {\rm d}^3r\, v_{\rm xc}(\boldsymbol{r})\boldsymbol{n}(\boldsymbol{r}) + E_{\rm xc}[\boldsymbol{n}]. \tag{D.22}$$

As in Hartree–Fock, Eq. (D.15), the sum of the eigenvalues counts the interaction twice and the other terms correct for it.

The eigenvalues are Lagrange multipliers, and Eq. (D.21) alone does not make them removal or addition energies. One exact statement connects them to energies. Let the occupations n_γ take fractional values and keep the orbitals self-consistent at each set of n_γ. Then E_U is a function of the occupations, and

$$\frac{\partial E_U}{\partial n_\gamma} = E_\gamma, \tag{D.23}$$

which is Janak's theorem [13]. The explicit dependence on n_γ, through the kinetic term and through $\boldsymbol{n}$, gives the expectation value of the Kohn–Sham Hamiltonian in ψ_γ, which is E_γ. The implicit dependence through the orbitals gives, by Eq. (D.21),

$$\sum_{\gamma'} n_{\gamma'}E_{\gamma'}\,\frac{\partial\langle\psi_{\gamma'}|\psi_{\gamma'}\rangle}{\partial n_\gamma} = 0,$$

because every orbital stays normalized. For the exact functional the ground-state energy of a fractional number of electrons is a straight line between integers [14]. Integrating Eq. (D.23) from $N_e - 1$ to N_e in the highest occupied level, whose eigenvalue is then constant, gives $E_{N_e} - E_{N_e-1} = E_{\rm HOMO}$, with $E_{\rm HOMO}$ the eigenvalue of the highest occupied molecular orbital (HOMO): the highest occupied Kohn–Sham eigenvalue is minus the ionization energy I. In the same way, the eigenvalue of the level that fills between N_e and N_e+1 is minus the electron affinity A. No other eigenvalue has an exact meaning.

The fundamental gap is $E_{\rm g} = I - A$, the difference of the two removal and addition energies at the edge. It is not the Kohn–Sham gap $E_{\rm g}^{\rm KS}$, the difference of the lowest empty and highest

occupied eigenvalues of the N_e-electron potential. The straight lines on either side of N_e have the slopes $-I$ and $-A$, so the chemical potential jumps at N_e by the fundamental gap. The derivative of the Kohn–Sham kinetic energy with respect to the electron number jumps only by $E_{\mathrm{g}}^{\mathrm{KS}}$, since the added electron enters the lowest unoccupied molecular orbital (LUMO). The Hartree energy is smooth in n, so the rest of the jump belongs to E_{xc}. As the electron number passes through N_e, the derivative $\delta E_{\mathrm{xc}}/\delta n$ of the exact functional therefore jumps by a constant Δ_{xc}, the derivative discontinuity, and every eigenvalue jumps with it; the Hartree and kinetic parts do not jump. The level that fills above N_e therefore has the eigenvalue $E_{\mathrm{LUMO}} + \Delta_{\mathrm{xc}}$, and

$$E_{\mathrm{g}} = I - A = E_{\mathrm{g}}^{\mathrm{KS}} + \Delta_{\mathrm{xc}} \tag{D.24}$$

for the exact functional [15, 16]. Even exact Kohn–Sham bands have the wrong gap. The local and semilocal functionals below are smooth functions of the density and have no discontinuity; this is the band-gap problem of Sect. 3.4.

The local-density approximation

The local-density approximation takes the exchange–correlation energy per electron of the uniform gas, $\varepsilon_{\mathrm{xc}}(n)$, at the local density,

$$E_{\mathrm{xc}}^{\mathrm{LDA}}[n] = \int \mathrm{d}^3 r\, n(\boldsymbol{r})\, \varepsilon_{\mathrm{xc}}(n(\boldsymbol{r})), \qquad v_{\mathrm{xc}}^{\mathrm{LDA}}(\boldsymbol{r}) = \left.\frac{\mathrm{d}[n\,\varepsilon_{\mathrm{xc}}(n)]}{\mathrm{d}n}\right|_{n=n(\boldsymbol{r})}. \tag{D.25}$$

It is exact for a uniform density and accurate where the density varies slowly on the scale of the local $1/k_{\mathrm{F}}$ [10]. The exchange part is Eq. (D.18), $\varepsilon_{\mathrm{x}}(n) = -3e^2 k_{\mathrm{F}}/16\pi^2$ with the local $k_{\mathrm{F}} = (3\pi^2 n)^{1/3}$, which follows from Eq. (2.71). Since $n\,\varepsilon_{\mathrm{x}} \propto n^{4/3}$, the exchange potential is

$$v_{\mathrm{x}}^{\mathrm{LDA}} = \frac{4}{3}\,\varepsilon_{\mathrm{x}} = -\frac{e^2 k_{\mathrm{F}}}{4\pi^2} = \Sigma_{\mathrm{x}}(k_{\mathrm{F}}),$$

the Hartree–Fock exchange of an electron at the Fermi surface of the gas, Eq. (D.17) at $F_{\mathrm{L}}(1) = \frac{1}{2}$. Slater had earlier proposed to replace the Fock term by its average over the occupied states [17], which in the gas is $\langle \Sigma_{\mathrm{x}} \rangle = \frac{3}{4}\Sigma_{\mathrm{x}}(0) = \frac{3}{2} v_{\mathrm{x}}^{\mathrm{LDA}}$; Kohn and Sham pointed out that the variational principle leads to two thirds of Slater's potential [10]. At the valence density of silicon, $v_{\mathrm{x}}^{\mathrm{LDA}} = -8.29\,\mathrm{eV}$ and Slater's average is $-12.4\,\mathrm{eV}$. The correlation part of $\varepsilon_{\mathrm{xc}}$ has no closed form. It is taken from quantum Monte Carlo energies of the gas [18], fitted as a function of r_s [19]. Unlike the Fock term, the local exchange potential does not cancel the self-repulsion of an electron in v_{H} exactly, and the residual self-interaction is one source of the error of the local-density approximation in localized states [19]. The generalized-gradient approximations add a dependence on $|\nabla n|$, and hybrid functionals replace part of the local exchange by the Fock term built from the Kohn–Sham orbitals; Appendix E.1 describes the functionals in use and their accuracy for silicon.

D.6 The f-sum rule

This section carries out the commutator algebra behind the f-sum rule Eq. (4.30), its species-resolved form Eq. (4.34), and the corrections from nonlocal potentials, Eq. (4.32), and from the mass–velocity term, Eq. (4.33). The results are stated and used in Sect. 4.3; the treatment of the local case follows Refs. [8, Sect. 3.4] and [20, Chap. 2].

One particle and a plane wave

Write $A = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ for one particle, with position $\boldsymbol{r}$ and momentum $\boldsymbol{p}$. Since $[p_\alpha, f(\boldsymbol{r})] = -\mathrm{i}\,\partial_\alpha f$, we have $[p_\alpha, A] = q_\alpha A$, and therefore $A^\dagger p_\alpha A = p_\alpha + q_\alpha$. Conjugation by a unitary operator preserves products and sums, so for any function $F(\boldsymbol{p})$ that is a polynomial or a convergent power series,

$$A^\dagger F(\boldsymbol{p})\, A = F(\boldsymbol{p}+\boldsymbol{q}), \qquad A\, F(\boldsymbol{p})\, A^\dagger = F(\boldsymbol{p}-\boldsymbol{q}). \tag{D.26}$$

The commutator with A follows by factoring A out on the left, $[F(\boldsymbol{p}), A] = A\,(A^\dagger F A - F) = A\,G(\boldsymbol{p})$ with $G(\boldsymbol{p}) = F(\boldsymbol{p}+\boldsymbol{q}) - F(\boldsymbol{p})$. Commuting once more with $A^\dagger$, and using $A^\dagger A = 1$ and Eq. (D.26),

$$[A^\dagger, A\,G(\boldsymbol{p})] = G(\boldsymbol{p}) - A\,G(\boldsymbol{p})\,A^\dagger = G(\boldsymbol{p}) - G(\boldsymbol{p}-\boldsymbol{q}).$$

Inserting G gives the double commutator of any function of the momentum:

$$\big[A^\dagger, [F(\boldsymbol{p}), A]\big] = F(\boldsymbol{p}+\boldsymbol{q}) + F(\boldsymbol{p}-\boldsymbol{q}) - 2F(\boldsymbol{p}). \tag{D.27}$$

It is a second difference of F, and for the kinetic energy $F = \boldsymbol{p}^2/2m$ it is the number q^2/m. A function of the position alone commutes with A and contributes nothing.

The current

For the kinetic energy the single commutator is $[\boldsymbol{p}^2/2m, A] = A\,(2\boldsymbol{q}\cdot\boldsymbol{p} + q^2)/2m$. The symmetrized current has the same form. With Eq. (D.26), $\frac{1}{2}\{\boldsymbol{p}, A\} = \frac{1}{2}(\boldsymbol{p}A + A\boldsymbol{p}) = \frac{1}{2}A(\boldsymbol{p}+\boldsymbol{q}+\boldsymbol{p}) = A(\boldsymbol{p}+\boldsymbol{q}/2)$, so

$$\Big[\frac{\boldsymbol{p}^2}{2m}, A\Big] = \boldsymbol{q}\cdot\frac{1}{2}\Big\{\frac{\boldsymbol{p}}{m}, A\Big\},$$

with no remainder: the term $q^2/2m$ is part of the symmetrized current. Summed over the electrons, this is Eq. (4.29).

Many particles and several species

The densities are sums over particles, $n_s(\boldsymbol{q}) = \sum_{i\in s} A_i$ for each species s, electrons or the nuclei of species κ, with $A_i = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$ and $\boldsymbol{r}_i$ the position of particle i ($\boldsymbol{X}_I$ for a nucleus). Take a Hamiltonian

$$H = \sum_s \sum_{i\in s} F_s(\boldsymbol{p}_i) + U(\{\boldsymbol{r}_i\}), \tag{D.28}$$

where F_s is a function of the momentum of one particle, $\boldsymbol{p}^2/2m_s$ in the nonrelativistic case, and U is any function of all positions, and of spins, if it does not involve momenta. The Coulomb interactions of Sect. 2.3, external static potentials, the Zeeman energy and the contact hyperfine interaction are of this kind. U commutes with every A_i, and so does $F_s(\boldsymbol{p}_j)$ for $j \neq i$. Hence

$$[H, n_{s'}(\boldsymbol{q})] = \sum_{i\in s'} [F_{s'}(\boldsymbol{p}_i), A_i],$$

an operator built from the positions and momenta of species s' alone. It commutes with $A_j^\dagger$ for every particle j not of species s', and within species s' only the terms with $j = i$ survive. By Eq. (D.27),

$$[n_s(\boldsymbol{q})^\dagger, [H, n_{s'}(\boldsymbol{q})]] = \delta_{ss'} \sum_{i\in s} [F_s(\boldsymbol{p}_i+\boldsymbol{q}) + F_s(\boldsymbol{p}_i-\boldsymbol{q}) - 2F_s(\boldsymbol{p}_i)]. \tag{D.29}$$

For $F_s = \boldsymbol{p}^2/2m_s$ each particle contributes q^2/m_s, the commutator is the number $N_s q^2/m_s$ in every state, and dividing by $\mathcal{V}$ gives Eq. (4.34). The derivation holds at any temperature, with the nuclei moving, and for any interactions in U.

The total charge density of Sect. 2.3, $n_Z = -n_e + \sum_\kappa Z_\kappa n_\kappa$, is a linear combination with real coefficients. The double commutator is bilinear and the mixed terms vanish, so

$$\mathscr{M}^1_{ZZ}(\boldsymbol{q}) = q^2\Big[\frac{\boldsymbol{n}_e}{m_e} + \sum_\kappa \frac{Z_\kappa^2\,\boldsymbol{n}_\kappa}{M_\kappa}\Big]. \tag{D.30}$$

In a neutral crystal $\boldsymbol{n}_e = \sum_\kappa Z_\kappa \boldsymbol{n}_\kappa$. The nuclear term relative to the electronic one is then $\sum_\kappa Z_\kappa^2 \boldsymbol{n}_\kappa m_e/M_\kappa$ divided by $\sum_\kappa Z_\kappa \boldsymbol{n}_\kappa$, which for a single species is $Z_\kappa m_e/M_\kappa$, or 2.73×10^{-4} for silicon.

Nonlocal potentials

A one-body potential that is not a function of position, such as the separable pseudopotential Eq. (E.8) or the Fock exchange operator, acts on each electron as an operator V_{NL} with matrix elements $\langle \boldsymbol{r}|V_{\mathrm{NL}}|\boldsymbol{r}'\rangle$ between different points. It does not commute with A, and since A is unitary,

$$[A^\dagger, [V_{\mathrm{NL}}, A]] = A^\dagger V_{\mathrm{NL}} A + A V_{\mathrm{NL}} A^\dagger - 2V_{\mathrm{NL}}. \tag{D.31}$$

Summed over electrons and averaged, this is the extra term of Eq. (4.32); its expectation value in one orbital is twice the δ_{NL} of Eq. (E.16). In the separable form, $A^\dagger V_{\mathrm{NL}} A$ is the same sum with the projector functions multiplied by $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, so the term is a sum of projections. At small q, expanding $A = 1 + \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r} - (\boldsymbol{q}\cdot\boldsymbol{r})^2/2 + \ldots$, the terms of first order cancel between $A^\dagger V_{\mathrm{NL}} A$ and $A V_{\mathrm{NL}} A^\dagger$, and

$$A^\dagger V_{\mathrm{NL}} A + A V_{\mathrm{NL}} A^\dagger - 2V_{\mathrm{NL}} = -[\boldsymbol{q}\cdot\boldsymbol{r}, [\boldsymbol{q}\cdot\boldsymbol{r}, V_{\mathrm{NL}}]] + O(q^4).$$

The kinetic energy gives $-[\boldsymbol{q}\cdot\boldsymbol{r}, [\boldsymbol{q}\cdot\boldsymbol{r}, \boldsymbol{p}^2/2m_e]] = q^2/m_e$ by the same expansion, and the sum of the two is $-\mathrm{i}\,[\boldsymbol{q}\cdot\boldsymbol{r}, \boldsymbol{q}\cdot\boldsymbol{v}]$ with the velocity $\boldsymbol{v} = \mathrm{i}[H, \boldsymbol{r}] = \boldsymbol{p}/m_e + \mathrm{i}[V_{\mathrm{NL}}, \boldsymbol{r}]$. The Thomas–Reiche–Kuhn rule of a nonlocal Hamiltonian is therefore a statement about the commutator of position and velocity, which equals $\mathrm{i}\delta_{\alpha\beta}/m_e$ only when V_{NL} commutes with $\boldsymbol{r}$.

Relativistic corrections

To order $1/m_e^2$, the one-body Hamiltonian of an electron in a potential $U_1(\boldsymbol{r})$ adds three terms to $\boldsymbol{p}^2/2m_e + U_1$ (Sect. 3.5):

$$H_{\mathrm{mv}} = -\frac{\boldsymbol{p}^4}{8m_e^3}, \qquad H_{\mathrm{Dar}} = \frac{\nabla^2 U_1}{8m_e^2}, \qquad H_{\mathrm{so}} = \frac{1}{4m_e^2}\,\boldsymbol{\sigma}\cdot(\nabla U_1 \times \boldsymbol{p}).$$

The Darwin term H_{Dar} is a function of position and drops out. The spin–orbit term is linear in the momentum, $H_{\mathrm{so}} = \sum_\alpha c_\alpha p_\alpha$, with coefficients c_α that depend on position and spin; the ordering of c_α and p_α does not matter, since $\varepsilon_{\gamma\alpha\beta}\partial_\alpha\partial_\beta U_1 = 0$. Its commutator with A is $\sum_\alpha c_\alpha q_\alpha A$, a function of position and spin, which commutes with $A^\dagger$: the double commutator vanishes. The mass–velocity term is of the form Eq. (D.27) with $F = -\boldsymbol{p}^4/8m_e^3$. Writing $\boldsymbol{p}^2 \pm 2\boldsymbol{p}\cdot\boldsymbol{q} + q^2$ for $(\boldsymbol{p}\pm\boldsymbol{q})^2$ and squaring,

$$(\boldsymbol{p}+\boldsymbol{q})^4 + (\boldsymbol{p}-\boldsymbol{q})^4 - 2\boldsymbol{p}^4 = 2(\boldsymbol{p}^2+q^2)^2 + 8(\boldsymbol{p}\cdot\boldsymbol{q})^2 - 2\boldsymbol{p}^4 = 4p^2q^2 + 8(\boldsymbol{p}\cdot\boldsymbol{q})^2 + 2q^4.$$

The expectation value of $(\boldsymbol{p}\cdot\boldsymbol{q})^2 = q_\alpha q_\beta\, p_\alpha p_\beta$ involves the tensor $\langle p_\alpha p_\beta\rangle$. In an atom, and in a crystal of cubic symmetry, a symmetric second-rank tensor is a multiple of the unit tensor, so

$\langle p_\alpha p_\beta\rangle = \delta_{\alpha\beta}\langle p^2\rangle/3$ and $\langle(\boldsymbol{p}\cdot\boldsymbol{q})^2\rangle = q^2\langle p^2\rangle/3$. Per electron, the mass–velocity term then adds

$$-\frac{1}{8m_e^3}\Big[\Big(4+\frac{8}{3}\Big)q^2\langle p^2\rangle + 2q^4\Big] = -\frac{q^2}{m_e}\Big[\frac{5}{3}\,\frac{\langle p^2\rangle/2m_e}{m_e} + \frac{q^2}{4m_e^2}\Big]$$

to the q^2/m_e of the kinetic energy. Averaged over the electrons, $\langle p^2\rangle/2m_e$ is the mean kinetic energy $\bar{K}$, and dividing the sum over electrons by $\mathcal{V}$ gives Eq. (4.33). For a Coulomb Hamiltonian the virial theorem gives $2\langle T\rangle = -\langle U\rangle$ for the total kinetic and potential energies, so $\langle T\rangle = -E$ and $\bar{K} = -E/Z$ per electron of an atom. The relativistic corrections also change the current: the velocity acquires the term $-\boldsymbol{p}^2\boldsymbol{p}/2m_e^3$ from H_{mv} and a spin-dependent term from H_{so}, and the continuity equation Eq. (4.29) holds with that current. The two-body Breit terms of the electron–electron interaction contain the momenta of pairs of electrons, contribute to the double commutator at the same order in $1/m_e^2$, and are not evaluated here.

D.7 The dielectric matrix

This section carries out the matrix algebra behind the dielectric matrix Eq. (4.43) and the macroscopic dielectric function Eq. (4.44) of Sect. 4.4. It covers the inversion of the matrix Dyson equation, the macroscopic average, the equivalent form of $\varepsilon_{\mathrm{mac}}$ with the long-range Coulomb term removed, the limit $\boldsymbol{q}\to 0$, and the two conventions for the matrix. The treatment follows Refs. [21, 22] and [23].

Inversion

Fix $\boldsymbol{q}$ in the first zone and ω, and write every quantity as a matrix in the reciprocal-lattice vectors: the improper response $R = (R_{\boldsymbol{G}\boldsymbol{G}'})$ of Eq. (4.42), a proper response $\tilde{R}$, and the diagonal Coulomb matrix v with elements $v_{\boldsymbol{G}} = v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)$. An external potential with components $\phi_{\mathrm{ext},\boldsymbol{G}'}$ induces the density $\delta n_{e,\boldsymbol{G}} = \sum_{\boldsymbol{G}'} R_{\boldsymbol{G}\boldsymbol{G}'}\phi_{\mathrm{ext},\boldsymbol{G}'}$. Each component of the induced density adds the Hartree potential $v_{\boldsymbol{G}}\,\delta n_{e,\boldsymbol{G}}$ at its own wavevector, since Coulomb's law is diagonal in momentum, and the total potential is $\phi_{\mathrm{tot}} = \phi_{\mathrm{ext}} + v\,\delta n_e$ as a vector. The proper response is defined by $\delta n_e = \tilde{R}\,\phi_{\mathrm{tot}}$. Eliminating ϕ_{tot} for arbitrary ϕ_{ext},

$$R = \tilde{R} + \tilde{R}\,v\,R, \qquad R = (\mathbf{1}-\tilde{R}v)^{-1}\tilde{R}, \tag{D.32}$$

the solution of Eq. (C.59) with $\mathbf{X}_0 = \tilde{R}$ and $\mathbf{K} = v$. The dielectric matrix and its inverse are defined by $\phi_{\mathrm{ext}} = \varepsilon\,\phi_{\mathrm{tot}}$ and $\phi_{\mathrm{tot}} = \varepsilon^{-1}\phi_{\mathrm{ext}}$, so that

$$\varepsilon = \mathbf{1} - v\tilde{R}, \qquad \varepsilon^{-1} = \mathbf{1} + vR. \tag{D.33}$$

That these are inverse matrices follows from the push-through identity $v(\mathbf{1}-\tilde{R}v)^{-1} = (\mathbf{1}-v\tilde{R})^{-1}v$, which holds because $(\mathbf{1}-v\tilde{R})\,v = v\,(\mathbf{1}-\tilde{R}v)$. With Eq. (D.32),

$$\mathbf{1} + vR = \mathbf{1} + (\mathbf{1}-v\tilde{R})^{-1}v\tilde{R} = (\mathbf{1}-v\tilde{R})^{-1}\big[(\mathbf{1}-v\tilde{R}) + v\tilde{R}\big] = (\mathbf{1}-v\tilde{R})^{-1}.$$

The order of the factors matters. In element form, $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\boldsymbol{G}}\tilde{R}_{\boldsymbol{G}\boldsymbol{G}'}$: the Coulomb factor carries the row index, the wavevector at which the potential is produced. This is Eq. (4.43).

The macroscopic average

A field that varies slowly on the scale of the cell has, at wavevector $\boldsymbol{q}\to 0$, only the component $\boldsymbol{G}' = 0$. The total potential it produces has components at every $\boldsymbol{G}$, $\phi_{\mathrm{tot},\boldsymbol{G}} = (\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{0}}\,\phi_{\mathrm{ext},\boldsymbol{0}}$,

which describe how the field inside the cell follows the electron density. The average over a cell of $\mathrm{e}^{-\mathrm{i}(\boldsymbol{q}+\boldsymbol{G})\cdot\boldsymbol{r}}$ vanishes for $\boldsymbol{G} \neq 0$ (Eq. (A.12)), so the macroscopic total potential is the component $\boldsymbol{G} = 0$ alone. The macroscopic dielectric function, the ratio of external to macroscopic total potential, is therefore

$$\varepsilon_{\mathrm{mac}} = \frac{\phi_{\mathrm{ext},\mathbf{0}}}{\phi_{\mathrm{tot},\mathbf{0}}} = \frac{1}{(\varepsilon^{-1})_{\mathbf{00}}}. \tag{D.34}$$

The head $\varepsilon_{\mathbf{00}}$ answers a different question: the external potential needed at $\boldsymbol{G} = 0$ to produce a total potential that has only a $\boldsymbol{G} = 0$ component. No macroscopic external field produces such a total potential in a crystal, and $\varepsilon_{\mathbf{00}}$ is not what an experiment measures. The head of the inverse is obtained from the block inversion Eq. (C.56) with the upper-left block of size one. By Eq. (C.57),

$$\varepsilon_{\mathrm{mac}} = \varepsilon_{\mathbf{00}} - \sum_{\boldsymbol{G},\boldsymbol{G}'\neq 0} \varepsilon_{\mathbf{0}\boldsymbol{G}} \left[(\varepsilon_{\neq 0})^{-1}\right]_{\boldsymbol{G}\boldsymbol{G}'} \varepsilon_{\boldsymbol{G}'\mathbf{0}}, \tag{D.35}$$

with $\varepsilon_{\neq 0}$ the block with $\boldsymbol{G}, \boldsymbol{G}' \neq 0$. The sum is the local-field correction. It vanishes when the wings $\varepsilon_{\mathbf{0}\boldsymbol{G}}$ and $\varepsilon_{\boldsymbol{G}\mathbf{0}}$ vanish, that is, when a uniform potential induces no density modulated on the scale of the cell, as in a homogeneous system.

The long-range term removed

A second form of $\varepsilon_{\mathrm{mac}}$ isolates the one element of v that is singular as $\boldsymbol{q} \to 0$. Split the Coulomb matrix as $v = v_0 + \bar{v}$, where v_0 has the single nonzero element $v_{\mathbf{0}} = v_{\mathrm{C}}(q)$ in the head and $\bar{v}$ holds all the others. Define $\bar{R}$ by the Dyson equation with $\bar{v}$ alone,

$$\bar{R} = \tilde{R} + \tilde{R}\,\bar{v}\,\bar{R}. \tag{D.36}$$

It is the response to the total potential with the Hartree potential at $\boldsymbol{G} = 0$ left out. Equivalently, it is the response of the electrons with the Coulomb kernel at the single wavevector $\boldsymbol{q}$ removed from their Hamiltonian. The full response follows from $\bar{R}$ by a second Dyson equation with v_0. From $R^{-1} = \tilde{R}^{-1} - v$ and $\bar{R}^{-1} = \tilde{R}^{-1} - \bar{v}$ (Appendix C.7) we have $R^{-1} = \bar{R}^{-1} - v_0$, which is

$$R = \bar{R} + \bar{R}\,v_0\,R.$$

Since v_0 connects only the index $\mathbf{0}$ to itself, the head of this equation closes on itself: $(\bar{R}v_0R)_{\mathbf{00}} = \bar{R}_{\mathbf{00}}\,v_{\mathrm{C}}(q)\,R_{\mathbf{00}}$, so $R_{\mathbf{00}} = \bar{R}_{\mathbf{00}}/[1 - v_{\mathrm{C}}(q)\bar{R}_{\mathbf{00}}]$. Then $(\varepsilon^{-1})_{\mathbf{00}} = 1 + v_{\mathrm{C}}(q)R_{\mathbf{00}} = 1/[1 - v_{\mathrm{C}}(q)\bar{R}_{\mathbf{00}}]$, and

$$\varepsilon_{\mathrm{mac}}(\boldsymbol{q},\omega) = 1 - v_{\mathrm{C}}(q)\,\bar{R}_{\mathbf{00}}(\boldsymbol{q},\omega). \tag{D.37}$$

The macroscopic function has the form of Eq. (4.41) for a homogeneous system, with $\bar{R}$ in place of $\tilde{R}$. The local fields are now inside $\bar{R}$, as the microscopic Hartree potentials $\bar{v}$; the exchange and correlation effects are inside $\tilde{R}$. Since $\bar{R}$ is a response of the electrons to a potential, causal and with a local Hamiltonian, the absorptive part $\varepsilon_2 = -v_{\mathrm{C}}(q)\,\mathrm{Im}\,\bar{R}_{\mathbf{00}}$ obeys the f-sum rule Eq. (4.47) with the local fields included [23]. In the absorption spectrum computed from Eq. (D.37) the long-range term is absent, and the Coulomb singularity appears only in the loss function.

The limit $\boldsymbol{q} \to 0$

At small q the head $v_{\mathrm{C}}(q) = e^2/q^2$ diverges, while $v_{\boldsymbol{G}}$ stays finite for $\boldsymbol{G} \neq 0$. For an insulator the proper response is analytic at $\boldsymbol{q} = 0$, and the continuity equation fixes how its elements vanish there. Taking matrix elements of the continuity equation Eq. (4.29) between eigenstates, $(E_f - E_i)\langle f|n_e(\boldsymbol{q})|i\rangle = \langle f|\boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q})|i\rangle$, so $\langle f|n_e(\boldsymbol{q})|i\rangle = \boldsymbol{q}\cdot\langle f|\boldsymbol{j}_e(\boldsymbol{q})|i\rangle/(E_f - E_i)$, which is $O(q)$

because the gap keeps the denominator away from zero. Every element of $\tilde{R}$ with an index $\mathbf{0}$ carries one such factor, and

$$\tilde{R}_{\mathbf{00}} = O(q^2), \qquad \tilde{R}_{\mathbf{0}\boldsymbol{G}},\ \tilde{R}_{\boldsymbol{G}\mathbf{0}} = O(q), \qquad \tilde{R}_{\boldsymbol{G}\boldsymbol{G}'} = O(1).$$

The head of the dielectric matrix, $\varepsilon_{\mathbf{00}} = 1 - (e^2/q^2)\tilde{R}_{\mathbf{00}}$, is then finite. Of the wings, $\varepsilon_{\mathbf{0}\boldsymbol{G}} = -(e^2/q^2)\tilde{R}_{\mathbf{0}\boldsymbol{G}}$ diverges as $1/q$ and $\varepsilon_{\boldsymbol{G}\mathbf{0}} = -v_{\boldsymbol{G}}\tilde{R}_{\boldsymbol{G}\mathbf{0}}$ vanishes as q, and their product in Eq. (D.35) is finite. So is $\varepsilon_{\mathrm{mac}}$, as Eq. (D.37) shows directly, since $\bar{R}$ contains no singular kernel. The finite limits depend on the direction $\hat{\boldsymbol{q}}$, through the factors $\hat{\boldsymbol{q}} \cdot \langle f|\boldsymbol{j}_e(\mathbf{0})|i\rangle$. This is the longitudinal projection $\hat{\boldsymbol{q}} \cdot \varepsilon \cdot \hat{\boldsymbol{q}}$ of the optical tensor of Sect. 4.4, which in a cubic crystal does not depend on $\hat{\boldsymbol{q}}$ (Appendix C.3). A code computes the head and wings analytically in $\hat{\boldsymbol{q}}$ from velocity matrix elements, the optical limit of Appendix E.6, rather than at a small finite q. A metal differs: its intraband transitions have $E_f - E_i \to 0$, which makes $\tilde{R}_{\mathbf{00}}$ tend to a constant at $\omega = 0$, and $\varepsilon_{\mathrm{mac}}$ diverges as $1/q^2$, the perfect screening of Sect. 4.4.

Symmetrized and unsymmetrized matrices

Many codes and papers use the symmetrized matrix $\varepsilon^{\mathrm{s}} = \mathbf{1} - v^{1/2}\tilde{R}\,v^{1/2}$, with $v^{1/2}$ the diagonal matrix of $v_{\boldsymbol{G}}^{1/2}$ [24, footnote 3]. It is a similarity transform of the matrix used here: $\varepsilon^{\mathrm{s}} = v^{-1/2}\,\varepsilon\,v^{1/2}$, since $v^{-1/2}(v\tilde{R})v^{1/2} = v^{1/2}\tilde{R}v^{1/2}$. Inverting, $(\varepsilon^{\mathrm{s}})^{-1} = v^{-1/2}\varepsilon^{-1}v^{1/2}$, and its elements are $(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'}(v_{\boldsymbol{G}'}/v_{\boldsymbol{G}})^{1/2}$. The diagonal elements, which give $\varepsilon_{\mathrm{mac}}$ and the loss function at every $\boldsymbol{G}_q$, are the same in both conventions; the off-diagonal elements differ by the factor $|\boldsymbol{q}+\boldsymbol{G}|/|\boldsymbol{q}+\boldsymbol{G}'|$. The symmetrized matrix is Hermitian when $\tilde{R}$ is, which is convenient for inversion, and its wings are finite as $\boldsymbol{q} \to 0$. Mixing off-diagonal elements from the two conventions is an error that no diagonal check reveals.

D.8 The Lindhard function at zero temperature

The density response of free electrons at $T = 0$ is a ratio of logarithms in ω and q, first written down by Lindhard [25]. Sect. 4.5 uses it for the valence electrons of silicon, where it fixes the static screening, the plasmon and the continuum of electron–hole pairs. This section evaluates it in closed form for any complex frequency in the upper half plane, takes the boundary value on the real axis, and checks the result against the structure factor of Worked example 4.3, the f-sum rule and the static limit of Appendix C.8. The derivation follows Ref. [8, Chap. 4] in the notation of Sects. 4.2 and 4.7.

The angular and radial integrals

The independent-electron response is Eq. (4.93) continued to the real axis, $R^0(\boldsymbol{q},\omega) = \mathscr{V}^{-1}\sum_{\boldsymbol{k}}[n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}}]/(\omega + \mathrm{i}0^+ + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}})$, with the spin in the label. We keep the frequency as a complex number z with $\operatorname{Im} z > 0$, so that every expression below is analytic, and set $z = \omega + \mathrm{i}0^+$ only at the end. At $T = 0$ the occupation is the Fermi sea of Eq. (2.71), $n_{\boldsymbol{k}} = \Theta(k_{\mathrm{F}} - k)$. In the term with $n_{\boldsymbol{k}+\boldsymbol{q}}$, relabel $\boldsymbol{k} \to -\boldsymbol{k} - \boldsymbol{q}$. The occupation becomes $n_{\boldsymbol{k}}$ and the denominator becomes $z + \xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}}$, since ξ depends on $|\boldsymbol{k}|$ only. With the two spins and $\sum_{\boldsymbol{k}} \to \mathscr{V}\int \mathrm{d}^3k/(2\pi)^3$,

$$R^0(q,z) = 2\int_{k<k_{\mathrm{F}}} \frac{\mathrm{d}^3k}{(2\pi)^3}\left[\frac{1}{z - \Delta_{\boldsymbol{k}}} - \frac{1}{z + \Delta_{\boldsymbol{k}}}\right], \qquad \Delta_{\boldsymbol{k}} = \frac{(\boldsymbol{k}+\boldsymbol{q})^2 - k^2}{2m_e} = \frac{kqu}{m_e} + \frac{q^2}{2m_e},$$

with u the cosine of the angle between $\boldsymbol{k}$ and $\boldsymbol{q}$. The resonant term, with $z - \Delta_{\boldsymbol{k}}$, is an electron leaving the sea at $\boldsymbol{k}$ for $\boldsymbol{k}+\boldsymbol{q}$, the transition that absorbs energy; the term with $z + \Delta_{\boldsymbol{k}}$ is

its antiresonant partner of Eq. (4.6). Only states inside the sphere appear, and the Pauli principle has moved into a cancellation between the two terms. A transition from $\boldsymbol{k}$ to an occupied $\boldsymbol{k}+\boldsymbol{q}$ appears in the first term at $\boldsymbol{k}$ and, with the opposite sign, in the second term at $-\boldsymbol{k}-\boldsymbol{q}$, which is also inside the sphere, and the two cancel.

The variables that make the result simple measure frequency and momentum in units set by the Fermi surface,

$$\nu = \frac{z}{q v_{\mathrm{F}}}, \qquad y = \frac{q}{2k_{\mathrm{F}}}, \qquad \tilde{k} = \frac{k}{k_{\mathrm{F}}}, \qquad \nu_{\pm} = \nu \pm y,$$

with $v_{\mathrm{F}} = k_{\mathrm{F}}/m_e$. Then $z \mp \Delta_{\boldsymbol{k}} = q v_{\mathrm{F}}(\nu \mp y \mp \tilde{k}u)$. The angular integrals are elementary, $\int_{-1}^{1} \mathrm{d}u\,(w - \tilde{k}u)^{-1} = \tilde{k}^{-1} \ln[(w+\tilde{k})/(w-\tilde{k})]$ for complex w off the real segment $[-\tilde{k}, \tilde{k}]$, with the principal branch of the logarithm. For $\operatorname{Im} w > 0$ the ratio $(w+\tilde{k})/(w-\tilde{k}) = 1 + 2\tilde{k}/(w-\tilde{k})$ lies in the lower half plane, away from the cut of the principal branch, so the logarithm is analytic there. With $\mathrm{d}^3k = 2\pi k^2\,\mathrm{d}k\,\mathrm{d}u$,

$$R^0(q,z) = \frac{m_e k_{\mathrm{F}}^2}{2\pi^2 q} \int_0^1 \mathrm{d}\tilde{k}\;\tilde{k} \left[\ln \frac{\nu_- + \tilde{k}}{\nu_- - \tilde{k}} - \ln \frac{\nu_+ + \tilde{k}}{\nu_+ - \tilde{k}} \right].$$

The radial integral is the one met in the exchange energy of Appendix D.4. The antiderivative found there, $\frac{1}{2}(\tilde{k}^2 - w^2)\ln[(w+\tilde{k})/(w-\tilde{k})] + w\tilde{k}$, holds for complex w, since the derivative was taken without using reality. Evaluated between 0 and 1 it gives

$$\mathcal{I}(w) = \int_0^1 \mathrm{d}\tilde{k}\;\tilde{k} \ln \frac{w+\tilde{k}}{w-\tilde{k}} = w + \frac{1-w^2}{2}\,\ell(w), \qquad \ell(w) = \ln \frac{w+1}{w-1},$$

and $R^0 = (m_e k_{\mathrm{F}}^2/2\pi^2 q)[\mathcal{I}(\nu_-) - \mathcal{I}(\nu_+)]$. Since $\nu_- - \nu_+ = -2y = -q/k_{\mathrm{F}}$, the linear terms of $\mathcal{I}$ give a constant. Collecting, with the density of states $\mathcal{N}_{\mathrm{F}} = m_e k_{\mathrm{F}}/2\pi^2$ of one spin at the Fermi level (Eq. (2.71)), we arrive at the Lindhard function,

$$R^0(q,z) = -2\mathcal{N}_{\mathrm{F}} \left\{ \frac{1}{2} + \frac{k_{\mathrm{F}}}{4q} \Big[(1-\nu_+^2)\,\ell(\nu_+) - (1-\nu_-^2)\,\ell(\nu_-) \Big] \right\}, \qquad \nu_{\pm} = \frac{z}{q v_{\mathrm{F}}} \pm \frac{q}{2k_{\mathrm{F}}}. \tag{D.38}$$

It is analytic in the upper half plane of z, as a retarded function must be (Appendix C.5), and it depends on q and z only through ν and y, apart from the prefactor $2\mathcal{N}_{\mathrm{F}}$.

Real and imaginary parts

On the real axis, $z = \omega + \mathrm{i}0^+$ with $\omega > 0$, and both $\nu_{\pm}$ acquire a positive imaginary part. By Eq. (C.75), $\ell(\nu_{\pm} + \mathrm{i}0^+) = \ln|(\nu_{\pm}+1)/(\nu_{\pm}-1)| - \mathrm{i}\pi\,\Theta(1-|\nu_{\pm}|)$. The real part of Eq. (D.38) keeps the form of the equation with every logarithm replaced by the logarithm of the modulus, now with $\nu_{\pm} = \omega/q v_{\mathrm{F}} \pm q/2k_{\mathrm{F}}$ real,

$$\operatorname{Re} R^0(q,\omega) = -2\mathcal{N}_{\mathrm{F}} \left\{ \frac{1}{2} + \frac{k_{\mathrm{F}}}{4q} \left[(1-\nu_+^2) \ln\Big|\frac{\nu_+ + 1}{\nu_+ - 1}\Big| - (1-\nu_-^2) \ln\Big|\frac{\nu_- + 1}{\nu_- - 1}\Big| \right] \right\}, \tag{D.39}$$

and the imaginary part collects the steps,

$$\operatorname{Im} R^0(q,\omega) = -\frac{\pi k_{\mathrm{F}} \mathcal{N}_{\mathrm{F}}}{2q} \Big[(1-\nu_-^2)\,\Theta(1-|\nu_-|) - (1-\nu_+^2)\,\Theta(1-|\nu_+|) \Big]. \tag{D.40}$$

For $\omega < 0$ the imaginary part changes sign, since $R^0(q,-\omega) = R^0(q,\omega)^*$ by Eq. (4.5) for a gas with $R^0(-\boldsymbol{q}) = R^0(\boldsymbol{q})$.

The steps divide the (q,ω) plane into a region where both are on, a region where only the step of ν_- is on, and a region outside the continuum. Since $\omega>0$, $\nu_+>\nu_-$ and $\nu_+>y>0$. The condition $|\nu_+|<1$ is $\omega<qv_\mathrm{F}-q^2/2m_e$, which needs $q<2k_\mathrm{F}$. The condition $|\nu_-|<1$ is $|m_e\omega/q-q/2|<k_\mathrm{F}$, which is the band $\omega_-<\omega<\omega_+$ with $\omega_+=qv_\mathrm{F}+q^2/2m_e$ and ω_- equal to $q^2/2m_e-qv_\mathrm{F}$ for $q>2k_\mathrm{F}$ and to zero otherwise. Below the line $\omega=qv_\mathrm{F}-q^2/2m_e$ both steps are on. The bracket is then $\nu_+^2-\nu_-^2=4\nu y=2m_e\omega/k_\mathrm{F}^2$, and

$$\operatorname{Im}R^0=-\frac{m_e^2\,\omega}{2\pi q}\qquad(\omega<qv_\mathrm{F}-q^2/2m_e).$$

Above it, up to ω_+, only the step of ν_- is on, and with $k_\mathrm{F}^2(1-\nu_-^2)=k_\mathrm{F}^2-(m_e\omega/q-q/2)^2$,

$$\operatorname{Im}R^0=-\frac{m_e}{4\pi q}\Big[k_\mathrm{F}^2-\Big(\frac{m_e\omega}{q}-\frac{q}{2}\Big)^2\Big]\qquad(\text{only }|\nu_-|<1).$$

Outside the band $\omega_-<\omega<\omega_+$ the imaginary part vanishes: no electron–hole pair has that energy at that momentum. The two forms are $-\frac12$ times the linear and parabolic forms of the structure factor S^0 in Worked example 4.3, which is the fluctuation–dissipation theorem Eq. (4.18) at $T=0$, $S^0=-2\operatorname{Im}R^0$ for $\omega>0$. The two calculations are independent: Worked example 4.3 counted the transitions with a delta function, and here they appear as the jump of a logarithm across its cut.

Static, small-q and high-frequency limits

Static response. At $\omega=0$, $\nu_\pm=\pm y$. The logarithm of the modulus is odd, $\ln|(1-y)/(-1-y)|=-\ln|(1+y)/(1-y)|$, so the two terms in the bracket of Eq. (D.39) add, and $(k_\mathrm{F}/4q)\cdot 2=1/4y$:

$$R^0(q,0)=-2\mathcal{N}_\mathrm{F}\left[\frac12+\frac{1-y^2}{4y}\ln\Big|\frac{1+y}{1-y}\Big|\right]=-2\mathcal{N}_\mathrm{F}\,F_\mathrm{L}\Big(\frac{q}{2k_\mathrm{F}}\Big),\tag{D.41}$$

with F_L of Eq. (C.74). The static response is negative, as it must be, since an attractive potential piles up electrons. As $q\to0$, $F_\mathrm{L}\to1$ and $R^0\to-2\mathcal{N}_\mathrm{F}$: a slowly varying potential shifts the local Fermi level, and the density follows with the full density of states of both spins. The Thomas–Fermi screening of Sect. 4.4 follows, $1-v_\mathrm{C}R^0=1+k_\mathrm{TF}^2/q^2$ with $k_\mathrm{TF}^2=2e^2\mathcal{N}_\mathrm{F}$. At $q=2k_\mathrm{F}$ the derivative of F_L diverges logarithmically (Appendix C.8).

Small momentum at fixed ω/qv_F. For $y\to0$ at fixed ν, $\mathcal{I}(\nu-y)-\mathcal{I}(\nu+y)=-2y\,\mathcal{I}'(\nu)+O(y^3)$. The derivative is $\mathcal{I}'(w)=1-w\,\ell(w)+\frac12(1-w^2)\ell'(w)$, and with $\ell'(w)=-2/(w^2-1)$ the last term is 1, so $\mathcal{I}'(w)=2-w\,\ell(w)$. Hence

$$R^0(q,z)=-2\mathcal{N}_\mathrm{F}\left[1-\frac{\nu}{2}\ln\frac{\nu+1}{\nu-1}\right]+O(y^2),\qquad\nu=\frac{z}{qv_\mathrm{F}}.\tag{D.42}$$

This is the response of a Fermi surface to a long-wavelength disturbance, the semiclassical limit in which the recoil $q^2/2m_e$ is negligible against qv_F. For $0<\omega<qv_\mathrm{F}$ the imaginary part is $-\pi\mathcal{N}_\mathrm{F}\,\omega/(qv_\mathrm{F})$, which equals $-m_e^2\omega/2\pi q$, the linear form above; above qv_F it vanishes.

High frequency. For $|\nu|\gg1$, outside the continuum, the logarithm has the expansion $\ell(w)=2(w^{-1}+w^{-3}/3+w^{-5}/5+\dots)$, and the powers w^{+1} and w^{-1} cancel in $\mathcal{I}$: $\mathcal{I}(w)=\frac23w^{-1}+\frac{2}{15}w^{-3}+O(w^{-5})$. With $(\nu-y)^{-1}-(\nu+y)^{-1}=(2y/\nu^2)(1+y^2/\nu^2+\dots)$ and $(\nu-y)^{-3}-(\nu+y)^{-3}=6y/\nu^4+\dots$,

$$R^0(q,z)=\frac{n_e q^2}{m_e z^2}\left[1+\frac{\frac35q^2v_\mathrm{F}^2+(q^2/2m_e)^2}{z^2}+O(z^{-4})\right],\tag{D.43}$$

where $n_e = k_F^3/3\pi^2$ has been used in $m_e k_F/(3\pi^2\nu^2) = n_e q^2/(m_e z^2)$. The leading term is the free-particle tail of Eq. (C.37), whose coefficient is fixed by the f-sum rule: at high frequency every electron responds as if it were alone. The correction holds the mean square velocity of the sea, $\frac{3}{5}v_F^2$, and the recoil, and in the random-phase approximation it is what makes the plasmon disperse (Sect. 4.5).

The f-sum rule. The tail Eq. (D.43) and the analyticity of Eq. (D.38) already imply $\int_0^\infty \mathrm{d}\omega\, \omega\, \mathrm{Im}\, R^0 = -\pi n_e q^2/2m_e$, by the argument of Appendix C.5. For $q > 2k_F$ the integral is also elementary. Only the parabolic form contributes, and with $s = m_e\omega/q - q/2$, which runs from $-k_F$ to k_F,

$$\int_0^\infty \mathrm{d}\omega\, \omega\, \mathrm{Im}\, R^0 = -\frac{q}{4\pi m_e}\int_{-k_F}^{k_F} \mathrm{d}s \left(s + \frac{q}{2}\right)(k_F^2 - s^2) = -\frac{q^2 k_F^3}{6\pi m_e} = -\frac{\pi n_e q^2}{2m_e},$$

since the odd part of the integrand drops out.

D.9 The Mermin relaxation-time dielectric function

Electrons in a real metal or a doped semiconductor scatter off impurities and phonons, at a rate γ_{rel}, and the sharp edges of the Lindhard function of Appendix D.8 are smeared on that scale. The substitution $\omega \to \omega + \mathrm{i}\gamma_{\mathrm{rel}}$ in the independent-electron response does not conserve the number of electrons (Sect. 4.5). Mermin [26] let the collisions relax the electrons toward a local equilibrium with the same density instead, and found the response that conserves particle number. This section derives it for Sect. 4.5 and shows what the difference does to screening and to the conductivity.

The equation of motion

Take independent electrons, $H_0 = \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}} c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}$ with the spin in the label, in equilibrium with occupations $n_{\boldsymbol{k}} = n_F(\xi_{\boldsymbol{k}})$, and couple a potential to the density as in Eq. (4.3), $H(t) = H_0 + n_e(\boldsymbol{q})\,\phi_{\mathrm{ext}}(t)$, with $\boldsymbol{q} \neq 0$. In the random-phase approximation of Sect. 4.5 the potential becomes the total one, external plus induced; here it is simply the potential the electrons feel. The response is carried by the coherences

$$\delta\varrho_{\boldsymbol{k}}(t) = \langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}\rangle(t),$$

the matrix elements of the one-electron density matrix between $\boldsymbol{k}+\boldsymbol{q}$ and $\boldsymbol{k}$, which vanish in equilibrium. Their sum is the induced density, $\delta n(t) \equiv \mathcal{V}^{-1}\sum_{\boldsymbol{k}} \delta\varrho_{\boldsymbol{k}} = \mathcal{V}^{-1}\delta\langle n_e(\boldsymbol{q})^\dagger\rangle$, since $n_e(\boldsymbol{q})^\dagger = \sum_{\boldsymbol{k}} c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}$ (Eq. (2.38)). By Eq. (2.33), $[c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}, H_0] = (\xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}})\, c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}$ and $[c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}, n_e(\boldsymbol{q})] = c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}} - c_{\boldsymbol{k}+\boldsymbol{q}}^\dagger c_{\boldsymbol{k}+\boldsymbol{q}}$. To first order in the potential the second commutator can be averaged in equilibrium, and $\mathrm{i}\,\partial_t\langle A\rangle = \langle [A, H]\rangle$ gives

$$\mathrm{i}\,\partial_t\,\delta\varrho_{\boldsymbol{k}} = -D_{\boldsymbol{k}}\,\delta\varrho_{\boldsymbol{k}} + (n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}})\,\phi_{\mathrm{ext}}(t), \qquad D_{\boldsymbol{k}} = \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}}.$$

For $\phi_{\mathrm{ext}} \propto \mathrm{e}^{-\mathrm{i}\omega t}$ the solution is $\delta\varrho_{\boldsymbol{k}} = (n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}})\,\phi_{\mathrm{ext}}/(\omega + \mathrm{i}0^+ + D_{\boldsymbol{k}})$, and summing over $\boldsymbol{k}$ gives back $\delta n = R^0(\boldsymbol{q},\omega)\,\phi_{\mathrm{ext}}$ with the bubble of Eq. (4.93). The equation is the one-electron Liouville equation, and collisions add a relaxation term to it.

Relaxation toward local equilibrium

In the relaxation-time approximation, collisions pull the density matrix toward a reference state at the rate γ_{rel}:

$$\mathrm{i}\,\partial_t\,\delta\varrho_{\boldsymbol{k}} = -D_{\boldsymbol{k}}\,\delta\varrho_{\boldsymbol{k}} + (n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}})\,\phi_{\mathrm{ext}} - \mathrm{i}\gamma_{\mathrm{rel}}(\delta\varrho_{\boldsymbol{k}} - \delta\varrho_{\boldsymbol{k}}^{\mathrm{loc}}).$$

With the global equilibrium as the reference, $\delta\varrho_{\boldsymbol{k}}^{\rm loc}=0$, the solution is that of the previous paragraph with ω replaced by $\omega+\mathrm{i}\gamma_{\rm rel}$, the naive substitution. Mermin's reference is the equilibrium state in which the density already has its instantaneous value, since collisions conserve the number of electrons (Sect. 4.5). A local equilibrium with a local chemical potential $\mu_{\rm ch}+\delta\mu(\boldsymbol{r},t)$ is the equilibrium state in the static potential $-\delta\mu$, and we write this potential as $\phi_{\rm loc}(t)\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}+\text{c.c.}$ Its density matrix is the time-independent solution of the collisionless equation in a static potential,

$$\delta\varrho_{\boldsymbol{k}}^{\rm loc}=\frac{n_{\boldsymbol{k}}-n_{\boldsymbol{k}+\boldsymbol{q}}}{D_{\boldsymbol{k}}}\,\phi_{\rm loc}(t),\qquad \frac{1}{\mathcal{V}}\sum_{\boldsymbol{k}}\delta\varrho_{\boldsymbol{k}}^{\rm loc}=R^0(\boldsymbol{q},0)\,\phi_{\rm loc}(t),$$

since $(1/\mathcal{V})\sum_{\boldsymbol{k}}(n_{\boldsymbol{k}}-n_{\boldsymbol{k}+\boldsymbol{q}})/D_{\boldsymbol{k}}$ is the static bubble. The static response that enters here is the isothermal one, the value of the Matsubara response at $\omega_0^{\rm B}=0$ in Sect. 4.7, which for free electrons at $\boldsymbol{q}\neq0$ coincides with $R^0(\boldsymbol{q},\omega\to0)$. The amplitude $\phi_{\rm loc}$ is fixed by requiring the local-equilibrium state to have the actual density,

$$\frac{1}{\mathcal{V}}\sum_{\boldsymbol{k}}\delta\varrho_{\boldsymbol{k}}^{\rm loc}=\frac{1}{\mathcal{V}}\sum_{\boldsymbol{k}}\delta\varrho_{\boldsymbol{k}},\qquad\text{that is}\qquad\phi_{\rm loc}=\frac{\delta n}{R^0(\boldsymbol{q},0)}.\tag{D.44}$$

For a field at frequency ω, the equation of motion becomes algebraic,

$$(\omega+\mathrm{i}\gamma_{\rm rel}+D_{\boldsymbol{k}})\,\delta\varrho_{\boldsymbol{k}}=(n_{\boldsymbol{k}}-n_{\boldsymbol{k}+\boldsymbol{q}})\Big[\phi_{\rm ext}+\frac{\mathrm{i}\gamma_{\rm rel}\,\phi_{\rm loc}}{D_{\boldsymbol{k}}}\Big].$$

Dividing by the bracket on the left and using the partial fractions $1/[D(\omega+\mathrm{i}\gamma_{\rm rel}+D)]=(\omega+\mathrm{i}\gamma_{\rm rel})^{-1}[1/D-1/(\omega+\mathrm{i}\gamma_{\rm rel}+D)]$, the sum over $\boldsymbol{k}$ produces two bubbles, one at the complex frequency $\omega+\mathrm{i}\gamma_{\rm rel}$ and one static. With the shorthand $R^0_\gamma=R^0(\boldsymbol{q},\omega+\mathrm{i}\gamma_{\rm rel})$ and $R^0_{\rm st}=R^0(\boldsymbol{q},0)$,

$$\delta n=R^0_\gamma\,\phi_{\rm ext}+\frac{\mathrm{i}\gamma_{\rm rel}}{\omega+\mathrm{i}\gamma_{\rm rel}}(R^0_{\rm st}-R^0_\gamma)\,\phi_{\rm loc}.$$

Inserting Eq. (D.44) and collecting the terms in δn on the left,

$$\frac{\omega+\mathrm{i}\gamma_{\rm rel}R^0_\gamma/R^0_{\rm st}}{\omega+\mathrm{i}\gamma_{\rm rel}}\,\delta n=R^0_\gamma\,\phi_{\rm ext},$$

and the response $\delta n/\phi_{\rm ext}$ is

$$R^{\rm Mer}(\boldsymbol{q},\omega)=\frac{(1+\mathrm{i}\gamma_{\rm rel}/\omega)\,R^0(\boldsymbol{q},\omega+\mathrm{i}\gamma_{\rm rel})}{1+(\mathrm{i}\gamma_{\rm rel}/\omega)\,R^0(\boldsymbol{q},\omega+\mathrm{i}\gamma_{\rm rel})/R^0(\boldsymbol{q},0)}.\tag{D.45}$$

In the random-phase approximation the Mermin dielectric function is $\varepsilon_{\rm Mer}=1-v_{\rm C}R^{\rm Mer}$. With $\varepsilon_{\rm Lin}=1-v_{\rm C}R^0$ for the Lindhard dielectric function, so that $R^0_\gamma/R^0_{\rm st}=[\varepsilon_{\rm Lin}(\boldsymbol{q},\omega+\mathrm{i}\gamma_{\rm rel})-1]/[\varepsilon_{\rm Lin}(\boldsymbol{q},0)-1]$, it takes the form given by Mermin [26],

$$\varepsilon_{\rm Mer}(\boldsymbol{q},\omega)=1+\frac{(1+\mathrm{i}\gamma_{\rm rel}/\omega)\,[\varepsilon_{\rm Lin}(\boldsymbol{q},\omega+\mathrm{i}\gamma_{\rm rel})-1]}{1+(\mathrm{i}\gamma_{\rm rel}/\omega)\,[\varepsilon_{\rm Lin}(\boldsymbol{q},\omega+\mathrm{i}\gamma_{\rm rel})-1]/[\varepsilon_{\rm Lin}(\boldsymbol{q},0)-1]}.\tag{D.46}$$

Nothing in the derivation used $T=0$; at finite temperature R^0 and $\varepsilon_{\rm Lin}$ are taken at that temperature.

Causality

The bubble at the shifted frequency, $R^0(\boldsymbol{q}, z + \mathrm{i}\gamma_{\mathrm{rel}})$, is analytic for $\operatorname{Im} z > -\gamma_{\mathrm{rel}}$. That alone does not make R^{Mer} causal, since the denominator of Eq. (D.45), which is proportional to $z + \mathrm{i}\gamma_{\mathrm{rel}} R^0_\gamma / R^0_{\mathrm{st}}$, could vanish in the upper half plane, and a pole there would be a response that grows after the field is gone. The relaxation equation has no such mode, and the proof uses only its structure.

A zero of the denominator is a free oscillation. With $\phi_{\mathrm{ext}} = 0$ and the time dependence $\mathrm{e}^{-\mathrm{i}zt}$, the collected equation above reads $(z + \mathrm{i}\gamma_{\mathrm{rel}} R^0_\gamma / R^0_{\mathrm{st}})\, \delta n = 0$, so a nonzero density, and with it the nonzero coherences $\delta\varrho_{\boldsymbol{k}} = (n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}})\, \mathrm{i}\gamma_{\mathrm{rel}} \phi_{\mathrm{loc}} / [D_{\boldsymbol{k}}(z + \mathrm{i}\gamma_{\mathrm{rel}} + D_{\boldsymbol{k}})]$, exist exactly where the denominator vanishes. A zero with $\operatorname{Im} z > 0$ would be a solution of the equation of motion without a field that grows as $\mathrm{e}^{(\operatorname{Im} z)t}$.

No such solution exists. Write $w_{\boldsymbol{k}} = (n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}})/(\xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}})$, which is positive wherever the two occupations differ, because n_{F} falls with energy, and zero where they are equal. Let $\varsigma = \sum_{\boldsymbol{k}} \delta\varrho_{\boldsymbol{k}}$ and $w_{\mathrm{tot}} = \sum_{\boldsymbol{k}} w_{\boldsymbol{k}}$. Since $R^0_{\mathrm{st}} = -w_{\mathrm{tot}}/\mathscr{V}$, the condition Eq. (D.44) makes the reference state $\delta\varrho^{\mathrm{loc}}_{\boldsymbol{k}} = w_{\boldsymbol{k}}\, \varsigma / w_{\mathrm{tot}}$, and the equation of motion without a field is

$$\partial_t\, \delta\varrho_{\boldsymbol{k}} = \mathrm{i} D_{\boldsymbol{k}}\, \delta\varrho_{\boldsymbol{k}} - \gamma_{\mathrm{rel}} \Big(\delta\varrho_{\boldsymbol{k}} - w_{\boldsymbol{k}} \frac{\varsigma}{w_{\mathrm{tot}}} \Big).$$

A coherence with $w_{\boldsymbol{k}} = 0$ obeys $\partial_t\, \delta\varrho_{\boldsymbol{k}} = (\mathrm{i}D_{\boldsymbol{k}} - \gamma_{\mathrm{rel}})\, \delta\varrho_{\boldsymbol{k}}$ and decays at the rate γ_{rel}, so a solution $\propto \mathrm{e}^{-\mathrm{i}zt}$ with $\operatorname{Im} z > 0$ has all these components zero, and ς is a sum over the others. For them take the weighted norm

$$\Phi_\varrho(t) = \sum_{w_{\boldsymbol{k}} > 0} \frac{|\delta\varrho_{\boldsymbol{k}}|^2}{w_{\boldsymbol{k}}} \geq 0.$$

The term $\mathrm{i}D_{\boldsymbol{k}}$ rotates each phase and does not change Φ_ϱ, and

$$\frac{\mathrm{d}\Phi_\varrho}{\mathrm{d}t} = -2\gamma_{\mathrm{rel}} \Big[\Phi_\varrho - \frac{|\varsigma|^2}{w_{\mathrm{tot}}} \Big] \leq 0,$$

because the Cauchy–Schwarz inequality gives $|\varsigma|^2 = |\sum_{\boldsymbol{k}} (\delta\varrho_{\boldsymbol{k}} / w_{\boldsymbol{k}}^{1/2})\, w_{\boldsymbol{k}}^{1/2}|^2 \leq \Phi_\varrho w_{\mathrm{tot}}$. A growing solution would make $\Phi_\varrho \propto \mathrm{e}^{2(\operatorname{Im} z)t}$ grow, so none exists, and the denominator of Eq. (D.45) has no zero with $\operatorname{Im} z > 0$. At $z = 0$ the numerator and the denominator both contain $\mathrm{i}\gamma_{\mathrm{rel}}/z$, the singularity is removable, and $R^{\mathrm{Mer}} \to R^0(\boldsymbol{q}, 0)$. With the decay at large $|z|$ shown below, R^{Mer} is analytic in the upper half plane and falls off at infinity, a causal response in the sense of Appendix C.5. For the naive substitution, with $\delta\varrho^{\mathrm{loc}} = 0$, every coherence decays at the rate γ_{rel}, and causality is immediate.

Particle conservation

The sum of the equation of motion over $\boldsymbol{k}$ is the continuity equation, and the reference state decides whether it holds. In the term with the potential, $\sum_{\boldsymbol{k}} (n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}}) = 0$, because the two sums run over the same occupations. In the term with $D_{\boldsymbol{k}}$, the free dispersion gives $-D_{\boldsymbol{k}} = \xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}} = \boldsymbol{q} \cdot (\boldsymbol{k} + \boldsymbol{q}/2)/m_e$. The factor $(\boldsymbol{k} + \boldsymbol{q}/2)/m_e$ is the plane-wave matrix element $\langle \boldsymbol{k} + \boldsymbol{q} | \tfrac{1}{2}\{\boldsymbol{p}/m_e, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\} | \boldsymbol{k} \rangle$ of the paramagnetic current $\boldsymbol{j}_e(\boldsymbol{q}) = \sum_i \tfrac{1}{2}\{\boldsymbol{p}_i/m_e, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}\}$, the velocity of the electrons weighted by the plane wave (Sect. 4.3). By Eq. (2.36), $\boldsymbol{j}_e(\boldsymbol{q})^\dagger = \sum_{\boldsymbol{k}} [(\boldsymbol{k} + \boldsymbol{q}/2)/m_e]\, c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}+\boldsymbol{q}}$, so that $-\sum_{\boldsymbol{k}} D_{\boldsymbol{k}} \delta\varrho_{\boldsymbol{k}} = \boldsymbol{q} \cdot \delta\langle \boldsymbol{j}_e(\boldsymbol{q})^\dagger \rangle$. At frequency ω the summed equation reads

$$\omega\, \delta\langle n_e(\boldsymbol{q})^\dagger \rangle - \boldsymbol{q} \cdot \delta\langle \boldsymbol{j}_e(\boldsymbol{q})^\dagger \rangle = -\mathrm{i}\gamma_{\mathrm{rel}} \sum_{\boldsymbol{k}} (\delta\varrho_{\boldsymbol{k}} - \delta\varrho^{\mathrm{loc}}_{\boldsymbol{k}}). \tag{D.47}$$

Without collisions the right-hand side is zero, and Eq. (D.47) is the Fourier transform of $\partial_t \boldsymbol{n}_e + \nabla \cdot \boldsymbol{j} = 0$: the density at a crest changes only as fast as the current carries electrons in or out. The naive substitution, with $\delta\varrho^{\text{loc}} = 0$, leaves $-\mathrm{i}\gamma_{\text{rel}}\,\delta\langle n_e(\boldsymbol{q})^\dagger\rangle$ on the right, a density that decays at the rate γ_{rel} whatever the current does. Mermin's condition Eq. (D.44) sets the right-hand side to zero exactly, at every frequency and momentum, and this is the sense in which R^{Mer} conserves particle number. Collisions now change only the current, and through the current the density.

Static and long-wavelength limits

The response to a static potential, the long-wavelength response and the static screening of a metal all tell Mermin's function apart from the naive substitution.

Static potential. For $\omega \to 0$ at fixed γ_{rel}, the factors $\mathrm{i}\gamma_{\text{rel}}/\omega$ dominate the numerator and the denominator of Eq. (D.45), and $R^{\text{Mer}} \to R^0(\boldsymbol{q}, 0)$. A static potential leaves the electrons in equilibrium in the potential, with or without collisions, and the response is the static one. The naive substitution leads to $R^0(\boldsymbol{q}, \mathrm{i}\gamma_{\text{rel}})$ instead, which differs from it at every q.

Long wavelength. For $q \to 0$ at fixed ω and γ_{rel}, the frequency $\omega + \mathrm{i}\gamma_{\text{rel}}$ lies far outside the continuum, $|\omega + \mathrm{i}\gamma_{\text{rel}}| \gg q v_{\text{F}}$, and the high-frequency form Eq. (D.43) gives $R^0_\gamma \to \boldsymbol{n}_e q^2/[m_e(\omega + \mathrm{i}\gamma_{\text{rel}})^2]$, while R^0_{st} stays finite. The ratio $R^0_\gamma/R^0_{\text{st}}$ vanishes as q^2, the denominator of Eq. (D.45) tends to one, and

$$R^{\text{Mer}} \to \frac{\boldsymbol{n}_e q^2}{m_e\,\omega(\omega + \mathrm{i}\gamma_{\text{rel}})}, \qquad \varepsilon_{\text{Mer}} \to 1 - \frac{\omega_p^2}{\omega(\omega + \mathrm{i}\gamma_{\text{rel}})} = 1 + \frac{\mathrm{i}\sigma_{\text{cond}}(\omega)}{\omega},$$
$$\sigma_{\text{cond}}(\omega) = \frac{\omega_p^2}{\gamma_{\text{rel}} - \mathrm{i}\omega},$$

with $\omega_p^2 = e^2\boldsymbol{n}_e/m_e$. This is the Drude dielectric function, with the dc conductivity $\omega_p^2/\gamma_{\text{rel}}$, in the relation between ε and σ_{cond} of Appendix A.7. The naive substitution has $1 - \omega_p^2/(\omega + \mathrm{i}\gamma_{\text{rel}})^2$, which tends to $1 + \omega_p^2/\gamma_{\text{rel}}^2$ as $\omega \to 0$: no pole, no dc current, and an electron gas that behaves as an insulator.

Perfect screening. A metal screens a static charge completely at long wavelength, $\varepsilon(\boldsymbol{q} \to 0, 0) \to \infty$ (Appendix C.5). Mermin's function inherits it from the static limit: $\varepsilon_{\text{Mer}}(\boldsymbol{q}, 0) = \varepsilon_{\text{Lin}}(\boldsymbol{q}, 0) = 1 + k_{\text{TF}}^2/q^2$ as $q \to 0$, by Eq. (D.41). The naive substitution leads to $\varepsilon(\boldsymbol{q} \to 0, 0) = 1 - v_{\text{C}} R^0(\boldsymbol{q}, \mathrm{i}\gamma_{\text{rel}}) \to 1 + \omega_p^2/\gamma_{\text{rel}}^2$, finite, the static dielectric constant of the same insulator.

The f-sum rule does not distinguish the two. At large ω, $R^{\text{Mer}} \to \boldsymbol{n}_e q^2/m_e\omega^2$, since $(1 + \mathrm{i}\gamma_{\text{rel}}/\omega)/(\omega + \mathrm{i}\gamma_{\text{rel}})^2 = 1/[\omega(\omega + \mathrm{i}\gamma_{\text{rel}})]$ and the denominator tends to one, and the naive substitution has the same leading tail. Both are analytic in the upper half plane, and by Eq. (C.37) both obey $\int_0^\infty \mathrm{d}\omega\,\omega\,\operatorname{Im} R = -\pi\boldsymbol{n}_e q^2/2m_e$. The f-sum rule constrains the first frequency moment, a property of the high-frequency response, while conservation is a property at low frequency and long wavelength, where the collisions act.

D.10 Imaginary-time formalism

This section supplies the steps behind the imaginary-time results of Sect. 4.7: the antiperiodicity of the Matsubara Green's function and its Fourier series, the occupation as a frequency sum, and the response of independent electrons written as a sum over a product of two Green's functions and evaluated. The residue formulas for Matsubara sums, Eq. (C.42), and the uniqueness of the analytic continuation are derived in Appendix C.5, and we use them here without repeating the arguments. The notation is that of Sect. 4.7: operators evolve in imaginary time with $H_\mu = H - \mu_{\text{ch}}\hat{N}_e$, $O(\tau) = \mathrm{e}^{\tau H_\mu} O\,\mathrm{e}^{-\tau H_\mu}$, and averages are taken in $\hat{\varrho} = \mathrm{e}^{-\beta H_\mu}/\mathcal{Z}$.

Antiperiodicity and the Fourier series

Take $0 < \tau < \beta$, so that $-\beta < \tau - \beta < 0$. At the negative argument the time ordering in Eq. (4.86) puts $c_{\boldsymbol{k}}^\dagger$ on the left with a minus sign, and with $c_{\boldsymbol{k}}(\tau-\beta) = \mathrm{e}^{(\tau-\beta)H_\mu} c_{\boldsymbol{k}}\, \mathrm{e}^{-(\tau-\beta)H_\mu}$,

$$\mathcal{Z}\,\mathsf{G}(\boldsymbol{k},\tau-\beta) = \mathrm{Tr}\big[\mathrm{e}^{-\beta H_\mu}\, c_{\boldsymbol{k}}^\dagger\, \mathrm{e}^{(\tau-\beta)H_\mu} c_{\boldsymbol{k}}\, \mathrm{e}^{(\beta-\tau)H_\mu}\big] = \mathrm{Tr}\big[\mathrm{e}^{-\tau H_\mu} c_{\boldsymbol{k}}^\dagger\, \mathrm{e}^{\tau H_\mu}\, \mathrm{e}^{-\beta H_\mu} c_{\boldsymbol{k}}\big],$$

where the cyclic property of the trace moved the last factor $\mathrm{e}^{(\beta-\tau)H_\mu}$ to the front, and it combined with $\mathrm{e}^{-\beta H_\mu}$ into $\mathrm{e}^{-\tau H_\mu}$. Cycling $\mathrm{e}^{-\beta H_\mu} c_{\boldsymbol{k}}$ to the front in the same way gives $\mathrm{Tr}[\mathrm{e}^{-\beta H_\mu} c_{\boldsymbol{k}}\, \mathrm{e}^{-\tau H_\mu} c_{\boldsymbol{k}}^\dagger\, \mathrm{e}^{\tau H_\mu}]$, and moving $\mathrm{e}^{\tau H_\mu}$ to the front, where it commutes with $\mathrm{e}^{-\beta H_\mu}$, gives $\mathrm{Tr}[\mathrm{e}^{-\beta H_\mu} c_{\boldsymbol{k}}(\tau)\, c_{\boldsymbol{k}}^\dagger] = -\mathcal{Z}\,\mathsf{G}(\boldsymbol{k},\tau)$. Hence

$$\mathsf{G}(\boldsymbol{k},\tau-\beta) = -\mathsf{G}(\boldsymbol{k},\tau), \qquad 0 < \tau < \beta. \tag{D.48}$$

For two bosonic operators the time ordering carries no sign, and the same steps give $R_{ab}(\boldsymbol{q},\tau-\beta) = +R_{ab}(\boldsymbol{q},\tau)$. The traces converge for $-\beta < \tau < \beta$, where $\mathrm{e}^{-\beta H_\mu}\mathrm{e}^{\pm\tau H_\mu}$ still decays at large energies, and this is the whole range on which the functions are defined.

A function on $-\beta < \tau < \beta$ that changes sign under $\tau \to \tau - \beta$ has period 2β, and its Fourier series runs over the frequencies $n\pi/\beta = n\pi T$ with integer n. The coefficient of $\mathrm{e}^{-\mathrm{i}n\pi T\tau}$ is $(2\beta)^{-1}\int_{-\beta}^{\beta} \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}n\pi T\tau}\mathsf{G}(\boldsymbol{k},\tau)$. Splitting the integral at $\tau = 0$ and shifting the negative half by β with Eq. (D.48), it is $(2\beta)^{-1}[1-\mathrm{e}^{-\mathrm{i}n\pi}]\int_0^\beta \mathrm{d}\tau\, \mathrm{e}^{\mathrm{i}n\pi T\tau}\mathsf{G}(\boldsymbol{k},\tau)$. The bracket is 0 for even n and 2 for odd $n = 2j+1$, which leaves the fermionic frequencies ω_j^{F} and the pair Eq. (4.87). For a periodic function the bracket is $1+\mathrm{e}^{-\mathrm{i}n\pi}$, and only the bosonic frequencies ω_j^{B} survive.

The occupation as a frequency sum

By Eq. (4.87) at $\tau = -0^+$, $\langle c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}\rangle = \mathsf{G}(\boldsymbol{k},-0^+) = T\sum_j \mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}}0^+}\mathsf{G}(\boldsymbol{k},\mathrm{i}\omega_j^{\mathrm{F}})$, which is Eq. (4.89). The factor $\mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}}0^+}$ cannot be dropped. Since $\mathsf{G} \to 1/(\mathrm{i}\omega_j^{\mathrm{F}})$ at large j, the sum without it converges only conditionally, and its symmetric value is $\frac{1}{2}(\mathsf{G}(\boldsymbol{k},0^+)+\mathsf{G}(\boldsymbol{k},0^-))$, the average of the two sides of the jump at $\tau = 0$, which is $\langle c^\dagger c\rangle - \frac{1}{2}$.

To evaluate the sum, let $g(z) = \mathrm{e}^{z0^+}\mathsf{G}(\boldsymbol{k},z)$ with $\mathsf{G}(\boldsymbol{k},z) = \int \mathrm{d}\omega'\, \mathsf{A}(\boldsymbol{k},\omega')/(z-\omega')$. Near each pole $\mathrm{i}\omega_j^{\mathrm{F}}$ of $n_{\mathrm{F}}(z) = 1/(\mathrm{e}^{\beta z}+1)$ the residue is $-T$ (Appendix C.5), so

$$T\sum_j g(\mathrm{i}\omega_j^{\mathrm{F}}) = -\oint_C \frac{\mathrm{d}z}{2\pi\mathrm{i}}\, n_{\mathrm{F}}(z)\, g(z),$$

with C the small counterclockwise circles of Fig. 4.13a. The function g is analytic off the real axis. In the upper half plane the circles can therefore be merged into the boundary of the half plane, traversed counterclockwise: the line $\omega + \mathrm{i}0^+$ from $-\infty$ to $+\infty$ and an arc at infinity. In the lower half plane they merge into the line $\omega - \mathrm{i}0^+$ from $+\infty$ to $-\infty$ and another arc. On the arcs $|g| \sim 1/|z|$, while $n_{\mathrm{F}}(z)\,\mathrm{e}^{z0^+}$ decays exponentially for $\mathrm{Re}\, z \to +\infty$ through n_{F} and for $\mathrm{Re}\, z \to -\infty$ through the convergence factor, so the arcs contribute nothing. What remains is

$$T\sum_j \mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}}0^+}\mathsf{G}(\boldsymbol{k},\mathrm{i}\omega_j^{\mathrm{F}}) = -\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega}{2\pi\mathrm{i}}\, n_{\mathrm{F}}(\omega)\big[\mathsf{G}(\boldsymbol{k},\omega+\mathrm{i}0^+)-\mathsf{G}(\boldsymbol{k},\omega-\mathrm{i}0^+)\big] = \int \mathrm{d}\omega\, n_{\mathrm{F}}(\omega)\,\mathsf{A}(\boldsymbol{k},\omega), \tag{D.49}$$

where the jump $\mathsf{G}(\boldsymbol{k},\omega+\mathrm{i}0^+) - \mathsf{G}(\boldsymbol{k},\omega-\mathrm{i}0^+) = -2\pi\mathrm{i}\mathsf{A}(\boldsymbol{k},\omega)$ follows from the Sokhotski–Plemelj formula Eq. (2.58) applied under the integral that defines $\mathsf{G}(\boldsymbol{k},z)$. This is Eq. (4.78), obtained without the Lehmann form. For a free electron, $\mathsf{A}_0 = \delta(\omega-\xi_{\boldsymbol{k}})$ and the sum is $n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$; equivalently, Eq. (C.42) with the single pole of g at $z = \xi_{\boldsymbol{k}}$ gives the residue $n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$ directly.

The bubble of independent electrons

Take free electrons, $H_\mu = \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}}$ with the spin in the label, and $O_a = O_b = n_e(\boldsymbol{q}) = \sum_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}+\boldsymbol{q}} c_{\boldsymbol{k}}$ (Eq. (2.38)), with $\boldsymbol{q} \neq 0$. In imaginary time $c_{\boldsymbol{k}}(\tau) = \mathrm{e}^{-\xi_{\boldsymbol{k}}\tau} c_{\boldsymbol{k}}$ and $c^\dagger_{\boldsymbol{k}}(\tau) = \mathrm{e}^{+\xi_{\boldsymbol{k}}\tau} c^\dagger_{\boldsymbol{k}}$, by the steps that gave the real-time form in Sect. 2.3. For $0 < \tau < \beta$, with $n_e(\boldsymbol{q})^\dagger = \sum_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}+\boldsymbol{q}}$,

$$R^0(\boldsymbol{q},\tau) = -\frac{1}{\mathscr{V}} \sum_{\boldsymbol{k},\boldsymbol{k}'} \mathrm{e}^{(\xi_{\boldsymbol{k}}-\xi_{\boldsymbol{k}+\boldsymbol{q}})\tau} \, \langle c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}+\boldsymbol{q}} \, c^\dagger_{\boldsymbol{k}'+\boldsymbol{q}} c_{\boldsymbol{k}'} \rangle .$$

The Gibbs state of independent electrons is a product over orbitals (Sect. 2.5), and an average of operators is nonzero only if every orbital that loses an electron also gains one. With $\boldsymbol{q} \neq 0$ that requires $\boldsymbol{k}' = \boldsymbol{k}$, and then $\langle c^\dagger_{\boldsymbol{k}} c_{\boldsymbol{k}+\boldsymbol{q}} c^\dagger_{\boldsymbol{k}+\boldsymbol{q}} c_{\boldsymbol{k}} \rangle = n_{\boldsymbol{k}}(1 - n_{\boldsymbol{k}+\boldsymbol{q}})$ with $n_{\boldsymbol{k}} = n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$, as in Eq. (2.70). The transform is

$$\int_0^\beta \mathrm{d}\tau \, \mathrm{e}^{(\mathrm{i}\omega_j^{\mathrm{B}} + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}})\tau} = \frac{\mathrm{e}^{\beta(\xi_{\boldsymbol{k}}-\xi_{\boldsymbol{k}+\boldsymbol{q}})} - 1}{\mathrm{i}\omega_j^{\mathrm{B}} + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}}} .$$

Since $n_{\boldsymbol{k}} \mathrm{e}^{\beta\xi_{\boldsymbol{k}}} = 1 - n_{\boldsymbol{k}}$ and $(1 - n_{\boldsymbol{k}+\boldsymbol{q}})\mathrm{e}^{-\beta\xi_{\boldsymbol{k}+\boldsymbol{q}}} = n_{\boldsymbol{k}+\boldsymbol{q}}$, the numerator times $n_{\boldsymbol{k}}(1 - n_{\boldsymbol{k}+\boldsymbol{q}})$ is $(1 - n_{\boldsymbol{k}})n_{\boldsymbol{k}+\boldsymbol{q}} - n_{\boldsymbol{k}}(1 - n_{\boldsymbol{k}+\boldsymbol{q}}) = n_{\boldsymbol{k}+\boldsymbol{q}} - n_{\boldsymbol{k}}$, and

$$R^0(\boldsymbol{q}, \mathrm{i}\omega_j^{\mathrm{B}}) = \frac{1}{\mathscr{V}} \sum_{\boldsymbol{k}} \frac{n_{\mathrm{F}}(\xi_{\boldsymbol{k}}) - n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}})}{\mathrm{i}\omega_j^{\mathrm{B}} + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}}} , \tag{D.50}$$

the right-hand side of Eq. (4.93). Continued to $\omega + \mathrm{i}0^+$, its imaginary part is $-(\pi/\mathscr{V}) \sum_{\boldsymbol{k}} [n_{\boldsymbol{k}} - n_{\boldsymbol{k}+\boldsymbol{q}}] \, \delta(\omega - \xi_{\boldsymbol{k}+\boldsymbol{q}} + \xi_{\boldsymbol{k}})$, which by Eq. (4.7) is the difference of the independent-electron transition sums of Eq. (2.70).

The same result comes out of a frequency sum. Consider

$$T \sum_{j'} \mathsf{G}_0(\boldsymbol{k}, \mathrm{i}\omega_{j'}^{\mathrm{F}}) \, \mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q}, \mathrm{i}\omega_{j'}^{\mathrm{F}} + \mathrm{i}\omega_j^{\mathrm{B}}) = T \sum_{j'} \frac{1}{(\mathrm{i}\omega_{j'}^{\mathrm{F}} - \xi_{\boldsymbol{k}})(\mathrm{i}\omega_{j'}^{\mathrm{F}} + \mathrm{i}\omega_j^{\mathrm{B}} - \xi_{\boldsymbol{k}+\boldsymbol{q}})} .$$

The summand falls off as $1/|z|^2$, so no convergence factor is needed, and the fermionic form of Eq. (C.42) applies with $g(z) = [(z - \xi_{\boldsymbol{k}})(z + \mathrm{i}\omega_j^{\mathrm{B}} - \xi_{\boldsymbol{k}+\boldsymbol{q}})]^{-1}$. Its pole at $z = \xi_{\boldsymbol{k}}$ has the residue $n_{\mathrm{F}}(\xi_{\boldsymbol{k}})/(\xi_{\boldsymbol{k}} + \mathrm{i}\omega_j^{\mathrm{B}} - \xi_{\boldsymbol{k}+\boldsymbol{q}})$. Its pole at $z = \xi_{\boldsymbol{k}+\boldsymbol{q}} - \mathrm{i}\omega_j^{\mathrm{B}}$ has the residue $n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}} - \mathrm{i}\omega_j^{\mathrm{B}})/(\xi_{\boldsymbol{k}+\boldsymbol{q}} - \mathrm{i}\omega_j^{\mathrm{B}} - \xi_{\boldsymbol{k}})$, and since $\mathrm{e}^{-\mathrm{i}\beta\omega_j^{\mathrm{B}}} = 1$ the Fermi function is $n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}})$. The sum of the two residues is

$$T \sum_{j'} \mathsf{G}_0(\boldsymbol{k}, \mathrm{i}\omega_{j'}^{\mathrm{F}}) \, \mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q}, \mathrm{i}\omega_{j'}^{\mathrm{F}} + \mathrm{i}\omega_j^{\mathrm{B}}) = \frac{n_{\mathrm{F}}(\xi_{\boldsymbol{k}}) - n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}})}{\mathrm{i}\omega_j^{\mathrm{B}} + \xi_{\boldsymbol{k}} - \xi_{\boldsymbol{k}+\boldsymbol{q}}} , \tag{D.51}$$

term by term the summand of Eq. (D.50), which proves Eq. (4.93). Two features recur in every diagram of Sect. 4.8. The bosonic frequency was simplified in the Fermi function before any continuation, $n_{\mathrm{F}}(\xi - \mathrm{i}\omega_j^{\mathrm{B}}) = n_{\mathrm{F}}(\xi)$; doing it afterwards would give a function of ω that is not analytic. And the occupation factors appear as differences, $n_{\mathrm{F}}(\xi_{\boldsymbol{k}}) - n_{\mathrm{F}}(\xi_{\boldsymbol{k}+\boldsymbol{q}})$, so that a transition between two occupied or two empty orbitals drops out, which is the Pauli blocking of Sect. 2.5.

The continuation

Eq. (D.50) is a ratio of finite sums in which $\mathrm{i}\omega_j^{\mathrm{B}}$ appears only in the denominators. Putting z in its place gives a function that is analytic off the real axis, takes the values Eq. (D.50) at every $\mathrm{i}\omega_j^{\mathrm{B}}$ with $j > 0$, and falls off as $1/|z|$. By the uniqueness argument of Appendix C.5 [27]

it is the only such function, and $z = \omega + \mathrm{i}0^+$ gives the retarded response. The general case is Eq. (4.91). At $j = 0$ a degenerate pair, $E_f = E_i$, contributes the term $-\beta p_i \langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle/\mathcal{V}$ instead of a ratio, a value at one point that no function falling off at infinity can carry over to the real axis. For data known only numerically at the Matsubara points, the continuation is no longer a substitution, and Appendix C.5 describes the difficulty.

D.11 Wick's theorem and the Feynman rules

Every diagram of Sect. 4.8 is one term of an expansion of the Green's function in powers of the interaction. Each term is an average of a long string of free creation and annihilation operators, and Wick's theorem reduces that average to a sum of products of free Green's functions, one product for each way of pairing the operators. This section derives the expansion in imaginary time, where it is exact at any temperature, and at $T = 0$ in real time, where it rests on the adiabatic theorem of Gell-Mann and Low [28]. It then proves Wick's theorem [29] by Gaudin's method [30], shows that the disconnected terms cancel, and derives the Feynman rules for the Coulomb interaction and for the exchange of a boson, with every sign and every combinatorial factor. The conventions for G are those of Sect. 4.7: a bare G with a real argument is the time-ordered function, and with an imaginary argument the Matsubara function. The treatment follows Refs. [3, Chaps. 3 and 7] and [2].

The expansion in imaginary time

Split $H_\mu = H_0 + H_{\mathrm{int}}$, with $H_0 = \sum_{\boldsymbol{k}} \xi_{\boldsymbol{k}} c_{\boldsymbol{k}}^\dagger c_{\boldsymbol{k}}$, plus $\sum_{\boldsymbol{Q}} \omega_{\boldsymbol{Q}} d_{\boldsymbol{Q}}^\dagger d_{\boldsymbol{Q}}$ when bosons are present. Operators in the interaction picture evolve with H_0 alone, $O_{\mathrm{I}}(\tau) = \mathrm{e}^{\tau H_0} O\, \mathrm{e}^{-\tau H_0}$, so that $c_{\boldsymbol{k},\mathrm{I}}(\tau) = \mathrm{e}^{-\xi_{\boldsymbol{k}}\tau} c_{\boldsymbol{k}}$ and $c_{\boldsymbol{k},\mathrm{I}}^\dagger(\tau) = \mathrm{e}^{\xi_{\boldsymbol{k}}\tau} c_{\boldsymbol{k}}^\dagger$. Define $U_{\mathrm{I}}(\tau, 0) = \mathrm{e}^{\tau H_0}\mathrm{e}^{-\tau H_\mu}$. Its derivative is $\mathrm{e}^{\tau H_0}(H_0 - H_\mu)\mathrm{e}^{-\tau H_\mu} = -H_{\mathrm{int,I}}(\tau)\, U_{\mathrm{I}}(\tau, 0)$, which has the form of Eq. (2.49) with $\mathrm{i}\,\mathrm{d}t$ replaced by $-\mathrm{d}\tau$. Integrating and iterating as in Eq. (2.50),

$$\begin{aligned} U_{\mathrm{I}}(\tau, 0) &= \sum_{n=0}^{\infty} (-1)^n \int_0^\tau \mathrm{d}\tau_1 \int_0^{\tau_1} \mathrm{d}\tau_2 \cdots \int_0^{\tau_{n-1}} \mathrm{d}\tau_n\, H_{\mathrm{int,I}}(\tau_1) \cdots H_{\mathrm{int,I}}(\tau_n) \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_0^\tau \mathrm{d}\tau_1 \cdots \mathrm{d}\tau_n\, \mathrm{T}_\tau \big[H_{\mathrm{int,I}}(\tau_1) \cdots H_{\mathrm{int,I}}(\tau_n) \big]. \end{aligned}$$

In the second form each integral runs over the whole interval. The $n!$ orderings of the times give the same ordered product, since T_τ restores the order and H_{int} contains fermion operators in pairs, so that no sign arises. The same steps give $U_{\mathrm{I}}(\tau, \tau') = \mathrm{e}^{\tau H_0}\mathrm{e}^{-(\tau-\tau')H_\mu}\mathrm{e}^{-\tau' H_0}$ as the ordered exponential over $[\tau', \tau]$.

The Boltzmann factor and the Heisenberg operators of Sect. 4.7 are both expressed through U_{I}. The Boltzmann factor is $\mathrm{e}^{-\beta H_\mu} = \mathrm{e}^{-\beta H_0} U_{\mathrm{I}}(\beta, 0)$, and the full imaginary-time operator is

$$c_{\boldsymbol{k}}(\tau) = \mathrm{e}^{\tau H_\mu} c_{\boldsymbol{k}} \mathrm{e}^{-\tau H_\mu} = U_{\mathrm{I}}(\tau, 0)^{-1}\, c_{\boldsymbol{k},\mathrm{I}}(\tau)\, U_{\mathrm{I}}(\tau, 0).$$

For $0 < \tau < \beta$, inserting both into Eq. (4.86) and using $U_{\mathrm{I}}(\beta, 0) U_{\mathrm{I}}(\tau, 0)^{-1} = U_{\mathrm{I}}(\beta, \tau)$,

$$\mathrm{Tr}\big[\mathrm{e}^{-\beta H_\mu} c_{\boldsymbol{k}}(\tau) c_{\boldsymbol{k}}^\dagger\big] = \mathrm{Tr}\big[\mathrm{e}^{-\beta H_0}\, U_{\mathrm{I}}(\beta, \tau)\, c_{\boldsymbol{k},\mathrm{I}}(\tau)\, U_{\mathrm{I}}(\tau, 0)\, c_{\boldsymbol{k},\mathrm{I}}^\dagger(0)\big].$$

Every factor on the right stands to the left of every factor with an earlier imaginary time, so the product is the time-ordered product of $U_{\mathrm{I}}(\beta, 0)$ with the two operators. The case $\tau < 0$ works the same way, and the partition function is $\mathcal{Z} = \mathcal{Z}_0 \langle U_{\mathrm{I}}(\beta, 0)\rangle_0$, where $\langle \cdots \rangle_0 = \mathrm{Tr}[\mathrm{e}^{-\beta H_0} \cdots]/\mathcal{Z}_0$ is the average in the free grand-canonical state. Hence

$$\mathsf{G}(\boldsymbol{k}, \tau) = -\frac{\big\langle \mathrm{T}_\tau\, U_{\mathrm{I}}(\beta, 0)\, c_{\boldsymbol{k},\mathrm{I}}(\tau)\, c_{\boldsymbol{k},\mathrm{I}}^\dagger(0)\big\rangle_0}{\big\langle U_{\mathrm{I}}(\beta, 0)\big\rangle_0}. \tag{D.52}$$

Nothing has been assumed beyond the convergence of the series. The density response Eq. (4.90) has the same expansion with $n_e(\boldsymbol{q})^\dagger$ and $n_e(\boldsymbol{q})$ in place of $c_{\boldsymbol{k}}$ and $c_{\boldsymbol{k}}^\dagger$, and a factor $1/\mathscr{V}$.

Inside $H_{\mathrm{int,I}}(\tau)$ creation and annihilation operators share one time, and the ordering symbol needs a rule for them. We keep them in the order in which they are written in H_{int}, creation operators to the left; equivalently, every creation operator in H_{int} is taken at $\tau + 0^+$.

The expansion at zero temperature

At $T = 0$ the time-ordered function Eq. (4.79) is an average in the interacting ground state $|0\rangle$, which is not known. Switch the interaction on and off slowly, $H_{\mathrm{int}}\mathrm{e}^{-\eta_{\mathrm{b}}|t|}$ with $\eta_{\mathrm{b}} \to 0^+$, and start from the ground state $|\Phi_0\rangle$ of H_0, the Fermi sea. The theorem of Gell-Mann and Low [28] states that $U_{\mathrm{I}}(0,-\infty)|\Phi_0\rangle / \langle\Phi_0|U_{\mathrm{I}}(0,-\infty)|\Phi_0\rangle$, with U_{I} the real-time operator of Eq. (2.50) with $H' = H_{\mathrm{int}}$, tends to an eigenstate of H_μ. It is the ground state when that state is nondegenerate and is reached from $|\Phi_0\rangle$ without a level crossing as the interaction is switched on, which we assume. Switching off toward $t = +\infty$ leads to the same state, up to a phase, so $\langle\Phi_0|U_{\mathrm{I}}(\infty,0)$ is proportional to $\langle 0|$. With $c_{\boldsymbol{k}}(t) = U_{\mathrm{I}}(t,0)^{-1} c_{\boldsymbol{k},\mathrm{I}}(t)\, U_{\mathrm{I}}(t,0)$ the steps of the previous paragraph go through with real times, and the unknown phases and normalizations cancel in the ratio:

$$\mathsf{G}(\boldsymbol{k},t) = -\mathrm{i}\,\frac{\langle\Phi_0|\mathrm{T}\, U_{\mathrm{I}}(\infty,-\infty)\, c_{\boldsymbol{k},\mathrm{I}}(t)\, c_{\boldsymbol{k},\mathrm{I}}^\dagger(0)|\Phi_0\rangle}{\langle\Phi_0|U_{\mathrm{I}}(\infty,-\infty)|\Phi_0\rangle}, \tag{D.53}$$

with $U_{\mathrm{I}}(\infty,-\infty) = \sum_n (-\mathrm{i})^n (n!)^{-1} \int \mathrm{d}t_1 \cdots \mathrm{d}t_n\, \mathrm{T}[H_{\mathrm{int,I}}(t_1)\cdots H_{\mathrm{int,I}}(t_n)]$. At $T > 0$ no expansion of this form exists for real times, since a thermal state is not reached adiabatically from a single free state. A real-time expansion at finite temperature needs a time contour that runs forward and back, and the imaginary-time form Eq. (D.52) replaces it here (Sect. 4.7).

Wick's theorem

Let $A_1, \ldots, A_{2n}$ be operators of the free problem, each a creation or an annihilation operator of one mode, multiplied by a number. In imaginary time $c_{\boldsymbol{k},\mathrm{I}}(\tau)$ is of this kind, and so is $c_{\boldsymbol{k},\mathrm{I}}(t)$ in real time. For fermions, the claim is

$$\langle A_1 A_2 \cdots A_{2n}\rangle_0 = \sum_{j=2}^{2n} (-1)^j \,\langle A_1 A_j\rangle_0 \,\langle A_2 \cdots \widehat{A_j} \cdots A_{2n}\rangle_0, \tag{D.54}$$

where the hat removes A_j from the string. Applied repeatedly, it writes the average as a sum over all ways of splitting the $2n$ operators into pairs. Each pairing contributes the product of its pair averages, with the pair kept in its original order, and the sign of the permutation that brings the members of every pair next to each other. An average of an odd number of operators vanishes, and so does a pair average unless it holds one creation and one annihilation operator of the same mode.

The proof uses only the anticommutators and the cyclic property of the trace [30]. Move A_1 to the right end, one step at a time. Each step past A_j replaces $A_1 A_j$ by $\{A_1, A_j\} - A_j A_1$, and the anticommutator is a number. After $2n-1$ steps,

$$A_1 A_2 \cdots A_{2n} = \sum_{j=2}^{2n} (-1)^j \{A_1, A_j\}\, A_2 \cdots \widehat{A_j} \cdots A_{2n} - A_2 \cdots A_{2n} A_1.$$

In the average of the last term, the trace moves A_1 back to the front, past the Boltzmann factor: $\mathrm{Tr}[\mathrm{e}^{-\beta H_0} A_2 \cdots A_{2n} A_1] = \mathrm{Tr}[A_1 \mathrm{e}^{-\beta H_0} A_2 \cdots A_{2n}]$. If A_1 annihilates a fermion of energy

ξ, then $A_1 \mathrm{e}^{-\beta H_0} = \mathrm{e}^{-\beta\xi}\mathrm{e}^{-\beta H_0} A_1$, because removing the fermion lowers H_0 by ξ; if it creates one, the factor is $\mathrm{e}^{+\beta\xi}$. Writing $\mathrm{e}^{-\beta\xi_1}$ for either case, with $\xi_1 = \xi$ for an annihilation and $\xi_1 = -\xi$ for a creation operator, the last term averages to $-\mathrm{e}^{-\beta\xi_1}\langle A_1 \cdots A_{2n}\rangle_0$, and

$$\langle A_1 \cdots A_{2n}\rangle_0 = \sum_{j=2}^{2n} (-1)^j \, \frac{\{A_1, A_j\}}{1+\mathrm{e}^{-\beta\xi_1}} \, \langle A_2 \cdots \widehat{A_j} \cdots A_{2n}\rangle_0 .$$

For $n = 1$ the same equation reads $\langle A_1 A_j\rangle_0 = \{A_1, A_j\}/(1+\mathrm{e}^{-\beta\xi_1})$; for example, $\langle c_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}}\rangle_0 = 1/(1+\mathrm{e}^{-\beta\xi_{\boldsymbol{k}}}) = 1 - n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$. Inserting this in the general case gives Eq. (D.54). For bosons every anticommutator becomes a commutator, no signs appear, and the denominator is $1-\mathrm{e}^{-\beta\xi_1}$, which gives $\langle d_{\boldsymbol{Q}} d^\dagger_{\boldsymbol{Q}}\rangle_0 = 1 + n_{\mathrm{B}}(\omega_{\boldsymbol{Q}})$. Fermion and boson operators commute and H_0 is a sum of a fermion and a boson part, and a mixed average factorizes into a fermion average and a boson average. Linear combinations of single-mode operators, such as $d_{\boldsymbol{Q}} + d^\dagger_{-\boldsymbol{Q}}$, are covered by linearity.

The expansions need the theorem for time-ordered strings and at $T = 0$. Inside T_τ the operators are first put in time order, at the cost of the sign of that permutation, and Eq. (D.54) is applied to the ordered string. Each pair then appears in time order, as $\langle \mathrm{T}_\tau A_i A_j\rangle_0$, and the sign of the time ordering combines with the sign of the pairing in the ordered string into the sign of the same pairing in the original string. Hence the time-ordered average is the sum over pairings of products of time-ordered pair averages, the contractions, with the sign of the pairing as written. The free contractions are, by Eq. (4.86) and Eq. (4.79),

$$\langle \mathrm{T}_\tau \, c_{\boldsymbol{k},\mathrm{I}}(\tau) \, c^\dagger_{\boldsymbol{k},\mathrm{I}}(\tau')\rangle_0 = -\mathsf{G}_0(\boldsymbol{k}, \tau-\tau'), \qquad \langle \Phi_0 | \mathrm{T} \, c_{\boldsymbol{k},\mathrm{I}}(t) \, c^\dagger_{\boldsymbol{k},\mathrm{I}}(t') | \Phi_0\rangle = \mathrm{i}\, \mathsf{G}_0(\boldsymbol{k}, t-t'). \quad \text{(D.55)}$$

At $T = 0$, the ground state $|\Phi_0\rangle$ of H_0 is the limit $\beta \to \infty$ of the free grand-canonical state, every average in Eq. (D.54) has a limit, and the theorem holds for $\langle \Phi_0 | \cdots | \Phi_0\rangle$, in real time as in imaginary time.

Disconnected terms cancel

At order n, each pairing in the numerator of Eq. (D.52) joins the two external operators to some set of m of the n factors H_{int}, the connected part, and joins the remaining $n-m$ factors only among themselves, the vacuum part. The value of the pairing is the product of the two parts. Since every H_{int} is even in fermion operators, its position under T_τ costs no sign, and the sign of the pairing is the product of the signs of the two parts. There are $n!/[m!\,(n-m)!]$ ways to choose which of the n identical, integrated factors form the connected part, and each gives the same value. With X_m the sum of connected pairings at order m, including the external operators, and Y_l the sum of all pairings of l factors H_{int} alone, the numerator is

$$\sum_n \frac{(-1)^n}{n!} \sum_{m=0}^{n} \frac{n!}{m!\,(n-m)!} X_m \, Y_{n-m} = \Big[\sum_m \frac{(-1)^m}{m!} X_m\Big]\Big[\sum_l \frac{(-1)^l}{l!} Y_l\Big].$$

The second factor is $\langle U_{\mathrm{I}}(\beta, 0)\rangle_0$, which cancels the denominator. Hence G is -1 times the sum of the connected pairings alone, the linked-cluster theorem. The same argument applies to Eq. (D.53) with $(-\mathrm{i})^n$ in place of $(-1)^n$.

Feynman rules for the Coulomb interaction

Take the electron gas of Eq. (2.47), whose interaction is

$$H_{\mathrm{int}} = \frac{1}{2\mathcal{V}} \sum_{\boldsymbol{k}\neq 0} v_{\mathrm{C}}(k) \sum_{\boldsymbol{p}\boldsymbol{p}' s s'} c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s} c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'} c_{\boldsymbol{p}'s'} c_{\boldsymbol{p}s}.$$

Under T_τ, with the creation operators at $\tau + 0^+$, the order of the four operators is fixed by the ordering symbol, and they may be regrouped with the sign of the permutation. Moving $c_{\boldsymbol{p}s}$ two places to the left costs no sign, and each factor is the product of two bilinears,

$$H_{\mathrm{int,I}}(\tau) = \frac{1}{2\mathscr{V}} \sum_{\boldsymbol{k}\neq 0} v_{\mathrm{C}}(k) \sum_{\boldsymbol{p}\boldsymbol{p}'ss'} \left[c^\dagger_{\boldsymbol{p}+\boldsymbol{k},s} c_{\boldsymbol{p}s}\right]\left[c^\dagger_{\boldsymbol{p}'-\boldsymbol{k},s'} c_{\boldsymbol{p}'s'}\right](\tau) \qquad (\text{inside } \mathrm{T}_\tau),$$

each bilinear a density component. We draw each factor as a dashed interaction line carrying the momentum $\boldsymbol{k}$, whose two ends are the bilinears, and each contraction Eq. (D.55) as a directed solid line from a creation operator to the annihilation operator it is paired with. At order n there are $2n$ bilinears and the two external operators, which makes $2n+1$ contractions.

The sign. Follow a fermion line from the external $c_{\boldsymbol{k}}(\tau)$ through the bilinears it visits to $c^\dagger_{\boldsymbol{k}}(0)$. Arrange the string so that the bilinears it visits stand in the order of the line, $c(\tau)\,[c^\dagger_1 c_1]\,[c^\dagger_2 c_2]\cdots[c^\dagger_m c_m]\,c^\dagger(0)$; moving whole bilinears costs no sign. The contractions $c(\tau)c^\dagger_1$, $c_1 c^\dagger_2$, …, $c_m c^\dagger(0)$ are then all between neighbors, the pairing sign is $+1$, and each contraction is $-\mathsf{G}_0$, which makes $(-1)^{m+1}$ for the $m+1$ propagators of the open line. A closed loop through m bilinears, $[c^\dagger_1 c_1]\cdots[c^\dagger_m c_m]$, is paired as $c_1 c^\dagger_2$, …, $c_{m-1}c^\dagger_m$ and finally $c_m c^\dagger_1$. All pairs are neighbors except the last. Bringing $c^\dagger_1$ from the front to the end passes $2m-1$ operators and costs a factor -1, in addition to $(-1)^m$ from the m propagators of the loop. With F closed loops, the $2n+1$ propagators therefore carry $(-1)^{2n+1}(-1)^F = -(-1)^F$. The definition Eq. (D.52) contributes -1 and the expansion $(-1)^n$. The total sign of a diagram of order n is $(-1)^n(-1)^F$.

The factors $1/n!$ *and* $1/2^n$. Relabeling the n integrated interaction factors among themselves, and exchanging the two ends of any interaction line, which is a relabeling $\boldsymbol{k}\to-\boldsymbol{k}$, $(\boldsymbol{p}s)\leftrightarrow(\boldsymbol{p}'s')$ under which v_{C} is unchanged, maps one pairing onto another with the same value. For a connected diagram with two external points the $n!\,2^n$ relabelings give $n!\,2^n$ different pairings, since the external points fix every line of the diagram. Summing over pairings is then the same as summing over topologically distinct diagrams with weight $n!\,2^n$, which cancels the $1/n!$ of the expansion and the $\frac{1}{2}$ in each H_{int}.

Frequencies. Each propagator is written as $\mathsf{G}_0(\boldsymbol{p},\tau_a-\tau_b) = T\sum_j \mathrm{e}^{-\mathrm{i}\omega^{\mathrm{F}}_j(\tau_a-\tau_b)}\mathsf{G}_0(\boldsymbol{p},\mathrm{i}\omega^{\mathrm{F}}_j)$ by Eq. (4.87). The four ends of an interaction line share one time τ_v. The integral $\int_0^\beta \mathrm{d}\tau_v$ of the product of their exponentials is β times a Kronecker delta that conserves frequency along the line, which therefore carries the bosonic difference of the two fermionic frequencies at either end. The transform $\int_0^\beta \mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}\omega^{\mathrm{F}}_j\tau}$ of the external time supplies one more β. Of the $2n+1$ factors T, $n+1$ are cancelled by these factors β, and n remain, one for each independent internal frequency. Momenta are conserved at each end of a line in the same way, and the n factors $1/\mathscr{V}$ go with the n independent internal momentum sums.

Collecting, the Matsubara Green's function $\mathsf{G}(\boldsymbol{k},\mathrm{i}\omega^{\mathrm{F}}_j)$ at order n is the sum over connected, topologically distinct diagrams with n interaction lines, each diagram evaluated as follows:

1. each solid line carrying $(\boldsymbol{p},\mathrm{i}\omega^{\mathrm{F}})$ is $\mathsf{G}_0(\boldsymbol{p},\mathrm{i}\omega^{\mathrm{F}}) = 1/(\mathrm{i}\omega^{\mathrm{F}}-\xi_{\boldsymbol{p}})$, and spin is conserved along it;

2. each interaction line carrying $(\boldsymbol{k},\mathrm{i}\omega^{\mathrm{B}})$ is $-v_{\mathrm{C}}(k)/\mathscr{V}$, independent of ω^{B}, with momentum and frequency conserved at both ends;

3. each closed fermion loop gives -1 and a sum over its spin;

4. each independent internal momentum is summed, and each independent internal frequency carries $T\sum_j$;

5. a solid line that begins and ends on the same interaction line carries the factor $\mathrm{e}^{\mathrm{i}\omega^{\mathrm{F}}0^+}$, from the creation operator at $\tau + 0^+$.

The self-energy $\Sigma(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}})$ is the sum of the diagrams that cannot be cut into two by cutting one solid line, with the two external propagators removed. The Dyson equation of Sect. 4.7 then sums all chains of them.

At first order the self-energy has a Hartree and a Fock diagram. In the Hartree diagram the interaction line carries $\boldsymbol{k} = 0$ into a closed loop; it is $(-1)(-1)\,[v_{\mathrm{C}}(0)/\mathcal{V}]$ $\sum_{\boldsymbol{p}s} T \sum_j \mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}}0^+} \mathsf{G}_0(\boldsymbol{p}, \mathrm{i}\omega_j^{\mathrm{F}}) = v_{\mathrm{C}}(0) N_e/\mathcal{V}$ by Eq. (4.89), the electrostatic potential of the uniform density, and it is absent in jellium, where the background removes $\boldsymbol{k} = 0$. In the Fock diagram the line carries $\boldsymbol{k} - \boldsymbol{p}$ and there is no loop:

$$\Sigma_{\mathrm{x}}(\boldsymbol{k}) = -\frac{1}{\mathcal{V}} \sum_{\boldsymbol{p}} v_{\mathrm{C}}(|\boldsymbol{k} - \boldsymbol{p}|)\, T \sum_j \mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{F}}0^+} \mathsf{G}_0(\boldsymbol{p}, \mathrm{i}\omega_j^{\mathrm{F}}) = -\frac{1}{\mathcal{V}} \sum_{\boldsymbol{p}} v_{\mathrm{C}}(|\boldsymbol{k} - \boldsymbol{p}|)\, n_{\mathrm{F}}(\xi_{\boldsymbol{p}}), \quad \text{(D.56)}$$

with $\boldsymbol{p}$ of the same spin as $\boldsymbol{k}$. At $T = 0$ this is the exchange self-energy of Eq. (D.17), and both signs agree with Appendix D.4: the Hartree term repels and exchange attracts. The rules are thus checked at first order against a calculation that used no diagrams.

Exchange of a boson

Let the electrons couple to a bosonic field with modes $\boldsymbol{Q}$ and energies $\omega_{\boldsymbol{Q}} = \omega_{-\boldsymbol{Q}}$,

$$H_{\mathrm{int}} = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\boldsymbol{k}\boldsymbol{Q}s} g(\boldsymbol{Q})\, c^\dagger_{\boldsymbol{k}+\boldsymbol{Q},s} c_{\boldsymbol{k}s}\, \varphi_{\boldsymbol{Q}}, \qquad \varphi_{\boldsymbol{Q}} = d_{\boldsymbol{Q}} + d^\dagger_{-\boldsymbol{Q}},$$

with $g(-\boldsymbol{Q}) = g(\boldsymbol{Q})^*$, so that H_{int} is Hermitian, since $\varphi^\dagger_{\boldsymbol{Q}} = \varphi_{-\boldsymbol{Q}}$. For phonons g becomes the vertex $g_{mn\nu}$ of Sect. 6.2 and $d_{\boldsymbol{Q}}$ becomes $b_{\boldsymbol{Q}\nu}$. The boson contraction defines the free boson propagator, $\mathsf{D}_0(\boldsymbol{Q}, \tau) = -\langle \mathrm{T}_\tau\, \varphi_{\boldsymbol{Q},\mathrm{I}}(\tau)\, \varphi_{-\boldsymbol{Q},\mathrm{I}}(0)\rangle_0$. For $0 < \tau < \beta$, with $d_{\boldsymbol{Q},\mathrm{I}}(\tau) = \mathrm{e}^{-\omega_{\boldsymbol{Q}}\tau} d_{\boldsymbol{Q}}$ and the averages of Eq. (2.69), it is $-[(1+n_{\mathrm{B}})\mathrm{e}^{-\omega_{\boldsymbol{Q}}\tau} + n_{\mathrm{B}}\mathrm{e}^{\omega_{\boldsymbol{Q}}\tau}]$ with $n_{\mathrm{B}} = n_{\mathrm{B}}(\omega_{\boldsymbol{Q}})$. Since $\mathrm{e}^{\mathrm{i}\omega_j^{\mathrm{B}}\beta} = 1$, the transform is $\int_0^\beta \mathrm{d}\tau\, \mathrm{e}^{(\mathrm{i}\omega_j^{\mathrm{B}} \mp \omega_{\boldsymbol{Q}})\tau} = (\mathrm{e}^{\mp\beta\omega_{\boldsymbol{Q}}} - 1)/(\mathrm{i}\omega_j^{\mathrm{B}} \mp \omega_{\boldsymbol{Q}})$, and with $(1+n_{\mathrm{B}})(1-\mathrm{e}^{-\beta\omega_{\boldsymbol{Q}}}) = 1$ and $n_{\mathrm{B}}(\mathrm{e}^{\beta\omega_{\boldsymbol{Q}}} - 1) = 1$,

$$\mathsf{D}_0(\boldsymbol{Q}, \mathrm{i}\omega_j^{\mathrm{B}}) = \frac{1}{\mathrm{i}\omega_j^{\mathrm{B}} - \omega_{\boldsymbol{Q}}} - \frac{1}{\mathrm{i}\omega_j^{\mathrm{B}} + \omega_{\boldsymbol{Q}}} = \frac{2\omega_{\boldsymbol{Q}}}{(\mathrm{i}\omega_j^{\mathrm{B}})^2 - \omega_{\boldsymbol{Q}}^2}. \quad \text{(D.57)}$$

The temperature has dropped out, as it did for G_0 in Sect. 4.7: thermal factors appear only after a frequency sum.

The interaction is odd in the boson operators, so only even orders $2m$ survive the boson average, and every pairing joins the $2m$ vertices into m boson lines. The counting runs as for the Coulomb case. The expansion gives $(-1)^{2m}/(2m)! = 1/(2m)!$, and the $(2m)!$ relabelings of the vertices, which include the exchange of the two ends of each boson line, cancel the factorial. The fermion signs are those of $2m$ bilinears, $-(-1)^F$, and the definition of G adds -1. Each boson line contributes $-\mathsf{D}_0$ from its contraction and $g(\boldsymbol{Q})g(-\boldsymbol{Q})/\mathcal{V} = |g(\boldsymbol{Q})|^2/\mathcal{V}$ from its two vertices. The total is $(-1)^m(-1)^F \prod_{\mathrm{lines}} |g|^2 \mathsf{D}_0/\mathcal{V}$, which is the Coulomb result with each line replaced:

$$-\frac{v_{\mathrm{C}}(k)}{\mathcal{V}} \;\longrightarrow\; -\frac{|g(\boldsymbol{Q})|^2\, \mathsf{D}_0(\boldsymbol{Q}, \mathrm{i}\omega^{\mathrm{B}})}{\mathcal{V}}. \quad \text{(D.58)}$$

The exchange of a boson is an interaction that depends on frequency. At zero frequency, $|g|^2 \mathsf{D}_0(\boldsymbol{Q}, 0) = -2|g|^2/\omega_{\boldsymbol{Q}}$, an attraction. Continued to real frequencies it is $2\omega_{\boldsymbol{Q}}|g|^2/(\omega^2 - \omega_{\boldsymbol{Q}}^2)$, attractive for $|\omega| < \omega_{\boldsymbol{Q}}$ and repulsive above. An electron polarizes the field, the polarization

lags behind it, and a second electron that arrives within a time $\sim 1/\omega_{\boldsymbol{Q}}$ is attracted to it. For phonons this is the attraction behind superconductivity (Sect. 7.2). The first diagram of the self-energy is the Fock diagram with a boson line,

$$\Sigma(\boldsymbol{k}, \mathrm{i}\omega_j^{\mathrm{F}}) = -\frac{1}{\mathscr{V}} \sum_{\boldsymbol{Q}} |g(\boldsymbol{Q})|^2 \, T \sum_{j'} \mathsf{D}_0(\boldsymbol{Q}, \mathrm{i}\omega_{j'}^{\mathrm{B}}) \, \mathsf{G}_0(\boldsymbol{k}-\boldsymbol{Q}, \mathrm{i}\omega_j^{\mathrm{F}} - \mathrm{i}\omega_{j'}^{\mathrm{B}}),$$

which Sect. 4.8 evaluates for an Einstein boson. Its Hartree partner carries $\boldsymbol{Q} = 0$ and is a static shift of all levels, which vanishes when $g(\boldsymbol{0}) = 0$.

The rules at zero temperature

With Eq. (D.53), the counting is the same with three changes: the expansion gives $(-\mathrm{i})^n$, each contraction is $\mathrm{i}\mathsf{G}_0$ by Eq. (D.55), and the definition of G carries $-\mathrm{i}$. The total factor of a diagram of order n is $-\mathrm{i}\,(-\mathrm{i})^n\,\mathrm{i}^{2n+1}(-1)^F = \mathrm{i}^n(-1)^F$. Transforming with $\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}$ and $\mathsf{G}_0(\boldsymbol{p}, t) = \int \mathrm{d}\omega\, (2\pi)^{-1} \mathrm{e}^{-\mathrm{i}\omega t} \mathsf{G}_0(\boldsymbol{p}, \omega)$ turns every time integral into $2\pi\delta$ of a frequency, and n integrals $\int \mathrm{d}\omega/2\pi$ remain. The rules for $\mathsf{G}(\boldsymbol{k}, \omega)$ are those listed above with the free time-ordered propagator $\mathsf{G}_0(\boldsymbol{p}, \omega) = 1/(\omega - \xi_{\boldsymbol{p}} + \mathrm{i}0^+ \operatorname{sgn} \xi_{\boldsymbol{p}})$ of Eq. (4.80), each interaction line $+\mathrm{i}v_{\mathrm{C}}(k)/\mathscr{V}$, and each internal frequency $\int \mathrm{d}\omega/2\pi$. The Fock diagram checks the signs once more: $\mathrm{i}\int \mathrm{d}\omega\, (2\pi)^{-1} \mathrm{e}^{\mathrm{i}\omega 0^+} \mathsf{G}_0(\boldsymbol{p}, \omega)$ is evaluated by closing the contour in the upper half plane, where the pole $\xi_{\boldsymbol{p}} + \mathrm{i}0^+$ of an occupied state lies, and it equals $\mathrm{i} \cdot \mathrm{i}\,\Theta(-\xi_{\boldsymbol{p}}) = -\Theta(-\xi_{\boldsymbol{p}})$. With the line $v_{\mathrm{C}}/\mathscr{V}$ this is Eq. (D.56) at $T = 0$. The time-ordered density response, $-(\mathrm{i}/\mathscr{V})\langle \mathrm{T}\, n_e(\boldsymbol{q}, t)^\dagger n_e(\boldsymbol{q}, 0)\rangle$, carries $-\mathrm{i}/\mathscr{V}$ in its definition, and its $2n+2$ contractions give the factor $(-\mathrm{i}/\mathscr{V})(-\mathrm{i})^n\,\mathrm{i}^{2n+2}(-1)^F = (\mathrm{i}/\mathscr{V})\,\mathrm{i}^n(-1)^F$: with the factor i^n in the lines, the response is $+\mathrm{i}/\mathscr{V}$ times its diagrams, where the imaginary-time rule has $-1/\mathscr{V}$. For the bubble this is $(\mathrm{i}/\mathscr{V})(-1)\sum_{\boldsymbol{k}\boldsymbol{s}} \int \mathrm{d}\omega\, (2\pi)^{-1} \mathsf{G}_0(\boldsymbol{k}, \omega) \mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q}, \omega+\Omega)$, whose frequency integral, done by closing the contour, gives the bubble Eq. (4.93) at $T = 0$ with time-ordered denominators. For a boson the same replacement Eq. (D.58) holds, with the time-ordered propagator $\mathsf{D}_0(\boldsymbol{Q}, t) = -\mathrm{i}\langle \Phi_0 | \mathrm{T}\, \varphi_{\boldsymbol{Q},\mathrm{I}}(t) \varphi_{-\boldsymbol{Q},\mathrm{I}}(0) | \Phi_0 \rangle = -\mathrm{i}\, \mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{Q}}|t|}$, whose transform is $\mathsf{D}_0(\boldsymbol{Q}, \omega) = 2\omega_{\boldsymbol{Q}}/(\omega^2 - \omega_{\boldsymbol{Q}}^2 + \mathrm{i}0^+)$. The imaginary-time rules are the ones used in practice at finite temperature, and the zero-temperature rules are their limit after the frequency sums have been done and continued to the real axis (Sect. 4.7).

D.12 The Bethe–Salpeter equation from the ladder

An electron lifted across the gap of silicon leaves a hole behind, and the two attract each other through the screened Coulomb interaction. In the bubble of Sects. 4.7 and 4.8 the electron and the hole propagate independently between the two density vertices. The ladder diagrams let them scatter off each other any number of times in between, and the Bethe–Salpeter equation of Sect. 4.9 is their sum. This section derives it with the rules of Appendix D.11. The first-order diagrams fix the exchange and direct kernels and their signs. A statically screened interaction then makes the sum close as a matrix equation in the space of transitions, and that equation becomes an eigenvalue problem for the exciton energies E_λ and envelopes $A^\lambda_{cv\boldsymbol{k}}$. Its eigenstates give the density response, the structure factor and the macroscopic dielectric function. The derivation follows the many-body treatments of Hanke and Sham [31] and of Strinati [32]; the first-principles form is that of Refs. [23, 33, 34].

Transitions and the pair propagator

Take a basis of orbitals $\psi_m(\boldsymbol{r})$ that diagonalize the one-body part of the problem, with energies ξ_m measured from μ_{ch} and the spin in the label. In a crystal m stands for a band,

a crystal momentum and a spin. The density operator is $n_e(\boldsymbol{q}) = \sum_{mn} \rho_{mn}(\boldsymbol{q})\, c_m^\dagger c_n$ with $\rho_{mn}(\boldsymbol{q}) = \langle m|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|n\rangle$, by Eq. (2.36). Each term moves one electron from n to m, and we call the pair (mn) a transition. The density response is then a combination of the correlation functions of the pair operators $c_m^\dagger c_n$. Define, in imaginary time,

$$\mathfrak{L}_{mn,m'n'}(\tau) = -\big\langle \mathrm{T}_\tau\, c_n^\dagger(\tau) c_m(\tau)\; c_{m'}^\dagger(0) c_{n'}(0)\big\rangle, \tag{D.59}$$

the pair propagator, written in fraktur; it is the two-particle correlation function with both creation and both annihilation times set equal in pairs. Since $n_e(\boldsymbol{q})^\dagger = \sum_{mn} \rho_{mn}(\boldsymbol{q})^* c_n^\dagger c_m$, the response of Eq. (4.90) with $O_a = n_e(\boldsymbol{q}+\boldsymbol{G})$ and $O_b = n_e(\boldsymbol{q}+\boldsymbol{G}')$ is

$$R_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}, \mathrm{i}\omega_j^{\mathrm{B}}) = \frac{1}{\mathscr{V}} \sum_{mn,m'n'} \rho_{mn}(\boldsymbol{q}+\boldsymbol{G})^*\, \mathfrak{L}_{mn,m'n'}(\mathrm{i}\omega_j^{\mathrm{B}})\, \rho_{m'n'}(\boldsymbol{q}+\boldsymbol{G}'), \tag{D.60}$$

where we take $\boldsymbol{q}$ not equal to a reciprocal-lattice vector, so that $\langle n_e(\boldsymbol{q}+\boldsymbol{G})\rangle = 0$ and only connected contractions contribute. Everything the target does to the pair is in $\mathfrak{L}$.

For independent quasiparticles, Wick's theorem pairs $c_m(\tau)$ with $c_{m'}^\dagger(0)$ and $c_{n'}(0)$ with $c_n^\dagger(\tau)$, which forces $m' = m$ and $n' = n$. The two contractions form one closed loop through the two pair operators, and by the counting of Appendix D.11 the result is $\mathsf{G}_m(\tau)\mathsf{G}_n(-\tau)$. Its transform is the frequency sum Eq. (D.51),

$$\mathfrak{L}^0_{mn,m'n'}(\mathrm{i}\omega_j^{\mathrm{B}}) = \delta_{mm'}\delta_{nn'}\, T \sum_{j'} \mathsf{G}_m(\mathrm{i}\omega_{j'}^{\mathrm{F}} + \mathrm{i}\omega_j^{\mathrm{B}})\, \mathsf{G}_n(\mathrm{i}\omega_{j'}^{\mathrm{F}}) = \delta_{mm'}\delta_{nn'}\, \frac{n_{\mathrm{F}}(\xi_n) - n_{\mathrm{F}}(\xi_m)}{\mathrm{i}\omega_j^{\mathrm{B}} - (\xi_m - \xi_n)}, \tag{D.61}$$

diagonal in transitions, with a pole at the transition energy $\xi_m - \xi_n$ and the Pauli factor $n_{\mathrm{F}}(\xi_n) - n_{\mathrm{F}}(\xi_m)$. With plane waves in Eq. (D.60) it is the bubble Eq. (4.93).

First-order contractions

The interaction is the Coulomb operator of Sect. 2.3 in the orbital basis. Under T_τ, with the creation operators at $\tau + 0^+$, it is a product of two bilinears, as in Appendix D.11,

$$H_{\mathrm{int}} = \frac{1}{2} \sum_{abcd} v_{ab,cd}\, [c_a^\dagger c_c]\, [c_b^\dagger c_d], \qquad v_{ab,cd} = \int \mathrm{d}^3r\, \mathrm{d}^3r'\, \psi_a^*(\boldsymbol{r})\psi_c(\boldsymbol{r})\, v_{\mathrm{C}}(\boldsymbol{r},\boldsymbol{r}')\, \psi_b^*(\boldsymbol{r}')\psi_d(\boldsymbol{r}'),$$

with $v_{\mathrm{C}}(\boldsymbol{r},\boldsymbol{r}') = e^2/(4\pi|\boldsymbol{r}-\boldsymbol{r}'|)$ the transform of Eq. (A.3); spin is conserved at each bilinear. The rules of Appendix D.11 were derived for plane waves, but their proof used only this bilinear structure, and they hold in any orbital basis with $v_{ab,cd}$ in place of $v_{\mathrm{C}}(k)/\mathscr{V}$. The expansion Eq. (D.52), applied to Eq. (D.59), gives at first order $+\int_0^\beta \mathrm{d}\tau_1\, \langle \mathrm{T}_\tau\, c_n^\dagger c_m(\tau)\, H_{\mathrm{int}}(\tau_1)\, c_{m'}^\dagger c_{n'}(0)\rangle_0$, connected part only. The two bilinears of H_{int} can be contracted with the external ones in three topologically different ways, which we call the exchange, the direct and the self-energy contractions.

In the exchange contraction, each bilinear of H_{int} closes a loop with one external pair operator: $[c_a^\dagger c_c]$ with $c_n^\dagger c_m(\tau)$, which needs $a = m$, $c = n$, and $[c_b^\dagger c_d]$ with $c_{m'}^\dagger c_{n'}(0)$, which needs $b = n'$, $d = m'$. Bilinears are even in fermion operators and can be regrouped at no cost, and the average factorizes into two free pair averages, each equal to $-\mathfrak{L}^0$. The product is $+\mathfrak{L}^0\mathfrak{L}^0$. The two choices of which bilinear goes with which end cancel the $\frac{1}{2}$, and the term is $+\mathfrak{L}^0_{mn}(\tau-\tau_1)\, v_{mn',nm'}\, \mathfrak{L}^0_{m'n'}(\tau_1)$, a convolution in τ. In the direct contraction, both bilinears sit on the single loop that runs through the two pair operators. One of them is on the electron line, which enters at $c_{m'}^\dagger$ and leaves at c_m, and requires $a = m$, $c = m'$. The other is on the hole line and requires $b = n'$, $d = n$. The diagram has the same four propagators and the same interaction line as the exchange contraction, but one closed loop instead of two, and by

the loop rule of Appendix D.11 it carries the opposite sign: $-\mathfrak{L}^0_{mn}(\tau-\tau_1)\,v_{mn',m'n}\,\mathfrak{L}^0_{m'n'}(\tau_1)$. In the self-energy contractions, both bilinears of $H_{\rm int}$ act on one propagator, or one of them closes a loop by itself; these are the Hartree and Fock self-energies of Appendix D.11 inserted into the electron or the hole line, and they shift the orbital energies.

The exchange and direct contractions define the kernels of the Bethe–Salpeter equation:

$$\begin{aligned} K^{\rm x}_{{\rm full},mn,m'n'} &= \int \mathrm{d}^3r\,\mathrm{d}^3r'\;\psi^*_m(\boldsymbol{r})\psi_n(\boldsymbol{r})\;v_{\rm C}(\boldsymbol{r},\boldsymbol{r}')\;\psi_{m'}(\boldsymbol{r}')\psi^*_{n'}(\boldsymbol{r}'),\\ K^{\rm d}_{mn,m'n'} &= \int \mathrm{d}^3r\,\mathrm{d}^3r'\;\psi^*_m(\boldsymbol{r})\psi_{m'}(\boldsymbol{r})\;W^{\rm scr}(\boldsymbol{r},\boldsymbol{r}')\;\psi^*_{n'}(\boldsymbol{r}')\psi_n(\boldsymbol{r}'). \end{aligned} \tag{D.62}$$

The exchange kernel $K^{\rm x}_{\rm full}$ carries the whole Coulomb interaction, including the long-range term that the transition-basis section below splits off. In it the pair annihilates at one point and is recreated at another: the electron and the hole of each transition sit at the same position, $\psi^*_m\psi_n$ at $\boldsymbol{r}$. It is the term of the random-phase approximation, and its name refers to the exchange of the electron and the hole between the two transitions. In the direct kernel $K^{\rm d}$ the electron stays at $\boldsymbol{r}$ and scatters from m' to m, while the hole stays at $\boldsymbol{r}'$ and scatters from n' to n. It is the Coulomb interaction between the two particles of the pair, written above already with the screened interaction that the next step supplies. A diagonal element $K^{\rm d}_{mn,mn}=\int\!\int|\psi_m(\boldsymbol{r})|^2W^{\rm scr}(\boldsymbol{r},\boldsymbol{r}')|\psi_n(\boldsymbol{r}')|^2$ is positive, and it enters with a minus sign. The electron–hole interaction is therefore an attraction, as it must be between charges of opposite sign.

The statically screened ladder

The Bethe–Salpeter equation keeps all diagrams built by repeating these two insertions, with the following choices, which define the approximation. The orbital energies ξ_m are the quasiparticle energies of Sect. 4.7, so that the self-energy contractions are summed into the propagators, with the quasiparticle widths neglected and $z_{\rm qp}$ set to one. A direct interaction line is screened: a chain of bubbles inserted into it turns $v_{\rm C}$ into $v_{\rm C}+v_{\rm C}Rv_{\rm C}=\varepsilon^{-1}v_{\rm C}=W^{\rm scr}$, the screened interaction of Sect. 4.8. An exchange line is not screened, since a bubble chain on it is already counted as repeated exchange. These choices are consistent with the GW self-energy of Sect. 4.8: the kernel is the change of the Hartree potential plus the GW self-energy when G changes, with the change of $W^{\rm scr}$ itself neglected [31, 32]. The Hartree term gives $K^{\rm x}_{\rm full}$ and the GW term gives $-K^{\rm d}$.

A screened direct line carries a frequency. In the direct diagram the electron line has the fermionic frequency $\mathrm{i}\omega^{\rm F}_{j_1}+\mathrm{i}\omega^{\rm B}_j$ on one side of the line and $\mathrm{i}\omega^{\rm F}_{j_2}+\mathrm{i}\omega^{\rm B}_j$ on the other, the hole line correspondingly, and the line is $W^{\rm scr}$ at the bosonic difference $\mathrm{i}\omega^{\rm F}_{j_1}-\mathrm{i}\omega^{\rm F}_{j_2}$. The two internal frequency sums are then coupled through the screened line, and $\mathfrak{L}$ does not close on itself: the pair splits into an electron and a hole at different times. The static approximation replaces $W^{\rm scr}(\mathrm{i}\omega^{\rm B})$ by $W^{\rm scr}(\omega=0)$ at all frequencies, an interaction that is instantaneous in τ. With it, the diagram with one direct line is

$$\begin{aligned} &T\sum_{j_1}\mathsf{G}_m(\mathrm{i}\omega^{\rm F}_{j_1}+\mathrm{i}\omega^{\rm B}_j)\mathsf{G}_n(\mathrm{i}\omega^{\rm F}_{j_1})\;(-K^{\rm d}_{mn,m'n'})\;T\sum_{j_2}\mathsf{G}_{m'}(\mathrm{i}\omega^{\rm F}_{j_2}+\mathrm{i}\omega^{\rm B}_j)\mathsf{G}_{n'}(\mathrm{i}\omega^{\rm F}_{j_2})\\ &\qquad=\mathfrak{L}^0_{mn}\,(-K^{\rm d}_{mn,m'n'})\,\mathfrak{L}^0_{m'n'}, \end{aligned}$$

since each sum is Eq. (D.51) on its own. The exchange line is instantaneous anyway and carries $\mathrm{i}\omega^{\rm B}_j$. Every further direct line repeats the step, and the sum of the ladder is the geometric series of Eq. (C.59),

$$\mathfrak{L}(\mathrm{i}\omega^{\rm B}_j)=\mathfrak{L}^0(\mathrm{i}\omega^{\rm B}_j)+\mathfrak{L}^0(\mathrm{i}\omega^{\rm B}_j)\,[K^{\rm x}_{\rm full}-K^{\rm d}]\,\mathfrak{L}(\mathrm{i}\omega^{\rm B}_j), \tag{D.63}$$

a matrix equation in the space of transitions at a single frequency. This is the Bethe–Salpeter equation in the static ladder approximation. The approximation is good when the energies that matter, the exciton binding and the spread of the transition energies near the gap, are small against the frequencies at which ε^{-1} changes, set in silicon by the plasma frequency of 16.6 eV (Appendix A.2). What it loses is the dynamical part of the screening, the retarded response of the electrons to the moving pair.

Spin structure of the kernel

For a spin-unpolarized target the orbitals come in pairs $m\uparrow$, $m\downarrow$ with the same spatial part. The exchange kernel couples a transition of either spin to a transition of either spin, because the density at $\boldsymbol{r}'$ sums both. The direct kernel conserves the spin of the electron and of the hole separately, and with $\psi_m^*\psi_n$ requiring equal spins it couples only transitions of the same spin. In the combination $(mn\uparrow + mn\downarrow)/\sqrt{2}$ the kernel is therefore $g_{\mathrm{deg}}K^{\mathrm{x}}_{\mathrm{full}} - K^{\mathrm{d}}$, with $g_{\mathrm{deg}} = 2$ and the kernels now written for spatial orbitals; in $(mn\uparrow - mn\downarrow)/\sqrt{2}$ it is $-K^{\mathrm{d}}$. These are the singlet and triplet excitons. The density vertex is the same for both spins, so $n_e(\boldsymbol{q})$ creates only singlets, with the weight $\sqrt{g_{\mathrm{deg}}}$ per transition. From here on the labels are spatial, the kernel is the singlet one, and g_{deg} appears in front of the response.

The transition-basis Hamiltonian

At $T = 0$, in an insulator, the Pauli factor of Eq. (D.61) is $+1$ for a transition from an occupied band v to an empty band c. These are the resonant transitions; the reverse ones, with the factor -1, are antiresonant, and we mark them with a bar. Transitions between two occupied or two empty orbitals have $\mathfrak{L}^0 = 0$ and drop out. In the matrix notation of Appendix C.7, let $\mathbf{F}$ be the diagonal matrix of the Pauli factors, with $\mathbf{F}^2 = \mathbf{1}$, and $\mathbf{D}$ the diagonal matrix of the transition energies; the inverse of Eq. (D.61) is then $\mathbf{F}(\mathrm{i}\omega_j^{\mathrm{B}} - \mathbf{D})$. Eq. (D.63) reads $\mathfrak{L}^{-1} = (\mathfrak{L}^0)^{-1} - \mathbf{K}$ with $\mathbf{K} = g_{\mathrm{deg}}K^{\mathrm{x}}_{\mathrm{full}} - K^{\mathrm{d}}$, and

$$\mathfrak{L} = \left[\mathbf{F}(\mathrm{i}\omega_j^{\mathrm{B}} - \mathbf{D}) - \mathbf{K}\right]^{-1} = \left(\mathrm{i}\omega_j^{\mathrm{B}} - \mathbf{D} - \mathbf{F}\mathbf{K}\right)^{-1}\mathbf{F}. \tag{D.64}$$

The poles are the eigenvalues of $\mathbf{D} + \mathbf{F}\mathbf{K}$, which in blocks of resonant and antiresonant transitions is

$$\mathbf{D} + \mathbf{F}\mathbf{K} = \begin{pmatrix} H^{\mathrm{exc}}_{\mathrm{full}} & K_{\mathrm{r\bar{r}}} \\ -K^*_{\mathrm{r\bar{r}}} & -H^{\mathrm{exc}*}_{\mathrm{full}} \end{pmatrix}, \qquad H^{\mathrm{exc}}_{\mathrm{full},cv,c'v'} = (\xi_c - \xi_v)\,\delta_{cc'}\delta_{vv'} + g_{\mathrm{deg}}K^{\mathrm{x}}_{\mathrm{full},cv,c'v'} - K^{\mathrm{d}}_{cv,c'v'}. \tag{D.65}$$

The lower blocks follow from Eq. (D.62). Exchanging m and n in both transitions conjugates $K^{\mathrm{x}}_{\mathrm{full}}$, since v_{C} is real. It also conjugates K^{d} when $W^{\mathrm{scr}}(\boldsymbol{r}, \boldsymbol{r}') = W^{\mathrm{scr}}(\boldsymbol{r}', \boldsymbol{r})$ is real, as the static screened interaction of a target with time-reversal symmetry is. The factor -1 of $\mathbf{F}$ on the antiresonant rows supplies the signs. The same two properties make $H^{\mathrm{exc}}_{\mathrm{full}}$ Hermitian. In the resonant block the exchange kernel carries the singlet factor g_{deg} [35, Eqs. (18) and (20)].

The coupling block $K_{\mathrm{r\bar{r}}}$ connects the creation of a pair with its annihilation. In a semiconductor its effect on the eigenvalues is of second order in the kernel divided by twice the gap, and the Tamm–Dancoff approximation drops it.

The $\boldsymbol{G} = 0$ term of the exchange kernel is the long-range macroscopic field and is treated apart. In a crystal the Coulomb line of $K^{\mathrm{x}}_{\mathrm{full}}$ carries $\boldsymbol{q} + \boldsymbol{G}$ for every reciprocal-lattice vector $\boldsymbol{G}$, and its term at $\boldsymbol{G} = 0$, which we call K^{x}_0, is the long-range macroscopic field. The rest is the exchange kernel proper, $K^{\mathrm{x}} = K^{\mathrm{x}}_{\mathrm{full}} - K^{\mathrm{x}}_0$, the local-field kernel. We write H^{exc} for $H^{\mathrm{exc}}_{\mathrm{full}}$ with K^{x} in place of $K^{\mathrm{x}}_{\mathrm{full}}$. With the band and crystal-momentum labels of a crystal written

out, a transition at fixed $\boldsymbol{q}$ takes an electron from $v\boldsymbol{k}$ to $(c,\boldsymbol{k}+\boldsymbol{q})$, and the resonant problem is

$$\sum_{c'v'\boldsymbol{k}'} H^{\mathrm{exc}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'}\,A^{\lambda}_{c'v'\boldsymbol{k}'} = E_\lambda\,A^{\lambda}_{cv\boldsymbol{k}}, \qquad \sum_{cv\boldsymbol{k}} |A^{\lambda}_{cv\boldsymbol{k}}|^2 = 1,$$
$$H^{\mathrm{exc}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} = (\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}})\delta_{cc'}\delta_{vv'}\delta_{\boldsymbol{k}\boldsymbol{k}'} + g_{\mathrm{deg}} K^{\mathrm{x}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} - K^{\mathrm{d}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'}, \tag{D.66}$$

a Hermitian eigenvalue problem, whose eigenvalues E_λ and envelopes A^λ depend on $\boldsymbol{q}$. It is Eq. (E.41). The same problem with $H^{\mathrm{exc}}_{\mathrm{full}}$ has the eigenpairs $E^{\mathrm{full}}_\lambda$ and $A^\lambda_{\mathrm{full}}$. In the crystal the kernels Eq. (D.62) become sums over reciprocal-lattice vectors with the convention of Appendix A.3:

$$\begin{aligned}
K^{\mathrm{x}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} &= \frac{1}{\mathscr{V}}\sum_{\boldsymbol{G}\neq 0} v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\,\rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})\,\rho_{c'v'}(\boldsymbol{k}',\boldsymbol{q}+\boldsymbol{G})^*,\\
K^{\mathrm{x}}_{\mathrm{full},cv\boldsymbol{k},c'v'\boldsymbol{k}'} &= K^{\mathrm{x}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} + \frac{1}{\mathscr{V}}\,v_{\mathrm{C}}(q)\,\rho_{cv}(\boldsymbol{k},\boldsymbol{q})\,\rho_{c'v'}(\boldsymbol{k}',\boldsymbol{q})^*,\\
K^{\mathrm{d}}_{cv\boldsymbol{k},c'v'\boldsymbol{k}'} &= \frac{1}{\mathscr{V}}\sum_{\boldsymbol{G}\boldsymbol{G}'} \rho_{cc'}(\boldsymbol{k}'+\boldsymbol{q},\boldsymbol{p}+\boldsymbol{G})\,W^{\mathrm{scr}}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p},0)\,\rho_{vv'}(\boldsymbol{k}',\boldsymbol{p}+\boldsymbol{G}')^*, \qquad \boldsymbol{p}=\boldsymbol{k}-\boldsymbol{k}',
\end{aligned} \tag{D.67}$$

with $W^{\mathrm{scr}}_{\boldsymbol{G}\boldsymbol{G}'} = (\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'} v_{\mathrm{C}}(|\boldsymbol{p}+\boldsymbol{G}'|)$. The direct kernel follows from Eq. (D.62) with the expansion $W^{\mathrm{scr}}(\boldsymbol{r},\boldsymbol{r}') = \mathscr{V}^{-1}\sum_{\boldsymbol{p}\boldsymbol{G}\boldsymbol{G}'} \mathrm{e}^{\mathrm{i}(\boldsymbol{p}+\boldsymbol{G})\cdot\boldsymbol{r}}\,W^{\mathrm{scr}}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p})\,\mathrm{e}^{-\mathrm{i}(\boldsymbol{p}+\boldsymbol{G}')\cdot\boldsymbol{r}'}$: the integral over $\boldsymbol{r}$ gives $\langle c,\boldsymbol{k}+\boldsymbol{q}|\mathrm{e}^{\mathrm{i}(\boldsymbol{p}+\boldsymbol{G})\cdot\boldsymbol{r}}|c',\boldsymbol{k}'+\boldsymbol{q}\rangle$, nonzero for $\boldsymbol{p}=\boldsymbol{k}-\boldsymbol{k}'$, and the integral over $\boldsymbol{r}'$ the complex conjugate of $\langle v\boldsymbol{k}|\mathrm{e}^{\mathrm{i}(\boldsymbol{p}+\boldsymbol{G}')\cdot\boldsymbol{r}}|v'\boldsymbol{k}'\rangle$. No symmetry of the crystal is needed. The exchange kernels carry the momentum $\boldsymbol{q}$ of the pair and the direct kernel the momentum $\boldsymbol{p}$ exchanged between the electron and the hole. All three scale as $1/\mathscr{V} = 1/(N\Omega)$, and the sum over $\boldsymbol{k}'$ in Eq. (D.66) makes H^{exc} act as an integral operator of order one, as the count of Appendix A.3 requires.

Response, structure factor and dielectric function

Continued to $\omega + \mathrm{i}0^+$ (Eq. (4.92)), the resonant block of Eq. (D.64) in the Tamm–Dancoff approximation is the resolvent of the pair Hamiltonian, $(\omega - H^{\mathrm{exc}} + \mathrm{i}0^+)^{-1} = \sum_\lambda A^\lambda A^{\lambda\dagger}/(\omega - E_\lambda + \mathrm{i}0^+)$, and likewise for $H^{\mathrm{exc}}_{\mathrm{full}}$. For the problem Eq. (D.66), Eq. (D.60) with the singlet weight gives

$$\bar{R}_{00}(\boldsymbol{q},\omega) = \frac{g_{\mathrm{deg}}}{\mathscr{V}}\sum_\lambda \frac{\left|\sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}\,\rho_{cv}(\boldsymbol{k},\boldsymbol{q})^*\right|^2}{\omega - E_\lambda(\boldsymbol{q}) + \mathrm{i}0^+} + (\text{antiresonant}), \tag{D.68}$$

where the antiresonant part has its poles at $-E_\lambda$. Since K^{x} lacks the term at $\boldsymbol{G}=0$, this is the head of the response that is proper with respect to the long-range Coulomb term. The complete response follows from it, because the difference of the two kernels, $g_{\mathrm{deg}}K^{\mathrm{x}}_0$ with $K^{\mathrm{x}}_0 = \mathscr{V}^{-1}v_{\mathrm{C}}(q)\rho(\boldsymbol{q})\rho(\boldsymbol{q})^\dagger$ in transition space, has rank one. Let $\bar{\mathfrak{L}}$ solve Eq. (D.63) with K^{x} in place of $K^{\mathrm{x}}_{\mathrm{full}}$. Then $\mathfrak{L}^{-1} = \bar{\mathfrak{L}}^{-1} - g_{\mathrm{deg}}K^{\mathrm{x}}_0$, which is $\mathfrak{L} = \bar{\mathfrak{L}} + \bar{\mathfrak{L}}\,g_{\mathrm{deg}}K^{\mathrm{x}}_0\,\mathfrak{L}$, and its head in Eq. (D.60) closes on itself:

$$R_{00} = \bar{R}_{00} + \bar{R}_{00}\,v_{\mathrm{C}}(q)\,R_{00}, \qquad R_{00} = \frac{\bar{R}_{00}}{1 - v_{\mathrm{C}}(q)\bar{R}_{00}}.$$

The same R_{00} is Eq. (D.68) with the eigenpairs $E^{\mathrm{full}}_\lambda$ and $A^\lambda_{\mathrm{full}}$ of the full kernel. It is the complete density response, the one whose imaginary part enters the dark-matter rate. At $T=0$ and $\omega>0$ only the resonant part contributes to $\mathrm{Im}\,R_{00}$, and the fluctuation–dissipation theorem Eq. (4.18) gives the exciton contribution to the structure factor,

$$S(\boldsymbol{q},\omega) = \frac{2\pi g_{\mathrm{deg}}}{\mathscr{V}}\sum_\lambda \Big|\sum_{cv\boldsymbol{k}} A^\lambda_{\mathrm{full},cv\boldsymbol{k}}(\boldsymbol{q})\,\rho_{cv}(\boldsymbol{k},\boldsymbol{q})^*\Big|^2\,\delta\big(\omega - E^{\mathrm{full}}_\lambda(\boldsymbol{q})\big). \tag{D.69}$$

The independent-particle weights $|\rho_{cv}|^2$ of the transitions are replaced by the squared coherent sum over the transitions that make up the exciton. A bound exciton collects the weight of many transitions near the gap into one line below the continuum. Without kernels, A^λ_{full} is a single transition and Eq. (D.69) is the independent-particle structure factor of Sect. 4.2, with $g_{\text{deg}} = 2$ for the spin.

The macroscopic dielectric function needs the proper head. Since $(\varepsilon^{-1})_{00} = 1 + v_{\text{C}} R_{00}$ (Sect. 4.4) and $1 + v_{\text{C}} R_{00} = 1/(1 - v_{\text{C}} \bar{R}_{00})$,

$$\varepsilon_{\text{mac}}(\boldsymbol{q}, \omega) = \frac{1}{(\varepsilon^{-1})_{00}} = 1 - v_{\text{C}}(q)\, \bar{R}_{00}(\boldsymbol{q}, \omega). \tag{D.70}$$

The macroscopic dielectric function is the head of the response built with the exchange kernel stripped of its long-range term. The block inversion Eq. (C.57) does the same for the dielectric matrix of the random-phase approximation; here the local fields of Sect. 4.6 and the electron–hole attraction are treated in one equation. In the optical limit, with $\rho_{cv} \to \boldsymbol{q} \cdot \boldsymbol{v}_{cv}/(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}})$ by Eq. (3.43), the imaginary part is

$$\varepsilon_2(\omega) = \frac{\pi g_{\text{deg}} e^2}{V q^2} \sum_\lambda \Big| \sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}\, \rho_{cv}(\boldsymbol{k}, \boldsymbol{q})^* \Big|^2 \delta(\omega - E_\lambda), \tag{D.71}$$

with the eigenpairs of Eq. (D.66). Without kernels, and with $\omega = \varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}$ on the delta function, the prefactor becomes $8\pi^2\alpha/(\Omega\omega^2)$ times $N^{-1} \sum_{cv\boldsymbol{k}} |\hat{\boldsymbol{q}} \cdot \boldsymbol{v}_{cv}|^2$, the independent-particle form of Sect. 4.4. The difference between E^{full}_λ and E_λ comes from the long-range exchange term alone. At $\boldsymbol{q} \to 0$ it splits longitudinal from transverse excitons, the electronic analog of the splitting between longitudinal and transverse optical phonons that the macroscopic field causes in a polar crystal (Appendix D.14).

D.13 The Born–Oppenheimer expansion

A silicon nucleus is 5.12×10^4 times heavier than an electron, with $M = 28.0855\,\text{u} = 26.16\,\text{GeV}$ for the natural isotope mixture, as in Sect. 5.3 and Appendix C.6. Held by force constants of the same electronic order, it vibrates $(M/m_e)^{1/2} \approx 226$ times more slowly than the electrons move, and the electrons have time to settle into their ground state for every position of the nuclei. Born and Oppenheimer turned this into an expansion in powers of $\kappa_{\text{BO}} = (m_e/M)^{1/4}$ [36], in the form that Born and Huang gave it [37]. Here we derive the equations that Sects. 5.1 and 6.1 state, count every term in powers of κ_{BO}, and test the counting on a model that can be solved exactly.

The coupled equations

We use the Hamiltonian Eq. (5.2) and the adiabatic states Eq. (5.3) of Sect. 5.1, orthonormal at every u and chosen real, and assume that the levels do not cross in the range of u the nuclei explore. The adiabatic states are the states of the clamped-nuclei problem of Chaps. 3 and 4, and the ground-state energy $E_0(u)$ is the energy surface E^{ad}. Any state of electrons and nuclei can be expanded as $\Psi(r, u) = \sum_i \Psi^{\text{nuc}}_i(u)\, \Psi_i(r; u)$, since the adiabatic states are complete in r at each u. The nuclear Laplacian acts on both factors, $\nabla^2(\Psi^{\text{nuc}}_i \Psi_i) = (\nabla^2 \Psi^{\text{nuc}}_i)\Psi_i + 2\nabla\Psi^{\text{nuc}}_i \cdot \nabla\Psi_i + \Psi^{\text{nuc}}_i \nabla^2 \Psi_i$, while H_{el} acting on Ψ_i yields $E_i(u)\Psi_i$. We insert the expansion into $H\Psi = E\Psi$, multiply by Ψ_j and integrate over the electron coordinates at fixed u. By orthonormality the first term of the Laplacian becomes $\delta_{ji} \nabla^2 \Psi^{\text{nuc}}_i$; the other two leave brackets that depend on u:

$$\int \mathrm{d}r\; \Psi_j\, \nabla^2_{\boldsymbol{R}\kappa} (\Psi^{\text{nuc}}_i \Psi_i) = \delta_{ji}\, \nabla^2_{\boldsymbol{R}\kappa} \Psi^{\text{nuc}}_i + 2\, \langle \Psi_j | \nabla_{\boldsymbol{R}\kappa} \Psi_i \rangle \cdot \nabla_{\boldsymbol{R}\kappa} \Psi^{\text{nuc}}_i + \langle \Psi_j | \nabla^2_{\boldsymbol{R}\kappa} \Psi_i \rangle\, \Psi^{\text{nuc}}_i .$$

Multiplied by $-1/2M_\kappa$, summed over the nuclei and over i, and added to $E_j(u)\Psi_j^{\rm nuc}$ from $H_{\rm el}$, this is the set of coupled equations Eq. (5.4), with the nonadiabatic operators $\mathcal{N}_{ji}$ of Eq. (5.5). The bracket with one gradient is an operator that acts on the nuclear wavefunction through $\nabla_{\boldsymbol{R}\kappa}$; the bracket with the Laplacian is a function of u that multiplies it. They are the only terms that connect one electronic state to another, and setting them all to zero is the Born–Oppenheimer approximation Eq. (5.6).

The adiabatic approximation of Born and Huang keeps the diagonal term $\mathcal{N}_{00}$ in addition [37]. For real states the derivative coupling has no diagonal element, since $\langle\Psi_0|\nabla\Psi_0\rangle = \frac{1}{2}\nabla\langle\Psi_0|\Psi_0\rangle = 0$. Differentiating this once more yields $\langle\Psi_0|\nabla^2\Psi_0\rangle = -\langle\nabla\Psi_0|\nabla\Psi_0\rangle$, so $\mathcal{N}_{00} = \sum_{\boldsymbol{R}\kappa}\langle\nabla_{\boldsymbol{R}\kappa}\Psi_0|\nabla_{\boldsymbol{R}\kappa}\Psi_0\rangle/2M_\kappa$, a small positive potential added to $E^{\rm ad}$.

The force and the derivative couplings

Both the force on a nucleus and the derivative couplings follow from one identity. Differentiate Eq. (5.3) with respect to $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ and project on Ψ_j:

$$\langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}H_{\rm el}|\Psi_i\rangle + E_j\,\langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}\Psi_i\rangle = \delta_{ij}\,\nabla_{\boldsymbol{R}\kappa}E_i + E_i\,\langle\Psi_j|\nabla_{\boldsymbol{R}\kappa}\Psi_i\rangle,$$

where $H_{\rm el}$ acting to the left on Ψ_j gave E_j. For $j = i$ the derivative couplings cancel, and with $i = 0$ what is left is the Hellmann–Feynman theorem Eq. (5.10), $-\nabla_{\boldsymbol{R}\kappa}E^{\rm ad} = -\langle\Psi_0|\nabla_{\boldsymbol{R}\kappa}(V_{eN}+V_{NN})|\Psi_0\rangle$, since only V_{eN} and V_{NN} depend on u. For $j \neq i$ the term δ_{ij} is absent, and V_{NN} drops out because it is a number at fixed u and $\langle\Psi_j|\Psi_i\rangle = 0$. Solving for the bracket yields Eq. (5.11): the derivative coupling is the matrix element of the force that the nucleus exerts on the electrons, divided by an electronic excitation energy $E_i - E_j$. Its numerator is the bare electron–phonon vertex, a one-body operator, $\nabla_{\boldsymbol{R}\kappa}V_{eN} = \sum_e \nabla_{\boldsymbol{R}\kappa}v_\kappa(\boldsymbol{r}_e - \boldsymbol{R} - \boldsymbol{\tau}_\kappa - \boldsymbol{u}_{\boldsymbol{R}\kappa})$ with v_κ the electron–nucleus potential. Its denominator decides when the adiabatic picture fails: the coupling is small when the excitation energies are large, and it diverges where two adiabatic levels meet.

Power counting

We measure lengths in a_0 and energies in $\alpha^2 m_e$, as in Sect. 5.1. The balance Eq. (5.8) then makes the zero-point amplitude the fraction $\kappa_{\rm BO}$ of the Bohr radius, $\ell_{\rm zp} \sim \kappa_{\rm BO}a_0$, and a vibrational quantum the fraction $\kappa_{\rm BO}^2$ of an electronic energy; Worked example 5.1 gives the numbers for silicon. If the displacement is written in units of $\kappa_{\rm BO}a_0$, the nuclear kinetic energy and the harmonic potential are both $\sim \kappa_{\rm BO}^2\alpha^2 m_e$, and every other term is smaller by further powers of $\kappa_{\rm BO}$.

The remaining terms follow by the substitution $u \sim \kappa_{\rm BO}a_0$ and $\nabla \sim 1/(\kappa_{\rm BO}a_0)$ acting on a nuclear wavefunction, with force constants $\Phi^{(n)} \sim \alpha^2 m_e/a_0^n$ and $1/M = \kappa_{\rm BO}^4/m_e$. The cubic term of the energy surface is $\sim \alpha^2 m_e\,\kappa_{\rm BO}^3$, smaller than a phonon energy by one power of $\kappa_{\rm BO}$, which is the anharmonic small parameter of Sect. 5.7. The quartic term is $\sim \alpha^2 m_e\,\kappa_{\rm BO}^4$, and so is the cubic term at second order, $(\kappa_{\rm BO}^3)^2/\kappa_{\rm BO}^2$, since its energy denominator is a phonon energy: the anharmonic shifts of the vibrational levels are $\sim \alpha^2 m_e\,\kappa_{\rm BO}^4$. In the nonadiabatic operators Eq. (5.5), the electronic brackets vary on the scale a_0, $\langle\Psi_j|\nabla\Psi_i\rangle \sim 1/a_0$ and $\langle\Psi_j|\nabla^2\Psi_i\rangle \sim 1/a_0^2$. The first-derivative coupling is then $(1/M)(1/a_0)(1/\kappa_{\rm BO}a_0) = \alpha^2 m_e\,\kappa_{\rm BO}^3$, and the second-derivative coupling $\alpha^2 m_e\,\kappa_{\rm BO}^4$. The diagonal correction $\mathcal{N}_{00}$ is $\sim \kappa_{\rm BO}^4$ as well, but most of it is a constant. Its variation over a zero-point amplitude changes the curvature of the surface by the fraction $\kappa_{\rm BO}^4$, and the vibrational levels by $\alpha^2 m_e\,\kappa_{\rm BO}^6$. The derivative coupling connects the electronic ground state to excited states at an energy $\sim \alpha^2 m_e$ above it, and it therefore shifts the vibrational levels only at second order, by $(\kappa_{\rm BO}^3)^2\alpha^2 m_e = \alpha^2 m_e\,\kappa_{\rm BO}^6$. Every nonadiabatic correction to the vibrational energies is therefore of relative order $\kappa_{\rm BO}^4 = m_e/M$, two powers of

$\kappa_{\rm BO}$ beyond the anharmonic ones. The electronic energy, the vibrations and the anharmonic corrections appear at orders $\kappa_{\rm BO}^0$, $\kappa_{\rm BO}^2$ and $\kappa_{\rm BO}^4$, and the coupling of the electronic and nuclear motion first at $\kappa_{\rm BO}^6$ [36].

The counting assumed that the electronic excitation energies ΔE are of order $\alpha^2 m_e$. When they are smaller, the derivative coupling Eq. (5.11) grows as $1/\Delta E$. Its matrix element between vibronic states is $(1/M)\langle\Psi_j|\nabla\Psi_0\rangle\cdot\nabla \sim (1/M)(F_{j0}/\Delta E)(1/\ell_{\rm zp})$, with F_{j0} the matrix element of the force in the numerator of Eq. (5.11); with $1/M = \omega_{\rm ph}\ell_{\rm zp}^2$ for $\ell_{\rm zp}^2 = 1/(M\omega_{\rm ph})$ it is $\omega_{\rm ph}\,\ell_{\rm zp}F_{j0}/\Delta E$. Divided by the energy ΔE that separates the two vibronic states, it gives the amplitude with which an excited electronic state is mixed into the ground state,

$$\frac{\omega_{\rm ph}}{\Delta E}\,\frac{\ell_{\rm zp}F_{j0}}{\Delta E}, \tag{D.72}$$

which is Eq. (5.9). At $\Delta E \sim \alpha^2 m_e$ the two factors are $\kappa_{\rm BO}^2$ and $\kappa_{\rm BO}$, the $\kappa_{\rm BO}^3$ above.

Clamped-nuclei states

The adiabatic states follow the nuclei, and the price is the derivative couplings. The alternative is to expand in the electronic states at the equilibrium configuration, $\Psi(r,u) = \sum_i \tilde\Psi_i^{\rm nuc}(u)\,\Psi_i(r;0)$, which do not depend on u and have no derivative couplings. The coupling then sits in the Hamiltonian, $H_{\rm el}(u) = H_{\rm el}(0) + [V_{eN}(u) - V_{eN}(0)] + [V_{NN}(u) - V_{NN}(0)]$, and its expansion in the displacements is

$$V_{eN}(u)-V_{eN}(0) = \sum_{\boldsymbol{R}\kappa} \boldsymbol{u}_{\boldsymbol{R}\kappa}\cdot\nabla_{\boldsymbol{R}\kappa}V_{eN}\Big|_0 + \frac{1}{2}\sum_{\boldsymbol{R}\kappa,\boldsymbol{R}'\kappa'} \boldsymbol{u}_{\boldsymbol{R}\kappa}\cdot\big(\nabla_{\boldsymbol{R}\kappa}\nabla_{\boldsymbol{R}'\kappa'}V_{eN}\big)\Big|_0\cdot\boldsymbol{u}_{\boldsymbol{R}'\kappa'} + \cdots . \tag{D.73}$$

The linear term is a one-body electron operator times a displacement. Written in second quantization over the Bloch states of the clamped crystal (Eq. (2.36)), with $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ expanded in phonons by Eq. (A.21), it has the form Eq. (A.25) of the electron–phonon Hamiltonian. Its vertex is Eq. (A.26) with the bare derivative ∇V_{eN} in place of $\partial_{\kappa\alpha,\boldsymbol{Q}}V_{\rm KS}$. The other electrons screen this bare vertex, and in density-functional theory the self-consistent change of the Kohn–Sham potential replaces it (Sect. 6.2). The quadratic term is the second-order (Allen–Heine) coupling of Sect. 6.3. The clamped-nuclei approximation of Chaps. 3 and 4 keeps $H_{\rm el}(0)$ alone.

The two expansions are equivalent order by order in u. At first order, the linear term of Eq. (D.73) mixes into $\Psi_i(r;0)$ the states $\Psi_j(r;0)$ with amplitude $\boldsymbol{u}\cdot\langle\Psi_j|\nabla V_{eN}|\Psi_i\rangle/(E_i - E_j)$, and by Eq. (5.11) this is $\boldsymbol{u}\cdot\langle\Psi_j|\nabla\Psi_i\rangle$, the first term of the Taylor series of $\Psi_i(r;u)$. The adiabatic states are the clamped states dressed by the electron–phonon coupling to all orders in u, with the nuclei held still; the nonadiabatic operators are what remains of the coupling when the nuclei move.

A solvable harmonic model

One electron bound harmonically to one nucleus, which is itself bound to a fixed point, tests the counting without approximation. In one dimension,

$$H = \frac{p^2}{2m_e} + \frac{P^2}{2M} + \frac{k_{\rm e}}{2}(x-X)^2 + \frac{K_{\rm N}}{2}X^2,$$

with x and p the electron's coordinate and momentum, X and P the nucleus's, and two spring constants of the same electronic order. The adiabatic states are oscillator states centered on the nucleus, with the excitation energy $\omega_{\rm e} = (k_{\rm e}/m_e)^{1/2}$ and the surface $E_0(X) = \omega_{\rm e}/2 + K_{\rm N}X^2/2$, so the Born–Oppenheimer frequency is $\omega_{\rm BO} = (K_{\rm N}/M)^{1/2}$. The diagonal correction $\mathcal{N}_{00} =$

$m_e\omega_{\mathrm{e}}/(4M)$ is a constant, since the ground state only translates with X, and it leaves the frequency unchanged. The exact frequencies are those of two coupled oscillators. With the force constants as a matrix, the equation of motion requires $\det\left[\begin{smallmatrix} k_{\mathrm{e}}-m_e\omega^2 & -k_{\mathrm{e}} \\ -k_{\mathrm{e}} & k_{\mathrm{e}}+K_{\mathrm{N}}-M\omega^2 \end{smallmatrix}\right] = 0$, that is $m_eM\omega^4 - [m_e(k_{\mathrm{e}}+K_{\mathrm{N}}) + Mk_{\mathrm{e}}]\,\omega^2 + k_{\mathrm{e}}K_{\mathrm{N}} = 0$. Its lower root, expanded in m_e/M, is

$$\omega^2 = \frac{K_{\mathrm{N}}}{M}\Big[1 - \frac{m_e}{M} + \Big(1 - \frac{K_{\mathrm{N}}}{k_{\mathrm{e}}}\Big)\frac{m_e^2}{M^2} + O\Big(\frac{m_e^3}{M^3}\Big)\Big]. \tag{D.74}$$

At first order the nucleus vibrates as if its mass were $M + m_e$: the electron is dragged along and adds its mass. Its lag behind the nucleus, which depends on the ratio of the two springs, enters only at the next order. The relative correction is $m_e/M = \kappa_{\mathrm{BO}}^4$ for ω^2, and $m_e/2M$ for the frequency itself, and it comes entirely from the off-diagonal coupling at second order, as the counting predicts. For $K_{\mathrm{N}} = k_{\mathrm{e}}/2$ and $m_e/M = 10^{-2}$, 10^{-3} and 10^{-4}, the exact ω^2 differs from $(K_{\mathrm{N}}/M)(1 - m_e/M)$ by 5.1×10^{-5}, 5.0×10^{-7} and 5.0×10^{-9} of its value, the m_e^2/M^2 term of Eq. (D.74) plus, at the largest mass ratio, a visible m_e^3/M^3 term.

D.14 Polar crystals

When the gallium and arsenic sublattices of GaAs move against each other, every cell acquires a dipole. In a transverse optical wave the dipoles line up across the direction of propagation and their fields cancel on average; in a longitudinal wave they add up to a macroscopic polarization, whose field pulls the ions back and stiffens the mode. That field is long-ranged. It makes the dynamical matrix depend on the direction from which $\boldsymbol{Q} \to 0$ is approached, and it splits the longitudinal from the transverse optical frequency. The Lyddane–Sachs–Teller relation [38] ties that splitting to the static and electronic dielectric constants. The same field couples the longitudinal phonon to an electron with a strength that diverges as $1/Q$, the Fröhlich vertex [39]. This section derives the direction-dependent dynamical matrix, the splitting, the Lyddane–Sachs–Teller relation and the Fröhlich vertex from one energy function of the displacements and the field, in the form Sects. 5.5 and 6.2 use. The approach is that of Huang's model as presented by Born and Huang [37] and Baroni et al. [40, Sect. II.D.2], generalized to anisotropic tensors [41, 42].

Energy at long wavelength

At long wavelength, the displacements and the field vary slowly from cell to cell, and within a region of many cells we may treat them as uniform: a displacement $\boldsymbol{u}_\kappa$ of every nucleus κ and a macroscopic field $\boldsymbol{\mathcal{E}}$. The electrons follow both adiabatically (Appendix D.13). The starting point is the energy per cell at fixed macroscopic field, Eq. (5.47). Its second derivatives are the analytic zone-center force constants $C^{\mathrm{an}}_{\kappa\kappa'}$, the Born charges Z^*_κ and the electronic dielectric tensor ε_∞; its first derivatives are the force $\boldsymbol{F}_\kappa$ on each sublattice and the polarization $\boldsymbol{P}$ of Eq. (5.48). The displacement field is $\boldsymbol{D} = \boldsymbol{\mathcal{E}} + \boldsymbol{P}$ in Heaviside–Lorentz units (Appendix A.2), and the Born charges obey the neutrality Eq. (5.50). The coupling term of Eq. (5.47), $-e\sum_\kappa \boldsymbol{u}_\kappa \cdot Z^*_\kappa \cdot \boldsymbol{\mathcal{E}}$, is the energy of a polar mode in a uniform field, and it is the form through which a dark photon drives optical phonons (Chap. 11).

The nonanalytic dynamical matrix

The field in Eq. (5.48) is not an independent variable: the ions create it. Let the displacements and the field vary as $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}}$ with $|\boldsymbol{Q}|$ small compared with the reciprocal-lattice vectors but large compared with ω/c, so that retardation can be neglected and the electrostatic equations

apply. Then $\nabla \times \boldsymbol{\mathcal{E}} = 0$ makes the field longitudinal, $\boldsymbol{\mathcal{E}} = \hat{\boldsymbol{Q}}\,\mathcal{E}_\parallel$, and the absence of free charge, $\nabla \cdot \boldsymbol{D} = 0$, requires $\hat{\boldsymbol{Q}} \cdot \boldsymbol{D} = 0$. With Eq. (5.48),

$$\hat{\boldsymbol{Q}} \cdot \varepsilon_\infty \cdot \hat{\boldsymbol{Q}}\, \mathcal{E}_\parallel + \frac{e}{\Omega} \sum_{\kappa'} \boldsymbol{u}_{\kappa'} \cdot Z^*_{\kappa'} \cdot \hat{\boldsymbol{Q}} = 0.$$

Solving for $\mathcal{E}_\parallel$ and inserting it in the force,

$$\boldsymbol{F}_\kappa = -\sum_{\kappa'} [C^{\text{an}}_{\kappa\kappa'} + C^{\text{na}}_{\kappa\kappa'}(\hat{\boldsymbol{Q}})] \cdot \boldsymbol{u}_{\kappa'}, \qquad C^{\text{na},\alpha\beta}_{\kappa\kappa'}(\hat{\boldsymbol{Q}}) = \frac{e^2}{\Omega} \frac{(Z^*_\kappa \cdot \hat{\boldsymbol{Q}})^\alpha\, (Z^*_{\kappa'} \cdot \hat{\boldsymbol{Q}})^\beta}{\hat{\boldsymbol{Q}} \cdot \varepsilon_\infty \cdot \hat{\boldsymbol{Q}}}, \tag{D.75}$$

with $(Z^*_\kappa \cdot \hat{\boldsymbol{Q}})^\alpha = \sum_\beta Z^{*,\alpha\beta}_\kappa \hat{Q}^\beta$. Divided by $(M_\kappa M_{\kappa'})^{1/2}$ as in Eq. (A.23), this is the nonanalytic part of the dynamical matrix, Eq. (E.24), and Ref. [40, Eq. (98)] after the substitution $4\pi e_{\text{G}}^2 \to e^2$ of Eq. (A.5). The analytic part C^{an} is the response with the $\boldsymbol{G} = 0$ term of the Coulomb interaction removed [43], which is why it can be computed at zero macroscopic field and interpolated in $\boldsymbol{Q}$, with the nonanalytic part added back from Z^*_κ and ε_∞ alone [42]. For a transverse displacement, $Z^*_\kappa \cdot \hat{\boldsymbol{Q}}$ is orthogonal to the displacement pattern in a cubic crystal, the field vanishes, and C^{na} does not act.

The LO–TO splitting

Take a cubic crystal with two atoms per cell, such as GaAs. By symmetry ε_∞ and the Born charges are multiples of the identity, and by Eq. (5.50) $Z^*_1 = -Z^*_2 = Z^*$. At $\boldsymbol{Q} \to 0$ the acoustic sum rule of Sect. 5.2, $\sum_{\kappa'} C^{\text{an}}_{\kappa\kappa'} = 0$, leaves one scalar constant, $C^{\text{an}}_{11} = C^{\text{an}}_{22} = -C^{\text{an}}_{12} = C_0$. The equations of motion of the two sublattices, $M_1\ddot{\boldsymbol{u}}_1 = \boldsymbol{F}_1$ and $M_2\ddot{\boldsymbol{u}}_2 = \boldsymbol{F}_2$, then involve only the relative displacement $\boldsymbol{u}_{\text{rel}} = \boldsymbol{u}_1 - \boldsymbol{u}_2$. Dividing each by its mass and subtracting, with $\boldsymbol{F}_1 = -C_0\boldsymbol{u}_{\text{rel}} + eZ^*\boldsymbol{\mathcal{E}} = -\boldsymbol{F}_2$ from Eq. (5.48),

$$\mu_{\text{red}}\, \ddot{\boldsymbol{u}}_{\text{rel}} = -C_0\, \boldsymbol{u}_{\text{rel}} + eZ^*\boldsymbol{\mathcal{E}}, \qquad \boldsymbol{P} = \frac{eZ^*}{\Omega}\, \boldsymbol{u}_{\text{rel}} + (\varepsilon_\infty - 1)\boldsymbol{\mathcal{E}},$$

with $\mu_{\text{red}} = M_1M_2/(M_1 + M_2)$ the reduced mass of the pair. For a transverse mode the field vanishes, and $\omega^2_{\text{T}} = C_0/\mu_{\text{red}}$. For a longitudinal mode, $\hat{\boldsymbol{Q}} \cdot \boldsymbol{D} = \varepsilon_\infty \mathcal{E}_\parallel + eZ^*\hat{\boldsymbol{Q}} \cdot \boldsymbol{u}_{\text{rel}}/\Omega = 0$ makes the field $\boldsymbol{\mathcal{E}} = -eZ^*\boldsymbol{u}_{\text{rel}}/(\Omega\varepsilon_\infty)$, a restoring field, which adds $e^2Z^{*2}/(\Omega\varepsilon_\infty)$ to C_0. The longitudinal frequency is therefore Eq. (5.52), $\omega^2_{\text{L}} = \omega^2_{\text{T}} + e^2Z^{*2}/(\Omega\varepsilon_\infty\mu_{\text{red}})$: the longitudinal mode is stiffer by the electrostatic energy of its own polarization, screened by the electrons.

The Lyddane–Sachs–Teller relation

A uniform field at frequency ω, $\boldsymbol{\mathcal{E}}\,\mathrm{e}^{-\mathrm{i}\omega t}$ with no longitudinal constraint, drives the relative displacement to the steady response $\boldsymbol{u}_{\text{rel}} = eZ^*\boldsymbol{\mathcal{E}}/[\mu_{\text{red}}(\omega^2_{\text{T}} - \omega^2)]$. The electronic part of the response stays at ε_∞ as long as ω is far below the gap, which is the adiabatic assumption of Appendix D.13. Inserting $\boldsymbol{u}_{\text{rel}}$ into $\boldsymbol{P}$, the dielectric function $\varepsilon(\omega) = D/\mathcal{E} = 1 + P/\mathcal{E}$ is $\varepsilon_\infty + e^2Z^{*2}/[\Omega\mu_{\text{red}}(\omega^2_{\text{T}} - \omega^2)]$, and replacing $e^2Z^{*2}/(\Omega\mu_{\text{red}})$ by $\varepsilon_\infty(\omega^2_{\text{L}} - \omega^2_{\text{T}})$ from Eq. (5.52) turns it into $\varepsilon_\infty(\omega^2_{\text{L}} - \omega^2)/(\omega^2_{\text{T}} - \omega^2)$. These are the two forms of Eq. (5.53), and at $\omega = 0$ the second is the Lyddane–Sachs–Teller relation Eq. (5.54) [38]. The dielectric function has a pole at the transverse frequency and a zero at the longitudinal one. This is general. A longitudinal mode has $\boldsymbol{\mathcal{E}} \parallel \hat{\boldsymbol{Q}}$ and $\hat{\boldsymbol{Q}} \cdot \boldsymbol{D} = \varepsilon(\omega)\mathcal{E}_\parallel = 0$; a longitudinal mode with a nonzero field can exist only where $\varepsilon(\omega) = 0$. A transverse mode needs no field and occurs where the response to a field diverges.

With several polar modes the same steps give the general form. Let $\omega_{\text{T}\nu}$ and $\boldsymbol{e}_{\kappa\nu}$ be the frequencies and polarization vectors of the analytic problem at $\boldsymbol{Q} = 0$, the eigenpairs

of $C^{\mathrm{an}}_{\kappa\kappa'}/(M_\kappa M_{\kappa'})^{1/2}$, chosen real. Expanding $\boldsymbol{u}_\kappa = \sum_\nu \boldsymbol{e}_{\kappa\nu} a_\nu/\sqrt{M_\kappa}$, each normal coordinate obeys $\ddot{a}_\nu = -\omega^2_{\mathrm{T}\nu} a_\nu + e\,\boldsymbol{z}_\nu \cdot \boldsymbol{\mathcal{E}}$, with the mode charge vector $\boldsymbol{z}_\nu = \sum_\kappa \boldsymbol{e}_{\kappa\nu} \cdot Z^*_\kappa/\sqrt{M_\kappa}$, and it contributes $(e/\Omega)\, a_\nu \boldsymbol{z}_\nu$ to the polarization. In a cubic crystal, along any direction $\hat{\boldsymbol{e}}$,

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{e^2}{\Omega} \sum_\nu \frac{(\boldsymbol{z}_\nu \cdot \hat{\boldsymbol{e}})^2}{\omega^2_{\mathrm{T}\nu} - \omega^2}.$$

This is a rational function of ω^2 that tends to ε_∞ at large ω, has simple poles at the polar transverse frequencies, and, by the argument above, vanishes at the longitudinal frequencies $\omega_{\mathrm{L}\nu}$. Between two consecutive poles it rises monotonically from $-\infty$ to $+\infty$, and above the highest pole it rises from $-\infty$ toward ε_∞, so there is exactly one zero above each pole and below the next. A rational function is fixed by its poles, its zeros and its limit at infinity, so $\varepsilon(\omega) = \varepsilon_\infty \prod_\nu (\omega^2_{\mathrm{L}\nu} - \omega^2)/(\omega^2_{\mathrm{T}\nu} - \omega^2)$, and at $\omega = 0$

$$\frac{\varepsilon_0}{\varepsilon_\infty} = \prod_\nu \frac{\omega^2_{\mathrm{L}\nu}}{\omega^2_{\mathrm{T}\nu}}, \tag{D.76}$$

the generalized relation of Cochran and Cowley [41].

The Fröhlich vertex

The field of a longitudinal phonon acts on the electrons as well. Take one mode $(\boldsymbol{Q}, \nu)$ at small $\boldsymbol{Q}$ in the expansion Eq. (A.21), $\boldsymbol{u}_{\boldsymbol{R}\kappa} = (2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{-1/2}\, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} B_{\boldsymbol{Q}\nu}$, and replace the cell position $\boldsymbol{R}$ by a continuous $\boldsymbol{r}$, which is allowed for $|\boldsymbol{Q}|$ small compared with the reciprocal-lattice vectors. The condition $\hat{\boldsymbol{Q}} \cdot \boldsymbol{D} = 0$ of the previous subsection gives the field of the wave,

$$\boldsymbol{\mathcal{E}}(\boldsymbol{r}) = -\hat{\boldsymbol{Q}}\, \frac{e}{\Omega} \sum_\kappa \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) \cdot Z^*_\kappa \cdot \hat{\boldsymbol{Q}}}{(2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{1/2}\, \hat{\boldsymbol{Q}} \cdot \varepsilon_\infty \cdot \hat{\boldsymbol{Q}}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}} B_{\boldsymbol{Q}\nu}.$$

An electron, with charge $-e$, feels the force $-e\boldsymbol{\mathcal{E}}$, and its potential energy $V^{\mathrm{L}}(\boldsymbol{r})$ obeys $\nabla V^{\mathrm{L}} = e\boldsymbol{\mathcal{E}}$. For a wave $\propto \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}}$ the gradient is $\mathrm{i}\boldsymbol{Q}$. The potential energy is then $V^{\mathrm{L}} = e\,\hat{\boldsymbol{Q}} \cdot \boldsymbol{\mathcal{E}}/(\mathrm{i}Q)$, or

$$V^{\mathrm{L}}_{\boldsymbol{Q}\nu}(\boldsymbol{r}) = \mathrm{i}\, \frac{e^2}{\Omega} \sum_\kappa \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) \cdot Z^*_\kappa \cdot \boldsymbol{Q}}{(2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{1/2}\, \boldsymbol{Q} \cdot \varepsilon_\infty \cdot \boldsymbol{Q}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}} B_{\boldsymbol{Q}\nu}.$$

Summed over modes and written in second quantization over Bloch states with Eq. (2.36), it has the form Eq. (A.25), with the matrix element of $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}}$ the density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{Q})$ of Eq. (A.17). Reading off the factor of $N^{-1/2} c^\dagger_{m,\boldsymbol{k}+\boldsymbol{Q}} c_{n\boldsymbol{k}} B_{\boldsymbol{Q}\nu}$, the long-range vertex is

$$g^{\mathrm{L}}_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q}) = \mathrm{i}\, \frac{e^2}{\Omega} \sum_\kappa \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) \cdot Z^*_\kappa \cdot \boldsymbol{Q}}{(2M_\kappa\omega_{\boldsymbol{Q}\nu})^{1/2}\, \boldsymbol{Q} \cdot \varepsilon_\infty \cdot \boldsymbol{Q}}\, \rho_{mn}(\boldsymbol{k}, \boldsymbol{Q}). \tag{D.77}$$

It is the $\boldsymbol{G} = 0$ term of Ref. [44, Eq. (4)], with $e^2_{\mathrm{SI}}/\varepsilon_0^{\mathrm{SI}} \to e^2$, the factor $N^{-1/2}$ moved into Eq. (A.25) and Z^* transposed to our index order. The terms with $\boldsymbol{G} \neq 0$ of that equation come from the periodic arrangement of the dipoles and are regular at $\boldsymbol{Q} \to 0$. Eq. (D.77) obeys the Hermiticity condition Eq. (A.27). Under $\boldsymbol{Q} \to -\boldsymbol{Q}$ and $(m, \boldsymbol{k}) \leftrightarrow (n, \boldsymbol{k}+\boldsymbol{Q})$, the polarization vector becomes $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$ by Eq. (A.22), the factor $\boldsymbol{Q}$ in the numerator changes sign, and $\rho_{nm}(\boldsymbol{k}+\boldsymbol{Q}, -\boldsymbol{Q}) = \rho_{mn}(\boldsymbol{k}, \boldsymbol{Q})^*$; the product is the complex conjugate of Eq. (D.77), since $\mathrm{i} \to -\mathrm{i}$ is supplied by the sign of $\boldsymbol{Q}$. Only longitudinal modes couple, through $Z^*_\kappa \cdot \boldsymbol{Q}$, and the vertex grows as $1/Q$ at small Q.

For the cubic two-atom crystal the longitudinal optical mode at small $\boldsymbol{Q}$ has $\boldsymbol{e}_1 = (M_2/M_{\rm cell})^{1/2}\hat{\boldsymbol{Q}}$ and $\boldsymbol{e}_2 = -(M_1/M_{\rm cell})^{1/2}\hat{\boldsymbol{Q}}$, which satisfy Eq. (A.22). The sum in Eq. (D.77) is then $Z^*Q\,[(M_2/M_{\rm cell}M_1)^{1/2} + (M_1/M_{\rm cell}M_2)^{1/2}] = Z^*Q/\sqrt{\mu_{\rm red}}$, and

$$g^{\rm L} = \mathrm{i}\,\frac{e^2 Z^*}{\Omega\,\varepsilon_\infty\,Q\,(2\omega_{\rm L}\mu_{\rm red})^{1/2}}\;\rho_{mn}(\boldsymbol{k},\boldsymbol{Q}).$$

By Eqs. (5.52) and (5.54), $e^2Z^{*2}/(\Omega\varepsilon_\infty\mu_{\rm red}) = \omega_{\rm L}^2 - \omega_{\rm T}^2 = \omega_{\rm L}^2(1-\varepsilon_\infty/\varepsilon_0)$, and the Born charge and the masses drop out:

$$g^{\rm L}_{mn}(\boldsymbol{k},\boldsymbol{Q}) = \frac{\mathrm{i}}{Q}\left[\frac{e^2\omega_{\rm L}}{2\Omega}\Big(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0}\Big)\right]^{1/2}\rho_{mn}(\boldsymbol{k},\boldsymbol{Q}), \tag{D.78}$$

for $Z^* > 0$, with an overall minus sign otherwise. This is Fröhlich's vertex [39, 44, Eq. (1)], with the unit and index changes listed above, and with the density vertex in place of the plane-wave overlap δ_{mn} of a single parabolic band. The combination $1/\varepsilon_\infty - 1/\varepsilon_0$ is the part of the screening due to the ions. The electrons see the field of the displaced ions screened by ε_∞; a static charge would be screened by ε_0, and the difference is the field that the lattice polarization adds. In a nonpolar crystal $\varepsilon_0 = \varepsilon_\infty$, $Z^* = 0$, and the vertex vanishes. Sect. 6.2 writes Eq. (D.78) through Fröhlich's coupling constant and evaluates it for GaAs.

D.15 The Debye–Waller factor

A nucleus in a crystal is never exactly at its lattice site. Its displacement $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ fluctuates, at zero temperature and even more at finite temperature, and a momentum transfer $\boldsymbol{q}$ to that nucleus acts through the operator $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}}$ (Appendix C.6). Its thermal average, the Debye–Waller factor $\mathrm{e}^{-W_\kappa(\boldsymbol{q})}$, controls the elastic weight of every lattice response at finite momentum, and the correlation of two such exponentials at different times generates the multiphonon response of Sect. 5.6. The average of the exponential of an operator is in general not the exponential of anything simple. For a harmonic crystal it is, by Bloch's identity Eq. (C.53), $\langle \mathrm{e}^{\hat{A}}\rangle = \mathrm{e}^{\langle\hat{A}^2\rangle/2}$ for $\hat{A}$ linear in the phonon operators. Here we prove the identity, write the displacement tensor B_κ and the displacement correlation $\mathcal{C}_{\kappa\kappa'}$ as sums over phonon modes, derive the generating formula used in Sect. 5.6, and evaluate $2W_\kappa$ in the Debye model at any temperature.

Proof for one mode

Take one oscillator, $H = \omega_0(d^\dagger d + \frac{1}{2})$ as in Appendix C.6, in thermal equilibrium at temperature $T = 1/\beta$, with $n_{\rm B} = n_{\rm B}(\omega_0)$, and let $\hat{A} = \eta_1 d + \eta_2 d^\dagger$ with arbitrary complex η_1 and η_2, so that $\hat{A}$ need not be Hermitian. Consider $F(\theta) = \langle \mathrm{e}^{\theta\hat{A}}\rangle$ as a function of a real parameter θ. The proof uses two facts. The commutators $[\hat{A}, d] = -\eta_2$ and $[\hat{A}, d^\dagger] = \eta_1$ are numbers. By Eq. (C.48), the transformation by $\mathrm{e}^{\theta\hat{A}}$ therefore only shifts the ladder operators,

$$d\,\mathrm{e}^{\theta\hat{A}} = \mathrm{e}^{\theta\hat{A}}\,(d + \theta\eta_2), \qquad d^\dagger\,\mathrm{e}^{\theta\hat{A}} = \mathrm{e}^{\theta\hat{A}}\,(d^\dagger - \theta\eta_1).$$

The cyclic property of the trace, in turn, moves a ladder operator past the Boltzmann factor at a known cost, as in the proof of Wick's theorem in Appendix D.11: $d\,\mathrm{e}^{-\beta H} = \mathrm{e}^{-\beta\omega_0}\mathrm{e}^{-\beta H}d$, since removing a quantum lowers H by ω_0, and $d^\dagger\mathrm{e}^{-\beta H} = \mathrm{e}^{+\beta\omega_0}\mathrm{e}^{-\beta H}d^\dagger$. As a result, $\langle \mathrm{e}^{\theta\hat{A}}d\rangle = \mathrm{Tr}[d\,\mathrm{e}^{-\beta H}\mathrm{e}^{\theta\hat{A}}]/\mathcal{Z} = \mathrm{e}^{-\beta\omega_0}\langle d\,\mathrm{e}^{\theta\hat{A}}\rangle$, and combined with the shift,

$$\langle d\,\mathrm{e}^{\theta\hat{A}}\rangle = \mathrm{e}^{-\beta\omega_0}\langle d\,\mathrm{e}^{\theta\hat{A}}\rangle + \theta\eta_2 F(\theta), \qquad \langle d\,\mathrm{e}^{\theta\hat{A}}\rangle = \frac{\theta\eta_2 F(\theta)}{1-\mathrm{e}^{-\beta\omega_0}} = \theta\eta_2\,(n_{\rm B}+1)\,F(\theta).$$

The same steps for $d^\dagger$ give $\langle d^\dagger \mathrm{e}^{\theta\hat{A}}\rangle = \mathrm{e}^{\beta\omega_0}\langle d^\dagger \mathrm{e}^{\theta\hat{A}}\rangle - \theta\eta_1 F(\theta)$, and $\langle d^\dagger \mathrm{e}^{\theta\hat{A}}\rangle = \theta\eta_1\, n_\mathrm{B}\, F(\theta)$. The derivative of F is then

$$F'(\theta) = \eta_1\langle d\,\mathrm{e}^{\theta\hat{A}}\rangle + \eta_2\langle d^\dagger \mathrm{e}^{\theta\hat{A}}\rangle = \theta\,\eta_1\eta_2\,(2n_\mathrm{B}+1)\,F(\theta) = \theta\,\langle\hat{A}^2\rangle\,F(\theta),$$

since $\langle\hat{A}^2\rangle = \eta_1\eta_2\langle dd^\dagger + d^\dagger d\rangle = \eta_1\eta_2(2n_\mathrm{B}+1)$, the averages of d^2 and $d^{\dagger 2}$ being zero. With $F(0)=1$ the solution is $F(\theta) = \mathrm{e}^{\theta^2\langle\hat{A}^2\rangle/2}$, and at $\theta = 1$

$$\langle \mathrm{e}^{\hat{A}}\rangle = \mathrm{e}^{\langle\hat{A}^2\rangle/2}. \tag{D.79}$$

For several independent modes in thermal equilibrium, $\hat{A} = \sum_\gamma \hat{A}_\gamma$ is a sum of commuting pieces, one per mode, the exponential factorizes, and the average in the Gibbs state of Sect. 2.5 factorizes too. The exponents add, and $\sum_\gamma\langle\hat{A}_\gamma^2\rangle = \langle\hat{A}^2\rangle$, because the cross terms $\langle\hat{A}_\gamma\hat{A}_{\gamma'}\rangle = \langle\hat{A}_\gamma\rangle\langle\hat{A}_{\gamma'}\rangle$ vanish for $\gamma \neq \gamma'$. Eq. (D.79) therefore holds for any $\hat{A}$ linear in the operators of a set of harmonic modes. Mermin gave a short evaluation of averages of this form by a different route [45]. The proof here is Wick's theorem of Appendix D.11 in generating-function form: expanded in powers of $\hat{A}$, both sides count the pairings of $2n$ linear operators, $(2n)!/(2^n n!)$ of them, each worth $\langle\hat{A}^2\rangle^n$.

The hypotheses are the ones the proof used. The operator must be linear in d and $d^\dagger$, so that its commutators are numbers, and the state must be the thermal state of a quadratic Hamiltonian, so that the trace moves each ladder operator at the cost of a number. In an anharmonic crystal neither holds exactly. The logarithm of $\langle\mathrm{e}^{\hat{A}}\rangle$ is then a series in the cumulants of $\hat{A}$, of which the harmonic crystal has only the second. The anharmonic terms of Sect. 5.7 make the fourth cumulant nonzero, and the Debye–Waller exponent acquires a correction that grows with temperature faster than T.

The displacement tensor

For a crystal, the operator is $\hat{A} = \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}$ with the displacement of Eq. (A.21). The thermal averages of the displacement fields are $\langle B_{\boldsymbol{Q}\nu}B_{\boldsymbol{Q}'\nu'}\rangle = \delta_{\boldsymbol{Q}',-\boldsymbol{Q}}\delta_{\nu\nu'}(2n_{\boldsymbol{Q}\nu}+1)$, since $\langle b_{\boldsymbol{Q}\nu}b^\dagger_{\boldsymbol{Q}\nu}\rangle + \langle b^\dagger_{-\boldsymbol{Q}\nu}b_{-\boldsymbol{Q}\nu}\rangle = (n_{\boldsymbol{Q}\nu}+1) + n_{\boldsymbol{Q}\nu}$ and $n_{-\boldsymbol{Q}\nu} = n_{\boldsymbol{Q}\nu}$. With $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$ from Eq. (A.22), the phases $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$ cancel, and the displacement tensor of Eq. (A.29) is

$$B_\kappa^{\alpha\beta} = \langle u^\alpha_{\boldsymbol{R}\kappa}u^\beta_{\boldsymbol{R}\kappa}\rangle = \frac{1}{N}\sum_{\boldsymbol{Q}\nu}\frac{2n_{\boldsymbol{Q}\nu}+1}{2M_\kappa\,\omega_{\boldsymbol{Q}\nu}}\,e^\alpha_{\kappa\nu}(\boldsymbol{Q})\,e^\beta_{\kappa\nu}(\boldsymbol{Q})^*, \tag{D.80}$$

the same in every cell. The terms at $\boldsymbol{Q}$ and $-\boldsymbol{Q}$ are complex conjugates of each other, which makes B_κ real and symmetric, and positive since every term of the diagonal is. By Eq. (D.79),

$$\left\langle \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}}\right\rangle = \mathrm{e}^{-\langle(\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa})^2\rangle/2} = \mathrm{e}^{-W_\kappa(\boldsymbol{q})}, \qquad 2W_\kappa(\boldsymbol{q}) = \boldsymbol{q}\cdot B_\kappa\cdot\boldsymbol{q}. \tag{D.81}$$

The amplitude for a nucleus to absorb the momentum $\boldsymbol{q}$ without changing the state of the lattice is reduced by e^{-W_κ}, and the probability by e^{-2W_κ}, the Debye–Waller factor. The two limits of Eq. (D.80) are those of a single oscillator. At $T=0$, $2n+1=1$, and B_κ is the zero-point tensor, a sum of $1/(2M_\kappa\omega_{\boldsymbol{Q}\nu})$ over the modes. For T above every phonon energy, $2n+1 \to 2T/\omega$, and $B_\kappa = (T/N)\sum_{\boldsymbol{Q}\nu}\boldsymbol{e}_{\kappa\nu}\boldsymbol{e}^*_{\kappa\nu}/(M_\kappa\omega^2_{\boldsymbol{Q}\nu})$ grows linearly with T, the classical equipartition value. The acoustic modes have $\omega_{\boldsymbol{Q}\nu}\to 0$ at small Q, and their terms grow as $1/\omega$ at $T=0$ and as $1/\omega^2$ at $T>0$. In three dimensions the number of modes per unit frequency vanishes as ω^2, and the sum stays finite at any temperature.

The generating formula

In a harmonic crystal the Heisenberg operators evolve as $b_{\boldsymbol{Q}\nu}(t) = b_{\boldsymbol{Q}\nu}\mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}$, and $\boldsymbol{u}_{\boldsymbol{R}\kappa}(t)$ stays linear in the ladder operators. Consider the correlation of the displacement of nucleus κ in cell $\boldsymbol{R}$ at time t with that of nucleus κ' in the cell at the origin at time zero. It follows from the same averages as Eq. (D.80), now with $\langle b_{\boldsymbol{Q}\nu} b^\dagger_{\boldsymbol{Q}\nu}\rangle \mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}$ and $\langle b^\dagger_{-\boldsymbol{Q}\nu} b_{-\boldsymbol{Q}\nu}\rangle \mathrm{e}^{+\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}$:

$$\mathcal{C}^{\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R},t) = \langle u^\alpha_{\boldsymbol{R}\kappa}(t)\, u^\beta_{\boldsymbol{0}\kappa'}(0)\rangle = \frac{1}{N}\sum_{\boldsymbol{Q}\nu} \frac{e^\alpha_{\kappa\nu}(\boldsymbol{Q})\, e^\beta_{\kappa'\nu}(\boldsymbol{Q})^*\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}}{2\omega_{\boldsymbol{Q}\nu}\sqrt{M_\kappa M_{\kappa'}}} \Big[(n_{\boldsymbol{Q}\nu}+1)\,\mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{Q}\nu}t} + n_{\boldsymbol{Q}\nu}\,\mathrm{e}^{\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}\Big]. \tag{D.82}$$

The term with $n+1$ creates a phonon at time zero and removes it at time t; the term with n does the reverse, and it vanishes at $T=0$. At equal times and on the same nucleus, $\mathcal{C}_{\kappa\kappa}(\boldsymbol{0},0) = B_\kappa$. Where the cell and the time are clear from the context, Sect. 5.6 writes $\mathcal{C}_{\kappa\kappa'}(t)$.

The product formula Eq. (C.54), which follows from Eq. (D.79) and the Baker–Campbell–Hausdorff formula, applies with $\hat{A} = -\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(t)$ and $\hat{B} = \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{0}\kappa'}(0)$, both linear. The averages of their squares are $-2W_\kappa(\boldsymbol{q})$ and $-2W_{\kappa'}(\boldsymbol{q})$, since the state is stationary, and the average of their product is $\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa'}(\boldsymbol{R},t)\cdot\boldsymbol{q}$. Hence

$$\big\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(t)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{0}\kappa'}(0)}\big\rangle = \mathrm{e}^{-W_\kappa(\boldsymbol{q})-W_{\kappa'}(\boldsymbol{q})}\, \mathrm{e}^{\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa'}(\boldsymbol{R},t)\cdot\boldsymbol{q}}, \tag{D.83}$$

and for one nucleus, $\boldsymbol{R}=0$ and $\kappa'=\kappa$, this is $\mathrm{e}^{-2W_\kappa}\mathrm{e}^{\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa}(t)\cdot\boldsymbol{q}}$, the generating formula of Sect. 5.6. At $t=0$ the two exponentials of Eq. (D.83) for one nucleus multiply to one, as they must, since $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}} = 1$. Expanding $\mathrm{e}^{\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa'}\cdot\boldsymbol{q}}$ in powers of $\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa'}\cdot\boldsymbol{q}$ sorts the response by the number of phonons. The nth term contains products of n factors $\mathrm{e}^{\mp\mathrm{i}\omega_{\boldsymbol{Q}\nu}t}$, and its Fourier transform puts weight at energies that are sums of n phonon energies, each created with $n_{\boldsymbol{Q}\nu}+1$ or absorbed with $n_{\boldsymbol{Q}\nu}$. The term $n=0$ is the elastic weight e^{-2W_κ}, and the term $n=1$ is the single-phonon response. The total weight of all terms is one, the value at $t=0$.

The Debye model at any temperature

For a crystal of one species, of nuclear mass M, with cubic site symmetry and all atoms of the cell equivalent, B_κ is a multiple of the identity, $B_\kappa = \frac{1}{3}\langle u^2\rangle\,\mathbf{1}$, the same for every κ. Averaging the trace of Eq. (D.80) over the n_{at} atoms of the cell, $\sum_\kappa |\boldsymbol{e}_{\kappa\nu}|^2 = 1$ removes the polarization vectors, and the sum over modes becomes an integral over the density of states per atom, $g_{\mathrm{ph}}/n_{\mathrm{at}}$, normalized to three (Eq. (5.36)). In the Debye model it is $9\omega^2/\omega_{\mathrm{D}}^3$ below the Debye frequency ω_{D} (Eq. (5.24)). With $2n_{\mathrm{B}}(\omega)+1 = \coth(\omega/2T)$,

$$\langle u^2\rangle = \int_0^{\omega_{\mathrm{D}}} \mathrm{d}\omega\, \frac{9\omega^2}{\omega_{\mathrm{D}}^3}\, \frac{\coth(\omega/2T)}{2M\omega} = \frac{9}{4M\omega_{\mathrm{D}}}\Big[1 + 4\Big(\frac{T}{\omega_{\mathrm{D}}}\Big)^2 \int_0^{\omega_{\mathrm{D}}/T} \frac{x\,\mathrm{d}x}{\mathrm{e}^x-1}\Big],$$

where $\coth = 1 + 2n_{\mathrm{B}}$ split the integral into its zero-point and thermal parts and $x = \omega/T$. With $2W_\kappa = q^2\langle u^2\rangle/3$,

$$2W_\kappa(\boldsymbol{q}) = \frac{3q^2}{4M\omega_{\mathrm{D}}}\Big[1 + 4\Big(\frac{T}{\omega_{\mathrm{D}}}\Big)^2 \int_0^{\omega_{\mathrm{D}}/T} \frac{x\,\mathrm{d}x}{\mathrm{e}^x-1}\Big]. \tag{D.84}$$

At $T=0$ the exponent is $\frac{3}{2}E_R/\omega_{\mathrm{D}}$, with $E_R = q^2/2M$ the free recoil energy: the Debye–Waller factor is small once the recoil energy exceeds the phonon energies, which is the condition for a nucleus to recoil as if free (Appendix C.6). At high temperature, the integral is $y - y^2/4 + y^3/36 - \cdots$ with $y = \omega_{\mathrm{D}}/T$, and

$$2W_\kappa(\boldsymbol{q}) = \frac{3q^2T}{M\omega_{\mathrm{D}}^2}\Big[1 + \frac{\omega_{\mathrm{D}}^2}{36\,T^2} + O\Big(\frac{\omega_{\mathrm{D}}^4}{T^4}\Big)\Big],$$

linear in T, with the zero-point motion as the leading correction. At $T = \omega_{\mathrm{D}}$ the bracket of Eq. (D.84) is 4.11: the exponent is four times its zero-point value. A crystal with several atoms per cell or several species needs the full Eq. (D.80), with the polarization vectors of each branch, and Sect. 5.6 computes it for silicon.

D.16 The Allen–Heine–Cardona self-energy

A band energy shifts at second order in the nuclear displacements through two terms of the same order. One is the first-order vertex g of Eq. (A.26) used twice, with a virtual phonon emitted and reabsorbed: the Fan–Migdal self-energy Σ^{FM}. The other is the second-order vertex g^{AH} used once, with its two displacements averaged over the thermal state of the lattice: the Allen–Heine self-energy Σ^{AH}. This section derives both with the rules of Appendix D.11 and continues them to real frequency. Their adiabatic limit is the formula of Allen and Heine [46]. In the rigid-ion approximation, the relation of Allen and Cardona [47] then expresses Σ^{AH} through first-order matrix elements. The field-theoretic forms follow Ref. [48, Sects. V.B and IX.A]; the steps are written out here in the conventions of Appendix A.6. These results are used in Sect. 6.3.

Second-order coupling

Let $V(\boldsymbol{r}; \{\boldsymbol{u}\})$ be the one-electron potential, the Kohn–Sham potential of Appendix D.5, as a function of the displacements $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ of all nuclei from equilibrium. To second order,

$$V = V^{(0)} + \sum_a \frac{\partial V}{\partial u_a}\, u_a + \frac{1}{2} \sum_{ab} \frac{\partial^2 V}{\partial u_a\, \partial u_b}\, u_a u_b,$$

where $a = (\boldsymbol{R}\kappa\alpha)$ collects the cell, the species and the Cartesian index, and the derivatives are total derivatives at equilibrium, which include the self-consistent response of the electrons. Inserting the normal-mode expansion Eq. (A.21) into the linear term and taking its matrix elements between Bloch states gives H_{ep} of Eq. (A.25) with the vertex Eq. (A.26). The same insertion into the quadratic term gives

$$H^{(2)}_{\mathrm{ep}} = \frac{1}{N} \sum_{\boldsymbol{k}\boldsymbol{Q}\boldsymbol{Q}'} \sum_{mn\nu\nu'} g^{\mathrm{AH}}_{mn\nu\nu'}(\boldsymbol{k}, \boldsymbol{Q}, \boldsymbol{Q}')\, c^\dagger_{m,\boldsymbol{k}+\boldsymbol{Q}+\boldsymbol{Q}'} c_{n\boldsymbol{k}}\, B_{\boldsymbol{Q}\nu} B_{\boldsymbol{Q}'\nu'},$$

$$g^{\mathrm{AH}}_{mn\nu\nu'}(\boldsymbol{k}, \boldsymbol{Q}, \boldsymbol{Q}') = \frac{1}{2} \sum_{\kappa\alpha,\kappa'\beta} \frac{e^\alpha_{\kappa\nu}(\boldsymbol{Q})\, e^\beta_{\kappa'\nu'}(\boldsymbol{Q}')}{\sqrt{2M_\kappa \omega_{\boldsymbol{Q}\nu}\, 2M_{\kappa'} \omega_{\boldsymbol{Q}'\nu'}}} \left\langle \psi_{m,\boldsymbol{k}+\boldsymbol{Q}+\boldsymbol{Q}'} \middle| \partial^2_{\kappa\alpha\boldsymbol{Q},\kappa'\beta\boldsymbol{Q}'} V \middle| \psi_{n\boldsymbol{k}} \right\rangle, \qquad \text{(D.85)}$$

with $\partial^2_{\kappa\alpha\boldsymbol{Q},\kappa'\beta\boldsymbol{Q}'} V = \sum_{\boldsymbol{R}\boldsymbol{R}'} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R} + \mathrm{i}\boldsymbol{Q}'\cdot\boldsymbol{R}'}\, \partial^2 V / \partial u^\alpha_{\boldsymbol{R}\kappa} \partial u^\beta_{\boldsymbol{R}'\kappa'}$. The two factors $1/\sqrt{N}$ of Eq. (A.21) combine into the single $1/N$, and the matrix element is of order one in N for the same reason as in Eq. (A.26). This is the normalization of Ref. [48, Eqs. (39) and (40)], where the vertex is called g^{DW}. Both H_{ep} and $H^{(2)}_{\mathrm{ep}}$ are of second order in the zero-point amplitude when they enter an energy: H_{ep} twice, $H^{(2)}_{\mathrm{ep}}$ once. A consistent second-order theory keeps both. The term of H_{ep} with $\boldsymbol{Q} = 0$ and one boson line attached to a closed electron loop, the tadpole, would describe a static shift of the equilibrium positions; it vanishes because the expansion is made about the minimum of the Born–Oppenheimer surface, where $\langle B_{\boldsymbol{0}\nu} \rangle = 0$.

The Fan–Migdal self-energy

The second-order self-energy from H_{ep} is the boson-exchange diagram of Appendix D.11, with two changes. The vertex of Eq. (A.25) carries $1/\sqrt{N}$ instead of $1/\sqrt{\mathcal{V}}$, so the replacement

Eq. (D.58) reads $-|g|^2 \mathsf{D}_0/N$. And the vertex carries band indices: an electron in $(n,\boldsymbol{k})$ goes to $(m,\boldsymbol{k}+\boldsymbol{Q})$ with $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})\,B_{\boldsymbol{Q}\nu}$ and returns with $g_{nm\nu}(\boldsymbol{k}+\boldsymbol{Q},-\boldsymbol{Q})\,B_{-\boldsymbol{Q}\nu}$, which is $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})^*$ by Eq. (A.27). Neglecting the off-diagonal elements in n, which is exact for a nondegenerate level and, once summed over the zone, for a level whose partners form one irreducible representation (Sect. 6.3),

$$\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\mathrm{i}\omega^{\mathrm{F}}_j) = -\frac{1}{N}\sum_{\boldsymbol{Q}\nu m} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2\, T\sum_{j'} \mathsf{D}_0(\boldsymbol{Q}\nu,\mathrm{i}\omega^{\mathrm{B}}_{j'})\,\mathsf{G}_0(m,\boldsymbol{k}+\boldsymbol{Q},\mathrm{i}\omega^{\mathrm{F}}_j-\mathrm{i}\omega^{\mathrm{B}}_{j'}), \tag{D.86}$$

with $\mathsf{D}_0(\boldsymbol{Q}\nu,z) = 1/(z-\omega_{\boldsymbol{Q}\nu}) - 1/(z+\omega_{\boldsymbol{Q}\nu})$ from Eq. (D.57) and $\mathsf{G}_0(m,\boldsymbol{p},z) = 1/(z-\xi_{m\boldsymbol{p}})$, where $\xi_{m\boldsymbol{p}} = \varepsilon_{m\boldsymbol{p}} - \mu_{\mathrm{ch}}$.

The frequency sum is done with the bosonic form of Eq. (C.42). Write $\omega_0 = \omega_{\boldsymbol{Q}\nu}$, $\xi = \xi_{m,\boldsymbol{k}+\boldsymbol{Q}}$ and $z_0 = \mathrm{i}\omega^{\mathrm{F}}_j - \xi$. The function $g(z) = \mathsf{D}_0(z)\,\mathsf{G}_0(\mathrm{i}\omega^{\mathrm{F}}_j - z)$ falls off as $1/|z|^3$ and has three simple poles. At $z=\omega_0$ the residue of $n_{\mathrm{B}}(z)g(z)$ is $n_{\mathrm{B}}(\omega_0)/(z_0-\omega_0)$. At $z=-\omega_0$ it is $-n_{\mathrm{B}}(-\omega_0)/(z_0+\omega_0) = [1+n_{\mathrm{B}}(\omega_0)]/(z_0+\omega_0)$. At $z=z_0$, where $\mathsf{G}_0(\mathrm{i}\omega^{\mathrm{F}}_j - z) = -1/(z-z_0)$, it is $-n_{\mathrm{B}}(z_0)\,\mathsf{D}_0(z_0)$, and since $\mathrm{e}^{\mathrm{i}\beta\omega^{\mathrm{F}}_j} = -1$, $n_{\mathrm{B}}(\mathrm{i}\omega^{\mathrm{F}}_j - \xi) = -1/(\mathrm{e}^{-\beta\xi}+1) = -[1-n_{\mathrm{F}}(\xi)]$. The three residues add up to $[n_{\mathrm{B}}+1-n_{\mathrm{F}}]/(z_0-\omega_0) + [n_{\mathrm{B}}+n_{\mathrm{F}}]/(z_0+\omega_0)$, and the sum is minus this. The two minus signs cancel in Eq. (D.86). Continuing $\mathrm{i}\omega^{\mathrm{F}}_j \to \omega + \mathrm{i}0^+$ by the argument of Appendix D.10, and measuring ω from μ_{ch} as in Sect. 4.7,

Fan–Migdal self-energy

$$\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\omega) = \frac{1}{N}\sum_{\boldsymbol{Q}\nu m} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2 \Big[\frac{n_{\boldsymbol{Q}\nu}+1-f_{m,\boldsymbol{k}+\boldsymbol{Q}}}{\omega-\xi_{m,\boldsymbol{k}+\boldsymbol{Q}}-\omega_{\boldsymbol{Q}\nu}+\mathrm{i}0^+} + \frac{n_{\boldsymbol{Q}\nu}+f_{m,\boldsymbol{k}+\boldsymbol{Q}}}{\omega-\xi_{m,\boldsymbol{k}+\boldsymbol{Q}}+\omega_{\boldsymbol{Q}\nu}+\mathrm{i}0^+}\Big], \tag{D.87}$$

with $f_{m\boldsymbol{p}} = n_{\mathrm{F}}(\xi_{m\boldsymbol{p}})$ and $n_{\boldsymbol{Q}\nu} = n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$. This is Ref. [48, Eqs. (157) and (158)]. The term weighted by $n+1-f$ describes the emission of a phonon, weighted by $n+1$ and blocked by an occupied final state. The term weighted by $n+f$ describes the absorption of a phonon, weighted by n, together with an electron in an occupied state $(m,\boldsymbol{k}+\boldsymbol{Q})$ that emits a phonon and fills the hole, weighted by f. At $T=0$ an empty conduction state can only emit, Σ^{FM} has poles only at $\omega = \xi_m + \omega_{\boldsymbol{Q}\nu}$ for empty m and at $\xi_m - \omega_{\boldsymbol{Q}\nu}$ for occupied m, and the zero-point motion enters through the 1 of $n+1$ and through f.

The imaginary part follows from Eq. (2.58):

$$\mathrm{Im}\,\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\omega) = -\frac{\pi}{N}\sum_{\boldsymbol{Q}\nu m} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2 \Big[(n_{\boldsymbol{Q}\nu}+1-f_{m,\boldsymbol{k}+\boldsymbol{Q}})\,\delta(\omega-\xi_{m,\boldsymbol{k}+\boldsymbol{Q}}-\omega_{\boldsymbol{Q}\nu}) + (n_{\boldsymbol{Q}\nu}+f_{m,\boldsymbol{k}+\boldsymbol{Q}})\,\delta(\omega-\xi_{m,\boldsymbol{k}+\boldsymbol{Q}}+\omega_{\boldsymbol{Q}\nu})\Big]. \tag{D.88}$$

Both weights are nonnegative, since $n+1-f \geq 0$ and $n+f \geq 0$, so $\mathrm{Im}\,\Sigma^{\mathrm{FM}} \leq 0$, as Eq. (4.82) requires. At $\omega = \xi_{n\boldsymbol{k}}$, Eq. (4.83) turns it into the phonon-limited decay rate $\Gamma_{n\boldsymbol{k}} = -2z_{\mathrm{qp}}\,\mathrm{Im}\,\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\xi_{n\boldsymbol{k}})$, which is the golden rule Eq. (2.67) summed over the emission and absorption of one phonon with the weights Eq. (A.28). An electron at the bottom of the conduction band of an insulator at $T=0$ has no state below it to emit into, and its rate vanishes.

The Allen–Heine self-energy

The second-order vertex enters at first order. To lowest order in the coupling, the phonons in $H^{(2)}_{\mathrm{ep}}$ may be replaced by their thermal average in the free lattice. The operators $B_{\boldsymbol{Q}\nu}$ and

$B_{-\boldsymbol{Q}\nu} = B^\dagger_{\boldsymbol{Q}\nu}$ commute, so their product needs no ordering rule, and by Eq. (2.69)

$$\langle B_{\boldsymbol{Q}\nu} B_{\boldsymbol{Q}'\nu'} \rangle = \delta_{\boldsymbol{Q}',-\boldsymbol{Q}}\, \delta_{\nu\nu'}\, \langle b_{\boldsymbol{Q}\nu} b^\dagger_{\boldsymbol{Q}\nu} + b^\dagger_{-\boldsymbol{Q}\nu} b_{-\boldsymbol{Q}\nu} \rangle = \delta_{\boldsymbol{Q}',-\boldsymbol{Q}}\, \delta_{\nu\nu'}\, (2n_{\boldsymbol{Q}\nu} + 1).$$

The average of $H^{(2)}_{\mathrm{ep}}$ is then a one-electron operator that conserves $\boldsymbol{k}$. A static one-body term enters the Dyson equation directly as a self-energy, the first-order rule of Appendix D.11 for a potential that carries no interaction line, so

Allen–Heine self-energy

$$\Sigma^{\mathrm{AH}}_{n\boldsymbol{k}} = \frac{1}{N} \sum_{\boldsymbol{Q}\nu} g^{\mathrm{AH}}_{nn\nu\nu}(\boldsymbol{k}, \boldsymbol{Q}, -\boldsymbol{Q})\, (2n_{\boldsymbol{Q}\nu} + 1). \tag{D.89}$$

This is Ref. [48, Eqs. (160) and (161)]. It is real and independent of frequency. Its diagram is a phonon line that starts and ends at the same point of the electron line. The thermal weight $2n + 1$ is that of the mean-square displacement, Worked example 2.7: Σ^{AH} is the average of the second-order potential over the zero-point and thermal motion of the nuclei. The literature calls it the Debye–Waller self-energy. It is not the factor e^{-2W_κ} of Sect. 5.6, which reduces the weight of a scattering process; Σ^{AH} shifts an energy level (Appendix A.6).

The adiabatic limit

The band energy renormalized to second order is $\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}} = \varepsilon_{n\boldsymbol{k}} + \mathrm{Re}[\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\omega) + \Sigma^{\mathrm{AH}}_{n\boldsymbol{k}}]$ at the pole of Eq. (4.81). Evaluating the self-energy at the unperturbed energy and dropping the phonon energies in the denominators gives the formula of Allen and Heine. Setting $\omega = \xi_{n\boldsymbol{k}}$ is Rayleigh–Schrödinger perturbation theory. Dropping $\pm\omega_{\boldsymbol{Q}\nu}$ against $\xi_{n\boldsymbol{k}} - \xi_{m,\boldsymbol{k}+\boldsymbol{Q}}$ is the adiabatic approximation, valid when the electronic energy differences that dominate the sum are large compared with phonon energies; it fails for $m = n$ at small Q and for degenerate states, where the denominator itself is of order $\omega_{\boldsymbol{Q}\nu}$. The two terms of Eq. (D.87) then have the same denominator, the occupations cancel in the sum of numerators, $(n+1-f)+(n+f) = 2n+1$, and

$$\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}} = \frac{1}{N} \sum_{\boldsymbol{Q}\nu} (2n_{\boldsymbol{Q}\nu} + 1) \left[\sum_m \frac{|g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q})|^2}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}} + g^{\mathrm{AH}}_{nn\nu\nu}(\boldsymbol{k}, \boldsymbol{Q}, -\boldsymbol{Q}) \right], \tag{D.90}$$

which is Ref. [48, Eq. (167)]. The occupations have dropped out, and every band m, occupied or empty, contributes. At $T = 0$ the weight is 1, and Eq. (D.90) is the zero-point renormalization.

The same formula follows without diagrams, which was the route of Allen and Heine [46]. Freeze the nuclei at displacements u_a. To second order in u, nondegenerate perturbation theory gives

$$\begin{aligned} \varepsilon_{n\boldsymbol{k}}(\{u\}) - \varepsilon_{n\boldsymbol{k}} = {} & \sum_a \langle n\boldsymbol{k} | \partial_a V | n\boldsymbol{k} \rangle\, u_a + \frac{1}{2} \sum_{ab} \Big[\langle n\boldsymbol{k} | \partial_a \partial_b V | n\boldsymbol{k} \rangle \\ & + 2\,\mathrm{Re} \sum_{m\boldsymbol{k}'}{}' \frac{\langle n\boldsymbol{k} | \partial_a V | m\boldsymbol{k}' \rangle \langle m\boldsymbol{k}' | \partial_b V | n\boldsymbol{k} \rangle}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}'}} \Big] u_a u_b, \end{aligned} \tag{D.91}$$

where $\partial_a = \partial/\partial u_a$ and the prime removes the states degenerate with $(n, \boldsymbol{k})$. Averaging over the harmonic lattice removes the linear term and replaces $u_a u_b$ by $\langle u_a u_b \rangle$. With Eq. (A.21), the average couples $\boldsymbol{Q}$ only to $-\boldsymbol{Q}$ with the weight $(2n_{\boldsymbol{Q}\nu} + 1)/(2N\sqrt{M_\kappa M_{\kappa'}}\,\omega_{\boldsymbol{Q}\nu})$ times the polarization vectors, and the two brackets become, term by term, the Fan sum and g^{AH} of Eq. (D.90). The diagrammatic form adds what the frozen picture misses: the phonon energies in the denominators, the separate weights of emission and absorption, and the imaginary part.

Translation invariance and the rigid-ion approximation

The second derivatives of V in g^{AH} require second-order density-functional perturbation theory. Translation invariance trades them for products of first derivatives, which a phonon calculation already supplies. Displace the whole crystal rigidly by $\boldsymbol{t}$, on top of an arbitrary configuration $\{u\}$. The potential follows, $V(\boldsymbol{r};\{\boldsymbol{u}+\boldsymbol{t}\}) = V(\boldsymbol{r}-\boldsymbol{t};\{\boldsymbol{u}\})$, and the eigenvalues do not change. Hence $\sum_{\boldsymbol{R}\kappa} \partial\varepsilon_{n\boldsymbol{k}}/\partial u^{\alpha}_{\boldsymbol{R}\kappa} = 0$ for every configuration. At equilibrium this is the first sum rule,

$$\sum_{\boldsymbol{R}\kappa} \langle n\boldsymbol{k}|\partial_{\boldsymbol{R}\kappa\alpha} V|n\boldsymbol{k}\rangle = \sum_{\kappa} \langle n\boldsymbol{k}|\partial_{\kappa\alpha,\boldsymbol{0}} V|n\boldsymbol{k}\rangle = 0, \tag{D.92}$$

by the Hellmann–Feynman theorem, with $\partial_{\kappa\alpha,\boldsymbol{Q}} V$ of Appendix A.6; it states that the diagonal vertex of a uniform translation vanishes, the acoustic sum rule of Sect. 6.2. Differentiating the identity once more with respect to $u^{\beta}_{\boldsymbol{R}'\kappa'}$ and using the second-order coefficient of Eq. (D.91) gives the second sum rule,

$$\sum_{\boldsymbol{R}\kappa} \langle n\boldsymbol{k}|\partial_{\boldsymbol{R}\kappa\alpha}\partial_{\boldsymbol{R}'\kappa'\beta} V|n\boldsymbol{k}\rangle = -2\,\mathrm{Re}\sum_{\kappa}\sum_{m}{}' \frac{\langle n\boldsymbol{k}|\partial_{\kappa\alpha,\boldsymbol{0}} V|m\boldsymbol{k}\rangle\,\langle m\boldsymbol{k}|\partial_{\boldsymbol{R}'\kappa'\beta} V|n\boldsymbol{k}\rangle}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}}}, \tag{D.93}$$

where $\partial_{\kappa\alpha,\boldsymbol{0}} V = \sum_{\boldsymbol{R}} \partial_{\boldsymbol{R}\kappa\alpha} V$ conserves $\boldsymbol{k}$, so that only $\boldsymbol{k}' = \boldsymbol{k}$ survives. Both rules are exact; they are Ref. [48, Eqs. (193) and (194)].

The left side of Eq. (D.93) is a sum over all sites, and it does not yet give the on-site element that g^{AH} needs. The rigid-ion approximation closes the gap. It writes the potential as a sum of atom-centered terms that move with their nuclei, $V(\boldsymbol{r}) = \sum_{\boldsymbol{R}\kappa} v_{\kappa}(\boldsymbol{r} - \boldsymbol{R} - \boldsymbol{\tau}_{\kappa} - \boldsymbol{u}_{\boldsymbol{R}\kappa})$, with $\boldsymbol{\tau}_{\kappa}$ the position of species κ in the cell. Then $\partial^2 V/\partial u_{\boldsymbol{R}\kappa\alpha}\partial u_{\boldsymbol{R}'\kappa'\beta}$ vanishes unless $(\boldsymbol{R}\kappa) = (\boldsymbol{R}'\kappa')$, and the left side of Eq. (D.93) is the on-site element $\langle n\boldsymbol{k}|\partial_{\alpha}\partial_{\beta} v_{\kappa'}(\boldsymbol{r} - \boldsymbol{R}' - \boldsymbol{\tau}_{\kappa'})|n\boldsymbol{k}\rangle$. On the right, the matrix element of the single-site operator $\partial_{\boldsymbol{R}'\kappa'\beta} V$ between two states of the same $\boldsymbol{k}$ is the same for every cell $\boldsymbol{R}'$, by the translation symmetry of Bloch states, and equals $N^{-1}\langle m\boldsymbol{k}|\partial_{\kappa'\beta,\boldsymbol{0}} V|n\boldsymbol{k}\rangle$. Hence

$$\langle n\boldsymbol{k}|\partial_{\alpha}\partial_{\beta} v_{\kappa'}|n\boldsymbol{k}\rangle = -\frac{2}{N}\,\mathrm{Re}\sum_{\kappa}\sum_{m}{}' \frac{\langle n\boldsymbol{k}|\partial_{\kappa\alpha,\boldsymbol{0}} V|m\boldsymbol{k}\rangle\,\langle m\boldsymbol{k}|\partial_{\kappa'\beta,\boldsymbol{0}} V|n\boldsymbol{k}\rangle}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}}}, \tag{D.94}$$

for every site of species κ'. The second derivative of the potential has been expressed through first derivatives at $\boldsymbol{Q} = 0$.

In the same approximation the Allen–Heine term is the on-site element averaged with the displacement tensor $B_{\kappa} = \langle \boldsymbol{u}_{\boldsymbol{R}\kappa}\boldsymbol{u}_{\boldsymbol{R}\kappa}\rangle$ of Eq. (A.29). Only the diagonal terms $(\boldsymbol{R}\kappa) = (\boldsymbol{R}'\kappa')$ of the quadratic potential survive, and by Eqs. (A.21) and (A.22),

$$\Sigma^{\mathrm{AH}}_{n\boldsymbol{k}} = \frac{1}{2}\sum_{\boldsymbol{R}\kappa'}\sum_{\alpha\beta} B^{\alpha\beta}_{\kappa'}\,\langle n\boldsymbol{k}|\partial_{\alpha}\partial_{\beta} v_{\kappa'}|n\boldsymbol{k}\rangle, \qquad B^{\alpha\beta}_{\kappa} = \frac{1}{N}\sum_{\boldsymbol{Q}\nu} \frac{2n_{\boldsymbol{Q}\nu}+1}{2M_{\kappa}\omega_{\boldsymbol{Q}\nu}}\, e^{\alpha}_{\kappa\nu}(\boldsymbol{Q})\, e^{\beta}_{\kappa\nu}(\boldsymbol{Q})^{*}.$$

The sum over cells gives a factor N. We insert Eq. (D.94) and symmetrize between $(\kappa\alpha)$ and $(\kappa'\beta)$, which leaves the real part of the product unchanged:

Allen–Heine term in the rigid-ion approximation

$$\Sigma^{\mathrm{AH}}_{n\boldsymbol{k}} = -\frac{1}{2}\sum_{\kappa\kappa'}\sum_{\alpha\beta}(B^{\alpha\beta}_{\kappa} + B^{\alpha\beta}_{\kappa'})\,\mathrm{Re}\sum_{m}{}' \frac{\langle n\boldsymbol{k}|\partial_{\kappa\alpha,\boldsymbol{0}} V|m\boldsymbol{k}\rangle\,\langle m\boldsymbol{k}|\partial_{\kappa'\beta,\boldsymbol{0}} V|n\boldsymbol{k}\rangle}{\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}}}. \tag{D.95}$$

Writing B_{κ} as its sum over modes gives Ref. [48, Eqs. (195)–(198)], where the $\boldsymbol{Q} = 0$ matrix elements are rebuilt from the vertices $g_{mn\nu}(\boldsymbol{k}, \boldsymbol{0})$ of the optical modes. This is the Allen–Cardona form for crystals with several atoms in the cell [47].

The Allen–Heine term Eq. (D.95) has the structure of a Fan term at $\boldsymbol{Q} = 0$ with the opposite sign, and the two partly cancel. Translation invariance requires this: for a uniform translation, the acoustic mode at $\boldsymbol{Q} = 0$, the two terms cancel exactly and no level moves. The translational modes at $\boldsymbol{Q} = 0$ themselves, with $\omega_{\boldsymbol{Q}\nu} = 0$, are excluded from all sums, consistently in both terms. And the rule Eq. (D.93) holds for any potential, while Eq. (D.94) needs the rigid-ion form. A self-consistent potential has off-site second derivatives, because moving two nuclei together changes the screening charge differently from moving each alone; Eq. (D.95) drops them. The error need not be small. In a one-dimensional crystal of Gaussian wells, adding a well between two atoms whose depth depends on their separation leaves Eq. (D.93) exact and makes Eq. (D.94) wrong by 66 %. In crystals the neglected term is usually taken to be small; Ref. [49] reports that it is large in molecules.

D.17 Phonon self-energy and the Eliashberg function

A phonon that can create an electron–hole pair of its own energy decays into it, and a phonon that can only do so virtually is shifted. Both effects come from one diagram, the electron bubble with two vertices g inserted in a phonon line. This section derives the bubble from the expansion of the phonon Green's function and sums it into a Dyson equation. From it follow the linewidth $\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}$, its form for metals due to Allen [50], the Eliashberg function $\alpha^2F(\omega)$ and the coupling λ_{ep}, and the identities that connect them. The conventions are those of Appendices A.6, D.11 and D.16; the results are used in Sects. 6.4 and 7.2.

The bubble

The phonon Green's function in imaginary time is $\mathsf{D}_\nu(\boldsymbol{Q},\tau) = -\langle \mathrm{T}_\tau\, B_{\boldsymbol{Q}\nu}(\tau)\, B_{-\boldsymbol{Q}\nu}(0)\rangle$, with the free form $\mathsf{D}_0(\boldsymbol{Q}\nu, \mathrm{i}\omega_j^{\mathrm{B}}) = 2\omega_{\boldsymbol{Q}\nu}/[(\mathrm{i}\omega_j^{\mathrm{B}})^2 - \omega^2_{\boldsymbol{Q}\nu}]$ of Eq. (D.57). Its expansion in H_{ep} of Eq. (A.25) is Eq. (D.52) with $B_{\boldsymbol{Q}\nu}$ and $B_{-\boldsymbol{Q}\nu}$ as the external operators. The first nonzero correction is of second order. The expansion supplies $(-1)^2/2!$. The external $B_{\boldsymbol{Q}\nu}(\tau)$ is contracted with the boson of one of the two factors H_{ep}, and $B_{-\boldsymbol{Q}\nu}(0)$ with the other; the two choices are equal after relabeling and cancel the $1/2!$. Each boson contraction is $-\mathsf{D}_0$. The vertex at τ_1 must carry $B_{-\boldsymbol{Q}\nu}$, and the one at τ_2 carries $B_{\boldsymbol{Q}\nu}$, so the vertices are $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})^*$ and $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})$ by Eq. (A.27), with the factor $1/N$ from the two $1/\sqrt{N}$. The two electron bilinears close into one loop, which gives -1 (rule 3 of Appendix D.11) and a sum over two spin orientations. With the -1 of the definition, the total sign is $(-1)(-1)^2(-1) = +1$. In frequency space the correction is $\mathsf{D}_0\,\Pi^{\mathrm{b}}\,\mathsf{D}_0$ with

$$\Pi^{\mathrm{b}}_\nu(\boldsymbol{Q}, \mathrm{i}\omega_j^{\mathrm{B}}) = \frac{2}{N}\sum_{\boldsymbol{k}mn} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2\, T\sum_{j'} \mathsf{G}_0(n,\boldsymbol{k},\mathrm{i}\omega_{j'}^{\mathrm{F}})\, \mathsf{G}_0(m,\boldsymbol{k}+\boldsymbol{Q},\mathrm{i}\omega_{j'}^{\mathrm{F}} + \mathrm{i}\omega_j^{\mathrm{B}}).$$

The frequency sum is Eq. (D.51) with band labels, and

$$\Pi^{\mathrm{b}}_\nu(\boldsymbol{Q}, \mathrm{i}\omega_j^{\mathrm{B}}) = \frac{2}{N}\sum_{\boldsymbol{k}mn} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2\, \frac{f_{n\boldsymbol{k}} - f_{m,\boldsymbol{k}+\boldsymbol{Q}}}{\mathrm{i}\omega_j^{\mathrm{B}} + \varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}}.$$

This is the density bubble of Eq. (4.93) with the density vertex replaced by the electron–phonon vertex. Bubbles that connect ν to another branch at the same $\boldsymbol{Q}$ are dropped (Sect. 6.4).

The Dyson equation and the self-energy

Repeating the bubble along the phonon line gives a geometric series, $\mathsf{D} = \mathsf{D}_0 + \mathsf{D}_0\Pi^{\mathrm{b}}\mathsf{D}_0 + \cdots = \mathsf{D}_0 + \mathsf{D}_0\Pi^{\mathrm{b}}\mathsf{D}$, so that $\mathsf{D}^{-1} = \mathsf{D}_0^{-1} - \Pi^{\mathrm{b}}$. With $\mathsf{D}_0^{-1} = [(\mathrm{i}\omega_j^{\mathrm{B}})^2 - \omega^2_{\boldsymbol{Q}\nu}]/2\omega_{\boldsymbol{Q}\nu}$ it is natural to define the self-energy in units of a squared frequency, $\Pi_\nu = 2\omega_{\boldsymbol{Q}\nu}\Pi^{\mathrm{b}}_\nu$. Continuing to $\omega + \mathrm{i}0^+$,

Phonon self-energy and Dyson equation

$$\mathsf{D}_\nu(\boldsymbol{Q},\omega) = \frac{2\omega_{\boldsymbol{Q}\nu}}{\omega^2 - \omega_{\boldsymbol{Q}\nu}^2 - \Pi_\nu(\boldsymbol{Q},\omega)},$$
$$\Pi_\nu(\boldsymbol{Q},\omega) = \frac{4\omega_{\boldsymbol{Q}\nu}}{N}\sum_{\boldsymbol{k}mn} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2 \frac{f_{n\boldsymbol{k}} - f_{m,\boldsymbol{k}+\boldsymbol{Q}}}{\omega + \varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} + \mathrm{i}0^+}. \tag{D.96}$$

The poles of D are the renormalized phonons, $\omega^2 = \omega_{\boldsymbol{Q}\nu}^2 + \Pi_\nu(\boldsymbol{Q},\omega)$, and in an insulator every term of $\Pi_\nu(\boldsymbol{Q},0)$ is negative (Sect. 6.4). This form, with the screened vertex g in both places and noninteracting electrons, is Ref. [48, Eq. (146) and the text before it] up to the normalization of Π; the more complete expression uses the bare vertex at one end [48, Eq. (145)].

The static part of Π_ν needs care. The phonons of a density-functional calculation are second derivatives of the Born–Oppenheimer energy with the electrons relaxed to their ground state at each nuclear configuration (Sect. 5.4). They already contain the response of the electrons at zero frequency, which is $\Pi_\nu(\boldsymbol{Q},0)$ to this order. Starting from those frequencies and adding $\Pi_\nu(\boldsymbol{Q},\omega)$ would count the static response twice. Only the nonadiabatic remainder may be added [51],

$$\left(\omega_{\boldsymbol{Q}\nu}^{\mathrm{ren}}\right)^2 = \omega_{\boldsymbol{Q}\nu}^2 + \mathrm{Re}\left[\Pi_\nu(\boldsymbol{Q},\omega_{\boldsymbol{Q}\nu}^{\mathrm{ren}}) - \Pi_\nu(\boldsymbol{Q},0)\right], \tag{D.97}$$

with $\omega_{\boldsymbol{Q}\nu}$ the adiabatic frequency and $\omega_{\boldsymbol{Q}\nu}^{\mathrm{ren}}$ the renormalized one; this is the subtraction $W_e(\omega) - W_e(0)$ of Ref. [48, Eq. (140)]. In an insulator with $\omega_{\boldsymbol{Q}\nu} \ll E_{\mathrm{g}}$ the difference is of relative order $(\omega_{\boldsymbol{Q}\nu}/E_{\mathrm{g}})^2$, since Π_ν is even in ω off resonance, and the adiabatic phonons are accurate. In a metal the intraband terms with $|\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}| \lesssim \omega$ make the difference of order one for those pairs; for small Q it can be large. The static bubble also inherits from the electron gas the singularity of the Lindhard function at $Q = 2k_{\mathrm{F}}$ (Appendix D.8), and the adiabatic frequencies of a metal show a kink there, the Kohn anomaly.

The linewidth

Near the pole, Π_ν is small compared with $\omega_{\boldsymbol{Q}\nu}^2$, and $\omega^2 - \omega_{\boldsymbol{Q}\nu}^2 - \Pi_\nu \simeq 2\omega_{\boldsymbol{Q}\nu}[\omega - \omega_{\boldsymbol{Q}\nu} - \Pi_\nu(\boldsymbol{Q},\omega_{\boldsymbol{Q}\nu})/2\omega_{\boldsymbol{Q}\nu}]$. Then $\mathsf{D}_\nu \simeq 1/(\omega - \tilde{\omega} + \mathrm{i}\gamma/2)$, a Lorentzian of full width γ at half maximum, centered at $\tilde{\omega} = \omega_{\boldsymbol{Q}\nu} + \mathrm{Re}\,\Pi_\nu/2\omega_{\boldsymbol{Q}\nu}$. With Eq. (2.58) in Eq. (D.96),

Electron–phonon linewidth

$$\gamma_{\boldsymbol{Q}\nu}^{\mathrm{ep}} = -\frac{\mathrm{Im}\,\Pi_\nu(\boldsymbol{Q},\omega_{\boldsymbol{Q}\nu})}{\omega_{\boldsymbol{Q}\nu}} = \frac{4\pi}{N}\sum_{\boldsymbol{k}mn} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2 \left(f_{n\boldsymbol{k}} - f_{m,\boldsymbol{k}+\boldsymbol{Q}}\right)\delta\left(\varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{n\boldsymbol{k}} - \omega_{\boldsymbol{Q}\nu}\right), \tag{D.98}$$

a full width, as for $\gamma_{\boldsymbol{Q}\nu}$ in Sect. 5.7, with the spin sum written as the factor in $4\pi = 2 \times 2\pi$. It is the golden-rule rate Eq. (2.67) at which a phonon is absorbed by creating an electron–hole pair, minus the rate of the inverse process; $f_n(1-f_m) - f_m(1-f_n) = f_n - f_m$. Ref. [48, Eq. (146)] writes the same rate as an inverse lifetime, with the spin factor 2 written explicitly, and its $1/\tau$ equals $\gamma_{\boldsymbol{Q}\nu}^{\mathrm{ep}}$. In an undoped insulator at $T = 0$ the delta function needs $\varepsilon_m - \varepsilon_n = \omega_{\boldsymbol{Q}\nu}$ between an occupied and an empty state, which is impossible for $\omega_{\boldsymbol{Q}\nu} < E_{\mathrm{g}}$: then $\gamma_{\boldsymbol{Q}\nu}^{\mathrm{ep}} = 0$, and phonons of an insulator are broadened only by anharmonicity and by defects. At $T > 0$

the thermally excited carriers add a width proportional to their density, of order $\mathrm{e}^{-E_\mathrm{g}/2T}$ in an intrinsic crystal.

In a metal the electronic states near the Fermi level change on a scale far larger than $\omega_{\boldsymbol{Q}\nu}$. For $T \ll E_\mathrm{F}$, the occupation difference at $\varepsilon_m = \varepsilon_n + \omega_{\boldsymbol{Q}\nu}$ is $f(\varepsilon_n) - f(\varepsilon_n + \omega) \simeq -\omega\, \partial f/\partial\varepsilon$, which tends to $\omega\, \delta(\xi_{n\boldsymbol{k}})$ for $T \to 0$. Dropping $\omega_{\boldsymbol{Q}\nu}$ also in the remaining delta function gives Allen's form [50],

$$\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu} \simeq \frac{4\pi\omega_{\boldsymbol{Q}\nu}}{N} \sum_{\boldsymbol{k}mn} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2\, \delta(\xi_{n\boldsymbol{k}})\, \delta(\xi_{m,\boldsymbol{k}+\boldsymbol{Q}}), \tag{D.99}$$

which is positive term by term and has lost its temperature dependence. It is Ref. [48, Eq. (185)], where γ denotes the half width and the spin sum is included. Its range of validity is discussed in Sect. 6.4.

The Eliashberg function

The double sum in Eq. (D.99) is the Fermi-surface average of $|g|^2$ for the phonon $(\boldsymbol{Q},\nu)$. Weighting these averages over all phonons at energy ω defines the Eliashberg function,

$$\alpha^2F(\omega) = \frac{1}{\mathcal{N}_\mathrm{F}\Omega\, N^2} \sum_{\boldsymbol{k}\boldsymbol{Q}} \sum_{mn\nu} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2\, \delta(\xi_{n\boldsymbol{k}})\, \delta(\xi_{m,\boldsymbol{k}+\boldsymbol{Q}})\, \delta(\omega - \omega_{\boldsymbol{Q}\nu}), \tag{D.100}$$

with Ω the cell volume, so that $\mathcal{N}_\mathrm{F}\Omega$ is the density of states per spin and per cell at the Fermi level, $N\mathcal{N}_\mathrm{F}\Omega = \sum_{n\boldsymbol{k}} \delta(\xi_{n\boldsymbol{k}})$. The prefactor makes α^2F an average over initial states on the Fermi surface and a sum over final states, and it is dimensionless. This is Ref. [48, Eq. (205)], with $\int \mathrm{d}\boldsymbol{k}/V_\mathrm{BZ} \to N^{-1}\sum_{\boldsymbol{k}}$. Comparing Eqs. (D.99) and (D.100) term by term,

Eliashberg function, coupling constant and linewidths

$$\begin{aligned} \alpha^2F(\omega) &= \frac{1}{4\pi\mathcal{N}_\mathrm{F}\Omega\, N} \sum_{\boldsymbol{Q}\nu} \frac{\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}}{\omega_{\boldsymbol{Q}\nu}}\, \delta(\omega - \omega_{\boldsymbol{Q}\nu}), \\ \lambda_\mathrm{ep} &= 2\int_0^\infty \mathrm{d}\omega\, \frac{\alpha^2F(\omega)}{\omega} = \frac{1}{2\pi\mathcal{N}_\mathrm{F}\Omega\, N} \sum_{\boldsymbol{Q}\nu} \frac{\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}}{\omega^2_{\boldsymbol{Q}\nu}}. \end{aligned} \tag{D.101}$$

The coupling constant is a sum of mode contributions $\lambda_{\boldsymbol{Q}\nu} = \gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}/(2\pi\mathcal{N}_\mathrm{F}\Omega\,\omega^2_{\boldsymbol{Q}\nu})$ averaged over the zone. Soft modes with large widths dominate it, and a measured linewidth gives the coupling of each mode directly, which was the point of Allen's relation [50]. The numerical factors in Eq. (D.101) depend on whether the width is full or half and on whether $\mathcal{N}_\mathrm{F}$ counts one spin or two; with the full width and one spin they are as written.

Mass enhancement

The same function controls the electron self-energy near the Fermi surface, which is where the name coupling constant comes from. Average the Fan–Migdal self-energy Eq. (D.87) of a state $(n,\boldsymbol{k})$ over the Fermi surface, and insert $1 = \int \mathrm{d}\omega'\, \delta(\omega' - \omega_{\boldsymbol{Q}\nu}) \int \mathrm{d}\varepsilon\, \delta(\varepsilon - \xi_{m,\boldsymbol{k}+\boldsymbol{Q}})$. Suppose that the density of final states and the vertices change little over the energy range ω_D of the phonons. Then $\delta(\varepsilon - \xi_{m,\boldsymbol{k}+\boldsymbol{Q}})$ may be replaced by $\delta(\xi_{m,\boldsymbol{k}+\boldsymbol{Q}})$ under the average, and the weight of each (ω',ε) is $\alpha^2F(\omega')$ by Eq. (D.100):

$$\langle\Sigma^\mathrm{FM}\rangle_\mathrm{FS}(\omega) = \int_0^\infty \mathrm{d}\omega'\, \alpha^2F(\omega') \int_{-E_\mathrm{band}}^{E_\mathrm{band}} \mathrm{d}\varepsilon \left[\frac{n_\mathrm{B}(\omega') + 1 - n_\mathrm{F}(\varepsilon)}{\omega - \varepsilon - \omega' + \mathrm{i}0^+} + \frac{n_\mathrm{B}(\omega') + n_\mathrm{F}(\varepsilon)}{\omega - \varepsilon + \omega' + \mathrm{i}0^+} \right],$$

where the cutoff $E_{\text{band}} \gg \omega_{\text{D}}$ is of the order of the band width. At $T = 0$ only $1-\Theta(-\varepsilon) = \Theta(\varepsilon)$ survives in the first numerator and $\Theta(-\varepsilon)$ in the second. The ε integrals are logarithms, $-\ln[(E_{\text{band}}+\omega'-\omega)/(\omega'-\omega)]$ and $\ln[(E_{\text{band}}+\omega'+\omega)/(\omega'+\omega)]$, whose derivatives at $\omega = 0$ are $-1/\omega' + 1/(E_{\text{band}}+\omega')$ each and tend to $-1/\omega'$ for $E_{\text{band}} \to \infty$; a finite band reduces each by the fraction $\omega'/(E_{\text{band}}+\omega')$, 1 % for $E_{\text{band}} = 100\,\omega'$. Hence

$$-\frac{\partial\,\mathrm{Re}\langle\Sigma^{\text{FM}}\rangle_{\text{FS}}}{\partial\omega}\bigg|_{\omega=0} = 2\int_0^\infty \mathrm{d}\omega'\,\frac{\alpha^2F(\omega')}{\omega'} = \lambda_{\text{ep}}. \tag{D.102}$$

By Eq. (4.83), quasiparticles within ω_{D} of the Fermi level have the residue $z_{\text{qp}} = 1/(1+\lambda_{\text{ep}})$, their velocity is reduced by the same factor, and the effective mass and the electronic specific heat are enhanced by $1+\lambda_{\text{ep}}$. The Allen–Heine term of Eq. (D.89) does not depend on ω and does not contribute.

Vertex corrections

The bubble keeps the vertex undressed at both ends. A vertex correction inserts a second phonon line across one vertex, and the counting of Migdal's theorem in Sect. 6.3 applies to it unchanged: the correction is of relative order $\lambda_{\text{ep}}\,\omega_{\text{D}}/E_{\text{F}}$ [52].

The electron gas

The electron gas makes the passage from Eq. (D.98) to Eq. (D.99) explicit. For one parabolic band and a constant vertex, the angular integral in $(1/\mathcal{V})\sum_{\boldsymbol{k}}(f_{\boldsymbol{k}} - f_{\boldsymbol{k}+\boldsymbol{Q}})\,\delta(\varepsilon_{\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{\boldsymbol{k}} - \omega)$ gives $m_e/(kQ)$ wherever the delta function can be satisfied, and with $k\,\mathrm{d}k = m_e\,\mathrm{d}\varepsilon$ the radial integral becomes $\int \mathrm{d}\varepsilon\,[f(\varepsilon) - f(\varepsilon+\omega)] = \omega$. The result, $m_e^2\omega/(4\pi^2 Q)$ per spin and volume, holds at any $T \ll E_{\text{F}}$ for $2k_{\text{F}} > Q \gg \omega/v_{\text{F}}$, and it is exactly what the double delta function of Eq. (D.99) gives, $\omega\int \mathrm{d}^3k\,(2\pi)^{-3}\delta(\xi_{\boldsymbol{k}})\,\delta(\xi_{\boldsymbol{k}+\boldsymbol{Q}}) = m_e^2\omega/(4\pi^2Q)$.

D.18 Phonon-assisted absorption

A photon carries almost no momentum on the scale of the Brillouin zone, so it can only lift an electron vertically, from $(v,\boldsymbol{k})$ to $(c,\boldsymbol{k})$. Across an indirect gap, where the conduction minimum sits at $\boldsymbol{k}_{\text{c}} \neq 0$ (Sect. 3.3), the lowest vertical transition lies well above E_{g}, and absorption between E_{g} and the direct gap needs a second process that supplies the momentum. A phonon does: at second order the photon and the phonon act in either order, through virtual intermediate states. This section derives the rate, writes it as the phonon-assisted dielectric function $\varepsilon_2^{\text{PA}}$ and as an absorption coefficient, and derives the quadratic edge of Hall, Bardeen and Blatt [53]. The results are used in Sect. 6.5 and, with the photon replaced by dark matter, in Chap. 13.

Absorption at first order

Take a uniform field $\boldsymbol{\mathcal{E}}(t) = -\mathcal{E}\,\hat{\boldsymbol{e}}_\gamma\cos\omega t$ with unit polarization vector $\hat{\boldsymbol{e}}_\gamma$, described by the vector potential $\boldsymbol{A}(t) = (\mathcal{E}/\omega)\,\hat{\boldsymbol{e}}_\gamma\sin\omega t$. To first order it couples to the electrons through $H_A = e\boldsymbol{A}\cdot\hat{\boldsymbol{V}}$, with $\hat{\boldsymbol{V}}$ the total velocity of the electrons, $\sum_j \boldsymbol{p}_j/m_e$ for a local potential, and the sign of the charge irrelevant for rates. In second quantization over Bloch states, $\hat{\boldsymbol{V}} = \sum_{\boldsymbol{k}mn}\boldsymbol{v}_{mn}(\boldsymbol{k})\,c^\dagger_{m\boldsymbol{k}}c_{n\boldsymbol{k}}$, with the velocity matrix elements of Eq. (3.38); it conserves $\boldsymbol{k}$ because $\boldsymbol{p}$ commutes with lattice translations. The part of H_A that oscillates as $\mathrm{e}^{-\mathrm{i}\omega t}$ absorbs energy ω, with the amplitude $(e\mathcal{E}/2\omega)\,\hat{\boldsymbol{e}}_\gamma\cdot\hat{\boldsymbol{V}}$ up to a phase. The golden rule Eq. (2.67) gives the absorption rate between many-body states i and f, and the absorbed power is ω times

the net rate, absorption minus stimulated emission. Equating the time-averaged power per volume with $\sigma_1\mathcal{E}^2/2$ and using $\sigma_1 = \omega\varepsilon_2$ (Sect. 4.4),

$$\varepsilon_2(\omega) = \frac{\pi e^2}{\omega^2 \mathcal{V}} \sum_{i,f} (p_i - p_f) \left| \hat{\boldsymbol{e}}_\gamma \cdot \langle f | \hat{\boldsymbol{V}} | i \rangle \right|^2 \delta(E_f - E_i - \omega). \tag{D.103}$$

For independent electrons at $T \ll E_\mathrm{g}$ the final states are one electron in $(c, \boldsymbol{k})$ and one hole in $(v, \boldsymbol{k})$, and Eq. (D.103) is the familiar direct-gap formula with $2\sum_{\boldsymbol{k}cv} |\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{cv}(\boldsymbol{k})|^2$, the 2 counting spin. In the frozen lattice nothing absorbs below the direct gap. The absorption coefficient is the inverse length over which the intensity falls by e. Write $\varepsilon = (n_r + \mathrm{i}\kappa_\mathrm{ext})^2$, with n_r the refractive index and κ_ext the extinction coefficient. The intensity decays as $\mathrm{e}^{-2\omega\kappa_\mathrm{ext} x}$, and $\varepsilon_2 = 2n_r\kappa_\mathrm{ext}$, so

$$\alpha_\mathrm{abs}(\omega) = \frac{\omega\, \varepsilon_2(\omega)}{n_r(\omega)}. \tag{D.104}$$

The second-order amplitude

Add H_ep of Eq. (A.25) and include the phonons in the states. The initial state $|i\rangle$ has filled valence bands, empty conduction bands and $n_{\boldsymbol{Q}\nu}$ phonons in each mode. A final state that the first-order term cannot reach has an electron in $(c, \boldsymbol{k}+\boldsymbol{Q})$, a hole in $(v, \boldsymbol{k})$, and one phonon fewer in $(\boldsymbol{Q}, \nu)$ or one more in $(-\boldsymbol{Q}, \nu)$; we write $|f\rangle = c^\dagger_{c,\boldsymbol{k}+\boldsymbol{Q}} c_{v\boldsymbol{k}} |i'\rangle$ with $|i'\rangle$ the initial state with the phonon changed. We collect the terms of the second-order amplitude Eq. (2.51) that are linear in the field and linear in H_ep and take the long-time limit as in Sect. 2.4. The result is the rate $2\pi|T_{fi}|^2\delta(E_f - E_i - \omega)$ with the two time orderings of Eq. (2.54):

$$T_{fi} = \frac{e\mathcal{E}}{2\omega} \sum_m \left[\frac{\langle f | H_\mathrm{ep} | m \rangle \langle m | \hat{\boldsymbol{e}}_\gamma \cdot \hat{\boldsymbol{V}} | i \rangle}{E_i + \omega - E_m} + \frac{\langle f | \hat{\boldsymbol{e}}_\gamma \cdot \hat{\boldsymbol{V}} | m \rangle \langle m | H_\mathrm{ep} | i \rangle}{E_i - E_m} \right].$$

The photon energy appears in the denominator of the photon-first ordering, where it has been absorbed in the intermediate state, and not in the phonon-first one.

Four kinds of intermediate state contribute; we drop the common factors $e\mathcal{E}/2\omega$, $N^{-1/2}$ and the phonon matrix element $(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})^{1/2}$ of Eq. (A.28), with the upper sign for emission throughout.

1. The photon lifts the electron from $(v, \boldsymbol{k})$ to an empty state $(p, \boldsymbol{k})$, and the phonon scatters it to $(c, \boldsymbol{k}+\boldsymbol{Q})$. The intermediate energy exceeds $E_i + \omega$ by $\varepsilon_{p\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}} - \omega$. The operator string is $c^\dagger_c c_p\, c^\dagger_p c_v$, and since p is empty it equals $c^\dagger_c c_v$. The term is $g_{cp\nu}(\boldsymbol{k}, \boldsymbol{Q}) \hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{pv}(\boldsymbol{k}) / (\omega - \varepsilon_{p\boldsymbol{k}} + \varepsilon_{v\boldsymbol{k}})$.

2. The phonon scatters the electron from $(v, \boldsymbol{k})$ to an empty $(p, \boldsymbol{k}+\boldsymbol{Q})$, and the photon lifts it to $(c, \boldsymbol{k}+\boldsymbol{Q})$. The intermediate state holds the pair and the changed phonon, $E_m - E_i = \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$, and the term is $\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{cp}(\boldsymbol{k}+\boldsymbol{Q})\, g_{pv\nu}(\boldsymbol{k}, \boldsymbol{Q}) / (\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} \mp \omega_{\boldsymbol{Q}\nu})$.

3. The photon creates the electron in $(c, \boldsymbol{k}+\boldsymbol{Q})$ from an occupied $(p, \boldsymbol{k}+\boldsymbol{Q})$, and the phonon then moves an electron from $(v, \boldsymbol{k})$ into the hole. The string $c^\dagger_p c_v\, c^\dagger_c c_p$ reorders to $-c^\dagger_c\, c^\dagger_p c_p\, c_v$, which is $-c^\dagger_c c_v$ on a state with p occupied. The term is $-\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{cp}(\boldsymbol{k}+\boldsymbol{Q})\, g_{pv\nu}(\boldsymbol{k}, \boldsymbol{Q}) / (\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} + \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}})$.

4. The phonon creates the electron in $(c, \boldsymbol{k}+\boldsymbol{Q})$ from an occupied $(p, \boldsymbol{k})$, and the photon moves an electron from $(v, \boldsymbol{k})$ into the hole. The same reordering gives -1, and the term is $-g_{cp\nu}(\boldsymbol{k}, \boldsymbol{Q}) \hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{pv}(\boldsymbol{k}) / (\varepsilon_{p\boldsymbol{k}} - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} \mp \omega_{\boldsymbol{Q}\nu})$.

In the first two paths the electron propagates in the intermediate state, in the last two the hole. On the energy shell, $\omega = \varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$, the denominator of path 3 becomes $\varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$, and its term takes the form of path 2 with an occupied p. The denominator of path 4 becomes $\varepsilon_{p\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}} - \omega$, and its term takes the form of path 1 with an occupied p. The fermion sign is what makes the hole paths fit. The four paths therefore combine into the one-electron amplitude summed over every band p, occupied or empty,

$$M^{\pm}_{cv\nu}(\boldsymbol{k},\boldsymbol{Q}) = \sum_{p} \left[\frac{g_{cp\nu}(\boldsymbol{k},\boldsymbol{Q})\, \hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{pv}(\boldsymbol{k})}{\omega - \varepsilon_{p\boldsymbol{k}} + \varepsilon_{v\boldsymbol{k}}} + \frac{\hat{\boldsymbol{e}}_\gamma \cdot \boldsymbol{v}_{cp}(\boldsymbol{k}+\boldsymbol{Q})\, g_{pv\nu}(\boldsymbol{k},\boldsymbol{Q})}{\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} \mp \omega_{\boldsymbol{Q}\nu}} \right], \tag{D.105}$$

which includes $p = v$ and $p = c$. Ref. [48, Eq. (200)] writes the same sum with the denominators of opposite sign.

The phonon-assisted dielectric function

We insert T_{fi} into the rate, the power and ε_2 as for Eq. (D.103). The final states are labeled by $\boldsymbol{k}$, $\boldsymbol{Q}$, ν, c, v, the spin and the sign, and summing over them gives

Phonon-assisted dielectric function

$$\varepsilon_2^{\mathrm{PA}}(\omega) = \frac{2\pi e^2}{\omega^2\, VN} \sum_{\boldsymbol{k}\boldsymbol{Q}\nu} \sum_{cv} \sum_{\pm} (n_{\boldsymbol{Q}\nu} + \tfrac{1}{2} \pm \tfrac{1}{2})\, \left| M^{\pm}_{cv\nu}(\boldsymbol{k},\boldsymbol{Q}) \right|^2 \delta(\varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu} - \omega), \tag{D.106}$$

for $T \ll E_{\mathrm{g}}$, where the inverse process from thermally excited pairs is negligible. The factor $2\pi = 2 \times \pi$ counts the spin. With $\mathscr{V} = N\Omega$ and the average $\langle\!\langle \cdots \rangle\!\rangle = N^{-2} \sum_{\boldsymbol{k}\boldsymbol{Q}}$ it is Eq. (6.42). With Eq. (D.104) it is Ref. [48, Eq. (200)], whose prefactor leaves the spin sum implicit, and the form used in first-principles calculations [54, Eq. (1)], which writes the spin factor explicitly. Three sums appear: over intermediate bands p inside $M^{\pm}$, over phonon modes, and over $\boldsymbol{k}$ and $\boldsymbol{Q}$ independently in the whole zone, constrained by one delta function; their convergence is the subject of Appendices E.5 and E.8. At $T = 0$ only emission survives. The phonon energy enters twice: as $\pm\omega_{\boldsymbol{Q}\nu}$ in the delta function, which shifts the threshold, and in the phonon-first denominator of Eq. (D.105), where it is usually negligible against electronic energies.

The Hall–Bardeen–Blatt edge

Near the indirect edge, the final hole lies near the valence maximum at $\boldsymbol{k} = 0$ and the final electron near one of the conduction minima $\boldsymbol{k}_{\mathrm{c}}$. Measure their crystal momenta from the extrema, $\boldsymbol{k} = \boldsymbol{\kappa}_{\mathrm{h}}$ and $\boldsymbol{k}+\boldsymbol{Q} = \boldsymbol{k}_{\mathrm{c}}+\boldsymbol{\kappa}_{\mathrm{e}}$, with energies $\varepsilon_{v\boldsymbol{k}} = -\kappa_{\mathrm{h}}^2/2m_{\mathrm{h}}$ and $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{Q}} = E_{\mathrm{g}} + \kappa_{\mathrm{e}}^2/2m_{\mathrm{d}}^*$ for one valley and one valence band, with $m_{\mathrm{d}}^* = (m_\parallel^* m_\perp^{*2})^{1/3}$ the density-of-states mass of the valley (Sect. 6.5). The sums become elementary because the phonon wavevector stays near $\boldsymbol{k}_{\mathrm{c}}$ and the denominators stay far from resonance. The phonon wavevector is $\boldsymbol{Q} = \boldsymbol{k}_{\mathrm{c}} + \boldsymbol{\kappa}_{\mathrm{e}} - \boldsymbol{\kappa}_{\mathrm{h}}$, close to $\boldsymbol{k}_{\mathrm{c}}$, so $\omega_{\boldsymbol{Q}\nu}$ and the vertices may be taken at $\boldsymbol{Q} = \boldsymbol{k}_{\mathrm{c}}$. And the denominators of Eq. (D.105) are electronic energy differences of order 1 eV away from any resonance, since the intermediate states are vertical transitions far above the edge; they change little over the small pockets. Hence $|M^{\pm}|^2$ may be replaced by its value $|M^{\pm}_\nu|^2$ at the extrema.

The sums over $\boldsymbol{k}$ and $\boldsymbol{Q}$ then become independent sums over $\boldsymbol{\kappa}_{\mathrm{h}}$ and $\boldsymbol{\kappa}_{\mathrm{e}}$. With $(VN)^{-1} \sum_{\boldsymbol{\kappa}_{\mathrm{h}}\boldsymbol{\kappa}_{\mathrm{e}}} = \Omega \int \mathrm{d}^3\kappa_{\mathrm{h}} (2\pi)^{-3} \int \mathrm{d}^3\kappa_{\mathrm{e}} (2\pi)^{-3}$ and $\int \mathrm{d}^3\kappa\, (2\pi)^{-3} = \int \mathrm{d}\varepsilon\, (2m)^{3/2}\varepsilon^{1/2}/4\pi^2$ for

each parabolic band,

$$\frac{1}{VN}\sum_{\boldsymbol{\kappa}_\mathrm{h}\boldsymbol{\kappa}_\mathrm{e}}\delta(x-E_\mathrm{e}-E_\mathrm{h})=\frac{\Omega\,(2m_\mathrm{d}^*)^{3/2}(2m_\mathrm{h})^{3/2}}{16\pi^4}\int_0^x \mathrm{d}E_\mathrm{e}\,E_\mathrm{e}^{1/2}(x-E_\mathrm{e})^{1/2}=\frac{\Omega\,(m_\mathrm{d}^*m_\mathrm{h})^{3/2}}{16\pi^3}\,x^2,$$

for $x\geq 0$, where E_e and E_h are the kinetic energies of the electron and the hole and the integral is $\pi x^2/8$. Each branch, valley and valence band adds one such term, with $x=\omega-E_\mathrm{g}\mp\omega_\nu$:

Hall–Bardeen–Blatt edge

$$\varepsilon_2^{\mathrm{PA}}(\omega)=\frac{e^2\,\Omega\,(m_\mathrm{d}^*m_\mathrm{h})^{3/2}}{8\pi^2\omega^2}\sum_{\text{valleys}}\sum_{\nu}\sum_{\pm}\left(n_\nu+\tfrac{1}{2}\pm\tfrac{1}{2}\right)|M_\nu^{\pm}|^2\,(\omega-E_\mathrm{g}\mp\omega_\nu)^2\,\Theta(\omega-E_\mathrm{g}\mp\omega_\nu),\tag{D.107}$$

with $\omega_\nu=\omega_{\boldsymbol{k}_\mathrm{c}\nu}$ and $n_\nu=n_\mathrm{B}(\omega_\nu)$, and one such sum for each valence band, heavy and light holes with their own m_h. By Eq. (D.104), $\alpha_\mathrm{abs}(\omega)$ carries one more factor ω/n_r; over the width of the edge, a few times ω_ν, the prefactors vary slowly and

$$\alpha_\mathrm{abs}(\omega)\simeq\sum_\nu\Big[A_\nu\,n_\nu\,(\omega-E_\mathrm{g}+\omega_\nu)^2\,\Theta(\omega-E_\mathrm{g}+\omega_\nu)+A_\nu\,(n_\nu+1)\,(\omega-E_\mathrm{g}-\omega_\nu)^2\,\Theta(\omega-E_\mathrm{g}-\omega_\nu)\Big],\tag{D.108}$$

with A_ν constant across the edge. The form is that of Hall, Bardeen and Blatt [53], who wrote it for germanium; we have assumed $|M_\nu^+|=|M_\nu^-|$, which holds when ω_ν is negligible in the denominators.

By Eq. (D.108), each component is a straight line in a plot of $\sqrt{\alpha_\mathrm{abs}}$ against ω, while their sum bends at every threshold. Each branch that couples adds a threshold at $E_\mathrm{g}-\omega_\nu$, where phonon absorption sets in, and another at $E_\mathrm{g}+\omega_\nu$, where emission does. The thresholds measure E_g and the phonon energies at $\boldsymbol{k}_\mathrm{c}$, and their midpoint is E_g. And the ratio of the coefficients of the two components of a branch, $n_\nu/(n_\nu+1)=\mathrm{e}^{-\omega_\nu/T}$, is a thermometer of the lattice: as $T\to 0$ the absorption component vanishes, and the edge sits at $E_\mathrm{g}+\omega_\nu$ for the softest branch that couples. The quadratic law holds when $M^\pm$ is constant, that is when the transition is allowed at the extrema. If symmetry forces $M^\pm$ to vanish there, it grows linearly in κ_e or κ_h, and the integral gives x^3 instead of x^2, the forbidden indirect edge. Electron–hole attraction, neglected here, adds excitonic thresholds below $E_\mathrm{g}\pm\omega_\nu$ (Sect. 4.9).

The structure of Eq. (D.105) carries over unchanged when the photon is replaced by any other probe that couples to the electrons through a one-body operator, with $\hat{\boldsymbol{e}}_\gamma\cdot\boldsymbol{v}_{ab}$ replaced by the matrix element of that operator between Bloch states. For a dark-matter particle that transfers momentum $\boldsymbol{q}$, the operator is the density $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, which does not conserve $\boldsymbol{k}$, and the phonon then supplies the part of $\boldsymbol{q}$ that the crystal does not; Chap. 13 develops this.

D.19 Antiferromagnetic spin waves

In a collinear antiferromagnet the classical ground state has the spins of one sublattice up and those of the other down. A spin flip on either sublattice is not an eigenstate, because the exchange term $S_i^+S_j^-$ between opposite sublattices creates two flips at once. The spin waves are therefore Bogoliubov mixtures of flips on the two sublattices, and their energy grows linearly at small wavevector instead of quadratically as in a ferromagnet. This section derives the dispersion with the Holstein–Primakoff transformation [55] at linear order, for exchange of any range within and between the sublattices. The result, Eq. (D.113), is used in Sect. 7.1 and Appendix E.9; Sect. 14.1 computes the spin weights from it.

The two-sublattice Heisenberg model

The Hamiltonian is the Heisenberg model Eq. (7.6), $H_{\mathrm{H}} = \sum_{\langle ij\rangle} J_{ij}^{\mathrm{ex}} \boldsymbol{S}_i \cdot \boldsymbol{S}_j$, with each pair counted once and $J_{ij}^{\mathrm{ex}} > 0$ antiferromagnetic. The magnetic ions form two interpenetrating sublattices A and B with N_{s} sites each, related by a translation or an inversion that exchanges them, and every ion has spin length $|\boldsymbol{S}|$. In the classical reference state the spins of A point along $+\hat{\boldsymbol{z}}$ and those of B along $-\hat{\boldsymbol{z}}$. Exchange of any range enters through two lattice sums, over the neighbors of a site of A on its own sublattice and on the other,

$$J_{\mathrm{AA}}^{\mathrm{ex}}(\boldsymbol{k}) = \sum_{\boldsymbol{\delta}\in\mathrm{A},\,\boldsymbol{\delta}\neq 0} J_{0\boldsymbol{\delta}}^{\mathrm{ex}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}}, \qquad J_{\mathrm{AB}}^{\mathrm{ex}}(\boldsymbol{k}) = \sum_{\boldsymbol{\delta}\in\mathrm{B}} J_{0\boldsymbol{\delta}}^{\mathrm{ex}}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}}, \tag{D.109}$$

where $\boldsymbol{\delta}$ runs over the positions of the neighbors relative to the site. The symmetry that exchanges A and B gives the same sums for a site of B. When the set of neighbors is symmetric under $\boldsymbol{\delta} \to -\boldsymbol{\delta}$, as it is for every lattice of this section, both sums are real and even in $\boldsymbol{k}$. In the type-II order of NiO, ferromagnetic (111) planes of the fcc nickel lattice alternate in sign (Appendix E.9); all 6 next-nearest neighbors and 6 of the 12 nearest neighbors lie on the other sublattice, and the other 6 nearest neighbors on the same one.

Holstein–Primakoff bosons

On a site of A, the operators

$$S_i^z = |\boldsymbol{S}| - d_{\mathrm{A}i}^\dagger d_{\mathrm{A}i}, \qquad S_i^+ = \left(2|\boldsymbol{S}| - d_{\mathrm{A}i}^\dagger d_{\mathrm{A}i}\right)^{1/2} d_{\mathrm{A}i}, \qquad S_i^- = (S_i^+)^\dagger, \tag{D.110}$$

obey the spin commutation relations when $d_{\mathrm{A}i}$ is a boson, $[d_{\mathrm{A}i}, d_{\mathrm{A}i}^\dagger] = 1$, on the physical states $d_{\mathrm{A}i}^\dagger d_{\mathrm{A}i} \le 2|\boldsymbol{S}|$ [55]. The boson number counts the deviation from the fully polarized site. On a site of B, where the reference spin points down, the roles of raising and lowering are exchanged: $S_j^z = -|\boldsymbol{S}| + d_{\mathrm{B}j}^\dagger d_{\mathrm{B}j}$ and $S_j^- = (2|\boldsymbol{S}| - d_{\mathrm{B}j}^\dagger d_{\mathrm{B}j})^{1/2} d_{\mathrm{B}j}$. Linear spin-wave theory keeps the leading order in $1/|\boldsymbol{S}|$: it replaces the square roots by $(2|\boldsymbol{S}|)^{1/2}$ and drops all terms of the Hamiltonian beyond the second order in the bosons.

For a pair on opposite sublattices, $i \in \mathrm{A}$ and $j \in \mathrm{B}$,

$$\boldsymbol{S}_i \cdot \boldsymbol{S}_j = S_i^z S_j^z + \tfrac{1}{2}(S_i^+ S_j^- + S_i^- S_j^+) \simeq -|\boldsymbol{S}|^2 + |\boldsymbol{S}|(d_{\mathrm{A}i}^\dagger d_{\mathrm{A}i} + d_{\mathrm{B}j}^\dagger d_{\mathrm{B}j}) + |\boldsymbol{S}|(d_{\mathrm{A}i} d_{\mathrm{B}j} + d_{\mathrm{A}i}^\dagger d_{\mathrm{B}j}^\dagger),$$

since $S_i^+ S_j^- \simeq 2|\boldsymbol{S}|\, d_{\mathrm{A}i} d_{\mathrm{B}j}$. Each deviation raises the energy of the antiparallel bond, and the transverse term creates or destroys two deviations together. For a pair on the same sublattice, both up,

$$\boldsymbol{S}_i \cdot \boldsymbol{S}_j \simeq |\boldsymbol{S}|^2 - |\boldsymbol{S}|(d_i^\dagger d_i + d_j^\dagger d_j) + |\boldsymbol{S}|(d_i^\dagger d_j + d_j^\dagger d_i),$$

the ferromagnetic form, in which deviations hop; the same holds on B. The quartic term $d^\dagger d\, d^\dagger d$ of $S_i^z S_j^z$ has been dropped.

Fourier transform

Sum over pairs. Every site of A has the antiparallel bonds of total strength $J_{\mathrm{AB}}^{\mathrm{ex}}(0)$ and the parallel bonds $J_{\mathrm{AA}}^{\mathrm{ex}}(0)$, and likewise every site of B. Introduce

$$d_{\mathrm{A}i} = N_{\mathrm{s}}^{-1/2} \sum_{\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}_i}\, d_{\mathrm{A}\boldsymbol{k}}, \qquad d_{\mathrm{B}j} = N_{\mathrm{s}}^{-1/2} \sum_{\boldsymbol{k}} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}_j}\, d_{\mathrm{B}\boldsymbol{k}},$$

with $\boldsymbol{k}$ in the Brillouin zone of one sublattice. The opposite sign on B is a convenience: $d_{\mathrm{B}\boldsymbol{k}}$ destroys a deviation of crystal momentum $-\boldsymbol{k}$, so that the pair terms couple $d_{\mathrm{A}\boldsymbol{k}}$ with

$d_{\mathrm{B}\boldsymbol{k}}$. With $\sum_i \mathrm{e}^{\mathrm{i}(\boldsymbol{k}-\boldsymbol{k}')\cdot\boldsymbol{r}_i} = N_\mathrm{s}\delta_{\boldsymbol{k}\boldsymbol{k}'}$, the hopping terms on A give $|\boldsymbol{S}|\sum_{\boldsymbol{k}} J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{k})\, d^\dagger_{\mathrm{A}\boldsymbol{k}} d_{\mathrm{A}\boldsymbol{k}}$, where each pair counted once contributes $J^{\mathrm{ex}}_{0\boldsymbol{\delta}}$ to both $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}}$ and $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}}$, and those on B give the same with $\boldsymbol{k}\to-\boldsymbol{k}$, which changes nothing. The pair terms give $\sum_i \sum_{\boldsymbol{\delta}\in\mathrm{B}} J^{\mathrm{ex}}_{0\boldsymbol{\delta}}\, d_{\mathrm{A}i} d_{\mathrm{B},i+\boldsymbol{\delta}} = \sum_{\boldsymbol{k}} J^{\mathrm{ex}}_{\mathrm{AB}}(-\boldsymbol{k})\, d_{\mathrm{A}\boldsymbol{k}} d_{\mathrm{B}\boldsymbol{k}}$. Collecting,

$$\begin{gathered} H_\mathrm{H} \simeq E_\mathrm{cl} + |\boldsymbol{S}| \sum_{\boldsymbol{k}} \Big[A_{\boldsymbol{k}} (d^\dagger_{\mathrm{A}\boldsymbol{k}} d_{\mathrm{A}\boldsymbol{k}} + d^\dagger_{\mathrm{B}\boldsymbol{k}} d_{\mathrm{B}\boldsymbol{k}}) + B_{\boldsymbol{k}} (d_{\mathrm{A}\boldsymbol{k}} d_{\mathrm{B}\boldsymbol{k}} + d^\dagger_{\mathrm{B}\boldsymbol{k}} d^\dagger_{\mathrm{A}\boldsymbol{k}}) \Big], \\ A_{\boldsymbol{k}} = J^{\mathrm{ex}}_{\mathrm{AB}}(0) - J^{\mathrm{ex}}_{\mathrm{AA}}(0) + J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{k}), \qquad B_{\boldsymbol{k}} = J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{k}), \end{gathered} \tag{D.111}$$

with the classical energy $E_\mathrm{cl} = N_\mathrm{s}|\boldsymbol{S}|^2[J^{\mathrm{ex}}_{\mathrm{AA}}(0) - J^{\mathrm{ex}}_{\mathrm{AB}}(0)]$, which counts each bond once. The coefficient $A_{\boldsymbol{k}}$ is the cost of one deviation, the field of the antiparallel neighbors minus that of the parallel ones, plus the ferromagnetic hopping on its own sublattice. For $J^{\mathrm{ex}}_{\mathrm{AB}} = 0$ and $J^{\mathrm{ex}}_{\mathrm{AA}} < 0$, Eq. (D.111) is a ferromagnet with $|\boldsymbol{S}|[J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{k}) - J^{\mathrm{ex}}_{\mathrm{AA}}(0)]$ per mode, the one-magnon energy of Sect. 7.1.

Bogoliubov transformation

At each $\boldsymbol{k}$, Eq. (D.111) is the two-mode Hamiltonian Eq. (C.61) with $d_1 = d_{\mathrm{A}\boldsymbol{k}}$, $d_2 = d_{\mathrm{B}\boldsymbol{k}}$, the diagonal energy $|\boldsymbol{S}|A_{\boldsymbol{k}}$ and the pairing amplitude $g_0 = |\boldsymbol{S}|B_{\boldsymbol{k}}$. Appendix C.7 diagonalizes it with the boson operators

$$a_{\boldsymbol{k}1} = u^\mathrm{M}_{\boldsymbol{k}}\, d_{\mathrm{A}\boldsymbol{k}} + v^\mathrm{M}_{\boldsymbol{k}}\, d^\dagger_{\mathrm{B}\boldsymbol{k}}, \qquad a_{\boldsymbol{k}2} = u^\mathrm{M}_{\boldsymbol{k}}\, d_{\mathrm{B}\boldsymbol{k}} + v^\mathrm{M}_{\boldsymbol{k}}\, d^\dagger_{\mathrm{A}\boldsymbol{k}}, \tag{D.112}$$

with $u^\mathrm{M}_{\boldsymbol{k}} = \cosh\theta_{\boldsymbol{k}}$, $v^\mathrm{M}_{\boldsymbol{k}} = \sinh\theta_{\boldsymbol{k}}$ and $\tanh 2\theta_{\boldsymbol{k}} = B_{\boldsymbol{k}}/A_{\boldsymbol{k}}$. Equivalently,

$$(u^\mathrm{M}_{\boldsymbol{k}})^2 = \frac{1}{2}\Big(\frac{A_{\boldsymbol{k}}}{\sqrt{A^2_{\boldsymbol{k}} - B^2_{\boldsymbol{k}}}} + 1\Big), \qquad (v^\mathrm{M}_{\boldsymbol{k}})^2 = \frac{1}{2}\Big(\frac{A_{\boldsymbol{k}}}{\sqrt{A^2_{\boldsymbol{k}} - B^2_{\boldsymbol{k}}}} - 1\Big), \qquad u^\mathrm{M}_{\boldsymbol{k}} v^\mathrm{M}_{\boldsymbol{k}} = \frac{B_{\boldsymbol{k}}}{2\sqrt{A^2_{\boldsymbol{k}} - B^2_{\boldsymbol{k}}}},$$

from $\cosh 2\theta = A/(A^2 - B^2)^{1/2}$ and $\sinh 2\theta = B/(A^2 - B^2)^{1/2}$. By Eq. (C.62),

$$\begin{gathered} H_\mathrm{H} \simeq E_\mathrm{gs} + \sum_{\boldsymbol{k}} \omega^\mathrm{M}_{\boldsymbol{k}} (a^\dagger_{\boldsymbol{k}1} a_{\boldsymbol{k}1} + a^\dagger_{\boldsymbol{k}2} a_{\boldsymbol{k}2}), \qquad \omega^\mathrm{M}_{\boldsymbol{k}} = |\boldsymbol{S}|(A^2_{\boldsymbol{k}} - B^2_{\boldsymbol{k}})^{1/2}, \\ E_\mathrm{gs} = E_\mathrm{cl} + \sum_{\boldsymbol{k}} (\omega^\mathrm{M}_{\boldsymbol{k}} - |\boldsymbol{S}|A_{\boldsymbol{k}}). \end{gathered} \tag{D.113}$$

With $A_{\boldsymbol{k}}$ and $B_{\boldsymbol{k}}$ from Eq. (D.111), this is Eq. (7.11), and also Eq. (E.64). The two branches are degenerate without anisotropy. They differ in the spin they carry: $a^\dagger_{\boldsymbol{k}1} = u^\mathrm{M}_{\boldsymbol{k}} d^\dagger_{\mathrm{A}\boldsymbol{k}} + v^\mathrm{M}_{\boldsymbol{k}} d_{\mathrm{B}\boldsymbol{k}}$ adds a deviation on A or removes one from B, and either lowers the total S^z by one, while $a^\dagger_{\boldsymbol{k}2}$ raises it by one. The ground-state energy lies below the classical one by $\sum_{\boldsymbol{k}}(|\boldsymbol{S}|A_{\boldsymbol{k}} - \omega^\mathrm{M}_{\boldsymbol{k}}) \geq 0$: the Néel state is not the ground state, and the zero-point motion of the spin waves lowers the energy. In the true ground state, annihilated by every $a_{\boldsymbol{k}\nu}$, a site of A carries the deviation $\langle d^\dagger_{\mathrm{A}\boldsymbol{k}} d_{\mathrm{A}\boldsymbol{k}}\rangle = (v^\mathrm{M}_{\boldsymbol{k}})^2$ per mode, by Appendix C.7; its sum over the zone is the zero-point reduction of the ordered moment that Sect. 14.1 evaluates.

The theory is consistent only if $A_{\boldsymbol{k}} \geq |B_{\boldsymbol{k}}|$ for every $\boldsymbol{k}$. Otherwise $\omega^\mathrm{M}_{\boldsymbol{k}}$ is imaginary, the deviations grow, and the assumed classical order is not a minimum of the classical energy. The expansion is in $1/|\boldsymbol{S}|$: the quartic terms that were dropped change the energies by a relative amount of order $1/(2|\boldsymbol{S}|)$ times a lattice sum of $(v^\mathrm{M}_{\boldsymbol{k}})^2$, small for three-dimensional lattices and for $|\boldsymbol{S}| \geq 1$.

Linear dispersion

At $\boldsymbol{k}=0$, $A_0=B_0=J^{\rm ex}_{\rm AB}(0)$ and $\omega^{\rm M}_0=0$: rotating all spins together costs nothing, the Goldstone mode of the broken spin-rotation symmetry. For small k, write $A^2_{\boldsymbol{k}}-B^2_{\boldsymbol{k}}=(A_{\boldsymbol{k}}-B_{\boldsymbol{k}})(A_{\boldsymbol{k}}+B_{\boldsymbol{k}})$. The sum tends to $2J^{\rm ex}_{\rm AB}(0)$, and the difference is quadratic, $A_{\boldsymbol{k}}-B_{\boldsymbol{k}}=[J^{\rm ex}_{\rm AB}(0)-J^{\rm ex}_{\rm AB}(\boldsymbol{k})]-[J^{\rm ex}_{\rm AA}(0)-J^{\rm ex}_{\rm AA}(\boldsymbol{k})]\simeq\frac{1}{2}\sum_{\boldsymbol{\delta}\in\rm B}J^{\rm ex}_{0\boldsymbol{\delta}}(\boldsymbol{k}\cdot\boldsymbol{\delta})^2-\frac{1}{2}\sum_{\boldsymbol{\delta}\in\rm A}J^{\rm ex}_{0\boldsymbol{\delta}}(\boldsymbol{k}\cdot\boldsymbol{\delta})^2$. Hence

$$\omega^{\rm M}_{\boldsymbol{k}}\simeq|\boldsymbol{S}|\Big\{J^{\rm ex}_{\rm AB}(0)\Big[\sum_{\boldsymbol{\delta}\in\rm B}J^{\rm ex}_{0\boldsymbol{\delta}}(\boldsymbol{k}\cdot\boldsymbol{\delta})^2-\sum_{\boldsymbol{\delta}\in\rm A}J^{\rm ex}_{0\boldsymbol{\delta}}(\boldsymbol{k}\cdot\boldsymbol{\delta})^2\Big]\Big\}^{1/2},\tag{D.114}$$

linear in k, with a velocity that is the geometric mean of the exchange field $|\boldsymbol{S}|J^{\rm ex}_{\rm AB}(0)$ and the stiffness term in brackets. A ferromagnet has no pair term and $\omega\propto k^2$; the antiferromagnet's linear law comes from the mixing of deviations on the two sublattices, which costs the large exchange field.

For the type-II order of the fcc lattice with only the next-nearest-neighbor exchange J_2, the 6 antiparallel neighbors sit at $\pm a\hat{\boldsymbol{x}}$, $\pm a\hat{\boldsymbol{y}}$, $\pm a\hat{\boldsymbol{z}}$ with a the cubic lattice constant, $J^{\rm ex}_{\rm AB}(\boldsymbol{k})=2J_2(\cos k_xa+\cos k_ya+\cos k_za)$ and $J^{\rm ex}_{\rm AA}=0$. Then $\omega^{\rm M}_{\boldsymbol{k}}=6J_2|\boldsymbol{S}|[1-\zeta^2_{\boldsymbol{k}}]^{1/2}$ with $\zeta_{\boldsymbol{k}}=\frac{1}{3}\sum_\alpha\cos k_\alpha a$. The maximum, $6J_2|\boldsymbol{S}|$, is reached where $\zeta_{\boldsymbol{k}}=0$, and the small-k law Eq. (D.114), with $\sum_{\boldsymbol{\delta}}(\boldsymbol{k}\cdot\boldsymbol{\delta})^2=2a^2k^2$, is $\omega^{\rm M}_{\boldsymbol{k}}\simeq2\sqrt{3}\,J_2|\boldsymbol{S}|\,ak$. The nearest-neighbor exchange J_1 adds 6 antiparallel bonds to $J^{\rm ex}_{\rm AB}$ and 6 parallel bonds to $J^{\rm ex}_{\rm AA}$ and changes these numbers at first order in J_1/J_2.

The spin operators

A probe that couples to the electron spin density, as in Sect. 7.1, needs the spin operators expressed through the magnon operators. With the Fourier sign of Sect. 7.1, $S^{\pm}(\boldsymbol{k})=\sum_j\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}_j}S^{\pm}_j$, the part on A is to linear order $S^+_{\rm A}(\boldsymbol{k})\equiv\sum_{i\in\rm A}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}_i}S^+_i=(2|\boldsymbol{S}|N_{\rm s})^{1/2}d_{{\rm A},-\boldsymbol{k}}$, by the lattice sum $\sum_i\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{k}')\cdot\boldsymbol{r}_i}=N_{\rm s}\delta_{\boldsymbol{k},-\boldsymbol{k}'}$. Inverting Eq. (D.112) as in Appendix C.7,

$$d_{{\rm A}\boldsymbol{k}}=u^{\rm M}_{\boldsymbol{k}}\,a_{\boldsymbol{k}1}-v^{\rm M}_{\boldsymbol{k}}\,a^\dagger_{\boldsymbol{k}2},\qquad d_{{\rm B}\boldsymbol{k}}=u^{\rm M}_{\boldsymbol{k}}\,a_{\boldsymbol{k}2}-v^{\rm M}_{\boldsymbol{k}}\,a^\dagger_{\boldsymbol{k}1}.\tag{D.115}$$

A transverse spin operator acting on the ground state therefore creates one magnon, with an amplitude $(2|\boldsymbol{S}|N_{\rm s})^{1/2}$ times $u^{\rm M}_{\boldsymbol{k}}$ or $v^{\rm M}_{\boldsymbol{k}}$. On B the raising operator is $S^+_j\simeq(2|\boldsymbol{S}|)^{1/2}d^\dagger_{{\rm B}j}$, and the same lattice sum turns $\sum_{j\in\rm B}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}_j}d^\dagger_{{\rm B}j}$ into $N_{\rm s}^{1/2}d^\dagger_{{\rm B},-\boldsymbol{k}}$. Since $A_{\boldsymbol{k}}$ and $B_{\boldsymbol{k}}$, and with them $u^{\rm M}_{\boldsymbol{k}}$ and $v^{\rm M}_{\boldsymbol{k}}$, are even in $\boldsymbol{k}$, the sum of both sublattices at small $\boldsymbol{k}$ is $S^+(\boldsymbol{k})\simeq(2|\boldsymbol{S}|N_{\rm s})^{1/2}(d_{{\rm A},-\boldsymbol{k}}+d^\dagger_{{\rm B},-\boldsymbol{k}})=(2|\boldsymbol{S}|N_{\rm s})^{1/2}(u^{\rm M}_{\boldsymbol{k}}-v^{\rm M}_{\boldsymbol{k}})(a_{-\boldsymbol{k}1}+a^\dagger_{-\boldsymbol{k}2})$. Near $\boldsymbol{k}=0$ both coefficients diverge as $(|\boldsymbol{S}|A_0/2\omega^{\rm M}_{\boldsymbol{k}})^{1/2}$, but $(u^{\rm M}_{\boldsymbol{k}}-v^{\rm M}_{\boldsymbol{k}})^2=[(A_{\boldsymbol{k}}-B_{\boldsymbol{k}})/(A_{\boldsymbol{k}}+B_{\boldsymbol{k}})]^{1/2}\to0$, so the response to a uniform transverse field is suppressed. Near the antiferromagnetic wavevector, where the phases of the two sublattices differ by π, the combination is $u^{\rm M}_{\boldsymbol{k}}+v^{\rm M}_{\boldsymbol{k}}$ and the response is enhanced. The longitudinal operator S^z is quadratic in the bosons and creates magnons in pairs. Sect. 14.1 assembles these weights.

The classical and the linear quantum theories agree at this order. On a periodic fcc cluster of 256 sites with type-II order and $J_1=0.1J_2$, the frequencies of the classical precession equations $\dot{\boldsymbol{S}}_i=-\boldsymbol{S}_i\times\sum_j J^{\rm ex}_{ij}\boldsymbol{S}_j$, linearized about the Néel state, coincide with Eq. (D.113) at all wavevectors of the cluster.

D.20 BCS gap and coherence factors

An attraction between electrons of opposite momentum and spin within $\omega_{\rm D}$ of the Fermi surface, however weak, makes the Fermi sea unstable. The electrons bind into pairs, and

a gap $\Delta_{\rm BCS}$ opens in the spectrum of single-particle excitations. This section solves the model of Bardeen, Cooper and Schrieffer [56] in mean field and derives the gap equation and its solutions at zero temperature and at T_c. Their ratio is $\pi \mathrm{e}^{-\gamma_{\rm E}} = 1.764$, with $\gamma_{\rm E}$ Euler's constant. We then compute the matrix elements of the density and the spin density between the ground state and the excited states, the coherence factors. The interaction is the one that Sect. 7.2 derives from phonon exchange; the Bogoliubov algebra is that of Appendix C.7.

The model and its mean field

The reduced Hamiltonian keeps only the pairs $(\boldsymbol{k}\uparrow, -\boldsymbol{k}\downarrow)$ of zero total momentum:

$$H_{\rm BCS} = \sum_{\boldsymbol{k}\sigma} \xi_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}\sigma} c_{\boldsymbol{k}\sigma} - \frac{V_{\rm BCS}}{\mathscr{V}} \sum_{\boldsymbol{k}\boldsymbol{k}'}{}' c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow} c_{-\boldsymbol{k}'\downarrow} c_{\boldsymbol{k}'\uparrow}, \tag{D.116}$$

with $\xi_{\boldsymbol{k}} = \xi_{-\boldsymbol{k}}$ the band energy from the chemical potential, $V_{\rm BCS} > 0$ the strength of the attraction, and the prime restricting both $|\xi_{\boldsymbol{k}}|$ and $|\xi_{\boldsymbol{k}'}|$ to less than $\omega_{\rm D}$. Define the pair amplitude $b_{\boldsymbol{k}} = \langle c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow} \rangle$ and the gap

$$\Delta_{\rm BCS} = \frac{V_{\rm BCS}}{\mathscr{V}} \sum_{\boldsymbol{k}}{}' b_{\boldsymbol{k}}, \tag{D.117}$$

which we take real by a choice of the overall phase of the state. Write each pair operator as its average plus a fluctuation, $c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow} = b_{\boldsymbol{k}} + (c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow} - b_{\boldsymbol{k}})$, and drop the product of two fluctuations. For the reduced interaction, in which one pair couples to all $\sim N$ others with strength $\propto 1/\mathscr{V}$, the dropped term vanishes relative to the kept ones as $\mathscr{V} \to \infty$. What remains is

$$H_{\rm MF} = \sum_{\boldsymbol{k}}{}' \Big[\xi_{\boldsymbol{k}} (c^\dagger_{\boldsymbol{k}\uparrow} c_{\boldsymbol{k}\uparrow} + c^\dagger_{-\boldsymbol{k}\downarrow} c_{-\boldsymbol{k}\downarrow}) - \Delta_{\rm BCS} (c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow} + c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow}) \Big] + \frac{\mathscr{V} \Delta^2_{\rm BCS}}{V_{\rm BCS}},$$

plus the unpaired states outside the shell. For each $\boldsymbol{k}$ this is Eq. (C.63) with $c_1 = c_{\boldsymbol{k}\uparrow}$, $c_2 = c_{-\boldsymbol{k}\downarrow}$, $\xi_0 = \xi_{\boldsymbol{k}}$ and $g_0 = \Delta_{\rm BCS}$.

Bogoliubov quasiparticles

By Appendix C.7, the operators

$$\gamma_{\boldsymbol{k}1} = u^{\rm B}_{\boldsymbol{k}}\, c_{\boldsymbol{k}\uparrow} - v^{\rm B}_{\boldsymbol{k}}\, c^\dagger_{-\boldsymbol{k}\downarrow}, \qquad \gamma_{\boldsymbol{k}2} = u^{\rm B}_{\boldsymbol{k}}\, c_{-\boldsymbol{k}\downarrow} + v^{\rm B}_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}\uparrow}, \tag{D.118}$$

with $u^{\rm B}_{\boldsymbol{k}} = \cos\theta_{\boldsymbol{k}}$, $v^{\rm B}_{\boldsymbol{k}} = \sin\theta_{\boldsymbol{k}}$, are fermions, and the pair terms drop out of $H_{\rm MF}$ when $\tan 2\theta_{\boldsymbol{k}} = \Delta_{\rm BCS}/\xi_{\boldsymbol{k}}$. Then $\cos 2\theta_{\boldsymbol{k}} = \xi_{\boldsymbol{k}}/E_{\boldsymbol{k}}$, $\sin 2\theta_{\boldsymbol{k}} = \Delta_{\rm BCS}/E_{\boldsymbol{k}}$, and

$$\begin{aligned} H_{\rm MF} &= \sum_{\boldsymbol{k}}{}' E_{\boldsymbol{k}} (\gamma^\dagger_{\boldsymbol{k}1}\gamma_{\boldsymbol{k}1} + \gamma^\dagger_{\boldsymbol{k}2}\gamma_{\boldsymbol{k}2}) + \text{const}, \qquad E_{\boldsymbol{k}} = \left(\xi^2_{\boldsymbol{k}} + \Delta^2_{\rm BCS}\right)^{1/2}, \\ (u^{\rm B}_{\boldsymbol{k}})^2 &= \frac{1}{2}\Big(1 + \frac{\xi_{\boldsymbol{k}}}{E_{\boldsymbol{k}}}\Big), \quad (v^{\rm B}_{\boldsymbol{k}})^2 = \frac{1}{2}\Big(1 - \frac{\xi_{\boldsymbol{k}}}{E_{\boldsymbol{k}}}\Big). \end{aligned} \tag{D.119}$$

The quasiparticle $\gamma^\dagger_{\boldsymbol{k}1}$ carries momentum $\boldsymbol{k}$ and spin up, $\gamma^\dagger_{\boldsymbol{k}2}$ momentum $-\boldsymbol{k}$ and spin down. Far above the Fermi surface $u^{\rm B} \to 1$ and it is an electron, far below $v^{\rm B} \to 1$ and it is a hole, and at the Fermi surface it is an equal mixture with energy $\Delta_{\rm BCS}$. The ground state, annihilated by all γ, is the BCS state $\prod_{\boldsymbol{k}} (u^{\rm B}_{\boldsymbol{k}} + v^{\rm B}_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow})|0\rangle$. To see this, note that $\gamma_{\boldsymbol{k}1}$ commutes with the factors of the product at other wavevectors, which contain fermion operators in pairs,

and act on the factor at $\boldsymbol{k}$. With $u = u^{\mathrm{B}}_{\boldsymbol{k}}$, $v = v^{\mathrm{B}}_{\boldsymbol{k}}$, $c_{\boldsymbol{k}\uparrow}|0\rangle = 0$, $c_{\boldsymbol{k}\uparrow}c^\dagger_{\boldsymbol{k}\uparrow}c^\dagger_{-\boldsymbol{k}\downarrow}|0\rangle = c^\dagger_{-\boldsymbol{k}\downarrow}|0\rangle$ and $(c^\dagger_{-\boldsymbol{k}\downarrow})^2 = 0$,

$$(uc_{\boldsymbol{k}\uparrow} - vc^\dagger_{-\boldsymbol{k}\downarrow})(u + vc^\dagger_{\boldsymbol{k}\uparrow}c^\dagger_{-\boldsymbol{k}\downarrow})|0\rangle = uv\, c^\dagger_{-\boldsymbol{k}\downarrow}|0\rangle - vu\, c^\dagger_{-\boldsymbol{k}\downarrow}|0\rangle = 0.$$

In the same way $\gamma_{\boldsymbol{k}2}$ yields $-uv\, c^\dagger_{\boldsymbol{k}\uparrow}|0\rangle + vu\, c^\dagger_{\boldsymbol{k}\uparrow}|0\rangle = 0$, the minus sign from moving $c_{-\boldsymbol{k}\downarrow}$ past $c^\dagger_{\boldsymbol{k}\uparrow}$. Since $\mathrm{d}E/\mathrm{d}\xi = \xi/E$, each energy $E > \Delta_{\mathrm{BCS}}$ is reached from the two values $\xi = \pm(E^2 - \Delta^2_{\mathrm{BCS}})^{1/2}$, an electronlike and a holelike state. With $\mathcal{N}_{\mathrm{F}}$ constant over the shell, the quasiparticles of one spin orientation have the density of states per volume

$$\mathcal{N}_{\mathrm{qp}}(E) = 2\mathcal{N}_{\mathrm{F}}\, \frac{E}{\sqrt{E^2 - \Delta^2_{\mathrm{BCS}}}}\, \Theta(E - \Delta_{\mathrm{BCS}}), \tag{D.120}$$

which tends to $2\mathcal{N}_{\mathrm{F}}$, the density of electron and hole excitations of the normal metal at energy E, when $\Delta_{\mathrm{BCS}} \to 0$. The states removed from the gap pile up at its edge, where the density diverges as $(E - \Delta_{\mathrm{BCS}})^{-1/2}$.

The gap equation

The gap must reproduce itself. Inverting Eq. (D.118), $c_{\boldsymbol{k}\uparrow} = u\gamma_{\boldsymbol{k}1} + v\gamma^\dagger_{\boldsymbol{k}2}$ and $c_{-\boldsymbol{k}\downarrow} = u\gamma_{\boldsymbol{k}2} - v\gamma^\dagger_{\boldsymbol{k}1}$, with the subscripts on u and v omitted. In thermal equilibrium the quasiparticles are independent fermions of energy $E_{\boldsymbol{k}}$, so $\langle\gamma^\dagger\gamma\rangle = n_{\mathrm{F}}(E_{\boldsymbol{k}})$, and

$$b_{\boldsymbol{k}} = \langle c_{-\boldsymbol{k}\downarrow}c_{\boldsymbol{k}\uparrow}\rangle = uv[\langle\gamma_{\boldsymbol{k}2}\gamma^\dagger_{\boldsymbol{k}2}\rangle - \langle\gamma^\dagger_{\boldsymbol{k}1}\gamma_{\boldsymbol{k}1}\rangle] = uv\,[1 - 2n_{\mathrm{F}}(E_{\boldsymbol{k}})] = \frac{\Delta_{\mathrm{BCS}}}{2E_{\boldsymbol{k}}}\tanh\frac{E_{\boldsymbol{k}}}{2T}.$$

Inserting this into Eq. (D.117), dividing by $\Delta_{\mathrm{BCS}} \neq 0$, and replacing $\mathcal{V}^{-1}\sum_{\boldsymbol{k}}$ by $\mathcal{N}_{\mathrm{F}}\int \mathrm{d}\xi$ over the shell,

$$\frac{1}{\mathcal{N}_{\mathrm{F}}V_{\mathrm{BCS}}} = \int_0^{\omega_{\mathrm{D}}} \mathrm{d}\xi\, \frac{\tanh(\sqrt{\xi^2 + \Delta^2_{\mathrm{BCS}}}/2T)}{\sqrt{\xi^2 + \Delta^2_{\mathrm{BCS}}}}. \tag{D.121}$$

The right side decreases with both Δ_{BCS} and T. It diverges logarithmically as both go to zero, so a solution exists for every $V_{\mathrm{BCS}} > 0$: the instability of the Fermi sea needs no threshold, because the pair states near the Fermi surface have a constant density at arbitrarily small energy.

At $T = 0$, tanh = 1 and the integral is $\mathrm{arsinh}(\omega_{\mathrm{D}}/\Delta_{\mathrm{BCS}})$. Hence

$$\Delta_{\mathrm{BCS}}(0) = \frac{\omega_{\mathrm{D}}}{\sinh[1/(\mathcal{N}_{\mathrm{F}}V_{\mathrm{BCS}})]} \simeq 2\omega_{\mathrm{D}}\, \mathrm{e}^{-1/(\mathcal{N}_{\mathrm{F}}V_{\mathrm{BCS}})}, \tag{D.122}$$

the exponential form being valid for weak coupling, $\mathcal{N}_{\mathrm{F}}V_{\mathrm{BCS}} \ll 1$. The gap is nonanalytic in the coupling and could not be found at any finite order of perturbation theory.

The critical temperature

At T_c the gap closes, and Eq. (D.121) with $\Delta_{\mathrm{BCS}} = 0$ fixes T_c:

$$\frac{1}{\mathcal{N}_{\mathrm{F}}V_{\mathrm{BCS}}} = \int_0^{\omega_{\mathrm{D}}} \mathrm{d}\xi\, \frac{\tanh(\xi/2T_c)}{\xi}.$$

The integral is evaluated through the Matsubara expansion of the hyperbolic tangent. The fermionic form of Eq. (C.42) with $g(z) = 1/(x^2 - z^2)$, whose poles at $z = \pm x$ have residues $\mp 1/2x$, gives $T\sum_j 1/(x^2 + (\omega_j^{\mathrm{F}})^2) = [n_{\mathrm{F}}(-x) - n_{\mathrm{F}}(x)]/2x = \tanh(x/2T)/2x$. Then

$$\int_0^{\omega_{\mathrm{D}}} \mathrm{d}\xi\, \frac{\tanh(\xi/2T)}{\xi} = 2T \sum_j \int_0^{\omega_{\mathrm{D}}} \frac{\mathrm{d}\xi}{\xi^2 + (\omega_j^{\mathrm{F}})^2} = \sum_{j=0}^{\infty} \frac{2}{\pi}\, \frac{\arctan[X/(j+\frac{1}{2})]}{j+\frac{1}{2}}, \qquad X = \frac{\omega_{\mathrm{D}}}{2\pi T},$$

where the terms j and $-j-1$ were combined and $\omega_j^{\mathrm{F}} = 2\pi T(j+\frac{1}{2})$. Split $(2/\pi)\arctan(X/y) = \Theta(X-y) + r(y/X)$, with $r(t) = (2/\pi)\arctan(1/t) - \Theta(1-t)$. The step function cuts the sum at $j + \frac{1}{2} < X$. The remainder, for $X \gg 1$, becomes the integral $\int_0^\infty \mathrm{d}t\, r(t)/t$, which vanishes. On $t < 1$, $r(t) = -(2/\pi)\arctan t$; on $t > 1$, $r(t) = (2/\pi)\arctan(1/t)$; and the substitution $t \to 1/t$ maps one piece onto minus the other. The truncated sum is a sum over odd integers, $\sum_{j=0}^{J}(j+\frac{1}{2})^{-1} = 2(H_{2J+2} - \frac{1}{2}H_{J+1})$ with the harmonic numbers $H_n = \sum_{m=1}^{n} 1/m = \ln n + \gamma_{\mathrm{E}} + O(1/n)$, which gives $\ln J + \gamma_{\mathrm{E}} + 2\ln 2$ for $J \simeq X \gg 1$. Altogether the integral is $\ln(4\mathrm{e}^{\gamma_{\mathrm{E}}}X) = \ln(2\mathrm{e}^{\gamma_{\mathrm{E}}}\omega_{\mathrm{D}}/\pi T)$, with corrections of order $(T/\omega_{\mathrm{D}})^2$, and

$$T_c = \frac{2\mathrm{e}^{\gamma_{\mathrm{E}}}}{\pi}\, \omega_{\mathrm{D}}\, \mathrm{e}^{-1/(\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}})} = 1.134\, \omega_{\mathrm{D}}\, \mathrm{e}^{-1/(\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}})}. \tag{D.123}$$

Dividing Eq. (D.122) by Eq. (D.123), the unknown ω_{D} and the exponential cancel:

$$\frac{\Delta_{\mathrm{BCS}}(0)}{T_c} = \pi\, \mathrm{e}^{-\gamma_{\mathrm{E}}} = 1.764. \tag{D.124}$$

The ratio is universal within the model, independent of the coupling, the cutoff and the material; deviations measure the strength and the retardation of the real interaction, which Eliashberg theory treats (Appendix E.9). Already at $\mathcal{N}_{\mathrm{F}} V_{\mathrm{BCS}} = 0.2$ the numerical solution of Eq. (D.121) has $\Delta_{\mathrm{BCS}}(0)/T_c = 1.7639$.

Coherence factors

A probe couples to the electrons through a one-body operator, and the superconducting state answers with matrix elements that mix the electron and hole parts of the quasiparticles. Take the density, $n_e(\boldsymbol{q}) = \sum_{\boldsymbol{k}\sigma} c^\dagger_{\boldsymbol{k}+\boldsymbol{q},\sigma} c_{\boldsymbol{k}\sigma}$, and the z component of the spin density, $s^z_e(\boldsymbol{q}) = \frac{1}{2}\sum_{\boldsymbol{k}}(c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\uparrow} c_{\boldsymbol{k}\uparrow} - c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\downarrow} c_{\boldsymbol{k}\downarrow})$, Eq. (7.13) in second quantization. Write $u = u^{\mathrm{B}}_{\boldsymbol{k}}$, $v = v^{\mathrm{B}}_{\boldsymbol{k}}$ and primes for $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{q}$. With the inverted Eq. (D.118),

$$c^\dagger_{\boldsymbol{k}'\uparrow} c_{\boldsymbol{k}\uparrow} = u'u\, \gamma^\dagger_{\boldsymbol{k}'1}\gamma_{\boldsymbol{k}1} + u'v\, \gamma^\dagger_{\boldsymbol{k}'1}\gamma^\dagger_{\boldsymbol{k}2} + v'u\, \gamma_{\boldsymbol{k}'2}\gamma_{\boldsymbol{k}1} + v'v\, \gamma_{\boldsymbol{k}'2}\gamma^\dagger_{\boldsymbol{k}2}.$$

For the spin-down part, relabel $\boldsymbol{k} \to -\boldsymbol{k} - \boldsymbol{q}$, so that $\sum_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\downarrow} c_{\boldsymbol{k}\downarrow} = \sum_{\boldsymbol{k}} c^\dagger_{-\boldsymbol{k}\downarrow} c_{-\boldsymbol{k}'\downarrow}$, and expand:

$$c^\dagger_{-\boldsymbol{k}\downarrow} c_{-\boldsymbol{k}'\downarrow} = uu'\, \gamma^\dagger_{\boldsymbol{k}2}\gamma_{\boldsymbol{k}'2} - uv'\, \gamma^\dagger_{\boldsymbol{k}2}\gamma^\dagger_{\boldsymbol{k}'1} - vu'\, \gamma_{\boldsymbol{k}1}\gamma_{\boldsymbol{k}'2} + vv'\, \gamma_{\boldsymbol{k}1}\gamma^\dagger_{\boldsymbol{k}'1}.$$

Collect the terms by process. The term that creates two quasiparticles, $\gamma^\dagger_{\boldsymbol{k}'1}\gamma^\dagger_{\boldsymbol{k}2}$, comes with $u'v$ from spin up and, after $-\gamma^\dagger_{\boldsymbol{k}2}\gamma^\dagger_{\boldsymbol{k}'1} = \gamma^\dagger_{\boldsymbol{k}'1}\gamma^\dagger_{\boldsymbol{k}2}$, with uv' from spin down. The term that scatters a quasiparticle from $\boldsymbol{k}$ to $\boldsymbol{k}'$, $\gamma^\dagger_{\boldsymbol{k}'1}\gamma_{\boldsymbol{k}1}$, comes with uu' from spin up and, after $\gamma_{\boldsymbol{k}1}\gamma^\dagger_{\boldsymbol{k}'1} = -\gamma^\dagger_{\boldsymbol{k}'1}\gamma_{\boldsymbol{k}1}$ for $\boldsymbol{q} \neq 0$, with $-vv'$ from spin down. The density adds the two spins and the spin density subtracts them, so

$$\begin{aligned} n_e(\boldsymbol{q}):&\quad \text{creation } u'v + v'u, \qquad \text{scattering } uu' - vv', \\ s^z_e(\boldsymbol{q}):&\quad \text{creation } \tfrac{1}{2}(u'v - v'u), \qquad \text{scattering } \tfrac{1}{2}(uu' + vv'). \end{aligned} \tag{D.125}$$

The scattering of the second branch, $\gamma^\dagger_{\boldsymbol{k}2}\gamma_{\boldsymbol{k}'2}$, has the same factors. Expressed through the energies, with Eq. (D.119) and $2uv = \Delta_{\rm BCS}/E$,

$$(uu' \mp vv')^2 = \frac{1}{2}\Big(1 + \frac{\xi\xi' \mp \Delta^2_{\rm BCS}}{EE'}\Big), \qquad (uv' \pm vu')^2 = \frac{1}{2}\Big(1 - \frac{\xi\xi' \mp \Delta^2_{\rm BCS}}{EE'}\Big), \tag{D.126}$$

with the upper signs for the density, a perturbation even under time reversal (case I), and the lower signs for the spin density, odd under time reversal (case II), in the terminology of Ref. [57]. At $T = 0$ only creation acts on the ground state. Its weight for the density, at the threshold $\xi = \xi' = 0$, is $\frac{1}{2}(1 + 1) = 1$. Multiplied by the divergent density Eq. (D.120) of both quasiparticles, it makes the pair-breaking absorption of the density jump at $\omega = 2\Delta_{\rm BCS}$. For the spin density it is $\frac{1}{2}(1 - 1) = 0$, and the absorption rises continuously. Scattering needs thermally excited quasiparticles and is weighted by $n_{\rm F}(E)[1 - n_{\rm F}(E')]$.

The same algebra shows what goes wrong at $\boldsymbol{q} = 0$. There the creation factor of the density is $2uv = \Delta_{\rm BCS}/E_{\boldsymbol{k}}$, not zero: the operator $n_e(0) = \hat{N}_e$ appears to break pairs. The total number of electrons commutes with $H_{\rm BCS}$ and cannot excite anything, but it does not commute with $H_{\rm MF}$, which creates and destroys pairs. The mean-field states are not eigenstates of $\hat{N}_e$, and a response computed with independent quasiparticles violates charge conservation and the f-sum rule. The spin density at $\boldsymbol{q} = 0$ has creation factor $\frac{1}{2}(uv - vu) = 0$: total S^z commutes with $H_{\rm MF}$, and the bare quasiparticle response to a spin probe is conserving. Appendix D.21 repairs the density response.

D.21 Conserving BCS density response

The independent Bogoliubov quasiparticles of Appendix D.20 respond to a uniform potential: at $\boldsymbol{q} = 0$ the density creates pairs of them with weight $\Delta^2_{\rm BCS}/E^2_{\boldsymbol{k}}$, although the electron number is conserved and a uniform potential can excite nothing. The defect is not in the mean-field ground state but in the response computed from it. When the density perturbs the quasiparticles, the pair field $\Delta_{\rm BCS}$ must respond as well, and its phase fluctuations restore charge conservation. This section writes the problem in Nambu form [58] and shows that the bare bubble violates the continuity equation by a term proportional to $\Delta_{\rm BCS}$. The conserving response of Anderson [59] is then derived as the linear response of the time-dependent mean field. We prove that it satisfies the continuity equation and the f-sum rule, and state the Ward identity that its vertex obeys. The result is used in Sect. 7.2.

Nambu form

Collect the two operators that the pairing mixes into the spinor $\Psi_{\boldsymbol{k}} = (c_{\boldsymbol{k}\uparrow}, c^\dagger_{-\boldsymbol{k}\downarrow})^{\rm T}$, and use the matrices

$$\boldsymbol{\eta} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \mathsf{s}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathsf{s}_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix},$$

with $\boldsymbol{\eta}$ the matrix of Appendix C.7; together with the unit matrix they obey the Pauli algebra, $\mathsf{s}_1\mathsf{s}_2 = \mathrm{i}\boldsymbol{\eta}$ and cyclically. For a 2×2 matrix A, define $O_A(\boldsymbol{q}) = \sum_{\boldsymbol{k}} \Psi^\dagger_{\boldsymbol{k}+\boldsymbol{q}} A\, \Psi_{\boldsymbol{k}}$. Writing out the components, $O_{\boldsymbol{\eta}}(\boldsymbol{q}) = \sum_{\boldsymbol{k}}[c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\uparrow}c_{\boldsymbol{k}\uparrow} - c_{-\boldsymbol{k}-\boldsymbol{q}\downarrow}c^\dagger_{-\boldsymbol{k}\downarrow}] = n_e(\boldsymbol{q})$ for $\boldsymbol{q} \neq 0$, after anticommuting and relabeling $-\boldsymbol{k} - \boldsymbol{q} \to \boldsymbol{k}$ in the spin-down term. The density is the $\boldsymbol{\eta}$ vertex. The pair operators are $O_{\mathsf{s}_1}(\boldsymbol{q}) = B^\dagger_{\boldsymbol{q}} + B_{-\boldsymbol{q}}$ and $O_{\mathsf{s}_2}(\boldsymbol{q}) = -\mathrm{i}(B^\dagger_{\boldsymbol{q}} - B_{-\boldsymbol{q}})$, with $B^\dagger_{\boldsymbol{P}} = \sum_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}+\boldsymbol{P}\uparrow}c^\dagger_{-\boldsymbol{k}\downarrow}$ the creation operator of a pair of total momentum $\boldsymbol{P}$. For a real gap, s_1 fluctuations change the magnitude of the pair field and s_2 fluctuations its phase.

The mean-field Hamiltonian of Appendix D.20 is $H_{\rm MF} = \sum_{\boldsymbol{k}} \Psi^\dagger_{\boldsymbol{k}} h_{\boldsymbol{k}} \Psi_{\boldsymbol{k}}$ + const with $h_{\boldsymbol{k}} = \xi_{\boldsymbol{k}}\boldsymbol{\eta} - \Delta_{\rm BCS}\mathsf{s}_1$, whose eigenvalues are $\pm E_{\boldsymbol{k}}$, with projectors $P^\pm_{\boldsymbol{k}} = \frac{1}{2}(1 \pm h_{\boldsymbol{k}}/E_{\boldsymbol{k}})$. The

Matsubara Green's function is the 2×2 matrix $\mathsf{G}_0(\boldsymbol{k},\mathrm{i}\omega^{\mathrm{F}}) = (\mathrm{i}\omega^{\mathrm{F}} - h_{\boldsymbol{k}})^{-1} = \sum_{\pm} P^{\pm}_{\boldsymbol{k}}/(\mathrm{i}\omega^{\mathrm{F}} \mp E_{\boldsymbol{k}})$. The off-diagonal elements are the anomalous averages of the pair state.

The interaction in the pairing channel

To let the pair field respond at wavevector $\boldsymbol{q}$, the interaction must scatter pairs of every total momentum, not only $\boldsymbol{P}=0$ as in the reduced Hamiltonian Eq. (D.116). Take the contact attraction

$$H_{\mathrm{int}} = -\frac{V_{\mathrm{BCS}}}{\mathscr{V}} \sum_{\boldsymbol{P}} B^{\dagger}_{\boldsymbol{P}} B_{\boldsymbol{P}} = -V_{\mathrm{BCS}} \int \mathrm{d}^3 r\, \psi^{\dagger}_{\uparrow}\psi^{\dagger}_{\downarrow}\psi_{\downarrow}\psi_{\uparrow}(\boldsymbol{r}),$$

with the pair states restricted to the shell $|\xi| < \omega_{\mathrm{D}}$. Without the restriction it is a function of the local densities $\psi^{\dagger}_{\sigma}\psi_{\sigma}$ alone and commutes with the density. On a lattice it is the on-site term $-V_{\mathrm{BCS}}\sum_i n_{i\uparrow}n_{i\downarrow}$, which needs no shell; in the continuum the shell makes it slightly nonlocal, and its commutator with $n_e(\boldsymbol{q})$ is of relative order $v_{\mathrm{F}}q/\omega_{\mathrm{D}}$, negligible at the wavevectors of interest. The Hamiltonian $H_0 + H_{\mathrm{int}}$ therefore obeys the continuity equation Eq. (4.29), $[H, n_e(\boldsymbol{q})] = \boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q})$. With $B^{\dagger}_{\boldsymbol{P}} = \tfrac{1}{2}(O_{\mathsf{s}_1} + \mathrm{i}O_{\mathsf{s}_2})(\boldsymbol{P})$ and $B_{\boldsymbol{P}} = \tfrac{1}{2}(O_{\mathsf{s}_1} - \mathrm{i}O_{\mathsf{s}_2})(-\boldsymbol{P})$,

$$H_{\mathrm{int}} = -\frac{V_{\mathrm{BCS}}}{4\mathscr{V}} \sum_{\boldsymbol{P}} \sum_{a=1,2} O_{\mathsf{s}_a}(\boldsymbol{P})\, O_{\mathsf{s}_a}(-\boldsymbol{P}) + (\text{one-body terms}),$$

where the one-body terms come from reordering and shift the chemical potential. Replacing one factor of each product by its average, in the two possible ways, gives the mean-field term $-(V_{\mathrm{BCS}}/2\mathscr{V})\sum_a \langle O_{\mathsf{s}_a}(\boldsymbol{0})\rangle O_{\mathsf{s}_a}(\boldsymbol{0})$, and with $\langle O_{\mathsf{s}_1}(\boldsymbol{0})\rangle = 2\sum_{\boldsymbol{k}} b_{\boldsymbol{k}}$ this is $-\Delta_{\mathrm{BCS}} O_{\mathsf{s}_1}(\boldsymbol{0})$, the pair term of H_{MF}.

Ward-identity violation of the bare bubble

The quasiparticle bubble with vertices A and B follows from Wick's theorem for the Nambu field as in Appendix D.10. It carries a trace over the Nambu index:

$$R^0_{AB}(\boldsymbol{q},\mathrm{i}\omega^{\mathrm{B}}) = \frac{1}{\mathscr{V}} \sum_{\boldsymbol{k}} T \sum_{j} \mathrm{Tr}\big[A\, \mathsf{G}_0(\boldsymbol{k}+\boldsymbol{q}, \mathrm{i}\omega^{\mathrm{F}}_j + \mathrm{i}\omega^{\mathrm{B}})\, B\, \mathsf{G}_0(\boldsymbol{k}, \mathrm{i}\omega^{\mathrm{F}}_j)\big].$$

With the spectral form of G_0, the frequency sum is Eq. (D.51) with $\xi_{\boldsymbol{k}} \to s'E_{\boldsymbol{k}}$ and $\xi_{\boldsymbol{k}+\boldsymbol{q}} \to sE_{\boldsymbol{k}+\boldsymbol{q}}$. At $T=0$, $n_{\mathrm{F}}(-E)=1$ and $n_{\mathrm{F}}(E)=0$, and after continuation

$$R^0_{AB}(\boldsymbol{q},\omega) = \frac{1}{\mathscr{V}} \sum_{\boldsymbol{k}} \left[\frac{\mathrm{Tr}[A P^{+}_{\boldsymbol{k}+\boldsymbol{q}} B P^{-}_{\boldsymbol{k}}]}{\omega - E_{\boldsymbol{k}} - E_{\boldsymbol{k}+\boldsymbol{q}} + \mathrm{i}0^{+}} - \frac{\mathrm{Tr}[A P^{-}_{\boldsymbol{k}+\boldsymbol{q}} B P^{+}_{\boldsymbol{k}}]}{\omega + E_{\boldsymbol{k}} + E_{\boldsymbol{k}+\boldsymbol{q}} + \mathrm{i}0^{+}} \right]. \tag{D.127}$$

For $A = B = \boldsymbol{\eta}$, $\boldsymbol{\eta} h_{\boldsymbol{k}'} \boldsymbol{\eta} = \xi' \boldsymbol{\eta} + \Delta_{\mathrm{BCS}}\mathsf{s}_1$, and the trace is $\tfrac{1}{2}[1 - (\xi\xi' - \Delta^2_{\mathrm{BCS}})/EE']$, the case-I creation factor of Eq. (D.126), as it must be.

At $\boldsymbol{q}=0$ the traces follow from the Pauli algebra of the Nambu matrices. Write $\mathsf{s}_3 \equiv \boldsymbol{\eta}$, so that $h_{\boldsymbol{k}}/E_{\boldsymbol{k}} = \sum_a n_a \mathsf{s}_a$ with the unit vector $\boldsymbol{n} = (-\Delta_{\mathrm{BCS}}/E, 0, \xi/E)$. Multiplying out with $\mathsf{s}_a\mathsf{s}_b = \delta_{ab} + \mathrm{i}\epsilon_{abc}\mathsf{s}_c$,

$$\mathrm{Tr}[\mathsf{s}_a P^{+} \mathsf{s}_b P^{-}] = \delta_{ab} - n_a n_b - \mathrm{i}\epsilon_{abc} n_c, \qquad \mathrm{Tr}[\mathsf{s}_a P^{-} \mathsf{s}_b P^{+}] = \delta_{ab} - n_a n_b + \mathrm{i}\epsilon_{abc} n_c.$$

With $F(\omega) = \mathscr{V}^{-1}\sum_{\boldsymbol{k}} [E_{\boldsymbol{k}}(\omega^2 - 4E^2_{\boldsymbol{k}})]^{-1}$, the symmetric parts combine into $4E/(\omega^2 - 4E^2)$ and the antisymmetric parts into $2\omega/(\omega^2 - 4E^2)$, and

$$\begin{gathered} R^0_{\boldsymbol{\eta}\boldsymbol{\eta}}(0,\omega) = 4\Delta^2_{\mathrm{BCS}} F, \qquad R^0_{\mathsf{s}_2\mathsf{s}_2}(0,\omega) = \omega^2 F - \frac{2}{V_{\mathrm{BCS}}}, \\ R^0_{\mathsf{s}_2\boldsymbol{\eta}}(0,\omega) = -R^0_{\boldsymbol{\eta}\mathsf{s}_2}(0,\omega) = 2\mathrm{i}\Delta_{\mathrm{BCS}}\, \omega F. \end{gathered} \tag{D.128}$$

In $R^0_{\mathsf{s}_2\mathsf{s}_2}$ we wrote $4E/(\omega^2-4E^2) = [\omega^2-(\omega^2-4E^2)]/[E(\omega^2-4E^2)]$ and used the gap equation at $T=0$, $\mathscr{V}^{-1}\sum_{\boldsymbol{k}} 1/E_{\boldsymbol{k}} = 2/V_{\mathrm{BCS}}$ (Eq. (D.121)). The elements that involve s_1 with $\boldsymbol{\eta}$ or s_2 are odd in ξ and vanish when the density of states is symmetric about the Fermi level over the shell, which we assume; the amplitude channel then decouples.

The bare density response at $\boldsymbol{q}=0$ is therefore not zero. Its spectral weight sits at $\omega = 2E_{\boldsymbol{k}}$, with $S(0,\omega) = -2\,\mathrm{Im}\,R^0 = (2\pi/\mathscr{V})\sum_{\boldsymbol{k}}(\Delta^2_{\mathrm{BCS}}/E^2_{\boldsymbol{k}})\,\delta(\omega-2E_{\boldsymbol{k}})$, and its first moment is

$$\int_0^\infty \mathrm{d}\omega\,\omega\,S^0(0,\omega) = \frac{4\pi}{\mathscr{V}}\sum_{\boldsymbol{k}}\frac{\Delta^2_{\mathrm{BCS}}}{E_{\boldsymbol{k}}} = \frac{8\pi\Delta^2_{\mathrm{BCS}}}{V_{\mathrm{BCS}}}, \tag{D.129}$$

by the gap equation once more. The f-sum rule Eq. (4.30) requires $\pi n_e q^2/m_e$, which vanishes at $q=0$. The origin is visible in the commutator. With $[\mathsf{s}_1,\boldsymbol{\eta}] = -2\mathrm{i}\mathsf{s}_2$,

$$[H_{\mathrm{MF}}, n_e(\boldsymbol{q})] = \boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q}) + 2\mathrm{i}\Delta_{\mathrm{BCS}}\,O_{\mathsf{s}_2}(\boldsymbol{q}),$$

where the kinetic term gives $\sum_{\boldsymbol{k}}(\xi_{\boldsymbol{k}+\boldsymbol{q}}-\xi_{\boldsymbol{k}})(c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\uparrow}c_{\boldsymbol{k}\uparrow} + c^\dagger_{\boldsymbol{k}+\boldsymbol{q}\downarrow}c_{\boldsymbol{k}\downarrow}) = \boldsymbol{q}\cdot\boldsymbol{j}_e(\boldsymbol{q})$ and the pair term the rest. The mean-field Hamiltonian converts density into pair phase, and charge leaks into the condensate without coming back.

The conserving response

The response that conserves charge is the linear response of the time-dependent mean field. Let an external potential ϕ_{ext} couple to $n_e(\boldsymbol{q})$. The quasiparticles then see the external field and the change of the pair field it induces, $\delta H = -(V_{\mathrm{BCS}}/2\mathscr{V})\sum_a \delta\langle O_{\mathsf{s}_a}(-\boldsymbol{q})\rangle O_{\mathsf{s}_a}(\boldsymbol{q})$, which is the mean-field term of H_{int} at wavevector $\boldsymbol{q}$, and respond to both with the bare bubbles. In the channel basis $(\boldsymbol{\eta},\mathsf{s}_1,\mathsf{s}_2)$ this is a random-phase approximation with the kernel $K = \mathrm{diag}(0,-V_{\mathrm{BCS}}/2,-V_{\mathrm{BCS}}/2)$:

$$R = R^0 + R^0KR, \qquad R_{\boldsymbol{\eta}\boldsymbol{\eta}} = R^0_{\boldsymbol{\eta}\boldsymbol{\eta}} + \sum_{a,b=1,2} R^0_{\boldsymbol{\eta}\mathsf{s}_a}\Big[(1-KR^0)^{-1}K\Big]_{ab}R^0_{\mathsf{s}_b\boldsymbol{\eta}}, \tag{D.130}$$

with all bubbles from Eq. (D.127) and the inverse taken in the $(\mathsf{s}_1,\mathsf{s}_2)$ block. In diagrams, the correction term is the bubble with the ladder of pair propagators between its two vertices [59].

That this response obeys the continuity equation follows from a property of mean-field dynamics. Under $H_{\mathrm{MF}}+\delta H$ the state stays a quasi-free state, one for which Wick's theorem holds, at every time. For any one-body operator A, the average $\langle[H_0+H_{\mathrm{int}},A]\rangle$ in such a state, evaluated with Wick's theorem, equals $\langle[H_{\mathrm{MF}}+\delta H,A]\rangle$, provided the mean fields are the self-consistent ones: each pair of contractions of H_{int} is a mean field acting on the remaining two operators. The Hartree mean fields of H_{int}, which are density–density terms, commute with n_e and may be kept or dropped without changing the argument; we drop them, since their static part only shifts μ_{ch}. Hence $\mathrm{d}\langle n_e(\boldsymbol{q})\rangle/\mathrm{d}t = \mathrm{i}\langle[H_0+H_{\mathrm{int}},n_e(\boldsymbol{q})]\rangle = \mathrm{i}\boldsymbol{q}\cdot\langle\boldsymbol{j}_e(\boldsymbol{q})\rangle$, since H_{int} commutes with the density. The term $2\mathrm{i}\Delta_{\mathrm{BCS}}\delta\langle O_{\mathsf{s}_2}\rangle$ that the bare dynamics produces is cancelled, to linear order, by the commutator of n_e with δH, which contains $\langle O_{\mathsf{s}_1}(\boldsymbol{0})\rangle\delta\langle O_{\mathsf{s}_2}\rangle$ with the opposite sign. A response that obeys continuity at all times satisfies the f-sum rule, by the double-commutator argument of Sect. 4.3, with the averages taken in the mean-field ground state, and it vanishes at $\boldsymbol{q}=0$, where n_e is the conserved particle number.

The explicit algebra at $\boldsymbol{q}=0$ confirms the cancellation. With Eq. (D.128), the denominator is $1-K_{22}R^0_{\mathsf{s}_2\mathsf{s}_2} = 1+\tfrac12 V_{\mathrm{BCS}}(\omega^2F - 2/V_{\mathrm{BCS}}) = \tfrac12 V_{\mathrm{BCS}}\,\omega^2F$, and

$$R_{\boldsymbol{\eta}\boldsymbol{\eta}}(0,\omega) = 4\Delta^2_{\mathrm{BCS}}F + \frac{(-2\mathrm{i}\Delta_{\mathrm{BCS}}\omega F)(-\frac12 V_{\mathrm{BCS}})(2\mathrm{i}\Delta_{\mathrm{BCS}}\omega F)}{\frac12 V_{\mathrm{BCS}}\,\omega^2F} = 4\Delta^2_{\mathrm{BCS}}F - 4\Delta^2_{\mathrm{BCS}}F = 0,$$

for every ω. The phase fluctuations of the condensate absorb exactly the spurious pair-breaking weight. The cancellation used the gap equation; a response built on a gap that does not solve it would not conserve charge.

The Goldstone mode

The same denominator vanishes at $\omega = 0$ for $\boldsymbol{q} = 0$, because the gap equation makes $1 - K_{22}R^0_{\mathsf{s}_2\mathsf{s}_2}(0,0) = 0$: a uniform rotation of the phase of Δ_{BCS} costs no energy. At small $\boldsymbol{q}$ the zero moves to a pole at $\omega = c_{\mathrm{AB}}q$, the Anderson–Bogoliubov mode, a sound wave of the condensate phase that carries the density response of a neutral superconductor at $\omega < 2\Delta_{\mathrm{BCS}}$ and exhausts the f-sum rule as $q \to 0$. Anderson found $c_{\mathrm{AB}} = v_{\mathrm{F}}/\sqrt{3}$ [59]. The zero of $\det(1 - KR^0)$, evaluated numerically from Eq. (D.127) for a parabolic band at $q = 0.02\,\Delta_{\mathrm{BCS}}/v_{\mathrm{F}}$, lies at $c_{\mathrm{AB}} = 0.579\,v_{\mathrm{F}}$, against $1/\sqrt{3} = 0.577$. With the long-range Coulomb interaction, which adds the random-phase kernel $K_{\boldsymbol{\eta}\boldsymbol{\eta}} = v_{\mathrm{C}}(q)$ in the $\boldsymbol{\eta}$ channel, the mode is pushed up to the plasma frequency, and the low-frequency density response of a charged superconductor is screened as in the normal metal [59]. Adding that channel keeps the response conserving, since a density–density interaction commutes with the density.

The Ward identity

Apply the continuity equation inside the three-point function $\langle \mathrm{T}\,\Psi_{\boldsymbol{k}+\boldsymbol{q}}\Psi^\dagger_{\boldsymbol{k}}\,n_e(\boldsymbol{q})^\dagger\rangle$, the amplitude for the density to scatter an electron from $\boldsymbol{k}$ to $\boldsymbol{k}+\boldsymbol{q}$. Its time derivative brings down the current, and the equal-time commutators of n_e with the two fields give two terms with a single Green's function each. Multiply the whole by G^{-1} on each side, which removes the propagation of the incoming and the outgoing electron and leaves the vertex alone (the step called amputation). The result is an identity for the dressed density vertex $\mathsf{V}_{\boldsymbol{\eta}}$ and current vertex $\mathbf{V}_{\boldsymbol{j}}$:

$$\boldsymbol{q}\cdot\mathbf{V}_{\boldsymbol{j}}(\boldsymbol{k}+\boldsymbol{q},\boldsymbol{k}) - \omega\,\mathsf{V}_{\boldsymbol{\eta}}(\boldsymbol{k}+\boldsymbol{q},\boldsymbol{k}) = \boldsymbol{\eta}\,\mathsf{G}^{-1}(\boldsymbol{k},\nu) - \mathsf{G}^{-1}(\boldsymbol{k}+\boldsymbol{q},\nu+\omega)\,\boldsymbol{\eta}. \tag{D.131}$$

In the normal state, with $\mathsf{G}^{-1} = \nu - \xi_{\boldsymbol{k}}\boldsymbol{\eta}$, the right side is $-\omega\boldsymbol{\eta} + (\xi_{\boldsymbol{k}+\boldsymbol{q}} - \xi_{\boldsymbol{k}})$, and the bare vertices $\mathsf{V}_{\boldsymbol{\eta}} = \boldsymbol{\eta}$ and $\mathbf{V}_{\boldsymbol{j}} = (\boldsymbol{k}+\boldsymbol{q}/2)/m_e$ satisfy it for a parabolic band. In the superconductor, $\mathsf{G}^{-1} = \nu - h_{\boldsymbol{k}}$ adds $\Delta_{\mathrm{BCS}}(\boldsymbol{\eta}\mathsf{s}_1 - \mathsf{s}_1\boldsymbol{\eta}) = 2\mathrm{i}\Delta_{\mathrm{BCS}}\mathsf{s}_2$ to the right side, which no bare vertex supplies. The ladder supplies it. Its density vertex is $\mathsf{V}_{\boldsymbol{\eta}} = \boldsymbol{\eta} + \mathsf{c}_2\mathsf{s}_2$ with the phase-channel coefficient $\mathsf{c}_2 = K_{22}R^0_{\mathsf{s}_2\boldsymbol{\eta}}/(1 - K_{22}R^0_{\mathsf{s}_2\mathsf{s}_2})$, which at $\boldsymbol{q} = 0$ is, by Eq. (D.128), $(-\tfrac{1}{2}V_{\mathrm{BCS}})(2\mathrm{i}\Delta_{\mathrm{BCS}}\omega F)/(\tfrac{1}{2}V_{\mathrm{BCS}}\omega^2 F) = -2\mathrm{i}\Delta_{\mathrm{BCS}}/\omega$. Then $-\omega\mathsf{V}_{\boldsymbol{\eta}} = -\omega\boldsymbol{\eta} + 2\mathrm{i}\Delta_{\mathrm{BCS}}\mathsf{s}_2$, which is Eq. (D.131) at $\boldsymbol{q} = 0$. The pole of c_2 at $\omega = 0$ is the Goldstone mode; at finite $\boldsymbol{q}$ it moves to $\omega = c_{\mathrm{AB}}q$, and the vertex keeps satisfying the identity. The bare vertex is enough for the spin density s^z_e, whose Nambu vertex is the unit matrix: $\not\vdash h_{\boldsymbol{k}} - h_{\boldsymbol{k}} \not\vdash = 0$ in the gap term, so no vertex correction is needed, in agreement with Appendix D.20.

A numerical check on a square-lattice model with this interaction, 48×48 sites and $\Delta_{\mathrm{BCS}} = 0.30t$, confirms the derivation. At $\boldsymbol{q} = 0$ the conserving $R_{\boldsymbol{\eta}\boldsymbol{\eta}}$ vanishes at frequencies below, inside and above the pair continuum, while the bare one does not. At the two smallest nonzero wavevectors of the lattice, the coefficient of $1/\omega^2$ in the conserving response, which is π^{-1} times the first moment, reproduces the double commutator $\mathscr{V}^{-1}\sum_{\boldsymbol{k}\sigma}(\varepsilon_{\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{\boldsymbol{k}-\boldsymbol{q}} - 2\varepsilon_{\boldsymbol{k}})\langle n_{\boldsymbol{k}\sigma}\rangle$, the lattice form of $n_e q^2/m_e$. The bare bubble overestimates it by factors of 28 and 4, and the error grows as $q \to 0$ like the constant Eq. (D.129) divided by q^2.

D.22 Polarization of Dirac electrons

Near a Dirac point the bands are $\pm E_{\boldsymbol{k}}$ with $E_{\boldsymbol{k}} = (v_{\rm D}^2 k^2 + \Delta_{\rm D}^2)^{1/2}$, and an undoped material at $T = 0$ screens through interband pairs alone, whose energy $E_{\boldsymbol{k}} + E_{\boldsymbol{k}+\boldsymbol{q}}$ is at least $(v_{\rm D}^2 q^2 + 4\Delta_{\rm D}^2)^{1/2}$. The linear dispersion makes the Lindhard sum of Sect. 4.5 look like the vacuum polarization of quantum electrodynamics with $v_{\rm D}$ in place of the speed of light, and the sum diverges logarithmically at large momentum, so the answer depends on where the linear dispersion ends. This section first shows how the Kramers degeneracy brings the $\boldsymbol{k}\cdot\boldsymbol{p}$ block of Eq. (7.37) to the Dirac form, and then evaluates the polarization sum with a hard momentum cutoff $\Lambda_{\rm D}$ in closed form, for any $\boldsymbol{q}$ and ω in the regime $q, \omega/v_{\rm D}, \Delta_{\rm D}/v_{\rm D} \ll \Lambda_{\rm D}$. Both results are used in Sect. 7.3.

The Dirac form

Start from the block Hamiltonian Eq. (7.37), with $M(\delta\boldsymbol{k}) = \sum_i \delta k_i M_i$ the 2×2 matrix of the velocity between the conduction and the valence doublet. As Sect. 7.3 shows, the Kramers degeneracy of both bands at every $\delta\boldsymbol{k}$ requires $MM^\dagger \propto \mathbb{1}$ for every $\delta\boldsymbol{k}$. For $\delta\boldsymbol{k}$ along the axis i this says $M_i M_i^\dagger \propto \mathbb{1}$. For $\delta\boldsymbol{k}$ along the diagonal of the axes i and j it says $(M_i + M_j)(M_i + M_j)^\dagger \propto \mathbb{1}$, which with the first condition requires

$$M_i M_i^\dagger \propto \mathbb{1}, \qquad M_i M_j^\dagger + M_j M_i^\dagger \propto \mathbb{1}, \tag{D.132}$$

with real coefficients of proportionality. We write the Pauli matrices that act within a doublet as $\sigma_1^{\rm d}, \sigma_2^{\rm d}, \sigma_3^{\rm d}$, and $\boldsymbol{c}\cdot\boldsymbol{\sigma}^{\rm d} = \sum_l c_l \sigma_l^{\rm d}$.

Step 1: reduce to one direction. Assume $M_1 \neq 0$. By Eq. (D.132), $M_1 M_1^\dagger = m^2 \mathbb{1}$ with $m > 0$, so $W = M_1/m$ is unitary. Define $N_i = M_i M_1^{-1}$, so that $M_i = N_i M_1$ and $N_1 = \mathbb{1}$. Then $M_i M_1^\dagger = N_i M_1 M_1^\dagger = m^2 N_i$ and $M_i M_i^\dagger = N_i M_1 M_1^\dagger N_i^\dagger = m^2 N_i N_i^\dagger$, and Eq. (D.132) with $j = 1$ becomes

$$N_i + N_i^\dagger \propto \mathbb{1}, \qquad N_i N_i^\dagger \propto \mathbb{1}.$$

Step 2: the form of each N_i. Every 2×2 matrix is the sum of a Hermitian part $\frac{1}{2}(N + N^\dagger)$ and an anti-Hermitian part, which is i times a Hermitian matrix, and every Hermitian matrix is a real combination of $\mathbb{1}$ and the $\sigma_l^{\rm d}$. The first condition makes the Hermitian part a real multiple $x_i\mathbb{1}$, and the anti-Hermitian part is $\mathrm{i}(y_i\mathbb{1} + \boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\rm d})$ with real y_i and $\boldsymbol{c}_i$. Hence $N_i = z_i\mathbb{1} + \mathrm{i}\,\boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\rm d}$ with $z_i = x_i + \mathrm{i}y_i$. With $(\boldsymbol{c}\cdot\boldsymbol{\sigma}^{\rm d})^2 = c^2\mathbb{1}$,

$$N_i N_i^\dagger = (z_i\mathbb{1} + \mathrm{i}\,\boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\rm d})(z_i^*\mathbb{1} - \mathrm{i}\,\boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\rm d}) = (|z_i|^2 + c_i^2)\mathbb{1} + 2y_i\,\boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\rm d},$$

and the second condition requires $y_i = 0$ unless $\boldsymbol{c}_i = 0$. A direction with $\boldsymbol{c}_i = 0$ and $y_i \neq 0$ is excluded by Eq. (D.132) for a pair $i, j \neq 1$: if $N_i = z_i\mathbb{1}$ and $N_j = x_j\mathbb{1} + \mathrm{i}\,\boldsymbol{c}_j\cdot\boldsymbol{\sigma}^{\rm d}$ with $\boldsymbol{c}_j \neq 0$, then $N_i N_j^\dagger + N_j N_i^\dagger = 2x_j x_i\mathbb{1} + 2y_i\,\boldsymbol{c}_j\cdot\boldsymbol{\sigma}^{\rm d}$, which requires $y_i = 0$. The only escape is that all $\boldsymbol{c}_i$ vanish. Then $M = (\delta k_1 + z_2\delta k_2 + z_3\delta k_3)M_1$ is one complex number times M_1, the bands $\pm(\Delta_{\rm D}^2 + m^2|\delta k_1 + z_2\delta k_2 + z_3\delta k_3|^2)^{1/2}$ do not disperse along the line in $\delta\boldsymbol{k}$ on which that complex number vanishes, and we exclude this nongeneric case. Every N_i is therefore a real combination $x_i\mathbb{1} + \mathrm{i}\,\boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\rm d}$, described by the real four-vector $(x_i, \boldsymbol{c}_i)$, and N_1 is the four-vector $(1, 0, 0, 0)$.

Step 3: remove M_1. A change of basis within the valence doublet, $M \to MU_{\rm v}^\dagger$ with the unitary $U_{\rm v} = W$, turns $M_i = N_i M_1 = mN_i W$ into mN_i. Absorbing m into the velocities, $M = m\sum_i \delta k_i\,(x_i\mathbb{1} + \mathrm{i}\,\boldsymbol{c}_i\cdot\boldsymbol{\sigma}^{\rm d})$.

Step 4: rotate the four-vectors. A matrix $Q = x\mathbb{1} + \mathrm{i}\,\boldsymbol{c}\cdot\boldsymbol{\sigma}^{\rm d}$ with real $(x, \boldsymbol{c})$ has $\det Q = x^2 + c^2$, the squared length of its four-vector. A change of basis in both doublets, $Q \to U_{\rm c} Q U_{\rm v}^\dagger$ with $U_{\rm c}$ and $U_{\rm v}$ unitary of unit determinant, maps such matrices onto such matrices, linearly, and

preserves the determinant. It therefore acts on the four-vector as an orthogonal transformation, and continuity from $U_\mathrm{c} = U_\mathrm{v} = \mathbb{1}$ makes it a rotation. Every rotation of four-dimensional space arises in this way. Choosing $U_\mathrm{c} = U_\mathrm{v}$ gives the rotations of the three components $\boldsymbol{c}$ that leave x fixed. Choosing $U_\mathrm{c} = \mathrm{e}^{\mathrm{i}\phi\sigma_3^\mathrm{d}}$ and $U_\mathrm{v} = \mathrm{e}^{-\mathrm{i}\phi\sigma_3^\mathrm{d}}$ gives $Q \to \mathrm{e}^{\mathrm{i}\phi\sigma_3^\mathrm{d}} Q\, \mathrm{e}^{\mathrm{i}\phi\sigma_3^\mathrm{d}}$, which on $Q = \mathbb{1}$ yields $\cos 2\phi\, \mathbb{1} + \mathrm{i} \sin 2\phi\, \sigma_3^\mathrm{d}$, a rotation in the plane of x and c_3. Rotations of $\boldsymbol{c}$ together with rotations in one plane that contains the x axis generate all rotations of four-dimensional space. The three four-vectors $(x_i, \boldsymbol{c}_i)$ span at most three dimensions, and a rotation can bring that subspace into the one orthogonal to $(1,0,0,0)$, which removes every component along $\mathbb{1}$. A phase $-\mathrm{i}$ in the conduction doublet then removes the factor i, and

$$M = \sum_{i,l} \delta k_i\, A_{il}\, \sigma_l^\mathrm{d}$$

with a real 3×3 matrix A.

Step 5: principal axes. The singular-value decomposition writes $A = R\, \mathrm{diag}(v_1, v_2, v_3)\, R'^\mathrm{T}$ with two rotations R and R' and real v_i, one of which may be negative when both rotations are chosen proper. The rotation R is a choice of crystal axes for $\delta\boldsymbol{k}$. The rotation R' is undone by the same unitary U in both doublets, since $U \sigma_l^\mathrm{d} U^\dagger = \sum_{l'} R'_{ll'} \sigma_{l'}^\mathrm{d}$ for a suitable U, by the correspondence between unitary 2×2 matrices of unit determinant and rotations. In the resulting basis $M = \sum_i v_i\, \delta k_i\, \sigma_i^\mathrm{d}$, which is the block of the Dirac Hamiltonian Eq. (7.38).

The bubble

Take one isotropic Dirac cone, $H_{\boldsymbol{k}} = v_\mathrm{D} \boldsymbol{\alpha}_\mathrm{D} \cdot \boldsymbol{k} + \Delta_\mathrm{D} \beta_\mathrm{D}$, with $\boldsymbol{k}$ measured from the node and $\boldsymbol{\alpha}_\mathrm{D}$ and β_D the 4×4 Dirac matrices, which anticommute pairwise and square to one. Then $H_{\boldsymbol{k}}^2 = E_{\boldsymbol{k}}^2$, and each band $\pm E_{\boldsymbol{k}}$ is doubly degenerate, with the projector $P_{\boldsymbol{k}}^\pm = \frac{1}{2}(1 \pm H_{\boldsymbol{k}}/E_{\boldsymbol{k}})$ of rank two. The density matrix element between Bloch states near the cone is the overlap of the band spinors: for q small compared with the reciprocal lattice, the factor $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ acts on the slowly varying envelope and conserves the spinor index [60, App. A.3]. Summed over the two degenerate states of each band, the squared overlap is $\mathrm{Tr}[P_{\boldsymbol{k}'}^+ P_{\boldsymbol{k}}^-]$, and with $\mathrm{Tr}(\alpha_\mathrm{D}^i \alpha_\mathrm{D}^j) = 4\delta_{ij}$, $\mathrm{Tr}\, \beta_\mathrm{D}^2 = 4$ and $\mathrm{Tr}(\alpha_\mathrm{D}^i \beta_\mathrm{D}) = 0$,

$$\mathcal{F}(\boldsymbol{k}, \boldsymbol{k}') = \frac{1}{2}\, \mathrm{Tr}\big[P_{\boldsymbol{k}'}^+ P_{\boldsymbol{k}}^-\big] = \frac{1}{2}\Big(1 - \frac{v_\mathrm{D}^2\, \boldsymbol{k}\cdot\boldsymbol{k}' + \Delta_\mathrm{D}^2}{E_{\boldsymbol{k}} E_{\boldsymbol{k}'}}\Big) \tag{D.133}$$

per state, in agreement with Ref. [60, Eq. (A.7)]. It vanishes at $\boldsymbol{k}' = \boldsymbol{k}$, because the two bands are orthogonal, and grows as q^2. Let g_deg count the degenerate states per band and wavevector, summed over cones: 2 for one four-component cone, and 2 times the number of cones in general. At $T = 0$, with the chemical potential in the gap, the occupations are 1 in the lower band and 0 in the upper. Of the bubble Eq. (4.93), continued to real frequency, the interband terms remain:

$$R_\mathrm{D}^0(\boldsymbol{q}, \omega) = \frac{g_\mathrm{deg}}{\mathscr{V}} \sum_{|\boldsymbol{k}| < \Lambda_\mathrm{D}} \mathcal{F}(\boldsymbol{k}, \boldsymbol{k}+\boldsymbol{q}) \left[\frac{1}{\omega - E_{\boldsymbol{k}} - E_{\boldsymbol{k}+\boldsymbol{q}} + \mathrm{i}0^+} - \frac{1}{\omega + E_{\boldsymbol{k}} + E_{\boldsymbol{k}+\boldsymbol{q}} + \mathrm{i}0^+}\right].$$

Reduction to one integral

The bracket depends on $\boldsymbol{k}$ only through the pair energy $S = E + E'$, with $E = E_{\boldsymbol{k}}$ and $E' = E_{\boldsymbol{k}+\boldsymbol{q}}$. Use $k = |\boldsymbol{k}|$ and $k' = |\boldsymbol{k}+\boldsymbol{q}|$ as coordinates; with the azimuth about $\boldsymbol{q}$ integrated out, $\mathrm{d}^3k = 2\pi k k'\, \mathrm{d}k\, \mathrm{d}k'/q$ over the region $|k - k'| \le q \le k + k'$. Since $k\, \mathrm{d}k = E\, \mathrm{d}E/v_\mathrm{D}^2$, this is $\mathrm{d}^3k = 2\pi E E'\, \mathrm{d}E\, \mathrm{d}E'/(v_\mathrm{D}^4 q)$. In the overlap, $2\boldsymbol{k}\cdot\boldsymbol{k}' = k^2 + k'^2 - q^2$ and $v_\mathrm{D}^2 k^2 = E^2 - \Delta_\mathrm{D}^2$, so

$$\mathcal{F} = \frac{v_\mathrm{D}^2 q^2 - (E - E')^2}{4EE'}.$$

The factors EE' cancel. With $S = E + E'$ and $D = E - E'$, $\mathrm{d}E\,\mathrm{d}E' = \frac{1}{2}\mathrm{d}S\,\mathrm{d}D$, and the boundary of the region follows from $(k^2 + k'^2 - q^2)^2 \le 4k^2k'^2$. Rewritten with the energies this is $D^2(S^2 - v_\mathrm{D}^2q^2) \le v_\mathrm{D}^2q^2(S^2 - v_\mathrm{D}^2q^2 - 4\Delta_\mathrm{D}^2)$. Write $u = S^2 - v_\mathrm{D}^2q^2$, the squared energy of the pair in its rest frame, and $D_\mathrm{m} = v_\mathrm{D}q(1 - 4\Delta_\mathrm{D}^2/u)^{1/2}$. Then $|D| \le D_\mathrm{m}$, pairs exist only for $u \ge 4\Delta_\mathrm{D}^2$, and

$$\int_{-D_\mathrm{m}}^{D_\mathrm{m}} \mathrm{d}D\,(v_\mathrm{D}^2q^2 - D^2) = 2D_\mathrm{m}\Big(v_\mathrm{D}^2q^2 - \frac{D_\mathrm{m}^2}{3}\Big) = \frac{4}{3}v_\mathrm{D}^3q^3\rho(u), \qquad \rho(u) = \Big(1 - \frac{4\Delta_\mathrm{D}^2}{u}\Big)^{1/2}\Big(1 + \frac{2\Delta_\mathrm{D}^2}{u}\Big).$$

Collecting the factors, $\mathscr{V}^{-1}\sum_{\boldsymbol{k}} \mathcal{F}\,h(S) = (q^2/24\pi^2v_\mathrm{D})\int \mathrm{d}S\,\rho(S^2 - v_\mathrm{D}^2q^2)\,h(S)$ for any function h. The bracket in R_D^0 is $2S/[(\omega + \mathrm{i}0^+)^2 - S^2]$, and with $\mathrm{d}u = 2S\,\mathrm{d}S$,

$$R_\mathrm{D}^0(\boldsymbol{q},\omega) = -\frac{g_\mathrm{deg}\,q^2}{24\pi^2v_\mathrm{D}}\int_{4\Delta_\mathrm{D}^2}^{4v_\mathrm{D}^2\Lambda_\mathrm{D}^2} \mathrm{d}u\,\frac{\rho(u)}{u - s - \mathrm{i}0^+\,\mathrm{sgn}\,\omega}, \qquad s = \omega^2 - v_\mathrm{D}^2q^2. \tag{D.134}$$

The cutoff $|\boldsymbol{k}| < \Lambda_\mathrm{D}$ has become $u < 4v_\mathrm{D}^2\Lambda_\mathrm{D}^2$, which differs from the exact boundary by terms of relative order q/Λ_D and $\Delta_\mathrm{D}^2/v_\mathrm{D}^2\Lambda_\mathrm{D}^2$. The response depends on $\boldsymbol{q}$ and ω through the factor q^2 and the invariant s alone, as Lorentz invariance with v_D in the role of the speed of light requires; the cutoff breaks the invariance in these neglected terms and nowhere else.

The dielectric function

The remote bands, far from the Dirac point, respond to small q and ω as a constant background ε_bg, and the Dirac electrons add their random-phase polarization, $\varepsilon = \varepsilon_\mathrm{bg} - v_\mathrm{C}(q)R_\mathrm{D}^0$ with $v_\mathrm{C} = e^2/q^2$ (Sect. 4.5). The factor q^2 in Eq. (D.134) cancels the Coulomb $1/q^2$, and with $e^2 = 4\pi\alpha$ and $\tilde{\alpha} = \alpha/(\varepsilon_\mathrm{bg}v_\mathrm{D})$,

$$\frac{\varepsilon(\boldsymbol{q},\omega)}{\varepsilon_\mathrm{bg}} = 1 + \frac{g_\mathrm{deg}\tilde{\alpha}}{6\pi}\int_{4\Delta_\mathrm{D}^2}^{U} \mathrm{d}u\,\frac{\rho(u)}{u - s - \mathrm{i}0^+\,\mathrm{sgn}\,\omega}, \qquad U = 4v_\mathrm{D}^2\Lambda_\mathrm{D}^2.$$

The imaginary part needs no cutoff. By Eq. (2.58), for $\omega > 0$,

$$\frac{\mathrm{Im}\,\varepsilon(\boldsymbol{q},\omega)}{\varepsilon_\mathrm{bg}} = \frac{g_\mathrm{deg}\tilde{\alpha}}{6}\Big(1 - \frac{4\Delta_\mathrm{D}^2}{s}\Big)^{1/2}\Big(1 + \frac{2\Delta_\mathrm{D}^2}{s}\Big)\Theta(s - 4\Delta_\mathrm{D}^2), \tag{D.135}$$

which is Ref. [60, Eq. (2.6)] with the degeneracy written separately from the coupling. Absorption starts where the invariant energy of a pair reaches $2\Delta_\mathrm{D}$, at $\omega = (v_\mathrm{D}^2q^2 + 4\Delta_\mathrm{D}^2)^{1/2}$, and far above it $\mathrm{Im}\,\varepsilon$ tends to the constant $\varepsilon_\mathrm{bg}g_\mathrm{deg}\tilde{\alpha}/6$, independent of q: the loss of a Dirac material grows with neither momentum nor frequency.

For the real part, split $\rho(u)/(u - s) = \rho(u)/u + s\rho(u)/[u(u - s)]$. The integral of ρ/u is elementary with $u = 4\Delta_\mathrm{D}^2/(1 - b^2)$, in which $\rho\,\mathrm{d}u/u = b^2(3 - b^2)(1 - b^2)^{-1}\mathrm{d}b = [b^2 - 2 + 2/(1 - b^2)]\mathrm{d}b$; integrated to $b_U = (1 - 4\Delta_\mathrm{D}^2/U)^{1/2}$ it gives $\frac{1}{3} - 2 + \ln[(1 + b_U)/(1 - b_U)] = \ln(U/\Delta_\mathrm{D}^2) - \frac{5}{3}$ for $U \gg \Delta_\mathrm{D}^2$. The subtracted integral converges, and its cutoff may be sent to infinity. Write $\ln(1 - s/a) = -s\int_a^\infty \mathrm{d}u/[u(u - s)]$ with $a = \Delta_\mathrm{D}^2/x(1 - x)$ and integrate over x with the weight $6x(1 - x)$. Exchanging the integrals, a given u contributes for $x(1 - x) \ge \Delta_\mathrm{D}^2/u$, between $x_\pm = \frac{1}{2}(1 \pm b)$ with $b = (1 - 4\Delta_\mathrm{D}^2/u)^{1/2}$, and $\int_{x_-}^{x_+} 6x(1 - x)\,\mathrm{d}x = b(3 - b^2)/2 = \rho(u)$. Hence $s\int \rho\,\mathrm{d}u/[u(u - s)] = -6\int_0^1 \mathrm{d}x\,x(1 - x)\ln[1 - x(1 - x)s/\Delta_\mathrm{D}^2]$, and

$$\begin{aligned}\frac{\varepsilon(\boldsymbol{q},\omega)}{\varepsilon_\mathrm{bg}} = 1 + \frac{g_\mathrm{deg}\tilde{\alpha}}{6\pi}\Big[&\ln\frac{4v_\mathrm{D}^2\Lambda_\mathrm{D}^2}{\Delta_\mathrm{D}^2} - \frac{5}{3} \\ &- 6\int_0^1 \mathrm{d}x\,x(1 - x)\ln\Big(1 - x(1 - x)\frac{\omega^2 - v_\mathrm{D}^2q^2}{\Delta_\mathrm{D}^2} - \mathrm{i}0^+\,\mathrm{sgn}\,\omega\Big)\Big],\end{aligned} \tag{D.136}$$

with the branch of the logarithm that gives $-\mathrm{i}\pi$ where its argument is negative, which reproduces Eq. (D.135). This is Ref. [60, Eq. (2.6)] with Λ_D replaced by $\mathrm{e}^{-5/6}\Lambda_\mathrm{D}$: the constant $-\frac{5}{3}$ is what a hard cutoff on $|\boldsymbol{k}|$ adds to the subtraction made there, which that reference absorbs into the cutoff. A hard cutoff and a subtraction differ in this constant and in nothing else. Either way, Λ_D is a model parameter, fixed by matching to the computed band structure.

Static and long-wavelength limits

At $q \to 0$ and $\omega = 0$ the integral in Eq. (D.136) vanishes, and the static long-wavelength constant of a gapped material is

$$\varepsilon(0,0) = \varepsilon_\mathrm{bg}\Big[1 + \frac{g_\mathrm{deg}\tilde{\alpha}}{3\pi}\Big(\ln\frac{2v_\mathrm{D}\Lambda_\mathrm{D}}{\Delta_\mathrm{D}} - \frac{5}{6}\Big)\Big]. \tag{D.137}$$

For $\Delta_\mathrm{D} \to 0$ at fixed s, the logarithm becomes $\ln[-x(1-x)s/\Delta_\mathrm{D}^2]$, and with $\int_0^1 x(1-x)\ln[x(1-x)]\,\mathrm{d}x = -5/18$ the constants cancel:

$$\frac{\varepsilon(\boldsymbol{q},\omega)}{\varepsilon_\mathrm{bg}} = 1 + \frac{g_\mathrm{deg}\tilde{\alpha}}{6\pi}\ln\frac{4v_\mathrm{D}^2\Lambda_\mathrm{D}^2}{|v_\mathrm{D}^2q^2 - \omega^2|} + \mathrm{i}\,\frac{g_\mathrm{deg}\tilde{\alpha}}{6}\,\Theta(\omega - v_\mathrm{D}q), \tag{D.138}$$

which is Ref. [60, Eq. (2.7)] for a hard cutoff. In the static gapless case, $\varepsilon(q,0) = \varepsilon_\mathrm{bg}[1 + (g_\mathrm{deg}\tilde{\alpha}/3\pi)\ln(2\Lambda_\mathrm{D}/q)]$. The screening grows logarithmically as $q \to 0$, no faster; a semimetal with no Fermi surface has no Thomas–Fermi screening length, because the density of states vanishes at the node. The logarithm also controls how strongly the material screens a dark-matter probe: $\varepsilon/\varepsilon_\mathrm{bg} - 1$ is of order $g_\mathrm{deg}\tilde{\alpha}$ times a logarithm, and perturbation theory in $\tilde{\alpha}$ is reliable for $g_\mathrm{deg}\tilde{\alpha}\ln(\Lambda_\mathrm{D}/q) \lesssim 1$.

Take an anisotropic cone with velocities v_x, v_y, v_z. The substitution $k_i \to k_i/v_i$ maps the sum onto the isotropic one with unit velocity, at the rescaled momentum $\tilde{\boldsymbol{q}} = (v_xq_x, v_yq_y, v_zq_z)$ and with the Jacobian $1/(v_xv_yv_z)$, so that R_D^0 is Eq. (D.134) with $v_\mathrm{D} = 1$, $q^2 \to \tilde{q}^2$ and an overall factor $1/(v_xv_yv_z)$ [60, Eq. (B.1)]; the cutoff is then an ellipsoid in $\boldsymbol{k}$.

D.23 Effective-mass theory of shallow donors

A phosphorus atom on a silicon site has one more valence electron than the host. Its core charge, screened by the static dielectric constant of the crystal, binds that electron in an orbit tens of ångströms across, many lattice constants, so the electron feels the crystal only through its conduction band near the minimum. This section derives the envelope equation of Kohn and Luttinger [61], solves it for an isotropic band, and solves it variationally for the anisotropic valleys of silicon. It then treats the coupling of the six valleys by the potential near the core, which splits the ground state. The results are used in Sect. 7.4 and Sect. 16.5; the hydrogenic solution itself is that of Sect. 4.9.

The envelope equation

The donor adds to the crystal Hamiltonian the potential $U(\boldsymbol{r})$ of its excess charge. Far from the core, at distances large compared with the lattice constant, it is the Coulomb potential of a unit charge, $-e^2/(4\pi r)$ in Heaviside–Lorentz units ($e^2 = 4\pi\alpha$). The host screens it by its static dielectric constant ε:

$$U(\boldsymbol{r}) = -\frac{e^2}{4\pi\varepsilon\, r} = -\frac{\alpha}{\varepsilon\, r}.$$

The dielectric constant is a pure number. For silicon, which being nonpolar has no lattice contribution, it is $\varepsilon = \varepsilon_0 = \varepsilon_\infty = 11.7$ (Sect. 5.5), and from here on we write it ε_0, the

static constant, as in Sect. 7.4. Expand the donor state in the Bloch states of Sect. 3.2, $\psi = \sum_{n\boldsymbol{k}} A_n(\boldsymbol{k})\,\psi_{n\boldsymbol{k}}$. The Schrödinger equation becomes

$$(\varepsilon_{n\boldsymbol{k}} - E)A_n(\boldsymbol{k}) + \sum_{n'\boldsymbol{k}'} \langle \psi_{n\boldsymbol{k}}|U|\psi_{n'\boldsymbol{k}'}\rangle\, A_{n'}(\boldsymbol{k}') = 0.$$

A slowly varying U simplifies the matrix element. Its Fourier components are concentrated at $|\boldsymbol{k}-\boldsymbol{k}'| \sim 1/a^*$, small compared with the zone, so for the cell-periodic parts $u_{n\boldsymbol{k}}$ of Eq. (3.20), $\langle \psi_{n\boldsymbol{k}}|U|\psi_{n'\boldsymbol{k}'}\rangle \simeq \tilde{U}(\boldsymbol{k}-\boldsymbol{k}')\,\langle u_{n\boldsymbol{k}}|u_{n'\boldsymbol{k}'}\rangle_{\text{cell}}$, where $\tilde{U}(\boldsymbol{p}) = \mathscr{V}^{-1}\int \mathrm{d}^3r\, \mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}}U(\boldsymbol{r})$ and the cell average is normalized to $\delta_{nn'}$ at $\boldsymbol{k}=\boldsymbol{k}'$. For $\boldsymbol{k}'$ close to $\boldsymbol{k}$ the overlap of different bands is of order $|\boldsymbol{k}-\boldsymbol{k}'|$ times a ratio of a velocity matrix element to an energy gap, by Eq. (3.42), and it couples bands only at the next order in $1/(a^*k_{\text{zone}})$. To leading order the bands decouple, and the conduction-band amplitude obeys

$$(\varepsilon_{c\boldsymbol{k}} - E)A(\boldsymbol{k}) + \sum_{\boldsymbol{k}'} \tilde{U}(\boldsymbol{k}-\boldsymbol{k}')\,A(\boldsymbol{k}') = 0.$$

Near one conduction minimum $\boldsymbol{k}_j$, write $\boldsymbol{k} = \boldsymbol{k}_j + \delta\boldsymbol{k}$ and define the envelope $F_j(\boldsymbol{r}) = \mathscr{V}^{-1/2}\sum_{\delta\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\delta\boldsymbol{k}\cdot\boldsymbol{r}}A(\boldsymbol{k}_j+\delta\boldsymbol{k})$. Multiplying by $\delta\boldsymbol{k}$ becomes $-\mathrm{i}\nabla$ acting on F_j, and the convolution with $\tilde{U}$ becomes multiplication by $U(\boldsymbol{r})$. With the band near the minimum from Sect. 3.3, $\varepsilon_{c,\boldsymbol{k}_j+\delta\boldsymbol{k}} = E_{\text{g}} + \delta k_\parallel^2/2m_\parallel^* + \delta k_\perp^2/2m_\perp^*$ (energies from the valence maximum), and z along the valley axis,

$$\Big[-\frac{1}{2m_\perp^*}\Big(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\Big) - \frac{1}{2m_\parallel^*}\frac{\partial^2}{\partial z^2} - \frac{\alpha}{\varepsilon_0 r}\Big]F_j(\boldsymbol{r}) = -E_{\text{b}}^{\text{D}}\,F_j(\boldsymbol{r}), \qquad \psi(\boldsymbol{r}) \simeq F_j(\boldsymbol{r})\,\psi_{c\boldsymbol{k}_j}(\boldsymbol{r}), \tag{D.139}$$

with $E = E_{\text{g}} - E_{\text{b}}^{\text{D}}$. The wavefunction is the Bloch function at the minimum, modulated by a hydrogen-like envelope. The derivation assumed three things: that the envelope extends over many cells, $a^* \gg a$; that the binding is small compared with the energy over which the band departs from its quadratic form; and that the potential is the screened Coulomb potential everywhere that matters. The third fails within a cell or two of the core, which is the central-cell region treated below.

Isotropic band

For $m_\parallel^* = m_\perp^* = m^*$, Eq. (D.139) is the hydrogen atom with $m_e \to m^*$ and $\alpha \to \alpha/\varepsilon_0$. The energies and the length scale follow from the hydrogenic ones by the same substitution: with $a_0 = 1/(\alpha m_e)$,

$$E_{\text{b}}^{\text{D}} = \frac{m^*\alpha^2}{2\varepsilon_0^2} = \frac{m^*/m_e}{\varepsilon_0^2}\,\text{Ry}, \qquad a^* = \frac{\varepsilon_0}{m^*\alpha} = \varepsilon_0\,\frac{m_e}{m^*}\,a_0, \tag{D.140}$$

with $\text{Ry} = 13.606\,\text{eV}$ and $a_0 = 0.529\,\text{Å}$ [62], and the excited levels at $E_{\text{b}}^{\text{D}}/n^2$. The factor $m^*/m_e\varepsilon_0^2$ is of order 10^{-3} in common semiconductors, and the orbit is $\varepsilon_0 m_e/m^*$ times larger than that of hydrogen, which justifies the envelope picture. For silicon the two masses of Table 3.1 differ by a factor of five, and the isotropic formula yields 19.0 meV with $m_\perp^* = 0.191\,m_e$ and 91.0 meV with $m_\parallel^* = 0.916\,m_e$ (Sect. 7.4).

Anisotropic valleys

With unequal masses, Eq. (D.139) does not separate, and Kohn and Luttinger solved it variationally with an envelope squeezed along the heavy axis [61],

$$F(\boldsymbol{r}) = (\pi r_\perp^2 r_\parallel)^{-1/2}\exp\Big[-\Big(\frac{x^2+y^2}{r_\perp^2} + \frac{z^2}{r_\parallel^2}\Big)^{1/2}\Big],$$

with the radii $r_\perp$ across the valley axis and $r_\parallel$ along it, normalized to one. In the scaled coordinates $(x/r_\perp, y/r_\perp, z/r_\parallel)$ it is the hydrogen ground state $\mathrm{e}^{-r}/\sqrt{\pi}$, for which each Cartesian component of $\langle|\nabla F|^2\rangle$ is $\frac{1}{3}$. The kinetic energy is therefore $(2/3)(2m^*_\perp r_\perp^2)^{-1} + (1/3)(2m^*_\parallel r_\parallel^2)^{-1}$. For the potential, write $r = R\,(r_\perp^2 \sin^2\theta + r_\parallel^2 \cos^2\theta)^{1/2}$ in the scaled spherical coordinates (R, θ). The radial integral $\int_0^\infty R\,\mathrm{e}^{-2R}\,\mathrm{d}R = \frac{1}{4}$ separates, and for $r_\parallel < r_\perp$

$$\left\langle \frac{1}{r} \right\rangle = \int_0^1 \frac{\mathrm{d}\mu}{\left[r_\perp^2 - (r_\perp^2 - r_\parallel^2)\mu^2\right]^{1/2}} = \frac{\arcsin e_{\mathrm{ecc}}}{r_\perp e_{\mathrm{ecc}}}, \qquad e_{\mathrm{ecc}} = \left(1 - \frac{r_\parallel^2}{r_\perp^2}\right)^{1/2},$$

Measuring lengths in units of $a^*_\perp = \varepsilon_0/(m^*_\perp \alpha)$ and energies in units of $\mathrm{Ry}^*_\perp = m^*_\perp \alpha^2/2\varepsilon_0^2$, with e_{ecc} the eccentricity of the spheroid, the energy is

$$\frac{E(r_\perp, r_\parallel)}{\mathrm{Ry}^*_\perp} = \frac{2}{3r_\perp^2} + \frac{\gamma_m}{3r_\parallel^2} - \frac{2\arcsin e_{\mathrm{ecc}}}{r_\perp e_{\mathrm{ecc}}}, \qquad \gamma_m = \frac{m^*_\perp}{m^*_\parallel}. \tag{D.141}$$

For $\gamma_m = 1$ the minimum is at $r_\perp = r_\parallel = 1$ with $E = -1$, the hydrogenic result. For silicon, $\gamma_m = 0.191/0.916 = 0.209$, $\mathrm{Ry}^*_\perp = 18.98\,\mathrm{meV}$ and $a^*_\perp = 32.4\,\text{Å}$. Minimizing Eq. (D.141) numerically gives $E = -1.564\,\mathrm{Ry}^*_\perp$ at $r_\perp = 24.3\,\text{Å}$ and $r_\parallel = 14.0\,\text{Å}$:

$$E_{\mathrm{b}}^{\mathrm{D}}(\mathrm{Si}) = 29.7\,\mathrm{meV} \qquad \text{(effective-mass theory, variational)}. \tag{D.142}$$

The orbit is an oblate spheroid, flattened along the heavy axis, with radii of four and a half and two and a half lattice constants. The variational energy is an upper bound on the exact effective-mass energy. It applies equally to every shallow donor in silicon, since nothing in Eq. (D.139) depends on the chemical species. The measured ground states of phosphorus, arsenic and antimony are all deeper and differ from one another (Sect. 7.4), which locates the missing physics near the core.

Valleys and the central cell

Silicon has six equivalent minima on the $\langle 100\rangle$ axes. At the order of Eq. (D.139) each valley carries its own envelope, rotated with its axis, and the ground state is sixfold degenerate (twelvefold with spin): $\psi = \sum_{j=1}^{6} c_j F_j(\boldsymbol{r}) \psi_{c\boldsymbol{k}_j}(\boldsymbol{r})$ with any coefficients c_j. The valleys couple through the part δU of the donor potential that the envelope equation leaves out, which differs from $-\alpha/\varepsilon_0 r$ only within about a lattice constant of the core, where the screening is incomplete and the ion has its own chemistry. Since δU is short-ranged and each envelope is nearly constant over its range, the matrix elements between valley states are $\langle j|\delta U|j'\rangle \simeq F_j(0)F_{j'}(0)\int \mathrm{d}^3 r\, \psi^*_{c\boldsymbol{k}_j}\, \delta U\, \psi_{c\boldsymbol{k}_{j'}}$. Their Fourier components at $\boldsymbol{k}_j - \boldsymbol{k}_{j'}$, a large fraction of the zone, are not small for a potential concentrated in one cell.

The tetrahedral symmetry of the site leaves three distinct elements: Δ_0 on the diagonal, Δ_{g} between opposite valleys and Δ_{f} between perpendicular ones. The 6×6 matrix has the eigenvectors

$$\begin{aligned}
A_1:&\quad (1,1,1,1,1,1)/\sqrt{6}, && E = \Delta_0 + \Delta_{\mathrm{g}} + 4\Delta_{\mathrm{f}},\\
E:&\quad (1,1,-1,-1,0,0)/2 \text{ and } (1,1,1,1,-2,-2)/\sqrt{12}, && E = \Delta_0 + \Delta_{\mathrm{g}} - 2\Delta_{\mathrm{f}},\\
T_2:&\quad (1,-1,0,0,0,0)/\sqrt{2} \text{ and its two partners}, && E = \Delta_0 - \Delta_{\mathrm{g}},
\end{aligned} \tag{D.143}$$

with the valleys ordered as $(+x, -x, +y, -y, +z, -z)$. In the T_2 states each pair of opposite valleys enters with opposite signs, and the four perpendicular couplings cancel. In the A_1 state all six valleys add in phase, so its density at the core is the largest, $|\psi(0)|^2 = 6|F(0)|^2 |u_c(0)|^2$,

and an attractive central-cell correction lowers it most. This is the ground state of the group-V donors in silicon, deeper than Eq. (D.142) by an amount that depends on the species. The E and T_2 levels have nodes at the core and lie close to the effective-mass value. The excited states with $l \geq 1$ envelopes, which vanish at the core, are almost unaffected by δU; the effective-mass theory accounts for them well, and the central-cell correction is a property of the s-like ground state alone.

D.24 The operator basis

The reduction of Sect. 8.1 takes an interaction written for a dark-matter particle and one free fermion and turns it into operators of the whole target. Four steps take it there: the invariants from which the operators are built, the non-relativistic limits of the Dirac bilinears between plane waves, the one-body operators that reproduce them in a many-body target, and the identities that relate the rows on shell. This section then treats nuclei, the spin traces of the dark-matter-side tensor, and the kinematics of absorption. Natural units are used, and $\boldsymbol{q}$ is the momentum the dark matter loses (Sect. 1.3).

D.24.1 Galilean invariants

The dark matter goes from $\boldsymbol{p}$ to $\boldsymbol{p}' = \boldsymbol{p} - \boldsymbol{q}$ and the target fermion, of mass m, from $\boldsymbol{k}$ to $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{q}$. A Galilean boost by a velocity $\boldsymbol{u}$ shifts $\boldsymbol{p}$ and $\boldsymbol{p}'$ by $m_\chi \boldsymbol{u}$, and $\boldsymbol{k}$ and $\boldsymbol{k}'$ by $m\boldsymbol{u}$. The momentum transfer $\boldsymbol{q}$ is invariant, and so is the difference of the mean velocities,

$$\boldsymbol{v}^\perp = \frac{\boldsymbol{p}+\boldsymbol{p}'}{2m_\chi} - \frac{\boldsymbol{k}+\boldsymbol{k}'}{2m} = \bar{\boldsymbol{v}}_\chi - \boldsymbol{v}_\mathrm{t}, \tag{D.144}$$

with $\boldsymbol{v}_\mathrm{t} \equiv (\boldsymbol{k}+\boldsymbol{k}')/2m$ (used only in this section). Of the four momenta, conservation leaves three and invariance two, $\boldsymbol{q}$ and $\boldsymbol{v}^\perp$. Exchanging the incoming and outgoing particles, which is what Hermitian conjugation does to a transition operator, sends $\boldsymbol{q} \to -\boldsymbol{q}$ and leaves $\boldsymbol{v}^\perp$ unchanged; since conjugation also sends $\mathrm{i} \to -\mathrm{i}$, the combination $\mathrm{i}\boldsymbol{q}$ is unchanged too. Operators built from $\mathrm{i}\boldsymbol{q}$, $\boldsymbol{v}^\perp$ and the spins are therefore Hermitian when their coefficients are real [63, Sect. 2.2].

For free particles, with the dark matter changing its mass by Δ_χ when it is inelastic, energy conservation reads $p^2/2m_\chi + k^2/2m = p'^2/2m_\chi + k'^2/2m + \Delta_\chi$. Since $p^2 - p'^2 = \boldsymbol{q}\cdot(\boldsymbol{p}+\boldsymbol{p}')$ and $k'^2 - k^2 = \boldsymbol{q}\cdot(\boldsymbol{k}+\boldsymbol{k}')$, this is

$$\boldsymbol{q}\cdot\boldsymbol{v}^\perp = \Delta_\chi, \tag{D.145}$$

and $\boldsymbol{v}^\perp$ is transverse to $\boldsymbol{q}$ for elastic scattering on any free target, whatever its initial momentum. In the rest frame of the target, $\boldsymbol{k} = 0$, this is $\boldsymbol{q}\cdot\boldsymbol{v} = q^2/2\mu_{\chi T}$ with $\mu_{\chi T}$ the reduced mass, as in Sect. 1.3.

D.24.2 Bilinears between plane waves

With the spinors Eq. (C.127), $u = (E+m)^{1/2}(\xi, \boldsymbol{\sigma}\cdot\boldsymbol{k}\,\xi/(E+m))$, and the states normalized to one particle per volume, each bilinear is divided by $(2E\cdot 2E')^{1/2}$. Write $\boldsymbol{S} = \boldsymbol{\sigma}/2$ and $\boldsymbol{K} = \boldsymbol{k}+\boldsymbol{k}'$. To second order in k/m, $[(E+m)/2E]^{1/2} = 1 - k^2/8m^2$ and $1/(E+m) = 1/2m$. With $(\boldsymbol{\sigma}\cdot\boldsymbol{k}')(\boldsymbol{\sigma}\cdot\boldsymbol{k}) = \boldsymbol{k}'\cdot\boldsymbol{k} + \mathrm{i}\boldsymbol{\sigma}\cdot(\boldsymbol{k}'\times\boldsymbol{k})$ and $\boldsymbol{k}'\times\boldsymbol{k} = \boldsymbol{q}\times\boldsymbol{K}/2$, the scalar and the time component of the vector are

$$\begin{aligned}
\frac{\bar{u}'u}{(4EE')^{1/2}} &= 1 - \frac{k^2+k'^2}{8m^2} - \frac{\boldsymbol{k}'\cdot\boldsymbol{k}}{4m^2} - \frac{\mathrm{i}\boldsymbol{S}\cdot(\boldsymbol{q}\times\boldsymbol{K})}{4m^2} = 1 - \frac{K^2}{8m^2} - \frac{\mathrm{i}\boldsymbol{S}\cdot(\boldsymbol{q}\times\boldsymbol{K})}{4m^2}, \\
\frac{\bar{u}'\gamma^0u}{(4EE')^{1/2}} &= 1 - \frac{k^2+k'^2}{8m^2} + \frac{\boldsymbol{k}'\cdot\boldsymbol{k}}{4m^2} + \frac{\mathrm{i}\boldsymbol{S}\cdot(\boldsymbol{q}\times\boldsymbol{K})}{4m^2} = 1 - \frac{q^2}{8m^2} + \frac{\mathrm{i}\boldsymbol{S}\cdot(\boldsymbol{q}\times\boldsymbol{K})}{4m^2}.
\end{aligned} \tag{D.146}$$

The second line is the vertex Eq. (C.134) found there by the Foldy–Wouthuysen transformation. The two differ at second order: the scalar loses $K^2/8m^2 = v_\mathrm{t}^2/2$, the vector time component only $q^2/8m^2$. The other bilinears are needed to first order. With the blocks of Eq. (C.125) and $\sigma^i\sigma^j = \delta^{ij} + \mathrm{i}\epsilon^{ijl}\sigma^l$,

$$\frac{\bar u'\gamma^i u}{2m} = \frac{K^i}{2m} - \frac{\mathrm{i}(\boldsymbol{q}\times\boldsymbol{S})^i}{m}, \qquad \frac{\bar u'\gamma^0\gamma^5 u}{2m} = \frac{2\boldsymbol{S}\cdot\boldsymbol{K}}{2m}, \qquad \frac{\bar u'\gamma^i\gamma^5 u}{2m} = 2S^i, \qquad \frac{\bar u'\mathrm{i}\gamma^5 u}{2m} = -\frac{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{S}}{m}, \tag{D.147}$$

with the two-component spinors understood on either side. For the spatial vector, $\bar u'\gamma^i u = u'^\dagger\alpha_\mathrm{D}^i u = \xi'^\dagger[\sigma^i(\boldsymbol{\sigma}\cdot\boldsymbol{k}) + (\boldsymbol{\sigma}\cdot\boldsymbol{k}')\sigma^i]\xi$ to this order, and the Pauli algebra gives $K^i - \mathrm{i}\epsilon^{ijl}q^j\sigma^l$. These agree with Ref. [64, Table I], whose $\boldsymbol{q}$ has the same sign. For the dark matter, whose momentum changes by $-\boldsymbol{q}$, replace $\boldsymbol{q}\to-\boldsymbol{q}$ and $\boldsymbol{K}/2m\to\bar{\boldsymbol{v}}_\chi$.

A massive vector mediator exchanged between two currents adds the spatial components to the potential of Sect. 1.4. The Proca field obeys $(\partial^2 + m_V^2)A'^\mu = y_\chi J_\chi^\mu$ for a source $-y_\chi A'_\mu J_\chi^\mu$, each component as the time component in Eq. (C.80). For $\omega \ll q$ its transform is $y_\chi J_\chi^\mu/(q^2+m_V^2)$, and a second current with coupling y has the energy $y(A'^0J^0 - \boldsymbol{A}'\cdot\boldsymbol{J})$. The potential is

$$\tilde V = \frac{y_\chi y}{q^2+m_V^2}\,(J_\chi^0 J^0 - \boldsymbol{J}_\chi\cdot\boldsymbol{J}), \tag{D.148}$$

with the currents the bilinears of Eqs. (D.146) and (D.147). Its leading term is $\mathcal{O}_1$, and the spatial product is the source of the corrections estimated in Worked example 8.1.

D.24.3 From one particle to many

Each operator of Table 8.1 has, between plane waves, a matrix element $f(\boldsymbol{k},\boldsymbol{k}')\,\delta_{\boldsymbol{k}',\boldsymbol{k}+\boldsymbol{q}}$ with f a polynomial of degree at most two in $\boldsymbol{K}$, times at most one spin. The one-body operator with these matrix elements is unique, and for the three monomials it is

$$1 \to \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}, \qquad \frac{\boldsymbol{K}}{2m} \to \frac{1}{2}\Big\{\frac{\boldsymbol{p}}{m}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\Big\}, \qquad \frac{K^2}{4m^2} \to \frac{1}{4m^2}\{\boldsymbol{p},\{\boldsymbol{p},\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\}\}. \tag{D.149}$$

The proof is one line each: $\langle\boldsymbol{k}'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\boldsymbol{k}\rangle = \delta_{\boldsymbol{k}',\boldsymbol{k}+\boldsymbol{q}}$, and a factor $\boldsymbol{p}$ on the left of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ contributes $\boldsymbol{k}'$ while one on the right contributes $\boldsymbol{k}$. The double anticommutator is $p^2X + 2\boldsymbol{p}\cdot X\boldsymbol{p} + Xp^2$ with $X = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, with element $k'^2 + 2\boldsymbol{k}'\cdot\boldsymbol{k} + k^2 = K^2$. Writing $K^2 = q^2 + 4\boldsymbol{k}\cdot\boldsymbol{k}'$ gives the second form of $\mathcal{T}_s$ in Eq. (8.3). The spin $\boldsymbol{S}$ commutes with $\boldsymbol{r}$ and $\boldsymbol{p}$ and simply multiplies. For a target of many particles of species s, the operator is the sum over particles, as for any one-body operator (Eq. (2.36)), and in second quantization it is $\sum_{\boldsymbol{k}} f(\boldsymbol{k},\boldsymbol{k}+\boldsymbol{q})\,c^\dagger_{\boldsymbol{k}+\boldsymbol{q}}c_{\boldsymbol{k}}$ with the spin indices of $\boldsymbol{S}$ between the operators. In position space, the density row is the transform $n_e(\boldsymbol{q}) = \int \mathrm{d}^3r\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,\hat n_e(\boldsymbol{r})$ of the density operator of Sect. 4.2, and the other rows are transforms of the corresponding densities. These are the rows of Eq. (8.3).

D.24.4 Reduction of the operators

With Eq. (D.144), each $\mathcal{O}_i$ is a polynomial in $\boldsymbol{v}_\mathrm{t}$ and $\boldsymbol{S}_e$ with coefficients built from $\bar{\boldsymbol{v}}_\chi$, $\boldsymbol{S}_\chi$ and $\boldsymbol{q}$. Expanding and replacing $1\to n$, $S_e^\beta\to s^\beta$, $v_\mathrm{t}^\alpha\to j^\alpha$, $v_\mathrm{t}^\alpha S_e^\beta\to\mathcal{J}^{\alpha\beta}$ and $v_\mathrm{t}^2\to(2/m)\mathcal{T}$ gives Table 8.1. Three cases show every pattern. For $\mathcal{O}_3$,

$$\mathrm{i}\boldsymbol{S}_e\cdot\Big(\frac{\boldsymbol{q}}{m}\times\boldsymbol{v}^\perp\Big) = \frac{\mathrm{i}}{m}\Big[(\boldsymbol{q}\times\bar{\boldsymbol{v}}_\chi)\cdot\boldsymbol{S}_e - \epsilon_{\alpha\beta\gamma}S_e^\alpha q^\beta v_\mathrm{t}^\gamma\Big] \;\to\; \frac{\mathrm{i}}{m}\Big[(\boldsymbol{q}\times\bar{\boldsymbol{v}}_\chi)\cdot\boldsymbol{s} - \epsilon_{\alpha\beta\gamma}q^\beta\mathcal{J}^{\gamma\alpha}\Big],$$

where the velocity index of $\mathcal{J}$ comes first. For $\mathcal{O}_{12}$, $(\boldsymbol{S}_e\times\boldsymbol{v}^\perp)^\alpha = \epsilon_{\alpha\beta\gamma}S_e^\beta(\bar v_\chi^\gamma - v_\mathrm{t}^\gamma)$, so $\boldsymbol{S}_\chi\cdot(\boldsymbol{S}_e\times\boldsymbol{v}^\perp)\to\boldsymbol{S}_\chi\cdot(\boldsymbol{s}\times\bar{\boldsymbol{v}}_\chi) - \epsilon_{\alpha\beta\gamma}S_\chi^\alpha\mathcal{J}^{\gamma\beta}$. For $\mathcal{O}_{15}$ the same cross product is dotted into $\boldsymbol{q}$, $(\boldsymbol{S}_e\times\boldsymbol{v}^\perp)\cdot\boldsymbol{q}\to\boldsymbol{q}\cdot(\boldsymbol{s}\times\bar{\boldsymbol{v}}_\chi) - \epsilon_{\alpha\beta\gamma}q^\alpha\mathcal{J}^{\gamma\beta}$, and multiplied by $-\boldsymbol{S}_\chi\cdot\boldsymbol{q}/m^2$. The operators without $\boldsymbol{S}_e$ need only n and $\boldsymbol{j}$, those linear in $\boldsymbol{S}_e$ without $\boldsymbol{v}^\perp$ only $\boldsymbol{s}$, and those with both need $\boldsymbol{s}$ and $\mathcal{J}$. Operator $\mathcal{O}_2$ is the one case of degree two, Eq. (8.9).

D.24.5 Longitudinal identities

Let the target Hamiltonian consist of the kinetic energies of its particles and of potentials that depend on their positions alone, as $H_{\rm C}$ of Eq. (2.48) does. Then $[H, n_s(\boldsymbol{q})] = \boldsymbol{q}\cdot\boldsymbol{j}_s(\boldsymbol{q})$ for each species, by the commutator of Eq. (4.29) with $m_e \to m_s$, and between eigenstates with $E_f - E_i = \omega$,

$$\langle f|\boldsymbol{q}\cdot\boldsymbol{j}_s(\boldsymbol{q})|i\rangle = \omega\,\langle f|n_s(\boldsymbol{q})|i\rangle. \tag{D.150}$$

If, in addition, H commutes with every spin $\boldsymbol{S}_j$ of the species, the same commutator multiplied by S_j^β gives $[H, s_s^\beta(\boldsymbol{q})] = q^\alpha \mathcal{J}_s^{\alpha\beta}(\boldsymbol{q})$, and $\langle f|q^\alpha\mathcal{J}_s^{\alpha\beta}|i\rangle = \omega\langle f|s_s^\beta|i\rangle$. Spin–orbit coupling, the hyperfine interaction and a Zeeman term break this second identity and leave the first intact. On the dark-matter side, Eq. (1.22) with $\bar{\boldsymbol{v}}_\chi = \boldsymbol{v} - \boldsymbol{q}/2m_\chi$ gives $\boldsymbol{q}\cdot\bar{\boldsymbol{v}}_\chi = \omega + \Delta_\chi$ on shell. Every occurrence of $\boldsymbol{q}\cdot\boldsymbol{v}^\perp$ in an operator therefore becomes, inside a rate,

$$\boldsymbol{q}\cdot\bar{\boldsymbol{v}}_\chi\, n_s - \boldsymbol{q}\cdot\boldsymbol{j}_s \;\to\; (\omega + \Delta_\chi - \omega)\, n_s = \Delta_\chi\, n_s, \tag{D.151}$$

which is Eq. (8.5), the many-body form of Eq. (D.145). For $\mathcal{O}_2$ the identity applies only to the cross term. The square of the longitudinal electron velocity is not fixed by ω in a bound target, because the potential energy changes along with the kinetic energy. The three rows of Eq. (8.9) are then needed as they stand, and the cancellation of the longitudinal parts used in Worked example 8.2 is exact only for free electrons.

D.24.6 Nuclei

Write the nucleons of nucleus I at $\boldsymbol{X}_I + \boldsymbol{y}_j$, with momenta $(m_N/M_\kappa)\boldsymbol{P}_I + \boldsymbol{\pi}_j$, where $\boldsymbol{P}_I$ is the momentum of the nucleus and $\sum_j \boldsymbol{\pi}_j = 0$. A nucleon operator $\sum_j c^{(N_j)} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{X}_I + \boldsymbol{y}_j)}$, with N_j the type of nucleon j, has, between nuclear ground states, the element $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}\langle 0|\sum_j c^{(N_j)}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{y}_j}|0\rangle$. For equal proton and neutron distributions this is $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}[Z_\kappa c^{(p)} + (A_\kappa - Z_\kappa)c^{(n)}]F_N(q)$ with F_N of Eq. (2.26), which is the first form of Eq. (8.8). The nucleon velocity $(\boldsymbol{k} + \boldsymbol{k}')/2m_N$ splits into $(\boldsymbol{P}_I + \boldsymbol{P}_I')/2M_\kappa$ and an internal part built from $\boldsymbol{\pi}_j$. The internal part has zero ground-state expectation value, and its contribution at first order in $\boldsymbol{q}\cdot\boldsymbol{y}_j$ involves the orbital and spin–orbit structure of the nucleus [63, Eqs. (26) and (27)]; dropping it leaves the rows $\boldsymbol{j}_\kappa$ and $\mathcal{T}_\kappa$ with the nuclear mass.

For the spin, $\boldsymbol{W} = \sum_j c^{(N_j)}\boldsymbol{S}_j$ is a vector operator. Within the multiplet of nuclear spin $J_{\rm nuc}$ the Wigner–Eckart theorem makes every vector operator proportional to $\boldsymbol{J}^{\rm nuc}$, $\langle M'|\boldsymbol{W}|M\rangle = w\,\langle M'|\boldsymbol{J}^{\rm nuc}|M\rangle$. Taking $M' = M = J_{\rm nuc}$ and the z components, $w\,J_{\rm nuc} = c^{(p)}\langle S_p\rangle + c^{(n)}\langle S_n\rangle$, where $\langle S_p\rangle = \langle J_{\rm nuc}, J_{\rm nuc}|\sum_{j\in p} S_j^z|J_{\rm nuc}, J_{\rm nuc}\rangle$ and similarly for neutrons. This is the second form of Eq. (8.8). Averaged over M and summed over M' and the components, $\sum_\alpha |\langle M'|J_{\rm nuc}^\alpha|M\rangle|^2$ gives $\mathrm{Tr}(\boldsymbol{J}_{\rm nuc}^2)/(2J_{\rm nuc}+1) = J_{\rm nuc}(J_{\rm nuc}+1)$, the factor used for ^{29}Si in Sect. 8.1. Its ratio to the free-nucleon value $\frac{3}{4}$ reproduces the spin weight $\frac{4}{3}(J_{\rm nuc}+1)J_{\rm nuc}^{-1}[c^{(p)}\langle S_p\rangle + c^{(n)}\langle S_n\rangle]^2$ of Appendix E.27.

D.24.7 Spin traces

The tensor Eq. (8.7) is Hermitian, since exchanging a and b conjugates the trace. It is positive semidefinite: for any complex numbers x_a, $\sum_{ab} x_a^*\mathfrak{D}_{ab}x_b = (2|\boldsymbol{S}_\chi|+1)^{-1}\,\mathrm{Tr}_\chi(Y^\dagger Y) \geq 0$ with $Y = \sum_b x_b\tilde{V}_b$. For a dark-matter spin S, $\mathrm{Tr}_\chi\,\boldsymbol{S}_\chi = 0$ and $\mathrm{Tr}_\chi(S_\chi^\alpha S_\chi^\beta) = \delta_{\alpha\beta}(2S+1)S(S+1)/3$, so the average of $S_\chi^\alpha S_\chi^\beta$ is $\delta_{\alpha\beta}/4$ for $S = \frac{1}{2}$. Products of three spin components have a trace proportional to $\epsilon_{\alpha\beta\gamma}$ for spin $\frac{1}{2}$, and they arise only when a coefficient quadratic in $\boldsymbol{S}_\chi$ meets a linear one, which none of the operators of Table 8.1 produces.

D.24.8 Absorption and the energy deposited

An absorbed boson enters through $\int \mathrm{d}^3x\,\phi(x)\,\mathcal{B}(\boldsymbol{x})$, with $\mathcal{B}$ a bilinear of the target fields and $\phi \propto \mathrm{e}^{-\mathrm{i}m_\phi t+\mathrm{i}\boldsymbol{p}_\phi\cdot\boldsymbol{x}}$. The spatial integral is the Fourier transform at $\boldsymbol{q}=\boldsymbol{p}_\phi$, so the plane-wave rules above apply with $\boldsymbol{q}=\boldsymbol{p}_\phi$. For the axion-like particle, $\hat\psi^\dagger\boldsymbol{\sigma}\cdot\overleftrightarrow{\nabla}\hat\psi=\hat\psi^\dagger\boldsymbol{\sigma}\cdot\nabla\hat\psi-(\nabla\hat\psi^\dagger)\cdot\boldsymbol{\sigma}\hat\psi$ has the element $\boldsymbol{\sigma}\cdot(\mathrm{i}\boldsymbol{k}+\mathrm{i}\boldsymbol{k}')=4\mathrm{i}m_e\,\boldsymbol{S}\cdot\boldsymbol{K}/2m_e$, since ∇ on the annihilated state gives $\mathrm{i}\boldsymbol{k}$ and on the created state $-\mathrm{i}\boldsymbol{k}'$. The second term of Eq. (1.40) is then $(\mathrm{i}g_{aee}/4m_e^2)(\partial_t\phi)\,4\mathrm{i}m_e\,\mathcal{J}_e^{\alpha\alpha}=-(g_{aee}/m_e)(\partial_t\phi)\,\mathcal{J}_e^{\alpha\alpha}$, and the first is $-(g_{aee}/m_e)(\nabla\phi)\cdot\boldsymbol{s}_e$; the interaction energy is minus the Lagrangian, which is Eq. (8.10). For the scalar, $\hat\psi^\dagger\overleftrightarrow{\nabla}^2\hat\psi$ has the element $-k^2-k'^2-2\boldsymbol{k}'\cdot\boldsymbol{k}=-K^2$, and $(1/8m_e^2)(-K^2)=-(1/m_e)(K^2/8m_e)$ is $-1/m_e$ times the element $K^2/8m_e$ of $\mathcal{T}_e$. This is the order-v^2 term of the scalar bilinear in Eq. (D.146), and with $H=-\mathcal{L}$ it gives Eq. (8.11).

The energy deposited is Eq. (8.12). For a non-relativistic outgoing particle of mass $m_{\rm out}=m_\chi+\Delta_\chi$,

$$\omega=\Big(m_\chi+\frac{p^2}{2m_\chi}\Big)-\Big(m_\chi+\Delta_\chi+\frac{|\boldsymbol{p}-\boldsymbol{q}|^2}{2m_{\rm out}}\Big)=\boldsymbol{q}\cdot\boldsymbol{v}-\frac{q^2}{2m_\chi}-\Delta_\chi+\frac{|\boldsymbol{p}-\boldsymbol{q}|^2}{2}\Big(\frac{1}{m_\chi}-\frac{1}{m_{\rm out}}\Big),$$

where $p^2-|\boldsymbol{p}-\boldsymbol{q}|^2=2\boldsymbol{q}\cdot\boldsymbol{p}-q^2$ was used. The first three terms are $\omega_{\boldsymbol{q}}$, and $1/m_\chi-1/m_{\rm out}=\Delta_\chi/(m_\chi m_{\rm out})$, which is Eq. (8.13). The same formula holds for fermionic absorption into a massive $\tilde\nu$ with $m_{\tilde\nu}=m_\chi+\Delta_\chi$. For a massless $\tilde\nu$, $E_{\rm out}=|\boldsymbol{p}-\boldsymbol{q}|$, which for $v\to0$ is q, and $\omega=m_\chi-q$, Eq. (8.14). With a free nucleus of mass M as the target, $\omega=q^2/2M$ and $q=m_\chi-q^2/2M$, so $q=m_\chi(1-m_\chi/2M+\dots)$ and the recoil energy is $m_\chi^2/2M$ to first order in m_χ/M.

D.25 Master rate formulas at finite temperature

The rates of Sect. 8.2 follow from the golden rule with the interaction of Eq. (8.6) and a target in thermal equilibrium. We derive the scattering rate with its normalization per unit target mass and the index order of the contraction. For absorption we derive the rate from a quantized and from a classical field, and the in-medium mixing of an absorbed dark photon. The last step relates the two directions of a dark-matter transition by detailed balance. Natural units are used; $\boldsymbol{q}$ is the momentum the dark matter loses and ω the energy the target gains.

D.25.1 Scattering

The dark matter is a plane wave of momentum $\boldsymbol{p}=m_\chi\boldsymbol{v}$ in the box of volume $\mathcal{V}$ (Appendix A.3), in the spin state $|s\rangle$, one of $2|\boldsymbol{S}_\chi|+1$. The target is in the eigenstate $|i\rangle$ with the Gibbs weight p_i. By Eq. (8.6), the matrix element to the dark-matter state $\boldsymbol{p}'=\boldsymbol{p}-\boldsymbol{q}$, $|s'\rangle$ and the target state $|f\rangle$ is

$$\mathcal{A}_{fi}=\frac{1}{\mathcal{V}}\sum_b\langle s'|\tilde V_b(\boldsymbol{q},\bar{\boldsymbol{v}}_\chi)|s\rangle\,\langle f|O_b(\boldsymbol{q})|i\rangle.$$

Energy conservation between the initial energy $E_i+E(\boldsymbol{p})$ and the final energy $E_f+E(\boldsymbol{p}')$ is $\delta(\omega_{\boldsymbol{q}}-E_f+E_i)$, with $\omega_{\boldsymbol{q}}=E(\boldsymbol{p})-E(\boldsymbol{p}')$ the energy of Eq. (1.22), mass splitting included. The golden rule Eq. (2.5), averaged over the initial states as in Eq. (2.64), and averaged over s and summed over s', gives

$$\Gamma=\sum_{\boldsymbol{p}'}\frac{2\pi}{2|\boldsymbol{S}_\chi|+1}\sum_{s,s'}\sum_{i,f}p_i\,|\mathcal{A}_{fi}|^2\,\delta(\omega_{\boldsymbol{q}}-E_f+E_i).$$

Squaring the sum over b produces the pairs (a,b), with the conjugate of the term a:

$$|\mathcal{A}_{fi}|^2 = \frac{1}{\mathscr{V}^2}\sum_{a,b}\langle s'|\tilde{V}_a|s\rangle^*\langle s'|\tilde{V}_b|s\rangle\ \langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle.$$

Since $\langle s'|\tilde{V}_a|s\rangle^* = \langle s|\tilde{V}_a^\dagger|s'\rangle$, the sum over s' is a completeness relation and the sum over s a trace:

$$\frac{1}{2|\boldsymbol{S}_\chi|+1}\sum_{s,s'}\langle s|\tilde{V}_a^\dagger|s'\rangle\langle s'|\tilde{V}_b|s\rangle = \frac{\mathrm{Tr}_\chi[\tilde{V}_a^\dagger\tilde{V}_b]}{2|\boldsymbol{S}_\chi|+1} = \mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v}),$$

the tensor Eq. (8.7). It depends on $\boldsymbol{v}$ and $\boldsymbol{q}$ through $\bar{\boldsymbol{v}}_\chi = \boldsymbol{v} - \boldsymbol{q}/2m_\chi$. The final momenta are $\boldsymbol{p}' = \boldsymbol{p} - \boldsymbol{q}$, and the sum over them is $\sum_{\boldsymbol{p}'} = \mathscr{V}\int \mathrm{d}^3q/(2\pi)^3$ (Eq. (2.10)). Collecting the factors,

$$\Gamma(\boldsymbol{v}) = \int \frac{\mathrm{d}^3q}{(2\pi)^3}\sum_{a,b}\mathfrak{D}_{ab}(\boldsymbol{q},\boldsymbol{v})\ \frac{2\pi}{\mathscr{V}}\sum_{i,f}p_i\,\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle\,\delta(\omega_{\boldsymbol{q}} - E_f + E_i),$$

and the last factor is $S_{ab}(\boldsymbol{q},\omega_{\boldsymbol{q}})$ of Eq. (4.13). This is Eq. (8.15). One power of $1/\mathscr{V}$ from the squared amplitude cancels against the sum over final momenta, the other is absorbed in S_{ab}, and the rate of one particle in the box is independent of $\mathscr{V}$.

The box holds $\rho_\chi\mathscr{V}/m_\chi$ dark-matter particles with velocities distributed as $f_\chi(\boldsymbol{v})$, and a target mass $\rho_T\mathscr{V}$. The rate per unit target mass is then $(\rho_\chi/m_\chi\rho_T)\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\,\Gamma(\boldsymbol{v})$. Inserting $1 = \int \mathrm{d}\omega\,\delta(\omega - \omega_{\boldsymbol{q}})$ under the $\boldsymbol{q}$ integral and replacing $\omega_{\boldsymbol{q}}$ by ω in S_{ab} gives the spectrum Eq. (8.16). For a single density row, $\mathfrak{D}_{n_en_e} = |\tilde{V}|^2$, it is Eq. (A.43), and with the angular step of Eq. (A.42) it is Eq. (1.47).

D.25.2 Index order of the contraction

The squared amplitude pairs $\tilde{V}_a^\dagger$ with $O_a^\dagger$, so the contraction is $\sum_{ab}\mathfrak{D}_{ab}S_{ab} = \mathrm{Tr}[\mathfrak{D}S^{\mathrm{T}}]$. Both matrices are Hermitian. Write each as a real symmetric part plus i times a real antisymmetric part, $\mathfrak{D} = \mathfrak{D}_{\mathrm{R}} + \mathrm{i}\mathfrak{D}_{\mathrm{I}}$ and $S = S_{\mathrm{R}} + \mathrm{i}S_{\mathrm{I}}$. Then $S^{\mathrm{T}} = S_{\mathrm{R}} - \mathrm{i}S_{\mathrm{I}}$, and since the trace of a symmetric matrix times an antisymmetric one vanishes,

$$\mathrm{Tr}[\mathfrak{D}S^{\mathrm{T}}] = \mathrm{Tr}[\mathfrak{D}_{\mathrm{R}}S_{\mathrm{R}}] + \mathrm{Tr}[\mathfrak{D}_{\mathrm{I}}S_{\mathrm{I}}], \qquad \mathrm{Tr}[\mathfrak{D}S] = \mathrm{Tr}[\mathfrak{D}_{\mathrm{R}}S_{\mathrm{R}}] - \mathrm{Tr}[\mathfrak{D}_{\mathrm{I}}S_{\mathrm{I}}].$$

The two orders differ by $2\,\mathrm{Tr}[\mathfrak{D}_{\mathrm{I}}S_{\mathrm{I}}]$, which vanishes when either matrix is real. Both forms are real. Only $\mathrm{Tr}[\mathfrak{D}S^{\mathrm{T}}]$ is the rate, and only it is guaranteed nonnegative: $S^{\mathrm{T}} = S^*$ is positive semidefinite when S is, and the trace of a product of two positive semidefinite matrices is nonnegative.

D.25.3 Absorption from a quantized field

Quantize a real scalar field in the box,

$$\phi(\boldsymbol{x}) = \sum_{\boldsymbol{k}}\frac{1}{\sqrt{2\omega_{\boldsymbol{k}}\mathscr{V}}}(a_{\boldsymbol{k}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}} + a_{\boldsymbol{k}}^\dagger\,\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}), \qquad \omega_{\boldsymbol{k}} = (m_\phi^2 + k^2)^{1/2},$$

with $[a_{\boldsymbol{k}}, a_{\boldsymbol{k}'}^\dagger] = \delta_{\boldsymbol{k}\boldsymbol{k}'}$; a vector field carries in addition a polarization vector for each mode. The coupling integrates the field against target bilinears, and the plane wave $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}$ turns each bilinear into the row at $\boldsymbol{q} = \boldsymbol{k}$ (Sect. 8.1.5); for the density, $\int \mathrm{d}^3x\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}\hat{\boldsymbol{n}}_e(\boldsymbol{x}) = n_e(\boldsymbol{k})$. With g_a^ϕ the coefficient of row a for the mode $\boldsymbol{k}$, derivatives of the field included,

$$H_{\mathrm{int}} = \sum_{\boldsymbol{k}}\frac{1}{\sqrt{2\omega_{\boldsymbol{k}}\mathscr{V}}}\Big[a_{\boldsymbol{k}}\sum_a g_a^\phi\,O_a(\boldsymbol{k}) + a_{\boldsymbol{k}}^\dagger\sum_a g_a^{\phi\,*}\,O_a(\boldsymbol{k})^\dagger\Big].$$

From the state with N_ϕ quanta in mode $\boldsymbol{k}$ and the target in $|i\rangle$, absorption leads to $N_\phi-1$ quanta and $|f\rangle$, with the matrix element $\sqrt{N_\phi/2\omega_{\boldsymbol{k}}\mathcal{V}}\,\sum_b g_b^\phi\langle f|O_b(\boldsymbol{k})|i\rangle$ and energy conservation $\delta(\omega_{\boldsymbol{k}}-E_f+E_i)$. The golden rule, thermally averaged and summed over f, gives

$$\Gamma_{N_\phi\to N_\phi-1}=\frac{N_\phi}{2\omega_{\boldsymbol{k}}}\sum_{a,b}g_a^{\phi\,*}g_b^\phi\,\frac{2\pi}{\mathcal{V}}\sum_{i,f}p_i\,\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle\,\delta(\omega_{\boldsymbol{k}}-E_f+E_i)=N_\phi\,\Gamma_{\rm abs},$$

with $\Gamma_{\rm abs}$ of Eq. (8.19) at $\boldsymbol{p}_\phi=\boldsymbol{k}$. The rate is N_ϕ times the rate per quantum.

Emission leads to $N_\phi+1$ quanta, with the matrix element $\sqrt{(N_\phi+1)/2\omega_{\boldsymbol{k}}\mathcal{V}}\,\sum_b g_b^{\phi\,*}\langle f|O_b^\dagger|i\rangle$ and $\delta(\omega_{\boldsymbol{k}}+E_f-E_i)$. Exchanging the names of i and f, as in the proof of Eq. (2.68), the weight becomes $p_f=p_i\,\mathrm{e}^{-\beta\omega_{\boldsymbol{k}}}$ and the squared element becomes $|\sum_b g_b^\phi\langle f|O_b|i\rangle|^2$, so that $\Gamma_{N_\phi\to N_\phi+1}=(N_\phi+1)\,\mathrm{e}^{-\beta\omega_{\boldsymbol{k}}}\Gamma_{\rm abs}$. The field loses quanta at the net rate

$$\Gamma_{N_\phi\to N_\phi-1}-\Gamma_{N_\phi\to N_\phi+1}=N_\phi\,\Gamma_{\rm abs}\Big[1-\Big(1+\frac{1}{N_\phi}\Big)\mathrm{e}^{-\beta\omega_{\boldsymbol{k}}}\Big].$$

The term $1/N_\phi$ is spontaneous emission. For a halo field it is negligible, and per quantum the net rate is $\Gamma_{\rm net}=(1-\mathrm{e}^{-\beta\omega_{\boldsymbol{k}}})\Gamma_{\rm abs}$. By the fluctuation–dissipation theorem Eq. (4.18), with $x_a=g_a^\phi$,

$$\begin{aligned}(1-\mathrm{e}^{-\beta\omega})\sum_{a,b}x_a^*x_bS_{ab}&=\mathrm{i}\sum_{a,b}x_a^*x_b(R_{ab}-R_{ba}^*)\\&=\mathrm{i}\Big[\sum_{a,b}x_a^*x_bR_{ab}-\Big(\sum_{a,b}x_a^*x_bR_{ab}\Big)^*\Big]=-2\,\mathrm{Im}\sum_{a,b}x_a^*x_bR_{ab},\end{aligned}$$

where the middle step relabels $a\leftrightarrow b$ in the term with R_{ba}^*. Dividing by $2\omega_{\boldsymbol{k}}$ gives the second form of Eq. (8.19). The modes are occupied with N_ϕ quanta, and their number density $\sum_{\boldsymbol{k}}N_\phi/\mathcal{V}$ is ρ_χ/m_ϕ up to corrections of order v^2, distributed in $\boldsymbol{k}=m_\phi\boldsymbol{v}$ as f_χ. The event rate per unit target mass is $(\rho_\chi/m_\phi\rho_T)\langle\Gamma_{\rm abs}\rangle$.

D.25.4 Absorption from a classical field

For $N_\phi\gg1$ the field can be treated as classical. Write it as a sum of modes with complex amplitudes,

$$\phi(\boldsymbol{x},t)=\sum_{\boldsymbol{k}}\big[\phi_{\boldsymbol{k}}\,\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega_{\boldsymbol{k}}t)}+\phi_{\boldsymbol{k}}^*\,\mathrm{e}^{-\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega_{\boldsymbol{k}}t)}\big].$$

A virialized halo is a superposition of many such waves with independent, uniformly distributed phases. By the central limit theorem the field at any point is then a Gaussian random variable, and all its correlations follow from $\langle\phi_{\boldsymbol{k}}\phi_{\boldsymbol{k}'}^*\rangle=\delta_{\boldsymbol{k}\boldsymbol{k}'}\langle|\phi_{\boldsymbol{k}}|^2\rangle$, with $\langle\phi_{\boldsymbol{k}}\phi_{\boldsymbol{k}'}\rangle=0$. The energy density, $\frac12\dot\phi^2+\frac12(\nabla\phi)^2+\frac12m_\phi^2\phi^2$, has the average $\sum_{\boldsymbol{k}}(\omega_{\boldsymbol{k}}^2+k^2+m_\phi^2)\langle|\phi_{\boldsymbol{k}}|^2\rangle=\sum_{\boldsymbol{k}}2\omega_{\boldsymbol{k}}^2\langle|\phi_{\boldsymbol{k}}|^2\rangle$. The number density of quanta in mode $\boldsymbol{k}$ is its energy density divided by $\omega_{\boldsymbol{k}}$,

$$n_{\boldsymbol{k}}=2\omega_{\boldsymbol{k}}\,\langle|\phi_{\boldsymbol{k}}|^2\rangle,\qquad\sum_{\boldsymbol{k}}n_{\boldsymbol{k}}\simeq\frac{\rho_\chi}{m_\phi}.$$

For a single mode, $\phi=\phi_0\cos(m_\phi t)$ has $|\phi_{\boldsymbol{0}}|=\phi_0/2$ and $\rho_\chi=\frac12m_\phi^2\phi_0^2$, as in Sect. 2.1.

The interaction is now a sum of harmonic perturbations,

$$H'(t)=\sum_{\boldsymbol{k}}\big[\mathrm{e}^{-\mathrm{i}\omega_{\boldsymbol{k}}t}\,\phi_{\boldsymbol{k}}\,\hat O_{\boldsymbol{k}}+\mathrm{e}^{\mathrm{i}\omega_{\boldsymbol{k}}t}\,\phi_{\boldsymbol{k}}^*\,\hat O_{\boldsymbol{k}}^\dagger\big],\qquad\hat O_{\boldsymbol{k}}=\sum_a g_a^\phi\,O_a(\boldsymbol{k}).$$

The first-order amplitude of Sect. 2.1 is a sum over modes, and its square contains products $\phi_{\boldsymbol{k}}\phi_{\boldsymbol{k}'}^*$. The Gaussian field average sets those with $\boldsymbol{k}\neq\boldsymbol{k}'$ to zero; a single realization does

the same in time, once the observation time exceeds the inverse spread of the frequencies $\omega_{\boldsymbol{k}}$, which is the coherence time $t_{\rm coh}$. What remains is the sum over modes of the harmonic rate Eq. (2.8), with $\hat{O} = \phi_{\boldsymbol{k}}\hat{O}_{\boldsymbol{k}}$. Summed over f and thermally averaged, the absorption term gives the rate in the box

$$\sum_{\boldsymbol{k}} 2\pi \,\langle|\phi_{\boldsymbol{k}}|^2\rangle \sum_{i,f} p_i \,|\langle f|\hat{O}_{\boldsymbol{k}}|i\rangle|^2 \,\delta(\omega_{\boldsymbol{k}} - E_f + E_i) = \mathscr{V} \sum_{\boldsymbol{k}} \frac{n_{\boldsymbol{k}}}{2\omega_{\boldsymbol{k}}} \sum_{a,b} g_a^{\phi\,*} g_b^{\phi}\, S_{ab}(\boldsymbol{k},\omega_{\boldsymbol{k}})$$
$$= \mathscr{V} \sum_{\boldsymbol{k}} n_{\boldsymbol{k}}\, \Gamma_{\rm abs}(\boldsymbol{k}).$$

Each quantum per unit volume is absorbed at the rate $\Gamma_{\rm abs}$, as in the quantized calculation. The emission term gives $\mathrm{e}^{-\beta\omega_{\boldsymbol{k}}}$ times the same, with no spontaneous part. The two descriptions agree for $N_\phi \gg 1$. The classical one shows what the golden rule needs: $t \gg t_{\rm coh}$, so that the modes add incoherently, and t long compared with the inverse width of the target's response, so that Eq. (2.8) holds for each mode. With $t_{\rm coh} \sim 1/(m_\phi v_0^2)$ this is about 1 ns at $m_\phi = 1\,\mathrm{eV}$ and grows as $1/m_\phi$.

D.25.5 In-medium mixing of the dark photon

A dark photon A' couples to the charges of the target with ζe in place of e, so the charges respond to the vector potential $\zeta\boldsymbol{A'} + \boldsymbol{A}$, where $\boldsymbol{A}$ is the ordinary electromagnetic field that the target itself produces. Take a transverse mode of frequency ω and wavevector $\boldsymbol{k}$ with $\omega^2 - k^2 = m_\phi^2$. The induced current is fixed by the photon self-energy of Sect. 4.4, $\boldsymbol{J} = -\Pi_{\rm em}(\zeta\boldsymbol{A'} + \boldsymbol{A})$, the relation whose homogeneous case is the medium's wave equation $(\omega^2 - k^2 - \Pi_{\rm em})\boldsymbol{A} = 0$. The current sources the ordinary field through $(k^2 - \omega^2)\boldsymbol{A} = \boldsymbol{J}$, so that $\boldsymbol{A} = -\boldsymbol{J}/m_\phi^2$ at this frequency and wavevector. Eliminating $\boldsymbol{A}$,

$$\boldsymbol{J} = -\zeta\,\frac{m_\phi^2\,\Pi_{\rm em}}{m_\phi^2 - \Pi_{\rm em}}\,\boldsymbol{A'}.$$

The dark photon therefore sees the self-energy $\zeta^2 m_\phi^2\Pi_{\rm em}/(m_\phi^2 - \Pi_{\rm em})$ instead of $\zeta^2\Pi_{\rm em}$, and its absorption rate is $-1/\omega$ times the imaginary part. Since $\mathrm{Im}[\Pi_{\rm em}(m_\phi^2 - \Pi_{\rm em}^*)] = m_\phi^2\,\mathrm{Im}\,\Pi_{\rm em}$,

$$\Gamma_{\rm net} = -\frac{\zeta^2}{\omega}\,\frac{m_\phi^4\,\mathrm{Im}\,\Pi_{\rm em}}{|m_\phi^2 - \Pi_{\rm em}|^2} = \zeta_{\rm eff}^2\,\sigma_1(\omega),$$

with $\mathrm{Im}\,\Pi_{\rm em} = -\omega\sigma_1$ from $\Pi_{\rm em} = -\mathrm{i}\omega\sigma_{\rm cond}$ and $\zeta_{\rm eff}$ of Sect. 1.4 [65, Eqs. (5) and (6), 66, Eqs. (20)–(22)]. To first order in $\Pi_{\rm em}/m_\phi^2$, the factor is one and the rate is $\zeta^2\sigma_1$, the result of the proper response alone. The photon-reducible part matters when m_ϕ is comparable to the plasma frequency, 16.6 eV in silicon (Sect. 4.3).

D.25.6 Detailed balance for the dark matter

Let the dark matter go from $\boldsymbol{p}$ to one final momentum $\boldsymbol{p'}$, depositing $\omega = E(\boldsymbol{p}) - E(\boldsymbol{p'})$, where the two states may belong to different mass eigenstates χ_1 and χ_2 of equal spin. By the golden rule and the steps that led to Eq. (8.15), the rate into $\boldsymbol{p'}$ is

$$\Gamma(\boldsymbol{p} \to \boldsymbol{p'}) = \frac{2\pi}{\mathscr{V}^2} \sum_{a,b} \mathfrak{D}_{ab} \sum_{i,f} p_i \,\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle \,\delta(\omega - E_f + E_i).$$

The reverse transition, from $\boldsymbol{p'}$ and the target state $|f\rangle$ to $\boldsymbol{p}$ and $|i\rangle$, has the matrix element of the Hermitian interaction between the exchanged states, the complex conjugate of the

forward one:

$$\langle \boldsymbol{p}, s; i|H_{\rm int}|\boldsymbol{p}', s'; f\rangle = \frac{1}{\mathscr{V}}\sum_b \langle s|\tilde{V}_b^\dagger|s'\rangle\,\langle i|O_b^\dagger|f\rangle.$$

Squaring, averaging over s' and summing over s gives

$$\sum_{a,b}\frac{\mathrm{Tr}_\chi[\tilde{V}_a\tilde{V}_b^\dagger]}{2|\boldsymbol{S}_\chi|+1}\,\langle f|O_a|i\rangle\langle i|O_b^\dagger|f\rangle,$$

and the cyclic trace is $\mathrm{Tr}_\chi[\tilde{V}_b^\dagger\tilde{V}_a] = (2|\boldsymbol{S}_\chi|+1)\,\mathfrak{D}_{ba}$. The initial target state is now $|f\rangle$, with weight p_f, and energy conservation is $\delta(-\omega - E_i + E_f)$, the same delta function. Relabeling $a \leftrightarrow b$,

$$\Gamma(\boldsymbol{p}' \to \boldsymbol{p}) = \frac{2\pi}{\mathscr{V}^2}\sum_{a,b}\mathfrak{D}_{ab}\sum_{i,f} p_f\,\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle\,\delta(\omega - E_f + E_i),$$

and on the support of the delta function $p_f = p_i\,\mathrm{e}^{-\beta\omega}$. This is Eq. (8.20). Neither the target's symmetries nor the form of the coupling entered; only the Hermiticity of the interaction and the Gibbs weights did. If the dark-matter states had different spins, the ratio would carry in addition the ratio of their multiplicities from the spin averages.

D.26 Moment matrices and Ward identities

This section derives the relations of Sect. 8.3 beyond the first moment. The species-resolved f-sum rule, Eq. (4.34), and the algebra behind it are those of Appendix D.6, which we use without repeating. The Hamiltonian is that of electrons and nuclei, $H = \sum_i \boldsymbol{p}_i^2/2m_i + U$, with U a function of the positions and spins of all particles; species are labeled s, particles i, j, and nuclei of species κ also I, with positions $\boldsymbol{X}_I$.

D.26.1 Moments as commutator brackets

For operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$ and a state diagonal in the energy basis with weights p_i, insert a complete set $|f\rangle$ and use $\langle f|[H,X]_k|i\rangle = \omega_{fi}^k\langle f|X|i\rangle$, $\omega_{fi} = E_f - E_i$:

$$\frac{1}{\mathscr{V}}\langle O_a^\dagger[H,O_b]_k\rangle = \int\frac{\mathrm{d}\omega}{2\pi}\,\omega^k S_{ab}(\boldsymbol{q},\omega), \qquad \frac{1}{\mathscr{V}}\langle [H,O_b]_k O_a^\dagger\rangle = (-1)^k\int\frac{\mathrm{d}\omega}{2\pi}\,\omega^k S_{ba}(-\boldsymbol{q},\omega),$$

the second after relabeling $i \leftrightarrow f$, which turns ω_{fi}^k into $(-\omega)^k$ and the matrix elements into those of $S_{ba}(-\boldsymbol{q},\omega)$ (Sect. 4.3). The difference of the two lines is the commutator, the sum the anticommutator, and $\int\omega^k[S_{ab}(\boldsymbol{q}) + S_{ba}(-\boldsymbol{q})]$ is the commutator for odd k and the anticommutator for even k: Eq. (8.22). The same expansion as in Eq. (4.11), carried to higher order, makes the coefficient of $z^{-(n+1)}$ in $R_{ab}(\boldsymbol{q},z)$ equal to $\mathscr{V}^{-1}\langle[O_a^\dagger,[H,O_b]_n]\rangle$ for every n; for odd n this is $\mathscr{M}_{ab}^n$.

In a stationary state the average of $[H,Y]$ vanishes for any Y, so by the Jacobi identity $\langle[X,[H,Y]]\rangle = \langle[[X,H],Y]\rangle$. For the third moment this moves one commutator to the left,

$$\mathscr{M}_{ab}^3 = \frac{1}{\mathscr{V}}\big\langle\,[[O_a^\dagger,H],\,[H,[H,O_b]]]\,\big\rangle. \tag{D.152}$$

D.26.2 Commutators of a density

For one particle of mass m, write $A = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$. The shift $\boldsymbol{p}A = A(\boldsymbol{p}+\boldsymbol{q})$ (Appendix D.6) gives, for any function $h(\boldsymbol{p})$,

$$\Big[\frac{\boldsymbol{p}^2}{2m},\,A\,h(\boldsymbol{p})\Big] = A\,f(\boldsymbol{p})\,h(\boldsymbol{p}), \qquad f(\boldsymbol{p}) = \frac{(\boldsymbol{p}+\boldsymbol{q})^2-\boldsymbol{p}^2}{2m} = \frac{\boldsymbol{q}\cdot\boldsymbol{p}+q^2/2}{m}, \tag{D.153}$$

where f is the energy a free particle of momentum $\boldsymbol{p}$ gains from the kick $\boldsymbol{q}$. With $h=1$ this is the continuity equation, $[H,A]=Af=\frac{1}{2}\{\boldsymbol{q}\cdot\boldsymbol{p}/m,A\}$. Applied once more, with the potential energy, which commutes with A but not with f,

$$[H,Af]=Af^2+\frac{\mathrm{i}}{m}\,A\,\boldsymbol{q}\cdot\nabla U, \tag{D.154}$$

since $[U,\boldsymbol{q}\cdot\boldsymbol{p}]=\mathrm{i}\boldsymbol{q}\cdot\nabla U$. Summed over the particles of species s, $[H,n_s]=\boldsymbol{q}\cdot\boldsymbol{j}_s=\sum_i A_i f_i$, and $[H,[H,n_s]]=\sum_i[A_i f_i^2+(\mathrm{i}/m_s)A_i\boldsymbol{q}\cdot\nabla_i U]$. The adjoint is $(Af)^\dagger=A^\dagger g(\boldsymbol{p})$, with $g(\boldsymbol{p})=(\boldsymbol{q}\cdot\boldsymbol{p}-q^2/2)/m$, and $[n_s^\dagger,H]=\sum_i A_i^\dagger g_i=\boldsymbol{q}\cdot\boldsymbol{j}_s^\dagger$.

D.26.3 Third moments

By Eq. (D.152), $\mathscr{M}^3_{ss'}=\mathscr{V}^{-1}\langle[\sum_{i\in s}A_i^\dagger g_i,\,\sum_{j\in s'}(A_j f_j^2+(\mathrm{i}/m_{s'})A_j\boldsymbol{q}\cdot\nabla_j U)]\rangle$. Three kinds of terms survive.

Same particle, kinetic. Using $A^\dagger g(\boldsymbol{p})A=g(\boldsymbol{p}+\boldsymbol{q})=f(\boldsymbol{p})$ and $Af(\boldsymbol{p})A^\dagger=f(\boldsymbol{p}-\boldsymbol{q})=g(\boldsymbol{p})$,

$$[A^\dagger g,Af^2]=f^3-g^3=(f-g)(f^2+fg+g^2)=\frac{q^2}{m}\Big[\frac{3(\boldsymbol{q}\cdot\boldsymbol{p})^2}{m^2}+\Big(\frac{q^2}{2m}\Big)^2\Big].$$

Same particle, potential. $[A^\dagger g,(\mathrm{i}/m)A\,\boldsymbol{q}\cdot\nabla U]=(\mathrm{i}/m)\{f\,\boldsymbol{q}\cdot\nabla U-\boldsymbol{q}\cdot\nabla U\,g\}=(\mathrm{i}/m)\{[f,\boldsymbol{q}\cdot\nabla U]+\boldsymbol{q}\cdot\nabla U(f-g)\}$. The commutator is $-(\mathrm{i}/m)(\boldsymbol{q}\cdot\nabla)^2U$, and the last term, $(\mathrm{i}q^2/m^2)\boldsymbol{q}\cdot\nabla U$, has zero average, because $\nabla_i U=-\dot{\boldsymbol{p}}_i$ and $\langle\dot{\boldsymbol{p}}_i\rangle=0$ in a stationary state. What remains is $m^{-2}(\boldsymbol{q}\cdot\nabla_i)^2U$.

Different particles. Kinetic terms of different particles commute. In the potential term, g_i fails to commute only with the dependence of $\nabla_j U$ on $\boldsymbol{r}_i$: $[A_i^\dagger g_i,(\mathrm{i}/m_{s'})A_j\boldsymbol{q}\cdot\nabla_j U]=(m_s m_{s'})^{-1}A_i^\dagger A_j(\boldsymbol{q}\cdot\nabla_i)(\boldsymbol{q}\cdot\nabla_j)U$.

Collecting, with $A_i^\dagger A_j=\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}_i-\boldsymbol{r}_j)}$ and the same-particle potential term included as $i=j$,

$$\mathscr{M}^3_{ss'}=\delta_{ss'}\,\frac{\boldsymbol{n}_s q^2}{m_s}\Big[\Big(\frac{q^2}{2m_s}\Big)^2+\frac{3q^2\langle(\hat{\boldsymbol{q}}\cdot\boldsymbol{p})^2\rangle_s}{m_s^2}\Big]+\frac{1}{m_s m_{s'}\mathscr{V}}\Big\langle\sum_{i\in s}\sum_{j\in s'}\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}_i-\boldsymbol{r}_j)}(\boldsymbol{q}\cdot\nabla_i)(\boldsymbol{q}\cdot\nabla_j)U\Big\rangle, \tag{D.155}$$

which is Eq. (8.27) for $s=s'$. For $s=e$ and $s'=\kappa$ only the part of U that depends on both an electron and a nucleus survives, $\sum_{j,I}w_\kappa(\boldsymbol{r}_j-\boldsymbol{X}_I)$ with $w_\kappa(\boldsymbol{y})=-Z_\kappa\alpha/y$, for which $(\boldsymbol{q}\cdot\nabla_j)(\boldsymbol{q}\cdot\nabla_I)w_\kappa=-(\boldsymbol{q}\cdot\nabla)^2w_\kappa$. Defining the conditional density by $N_\kappa\boldsymbol{n}_e^{(\kappa)}(\boldsymbol{y})=\langle\sum_j\sum_{I\in\kappa}\delta^3(\boldsymbol{r}_j-\boldsymbol{X}_I-\boldsymbol{y})\rangle$, with N_κ the number of nuclei of species κ, the average becomes an integral over $\boldsymbol{y}$, and this is Eq. (8.24). A spring $U=\frac{1}{2}K(\boldsymbol{r}-\boldsymbol{X})^2$ between one electron and one nucleus has $(\boldsymbol{q}\cdot\nabla_r)(\boldsymbol{q}\cdot\nabla_X)U=-Kq^2$, which gives $\mathscr{M}^3_{e\kappa}=-Kq^2\langle\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{X})}\rangle/(m_eM_\kappa\mathscr{V})\to-Kq^2/(m_eM_\kappa\mathscr{V})$ at small q. The same number follows from the spectrum. In the relative mode, at $\omega_\mathrm{r}^2=K/\mu$ with $\mu=m_eM_\kappa/(m_e+M_\kappa)$, the displacements are $M_\kappa/(m_e+M_\kappa)$ and $-m_e/(m_e+M_\kappa)$ times the relative one, whose squared transition amplitude is $1/2\mu\omega_\mathrm{r}$. The cross weight $-q^2[m_eM_\kappa/(m_e+M_\kappa)^2]/2\mu\omega_\mathrm{r}$, times $2\omega_\mathrm{r}^3/\mathscr{V}$ for the two terms of Eq. (8.22), is $-Kq^2/(m_eM_\kappa\mathscr{V})$; the center-of-mass mode, in which the pair recoils as a whole with $\omega=q^2/2(m_e+M_\kappa)$, contributes at order q^6.

D.26.4 The mixed moment in a crystal

In Fourier space, $(\boldsymbol{q}\cdot\nabla)^2w_\kappa$ has the transform $-(\boldsymbol{q}\cdot\boldsymbol{k})^2\tilde{w}_\kappa(\boldsymbol{k})$, with $\tilde{w}_\kappa(\boldsymbol{k})=-Z_\kappa e^2/k^2$. For a static crystal, $\boldsymbol{n}_e^{(\kappa)}(\boldsymbol{y})=\sum_{\boldsymbol{G}}\boldsymbol{n}_{\boldsymbol{G}}^{(\kappa)}\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{y}}$ with $\boldsymbol{n}_{\boldsymbol{0}}^{(\kappa)}=\boldsymbol{n}_e$ and $\sum_{\boldsymbol{G}}\boldsymbol{n}_{\boldsymbol{G}}^{(\kappa)}=\boldsymbol{n}_e^{(\kappa)}$, and the integral in Eq. (8.24) picks $\boldsymbol{k}=\boldsymbol{q}-\boldsymbol{G}$:

$$\mathscr{M}^3_{e\kappa}(\boldsymbol{q})=-\frac{Z_\kappa e^2\,\boldsymbol{n}_\kappa}{m_eM_\kappa}\sum_{\boldsymbol{G}}\boldsymbol{n}_{\boldsymbol{G}}^{(\kappa)}\,\frac{[\boldsymbol{q}\cdot(\boldsymbol{q}-\boldsymbol{G})]^2}{|\boldsymbol{q}-\boldsymbol{G}|^2}. \tag{D.156}$$

The term $\boldsymbol{G}=0$ is exactly $\boldsymbol{n}_e q^2$ at every q; it does not vanish faster than q^2 as $q\to 0$, because the Coulomb transform $1/k^2$ is evaluated at $k=q$. For $\boldsymbol{G}\neq 0$ and $q\ll G$, the weight is $(\boldsymbol{q}\cdot\hat{\boldsymbol{G}})^2[1+O(q/G)]$. On a site of cubic symmetry the coefficients are equal over each star of $\boldsymbol{G}$, and the average of $(\boldsymbol{q}\cdot\hat{\boldsymbol{G}})^2$ over a cubic star is $q^2/3$. The sum is then $q^2[\boldsymbol{n}_e+(\boldsymbol{n}_e^{(\kappa)}-\boldsymbol{n}_e)/3]$, which is Eq. (8.25). The correction of first order in q/G is odd in $\boldsymbol{G}$; it cancels between $\boldsymbol{G}$ and $-\boldsymbol{G}$ when the site has inversion symmetry, $\boldsymbol{n}^{(\kappa)}_{-\boldsymbol{G}}=\boldsymbol{n}^{(\kappa)}_{\boldsymbol{G}}$, and survives otherwise. Since the density is real, $\boldsymbol{n}^{(\kappa)}_{-\boldsymbol{G}}=(\boldsymbol{n}^{(\kappa)}_{\boldsymbol{G}})^*$, and the surviving odd part is $\mathrm{i}\,\mathrm{Im}\,\boldsymbol{n}^{(\kappa)}_{\boldsymbol{G}}$: the first-order correction is purely imaginary and enters $\mathrm{Im}\,\mathscr{M}^3_{e\kappa}$ alone, while the real part is corrected at order $(q/G)^2$ at every cubic site. When q exceeds G for some stars, their weight tends to q^2 instead of $q^2/3$, which is the middle regime of Sect. 8.3.3.

For a density with spherical symmetry about the nucleus the angular integral can be done in real space. Writing $(\boldsymbol{q}\cdot\nabla)^2(1/y)=-(4\pi/3)q^2\delta^3(\boldsymbol{y})+2q^2P_2(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{y}})/y^3$, with P_2 the Legendre polynomial, the angular integral of the plane wave against P_2 is $\int \mathrm{d}\Omega_y\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{y}}P_2(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{y}})=-4\pi j_2(qy)$. With $w_\kappa=-Z_\kappa e^2/(4\pi y)$, Eq. (8.24) becomes

$$\mathscr{M}^3_{e\kappa}=-\frac{\boldsymbol{n}_\kappa}{m_e M_\kappa}\Big(-\frac{Z_\kappa e^2}{4\pi}\Big)\Big[-\frac{4\pi}{3}q^2\bar{\boldsymbol{n}}_e^{(\kappa)}(0)-8\pi q^2\int_0^\infty \mathrm{d}y\,\frac{\bar{\boldsymbol{n}}_e^{(\kappa)}(y)\,j_2(qy)}{y}\Big],$$

which is Eq. (8.26). For a uniform density the integral is $\boldsymbol{n}_e\int_0^\infty \mathrm{d}x\,j_2(x)/x=\boldsymbol{n}_e/3$, which reproduces the $\boldsymbol{G}=0$ term of Eq. (D.156), $\boldsymbol{n}_e q^2$ in total. The same check runs in momentum space: the spherical form equals $\int\frac{\mathrm{d}^3p}{(2\pi)^3}A(p/q)\,\tilde{\boldsymbol{n}}(p)$ with $A(x)=\frac12\int_{-1}^1\mathrm{d}u\,(1-xu)^2/(1+x^2-2xu)$, which tends to 1 for structure much smoother than $1/q$ and to $1/3$ for structure much sharper.

D.26.5 Existence of the fourth moment

The even moment $\mathscr{M}^4_{ss}$ is $\mathscr{V}^{-1}\langle X^\dagger X+XX^\dagger\rangle$ with $X=[H,[H,n_s]]$, a sum of nonnegative terms. By Eq. (D.154), X contains $(\mathrm{i}/m_s)\sum_i A_i\boldsymbol{q}\cdot\nabla_i U$, whose square contains the mean square of $\boldsymbol{q}\cdot\nabla_i U$. For the Coulomb attraction of a nucleus, $|\nabla w_\kappa|^2=(Z_\kappa\alpha)^2/y^4$, and against an electron density that is finite at $y=0$ its average $\int 4\pi y^2\mathrm{d}y\,\boldsymbol{n}(y)\,(Z_\kappa\alpha)^2/y^4$ diverges at small y. The kinetic part of X, $\sum_i A_i f_i^2$, has a finite mean square, set by the average of p^4, which is finite even with the cusp of the wavefunction at the nucleus. By the triangle inequality for the norm $\langle X^\dagger X\rangle^{1/2}$ it cannot cancel the divergence, and $\mathscr{M}^4$ is infinite for both electrons and nuclei. The third moment involves $\nabla^2 w_\kappa$ only, which is a delta function at the nucleus, and stays finite. In the Born–Oppenheimer approximation U is replaced, for the nuclei, by a smooth surface, and the force on a nucleus is bounded.

D.26.6 Momentum flux

Let $\boldsymbol{g}(\boldsymbol{q})=\sum_i\frac12\{\boldsymbol{p}_i,A_i\}=\sum_i A_i(\boldsymbol{p}_i+\boldsymbol{q}/2)$. The kinetic part of its commutator with H follows from Eq. (D.153) with $h=p^\alpha+q^\alpha/2$:

$$\Big[\sum_j\frac{\boldsymbol{p}_j^2}{2m_j},\,g^\alpha\Big]=q_\beta\sum_i\frac{A_i}{m_i}\Big(p_i+\frac{q}{2}\Big)^\alpha\Big(p_i+\frac{q}{2}\Big)^\beta.$$

For a pair potential $U=\sum_{i<j}w_{ij}(\boldsymbol{r}_i-\boldsymbol{r}_j)$ the potential part is

$$\mathrm{i}\sum_i\partial_i^\alpha U\,A_i=\mathrm{i}\sum_{i<j}\partial^\alpha w_{ij}(\boldsymbol{r}_{ij})\,(A_i-A_j),$$

since the force of j on i is minus that of i on j. With $\boldsymbol{r}_{ij} = \boldsymbol{r}_i - \boldsymbol{r}_j$,

$$A_i - A_j = \mathrm{i} q_\beta r_{ij}^\beta \int_0^1 \mathrm{d}\lambda \, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}_j + \lambda \boldsymbol{r}_{ij})},$$

which contains the plane wave averaged along the segment between the two particles. Both parts carry a factor q_β, and

$$\mathsf{T}^{\alpha\beta}(\boldsymbol{q}) = \sum_i \frac{A_i}{m_i} \Big(p_i + \frac{q}{2}\Big)^\alpha \Big(p_i + \frac{q}{2}\Big)^\beta - \sum_{i<j} \partial^\alpha w_{ij}(\boldsymbol{r}_{ij})\, r_{ij}^\beta \int_0^1 \mathrm{d}\lambda \, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}_j + \lambda \boldsymbol{r}_{ij})} \tag{D.157}$$

is the momentum flux of Eq. (8.30): the momentum carried by the particles plus the stress transmitted along each pair. Two hypotheses enter. There must be no external potential, which would add a force $\sum_i \nabla_i V_{\mathrm{ext}}$ that no factor of q cancels. The interactions must be pairwise, or more generally translation invariant with forces that sum to zero; spin-dependent interactions that are functions of positions and spins satisfy this.

The soft acoustic limit follows in two steps. Between eigenstates, Eq. (8.28) gives $\omega_{fi}\langle f|n_{\mathrm{M}}|i\rangle = q_\alpha \langle f|g^\alpha|i\rangle$ and Eq. (8.30) gives $\omega_{fi}\langle f|g^\alpha|i\rangle = q_\beta \langle f|\mathsf{T}^{\alpha\beta}|i\rangle$, so for $\omega_{fi} \neq 0$

$$\langle f|n_{\mathrm{M}}(\boldsymbol{q})|i\rangle = \frac{q_\alpha q_\beta}{\omega_{fi}^2} \, \langle f|\mathsf{T}^{\alpha\beta}(\boldsymbol{q})|i\rangle.$$

Inserted into Eq. (4.13) in the second slot, or in both, this is Eq. (8.31). The matrix elements of $\mathsf{T}^{\alpha\beta}(\boldsymbol{q})$ between the ground state and a state of finite energy are finite as $q \to 0$, those of the total stress, so the mass-density amplitude to such a state is of order q^2.

D.26.7 The boost from momentum conservation

In a crystal of one element, $n_{\mathrm{M}} = m_e n_e + M_\kappa \sum_\kappa n_\kappa$. For a final state f at a finite energy above the ground state, the amplitude of n_{M} is of order q^2, while that of n_e to an electronic excitation is of order q (the dipole term). Dividing by M_κ gives Eq. (8.32). The same result follows from the superadiabatic amplitude Eq. (8.53) summed over all nuclei at small q, where $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I} \to 1$ inside the electronic factor. The adiabatic states move rigidly with a rigid translation of all nuclei, so $\sum_I \nabla_I \Psi_j = -\sum_i \nabla_i \Psi_j$, and the summed derivative coupling is $\sum_I \boldsymbol{d}_{j0}^I = -\langle \Psi_j | \sum_i \nabla_i | \Psi_0 \rangle = -\mathrm{i}\langle \Psi_j | \boldsymbol{P}_e | \Psi_0 \rangle$, with $\boldsymbol{P}_e$ the total electron momentum. Since $\boldsymbol{P}_e = \mathrm{i} m_e [H_e, \sum_i \boldsymbol{r}_i]$, $\langle \Psi_j | \boldsymbol{P}_e | \Psi_0 \rangle = \mathrm{i} m_e (E_j - E_0) \langle \Psi_j | \sum_i \boldsymbol{r}_i | \Psi_0 \rangle$. The summed amplitude Eq. (8.53) is then $-\mathrm{i}\boldsymbol{q} \cdot \sum_I \boldsymbol{d}_{j0}^I / [M_\kappa (E_j - E_0)] = -(m_e/M_\kappa) \langle \Psi_j | \mathrm{i}\boldsymbol{q} \cdot \sum_i \boldsymbol{r}_i | \Psi_0 \rangle$, which is $-(m_e/M_\kappa)\langle \Psi_j | n_e(\boldsymbol{q}) | \Psi_0 \rangle$ to first order in q. The two routes agree.

D.26.8 The electron third moment

Multiply Eq. (8.29), with $O_B = n_e$, by ω^3, add the term with $-\boldsymbol{q}$, and integrate. By Eq. (8.30) the right side is $q_\alpha q_\beta$ times the first moment of n_e and $\mathsf{T}^{\alpha\beta}$, $\mathscr{V}^{-1}\langle [n_e^\dagger, [H, \mathsf{T}^{\alpha\beta}]] \rangle$, and the outer commutator with $n_e^\dagger$ brings one more factor of q. Hence $\sum_{s'} m_{s'} \mathscr{M}^3_{es'} = O(q^3)$, and the terms of order q^2 of $m_e \mathscr{M}^3_{ee}$ and $\sum_\kappa M_\kappa \mathscr{M}^3_{e\kappa}$ must cancel.

We check this directly. In Eq. (D.155) with $s = s' = e$, the kinetic bracket is of order q^4. In the potential term, the electron–nucleus energy contributes only at $i = j$, $\sum_{i,I} (\boldsymbol{q}\cdot\nabla)^2 w_\kappa(\boldsymbol{r}_i - \boldsymbol{X}_I)$, whose average over a cubic crystal keeps the trace, $q^2 \nabla^2 w_\kappa / 3 = Z_\kappa e^2 q^2 \delta^3 / 3$, and gives $(e^2 q^2 / 3 m_e^2) \sum_\kappa Z_\kappa \boldsymbol{n}_\kappa \boldsymbol{n}_e^{(\kappa)}$. The electron–electron energy $\sum_{i<j} e^2/(4\pi r_{ij})$ contributes at $i = j$ and at $i \neq j$, which combine into $\sum_{i \neq j} (\boldsymbol{q}\cdot\nabla)^2 w_{ee}(\boldsymbol{r}_{ij}) [1 - \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_{ij}}]$. The bracket vanishes at $\boldsymbol{r}_{ij} = 0$, so the contact term and the short-range exchange–correlation hole contribute only at higher order in q. Against the uniform density $\boldsymbol{n}_e$, the term 1 gives

$n_e \int \mathrm{d}^3 r\, (\boldsymbol{q}\cdot\nabla)^2 w_{ee} = -e^2 q^2 n_e/3$, since the quadrupolar part averages out over angles and the contact part remains. The phase term gives $-n_e \int \mathrm{d}^3 r\, \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} (\boldsymbol{q}\cdot\nabla)^2 w_{ee} = n_e q^4 \tilde{w}_{ee}(q) = +e^2 q^2 n_e$, with $\tilde{w}_{ee} = e^2/q^2$. Together the electron pairs contribute $\frac{2}{3} e^2 q^2 n_e$ per electron. The periodic part of the density cancels between the two terms on cubic sites, star by star as in Eq. (D.156). Altogether

$$m_e \mathscr{M}^3_{ee} = \frac{e^2 q^2}{3 m_e} \Big[\sum_\kappa Z_\kappa n_\kappa n_e^{(\kappa)} + 2 n_e^2 \Big] + O(q^3),$$

which is Eq. (8.33). By Eq. (8.25) and neutrality, $\sum_\kappa M_\kappa \mathscr{M}^3_{e\kappa} = -(e^2 q^2/3m_e)[\sum_\kappa Z_\kappa n_\kappa n_e^{(\kappa)} + 2 n_e^2]$, and the sum vanishes at order q^2, as it must. In jellium the nuclei are a rigid uniform background with $\nabla^2 U_{\mathrm{bg}} = e^2 n_e$ at every electron, the contact sum becomes n_e^2, and $\mathscr{M}^3_{ee} = (n_e q^2/m_e)(e^2 n_e/m_e) = n_e q^2 \omega_p^2/m_e$ at small q, the third moment of the electron gas.

D.27 Screening for every external coupling

This section derives the screened response Eq. (8.35) for any pair of target rows, its total-charge and electron-density forms, the exactness theorem Eq. (8.38) in full, and the classification of Coulomb lines of Sect. 8.4.4. The notation is that of Sects. 4.4 and 8.4: $\boldsymbol{q}$ lies in the first zone, the reciprocal-lattice vectors $\boldsymbol{G}$ label the channel $\{\boldsymbol{q}+\boldsymbol{G}\}$, $v_{\boldsymbol{G}} \equiv v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)$, and the dark-matter vertex sits at the row $\boldsymbol{G}_q$ of the unreduced momentum (Eq. (A.13)).

D.27.1 The screened response

The Coulomb Hamiltonian Eq. (2.48) couples the species only through $n_Z = -n_e + \sum_\kappa Z_\kappa n_\kappa$, by the term $(2\mathscr{V})^{-1} \sum_{\boldsymbol{k}\neq 0} v_{\mathrm{C}}(k)\, n_Z(\boldsymbol{k})^\dagger n_Z(\boldsymbol{k})$ (the self-energies of the point charges aside). In the presence of a weak external field the induced charge-number density at $\boldsymbol{q}+\boldsymbol{G}$, $\delta n_{Z,\boldsymbol{G}} = \delta\langle n_Z(\boldsymbol{q}+\boldsymbol{G})^\dagger\rangle/\mathscr{V}$, adds to the Hamiltonian the mean-field term $v_{\boldsymbol{G}}\, \delta n_{Z,\boldsymbol{G}}\, n_Z(\boldsymbol{q}+\boldsymbol{G})$ plus its conjugate, which is a field of amplitude $v_{\boldsymbol{G}} \delta n_{Z,\boldsymbol{G}}$ on the row $n_Z(\boldsymbol{q}+\boldsymbol{G})$. The proper response of the channel is defined by Eq. (8.34) as the response to the external fields ϕ_b and to these Hartree fields together. Everything else that the Coulomb interaction does, including the Coulomb lines at momenta outside the channel and the fluctuations of the channel components beyond their mean, stays inside $\tilde{R}$. By the argument of Sect. 4.8.3, $\tilde{R}$ is the response of the Hamiltonian with the Coulomb terms of the channel removed, in the thermodynamic limit.

Write the induced charge as a column vector δn_Z over $\boldsymbol{G}$, the proper charge–charge block as the matrix $P_{\boldsymbol{G}\boldsymbol{G}'} = \tilde{R}_{Z_{\boldsymbol{G}} Z_{\boldsymbol{G}'}}$, and v as the diagonal matrix of $v_{\boldsymbol{G}}$. For a field on row b alone, Eq. (8.34) with $a = Z_{\boldsymbol{G}}$ reads

$$\delta n_Z = \tilde{R}_{Zb}\, \phi_b + P\, v\, \delta n_Z, \qquad \delta n_Z = (1 - Pv)^{-1} \tilde{R}_{Zb}\, \phi_b, \tag{D.158}$$

with $\tilde{R}_{Zb}$ the column of $\tilde{R}_{Z_{\boldsymbol{G}} b}$. Inserting this into Eq. (8.34) for a general row a,

$$\delta O_a = [\tilde{R}_{ab} + \tilde{R}_{aZ}\, v\, (1 - Pv)^{-1} \tilde{R}_{Zb}]\, \phi_b,$$

with $\tilde{R}_{aZ}$ the row vector of $\tilde{R}_{a Z_{\boldsymbol{G}}}$. The push-through identity $v(1-Pv)^{-1} = (1-vP)^{-1}v$, which follows from $(1-vP)v = v(1-Pv)$ by multiplying with the two inverses, turns the bracket into

$$R_{ab} = \tilde{R}_{ab} + \tilde{R}_{aZ}\, \varepsilon^{-1} v\, \tilde{R}_{Zb}, \qquad \varepsilon = 1 - vP, \tag{D.159}$$

which is Eq. (8.35) with $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\boldsymbol{G}} \tilde{R}_{Z_{\boldsymbol{G}} Z_{\boldsymbol{G}'}}$. The same steps give the Dyson form $R_{ab} = \tilde{R}_{ab} + \tilde{R}_{aZ}\, v\, R_{Zb}$, since $R_{Zb} = (1-Pv)^{-1}\tilde{R}_{Zb}$ by Eq. (D.158). For the electron density

alone, $a = b = e$ and $n_Z = -n_e$, this is Eq. (4.40) and Eq. (4.43), and the one-species algebra of Appendix D.7 is the special case. The matrix ε is invertible at every real ω except at the self-sustained longitudinal oscillations of the target, plasmons and longitudinal optical phonons, where R itself has its poles.

With a single $\boldsymbol{G}$, P and v are numbers, and Eq. (D.159) is Eq. (8.36). Written as a matrix over the rows a, b of the species, the screening term $v\tilde{R}_{aZ}\tilde{R}_{Zb}/\varepsilon$ is a rank-one correction of $\tilde{R}$, and Eq. (8.36) is the Sherman–Morrison form of the inverse in species space.

Total charge. Take $a = b = Z$, so that $\tilde{R}_{aZ} = \tilde{R}_{Zb} = P$. Since $vP = 1 - \varepsilon$ and $1 - \varepsilon$ commutes with ε^{-1},

$$vR_{ZZ} = vP + vP\,\varepsilon^{-1}\,vP = (1-\varepsilon)\big[1 + \varepsilon^{-1}(1-\varepsilon)\big] = (1-\varepsilon)\,\varepsilon^{-1} = \varepsilon^{-1} - 1. \tag{D.160}$$

Its element at $\boldsymbol{G} = \boldsymbol{G}' = \boldsymbol{G}_q$ is $v_{\mathrm{C}}(q)R_{ZZ}(\boldsymbol{q},\omega) = (\varepsilon^{-1})_{00}(\boldsymbol{q},\omega) - 1$. With the fluctuation–dissipation theorem Eq. (4.18), $S_{ZZ} = -2\,\mathrm{Im}\,R_{ZZ}/(1 - \mathrm{e}^{-\beta\omega}) = (2q^2/e^2)\,L/(1 - \mathrm{e}^{-\beta\omega})$, with $L = \mathrm{Im}[-(\varepsilon^{-1})_{00}]$ of the total dielectric matrix. No approximation enters: the relation holds for moving nuclei, any temperature and any ω.

Electron density. The electron density is $n_e = -n_Z + \sum_\kappa Z_\kappa n_\kappa$, and the response is bilinear in its two rows. Hence

$$R_{ee} = R_{ZZ} - \sum_\kappa Z_\kappa(R_{Z\kappa} + R_{\kappa Z}) + \sum_{\kappa\kappa'} Z_\kappa Z_{\kappa'} R_{\kappa\kappa'}. \tag{D.161}$$

With the nuclei clamped, each n_κ is a c-number Bragg sum, its commutators vanish, and every term with a nuclear row drops out: $R_{ee} = R_{ZZ}$, and the electron density is screened by the loss function of the total dielectric matrix, which is then the electronic one. With moving nuclei the nuclear terms describe the charge that the nuclei carry and the electrons they drag; they are what separates the electron-channel function $\varepsilon_e^{-1} = 1 + v_{\mathrm{C}}R_{ee}$ of Appendix A.7 from ε^{-1}.

D.27.2 Exactness of the loss-function form

Let the coupling be $O_c = \sum_s c_s n_s(\boldsymbol{q})$, with $c_s = c_1^{(s)}$ of Sect. 8.1, real, independent of $\boldsymbol{q}$ (nuclear form factors set to one), and S_{cc} its structure factor. By Eq. (D.160), Eq. (8.38) is the statement $S_{cc} = \lambda_Z^2 S_{ZZ}$ at every $\boldsymbol{q}$ and every $\omega \neq 0$; at $\omega = 0$ the static Bragg terms of a clamped lattice differ, and only the charge itself, with $D = 0$ below, holds there too. The target is a crystal: the small-momentum step below uses its long-wavelength longitudinal acoustic phonon.

Sufficiency. Suppose $O_c - \lambda_Z n_Z = D$ has no matrix elements between states of different energy. In the Lehmann form Eq. (4.13), $\langle f|O_c|i\rangle = \lambda_Z\langle f|n_Z|i\rangle$ for every pair with $E_f \neq E_i$, and $S_{cc} = \lambda_Z^2 S_{ZZ}$ at every $\omega \neq 0$. For $c_e = -\lambda_Z$, $c_\kappa = \lambda_Z Z_\kappa$, $D = 0$. With clamped nuclei, $D = (c_e + \lambda_Z)n_e + \sum_\kappa (c_\kappa - \lambda_Z Z_\kappa)n_\kappa$ is static exactly when $c_e = -\lambda_Z$, for any c_κ; the sign of λ_Z is free.

Necessity, large momentum. Let the nuclei move. For q large compared with the inverse size of every electron orbital and with $q^2/2M_\kappa$ large compared with the phonon energies, the impulse regime of Sect. 8.5, every particle recoils as if free during the time $1/\omega$ that the response probes. The self term of each particle in $\langle O_c(\boldsymbol{q},t)^\dagger O_c(\boldsymbol{q},0)\rangle$ then gives a recoil peak centered at $q^2/2m_s$. Its width is of order $q\,p_s/m_s$, with p_s the root-mean-square momentum of the particle in the target, and its weight is $\int \mathrm{d}\omega\, S_{cc}/2\pi = c_s^2 \boldsymbol{n}_s$ per species, by the zeroth moment of the self term. The terms between two different particles involve $\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i(t)}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j(0)}\rangle$, the Fourier transform of a pair correlation that is smooth on the scale $1/q$, and vanish as $q \to \infty$; in particular the electrons no longer follow the nuclei, and the electron–nucleus interference disappears. The distance between the peaks of two masses grows as q^2 and their

widths as q, so at large enough q each mass has a separate peak. Group the nuclear species by mass. Since nuclei of different atomic number have different masses, the nuclei of one group share an atomic number Z_g; let n_g be their total density. Equality of S_{cc} and $\lambda_Z^2 S_{ZZ}$ peak by peak requires

$$c_e^2 = \lambda_Z^2, \qquad \sum_{\kappa \in g} c_\kappa^2 n_\kappa = \lambda_Z^2 Z_g^2 \, n_g \quad \text{for every mass group } g. \tag{D.162}$$

Necessity, small momentum. At $\boldsymbol{q} \to 0$ and the frequency of a longitudinal acoustic phonon, the target is displaced locally as a whole: a long-wavelength acoustic phonon is a slow, local translation of the crystal, and the electrons go with it. A displacement field $\boldsymbol{U}(\boldsymbol{r})$ that varies on the scale $1/q$ moves each particle from $\boldsymbol{r}_j$ to $\boldsymbol{r}_j + \boldsymbol{U}(\boldsymbol{r}_j)$, and $n_s(\boldsymbol{q}) \to \sum_{j \in s} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}[1 + \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{U}(\boldsymbol{r}_j)]$. For the mode of wavevector $\boldsymbol{q}$, $\boldsymbol{U}(\boldsymbol{r}) \propto \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, and the amplitude to create the phonon is proportional to $q \sum_s c_s N_s$, with N_s the number of particles of species s, up to corrections of relative order q. The weight of the acoustic peak in S_{cc} is therefore $q^2 (\sum_s c_s n_s)^2$ times a factor common to all couplings. For n_Z the sum is the net charge-number density, which vanishes by neutrality, $n_e = \sum_g Z_g n_g$, and the acoustic weight of S_{ZZ} starts at order q^4, from the charge that a long-wavelength strain induces. Equality as $q \to 0$ requires

$$c_e n_e + \sum_\kappa c_\kappa n_\kappa = 0. \tag{D.163}$$

Conclusion. Choose the sign of λ_Z so that $c_e = -\lambda_Z$ with $\lambda_Z > 0$; Eq. (D.163) becomes $\sum_\kappa c_\kappa n_\kappa = \lambda_Z n_e = \lambda_Z \sum_g Z_g n_g$. By the Cauchy–Schwarz inequality with the weights n_κ, and by Eq. (D.162),

$$\sum_{\kappa \in g} c_\kappa n_\kappa \le \Big(\sum_{\kappa \in g} c_\kappa^2 n_\kappa \Big)^{1/2} n_g^{1/2} = \lambda_Z Z_g \, n_g,$$

with equality only if $c_\kappa = \lambda_Z Z_g$ for every species of the group. Summed over the groups, the left-hand side reaches $\lambda_Z \sum_g Z_g n_g$ only if equality holds in every group. Hence $c_e = -\lambda_Z$ and $c_\kappa = \lambda_Z Z_\kappa$ for every κ: $O_c = \lambda_Z n_Z$. With the nuclei clamped there are no nuclear recoil peaks and no acoustic phonon, and Eq. (D.162) leaves only $c_e^2 = \lambda_Z^2$, which sufficiency already covers. This proves Eq. (8.38).

The hypothesis on the masses excludes isobars of different atomic number in one target, which no crystal of this book contains. Without it, two species of equal mass and different Z would share a recoil peak, and Eq. (D.162) would constrain only a combination of their couplings.

D.27.3 Coulomb-line classes

The classification of Sect. 8.4.4 is a statement about the diagrams of the response at the random-phase level, also called time-dependent Hartree, at order g^2 in the rate with one phonon line. In this approximation every induced density acts on the electrons only through its Hartree field. A diagram is then built from fermion loops, each a Kohn–Sham response of some order. A loop takes its input fields, external, Hartree or ionic, and returns one induced density. That density reaches the rest of the diagram through one Coulomb line, or it is the output of the response. Each loop therefore has exactly one line toward the output, and the loops and Coulomb lines form a tree. Lines inside a loop, the exchange and ladder diagrams of Sect. 4.8, are absent at this level.

Now add the phonon line, which joins two ion vertices on two loops of the diagram. Two cases arise.

The diagram without the phonon line is connected. Then it is a single tree, and adding one line to a tree closes exactly one cycle. Every Coulomb line off the cycle is a bridge:

cutting it separates a piece that is attached to the rest only through that line. Momentum conservation fixes its momentum to the total momentum entering that piece. The piece contains no part of the cycle and at most the two external vertices, so the momentum is $\boldsymbol{q}$ up to a reciprocal-lattice vector and the frequency is ω. These are the external lines, and their chains are resummed by Eq. (D.159). On the cycle, let A and B be the loops at which the paths from the two external vertices meet it; a loop that carries an external vertex and lies on the cycle is its own meeting point. The cycle consists of the phonon line and a path from one ion vertex to the other that passes through A and B. Momentum is conserved at every loop, and the only momenta that enter the cycle from outside are $\boldsymbol{q}$ at A and $-\boldsymbol{q}$ at B (or the reverse). Follow the cycle from the ion vertex at one end, with $\boldsymbol{Q}$ the momentum of the phonon line. The momentum is $\boldsymbol{Q}+\boldsymbol{G}$ up to the first of A and B, it is $\boldsymbol{Q}+\boldsymbol{q}+\boldsymbol{G}$ between A and B, and it is $\boldsymbol{Q}+\boldsymbol{G}$ again from the second of them to the other ion vertex. The frequencies follow in the same way: the frequency ω' of the phonon line on the segments next to the ion vertices, $\omega-\omega'$ or $\omega+\omega'$ between A and B, and $\omega'=\pm\omega_{\boldsymbol{Q}\nu}$ when the phonon is on shell in the final state. The segments next to the ion vertices are the vertex lines, and the segment between A and B is the pair segment; when $A=B$, it is empty. Every Coulomb line is thus external, vertex or pair, and only one of these.

The chains on the vertex segments, summed over all diagrams, dress the bare potential of the displaced ion with every Coulomb chain at $(\boldsymbol{Q}+\boldsymbol{G},\omega')$ before it reaches the first loop that carries an external vertex. This is the screened potential $\varepsilon^{-1}(\boldsymbol{Q},\omega')$ acting on the bare ionic potential, Eq. (6.10) at the frequency ω'. The chains on the pair segment between two loops sum to $v+vRv=\varepsilon^{-1}v$ at $(\boldsymbol{q}+\boldsymbol{Q},\omega\mp\omega_{\boldsymbol{Q}\nu})$, the screened interaction W^{scr} of Sect. 4.8.

The diagram without the phonon line falls into two trees. Then the phonon line is a bridge, each tree contains one external vertex, and the phonon line carries $(\boldsymbol{q}+\boldsymbol{G},\omega)$. It is not a Coulomb line but the ionic polarization inserted into the external chain, and summing such insertions turns the electronic dielectric matrix into the total one. This case does not describe a pair with a phonon in the final state: its phonon line is virtual and screens the dark-matter vertex. In the amplitude of the phonon-assisted process it would be a second phonon line, at $(\boldsymbol{q}+\boldsymbol{G},\omega)$, beside the real phonon at $\boldsymbol{Q}$, and it lies outside the one-phonon diagrams of the first case. Using the total ε of Eq. (D.159) in the external and pair chains includes it once.

The assignment uses only the position of a line relative to the cycle and the external vertices, not the values of the momenta. It remains unique when the momenta coincide modulo $\boldsymbol{G}$, at $\boldsymbol{q}\to 0$, where the pair and vertex segments both carry $\boldsymbol{Q}$, or at $\boldsymbol{Q}\to 0$, where the pair segment and the external lines both carry $\boldsymbol{q}$.

The static vertex. Density-functional perturbation theory evaluates the vertex chains at $\omega'=0$. For densities the reality of the response in real space gives $R_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{Q},\omega')^*=R_{-\boldsymbol{G},-\boldsymbol{G}'}(-\boldsymbol{Q},-\omega')$, and time reversal of the equilibrium state gives $R_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{Q},\omega')=R_{-\boldsymbol{G}',-\boldsymbol{G}}(-\boldsymbol{Q},\omega')$. Together, $R(\boldsymbol{Q},-\omega')=R(\boldsymbol{Q},\omega')^\dagger$. Below the gap the response has no absorptive part, its anti-Hermitian part vanishes, and $R(\boldsymbol{Q},-\omega')=R(\boldsymbol{Q},\omega')$; the same holds for the proper response and for ε. The function is analytic for $|\omega'|<E_{\mathrm{g}}$, since its poles lie at the excitation energies, and an even analytic function differs from its value at zero by a term of order ω'^2. With the scale E_{g} set by the nearest poles, the static vertex errs by a relative amount of order $(\omega_{\boldsymbol{Q}\nu}/E_{\mathrm{g}})^2$. Without time reversal, in a magnetic insulator, only the reality relation remains, which ties ω' to $-\omega'$ at $-\boldsymbol{Q}$; the linear term survives, and the error is of order $\omega_{\boldsymbol{Q}\nu}/E_{\mathrm{g}}$.

D.28 The decomposition theorem

A response summed over sectors of final states equals the full response only if the sector projectors commute with the Hamiltonian, and none of the projectors of Sect. 8.6 does so exactly. This section bounds the error, constructs the electron-following projector, derives the amplitude by which a kicked nucleus reaches the electronic sectors, proves the partial sum rules, and expands the resonant amplitude into the Dyson convention with its Fano term. The notation is that of Sect. 8.6; the adiabatic states, surfaces and couplings are those of Sects. 5.1 and 6.1.

D.28.1 Sectors and the bin bound

Let $\mathcal{P}_A$ be orthogonal projectors with $\sum_A \mathcal{P}_A = 1$, write $H = H_\mathrm{d} + H_\mathrm{off}$ with $H_\mathrm{d} = \sum_A \mathcal{P}_A H \mathcal{P}_A$, and fix one initial state $|i\rangle$ with $\psi_b = O_b|i\rangle$. A measured spectrum is never resolved better than some bin, and we describe the bin by a shape $w_\mathrm{b}(x) \geq 0$ with $\int \mathrm{d}x\, w_\mathrm{b} = 1$, its width $\Delta\omega_\mathrm{bin}$, the kernel $f_\Delta(x) = w_\mathrm{b}(x/\Delta\omega_\mathrm{bin})/\Delta\omega_\mathrm{bin}$ and the transform $\hat{w}_\mathrm{b}(s) = \int \mathrm{d}x\, w_\mathrm{b}(x)\, \mathrm{e}^{-\mathrm{i}sx}$. Without the factor $2\pi/\mathcal{V}$, the binned response of one initial state is $S^\mathrm{bin}_{ab}(\omega) = \langle i|O_a^\dagger f_\Delta(\omega + E_i - H) O_b|i\rangle$. Since $f_\Delta(\omega + E_i - H_\mathrm{d})$ commutes with every $\mathcal{P}_A$, it has no blocks between different sectors, and

$$S^\mathrm{bin}_{ab} = \sum_A \langle i|O_a^\dagger\, \mathcal{P}_A\, f_\Delta(\omega + E_i - H_\mathrm{d})\, \mathcal{P}_A\, O_b|i\rangle + C^\mathrm{cross}_{ab},$$
$$C^\mathrm{cross}_{ab} = \langle O_a i|\big[f_\Delta(\omega + E_i - H) - f_\Delta(\omega + E_i - H_\mathrm{d})\big]|\psi_b\rangle.$$

Each sector term is a restriction by orthogonal projectors, positive for $a = b$ and positive semidefinite as a matrix. To bound C^cross_{ab}, write the kernel as a time integral, $f_\Delta(\omega + E_i - H) = \int (\mathrm{d}t/2\pi)\, \hat{w}_\mathrm{b}(\Delta\omega_\mathrm{bin} t)\, \mathrm{e}^{\mathrm{i}(\omega + E_i)t} \mathrm{e}^{-\mathrm{i}Ht}$, and compare the two evolutions with the Duhamel formula, which follows by differentiating $\mathrm{e}^{\mathrm{i}Ht}\mathrm{e}^{-\mathrm{i}H_\mathrm{d}t}$:

$$\big(\mathrm{e}^{-\mathrm{i}Ht} - \mathrm{e}^{-\mathrm{i}H_\mathrm{d}t}\big)\psi_b = -\mathrm{i}\int_0^t \mathrm{d}t'\, \mathrm{e}^{-\mathrm{i}H(t-t')}\, H_\mathrm{off}\, \mathrm{e}^{-\mathrm{i}H_\mathrm{d}t'}\, \psi_b.$$

With $\mathfrak{h}_\mathrm{off} = \sup_{t'} \|H_\mathrm{off} \mathrm{e}^{-\mathrm{i}H_\mathrm{d}t'} \psi_b\| / \|\psi_b\|$ the norm of the right side is at most $|t|\, \mathfrak{h}_\mathrm{off} \|\psi_b\|$. Inserting it in the time integral and substituting $y = \Delta\omega_\mathrm{bin} t$, the Cauchy–Schwarz inequality gives

$$|C^\mathrm{cross}_{ab}| \leq \mathfrak{c}_\mathrm{b}\, \frac{\mathfrak{h}_\mathrm{off}}{\Delta\omega^2_\mathrm{bin}}\, \big\|O_a|i\rangle\big\|\, \big\|O_b|i\rangle\big\|, \qquad \mathfrak{c}_\mathrm{b} = \frac{1}{2\pi}\int \mathrm{d}y\, |y|\, |\hat{w}_\mathrm{b}(y)|. \tag{D.164}$$

For a Gaussian bin of standard deviation $\Delta\omega_\mathrm{bin}$, $w_\mathrm{b}(x) = \mathrm{e}^{-x^2/2}/\sqrt{2\pi}$, the transform is $\mathrm{e}^{-y^2/2}$, the integral $\int \mathrm{d}y\, |y|\, \mathrm{e}^{-y^2/2} = 2$, and $\mathfrak{c}_\mathrm{b} = 1/\pi$, the constant of Eq. (8.63). A rectangular bin has a transform that falls only as $1/y$, and $\mathfrak{c}_\mathrm{b}$ diverges: the bound requires bins with smooth edges. Only the state ψ_b that the probe creates enters $\mathfrak{h}_\mathrm{off}$, not the norm of H_off over the whole space, which is proportional to the volume for a local coupling. Relative to a typical binned value $\|\psi_b\|^2/\Delta\omega_\mathrm{bin}$, the cross terms are of order $\mathfrak{h}_\mathrm{off}/\Delta\omega_\mathrm{bin}$.

D.28.2 The electron-following projector

The hypotheses of Sect. 8.6 are those under which the space-adiabatic theorem applies. The electronic Hamiltonian $H_\mathrm{el}(u)$ must be a smooth function of the nuclear positions; with nuclei of finite size, rigid charge distributions $Z_\kappa F^\mathrm{ch}_\kappa$ with F^ch_κ the nuclear charge form factor, the electron–nucleus potential is smooth and Spohn and Teufel's regularity condition holds, whereas for point nuclei it fails [67, Sect. 2]. The electron–electron repulsion does not depend on u and does not enter. The ground-state energy $E_0(u)$ must be nondegenerate and separated

from the rest of the spectrum of $H_{\rm el}(u)$ by a gap, here at least $E_{\rm g}$, on the configurations that the vibrational state visits, up to exponentially small tails. The lattice must be harmonic with perturbative anharmonicity, and $T \ll E_{\rm g}$.

In units where the electronic scales are one and the nuclear momenta are measured in units of $(M/m_e)^{1/2}$, the nuclear kinetic energy is $-\kappa_{\rm BO}^4\nabla^2/2$, the small parameter of the semiclassical expansion is $(m_e/M)^{1/2} = \kappa_{\rm BO}^2$, and the full Hamiltonian is the quantization of the operator-valued symbol $\frac{1}{2}p^2 + H_{\rm el}(u)$ [68, Eq. (38)]. Theorem 2.1 of Panati, Spohn and Teufel then provides an orthogonal projector Π whose symbol is a series $\sum_j \kappa_{\rm BO}^{2j}\pi_j$ with leading term the spectral projector of the electronic ground state, $\pi_0(u) = |\Psi_0(u)\rangle\langle\Psi_0(u)|$, and with $[H,\Pi]$ smaller than any power of $\kappa_{\rm BO}$ [68, Thm. 2.1]. The Born–Oppenheimer Hamiltonian grows quadratically in the nuclear momentum while the gap does not, and the theorem is applied below an arbitrary fixed energy through an auxiliary symbol equal to the true one on the region of phase space where the classical energy $\frac{1}{2}p^2 + E_0(u)$ lies below it [68, Sect. 4.5]. Truncating the series after $j = n$ defines $\mathcal{P}_{\rm ad}$, and since $(1-\mathcal{P}_{\rm ad})H\mathcal{P}_{\rm ad} = (1-\mathcal{P}_{\rm ad})[H,\mathcal{P}_{\rm ad}]\mathcal{P}_{\rm ad}$, the leak is of the order of the first omitted term, $\kappa_{\rm BO}^{2(n+1)}$, in operator norm on the states below that energy. This is Eq. (8.47). Low-lying vibrational states lie below any such energy above the zero-point energy, so the bound applies to them with the same order. The constants depend on the number of slow coordinates, which is why the proof covers a finite crystal.

The first correction to the state follows more simply from the coupled equations Eq. (5.4). Let a state on the ground surface have the nuclear wavefunction $\Psi_0^{\rm nuc}$ and the energy E, and let $\Psi_j^{\rm nuc}$ be its component on the excited adiabatic state Ψ_j, as in Sect. 5.1. To first order in the coupling, $[T_N + E_j(u) - E]\Psi_j^{\rm nuc} = -\mathcal{N}_{j0}\Psi_0^{\rm nuc}$, with T_N the kinetic energy of the nuclei. The derivative coupling dominates $\mathcal{N}_{j0}\Psi_0^{\rm nuc}$, which is $-M_\kappa^{-1}\boldsymbol{d}_{j0}\cdot\nabla\Psi_0^{\rm nuc}$ by Eqs. (5.5) and (6.1), and the operator on the left is $E_j(u) - E_0(u) \equiv \Delta E_j$ up to vibrational energies. Hence

$$\Psi_j^{\rm nuc} = \frac{\boldsymbol{d}_{j0}\cdot\nabla\Psi_0^{\rm nuc}}{M_\kappa\,\Delta E_j} = \frac{\mathrm{i}\,\boldsymbol{d}_{j0}\cdot\boldsymbol{p}\,\Psi_0^{\rm nuc}}{M_\kappa\,\Delta E_j}, \tag{D.165}$$

with $\boldsymbol{p} = -\mathrm{i}\nabla$ the nuclear momentum, a sum over nuclei being understood. It is the first-order term of the unitary of Panati, Spohn and Teufel, written in physical units [68, below Eq. (39)].

The same order of the expansion changes the mass. The second-order effective Hamiltonian on the ground surface is [68, Eq. (40)], for a real nondegenerate state with vanishing Berry connection,

$$h_{\rm BO} = \frac{p^2}{2M} + E_0(u) + \frac{1}{2M}\sum_{j\neq 0}|\boldsymbol{d}_{j0}|^2 - \sum_{j\neq 0}\frac{(\boldsymbol{p}\cdot\boldsymbol{d}_{0j})(\boldsymbol{d}_{j0}\cdot\boldsymbol{p})}{M^2\,\Delta E_j},$$

in physical units for one nucleus. The third term is the diagonal correction of Born and Huang (Appendix D.13). The last term is quadratic in the momentum. With $\boldsymbol{d}_{j0} = \boldsymbol{F}_{j0}/(E_0 - E_j)$ from Eq. (5.11) it equals $-\boldsymbol{p}\cdot\delta M\cdot\boldsymbol{p}/2M^2$ with

$$\delta M = 2\sum_{j\neq 0}\frac{\mathrm{Re}[\boldsymbol{F}_{0j}\boldsymbol{F}_{j0}]}{\Delta E_j^3}, \tag{D.166}$$

a positive tensor, and together with $p^2/2M$ it is the kinetic energy $\frac{1}{2}\boldsymbol{p}\cdot(M+\delta M)^{-1}\cdot\boldsymbol{p}$ to first order. For an atom the adiabatic states depend on the position $\boldsymbol{X}$ of the nucleus only through $\boldsymbol{r}_e - \boldsymbol{X}$, and Sect. 6.1 showed that $\boldsymbol{d}_{j0} = m_e\Delta E_j\langle\Psi_j|\sum_e \boldsymbol{r}_e|\Psi_0\rangle$. Then $\boldsymbol{F}_{j0} = -m_e\Delta E_j^2\langle\Psi_j|\sum_e \boldsymbol{r}_e|\Psi_0\rangle$, and a diagonal element of Eq. (D.166) is $2m_e^2\sum_j \Delta E_j\,|\langle\Psi_j|\sum_e x_e|\Psi_0\rangle|^2 = m_e\sum_j f_{j0}^{\rm osc}$ with the oscillator strengths of Eq. (4.36). By the Thomas–Reiche–Kuhn rule Eq. (4.37), $\delta M = N_e m_e$ for an atom with N_e electrons: the whole cloud is dragged, and the atom moves with its full mass, as in the solvable model of Appendix D.13.

D.28.3 The kick and the electronic sectors

Consider one nucleus at $\boldsymbol{X}$ and the kick $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}$, which is diagonal in the electronic label at each $\boldsymbol{X}$. The tilted initial state is $\sum_k \Psi_k^{\mathrm{nuc}}\Psi_k$, with Ψ_k^{nuc} from Eq. (D.165). The tilted final state on surface j is $\sum_k \Psi_k^{\mathrm{nuc}\prime}\Psi_k$, dominated by its vibrational state $\Psi_j^{\mathrm{nuc}\prime}$, and its component on the ground surface is, by the same argument with j and 0 exchanged, $\Psi_0^{\mathrm{nuc}\prime} = \boldsymbol{d}_{0j}\cdot\nabla\Psi_j^{\mathrm{nuc}\prime}/[M(E_0-E_j)] = \boldsymbol{d}_{j0}^{*}\cdot\nabla\Psi_j^{\mathrm{nuc}\prime}/(M\Delta E_j)$, using $\boldsymbol{d}_{0j} = -\boldsymbol{d}_{j0}^{*}$, the anti-Hermiticity of the derivative coupling; the complex conjugate of $\Psi_0^{\mathrm{nuc}\prime}$ that enters the overlap below then carries $\boldsymbol{d}_{j0}$ itself. The leading term $\langle\Psi_j^{\mathrm{nuc}\prime}\Psi_j|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|\Psi_0^{\mathrm{nuc}}\Psi_0\rangle$ vanishes by orthogonality of the adiabatic states at each $\boldsymbol{X}$, and to first order the amplitude is

$$\langle\Psi_j^{\mathrm{nuc}\prime}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|\Psi_j^{\mathrm{nuc}}\rangle + \langle\Psi_0^{\mathrm{nuc}\prime}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|\Psi_0^{\mathrm{nuc}}\rangle = \frac{1}{M\Delta E_j}\int \mathrm{d}\boldsymbol{X}\,\Big[\Psi_j^{\mathrm{nuc}\prime *}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}\,\boldsymbol{d}_{j0}\cdot\nabla\Psi_0^{\mathrm{nuc}} + (\boldsymbol{d}_{j0}\cdot\nabla\Psi_j^{\mathrm{nuc}\prime *})\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}\Psi_0^{\mathrm{nuc}}\Big].$$

In the second term we integrate by parts. Treating $\boldsymbol{d}_{j0}$ and ΔE_j as constant over the extent of the nuclear wavefunctions, which neglects corrections of relative order $\ell_{\mathrm{zp}}|\nabla\ln\boldsymbol{d}_{j0}| \sim \kappa_{\mathrm{BO}}$, the gradient falls on $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}\Psi_0^{\mathrm{nuc}}$ and gives $-\Psi_j^{\mathrm{nuc}\prime *}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}(\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{d}_{j0}\Psi_0^{\mathrm{nuc}} + \boldsymbol{d}_{j0}\cdot\nabla\Psi_0^{\mathrm{nuc}})$. The gradients of Ψ_0^{nuc} cancel between the two terms, and

$$\langle\widetilde{\Psi^{\mathrm{nuc}\prime}\Psi}_j|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|\widetilde{\Psi^{\mathrm{nuc}}\Psi}_0\rangle = -\mathrm{i}\,\frac{\boldsymbol{q}\cdot\boldsymbol{d}_{j0}}{M\Delta E_j}\,\langle\Psi_j^{\mathrm{nuc}\prime}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|\Psi_0^{\mathrm{nuc}}\rangle, \tag{D.167}$$

which is Eq. (8.53). The cancellation is what makes the result independent of how the nucleus moved before the kick: only the momentum transferred appears. Summing the squared modulus over the complete set of vibrational states $\Psi_j^{\mathrm{nuc}\prime}$ of surface j removes the nuclear factor, since $\sum|\langle\Psi_j^{\mathrm{nuc}\prime}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|\Psi_0^{\mathrm{nuc}}\rangle|^2 = 1$, and gives P_{Mig} of Eq. (8.55).

D.28.4 Partial sum rules

At $T \ll E_{\mathrm{g}}$ the initial states lie in the electron-following sector. For a projector that commutes with H, the response of sector ad is the full response of the restricted operators $\bar{O} = \mathcal{P}_{\mathrm{ad}}O\mathcal{P}_{\mathrm{ad}}$ under $H_{\mathrm{ad}} = \mathcal{P}_{\mathrm{ad}}H\mathcal{P}_{\mathrm{ad}}$, and the first-moment identity Eq. (4.28) applies to it inside the sector: $\mathscr{M}^1_{ab}|_{\mathrm{ad}} = \mathscr{V}^{-1}\langle[\bar{O}_a^{\dagger},[H_{\mathrm{ad}},\bar{O}_b]]\rangle$. Rotated by $U_n^{\dagger}$, the sector becomes functions of u times $\Psi_0(u)$, H_{ad} becomes h_{BO}, and the electron density becomes $\bar{\rho}_e(\boldsymbol{q};u)$ of Eq. (8.51) up to the tilt, of relative order κ_{BO}^2. A function $F(u)$ commutes with $E_0(u)$, and the double commutator with the kinetic energy of the nuclei is $[F^{\dagger},[p^2/2M,F]] = |\nabla F|^2/M$, from $[p^2,F] = -\nabla^2F - 2\mathrm{i}(\nabla F)\cdot\boldsymbol{p}$. Summed over nuclei,

$$\mathscr{M}^1_{ee}\big|_{\mathrm{ad}} = \frac{1}{\mathscr{V}}\Big\langle\sum_{\boldsymbol{R}\kappa}\frac{|\nabla_{\boldsymbol{R}\kappa}\bar{\rho}_e|^2}{M_\kappa}\Big\rangle, \qquad \mathscr{M}^1_{e\kappa}\big|_{\mathrm{ad}} = \frac{1}{\mathscr{V}}\Big\langle\sum_{I\in\kappa}\frac{\nabla_I\bar{\rho}_e^{\dagger}\cdot\nabla_I\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}}{M_\kappa}\Big\rangle,$$

with $\nabla_I\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I} = \mathrm{i}\boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$. The mass correction changes these by the relative amount $\delta M/M \sim \kappa_{\mathrm{BO}}^4$. For rigid clouds, $\bar{\rho}_e = \sum_I f_{\kappa(I)}(\boldsymbol{q})\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$, the gradient with respect to $\boldsymbol{X}_I$ is $\mathrm{i}\boldsymbol{q}f_\kappa\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$, and the averages give $q^2\sum_\kappa \boldsymbol{n}_\kappa|f_\kappa|^2/M_\kappa$ and $q^2\boldsymbol{n}_\kappa f_\kappa^{*}/M_\kappa$, the forms quoted in Eqs. (8.54) and (8.57). The total of $\mathscr{M}^1_{e\kappa}$ over all sectors is zero by Eq. (4.34), and the electronic sectors carry the negative of the electron-following part.

For the nuclear density the electronic share follows from Eq. (D.167). Each final state lies above the initial one by ΔE_j plus the vibrational energy of $\Psi_j^{\mathrm{nuc}\prime}$ relative to Ψ_0^{nuc}. Weighted with the squared amplitude Eq. (D.167) and summed over the vibrational states, the electronic energies give $\sum_j|\boldsymbol{q}\cdot\boldsymbol{d}_{j0}|^2/(M^2\Delta E_j) = \sum_j|\boldsymbol{q}\cdot\boldsymbol{F}_{j0}|^2/(M^2\Delta E_j^3) = \boldsymbol{q}\cdot\delta M\cdot\boldsymbol{q}/2M^2$, and the vibrational energies give the first moment of the kick on the excited surface, E_R by the

argument of Eq. (5.61), times P_{Mig}. The first moment summed over $\pm\boldsymbol{q}$ is twice the one-sided one, and per nucleus the electronic sectors carry $\mathscr{M}^1|_{0+} = \boldsymbol{\nu}_\kappa[P_{\mathrm{Mig}} q^2/M + \boldsymbol{q}\cdot\delta M\cdot\boldsymbol{q}/M^2]$. The total is $\boldsymbol{\nu}_\kappa q^2/M$, and the electron-following sector carries the difference, $\boldsymbol{\nu}_\kappa[(1-P_{\mathrm{Mig}})q^2/M - \boldsymbol{q}\cdot\delta M\cdot\boldsymbol{q}/M^2]$. To first order in the two small quantities this is $\boldsymbol{\nu}_\kappa(1-P_{\mathrm{Mig}})\,\boldsymbol{q}\cdot(M+\delta M)^{-1}\cdot\boldsymbol{q}$, the same result that follows from h_{BO} with the depleted weight $1-P_{\mathrm{Mig}}$ of $\bar{n}_\kappa$. This is Eq. (8.56).

D.28.5 Labels inside the off-shell window

Let H_0 contain the electronic excitations at clamped, renormalized bands and the harmonic phonons, and let H_{ep} couple them; the eigenstates of H_0 carry a definite phonon number. Take ω inside the off-shell window, so that every zero-phonon state $|m\rangle$ at the same total momentum has $E_m \geq \omega + \delta_{\mathrm{w}}$. An exact final state at ω grows out of a one-phonon state $|f\rangle$ of H_0. The Lippmann–Schwinger equation $|\tilde{f}\rangle = |f\rangle + (E_f - H_0 + \mathrm{i}0^+)^{-1} H_{\mathrm{ep}}|\tilde{f}\rangle$, projected on the zero-phonon states, has denominators $E_f - E_m \leq -\delta_{\mathrm{w}}$ that never vanish, and its iteration converges for $|g| \ll \delta_{\mathrm{w}}$. The zero-phonon component is $\sum_m |m\rangle g_{mf}/(E_f - E_m) + O(g^3/\delta_{\mathrm{w}}^3)$, with squared norm $\sum_m |g_{mf}|^2/(E_f-E_m)^2 \leq \theta_{\mathrm{ep}}$ when δ_{off} in Eq. (8.40) is taken as the smallest denominator, δ_{w}. Any two definitions of the label that agree at $g=0$ and are continuous in g, the phonon number of the leading component, a Schrieffer–Wolff number, the number of phonons in the outgoing state, assign $|\tilde{f}\rangle$ to $\mathcal{P}_+$ and differ by at most this weight. The response at ω is $(2\pi/\mathscr{V})\sum_f |\langle\tilde{f}|O_b|i\rangle|^2\delta(\omega - E_f)$, all of it in $\mathcal{P}_+$ to relative order θ_{ep}. To leading order $\langle\tilde{f}|O_b|i\rangle = g_{fm}[\mathcal{A}_f^{\mathrm{off}} + \rho_m/(\omega - E_m)]$, summed over m, which is Eq. (8.59) with no delta term, since $\omega - \tilde{E}_m$ never vanishes on the support. The bare projector $\mathcal{P}_0^{(0)}$, by contrast, gives $\langle\tilde{f}|\mathcal{P}_0^{(0)}O_b|i\rangle = \sum_m g_{mf}^*\rho_m/(E_f - E_m)$, the resonant part of the same amplitude, and it would assign to $\mathcal{P}_0$ a weight of the size of the rate.

D.28.6 The resonant amplitude

Outside the window, keep one zero-phonon pair $|m\rangle$ at energy E_m and the one-phonon states $|f\rangle$ it couples to, with $H = E_m|m\rangle\langle m| + \sum_f E_f|f\rangle\langle f| + \sum_f (g_{fm}|f\rangle\langle m| + g_{fm}^*|m\rangle\langle f|)$; nonresonant pairs and the paths through occupied states only enter the amplitudes $\mathcal{A}_f^{\mathrm{off}}$, and degenerate pairs need the matrix form of the same algebra. The partitioning Eq. (C.58) with $P = |m\rangle\langle m|$ gives the level's Green's function $\mathsf{G}_m(z) = [z - E_m - \Sigma_m(z)]^{-1}$ with $\Sigma_m(z) = \sum_f |g_{fm}|^2/(z - E_f)$, as in Worked example 4.12. The scattering states $|f^-\rangle$ of the continuum satisfy $\langle f^-|m\rangle = g_{fm}\mathsf{G}_m(E_f + \mathrm{i}0^+)$ and $\langle f^-|f'\rangle = \delta_{ff'} + g_{fm}\mathsf{G}_m(E_f+\mathrm{i}0^+)\,g_{f'm}^*/(E_f - E_{f'} + \mathrm{i}0^+)$, the Lippmann–Schwinger solution of the model [69]. If no bound state splits off, they are complete, and the response of the source $O_b|i\rangle = \rho_m|m\rangle + \sum_f g_{fm}\mathcal{A}_f^{\mathrm{off}}|f\rangle$ is $(2\pi/\mathscr{V})\sum_f |\langle f^-|O_b|i\rangle|^2\delta(\omega - E_f)$. The amplitude is Eq. (8.58).

Let $\mathcal{A}_f^{\mathrm{off}}$ take the value $\mathcal{A}_\nu^{\mathrm{off}}$ on the energy shell of branch ν, and use $\sum_{f\in\nu}|g_{fm}|^2\delta(\omega - E_f) = \varGamma_m^{(\nu)}(\omega)/2\pi$. The response in branch ν is $(2\pi/\mathscr{V})(\varGamma_m^{(\nu)}/2\pi)|\mathcal{A}_\nu^{\mathrm{off}} + \mathsf{G}_m(\rho_m + \Upsilon_m)|^2$, and we expand its three terms near $\tilde{E}_m$, with $x = \omega - \tilde{E}_m$, against a test function smooth on δ_{F}. Near the pole $\mathsf{G}_m \simeq z_{\mathrm{qp}}/(x + \mathrm{i}z_{\mathrm{qp}}\varGamma_m/2)$, and Eq. (2.60) with the width $z_{\mathrm{qp}}\varGamma_m$ gives $|\mathsf{G}_m|^2 = (2\pi z_{\mathrm{qp}}/\varGamma_m)\delta(x) + \mathrm{fp}(1/x^2) + O(g^2)$. In the cross term $2\,\mathrm{Re}[\mathcal{A}_\nu^{\mathrm{off}*}\mathsf{G}_m\rho_m]$ the propagator becomes $\mathrm{PV}(1/x) - \mathrm{i}\pi\delta(x)$ by Eq. (C.27), which yields $2\,\mathrm{Re}(\mathcal{A}_\nu^{\mathrm{off}*}\rho_m)\,\mathrm{PV}(1/x) + 2\pi\,\mathrm{Im}(\mathcal{A}_\nu^{\mathrm{off}*}\rho_m)\delta(x)$. In $|\rho_m + \Upsilon_m|^2 = |\rho_m|^2 + 2\,\mathrm{Re}(\rho_m^*\Upsilon_m) + O(g^4)$, the vertex correction at the pole is $\Upsilon_m = \mathrm{PV}\,\Upsilon_m - (\mathrm{i}/2)\sum_\mu \varGamma_m^{(\mu)}\mathcal{A}_\mu^{\mathrm{off}}$, and $2\,\mathrm{Re}(\rho_m^*\Upsilon_m) = 2\,\mathrm{Re}(\rho_m^*\,\mathrm{PV}\,\Upsilon_m) + \sum_\mu \varGamma_m^{(\mu)}\,\mathrm{Im}(\rho_m^*\mathcal{A}_\mu^{\mathrm{off}})$. With $c_\mu = \mathrm{Im}(\rho_m^*\mathcal{A}_\mu^{\mathrm{off}})$ and $\bar{c} = \sum_\mu \varGamma_m^{(\mu)}c_\mu/\varGamma_m$, the coefficient of $(2\pi/\mathscr{V})\delta(x)$ in branch ν collects to

$$\frac{\varGamma_m^{(\nu)}}{\varGamma_m}\Big[z_{\mathrm{qp}}|\tilde{\rho}_m|^2 + \varGamma_m(\bar{c} - c_\nu)\Big] = z_{\mathrm{qp}}|\tilde{\rho}_m|^2\,\frac{\varGamma_m^{(\nu)}}{\varGamma_m}(1 + \bar{c}^{\mathrm{F}} - c_\nu^{\mathrm{F}}) + O(g^4), \tag{D.168}$$

with $c_\nu^{\mathrm{F}} = \Gamma_m c_\nu/|\rho_m|^2$. Summed over ν with the weights $\Gamma_m^{(\nu)}/\Gamma_m$, which add up to one, the Fano terms cancel, and the delta part is S^0 of Eq. (8.59). What remains, $(\Gamma_m^{(\nu)}/2\pi)[|\rho_m|^2\,\mathrm{fp}(1/x^2) + 2\,\mathrm{Re}(\mathcal{A}_\nu^{\mathrm{off}*}\rho_m)\,\mathrm{PV}(1/x) + |\mathcal{A}_\nu^{\mathrm{off}}|^2]$, summed over ν, is S^+. The neglected terms are of three kinds: order g^4, from $z_{\mathrm{qp}}^2 \to 1$ in the finite part and from $|\Upsilon_m|^2$; the term $-\pi\Gamma_m\delta''(x)/4$ of Eq. (2.60), which against a test function smooth on δ_{F} is of relative order $\Gamma_m^2/\delta_{\mathrm{F}}^2$; and the shift of the finite part from E_m to $\tilde{E}_m$, of order g^4. The totals therefore agree with the exact response to $O(g^4, \Gamma_m^2/\delta_{\mathrm{F}}^2)$. If the pair also decays by channels other than phonon emission, impact ionization for instance, the same algebra holds with ν and μ running over every channel. Each channel has its own off-shell amplitude $\mathcal{A}_\nu^{\mathrm{off}}$, the self-energy and the vertex correction Υ_m sum over all of them, the Lorentzian carries the total width $\Gamma_m^{\mathrm{tot}} = \sum_\nu \Gamma_m^{(\nu)}$, and Eq. (D.168) becomes Eq. (8.62). The deviations $c_\nu^{\mathrm{F}} - \bar{c}^{\mathrm{F}}$ then average to zero over all channels; restricted to the phonon branches but normalized by Γ_m^{tot}, they would not, and the number of zero-phonon events would change. As before, the delta function itself needs no width, so the response calculation does not need Γ_m^{tot}, only the cascade does. A cascade that lets the zero-phonon pair emit its first phonon with these branching ratios, Fano terms included, reproduces the exact distribution of the first emission at order g^2.

The residue identity Eq. (8.60) follows from the Kramers–Kronig form $\mathrm{Re}\,\Sigma_m(\omega) = (2\pi)^{-1}\,\mathrm{PV}\!\int \mathrm{d}\omega'\,\Gamma_m(\omega')/(\omega - \omega')$ and Eq. (C.31). Its use for the ambiguity is a bound on $1 - z_{\mathrm{qp}}^{-1} = \partial_\omega\,\mathrm{Re}\,\Sigma_m$. Split the finite-part integral at $|\omega' - \tilde{E}_m| = \delta_{\mathrm{F}}$. Outside, $|\int \mathrm{d}\omega'\,\Gamma_m/(\omega' - \tilde{E}_m)^2| \le 2\hat{\Gamma}_m/\delta_{\mathrm{F}}$, with $\hat{\Gamma}_m$ the largest width. Inside, $\Gamma_m(\tilde{E}_m)$ times $\mathrm{fp}\int_{-\delta_{\mathrm{F}}}^{\delta_{\mathrm{F}}} \mathrm{d}y/y^2 = -2/\delta_{\mathrm{F}}$ contributes $2\Gamma_m/\delta_{\mathrm{F}}$ in modulus, the linear term of Γ_m drops out by symmetry, and the curvature contributes at most $\delta_{\mathrm{F}}\|\Gamma_m''\|$, which is $\hat{\Gamma}_m/\delta_{\mathrm{F}}$ if Γ_m varies on the scale δ_{F}. Hence $|1 - z_{\mathrm{qp}}^{-1}| \le (5/2\pi)\hat{\Gamma}_m/\delta_{\mathrm{F}}$. Raising the residue to a power between 1 and 2 changes the zero-phonon weight by at most this amount, relative to w_0, which is the first term of Eq. (8.61). Replacing the delta function by a normalized kernel of width Γ_m moves the tail weight $2\int_{x_{\mathrm{w}}}^\infty \mathrm{d}x\,(\Gamma_m/2\pi)/x^2 = \Gamma_m/\pi x_{\mathrm{w}}$ outside a window of half-width x_{w}, the second term. Assigning to the line all weight within $|x| < x_{\mathrm{w}}$ moves into it the finite-part, principal-value and $|\mathcal{A}^{\mathrm{off}}|^2$ terms of S^+ there. Their pole parts reproduce the second term; their smooth parts are bounded by $\hat{\Gamma}_m x_{\mathrm{w}}/\delta_{\mathrm{F}}^2$ times $|\rho_m|^2$ for the finite part, $2|\rho_m||\mathcal{A}^{\mathrm{off}}|\delta_{\mathrm{F}}$ for the principal value and $|\mathcal{A}^{\mathrm{off}}|^2\delta_{\mathrm{F}}^2$ for the last term. Divided by $w_0 \simeq |\rho_m|^2$ they add up to the third term. Each step is first order in $\Gamma_m/\delta_{\mathrm{F}}$, and the bound holds to that order.

D.28.7 Real amplitudes under inversion and time reversal

Let PT be the product of inversion and time reversal. It is antiunitary and commutes with H if both symmetries hold. On states of N electrons $\mathsf{T}^2 = (-1)^N$, $\mathsf{P}^2 = 1$ and the two commute, so $(\mathsf{PT})^2 = (-1)^N$; on phonons $(\mathsf{PT})^2 = 1$. For even N, then, $(\mathsf{PT})^2 = 1$ on the whole space. An antiunitary involution that commutes with a Hermitian operator leaves each eigenspace invariant, and every eigenspace has an orthonormal basis of invariant vectors: for any v in it, $v + \mathsf{PT}v$ and $\mathrm{i}(v - \mathsf{PT}v)$ are invariant, and v is a combination of them. Choose such a basis for the eigenstates of H_0 within each sector, which is possible because PT commutes with the phonon number. For the one-phonon states this fixes the phase of each polarization vector $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$, since inversion and time reversal each reverse $\boldsymbol{Q}$ and PT maps the mode $(\boldsymbol{Q}, \nu)$ to itself. For two invariant states a, b and an invariant operator O, antiunitarity gives $\langle a|O|b\rangle = \langle \mathsf{PT}a|\mathsf{PT}Ob\rangle^* = \langle a|Ob\rangle^*$, a real number. The density is invariant: inversion maps $n_e(\boldsymbol{q})$ to $n_e(-\boldsymbol{q})$, and time reversal, which conjugates $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, maps it back. So is H_{ep}, as part of H. Every ρ_m and g_{fm} is real, and so is every off-shell amplitude $\mathcal{A}_f^{\mathrm{off}}$, a sum of products of real matrix elements and real denominators. Hence $c_\nu = 0$ and the Fano coefficients vanish, provided the invariant basis of a degenerate zero-phonon manifold also diagonalizes its level-shift matrix. One-body self-energies are a multiple of the identity on each degenerate

manifold, a Kramers pair or an irreducible representation, so any invariant basis does. Level shifts from the electron–hole vertex can split the manifold in a basis that is not invariant; then the c_ν need not vanish branch by branch, and only the weighted sum of the Fano terms, which cancels in every crystal, is zero. With spin–orbit coupling the single-electron states form Kramers doublets with $(\mathsf{PT})^2 = -1$, but the argument concerns many-electron states with even N and needs no real single-particle gauge.

D.28.8 Gapless targets

For a state ψ_b that decays by the golden rule into a continuum of final states f with a flat density $\mathcal{D}_f$ over a width ΔE_c, $\Gamma = 2\pi|\langle f|H_\mathrm{off}|\psi_b\rangle|^2\mathcal{D}_f/\|\psi_b\|^2$ and $\|H_\mathrm{off}\psi_b\|^2 = \sum_f |\langle f|H_\mathrm{off}|\psi_b\rangle|^2 = \Gamma\,\Delta E_\mathrm{c}\|\psi_b\|^2/2\pi$, which is the estimate of Sect. 8.6. Without any dressing, $U_\mathrm{qp} = 1$, the estimate does not become a bound. The diagonal part H_d conserves the bare numbers, but the probe's quasiparticle sits on the filled Fermi sea, and H_off also acts on the sea, creating a phonon and a pair away from the probe's excitations. The squared norm of this disconnected term grows with the number of cells, and so does $\mathfrak{h}_\mathrm{off}$. The connected part, in which H_off acts on the quasiparticle $|n\boldsymbol{k}\rangle$ itself, is what the estimate describes. For the electron–phonon part, the state $H_\mathrm{ep}|n\boldsymbol{k}\rangle$ contains one phonon emitted or absorbed, and its squared norm is $N^{-1}\sum_{m\nu\boldsymbol{Q}} |g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2$ times $n_{\boldsymbol{Q}\nu}+1-f_{m,\boldsymbol{k}+\boldsymbol{Q}}$ for emission and $n_{\boldsymbol{Q}\nu}+f_{m,\boldsymbol{k}+\boldsymbol{Q}}$ for absorption, at most $2n_{\boldsymbol{Q}\nu}+1$ in total. This is $-\pi^{-1}\int \mathrm{d}\omega\ \mathrm{Im}\,\Sigma^\mathrm{FM}_{n\boldsymbol{k}}(\omega)$ from Eq. (6.27), the zeroth moment of the Fan–Migdal spectral function. The electron–electron part adds the analogous sum over final states with two particles and a hole. For a pair source, the electron and hole contributions add. Taken over the bands, these moments form a matrix whose largest eigenvalue bounds the square of the connected part, estimated at about 0.2 eV in aluminum. Separating the connected part from the disconnected one needs the dressed vacuum U_qp and a commutator bound uniform in N, which is open (Sect. 8.6).

D.29 The crystal form-factor rate

The electron-recoil rate of a crystal at the base point $\mathfrak{T}_0$ follows from the master formula Eq. (8.16) in three steps. We compute the structure factor of filled bands, average it over directions, and regroup the average into the crystal form factor of Ref. [1]. The last subsection converts the rate to the normalizations in use.

D.29.1 The structure factor of filled bands

At $\mathfrak{T}_0$ the target is a Slater determinant $|0\rangle$ in which every valence state $v\boldsymbol{k}$ is filled with both spins and every conduction state is empty, and its excited states are again determinants of the same one-electron states. The one-electron energies are Kohn–Sham energies with the conduction bands raised by the scissor shift, Eq. (3.66), which leaves the states unchanged. The temperature is zero, so only $|0\rangle$ enters the thermal sum of Eq. (4.13).

In the Bloch basis the density operator is Eq. (3.37). Acting on $|0\rangle$, a term $c^\dagger_{m,\boldsymbol{k}+\boldsymbol{q},s}c_{n\boldsymbol{k}s}$ vanishes if n is a conduction band, since there is no electron to remove, and if m is a valence band, since the state is already occupied, unless it is the state just emptied. That exception requires $\boldsymbol{k}+\boldsymbol{q} \equiv \boldsymbol{k}$, that is $\boldsymbol{q} = \boldsymbol{G}$, and it returns $|0\rangle$ itself with $\omega = 0$, the elastic Bragg term that no detector sees. For $\omega > 0$,

$$n_e(\boldsymbol{q})|0\rangle = \sum_s \sum_{\boldsymbol{k}\in\mathrm{BZ}} \sum_{v,c} \rho_{cv}(\boldsymbol{k},\boldsymbol{q})\,|c,\boldsymbol{k}+\boldsymbol{q},s;\,v\boldsymbol{k}s\rangle, \tag{D.169}$$

where the ket is $|0\rangle$ with the electron moved from $v\boldsymbol{k}s$ to $c,\boldsymbol{k}+\boldsymbol{q},s$. These pair states are orthonormal, and their energies above the ground state are $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}}$, with $\boldsymbol{k}+\boldsymbol{q}$ folded into

the first zone by the Umklapp vector $\boldsymbol{G}_{\mathrm{U}}$ of Eq. (3.34). Each pair appears once in Eq. (D.169): at fixed $\boldsymbol{q}$ the hole $v\boldsymbol{k}s$ fixes the crystal momentum of the electron. Inserting Eq. (D.169) into Eq. (4.13) with $a = b = n_e$, the two spins give equal terms, and

$$S^0(\boldsymbol{q},\omega) = \frac{2\pi}{\mathcal{V}}\, 2 \sum_{\boldsymbol{k}\in\mathrm{BZ}} \sum_{v,c} |\rho_{cv}(\boldsymbol{k},\boldsymbol{q})|^2\, \delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}).$$

The crystal has N cells and $\mathcal{V} = N\Omega$, and the N allowed crystal momenta fill the zone uniformly, one per volume $(2\pi)^3/\mathcal{V}$ (Sect. 3.1), so that $\sum_{\boldsymbol{k}} = \mathcal{V} \int_{\mathrm{BZ}} \mathrm{d}^3k/(2\pi)^3$. The volume cancels:

$$S^0(\boldsymbol{q},\omega) = 4\pi \sum_{v,c} \int_{\mathrm{BZ}} \frac{\mathrm{d}^3k}{(2\pi)^3}\, |\rho_{cv}(\boldsymbol{k},\boldsymbol{q})|^2\, \delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}). \tag{D.170}$$

The factor 4π is 2π from the definition of S times 2 for spin. The same expression is $-2\,\mathrm{Im}\, R^0_{\boldsymbol{00}}(\boldsymbol{q},\omega)$ for the bubble Eq. (4.63) at $T = 0$ and $\omega > 0$, where only the resonant term has an imaginary part, $-\pi\delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}})$, as the fluctuation–dissipation theorem Eq. (A.33) requires.

Two checks fix the normalization. For a local one-electron potential, the double commutator of Sect. 4.3 taken for one electron gives $\sum_m (\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{n\boldsymbol{k}})|\rho_{mn}(\boldsymbol{k},\boldsymbol{q})|^2 = q^2/2m_e$ for every state $n\boldsymbol{k}$, the sum running over all bands m. The terms in which m is another valence band are absent from Eq. (D.170), but summed over $\boldsymbol{k}$ they cancel: each transition at $-\boldsymbol{q}$ is the reverse of one at $\boldsymbol{q}$, so the valence–valence sum at $-\boldsymbol{q}$ is minus the sum at $\boldsymbol{q}$, while $S^0(\boldsymbol{q},\omega) = S^0(-\boldsymbol{q},\omega)$ in silicon makes the two sums equal, and a sum equal to its own negative vanishes. The first moment of Eq. (D.170) is then $4\pi \cdot 4 \cdot \Omega^{-1} \cdot q^2/2m_e = \pi \boldsymbol{n}_e q^2/m_e$ with $\boldsymbol{n}_e = 8/\Omega$, the f-sum rule Eq. (4.30) for the valence electrons, since $\int_{\mathrm{BZ}} \mathrm{d}^3k/(2\pi)^3 = 1/\Omega$. The zeroth moment, by the closure relation Eq. (3.36), is at most $4\pi \cdot 4/\Omega$, four valence bands of unit weight, a bound reached only when every transition leads into a conduction band.

D.29.2 The rate per unit mass

For a target whose response is averaged over the directions of $\boldsymbol{q}$, the rate is Eq. (A.44) with $\bar{S}^0(q,\omega) = \int \mathrm{d}^2\hat{\boldsymbol{q}}\, S^0(q\hat{\boldsymbol{q}},\omega)/4\pi$. Substituting $\bar{S}^0 = 4\pi\alpha m_e^2 |f_{\mathrm{crys}}|^2/q^3\Omega$ from Eq. (A.45) and using $\rho_T \Omega = M_{\mathrm{cell}}$,

$$\begin{aligned} \frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} &= \frac{\rho_\chi\, \bar{\sigma}_e\, \alpha\, m_e^2}{m_\chi\, M_{\mathrm{cell}}\, \mu_{\chi e}^2} \\ &\quad \times \int \frac{\mathrm{d}q}{q^2}\, \eta\big(v_{\mathrm{min}}(q,\omega)\big)\, F_{\mathrm{DM}}(q)^2\, |f_{\mathrm{crys}}(q,\omega)|^2. \end{aligned} \tag{D.171}$$

Multiplying by ω turns $\mathrm{d}q/q^2$ into $\mathrm{d}\ln q\, (\omega/q)$ and the left side into $\mathrm{d}\mathcal{R}/\mathrm{d}\ln\omega$. Multiplying by the target mass $N_{\mathrm{cell}} M_{\mathrm{cell}}$ gives the rate of a whole target, which is Ref. [1, Eq. (3.13)] term by term. The cell volume has dropped out: the rate per unit mass needs only the cell mass, because $|f_{\mathrm{crys}}|^2$ is defined per cell.

D.29.3 The crystal form factor

What remains is to show that $q^3\Omega\bar{S}^0/4\pi\alpha m_e^2$ is the crystal form factor as Ref. [1, Eq. (3.17)] defines it, with the angle average written as an integral over a second crystal momentum. For any function $g(\boldsymbol{q})$,

$$\int \frac{\mathrm{d}^2\hat{\boldsymbol{q}}}{4\pi}\, g(q\hat{\boldsymbol{q}}) = \frac{1}{4\pi q^2} \int \mathrm{d}^3q'\, \delta(q - |\boldsymbol{q}'|)\, g(\boldsymbol{q}'),$$

since $\mathrm{d}^3q' = q'^2 \mathrm{d}q'\,\mathrm{d}^2\hat{\boldsymbol{q}}'$. At fixed $\boldsymbol{k}$, every $\boldsymbol{q}'$ is reached exactly once as $\boldsymbol{q}' = \boldsymbol{k}' - \boldsymbol{k} + \boldsymbol{G}'$ with $\boldsymbol{k}'$ in the first zone and $\boldsymbol{G}'$ a reciprocal-lattice vector, because the zones translated by all $\boldsymbol{G}'$ tile space. Then $\int \mathrm{d}^3q' = \sum_{\boldsymbol{G}'} \int_{\mathrm{BZ}} \mathrm{d}^3k'$, the final crystal momentum is $\boldsymbol{k}'$, and $\boldsymbol{G}'$ is the Umklapp vector of the transition. Applied to Eq. (D.170),

$$\bar{S}^0(q,\omega) = \frac{(2\pi)^3}{q^2} \sum_{v,c} \int_{\mathrm{BZ}} \frac{\mathrm{d}^3k}{(2\pi)^3} \int_{\mathrm{BZ}} \frac{\mathrm{d}^3k'}{(2\pi)^3} \sum_{\boldsymbol{G}'} \delta(q - |\boldsymbol{k}' - \boldsymbol{k} + \boldsymbol{G}'|) \times |\rho_{cv}(\boldsymbol{k}, \boldsymbol{k}' - \boldsymbol{k} + \boldsymbol{G}')|^2\, \delta(\omega - \varepsilon_{c\boldsymbol{k}'} + \varepsilon_{v\boldsymbol{k}}).$$

Multiplying by $q^3\Omega/4\pi\alpha m_e^2$ and writing $(2\pi)^3/4\pi = 2\pi^2$,

$$|f_{\mathrm{crys}}(q,\omega)|^2 = \frac{2\pi^2}{\alpha m_e^2 \Omega} \sum_{v,c} \int_{\mathrm{BZ}} \frac{\Omega\,\mathrm{d}^3k}{(2\pi)^3} \int_{\mathrm{BZ}} \frac{\Omega\,\mathrm{d}^3k'}{(2\pi)^3} \sum_{\boldsymbol{G}'} q\,\delta(q - |\boldsymbol{k}' - \boldsymbol{k} + \boldsymbol{G}'|) \times |\rho_{cv}(\boldsymbol{k}, \boldsymbol{k}' - \boldsymbol{k} + \boldsymbol{G}')|^2\, \delta(\omega - \varepsilon_{c\boldsymbol{k}'} + \varepsilon_{v\boldsymbol{k}}). \tag{D.172}$$

This is Ref. [1, Eq. (3.17)]: their prefactor $2\pi^2(\alpha m_e^2 V_{\mathrm{cell}})^{-1}E_e^{-1}$ times their $E_e\,\delta(E_e - \cdots)$ is our $2\pi^2/\alpha m_e^2\Omega$ times the energy delta function, their $f_{[i\boldsymbol{k},i'\boldsymbol{k}',\boldsymbol{G}']}$ is $\rho_{cv}(\boldsymbol{k}, \boldsymbol{k}' - \boldsymbol{k} + \boldsymbol{G}')$ in the plane-wave form Eq. (3.35), and the spin factor 2 sits in their $2\pi^2$ as it sits in our 4π [1, Eq. (A.29)]. The prefactor $2\pi^2/\alpha m_e^2\Omega$ is 1.99 eV for silicon, with $\Omega = 40.05\,\text{Å}^3$.

D.29.4 Conversions between normalizations

The literature states the same rate in several normalizations, and each converts to $\bar{S}^0$ in one line.

- *Structure factor, form factor and dielectric function.* $\Omega\bar{S}^0 = 4\pi\alpha m_e^2 |f_{\mathrm{crys}}|^2/q^3$ and $|f_{\mathrm{crys}}|^2 = q^5\Omega\,\varepsilon_2/8\pi^2\alpha^2 m_e^2$, Eq. (A.45); the second uses the unscreened $\varepsilon_2 = 2\pi\alpha\bar{S}^0/q^2$ of Eq. (A.37) at $T = 0$, and agrees with Ref. [24, Eq. (16)].

- *Per target or per unit mass.* A rate for a target of N_{cell} cells [1, Eq. (3.13), 70, Eq. (36)] is $N_{\mathrm{cell}}M_{\mathrm{cell}}$ times Eq. (D.171).

- *Sums over states.* Ref. [71, Eqs. (3.6) and (3.12)] writes the rate with the transition form factor $f_{i\to f}(\boldsymbol{q}) = \langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|i\rangle$ between unit-normalized spatial orbitals and a separate spin factor. Then $(2\pi/\mathcal{V})\sum_{i,f} |f_{i\to f}|^2 \delta(\omega - E_f + E_i) = S^0/2$. The transition matrix element of the density in Ref. [72, Eqs. (11) and (17)] is the vertex $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q})$ of Eq. (3.34).

- *Crystal-momentum weights.* Plane-wave codes replace $\int_{\mathrm{BZ}} \Omega\,\mathrm{d}^3k/(2\pi)^3$ by $\frac{1}{2}\sum_{\boldsymbol{k}} w_{\boldsymbol{k}}$ with weights that sum to 2 [1, Eq. (4.2)]. The 2 is the spin factor already present in the prefactor, and it must not be counted twice.

- *Bins.* A tabulated form factor is an average over bins of width Δq and $\Delta\omega$ [1, Eq. (4.1)], and Eq. (D.171) becomes a double sum with $\mathrm{d}q\,\mathrm{d}\omega \to \Delta q\,\Delta\omega$. A table that stores the sum over a bin rather than the average differs by the bin widths. The f-sum rule is the test that catches such a factor: at $q \approx q_0$, where the free recoil energy $q^2/2m_e = 13.6\,\mathrm{eV}$ lies well inside the tabulated range, $\int \mathrm{d}\omega\,\omega\,\bar{S}^0$ must come within some ten percent of $\pi n_e q^2/m_e$. A rigid scissor shift Δ_{scis} raises it by $\Delta_{\mathrm{scis}} \int \mathrm{d}\omega\,\bar{S}^0$, since it moves the weight without changing the vertices.

D.30 Multi-operator responses in the medium

The rate of a velocity-dependent coupling needs more rows of Eq. (8.3) than the density. We write it with those rows, reduce them to a minimal set, relate the set to the published crystal responses, screen each row, and use Eq. (8.21) to bound a rate whose cross term has not been computed; Sect. 10.4 uses each result. The target is an insulator at $T = 0$ in its ground state $|0\rangle$, with a Hamiltonian whose potentials are local, without spin–orbit coupling and without magnetic order, unless stated otherwise. All energies deposited are positive, $\omega > 0$.

D.30.1 The contraction for one velocity-dependent operator

Take $\mathcal{O}_8 = \boldsymbol{S}_\chi \cdot \boldsymbol{v}^\perp$ with coefficient $c_8 \equiv c_8^{(e)}(q)$ for dark matter of spin $\frac{1}{2}$. Its reduction Eq. (8.4) gives the potentials $\tilde{V}_{n_e} = c_8\, \boldsymbol{S}_\chi \cdot \bar{\boldsymbol{v}}_\chi$ and $\tilde{V}_{j_e^\alpha} = -c_8 S_\chi^\alpha$ on the density and the three components of the current. With $\mathrm{Tr}_\chi(S_\chi^\alpha S_\chi^\beta)/2 = \delta_{\alpha\beta}/4$, the tensor Eq. (8.7) has the elements $\mathfrak{D}_{nn} = c_8^2 |\bar{\boldsymbol{v}}_\chi|^2/4$, $\mathfrak{D}_{nj^\alpha} = \mathfrak{D}_{j^\alpha n} = -c_8^2 \bar{v}_\chi^\alpha/4$ and $\mathfrak{D}_{j^\alpha j^\beta} = c_8^2 \delta_{\alpha\beta}/4$. The contraction of Eq. (8.15) is then

$$\sum_{a,b} \mathfrak{D}_{ab} S_{ab} = \frac{c_8^2}{4}\Big[|\bar{\boldsymbol{v}}_\chi|^2 S_{nn} - 2\,\mathrm{Re}(\bar{\boldsymbol{v}}_\chi \cdot \boldsymbol{S}_{n\boldsymbol{j}}) + \sum_\alpha S_{j^\alpha j^\alpha}\Big], \tag{D.173}$$

where $\boldsymbol{S}_{n\boldsymbol{j}}$ is the vector of elements S_{nj^α} and we used $S_{j^\alpha n} = S_{nj^\alpha}^*$, which holds because the response matrix is Hermitian (Eq. (4.20)).

Now split every vector into its component along $\hat{\boldsymbol{q}}$ and its components across. With two unit vectors $\hat{\boldsymbol{e}}_1$, $\hat{\boldsymbol{e}}_2$ perpendicular to $\hat{\boldsymbol{q}}$ and to each other, write $j_\mathrm{L} = \hat{\boldsymbol{q}} \cdot \boldsymbol{j}_e(\boldsymbol{q})$ and $j_{\mathrm{T}i} = \hat{\boldsymbol{e}}_i \cdot \boldsymbol{j}_e(\boldsymbol{q})$, and likewise $\bar{v}_\mathrm{L} = \hat{\boldsymbol{q}} \cdot \bar{\boldsymbol{v}}_\chi$ and $\bar{\boldsymbol{v}}_\mathrm{T} = \bar{\boldsymbol{v}}_\chi - \bar{v}_\mathrm{L}\hat{\boldsymbol{q}}$. For a local Hamiltonian the continuity equation Eq. (4.29) gives, between eigenstates, $\langle f|j_\mathrm{L}|0\rangle = (\omega/q)\langle f|n_e|0\rangle$ with $\omega = E_f - E_0$. Every element of the response matrix that carries j_L is a multiple of one without it:

$$S_{nj_\mathrm{L}} = \frac{\omega}{q} S_{nn}, \qquad S_{j_\mathrm{L} j_\mathrm{L}} = \frac{\omega^2}{q^2} S_{nn}, \qquad S_{j_\mathrm{L} j_{\mathrm{T}i}} = \frac{\omega}{q} S_{n j_{\mathrm{T}i}}. \tag{D.174}$$

On the dark-matter side, energy conservation Eq. (1.22) gives $\boldsymbol{q} \cdot \bar{\boldsymbol{v}}_\chi = \omega + \Delta_\chi$, so $\bar{v}_\mathrm{L} = (\omega + \Delta_\chi)/q$ (Eq. (8.5)). Inserting both into Eq. (D.173), the longitudinal terms collect into $(\bar{v}_\mathrm{L} - \omega/q)^2 S_{nn} = (\Delta_\chi/q)^2 S_{nn}$, and

$$\sum_{a,b} \mathfrak{D}_{ab} S_{ab} = \frac{c_8^2}{4}\Big[\Big(|\bar{\boldsymbol{v}}_\mathrm{T}|^2 + \frac{\Delta_\chi^2}{q^2}\Big) S_{nn} - 2\,\mathrm{Re}(\bar{\boldsymbol{v}}_\mathrm{T} \cdot \boldsymbol{S}_{n\boldsymbol{j}_\mathrm{T}}) + \mathrm{Tr}\, S_{\boldsymbol{j}_\mathrm{T}\boldsymbol{j}_\mathrm{T}}\Big], \tag{D.175}$$

with $\mathrm{Tr}\, S_{\boldsymbol{j}_\mathrm{T}\boldsymbol{j}_\mathrm{T}} = S_{j_{\mathrm{T}1} j_{\mathrm{T}1}} + S_{j_{\mathrm{T}2} j_{\mathrm{T}2}}$. For elastic dark matter only the transverse parts survive, on both sides. The first term alone is the rate of a static electron, which sees the dark matter's transverse velocity and nothing else. The last term is the electron's own transverse motion, and the middle term their interference. The same steps apply to $\mathcal{O}_5$, whose target operator $\boldsymbol{q} \times \boldsymbol{j}_e$ is already transverse, and to the current part $-2\bar{\boldsymbol{v}}_\chi \cdot \boldsymbol{j}_e$ of $\mathcal{O}_2$.

D.30.2 The kinematic kernel

The rate per unit target mass needs Eq. (D.175) under the integral of Eq. (8.16). Suppose the responses have been averaged over the directions of $\boldsymbol{q}$, as in Appendix A.8. The averaged cross block $\bar{\boldsymbol{S}}_{n\boldsymbol{j}_\mathrm{T}}$ is then a vector perpendicular to $\hat{\boldsymbol{q}}$ that cannot depend on the orientation of $\hat{\boldsymbol{e}}_1$ and $\hat{\boldsymbol{e}}_2$, since rotating them about $\hat{\boldsymbol{q}}$ is part of the average; it vanishes, and the cross term drops out. For the two diagonal terms the velocity integral at fixed q and ω is done as in Appendix A.8. With $\boldsymbol{v}$ at polar angle θ to $\hat{\boldsymbol{q}}$, the delta function fixes $v\cos\theta = v_\mathrm{min}$, so that

$|\bar{\boldsymbol{v}}_{\rm T}|^2 = v^2\sin^2\theta = v^2 - v_{\rm min}^2$ (the shift $\boldsymbol{q}/2m_\chi$ of $\bar{\boldsymbol{v}}_\chi$ is longitudinal). The angular integral is $\Theta(v - v_{\rm min})/(2qv)$ as before, and per unit $\mathrm{d}^3q/(2\pi)^3$, for an isotropic f_χ or after the average over $\hat{\boldsymbol{q}}$ of Appendix A.8,

$$\begin{aligned}\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\,\delta(\omega-\omega_{\boldsymbol{q}})\,\{1,\ |\bar{\boldsymbol{v}}_{\rm T}|^2\} &= \frac{1}{2q}\{\eta(v_{\rm min}),\ \eta_2(v_{\rm min})\},\\ \eta_2(v_{\rm min}) &= \int \mathrm{d}^3v\, f_\chi\,\frac{v^2-v_{\rm min}^2}{v}\,\Theta(v-v_{\rm min}),\end{aligned} \tag{D.176}$$

where the second form of η_2 equals the definition $2\int_{v_{\rm min}}^\infty \mathrm{d}v'\,v'\eta(v')$ of Appendix A.8: exchanging the order of the integrals, $\int_{v_{\rm min}}^{v} 2v'\,\mathrm{d}v' = v^2 - v_{\rm min}^2$. As for η, only the average of the halo over directions enters once the response is averaged. For elastic dark matter the rate of $\mathcal{O}_8$ per unit target mass is then the sum of a density term weighted by η_2 and a current term weighted by η. The ratio that Sect. 10.4 calls r is

$$r = \frac{\Gamma_B}{\Gamma_A} = \frac{\int \mathrm{d}\omega \int \mathrm{d}q\, q\, c_8^2\,\eta(v_{\rm min})\ \mathrm{Tr}\,\bar{S}_{\boldsymbol{j}_{\rm T}\boldsymbol{j}_{\rm T}}(q,\omega)}{\int \mathrm{d}\omega \int \mathrm{d}q\, q\, c_8^2\,\eta_2(v_{\rm min})\,\bar{S}_{nn}(q,\omega)}, \tag{D.177}$$

both integrals running over the kinematically allowed region above threshold.

D.30.3 The minimal set of rows

Three symmetries reduce the rows of Eq. (8.3) for the electrons.

Continuity. Eq. (D.174) removes $j_{\rm L}$. If the Hamiltonian also conserves the spin of each electron, the spin current obeys its own continuity equation, $\partial_t s_e^\beta(\boldsymbol{q},t) = \mathrm{i}q^\alpha \mathcal{J}_e^{\alpha\beta}(\boldsymbol{q},t)$, by the same commutator with the kinetic energy that leads to Eq. (4.29), and $\hat{q}^\alpha\mathcal{J}_e^{\alpha\beta}$ acts as $(\omega/q)s_e^\beta$. What remains of the currents are the transverse components $j_{{\rm T}i}$ and $\mathcal{J}_e^{{\rm T}i,\beta} \equiv \hat{e}_i^\alpha \mathcal{J}_e^{\alpha\beta}$.

Spin rotations. Without spin–orbit coupling and without magnetic order, the Hamiltonian commutes with the total spin and the ground state of an insulator with filled bands is a singlet. The density, the current and the kinetic-energy density are scalars under spin rotations; $\boldsymbol{s}_e$ and $\mathcal{J}_e^{\alpha\cdot}$ are vectors in their spin index. A scalar acting on the singlet produces singlets, and a vector produces triplets, which are orthogonal to them. Every element of the response matrix between a scalar row and a vector row vanishes, and by the Wigner–Eckart theorem the vector–vector elements are $\delta_{\beta\beta'}$ times one function. The rows fall into a singlet block, $\{n_e, j_{\rm T1}, j_{\rm T2}\}$ plus $\mathcal{T}_e$, and a triplet block, $\{s_e^z, \mathcal{J}_e^{{\rm T1},z}, \mathcal{J}_e^{{\rm T2},z}\}$, the same for each spin component.

For independent particles the two blocks are related. The determinant of filled bands has both spins of every valence orbital occupied. A singlet row O has the one-electron vertex $o_{cv}\,\delta_{\sigma\sigma'}$ between orbitals with spins σ and σ'; its triplet partner $O\otimes\sigma^z/2$ has $o_{cv}\,(\sigma^z)_{\sigma'\sigma}/2$. Summing the products of two vertices over the four spin combinations gives $\sum_{\sigma\sigma'}\delta_{\sigma\sigma'} = 2$ for the singlet and $\sum_{\sigma\sigma'}|(\sigma^z)_{\sigma'\sigma}|^2/4 = \frac{1}{2}$ for the triplet, and the spatial sums are the same. The x and y components give $\mathrm{Tr}(\sigma^x\sigma^x)/4 = \frac{1}{2}$ too. Hence

$$\begin{aligned}S^0_{s^\beta s^{\beta'}} &= \frac{\delta_{\beta\beta'}}{4}\,S^0_{nn}, \qquad S^0_{s^\beta \mathcal{J}^{{\rm T}i,\beta'}} = \frac{\delta_{\beta\beta'}}{4}\,S^0_{n j_{{\rm T}i}},\\ S^0_{\mathcal{J}^{{\rm T}i,\beta}\mathcal{J}^{{\rm T}k,\beta'}} &= \frac{\delta_{\beta\beta'}}{4}\,S^0_{j_{{\rm T}i}j_{{\rm T}k}},\end{aligned} \tag{D.178}$$

with the superscript 0 for independent particles. Beyond independent particles the blocks differ, because the Coulomb interaction acts on the total density $n_\uparrow + n_\downarrow$ and not on the spin density $n_\uparrow - n_\downarrow$. The Hartree kernel and its local fields enter the singlet block only. In the language of Sect. 4.9 the triplet keeps the direct kernel $-K^{\rm d}$ and loses the exchange kernel $K^{\rm x}$, and in density-functional language its kernel is the spin part of $f_{\rm xc}$, $(f_{\uparrow\uparrow} - f_{\uparrow\downarrow})/2$, without $v_{\rm C}$.

Inversion with time reversal. In silicon and germanium the antiunitary product PT of inversion and time reversal is a symmetry. Time reversal maps $n_e(\boldsymbol{q})$ to $n_e(-\boldsymbol{q})$ and $\boldsymbol{j}_e(\boldsymbol{q})$ to $-\boldsymbol{j}_e(-\boldsymbol{q})$, since it reverses momenta; inversion maps $\boldsymbol{q}$ back to $-\boldsymbol{q}$ and reverses $\boldsymbol{p}$ again. Both rows are even under PT at the same $\boldsymbol{q}$, and so is $\mathcal{T}_e$. An antiunitary symmetry of the ground state and of H maps $\langle 0|O_a^\dagger|f\rangle\langle f|O_b|0\rangle$, summed over a degenerate set of final states, to its complex conjugate, and every element of the singlet block is real. The same holds for the triplet block: its rows change sign under PT, since time reversal also flips the spin, and the products of two of them stay real. For each $(\boldsymbol{q},\omega)$ the singlet block on $\{n_e, j_{\mathrm{T}1}, j_{\mathrm{T}2}\}$ is then a real symmetric 3×3 matrix, six real functions: S_{nn}, two cross elements $S_{nj_{\mathrm{T}i}}$ and three current elements. The calculation of Appendix E.16 computes them twice, screened and for independent particles, and the independent-particle set fixes the independent-particle triplet block by Eq. (D.178): twelve functions. The kinetic-energy row adds $S_{\mathcal{T}\mathcal{T}}$, $S_{n\mathcal{T}}$ and two elements $S_{\mathcal{T}j_{\mathrm{T}i}}$, four more in each version.

Symmetry axes. When $\hat{\boldsymbol{q}}$ lies on a two-, three- or fourfold axis of the cubic group, the symmetries that leave $\boldsymbol{q}$ fixed include a rotation about $\hat{\boldsymbol{q}}$ by π, $2\pi/3$ or $\pi/2$. Such a rotation leaves $n_e(\boldsymbol{q})$ unchanged and rotates $(j_{\mathrm{T}1}, j_{\mathrm{T}2})$ as a vector in the transverse plane. A vector in that plane that is unchanged by the rotation is zero, so $S_{nj_{\mathrm{T}i}} = 0$. On the three- and fourfold axes the current block is also a multiple of the unit matrix.

D.30.4 Response functions $\overline{W}_1$–$\overline{W}_5$

Catena and collaborators write the general rate with the density vertex $f = \langle i'\boldsymbol{k}'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}|i\boldsymbol{k}\rangle$ and the vector vertex $\boldsymbol{f} = \langle i'\boldsymbol{k}'|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}\,\mathrm{i}\nabla/m_e|i\boldsymbol{k}\rangle$ [70, Eqs. (10), (11)]. They use five bilinears, $B_1 = |f|^2$, $B_2 = (\boldsymbol{q}/m_e)\cdot f\boldsymbol{f}^*$, $B_3 = |\boldsymbol{f}|^2$, $B_4 = |(\boldsymbol{q}/m_e)\cdot\boldsymbol{f}|^2$ and $B_5 = \mathrm{i}(\boldsymbol{q}/m_e)\cdot(\boldsymbol{f}\times\boldsymbol{f}^*)$, each summed over pairs into a response $\overline{W}_l(q,\omega)$ [70, Eqs. (37)–(42), (47)]. Since $\mathrm{i}\nabla = -\boldsymbol{p}$, a plane wave $\boldsymbol{k}$ gives $\boldsymbol{f}$ the element $-\boldsymbol{k}/m_e$, while the current vertex of Eq. (8.3) is $(\boldsymbol{k}+\boldsymbol{k}')/2m_e = \boldsymbol{k}/m_e + \boldsymbol{q}/2m_e$. Between any two states $\boldsymbol{f} = (\boldsymbol{q}/2m_e)f - \boldsymbol{j}$, with $\boldsymbol{j}$ the current vertex. The continuity equation reads $\boldsymbol{q}\cdot\boldsymbol{j} = \omega f$, and

$$\boldsymbol{q}\cdot\boldsymbol{f} = -\Big(\omega - \frac{q^2}{2m_e}\Big)f, \qquad \hat{\boldsymbol{e}}_i\cdot\boldsymbol{f} = -j_{\mathrm{T}i}. \tag{D.179}$$

Hence $\overline{W}_2 = -(\omega - q^2/2m_e)\overline{W}_1/m_e$ and $\overline{W}_4 = (\omega - q^2/2m_e)^2\overline{W}_1/m_e^2$; the transverse combination $\overline{W}_3 - (m_e^2/q^2)\overline{W}_4$ is the sum of $|j_{\mathrm{T}i}|^2$; and $\overline{W}_5 = 0$ and $\operatorname{Im}\overline{W}_2 = 0$ when the vertices are real, as PT makes them; nonzero values in a calculation for silicon or germanium are numerical. Of the five responses two are independent. Comparing B_1 with Eq. (10.1), and using the relation $\overline{W}_1 = (8\omega\alpha m_e^2/q^3)|f_{\mathrm{crys}}|^2$ [70, Eq. (D2)] with Eq. (10.3), the dictionary is $\overline{W}_1 = (2\Omega\omega/\pi)\bar{S}^0_{nn}$ and, by the same sum over pairs, $\overline{W}_3 - (m_e^2/q^2)\overline{W}_4 = (2\Omega\omega/\pi)\operatorname{Tr}\bar{S}^0_{\boldsymbol{j}_{\mathrm{T}}\boldsymbol{j}_{\mathrm{T}}}$. A nonlocal pseudopotential breaks Eq. (D.179) through the commutator $[V_{\mathrm{NL}}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}]$, and the current vertex then needs the correction Eq. (E.46).

D.30.5 Screening of each row

The screened response of any rows follows from Eq. (8.35). Work in the reciprocal-lattice basis of Sect. 4.6, with the Coulomb interaction acting on the densities $n_e(\boldsymbol{q}+\boldsymbol{G})$ and the rows a, b entering only as external vertices. The random-phase approximation with local fields gives

$$R_{ab}(\boldsymbol{q},\omega) = \tilde{R}_{ab} + \sum_{\boldsymbol{G}\boldsymbol{G}'} \tilde{R}_{a,\boldsymbol{G}}\,\big[v_{\mathrm{C}}(1-\tilde{R}\,v_{\mathrm{C}})^{-1}\big]_{\boldsymbol{G}\boldsymbol{G}'}\,\tilde{R}_{\boldsymbol{G}',b}, \qquad S_{ab} = \mathrm{i}(R_{ab} - R^*_{ba}), \tag{D.180}$$

where v_{C} is the diagonal matrix $e^2/|\boldsymbol{q}+\boldsymbol{G}|^2$, $\tilde{R}_{a,\boldsymbol{G}}$ is the proper response of row a to the density at $\boldsymbol{q}+\boldsymbol{G}$, and $\tilde{R} = R^0$ at this level. The second relation is the anti-Hermitian part of

R at $T = 0$, the generalization of $S = -2\,\mathrm{Im}\,R$ to off-diagonal elements. The correction to row a is set by its proper correlation with the densities. For $a = n_e(\boldsymbol{q})$ itself it is the screening of Sect. 10.2, with the head $\boldsymbol{G} = 0$ dominant at small q. For $a = j_{\mathrm{T}i}$ with $\hat{\boldsymbol{q}}$ on a symmetry axis, $\tilde{R}_{j_{\mathrm{T}i},\boldsymbol{0}} = 0$ by the argument of the previous subsection, applied to the proper response, so only $\boldsymbol{G} \neq 0$ contributes: the transverse current is screened through local fields alone. For $a = \mathcal{T}_e$, a scalar under the point group, $\tilde{R}_{\mathcal{T},\boldsymbol{0}} \neq 0$ and the head acts. For the triplet rows every $\tilde{R}_{a,\boldsymbol{G}}$ vanishes by spin rotation, and Eq. (D.180) leaves them unscreened.

The transverse block has a measurable limit. At $\boldsymbol{q} \to 0$ the dielectric tensor of Sect. 4.4 is $\varepsilon^{\alpha\beta} = \delta^{\alpha\beta} - (e^2/\omega^2)[\boldsymbol{n}_e\delta^{\alpha\beta}/m_e + \tilde{R}^{\alpha\beta}_{jj}]$, so that $\mathrm{Im}\,\varepsilon^{\mathrm{T}} = -(e^2/\omega^2)\,\mathrm{Im}\,\tilde{R}_{j_{\mathrm{T}}j_{\mathrm{T}}}$, and with $S = -2\,\mathrm{Im}\,R$ and $R_{j_{\mathrm{T}}j_{\mathrm{T}}} = \tilde{R}_{j_{\mathrm{T}}j_{\mathrm{T}}}$ on a symmetry axis without local fields,

$$\mathrm{Tr}\,S_{\boldsymbol{j}_{\mathrm{T}}\boldsymbol{j}_{\mathrm{T}}} = \frac{4\omega^2}{e^2}\,\mathrm{Im}\,\varepsilon^{\mathrm{T}}(\boldsymbol{q},\omega), \qquad S_{nn} = \frac{2q^2}{e^2}\,\mathrm{Im}\Big[-\frac{1}{\varepsilon(\boldsymbol{q},\omega)}\Big], \qquad \frac{\mathrm{Tr}\,S_{\boldsymbol{j}_{\mathrm{T}}\boldsymbol{j}_{\mathrm{T}}}}{S_{nn}} \to \frac{2\omega^2}{q^2}\,|\varepsilon(\omega)|^2, \tag{D.181}$$

the last form at $q \to 0$ in a cubic crystal, where $\varepsilon^{\mathrm{T}} \to \varepsilon$ (Eq. (4.52)). The second relation is Eq. (4.45). For independent particles the same steps give $2(\omega/q)^2$ without the factor $|\varepsilon|^2$: the density row is screened and the transverse current is not.

D.30.6 Interference band on a mixed block

Group A is the density row with its dark-matter weight $\bar{\boldsymbol{v}}_{\mathrm{T}}$, and group B the transverse current, with rates Γ_A and Γ_B of each group alone. For any target, any halo and any binning, Eq. (8.21) gives

$$\big(\sqrt{r} - 1\big)^2 \le \frac{\Gamma}{\Gamma_A} \le \big(\sqrt{r} + 1\big)^2, \qquad r = \frac{\Gamma_B}{\Gamma_A}. \tag{D.182}$$

The band is reached only if the two rows are proportional on every final state that the rate samples, with the phase that makes them cancel or add. For $r \ge 1$ the lower edge rises with r and exceeds the static-electron rate Γ_A once $r > 4$; at $r = 20$ it is $(\sqrt{20} - 1)^2 = 12.1$. The ratio of the half-width of the band, $2\sqrt{r}$, to the rate without a cross term, $1 + r$, is $2\sqrt{r}/(1 + r)$, below 0.1 only for $r < 2.5 \times 10^{-3}$ or $r > 4.0 \times 10^2$. Between those values the cross term matters unless it is computed or vanishes by symmetry, as it does for responses averaged over directions.

In the medium, the screened rates enter Eq. (D.182), and the ratio becomes $r_{\mathrm{med}} = \Gamma_B^{\mathrm{med}}/\Gamma_A^{\mathrm{med}}$. Let screening lower the density rate by the factor $\Gamma_A/\Gamma_A^{\mathrm{med}}$ and change the current rate by the factor $1 + \delta^j_{\mathrm{LF}}$, the local-field correction of the transverse current averaged over the rate. Then

$$r_{\mathrm{med}} = r_{\mathrm{IP}}\,\frac{\Gamma_A}{\Gamma_A^{\mathrm{med}}}\,(1 + \delta^j_{\mathrm{LF}}), \tag{D.183}$$

with r_{IP} the ratio for independent particles, and $r_{\mathrm{med}} \ge r_{\mathrm{IP}}$ whenever the product of the two factors is at least one. Since the lower edge of Eq. (D.182) increases with r for $r \ge 1$, $(\sqrt{r_{\mathrm{med}}} - 1)^2 \ge (\sqrt{r_{\mathrm{IP}}} - 1)^2$ under the same condition.

D.31 Nuclear response at all phonon orders

A coupling that multiplies the density of each species κ by a number c_κ probes the crystal through $O(\boldsymbol{q}) = \sum_\kappa c_\kappa n_\kappa(\boldsymbol{q})$, with $n_\kappa(\boldsymbol{q}) = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_{\boldsymbol{R}\kappa}}$ and $\boldsymbol{X}_{\boldsymbol{R}\kappa} = \boldsymbol{R} + \boldsymbol{\tau}_\kappa + \boldsymbol{u}_{\boldsymbol{R}\kappa}$. We derive its response in a harmonic crystal in thermal equilibrium, Eq. (11.24), and its terms of order n in the phonons, Eq. (11.25). Then we prove the statements of Sect. 11.3 about them: the first moment of each order, the average over a zone, the ω-integrated weight with the decay of the static correlation, the pointwise band, and the bound on the second-order thermal diffuse

scattering. The notation is that of Sect. 5.6 and Appendix D.15; N is the number of cells, $\mathscr{V} = N\Omega$ the volume, and the phonon wavevectors $\boldsymbol{Q}$ run over the N points of the first zone.

D.31.1 Sum over pairs

By Eq. (4.13), $S_A(\boldsymbol{q},\omega) = \mathscr{V}^{-1}\int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\langle O(\boldsymbol{q},t)^\dagger O(\boldsymbol{q},0)\rangle$, and with $O^\dagger = \sum_\kappa c_\kappa \sum_{\boldsymbol{R}} \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_{\boldsymbol{R}\kappa}}$ the correlation is a double sum over nuclei $(\boldsymbol{R},\kappa)$ and $(\boldsymbol{R}',\kappa')$ of

$$\big\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_{\boldsymbol{R}\kappa}(t)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_{\boldsymbol{R}'\kappa'}(0)}\big\rangle = \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{R}-\boldsymbol{R}'+\boldsymbol{\tau}_\kappa-\boldsymbol{\tau}_{\kappa'})}\,\big\langle \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}\kappa}(t)}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}_{\boldsymbol{R}'\kappa'}(0)}\big\rangle.$$

The displacements are linear in the phonon operators, and Eq. (D.83) gives the average as $\mathrm{e}^{-W_\kappa - W_{\kappa'}}\mathrm{e}^{\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa'}(\boldsymbol{R}-\boldsymbol{R}',t)\cdot\boldsymbol{q}}$. The displacement correlation depends on the two cells only through $\Delta\boldsymbol{R} = \boldsymbol{R} - \boldsymbol{R}'$, because the eigenvectors carry the cell phase $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$ (Eq. (5.32)) and the thermal state is invariant under lattice translations. At fixed $\Delta\boldsymbol{R}$ the sum over $\boldsymbol{R}'$ therefore contributes the factor N, and $N/\mathscr{V} = 1/\Omega$. This is Eq. (11.24), with $\Delta\boldsymbol{R}_{\kappa\kappa'} = \Delta\boldsymbol{R} + \boldsymbol{\tau}_\kappa - \boldsymbol{\tau}_{\kappa'}$. No approximation beyond the harmonic Hamiltonian and the Gibbs state has entered; the expansion in phonon number below is a reorganization of the same exact expression.

D.31.2 The n-phonon term in reciprocal space

In Eq. (5.57), write the absorption term at $-\boldsymbol{Q}$ instead of $\boldsymbol{Q}$. With $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$ and frequencies and occupations even in $\boldsymbol{Q}$, both terms take the same form,

$$\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa'}(\Delta\boldsymbol{R},t)\cdot\boldsymbol{q} = \frac{1}{N}\sum_{\boldsymbol{Q}\nu}\sum_{s=\pm1}\frac{n_{\boldsymbol{Q}\nu}+\frac{1}{2}(1+s)}{2\omega_{\boldsymbol{Q}\nu}\sqrt{M_\kappa M_{\kappa'}}}\; y^s_\kappa\,(y^s_{\kappa'})^*\;\mathrm{e}^{\mathrm{i}s\boldsymbol{Q}\cdot\Delta\boldsymbol{R}}\,\mathrm{e}^{-\mathrm{i}s\omega_{\boldsymbol{Q}\nu}t}, \tag{D.184}$$

with $y^+_\kappa = \boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$ and $y^-_\kappa = (y^+_\kappa)^*$. The term $s=+1$ creates the phonon $\boldsymbol{Q}\nu$, the term $s=-1$ absorbs it. The nth power of Eq. (D.184), divided by $n!$, is the term of order n of the exponential in Eq. (11.24). It is a sum over n phonons $(\boldsymbol{Q}_i,\nu_i,s_i)$ with the phase $\mathrm{e}^{\mathrm{i}\sum_i s_i\boldsymbol{Q}_i\cdot\Delta\boldsymbol{R}}$. Together with $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\Delta\boldsymbol{R}}$, the sum over the N lattice vectors gives

$$\sum_{\Delta\boldsymbol{R}}\mathrm{e}^{-\mathrm{i}(\boldsymbol{q}-\sum_i s_i\boldsymbol{Q}_i)\cdot\Delta\boldsymbol{R}} = N\sum_{\boldsymbol{G}}\delta_{\boldsymbol{q}-\sum_i s_i\boldsymbol{Q}_i,\,\boldsymbol{G}},$$

the conservation of crystal momentum. The time integral of $\mathrm{e}^{\mathrm{i}\omega t}\prod_i \mathrm{e}^{-\mathrm{i}s_i\omega_i t}$ is $2\pi\delta(\omega - \sum_i s_i\omega_i)$. What remains of the sum over κ and κ' is

$$\sum_{\kappa\kappa'} c_\kappa c_{\kappa'}\,\mathrm{e}^{-W_\kappa - W_{\kappa'}}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{\tau}_\kappa-\boldsymbol{\tau}_{\kappa'})}\,\frac{\prod_i y^{s_i}_\kappa}{M_\kappa^{n/2}}\left(\frac{\prod_i y^{s_i}_{\kappa'}}{M_{\kappa'}^{n/2}}\right)^* = \Big|\sum_\kappa c_\kappa\,\mathrm{e}^{-W_\kappa}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}\,\frac{\prod_i (y^{s_i}_\kappa)^*}{M_\kappa^{n/2}}\Big|^2,$$

and $(y^s_\kappa)^* = \boldsymbol{q}\cdot\boldsymbol{e}^s_{\kappa\nu}$ with $\boldsymbol{e}^+ = \boldsymbol{e}^*$ and $\boldsymbol{e}^- = \boldsymbol{e}$. The prefactor is $(1/\Omega)(1/n!)N^{-n}\cdot N$, and the Kronecker delta removes one of the n sums over wavevectors, leaving the average $N^{-(n-1)}\sum$ over the others. This is Eq. (11.25). The factor $1/n!$ comes from the exponential and needs no counting of final states; it also gives the right weight when two phonons coincide, which a sum over unordered final states would have to supply separately. The phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}$ carries the unreduced $\boldsymbol{q}$, since it came from the static positions, not from the phonons.

D.31.3 First moments

For any function $f(t)$ with $S(\omega) = \Omega^{-1}\int \mathrm{d}t\,\mathrm{e}^{\mathrm{i}\omega t}f(t)$, $\int \mathrm{d}\omega\,\omega S(\omega) = (2\pi/\Omega)\,\mathrm{i}f'(0)$. We need the time derivative of Eq. (D.184) at $t=0$. The thermal factors cancel between $s=+1$ and $s=-1$, as in Eq. (5.61), and

$$\partial_t\,\mathcal{C}_{\kappa\kappa'}(\Delta\boldsymbol{R},t)\big|_{t=0} = -\frac{\mathrm{i}}{2\sqrt{M_\kappa M_{\kappa'}}}\,\frac{1}{N}\sum_{\boldsymbol{Q}}\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\Delta\boldsymbol{R}}\sum_\nu \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\,\boldsymbol{e}_{\kappa'\nu}(\boldsymbol{Q})^\dagger = -\frac{\mathrm{i}}{2M_\kappa}\,\delta_{\kappa\kappa'}\,\delta_{\Delta\boldsymbol{R},\boldsymbol{0}}\,\mathbf{1},$$

by the completeness of the polarization vectors at each $\boldsymbol{Q}$, Eq. (5.17), and $N^{-1}\sum_{\boldsymbol{Q}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\Delta\boldsymbol{R}} = \delta_{\Delta\boldsymbol{R},\boldsymbol{0}}$. The term of order n of a pair is $f_n(t) = c_\kappa c_{\kappa'} \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\Delta\boldsymbol{R}_{\kappa\kappa'}} \mathrm{e}^{-W_\kappa - W_{\kappa'}} [\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa'}\cdot\boldsymbol{q}]^n/n!$, and its derivative at $t=0$ is proportional to $\boldsymbol{q}\cdot\partial_t\mathcal{C}_{\kappa\kappa'}\cdot\boldsymbol{q}$, which vanishes for every distinct pair. Every distinct term thus has zero first moment, at every order and temperature. For the self terms $\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa}(\boldsymbol{0},0)\cdot\boldsymbol{q} = 2W_\kappa$ and

$$\mathrm{i}f_n'(0) = c_\kappa^2\,\mathrm{e}^{-2W_\kappa}\,\frac{n(2W_\kappa)^{n-1}}{n!}\,\frac{q^2}{2M_\kappa} = c_\kappa^2\,E_{R,\kappa}\,\mathrm{e}^{-2W_\kappa}\,\frac{(2W_\kappa)^{n-1}}{(n-1)!},$$

with $E_{R,\kappa} = q^2/2M_\kappa$. Summed over n, the Poisson weights add to one and the first moment is $(2\pi/\Omega)\sum_\kappa c_\kappa^2 E_{R,\kappa}$, the recoil sum rule Eq. (5.61) for the crystal. For one species the share of order n is Eq. (11.27); with several species the shares of the species are averaged with the weights $c_\kappa^2 E_{R,\kappa} \propto c_\kappa^2/M_\kappa$.

D.31.4 The average over a zone

Every term of Eq. (11.24) depends on $\boldsymbol{q}$ in two ways. The phase $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\Delta\boldsymbol{R}_{\kappa\kappa'}}$ changes by order one when $\boldsymbol{q}$ moves by $2\pi/|\Delta\boldsymbol{R}_{\kappa\kappa'}|$. The vertex, the factors $[\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa'}\cdot\boldsymbol{q}]^n$ and $\mathrm{e}^{-W_\kappa - W_{\kappa'}}$, changes under a shift $\delta\boldsymbol{q}$ only by relative amounts $\sim 2n\,|\delta\boldsymbol{q}|/q$ and $4W_\kappa|\delta\boldsymbol{q}|/q$. Average $\boldsymbol{q}$ over a cell of reciprocal space with the shape and volume of the first zone, $\boldsymbol{q}\to\boldsymbol{q}+\boldsymbol{p}$ with $\boldsymbol{p}$ in the zone, holding the vertex at $\boldsymbol{q}$. The phase averages to

$$j_{\mathrm{BZ}}(\boldsymbol{r}) = \frac{\Omega}{(2\pi)^3}\int_{\mathrm{BZ}} \mathrm{d}^3p\;\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}}, \qquad \boldsymbol{r} = \Delta\boldsymbol{R}_{\kappa\kappa'}. \tag{D.185}$$

For a lattice vector $\boldsymbol{r} = \Delta\boldsymbol{R}$ the plane waves $\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\Delta\boldsymbol{R}}$ are orthogonal on the zone and $j_{\mathrm{BZ}}(\Delta\boldsymbol{R}) = \delta_{\Delta\boldsymbol{R},\boldsymbol{0}}$. In a lattice with one atom per cell every pair is separated by a lattice vector, only the self terms survive, and the zone-averaged response is the incoherent one, Eq. (11.26). The vertex held at $\boldsymbol{q}$ is the approximation. Letting it vary adds, for the self terms, a correction linear in $\boldsymbol{p}$, which averages to zero over the inversion-symmetric zone, and one $\sim (nq_{\mathrm{BZ}}/q)^2$. For the distinct terms the linear correction carries $\int_{\mathrm{BZ}}\mathrm{d}^3p\,\boldsymbol{p}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\Delta\boldsymbol{R}}$, which does not vanish at lattice vectors; it is $\sim nq_{\mathrm{BZ}}/q$ times the size of those terms. The statement is therefore asymptotic: the zone average of the exact response tends to the incoherent response for $q \gg \max(n, 2W_\kappa)\,q_{\mathrm{BZ}}$.

With a basis, a pair on two sublattices is separated by $\Delta\boldsymbol{R} + \boldsymbol{\tau}_\kappa - \boldsymbol{\tau}_{\kappa'}$, which is never a lattice vector, and j_{BZ} does not vanish there. Summed over all $\Delta\boldsymbol{R}$ it is one, since $\sum_{\Delta\boldsymbol{R}}\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\Delta\boldsymbol{R}} = N\delta_{\boldsymbol{p},\boldsymbol{0}}$ inside the zone. For a zone replaced by the sphere of equal volume, of radius q_{BZ}, $j_{\mathrm{BZ}}(\boldsymbol{r}) = 3j_1(q_{\mathrm{BZ}}r)/q_{\mathrm{BZ}}r$, with j_1 the spherical Bessel function; for the bond of silicon, $r = 2.352$ Å and $q_{\mathrm{BZ}}r = 2.68$, it is 0.44. The four bonds of an atom alone give 1.78, and the farther pairs make up the difference to one with the opposite sign. The cross-sublattice terms therefore survive the zone average, reduced only by the decay of the correlations with distance, and their size has to be computed; Eq. (11.29) does so for the ω-integrated response. An average over the directions of $\boldsymbol{q}$ at fixed $|\boldsymbol{q}|$ acts on each pair as $\langle\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}P(\hat{\boldsymbol{q}})\rangle$ with P a polynomial of degree $2n$ in $\hat{\boldsymbol{q}}$. In the frame with $\boldsymbol{r}$ along the polar axis, the average of P over the azimuth is a polynomial in $\mu = \cos\theta$ of degree $2n$, with an expansion $\sum_l a_l P_l(\mu)$ in Legendre polynomials. Since $\frac{1}{2}\int_{-1}^{1}\mathrm{d}\mu\,\mathrm{e}^{-\mathrm{i}x\mu}P_l(\mu) = (-\mathrm{i})^l j_l(x)$, $\langle\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}P\rangle = \sum_l(-\mathrm{i})^l a_l j_l(qr)$: each pair is reduced by spherical Bessel functions of $q|\Delta\boldsymbol{R}_{\kappa\kappa'}|$, which fall as $1/qr$. The curves of Fig. 11.11 use this form.

D.31.5 The ω-integrated weight

Since $\int\mathrm{d}\omega\,\mathrm{e}^{\mathrm{i}\omega t}/2\pi = \delta(t)$, the integral over ω of the term of order n of Eq. (11.24) sets $t=0$, where the two terms of Eq. (5.57) add to $2n_{\boldsymbol{Q}\nu}+1$ and $\mathcal{C}_{\kappa\kappa'}(\Delta\boldsymbol{R},0) = B_{\kappa\kappa'}(\Delta\boldsymbol{R})$ of Eq. (11.28).

Multiplying by Ω gives Eq. (11.29).

Its convergence is set by the acoustic modes at small $\boldsymbol{Q}$, which dominate $B_{\kappa\kappa'}$ at large separations. There $\boldsymbol{e}_{\kappa\nu} \to (M_\kappa/M_{\text{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$, so the factor $\boldsymbol{e}\boldsymbol{e}^\dagger/\sqrt{M_\kappa M_{\kappa'}}$ becomes $\hat{\boldsymbol{e}}_\nu\hat{\boldsymbol{e}}_\nu/M_{\text{cell}}$ for every pair, and $N^{-1}\sum_{\boldsymbol{Q}} \to \Omega\int \mathrm{d}^3Q/(2\pi)^3$. At $T=0$ the weight is $1/2\omega = 1/(2c_\nu Q)$, and since $\int \mathrm{d}^3Q\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}}/Q = 4\pi/r^2$, the correlation falls as $\Omega/(M_{\text{cell}} c\, r^2)$ up to a factor of order one that depends on the direction of $\boldsymbol{r}$ and the anisotropy of the speeds. The terms of order n fall as r^{-2n} against $4\pi r^2\,\mathrm{d}r/\Omega$ pairs in a shell, and the sum converges absolutely for $n \geq 2$; at $n=1$ it does not, and the phase makes it a distribution in $\boldsymbol{q}$, which reciprocal space evaluates as Eq. (11.1). At $T>0$ the weight of the soft modes is $(2n+1)/2\omega \approx T/(c_\nu Q)^2$, the Fourier transform of $1/Q^2$ is $2\pi^2/r$, and $B_{\kappa\kappa'}$ falls only as T/r; at $n=2$ the shells then contribute $\propto r^2 \cdot r^{-2}$ each, and the sum diverges linearly. The divergence is an artifact of summing in real space, and the last subsection removes it.

D.31.6 The pointwise band

In Eq. (11.25) each term of the average over phonons is $|\sum_\kappa a_\kappa|^2$ times nonnegative factors, with n_{at} amplitudes a_κ. The Cauchy–Schwarz inequality gives $0 \leq |\sum_\kappa a_\kappa|^2 \leq n_{\text{at}} \sum_\kappa |a_\kappa|^2$ term by term, and with $\mathfrak{P}^{(n)}$ defined as Eq. (11.25) with $\sum_\kappa |a_\kappa|^2$ in place of $|\sum_\kappa a_\kappa|^2$, the inequality of Eq. (11.30) follows. For $n_{\text{at}} = 1$ the two expressions are the same. For the average, shift $\boldsymbol{q}$ by $\boldsymbol{p}$ in the Kronecker delta alone and average $\boldsymbol{p}$ over the N wavevectors of the zone. For each set of phonons exactly one $\boldsymbol{p}$ satisfies the constraint, so the average replaces the constraint by the factor $1/N$, and the average over $\boldsymbol{Q}_1, \ldots, \boldsymbol{Q}_{n-1}$ becomes an unconstrained average over all n wavevectors. For each nucleus the summand is then a product of n independent single-phonon factors,

$$\frac{1}{N}\sum_{\boldsymbol{Q}\nu,s}\Big[n_{\boldsymbol{Q}\nu} + \tfrac{1}{2}(1+s)\Big]\frac{|\boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2}{2M_\kappa\omega_{\boldsymbol{Q}\nu}}\,\delta(\omega_i - s\,\omega_{\boldsymbol{Q}\nu}) = 2W_\kappa(\boldsymbol{q})\,P_1^\kappa(\hat{\boldsymbol{q}},\omega_i),$$

convolved in energy, which is $(2W_\kappa)^n P_n^\kappa$. With $c_\kappa^2 \mathrm{e}^{-2W_\kappa}(2\pi/\Omega)/n!$ in front this is the term of order n of Eq. (11.26), the equality of Eq. (11.30). The self density keeps the momentum constraint of the exact response but drops the interference between the nuclei of a cell; the zone average removes the constraint.

D.31.7 Second-order thermal diffuse scattering

Summing Eq. (11.25) at $n=2$ over s_1 and s_2 and integrating over ω, we relabel $\boldsymbol{Q}_i \to -\boldsymbol{Q}_i$ in every absorbed phonon. Its factor $\boldsymbol{q}\cdot\boldsymbol{e}^-(-\boldsymbol{Q}_i) = \boldsymbol{q}\cdot\boldsymbol{e}(\boldsymbol{Q}_i)^*$ is then that of an emitted phonon at $\boldsymbol{Q}_i$, the constraint becomes $\boldsymbol{Q}_1 + \boldsymbol{Q}_2 = \boldsymbol{q} - \boldsymbol{G}$ for all four sign choices, and the thermal factors add to $2n+1$:

$$\mathfrak{W}^{(2)}(\boldsymbol{q}) = \frac{\Omega}{2}\int_{\text{BZ}}\frac{\mathrm{d}^3Q_1}{(2\pi)^3}\sum_{\nu_1\nu_2}\frac{(2n_1+1)(2n_2+1)}{4\omega_1\omega_2}\Big|\sum_\kappa c_\kappa\,\mathrm{e}^{-W_\kappa}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}\,\frac{(\boldsymbol{q}\cdot\boldsymbol{e}^*_{\kappa\nu_1})(\boldsymbol{q}\cdot\boldsymbol{e}^*_{\kappa\nu_2})}{M_\kappa}\Big|^2, \tag{D.186}$$

with $\boldsymbol{Q}_2 = \boldsymbol{q} - \boldsymbol{G} - \boldsymbol{Q}_1$ and the labels $1,2$ for $(\boldsymbol{Q}_1,\nu_1)$ and $(\boldsymbol{Q}_2,\nu_2)$. This is the momentum-space form of Eq. (11.29) at $n=2$, the two being related by Poisson summation, and it converges at any temperature: each factor $(2n+1)/2\omega$ grows at most as $T/(cQ)^2$, integrable in three dimensions. It is singular only where both factors are singular together, $\boldsymbol{Q}_1 \to 0$ and $\boldsymbol{Q}_2 \to 0$, which needs $\boldsymbol{k} = \boldsymbol{q} - \boldsymbol{G} \to 0$.

Split the integral at a radius Q_s below which the acoustic branches are linear. Where $|\boldsymbol{Q}_1| > Q_s$ or $|\boldsymbol{Q}_2| > Q_s$, at most one factor is singular, and the integral is bounded uniformly in $\boldsymbol{k}$ and continuous at $\boldsymbol{k}=0$; these are the regular terms. Where both are soft, only acoustic

branches matter, $\boldsymbol{e}_{\kappa\nu} = (M_\kappa/M_{\rm cell})^{1/2}\hat{\boldsymbol{e}}_\nu$, and the amplitude becomes $(\boldsymbol{q}\cdot\hat{\boldsymbol{e}}_1)(\boldsymbol{q}\cdot\hat{\boldsymbol{e}}_2)/M_{\rm cell}$ times $\sum_\kappa c_\kappa e^{-W_\kappa} e^{i\boldsymbol{G}\cdot\boldsymbol{\tau}_\kappa}$, up to a factor $1+O(ka)$ from $e^{i\boldsymbol{k}\cdot\boldsymbol{\tau}_\kappa}$. With $\coth x \le 1/x + x/3$, the thermal factors are $(2n+1)/2\omega \le T/\omega^2 + 1/12T$. The part of the product second order in T/ω, $T^2/\omega_1^2\omega_2^2$, is the singular one. With $\omega_\nu \ge c_{\min}Q$ and the completeness of the three acoustic polarizations, $\sum_\nu(\boldsymbol{q}\cdot\hat{\boldsymbol{e}}_\nu)^2/\omega_\nu^2 \le q^2/(c_{\min}Q)^2$ for each phonon. The remaining integral, extended over all space, is

$$\int \frac{\mathrm{d}^3Q}{Q^2\,|\boldsymbol{k}-\boldsymbol{Q}|^2} = (2\pi)^3 \int \mathrm{d}^3r\; e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} \Big(\frac{1}{4\pi r}\Big)^2 = \frac{(2\pi)^3}{16\pi^2}\,\frac{2\pi^2}{k} = \frac{\pi^3}{k}, \tag{D.187}$$

by the convolution theorem and the two transforms

$$\int \frac{\mathrm{d}^3Q}{(2\pi)^3}\,\frac{e^{i\boldsymbol{Q}\cdot\boldsymbol{r}}}{Q^2} = \frac{1}{4\pi r}, \qquad \int \mathrm{d}^3r\,\frac{e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}}{r^2} = 4\pi\int_0^\infty \mathrm{d}r\,\frac{\sin kr}{kr} = \frac{2\pi^2}{k}.$$

Collecting $\Omega/2$, $1/(2\pi)^3$, $T^2q^4/(M_{\rm cell}^2c_{\min}^4)$ and π^3/k gives Eq. (11.31). The cross terms of first order in T/ω carry T/ω_1^2 times $1/\omega_2$, whose integral grows only as $\ln(Q_s/k)$, and the zero-point term is finite at $\boldsymbol{k}=0$. In the classical limit with no cut the singular part is exactly $1/k$ times $|\sum_\kappa c_\kappa e^{-W_\kappa} e^{i\boldsymbol{G}\cdot\boldsymbol{\tau}_\kappa}|^2 \Omega T^2 J/(16\pi^3 M_{\rm cell}^2)$, with $J = k\int \mathrm{d}^3Q_1\, h(\hat{\boldsymbol{Q}}_1)h(\hat{\boldsymbol{Q}}_2)/(Q_1^2Q_2^2)$ and $h = \sum_\nu(\boldsymbol{q}\cdot\hat{\boldsymbol{e}}_\nu)^2/c_\nu^2$, a number that depends only on the directions of $\boldsymbol{k}$ and $\boldsymbol{q}$; the bound replaces h by $q^2/c_{\min}^2$, for which $J = \pi^3q^4/c_{\min}^4$.

D.32 Impulse limit and Doppler width

A nucleus of mass M_κ at position $\boldsymbol{X}$ with momentum $\boldsymbol{p}$ takes up the momentum $\boldsymbol{q}$ through $\hat{O} = e^{i\boldsymbol{q}\cdot\boldsymbol{X}}$. We derive the moments of its self response for any potential, its impulse limit, the harmonic width, the free-recoil rate of Eq. (11.35) and the absorption rate Eq. (11.36). Throughout, $E_R = q^2/2M_\kappa$ and $p_{\hat{q}} = \hat{\boldsymbol{q}}\cdot\boldsymbol{p}$.

D.32.1 The self correlation

The self part of Eq. (11.24) for one nucleus is $c_\kappa^2\Omega^{-1}\int \mathrm{d}t\, e^{i\omega t} I(t)$ with $I(t) = \langle e^{-i\boldsymbol{q}\cdot\boldsymbol{X}(t)} e^{i\boldsymbol{q}\cdot\boldsymbol{X}}\rangle$. Write $e^{-i\boldsymbol{q}\cdot\boldsymbol{X}(t)} = e^{iHt}e^{-i\boldsymbol{q}\cdot\boldsymbol{X}}e^{-iHt}$ and move the kick through the second evolution operator,

$$e^{-i\boldsymbol{q}\cdot\boldsymbol{X}}\,e^{-iHt}\,e^{i\boldsymbol{q}\cdot\boldsymbol{X}} = e^{-iH_{\boldsymbol{q}}t}, \qquad H_{\boldsymbol{q}} = e^{-i\boldsymbol{q}\cdot\boldsymbol{X}} H\, e^{i\boldsymbol{q}\cdot\boldsymbol{X}}.$$

The translation $e^{-i\boldsymbol{q}\cdot\boldsymbol{X}}\boldsymbol{p}\,e^{i\boldsymbol{q}\cdot\boldsymbol{X}} = \boldsymbol{p}+\boldsymbol{q}$ changes the kinetic energy $\boldsymbol{p}^2/2M_\kappa$ and nothing else, provided the potential energy depends on $\boldsymbol{X}$ but not on $\boldsymbol{p}$. All Coulomb interactions are of this kind. Then

$$I(t) = \big\langle e^{iHt}\, e^{-i(H+D)t}\big\rangle, \qquad D = H_{\boldsymbol{q}} - H = E_R + \frac{\boldsymbol{q}\cdot\boldsymbol{p}}{M_\kappa}, \tag{D.188}$$

exactly, for any potential and any stationary state. The operator D is the energy that a kick $\boldsymbol{q}$ would give a free particle of momentum $\boldsymbol{p}$.

D.32.2 Moments of the recoil spectrum

The moments of the spectrum are the derivatives of I at $t=0$, $\int \mathrm{d}\omega\,\omega^k S/\int \mathrm{d}\omega\, S = i^k\,\mathrm{d}^kI/\mathrm{d}t^k|_0$ with the sign of Eq. (4.13). The first derivative gives $\langle (H+D)-H\rangle = \langle D\rangle = E_R$, since $\langle \boldsymbol{p}\rangle = 0$; this is Eq. (5.61). The second derivative gives $\langle H^2 - 2H(H+D) + (H+D)^2\rangle = \langle D^2\rangle + \langle [D,H]\rangle$, and the average of a commutator with H vanishes in a stationary state. Hence the variance

$$\sigma_\omega^2 = \langle D^2\rangle - E_R^2 = \frac{q^2\langle p_{\hat{q}}^2\rangle}{M_\kappa^2} = 4E_R\,E_\kappa^{\rm kin}(\hat{\boldsymbol{q}}), \qquad E_\kappa^{\rm kin}(\hat{\boldsymbol{q}}) = \frac{\langle p_{\hat{q}}^2\rangle}{2M_\kappa}, \tag{D.189}$$

holds at any temperature and for any potential local in $\boldsymbol{X}$. The width of the response measures the kinetic energy of the nucleus along $\hat{\boldsymbol{q}}$. At $T = 0$, with $|0\rangle$ the ground state, the third moment is $\langle 0|(H - E_0 + D)^3|0\rangle$. Every term with $H - E_0$ at either end vanishes, which leaves $\langle D^3\rangle + \langle D(H - E_0)D\rangle$. The first is $E_R^3 + 3E_R\sigma_\omega^2$, because $\langle p_{\hat{q}}^3\rangle = 0$ on a site with inversion symmetry or by time reversal. The second is half the double commutator $\langle [D, [H, D]]\rangle$, and with $[V, p_{\hat{q}}] = \mathrm{i}\,\partial_{\hat{q}} V$ this is $(q^2/2M_\kappa^2)\langle \partial_{\hat{q}}^2 V\rangle$, the mean curvature of the potential along $\hat{\boldsymbol{q}}$. Subtracting the powers of the lower moments, the third cumulant is

$$c_3 = \frac{q^2}{2M_\kappa^2}\,\langle \partial_{\hat{q}}^2 V\rangle. \tag{D.190}$$

For a nucleus whose electrons follow it, at $n_B = 0$, V is the Born–Oppenheimer surface of Sect. 5.1. Taken with the full Coulomb Hamiltonian instead, V contains the attraction of the electrons, whose curvature at the nucleus is the electron density there; that third moment includes the electronic excitations that the kick causes and is the subject of Sect. 12.5.

D.32.3 The moments at any temperature

In thermal equilibrium the response is the average over the initial eigenstates $|i\rangle$ of H with the Boltzmann weights p_i. From one eigenstate, the moments of the spectrum of $\omega = E_f - E_i$ are $\sum_f |\langle f|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}|i\rangle|^2 (E_f - E_i)^k = \langle i|(A_i + D)^k|i\rangle$, with $A_i = H - E_i$, by the same translation of H as above. For a Hamiltonian invariant under time reversal the eigenfunctions can be chosen real, and then $\langle i|p_{\hat{q}}|i\rangle = \langle i|p_{\hat{q}}^3|i\rangle = 0$: in a real state the expectation value of $p_{\hat{q}} = -\mathrm{i}\partial_{\hat{q}}$, or of its cube, is i times a real integral, while it must be real. The mean is therefore E_R from every $|i\rangle$, and the central moments are $\langle i|(A_i + \delta D)^k|i\rangle$ with $\delta D = (q/M_\kappa)p_{\hat{q}}$. Since $A_i|i\rangle = 0$, every term with A_i at either end vanishes, exactly as for the ground state. The second moment is $\langle i|\delta D^2|i\rangle$. The third is $\langle i|\delta D^3|i\rangle + \langle i|\delta D\, A_i\, \delta D|i\rangle$; the first term vanishes, and the second is $\frac{1}{2}\langle i|[\delta D, [H, \delta D]]|i\rangle = (q^2/2M_\kappa^2)\langle i|\partial_{\hat{q}}^2 V|i\rangle$. The fourth is $\langle i|\delta D^4|i\rangle + \|A_i\,\delta D|i\rangle\|^2$ plus the mixed term $\langle i|\delta D\, A_i\, \delta D^2|i\rangle$ and its complex conjugate. Since $A_i\,\delta D|i\rangle = [H, \delta D]|i\rangle = \mathrm{i}(q/M_\kappa)\,\partial_{\hat{q}} V|i\rangle$, the norm is $(q/M_\kappa)^2\langle i|(\partial_{\hat{q}} V)^2|i\rangle$; the mixed term is $\mathrm{i}(q/M_\kappa)^3\langle i|p_{\hat{q}}(p_{\hat{q}}\partial_{\hat{q}} V + \partial_{\hat{q}} V\, p_{\hat{q}})|i\rangle$, i times a real number in a real state, and cancels against its conjugate. Averaging over i with the weights p_i gives the moments Eq. (11.37) with thermal averages in place of ground-state ones: all five hold at any temperature. For one harmonic oscillator of frequency ω_0 the fourth cumulant that results is $E_R\omega_0^3 \coth(\omega_0/2T)$, the even case of Eq. (D.193) below.

D.32.4 The impulse approximation

If D commuted with H, Eq. (D.188) would be $I(t) = \langle \mathrm{e}^{-\mathrm{i}Dt}\rangle$. For a free particle this is exact. For a bound one the commutator $[H, D] = \mathrm{i}(q/M_\kappa)\,\partial_{\hat{q}} V$ is the force along $\hat{\boldsymbol{q}}$, and dropping it is the impulse approximation: the nucleus is treated as free during the kick. Writing the average over the momentum distribution $\mathfrak{f}_\kappa(\boldsymbol{p})$ of the nucleus, normalized to one,

$$I_{\mathrm{IA}}(t) = \int \mathrm{d}^3p\, \mathfrak{f}_\kappa(\boldsymbol{p})\, \mathrm{e}^{-\mathrm{i}(E_R + \boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa)t}, \qquad \int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} I_{\mathrm{IA}}(t) = 2\pi \int \mathrm{d}^3p\, \mathfrak{f}_\kappa(\boldsymbol{p})\, \delta\Big(\omega - E_R - \frac{\boldsymbol{q}\cdot\boldsymbol{p}}{M_\kappa}\Big). \tag{D.191}$$

Its mean and variance are those of Eq. (D.189), exactly, and its third cumulant vanishes. The approximation therefore errs first in c_3. In the scaled variable $(\omega - E_R)/\sigma_\omega$ the omitted skewness is c_3/σ_ω^3. With $c_3 \propto q^2$ and $\sigma_\omega \propto q$, it falls as $1/q$, while every cumulant that Eq. (D.191) keeps scales with the width. The impulse approximation is the leading term of an expansion in $1/q$ at fixed $(\omega - E_R)/\sigma_\omega$. The time scale makes the same point: the integral over t is cut off at $t \sim 1/\sigma_\omega$, and the force changes the momentum in that time by an amount small compared with its spread when σ_ω is large compared with the vibration frequencies.

D.32.5 The harmonic crystal

In a harmonic crystal $\mathfrak{f}_\kappa$ is Gaussian. From Eq. (5.57),

$$\langle p^\alpha p^\beta\rangle = M_\kappa^2\langle \dot u^\alpha \dot u^\beta\rangle = \frac{M_\kappa}{2N}\sum_{\boldsymbol{Q}\nu}\omega_{\boldsymbol{Q}\nu}\, e^\alpha_{\kappa\nu}(\boldsymbol{Q})\, e^\beta_{\kappa\nu}(\boldsymbol{Q})^*\coth\frac{\omega_{\boldsymbol{Q}\nu}}{2T}.$$

On a site of cubic symmetry the tensor is diagonal, and with the projected density of states $g_{\mathrm{ph},\kappa}/3$ of Sect. 11.3, normalized to one,

$$E^{\mathrm{kin}}_\kappa = \frac14\int \mathrm{d}\omega\,\frac{g_{\mathrm{ph},\kappa}(\omega)}{3}\,\omega\coth\frac{\omega}{2T}\ \xrightarrow{T=0}\ \frac{\langle\omega\rangle}{4}, \tag{D.192}$$

half the zero-point energy of each mode, averaged. The impulse spectrum Eq. (D.191) is then a Gaussian of mean E_R and variance $E_R\langle\omega\coth(\omega/2T)\rangle$. The same Gaussian follows from the generating formula Eq. (5.58) directly. Its exponent $\boldsymbol{q}\cdot\mathcal{C}_{\kappa\kappa}(t)\cdot\boldsymbol{q}$, expanded to second order in t, is $2W_\kappa - \mathrm{i}E_R t - \frac12 E_R\langle\omega\coth(\omega/2T)\rangle t^2$; the constant cancels the Debye–Waller factor, and the time integral is Gaussian. This is the saddle-point evaluation of Campbell-Deem and collaborators [73, Eqs. (37)–(39) and App. B]. Every order of the expansion is available: the k-th derivative of the exponent at $t=0$ gives the cumulant

$$c_k = E_R\,\langle\omega^{k-1}\rangle\ (k\text{ odd}),\qquad c_k = E_R\,\Big\langle\omega^{k-1}\coth\frac{\omega}{2T}\Big\rangle\ (k\text{ even}), \tag{D.193}$$

which at $T=0$ is Eq. (8.42). The odd cumulants do not depend on temperature. For $k=3$, Eq. (D.190) agrees, since the on-site force constant of a harmonic crystal is $M_\kappa\langle\omega^2\rangle$ by completeness of the polarization vectors.

D.32.6 Free recoil and the standard rate

As $x_R = E_R/\langle\omega\rangle$ grows, $\sigma_\omega/E_R = x_R^{-1/2}$ at $T=0$ and the Gaussian tends to $\delta(\omega-E_R)$ [73, Eq. (42)]. For a coupling to nucleons the vertex of each nucleus is $A_\kappa F_N(q)$, the elastic form factor Eq. (2.26); elastic, because the momenta of the impulse regime stay far below the energies that excite the nucleus internally. For one species the response is then $S_A = 2\pi\boldsymbol{n}_N A^2 F_N(q)^2\delta(\omega-q^2/2M)$, with $\boldsymbol{n}_N = n_{\mathrm{at}}/\Omega$ the number density of nuclei. Insert it into Eq. (8.16) with one row, $\mathfrak{D}=|\tilde V_N|^2$. The response does not depend on $\hat{\boldsymbol{q}}$, so the angle between $\boldsymbol{q}$ and $\boldsymbol{v}$ can be integrated first: $\int \mathrm{d}^3q/(2\pi)^3$ of $\delta(\omega-\boldsymbol{q}\cdot\boldsymbol{v}+q^2/2m_\chi)$ gives $\int q\,\mathrm{d}q\,\Theta(v-v_{\min})/(4\pi^2 v)$, as in Sect. 1.3, and the average over f_χ turns Θ/v into $\eta(v_{\min})$ of Eq. (1.28). With $\rho_T = \boldsymbol{n}_N M$,

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{\rho_\chi}{m_\chi\,\boldsymbol{n}_N M}\int\frac{q\,\mathrm{d}q}{4\pi^2}\,|\tilde V_N|^2\,2\pi\boldsymbol{n}_N A^2F_N^2\,\delta\Big(\omega-\frac{q^2}{2M}\Big)\,\eta(v_{\min}) = \frac{\rho_\chi A^2|\tilde V_N|^2}{2\pi m_\chi}\,F_N(q)^2\,\eta(v_{\min}),$$

at $q=(2M\omega)^{1/2}$, and Eq. (1.23) with $\Delta_\chi=0$ gives $v_{\min} = q/2M + q/2m_\chi = q/2\mu_{\chi N}$. A heavy mediator, $|\tilde V_N|^2 = \pi\sigma_p/\mu_{\chi p}^2$, with $\sigma_N = \sigma_p A^2\mu_{\chi N}^2/\mu_{\chi p}^2$ from Eq. (2.27), gives Eq. (11.35). The same steps applied to the impulse Gaussian give the rate with the width, which is what Fig. 11.15 evaluates.

D.32.7 Absorption on a nucleus

For a fermion absorbed at rest the momentum transfer is $q = m_\chi - \omega$ (Eq. (8.14)), and the vector rate of Eq. (10.60) is $(\sigma_F/2\pi)\bar S_{nn}(q,\omega)$, where the density now counts nucleons. The two operators differ in the spin trace. With the dark matter at rest, $p=(m_\chi,\boldsymbol{0})$, a massless outgoing fermion of four-momentum $(q,-\boldsymbol{q})$ and the nucleon current $J=(\rho,(\omega/q)\rho\,\hat{\boldsymbol{q}})$ fixed

by continuity, the spin-summed contraction is $m_\chi^3\rho^2/q$ for the vector operator and $m_\chi q\rho^2$ for the scalar density; their ratio is q^2/m_χ^2. Continuity holds for the nucleon current exactly, and the transverse current of a nucleus is of relative order $E_\kappa^{\rm kin}/M$ or q^2/M^2 and is dropped. For a free nucleus,

$$\Gamma_V = \sigma_F\, \boldsymbol{n}_N A^2 \int \mathrm{d}\omega\; F_N(m_\chi-\omega)^2\, \delta\Big(\omega - \frac{(m_\chi-\omega)^2}{2M}\Big) = \frac{\sigma_F\, \boldsymbol{n}_N A^2 F_N(q_{\rm abs})^2}{1+q_{\rm abs}/M},$$

since the argument of the delta function has slope $1+(m_\chi-\omega)/M$, with $q_{\rm abs}$ the root of $m_\chi = q_{\rm abs} + q_{\rm abs}^2/2M$. Expanding, $q_{\rm abs} = m_\chi(1 - m_\chi/2M + \dots)$ and $\Gamma_V = \sigma_F \boldsymbol{n}_N A^2 F_N^2(1 - m_\chi/M + \dots)$. Since $\sigma_F \propto m_\chi^2$, the rate carries $m_\chi^2(1-m_\chi/M)$. The scalar operator weights $\bar S_{nn}$ by $q^2/m_\chi^2 = 1 - m_\chi/M + \dots$ in addition, and gives $m_\chi^2(1-2m_\chi/M)$: the density row alone, without the longitudinal current that continuity adds, has twice the correction.

In a crystal the free-recoil peak is replaced by the response of Sect. 11.3. Its zero-phonon term carries $\delta(\omega)$ and conserves crystal momentum, $\boldsymbol{q} = \boldsymbol{G}$. With $\omega = 0$ the absorption requires $|\boldsymbol{q}| = m_\chi$ for a dark-matter particle at rest, and with speed v the condition $|\boldsymbol{G}| = m_\chi$ is relaxed only by a relative amount of order v: zero-phonon absorption is confined to masses within $\sim 10^{-3}$ of the length of a reciprocal-lattice vector. At any other mass only the terms $n \geq 1$ absorb, and in the incoherent approximation Eq. (11.26) their weight is $1 - \mathrm{e}^{-2W_\kappa(q)}$. Along the line $q = m_\chi - \omega$ the momentum changes by ω, a fraction $E_R/m_\chi = m_\chi/2M$ of itself, which rebuilds the factor $1/(1+q_{\rm abs}/M)$ in the impulse regime and is negligible below it. Hence Eq. (11.36).

D.33 The joint electron–lattice response

We derive the properties of the joint response Eq. (12.1), the Migdal entry Eq. (12.6), and its reductions to the four published forms of Table 12.1. The notation is that of Sect. 8.6 and Appendix D.28: adiabatic electronic states $\Psi_j(u)$ with energies $E_j(u)$, $\Delta E_j = E_j - E_0$, the derivative coupling $\boldsymbol{d}^I_{j0} = \langle\Psi_j|\nabla_{\boldsymbol{X}_I}\Psi_0\rangle$ and the force matrix element $\boldsymbol{F}^I_{j0} = \langle\Psi_j|\nabla_{\boldsymbol{X}_I}V_{eN}|\Psi_0\rangle$.

D.33.1 Labels, marginal and positivity

Let $H_{\rm el} = \sum_\Lambda E^{\rm el}_\Lambda P_\Lambda$ be the operator that multiplies each final-state class Λ of Sect. 8.7.1 by its electronic energy, and $H_{\rm lat} = H - H_{\rm el}$. Where the classes are good labels, $[P_\Lambda, H] = 0$, and then $H_{\rm el}$ and $H_{\rm lat}$ commute with H and with each other. Their joint eigenstates are the eigenstates $|f\rangle$ of H, with eigenvalues $E^{\rm el}_f$ and $E_f - E^{\rm el}_f$, and Eq. (12.1) can be written without reference to a basis,

$$S_{ab}(\boldsymbol{q};\omega_l,\omega_e) = \frac{2\pi}{\mathcal{V}} \sum_i p_i\, \langle i|O_a^\dagger\, \delta\big(\omega_e + E_i^{\rm el} - H_{\rm el}\big)\, \delta\big(\omega_l + E_i - E_i^{\rm el} - H_{\rm lat}\big)\, O_b|i\rangle.$$

Integrating over ω_e at fixed $\omega = \omega_l + \omega_e$ turns the product of the two delta functions into $\delta(\omega + E_i - H_{\rm el} - H_{\rm lat}) = \delta(\omega + E_i - H)$, since the two operators commute, and leaves Eq. (4.13); this is Eq. (12.2). For any complex vector c_a, the contraction $\sum_{ab} c_a^* S_{ab} c_b$ at a point of the plane is $(2\pi/\mathcal{V})\sum_{i,f} p_i |\langle f|\sum_b c_b O_b|i\rangle|^2$ times the two delta functions, a sum of nonnegative terms. The matrix is therefore positive semidefinite at every (ω_l,ω_e), and the bound $|S_{e\kappa}|^2 \leq S_{ee}S_{\kappa\kappa}$ of Sect. 12.1.1 is its two-by-two minor. Where the labels commute with H only to the accuracy of Eq. (8.47) or of the off-shell window, these statements hold on bins of the plane, with the remainders of Eqs. (8.50) and (8.63).

At $T = 0$ the initial state is the vibrational ground state of the ground surface, with $E_i^{\rm el} = 0$. A final state in $\mathcal{P}_{\rm ad}$ has $E_f^{\rm el} = 0$, which places the electron-following sector on the line

$\omega_e = 0$. A final state in $\mathcal{P}_0$ has the zero-phonon energy of its excitation as its total excitation energy, $E_f - E_i = E_f^{\mathrm{el}}$, which places it on the line $\omega_l = 0$. Every other final state has $\omega_e > 0$ and $\omega_l > 0$. This is Eq. (12.3).

D.33.2 The force operator and the amplitude

Let the electrons feel nucleus I, of species κ, through a potential that moves rigidly with it, $V_{eN} \ni \sum_e v_\kappa(\boldsymbol{r}_e - \boldsymbol{X}_I)$. With the transform Eq. (A.6), $v_\kappa(\boldsymbol{r}) = \int \mathrm{d}^3p\,(2\pi)^{-3}\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}}v_\kappa(\boldsymbol{p})$, and with $v_\kappa(-\boldsymbol{p}) = v_\kappa(\boldsymbol{p})$ for a spherical potential,

$$\sum_e v_\kappa(\boldsymbol{r}_e - \boldsymbol{X}_I) = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\, v_\kappa(\boldsymbol{p})\,\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{X}_I}\sum_e \mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}_e} = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\, v_\kappa(\boldsymbol{p})\,\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{X}_I}\, n_e(\boldsymbol{p}),$$

after the substitution $\boldsymbol{p} \to -\boldsymbol{p}$, with $n_e(\boldsymbol{p})$ of Eq. (A.8). Only this term of V_{eN} depends on $\boldsymbol{X}_I$. Its gradient gives Eq. (12.4), and its matrix element between adiabatic states gives

$$\boldsymbol{q}\cdot\boldsymbol{F}^I_{j0} = -\mathrm{i}\int \frac{\mathrm{d}^3p}{(2\pi)^3}\,(\boldsymbol{q}\cdot\boldsymbol{p})\,v_\kappa(\boldsymbol{p})\,\langle\Psi_j(u)|\,n_e(\boldsymbol{p})\,|\Psi_0(u)\rangle\,\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{X}_I}. \tag{D.194}$$

Both $\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{X}_I}$ and the transition density depend on the nuclear positions, and the product is a function of u that multiplies the nuclear wavefunction.

The boost amplitude Eq. (8.53) is $-\mathrm{i}\,\boldsymbol{q}\cdot\boldsymbol{d}^I_{j0}/(M_\kappa\Delta E_j)$ times $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$ between the vibrational states. By Eq. (5.11), $\boldsymbol{d}^I_{j0} = \boldsymbol{F}^I_{j0}/(E_0 - E_j) = -\boldsymbol{F}^I_{j0}/\Delta E_j$, so the electronic factor is $\mathrm{i}\,\boldsymbol{q}\cdot\boldsymbol{F}^I_{j0}/(M_\kappa\Delta E_j^2)$. The derivation of Eq. (D.167) treats the derivative coupling as a constant over the vibrational wavefunctions. Keeping it inside the nuclear matrix element, as a function of u, gives the same leading term together with the Herzberg–Teller corrections. With Eq. (D.194) and $\mathrm{i}\cdot(-\mathrm{i}) = 1$,

$$A_{fj} = \sum_{I\in\kappa}\int \frac{\mathrm{d}^3p}{(2\pi)^3}\,\frac{\boldsymbol{q}\cdot\boldsymbol{p}}{M_\kappa\Delta E_j^2}\,v_\kappa(\boldsymbol{p})\,\langle\Psi^{\mathrm{nuc}\prime}|\,\langle\Psi_j(u)|n_e(\boldsymbol{p})|\Psi_0(u)\rangle\,\mathrm{e}^{\mathrm{i}(\boldsymbol{q}-\boldsymbol{p})\cdot\boldsymbol{X}_I}\,|\Psi^{\mathrm{nuc}}\rangle.$$

The Condon approximation in the frame of the crystal replaces the transition density by its value at $u = 0$, which leaves the operator $\mathrm{e}^{\mathrm{i}(\boldsymbol{q}-\boldsymbol{p})\cdot\boldsymbol{X}_I}$ between the vibrational states and gives Eq. (12.5). Its first correction is the gradient of the transition density with respect to the displacements, the Herzberg–Teller term of Eq. (8.52).

If instead the electrons move rigidly with nucleus I, as in an isolated atom, the transition density at nuclear position $\boldsymbol{X}_I$ is that at the origin times $\mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{X}_I}$, since translating every electron by $\boldsymbol{X}_I$ multiplies $n_e(\boldsymbol{p})$ by this phase. The phase cancels the factor $\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{X}_I}$ of Eq. (D.194), and the operator between the vibrational states becomes $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}$, a recoil with the full momentum $\boldsymbol{q}$. The two cases differ by the dependence of the transition density on u that the frame of the crystal neglects and the frame of the nucleus overstates. Expanded in $\boldsymbol{p}\cdot\boldsymbol{u}_I$, the two lattice factors agree at zeroth order and differ at first order: the lattice keeps $\boldsymbol{q}-\boldsymbol{p}$ in one frame and $\boldsymbol{q}$ in the other, which matters for the lattice spectrum when p is not small against q (Sect. 12.4).

D.33.3 Squared amplitude and crystal momentum

In a box of volume $\mathcal{V}$, $\int \mathrm{d}^3p\,(2\pi)^{-3} \to \mathcal{V}^{-1}\sum_{\boldsymbol{p}}$. The squared amplitude summed over the nuclei of species κ is

$$|A_{fj}|^2 = \frac{1}{M_\kappa^2\Delta E_j^4\mathcal{V}^2}\sum_{\boldsymbol{p}\boldsymbol{p}'}(\boldsymbol{q}\cdot\boldsymbol{p})(\boldsymbol{q}\cdot\boldsymbol{p}')\,v_\kappa(\boldsymbol{p})^*v_\kappa(\boldsymbol{p}')\,\langle\Psi_0|n_e(\boldsymbol{p})^\dagger|\Psi_j\rangle\langle\Psi_j|n_e(\boldsymbol{p}')|\Psi_0\rangle$$
$$\times\,\langle\Psi^{\mathrm{nuc}}|n_\kappa(\boldsymbol{q}-\boldsymbol{p})^\dagger|\Psi^{\mathrm{nuc}\prime}\rangle\langle\Psi^{\mathrm{nuc}\prime}|n_\kappa(\boldsymbol{q}-\boldsymbol{p}')|\Psi^{\mathrm{nuc}}\rangle,$$

with $n_\kappa(\boldsymbol{Q}) = \sum_{I\in\kappa} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{X}_I}$, the transition densities at $u = 0$, and the real factor $\boldsymbol{q}\cdot\boldsymbol{p}$. The final state has $\omega_e = \Delta E_j$ and ω_l equal to the energy of $\Psi^{\mathrm{nuc}\prime}$ above Ψ^{nuc}. Multiplying by $(2\pi/\mathscr{V})p_i\,\delta(\omega_e - \Delta E_j)\,\delta(\omega_l - E^{\mathrm{nuc}\prime} + E^{\mathrm{nuc}})$ and summing over $j \neq 0$, over $\Psi^{\mathrm{nuc}\prime}$ and over the thermal initial states, the electronic sum is $(\mathscr{V}/2\pi)$ times the density response between $n_e(\boldsymbol{p})$ and $n_e(\boldsymbol{p}')$ at clamped nuclei, and the nuclear sum is $(\mathscr{V}/2\pi)$ times the electron-following response between $n_\kappa(\boldsymbol{q}-\boldsymbol{p})$ and $n_\kappa(\boldsymbol{q}-\boldsymbol{p}')$. In a crystal the clamped electrons conserve crystal momentum, and their response vanishes unless $\boldsymbol{p}$ and $\boldsymbol{p}'$ differ by a reciprocal-lattice vector. Writing $\boldsymbol{p} = \boldsymbol{k} + \boldsymbol{G}$ and $\boldsymbol{p}' = \boldsymbol{k} + \boldsymbol{G}'$ with $\boldsymbol{k}$ in the zone, the double sum becomes $\sum_{\boldsymbol{k}}\sum_{\boldsymbol{G}\boldsymbol{G}'}$, and $\mathscr{V}^{-1}\sum_{\boldsymbol{k}} \to \int_{\mathrm{BZ}} \mathrm{d}^3k\,(2\pi)^{-3}$. Collecting the factors, $(2\pi/\mathscr{V})(\mathscr{V}/2\pi)^2\mathscr{V}^{-2}\cdot\mathscr{V} = (2\pi)^{-1}$, and with $\Delta E_j = \omega_e$ under the delta function we obtain Eq. (12.6). The units check: the joint response has the units of $S_{ab}(\boldsymbol{q},\omega)$ per unit energy.

Of the approximations of Sect. 12.1.2, the ground-surface final states, which make the lattice factor the electron-following response and ΔE_j the vertical energy, can be lifted by projecting on the vibrational states of surface j, which Worked example 12.1 does for one oscillator. For a pair delocalized over N cells the displacement of the excited surface is of order N^{-1} per nucleus (Sect. 8.6), and the two choices agree.

D.33.4 Diagonal and incoherent reduction

Keep only $\boldsymbol{G} = \boldsymbol{G}'$ and the self part of the lattice factor. The self part is exact for the phonon-emitting terms, which are incoherent, and, averaged over the zone, for the zero-phonon term at $q \gtrsim 2q_{\mathrm{BZ}}$. At $q \ll q_{\mathrm{BZ}}$ the coherent zero-phonon part keeps the cross-site terms of the lattice factor and the full matrix $\boldsymbol{G} \neq \boldsymbol{G}'$, and obeys Eq. (8.32) instead (Sect. 12.4). For one nucleus per site the self response is $S^{\mathrm{ad}}_{\kappa;\boldsymbol{G}\boldsymbol{G}}(\boldsymbol{q}-\boldsymbol{k},\omega_l) = 2\pi\boldsymbol{n}_\kappa\, s^{\mathrm{inc}}_\kappa(\boldsymbol{q}-\boldsymbol{k}-\boldsymbol{G},\omega_l)$, with $s^{\mathrm{inc}}_\kappa(\boldsymbol{Q},\omega)$ the spectrum of one nucleus normalized to $\int \mathrm{d}\omega\, s^{\mathrm{inc}}_\kappa = 1$, zero-phonon line included, as in Eq. (11.26). At $T = 0$ the diagonal electronic factor is fixed by the fluctuation–dissipation theorem of Sect. 10.2, $S^{ee}_{\boldsymbol{G}\boldsymbol{G}}(\boldsymbol{k},\omega) = (2|\boldsymbol{p}|^2/e^2)\,L(\boldsymbol{p},\omega)$, with $\boldsymbol{p} = \boldsymbol{k} + \boldsymbol{G}$ and $L(\boldsymbol{p},\omega) = \mathrm{Im}[-(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}}(\boldsymbol{k},\omega)]$ the loss function at the unreduced momentum. For a vertex $v_\kappa(\boldsymbol{p}) = -Z_\kappa e^2/p^2$, the sum over $\boldsymbol{G}$ and the integral over the zone combine into an integral over all $\boldsymbol{p}$, and dividing by the weight $2\pi\boldsymbol{n}_\kappa$ of one nucleus,

$$\frac{\mathrm{d}^2P}{\mathrm{d}\omega_l\,\mathrm{d}\omega_e} = 4\alpha\, Z_\kappa^2 \int \frac{\mathrm{d}^3p}{(2\pi)^3}\,\frac{(\boldsymbol{v}_N\cdot\boldsymbol{p})^2}{\omega_e^4\,p^2}\,L(\boldsymbol{p},\omega_e)\,s^{\mathrm{inc}}_\kappa(\boldsymbol{q}-\boldsymbol{p},\omega_l), \tag{D.195}$$

with $\boldsymbol{v}_N = \boldsymbol{q}/M_\kappa$ and $e^2 = 4\pi\alpha$: the factors are $(2\pi)^{-1}\cdot Z_\kappa^2 e^4 p^{-4}\cdot 2p^2e^{-2} = Z_\kappa^2e^2/(\pi p^2)$. Fig. 12.2 evaluates Eq. (D.195) for silicon with $Z_\kappa = 4$. Let the momentum taken by the electrons be small against the kick, $p \ll q$. The lattice factor then depends on $\boldsymbol{q}$ alone, its integral over ω_l is one, and

$$\frac{\mathrm{d}P}{\mathrm{d}\omega_e} = 4\alpha \int \frac{\mathrm{d}^3p}{(2\pi)^3}\,Z_\kappa^2\,\frac{(\boldsymbol{v}_N\cdot\boldsymbol{p})^2}{\omega_e^4\,p^2}\,L(\boldsymbol{p},\omega_e). \tag{D.196}$$

With the charge Z_κ replaced by the momentum-dependent ion charge $Z_{\mathrm{ion}}(p)$ of the nucleus and its frozen core, this is the soft-limit formula of Knapen, Kozaczuk and Lin [74, Eq. (6)], coefficient 4α included. Their full expression [74, Eq. (3)] keeps the impulse lattice factor at $\boldsymbol{q}-\boldsymbol{p}$, a free recoil smeared by the motion of the nucleus in its well, which is Eq. (D.195) with the impulse form of s^{inc}_κ (Sect. 11.4). Its energy bracket carries the final momentum of the nucleus in the numerator and drops $p^2/2M_\kappa$ in both denominators, which is consistent only in the soft limit; the exact free-recoil bracket, with $p^2/2M_\kappa$ in both denominators and the numerator $\boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa$, is Eq. (12.24).

D.33.5 The isolated atom

For an atom the electrons move with the nucleus, and by Appendix D.33.2 the lattice factor is $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}}$. The electronic factor $\boldsymbol{q}\cdot\boldsymbol{F}_{j0}/(M\Delta E_j^2)$ is then evaluated with the atom at the origin, where the adiabatic states depend on $\boldsymbol{X}$ only through $\boldsymbol{r}_e-\boldsymbol{X}$. Sect. 6.1 showed that this makes the derivative coupling a dipole matrix element, $\boldsymbol{d}_{j0}=m_e\Delta E_j\langle\Psi_j|\sum_e\boldsymbol{r}_e|\Psi_0\rangle$. The amplitude becomes $-\mathrm{i}m_e\boldsymbol{v}_N\cdot\langle\Psi_j|\sum_e\boldsymbol{r}_e|\Psi_0\rangle$, the first-order term of Migdal's boost Eq. (6.8). The boost itself is exact for a free atom in the sudden limit: the electrons are left in Ψ_0 while the nucleus moves with $\boldsymbol{v}_N$, and the probability of the excited state j is $|\langle\Psi_j|\exp(-\mathrm{i}m_e\boldsymbol{v}_N\cdot\sum_e\boldsymbol{r}_e)|\Psi_0\rangle|^2$ to all orders in v_N. This is the electron-cloud transition factor of Ibe and collaborators [75, Eqs. (55) and (57)], whose argument is m_e times the final velocity of the atom [75, Eq. (58)], $\boldsymbol{v}_N$ up to corrections of order m_e/M_κ. The lattice factor is the free recoil of the atom. Its mass is that of the atom, not of the bare nucleus, since the drag mass of an atom is $\delta M=N_em_e$ (Sect. 8.6); this is the boost mass m_A of Sect. 12.2. With the free recoil the energy is $\omega_l=q^2/2m_A$ for every Migdal event, and the joint response is a line in the plane.

D.33.6 Bound nuclei and the virtual phonon

Below the impulse onset the lattice factor of Eq. (12.6) is the harmonic multiphonon response at $\boldsymbol{q}-\boldsymbol{k}-\boldsymbol{G}$, Eq. (11.24) or its incoherent form Eq. (11.26). With the diagonal of the loss function and the vertex of a pseudo-ion, Eq. (12.6) then has the form of the rate of Berghaus and collaborators [76, Eq. (3.8)], whose effective interaction $\nabla_I H_{\chi L}\cdot\nabla_I H_{eL}/(m_N\omega^2)$ [76, Eq. (3.6)] is the product of the kick and the force Eq. (12.4) divided by $M_\kappa\omega_e^2$. Their lattice structure factor is evaluated at $\boldsymbol{q}-\boldsymbol{k}_e-\boldsymbol{K}$, the frame of the crystal of Appendix D.33.2.

The energy denominator $M_\kappa\omega_e^2$ assumes a nucleus that is free during the electronic time $1/\omega_e$. A nucleus bound in the lattice is pulled back by its neighbors, and Sect. 6.1 found that the sudden amplitude of a nucleus kicked in a harmonic well of frequency ω_{ph} acquires the factor $[1-(\omega_{\mathrm{ph}}/\omega_e)^2]^{-1}$. Resolved into phonon branches, the inertial response $1/(M_\kappa\omega_e^2)$ of the kicked nucleus becomes the phonon propagator

$$\frac{\boldsymbol{q}\cdot\boldsymbol{p}}{M_\kappa\omega_e^2}\;\to\;\frac{1}{N}\sum_{\boldsymbol{Q}\nu}\frac{(\boldsymbol{q}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}))\,(\boldsymbol{p}\cdot\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*)}{M_\kappa\,(\omega_e^2-\omega_{\boldsymbol{Q}\nu}^2)}=\frac{\boldsymbol{q}\cdot\boldsymbol{p}}{M_\kappa\omega_e^2}\Big[1+O\Big(\frac{\omega_{\mathrm{ph}}^2}{\omega_e^2}\Big)\Big],\tag{D.197}$$

where the equality uses the completeness of the polarization vectors of each $\boldsymbol{Q}$, $\sum_\nu\boldsymbol{e}_{\kappa\nu}\boldsymbol{e}_{\kappa\nu}^*=\mathbf{1}$, on the leading term. This is the virtual phonon of Liang and collaborators [77, Eqs. (3.1) and (3.2)], which they reduce to $1/(\varepsilon_i-\varepsilon_j)^2$ for gaps large against the phonon energies, and whose multiphonon sum [77, Eqs. (3.3) and (3.4)] is the incoherent lattice factor with its zero-phonon term. Averaged over the phonon density of states, the correction in the probability is $2\langle\omega_{\mathrm{ph}}^2\rangle/\omega_e^2$, which in silicon at the gap, with $\langle\omega_{\mathrm{ph}}^2\rangle^{1/2}=44.7\,\mathrm{meV}$, is 2.9×10^{-3}, of the order of the Condon and screening corrections that the same formulas neglect. Sect. 12.4 shows that for a harmonic lattice the propagator form is exact at first order in $\boldsymbol{p}$, treats it below the impulse onset and splits it into its zero-phonon and phonon-emitting terms.

D.34 Moments of the Migdal spectrum

This section derives the moments of Sect. 12.5: the inverse force moments, the trace rule, the cross and third moments, and the Dirac contact density. The notation is that of Sects. 8.6, 12.1 and 12.5: $|0\rangle$ and $|a\rangle$ are the ground and excited states of the clamped electrons, $\Delta_a=E_a-E_0$, and $\boldsymbol{F}_{a0}=\langle a|\nabla_{\boldsymbol{R}_\kappa}V_{eN}|0\rangle$.

D.34.1 Moments from the Migdal entry

At leading order the kick of a nucleus reaches the excited surface a with the amplitude Eq. (D.167), $-\mathrm{i}\,\boldsymbol{q}\cdot\boldsymbol{d}_{a0}/(M_\kappa\Delta_a)$ times a nuclear overlap. Summing the squared modulus over the complete set of vibrational states of that surface removes the overlap, and the electronic energy of the final state is Δ_a in the Condon approximation with vertical energies (Sect. 12.1.2). With $\boldsymbol{d}_{a0}=-\boldsymbol{F}_{a0}/\Delta_a$ from Eq. (5.11),

$$\frac{\mathrm{d}P_{\mathrm{Mig}}}{\mathrm{d}\omega_e}=\sum_{a\neq0}\frac{|\boldsymbol{q}\cdot\boldsymbol{F}_{a0}|^2}{M_\kappa^2\Delta_a^4}\,\delta(\omega_e-\Delta_a)=\frac{q^2}{M_\kappa^2\,\omega_e^4}\,\frac{\mathrm{d}\sigma_\kappa}{\mathrm{d}\Delta}\bigg|_{\Delta=\omega_e},$$

which is Eq. (12.39). The same function is the electronic marginal of the Migdal entry: integrating Eq. (12.6) over ω_l with the incoherent lattice factor, whose weight is $2\pi n_\kappa$ at every momentum (the coherent zero-phonon process for $q\gtrsim 2q_{\mathrm{BZ}}$ on zone average; at $q\ll q_{\mathrm{BZ}}$ that process obeys Eq. (8.32), Appendix D.33.4), and summing the electronic factor over final states with Eq. (12.4), gives $2\pi n_\kappa$ times this expression (Appendix D.33.3). The kth moment is $(q^2/M_\kappa^2)\int\Delta^{k-4}\mathrm{d}\sigma_\kappa=q^2\mathcal{I}_{4-k}/M_\kappa^2$. For $k=1$, the projection of Eq. (D.166) on $\hat{\boldsymbol{q}}$ is $2\sum_a|\hat{\boldsymbol{q}}\cdot\boldsymbol{F}_{a0}|^2/\Delta_a^3=2\mathcal{I}_3$, so that $E_1^{\mathrm{Mig}}=q^2\delta M_\kappa/(2M_\kappa^2)=E_R\,\delta M_\kappa/M_\kappa$. For $k=2$, the diagonal correction of Born and Huang in the effective Hamiltonian of Appendix D.28 is $\sum_a|\boldsymbol{d}_{a0}|^2/2M_\kappa$; its part along $\hat{\boldsymbol{q}}$ is $E_{\mathrm{BH}}=\mathcal{I}_2/2M_\kappa$, and $E_2^{\mathrm{Mig}}=q^2\mathcal{I}_2/M_\kappa^2=4E_RE_{\mathrm{BH}}$.

For the valence electrons in the soft limit, Eq. (12.21) and Eq. (12.39) describe the same probability, so

$$\frac{\mathrm{d}\sigma_\kappa}{\mathrm{d}\omega}=\frac{M_\kappa^2\omega^4}{q^2}\,\frac{\mathrm{d}P_{\mathrm{Mig}}}{\mathrm{d}\omega}=4\alpha\int\frac{\mathrm{d}^3p}{(2\pi)^3}\,Z_{\mathrm{ion}}(p)^2\,\frac{(\hat{\boldsymbol{q}}\cdot\boldsymbol{p})^2}{p^2}\,L(\boldsymbol{p},\omega),$$

with $\boldsymbol{v}_N=\boldsymbol{q}/M_\kappa$. Then $\delta M_\kappa^{\mathrm{val}}=2\int\omega^{-3}\mathrm{d}\sigma_\kappa$, and for a direction-averaged loss function the angular average of $(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{p}})^2$ is $1/3$, which gives Eq. (12.41).

D.34.2 Dynamical force constant

Displace nucleus κ by $u\,\hat{\boldsymbol{q}}\cos\omega t$ at a frequency below the gap. The electrons respond linearly, and the force they exert back on the nucleus is $-\Phi^{\mathrm{el}}_{\kappa\kappa}(\omega)u$, with the static term $\langle\partial^2_{\hat q}V_{eN}\rangle$ and the retarded response of the operator $\partial_{\hat q}V_{eN}$ to itself,

$$\Phi^{\mathrm{el}}_{\kappa\kappa}(\omega)=\langle\partial^2_{\hat q}V_{eN}\rangle-\sum_{a\neq0}|\hat{\boldsymbol{q}}\cdot\boldsymbol{F}_{a0}|^2\Big[\frac{1}{\Delta_a-\omega}+\frac{1}{\Delta_a+\omega}\Big]=\langle\partial^2_{\hat q}V_{eN}\rangle-2\int\mathrm{d}\sigma_\kappa(\Delta)\,\frac{\Delta}{\Delta^2-\omega^2}.$$

For $\omega<\Delta$ on the whole support, $\Delta/(\Delta^2-\omega^2)=\Delta^{-1}+\omega^2\Delta^{-3}+O(\omega^4\Delta^{-5})$, and

$$\Phi^{\mathrm{el}}_{\kappa\kappa}(\omega)=\big[\langle\partial^2_{\hat q}V_{eN}\rangle-2\mathcal{I}_1\big]-\omega^2\,2\mathcal{I}_3+O(\omega^4)=\Phi^{\mathrm{el}}_{\kappa\kappa}(0)-\omega^2\delta M_\kappa+O(\omega^4).$$

The equation of motion $M_\kappa\ddot u=-\Phi u$ at frequency ω becomes $[\Phi^{\mathrm{el}}_{\kappa\kappa}(0)-\omega^2(M_\kappa+\delta M_\kappa)]u=0$ to this order. At imaginary frequency the sign of the ω^2 term changes, $\Phi^{\mathrm{el}}_{\kappa\kappa}(\mathrm{i}\omega)=\Phi^{\mathrm{el}}_{\kappa\kappa}(0)+\omega^2\delta M_\kappa+O(\omega^4)$, which is the form Appendix E.18 fits.

D.34.3 The trace rule

Write $V_{eN}=\sum_{i,J}v_J(\boldsymbol{r}_i-\boldsymbol{R}_J)$, with i the electrons and J the nuclei. Since $\nabla_{\boldsymbol{R}_J}v_J(\boldsymbol{r}_i-\boldsymbol{R}_J)=-\nabla_{\boldsymbol{r}_i}v_J(\boldsymbol{r}_i-\boldsymbol{R}_J)$, the force summed over all nuclei is minus the force on all electrons, $\sum_J\nabla_{\boldsymbol{R}_J}V_{eN}=-\sum_i\nabla_{\boldsymbol{r}_i}V_{eN}$. The electron–electron interaction depends on differences of

electron positions, so $\sum_i \nabla_{\boldsymbol{r}_i} V_{ee} = 0$, and with the total electronic momentum $\boldsymbol{P}_e = -\mathrm{i}\sum_i \nabla_{\boldsymbol{r}_i}$ and the clamped Hamiltonian H_{el},

$$[\boldsymbol{P}_e, H_{\mathrm{el}}] = -\mathrm{i}\sum_i \nabla_{\boldsymbol{r}_i} V_{eN},$$

$$\sum_J \langle a|\nabla_{\boldsymbol{R}_J} V_{eN}|0\rangle = -\mathrm{i}\,\langle a|[\boldsymbol{P}_e, H_{\mathrm{el}}]|0\rangle = \mathrm{i}\Delta_a\,\langle a|\boldsymbol{P}_e|0\rangle.$$

Take a finite piece of an insulator, so that the total position $\boldsymbol{X}_e = \sum_i \boldsymbol{r}_i$ is a bounded operator on its states, and use $\boldsymbol{P}_e = \mathrm{i}m_e[H_{\mathrm{el}}, \boldsymbol{X}_e]$, whose matrix element is $\mathrm{i}m_e\Delta_a\langle a|\boldsymbol{X}_e|0\rangle$. Summing Eq. (D.166) over both nuclear indices,

$$\sum_{I,J} \hat{\boldsymbol{q}}\cdot\delta M_{IJ}\cdot\hat{\boldsymbol{q}} = 2\sum_{a\neq 0}\frac{\Delta_a^2|\hat{\boldsymbol{q}}\cdot\boldsymbol{P}_{a0}|^2}{\Delta_a^3} = 2m_e^2\sum_{a\neq 0}\Delta_a\,|\langle a|\hat{\boldsymbol{q}}\cdot\boldsymbol{X}_e|0\rangle|^2 = N_e m_e,$$

by the Thomas–Reiche–Kuhn rule Eq. (4.37) for the N_e electrons of the piece. The result holds for interacting electrons, since only the double commutator of $\boldsymbol{X}_e$ with the kinetic energy enters. In the bulk of a crystal with one atom per cell every nucleus is equivalent, so $\sum_J \delta M_{IJ}$ is independent of I and equals $Z_\kappa m_e$. With several atoms per cell the same sum, written with the sublattice displacements of wavevector $\boldsymbol{Q} = 0$, is $N_e m_e$ per cell. The surface of the piece changes these numbers by a fraction that vanishes as the piece grows. Both forms are Eq. (12.42). The single element δM_{II} contains only $\boldsymbol{F}^I_{a0}$, not the sum, and no commutator turns it into a ground-state expectation value; this is why Eq. (12.43) leaves it free.

In the one-dimensional model of Sect. 12.5.2, with independent electrons, the sum over excited states runs over pairs $v \to c$ from the filled band to the empty ones, and $\delta M_{IJ} = 4\sum_{vc}\mathrm{Re}[\langle v|\partial_{X_I}V|c\rangle\langle c|\partial_{X_J}V|v\rangle]/(\varepsilon_c-\varepsilon_v)^3$, the factor 4 being 2 from Eq. (D.166) and 2 from spin. The model is solved in plane waves on a supercell of up to 32 wells, which samples the zone of the primitive cell at 32 points. The trace equals $2\,m_e$ per well to 10^{-4}, and a single well in a large box gives $\delta M = 2\,m_e$, the Thomas–Reiche–Kuhn value.

D.34.4 Cross first moment of an atom

With $\boldsymbol{\rho}_e = \boldsymbol{r}_e - \boldsymbol{X}$, the center of mass $\boldsymbol{R}_{\mathrm{cm}}$ and $\boldsymbol{q}_e = m_e\boldsymbol{q}/m_A$, Eq. (12.10) gives $n_\kappa(\boldsymbol{q}) = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}} = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_{\mathrm{cm}}}A_\kappa$ with $A_\kappa = \mathrm{e}^{-\mathrm{i}\boldsymbol{q}_e\cdot\sum_{e'}\boldsymbol{\rho}_{e'}}$. Since $\boldsymbol{r}_e = \boldsymbol{R}_{\mathrm{cm}} + \boldsymbol{\rho}_e - (m_e/m_A)\sum_{e'}\boldsymbol{\rho}_{e'}$, the electron density is $n_e(\boldsymbol{q}) = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_{\mathrm{cm}}}A_e$ with $A_e = \sum_e \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\rho}_e}A_\kappa$. Both operators move the center of mass by $\boldsymbol{q}$, and a final state is a plane wave of the center of mass times an internal state $|f\rangle$ of energy E_f, with the total energy transfer $\omega = q^2/2m_A + E_f - E_0$. For a gas of $\boldsymbol{n}_\kappa$ atoms per volume at rest,

$$S_{e\kappa}(\boldsymbol{q},\omega) = 2\pi\boldsymbol{n}_\kappa\sum_f \langle 0|A_e^\dagger|f\rangle\langle f|A_\kappa|0\rangle\,\delta(\omega - q^2/2m_A - E_f + E_0).$$

The total first moment $\mathscr{M}^1_{e\kappa} = \int(\mathrm{d}\omega/2\pi)\,\omega[S_{e\kappa}(\boldsymbol{q},\omega)+S_{\kappa e}(-\boldsymbol{q},\omega)]$ vanishes, Eq. (4.34), because n_e commutes with the nuclear kinetic energy and n_κ with everything else. The elastic term $f = 0$ sits at $\omega = q^2/2m_A$. For a real ground state the elements at $-\boldsymbol{q}$ are the complex conjugates of those at $\boldsymbol{q}$, so the two terms of the moment are equal, and the elastic part is

$$\mathscr{M}^1_{e\kappa}\big|_{f=0} = 2\boldsymbol{n}_\kappa\,\frac{q^2}{2m_A}\,\mathrm{Re}\big[\langle 0|A_e|0\rangle^*\langle 0|A_\kappa|0\rangle\big] = \boldsymbol{n}_\kappa\,\frac{q^2}{m_A}\,F_{\mathrm{cr}}(q).$$

The excited internal states, bound and ionized, carry the rest of the vanishing total, which is Eq. (12.45). No expansion in q_e has been made. To lowest order, $\langle 0|A_\kappa|0\rangle = 1 - O(q_e^2\langle\rho^2\rangle)$

and $\langle 0|A_e|0\rangle = f(q)$, the form factor of the electron cloud, so that $F_{\rm cr} = f(q)$ up to those corrections and terms of relative order m_e/m_A.

In a crystal at leading order in $\kappa_{\rm BO}$, Eq. (8.57) gives the electron-following part as $\boldsymbol{n}_\kappa q^2 F^*_{\rm cr}/M_\kappa$ with $F_{\rm cr}(\boldsymbol{q}) = (N_\kappa q^2)^{-1}\langle \sum_{I\in\kappa} \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{X}_I}\,(-\mathrm{i}\boldsymbol{q})\cdot\nabla_I \bar\rho_e(\boldsymbol{q};u)\rangle$, and the electronic sectors carry its negative. At small q the derivative of the adiabatic density along a displacement of nucleus I is $\mathrm{i}\boldsymbol{q}$ times the charge that follows it, $(Z_\kappa \mathbf{1} - Z^*_\kappa)$ at zero macroscopic field (Sect. 8.6), so $F_{\rm cr} \to \hat{\boldsymbol{q}}\cdot(Z_\kappa\mathbf{1} - Z^*_\kappa)\cdot\hat{\boldsymbol{q}}$.

D.34.5 The third moment

Let the nucleus κ be displaced by $u\hat{\boldsymbol{q}}$ in the clamped Hamiltonian. Second-order perturbation theory in u gives the curvature of the ground-state energy,

$$\partial^2_{\hat q} E_0 = \langle 0|\partial^2_{\hat q} V|0\rangle - 2\sum_{a\neq 0} \frac{|\hat{\boldsymbol{q}}\cdot\boldsymbol{F}_{a0}|^2}{\Delta_a} = \langle \partial^2_{\hat q} V\rangle - 2\mathcal{I}_1,$$

with $V = V_{eN} + V_{NN}$; the nucleus–nucleus term has no electronic matrix elements and enters only the first term. With Eq. (12.40), $E_3^{\rm Mig} = q^2\mathcal{I}_1/M^2_\kappa = (q^2/2M^2_\kappa)[\langle\partial^2_{\hat q}V\rangle - \partial^2_{\hat q}E_0]$. For a point nucleus, $v_\kappa(\boldsymbol{r}) = -Z_\kappa e^2/(4\pi r)$ and $\nabla^2(1/4\pi r) = -\delta^3(\boldsymbol{r})$, so $\nabla^2_{\boldsymbol{R}} V_{eN} = Z_\kappa e^2 \sum_i \delta^3(\boldsymbol{r}_i - \boldsymbol{R}_\kappa)$, whose expectation value is $Z_\kappa e^2 \boldsymbol{n}_e^{(\kappa)}$. The nucleus–nucleus potential is harmonic at $\boldsymbol{R}_\kappa$, $\nabla^2_{\boldsymbol{R}} V_{NN} = 0$. On a site of cubic symmetry the tensor $\langle\partial_\alpha\partial_\beta V\rangle$ is proportional to the unit tensor, so each direction holds a third of the trace, $\langle\partial^2_{\hat q}V\rangle = Z_\kappa e^2 \boldsymbol{n}_e^{(\kappa)}/3$. Averaging over the zero-point motion of the nuclei and writing $c_3 = (q^2/2M^2_\kappa)\langle\partial^2_{\hat q}E_0\rangle$ gives Eq. (12.46). The leading corrections are those of the leading-order boost: relative order $\kappa^2_{\rm BO}$, and the ratio of the mean phonon energy to the electronic energies that dominate the moment, which are of the order of the $1s$ binding.

For a nucleus of finite size with charge density $\rho_{\rm nuc}$, normalized to one, the potential is the Coulomb potential folded with $\rho_{\rm nuc}$, and $\nabla^2_{\boldsymbol{R}} V_{eN} = Z_\kappa e^2 \sum_i \rho_{\rm nuc}(\boldsymbol{r}_i - \boldsymbol{R}_\kappa)$: the contact density becomes $\int \mathrm{d}^3 r\, \boldsymbol{n}_e(\boldsymbol{r})\rho_{\rm nuc}(\boldsymbol{r} - \boldsymbol{R}_\kappa)$. For Dirac electrons the Hamiltonian depends on $\boldsymbol{R}_\kappa$ only through V_{eN}, the kinetic term $\boldsymbol{\alpha}_{\rm D}\cdot\boldsymbol{p} + \beta_{\rm D} m_e$ of Appendix C.15 being independent of it, so the steps above hold unchanged with the relativistic density.

D.34.6 Dirac density at a finite nucleus

The $1s$ solution of the Dirac equation Eq. (C.126) in the Coulomb field of a point charge Z has large and small radial components $g(r) = N r^{\gamma_{\rm D}-1}\mathrm{e}^{-\lambda r}$ and $f(r) = -[(1-\gamma_{\rm D})/(1+\gamma_{\rm D})]^{1/2} g(r)$, with $\gamma_{\rm D} = [1-(Z\alpha)^2]^{1/2}$, energy $E = m_e\gamma_{\rm D}$ and $\lambda = (m_e^2 - E^2)^{1/2} = Z\alpha m_e = Z/a_0$ [78]. These satisfy the radial equations: inserting the power law, the coefficients of $r^{\gamma_{\rm D}-2}$ require $\gamma^2_{\rm D} = 1-(Z\alpha)^2$ and those of $r^{\gamma_{\rm D}-1}$ fix λ and the ratio f/g. The density, summed over both components and divided by 4π for the angles, is

$$\boldsymbol{n}(r) = \frac{g^2+f^2}{4\pi} = \frac{N^2}{4\pi}\,\frac{2}{1+\gamma_{\rm D}}\, r^{2\gamma_{\rm D}-2}\mathrm{e}^{-2\lambda r},$$

and the normalization $\int_0^\infty \mathrm{d}r\, r^2(g^2+f^2) = 1$, with $\int_0^\infty \mathrm{d}r\, r^{2\gamma_{\rm D}}\mathrm{e}^{-2\lambda r} = \Gamma(2\gamma_{\rm D}+1)/(2\lambda)^{2\gamma_{\rm D}+1}$, gives $\boldsymbol{n}(r) = (2\lambda)^3(2\lambda r)^{2\gamma_{\rm D}-2}\mathrm{e}^{-2\lambda r}/[4\pi\,\Gamma(2\gamma_{\rm D}+1)]$. The nonrelativistic contact density is λ^3/π, Appendix C.15, so the ratio is $2(2\lambda r)^{2\gamma_{\rm D}-2}\mathrm{e}^{-2\lambda r}/\Gamma(2\gamma_{\rm D}+1)$, which equals one at $\gamma_{\rm D} = 1$ and diverges weakly at $r = 0$ otherwise. Inside a nucleus $2\lambda r \le 2ZR_{\rm nuc}/a_0$, which is 1.8×10^{-3} for silicon and 5.8×10^{-3} for germanium, and we set $\mathrm{e}^{-2\lambda r} = 1$ there; this overstates the average by 0.14 % in silicon and 0.43 % in germanium. We keep the point-charge orbital, whose shape outside the nucleus the finite size changes only at higher order

in $2ZR_{\rm nuc}/a_0$. Averaging over a uniform sphere of radius $R_{\rm nuc}$, $(3/R_{\rm nuc}^3)\int_0^{R_{\rm nuc}} {\rm d}r\, r^2\, r^{2\gamma_{\rm D}-2} = 3R_{\rm nuc}^{2\gamma_{\rm D}-2}/(2\gamma_{\rm D}+1)$, gives the first form of Eq. (12.47). With $2\gamma_{\rm D}-2 = -(Z\alpha)^2 + O((Z\alpha)^4)$, the power is $\exp[(Z\alpha)^2 \ln(a_0/2ZR_{\rm nuc})]$, and the prefactor is $1+O((Z\alpha)^2)$ without a logarithm; this is the second form, with that non-logarithmic remainder written $c_{\rm rel}(Z\alpha)^2$. The point-charge orbital inside the nucleus fixes the logarithm but not $c_{\rm rel}$, which depends on the model of the orbital there. The small components of the $p_{1/2}$ orbitals are s-like at the origin and add contact density of relative order $(Z\alpha)^2$, which the $1s$ factor does not contain. The density at the nucleus of every s shell has the same radial dependence $r^{2\gamma_{\rm D}-2}$ near the origin, set by the nuclear charge alone, and Sect. 12.5.5 applies the $1s$ ratio to all of them.

D.35 The phonon-assisted band sum

Two one-body vertices acting on a Slater determinant pass only through intermediate states with one electron–hole pair, and on the energy shell the paths through a hole in a filled band take exactly the form of the Pauli-blocked electron paths. We prove both statements, which Sect. 13.2 uses, together with the signs of the hole paths and the reduction of the amplitude to the derivative of a frozen-phonon transition density. The proof of the band-sum theorem does not use the form of the vertices, so it holds for light (Appendix D.18) as well as for dark matter.

Intermediate states and fermion signs

Let $|\Phi_0\rangle$ be a Slater determinant of one-electron states $|a\rangle$ with energies ε_a: the filled set $\mathcal{F}$ (the valence bands) and the empty set $\mathcal{E}$ (the conduction bands). Let

$$X = \sum_{ab} \mathsf{x}_{ab}\, c_a^\dagger c_b, \qquad Y = \sum_{ab} \mathsf{y}_{ab}\, c_a^\dagger c_b,$$

be two one-body operators. For phonon-assisted electron recoil x_{ab} is the density vertex ρ and y_{ab} is $N^{-1/2}g$ times the phonon matrix element $(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})^{1/2}$; for light x_{ab} is the velocity vertex. The process absorbs the energy ω at the vertex X and changes the energy of the bosons by $\pm\omega_{\boldsymbol{Q}\nu}$ at the vertex Y. The final state is the single pair

$$|\Phi_f\rangle = c_c^\dagger c_v |\Phi_0\rangle, \qquad c \in \mathcal{E},\ v \in \mathcal{F},$$

with $E_f - E_0 = \varepsilon_c - \varepsilon_v \pm \omega_{\boldsymbol{Q}\nu}$, the boson energy included. The second-order amplitude is

$$T = \sum_m \left[\frac{\langle\Phi_f|Y|m\rangle\langle m|X|\Phi_0\rangle}{E_0 + \omega - E_m} + \frac{\langle\Phi_f|X|m\rangle\langle m|Y|\Phi_0\rangle}{E_0 - E_m \mp \omega_{\boldsymbol{Q}\nu}} \right], \tag{D.198}$$

with E_m the electronic energy of the intermediate state; in the second ordering the boson has already been emitted or absorbed. This is Eq. (13.7) with the boson energy written out.

Intermediate states. Acting on $|\Phi_0\rangle$, $c_a^\dagger c_b$ is nonzero only for $b \in \mathcal{F}$, and then either $a = b$, which returns $|\Phi_0\rangle$, or $a \in \mathcal{E}$, which creates the pair $(a;b)$. So

$$X|\Phi_0\rangle = \langle X\rangle_0\, |\Phi_0\rangle + \sum_{a\in\mathcal{E},\, b\in\mathcal{F}} \mathsf{x}_{ab}\, c_a^\dagger c_b |\Phi_0\rangle, \qquad \langle X\rangle_0 = \sum_{b\in\mathcal{F}} \mathsf{x}_{bb},$$

and the same for Y. The second vertex must turn the intermediate state into $|\Phi_f\rangle$ by moving one electron. From the pair $(a;b)$ it can move the electron a into c, if $b = v$, or move the electron of v into the hole b, if $a = c$; any other move leaves a pair different from $(c;v)$ or creates a second pair. From $|\Phi_0\rangle$ it must create $(c;v)$ itself. The intermediate states are

therefore of three kinds. One is an electron in an empty band with the final hole, $(p;v)$ with $p\in\mathcal{E}$. Another is the final electron with a hole in a filled band, $(c;p)$ with $p\in\mathcal{F}$. The third is $|\Phi_0\rangle$ or $|\Phi_f\rangle$ itself, reached through the diagonal parts of the vertices. Crystal momentum fixes where p sits: in the density-first ordering at $\boldsymbol{k}+\boldsymbol{q}$ for an electron and at $\boldsymbol{k}+\boldsymbol{Q}$ for a hole, and the reverse in the phonon-first ordering.

Signs. For the electron paths, $c_c^\dagger c_p\, c_p^\dagger c_v|\Phi_0\rangle = c_c^\dagger(1-n_p)c_v|\Phi_0\rangle = c_c^\dagger c_v|\Phi_0\rangle$, since p is empty and differs from v. For the hole paths, with p filled and $p\neq v$, $c\neq p$,

$$c_p^\dagger c_v\, c_c^\dagger c_p = -c_p^\dagger\, c_c^\dagger\, c_v\, c_p = c_c^\dagger\, c_p^\dagger\, c_v\, c_p = -c_c^\dagger\, c_p^\dagger c_p\, c_v,$$

and $c_p^\dagger c_p = n_p = 1$ on $c_v|\Phi_0\rangle$, so the product is $-c_c^\dagger c_v$ on $|\Phi_0\rangle$. Each hole path carries a factor -1.

Diagonal terms. For the density, $\langle X\rangle_0$ is the Fourier component $\bar\rho_e(\boldsymbol{q})$ of the ground-state density, nonzero only at $\boldsymbol{q}=\boldsymbol{G}$; for the electron–phonon vertex it needs $\boldsymbol{Q}=0$. When it is present, it enters Eq. (D.198) twice. With $m=\Phi_0$ in the density-first ordering the term is $\mathsf{y}_{cv}\langle X\rangle_0/\omega$. With $m=\Phi_f$ in the phonon-first ordering it is $\langle\Phi_f|X|\Phi_f\rangle\,\mathsf{y}_{cv}/(E_0-E_m\mp\omega_{\boldsymbol{Q}\nu})$, where $E_m=E_f\mp\omega_{\boldsymbol{Q}\nu}$ is the electronic energy of Φ_f, so that the denominator is $E_0-E_f=-\omega$ and the term is $-(\langle X\rangle_0+\mathsf{x}_{cc}-\mathsf{x}_{vv})\,\mathsf{y}_{cv}/\omega$ on the shell $\omega=\varepsilon_c-\varepsilon_v\pm\omega_{\boldsymbol{Q}\nu}$. The ground-state expectation value cancels between the two, as the linked-cluster theorem requires, and what remains, $(\mathsf{x}_{vv}-\mathsf{x}_{cc})\,\mathsf{y}_{cv}/\omega$, is the $p=v$ electron term of the first ordering plus the $p=c$ electron term of the second, which the band sum below already contains. The diagonal part of Y cancels in the same way.

The band-sum theorem

Collecting the four kinds of intermediate pair with their signs and energies,

$$T=\sum_{p\in\mathcal{E}}\left[\frac{\mathsf{y}_{cp}\,\mathsf{x}_{pv}}{\omega-\varepsilon_p+\varepsilon_v}+\frac{\mathsf{x}_{cp}\,\mathsf{y}_{pv}}{\varepsilon_v-\varepsilon_p\mp\omega_{\boldsymbol{Q}\nu}}\right]$$
$$-\sum_{p\in\mathcal{F}}\left[\frac{\mathsf{x}_{cp}\,\mathsf{y}_{pv}}{\omega-\varepsilon_c+\varepsilon_p}+\frac{\mathsf{y}_{cp}\,\mathsf{x}_{pv}}{\varepsilon_p-\varepsilon_c\mp\omega_{\boldsymbol{Q}\nu}}\right],$$

where the first bracket holds the electron paths A and B of Eq. (13.8) and the second the hole paths C and D. On the energy shell, $\omega=\varepsilon_c-\varepsilon_v\pm\omega_{\boldsymbol{Q}\nu}$, the hole denominators are

$$\omega-\varepsilon_c+\varepsilon_p=-(\varepsilon_v-\varepsilon_p\mp\omega_{\boldsymbol{Q}\nu}),\qquad \varepsilon_p-\varepsilon_c\mp\omega_{\boldsymbol{Q}\nu}=-(\omega-\varepsilon_p+\varepsilon_v),$$

the negatives of the electron denominators with the same p. The two minus signs cancel, and the filled-band terms take the form of the empty-band terms:

$$T=\sum_{p\in\mathcal{E}\cup\mathcal{F}}\left[\frac{\mathsf{y}_{cp}\,\mathsf{x}_{pv}}{\omega-\varepsilon_p+\varepsilon_v}+\frac{\mathsf{x}_{cp}\,\mathsf{y}_{pv}}{\varepsilon_v-\varepsilon_p\mp\omega_{\boldsymbol{Q}\nu}}\right]=\langle c|\,\mathsf{y}\,\frac{1}{\varepsilon_v+\omega-h}\,\mathsf{x}+\mathsf{x}\,\frac{1}{\varepsilon_v\mp\omega_{\boldsymbol{Q}\nu}-h}\,\mathsf{y}\,|v\rangle,\tag{D.199}$$

where h is the one-electron Hamiltonian and x, y are the one-electron operators with matrix elements x_{ab}, y_{ab}. The many-electron amplitude on the shell is the amplitude of one electron that starts in v in an otherwise empty crystal, with the full one-electron resolvent and no projector on empty states. The terms $p=v$ in the first resolvent and $p=c$ in the second are finite, with denominators ω and $-\omega$. With $\mathsf{x}=\rho$, $\mathsf{y}=g$ (the factor $N^{-1/2}(n_{\boldsymbol{Q}\nu}+\frac12\pm\frac12)^{1/2}$ taken out) and the crystal momenta written out, Eq. (D.199) is Eq. (13.9); with x the velocity vertex it is Eq. (D.105).

Off the shell, at an energy ω that differs from E_f-E_0 by Δ, the hole denominators become $-(\varepsilon_v-\varepsilon_p\mp\omega_{\boldsymbol{Q}\nu})+\Delta$ and $-(\omega-\varepsilon_p+\varepsilon_v)+\Delta$, and the identification fails at first order in Δ. A quantity built from the amplitude away from the shell keeps the four paths separately.

The adiabatic form

Replace the boson operator $B_{\boldsymbol{Q}\nu}$ in Eq. (A.25) by a complex number b and $B_{-\boldsymbol{Q}\nu} = B^\dagger_{\boldsymbol{Q}\nu}$ by b^*. The electrons then move in the static one-body potential $N^{-1/2}(b\,V_{\boldsymbol{Q}} + b^* V^\dagger_{\boldsymbol{Q}})$, with $\langle\psi_{m,\boldsymbol{k}+\boldsymbol{Q}}|V_{\boldsymbol{Q}}|\psi_{n\boldsymbol{k}}\rangle = g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})$ and the Hermiticity relation Eq. (A.27). To first order, with the states degenerate with the unperturbed one excluded,

$$|\psi_{v\boldsymbol{k}}(b)\rangle = |\psi_{v\boldsymbol{k}}\rangle + \frac{b}{\sqrt{N}}\sum_p |\psi_{p,\boldsymbol{k}+\boldsymbol{Q}}\rangle\,\frac{g_{pv\nu}(\boldsymbol{k},\boldsymbol{Q})}{\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}}} + \frac{b^*}{\sqrt{N}}\sum_p |\psi_{p,\boldsymbol{k}-\boldsymbol{Q}}\rangle(\cdots),$$

$$\langle\psi_{c\boldsymbol{k}'}(b)| = \langle\psi_{c\boldsymbol{k}'}| + \frac{b}{\sqrt{N}}\sum_p \frac{g_{cp\nu}(\boldsymbol{k}'-\boldsymbol{Q},\boldsymbol{Q})}{\varepsilon_{c\boldsymbol{k}'} - \varepsilon_{p,\boldsymbol{k}'-\boldsymbol{Q}}}\,\langle\psi_{p,\boldsymbol{k}'-\boldsymbol{Q}}| + \frac{b^*}{\sqrt{N}}\sum_p(\cdots)\langle\psi_{p,\boldsymbol{k}'+\boldsymbol{Q}}|.$$

The bra of the conduction state is the conjugate of its ket, whose b^* term, from $V^\dagger_{\boldsymbol{Q}}$, holds the states at $\boldsymbol{k}'-\boldsymbol{Q}$ with the matrix elements $\langle\psi_{p,\boldsymbol{k}'-\boldsymbol{Q}}|V^\dagger_{\boldsymbol{Q}}|\psi_{c\boldsymbol{k}'}\rangle = g_{cp\nu}(\boldsymbol{k}'-\boldsymbol{Q},\boldsymbol{Q})^*$. With $\boldsymbol{k}' = \boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}$, only the displayed terms connect to $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ between the two states: $\langle\psi_{p,\boldsymbol{k}+\boldsymbol{q}}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_{v\boldsymbol{k}}\rangle = \rho_{pv}(\boldsymbol{k},\boldsymbol{q})$ and $\langle\psi_{c\boldsymbol{k}'}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_{p,\boldsymbol{k}+\boldsymbol{Q}}\rangle = \rho_{cp}(\boldsymbol{k}+\boldsymbol{Q},\boldsymbol{q})$. The terms with b^* reach crystal momenta $\boldsymbol{k}\pm 2\boldsymbol{Q}$ away from the pair and drop out unless $2\boldsymbol{Q}$ is a reciprocal-lattice vector. Hence

$$\rho_{cv}(\boldsymbol{k},\boldsymbol{q};b) = \frac{b}{\sqrt{N}}\sum_p\left[\frac{g_{cp\nu}(\boldsymbol{k}+\boldsymbol{q},\boldsymbol{Q})\,\rho_{pv}(\boldsymbol{k},\boldsymbol{q})}{\varepsilon_{c\boldsymbol{k}'} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{q}}} + \frac{\rho_{cp}(\boldsymbol{k}+\boldsymbol{Q},\boldsymbol{q})\,g_{pv\nu}(\boldsymbol{k},\boldsymbol{Q})}{\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}}}\right] + O(|b|^2), \tag{D.200}$$

which is $N^{-1/2}b\,M^{\mathrm{ad}}_{cv\nu}$ of Eq. (13.12). The sum runs over filled and empty bands because the first-order change of one orbital is a sum over all others. Restoring $b \to B_{\boldsymbol{Q}\nu}$ and taking the phonon matrix element gives Eq. (13.13).

Error. The full amplitude Eq. (13.9) differs from Eq. (D.200) only by $\pm\omega_{\boldsymbol{Q}\nu}$ in the denominators. On the shell the first denominator is $D_{\mathrm{A}} \pm \omega_{\boldsymbol{Q}\nu}$ with $D_{\mathrm{A}} = \varepsilon_{c\boldsymbol{k}'} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{q}}$, and the second $D_{\mathrm{B}} \mp \omega_{\boldsymbol{Q}\nu}$ with $D_{\mathrm{B}} = \varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}}$. With $1/(D\pm\omega) = 1/D \mp \omega/D^2 + \omega^2/D^3 \mp \cdots$,

$$M^{\pm} - M^{\mathrm{ad}} = \mp\,\omega_{\boldsymbol{Q}\nu}\sum_p\left[\frac{g_{cp\nu}\,\rho_{pv}}{D_{\mathrm{A}}^2} - \frac{\rho_{cp}\,g_{pv\nu}}{D_{\mathrm{B}}^2}\right] + \omega^2_{\boldsymbol{Q}\nu}\sum_p\left[\frac{g_{cp\nu}\,\rho_{pv}}{D_{\mathrm{A}}^3} + \frac{\rho_{cp}\,g_{pv\nu}}{D_{\mathrm{B}}^3}\right] + \cdots,$$

which is Eq. (13.14). Let δ_{off} be the smallest of the full denominators $|D_{\mathrm{A}}\pm\omega_{\boldsymbol{Q}\nu}|$ and $|D_{\mathrm{B}}\mp\omega_{\boldsymbol{Q}\nu}|$, bounded below by Eq. (13.11). Each adiabatic denominator then exceeds $\delta_{\mathrm{off}} - \omega_{\boldsymbol{Q}\nu}$, the nth term of the series is bounded by $(\omega_{\boldsymbol{Q}\nu}/(\delta_{\mathrm{off}}-\omega_{\boldsymbol{Q}\nu}))^n$ times $\sum_p(|g_{cp\nu}\rho_{pv}|/|D_{\mathrm{A}}| + |\rho_{cp}g_{pv\nu}|/|D_{\mathrm{B}}|)$, and

$$|M^{\pm} - M^{\mathrm{ad}}| \le \frac{\omega_{\boldsymbol{Q}\nu}}{\delta_{\mathrm{off}} - 2\omega_{\boldsymbol{Q}\nu}}\sum_p\left[\frac{|g_{cp\nu}\,\rho_{pv}|}{|D_{\mathrm{A}}|} + \frac{|\rho_{cp}\,g_{pv\nu}|}{|D_{\mathrm{B}}|}\right],$$

for $\delta_{\mathrm{off}} > 2\omega_{\boldsymbol{Q}\nu}$. The sum on the right is $|M^{\mathrm{ad}}|$ when its terms do not cancel, and the relative error is then of order $\omega_{\boldsymbol{Q}\nu}/\delta_{\mathrm{off}}$. When they cancel strongly, as for $\boldsymbol{q}\to 0$ where $M^{\pm}$ and M^{ad} both vanish by Eq. (13.10), the bound is on the absolute error only.

The rate and the loss function

The thermal state of the lattice is a product over modes of $\sum_n p_n|n\rangle\langle n|$ with $p_n = (1 - \mathrm{e}^{-\beta\omega_{\boldsymbol{Q}\nu}})\,\mathrm{e}^{-n\beta\omega_{\boldsymbol{Q}\nu}}$. The operator $B_{\boldsymbol{Q}\nu} = b_{\boldsymbol{Q}\nu} + b^\dagger_{-\boldsymbol{Q}\nu}$ changes one occupation by one, so final states that differ in which mode changed are orthogonal, and the squared matrix elements are averaged mode by mode:

$$\sum_n p_n\,|\langle n+1|b^\dagger|n\rangle|^2 = \sum_n p_n\,(n+1) = n_{\boldsymbol{Q}\nu} + 1, \qquad \sum_n p_n\,|\langle n-1|b|n\rangle|^2 = n_{\boldsymbol{Q}\nu},$$

with $n_{\boldsymbol{Q}\nu} = n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$. These are the factors $n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2}$ of Eq. (13.9), now averaged over the thermal state. For the electrons, the transition from $(v, \boldsymbol{k})$ to $(c, \boldsymbol{k}')$ needs the first state filled and the second empty, with the weight $f_v(1 - f_c)$ of the Fermi occupations; with the chemical potential in the gap, it differs from one by terms of order $\mathrm{e}^{-E_{\mathrm{g}}/2T}$. Inserting the squared amplitude into Eq. (2.67), summing over $\boldsymbol{k}$, $\boldsymbol{Q}$, the bands, the branches, the sign and the two spin states, and using $\mathscr{V} = N\Omega$ with Eq. (A.11), gives Eq. (13.15) with the extra factor $f_v(1 - f_c)$ inside the average.

The loss function is the dissipative part, the rate of each transition minus the rate of its inverse. For emission, the inverse removes the pair and absorbs a phonon, with the weight $f_c(1-f_v)\, n_{\boldsymbol{Q}\nu}$. In equilibrium $f_c(1-f_v) = f_v(1-f_c)\,\mathrm{e}^{-\beta(\varepsilon_{c\boldsymbol{k}'} - \varepsilon_{v\boldsymbol{k}})}$ and $n_{\boldsymbol{Q}\nu} = (n_{\boldsymbol{Q}\nu}+1)\,\mathrm{e}^{-\beta\omega_{\boldsymbol{Q}\nu}}$, so on the energy shell

$$f_v(1 - f_c)(n_{\boldsymbol{Q}\nu} + 1) - f_c(1 - f_v)\, n_{\boldsymbol{Q}\nu} = f_v(1 - f_c)(n_{\boldsymbol{Q}\nu} + 1)(1 - \mathrm{e}^{-\beta\omega}),$$

and in the same way for absorption with $n_{\boldsymbol{Q}\nu}$ in place of $n_{\boldsymbol{Q}\nu} + 1$. The factor $1 - \mathrm{e}^{-\beta\omega}$ of Eq. (4.45) is therefore carried by each transition, and L^{PA} is Eq. (13.16) with the weights $w_{\pm} f_v(1 - f_c)$. Both weights are nonnegative for $\omega > 0$, and so is L^{PA}. For $T \ll E_{\mathrm{g}}$ the Fermi factors are one.

D.36 Screening of the phonon-assisted amplitude

The screened phonon-assisted amplitude carries exactly one factor $1/\varepsilon_{\mathrm{mac}}(\boldsymbol{q}, \omega)$, and the screening of the final pair adds a term, the pair channel, screened at the pair's own momentum and energy. We derive both, and the screened loss function that Sect. 13.4 uses, from second-order time-dependent mean-field theory; the band-sum theorem follows as a by-product. The section then obtains the rate from the optical theorem, works two solvable limits, and adds the extra source term of an adiabatic local-density kernel. The level is the random-phase approximation, also called time-dependent Hartree, with an exchange–correlation kernel added where stated; lines inside a loop, the electron–hole ladder of Sect. 4.9, are left out.

Second-order mean-field response

The Kohn–Sham electrons move in the one-electron Hamiltonian h, with eigenstates $|a\rangle$, energies ε_a and occupations $f_a \in \{0, 1\}$; each label a stands for a band and a crystal momentum. Their density matrix $\varrho(t)$ obeys

$$\mathrm{i}\,\partial_t \varrho = \big[h + U(t),\, \varrho\big], \qquad U(t) = U_{\mathrm{ext}}(t) + f_{\mathrm{Hxc}}\,\delta n(t),$$

where U_{ext} is the applied potential and the second term the Hartree and exchange–correlation potential of the induced density δn, linear in it at this level. Two fields are applied. The dark matter acts at $(\boldsymbol{q}, \omega)$ through the one-body operator x, and the phonon through y at frequency $\omega_2 = \mp\omega_{\boldsymbol{Q}\nu}$, the upper sign for emission; the matrix element $N^{-1/2}(n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2})^{1/2}$ of the phonon operator is taken out, as in Eq. (13.9). Each field is first dressed by its own induced density. For the phonon this gives the vertex of density-functional perturbation theory, $\mathsf{y}_{ab} = g_{ab\nu}$ (Eq. (6.10)), evaluated statically with an error of order $(\omega_{\boldsymbol{Q}\nu}/E_{\mathrm{g}})^2$ (Appendix D.27). For the dark matter it gives the screened density vertex $\mathsf{x}_{ab} = \rho^{\mathrm{scr}}_{ab}$ of Eq. (13.20), the bare vertex of the total potential at $\boldsymbol{q} + \boldsymbol{G}$. With $\varrho = \varrho^{(0)} + \varrho^{(1)} + \varrho^{(2)} + \cdots$ and $\varrho^{(0)}_{ab} = f_a \delta_{ab}$, the first order in a field F of frequency ω_i is

$$(\omega_i - \varepsilon_a + \varepsilon_b)\, \varrho^F_{ab} = F_{ab}\,(f_b - f_a), \qquad \varrho^F_{ab} = \frac{F_{ab}\,(f_b - f_a)}{\omega_i - \varepsilon_a + \varepsilon_b + \mathrm{i}0^+},$$

the retarded solution. The part of $\varrho^{(2)}$ bilinear in the two fields oscillates at $\omega_f = \omega + \omega_2 = \omega \mp \omega_{\boldsymbol{Q}\nu}$ and obeys

$$(\omega_f - \varepsilon_a + \varepsilon_b)\, \varrho^{(2)}_{ab} = \mathfrak{s}^{(2)}_{ab} + U^{(2)}_{ab}\, (f_b - f_a), \qquad \mathfrak{s}^{(2)}_{ab} = \sum_m \big[\mathsf{x}_{am}\varrho^{\mathsf{y}}_{mb} - \varrho^{\mathsf{y}}_{am}\mathsf{x}_{mb} + \mathsf{y}_{am}\varrho^{\mathsf{x}}_{mb} - \varrho^{\mathsf{x}}_{am}\mathsf{y}_{mb}\big], \tag{D.201}$$

where $\mathfrak{s}^{(2)}$ is the source that each field builds on the first-order response to the other, and $U^{(2)}$ the potential of the second-order density. The sum over m runs over every state, and $\mathfrak{s}^{(2)}_{ab}$ is nonzero for every kind of pair (a, b): empty and filled, filled and filled, empty and empty.

The band-sum theorem again. Take $a = c$ empty and $b = v$ filled. For m empty, $\varrho^{\mathsf{y}}_{mv} = \mathsf{y}_{mv}/(\omega_2 + \varepsilon_v - \varepsilon_m)$ and $\varrho^{\mathsf{y}}_{cm} = 0$, and likewise for x; the source collects $\mathsf{x}_{cm}\mathsf{y}_{mv}/(\omega_2 + \varepsilon_v - \varepsilon_m) + \mathsf{y}_{cm}\mathsf{x}_{mv}/(\omega + \varepsilon_v - \varepsilon_m)$, the electron paths. For m filled, $\varrho^{\mathsf{y}}_{mv} = 0$ and $\varrho^{\mathsf{y}}_{cm} = \mathsf{y}_{cm}/(\omega_2 + \varepsilon_m - \varepsilon_c)$, and the source collects $-\mathsf{y}_{cm}\mathsf{x}_{mv}/(\omega_2 + \varepsilon_m - \varepsilon_c) - \mathsf{x}_{cm}\mathsf{y}_{mv}/(\omega + \varepsilon_m - \varepsilon_c)$, the hole paths with their sign. On the shell $\omega_f = \varepsilon_c - \varepsilon_v$ the hole denominators are $\omega_2 + \varepsilon_m - \varepsilon_c = -(\omega + \varepsilon_v - \varepsilon_m)$ and $\omega + \varepsilon_m - \varepsilon_c = -(\omega_2 + \varepsilon_v - \varepsilon_m)$, and

$$\mathfrak{s}^{(2)}_{cv} = \Big\langle c \Big| \,\mathsf{y}\, \frac{1}{\varepsilon_v + \omega - h}\, \mathsf{x} + \mathsf{x}\, \frac{1}{\varepsilon_v + \omega_2 - h}\, \mathsf{y} \,\Big| v \Big\rangle,$$

which is Eq. (D.199): the source of the pair is the four-path amplitude with the unrestricted band sum, here evaluated with the screened vertex $\mathsf{x} = \rho^{\mathrm{scr}}$. The terms $m = v$ and $m = c$ are included, with the denominators ω and $-\omega$ of Appendix D.35. The density matrix reaches the theorem without Fock space: the filled-band terms enter through the occupation differences.

The pair channel

The second-order density at the wavevector $\boldsymbol{q} + \boldsymbol{Q} + \boldsymbol{G}$ is the trace of $\varrho^{(2)}$ with $\mathrm{e}^{-\mathrm{i}(\boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G})\cdot\boldsymbol{r}}$, two spins included. With b at $\boldsymbol{k}$ and a at $\boldsymbol{k} + \boldsymbol{q} + \boldsymbol{Q}$, the matrix element of the plane wave is $\rho_{ab}(\boldsymbol{k}, \boldsymbol{q} + \boldsymbol{Q} + \boldsymbol{G})^*$. Eq. (D.201) then gives two terms:

$$\delta n^{(2)}_{\mathrm{tot},\boldsymbol{G}} = \delta n^{(2)}_{\boldsymbol{G}} + \sum_{\boldsymbol{G}'} R^0_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q} + \boldsymbol{Q}, \omega_f)\, U^{(2)}_{\boldsymbol{G}'},$$

where R^0 is the bubble Eq. (4.63), since $U^{(2)}_{ab} = \sum_{\boldsymbol{G}'} U^{(2)}_{\boldsymbol{G}'} \rho_{ab}(\boldsymbol{k}, \boldsymbol{q} + \boldsymbol{Q} + \boldsymbol{G}')$, and

$$\delta n^{(2)}_{\boldsymbol{G}}(\boldsymbol{q} + \boldsymbol{Q}) = \frac{2}{\mathscr{V}} \sum_{\boldsymbol{k}} \sum_{ab} \frac{\rho_{ab}(\boldsymbol{k}, \boldsymbol{q} + \boldsymbol{Q} + \boldsymbol{G})^*\, \mathfrak{s}^{(2)}_{ab}(\boldsymbol{k})}{\omega_f + \varepsilon_{b\boldsymbol{k}} - \varepsilon_{a,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}} + \mathrm{i}0^+} \tag{D.202}$$

is the density that the source alone creates in the Kohn–Sham electrons, summed over all pairs of states. The potential closes the loop, $U^{(2)} = f_{\mathrm{Hxc}}\, \delta n^{(2)}_{\mathrm{tot}}$, a matrix product at $(\boldsymbol{q} + \boldsymbol{Q}, \omega_f)$. Eliminating $\delta n^{(2)}_{\mathrm{tot}}$,

$$U^{(2)} = f_{\mathrm{Hxc}}(1 - R^0 f_{\mathrm{Hxc}})^{-1} \delta n^{(2)} = (1 - f_{\mathrm{Hxc}} R^0)^{-1} f_{\mathrm{Hxc}}\, \delta n^{(2)} = \tilde{\varepsilon}^{-1} f_{\mathrm{Hxc}}\, \delta n^{(2)},$$

with $\tilde{\varepsilon}^{-1} = (1 - f_{\mathrm{Hxc}} R^0)^{-1} = 1 + f_{\mathrm{Hxc}} R$ the inverse dielectric matrix of a test electron (Sect. 8.7), retarded, at $(\boldsymbol{q} + \boldsymbol{Q}, \omega_f)$. In the random-phase approximation $\tilde{\varepsilon}^{-1} f_{\mathrm{Hxc}} = \varepsilon^{-1} v_{\mathrm{C}}$ is the retarded screened interaction $W^{\mathrm{scr,R}}(\boldsymbol{q} + \boldsymbol{Q}, \omega_f)$ of Eq. (4.99).

For the final pair, $a = c$ at $\boldsymbol{k} + \boldsymbol{q} + \boldsymbol{Q}$ and $b = v$ at $\boldsymbol{k}$, Eq. (D.201) reads $(\omega_f - \varepsilon_c + \varepsilon_v)\varrho^{(2)}_{cv} = \mathfrak{s}^{(2)}_{cv} + U^{(2)}_{cv}$. This is the equation of a first-order coherence driven at ω_f by the one-body potential $\mathfrak{s}^{(2)} + U^{(2)}$, and its numerator is the amplitude for creating the pair, exactly as F_{cv}

is for a single field. The bare pairs are the asymptotic states of the mean-field evolution, and the potential that creates them is the retarded one (Sect. 8.7). The amplitude is therefore

$$\mathfrak{s}^{(2)}_{cv} + U^{(2)}_{cv} = \mathfrak{s}^{(2)}_{cv} + \sum_{\boldsymbol{G}\boldsymbol{G}'} \rho_{cv}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G}) \left[\tilde{\varepsilon}^{-1} f_{\mathrm{Hxc}}\right]_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}+\boldsymbol{Q}, \omega_f)\, \delta n^{(2)}_{\boldsymbol{G}'}(\boldsymbol{q}+\boldsymbol{Q}), \quad \text{(D.203)}$$

with $\mathfrak{s}^{(2)}_{cv}$ the four-path amplitude of the screened vertex. The pair channel is the second term. Its momentum is fixed by the final pair, and it depends on $\boldsymbol{q}$ only through $\delta n^{(2)}$.

The macroscopic factor

Every term of Eq. (D.203) is linear in $\mathsf{x} = \rho^{\mathrm{scr}}$, since $\mathfrak{s}^{(2)}$ is. The external screening therefore factors out of the whole amplitude once it factors out of ρ^{scr}. Split the kernel into its macroscopic Coulomb term, $v_0 = v_{\mathrm{C}}(q)$ at $\boldsymbol{G} = \boldsymbol{G}' = 0$, and the rest, $\bar{f} = f_{\mathrm{Hxc}} - v_0 = \bar{v} + f_{\mathrm{xc}}$ with $\bar{v}$ the Coulomb kernel without its head (Sect. 4.6). Let $\bar{R}$ be the response to $\bar{f}$ alone, $\bar{R} = R^0 + R^0 \bar{f} \bar{R}$, the response with the macroscopic Coulomb term removed of Sect. 4.4. Then $R = \bar{R} + \bar{R} v_0 R$, so that $R_{\boldsymbol{G}0} = \bar{R}_{\boldsymbol{G}0}(1 + v_0 R_{00})$, and

$$\left(\tilde{\varepsilon}^{-1}\right)_{\boldsymbol{G}0} = \delta_{\boldsymbol{G}0} + v_0 \delta_{\boldsymbol{G}0} R_{00} + \left(\bar{f}R\right)_{\boldsymbol{G}0} = \left(1 + v_0 R_{00}\right)\left[\delta_{\boldsymbol{G}0} + \left(\bar{f}\bar{R}\right)_{\boldsymbol{G}0}\right].$$

The prefactor is $(\varepsilon^{-1})_{00} = 1/\varepsilon_{\mathrm{mac}}$ with $\varepsilon_{\mathrm{mac}} = 1 - v_0 \bar{R}_{00}$ (Eq. (4.44)). In the random-phase approximation $\bar{f}\bar{R}$ has the elements $v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{G}|)\bar{R}_{\boldsymbol{G}0}$ for $\boldsymbol{G} \neq 0$ and none at $\boldsymbol{G} = 0$, which gives Eq. (13.21): $\rho^{\mathrm{scr}} = \rho^{\mathrm{LF}}/\varepsilon_{\mathrm{mac}}(\boldsymbol{q}, \omega)$ exactly. The whole amplitude is $\tilde{M}^{\pm}/\varepsilon_{\mathrm{mac}}(\boldsymbol{q}, \omega)$, with $\tilde{M}^{\pm}$ the amplitude Eq. (D.203) built with $\mathsf{x} = \rho^{\mathrm{LF}}$; that is Eq. (13.23). With a kernel, $\bar{f}$ adds the elements of f_{xc}, including its head, and ρ^{LF} gains the corresponding terms; the factorization is unchanged.

The rate

The transition rate into the pair is the golden rule with the amplitude Eq. (D.203), summed over final states as in Sect. 13.3; this gives Eq. (13.16) with $M^{\pm} \to \tilde{M}^{\pm}/\varepsilon_{\mathrm{mac}}$. That the sum over bare pairs is the whole absorptive part of the one-phonon response, with nothing counted twice, is the optical theorem of the mean-field evolution. The one-phonon response at $(\boldsymbol{q}, \omega)$ with its external chains removed is a sum of two kinds of diagram (Sect. 8.4.4): those with both dark-matter vertices on one loop, whose cut at the pair poles gives $\sum_f |\mathfrak{s}^{(2)}_f|^2$, and those with a pair segment, two second-order densities Eq. (D.202) joined by $\tilde{\varepsilon}^{-1} f_{\mathrm{Hxc}}$. The absorptive part of the second kind has three pieces. The cut of either density at a pair pole gives $\rho^*_f \mathfrak{s}^{(2)}_f$; the absorptive part of the screened interaction is $W^{\mathrm{scr,R}\dagger}\left[\frac{1}{2\mathrm{i}}(R^0 - R^{0\dagger})\right] W^{\mathrm{scr,R}}$, by Eq. (8.67) multiplied by f_{Hxc} on both sides, and the cut of R^0 is again a sum over bare pairs. The cross pieces and the square of the pair term complete the square $|\mathfrak{s}^{(2)}_f + U^{(2)}_f|^2$ pair by pair. With the retarded interaction this holds for each pair; the advanced one would give the same sum with a different share per pair, as in Sect. 8.7.

Inside the window the external chains are not absorptive at this order. No zero-phonon pair exists at $(\boldsymbol{q}, \omega)$, so $R^0(\boldsymbol{q}, \omega)$ is Hermitian, $\varepsilon_{\mathrm{mac}}(\boldsymbol{q}, \omega)$ is real up to terms of order g^2, and the retarded and advanced external weights coincide. With $|\varepsilon_{\mathrm{mac}}|^2 = (\varepsilon^0_1)^2[1 + O(g^2)]$ and $|\tilde{M}^{\pm}|^2 = O(g^2)$, Eq. (13.24) follows with an error $O(g^4)$. Outside the window $\varepsilon^0_2 \neq 0$, and both statements fail (Sect. 13.6).

Two solvable limits

A source proportional to a potential. Suppose the source acts on every pair at $\boldsymbol{q}+\boldsymbol{Q}$ as an applied potential Φ would, $\mathfrak{s}^{(2)}_{ab} = \Phi\, \rho_{ab}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{Q})(f_b - f_a)$ with $\boldsymbol{G} = 0$ only. Then

$\delta n^{(2)}_{\boldsymbol{G}} = R^0_{\boldsymbol{G}0}\Phi$, and with $1+\tilde\varepsilon^{-1}f_{\mathrm{Hxc}}R^0=\tilde\varepsilon^{-1}$,

$$\mathfrak{s}^{(2)}_{cv}+U^{(2)}_{cv}=\Phi\sum_{\boldsymbol{G}}\rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G})\,(\tilde\varepsilon^{-1})_{\boldsymbol{G}0}(\boldsymbol{q}+\boldsymbol{Q},\omega_f),$$

the source screened like an external potential at the pair's momentum and energy. Without local fields this is $\mathfrak{s}^{(2)}_{cv}/\varepsilon(\boldsymbol{q}+\boldsymbol{Q},\omega_f)$.

A rigid translation. At $\boldsymbol{Q}=0$ a uniform displacement $u(t)$ of all nuclei couples through $\mathsf{y}=u[\mathcal{A},h]$ with $\mathcal{A}=-\mathrm{i}\ell\,\hat{\boldsymbol{e}}\cdot\hat{\boldsymbol{p}}$, the generator of translations of the electrons (Eq. (6.12)), and $[\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}},\mathcal{A}]=c_{\mathcal{A}}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ with $c_{\mathcal{A}}=\mathrm{i}\ell\,\hat{\boldsymbol{e}}\cdot\boldsymbol{q}$. In the frame that moves with the nuclei, $\mathrm{e}^{u\mathcal{A}}$ removes y to first order in u, multiplies the dark-matter potential by $1+c_{\mathcal{A}}u$, and adds $-\mathrm{i}\dot u\mathcal{A}$. Since $\mathcal{A}$ commutes with the Coulomb interaction and with the Hartree potential, the term $c_{\mathcal{A}}u\,\phi\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ at frequency ω_f is a perturbation of the undisplaced crystal, screened by $\varepsilon(\boldsymbol{q},\omega_f)$; the term in $\dot u$ is of order ω_2. For pairs at $(\boldsymbol{q},\omega_f)$, without local fields, the amplitude is $c_{\mathcal{A}}\rho_{cv}(\boldsymbol{k},\boldsymbol{q})/\varepsilon(\boldsymbol{q},\omega_f)+O(\omega_2)$. The four-path term alone gives something else. With $\mathsf{y}_{ab}=u(\varepsilon_b-\varepsilon_a)\mathcal{A}_{ab}$, write $\varepsilon_p-\varepsilon_c=(\varepsilon_p-\varepsilon_v-\omega)-\omega_2$ in the first term of Eq. (D.199) and $\varepsilon_v-\varepsilon_p=(\varepsilon_v-\varepsilon_p+\omega_2)-\omega_2$ in the second; the parts without ω_2 cancel their denominators and sum to $u[\mathsf{x},\mathcal{A}]_{cv}=c_{\mathcal{A}}u\,\mathsf{x}_{cv}$, so $\mathfrak{s}^{(2)}_{cv}=c_{\mathcal{A}}u\,\rho_{cv}/\varepsilon(\boldsymbol{q},\omega)+O(\omega_2)$, screened at the energy deposited. The Kohn–Sham density of the source is the same frame argument without self-consistency: the response $c_{\mathcal{A}}u\,\phi' R^0(\boldsymbol{q},\omega_f)$ to the phase factor, minus the first-order density $c_{\mathcal{A}}u\,\phi' R^0(\boldsymbol{q},\omega)$ carried along by the translation, with $\phi'=\phi/\varepsilon(\boldsymbol{q},\omega)$ the screened potential. The pair term of Eq. (D.203) is then

$$\rho_{cv}\,\frac{v_{\mathrm{C}}(q)}{\varepsilon(\boldsymbol{q},\omega_f)}\,\frac{c_{\mathcal{A}}u\big[R^0(\boldsymbol{q},\omega_f)-R^0(\boldsymbol{q},\omega)\big]}{\varepsilon(\boldsymbol{q},\omega)}=c_{\mathcal{A}}u\,\rho_{cv}\Big[\frac{1}{\varepsilon(\boldsymbol{q},\omega_f)}-\frac{1}{\varepsilon(\boldsymbol{q},\omega)}\Big],$$

using $v_{\mathrm{C}}R^0=1-\varepsilon$, and the two terms add to the frame result. The pair channel moves the screening of the displaced density from the energy deposited to the energy of the final pair. This limit needs a zero-phonon pair at $(\boldsymbol{q},\omega_f)$, which the window excludes; it is a check of Eq. (D.203), not a prescription.

The adiabatic local-density source

An adiabatic local kernel takes v_{xc} as a function of the local density (Eq. (4.71)), and to second order

$$v_{\mathrm{xc}}(n_{\mathrm{gs}}+\delta n)=v_{\mathrm{xc}}(n_{\mathrm{gs}})+f_{\mathrm{xc}}\,\delta n+\tfrac12 k_{\mathrm{xc}}\,\delta n^2+\cdots,\qquad k_{\mathrm{xc}}=\frac{\mathrm{d}^3[n\,\varepsilon_{\mathrm{xc}}(n)]}{\mathrm{d}n^3}\Big|_{n_{\mathrm{gs}}(\boldsymbol{r})}.$$

With $\delta n=\delta n_{\mathsf{x}}+\delta n_{\mathsf{y}}+\delta n^{(2)}_{\mathrm{tot}}$, the first-order densities induced by the dark matter at $(\boldsymbol{q},\omega)$ and by the phonon at $(\boldsymbol{Q},\omega_2)$, the part bilinear in the two fields is $f_{\mathrm{xc}}\delta n^{(2)}_{\mathrm{tot}}+k_{\mathrm{xc}}\,\delta n_{\mathsf{x}}\delta n_{\mathsf{y}}$; the cross term of $\frac12(\delta n_{\mathsf{x}}+\delta n_{\mathsf{y}})^2$ has coefficient one. The first is in $U^{(2)}$ already. The second is a new potential at $(\boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G},\omega_f)$, the Fourier component of a product in real space, which the electrons screen as any applied potential:

$$U^{(2)}=\tilde\varepsilon^{-1}\big[f_{\mathrm{Hxc}}\,\delta n^{(2)}+(k_{\mathrm{xc}}\,\delta n_{\mathsf{x}}\delta n_{\mathsf{y}})\big],$$

which adds $\sum_{\boldsymbol{G}}\rho_{cv}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{Q}+\boldsymbol{G})\,[\tilde\varepsilon^{-1}(k_{\mathrm{xc}}\delta n_{\mathsf{x}}\delta n_{\mathsf{y}})]_{\boldsymbol{G}}$ to Eq. (D.203). The random-phase approximation has no such term, since the Hartree potential is linear in the density. For exchange alone, $n\varepsilon_{\mathsf{x}}=-C_{\mathsf{x}}n^{4/3}$ gives $f_{\mathsf{x}}=-\frac49 C_{\mathsf{x}}n^{-2/3}$ and $k_{\mathsf{x}}=\frac{8}{27}C_{\mathsf{x}}n^{-5/3}=-\frac23 f_{\mathsf{x}}/n$. Against the Hartree part of the pair term, $v_{\mathrm{C}}\,\delta n^{(2)}$, the new term is then

$$\frac{|k_{\mathsf{x}}\,\delta n_{\mathsf{x}}\delta n_{\mathsf{y}}|}{|v_{\mathrm{C}}\,\delta n^{(2)}|}=\frac23\Big|\frac{f_{\mathsf{x}}}{v_{\mathrm{C}}(|\boldsymbol{q}+\boldsymbol{Q}|)}\Big|\,\frac{|\delta n_{\mathsf{x}}\delta n_{\mathsf{y}}|}{n\,|\delta n^{(2)}|}=\frac23\Big(\frac{|\boldsymbol{q}+\boldsymbol{Q}|}{2k_{\mathrm{F}}}\Big)^2\frac{|\delta n_{\mathsf{x}}\delta n_{\mathsf{y}}|}{n\,|\delta n^{(2)}|},$$

with $f_{\mathrm{x}} = -e^2/4k_{\mathrm{F}}^2$ of the gas (Sect. 4.6). The last ratio is of order one in the electron gas, whose second-order response is of order R^2/n.

D.37 The soft acoustic limit

For a long-wavelength acoustic phonon the phonon-assisted amplitude Eq. (13.9) tends to the limit Eq. (13.31), of order $\ell_{\boldsymbol{Q}\nu} \propto |\boldsymbol{Q}|^{-1/2}$, although its intraband terms are of the form 0/0 and taken alone would suggest $\ell_{\boldsymbol{Q}\nu}/|\boldsymbol{Q}|$. The derivation has three steps. An exact identity, valid at every $\boldsymbol{Q}$ on the energy shell, trades a part of the electron–phonon vertex for a contact term. A particular choice of that part, the displacement that carries each electron along with the lattice, removes the rigid translation from the vertex. What is left vanishes at $\boldsymbol{Q} = 0$, and its first order in $\boldsymbol{Q}$ is the deformation potential and a velocity term, which give the limit.

A Ward identity for the amplitude

Let $\hat{\mathrm{t}}_{\boldsymbol{Q}}$ be any one-electron operator that raises the crystal momentum by $\boldsymbol{Q}$, with matrix elements $\mathrm{t}_{mn}(\boldsymbol{\kappa}, \boldsymbol{Q}) = \langle m, \boldsymbol{\kappa}+\boldsymbol{Q}|\hat{\mathrm{t}}_{\boldsymbol{Q}}|n, \boldsymbol{\kappa}\rangle$, and define the vertex

$$\delta g_{mn}(\boldsymbol{\kappa}, \boldsymbol{Q}) = \left(\varepsilon_{m,\boldsymbol{\kappa}+\boldsymbol{Q}} - \varepsilon_{n\boldsymbol{\kappa}} \pm \omega_{\boldsymbol{Q}\nu}\right) \mathrm{t}_{mn}(\boldsymbol{\kappa}, \boldsymbol{Q}), \tag{D.204}$$

with the sign of emission (upper) or absorption (lower) as in Eq. (13.9). It is the matrix element of the commutator of $\hat{\mathrm{t}}_{\boldsymbol{Q}}$, multiplied by the phonon operator, with the Hamiltonian of free electrons and phonons. Insert δg in place of g into Eq. (13.9). In the first term, with p at $\boldsymbol{k}+\boldsymbol{q}$, the vertex is $\delta g_{cp}(\boldsymbol{k}+\boldsymbol{q}, \boldsymbol{Q}) = (\varepsilon_{c\boldsymbol{k}'} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{q}} \pm \omega_{\boldsymbol{Q}\nu})\mathrm{t}_{cp}$, and on the energy shell, $\omega = \varepsilon_{c\boldsymbol{k}'} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu}$, the denominator $\omega - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}$ is the same number. Their ratio is exactly 1. In the second term the vertex is $\delta g_{pv}(\boldsymbol{k}, \boldsymbol{Q}) = (\varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}} \pm \omega_{\boldsymbol{Q}\nu})\mathrm{t}_{pv}$ and the denominator its negative, so the ratio is -1. Both statements hold band by band, including the intraband terms $p = c$ in the first sum and $p = v$ in the second, whose denominators are of order $|\boldsymbol{Q}|$: the vertex vanishes with them, and no quotient 0/0 ever has to be taken. The sums over all bands p are then complete, and

$$M^{\pm}[\delta g] = \sum_p \left[\mathrm{t}_{cp}(\boldsymbol{k}+\boldsymbol{q}, \boldsymbol{Q})\, \rho_{pv}(\boldsymbol{k}, \boldsymbol{q}) - \rho_{cp}(\boldsymbol{k}+\boldsymbol{Q}, \boldsymbol{q})\, \mathrm{t}_{pv}(\boldsymbol{k}, \boldsymbol{Q})\right] = \langle c, \boldsymbol{k}'|\, [\hat{\mathrm{t}}_{\boldsymbol{Q}}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}]\, |v, \boldsymbol{k}\rangle. \tag{D.205}$$

Since $M^{\pm}$ is linear in the vertex, for any $\hat{\mathrm{t}}_{\boldsymbol{Q}}$

$$M^{\pm}[g] = M^{\pm}[g - \delta g] + \langle c, \boldsymbol{k}'|\, [\hat{\mathrm{t}}_{\boldsymbol{Q}}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}]\, |v, \boldsymbol{k}\rangle. \tag{D.206}$$

This is exact on the energy shell, at every $\boldsymbol{Q}$, for every band and branch. It is a Ward identity in the sense of Sect. 8.3: a transformation of the states generated by $\hat{\mathrm{t}}_{\boldsymbol{Q}}$ changes the electron–phonon vertex by a commutator with the unperturbed Hamiltonian and the density operator by a commutator with $\hat{\mathrm{t}}_{\boldsymbol{Q}}$, and the physical amplitude does not change. Off the shell the ratios are no longer ± 1 and the identity fails, as the band-sum theorem of Appendix D.35 does.

The comoving vertex

A long acoustic wave displaces the lattice near $\boldsymbol{r}$ by $\boldsymbol{u}(\boldsymbol{r}) = (\ell_{\boldsymbol{Q}\nu}/\sqrt{N})\, \hat{\boldsymbol{e}}_\nu\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}} B_{\boldsymbol{Q}\nu}$ (Sect. 6.2), and an orbital carried along with it changes by $-\boldsymbol{u}\cdot\nabla\psi = -\mathrm{i}\,\boldsymbol{u}\cdot\hat{\boldsymbol{p}}\,\psi$. Take $\hat{\mathrm{t}}_{\boldsymbol{Q}}$ to be minus this change per unit of $N^{-1/2}B_{\boldsymbol{Q}\nu}$, symmetrized so that it is defined for a $\boldsymbol{u}$ that varies in space,

$$\hat{\mathrm{t}}_{\boldsymbol{Q}} = \mathrm{i}\,\ell_{\boldsymbol{Q}\nu}\, \tfrac{1}{2}\{\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}},\, \hat{\boldsymbol{e}}_\nu\cdot\hat{\boldsymbol{p}}\}. \tag{D.207}$$

With a nonlocal pseudopotential, $\hat{\boldsymbol{p}}$ is the generator of translations of Sect. 6.2, which moves the nonlocal projectors with the ions. The choice produces a recoil term, removes the rigid translation from the vertex, and leaves a strain and a velocity coupling at first order in $\boldsymbol{Q}$.

Contact term. Since $[\hat{\boldsymbol{p}}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}] = \boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, the commutator in Eq. (D.206) is $\mathrm{i}\ell_{\boldsymbol{Q}\nu}(\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{q})\,\mathrm{e}^{\mathrm{i}(\boldsymbol{q}+\boldsymbol{Q})\cdot\boldsymbol{r}}$, and

$$\langle c, \boldsymbol{k}'|\,[\hat{\mathrm{t}}_{\boldsymbol{Q}}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}]\,|v, \boldsymbol{k}\rangle = \mathrm{i}\,\ell_{\boldsymbol{Q}\nu}\,(\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{q})\,\rho_{cv}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{Q}). \tag{D.208}$$

It is the term of first order in the displacement of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{r}-\boldsymbol{u})}$, the density of electrons that ride on the displaced lattice: the dark matter kicks an electron, and the lattice under it recoils. The commutator of $\hat{\boldsymbol{p}}$ with $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, which is $\boldsymbol{q}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, is the electron part of the translation algebra behind Eq. (8.28): the density at wavevector $\boldsymbol{q}$ carries momentum $\boldsymbol{q}$.

No rigid translation. At $\boldsymbol{Q} = 0$, since $\omega_{\boldsymbol{Q}\nu} \to 0$ on an acoustic branch,

$$\mathrm{t}_{mn}(\boldsymbol{\kappa}, 0) = \mathrm{i}\ell_{\boldsymbol{0}\nu}\,\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{p}_{mn}(\boldsymbol{\kappa}),$$
$$\delta g_{mn}(\boldsymbol{\kappa}, 0) = \mathrm{i}\ell_{\boldsymbol{0}\nu}\,(\varepsilon_{m\boldsymbol{\kappa}} - \varepsilon_{n\boldsymbol{\kappa}})\,\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{p}_{mn}(\boldsymbol{\kappa}).$$

For the acoustic polarization $(M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$, the vertex Eq. (A.26) at $\boldsymbol{Q} = 0$ is $\ell\,\hat{\boldsymbol{e}}_\nu \cdot \sum_\kappa \langle m|\partial_{\kappa,\boldsymbol{0}} V_{\mathrm{KS}}|n\rangle$, which Eq. (6.12) evaluates to the same expression. The comoving vertex

$$g^{\mathrm{cm}}_{mn\nu}(\boldsymbol{\kappa}, \boldsymbol{Q}) = g_{mn\nu}(\boldsymbol{\kappa}, \boldsymbol{Q}) - \delta g_{mn}(\boldsymbol{\kappa}, \boldsymbol{Q}) \tag{D.209}$$

therefore vanishes at $\boldsymbol{Q} = 0$ for every pair of bands, interband as well as intraband. The lab-frame vertex does not: its interband elements at $\boldsymbol{Q} \to 0$ are those of a rigid translation, of order $\ell_{\boldsymbol{Q}\nu} \propto |\boldsymbol{Q}|^{-1/2}$.

First order in $\boldsymbol{Q}$. Write Eq. (D.209) as $[g - (\varepsilon_{m,\boldsymbol{\kappa}+\boldsymbol{Q}} - \varepsilon_{n\boldsymbol{\kappa}})\mathrm{t}_{mn}] \mp \omega_{\boldsymbol{Q}\nu}\mathrm{t}_{mn}$. The bracket is the potential seen by an electron in coordinates that move with the lattice. Over a region short compared with $2\pi/|\boldsymbol{Q}|$ the wave is a uniform strain there, $u_{\alpha\beta} = \frac{\mathrm{i}}{2}(Q_\alpha \hat{e}_{\nu\beta} + Q_\beta \hat{e}_{\nu\alpha})\ell_{\boldsymbol{Q}\nu}$ per unit amplitude, and in comoving coordinates the reduced wavevector of a Bloch state does not change. The intraband element of the bracket is therefore the band shift under a uniform strain at fixed reduced wavevector, $\sum D^n_{\alpha\beta} u_{\alpha\beta}$, which is the Bardeen–Shockley argument of Sect. 6.2 and defines the deformation-potential tensor D^n of Eq. (6.13). The last term is $\mp c_\nu|\boldsymbol{Q}|$ times $\mathrm{t}_{mn} \to \mathrm{i}\ell\,\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{p}_{mn}$: the coupling to the velocity of the moving lattice. To first order in $\boldsymbol{Q}$, with $\hat{\boldsymbol{Q}} = \boldsymbol{Q}/|\boldsymbol{Q}|$,

$$g^{\mathrm{cm}}_{nn\nu}(\boldsymbol{\kappa}, \boldsymbol{Q}) = \mathrm{i}\,\ell_{\boldsymbol{Q}\nu}|\boldsymbol{Q}|\,[\hat{\boldsymbol{Q}} \cdot D^n \cdot \hat{\boldsymbol{e}}_\nu \mp c_\nu\,\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{p}_{nn}(\boldsymbol{\kappa})], \tag{D.210}$$

and the interband elements of g^{cm} are also of order $\ell_{\boldsymbol{Q}\nu}|\boldsymbol{Q}|$. Adding $\delta g_{nn} = (|\boldsymbol{Q}|\,\hat{\boldsymbol{Q}} \cdot \boldsymbol{v}^{\mathrm{g}}_n \pm c_\nu|\boldsymbol{Q}|)\,\mathrm{i}\ell\,\hat{\boldsymbol{e}}_\nu \cdot \boldsymbol{p}_{nn}$ back, the velocity terms cancel, and the intraband vertex in the laboratory frame is

$$g_{nn\nu}(\boldsymbol{\kappa}, \boldsymbol{Q}) = \mathrm{i}\,\ell_{\boldsymbol{Q}\nu}|\boldsymbol{Q}|\;\hat{\boldsymbol{Q}} \cdot (D^n + \boldsymbol{v}^{\mathrm{g}}_n \otimes \boldsymbol{p}_{nn}) \cdot \hat{\boldsymbol{e}}_\nu, \tag{D.211}$$

with $\boldsymbol{v}^{\mathrm{g}}_n$ the group velocity at $\boldsymbol{\kappa}$. At a band extremum $\boldsymbol{v}^{\mathrm{g}}_n = 0$, and Eq. (D.211) reduces to Eq. (6.13). Away from it the laboratory vertex carries the Galilean term $\boldsymbol{v}^{\mathrm{g}}_n \otimes \boldsymbol{p}_{nn}$, equal to $\boldsymbol{p}_{nn} \otimes \boldsymbol{p}_{nn}/m_e$ for a local potential, which the frozen-strain calculation of Appendix E.17 does not produce, because it computes D^n alone; a nonlocal potential, with $\boldsymbol{v}^{\mathrm{g}}_n \neq \boldsymbol{p}_{nn}/m_e$, adds a part that is not symmetric in the two indices.

The limit

Apply Eq. (D.206) with the comoving choice. The contact term is Eq. (D.208), with $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{Q}) \to \rho_{cv}(\boldsymbol{k}, \boldsymbol{q})$ as $\boldsymbol{Q} \to 0$. In $M^{\pm}[g^{\mathrm{cm}}]$, the interband terms have vertices of order $\ell|\boldsymbol{Q}|$ and denominators that stay finite, and they are of order $\ell|\boldsymbol{Q}|$. Two terms are not: the intraband

terms, $p = c$ in the first sum and $p = v$ in the second, whose denominators vanish with $\boldsymbol{Q}$. On the shell they are

$$\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}} = \varepsilon_{c\boldsymbol{k}'} - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} \pm \omega_{\boldsymbol{Q}\nu} = |\boldsymbol{Q}|(\hat{\boldsymbol{Q}}\cdot\boldsymbol{v}_c^{\mathrm{g}} \pm c_\nu) + O(Q^2),$$

with $\boldsymbol{v}_c^{\mathrm{g}}$ at $\boldsymbol{k}+\boldsymbol{q}$, and $\varepsilon_{v\boldsymbol{k}} - \varepsilon_{v,\boldsymbol{k}+\boldsymbol{Q}} \mp \omega_{\boldsymbol{Q}\nu} = -|\boldsymbol{Q}|(\hat{\boldsymbol{Q}}\cdot\boldsymbol{v}_v^{\mathrm{g}} \pm c_\nu) + O(Q^2)$, with $\boldsymbol{v}_v^{\mathrm{g}}$ at $\boldsymbol{k}$. Divided into Eq. (D.210), the factor $|\boldsymbol{Q}|$ cancels and each term stays of order $\ell_{\boldsymbol{Q}\nu}$. Collecting,

$$M^{\pm}_{cv\nu}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q}) = \mathrm{i}\,\ell_{\boldsymbol{Q}\nu}\,\rho_{cv}(\boldsymbol{k},\boldsymbol{q})\Big[\hat{\boldsymbol{e}}_\nu\cdot\boldsymbol{q} + \frac{\hat{\boldsymbol{Q}}\cdot D^c\cdot\hat{\boldsymbol{e}}_\nu \mp c_\nu\,\hat{\boldsymbol{e}}_\nu\cdot\boldsymbol{p}_{cc}}{\hat{\boldsymbol{Q}}\cdot\boldsymbol{v}_c^{\mathrm{g}} \pm c_\nu} - \frac{\hat{\boldsymbol{Q}}\cdot D^v\cdot\hat{\boldsymbol{e}}_\nu \mp c_\nu\,\hat{\boldsymbol{e}}_\nu\cdot\boldsymbol{p}_{vv}}{\hat{\boldsymbol{Q}}\cdot\boldsymbol{v}_v^{\mathrm{g}} \pm c_\nu}\Big] + O(\ell_{\boldsymbol{Q}\nu}|\boldsymbol{Q}|), \tag{D.212}$$

which is Eq. (13.31). The leading term is of order $|\boldsymbol{Q}|^{-1/2}$ and the correction of order $|\boldsymbol{Q}|^{1/2}$, a relative correction of order $|\boldsymbol{Q}|$. The expansion of the intraband detunings to first order requires $|\boldsymbol{Q}|$ small compared with the scale on which the group velocity changes, $|\hat{\boldsymbol{Q}}\cdot\boldsymbol{v}_n^{\mathrm{g}} \pm c_\nu|$ divided by the band curvature, which at an extremum is $|\boldsymbol{Q}| \ll 2m^* c_\nu$ for a band of mass m^*. For a degenerate band, such as the valence maximum of silicon, the intraband terms are matrices in the degenerate partners, and the $O(Q)$ splitting along $\hat{\boldsymbol{Q}}$ is diagonalized first. Worked example 13.6 checks each step in a crystal of rigid ions in one dimension, where the laboratory vertex is computed directly from the displaced ions.

The same result follows from Eq. (13.9) in the laboratory frame, without Eq. (D.206). There the rigid-translation part of the interband vertices, $\mathrm{i}\ell(\varepsilon_m - \varepsilon_n)\hat{\boldsymbol{e}}_\nu\cdot\boldsymbol{p}_{mn}$, combines over all bands into $\mathrm{i}\ell[\hat{\boldsymbol{e}}_\nu\cdot\boldsymbol{q} - \hat{\boldsymbol{e}}_\nu\cdot(\boldsymbol{p}_{cc} - \boldsymbol{p}_{vv})]\rho_{cv}$, by the same completeness as in Eq. (D.205), and the intraband terms with the vertex Eq. (D.211) give $\mathrm{i}\ell\rho_{cv}\,\hat{\boldsymbol{Q}}\cdot(D^c + \boldsymbol{v}_c^{\mathrm{g}}\otimes\boldsymbol{p}_{cc})\cdot\hat{\boldsymbol{e}}_\nu/(\hat{\boldsymbol{Q}}\cdot\boldsymbol{v}_c^{\mathrm{g}} \pm c_\nu)$ and its valence partner. Since $(\hat{\boldsymbol{Q}}\cdot\boldsymbol{v}^{\mathrm{g}})/(\hat{\boldsymbol{Q}}\cdot\boldsymbol{v}^{\mathrm{g}} \pm c_\nu) - 1 = \mp c_\nu/(\hat{\boldsymbol{Q}}\cdot\boldsymbol{v}^{\mathrm{g}} \pm c_\nu)$, the $\boldsymbol{p}$ terms combine into those of Eq. (D.212). The route shows what is lost when the interband vertices of an acoustic mode are set to zero near $\boldsymbol{Q} = 0$: the recoil term and the bare Galilean terms, $\mathrm{i}\ell[\hat{\boldsymbol{e}}_\nu\cdot(\boldsymbol{q} - \boldsymbol{p}_{cc} + \boldsymbol{p}_{vv})]\rho_{cv}$.

Inside and outside the window

On the energy shell, $\boldsymbol{Q}$ cannot be small inside the off-shell window. A final state with $\omega \le E_{\min}(\boldsymbol{q}) - \delta_{\mathrm{w}}$ and an electron at $\boldsymbol{k}' = \boldsymbol{k}+\boldsymbol{q}+\boldsymbol{Q}$ has $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}} \ge E_{\min}(\boldsymbol{q})$, so $\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{c\boldsymbol{k}'} \ge \delta_{\mathrm{w}} \pm \omega_{\boldsymbol{Q}\nu}$. Emission gives $|\boldsymbol{Q}| \ge \delta_{\mathrm{w}}/(v^{\mathrm{g}}_{\max} - c_\nu)$ and absorption the weaker bound, which holds for either sign,

$$|\boldsymbol{Q}| \ge \frac{\delta_{\mathrm{w}}}{v^{\mathrm{g}}_{\max} + c_\nu}, \tag{D.213}$$

with $v^{\mathrm{g}}_{\max}$ the largest conduction-band group velocity. The intraband denominators are then the zero-phonon detunings $\omega - E_{cv}(\boldsymbol{k},\boldsymbol{q})$ and $E_{cv}(\boldsymbol{k}+\boldsymbol{Q},\boldsymbol{q}) - \omega$ of the pairs at $\boldsymbol{q}$, both at least δ_{w} in size, and the amplitude has no feature at small $\boldsymbol{Q}$ at any temperature. Outside the window the zero-phonon pair at $\boldsymbol{q}$ is real and $\boldsymbol{Q}$ can go to zero. Then Eq. (D.212) is the amplitude for the pair to emit a soft phonon. Its denominators vanish on the cone $\hat{\boldsymbol{Q}}\cdot\boldsymbol{v}_c^{\mathrm{g}} = \mp c_\nu$, where the electron moves faster than sound along $-\hat{\boldsymbol{Q}}$; this is the Cherenkov emission of Eq. (6.28), a real decay of the pair, which the width and the Dyson convention of Sect. 8.6 handle like any other resonance. Away from the cone, $|M^{\pm}|^2 \propto \ell^2_{\boldsymbol{Q}\nu} \propto 1/|\boldsymbol{Q}|$, and with $n_{\boldsymbol{Q}\nu} + 1$ and the phase space $\mathrm{d}^3Q \propto Q^2\mathrm{d}Q$ the emission rate behaves as $Q\,\mathrm{d}Q$ at $T = 0$ and, with $n_{\boldsymbol{Q}\nu} \simeq T/c_\nu|\boldsymbol{Q}|$, as $\mathrm{d}Q$ at $T > 0$. Both are integrable at $Q = 0$, since the squared amplitude grows only as $1/|\boldsymbol{Q}|$.

D.38 Linear spin-wave weights of a two-sublattice antiferromagnet

The one-magnon response of an antiferromagnet at an arbitrary momentum transfer, its total weight, and the weight that the longitudinal spin puts into magnon pairs follow at linear spin-wave order from the operators of Appendix D.19; with the exact spin sum rule evaluated in the same ground state they give Eqs. (14.1) and (14.6). The notation is that of Appendix D.19: sublattices A and B of N_s sites each, Holstein–Primakoff bosons $d_{\mathrm{A}i}$ and $d_{\mathrm{B}j}$, the coefficients $A_{\boldsymbol{k}}$ and $B_{\boldsymbol{k}}$ of Eq. (D.111), the magnon operators $a_{\boldsymbol{k}1}$ and $a_{\boldsymbol{k}2}$ of Eq. (D.112), and the ground state $|0\rangle$ that they annihilate. We take B to be A translated by a fixed vector $\boldsymbol{\delta}_0$, as in NiO and in the simple-cubic antiferromagnet; then $2\boldsymbol{\delta}_0$ is a vector of sublattice A.

Transverse matrix elements

A momentum transfer $\boldsymbol{q}$ outside the Brillouin zone of one sublattice is split as $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}$, with $\boldsymbol{q}_r$ in that zone and $\boldsymbol{G}$ a reciprocal-lattice vector of the sublattice, as in Eq. (3.16). The sites of A sit on that lattice, so $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}_i} = 1$ on A, while on B, $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}_j} = \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\delta}_0} \equiv \epsilon_{\boldsymbol{G}}$. Since $2\boldsymbol{\delta}_0$ lies on A, $\epsilon_{\boldsymbol{G}}^2 = 1$ and $\epsilon_{\boldsymbol{G}} = \pm 1$. The same phase relates the lattice sums of Eq. (D.109) at $\boldsymbol{q}$ and at $\boldsymbol{q}_r$:

$$J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{q}) = J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{q}_r), \qquad J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{q}) = \epsilon_{\boldsymbol{G}}\, J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{q}_r),$$

because every neighbor on A differs from the origin by a vector of A, and every neighbor on B by $\boldsymbol{\delta}_0$ plus such a vector. Defined by the same formulas at the full $\boldsymbol{q}$, the coefficients of Eq. (D.111) are therefore $A_{\boldsymbol{q}} = A_{\boldsymbol{q}_r}$ and $B_{\boldsymbol{q}} = \epsilon_{\boldsymbol{G}} B_{\boldsymbol{q}_r}$.

At linear order $S^+_i = (2|\boldsymbol{S}|)^{1/2} d_{\mathrm{A}i}$ on A and $S^+_j = (2|\boldsymbol{S}|)^{1/2} d^\dagger_{\mathrm{B}j}$ on B (Appendix D.19). We insert the Fourier expansions of Appendix D.19 into $S^+(\boldsymbol{q}) = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j} S^+_j$. The lattice sum $\sum_{i\in\mathrm{A}} \mathrm{e}^{\mathrm{i}(\boldsymbol{q}+\boldsymbol{k})\cdot\boldsymbol{r}_i}$ is N_s for $\boldsymbol{k} = -\boldsymbol{q}_r$ and zero otherwise, and the sum over B carries the extra factor $\epsilon_{\boldsymbol{G}}$. Hence

$$S^+(\boldsymbol{q}) = (2|\boldsymbol{S}|N_\mathrm{s})^{1/2}(d_{\mathrm{A},-\boldsymbol{q}_r} + \epsilon_{\boldsymbol{G}}\, d^\dagger_{\mathrm{B},-\boldsymbol{q}_r}), \qquad S^-(\boldsymbol{q}) = (2|\boldsymbol{S}|N_\mathrm{s})^{1/2}(d^\dagger_{\mathrm{A}\boldsymbol{q}_r} + \epsilon_{\boldsymbol{G}}\, d_{\mathrm{B}\boldsymbol{q}_r}).$$

The inverse transformation Eq. (D.115), with $u^{\mathrm{M}}_{\boldsymbol{k}}$ and $v^{\mathrm{M}}_{\boldsymbol{k}}$ real and even in $\boldsymbol{k}$, turns these into magnon operators:

$$\begin{aligned} S^+(\boldsymbol{q}) &= (2|\boldsymbol{S}|N_\mathrm{s})^{1/2}\big[(u^{\mathrm{M}} - \epsilon_{\boldsymbol{G}} v^{\mathrm{M}})\, a_{-\boldsymbol{q}_r 1} + (\epsilon_{\boldsymbol{G}} u^{\mathrm{M}} - v^{\mathrm{M}})\, a^\dagger_{-\boldsymbol{q}_r 2}\big], \\ S^-(\boldsymbol{q}) &= (2|\boldsymbol{S}|N_\mathrm{s})^{1/2}\big[(u^{\mathrm{M}} - \epsilon_{\boldsymbol{G}} v^{\mathrm{M}})\, a^\dagger_{\boldsymbol{q}_r 1} + (\epsilon_{\boldsymbol{G}} u^{\mathrm{M}} - v^{\mathrm{M}})\, a_{\boldsymbol{q}_r 2}\big], \end{aligned}$$

with the coefficients at $\boldsymbol{q}_r$. On the ground state, $S^-(\boldsymbol{q})$ creates the magnon $a^\dagger_{\boldsymbol{q}_r 1}$ and $S^+(\boldsymbol{q})$ the magnon $a^\dagger_{-\boldsymbol{q}_r 2}$, both of energy $\omega^{\mathrm{M}}_{\boldsymbol{q}_r}$, with amplitudes c and $\epsilon_{\boldsymbol{G}} c$, where $c = (2|\boldsymbol{S}|N_\mathrm{s})^{1/2}(u^{\mathrm{M}}_{\boldsymbol{q}_r} - \epsilon_{\boldsymbol{G}} v^{\mathrm{M}}_{\boldsymbol{q}_r})$. The expressions of Appendix D.19 for $(u^{\mathrm{M}})^2$, $(v^{\mathrm{M}})^2$ and $u^{\mathrm{M}}v^{\mathrm{M}}$ give

$$w^{\mathrm{M}}_{\boldsymbol{q}} \equiv \big(u^{\mathrm{M}}_{\boldsymbol{q}_r} - \epsilon_{\boldsymbol{G}} v^{\mathrm{M}}_{\boldsymbol{q}_r}\big)^2 = \frac{A_{\boldsymbol{q}_r} - \epsilon_{\boldsymbol{G}} B_{\boldsymbol{q}_r}}{(A^2_{\boldsymbol{q}_r} - B^2_{\boldsymbol{q}_r})^{1/2}} = \frac{|\boldsymbol{S}|\,(A_{\boldsymbol{q}} - B_{\boldsymbol{q}})}{\omega^{\mathrm{M}}_{\boldsymbol{q}}}, \tag{D.214}$$

where the last form uses the coefficients at the full momentum transfer, which absorb the phase. For $\epsilon_{\boldsymbol{G}} = \pm 1$ it is the weight $(u^{\mathrm{M}} \mp v^{\mathrm{M}})^2$ of Eq. (7.16), near Γ and near the ordering wavevector.

The spin density is $\boldsymbol{s}_e(\boldsymbol{q}) = F^{\mathrm{mag}}(q) \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j} \boldsymbol{S}_j$ (Eq. (7.14)), with $s_{e,x} = F^{\mathrm{mag}}(S^+ + S^-)/2$ and $s_{e,y} = F^{\mathrm{mag}}(S^+ - S^-)/2\mathrm{i}$, and S^z creates no single magnon. For the magnon of branch 1 the matrix elements are $\langle s_{e,x}\rangle = F^{\mathrm{mag}} c/2$ and $\langle s_{e,y}\rangle = \mathrm{i}F^{\mathrm{mag}} c/2$, the polarization of the ferromagnet in Sect. 7.1. For branch 2 they are $F^{\mathrm{mag}}\epsilon_{\boldsymbol{G}} c/2$ and $-\mathrm{i}F^{\mathrm{mag}}\epsilon_{\boldsymbol{G}} c/2$, the opposite

sense of rotation. In Eq. (4.13) at $T = 0$, with $\mathscr{V} = 2N_\mathrm{s}\Omega_\mathrm{ion}$ and Ω_ion the volume per magnetic ion, the two branches contribute equal diagonal elements $(2\pi/\mathscr{V})|F^\mathrm{mag}c|^2/4$ in x and y, while their off-diagonal elements $\pm\mathrm{i}(2\pi/\mathscr{V})|F^\mathrm{mag}c|^2/4$ cancel. With $|c|^2 = 2|\boldsymbol{S}|N_\mathrm{s}w_{\boldsymbol{q}}^\mathrm{M}$ the sum is Eq. (14.1). Replacing $\hat{\boldsymbol{z}}$ by the local ordering axis, the result holds for any orientation of the moments.

Transverse weight and zero-point reduction

The equal-time weight of a response is its integral over ω, and summed over $\boldsymbol{q}$ in the zone it becomes a correlation of one site. For the transverse spin the site correlation is direct: on A, $\frac{1}{2}\langle S_i^+ S_i^- + S_i^- S_i^+\rangle = |\boldsymbol{S}|\langle d_{\mathrm{A}i}d_{\mathrm{A}i}^\dagger + d_{\mathrm{A}i}^\dagger d_{\mathrm{A}i}\rangle$ at linear order. The deviation per site follows from Eq. (D.115): $\langle d_{\mathrm{A}\boldsymbol{k}}^\dagger d_{\mathrm{A}\boldsymbol{k}}\rangle = (v_{\boldsymbol{k}}^\mathrm{M})^2$, since only the term $(v_{\boldsymbol{k}}^\mathrm{M})^2 a_{\boldsymbol{k}2}a_{\boldsymbol{k}2}^\dagger$ survives in the vacuum. Hence

$$\begin{gathered}\Delta S \equiv \langle d_{\mathrm{A}i}^\dagger d_{\mathrm{A}i}\rangle = \frac{1}{N_\mathrm{s}}\sum_{\boldsymbol{k}}(v_{\boldsymbol{k}}^\mathrm{M})^2, \qquad \langle S_i^z\rangle = |\boldsymbol{S}| - \Delta S, \\ W_\perp \equiv \langle S_i^x S_i^x + S_i^y S_i^y\rangle = |\boldsymbol{S}|\,(1 + 2\Delta S),\end{gathered} \tag{D.215}$$

the same on B. The zero-point reduction ΔS does not depend on $|\boldsymbol{S}|$, since $A_{\boldsymbol{k}}$ and $B_{\boldsymbol{k}}$ do not. The same $W_\perp$ follows from Eq. (D.214). As $\boldsymbol{q}$ runs over a full period of the reciprocal lattice of all the magnetic ions, A and B together, each $\boldsymbol{q}_r$ appears once with $\epsilon_{\boldsymbol{G}} = +1$ and once with -1, so the average of $w_{\boldsymbol{q}}^\mathrm{M}$ is $(u^\mathrm{M})^2 + (v^\mathrm{M})^2 = 1 + 2(v^\mathrm{M})^2$, averaged over the zone, $1 + 2\Delta S$. The integral of $S_{xx} + S_{yy}$ in Eq. (14.1) over $\omega/2\pi$, times Ω_ion and averaged over $\boldsymbol{q}$, is then $|\boldsymbol{S}|(1 + 2\Delta S)$ at $F^\mathrm{mag} = 1$. The one-magnon line carries all of $W_\perp$ at this order.

Longitudinal weight

On A, $S_i^z - \langle S_i^z\rangle = -(n_i - \Delta S)$ with $n_i = d_{\mathrm{A}i}^\dagger d_{\mathrm{A}i}$. By Eq. (D.115), n_i contains terms $a_1^\dagger a_2^\dagger$, which acting on $|0\rangle$ create two magnons, and terms that annihilate it, apart from the constant ΔS. The connected longitudinal weight is therefore the two-magnon weight. The ground state of a quadratic boson Hamiltonian is Gaussian, and Wick's theorem gives

$$\langle n_i^2\rangle = \langle d^\dagger d\rangle^2 + \langle d^\dagger d\rangle\langle dd^\dagger\rangle + \langle d^\dagger d^\dagger\rangle\langle dd\rangle = \Delta S^2 + \Delta S(1 + \Delta S),$$

since the anomalous average on one site vanishes: $\langle d_{\mathrm{A}\boldsymbol{k}}d_{\mathrm{A}\boldsymbol{k}'}\rangle = -u_{\boldsymbol{k}}^\mathrm{M}v_{\boldsymbol{k}'}^\mathrm{M}\langle a_{\boldsymbol{k}1}a_{\boldsymbol{k}'2}^\dagger\rangle = 0$, the two branches being independent. The pairing of Eq. (D.111) correlates a deviation on A only with one on B. Hence

$$W_\parallel \equiv \langle (S_i^z)^2\rangle - \langle S_i^z\rangle^2 = \Delta S\,(1 + \Delta S), \qquad \frac{W_\parallel}{W_\perp} = \frac{\Delta S\,(1 + \Delta S)}{|\boldsymbol{S}|\,(1 + 2\Delta S)}, \tag{D.216}$$

which is Eq. (14.6). A ferromagnet at $T = 0$ has $v_{\boldsymbol{k}}^\mathrm{M} = 0$, hence $\Delta S = W_\parallel = 0$: its fully polarized ground state is exact and S^z cannot change it.

The spin sum rule

The identity $\boldsymbol{S}_i^2 = |\boldsymbol{S}|(|\boldsymbol{S}| + 1)$ holds on every state, so the three weights and the ordered moment obey $W_\perp + W_\parallel + \langle S_i^z\rangle^2 = |\boldsymbol{S}|(|\boldsymbol{S}| + 1)$ exactly. Inserting Eqs. (D.215) and (D.216), the left side is $|\boldsymbol{S}|(|\boldsymbol{S}| + 1) + \Delta S(1 + 2\Delta S)$: linear spin-wave theory overcounts the total weight by $\Delta S(1 + 2\Delta S)$ per site. With $\langle S_i^z\rangle$ and $W_\parallel$ as above, the identity requires

$$W_\perp^\mathrm{sr} = |\boldsymbol{S}|(|\boldsymbol{S}| + 1) - (|\boldsymbol{S}| - \Delta S)^2 - \Delta S(1 + \Delta S) = (|\boldsymbol{S}| - \Delta S)(1 + 2\Delta S), \tag{D.217}$$

a fraction $1-\Delta S/|\boldsymbol{S}|$ of the linear spin-wave value. The excess sits in the square roots of Eq. (D.110), which linear spin-wave theory replaces by $(2|\boldsymbol{S}|)^{1/2}$. Kept whole and evaluated in the same Gaussian state, with $f(n)d = d\,f(n-1)$ as in Sect. 7.1,

$$\langle S_i^+ S_i^- \rangle = \langle (2|\boldsymbol{S}| - n_i)(n_i+1) \rangle, \qquad \langle S_i^- S_i^+ \rangle = \langle n_i (2|\boldsymbol{S}| + 1 - n_i) \rangle,$$

and with $\langle n_i \rangle = \Delta S$ and $\langle n_i^2 \rangle = \Delta S(1+2\Delta S)$ their half-sum is $|\boldsymbol{S}| + 2|\boldsymbol{S}|\Delta S - \Delta S - 2\Delta S^2$, which is Eq. (D.217). The first omitted term of the transverse operator, $-(2|\boldsymbol{S}|)^{1/2} n_i d_{\mathrm{A}i}/4|\boldsymbol{S}|$, removes the excess to this order. It also creates three magnons, which lowers the one-magnon part of $W_\perp^{\mathrm{sr}}$ further. The change of the ground state at the same order, which moves $\langle S_i^z \rangle$ and $W_\parallel$, and the shift of the magnon energies by the quartic terms are not included, so $\Delta S/|\boldsymbol{S}|$ is an estimate of the leading relative correction to the one-magnon weight, not a bound. The one-magnon and two-magnon weights together, $W_\perp^{\mathrm{sr}} + W_\parallel = |\boldsymbol{S}|(1+2\Delta S) - \Delta S^2$, differ from the linear spin-wave one-magnon weight only at second order in ΔS.

The value for NiO, $\Delta S = 0.0784$, is computed in Worked example 14.1. Because $(v_{\boldsymbol{k}}^{\mathrm{M}})^2$ diverges as $1/k$ at the Goldstone points, such grid sums converge as the inverse square of the grid size, and the NiO value extrapolates to 0.0785.

D.39 The pair function and the resummed loss function at the kinematic edge

This section collects the algebra behind Sects. 15.1 and 15.2: the closed form of the pair function F_μ and the bubbles of Eq. (15.5), the single quadrature formula for the conserving response Eq. (15.6), the resummation that gives Eqs. (15.10) and (15.11), the evaluation of the Kramers–Kronig shift $\mathfrak{s}_{\mathrm{BCS}}$, and the sum rule that leaves no loss weight below the gap. The hypotheses are those of Sect. 15.1.2: $T = 0$, a constant density of states $\mathcal{N}_{\mathrm{F}}$ per spin orientation, a linearized band with $q \ll k_{\mathrm{F}}$, and no impurities. With the "1" of ε dropped, $\varepsilon = -(k_{\mathrm{TF}}^2/q^2)R/2\mathcal{N}_{\mathrm{F}}$, and every ratio below is a function of $y = \omega/\Delta_{\mathrm{BCS}}$ and $qv_{\mathrm{F}}/\Delta_{\mathrm{BCS}}$.

The pair function

The combination $(E+E')^2 - v_{\mathrm{F}}^2 q^2 \mu^2$ of Eq. (15.3) is never smaller than $4\Delta_{\mathrm{BCS}}^2$. Write $\xi = \Delta_{\mathrm{BCS}} \sinh a$ and $\xi' = \Delta_{\mathrm{BCS}} \sinh b$, so that $E + E' = 2\Delta_{\mathrm{BCS}} \cosh A \cosh B$ and $\xi' - \xi = v_{\mathrm{F}} q\mu = 2\Delta_{\mathrm{BCS}} \cosh A \sinh B$ with $A = (a+b)/2$ and $B = (b-a)/2$. Then $(E+E')^2 - v_{\mathrm{F}}^2 q^2 \mu^2 = 4\Delta_{\mathrm{BCS}}^2 \cosh^2 A$. At fixed μ, that is at fixed $\xi' - \xi$, the relation $\cosh b\, \mathrm{d}b = \cosh a\, \mathrm{d}a$ gives $\mathrm{d}A = \mathrm{d}a\,(\cosh a + \cosh b)/2\cosh b$; with $\mathrm{d}\xi = \Delta_{\mathrm{BCS}} \cosh a\, \mathrm{d}a$ the measure becomes $\mathrm{d}\xi\,(E+E')/EE' = 2\,\mathrm{d}A$. The same substitution gives $\int \mathrm{d}\xi\,(E+E')/\{EE'[(E+E')^2 - v_{\mathrm{F}}^2 q^2 \mu^2]\} = \int \mathrm{d}A/(2\Delta_{\mathrm{BCS}}^2 \cosh^2 A) = 1/\Delta_{\mathrm{BCS}}^2$ for every μ. For the pair function of Eq. (15.4) it gives $F_\mu = (\mathcal{N}_{\mathrm{F}}/4\Delta_{\mathrm{BCS}}^2) \int \mathrm{d}A/(\mathfrak{p}_\mu - \cosh^2 A)$, and with $t = \tanh A$ the integral becomes $\int_{-1}^{1} \mathrm{d}t/(\mathfrak{p}_\mu - 1 - \mathfrak{p}_\mu t^2)$, which is elementary. It gives Eq. (15.4) for $0 < \mathfrak{p}_\mu < 1$, and for the other ranges

$$\frac{\arcsin\sqrt{\mathfrak{p}}}{\sqrt{\mathfrak{p}(1-\mathfrak{p})}} \to \frac{\operatorname{arsinh}\sqrt{|\mathfrak{p}|}}{\sqrt{|\mathfrak{p}|(1+|\mathfrak{p}|)}} \quad (\mathfrak{p} < 0), \qquad \to -\frac{\operatorname{arcosh}\sqrt{\mathfrak{p}} - \mathrm{i}\pi/2}{\sqrt{\mathfrak{p}(\mathfrak{p}-1)}} \quad (\mathfrak{p} > 1), \tag{D.218}$$

where the sign of the imaginary part is that of the retarded function, $\omega + \mathrm{i}0^+$.

The density bubble has the integrand $|M_{\boldsymbol{\eta}}^{\mathrm{N}}|^2 \cdot 2(E+E')/[\omega^2 - (E+E')^2]$. With $W = E + E'$ the partial fractions

$$\frac{W^2}{(W^2 - v_{\mathrm{F}}^2 q^2 \mu^2)(\omega^2 - W^2)} = \frac{1}{\omega_\mu^2}\Big[\frac{\omega^2}{\omega^2 - W^2} + \frac{v_{\mathrm{F}}^2 q^2 \mu^2}{W^2 - v_{\mathrm{F}}^2 q^2 \mu^2}\Big],$$

split it into an F_μ term and a term that the integral identity above evaluates; this is $R^0_{\eta\eta}$ of Eq. (15.5). The mixed and phase bubbles follow from the products of Sect. 15.1.2 with F_μ alone: $R^0_{\mathfrak{s}_2\eta} = 2\mathrm{i}\Delta_{\mathrm{BCS}}\omega\langle F_\mu\rangle_\mu$, and $R^0_{\mathfrak{s}_2\mathfrak{s}_2}(q,\omega) - R^0_{\mathfrak{s}_2\mathfrak{s}_2}(0,0) = \langle\omega_\mu^2 F_\mu\rangle_\mu$. The logarithmic dependence of $R^0_{\mathfrak{s}_2\mathfrak{s}_2}$ on the cutoff ω_{D} cancels in that difference: the cutoff-dependent part is $\mathcal{N}_{\mathrm{F}}\int \mathrm{d}\xi/E$ over the shell $|\xi| < \omega_{\mathrm{D}}$, and the Doppler shift $\xi' - \xi = v_{\mathrm{F}}q\mu$ moves its limits by $v_{\mathrm{F}}q\mu$, a change of relative order $v_{\mathrm{F}}q/\omega_{\mathrm{D}}$. The gap equation fixes $R^0_{\mathfrak{s}_2\mathfrak{s}_2}(0,0) = -2/V_{\mathrm{BCS}}$, which gives the last of Eq. (15.5).

Quadrature of the conserving response

The factor F_μ has an integrable inverse-square-root edge where $\omega_\mu = 2\Delta_{\mathrm{BCS}}$, at $v_{\mathrm{F}}q\mu = s_* = (\omega^2 - 4\Delta_{\mathrm{BCS}}^2)^{1/2}$. The substitution $v_{\mathrm{F}}q\mu = s_* \mp t^2$ on the two sides removes it, after which adaptive quadrature of the three averages $\langle F_\mu\rangle_\mu$, $\langle\omega_\mu^2 F_\mu\rangle_\mu$ and the bubble of Eq. (15.6) converges to 10^{-11}. This one quadrature gives every value of R^{cons}, $\mathfrak{v}$ and δ in Sects. 15.1 and 15.2. With R^0 the Lindhard bubble, $\mathfrak{g} = \operatorname{Im}R^{\mathrm{cons}}/\operatorname{Im}R^0 = \mathfrak{g}_{\mathrm{I}}(1+\mathfrak{v})$, since $\operatorname{Im}R^0_{\eta\eta} = \mathfrak{g}_{\mathrm{I}}\operatorname{Im}R^0$, and $1+\delta = \operatorname{Re}R^{\mathrm{cons}}/\operatorname{Re}R^0$, the quantities of Eq. (15.11). On the kinematic edge at $y = 3$ this gives $\mathfrak{v} = 0.214$ and $1+\delta = 0.596$; the bare bubble computed in the same way has $1+\delta = 1.329$, against $1+\delta_{\mathfrak{s}} = 1.318$ from $\mathfrak{s}_{\mathrm{BCS}}$.

The resummation

Let a response have $\operatorname{Im}\varepsilon = \mathfrak{g}\operatorname{Im}\varepsilon_{\mathrm{Lin}}$ and $\operatorname{Re}\varepsilon = (1+\delta)\operatorname{Re}\varepsilon_{\mathrm{Lin}}$ at some (q,ω), with $\mathfrak{g}$ and δ real. Dividing $L = \operatorname{Im}\varepsilon/[(\operatorname{Re}\varepsilon)^2 + (\operatorname{Im}\varepsilon)^2]$ by the same expression for $\varepsilon_{\mathrm{Lin}}$, and then numerator and denominator by $(\operatorname{Re}\varepsilon_{\mathrm{Lin}})^2$,

$$\frac{L}{L^{\mathrm{nor}}} = \frac{\mathfrak{g}\operatorname{Im}\varepsilon_{\mathrm{Lin}}}{(1+\delta)^2(\operatorname{Re}\varepsilon_{\mathrm{Lin}})^2 + \mathfrak{g}^2(\operatorname{Im}\varepsilon_{\mathrm{Lin}})^2}\cdot\frac{(\operatorname{Re}\varepsilon_{\mathrm{Lin}})^2 + (\operatorname{Im}\varepsilon_{\mathrm{Lin}})^2}{\operatorname{Im}\varepsilon_{\mathrm{Lin}}} = \mathfrak{g}\,\frac{1+\mathfrak{r}_\varepsilon^2}{(1+\delta)^2 + \mathfrak{g}^2\mathfrak{r}_\varepsilon^2}, \tag{D.219}$$

with $\mathfrak{r}_\varepsilon = \operatorname{Im}\varepsilon_{\mathrm{Lin}}/\operatorname{Re}\varepsilon_{\mathrm{Lin}} = \pi\mathfrak{w}/2\operatorname{Re}\mathfrak{f}_{\mathrm{L}}(\mathfrak{w})$ from Eq. (15.9). No approximation enters. The ratio is positive, and for $\delta \geq 0$ it is at most $\mathfrak{g}(1+\mathfrak{r}_\varepsilon^2)/(1+\mathfrak{g}^2\mathfrak{r}_\varepsilon^2)$.

The bare bubble has $\mathfrak{g} = \mathfrak{g}_{\mathrm{I}}(y)$ for $\omega < qv_{\mathrm{F}}$ by Eq. (15.7). Its shift follows from Eq. (15.8), $\operatorname{Re}\varepsilon^{\mathrm{bare}} - \operatorname{Re}\varepsilon_{\mathrm{Lin}} = (k_{\mathrm{TF}}^2/q^2)(\Delta_{\mathrm{BCS}}/qv_{\mathrm{F}})\mathfrak{s}_{\mathrm{BCS}}$; dividing by $\operatorname{Re}\varepsilon_{\mathrm{Lin}} = (k_{\mathrm{TF}}^2/q^2)\operatorname{Re}\mathfrak{f}_{\mathrm{L}}(\mathfrak{w})$ gives $\delta = \delta_{\mathfrak{s}}$ of Eq. (15.10). To first order in δ and in $\mathfrak{r}_\varepsilon^2$, Eq. (D.219) is $\mathfrak{g}[1 - 2\delta - (\mathfrak{g}^2-1)\mathfrak{r}_\varepsilon^2]$. As $\omega/qv_{\mathrm{F}} \to 0$, $\mathfrak{r}_\varepsilon \to 0$ and $\mathfrak{f}_{\mathrm{L}} \to 1$, and this becomes $\mathfrak{g}_{\mathrm{I}}[1 - 2(\Delta_{\mathrm{BCS}}/qv_{\mathrm{F}})\mathfrak{s}_{\mathrm{BCS}}]$, the first-order form Eq. (15.8). On the kinematic edge at $y = 3$ the two first-order terms are $2\delta_{\mathfrak{s}} = 0.64$ and $(\mathfrak{g}_{\mathrm{I}}^2 - 1)\mathfrak{r}_\varepsilon^2 = 0.41$, so neither expansion holds there.

The Kramers–Kronig shift

The function $\mathfrak{s}_{\mathrm{BCS}}(y) = \mathrm{PV}\int_0^\infty \mathrm{d}y'\, y'^2\delta\mathfrak{g}(y')/(y'^2 - y^2)$, with $\delta\mathfrak{g} = \mathfrak{g}_{\mathrm{I}}\Theta(y'-2) - 1$, is evaluated as follows. With $\mathfrak{h}(y') = y'^2\delta\mathfrak{g}(y')/(y'+y)$ the integrand is $\mathfrak{h}(y')/(y'-y)$, and for an upper limit $Y > y$

$$\mathrm{PV}\int_0^Y \mathrm{d}y'\,\frac{\mathfrak{h}(y')}{y'-y} = \int_0^Y \mathrm{d}y'\,\frac{\mathfrak{h}(y') - \mathfrak{h}(y)}{y'-y} + \mathfrak{h}(y)\ln\frac{Y-y}{y},$$

whose first integrand is regular at $y' = y$. For $Y \to \infty$ the large-y' form $\mathfrak{g}_{\mathrm{I}} - 1 = [2\ln(2y') - 1]/y'^2 + O(\ln y'/y'^4)$ makes the integrand of the original integral $[2\ln(2y') - 1]/y'^2$, whose integral from Y_{tail} to infinity is $[2\ln(2Y_{\mathrm{tail}}) + 1]/Y_{\mathrm{tail}}$; we integrate numerically to $Y_{\mathrm{tail}} = 10^4$ and add this tail. The cut prescription of Sect. 15.2.1 sets $Y = qv_{\mathrm{F}}/\Delta_{\mathrm{BCS}}$, which on the edge is $y\, v_{\mathrm{F}}/v_{\max}$. Near $y = 2$ the function $\mathfrak{h}$ steps by $(\pi/2)\cdot 4/(2+y) \to \pi/2$ at $y' = 2$, and a step J in $\mathfrak{h}$ at $y' = 2$ adds $J\ln[1/|y-2|]$ to the principal value; hence $\mathfrak{s}_{\mathrm{BCS}} \simeq (\pi/2)\ln[1/|y-2|] + O(1)$. With $\delta_{\mathfrak{s}} \propto \ln[1/(y-2)]$, Eq. (D.219) falls as $(1+\delta_{\mathfrak{s}})^{-2}$ toward the threshold once $\delta_{\mathfrak{s}}$ exceeds one.

No loss weight below the gap

Let $\mathfrak{k}(\omega) = 2\mathcal{N}_\mathrm{F}/R(q,\omega)$, which is analytic in the upper half plane where R has no zero, with $\operatorname{Im}\mathfrak{k} \geq 0$ on the real axis because $\operatorname{Im} R \leq 0$. For the normal metal the moments Eq. (4.58) at $q \ll k_\mathrm{F}$ give $R^0 = (n_e q^2/m_e\omega^2)[1 + \frac{3}{5}(qv_\mathrm{F}/\omega)^2 + \dots]$, so $\mathfrak{k} = 3\omega^2/(qv_\mathrm{F})^2 - \frac{9}{5} + O(\omega^{-2})$, and $\mathfrak{k}(0) = -1$ by the static limit. The function $\tilde{\mathfrak{k}} = \mathfrak{k} - 3\omega^2/(qv_\mathrm{F})^2$ tends to a constant, and the half-range Kramers–Kronig relation Eq. (4.25) for $\tilde{\mathfrak{k}}$ at $\omega = 0$ reads

$$\frac{2}{\pi}\int_0^\infty \mathrm{d}\omega\, \frac{\operatorname{Im}\mathfrak{k}(\omega)}{\omega} = \tilde{\mathfrak{k}}(0) - \tilde{\mathfrak{k}}(\infty) = -1 + \frac{9}{5} = \frac{4}{5}.$$

A zero of R on the real axis below $2\Delta_\mathrm{BCS}$, which is a zero of ε and a δ-function in L, would contribute to the left side. The conserving response has the same two limits: numerically $\tilde{\mathfrak{k}} = -1.8000$ at $\omega = 10^3\Delta_\mathrm{BCS}$ for $qv_\mathrm{F} = 5$ and $12\,\Delta_\mathrm{BCS}$, and $\mathfrak{k}(0) = -1$ by Sect. 15.1.2. Its integral over $\omega > 2\Delta_\mathrm{BCS}$ alone is 0.8000 at both momenta, as is that of the normal metal. The Anderson–Bogoliubov pole of R^cons is a zero of $\mathfrak{k}$ and does not enter.

D.40 Absorption by the uniform current and spin current of a superconductor

A boson absorbed at $\boldsymbol{q} \to 0$ couples to a superconducting metal through the total velocity of its electrons, if it is a dark photon or a photon, and through their spin current, if it is an axion-like particle. This section derives what Sect. 15.4 uses: the Nambu form of both operators, the absorption by the spin current of a clean superconductor, the exact second-order amplitude of phonon-assisted absorption and its conductivity, the limit of a Debye transport function, and impurity-assisted absorption in the dirty limit. The notation is that of Sect. 7.2 and Appendix D.20: the band energy $\xi_{\boldsymbol{k}}$ from the chemical potential, the gap Δ_BCS, the quasiparticle energy $E_{\boldsymbol{k}}$ and coherence factors $u^\mathrm{B}_{\boldsymbol{k}}$, $v^\mathrm{B}_{\boldsymbol{k}}$ of Eq. (7.23), the quasiparticle operators $\gamma_{\boldsymbol{k}1}$, $\gamma_{\boldsymbol{k}2}$ of Eq. (7.22), and the Nambu spinor $\Psi_{\boldsymbol{k}} = (c_{\boldsymbol{k}\uparrow}, c^\dagger_{-\boldsymbol{k}\downarrow})^\mathrm{T}$ with the matrices of Eq. (7.32). Throughout, $T = 0$, one band crosses the Fermi level, and the crystal is symmetric under time reversal, so that $\xi_{-\boldsymbol{k}} = \xi_{\boldsymbol{k}}$ and the band velocity $\boldsymbol{v}_{\boldsymbol{k}} = \partial\xi_{\boldsymbol{k}}/\partial\boldsymbol{k}$ obeys $\boldsymbol{v}_{-\boldsymbol{k}} = -\boldsymbol{v}_{\boldsymbol{k}}$.

Uniform one-body operators in Nambu form

Take a uniform operator that is diagonal in $\boldsymbol{k}$ and in spin, with a weight $f_{\boldsymbol{k}}$ that is odd, $f_{-\boldsymbol{k}} = -f_{\boldsymbol{k}}$, and a spin factor s_σ: $O = \sum_{\boldsymbol{k}\sigma} s_\sigma f_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}\sigma} c_{\boldsymbol{k}\sigma}$. In the spin-down term relabel $\boldsymbol{k} \to -\boldsymbol{k}$, so that $\sum_{\boldsymbol{k}} f_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\downarrow} = -\sum_{\boldsymbol{k}} f_{\boldsymbol{k}}\, c^\dagger_{-\boldsymbol{k}\downarrow} c_{-\boldsymbol{k}\downarrow}$. With $c^\dagger_{-\boldsymbol{k}\downarrow} c_{-\boldsymbol{k}\downarrow} = 1 - c_{-\boldsymbol{k}\downarrow} c^\dagger_{-\boldsymbol{k}\downarrow}$, and since $\sum_{\boldsymbol{k}} f_{\boldsymbol{k}} = 0$,

$$O = \sum_{\boldsymbol{k}} f_{\boldsymbol{k}} \big[s_\uparrow c^\dagger_{\boldsymbol{k}\uparrow} c_{\boldsymbol{k}\uparrow} + s_\downarrow c_{-\boldsymbol{k}\downarrow} c^\dagger_{-\boldsymbol{k}\downarrow} \big]. \tag{D.220}$$

For the current, $s_\uparrow = s_\downarrow = 1$, and the bracket is $\Psi^\dagger_{\boldsymbol{k}}\Psi_{\boldsymbol{k}}$, the unit matrix in Nambu space. For the z component of the spin current, $s_\sigma = \pm\frac{1}{2}$, and the bracket is $\frac{1}{2}\Psi^\dagger_{\boldsymbol{k}}\boldsymbol{\eta}\Psi_{\boldsymbol{k}}$, the matrix of the density. For an even weight the two assignments are exchanged: the density has $\boldsymbol{\eta}$ (Appendix D.21), the spin density the unit matrix (Sect. 7.2).

The inverse of Eq. (7.22) is $c_{\boldsymbol{k}\uparrow} = u\gamma_{\boldsymbol{k}1} + v\gamma^\dagger_{\boldsymbol{k}2}$ and $c^\dagger_{-\boldsymbol{k}\downarrow} = -v\gamma_{\boldsymbol{k}1} + u\gamma^\dagger_{\boldsymbol{k}2}$, with $u = u^\mathrm{B}_{\boldsymbol{k}}$ and $v = v^\mathrm{B}_{\boldsymbol{k}}$. Inserting it and using $\gamma_{\boldsymbol{k}2}\gamma^\dagger_{\boldsymbol{k}2} = 1 - \gamma^\dagger_{\boldsymbol{k}2}\gamma_{\boldsymbol{k}2}$, $u^2 + v^2 = 1$, $u^2 - v^2 = \xi_{\boldsymbol{k}}/E_{\boldsymbol{k}}$ and $2uv = \Delta_\mathrm{BCS}/E_{\boldsymbol{k}}$,

$$\begin{aligned} \Psi^\dagger_{\boldsymbol{k}}\Psi_{\boldsymbol{k}} &= \gamma^\dagger_{\boldsymbol{k}1}\gamma_{\boldsymbol{k}1} - \gamma^\dagger_{\boldsymbol{k}2}\gamma_{\boldsymbol{k}2} + 1, \\ \Psi^\dagger_{\boldsymbol{k}}\boldsymbol{\eta}\Psi_{\boldsymbol{k}} &= \frac{\xi_{\boldsymbol{k}}}{E_{\boldsymbol{k}}}\big(\gamma^\dagger_{\boldsymbol{k}1}\gamma_{\boldsymbol{k}1} + \gamma^\dagger_{\boldsymbol{k}2}\gamma_{\boldsymbol{k}2} - 1\big) + \frac{\Delta_\mathrm{BCS}}{E_{\boldsymbol{k}}}\big(\gamma^\dagger_{\boldsymbol{k}1}\gamma^\dagger_{\boldsymbol{k}2} + \gamma_{\boldsymbol{k}2}\gamma_{\boldsymbol{k}1}\big). \end{aligned} \tag{D.221}$$

The current $\hat{\boldsymbol{e}}\cdot\hat{\boldsymbol{V}}=\sum_{\boldsymbol{k}}(\hat{\boldsymbol{e}}\cdot\boldsymbol{v}_{\boldsymbol{k}})\Psi^\dagger_{\boldsymbol{k}}\Psi_{\boldsymbol{k}}$ is therefore diagonal in the quasiparticles and commutes with $H_{\rm MF}$. The quasiparticle $\gamma^\dagger_{\boldsymbol{k}1}$, an excitation of $(\boldsymbol{k}\uparrow)$, carries $\hat{\boldsymbol{e}}\cdot\boldsymbol{v}_{\boldsymbol{k}}$, and $\gamma^\dagger_{\boldsymbol{k}2}$, an excitation of $(-\boldsymbol{k}\downarrow)$, carries $-\hat{\boldsymbol{e}}\cdot\boldsymbol{v}_{\boldsymbol{k}}=\hat{\boldsymbol{e}}\cdot\boldsymbol{v}_{-\boldsymbol{k}}$; the ground state carries $\sum_{\boldsymbol{k}}\hat{\boldsymbol{e}}\cdot\boldsymbol{v}_{\boldsymbol{k}}=0$. The spin current is not diagonal. Its last term creates the pair $\gamma^\dagger_{\boldsymbol{k}1}\gamma^\dagger_{\boldsymbol{k}2}|\mathrm{BCS}\rangle$ of energy $2E_{\boldsymbol{k}}$ and total momentum zero.

Spin current without impurities

With $\hat{\boldsymbol{e}}=\hat{\boldsymbol{z}}$, the spin current $\mathcal{J}_e^{zz}=\frac{1}{2}\sum_{\boldsymbol{k}}v^z_{\boldsymbol{k}}\Psi^\dagger_{\boldsymbol{k}}\boldsymbol{\eta}\Psi_{\boldsymbol{k}}$ connects $|\mathrm{BCS}\rangle$ to the pair state at $\boldsymbol{k}$ with the amplitude $\frac{1}{2}v^z_{\boldsymbol{k}}\Delta_{\rm BCS}/E_{\boldsymbol{k}}$. Without spin–orbit coupling the Hamiltonian and the singlet ground state are invariant under rotations of the spin alone, so the three components $\mathcal{J}_e^{\alpha\alpha}$ of $\mathcal{J}_e$ do not interfere and each has the response of $\mathcal{J}_e^{zz}$ in a cubic crystal: $S_{\mathcal{J}\mathcal{J}}=3S_{\mathcal{J}^{zz}\mathcal{J}^{zz}}$. The conductivity $\sigma_1^{\mathcal{J}}=(2e^2/3\omega)S_{\mathcal{J}\mathcal{J}}$ of Eq. (15.26) is then

$$\sigma_1^{\mathcal{J}}(\omega)=\frac{2e^2}{\omega}\,\frac{2\pi}{\mathscr{V}}\sum_{\boldsymbol{k}}\frac{1}{4}(v^z_{\boldsymbol{k}})^2\frac{\Delta^2_{\rm BCS}}{E^2_{\boldsymbol{k}}}\,\delta(\omega-2E_{\boldsymbol{k}})=\frac{\pi e^2}{\omega}\,\mathcal{N}_{\rm F}\langle(v^z)^2\rangle_{\rm FS}\int\mathrm{d}\xi\,\frac{\Delta^2_{\rm BCS}}{E^2}\,\delta(\omega-2E),$$

where the sum over $\boldsymbol{k}$ has become $\mathscr{V}\mathcal{N}_{\rm F}\int\mathrm{d}\xi$ times the Fermi-surface average, each $\boldsymbol{k}$ labeling one pair state of both spins. The delta function fixes $E=\omega/2$ and is reached from $\xi=\pm(\omega^2/4-\Delta^2_{\rm BCS})^{1/2}$, each with the weight $E/2|\xi|$. With the intraband plasma frequency $(\omega_p^{\rm intra})^2=2e^2\mathcal{N}_{\rm F}\langle(v^z)^2\rangle_{\rm FS}$ this gives Eq. (15.27),

$$\sigma_1^{\mathcal{J}}(\omega)=\frac{2\pi(\omega_p^{\rm intra})^2\Delta^2_{\rm BCS}}{\omega^2\,(\omega^2-4\Delta^2_{\rm BCS})^{1/2}}\,\Theta(\omega-2\Delta_{\rm BCS}).\tag{D.222}$$

Its integral follows with $\omega=2\Delta_{\rm BCS}\cosh t$: $\mathrm{d}\omega=2\Delta_{\rm BCS}\sinh t\,\mathrm{d}t$ and $(\omega^2-4\Delta^2_{\rm BCS})^{1/2}=2\Delta_{\rm BCS}\sinh t$, so that $\int_{2\Delta}^{\omega}\sigma_1^{\mathcal{J}}\mathrm{d}\omega'=\frac{\pi}{2}(\omega_p^{\rm intra})^2\int_0^t\mathrm{d}t'/\cosh^2t'=\frac{\pi}{2}(\omega_p^{\rm intra})^2\tanh t$. The total is $\frac{\pi}{2}(\omega_p^{\rm intra})^2$, the intraband f-sum weight. In the clean normal metal the spin current commutes with the one-body band Hamiltonian, and in that model the same weight sits at $\omega=0$. It does not commute with the interactions: a two-body force acts on it with the weight $\sigma_1^z-\sigma_2^z$, which cancels only for the charge current, so electron–electron collisions (spin Coulomb drag) and normal phonon processes broaden the normal-state weight into a narrow Drude peak.

The total weight, summed over all frequencies, does not depend on the mean-field treatment. For a Hamiltonian H that does not act on spin, the z spin current is the time derivative of the spin dipole $D_s=\sum_i\sigma_i^z z_i/2$, $\mathcal{J}_e^{zz}=\mathrm{i}[H,D_s]$, and the double commutator $[[H,D_s],D_s]$ is a quarter of the one-body constant of the charge dipole, the factor that the normalization of Eq. (15.26) compensates. The spin-current absorption therefore obeys the f-sum rule of Sect. 4.3 exactly, with interactions: for the full Hamiltonian the total weight, interband absorption included, is $(\pi/2)\omega_p^2$ with the free-electron ω_p, and for the conduction band alone (a one-band Hamiltonian with spin-independent interactions, the model of this section) it is $(\pi/2)(\omega_p^{\rm intra})^2$. Since D_s commutes with every spin-independent term, this holds although $\mathcal{J}_e^{zz}$ itself is not conserved by H. The mean-field Hamiltonian respects it too: a contact singlet pairing term $\psi^\dagger_\uparrow\psi^\dagger_\downarrow$ commutes with the local spin density, so $H_{\rm MF}$ conserves spin locally, and the s-wave ladder of Eq. (7.34) does not dress the vertex $v^z_{\boldsymbol{k}}$, which is odd in $\boldsymbol{k}$. At $T=0$ the paramagnetic part of the static response, $(2/\pi)\int\sigma_1^{\mathcal{J}}\mathrm{d}\omega=(\omega_p^{\rm intra})^2$, cancels the diamagnetic term exactly, so the singlet carries no spin supercurrent, the counterpart of its vanishing Pauli susceptibility. Beyond mean field the same holds: a uniform static spin twist is a gauge transformation of a spin-independent Hamiltonian that leaves $\psi_\uparrow\psi_\downarrow$ unchanged, so the spin stiffness of the singlet vanishes.

The weight near $2\Delta_{\rm BCS}$ is the coherent part of the total. Two corrections set it. The self-energy gives the quasiparticle residue $1/(1+\lambda_{\rm ep})$ of Eq. (6.36) (with F_1^s entering through the

mass); the phonon interaction does not depend on spin and adds nothing in the spin channel. The interaction in the $\ell = 1$ spin channel, the Landau parameter F_1^a, gives the quasiparticle spin current $(1+F_1^a/3)\boldsymbol{v}^*$. Near the edge the weight is therefore $\frac{\pi}{2}(\omega_p^{\rm intra})^2(1+F_1^a/3)/(1+\lambda_{\rm ep})$, and the remainder of the sum rule lies near $\omega_{\rm D}$ and at the scale of the Fermi energy. With $f(\omega)$ the bare bubble of Eq. (D.222) normalized to $f(0) = 1$, $\operatorname{Im} f = 2\pi\Delta_{\rm BCS}^2/[\omega(\omega^2 - 4\Delta_{\rm BCS}^2)^{1/2}]$, the dressed kernel has the form of Leggett,

$$K_s(\omega) = \frac{(\omega_p^{\rm intra})^2}{1+\lambda_{\rm ep}}\,\frac{(1+F)\,[1-f(\omega)]}{1+Ff(\omega)}, \qquad F = \frac{F_1^a}{3}, \qquad \sigma_1^{\mathcal{J}} = -\frac{\operatorname{Im} K_s}{\omega},$$

which vanishes at $\omega = 0$ (no spin stiffness) and tends to the normal-state spin Drude weight for $f \to 0$. At $4\Delta_{\rm BCS}$ it multiplies the bare $\sigma_1^{\mathcal{J}}$ by 1.31, 1.15, 0.87 and 0.75 for $F = 0.1$, 0.05, -0.05 and -0.1. Below the edge $\operatorname{Re} f$ grows as $(2\Delta_{\rm BCS} - \omega)^{-1/2}$, so for $F_1^a < 0$ the denominator has a root below $2\Delta_{\rm BCS}$: a bound spin-one mode with $\ell = 1$, bound by 5.6×10^{-3}, 2.1×10^{-2} and $7.4\times10^{-2}\Delta_{\rm BCS}$ at $F = -0.05$, -0.1 and -0.2, which takes weight from the continuum. A triplet p-wave attraction acts in the same channel.

The phonon-assisted amplitude

Let the band at the Fermi level couple to the phonons through Eq. (A.25) with the vertex $g_\nu(\boldsymbol{k},\boldsymbol{Q})$, band indices dropped, and assume the phase convention in which time reversal gives $g_\nu(-\boldsymbol{k}-\boldsymbol{Q},\boldsymbol{Q}) = g_\nu(\boldsymbol{k},\boldsymbol{Q})$, as for any vertex of a perturbation even under time reversal. The final states with one phonon emitted are

$$|f\rangle = \gamma^\dagger_{\boldsymbol{k}+\boldsymbol{Q},1}\,\gamma^\dagger_{\boldsymbol{k}2}\,b^\dagger_{-\boldsymbol{Q}\nu}|{\rm BCS}\rangle, \qquad E_f - E_i = E_{\boldsymbol{k}+\boldsymbol{Q}} + E_{\boldsymbol{k}} + \omega_{\boldsymbol{Q}\nu},$$

an excitation of $(\boldsymbol{k}+\boldsymbol{Q}\uparrow)$, one of $(-\boldsymbol{k}\downarrow)$ and the phonon; the other spin assignment is the same family with $\boldsymbol{k} \to -\boldsymbol{k}-\boldsymbol{Q}$. The phonon vertex creates them from $|{\rm BCS}\rangle$ through its spin-up term $c^\dagger_{\boldsymbol{k}+\boldsymbol{Q}\uparrow}c_{\boldsymbol{k}\uparrow}$, with the coefficient $u^{\rm B}_{\boldsymbol{k}+\boldsymbol{Q}}v^{\rm B}_{\boldsymbol{k}}$, and through its spin-down term $c^\dagger_{-\boldsymbol{k}\downarrow}c_{-\boldsymbol{k}-\boldsymbol{Q}\downarrow}$, with $v^{\rm B}_{\boldsymbol{k}+\boldsymbol{Q}}u^{\rm B}_{\boldsymbol{k}}$, exactly as the density does in Eq. (7.28); the time-reversal condition makes the two vertices equal, so that

$$\langle f|H_{\rm ep}|{\rm BCS}\rangle = \frac{1}{\sqrt{N}}\,g_\nu(\boldsymbol{k},\boldsymbol{Q})\,(u^{\rm B}_{\boldsymbol{k}+\boldsymbol{Q}}v^{\rm B}_{\boldsymbol{k}} + v^{\rm B}_{\boldsymbol{k}+\boldsymbol{Q}}u^{\rm B}_{\boldsymbol{k}}).$$

The amplitude to reach $|f\rangle$ from $|i\rangle = |{\rm BCS}\rangle$ with one current vertex $O = \hat{\boldsymbol{e}}\cdot\hat{\boldsymbol{V}}$ and one phonon vertex is Eq. (2.51) with the two orderings of Eq. (13.7),

$$T_{fi} = \sum_m \left[\frac{\langle f|H_{\rm ep}|m\rangle\langle m|O|i\rangle}{E_i+\omega-E_m} + \frac{\langle f|O|m\rangle\langle m|H_{\rm ep}|i\rangle}{E_i-E_m}\right], \qquad \omega = E_f - E_i,$$

with $|m\rangle$ the eigenstates of $H_{\rm MF}$ plus the free phonons. Since O is diagonal in that basis, $\langle m|O|i\rangle = J_i\delta_{mi}$ and $\langle f|O|m\rangle = J_f\delta_{fm}$, with J_i and J_f the eigenvalues of O. The sums collapse:

$$T_{fi} = \frac{J_i\,\langle f|H_{\rm ep}|i\rangle}{\omega} + \frac{J_f\,\langle f|H_{\rm ep}|i\rangle}{E_i-E_f} = -\frac{J_f-J_i}{\omega}\,\langle f|H_{\rm ep}|i\rangle. \tag{D.223}$$

No intermediate state needs to be far from the energy shell; the only input is that O commutes with $H_{\rm MF}$ and with the phonon Hamiltonian. By Eq. (D.221), $J_i = 0$ and $J_f = \hat{\boldsymbol{e}}\cdot(\boldsymbol{v}_{\boldsymbol{k}+\boldsymbol{Q}} - \boldsymbol{v}_{\boldsymbol{k}})$, which is Eq. (15.23). The same steps with the density in place of the current give zero, since its $\boldsymbol{q} = 0$ eigenvalue is the conserved electron number, as in Eq. (13.10).

The phonon-assisted conductivity

The absorbed power of a field $\hat{\boldsymbol{e}}\mathcal{E}\cos\omega t$ coupled through Eq. (6.38) gives the conductivity as in Eq. (6.39), $\sigma_1 = (\pi e^2/\omega\mathcal{V})\sum_f |T_{fi}|^2\delta(E_f - E_i - \omega)$, where the sum over f runs over $\boldsymbol{k}$, $\boldsymbol{Q}$ and ν. The squared coherence factor of $\langle f|H_{\rm ep}|i\rangle$ is the case-I factor of Eq. (7.29), $\frac{1}{2}[1 - (\xi\xi' - \Delta_{\rm BCS}^2)/EE']$, with primes at $\boldsymbol{k}+\boldsymbol{Q}$. In the Fermi-surface approximation g_ν and $\boldsymbol{v}$ depend on the positions of $\boldsymbol{k}$ and $\boldsymbol{k}+\boldsymbol{Q}$ on the Fermi surface and not on ξ and ξ' over the range ω, so the sum over $\boldsymbol{k}$ and $\boldsymbol{Q}$ factorizes into a Fermi-surface sum, carrying $\delta(\xi_{\boldsymbol{k}})\delta(\xi_{\boldsymbol{k}+\boldsymbol{Q}})$, times $\int \mathrm{d}\xi \int \mathrm{d}\xi'$. The term odd in $\xi\xi'$ integrates to zero. Each of ξ and ξ' reaches the quasiparticle energy E from both signs, with $\mathrm{d}\xi = E\,\mathrm{d}E/(E^2 - \Delta_{\rm BCS}^2)^{1/2}$, and at fixed pair energy $x = \omega - \omega_{\boldsymbol{Q}\nu}$

$$\int \mathrm{d}\xi \int \mathrm{d}\xi' \,\frac{1}{2}\Big(1 + \frac{\Delta_{\rm BCS}^2}{EE'}\Big)\delta(x - E - E') = 2\int_{\Delta_{\rm BCS}}^{x-\Delta_{\rm BCS}} \mathrm{d}E\,\frac{E(x-E) + \Delta_{\rm BCS}^2}{(E^2 - \Delta_{\rm BCS}^2)^{1/2}[(x-E)^2 - \Delta_{\rm BCS}^2]^{1/2}} = 2x\,\mathfrak{g}_{\rm I}(x/\Delta_{\rm BCS}),$$

by Eq. (7.30); in the normal state it is $2x$, the two spin orientations of an electron–hole pair of energy x. The Fermi-surface sum of $|g_\nu|^2[\hat{\boldsymbol{e}}\cdot(\boldsymbol{v}_{\boldsymbol{k}+\boldsymbol{Q}} - \boldsymbol{v}_{\boldsymbol{k}})]^2\delta(\xi)\delta(\xi')\delta(\omega' - \omega_{\boldsymbol{Q}\nu})$ is $2\langle(\hat{\boldsymbol{e}}\cdot\boldsymbol{v})^2\rangle_{\rm FS}\mathcal{N}_{\rm F}\Omega N^2\,\alpha^2F_{\rm tr}(\omega')$ by the definition Eq. (15.24). With $\mathcal{V} = N\Omega$,

$$\sigma_1^{\rm S}(\omega) = \frac{\pi e^2}{\omega^3}\,2\langle(\hat{\boldsymbol{e}}\cdot\boldsymbol{v})^2\rangle_{\rm FS}\mathcal{N}_{\rm F}\int \mathrm{d}\omega'\,\alpha^2F_{\rm tr}(\omega')\,2(\omega-\omega')\,\mathfrak{g}_{\rm I}\Big(\frac{\omega-\omega'}{\Delta_{\rm BCS}}\Big),$$

which is Eq. (15.25). The pair energy $\omega - \omega'$ must exceed $2\Delta_{\rm BCS}$, which bounds ω' by $\omega - 2\Delta_{\rm BCS}$.

For the Debye form $\alpha^2F_{\rm tr} = 2\lambda_{\rm tr}\omega'^4/\omega_{\rm D}^4$ below $\omega_{\rm D}$, which satisfies $\lambda_{\rm tr} = 2\int \alpha^2F_{\rm tr}/\omega'\,\mathrm{d}\omega'$, the normal-state rate $\gamma_{\rm H}$ of Eq. (15.25) needs $\int_0^\omega(\omega-\omega')\omega'^4\mathrm{d}\omega' = \omega^6/30$ for $\omega < \omega_{\rm D}$ and $\int_0^{\omega_{\rm D}}(\omega-\omega')\omega'^4\mathrm{d}\omega' = \omega\omega_{\rm D}^5/5 - \omega_{\rm D}^6/6$ above, so that

$$\gamma_{\rm H}(\omega) = \frac{2\pi}{15}\,\lambda_{\rm tr}\,\frac{\omega^5}{\omega_{\rm D}^4}\quad(\omega<\omega_{\rm D}),\qquad \gamma_{\rm H}(\omega) = \frac{4\pi}{5}\,\lambda_{\rm tr}\,\omega_{\rm D}\Big(1 - \frac{5\omega_{\rm D}}{6\omega}\Big)\quad(\omega\ge\omega_{\rm D}),$$

the two branches of Ref. [79, Eq. (16)]. The ratio of Eq. (15.25) to its normal-state value is then the ratio of $\int_0^{\omega-2\Delta}\mathrm{d}\omega'\,\omega'^4(\omega-\omega')\mathfrak{g}_{\rm I}$ to $\omega^6/30$, the form of Ref. [79, Eq. (17)], whose elliptic integral is $\mathfrak{g}_{\rm I} = \mathrm{E}((1 - 4\Delta_{\rm BCS}^2/x^2)^{1/2})$, the Landen transform (a change of modulus that maps one elliptic integral onto another) of the form below.

The coherence integrals in closed form

With $y = x/\Delta_{\rm BCS}$ and $\Delta_{\rm BCS} = 1$, write $E = y/2 + s$. Then $E(y-E) = y^2/4 - s^2$ and $(E^2-1)[(y-E)^2-1] = (a^2-s^2)(c^2-s^2)$ with $a = y/2 - 1$ and $c = y/2 + 1$, so that the integrals of Eq. (7.30) are $(2/y)\int_0^a \mathrm{d}s\,(y^2/4 \pm 1 - s^2)/[(a^2-s^2)(c^2-s^2)]^{1/2}$. The substitution $s = a\sin\phi$ and the modulus $k = a/c = (y-2)/(y+2)$ (a number, not a momentum) turn them into $(2/yc)\int_0^{\pi/2}\mathrm{d}\phi\,(y^2/4 \pm 1 - a^2\sin^2\phi)/(1 - k^2\sin^2\phi)^{1/2}$. With $\int_0^{\pi/2}\sin^2\phi\,(1-k^2\sin^2\phi)^{-1/2}\mathrm{d}\phi = (\mathrm{K}-\mathrm{E})/k^2$, with the complete elliptic integrals $\mathrm{K}(k)$ and $\mathrm{E}(k)$, and $a^2/k^2 = c^2$, the bracket becomes $(y^2/4 \pm 1 - c^2)\mathrm{K} + c^2\mathrm{E}$, where $y^2/4 + 1 - c^2 = -y$ and $y^2/4 - 1 - c^2 = -(y+2)$. Hence

$$\mathfrak{g}_{\rm I}(y) = \Big(1 + \frac{2}{y}\Big)\mathrm{E}(k) - \frac{4}{y+2}\,\mathrm{K}(k),\qquad \mathfrak{g}_{\rm II}(y) = \Big(1 + \frac{2}{y}\Big)\mathrm{E}(k) - \frac{4}{y}\,\mathrm{K}(k),\tag{D.224}$$

for $y \ge 2$. At $y = 2$, $k = 0$ and $\mathrm{E} = \mathrm{K} = \pi/2$, so that $\mathfrak{g}_{\rm I} = \pi/2$ and $\mathfrak{g}_{\rm II} = 0$; as $y \to \infty$ both tend to one. At $y = 4$ they are 1.211 and 0.672, the values of Sect. 7.2, and both forms agree with direct quadrature of Eq. (7.30) to 10^{-7}.

Impurity-assisted absorption

In a disordered normal metal the one-electron eigenstates α have energies ξ_α and no definite momentum; time reversal pairs each with a partner $\bar\alpha$ of the same energy. The paired state is built from $(\alpha\uparrow, \bar\alpha\downarrow)$ as Eq. (7.24) is built from $(\boldsymbol{k}\uparrow, -\boldsymbol{k}\downarrow)$, with the same gap for nonmagnetic disorder (Sect. 15.3) and with $u^{\rm B}_\alpha$, $v^{\rm B}_\alpha$ from Eq. (7.23) at ξ_α. The current $\sum_{\alpha\beta\sigma} V^e_{\alpha\beta} c^\dagger_{\alpha\sigma} c_{\beta\sigma}$, with $V^e_{\alpha\beta} = \langle\alpha|\hat{\boldsymbol{e}}\cdot\hat{\boldsymbol{v}}|\beta\rangle$, is odd under time reversal, $V^e_{\bar\alpha\bar\beta} = -V^e_{\beta\alpha}$. The step of Eq. (D.220), with the relabeling $\alpha \to \bar\alpha$ in place of $\boldsymbol{k} \to -\boldsymbol{k}$, turns its spin-down term into $\sum V^e_{\alpha\beta} c_{\bar\alpha\downarrow} c^\dagger_{\bar\beta\downarrow}$, and the operator into $\sum_{\alpha\beta} V^e_{\alpha\beta} \Psi^\dagger_\alpha \Psi_\beta$, again the unit matrix. Its pair-creation amplitude from the ground state into $\gamma^\dagger_{\alpha1}\gamma^\dagger_{\beta2}$ is then $V^e_{\alpha\beta}(u^{\rm B}_\alpha v^{\rm B}_\beta - v^{\rm B}_\alpha u^{\rm B}_\beta)$, the case-II combination of Eq. (7.28), with the squared form $\frac{1}{2}[1 - (\xi_\alpha\xi_\beta + \Delta^2_{\rm BCS})/E_\alpha E_\beta]$. A spin current, even under time reversal, carries $\boldsymbol{\eta}$ instead and the case-I combination.

In the dirty limit the relaxation rate $\gamma_{\rm rel}$ of the electrons far exceeds $\Delta_{\rm BCS}$. The average of $|V^e_{\alpha\beta}|^2$ over states at fixed ξ_α and ξ_β then changes only when $\xi_\beta - \xi_\alpha$ changes by about $\gamma_{\rm rel}$, and over the few $\Delta_{\rm BCS}$ where the coherence factors differ from their normal values it is a constant, the same in the normal and the paired state. With it taken out, the sum over α and β at fixed energy is the double integral over independent ξ_α and ξ_β that led to Eq. (7.30), and the ratio of the paired to the normal conductivity is the case-II integral $\mathfrak{g}_{\rm II}(\omega/\Delta_{\rm BCS})$: Eq. (15.22). For a probe that is even under time reversal the same steps give $\mathfrak{g}_{\rm I}$. The assumption fails when the mean free path approaches the size of a pair, $\gamma_{\rm rel} \sim \Delta_{\rm BCS}$, where the correlation of α and β through their momenta survives; the clean limit is Eq. (D.221).

D.41 Background screening of Dirac materials

A Dirac target screens a dark-photon field twice, through its remote bands and through its Dirac electrons, and the scattering and absorption rates depend on where the remote bands enter. With the background in the total dielectric function the rates are smaller than in the form in which the background screens only the Dirac electrons, by $\varepsilon^2_{\rm bg}$ for scattering and $\varepsilon_{\rm bg}$ for absorption, the factors used in Sect. 16.1. The notation is that of Sect. 7.3 and Appendix D.22: the half-gap $\Delta_{\rm D}$, the Dirac velocity $v_{\rm D}$ (principal velocities v_i), the cutoff $\Lambda_{\rm D}$, the degeneracy $g_{\rm deg}$, the independent-pair response $R^0_{\rm D}$ of Eq. (D.134) and the effective coupling $\tilde\alpha = \alpha/(\varepsilon_{\rm bg} v_{\rm D})$.

The total dielectric function

The longitudinal dielectric function is $\varepsilon = 1 - v_{\rm C}\tilde R$, with $\tilde R$ the proper density response of all the charges (Sect. 8.4.1). Split $\tilde R = \tilde R_{\rm D} + \tilde R_{\rm bg}$ into the Dirac bands inside the cutoff and everything else, the remote bands and, below the optical phonons, the ionic polarization. At energies far below the gap of the remote bands and momenta small compared with the reciprocal lattice, the background responds locally and instantaneously, $1 - v_{\rm C}\tilde R_{\rm bg} \to \varepsilon_{\rm bg}$, real and independent of $\boldsymbol{q}$ and ω, with local fields neglected. In the random-phase approximation the proper Dirac response is the independent-pair bubble, $\tilde R_{\rm D} = R^0_{\rm D}$, and

$$\varepsilon(\boldsymbol{q},\omega) = \varepsilon_{\rm bg} + \varepsilon_{\rm D}(\boldsymbol{q},\omega), \qquad \varepsilon_{\rm D} = -v_{\rm C}(q)\, R^0_{\rm D}(\boldsymbol{q},\omega), \tag{D.225}$$

which is Eq. (7.42). The Dirac part $\varepsilon_{\rm D}$ does not contain $\varepsilon_{\rm bg}$: by Eq. (D.134) it is $g_{\rm deg}\alpha/(6\pi v_{\rm D})$ times the bracket of Eq. (D.136), and its imaginary part deep in the continuum is $g_{\rm deg}\alpha/6v_{\rm D}$. Take an anisotropic cone and a background tensor with the same principal axes. The background enters the longitudinal function as $\hat{\boldsymbol{q}}\cdot\varepsilon_{\rm bg}\cdot\hat{\boldsymbol{q}} = \sum_i \varepsilon_{{\rm bg},ii}\hat q_i^2$. The rescaling at the

end of Appendix D.22 turns q^2/v_D in the overlap into $\tilde{q}^2/(v_1v_2v_3)$ with $\tilde{\boldsymbol{q}} = (v_1q_1, v_2q_2, v_3q_3)$, so that

$$\varepsilon(\boldsymbol{q},\omega) = \sum_i \varepsilon_{\mathrm{bg},ii}\hat{q}_i^2 + \sum_i \hat{q}_i^2\,\varepsilon_{\mathrm{D},i}(\omega^2-\tilde{q}^2), \qquad \varepsilon_{\mathrm{D},i} = \frac{g_\mathrm{deg}\alpha}{6\pi}\,\frac{v_i^2}{v_1v_2v_3}\,B_\mathrm{D}(\omega^2-\tilde{q}^2), \quad \text{(D.226)}$$

with B_D the bracket of Eq. (7.44) and $v_\mathrm{D}\Lambda_\mathrm{D}$ replaced by the energy cutoff $\tilde{\Lambda}$ of Eq. (7.47). Both terms of Eq. (D.226) carry the factor $\hat{q}_i^2$ of a longitudinal function; the background is not divided into the Dirac term.

Scattering

A kinetically mixed dark photon couples to the total charge, and by Eq. (8.38) its structure factor is $(2q^2/e^2)L/(1-\mathrm{e}^{-\beta\omega})$ with $L = \mathrm{Im}(-1/\varepsilon)$ of the total dielectric function. Since ε_bg is real,

$$L = \frac{\mathrm{Im}\,\varepsilon}{|\varepsilon|^2} = \frac{\mathrm{Im}\,\varepsilon_\mathrm{D}}{|\varepsilon_\mathrm{bg}+\varepsilon_\mathrm{D}|^2}. \quad \text{(D.227)}$$

The numerator counts the Dirac pairs. With the spectral form of R_D^0 (Eq. (2.58)), $\mathrm{Im}\,\varepsilon_\mathrm{D} = (\pi e^2/q^2)\mathcal{V}^{-1}\sum_f |\langle f|n_\mathrm{D}(\boldsymbol{q})|0\rangle|^2\delta(\omega-\omega_f)$ for $\omega>0$, the golden-rule sum of pair transitions with a vertex of unit charge, and the denominator is the screening of the field by every charge in the crystal. Eq. (D.227) is exact for the total charge within the two approximations of Eq. (D.225), the local background and the random-phase Dirac response, and it holds for any size of ε_D.

The published Dirac rates [60, 80] are assembled differently. The reference cross-section $\bar{\sigma}_e$ is defined in vacuum, from the free amplitude [60, Eqs. (3.8), (3.10), (3.11)], [80, Eqs. (2.3), (2.4)]. The pair transitions are summed with the same unit-charge vertex, through the overlap of the band spinors. The medium enters through the in-medium form factor $F_\mathrm{med} = 1/\varepsilon_\mathrm{pub}$, where $\varepsilon_\mathrm{pub} = 1+\varepsilon_\mathrm{D}/\varepsilon_\mathrm{bg}$ is the polarization of the Dirac electrons alone, computed with the charge $e/\varepsilon_\mathrm{bg}^{1/2}$ in the loop and the vacuum photon propagator outside it [60, Eqs. (2.6), (3.7), (3.11), (A.8)], [80, Eqs. (2.5)–(2.7), (3.3)]. Since $\varepsilon_\mathrm{pub} = \varepsilon/\varepsilon_\mathrm{bg}$, the loss function of that construction is

$$L_\mathrm{pub} = \frac{\mathrm{Im}\,\varepsilon_\mathrm{D}}{|\varepsilon/\varepsilon_\mathrm{bg}|^2} = \varepsilon_\mathrm{bg}^2\,L, \quad \text{(D.228)}$$

at every $(\boldsymbol{q},\omega)$ and for any ε_D. The published scattering rate is the total-charge rate times $\varepsilon_\mathrm{bg}^2$ exactly.

For an anisotropic target the in-medium factor of Ref. [80, Eq. (3.14)] divides each component of the Dirac polarization by its own background constant, $\varepsilon_\mathrm{pub} = 1+\sum_i \hat{q}_i^2\varepsilon_{\mathrm{D},i}/\varepsilon_{\mathrm{bg},ii}$, and the ratio of the two loss functions is

$$\frac{L_\mathrm{pub}}{L} = \frac{|\sum_i \hat{q}_i^2(\varepsilon_{\mathrm{bg},ii}+\varepsilon_{\mathrm{D},i})|^2}{|1+\sum_i \hat{q}_i^2\varepsilon_{\mathrm{D},i}/\varepsilon_{\mathrm{bg},ii}|^2}. \quad \text{(D.229)}$$

(The term of the x axis is read with $q_x^2v_x^2$, where that equation prints $q_x^2v_z^2$.) Along a principal axis it is $\varepsilon_{\mathrm{bg},ii}^2$. Off the axes it is not a power of any single background constant: for $\varepsilon_\mathrm{D}\to 0$ it tends to $(\hat{\boldsymbol{q}}\cdot\varepsilon_\mathrm{bg}\cdot\hat{\boldsymbol{q}})^2$, and for finite ε_D it depends on $\hat{\boldsymbol{q}}$ through both sums. The event rates then differ by the ratio of their integrals over directions and energies, which the reach of an anisotropic target must be computed with.

Eq. (D.228) has an exact physical meaning. Let a field couple to the Dirac electrons and to no other charge. Its potential induces a Dirac charge, whose Coulomb field, screened by the background, acts back on the Dirac electrons with $v_\mathrm{C}/\varepsilon_\mathrm{bg}$; the random-phase sum of this chain is $R_\mathrm{DD} = R_\mathrm{D}^0/(1-v_\mathrm{C}R_\mathrm{D}^0/\varepsilon_\mathrm{bg}) = R_\mathrm{D}^0/\varepsilon_\mathrm{pub}$. Because $R_\mathrm{D}^0\varepsilon_\mathrm{pub}^* = R_\mathrm{D}^0 - v_\mathrm{C}|R_\mathrm{D}^0|^2/\varepsilon_\mathrm{bg}$

has the imaginary part $\operatorname{Im} R_{\mathrm{D}}^0$, $-v_{\mathrm{C}} \operatorname{Im} R_{\mathrm{DD}} = \operatorname{Im} \varepsilon_{\mathrm{D}}/|\varepsilon_{\mathrm{pub}}|^2 = L_{\mathrm{pub}}$. The published rate is therefore the rate of a mediator that couples to the Dirac electrons alone. A kinetically mixed dark photon couples to the electromagnetic current, remote bands included [60, Eq. (2.1)], and a mediator that couples to the electron number acts on the remote-band electrons as well; with the nuclei clamped its rate is again the loss function of the total ε (Eq. (8.38)).

The factor $\varepsilon_{\mathrm{bg}}^2$ is not an artifact of the cutoff. For a gapless cone with a hard cutoff, Eq. (D.138) gives $\varepsilon/\varepsilon_{\mathrm{bg}} = 1 + (g_{\mathrm{deg}}\tilde{\alpha}/6\pi)[\ln(4v_{\mathrm{D}}^2\Lambda_{\mathrm{D}}^2/|s|) + \mathrm{i}\pi\Theta(s)]$ with $s = \omega^2 - v_{\mathrm{D}}^2 q^2$. The constant 1 can be absorbed into the cutoff, $\Lambda' = \Lambda_{\mathrm{D}} \exp[3\pi/(g_{\mathrm{deg}}\tilde{\alpha})]$, and then $\varepsilon/\varepsilon_{\mathrm{bg}} = (g_{\mathrm{deg}}\tilde{\alpha}/6\pi)\mathcal{L}'$ with $\mathcal{L}' = \ln(4v_{\mathrm{D}}^2\Lambda'^2/|s|) + \mathrm{i}\pi\Theta(s)$, an identity. It holds in either subtraction scheme, since the constant $\frac{5}{3}$ by which the schemes differ is absorbed into Λ' in the same way. It gives

$$\frac{1}{|\varepsilon|^2} = \Big(\frac{6\pi v_{\mathrm{D}}}{g_{\mathrm{deg}}\alpha}\Big)^2 \frac{1}{|\mathcal{L}'|^2}, \qquad \frac{1}{|\varepsilon_{\mathrm{pub}}|^2} = \Big(\frac{6\pi \varepsilon_{\mathrm{bg}} v_{\mathrm{D}}}{g_{\mathrm{deg}}\alpha}\Big)^2 \frac{1}{|\mathcal{L}'|^2}. \tag{D.230}$$

In the total-ε rate the prefactor $g_{\mathrm{deg}}\tilde{\alpha}\varepsilon_{\mathrm{bg}} = g_{\mathrm{deg}}\alpha/v_{\mathrm{D}}$ does not depend on the background; in the published rate it carries $\varepsilon_{\mathrm{bg}}^2$ explicitly. The rescaled cutoff is that of Ref. [60, text after Eq. (3.13)], $\Lambda' = \Lambda \exp[12\pi^2\varepsilon_{\mathrm{bg}} v_{\mathrm{D}}/(g_{\mathrm{deg}} e^2)]$, and the prefactor $288\pi^4\varepsilon_{\mathrm{bg}}^2 v_{\mathrm{D}}^2/(e^4 g_{\mathrm{deg}}^2)$ of its Eq. (3.13) is $\frac{1}{2}(6\pi\varepsilon_{\mathrm{bg}} v_{\mathrm{D}}/g_{\mathrm{deg}}\alpha)^2$: the explicit $\varepsilon_{\mathrm{bg}}^2$ in it is the omitted $1/\varepsilon_{\mathrm{bg}}^2$. At large $\varepsilon_{\mathrm{bg}}$, $|\mathcal{L}'| \approx 6\pi/(g_{\mathrm{deg}}\tilde{\alpha}) \propto \varepsilon_{\mathrm{bg}}$, so the published rate tends to a constant, the unscreened rate of the Dirac pairs, while the total-ε rate falls as $1/\varepsilon_{\mathrm{bg}}^2$.

Absorption

A dark photon at rest, $\omega = m_\phi$ and $\boldsymbol{q} \to 0$, mixes with the photon of the medium through the self-energy of all charges, $\Pi_{\mathrm{em}} = \omega^2(1 - \varepsilon)$, the same for both polarizations at $\boldsymbol{q} \to 0$ (Sect. 8.4.5). By Eq. (8.39), $\zeta_{\mathrm{eff}}^2 = \zeta^2 m_\phi^4/|m_\phi^2 - \Pi_{\mathrm{em}}|^2 = \zeta^2/|\varepsilon|^2$, and the absorption rate per quantum is

$$\Gamma = -\frac{\zeta_{\mathrm{eff}}^2}{\omega} \operatorname{Im} \Pi_{\mathrm{em}} = \zeta^2 m_\phi \frac{\operatorname{Im} \varepsilon}{|\varepsilon|^2} = \zeta^2 m_\phi L(0, m_\phi), \tag{D.231}$$

which is Eq. (10.31). The published absorption rates use the self-energy of the Dirac electrons with the charge $e/\varepsilon_{\mathrm{bg}}^{1/2}$, $\Pi = \omega^2(1 - \varepsilon_{\mathrm{pub}})$ [60, Eqs. (4.4), (4.5)], [80, Eqs. (2.25)–(2.27)]. Then $\zeta_{\mathrm{eff}}^2 = \zeta^2/|\varepsilon_{\mathrm{pub}}|^2$ and $-\operatorname{Im}\Pi/\omega = m_\phi \operatorname{Im} \varepsilon_{\mathrm{pub}} = m_\phi \operatorname{Im} \varepsilon_{\mathrm{D}}/\varepsilon_{\mathrm{bg}}$, so

$$\Gamma_{\mathrm{pub}} = \zeta^2 m_\phi \frac{\operatorname{Im} \varepsilon_{\mathrm{pub}}}{|\varepsilon_{\mathrm{pub}}|^2} = \zeta^2 m_\phi \frac{\varepsilon_{\mathrm{bg}} \operatorname{Im} \varepsilon_{\mathrm{D}}}{|\varepsilon|^2} = \varepsilon_{\mathrm{bg}} \Gamma. \tag{D.232}$$

The power is one, not two. The absorptive part of the published self-energy already carries one factor $1/\varepsilon_{\mathrm{bg}}$, the square of the charge in the loop, whereas the scattering numerator of Eq. (D.228) is the bare pair spectrum with the vacuum charge. For a dark photon polarized along the principal axis i of an anisotropic target, ε and $\varepsilon_{\mathrm{pub}}$ are the ii components and the factor is $\varepsilon_{\mathrm{bg},ii}$. An unpolarized field averages the three rates, and the factor of the averaged rate lies between the smallest and the largest $\varepsilon_{\mathrm{bg},ii}$.

Reach

In a background-free search the excluded cross-section at a given mass is inversely proportional to the expected number of events, and the excluded mixing to its square root. With Eqs. (D.228) and (D.232), the reach curves of an isotropic model correct by exact vertical shifts,

$$\bar{\sigma}_e^{\mathrm{lim}}(m_\chi) \to \varepsilon_{\mathrm{bg}}^2 \, \bar{\sigma}_e^{\mathrm{lim}}(m_\chi), \qquad \zeta^{\mathrm{lim}}(m_\phi) \to \varepsilon_{\mathrm{bg}}^{1/2} \, \zeta^{\mathrm{lim}}(m_\phi), \tag{D.233}$$

at every mass. For an anisotropic target the shift is the ratio of the event integrals with Eq. (D.229) inside, which depends on the mass and on the orientation of the crystal. The published curves keep the real part of $\varepsilon_{\rm D}$ in $|\varepsilon_{\rm pub}|^2$, so Eq. (D.233) needs no further factor. A recomputation that keeps only the leading term of Eq. (D.227) at weak screening, $L \to \mathrm{Im}\,\varepsilon_{\rm D}/\varepsilon_{\rm bg}^2$, must restore the factor $|1+\varepsilon_{\rm D}/\varepsilon_{\rm bg}|^{-2}$ instead, and applying both would count the screening by the Dirac electrons twice.

D.42 Two-level relaxation by phonons and electrons

For the coupling $\frac{1}{2}\sigma_z X$ of a two-level system to a bath coordinate X, Eq. (16.68) needs only the spectral density $J_{\rm s}(\omega)$ of Eq. (C.170). We derive it for long acoustic phonons and for the conduction electrons of a metal, with the Born-limit coupling $\alpha_{\rm ohm} = 4K_{\rm b}$, its exact form for one contact channel, and the factor by which pairing suppresses the electron rate. The notation is that of Appendix C.20 and Sect. 16.8: asymmetry $\Delta_{\rm as}$, tunneling amplitude Δ_0, splitting $\Delta_{\rm tl}$, and $\sigma_z = |{\rm L}\rangle\langle{\rm L}| - |{\rm R}\rangle\langle{\rm R}|$.

The golden-rule rate

Write the eigenstates as $|\pm\rangle$ with energies $\pm\frac{1}{2}\Delta_{\rm tl}$, and define the mixing angle by $\cos\theta = \Delta_{\rm as}/\Delta_{\rm tl}$, $\sin\theta = \Delta_0/\Delta_{\rm tl}$. Then $\sigma_z = \cos\theta\,(|+\rangle\langle+| - |-\rangle\langle-|) - \sin\theta\,(|+\rangle\langle-| + |-\rangle\langle+|)$, up to the sign of $\sin\theta$, which no rate depends on. With $X = \sum_k \lambda_k (d_k + d_k^\dagger)$ the matrix element for the defect to drop from $|+\rangle$ to $|-\rangle$ while mode k gains one quantum is $\frac{1}{2}\lambda_k \sin\theta\,(n_k+1)^{1/2}$, and the golden rule gives the downward rate $\Gamma_\downarrow = 2\pi \sum_k \frac{1}{4}\lambda_k^2 \sin^2\theta\,(n_k+1)\,\delta(\Delta_{\rm tl} - \omega_k) = \frac{1}{2}\sin^2\theta\, J_{\rm s}(\Delta_{\rm tl})[n_{\rm B}(\Delta_{\rm tl})+1]$. The upward rate has $n_{\rm B}$ in place of $n_{\rm B}+1$, so $\Gamma_\uparrow/\Gamma_\downarrow = {\rm e}^{-\Delta_{\rm tl}/T}$. The deviation of the upper population from equilibrium decays as ${\rm e}^{-(\Gamma_\downarrow+\Gamma_\uparrow)t}$, and $\Gamma_1 = \Gamma_\downarrow + \Gamma_\uparrow$ is Eq. (16.68). The diagonal part $\cos\theta$ of σ_z does not change the state; it shifts the splitting by the fluctuating bath and contributes only to the dephasing.

Acoustic phonons

Let a strain $u_{\alpha\beta}$ shift the asymmetry by $\delta\Delta_{\rm as} = 2\sum_{\alpha\beta}\gamma_{\alpha\beta}u_{\alpha\beta}$, so that the coupling is $\sigma_z \sum_{\alpha\beta}\gamma_{\alpha\beta}u_{\alpha\beta}$, with the deformation-potential tensor $\gamma_{\alpha\beta}$ of the defect. This is the convention of [81, Eq. (16)] and of the standard tunneling model, in which the relaxation rate takes the form Eq. (16.53). The text of Meißner et al. states $\delta\varepsilon = \gamma u_{\rm d}$ for the energy shift in a strain component $u_{\rm d}$, but their one-phonon rate [82, Eq. (7)] holds for $\delta\varepsilon = 2\gamma u_{\rm d}$; with $\delta\Delta_{\rm as} = \gamma S$ it would be four times smaller. In an isotropic solid of density ρ_T and volume $\mathcal{V}$ the displacement field at long wavelength is the continuum limit of Sect. 5.3,

$$\boldsymbol{u}(\boldsymbol{r}) = \sum_{\boldsymbol{Q}\nu} \frac{1}{(2\rho_T \mathcal{V}\,\omega_{\boldsymbol{Q}\nu})^{1/2}}\,\hat{\boldsymbol{e}}_{\boldsymbol{Q}\nu}\big(d_{\boldsymbol{Q}\nu}{\rm e}^{{\rm i}\boldsymbol{Q}\cdot\boldsymbol{r}} + d_{\boldsymbol{Q}\nu}^\dagger {\rm e}^{-{\rm i}\boldsymbol{Q}\cdot\boldsymbol{r}}\big), \qquad \omega_{\boldsymbol{Q}\nu} = c_\nu Q,$$

with one longitudinal branch, $\hat{\boldsymbol{e}} = \hat{\boldsymbol{Q}}$, and two transverse ones. At the defect, placed at the origin, the strain $u_{\alpha\beta} = \frac{1}{2}(\partial_\alpha u_\beta + \partial_\beta u_\alpha)$ of one mode is ${\rm i}Q\,\frac{1}{2}(\hat{Q}_\alpha\hat{e}_\beta + \hat{Q}_\beta\hat{e}_\alpha)(2\rho_T\mathcal{V}\omega_{\boldsymbol{Q}\nu})^{-1/2}(d_{\boldsymbol{Q}\nu} - d_{\boldsymbol{Q}\nu}^\dagger)$. The phase i is removed by redefining $d_{\boldsymbol{Q}\nu} \to -{\rm i}d_{\boldsymbol{Q}\nu}$, and the coupling takes the form $\frac{1}{2}\sigma_z\lambda_{\boldsymbol{Q}\nu}(d_{\boldsymbol{Q}\nu} + d_{\boldsymbol{Q}\nu}^\dagger)$ with

$$\lambda_{\boldsymbol{Q}\nu} = \frac{2Q\,\gamma_{\boldsymbol{Q}\nu}}{(2\rho_T\mathcal{V}\omega_{\boldsymbol{Q}\nu})^{1/2}}, \qquad \gamma_{\boldsymbol{Q}\nu} = \sum_{\alpha\beta}\gamma_{\alpha\beta}\hat{Q}_\alpha\hat{e}_\beta,$$

where the symmetry of $\gamma_{\alpha\beta}$ has been used. The squares $\gamma^2_{\boldsymbol{Q}\nu}$ depend on the direction of $\boldsymbol{Q}$ relative to the defect. Averaged over the orientations of the defects of a glass, or equivalently

over $\hat{\boldsymbol{Q}}$ for one defect, they define $\gamma_\mathrm{l}^2 = \langle(\hat{\boldsymbol{Q}}\cdot\gamma\cdot\hat{\boldsymbol{Q}})^2\rangle$ and, for each transverse polarization, $\gamma_\mathrm{t}^2 = \langle(\hat{\boldsymbol{Q}}\cdot\gamma\cdot\hat{\boldsymbol{e}}_\mathrm{t})^2\rangle$. Then $\lambda_{\boldsymbol{Q}\nu}^2 = 2\gamma_\nu^2\omega/(\rho_T\mathcal{V}c_\nu^2)$ at $\omega = c_\nu Q$, and with $\sum_{\boldsymbol{Q}} \to \mathcal{V}\int \mathrm{d}^3Q/(2\pi)^3$,

$$J_\mathrm{s}^\mathrm{ph}(\omega) = \pi\sum_\nu \int \frac{\mathcal{V}\,4\pi Q^2\mathrm{d}Q}{(2\pi)^3}\,\frac{2\gamma_\nu^2\,c_\nu Q}{\rho_T\mathcal{V}c_\nu^2}\,\delta(\omega - c_\nu Q) = \sum_\nu \frac{\gamma_\nu^2\,\omega^3}{\pi\rho_T c_\nu^5},$$

which with one longitudinal and two transverse branches is Eq. (16.69). In Eq. (16.68) it gives Eq. (16.53). The form of Sahling et al., $[A\Delta_\mathrm{tl}\Delta_0^2\coth(\Delta_\mathrm{tl}/2T)]$ with $A = (8\pi^3/\rho h^4)(\gamma_\mathrm{l}^2/v_\mathrm{l}^5 + 2\gamma_\mathrm{t}^2/v_\mathrm{t}^5)$ [83, Eqs. (4), (5)], is the same with $h = 2\pi$ and $k_\mathrm{B} = 1$. The long-wavelength form needs Δ_tl/c_ν small compared with the inverse size of the defect, which at 0.1 meV and 4 km s^{-1} is a wavelength of 165 nm against a defect of a few ångströms.

Conduction electrons in the Born limit

Let the defect act on the conduction electrons as a contact potential that differs between its configurations by $2V_\mathrm{d}$, $\pm V_\mathrm{d}$ over a volume Ω around the defect, as in Appendix E.10. With field operators ψ_σ the coupling is $\sigma_z V_\mathrm{d}\Omega\sum_\sigma \psi_\sigma^\dagger(0)\psi_\sigma(0)$, so the bath coordinate is $X = 2V_\mathrm{d}\Omega\sum_\sigma \psi_\sigma^\dagger(0)\psi_\sigma(0)$, measured from its mean. For a bath of oscillators, Eq. (C.170) gives at $T = 0$ the spectral function $\int \mathrm{d}t\,\mathrm{e}^{\mathrm{i}\omega t}\langle X(t)X(0)\rangle = 2J_\mathrm{s}(\omega)$ for $\omega > 0$, since each mode contributes $2\pi\lambda_k^2\delta(\omega - \omega_k)$. For the electrons the same spectral function, to second order in V_d, is a sum over electron–hole pairs. With $\psi_\sigma(0) = \mathcal{V}^{-1/2}\sum_{\boldsymbol{k}} c_{\boldsymbol{k}\sigma}$, a pair with the electron at ε' above the Fermi level and the hole at ε below it has the matrix element $2V_\mathrm{d}\Omega/\mathcal{V}$, and the sums over states become $\mathcal{V}\mathcal{N}_\mathrm{F}\mathrm{d}\varepsilon$ each:

$$2J_\mathrm{s}^\mathrm{el}(\omega) = 2\pi\sum_\sigma (2V_\mathrm{d}\Omega)^2\mathcal{N}_\mathrm{F}^2\int_{-\infty}^0 \mathrm{d}\varepsilon\int_0^\infty \mathrm{d}\varepsilon'\,\delta(\omega - \varepsilon' + \varepsilon) = 16\pi(\mathcal{N}_\mathrm{F}V_\mathrm{d}\Omega)^2\,\omega.$$

Comparing with $J_\mathrm{s} = 2\pi\alpha_\mathrm{ohm}\omega$ of Eq. (C.171) gives $\alpha_\mathrm{ohm} = 4(\mathcal{N}_\mathrm{F}V_\mathrm{d}\Omega)^2 = 4K_\mathrm{b}$, Eq. (16.71), and Eq. (16.68) gives Eq. (16.70). The factor four is two from the spin sum and two from the difference $2V_\mathrm{d}$ between the configurations. The density of states is per spin: written with the density of states of both spins, $n_0 = 2\mathcal{N}_\mathrm{F}$, and $K = (n_0V_\mathrm{d}\Omega)^2$, the same result reads $\alpha_\mathrm{ohm} = K$, and taking $\alpha_\mathrm{ohm} = 4K$ with that K overstates the coupling fourfold. The rate of Maillet et al. [84, Eq. (4)], $4\pi K\Delta_0^2\coth(\Delta_\mathrm{tl}/2T)/\Delta_\mathrm{tl}$, is Eq. (16.70) with $K = K_\mathrm{b}$, and their derivation uses the single-spin density of states.

The pair excitations of a Fermi sea are not oscillators, and the Gaussian form fails at higher order in V_d. The exact long-time decay of the overlap of the two Fermi seas, the orthogonality catastrophe, matches the ohmic form with the coupling Eq. (E.73), $\alpha_\mathrm{ohm} = (1/2\pi^2)\sum_\sigma\sum_\nu \tilde{\delta}_\nu^2$, with $\tilde{\delta}_\nu$ the differences of the scattering phases at the Fermi level (Appendix E.10). For a contact potential V the scattering in the s channel is the geometric series $t = V\Omega + V\Omega\,G(0)\,V\Omega + \cdots = V\Omega/(1 - V\Omega\,G(0))$, with $G(0) = -\mathrm{i}\pi\mathcal{N}_\mathrm{F}$ the imaginary part of the local Green's function at the Fermi level; the real part, which depends on the band away from the Fermi level, renormalizes V and is absorbed into it. Unitarity, $S = 1 - 2\pi\mathrm{i}\mathcal{N}_\mathrm{F}t = \mathrm{e}^{2\mathrm{i}\delta}$, then gives

$$\mathrm{e}^{2\mathrm{i}\delta} = \frac{1 - \mathrm{i}\pi\mathcal{N}_\mathrm{F}V\Omega}{1 + \mathrm{i}\pi\mathcal{N}_\mathrm{F}V\Omega}, \qquad \tan\delta = -\pi\mathcal{N}_\mathrm{F}V\Omega.$$

The two configurations, $V = \pm V_\mathrm{d}$, have phases $\mp\arctan(\pi\mathcal{N}_\mathrm{F}V_\mathrm{d}\Omega)$, so $\tilde{\delta} = 2\arctan(\pi K_\mathrm{b}^{1/2})$ for each spin, and

$$\alpha_\mathrm{ohm} = \frac{4}{\pi^2}\arctan^2\!\big(\pi K_\mathrm{b}^{1/2}\big) = 4K_\mathrm{b}\Big(1 - \frac{2\pi^2}{3}K_\mathrm{b} + O(K_\mathrm{b}^2)\Big). \tag{D.234}$$

The Born limit is the first term. The coupling saturates at $\alpha_\mathrm{ohm} = 1$ when $\tilde{\delta} = \pi$, the largest a single channel per spin can give; larger values need several channels, as for a hopping scatterer, whose displacement mixes the angular momenta about any fixed center (Appendix E.22).

The superconducting factor

In a superconductor the contact coupling scatters Bogoliubov quasiparticles. For $\Delta_{\rm tl} < 2\Delta_{\rm BCS}$ it cannot create a pair, and the only processes are the scattering of a quasiparticle of energy E to $E' = E \pm \Delta_{\rm tl}$. The potential is even under time reversal, so the matrix element between $\boldsymbol{k}$ and $\boldsymbol{k}'$ carries the case-I factor $(uu' - vv')^2 = \frac{1}{2}[1 + (\xi\xi' - \Delta^2_{\rm BCS})/EE']$ of Eq. (7.29). Each energy $E > \Delta_{\rm BCS}$ comes from the two values $\xi = \pm(E^2 - \Delta^2_{\rm BCS})^{1/2}$, and summing the factor over the four sign combinations of ξ and ξ' cancels the term in $\xi\xi'$ and leaves $2(1 - \Delta^2_{\rm BCS}/EE')$. Changing variables from ξ to E brings the density-of-states factor $\rho_{\rm s}(E) = E/(E^2 - \Delta^2_{\rm BCS})^{1/2}$ for each quasiparticle. In the normal state the same golden rule, with constant matrix elements, gives $\int_{-\infty}^{\infty} {\rm d}\xi\,[f(\xi)(1 - f(\xi + \Delta_{\rm tl})) + f(\xi + \Delta_{\rm tl})(1 - f(\xi))] = \Delta_{\rm tl} \coth(\Delta_{\rm tl}/2T)$ for emission plus absorption, with f the Fermi function. The ratio of the two rates is therefore

$$\frac{\Gamma_1^{\rm s}}{\Gamma_1^{\rm el}} = \frac{2}{\Delta_{\rm tl} \coth(\Delta_{\rm tl}/2T)} \int_{\Delta_{\rm BCS}}^{\infty} {\rm d}E\, \frac{EE' - \Delta^2_{\rm BCS}}{(E^2 - \Delta^2_{\rm BCS})^{1/2}(E'^2 - \Delta^2_{\rm BCS})^{1/2}} \times \big[f(E)(1 - f(E')) + f(E')(1 - f(E))\big], \tag{D.235}$$

with $E' = E + \Delta_{\rm tl}$. The singularity at $E = \Delta_{\rm BCS}$ is integrable and is removed by $E = \Delta_{\rm BCS} + x^2$. Figs. 16.22 and 16.23 use this integral with the weak-coupling gap $\Delta_{\rm BCS}(T)$.

Two limits are closed. As $\Delta_{\rm tl} \to 0$ the coherence fraction tends to one, the occupation bracket to $2f(E)(1 - f(E)) = -2T\,\partial f/\partial E$, and $\Delta_{\rm tl} \coth(\Delta_{\rm tl}/2T) \to 2T$, so that the ratio becomes $2\int_{\Delta_{\rm BCS}}^{\infty} {\rm d}E\,(-\partial f/\partial E) = 2f(\Delta_{\rm BCS}) = 2/(1 + {\rm e}^{\Delta_{\rm BCS}/T})$, the factor of Meißner et al. [82, Eq. (9)]. For T and $\Delta_{\rm tl}$ both small compared with $\Delta_{\rm BCS}$, but in any ratio to each other, write $E = \Delta_{\rm BCS} + y$ with $y \sim T$. To leading order in $y/\Delta_{\rm BCS}$ and $\Delta_{\rm tl}/\Delta_{\rm BCS}$ the coherence fraction is $(2y + \Delta_{\rm tl})/2[y(y + \Delta_{\rm tl})]^{1/2}$, and the occupations are Boltzmann factors, so the bracket is ${\rm e}^{-(\Delta_{\rm BCS}+y)/T}(1 + {\rm e}^{-\Delta_{\rm tl}/T})$. With $a = \Delta_{\rm tl}/2$ and $y = a(\cosh\vartheta - 1)$, so that $y(y + 2a) = a^2 \sinh^2\vartheta$ and $y + a = a\cosh\vartheta$,

$$\int_0^{\infty} {\rm d}y\, {\rm e}^{-y/T}\, \frac{y + a}{[y(y + 2a)]^{1/2}} = a\, {\rm e}^{a/T} \int_0^{\infty} {\rm d}\vartheta\, {\rm e}^{-(a/T)\cosh\vartheta} \cosh\vartheta = a\, {\rm e}^{a/T} K_1(a/T),$$

the integral representation of the modified Bessel function K_1. With $u = \Delta_{\rm tl}/2T$, $1 + {\rm e}^{-2u} = 2{\rm e}^{-u}\cosh u$ and $\coth u = \cosh u/\sinh u$, Eq. (D.235) becomes

$$\frac{\Gamma_1^{\rm s}}{\Gamma_1^{\rm el}} = 2{\rm e}^{-\Delta_{\rm BCS}/T}\, \sinh u\, K_1(u),$$

which is Eq. (16.72) to the same order, since $2/(1 + {\rm e}^{\Delta_{\rm BCS}/T}) = 2{\rm e}^{-\Delta_{\rm BCS}/T}$ there. From $K_1(u) = 1/u + (u/2)[\ln(u/2) + \gamma_{\rm E}] - u/4 + O(u^3 \ln u)$ the Bessel factor is $1 + (u^2/2)[\ln(u/2) + \gamma_{\rm E} - \frac{1}{6}] + \ldots$, the leading correction to the published factor; for $u \gg 1$ it falls as $(\pi/8u)^{1/2}$. At $T = 0.02\,\Delta_{\rm BCS}$ and $\Delta_{\rm tl} = 5T$, where $\Delta_{\rm tl}/\Delta_{\rm BCS}$ is no longer small, the integral Eq. (D.235) departs from the closed form by 2 %.

D.43 The cascade generating functional

The backward equation Eq. (18.4) gives every statistic of the deposits of a hot-carrier cascade from the kernels of its collisions. We derive it by first-step analysis, show that the minimal solution is the physical one, and write the recursions for the first and second moments, with covariances between deposits. Under the energy-only reduction of Sect. 18.1.4 the recursions become one-dimensional convolutions. The energy identity Eq. (18.5) then turns the first two moments of the pair number into the mean energy per pair and the Fano factor, and splits the Fano factor into the three variance terms of Sect. 18.2. The notation is that of Sect. 18.1.3; Appendix C.11 gives the discrete branching process on which the argument is modeled.

D.43.1 The process

A particle in a state x that is not terminal waits a time drawn from $\Gamma_x \mathrm{e}^{-\Gamma_x t}$, with $\Gamma_x = \sum_c \int \mathscr{K}_c$, and then collides. The collision is in channel c and produces the particles $x'_1, \dots, x'_m$ with probability density $\mathscr{K}_c(x \to x'_1 \dots x'_m)/\Gamma_x$, and it adds the deposit $\delta \boldsymbol{N}_c$. A particle in a terminal state adds $\boldsymbol{N}_{\mathrm{term}}(x)$ and does nothing more. Two hypotheses of Sect. 18.1.3 enter. The process is Markov: the outcome of a collision depends only on the state of the colliding particle. And different particles evolve independently: the kernels of one particle do not depend on the others. The total deposit $\boldsymbol{N}$ of the cascade started by x is the sum of the deposits of all its collisions and terminal states. Its entries are nonnegative: pair numbers, heat, numbers of phonons.

D.43.2 First-step analysis

Condition on the first collision of the particle at x, of channel c with products $x'_1, \dots, x'_m$. By the Markov property, each product then starts a cascade whose law is that of a cascade started at x'_j, whatever happened before. By independence, these m cascades are independent of each other. Hence

$$\boldsymbol{N} = \delta \boldsymbol{N}_c + \sum_{j=1}^{m} \boldsymbol{N}^{(j)},$$

with $\boldsymbol{N}^{(j)}$ independent and distributed as the deposit of a cascade started at x'_j. For a vector $\boldsymbol{\lambda}$ with real, nonpositive entries, $0 < \mathrm{e}^{\boldsymbol{\lambda}\cdot\boldsymbol{N}} \le 1$, so every average below exists. The conditional average of $\mathrm{e}^{\boldsymbol{\lambda}\cdot\boldsymbol{N}}$ factorizes,

$$\left\langle \mathrm{e}^{\boldsymbol{\lambda}\cdot\boldsymbol{N}} \,\middle|\, c;\, x'_1 \dots x'_m \right\rangle = \mathrm{e}^{\boldsymbol{\lambda}\cdot\delta\boldsymbol{N}_c} \prod_{j=1}^{m} \mathfrak{G}_{x'_j}(\boldsymbol{\lambda}),$$

and averaging over the first collision with its density $\mathscr{K}_c/\Gamma_x$ gives

$$\mathfrak{G}_x(\boldsymbol{\lambda}) = \sum_c \int \mathrm{d}x'_1 \cdots \mathrm{d}x'_m \, \frac{\mathscr{K}_c(x \to x'_1 \dots x'_m)}{\Gamma_x} \, \mathrm{e}^{\boldsymbol{\lambda}\cdot\delta\boldsymbol{N}_c} \prod_{j=1}^{m} \mathfrak{G}_{x'_j}(\boldsymbol{\lambda}). \tag{D.236}$$

Multiplied by Γ_x this is Eq. (18.4). For an integer deposit and $s = \mathrm{e}^{\lambda}$ it is Eq. (C.102), with p_c the integrated branching ratio and the average over products written as an integral. The waiting time has dropped out, since $\boldsymbol{N}$ does not depend on when the collisions happen. If only deposits made before a time t are counted, the same conditioning, now on whether the first collision comes before t, gives the backward Kolmogorov equation

$$\partial_t \mathfrak{G}_x(\boldsymbol{\lambda}, t) = -\Gamma_x \mathfrak{G}_x(\boldsymbol{\lambda}, t) + \sum_c \int \mathrm{d}x'_1 \cdots \mathrm{d}x'_m \, \mathscr{K}_c \, \mathrm{e}^{\boldsymbol{\lambda}\cdot\delta\boldsymbol{N}_c} \prod_{j} \mathfrak{G}_{x'_j}(\boldsymbol{\lambda}, t),$$

with the initial value $\mathfrak{G}_x(\boldsymbol{\lambda}, 0) = 1$, since nothing has been deposited at $t = 0$. Its stationary limit is Eq. (D.236). The time-resolved form is the one the kinetics of Sect. 18.5 needs; for the final pair number the stationary form suffices.

D.43.3 The minimal solution

Write Eq. (D.236) as $\mathfrak{G} = \mathcal{T}[\mathfrak{G}]$ on the nonterminal states, with $\mathfrak{G}_x = \mathrm{e}^{\boldsymbol{\lambda}\cdot\boldsymbol{N}_{\mathrm{term}}(x)}$ on the terminal ones. For real $\boldsymbol{\lambda} \le 0$ the map $\mathcal{T}$ takes functions with values in $[0, 1]$ to functions with values in $[0, 1]$, and it is monotone: if $\mathfrak{G} \le \mathfrak{G}'$ at every state, then $\mathcal{T}[\mathfrak{G}] \le \mathcal{T}[\mathfrak{G}']$, since

the right side of Eq. (D.236) is a sum of products of nonnegative factors. Start the iteration from $\mathfrak{G}_x^{(0)} = 0$ on the nonterminal states and set $\mathfrak{G}^{(n+1)} = \mathcal{T}[\mathfrak{G}^{(n)}]$. By induction on n,

$$\mathfrak{G}_x^{(n)}(\boldsymbol{\lambda}) = \langle \mathrm{e}^{\boldsymbol{\lambda}\cdot\boldsymbol{N}} \text{; every line of the tree from } x \text{ ends within } n \text{ collisions} \,|\, x \rangle,$$

with the average restricted to the trees that satisfy the condition. It holds at $n = 0$, where only a terminal start contributes. If it holds at n, the tree from x ends within $n + 1$ collisions on every line exactly when the subtree of each product of the first collision ends within n, and the first-step factorization gives it at $n + 1$. The sequence increases with n and converges to the average over all finite trees. If $\mathfrak{G}^*$ is any solution with values in $[0, 1]$, then $\mathfrak{G}^{(0)} \le \mathfrak{G}^*$, and monotonicity gives $\mathfrak{G}^{(n)} \le \mathcal{T}^n[\mathfrak{G}^*] = \mathfrak{G}^*$ for all n. The limit is therefore the smallest solution in $[0, 1]$.

The physical functional is the average over all trees. It equals the minimal solution exactly when the trees are finite with probability one, the termination hypothesis. Otherwise the minimal solution misses the weight of the infinite trees, as in the toy law $1 - p + ps^2$ of Appendix C.11 with $p > 1/2$, whose iteration converges to $(1 - p)/p$ at $s = 1$. A carrier cascade terminates for a simple reason. Order the states by the kinetic energy of the particle. An ionization at x leaves products with total kinetic energy $x - E_\mathrm{g}$, and an emission leaves $x - \omega_{\boldsymbol{Q}\nu}$. If every collision with a nonzero rate removes at least $\delta_\mathrm{min} > 0$ from the total kinetic energy of the nonterminal particles, the number of collisions in a cascade of energy ω is at most $\omega/\delta_\mathrm{min}$, and every tree is finite. Acoustic emission with $\omega_{\boldsymbol{Q}\nu} \to 0$ violates the uniform bound; finiteness then holds with probability one as long as the mean rate of energy loss of every nonterminal state is positive, which is true for a hot carrier at temperatures where phonon absorption is negligible. In the energy ordering, the right side of Eq. (D.236) refers only to states of lower energy than x, and the equation is an explicit recursion. Solved upward in energy from the terminal states, it constructs the minimal solution directly, and no other solution can appear.

D.43.4 Moment recursions

The deposits of a cascade of finite energy are bounded, so $\mathfrak{G}_x$ is smooth at $\boldsymbol{\lambda} = 0$ and its derivatives there are the moments. Write $\boldsymbol{m}_x = \langle \boldsymbol{N} \,|\, x \rangle$ for the mean vector, $\mathbf{M}_x = \langle \boldsymbol{N}\boldsymbol{N}^\mathrm{T} \,|\, x \rangle$ for the matrix of second moments, and abbreviate $\boldsymbol{m}_j = \boldsymbol{m}_{x'_j}$, $\mathbf{M}_j = \mathbf{M}_{x'_j}$ and $\langle \cdots \rangle_c = \int \mathrm{d}x'_1 \cdots \mathrm{d}x'_m \, \mathscr{K}_c(\cdots)$. One derivative of the product $\mathrm{e}^{\boldsymbol{\lambda}\cdot\delta\boldsymbol{N}_c} \prod_j \mathfrak{G}_{x'_j}$ at $\boldsymbol{\lambda} = 0$, where every factor is 1, gives $\delta\boldsymbol{N}_c + \sum_j \boldsymbol{m}_j$. Two derivatives give the second derivative of each factor plus the products of first derivatives of two different factors. Hence

$$\Gamma_x \, \boldsymbol{m}_x = \sum_c \Big\langle \delta\boldsymbol{N}_c + \sum_j \boldsymbol{m}_j \Big\rangle_c , \tag{D.237}$$

$$\Gamma_x \, \mathbf{M}_x = \sum_c \Big\langle \delta\boldsymbol{N}_c \delta\boldsymbol{N}_c^\mathrm{T} + \sum_j (\delta\boldsymbol{N}_c \boldsymbol{m}_j^\mathrm{T} + \boldsymbol{m}_j \delta\boldsymbol{N}_c^\mathrm{T}) + \sum_j \mathbf{M}_j + \sum_{i \ne j} \boldsymbol{m}_i \boldsymbol{m}_j^\mathrm{T} \Big\rangle_c , \tag{D.238}$$

the vector form of Eqs. (C.103) and (C.104), with covariances between different deposits in the off-diagonal entries.

The covariance matrix $\mathbf{C}_x = \mathbf{M}_x - \boldsymbol{m}_x \boldsymbol{m}_x^\mathrm{T}$ obeys a form without cancellations. Let $\boldsymbol{Z}_c = \delta\boldsymbol{N}_c + \sum_j \boldsymbol{m}_j$ be the mean deposit given the first collision. Expand $\langle (\boldsymbol{Z}_c - \boldsymbol{m}_x)(\boldsymbol{Z}_c - \boldsymbol{m}_x)^\mathrm{T} + \sum_j \mathbf{C}_j \rangle_c$, summed over c and divided by Γ_x. The terms linear in $\boldsymbol{m}_x$ average to $-2\boldsymbol{m}_x \boldsymbol{m}_x^\mathrm{T}$ by Eq. (D.237), the constant term is $\boldsymbol{m}_x \boldsymbol{m}_x^\mathrm{T}$, and $\boldsymbol{Z}_c \boldsymbol{Z}_c^\mathrm{T} + \sum_j (\mathbf{M}_j - \boldsymbol{m}_j \boldsymbol{m}_j^\mathrm{T})$ reproduces the bracket of Eq. (D.238), since $\sum_{i,j} \boldsymbol{m}_i \boldsymbol{m}_j^\mathrm{T} - \sum_j \boldsymbol{m}_j \boldsymbol{m}_j^\mathrm{T} = \sum_{i \ne j} \boldsymbol{m}_i \boldsymbol{m}_j^\mathrm{T}$. So

$$\Gamma_x \, \mathbf{C}_x = \sum_c \Big\langle \sum_j \mathbf{C}_j + (\boldsymbol{Z}_c - \boldsymbol{m}_x)(\boldsymbol{Z}_c - \boldsymbol{m}_x)^\mathrm{T} \Big\rangle_c , \tag{D.239}$$

the law of total covariance: the covariance of the cascade is the average covariance of the subtrees plus the covariance of the first step's conditional mean. For one deposit both terms are nonnegative, which keeps the recursion stable on a grid; Appendix E.24 uses the same split for its estimator. For a seed of independent particles, such as the pair of Eq. (18.9), the means and covariances of its particles add, plus the deposit of the seed itself.

D.43.5 The energy-only recursions

Under the energy-only reduction the state is the type t and the kinetic energy x, and there are two channels. Impact ionization, at the iso-energy rate $\Gamma_{\mathrm{ii},t}(x)$, leads to (t, x_1), (t, x_2), $(\bar{t}, x_3)$ on the plane $x_1 + x_2 + x_3 = x - E_{\mathrm{g}}$. Emission of a phonon of energy ω_{AB}, at the rate $\Gamma_{\mathrm{ph},t}(x)$, leads to $(t, x - \omega_{\mathrm{AB}})$. Let $p_t(x) = \Gamma_{\mathrm{ii},t}/(\Gamma_{\mathrm{ii},t} + \Gamma_{\mathrm{ph},t})$ be the probability that the next collision ionizes. For one entry Y of $\boldsymbol{N}$, with deposits d_{ii} per ionization, d_{ph} per emission and $d_<(x)$ at the terminal state, write $m_t(x)$ and $M_t(x)$ for its first and second moments. In the random-k form the products have the density $\rho_t(x_1)\rho_t(x_2)\rho_{\bar{t}}(x_3)/\mathfrak{P}_t(y)$ of Eq. (18.7), with $y = x - E_{\mathrm{g}}$. With $P_t = \rho_t m_t$, $S_t = \rho_t M_t$ and $*$ the convolution on $[0, \infty)$, the averages over the products are convolutions:

$$\Big\langle \sum_j m_j \Big\rangle = \frac{[2P_t * \rho_t * \rho_{\bar{t}} + \rho_t * \rho_t * P_{\bar{t}}](y)}{\mathfrak{P}_t(y)}, \qquad \Big\langle \sum_j M_j \Big\rangle = \frac{[2S_t * \rho_t * \rho_{\bar{t}} + \rho_t * \rho_t * S_{\bar{t}}](y)}{\mathfrak{P}_t(y)},$$

$$\Big\langle \sum_{i<j} m_i m_j \Big\rangle = \frac{[P_t * P_t * \rho_{\bar{t}} + 2P_t * \rho_t * P_{\bar{t}}](y)}{\mathfrak{P}_t(y)}.$$

The pair sum has one term for the two products of type t and two for the mixed pairs. The scalar forms of Eqs. (D.237) and (D.238) then read, for $x \geq \theta_{\mathrm{ii}} E_{\mathrm{g}}$,

$$m_t(x) = p_t\Big[d_{\mathrm{ii}} + \Big\langle \sum_j m_j \Big\rangle\Big] + (1 - p_t)\,[d_{\mathrm{ph}} + m_t(x - \omega_{\mathrm{AB}})], \tag{D.240}$$

$$M_t(x) = p_t\Big[d_{\mathrm{ii}}^2 + 2d_{\mathrm{ii}}\Big\langle \sum_j m_j \Big\rangle + \Big\langle \sum_j M_j \Big\rangle + 2\Big\langle \sum_{i<j} m_i m_j \Big\rangle\Big] + (1 - p_t)\,[d_{\mathrm{ph}}^2 + 2d_{\mathrm{ph}} m_t(x - \omega_{\mathrm{AB}}) + M_t(x - \omega_{\mathrm{AB}})], \tag{D.241}$$

and $m_t = d_<$, $M_t = d_<^2$ below threshold. Every term on the right is known at energies below x, so one sweep upward on a grid of K points solves both at a cost of order K^2. The grid must contain ω_{AB}, E_{g} and $\theta_{\mathrm{ii}} E_{\mathrm{g}}$ as whole numbers of steps, since the recursion reads m_t at $x - \omega_{\mathrm{AB}}$ and starts ionizing at $\theta_{\mathrm{ii}} E_{\mathrm{g}}$. When ω_{AB} is not a whole number of steps, the recursions are solved at the three commensurate phonon energies nearest it and interpolated to its value. With phonons the discretization error falls as $h^{3/2}$ in the step h, so values at the steps h and $h/2$ extrapolate to zero step as $f(h/2) + [f(h/2) - f(h)]/(2^{3/2} - 1)$; the silicon numbers of Sects. 18.2 and 18.4 are extrapolated from $h = E_{\mathrm{g}}/74$ and $E_{\mathrm{g}}/148$ (or $E_{\mathrm{g}}/73$ and $E_{\mathrm{g}}/146$). Covariances between two entries follow from the second moment of their sum, or of their difference, which gives a second, independent route.

D.43.6 Pair energy and Fano factor

Take the deposit vector $\boldsymbol{N} = (N_{\mathrm{eh}}, E_<, E_>)$, with $E_<$ the kinetic energy each carrier holds when it falls below threshold for good, summed over carriers, and $E_>$ the phonon energy emitted above threshold. The deposits are $d_{\mathrm{ii}} = (1, 0, 0)$, $d_{\mathrm{ph}} = (0, 0, \omega_{\mathrm{AB}})$ and $d_<(x) = (0, x, 0)$, and the heat of Eq. (18.5) is $E_{\mathrm{heat}} = E_< + E_>$. For a cascade of total energy ω, with the seed pair

counted in N_{eh}, the identity $\omega = N_{\mathrm{eh}}E_{\mathrm{g}} + E_< + E_>$ holds on every path. Its mean gives the mean pair number $\bar{n} = \langle N_{\mathrm{eh}} \rangle$ and with it

$$\varepsilon_{\mathrm{eh}} = \frac{\omega}{\bar{n}}, \qquad F_{\mathrm{eh}} = \frac{\mathrm{Var}\, N_{\mathrm{eh}}}{\bar{n}} = \frac{\mathrm{Var}(E_< + E_>)}{E_{\mathrm{g}}^2\, \bar{n}}, \tag{D.242}$$

since $N_{\mathrm{eh}} = (\omega - E_< - E_>)/E_{\mathrm{g}}$ and ω is fixed. Expanding the variance of the sum,

$$F_{\mathrm{eh}} = \mathcal{V}_< + \mathcal{V}_> + 2\mathcal{V}_{<>}, \qquad \mathcal{V}_< = \frac{\mathrm{Var}\, E_<}{E_{\mathrm{g}}^2\, \bar{n}}, \quad \mathcal{V}_> = \frac{\mathrm{Var}\, E_>}{E_{\mathrm{g}}^2\, \bar{n}}, \quad \mathcal{V}_{<>} = \frac{\mathrm{Cov}(E_<, E_>)}{E_{\mathrm{g}}^2\, \bar{n}}. \tag{D.243}$$

All three terms are entries of $\mathbf{C}$ in Eq. (D.239), computed with the deposits above, and Sect. 18.2 evaluates them for silicon. The identity provides two checks on any solution. The means must satisfy $E_{\mathrm{g}}\bar{n} + \langle E_< \rangle + \langle E_> \rangle = \omega$, and the variance of N_{eh} computed with $d_{\mathrm{ii}} = 1$ alone must equal $\mathrm{Var}(E_< + E_>)/E_{\mathrm{g}}^2$ computed from the heat deposits.

D.44 Renewal theory of the mean yield

The mean pair number of a cascade grows in proportion to the deposited energy, $\bar{n}(\omega) = \omega/\varepsilon_{\mathrm{eh}} + o(\omega)$ (Eq. (18.15)). For a single chain of independent losses this is the key renewal theorem, Eq. (C.108). A cascade splits its energy at every ionization and has no common loss law, and we prove the leading term by following one unit of energy through the tree. The argument needs energy conservation and the Markov property of Sect. 18.1.3; it does not need the independence of siblings.

D.44.1 The tagged-unit identity

Start a cascade with one carrier of kinetic energy E and let $N_{\mathrm{sub}}(E)$ be the number of ionizations in it. Represent the energy of the first carrier by the interval $[0, E]$. At every collision, divide the interval of the colliding carrier, of length x, into consecutive pieces. An ionization makes pieces of lengths x_1, x_2, x_3 for the three products and one of length E_{g} for the new pair. A phonon emission makes a piece of length $x - \omega_{\mathrm{AB}}$ for the carrier and one of length ω_{AB} for the phonon. Energy conservation makes the pieces fill the interval exactly. A terminal carrier keeps its piece. Every point $u \in [0, E]$ then has a line: the sequence of carriers whose pieces contain u, ending when u falls into a pair or phonon piece or its carrier becomes terminal. Let $x_1 = E, x_2, \ldots$ be the kinetic energies of the carriers on the line of u at its successive collisions, ionizations and phonon emissions alike; only the ionizations carry weight below.

Each ionization of a carrier with kinetic energy x lies on the lines of exactly the x units of its piece. Giving each of them the weight $1/x$ therefore counts the ionization once:

$$N_{\mathrm{sub}}(E) = \int_0^E \mathrm{d}u\, S(u), \qquad S(u) = \sum_{j \,\in\, \mathrm{ii}(u)} \frac{1}{x_j},$$

with $\mathrm{ii}(u)$ the collisions on the line of u that are ionizations. Taking the mean and exchanging it with the integral,

$$\langle N_{\mathrm{sub}}(E) \rangle = E\, \langle S \rangle_u, \tag{D.244}$$

with u uniform on $[0, E]$ and the mean taken over u and the cascade. This is Eq. (18.14).

The line of a uniform unit is a Markov chain. Suppose that its carrier has kinetic energy x and that, given everything before, u is uniform on the carrier's piece. This holds for the first carrier. At the next collision the carrier's outcome is drawn from its kernel, independently of

u by the Markov property. The unit u then lies in the piece of product i with probability x_i/x, in the pair piece with probability E_g/x, or in the phonon piece with probability ω_AB/x, and given that it lies in a piece it is uniform on it. The uniformity therefore propagates, and the line moves from x to a product of energy x_i with probability x_i/x times the probability of that outcome, or leaves the chain. Write $\xi = \ln(x_{j+1}/x_j)$ for one step of $\ln x$ and K_x for its law, with an extra state for leaving. Every step has $\xi \le 0$, because a product never carries more energy than its parent.

For a check, let a piece of length $x \ge 1$ split at a uniform point into two pieces, with no pair and no phonon, until every piece is shorter than 1. The mean number of splits $M(x)$ obeys $M(x) = 1 + (2/x)\int_1^x M(y)\,\mathrm{d}y$ for $x \ge 1$, and $M(x) = 2x - 1$ solves it, since $(2/x)\int_1^x (2y-1)\,\mathrm{d}y = 2(x-1)$. A unit lands in a piece of share v with the size-biased density $2v$ on $[0,1]$, so the walk of $\ln x$ has mean step $\int_0^1 2v \ln v\,\mathrm{d}v = -\frac{1}{2}$, and Eq. (D.244) gives $\langle S\rangle_u = M(x)/x = 2 - 1/x$, which tends to 2: one split per half unit of energy.

D.44.2 Hypotheses

Let x_cut be a cut energy and τ the first collision of the line at or below x_cut. Four conditions are needed.

1. *Scale-free limit, approached fast enough.* There is a law K_∞ of ξ, without leaving, such that $\|K_x - K_\infty\|_\mathrm{TV} \le C(E_\mathrm{g}/x)^\varkappa$ for some $C, \varkappa > 0$, with $\|\cdot\|_\mathrm{TV}$ the total-variation distance.
2. *No lattice.* K_∞ is not concentrated on a lattice $\{0, -d, -2d, \dots\}$.
3. *Finite mean.* K_∞ has a finite mean $-\mu_\mathrm{ren}$; by the second condition and $\xi \le 0$, $\mu_\mathrm{ren} > 0$.
4. *Regularity near threshold.* The mean weight collected at or below the cut, $h_\mathrm{c}(z) = \langle \sum_{j \ge \tau,\, j \in \mathrm{ii}} 1/x_j \,|\, x_\tau = z\rangle$, is bounded uniformly in x_cut and continuous in z.

In the random-k cascade of Sect. 18.1.4 the shares of the products tend to a law independent of x: the Dirichlet law with parameters 3/2 without a hole cutoff, and, with the hole density of states cut at B_h, the Beta law with parameters 3/2, 3/2 for the two electrons, since then $\mathfrak{P}_\mathrm{e}(y) \propto y^2$ at $y \gg B_\mathrm{h}$ and the hole's share tends to zero. The probability of leaving into a pair is E_g/x, and the phonon branch falls as x^{-3} without the cutoff and as $x^{-3/2}$ with it; in both cases the leaving probability dominates and the first condition holds with $\varkappa = 1$. Both limit laws are continuous, which gives the second, and the mean of the logarithm of a share is finite for it, which gives the third. The fourth needs a finite mean number of collisions between E_g and x_cut and moments that are continuous in the energy of entry; both hold for the recursions Eqs. (D.240) and (D.241), whose ionization probability vanishes continuously at threshold.

D.44.3 Proof

Split $S = S_> + S_\le$ into the ionizations above and at or below x_cut.

Visits. By the first and third conditions there are $\delta, p > 0$ and x_1 such that $P(\xi \le -\delta) \ge p$ under K_x for every $x \ge x_1$. Since $\ln x_j$ never increases, the collisions the line spends in any interval of $\ln x$ of unit length above $\ln x_1$ form at most $1/\delta + 1$ runs, each of geometrically distributed length with parameter p. Their mean number is therefore at most a constant C_v independent of E. For $x_\mathrm{cut} \ge x_1$, summing over the intervals $[\ln x_\mathrm{cut} + k, \ln x_\mathrm{cut} + k + 1)$, in which each term of $S_>$ is at most $\mathrm{e}^{-k}/x_\mathrm{cut}$,

$$\langle S_>\rangle \le C_\mathrm{v} \sum_{k \ge 0} \frac{\mathrm{e}^{-k}}{x_\mathrm{cut}} = O\Big(\frac{1}{x_\mathrm{cut}}\Big),$$

uniformly in E.

Coupling. Run the true line and a walk with steps drawn from K_∞ from the same start, coupled at each step so that they take the same step with probability $1 - \|K_{x_j} - K_\infty\|_{\mathrm{TV}}$. The K_∞ walk obeys the same visit bound. The probability that the two separate before reaching x_{cut} is at most the mean of $\sum_{x_j > x_{\mathrm{cut}}} \|K_{x_j} - K_\infty\|_{\mathrm{TV}}$, which by the first condition and the visit bound is $O((E_{\mathrm{g}}/x_{\mathrm{cut}})^{\varkappa})$.

Undershoot. Under K_∞, $-\ln x_j$ is a renewal chain with independent, identically distributed, non-lattice increments of finite mean μ_{ren}. The undershoot $\ln x_{\mathrm{cut}} - \ln x_\tau$ is the overshoot of that chain over the level $\ln(E/x_{\mathrm{cut}})$. By the key renewal theorem, Eq. (C.106), its law converges to the limit of Eq. (C.107) as E/x_{cut} grows without bound. By the fourth condition h_{c} is bounded and continuous, so the mean of $h_{\mathrm{c}}(x_\tau)$ under K_∞ converges.

Limit. At τ the line restarts from x_τ, so $\langle S_{\le} \rangle = \langle h_{\mathrm{c}}(x_\tau) \rangle$. For the true line this differs from the K_∞ value by at most $2 \sup h_{\mathrm{c}}$ times the separation probability. Together,

$$\limsup_{E \to \infty} \langle S \rangle_u - \liminf_{E \to \infty} \langle S \rangle_u \le O\Big(\frac{1}{x_{\mathrm{cut}}}\Big) + O\Big(\Big(\frac{E_{\mathrm{g}}}{x_{\mathrm{cut}}}\Big)^{\varkappa}\Big),$$

for every x_{cut}. Letting $x_{\mathrm{cut}} \to \infty$, the limit of $\langle S \rangle_u$ exists. It is finite by the visit bound and positive whenever the cascade ionizes at all; call it $1/\varepsilon_{\mathrm{eh}}$. By Eq. (D.244), $\langle N_{\mathrm{sub}}(E) \rangle = E/\varepsilon_{\mathrm{eh}} + o(E)$, and with the seed pair counted and $E = \omega - E_{\mathrm{g}}$ for a seed electron, $\bar{n}(\omega) = \omega/\varepsilon_{\mathrm{eh}} + o(\omega)$.

D.44.4 Constant term, variance and lattice energies

The proof claims no rate for the $o(\omega)$. The constant term b_N of $\bar{n}$, and the linear growth of $\mathrm{Var}\, N_{\mathrm{eh}}$, need second-order renewal estimates for the walk and a bound on how fast the cascade forgets its seed, which are open (Sect. 18.2.6). The lattice of energy losses in a one-phonon model does not affect the leading term, because the steps of K_∞ come from continuous shares. It enters the constant term through the threshold region, where the losses E_{g} and ω_{AB} take only a few values, and as for the deterministic chain of Appendix C.11 the constant then carries a periodic remainder. A cascade whose shares are fixed proportions violates the second condition. Then $\ln x_j$ moves on a lattice, the undershoot has no limit, and $\langle S \rangle_u$ oscillates in $\ln E$ without converging, so that $\bar{n}/\omega$ has no limit either.

D.45 Energy balance and calibration algebra

This section proves the balance law Eq. (19.27), derives the bias work Eq. (19.28), and carries out the algebra behind Eqs. (19.33) to (19.37) and (19.39).

D.45.1 Proof of the balance law

Take as the system the crystal with its electrons and nuclei, the sensor films, holders and other bodies that absorb phonons, the radiation field and the electrodes, held at fixed potentials V_k by an external source. The Hamiltonian of this system is conserved except for the power the source delivers. A charge δQ_k that enters electrode k from the source receives the work $V_k\, \delta Q_k$, so

$$\langle H_{\mathrm{tot}} \rangle(t) - \langle H_{\mathrm{tot}} \rangle(0^-) = \omega + \sum_k V_k\, \Delta Q_k(t) \equiv \omega + W_{\mathrm{bias}}(t),$$

where ω is the energy the dark-matter particle leaves at $t = 0$ and $\Delta Q_k(t)$ the charge that has flowed into electrode k since then. The proof consists in writing the left side as the sum of the reservoirs.

Between collisions the state of the crystal has a definite content of carriers, excitons, trapped carriers, defects and phonons. Its energy above the ground state is the sum of four terms. The first is the band energy of each carrier, counted from the Fermi level of the electrodes (see below), with $E_{\rm g} - E_x^b$ per exciton and $E_j - \mu_F$ or $\mu_F - E_j$ per trapped carrier. The second is the Coulomb energy of the charges in the crystal. With the electrodes at fixed potential it is

$$E_{\rm es} = \frac{1}{2} \sum_{i,j} q_i q_j \, G(\boldsymbol{r}_i, \boldsymbol{r}_j),$$

with G the Green's function of the Poisson equation in the dielectric that vanishes on the electrodes. The terms $i = j$, made finite by the size of the carrier wave packet, are the image energies, and G includes the screening by the dielectric. The third is the configurational energy of the nuclear arrangement relative to the perfect crystal, $E_{\rm st}$ of Eq. (19.4). The fourth is the vibrational energy above the thermal bath, which includes the zero-point shift $\Delta E_{\rm zp}$ of the occupied basins. The interactions of the excited carriers among themselves enter in two ways. Their long-range part is $E_{\rm es}$ and the binding energy of excitons. Their short-range part acts only while two particles are within a collision distance, and its expectation value between collisions is smaller than the sum of the others by the ratio of the collision time to the time between collisions. Put together, the crystal holds $E_{\rm pairs} + E_{\rm phon} + E_{\rm st}$, with $E_{\rm es}$ inside $E_{\rm pairs}$. The sensors and holders hold the energy absorbed at the surfaces, $E_{\rm sens}$, and the radiation field holds $E_{\rm esc}$. Equating the two expressions for $\langle H_{\rm tot} \rangle$ gives Eq. (19.27).

The argument applies to each realization of the random history, the sequence of captures, hops, emissions and absorptions in a given event. Each step is a transition between states of definite content that conserves the total energy, or a drift whose energy exchange the bias work and $E_{\rm es}$ record. It applies in the mean by averaging. Its physical content is that the list of reservoirs is complete: an energy that is in none of them, such as the electrostatic energy of a trapped charge if $E_{\rm es}$ is omitted, breaks the law.

The energy reference. Count the carrier energies from an arbitrary level μ: a free electron holds $E_c - \mu$, a free hole $\mu - E_v$. Then

$$\frac{\partial E_{\rm pairs}}{\partial \mu} = N_{\rm h} - N_{\rm e},$$

the number of uncollected holes minus that of uncollected electrons, trapped or free. A carrier that enters an electrode thermalizes to the Fermi level μ_F of the metal and gives up, as phonons at the contact, $E_c - \mu_F$ for an electron and $\mu_F - E_v$ for a hole. If $\mu \neq \mu_F$, the carrier removes $E_c - \mu$ from $E_{\rm pairs}$ and adds $E_c - \mu_F$ to the phonons, and the difference has to be found elsewhere. It is zero only while $N_{\rm h} = N_{\rm e}$, that is, until the first carrier whose partner stays behind is collected. The level $\mu = \mu_F$ is the only one for which every step conserves the sum of the reservoirs. With a bias, each electrode has its own Fermi level and the difference between them is $eV_{\rm bias}$, which the bias work accounts for.

Bias work and Luke heat. Let the two electrodes be planes a distance L apart, at potentials 0 and $V_{\rm bias}$. A charge q at height z induces on the electrode at $z = L$ the charge $-qz/L$ (Ramo's theorem, with the weighting potential z/L of this geometry). Moving it from z_1 to z_2 therefore passes the charge $|q|\,|z_2 - z_1|/L$ through the source, and for motion in the direction of the electric force the source does the work $|q|V_{\rm bias}|z_2 - z_1|/L$. A carrier that crosses the fraction s of the gap in the direction of the force contributes $eV_{\rm bias}s$, and summing over carriers gives Eq. (19.28) with $N_{\rm ind} = \sum_i s_i$. For each pair that is fully collected, the electron and the hole together cross the whole gap, whatever the point of creation, and contribute

$eV_{\rm bias}$. In a quasistatic move the work of the source equals the change of $E_{\rm es}$ plus the work of the field on the carrier. At constant drift velocity the carrier's kinetic energy does not change, and the work of the field goes to phonons. The Luke heat is therefore $W_{\rm bias} - \Delta E_{\rm es}$. For pairs created neutral and fully collected, $\Delta E_{\rm es} = 0$ at the end and the Luke heat is $eV_{\rm bias}N_{\rm eh}$, Eq. (1.44).

D.45.2 The phonon-yield ratio

With the signals Eq. (19.32), the electron-recoil calibration multiplies every phonon signal by the factor $k_{\rm cal}$ for which an electron recoil reads $\omega(1+g_{\rm cal})$, with $g_{\rm cal} = eV_{\rm bias}/\varepsilon_{\rm cal}$ the gain of the reconstruction and $g_{\rm Luke} = eV_{\rm bias}/\varepsilon_{\rm eh}$ the true Luke gain:

$$k_{\rm cal}\, S_{\rm ER} = \omega(1+g_{\rm cal}) \quad\Longrightarrow\quad k_{\rm cal} = \frac{1+g_{\rm cal}}{\epsilon^{\rm ER}_{\rm prim}\,(1-d_{\rm ER}+\mathfrak{r}_{\rm coll}g_{\rm Luke})}.$$

The nuclear recoil is assigned $k_{\rm cal}S_{\rm NR} - f_{\rm ion}g_{\rm cal}\omega$, and dividing by ω gives Eq. (19.33). Write $A = (1-\delta_{\rm coll})(1-d_{\rm NR})$ and $B = 1-d_{\rm ER}$. Multiplying Eq. (19.33) by $B + \mathfrak{r}_{\rm coll}g_{\rm Luke}$,

$$(Y_{\rm ph} + f_{\rm ion}g_{\rm cal})(B + \mathfrak{r}_{\rm coll}g_{\rm Luke}) = (1+g_{\rm cal})(A + \mathfrak{r}_{\rm coll}g_{\rm Luke}f_{\rm ion}),$$

and the terms in $\mathfrak{r}_{\rm coll}g_{\rm Luke}g_{\rm cal}f_{\rm ion}$ cancel, leaving

$$(1+g_{\rm cal})A - (Y_{\rm ph} + f_{\rm ion}g_{\rm cal})B = \mathfrak{r}_{\rm coll}g_{\rm Luke}(Y_{\rm ph} - f_{\rm ion}). \tag{D.245}$$

Eq. (D.245) is homogeneous in $(A, B, \mathfrak{r}_{\rm coll})$: the measurement determines only ratios, A/B and $\mathfrak{r}_{\rm coll}/B$. At zero bias it reads $Y_{\rm ph}(0) = A/B$, the heat quenching Eq. (19.29). Dividing by B and solving for $Y_{\rm ph}$ with $\mathfrak{r}' = \mathfrak{r}_{\rm coll}/B$ gives Eq. (19.34); written for the gains of bias i, the same division gives Eq. (19.39). Both gains are proportional to the bias, so their ratio $\varepsilon_{\rm cal}/\varepsilon_{\rm eh}$ is the same at every bias.

First order. At $A = B = 1$, Eq. (19.33) is

$$\begin{aligned} Y_{\rm ph} &= \frac{(1+g_{\rm cal})(1+\mathfrak{r}_{\rm coll}g_{\rm Luke}f_{\rm ion}) - f_{\rm ion}g_{\rm cal}(1+\mathfrak{r}_{\rm coll}g_{\rm Luke})}{1+\mathfrak{r}_{\rm coll}g_{\rm Luke}} \\ &= \frac{1+g_{\rm cal}+f_{\rm ion}(\mathfrak{r}_{\rm coll}g_{\rm Luke} - g_{\rm cal})}{1+\mathfrak{r}_{\rm coll}g_{\rm Luke}}, \end{aligned}$$

and $1 - Y_{\rm ph} = (\mathfrak{r}_{\rm coll}g_{\rm Luke} - g_{\rm cal})(1-f_{\rm ion})/(1+\mathfrak{r}_{\rm coll}g_{\rm Luke})$, the collection term. The derivatives of Eq. (19.33) at that point are

$$\frac{\partial Y_{\rm ph}}{\partial A} = \frac{1+g_{\rm cal}}{1+\mathfrak{r}_{\rm coll}g_{\rm Luke}}, \qquad \frac{\partial Y_{\rm ph}}{\partial B} = -\frac{(1+g_{\rm cal})(1+\mathfrak{r}_{\rm coll}g_{\rm Luke}f_{\rm ion})}{(1+\mathfrak{r}_{\rm coll}g_{\rm Luke})^2},$$

and with $A - 1 = -(\delta_{\rm coll} + d_{\rm NR}) + \delta_{\rm coll}d_{\rm NR}$ and $B - 1 = -d_{\rm ER}$ they give Eq. (19.35) with $C_{\rm cal} = (1+\mathfrak{r}_{\rm coll}g_{\rm Luke}f_{\rm ion})/(1+\mathfrak{r}_{\rm coll}g_{\rm Luke})$. Since $Y_{\rm ph}$ is linear in A, the second-order terms are $\delta_{\rm coll}d_{\rm NR}$ and those in $d_{\rm ER}$.

D.45.3 The bound on the stored energy

Put $d_{\mathrm{NR}} = e_{\mathrm{st}} + x_{\mathrm{NR}}$ with $e_{\mathrm{st}} = E_{\mathrm{st}}/\omega$, and $d_{\mathrm{ER}} = x_{\mathrm{ER}}$. Multiply $1 - Y_{\mathrm{ph}} - e_{\mathrm{st}}$ by the denominator $D_{\mathrm{c}} = 1 - x_{\mathrm{ER}} + \mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}}$ of Eq. (19.33) and abbreviate $a = 1 - e_{\mathrm{st}}$:

$$\begin{aligned}(1 - Y_{\mathrm{ph}} - e_{\mathrm{st}})D_{\mathrm{c}} &= aD_{\mathrm{c}} + f_{\mathrm{ion}} g_{\mathrm{cal}} D_{\mathrm{c}} \\ &\quad - (1 + g_{\mathrm{cal}})\big[(1 - \delta_{\mathrm{coll}})(a - x_{\mathrm{NR}}) + \mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} f_{\mathrm{ion}}\big] \\ &= a(1 + \mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}}) - a(1 + g_{\mathrm{cal}}) + (1 + g_{\mathrm{cal}})\delta_{\mathrm{coll}} a \\ &\quad + (1 + g_{\mathrm{cal}})(1 - \delta_{\mathrm{coll}})x_{\mathrm{NR}} - (a + f_{\mathrm{ion}} g_{\mathrm{cal}})x_{\mathrm{ER}} \\ &\quad + f_{\mathrm{ion}} g_{\mathrm{cal}}(1 + \mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}}) - (1 + g_{\mathrm{cal}})\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} f_{\mathrm{ion}}.\end{aligned}$$

The terms in a free of δ_{coll} and x combine to $a(\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} - g_{\mathrm{cal}})$ and the terms free of a and x to $-f_{\mathrm{ion}}(\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} - g_{\mathrm{cal}})$, so

$$\begin{aligned}(1 - Y_{\mathrm{ph}} - e_{\mathrm{st}})D_{\mathrm{c}} &= (\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} - g_{\mathrm{cal}})(1 - e_{\mathrm{st}} - f_{\mathrm{ion}}) \\ &\quad + (1 + g_{\mathrm{cal}})\big[\delta_{\mathrm{coll}}(1 - e_{\mathrm{st}}) + (1 - \delta_{\mathrm{coll}})x_{\mathrm{NR}}\big] - (1 - e_{\mathrm{st}} + f_{\mathrm{ion}} g_{\mathrm{cal}})x_{\mathrm{ER}},\end{aligned}$$

which is Eq. (19.36). Nothing was expanded. With $D_{\mathrm{c}} > 0$ and $e_{\mathrm{st}} < 1$, the right side is non-negative term by term under conditions (a), (b), (d) and (e) of Sect. 19.5.3, since (b) and (e) give $\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} \geq g_{\mathrm{Luke}} \geq g_{\mathrm{cal}}$, together with (c), which makes the x terms non-negative together. Condition (c) reduces to $x_{\mathrm{NR}} \geq [(1 + f_{\mathrm{ion}} g_{\mathrm{cal}})/(1 + g_{\mathrm{cal}})]\, x_{\mathrm{ER}}$ as $e_{\mathrm{st}}, \delta_{\mathrm{coll}} \to 0$; it holds for every $x_{\mathrm{NR}} \geq 0$ when $x_{\mathrm{ER}} \leq 0$, and it allows deficits of either sign.

On the branch $\mathfrak{r}_{\mathrm{coll}} < 1$, set $\delta_{\mathrm{coll}} = x_{\mathrm{NR}} = x_{\mathrm{ER}} = 0$. The identity becomes $(1 - Y_{\mathrm{ph}} - e_{\mathrm{st}})(1 + \mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}}) = (\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} - g_{\mathrm{cal}})(1 - e_{\mathrm{st}} - f_{\mathrm{ion}})$, linear in e_{st}. Collecting the terms in e_{st} gives $e_{\mathrm{st}}(1 + g_{\mathrm{cal}}) = (1 - Y_{\mathrm{ph}})(1 + \mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}}) - (\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} - g_{\mathrm{cal}})(1 - f_{\mathrm{ion}})$, and writing $1 + \mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} = (1 + g_{\mathrm{cal}}) + (\mathfrak{r}_{\mathrm{coll}} g_{\mathrm{Luke}} - g_{\mathrm{cal}})$ gives Eq. (19.37).

D.45.4 Two biases

Eq. (19.39) at two biases, with gains $g_{\mathrm{Luke},i}$, $g_{\mathrm{cal},i}$ and measured ratios Y_i, is the linear system

$$\begin{pmatrix} 1 + g_{\mathrm{cal},1} & -g_{\mathrm{Luke},1}(Y_1 - f_{\mathrm{ion}}) \\ 1 + g_{\mathrm{cal},2} & -g_{\mathrm{Luke},2}(Y_2 - f_{\mathrm{ion}}) \end{pmatrix} \begin{pmatrix} Y_{\mathrm{ph}}(0) \\ \mathfrak{r}' \end{pmatrix} = \begin{pmatrix} Y_1 + f_{\mathrm{ion}} g_{\mathrm{cal},1} \\ Y_2 + f_{\mathrm{ion}} g_{\mathrm{cal},2} \end{pmatrix}.$$

Its determinant is $\Delta_2 = g_{\mathrm{Luke},1}(1 + g_{\mathrm{cal},2})(Y_1 - f_{\mathrm{ion}}) - g_{\mathrm{Luke},2}(1 + g_{\mathrm{cal},1})(Y_2 - f_{\mathrm{ion}})$. When $Y_1 = Y_2 = Y$ it equals $(Y - f_{\mathrm{ion}})(g_{\mathrm{Luke},1} - g_{\mathrm{Luke},2})$, since $g_{\mathrm{Luke},1} g_{\mathrm{cal},2} = g_{\mathrm{Luke},2} g_{\mathrm{cal},1}$. By Cramer's rule,

$$\begin{aligned}\mathfrak{r}' &= \frac{(1 + g_{\mathrm{cal},1})(Y_2 + f_{\mathrm{ion}} g_{\mathrm{cal},2}) - (1 + g_{\mathrm{cal},2})(Y_1 + f_{\mathrm{ion}} g_{\mathrm{cal},1})}{\Delta_2}, \\ Y_{\mathrm{ph}}(0) &= \frac{g_{\mathrm{Luke},1}(Y_1 - f_{\mathrm{ion}})(Y_2 + f_{\mathrm{ion}} g_{\mathrm{cal},2}) - g_{\mathrm{Luke},2}(Y_2 - f_{\mathrm{ion}})(Y_1 + f_{\mathrm{ion}} g_{\mathrm{cal},1})}{\Delta_2}.\end{aligned} \tag{D.246}$$

The inversion is exact within the signal model Eq. (19.32); it needs no expansion in the deficits and no assumption on the sign of $\mathfrak{r}' - 1$. Equal measured ratios give $\mathfrak{r}' = \varepsilon_{\mathrm{eh}}/\varepsilon_{\mathrm{cal}}$ and $Y_{\mathrm{ph}}(0) = Y$: a ratio that does not change with bias means that the Luke phonons are collected so that $\mathfrak{r}' g_{\mathrm{Luke}} = g_{\mathrm{cal}}$, and that the deficit is in the heat itself. As the bias grows, Eq. (19.34) tends to $f_{\mathrm{ion}} + [Y_{\mathrm{ph}}(0) - f_{\mathrm{ion}}]\varepsilon_{\mathrm{eh}}/(\varepsilon_{\mathrm{cal}}\mathfrak{r}')$.

D.46 Halo-independent bounds by linear programming

A single stream at galactic velocity $\boldsymbol{w}$ leaves in bin i of a crystal fixed in the laboratory the count $\mathsf{k}_i(\boldsymbol{w})$. Here we derive that stream kernel and its multipole form Eq. (21.13), and we prove the halo-independent bound Eq. (21.12) of a response that depends only on the speed, the bound with a defective certificate Eq. (21.14), and the leakage of a final-state split, Eq. (21.15). The linear-programming facts it uses, weak duality, the mass bound and the discretization bracket, are those of Appendix C.12.

D.46.1 The stream kernel

By Eq. (21.3), a target of mass $M_{\rm det}$ counts in bin i over a live time $t_{\rm exp}$ on average

$$N_i = \frac{M_{\rm det} t_{\rm exp}\, \rho_\chi}{m_\chi\, \rho_T} \int \mathrm{d}^3 w\, f_{\rm gal}(\boldsymbol{w})\, \big\langle \Gamma_i(\boldsymbol{w} - \boldsymbol{v}_{\rm E}(t)) \big\rangle_t ,$$

with $\boldsymbol{w}$ the galactic-frame velocity, Γ_i the rate kernel Eq. (21.2) with $P_{\rm casc}(N_{\rm eh}|\Lambda)$ replaced by the probability $P(i|\Lambda)$ that a final state of class Λ ends in bin i, and its argument rotated into crystal axes at each time. For a coupling through the density, $\mathfrak{D}_{ee} = \pi \bar\sigma_e F_{\rm DM}^2/\mu_{\chi e}^2$ (Eq. (1.34)) is proportional to $\bar\sigma_e$, and so is Γ_i. We therefore set

$$\mathrm{d}\Xi(\boldsymbol{w}) = \frac{\rho_\chi \bar\sigma_e}{m_\chi}\, f_{\rm gal}(\boldsymbol{w})\, \mathrm{d}^3 w, \qquad \mathsf{k}_i(\boldsymbol{w}) = \frac{M_{\rm det} t_{\rm exp}}{\rho_T\, \bar\sigma_e} \big\langle \Gamma_i(\boldsymbol{w} - \boldsymbol{v}_{\rm E}(t)) \big\rangle_t , \tag{D.247}$$

so that $N_i = \int \mathsf{k}_i\, \mathrm{d}\Xi$. The measure Ξ is nonnegative and is supported on the ball $\mathcal{B} = \{|\boldsymbol{w}| \le v_{\rm esc}\}$; its total mass, $\rho_\chi \bar\sigma_e/m_\chi$, is not known and is left free. The kernel does not contain $\bar\sigma_e$, and $\mathsf{k}_i(\boldsymbol{w})$ is the count that a single stream of unit Ξ-mass at $\boldsymbol{w}$ would produce. In a laboratory speed $v = |\boldsymbol{w} - \boldsymbol{v}_{\rm E}|$, the mean inverse speed of Eq. (1.28) becomes $\int \Theta(v - u_0)\, v^{-1}\, \mathrm{d}\Xi = \rho_\chi \bar\sigma_e \eta(u_0)/m_\chi = \tilde\eta(u_0)$, the combination of Appendix A.8.

The kernel of one stream follows from Eq. (21.2). Write the response, contracted with $\mathfrak{D}$ and weighted with $P(i|\Lambda)$, as $s_i(\boldsymbol{q},\omega) = \sum_\Lambda \sum_{a,b} \mathfrak{D}_{ab} S_{ab}(\boldsymbol{q},\omega;\Lambda) P(i|\Lambda)$, for a coupling that depends on $|\boldsymbol{q}|$ only, and expand it in real spherical harmonics in crystal axes,

$$s_i(\boldsymbol{q},\omega) = \bar s_i(q,\omega) \Big[1 + \sum_{\ell \ge 1} \sum_m a^i_{\ell m}(q,\omega)\, Y_{\ell m}(\hat{\boldsymbol{q}}) \Big],$$

with $\bar s_i$ its average over directions. For a stream of velocity $\boldsymbol{v}$ in crystal axes, the energy delta function $\delta(\omega - \boldsymbol{q}\cdot\boldsymbol{v} + q^2/2m_\chi)$, written at fixed $|\boldsymbol{q}|$, is $(qv)^{-1}\delta(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{v}} - v_{\rm min}/v)$ with $v_{\rm min} = \omega/q + q/2m_\chi$ (Eq. (1.23)). It restricts $\hat{\boldsymbol{q}}$ to a circle of half-opening $\arccos(v_{\rm min}/v)$ about $\hat{\boldsymbol{v}}$. The integral of a harmonic over that circle follows from the Legendre expansion of the delta function, $\delta(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{v}} - x) = \sum_\ell \frac{1}{2}(2\ell+1) P_\ell(x) P_\ell(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{v}})$, and the projection $\int \mathrm{d}^2\hat{\boldsymbol{q}}\, P_\ell(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{v}}) Y_{\ell' m}(\hat{\boldsymbol{q}}) = [4\pi/(2\ell+1)]\delta_{\ell\ell'} Y_{\ell m}(\hat{\boldsymbol{v}})$ of the addition theorem Eq. (C.21) (Sect. 11.7):

$$\int \mathrm{d}^2\hat{\boldsymbol{q}}\, \delta(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{v}} - x)\, Y_{\ell m}(\hat{\boldsymbol{q}}) = 2\pi\, P_\ell(x)\, Y_{\ell m}(\hat{\boldsymbol{v}}) \qquad (|x| \le 1). \tag{D.248}$$

For $\ell = 0$ it is the factor 2π of the angular average in Sect. 1.3. With $\int \mathrm{d}^3 q/(2\pi)^3 = \int q^2 \mathrm{d}q\, \mathrm{d}^2\hat{\boldsymbol{q}}/8\pi^3$, the kernel of one stream is

$$\Gamma_i(\boldsymbol{v}) = \frac{1}{v} \int \mathrm{d}\omega \int \frac{q\, \mathrm{d}q}{4\pi^2}\, \bar s_i(q,\omega) \Big[1 + \sum_{\ell \ge 1} \sum_m a^i_{\ell m}(q,\omega)\, P_\ell\Big(\frac{v_{\rm min}}{v}\Big) Y_{\ell m}(\hat{\boldsymbol{v}}) \Big] \Theta(v - v_{\rm min}),$$

which for $a^i_{\ell m} = 0$ is Eq. (21.4). For a cubic crystal $\sum_m a^i_{\ell m} Y_{\ell m} = a^i_\ell Y^{\rm c}_\ell$, with $Y^{\rm c}_\ell$ the cubic harmonics of Appendix C.3, $\ell = 0, 4, 6, 8, 10$ (one harmonic per ℓ below $\ell = 12$; from $\ell = 12$

on, two or more harmonics of each ℓ carry their own coefficients), and inserting the kernel in Eq. (D.247) gives Eq. (21.13). A pair with $v_{\min}$ close to v is reached only by momentum transfers almost parallel to the stream, $P_\ell \to 1$, and carries the full anisotropy of the response. A pair with $v_{\min} \ll v$ is reached on a wide circle, and its anisotropy is averaged by $P_\ell(v_{\min}/v)$.

D.46.2 The step column and the bound

Let the response be independent of $\hat{\boldsymbol{q}}$, let $\boldsymbol{v}_{\rm E}$ be constant over the exposure, and write the stream in the laboratory as $\boldsymbol{v} = \boldsymbol{w} - \boldsymbol{v}_{\rm E}$. Then k_i depends only on $v = |\boldsymbol{v}|$,

$$\mathsf{k}_i(v) = \frac{\mathcal{A}_i(v)}{v}, \qquad \mathcal{A}_i(v) = \frac{M_{\rm det} t_{\rm exp}}{\rho_T\, \bar{\sigma}_e} \int \mathrm{d}\omega \int \frac{q\,\mathrm{d}q}{4\pi^2}\, \bar{s}_i(q,\omega)\, \Theta(v - v_{\min}(q,\omega)). \tag{D.249}$$

The *step column* $\mathcal{A}_i(v)$ is the count in bin i for the step $\tilde{\eta}(u) = \Theta(v-u)$, one unit of $\tilde{\eta}$ below the speed v; it is $M_{\rm det} t_{\rm exp}\, v \Gamma_i(v)/\rho_T \bar{\sigma}_e$ in the notation of Sect. 21.1. Its derivative replaces Θ by $\delta(v - v_{\min})$,

$$\frac{\mathrm{d}\mathcal{A}_i}{\mathrm{d}v} = \frac{M_{\rm det} t_{\rm exp}}{\rho_T\, \bar{\sigma}_e} \int \mathrm{d}\omega \int \frac{q\,\mathrm{d}q}{4\pi^2}\, \bar{s}_i(q,\omega)\, \delta\big(v - v_{\min}(q,\omega)\big) \geq 0, \tag{D.250}$$

nonnegative wherever $\bar{s}_i \geq 0$, which holds for a positive response and a probability $P(i|\Lambda)$.

The objective is $\tilde{\eta}(u_0) = \int \mathsf{f}_{u_0} \mathrm{d}\Xi$ with $\mathsf{f}_{u_0}(\boldsymbol{w}) = \Theta(v - u_0)/v$, a function of the same laboratory speed. The shifted ball $\{|\boldsymbol{v} + \boldsymbol{v}_{\rm E}| \leq v_{\rm esc}\}$ contains streams of every speed from 0 to $v_{\max} = v_{\rm esc} + |\boldsymbol{v}_{\rm E}|$, the fastest antiparallel to $\boldsymbol{v}_{\rm E}$. Let $0 < u_0 \leq v_{\max}$ and $\max_i \mathcal{A}_i(u_0) > 0$, and let j be a bin that minimizes $N_i^{\rm up}/\mathcal{A}_i(u_0)$. Then

$$\tilde{\eta}_{\max}(u_0) = \frac{N_j^{\rm up}}{\mathcal{A}_j(u_0)} = \min_i \frac{N_i^{\rm up}}{\mathcal{A}_i(u_0)}. \tag{D.251}$$

Primal. One stream at a laboratory speed u_0, in any allowed direction, with Ξ-mass $\xi_1 = u_0 N_j^{\rm up}/\mathcal{A}_j(u_0)$, produces $\xi_1 \mathsf{k}_i(u_0) = N_j^{\rm up} \mathcal{A}_i(u_0)/\mathcal{A}_j(u_0) \leq N_i^{\rm up}$ counts in every bin, with equality in bin j, and has $\tilde{\eta}(u_0) = \xi_1/u_0 = N_j^{\rm up}/\mathcal{A}_j(u_0)$. *Dual.* Take $\lambda_j = 1/\mathcal{A}_j(u_0)$ and $\lambda_i = 0$ for $i \neq j$. The dual function $\mathsf{g}_\lambda(v) = \mathcal{A}_j(v)/[v\,\mathcal{A}_j(u_0)] - \Theta(v-u_0)/v$ is nonnegative below u_0, where the subtracted term vanishes, and at and above u_0, where $\mathcal{A}_j(v) \geq \mathcal{A}_j(u_0)$ by Eq. (D.250). So $\boldsymbol{\lambda}$ is a certificate, and its value $\lambda_j N_j^{\rm up}$ equals the primal value. By weak duality, Eq. (C.112), both are optimal. The optimal stream sits where $\mathsf{g}_\lambda = 0$, at $v = u_0$, and only the saturated bin carries a dual weight, as complementary slackness requires (Appendix C.12).

Two hypotheses carry the result. Monotonicity is used only in the dual. For a column that decreases somewhere, the single-bin value is $N_i^{\rm up}/\inf_{v \geq u_0} \mathcal{A}_i(v)$ by Eq. (C.114), and with several bins the minimum over bins of these values is only an upper estimate of the optimum, which the program then has to supply. The common frame is the second hypothesis. The kernel is a function of the laboratory speed, which in galactic coordinates is $|\boldsymbol{w} - \boldsymbol{v}_{\rm E}|$; an objective written with the galactic speed $|\boldsymbol{w}|$ is a different functional, and Eq. (D.251) does not bound it. With a time-dependent $\boldsymbol{v}_{\rm E}(t)$ a stream has no single laboratory speed, and the reduction holds bin by bin in time only where $|\boldsymbol{v}_{\rm E}|$ changes little over a bin.

D.46.3 A certificate with a defect

Let the kernels k_i be nonnegative and continuous, let $\mathsf{f} \geq 0$ be continuous on its support $\operatorname{supp} \mathsf{f}$, and let $\kappa_{\rm v} = \min_{\operatorname{supp} \mathsf{f}} \max_i \mathsf{k}_i > 0$ as in Appendix C.12. Suppose $\boldsymbol{\lambda} \geq 0$ satisfies

$$\sum_i \lambda_i \mathsf{k}_i(\boldsymbol{w}) \geq \mathsf{f}(\boldsymbol{w}) - \delta_{\rm cert} \qquad \text{for all } \boldsymbol{w} \in \operatorname{supp} \mathsf{f},$$

with some defect $\delta_{\mathrm{cert}} \geq 0$. Take a feasible Ξ and its restriction Ξ' to $\operatorname{supp} \mathsf{f}$. Since $\mathsf{k}_i \geq 0$, Ξ' is feasible as well, and it has the same objective. Then

$$\begin{aligned} \int \mathsf{f}\,\mathrm{d}\Xi = \int \mathsf{f}\,\mathrm{d}\Xi' &\leq \int \Big(\sum_i \lambda_i \mathsf{k}_i + \delta_{\mathrm{cert}}\Big)\mathrm{d}\Xi' \\ &\leq \sum_i \lambda_i N_i^{\mathrm{up}} + \delta_{\mathrm{cert}}\,\Xi(\operatorname{supp}\mathsf{f}) \leq \sum_i \lambda_i N_i^{\mathrm{up}} + \frac{\delta_{\mathrm{cert}}\,N_{\mathrm{tot}}}{\kappa_{\mathrm{v}}}, \end{aligned} \tag{D.252}$$

the last step by the mass bound Eq. (C.113). A grid dual violates the certificate by at most $L_h h$ between grid points, and $\delta_{\mathrm{cert}} = L_h h$ gives Eq. (C.115).

The same inequality handles a kernel that is computed with negative pieces. Let $\tilde{\mathsf{k}}_i$ be the computed kernel, of either sign, and $\mathsf{k}_i \geq 0$ a kernel within $\mathsf{l}_i \geq 0$ of it, $|\mathsf{k}_i - \tilde{\mathsf{k}}_i| \leq \mathsf{l}_i$. If the dual of the computed program satisfies $\sum_i \lambda_i \tilde{\mathsf{k}}_i \geq \mathsf{f} - \delta_h$ on $\operatorname{supp}\mathsf{f}$, then

$$\sum_i \lambda_i \mathsf{k}_i \geq \mathsf{f} - \delta_h - \sum_i \lambda_i \mathsf{l}_i,$$

so Eq. (D.252) holds for the nonnegative kernel with $\delta_{\mathrm{cert}} = \delta_h + \sup_{\operatorname{supp}\mathsf{f}} \sum_i \lambda_i \mathsf{l}_i$, and with κ_{v} replaced by its lower estimate $\min_{\operatorname{supp}\mathsf{f}} \max_i(\tilde{\mathsf{k}}_i - \mathsf{l}_i)$. This is Eq. (21.14).

D.46.4 Leakage of a final-state split

Outside the off-shell window of Sect. 8.6 the electronic response of a crystal is split by the Dyson convention Eq. (8.59) into a zero-phonon part and a phonon-dressed part. Their sum is a positive spectrum, each alone need not be. Write $s^0(\boldsymbol{q},\omega)$ and $s^+(\boldsymbol{q},\omega)$ for the two parts contracted with $\mathfrak{D}$, so that $s^0 + s^+ \geq 0$. A zero-phonon pair carries the electronic energy ω into the cascade, a phonon-dressed pair $\omega - \omega_{\mathrm{ph}}$, with $0 < \omega_{\mathrm{ph}} \leq \omega_{\max}$ the energy of the phonon. If the bin probability depends on the electronic energy E alone, $P_i(E)$, the bin-weighted response is

$$s_i(\omega) = P_i(\omega)\, s^0 + P_i(\omega - \omega_{\mathrm{ph}})\, s^+,$$

at each $\boldsymbol{q}$, with one phonon energy for brevity; a spread of phonon energies replaces ω_{ph} by each value in turn, and every bound below holds with $\omega_{\max}$. Let $\Delta P_i(\omega) = P_i(\omega - \omega_{\mathrm{ph}}) - P_i(\omega)$. Where $\Delta P_i \geq 0$, $s_i = P_i(\omega)(s^0 + s^+) + \Delta P_i\, s^+ \geq -\Delta P_i [s^+]_-$; where $\Delta P_i < 0$, $s_i = P_i(\omega - \omega_{\mathrm{ph}})(s^0 + s^+) + |\Delta P_i|\, s^0 \geq -|\Delta P_i|\,[s^0]_-$, with $[x]_- = \max(-x, 0)$. Hence

$$[s_i]_- \leq |\Delta P_i(\omega)|\, \max\big([s^0]_-, [s^+]_-\big).$$

Both parts are needed, since ΔP_i takes both signs across an interior bin, where $P_i(E)$ first rises and then falls. Seeding the phonon-dressed pair with the probability of its zero-phonon partner, $s_i^{\mathrm{c}} = P_i(\omega)(s^0 + s^+)$, gives a response that is never negative, and it differs from s_i by $|\Delta P_i|\,|s^+|$. That difference, inserted in the stream kernel in place of s_i, is the l_i of the previous subsection.

Its size follows from the variation of P_i. For a function of bounded variation and $0 < h \leq \omega_{\max}$, $|P_i(E) - P_i(E-h)| \leq \int_{E-\omega_{\max}}^{E} |\mathrm{d}P_i|$; integrating over E and exchanging the integrals,

$$\int \mathrm{d}\omega\, |\Delta P_i(\omega)| \leq \omega_{\max}\, \mathrm{TV}(P_i) \leq 2\,\omega_{\max}, \tag{D.253}$$

where TV is the total variation. The last step assumes that the number of pairs is stochastically ordered in the energy of the seed: $P(N_{\mathrm{eh}} \geq n|E)$ does not decrease with E for any n. Then $P_i(E) = P(N_{\mathrm{eh}} \geq n_i|E) - P(N_{\mathrm{eh}} > n_i'|E)$ for a bin of pair numbers $n_i \leq N_{\mathrm{eh}} \leq n_i'$ is a difference of two nondecreasing functions with values in $[0,1]$, and its total variation is at

most 2, exactly 2 for the step model Eq. (1.43). Let the bin have the width $\Delta_i = \int \mathrm{d}\omega\, P_i(\omega)$, which is $\varepsilon_{\mathrm{eh}}$ for the step model and, for the cascade of Sect. 18.3, tends to it for bins well above threshold, and let its signal be $\int \mathrm{d}\omega\, P_i(s^0 + s^+) = \Delta_i \langle s^0 + s^+ \rangle_i$, with $\langle\,\rangle_i$ the average weighted with P_i. Since ΔP_i vanishes outside the support of P_i widened by $\omega_{\max}$, Eq. (D.253) bounds the misassigned weight at fixed $\boldsymbol{q}$ by

$$\frac{\int \mathrm{d}\omega\, |s_i - s_i^{\mathrm{c}}|}{\Delta_i \langle s^0 + s^+ \rangle_i} \leq \frac{2\,\omega_{\max}}{\Delta_i}\, \varrho_i^{\mathrm{leak}}, \qquad \varrho_i^{\mathrm{leak}} = \frac{\sup |s^+|}{\langle s^0 + s^+ \rangle_i}, \tag{D.254}$$

with the supremum over that widened support. and the same with $\max([s^0]_-, [s^+]_-)$ in place of $|s^+|$ for the negative part of s_i alone. For a response averaged over the directions of $\boldsymbol{q}$, the stream kernel at fixed q is an integral of s_i over the window of ω that the speed reaches, Eq. (D.249), and Eq. (D.254) bounds l_i relative to k_i wherever the window covers the bin. For a directional response the delta function fixes ω on each $\boldsymbol{q}$, and l_i is computed pointwise, in the same pass over the response as the kernel itself (Appendix E.27).

D.47 Theory nuisances in a profile likelihood

An exclusion limit computed with every truncation at its central value moves when a truncation moves, and the profile likelihood of Eq. (8.78) penalizes that motion through its constraint terms. Here we derive the three statements Sect. 21.4.2 uses: how far the limit moves per unit of each nuisance, Eq. (21.23); the theory band that follows when the nuisances are independent and standard normal, Eq. (21.24); and the cost of profiling them in the Neyman construction of Sect. 21.4.2, of second order in their sizes, Eq. (21.25). The last two subsections treat nuisances of known sign and the removal of cells from the signal.

D.47.1 The limit at fixed nuisances

In bin j the expected count is $N_j(\bar\sigma_e, \vartheta) = \bar\sigma_e s_j(\vartheta) + N_{\mathrm{bkg},j}$, with $s_j = N_{\mathrm{sig},j}/\bar\sigma_e$ the signal per unit cross-section of Eq. (21.22). The theory nuisances act only on s_j. Up to terms independent of the parameters, the Poisson part of $-2\ln \mathsf{L}$ is $\sum_j \ell_j(N_j)$ with

$$\ell_j(N) = 2\big(N - N_{\mathrm{obs},j} \ln N\big), \qquad \ell_j'(N) = 2\Big(1 - \frac{N_{\mathrm{obs},j}}{N}\Big).$$

We consider data that show no excess, so that the best fit lies at $\hat{\bar\sigma}_e = 0$, the case of an exclusion. Then the best fit does not depend on the theory nuisances, which multiply a signal that vanishes there, and at fixed ϑ the one-sided statistic of Appendix C.14 is

$$\mathsf{t}_\vartheta(\bar\sigma_e) = \sum_j \Big[\ell_j\big(\bar\sigma_e s_j(\vartheta) + N_{\mathrm{bkg},j}\big) - \ell_j\big(N_{\mathrm{bkg},j}\big)\Big]. \tag{D.255}$$

The limit $\bar\sigma_e^{\mathrm{up}}(\vartheta)$ is where t_ϑ reaches the threshold t^* of the construction. Differentiating $\mathsf{t}_\vartheta(\bar\sigma_e^{\mathrm{up}}(\vartheta)) = \mathsf{t}^*$ with respect to ϑ_k at fixed threshold gives

$$\frac{\partial \ln \bar\sigma_e^{\mathrm{up}}}{\partial \vartheta_k} = -\frac{\partial \mathsf{t}_\vartheta / \partial \vartheta_k}{\partial \mathsf{t}_\vartheta / \partial \ln \bar\sigma_e} = -\frac{\sum_j \ell_j'\, \bar\sigma_e s_j\, \partial \ln s_j / \partial \vartheta_k}{\sum_j \ell_j'\, \bar\sigma_e s_j}.$$

In the power form Eq. (8.72), the bin signal is an integral of the spectrum times $\prod_k (1+\delta_k)^{\vartheta_k}$, so at $\vartheta = 0$ the logarithmic derivative $\partial \ln s_j / \partial \vartheta_k$ is the average of $\ln(1+\delta_k)$ over the signal

in the bin, which we write $\ln(1+\bar{\delta}_{k,j})$. For a normalization template it is exact; for a tilt it holds at first order, and a finite step is used in practice. Hence Eq. (21.23), with the weights

$$a_j = \left.\frac{\ell'_j\, N_{\mathrm{sig},j}}{\sum_i \ell'_i\, N_{\mathrm{sig},i}}\right|_{\bar{\sigma}_e=\bar{\sigma}_e^{\mathrm{up}}(0)}, \qquad \sum_j a_j = 1. \tag{D.256}$$

A bin with no events has $\ell'_j = 2$, so for zero counts a_j is the share of bin j in the signal. A bin with fewer events than background, $N_{\mathrm{obs},j} \le N_{\mathrm{bkg},j}$, has $\ell'_j \ge 0$ at every $\bar{\sigma}_e \ge 0$ and a nonnegative weight. A bin with an excess can carry a negative weight: raising the signal there improves the fit and weakens the limit.

D.47.2 The theory band

To first order in the nuisances, $\ln \bar{\sigma}_e^{\mathrm{up}}(\vartheta) = \ln \bar{\sigma}_e^{\mathrm{up}}(0) + \sum_k \beta_k \vartheta_k$. With independent standard-normal ϑ_k the right side is Gaussian with mean $\ln \bar{\sigma}_e^{\mathrm{up}}(0)$ and variance $\sigma_\beta^2 = \sum_k \beta_k^2$, which is Eq. (21.24). The envelope that sets one nuisance at a time to ± 1 moves the limit by $\mathrm{e}^{\pm|\beta_k|}$.

A template enters β_k as a weighted sum over bins with one parameter, $\beta_k = -\sum_j a_j \ln(1+\bar{\delta}_{k,j})$. Suppose instead that the same template were given an independent parameter in each bin. Then the variance would be $\sum_j a_j^2 \ln^2(1+\bar{\delta}_{k,j})$. When all terms $a_j \ln(1+\bar{\delta}_{k,j})$ have one sign, the square of their sum exceeds the sum of their squares, and the independent treatment understates the band. When they change sign, as a many-body template must across a spectrum that reaches its compensating energies (Eq. (8.76)), the coherent parameter partly cancels and the independent treatment overstates it. Either way the independent treatment describes no calculation.

D.47.3 The cost of profiling

With standard-normal constraints the profiled statistic is

$$\mathtt{t}_{\mathrm{p}}(\bar{\sigma}_e) = \min_{\vartheta} \Big[\mathtt{t}_\vartheta(\bar{\sigma}_e) + \sum_k \vartheta_k^2 \Big],$$

since the overall best fit, at $\hat{\bar{\sigma}}_e = 0$ and $\vartheta = 0$, contributes zero. Let $x = \ln \bar{\sigma}_e$ and $Q(x) = \mathtt{t}_0(\mathrm{e}^x)$. To first order in the templates, $\partial \mathtt{t}_\vartheta / \partial \vartheta_k = -\beta_k Q'(x)$ by the result above, so that $\mathtt{t}_\vartheta = Q(x - \sum_k \beta_k \vartheta_k)$, with β evaluated at x. For zero counts or one bin the weights a_j, and hence β, do not depend on x; otherwise their dependence on x enters the result at $O(\sigma_\beta^4)$. Only the combination $y = \sum_k \beta_k \vartheta_k$ enters. At fixed y the smallest $\sum_k \vartheta_k^2$ is y^2/σ_β^2, reached for ϑ parallel to β, and

$$\mathtt{t}_{\mathrm{p}}(\bar{\sigma}_e) = \min_{y} \Big[Q(x-y) + \frac{y^2}{\sigma_\beta^2} \Big].$$

To first order in the templates, any number of nuisances thus acts like one nuisance of logarithmic width σ_β. The minimum lies at $Q'(x-y) = 2y/\sigma_\beta^2$, the conditional best fit

$$y = \tfrac{1}{2}\sigma_\beta^2\, Q'(x) + O(\sigma_\beta^4), \tag{D.257}$$

a shift of the signal toward smaller values, since the data show no excess.

The Neyman construction of Sect. 21.4.2 evaluates the probability of the outcomes at least as extreme as the observed one with the nuisances at this best fit. The expected counts are then those of the nominal model at the shifted cross-section e^{x-y}, and the probability equals $1-\mathrm{CL}$ where the shifted cross-section is the nominal limit, $x - y = x_0$. With Eq. (D.257) at $x_{\mathrm{p}} = x_0 + O(\sigma_\beta^2)$,

$$x_{\mathrm{p}} - x_0 = \tfrac{1}{2}\sigma_\beta^2\, Q'(x_0) + O(\sigma_\beta^4),$$

which is Eq. (21.25) with $\mathsf{t}' = Q'(x_0) = \sum_j \ell'_j N_{\mathrm{sig},j}$ at the nominal limit. For one bin the result is exact. There the best fit obeys $\vartheta_k = \frac{1}{2}\beta_k \ell'(N)\,\bar\sigma_e s(\vartheta)$ exactly, and at the limit the counts N and the signal $\bar\sigma_e s(\vartheta)$ are those of the classical limit, so $y = \frac{1}{2}\sigma_\beta^2 \mathsf{t}'$ with no higher orders. With no events $\mathsf{t}' = 2N^{\mathrm{up}}$, and the profiled limit is $\mathrm{e}^{\sigma_\beta^2 N^{\mathrm{up}}}$ times the nominal one for any background: 1.427 at $\sigma_\beta = 0.393$ without background. A known background lowers t': with no event and $N_{\mathrm{bkg}} = 1$ the classical limit is $N^{\mathrm{up}} = 1.303$ and $\mathsf{t}' = 2N^{\mathrm{up}} = 2.61$, against 4.61 without background. One event on a background of one gives $\mathsf{t}' = 4.29$ and, at $\sigma_\beta = 0.393$, a factor 1.393.

A fixed threshold gives half of this. If instead the profiled statistic is compared with the threshold t^* that reproduces the classical limit without nuisances, as in the illustration of Eq. (8.79), expanding $Q(x - y) = Q(x) - yQ'(x) + O(y^2)$ gives

$$\mathsf{t}_{\mathrm{p}}(\bar\sigma_e) = Q(x) - \frac{\sigma_\beta^2}{4}\,Q'(x)^2 + O(\sigma_\beta^4),$$

and subtracting $Q(x_0) = \mathsf{t}^*$ from $\mathsf{t}_{\mathrm{p}}(x_{\mathrm{p}}) = \mathsf{t}^*$ gives $x_{\mathrm{p}} - x_0 = \frac{1}{4}\sigma_\beta^2 Q'(x_0)$. The fixed threshold lets the profiled statistic fall by the constraint it saves, while the Neyman construction charges the whole shift y of the signal. For one bin with no events and no background, $Q(x) = 2s\,\mathrm{e}^x$ and $\mathsf{t}' = \mathsf{t}^*$, and $\sigma_\beta^2 \mathsf{t}^*/4$ is the expansion of Eq. (8.79): at $\sigma_\beta = 0.3$ and $\mathsf{t}^* = 4.605$ the exact ratio is 1.104 and the second-order one 1.109, against 1.230 in the Neyman construction.

D.47.4 Nuisances of known sign

Let nuisance k have a template of known sign, of one sign in every bin, constrained to $\vartheta_k \geq 0$ with the weight $\mathrm{e}^{-\vartheta_k^2/2}$ there, a half-normal constraint. Assume every weight a_j is nonnegative at every $\bar\sigma_e$, as in data without an excess. If $\vartheta_k > 0$ raises the signal, so that $\beta_k < 0$, then $\mathsf{t}_\vartheta(\bar\sigma_e)$ increases with ϑ_k at every $\bar\sigma_e$, and so does the constraint term. The minimum over $\vartheta_k \geq 0$ sits on the boundary $\vartheta_k = 0$ at every $\bar\sigma_e$, the nuisance drops out of the profile, and the profiled limit equals the limit without it. The limit at $\vartheta_k = 0$ is then the conservative end. If $\vartheta_k > 0$ lowers the signal, the unconstrained minimum already lies at $\vartheta_k > 0$, and the half-normal constraint costs as much as a two-sided one.

D.47.5 Removing cells from the signal

Cells of the $(\boldsymbol{q}, \omega)$ plane where a response has no controlled central value can be removed from the signal. Removing them lowers each s_j by a nonnegative amount. From Eq. (D.255), $\partial \mathsf{t}_\vartheta / \partial s_j = \bar\sigma_e\,\ell'_j(N_j)$, and for $N_{\mathrm{obs},j} \leq N_{\mathrm{bkg},j}$ this is nonnegative at every $\bar\sigma_e \geq 0$, since $N_j \geq N_{\mathrm{bkg},j}$. Lowering any s_j of such a bin therefore lowers $\mathsf{t}_\vartheta(\bar\sigma_e)$ at every $\bar\sigma_e$, and the point where it reaches t^* moves up. The same holds for t_{p}, a minimum of functions that each decrease, and for the classical limit of one bin, $\bar\sigma_e^{\mathrm{up}} = N^{\mathrm{up}}/s$. Removing signal therefore never strengthens a limit set from bins without an excess. In a bin with $N_{\mathrm{obs},j} > N_{\mathrm{bkg},j}$, ℓ'_j is negative for $\bar\sigma_e s_j < N_{\mathrm{obs},j} - N_{\mathrm{bkg},j}$, and the statement can fail there. For one bin the limit rises by the factor $1/(1 - f)$, with f the removed share of the signal.

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Appendix E

Ab initio methods

A dark-matter rate in silicon needs a few dozen numbers about the crystal: a gap good to tens of meV, a loss function out to 100 eV, density vertices at momentum transfers of tens of keV, electron–phonon couplings on grids of 10^9 points, and the energy a struck atom leaves behind. Almost none of them is measured everywhere it is needed. Each is computed by a method with its own approximation, numerical parameters and typical failure. Appendices E.1 to E.13 follow each method through one calculation, almost always of silicon, so that a number in the text can be traced to what it neglects and to the check that would expose it. Appendices E.14 to E.27 go the other way and specify, quantity by quantity, the calculation behind each number the chapters use, with its inputs, its method and its convergence targets. Every method is placed on the truncation label $\mathfrak{T}$ of Sect. 8.5, whose components include the Born–Oppenheimer order n_B, the electron–phonon order n_g, the anharmonic order n_A, the thermal order n_T, the many-body approximation $\mathcal{X}$, the lattice regime $\mathcal{K}$ (one phonon, multiphonon or impulse) and the Migdal regime $\mathfrak{M}$; a calculation at clamped nuclei, with independent particles and no phonons, sits at its base point $\mathfrak{T}_0$.

E.1 Density-functional theory in practice

A density-functional calculation of silicon at its measured lattice constant returns, in seconds on a few processor cores, a total energy, an electron density and a set of Kohn–Sham bands with an indirect gap near 0.6 eV. The measured gap at low temperature is 1.17 eV [1, Eq. (11)]. Both numbers matter for a dark-matter rate, and the work of a calculation lies in knowing which outputs to trust, how far, and what each numerical parameter does to them. Sect. 3.4 stated the Kohn–Sham equations, derived in Appendix D.5, and explained why their eigenvalues are not quasiparticle energies. We follow one calculation of silicon from its input to its bands and masses, and test which of its numbers can be used as they stand: the PBE functional, the norm-conserving silicon pseudopotential of the SG15 library [2] (four valence electrons per atom, Appendix E.2), and the diamond structure with two atoms per cell.

E.1.1 Computed quantities

The Kohn–Sham equations of Sect. 3.4 are solved for the periodic parts $u_{n\boldsymbol{k}}$ of the Bloch states on a grid of crystal momenta. The states define the density

$$n(\boldsymbol{r}) = \frac{2}{n_k} \sum_{\boldsymbol{k}} \sum_{n} f_{n\boldsymbol{k}} \, |u_{n\boldsymbol{k}}(\boldsymbol{r})|^2, \tag{E.1}$$

in which the factor 2 counts spin, n_k is the number of grid points and $f_{n\boldsymbol{k}}$ are the occupations. Since $u_{n\boldsymbol{k}}$ is normalized over the cell (Appendix A.5), n integrates over Ω to the number of

valence electrons per cell. The density fixes the Kohn–Sham potential V_{KS}, the potential fixes the states, and the loop is iterated until input and output agree. The outputs deserve different levels of trust: ground-state quantities, eigenvalues and states, and the responses built from both. The total energy, the density, and the forces and stresses derived from the energy are ground-state quantities, exact in principle for the exact functional (Sect. 3.4); their error is that of the approximate functional and of the numerics. The eigenvalues $\varepsilon_{n\boldsymbol{k}}$ and states $\psi_{n\boldsymbol{k}}$ are auxiliary. Apart from the highest occupied level, they have no exact physical meaning, and the band gap they give is systematically too small. Part of the difference would remain even with the exact functional: the exact exchange–correlation potential jumps by a constant when an electron is added across the gap, and the Kohn–Sham gap leaves that derivative discontinuity out. The response quantities built from the states, such as ε_∞, the phonons of Appendix E.4 or the density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ of Appendix E.7, inherit the errors of both.

E.1.2 Place on the hierarchy

On the expansion hierarchy of Sect. 8.5, a ground-state calculation sits at clamped nuclei, $n_B = 0$, and at zeroth thermal order, $n_T = 0$, with no thermal occupation of phonons: the nuclei are fixed at their equilibrium positions, and neither phonons nor zero-point motion enter. The Kohn–Sham states are the independent particles, $\mathcal{X} = \mathrm{IP}$, of the base point $\mathfrak{T}_0$, and every further many-body approximation (RPA, local fields, TDDFT, BSE; Appendix E.6) is built on them. The exchange–correlation functional, however, is not a component of $\mathfrak{T}$. It is not an expansion in a small number, and no next order exists that would estimate its error. As Sect. 8.5 explains for the many-body approximation $\mathcal{X}$, the error is estimated instead by comparing functionals, by comparing with a quasiparticle calculation (Appendix E.3), and by checking measured quantities such as the lattice constant, the phonon frequencies and the gap. Sect. 8.8 turns that comparison into a nuisance parameter.

E.1.3 Functionals

A calculation chooses one approximation to the exchange–correlation energy E_{xc}, and the choice sets the accuracy of everything downstream. The local-density approximation (LDA) takes the exchange–correlation energy per electron of the homogeneous electron gas at the local density, from the quantum Monte Carlo energies of Ceperley and Alder [3] in the parametrization of Perdew and Zunger [4]. The generalized-gradient approximation of Perdew, Burke and Ernzerhof (PBE) [5] adds a dependence on $|\nabla n|/n^{4/3}$, built to satisfy exact constraints of the electron gas. Hybrid functionals mix in a fraction of Hartree–Fock exchange computed with the Kohn–Sham orbitals. The screened hybrid of Heyd, Scuseria and Ernzerhof (HSE) [6] takes one quarter of the exchange from Hartree–Fock, but only at short range. The Coulomb kernel $e^2/4\pi r$ is split, with an error function, into a part that decays on a length of a few Å and a long-range remainder, and the exact exchange acts with the short-range part only. The screening reflects what happens in a semiconductor, where the exchange interaction between distant electrons is screened by the others (Sect. 4.8).

The trends are systematic. LDA binds too strongly and gives lattice constants that are slightly too small, by 0.2 % to 0.6 % for the materials of Table E.1; PBE binds too weakly and gives them too large, by 0.9 % to 2.1 %. Over the elemental crystals, PBE equilibrium volumes are too large by 3.6 % on average [7, Table IV], about 1.2 % in the lattice constant. The semilocal functionals underestimate gaps, LDA and PBE alike, and close the small gap of germanium altogether; HSE recovers most of the gap. Against the measured gaps of 41 semiconductors and insulators, HSE has a mean absolute error of 0.38 eV, falling to 0.18 eV for the typical semiconductors of the set, whose gaps are measured at low temperature [8,

Table E.1: Lattice constants and band gaps of silicon, germanium and gallium arsenide for three functionals, without spin–orbit coupling. Lattice constants from LDA and PBE are from Ref. [9, Table II], those from HSE from Ref. [10, Table I]. Gaps are evaluated at each functional's own relaxed lattice constant, from Ref. [10, Table I]; a zero gap means that the functional makes the crystal a metal. The static-lattice constants are the measured ones extrapolated to zero temperature with the zero-point anharmonic expansion removed [9, Table II]; they are not measured at any temperature, and they lie below the lattice constants measured at low temperature. They are used only inside calculations, as the cell of a static lattice. The room-temperature lattice constants are measured: silicon [11, lattice parameter of silicon], germanium [12, Sect. II A] and GaAs [13]; they are the values for observables and the cell of the shared lattice setting of Appendix E.17. The measured gaps are low-temperature values [10, Table I].

	lattice constant (Å)					gap (eV)		
	LDA	PBE	HSE	static lattice	room T (expt.)	LDA	HSE	measured
Si	5.403	5.466	5.451	5.416	5.431	0.51	1.22	1.17
Ge	5.623	5.759	5.706	5.640	5.658	0.00	0.54	0.74
GaAs	5.605	5.746	5.711	5.638	5.653	0.43	1.18	1.52

Table I and text]. Table E.1 collects the numbers for the book's running materials. In silicon the hybrid lattice constant is 0.6 % too large and its gap 0.05 eV too large. In germanium, where LDA gives a metal, the hybrid reopens the gap but leaves it 0.2 eV too small.

The price of a hybrid is the exact exchange. A semilocal potential acts on a state $\psi_{n\boldsymbol{k}}$ by a multiplication in real space. The Fock operator instead needs, for every pair of the state with an occupied state $\psi_{m\boldsymbol{k}'}$, the potential of the pair density $\psi^*_{m\boldsymbol{k}'}\psi_{n\boldsymbol{k}}$. Each such potential is one Poisson solve, a fast Fourier transform (FFT) of the pair density, a division by $|\boldsymbol{k}-\boldsymbol{k}'+\boldsymbol{G}|^2$ and an FFT back. With N_{occ} occupied bands, applying the Fock operator to all N_{b} bands at one $\boldsymbol{k}$ therefore costs about $n_k N_{\mathrm{occ}} N_{\mathrm{b}}$ FFTs, against N_{b} for a semilocal potential, a factor $n_k N_{\mathrm{occ}}$ more. For silicon on an 8^3 grid this factor is $512 \times 4 \approx 2 \times 10^3$. In practice the inner sum over $\boldsymbol{k}'$ runs over a coarser grid than the outer one, which reduces the factor by one or two orders of magnitude. A hybrid calculation of a two-atom cell therefore costs 10 to 10^3 times a semilocal one, minutes to hours where PBE takes seconds.

In this book the choice is usually PBE (or LDA) for structures, phonons and electron–phonon couplings, where semilocal functionals are accurate to a few percent (Appendix E.4). Wherever an excitation energy sets a threshold, a gap correction, a hybrid functional or a quasiparticle calculation is used instead (Appendix E.3). Note that the two choices do not mix freely: a phonon computed with PBE and a threshold computed with HSE describe two slightly different crystals, and the lattice constant at which both are evaluated must be stated.

E.1.4 Plane waves and the cutoff

The periodic part of each Bloch state is expanded in plane waves,

$$u_{n\boldsymbol{k}}(\boldsymbol{r}) = \Omega^{-1/2} \sum_{\boldsymbol{G}} c_{n\boldsymbol{k}}(\boldsymbol{G})\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}, \qquad \frac{|\boldsymbol{k}+\boldsymbol{G}|^2}{2m_e} \le E_{\mathrm{cut}},$$

truncated at the kinetic-energy cutoff E_{cut} (Sect. 3.5). Codes quote E_{cut} in rydberg (1 Ry = 13.606 eV) or hartree (two rydberg). The number of plane waves follows from counting reciprocal-lattice points in a sphere of radius $k_{\mathrm{cut}} = (2m_e E_{\mathrm{cut}})^{1/2}$. Each point occupies a

Table E.2: Convergence of the total energy and pressure with the plane-wave cutoff in the representative PBE calculations of silicon and germanium at their measured room-temperature lattice constants, 5.431 Å and 5.658 Å (Table E.1). The norm-conserving pseudopotentials carry four (Si) and fourteen (Ge, with 3d) valence electrons, and the grid has 8^3 points $\boldsymbol{k}$. ΔE is the total energy per atom minus its value at 120 Ry.

	Si		Ge	
$E_{\rm cut}$ (Ry)	ΔE (meV)	P (kbar)	ΔE (meV)	P (kbar)
30	8.36	23.6	2880	−1615
40	2.52	23.4	11.7	27.5
50	0.71	23.6	4.42	37.6
60	0.14	23.7	3.10	37.2
70	0.02	23.7	0.49	37.7
80	0.01	23.7	0.22	38.0
100	0.00	23.7	0.04	38.0

volume $(2\pi)^3/\Omega$ of reciprocal space, so

$$N_{\rm PW} \simeq \frac{4\pi k_{\rm cut}^3/3}{(2\pi)^3/\Omega} = \frac{\Omega\, k_{\rm cut}^3}{6\pi^2}. \tag{E.2}$$

For silicon, $\Omega = 40.05\,\text{Å}^3$ (Sect. 3.1) is 270.3 cubic bohr. At $E_{\rm cut} = 60$ Ry, with $\hbar^2/2m_e = 1\,\text{Ry bohr}^2$, $k_{\rm cut} = \sqrt{60}$ inverse bohr and Eq. (E.2) gives 2121 plane waves per state. The density is a product of two states and has Fourier components up to $2k_{\rm cut}$, eight times as many, about 1.7×10^4. These set the size of the FFT grid on which the potential is applied. A component $\boldsymbol{G}$ has integer coordinates $m_i = \boldsymbol{G}\cdot\boldsymbol{a}_i/2\pi$ with $|m_i| \le 2k_{\rm cut}|\boldsymbol{a}_i|/2\pi$, which is 17.9 for $|\boldsymbol{a}_i| = a/\sqrt{2}$. The indices then run from −17 to 17, so the grid needs at least 35 points along each primitive vector, in practice 36, the next size that the FFT handles efficiently, and $36^3 \approx 4.7 \times 10^4$ points in all.

The cutoff is a variational parameter. For a fixed pseudopotential, enlarging the basis can only lower the total energy, so the energy converges from above and monotonically. The change between successive cutoffs estimates the error only once the scan has reached its asymptotic regime: in germanium the change from 50 to 60 Ry, 1.3 meV, is less than half the error that remains at 60 Ry, 3.1 meV. Table E.2 shows the scan for silicon and germanium. Silicon converges to 1 meV per atom at 50 Ry, and to 0.1 meV at about 60 Ry. Germanium, whose pseudopotential carries the ten 3d electrons in the valence (Appendix E.2), needs 80 Ry for 0.2 meV, and at 30 Ry it is wrong by almost 3 eV per atom. The pressure behaves differently in the two cases: in silicon it is stable to 0.3 kbar from 30 Ry, while in germanium it is not even of the right sign at 30 Ry. The positive pressures at the measured lattice constant, 24 kbar and 38 kbar, are the PBE tendency to expand the lattice seen from the other side.

Two things go wrong with the cutoff: different quantities converge at different rates, and the stress picks up a Pulay term. Energy differences converge faster than energies, since most of the cutoff error sits near the nuclei and is the same in both structures, whereas absolute quantities, and quantities that probe the region near the nuclei, converge more slowly. The density vertex at large momentum transfer is the extreme case, treated in Appendix E.2. At a fixed cutoff the number of plane waves changes with the cell volume, so the basis is not the same at two lattice constants, and the computed stress carries a spurious negative contribution that vanishes only at convergence. A structural relaxation must therefore be done at a cutoff where the stress, and not only the energy, is converged.

E.1.5 Brillouin-zone sampling

Sums over the zone become averages over a uniform grid of $n_k = n_1 n_2 n_3$ points, possibly shifted from the origin as proposed by Monkhorst and Pack [14]. Appendix C.2 derived the error of such a grid, Eq. (C.16): it is the sum of the real-space Fourier components of the integrand at the superlattice vectors $\boldsymbol{R}$ of the $n_1 \times n_2 \times n_3$ block of cells that the grid implies. For an insulator these decay exponentially, so a modest grid suffices. In the representative silicon calculation at 60 Ry, the total energy changes by 1.6 meV per atom from 8^3 to 10^3, by 0.3 meV from 10^3 to 12^3 and by 0.06 meV from 12^3 to 14^3. Symmetry reduces the work: on a Γ-centered 12^3 grid, the 48 operations of the diamond structure leave 72 inequivalent points.

The grid needed depends on the quantity, however, and a converged total energy says little about a converged response. The electronic dielectric constant ε_∞ (Sect. 4.4), computed with density-functional perturbation theory (Appendix E.4) at the PBE lattice constant, shows this. It is 14.36 on an 8^3 grid, 13.57 on 10^3, 13.29 on 12^3 and 13.20 on 14^3; at the measured lattice constant it reaches 13.04 on 18^3. The total energy has converged to 1 meV per atom long before ε_∞ converges to one percent. The reason is that a response involves transitions between occupied and empty states, energy denominators $1/(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}})$ that are largest near the minimum direct gap, and a grid must resolve the region of the zone where they are largest. The same holds, more severely, for any quantity that probes a small region of the zone, such as the conduction-band minimum of silicon at 0.84 of the way to X or the valley edges of a phonon-assisted window (Appendix E.5).

E.1.6 Metals and smearing

In a metal the occupations jump from one to zero at the Fermi surface. The integrand of Eq. (C.16) then has a discontinuity, its Fourier components decay only as a power of $|\boldsymbol{R}|$, and a grid converges slowly and erratically. The remedy is to replace the step by a smooth function of width σ_{sm}, $f_{n\boldsymbol{k}} = \tilde{f}\big((\mu_{\mathrm{ch}} - \varepsilon_{n\boldsymbol{k}})/\sigma_{\mathrm{sm}}\big)$, which makes the integrand smooth on the scale of the grid. The smearing changes the answer, and the change must be controlled. The cleanest case is Fermi–Dirac smearing, $\tilde{f}(x) = 1/(\mathrm{e}^{-x} + 1)$, where σ_{sm} is a fictitious temperature. The quantity that a self-consistent calculation minimizes is then the free energy $F_{\mathrm{sm}} = E - \sigma_{\mathrm{sm}} S_{\mathrm{sm}}$, with the entropy S_{sm} of the occupations. Let the density of states be smooth on the scale of σ_{sm}, with $\mathcal{N}_{\mathrm{F}}$ per spin and volume at the Fermi level (Sect. 2.5), so $2\Omega\mathcal{N}_{\mathrm{F}}$ per cell. The Sommerfeld expansion then gives

$$E(\sigma_{\mathrm{sm}}) = E(\sigma_{\mathrm{sm}} = 0) + \frac{\pi^2}{3}\,\Omega\mathcal{N}_{\mathrm{F}}\,\sigma_{\mathrm{sm}}^2, \qquad F_{\mathrm{sm}}(\sigma_{\mathrm{sm}}) = E(\sigma_{\mathrm{sm}} = 0) - \frac{\pi^2}{3}\,\Omega\mathcal{N}_{\mathrm{F}}\,\sigma_{\mathrm{sm}}^2, \tag{E.3}$$

up to terms of order σ_{sm}^4, with $E(\sigma_{\mathrm{sm}} = 0)$ the energy per cell of the unsmeared metal. The first relation is the electronic heat of a metal, with σ_{sm} in the role of the temperature; the second follows from the electronic entropy at the same order, $S_{\mathrm{sm}} = (2\pi^2/3)\Omega\mathcal{N}_{\mathrm{F}}\sigma_{\mathrm{sm}}$. The errors of E and F_{sm} are equal and opposite, and the average $\frac{1}{2}(E + F_{\mathrm{sm}})$ is correct to order σ_{sm}^4. The forces, however, are derivatives of F_{sm}, not of E, so they carry the full σ_{sm}^2 error.

Other smearing functions remove the σ_{sm}^2 term from the start. The functions of Methfessel and Paxton [15] are Gaussians multiplied by Hermite polynomials, whose low moments vanish, at the price of occupations that are not confined to $[0, 1]$. The "cold smearing" of Marzari and Vanderbilt [16, Eq. (1)],

$$\frac{\mathrm{d}\tilde{f}}{\mathrm{d}x} = \frac{1}{\sqrt{\pi}}\,\exp\Big[-\Big(x - \frac{1}{\sqrt{2}}\Big)^2\Big]\,(2 - \sqrt{2}\,x), \tag{E.4}$$

is the broadening that replaces the delta function. The occupations it integrates to are never negative, although they can exceed one, and it is chosen so that the free energy

has no quadratic dependence on σ_{sm}, and neither do its derivatives, forces and stresses included [16]. A width of a few tenths of an eV is common for a structural calculation of a simple metal. Such a smearing must not be confused with a physical temperature. For a quantity that depends on the occupations near the Fermi surface, such as the density of states at μ_{ch} of aluminum (Sect. 7.2) or a Fermi-surface average of the electron–phonon coupling (Appendix E.5), the smearing and the grid must be converged together. The tetrahedron method of Appendix C.2 [17] avoids smearing altogether. For an insulator such as silicon, all of this is unnecessary: occupations are fixed at one and zero, and smearing an insulator is an error, discussed at the end of this section.

E.1.7 Self-consistency

The density that goes into V_{KS} and the density that comes out of the states must agree. A calculation starts from a guess, usually a superposition of atomic densities, and iterates. Feeding the output straight back in diverges in all but the smallest systems, and the reason is screening. Let the input density differ from the self-consistent one by δn_{in}. Its Hartree potential $v_{\mathrm{C}}\,\delta n_{\mathrm{in}}$ changes the states, and the output density differs by $\delta n_{\mathrm{out}} = R^0 v_{\mathrm{C}}\,\delta n_{\mathrm{in}}$, with R^0 the static independent-particle response of Sect. 4.6 (we drop the exchange–correlation kernel, which does not change the argument). The dielectric matrix of the random-phase approximation is $\varepsilon = 1 - v_{\mathrm{C}} R^0$ (Sect. 4.4); the same product in the other order, $\varepsilon^{\mathrm{T}} = 1 - R^0 v_{\mathrm{C}}$, acts on densities and has the same eigenvalues. The output error is $(1 - \varepsilon^{\mathrm{T}})\,\delta n_{\mathrm{in}}$. Linear mixing takes the new input as $n_{\mathrm{in}} + \beta_{\mathrm{mix}}(n_{\mathrm{out}} - n_{\mathrm{in}})$, and the error then evolves as

$$\delta n_{\mathrm{in}}^{(i+1)} = (1 - \beta_{\mathrm{mix}}\,\varepsilon^{\mathrm{T}})\,\delta n_{\mathrm{in}}^{(i)}. \tag{E.5}$$

The iteration converges if $|1 - \beta_{\mathrm{mix}}\varepsilon_\lambda| < 1$ for every eigenvalue ε_λ of the static dielectric matrix, that is for $\beta_{\mathrm{mix}} < 2/\max_\lambda \varepsilon_\lambda$. In an insulator the largest eigenvalue stays finite as the cell grows, of the order of ε_∞, and a fixed mixing parameter works. In a metal, the Thomas–Fermi form $\varepsilon(\boldsymbol{G}) = 1 + k_{\mathrm{TF}}^2/G^2$ (Sect. 4.4) grows without limit at small G, and the smallest G of a cell of length L is $2\pi/L$. A large metallic cell therefore needs $\beta_{\mathrm{mix}} \lesssim 2(2\pi/k_{\mathrm{TF}}L)^2$, and long-wavelength charge oscillates back and forth between iterations, a failure known as charge sloshing. The cure of Kerker [18] is to mix each Fourier component with $\beta_{\mathrm{mix}} G^2/(G^2 + k_{\mathrm{TF}}^2)$, with a wavevector of the order of k_{TF} in the denominator, which cancels the $1/G^2$ of the Thomas–Fermi dielectric function, so that $\beta_{\mathrm{mix}}\varepsilon(\boldsymbol{G}) \approx \beta_{\mathrm{mix}}$ for all $\boldsymbol{G}$. Production codes combine such a preconditioner with a quasi-Newton update that uses the history of previous iterations to build an approximate inverse of ε, the modified Broyden method of Johnson [19] or the closely related Pulay mixing.

The convergence threshold of the loop must match the use of the result. A total energy to 0.1 meV needs a loose threshold, since the energy is variational and its error is quadratic in the density error. A response calculation started from the converged ground state needs a tight one, since the first-order change of the states is computed against the ground-state Hamiltonian, and a residual error in the density appears linearly in the response (Appendix E.4).

E.1.8 Cost

The work of an iteration is dominated by applying the Hamiltonian, local and nonlocal parts, to the N_{b} bands at each of the $n_{\boldsymbol{k}}^{\mathrm{irr}}$ inequivalent $\boldsymbol{k}$ points, and by keeping those bands orthonormal. Applying the local part of the Hamiltonian to one band takes two FFTs, of order $N_{\mathrm{PW}} \log N_{\mathrm{PW}}$ each. The nonlocal projectors of the pseudopotential (Appendix E.2) take of order $N_{\mathrm{PW}} N_{\mathrm{proj}}$, with N_{proj} the number of projectors in the cell. Keeping the bands orthonormal, with the

diagonalization in the subspace they span, takes of order $N_{\mathrm{b}}^2 N_{\mathrm{PW}} + N_{\mathrm{b}}^3$ [20]. The cost per iteration is therefore

$$t \propto n_k^{\mathrm{irr}} \left(N_{\mathrm{b}} N_{\mathrm{PW}} \log N_{\mathrm{PW}} + N_{\mathrm{b}} N_{\mathrm{PW}} N_{\mathrm{proj}} + N_{\mathrm{b}}^2 N_{\mathrm{PW}} \right). \tag{E.6}$$

Each term grows with the size of the cell. The number of bands and of projectors is proportional to the number of atoms N_{at}, and so is N_{PW} at fixed cutoff, by Eq. (E.2). The last two terms therefore grow as N_{at}^3 and dominate in large cells, while n_k^{irr} falls as $1/N_{\mathrm{at}}$ for a fixed sampling density. Raising the cutoff increases N_{PW} as $E_{\mathrm{cut}}^{3/2}$, which is why the 3d electrons of germanium, at 80 Ry against 60 Ry, cost a factor $(4/3)^{3/2} \approx 1.5$ in plane waves on top of the larger number of bands. Spin–orbit coupling doubles both the length of each state, now a two-component spinor, and the number of bands, and multiplies the $N_{\mathrm{b}}^2 N_{\mathrm{PW}}$ term by eight. For a two-atom cell all this is small: the self-consistent calculation of silicon at 60 Ry on a 12^3 grid converges in eight iterations and about four seconds on four cores. Cost becomes a constraint only for supercells (Appendix E.10), hybrids, the many empty states of a quasiparticle calculation (Appendix E.3), and the dense grids of response and electron–phonon calculations (Appendices E.4 and E.5).

E.1.9 Bands and effective masses

Once the density is converged, it is held fixed and the Kohn–Sham Hamiltonian is diagonalized at any set of $\boldsymbol{k}$ points, on lines through the zone for a band structure, or on a fine grid around an extremum. This non-self-consistent step is cheap, because no iteration is needed, and it adds no approximation: the Kohn–Sham Hamiltonian is fixed by the converged density, and once the density is known it can be evaluated at any $\boldsymbol{k}$.

An effective mass is the inverse curvature of a band at an extremum, $(m^{*-1})_{\alpha\beta} = \partial^2 \varepsilon_{n\boldsymbol{k}} / \partial k_\alpha \partial k_\beta$ (Sect. 3.3). It is computed by finite differences, by $\boldsymbol{k} \cdot \boldsymbol{p}$ theory or by Wannier interpolation. The three-point finite-difference formula along a direction,

$$\frac{1}{m^*} \simeq \frac{\varepsilon(k_0 + h) - 2\varepsilon(k_0) + \varepsilon(k_0 - h)}{h^2}, \tag{E.7}$$

has a truncation error $h^2 \varepsilon''''/12$ from the next term of the Taylor series, and an error of about $4\delta\varepsilon/h^2$ if the eigenvalues carry an error $\delta\varepsilon$. A small step amplifies the noise of the eigenvalues, a large step samples the departure from a parabola, and the step must be chosen between the two. A least-squares parabola through several points around the extremum is more robust than Eq. (E.7), and varying the fitting range tests both errors at once. The $\boldsymbol{k} \cdot \boldsymbol{p}$ route (Sect. 3.3) uses perturbation theory with the computed velocity matrix elements, which gives the full tensor in one calculation at the extremum, provided enough bands are included and the velocities are corrected for the nonlocal pseudopotential (Appendices A.5 and E.2). Wannier interpolation (Sect. 3.6 and Appendix E.5) yields the band at any $\boldsymbol{k}$ and its derivatives analytically.

The masses inherit the errors of the functional. In $\boldsymbol{k} \cdot \boldsymbol{p}$ theory the curvature at an extremum comes from the coupling to the other bands at the same $\boldsymbol{k}$, each term weighted by the inverse energy separation. A mass set by a small direct gap, such as the conduction-band mass of GaAs at Γ, therefore follows the gap. A semilocal functional that puts the gap too low gives a mass that is too light, and a method that overestimates the gap gives one that is too heavy (Appendix E.3). Masses are checked against cyclotron resonance, and gaps against optical absorption or photoemission, before they are used in a rate.

Worked example E.1: Silicon from input to bands

The representative calculation uses the PBE functional, the SG15 norm-conserving pseudopotential with four valence electrons, a cutoff of 60 Ry (816 eV) for the states and 240 Ry for the density, fixed occupations of the four valence bands, a Γ-centered 12^3 grid (72 points by symmetry), scalar-relativistic potentials without spin–orbit coupling, and a stopping criterion of 10^{-12} Ry on the change of the total energy. A relaxation of the lattice at this cutoff gives $a = 5.477$ Å, 0.85 % above the measured room-temperature 5.431 Å (Sect. 3.1; the zero-temperature value of Table E.1, with the zero-point expansion removed, is 5.416 Å), in line with the PBE row of that table. At the relaxed lattice constant the valence-band maximum is the threefold Γ state, and the conduction-band minimum lies on the line from Γ to X. A scan of that line in steps of $0.005 \times 2\pi/a$ puts the minimum at $k_c = (0.840 \pm 0.003) \times 2\pi/a$, against the measured position near 0.85 (Sect. 3.3), and the indirect gap at 0.602 eV, about half of the measured 1.17 eV. The direct gap at Γ is 2.55 eV. At the measured lattice constant, with the same pseudopotential at 40 Ry, where band energies are converged to about 2 meV, the indirect gap is 0.557 eV: the 0.85 % expansion moves it by about 45 meV.

The longitudinal mass of the conduction valley follows from the same scan. A parabola fitted to the nine eigenvalues within $\pm 0.02 \times 2\pi/a$ of the minimum gives $m^*_{\parallel} = 0.955\, m_e$; widening the window to 13 and 17 points gives 0.958 and 0.951. The spread, below one percent, is the combined truncation and rounding error of the fit, and $m^*_{\parallel} = 0.95\, m_e$ is the PBE value at this lattice constant. The measured value is $0.92\, m_e$ [21]. The PBE mass is 4 % too heavy, while the gap is 49 % too small: the longitudinal mass of the silicon valleys is insensitive to the error of the fundamental gap, and the functional error that matters for this material is the gap itself.

E.1.10 Pitfalls

A semilocal functional can get the order of the bands wrong, and no later correction of the gap alone repairs that. In germanium, PBE without spin–orbit coupling at the measured lattice constant places the Γ conduction state 0.047 eV above the valence-band maximum and the L state at 0.191 eV, while in experiment the L valley is lowest, at 0.74 eV [10, Table I], and Γ lies above it. At the larger PBE lattice constant the gap closes and the crystal becomes a metal. A rigid upward shift of the conduction bands (Appendix E.3) keeps the wrong order, so any germanium band input needs bands that pass the check of Appendix E.14 (valley order, both gaps, masses at L and Γ), with spin–orbit coupling. A hybrid at the standard exchange fraction inverts Γ and L even with spin–orbit coupling, and a shift of the 4s-like orbitals fitted to both gaps gets the masses wrong (Appendix E.14); a quasiparticle calculation is the next route under that check. The spin–orbit splitting of the valence-band maximum is 0.296 eV in germanium against 0.044 eV in silicon (Sect. 3.3). The scalar-relativistic potentials of the worked example omit the splitting altogether; a calculation that includes it needs fully relativistic pseudopotentials and two-component spinors.

The gap error does not stay in the band structure: the dielectric constant of the representative calculation, 13.04, is 11 % above the measured 11.7, the overscreening of Appendix E.4. In germanium the bias becomes a failure: the near-metallic PBE state gives a dielectric constant of order 10^4 that does not converge with the grid. Either a corrected gap or the measured screening is used in such cases.

Smearing hides a metal. A semiconductor computed with smeared occupations, as some codes do by default, converges smoothly even when the functional has closed its gap, and the

result looks normal unless the band structure is inspected. With fixed occupations, the same calculation struggles to converge, which is the useful symptom.

The pseudopotential behind every number here is tested first, in Appendix E.2; the first correction the bands need, for most dark-matter thresholds, is to the gap, which Appendix E.3 supplies.

E.2 Pseudopotentials and all-electron reconstruction

The 1s orbital of silicon has a radius of about $a_0/Z = 0.038\,\text{Å}$. To resolve it, a plane-wave basis needs wavevectors up to a few times Z/a_0. With $3Z/a_0 = 42$ inverse bohr, Eq. (E.2) asks for a cutoff near 1800 Ry and 3×10^5 plane waves per state, 160 times the 2121 of the converged calculation of Appendix E.1, and the cost of each band grows at least in proportion. The ten core electrons do none of the bonding, and the shallowest of them, the 2p shell, is bound by about 100 eV (Appendix E.7). A pseudopotential removes them, together with the rapid oscillations that the valence states make in the core region to stay orthogonal to them, and replaces the nucleus and core by a soft effective potential that acts on smooth valence states. Sect. 3.5 introduced the idea, the norm-conservation conditions and the projector-augmented-wave (PAW) transformation. What the pseudopotential gives up is the true valence wavefunction near the nucleus, exactly where the density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})$ of a heavy-mediator rate is evaluated, at momentum transfers of tens of keV. There a pseudopotential that is excellent for total energies, bands and phonons can fail without any sign in those quantities.

E.2.1 The frozen core

A pseudopotential acts on the valence states through a local radial potential $V_{\text{loc}}(r)$ and a nonlocal part in the separable form of Kleinman and Bylander [22],

$$V_{\text{NL}} = \sum_{\kappa} \sum_{ij} |\tilde{p}_{\kappa i}\rangle D_{\kappa,ij} \langle \tilde{p}_{\kappa j}|. \tag{E.8}$$

The projector functions $\tilde{p}_{\kappa i}(\boldsymbol{r})$, the nonlocal counterparts of the PAW projectors of Sect. 3.5, are localized within a core radius r_c of nucleus κ and carry an angular momentum l each. The coefficients $D_{\kappa,ij}$ form a small matrix. V_{NL} is the nonlocal pseudopotential of Appendix A.5. The pseudo-wavefunctions it produces equal the all-electron valence wavefunctions outside the spheres of radius r_c and are smooth and nodeless inside them. For a norm-conserving pseudopotential [23] the charge of each reference state inside r_c is also the same, which fixes the scattering properties of the potential to first order in energy around the reference energies (Sect. 3.5).

The approximation behind all of this is the frozen core. The core electrons are assumed to keep the orbitals they have in the isolated atom, unaffected by the crystal, the valence electrons, or a displacement of the nucleus. It is not a component of the truncation label $\mathfrak{T}$ of Sect. 8.5, since it is not an expansion in a small parameter, and its errors are of three kinds: transferability, the missing core region of the valence states, and the missing core states. Transferability means that a potential built for the atom must describe the atom in the crystal, at other bond lengths and charge states. Tests control this error. The missing core region of the valence states leaves total energies and bands almost untouched but changes matrix elements of operators that probe short distances, such as $e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$ at large q. The all-electron reconstruction of this section repairs it. The core states themselves are absent from the spectrum, so a pseudopotential calculation of a response has no core edges. They are added back as atomic orbitals (Appendix E.7 and Sect. 10.6).

E.2.2 Construction

A pseudopotential is built from an all-electron calculation of the isolated atom, followed by the construction of pseudo-wavefunctions, the removal of the valence screening and the separable form. The atomic calculation is scalar-relativistic (that is, with the mass–velocity and Darwin terms of Sect. 3.5 but without spin–orbit coupling) and uses a reference configuration such as $[\mathrm{Ne}]\,3s^2 3p^2$ for silicon. The choice of valence comes with it. Silicon has an obvious split, four valence electrons above a core bound by about 100 eV and more. Germanium does not: its 3d shell lies about 29 eV below the valence-band maximum in the crystal [24, Table 1-1], overlaps the 4s and 4p orbitals in space, and is polarized by the crystal. A germanium potential either freezes the 3d shell, with four valence electrons, or puts it in the valence, with fourteen.

The pseudo-wavefunction is constructed for each angular momentum l. It equals the all-electron one beyond r_c, joins it smoothly at r_c, and is nodeless inside. The radius sets the trade-off. A large r_c gives a smooth function and a low cutoff, but a potential that must describe a large region of the atom with few parameters, and that transfers worse. Within these constraints, the shape inside r_c is chosen to make the plane-wave expansion converge fast. In the optimized norm-conserving construction of Hamann [25], the pseudo-wavefunction is expanded in a few spherical Bessel functions inside r_c. The coefficients minimize the residual kinetic energy above a wavevector q_{opt} [25, Eq. (1)],

$$E^{\mathrm{res}}(q_{\mathrm{opt}}) = \int_{q_{\mathrm{opt}}}^{\infty} \mathrm{d}q\, q^4\, |\phi(q)|^2, \qquad \phi(q) = 4\pi \int_0^{\infty} \mathrm{d}r\, r^2 j_l(qr)\, \phi(r), \tag{E.9}$$

subject to norm conservation and continuity at r_c. Here $\phi(r)$ is the radial pseudo-wavefunction and j_l the spherical Bessel function of Appendix C.3. Up to a constant, $q^4|\phi(q)|^2$ is the kinetic-energy density in momentum space, $q^2/2m_e$ times the radial measure $q^2|\phi(q)|^2$, so E^{res} is the kinetic energy missed by a basis truncated at q_{opt}. The cutoff of the solid is then set by q_{opt}, and the tail above it decays fast. The same construction uses two projectors per angular momentum, which reproduce the scattering of the all-electron atom at two energies, typically a bound state and a scattering state above it, rather than one.

Next, the valence part of the potential is removed. The potential that reproduces the pseudo-wavefunctions is a screened one: it contains the Hartree and exchange–correlation potentials of the valence density of the reference atom, which must be subtracted, since the crystal supplies its own. The Hartree potential is linear in the density and subtracts cleanly. The exchange–correlation potential is not, and where the core and valence densities overlap, subtracting the valence part alone leaves an error that changes with the valence configuration. The nonlinear core correction of Louie, Froyen and Cohen [26] keeps a smoothed partial core density in the exchange–correlation term for this reason. Finally, the semilocal potential is cast into the separable form Eq. (E.8), which reduces the cost of applying it from a double sum over plane waves to a few projections (Eq. (E.6)).

Ultrasoft pseudopotentials and the PAW method relax these conditions to lower the cutoff. Ultrasoft pseudopotentials [27] drop norm conservation and restore the missing charge through augmentation functions localized in the spheres, at the price of a generalized eigenvalue problem. The PAW method [28] keeps an explicit linear map from the pseudo-wavefunctions to the all-electron ones (Sect. 3.5), so that all-electron quantities can be evaluated after the calculation; Kresse and Joubert showed how ultrasoft potentials follow from it as an approximation [29]. Norm-conserving potentials remain the simplest to use in density-functional perturbation theory, in electron–phonon calculations and in many-body perturbation theory (Appendices E.3 to E.5), where every extra term of the ultrasoft and PAW formalisms has to be carried through each derivation [25].

The potentials of the representative calculations of Appendix E.1 come from the SG15 library of Schlipf and Gygi [2], generated with Hamann's construction, with parameters

optimized automatically against all-electron results. The construction parameters of the two potentials used here, as listed in their files, are inputs of the book's calculation. The silicon potential has four valence electrons, s and p channels with $r_c = 2.40$ and 3.56 bohr (1.27 Å and 1.88 Å), two projectors per channel, $q_{\text{opt}} = 5.11$ and 3.50 inverse bohr, and no nonlinear core correction. The germanium potential has fourteen valence electrons, s, p and d channels with r_c between 2.54 and 2.67 bohr, two projectors each, and q_{opt} up to 6.36 inverse bohr for the 3d channel. The squares of these wavevectors, 26 Ry and 12 Ry for silicon and 40 Ry for the germanium 3d, are the scale at which the residual kinetic energy has been made small. They are below the cutoffs at which the total energy converges in Table E.2, 60 Ry and 80 Ry, because the residual Eq. (E.9) decays above q_{opt} but does not vanish.

E.2.3 Testing

A pseudopotential is tested on the atom, for transferability, and in the solid before it is used. The logarithmic derivative $\mathrm{d}\ln R_l(r)/\mathrm{d}r$ of the radial solution at a radius just beyond r_c, plotted against energy for the all-electron and the pseudo atom, measures how well the potential scatters at energies other than its reference energies. The two curves must agree over the whole window of energies the calculation will use, which for a response calculation includes empty states tens of eV above the gap. A separable potential can also bind a spurious state, a ghost, below or among the physical ones, because the eigenstates of Eq. (E.8) are not ordered in energy by their number of nodes [25]. A ghost appears as a spurious step in the logarithmic derivative, or in the crystal as a flat band. The PseudoDojo tables were scanned for ghosts up to about 200 eV above the Fermi level, a range chosen for optical and *GW* calculations, and a few heavy elements still show one in that range, between 20 and 80 eV [30, Table 8]. Ghosts at still higher energies are unavoidable [30].

For transferability, the potential, built in one configuration, is applied to excited and ionized configurations of the atom, and the excitation energies are compared with all-electron values. In the solid, the most systematic test is the Δ gauge of Lejaeghere and coworkers. It is the root-mean-square difference between two equations of state $E(\mathcal{V})$ of the same crystal [7, Eq. (15)],

$$\Delta_{\text{gauge}} = \left[\frac{1}{\Delta\mathcal{V}} \int_{\mathcal{V}_0-\Delta\mathcal{V}/2}^{\mathcal{V}_0+\Delta\mathcal{V}/2} \mathrm{d}\mathcal{V} \left(E_1(\mathcal{V}) - E_2(\mathcal{V}) \right)^2 \right]^{1/2}, \qquad \Delta\mathcal{V} = 0.12\,\mathcal{V}_0, \tag{E.10}$$

with both curves set to zero at their minima. The window spans $\pm 6\,\%$ of the reference volume $\mathcal{V}_0$. It folds the errors of the volume, the bulk modulus and its pressure derivative into one number in meV per atom. Different all-electron codes agree with each other to 0.3–0.5 meV per atom on average over the periodic table. The norm-conserving PseudoDojo tables reach a mean of about 0.7 meV per atom against an all-electron reference (excluding the magnetic Cr, Mn and Fe), against 1.4 meV for earlier norm-conserving tables [30]. Direct comparisons with all-electron lattice constants and bulk moduli are the older form of the same test. For silicon in LDA, Hamann's optimized potential reproduces the all-electron lattice constant to the precision quoted, 10.21 bohr, and the bulk modulus to 1.3 %, 95.8 GPa against 97.1 GPa [25, Table I]. A potential must also be used with the functional with which it was generated; Hamann found that using a PBE-generated potential in an LDA calculation increased the differences from the all-electron results [25].

The standard tests are necessary but not sufficient for a dark-matter calculation. They probe the total energy and its derivatives with respect to volume, and the bands near the gap. Such a calculation also needs phonons (Appendix E.4), which probe the potential under small displacements of the nucleus. It needs empty states far above the gap, which probe the logarithmic derivatives at high energy. And it needs the density vertex at momentum

transfers up to tens of keV, which probes the core region. The last of these is not part of any standard test.

E.2.4 Semicore states

When a shell sits between the deep core and the valence, its treatment changes both the accuracy and the cost. The germanium 3d shell is the case the book needs. Keeping it in the valence improves transferability, since the shell responds to its environment; it puts the 3d states into the band structure, where they contribute to the response above an energy deposit of about 30 eV (Sect. 10.6); and it costs. The fourteen electrons need seven occupied bands per atom instead of two, and the localized d states need a higher cutoff. In Table E.2 germanium with the 3d in valence needs 80 Ry where silicon needs 60 Ry, and a cutoff of 30 Ry, enough to give silicon to 8 meV per atom, leaves the germanium energy almost 3 eV per atom too high. Gallium arsenide, with the Ga 3d in the valence, is harder still: in the same kind of calculation the energy still changes by 25 meV per atom between 100 and 120 Ry. The alternative, a frozen 3d shell with a nonlinear core correction, gives a cheaper calculation of the valence and conduction bands and a response without the 3d channel, which must then be added as core states.

E.2.5 The density vertex at large momentum

The crystal form factors of Sect. 10.1 are built from the density vertex

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) = \langle \psi_{m,\boldsymbol{k}+\boldsymbol{q}} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \psi_{n\boldsymbol{k}} \rangle, \tag{E.11}$$

between all-electron states. A pseudopotential calculation supplies the pseudo-states $\tilde{\psi}_{n\boldsymbol{k}}$ instead. Within PAW, the two are related by the transformation of Sect. 3.5 [28],

$$|\psi_{n\boldsymbol{k}}\rangle = |\tilde{\psi}_{n\boldsymbol{k}}\rangle + \sum_{\kappa j} \big(|\phi^{\mathrm{AE}}_{\kappa j}\rangle - |\phi^{\mathrm{PS}}_{\kappa j}\rangle\big)\langle \tilde{p}_{\kappa j} | \tilde{\psi}_{n\boldsymbol{k}}\rangle. \tag{E.12}$$

Here $\phi^{\mathrm{AE}}_{\kappa j}$ and $\phi^{\mathrm{PS}}_{\kappa j}$ are all-electron and pseudo partial waves that coincide outside the sphere of atom κ, and the projectors $\tilde{p}_{\kappa j}$ are dual to the pseudo partial waves inside it, $\langle \tilde{p}_{\kappa i} | \phi^{\mathrm{PS}}_{\kappa j}\rangle = \delta_{ij}$. The same map, with the partial waves of the atom the potential was built from, serves as an approximate reconstruction for a norm-conserving potential.

Inserting Eq. (E.12) into Eq. (E.11) gives the vertex as a pseudo part plus one-center corrections. The result is compact because the partial-wave differences vanish outside the spheres and the pseudo partial waves are complete inside them. The differences vanishing outside means that cross terms between a difference and the smooth part of the other state live inside the spheres only. Completeness inside a sphere means $\sum_j |\phi^{\mathrm{PS}}_{\kappa j}\rangle\langle \tilde{p}_{\kappa j}| = 1$ when acting on $\tilde{\psi}$, which lets the smooth part of each state be replaced by its partial-wave expansion there. Collecting terms [28],

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) = \langle \tilde{\psi}_{m,\boldsymbol{k}+\boldsymbol{q}} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \tilde{\psi}_{n\boldsymbol{k}}\rangle + \sum_{\kappa ij} \langle \tilde{\psi}_{m,\boldsymbol{k}+\boldsymbol{q}} | \tilde{p}_{\kappa i}\rangle \, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa} \, \Delta_{\kappa,ij}(\boldsymbol{q}) \, \langle \tilde{p}_{\kappa j} | \tilde{\psi}_{n\boldsymbol{k}}\rangle, \tag{E.13}$$

with the one-center matrix

$$\Delta_{\kappa,ij}(\boldsymbol{q}) = \langle \phi^{\mathrm{AE}}_{\kappa i} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}} | \phi^{\mathrm{AE}}_{\kappa j}\rangle - \langle \phi^{\mathrm{PS}}_{\kappa i} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}} | \phi^{\mathrm{PS}}_{\kappa j}\rangle, \tag{E.14}$$

where $\boldsymbol{x} = \boldsymbol{r} - \boldsymbol{\tau}_\kappa$ is measured from the nucleus. The sum over κ runs over the atoms of one cell: the states are normalized over the cell (Appendix A.5), by Bloch's theorem every cell contributes the same term, and the phase is written for the atom in the reference cell. The

plane wave about the nucleus expands in spherical harmonics and spherical Bessel functions, $e^{i\boldsymbol{q}\cdot\boldsymbol{x}} = 4\pi \sum_{lm} i^l j_l(qx) Y^*_{lm}(\hat{\boldsymbol{q}}) Y_{lm}(\hat{\boldsymbol{x}})$ (Appendix C.3), so each element of $\Delta_{\kappa,ij}(\boldsymbol{q})$ is a sum of radial integrals of $j_l(qx)$ against products of partial waves, times Gaunt coefficients (integrals of three spherical harmonics) from the angular parts. These are computed once per species and per $|q|$.

The size of the correction follows from the small-q expansion of Eq. (E.14). Its zeroth-order term is the difference of overlaps inside the sphere, which vanishes for norm-conserving potentials and is the augmentation charge of an ultrasoft or PAW potential. The next terms are differences of dipole and quadrupole moments of the partial-wave products inside the sphere, of relative size qr_c and $(qr_c)^2$, weighted by the fraction of the state that lies inside the sphere. The pseudo vertex is therefore accurate for $qr_c \ll 1$ and must be checked beyond. In the opposite limit, $q \gg q_{\text{opt}}$, the pseudo-wavefunctions have almost no Fourier components by construction, Eq. (E.9). The pseudo vertex then falls off much faster than the true one, whose large-q tail comes from the nodes and the cusp of the valence states near the nucleus. In between lies the range where most electron-recoil rates are decided.

A plane-wave calculation adds one more limit, which is exact. A state at $\boldsymbol{k}$ has components only for $|\boldsymbol{k}+\boldsymbol{G}| \le k_{\text{cut}}$, so the overlap of $\tilde{\psi}_{m,\boldsymbol{k}+\boldsymbol{q}}$ with $e^{i\boldsymbol{q}\cdot\boldsymbol{r}}\tilde{\psi}_{n\boldsymbol{k}}$ needs a pair of plane waves that both lie inside the cutoff sphere and differ by $\boldsymbol{q}$. Two points of a sphere of radius k_{cut} are at most $2k_{\text{cut}}$ apart, and the pseudo vertex vanishes identically for $|\boldsymbol{q}| > 2k_{\text{cut}}$. At 60 Ry this limit is 58 keV. Well below it, the vertex is already unconverged, and Sect. 3.5 gives the cutoff needed for a given q. Increasing the cutoff of a pseudopotential calculation makes the pseudo vertex converge to the pseudo answer, not to the all-electron one: beyond $q \sim 1/r_c$ only the reconstruction Eq. (E.13) moves the result toward the true vertex.

Griffin and coworkers carried out the reconstruction for silicon and germanium with PAW potentials [31]. They computed the pseudo-states with a cutoff of 1 keV (73.5 Ry) on a 10^3 grid and reconstructed the all-electron states up to 2 keV (147 Ry) [31, Sect. 2.1], which holds momentum components up to $(2m_e \times 2\,\text{keV})^{1/2} \approx 45\,\text{keV}$. The reconstructed states have large-momentum components that the pseudo-states lack. For a heavy mediator, which weights large q, the reconstruction makes the differential rate much higher than the pseudo result at energy deposits above about 15 eV, so that the rate integrated above a threshold is enhanced even for thresholds as low as about 10 eV; in germanium it is what makes the 3d electrons dominate the heavy-mediator rate. For a light mediator, dominated by small q, the effect is smaller [31, Sects. 1 and 3.1]. The deeper core states, 2p and below in silicon and 3d and below in germanium in their treatment, were modeled as atomic orbitals in Bloch sums, and the atomic description agreed with the reconstructed crystal states up to the reconstruction cutoff [31, Sect. 2.2].

E.2.6 Nonlocal sum rules

A nonlocal potential changes the commutators on which sum rules rest, and a test of a pseudopotential calculation against a sum rule must account for it. For a local potential, the double commutator of the Hamiltonian with $e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$ involves only the kinetic energy and gives the f-sum rule of Sect. 4.3. The plane wave $e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$ is unitary. Inserting a complete set of states $|m\rangle$ of the one-electron Hamiltonian H gives $\sum_m (\varepsilon_m - \varepsilon_n)|\langle m|e^{i\boldsymbol{q}\cdot\boldsymbol{r}}|n\rangle|^2 = \langle n|e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}[H, e^{i\boldsymbol{q}\cdot\boldsymbol{r}}]|n\rangle = \langle n|e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}He^{i\boldsymbol{q}\cdot\boldsymbol{r}} - H|n\rangle$, and the same with $-\boldsymbol{q}$ in place of $\boldsymbol{q}$. Adding the two,

$$\tfrac{1}{2}\sum_m (\varepsilon_m - \varepsilon_n)\left(|\langle m|e^{i\boldsymbol{q}\cdot\boldsymbol{r}}|n\rangle|^2 + |\langle m|e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}|n\rangle|^2\right) = \tfrac{1}{2}\langle n|e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}He^{i\boldsymbol{q}\cdot\boldsymbol{r}} + e^{i\boldsymbol{q}\cdot\boldsymbol{r}}He^{-i\boldsymbol{q}\cdot\boldsymbol{r}} - 2H|n\rangle. \tag{E.15}$$

The kinetic energy is shifted by the plane wave, $e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\boldsymbol{p}\,e^{i\boldsymbol{q}\cdot\boldsymbol{r}} = \boldsymbol{p}+\boldsymbol{q}$, so its contribution is $\tfrac{1}{2}\langle(\boldsymbol{p}+\boldsymbol{q})^2 + (\boldsymbol{p}-\boldsymbol{q})^2 - 2\boldsymbol{p}^2\rangle/2m_e = q^2/2m_e$. A local potential commutes with the plane wave

and drops out. A nonlocal one does not, and leaves

$$\frac{q^2}{2m_e} + \delta_{\mathrm{NL}}(\boldsymbol{q}), \qquad \delta_{\mathrm{NL}}(\boldsymbol{q}) = \tfrac{1}{2}\langle n|\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} V_{\mathrm{NL}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} + \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} V_{\mathrm{NL}} \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} - 2V_{\mathrm{NL}}|n\rangle, \tag{E.16}$$

on the right-hand side of Eq. (E.15). In the separable form Eq. (E.8), the conjugated potential $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} V_{\mathrm{NL}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ has the projectors $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\tilde{p}_{\kappa i}$, so δ_{NL} is a sum of projections and costs almost nothing to evaluate. At small q it is of order q^2, like the kinetic term, and it does not vanish when all bands are summed: the pseudo-Hamiltonian obeys a modified f-sum rule. The all-electron Hamiltonian, whose potential is local, obeys the unmodified one. For a single Bloch state the same sum rule, without the symmetrization in $\pm\boldsymbol{q}$, keeps a term linear in q set by the group velocity; Eq. (E.50) gives that form, and summed over a filled band it reduces to Eq. (E.15).

The representative silicon calculation shows the size of the effect. With the SG15 potential and the states at $\boldsymbol{k} = 0$, the left-hand side of Eq. (E.15), divided by $q^2/2m_e$, is 0.81 with 16 bands, 0.85 with 30, 0.86 with 50 and 0.88 with 100 at $|\boldsymbol{q}| = 0.1\,\text{Å}^{-1}$ along [100], and nearly the same at $0.5\,\text{Å}^{-1}$. The sum still rises slowly at 100 bands, so part of the deficit is band truncation, and part is δ_{NL}. The two must be separated, by computing Eq. (E.16) directly, before the sum rule can be used as a convergence test of the number of bands (Appendices E.6 and E.7).

The limit $q \to 0$ of the same algebra gives the velocity, which carries the nonlocal commutator $\mathrm{i}[V_{\mathrm{NL}}, \boldsymbol{r}]$ of Appendix A.5; Eq. (E.46) gives its plane-wave form. The density vertex, on the other hand, needs no such correction: a scalar potential couples to the density by multiplication in their result, as in a local Hamiltonian [32, Eq. (5)]. The pseudo vertex is therefore the correct vertex of the pseudo-Hamiltonian, and its only error is the missing core region, which Eq. (E.13) restores.

E.2.7 Silicon and the vertex

The SG15 silicon potential, with $r_c = 2.40$ and 3.56 bohr, has $1/r_c = 1.6$ and 1.0 keV, and its optimization wavevectors q_{opt} correspond to 19 and 13 keV. A dark-matter particle of 10 MeV at the maximum speed in the Earth frame, $v_{\max} = 2.65 \times 10^{-3}$ (Sect. 1.3), transfers at most $2m_\chi v_{\max} \approx 53$ keV, and the typical momentum transfer of electron recoil is of the order of $\alpha m_e = 3.7$ keV (Sect. 1.3). Much of the rate therefore comes from momentum transfers between $1/r_c$ and q_{opt}, where the pseudo vertex of this potential is neither guaranteed accurate nor negligible, and a heavy-mediator rate weights the upper end of that range. A potential with a smaller r_c pushes the problem to larger q at the price of a higher cutoff. Griffin and coworkers handle this range by reconstructing the all-electron states up to a cutoff set by the largest q needed, checking them against atomic orbitals for the shallowest core shell, and treating the deeper shells as atomic orbitals.

E.2.8 Pitfalls

Most failures of a pseudopotential in a dark-matter calculation come from using it outside the conditions it was tested for. One consequence of the frozen core lies outside the electronic response: core polarization by a displaced nucleus, which enters the large-q lattice response, is identically zero in a frozen-core calculation and has to be computed separately (Appendix E.4). With the representation of the states settled, the remaining error of the bands near the gap is that of the functional itself, the subject of Appendix E.3.

E.3 Quasiparticle corrections

A one-shot *GW* calculation started from LDA puts the silicon gap at 1.24 to 1.29 eV in the calculations of the 1980s and 1990s [33, Table I], and at 1.10 eV once it is converged with current methods [34, Sect. IV C], where LDA gives about 0.5 eV and the measured gap at low temperature is 1.17 eV [1, Eq. (11)]. For an electronic threshold this correction of more than half an eV is the largest a calculation makes. Its effect on the minimum mass a target can detect, the reason Kohn–Sham gaps are too small, and what a rigid shift of the conduction bands does to the velocities are in Sect. 3.4; the *GW* self-energy $\Sigma = \mathrm{i}\mathsf{G}W^{\mathrm{scr}}$, with $W^{\mathrm{scr}} = \varepsilon^{-1}v_{\mathrm{C}}$, and the electron–electron lifetime from $\operatorname{Im}\Sigma$ are defined in Sect. 4.8. This section computes the corrections in practice, from the simplest, a scissor shift, to a full one-shot G_0W_0 calculation and its self-consistent successors, and shows what they do to the energies, the lifetimes and the matrix elements of a dark-matter rate and which parameters must be converged, at what cost.

E.3.1 Computed quantities

The quantities are the quasiparticle energies $E_{\mathrm{qp}}(n\boldsymbol{k})$, the positions of the poles of the one-particle Green's function G (Sect. 4.7), with their residues z_{qp} and their widths. In practice they are computed as corrections to the Kohn–Sham energies of Appendix E.1, and the Kohn–Sham states are kept. A quasiparticle calculation changes the inputs of every later step (the threshold $E_{\min}(\boldsymbol{q})$ of Sect. 3.3, the energy denominators of a response, the screening) but not its approximation level.

E.3.2 Place on the hierarchy

On the hierarchy of Sect. 8.5, a response built from quasiparticle energies and Kohn–Sham states is still at $\mathcal{X} = \mathrm{IP}$ (or RPA, when the Coulomb interaction is resummed): the electron and the hole it leaves behind do not interact. Their attraction, the exciton, enters at $\mathcal{X} = \mathrm{BSE}$ (Sect. 4.9 and Appendix E.6), which uses the quasiparticle energies as input. The *GW* approximation is itself not an expansion in a small number. It is the first term of Hedin's equations [35], with the vertex function set to one [33, Eqs. (43)–(46) and (60)], and like the exchange–correlation functional its error is estimated by comparison, with measured gaps, with self-consistent variants and with vertex corrections, rather than by a next order.

E.3.3 The scissor shift

The simplest correction shifts every conduction band rigidly upward by Δ_{scis}, chosen so that the computed gap equals a measured one. For silicon in the PBE calculation of Worked example E.1, $\Delta_{\mathrm{scis}} = 1.170 - 0.602 = 0.568$ eV at the PBE lattice constant, and 0.613 eV at the measured one: the shift depends on the calculation it corrects. The approximation is better than it looks in silicon. A rigid shift reproduces the full *GW* correction of silicon to about 0.1 eV across the bands, and those of GaAs and diamond to 0.2 and 0.4 eV [36] [33, Sect. V B]. The accuracy in silicon is partly fortuitous: the energy dependence of Σ and its nonlocality would each make the correction strongly state-dependent, and in silicon the two almost cancel [33, Sect. V B].

The scissor operator $H + \Delta_{\mathrm{scis}} \sum_c |\psi_{c\boldsymbol{k}}\rangle\langle\psi_{c\boldsymbol{k}}|$, summed over the conduction states, has the eigenstates of H and shifted eigenvalues, and that fixes what it does to each ingredient of a rate. The threshold $E_{\min}(\boldsymbol{q})$ moves up by Δ_{scis}, and so does the energy of every transition. The density vertex $\rho_{cv}(\boldsymbol{k},\boldsymbol{q})$ is unchanged, since the states are. The velocity matrix elements are not: the operator is nonlocal, it adds $\mathrm{i}[\Delta_{\mathrm{scis}} \sum_c |\psi_c\rangle\langle\psi_c|, \boldsymbol{r}]$ to the velocity, and Sect. 3.4

showed that interband velocities are rescaled by $(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}} + \Delta_{\mathrm{scis}})/(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}})$. The f-sum rule is changed in the same way. With ρ_{cv} fixed and every interband energy raised, the sum $\sum_c (\varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}} + \Delta_{\mathrm{scis}})|\rho_{cv}(\boldsymbol{k},\boldsymbol{q})|^2$ exceeds its unshifted value by $\Delta_{\mathrm{scis}} \sum_c |\rho_{cv}|^2$, the extra term Eq. (E.16) generated by a nonlocal operator. A scissor-corrected response therefore violates the f-sum rule unless the transition strengths are rescaled with the energies, and a response used in an f-sum check must be built consistently (Appendix E.6).

The shift also changes the screening. In the independent-particle response each transition contributes $|\rho_{cv}|^2$ divided by its energy, so with ρ_{cv} fixed and the energies raised, $\varepsilon_\infty - 1$ decreases. The PBE value of silicon, 13.04 against the measured 11.7 (Appendix E.1), moves in the direction of the measurement. Whether it lands above or below 11.7 depends on the rest of the calculation, local fields included (Appendix E.6), so a scissored screening is not guaranteed to be closer to the measured one.

A scissor correction stands or falls with the target gap, the kind of gap and the band order. A measured gap at temperature T contains the renormalization of the gap by the lattice vibrations, including their zero-point motion. Shifting a static-lattice calculation to the measured gap and then adding the electron–phonon renormalization of Sect. 13.7 and Appendix E.8 counts it twice; in silicon the zero-point part is about 60 meV (Worked example E.7), small next to the gap but not against the window widths of Chap. 13. As for the kind of gap, an optical absorption edge sits below the fundamental gap by the exciton binding energy (Sect. 4.9), and a scissor to the optical edge puts the free-pair continuum too low. Finally, a scissor keeps the order of the Kohn–Sham bands, and where the functional gets it wrong, as PBE does in germanium (Appendix E.1), no shift repairs it. There the quasiparticle calculation itself is needed.

E.3.4 One-shot G_0W_0

The workhorse for semiconductors is the one-shot G_0W_0 method of Hybertsen and Louie [37]: G and W^{scr} are built once from the Kohn–Sham states, and the energies are corrected to first order.

The input is a Kohn–Sham calculation as in Appendix E.1, extended to many empty bands, of the order of a hundred per atom (the count is made below). The screening is built from the static independent-particle response $R^0_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega=0)$ of Sect. 4.6, a sum over pairs of occupied and empty states of products of the plane-wave matrix elements

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G}) = \langle \psi_{m,\boldsymbol{k}+\boldsymbol{q}} | \mathrm{e}^{\mathrm{i}(\boldsymbol{q}+\boldsymbol{G})\cdot\boldsymbol{r}} | \psi_{n\boldsymbol{k}} \rangle, \tag{E.17}$$

divided by the energy differences [38, Eqs. (7) and (8)]. These are the density vertices of Sect. 3.2 at the unreduced momentum $\boldsymbol{q}+\boldsymbol{G}$ (in the periodic gauge, $\psi_{m,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{G}} = \psi_{m,\boldsymbol{k}+\boldsymbol{q}}$). The same objects that enter a dark-matter rate enter the screening of the electron's own interaction, and the large-q caveats of Appendix E.2 apply to both. From R^0 follow the dielectric matrix of the random-phase approximation, $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\mathrm{C}}(\boldsymbol{q}+\boldsymbol{G}) R^0_{\boldsymbol{G}\boldsymbol{G}'}$ (Sect. 4.6), its inverse, and the screened interaction $W^{\mathrm{scr}}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}) = (\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q})\, v_{\mathrm{C}}(\boldsymbol{q}+\boldsymbol{G}')$ [38, Eq. (12)].

The frequency dependence of W^{scr} comes from a model or from a frequency grid. In the generalized plasmon-pole model of Hybertsen and Louie, each element of $\mathrm{Im}(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega)$ at $\omega > 0$ is a single delta function at a frequency $\tilde{\omega}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q})$, and the real part follows by the Kramers–Kronig relation. The two parameters of each element, the weight and the frequency of its pole, are fixed by two exact constraints: the static value $(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},0)$ just computed, and the f-sum rule for ε^{-1}, $\int_0^\infty \mathrm{d}\omega\, \omega\, \mathrm{Im}(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'} = -(\pi/2)\,\omega_p^2\,[(\boldsymbol{q}+\boldsymbol{G})\cdot(\boldsymbol{q}+\boldsymbol{G}')/|\boldsymbol{q}+\boldsymbol{G}|^2]\, n(\boldsymbol{G}-\boldsymbol{G}')/n(0)$, with $n(\boldsymbol{G})$ the Fourier components of the density [33, Eqs. (83)–(86)]. No parameter is adjusted. The model works because in sp semiconductors most of the spectral weight of $\mathrm{Im}\,\varepsilon^{-1}$ sits in the plasmon, near the free-electron plasma frequency of the valence electrons,

Table E.3: Minimum band gaps of silicon, germanium and GaAs in LDA, in one-shot *GW* calculations started from LDA, and from experiment, in eV, as collected in Ref. [33, Table I]. The *GW* column gives the range of the independent calculations of Refs. [36, 37, 39, 40]. The LDA gap of germanium is negative (the crystal is a metal); for GaAs the LDA gap is the one of Table E.1, 0.43 eV. The last two columns are the *GW* calculations of Shirley, Zhu and Louie [41], in which the core–valence exchange and correlation are treated with the local-density approximation ("core in LDA") or with a core-polarization potential, as collected in Ref. [33, Table II]: the silicon gap from Γ to the conduction minimum at 0.85 X, and the direct (Γ) and indirect (L) gaps of germanium, with spin–orbit coupling. The measured direct gap of germanium is 0.89 eV [33, Table II].

	LDA	*GW*	measured	*GW*, core in LDA	*GW*, core polarization
Si	0.52	1.24–1.29	1.17	0.85 X 1.29	0.85 X 1.16
Ge	< 0	0.65–0.75	0.744	Γ 0.53, L 0.58	Γ 0.85, L 0.73
GaAs	–	1.22–1.58	1.52	–	–

which for the eight valence electrons in the 40.05 Å^3 cell of silicon is $\omega_p = 16.6$ eV (Sect. 4.3). The alternative is to compute ε^{-1} at many frequencies, on the real axis or on the imaginary axis followed by an analytic continuation. The continuation is usually a rational (Padé) fit, and Appendix C.5 explains why such fits are unstable and must be checked against sum rules.

The self-energy is a sum over bands, $\boldsymbol{q}$ and $\boldsymbol{G}$ of products of the matrix elements Eq. (E.17), weighted by v_C and by the frequency structure of ε^{-1}. It is organized as a screened-exchange part, a sum over occupied states only, and a Coulomb-hole part, a sum over all bands, occupied and empty [38, Eqs. (19)–(22)]. Codes evaluate separately the bare exchange Σ_x, the Fock exchange of Sect. 3.4 built from the occupied Kohn–Sham states, and the correlation part Σ_c that $W^\text{scr} - v_\text{C}$ adds to it.

The quasiparticle equation, which corrects the energies, replaces v_xc in the Kohn–Sham equation by $\Sigma(E)$, evaluated at the quasiparticle energy itself. Keeping only the diagonal matrix elements in the Kohn–Sham basis, which assumes that the Kohn–Sham states are close to the quasiparticle states, and expanding $\Sigma(E)$ to first order about $\varepsilon_{n\boldsymbol{k}}$ gives [33, Eqs. (57) and (58)]

$$E_\text{qp}(n\boldsymbol{k}) = \varepsilon_{n\boldsymbol{k}} + z_\text{qp}\ \text{Re}\langle\psi_{n\boldsymbol{k}}|\Sigma(\varepsilon_{n\boldsymbol{k}}) - v_\text{xc}|\psi_{n\boldsymbol{k}}\rangle, \qquad z_\text{qp} = \left[1 - \frac{\partial\,\text{Re}\langle\psi_{n\boldsymbol{k}}|\Sigma(\omega)|\psi_{n\boldsymbol{k}}\rangle}{\partial\omega}\right]^{-1}_{\omega=\varepsilon_{n\boldsymbol{k}}}. \tag{E.18}$$

The residue z_qp is 0.7 to 0.8 in semiconductors [33, Sect. V B]. The slope of $\text{Re}\,\Sigma$ is negative, so $z_\text{qp} < 1$ reduces the correction: neglecting the energy dependence of Σ, as the static Coulomb-hole plus screened-exchange (COHSEX) approximation does, overestimates the gap correction [33, Sect. V B]. The off-diagonal elements, when kept, turn Eq. (E.18) into a small matrix diagonalization in the Kohn–Sham basis [38, Eqs. (17) and (18)], which changes the states as well as the energies.

Table E.3 shows what these G_0W_0 calculations achieve for the book's running materials. In silicon they correct most of the LDA error and overshoot the measured gap by 0.07 to 0.12 eV. In germanium and GaAs the independent calculations scatter by 0.1 and 0.36 eV, a measure of the sensitivity of the method to its numerical and physical choices rather than of its intrinsic accuracy. One such choice is the treatment of the core. The calculation of Shirley, Zhu and Louie [41] includes spin–orbit coupling and resolves both gaps. In it, approximating the exchange and correlation of the valence electrons with the 3d and deeper shells of germanium within LDA puts the direct gap below the indirect one, the wrong order. Treating the core

polarization explicitly restores the order and brings both gaps within 0.04 eV of experiment, and it lowers the silicon gap from 1.29 to 1.16 eV (Table E.3, last two columns). The core electrons must therefore be treated beyond LDA in the self-energy, either in the valence or through a core-polarization model, and not only in the density.

E.3.5 Self-consistency and the starting point

A one-shot calculation inherits its starting point. G_0W_0 on LDA states underestimates gaps systematically once core states are treated adequately, and it can fail qualitatively in transition-metal compounds [42]. Updating the energies in G and W^{scr} until they are self-consistent (eigenvalue self-consistency) removes part of the dependence. The quasiparticle self-consistent *GW* method of van Schilfgaarde, Kotani and Faleev goes further and determines the best static, Hermitian one-particle Hamiltonian from the self-energy itself, so that the result no longer depends on the starting functional [42]. Its errors are small and systematic. It overestimates the gaps of sp semiconductors by about 0.2 eV for transitions between Γ points and 0.1 eV for the others, and it slightly underestimates ε_∞. The reason given is that the random-phase approximation of W^{scr} omits the electron–hole attraction, which makes the screening too weak and the gap too large [42]. A starting point that is metallic, LDA or PBE germanium, is the case where the choice matters most: the first R^0 then contains intraband transitions that an insulator does not have, and either a hybrid-functional start or a self-consistent scheme is used.

Current practice differs from the calculations of Table E.3 in three ways [43, Sects. IV C–F, V and XI D]. The quasiparticle energies converge only as the inverse of the number of plane waves in the basis for the empty states and the response, so that the few hundred empty bands per atom of the early calculations can leave errors of tenths of an eV. Modern calculations extrapolate in $1/N_{\mathrm{PW}}$, or avoid the sum over empty states with a Sternheimer equation or a static remainder; converged in this way, G_0W_0 started from LDA gives 1.10 eV for silicon [34, Sect. IV C], below the measured gap rather than above it. The frequency integral is done by contour deformation or on the imaginary axis rather than with a plasmon pole. And the missing electron–hole attraction in W^{scr} is put back as a vertex correction: adding the ladder diagrams of the Bethe–Salpeter equation (Sect. 4.9) to the polarizability of quasiparticle self-consistent *GW* removes most of its overestimate of the gaps and its underestimate of ε_∞, and gives 1.08 eV for the silicon gap near X against the measured 1.17 eV [44, Table 1].

E.3.6 Lifetimes

The imaginary part of the self-energy gives the width of the quasiparticle. Near the pole, the linearized quasiparticle energy of Eq. (E.18) acquires an imaginary part $z_{\mathrm{qp}}\,\mathrm{Im}\,\Sigma$, and the probability that the electron is still in the state decays as $\exp(-2z_{\mathrm{qp}}|\,\mathrm{Im}\,\Sigma|\,t)$. The decay rate of the electron, its inverse lifetime, is therefore $2z_{\mathrm{qp}}|\,\mathrm{Im}\,\Sigma(E_{\mathrm{qp}}(n\boldsymbol{k}))|$. By the cutting rule of Sect. 4.8, $\mathrm{Im}\,\Sigma$ in the *GW* approximation is the golden-rule rate of the decay of the electron into a lower state with the emission of an excitation of the screened interaction: an electron–hole pair or a plasmon. An electron in the conduction band of an insulator can create an electron–hole pair only if it has the energy to cross the gap. Its energy above the conduction-band minimum must exceed E_{g}, and the conservation of crystal momentum raises the true threshold above this bound. Below it the electron–electron width vanishes, and only phonon emission (Sect. 6.3) relaxes the electron. Above it, impact ionization opens, and its rate grows with the phase space of the final pairs. These two rates, as functions of energy, are the input of the hot-carrier cascades that set the mean energy per electron–hole pair ε_{eh} (Appendix E.13).

The plasmon-pole model, with all the weight of $\operatorname{Im}\varepsilon^{-1}$ in delta functions at the plasmon energies, gives $\operatorname{Im}\Sigma = 0$ except at those energies, and it cannot produce lifetimes [33, Sect. III C]. A lifetime needs the full frequency dependence of ε^{-1} near the energies of the pairs that are created, which for silicon starts at the gap. A full-frequency calculation costs more than a plasmon-pole one by about the number of frequencies at which ε^{-1} is computed.

E.3.7 Matrix elements

The diagonal approximation of Eq. (E.18) keeps the Kohn–Sham states and changes the energies only, and its effect on matrix elements follows from that alone. The density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ is unchanged. The velocity changes, and its change can be computed exactly within the approximation. For eigenstates of a Hamiltonian H, the interband velocity is $\boldsymbol{v}_{mn} = \mathrm{i}\langle m|[H,\boldsymbol{r}]|n\rangle = \mathrm{i}(\varepsilon_m - \varepsilon_n)\langle m|\boldsymbol{r}|n\rangle$ for $m \neq n$. Between Bloch states at the same $\boldsymbol{k}$ the interband position matrix element is well defined, and this relation, applied to the Kohn–Sham Hamiltonian, gives it as $\langle m|\boldsymbol{r}|n\rangle = -\mathrm{i}\boldsymbol{v}^{\mathrm{KS}}_{mn}/(\varepsilon_m - \varepsilon_n)$. A quasiparticle Hamiltonian that is diagonal in the Kohn–Sham basis has the same $\langle m|\boldsymbol{r}|n\rangle$ and the energies E_{qp}, so

$$\boldsymbol{v}^{\mathrm{qp}}_{mn}(\boldsymbol{k}) = \boldsymbol{v}^{\mathrm{KS}}_{mn}(\boldsymbol{k})\,\frac{E_{\mathrm{qp}}(m\boldsymbol{k}) - E_{\mathrm{qp}}(n\boldsymbol{k})}{\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}}, \qquad m \neq n. \tag{E.19}$$

For a rigid shift this is the scissor rescaling of Sect. 3.4. For a state-dependent correction it rescales each transition by its own factor. The velocity vertex enters the absorption of dark matter at small momentum transfer and the optical limit of ρ_{mn} (Sect. 3.2). Eq. (E.19) keeps the optical limit of the density vertex, $\boldsymbol{q}\cdot\boldsymbol{v}_{mn}/(E_m - E_n)$, equal to that of the uncorrected calculation, as it must be when the states are unchanged. The intraband velocities, the group velocities, follow from the dispersion of the corrected bands. Since the corrections are computed on a coarse grid, they are interpolated to fine grids with Wannier functions (Appendix E.5), with the same gauge as every other interpolated quantity (Sect. 3.6).

Effective masses change with the dispersion. The $\boldsymbol{k}\cdot\boldsymbol{p}$ argument of Appendix E.1 connects a mass at a direct gap to the gap, and the quasiparticle self-consistent calculation of GaAs shows the connection: it overestimates the gap, 1.77 eV against 1.52 eV, and the conduction-band mass at Γ, $0.077\,m_e$ [42, Fig. 2 and text] against the measured $0.067\,m_e$ (Sect. 3.3).

E.3.8 Convergence and cost

A G_0W_0 calculation is converged in the number of empty bands, the dielectric cutoff, the grid of crystal momenta and the frequency treatment, and these parameters are coupled. The number of empty bands N_{b} enters the sums for R^0 and for Σ_{c}. The dielectric cutoff is the kinetic energy $E^{\varepsilon}_{\mathrm{cut}}$ below which the vectors $\boldsymbol{q}+\boldsymbol{G}$ of the matrix $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}$ are kept. The two are not independent. The highest empty band included should have an energy of about the dielectric cutoff, since a plane wave $\boldsymbol{q}+\boldsymbol{G}$ resolves transitions into states up to that energy [38, Sect. 4]. A free-electron count makes the link concrete: each plane wave with kinetic energy below E contributes one band below E, so the number of bands below E is about $N_{\mathrm{PW}}(E)$ of Eq. (E.2). For silicon, a dielectric cutoff of 10 Ry keeps 144 vectors $\boldsymbol{G}$ and asks for about 150 bands; 15 Ry keeps 265 and asks for about 270. The sum for the Coulomb-hole part of Σ_{c} converges much more slowly with N_{b} than the sum for R^0. A test that converges R^0 alone, or one parameter with the other held fixed, can therefore stop at a value far from the converged one [38, Sect. 4 and Fig. 8]. The grid of $\boldsymbol{k}$ and $\boldsymbol{q}$ points is the same for both. The divergence of $v_{\mathrm{C}}(\boldsymbol{q})$ at $\boldsymbol{q}\to 0$ is handled by averaging v_{C} and W^{scr} over the small cell around $\boldsymbol{q}=0$ and by computing the head and wings of $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}$ (its $\boldsymbol{G}=\boldsymbol{G}'=0$ element and the rest of the $\boldsymbol{G}=0$ row and column) at $\boldsymbol{q}\to 0$ from the optical limit of the vertex (Appendix E.6). The frequency

treatment is either a plasmon pole or a frequency grid. The bare exchange Σ_x is a separate case: it involves the unscreened interaction, whose $1/|\boldsymbol{q}+\boldsymbol{G}|^2$ falls off slowly, and its sum over $\boldsymbol{G}$ runs to the cutoff of the pair densities rather than to the dielectric cutoff.

The cost follows from the structure of the sums [38, Sect. 4]. The number of occupied and of empty bands, and the number of plane waves, each grow in proportion to the number of atoms N_{at}. Computing the matrix elements Eq. (E.17) for all pairs of bands is N_{at}^2 pairs times an FFT of size $N_{\mathrm{at}} \log N_{\mathrm{at}}$, so $N_{\mathrm{at}}^3 \log N_{\mathrm{at}}$. Summing them into $R^0_{\boldsymbol{G}\boldsymbol{G}'}$ is N_{at}^2 pairs for each of N_{at}^2 pairs of $\boldsymbol{G}$ vectors, so N_{at}^4, done as one dense matrix product; the inversion of ε is N_{at}^3 per $\boldsymbol{q}$ and per frequency. The self-energy of one state is N_{at}^3, and all states in an energy window add a factor N_{at}. A G_0W_0 calculation thus scales formally as the fourth power of the system size, against the third power of Eq. (E.6), with a prefactor set by the hundreds of empty bands. For the two-atom cell of silicon all of this remains modest. For the supercells of defects (Appendix E.10) and for the dense grids of a finite-momentum exciton calculation (Appendix E.6), it is the step that limits what can be computed.

Worked example E.2: The silicon gap from three corrections

A scissor, a hybrid functional and a G_0W_0 calculation each lead from the PBE calculation of Worked example E.1 to a gap for a silicon threshold. The scissor to the measured gap at zero temperature uses $\Delta_{\mathrm{scis}} = 0.568\,\mathrm{eV}$ at the PBE lattice constant, or 0.613 eV at the measured one. It reproduces the measured gap by construction, includes the zero-point renormalization through the measurement, and is accurate to about 0.1 eV for other bands [36]. The hybrid functional HSE gives 1.22 eV at its own lattice constant (Table E.1). A one-shot G_0W_0 calculation from LDA gives 1.24 to 1.29 eV in the older calculations of Table E.3 and 1.10 eV converged, from a static lattice. Two corrections separate it from the measurement: spin–orbit coupling, which these calculations omit, and the zero-point renormalization. Spin–orbit coupling splits the sixfold valence-band maximum (with spin) into a fourfold level at $+\Delta_{\mathrm{SO}}/3$ and a twofold level at $-2\Delta_{\mathrm{SO}}/3$; the weighted mean, $4\times\Delta_{\mathrm{SO}}/3 - 2\times 2\Delta_{\mathrm{SO}}/3 = 0$, is the level without spin–orbit coupling. The gap therefore closes by $\Delta_{\mathrm{SO}}/3$, 15 meV in silicon and, with $\Delta_{\mathrm{SO}} = 0.296\,\mathrm{eV}$, 99 meV in germanium (Sect. 3.3). The zero-point renormalization lowers the gap of a real crystal relative to the static lattice by about 60 meV and is computed separately (Appendix E.8). For a dark-matter threshold in silicon, the three agree to about 0.1 eV. That spread is the theory uncertainty of the threshold if the scissor is not anchored to a measurement, and Sect. 3.4 showed what it does to the minimum mass.

What a quasiparticle calculation does not include, the interaction of the excited electron with its hole, is the subject of the Bethe–Salpeter equation of Sect. 4.9 and of Appendix E.6.

E.4 Density-functional perturbation theory in practice

Every number that Sects. 5.2 to 5.6 attach to the lattice of silicon, from the 15.5 THz optical phonon at the zone center [45] to the Debye–Waller exponent at a momentum transfer of tens of keV, comes out of one kind of calculation: the first-order change of the Kohn–Sham potential when the nuclei of one species are displaced in a wave of crystal momentum $\boldsymbol{Q}$. Sect. 5.4 derived the equations of density-functional perturbation theory (DFPT). Here we use them: the same linear-response equations that give a phonon also give, with an electric field or a strain in place of the displacement, the Born charges, ε_∞ and the elastic constants, and, extended slightly, the changes of the empty states that a phonon-assisted amplitude needs (Sect. 13.2).

E.4.1 Outputs of a linear-response run

One perturbation is labeled by a phonon wavevector $\boldsymbol{Q}$, a nucleus κ and a Cartesian direction α. It is the bare electron–nucleus potential change $\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\text{ext}}$ of a displacement wave $u^{\alpha}_{\boldsymbol{R}\kappa} \propto \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$. DFPT solves, for every occupied state $\psi_{v\boldsymbol{k}}$, the projected Sternheimer equation of Sect. 5.4,

$$(H_{\text{KS}} + a_{\text{sh}} \hat{P}^{\text{occ}}_{\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}})\, |\partial\psi_{v\boldsymbol{k}}\rangle = -(1 - \hat{P}^{\text{occ}}_{\boldsymbol{k}+\boldsymbol{Q}})\, \partial_{\kappa\alpha,\boldsymbol{Q}} V_{\text{KS}}\, |\psi_{v\boldsymbol{k}}\rangle, \tag{E.20}$$

where $\hat{P}^{\text{occ}}_{\boldsymbol{k}+\boldsymbol{Q}} = \sum_{v'} |\psi_{v',\boldsymbol{k}+\boldsymbol{Q}}\rangle\langle\psi_{v',\boldsymbol{k}+\boldsymbol{Q}}|$ projects on the occupied states at $\boldsymbol{k}+\boldsymbol{Q}$, and the shift a_{sh}, any energy larger than the valence bandwidth, makes the operator on the left positive definite without changing the solution [46, Eqs. (30)–(31)], [47, Eq. (48)]. The solution $|\partial\psi_{v\boldsymbol{k}}\rangle$ has crystal momentum $\boldsymbol{k}+\boldsymbol{Q}$ and lies entirely in the empty manifold. From it the first-order density $\partial n_{\kappa\alpha,\boldsymbol{Q}}(\boldsymbol{r})$ is a sum of $\psi^{*}_{v\boldsymbol{k}}\, \partial\psi_{v\boldsymbol{k}}$ over occupied states and $\boldsymbol{k}$ [47, Eq. (49)], and the screened potential is

$$\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\text{KS}} = \partial_{\kappa\alpha,\boldsymbol{Q}} V_{\text{ext}} + v_{\text{C}} \star \partial n_{\kappa\alpha,\boldsymbol{Q}} + \frac{\delta v_{\text{xc}}}{\delta n}\, \partial n_{\kappa\alpha,\boldsymbol{Q}}, \tag{E.21}$$

with $v_{\text{C}}\star$ the convolution with the Coulomb kernel and $\delta v_{\text{xc}}/\delta n$ the static exchange–correlation kernel of the ground-state functional. Eqs. (E.20) and (E.21) are iterated to self-consistency, exactly as a ground-state calculation is, and the cost of one perturbation is of the same order as that of the ground-state density [46, Sect. II.C].

A converged run returns the screened potential, from which the vertex follows; the dynamical matrix; and, at $\boldsymbol{Q} = 0$, the response to a uniform field. The screened potential $\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\text{KS}}$ comes for all $3N_{\text{at}}$ perturbations at this $\boldsymbol{Q}$, with N_{at} the number of atoms per cell, and its matrix elements between Bloch states, weighted by the polarization vectors, are the electron–phonon vertex $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})$ of Eq. (A.26). The dynamical matrix $D^{\text{dyn}}(\boldsymbol{Q})$ of Eq. (A.23) is a second derivative of the energy that needs only the first-order density: by the $2n+1$ theorem of Sect. 5.4, the first-order wavefunctions fix the energy to third order. Diagonalizing it yields $\omega_{\boldsymbol{Q}\nu}$ and $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$. The response to a uniform electric field gives the electronic dielectric tensor ε_{∞} and the Born effective charges Z^{*}_{κ}, which we take up next.

E.4.2 Place on the hierarchy

On the truncation label $\mathfrak{T}$ of Sect. 8.5, a DFPT phonon is a point with $n_B = 0$ and $n_A = 0$. The nuclei are clamped while the electrons relax: the Born–Oppenheimer surface is expanded to second order about its minimum, and the electrons respond statically, as if the displacement were frozen. The electronic part of the response is the static limit of the time-dependent density-functional response of Sect. 4.6, with the adiabatic kernel of whatever functional the ground state used. Within density-functional theory this static response is formally exact for the exact functional, so the error of a DFPT phonon is the error of the functional, not of a perturbative truncation. As a result, DFPT frequencies inherit the strengths and the failures of the ground state (Appendix E.1). The calculation is at zeroth thermal order, $n_T = 0$, in the sense that the lattice constant, the force constants and the electronic screening are those of the static lattice; thermal expansion and the temperature dependence of the frequencies need the quasi-harmonic and anharmonic methods of Appendix E.8. The vertex g from Eq. (E.21) is the first-order coupling, $n_g = 1$ in the hierarchy; a rate that contains $|g|^2$ sits at $n_g = 2$.

E.4.3 Born charges and dielectric tensors

A uniform electric field $\boldsymbol{\mathcal{E}}$ cannot be written as a periodic potential, since $-e\boldsymbol{\mathcal{E}}\cdot\boldsymbol{r}$ grows without bound. DFPT avoids the problem because Eq. (E.20) needs only matrix elements of

the perturbation between different eigenstates, and those of $\boldsymbol{r}$ are well defined through the commutator,

$$\langle \psi_{c\boldsymbol{k}} | \boldsymbol{r} | \psi_{v\boldsymbol{k}} \rangle = \frac{\langle \psi_{c\boldsymbol{k}} | [H_{\mathrm{KS}}, \boldsymbol{r}] | \psi_{v\boldsymbol{k}} \rangle}{\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}}, \tag{E.22}$$

which is the relation between the position and velocity matrix elements of Sect. 3.2 [46, Eq. (37)]. For a local potential $[H_{\mathrm{KS}}, \boldsymbol{r}] = -\mathrm{i}\boldsymbol{p}/m_e$; a nonlocal pseudopotential (Sect. 3.5 and Appendix E.2) adds its own commutator, and omitting it biases every quantity built from the optical limit, the dielectric constant first among them. The projected state $\hat{P}^{\mathrm{c}}\boldsymbol{r}|\psi_{v\boldsymbol{k}}\rangle$ is itself the solution of a Sternheimer equation with $[H_{\mathrm{KS}}, \boldsymbol{r}]|\psi_{v\boldsymbol{k}}\rangle$ on the right-hand side [46, Eq. (44)]. With it, the field enters Eq. (E.20) like any other perturbation, and its self-consistent solution is the induced macroscopic polarization $\boldsymbol{P}$.

The macroscopic quantities then follow by differentiation. In the Heaviside–Lorentz units of the book (Appendix A.2), with $\boldsymbol{D} = \boldsymbol{\mathcal{E}} + \boldsymbol{P}$,

$$\varepsilon_\infty^{\alpha\beta} = \delta_{\alpha\beta} + \left. \frac{\partial P^\alpha}{\partial \mathcal{E}^\beta} \right|_{\boldsymbol{u}=0}, \qquad e\, Z_\kappa^{*,\alpha\beta} = \Omega \left. \frac{\partial P^\beta}{\partial u_\kappa^\alpha} \right|_{\boldsymbol{\mathcal{E}}=0} = \left. \frac{\partial F_\kappa^\alpha}{\partial \mathcal{E}^\beta} \right|_{\boldsymbol{u}=0}, \tag{E.23}$$

where Ω is the cell volume and $\boldsymbol{u}_\kappa$ a uniform displacement of sublattice κ; Ref. [46, Eqs. (94)–(96)] writes the same relations in Gaussian units, with 4π in front of $\partial P/\partial\mathcal{E}$. The equality of the two forms of Z_κ^* in Eq. (E.23), as the polarization per displacement and as the force per field, holds because both are the mixed second derivative of one energy, so a code may compute whichever is cheaper. The quantities ε_∞ and Z_κ^* are all that the nonanalytic part of the dynamical matrix needs (Sect. 5.5): in Heaviside–Lorentz units

$$D_{\kappa\kappa'}^{\mathrm{dyn,na},\alpha\beta}(\boldsymbol{Q} \to 0) = \frac{e^2}{\Omega\sqrt{M_\kappa M_{\kappa'}}} \frac{(Z_\kappa^* \cdot \boldsymbol{Q})_\alpha \, (Z_{\kappa'}^* \cdot \boldsymbol{Q})_\beta}{\boldsymbol{Q} \cdot \varepsilon_\infty \cdot \boldsymbol{Q}}, \tag{E.24}$$

which is Ref. [46, Eq. (98)] with $4\pi e_{\mathrm{G}}^2 \to e^2$ and the mass factors of Eq. (A.23). The same two tensors build the long-range vertex g^{L} of Sect. 6.2 and of Appendix E.5.

Measured frequencies give an independent check on Z^*. For a cubic crystal with two atoms per cell, $Z_1^* = -Z_2^* = Z^*$ times the identity, and the optical mode at $\boldsymbol{Q} \to 0$ has $\boldsymbol{e}_1 = (M_2/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}$ and $\boldsymbol{e}_2 = -(M_1/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}$. Its expectation value of Eq. (E.24) is $(e^2 Z^{*2}/\Omega\varepsilon_\infty)\,[e_1/\sqrt{M_1} - e_2/\sqrt{M_2}]^2 = e^2 Z^{*2}/(\Omega\varepsilon_\infty\mu_{\mathrm{red}})$, with $\mu_{\mathrm{red}} = M_1 M_2/M_{\mathrm{cell}}$ the reduced mass of the pair, for a longitudinal mode and zero for a transverse one. Hence

$$\omega_{\mathrm{L}}^2 - \omega_{\mathrm{T}}^2 = \frac{e^2 Z^{*2}}{\Omega\,\varepsilon_\infty\,\mu_{\mathrm{red}}}, \tag{E.25}$$

and with the measured ω_{L}, ω_{T} and ε_∞ (or ε_0, through the Lyddane–Sachs–Teller relation of Sect. 5.5) it fixes $|Z^*|$. Where a rate depends on a near-cancellation, as the electron-coupled scalar in GaAs does through the vertex $Z_\kappa - Z_\kappa^*$ (Sect. 11.8), this check matters more than any convergence parameter. A rate $\propto (Z_\kappa - Z_\kappa^*)^2$ has relative error $2\,\delta Z_\kappa^*/|Z_\kappa - Z_\kappa^*|$, so a one-percent error in Z_κ^* becomes a large error in the rate when $|Z_\kappa - Z_\kappa^*|$ is a small fraction of Z_κ^*. The computed Born charge and ε_∞ must then be tested against Eq. (E.25) with measured frequencies before they are used.

The Berry-phase theory of polarization (Sect. 5.5) is the other route to Z_κ^*. The electronic polarization of an insulator is

$$\boldsymbol{P}_{\mathrm{el}} = -\frac{e\,g_{\mathrm{deg}}}{(2\pi)^3} \sum_v \int_{\mathrm{BZ}} \mathrm{d}^3k \, \langle u_{v\boldsymbol{k}} | \mathrm{i}\nabla_{\boldsymbol{k}} | u_{v\boldsymbol{k}} \rangle, \tag{E.26}$$

with $g_{\mathrm{deg}} = 2$ for spin, the integrand a Berry connection of the periodic parts $u_{v\boldsymbol{k}}$, and the result equal to $-e g_{\mathrm{deg}}/\Omega$ times the sum of the Wannier centers in one cell [48, Eqs. (89)–(90)].

On a grid of $\boldsymbol{k}$ the derivative is replaced by the phase of a product of overlaps $\langle u_{v\boldsymbol{k}}|u_{v'\boldsymbol{k}+\Delta\boldsymbol{k}}\rangle$ along strings of points, which is gauge invariant. Since the choice of which Wannier center belongs to the home cell is arbitrary, $\boldsymbol{P}_{\mathrm{el}}$ is defined only modulo the quantum $eg_{\mathrm{deg}}\boldsymbol{R}/\Omega$ for a lattice vector $\boldsymbol{R}$ [48, Sect. V.A.1]. A Born charge is a difference of two polarizations at small displacements of opposite sign, $\pm\delta u$, and the two must sit on the same branch of the multivalued function; with δu a small fraction of a bohr this is automatic. Linear response and finite differences of Eq. (E.26) agree within numerical accuracy [46, Sect. II.D.2], and the Berry-phase route is the one that also works at finite fields and for ferroelectrics.

The acoustic sum rule for charges and the measured ε_∞ cost nothing to check. The sum rule, $\sum_\kappa Z^*_\kappa = 0$ (Sect. 5.5), holds exactly only at convergence; its violation on a finite $\boldsymbol{k}$ grid is the cleanest measure of how well ε_∞ and Z^* are converged. In the comparison of ε_∞ with experiment the functional dominates. Local and semilocal functionals put the gap too low (Sect. 3.4), the energy denominators of Eq. (E.22) come out too small, and the material is overscreened. The direction follows from the static limit of the Kramers–Kronig relation of Sect. 4.3,

$$\varepsilon_1(\omega=0) - 1 = \frac{2}{\pi}\int_0^\infty \frac{\varepsilon_2(\omega')}{\omega'}\,\mathrm{d}\omega' : \tag{E.27}$$

shifting the absorption spectrum ε_2 down in energy by a fraction of an eV, at roughly fixed weight, increases the integral. For silicon, a DFT calculation finds $\varepsilon_\infty = 12.89$ [49], against the measured room-temperature value of 11.7 [50] (Sect. 4.4) used throughout this book. When ε_∞ enters a rate through Eq. (E.24) or the Fröhlich vertex, the measured value is usually the better input; Ref. [49] rescales the screening of the electron–phonon vertex of silicon to the measured value it adopts, 11.94, for exactly this reason.

E.4.4 Force-constant interpolation

DFPT gives $D^{\mathrm{dyn}}(\boldsymbol{Q})$ at any $\boldsymbol{Q}$, but a dispersion, a density of states or a Debye–Waller sum needs it at thousands of points. The standard route computes it on a uniform $N_Q \times N_Q \times N_Q$ grid, transforms to real-space force constants $\Phi^{(2)}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})$ by the inverse of Eq. (A.23),

$$\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R}) = \frac{\sqrt{M_\kappa M_{\kappa'}}}{N_Q^3}\sum_{\boldsymbol{Q}\in\mathrm{grid}} D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q})\,\mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}, \tag{E.28}$$

and transforms back at any $\boldsymbol{Q}$ [46, Eq. (99)]. The discrete transform of Eq. (E.28) returns force constants that are periodic on the N_Q^3 supercell, so it is exact only if the true constants vanish beyond that supercell. In practice the vectors $\boldsymbol{R}$ are folded into the Wigner–Seitz cell of the supercell, with weights shared among equidistant images, and the grid is chosen large enough that the force constants have decayed at its edge. A rule of thumb is to include neighbors out to two or three bond lengths and to check the interpolated frequencies against direct DFPT runs at a few $\boldsymbol{Q}$ off the grid [46, Sect. II.D.3]. In diamond and zincblende crystals the force constants are long ranged along the $\langle 110\rangle$ chains of bonds [46, Sect. V.A.1], so silicon needs larger grids than its short bonds suggest.

Polar crystals need one more step. The nonanalytic term Eq. (E.24) decays in real space as R^{-3}, the dipole–dipole interaction, which no finite supercell captures. It is subtracted from the computed $D^{\mathrm{dyn}}(\boldsymbol{Q})$ in its Ewald-summed form, the short-range remainder is interpolated, and the dipole term is added back analytically at the target $\boldsymbol{Q}$; this restores the LO–TO splitting and its dependence on the direction of $\boldsymbol{Q}\to 0$ (Sect. 5.5).

The acoustic sum rule of Sect. 5.2, $\sum_{\boldsymbol{R}\kappa'} \Phi^{(2)}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R}) = 0$, holds for the exact force constants but not for the computed ones. The exchange–correlation potential is evaluated on a real-space grid that is not invariant under arbitrary translations, so a rigid shift of the crystal changes

the energy slightly. Codes impose the rule after the fact, by subtracting the violation from the on-site constants or by a minimal projection. The size of the correction is a diagnostic: the acoustic frequencies at Γ before it should be small compared with the lowest frequency the application needs, and a value that grows instead of shrinking with the cutoff signals an unconverged real-space grid. The correction is applied to D^{dyn} and to Z^*, never as a zeroing of g at $\boldsymbol{Q} \to 0$: the acoustic vertex vanishes there by its own sum rule (Sect. 6.2), and its leading term, linear in $\boldsymbol{Q}$, is physical.

The cost argument for DFPT over frozen phonons is a scaling law. A frozen-phonon calculation of the force constants out to a range R_{IFC} (in units of the cell size) needs supercells of $\sim R_{\mathrm{IFC}}^3$ cells, $N_{\mathrm{at}} R_{\mathrm{IFC}}^3$ atoms; with ground-state cost cubic in the number of atoms and $3N_{\mathrm{at}}$ independent displacements, the total scales as $3N_{\mathrm{at}}^4 R_{\mathrm{IFC}}^9$. DFPT works in the primitive cell on a grid of $\sim R_{\mathrm{IFC}}^3$ wavevectors, each costing $\sim N_{\mathrm{at}}^3$ per perturbation, which totals $\sim 3N_{\mathrm{at}}^4 R_{\mathrm{IFC}}^3$ [46, Sect. IV.B], a factor R_{IFC}^6 less. For a two-atom cell and force constants that extend several shells, the difference is decisive.

E.4.5 Displacement correlations from DFPT phonons

The Debye–Waller exponent $2W_\kappa(\boldsymbol{q}) = \boldsymbol{q} \cdot B_\kappa \cdot \boldsymbol{q}$ and the displacement correlation $\mathcal{C}_{\kappa\kappa'}(t)$ of Sect. 5.6 are sums over all phonons of the zone. Inserting the mode expansion Eq. (A.21) into $B_\kappa = \langle \boldsymbol{u}_\kappa \boldsymbol{u}_\kappa \rangle$ and using $\langle B_{\boldsymbol{Q}\nu} B_{-\boldsymbol{Q}\nu} \rangle = 2n_{\boldsymbol{Q}\nu} + 1$ in the harmonic thermal state,

$$B_\kappa^{\alpha\beta} = \frac{1}{N} \sum_{\boldsymbol{Q}\nu} \frac{2n_{\boldsymbol{Q}\nu} + 1}{2M_\kappa\, \omega_{\boldsymbol{Q}\nu}}\, e_{\kappa\nu}^{\alpha}(\boldsymbol{Q})\, e_{\kappa\nu}^{\beta}(\boldsymbol{Q})^*, \tag{E.29}$$

and $\mathcal{C}_{\kappa\kappa'}(t)$ is the same sum with the phases $\mathrm{e}^{\mp \mathrm{i}\omega_{\boldsymbol{Q}\nu} t}$ on the emission and absorption parts and the cross products $e_{\kappa\nu} e_{\kappa'\nu}^*$. These are post-processing sums: the frequencies and polarization vectors come from the interpolated force constants on a fine grid, and the cost is negligible. The approximation is $n_A = 0$, with the thermal occupation exact for the harmonic crystal; at high temperature or for light, soft lattices the anharmonic methods of Appendix E.8 replace the harmonic phonons in Eq. (E.29).

The catch is the zone center, where the acoustic frequencies vanish as $\omega = c_\nu Q$ and the summand of Eq. (E.29) diverges. The divergence is integrable in three dimensions, but a uniform grid samples it badly. How badly follows from replacing the sum over the grid cell around Γ by an integral over a sphere of the same volume, of radius Q_c with $(4\pi/3)Q_c^3 = (2\pi)^3/(N_Q^3 \Omega)$. With the acoustic polarization $\boldsymbol{e}_{\kappa\nu} = (M_\kappa/M_{\mathrm{cell}})^{1/2} \hat{\boldsymbol{e}}_\nu$ of Appendix A.6, an isotropic average over directions, and $\sum_\nu$ over the three acoustic branches, the contribution of that sphere to one diagonal component is

$$\Delta B^{\alpha\alpha} = \frac{\Omega}{2\pi^2} \int_0^{Q_c} Q^2 \,\mathrm{d}Q\, \frac{1}{3} \sum_\nu \frac{2n_{\boldsymbol{Q}\nu} + 1}{2M_{\mathrm{cell}} c_\nu Q} \simeq \begin{cases} \dfrac{\Omega\, Q_c^2}{24\pi^2 M_{\mathrm{cell}}} \displaystyle\sum_\nu \frac{1}{c_\nu}, & T = 0, \\[2ex] \dfrac{\Omega\, Q_c\, T}{6\pi^2 M_{\mathrm{cell}}} \displaystyle\sum_\nu \frac{1}{c_\nu^2}, & T \gg c_\nu Q_c, \end{cases} \tag{E.30}$$

where the high-temperature form uses $2n + 1 \simeq 2T/\omega$. Since $Q_c \propto 1/N_Q$, the error of a calculation that simply omits the Γ cell falls as N_Q^{-2} at zero temperature but only as N_Q^{-1} at room temperature. For silicon, with $\Omega = a^3/4 = 40.05\,\text{Å}^3$, M_{cell} two silicon masses, a representative sound speed of $6\,\mathrm{km\,s^{-1}}$ for all three branches (the range is 3 to $10\,\mathrm{km\,s^{-1}}$ across materials, Sect. 1.3) and an 8^3 grid, $Q_c = 0.142\,\text{Å}^{-1}$ and the room-temperature term is $\Delta B^{\alpha\alpha} \approx 3.6 \times 10^{-4}\,\text{Å}^2$, while the zero-temperature term is twenty times smaller, $1.9 \times 10^{-5}\,\text{Å}^2$. (The high-temperature form applies since $c\,Q_c$ corresponds to 65 K.) The remedy is to add

Eq. (E.30), evaluated with the computed sound speeds, as an analytic correction, or to use a grid fine enough that the residual is below the target accuracy. The same care applies to $\mathcal{C}_{\kappa\kappa'}(t)$ at long times, which are controlled by the smallest frequencies.

E.4.6 Wavefunction changes beyond the density

The vertex of Eq. (A.26) needs only $\partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}}$ and two Bloch states, so every DFPT run already contains it [47, Sect. III.B.3]. Codes write the potentials to disk for each irreducible $\boldsymbol{Q}$ of the coarse grid, one complex function on the real-space grid per perturbation, and a later step forms the matrix elements and interpolates them (Appendix E.5). The storage is modest by modern standards: for a two-atom cell, six perturbations per $\boldsymbol{Q}$, each a complex function on a real-space grid of $\sim 10^5$ points, are about 10 MB per wavevector, so a few hundred irreducible wavevectors fit in a few gigabytes.

A transition amplitude needs more than the density does. The density vertex of Sect. 3.2, $\rho_{cv}(\boldsymbol{k},\boldsymbol{q}) = \langle \psi_{c,\boldsymbol{k}+\boldsymbol{q}} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \psi_{v\boldsymbol{k}} \rangle$, changes under a frozen displacement of crystal momentum $\boldsymbol{Q}$ by

$$\partial\rho_{cv} = \langle \partial\psi_c | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \psi_v \rangle + \langle \psi_c | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \partial\psi_v \rangle, \tag{E.31}$$

which is the adiabatic (Herzberg–Teller) form of the phonon-assisted amplitude of Sect. 13.2, valid to relative order $\omega_{\boldsymbol{Q}\nu}/\delta_{\mathrm{off}}$, with δ_{off} the energy by which the intermediate states are off shell (Sect. 13.2). Both terms need the complete first-order change of an individual state, while Eq. (E.20) returns only part of one of them. The full change of an occupied state is

$$|\partial\psi_{v\boldsymbol{k}}\rangle = \sum_{m \notin \deg(v\boldsymbol{k})} |\psi_{m,\boldsymbol{k}+\boldsymbol{Q}}\rangle \, \frac{\langle \psi_{m,\boldsymbol{k}+\boldsymbol{Q}} | \partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}} | \psi_{v\boldsymbol{k}} \rangle}{\varepsilon_{v\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}}, \tag{E.32}$$

with the states degenerate with $\psi_{v\boldsymbol{k}}$ excluded. DFPT computes the part with m empty, by Eq. (E.20), without ever forming the sum. The part with m occupied, $\hat{P}^{\mathrm{occ}}_{\boldsymbol{k}+\boldsymbol{Q}} |\partial\psi_{v\boldsymbol{k}}\rangle$, is left out on purpose: these terms drop out of the density change once the $\boldsymbol{Q}$ and $-\boldsymbol{Q}$ components of the Hermitian perturbation are combined, since exchanging the roles of the two occupied states flips the sign of the denominator [46, Sect. II.C], [47, Eq. (48)]. In Eq. (E.31) they do not cancel, because ψ_c singles out one state. They are, however, cheap. The sum runs over the few occupied bands at $\boldsymbol{k}+\boldsymbol{Q}$, and the numerators are the occupied–occupied matrix elements of the potential that DFPT has already stored. Two cautions apply. The denominators $\varepsilon_{v\boldsymbol{k}} - \varepsilon_{v',\boldsymbol{k}+\boldsymbol{Q}}$ vanish wherever two valence bands at $\boldsymbol{k}$ and $\boldsymbol{k}+\boldsymbol{Q}$ cross, and near such crossings the adiabatic form fails: there the dynamical denominators $\varepsilon_{v\boldsymbol{k}} - \varepsilon_{v',\boldsymbol{k}+\boldsymbol{Q}} \pm \omega_{\boldsymbol{Q}\nu}$ of the full amplitude (Sects. 13.2 and 13.6) must be used instead. And at $\boldsymbol{Q} = 0$ the degenerate subspace has to be treated in the same gauge as the states that enter ρ_{cv} (Appendix A.5).

The change of an empty state is not produced by standard DFPT at all, since empty states do not contribute to the density. Its equation is a Sternheimer equation with the converged potential on the right,

$$(H_{\mathrm{KS}} - \varepsilon_{c\boldsymbol{k}}) |\partial\psi_{c\boldsymbol{k}}\rangle = -(1 - \hat{P}^{\mathrm{deg}}_{c}) \, \partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}} \, |\psi_{c\boldsymbol{k}}\rangle, \tag{E.33}$$

with $\hat{P}^{\mathrm{deg}}_{c}$ the projector on the states at $\boldsymbol{k}+\boldsymbol{Q}$ degenerate with $\psi_{c\boldsymbol{k}}$. No self-consistency is needed, because ∂V_{KS} is already the screened potential of the occupied manifold, and that is the main saving. The operator on the left, however, is indefinite: $\varepsilon_{c\boldsymbol{k}}$ lies inside the spectrum of H_{KS}, with all the valence states and part of the conduction bands below it, and the conjugate-gradient solvers used for Eq. (E.20) do not converge on an indefinite operator. One way out splits the space: a window $\mathcal{W}$ of bands at $\boldsymbol{k}+\boldsymbol{Q}$, containing every state below $\varepsilon_{c\boldsymbol{k}}$ and the conduction bands up to some energy above it, is summed explicitly

with Eq. (E.32), where the Wannier-interpolated vertices of Appendix E.5 make the terms cheap. The remainder obeys

$$\hat{P}^{\perp}_{\mathcal{W}}(H_{\mathrm{KS}}-\varepsilon_{c\boldsymbol{k}})\hat{P}^{\perp}_{\mathcal{W}}\,|\partial\psi^{\perp}_{c\boldsymbol{k}}\rangle=-\hat{P}^{\perp}_{\mathcal{W}}\,\partial_{\kappa,\boldsymbol{Q}}V_{\mathrm{KS}}\,|\psi_{c\boldsymbol{k}}\rangle, \tag{E.34}$$

with $\hat{P}^{\perp}_{\mathcal{W}}$ the projector on the complement of the window. Every state left in the complement lies above $\varepsilon_{c\boldsymbol{k}}$, so the projected operator is positive definite and the standard solvers apply. The remainder carries the high-energy intermediate states, which matter for large momentum transfers, where $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ couples to bands far above the gap. The window is a convergence parameter: the sum of the explicit and the Sternheimer parts must not depend on it.

The same equations at finite frequency give dynamical response without empty states. For a perturbation oscillating as $\mathrm{e}^{-\mathrm{i}\omega t}$, first-order time-dependent perturbation theory (Sect. 2.4) leads to two components of the change of each occupied state,

$$(H_{\mathrm{KS}}-\varepsilon_{v\boldsymbol{k}}\mp\omega)|\partial\psi^{\pm}_{v\boldsymbol{k}}\rangle=-(1-\hat{P}^{\mathrm{occ}})\,\partial V_{\mathrm{KS}}(\pm\omega)\,|\psi_{v\boldsymbol{k}}\rangle, \tag{E.35}$$

with the density change built from both and the potential updated as in Eq. (E.21) with an adiabatic kernel. Below the gap the operators remain positive definite on the empty manifold, because $\varepsilon_c-\varepsilon_v-\omega>0$ for every pair, and the solution is as cheap as a static one. Above the gap a small broadening η_{b}, $\omega\to\omega+\mathrm{i}\eta_{\mathrm{b}}$, is added and the linear systems become complex and non-Hermitian. Frequency-dependent Sternheimer calculations of this kind give the inverse frequency moments and the drag mass of Sect. 12.5, and the screened interaction in Sternheimer formulations of the methods of Appendix E.3.

E.4.7 Elastic constants

The sound speeds of Sect. 11.1 follow from the elastic tensor C_{ijkl} through the Christoffel equation of Appendix C.18; for a cubic crystal along [100] it reduces to $\rho_T c^2_{\mathrm{LA}}=C_{11}$ and $\rho_T c^2_{\mathrm{TA}}=C_{44}$ in Voigt notation. Three routes give C_{ijkl}. The slopes of the interpolated acoustic branches are the cheapest and the least accurate, since the small-Q part of the dispersion is precisely where the Fourier interpolation of Eq. (E.28) is weakest, and the slopes converge only slowly with N_Q. Finite strains are the most reliable: strains of a few tenths of a percent of both signs are applied, the internal coordinates are relaxed at each strain, and the stress is fitted. The internal relaxation matters for any structure with internal degrees of freedom, and in the diamond structure it affects C_{44} because the two sublattices shift against each other under shear. Since relaxation can only lower the energy at fixed strain, the relaxed constants are never larger than the clamped ones. DFPT with strain as the perturbation returns the clamped-ion tensor and the internal-strain couplings in one run, from which the relaxed tensor follows by eliminating the internal displacements [46, Sect. II.E.1].

Codes differ in the order of the shear rows. Voigt notation assigns $4\leftrightarrow yz$, $5\leftrightarrow xz$, $6\leftrightarrow xy$, but several codes and databases store the shear components in the order xy, yz, zx. For a cubic crystal the error is invisible, since $C_{44}=C_{55}=C_{66}$. For a lower-symmetry crystal the permuted tensor still looks reasonable, but the sound speeds along the low-symmetry directions come out wrong. A cheap check is to compute $\rho_T c^2$ from the tensor along one direction and compare it with the slope of the computed dispersion along the same direction.

E.4.8 Convergence parameters and cost

Five numerical parameters control a DFPT calculation. Each must be converged on the quantity of interest, not on the total energy, and with a tolerance fixed before the scan. The plane-wave cutoff of Appendix E.1 sets the basis; the test is the frequency at Γ and at one

zone-boundary point at two cutoffs, not the total energy. The $\boldsymbol{k}$ grid sets the Brillouin-zone integrals in Eq. (E.20); for an insulator the density and the frequencies converge on modest grids, but ε_∞ and Z^* need noticeably finer ones, because the response to a field involves the velocity matrix elements and energy denominators of Eq. (E.22) at every $\boldsymbol{k}$, and these converge slowly where the direct gap is small [46, Sect. II.C.2]. Metals need smearing of the occupations (Appendix E.1) and a correspondingly finer grid. The $\boldsymbol{Q}$ grid sets the range of the interpolated force constants and is tested by direct runs off the grid. The self-consistency threshold of Eq. (E.21) must be much tighter than in a ground-state run. An error in ∂n enters the force constants linearly, and where a frequency is small, as for the acoustic modes near Γ or a soft mode, a small error in the force constant is a large relative error in the frequency, or makes it imaginary.

The lattice constant is a parameter in its own right. Phonons are usually computed at the theoretical equilibrium of the functional. In diamond and zincblende crystals the forces vanish by symmetry at any lattice constant, but only at the equilibrium does the stress vanish, so that the second derivatives are taken at a minimum of the same energy surface. Each frequency moves with the lattice constant through its Grüneisen parameter (Appendix E.8); the generalized-gradient approximation, which expands lattice constants, lowers the branches with positive Grüneisen parameter, and for silicon the local-density approximation already reproduces the measured dispersion closely while the generalized-gradient approximation slightly worsens it [46, Sect. V.A].

The cost of a complete phonon calculation is the number of irreducible wavevectors on the $\boldsymbol{Q}$ grid, times $3N_{\mathrm{at}}$ perturbations (fewer, by symmetry, at high-symmetry wavevectors), times the cost of a ground-state calculation with the reduced symmetry of the perturbed crystal. That reduced symmetry is the hidden factor: a perturbation at a general $\boldsymbol{Q}$ leaves few symmetry operations, so the $\boldsymbol{k}$ sum runs over many more points than in the ground state. An N_Q^3 grid has about $N_Q^3/48$ irreducible points for a cubic crystal when N_Q is large; an 8^3 grid for silicon has 29, a count that follows from applying the 48 operations of the cubic group to the grid points and counting the distinct orbits. With six perturbations per wavevector, fewer at the high-symmetry points, and the three field perturbations at Γ, a complete silicon calculation on that grid costs at most about 180 self-consistent linear-response solutions, each of the order of one ground-state calculation at reduced symmetry. The cost grows with the fourth power of the number of atoms per cell [46, Sect. IV.B].

Worked example E.3: A DFPT calculation of silicon

We assemble the settings of two published silicon calculations and the checks they pass. The first used the local-density approximation with plane-wave pseudopotentials, a 35 Ry cutoff and a 14^3 grid for the ground-state density, and computed the dynamical matrices and the potential changes on a 6^3 grid of wavevectors [51]. The second used the generalized-gradient approximation at 35 Ry (45 Ry for the local-density approximation), again a 6^3 wavevector grid, and found that an 18^3 grid of $\boldsymbol{k}$ points was needed for the Born charges to vanish [49]. A 35 Ry cutoff corresponds to plane waves up to $|\boldsymbol{G}| = (2m_e \times 476\,\mathrm{eV})^{1/2} = 22\,\mathrm{keV}$, between five and six times the shortest reciprocal-lattice vector $|\boldsymbol{G}_{111}| = \sqrt{3}\,2\pi/a = 3.95\,\mathrm{keV}$ (Sect. 1.3), which is ample for phonons but, as Appendices E.6 and E.7 explain, not automatically for a density vertex at large momentum transfer.

The vanishing of the Born charges is a symmetry statement that the numerics must reproduce. The two atoms of the diamond structure are exchanged by inversion through the bond center. Z^*_κ is the derivative of a polar vector with respect to a polar vector, which is even under inversion, so $Z^*_1 = Z^*_2$. Each atom sits at a site of tetrahedral symmetry, so each tensor is a multiple of the identity. Charge neutrality, $Z^*_1 + Z^*_2 = 0$, then forces

both to vanish. On a coarse $\boldsymbol{k}$ grid they do not, and since Z^* enters Eq. (E.24), a residual Born charge produces a spurious LO–TO splitting in a crystal that has none. The $\boldsymbol{k}$ grid is therefore fixed by the Born charge, not by the frequencies.
The dielectric constant is the second check, and here the functional limits the result: DFT gives $\varepsilon_\infty = 12.89$ [49], the measurement 11.7 [50]. The PBE calculation of Worked example E.1 reaches 13.04 at the measured lattice constant on an 18^3 grid, within about one percent of that value. The overestimate, 10 % for 12.89 and 11 % for 13.04, is the overscreening of Eq. (E.27). It is not a convergence error, and it does not shrink with any numerical parameter. The frequencies are the third check. The measured optical frequency at Γ is 15.5 THz [45], and the local-density dispersion agrees with neutron scattering across the zone [46, Sect. V.A]. A calculation at the generalized-gradient lattice constant comes out soft for the branches with positive Grüneisen parameter and hard for those with negative ones.

The vertices for the coarse grids are the input of the Wannier interpolation in Appendix E.5, which turns these few hundred potential changes into the millions of matrix elements that a rate needs.

E.5 Wannier functions in practice

The first-principles phonon-assisted absorption spectrum of silicon was converged to an energy resolution of 14 meV with $40 \times 40 \times 40$ grids for both the electron wavevector $\boldsymbol{k}$ and the phonon wavevector $\boldsymbol{Q}$ [51]. That is $64{,}000^2 \approx 4 \times 10^9$ pairs $(\boldsymbol{k}, \boldsymbol{Q})$, each with a vertex $g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q})$ for every pair of bands and every branch. A direct DFPT calculation (Appendix E.4) costs about one ground-state calculation per perturbation and wavevector, so 64,000 wavevectors are already out of reach, and the matrix elements between all those states even more so. The same study computed the vertices directly on a 6^3 grid and produced the rest by Wannier interpolation. Sect. 3.6 showed why such an interpolation is exact for quantities of finite range in the Wannier basis, and Sect. 6.2 why the electron–phonon vertex is such a quantity. In practice the work is in choosing and testing the Wannier functions and in keeping every interpolated quantity in one gauge; the result is a rate whose fine-grid vertices, 270 TB of them, are never stored.

E.5.1 Computed quantities

A Wannier calculation turns quantities known on a coarse uniform grid into real-space matrices, and those back into the same quantities anywhere in the zone. For the bands, the coarse-grid Hamiltonian in the rotated (Wannier) gauge, $U(\boldsymbol{k})^\dagger \operatorname{diag}(\varepsilon_{n\boldsymbol{k}})\, U(\boldsymbol{k})$, is transformed to the real-space Hamiltonian $H^W(\boldsymbol{R})$ of Sect. 3.6, and at any $\boldsymbol{k}$ the matrix $\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} H^W(\boldsymbol{R})$ is diagonalized. Its eigenvalues are the interpolated bands, and its eigenvectors define $U(\boldsymbol{k})$ on the fine grid. For the electron–phonon vertex, the coarse-grid vertices are rotated with the coarse-grid U matrices and the phonon displacement patterns are undone, which leaves a matrix in two lattice vectors, one for the electron hop and one for the displaced atom [47, Eqs. (177)–(179)]. Back on a fine grid,

$$g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q}) = \sum_{\boldsymbol{R}_e, \boldsymbol{R}_p} \mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{R}_e + \boldsymbol{Q}\cdot\boldsymbol{R}_p)} \sum_{m'n'\kappa\alpha} U_{mm'}(\boldsymbol{k}+\boldsymbol{Q})\, g^W_{m'n'\kappa\alpha}(\boldsymbol{R}_e, \boldsymbol{R}_p)\, U^\dagger_{n'n}(\boldsymbol{k})\, \frac{e^\alpha_{\kappa\nu}(\boldsymbol{Q})}{\sqrt{2M_\kappa \omega_{\boldsymbol{Q}\nu}}}, \tag{E.36}$$

in the normalization of Eq. (A.26). The phonon frequencies and polarization vectors on the fine grid come from the interpolated force constants of Appendix E.4. The same construction

interpolates the velocity matrix elements of Sect. 3.2, which the optical limit needs, and, with a restriction discussed below, the density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})$.

E.5.2 Place on the hierarchy

The interpolation is a numerical device, not a physical approximation, and it has no component on the truncation label $\mathfrak{T}$ of Sect. 8.5. What the label records is the physics of the quantity being interpolated. The bands are independent-particle bands, $\mathcal{X} = \mathrm{IP}$, unless quasiparticle corrections are interpolated in their place (Appendix E.3). The vertex is first order in the displacement, $n_g = 1$. Phonon-assisted absorption is $n_g = 2$ with $n_T = 1$, first order in the thermal phonon occupations $n_{\boldsymbol{Q}\nu}$ that weight its absorption branch, and $\mathcal{X} = \mathrm{IP}$, as Sect. 8.5 reads it off from the assumptions of Sect. 6.5, and the phonon linewidth of Sect. 6.4 is the lowest order in g of the phonon self-energy. The interpolation error belongs to the numerical part of the error budget of Sect. 8.8, and it is tested and reported separately from the truncation.

E.5.3 Projections and disentanglement

Silicon has two textbook choices. The four valence bands form an isolated group, separated from the conduction bands by the gap at every $\boldsymbol{k}$, and they can be rotated into four Wannier functions centered on the bonds, each a bonding combination of two sp^3 hybrids on neighboring atoms [48, Fig. 2]. No disentanglement is needed, and these functions are the natural choice for the valence part of a polarization calculation (Eq. (E.26)). A dark-matter rate needs the conduction bands as well. Eight sp^3 hybrids, four per atom, describe the valence bands and the lowest conduction bands, but the conduction bands of silicon cross higher bands of other character, so the eight-dimensional subspace must be extracted from a larger set at every $\boldsymbol{k}$. This is disentanglement [52].

Two energy windows control it. The outer (disentanglement) window contains every band that may contribute to the subspace. The inner (frozen) window contains the states that must be reproduced exactly; inside it the Bloch states are kept unchanged, and only outside it does the procedure choose, at each $\boldsymbol{k}$, the combination of bands that varies most smoothly across the zone. Projection onto atomic sp^3 orbitals alone reproduces the bands of silicon well except where higher bands with some sp^3 character mix in; forcing the whole valence manifold into the frozen window removes those deviations [48, Fig. 5]. For an electron-recoil calculation the inner window should contain every state that can be a final state at the energies of interest, plus a margin, since a conduction state outside the frozen window is reproduced only approximately.

The number of Wannier functions grows with the energy range that must be described. For phonon-assisted absorption up to the direct gap and beyond, Ref. [51] extracted 26 Wannier functions from 34 bands, reproducing the band structure from 10 eV below to 30 eV above the Fermi level. The range is set by the intermediate states. A second-order amplitude sums over all bands m (Sects. 6.5 and 13.2), and the Wannier manifold truncates that sum; in Appendix E.4 we saw how the remainder above the manifold can be added by a projected Sternheimer equation.

Every set of Wannier functions passes four checks before use. The spread functional Ω_W of Sect. 3.6 must be converged and the individual spreads small compared with the cell. The centers must sit where the symmetry says, on the bonds or the atoms. For a time-reversal-symmetric crystal the maximally localized functions can be chosen real, and a large ratio of imaginary to real parts signals a poor minimum. Above all, the interpolated bands must agree with a direct calculation at points off the coarse grid, inside the frozen

window, to the tolerance the application needs. In silicon the last check has a twist: the conduction minimum lies on the Δ line at about 0.85 of the way from Γ to X, not at a high-symmetry point, so a coarse grid that does not resolve it misplaces the valley. Ref. [49] needed a 12^3 coarse $\boldsymbol{k}$ grid for the conduction bands for this reason, while 6^3 was enough for the phonon wavevectors.

E.5.4 Interpolating vertices, and the long-range parts

The decay of $g^W(\boldsymbol{R}_e, \boldsymbol{R}_p)$ decides whether Eq. (E.36) converges. In the electron vector $\boldsymbol{R}_e$ it decays at least as fast as the Wannier functions themselves. In the atom vector $\boldsymbol{R}_p$ it decays as the screened potential of a displaced atom. In a nonpolar semiconductor or insulator that potential falls off at least as fast as a quadrupole, R_p^{-3}, so the vertices in reciprocal space stay finite as $\boldsymbol{Q} \to 0$ and the interpolation converges [47, Sect. VI.A.2]. For diamond, coarse grids of 4^3, 6^3 and 8^3 points interpolate the vertex of the highest optical branch along high-symmetry lines onto the direct DFPT values, with the 4^3 grid already close [47, Fig. 6]. For a new material, the same comparison, of g from direct DFPT at a few $(\boldsymbol{k}, \boldsymbol{Q})$ off the coarse grid with the interpolated values, is the only evidence that the interpolation converges.

In a polar crystal the displaced atom carries a dipole, its potential decays as R_p^{-2}, and the vertex of the longitudinal optical branch diverges as $1/Q$: the Fröhlich coupling of Sect. 6.2. No finite real-space sum reproduces a divergence, so the long-range part is split off analytically. With the book's normalization,

$$g^{\mathrm{L}}_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q}) = \mathrm{i}\,\frac{e^2}{\Omega} \sum_{\kappa} \frac{1}{\sqrt{2M_\kappa \omega_{\boldsymbol{Q}\nu}}} \sum_{\boldsymbol{G} \neq -\boldsymbol{Q}} \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) \cdot Z^*_\kappa \cdot (\boldsymbol{Q}+\boldsymbol{G})}{(\boldsymbol{Q}+\boldsymbol{G}) \cdot \varepsilon_\infty \cdot (\boldsymbol{Q}+\boldsymbol{G})} \, \langle \psi_{m,\boldsymbol{k}+\boldsymbol{Q}} | \mathrm{e}^{\mathrm{i}(\boldsymbol{Q}+\boldsymbol{G})\cdot(\boldsymbol{r}-\boldsymbol{\tau}_\kappa)} | \psi_{n\boldsymbol{k}} \rangle, \tag{E.37}$$

with $\boldsymbol{\tau}_\kappa$ the position of atom κ in the cell. It is the potential of the bound charge $-\nabla \cdot \boldsymbol{P}$ of the displacement wave, with $\boldsymbol{P} = e\,\boldsymbol{u}_\kappa \cdot Z^*_\kappa / \Omega$ per cell, screened by ε_∞ and acting on an electron of charge $-e$, for the displacement per unit $B_{\boldsymbol{Q}\nu}$ of Eq. (A.21); it is Ref. [47, Eq. (180)] in Heaviside–Lorentz units and the normalization of Appendix A.6 [53]. The procedure subtracts g^{L} from the coarse-grid vertices, interpolates the smooth short-range remainder g^{S}, and adds g^{L} back on the fine grid. The overlap in Eq. (E.37) is needed at the fine-grid points; for small $\boldsymbol{Q}+\boldsymbol{G}$ it is $[U(\boldsymbol{k}+\boldsymbol{Q})U^\dagger(\boldsymbol{k})]_{mn}$ in the Wannier gauge [47, Sect. VI.A.3], and the sum over $\boldsymbol{G}$ is cut off by a Gaussian factor in practice. The inputs Z^*_κ and ε_∞ come from Appendix E.4, with the caveat about the functional's ε_∞ stated there.

Besides its dipole, a displaced atom carries a dynamical quadrupole, which is nonzero even in silicon, where every Born charge vanishes, and its potential gives vertices that are finite as $\boldsymbol{Q} \to 0$ but depend on the direction of $\boldsymbol{Q}$. Such a nonanalytic angular dependence is long-ranged in real space and is not removed by a finer coarse grid: densifying the grid confines the spurious oscillations of the interpolated potential to a smaller region around Γ without reducing their amplitude [54]. In silicon, omitting the quadrupole term changes the computed electron mobility by about 10 %, and in GaAs by about 30 % [54]. Codes that support it subtract the quadrupolar term along with the dipole; where the code does not, the error must be estimated from the small-Q behavior of directly computed vertices.

One rule ties all of this together, the rule of Appendix A.5: every quantity that enters one amplitude must be interpolated with the same $U(\boldsymbol{k})$. The vertex g, the velocity, the density vertex and the energies of a phonon-assisted amplitude are products of matrices in band space, and a gauge mismatch between two of them leaves the eigenvalues intact while scrambling the amplitude. The safest practice is to compute all coarse-grid quantities from one set of wavefunctions, rotate them with one set of U matrices, and interpolate them with one $H^W(\boldsymbol{R})$. Quasiparticle corrections (Appendix E.3) are interpolated on the same Wannier

basis, and since they change the energy denominators but not the wavefunctions, velocity matrix elements must be rescaled by the ratio of quasiparticle to Kohn–Sham transition energies, Eq. (E.19) [51].

E.5.5 Density vertices at large momentum

The density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ is interpolated by the same logic, but with a restriction that matters for dark matter. In the Wannier gauge it is the matrix $\langle w_{m'\boldsymbol{0}}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|w_{n'\boldsymbol{R}}\rangle$, Fourier transformed in $\boldsymbol{R}$. For small $\boldsymbol{q}$ one may expand $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \approx 1 + \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}$, and the vertex follows from the overlaps and the position matrix elements already computed for the Wannier functions; this is the small-$\boldsymbol{q}$ form used for g^{L} above. The expansion is controlled by $|\boldsymbol{q}|$ times the spatial extent of a Wannier function, which is of the order of a bond length, 2.35 Å in silicon. At the momentum transfers of electron recoil, a keV and above, $|\boldsymbol{q}| \times 2.35\,\text{Å} \approx 1.2$ at 1 keV and grows linearly beyond, so the expansion fails. The momentum transfer is a parameter of the vertex, not a lattice wavevector, and for each $\boldsymbol{q}$ of interest the real-space matrix must be built anew.

The practical route therefore computes $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ directly on the coarse $\boldsymbol{k}$ grid for each target $\boldsymbol{q}$, from the plane-wave coefficients of the states at $\boldsymbol{k}$ and at $\boldsymbol{k}+\boldsymbol{q}$ (Appendix E.7), rotates it with the same U matrices into the Wannier gauge, transforms it to real space in $\boldsymbol{R}$, and interpolates in $\boldsymbol{k}$ alone. The cost is one non-self-consistent calculation at the shifted grid $\boldsymbol{k}+\boldsymbol{q}$ per target $\boldsymbol{q}$, and the number of targets is set by the resolution needed in $|\boldsymbol{q}|$ and in direction. For a cubic crystal the directions can be restricted to the irreducible wedge.

E.5.6 Counting matrix elements and memory

Worked example E.4: Matrix elements of a converged silicon rate

The setup of Ref. [51] has $N_W = 26$ Wannier functions, coarse 6^3 grids for $\boldsymbol{k}$ and $\boldsymbol{Q}$, and fine 40^3 grids for both, with two atoms and six branches. We count what a phonon-assisted rate needs and where it must live.

Fine-grid vertices. There are $N_k N_Q = 64{,}000^2 = 4.1 \times 10^9$ pairs $(\boldsymbol{k},\boldsymbol{Q})$, and at each pair $N_W^2 \times 6 = 4{,}056$ complex vertices. At 16 bytes per complex number this is 1.7×10^{13} numbers, or 270 TB. Nobody stores it. Codes loop over $\boldsymbol{Q}$ (or over $\boldsymbol{k}$), interpolate the vertices for all partners at once, accumulate the rate, and discard them. Even one $\boldsymbol{Q}$ with all 64,000 partners and all band pairs is 4 GB, which is why the band indices are restricted at once to the bands that can be initial, final or intermediate states at the energies of interest.

Real-space vertices. The matrix $g^W(\boldsymbol{R}_e,\boldsymbol{R}_p)$ of Eq. (E.36) has one entry for each pair of lattice vectors in the Wigner–Seitz supercells of the two coarse grids, about $6^3 = 216$ vectors each, times N_W^2 band pairs and six displacement directions: $216^2 \times 676 \times 6 \approx 1.9 \times 10^8$ complex numbers, 3 GB. This is the object that is kept, and it must fit in the memory of one node, or be distributed over several.

Interpolation work. Eq. (E.36) is done in two stages. For each $\boldsymbol{Q}$ the sum over $\boldsymbol{R}_p$ costs $216 \times 216 \times 676 \times 6 \approx 1.9 \times 10^8$ complex multiply–adds; for each $\boldsymbol{k}$ at that $\boldsymbol{Q}$ the sum over $\boldsymbol{R}_e$ costs $216 \times 676 \times 6 \approx 8.8 \times 10^5$, plus the U rotations, which are smaller. Over all pairs the second stage dominates: $4.1 \times 10^9 \times 8.8 \times 10^5 \approx 3.6 \times 10^{15}$ multiply–adds, about 3×10^{16} floating-point operations. Restricting $\boldsymbol{Q}$ to the irreducible wedge divides this by the 48 operations of the cubic group, to 6×10^{14}, and a pair of energy windows does far more.

Energy windows. Near the absorption edge only valence states within a few tenths of an eV of the valence maximum and conduction states within the same range of the six

conduction minima contribute. For a parabolic band of mass m^*, the states within δ of the edge fill a sphere of radius $k_\delta = (2m^*\delta)^{1/2}$, a fraction $(4\pi/3)k_\delta^3/V_{\mathrm{BZ}}$ of the zone, with $V_{\mathrm{BZ}} = (2\pi)^3/\Omega = 6.19\,\text{Å}^{-3}$. With $\delta = 0.3\,\mathrm{eV}$ and masses of a few tenths of m_e (Sect. 3.3), say $0.5\,m_e$ for the holes and, per conduction valley, the density-of-states mass $(m_\parallel^* m_\perp^{*2})^{1/3} = (0.92 \times 0.19^2)^{1/3}\,m_e = 0.32\,m_e$, the fractions are 5.3×10^{-3} for one valence band and $6 \times 2.7 \times 10^{-3} = 1.6 \times 10^{-2}$ for the six valleys. On the 40^3 grid this leaves about 340 valence and 1,040 conduction points per band, so 3.5×10^5 pairs instead of 4.1×10^9, four orders of magnitude fewer. The window, not the interpolation, is what makes fine grids affordable near an edge.

E.5.7 Phonon linewidths and the Eliashberg function

The phonon linewidth $\gamma_{\boldsymbol{Q}\nu}^{\mathrm{ep}}$ and the Eliashberg function $\alpha^2F(\omega)$ of Sect. 6.4 are Fermi-surface averages of $|g|^2$: double sums over $\boldsymbol{k}$ with two delta functions that place both the initial and the final electron on the Fermi surface. They are the hardest integrals in this section, because a delta function in three dimensions samples a two-dimensional surface, and a product of two samples the intersection of two surfaces. In practice each delta function is replaced by a Gaussian of width σ. The width must be larger than the spacing of the fine-grid energies near the Fermi surface, of order $|\nabla_{\boldsymbol{k}}\varepsilon|\,\Delta k$ with $\Delta k = |\boldsymbol{b}_i|/N_{\mathrm{fine}}$ the fine-grid step and N_{fine} the number of fine-grid points per direction, or the sum is noise. It must also be smaller than the energy scale on which $|g|^2$ and the density of states vary, or the result is biased. Convergence means decreasing σ while refining the grid until the result is independent of both.

For metals with complicated Fermi surfaces the requirement is severe. For lead, the DFPT phonons needed 14^3 $\boldsymbol{k}$ points and a 10^3 wavevector grid with a Gaussian broadening of 0.025 Ry, the electron–phonon vertices were interpolated from 6^3 coarse grids, and the coupling strength λ_{ep} converged only with 400,000 quasi-random $\boldsymbol{k}$ points and 27,000 random wavevectors [55, Fig. 12]. Quasi-random (Sobol) sampling converged faster than uniform grids in that study [55], since it avoids the systematic aliasing of a regular grid with the Fermi surface. For an undoped insulator at $T = 0$ the electron–phonon linewidth below the gap vanishes (Sect. 6.4), and the calculation is needed only for metals, superconductors (Appendix E.9) and doped semiconductors.

E.5.8 Convergence of the phonon-assisted dielectric function

The phonon-assisted $\varepsilon_2^{\mathrm{PA}}(\omega)$ of Sect. 6.5 is a double sum over $\boldsymbol{k}$ and $\boldsymbol{Q}$ of the squared two-path amplitude, with one energy delta function [51, Eqs. (1)–(3)]. Its convergence parameters are the two fine grids, the broadening of the delta function, the imaginary part Γ_{w} given to the intermediate states, and the number of intermediate bands. The energy resolution is set by the grid: in silicon, 40^3 grids for both sums resolve features of 14 meV, which is what the onset needs, while the smoother spectrum between the indirect and direct gaps converges at 24^3 with a resolution of 30 meV [51]. In that range the result is insensitive to the intermediate-state width, set to a constant 100 meV [51], because the intermediate states are far from resonance; near the direct gap they are not, and the width matters (Sect. 13.6).

The comparison with experiment is where the physics outside the calculation shows. The computed spectra of Ref. [51] had to be shifted rigidly by 0.15 eV to 0.23 eV to match the measured onsets. The shift absorbs the remaining error of the quasiparticle gap, of order 0.1 eV, and the electron–phonon renormalization of the gap, 0.05 eV to 0.1 eV between 0 and 400 K, which the calculation did not include [51]. The second effect is the subject of Sect. 13.7

and Appendix E.8. A computed absorption edge without such a shift is accurate only to the accuracy of the gap.

E.5.9 A phonon-assisted electron-recoil calculation

A phonon-assisted electron recoil (Sect. 13.2) replaces the photon of Sect. 6.5 by the dark-matter density vertex at finite $\boldsymbol{q}$, and the four paths of its amplitude contain ρ, g and the energies at three crystal momenta. The calculation chains the steps above, and each link has its own check. It starts from the ground state and DFPT on a coarse wavevector grid (Appendix E.4), with the frequencies checked against neutron data and the Born charges against zero. Bands on a coarse $\boldsymbol{k}$ grid fine enough to place the conduction minimum fix the Wannier manifold, whose frozen window covers the final and intermediate states and whose bands are tested off the grid. The coarse-grid vertices g are rotated into the Wannier gauge with their dipole and quadrupole parts removed, and tested against direct DFPT off the grid. For each target $\boldsymbol{q}$, $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ is computed from the plane-wave coefficients and rotated with the same $U(\boldsymbol{k})$, and it too is tested against direct values off the grid. For each $\boldsymbol{q}$ the window is the range $\omega < E_{\min}(\boldsymbol{q})$ with no zero-phonon final state (Sect. 13.1), cut to the fine-grid pockets of states that can contribute; its edges are in Sect. 13.9. In that window the four-path amplitude is summed over the Wannier manifold, with the remainder above it from the projected Sternheimer equation Eq. (E.34), into the phonon-assisted loss function on a grid in $(|\boldsymbol{q}|,\omega)$.

Three controls close the chain. As $\boldsymbol{q}\to 0$ the density vertex reduces to its optical limit (Sect. 3.2), and the computed loss function must reproduce the phonon-assisted $\varepsilon_2^{\mathrm{PA}}$ of the same code and, after the gap shift above, the measured indirect absorption (Sect. 13.5). With g set to zero the phonon-assisted rate must vanish identically, which catches any zero-phonon contamination of the window. The third is the f-sum rule for the density vertices of one occupied state in its nonlocal form, Eq. (E.50), whose shortfall measures the truncation of the intermediate-state sum once the nonlocal term δ_{NL} is included.

The cost of a silicon run follows from the counts above. The DFPT stage is the one of Worked example E.3, a few hundred linear-response solutions. The density vertices need one non-self-consistent calculation per target $\boldsymbol{q}$; a grid of $|\boldsymbol{q}|$ in steps of $\sim 0.1\,\mathrm{keV}$ up to a few keV and a few tens of directions in the irreducible wedge gives several hundred targets, and this stage dominates the plane-wave part of the work. The amplitude stage is a sum over pocket pairs. Near the edge the relevant states lie within $\sim 0.1\,\mathrm{eV}$ of the band extrema, and the window fractions of Worked example E.4, which scale as $\delta^{3/2}$, drop by a factor $3^{3/2}\approx 5$ from their values at 0.3 eV. On a 120^3 fine grid this leaves about 1.8×10^3 valence points per band and 5.4×10^3 conduction points. Three valence bands lie within 0.1 eV of the maximum, the heavy-hole, light-hole and split-off bands (44 meV below the others, Sect. 3.3), so there are $\sim 3\times10^7$ pairs per $\boldsymbol{q}$. Momentum conservation fixes the phonon wavevector of each pair, $\boldsymbol{Q}=\boldsymbol{k}'-\boldsymbol{k}-\boldsymbol{q}$ up to a reciprocal-lattice vector, so each pair needs its own interpolated vertices. Restricted to one external band and N_W intermediate bands, the four paths need about $4\times216\times26\times6\approx1.3\times10^5$ complex multiply–adds of interpolation per pair, plus a smaller amount for ρ, which is $\sim 1\times10^6$ floating-point operations. That count assumes the $\boldsymbol{R}_p$ stage of Eq. (E.36) is already done for the pair's $\boldsymbol{Q}$; it runs once per distinct $\boldsymbol{Q}$ of the pocket-difference set, at $216^2\times26\times6\approx7\times10^6$ multiply–adds per external band, and stays below the $\boldsymbol{R}_e$ stage for up to $\sim10^5$ distinct $\boldsymbol{Q}$. Per $\boldsymbol{q}$ the total is $\sim 3\times10^{13}$ operations, and for several hundred targets $\sim 1\times10^{16}$. At a sustained 10^{13} operations per second, typical of one modern accelerator, that is $10^3\,\mathrm{s}$, a quarter of an hour; at 5×10^{10}, typical of a few processor cores, it is two to three days. The estimate scales as δ^3 with the energy range above the edge and as N_{fine}^6 with the fine grid, so these two choices, not the plane-wave stages, decide

the budget.

E.5.10 Pitfalls

A coarse grid that looks converged for bands and phonons can still be too coarse for vertices, and a fine grid that resolves a 30 meV spectrum can be too coarse for a 14 meV onset. Each convergence statement belongs to the quantity, the energy range and the resolution for which it was made. Appendix E.6 turns from the vertices to the many-body response that screens them.

E.6 Dielectric response from first principles

The valence electrons of silicon, eight per primitive cell of volume $\Omega = a^3/4 = 40.05\,\text{Å}^3$, have a plasma frequency $\omega_p = (4\pi\alpha\, n_e/m_e)^{1/2} = 16.6\,\text{eV}$, and the loss function $L(\boldsymbol{q},\omega)$ of Sect. 4.4 has its main peak near that energy at small momentum. Where exactly the peak sits, how sharp it is, and how much spectral weight lies near the gap, where the dark-matter rate is largest for light masses, depend on how the response is computed. Sects. 4.6 and 4.9 built the hierarchy of many-body approximations: the independent-particle bubble, the random-phase approximation with and without local fields, time-dependent density-functional kernels and the Bethe–Salpeter equation. Each becomes a calculation with its own inputs and cost, and the approximation chosen moves both the peak and the weight near the gap that light dark matter sees.

E.6.1 Computed quantities

Every approximation starts from the crystal bubble of Sect. 4.6, the Adler–Wiser form of the independent-particle response [56, 57],

$$R^0_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega) = \frac{g_{\text{deg}}}{N\Omega}\sum_{\boldsymbol{k}}\sum_{n,m}(f_{n\boldsymbol{k}} - f_{m,\boldsymbol{k}+\boldsymbol{q}})\,\frac{\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})^*\,\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G}')}{\omega + \varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}} + \mathrm{i}\eta_{\text{b}}}, \tag{E.38}$$

with $\boldsymbol{q}$ in the first zone, N the number of $\boldsymbol{k}$ points, $g_{\text{deg}} = 2$ for spin, the density vertices ρ_{mn} of Sect. 3.2, and a broadening η_{b} that tends to 0^+ in the exact expression and is finite on a grid (below). The random-phase dielectric matrix is $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\text{C}}(\boldsymbol{q}+\boldsymbol{G})\,R^0_{\boldsymbol{G}\boldsymbol{G}'}$, the proper response being approximated by the bubble (Sect. 4.4). Time-dependent density-functional theory replaces it by the solution of the matrix Dyson equation

$$R = R^0 + R^0 f_{\text{Hxc}} R, \qquad (f_{\text{Hxc}})_{\boldsymbol{G}\boldsymbol{G}'} = v_{\text{C}}(\boldsymbol{q}+\boldsymbol{G})\,\delta_{\boldsymbol{G}\boldsymbol{G}'} + (f_{\text{xc}})_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}), \tag{E.39}$$

from which $\varepsilon^{-1} = 1 + v_{\text{C}}R$ in matrix form; with $f_{\text{xc}} = 0$ it reduces to the random-phase result. From the inverse matrix come the macroscopic dielectric function $\varepsilon_{\text{mac}} = 1/(\varepsilon^{-1})_{\boldsymbol{0}\boldsymbol{0}}$ and the loss function. For a momentum transfer outside the first zone, $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$ in the notation of Sect. 3.1, the loss function that enters the rate is the diagonal element of the inverse at $\boldsymbol{G}_q$,

$$L(\boldsymbol{q},\omega) = \mathrm{Im}\big[-(\varepsilon^{-1})_{\boldsymbol{G}_q\boldsymbol{G}_q}(\boldsymbol{q}_r,\omega)\big], \tag{E.40}$$

so a calculation at dark-matter momentum transfers of several keV needs the matrix at reciprocal vectors far from the origin, not just its head (Sect. 10.2). The Bethe–Salpeter equation leaves the two-point language altogether; we treat it separately below.

E.6.2 Place on the hierarchy

On the truncation label $\mathfrak{T}$ of Sect. 8.5, these calculations differ only in the many-body approximation $\mathcal{X}$: IP for Eq. (E.38) without screening, RPA for the diagonal of $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}$, RPA with local fields for the full matrix, TDDFT for Eq. (E.39) with a kernel, and BSE. All are at $n_B = 0$, since the nuclei are clamped and ε_∞ is the static-lattice constant (the lattice part of ε_0 in a polar crystal comes from Appendix E.4), and at zeroth thermal order, $n_T = 0$, with zero-temperature occupations on the static lattice; the temperature dependence of the gap enters only through a shift (Appendix E.8). As Sects. 4.6 and 8.5 stress, $\mathcal{X}$ is not an expansion in a small number. Its error is estimated from the spread between approximations, and for that estimate to mean anything, every approximation must be converged with the same numerical parameters.

E.6.3 Convergence parameters

The wavefunction cutoff, the local-field cutoff, the band count, the $\boldsymbol{k}$ grid and the gap control Eqs. (E.38) and (E.39), and they interact.

The wavefunction cutoff limits the largest momentum transfer at which ρ_{mn} can be trusted. A density vertex at $\boldsymbol{q}+\boldsymbol{G}$ is an overlap of two sets of plane-wave coefficients displaced by $\boldsymbol{q}+\boldsymbol{G}$, and when the displacement approaches the cutoff momentum, $k_{\mathrm{cut}} = (2m_e E_{\mathrm{cut}})^{1/2}$, few coefficients overlap. For a cutoff of 500 eV, $k_{\mathrm{cut}} = 22.6\,\mathrm{keV}$. Within that range the vertex is still only as good as the pseudo-wavefunction is near the nucleus, which is the subject of Appendices E.2 and E.7.

The local-field cutoff sets the size N_G of the matrix $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}$, through a separate, smaller energy cutoff on $|\boldsymbol{q}+\boldsymbol{G}|$. Building the matrix costs $\propto N_G^2$ per transition and inverting it $\propto N_G^3$ per $(\boldsymbol{q},\omega)$. The test is the change of the quantity of interest, $\varepsilon_{\mathrm{mac}}$ near the gap or L at the target $\boldsymbol{q}$, as N_G grows. At large $|\boldsymbol{q}|$ the matrix must contain reciprocal vectors around $\boldsymbol{G}_q$, not only around the origin.

The number of empty bands truncates the sum over m in Eq. (E.38). Its test is a sum rule: $\int_0^\infty \mathrm{d}\omega\,\omega\,\varepsilon_2 = (\pi/2)\omega_p^2$ and the same holds for ωL (Sect. 4.4), and a truncated band sum misses the weight at high energy. A free-electron count gives the scale. Above the bottom of the valence band, N_b bands per spin fill a sphere of radius k with $\Omega k^3/(6\pi^2) = N_\mathrm{b}$, so the highest included band lies at $k^2/2m_e$ above the bottom. In silicon 70 bands reach $k = 4.7\,\text{Å}^{-1}$, which is 84 eV above the bottom of the valence band. The weight above that is small but not zero, and the loss function at large ω falls short of the measured one for exactly this reason [58].

The $\boldsymbol{k}$ grid is usually the bottleneck in a semiconductor [59]. The bands disperse strongly, and Eq. (E.38) is a sum of sharp poles that must merge into a continuum. The broadening η_b must exceed the spacing of the transition energies on the grid, of order $|\nabla_{\boldsymbol{k}}(\varepsilon_m - \varepsilon_n)|\,\Delta k$; a spectrum computed with too small an η_b for its grid shows spurious peaks. For finite $\boldsymbol{q}$, the first-zone part $\boldsymbol{q}_r$ must be a difference of grid points, so the $\boldsymbol{k}$ grid also fixes the momentum resolution. A coarse grid with a moderate η_b suffices where the spectrum is smooth on the scale of η_b; the region near the gap is not, and it needs much finer sampling, often on a shifted or locally refined grid.

The gap enters every denominator. Kohn–Sham gaps are too small (Sect. 3.4), and the usual remedies are a rigid scissor shift Δ_{scis} of the empty bands, quasiparticle energies from Appendix E.3, or a functional designed for gaps. A scissor keeps the states and raises the transition energies, and its effect on the f-sum check follows from that. The density vertices are unchanged (Appendix E.3), so in the optical limit the spectrum moves rigidly, $\varepsilon_2(\omega) = \varepsilon_2^{\mathrm{KS}}(\omega - \Delta_{\mathrm{scis}})$, and its first moment becomes $\int \mathrm{d}\omega\,\omega\,\varepsilon_2^{\mathrm{KS}}(\omega-\Delta_{\mathrm{scis}}) = (\pi/2)\omega_p^2 + \Delta_{\mathrm{scis}}\int \mathrm{d}\omega\,\varepsilon_2^{\mathrm{KS}}$,

larger by $\Delta_{\mathrm{scis}} \int \mathrm{d}\omega\, \varepsilon_2^{\mathrm{KS}}$. The excess is the f-sum rule of a nonlocal operator, the scissor, and a band-convergence test on a scissored spectrum subtracts it. Codes that build the optical limit from velocity matrix elements must rescale them as in Eq. (E.19); otherwise the spectrum acquires a spurious factor $[(\omega - \Delta_{\mathrm{scis}})/\omega]^2$ and is no longer the rigid shift that unchanged states imply. A density vertex computed directly from plane-wave coefficients at finite $\boldsymbol{q}$ needs no such care, since it depends on the states and not on their energies.

Semicore and core states matter at large ω and large $\boldsymbol{q}$. The silicon $2p$ and germanium $3d$ shells are frozen into the core by most pseudopotentials. Their contribution to L starts at their binding energies, and their contribution to the f-sum adds to the valence one. A loss function for ω above a few tens of eV, or at $|\boldsymbol{q}|$ where core orbitals overlap the probe, needs them in the valence or an all-electron treatment (Sect. 10.6 and Appendix E.7).

E.6.4 Local fields and exchange–correlation kernels

Local fields are the difference between the inverse of the averaged matrix and the average of the inverse (Sect. 4.6). Computationally they cost one matrix inversion per $(\boldsymbol{q}, \omega)$, and they change the spectrum where the response is inhomogeneous on the scale of the cell. In the optical spectrum of silicon they lower and shift the two main peaks to higher energy, which moves the random-phase result further from experiment [59]. In the dark-matter rate of silicon and germanium, by contrast, the difference between the loss function with and without local fields is small [58], because the rate integrates over $\boldsymbol{q}$ and ω and is dominated by energies and momenta where the response is smooth.

The adiabatic local-density kernel adds little to the random-phase result for the absorption of semiconductors: its spectra are extremely close to those of the random-phase approximation with local fields [60]. For energy-loss spectra of solids the same kernel does well, but largely because the random-phase approximation with local fields already agrees with experiment there [60], which is good news for the loss function of a dark-matter rate. The reason is that the kernel is short-ranged, while the electron–hole attraction that produces excitons is long-ranged. A kernel of the form $f_{\mathrm{xc}}(\boldsymbol{r}, \boldsymbol{r}') = -\alpha_{\mathrm{xc}}/(4\pi|\boldsymbol{r} - \boldsymbol{r}'|)$, which behaves as $-\alpha_{\mathrm{xc}}/|\boldsymbol{q} + \boldsymbol{G}|^2$ in reciprocal space, fits the measured optical spectrum of silicon almost as well as the Bethe–Salpeter equation, with quasiparticle energies in the bubble and the empirical value $\alpha_{\mathrm{xc}} = 0.2$ [60], a fit for silicon rather than a universal constant. (Ref. [60] writes α; we add the subscript to keep the fine-structure constant unambiguous.) The cost of any two-point kernel is that of the random-phase calculation: the matrices have the same size, and only the Dyson equation changes. This is the attraction of TDDFT for dark-matter loss functions on fine $(\boldsymbol{q}, \omega)$ grids, and the reason its kernels are worth testing against the Bethe–Salpeter equation on a few points.

E.6.5 Optical conductivity and large momentum

Bosonic absorption (Sect. 10.7) and the crystal Migdal response (Sect. 12.3) take opposite limits of one calculation. For absorption of a bosonic dark-matter particle (Sect. 10.7) the relevant quantity is the optical conductivity $\sigma_1 = \omega\varepsilon_2$ at $\boldsymbol{q} \to 0$, where the transverse and longitudinal dielectric functions coincide (Sect. 4.4). Computing it needs the head of Eq. (E.38) in the optical limit, with the velocity matrix elements and their nonlocal commutator (Appendices E.2 and E.7), a fine $\boldsymbol{k}$ grid near the absorption edge, and the approximation that the energy range demands: near the gap the Bethe–Salpeter equation, well above it the random-phase approximation with local fields. Measured optical data are available for most candidate targets, and where they are, they fix σ_1 better than any calculation. The calculation is then needed where the data stop: at low temperature, below the measured range, or for anisotropic

crystals where only some tensor components were measured.

The opposite limit is the loss function at momenta far beyond the zone, up to tens of keV, and energies up to about 100 eV, which the crystal Migdal response of Sect. 12.3 convolves with the nuclear motion. Every convergence parameter above is pushed at once. The cutoff must hold momentum components well above the largest $|\boldsymbol{q}|$, the band sum must reach 100 eV above the gap and beyond, and the semicore shells must be in the valence. In addition, the local-field matrix must be centered on $\boldsymbol{G}_q$, which at $|\boldsymbol{q}| = 30\,\mathrm{keV}$ lies more than seven shortest reciprocal vectors from the origin. The pseudo-wavefunctions are no longer adequate there, and the all-electron reconstruction of Appendices E.2 and E.7 becomes part of the response calculation. The cost grows with all of these at once, and in practice such a calculation is done on a coarse grid of $|\boldsymbol{q}|$ and a few directions, with the convergence tested on the f-sum rule at each $\boldsymbol{q}$.

E.6.6 The Bethe–Salpeter equation

The Bethe–Salpeter equation of Sect. 4.9 is an eigenvalue problem in the space of transitions $(v, c, \boldsymbol{k})$. Its Hamiltonian, in the resonant (Tamm–Dancoff) form, is

$$H^{\mathrm{exc}}_{vc\boldsymbol{k},v'c'\boldsymbol{k}'} = (\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}})\,\delta_{vv'}\delta_{cc'}\delta_{\boldsymbol{k}\boldsymbol{k}'} + g_{\mathrm{deg}}\,K^{\mathrm{x}}_{vc\boldsymbol{k},v'c'\boldsymbol{k}'} - K^{\mathrm{d}}_{vc\boldsymbol{k},v'c'\boldsymbol{k}'}, \tag{E.41}$$

with the energies from quasiparticle bands (Appendix E.3), an exchange kernel K^{x} built from the Coulomb kernel without its $\boldsymbol{G} = 0$ component, and a direct kernel K^{d} built from the statically screened interaction W^{scr} at the momentum $\boldsymbol{k} - \boldsymbol{k}'$ of the scattered pair (Sect. 4.9); the factor g_{deg} holds for singlets. Its eigenvalues are the exciton energies E_λ and its eigenvectors the envelopes $A^{\lambda}_{cv\boldsymbol{k}}$, and the macroscopic dielectric function is a sum over λ of oscillator strengths $|\sum_{cv\boldsymbol{k}} A^{\lambda}_{cv\boldsymbol{k}}\rho_{cv}(\boldsymbol{k},\boldsymbol{q})^*|^2$ over $\omega - E_\lambda$. At finite $\boldsymbol{q}$ the transitions are $(v\boldsymbol{k}) \to (c, \boldsymbol{k}+\boldsymbol{q})$ and the same structure gives the excitonic loss function (Sect. 10.5).

The cost has two parts. Building the kernel requires the screened interaction $\varepsilon^{-1}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k}-\boldsymbol{k}')$, which is a random-phase calculation of its own, and the matrix elements of K^{d} between all pairs of transitions, $\propto (N_v N_c N_k)^2$ entries, each a sum over N_G^2 reciprocal vectors when the full inverse is used. Solving it costs $\propto (N_v N_c N_k)^3$ for a full diagonalization, or $\propto (N_v N_c N_k)^2$ per step of an iterative (Lanczos–Haydock) method that delivers the spectrum without the eigenvectors. The transition-basis dimension $N_v N_c N_k$ is the number to watch. For the optical spectrum of silicon, Ref. [59] used norm-conserving pseudopotentials at 15 Ry, quasiparticle corrections interpolated onto the fine grid, and up to 2,048 $\boldsymbol{k}$ points in the full zone. On an 864-point grid four valence and two conduction bands proved sufficient below 4.5 eV, and neglecting the off-diagonal elements of ε^{-1} in K^{d} changed nothing visible, so the production run used six bands and a diagonal ε^{-1} [59]. The dimension was then $2{,}048 \times 4 \times 2 \approx 1.6 \times 10^4$. A dense diagonalization of that size takes $\sim 10^{13}$ operations, trivial today, and the authors identified the $\boldsymbol{k}$ grid, not the cutoff, as the bottleneck [59].

Two lessons of that calculation carry over: symmetry rotations must keep the phases of the states consistent, and a finite grid exaggerates individual structures. Symmetry can be used to reduce the zone, but the spectrum turned out to be extremely sensitive to inconsistent phases when wavefunctions are rotated, especially for degenerate bands, and only a subgroup of the operations was used safely [59]. This is the gauge rule of Appendix A.5 once more. And a finite $\boldsymbol{k}$ grid overestimates individual structures; the shoulder at 3.8 eV in the silicon spectrum came out slightly too strong for that reason [59].

Bound excitons near the gap are harder still, because their envelopes are wide in real space and narrow in $\boldsymbol{k}$. The Wannier–Mott relation of Sect. 4.9, $E_x^b = \mu_x \alpha^2/(2\varepsilon_\infty^2)$, with the measured binding energy of 14.7 meV and $\varepsilon_\infty = 11.7$, gives $\mu_x \approx 0.15\,m_e$ and a radius $a_x = \varepsilon_\infty a_0 m_e/\mu_x \approx 4.2\,\mathrm{nm}$. Resolving an envelope of that size needs a $\boldsymbol{k}$ spacing well below

$1/a_x \approx 0.024\,\text{Å}^{-1}$, against a zone of width $|\boldsymbol{b}_i| = \sqrt{3}\,2\pi/a = 2.0\,\text{Å}^{-1}$: close to a hundred points per direction near the band edges, and so of order 10^6 points if the grid were uniform. Calculations of such excitons therefore use grids refined only around the band extrema, or the effective-mass limit, with the Bethe–Salpeter equation reserved for the continuum and the finite-momentum structure (Sect. 10.5).

E.6.7 Molecular *GW* and Bethe–Salpeter calculations

Molecular crystals (Sect. 16.4) are often treated molecule by molecule, with the environment added afterwards. The same *GW* and Bethe–Salpeter equations are then solved in a basis of atom-centered Gaussian orbitals instead of plane waves. The *GW* quasiparticle energies of molecules improve the Kohn–Sham gaps to within a few tenths of an eV at a cost that scales as the fourth power of the system size, and the Bethe–Salpeter singlet excitation energies that follow have typical errors of 0.1 eV to 0.3 eV, with a mean error of 0.2 eV on a standard benchmark set of about 200 excitations [61]. An iterative search for the lowest eigenstates costs the same as in TDDFT, $O(N_{\text{orb}}^4)$ operations with N_{orb} the number of orbitals, when resolution-of-the-identity techniques are used to build the kernel [61]. The convergence parameter specific to this setting is the Gaussian basis: diffuse functions are needed for the empty states and for the screening, and results are tested along a family of basis sets of increasing size. The crystal environment enters through its dielectric screening, as a reaction field added to the bare Coulomb interaction [61], or through a periodic calculation of the full crystal. Sect. 16.4 compares the periodic result with measurement for anthracene: 3.3 eV against the measured 3.1 eV for the lowest singlet of the crystal.

E.6.8 Silicon

Worked example E.5: The loss function of silicon from Kohn–Sham states

A published dark-matter calculation of the silicon loss function used a plane-wave basis with a 500 eV cutoff, a Γ-centered $8 \times 8 \times 8$ grid of $\boldsymbol{k}$ points, 70 bands per spin, a functional designed for band gaps with a scissor shift to the measured gap, and the random-phase dielectric matrix with local fields at all $\boldsymbol{q}$ of the grid [58]. For the rate, the matrix was reduced to an angular average of its diagonal elements over all $\boldsymbol{q}_r + \boldsymbol{G}$ of equal length [58, Eq. (24)]. We check each setting against the estimates of this section.

The cutoff corresponds to $k_{\text{cut}} = (2m_e \times 500\,\text{eV})^{1/2} = 22.6\,\text{keV}$, the range quoted by the authors [58]. Kinematics allows momentum transfers up to $2m_\chi v_{\max}$, 5.3 keV per MeV of dark-matter mass (Sect. 1.3), but the valence density vertices fall off beyond a few times $\alpha m_e = 3.7\,\text{keV}$ (Sects. 3.5 and 10.1), and the cutoff covers that range, provided the pseudo-wavefunctions are accurate there (Appendix E.7). The 70 bands reach 84 eV above the bottom of the valence band by the free-electron count above, about 70 eV above the valence maximum since the occupied bands are about 12 eV wide (Sect. 3.3). The authors traced the shortfall of the computed loss function at large energy to this truncation [58].

An 8^3 grid has spacing $|\boldsymbol{b}_i|/8 = 0.25\,\text{Å}^{-1}$, or 0.49 keV, which is the resolution in $\boldsymbol{q}_r$; it is adequate for a loss function dominated by energies of several eV, and coarse for the structure within a few tenths of an eV of the gap, where excitonic and phonon-assisted effects live (Sect. 13.1).

The f-sum rule is the cleanest single check. With $\omega_p = 16.6\,\text{eV}$ from the valence density, the integral $\int_0^\infty \mathrm{d}\omega\, \omega L(\boldsymbol{q}, \omega) = (\pi/2)\omega_p^2$ at small $\boldsymbol{q}$ is $433\,\text{eV}^2$. A calculation that recovers, say, 0.9 of it is missing a tenth of the weight, all at high energy, and that missing weight is a bound on what the band truncation can do to a rate at large ω.

The optical spectrum gives the complementary test near the gap. On top of quasiparticle energies, the random-phase spectrum of silicon misplaces the weight of the two main peaks, local fields alone make the agreement worse, and only the electron–hole attraction of the Bethe–Salpeter equation restores the measured line shape and absolute intensity [59]. The same hierarchy matters for a dark-matter rate in proportion to the weight it gives to energies within an eV or two of the gap, which is why the many-body spread of Sects. 4.6 and 4.9 is quoted as an uncertainty for light masses and becomes negligible for heavier ones.

E.6.9 Pitfalls

Three settings are easy to get wrong and hard to spot afterwards: the head element, the reciprocal-lattice index of the momentum transfer, and the broadening. The head element $\varepsilon_{\mathbf{00}}$ used in place of $1/(\varepsilon^{-1})_{\mathbf{00}}$ drops the local fields, and the element at $\boldsymbol{G} = 0$ used in place of $\boldsymbol{G}_q$ for a momentum transfer beyond the first zone computes the response at the wrong momentum. A broadening too large for the grid smears the onset, which is the part of the spectrum that light dark matter sees. The response computed here screens the density vertex, the electron–phonon vertex of Appendix E.5 and, through W^{scr}, the quasiparticle energies of Appendix E.3, so every one of these errors propagates into the rates of the later chapters.

E.7 Crystal form factors

An electron-recoil rate of silicon at independent-particle level is a sum of $|\rho_{cv}(\boldsymbol{k}, \boldsymbol{q})|^2$ over valence states $v\boldsymbol{k}$, conduction states $c\boldsymbol{k}'$ and momentum transfers $\boldsymbol{q}$, each term weighted by an energy delta function. With four valence bands, 52 conduction bands, a $10 \times 10 \times 10$ grid of crystal momenta and every reciprocal-lattice vector up to $q = 30\,\mathrm{keV}$, the sum has about 5×10^{11} terms, as we count below. Each term is an overlap of two sets of plane-wave coefficients. The work is in restoring the core region that the pseudopotential removed, adding the core shells, and knowing when the sum has converged; the bands come from the calculation of Appendix E.1, and the screening that turns the bare vertex into a measured loss function from Appendix E.6.

E.7.1 The density vertex

The object is the density vertex of Eq. (A.17),

$$\rho_{mn}(\boldsymbol{k}, \boldsymbol{q}) = \langle \psi_{m,\boldsymbol{k}+\boldsymbol{q}} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \psi_{n\boldsymbol{k}} \rangle,$$

with $\boldsymbol{q}$ never reduced to the first zone (Appendices A.3 and A.4). From it the independent-particle structure factor follows as a sum over pairs (Sect. 10.1); the crystal form factor f_{crys} of the literature is the same object in another normalization, Eq. (A.45); and the vertices of the spin and current couplings of Sect. 10.4 are built the same way, with $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ multiplied by a Pauli matrix or by the velocity.

E.7.2 Place on the hierarchy

On the truncation label $\mathfrak{T}$ of Sect. 8.5, a form-factor calculation sits at the base point $\mathfrak{T}_0$. The nuclei are clamped at their equilibrium positions, $n_B = 0$; there is no electron–phonon insertion, $n_g = 0$, and no anharmonic or thermal correction, $n_A = 0$ and zeroth thermal order, $n_T = 0$; the many-body approximation is that of independent particles, $\mathcal{X} = \mathrm{IP}$. The plane wave is kept whole, so the result is exact to all orders in the multipole parameter ξ_D, and the dipole form $\boldsymbol{q} \cdot \boldsymbol{v}_{mn}/(\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}})$ of Eq. (A.18) is only a check at small q. The

Kohn–Sham approximation and the pseudopotential sit inside $\mathcal{X} = \mathrm{IP}$ and are not part of the label. The states are Kohn–Sham states rather than quasiparticles (Sect. 3.4), with a scissor shift Δ_{scis} on the conduction energies or with GW energies (Appendix E.3); and they are pseudo-wavefunctions unless the region inside the core radius is reconstructed. What the base point leaves out is screening and local fields ($\mathcal{X} = \mathrm{RPA}$ and beyond, Appendix E.6), excitons ($\mathcal{X} = \mathrm{BSE}$), and the lattice ($n_g \geq 1$, Appendices E.5 and E.8). Screening divides the squared potential by $|\varepsilon(\boldsymbol{q},\omega)|^2$, which is ε_∞^2 as $q \to 0$ and approaches one only at large q (Sect. 4.4), so a form factor at $\mathfrak{T}_0$ is an input to that correction, not a substitute for it.

E.7.3 Plane-wave evaluation

A plane-wave code returns, for each band n and each $\boldsymbol{k}$ of its grid, the coefficients $c_{n\boldsymbol{k}}(\boldsymbol{G})$ of

$$u_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{\Omega}} \sum_{|\boldsymbol{k}+\boldsymbol{G}|^2 \leq 2m_e E_{\mathrm{cut}}} c_{n\boldsymbol{k}}(\boldsymbol{G})\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}, \qquad \sum_{\boldsymbol{G}} |c_{n\boldsymbol{k}}(\boldsymbol{G})|^2 = 1, \tag{E.42}$$

which is the normalization of Eq. (A.15) and the code convention of Appendix A.3 (coefficients of $\mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$, the opposite of the book's lattice Fourier components). The cutoff E_{cut} is the one of Sect. 3.5. With $\boldsymbol{k}+\boldsymbol{q}$ folded back into the zone, $\boldsymbol{k}+\boldsymbol{q} = \boldsymbol{k}' + \boldsymbol{G}_{\mathrm{U}}$ with $\boldsymbol{k}'$ a point of the grid and $\boldsymbol{G}_{\mathrm{U}}$ a reciprocal-lattice vector, and by Eq. (A.16) the vertex is $\int_\Omega u^*_{m\boldsymbol{k}'} \mathrm{e}^{\mathrm{i}\boldsymbol{G}_{\mathrm{U}}\cdot\boldsymbol{r}} u_{n\boldsymbol{k}}$, and inserting Eq. (E.42) the cell integral of $\mathrm{e}^{\mathrm{i}(\boldsymbol{G}_{\mathrm{U}}+\boldsymbol{G}-\boldsymbol{G}')\cdot\boldsymbol{r}}$ is $\Omega\,\delta_{\boldsymbol{G}',\boldsymbol{G}+\boldsymbol{G}_{\mathrm{U}}}$:

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) = \sum_{\boldsymbol{G}} c^*_{m\boldsymbol{k}'}(\boldsymbol{G}+\boldsymbol{G}_{\mathrm{U}})\, c_{n\boldsymbol{k}}(\boldsymbol{G}), \qquad \boldsymbol{q} = \boldsymbol{k}' - \boldsymbol{k} + \boldsymbol{G}_{\mathrm{U}}. \tag{E.43}$$

The sum runs over the vectors present in both basis sets. Coefficients are matched by their Miller indices, the integer coordinates of $\boldsymbol{G}$ in the basis $\boldsymbol{b}_i$, since the two states are stored on different lists of $\boldsymbol{G}$. A term with $\boldsymbol{G}+\boldsymbol{G}_{\mathrm{U}}$ outside the basis of $\boldsymbol{k}'$ is zero in the representation, so Eq. (E.43) is exact for the functions the code represents; its error is the basis truncation itself, which we discuss below. This is the expression of Ref. [62, Eq. (3.16)], in the notation of Appendix A.5.

Each value of Eq. (E.43) costs one dot product of length N_{PW}, the number of plane waves per state. All the $\boldsymbol{G}_{\mathrm{U}}$ for one pair $(m\boldsymbol{k}', n\boldsymbol{k})$ come at once from a fast Fourier transform: the product $u^*_{m\boldsymbol{k}'}(\boldsymbol{r})\, u_{n\boldsymbol{k}}(\boldsymbol{r})$ is formed on the real-space grid and transformed, and the component at $\boldsymbol{G}_{\mathrm{U}}$ is ρ_{mn} for $\boldsymbol{q} = \boldsymbol{k}' - \boldsymbol{k} + \boldsymbol{G}_{\mathrm{U}}$. The cost per pair drops from $N_{\mathrm{PW}} N_{\boldsymbol{G}}$ to of order $N_{\mathrm{FFT}} \log N_{\mathrm{FFT}}$, where $N_{\boldsymbol{G}}$ is the number of $\boldsymbol{G}_{\mathrm{U}}$ needed and N_{FFT} the number of grid points [31, below Eq. (3.15)]. The real-space grid has to resolve the product, whose components reach twice the wavefunction cutoff. For a cubic cell it covers a cube of side $4\sqrt{2m_e E_{\mathrm{cut}}}$ in reciprocal space, which holds $8\Omega(2m_e E_{\mathrm{cut}})^{3/2}/\pi^3$ points, while the sphere of Eq. (E.42) holds $N_{\mathrm{PW}} = \Omega(2m_e E_{\mathrm{cut}})^{3/2}/(6\pi^2)$, so $N_{\mathrm{FFT}} \approx (48/\pi) N_{\mathrm{PW}}$. The fcc grid runs along the primitive vectors, whose lengths multiply to $\sqrt{2}\,\Omega$, and holds $\sqrt{2}$ times more, $N_{\mathrm{FFT}} \approx (48\sqrt{2}/\pi) N_{\mathrm{PW}} \approx 22\, N_{\mathrm{PW}}$: about 36^3 points at 60 Ry, as in Appendix E.1.

Which $\boldsymbol{q}$ a grid can reach. Eq. (E.43) needs the state at $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{q} - \boldsymbol{G}_{\mathrm{U}}$. If both $\boldsymbol{k}$ and $\boldsymbol{k}'$ are taken from one uniform grid of $n_k = n_1 n_2 n_3$ points (Appendix C.2), the first-zone part of $\boldsymbol{q}$ is a difference of grid vectors, and $\boldsymbol{q}$ is resolved in steps of $|\boldsymbol{b}_i|/n_i$. A form factor tabulated on a finer or arbitrary set of $\boldsymbol{q}$ needs a second, shifted set of states: one non-self-consistent calculation at the points $\boldsymbol{k}+\boldsymbol{q}$, as density-functional perturbation theory does for phonon wavevectors (Appendix E.4). The states at $\boldsymbol{k}'$ must be computed, or unfolded from the irreducible wedge by a symmetry operation, with the phases that the operation carries. In the diamond structure some operations include a fractional translation $\boldsymbol{t}_{\mathrm{f}}$, and each coefficient

of a rotated Bloch state picks up a phase $e^{-i\boldsymbol{k}_{\rm rot}\cdot\boldsymbol{t}_{\rm f}}$, with $\boldsymbol{k}_{\rm rot}$ the rotated wavevector $\boldsymbol{k}+\boldsymbol{G}$. Computing the full grid without symmetry avoids the bookkeeping at a cost of the order of the symmetry factor, 48 for silicon.

Gauge and degeneracy. Each Bloch state carries an arbitrary phase, and ρ_{mn} picks up the phase difference of its two states, while $|\rho_{mn}|^2$ does not. A rate that sums $|\rho_{mn}|^2$ is therefore gauge independent, provided that the sum over m and n covers complete degenerate sets: a degenerate subspace may be rotated by any unitary matrix, and only the trace over it is fixed. At the valence maximum of silicon the threefold degeneracy at Γ makes this concrete; a sum restricted to one of the three states there is meaningless. Amplitudes that interfere, such as the second-order amplitudes of phonon-assisted processes in which ρ_{mn} multiplies an electron–phonon vertex, need ρ and g in one gauge, computed from the same stored states (Appendices A.5 and E.5).

E.7.4 Velocity matrix elements and nonlocal potentials

The momentum matrix element follows from Eq. (E.42) in the same way,

$$\boldsymbol{p}_{mn}(\boldsymbol{k}) = \sum_{\boldsymbol{G}} c^*_{m\boldsymbol{k}}(\boldsymbol{G})\,(\boldsymbol{k}+\boldsymbol{G})\,c_{n\boldsymbol{k}}(\boldsymbol{G}), \tag{E.44}$$

but the velocity is not $\boldsymbol{p}_{mn}/m_e$ in a pseudopotential calculation. The velocity operator is $\boldsymbol{v} = i[H,\boldsymbol{r}]$. For $H = \boldsymbol{p}^2/2m_e + V_{\rm loc}(\boldsymbol{r}) + V_{\rm NL}$, with $V_{\rm NL}$ the nonlocal part of the pseudopotential (Appendix A.5), the local potential commutes with $\boldsymbol{r}$ and

$$\boldsymbol{v} = \frac{\boldsymbol{p}}{m_e} + i[V_{\rm NL},\boldsymbol{r}]. \tag{E.45}$$

The commutator is evaluated in the $\boldsymbol{k}$ representation. Define $V_{\rm NL}(\boldsymbol{k}) \equiv e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} V_{\rm NL} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$, the operator that acts on $u_{n\boldsymbol{k}}$. Differentiating, $\nabla_{\boldsymbol{k}} V_{\rm NL}(\boldsymbol{k}) = e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\, i[V_{\rm NL},\boldsymbol{r}]\, e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$, so that

$$\boldsymbol{v}_{mn}(\boldsymbol{k}) = \frac{\boldsymbol{p}_{mn}(\boldsymbol{k})}{m_e} + \langle u_{m\boldsymbol{k}}|\nabla_{\boldsymbol{k}} V_{\rm NL}(\boldsymbol{k})|u_{n\boldsymbol{k}}\rangle. \tag{E.46}$$

For the separable form Eq. (E.8) that plane-wave codes use, with projectors $\tilde{p}_{\kappa i}$ centered on the atoms and coefficients $D_{\kappa,ij}$, the gradient acts only on the Fourier transforms of the projectors, $\tilde{p}_{\kappa i}(\boldsymbol{k}+\boldsymbol{G})$, and costs about as much as applying $V_{\rm NL}$ once. The correction is not small. In diamond, the absorption spectrum computed with $\boldsymbol{p}/m_e$ instead of $\boldsymbol{v}$ differs by 15 to 20 percent [32, Fig. 1], and Ref. [32, Eq. (20)] shows that a long-wavelength field couples to $\boldsymbol{v}$, not to $\boldsymbol{p}/m_e$.

For the dark-matter vertices the correction has opposite consequences for the density and for the current. The density vertex needs none. The operator $e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$ is local, and Eq. (E.43) evaluated at small q reproduces $\boldsymbol{q}\cdot\boldsymbol{v}_{mn}/(\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}})$ with the full velocity of Eq. (E.46), since the derivation of Eq. (A.18) used only $i[H,\boldsymbol{r}] = \boldsymbol{v}$. The comparison of the two sides at $q \sim 10^{-3}\,|\boldsymbol{b}_i|$ is a test that catches a missing nonlocal term in a velocity routine. A coupling to the electron current, on the other hand, is ambiguous with a nonlocal Hamiltonian. The paramagnetic current $j_e(\boldsymbol{q})$ of Sect. 4.3 is built from $\boldsymbol{p}/m_e$; gauge invariance requires an additional nonlocal term, which Ref. [32, Eqs. (18) and (19)] derives as a phase $\exp(i\int \boldsymbol{A}\cdot d\boldsymbol{x})$ along the straight line between the two arguments of $V_{\rm NL}(\boldsymbol{r},\boldsymbol{r}')$. Its long-wavelength limit is the commutator of Eq. (E.45). At finite q the term depends on $\boldsymbol{q}$ through the extent of the projectors, of order $q\,r_c$, and codes that compute current responses at finite q either include it or restrict themselves to $q\,r_c \ll 1$.

The scissor shift. The scissor operator $\Delta_{\text{scis}} \sum_c |\psi_c\rangle\langle\psi_c|$ is nonlocal too. It rescales each interband velocity by $(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}} + \Delta_{\text{scis}})/(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}})$, the rigid-shift case of Eq. (E.19), and leaves the density vertex and its optical limit unchanged. An optical spectrum built from $|\boldsymbol{v}_{cv}|^2$ with scissored energies but unscaled velocities violates the f-sum rule; first-principles absorption calculations therefore carry the nonlocality of the scissor into the oscillator strengths [63, Sect. VII], where the inverse factor appears because it multiplies their oscillator-strength form rather than the velocity. For silicon in the local-density approximation, with $\Delta_{\text{scis}} = 0.74\,\text{eV}$ [63, Sect. VII] and a direct gap at Γ of 2.57 eV [64, Table V], the factor is 1.29 at Γ, and $|\boldsymbol{v}_{cv}|^2$ changes by 66 percent.

E.7.5 All-electron reconstruction

A pseudo-wavefunction agrees with the all-electron one outside a sphere of radius r_c around each nucleus and is smooth and nodeless inside it (Sect. 3.5). The two differ only inside the sphere, so their difference has Fourier components at momenta of order π/r_c and above, and the pseudo vertex is reliable only for $q\,r_c \ll 1$. The PAW map Eq. (E.12) restores the difference, in one of two ways. The first route expands the all-electron state itself in plane waves up to a second, larger cutoff and then applies Eq. (E.43) unchanged. Ref. [31, Sect. 2.1.1] does this for silicon and germanium with a second cutoff of 2 keV, which by $q_{\max} = (2m_e E)^{1/2}$ reaches $q \approx 45\,\text{keV}$. The second route works with the matrix elements, Eq. (E.13). With $\boldsymbol{r} = \boldsymbol{\tau}_\kappa + \boldsymbol{x}$ in each sphere, the Rayleigh expansion Eq. (C.23) of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}$ reduces each element of the one-center matrix Eq. (E.14) to Gaunt coefficients times the radial integrals

$$\int_0^{r_c} \mathrm{d}x\, x^2\, j_l(qx)\, [\phi^{\text{AE}}_{\kappa i}(x)\phi^{\text{AE}}_{\kappa j}(x) - \phi^{\text{PS}}_{\kappa i}(x)\phi^{\text{PS}}_{\kappa j}(x)], \tag{E.47}$$

with $\phi(x)$ the radial parts of the partial waves of Eq. (E.12) and j_l the spherical Bessel functions of Appendix C.3. The integrals live on the radial grid of the atomic calculation, which resolves the nodes of the all-electron partial waves, so the correction has no plane-wave cutoff at all; it is limited by the completeness of the partial waves, usually two per angular momentum. They are computed once per species and per q, and the per-state work is in the projections $\langle\tilde{p}_{\kappa j}|\tilde{\psi}_{n\boldsymbol{k}}\rangle$, which the code already stores.

How much the reconstruction matters depends on which q the rate weights. For a heavy mediator, $F_{\text{DM}} = 1$, and the factor q of the measure in Eq. (A.44) gives large momentum transfers more weight than any lighter mediator does, and the reconstruction raises the silicon rate even above low thresholds (Appendix E.2). For a light mediator, $F^2_{\text{DM}} \propto q^{-4}$ weights the smallest momentum transfers, and the reconstruction changes the rate little except at high thresholds, where only large q is kinematically allowed. Norm-conserving pseudopotentials can be reconstructed in the same way when the generator writes out its all-electron and pseudo partial waves; without them the pseudo vertex is all there is, and its validity is limited to $q\,r_c \ll 1$.

E.7.6 Core and semicore states

Below the valence bands of silicon lie the atomic shells $2p$, $2s$ and $1s$, with measured binding energies relative to the valence maximum of 99.4 eV ($2p_{3/2}$; $2p_{1/2}$ at 99.8 eV), 150 eV and 1839 eV; in germanium the $3d$ shell sits only 29 eV below it [24, Table 1-1]. Kinematics decides when they matter. A transition from a shell at depth E_{core} needs $\omega \geq E_{\text{g}} + E_{\text{core}}$, and since $v_{\min} = \omega/q + q/2m_\chi \leq v_{\max}$ it also needs $q \geq \omega/v_{\max}$ and $m_\chi \geq 2\omega/v^2_{\max}$ (Sect. 1.3). With $v_{\max} = 2.650 \times 10^{-3}$, the silicon $2p$ shell opens at $m_\chi = 29\,\text{MeV}$ and needs $q \geq 38\,\text{keV}$, and the $2s$ shell at 43 MeV with $q \geq 57\,\text{keV}$; the germanium $3d$ shell opens at 8.5 MeV and needs

$q \geq 11\,\mathrm{keV}$. Silicon $2p$ transitions therefore probe the wavefunctions at momenta beyond the reach of a typical plane-wave basis (Sect. 3.5), and a pseudopotential calculation does not contain the shell at all.

Such shells are either kept in the pseudopotential or taken from the atom. A *semicore* shell, such as germanium $3d$, can be kept in the valence of the pseudopotential, at the price of a higher cutoff, and is then computed with the other bands and reconstructed as above. A deeper shell is taken from the isolated atom. Its Bloch states are the lattice sums

$$\psi_{\kappa nlm,\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_\kappa)}\, \psi^{\mathrm{at}}_{\kappa nlm}(\boldsymbol{r} - \boldsymbol{R} - \boldsymbol{\tau}_\kappa), \tag{E.48}$$

of the atomic orbitals $\psi^{\mathrm{at}}_{\kappa nlm}$ [31, Eq. (2.8)], labeled by the atomic quantum numbers instead of a band index; the orbitals barely overlap, so the band is flat and its energy is the atomic level. The atomic approximation can be tested where both treatments exist. Ref. [31, Sect. 2.2] computes the silicon $2p$ and germanium $3d$ shells both ways and finds that the atomic orbitals reproduce the reconstructed crystal wavefunctions up to the largest momentum the crystal calculation resolves, about 45 keV. The vertex between a core state and a conduction state is then $\int \mathrm{d}^3 r\, \psi^*_{c,\boldsymbol{k}+\boldsymbol{q}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \psi_{\kappa nlm,\boldsymbol{k}}$, with the conduction state from the crystal calculation; above the highest computed band the final states are approximated by plane waves, in the spirit of Eq. (2.24), with or without a Coulomb factor [31, Sects. 2.3 and 3.4]. Relativistic corrections to the core orbitals are of relative size $(Z\alpha)^2$, about 1 % for the silicon $1s$ shell and 5 % for germanium $1s$. That is the size of each term; the net shift of the $1s$ level is about a quarter of it, 0.26 % for silicon and 1.46 % for germanium (Worked example 3.9). The approximation leaves out the relaxation of the other electrons around the core hole, which shifts and reshapes the edge; that final-state effect lies beyond $\mathcal{X} = \mathrm{IP}$.

E.7.7 Atomic ionization form factors

For a target made of isolated atoms, such as a noble liquid, the rate of Sect. 2.2 needs the ionization form factor $|f_{\mathrm{ion}}(p_{\mathrm{ej}}, q)|^2$ of Eq. (2.22) for each occupied shell. The initial orbitals are Hartree–Fock orbitals (Sect. 3.4), almost always the Roothaan–Hartree–Fock solutions of Bunge and collaborators [65], which expand each radial function in Slater-type orbitals,

$$R^{\mathrm{at}}_{nl}(r) = a_0^{-3/2} \sum_j C_{jln}\, \frac{(2Z_{jl})^{\tilde{n}_{jl}+1/2}}{\sqrt{(2\tilde{n}_{jl})!}} \Big(\frac{r}{a_0}\Big)^{\tilde{n}_{jl}-1} \mathrm{e}^{-Z_{jl} r/a_0}, \tag{E.49}$$

with tabulated coefficients C_{jln}, exponents Z_{jl} and integers $\tilde{n}_{jl}$ [66, Eq. (B33)]. The final state is a continuum state of energy $E_{\mathrm{ej}} = p^2_{\mathrm{ej}}/2m_e$, expanded in partial waves l'. In the simplest and most common choice it is the exact Coulomb wave of a hydrogenic potential $-Z_{\mathrm{eff}}\alpha/r$, with $Z_{\mathrm{eff}} = n\,(E_B/13.6\,\mathrm{eV})^{1/2}$ fixed by the binding energy E_B of the shell [66, Eqs. (B34) and (B35)]; a better choice solves the radial equation in the self-consistent potential of the ion. The radial integrals of $j_L(qr)$ between the two radial functions are then evaluated for every (l, l', L) allowed by the Gaunt coefficients.

The convergence parameters are the partial-wave cutoff l'_{max}, the grids in q and E_{ej}, and the accuracy of the radial integrals. Ref. [66, footnote 11] sums to $l'_{\mathrm{max}} = 7$; the number needed grows with q, since a plane wave of momentum q reaches angular momenta up to about q times the orbital radius, and has to be checked at the largest q of the table. The q grid must extend to the kinematic maximum $2m_\chi v_{\mathrm{max}}$ of the heaviest mass considered, 5.3 MeV at 1 GeV, and is best logarithmic. The radial integrands oscillate rapidly at large q, and an integration split into intervals can misread an accidental cancellation over one interval as convergence; Ref. [66, Appendix B 5] therefore requires two consecutive intervals to agree

and switches to a series representation at large q. The cost is negligible next to any crystal calculation: a few shells, a few partial waves, and one-dimensional integrals.

The approximation behind the table is that of frozen orbitals. The binding energies are single-particle (Koopmans) energies, the other electrons do not relax when one is removed, and the final state does not feel the polarization of the ion. It is also nonrelativistic, which matters for the inner shells of xenon, where $(Z\alpha)^2 \approx 0.16$. A subtler problem is orthogonality. A continuum state of the model potential $-Z_{\rm eff}\alpha/r$ is not an eigenstate of the Hartree–Fock Hamiltonian and is not orthogonal to the bound orbital, so that $f_{\rm ion}$ does not vanish as $q \to 0$, as it must for $\langle f|i\rangle = 0$. Orthogonalizing the continuum state to the occupied orbitals removes the spurious small-q weight, which a light mediator would otherwise amplify. For silicon these atomic tables play a second role: the deep shells of the crystal in Eq. (E.48) are built from the same orbitals.

E.7.8 Convergence parameters and checks

A crystal form factor is controlled by the cutoff, the number of conduction bands, the $\boldsymbol{k}$ grid and the treatment of the energy delta function. The cutoff $E_{\rm cut}$ sets the largest momentum the vertex contains, $(2m_e E_{\rm cut})^{1/2}$, which is 31 keV at 70 Ry; Ref. [62, Appendix C] varies it from 30 to 110 Ry and finds the silicon rate at $m_\chi = 1\,\text{GeV}$ changed by less than one percent above 70 Ry. The number of conduction bands N_c sets the largest energy deposition the spectrum contains; with 52 conduction bands the lowest band left out lies about 57 eV above the gap, and the computed spectrum ends there [62, Sect. 4.2]. A spectrum above the highest band is zero, not small, and a rate integrated over ω is missing everything beyond it. The $\boldsymbol{k}$ grid sets both the resolution in q and the sampling of the onset near $E_{\rm min}(q)$; Ref. [62, Appendix C] finds the grid matters only at low masses and low thresholds, where few transitions are kinematically allowed. It tests the grid by shaking every grid point randomly by up to half the grid spacing and comparing ten such grids. Finally, the delta function of energy needs a broadening or a tetrahedron integration, with the width tied to the grid spacing as described in Appendix C.2, or a binning: Ref. [62, Sect. 4.2] uses bins of 0.1 eV in ω and $0.02\,\alpha m_e = 75\,\text{eV}$ in q.

Most errors are caught by the small-q limit and by a sum rule. The small-q check is the normalization $\rho_{nn}(\boldsymbol{k}, 0) = 1$ with $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})$ for $m \neq n$ vanishing linearly in q, with the slope given by the velocity, as discussed above. The sum rule holds for each occupied state and measures the effect of truncating the bands. Since $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q}) = \langle u_{m,\boldsymbol{k}+\boldsymbol{q}}|u_{n\boldsymbol{k}}\rangle$ (Eq. (A.17)), and the $u_{m,\boldsymbol{k}+\boldsymbol{q}}$ form a complete set of eigenstates of $H(\boldsymbol{k}+\boldsymbol{q}) = \mathrm{e}^{-\mathrm{i}(\boldsymbol{k}+\boldsymbol{q})\cdot\boldsymbol{r}} H \mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{q})\cdot\boldsymbol{r}}$,

$$\sum_m (\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{n\boldsymbol{k}})\,|\rho_{mn}(\boldsymbol{k},\boldsymbol{q})|^2 = \langle u_{n\boldsymbol{k}}|H(\boldsymbol{k}+\boldsymbol{q}) - H(\boldsymbol{k})|u_{n\boldsymbol{k}}\rangle = \boldsymbol{q}\cdot\boldsymbol{v}_n^{\rm g}(\boldsymbol{k}) + \frac{q^2}{2m_e} + \delta_{\rm NL}, \quad \text{(E.50)}$$

where the sum over m runs over all bands and $\delta_{\rm NL}$ is the part of $\langle u_{n\boldsymbol{k}}|V_{\rm NL}(\boldsymbol{k}+\boldsymbol{q}) - V_{\rm NL}(\boldsymbol{k})|u_{n\boldsymbol{k}}\rangle$ beyond first order in q. The kinetic part of $H(\boldsymbol{k}) = (\boldsymbol{p}+\boldsymbol{k})^2/2m_e + V_{\rm loc} + V_{\rm NL}(\boldsymbol{k})$ gives $\boldsymbol{q}\cdot(\boldsymbol{p}+\boldsymbol{k})/m_e + q^2/2m_e$, and the first-order part of the nonlocal term combines with it into the group velocity, by the Hellmann–Feynman relation $\nabla_{\boldsymbol{k}}\varepsilon_{n\boldsymbol{k}} = \langle u_{n\boldsymbol{k}}|\nabla_{\boldsymbol{k}} H(\boldsymbol{k})|u_{n\boldsymbol{k}}\rangle$. Summed over a filled band the linear term cancels. Transitions between two occupied states cancel in pairs, since time reversal gives $|\rho_{nm}(-\boldsymbol{k}-\boldsymbol{q}, \boldsymbol{q})| = |\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})|$ with the opposite energy difference. What remains is the valence f-sum rule of a pseudo-Hamiltonian, the band-summed form of Eq. (E.15): $q^2/2m_e$ per valence electron, plus the nonlocal term, which is absent for the all-electron Hamiltonian of Eq. (A.39). The right side of Eq. (E.50) is a ground-state expectation value and costs nothing; the left side with N_c conduction bands falls short of it by the weight of the bands left out. With a nonlocal pseudopotential, a test that compares the truncated sum with $q^2/2m_e$ alone confuses the two effects, and the nonlocal term must be

included before the shortfall is read as band truncation. The shortfall falls slowly with N_c, because at momentum transfer q much of the missing weight sits near the free-electron energy $q^2/2m_e$.

E.7.9 Cost and pitfalls

The count of vertices sets the cost. With both $\boldsymbol{k}$ and $\boldsymbol{k}'$ on a grid of n_k points, N_v valence and N_c conduction bands, and $N_{\boldsymbol{G}}$ reciprocal-lattice vectors within the largest q, there are $n_k^2 N_v N_c N_{\boldsymbol{G}}$ vertices, computed with $n_k^2 N_v N_c$ fast Fourier transforms of N_{FFT} points each. The number of reciprocal-lattice vectors within $q_{\max}$ is $\frac{4\pi}{3} q_{\max}^3 \,\Omega/(2\pi)^3$. Directional averages can be taken over the irreducible wedge of $\boldsymbol{k}$, with the full set of $\boldsymbol{k}'$. The vertices themselves are far too many to store, and they are accumulated into bins of (q, ω), or of $(\boldsymbol{q}, \omega)$ for a directional response, as they are computed; a tabulated crystal form factor is that binned sum. The all-electron correction adds, per state, the projections onto the atomic projectors, and per q the radial integrals of Eq. (E.47).

Three errors are not caught by the checks above. A $\boldsymbol{q}$ reduced to the first zone assigns an Umklapp vertex to the wrong momentum transfer (Appendix A.3). A spectrum cut by the highest band, or a q range cut by E_{cut}, looks converged in its own window and silently loses everything outside it. And normalizations differ between codes and papers: plane-wave codes often fold the spin factor 2 into the $\boldsymbol{k}$ weights [62, Eq. (4.2)], and the conversions between f_{crys} and $\bar{S}$ are in Eq. (A.45). For germanium, which semilocal functionals leave without a gap [31, Sect. 2.1], the band order must be fixed before any rigid shift (Appendix E.1).

Worked example E.6: The silicon form factor

We set up a valence-to-conduction calculation for silicon at $\mathfrak{T}_0$ and count its cost. The cell volume is $\Omega = a^3/4 = 40.05\,\text{Å}^3$ at $a = 5.431\,\text{Å}$. A cutoff of 70 Ry, 952 eV, reaches $(2m_e E_{\mathrm{cut}})^{1/2} = 31.2\,\mathrm{keV} = 15.8\,\text{Å}^{-1}$ and holds $N_{\mathrm{PW}} = \Omega(2m_e E_{\mathrm{cut}})^{3/2}/6\pi^2 \approx 2700$ plane waves per state. The non-self-consistent run computes 4 valence and 52 conduction bands on a $10 \times 10 \times 10$ grid without symmetry, $n_k = 1000$; the coefficients take $1000 \times 56 \times 2700$ complex numbers, 2.4 GB in double precision, which fits in memory. The Kohn–Sham gap is about 0.6 eV in the representative PBE calculation (Worked example E.1), and a scissor $\Delta_{\mathrm{scis}} = E_{\mathrm{g}} - E_{\mathrm{g}}^{\mathrm{KS}}$ moves the conduction bands up to the measured low-temperature gap, 1.17 eV (Sect. 1.3).

The reciprocal-lattice vectors within $q_{\max} = 30\,\mathrm{keV}$ number $\frac{4\pi}{3} q_{\max}^3 \Omega/(2\pi)^3 \approx 2400$, since the zone of silicon has volume $(2\pi)^3/\Omega = 47.6\,\mathrm{keV}^3$. The vertex count is $n_k^2 N_v N_c N_{\boldsymbol{G}} = 10^6 \times 4 \times 52 \times 2400 \approx 5 \times 10^{11}$, the number quoted at the start of this section. They come from $n_k^2 N_v N_c = 2.1 \times 10^8$ transforms on grids of $N_{\mathrm{FFT}} \approx 22 N_{\mathrm{PW}} \approx 6 \times 10^4$ points. At about $5 N_{\mathrm{FFT}} \log_2 N_{\mathrm{FFT}} \approx 5 \times 10^6$ floating-point operations per transform, the total is 1×10^{15} operations, about three hours on a node that sustains 10^{11} operations per second, and less with the irreducible wedge. The calculation is cheap; the binned output, at the bin widths quoted above, has about 500×400 entries up to 30 keV (the published table of Ref. [62] has 500×900 and reaches 67 keV).

The calculation is then tested at small q, against the sum rule and against the cutoff. At $q = 10^{-3}\,|\boldsymbol{b}_i|$ the ratio $|\rho_{cv}|/q$ must equal $|\hat{\boldsymbol{q}} \cdot \boldsymbol{v}_{cv}|/(\varepsilon_c - \varepsilon_v)$ with the velocity of Eq. (E.46). For each valence state the truncated left side of Eq. (E.50), with its nonlocal term on the right, measures the band truncation. And the rate at $m_\chi = 1\,\mathrm{GeV}$ for a heavy mediator, recomputed at a higher cutoff and with all-electron reconstruction, fixes where the pseudo vertex stops being adequate. Kinematics tells us where to look: at 10 MeV, with ω just above the gap, momentum transfers between 0.45 keV and 53 keV are allowed, but most

of the valence weight lies at a few keV, and the cutoff test decides how much of the rate comes from beyond 30 keV; the $2p$ shell, which would need $q \geq 38$ keV, is still closed below 29 MeV.

The independent-particle form factor is where every electronic rate of Chap. 10 starts. The next corrections, screening and local fields on the many-body axis $\mathcal{X}$, use the same vertices as input; Appendix E.6 treated them.

E.8 Anharmonic and finite-temperature methods

The band gap of silicon near zero temperature is about 60 meV smaller than the gap of the same crystal with its nuclei clamped at their equilibrium positions: the zero-point motion of the lattice closes it, and the measured values are 62 and 64 meV [64, Table VI]. Between zero and room temperature the gap closes by a further 50 meV (Sect. 1.3), and the phonons themselves shift and broaden. A detector at 10 mK sits deep in the zero-point regime, where none of these effects are thermal and all of them are quantum. We will see how a gap shift of tens of meV and the phonon shifts behind it are computed: some methods average the energy surface itself, and others average an electronic property over the zero-point distribution of the nuclei. All of them start from the harmonic phonons, Born charges and electron–phonon potentials of density-functional perturbation theory (DFPT, Appendix E.4).

E.8.1 Place on the hierarchy

Free energies, effective force constants, band energies and absorption spectra at temperature T are all averages over the distribution of nuclear positions, either of the energy surface itself or of an electronic property computed at fixed nuclei. In the language of the truncation label $\mathfrak{T}$ of Sect. 8.5, the lattice-side methods move the anharmonic order n_A and the thermal order n_T away from the base point $\mathfrak{T}_0$, while keeping the Born–Oppenheimer order at $n_B = 0$. The quasi-harmonic approximation keeps every mode harmonic and lets the frequencies depend on volume, the lowest-order effect of the cubic terms on the free energy. The self-consistent harmonic approximation resums the even-order anharmonic terms into an effective harmonic problem and is not a fixed order in n_A, and the temperature-dependent effective potential fits effective second- and third-order force constants to a sampled ensemble. The electron-side methods move the electron–phonon order: AHC theory is the $n_g = 2$ correction to the band energies, with two first-order vertices (the Fan–Migdal term) or one second-order vertex (the Allen–Heine term), in the harmonic approximation. The special-displacement method evaluates the adiabatic average of the electronic property to all orders in the displacements, at $n_A = 0$ and in the adiabatic limit. The anharmonic small parameter η_{anh} of Sect. 5.7 is small for silicon, and the lattice-side corrections are correspondingly small; the electron-side ones are not.

E.8.2 Quasi-harmonic approximation

In the quasi-harmonic approximation the free energy per cell, $\mathcal{F}$, at cell volume Ω is the harmonic one, with frequencies computed at that volume [67, Eq. (4)],

$$\mathcal{F}(\Omega, T) = E_0^{\mathrm{ad}}(\Omega) + \frac{1}{N}\sum_{\boldsymbol{Q}\nu} \frac{\omega_{\boldsymbol{Q}\nu}(\Omega)}{2} + \frac{T}{N}\sum_{\boldsymbol{Q}\nu} \ln\!\left[1 - \mathrm{e}^{-\omega_{\boldsymbol{Q}\nu}(\Omega)/T}\right], \tag{E.51}$$

where $E_0^{\mathrm{ad}}(\Omega)$ is the clamped-nucleus energy per cell, the minimum of the energy surface of Sect. 5.1 at that volume, and the sums run over the N points of the phonon grid. The

equilibrium volume minimizes $\mathcal{F}$ at each T. Differentiating, the volume thermal expansion coefficient is

$$\alpha_{\mathrm{th}} = \frac{1}{NB\Omega} \sum_{\boldsymbol{Q}\nu} \omega_{\boldsymbol{Q}\nu}\, \gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu}\, \frac{\partial n_{\boldsymbol{Q}\nu}}{\partial T}, \qquad \gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu} = -\frac{\Omega}{\omega_{\boldsymbol{Q}\nu}} \frac{\partial \omega_{\boldsymbol{Q}\nu}}{\partial \Omega}, \tag{E.52}$$

with B the isothermal bulk modulus and $\gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu}$ the mode Grüneisen parameters [67, Eqs. (6) and (9)]. A perfectly harmonic crystal has $\gamma^{\mathrm{G}} = 0$ and no thermal expansion. The calculation is a set of harmonic calculations: DFPT phonons at five to seven volumes around the minimum, a fit of $E_0^{\mathrm{ad}}(\Omega)$ and of each $\omega_{\boldsymbol{Q}\nu}(\Omega)$, and a minimization of Eq. (E.51). The cost is that number of DFPT runs. In silicon the transverse acoustic branch has negative Grüneisen parameters, and its large weight in the density of states at low frequency makes the thermal expansion negative at low temperature, a behavior the quasi-harmonic approximation reproduces [67, Sect. 5.1].

The approximation holds as long as the frequencies depend on the volume but not on the amplitude of vibration, that is, as long as the phonons are well-defined harmonic excitations at each volume. It fails for modes that are unstable in the harmonic approximation, near structural transitions and near melting, which is where the next two methods take over. The zero-point expansion and the sensitivity of the phonons to the lattice constant matter even at 10 mK. The zero-point term of Eq. (E.51) expands the lattice at $T = 0$, and this expansion changes the gap: for silicon the zero-point lattice expansion alone shifts the gap by +9 meV [68, Table S2], opposite to the electron–phonon renormalization. And the same Grüneisen parameters set how sensitive a phonon calculation is to the lattice constant it uses: by Eq. (E.52), a relative error $\delta a/a$ in the lattice constant shifts each frequency by $\delta\omega_{\boldsymbol{Q}\nu}/\omega_{\boldsymbol{Q}\nu} = -3\gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu}\, \delta a/a$. In silicon the transverse acoustic modes, with $\gamma^{\mathrm{G}} < 0$, and the optical modes, with $\gamma^{\mathrm{G}} > 0$, move in opposite directions, so that no single choice between the theoretical and the measured lattice constant improves both. A calculation that needs the transverse acoustic branch accurately, as the phonon-assisted thresholds of Chap. 13 do, has to test both lattice constants and quote the spread.

E.8.3 Self-consistent harmonic approximation

When the anharmonicity is not small, the harmonic expansion about the minimum is the wrong starting point, and the self-consistent harmonic approximation (SCHA) replaces it by the best harmonic description of the actual distribution of nuclei. Its basis is the Gibbs–Bogoliubov inequality: for any trial density matrix $\tilde{\varrho}$, the functional $\mathcal{F}[\tilde{\varrho}] = \langle H \rangle_{\tilde{\varrho}} - T S_{\mathrm{ent}}[\tilde{\varrho}]$, with S_{ent} the entropy of $\tilde{\varrho}$, is an upper bound on the true free energy [69, Eq. (2)]. The trial density matrices are those of harmonic Hamiltonians,

$$H_{\mathrm{trial}} = (\text{nuclear kinetic energy}) + \frac{1}{2} \sum_{ab} (X_a - \bar{X}_a)\, \Phi^{(2)}_{\mathrm{aux},ab}\, (X_b - \bar{X}_b), \tag{E.53}$$

with X_a the $3N_{\mathrm{at}}$ Cartesian coordinates of the nuclei of a supercell of N_{at} atoms, and the centroids $\bar{X}_a$ and the auxiliary force constants $\Phi^{(2)}_{\mathrm{aux}}$ as variational parameters [69, Eqs. (4) and (7)]. Minimizing $\mathcal{F}$ over $\bar{X}$ and $\Phi^{(2)}_{\mathrm{aux}}$ gives the SCHA free energy, the equilibrium positions at temperature T, and an effective harmonic problem whose quantum zero-point motion is included by construction. At the minimum $\Phi^{(2)}_{\mathrm{aux}}$ satisfies the self-consistency condition of Sect. 5.7: it equals the average of the true second derivatives of the energy surface over the Gaussian distribution it generates.

The averages are computed stochastically. A population of N_{conf} configurations is drawn in a supercell from the Gaussian distribution of the current trial Hamiltonian, each together with its mirror image about the centroids to reduce the statistical error; DFT supplies the energy,

forces and stress of each; and the gradients of $\mathcal{F}$ with respect to $\bar{X}$ and $\Phi^{(2)}_{\text{aux}}$ follow from these forces. As the parameters move, the same configurations are reweighted by the ratio of the new to the old Gaussian density, until the effective number of samples, $N_{\text{eff}} = (\sum_j w_j)^2 / \sum_j w_j^2$ for weights w_j, drops below about half of N_{conf}; then a new population is drawn [69, Eqs. (21), (38) and (39)]. The cost is entirely in the DFT forces: a population holds several hundred to several thousand configurations, and most materials converge within three populations [69, Sect. III B]. The convergence parameters are the supercell size, N_{conf}, and the threshold on $N_{\text{eff}}/N_{\text{conf}}$.

The auxiliary frequencies of $\Phi^{(2)}_{\text{aux}}$ are not the physical phonon energies. They describe the Gaussian distribution of the nuclei; the frequencies measured by a probe at small energy follow from the curvature of the free energy with respect to the centroids, which adds a term with third-order force constants [69, Sect. IV]. Mistaking one for the other is the most common error in reading SCHA output. For silicon the SCHA changes the phonons little, since η_{anh} is small; it is needed for light atoms, soft modes and quantum crystals, the targets of Sect. 11.5.

E.8.4 Temperature-dependent effective potential

The temperature-dependent effective potential (TDEP) method fits effective force constants to forces sampled at temperature T. Molecular dynamics in a supercell of N_{at} atoms supplies, at each of N_t time steps, the displacements $\boldsymbol{u}_i$ and the DFT forces $\boldsymbol{F}_i$ on every atom i of the supercell. The effective second-order force constants $\tilde{\Phi}^{(2)}_{ij}$ are those that minimize the mismatch

$$\Delta_{\text{fit}} = \frac{1}{N_t} \sum_{t,i} \Big| \boldsymbol{F}_i(t) + \sum_j \tilde{\Phi}^{(2)}_{ij}\, \boldsymbol{u}_j(t) \Big|^2 , \tag{E.54}$$

since a harmonic energy surface gives $\boldsymbol{F}_i = -\sum_j \tilde{\Phi}^{(2)}_{ij} \boldsymbol{u}_j$. This is a linear least-squares problem, overdetermined once $N_t > 3N_{\text{at}}$ [70, Eq. (3), with the opposite sign convention for the force constants]. Symmetry reduces the unknowns to a few per coordination shell, and the same fit extends to third and fourth order. The fitted constants absorb the anharmonicity sampled at that temperature into effective harmonic ones; the phonons and free energy follow as in the harmonic case [71]. The convergence parameters are the number of time steps, which Ref. [70, Fig. 1] tracks through the change in the fitted constants, the supercell size, and the range of the force constants kept in the fit.

The method inherits the statistics of its sampling. Classical molecular dynamics samples a classical distribution, with mean squared displacement $T/(M_\kappa \omega^2_{\boldsymbol{Q}\nu})$ per mode, while the quantum harmonic value is $(2n_{\boldsymbol{Q}\nu} + 1)/(2M_\kappa \omega_{\boldsymbol{Q}\nu})$. Their ratio is $2T/\omega_{\boldsymbol{Q}\nu}$ for $T \ll \omega_{\boldsymbol{Q}\nu}$. For silicon phonons of 10 to 60 meV at 10 mK, the ratio is below 2×10^{-4}, so a classical sample sees an almost static lattice and misses the zero-point motion that sets every renormalization at detector temperatures. At low temperature the configurations must be drawn from the quantum distribution instead, as the SCHA does, or from path-integral molecular dynamics (Appendix E.11). The cost of TDEP is that of the molecular dynamics, N_t force calculations of which only those separated by more than a correlation time are independent.

Third-order force constants also give the three-phonon linewidths $\gamma^{\text{anh}}_{\boldsymbol{Q}\nu}$ of Sect. 5.7. They are fitted in the same way, or computed by finite differences in supercells, one supercell calculation per pair of displaced atoms allowed by symmetry, with a cutoff radius on the interacting triplets. The linewidth then needs a sum over pairs of final phonons on a dense $\boldsymbol{Q}$ grid with an energy delta function, and the same rules for broadening and grid spacing apply as for electronic spectra (Appendix C.2).

E.8.5 Allen–Heine–Cardona calculations

The AHC theory of Sect. 6.3 gives the shift of a band energy at second order in the electron–phonon coupling, summed over all phonons with the thermal weight $2n_{\boldsymbol{Q}\nu}+1$. In the adiabatic form, which neglects the phonon energy next to electronic energy differences, the Fan–Migdal part is

$$\Delta\varepsilon^{\mathrm{FM}}_{n\boldsymbol{k}}(T) = \frac{1}{N}\sum_{\boldsymbol{Q}\nu}(2n_{\boldsymbol{Q}\nu}+1)\sum_{m}\frac{|g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2}{\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}}, \tag{E.55}$$

with the vertex of Eq. (A.26), and the Allen–Heine part adds the diagonal element of the second-order vertex g^{AH} with the same weight. In the rigid-ion approximation the second-order vertex is rewritten through first-order matrix elements at $\boldsymbol{Q}=0$, using the translation invariance of the crystal, so that no second derivative of the potential is computed [64, Eqs. (11) and (15)] (Sect. 6.3). The AHC literature calls this term the Debye–Waller term; in this book Debye–Waller refers only to e^{-2W_κ} (Appendix A.6). The nonadiabatic form keeps $\pm\omega_{\boldsymbol{Q}\nu}$ in the denominators of Eq. (E.55), separately for phonon emission and absorption, and gives the renormalization a lifetime as well [64, Eq. (16)].

The sum over m runs over all bands. In practice it is split: the bands within an active window are summed explicitly, and the contribution of all higher bands is obtained from a Sternheimer equation for the first-order change of the wavefunction projected on the space above the window, as DFPT does for phonons (Appendix E.4). The split is exact in the adiabatic limit; in the nonadiabatic form the Sternheimer part drops the phonon energy, and the active window must be large enough for that to be harmless [64, below Eq. (16)].

The sum over $\boldsymbol{Q}$ is the hard part. The Fan–Migdal denominators vanish when $\varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}=\varepsilon_{n\boldsymbol{k}}$, which happens for $m=n$ at $\boldsymbol{Q}\to 0$ and wherever another band is degenerate with the state. Near these points the integrand is singular, and the result converges slowly with the number N_Q^3 of grid points, at a rate that depends on the band and on the form of the theory; a small imaginary part $\mathrm{i}\delta$ in the denominators smooths the integrand, at the price of a bias that must be extrapolated to $\delta\to 0$ together with $1/N_Q\to 0$ [64, Sect. III and Table II]. A nonzero Born effective charge makes the vertex diverge as $1/Q$ at small Q. In a nonpolar crystal the charges vanish by symmetry, but a DFPT calculation always leaves small residual charges that violate the charge-neutrality sum rule, and these must be removed before the $\boldsymbol{Q}$ sum, or the integral diverges at band extrema [64, Sect. III]. In a polar crystal the divergence is physical: the adiabatic integral does not converge at all, and only the nonadiabatic form gives a finite result [64]. For polar materials the nonadiabatic, Fröhlich-like part often dominates the zero-point renormalization [68].

The cost is that of DFPT at every $\boldsymbol{Q}$ of a dense grid: for silicon, Ref. [64] goes up to a $100\times100\times100$ grid, 22 776 irreducible points, to converge the values of its Table VI. Wannier interpolation of the vertices (Appendix E.5) reduces this to DFPT on a coarse grid. The results for silicon are in Worked example E.7 below. The calculations use Kohn–Sham states; since a semilocal functional underestimates the gap, it overscreens the electron–phonon interaction, and DFT-based renormalizations come out smaller than GW-based ones [72, Sect. VII B].

E.8.6 Special displacements

The perturbative route needs vertices and a converged sum over phonon wavevectors. The nonperturbative alternative computes the electronic property directly in a supercell whose nuclei are displaced, and averages over displacements. In the adiabatic and harmonic approximations, the Williams–Lax expression for a temperature-dependent property, here the imaginary part of

the dielectric function, is a Gaussian average over the normal coordinates x_ν of the supercell,

$$\varepsilon_2(\omega;T) = \prod_{\boldsymbol{Q}\nu} \int \mathrm{d}x_{\boldsymbol{Q}\nu}\, \frac{\mathrm{e}^{-x_{\boldsymbol{Q}\nu}^2/2\sigma_{\boldsymbol{Q}\nu}^2}}{\sqrt{2\pi}\,\sigma_{\boldsymbol{Q}\nu}}\, \varepsilon_2(\omega;\{x\}), \qquad \sigma_{\boldsymbol{Q}\nu}^2 = \frac{2n_{\boldsymbol{Q}\nu}+1}{2\omega_{\boldsymbol{Q}\nu}}, \tag{E.56}$$

where the $x_{\boldsymbol{Q}\nu}$ are the mass-weighted normal coordinates of the supercell, labeled by the wavevectors $\boldsymbol{Q}$ of the primitive-cell phonons it contains, and $\varepsilon_2(\omega;\{x\})$ is computed with the nuclei frozen at the displaced positions [63, Eqs. (3) and (4), with mass-weighted coordinates]. The same formula holds for a band energy or any other property computed at fixed nuclei. Earlier calculations evaluated it by Monte Carlo over configurations; for the optical spectrum of silicon fewer than ten samples sufficed, and a single sample came close [63, Sect. II].

Zacharias and Giustino explained why, and replaced the sampling by one deterministic configuration. In a large supercell, the average of a smooth function over the Gaussian distribution of Eq. (E.56) is dominated by configurations in which every normal coordinate has the magnitude $\sigma_{\boldsymbol{Q}\nu}$, and the cross terms between pairs of modes cancel if the signs alternate in a suitable order [63, Sect. III]. The resulting special, or ZG, displacement of nucleus κ in cell $\boldsymbol{R}$ of a supercell of N primitive cells is

$$\boldsymbol{u}_{\boldsymbol{R}\kappa}^{\mathrm{ZG}} = \sum_{\boldsymbol{Q}\in\mathcal{B},\nu} \varsigma_{\boldsymbol{Q}\nu} \left[\frac{2n_{\boldsymbol{Q}\nu}+1}{2NM_\kappa\omega_{\boldsymbol{Q}\nu}}\right]^{1/2} 2\,\mathrm{Re}\big[\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\,\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\big], \tag{E.57}$$

in the conventions of Eq. (A.21) [72, Eq. (2)]. The set $\mathcal{B}$ contains one wavevector from each pair $\pm\boldsymbol{Q}$ of the grid commensurate with the supercell, and the signs $\varsigma_{\boldsymbol{Q}\nu} = \pm 1$ alternate from one mode to the next in a prescribed order. Wavevectors equal to their own time-reversal partners, such as $\boldsymbol{Q} = 0$, are left out: they describe standing waves that break the crystal symmetry, and their weight vanishes as the supercell grows [72, Sects. II and VII B]. The mean squared displacement of each atom, averaged over the supercell, then equals its thermal value from Eq. (A.21). The method needs the phonons of the primitive cell on the commensurate grid, from DFPT, and one DFT calculation of the displaced supercell with an electronic $\boldsymbol{k}$ sampling appropriate to its size.

The convergence parameter is the supercell size. Including or excluding the wavevectors with $\boldsymbol{Q} = -\boldsymbol{Q}$ modulo $\boldsymbol{G}$ brackets the converged value from above and below, and the width of the bracket is a direct error estimate [72, Sect. VII B]. The phases of the eigenvectors must be fixed consistently from mode to mode before the signs are assigned; Ref. [63, Sect. II] chooses the first nonzero component of each eigenvector positive. Since the displaced supercell is a static calculation, it needs no sum over phonon wavevectors with vanishing denominators, and the adiabatic divergence of polar materials is absent: for GaAs the zero-point renormalization converges smoothly with supercell size, though a $6\times6\times6$ supercell, 432 atoms, is needed to come within ten percent [72, Sect. VII]. The cost is one supercell calculation, which scales as the cube of the number of atoms for the diagonalization; an $8\times8\times8$ supercell of silicon holds 1024 atoms and costs about $8^3 = 512$ times more than a $4\times4\times4$ one at fixed accuracy per atom.

Phonon-assisted absorption. With Eq. (E.56) applied to ε_2, a single ZG supercell gives the temperature-dependent absorption spectrum including indirect transitions. The displaced supercell breaks translational symmetry, so that a photon can connect states that differ in crystal momentum by a phonon wavevector of the supercell grid. For silicon at room temperature, an $8\times8\times8$ supercell with 40 random $\boldsymbol{k}$ points, a scissor of 0.74 eV and a Gaussian broadening of 30 meV reproduces the measured absorption coefficient over seven orders of magnitude [63, Sect. II and VII]. Being adiabatic, the method does not distinguish

the absorption of a phonon from its emission; it cannot reproduce the two thresholds $E_\mathrm{g} \pm \omega_{\boldsymbol{Q}\nu}$ of the Hall–Bardeen–Blatt theory near the onset [63, Sect. II], which is precisely the region that sets the phonon-assisted thresholds of Chap. 13. At 10 mK the phonon-absorption branch is absent, and the onset is at $E_\mathrm{g} + \omega_{\boldsymbol{Q}\nu}$ (Sect. 6.5). Special displacements therefore give the spectrum and its renormalization away from the edge, while the edge itself needs the nonadiabatic treatment with explicit phonon energies of Appendix E.5 (Sect. 13.7).

Zero-point renormalization from special displacements. The band energies of the displaced supercell, unfolded onto the primitive zone as a spectral function, give the renormalized bands $\varepsilon^\mathrm{ren}_{n\boldsymbol{k}}(T)$ directly. Unlike the perturbative sum, this includes all orders in the displacements, within the adiabatic approximation, and needs no electron–phonon vertices. It misses the nonadiabatic correction; in silicon the adiabatic and nonadiabatic AHC renormalizations differ by 8 meV [64, Table VI].

Worked example E.7: The zero-point renormalization of the silicon gap

Perturbative AHC theory, the ZG displacement and experiment each give the indirect-gap renormalization of silicon at $T = 0$. Perturbative AHC theory with the local-density approximation, Sternheimer equations for the high bands and grids up to $100 \times 100 \times 100$ phonon wavevectors, extrapolated to $\delta \to 0$, gives -64.3 meV in the adiabatic form and -56.2 meV in the nonadiabatic one [64, Table VI]. The ZG displacement in an $8 \times 8 \times 8$ supercell, 1024 atoms, gives -57 meV, made of a rise of the valence maximum by 35 meV and a drop of the conduction minimum, near 0.87 of the way from Γ to X, by 22 meV; a $10 \times 10 \times 10$ supercell brackets the converged value between 55 and 65 meV [72, Sect. VII B]. A $4 \times 4 \times 4$ supercell, 128 atoms, is already within ten percent of the converged value [72, Sect. VII]. The measured renormalization is -62 or -64 meV, depending on how the data are extrapolated [64, Table VI], and the zero-point lattice expansion contributes $+9$ meV of the opposite sign [68, Table S2].

The measured value contains both the electron–phonon renormalization and the lattice expansion. The lattice term is therefore added to the computed fixed-volume value before the two are compared [68]: $-56 + 9 = -47$ meV for the nonadiabatic AHC result and $-57 + 9 = -48$ meV for the ZG displacement, against -62 to -64 meV. Equivalently, the electron–phonon part implied by the measurement is -71 to -73 meV, and the calculations underestimate it by about 20 %. The shortfall is consistent with the overscreening of the vertex by the small Kohn–Sham gap: a calculation of the same kind on *GW* bands gives 66 meV [72, Sect. VII B]. The cost differs by orders of magnitude: tens of thousands of DFPT calculations in the primitive cell for the perturbative route, against DFPT phonons on an $8 \times 8 \times 8$ grid and one 1024-atom supercell for the special displacement. A measured low-temperature gap, such as the 1.17 eV of silicon (Sect. 1.3), already contains the zero-point renormalization. A calculation that scissors Kohn–Sham bands to it and then adds a computed renormalization counts it twice; the renormalization is added only to a clamped-nucleus gap, from *GW* or from a hybrid functional. So the window and threshold calculations of Chap. 13 take the measured gap and do not add the renormalization again. Between 10 mK and 1 K all occupations $n_{\boldsymbol{Q}\nu}$ of silicon are of order $\mathrm{e}^{-\omega_{\boldsymbol{Q}\nu}/T}$ with $\omega_{\boldsymbol{Q}\nu}/T \gtrsim 10^2$ for every phonon above 10 meV, and the thermal part of the renormalization is negligible; only the acoustic modes at small Q are thermally occupied, and their contribution to Eq. (E.55) vanishes with their phase space. What remains temperature dependent at those temperatures is the population of the final states, not their energies.

E.8.7 Pitfalls

Three numerical effects are easy to mistake for physics: the band splitting from the $\boldsymbol{Q} = -\boldsymbol{Q}$ modes, the imaginary broadening of the AHC denominators, and the missing edges of phonon-assisted absorption. Special displacements with the $\boldsymbol{Q} = -\boldsymbol{Q}$ modes included split degenerate bands, such as the threefold valence maximum of silicon, and the splitting is an artifact [72, Sect. VII B]. An imaginary broadening $\mathrm{i}\delta$ in the AHC denominators is a numerical device; its effect is extrapolated to $\delta \to 0$, never quoted as a lifetime. The adiabatic special displacements also cannot give the edges of phonon-assisted absorption, which is why the edges of Chap. 13 use the Wannier route of Appendix E.5.

E.9 Magnetic and superconducting parameters

The magnon band of NiO is set almost entirely by one number, the antiferromagnetic exchange between next-nearest-neighbor nickel ions, and the density-functional values of it collected in one comparison span a factor of two and a half, from the same crystal structure, depending on the functional and on the Hubbard correction used [73, Table III]. The superconducting gap of aluminum, by contrast, is computed from first principles to within a few percent of the measured 0.18 meV [74, Table III], but only after one parameter, the Coulomb pseudopotential, is fixed by the measured critical temperature. These two cases mark the range of what first-principles methods deliver for the magnetic and superconducting targets of Sects. 7.1 and 7.2 and of Chaps. 14 and 15. We will see how DFT $+U$ total energies, spin spirals and Green's functions lead to the exchange constants J_{ij}^{ex} of the Heisenberg model, and how the Eliashberg equations turn the spectral function $\alpha^2F(\omega)$ of Sect. 6.4, built from the vertices of Appendices E.4 and E.5, into a gap and a critical temperature. Silicon, the running example of the book, has neither magnetic order nor a Fermi surface; a spin-polarized calculation returns zero moment, and α^2F vanishes identically for an undoped insulator at $T = 0$ (Sect. 6.4). Both statements are themselves computed results, and the worked cases here are NiO and aluminum.

E.9.1 Place on the hierarchy

Both calculations map a first-principles electronic structure onto a model whose parameters are then used elsewhere in the book. For magnets the model is the Heisenberg Hamiltonian, which assumes a rigid local moment on each magnetic ion and keeps only the energy of their relative orientation. Its parameters are static: they come from total energies or from static response, with the electrons treated in a mean field corrected on the d shell. On the many-body axis $\mathcal{X}$ of Sect. 8.5, DFT $+U$ is a static, Hartree–Fock-like treatment of the localized shell, below the dynamical treatments that would describe the paramagnetic phase or the multiplet structure of the ion. The spin-wave theory that uses the constants is the lowest order in $1/|\boldsymbol{S}_j|$ (Sect. 7.1). For superconductors, Eliashberg theory is the self-consistent electron self-energy at second order in the electron–phonon vertex, $n_g = 2$, summed to all orders in Nambu space, with vertex corrections dropped by Migdal's theorem (Sects. 6.3 and 6.4) and with the Coulomb repulsion reduced to one parameter. The weak-coupling BCS theory of Sect. 7.2 is its limit for $\lambda_{\mathrm{ep}} \ll 1$ and a constant interaction.

E.9.2 DFT $+U$ and the Hubbard parameter

Semilocal functionals spread the d electrons of transition-metal oxides over too many orbitals. In NiO they put nickel d states at the top of the valence band and give a small gap of the wrong character, while DFT $+U$ puts oxygen p states there and gives the charge-transfer gap that experiments see [75, Sect. V]. The correction acts on the occupation matrices

$n^{I\sigma}_{mm'} = \sum_{v\boldsymbol{k}} f_{v\boldsymbol{k}} \langle \psi^{\sigma}_{v\boldsymbol{k}} | P^{I}_{mm'} | \psi^{\sigma}_{v\boldsymbol{k}} \rangle$ of the localized orbitals of site I with spin σ, where $P^{I}_{mm'}$ projects on the atomic orbitals m and m' of the shell [75, Eqs. (5) and (8)]. In its simplest rotationally invariant form it adds

$$E_U = \frac{U_{\mathrm{H}}}{2} \sum_{I\sigma} \mathrm{Tr}\big[n^{I\sigma} (1 - n^{I\sigma}) \big] \tag{E.58}$$

to the energy functional [75, Eq. (9)]. In the basis that diagonalizes $n^{I\sigma}$ the trace is $\sum_i \lambda_i (1-\lambda_i)$ over its eigenvalues $0 \le \lambda_i \le 1$, so the correction penalizes fractional occupations and drives each orbital toward full or empty. The physical content is a restoration of the piecewise linearity of the energy as a function of the occupation of the shell, which semilocal functionals violate.

The parameter U_{H} can be computed rather than fitted. In the linear-response method of Cococcioni and de Gironcoli, a small potential shift δV^{U}_{I} is applied to the localized orbitals of one site in a supercell, and the response of the occupations n_J is computed twice: once self-consistently, giving the response matrix $R^{U}_{IJ} = \partial n_I / \partial (\delta V^{U}_{J})$, and once at fixed Kohn–Sham potential, giving the bare response $R^{U,0}_{IJ}$. The Hubbard parameter is the difference of their inverses,

$$U_{\mathrm{H}} = \big[(R^{U,0})^{-1} - (R^{U})^{-1} \big]_{II}, \tag{E.59}$$

which removes the curvature of the energy that comes from rehybridization of the noninteracting bands [75, Eqs. (15), (18) and (19)]. The convergence parameter is the supercell size, since the perturbation of one site must not interact with its images. Density-functional perturbation theory replaces the supercells by a sum over wavevectors in the primitive cell and reaches the same U_{H} with better numerical control: for nickel in NiO, the two routes agree to $5 \times 10^{-3}\,\mathrm{eV}$, and the value converges from 7.90 to 8.17 eV as the supercell grows from $2 \times 2 \times 2$ to $4 \times 4 \times 4$ or the equivalent $\boldsymbol{q}$ grid is refined [76, Table I].

The value is not unique. It depends on the projectors that define $n^{I\sigma}$, on the functional and on the pseudopotential [76, Sect. V], and on whether U_{H} is recomputed self-consistently from the DFT $+U$ ground state. For the same nickel shell in NiO, published linear-response values include 4.6 eV with nonorthogonal atomic projectors [75, Sect. V], 8.17 eV with Löwdin-orthogonalized projectors from a semilocal ground state [76, Table I], and 6.26 eV computed self-consistently [77, Sect. II]. A U_{H} is meaningful only together with its projectors, and a calculation that takes U_{H} from one code and uses it in another changes the physics.

E.9.3 Exchange constants

We write the Heisenberg model as

$$H_{\mathrm{H}} = \sum_{\langle ij \rangle} J^{\mathrm{ex}}_{ij}\, \boldsymbol{S}_i \cdot \boldsymbol{S}_j, \tag{E.60}$$

with each pair of ions counted once and $J^{\mathrm{ex}}_{ij} > 0$ antiferromagnetic. The literature uses every combination of sign, of single or double counting of pairs, and of spin vectors of length $|\boldsymbol{S}|$ or of unit length. The conversions between them are factors of 2 and of $|\boldsymbol{S}|^2$ [78, Table 1], and a value of J^{ex} quoted without its Hamiltonian is not a number.

Energy mapping. The most direct route computes the total energies of several collinear magnetic configurations in DFT $+U$ and equates their differences to those of Eq. (E.60) with classical spins. In the fcc lattice of nickel ions in NiO, with J_1 between the 12 nearest neighbors and J_2 between the 6 next-nearest neighbors, three configurations fix both. In the ferromagnet each ion has all its neighbors parallel. In the antiferromagnet of type I, with ferromagnetic

(001) planes alternating in sign, 4 of the 12 nearest neighbors are parallel and 8 antiparallel, and all 6 next-nearest neighbors, at $\pm a$ along the cube axes, lie in planes of the same sign. In the type-II structure of the ground state, with ferromagnetic (111) planes alternating, half of the nearest neighbors are parallel and all next-nearest neighbors antiparallel. Counting each pair once, the energies per ion are

$$E_{\mathrm{FM}} = (6J_1 + 3J_2)|\boldsymbol{S}|^2, \qquad E_{\mathrm{AFI}} = (-2J_1 + 3J_2)|\boldsymbol{S}|^2, \qquad E_{\mathrm{AFII}} = -3J_2|\boldsymbol{S}|^2, \tag{E.61}$$

relative to a common reference, and therefore

$$J_1 = \frac{E_{\mathrm{FM}} - E_{\mathrm{AFI}}}{8|\boldsymbol{S}|^2}, \qquad J_2 = \frac{E_{\mathrm{FM}} + 3E_{\mathrm{AFI}} - 4E_{\mathrm{AFII}}}{24|\boldsymbol{S}|^2}. \tag{E.62}$$

These are Ref. [73, Eqs. (6) and (7)] converted to Eq. (E.60): that reference uses unit vectors, counts each pair twice and puts a minus sign in front, so that its J_i equal $-J_i/2$ of ours for $|\boldsymbol{S}| = 1$. The two measured values for NiO that it collects, $J_2 = -8.66\,\mathrm{meV}$ (Shanker and Singh) and $-9.51\,\mathrm{meV}$ (Hutchings and Samuelsen [79]) in its convention, are $J_2 = 17.3$ and $19.0\,\mathrm{meV}$ in ours.

More configurations resolve more neighbors. A cleaner variant flips one chosen pair of spins in a large supercell, in the four combinations $\uparrow\uparrow$, $\uparrow\downarrow$, $\downarrow\uparrow$ and $\downarrow\downarrow$, and isolates the coupling of that pair from all others; Ref. [77, Sect. II] uses it for NiO in $4\times4\times4$ supercells. The mapping has three weaknesses. The magnetic moment computed in DFT is not $2|\boldsymbol{S}|$ Bohr magnetons, and the classical $|\boldsymbol{S}|^2$ in Eq. (E.61) must be replaced by a choice, $|\boldsymbol{S}|^2$ or $|\boldsymbol{S}|(|\boldsymbol{S}|+1)$, that changes the constants by up to a factor of two for $|\boldsymbol{S}| = 1$. The moments may change size from one configuration to the next, which a Heisenberg model cannot describe. And the result inherits the full dependence on U_{H}: in NiO the superexchange scales roughly as $t_{\mathrm{hop}}^2/U_{\mathrm{H}}$ (Sect. 7.1), so the spread of U_{H} above becomes a spread of J_2, and the density-functional values of Ref. [73, Table III] range from 11 to 27 meV in our convention (the unrestricted Hartree–Fock value in the same table is 4.6 meV).

Spin spirals. A spin spiral of wavevector $\boldsymbol{Q}$, with every moment tilted by a cone angle θ from a fixed axis and rotated in phase by $\boldsymbol{Q}\cdot\boldsymbol{R}$ from cell to cell, can be computed in the primitive cell when spin–orbit coupling is neglected, by a generalized Bloch theorem. For a lattice with one magnetic ion per cell, Eq. (E.60) has $\boldsymbol{S}_{\boldsymbol{R}}\cdot\boldsymbol{S}_0 = |\boldsymbol{S}|^2[\sin^2\theta\cos(\boldsymbol{Q}\cdot\boldsymbol{R}) + \cos^2\theta]$ and therefore

$$E(\boldsymbol{Q},\theta) - E(0,\theta) = \frac{|\boldsymbol{S}|^2\sin^2\theta}{2}\,[J^{\mathrm{ex}}(\boldsymbol{Q}) - J^{\mathrm{ex}}(0)], \qquad J^{\mathrm{ex}}(\boldsymbol{Q}) = \sum_{\boldsymbol{R}\neq0} J^{\mathrm{ex}}_{0\boldsymbol{R}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}, \tag{E.63}$$

per ion, the single-sublattice case of Ref. [80, Eq. (7)]. A set of spirals on a grid of $\boldsymbol{Q}$ maps out $J^{\mathrm{ex}}(\boldsymbol{Q})$, and a Fourier transform gives the real-space constants out to a range set by the grid. Small cone angles keep the perturbation of the ground state small, which reduces the change of the moments that plagues collinear mapping [80, Sect. II]. For a ferromagnet, linear spin-wave theory in the convention of Eq. (E.60) has $\omega(\boldsymbol{Q}) = |\boldsymbol{S}|[J^{\mathrm{ex}}(\boldsymbol{Q}) - J^{\mathrm{ex}}(0)]$, so the spiral energy is the magnon energy directly: $\omega(\boldsymbol{Q}) = 2[E(\boldsymbol{Q},\theta) - E(0,\theta)]/(|\boldsymbol{S}|\sin^2\theta)$, the frozen-magnon relation. For a collinear antiferromagnet with two sublattices, the same theory has

$$\omega(\boldsymbol{Q}) = |\boldsymbol{S}|\Big\{\big[J^{\mathrm{ex}}_{\mathrm{AB}}(0) - J^{\mathrm{ex}}_{\mathrm{AA}}(0) + J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{Q})\big]^2 - |J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{Q})|^2\Big\}^{1/2}, \tag{E.64}$$

with $J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{Q})$ and $J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{Q})$ the sums of $J^{\mathrm{ex}}_{0\boldsymbol{R}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$ over the sites $\boldsymbol{R}\neq0$ of the same and of the other sublattice; with $J^{\mathrm{ex}}_{\mathrm{AB}} = 0$ it reduces to the ferromagnetic form. The spirals then need one cone angle per sublattice [80, Eq. (8)], and the two sums follow from their energies.

Green's functions. The third route needs only the ground state. Rotating two moments by small angles changes the energy, by the magnetic force theorem, by the change of the single-particle energies at fixed potential, which second-order perturbation theory expresses through the Green's function of the ground state. The result is the formula of Liechtenstein, Katsnelson, Antropov and Gubanov, which in a Wannier basis reads

$$J_{ij}^{\mathrm{LKAG}} = -\frac{1}{4\pi}\int_{-\infty}^{\mu_{\mathrm{ch}}} \mathrm{d}\varepsilon\ \mathrm{Im}\,\mathrm{Tr}\big[\Delta_i\,\mathsf{G}_{ij}^{\uparrow}(\varepsilon)\,\Delta_j\,\mathsf{G}_{ji}^{\downarrow}(\varepsilon)\big] \tag{E.65}$$

in the convention of its source [78, Eq. (19) and Table 1], $H = -\sum_{i\neq j} J_{ij}^{\mathrm{LKAG}}\,\hat{\boldsymbol{e}}_i\cdot\hat{\boldsymbol{e}}_j$ with unit vectors and each pair counted twice, so that $J_{ij}^{\mathrm{ex}} = -2J_{ij}^{\mathrm{LKAG}}/|\boldsymbol{S}|^2$ in the convention of Eq. (E.60). Here Δ_i is the exchange splitting of the on-site Hamiltonian of ion i, G_{ij}^{σ} the block of the spin-σ Green's function between the Wannier orbitals of ions i and j, the trace runs over orbitals, and the integral runs up to the chemical potential μ_{ch} (Sect. 2.5). One ground-state calculation yields every pair at once, ferromagnetic or antiferromagnetic, and decomposes each constant by orbital. The formula describes infinitesimal rotations about the ground state, while energy mapping measures large ones, and the two differ where the moments are not rigid. For NiO, linear spin-wave theory with constants from energy mapping overestimates the measured magnon energies by about 14 percent, and with constants from the Green's-function formula by up to 40 percent [77, Sect. III]. The convergence parameters of Eq. (E.65) are the Wannier basis, which must contain the magnetic d orbitals and the ligand p orbitals that mediate superexchange, and the $\boldsymbol{k}$ grid of the Green's function, which sets the range of $|\boldsymbol{R}_i - \boldsymbol{R}_j|$ that is free of images.

Spin response. The most complete route skips the Heisenberg model. Time-dependent density-functional perturbation theory computes the transverse spin susceptibility of the DFT $+U$ ground state as a function of $(\boldsymbol{Q},\omega)$, and its poles are the magnons; constants are then fitted to the dispersion only for comparison [77, Sect. II]. This is the spin analogue of the dielectric response of Appendix E.6, and it costs correspondingly more.

E.9.4 Eliashberg theory

The gap and critical temperature of a phonon-mediated superconductor follow from the Eliashberg equations, the Nambu–Gor'kov form of the electron self-energy with the phonon propagator of Sect. 6.4 and the Coulomb repulsion. On the imaginary axis, at the fermionic Matsubara frequencies ω_j^{F} of Sect. 4.7, and for an isotropic gap restricted to the Fermi surface, they are two coupled equations for a mass-renormalization function $Z^{\mathrm{El}}(\mathrm{i}\omega_j^{\mathrm{F}})$ and a gap function $\Delta^{\mathrm{El}}(\mathrm{i}\omega_j^{\mathrm{F}})$,

$$Z^{\mathrm{El}}(\mathrm{i}\omega_j^{\mathrm{F}}) = 1 + \frac{\pi T}{\omega_j^{\mathrm{F}}}\sum_{j'}\frac{\omega_{j'}^{\mathrm{F}}}{\sqrt{(\omega_{j'}^{\mathrm{F}})^2 + \Delta^{\mathrm{El}}(\mathrm{i}\omega_{j'}^{\mathrm{F}})^2}}\,\lambda_{\mathrm{ep}}(\omega_j^{\mathrm{F}} - \omega_{j'}^{\mathrm{F}}), \tag{E.66}$$

$$Z^{\mathrm{El}}(\mathrm{i}\omega_j^{\mathrm{F}})\,\Delta^{\mathrm{El}}(\mathrm{i}\omega_j^{\mathrm{F}}) = \pi T\sum_{j'}\frac{\Delta^{\mathrm{El}}(\mathrm{i}\omega_{j'}^{\mathrm{F}})}{\sqrt{(\omega_{j'}^{\mathrm{F}})^2 + \Delta^{\mathrm{El}}(\mathrm{i}\omega_{j'}^{\mathrm{F}})^2}}\Big[\lambda_{\mathrm{ep}}(\omega_j^{\mathrm{F}} - \omega_{j'}^{\mathrm{F}}) - \mu_{\mathrm{C}}^{*}\Big], \tag{E.67}$$

with the frequency-dependent coupling

$$\lambda_{\mathrm{ep}}(\omega^{\mathrm{B}}) = \int_0^{\infty}\mathrm{d}\omega\,\frac{2\omega\,\alpha^2F(\omega)}{\omega^2 + (\omega^{\mathrm{B}})^2}, \tag{E.68}$$

evaluated at the bosonic differences $\omega^{\mathrm{B}} = \omega_j^{\mathrm{F}} - \omega_{j'}^{\mathrm{F}}$, so that $\lambda_{\mathrm{ep}}(0)$ is the coupling constant λ_{ep} of Sect. 6.4. These are the isotropic forms of the Fermi-surface-restricted equations of Ref. [81,

Eqs. (41), (42) and (48)]; the literature writes Z, Δ, λ and μ^*, symbols that this book uses for other things. The highest temperature at which Eq. (E.67) has a nonzero solution is T_c, and the gap on the real axis follows by analytic continuation of $\Delta^{\mathrm{El}}(\mathrm{i}\omega_j^{\mathrm{F}})$, usually with Padé approximants (Appendix C.5).

The Coulomb pseudopotential μ_{C}^* is the weak point. It replaces the screened Coulomb repulsion between paired electrons, renormalized down from its bare Fermi-surface value by the retardation of the phonon interaction, and in most calculations it is not computed but chosen, with values from 0.1 to 0.2 giving reasonable agreement with experiment [81, Sect. 4]. Since T_c depends on it exponentially, a computed T_c with a chosen μ_{C}^* is a consistency check on α^2F rather than a prediction. Two alternatives remove the parameter: computing μ_{C}^* as a Fermi-surface average of the screened interaction, and density-functional theory for superconductors [81, Sect. 4].

A quick estimate replaces the equations by the McMillan formula in the form of Allen and Dynes,

$$T_c = \frac{\omega_{\log}}{1.2} \exp\left[-\frac{1.04(1+\lambda_{\mathrm{ep}})}{\lambda_{\mathrm{ep}} - \mu_{\mathrm{C}}^*(1+0.62\lambda_{\mathrm{ep}})} \right], \qquad \omega_{\log} = \exp\left[\frac{2}{\lambda_{\mathrm{ep}}} \int_0^\infty \mathrm{d}\omega \, \frac{\alpha^2F(\omega)}{\omega} \ln\omega \right], \tag{E.69}$$

which is Ref. [81, Eq. (30)], a fit to numerical solutions of the Eliashberg equations useful for preliminary estimates.

Convergence and cost. The input α^2F is a double Fermi-surface average of $|g|^2$, with two energy delta functions, and it converges only on fine grids of $\boldsymbol{k}$ and $\boldsymbol{Q}$, which in practice means Wannier interpolation of the vertices (Appendix E.5). The anisotropic equations, with a gap that varies over the Fermi surface, need the vertices on commensurate fine grids, restricted to an energy window of a few times the largest phonon energy around the Fermi level [81, Sect. 4]. The Matsubara sum needs frequencies up to a cutoff, commonly ten times the largest phonon frequency, with convergence tests recommended [81, Sect. 4]. Since the spacing is $2\pi T$, the number of frequencies grows as $1/T$, and low-T_c superconductors are expensive: for aluminum, with phonons up to 40.5 meV ($\omega_{\max} \approx 470\,\mathrm{K}$ [74, Sect. III]), a cutoff of ten times that holds about 750 positive frequencies at 1 K and 7500 at 0.1 K. Sparse sampling of the Matsubara axis reduces the count [81, Sect. 4]. The analytic continuation to the real axis is the last numerical hazard: Padé approximants can produce spurious structure near the gap edge, which is where the pair-breaking absorption of Chap. 15 begins, and a real-axis solution or a check against the imaginary-axis data is needed there.

Worked example E.8: Aluminum from its Eliashberg function

Linear-response calculations of aluminum give $\lambda_{\mathrm{ep}} = 0.44$, against 0.42 from tunneling, and a logarithmic phonon frequency $\omega_{\log} = 270\,\mathrm{K}$ [74, Tables II and III]. Solving the Eliashberg equations with this α^2F, the measured $T_c = 1.18\,\mathrm{K}$ is reproduced with $\mu_{\mathrm{C}}^* = 0.12$, and the zero-temperature gap is then 0.18 meV, equal to the measured value [74, Table III]. The weak-coupling ratio of Sect. 7.2 agrees, $\Delta_{\mathrm{BCS}} = 1.764\,T_c = 0.179\,\mathrm{meV}$, as it should for $\lambda_{\mathrm{ep}} = 0.44$.

Eq. (E.69) with the same three numbers gives $T_c = (270/1.2)\,\mathrm{e}^{-5.213}\,\mathrm{K} = 1.22\,\mathrm{K}$, within four percent of the full solution. Its sensitivity to μ_{C}^* is the logarithmic derivative of the exponent, $\partial \ln T_c/\partial\mu_{\mathrm{C}}^* = -1.04(1+\lambda_{\mathrm{ep}})(1+0.62\lambda_{\mathrm{ep}})/[\lambda_{\mathrm{ep}} - \mu_{\mathrm{C}}^*(1+0.62\lambda_{\mathrm{ep}})]^2 = -23$. A change of μ_{C}^* by ± 0.02, well inside the range used in the literature, moves T_c by a factor of 0.60 or 1.53, from 0.74 K at $\mu_{\mathrm{C}}^* = 0.14$ to 1.87 K at 0.10. The gap and T_c of aluminum are measured, and the first-principles α^2F enters where a measured quantity is not available,

in the frequency dependence of the phonon-assisted pair-breaking absorption of Chap. 15.

Exchange constants and Eliashberg gaps carry uncertainties of different kinds. Exchange constants depend on the treatment of the localized shell, and their uncertainty is a spread over choices of U_{H} and of mapping; Eliashberg gaps depend on the electron–phonon vertices of Appendix E.5, which are well controlled, and on one phenomenological parameter. Wherever a magnetic or superconducting target is ranked by a computed parameter, the corresponding spread is the theory uncertainty to carry (Sect. 8.8).

E.10 Defects

A carbon pair with a bound hydrogen, the T center of silicon, emits its zero-phonon line at 935 meV; a hybrid-functional calculation in a 512-atom supercell puts it at 985 meV [82]. Displacing a silicon atom from its site takes between 17.5 and 63.5 eV, depending on the direction in which it is struck [83]. Both numbers are defect properties that enter the book: the first in the color centers and dopants that serve as targets and readouts (Sects. 7.4 and 16.6), the second in the energy that a nuclear recoil stores in the lattice (Chap. 19). Both come from supercells whose defect must be isolated from its own images, electrostatically for charged states and dynamically for a recoil; the effective-mass theory of shallow donors, which no supercell reaches, is in Sect. 7.4 and Appendix D.23.

E.10.1 Place on the hierarchy

A defect calculation is a calculation for a crystal whose translational symmetry is broken, and periodic codes restore it by repeating the defect in a supercell of side L_{sc}. Each defect configuration is then an ordinary clamped-nucleus calculation at the base point $\mathfrak{T}_0$ of Sect. 8.5, with one control that the perfect crystal does not need: the result must converge in L_{sc}, and for charged defects it converges only after an electrostatic correction. The ground state of each charge state is a DFT ground state; excited states of color centers are obtained from constrained occupations, which go beyond ground-state DFT without reaching a correlated treatment of the many-body approximation $\mathcal{X}$. The vibronic side of an optical transition is computed in the harmonic approximation about each minimum, as the displaced-oscillator problem of Sect. 5.6, and summed to all orders in the Huang–Rhys factor, which is the multiphonon end of the lattice regime $\mathcal{K}$. Displacement thresholds, finally, come from molecular dynamics with classical nuclei on the Born–Oppenheimer surface, $n_B = 0$, with the electronic excitations that a fast recoil produces left out.

E.10.2 Formation energies and charge states

The formation energy of a defect D in charge state z_{D}, in units of e, compares two supercells, with and without the defect,

$$E_{\mathrm{D}}^{\mathrm{f}}(z_{\mathrm{D}}) = E_{\mathrm{D}}(z_{\mathrm{D}}) - E_{\mathrm{host}} - \sum_s n_s\,\mu_s^{\mathrm{at}} + z_{\mathrm{D}}\big(\varepsilon_{\mathrm{VBM}} + \mu_{\mathrm{ch}}\big) + E_{\mathrm{corr}}, \tag{E.70}$$

where E_{D} and E_{host} are the total energies of the defective and the host supercell, n_s atoms of species s are added ($n_s > 0$) or removed ($n_s < 0$) and exchanged with a reservoir at chemical potential μ_s^{at}, the electron chemical potential μ_{ch} is measured from the valence maximum $\varepsilon_{\mathrm{VBM}}$, and E_{corr} is the finite-size correction of the next subsection [84, Eq. (3)]. The atomic chemical potentials are bounded by the stability of the host against its elements and against competing phases, and μ_{ch} ranges over the gap [84, Sect. 3]. Where two charge states have

equal formation energy, $E^{\mathrm{f}}_{\mathrm{D}}(z_{\mathrm{D}}) = E^{\mathrm{f}}_{\mathrm{D}}(z'_{\mathrm{D}})$, Eq. (E.70) gives the thermodynamic transition level

$$\mu_{\mathrm{ch}}(z_{\mathrm{D}}/z'_{\mathrm{D}}) = \frac{E^{\mathrm{f}}_{\mathrm{D}}(z_{\mathrm{D}})\big|_{\mu_{\mathrm{ch}}=0} - E^{\mathrm{f}}_{\mathrm{D}}(z'_{\mathrm{D}})\big|_{\mu_{\mathrm{ch}}=0}}{z'_{\mathrm{D}} - z_{\mathrm{D}}}, \tag{E.71}$$

the electron chemical potential at which the defect changes charge with full relaxation of the lattice. It is the quantity that photoluminescence quenching and capture experiments measure, and it differs from a Kohn–Sham eigenvalue, which carries no relaxation and no correction.

The defect is relaxed in a supercell as close to cubic as possible, built from the conventional cell, at the volume of the host [84, Sect. 3]. The work lies in the number of charge and spin states, which must all be tried, since a defect in a nonmagnetic host may have a spin-polarized ground state. The gap problem of semilocal functionals is more serious for defects than for bulk bands, because defect levels sit inside the gap and their position relative to both band edges matters; hybrid functionals, which give silicon a gap of 1.114 eV [82], are the standard for levels and optical energies.

E.10.3 Charged defects in a periodic cell

A charged defect in a supercell is a periodic array of charges, neutralized by a uniform background so that the electrostatic energy is finite. Both the interaction with the images and the background are artifacts. The leading term is the energy of a simple-cubic lattice of point charges $z_{\mathrm{D}}e$ in a compensating background and a dielectric medium. In Heaviside–Lorentz units, with $e^2 = 4\pi\alpha$, the Coulomb energy of two charges e at distance r in vacuum is α/r, and the Madelung energy per charge of the array is $-z_{\mathrm{D}}^2 \alpha\, \alpha_{\mathrm{M}}/(2\varepsilon L_{\mathrm{sc}})$, with $\alpha_{\mathrm{M}} = 2.8373$ the Madelung constant of the simple-cubic lattice [85, Sect. II]. The correction removes it:

$$E_{\mathrm{corr}} \simeq \frac{z_{\mathrm{D}}^2\, \alpha\, \alpha_{\mathrm{M}}}{2\varepsilon\, L_{\mathrm{sc}}} + O(L_{\mathrm{sc}}^{-3}), \tag{E.72}$$

where the next term, of order $z_{\mathrm{D}} L_{\mathrm{sc}}^{-3}$ times the quadrupole moment of the defect charge, comes from the shape of that charge; this is the Makov–Payne form, quoted in Gaussian units in Ref. [84, Eq. (8)]. The dielectric constant is ε_0 for a relaxed defect, whose screening includes the ions, and ε_∞ for a vertical transition at frozen nuclei; in silicon the two are equal, 11.7. The bare point-charge term often overcorrects: for the doubly charged carbon vacancy in diamond it does so clearly [85, Sect. III B], because the defect charge is not a point. The quadrupole term is hard to determine, and a second artifact remains: the average electrostatic potential of a charged cell is set by convention, so that band edges and defect levels of different cells must be aligned. The schemes in current use, by Freysoldt, Neugebauer and Van de Walle and by Kumagai and Oba, model the defect charge as a Gaussian, compute its periodic energy analytically, and align the potential far from the defect, read from the plane-averaged or the atomic-site potentials, with the model potential [86, Sect. II and Eq. (1)]. The Kumagai–Oba version also handles an anisotropic dielectric tensor, and shows that the conventional potential-alignment term is already contained in the image-charge correction [86, Sect. IV]. Without a correction, formation energies of charged defects converge as $1/L_{\mathrm{sc}}$, and their spread between correction schemes translates into defect densities that differ by orders of magnitude [84, Sect. 3].

E.10.4 Color centers

The optical transition of a color center connects two states of a few localized electrons. Its zero-phonon energy E_{ZPL} is the difference of the total energies of the relaxed excited and ground states. When both states are well described by single Slater determinants, the excited

state is computed by constraining the occupations, promoting an electron from one Kohn–Sham orbital to another and relaxing the structure in that configuration; this is the ΔSCF method. For the negatively charged nitrogen–vacancy center of diamond, with its triplet transition between single-determinant states, a 512-site supercell gives $E_{\mathrm{ZPL}} = 1.757\,\mathrm{eV}$ with a semilocal functional and 2.035 eV with the HSE hybrid functional, against the measured 1.945 eV [87, Sect. 4.1]. For the T center of silicon, the same approach in a 512-atom supercell with HSE and Γ-point sampling gives 985 meV against 935 meV, and the $(0/-1)$ transition level at 1.07 eV above the valence maximum [82]. The radiative lifetime, from the transition dipole of the two Kohn–Sham orbitals and the Wigner–Weisskopf rate, comes out as 1.3 to 5 µs, against a measured 0.9 µs [82].

The same two relaxed geometries give the vibronic coupling. The Huang–Rhys factor S_{HR} of Sect. 5.6 is the number of phonons emitted on average when the lattice relaxes after the transition, $S_{\mathrm{HR}} = \sum_k \omega_k \Delta Q_k^2/2$, with ΔQ_k the projection of the mass-weighted displacement between the two minima on the vibrational mode k of frequency ω_k; the zero-phonon fraction of the emission is $\mathrm{e}^{-S_{\mathrm{HR}}}$. Its convergence with supercell size is slow, because the relaxation reaches into long-wavelength acoustic modes that a small cell does not contain. For the nitrogen–vacancy center, a 64-site supercell gives $S_{\mathrm{HR}} = 3.02$, 20 percent below the converged 3.67, and the converged value needs an embedding scheme: the forces of the transition from a 512-site hybrid-functional calculation, and the vibrational modes of supercells of up to 10 648 sites from force constants computed with a semilocal functional [87, Sects. 2.3 and 4.2]. The measured value is 3.85 ± 0.05 [87, Sect. 3].

ΔSCF has a clear limit. States that are not single determinants, such as the singlet states of the nitrogen–vacancy center, need a correlated treatment of an active space of defect orbitals, embedded in the screened host, and the density transition matrix elements that a dark-matter excitation needs, $|\langle F|n_e(\boldsymbol{q})|I\rangle|^2$ over keV momenta, need the many-body states themselves rather than their energies (Sect. 16.6). Shallow donors are the opposite limit: their effective-mass envelope extends over several nanometers (Sect. 7.4), far beyond any supercell, and a supercell calculation can only supply the central-cell correction to the effective-mass theory of Appendix D.23.

E.10.5 Displacement thresholds

The displacement threshold E_{d} of Sect. 7.4 is the smallest kinetic energy that, given to one atom in a given direction, leaves a stable defect. It is computed by molecular dynamics over many directions: for silicon it ranges from 17.5 to 63.5 eV with the Stillinger–Weber potential, which of three common silicon potentials reproduces the density-functional thresholds best [83]. Appendix E.11 gives the protocol, its cost and its approximations, among them the electronic excitation that Born–Oppenheimer dynamics leaves out (Chap. 19).

E.10.6 Tunneling defects in metals

A defect that tunnels between two configurations L and R, such as a light interstitial or an atom in a disordered film, couples to the conduction electrons of a metal through the change of their scattering potential. That coupling damps the tunneling and is described by the ohmic constant α_{ohm} of Sect. 16.8. In the Born limit, $\alpha_{\mathrm{ohm}} = 4K_{\mathrm{b}}$ with $K_{\mathrm{b}} = (\mathcal{N}_{\mathrm{F}} V_{\mathrm{d}} \Omega)^2$, where $\mathcal{N}_{\mathrm{F}}$ is the density of states per spin and volume at the Fermi level and V_{d} half the difference of the contact potentials of the two configurations, which sit at $\pm V_{\mathrm{d}}$ [88, Eq. (4)]. Beyond the Born limit the change of potential is described by the scattering matrices S_L and S_R of the conduction electrons at the Fermi energy in the two configurations, with $S = \mathrm{e}^{2\mathrm{i}\delta}$ for a channel of phase shift δ, and by the eigenvalues $\mathrm{e}^{2\mathrm{i}\tilde{\delta}_\nu}$ of $S_R S_L^\dagger$, so that $\tilde{\delta}_\nu$ is the difference of the phase

shifts in channel ν. A channel whose phase shift changes by $\tilde{\delta}$ reduces the long-time overlap of the two Fermi seas as $t^{-(\tilde{\delta}/\pi)^2}$, the orthogonality exponent of Anderson [89], and an ohmic bath of coupling α_{ohm} produces $t^{-2\alpha_{\mathrm{ohm}}}$. Equating the two over all channels and both spins,

$$\alpha_{\mathrm{ohm}} = \frac{1}{2\pi^2} \sum_{\sigma} \sum_{\nu} \tilde{\delta}_{\nu}^{2}, \tag{E.73}$$

which in the Born limit, with one channel and phase shifts $\mp\pi\mathcal{N}_{\mathrm{F}}V_{\mathrm{d}}\Omega$ in the two configurations, so that $\tilde{\delta} = 2\pi\mathcal{N}_{\mathrm{F}}V_{\mathrm{d}}\Omega$, returns $\alpha_{\mathrm{ohm}} = 4(\mathcal{N}_{\mathrm{F}}V_{\mathrm{d}}\Omega)^2 = 4K_{\mathrm{b}}$. A first-principles calculation needs the two relaxed configurations, from a nudged-elastic-band search (Appendix E.11), identical relaxation of the host around both, and the scattering phases at the Fermi energy, either from a Green's-function method or from the overlap of the two many-electron ground states of a supercell as a function of its size. The orthogonality exponent appears only with fixed occupations: $\boldsymbol{k}$ sampling and smearing of the Fermi surface blur the logarithm on which it rests and must be removed or extrapolated.

Worked example E.9: A charged defect in a silicon supercell

We estimate the image-charge error for a singly charged defect, $z_{\mathrm{D}} = \pm 1$, in cubic supercells of silicon built from the conventional cell, $L_{\mathrm{sc}} = na$ with $a = 5.431\,\text{Å}$, and with $\alpha\hbar c = 14.40\,\text{eV Å}$ and $\varepsilon_0 = 11.7$. Eq. (E.72) gives

$$E_{\mathrm{corr}} = \frac{2.8373 \times 14.40\,\text{eV Å}}{2 \times 11.7 \times L_{\mathrm{sc}}} = 0.161\,\text{eV},\ 0.107\,\text{eV},\ 0.080\,\text{eV}$$

for 64, 216 and 512 atoms ($n = 2$, 3, 4). Even the 512-atom cell, the size used for the T center, carries a point-charge correction of 80 meV, larger than the 50 meV by which its computed zero-phonon line misses the measurement. Since the point-charge term can overcorrect (above), production calculations use the Gaussian-charge schemes; what the estimate shows is that the error does not become negligible at any affordable size. It falls only as $1/L_{\mathrm{sc}}$, that is, as $N_{\mathrm{at}}^{-1/3}$, so that halving it means eight times more atoms. The cost of a hybrid-functional calculation grows faster than the cube of the number of atoms, and a 4096-atom cell is out of reach where a 512-atom cell is routine; the correction, not the cell, is what converges the result. The neutral T center needs none of this, and its error budget is the functional; its $(0/-1)$ level at 1.07 eV does, since the charged state enters it [82].

Appendix E.11 extends these calculations to millions of atoms and to the statistics of nuclear-recoil cascades.

E.11 Recoil molecular dynamics

A silicon atom struck with 100 eV leaves its site at 26 km/s, comes within about 1 Å of a neighbor, and within a few picoseconds has either fallen back into the lattice or left behind a vacancy and an interstitial that store several eV between them. Nothing in this sequence is a small perturbation of the harmonic crystal. The atoms leave the region where the force constants of Sect. 5.2 apply, bonds break and re-form, and the outcome is decided by the shape of the Born–Oppenheimer energy surface far from equilibrium. Sect. 19.1 hands the recoil over to this dynamics, and Sects. 19.4 and 19.6 need four of its products: the displacement threshold E_{d} as a function of recoil direction (Sect. 7.4), the distribution of the stored energy E_{st} at a given recoil energy E_R, the configurations of the defects that survive, and the barriers

along the paths by which they recombine or move. Each comes from many trajectories on an energy surface fitted to density-functional theory, with the close collisions patched in by a screened Coulomb repulsion, and from the statistics of those trajectories over recoil directions.

E.11.1 Computed quantities

Molecular dynamics integrates Newton's equations for the nuclei on a single energy surface,

$$M_\kappa \, \ddot{\boldsymbol{R}}_I = -\frac{\partial E^{\mathrm{ad}}(\{\boldsymbol{R}\})}{\partial \boldsymbol{R}_I}, \tag{E.74}$$

where $\boldsymbol{R}_I$ is the position of nucleus I of species κ and $E^{\mathrm{ad}}(\{\boldsymbol{R}\})$ is the electronic ground-state energy at fixed nuclear positions, the Born–Oppenheimer surface of Sect. 5.1. A recoil simulation starts from a crystal equilibrated at the base temperature, gives one atom (the primary knock-on atom) the kinetic energy E_R in a chosen direction $\hat{\boldsymbol{r}}_R$, integrates Eq. (E.74) until the excess energy has spread through the cell, and records what changed. The quantities read off a single trajectory are whether a defect survived, which one, and the potential energy it stores. Repeated over directions and energies, the trajectories give the probability $p_{\mathrm{FP}}(E_R, \hat{\boldsymbol{r}}_R)$ that a Frenkel pair survives, the threshold surface $E_{\mathrm{d}}(\hat{\boldsymbol{r}}_R)$, and the distribution of E_{st} that Sect. 19.4 bounds with calibration data.

Everything therefore rests on E^{ad}. Density-functional theory (Appendix E.1) gives E^{ad} and its gradients at a cost that grows as the cube of the number of atoms. A plane-wave calculation with N_{at} atoms holds a number of occupied bands and a number of plane waves that are both proportional to N_{at}, and orthogonalizing the bands costs their number squared times the basis size. A recoil cascade needs thousands of atoms, to keep the recoil away from its periodic images, and thousands of trajectories, to sample directions. Direct density-functional molecular dynamics can afford a few directions in a cell of a few hundred atoms, and serves as the reference; the production runs use an interatomic potential that reproduces E^{ad} at a cost linear in N_{at}.

E.11.2 Place on the hierarchy

In the language of the truncation label $\mathfrak{T}$ of Sect. 8.5, Eq. (E.74) is the adiabatic limit, $n_B = 0$: the electrons stay in their ground state for every nuclear configuration and never take energy from the nuclei. It keeps the anharmonicity of E^{ad} to all orders, which the expansion of Sect. 5.7 cannot, and it sits beyond the impulse regime of the lattice label $\mathcal{K}$, since the struck atom does not merely recoil from its site but collides with its neighbors. Two approximations come with it. The nuclei are classical, so zero-point motion is absent and the lattice starts at rest apart from thermal motion. For recoils above 10 eV this does not matter for the mean threshold, since the kinetic energies exceed the vibrational energies of a few tens of meV by orders of magnitude [90, Sect. II A]; it does set the width of the threshold step, 0.7 eV at 12.5 eV (Sect. 19.1.1). It does matter for the slow processes at millikelvin, where barriers are crossed by tunneling rather than by thermal activation; we return to this when we discuss defect paths. The second approximation is that energy given to the electrons is lost from the description. At 1 keV a silicon recoil already transfers part of its energy to electronic excitations (Sect. 19.2), and that channel needs the nonadiabatic treatment of Sect. 19.2; molecular dynamics then supplies the nuclear trajectory on which the electronic losses are evaluated.

E.11.3 Interatomic potentials

An interatomic potential replaces E^{ad} by a sum of local terms,

$$E^{\mathrm{ad}}(\{\boldsymbol{R}\}) \approx \sum_{I} E^{\mathrm{at}}_{\kappa}(\mathcal{N}_I), \tag{E.75}$$

where $\mathcal{N}_I$ is the set of positions of the neighbors of atom I within a cutoff radius r_{c}, measured from $\boldsymbol{R}_I$. Locality is what makes the cost linear in N_{at}. It holds for covalent and metallic solids because the density matrix decays with distance, and it fails for long-range electrostatics in ionic materials, which are then added as an explicit Coulomb term.

The classical forms write E^{at}_{κ} as a handful of terms with physically motivated shapes. The Stillinger–Weber potential for silicon is a pair term plus a three-body term that penalizes deviations from the tetrahedral angle [91]; Tersoff-type potentials let the strength of a bond depend on its local coordination. With some ten parameters fitted to a few properties, they reproduce the crystal well and the configurations far from it less well. For the defect energies and planar faults of silicon, the analytical potentials differ from density-functional values by tens of percent for at least some of the defects [92, Sect. III E].

A machine-learned potential keeps Eq. (E.75) but lets E^{at}_{κ} be a flexible function of the neighborhood, fitted to density-functional data. The neighborhood is first encoded in a descriptor that is invariant under rotations, reflections and permutations of like atoms. Examples are symmetry functions of the neighbor distances and angles feeding a neural network [93], the smooth overlap of atomic positions feeding a kernel regression [94], or a systematic expansion in body-ordered invariants, the atomic cluster expansion [95]. Equivariant message-passing networks build the descriptor from features that rotate with the structure and are passed between neighbors over several layers [96, 97]. The families differ in cost and in how much data they need; they share Eq. (E.75) and therefore share its failure modes.

The Gaussian approximation potential for silicon [92] is a well-documented example and the one we use as the silicon case. Its energy is a fixed pair term plus a kernel sum,

$$E^{\mathrm{ad}} = \sum_{I<J} V^{(2)}(|\boldsymbol{R}_I - \boldsymbol{R}_J|) + \sum_{I} \sum_{s=1}^{M_{\mathrm{rep}}} \alpha_s \, K_{\mathrm{GAP}}(\mathcal{N}_I, \mathcal{N}_s), \tag{E.76}$$

with K_{GAP} a similarity kernel between the neighborhood of atom I and M_{rep} representative neighborhoods $\mathcal{N}_s$ drawn from the training data, and coefficients α_s fitted by regularized least squares [92, Eq. (1)]. The pair term $V^{(2)}$ is purely repulsive and is fitted to the density-functional energy of an isolated pair of silicon atoms; it carries the steep exchange repulsion at short distance and leaves the bonding to the kernel term [92, Sect. II A]. The cutoff is 5 Å.

E.11.4 Training sets

The fit is only as good as the configurations it has seen. Each density-functional calculation in the training set yields a total energy, six components of the stress and three force components per atom, and all of them enter the fit with tolerances that set their relative weight. For silicon the database holds 2475 periodic cells with 171 815 atomic environments [92, Table I]. They cover the diamond crystal at strains and temperatures up to melting, other crystal phases (β-tin, simple hexagonal, bc8, fcc, hcp and others), liquid and amorphous cells, low-index surfaces with their reconstructions and decohesion paths, the vacancy, the divacancy and the interstitials in cells of 63 to 217 atoms, crack tips and a screw dislocation. The reference is the PW91 functional, with a plane-wave cutoff of 250 eV [92, Sect. II B].

What a recoil needs from this list is the defect region and the short-distance region. The defects are there by design. The short distances are not, and thermal sampling cannot put

them there. At 1.0 Å two silicon atoms repel each other with about 51 eV (Eq. (E.77) below), while the hottest training configurations, liquid silicon near 2000 K, have $T \approx 0.17$ eV. A thermal ensemble visits such a pair separation with a Boltzmann weight of order $\mathrm{e}^{-51/0.17}$, which is zero for every practical purpose. A training set assembled from equilibrium and thermal configurations therefore leaves the potential unconstrained exactly where a recoil cascade begins, and the cure is the splice described next.

E.11.5 Close collisions and the Ziegler–Biersack–Littmark splice

At separations well below a bond length the energy of two atoms is dominated by the Coulomb repulsion of the nuclei, screened by the electrons that remain between them. The standard description is the universal screened potential of Ziegler, Biersack and Littmark (ZBL) [98, Chap. 2],

$$V_{\mathrm{ZBL}}(r) = \frac{Z_1 Z_2\, \alpha}{r}\, \phi\!\left(\frac{r}{a_{\mathrm{U}}}\right), \qquad a_{\mathrm{U}} = \frac{0.8854\, a_0}{Z_1^{0.23} + Z_2^{0.23}}, \tag{E.77}$$

with $\phi(x) = 0.1818\,\mathrm{e}^{-3.2x} + 0.5099\,\mathrm{e}^{-0.9423x} + 0.2802\,\mathrm{e}^{-0.4029x} + 0.02817\,\mathrm{e}^{-0.2016x}$, a_0 the Bohr radius and $Z_{1,2}$ the nuclear charges. It is a fit to the repulsion of many atom pairs computed from their free-atom charge densities, which makes it universal in the sense that one screening function serves all pairs once lengths are measured in a_{U}. For silicon on silicon, $a_{\mathrm{U}} = 0.128$ Å, and the unscreened Coulomb energy $Z^2\alpha/r = 2822\,\mathrm{eV\,Å}/r$ is reduced by ϕ to 51 eV at 1.0 Å and 9.6 eV at 1.5 Å.

In a head-on collision of the recoiling atom with a neighbor at rest, the two equal masses share the kinetic energy $E_R/2$ of the center-of-mass frame, and the turning point $r_{\min}$ solves $V_{\mathrm{ZBL}}(r_{\min}) = E_R/2$. Solved numerically with Eq. (E.77) for silicon, it is 1.49, 1.20, 1.00, 0.83, 0.49 and 0.18 Å at $E_R = 20$, 50, 100, 200, 1000 and 10 000 eV (Table E.4), against the bond length of 2.35 Å. Head-on collisions are the extreme case, and glancing ones stay farther apart. Near the threshold, $E_R \approx 20$ eV, the atoms come to about 1.5 Å, inside the bonding region that a good training set covers, while above 100 eV they reach the core region that no thermal training set covers.

The two descriptions are joined by a smooth switch,

$$E^{\mathrm{ad}} = \sum_{I<J} \big[1 - f_{\mathrm{sw}}(r_{IJ})\big]\, V_{\mathrm{ZBL}}(r_{IJ}) + f_{\mathrm{sw}}(r_{IJ})\, E^{\mathrm{ad}}_{\mathrm{ML}}, \qquad f_{\mathrm{sw}}(r) = \frac{1}{1 + \mathrm{e}^{-b_{\mathrm{sw}}(r - r_{\mathrm{sw}})}}, \tag{E.78}$$

written here schematically: in practice the switch multiplies the pair part of the machine-learned energy and the many-body part is left to vanish at short distance by its training. The center r_{sw} and steepness b_{sw} are chosen so that the potential is pure V_{ZBL} below about 1 Å and pure machine-learned energy above about 1.5 Å. The design rule follows from the turning points. Threshold dynamics must not feel the splice, so the machine-learned energy must be fully restored above the $r_{\min}$ of a 20 eV collision, 1.49 Å. It also must not leave a minimum, a shoulder or a jump in the force anywhere in between, which would deflect every collision that passes through it. The splice is checked against density-functional energies of a silicon pair, and of a crystal with one atom pushed toward a neighbor, at separations between 0.8 and 1.8 Å. A potential whose own pair term was already fitted to the density-functional dimer, as in Eq. (E.76), is usually close to V_{ZBL} near 1 Å, and the check then confirms rather than repairs.

E.11.6 Integrating a recoil

The equations of motion are integrated with the velocity Verlet scheme, which is time-reversible and conserves a shadow energy close to the true one, so that the total energy of an isolated

cell drifts only through round-off and through forces that are not smooth. The time step is set by the fastest atom. A silicon atom with 100 eV moves at $v = (2E_R/M_\kappa)^{1/2} = 26.2\,\mathrm{km/s}$, which is 0.26 Å per femtosecond; at 1 keV it moves 0.83 Å/fs. Resolving a collision whose turning point lies within a fraction of an Å of the steep part of V_{ZBL} requires steps in which no atom moves more than a few hundredths of an Å: 0.05 Å per step means 0.19 fs at 100 eV and 0.06 fs at 1 keV. After a few hundred femtoseconds the fastest atom carries a few eV, and the step can grow to the 1 fs that thermal silicon allows, about a sixty-fifth of the shortest phonon period (65 fs, from the 64.1 meV optical phonon). Recoil codes therefore adapt the step to the current maximum velocity and force, a practice the silicon and germanium threshold studies follow [90, Table I]. The check is energy conservation: with the thermostat switched off, the total energy of the cell must stay constant to well below the stored energies being measured, which for this application means a small fraction of an eV over the whole run.

The cell has to absorb the recoil. Its excess energy, E_R minus whatever is stored, is spread over the cell as heat, and it must leave through a boundary layer held at the base temperature by a thermostat. The silicon and germanium studies use cells of $8 \times 8 \times 8$ conventional cubes (4096 atoms), a thermostatted border 6 Å thick, and runs of 20 ps [90, Sect. II A and Table I]. Three things set these numbers. The recoil must stop well inside the cell, so that it neither crosses the periodic boundary nor reaches the thermostat while still energetic; at 100 eV the primary travels a few Å to about a nanometer, and $8a = 4.3\,\mathrm{nm}$ is ample. The heat must be removed without reflecting back from the thermostat as a coherent pulse, which a too thin or too stiff border does. And the run must last until the potential energy of the cell has settled; the same study fixed its run lengths from test runs longer than 5 ps, by the time the potential energy took to come within 1 eV of its final value [90, Sect. II A].

E.11.7 Cascade statistics

A single trajectory answers a yes-or-no question for one direction and one energy. The physical quantities are averages over the direction of the recoil, which a dark-matter or neutron scattering event draws from its own angular distribution, and over the position of the struck atom within the unit cell. The standard protocol draws N_{dir} directions uniformly on the sphere and, for each, increases E_R in steps of about 1 eV through the threshold region and more coarsely above it. The published silicon and germanium scans use more than 1000 directions [90, Sect. II A], and 1003 for silicon in the earlier stored-energy study [99].

Whether a defect survived is decided by an occupation count: each atom of the final configuration is assigned to the Wigner–Seitz cell of the nearest perfect-lattice site, and any site with occupation other than one marks a vacancy or an interstitial. The method has no free parameter and does not count the bond-switching defects that relax away [90, Sect. II A]. From the counts we obtain $p_{\mathrm{FP}}(E_R)$, the fraction of directions with a surviving defect, with the binomial error $[p_{\mathrm{FP}}(1 - p_{\mathrm{FP}})/N_{\mathrm{dir}}]^{1/2}$, at most 0.016 for 1000 directions. The threshold surface needs more care than the word "threshold" suggests. Along many directions the defect appears at one energy, disappears at a slightly higher energy because the interstitial now lands close enough to its vacancy to recombine, and reappears above that. Silicon and germanium show some of this recombination; sapphire shows so much that a direction-resolved threshold is ill-defined [90, Sect. II B]. The quantity the rate formulas of Sect. 19.4 need is therefore $p_{\mathrm{FP}}(E_R, \hat{\boldsymbol{r}}_R)$ itself, and $E_{\mathrm{d}}(\hat{\boldsymbol{r}}_R)$ is quoted as its lowest crossing. With the Stillinger–Weber potential the minimum threshold over all directions is 18 eV in silicon and 13 eV in germanium (with a modified Stillinger–Weber potential); above it the defect probability rises steeply, though more slowly than in diamond, because of recombination [90, Table II, Fig. 1 and Sect. II B]. Both numbers are properties of the potentials; Sect. 19.4 discusses why the germanium value is model-conditional.

E.11.8 Stored energy and the quench protocol

The stored energy of a trajectory is the potential energy the cell keeps once all motion has died out. Taking the difference of potential energies at the end and at the start of a finite run, as the published scans do [90, Sect. II A], includes whatever heat has not yet left the cell. The size of that error follows from equipartition: in the harmonic crystal the potential energy is $\frac{3}{2}N_{\mathrm{at}}T$, so a residual temperature rise of 1 K in 4096 atoms adds $\frac{3}{2}\times 4096\times 8.6\times 10^{-5}\,\mathrm{eV} = 0.53\,\mathrm{eV}$, comparable to the uncertainty of the stored energy itself. A quench removes it. The final configuration is relaxed to its local minimum by energy minimization: conjugate gradients, or damped dynamics with the velocities set to zero whenever the power $-\nabla E^{\mathrm{ad}}\cdot\dot{\boldsymbol{R}}$ turns negative. The perfect crystal is relaxed the same way, and

$$E_{\mathrm{st}} = E^{\mathrm{ad}}_{\mathrm{min}}(\mathrm{final}) - E^{\mathrm{ad}}_{\mathrm{min}}(\mathrm{perfect}). \tag{E.79}$$

The minimum reached is the inherent structure of the final configuration: thermal motion is removed and the defects are kept, as long as the barriers between them exceed the small forces of the minimizer. For the experiments of Sect. 19.4, which measure heat at millikelvin long after the event, Eq. (E.79) is the relevant quantity.

A second protocol choice concerns the reference energies. A cascade census produces vacancies, interstitials of several types, close Frenkel pairs and small clusters, each with a formation energy that the potential predicts with some error. The stored energy of a trajectory is roughly the sum of its defects' energies plus their interactions. To make results from different potentials, or from potential and density functional, comparable, one table of configurational energies of the recurring defects is computed once with density-functional theory in large relaxed cells, and every census is checked against it. For the silicon potential of Eq. (E.76), the formation energies of the vacancy and of the hexagonal, tetrahedral and dumbbell interstitials lie within 7 % of the density-functional reference [92, Sect. III E]. A defect storing several eV then carries an error of a few tenths of an eV from the potential alone, which is the floor on how precisely a single-pair stored energy can be quoted.

E.11.9 Defect paths and barriers

Once defects survive, Sect. 19.6 needs to know how they move and recombine. The tool is the nudged elastic band: a chain of N_{im} replicas of the cell, between two relaxed end states, is relaxed with spring forces along the chain and the true force perpendicular to it, until the chain lies on the minimum-energy path. In the climbing-image variant the highest replica is driven uphill along the path and converges onto the saddle point, whose energy relative to the initial minimum is the barrier V_{b} [100]. End states come from the cascade census, which at $E_R = 100\,\mathrm{eV}$ already supplies close pairs at many separations and orientations. Paths are first mapped with the machine-learned potential, at a cost of a few thousand force calls each, and the saddle points that matter are recomputed with density-functional theory in cells of 64 to 216 atoms. The silicon potential reproduces the density-functional ordering of the hexagonal and tetrahedral interstitials, which are related by a short displacement along a diffusion path, though it underestimates their energy difference; the analytical potentials make larger errors, and many reverse the order [92, Sect. III E].

A barrier becomes a rate through harmonic transition-state theory, $\Gamma = \Gamma_0\,\mathrm{e}^{-V_{\mathrm{b}}/T}$, with an attempt rate Γ_0 from the vibrational frequencies at the minimum and at the saddle. At the operating temperature of a cryogenic detector this is useless for any barrier of chemical size: at 10 mK, $T = 0.86\,\mu\mathrm{eV}$, and a barrier of 0.1 eV gives a Boltzmann factor of $\mathrm{e}^{-1.2\times 10^{5}}$. What moves defects at millikelvin is tunneling through low barriers, or activation by the athermal phonons of a later event. The first needs the barrier shape and the tunneling mass along the

path, which the nudged elastic band provides and Sect. 19.6 turns into a rate; the second needs the energy that reaches the defect, which the transport of Appendix E.12 provides.

E.11.10 Convergence parameters and cost

The parameters a recoil study has to converge are the following. For the potential: the size and coverage of the training set, the cutoff r_{c}, the descriptor resolution, and the splice $(r_{\mathrm{sw}}, b_{\mathrm{sw}})$. For the dynamics: the time-step criterion, the cell size, the border thickness and the run length. For the statistics: N_{dir}, the energy grid, and the sampling of the struck atom over the inequivalent sites of the cell. For the stored energy: the quench tolerance. Each is converged by varying it and checking that p_{FP} and the mean E_{st} move by less than their statistical errors.

The cost is set by the number of force evaluations. The silicon Gaussian approximation potential costs about 100 ms per atom per force call on one core [92, Sect. V]. A 4096-atom cell then needs 410 s per step, and a trajectory of 2×10^4 steps (~ 20 ps at a typical 1 fs step after the collision phase) needs 8×10^6 core-seconds, about 2×10^3 core-hours. A scan of 1000 directions at 20 energies is 2×10^4 trajectories, or 5×10^7 core-hours. This is why the published scans use the cheaper analytical potentials, and why a production study with a machine-learned potential has to use one of the faster descriptor families, on graphics processors, and has to spend its trajectories where p_{FP} changes fastest. A density-functional reference trajectory in a 216-atom cell costs far more per atom, but only a few directions are needed to validate the potential.

E.11.11 Pitfalls

The most common failure is a potential that is accurate at equilibrium and wrong in the region a recoil explores. Two potentials can agree on elastic constants and defect energies and still predict thresholds that differ by several eV, because the threshold is controlled by the energy along the path of the struck atom between its neighbors, not at the end points. The remedy is validation on the quantity itself: thresholds along a few directions from density-functional molecular dynamics, the approach of the density-functional reference study for silicon [101]. Two physical omissions of Eq. (E.74) remain: electronic energy losses, which grow with E_R and which Sect. 19.2 treats, and the quantum motion of light defects at millikelvin, which classical dynamics cannot describe.

E.11.12 Silicon

Table E.4 collects the silicon numbers of this section. The analytical potentials are used for the large scans, the Gaussian approximation potential for validation and for paths, and density-functional theory as the reference for both.

The machine-learned potential removes the main systematic of the published silicon scans, the analytical potential, but at the 5×10^7 core-hours estimated above for a full scan; that trade decides how far the stored-energy predictions of Sect. 19.4 can be pushed.

Table E.4: Recoil molecular dynamics in silicon: the inputs and the quantities a production study converges.

Quantity	Silicon
Bond length; shortest phonon period	2.35 Å; 65 fs
Head-on turning point $r_{\min}$ at $E_R =$ 20, 50, 100, 200, 1000, 10 000 eV	1.49, 1.20, 1.00, 0.83, 0.49, 0.18 Å (Eq. (E.77))
Speed of the struck atom at 100 and 1000 eV	0.26 and 0.83 Å/fs
Training set of the machine-learned potential	2475 cells, 171 815 environments, PW91 [92]
Point-defect formation energies vs density-functional theory	within 7 % [92]
Cost per atom per force call	100 ms on one core [92]
Cell, border, run length	4096 atoms, 6 Å, 20 ps [90]
Directions sampled	> 1000 [90]
Minimum threshold (Stillinger–Weber)	18 eV [90]
Residual heat per K in 4096 atoms	0.53 eV (equipartition)

E.12 Phonon and charge transport Monte Carlo

A 100 eV deposit in a silicon crystal ends, after the cascades of Chaps. 18 and 19, as some 10^4 acoustic phonons of a few meV and, in a biased detector, a few tens of electrons and holes drifting toward the electrodes. The sensors on the surface see neither the deposit nor the cascade. They see phonons that arrive over microseconds after scattering off isotopes, decaying, bouncing off surfaces and crossing into metal films, and charges that arrive after drifting, emitting phonons of their own, and sometimes getting trapped. Sects. 20.1 to 20.5 derive the physics of each step; combined in one simulation of a whole detector, they turn an event into the arrival distributions that Sect. 20.5 calls collection kernels. The method is Monte Carlo solution of the Boltzmann equation for phonons and carriers, and the silicon case is worked with the rate constants and code conventions of the public detector-physics toolkit G4CMP [102, 103].

E.12.1 Computed quantities

The state of a phonon is its wavevector $\boldsymbol{Q}$, its branch ν and its position; the state of a carrier is its crystal momentum, its valley or band, and its position. Between events each particle moves on a straight line with its group velocity (a carrier also accelerates in the applied field). Events are drawn at random with the rates of the scattering processes, and each event replaces the particle by its products. With many particles, the procedure samples the solution of the linear Boltzmann equation

$$\partial_t f_i + \boldsymbol{v}_i \cdot \nabla_{\boldsymbol{r}} f_i + \boldsymbol{F}_i \cdot \nabla_{\boldsymbol{k}} f_i = \left(\frac{\partial f_i}{\partial t}\right)_{\text{coll}}, \tag{E.80}$$

for the occupation $f_i(\boldsymbol{r}, \boldsymbol{k}, t)$ of each species i, with $\boldsymbol{v}_i$ its group velocity, $\boldsymbol{F}_i$ the force on it ($\pm e\boldsymbol{\mathcal{E}}$ for carriers in a field $\boldsymbol{\mathcal{E}}$, zero for phonons) and a collision term built from golden-rule rates [104]. The equation is linear because the athermal populations are dilute: phonons scatter off the static disorder and decay spontaneously, carriers emit into an empty phonon bath, and no process depends on the occupation of the other particles. The outputs are tallies

at the boundaries: for every sensor, the energy and time of each phonon it absorbs, and for every electrode, the charge induced on it.

E.12.2 Place on the hierarchy

Every rate in the collision term is a lowest-order golden-rule rate. Isotope scattering is the second-order Born rate of a phonon off the random mass defects of Sect. 11.6. Anharmonic decay is the rate of the cubic term of the lattice potential (Sect. 5.7), at the lowest order at which it does not vanish, the cubic vertex squared. Neganov–Trofimov–Luke emission and intervalley scattering are one-phonon emission by a carrier through a deformation potential, $n_g = 2$ in the notation of the truncation label $\mathfrak{T}$ of Sect. 8.5, with $n_B = 0$. The transport itself is incoherent: each particle is a wavepacket with a well-defined momentum, and interference between successive events is dropped. This holds when the scattering rate is small compared with the particle's frequency. For a silicon phonon at 1 THz the isotope rate is $2.4 \times 10^6\,\mathrm{s}^{-1}$ (Eq. (E.81) below) against $2\pi \times 10^{12}\,\mathrm{s}^{-1}$; even at the top of the acoustic band, 15 THz, the ratio is $\sim 10^{-3}$. It also needs mean free paths long compared with the wavelength, which they are by a factor 4×10^5 at 1 THz (a wavelength of 5.4 nm against a mean free path of 2.2 mm). The Boltzmann description is therefore safe for the bulk; its approximations are concentrated in the rates and in the surface models, which is where the pitfalls below lie.

E.12.3 Phonon rates

Isotope scattering and anharmonic decay remove a phonon from its ballistic path. Isotope scattering redirects it, and may change its branch, at the rate [105], [103, Eq. (6)]

$$\Gamma_{\mathrm{iso}} = B_{\mathrm{iso}} \left(\frac{E_{\mathrm{ath}}}{h} \right)^4, \qquad B_{\mathrm{iso}} = 4\pi^3\, g_{\mathrm{md}}\, \Omega_{\mathrm{at}}\, \langle c^{-3} \rangle, \tag{E.81}$$

where E_{ath}/h is the phonon frequency, Ω_{at} the volume per atom, $\langle c^{-3} \rangle = \frac{1}{3} \sum_\nu c_\nu^{-3}$ the average of the inverse cubed sound speed over the three acoustic branches ν (the source writes $1/\langle c^3 \rangle$ with the polarization-averaged cubed speed; the two agree when the speeds are equal, and $\langle c^{-3} \rangle$ is the average the Debye density of states contains, Sect. 20.2), and $g_{\mathrm{md}} = \langle (M - \bar{M})^2 \rangle / \bar{M}^2$ the mass variance of the isotope mixture over the squared mean mass [106, Eqs. (3)–(5)]. It is the Rayleigh law of a point scatterer, valid as long as the wavelength is long compared with the interatomic spacing. For natural silicon, $g_{\mathrm{md}} = 2.02 \times 10^{-4}$, $\Omega_{\mathrm{at}} = \Omega/2 = 20.0\,\text{Å}^3$ and, with $c_{\mathrm{LA}} = 9.0\,\mathrm{km/s}$ and $c_{\mathrm{T}} = 5.4\,\mathrm{km/s}$, $\langle c^{-3} \rangle = 1/(2.13 \times 10^{11}\,\mathrm{m^3\,s^{-3}}) = 4.69 \times 10^{-12}\,\mathrm{s^3\,m^{-3}}$, and Eq. (E.81) gives $B_{\mathrm{iso}} = 124.0 \times 2.02 \times 10^{-4} \times 2.00 \times 10^{-29}\,\mathrm{m^3} \times 4.69 \times 10^{-12}\,\mathrm{s^3\,m^{-3}} = 2.35 \times 10^{-42}\,\mathrm{s^3}$, in line with the tabulated $2.33 \times 10^{-42}\,\mathrm{s^3}$ and with the value $2.43 \times 10^{-42}\,\mathrm{s^3}$ that most transport codes use [106, Table 4], [103, Table I]. The direction and branch of the scattered phonon are then drawn with the weight $|\hat{\boldsymbol{e}} \cdot \hat{\boldsymbol{e}}'|^2$ of the polarization vectors of the initial and final modes, divided by the cube of the final frequency [103, Eq. (7)]. Repeated isotope scattering drives the branch populations of silicon to slow transverse 55 %, fast transverse 35 % and longitudinal 10 % [103, Sect. III E].

Anharmonic decay splits a longitudinal phonon into two of lower energy, at the rate [107], [103, Eq. (8)]

$$\Gamma_{\mathrm{anh}} = A_{\mathrm{anh}} \left(\frac{E_{\mathrm{ath}}}{h} \right)^5, \tag{E.82}$$

with A_{anh} fixed by the second- and third-order elastic constants in the isotropic continuum [103, Eq. (8)]. Transverse phonons hardly decay at low frequency. The two channels, $\mathrm{L} \to \mathrm{L} + \mathrm{T}$ and $\mathrm{L} \to \mathrm{T} + \mathrm{T}$, have energy distributions of the daughters that the isotropic approximation fixes in closed form. For $\mathrm{L} \to \mathrm{L}' + \mathrm{T}'$, with $x = E_{\mathrm{L}'}/E_{\mathrm{L}}$ and $\delta = c_{\mathrm{LA}}/c_{\mathrm{T}}$, the rate is distributed as

Table E.5: Bulk phonon rates in natural silicon from Eqs. (E.81) and (E.82) with $B_{\mathrm{iso}} = 2.43 \times 10^{-42}\,\mathrm{s}^3$ and $A_{\mathrm{anh}} = 7.41 \times 10^{-56}\,\mathrm{s}^4$ [103]. The anharmonic rate applies to longitudinal phonons. The isotope mean free path uses the transverse speed 5.4 km/s [103].

E_{ath}/h (THz)	E_{ath} (meV)	Γ_{iso} (s^{-1})	isotope mean free path	Γ_{anh} (s^{-1})
0.5	2.07	1.5×10^5	3.6 cm	2.3×10^3
1	4.14	2.4×10^6	2.2 mm	7.4×10^4
3	12.4	2.0×10^8	27 µm	1.8×10^7
10	41.4	2.4×10^{10}	0.22 µm	7.4×10^9

$x^{-2}(1-x^2)^2[(1+x)^2-\delta^2(1-x)^2][1+x^2-\delta^2(1-x)^2]^2$ over $(\delta-1)/(\delta+1) < x < 1$, and the T + T channel has a polynomial distribution of the same kind with coefficients set by the elastic constants [103, Eqs. (9) and (12)]. The daughter directions then follow from energy and momentum conservation; for the longitudinal daughter, $\cos\theta_{\mathrm{L}'} = [1+x^2-\delta^2(1-x)^2]/(2x)$ [103, Eq. (10)]. For silicon, Tamura's isotropic calculation gives $A_{\mathrm{anh}} = 7.41 \times 10^{-56}\,\mathrm{s}^4$, as tabulated by Leman [103, Table I]; a recent recalculation from cryogenic elastic constants gives $1.15 \times 10^{-55}\,\mathrm{s}^4$ [106, Table 3]. No direct measurement exists for silicon, so A_{anh} is uncertain by at least the factor 1.5 between the two.

Table E.5 evaluates Eqs. (E.81) and (E.82) for silicon. The ratio of the two rates, $\Gamma_{\mathrm{anh}}/\Gamma_{\mathrm{iso}} = (A_{\mathrm{anh}}/B_{\mathrm{iso}})(E_{\mathrm{ath}}/h)$, equals $(E_{\mathrm{ath}}/h)/33\,\mathrm{THz}$ and is below one everywhere in the acoustic band. Decay is nevertheless what lowers the frequency, since isotope scattering conserves energy. A 10 THz longitudinal phonon decays within 0.13 ns. Its daughters near 3 THz live some 55 ns before decaying again, scattering off isotopes about ten times in between, and below 1 THz the decay time exceeds 10 µs, longer than the time a phonon needs to reach a surface of a centimeter-sized crystal.

E.12.4 Free flight, focusing and surfaces

Between events a phonon moves with the group velocity of its branch, obtained from the Christoffel equation of continuum elasticity, $\rho_T \omega^2 \hat{e}_i = C_{ijkl} Q_j Q_k \hat{e}_l$, for the unit polarization $\hat{\boldsymbol{e}}$ of the elastic wave and the elastic tensor C_{ijkl} (Appendix C.18). In a cubic crystal the group velocity is not parallel to $\boldsymbol{Q}$, and an isotropic distribution of wavevectors becomes a strongly peaked distribution of energy flux, the phonon focusing that concentrates ballistic phonons along certain crystal directions [106, Sect. 2.4]. Using the continuum dispersion is an approximation for the phonons near the top of the acoustic band, whose dispersion bends over; it is good for the few-THz phonons that dominate after the first decays.

A phonon that reaches a surface is reflected specularly, reflected diffusely, down-converted into lower-energy phonons, or transmitted into whatever is attached. The bare crystal surface is described by the probabilities of these outcomes, which are inputs of the simulation, fitted to measured pulse shapes for a given surface preparation. An interface to a sensor film is described by a transmission probability. The acoustic mismatch model treats the interface as a welded plane between two elastic media and computes the transmission of each incident mode from continuity of displacement and stress. The diffuse mismatch model assumes that a phonon forgets its direction at the interface and is transmitted with a probability fixed by the ratio of the phonon densities of states on the two sides [108]. Both obey the reciprocity derived in Sect. 20.3. Measured silicon-to-aluminum values, $\bar{t}_{12} P_{\mathrm{abs}} = 0.3$–$0.55$, are absorption probabilities and not transmissions [109], as Sect. 20.3 shows, and a simulation that uses them must use them in that role.

E.12.5 Films

Inside a superconducting film the phonon breaks Cooper pairs, the resulting quasiparticles relax by emitting phonons, and phonons with energy above $2\Delta_{\mathrm{BCS}}$ break further pairs; this is the Kaplan cascade of Sect. 20.4 [110]. A full simulation tracks quasiparticles and phonons in the film. The lumped model used in detector codes treats the film as a single object with a thickness d_{f}, a gap Δ_{BCS} and an energy-dependent phonon mean free path $\ell_{\mathrm{f}}(E_{\mathrm{ath}})$. An incident phonon escapes back into the substrate unabsorbed with probability $\exp[-4d_{\mathrm{f}}/\ell_{\mathrm{f}}(E_{\mathrm{ath}})]$, the factor 4 being twice the thickness, for the round trip, times two from the average over incidence angles [102, Eq. (17)]. Otherwise its energy goes to two quasiparticles, which share it with the BCS density of states weight; quasiparticles below $3\Delta_{\mathrm{BCS}}$ are counted as collected, and those above emit a phonon and continue [102, Sect. 2.4]. Phonons below $2\Delta_{\mathrm{BCS}}$ cannot break pairs and are either absorbed directly with a set probability or returned to the substrate. The output is the quasiparticle yield $Y_{\mathrm{qp}}(E_{\mathrm{ath}})$ of Sect. 20.4, sampled event by event, and the phonons the film sends back into the crystal.

E.12.6 Carrier transport

In a biased detector the electrons and holes of the cascade drift in the field $\boldsymbol{\mathcal{E}}$, computed beforehand for the electrode geometry by a finite-element solution of the Laplace equation. A carrier accelerates until it moves faster than the longitudinal sound speed, and then emits acoustic phonons, much as a charge faster than light in a medium emits Cherenkov radiation. These are the Neganov–Trofimov–Luke phonons, and they carry the work eV_{bias} that the bias does on each pair, the Luke gain of Sect. 19.5 [111]. For a carrier with an isotropic mass m and momentum k, and the acoustic deformation potential D^n of its band (Sect. 6.2) taken isotropic, the golden-rule emission rate is (Eq. (20.10))

$$\Gamma_{\mathrm{L}}(k) = \frac{c_{\mathrm{LA}}}{3\,l_0}\left(\frac{k}{k_{\mathrm{s}}}\right)^2\left(1-\frac{k_{\mathrm{s}}}{k}\right)^3 \quad (k > k_{\mathrm{s}}), \qquad k_{\mathrm{s}} = m\,c_{\mathrm{LA}}, \qquad l_0 = \frac{\pi\rho_T}{2m^3(D^n)^2}, \tag{E.83}$$

in natural units, with k_{s} the momentum of a carrier moving at the sound speed and l_0 a characteristic scattering length; the polar angle θ of the emitted phonon, measured from $\boldsymbol{k}$, is drawn from $P(\theta) \propto (\cos\theta - k_{\mathrm{s}}/k)^2 \sin\theta$ for $0 \le \theta \le \arccos(k_{\mathrm{s}}/k)$ [102, Eq. (9)], a law that integrates to Eq. (E.83); the total rate printed beside it [102, Eq. (8)] is lower by the factor k_{s}/k. The rate vanishes below k_{s}, which is the Cherenkov condition, and grows as k^2 at large k. Eq. (E.83) counts longitudinal phonons only; a carrier faster than the transverse sound speed, which is lower, can also emit transverse ones, and a model that omits them shifts the Luke phonons toward the longitudinal branch.

Holes in silicon and germanium are close to isotropic, and Eq. (E.83) applies directly. Electrons sit in anisotropic valleys, six along $\langle 100\rangle$ in silicon and four along $\langle 111\rangle$ in germanium, each with the mass tensor of components $m^*_{\parallel}$ along the valley axis and $m^*_{\perp}$ across it (Sect. 3.3). A Herring–Vogt transformation rescales the momentum along each principal axis by $(m_{\mathrm{c}}/m^*_i)^{1/2}$, with the conductivity mass $3/m_{\mathrm{c}} = 1/m^*_{\parallel} + 2/m^*_{\perp}$, and turns the valley into an isotropic one. The field and the emission are handled there as for holes, and the results are transformed back [112], [102, Eqs. (11)–(12)]. The transformation is not unitary, and the emitted phonon's momentum is transformed back by keeping its magnitude, which conserves energy, and transforming only its direction [103, Sect. V C]. Electrons also scatter between valleys by emitting or absorbing intervalley phonons, which turns their trajectories toward the field direction. The intervalley rates are parametrized, from measurements in silicon and from a model in germanium, rather than computed from first principles [102, Sect. 2.3]; the drift speed against field is then a check of the whole carrier model, which reproduces measurements in germanium [102, Fig. 7].

Impurities enter as two processes, each with a mean free path that is fitted to the device. One is trapping, in which a carrier is captured by a neutral impurity and stops, and the other is impact ionization of an impurity, in which a carrier frees a second charge from a charged one [102, Eqs. (13)–(16)]. Trapping reduces the collected charge and the Luke phonons, impact ionization increases them, and both fill the regions between the single-pair peaks of a charge-resolving silicon detector, from whose shape the four or six mean free paths are fitted [102, Fig. 6]. In a uniform field with trapping alone, the simulation must reproduce the Hecht relation of Sect. 20.1; that is the first check of a trapping model.

The step size for carriers is limited by the requirement that the rate stays nearly constant during a step. Taking a step of half the current mean free time conserves energy, in the sense that the energy of the Luke phonons emitted equals the work done by the field, and longer steps produce too few Luke phonons [103, Sect. V D]. The number of Luke phonons is large: each pair crossing a bias V_{bias} releases eV_{bias}, and a code that assumes 2 meV per Luke phonon in estimating its event size [102, Sect. 3.4] expects $100\,\mathrm{eV}/2\,\mathrm{meV} = 5 \times 10^4$ phonons per pair at 100 V.

E.12.7 Collection kernels

The quantity a detector analysis needs is not the history of each phonon but the probability that energy deposited at $\boldsymbol{r}_{\mathrm{ev}}$, in phonons of energy E_{ath}, is absorbed in sensor s within a time window. The collection kernel of Sect. 20.5 is this probability, tallied by launching phonons of each energy from a grid of event positions and counting where they end. For charge, the analogous kernel is the induced charge on each electrode, computed from the trajectories with Ramo's theorem. A charge q moving from $\boldsymbol{r}_1$ to $\boldsymbol{r}_2$ induces $q[\varphi_s(\boldsymbol{r}_2) - \varphi_s(\boldsymbol{r}_1)]$ on electrode s, where φ_s is the weighting potential, the potential with electrode s at unit voltage and the others grounded (Sect. 20.1). Kernels separate the transport from the event: once tabulated, they are convolved with the deposit and the phonon spectrum of any event, from the created-phonon spectra of Chap. 19 to the down-converted spectra of Sect. 20.2, without rerunning the transport.

The statistical error of a kernel bin with n absorbed out of N_{l} launched phonons is binomial, $[n(1 - n/N_{\mathrm{l}})]^{1/2}/N_{\mathrm{l}}$. For the collection contrasts of Sect. 20.5, which are differences between kernels of similar size, the useful technique is to run the two cases with common random numbers, so that their statistical fluctuations are correlated and cancel in the difference.

E.12.8 Validation and the closure checks

A transport run is only as good as its checks, and four of them cost nothing: energy, charge, rate and step closure. For energy closure, The deposit, plus the work of the bias on the carriers, must equal the energy absorbed in the sensors, plus the energy lost at bare surfaces, plus the energy in trapped carriers and in particles still in flight at the end, to the round-off of the tallies. A missing term in this balance signals a process that loses energy without booking it, which is the most common silent bug in such codes. For charge closure, the charge induced on all electrodes, summed with the trapped charges weighted by their weighting potentials, must equal the charge created. For rate closure, in a large block with surfaces removed and one process switched on at a time, the measured mean free paths must follow Eqs. (E.81) and (E.82) with the input constants, with the slopes E_{ath}^{-4} and E_{ath}^{-5}; this has been done for the materials of the recent rate tabulation [106, Sect. 3]. For step closure, halving every time step must leave the kernels unchanged within their statistical errors.

Beyond these internal checks, a run must report its agreement with external benchmarks that test the physics rather than the bookkeeping. They are carrier drift velocities against

field, the Hecht curve for a known trap density, the ballistic focusing pattern of a thin crystal, the quasiparticle yield $Y_{\mathrm{qp}}(E_{\mathrm{ath}})$ against the Kaplan cascade of Sect. 20.4, and, where it exists, a measured pulse shape. A run that reports kernels without these numbers cannot be distinguished from one with an energy leak.

E.12.9 Cost

The cost is set by the number of particle steps. A 10 keV deposit in silicon that has down-converted to 1 THz phonons (4.1 meV) holds 2.4×10^6 phonons. In a crystal a centimeter across, each phonon scatters off isotopes about every 2 mm (Table E.5) and bounces off the surfaces many times before a sensor covering a few percent of the surface absorbs it. At a hundred steps per phonon the event needs $\sim 10^8$–10^9 steps, and a biased detector adds the 5×10^4 Luke phonons per pair per 100 V. With full tracking, a few-keV event takes a day or more of central processing unit (CPU) time, and with the down-sampling described next about an hour per event [102, Sect. 3.4 and Fig. 8]. Collection kernels are much cheaper than events, since each launched phonon is an independent sample and no event needs to be complete.

Down-sampling generates only a fraction f of the phonons and carriers and gives each the weight $1/f$ [102, Sect. 3.4]. It conserves the mean of every tally. It does not conserve the fluctuations: a weighted event has a variance $1/f$ times too large for the collected energy, and a variance computed from weighted samples is not the physical variance. Kernels and means may be computed with down-sampling; the measured-charge Fano factor F_{meas} of Sect. 20.1, and any statement about event-by-event resolution, may not.

E.12.10 Pitfalls

No closure check sees the errors in three inputs: the sound speed used to convert a decay calculation to the fifth-power law, the starting phonon spectrum, and the surface, intervalley and trapping parameters. The fifth-power law of Eq. (E.82) amplifies any inconsistency in that sound speed, as the recalculation found for sapphire [106, Sect. 3.1]. The transport is usually started from phonons at the Debye frequency, on the grounds that the first decays erase the details [103, Sect. III B], which is safe for kernels at long times but not for the prompt signal near the event, the subject of the memory-loss arguments of Sect. 20.2. The surface parameters are fits to the device in which they were measured, so that transferring them to a crystal from a different boule is an assumption to be stated.

E.12.11 Silicon

Table E.6 lists the silicon inputs of a phonon and charge transport run, with the checks that a run must report alongside its kernels.

What the transport adds to the physics of Chap. 20 is the geometry: the same rates in a crystal of a different shape, with sensors elsewhere, give different kernels, and every statement about what a readout measures is a statement about a detector, not about silicon.

Table E.6: Phonon and charge transport in silicon: inputs, the quantities they control, and the checks a run must report. Rates are from Eqs. (E.81) and (E.82) with the cited constants.

Input or check	Silicon
B_{iso}; A_{anh}	$2.43 \times 10^{-42}\,\mathrm{s}^3$; $7.41 \times 10^{-56}\,\mathrm{s}^4$ to $1.15 \times 10^{-55}\,\mathrm{s}^4$ [103, 106]
Sound speeds L, T	9.0, 5.4 km/s [103]
Isotope mean free path at 1 THz	2.2 mm
Equilibrium branch populations slow transverse (ST) : fast transverse (FT) : longitudinal	55 : 35 : 10 [103]
Si $\to$ Al absorption probability $\bar{t}_{12}P_{\mathrm{abs}}$	0.3–0.55 [109]
Luke phonons per pair at 100 V	5×10^4 (at 2 meV each)
Carrier model	six $\langle 100 \rangle$ electron valleys, Herring–Vogt; isotropic holes; fitted intervalley, trapping and impact-ionization mean free paths
Closure checks	energy, induced charge, rate slopes E_{ath}^{-4} and E_{ath}^{-5}, time step
Benchmarks	drift velocity, Hecht curve, focusing pattern, $Y_{\mathrm{qp}}(E_{\mathrm{ath}})$, pulse shape

E.13 Hot-carrier cascades from first principles

An electron given 50 eV in silicon ends, a picosecond later, as about 13 electron–hole pairs at the band edges and some 35 eV of phonons. Which share goes where, and how much the pair number fluctuates from one event to the next, is decided by a competition between two processes at every energy along the way. One is impact ionization, in which the hot carrier gives part of its energy to a valence electron and lifts it across the gap; the other is phonon emission, in which it gives a few tens of meV to the lattice. Sect. 18.1 writes the cascade as a branching process whose kernels are master-formula rates with the carrier as the projectile, and Sect. 18.2 defines the mean energy per pair $\varepsilon_{\mathrm{eh}}$ and the Fano factor F_{eh} as its first two moments. Here we explain how those kernels are computed from the band structure, the screened interaction and the electron–phonon vertex, how the cascade is then sampled, and what each step costs. Silicon serves as the case where both measurements and a calibrated phenomenological model exist to compare against.

E.13.1 Computed quantities

For a seed class Λ, the set of initial electron and hole states that the primary excitation produces (Sect. 18.3), the calculation produces the distribution $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ of the final number of pairs. Its moments are the measured quantities,

$$\varepsilon_{\mathrm{eh}} = \frac{E_0}{\langle N_{\mathrm{eh}} \rangle}, \qquad F_{\mathrm{eh}} = \frac{\langle N_{\mathrm{eh}}^2 \rangle - \langle N_{\mathrm{eh}} \rangle^2}{\langle N_{\mathrm{eh}} \rangle}, \tag{E.84}$$

for a seed of total energy E_0, in the limit of large E_0 where both become independent of it. The cascade needs the quasiparticle bands $\varepsilon_{n\boldsymbol{k}}$ from the band edges up to the highest energy a carrier reaches, tens of eV for the cascades of Chap. 18; the impact-ionization rate $\Gamma_{\mathrm{ii}}(n\boldsymbol{k})$ of a carrier in state $n\boldsymbol{k}$, with the distribution of its three products over final states; and the

phonon emission and absorption rate $\Gamma_{\mathrm{ph}}(n\boldsymbol{k})$, with its final states and phonon branches. The cascade Monte Carlo then follows each carrier from state to state until every carrier is below the impact-ionization threshold $\theta_{\mathrm{ii}}E_{\mathrm{g}}$ (Sect. 18.1), after which it can only lose energy to phonons and cannot add a pair.

E.13.2 Place on the hierarchy

Each kernel is a golden-rule rate at a definite place on the truncation label $\mathfrak{T}$ of Sect. 8.5. The impact-ionization rate is the decay of a quasiparticle into a lower-energy quasiparticle and an electron–hole pair through the dynamically screened interaction W^{scr} of Sect. 4.8; with W^{scr} in the random-phase approximation, its many-body approximation is $\mathcal{X} = \mathrm{RPA}$, applied to quasiparticle rather than Kohn–Sham energies. The phonon rate is one-phonon emission or absorption, $n_g = 2$, with $n_B = 0$. The cascade itself is a sequence of independent golden-rule events: the carrier is on shell between events, and interference between successive events is dropped. This is the same incoherent limit as the transport of Appendix E.12, and it holds when the total scattering rate of a carrier is small compared with its energy above the band edge, the condition that the off-shell parameter of Sect. 8.5 be small. A rate of $10^{14}\,\mathrm{s}^{-1}$ corresponds to a width $\Gamma = 0.066\,\mathrm{eV}$ in natural units, so a carrier 5 eV above the conduction-band minimum meets the condition even at rates ten times larger. A carrier a few tens of meV above the impact-ionization threshold, whose energy relative to that threshold is comparable to the width, does not, and the threshold region is where the incoherent picture is least reliable.

E.13.3 Quasiparticle bands

The impact-ionization threshold is set by the gap, and the rate just above it rises steeply with the energy above threshold, so the gap must be right before anything else is. Semilocal density-functional theory places the conduction bands of silicon too low, with an indirect gap of about half the measured 1.17 eV (Appendix E.1). Quasiparticle corrections in the G_0W_0 approximation (Appendix E.3) repair the gap and the ordering [37]. A self-consistent variant gives 1.23 eV for silicon without spin–orbit coupling, and a single empirical scaling of the self-energy correction brings the gap to experiment and the other tabulated conduction-band energies of silicon to within 0.2 eV of measurement [113, Table I]. A rigid scissor shift fixes the gap by assuming that the correction is the same for every conduction state, which has to be checked over the tens of eV that the cascade spans.

The band window is large. The number of bands below an energy E above the band bottom is about the number of plane waves in a sphere of radius $k_E = (2m_eE)^{1/2}$ in the unit cell, $\Omega k_E^3/(6\pi^2)$. For silicon, with $\Omega = a^3/4 = 40.05\,\text{Å}^3$, this is about 8 bands up to 20 eV, 32 up to 50 eV and 90 up to 100 eV. All of these bands, on a dense $\boldsymbol{k}$ grid, enter both kernels.

E.13.4 The impact-ionization kernel

The rate at which a quasiparticle in an empty state $n\boldsymbol{k}$ decays is set by the imaginary part of its self-energy Σ (Sect. 4.7). In the GW approximation, $\Sigma = \mathrm{i}\mathsf{G}W^{\mathrm{scr}}$, the imaginary part for an unoccupied state is [113, Eqs. (1)–(2)]

$$\Gamma_{\mathrm{ii}}(n\boldsymbol{k}) = 2z_{\mathrm{qp}}\,|\,\mathrm{Im}\,\Sigma_{n\boldsymbol{k}}|,$$
$$\mathrm{Im}\,\Sigma_{n\boldsymbol{k}} = -\pi \sum_{n'\boldsymbol{k}'} \iint \mathrm{d}^3r\,\mathrm{d}^3r'\,\psi^*_{n\boldsymbol{k}}(\boldsymbol{r})\psi_{n'\boldsymbol{k}'}(\boldsymbol{r})$$
$$\times\,\mathrm{Im}\,W^{\mathrm{scr}}(\boldsymbol{r},\boldsymbol{r}',\varepsilon_{n\boldsymbol{k}}-\varepsilon_{n'\boldsymbol{k}'})\,\psi^*_{n'\boldsymbol{k}'}(\boldsymbol{r}')\psi_{n\boldsymbol{k}}(\boldsymbol{r}'), \tag{E.85}$$

with the sum over empty final states $n'\boldsymbol{k}'$ between the Fermi level and $\varepsilon_{n\boldsymbol{k}}$, and z_{qp} the quasiparticle residue of Eq. (E.18), typically about 0.8 [113, Sect. II]. The physical content is in $\operatorname{Im} W^{\mathrm{scr}}$: at energy transfer $\omega = \varepsilon_{n\boldsymbol{k}} - \varepsilon_{n'\boldsymbol{k}'}$ it is the spectral weight of the excitations the carrier can create, which below the plasmon energy are electron–hole pairs and above it are mostly plasmons. In silicon the plasmon is at 16.6 eV [1, Table I], and for a carrier more than that above the band edge Eq. (E.85) includes plasmon emission automatically; the plasmon then decays into a pair. A cascade code that takes its rates from Eq. (E.85) must therefore not add a separate plasmon channel.

The older route writes the rate directly as a golden-rule sum over the two electrons and one hole of the final state. It uses the matrix element of the screened interaction between the initial and final pairs, direct and exchange terms combined as $|M|^2 = \frac{1}{2}(|M_{\mathrm{D}}|^2 + |M_{\mathrm{E}}|^2 + |M_{\mathrm{D}} - M_{\mathrm{E}}|^2)$, and momentum conservation $\boldsymbol{k} = \boldsymbol{k}' + \boldsymbol{k}_1 - \boldsymbol{k}_2$ [114], [113, Eq. (5)]. For weak $\operatorname{Im} W^{\mathrm{scr}}$ the two expressions differ by the factor z_{qp} and by the exchange term, which Eq. (E.85) omits; together these make the two rates differ by a factor between 0.5 and 1 [113, Sect. II]. Omitting exchange alone multiplies the rate by no less than 0.75 [113, Sect. II]. On the logarithmic scale on which impact-ionization rates are usually compared this is a small difference. For F_{eh}, which depends on how the rate rises above threshold relative to the phonon rate, it is not negligible, and the choice belongs among the stated approximations of a calculation.

The cascade needs more than the total rate. When an ionization event is selected, the energies and momenta of the three products are drawn from the integrand of the rate, resolved by final state. The kernel is then a table of probabilities. For each initial $(n, \boldsymbol{k})$ it holds the probabilities of the final $(n', \boldsymbol{k}')$ of the scattered carrier and of the transfer $(\boldsymbol{q}, \omega)$ to the pair; the pair's electron and hole are then drawn with the weight of the joint density of states at that $(\boldsymbol{q}, \omega)$. The screened interaction enters at all transfers up to the carrier energy, so W^{scr} is needed with full frequency dependence from 0 to tens of eV and with local fields, which is the dielectric calculation of Appendix E.6 carried to high frequency.

E.13.5 The electron–phonon kernel

The phonon rate of a carrier in state $n\boldsymbol{k}$ is the golden-rule rate of Sect. 6.2,

$$\Gamma_{\mathrm{ph}}(n\boldsymbol{k}) = 2\pi \sum_{m\nu} \int \frac{\mathrm{d}^3 Q}{V_{\mathrm{BZ}}} \, |g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q})|^2 \Big[(n_{\boldsymbol{Q}\nu} + 1)\, \delta(\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} - \omega_{\boldsymbol{Q}\nu}) + n_{\boldsymbol{Q}\nu}\, \delta(\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} + \omega_{\boldsymbol{Q}\nu}) \Big], \tag{E.86}$$

with V_{BZ} the zone volume, the emission term carrying $n_{\boldsymbol{Q}\nu} + 1$ and the absorption term $n_{\boldsymbol{Q}\nu}$. At the millikelvin temperatures of a detector, $n_{\boldsymbol{Q}\nu} = 0$ for the optical and zone-boundary phonons that dominate, and only emission survives. The vertex $g_{mn\nu}$ comes from density-functional perturbation theory (Appendix E.4) and is interpolated to fine grids with Wannier functions (Appendix E.5) [47, 55]. First-principles rates of this kind, combined with GW electron–electron rates, have been used to follow hot carriers in silicon through their first picosecond [115].

The catch for a cascade is the energy range. The Wannier set that interpolates the band edges, eight sp^3-like orbitals per silicon cell spanning the valence bands and the lowest four conduction bands (Appendix E.5), does not represent states tens of eV up, and its interpolated g is meaningless there. Either the Wannier manifold is enlarged, with disentanglement over a window that contains tens of entangled bands, or the vertex is computed directly on a coarser grid at high energy, where the rates are smoother functions of $\boldsymbol{k}$ because the density of final states averages over many bands. Above a few eV the phonon rate is dominated by the emission of optical and intervalley phonons of about 60 meV [1, Sect. II B], so the energy

lost per phonon event is nearly fixed, and what the calculation must get right is the rate, not the detailed spectrum.

E.13.6 Full-band cascade Monte Carlo

With the two kernels tabulated, the cascade is sampled as in full-band device simulation [116], without a field. Each carrier is a state $(n, \boldsymbol{k})$; its total rate is $\Gamma_{\mathrm{tot}} = \Gamma_{\mathrm{ii}} + \Gamma_{\mathrm{ph}}$. The time to its next event is drawn by self-scattering: a constant ceiling $\Gamma_{\max} \geq \Gamma_{\mathrm{tot}}$ is chosen, a time is drawn from $\Gamma_{\max} \mathrm{e}^{-\Gamma_{\max} t}$, and at that time a real event is accepted with probability $\Gamma_{\mathrm{tot}}/\Gamma_{\max}$, otherwise nothing happens [104]. Without a field the times only order the events and do not change the outcome, but they are kept because the same code with a field gives the drift of Appendix E.12. An accepted event is impact ionization with probability $\Gamma_{\mathrm{ii}}/\Gamma_{\mathrm{tot}}$, which replaces the carrier by three, or phonon emission, which lowers its energy by $\omega_{\boldsymbol{Q}\nu}$ and moves it to $m, \boldsymbol{k} + \boldsymbol{Q}$. Holes are followed in the valence bands with their own kernels, and the hole band width, about 12 eV in silicon, limits the energy a hole can take [1, Sect. II A]. A carrier that falls below $\theta_{\mathrm{ii}} E_{\mathrm{g}}$ is removed from the list with its pair counted. The cascade ends when the list is empty.

The seed matters at low energy. Sect. 18.1 explains why the cascade must start from the final states of the primary process, with their momentum and band structure, rather than from a carrier of given energy. The energy sharing between the first electron and hole sets how much energy is above threshold, and below about 20 eV in silicon different sharing assumptions give different pair yields [1, Sect. IV B]. The seed distribution of Sect. 18.3 is drawn into the initial carrier list, and $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ inherits the seed class Λ.

E.13.7 The single-parameter model

Before the first-principles cascade, silicon already has a cascade model that reproduces the measured pair energy, and it is the natural benchmark. The model of Alig and Bloom, as implemented by Ramanathan and Kurinsky [1], replaces both kernels by free-particle forms with a single optical phonon of energy $\omega_{\mathrm{AB}} = 63$ meV. For a carrier at energy E above its band edge, the ratio of the phonon and impact-ionization rates is then [1, Eq. (8)]

$$\frac{\Gamma_{\mathrm{ph}}(E)}{\Gamma_{\mathrm{ii}}(E)} = A_{\mathrm{AB}} \, \frac{105}{2\pi} \, \frac{(E - \omega_{\mathrm{AB}})^{1/2}}{(E - E_{\mathrm{g}})^{7/2}}, \tag{E.87}$$

with energies in eV and one constant A_{AB}, the ratio of the squared phonon and Coulomb matrix elements times phase-space factors, and the probability that the next event is an ionization is $[1 + \Gamma_{\mathrm{ph}}/\Gamma_{\mathrm{ii}}]^{-1}$ [1, Eq. (10)]. The value $A_{\mathrm{AB}} = 5.2$, in the units of the source (quoted there as eV^2, with which the ratio in Eq. (E.87) carries a leftover eV^{-1}), was chosen in the original model to reproduce the measured pair energy at high energy, and hot-electron injection data confirm the model up to 5 eV [1, Sect. II and Table 1]. It makes the ionization probability per event 0.05 at 3 eV, 0.36 at 5 eV, 0.77 at 8 eV and 0.88 at 10 eV, from Eq. (E.87) with $E_{\mathrm{g}} = 1.17$ eV. Phonon emission wins below about 6 eV, where the two probabilities cross, and the many 63 meV losses a carrier suffers on its way down from there are what puts $\varepsilon_{\mathrm{eh}}$ at three times the gap.

Simulated with this model, the high-energy moments follow the linear fits $\varepsilon_{\mathrm{eh}} = 1.7 E_{\mathrm{g}} + 0.084 A_{\mathrm{AB}} + 1.3$ and $F_{\mathrm{eh}} = -0.028 E_{\mathrm{g}} + 0.0015 A_{\mathrm{AB}} + 0.14$, with energies in eV and A_{AB} in the same units [1, Eqs. (12)–(13)]. At 123 K, where the gap is 1.160 eV [1, Eq. (11)], they give $\varepsilon_{\mathrm{eh}} = 3.71$ eV and $F_{\mathrm{eh}} = 0.115$. The measurement at the same temperature with a charge-resolving silicon sensor is $\varepsilon_{\mathrm{eh}} = (3.749 \pm 0.001)$ eV and $F_{\mathrm{eh}} = 0.119 \pm 0.002$ [117]. The pair energy agrees to 1 % because A_{AB} was tuned to it; the Fano factor is a prediction and

falls two standard deviations low. This is the comparison the first-principles cascade must improve on. Since its kernels contain no fitted constant, matching $\varepsilon_{\mathrm{eh}}$ is a test of the kernels, and F_{eh} is a second, independent one.

E.13.8 Convergence parameters

The parameters fall into three groups. For the bands: the quasiparticle correction, the number of bands (Eq. (E.84) needs every band a carrier can reach) and the $\boldsymbol{k}$ grid. For W^{scr}: the plane-wave cutoff of the dielectric matrix, which must resolve momentum transfers up to about twice the largest carrier momentum, the number of bands in the polarizability, the frequency grid from 0 to the largest carrier energy, and the $\boldsymbol{q}$ grid. For the kernels: the $\boldsymbol{k}$ grid on which Γ_{ii} and Γ_{ph} are tabulated, and the broadening of the energy-conserving delta functions. Near threshold the impact-ionization rate is a sum over few final states that conserve both energy and momentum. On a finite grid it scatters strongly from one $\boldsymbol{k}$ to the next; with 1728 $\boldsymbol{k}$ points in the zone the numerical error is about a factor two at rates below $10^{10}\,\mathrm{s}^{-1}$ [113, Fig. 2]. The scatter is a physical feature of the band structure smeared by the grid, and it averages out at higher energies, where many final states contribute [113, Sect. III].

The statistical parameter is the number of simulated cascades N_{casc}. The mean $\langle N_{\mathrm{eh}}\rangle$ converges quickly, since its relative error is $[F_{\mathrm{eh}}/(\langle N_{\mathrm{eh}}\rangle N_{\mathrm{casc}})]^{1/2}$. The variance converges slowly: for a nearly Gaussian pair number, the relative error of a sample variance is $(2/N_{\mathrm{casc}})^{1/2}$. Resolving F_{eh} to ± 0.002 at 0.119, the precision of the silicon measurement, needs a relative error of 1.7 % and therefore $N_{\mathrm{casc}} \approx 2/0.017^2 \approx 7\times 10^3$ cascades at each seed energy, and more if the comparison is to be made at the two-standard-deviation level.

E.13.9 Cost

The dominant cost is the full-frequency screened interaction. For carriers up to 50 eV in silicon, the momentum transfers reach $2k_E = 7.2\,\text{Å}^{-1}$, and a dielectric matrix that resolves them holds $N_G \approx \Omega(2k_E)^3/(6\pi^2) \approx 260$ plane waves. Stored at 200 frequencies on the 29 irreducible points of an 8^3 $\boldsymbol{q}$ grid, in double-precision complex numbers, the matrices $W^{\mathrm{scr}}_{\boldsymbol{GG}'}(\boldsymbol{q},\omega)$ occupy $29\times 260^2\times 200\times 16\,\mathrm{byte} \approx 6\,\mathrm{GB}$, and computing them requires the polarizability summed over about 30 empty bands on a $\boldsymbol{k}$ grid of similar density. For carriers up to 100 eV the band count triples, N_G grows to about 730 and N_G^2 eightfold. The kernel tables add a sum over final states for every tabulated initial state: with N_k points and N_b bands it scales as $N_k^2 N_b^2$ before symmetry, which for the 1728-point grid above and 30 bands is about 3×10^9 terms, each a contraction with W^{scr}. The phonon kernel is a standard Wannier-interpolated calculation (Appendix E.5), cheaper than the electronic one once the high-energy vertex problem is solved. The cascade Monte Carlo is cheap: a 5 keV seed makes some 1300 pairs and a few times 10^4 phonon events, so that 10^4 cascades amount to some 6×10^8 events, negligible next to the tabulation of the kernels.

E.13.10 Pitfalls

The gap must match the detector temperature, 1.17 eV at millikelvin against 1.12 eV at room temperature [1, Table I], since every rate near threshold depends on the energy above $\theta_{\mathrm{ii}}E_{\mathrm{g}}$. Two double countings are easy to make: the GW rate already contains plasmon emission, so no plasmon channel is added; and the first-principles phonon rate of Eq. (E.86) replaces, and is not added to, a phenomenological one. Two models that reproduce the same $\varepsilon_{\mathrm{eh}}$ do not validate each other's F_{eh}, and the agreement of the cascade of Sect. 18.4 with the single-parameter model is agreement of two models, not independent support.

Table E.7: The hot-carrier cascade in silicon. Band and matrix sizes are free-electron estimates from the text; the model values follow from Eq. (E.87) and its fits at 123 K [1]; the measurement is at 123 K and 5.9 keV [117].

Quantity	Silicon
Gap at low temperature; quasiparticle gap without spin–orbit	1.17 eV; 1.23 eV [113]
Plasmon energy; hole band width	16.6 eV; 12 eV [1]
Bands up to 20, 50, 100 eV	8, 32, 90
N_G and storage of W^{scr} for 50 eV carriers	260; 6 GB (29 $\boldsymbol{q}$, 200 frequencies)
$\boldsymbol{k}$ grid; threshold-region error	1728 points; factor 2 below $10^{10}\,\mathrm{s}^{-1}$ [113]
Ionization probability per event at 3, 5, 8 eV	0.05, 0.36, 0.77 (Eq. (E.87))
$\varepsilon_{\mathrm{eh}}$, F_{eh}: model	3.71 eV, 0.115
$\varepsilon_{\mathrm{eh}}$, F_{eh}: measured	(3.749 ± 0.001) eV, 0.119 ± 0.002
Cascades for $\delta F_{\mathrm{eh}} = 0.002$	7×10^3 per seed energy

E.13.11 Silicon

Table E.7 summarizes the silicon inputs, the reference values and the sizes derived in this section.

What a first-principles cascade adds is a prediction for the materials of Chap. 17, where no measurement exists, with kernels that were not tuned to any of them; silicon, whose $\varepsilon_{\mathrm{eh}}$ the measurement fixes, is where that prediction is first put to the test.

E.14 Phonon-assisted electron recoil from first principles

Between about 0.333 and 0.405 MeV, dark matter can create an electron–hole pair in silicon only by creating a phonon with it, and over that mass band the phonon-assisted rate is the only electron-recoil channel (Sect. 13.9). That rate is known only to a factor of ten. The estimate behind it takes the measured indirect absorption of silicon for the optical limit and multiplies it by a computed form factor of the zero-phonon vertex for the fall-off at finite q; it never evaluates the four-path amplitude of Sect. 13.2. The amplitude is hard for three reasons: the band sum over intermediate states is unrestricted and cannot stop at the top of a Wannier manifold; the density vertex is needed at momentum transfers of one to a few keV, where the small-q expansion in Wannier position matrix elements fails (Appendix E.5); and the external screening carries local fields, which the macroscopic dielectric function does not. Everything else follows the standard steps of Appendix E.5, including its operation count.

E.14.1 The phonon-assisted loss-function grid

What a rate needs is the phonon-assisted loss function $L^{\mathrm{PA}}(\boldsymbol{q}, \omega)$ of Sect. 13.3, a dimensionless number per bin of $(|\boldsymbol{q}|, \hat{\boldsymbol{q}}, \omega)$, built from the four-path amplitude of Sect. 13.2 and screened as in Sect. 13.4. The bins are those of the estimate in Sect. 13.9, so that the two can be compared bin by bin. They are $|\boldsymbol{q}| = 0.25$ to $1.25\,\text{Å}^{-1}$ (0.49 to 2.47 keV) in steps of $0.05\,\text{Å}^{-1}$, 30 directions in the irreducible wedge of the cubic group, and ω from $E_{\mathrm{g}} + \omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$ up to the pointwise window edge $E_{\mathrm{min}}(\boldsymbol{q}) - \delta_w$ of Sect. 13.6. From it follow the rate $\mathrm{d}\mathcal{R}/\mathrm{d}\omega$ and the reach of Sect. 13.3, for the heavy and the light mediator. On the truncation label $\mathfrak{T}$ (Sect. 8.5) the calculation sits at $n_B = 0$, $n_g = 2$ (one density vertex and one electron–phonon vertex, all

four time orderings), $n_T = 0$, lattice regime $\mathcal{K}$ = one phonon, and $\mathcal{X}$ = RPA with local fields for the external screening. The free-pair amplitude has no weight below $E_\mathrm{g} + \omega_\mathrm{TA}$; the replica shell below it (0.3332 to 0.3374 MeV in mass) is the exciton envelope of Appendix E.15 times the same amplitude, matched at $E_\mathrm{g} + \omega_\mathrm{TA}$.

The ground state and the phonons are the shared harmonic base of Appendix E.17: PBE, SG15 at 60 Ry, the measured lattice constant, a 12^3 $\boldsymbol{k}$ grid and an 8^3 wavevector grid. The 8^3 grid matters here. On a 4^3 grid the interpolated transverse-acoustic frequency at 0.85 X comes out above its value at X, an artifact of up to 1.5 meV on exactly the branch that sets the window. The self-consistent potential changes are kept for every wavevector, since the vertices are built from them, and the long-range quadrupolar part of g is split off before interpolation (Appendix E.5). An 18-function Wannier manifold (sp^3 on each atom plus d-like functions) is disentangled from about 60 bands on the full 12^3 grid without symmetry. The conduction states to 10 eV above the gap then sit inside the frozen window, and the size of the manifold is tested by repeating the amplitude with 26 functions.

Since the band sum multiplies matrices in band space, g, the density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})$ and the energies have to be rotated with the same $U(\boldsymbol{k})$ (Appendix A.5); with a mismatch every band energy is still correct but the phases of the sum are wrong. The acoustic sum rule is imposed on the dynamical matrix, never by setting g to zero as $\boldsymbol{Q} \to 0$, because the part of g linear in $\boldsymbol{Q}$ is physical and is exactly what the soft acoustic limit of Sect. 13.6 tests.

The density vertex is computed for each target $\boldsymbol{q}$ from the plane-wave coefficients of one non-self-consistent run at the shifted points $\boldsymbol{k} + \boldsymbol{q}$, with $\boldsymbol{q}$ kept unreduced (Appendix A.3); a vertex at a reduced $\boldsymbol{q}$ belongs to a different Umklapp class. With 21 values of $|\boldsymbol{q}|$ and 30 directions there are 630 targets in the irreducible wedge, about 1800 once equivalent pockets are unfolded, and each needs the full 12^3 grid without symmetry, about 10^6 diagonalizations in all. The amplitude is then the four-path sum with an unrestricted band sum: explicit inside the Wannier manifold, and above it the projected Sternheimer solution of Eq. (E.34) contracted with $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, one linear solve per external state and perturbation. That remainder matters most on the Bethe ridge, where $q^2/2m_e$ approaches the energy deposited. Below $|\boldsymbol{q}| \approx 0.5\,\alpha m_e \approx 1.9$ keV it is a correction to the Wannier sum; on the upper bins, 1.9 to 2.5 keV, it is carried in full. The external screening is the column $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q}, \omega)$ of an RPA calculation with local fields (Appendix E.6). The pair channel of Sect. 13.4 adds $W^\mathrm{scr}(\boldsymbol{q} + \boldsymbol{Q}, \omega_f)$ at the pair momentum $\boldsymbol{q} + \boldsymbol{Q}$, which kinematics pins near 0.85 X, so one dielectric matrix serves every bin. The two corrections are reported as ratios δ_LF and δ_pair, to be set against the screening correction of the estimate of Sect. 13.9, 0.3 to 2.6 relative to the optical calibration for the external factor and the local-field wings. The pair channel beyond that calibration has no estimate; Appendix E.16 gives the supercell route that sizes it. The sums over pocket pairs run on fine grids of 60^3, refined to 120^3, with tetrahedra and no broadening.

Quantum ESPRESSO [118], Wannier90 [119], EPW [55] and BerkeleyGW [38] carry most of this, and a long-wave DFPT code the quadrupoles (Appendix E.5). The density vertex in the Wannier gauge and the second-order amplitude with its Sternheimer remainder are in none of them; they are new code, a few hundred lines on top of EPW. Table E.8 collects parameters and costs. The plane-wave stages total about 350 to 450 core-hours, dominated by the density vertices; the amplitude, at the 10^{16} operations that Appendix E.5 counts, adds a quarter of an hour to a few hours on one accelerator.

E.14.2 Physical controls

A bin converged to 5 % under the refinements of Table E.8 need not be correct, however, and each control below tests the physics against something the calculation was not fitted to.

The optical limit tests the vertices against data. At $|\boldsymbol{q}| = 10^{-3}\,\text{Å}^{-1}$ the density vertex

Table E.8: Parameters and costs of the phonon-assisted loss function of silicon. The cutoff refers to the SG15 norm-conserving pseudopotential [2] with four valence electrons. Fine grids are those of the pockets within 0.1 eV of the band extrema, quoted as their equivalent uniform density. Costs are in core-hours unless stated; "base" is the shared DFPT run of Appendix E.17.

stage	grids and parameters	acceptance test	cost
ground state, phonons	PBE, $a = 5.431$ Å, 60 Ry, 12^3 $\boldsymbol{k}$, 8^3 $\boldsymbol{Q}$ (29 irreducible)	neutron data per branch (Appendix E.17)	base, ≈ 100
long-range g	dipoles (zero in Si), quadrupoles	small-$\boldsymbol{Q}$ angular dependence	minutes
Wannier gauge	18 functions (sp^3, d); window to 40 eV	bands ≤ 10 meV to 10 eV above the gap	3–5
density vertex	630 targets, 12^3 without symmetry	direct vs Wannier at 5 off-grid $\boldsymbol{k}$, $\leq 2\,\%$	200–300
vertex g	interpolated	direct DFPT at 3 off-grid $\boldsymbol{Q}$, $\leq 5\,\%$	10
amplitude	$60^3 \rightarrow 120^3$, tetrahedra; Wannier sum plus Sternheimer remainder (10^3 solves per target)	5 % per bin in L^{PA}	accelerator hours (Appendix E.5)
screening	RPA, $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q},\omega)$; W^{scr} at $\boldsymbol{q}+\boldsymbol{Q}$	δ_{LF}, δ_{pair} reported	tens

reduces to its dipole form (Sect. 3.2), and L^{PA} has to reproduce the phonon-assisted $\varepsilon_2^{\mathrm{PA}}$ that EPW computes from the same vertices. After the gap shift of Appendix E.5 it also has to reproduce the measured absorption edge at the temperature of each measurement: silicon at 20 K, where the transverse-acoustic and transverse-optical replicas are resolved [120], germanium against its edge data at 77 to 80 K, and GaAs at the temperature of its data. The comparison is made branch by branch, with $\varepsilon_2^{\mathrm{PA}}$ resolved by phonon branch, since the transfer of a measured edge to millikelvin needs the branch weights and a measured spectrum gives only their sum (Sect. 13.5). The factor-ten estimate used these data as input; the full amplitude is tested by them. Switching g off tests the labeling of final states: the phonon-assisted rate then vanishes identically, and any residue is zero-phonon weight leaking into the window through a mislabeled state.

A frozen phonon tests the adiabatic form of the amplitude derived in Sect. 13.2. At $\boldsymbol{Q} = \mathrm{X}$ an eight-atom cell holds the phonon exactly; displacing the atoms by ± 0.005 Å along a mode and differencing the zero-phonon transition densities gives the adiabatic amplitude directly, which agrees with the second-order sum up to the expected $\omega_{\boldsymbol{Q}\nu}/\delta_{\mathrm{off}}$. The soft acoustic limit of Sect. 13.6, checked at $|\boldsymbol{Q}| = 10^{-2}$, 10^{-3} and 10^{-4} of $2\pi/a$ in three directions, fails at once for a vertex whose small-$\boldsymbol{Q}$ part was zeroed. Symmetry gives two exact zeros: on the valley axis the longitudinal-acoustic path of the class without Umklapp vanishes, and so does the longitudinal-optical path of the $(\bar{2}\,0\,0)$ class (Sect. 13.9); both come out at the level of numerical noise. The direct-gap control, GaAs, is described with that crystal below.

The form-factor estimate is not a lower bound on L^{PA}, since its intraband terms, which cancel at $q \rightarrow 0$, decorrelate at finite q. Between 0.333 and about 0.35 MeV the rate is a power of the distance from the onset (Sect. 13.3), so there the exponent is the meaningful output.

E.14.3 Zero-phonon rates on common bins

Sect. 13.9 places the end of the phonon-assisted lead at about 0.42 to 0.43 MeV, with a two-channel band to about 0.51 MeV. That is a ratio of two rates, settled only when both are computed on the same bins. We therefore extend the output grid past the window into three regions, which have to be kept apart. The first zero-phonon edge, 0.405 to 0.408 MeV with a best value of 0.407 MeV, is a joint minimum over the momentum transfer and the hole's crystal momentum and lies off the valley axis. Its state belongs to the symmetry-suppressed (glide) class, so its vertex is small and it does not end the lead. Around this mirror-plane optimum the bins are cones: $|\boldsymbol{q}| = 1.2$ to 1.9 keV, tilts $\vartheta \leq 0.4$ rad from the axis within the {001} and {011} planes and up to 0.15 rad out of plane, and ω within 0.3 eV of $E_{\min}(\boldsymbol{q})$. The second region is the cone of the second heavy hole on the axis, the mirror-even band that opens from 0.415 to 0.420 MeV. The third is the X-face edge of the allowed $(\bar{2}\,0\,0)$ class, where the zero-phonon line of allowed size opens near 0.50 MeV. Above $E_{\min}(\boldsymbol{q})$ the phonon-assisted value follows the convention of Sect. 13.6; the shell just below the edge is reported separately, and the bins contain the joint optimum at 0.407 MeV, so that the edge is the one the rate sees.

The zero-phonon side is cheap, under one core-hour. It is a Wannier–Mott continuum (Sect. 4.9) with the anisotropic reduced-mass tensor of the pair at each bin, from the Hessians of the interpolated bands at the pair's two crystal momenta. It uses a static ε_∞ and density vertices computed directly from the plane-wave states of both top heavy-hole bands. The hole is placed where the pair is lowest at that $\boldsymbol{q}$; at the off-axis optimum it sits about 0.06 eV below the valence maximum, and placing it at Γ misses the optimum. At the class extremum the allowed $(\bar{2}\,0\,0)$ vertex is $\sum_v |\rho_{cv}|^2 = 0.150$, the size of an ordinary allowed vertex (Sect. 13.9); the comparison needs it on every bin, with spin–orbit coupling in the bands, which shifts the first-hole bins by about 15 meV. The vertices come from matching Miller indices on a three-point stencil around each bin. They are converged when halving the stencil step from 0.02 to $0.01 \times 2\pi/a$ changes $|\rho|^2$ by less than 2 %. Halving the angular step from 0.01 to 0.005 rad has to change the rate by less than 1 %, and the ratio of the two rates per bin has to be stable to 5 % when the number of bins grows by half. Off the axis the band index no longer orders the states by mirror parity, and each state is assigned by its parity; on the axis the degenerate pair is summed.

E.14.4 Verification of the onset

The onset of the phonon-assisted rate, $m_{\min}^{\mathrm{PA}} = 0.3332$ MeV for the exciton replica and 0.3374 MeV for the free pair (Sect. 13.9), follows from three numbers. They are the measured gap $E_{\mathrm{g}}(0) = 1.170$ eV, the exciton binding $E_x^b = 14.7$ meV [121], and the lowest phonon energy that supplies the missing crystal momentum, $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*) = 14.79$ meV at $|\boldsymbol{Q}^*| = 0.53\,\text{Å}^{-1}$, from the PBE force constants of the shared base at the measured lattice constant, with the measured valley momentum $k_{\mathrm{c}} = 1.94$ keV (Sect. 13.1). At the PBE lattice constant, where $k_{\mathrm{c}} = 1.906$ keV, the same chain gives 14.72 meV at $0.510\,\text{Å}^{-1}$ and the same onsets to four digits. Only the last is computed, and it is the one a verification has to reproduce. Since the onset mass is $2(E_{\mathrm{g}} + \omega_{\mathrm{TA}})/v_{\max}^2$ at the optimal q (Sect. 13.1), an error $\delta\omega$ moves it by $2\,\delta\omega/v_{\max}^2$, which is 8.5×10^{-5} MeV for $\delta\omega = 0.3$ meV. The onset is robust to everything except a systematic error in the transverse-acoustic branch.

That systematic is real. Silicon's transverse-acoustic modes have negative Grüneisen parameters: at the measured lattice constant of the shared base they are 8 % low at X and L (Sect. 5.2), while at the PBE lattice constant every branch is soft by about 3.5 % [45]. We therefore compute $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$ directly by DFPT at $\boldsymbol{Q}^*$, not by interpolation, at both lattice constants and with an independent pseudopotential from the PseudoDojo table [30], and

add a third variant with the whole branch scaled by the ratio of the neutron frequency to the computed one at X, as in Sect. 13.1. The test is that the direct values bracket the interpolated one within its 0.3 meV grid tolerance. The neutron-scaled variant gives the systematic: $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*) = 16.1\,\mathrm{meV}$, and by the rule above onsets of 0.3336 and 0.3378 MeV. Whether the replica opens above or below the free-pair floor $2E_{\mathrm{g}}/v_{\max}^2 = 0.3332\,\mathrm{MeV}$ is decided by the sign of $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*) - E_x^b$. It is positive in both variants, +0.1 meV with the computed branch and +1.4 meV with the neutron-scaled one, so the replica opens at most 0.0004 MeV above the floor. One DFPT wavevector at the base settings costs about three core-hours, and the whole verification a few tens.

E.14.5 Germanium and gallium arsenide

Germanium cannot follow silicon, because PBE gets its band order wrong. It puts the conduction state at Γ 0.047 eV above the valence maximum and the L valley above it, the reverse of the measured order, in which the indirect gap at L (0.744 eV) lies below the direct gap at Γ (0.89 eV). A scissor applied to that order gives a window that is wrong by the inversion of Γ and L. Any germanium band structure is therefore checked before it is used: L below Γ, both gaps within 0.1 eV of the measured ones, and the band masses at both valleys against measurement, the transverse mass at L within $0.010\,m_e$ of $m_\perp^* = 0.0815\,m_e$ (Table 3.1) and the mass at Γ within $0.005\,m_e$ of $0.038\,m_e$.

HSE06 with spin–orbit coupling at the standard exchange fraction of one quarter (Appendix E.1) fails this check. At the measured lattice constant it puts Γ at 0.64 to 0.69 eV and L at 0.79 to 0.83 eV above the valence maximum, so the gap at L passes while the order and the direct gap fail, and the inversion of 0.09 to 0.15 eV persists across $\boldsymbol{k}$ and exchange grids, Fock cutoffs of 160 and 320 Ry and two families of fully relativistic pseudopotentials. Abdurrazaq and coworkers find the same inversion without spin–orbit coupling, at the standard fraction and their optimized lattice constant: the direct gap at 0.775 eV below the indirect gap at 0.823 eV. Above a fraction of 0.28 the order is restored, but at 0.3 the direct and indirect gaps are still 34 % and 17 % off [122, Fig. 1(a)]. A correction that shifts only the 4s-like orbitals [122], with its single parameter fitted to both gaps, passes the energies, L at 0.749 eV below Γ at 0.918 eV. It fails the masses: the transverse mass at L comes out $0.101\,m_e$ and the mass at Γ $0.053\,m_e$, 24 % and 39 % heavy, so the valley energies are right and the curvatures are not. The next route is a quasiparticle *GW* correction (Appendix E.3) on top of the hybrid or PBE bands, subject to the same check of order, both gaps and the masses at L and Γ. Until bands pass, the zero-phonon edge of germanium stays the estimate of Sect. 13.1, 0.32 MeV on the L axis inside a bracket of 0.212 to 0.420 MeV.

On bands that pass, sixteen spinor sp^3 Wannier functions are extracted with the twenty 3d bands excluded and a frozen window to 3 eV above the valence maximum, and the interpolated edges and masses are compared with direct calculations at off-grid points. The zero-phonon edge then follows by the joint minimization over $(\boldsymbol{q}, \boldsymbol{k})$ for the L-valley class and its Umklapp copies. Whether that optimum leaves the axis is not known: the germanium heavy hole is less warped than that of silicon, and nothing transfers from the silicon result. The mirror rule of Sect. 13.9 becomes one for the diagonal mirror planes, which carry no glide phase; whether the first state is allowed is checked at the true optimum.

The germanium phonons are the shared base (Appendix E.17), with an 8^3 $\boldsymbol{k}$ grid and a 6^3 wavevector grid. On a 4^3 wavevector grid the transverse-acoustic frequencies are 13.5 % (X) and 11.6 % (L) below neutron data at 80 K [123], while the optical and longitudinal branches agree to 2.3 %. Either the nearly closed PBE gap overscreens the shear modes or the 8^3 $\boldsymbol{k}$ grid is too coarse for a 47 meV gap. One DFPT run at X on a 12^3 grid, about three core-hours, separates the two, and $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$ has to be stable to 0.3 meV between the 4^3 and 6^3 grids. If

the deficit survives, the neutron-scaled branch is used for the window; the phonon-assisted edge moves by at most 0.0003 MeV either way. The germanium rate then follows the silicon chain with spin–orbit coupling and the 3d shell in the valence (14 electrons per atom): 36 spinor Wannier functions, the sixteen sp^3 ones plus twenty for the 3d shell, and 120 spinor bands. Spinors double both the length of each state and the band count, which multiplies each plane-wave stage by four to eight; for germanium's smaller DFPT grid and targets the chain comes to 400 to 800 core-hours, or hours on one accelerator node.

Gallium arsenide is the direct-gap control. There the zero-phonon pair at small q is out of reach for any mass: the pair at Γ needs $E_{\mathrm{dir}} + q^2/2M_x \leq q v_{\max}$, and the right side minus the recoil is at most $M_x v_{\max}^2/2 \approx 1.04\,\mathrm{eV}$ for a pair mass $M_x \approx 0.58\,m_e$, below the direct gap of 1.52 eV. A phonon can absorb the momentum instead, and the phonon-assisted rate leads until the L-valley zero-phonon state opens, from about 0.43 MeV to between 0.52 and 0.56 MeV in a parabolic estimate with the L valley at 1.815 eV (Sects. 13.1 and 13.9). The chain, with the Fröhlich vertex (Eq. (E.37)), has to reproduce that band and give a small correction above the zero-phonon threshold; without the long-range part the vertex diverges at small $\boldsymbol{Q}$. The ground state is the shared base at $a = 5.653\,\text{Å}$ with the gallium 3d shell in the valence. PBE puts the direct gap at about 0.5 eV (Appendix E.18.1), so the bands come from HSE06 or from a $\boldsymbol{k}$-dependent scissor fitted to the low-temperature conduction minima at Γ (1.519 eV), L (1.815 eV) and X (1.981 eV) [13], with spin–orbit coupling for the holes. The Fröhlich vertex takes ε_∞ and Z^* from measurement through Eq. (E.96), since the PBE ε_∞ of GaAs is tens of percent too large (Appendix E.17). The window costs about 8 core-hours with the scissor and 40 to 80 with the hybrid; the full rate costs about as much as silicon's.

For the other indirect-gap candidates the window comes before any rate: $E_{\min}(\boldsymbol{q})$ on the joint $(\boldsymbol{q}, \boldsymbol{k})$ space, the band masses, $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$ and g at $\boldsymbol{Q}^*$ from one DFPT run. PBE with a scissor serves wherever PBE gets the band order right, checked against measured gaps at the high-symmetry points; narrow-gap and d-electron crystals need bands that pass the germanium check of order, gaps and masses. The kinematics are checked in every case by the limit $|\boldsymbol{q}| \to |\boldsymbol{K}_{\mathrm{pk}}|$, the crystal momentum of the conduction pocket, where $E_{\min}$ returns the gap.

E.14.6 Fano widths and detunings in GaAs

Near the edge of the off-shell window of Sect. 13.6, a zero-phonon final state of energy $\tilde{E}$ lies inside the phonon-assisted continuum and interferes with it, the Fano term of Sect. 8.6. The term vanishes in crystals invariant under inversion combined with time reversal with an even electron number, such as silicon and germanium, at the level of one-body self-energies, but not in GaAs. Level shifts from the electron–hole vertex can split a degenerate zero-phonon manifold in a basis that is not invariant, and then only the sum of the Fano terms over the branches vanishes. The Fano term is perturbative only where the width Γ of the resonance is small compared with its detuning δ_{F}, the distance of $\tilde{E}$ to the nearest feature of the continuum. With the Fano coefficient c_ν^F of each branch ν, the branch probability of the first emission is $(\Gamma^{(\nu)}/\Gamma^{\mathrm{tot}})(1 + \bar{c}^F - c_\nu^F)$, with $\bar{c}^F$ the width-weighted mean. In GaAs $|c_\nu^F| \lesssim 0.3\Gamma/\delta_{\mathrm{F}}$ to first order, which bounds the change of the pair-number distribution by $0.3\Gamma/\delta_{\mathrm{F}}$ and keeps the probabilities positive for $\Gamma/\delta_{\mathrm{F}} \leq 0.5$. What Sect. 13.6 needs is therefore a map of $\Gamma/\delta_{\mathrm{F}}$ over (q, ω).

The width of the pair state is the sum of the electron and hole widths $\Gamma_n(\boldsymbol{k})$. These are on-shell Fan–Migdal widths,

$$\Gamma_n(\boldsymbol{k}) = -2\,\mathrm{Im}\,\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\varepsilon_{n\boldsymbol{k}}),$$

computed at 10 K (Sect. 6.3), at $n_g = 2$ with a statically screened vertex. The detuning is the

smallest of the distances to the renormalized edge $E^{\mathrm{ren}}_{\mathrm{min}}(q)$, to the nearest intervalley threshold of the electron (L, X) or the split-off threshold of the hole, and the local scale of the joint density of states. The coefficient c^F_ν is the non-resonant remainder of the same second-order amplitude, evaluated on the final shell. Estimates give a Fröhlich LO width of a few meV from just above the conduction minimum to the Γ–L onset (about 3 meV at the onset, 0.3 eV above the minimum, with $\alpha_F = 0.068$ and $\omega_{\mathrm{LO}} = 36\,\mathrm{meV}$), and 20 to 60 meV above the onset, so $\Gamma/\delta_{\mathrm{F}}$ is 0.01 to 0.2 away from thresholds and diverges at each onset. The map excludes shells of $\pm 2\Gamma$ around the Γ–L onset and the split-off onset (0.34 eV below the valence maximum), and the shell within 3Γ of the window edge; inside them only totals from the unexpanded Dyson form are quoted.

Because the split-off band sets the hole widths, the electrons need fully relativistic pseudopotentials, at 90 Ry because the spinor calculation with the Ga 3d shell converges more slowly than the scalar base. Phonons, Born charges and ε_∞ come from the shared base on a 6^3 grid. The run uses an 8^3 $\boldsymbol{k}$ grid, 16 spinor sp^3 Wannier functions with the 3d bands excluded, the polar vertex, and self-energies on a 100^3 fine $\boldsymbol{Q}$ grid for the 13 824 $\boldsymbol{k}$ points of the final shell. The widths converge to 5 % from 80^3 to 100^3 and are stable as the smearing goes from 10 to 2.5 meV; c^F_ν converges to 10 % when 20 bands are added to the intermediate sum. Silicon, with eight sp^3 functions and a 60^3 grid, has to return $c^F_\nu = 0$ to numerical precision, which tests the chain against the symmetry theorem of Sect. 8.6. The self-energies cost about 10^4 core-hours (30 to 60 accelerator-hours), silicon a tenth of this. The map fixes the (q, ω) region in which Sect. 13.6 quotes the Fano term for GaAs.

E.14.7 The phonon-assisted kinetic-energy row

A scalar coupled to the electron's mass is absorbed below the direct gap of silicon only with a phonon, through the kinetic-energy density $\mathcal{T}_e$, and its rate is the structure factor $S^{\mathrm{PA}}_{\mathcal{T}\mathcal{T}}(0, m_\phi)$ of Eq. (13.49), for a point group without a polar vector (Sects. 10.7.4 and 13.8). No measurement supplies it, since its amplitude Eq. (13.48) contains no velocity vertex and is not proportional to $\varepsilon_2^{\mathrm{PA}}$. The chain above computes it with one change: the density vertex at $\boldsymbol{q}$ is replaced by the kinetic vertex $\langle m|p^2/2m_e|n\rangle$ at $\boldsymbol{q} = 0$, and the energy is $\omega = m_\phi$ from the indirect edge to the direct gap. The Wannier gauge, the vertices g, the Sternheimer remainder above the manifold and the fine pocket grids are those of the density row. The kinetic matrix element converges like the total kinetic energy and is weighted toward the cores, so it is taken from PAW or all-electron states as in Appendix E.16, at a cutoff of at least 80 Ry for the norm-conserving comparison. Since $\langle c|p^2/2m_e|v\rangle = -\langle c|V|v\rangle$ between eigenstates of different energy, Eq. (13.48) can be evaluated twice, once with the kinetic vertex and once with the matrix elements of the crystal potential, and the two must agree to the 5 % per bin of Table E.8; a disagreement points to the core region or to an incomplete band sum. The same chain with the density vertex at $\boldsymbol{q} \to 0$ reproduces $\varepsilon_2^{\mathrm{PA}}$ and the measured edge, which tests everything except the kinetic vertex. Exciton final states near the edge enter as for the density row, with the envelope of Appendix E.15. With one momentum in place of the 630 targets of the density row, the row costs a small fraction of that row, and it enters Sects. 10.7.4 and 13.8 and the scalar row of Sect. 21.7.

E.14.8 The f-sum check with its parts separated

The sum rule Eq. (E.50) tests whether the intermediate-state sum is complete. With norm-conserving pseudopotentials the band sum $\sum_m (\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}})|\rho_{mv}(\boldsymbol{k}, \boldsymbol{q})|^2$ at $\boldsymbol{k} = \Gamma$ reaches 0.81 of $q^2/2m_e$ with 16 bands and 0.89 with 100 bands, at $|\boldsymbol{q}| = 0.1$ to $0.5\,\text{Å}^{-1}$, still rising slowly (Sect. 13.9). That shortfall mixes the bands left out with the nonlocal term δ_{NL}, which

a comparison with $q^2/2m_e$ alone ignores, and the two can be separated exactly.

The truncation part needs no more bands. Let $P_N^{\perp}$ project out the lowest N eigenstates of $H(\boldsymbol{k}+\boldsymbol{q})$ and let $|\phi\rangle = P_N^{\perp}|u_{v\boldsymbol{k}}\rangle$. Since $\rho_{mv}(\boldsymbol{k},\boldsymbol{q}) = \langle u_{m,\boldsymbol{k}+\boldsymbol{q}}|u_{v\boldsymbol{k}}\rangle$ and $P_N^{\perp}$ commutes with $H(\boldsymbol{k}+\boldsymbol{q})$,

$$\sum_{m>N} (\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}})\,|\rho_{mv}(\boldsymbol{k},\boldsymbol{q})|^2 = \langle\phi|H(\boldsymbol{k}+\boldsymbol{q}) - \varepsilon_{v\boldsymbol{k}}|\phi\rangle, \tag{E.88}$$

one application of the Hamiltonian to a projected vector, the cost of a single eigensolver iteration. The nonlocal part is a ground-state expectation value, $\delta_{\mathrm{NL}} = \langle u_{v\boldsymbol{k}}|V_{\mathrm{NL}}(\boldsymbol{k}+\boldsymbol{q}) - V_{\mathrm{NL}}(\boldsymbol{k})|u_{v\boldsymbol{k}}\rangle$ minus its part linear in $\boldsymbol{q}$, from the same projectors as the Hamiltonian. The truncated sum plus Eq. (E.88) then equals $\boldsymbol{q}\cdot\boldsymbol{v}_v^{\mathrm{g}} + q^2/2m_e + \delta_{\mathrm{NL}}$ to the precision of the eigenvectors, a test of the code rather than of the physics. The ratio $\delta_{\mathrm{NL}}/(q^2/2m_e)$ against q says how much of the 0.89 shortfall is the nonlocal term rather than missing bands; an all-electron (PAW) calculation, which has no nonlocal term (Appendix E.2), checks it. The remainder also measures the intermediate-state weight above the Wannier manifold that the Sternheimer remainder supplies: with the nonlocal term, 60 bands should exhaust at least 95 % of the sum. The check runs at Γ and five off-grid points, for the three cubic directions, in minutes.

The loss function and the rate replace the factor-ten estimate in the table of the window of Sect. 13.9, and the reach replaces its curve. The ratio of the two rates on common bins fixes where the phonon-assisted rate stops leading, and the verified onset and the gallium arsenide control replace two of the three estimated entries of that table with computed ones. The germanium window replaces the third only once bands pass the germanium check above.

E.15 Excitons and core states

In silicon an electron and a hole attract each other strongly enough to bind 14.7 meV below the gap [121], and the same attraction enhances the continuum just above it. Every near-threshold rate that a random-phase calculation gives misses both effects. The finite-momentum Bethe–Salpeter equation (BSE) of Sect. 4.9 and Appendix E.6 contains them, and Sect. 10.5 shows how its output enters a rate. Published dark-matter calculations use its Tamm–Dancoff form, however, and none treats the bound lines of an indirect gap. We need the BSE with its coupling block, and with it the exciton series, the phonon replicas of Sect. 13.8, the exciton lines of the Migdal effect (Sect. 12.3) and the self-trapped exciton of an alkali halide; the semicore response and the spin–orbit mixing at the band edges (Sect. 10.6) complete the set.

E.15.1 Finite-momentum BSE with coupling

What a rate near the gap needs is the loss function $L(\boldsymbol{q},\omega)$ of Sect. 10.2 on bins of $(|\boldsymbol{q}|,\hat{\boldsymbol{q}},\omega)$ for $|\boldsymbol{q}| \le 10\,\mathrm{keV}$ and $\omega \le E_{\mathrm{g}} + 10\,\mathrm{eV}$, with the energies and weights of the bound lines listed separately. Outside that box the RPA with local fields of Appendix E.16 serves, and the two are matched on a strip where both are computed; their difference is the many-body nuisance of Sect. 8.8. On the truncation label $\mathfrak{T}$ (Sect. 8.5) the calculation sits at $n_B = 0$ and $n_g = 0$, with $\mathcal{X}$ = BSE on quasiparticle bands.

At finite q the coupling block between pairs and antipairs carries the long-range exchange $v_{\mathrm{C}}(\boldsymbol{q}+\boldsymbol{G})$, which turns the proper response into the screened loss function (Sect. 4.9). The largest error comes from a resonant-only response, which keeps one time ordering. For a Penn-type oscillator at E_0 with weight A, the full response gives $\varepsilon_1 - 1 = A[1/(E_0-\omega) + 1/(E_0+\omega)]$, while the resonant-only form keeps the first term, so

$$\frac{\varepsilon_1^{\mathrm{res}} - 1}{\varepsilon_1 - 1} = \frac{E_0+\omega}{2E_0}, \qquad \frac{L^{\mathrm{res}}}{L} \approx \Big(\frac{\varepsilon_1}{\varepsilon_1^{\mathrm{res}}}\Big)^2, \tag{E.89}$$

with ε_2 unchanged at $\omega > 0$. For silicon at $|\boldsymbol{q}| = 0.71\,\text{Å}^{-1}$ (1.39 keV) and $\omega = 1.3\,\text{eV}$, the tables of Sect. 10.2 give $\varepsilon_1 = 4.45$ and $\varepsilon_1^{\text{res}} = 3.05$, so the resonant-only form overstates the loss function by a factor of 2.1 in the momentum range of the phonon-assisted window (Sect. 10.5), and a single oscillator at $E_0 = 9.1\,\text{eV}$ gives 2.2; the factor tends to one as $q \to \infty$, where $\varepsilon_1 \to 1$. A retarded Tamm–Dancoff calculation has no such error, since it keeps both poles $\pm E_\lambda$ and takes ε_1 by Kramers–Kronig; what it drops is the coupling block itself, an error set by the ratio of the local-field exchange to the pair energy, about 10 % on the proper ε_2. The error is avoided by the full coupling, or by keeping the Tamm–Dancoff BSE for the proper ε_2 (line shapes and bound lines) and taking the external screening from the RPA with local fields, with the head removed from the BSE exchange so that it is not counted twice.

The bands come from Quantum ESPRESSO with PBE and G_0W_0 quasiparticle energies (Appendix E.3), and a code with a finite-momentum BSE, such as BerkeleyGW [38], solves the pair problem on them. For germanium the PBE starting point has Γ below L and a nearly closed gap, so the quasiparticle bands must first pass the check of Appendix E.14 (valley order, both gaps, masses at L and Γ); the germanium BSE, and the exciton dispersion along Γ–L below, wait for bands that pass. The pair basis has 4 valence bands (for germanium plus the 3d shell) and 12 conduction bands. On a coarse 8^3 grid the kernel holds $8^3 \times 4 \times 12 = 24{,}576$ pairs, about 10 GB per reduced momentum, and it is interpolated to a fine 24^3 grid of 6.6×10^5 pairs, whose dense kernel would hold 4.4×10^{11} complex numbers, 7 TB per momentum. That kernel is never stored: a Haydock (Lanczos) solver of about 200 iterations applies it to a vector on the fly. The momenta are the reduced points of the 24^3 set with reciprocal-lattice shells up to 10 keV, about 10^3 points after the cubic symmetry, each of order 10^{13} to 10^{14} floating-point operations, so the whole run takes about 10^3 node-hours, a few times 10^4 core-hours.

Even the fine grid cannot resolve the silicon exciton. Its radius is $a_x = 4.2\,\text{nm}$ (Sect. 10.5), and a hydrogenic envelope needs a $\boldsymbol{k}$ spacing below about $1/(3a_x) \approx 0.008\,\text{Å}^{-1}$, against $0.08\,\text{Å}^{-1}$ on the 24^3 grid. The bound lines are therefore added from the effective-mass model of Sect. 10.5, with ab initio masses, $\varepsilon_\infty = 11.7$ and the vertices and their gradients at the band extrema, and the BSE supplies the continuum and its Sommerfeld enhancement. The pair basis is evaluated around the off-axis pair optimum of Sect. 13.9, where the hole sits off Γ, not only at the band extrema.

The coupling block restores the f-sum rule of Sect. 4.3, which the scissor and the band cutoff also strain, so with the coupling on, the sum rule of the full loss function holds to better than 1 %. At $q < 3\,\alpha m_e$ the loss function is compared with inelastic x-ray and electron energy-loss data for silicon, and the bound-line model has to reproduce the free-exciton binding of silicon, 14.7 meV. The structure factor is reported three ways, with the full coupling, in the retarded Tamm–Dancoff form and in the resonant-only form, at $|\boldsymbol{q}| = 0.5$, 0.7, 1.0 and 1.4 keV; the Tamm–Dancoff route passes if it agrees with the full one to 15 %, and the resonant-only result tests Eq. (E.89).

E.15.2 Exciton dispersion and the hydrogenic series

A bound exciton of total momentum $\boldsymbol{Q}$ sits below the lowest pair at that momentum by its binding energy,

$$E_x(\boldsymbol{Q}) = E_{\min}(\boldsymbol{Q}) - E_x^b[\mu(\boldsymbol{Q}), \varepsilon_\infty], \tag{E.90}$$

where the reduced-mass tensor $\mu(\boldsymbol{Q})^{-1} = \nabla^2\varepsilon_c(\boldsymbol{k}_v + \boldsymbol{Q}) - \nabla^2\varepsilon_v(\boldsymbol{k}_v)$ is evaluated at the crystal momenta $\boldsymbol{k}_v(\boldsymbol{Q})$ and $\boldsymbol{k}_v(\boldsymbol{Q}) + \boldsymbol{Q}$ of the lowest pair. Since the hole is not at Γ near the zero-phonon optimum (Sect. 13.9), the mass changes along the dispersion. The Hessians of the Wannier-interpolated bands are taken by finite differences with a step of $0.005 \times 2\pi/a$, on the momentum grid of the zero-phonon edge along Γ–X on both sides of X and across

the {001} planes (along Γ–L and near Γ for germanium, on bands that pass the check of Appendix E.14). A variational anisotropic hydrogen problem in a basis of 30 anisotropic Gaussians, with the measured $\varepsilon_\infty = 11.7$, then gives $E_x(\boldsymbol{Q})$ with the weights of the 1s and 2p states, converged when E_x^b is stable to 0.1 meV in the basis size. Near Γ the heavy and light holes are degenerate, so scalar masses fail and the hole needs the six-band $\boldsymbol{k}\cdot\boldsymbol{p}$ form with spin–orbit coupling (split-off energy 44 meV in silicon). The calculation takes minutes; it enters Sect. 10.5, and its bound lines feed the BSE above.

E.15.3 Exciton–phonon replicas

The replicas of Sect. 13.8, lines at $E_x + \omega_\nu(\boldsymbol{Q}^*)$ below the free-pair edge, need the exciton–phonon vertex: the electron and hole vertices g weighted by the envelopes of the initial and final excitons. Their strength per phonon branch and their position are wanted at pair momenta near 0.85 X and, following Sect. 13.9, at the off-axis momenta where the hole sits off Γ. A finite-momentum BSE at that pocket set gives them, with the vertices g from EPW in the same Wannier gauge $U(\boldsymbol{k})$ as the exciton eigenvectors, since with a mismatch the phases of the sum are wrong; about 10^3 $\boldsymbol{k}$ points per pocket resolve an exciton of radius 4.2 nm, at about 10^4 core-hours. A cheaper route, a Wannier–Mott envelope multiplying the free-pair amplitude of Appendix E.14, is the cross-check, and both must reproduce the replica edge at 0.3332 MeV (Sect. 13.9).

Whether the replica formula of Sect. 13.8 is complete cannot be tested in a crystal. The band sum of Sect. 13.2 has one pair in every intermediate state; states with two pairs enter only through the interaction beyond the Tamm–Dancoff space of one pair (Sect. 13.8), and whether the two-pair sector reduces to the one-pair terms when the electron and hole interact is proven only without interaction. A model that can be solved exactly decides it. Take a one-dimensional two-band chain of 8 to 12 cells, with spinless electrons in two cosine bands of hopping t separated by $E_\mathrm{g} = 2t$. Add a contact attraction W between electron and hole, and one phonon mode of frequency $0.05\,E_\mathrm{g}$ coupled linearly with strength g, truncated at three phonons (checked at four). The Lanczos method diagonalizes it in the sectors with zero, one and two pairs. A linear fit in g between 10^{-3} and 10^{-2} extracts the exact first-order phonon-assisted amplitude, which is compared with the replica formula built from the model's exact one-pair excitons. At $W = 0$ the two must agree to 10^{-6}, as the theorem requires. For $W/E_\mathrm{g} = 0.05$ to 0.3 the deviation is the two-pair correction, to be compared with its estimated size $\sim E_x^b/E_\mathrm{g}$, about 1 % in silicon; it is extrapolated in the chain length (8, 10, 12) with a Lanczos residual below 10^{-10}. The two-pair sector must be in the basis, whose dimension grows as N^4, since without it the test is circular. It runs in minutes to an hour on one core.

E.15.4 Exciton lines in the Migdal response

A nuclear kick can leave the crystal in a bound exciton rather than an ionized pair (Sect. 12.3). The Migdal convolution of Appendix E.18 integrates the electronic momentum $\boldsymbol{p}$ against the lattice response at $\boldsymbol{q} - \boldsymbol{p}$ over the whole zone, so the exciton lines are needed everywhere, not only near $\boldsymbol{p} \to 0$: the line strengths $\rho_{\lambda 0}(\boldsymbol{p} + \boldsymbol{G})$, including the cross terms between different $\boldsymbol{G}$ and $\boldsymbol{G}'$, and the energies $E_\lambda(\boldsymbol{p})$. The boost operator carries the unreduced $\boldsymbol{p}$, so the Umklapp copies $\boldsymbol{p} + \boldsymbol{G}$ carry the same exciton band with different strengths, and a reduced $\boldsymbol{p}$ would merge them. The finite-momentum BSE above, in the resonant form for the proper lines with external screening from the RPA with local fields, supplies them on the 29 irreducible points of an 8^3 grid of $\boldsymbol{p}$ and their Umklapp copies to $|\boldsymbol{p}| \le 5$ keV, about 60 points, with the three lowest exciton bands per point, up to 20 lines and 50 reciprocal-lattice vectors. The

line strengths are interpolated in $\boldsymbol{p}$ and convolved with the lattice response of Appendix E.17. They must reproduce the optical exciton strength as $\boldsymbol{p} \to 0$, close the f-sum rule with the continuum to 1 % and approach the Wannier–Mott line weights in that limit, and change by less than 5 % when the $\boldsymbol{k}$ points per pocket go from 10^3 to 8×10^3 and when the bands are doubled. At about one node-hour per momentum the lines cost 60 to 100 node-hours, or 2 to 3×10^3 core-hours.

E.15.5 Self-trapped excitons

In an alkali halide such as NaI the exciton does not stay free. The hole localizes on a pair of anions, the lattice relaxes around it, and the energy of the self-trapped state differs from that of the free pair. Under linear coupling, a harmonic lattice and the Condon approximation, dark matter creates the exciton λ vertically, and its line is dressed by a Poisson series of phonons [124],

$$S_\lambda(\boldsymbol{q},\omega) = |\rho_{\lambda 0}(\boldsymbol{q})|^2 \sum_{n\geq 0} \mathrm{e}^{-S_{\mathrm{HR}}} \frac{S_{\mathrm{HR}}^n}{n!}\, 2\pi\delta(\omega - E_\lambda - n\bar{\omega}), \tag{E.91}$$

with the Huang–Rhys factor S_{HR} and a mean phonon energy $\bar{\omega}$; for $S_{\mathrm{HR}} \gg 1$ it is a Gaussian of mean $E_\lambda + S_{\mathrm{HR}}\bar{\omega}$ and variance $S_{\mathrm{HR}}\bar{\omega}^2$. The relaxation into the self-trapped state comes after the creation and belongs to the cascade of Sect. 18.5, not to the vertex.

The line strength, the line energy and the Huang–Rhys factor come from separate calculations. A finite-q BSE for NaI, with the Na 2p and I 4d semicore frozen and a 6^3 grid interpolated to 18^3, gives $\rho_{\lambda 0}(\boldsymbol{q})$ and E_λ. The relaxation energy comes from a constrained-occupation hybrid calculation of the self-trapped exciton (a hole on an anion pair plus an electron) in 64- to 216-atom supercells at the measured lattice constant, with exact-exchange fractions 0.25 and 0.40, Γ-point sampling checked on a 2^3 grid in the 64-atom cell, and the cutoff of the SG15 sodium and iodine potentials converged to 1 meV per atom; S_{HR} is that energy divided by $\bar{\omega}$, taken from the projected phonon density of states. A semilocal functional delocalizes the hole and must not be used, and the dependence on the exchange fraction is reported as the nuisance. Iodine's spin–orbit coupling, about 0.9 eV, splits the exciton and is included. The result is trusted when S_{HR} and $\bar{\omega}$ change by less than 10 % from 64 to 216 atoms (otherwise 512 are added), when constrained occupations and time-dependent DFT give relaxation energies within 20 % of each other, and when the computed luminescence energy of the self-trapped exciton matches the measured one to 0.2 eV. The hybrid relaxations of the 216-atom cell with iodine spin–orbit coupling dominate the cost, about 10^4 core-hours, and the time-dependent check doubles it; the factors enter Sect. 10.5.

E.15.6 Core levels with spin–orbit coupling

The semicore shells of germanium and gallium, 3d at about 30 eV below the valence maximum, and the silicon 2p shell contribute to $L(\boldsymbol{q},\omega)$ between 30 and 200 eV, and relativistic operators enter at order α^2 (Sect. 10.6). The core part of the loss function is wanted for $q \leq 30$ keV and $\omega \leq 200$ eV, resolved into the $3d_{5/2}$ and $3d_{3/2}$ components. An all-electron calculation with spin–orbit coupling in the zeroth-order regular approximation, in a full-potential code such as WIEN2k or exciting, or PAW with core reconstruction (Appendix E.2), followed by a core-level BSE at finite q on a 12^3 grid, provides it. No crystal code computes the four-component response with projection onto positive-energy states; the two-component calculation misses only terms of order $(Z_{\mathrm{eff}}\alpha)^4 \approx 10^{-6}$ for the germanium 3d shell and a projection error that is at most 2 % there, with a central estimate near 10^{-3} (Sect. 10.6). The binding energy of the 3d shell is computed with PBE, HSE06, G_0W_0 and total-energy differences, and their spread about the measured doublet, $3d_{5/2}$ at 29.2 eV and $3d_{3/2}$ at 29.8 eV with the centroid at 29.4 eV

(Sect. 10.6), is the nuisance. The decisive check is the mass–velocity shift of the f-sum rule of germanium, $-0.58\,\%$, the mass–velocity part of the total relativistic shift of $-0.76\,\%$ in Sect. 10.6: this crystal calculation and the atomic one of Appendix E.16 both compute it, and they have to agree to $0.1\,\%$. The step from 12^3 to 16^3 must change the shift by less than $0.05\,\%$ and $S(q,\omega)$ at 10 and 30 keV by less than $2\,\%$. By analogy with the valence BSE, each material costs about 10^3 core-hours.

E.15.7 Spin–orbit mixing at the band edges

Spin–orbit coupling mixes spin into the band edges and can lift the symmetry zeros of the density vertex. In silicon the on-axis zero is lifted to $|\rho|^2 \approx 3.6 \times 10^{-6}$ (summed over the four Γ_8 states) and the mirror-plane zero to 1.3×10^{-4} (Sect. 13.9), too small to move the window. For germanium the mixing near the L-valley edge is estimated at up to $10\,\%$ in amplitude, and it needs bands in the right order, which neither PBE with spin–orbit coupling nor HSE06 with it gives (Appendix E.14). The step therefore runs on germanium bands with spin–orbit coupling that pass the check of Appendix E.14 (valley order, both gaps, masses at L and Γ); quasiparticle GW bands are the next route under that check. On those bands a non-self-consistent run at the crystal momenta $(\boldsymbol{k}_v, \boldsymbol{k}_v + \boldsymbol{q})$ of the zero-phonon optimum gives the vertices from spinor wavefunctions matched by Miller index. Single spinor bands are gauge dependent, so only sums over Kramers pairs, or over degenerate quartets, are compared. Those sums are converged when they change by less than $2\,\%$ from 60 to 80 Ry and when the L and Γ energies of the underlying bands are stable to 10 meV in the number of bands and the screening cutoff. The silicon result, which must vanish as the spin–orbit strength is scaled to zero, is the control. Once bands that pass exist, the step costs less than a core-hour, and it replaces the estimate of Sect. 10.6.

E.16 Electronic response libraries

A spin-dependent coupling probes the spin density of the crystal, a velocity-dependent one its current, a scalar coupled to the electron's mass its kinetic-energy density, and each of these is screened differently from the charge density (Sects. 10.4 and 10.7). A rate for any of them needs a response function that most published calculations do not export: the full matrix of the independent-particle response, from which the screened responses to every operator follow by one Dyson equation. A library of these matrices, and the calculations that use it, follow the methods of Appendices E.6 and E.7.

E.16.1 The response library

For a crystal symmetric under the combination of inversion and time reversal, such as silicon or germanium, the response to the density $n_e(\boldsymbol{q})$ and the two transverse components of the current $j_{\mathrm{T}}(\boldsymbol{q})$ is fixed at each $(\boldsymbol{q},\omega)$ by 12 real functions (Sect. 10.4). Six belong to the screened singlet block: the density structure factor S_{nn}, two density–current cross terms, and three symmetric current–current terms. Six more give the same block for independent particles, which also fixes the independent-particle triplet block, a quarter of the singlet one. The library holds these 12 functions on a grid of $(|\boldsymbol{q}|, \hat{\boldsymbol{q}}, \omega)$ and, in addition, the full independent-particle matrix $R^0_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega)$, so that other kernels can be applied later without repeating the band calculation. For operator $\mathcal{O}_2$, whose velocity squared brings in the kinetic-energy density $\mathcal{T}_e$, the library adds four functions in each version, the kinetic rows below, so that it holds 20 functions in all (Sect. 21.6). On the truncation label $\mathfrak{T}$ (Sect. 8.5) it sits at $n_B = 0$, $n_g = 0$, with $\mathcal{X} = \mathrm{IP}$ and $\mathcal{X} = \mathrm{RPA}$ with local fields.

For silicon the ground state is the shared setting of Appendix E.17 (PBE, SG15 at 60 Ry, measured lattice constant) with a scissor to the measured gap, as in Appendix E.7. A scissor keeps the band order of PBE, which in germanium is wrong (Appendix E.1), so the germanium library is built only on germanium bands that pass the check of Appendix E.14 (valley order, both gaps, masses at L and Γ), with quasiparticle GW bands as the next route under that check; until such bands exist, no germanium version is released. Density vertices $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q}+\boldsymbol{G})$ come from matching Miller indices. The current vertex $\langle c|\frac{1}{2}\{\boldsymbol{p}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\}|v\rangle/m_e$ needs the correction for the nonlocal pseudopotential of Eq. (E.46); without it the transverse current of silicon is off by about 10 %. The Dyson equation of Eq. (E.39) is solved on the joint index set of the density at $\boldsymbol{q}+\boldsymbol{G}$, with $|\boldsymbol{q}+\boldsymbol{G}| \le 8\,\alpha m_e$, and the current, which enters only as an external row and column. The grids follow published crystal form-factor calculations: 10^3 $\boldsymbol{k}$ points without symmetry, $|\boldsymbol{q}| \le 25\,\alpha m_e$ in steps of $0.1\,\alpha m_e$, 30 directions in the irreducible wedge, $\omega \le 50\,\mathrm{eV}$ in steps of 0.05 eV, and 60 bands. The tables are converged when S_{nn} at 1 keV changes by less than 3 % from 8^3 to 12^3 $\boldsymbol{k}$ points and by less than 1 % when the local-field cutoff goes from 6 to $8\,\alpha m_e$. With the nonlocal term included, the f-sum rule of Eq. (E.50) has to hold to 2 % below the band cutoff, the tolerance at which Sect. 21.6 releases a version and on which the transfer rule of Worked example 22.1 relies; where the spectrum ends early, the fraction it holds is stored as a value per $|\boldsymbol{q}|$. Symmetry gives a check that costs nothing: in such a crystal the imaginary part of the density–current cross term and the antisymmetric part of the current–current block vanish, and numerically they stay below 10^{-3} of the real parts.

Storing the full matrix is impossible. At $|\boldsymbol{q}+\boldsymbol{G}| \le 8\,\alpha m_e$ ($15\,\text{Å}^{-1}$) it has about 2300 reciprocal-lattice vectors, 85 MB per $(\boldsymbol{q}, \omega)$, and on the full grid of 7.5×10^6 points it would fill some 600 TB. The 12 functions need only the screened column at $\boldsymbol{G} = 0$ and the current rows, so the Dyson equation is solved for those few right-hand sides by an iterative solver, about 10^9 operations per point, and never by inverting the matrix; the full $R^0_{\boldsymbol{G}\boldsymbol{G}'}$ is stored only on a coarse subset (25 values of $|\boldsymbol{q}|$, 3 directions, 100 frequencies) at a cutoff of $4\,\alpha m_e$, about 10 GB. Building the bubble then dominates, about 10^{16} operations, some 10^3 core-hours for silicon and four times that for germanium with spin–orbit coupling. The output is a set of versioned tables, which Sect. 21.6 uses as the reference electronic response.

The library also gives the screening of the current by local fields and the spin channel beyond independent particles. With the Dyson equation solved on the joint index set, the screened current–current response differs from its independent-particle value by a term built from the mixed blocks $R^0_{j_\mathrm{T},n(\boldsymbol{q}+\boldsymbol{G})}$, the density response to a transverse current. At $\boldsymbol{G} = 0$ this block vanishes when $\hat{\boldsymbol{q}}$ lies on a three- or fourfold axis, where a transverse current cannot drive a longitudinal density, so the screening of the current is purely a local-field effect, estimated at 10 to 30 %. Its value per $(|\boldsymbol{q}|, \omega)$ is what settles the in-medium factor for the velocity-dependent operator in Sect. 10.4. The spin channel has a Dyson equation that uses the matrix $R^0_{\boldsymbol{G}\boldsymbol{G}'}$ with a kernel made of the triplet exchange–correlation kernel and the screened electron–hole ladder, but without the exchange term that the charge channel carries. Since the exchange matrix element of a Wannier–Mott exciton is of relative order $(a/a_x)^3 \approx 10^{-3}$ in silicon, apart from its long-range part, which is the external screening and is absent from the spin channel by construction, the triplet ladder equals the singlet one to that accuracy. The ladder is that of the BSE run of Appendix E.15 with the exchange switched off, so it costs little on top of that run, and the limit without interaction, which must return a quarter of the independent-particle density response, controls it.

E.16.2 Anisotropy of the electronic response

A cubic crystal's response depends on the direction of $\boldsymbol{q}$, and that dependence is what turns the Earth's rotation into a daily modulation (Sect. 11.7). It is carried by the cubic-harmonic components $S_\ell(|\boldsymbol{q}|,\omega) = \int \mathrm{d}\hat{\boldsymbol{q}}\, Y_\ell^{\mathrm{c}}(\hat{\boldsymbol{q}})\, S(\boldsymbol{q},\omega)$ on the cubic harmonics Y_ℓ^{c} of Appendix C.3, for $\ell = 4$, 6 and 8, reported as S_ℓ/S_0. Odd ℓ vanish by time reversal and $\ell = 2$ by cubic symmetry. The loss function with local fields is computed at full $\boldsymbol{q}$ (GPAW or BerkeleyGW) on a Lebedev grid of directions and projected. Near the gap the response has structure on a scale of 0.1 rad: at $|\boldsymbol{q}| = 1.2$ to 1.48 keV the zero-phonon edge of silicon has its minima 0.13 to 0.22 rad from the valley axis in the {001} planes, 14 to 107 meV below the on-axis value, and shallower minima in the {011} planes (Sect. 13.9). A 110-point Lebedev rule integrates polynomials exactly only to degree 17, which is not enough once the response has components with $\ell \geq 10$. The grid is therefore refined from 110 to 230 to 434 points, exact to degrees 17, 25 and 35, and the test is that S_4, S_6 and S_8 change by less than 1 % between the last two. Near the edge the adaptive cone grid of Appendix E.14 is added. Time reversal ($\ell = 1, 3$) and cubic symmetry ($\ell = 2$) supply checks that cost nothing: those components stay below 10^{-3} of the isotropic part. Projecting a line that is sharp in ω before binning gives meaningless coefficients, so ω is binned first. The run shares its band calculation with the library and adds little cost.

E.16.3 Local fields in the phonon-assisted window

The rate estimate of silicon in Sect. 13.9 is normalized to the measured optical edge and carries a screening correction relative to that calibration: 0.3 to 2.6 for the external factor and the local-field wings, an estimate, and the pair channel of Sect. 13.4 beyond the calibration, which no estimate sizes. A model, a supercell calculation and a bound from zero-phonon vertices size the correction.

The model tests the method itself. A one-dimensional crystal with two cosine bands, each of width 2 and separated by a gap of 1 in model units, a contact Coulomb kernel v_0 for wavevectors below half the zone, a classical phonon field $u_0 \cos \omega_Q t$ at wavevector Q, and a probe at (q,ω) has a second-order induced density at the pair momentum $q+Q$ and frequency $\omega_f = \omega - \omega_Q$. It is computed twice: in the frequency domain, from the screened amplitude of Sect. 13.4, and in real time, by propagating the density matrix in the time-dependent Hartree approximation with both perturbations and Fourier transforming. The limits of three couplings are known: a rigid coupling, $g = -u\,\partial_x V$, gives a screening ratio $\mathfrak{r}_{\mathrm{scr}} = |\tilde{M}|^2/|M|^2$ of order ω_Q/ω; a coupling proportional to the density vertex gives $\tilde{M} = M/\varepsilon(q+Q,\omega_f)$ exactly; a generic intracell coupling gives $\mathfrak{r}_{\mathrm{scr}}$ of order $1 - 1/\varepsilon(q+Q)$. The model uses $\omega_Q = 0.05$, Q at a quarter of the zone, u_0 small enough that the response is linear in it (checked by halving), 2000 to 8000 k points, v_0 scanned so that $\varepsilon(q+Q)$ runs from 1.5 to 4, a fourth-order Runge–Kutta step of $0.01/E_{\mathrm{g}}$, a total time of at least $200/E_{\mathrm{g}}$ and an adiabatic switch-on. The two routes agree to 10^{-3}, stable under doubling the k points and halving the step. On-shell residues cannot be compared on a finite ring, so the comparison uses many k points and a damping extrapolated to zero. The model runs in minutes.

The silicon calculation uses a supercell. A supercell displaced along the special (zero-gradient) configuration of Appendix E.8 samples the phonon average deterministically, and the loss function of the displaced supercell, unfolded to the primitive momentum, contains the phonon-assisted response at the momenta the supercell resolves, and three runs of it separate the corrections. The supercell with RPA local fields in its own reciprocal-lattice basis gives the full result; the same supercell with independent particles and a scalar screening $1/\varepsilon(\boldsymbol{q},\omega)$ gives the reference, and the difference is $\delta_{\mathrm{LF}} + \delta_{\mathrm{pair}}$; a third run with local fields restricted to the primitive reciprocal lattice gives δ_{LF} alone. On $\mathfrak{T}$ this is $n_B = 0$, $n_g = 2$, $\mathcal{X}$ = RPA with local fields; the special displacements are the adiabatic quadrature, and the correction of order

ω_Q of Sect. 13.2 is applied afterwards. Cubic supercells of $n \times n \times n$ conventional cells resolve only certain momenta: $n = 3$ (216 atoms) reaches $|\boldsymbol{q}| = 0$, 0.76, 1.08, 1.32 and 1.52 keV, and $n = 4$ (512 atoms) reaches 0, 0.57, 0.81, 0.99 and 1.14 keV. The two meshes share no nonzero momentum below 2.28 keV, so a convergence test between them is impossible at fixed $|\boldsymbol{q}|$. The tests are instead the change from $n = 2$ (64 atoms) to $n = 4$ at their common 1.14 keV, below 5 %, and the agreement of $n = 3$ and $n = 4$ after interpolation in $|\boldsymbol{q}|$ between 0.76 and 1.52 keV, also within 5 %. At $n = 2$ the special configuration samples only a 2^3 phonon mesh, so the first test also measures phonon sampling; both antithetic configurations are reported. The ground state is PBE with a scissor; the screening uses Yambo or a Liouville–Lanczos solver, which scales better, with a local-field cutoff of 3 Ry and bands to 30 eV above the Fermi level, and Γ-only sampling checked against 2^3. The output is the screening ratio $\mathfrak{r}_{\mathrm{scr}}(\boldsymbol{q}) = 1 + \delta_{\mathrm{LF}} + \delta_{\mathrm{pair}}$, the full result over the reference, at the resolved momenta, and the test $\mathfrak{r}_{\mathrm{scr}}(\boldsymbol{q}) \approx \mathfrak{r}_{\mathrm{scr}}(0)$. The three cell sizes cost about 10^2, 2×10^3 and 2×10^4 core-hours, hours to a day on accelerators. A second pass with the adiabatic local-density kernel isolates its extra source term, estimated at a few percent of L^{PA} (Sect. 13.4).

The bound needs no amplitude. The external local-field correction alone can be bounded from the zero-phonon vertices at the same $(\boldsymbol{q}, \omega)$,

$$1 + \delta^{(0)}_{\mathrm{LF}}(\boldsymbol{q}, \omega) = \frac{\sum_{cv} \left| \sum_{\boldsymbol{G}} (\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q}, \omega)\, \rho_{cv}(\boldsymbol{q} + \boldsymbol{G}) \right|^2}{\sum_{cv} |\rho_{cv}(\boldsymbol{q})/\varepsilon_{\mathrm{mac}}|^2}, \qquad \varepsilon_{\mathrm{mac}} = 1/(\tilde{\varepsilon}^{-1})_{00}, \tag{E.92}$$

a ratio that is one without local fields. It needs the column $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q}, \omega)$ in the primitive cell for $\boldsymbol{q}$ on the library grid and ω from 1.2 to 1.6 eV, with 40 bands, a 12^3 grid and a cutoff of 4 Ry (about 37 reciprocal-lattice vectors), converged to 1 % between 3 and 5 Ry. It costs about 10 core-hours.

E.16.4 Loss functions of the candidate targets

The ranking of targets in Sect. 21.7 needs the loss function $L(\boldsymbol{k}, \omega) = \mathrm{Im}[-(\varepsilon^{-1})_{00}(\boldsymbol{k}, \omega)]$ of each candidate for $|\boldsymbol{k}|$ up to about 5 keV and ω from the gap to about five valence plasma energies, 75 to 150 eV. Its inverse fourth moment in ω sets the Migdal yield, and its weight near the gap sets the electron-recoil rate at low mass. The level is RPA with the full matrix $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}$, since L is not $\mathrm{Im}\, \varepsilon_{00}/|\varepsilon_{00}|^2$ once local fields matter, on a ground state with a scissor or quasiparticle gap matched to the measured one, with the adiabatic local-density kernel as the cross-check. A scissor serves only where PBE gets the band order right, checked against measured gaps at the high-symmetry points; elsewhere the bands must pass the check of order, gaps and masses of Appendix E.14, as for germanium. Semicore shells within five plasma energies of the gap are included: Ga 3d, Ge 3d, W 5p and 4f, Pb 5d, Te 4d, Cs 5p, I 4d, Mo 4p. A cubic crystal needs three directions, a uniaxial one the directions parallel and perpendicular to the axis and one oblique. Every directional average is over $\hat{\boldsymbol{k}}$, never one direction.

Since the inverse fourth moment weights the gap as E_{g}^{-5}, every grid test is run on that moment, not on L itself. The f-sum rule $\int \omega L \, \mathrm{d}\omega = (\pi/2)\omega_p^2$ holds to 2 % with the electrons counted (valence plus the semicore included), which also fixes the split between valence and core. The number of bands, the $\boldsymbol{k}$ grid and the local-field cutoff are each increased until $\int L/\omega^4 \, \mathrm{d}\omega$ changes by less than 1 %. The broadening is at most $E_{\mathrm{g}}/20$ and is extrapolated to zero, since the inverse fourth moment is weighted toward the gap. Missing semicore weight biases that moment low at a fixed f-sum, because the weight then sits at high ω. The scissor moves the onset, which the moment weights as E_{g}^{-5}, and the computed moment is compared with the bound $(\pi/2)\omega_p^2/E_{\mathrm{g}}^5$ that the f-sum rule gives. Excitonic weight below the gap, absent in the RPA, adds to the moment; the BSE of Appendix E.15 bounds it. Beyond the first zone, the element at $\boldsymbol{G} = 0$ must be taken at the unreduced $\boldsymbol{k}$ (Appendix E.6). We estimate

the cost from Worked example E.5 at tens to a hundred core-hours for a two-atom cell and several hundred for a ten-atom oxide.

E.16.5 Relativistic shift of the f-sum rule

The f-sum rule holds exactly only for a nonrelativistic Hamiltonian. At order α^2, the mass–velocity term reduces it, by 1.8×10^{-3} in silicon (Sect. 4.3), and so does the orbit–orbit (Breit) interaction between electron pairs, which Sect. 10.6 writes as an integral of the all-electron static structure factor over the Coulomb kernel with a transverse projector. The all-electron atoms of Sect. 10.6 put the total relativistic shift at small q at $-0.24\,\%$ for silicon and $-0.76\,\%$ for germanium, of which the mass–velocity part is $-0.18\,\%$ and $-0.58\,\%$; a Thomas–Fermi estimate gives -0.23 and $-0.73\,\%$. What is missing is the orbit–orbit shift as a function of q from 1 to 100 keV, and the total with the mass–velocity term. The cores dominate it, so the calculation is atomic: Hartree–Fock with the exact two-component (X2C) scalar-relativistic Hamiltonian in pyscf, in an uncontracted large basis (the atomic natural orbital (ANO) set for relativistic core correlation (RCC), ANO-RCC, or cc-pwCV5Z) with tight s and p exponents up to 10^7, gives the two-electron energy and, from the one-particle density matrix in momentum space, the static structure factor. A crystal correction for the valence comes from a PAW all-electron density with the electron-gas exchange–correlation hole. The two-electron energy is converged to 10^{-4} when the basis goes from quadruple to quintuple zeta, and the k integral to 10^{-3} with a radial grid out to $50/a_0$ for the germanium 1s shell; contracted valence basis sets would underestimate the core cusp by several percent. The orbit–orbit term is about 0.31 of the mass–velocity one in both atoms, so including only one misstates the total by about 24 %. The formula subtracts the uniform Hartree term itself, so it is never subtracted by hand, and for the valence, where q times the pair distance exceeds one, the full angular dependence is kept rather than its average. The atoms cost minutes, the crystal correction about 10 core-hours, and the result replaces the estimate in Sect. 10.6.

E.16.6 Static screening with lattice relaxation

A probe that couples only to electrons still moves the nuclei in a static measurement, because the electrons drag them. The static response of the electron channel with the lattice allowed to relax is the clamped-nuclei step plus a lattice term (Sect. 10.2),

$$1 - \varepsilon_e^{-1}(\boldsymbol{q},0) = [1 - \varepsilon_{\mathrm{cl}}^{-1}(\boldsymbol{q})] + \mathcal{G}_{\mathrm{rel}}(\boldsymbol{q}),$$
$$\mathcal{G}_{\mathrm{rel}}(\boldsymbol{q}) = \frac{e^2}{q^2 \Omega} \sum_{\nu} \frac{\left| \sum_{\kappa} \boldsymbol{q} \cdot Z_{\kappa}^{(e)}(\boldsymbol{q}) \cdot \boldsymbol{e}_{\kappa\nu}(\boldsymbol{q}_r)\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_{\kappa}} / \sqrt{M_{\kappa}} \right|^2}{\omega_{\nu}(\boldsymbol{q}_r)^2}, \tag{E.93}$$

at adiabatic, harmonic, one-phonon order. Here $Z_{\kappa}^{(e)}(\boldsymbol{q})$ is the electron charge that follows atom κ when it moves, which at $q \to 0$ in an insulator is $Z_{\kappa} - Z_{\kappa}^{*}$ with all electrons counted, 14 for silicon. As $q \to 0$ the acoustic branch dominates and $\mathcal{G}_{\mathrm{rel}} \to \Omega_{P,e}^2/(c_{\mathrm{LA}} q)^2$, with $\Omega_{P,e}^2 = e^2 n_e^2/\rho_T$ and c_{LA} the longitudinal sound speed: the step diverges as $1/q^2$ in every solid. For silicon $\Omega_{P,e} = 0.51\,\mathrm{eV}$, and the limiting form reaches one at 18 keV, above the zone boundary; it gives 3×10^2 at 1 keV, where neither the linear dispersion nor $Z^{(e)} \approx 14$ holds, so the true value there is only estimated at $\gtrsim 10^2$.

The calculation needs $Z_{\kappa}^{(e)}(\boldsymbol{q})$ at finite q. Frozen-phonon supercells commensurate with $\boldsymbol{q}$, for silicon $1 \times 1 \times N$ conventional cubic cells along [100] with $N = 2$ to 8, displace one sublattice by $u\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}}$; the Fourier component of the induced electron density at $\boldsymbol{q}$, at two amplitudes to check linearity, gives the charge. The DFPT induced density is equivalent. A pseudopotential calculation gives only the valence part; the frozen cores are added back

as rigid charges, valid for q well below the inverse core radius, about 30 keV for the silicon 1s shell. The phonons come from Appendix E.17, the clamped step from the loss functions above. $Z^{(e)}$ has to be stable to 1 % in cutoff and $\boldsymbol{k}$ grid and $\mathcal{G}_{\mathrm{rel}}$ to 2 % under refinement in q, and the small-q limit has to match c_{LA} from the computed elastic constants. A violated acoustic sum rule sends $\omega_\nu \to 0$ with a finite numerator and fakes a divergence, so the sum rule is enforced on the force constants. The momentum is kept unreduced, $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}$, so that the step follows the density to several zone momenta, with its Umklapp terms; above a Debye–Waller exponent $2W \approx 0.1$ multiphonon terms are added; and in GaAs the direction dependence of the sound speed and of the nonanalytic optical term must be kept. Silicon and GaAs together cost about 10^2 core-hours.

E.16.7 Kinetic-energy response

A scalar coupled to the electron's mass is absorbed through the kinetic-energy density $\mathcal{T}_e$, not the density (Sect. 10.7). For every crystal whose point group has no polar vector, the absorption rate is proportional to $-\operatorname{Im} R_{\mathcal{T}\mathcal{T}}(0, m_\phi)$ (Eq. (10.35)), whose independent-particle form is a joint density of states weighted by second-order momentum matrix elements; with $\mathcal{T}_e = \sum_j p_j^2/2m_e$ as in Sect. 8.1, the squared element carries $1/4m_e^2$,

$$\operatorname{Im} R^0_{\mathcal{T}\mathcal{T}}(0,\omega) = -\frac{2\pi}{\mathcal{V}} \sum_{cv\boldsymbol{k}} \frac{|\langle c\boldsymbol{k}|p^2|v\boldsymbol{k}\rangle|^2}{4m_e^2}\, \delta(\omega - \varepsilon_{c\boldsymbol{k}} + \varepsilon_{v\boldsymbol{k}}), \tag{E.94}$$

with spin summed. The matrix element is computed directly from the plane-wave coefficients,

$$\langle c\boldsymbol{k}|p^2|v\boldsymbol{k}\rangle = \sum_{\boldsymbol{G}} |\boldsymbol{k}+\boldsymbol{G}|^2\, c^*_{c\boldsymbol{k}}(\boldsymbol{G})\, c_{v\boldsymbol{k}}(\boldsymbol{G}),$$

plus the PAW one-center correction. It equals $\sum_m \boldsymbol{p}_{cm} \cdot \boldsymbol{p}_{mv}$ over all bands including the core, a sum that converges slowly. The kinetic matrix element converges like the total kinetic energy, so silicon needs at least 80 Ry with SG15; the grid needs at least 24^3 points, the delta function a broadening of at most 20 meV, and the gap a scissor. The f-sum rule is not a check here. Its analogue for $\mathcal{T}_e$ fixes the energy-weighted integral by $\langle|\nabla V|^2\rangle/2m_e$ per electron (Sect. 10.7), which the core region dominates, and the check uses that sum rule restricted to the valence. For the same reason, norm-conserving pseudo-wavefunctions, which lack the core oscillations, are compared with a PAW or all-electron calculation. $\mathcal{T}_e$ is the canonical kinetic operator; velocity matrix elements $\mathrm{i}[H, \boldsymbol{r}]$ are not a substitute. Since $\mathcal{T}_e$ is even under inversion, the direct edge of silicon at Γ is suppressed, and absorption inside the gap needs the phonon-assisted row of Appendix E.14. Polar crystals add a mixing term through the current response. Local fields, the next many-body approximation on $\mathfrak{T}$, are added at RPA level. Silicon takes minutes; a ten-atom oxide about one accelerator-hour.

Operator $\mathcal{O}_2$ needs the same row at finite momentum. Its four functions per version, $\mathcal{T}_e(\boldsymbol{q})$ with itself and with the density and the two transverse currents, enter the Dyson solve of the library as external rows and columns, like the current, so that they are screened by the same density block (Sect. 21.6). The vertex $\langle c, \boldsymbol{k}+\boldsymbol{q}|\mathcal{T}_e(\boldsymbol{q})|v\boldsymbol{k}\rangle$ is built from the plane-wave coefficients with the matrix element $(\boldsymbol{k}+\boldsymbol{k}')^2/8m_e$ between plane waves of Eq. (8.3), plus the PAW one-center correction, on the grid of the library, and as $\boldsymbol{q} \to 0$ it must return Eq. (E.94).

E.16.8 Cosmic-ray-boosted dark matter

For dark matter boosted to relativistic speeds, the density row of the response is contracted with the full current. The published treatment of cosmic-ray-boosted dark matter scattering in silicon and germanium [125] drops the factor $(1-\omega^2/\mathcal{Q}^2)^2$ of the exact density row, with $\mathcal{Q}$

the magnitude of the momentum transfer, and the transverse row (Sect. 10.9). Since $\omega \le v_\chi \mathcal{Q}$, the rate changes by at most 2 % for $v_\chi \lesssim 0.1$, but for a cosmic-ray-boosted flux with $v_\chi \to 1$ the bound is empty, and a transverse row, $(v_\chi^2 - \omega^2/\mathcal{Q}^2)\ \mathrm{Im}[-\mathcal{Q}^2/(\omega^2\varepsilon^{\mathrm{T}} - \mathcal{Q}^2)]$ per unit of the density normalization, adds to it. The recast re-implements the published rate with the exact density row and the transverse term: the source's random-phase longitudinal function, a transverse function equal to it as $\mathcal{Q} \to 0$ and of Lindhard form otherwise, the source's flux tables, and a three-dimensional quadrature in $\mathcal{Q}$, ω and the flux angle. The first step is to reproduce the published limit with the published rate to 5 %. The transverse term has a Cherenkov-type pole where $\omega^2 \,\mathrm{Re}\,\varepsilon^{\mathrm{T}} = \mathcal{Q}^2$, which needs a scan in the imaginary part of the frequency, and the mediator form factor is evaluated at $\mathcal{Q}^2 - \omega^2$, not at $\mathcal{Q}^2$. For a heavy mediator the exact density row changes the rate by 10^{-4} and the transverse row adds at most 10^{-4}, so that limit is unchanged; since the boosted flux is itself proportional to $\bar{\sigma}_e$, a change R of the detection rate moves a limit by $R^{-1/2}$ (Sect. 21.5). For a massless mediator the shift of $\bar{\sigma}_e$ is below one percent for $m_\chi \ge 10\,\mathrm{keV}$, and below about 3 keV it is undetermined: the published density row has a pole at $\mathcal{Q} = \omega$, whose weight depends on the treatment of the lowest kinetic energies of the flux, which the source does not state. Depending on that treatment the ratio of the density rows ranges from 0.05 to 0.93 at 1 keV, so the recast must state the treatment it adopts, and its light-mediator result stands only with that statement. The recast costs a few core-hours.

E.17 Lattice libraries

Every lattice channel of Sects. 11.1 to 11.4 uses the same few objects for a given crystal: force constants, the frequencies and polarization vectors they give, the per-atom phonon density of states, the displacement tensors B_κ, and, in a polar crystal, the Born charges and ε_∞. For silicon, germanium and gallium arsenide these come from one DFPT calculation each, and the multiphonon tables of Sect. 11.3 follow from them in minutes. For the candidate targets named in Sect. 17.5 most of the same numbers are not tabulated, and some of the tables need more than harmonic DFPT: all-electron densities at large momentum, path-integral momentum distributions, isotope-resolved Hessians and frequency-dependent Born charges. The DFPT method is that of Appendix E.4, the anharmonic and path-integral methods those of Appendix E.8.

E.17.1 The shared harmonic base

One harmonic lattice per material serves every lattice channel, and a mismatch between channels computed from different lattices is an error that no single channel reveals. The base consists of the force constants $\Phi_{\kappa\kappa'}(\Delta\boldsymbol{R})$, the eigenpairs $(\omega_{\boldsymbol{Q}\nu}, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}))$, the per-atom density of states $D_\kappa(\omega)$, the tensors B_κ and their off-site generalization $B_{\kappa\kappa'}(\Delta\boldsymbol{R})$ (Eq. (11.28)), and for GaAs Z^*_κ and ε_∞. On the truncation label $\mathfrak{T}$ (Sect. 8.5) it sits at $n_B = 0$ and $n_A = 0$, static DFPT on the harmonic Born–Oppenheimer surface. The setting is the one of the representative calculations of Appendix E.1 and Sect. 5.2, shared by every specification of this appendix: PBE, SG15 norm-conserving pseudopotentials at 60 Ry, and the measured room-temperature lattice constants of Table E.1, 5.431 Å (Si), 5.658 Å (Ge) and 5.653 Å (GaAs), with the GaAs frequencies also reported at the static-lattice constant 5.638 Å, at which Sect. 5.5 quotes them. The germanium and gallium pseudopotentials carry the 3d shell, which converges the total energy more slowly (Appendix E.1), so for those two the frequencies are checked once at 80 Ry. Silicon uses a 12^3 $\boldsymbol{k}$ grid and an 8^3 wavevector grid (29 irreducible points), which the phonon-assisted window needs (Appendix E.14); germanium and GaAs use 8^3 and 6^3. DFPT runs to a self-consistency threshold of 10^{-16}, with the

electric-field perturbation for GaAs, the potential changes kept on disk for the vertices, and Fourier interpolation with the acoustic sum rule imposed on the force constants. GaAs needs the nonanalytic term of Eq. (E.24), without which its longitudinal-optical branch is wrong at small $\boldsymbol{Q}$.

One low-symmetry wavevector of silicon at 40 Ry with an 8^3 $\boldsymbol{k}$ grid takes about eight minutes on four cores; scaled by $(60/40)^{3/2}$ for the cutoff and $(12/8)^3$ for the grid, the 29 wavevectors take about 24 h on four cores, roughly 100 core-hours, in less than 2 GB. Germanium takes 13 to 16 minutes per wavevector on four cores at its smaller grids, about 15 core-hours, and GaAs with the electric field about 20.

The frequencies change by less than 0.1 meV between the two coarsest wavevector grids, and they are compared with neutron data at Γ, X and L (silicon [45], germanium [123]) with a tolerance per branch, since the error of PBE at the measured lattice constant differs from branch to branch. The optical and longitudinal-acoustic branches pass at 3 %. The transverse-acoustic branches of silicon are 8 % soft at X and L in this setting (Sect. 5.2), and pass at 10 %; those of germanium are 11.6 to 13.5 % soft on the 4^3 grid (Appendix E.14) and pass at 15 % until the test of Appendix E.14 has separated the grid error from the functional. Where a transverse-acoustic frequency sets an edge, the neutron-scaled branch is the reference and the unscaled one is reported beside it (Appendix E.14). The phase convention matters for every coherent quantity downstream: the eigenvector carries the cell phase $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$, and the intracell phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}$ is taken at the unreduced $\boldsymbol{q}$ (Appendix A.3). The phonons of Appendix E.14 are this base, and so is the lattice input of the reference library of Sect. 21.6.

The deformation-potential tensors D^n of Eq. (6.13) belong with the base, since they set the soft acoustic limit of the phonon-assisted amplitude (Sect. 13.6 and Appendix D.37). They are computed by frozen strain: each of the six strain components is applied at ±0.5 % to the two-atom cell with the internal coordinate relaxed, and the band energies at the conduction minimum and at Γ are referenced to the average electrostatic potential. Only differences between the conduction and valence tensors enter at the band extrema, so the absolute alignment from a slab calculation is needed only for single-band checks against published deformation potentials [126]. At the extrema the group velocity vanishes, and the comoving intraband vertex of Eq. (D.211) reduces to the symmetric tensor that frozen strain computes; away from them it carries the Galilean term $\boldsymbol{v}_n^{\mathrm{g}} \otimes \boldsymbol{p}_{nn}$, which frozen strain does not produce and which is added from the band velocities and momentum matrix elements. Valley degeneracy is lifted under shear strain, and the bands must be followed by character, not by index. The run takes minutes, and its size decides which terms dominate the soft limit: with deformation potentials of order 9 eV and a longitudinal sound speed of 8.43 km s^{-1}, the ratio D/c_ν is 300 to 350 keV (Sect. 13.6), far above the momentum transfers of a few keV, so the deformation terms dominate the soft limit by two orders of magnitude.

E.17.2 Multiphonon tables and coherence

At momentum transfers far above the zone boundary a nuclear recoil excites many phonons, and in the incoherent approximation of Sect. 11.3 the response of each atom follows from its own density of states. On a cubic site the directional density of states is a third of $D_\kappa(\omega)$, so the per-atom density of states is the complete input. The library holds $S(\boldsymbol{q}, \hat{\boldsymbol{q}}, \omega)$ per cell, in eV^{-1}; the per-volume S of Sect. 4.2, in eV^2 in natural units, is this divided by the cell volume Ω. The library also holds the probability $P_{\mathrm{ph}}(n|\boldsymbol{q}, \omega)$ that n phonons were created, the branch content per atom, the coherent fraction and the coherence correction below, on a grid of 400 logarithmic points in q from 0.1 to 300 keV, ω from 0 to 1 eV in steps of 0.1 meV, and temperatures of 10 mK, 1 K and 77 K. Deep in the impulse regime, for $x_R > 16$, the table

switches to the impulse Gaussian with the third- and fourth-cumulant corrections whose inputs come from the path-integral calculation below. The level is $\mathcal{K}$ = multiphonon, incoherent, with the coherent one-phonon term added and $n_A = 0$.

At zero temperature the multiphonon expansion is a forward recursion in n; at finite temperature the time-domain form $\mathrm{e}^{-2W_\kappa}[\mathrm{e}^{\mathcal{C}_{\kappa\kappa}(t)} - 1]$ of Sect. 5.6 is Fourier transformed with a Gaussian regulator in time. The densities of states come from 48^3 grids with tetrahedra. The code is about 300 lines on top of the base and runs in minutes. Its validation is an Einstein site, whose n-phonon weights are known in closed form at any temperature and are reproduced to 10^{-5}. The convergence tests are the f-sum rule of the lattice, $\int \omega S \,\mathrm{d}\omega$ fixed by the recoil energy, to 10^{-4} at every q, and a time window $t_{\max} \geq 20\pi/\delta\omega$ for the bin width $\delta\omega$. The exponent $2W_\kappa$ uses B_κ with the thermal factor $2n_{\boldsymbol{Q}\nu} + 1$; the elastic Bragg peak is removed analytically; and GaAs needs a separate W_κ for each species. When the Migdal channel is computed separately, the lattice response is multiplied by $1 - P_{\mathrm{Mig}}$ and uses the dressed mass, as Appendix E.18 requires.

Below the zone boundary the incoherent approximation fails pointwise for two-phonon emission, because the two phonons from neighboring atoms interfere. The coherent two-phonon response is a Brillouin-zone sum over one phonon wavevector, the second fixed by momentum conservation, and its ω integral has an exact real-space form. With the static correlation tensor $B_{\kappa\kappa'}(\Delta\boldsymbol{R})$ of Eq. (11.28), whose on-site value is B_κ, the n-phonon weight is the lattice sum of Eq. (11.29),

$$\mathfrak{W}^{(n)}(\boldsymbol{q}) = \int \frac{\mathrm{d}\omega}{2\pi} S^{(n)}(\boldsymbol{q},\omega) = \frac{1}{n!} \sum_{\Delta\boldsymbol{R},\kappa,\kappa'} c_\kappa^* c_{\kappa'}\, \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\Delta\boldsymbol{R}_{\kappa\kappa'}}\, \mathrm{e}^{-W_\kappa - W_{\kappa'}}\, [\boldsymbol{q}\cdot B_{\kappa\kappa'}(\Delta\boldsymbol{R})\cdot\boldsymbol{q}]^n, \quad \text{(E.95)}$$

a dimensionless weight per cell, with $\Delta\boldsymbol{R}_{\kappa\kappa'} = \Delta\boldsymbol{R} + \boldsymbol{\tau}_\kappa - \boldsymbol{\tau}_{\kappa'}$ and c_κ the coupling of species κ (Sect. 11.3), with the per-cell S of the library in place of the per-volume S_A. The on-site terms are the incoherent weight, and the ratio $\delta_n^{\mathrm{coh}}(q) = \mathfrak{W}^{(n)}/\mathfrak{W}^{(n)}_{\mathrm{inc}} - 1$ is the exact coherence correction for bins that cover the whole n-phonon band. At zero temperature in three dimensions $B_{\kappa\kappa'}$ falls as ΔR^{-2}, so the sum converges absolutely for $n \geq 2$. The lattice sum Eq. (E.95) is not used at finite temperature, where the thermal factor spoils the convergence, nor at $n = 1$; at 1 K and 77 K the coherence correction comes from the pointwise zone sum below, which converges at any temperature. The pointwise response $S^{(2)}_{\mathrm{coh}}(\boldsymbol{q},\omega)$ is computed for silicon, then germanium and GaAs, on 40 logarithmic values of $|\boldsymbol{q}|$ from 0.2 to 10 keV, the 20 or so irreducible directions of a 590-point Lebedev grid, and bins of 0.5 meV, with $B_{\kappa\kappa'}$ to 40 Å by FFT from 48^3 eigenpairs and the zone sum on 48^3 with a Gaussian of 0.25 meV. The tests are a change below 2 % per bin from 32^3 through 48^3 to 64^3 and below 10^{-3} in δ_n^{coh} under the real-space radius. The acoustic point $\boldsymbol{Q} = 0$ has an integrable singularity that a mesh samples badly: it is excluded and a small sphere around it is added analytically. The whole run costs a few core-hours after the base.

The same tables, projected on cubic harmonics, give the anisotropy of the lattice response for Sect. 11.7. In the harmonic incoherent part a cubic site is isotropic at every phonon order, so the components with $\ell \geq 4$ vanish there to numerical noise, below 10^{-6} of the isotropic part; a larger value is a symmetry-breaking artifact of the mesh. The anisotropy comes only from the coherent terms, chiefly the one-phonon term. At long wavelength its $\ell = 4$ component in silicon is $S_4/S_0 = 0.028$ with the measured elastic constants and 0.025 with the PBE force constants at $q \leq 0.3\, q_{\mathrm{BZ}}$, and $|S_6/S_0|$ is below 10^{-3} (Sect. 11.7); the components are tabulated as S_ℓ/S_0 on the cubic harmonics Y^{c}_ℓ. The projection uses the 590-point Lebedev grid, exact to degree 41, with $\ell \leq 12$ for cubic crystals and $\ell \leq 8$ for sapphire and SiC, and ω bins wider than the change of the dispersion between neighboring directions.

E.17.3 The charge vertex at large momentum

Two corrections to the lattice vertex need all-electron densities: the valence charge deformation and the finite-q contact density. The first is the valence charge deformation of Sect. 11.3: whether the electron cloud follows a displaced nucleus at q far above the zone boundary. Split into a rigid neutral atom and a deformation, the charge vertex of atom κ is $q_\alpha[Z_\kappa - f_\kappa^{\mathrm{at}}(q)] - \Delta_{\kappa\alpha}(\boldsymbol{q})$, with f_κ^{at} the all-electron atomic form factor, and the deformations obey an exact sum rule: summed over the atoms of the cell at $\boldsymbol{Q} = 0$, the density changes give G_α times the Fourier component of the electron density at $\boldsymbol{G}$, the response to a uniform translation. The deformation $\Delta_{\kappa\alpha}(\boldsymbol{Q}+\boldsymbol{G})$ is wanted for Si, Ge and GaAs at $|\boldsymbol{Q}+\boldsymbol{G}| \le 20\,\mathrm{keV}$, about 725 reciprocal-lattice vectors per $\boldsymbol{Q}$ in silicon and 821 in germanium and GaAs. Up to about 10 keV it is the DFPT density change of the base, Fourier transformed per $\boldsymbol{Q}$, minus $q f_\kappa^{\mathrm{at}}$. At all q it is the symmetric difference of all-electron densities with one atom displaced by ± 0.01 Å in a 216-atom cell, from PAW with all-electron reconstruction or from a full-potential code. The PAW density on the FFT grid aliases its core-region components above about 10 keV, so those components are taken as the smooth-part FFT plus one-center radial transforms per atom. The density cutoff must reach 1.5 times the largest q, at least 65 Ry, or what the PAW dataset needs if that is more. The result must satisfy the sum rule to 10^{-3} and be linear in $\pm u$. A frozen-core calculation gives no core polarization at all; that part, the polarization of the core by the residual field at the moving nucleus, comes from a Sternheimer (Dalgarno–Lewis) calculation on the all-electron core orbitals, checked by the sum rule that its dipole moment equals the core polarizability. The DFPT part costs nothing extra, the supercells about 10^2 core-hours, the cores minutes.

The second correction is the finite-q factor of the electron density at the nucleus (Sect. 12.5), which enters the cross term between nuclear and electronic recoil. Its spherical atomic part is known from all-electron atoms; the crystal anisotropy and the relativistic enhancement are not. The crystal valence adds an anisotropy of tetrahedral symmetry, estimated at 1.3×10^{-3} of the factor in silicon; it is computed from the valence all-electron density (PAW minus the atomic core, or a full-potential code) to $|\boldsymbol{G}| \le 40/a_0$ on a Lebedev grid, converged to 10^{-5} of the density at the nucleus under the cutoff and the augmentation radius. The relativistic density averaged over the finite nucleus raises the contact density by the factor $\mathcal{R}_{\mathrm{rel}}$ of Eq. (12.47), 8 % in silicon and 40 to 45 % in Ge, Ga and As. It comes from a finite-nucleus Dirac calculation of the atom with a Fermi or uniform nuclear charge of the measured radius, with the innermost radial point below a hundredth of the nuclear radius; the choice of nuclear model changes it by less than 1 %. A relativistic point-nucleus density diverges at the origin, and its value at the first grid point is an artifact of the grid. Norm-conserving pseudo-densities are unusable here. The crystal part costs about 10 core-hours, each atom minutes.

E.17.4 Nuclear momentum distributions

The width and skewness of the impulse peak of Sect. 11.4 are set by the quantum motion of the nuclei, which harmonic theory gives only to leading order. Path-integral molecular dynamics with i-PI [127], driving a machine-learned potential (Appendix E.11) trained on PBE or SCAN (strongly constrained and appropriately normed) forces in 216-atom cells, gives the momentum distribution of each species to all orders in the anharmonicity: the centroid-virial estimator yields the kinetic energy along $\hat{\boldsymbol{q}}$, the open-path estimator the momentum distribution and the fourth moment, and the analytic Hessian averaged over beads the mean curvature of the Born–Oppenheimer surface, which is the third cumulant of the impulse peak. At 10 K silicon would need about 250 beads, from $\omega_{\max}/(k_B T P) \lesssim 0.3$, so we run at 50 to 100 K with a colored-noise thermostat, which brings the bead number to 16 to 32, and correct to zero temperature with the harmonic difference from the base. The kinetic energy anchors the

anharmonic template of Appendix E.19, and the anharmonic change of the kinetic energy of silicon is at most $\pm 0.2\,\%$ (Sect. 11.5); its error, statistical plus that of the harmonic-difference correction, must therefore be at most $0.2\,\%$. The potential is validated against DFT on displaced configurations out to three standard deviations of the sampled distribution, since extrapolation beyond the training set biases the fourth cumulant. The open-path estimator of the fourth moment is noisy and needs at least 10^6 steps, and the whole calculation costs about 10^3 core-hours per material.

E.17.5 Isotope-resolved Hessians

A vector coupled to baryon minus lepton number sees the isotope disorder of the lattice (Sect. 11.6), and its absorption rate $\Gamma_{\mathrm{iso}}(\omega)$ is computed exactly by diagonalizing the Hessian of a disordered supercell. For natural silicon the supercell has 4096 atoms ($8 \times 8 \times 8$ conventional cells, 12,288 degrees of freedom, about ten minutes per dense diagonalization), with the force constants of the base and ten random isotope configurations; the vertex is the mass-subtracted one, with the mean taken mass-weighted, otherwise a spurious acoustic term appears. The check is the exact single-site (Lifshitz) Green function, which gives the leading correlation correction between vertex and eigenvector; with the PBE density of states it is $+8.0\,\%$ at $50\,\mathrm{meV}$ (Sect. 11.6). The rate must change by less than $2\,\%$ from 10 to 20 configurations and from 4096 to 13,824 atoms, the larger cell by the kernel polynomial method; the whole calculation costs about 10 core-hours.

The competing electromagnetic absorption at the same frequency is two-phonon absorption, and the bound on the ratio in Sect. 11.6 uses room-temperature data where the millikelvin spectrum is needed. The zero-temperature two-phonon $\varepsilon_2(\omega)$ of silicon follows from the second-order dipole moments $\partial Z^*/\partial u$, by the first-principles route of Deinzer and Strauch [128]. The derivatives come from finite differences of DFPT Born charges in displaced 64- and 216-atom cells, with displacements of ± 0.005, ± 0.01 and $\pm 0.02\,\text{Å}$ and linearity to $2\,\%$ (in silicon $Z^* = 0$ at equilibrium, so the first order is linear in u), at $80\,\mathrm{Ry}$, with 4^3 and 2^3 $\boldsymbol{k}$ grids checked against 6^3 and 3^3. The two-phonon sum uses only the sum processes, which survive at zero temperature, on 32^3 and 48^3 grids with Gaussians of 0.25 to $1\,\mathrm{meV}$ extrapolated to zero. The test comes first: the same calculation with room-temperature occupations, sum and difference processes, and quasi-harmonic frequencies reproduces the measured room-temperature absorption to $20\,\%$ between 40 and $60\,\mathrm{meV}$, and only then is the zero-temperature curve used. The two cells cost about 10^2 and 2×10^3 core-hours.

E.17.6 Dynamical Born charges

A recoiling ion in a polar crystal radiates through its Born charge at the photon frequency, not at zero frequency (Sect. 12.6). What is needed is $Z^*_\kappa(\omega)$ at zero macroscopic field for ω from $0.05\,\mathrm{eV}$ to the gap, with the refractive index. At small ω, $Z^*_\kappa(\omega) = Z^*_\kappa(0) - \omega^2 C_\kappa + O(\omega^4)$, and a Cauchy–Schwarz bound on the same spectral sums gives for GaAs $|Z^*(\omega) - Z^*(0)| \le 0.9\,(\omega/1\,\mathrm{eV})^2$ against $Z^*(0) \approx 2.1$: at photon energies near $1\,\mathrm{eV}$ the dispersion is not negligible a priori and must be computed. Frequency-dependent DFPT computes it as a mixed response to an electric field and an atomic displacement, solved by shifted Sternheimer equations (Eq. (E.35)), at the RPA and then with the adiabatic local-density kernel. The PBE gap would place the resonance of $Z^*(\omega)$ far below the true edge, so the bands carry a scissor to the measured gap, and the response is taken at zero macroscopic field, since with the field the photon line is counted twice. Imaginary frequencies give C_κ by fits in ω^2; real frequencies near the gap use a damping of 10 to $20\,\mathrm{meV}$, extrapolated to zero. $Z^*(0)$ and ε_∞ must agree with static DFPT to $1\,\%$, and $\sum_\kappa Z^*_\kappa(\omega)$ must tend to 0 as $\omega \to 0$ and to $\sum_\kappa Z_\kappa$ as $\omega \to \infty$. Each frequency

Table E.9: Cutoffs and costs of the zone-center and phonon-grid DFPT runs for the candidates beyond the shared base, with SG15 norm-conserving pseudopotentials. Costs are in core-hours; on one accelerator each small-cell run takes minutes to an hour. The last four rows are batches.

material	cutoff (Ry)	cell	cost (core-hours)
diamond, 3C-SiC, LiF, NaF	90–100	2 atoms	2–20
4H-SiC	90	8 atoms	40–80
Al_2O_3	100	10 atoms	120–240
$CaWO_4$, Li_2MoO_4	W, Mo semicore	primitive cells	1.2×10^3–5.4×10^3
magnetic, DFT $+U$ (YIG dominant)	per material	4–80 atoms	1.1×10^4–10^5
23 further candidates	per material	small cells	10^3–5×10^3
thermal contraction, five volumes	as above	all candidates	5×10^3–2.5×10^4

batch costs about 50 core-hours for GaAs.

For a scalar coupled to electrons, the single-phonon rate of GaAs is a near-cancellation (Sect. 11.8). The vertex enters through $d = v_{\mathrm{Ga}}/M_{\mathrm{Ga}} - v_{\mathrm{As}}/M_{\mathrm{As}}$ with $v_\kappa = Z_\kappa - Z^*_\kappa/\varepsilon_\infty$, and since $Z_{\mathrm{Ga}}/M_{\mathrm{Ga}} = 0.4446$ and $Z_{\mathrm{As}}/M_{\mathrm{As}} = 0.4405$ (in units of e per atomic mass unit) nearly coincide, d is small: -1.21×10^{-3} per u for $Z^* = 2.11$ and $\varepsilon_\infty = 10.88$ (Sect. 11.8). A 10 % change in Z^*/ε_∞ changes d by 44 %, so a rate good to 20 % needs Z^*/ε_∞ to 2.3 %. PBE-DFPT cannot deliver this, since its ε_∞ for GaAs is tens of percent too large. The combination is taken instead from measured quantities through the Lyddane–Sachs–Teller relation [129],

$$\frac{Z^*}{\varepsilon_\infty} = \Big[\frac{\Omega\,\mu_{\mathrm{red}}(\omega_{\mathrm{L}}^2 - \omega_{\mathrm{T}}^2)}{e^2\varepsilon_\infty}\Big]^{1/2}, \tag{E.96}$$

in Heaviside–Lorentz units ($e^2 = 4\pi\alpha$), with μ_{red} the reduced mass of the pair and ω_{L}, ω_{T} the zone-center longitudinal and transverse optical frequencies, all at low temperature, since ε_∞ changes by 2 to 3 % between room temperature and 4 K. With ε_∞ to 1 % and the Raman frequencies to 0.1 % the requirement is met; DFPT supplies the sign of Z^*_{Ga} and a consistency check. The phonon widths come from three-phonon calculations (below) to 10 %. Rigid-ion charges change d by a factor of three and must not be used, and the gallium isotopes add a broadband channel of the isotope type above.

E.17.7 Data for the candidate targets

The comparison of Sect. 17.5 and the target table of Sect. 21.7 need, for each candidate, ε_∞ and ε_0, the Born-charge tensors, the zone-center transverse and longitudinal optical frequencies, the sound speeds along the principal directions, $\omega_{\max}$, the phonon density of states, the elastic tensor, phonon lifetimes and the thermal contraction to millikelvin. Where a measured value exists, it is used; DFPT fills the rest. Each candidate gets a relaxation, a zone-center DFPT run with the electric field (ε_∞, Z^*, the optical modes and, through the mode sum, ε_0), and 4^3 and 6^3 wavevector grids for the slopes, sound speeds, $\omega_{\max}$ and density of states. Table E.9 lists the cutoffs, set where the total energy converges to 1 meV per atom and the pressure to 1 kbar.

PBE overestimates ε_∞ systematically, by 11 % in silicon (Appendix E.4), so measured dielectric constants are used for ranking and the computed ones are diagnostics. The Lyddane–Sachs–Teller relation, $\varepsilon_0/\varepsilon_\infty = \prod_j \omega_{\mathrm{L},j}^2/\omega_{\mathrm{T},j}^2$, holds per principal axis and checks each run; a non-cubic crystal has a pair of dielectric constants per axis. In soft-mode crystals ($SrTiO_3$, the lead chalcogenides) the DFPT ε_0 is unusable, and the measured low-temperature value replaces it; an infrared mode below 20 cm^{-1} flags such a case. Heavy elements (Pb, Bi, Te, W)

need spin–orbit coupling at least as a check. The scintillating crystals $CaWO_4$ and Li_2MoO_4 are treated like the oxides, with the tungsten or molybdenum semicore in the valence and a measured elastic tensor where one exists. For the magnetic candidates (YIG, NiO, MnO, Cr_2O_3, $NiPS_3$) DFPT runs with a Hubbard U from linear response [76] on the magnetic ground state of Appendix E.9, with antiferromagnetic cell doubling for NiO and MnO; the dependence of ε and Z^* on U is reported as the nuisance, and measured moments and infrared modes are the checks.

Elastic tensors come from strain–stress calculations or DFPT. Each is positive definite and in the Voigt order; at least one public database lists the shear components in a different order, a trap when tables are merged. For a cubic crystal the longitudinal sound speed in any direction is at least $\sqrt{B/\rho_{\mathrm{T}}}$ (Sect. 11.1), which tests both computed and quoted values. Where a ranking turns on a computed tensor, as for TeO_2 against the measured $CaWO_4$ in Sect. 21.7, the measured tensor is the requirement and the offset of computed tensors, 4.9 % on average for silicon, germanium and diamond (Sect. 17.6.4), is the nuisance until it exists. CsI, whose sound speeds set its nuclear-recoil edge, enters with its measured elastic constants at 4.2 K, $C_{11} = 27.37$, $C_{12} = 7.93$ and $C_{44} = 8.25$ GPa; what it lacks is the density of the cold cell, which the thermal contraction below supplies.

Phonon lifetimes come from three-phonon perturbation theory, on-shell, which is $n_A = 2$ on the truncation label: third-order force constants from phono3py [130] on $3 \times 3 \times 3$ supercells, or from a machine-learned potential, converged in supercell size and interaction cutoff. The anharmonic decay constant of silicon [107] is the check; for germanium the published values spread by a factor of ten. The cost is 10^3 to 10^4 core-hours per material, and the lifetimes enter Sect. 11.8. The thermal contraction integrated from 300 K to millikelvin, which stores elastic energy between a crystal and its holder, comes from the quasi-harmonic approximation (Eq. (E.51)) with DFPT at about five volumes, with the PBE and PBEsol spread as the nuisance and measured expansion coefficients (Si, Ge, sapphire) as checks. The quasi-harmonic approximation fails near soft modes, and silicon expands on cooling below about 120 K, so the sign of the integrand must be checked.

E.18 Migdal and nonadiabatic quantities

The drag mass of silicon has two values that must not be confused. The trace value, $14\,m_e$ or $\delta M/M = 2.73 \times 10^{-4}$ for natural silicon ($M = 26.16$ GeV, Sect. 5.3), is the sum of the drag-mass tensor over all nuclei and counts every electron; it is exact, but a kick does not measure it. A kick measures the on-site element, which the tables of Sect. 12.5 put at $28.7\,m_e$, $\delta M_{II}/M = 5.60 \times 10^{-4}$: $10.1\,m_e$ from the cores and $18.6\,m_e$ from the valence, with the band 12 to $24\,m_e$ for the valence and the sign of the off-site coupling not fixed. The other nonadiabatic inputs of Sects. 12.3, 12.5 and 12.6 are likewise estimates from tables or models. The crystal Migdal rate is written there as a convolution of an electronic and a lattice response, and no single code has yet evaluated both pieces from first principles; below $2q_{\mathrm{BZ}}$ the coherent zero-phonon part of that rate, which carries nearly all of the silicon Migdal rate up to 1 MeV, is not computed at all (Sect. 12.4). The Wannier picture of the boost is exact only for a flat band. A scalar that couples to nuclei out of proportion to their mass needs electronic matrix elements at reciprocal-lattice momenta. The calculations below supply the drag masses, the crystal Migdal rate with its coherent zero-phonon part, the error of the Wannier picture, and the matrix elements of couplings not proportional to mass. All of them sit at the leading nonadiabatic order of the Born–Oppenheimer parameter on the truncation label $\mathfrak{T}$ (Sect. 8.5), harmonic ($n_A = 0$) and at zero temperature ($n_T = 0$). They differ in the many-body approximation $\mathcal{X}$, the lattice regime $\mathcal{K}$ and the Migdal regime $\mathfrak{M}$. Table E.10

collects their parameters.

E.18.1 Drag masses and inverse moments

The electronic spectral measure of the force on nucleus κ fixes every quantity of Sect. 12.5. With $|0\rangle$ the clamped-nuclei ground state, $|a\rangle$ its excited states and $\Delta_a = E_a - E_0$, we define

$$\mathrm{d}\sigma_\kappa(\Delta) = \sum_a |\hat{\boldsymbol{q}} \cdot \langle a|\nabla_{\boldsymbol{R}_\kappa} V|0\rangle|^2 \, \delta(\Delta - \Delta_a), \qquad \mathcal{I}_k \equiv \int \Delta^{-k} \, \mathrm{d}\sigma_\kappa(\Delta), \tag{E.97}$$

where V is the electron–nucleus potential. Sect. 12.5 shows that the drag mass along $\hat{\boldsymbol{q}}$ is $\delta M_\kappa = 2\mathcal{I}_3$ and that the Migdal probability is $P_{\mathrm{Mig}} = q^2\mathcal{I}_4/M_\kappa^2$. Because $\Delta \geq E_{\mathrm{g}}$ on the support of $\mathrm{d}\sigma_\kappa$, $\mathcal{I}_4 \leq \mathcal{I}_3/E_{\mathrm{g}}$, and so $P_{\mathrm{Mig}} \leq E_R \, \delta M_\kappa/(M_\kappa E_{\mathrm{g}})$ with $E_R = q^2/2M_\kappa$. With the on-site drag mass $28.7\, m_e$ and the gap of 1.17 eV this bound keeps P_{Mig} below 10^{-3} only up to $E_R \approx 2.1$ eV in silicon, and to less in germanium; the trace value $14\, m_e$ would give 4.3 eV, but it does not bound the probability (Sect. 12.5.3). Above that the computed $\mathcal{I}_4$ is needed.

The three moments follow from two Sternheimer solves per occupied state, the static form of Eq. (E.20). The first, $|\psi_v^{(1)}\rangle = -(H_{\mathrm{KS}} - \varepsilon_v)^{-1} \hat{P}^{\mathrm{c}} \, \partial_{\hat{\boldsymbol{q}}} V |\psi_v\rangle$, is the ordinary phonon response; the second, $|\psi_v^{(2)}\rangle = (H_{\mathrm{KS}} - \varepsilon_v)^{-1} \hat{P}^{\mathrm{c}} |\psi_v^{(1)}\rangle$, uses $|\psi_v^{(1)}\rangle$ as its source. Expanding both in conduction states $|c\rangle$ gives one power of Δ_{cv}^{-1} per solve, so

$$\mathcal{I}_2 = 2\sum_v \langle \psi_v^{(1)}|\psi_v^{(1)}\rangle, \qquad \mathcal{I}_3 = 2\sum_v \langle \psi_v^{(1)}|\psi_v^{(2)}\rangle, \qquad \mathcal{I}_4 = 2\sum_v \langle \psi_v^{(2)}|\psi_v^{(2)}\rangle, \tag{E.98}$$

with the factor 2 for spin and the sums over occupied bands and $\boldsymbol{k}$. These are sums over Kohn–Sham pairs. The interacting drag mass at $\mathcal{X} = \mathrm{RPA}$ comes instead from the frequency-dependent Sternheimer equation of Appendix E.4: the on-site electronic force constant at imaginary frequency obeys $\Phi^{\mathrm{el}}_{\kappa\kappa}(\mathrm{i}\omega) = \Phi^{\mathrm{el}}_{\kappa\kappa}(0) + \omega^2 \delta M_\kappa + O(\omega^4)$, so a fit in ω^2 through five points between 0.1 and 0.5 eV returns δM_κ. The difference between the pair sum and the RPA slope is the many-body nuisance of Sect. 8.8. We compute δM_κ both for the $\boldsymbol{Q} = 0$ sublattice displacement and on site (the Brillouin-zone average over an 8^3 grid of $\boldsymbol{Q}$), with the inputs of the shared harmonic base of Appendix E.17: PBE, SG15 pseudopotentials at 60 Ry, a 12^3 $\boldsymbol{k}$ grid and the measured lattice constant. The pseudopotential freezes the core, whose electrons move rigidly with the nucleus; their contribution, $N_{\mathrm{core}} m_e$ to δM_κ, is added exactly, and the core part of P_{Mig} is added shell by shell from the atomic calculation of Sect. 12.2.

The trace rule tests the result. Displacing the whole cell rigidly drags every electron, so the sum over all sites of the $\boldsymbol{Q} = 0$ tensor equals $N_e m_e$ per cell. It holds only with the core term included, and it tests the $\boldsymbol{k}$ grid and the band completeness of Eq. (E.98) at once. δM_κ must change by less than 1 % from 12^3 to 16^3 $\boldsymbol{k}$ points. Each perturbation needs one Sternheimer solve beyond those of the shared DFPT run, so the cost is about twice that run (Appendix E.17): some 200 core-hours for silicon, 30 for germanium and 40 for gallium arsenide, at a few gigabytes of memory, and 2.4 times as much for the 16^3 test. The Kohn–Sham gap of gallium arsenide, 0.5 eV in PBE against 1.52 eV measured, puts spurious resonances below the real gap and inflates $\mathcal{I}_3$ and $\mathcal{I}_4$, which carry Δ^{-3} and Δ^{-4}; a scissor or a hybrid functional is required (Appendix E.3). Germanium is worse: PBE at the measured lattice constant puts the conduction state at Γ only 0.047 eV above the valence maximum (Appendix E.1), so the inflation of $\mathcal{I}_3$ and $\mathcal{I}_4$ is far larger, and a scissor keeps the wrong valley order. The germanium drag masses, like the electronic piece of the germanium Migdal rate below, are computed on germanium bands that pass the check of Appendix E.14 (valley order, both gaps, masses at L and Γ), with quasiparticle GW bands as the next route under that check. In a polar crystal the $\boldsymbol{Q} = 0$ response must be taken at zero macroscopic field, without the nonanalytic term of Appendix E.4, because the drag mass belongs to the transverse

displacement. The results replace the table-based on-site values of Sect. 12.5 and fix the sign of the off-site coupling.

E.18.2 The crystal Migdal rate

The Migdal rate of a crystal is set by the Migdal entry $S_{\kappa\kappa}(\boldsymbol{q};\omega_l,\omega_e)$ of the joint response, Eq. (12.6), a density in two frequencies, in eV^{-2} per cell. It is a convolution of two pieces: the density response $S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p},\omega_e)$ at the momentum $\boldsymbol{p}$ that the boost gives the electrons, with local fields, and the multiphonon or impulse response of the struck nucleus. We want it for silicon and germanium with $p \le 30\,\mathrm{keV}$ and $\omega_e \le 100\,\mathrm{eV}$. On $\mathfrak{T}$ it sits at the leading nonadiabatic order with $\mathfrak{M}$ = sudden. It uses the Condon approximation with the lattice in its ground vibrational manifold, at $\mathcal{X}$ = RPA with local fields, and with $\mathcal{K}$ the multiphonon or impulse regime of the lattice tables of Appendix E.17.

The electronic piece is all-electron: GPAW with PAW reconstruction, or Yambo [131] and BerkeleyGW [38] with the 2p shell of silicon and the 3d shell of germanium in the valence; the deeper shells enter one at a time from the atomic calculation of Sect. 12.2. The convolution uses the exact impulse bracket of Sect. 12.3 wherever $|\boldsymbol{q}\cdot\boldsymbol{p}|/(M_\kappa\omega_e) > 10^{-2}$, and the soft limit elsewhere. The matrix is not computed on the Cartesian 50^3 grid of $\boldsymbol{p}$ per shell that the convolution integrates over; that would be some 10^{19} operations. It is computed at the 72 irreducible reduced momenta of a 12^3 grid, whose Umklapp copies $\boldsymbol{p}_{\mathrm{red}} + \boldsymbol{G}$ cover the unreduced momenta, and interpolated in $\boldsymbol{p}$ for the convolution. Local fields are converged from $|\boldsymbol{G}| \le 4$ to $6\,a_0^{-1}$, which is about 300 to 1000 plane waves in silicon. A $10^3 \times 10^3$ complex matrix takes 16 MB per $(\boldsymbol{p},\omega_e)$ and the 0.05 eV grid has 2000 frequencies, so a single $\boldsymbol{p}$ holds 32 GB; the frequencies are processed in blocks, contracted with the lattice response on the fly, and only the convolved $S_{\kappa\kappa}$ is stored. Building the independent-particle matrix costs about 3×10^{12} operations per $(\boldsymbol{p},\omega_e)$ on the 12^3 $\boldsymbol{k}$ grid, and the whole calculation about 10^4 core-hours. The same code, fed the same electronic and lattice tables, reports the rate at each truncation of Table 12.1, so that the steps between them are differences of one calculation and not of different inputs.

The calculation is accepted only if the third moment, the drag mass, the Migdal bound and a measured signal all check. The third electronic moment, summed over shells and valence, must equal the contact-density sum rule of Sect. 12.5. The ω_e^{-3} moment must reproduce the drag mass of Appendix E.18.1, and its trace $N_e m_e$. The computed spectrum must respect the bound $P_{\mathrm{Mig}} \le E_R\,\delta M_\kappa/(M_\kappa E_{\mathrm{g}})$. No measurement of the crystal Migdal effect exists yet; once the tagged-recoil neutron measurement proposed by Adams et al. [132] has been made on silicon, the computed rate is compared with it at the same recoil energies. The difference between the result with the full matrix and with its diagonal is reported as the local-field nuisance.

The Migdal term must not be added to a lattice response computed with the bare mass and unit weight (Sect. 12.3): the lattice part loses the weight P_{Mig} and moves with the mass $M_\kappa + \delta M_\kappa$, or the same recoil is counted twice. A response diagonal in $\boldsymbol{G}$ keeps the local fields on the diagonal of ε^{-1}, which lower the silicon probability by a factor 1.35 against a scalar dielectric function (Limit Check 12.3). It drops the elements $\boldsymbol{G} \neq \boldsymbol{G}'$, which raise it again by a factor of about 1.2 to 1.4 in an empirical-pseudopotential model that is not converged in the reciprocal-lattice sum (Sect. 12.3.2), and this calculation is what sizes them. A valence-only electronic piece misses the core-dominated ω_e^3 moment entirely. The results enter Sect. 12.3 as the crystal Migdal spectra of silicon and germanium.

E.18.3 Coherent zero-phonon Migdal response

The convolution above treats the lattice factor in its incoherent form, one nucleus at a time and diagonal in $\boldsymbol{G}$. A zero-phonon Migdal event is a Bragg process instead: the crystal takes a reciprocal-lattice vector $\boldsymbol{G}$ as a whole, with weight $\mathrm{e}^{-2W_\kappa(\boldsymbol{G})}$ and the structure factor of the cell, and the electrons take exactly $\boldsymbol{p} = \boldsymbol{q} - \boldsymbol{G}$ (Sect. 12.4). The amplitudes of different $\boldsymbol{G}$ interfere, so the rate is a double sum over $\boldsymbol{G}$ and $\boldsymbol{G}'$ with the cross-site terms $S_{\kappa\kappa'}$ and the full matrix $S^{ee}_{\boldsymbol{q}-\boldsymbol{G},\boldsymbol{q}-\boldsymbol{G}'}$. Above $2q_{\mathrm{BZ}}$ the direction-averaged double sum agrees with the incoherent form. Far below q_{BZ} it obeys the boost identity Eq. (8.32), which gives 1/14.9 of the diagonal form in silicon, with the median ω_e moved from 4.5 eV to the plasmon at 17.4 eV; a model puts the diagonal form within 15 % of the coherent one only for $q \leq 0.75$ keV. Between 0.75 keV and $2q_{\mathrm{BZ}}$, where nearly all of the silicon Migdal rate up to 1 MeV sits, 33 % at 2 MeV and 7 % at 3 MeV (Table 12.3), nothing fixes it.

The calculation evaluates the coherent zero-phonon response of silicon on that band of q, at $\omega_l = 0$, with every cross-site term and local fields at $\mathcal{X} = \mathrm{RPA}$. Each $\boldsymbol{q}$ is its own reduced momentum plus a reciprocal-lattice vector, so the matrices $S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p}_{\mathrm{red}}, \omega_e)$ of the crystal rate serve unchanged; only the head and wings, which are nonanalytic as $\boldsymbol{p}_{\mathrm{red}} \to 0$, need reduced momenta added on three lines toward Γ, at the cost per momentum of the crystal rate. The ion vertex is the all-electron or PAW one of the crystal rate, and the Bragg weights carry $\mathrm{e}^{-2W_\kappa(\boldsymbol{G})}$ from the base of Appendix E.17. The result is reported as the ratio of the coherent to the diagonal rate per bin of (q, ω_e), averaged over directions on the 590-point Lebedev grid of Appendix E.17 and kept per direction for the daily modulation. It has three tests: as q falls toward 0.75 keV it must approach the boost identity, the factor 1/14.9 and the plasmon median; above $2q_{\mathrm{BZ}}$ it must return the diagonal form; and at every q it must satisfy the partial sum rule Eq. (8.56). The ratio replaces the unsized entry of Table 12.3 and the Migdal band of Sect. 17.1.

E.18.4 The non-rigid Wannier ratio

The Wannier picture of Sect. 12.3 replaces the force operator of nucleus 0 by the electronic translation generator restricted, on the ket side, to the Wannier functions centered on site 0 [133]. It is exact for a flat manifold, such as a core shell, and the error for a band of width W at excitation energy ω_{cv} is bounded only in absolute terms, with coefficient $2W/\omega_{cv} + W^2/\omega_{cv}^2$. For the silicon valence band, $W \approx 12$ eV, the coefficient is 1.56 at $\omega_e = 20$ eV and 0.25 at 100 eV; no relative bound exists. What is computed is the ratio

$$\mathfrak{r}_W(\boldsymbol{q}, \omega_e) = \frac{|A^{\mathrm{DFPT}}|^2}{|A^{W}|^2}, \tag{E.99}$$

of the squared Migdal amplitudes with the true vertex and with the Wannier vertex, a dimensionless number per bin of $(\boldsymbol{q}, \omega_e)$, for silicon, germanium and diamond. On $\mathfrak{T}$ it sits where the crystal rate does, with clamped-nuclei final states.

The true vertex is the matrix element of the bare derivative $\partial_{\boldsymbol{R}_0} V$ between the many-body states, screened at the excitation energy. A static self-consistent DFPT potential is screened with $\varepsilon^{-1}(\omega = 0)$, the wrong screening for transitions at 20 to 100 eV. The calculation therefore takes only the bare perturbation (the ionic potential change of the displaced site, without the induced Hartree and exchange–correlation parts), evaluates $\langle c|\partial_{\boldsymbol{R}_0} V|v\rangle$ between Bloch states, and screens it with the same RPA local-field matrix $(\varepsilon^{-1})_{\boldsymbol{0}\boldsymbol{G}}(\boldsymbol{p}, \omega_e)$ as the electronic piece above. `ph.x` computes the bare change but does not write it out, so the code needs a small modification. The Wannier side uses maximally localized valence Wannier functions [119] from a 16^3 grid, in the same gauge $U(\boldsymbol{k})$ as the Bloch matrix elements, so that both amplitudes are

interpolated together (Appendix E.5). The static self-consistent vertex is kept as a labeled variant; its difference from the dynamically screened one is a screening nuisance, kept apart from $\mathfrak{r}_W$.

The acoustic and the drag-mass sum rules test it. The acoustic sum rule, $\sum_\kappa \partial_{\boldsymbol{R}_\kappa} V^{\mathrm{scf}} = -\sum_i \nabla_i V_{\mathrm{KS}}$ to 10^{-3}, holds for the static self-consistent response; it validates the gauge and the $\boldsymbol{k}$ grid, not the dynamical screening. The drag-mass trace of the first subsection must also hold. A nonlocal pseudopotential adds $\mathrm{i}[V_{\mathrm{NL}}, \boldsymbol{r}]$ to the velocity, and the Wannier amplitude must carry it (Appendix E.7). At about 10^3 core-hours the ratio enters Sect. 12.3 as the model nuisance of the Wannier picture.

E.18.5 Couplings not proportional to mass

A scalar that couples to nucleus κ with strength c_κ is absorbed at momentum $k \to 0$ through the acceleration form Eq. (12.54), with the electronic factor $\langle f|\nabla_{\boldsymbol{R}_\kappa} V|i\rangle$. If c_κ/M_κ is the same for every species, Sect. 12.6 reduces the amplitude exactly to the electron density and the rate to the loss function. Write $c_\kappa/M_\kappa = \tilde{c} + \delta\tilde{c}_\kappa$. The force operator of a point nucleus is $\nabla_{\boldsymbol{R}_I} V = \mathscr{V}^{-1} \sum_{\boldsymbol{p}} \mathrm{i}\boldsymbol{p}\, Z_\kappa v_{\mathrm{C}}(\boldsymbol{p})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{R}_I} n_e(\boldsymbol{p})$, and the sum over sites of $\delta\tilde{c}_\kappa\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_I}$ forces $\boldsymbol{p} = \boldsymbol{k} + \boldsymbol{G}$. The remainder of the rate is then

$$\begin{gathered} \Gamma \propto \frac{1}{\Omega^2\omega^4} \sum_{\boldsymbol{G}\boldsymbol{G}'} [\boldsymbol{k}\cdot(\boldsymbol{k}+\boldsymbol{G})][\boldsymbol{k}\cdot(\boldsymbol{k}+\boldsymbol{G}')]\, \Xi_{\boldsymbol{G}}(\boldsymbol{k}) \Xi^*_{\boldsymbol{G}'}(\boldsymbol{k})\, S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k}\to 0, \omega), \\ \Xi_{\boldsymbol{G}}(\boldsymbol{k}) = \sum_\kappa \delta\tilde{c}_\kappa Z_\kappa v_{\mathrm{C}}(\boldsymbol{k}+\boldsymbol{G})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_\kappa}, \end{gathered} \tag{E.100}$$

and every Umklapp term is $O(k)$, like the $\boldsymbol{G} = 0$ term. The whole effect therefore sits in the off-diagonal elements $S^{ee}_{\boldsymbol{0}\boldsymbol{G}}$ and $S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}$ at vanishing reduced momentum. For equal $\delta\tilde{c}$ on both sites of the diamond structure, $\Xi_{\boldsymbol{G}}$ is proportional to the basis structure factor, which vanishes for (200) and has $|S^{\mathrm{bas}}|^2 = 2$ and 4 for (111) and (220).

The matrix $S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k}\to 0, \omega)$ is needed for $|\boldsymbol{G}| \le 6(2\pi/a)$, about 230 vectors in silicon, and ω from E_{g} to 100 eV plus the core edges, contracted with Eq. (E.100) for each coupling model. On $\mathfrak{T}$ it is $\mathcal{X} = \mathrm{RPA}$ with local fields (then the adiabatic local-density kernel), at clamped nuclei because $\omega \gg \omega_{\max}$. Yambo or BerkeleyGW give $\chi_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}\to 0, \omega)$ directly. A diagonal dielectric matrix would miss every Umklapp term, so the local-field set is enlarged until Γ changes by less than 2 %, on a 12^3 grid; each frequency needs under 1 MB, and the cost is about 10^2 core-hours. The $\boldsymbol{G} = 0$ term uses the full S^{ee}, which already contains the factor $1/|\varepsilon|^2$, and screening it again would double count. The calculation is accepted when the (200) channel vanishes to numerical noise for equal $\delta\tilde{c}$, and when for $c_\kappa \propto M_\kappa$, where $\delta\tilde{c}_\kappa$ is only the mass defect of order 10^{-3}, the result reduces to the loss-function rate Eq. (12.56). The result enters Sect. 12.6.

Table E.10: Nonadiabatic calculations of silicon (and germanium, gallium arsenide or diamond where named): the computed quantity, the main convergence parameters, the acceptance test and the cost. N_e is the number of electrons per cell; the cost is in core-hours.

quantity	grids and parameters	acceptance test	cost
δM_κ, $\mathcal{I}_2$–$\mathcal{I}_4$	shared base (60 Ry); $\boldsymbol{k}$: $12^3 \to 16^3$ (1 %); 8^3 $\boldsymbol{Q}$; iω = 0.1–0.5 eV	trace $N_e m_e$ per cell	2× base: 200 (Si), 30 (Ge), 40 (GaAs)
$S_{\kappa\kappa}(\boldsymbol{q};\omega_l,\omega_e)$	$p \le 30$ keV from 72 reduced $\boldsymbol{p}$; $\omega_e \le 100$ eV in 0.05 eV; $\lvert\boldsymbol{G}\rvert \le 4 \to 6\,a_0^{-1}$; $\boldsymbol{k}$: $8^3 \to 12^3$	third moment, drag mass, P_{Mig} bound; neutron data when measured	10^4
coherent zero-phonon ratio	0.75 keV to $2q_{\text{BZ}}$; matrices of the row above plus reduced $\boldsymbol{p}$ toward Γ; 590 directions	boost limit (1/14.9); diagonal form above $2q_{\text{BZ}}$; Eq. (8.56)	per added $\boldsymbol{p}$ as above
$\mathfrak{r}_W(\boldsymbol{q},\omega_e)$	Wannier $\boldsymbol{k}$: 16^3; bare vertex screened at ω_e	acoustic sum rule to 10^{-3}; drag-mass trace	10^3
$S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k}\to 0,\omega)$	$\lvert\boldsymbol{G}\rvert \le 6(2\pi/a)$; $\boldsymbol{k}$: 12^3	(200) zero; mass-proportional limit	10^2

E.19 Anharmonic, disordered and quantum solids

Solid helium near melting has a Lindemann ratio of about a quarter: the zero-point amplitude of an atom is about a quarter of the distance to its neighbor (Sect. 11.5), and the harmonic lattice is not a valid starting point. Strontium titanate and potassium tantalate have soft modes whose frequencies depend on temperature at order one; lithium hydride has a light atom whose zero-point motion is large. For these targets the multiphonon response of Sect. 11.5 has no small parameter, and only exact static moments bound the answer. Glasses lack the lattice that the response tables assume, and a random medium can scatter a strongly interacting particle coherently over its whole volume. For Sects. 11.5, 14.5 and 16.7 the calculations below supply the moments and approximate dynamics of strongly anharmonic and quantum crystals, the anharmonic correction of an ordinary crystal, the glass models, the force from a random medium beyond the Born approximation, and the decay of chiral phonons in molecular frameworks.

E.19.1 Strongly anharmonic and quantum crystals

Every integer frequency moment of the self response is an equal-time correlation function. The moments $\mathscr{M}^k$, $k = 0, \dots, 4$, of the self response of Sect. 8.3 are therefore exact from a static path-integral simulation, for any anharmonicity, any temperature and any Lindemann ratio. This holds in the Born–Oppenheimer approximation, where the nuclei move on a smooth energy surface and every moment is finite; with point nuclei and the full Coulomb interaction the fourth moment diverges (Sect. 8.3). The Markov–Krein band of Appendix C.10 and Sect. 11.5 then brackets every response with these five moments; it bounds bin-integrated rates in the bulk of the response, and threshold rates only from above. The calculation therefore has two parts, exact moments and an approximate real-time response tested against them.

For $KTaO_3$, $SrTiO_3$ and LiH we want $\mathscr{M}^0$–$\mathscr{M}^4$ of the self response at $\lvert\boldsymbol{q}\rvert = 0.5$ to 20 keV, the band per $\boldsymbol{q}$ at the millikelvin operating temperature, and the self response $S_{\text{self}}(\boldsymbol{q},\omega)$ (per volume, in eV^2, the units of Sect. 4.2) at 50 and 100 K, where the real-time methods below hold. On the truncation label $\mathfrak{T}$ (Sect. 8.5) the moments are exact in n_A; the dynamics is approximate at all orders of n_A, and the moment test measures its error. The static

part is path-integral molecular dynamics in i-PI [127] with a machine-learned potential (Appendix E.11) trained on PBE and validated on displaced configurations to three standard deviations of the path-integral distribution. The cells are cubic $n \times n \times n$ supercells with $n = 4$, 6 and 8 (320, 1080 and 2560 atoms) for the perovskites, even n so that the soft mode at the R point is commensurate, and 4^3 to 6^3 conventional cells (512 to 1728 atoms) for LiH. The number of beads P satisfies $\beta\omega_{\max}/P \lesssim 0.3$; for $SrTiO_3$ at 50 K with $\omega_{\max} \approx 100$ meV that is $P \approx 80$, doubled until $\mathscr{M}^2$ changes by less than 1 %. Colored-noise thermostats may accelerate the static moments but must not be used in the dynamics, which they alter.

The dynamic part compares three approximate methods. Thermostatted ring-polymer molecular dynamics damps only the internal ring-polymer modes, with friction $\gamma_k = \lambda\omega_k$ at $\lambda = 1/2$ and checks at 1/4 and 1, which preserves the exact limits of the method [134]; the centroid is not thermostatted during the dynamics. Centroid molecular dynamics, with the time step reduced threefold, is the second method, and their difference is the method nuisance. The third is the stochastic self-consistent harmonic approximation [69] (Appendix E.8), first static, then with the dynamical bubble, then the full time-dependent form [135]; the bubble-to-full difference measures the order. The time step is at most $0.1/\omega_P$ of the stiffest free ring-polymer mode, typically 0.1 to 0.25 fs, with centroid energy drift below 10^{-4} per ps. Resolving $\delta\omega = 0.5$ meV needs trajectories of $20\pi/\delta\omega \approx 80$ ps at $q \lesssim q_{\mathrm{BZ}}$, shorter at large q where the correlation decays; 100 independent initial conditions give 1 % statistical error on bin integrals. Every approximate response must lie inside the band of its exact moments, and the response must change by less than 2 % from one cell size to the next while the finite-size gap $2\pi c_\nu/L$ stays below the lowest frequency bin. A response outside the band is flagged, not used.

Solid ^{4}He is computed the same way with the Aziz pair potential, which needs no fitting. The simulation is path-integral molecular dynamics at a molar volume of 21 $\mathrm{cm^3\,mol^{-1}}$ and 1 K, 180 to 432 atoms and $P = 64 \to 128$ beads (the Debye energy is 2 meV, so $\beta\omega_{\mathrm{D}} \approx 25$), with an open-path estimator for the momentum distribution. The band is computed for the self and the coherent response; the coherent moments are the static structure factor, the f-sum, the current correlation and the third-moment correlator, all equal-time. Bose exchange is negligible for the statics at this temperature but is checked once by permutation sampling. At 1 to 40 GPa, where the Lindemann ratio drops to about 0.1, the self-consistent harmonic force constants from a potential trained on dispersion-corrected PBE replace the static Hessian in the phonon response of Sect. 11.5, converged to 2 % in the frequencies and 3 % in the displacement correlations.

The pitfalls decide what may be quoted. Ring-polymer and centroid methods give Kubo-transformed correlations; converting to S multiplies by $\beta\omega/(1 - \mathrm{e}^{-\beta\omega})$, which amplifies their error by $\beta\omega$, so they are usable only for $T \gtrsim \omega/10$ and not at a 10 mK operating temperature. Their short-time error for the nonlinear operator $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}}$ is already $O(t^3)$. The time-dependent self-consistent method excites at most two phonons at a time, so at $q \gg q_{\mathrm{BZ}}$ it can supply only the dressed one-phonon propagator inside the Gaussian form. Tunneling splittings in double wells are missed by centroid-like methods. At millikelvin, therefore, the response of these targets is bounded by the band alone. The ring-polymer runs cost 10^3 to 10^4 core-hours per material and temperature, and centroid dynamics three times more. The self-consistent harmonic runs cost about 10^2 core-hours per material, path-integral helium at melting about 10^3, and high-pressure helium about 10^4 per pressure. The memory is a few gigabytes. The moments and bands enter Sect. 11.5.

E.19.2 Weak anharmonicity

In silicon the anharmonicity is small, and the question is the size of its first correction to the multiphonon response above the one-phonon band, where the low-mass reach is set. Time reversal fixes the structure. For a target invariant under time reversal $S_{\kappa\kappa}(\boldsymbol{q},\omega) = S_{\kappa\kappa}(-\boldsymbol{q},\omega)$, and a term of first order in the cubic force constants $\Phi^{(3)}$ is of third degree in $\boldsymbol{q}$, so it cancels. The leading correction is therefore of second order in $\Phi^{(3)}$ or first order in $\Phi^{(4)}$, the level $n_A = 2$ of Sect. 8.5: the one-phonon propagator dressed by the bubble and loop self-energies, the connected four-point function, and the square of the connected three-point function. We want the relative correction $\delta_{\mathrm{anh}}(\boldsymbol{q},\omega) = S/S^{\mathrm{harm}} - 1$ of Sect. 11.5, for silicon first, then germanium and gallium arsenide.

The cubic force constants come from finite displacements in a $4 \times 4 \times 4$ primitive supercell (128 atoms) with a pair cutoff of 5 Å, about 300 supercell DFT runs, through phono3py [130]; the quartic ones from hiPhive [136] or TDEP (Appendix E.8). The bubble self-energy gives the full spectral function on a 24^3 grid, not on-shell linewidths, and the quartic loop gives the shift. The three- and four-point functions are mode sums over four branch labels with contour kernels built from the thermal propagator, about 10^3 core-hours. The dense cubic constants of the 128-atom cell have 384^3 entries, 0.45 GB in double precision; the quartic set within the cutoff is smaller. The spread under the supercell from 3^3 to 4^3, the cutoff from 4 to 5 Å and the grid from 24^3 to 32^3 is the nuisance width. The first-order cubic term must vanish numerically, which checks the placement of the time arguments on the contour, and the correction must satisfy the exact linear constraints of Sect. 11.5, anchored by the kinetic energy of the struck atom from the path-integral calculation of Appendix E.17, whose error is at most 0.2 %. The total cost is 10^3 to 10^4 core-hours.

E.19.3 Glass models

Amorphous SiO_2, silicon and silicon nitride need structures before they need responses. The models are made by melt-quench molecular dynamics with a machine-learned potential (Appendix E.11), 648 to 5184 atoms and at least five independent quenches, validated on the measured density and pair distribution. From each model come the full Hessian and the Born charges Z_I^* of every atom, either by DFPT on 192- to 648-atom subcells (Appendix E.4) or from a charge-equilibration model fitted to them. The Hessian of 5184 atoms has 15552^2 entries, 1.9 GB in double precision, and its dense diagonalization costs about 4×10^{12} floating-point operations.

The infrared coefficient $C^{\mathrm{IR}}(\omega) = A^{\mathrm{IR}} + B^{\mathrm{IR}}\omega^2$ of glass absorption below the Ioffe–Regel frequency is the first quantity that enters Sect. 16.7. If the modes there are plane waves of branch ν with sound speed c_ν, the mode polarization is a Fourier component of the Born-charge field, the masses cancel, and with the Born-charge structure factor $S_\nu^Z(k) = N^{-1}\langle|\sum_I \hat{\boldsymbol{\epsilon}}\cdot Z_I^*\cdot\hat{\boldsymbol{e}}_\nu\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}_I}|^2\rangle$, averaged over directions,

$$C^{\mathrm{IR}}(\omega) = \frac{1}{\bar{M}}\sum_\nu \hat{w}_\nu S_\nu^Z(\omega/c_\nu), \qquad \hat{w}_\nu = \frac{c_\nu^{-3}}{\sum_{\nu'} c_{\nu'}^{-3}}, \tag{E.101}$$

where $\bar{M}$ is the mean atomic mass and $\hat{w}_\nu$ is the fraction of modes at fixed ω that belong to branch ν. Hence A^{IR} is the $k \to 0^+$ limit of S_ν^Z, and B^{IR} its curvature divided by $2c_\nu^2$. Both follow from static configurations and a charge model, without the Hessian, on wavevectors $k \le 0.3\,\text{Å}^{-1}$ of the periodic cell; the Hessian gives the full $C^{\mathrm{IR}}(\omega)$ as a cross-check up to the boson peak. The second quantity is the distinct part of the response at two and three phonons. It needs the configurational averages, at fixed pair distance r, of the longitudinal and transverse parts of the pair displacement tensor, its longitudinal–transverse vector and

its traceless transverse part, binned in $\Delta r = 0.02\,\text{Å}$ up to 15 Å; the third is the site-resolved phonon density of states. The ensemble is validated against the measured terahertz absorption of the same glass [137].

Size convergence and neutrality test the models. The pair distribution, the moments, the density of states and C^{IR} change by less than 5 % between 1728 and 5184 atoms; the box is at least 4 nm, so that $2\pi/L$ lies below the correlation length of the charges; and $\sum_I Z_I^* = 0$ in every configuration. The 5184-atom box of SiO_2 is 4.3 nm across; the 1728-atom box, 3.0 nm, serves only as the smaller member of the size test. A locally neutral charge model forces $A^{\mathrm{IR}} = 0$ by construction and must not be used; the model has to reproduce the site-charge fluctuations of the subcell DFPT. The finite-size gap $2\pi c_\nu/L$, 3 to 4 meV at 4 nm, removes the lowest modes, and the dependence on quench rate is a model systematic. The cost is 10^4 core-hours for the molecular dynamics, 10^3 for the subcell Born charges and 10^2 per model afterward.

E.19.4 Coherent force from a random medium

A dark-matter particle with a large cross-section on nuclei sees a solid as an optical potential $V_{\mathrm{opt}} = 2\pi b_{\chi N} n_0/\mu_{\chi N}$, with $b_{\chi N}$ the scattering length ($\sigma_{\chi N} = 4\pi b_{\chi N}^2$), n_0 the mean nuclear density and $\mu_{\chi N}$ the reduced mass. Sect. 16.7 shows that the Born treatment of scattering off density fluctuations holds only for $\sigma_{\chi N} \ll \sigma_{\mathrm{crit}}$; for a Ti–6Al–4V test mass and $m_\chi = 20\,\text{eV}$, $\sigma_{\mathrm{crit}} \approx 1.2 \times 10^{-38}\,\text{cm}^2$ at $v = 10^{-3}$, below the cross-sections where the published bound [138] sits. Above it the force on a test mass comes from reflection at its faces. A plane wave with normal velocity v_n meets the step with reflection probability $R(v_n) = |(k_n - k')/(k_n + k')|^2$, $k'^2 = k_n^2 - 2\mu_{\chi N} V_{\mathrm{opt}}$; a slab many de Broglie wavelengths thick reflects, averaged over its thickness, $R_{\mathrm{slab}} = 2R/(1+R)$. Each reflected particle returns momentum $2m_\chi v_n$, and the flux on a face of outward normal $\hat{\boldsymbol{n}}$ is ρ_χ/m_χ times the inward normal velocity, so the pressure is

$$P_{\mathrm{wall}}(\hat{\boldsymbol{n}}) = 2\rho_\chi \int \mathrm{d}^3 v\, f_\chi(\boldsymbol{v})\, (\boldsymbol{v}\cdot\hat{\boldsymbol{n}})^2 R_{\mathrm{slab}}(\boldsymbol{v}\cdot\hat{\boldsymbol{n}})\, \Theta(-\boldsymbol{v}\cdot\hat{\boldsymbol{n}}), \tag{E.102}$$

while transmitted particles leave with their initial velocity and transfer nothing on average. What the equivalence-principle test measures is the differential acceleration $\Delta a(\sigma_{\chi N}, m_\chi)$, in $\mathrm{m\,s^{-2}}$, of the two cylinders of the MICROSCOPE equivalence-principle test (a platinum alloy and Ti–6Al–4V), summed over faces and averaged over the standard halo with the Earth's velocity. The scale is set by $\rho_\chi v^2 \approx 5 \times 10^{-11}\,\text{Pa}$.

The calculation evaluates Eq. (E.102) per face of the axisymmetric geometry, with one-dimensional transfer matrices along the local normal (the curvature radius is far larger than the de Broglie wavelength, 60 µm at 20 eV), and a four-dimensional Monte Carlo over velocity and face. The code is about 300 lines and runs in CPU minutes. Below σ_{crit} the same code evaluates the distorted-wave Born form, with the density correlation $\langle \delta n\, \delta n \rangle$ taken from small-angle X-ray scattering on the actual alloy. As $V_{\mathrm{opt}}/E_{\mathrm{kin}} \to 0$ the result must reproduce the Born random-medium rate of Sect. 16.7. The criterion is applied over the velocity distribution, not at one speed, since $V_{\mathrm{opt}}/E_{\mathrm{kin}} \propto v^{-2}$. Roughness on the de Broglie scale makes the reflection diffuse, and the nested cylinders reflect particles back and forth; both are included in the Monte Carlo. Until this calculation exists, any bound from the channel is a conjecture.

E.19.5 Chiral phonons in molecular frameworks

A chiral phonon carries angular momentum $L^{\mathrm{ph}}_{\boldsymbol{Q}\nu} = -\mathrm{i} \sum_\kappa \boldsymbol{e}^*_{\kappa\nu}(\boldsymbol{Q}) \times \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$. The proposal of Sect. 14.5 reads out the phonons that end, after anharmonic decay, in a long-lived chiral band with $|L^{\mathrm{ph}}| > L_{\mathrm{th}}$ [139]. Its yield is the mean number of such phonons per phonon created

in mode $\lambda = (\boldsymbol{Q}\nu)$, averaged over the created spectrum. Although the decay branches, the expectations obey a linear system. Let $\mathsf{M}_{\lambda\lambda'}$ be the mean number of daughters in λ' per decay of λ: the three-phonon rates into unordered pairs, counted twice for identical daughters, plus the elastic isotope rate, all divided by the total rate of λ, which includes boundary loss and sensor capture. Let $b_\lambda = 1$ for modes in the terminal chiral set, those with $|L^{\mathrm{ph}}| > L_{\mathrm{th}}$ and a decay time longer than the observation time t_{obs}, and $b_\lambda = 0$ otherwise, with the rows of M zero for terminal modes. Conditioning on the first decay gives $\mathcal{N}^{\mathrm{ch}}_\lambda = b_\lambda + \sum_{\lambda'} \mathsf{M}_{\lambda\lambda'} \mathcal{N}^{\mathrm{ch}}_{\lambda'}$, so

$$\boldsymbol{\mathcal{N}}^{\mathrm{ch}} = (\mathbb{1} - \mathsf{M})^{-1} \boldsymbol{b}. \tag{E.103}$$

Without isotope scattering M is nilpotent in energy order. With it, $\mathbb{1} - \mathsf{M}$ is still invertible as long as every closed set of elastically connected modes has some decay or loss.

The first framework is InF_3(4,4′-bipyridine), whose chiral phonon bands Romao et al. computed [139]. The rates come from phono3py on the framework cell, at $n_A = 2$ (Fermi's golden rule with two cubic vertices) with harmonic L^{ph}, on a $10 \times 10 \times 20$ grid; Eq. (E.103) is a sparse solve. Cubic force constants from DFT cost about 10^5 core-hours for a metal–organic framework; a machine-learned potential validated on 50 DFT supercells brings this to 10^3. The grid, the cutoff of the cubic constants and the smearing of energy conservation are converged, and the sensitivity to t_{obs} is reported. A direct Monte Carlo of the branching process with the same rates must reproduce $\boldsymbol{\mathcal{N}}^{\mathrm{ch}}$ within its statistical error, and the energy in the terminal set plus the energy lost must equal the energy created. L^{ph} is not unique inside a degenerate subspace, so it is diagonalized there. It is not conserved in decay, so the label is recomputed for each daughter. And the code must state whether its rates are per ordered or per unordered pair, since the factor two moves the answer at order one.

Frameworks with lanthanide ions, whose host and ion Sect. 14.5 selects, add one-phonon amplitudes through the crystal field. Let $P_{\mathrm{K}}(\boldsymbol{u})$ be the projector on the ground Kramers doublet, of energy E_{K}, of the crystal-field Hamiltonian $H_{\mathrm{CF}}(\boldsymbol{u})$ of the $4f$ shell at displacement $\boldsymbol{u}$. Let O be the spin $\boldsymbol{s}_e$ or the orbital moment $\boldsymbol{L}$, and $R = \sum_{c \notin \mathrm{K}} |c\rangle\langle c| / (E_{\mathrm{K}} - E_c)$. First-order perturbation theory gives $\partial P_{\mathrm{K}} = R\, \partial H_{\mathrm{CF}}\, P_{\mathrm{K}} + P_{\mathrm{K}}\, \partial H_{\mathrm{CF}}\, R$, and the amplitude follows from $\partial(P_{\mathrm{K}} O P_{\mathrm{K}})$ contracted with the phonon eigenvector, $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) / \sqrt{2 N M_\kappa \omega_{\boldsymbol{Q}\nu}}$. The projector form avoids derivatives of individual doublet states, which depend on an arbitrary phase. The inputs are the derivatives of the 27 Stevens parameters ($k = 2, 4, 6$) and of the g tensor with respect to symmetry-adapted displacements of an embedded cluster (the ion, its first coordination shell and whole ligands, in Ewald point charges with model potentials on boundary cations), computed by central differences of ± 0.01 Å. The electronic structure is a state-averaged complete-active-space self-consistent field (CASSCF) calculation over the ground term with spin–orbit mixing, a perturbative correction of the crystal-field splittings, a scalar-relativistic Hamiltonian and ANO-type basis sets. The derivatives must be linear over 0.005 to 0.02 Å to 2 % and stable to 10 % under enlargement of the active space, with the cluster extended to its second shell and the basis enlarged as further checks. The roots must be tracked by overlap, because they can exchange order under displacement, and $\boldsymbol{s}_e$ and $\boldsymbol{L}$ are kept separate since dark matter couples to them differently. Each geometry costs 0.5 to 2 core-hours, 10^2 to 10^3 per inequivalent site. Both results enter Sect. 14.5.

E.20 Magnetic, superconducting and Dirac targets

Clean aluminum needs three numbers for its loss function near the pair-breaking edge, the density of states $\mathcal{N}_{\mathrm{F}}$ (free-electron value), the Fermi velocity $v_{\mathrm{F}} = 2.0 \times 10^6\,\mathrm{m\,s^{-1}}$ and the gap $\Delta_{\mathrm{BCS}} = 0.18\,\mathrm{meV}$, and no first-principles calculation at all; its conserving response is a two-dimensional integral that runs in minutes. Amorphous WSi, at the other extreme, needs

molecular dynamics, DFT, the RPA and a Bogoliubov–de Gennes solution on each quenched cell. The magnets and Dirac materials of Sects. 14.1, 14.2, 14.5 and 16.1 lie in between. Their rates are written in the book through a few model parameters (exchange constants, anisotropies, form factors, Dirac velocities and background screening), and the calculations below supply those parameters and the responses that go beyond the model. Table E.12 lists them with their costs.

E.20.1 The conserving response of aluminum

The conserving density response of Sect. 15.1 adds a phase-mode term to the bare quasiparticle bubble, and the two cancel at $\boldsymbol{q} = 0$. Its relative size is the vertex fraction $\mathfrak{v} = \mathrm{Im}(R^{\mathrm{cons}} - R^0_{\boldsymbol{\eta}\boldsymbol{\eta}})/\,\mathrm{Im}\,R^0_{\boldsymbol{\eta}\boldsymbol{\eta}}$, which we want on the grid of Sect. 15.1: $y = \omega/\Delta_{\mathrm{BCS}}$ from 2 to 50 and $qv_{\mathrm{F}}/\Delta_{\mathrm{BCS}}$ from its kinematic minimum $y\,v_{\mathrm{F}}/v_{\max}$ to 10^3, together with the full $\mathrm{Im}\,\varepsilon^{\mathrm{cons}}$, its Kramers–Kronig transform and the loss function L^{cons} in its resummed form at the edge. In the clean limit halo dark matter reaches only $q\xi_{\mathrm{coh}} \geq 2v_{\mathrm{F}}/(\pi v_{\max}) = 1.6$ at $\omega \geq 2\Delta_{\mathrm{BCS}}$, with $\xi_{\mathrm{coh}} = v_{\mathrm{F}}/(\pi\Delta_{\mathrm{BCS}})$ the coherence length and $v_{\max}/v_{\mathrm{F}} \approx 0.39$, so the small-$q$ region where the vertex dominates lies outside the kinematics. A film is not clean. At $y = 2.2$ the momentum transfer is $q = 0.15\,\mathrm{eV}$ whatever v_{F} is, and a film with $\mathfrak{l}_e = 30\,\mathrm{nm}$ has $q\mathfrak{l}_e = 0.023$ there, so its electrons are diffusive on the scale of the probe (Sect. 15.3). The clean table holds at $y \leq 2.5$ only for $\mathfrak{l}_e \gtrsim 10\,\mu\mathrm{m}$, not for typical films, and the calculation must therefore include impurity scattering. On the truncation label $\mathfrak{T}$ (Sect. 8.5) it is $\mathcal{X} = \mathrm{RPA}$ in the Coulomb channel with the pairing ladder, on the BCS reference, quasiclassical (constant $\mathcal{N}_{\mathrm{F}}$, spherical Fermi surface, $T = 0$), clean or with Born impurity scattering.

Each clean bubble is an integral over the angle μ between $\boldsymbol{k}$ and $\boldsymbol{q}$ and over the band energy ξ, with $\xi' = \xi + v_{\mathrm{F}}q\mu + q^2/2m_e$, $E = (\xi^2 + \Delta^2_{\mathrm{BCS}})^{1/2}$, and the pair-creation coherence factors $\frac{1}{2}[1 - (\xi\xi' - \Delta^2_{\mathrm{BCS}})/(EE')]$ for the density vertex and $\frac{1}{2}[1 + (\xi\xi' + \Delta^2_{\mathrm{BCS}})/(EE')]$ for the phase vertex. Gauss–Legendre quadrature in μ and adaptive quadrature in ξ evaluate it. The imaginary parts are taken at the roots of $E + E' = \omega$, solved analytically, with no broadening, and the real parts are principal values with the pole subtracted. The phase–phase bubble diverges logarithmically in the ξ cutoff, but only its difference from the value at $(0, 0)$ enters, so the cutoff is taken at $10^3\Delta_{\mathrm{BCS}}$ and the logarithm is cancelled analytically; the value at $(0, 0)$ itself must be computed with the same cutoff as the gap equation, where it equals the inverse pairing coupling, or the Goldstone cancellation fails and $\mathfrak{v}$ acquires a spurious constant. The Kramers–Kronig integral runs on a logarithmic grid to $10^3 qv_{\mathrm{F}}$, with the Lindhard normal state above qv_{F}. The sign of i0 on the antiresonant pole is that of the retarded function, with which the time-ordered form agrees only for $\omega > 0$ at $T = 0$, and the bare and conserving bubbles are never mixed between $\mathrm{Im}\,\varepsilon$ and $\mathrm{Re}\,\varepsilon$, since that breaks positivity.

Impurities enter in Nambu space through $\boldsymbol{\eta}$, the Nambu matrix of the density, so the Born self-energy is $\boldsymbol{\eta}\,\mathsf{G}\,\boldsymbol{\eta}$ averaged over the Fermi surface (Sect. 15.3). The same impurity ladder dresses the density vertex and the two pair vertices, which keeps the response conserving, and its normal state is the Mermin function Eq. (4.62) at the relaxation rate $\gamma_{\mathrm{rel}} = v_{\mathrm{F}}/\mathfrak{l}_e$. The table is computed in the clean limit and for $\mathfrak{l}_e = 1\,\mu\mathrm{m}$, 300 nm and 30 nm, with the edge values divided by $\mathfrak{g}_{\mathrm{I}}$ times that normal state, as in Table 15.1.

The conserving response must vanish at $q \to 0$ for every ω, and with the ladder switched off it must return the bare bubble with the density coherence factor [140, Eq. (7)]. $\mathfrak{v}$ must be stable to 10^{-3} under doubling of both grids, the Kramers–Kronig tail must stay below 10^{-3} of $\mathrm{Re}\,\varepsilon$, and the loss-function table must be converged to 1 %. With impurities the normalized edge ratio has to reproduce the conserving values of Sect. 15.3: 1.008 to 1.091 at $y \geq 3$ for every mean free path down to 30 nm, and at $y = 2.2$ the fall from 0.87 in the clean limit to 0.60, 0.38 and 0.19 at 1 µm, 300 nm and 30 nm; L must still vanish at $2\Delta_{\mathrm{BCS}}$. The table

replaces the exclusion of the cells nearest the edge in Sect. 15.1, for the mean free path of the film at hand.

E.20.2 Amorphous WSi

A W_xSi_{1-x} film has $k_F \ell_e$ of a few (1.8–5.5 from its measured transport), and at $q = 1$ to 5 keV, where q is comparable to both k_F and the screening wavenumber, no relaxation-time form of ε is valid (Sect. 15.3). The rate needs the configurational average $\langle L \rangle(\boldsymbol{q}, \omega)$ of the loss function, not the loss function of the averaged ε, for $q = 0.5$ to 5 keV and ω from $2\Delta_{BCS}$ to 50 eV, together with the distribution P_Δ of the local gap. On $\mathfrak{T}$ the normal state is $n_B = 0$, $\mathcal{X}$ = RPA with local fields, averaged over amorphous configurations, and the gap is a Bogoliubov–de Gennes mean field with an attractive interaction. The superconducting factor is applied locally with P_Δ, which Anderson's theorem justifies where the mean free path is far shorter than the coherence length.

A machine-learned potential (Appendix E.11) is trained on PBE molecular dynamics of 216-atom cells at the film composition, $x \approx 0.2$–0.25, and 512- to 1000-atom cells are melt-quenched at no more than $10^{13}\,\mathrm{K\,s^{-1}}$, at least five times, and validated on the measured density and the W–Si and W–W pair distributions. Each quench gets a DFT ground state (SG15, with the tungsten 5s5p semicore in the valence) at Γ plus a 2^3 grid, and a non-self-consistent run with at least three times as many bands as are occupied, so that the semicore states, which contribute above about 35 eV, are in the response. Its Kubo–Greenwood conductivity $\sigma_1(\omega)$ is the optical control, compared with film ellipsometry and sheet resistance. The RPA dielectric matrix (Yambo or BerkeleyGW, Appendix E.6) is computed directly at the reciprocal vectors of the supercell, which are $2\pi/L \approx 0.5$ keV apart for $L = 25$ Å, so no folding is needed above that, with the $\boldsymbol{G}$ cutoff converged to $q + 3$ keV and the bands to $\omega_{max} + 30$ eV. Local fields are large in a disordered cell, and the loss function is $\mathrm{Im}[-(\varepsilon^{-1})_{\mathbf{00}}]$, never $\mathrm{Im}[-1/\varepsilon_{\mathbf{00}}]$. A Wannier tight-binding model of each cell, solved in the Bogoliubov–de Gennes approximation with the attractive interaction fitted to the measured film T_c, gives $\Delta(\boldsymbol{r})$ and P_Δ. The f-sum per q must hold to 2 % and the measured σ_1 must be reproduced; the spread of $\langle L \rangle$ across quenches and from 512 to 1000 atoms, required below 5 %, is the error bar. The RPA matrices, at tens of gigabytes each for 1000 atoms, dominate the cost (Table E.12), and the result enters Sect. 15.3.

E.20.3 Holstein absorption with Eliashberg functions

Above $2\Delta_{BCS}$ a superconductor absorbs a dark photon by breaking a pair and emitting a phonon, with the conductivity $\sigma_1^S(\omega)$ of Eq. (15.25). With a Debye form of α^2F_{tr} it reduces to the result of Hochberg, Lin and Zurek [141, Eqs. (16)–(17)], but the rate within a few meV of the edge depends on the shape of $\alpha^2F_{tr}(\omega')$ below a few meV, where the free-electron model holds 1.25×10^4 times the weight of the ω'^4 law (Sect. 15.4). We therefore want $\sigma_1^S(\omega)/\sigma_n$ for aluminum and niobium with α^2F_{tr} from first principles, also in its strong-coupling form, where the pair weight becomes $\int \mathrm{d}E\,[N^{El}(E)N^{El}(E') + P^{El}(E)P^{El}(E')]$ with the Eliashberg density of states and pair amplitude of Sect. 15.4.4. On $\mathfrak{T}$ it is $n_g = 2$, one electron–phonon vertex and one current vertex, with isotropic Eliashberg dressing.

EPW [55] (Appendices E.5 and E.9) computes the vertices by DFPT on an 8^3 grid and interpolates them to fine grids of 60^3 $\boldsymbol{k}$ and 30^3 $\boldsymbol{Q}$ for aluminum and 40^3 and 20^3 for niobium, for both α^2F and the transport function with its weight $1 - \cos\theta$. The 30^3 grid does not resolve the ω'^4 regime, so the transport function below a few meV is recomputed on a refined grid of small $\boldsymbol{Q}$ around Γ, and it must follow the ω'^4 law there. That law holds only if the vertex obeys the acoustic sum rule Eq. (6.12), which makes it vanish for $\boldsymbol{Q} \to 0$ between states on the

Fermi surface, so the sum rule is checked on the interpolated vertex. The isotropic Eliashberg equations [142] are solved on the real axis, since Padé continuation produces artifacts near $2\Delta_{\mathrm{BCS}}$, and Eq. (15.25) is then integrated with a phonon step no larger than $\Delta_{\mathrm{BCS}}/20$. The transport coupling λ_{tr} must match the value 0.39 from the resistivity of aluminum, against which the free-electron model gives 0.32 (Sect. 15.4); T_c must come within 10 % of experiment with a single Coulomb pseudopotential μ^*, σ_1 must be stable to 5 % under doubling of the fine grids, and the normal-state σ_1 must lie within 20 % of the measured optical conductivity of aluminum between 0.2 and 3 eV. Which renormalization factors survive in the strong-coupling form is open in Sect. 15.4.4; in the normal metal the factor is $(1+\lambda_{\mathrm{tr}})^{-2} = 0.52$ in the rate, and the calculation reports the superconducting result with and without the factor $1/(1+\lambda_{\mathrm{tr}})$ in the current, so that their difference measures it.

E.20.4 Magnet inputs from DFT $+U$

The magnon rates of Sects. 14.1 and 14.2 need exchange constants J_{ij}^{ex}, single-ion anisotropies and g_{ion} tensors. For YIG, NiO, MnO, Cr_2O_3 and $NiPS_3$ these come from inelastic neutron or resonance fits where they exist, and otherwise from DFT $+U$ (Appendix E.9), in the Heisenberg convention of the book, $J^{\mathrm{ex}} > 0$ antiferromagnetic with each pair counted once. The Hubbard U is computed by linear response [76], and the exchange constants by four-state energy mapping or by the Green-function method of TB2J [78] on Wannier functions, in the magnetic unit cell at the measured lattice constant, shell by shell until a further shell no longer moves the computed magnon band by more than its spread over U. That spread is the main systematic, and the measured ordering temperature and, where available, the measured magnon band are the checks. The orbital moments of Cr^{3+} and Ni^{2+} enter g_{ion} and are computed with spin–orbit coupling. Spin–orbit coupling also gives the Dzyaloshinskii–Moriya constants of Cr_2O_3, by four-state mapping with the two spins of each bond placed in the plane normal to the component sought, which the spin model of Sect. 14.5 needs together with the exchange and anisotropy constants for $T \ll T_{\mathrm{CM}}$. For a target without ultrasonic data, the magnetoelastic constants B_1 and B_2 of Sect. 14.2 follow from the strain derivative of the magnetic anisotropy energy, computed with spin–orbit coupling at a few strained cells, with strains of ± 0.5 % as for the deformation potentials of Appendix E.17; where ultrasonic data exist, they check it.

The same linear spin-wave code that serves the two-magnon response below gives the uniform modes that absorption into magnons needs (Sect. 14.3). For a sample of finite size these carry the demagnetizing field of its shape, and in a body that is not an ellipsoid the dipolar lattice sums over the real sample spread the weight over several magnetostatic modes, so the code takes the sample shape and the dipolar sums as inputs. In the ferromagnetic limit the uniform mode must sit exactly at the Larmor frequency, and in the two-sublattice antiferromagnet the uniform transverse weight per antiparallel pair must equal $|\boldsymbol{S}|\omega_{\mathrm{AFMR}}/(2\omega_E + \omega_A)$. YIG, with 80 atoms in its magnetic cell, dominates the cost.

E.20.5 Two-magnon responses

At linear spin-wave order a ferromagnet at $T = 0$ has no two-magnon density response, while an antiferromagnet has a two-magnon continuum whose weight relative to the one-magnon line is 0.146 to 0.029 for $|\boldsymbol{S}| = 1/2$ to $5/2$ in a simple-cubic lattice (Sect. 14.1). The quartic terms of the Holstein–Primakoff expansion add a two-magnon interaction of relative order $1/|\boldsymbol{S}|$, which Sect. 14.2 resums as a ladder in the pair channel of total momentum $\boldsymbol{q}$; its poles below the continuum are two-magnon bound states. Whether NiO has such a state, and the interacting response $S^{(2m)}(\boldsymbol{q},\omega)$ in eV^2, are open; we want them, with the bound-state energies

in meV and their weights, for YIG, NiO, MnF_2 and, as an uncontrolled benchmark, α-$RuCl_3$. On $\mathfrak{T}$ the reference is linear spin-wave theory with the pair ladder, and for noncollinear order the cubic vertex is added.

Linear spin-wave theory with Colpa's diagonalization [143] (SpinW [144] or Sunny), with the inputs of Appendix E.20.4, gives the reference. The quartic vertex is derived symbolically from the spin Hamiltonian, and the Holstein–Primakoff and Dyson–Maleev vertices agree only on shell, so one of them is used throughout. The ladder is one linear solve per $(\boldsymbol{q}, \omega)$ on 30^3 to 40^3 grids, about 3×10^7 operations per $\boldsymbol{q}$, and exact diagonalization of the two-magnon sector in small cells checks it. The zero-point reduction ΔS, a sum over Bogoliubov coefficients, is converged to 10^{-3} on a 40^3 grid. The sum rule on $\langle \boldsymbol{S}^2 \rangle$ must hold to 1 %, and the bound-state energies must be stable to 1 % when the grid is doubled. YIG, with 20 iron ions per primitive cell, needs the full multi-sublattice transformation, and a matrix that is not positive definite signals a wrong classical ground state. The results enter Sect. 14.2.

E.20.6 Orbital form factors

Ions with unquenched orbital moments, the $4f$ ions and some $3d$ ions with strong spin–orbit coupling, scatter with form factors beyond the dipole form $\langle j_0 \rangle + c_2 \langle j_2 \rangle$ at $q \gtrsim 1/\langle r \rangle_{4f} \approx 3\,\mathrm{keV}$. Expanding $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ in spherical Bessel functions and applying the Wigner–Eckart theorem inside the ground J multiplet reduces the spin and orbital matrix elements to the radial integrals $\langle j_K(q) \rangle = \int r^2 \mathrm{d}r\, R_{n\ell}(r)^2 j_K(qr)$ with K even up to $2\ell = 6$, times angular coefficients. The integrals for $K = 0, \ldots, 6$ (dimensionless) and the multiplet-projected matrix elements are wanted for the ions of the magnetic targets of Sect. 14.1, with the crystal field treated as a perturbation within J; crystal-field mixing of J multiplets matters for the light rare earths. On $\mathfrak{T}$ this is a single-ion calculation in the ground multiplet at clamped nuclei, and the magnetic response that uses it is at linear spin-wave order. Dirac–Fock calculations with the General-purpose Relativistic Atomic Structure Package (GRASP) or fully relativistic DFT $+U$ Wannier projections give the radial functions. Dark matter that couples to the orbital moment couples to the orbital current, not to $\boldsymbol{\ell}$ times the density, and the operator is built that way. The integrals must satisfy $\langle j_0(0) \rangle = 1$ and the orbital-moment sum rules to 10^{-3}, and agree with the tabulated dipole form factors to 2 % for $q \leq 1\,\mathrm{keV}$.

E.20.7 Finite-momentum optical activity

In a chiral crystal the current response has an antisymmetric part odd in $\boldsymbol{q}$, $\mathrm{i}\epsilon_{\alpha\beta\gamma}\mathcal{G}^{\gamma}_{\mathrm{OA}}(\boldsymbol{q}, \omega)$ in $S_{j^\alpha j^\beta}$ (Eq. (14.35)), which changes sign between the two enantiomers and gives the difference of their rates in Sect. 14.5. DFPT gives only its slope at $q = 0$, the natural optical activity [145]. We want $\mathcal{G}_{\mathrm{OA}}(\boldsymbol{q}, \omega)$, with the units of $S_{j^\alpha j^\beta}$, and the full current tensor at $\pm\boldsymbol{q}$ for tellurium or α-quartz and for the magnetoelectric crystal Cr_2O_3, at $q \leq 10\,\mathrm{keV}$ and $\omega \leq 50\,\mathrm{eV}$. On $\mathfrak{T}$ the longitudinal part is at $\mathcal{X} = \mathrm{RPA}$ with local fields and the transverse part at independent particles with local fields, both with spin–orbit coupling. Noncollinear DFT with fully relativistic SG15 pseudopotentials gives the states. The current vertex $\langle \boldsymbol{k} + \boldsymbol{q} | \{ \boldsymbol{p} + \mathrm{i}[V_{\mathrm{NL}}, \boldsymbol{r}], \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \}/2 | \boldsymbol{k} \rangle$ needs the nonlocal commutator (Appendix E.7) and is evaluated on Wannier-interpolated states in the same gauge at $+\boldsymbol{q}$ and $-\boldsymbol{q}$, and Yambo adds the local fields of the longitudinal channel. The sign of the odd part follows the sign convention for $\boldsymbol{q}$ of Appendix A. The odd part must vanish to numerical noise in silicon, which is centrosymmetric, and its linear term must reproduce the DFPT optical activity; the grid goes from 16^3 to 24^3 until the odd part changes by less than 5 %.

E.20.8 $ZrTe_5$ and corrected Dirac reaches

The excitonic binding of a gapped Dirac material is set by $\mathrm{Ry_D} = \Delta_\mathrm{D}\tilde{\alpha}^2/4$, the Rydberg of a pair with reduced mass $\Delta_\mathrm{D}/(2v_\mathrm{D}^2)$ and coupling $\alpha/\varepsilon_\mathrm{bg}$, with $\tilde{\alpha} = \alpha/(\varepsilon_\mathrm{bg} v_\mathrm{D})$ the effective coupling of Eq. (7.43). For $ZrTe_5$ the central parameters are those of the measured crystal structure (Sect. 16.1.3, the first $ZrTe_5$ row of Table E.11): the half-gap $\Delta_\mathrm{D} = 11.8\,\mathrm{meV}$, the principal velocities 1.1×10^{-3}, 4.5×10^{-4} and 1.0×10^{-3} with geometric mean $v_\mathrm{D} = 7.9\times10^{-4}$, the energy cutoff $\tilde{\Lambda} = 0.191\,\mathrm{eV}$, $g_\mathrm{deg} = 2$, and $\varepsilon_\mathrm{bg} = 50.5$, the harmonic mean of the dielectric tensor. Then $\tilde{\alpha} \approx 0.18$, $\mathrm{Ry_D} \approx 0.10\,\mathrm{meV}$, and the exciton radius is $a_\mathrm{D} = 2v_\mathrm{D}/(\tilde{\alpha}\Delta_\mathrm{D}) \approx 1.4\times10^3\,\text{Å}$. The relaxed cell that Geilhufe et al. use in their rate [146, Tables 1 and 2], with $\Delta_\mathrm{D} = 15.6\,\mathrm{meV}$, gives $\tilde{\alpha} = 0.19$; the parameters of Hochberg et al. [147, Table 1] give 0.20, and their quoted 0.80 is $g_\mathrm{deg}\tilde{\alpha}$ with $g_\mathrm{deg} = 4$. A 24^3 grid over the Brillouin zone has a spacing of $1.9\times10^{-2}\,\text{Å}^{-1}$, 27 times $1/a_\mathrm{D}$, so an ab initio BSE cannot resolve the threshold. What is wanted instead is $L(\boldsymbol{q},\omega)$ of the anisotropic massive-Dirac model for $|\boldsymbol{q}| \le \Lambda_\mathrm{D}$ and $2\Delta_\mathrm{D} \le \omega \le 2\Delta_\mathrm{D} + 50\,\mathrm{meV}$ (at $\boldsymbol{q}\to0$; at finite $\boldsymbol{q}$ the invariant pair energy replaces ω), with the bound-state energies and weights. The box contains the region up to 1.33 times the edge where the Sommerfeld factor exceeds 1.5 (Sect. 16.1).

G_0W_0 with spin–orbit coupling on the 12-atom cell ($12\times12\times4$ grid, refined near Γ) fits the direction-resolved velocities $v_{\mathrm{D},i}$, the half-gap, the cutoffs $\Lambda_{\mathrm{D},i}$ and the background tensor $\varepsilon_{\mathrm{bg},ii}$. Since $\mathrm{Ry_D}$ lies far below the optical phonons, the right background is the static tensor with the ionic part, and the G_0W_0 stage computes it that way as its own check on the tabulated one; remote bands belong in ε_bg, not also in the GW screening. The threshold region, within about 10 $\mathrm{Ry_D}$ of $2\Delta_\mathrm{D}$, is the Elliott–Sommerfeld problem of an anisotropic hydrogen atom with masses $\Delta_\mathrm{D}/(2v_{\mathrm{D},i}^2)$ and dielectric tensor $\varepsilon_{\mathrm{bg},ii}$, solved on a radial grid. Away from threshold the BSE of the model, with the Coulomb interaction screened by ε_bg plus the RPA Dirac polarization, runs on a logarithmically graded radial grid from $k_\mathrm{min} \le 0.1/a_\mathrm{D}$ to Λ_D, about 200 radial points times 590 Lebedev directions, matched to the analytic solution over 5 to 10 $\mathrm{Ry_D}$; a uniform grid would give a spurious absence of bound states. The spread of $v_{\mathrm{D},i}$ and Δ_D between one-shot and eigenvalue-self-consistent GW, and between the measured-structure and relaxed rows, is propagated through both stages. On $\mathfrak{T}$ this is $\mathcal{X}$ = BSE with static screening, the controlled first correction in $\tilde{\alpha}$; beyond it no controlled method exists at $\tilde{\alpha}$ of order one, so that stage is a nuisance estimate. $\mathrm{Ry_D}$ and the first three bound states must be stable to 5 % when k_min is halved and the radial points doubled, the Sommerfeld factor must be continuous to 2 % across the matching window, and the model f-sum must hold to 10^{-3}. The model stages take minutes and the G_0W_0 about 10^4 core-hours; the same run gives ε_bg of Bi_2Se_3.

The published Dirac reaches [146, 147] left the background dielectric constant out of the total ε, which Sect. 16.1 shows overstates scattering reaches by $\varepsilon_\mathrm{bg}^2$ and absorption reaches by ε_bg in rate. For the isotropic model the corrected curves are vertical shifts, $\bar{\sigma}_e^\mathrm{lim} \to \varepsilon_\mathrm{bg}^2\bar{\sigma}_e^\mathrm{lim}$ and $\zeta^\mathrm{lim} \to \varepsilon_\mathrm{bg}^{1/2}\zeta^\mathrm{lim}$: factors 642 and 5.03 for Hochberg et al.'s $\varepsilon_\mathrm{bg} = 25.34$, and 1600 and 6.3 at $\varepsilon_\mathrm{bg} = 40$. Their curves keep the real part of the Dirac polarization, so the factor 0.54 to 0.61 between the full loss function and its leading form (Worked example 16.1) is already in them. For the anisotropic targets of Geilhufe et al. (Table E.11, one kilogram-year) the correction depends on $\hat{\boldsymbol{q}}$, and the corrected limit is the published one times the ratio of event counts $N_\mathrm{src}/N_\mathrm{corr}$, resolved in daily modulation. Along the axes of $ZrTe_5$ it is $\varepsilon_{\mathrm{bg},ii}^2$, that is 9.5×10^4, 431 and 1.6×10^4; averaged over the orientations of the crystal it is 5200 to 6400 for the central row and the book's halo, and 4800 to 6600 over both computed rows and two halos (Sect. 16.1.3). The recalculation re-implements their rate on a 590-direction Lebedev grid times a logarithmic grid in $|\boldsymbol{q}|$ and replaces their in-medium factor, which divides only the Dirac polarization by the background tensor, by $1/\varepsilon$ with $\varepsilon = \hat{\boldsymbol{q}}\cdot\varepsilon_\mathrm{bg}\cdot\hat{\boldsymbol{q}}$ minus the Dirac

Table E.11: Dirac parameters of the anisotropic targets [146, Tables 1 and 2]: half-gap, principal velocities (units of c), diagonal background dielectric tensor and degeneracy. For $ZrTe_5$ the first row is the measured crystal structure, the central row of Sect. 16.1.3; the second is the relaxed cell that Geilhufe et al. use in their rate, kept for the recalculation of their curve. The Yb_3PbO tensor has off-diagonal entries of -12.4 and 8.6.

material	Δ_{D} (meV)	$v_{\mathrm{D},x}, v_{\mathrm{D},y}, v_{\mathrm{D},z}$ (10^{-4})	$\varepsilon_{\mathrm{bg},xx}, \varepsilon_{\mathrm{bg},yy}, \varepsilon_{\mathrm{bg},zz}$	g_{deg}
$ZrTe_5$, measured structure	11.8	11, 4.5, 10	308.4, 20.75, 126.1	2
$ZrTe_5$, relaxed cell	15.6	11, 4.4, 9.1	308.4, 20.75, 126.1	2
Yb_3PbO	19.4	8.9, 8.9, 8.9	42.8, 42.8, 42.8	12
BNQ-TTF	5.0	2.0, 2.0, 1.0	18.7, 5.6, 10.3	8

polarization. It must first reproduce their published curve to 5 % with their own screening rule and their relaxed-cell row. Their Δ is the half-gap Δ_{D}, the anisotropic velocity factor along x is $q_x^2 v_{\mathrm{D},x}^2$ (a misprint in [146, Eq. (3.14)]), and the off-diagonal entries of the tensor are kept. The recalculation takes minutes and enters Sect. 16.1.

Table E.12: Magnetic, superconducting and Dirac calculations: targets, main grids, acceptance tests and cost in core-hours. KK, Kramers–Kronig.

quantity	targets	grids and parameters	acceptance test	cost
vertex fraction $\mathfrak{v}$, L^{cons}	Al, clean and $\ell_e = 1\,\mu$m, 300 nm, 30 nm	$y \in [2, 50]$; $qv_{\mathrm{F}}/\Delta_{\mathrm{BCS}} \le 10^3$; KK to $10^3 qv_{\mathrm{F}}$	$q \to 0$ cancellation; bare limit; edge values of Sect. 15.3	minutes
$\langle L \rangle$, P_Δ	a-W_xSi_{1-x}	512–1000 atoms, $\ge$5 quenches	f-sum 2 %; film σ_1	1.6×10^4–1.6×10^5
$\sigma_1^{\mathrm{S}}(\omega)$, α^2F_{tr}	Al, Nb	$\boldsymbol{k}$ 60^3, $\boldsymbol{Q}$ 30^3 (Al), refined small $\boldsymbol{Q}$	λ_{tr} 0.39; acoustic sum rule; T_c 10 %; optical σ_1	10^3
J^{ex}, anisotropy, Dzyaloshinskii–Moriya, B_1, B_2, g_{ion}	YIG, NiO, MnO, Cr_2O_3, $NiPS_3$	magnetic cell; strained cells	T_{N}, T_{C}; magnon bands; Larmor; two sublattices	10^3–5×10^3
$S^{(2m)}$, bound states	YIG, NiO, MnF_2	30^3–40^3	$\langle \boldsymbol{S}^2 \rangle$ 1 %; grid doubling	10^2
$\langle j_K(q) \rangle$	$4f$ and spin–orbit ions	$K \le 6$	$\langle j_0(0) \rangle = 1$; dipole tables	<10 per ion
$\mathcal{G}_{\mathrm{OA}}$	Te or α-SiO_2; Cr_2O_3	16^3–24^3	zero in Si	10^3–10^4
L near $2\Delta_{\mathrm{D}}$	$ZrTe_5$	log radial, 590 directions	Ry_{D} 5 %; f-sum	10^4
corrected reaches	$ZrTe_5$, BNQ-TTF, Yb_3PbO	$590 \times \log \lvert\boldsymbol{q}\rvert$	published curve 5 %	minutes

E.21 Dopants, color centers and small targets

The negatively charged nitrogen–vacancy center in diamond absorbs at its zero-phonon line, 1.945 eV, between two many-electron states, $^3A_2$ and ^3E, that no single Kohn–Sham determinant describes [148]. A density-functional transition density between them is therefore not even defined. The same problem, a final state that the band picture does not contain, recurs in donor clusters near the metal–insulator transition, molecular excitons with charge-transfer admixture, the ejected electron of a graphene sheet, the self-trapped exciton that becomes a Frenkel pair, and a molecule that dissociates in its cage (Sects. 16.2 to 16.6). Table E.13 summarizes the calculations.

E.21.1 Transition densities of the nitrogen–vacancy center

In an active-space description with orbitals $\{\varphi_i\}$ and many-body states $|I\rangle$, $|F\rangle$, the transition density is a contraction of the one-body transition matrix $\gamma_{ij}^{FI} = \langle F | c_i^\dagger c_j | I \rangle$ (spin-summed) with orbital form factors,

$$\langle F | n_e(\boldsymbol{q}) | I \rangle = \sum_{ij \in \mathrm{act}} \gamma_{ij}^{FI} \, \langle \varphi_i | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \varphi_j \rangle, \qquad F \neq I, \tag{E.104}$$

and the environment enters only through the screening $(\varepsilon^{-1}_{\mathrm{host}})_{\boldsymbol{G}\boldsymbol{0}}$ of Sect. 16.6, which is the hypothesis of the active space. The spin-flip transitions to the singlets ^1E and $^1A_1$ use the spin-density analogue of γ^{FI} and vanish linearly in q. We tabulate the dimensionless $|\langle F | \tilde{n}_e(\boldsymbol{q}) | I \rangle|^2$ between the four multiplets $^3A_2$, ^3E, $^1A_1$ and ^1E, on $|\boldsymbol{q}|$ from 0.5 to 30 keV times 590 Lebedev directions. On the truncation label $\mathfrak{T}$ (Sect. 8.5) this is $n_g = 0$, the Condon

limit, with a many-body approximation equivalent to the BSE within the active space; the vibronic sidebands of Sect. 16.6 are added separately.

Supercells of 512 and 1000 atoms are relaxed and solved with PBE, then HSE (Appendix E.10), with SG15 carbon and nitrogen at 80 Ry and Γ-point sampling, checked on a 2^3 grid in the 512-atom cell. Quantum defect embedding with WEST (Without Empty STates) or the constrained RPA builds an effective Hamiltonian on 4 to 6 defect orbitals, which full configuration interaction diagonalizes, and Eq. (E.104) is formed from γ^{FI} and the plane-wave form factors of the Kohn–Sham orbitals by FFT on a real-space grid with spacing $\Delta r \le \pi/q_{\max}$. The multiplet energies must reproduce the zero-phonon line and the measured 1.19 eV singlet transition [148] to 0.05 eV, and $|\langle F|\tilde{n}_e|I\rangle|^2$ must change by less than 5 % when the active space is enlarged and when the cell grows from 512 to 1000 atoms; if it changes more, a 1728-atom PBE cell is added as a third size. Embedding double counts the screened interaction unless the double-counting correction is consistent with the functional. The orbitals relax between $^3A_2$ and ^3E, so the two states need common orbitals or a biorthogonal correction. The SG15 pseudo-orbitals of carbon are unphysical above about 25 keV, so the top of the range needs the all-electron reconstruction of Appendix E.2, or the table stops there. The hybrid functional and the embedding dominate a cost of about 10^5 core-hours.

E.21.2 Disorder-averaged donor responses

Phosphorus donors in silicon at $n_D = 10^{18}$ and $1.8 \times 10^{18}\,\mathrm{cm}^{-3}$ sit at $n_D/n_c \approx 0.3$–0.6, where the dilute response of Sect. 16.5 fails at order one near the edge: donor pairs and clusters broaden the lines into an impurity band. What replaces it is the configurational average $\langle S^D(\boldsymbol{q},\omega)\rangle$ per donor, in eV^{-1}, the per-cell convention of Appendix E.17 with the donor in place of the cell, at $n_B = 0$ and $n_g = 0$ on $\mathfrak{T}$, with a Hartree mean field on the donor electrons and host screening. Each donor carries the six-valley orbitals of Sect. 16.5, split into A_1, E and T_2, at Poisson-random positions. The intervalley terms of the transition density sit at $\boldsymbol{q} = \boldsymbol{k}_\iota - \boldsymbol{k}_{\iota'} + \boldsymbol{G}$, with the valley minima at $k_0 = 0.85(2\pi/a) = 1.94$ keV: at 0.685 (after an umklapp), 3.60 and 3.88 keV for opposite valleys, and at 2.33 and 2.74 keV for perpendicular ones. The lightest particle that delivers the 11.70 meV of the $A_1 \to T_2$ line at 0.685 keV has 130 keV, and at 2.33 keV it has 441 keV (Sect. 16.5), so the $1s(A_1) \to 1s(E, T_2)$ lines, absent in optics, appear at finite q for MeV-scale dark matter.

A Hubbard–Anderson tight-binding model of the random donor ensemble carries the disorder. Its hopping comes from the overlaps of the multivalley envelopes (effective Bohr radius a^*, with the central-cell correction of Sect. 16.5 on each site), and it is solved for 10^4 to 10^5 sites by Lanczos or the kernel polynomial method and averaged over 100 realizations, for about 10^3 core-hours. DFT on 1000- to 4000-atom silicon clusters with two to four phosphorus atoms, at the shared 60 Ry setting of Appendix E.17, checks the central-cell part for about 10^4 more. The ensemble has to satisfy the Thomas–Reiche–Kuhn sum per donor, return the hydrogenic oscillator strength of Sect. 16.5 in the dilute limit, and give $\langle S^D\rangle$ stable to 5 % in the number of sites and realizations. The hopping oscillates with distance because of valley interference, so a single-valley envelope is wrong; compensation changes the occupation and must be stated. The Bloch functions at the conduction minima, on the same grid, also give the coefficients $\mathcal{U}_{j'j}(\boldsymbol{G})$ that set the strengths of the intervalley lines of Sect. 16.5, and with the dielectric matrix of the host from Appendix E.6 they give the local-field ratio $\mathfrak{f}_{\mathrm{D}}^{\mathrm{loc}}$ of Eq. (16.36), which Sect. 16.5 takes as one.

The oscillator-strength bracket of the dopant proposal of Du et al. [149] is printed with $3\tilde{q}$ for $3\tilde{q}^2$ (Sect. 16.5), which with the host ε held fixed changes the estimated heavy-mediator rate by factors of about 0.6 to 1.5 over 0.02 to 10 MeV and leaves absorption unaffected. Redoing the integral with their Lindhard $\varepsilon(q,\omega)$ gives the corrected curves; it takes minutes.

E.21.3 Molecular crystals

In anthracene and trans-stilbene the lowest excitons are Frenkel-like, localized on one molecule, but they mix with charge-transfer states in which the electron sits on a neighbor. A two-state model with transfer integral t_{CT} and energy splitting $E_{\mathrm{CT}} - E_{\mathrm{Fr}}$ gives a mixing angle $\tan 2\theta_{\mathrm{CT}} = 2t_{\mathrm{CT}}/(E_{\mathrm{CT}} - E_{\mathrm{Fr}})$ (Sect. 16.4), and the charge-transfer form factor carries the intermolecular phase, which no Frenkel model has; for orthogonal orbitals it vanishes at $q = 0$ and enters at order $q^2 d^2$, with d the intermolecular distance. The periodic Bethe–Salpeter response measures how much this matters: we tabulate its difference $\Delta S(\boldsymbol{q}, \omega)$ from the zero-overlap Frenkel response, the product $\Omega S(\boldsymbol{q}, \omega)$ on $|\boldsymbol{q}| \leq 30\,\mathrm{keV}$ times 590 directions, the charge-transfer weight θ^2_{CT}, and, as the summary, the largest relative change of $|f|^2$ over q for each state. The periodic calculation sits at $\mathcal{X} = \mathrm{BSE}$ with intermolecular overlap, and the Frenkel model is its zero-overlap truncation.

The periodic calculation is G_0W_0 plus BSE (Yambo or BerkeleyGW, Appendix E.6) on the measured crystal structures from the Cambridge Structural Database, with 50 to 100 atoms per cell and a 4^3 grid at finite q. Anthracene has 66 and trans-stilbene 136 occupied bands, so G_0W_0 needs about 1000 bands, or a Sternheimer formulation that needs no empty states; the BSE needs about 200. The molecular inputs of the Frenkel model come from TDDFT or molecular GW–BSE (PySCF or ORCA), and the Pariser–Parr–Pople model with standard parameters gives the semi-empirical comparison. The Frenkel code takes the transition densities on a real-space grid with $\Delta r \leq \pi/q_{\max} \approx 0.4\,a_0$ at 30 keV, Fourier transforms them, builds the intermolecular coupling by Ewald summation with the $\boldsymbol{G} = 0$ term removed (it belongs to ε), and diagonalizes the Frenkel Hamiltonian for about 30 states up to 10 eV. The exciton energies must be stable to 50 meV under doubling of bands and grid, the partial f-sum per molecule must hold to 2 %, and $|f|^2$ at $q = \alpha m_e$ must be stable to 5 % in the basis. Relaxed van der Waals geometries are not good enough, which is why measured structures are used; the BSE exchange term must carry local fields consistently; and the π-electron model fails at $q \gtrsim 10\,\mathrm{keV}$, where the σ electrons contribute. Each crystal costs 10^3 to 10^4 core-hours, and its tables about 10.

E.21.4 Low-dimensional response tables

Nanotubes, graphene sheets, layered stacks and quantum dots each screen with their own Coulomb kernel (Sect. 16.2). For sheets and stacks we tabulate $S(q_\parallel, q_z, \omega)$, for tubes of radius r_{t} $S(q_\parallel, q_\perp, \omega)$, directional and per unit area of sheet or length of tube in the normalization of Sect. 4.2 (with $2\pi\delta$ in S), and for dots $S^{\mathrm{QD}}(\boldsymbol{q}, \omega)$ in the dilute embedding of Sect. 16.3. Two inputs are reported with them: the proper density response $\tilde{R}^{\mathrm{2D}}$ of the monolayer per unit area, from Yambo with a two-dimensional Coulomb cutoff or from a Wannier bubble, and the Keldysh length r_0 defined by its static remote-band part, $-\tilde{R}^{\mathrm{2D}}_{\mathrm{bg}} = 2r_0 q_\parallel^2/e^2$ (Sect. 16.2). The tables sit at $\mathcal{X} = \mathrm{RPA}$ on the projected low-energy bands, with the remote bands entering only through r_0 as a static background. The RPA is solved in one and two dimensions with an arbitrary kernel, the inter-unit lattice sums and the transverse form factor, on $q_\parallel$ from 10^{-4} to 10 keV (200 logarithmic points) and ω up to v_D/r_{t} for tubes or 2 eV for sheets with at least 2000 points; stacks of period a_s keep q_z unreduced, with at least 64 points per $2\pi/a_s$. The interlayer spacing and the substrate $\varepsilon_b(\omega)$ with its polar phonons complete the input.

A semiconducting tube screens its electron–hole pairs weakly, so its lowest excitations are excitons, which the RPA does not contain. For such a tube the RPA step is followed by a ladder (Bethe–Salpeter) step on the same projected bands, with the interaction of the tube statically screened by the projected response, which gives the exciton energies and their form factors along $q_\parallel$. The lowest exciton energy must be stable to 10 meV when the bands and

the $q_\parallel$ grid are doubled and must reproduce the measured optical transition of the same tube, and the ladder adds 10^2 to 10^3 core-hours per chirality.

Analytic limits accept the response module. The band-projected f-sum of the linear-band model, $\int \omega S \, \mathrm{d}\omega = N_{\mathrm{ch}} v_D q^2$ per unit length of tube in that normalization, holds to 10^{-3}. The undoped Dirac sheet reproduces its closed-form bubble to 10^{-4}, a finite stack of about 50 layers converges to the infinite-stack form to 1 %, and the Yambo slab calculation with a Coulomb cutoff cross-checks one sheet. A metallic nanotube couples to dark matter through its charge mode only, with all four spin and valley channels; a one-channel model is low by a factor $N_{\mathrm{ch}} = 4$ at fixed Luttinger parameter. For dots, the envelope-function solver (spherical-well levels up to $\ell = 20$ and $n = 10$) and the dielectric-confinement factors of Sect. 16.3 are checked against tight-binding or DFT clusters of up to 2000 atoms. The model module runs in less than one core-hour; the DFT bubbles of real sheets cost 10^2 to 10^3 core-hours and the dot clusters 10^4.

E.21.5 Ejection factors of graphene and nanotubes

A plane-wave final state is not orthogonal to the occupied orbital it leaves, and the overlap adds a spurious monopole term to the ejection amplitude. Orthogonalizing the plane wave to the occupied states removes it (Sect. 16.3). For a carbon $2p_z$ orbital of Slater form with the exponent $1.568/a_0$ of Clementi and Raimondi, more diffuse than the hydrogenic orbital of Sect. 16.3, the ratio of the orthogonalized to the plain squared amplitude ranges from 0.4 to 4.3 at kinetic energies of 50 to 100 eV and reaches up to 740 at 10 eV, depending on q and on the direction. The large values occur where q far exceeds the final momentum and the plain amplitude has already fallen as $|\boldsymbol{p} - \boldsymbol{q}|^{-5}$, its square as $|\boldsymbol{p} - \boldsymbol{q}|^{-10}$, so a table of such ratios is not a rate correction. A rate needs the factor weighted by the halo,

$$R_{\mathrm{OPW}}(E_{\mathrm{kin}}) = \frac{\int \mathrm{d}q \, q \, \eta \, F_{\mathrm{DM}}^2 \sum_{\mathrm{occ}} |M_{\mathrm{OPW}}|^2}{\int \mathrm{d}q \, q \, \eta \, F_{\mathrm{DM}}^2 \sum_{\mathrm{occ}} |M_{\mathrm{PW}}|^2}, \tag{E.105}$$

dimensionless, per occupied orbital and in total, for graphene and nanotubes, with η the mean inverse speed of the standard halo. The factor is computed at $\mathcal{X} = \mathrm{IP}$ with frozen orbitals and a single continuum orthogonalized to all occupied states, the carbon 1s and the σ and π bands.

The occupied states come from a DFT ground state (SG15) as maximally localized Wannier functions (Appendix E.5), with the 1s orbital from an all-electron atomic calculation, since the pseudopotential has none. Their plane-wave transforms and mutual overlaps come by FFT, and the integrals of Eq. (E.105) use 2×10^4 final directions on a Fibonacci set and a logarithmic q grid from 1 to 60 keV, which covers heavy-mediator kinematics to $m_\chi \approx 10$ MeV. Orthogonality holds to 10^{-6}, and R_{OPW} must change by less than 1 % when the directions or the q grid are doubled and the Wannier spread is reminimized; the whole calculation takes CPU minutes. The final-state interaction with the hole is a separate factor, bounded above by the Fermi factor of an unscreened point charge, Eq. (2.25) (7.3, 3.4 and 2.6 at 10, 50 and 100 eV), and computed in CPU hours with scattering states of the sheet's Hartree potential plus the screened hole. An orthogonalized plane wave is not a scattering state of the hole, so multiplying it by a Fermi factor double counts. Both factors replace the bounds of Sect. 16.3.

E.21.6 Ionic color centers

In LiF, NaF, MgO and LiH an electronic deposit can be stored promptly as a Frenkel pair: a self-trapped exciton converts within picoseconds into an F center and an H center. The yield factorizes as in Sect. 16.6, and two of its factors have not been computed: the probability

$p_{\mathrm{FH}}(T)$ that a self-trapped exciton becomes a pair at millikelvin, and the survival p_{surv} of the close pair against tunneling recombination. The calculation gives these two probabilities, the direction-resolved displacement threshold $E_d(\hat{\boldsymbol{r}})$ of the anion sublattice, the charge levels that set p_q and the F-band cross-sections, tabulated as yields per recoil energy. On $\mathfrak{T}$ the nuclear-recoil seed is in the impulse regime of $\mathcal{K}$ and the electronic seed at $\mathcal{X} = \mathrm{RPA}$ with local fields; the yield factors themselves lie outside $\mathfrak{T}$, as properties of the final state, and come from the dynamics below.

The adiabatic surfaces of the self-trapped exciton and the pair come from constrained-occupation hybrid DFT (ΔSCF with HSE or PBE0, the hybrid built on the functional of Perdew, Burke and Ernzerhof; exact-exchange fraction 0.25 and 0.40, both reported) in 64- to 216-atom cells, large enough for pair separations up to the third neighbor (Appendix E.10). Direct surface hopping on hybrid surfaces would cost 10^6 to 10^7 core-hours, so a machine-learned surface is fitted to the ΔSCF points and validated on held-out ones to 50 meV, and decoherence-corrected surface hopping runs on it with at least 10^3 trajectories, which fixes p_{FH} to 10 %. The self-trapped-exciton and F–H pair energies are converged to 0.1 eV between the 64- and 216-atom cells. The threshold $E_d(\hat{\boldsymbol{r}})$ comes from recoil molecular dynamics (Appendix E.11) over 10^4 zero-point-sampled directions, converged to 1 eV when the directions are doubled, and p_{surv} from the tunneling-path calculation of Appendix E.22 applied to the close pair. The F-band cross-section comes from a BSE on the defect supercell. The charged components need the finite-size corrections of Appendix E.10, and at millikelvin only the athermal channel acts. The ΔSCF points dominate a cost of about 10^5 core-hours per material. The calculation has a direct control: tagged neutron scattering on LiF at 10 mK followed by single-center optical readout measures the product of the recoil factors, and 5.9 keV X-rays measure p_{FH} through the quenching of the exciton luminescence.

E.21.7 Bond breaking in molecular solids

A nuclear recoil on one atom of a molecule splits, for an internal coordinate independent of the intermolecular potential, into a center-of-mass recoil of the molecule and an internal excitation at the reduced momentum $(\mu_{12}/M_1)\boldsymbol{q}$, with μ_{12} the reduced mass of the bond and M_1 the mass of the struck atom. The response is their convolution (Sect. 16.6), dissociation is the internal response above the bond energy, and in the crystal the fragments also have to escape the cage. The first target is anthracene, the molecular crystal above, with a C–H bond as the internal coordinate, and the calculation gives the internal response S_{int}, the center-of-mass response S_{CM} and the escape probability p_{esc} per internal energy and direction. The internal motion is in the impulse regime of $\mathcal{K}$ with a bound center of mass; at the relevant momenta, about 10^2 keV, the recoil-free fraction is essentially zero, so the center-of-mass recoil is impulsive with a Doppler width from its zero-point momentum.

The internal response comes from Morse or ab initio potential curves, the center-of-mass response from phonopy or path-integral moments (Appendix E.8), and p_{esc} from molecular dynamics with a machine-learned potential whose training set contains the dissociative region (Appendix E.11). With 10^3 trajectories per internal energy and direction, p_{esc} is known to about 3 %, for about 10^3 core-hours in all. In the free-molecule limit, S_{CM} replaced by free recoil and $p_{\mathrm{esc}} = 1$, the calculation must reproduce the dissociation rate of Essig et al. [150], and S_{CM} must satisfy the f-sum for the whole molecular mass. Like the color-center yields, the calculation has a measured counterpart: the photodissociation quantum yield of the same molecular crystal at millikelvin, with fragments detected by infrared absorption, measures p_{esc}.

Table E.13: Dopant, color-center, molecular and low-dimensional calculations: targets, main parameters, acceptance tests and cost in core-hours. CNT, carbon nanotube; ML, machine-learned.

quantity	targets	parameters	acceptance	cost
$\lvert\langle F\rvert\tilde{n}_e(\boldsymbol{q})\rvert I\rangle\rvert^2$	nitrogen–vacancy (NV) center NV^-	512–1000 atoms; 4–6 orbitals; $q \leq 25\,\mathrm{keV}$ (pseudo)	multiplet energies 0.05 eV	10^5
$\langle S^D(\boldsymbol{q},\omega)\rangle$, $\mathcal{U}_{j'j}(\boldsymbol{G})$	Si:P	10^4–10^5 sites; 100 realizations	sum per donor; dilute limit	10^3–10^4
ΔS, θ^2_{CT}, tables	anthracene, t-stilbene	4^3, 200 bands; 590 directions	partial f-sum 2 %	10^3–10^4
S of tubes, sheets, stacks, dots; $\tilde{R}^{2\mathrm{D}}$, r_0	graphene, CNT, stacks, dots	200 $q_\parallel$; ≥2000 ω	projected f-sum 10^{-3}	<1 to 10^4
tube excitons	semiconducting CNT	ladder on projected bands	lowest exciton 10 meV	10^2–10^3
R_{OPW}	graphene, CNT	2×10^4 directions; q 1–60 keV	orthogonality 10^{-6}	minutes
p_{FH}, p_{surv}, $E_d(\hat{\boldsymbol{r}})$	LiF, NaF, MgO, LiH	64–216 atoms; ML surface; $\geq10^3$ trajectories	tagged neutrons, X-ray quenching	10^5
S_{int}, S_{CM}, p_{esc}	anthracene	10^3 trajectories per point	free-molecule limit	10^3

E.22 Defects and two-level systems

A group of atoms in fused silica that can sit in either of two configurations, a fraction of an ångström apart and equal in energy to within a meV, absorbs and scatters at micro-electronvolt energies. A close Frenkel pair in silicon stores a few eV and may give them back long after the event that made it. Sects. 16.7, 16.8 and 19.6 describe both kinds of defect with a few numbers. For defects in crystals they are the ohmic coupling α_{ohm} to the electrons of a metal, the tunneling exponents and Huang–Rhys factors of release paths, and the barriers and masses of crowdions and dumbbells. For glasses they are the density vertex of a two-level system and the distribution of two-level parameters. None of them has been computed for a real material at the level the text needs. Each calculation below is built from the methods of Appendices E.10 and E.11: relaxed supercells, nudged elastic bands, density-functional perturbation theory and molecular dynamics on machine-learned potentials. Table E.14 collects the targets, the tests and the costs.

E.22.1 Ohmic coupling of tunneling defects

A defect that hops between two configurations L and R in a normal metal changes the potential seen by the conduction electrons, and the Fermi sea has to rearrange at every hop. Sect. 16.8 turns this into a relaxation rate through the ohmic coupling α_{ohm}, and Appendix C.20 shows how it renormalizes the tunneling amplitude. We compute the dimensionless α_{ohm} of Eq. (E.73) from the phase-shift differences $\tilde{\delta}_\nu$, the eigenphases of $S_R S_L^\dagger$ at the Fermi energy. The targets are a crowdion in tungsten, hydrogen in niobium, and two-level systems in aluminum, its oxide and amorphous WSi. On the truncation label $\mathfrak{T}$ (Sect. 8.5) the calculation sits at the base point for the electrons, with Kohn–Sham scattering states at clamped nuclei, $n_B = 0$; the defect enters only through its two relaxed geometries.

Two routes give the phases. The first computes S_L and S_R directly, from the impurity Green's function of a Korringa–Kohn–Rostoker code (JuKKR or AKAIKKR, full potential,

angular momenta up to $l = 3$ or 4) with each relaxed configuration embedded in the host. The two matrices are expressed in one channel basis centered at the midpoint of the hop, and Eq. (E.73) is applied to the eigenphases. A basis centered on the defect is not enough: a hop displaces the scatterer, so for crowdions and hydrogen the eigenphase form holds and a single-center phase-shift sum does not.

The second route uses plane-wave supercells and the orthogonality catastrophe itself. The many-electron ground states of the two configurations in a cell of size L overlap as $|\langle\Psi_L|\Psi_R\rangle| \propto (k_\mathrm{F}L)^{-\frac{1}{2}\sum_{\sigma\nu}(\tilde{\delta}_\nu/\pi)^2}$ [89], and by Eq. (E.73) the exponent is $\frac{1}{2}\sum_{\sigma\nu}\tilde{\delta}_\nu^2/\pi^2 = \alpha_\mathrm{ohm}$. The slope of $\ln|\langle\Psi_L|\Psi_R\rangle|$ against $\ln L$ is therefore $-\alpha_\mathrm{ohm}$. The overlap is the determinant of the matrix of overlaps of the occupied Kohn–Sham orbitals of the two cells. For tungsten we take cubic cells of 54, 128 and 250 atoms ($3a$, $4a$ and $5a$) with the PBE functional, an optimized norm-conserving potential with the $5s$ and $5p$ semicore states and a cutoff of 60 Ry, all at Γ only. The length L of the slope is then the cell side itself; a $\boldsymbol{k}$ grid that shrinks as the cell grows would hold the Born–von Kármán length fixed and leave the slope undefined. A sparser fixed-density grid, the same in every cell, is the alternative, with L its Born–von Kármán length. The 250-atom cell holds about 2.1×10^5 plane waves and 2100 bands, about 7 GB per $\boldsymbol{k}$ point. Each defect costs about 10^3 hours on graphics processors.

The Born limit and one measurement accept the result. A contact coupling $\pm V_\mathrm{d}$ in a single channel returns $\alpha_\mathrm{ohm} = 4K_\mathrm{b}$ (Appendix E.10); the same code also reproduces the exponent of a one-dimensional tight-binding chain with one impurity site, whose phase shift is known in closed form. The measurement is the damping of aluminum-covered nanomechanical resonators, fitted with $K_\mathrm{b} = 0.07$ and 0.11 per spin [88], which the Born reading $\alpha_\mathrm{ohm} = 4K_\mathrm{b}$ turns into 0.28 and 0.44, with the spin counting of Appendix E.10. Beyond the Born limit the same contact potential gives $\alpha_\mathrm{ohm} = 0.19$ at $K_\mathrm{b} = 0.07$, and the measured rate needs a contact potential 32 % stronger than the Born estimate (Sect. 16.8). The computed α_ohm, which contains every order in the phase shifts, is therefore compared through the rate of Sect. 16.8 with the measured damping, not with the Born values 0.28 and 0.44. $\boldsymbol{k}$ sampling and smearing blur the Fermi surface on which the logarithm rests, so the overlaps must be taken with fixed occupations at a common Fermi level and the smearing extrapolated to zero. The host must be relaxed identically in both cells except near the defect, or the overlap measures the relaxation and not the hop. The computed couplings set the electron-assisted relaxation rates of Sect. 16.8 and, for tungsten, the coherence of crowdion motion in Sect. 19.6.

E.22.2 Release paths of eV-scale defects

Whether a close Frenkel pair in silicon or germanium, or a split interstitial in aluminum, releases its stored energy at millikelvin depends on the tunneling exponent through its barrier and on how many local vibrations accept the eV released in the hop (Sects. 16.8 and 19.6). Let $\Delta\boldsymbol{R}_I$ be the change of the position of atom I between the two quenched minima, and $\boldsymbol{e}_{k,I}$ the mass-weighted normal modes, of energies ω_k, at the final minimum. The displacement of mode k is $\Delta Q_k = \sum_I M_I^{1/2}\boldsymbol{e}_{k,I}\cdot\Delta\boldsymbol{R}_I$. A harmonic oscillator displaced by ΔQ_k stores $\frac{1}{2}\omega_k^2\Delta Q_k^2$, which in units of its quantum is the Huang–Rhys factor [124]

$$S_{\mathrm{HR},k} = \tfrac{1}{2}\,\omega_k\,\Delta Q_k^2, \qquad E_\mathrm{reorg} = \sum_k S_{\mathrm{HR},k}\,\omega_k. \tag{E.106}$$

These quantities lie on the Born–Oppenheimer surface, $n_B = 0$, harmonic about each minimum, with the hop treated either as a two-state transition or, when the reduction fails, by an instanton on the full surface. The reorganization energy E_reorg decides the regime. If it is small compared with the energy ε_hop released in the hop, the hop must put that energy into many quanta of weakly coupled modes, and the rate is suppressed by the energy-gap law. If $E_\mathrm{reorg} \approx \varepsilon_\mathrm{hop}$, the

reaction coordinate itself takes the energy and there is no such suppression. The tunneling exponent is $\lambda_{\mathrm{tun}} = \int \mathrm{d}s\,[2(V(s)-E)]^{1/2}$ along the mass-weighted minimum-energy path $V(s)$, and the zero-temperature ring-polymer instanton action equals $2\lambda_{\mathrm{tun}}$ when the path does not cut corners [151].

A release within a time t_{obs} needs the tunneling factor $\mathrm{e}^{-2\lambda_{\mathrm{tun}}}$ to beat $1/(\Gamma_0 t_{\mathrm{obs}})$, with Γ_0 the attempt rate of Appendix E.11. With $\Gamma_0 = 10^{13}\,\mathrm{s}^{-1}$ and one year, $\ln(\Gamma_0 t_{\mathrm{obs}}) = \ln(3.2\times10^{20}) = 47.2$, so in the strong-coupling regime, where no energy-gap factor is added, the pair relaxes only if $\lambda_{\mathrm{tun}} \le 23.6$. The two-parabola estimate $\lambda_{\mathrm{tun}} = (d/2)(2M_\kappa V)^{1/2}$ for a silicon atom crossing a barrier $V = 0.1\,\mathrm{eV}$ over $d = 1.5\,\text{Å}$ gives $(0.75/1973\,\mathrm{eV}^{-1})(2\times 26.2\,\mathrm{GeV}\times 0.1\,\mathrm{eV})^{1/2} = 27.5$. The pair is frozen unless the true exponent lies 14 % or more below this estimate, so λ_{tun} and $E_{\mathrm{reorg}}/\varepsilon_{\mathrm{hop}}$ are needed to about 5 %, for real paths.

A first calculation uses a 64-atom silicon cell with a $\langle 110\rangle$ split interstitial 2 to 4 Å from a vacancy. It uses Quantum ESPRESSO with PBE and optimized norm-conserving potentials, a cutoff of 40 Ry checked at 30 and 50 Ry, a 2^3 Monkhorst–Pack grid, and the cell fixed at the PBE lattice constant. The climbing-image nudged elastic band of Appendix E.11 runs with nine images to a force tolerance of 5 meV/Å. The Hessians at both minima come from finite displacements of 0.01 Å, $64\times3\times2 = 384$ calculations each. The path then gives $S_{\mathrm{HR},k}$, E_{reorg}, λ_{tun} and the line shape of the hop, and one ring-polymer instanton with 64 to 128 beads, run with i-PI on the machine-learned potential, checks the action against $2\lambda_{\mathrm{tun}}$. Its cost is set by the roughly 900 self-consistent calculations, about 120 core-hours, at under 0.1 GB per $\boldsymbol{k}$ point. The full catalogue needs 50 paths per material, mapped on the machine-learned potential and checked against density-functional theory. A kinetic Monte Carlo over the resulting rates at 10 to 100 mK, with 10^6 seeds, gives the release spectrum. The cost, mostly the instantons, is estimated at 10^4 to 10^5 hours on graphics processors.

The barrier and the exponent are converged in the cutoff at 30, 40 and 50 Ry, the grid at 2^3 and 3^3, and one path repeated in a 216-atom cell, since the 64-atom cell compresses the strain field of the pair. The catalogue is accepted by the release-path sum rule of Sect. 19.6: the energy released by all hops of the kinetic Monte Carlo, plus the energy still stored at the end, equals the initial stored energy, with every energy taken from the table of configurational energies of Appendix E.23. Charged close pairs need the finite-size corrections of Appendix E.10. A charge-state variant of the same path, computed by constrained occupations, gives the Huang–Rhys factor $S_{\mathrm{HR},R}$ and frequency ω_R of the reaction coordinate when a carrier is captured. The probability that a capture drives the reaction over a barrier E_{b} is then the tail above $n = E_{\mathrm{b}}/\omega_R$ of the Poisson series of Eq. (E.91). This is the athermal channel by which the carriers of a later event can release stored energy (Sect. 19.6).

E.22.3 Crowdions and dumbbells

A crowdion in tungsten is an extra atom spread along a $\langle 111\rangle$ row. It glides over a Peierls barrier of meV in atomistic calculations and of micro-electronvolts in the Frenkel–Kontorova model [152], and its kink mass M_{cr} is a fraction of an atomic mass (Appendix C.20). Sect. 19.6 asks whether it moves coherently at millikelvin. Coherent motion needs a band that survives three effects: the phonons that distinguish neighboring sites, the conduction electrons of the normal metal (bulk tungsten becomes superconducting only near 15 mK), and the strain field of the vacancy it left behind. The phonon dressing multiplies the hopping by $\mathrm{e}^{-S_{\mathrm{pol}}(T)}$, with

$$S_{\mathrm{pol}}(T) = \frac{1}{\pi}\int \mathrm{d}\omega\,\frac{J_{\mathrm{s}}(\omega)}{\omega^2}\coth\frac{\omega}{2T}, \tag{E.107}$$

for a linear coupling with spectral density J_{s} of Appendix C.20. For acoustic phonons $J_{\mathrm{s}} = J_{\mathrm{s},3}\omega^3$ up to the Debye energy ω_{D}, which gives a finite $S_{\mathrm{pol}}(0) = J_{\mathrm{s},3}\omega_{\mathrm{D}}^2/2\pi$ and a

thermal part $\pi J_{\mathrm{s},3} T^2/3$ (from $\int_0^\infty \mathrm{d}x\, x/(\mathrm{e}^x - 1) = \pi^2/6$). The electrons renormalize the hopping further by a power of α_{ohm} (Appendix C.20), with α_{ohm} computed as above for the crowdion geometry.

The crowdion sits on the Born–Oppenheimer surface, $n_B = 0$, with phonons and electrons coupled linearly to its position. The calculation for tungsten uses PBE, the semicore potential above, a 250-atom body-centered cubic cell, a cutoff of 50 Ry, a 3^3 grid and Marzari–Vanderbilt smearing of 0.01 Ry. A climbing-image nudged elastic band along $\langle 111 \rangle$ gives the Peierls profile $E(z_0)$ of the kink center. The Hessian of the crowdion core gives the phonon coupling and J_{s}. A fit of the profile to the Frenkel–Kontorova form gives the kink width and mass. The interaction energy with the parent vacancy as a function of separation gives the radius inside which the strain gradient across one spacing exceeds the band width, and the crowdion is bound. For aluminum the same steps give the migration of the $\langle 100 \rangle$ dumbbell, in cells of 108 to 256 atoms, and the one-dimensional glide of di- and tri-interstitials comes from machine-learned molecular dynamics at 10 to 100 K, extrapolated to low temperature with instanton rates. Each material costs about 10^5 core-hours.

The Peierls barrier is below a meV per cell, so the energies are differences at identical meshes and smearing, extrapolated to zero smearing and checked on a 4^3 grid, with noise below 0.1 meV per cell. The fit to the Frenkel–Kontorova form is valid only if the profile is sinusoidal; its higher harmonics measure the failure, and the kink width over the row spacing controls the continuum description (Appendix C.20). In Sect. 19.6 the barrier and mass of the kink replace the model values, and the polaron factor, the electron renormalization and the confinement radius decide whether the crowdion band is real.

E.22.4 Density vertex of a two-level system

Dark matter couples to a two-level system in a glass through the change of the electron density between its configurations,

$$f_{\mathrm{TLS}}(\boldsymbol{q}) = \langle \mathrm{R} | n_e(\boldsymbol{q}) | \mathrm{R} \rangle - \langle \mathrm{L} | n_e(\boldsymbol{q}) | \mathrm{L} \rangle, \tag{E.108}$$

which enters the scattering rate of Sect. 16.7 as $|f_{\mathrm{TLS}}|^2$. The estimate there moves one oxygen ion, eight electrons, rigidly by 1 Å, and misses the redistribution of the valence electrons, the screening of the dipole by the glass and the variety of real geometries. The vertex is a ground-state density at $n_B = 0$, with the adiabatic electrons following the atoms; the screening is that of the self-consistent density, and no further dielectric factor is applied.

The preferred method takes two self-consistent calculations of the same glass cell, relaxed in configurations L and R, and Fourier transforms the difference of the two densities at the reciprocal vectors $\boldsymbol{G}_{\mathrm{sc}}$ of the cell. It is exact to first order in the dark-matter coupling and keeps every order in the atomic displacement $\boldsymbol{d}$. That matters because qd reaches 0.5 to 1 at $q = 1$ to 2 keV for $d = 1$ Å, where a linearization errs at order $(qd)^2$. The alternative is density-functional perturbation theory at Γ of the supercell, with only the core atoms displaced. Every $\boldsymbol{G}_{\mathrm{sc}}$ is equivalent to Γ, so one response gives the density change at all of them, and the vertex is the sum over core atoms of the density response to each atom times its displacement from L to R. This route costs about ten times more and serves as a check on the relaxation field of the host.

The cells hold 216 atoms (72 formula units) of amorphous $\mathrm{SiO_2}$, a cube of side 15 Å. A two-level system found in the large glasses of the ensemble below is carved out with its environment to that size, the outer shell re-quenched with the core held fixed, and its two configurations re-relaxed by a nudged elastic band in the small cell; systems whose asymmetry moves by more than a meV in the transplant are discarded. The calculation uses PBE, optimized norm-conserving potentials, a wavefunction cutoff of 80 Ry checked at 70 and 90 Ry,

a density cutoff of 320 Ry, a self-consistency threshold of 10^{-10} and the Γ point only. The momentum transfers run from $2\pi/L = 0.42\,\text{Å}^{-1}$ (0.83 keV) to $5\,\text{Å}^{-1}$, and the 576 occupied bands take about 2.5 GB. Each self-consistent calculation costs about one hour on a graphics processor, so 100 two-level systems cost about 200 hours with the preferred method.

The vertex is converged when it changes by less than 2 % with the cutoff raised by 10 Ry and with the cell enlarged from 216 to 648 atoms for one system; the ensemble average $\langle |f_{\text{TLS}}|^2 \rangle$ needs at least 100 systems for 10 %. The benchmark is the rigid-ion value, which the computed vertex approaches at large q, where only the core electrons of the moving atoms contribute. Below $q = 2\pi/L$ the periodic cell forces the macroscopic field to vanish, so the cell gives the unscreened dipole of the core. The long-range part, the change of the electronic dipole screened by the high-frequency dielectric constant of the glass, must be added analytically at small q. Rebuilding the density change from Born effective charges places point charges where the vertex needs the valence density. Sect. 16.7 uses the ratio of the computed $\langle |f_{\text{TLS}}|^2 \rangle$ to the rigid-ion value, together with the interference between core and host for cores off the center of the defect, which vanishes only for symmetric cores.

E.22.5 Ensemble sampling of two-level systems

The standard tunneling model of Sect. 16.7 assumes a flat distribution of asymmetries and of tunneling exponents, with density $\bar{P}$, and Sect. 16.8 builds its relaxation rates on it. Testing that assumption for a specific glass needs the distribution itself, computed from the potential-energy surface of the glass. The targets are amorphous SiO_2, amorphous Al_2O_3 and amorphous WSi. For each we need the distributions of the asymmetry Δ_{as}, the tunneling amplitude Δ_0 of Appendix C.20, the deformation potential and the dipole moment. From them follow tables of the two-level response of Sect. 16.7 on at least 590 directions of $\boldsymbol{q}$, 60 logarithmic points in $|\boldsymbol{q}|$ from 0.1 eV to 10 keV, and 50 energies from 10^{-7} to 10^{-3} eV.

The two-level systems that matter, with asymmetries below a meV and tunneling exponents below about 20, are rare among pairs of neighboring minima. A study of the energy minima of a model glass found of order 10^3 active two-level systems after sampling some 10^6 minima [153]. The sampling is classical on the Born–Oppenheimer surface of the potential, $n_B = 0$, with tunneling added through the instanton path. The method follows that study. Glasses of 10^3 to 10^4 atoms are quenched with the machine-learned potential, at least 10^4 of them, with swap Monte Carlo or a comparable method to reach stable glasses. Molecular dynamics at low temperature visits the minima of each glass, pairs of adjacent minima with energy difference below 5 meV are kept, and a nudged elastic band connects each pair. The tunneling exponent comes from the instanton path; the deformation potential from the energy difference under a finite strain; the dipole moment from the Berry-phase polarization of the two configurations. To resolve at least 10^3 systems per decade of splitting, about 10^8 pairs of minima are screened, some 10^4 from each glass. The sampling costs 10^5 to 10^6 hours on graphics processors; the response tables, under ten core-hours.

The tables contain a principal value and a Hadamard finite part, which are implemented by subtracting the first two terms of the Taylor expansion of the integrand inside a unit interval. The ensemble average is converged to 1 %, and the finite part is stable to 10^{-3} when the inner interval is halved. The measured control is the acoustic loss. The computed tunneling strength $\bar{P}\gamma^2/\rho_T c^2$, with γ the deformation potential and c the sound speed, must reproduce the measured internal friction of the same glass [154]. The dielectric loss tangent $\tan\delta_0$ of Sect. 16.7 gives a second control through the dipole moments. The result depends most on how the glass is prepared. The density of two-level systems in the model glass fell by two orders of magnitude between its least and most stable preparations [153]. A glass quenched on the computer is far less stable than one cooled in the laboratory, so the computed $\bar{P}$ is

Table E.14: Defect and two-level calculations. Costs are estimates. GPU-h are hours on a graphics processing unit (GPU); KKR, Korringa–Kohn–Rostoker; NEB, nudged elastic band; SCF, self-consistent field; MD, molecular dynamics.

quantity	method and grid	acceptance	cost
α_{ohm} (W, Nb–H, Al, AlO_x, WSi)	KKR eigenphases; or supercell overlaps, 54–250 atoms at Γ	Born limit $4K_{\mathrm{b}}$; resonator damping (Al), beyond Born	10^3 GPU-h per defect
$S_{\mathrm{HR},k}$, E_{reorg}, λ_{tun} (Si, Ge pairs; Al)	NEB, Hessians, instantons; 64 and 216 atoms; 50 paths	release-path sum rule; λ_{tun} to 5 %	120 core-h (pilot); 10^4–10^5 GPU-h
Peierls profile, M_{cr}, S_{pol} (W); dumbbell glide (Al)	NEB, 250 atoms; MD at 10–100 K	noise < 0.1 meV per cell; sinusoidal profile	10^5 core-h per material
$\langle\lvert f_{\mathrm{TLS}}\rvert^2\rangle$ (a-SiO_2)	two SCFs per system, 216 atoms, 80 Ry	2 % per system; 100 systems for 10 %	200 GPU-h
Two-level ensemble (a-SiO_2, AlO_x, a-WSi)	10^8 pairs of minima; NEB; response tables	1 %; acoustic loss	10^5–10^6 GPU-h

meaningful only as a function of the stated preparation. In Sects. 16.7 and 16.8 the tables replace the flat distribution in the scattering and absorption rates, and the distributions of Δ_{as}, Δ_0 and the couplings test the universality that the standard tunneling model assumes.

E.23 Nuclear-recoil partition and charge yield

A silicon nucleus recoiling with 100 eV cannot create an electron–hole pair in a binary collision with a valence electron, since that needs a recoil above the binary threshold $E_{R,\mathrm{th}} \approx 352$ eV of Sect. 19.2. Yet the measured ionization yield at 100 eV is $0.102^{+0.034}_{-0.030}$, and the data show no sign of an ionization threshold above 100 eV [155, Table II]. In germanium a recoil of 254 eV, below its threshold of 399 eV, gives (64 ± 8) eV of electron-equivalent energy, a quenching factor of 25 ± 3 % [156]. The charge at these energies comes from processes that the Lindhard model of Sect. 19.3 does not contain, and the share of the recoil energy that stays in the lattice as stored energy is bounded by calibration data but not computed (Sect. 19.4). Single recoils need the electronic excitation in a close collision of two atoms, the partition of the recoil energy into heat, stored energy and charge, and the charge yield below the binary threshold; dense tracks need the interaction of their carriers, the recombination of the track plasma and the amorphous pockets left by recoils of keV. Molecular dynamics on machine-learned potentials (Appendix E.11) carries all of them except the excitation kernel and the track recombination. Table E.15 collects their grids, tests and costs.

E.23.1 Close-collision excitation kernel

Below $E_{R,\mathrm{th}}$, electrons are excited when two atoms approach so closely that the local gap of the distorted configuration closes. Levels of the quasi-molecule then cross, and an electron can be promoted at a crossing with a probability that falls exponentially with the square of the coupling between the crossing levels, divided by the velocity and the difference of their slopes [157, 158]. Sect. 19.2 writes this as a kernel, the distribution $\mathrm{d}P/\mathrm{d}\omega_e(b_{\mathrm{imp}}, E_{\mathrm{coll}})$ of the electronic excitation energy ω_e left behind when two Si or two Ge atoms collide at impact parameter b_{imp} and relative energy E_{coll}, in eV^{-1}. The grid has 10 logarithmic values of E_{coll} from 50 eV to 5 keV and 8 values of b_{imp} from 0 to 1.5 Å, 80 points in all.

The Born–Oppenheimer order is not small here, so the calculation sits off the perturbative axes of $\mathfrak{T}$ (Sect. 8.5): the nuclei are classical, the electrons are propagated in time. The Massey parameter $\xi_{\mathrm{M}} = E_{\mathrm{g}}\ell_{\mathrm{int}}/v_N$, with an interaction length $\ell_{\mathrm{int}} = 1\,\text{Å}$, measures adiabaticity. A silicon atom of 20 eV moves at $v_N = 11.7\,\mathrm{km/s}$ and has $\xi_{\mathrm{M}} = 1.17\,\mathrm{eV} \times 10^{-10}\,\mathrm{m}/(6.58 \times 10^{-16}\,\mathrm{eV\,s} \times 1.17 \times 10^{4}\,\mathrm{m/s}) \approx 15$, so nonadiabatic excitation is suppressed as e^{-15}. At 1 keV, $v_N = 83\,\mathrm{km/s}$ and $\xi_{\mathrm{M}} \approx 2$, and it is not.

Real-time time-dependent density-functional theory with Ehrenfest nuclei propagates the collision, in the real-space codes Octopus [159] and SALMON (Scalable Ab initio Light–Matter simulator for Optics and Nanoscience) or the plane-wave code Qb@ll. Cells hold 216 atoms (16.3 Å across) or more; the projectile is launched along random directions; and the excitation is counted by projecting the time-evolved orbitals on the conduction manifold of the ground state about 20 fs after the collision, 20 to 50 trajectories per point. The time step is at most 2 as near closest approach, the real-space grid at most 0.2 Å (or the equivalent plane-wave cutoff), and the pseudopotentials include semicore states and are hard enough for the close approach at $E_{\mathrm{coll}} \geq 1\,\mathrm{keV}$. The cell has to be at least 15 Å across, so that the wake does not re-enter; the 216-atom cell on a 0.2 Å grid has 5.4×10^5 points, and its 432 occupied orbitals take about 3.7 GB. A trajectory of about 30 fs costs some 250 core-hours, and the 1600 to 4000 trajectories of the grid about 10^6.

The adiabatic exchange–correlation functional misplaces level crossings that involve charge transfer, so the gap of the distorted configuration comes out too small. The theory uncertainty is therefore the spread between three treatments of a 10 % subset of trajectories, which costs a few times 10^6 core-hours more because the hybrid functional is some thirty times dearer per step: Ehrenfest with a hybrid functional (HSE), decoherence-corrected surface hopping [160], and Landau–Zener probabilities from diabatic couplings of the Si_2 or Ge_2 quasi-molecule computed by multireference configuration interaction. The hybrid arm serves silicon only. In germanium HSE06 at the standard exchange fraction of one quarter inverts the Γ and L valleys and fails the band check of Appendix E.14 (valley order, both gaps, masses at L and Γ), so a germanium hybrid arm would measure that defect together with the local-gap error; the germanium spread rests on the other two treatments until a functional passes the check. Ehrenfest dynamics gives the mean excitation, not its distribution, so the kernel is built from the projection statistics, never from the mean electronic energy. Above $E_{R,\mathrm{th}}$ the mean energy lost per unit path must approach the electronic stopping that the loss function gives (Sect. 19.3). The kernel supplies the charge channel of slow recoils in Sect. 19.2 and the stochastic excitation events of the next two subsections.

E.23.2 Partition tables

Sect. 19.1 splits the recoil energy into heat E_{heat}, all of the phonon energy, stored energy E_{st} and charge, and Sect. 19.4 bounds the stored share with calibration data. The calculation tabulates the joint distribution $P(E_{\mathrm{heat}}, E_{\mathrm{st}}, N_{\mathrm{eh}} | E_R, \hat{\boldsymbol{r}}_R, T)$ for silicon and germanium on E_R from 1 eV to 10 keV (40 logarithmic points), 100 recoil directions, and 100 initial conditions per direction, drawn from the zero-point and thermal distribution of the lattice. It also gives the threshold surface $E_{\mathrm{d}}(\hat{\boldsymbol{r}}_R)$ and the Frenkel-pair probability $p_{\mathrm{FP}}(E_R, \hat{\boldsymbol{r}}_R)$ of Appendix E.11, and the spectrum of the phonons the recoil creates. On $\mathfrak{T}$ the dynamics is Born–Oppenheimer, $n_B = 0$, fully anharmonic and classical for the nuclei. This treatment is controlled between the quantum–classical boundary at 0.41 eV (Sect. 19.1) and E_{d}, 12.5 to 36 eV by direction, where $\xi_{\mathrm{M}} \gg 1$; above, the close-collision kernel is attached as stochastic events.

The potential is a machine-learned interatomic potential (MLIP) of the fast linear atomic cluster expansion (ACE) family, spliced to the screened Coulomb repulsion as in Appendix E.11. It is trained on about 10^4 configurations computed with PBE and SCAN: dimers down to

0.5 Å, point defects, close pairs, amorphous and liquid structures, and snapshots of recoils. It is accepted at force errors below 50 meV/Å and energy errors below 3 meV per atom. E_{d} and p_{FP} follow the protocol of Appendix E.11, its directions, energy steps, cell, border and run length, with this potential. What differs is the quench at t_{pr} below and a check of twenty directions by Born–Oppenheimer molecular dynamics in density-functional theory, 512 atoms, about 10^5 core-hours. The partition tables use cells from 10^3 atoms below 100 eV to 10^5 atoms at 10 keV, run for 1 to 10 ps, and cost 10^4 to 10^5 hours on graphics processors per material.

The stored energy is defined by the quench of Appendix E.11, Eq. (E.79), at a time t_{pr} of 10 to 100 ps after the recoil, by steepest descent or the fast inertial relaxation engine (FIRE) [161] at fixed cell, without annealing. Its reference is the relaxed perfect cell, computed once for each potential, cell and boundary condition, and never the initial configuration of the run. The difference is large. The initial atoms are drawn from the zero-point distribution and carry half of the zero-point energy as potential energy. Sect. 5.3 gives a zero-point energy of 123 meV per cell of silicon, so the initial configuration lies 31 meV per atom, or 126 eV in a 4096-atom cell, above the minimum, 17 to 19 times the 6.8 to 7.5 eV that a close Frenkel pair stores (Sect. 19.1). The change of zero-point energy between the end basin and the perfect crystal comes from the Hessians at both minima; it is booked in the heat, not in E_{st}. One table of configurational energies, with one potential, cell and quench, serves the stored energy here, the end states of the release paths of Appendix E.22 and the pockets below; otherwise the end states and the stored energies disagree by the potential's error on each defect. The spread of its entries between the potential and density-functional theory is the potential's uncertainty on E_{st}.

The created-phonon spectrum is recorded at t_{pr}, before the quench. The excess lattice state is projected on the harmonic modes s of the quenched end state, $E_s = \frac{1}{2}(p_s^2 + \omega_s^2 u_s^2)$, and the same projection of the initial state is subtracted to remove the zero-point spectrum. The result is binned at 0.5 meV, with branch labels from the overlap of eigenvectors with those of the perfect crystal. The energy that the projection misses, the anharmonic residual, must stay below 10 %. The energy absorbed by the boundary thermostat is recorded with its spectrum, from the flux into the thermostat layer, or the spectrum is taken before the front reaches it. The acceptance test for the spectrum is the hand-over of Sect. 19.1: for recoils small enough to stay harmonic, the created minus absorbed spectrum must reproduce the phonon spectrum of the nuclear response of Sect. 11.3. The tables give the partition of Sect. 19.1 and the computed stored energy that Sect. 19.4 sets against its bounds.

E.23.3 Charge yield below the binary threshold

The measured silicon yields at 100 to 460 eV straddle $E_{R,\mathrm{th}}$, and no calculation yet predicts them (Sect. 19.2). Predicting them needs the full $P(N_{\mathrm{eh}}|E_R, \hat{\boldsymbol{r}}_R, T)$ for $N_{\mathrm{eh}} \leq 30$ on E_R from 20 eV to 1 keV, with at least 25 logarithmic points, dense near 352 and 399 eV. From it follow the yield $Y_q(E_R) = \langle N_{\mathrm{eh}} \rangle \varepsilon_{\mathrm{eh}} / E_R$, the probability $P(N_{\mathrm{eh}} = 0|E_R)$ of a recoil without charge, the Fano factor $F_{\mathrm{NR}}(E_R)$ of the nuclear-recoil charge, and the correlations of N_{eh} with the heat and with E_{st} at fixed E_R, which Sect. 19.5 needs.

The trajectories are those of the partition tables, with excitation events drawn from the close-collision kernel at every close approach and the electronic friction of the potential switched off where the kernel acts. Three channels are flagged separately: binary excitation above $E_{R,\mathrm{th}}$, Landau–Zener promotion, and linear-response excitation through the high-momentum tail of the valence density, which makes the binary threshold soft (Sect. 19.2). Each excitation ω_e is converted into pairs with the cascade distribution P_{casc} at that seed (Appendices E.13 and E.24). Slow-ion excitations lie near the gap, where the cascade is far from its asymptote, so dividing ω_e by the high-energy $\varepsilon_{\mathrm{eh}}$ is wrong. The Fano factor follows from the law of total

variance, $\operatorname{Var} N_{\mathrm{eh}} = \mathbb{E}[\operatorname{Var}(N_{\mathrm{eh}}|\mathrm{seeds})] + \operatorname{Var}(\mathbb{E}[N_{\mathrm{eh}}|\mathrm{seeds}])$, which separates the fluctuations of the cascade from those of the excitation events.

The number of trajectories per point is set by the variance. For a sample of n values with excess kurtosis κ, the relative error of the sample variance is $[2/(n-1) + \kappa/n]^{1/2}$; for 6 % at $\kappa = 2$ this needs $n \geq 1112$. The cost lies within the molecular-dynamics budget of the partition tables, about 10^3 to 5×10^3 hours on graphics processors. The result is validated, not fitted, against the silicon data: $Y_q = 0.102^{+0.034}_{-0.030}$ and $F_{\mathrm{NR}} = 0.9^{+0.7}_{-0.4}$ at 0.10 keV, $0.108^{+0.009}_{-0.010}$ and $0.5^{+0.2}_{-0.1}$ at 0.22 keV, $0.136^{+0.009}_{-0.008}$ and $1.8^{+0.6}_{-0.5}$ at 0.46 keV [155, Table II], and against the germanium point at 254 eV [156]. The silicon yields assume 3.8 eV per pair and a total phonon energy equal to the recoil energy plus the work of the pairs in the bias field, so that no energy is stored [155, Eq. (1)]. The computed yield is converted to the same definition before the comparison. The stored energy, at most a few percent of E_R (Sect. 19.4), lies inside their errors.

E.23.4 Dense tracks

The cascade of Sect. 18.1 treats the pairs of one event as independent, each subtree evolving alone. At keV, a nuclear-recoil track packs its carriers so densely that they interact: the independence is the zeroth order of an expansion in the pair density times the exciton volume, $n_{\mathrm{pair}} a_x^3$, and the first correction is a carrier–carrier collision term. In germanium the lattice itself can melt locally in a thermal spike, beyond the description of the ion as a sequence of binary collisions. The calculation gives the ratio of $P(N_{\mathrm{eh}}|E_R)$ with and without interactions and the change of F_{NR}, per recoil energy and species, which Sect. 19.1 uses to bound the independence of the subtrees.

The carriers are followed by ensemble Monte Carlo with molecular-dynamics Coulomb forces among all of them, computed by the particle–particle particle–mesh method, seeded with the end-of-cascade distribution and the geometry of the track, with screened impact ionization and Auger recombination as extra channels. The lattice is followed by machine-learned molecular dynamics of the recoil coupled to a two-temperature model of the electrons [162], with the electron–phonon coupling from the vertex of Appendix E.5. The time step is 0.1 fs, the Coulomb mesh at most 1 nm, the electron-temperature grid at most 1 nm, and at least 10^3 tracks are run per energy and species. The dynamics costs about 10^4 hours on graphics processors and the carriers 10^3 core-hours. In the dilute limit, with the carrier density scaled down, the calculation must recover the independent cascade. Three errors must be avoided: the electronic stopping friction of the potential and the close-collision kernel must not both act; carriers that leave the molecular-dynamics box must be continued in the cascade, not dropped; and the thermostat must not absorb the heat of the spike before it has spread.

E.23.5 Track recombination

A dense track is a plasma of electrons and holes that shields an applied field from its interior and loses carriers by Auger recombination, at a rate $C_{\mathrm{Aug}} n^3$ per volume. A first estimate shows which parameters matter. Take a cylindrical Gaussian track of λ_{tr} pairs per length and initial radius r_{tr}, spreading by ambipolar diffusion with constant D_{amb} at zero field. Its density is $n = \lambda_{\mathrm{tr}} \mathrm{e}^{-\rho^2/s}/\pi s$ with $s = r_{\mathrm{tr}}^2 + 4 D_{\mathrm{amb}} t$, and $\int \mathrm{d}^2\rho\, n^3 = \lambda_{\mathrm{tr}}^3/3\pi^2 s^2$. To first order in C_{Aug} the fraction lost is

$$f_{\mathrm{A}} = \frac{C_{\mathrm{Aug}} \lambda_{\mathrm{tr}}^2}{3\pi^2} \int_0^\infty \frac{\mathrm{d}t}{(r_{\mathrm{tr}}^2 + 4 D_{\mathrm{amb}} t)^2} = \frac{C_{\mathrm{Aug}} \lambda_{\mathrm{tr}}^2}{12 \pi^2 D_{\mathrm{amb}} r_{\mathrm{tr}}^2} . \tag{E.109}$$

The loss is set at early times, as $1/r_{\mathrm{tr}}^2$, so the initial radius from the dense-track and pocket calculations matters more than the late diffusion. In a biased detector the plasma screens the

field until diffusion erodes it; the time this takes, the plasma time, delays the induced current.

The full calculation solves drift–diffusion and Poisson equations with rate terms for Auger recombination, exciton formation and Shockley–Read–Hall capture on the geometry of the track, two-dimensional with axial symmetry, with a mesh of at most 1 nm near the core and fields from 0 to 250 V/cm. It gives $f_{\mathrm{A}}(E_R, \mathcal{E}_{\mathrm{b}})$, with $\mathcal{E}_{\mathrm{b}}$ the bias field, the plasma time and the delay of the induced current, for under 10^3 core-hours, and it must reproduce Eq. (E.109) for $f_{\mathrm{A}} \ll 0.1$; above $f_{\mathrm{A}} \approx 0.1$ the loss is nonlinear and only the full solution holds. The weak input is C_{Aug}, which has not been measured at millikelvin in silicon or germanium; it is carried as a stated parameter, scanned from a tenth to ten times its room-temperature value. Exciton formation in the track does not by itself remove gap energy, since a bound pair still holds $E_{\mathrm{g}} - E_x^b$; only loss channels, Auger recombination and exciton–exciton annihilation, do. The computed loss and its field dependence feed the dense-track charge of Sect. 19.1 and the kinetics of Appendix E.25.

E.23.6 Amorphous pockets

Recoils of keV in silicon and germanium leave small amorphous pockets as well as point defects. A pocket that recrystallizes later releases its stored energy at once, and Sect. 19.6 counts such releases among the candidate sources of the low-energy excess. A release is visible only if its energy, after collection, exceeds the threshold; with a recrystallization energy of order 0.1 eV per atom, an estimate, a threshold of tens of eV needs pockets of hundreds of atoms. The calculation gives the size distribution $P(N_a | E_R, \hat{\boldsymbol{r}}_R)$ of the number N_a of atoms in pockets at t_{pr}, and the spectrum of recrystallization barriers.

The calculation sits at the same place on $\mathfrak{T}$ as the partition tables. The cascades are run with the spliced machine-learned potential at E_R from 0.2 to 20 keV, at least 10^3 per energy and material, in cells of 10^5 to 10^6 atoms for 10 to 50 ps with a boundary thermostat, and quenched as above. A pocket is a connected set of atoms whose local order parameter, a bond-angle distribution or a Steinhardt parameter, falls below a crystalline cut. The barrier spectrum comes from nudged elastic bands and accelerated molecular dynamics on about 100 pockets. The cost is 10^4 to 10^5 hours on graphics processors. The definition of a pocket changes N_a by a factor of about two, so the distribution is reported for two cuts. Thermal spikes in germanium need the two-temperature coupling of the dense-track calculation. The pocket energies come from the configurational table above, so that pockets and point defects add up to the stored energy of the partition tables. The size distribution and barriers give the pocket term of the delayed-release spectrum of Sect. 19.6.

Table E.15: Nuclear-recoil calculations for silicon and germanium. Costs are estimates. GPU-h are hours on graphics processors; TDDFT, time-dependent density-functional theory; MLIP-MD, molecular dynamics with a machine-learned interatomic potential; MC, Monte Carlo.

quantity	method and grid	acceptance	cost
$\mathrm{d}P/\mathrm{d}\omega_e(b_{\mathrm{imp}}, E_{\mathrm{coll}})$	real-time TDDFT on 216 atoms, step 2 as, grid 0.2 Å; 10×8 points from 50 eV to 5 keV	stopping above $E_{R,\mathrm{th}}$; HSE (Si only), surface hopping, Landau–Zener spread	10^6 core-h; hybrid subset a few 10^6
$P(E_{\mathrm{heat}}, E_{\mathrm{st}}, N_{\mathrm{eh}})$, E_{d}, p_{FP}, phonon spectra	MLIP-MD, 40 energies $\times$ 100 directions $\times$ 100 seeds	DFT on 20 directions; harmonic hand-over	10^4–10^5 GPU-h
Y_q, F_{NR}, $P(N_{\mathrm{eh}} = 0)$	MD with excitation events, 20 eV–1 keV, ≥ 1112 runs per point	silicon data at 0.10, 0.22, 0.46 keV	10^3–5×10^3 GPU-h
Dense-track ratio, ΔF_{NR}	carrier MC with Coulomb forces; MD with two-temperature model	dilute limit	10^4 GPU-h
f_{A}, plasma time	drift–diffusion–Poisson, mesh ≤ 1 nm	Eq. (E.109) for $f_{\mathrm{A}} \ll 0.1$	10^3 core-h
$P(N_a\|E_R)$, barrier spectrum	MLIP-MD, 0.2–20 keV, 10^5–10^6 atoms	two cluster cuts; energy closure	10^4–10^5 GPU-h

E.24 Electron cascades

A full-band Monte Carlo of electron thermalization in silicon, with density-functional bands and phonons and a time-dependent density-functional loss function, gives 3.10 eV per pair, 14 % below the room-temperature value of about 3.6 eV; with a GW/Bethe–Salpeter loss function it gives 3.50 eV [163, Sect. III]. The full-band cascade of Sect. 18.3.3, with empirical pseudopotential bands and deformation-potential phonons, gives 4.15 eV, with the envelope [3.95, 4.73] eV of its variants and a systematic of at least ± 0.4 eV for its class of model, and even with matched ingredients 0.6 eV separate it from the 3.50 eV (Sect. 18.4). The single-parameter model of Appendix E.13 reproduces the measured $\varepsilon_{\mathrm{eh}}$ exactly only because its one constant is fitted to it, and its Fano factor, 0.114, lies 2.4σ below the measured 0.119 ± 0.002 (Sect. 18.4). No first-principles calculation yet reaches the precision of the silicon data, and none at all exists for most of the materials ranked in Chap. 17. Appendix E.13 explains the method, a cascade built from quasiparticle bands, an impact-ionization kernel from the screened interaction and an electron–phonon kernel, sampled carrier by carrier; the calculations below apply it, and Table E.16 collects their grids, tests and costs.

E.24.1 Silicon, germanium and GaAs

The output is the distribution $P_n(\boldsymbol{q}, \omega, T) = P(N_{\mathrm{eh}} = n|\boldsymbol{q}, \omega, T)$ of the pair number for a deposit of energy ω at momentum transfer $\boldsymbol{q}$, the distribution that Sect. 18.1 builds from the branching process and Sect. 18.4 compares with measurements. Its moments are $\varepsilon_{\mathrm{eh}}(\boldsymbol{q}, \omega, T)$ and $F_{\mathrm{eh}}(\boldsymbol{q}, \omega, T)$ of Eq. (E.84), and the variance is also reported split into the terms $\mathcal{V}_<$, $\mathcal{V}_>$ and $\mathcal{V}_{<>}$ of Sect. 18.2, from below and above the impact-ionization threshold and their covariance. For silicon the run also records the energies of the carriers at the end of the cascade and the separations of the pairs they form, which Sect. 18.5 compares with the field-separation radius $r_{\mathcal{E}}$ and over which the capture probability of the last subsection is averaged. The grid is ω from E_{g} to 50 eV in steps of 50 meV, $|\boldsymbol{q}|$ from 0 to $25\alpha m_e$ in steps of αm_e along three

directions plus the angular average, and five temperatures, 0, 50, 100, 150 and 300 K. Above 50 eV the moments are matched to their asymptotic forms, a mean $\omega/\varepsilon_{\mathrm{eh}}$ plus a constant offset and a variance $F_{\mathrm{eh}}\omega/\varepsilon_{\mathrm{eh}}$. On $\mathfrak{T}$ (Sect. 8.5) the bands are G_0W_0 quasiparticles, the screening is RPA with local fields ($\mathcal{X}$ = RPA), the phonon kernel is at $n_g = 2$ and $n_B = 0$, and the events are incoherent (Appendix E.13). Temperature enters three times: through the occupation factors of the phonon kernel, through the Bose factor $1 + n_{\mathrm{B}}(\omega)$ of the pair or plasmon excited at energy transfer ω in the impact-ionization kernel, and through the temperature-renormalized gap (Appendix E.8).

The bands come from Quantum ESPRESSO and G_0W_0 in BerkeleyGW [38] or Yambo [131], Wannier-interpolated up to about 15 eV above the conduction minimum. Silicon needs 16 to 20 disentangled sp^3d Wannier functions; germanium and GaAs need about as many, with spin–orbit coupling. A germanium band structure enters only if it passes the check of Appendix E.14: L below Γ, both gaps within 0.1 eV of the measured ones, and the transverse mass at L and the mass at Γ against measurement. PBE at the measured lattice constant reverses the valley order, HSE06 with spin–orbit coupling at the standard exchange fraction inverts Γ and L, and a correction of the 4s-like orbitals fitted to both gaps fails the masses (Appendix E.14). The next route is G_0W_0 with spin–orbit coupling on a PBE or hybrid ground state, and its quasiparticle bands pass the same check before any germanium cascade is run. Above 15 eV the states are taken as free-electron-like with *GW* lifetimes, which is where the free-particle form of the single-parameter model is accurate, and the matching point serves as a check. The screened interaction is the full-frequency $\varepsilon^{-1}_{\boldsymbol{GG'}}(\boldsymbol{p},\omega)$ for ω from 0 to 60 eV, on about 200 frequencies that are denser below 20 eV; carriers above 60 eV, up to the energy cap, use the free-electron kernel of Appendix E.13, so the grid of W^{scr} need not reach the cap. It uses 10^3 to 12^3 momenta $\boldsymbol{p}$, $N_G \approx 200$ to 300 and at least 200 empty bands, and $\mathrm{Im}\,\varepsilon^{-1}$ at the plasmon, whose loss peak lies at 16.4 eV in silicon (Sect. 4.4), is converged to 2 %. For GaAs the total dielectric function includes the ionic part built from the measured ε_∞ and the measured LO–TO splitting, which gives $Z^*/\varepsilon_\infty = 0.194$ (Sects. 5.5 and 11.9); a semilocal ε_∞ of GaAs is not a safe input. The impact-ionization kernel of Eq. (E.85) is tabulated with local fields on iso-energy surfaces, about 10^3 $\boldsymbol{k}$ points per 50 meV bin. The electron–phonon kernel comes from EPW [55], with 8^3 coarse grids, fine $\boldsymbol{k}$ grids of 80^3 to 100^3 and 50^3 phonon wavevectors sampled by Monte Carlo. GaAs needs the long-range dipole and quadrupole terms, and every material needs the same vertex and phonons as the response code (Appendix E.5).

The pair number is computed in two stages. The first generation, the primary carriers and the products of their first impact ionization, keeps its full $\boldsymbol{k}$ resolution, because the reduction to energy alone is least accurate there; it is seeded with the final states of the dark-matter response (Sect. 18.3). Later generations use the energy-only reduction of Sect. 18.1, solved deterministically by the moment equations on an energy grid of bin h, with kernels averaged over iso-energy surfaces. For N_s sampled first steps with subtree means M_i and variances V_i, the estimator is $\bar{n} = 1 + \mathrm{mean}_i\, M_i$ and $\mathrm{Var}\, N = \mathrm{mean}_i\, V_i + \mathrm{var}_i\, M_i$, the law of total variance, with errors from 20 batch means. The relative error of $\mathrm{var}_i\, M_i$ is $(2/N_s)^{1/2}$ for a near-Gaussian M, so a Fano factor to 0.5 % needs at most $2/0.005^2 = 8 \times 10^4$ samples; the run starts at 10^4 and doubles until the batch errors fall below 0.5 % on F_{eh} and 0.1 % on $\varepsilon_{\mathrm{eh}}$. The plasmon part of $\mathrm{Im}\, W^{\mathrm{scr}}$ is tagged and its share of the variance reported (Appendix E.13).

The grid is converged when the bin, the energy cap, the matching point, the fit window and the $\boldsymbol{k}$ mesh each pass. Halving h from at most 10 meV changes $\varepsilon_{\mathrm{eh}}$ by less than 10^{-3} and F_{eh} by less than 3×10^{-3}, relative. Raising the energy cap from 100 to 150 eV, and the free-electron matching point from 15 to 20 eV, change both by less than 10^{-3}. The asymptotic slopes fitted over $[X/2, X]$ and over $[3X/4, X]$, for a top energy X, agree to 10^{-4}. A second $\boldsymbol{k}$ mesh for the iso-energy averages changes both by less than 10^{-3}. Energy is conserved along every path

to 10^{-12}, and F_{eh} computed from the pair number and from the heat agree to 10^{-9}. Before the full kernels are switched on, the code must reproduce, with free-particle kernels in their place, the single-parameter model of Appendix E.13 within its Monte Carlo scatter of about 2 %. The validation, not used in any fit, is the silicon measurement $\varepsilon_{\mathrm{eh}} = (3.749 \pm 0.001)\,\mathrm{eV}$ and $F_{\mathrm{eh}} = 0.119 \pm 0.002$ at 123 K [117] and the silicon quantum yield of photons between 3 and 8 eV and between 50 and 1500 eV [1], with the 50 to 1500 eV data run at 300 K, their measurement temperature. For germanium no measurement at 77 K uses the same definition of the pair energy; until one does, the germanium value is an input, not a test. A photon seed at $q \approx 0$ and a dark-matter seed at q of 1 to 10 keV, run in one code, give the q dependence of the yield directly.

The cost per material is 10^4 to 10^5 core-hours for the screened interaction, as much for the impact-ionization kernel, and about 10^4 for the phonon kernel; the three materials together need 10^5 to 6×10^5 core-hours. Memory is set by W^{scr}: at $N_G = 300$, 200 frequencies and the 72 irreducible points of a 12^3 grid, $72 \times 300^2 \times 200 \times 16\,\mathrm{byte} \approx 21\,\mathrm{GB}$. Beyond the pitfalls of Appendix E.13, each phonon step must be an integer number of energy bins to better than 1 %, either by a bin commensurate with the dominant phonon or by interpolating the other branches, since an off-grid step biases every phonon loss the same way. The quasiparticle residue enters once, in the rate or in the weight, never in both. Kohn–Sham and quasiparticle energies must not be mixed between kernels. F_{eh} is a small difference of large terms (Sect. 18.2), so the three variance terms are reported with their errors. The results replace the model values of Sect. 18.4 for the three materials and fix the q dependence of the yield used in Sect. 18.1.

E.24.2 Wide-gap and polar crystals

Sapphire, $CaWO_4$, LiF and NaF, among the materials ranked in Chap. 17, have no measured Fano factor and no validated pair energy, and Sect. 18.4 shows that they lie outside the range in which the single-parameter model was validated. These four come first; 4H-SiC, diamond, 3C-SiC, GaN, AlN and TeO_2 follow. The quantities and grids are those of the silicon cascade. Polar crystals add the long-range Fröhlich vertex, built with the same Born charges and dielectric constants as the response code (Appendices E.4 and E.5). LiF and NaF have excitons bound by an eV or more, so the Bethe–Salpeter equation of Appendix E.6 is solved as well. The bookkeeping must then state whether a bound exciton counts as a pair. A carrier between $E_{\mathrm{g}} - E_x^b$ and the ionization threshold can also create a bound pair directly, and the cascade must include that channel. Tungsten in $CaWO_4$ needs spin–orbit coupling.

The cells are small, two atoms for LiF and NaF, 10 for sapphire and 12 for the primitive cells of $CaWO_4$ and TeO_2, but the oxides have many bands. The cost is estimated at about 10^5 core-hours for each oxide, 10^4 to 10^5 for each halide plus the Bethe–Salpeter step, and twice the silicon cost for diamond, the silicon carbides, GaN and AlN; the set needs 6×10^5 to 2.5×10^6 core-hours. The silicon validation comes first, because silicon alone has a measurement at the precision that tests a Fano factor, and the rest are run after it passes, reusing its kernels. Their benchmark is the full-band Monte Carlo with density-functional inputs, which gives 7.3 eV against the measured 7.8 eV for 4H-SiC, 6.6 eV against 6.9 eV for 3C-SiC and 16.7 eV against 18.0 eV for SiO_2 [163, Table II], a few to 15 % below measurement where measurements exist. The computed pair energies and the first ab initio Fano factors of these materials replace the estimates of Sect. 18.4.

E.24.3 Impact ionization beyond first order

Appendix E.13 gives two definitions of the impact-ionization rate, from the imaginary part of the GW self-energy with its residue z_{qp} and from the golden rule with exchange, which differ by a factor between 0.5 and 1. A constant factor does not matter once a model is calibrated on silicon. In the single-parameter model a factor $z_{\mathrm{qp}} = 0.8$ on the rate raises the fitted phonon-to-ionization constant $A_{\mathrm{AB}} = 5.703\,\mathrm{eV}^3$ by 25 %, $\Delta A_{\mathrm{AB}} = 1.43\,\mathrm{eV}^3$, and with the secant slope $\partial\varepsilon_{\mathrm{eh}}/\partial A_{\mathrm{AB}} = 0.076\,\mathrm{eV}^{-2}$ of Sect. 18.4 it raises $\varepsilon_{\mathrm{eh}}$ by 0.11 eV, 2.9 %, which recalibration removes. A factor that varies with energy cannot be removed, and it changes F_{eh}, which depends on how the rate rises above threshold (Sect. 18.5). We compute the ratio $r_{\mathrm{ii}}(E)$ of the impact-ionization rate beyond first order in W^{scr} to the first-order rate, on 50 meV bins from threshold to about 10 eV, for silicon.

The calculation evaluates $\mathrm{Im}\,\Sigma_{n\boldsymbol{k}}$ in G_0W_0 with and without z_{qp}, and adds the second-order screened exchange as the first vertex correction beyond GW. Vertex corrections of this kind shift the absolute band energies of solids by tenths of an eV [164], and the question here is how much they change the rate. The grids are those of the cascade: $N_G \approx 250$, 200 empty bands, the full-frequency ε^{-1} and 10^3 $\boldsymbol{k}$ points per bin. The ratio is converged when it changes by less than 2 % per bin with N_G raised to 350, the bands to 400 and the $\boldsymbol{k}$ points to 2×10^3; the second-order exchange is reported separately to the same precision. A variation below 2 % across the window counts as constant. The cascade is then rerun with $\Gamma_{\mathrm{ii}} \to r_{\mathrm{ii}}(E)\Gamma_{\mathrm{ii}}$, and the change of F_{eh}, for about 3×10^4 core-hours in all, is the uncertainty that the order of the rate adds to F_{eh} in Sect. 18.5.

E.24.4 Momentum memory between events

The energy-only reduction of Sect. 18.1 assumes that a carrier forgets its momentum on its iso-energy surface between two events that change its energy. Suppose momentum-randomizing scattering, by intervalley and acoustic phonons, is a Poisson process of rate Γ_k that leaves the carrier uniform on the surface, while energy-changing events occur at rate $\Gamma_E = \Gamma_{\mathrm{ii}} + \Gamma_{\mathrm{ph,loss}}$. The probability that the next event changes the energy before any randomization is

$$w_{\mathrm{mem}}(E) = \frac{\Gamma_E}{\Gamma_E + \Gamma_k}, \tag{E.110}$$

the ratio of competing Poisson rates. At the next energy-changing event the momentum distribution is then $1 - w_{\mathrm{mem}}$ times the uniform one plus w_{mem} times the remembered one. For any cascade observable $m_{\boldsymbol{k}}(E)$ that depends on the starting $\boldsymbol{k}$, such as the mean pair number or its square, the error of the reduction per generation is therefore at most $w_{\mathrm{mem}}(E)$ times the spread of $m_{\boldsymbol{k}}$ over the surface, $\max_{\boldsymbol{k}} m_{\boldsymbol{k}} - \min_{\boldsymbol{k}} m_{\boldsymbol{k}}$. Within about 1 eV of threshold, intervalley scattering at 10^{13} to $10^{14}\,\mathrm{s}^{-1}$ dominates and w_{mem} is small. Above 3 to 5 eV in silicon, impact ionization outpaces the phonons and w_{mem} can reach one half, so the bound is useful only if the spread is small.

The calculation gives the spread of the mean pair number and of its second moment, and $w_{\mathrm{mem}}(E)$, for silicon between 3 and 8 eV on 50 meV bins. They come from the $\boldsymbol{k}$-resolved second generation of the cascade, with at least 200 $\boldsymbol{k}$ samples per bin, weighted by area on the surface, and at least 10^3 cascades per sample. Γ_k comes from EPW, Γ_E from the impact-ionization and phonon rates of the same run. The spread is converged to 10 % when the $\boldsymbol{k}$ samples are doubled to 400 and the cascades quadrupled, for 10^3 to 10^4 core-hours. The maximum minus the minimum of n_k noisy estimates is biased upward by about $2\sigma(2\ln n_k)^{1/2}$, with σ the noise of one estimate, so the range between the 5th and 95th percentiles is used, or the noise is subtracted. The bound rests on the assumption that randomization leaves the

carrier uniform on the surface, itself a form of the energy-only reduction; it bounds the error under that assumption and is not an independent test of it.

E.24.5 A library of pair-number distributions

The response libraries of Sect. 21.6 need, for every final-state class Λ of every material, not just the moments but the full distribution $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ (Sect. 18.3). The reduced cascade gives it cheaply. With energy-only branching, random final momenta for impact ionization, one phonon energy ω_{ph} and one phonon-to-ionization constant, the generating function $\mathcal{G}_\Lambda(z) = \sum_N z^N P_{\mathrm{casc}}(N|\Lambda)$ obeys the recursion of Sect. 18.1 in the energies x_e and x_h of the seed electron and hole above their band edges. We evaluate it at the N_z roots of unity $z_j = \mathrm{e}^{2\pi i j/N_z}$. The discrete Fourier sum

$$P_{\mathrm{casc}}(N|\Lambda) = \frac{1}{N_z} \sum_{j=0}^{N_z-1} z_j^{-N}\, \mathcal{G}_\Lambda(z_j), \tag{E.111}$$

returns $\sum_{N'} P_{\mathrm{casc}}(N + N'N_z|\Lambda)$. That is the exact probability as long as N_z exceeds every pair number with appreciable weight, so N_z is taken as the smallest power of two above $\bar{N} + 8\sigma_N$, with σ_N the standard deviation of the pair number.

The energy grid has a step h of about $E_{\mathrm{g}}/50$, chosen so that it divides E_{g}, $\theta_{\mathrm{ii}}E_{\mathrm{g}}$ and ω_{ph}. With phonons the discretization error falls as $h^{3/2}$, so the values at h and $h/2$ are extrapolated to zero step as $f(h/2) + [f(h/2) - f(h)]/(2^{3/2} - 1)$ (Appendix D.43); the h^2 formula would leave a bias. The energies run to $20E_{\mathrm{g}}$, and beyond the table the distribution is the asymptotic lattice Gaussian of mean $\bar{N}$ and variance σ_N^2, sampled at the integers, since integrating it over unit bins adds $1/12$ to the variance, with the Edgeworth terms added. The Gaussian is approached slowly: with x the carrier energy in units of E_{g}, the absolute pointwise error falls as C/x and the relative skewness as $x^{-1/2}$, with $C \approx 0.24$ for silicon, so the match at $20E_{\mathrm{g}}$ carries an error of about 1.2×10^{-2} in each probability. The tests are internal: the probabilities sum to one to machine precision and are non-negative; the derivatives of $\mathcal{G}_\Lambda$ at $z = 1$ reproduce the moment recursion to 10^{-9}; and below the first impact-ionization threshold $P_{\mathrm{casc}}(N|\Lambda) = \delta_{N1}$. The reader that interpolates the tables must return that exact value below $\theta_{\mathrm{ii}}E_{\mathrm{g}}$ and interpolate only from the threshold upward, with a table row at the threshold itself: for $\theta_{\mathrm{ii}} > 1$ the distribution jumps there, and an interpolation across the jump puts weight on extra pairs below threshold.

The same tables carry the marks $\mathfrak{m}$ of Eq. (18.22) that a readout needs (Sect. 21.6): the phonon spectrum and the separate electron and hole counts, which follow from the same recursion with one further generating variable for each, the carrier positions, taken from the ranges between events in the $\boldsymbol{k}$-resolved cascade of silicon, and the stored energy, taken from the table of configurational energies of Appendix E.23. A table takes seconds per material and parameter set, so five parameter sets for every ranked material take minutes on one core. The phonon step must be commensurate with the energy grid. Rounding ω_{ph} to the grid changes it by up to $h/2$; with $h = 0.02E_{\mathrm{g}}$ in silicon that is up to 12 meV of a 63 meV phonon, and every phonon loss of the cascade inherits the error. Two smaller ones are aliasing in z, which Eq. (E.111) avoids only if N_z is large enough, and the seed dependence of the constant offset of $\bar{N}$. A seed that gives all the energy to one carrier and a seed that shares it have different offsets. Temperature beyond its effect on the gap, and excitons, are outside the reduced model. When the full-band kernels of the silicon cascade exist, they replace the free-particle forms in the same recursion.

E.24.6 Exciton and capture parameters

At millikelvin the pairs at the end of the cascade bind into free excitons, bind to impurities, are trapped, or recombine, and Sect. 18.5 writes these processes as rate equations whose rates are unknown in silicon and germanium. The exciton itself needs its binding energy E_x^b, radius a_x and oscillator strengths, and its binding to neutral donors and acceptors. Capture needs the cross-section $\sigma_{\mathrm{cap}}(E, T, \mathcal{E}_{\mathrm{b}})$ of a carrier of energy E on shallow centers in a bias field $\mathcal{E}_{\mathrm{b}}$, and multiphonon capture on deep levels. The kinetics needs the rates of exciton formation, trapping of free carriers, binding of excitons to each impurity species and decay of bound excitons, the luminescence fractions, the bimolecular formation coefficient, exciton–exciton annihilation and Auger recombination, all at 10, 20 and 50 mK. The targets are silicon and germanium, then LiF, NaF, sapphire and $CaWO_4$.

On $\mathfrak{T}$ (Sect. 8.5) the exciton parameters sit at $\mathcal{X}$ = BSE on quasiparticle bands, the bound excitons at effective-mass level with a density-functional central cell, and the capture rates at $n_g = 2$. The exciton parameters come from G_0W_0 and the Bethe–Salpeter equation (Appendix E.6) in BerkeleyGW or Yambo, which in silicon and germanium (for germanium on quasiparticle bands that pass the check of Appendix E.14) checks the effective-mass binding of Appendix E.15, with 4 to 8 valence and 4 to 8 conduction bands. In silicon and germanium the exciton radius spans many cells, and the $\boldsymbol{k}$ grid must resolve it: a spacing below $1/(3a_x)$ means 10^4 to 10^5 points, reached by double-grid interpolation, against 8^3 to 12^3 for the ionic crystals. When E_x^b approaches the longitudinal optical phonon energy, the lattice part of the screening enters through a polaronic correction. Donor- and acceptor-bound excitons come from effective-mass theory with central-cell corrections from density-functional calculations in 216- to 512-atom cells. Capture on shallow centers follows cascade-capture theory [165], a carrier captured if it emits phonons inside the Coulomb radius, with acoustic and optical emission rates from EPW and the vertex of the cascade. Deep levels use charge-transition levels and Huang–Rhys factors (Appendices E.10 and E.22) [87]. The Bethe–Salpeter step costs 10^3 to 10^4 core-hours for the semiconductors and about 10^4 for the oxides; capture about 10^5 core-hours per material; the six materials about 6.5×10^5.

The exciton levels of the ionic crystals are converged to 1 meV when the $\boldsymbol{k}$ mesh goes from 8^3 to 12^3, and those of silicon and germanium when the double-grid density near the band edges is doubled, with the bands from four to six on each side; central-cell binding energies to 10 % between 216 and 512 atoms. A rate that changes by more than 10 % between 10 and 50 mK is reported as a temperature dependence. The measured control is the free-exciton binding of silicon, 14.7 meV [121]. Formation of an exciton from two free carriers moves gap energy between pairs and excitons and releases only E_x^b, while exciton–exciton annihilation and Auger recombination remove gap energy, so the bimolecular channel must be named. The excitons cool by acoustic phonons slowly at millikelvin, so their temperature is not the lattice temperature. Capture at the end of the cascade changes the pair statistics. If every carrier is captured with the same probability f, the number surviving is binomial in N_{eh}, with mean $(1-f)\langle N_{\mathrm{eh}}\rangle$ and variance $(1-f)^2 \operatorname{Var} N_{\mathrm{eh}} + f(1-f)\langle N_{\mathrm{eh}}\rangle$. Then $\varepsilon_{\mathrm{eh}} \to \varepsilon_{\mathrm{eh}}/(1-f)$ and $F_{\mathrm{eh}} \to (1-f)F_{\mathrm{eh}} + f$. A capture probability $f(x)$ that depends on the final energy x of the carrier is instead averaged, with the law of total variance, over the end-of-cascade energy distribution that the silicon cascade records. The parameters close the rate equations and the end-of-cascade populations of Sect. 18.5, and the charge and exciton kinetics of Appendix E.25.

Table E.16: Electron-cascade calculations, with estimated costs in core-hours. GW+BSE, quasiparticle and Bethe–Salpeter calculation.

quantity	method and grid	acceptance	cost
$P_n(\boldsymbol{q},\omega,T)$, $\varepsilon_{\mathrm{eh}}$, F_{eh}, end-of-cascade energies and pair separations (Si, Ge, GaAs)	G_0W_0 bands (Ge after the Appendix E.14 check), full-frequency W^{scr} to 60 eV, EPW; k-resolved first generation	Si 3.749 eV, 0.119 at 123 K; quantum yield at 300 K	10^5–6×10^5
$\varepsilon_{\mathrm{eh}}$, F_{eh} (wide-gap, polar)	as above, Fröhlich vertex, BSE for halides	silicon validation first; DFT full-band benchmarks	6×10^5–2.5×10^6
$r_{\mathrm{ii}}(E)$, Si	G_0W_0 with and without z_{qp}, plus second-order exchange	2 % per 50 meV bin	3×10^4
$w_{\mathrm{mem}}(E)$ and spread, Si, 3–8 eV	k-resolved second generation, 200 $\boldsymbol{k}$ per bin	10 %; percentile range	10^3–10^4
$P_{\mathrm{casc}}(N\vert\Lambda)$ library, marks	reduced recursion, $h=E_{\mathrm{g}}/50$, Eq. (E.111), $h^{3/2}$ extrapolation	sum to one; moments; δ_{N1}	≤ 1
Exciton and capture set	GW+BSE, central-cell DFT, cascade capture	Si $E_x^b=14.7$ meV	6.5×10^5

E.25 Event-level transport

A 100 eV electron recoil in silicon ends its cascade with about 27 electron–hole pairs, at 3.75 eV per pair [117], and with the rest of its energy in phonons. What the detector reports is decided afterwards, in a readout window of a microsecond to a second. In that window the pairs drift, bind into excitons or get trapped, the gap energy they carry is released somewhere as phonons, and each sensor absorbs some share of each phonon population. Sects. 18.5 and 19.6 and Chap. 20 derive the physics of every step, and Appendix E.12 sets out the transport Monte Carlo that chains the steps. The chapters do not supply the rates, kernels and efficiencies of a stated detector, and the readout conclusions in the Summary of Chap. 20 stay conditional until these are computed. Every calculation below sits at the place on the truncation label $\mathfrak{T}$ (Sect. 8.5) that Appendix E.12 gives the transport: lowest-order golden-rule rates, with $n_g=2$ for phonon emission by carriers, $n_A=2$ for three-phonon decay, $n_B=0$, and incoherent propagation between events. Table E.17 at the end collects grids, acceptance tests and costs.

E.25.1 Charge and exciton kinetics

The measured charge and the measured heat of an event both depend on what the pairs do during the window t_{win}, and both follow from the joint distribution of the measured pair number N_{meas} and of the energy $E_{\mathrm{st}}(t_{\mathrm{win}})$ still stored at the end of the window. It is conditional on the recoil energy E_R, the recoil direction $\hat{\boldsymbol{r}}_R$, the applied field $\mathcal{E}$, the window and the temperature T. From it follow two tables that Sects. 19.5 and 19.6 use: the phonon-yield ratio $Y_{\mathrm{ph}}(t_{\mathrm{win}})$ of nuclear to electron recoils evaluated at a finite window, for $t_{\mathrm{win}}\in\{1\,\mu\mathrm{s},100\,\mu\mathrm{s},10\,\mathrm{ms},1\,\mathrm{s}\}$, and the same ratio at detector thresholds of 0.3, 1, 3 and 16 eV. The grid is 20 logarithmic recoil energies from 10 eV to 10 keV, fields of 0, 1, 4 and 40 V/cm, and the impurity census of the detector modeled (boron, phosphorus and the deep centers its growth certificate lists, at their stated concentrations).

That distribution comes from the rate equations of Sect. 18.5 for free pairs, free and bound excitons and, at high density, electron–hole droplets, coupled to drift; the luminescent share leaves the crystal as escaping energy E_{esc} (Sect. 19.5). The field enters in two ways:

pairs separated by drift faster than the binding time do not form excitons, and above $E_x^b/(ea_x) = 3.5 \times 10^4\,\mathrm{V/cm}$ excitons are field-ionized (Sect. 18.5). Drift, Luke emission, trapping and Ramo induction are those of the carrier Monte Carlo of Appendix E.12, with four additions: the trap occupations $n_j(t)$ carry over from event to event, trap energies are referred to the Fermi level, the electrostatic energy of the trapped charge is a separate share, and the drift during the prompt phase is kept. The exciton module is new code. The exciton and capture parameters come from Appendix E.24.

Two measured numbers serve as acceptance tests. The silicon phonon-yield ratio of the CDMS II detectors, $Y_{\mathrm{ph}} = 0.952^{+0.009}_{-0.007}$ (95 % confidence) at 4 V/cm [166, Sect. 4], and the defect energy loss of ^{206}Pb recoils in germanium, $(6.08 \pm 0.18)\,\%$ [167], must both be reproduced by one parameter set. The Hecht limit of Appendix E.12 is the numerical check. The calculation is meaningful only together with the collection efficiencies and the athermal emission rates specified below, since the measured yield mixes all three. A parameter set costs up to 10^3 core-hours; the full table over windows, thresholds and the two materials 10^3 to 10^4.

E.25.2 Athermal emission and space charge

At millikelvin temperatures thermal emission from traps is frozen out, and two effects remain that a constant trapping length cannot describe. Trapped charge is released athermally, by phonons and by hot carriers, and the charge that stays trapped distorts the drift field. The calculation gives the emission rates Γ_j^{em} of each trap class j, in s^{-1}, over a stated exposure history, and the field distortion $\xi_{\mathrm{SC}} = \max_z |\Delta\mathcal{E}|/\mathcal{E}$ of Sect. 20.1.

The field distortion follows from the one-dimensional Poisson equation. In a planar detector at fixed bias, a sheet of n_{sh} trapped charges per unit area produces a field jump $en_{\mathrm{sh}}/\varepsilon_0$ across it, with ε_0 the static dielectric constant (Heaviside–Lorentz units; in SI the vacuum permittivity multiplies it), and the largest field change lies between one half and the full jump, depending on the depth of the sheet (Sect. 20.1). With the liquid-helium value $\varepsilon_0 = 11.4$ of silicon (Sect. 7.4), $\xi_{\mathrm{SC}} = 1\,\%$ at 1 V/cm therefore needs $n_{\mathrm{sh}} = 6.3$ to $12.6 \times 10^4\,\mathrm{cm}^{-2}$. A trapping probability $p_{\mathrm{tr}} = 12.7\,\%$ is the value used to model a silicon charge-resolving detector [155]. With it, 100 eV events at 1 Hz on 1 cm^2 reach this distortion in 5 to 10 hours if one sign of charge dominates; electrons and holes trapped at different depths partly cancel. Dense calibration runs reach it in minutes. The grid satisfies $\mathrm{d}z \le d/200$ for a crystal of thickness d, and the event sequence runs until the distortion saturates or is reset by flashing with infrared light.

The emission rates need cross-sections that are not known at millikelvin. A neutral phosphorus donor in silicon is bound by 45 meV, and phonons above 10.9 THz live below a nanosecond and travel about a micrometer (Sect. 20.2), so phonon-driven emission from neutral shallow centers happens only next to the track of a later event. The Luke phonons, a few meV, cannot ionize neutral centers, but can ionize the over-charged centers D^- and A^+, which bind their second carrier by a few meV; that is the dominant athermal route once flashing has populated them. The phonon-ionization cross-sections follow from the golden rule for one-phonon absorption with the deformation-potential coupling of Sect. 6.2, between effective-mass envelopes that include the valley–orbit splitting (six valleys in silicon, four in germanium), since the splitting sets the threshold. For D^- and A^+ the bound state is a correlated two-electron envelope of Chandrasekhar type. Impact ionization by drifting carriers uses Born cross-sections on the same envelopes away from threshold and a distorted-wave calculation near it, where the hot tail of a Luke-limited distribution sits. The envelope basis is converged to 1 % in binding energies and cross-sections, on a threshold grid no coarser than a fiftieth of the binding energy. A 216-atom supercell (Appendix E.10), 1.6 nm across, cannot hold a donor envelope of radius 2 to 3 nm and supplies only the central-cell correction, at

about 10^3 core-hours with the shared 60 Ry setting; the envelopes and the event-sequence driver take minutes.

Ramo's theorem stays exact with space charge, but the heat dissipated by drifting carriers differs from the induced-charge work by the change of electrostatic energy, which the energy balance of Sect. 19.5 carries; and flashing creates the D^- and A^+ populations whose emission it was meant to remove.

E.25.3 The mark-sufficiency test

The two-stage factorization of Sects. 18.3 and 19.6 lets the detector stage see the cascade only through a few marks (the pair number, the centroid and spread of the charge cloud, the phonon spectrum), and it is exact if the marks are a sufficient statistic for the seed class. The test compares the detector stage run on full cascade microstates with the same stage run on microstates resampled to match only the marks, for a dark-matter seed class and a photon seed class. The test statistic is the difference ΔP between the two arms in the distributions of the measured charge and of the phonon signal. A difference resolved at two standard deviations needs $2p(1-p)(z/\Delta P)^2$ trajectories per arm, with $z = 2$, which for $\Delta P = 0.01$ at $p = 0.5$ is 2×10^4 per seed class. Common random numbers between the arms, as in Appendix E.12, lower this, and the test costs 10^2 to 10^3 core-hours. The test can fail but cannot prove: a null difference can hold while the omitted features differ, if the detector stage happens to ignore them.

E.25.4 Low-energy-excess inputs

The collective-release model of Sect. 19.6 lacks two inputs: the relaxable elastic energy U_{rel} stored at cooldown, with its spread of release rates $[\Gamma_{\mathrm{min}}, \Gamma_{\mathrm{max}}]$ for a given holder and glue, and the branching ratio b_{br}, the mean number of further releases triggered by the phonons of one release, which fixes the size law.

U_{rel} comes from a thermo-elastic finite-element calculation of the cooldown from 300 K to 10 mK for the crystal, the glue and the holder, with measured elastic tensors $C_{ij}(T)$ and expansion coefficients, or with the integrated thermal contraction of Appendix E.17 where no measurement exists. The elastic energy of each component is converged to 5 % under a doubling of the mesh. The relaxable fraction and its spread in $\ln \Gamma$ come from low-temperature dynamic-mechanical data of the glue, not from the simulation. The energy law of single release events comes from molecular dynamics of a crystal–epoxy interface with 10^6 atoms at 1 to 10 K, which needs a machine-learned potential for cured epoxy (Appendix E.11). None exists; one has to be trained on dispersion-corrected density-functional energies of the cured network and the interface, validated on elastic constants and thermal expansion, and extended to bond breaking by active learning. Event energies are read from energy drops, and the law is stable when the box grows from 10^6 to 4×10^6 atoms. Molecular dynamics reaches nanoseconds, so it gives the energy law and never the rates, which come from the finite-element step.

The branching ratio needs the release-path catalogue of Appendix E.22. The phonon pulse of one release is propagated by the transport Monte Carlo of Appendix E.12, and a catalogued site with barrier E_b triggers with probability $1 - \exp[-\int \mathrm{d}t\, \Gamma(t)]$, where $\Gamma(t)$ is its phonon-assisted rate in the absorbed flux above E_b. The triggering volume is the integral of this probability over space, and b_{br} is the density of sites with barriers below the phonon energy times that volume, averaged by kinetic Monte Carlo over 10^6 seeds. It is first-generation and linear in the flux, and it has to change by less than 10 % when the catalogue grows from 50 to 100 paths, the phonon packets from 10^5 to 4×10^5, and the box from L to $2L$. An incomplete catalogue, missing the low barriers, biases b_{br} low; a box smaller than the phonon mean free

path truncates the triggering volume; and the thermal bath and the pulse must not both drive the same rate. The finite-element part costs 10^2 to 10^3 core-hours, the molecular dynamics 10^3 to 10^4 GPU-hours. A fit of the model to the binned rates of a glued and an unglued crystal [168] then tests the predicted scaling with U_{rel} without free parameters.

E.25.5 Collection kernels of a stated detector

The collection contrasts ϖ_O and ϖ_{gap}, the residual position spread and the collection ratio $\mathfrak{r}_{\mathrm{coll}}$, defined in Sects. 19.5 and 20.5, are linear functionals of the phonon birth spectra, since phonon–phonon interactions among the signal phonons are negligible. A detector is therefore characterized by the collection kernels $\mathfrak{C}_s(E_{\mathrm{ath}}; \boldsymbol{r}_{\mathrm{ev}})$ of its sensors s, the fraction of the energy of phonons born with energy E_{ath} at $\boldsymbol{r}_{\mathrm{ev}}$ that ends as quasiparticles in sensor s; a population with birth spectrum u gives sensor s the energy $\int \mathrm{d}E_{\mathrm{ath}}\, \mathfrak{C}_s u$. Electron and nuclear recoils enter only through their birth spectra, never through the kernels.

No kernel has been computed for a real detector. Two geometries are specified: the $1 \times 1 \times 0.4\,\mathrm{cm}^3$ silicon crystal of Ref. [169], whose top face carries two sensor arrays, and a germanium cylinder 1 cm thick of the CDMS type. Table E.17 gives the grids: 40 logarithmic points in E_{ath} between 0.41 and 62 meV (0.1 to 15 THz), 8^3 event positions in the symmetry-reduced octant, and 10^4 phonons launched per cell as monochromatic sources, which is what makes the tally a kernel. Luke phonons are generated by the carrier transport at biases of 2, 4 and 10 V. The approximation level, including the lumped films, is that of Appendix E.12. Each geometry uses its actual sensor films; the mean-field estimate of Sect. 20.5, by contrast, takes one film with a niobium-like gap ($2\Delta_{\mathrm{BCS}} = 3.0\,\mathrm{meV}$) and one of aluminum ($2\Delta_{\mathrm{BCS}} = 0.36\,\mathrm{meV}$).

The kernels are accepted on numerical and physical grounds. ϖ_O must change by less than 0.01 when the energy grid is refined, and the statistical error per cell must fall below 1 % when the launch count grows from 10^4 to 4×10^4. In the limit of diffuse, grey surfaces and diffusion faster than absorption, the position-averaged kernels must reproduce the mean-field solution of Sect. 20.5, with a bare-surface loss of about 10^{-3} per bounce [103, Sect. III G]; there ϖ_O lies between 0.02 and 0.39 for mean Luke energies from 4.7 down to 1.5 meV and the position spread is zero by construction, so the computed spread is the first new number the kernels give. Three systematic scans make up the nuisance of the contrasts: the surface coefficients varied between the acoustic- and diffuse-mismatch values, the probability of specular reflection, which no code default fixes, and, for germanium, the anharmonic constant over its factor-of-ten range.

Luke-phonon birth spectra depend on the field, so they must come from the carrier transport at each bias; a sampler that assigns one phonon energy per emission fixes $\mathfrak{r}_{\mathrm{coll}}$ by construction. Phonons below $2\Delta_{\mathrm{BCS}}$ of aluminum are lost energy, not absorbed energy, unless a tungsten sensor or another film takes them, and that assignment sets the sign of $\mathfrak{r}_{\mathrm{coll}} - 1$ (Sect. 20.5); it must be stated with every result. Accelerated propagation modes that skip scattering events are checked against full tracking before use; and the transport code has to pass the film validation of Appendix E.25.9 first. The cost is about 2×10^{12} propagation steps, some 140 core-hours per geometry; a first pass at 10^3 phonons per cell takes under an hour at about 3 % per cell.

E.25.6 Gap-energy release and created-phonon spectra

Each pair created at zero field carries the gap energy E_{g}, which comes back as phonons when the pair recombines, and where and as what spectrum it comes back decides ϖ_{gap}. Its phonon-borne part has a birth density $N_{\mathrm{eh}} E_{\mathrm{g}} \sum_\ell \pi_\ell(\boldsymbol{r}, t)\, s_\ell(E_{\mathrm{ath}})$ summed over release

channels ℓ: bulk recombination of excitons by Auger or multiphonon processes, recombination at a surface, collection at an electrode, and quasiparticle injection into a film. Here π_ℓ is the channel flux in space and time and s_ℓ the spectrum emitted per channel. Then ϖ_{gap} is the sum over channels of the time-integrated channel fraction times the contrast of s_ℓ born at the release site, read from the kernels above. Bulk release of a hard spectrum, above the memory-loss scale of Sect. 20.2, contributes nothing.

Besides the kinetic rates of Appendix E.24, the release needs the exciton diffusion constant D_X, the surface recombination velocity S_{surf} and the spectra s_ℓ. The channel fluxes come from a one-dimensional implicit finite-difference solution of exciton diffusion with Robin faces, with $\mathrm{d}z \le L_X/20$ for the diffusion length $L_X = (D_X/\Gamma_X)^{1/2}$ and exciton decay rate Γ_X. An exciton born at depth z_0 recombines at a face with probability $[S_{\mathrm{surf}}/(S_{\mathrm{surf}} + D_X/L_X)]\,\mathrm{e}^{-z_0/L_X}$, which reduces to e^{-z_0/L_X} only for $S_{\mathrm{surf}} \to \infty$. The fluxes enter the transport as time-tagged sources; the released fraction is converged to 10^{-3} under grid and time step, with S_{surf} scanned from 10 to 10^4 cm/s. The surface and electrode spectra cannot be assumed: taking them as a hard line at the optical-phonon energy decides $\varpi_{\mathrm{gap}} = 0$ in advance. Until they are computed, they are scanned as mixtures $(1-\kappa_s)s_{\mathrm{hard}} + \kappa_s s_{\mathrm{soft}}$, with $\kappa_s \in \{0, 0.25, 0.5, 0.75, 1\}$ and s_{soft} a Debye spectrum cut at 2, 4 or 8 meV. Soft surface spectra are measured: in germanium at ± 3 V, recombination phonons from the surfaces carry 13 % of the phonon signal in a low-frequency, ballistic regime [103, Sect. V A]. The scan is replaced by the multiphonon line shape of the recombination center (a dangling bond at the Si/SiO_2 interface, or a state of a metal contact), with the Huang–Rhys factor and local modes from defect supercells (Appendix E.10) converged to 10 %, at about 10^3 core-hours per center. At millikelvin D_X and the binding rates depend on the impurity census of the actual crystal, droplets lie outside linear kinetics, and luminescence belongs to E_{esc}, not to ϖ_{gap}.

The primary phonons are treated the same way. The created-phonon spectra of electron and nuclear recoils at $E_R = 50$ eV, 100 eV, 300 eV and 1000 eV, branch-resolved on a 0.5 meV grid with the zero-point energy of the final state subtracted (Appendix E.23), are convolved with the kernels to give the arrival spectra and the observable difference between the two recoil types. Sect. 20.2 bounds that difference by the L^1 distance of the normalized birth spectra below the memory-loss scale plus a term Δ_{h} from the hard part; the bound reduces to the soft-fraction difference only when the spectral difference has one sign below that scale. The test is that refining the grid to 0.25 meV, or moving the memory-loss scale by a factor of two, changes the bound by no more than Δ_{h}. Thermal phonons absorbed during the recoil enter as negative sources, never clipped, and the energy the molecular-dynamics thermostat removed enters with its own spectrum. The convolution costs nothing; a direct transport check samples births from the tables at about 10^8 steps per seed set, minutes.

E.25.7 Collection efficiencies of the calibration detectors

The silicon phonon-yield ratio $Y_{\mathrm{ph}} = 0.952$ is degenerate between stored energy and collection (Sect. 19.5). Breaking the degeneracy needs the collection efficiencies $f_{\mathrm{coll}}^{(p)}$ of each population p, primary phonons of nuclear and of electron recoils, Luke phonons and gap-energy phonons, for the detectors that measured the data. These are a CDMS II silicon detector, 76 mm across and 10 mm thick at 4 V/cm [166, Sect. 2], and a SuperCDMS germanium detector at ± 2 V with a ^{206}Pb surface source [167]. The gap part of the collection offset δ_{coll} is fixed by ρ_g, the efficiency for gap-energy phonons over that for hard primaries; $\rho_g - 1 = 0.21$ alone would account for the silicon deficit $1 - Y_{\mathrm{ph}} = 0.048$ beyond the 0.006 that the pair energy of the reconstruction produces, so ρ_g must be known to ± 0.02. The two-bias inversion of Sect. 19.5, at 4 and 40 V, assumes that $\mathfrak{r}_{\mathrm{coll}}$, the ionized fraction f_{ion} and the collection offset do not change with bias and that the reconstruction keeps its pair energy; the run computes the

residual from the bias dependence of $\mathfrak{r}_{\mathrm{coll}}$, and a third bias tests all three.

The calculation adds charge transport to the kernels: Herring–Vogt valleys, Luke emission, intervalley scattering inside the field range of its fitted parameters, and trapping at constant cross-section (Appendix E.12), with event positions from source maps of a Geant4 simulation [170] of the neutron, gamma and lead sources. A carrier collected at an electrode releases its kinetic energy plus its energy relative to the Fermi level of that contact, supplied per contact; booking $E_{\mathrm{g}}/2$ per carrier with no phonons would set ρ_g by construction. Each population needs 10^5 events, and the inversion must be consistent between the two biases. The surface source of the germanium data makes the escape term of δ_{coll} the leading one there. The cost is about 10^{11} steps, some 8 core-hours, per geometry.

E.25.8 Anisotropic film interfaces

The escape time $t_l = 4d_{\mathrm{f}}/(v_s \bar{t}_{\mathrm{esc}})$ of a phonon from a sensor film (Sect. 20.3) needs the escape probability $\bar{t}_{\mathrm{esc}}$ of the actual interface. The isotropic acoustic- and diffuse-mismatch values differ by a factor 1.31 in t_l for aluminum on silicon. The welded interface between two anisotropic elastic media, with Christoffel-matched plane waves and the six-by-six boundary problem for displacement and traction, gives the transmissions $t_\nu(\theta, \phi)$ per mode and incidence direction for Al(111) on Si(001) and for tungsten on silicon, in both directions. Thin-film interference enters through a transfer matrix when the wavelength is comparable to the thickness, and $t_l(E_{\mathrm{ath}})$ is tabulated for d_{f} from 40 to 600 nm. The aluminum constants are the measured ones extrapolated to zero temperature, the set of Sect. 20.3, whose isotropic (Voigt) average gives $c_{\mathrm{LA}} = 6.63$ and $c_{\mathrm{T}} = 3.30\,\mathrm{km\,s^{-1}}$ at $2.70\,\mathrm{g\,cm^{-3}}$; finite-strain density-functional constants (Appendix E.4), half an hour for fcc aluminum at a 60 Ry cutoff, a 16^3 grid and 0.02 Ry smearing, serve only where no measurement exists, as for tungsten films.

The transmitted and reflected fluxes sum to one to 10^{-14}; in the isotropic limit the code returns the escape probabilities of Sect. 20.3 for aluminum on silicon, $\bar{t}_{\mathrm{esc}} = 0.368$ (acoustic mismatch) and 0.282 (diffuse mismatch); the angle averages satisfy the reciprocity $\mathfrak{n}_1 \bar{t}_{12} = \mathfrak{n}_2 \bar{t}_{21}$ of Sect. 20.3; and $\bar{t}$ is stable to 10^{-3} under refinement of the angular grid. The anisotropic result for Al(111) on Si(001) is compared with the sensitivity of Sect. 20.3, where giving isotropic silicon its [100] or [111] speeds moves $\bar{t}_{\mathrm{esc}}$ to 0.327 or 0.405; a value outside that range points to the treatment of the aluminum side. The film formula carries its own hypotheses. Its flux average of the per-angle absorption is $1 - 2E_3(2d_{\mathrm{f}}/\mathfrak{l}_{\mathrm{f}})$, with E_3 the exponential integral, which the lumped form $1 - \mathrm{e}^{-4d_{\mathrm{f}}/\mathfrak{l}_{\mathrm{f}}}$ of Appendix E.12 matches only to first order: with the escape probability set to one in both, the lumped form lies 3 % to 13 % above it for $d_{\mathrm{f}}/\mathfrak{l}_{\mathrm{f}}$ from 0.01 to 0.2. And t_l holds for a film population that is isotropized faster than it escapes: without mixing in the film, transverse modes beyond 37.7° are totally reflected at Al/Si, some 56 % of the film's phonon flux, and never escape. Real interfaces carry oxide or amorphous interlayers, and the fitted silicon-to-aluminum value 0.3 to 0.55 [109] is an absorption probability $\bar{t}\,P_{\mathrm{abs}}$, not an escape probability. The whole calculation costs 2 to 10 core-hours.

E.25.9 Validation of the film cascade

Every number that Chap. 20 takes from a lumped film rests on the quasiparticle yield $Y_{\mathrm{qp}}(E_{\mathrm{ath}})$ of Sect. 20.4, and the code that produces it is tested before any kernel is trusted. Without film loss, Y_{qp} must equal $2\Delta_{\mathrm{BCS}}/E_{\mathrm{ath}}$ to round-off below $4\Delta_{\mathrm{BCS}}$, reach the maximum of 0.600 at $5.46\,\Delta_{\mathrm{BCS}}$ and settle at the 0.590 that the Kaplan cascade of Sect. 20.4 gives, inside the range 0.57 to 0.6 of earlier calculations for bulk aluminum, niobium and tin [171, p. 2]; the test points are $4.5\Delta_{\mathrm{BCS}}$, $6\Delta_{\mathrm{BCS}}$ and 10 to $100\Delta_{\mathrm{BCS}}$. With the escape law of a 600 nm

Table E.17: Event-level transport: grids, acceptance tests and cost of each calculation. Energies of phonons are E_{ath}; core-hours are CPU core-hours unless marked GPU. Costs are per geometry or per parameter set as stated.

quantity	grid	acceptance	cost
Charge and exciton kinetics	t_{win}: 1 µs to 1 s; thresholds 0.3–16 eV	Y_{ph} = 0.952 (Si), 6.08 % (Ge), Hecht limit	$\leq 10^3$ core-h per set
Athermal emission, space charge	envelope basis to 1 %; $\mathrm{d}z \leq d/200$	saturation of the field distortion	$\sim 10^3$ core-h
Mark sufficiency	2×10^4 trajectories per arm	ΔP resolved to 0.01	10^2–10^3 core-h
Low-energy-excess inputs	mesh ×2; $10^6 \rightarrow 4 \times 10^6$ atoms; catalogue 50 → 100	elastic energy per component to 5 %; b_{br} to 10 %	10^2–10^3 core-h; 10^3–10^4 GPU-h
Collection kernels	40 energies, 0.41–62 meV; 8^3 positions; 10^4 phonons per cell	ϖ_O to 0.01; mean-field limit	~ 140 core-h
Gap-energy release	$\mathrm{d}z \leq L_X/20$; S_{surf} 10–10^4 cm/s	released fraction to 10^{-3}	seconds; 10^3 per center
Created-phonon spectra	0.5 meV → 0.25 meV	bound stable to Δ_{h}	minutes
Collection efficiencies	10^5 events per population	ρ_g to ±0.02; two biases	~ 8 core-h
Film interfaces	angular grid; d_{f} 40–600 nm	flux sum to 10^{-14}; isotropic 0.368, 0.282; reciprocity	2–10 core-h
Film cascade validation	3–350Δ_{BCS}	$2\Delta_{\mathrm{BCS}}/E_{\mathrm{ath}}$; 0.590; closure 10^{-14}	< 1 core-h

aluminum film, Y_{qp} falls with energy at 3, 5, 10 and 40Δ_{BCS}, because the fastest phonons escape before they break pairs (Sect. 20.4). Energy closes per event to 10^{-14} relative for incident phonons from 50 to 350Δ_{BCS}. The last range matters because germanium prompt phonons near 36 meV arrive at an aluminum film with some 200Δ_{BCS}. The closure counts escaped energy as phonons actually returned to the substrate, so that an escape bin cannot absorb lost energy. The mean Luke-phonon energy against field is compared with the carrier model of Appendix E.12. Each test runs in seconds and a benchmark of a few thousand events in minutes. Below 100Δ_{BCS} the yields agree with the loss-free forms within statistical error; above it the closure test decides.

E.26 Coherent and second-order absorption

An axion-like field of mass 1 µeV stays coherent for $t_{\mathrm{coh}} = 1/(m_\phi v_0^2) \approx 1.04$ ms, with v_0 the circular speed of the standard halo (Sect. 1.2), and over $1/(m_\phi v_0) \approx 249$ m, so a nuclear-spin sample or an acoustic resonator sees one classical drive, not a stream of quanta. A fermion of 100 keV absorbed in germanium deposits nearly its whole mass at a momentum transfer of the same size. A dark photon can be absorbed in pairs, or scattered into a photon. These processes lie outside the single-vertex golden rule of Appendices E.14 to E.16, and Sects. 10.8, 10.9, 11.8 and 14.6 derive them; what they need from the material is mostly a response that Appendices E.4 to E.7 already compute, read on a different line of the (q, ω) plane.

E.26.1 Coherent-regime material inputs

The first decision is the regime, and it is made before any material calculation. The target responds on a time $1/\Gamma_T$, which is T_2 for spins and the mode quality over the mode frequency for a resonator. For an integration time t_{int} long compared with the smaller of t_{coh} and $1/\Gamma_T$, the mean absorbed power equals the velocity-averaged golden rule, and the rate tables of the earlier sections apply. A single record matches that mean only for $t_{\mathrm{int}} \gg t_{\mathrm{coh}}$. Below t_{coh} the drive is a single Rayleigh-distributed amplitude, the signal is the windowed linear response of Sect. 14.6, and the sensitivity scales as $t_{\mathrm{int}}^{-1/2}$ instead of $(t_{\mathrm{int}} t_{\mathrm{coh}})^{-1/4}$. The material inputs differ by platform.

For nuclear-spin precession, the chemical shielding and electric-field-gradient tensors of every nuclear site give the inhomogeneous line as a site sum, and come from the gauge-including projector augmented-wave method [172]. The effective electric field at the nucleus that converts a nucleon electric dipole moment into an energy shift comes from density-functional theory, and T_1 and T_2 come from data. Prepolarized samples are out of equilibrium, with T_1 of hours, so their spin noise is evaluated at the spin temperature fixed by the polarization p through $\tanh(\beta_s \omega_{\mathrm{Lar}}/2) = p$, with ω_{Lar} the Larmor frequency, not at the lattice temperature [173]. Bulk acoustic resonators need the elastic and piezoelectric tensors from density-functional perturbation theory (Appendix E.4) and the mode frequencies and shapes from a finite-element solution of the body. They also need the orientation factor of the field in the sample (1 normal to a slab, $3\varepsilon_0/(\varepsilon_0 + 2)$ for a sphere, ε_0 parallel to a slab; Sect. 11.8), and the measured mode quality [174]. For resonant-mass detectors driven by a scalar that changes α or m_e, the inputs are $\mathrm{d}\ln a/\mathrm{d}\ln\alpha$ and $\mathrm{d}\ln a/\mathrm{d}\ln m_e$ of the lattice constant a [175]. The Coulomb Hamiltonian with infinite nuclear masses depends on α and m_e only through the length unit $1/(\alpha m_e)$, so a nonrelativistic density-functional calculation with scaled constants must return -1 for both to numerical precision; that is the benchmark. The physical deviation comes from what the Coulomb Hamiltonian leaves out (Sect. 11.8). Relativistic corrections add to $\mathrm{d}\ln a/\mathrm{d}\ln\alpha$ alone a term that fully relativistic calculations put at -1.97×10^{-2} for copper and -2.2×10^{-4} for silicon, and the finite nuclear mass adds a dependence on the proton-to-electron mass ratio, -1.12×10^{-3} per unit fractional change for copper and -0.86×10^{-3} for silicon, which enters through m_e [176, Table II]. A fully relativistic calculation of a new crystal is accepted once it reproduces these values for copper and silicon. For magnetoelectric multiferroics, linear spin-wave theory or time-dependent density-functional theory with spin–orbit coupling on a magnetic structure computed at $T = 0$ gives the response only far below the ordering temperature. Near it, where the proposed signal peaks, a finite-temperature spin-dynamics method is needed and nothing in this appendix supplies it.

The frequency grid resolves the one-sided dark-matter line, $\omega \in [m_\phi, (1 + 3.5 \times 10^{-6}) m_\phi]$, with at least 3000 points, and t_{int} runs from 10^{-2} to $10^{3}\, t_{\mathrm{coh}}$. The Asimov test statistic of Sect. 14.6 must change by less than 1 % when the frequency step is halved, and the resonator mode frequencies must be converged to 0.1 %. Two pitfalls are statistical rather than material: the random amplitude is either in the likelihood or in a penalty factor, never in both [177], and an analysis with fixed amplitude overestimates the reach, by a factor 1.99 for the median limit of a record shorter than t_{coh} and 2.7 for data at the detection threshold (up to 10 with other priors; Sect. 14.6). Once the responses are tabulated, a point in $(m_\phi, t_{\mathrm{int}})$ takes seconds. The gauge-including projector-augmented-wave (GIPAW) and DFPT inputs use the shared setting of Appendix E.17, converged to 1 % in cutoff and $\boldsymbol{k}$ grid, and cost 10^2 to 10^4 core-hours per target, the multiferroic response about 10^3.

E.26.2 Fermionic absorption

A fermion χ absorbed by the target turns into a light neutral fermion $\tilde{\nu}$ (Sect. 10.9), and for $v \to 0$ and a massless $\tilde{\nu}$ the target is probed along the line $\omega = m_\chi - q$, at $q \approx m_\chi$. For m_χ from E_g to 1 MeV this reaches momentum transfers far above the valence scale, into the cores. We need $\mathrm{d}\mathcal{R}/\mathrm{d}\omega$ for vector, scalar and axial operators. The scalar operator needs only the density row $S(q,\omega)$. The vector operator needs the density row and the transverse current row of S_{ab}: continuity turns the longitudinal current into (ω/q) times the density, so that the density enters with weight m_χ^2/q^2 and the transverse current with weight one. A light mediator multiplies the scalar rate by $(m_\mathrm{med}^2 - 2m_\chi\omega + m_\chi^2)^{-2}$, normalized to its contact value, where m_med is the mediator mass, since the four-momentum transfer is set by the relativistic $\tilde{\nu}$ leg and not by $-q^2$; its expansion, $1 - 2(m_\chi^2 - 2m_\chi\omega)/m_\mathrm{med}^2 + \ldots$, is the correction of relative order $(m_\chi/m_\mathrm{med})^2$ attached to the contact limit in Ref. [178], with an ω dependence that tilts the spectrum.

The responses are all-electron. In crystals they come from the plane-wave calculation of Appendix E.7 with augmentation reconstructed to high q (Appendix E.2); for $q \gtrsim m_e/3$, where the core shells dominate, from Dirac–Hartree–Fock orbitals of the atom. The grid is about 200 points in q per mass and 40 masses per decade. The electron spectra must change by less than 2 % when the augmentation is enlarged, and the nuclear line, an impulse response broadened by the zero-point momentum $\langle p_\kappa^2 \rangle$ from density-functional perturbation theory, must have its width converged to 1 %. The spin convention is the commonest error: the atomic ionization form factor that the literature uses counts both spins in its sum over occupied states [62, Eqs. (A.18)–(A.21)], and a table that does not state its convention in the header can be off by a factor of two. The calculation also sizes a defect of the literature formula for the vector operator, which multiplies the ionization form factor by the free-electron matrix element instead of using continuity [179, Eq. (9)]. At bound kinematics the two differ by an $O(1)$ term $-\omega^2/m_\chi^2$. Sect. 10.9 bounds the defect from one side: the current-conserving rate exceeds the free-vertex form by at least the factor $1/r$, 1.7 for the xenon $1s$ shell at $m_\chi = 56.5\,\mathrm{keV}$, the transverse current can only raise it further, and only the relativistic corrections to the bound states, of relative size $(Z\alpha)^2 = (54/137.04)^2 = 0.155$ for the xenon K shell, have an unknown sign. Computing the transverse current and the relativistic corrections with Dirac–Hartree–Fock states fixes the size of the defect. The cost is 10^4 to 10^5 core-hours per crystal; the nuclear part is negligible.

E.26.3 Pair, Compton and Barkas responses

Pair absorption, dark Compton scattering and the second-order Barkas term need responses at two insertions of the field (Sect. 10.8). Contact pair absorption reduces to single absorption at $\omega \approx 2m_\phi$ and $k \approx 0$, so it uses the absorption tables of Appendix E.16 (charge-like rows) or, for the pseudoscalar coupling, the axioelectric (spin-current) row of Eq. (10.32) at $\omega = 2m_\phi$ (Eq. (10.43)), contracted with the pair tensor of Sect. 10.8 [180, Eqs. (5)–(8)]. The pseudoscalar route inherits the caveats of that row: the measured σ_1 belongs to singlet pairs while the row creates triplets, spin–orbit coupling spoils the factor $\frac{3}{4}$ (Sect. 10.8), and in a superconductor the spin-current conductivity replaces σ_1 only under the conditions of Sect. 15.4.3. The tests below accept the pair tensor and its velocity moments, not the row. For the scalar and pseudoscalar couplings the tensor is constant; for the vector and axial couplings it carries the relative momentum of the two absorbed particles, and its velocity moments are computed over the product of the two velocity distributions on a 20^3 quadrature. Two tests accept it: for the scalar coupling the pair rate reduces to the single-absorption rate at $\omega = 2m_\phi$, $k \to 0$, times the constant tensor, and the velocity moments reproduce their analytic values for the standard halo to 10^{-3}. Bauer et al. take a local density of $0.39\,\mathrm{GeV\,cm^{-3}}$ [180]; the rate scales

as ρ_χ^2, so its curves move down by $(0.3/0.39)^2 = 0.59$ for the $0.3\,\mathrm{GeV\,cm^{-3}}$ of Sect. 1.2. Dark Compton scattering of a vector of mass m_ϕ on electrons bound with energies small compared with m_ϕ needs the shell Compton profiles, or the high-q density response. It also needs the refractive index $n_r(\omega')$ of the medium at the outgoing photon energy $\omega' \approx m_\phi$; the grid is 100 points in $\omega' \in m_\phi(1 \pm 0.1)$ and 32 angles. For a free electron at rest the rate must reproduce the Thomson-limit form $n_e \zeta^2 \sigma_{\mathrm{Th}}\, \omega'/[m_\phi(1+\omega'/m_e)]$, with $\sigma_{\mathrm{Th}} = e^4/(6\pi m_e^2)$ the Thomson cross-section and $\omega' + \omega'^2/2m_e = m_\phi$, which agrees with the tree-level result [181, Eq. (2)] through second order in m_ϕ/m_e; this is the acceptance test. The Barkas term needs the quadratic density response of the electron gas, tabulated for 50 densities with r_s from 1 to 6 and mapped onto the crystal by the local-density approximation on the (q,ω) grid of the loss functions of Appendix E.16. It is accepted when the electron gas returns the known Z_1^3 (Barkas) correction to the stopping of slow charges and the linear response when the quadratic kernel is switched off. The Barkas table is new code of about 10^2 core-hours; the other two calculations take minutes.

E.27 Halo, background and nuclear-spin inputs

A limit from a silicon crystal that is valid for every velocity distribution of the halo, not only the standard one, is the solution of a linear program whose columns are the counts that single streams of dark matter would produce in each bin. A crystal kept at sea level for a hundred days before it goes underground carries tritium whose β spectrum puts a fraction of an event per kilogram, day and keV below 1 keV. And a spin-dependent coupling to unpolarized nuclei produces no coherent phonon at all. These three facts belong to Sects. 11.6, 21.2 and 21.7, and each needs one computation, specified below.

E.27.1 The directional halo-independent bound

Sect. 21.2 writes the expected count in bin i as $\int_{\mathcal{B}} \mathsf{k}_i(\boldsymbol{w})\,\mathrm{d}\Xi(\boldsymbol{w})$, linear in the nonnegative measure Ξ of Eq. (21.9) on the ball $\mathcal{B} = \{|\boldsymbol{w}| \le v_{\mathrm{esc}}\}$ of galactic-frame velocities (v_{esc}, v_0 and ρ_χ as in Sect. 1.2), with the stream kernel $\mathsf{k}_i(\boldsymbol{w})$ the count that a single stream of unit Ξ-mass at $\boldsymbol{w}$ would leave in bin i; Appendix C.12 sets up the linear program in the same symbols. A crystal fixed in the laboratory sees the stream at velocity $\boldsymbol{w} - \boldsymbol{v}_E(t)$ and turns with the sidereal rotation. The kernel is then the time average over the bin of the master rate of Sect. 8.2 with the crystal's direction-dependent response $S(\boldsymbol{q},\omega;\Lambda)$ and the final-state weights of Sect. 8.7. The largest value of a linear functional $\int \mathsf{f}\,\mathrm{d}\Xi$ allowed by the measured upper counts N_i^{up} is

$$\max_{\Xi \ge 0}\Big\{\int \mathsf{f}\,\mathrm{d}\Xi : \int \mathsf{k}_i\,\mathrm{d}\Xi \le N_i^{\mathrm{up}}\Big\} = \min_{\boldsymbol{\lambda}\ge 0}\Big\{\sum_i \lambda_i N_i^{\mathrm{up}} : \sum_i \lambda_i \mathsf{k}_i(\boldsymbol{w}) \ge \mathsf{f}(\boldsymbol{w})\ \forall\, \boldsymbol{w} \in \mathcal{B}\Big\}, \quad \text{(E.112)}$$

where the λ_i are the nonnegative dual weights of Appendix C.12; the maximum is attained by at most as many streams as there are bins (Sect. 21.2). The functional f can be a halo integral above a speed in a direction, the rate of a second crystal or of a second orientation of the same crystal, or a modulation amplitude; a signed f is allowed provided some bin sees every stream on which $\mathsf{f} > 0$. The method extends the stream approach of nuclear-recoil analyses [182, 183] to electronic and directional responses.

The computation is a sweep followed by a small linear program. The response must be resolved in the direction of $\boldsymbol{q}$, not averaged over angles. The non-self-consistent band calculation behind it (Appendix E.1) therefore runs on the full zone without symmetry reduction, and the library sits at the truncation label $\mathfrak{T}$ of the target's response tables (Appendices E.14 and E.16). The stream grid has 768 directions on an equal-area pixelization

of the sphere [184], 55 galactic-frame speeds in steps of 10 km/s up to v_{esc}, and 24 sidereal phases, about 10^6 kernel columns. For each stream and phase the energy delta function of the master rate fixes ω on every vector of the $\boldsymbol{q}$ grid. Each column is then one pass over about 10^6 $\boldsymbol{q}$ vectors, some 10^{12} multiply-adds per mass, 5 to 50 GPU-hours for about 50 masses. The linear program has one constraint per bin and about 4×10^4 variables and is solved by a dual simplex code [185] in seconds.

The grid is refined around the few optimal streams by column generation, and the dual vector $\boldsymbol{\lambda}$ must satisfy $\sum_i \lambda_i \mathsf{k}_i \geq \mathsf{f}$ on the refined grid; that inequality certifies the bound, and the result is reported at two grid levels. In the laboratory frame, a kernel that depends only on the speed must reproduce the single-bin result $N_i^{\mathrm{up}} / \inf_{v \geq u_0} [v\, \mathsf{k}_i(v)]$ for the halo integral above u_0; the same statement in galactic speeds does not hold, because there the kernel depends on $|\boldsymbol{w} - \boldsymbol{v}_E(t)|$. And the kernel must be nonnegative for every stream. A response truncated at some order can have negative pieces. A dual vector that satisfies the constraint of Eq. (E.112) still bounds the objective by weak duality, but the mass bound behind the grid bracket fails for the computed kernel, and the leakage-corrected bound of Sect. 21.2.4 applies instead. The support constraint is imposed in the galactic frame, a per-bin confidence level needs the Bonferroni correction across bins, and slow streams alias on a coarse $\boldsymbol{q}$ grid, where the energy delta function falls between grid points.

E.27.2 Cosmogenic production

A nuclide produced at a rate R_k during exposure k of length t_k and decaying with constant λ_{dec} has, after the whole history, the activity $\mathcal{A}_{\mathrm{act}} = \sum_k R_k (1 - \mathrm{e}^{-\lambda_{\mathrm{dec}} t_k})\, \mathrm{e}^{-\lambda_{\mathrm{dec}} t'_k}$, where t'_k is the time from the end of exposure k to the measurement; at saturation $\mathcal{A}_{\mathrm{act}} = R$. The sub-keV rate density is $\mathcal{A}_{\mathrm{act}}$ times the fraction of the β spectrum below 1 keV, per keV, plus the lines of electron capture. L- and M-shell captures produce lines near 1 and 0.1 keV [186, Sect. 1], so capture nuclides land in the sub-keV region through their L and M fractions. Tritium in silicon is produced at sea level at 124 ± 25 atoms per kilogram and day from all cosmic-ray components, 112 ± 24 of them from neutrons [187], has a half-life of 12.3 years, and puts 8.9 % of its β spectrum below 1 keV; a hundred days of exposure then give $0.17\,\mathrm{events\,kg^{-1}\,d^{-1}\,keV^{-1}}$ below 1 keV, and saturation gives 11.

What is missing is the sea-level production rate of each relevant nuclide, in atoms per kilogram and day, for every target retained in Sect. 21.7 (the survivors) that has no measured or published rate. The semi-empirical cross-sections of ACTIVIA [188], with measured cross-sections where they exist, are folded with the measured sea-level neutron spectrum, scaled to New York City at the midpoint of solar modulation, and with the measured sea-level proton spectra, the inputs of the silicon measurement [187]. For the light nuclides tritium and ^{7}Be, where the semi-empirical systematics fail, cross-sections from a Geant4 hadronic list [170] or from TALYS are folded with the same spectra. The inputs are the composition and isotopic abundances, the spectrum parameterization and the exposure history. The calculation must reproduce the measured rates of silicon [187] and germanium [186] within their errors before any other material is computed. The models disagree most for tritium: in germanium they span 27.7 to 210 nuclei per kilogram and day [186, Sect. 6.3], so the light nuclides carry the largest error. Tritium also diffuses out of silicon, which makes its activity in the crystal depend on the thermal history, and the K:L:M capture fractions are needed per nuclide. ACTIVIA takes seconds per material and Geant4 with 10^8 primaries 1 to 3 CPU-hours, a few tens to a hundred core-hours for the survivor pool.

E.27.3 Spin-dependent nucleon inputs

For unpolarized nuclear spins the spin correlation of two nuclei averages to $\delta_{\kappa\kappa'}\delta^{ab}J_{\mathrm{nuc}}(J_{\mathrm{nuc}}+1)/3$, with J_{nuc} the nuclear spin, so the coherent one-phonon amplitude of a spin-dependent coupling vanishes and only the incoherent response survives (Sect. 11.6) [189]. It is the self response of the nuclei that carry a spin, Eq. (11.42), weighted by $J_{\mathrm{nuc},i}(J_{\mathrm{nuc},i}+1)/3$ for isotope i. For the spin-dependent operator $\mathcal{O}_4$ the rate contracts it as $\sum_i |c_4^{(i)}|^2\,|\boldsymbol{S}_\chi|(|\boldsymbol{S}_\chi|+1)\,J_{\mathrm{nuc},i}(J_{\mathrm{nuc},i}+1)\,S_i^{\mathrm{self}}/3$, with the nuclear coefficient $c_4^{(i)}=[c_4^{(p)}\langle S_p\rangle+c_4^{(n)}\langle S_n\rangle]/J_{\mathrm{nuc},i}$ of Eq. (8.8), where $\langle S_p\rangle$ and $\langle S_n\rangle$ are the expectation values of the proton and neutron spins in the nuclear ground state; the weight is that of the nucleus, not of a free nucleon. At $T=0$ the one-phonon term on a cubic site is $S_i^{\mathrm{self}}=(2\pi/\mathcal{V})N_i\,\mathrm{e}^{-2W_i}\,(q^2/2M_i\omega)\,g_i(\omega)/3$, with N_i the number of nuclei of isotope i, the factor $1/3$ projecting the density of states on $\hat{\boldsymbol{q}}$, and g_i the local density of states of the isotope, Eq. (11.44), which differs from the host density of states of species κ near the edges of the optical band. A coherent piece proportional to the squared polarization returns only in a polarizing field.

Two inputs are missing for the survivors. The spin expectation values come from large-scale shell-model calculations: $\langle S_n\rangle=0.156$ and $\langle S_p\rangle=0.016$ for ^{29}Si, and $\langle S_n\rangle=0.439$ and $\langle S_p\rangle=0.031$ for ^{73}Ge, where a second interaction gives 0.450 and 0.006 [190, Table III]. Each odd-A isotope of the survivors needs values from at least two interactions, and their spread is the nuisance. The host density of states $g_{\mathrm{ph},\kappa}$ comes from density-functional perturbation theory (Appendices E.4 and E.17), interpolated to a fine grid, with the $\boldsymbol{Q}$ grid refined until $\langle\omega^{-1}\rangle$ over $g_{\mathrm{ph},\kappa}$ changes by less than 1 % and an energy bin no wider than a quarter of the threshold; the local g_i of each isotope follows from it by Eq. (11.44). Reusing the coherent one-phonon response with spin weights in place of the scalar couplings gives zero for the wrong reason, because the coherent amplitude averages out. Even–even isotopes have no spin and contribute nothing, so an element's weight is set by its odd-A abundance alone. The density-of-states runs cost 10 to 100 core-hours over the survivors.

Summary

Every method in this appendix starts from one base calculation, at clamped nuclei with independent particles, and moves one entry of the truncation label at a time. Every numerical parameter is converged on the quantity that is used, not on the total energy: the cutoff that converges the energy of silicon to 0.1 meV per atom leaves the density vertex at large momentum transfer unconverged, and the 8^3 grid that converges the energy to 2 meV per atom leaves ε_∞ nine percent too high. Three rules of bookkeeping recur: a measured gap already contains the zero-point renormalization; every quantity that enters one amplitude is computed in one gauge; and a sum rule tested with a nonlocal Hamiltonian carries its nonlocal term. For silicon the dominant theory uncertainties are few: the gap (about 0.1 eV), the screening (ε_∞ of 13.0 computed against 11.7 measured), the vertex beyond $q\sim 1/r_c$, and, for recoils, the interatomic potential. The rest are controlled numerically. The specifications of Appendices E.14 to E.27 share one rule: each calculation is accepted against an exact limit or sum rule and, wherever one exists, a measurement made at the same kinematics, never against the estimate it replaces. Their costs run from seconds (the pair-number library, the corrected Dirac reaches) to about 10^6 core-hours (the close-collision kernel), and several of the cheapest, the verified onset of silicon, the boosted-dark-matter recast and the published-limit corrections, decide numbers the chapters quote today.

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Appendix F

Literature formulas as limits

Each row of Table F.1 records a published formula that the book recovers as a limit, together with four items that its Limit Check does not state: the source equation, the approximation chain, the conditions of validity and the leading correction. The first column names the formula, its equation in the book and its Limit Check, or marks the row "(no box)"; Sect. 9.4 explains how a box is read and places the formulas of Chaps. 10 to 17 on the truncation label. The rows follow the order of the chapters, and the last column gives each formula's home section, where it is derived and where its box, if it has one, stands. The Leading-correction column opens with the size of that correction in bold, taken from the text, the Limit Check or its figure; **unsized** marks a correction that the book does not yet size.

Table F.1: Literature formulas recovered as limits: source equation, approximation chain, validity and leading correction.

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Elastic nuclear recoil (Limit Check 1.1), Eqs. (1.21) and (1.23)	[1, Eqs. (77), (78)]	Eq. (1.22) with $\Delta_\chi = 0$ and a free nucleus at rest, $\omega = q^2/2M$, reduces to Eq. (1.20); maximizing over the angle gives $q \leq 2\mu_{\chi N} v$ and Eq. (1.21). Eq. (1.23) at $\omega = E_R = q^2/2M$ gives $v_{\min} = q/2\mu_{\chi N}$.	Impulse approximation: the nucleus recoils as if free and at rest. In the harmonic model with $\omega_0 = 60\,\text{meV}$ the rms width of the deposit is below $E_R/3$ for $E_R \geq 9\omega_0 = 0.54\,\text{eV}$ in silicon. Transfer below the inverse nuclear radius, so the nucleus recoils as a whole.	**relative width $(\omega_0/E_R)^{1/2} \leq 1/3$ for $E_R \geq 0.54\,\text{eV}$ (Si)**. Binding to the lattice replaces $\delta(\omega - q^2/2M)$ by a peak of mean E_R and rms width $(E_R\omega_0)^{1/2}$ [2, Sect. IV A, Eqs. (90), (91)]; for $E_R \lesssim 3\omega_0$ it breaks into single- and few-phonon lines (Chap. 5).	Sect. 1.3
Heavy and light mediators, $F_{\text{DM}} = 1$ and $(q_0/q)^2$ (Limit Check 1.2), Eq. (1.35)	[3, Eq. (15) and the display above it]	Eq. (1.35) with $m_V \gg q, q_0$ for every contributing q gives $F_{\text{DM}} = 1$ and $\bar\sigma_e = \mu_{\chi e}^2 y_\chi^2 y_e^2/(\pi m_V^4)$; with $m_V \ll q, q_0$ it gives $F_{\text{DM}} = (q_0/q)^2$ and $\bar\sigma_e = \mu_{\chi e}^2 y_\chi^2 y_e^2/(\pi q_0^4)$.	Heavy: m_V above the largest transfer, $2m_\chi v_{\max}$ (0.5 MeV at $m_\chi = 100\,\text{MeV}$); within 10 % for $0.11\,q_0 \leq q \leq q_0$ when $m_V \geq 4.5\,q_0$. Light: m_V below the smallest transfer, $\omega/v_{\max} \approx 0.11\,q_0$ for the 1.11 eV gap; within 10 % for all $q \geq 0.11\,q_0$ when $m_V \leq 0.026\,q_0$.	**≤ 10 % for $m_V \geq 4.5\,q_0$ (heavy) or $m_V \leq 0.026\,q_0$ (light)**. Heavy, $F_{\text{DM}} \simeq 1 + (q_0^2 - q^2)/m_V^2$; light, $F_{\text{DM}} \simeq (q_0/q)^2(1 + m_V^2/q_0^2 - m_V^2/q^2)$.	Sect. 1.4
Atomic ionization rate (Limit Check 2.1), Eqs. (2.22) and (2.23)	[3, Eqs. (5), (6), Eq. (1) for the Fermi factor]	Eq. (2.20) summed over the occupied states of the shell; outgoing waves normalized to $(2\pi)^3\delta_{l'l}\delta_{m'm}\delta(p'-p)/p^2$, so $\sum_f = \sum_{l'm'}\int p_{\text{ej}}^2 \mathrm{d}p_{\text{ej}}/(2\pi)^3$ and $p_{\text{ej}}^2\mathrm{d}p_{\text{ej}} = (p_{\text{ej}}^3/2)\mathrm{d}\ln E_{\text{ej}}$; with $\lvert\bar V\rvert^2 = \pi\bar\sigma_e F_{\text{DM}}^2/\mu_{\chi e}^2$ the prefactor is $\bar\sigma_e/(8\mu_{\chi e}^2)$.	First Born approximation in the dark-matter potential; one electron at a time in a fixed atomic potential (no correlation, no recoil of the ion).	**about 30 % (outer shells of heavy atoms); $\times$0.46 (H 1s, 1 GeV)**. Distortion of the outgoing wave by the ion, growing as E_{ej} falls. Plane wave times the Fermi factor with $Z_{\text{eff}} = 1$: within about 30 % of exact continuum states for outer shells of heavy atoms [3, text after Eq. (6)]; for hydrogen 1s, 0.46 of the exact rate at $m_\chi = 1\,\text{GeV}$ (Fig. 2.4).	Sect. 2.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Thomas–Fermi screening (Limit Check 4.1), Eq. (4.53)	[4, Eq. (S.14)]	Random-phase approximation Eq. (4.59) with the static Lindhard function Eq. (4.57), $\varepsilon_{\mathrm{Lin}}(q,0) = 1 + (k_{\mathrm{TF}}^2/q^2)F_{\mathrm{L}}(q/2k_{\mathrm{F}})$; $F_{\mathrm{L}} \to 1$ as $q \to 0$; $k_{\mathrm{TF}}^2 = 2e^2\mathcal{N}_{\mathrm{F}}$ (Eq. (4.53)).	Static potential in a homogeneous electron gas, $q \ll k_{\mathrm{F}}$: within 2 percent of the Lindhard value for $q < 0.49\,k_{\mathrm{F}}$, 10 percent at $q = 1.06\,k_{\mathrm{F}}$ (Fig. 4.8). Not for an insulator at q below about $k_{\mathrm{TF}}/(\varepsilon_\infty - 1)^{1/2}$ (Worked example 4.8).	$\leq 2\,\%$ **for $q < 0.49\,k_{\mathrm{F}}$;** $10\,\%$ **at $1.06\,k_{\mathrm{F}}$.** $F_{\mathrm{L}} = 1 - q^2/12k_{\mathrm{F}}^2 + O(q^4)$ (Eq. (C.74)); exchange–correlation corrections to the proper response (Sect. 4.6).	Sect. 4.5
Plasmon dispersion and cutoff (Limit Check 4.2), Eq. (4.60)	[2, Eq. (171) and the text after it]	Zero of $\varepsilon_{\mathrm{Lin}} = 1 - v_{\mathrm{C}}R^0$ above the continuum; R^0 expanded in $1/z^2$ through the odd moments of the pair energies over the Fermi sphere (Eq. (4.58)); iteration of $\omega^2 = \omega_p^2(1 + A/\omega^2 + B/\omega^4)$; cutoff where ω_{pl} meets $\omega_+ = qv_{\mathrm{F}} + q^2/2m_e$, estimated with $\omega_+ \simeq qv_{\mathrm{F}}$.	Random-phase approximation, $T = 0$, no collisions; $q \ll q_{\mathrm{c}}$. At $r_s = 2$ the quadratic formula is within 1 percent of the exact root for $q < 0.39\,k_{\mathrm{F}}$; exact $q_{\mathrm{c}} = 0.730\,k_{\mathrm{F}}$ against the estimate $0.666\,k_{\mathrm{F}}$ (Fig. 4.9).	$\leq 1\,\%$ **for $q < 0.39\,k_{\mathrm{F}}$ ($r_s = 2$); cutoff estimate** $9\,\%$ **low.** $\left[1/4m_e^2 + (12/175)v_{\mathrm{F}}^4/\omega_p^2\right]q^4$ in ω_{pl}^2 (Eq. (4.60)); local-field (exchange–correlation) corrections (Sect. 4.6); damping by collisions (Eq. (4.62)) or interband decay (Worked example 4.10).	Sect. 4.5
Gapless Dirac dielectric function (Limit Check 7.1), Eq. (D.138)	[5, Eq. (2.7)], whose cutoff is a hard cutoff; g_{deg} taken out of the coupling	Interband RPA bubble of one isotropic cone at $T = 0$, chemical potential in the gap, hard cutoff $\lvert\boldsymbol{k}\rvert < \Lambda_{\mathrm{D}}$ (Eq. (D.134)); background added as $\varepsilon_{\mathrm{bg}}$ (Eq. (7.42)); gapped result Eq. (D.136); logarithm expanded for $\lvert\omega^2 - v_{\mathrm{D}}^2q^2\rvert \gg \Delta_{\mathrm{D}}^2$ with $\int_0^1 x(1-x)\ln[x(1-x)]\,\mathrm{d}x = -5/18$ (Eq. (D.138))	$\lvert v_{\mathrm{D}}^2q^2 - \omega^2\rvert \gg 4\Delta_{\mathrm{D}}^2$; $q, \omega/v_{\mathrm{D}} \ll \Lambda_{\mathrm{D}}$; $g_{\mathrm{deg}}\tilde{\alpha}\ln(\Lambda_{\mathrm{D}}/q) \lesssim 1$. Static: within 0.1 (in units of $g_{\mathrm{deg}}\tilde{\alpha}/6\pi$) of the gapped result for $v_{\mathrm{D}}q > 7.5\,\Delta_{\mathrm{D}}$ (Fig. 7.4)	$\leq$ **0.1 (units of $g_{\mathrm{deg}}\tilde{\alpha}/6\pi$) for $v_{\mathrm{D}}q > 7.5\,\Delta_{\mathrm{D}}$.** $-6\Delta_{\mathrm{D}}^2/(v_{\mathrm{D}}^2q^2 - \omega^2)$ in the bracket (residual 3×10^{-3} at $v_{\mathrm{D}}q = 10\,\Delta_{\mathrm{D}}$); anisotropy by rescaling (Eq. (7.47)); higher orders in $\tilde{\alpha}$	Sect. 7.3

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Second Born correction for a Yukawa potential (Limit Check 8.1), Eq. (8.18)	[6, Chap. 6] (conditions $\lvert f\rvert m_V \ll 1$ and $y_\chi y_T/\,4\pi v \ll 1$, Sect. 2.2)	Born series of Eq. (2.14) to second order in $\tilde{V}$; zero-energy radial equation, $a = a_{\text{Born}}(1 \pm \lvert f(0)\rvert m_V/2)$, $\sigma = 4\pi a^2$; partial waves by the variable-phase method at $\lvert f(0)\rvert m_V = 10^{-3}$, $p_{\text{rel}}/m_V = 0.1$–10; coupling from $\bar{\sigma}_e$ or σ_p with F_{DM} normalized at q_0 (Eq. (1.35)); $p_{\text{rel}} = \mu_{\chi T} v$, $v = 10^{-3}$.	Exact coefficient for $p_{\text{rel}} \ll m_V$; within 30 % of the largest partial-wave correction (total and $\mathrm{d}\sigma/\mathrm{d}\Omega$, $q \le 2p_{\text{rel}}$) for $p_{\text{rel}} \le m_V$, above it for $p_{\text{rel}} \ge 3m_V$ (0.15 of it at $10\,m_V$). Plot (Fig. 8.2): $x = m_\chi$, 1 keV–1 GeV; $y = m_V$, 1 eV–10 GeV; contours of the cross-section at $\lvert\delta_{\text{Born}}\rvert = 0.1$, electrons and silicon nuclei; gray where that needs couplings above 4π. At $\bar{\sigma}_e = 10^{-37}\,\mathrm{cm}^2$, $\sigma_p = 10^{-40}\,\mathrm{cm}^2$: $\lvert\delta_{\text{Born}}\rvert \le 4.5 \times 10^{-5}$, 8.0×10^{-5}.	**$\le 30\,\%$ of $\boldsymbol{\delta_{\text{Born}}}$ for $\boldsymbol{p_{\text{rel}} \le m_V}$; $\boldsymbol{\lvert\delta_{\text{Born}}\rvert \le 8.0 \times 10^{-5}}$ at the benchmark cross-sections**. Order $(y_\chi y_T)^2$ relative: resonances of the attractive potential for $\lvert f(0)\rvert m_V \gtrsim 1$; for $p_{\text{rel}} \gg m_V$ the correction falls below $y_\chi y_T/4\pi v$ (0.28 of it at $q = 2p_{\text{rel}} = 20\,m_V$); the speed of a bound electron raises p_{rel} and lowers the correction.	Sect. 8.2
Crystal form-factor rate (Limit Check 10.1), Eq. (10.3)	[7, Eqs. (3.13) and (3.17)], divided by the target mass $N_{\text{cell}} M_{\text{cell}}$	Density row of Eq. (8.16) with $\mathfrak{D}_{ee} = \pi\bar{\sigma}_e F_{\text{DM}}^2/\mu_{\chi e}^2$; filled-band determinant at $T = 0$, $S = S^0$ (Eq. (10.1), Eq. (D.170)); angle average and isotropic kernel $\eta/2q$ (Eq. (A.42)); $\rho_T \Omega = M_{\text{cell}}$ and Essig's integral form (Eq. (D.172)). $\mathfrak{T}$: $\mathfrak{T}_0$, unscreened.	Independent particles, clamped nuclei, $T = 0$, valence to conduction. Plot (Fig. 10.2): silicon, $m_\chi = 10\,\mathrm{MeV}$, heavy mediator, $\omega = 5\,\mathrm{eV}$; $x = q/q_0$; $y = q\,\eta\,S$ for IP, RPA (head) and RPA + local fields in one empirical-pseudopotential model, $\boldsymbol{q}$ along $(5, 3, 1)$. No $q \le 2.8\,q_0$ within 10 % of RPA + LF (nothing shaded); the head-only screening is within 10 % above $2.2\,q_0$.	**screening $\times$0.47 (q-integrated, 5 eV); 0.05–0.61 pointwise**. Screening, $\mathcal{X}$ = RPA + LF: response 0.05 of the unscreened value at the kinematic edge ($0.55\,q_0$), 0.22 at q_0, 0.54–0.61 at 2–$2.8\,q_0$ (local fields); q-integrated 0.47 at 5 eV (Sect. 10.2). Then excitons (Sect. 10.5), core states (Sect. 10.6), all-electron vertices at large q (Appendix E.2).	Sect. 10.1

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Loss-function rate (Limit Check 10.2), Eq. (10.4)	[8, Eqs. (6) and (13)]; [4, Eq. (1)]	Density row of Eq. (8.16) with $\mathfrak{D}_{ee} = \pi\bar{\sigma}_e F_{\mathrm{DM}}^2/\mu_{\chi e}^2$; $S = (2q^2/e^2)L/(1-\mathrm{e}^{-\beta\omega})$ (Eq. (4.45)), exact by Eq. (8.38) for the total charge at any lattice and for the electron density at clamped nuclei; isotropic kernel $\eta/2q$ (Eq. (A.42)). $\mathfrak{T}$: $\mathcal{X}$ = RPA + LF (any many-body approximation in $\bar{L}$), $n_B = 0$.	Electron-only coupling: ω above the phonon band, to Migdal order in κ_{BO}; total charge: all ω. Plot (Fig. 10.4): silicon, $m_\chi = 1\,\mathrm{MeV}$, heavy mediator; $x = \omega/\omega_{\max}$; $y = \mathrm{d}\mathcal{R}/\mathrm{d}\omega$ of the loss-function rate (dark photon, whole rate) against the electron-only coupling with the lattice free (model acoustic peak added); shaded $\omega > \omega_{\max}$.	**inside the phonon band $> 10^4\times$ the electronic rate (electron-only coupling, 1 MeV, model)**. Inside the phonon band, the acoustic peak of S_{ee} carried by the electrons that follow the nuclei, f-sum share $Z_\kappa m_e/M_\kappa = 2.73\times10^{-4}$ but rate more than 10^4 times the electronic one at 1 MeV in the model; above it, Migdal amplitudes (Sect. 8.6); for a crystal at fixed time, the direction dependence of L (Appendix A.8).	Sect. 10.2
Sum-rule bounds: f-sum and screening sum rules, rate bound (Limit Check 10.3), Eq. (10.7)	[9, Eqs. (1), (8) and (9)]; [10, Eqs. (1), (2), (8) and (9)]; non-density couplings [11, Eqs. (4.16)–(4.20)]	Eq. (4.46) and Eq. (C.40) at $\omega = 0$; Hölder with $r = 2/(1+n)$, $s = 2/(1-n)$ (Eq. (10.6)); $\eta \le \omega^n \eta_n^{\max}$ and $L \ge 0$. Step ≤ 1: proven at RPA + LF for insulators (eigenvalues of $X(1-X)^{-1}$), beyond it if $(-R^0)^{-1} + f_{\mathrm{xc}} > 0$. $\mathfrak{T}$: $\mathcal{X}$ exact under that proviso, clamped nuclei; the relaxed-lattice term (Eq. (10.8)) at $n_B = 0$, $\mathcal{K}$ = one-phonon.	Total charge at any lattice; electron density at clamped nuclei, or with ω_{th} above the phonon band. Plot (Fig. 10.6): silicon, $m_\chi = 1\,\mathrm{MeV}$, heavy mediator, $\omega_{\mathrm{th}} = 1\,\mathrm{meV}$; $x = q$ (keV); $y = \mathrm{d}\mathcal{R}/\mathrm{d}\ln q$ of the computed RPA + LF rate, the clamped $n = \pm1$ bounds, the model acoustic peak of an electron-only coupling and the $n = -1$ bound with the lattice term; shaded $q \ge E_{\mathrm{g}}/v_{\max}$. At 1 MeV the clamped bounds exceed the computed rate by 3.3×10^3 ($n=-1$) and 4×10^6 ($n=1$) (Worked example 10.2).	**relaxed-lattice term $\mathcal{G}_{\mathrm{rel}} = 1$ at 18 keV (Si), growing as q^{-2} below**. Electron-only coupling with the lattice relaxed: $1-\varepsilon_e^{-1}(q,0) = [1-\varepsilon_{\mathrm{cl}}^{-1}] + \mathcal{G}_{\mathrm{rel}}$, $\mathcal{G}_{\mathrm{rel}} \to \Omega_{P,e}^2/(c_{\mathrm{LA}}q)^2$ ($\Omega_{P,e} = 0.51\,\mathrm{eV}$ in silicon, one at 18 keV); every $n < 1$ fails throughout the zone, the f-sum bound holds; finite-q form Eq. (E.93).	Sect. 10.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Electron-gas (Lindhard, Mermin) loss-function rate (Limit Check 10.4), Eqs. (4.56) and (10.9)	[8, Eqs. (13), (17)–(19), oscillator form Eqs. (20), (21)]; [12]	Eq. (10.4); jellium with plane-wave states, $\tilde{R} \approx R^0$ (Eq. (4.59)), R^0 of Eq. (4.56); isotropic, no direction average; $3\omega_p^2/v_\mathrm{F}^2 = k_\mathrm{TF}^2$. Mermin: number-conserving relaxation (Eq. (4.62)), not a many-body approximation. $\mathfrak{T}$: $\mathcal{X}$ = RPA for a homogeneous medium, $n_B = 0$. Leading correction on a qualifier, the target model (Sect. 9.4.2): electron gas → crystal, before any step of $\mathcal{X}$.	Valence electrons free: ω well above the absorption peaks and $q \lesssim k_\mathrm{F}$, which along the halo reach is $4.2\,\mathrm{eV} \lesssim \omega \lesssim k_\mathrm{F} v_\mathrm{max} = 9.5\,\mathrm{eV}$ in silicon. Plot (Fig. 10.7): silicon, $m_\chi = 100\,\mathrm{MeV}$, heavy mediator; $x = \omega/E_\mathrm{g}$ ($E_\mathrm{g} = 1.11\,\mathrm{eV}$ of the tables); $y = \mathrm{d}\mathcal{R}/\mathrm{d}\omega$ for Lindhard ($r_s = 2$), Mermin ($\gamma_\mathrm{rel} = 3.66\,\mathrm{eV}$) and the silicon RPA + LF tables, same momenta (0.49–22.5 keV); shaded 4.2–9.5 eV, where Lindhard/silicon = 0.70–1.48. Integrated Lindhard/silicon: 45 (heavy), 70 (light) at 1 MeV; 1.42, 2.6 at 10 MeV; 1.07, 2.1 at 100 MeV.	**Lindhard/silicon 0.70–1.48 at 4.2–9.5 eV; ×24 at** 2 eV. Target model, electron gas → crystal: the gap and the interband absorption peaks (excess of the gas below 4 eV, a factor 24 at 2 eV); the bound momentum distribution (the gas is empty below $\omega_- = q^2/2m_e - q v_\mathrm{F}$, beyond the halo reach for $q > 2m_e(v_\mathrm{F} + v_\mathrm{max}) = 9.85\,\mathrm{keV}$); then local fields (Sect. 10.2).	Sect. 10.3
Plasmon production from the pole of $1/\varepsilon$, light mediator (Limit Check 10.5), Eq. (10.10)	[13, Eqs. (5), (7)]; [14, Eqs. (4), (7), (13), (14)]	Eq. (4.61) (Sokhotski–Plemelj at the zero of ε_1); high-frequency expansion of ε_Lin to $O(q^2)$; residue $\pi/\partial_\omega\varepsilon_1 = (\pi\omega_\mathrm{pl}/2)Z_L$; delta function integrated in Eq. (10.4); light mediator $q^3 F_\mathrm{DM}^2 = q_0^4/q$. Threshold $v_\mathrm{th} \simeq v_\mathrm{F} + q_\mathrm{c}/2m_e$ (Eq. (10.11)). $\mathfrak{T}$: $\mathcal{X}$ = RPA (pole term), $n_B = 0$.	$q < q_\mathrm{c}$: the formula's own cutoff, where the quadratic ω_pl meets ω_+, is 2.24 keV; the fuller result (exact zero of ε_Lin) ends at $q_\mathrm{c} = 2.60\,\mathrm{keV}$; the optical cutoff $2\pi/a \sim 5\,\mathrm{keV}$ of the source is 2.28 keV for silicon. Halo: $v_\mathrm{th} = 9.18\text{–}9.54 \times 10^{-3} = 3.5\text{–}3.6\,v_\mathrm{max}$, rate zero. Plot (Fig. 10.8): $x = q/q_\mathrm{c}$; y = share of the valence f-sum: pole formula, optical line shape to 5 keV, exact Lindhard pole, and the weight below $\omega = q v_\mathrm{max}$ for the gas and for the silicon tables; shaded $q < q_\mathrm{c}$.	**pole share 0.99 at $q_\mathrm{c}/2$, 0.77 at 0.9 q_c; cutoff choice ×1.24.** The exact pole weight falls to zero at q_c (logarithmic slope of ε_1 at ω_+): share 0.99 at $q_\mathrm{c}/2$, 0.77 at $0.9\,q_\mathrm{c}$; interband damping in silicon (width 3.7 eV at $q \to 0$). Cutoff 5 keV instead of 2.60 keV raises the light-mediator integral at $v = 0.1$ by 1.24 (logarithmic, not $(5/2.6)^2$).	Sect. 10.3

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Velocity coupling $\mathcal{O}_8$ with the electron at rest; crystal responses $\overline{W}_1$–$\overline{W}_5$ (Limit Check 10.6), Eq. (10.14)	[15, Eqs. (5), (6)]; [16, Eqs. (36)–(42)]; [17, Eqs. (18), (23), App. B]; [18, Table III]	Source forms: $\mathrm{d}\Gamma_8/\mathrm{d}^3q = (\lvert\bar{\boldsymbol{v}}_\mathrm{T}\rvert^2/4)\,\mathrm{d}\Gamma_1/\mathrm{d}^3q$, the density response screened through the loss function, the current row dropped; $\overline{W}_1$–$\overline{W}_5$ for independent particles, unscreened; the density row of a dark photon screened by $1/(\hat{\boldsymbol{q}}\cdot\varepsilon\cdot\hat{\boldsymbol{q}})$, the spin current unscreened. Rows of Eq. (8.3) contracted with $\mathfrak{D}_{ab}$ (Eq. (D.173)); continuity removes j_L and $\bar{v}_\mathrm{L}$ (Eq. (D.175)); spin rotation splits singlet and triplet blocks, triplet $=\frac{1}{4}$ singlet for independent particles (Eq. (D.178)); PT makes both real, 12 functions (Table 10.1); halo kernel η_2 for $\lvert\bar{\boldsymbol{v}}_\mathrm{T}\rvert^2$ (Eq. (D.176)); screening by Eq. (D.180): density at the head and through local fields, j_T through local fields only on symmetry axes, $\mathcal{T}_e$ at the head, triplet rows not at all. $\mathfrak{T}$: $\mathcal{X}$ = IP for the current and spin rows, RPA + LF for the density row; $n_B = 0$.	The formula is within 10 % of every rate allowed by the band Eq. (10.15) for $r = \Gamma_B/\Gamma_A \le 2.4\times10^{-3}$, $(\sqrt{r}+1)^2 \le 1.1$. Plot (Fig. 10.10): $x = r$; $y = \Gamma/\Gamma_A$: the band $(\sqrt{r}\mp1)^2$, the direction-averaged value $1+r$ and the formula; silicon at $m_\chi = 10\,\mathrm{MeV}$, heavy mediator, $r_\mathrm{IP} = 25.3$ (empirical pseudopotential, threshold 1.11 eV) and $r_\mathrm{med} = r_\mathrm{IP}\times2.31\times(1\pm0.3)$ (density screening from the tables of Sect. 10.2); shaded $r \le 2.4\times10^{-3}$.	**dropped transverse term r_IP = 22–56 times the kept one (Si, 3–30 MeV)**. The transverse motion of the electron, $\eta\,\mathrm{Tr}\,\bar{S}_{\boldsymbol{j}_\mathrm{T}\boldsymbol{j}_\mathrm{T}}$ in Eq. (10.14), larger than the term kept by $r_\mathrm{IP} = 22$–56 in silicon between 30 and 3 MeV, and the spin-current row $\mathcal{J}_\mathrm{T}$ of $\mathcal{O}_3$, $\mathcal{O}_7$, $\mathcal{O}_{12}$–$\mathcal{O}_{15}$ (one quarter of the current block for independent particles); then the cross term, half-width $2\sqrt{r}$ relative to $1+r$, zero after direction averaging; then the local-field change of the current row (model −4 to −15 percent) and the electron–hole attraction in both blocks (Sect. 10.5).	Sect. 10.4
Bethe–Salpeter exciton rate (Limit Check 10.7), Eq. (10.17)	[19, Eqs. (1) and (2)]	Eq. (10.4); S of the singlet excitons Eq. (4.112) from Eq. (4.104) (static screened ladder, Appendix D.12); $\langle\lambda\lvert n_e\rvert0\rangle = \sqrt{g_\mathrm{deg}}\rho_{\lambda0}$ (Eq. (10.16)); line form valid when no line moves ε_mac near its energy. Real part with both time orderings (Eq. (4.109), Eq. (10.20)). Near the pair minimum: Wannier lines and Fermi factor (Eq. (10.18)), second class Eq. (10.19). $\mathfrak{T}$: $\mathcal{X}$ = BSE (static screened interaction, Tamm–Dancoff), $n_B = 0$.	Resonant-only real part: $L^\mathrm{res}/L = (\varepsilon_1^2+\varepsilon_2^2)/((\varepsilon_1^\mathrm{res})^2+\varepsilon_2^2)$, within 15 % of one only for $\lvert\boldsymbol{q}\rvert \ge 6.7\,\mathrm{keV}$ in silicon at 1.3 eV; 2.12 at $0.71\,\text{Å}^{-1}$. Plot (Fig. 10.12): $x = \lvert\boldsymbol{q}\rvert$; $y = L/L^\mathrm{full}$: resonant-only at 1.3 and 3 eV by Eq. (10.20) from the random-phase tables with local fields, one-oscillator Eq. (10.21), both orderings with a ±10 % coupling band; shaded $\lvert\boldsymbol{q}\rvert \ge 6.7\,\mathrm{keV}$.	**about 10 % on ε_2 (coupling block)**. The coupling block, first order in the kernel left in it, $K/(\Delta+K)$ (Eq. (10.22)), about 10 % on the proper ε_2 when the head is taken from the full response; the bound lines of an indirect gap, unresolved on uniform grids (Eq. (10.18), Eq. (10.19)); dynamical screening of the electron–hole interaction; phonon replicas (Sect. 13.8).	Sect. 10.5

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Absorption with in-medium mixing: dark photon; axion-like particle in a field (Limit Check 10.8), Eqs. (10.31) and (10.33)	[20, Eqs. (6) and (7), 21, Eqs. (2), (18) and (20), 22, Eqs. (38)–(40), 23, Eqs. (3), (6) and (7)]	Eq. (8.19) with the current row; proper transverse response $\to \sigma_1$ (Sect. 8.2.3); photon-reducible chain for each polarization (Appendix D.25) $\to \zeta^2 m_\phi^4/\lvert m_\phi^2 - \Pi_{\mathrm{T,L}}\rvert^2$ with $m_\phi^2 - \Pi_\mathrm{L} = m_\phi^2\varepsilon$ and $m_\phi^2 - \Pi_\mathrm{T} = \omega^2\varepsilon^\mathrm{T} - p_\phi^2$ (Eq. (10.30)); polarization average at $p_\phi \to 0$. Field: source current $g_{a\gamma\gamma}(\partial_t\phi)\boldsymbol{B}_0$, Ampère–Maxwell at $p_\phi \to 0$, dissipated power over the number density. $\mathfrak{T}$: $\mathcal{X}$ exact with measured ε, at the temperature of the data; above E_dir $n_g = 0$; between the indirect and the direct gap the measured σ_1 is $n_g = 2$, $n_T = 1$, and a detector needs $n_T = 0$ (Sect. 13.5).	$v^2 \ll 1$: $\Gamma_\mathrm{T}/\Gamma_\mathrm{L} - 1 = v^2[2\,\mathrm{Re}(1/\varepsilon) - 1]$, -1.7×10^{-6} at 10 eV; sample larger than the skin depth $(m_\phi\,\mathrm{Im}\,\sqrt{\varepsilon})^{-1}$. Vacuum mixing within 10 % only above 63 eV in silicon; factor 4×10^{-4} at 3.35 eV, 20 at 16.6 eV. Isotropic field formula exact for cubic crystals and on principal axes; sapphire at 45°: within 10 % on 37 % of 40–120 meV, low by up to 560 at a longitudinal resonance (Eq. (10.34)). Plot (Fig. 10.15): (a) $x = m_\phi$, 1–10^4 eV; $y = 1/\lvert\varepsilon\rvert^2$ against 1 and $1 + 2(1 - \varepsilon_1)$, measured ε of silicon at 300 K and atomic scattering factors above 40 eV; (b) $x = m_\phi$, 40–120 meV; $y = \Gamma m_\phi/(g_{a\gamma\gamma}B_0)^2$, tensor and projected forms at 45°, damped factorized ε of sapphire.	**$O(v^2)$: -1.7×10^{-6} at 10 eV; vacuum mixing off by 4×10^{-4} to 20 below** 63 eV. First order in Π_em/m_ϕ^2, $1 + 2(1 - \varepsilon_1)$, useful only above about 36 eV; $O(v^2)$ polarization terms; temperature transfer of measured σ_1 near the indirect edge, more than a factor 2 within 20 meV (Sect. 13.5); the field's action on the target, $\mu_\mathrm{B}B_0 = 0.58$ meV at 10 T.	Sect. 10.7
Scalar absorption through the kinetic-energy row and its sum rule (Limit Check 10.9), Eqs. (10.35) and (10.37)	[22, Eqs. (18), (55) and (56)], with $\bar{v}^2 = \mathcal{T}_e/m_e$ and $g_{\phi ee} = d_{\phi ee}\sqrt{4\pi} \times m_e/M_\mathrm{Pl}$	Eq. (8.11); density row of order p_ϕ; $\tilde{R}_{\mathcal{T}_e n_e} = \boldsymbol{p}_\phi \cdot \boldsymbol{F} + O(p_\phi^2)$ with $\boldsymbol{F} = 0$ for point groups without a polar vector, so the photon-reducible part is $O(p_\phi^2)$; Eq. (8.19) with $g^\phi_{\mathcal{T}_e} = g_{\phi ee}/m_e$. Sum rule: Eq. (4.28) at $\boldsymbol{q} = 0$, $\mathcal{T}_e = H - V$, eigenstate, $[[p^2, V], V] = -2\lvert\nabla V\rvert^2$. $\mathfrak{T}$: any $\mathcal{X}$ for the rate (the independent-particle form is Eq. (10.36)); $n_g = 0$ above E_dir, the phonon-assisted kinetic row below it.	$v^2 \ll 1$ and no polar vector. Density row alone (model kinetic row, measured L, $v = 10^{-3}$): 3×10^{-3} of the kinetic rate at 5 eV, 2×10^{-2} at 10 eV, about one at 16–18 eV. Sum rule: local, one-body and regular potentials only. Plot (Fig. 10.16): one s electron; $x = \Lambda$, 1–2000 hartree; $y = \sum_{E_f - E_0 < \Lambda}(E_f - E_0)\lvert\langle f\rvert T\rvert 0\rangle\rvert^2$ for $-1/(r^2 + r_\mathrm{c}^2)^{1/2}$, $r_\mathrm{c} = 0.3$ and $0.1\,a_0$ (90 % of the sum rule above 25 and 176 hartree), and for $-1/r$, which grows as $(4/\pi)\sqrt{2\Lambda}$ (0.58, 0.74, 0.83 of it at 100, 500, 2000 hartree).	**density row 3×10^{-3} (5 eV) to about 1 (16–18 eV) of the kinetic rate**. The $O(p_\phi^2)$ terms of Ref. [22, Eq. (55)]: screened density row, density–kinetic cross term, mixing through $\tilde{R}_{\mathcal{T}_e Z}$; in polar crystals a mixing term at $p_\phi \to 0$. Sum rule: nonlocal pseudopotentials, $-\langle[[\mathcal{T}_e, V_\mathrm{NL}], V_\mathrm{NL}]\rangle$ (cf. Eq. (4.32)); electron–electron forces; divergence for the Coulomb potential of a nucleus (s character at the nucleus).	Sect. 10.7

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Pair absorption of a complex scalar (Limit Check 10.10), Eq. (10.41)	[24, Eqs. (3) and (5)–(8)]; $\xi = 44.8$ gives $\rho_\chi/2m_\phi$ per species	First-order term of T with the contact operators Eq. (10.39); pair factor $1/(\mathcal{V}\sqrt{4E\bar{E}})$; golden rule, Eq. (10.40); S_{ab} at $(\mathbf{0}, 2m_\phi)$. Rows: S $\to$ kinetic row, $g_{\text{eff}}^2 = (\rho_\chi/2m_\phi)m_e^2G^2/m_\phi$ (Eq. (10.42)); P $\to$ axioelectric row at $2m_\phi$ (Eq. (10.43)); V $\to$ dark-photon row with polarization $(p-\bar{p})^\mu$ and $L(2m_\phi)$ (Eq. (10.44)); A $\to$ order v^4 without spin–orbit coupling or magnetic order. $\mathfrak{T}$: $\mathcal{X}$ as for single absorption (Sect. 10.7), $n_g = 0$, $n_T = 0$; rate second order in n_χ.	Response smooth on the scale of the pair kinetic energy $\bar{K} = m_\phi\langle v^2\rangle = 1.60 \times 10^{-6}\, m_\phi$; $qa < 10^{-4}$. Line of half-width γ: peak within 10 % for $\gamma \geq 1.5\,\bar{K}$, on-line rate for $\gamma \geq 3.8\,\bar{K}$. Threshold $2m_\phi \geq 1.17\,\text{eV}$ in silicon. Plot (Fig. 10.18): $x = \gamma/\bar{K}$, 10^{-2}–10^3; $y =$ peak of the halo-averaged line over the formula's peak, and the ratio at $2m_\phi = \Delta E$; one Lorentzian line (model target), halo of Sect. 1.2; shaded $\gamma \geq 1.5\,\bar{K}$.	$\mathbf{Var(K)/\gamma^2}$**:** $\leq 10\,\%$ **for** $\boldsymbol{\gamma \geq 1.5\,\bar{K}}$. Broad lines: $1 - \text{Var}(K)/\gamma^2$; narrow lines: $\pi\gamma \max p_K$, the discrete-transition form of Ref. [24]; then the q dependence of S_{ab}, relative $(qa)^2$; quantum statistics of the pair; interference with second-order amplitudes (Sect. 10.8.4).	Sect. 10.8
Dark Compton scattering of a vector on a free electron at rest (Limit Check 10.11), Eqs. (10.47) and (10.48)	[25, Eq. (2) and App. Eqs. (7)–(12)], their ε = our ζ	Transition operator Eq. (10.45) (seagull plus two orderings of the current rows), Eq. (10.46); free electron at rest: orderings vanish, $S_{nn} = 2\pi n_e\delta(\omega - q^2/2m_e)$, polarization average $\frac{2}{3}$ $\to$ Eq. (10.47). Relativistic: two Compton diagrams, Dirac traces, $k_\mu\mathcal{M}^\mu = 0$, two-body threshold kinematics $\to$ Eq. (10.48), exact at tree level for every x (checked to 10^{-12}). $\mathfrak{T}$: $\mathcal{X}$ = free electrons (impulse limit), $n_g = 0$, $n_T = 0$.	Electrons with binding $\ll \omega_C = m_\phi^2/2(m_\phi + m_e)$: all 14 of silicon above 45 keV. Seagull within 1 % of the tree-level rate for $m_\phi < 165\,\text{keV}$, 10 % below 517 keV. Plot (Fig. 10.19a): $x = m_\phi/m_e$, 10^{-2}–4; $y =$ seagull rate (nonrelativistic (NR) recoil and exact ω_γ) over Eq. (10.48), with $1 - \frac{5}{8}x^3$; shaded $x < 0.32$.	$\boldsymbol{\frac{5}{8}x^3}$**:** $\leq 1\,\%$ **for** $\boldsymbol{m_\phi} < 165\,\text{keV}$**,** 10 % **below** 517 keV. $\frac{5}{8}x^3$ (Eq. (10.49)): $\frac{1}{8}x^3$ kinematic, $\frac{1}{2}x^3$ from the Dirac amplitude (spin, Klein–Nishina structure); in a crystal, binding through the second-order terms of Eq. (10.45) and the shell Compton profiles; $n_r(\omega_\gamma) - 1 \approx -\omega_p^2/2\omega_\gamma^2$.	Sect. 10.8

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Dark Compton scattering of an axion-like particle (Limit Check 10.11): published rate, against Eq. (10.51)	[25, Eqs. (3) and (17)]; squared amplitude [25, Eq. (15)]	Published form: $\Gamma = n_e(e^2 g_{aee}^2/16\pi m_e^2) \times m_\phi^2 (m_\phi + 2m_e)^2/(m_\phi + m_e)^4$. Tree level with $g_{aee}\phi\bar{\psi}\mathrm{i}\gamma^5\psi$, two Compton diagrams: Eq. (10.51), smaller by $(m_\phi + 2m_e)/(m_\phi + m_e)$ (Eq. (10.52)); the printed squared amplitude agrees with the trace. Non-relativistic check: spin seagull $(eg_{aee}/m_e^2)(\partial_t\phi)\boldsymbol{A}\cdot\boldsymbol{s}_e$ from Eq. (1.40), Eq. (10.50), equal to Eq. (10.51) at leading order in x. $\mathfrak{T}$: as the vector row.	None: the published form exceeds the tree-level rate by a factor between 2 ($x \to 0$) and 1 ($x \to \infty$); 1.836 at 100 keV (Worked example 10.8). Plot (Fig. 10.19b): $x = m_\phi/m_e$, 10^{-2}–4; y = published rate and spin seagull over Eq. (10.51); no shading.	**published form high by $\times$1.836 at** 100 keV **($\times$2 as $x \to 0$); spin seagull relative $O(x)$.** Spin seagull: relative order x from the Pauli coupling of the photon ($1 - \frac{5}{2}x$ against $1 - \frac{3}{2}x$); binding as for the vector.	Sect. 10.8
Inelastic electron recoil in crystals (Limit Check 10.12), Eqs. (10.53) and (10.54)	[26, Eqs. (10), (11), (15), (16) and (20)], with the factor q^{-2} of [7, Eq. (3.13)] restored	Density row of Eq. (8.16) with $\omega_{\boldsymbol{q}}$ carrying $-\Delta_\chi$ (Eq. (1.22)); off-diagonal current reduces to the elastic coefficient up to $O(\Delta_\chi/m_\chi)$; angular integral with the absolute value in $v_{\min}$ (Eq. (A.42)); then as Sect. 10.1 (Eq. (D.171)). Shell: $\lvert\boldsymbol{q} - m_\chi\boldsymbol{v}\rvert = k_*$, sphere average, $O(v)$ term cancels. $\mathfrak{T}$: $\mathfrak{T}_0$, unscreened.	Screening: at $m_\chi = 1$ MeV in silicon the unscreened rate exceeds the screened one by 9.4 (elastic), 18.5 ($\Delta_\chi = -5$ eV), 2.5 (-20 eV), within 10 % for $\lvert\Delta_\chi\rvert \geq 57.5$ eV (heavy mediator); light mediator up to 317 at -5 eV, 0.56 at -20 eV. Shell form: $v_{\max} \ll v_\Delta$, $m_\chi \ll 2\lvert\Delta_\chi\rvert/v_{\max}^2$. Plot (Fig. 10.21): $x = \Delta_\chi$, -85 to 2.3 eV, $m_\chi = 1$ MeV; $y = \mathcal{R}_{\mathrm{IP}}/\mathcal{R}_{\mathrm{RPA+LF}}$, heavy and light mediators, tables of Sect. 10.2; shaded $\lvert\Delta_\chi\rvert \geq 57.5$ eV.	**screening $\times$9.4–18.5 at** 1 MeV **(heavy), up to 317 (light); shell form $\approx 2.5\langle v^2\rangle/v_\Delta^2$.** Screening and local fields ($\mathcal{X}$); shell form $O(v^2/v_\Delta^2)$, about $2.5\langle v^2\rangle/v_\Delta^2$ for a heavy mediator; light-mediator end point regularized within $\frac{1}{2}m_\chi v^2$ of $\lvert\Delta_\chi\rvert$ (Eq. (10.55)); the $2p$ shell above 99 eV (Sect. 10.6).	Sect. 10.9

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Boosted dark matter, relativistic density row (Limit Check 10.13), Eqs. (10.57) and (10.59)	[27, Eqs. (4)–(6), 28, Eqs. (9)–(11), 29, Eq. (2)]	Vector mediator with conserved currents, full propagator (Eq. (10.56)); continuity for target and Dirac current; spin trace $(2E_\chi - \omega)^2 - \mathcal{Q}^2$; normalization $4E_\chi E'_\chi$ (Eq. (10.57)); transverse row with ε^{T} (Eq. (10.58)); convexity of $E_\chi(p)$ gives $\omega \leq v\mathcal{Q}$. $\mathfrak{T}$: target in any many-body approximation; the published row departs on the dark-matter side. The missing factor $(1-\omega^2/\mathcal{Q}^2)^2$ in Ref. [27, Eq. (5)] follows from the derivation of Sect. 10.9.	$v_{\chi,\max} \lesssim 0.1$: within 2 % for any target. Silicon, speed v: heavy mediator within 10^{-4} at all v; massless mediator 0.997 ($v = 0.1$), 0.94 (0.5), 0.65 (0.9), 0.17 (0.99). Plot (Fig. 10.22): $x = v$, 0–1; $y = \mathcal{R}/\mathcal{R}_{\mathrm{source}}$ for the exact density row (heavy, massless) and with the transverse row; band $[(1-v^2)^2, 1]$; shaded $v \leq 0.1$.	**transverse row 2.7×10^{-5} ($v = 0.1$) to 5.5 % ($v = 0.99$) of the silicon rate**. Transverse row, $(v^2 - \omega^2/\mathcal{Q}^2)$ times the transverse loss function: with measured ε a Cherenkov pole for $v > \varepsilon_1^{-1/2}$ (0.28 at 1.2 eV); share of the silicon rate 2.7×10^{-5} ($v = 0.1$), 1 % (0.5), 5.5 % (0.99), above the density row below 2 eV for $v \geq 0.3$; anisotropic fluxes; effect on the published limits open (Appendix E.16).	Sect. 10.9
Fermionic absorption on electrons (Limit Check 10.14), Eq. (10.60)	[30, Eqs. (7)–(10) and (30), 7, Eqs. (A.18)–(A.21), 31, Eqs. (1), (2), (14) and (15)]	Master formula with the sum over $\tilde{\nu}$ momenta (Sect. 8.2.3), $\omega = m_\chi - q$ (Eq. (8.14)); spin traces with chirality projectors; normalization $4m_\chi q$; continuity for the vector current, $(m_\chi/q)^2 S_{nn} + \mathrm{Tr}\, S_{j_{\mathrm{T}} j_{\mathrm{T}}}$; response per atom $\pi\lvert f_{\mathrm{ion}}\rvert^2/2E_{\mathrm{ej}}$ with both spins (Eq. (2.22)). $\mathfrak{T}$: $\mathcal{X}$ = IP (atomic); the vector form of the source is a free-vertex ansatz, not current-conserving at bound kinematics (ratio Eq. (10.61)).	Free kinematics, $\omega \ll m_\chi$; scalar operator exact for the density row. Hydrogen-like $1s$ model: the two vector forms agree within 10 % for $m_\chi \geq 4.4E_B$ (silicon K shell, 8.1 keV); xenon $1s$ 0.59 at 56.5 keV, 0.89 at 100 keV. Plot (Fig. 10.23): $x = m_\chi/E_B$, 1–100; $y = \langle\sigma v\rangle/N_e\sigma_F$, scalar and the two vector forms; shaded $m_\chi \geq 4.4E_B$.	**unsized**. The transverse-current row of the vector operator, $O((Z\alpha)^2)$ for inner shells, and Dirac corrections of the same size; the size of the defect is open (Appendix E.26); light scalar mediator $[m_{\mathrm{med}}^2/(m_{\mathrm{med}}^2 + m_\chi^2 - 2m_\chi\omega)]^2$; massive $\tilde{\nu}$ reduces to inelastic scattering.	Sect. 10.9
In-medium decay of an excited state, $\tau^{-1} \sim \pi\lvert\Delta_\chi\rvert^3/\Lambda_d^2$ (Limit Check 10.15), Eqs. (10.62) and (10.63)	[32, Eqs. (2.1) and (2.2)]	Golden rule with the magnetic field of the medium as final state; transverse propagator $1/(\mathcal{Q}^2 - \omega^2\varepsilon^{\mathrm{T}})$; transparent medium n_r^3 (magnetic dipole), n_r (electric dipole); absorbing medium: large-$\mathcal{Q}$ part is the master formula with $\Delta_\chi < 0$ and the magnetic-dipole operators. $\mathfrak{T}$: interface, no target response in the radiative part.	$\lvert\Delta_\chi\rvert \gg \omega_p$: silicon within 10 % above 55 eV (3.3 ω_p); $n_r^3 = 41$–44 below the gap; non-radiative between 3.2 and 22.5 eV. Plot (Fig. 10.24): $x = \lvert\Delta_\chi\rvert/\omega_p$, $\omega_p = 16.6$ eV; $y = (\mathrm{Re}\, n_r)^3$ and $\mathrm{Re}\, n_r$ from measured ε; hatched where $\mathrm{Im}\, n_r > 0.1\, \mathrm{Re}\, n_r$; shaded above 55 eV.	**unsized**. Local field at the particle; finite-$\mathcal{Q}$ ε^{T} for the non-radiative part (Appendix E.16).	Sect. 10.9

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Long-wavelength acoustic structure factor (Limit Check 11.1), Eq. (11.3)	[33, Eq. (30) with Eq. (27)], longitudinal term, with m_p for m_u and the cutoff $\Theta(\omega_{\mathrm{LO}}-\omega)$; prior art [34, Eq. (10)]	One-phonon term of the harmonic nuclear response, Eq. (11.1), at $\boldsymbol{G}=0$, $\mathrm{e}^{-W_\kappa}=1$, $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}=1$, acoustic polarizations $(M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$, linear dispersion; coupling $A_\kappa\tilde{V}_N$; $M_\kappa=A_\kappa m_u$. $\mathfrak{T}$: $\mathcal{K}$ = one-phonon, $n_A=0$, $n_T=0$, $n_B=0$.	$q\ll q_{\mathrm{BZ}}$. Silicon (PBE phonons, direction average): within 10 % of the full one-phonon response for $q<0.48\,q_{\mathrm{BZ}}=1.08\,\mathrm{keV}$, 5 % for $q<0.34\,q_{\mathrm{BZ}}$; 26 % low at q_{BZ}. Plot (Fig. 11.2): $x=q/q_{\mathrm{BZ}}$; $y=$ full ω-integrated response / formula.	$\leq 10\,\%$ **for $q<0.48\,q_{\mathrm{BZ}}$;** 26 % **at q_{BZ}.** Acoustic dispersion below $c_\nu q$, relative $O((q/q_{\mathrm{BZ}})^2)$; polarization and phase changes $O((qa)^2)$; Debye–Waller $2W=q^2\langle u_x^2\rangle$ (3×10^{-3} at q_{BZ}); beyond q_{BZ} the incoherent q^2 form (Sect. 11.3).	Sect. 11.1
Light-mediator single-phonon rate per mass (Limit Check 11.2), Eq. (11.6)	[35, Eq. (35)], with their $\bar{\sigma}_n$ at the reference momentum $m_\chi v_0$	Eq. (11.4) with $F_{\mathrm{DM}}=(q_0/q)^2$; the lower limit $q=\omega_{\mathrm{th}}/c_\nu$ dominates; c_ν cancels between the weight $1/c_\nu$ and the integral; completeness $\sum_\nu(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2=1$. $\mathfrak{T}$ as the row above.	$m_\chi\gg(1+\kappa_\eta)m_{\mathrm{min}}^{\mathrm{ac}}$: within 10 % for $m_\chi\geq44\,m_{\mathrm{min}}^{\mathrm{ac}}$ (silicon 0.28 MeV at $\omega_{\mathrm{th}}=1\,\mathrm{meV}$). Plot (Fig. 11.4): $x=m_\chi/m_{\mathrm{min}}^{\mathrm{ac}}$; $y=$ full rate / formula, one branch and silicon.	$\boldsymbol{(1+\kappa_\eta)m_{\mathrm{min}}^{\mathrm{ac}}/m_\chi}$**:** $\leq10\,\%$ **for** $\boldsymbol{m_\chi\geq44\,m_{\mathrm{min}}^{\mathrm{ac}}}$. Factor $1-(1+\kappa_\eta)m_{\mathrm{min}}^{\mathrm{ac}}/m_\chi$, $\kappa_\eta=3.14$ (within 10 % of the full rate for $m_\chi\geq9\,m_{\mathrm{min}}^{\mathrm{ac}}$); above $q_{\mathrm{BZ}}/2v_{\mathrm{max}}$ the zone cut $1-\omega_{\mathrm{th}}/(c\,q_{\mathrm{BZ}})$; shift of η by c_ν, $O(c^2/v_0^2)$.	Sect. 11.1
Sphere-averaged longitudinal speed $\bar{c}_{\mathrm{LA}}$ (Limit Check 11.3), Eq. (11.10)	[35, App. A 3] (there from density-functional ω_{LA}/q on a sphere of radius 10 eV)	Largest eigenvalue of the Christoffel matrix of the Voigt-ordered tensor (Eq. (C.156)), averaged over directions; replaces $\langle\sum_\nu(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2/c_\nu\rangle$ of Eq. (11.9) by $1/\bar{c}_{\mathrm{LA}}$. $\mathfrak{T}$: $\mathcal{K}$ = one-phonon, harmonic, room-temperature tensor.	Near-isotropic cubic crystals: within 1 % for $0.50\leq A_{\mathrm{Z}}\leq1.70$ (silicon 0.67, germanium 0.92, diamond 0.21 %). Plot (Fig. 11.5): $x=A_{\mathrm{Z}}$ at silicon's C_{11}, C_{12}; $y=\bar{c}_{\mathrm{LA}}\langle\sum_\nu(\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{e}}_\nu)^2/c_\nu\rangle$.	$\leq1\,\%$ **(Si 0.67, Ge 0.92, diamond** 0.21 %**).** Second order in $\Delta_C=C_{11}-C_{12}-2C_{44}$: longitudinal weight of the quasi-transverse branches, $\Delta_C^2 p_Q/(C_{12}+C_{44})^2$, and the Jensen gap of $1/c_{\mathrm{LA}}$; the edge needs $c_{\mathrm{LA,max}}$ (3.7 % above $\bar{c}_{\mathrm{LA}}$ in silicon).	Sect. 11.1

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Born-charge one-phonon structure factor of a total-charge coupling (Limit Check 11.4), Eq. (11.15)	[36, Eqs. (92), (105), (106), 37, Eqs. (19), (23), v1 numbering, 18, Eq. (61)]; one branch: [38, Eqs. (4)–(7)], whose $C_F^2 \propto 1/V_{\text{cell}}$ must be read with the normalization volume, which cancels	Total-charge row, $S_{ZZ} = (2q^2/e^2)L/(1-\mathrm{e}^{-\beta\omega})$ (Sect. 8.4); rank-one total ε below the gap, ionic part from the analytic zone-center modes (Eq. (11.13)); Sherman–Morrison (Eq. (11.14)) turns $1/\varepsilon$ into a sum over the eigenmodes of the nonanalytic dynamical matrix Eq. (5.51); finite q inside the zone through $D(\boldsymbol{Q})$ and the site phases. $\mathfrak{T}$: $\mathcal{K}$ = one-phonon, $n_B = n_g = n_A = n_T = 0$; $\mathcal{X}$: static DFPT for Z^*, ε_∞ (or measured ε_∞).	$q \ll 2\pi/a$, $\omega \ll E_{\text{g}}$, harmonic, $T \ll \omega_{\text{L}}$. GaAs: optical weight within 10^{-4} of the zone-center LO and acoustic weight at most about 1 % for $q < 88$ eV, 0.4 % at 450 eV. Plot (Fig. 11.6): $x = q/(2\pi/a)$ along [100], [110], [111]; y = optical and acoustic one-phonon weight over $Z^{*2}/2\mu_{\text{red}}\omega_{\text{L}}$, DFPT phonons at $\boldsymbol{q}$, rigid Born charges. The dotted acoustic curves estimate a term that the rigid Born charges fix only in part; full curve not computed (finite-q charge vertex).	**$O(qa)$:** 1 % **at** 90 eV**,** 5 % **at** 450 eV **(GaAs, [111])**. Finite-q charge vertex, relative $O(qa)$: dynamical quadrupoles and the piezoelectric charge of the acoustic branches (rigid-Z^* estimate 1 % at 90 eV, 5 % at 450 eV along [111], zero along [100]); static vertex $(\omega_{\text{L}}/E_{\text{g}})^2 = 6\times10^{-4}$ (GaAs); Debye–Waller 8×10^{-8} at 13.7 eV; anharmonic width of the LO line.	Sect. 11.2
One polar branch and the band-integral rate (Limit Check 11.5), Eqs. (11.18), (11.20) and (11.21)	[37, Eqs. (21), (22), (24), v1 numbering] (prior art for the one-branch formula; the band-integral form is Eq. (11.20))	Eq. (11.15) with the zone-center LO eigenvector of Appendix D.14 and Eqs. (5.52) and (5.54) → Eq. (11.17); $\tilde{V} = ee'/q^2$ and Eq. (11.16) at one speed → Eq. (11.18); inverse moment of $1/\varepsilon$ over the phonon band (Eq. (11.19)) → $\sum_\nu w_\nu^{\text{pol}} = 1/\varepsilon_\infty - 1/\varepsilon_0$; zeroth moment → $\mathcal{I}_{\text{pol}}$ and its bracket. $\mathfrak{T}$: as the row above.	Rank-one ε, $q \ll 2\pi/a$, $\omega \ll E_{\text{g}}$; lossless factorized ε per principal axis, with ε_0, ε_∞ of that axis and model (interlacing poles and zeros); band-uniform kinematic weight for $\mathcal{R} \propto \mathcal{I}_{\text{pol}}$: sapphire ordinary axis within 7 % above 64 keV, factor 2.1 at 40 keV. Plot (Fig. 11.8): $x = m_\chi$; y = exact per-branch rate over $\mathcal{I}_{\text{pol}} K_\eta(\bar{\omega})$ and over the one-branch form at $\omega_{\text{L,max}}$.	≤ 7 % **above** 64 keV **(sapphire);** **×2.1 at** 40 keV. Per-branch kinematic weights $K_\eta(\omega_{\text{L}\nu}, m_\chi)$ near the edges; damping, $O(\gamma_{\text{L}}/\omega_{\text{L}})$ (1.8 % for the top LO of sapphire at 77 K); the finite-q corrections of the row above.	Sect. 11.2
Static dielectric constant of one polar branch (Limit Check 11.6), Eq. (11.22)	[39, Chap. II, 37, Eq. (B14) at $\omega = 0$, v1 numbering]	Eq. (5.53) at $\omega = 0$; with several branches $\varepsilon_0 - \varepsilon_\infty = (e^2/\Omega)\sum_\nu (\boldsymbol{z}_\nu \cdot \hat{\boldsymbol{e}})^2/\omega_{\text{T}\nu}^2$ (Appendix D.14). $\mathfrak{T}$: $n_B = n_A = n_T = 0$, harmonic ions, adiabatic electrons; no final state ($\mathcal{K}$ not applicable).	One polar transverse branch per axis (exact for cubic diatomic crystals). Plot (Fig. 11.9): for each branch of sapphire (ordinary and extraordinary axes), x = its share of $1/\varepsilon_\infty - 1/\varepsilon_0$, y = its share of $\varepsilon_0 - \varepsilon_\infty$; GaAs at (1, 1).	**exact for cubic diatomics; in sapphire the rate branch holds** 2.7 % **of $\boldsymbol{\varepsilon_0 - \varepsilon_\infty}$.** Sum over the other polar branches; in sapphire the branch that dominates the rate carries 2.7 % of $\varepsilon_0 - \varepsilon_\infty$; damping and temperature dependence of ε_0 at the percent level.	Sect. 11.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Born charge over ε_∞ from the LO–TO splitting (Limit Check 11.7), Eq. (11.23)	[39, Chap. II, 37, Eqs. (B11), (B15), v1 numbering]; [40]	Eq. (5.52) solved for Z^* and divided by ε_∞; GaAs, Worked example 5.6: $Z^* = 2.11$, $Z^*/\varepsilon_\infty = 0.194$. Error budget: $-\frac{1}{2}\delta\varepsilon_\infty/\varepsilon_\infty + (\omega_\mathrm{L}\delta\omega_\mathrm{L} - \omega_\mathrm{T}\delta\omega_\mathrm{T})/(\omega_\mathrm{L}^2 - \omega_\mathrm{T}^2)$. $\mathfrak{T}$: as the row above.	One polar branch; measured ω_L, ω_T, low-temperature ε_∞. Independent frequency errors amplified by $\omega^2/(\omega_\mathrm{L}^2 - \omega_\mathrm{T}^2) = 6.9$ (GaAs): 1.0 % for 0.1 % on each frequency and 1 % on ε_∞. Plot (Fig. 11.10): x = residual r of the Lyddane–Sachs–Teller relation of the inputs; y = static-route over splitting-route Z^*, GaAs and LiF.	1.0 % **for** 0.1 % **frequency and** 1 % $\boldsymbol{\varepsilon_\infty}$ **errors (GaAs)**. The static route amplifies r by $\omega_\mathrm{L}^2/2(\omega_\mathrm{L}^2 - \omega_\mathrm{T}^2)$ (3.5 for GaAs, 0.6 for LiF); temperature dependence of ε_∞ at the percent level.	Sect. 11.2
Multiphonon incoherent approximation and Poisson tail (Limit Check 11.8), Eq. (11.26)	[33, Eqs. (15), (23), (24), (36), 41, Eqs. (7), (10), 34, Eq. (12)]	Harmonic response at all orders Eq. (11.24) (pair form of Eq. (D.83)); drop every pair with $\Delta\boldsymbol{R}_{\kappa\kappa'} \neq 0$ (zone average, exact for one atom per cell up to $O(nq_\mathrm{BZ}/q)$, Appendix D.31); projected density of states, $g_{\mathrm{ph},\kappa}/3$ on cubic sites; $T = 0$; Einstein site $P_n = \delta(\omega - n\omega_0)$. Map: $\bar{f}_d \to c_\kappa$, $m_d \to M_\kappa$, $D_d \to g_{\mathrm{ph},\kappa}/3$, $\Omega_c \to \Omega$. First moment of order n exact including coherent terms, Eq. (11.27). $\mathfrak{T}$: $\mathcal{K}$ = multiphonon (incoherent), $n_A = 0$, $n_B = 0$, any n_T through the Bose factors ($n_T = 0$ at 10 mK).	$q \gg \max(n, 2W_\kappa)\, q_\mathrm{BZ}$ for each order $n \geq 2$; ω-integrated two- and three-phonon weights of silicon within 10 % for $q \geq 2.95\, q_\mathrm{BZ} = 6.6$ keV (three phonons at every q), one phonon only above $7.9\, q_\mathrm{BZ}$; pointwise in ω up to 17 % at 10 keV. Plot (Fig. 11.11): $x = q/q_\mathrm{BZ}$ (log); $y = \mathfrak{W}^{(n)}/\mathfrak{W}^{(n)}_\mathrm{inc}$ for $n = 1, 2, 3$, Si PBE, $T = 0$, direction-averaged; dotted nearest-neighbor pairs; shaded $q \geq 2.95\, q_\mathrm{BZ}$.	$\boldsymbol{\delta_2^\mathrm{coh} = +0.35\ (q \to 0),\ -0.11\ (q_\mathrm{BZ}),\ -0.018}$ **(10 keV)**. Distinct pairs, exactly δ_n^coh of Eq. (11.29) in the weight, led by the nearest-neighbor shell; pointwise the certified band Eq. (11.30); anharmonic terms (Sect. 11.5); for a charge coupling, the valence deformation of the vertex above q_BZ.	Sect. 11.3
Second-order thermal diffuse scattering (Limit Check 11.9), Eq. (11.31)	[42] (classical $1/\lvert\boldsymbol{q} - \boldsymbol{G}\rvert$ law)	Two-phonon term of Eq. (11.25) summed over creation and absorption and integrated over ω (Eq. (D.186)); both phonons soft and acoustic, $\boldsymbol{e}_{\kappa\nu} = (M_\kappa/M_\mathrm{cell})^{1/2}\hat{\boldsymbol{e}}_\nu$; classical occupations $(2n+1)/2\omega \to T/\omega^2$; completeness of the acoustic polarizations, $\omega_\nu \geq c_\mathrm{min} Q$; $\int \mathrm{d}^3Q/(Q^2\lvert\boldsymbol{k} - \boldsymbol{Q}\rvert^2) = \pi^3/k$ (Eq. (D.187)). $\mathfrak{T}$: $\mathcal{K}$ = multiphonon (two phonons, coherent), $n_T = 2$, $n_A = 0$, $n_B = 0$.	Soft pair classical, $c_\nu\lvert\boldsymbol{q} - \boldsymbol{G}\rvert \ll T$, and $\lvert\boldsymbol{q} - \boldsymbol{G}\rvert \ll q_\mathrm{BZ}$. Isotropic model of silicon, $\boldsymbol{q} - \boldsymbol{G} \parallel \boldsymbol{q}$ at $\boldsymbol{G}_{111}$: classical coefficient 0.46 of the bound; within 10 % of the law for $\lvert\boldsymbol{q} - \boldsymbol{G}\rvert \leq 0.13\, q_\mathrm{BZ}$ (293 K) and $0.065\, q_\mathrm{BZ}$ (77 K). Plot (Fig. 11.13): $x = \lvert\boldsymbol{q} - \boldsymbol{G}\rvert/q_\mathrm{BZ}$ (log); $y = \lvert\boldsymbol{q} - \boldsymbol{G}\rvert\, \mathfrak{W}^{(2),\mathrm{th}}$ over the bound coefficient, model at 77 and 293 K, classical law dashed, boundaries T/c_T marked.	**classical coefficient 0.46 of the bound;** $\boldsymbol{\leq 10}$ **% for** $\boldsymbol{\lvert q - G\rvert \leq 0.13\, q_\mathrm{BZ}}$ **(293 K)**. Quantum occupations of the pair (fall below the law for $c_\nu\lvert\boldsymbol{q} - \boldsymbol{G}\rvert \gtrsim 0.2$–$0.4\,T$); terms linear in $n_{\boldsymbol{Q}\nu}$ ($n_T = 1$, logarithmic in $\lvert\boldsymbol{q} - \boldsymbol{G}\rvert$); dispersion and optical pairs at the zone scale; $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\tau}_\kappa}$ at order ka. At 10 mK: boundary $2 \times 10^{-5}\, q_\mathrm{BZ}$, coefficient 10^{-9} of its room-temperature value.	Sect. 11.3

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Free nuclear recoil with the Helm form factor (Limit Check 11.10), Eqs. (2.28) and (11.35)	[33, Eqs. (37)–(39), (42), 1, Eq. (104), 43]; [2, Eqs. (90)–(92), 41, Eqs. (4), (12)]	Self part of Eq. (11.24) in the exact form $I(t) = \langle e^{iHt} e^{-i(H+D)t} \rangle$, $D = E_R + \boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa$ (Eq. (D.188)); mean E_R and variance $4E_R E_\kappa^{\text{kin}}$ exact for any local potential (Eq. (D.189)); drop $[H, D]$ (impulse approximation, Eq. (11.33)); harmonic Gaussian Eq. (11.34), $\sigma_\omega^2 = E_R\langle\omega\coth(\omega/2T)\rangle$, $= \Delta_d^2$ of the source at $T = 0$; $x_R \to \infty$; vertex $A_\kappa F_N(q)$; Eq. (8.16) (one row) with the orientation-independent response, $v_{\min}$ from Eq. (1.23) at $\omega = q^2/2M$ (Appendix D.32). $\mathfrak{T}$: $\mathcal{K}$ = impulse (zero width), $n_B = 0$, mean exact at every n_A and n_T.	$x_R = E_R/\langle\omega\rangle \gg 1$ and a rate weight smooth across σ_ω. Silicon, heavy mediator, $m_\chi = 1\,\text{GeV}$: within 10 % for $x_R \geq 2.3$ ($q \geq 70\,\text{keV}$), 2 % for $x_R \geq 4.3$; $m_\chi = 50\,\text{MeV}$: free form 5–21 % high for $3 < x_R < 8$ and more than 10 % low from $4.9\,\sigma_\omega$ below the endpoint. Onset ($x_R = 4$): Si 92.5 keV, 164 meV; Ge 113 keV, 95 meV (estimated $\langle\omega\rangle = 23.6\,\text{meV}$). Plot (Fig. 11.15): $x = x_R$ (log); y = crystal rate over Eq. (11.35) at $E_R = \omega$, Si PBE density of states (DOS), $T = 0$, $m_\chi = 1\,\text{GeV}$ and 50 MeV; dotted impulse Gaussian; shaded $x_R \geq 2.3$.	**relative width $x_R^{-1/2}$: ≤ 10 % for $x_R \geq 2.3$, 2 % for $x_R \geq 4.3$ (Si, 1 GeV).** Doppler width $\sigma_\omega = (4E_R E_\kappa^{\text{kin}})^{1/2}$, relative $x_R^{-1/2}$ at $T = 0$ (Eq. (11.32)); then the skewness $c_3 = (q^2/2M_\kappa^2)\langle\partial_{\hat{q}}^2 V\rangle$ (Eq. (D.190)), relative $(\langle\omega^2\rangle/\langle\omega\rangle^2)x_R^{-1/2}$; anharmonic E_κ^{kin} (Sect. 11.5); Migdal excitation of the electrons (Chap. 12).	Sect. 11.4
Fermionic absorption on nuclei (Limit Check 11.11), Eq. (11.36)	[44, Eqs. (1), (3), (4)] (their σ_{NC}, $\rho_\chi \simeq 0.4\,\text{GeV}\,\text{cm}^{-3}$)	Vector rate of Eq. (10.60) with the nucleon density; continuity exact for the nucleon current; transverse current dropped ($O(E_\kappa^{\text{kin}}/M, q^2/M^2)$); free nucleus at rest, line integral of $\delta(\omega - (m_\chi - \omega)^2/2M)$ gives $1/(1 + q_{\text{abs}}/M)$; density row alone (scalar) $m_\chi^2(1 - 2m_\chi/M)$, with continuity $m_\chi^2(1 - m_\chi/M)$. Crystal: zero-phonon term only at $\lvert\boldsymbol{G}\rvert = m_\chi$ within $O(v)$; incoherent weight $1 - e^{-2W_\kappa(q_{\text{abs}})}$ (Appendix D.32). $\mathfrak{T}$: $n_B = 0$, $n_T = 0$, $\mathcal{K}$ = multiphonon (incoherent) below the impulse onset, impulse above.	Silicon, vector operator, form factor divided out: within 10 % for $61\,\text{keV} \leq m_\chi \leq 3.0\,\text{GeV}$ (density row alone to 1.4 GeV). Line width $x_R^{-1/2} = 46.3\,\text{keV}/m_\chi$ exceeds the halo's 1.5×10^{-3} below $m_\chi = 32\,\text{MeV}$. Plot (Fig. 11.16): $x = m_\chi$ (log, 3 keV to 10 GeV); y = crystal over free-nucleus rate, vector (solid) and density row alone (dotted); shaded 61 keV–3.0 GeV.	**≤ 10 % for** 61 keV–3.0 GeV**; zero-phonon weight 0.94 at** 10 keV. Low mass: zero-phonon weight e^{-2W_κ} absent (0.94 at 10 keV); coherence corrections of Sect. 11.3 below about 7–18 keV. High mass: recoil factor $1 - m_\chi/M$ and F_N at q_{abs} instead of m_χ; nuclear excitation for $q \gtrsim 1/R_{\text{nuc}}$.	Sect. 11.4

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Anharmonic corrections to multiphonon weights (Limit Check 11.12), Eqs. (11.38) to (11.40)	[45, Eqs. (26), (39), (45)–(48), (65)]; [34, Eq. (13)]	Self response at q above a few q_{BZ}; expand $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}}$ to $O(q^2)$: displacement correlation $= -2\,\mathrm{Im}\,\mathsf{D}_\nu$ (Eq. (6.29)) with the cubic bubble; first order in Π_ν off the poles, $-\mathrm{Im}\,\Pi_\nu = \omega_{\boldsymbol{Q}\nu}\gamma^{\mathrm{anh}}_{\boldsymbol{Q}\nu}(\omega)$ (Eq. (5.70)). First order in $\Phi^{(3)}$ vanishes by time reversal. Exact moments Eq. (11.37) give three linear constraints on δ_{anh}. $\mathfrak{T}$: $n_A = 2$ (rate order; planned as $n_A = 1$), $\mathcal{K}$ = multiphonon, $n_B = 0$, $n_T = 0$.	$4\lambda_3 \approx \eta_{\mathrm{anh}}$ (site) $\lesssim 0.04$; $q \gtrsim q_{\mathrm{BZ}}$ (below, translation invariance forces q^4, Eq. (8.31)). Silicon model ($\lambda_3 = 0.01$, $\lambda_4 = 2.5 \times 10^{-4}$, $\omega_0 = 30.4\,\mathrm{meV}$): leading terms for $n = 2, 3$ within 10 % of the exact model for $q \geq 2\,\mathrm{keV}$; harmonic weights within 10 % only above 5, 12, 20 keV for $n = 2, 3, 4$. Plot (Fig. 11.17): $x = k$ (log), $y =$ exact model weight over harmonic for $n = 2, 3, 4$; dashed $1 + 16\lambda_3^2/k^2$, $1 + 108\lambda_3^2/k^2$; shaded $k \geq 0.05$. Model, labelled.	$\leq 10\,\%$ **for $\boldsymbol{q} \geq 2\,\mathrm{keV}$ (Si model); impulse width −0.14 to +0.08 %.** $\lambda_3^4 k^2$ term in the three-phonon weight (dominant for $k \lesssim 0.05$); branch mixing of Π; impulse width change −0.14 to +0.08 % (silicon).	Sect. 11.5
Solid helium, single-phonon rate, quasi-harmonic and self-consistent (Limit Check 11.14), Eqs. (5.75) and (11.1)	[46, Eqs. (2), (4), (5)]; [47, Table I] (reference Lindemann ratios)	Eq. (8.16) (one row) with the one-phonon response; quasi-harmonic: harmonic expansion about the static lattice at the compressed volume (PBE + dispersion correction in the source); self-consistent: Gaussian-averaged curvature Eq. (5.75). $\mathfrak{T}$: $n_A = 0$ (quasi-harmonic) or resummed even orders (SCHA), $\mathcal{K}$ = one phonon.	Lindemann ratio r_{L}. Cell model (12 fixed fcc neighbors, Morse pair with the Aziz well, exact): quasi-harmonic one-phonon weight within 10 % for $r_{\mathrm{L}} \leq 0.115$ ($V_m \leq 6.5\,\mathrm{cm}^3\,\mathrm{mol}^{-1}$, $\approx 2\,\mathrm{GPa}$); static curvature negative for $r_{\mathrm{L}} \geq 0.20$; SCHA within 10 % for $r_{\mathrm{L}} \leq 0.227$. Plot (Fig. 11.20): $x = r_{\mathrm{L}}$; $y =$ approximate over exact weight (solid) and frequency over excitation energy (dashed); shaded $r_{\mathrm{L}} \leq 0.115$. Model, labelled.	**Gaussian (SCHA) miss $\leq 10\,\%$ for $\boldsymbol{r_{\mathrm{L}} \leq 0.227}$, 0.896 of exact at $\boldsymbol{r_{\mathrm{L}} = 0.230}$ (cell model).** Short-range correlations that a Gaussian misses near melting; cubic bubble of the physical frequencies (Appendix E.8); exact moments and the band (Limit Check 11.13).	Sect. 11.5

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Markov–Krein band and its tail (Limit Check 11.13), Eqs. (C.91), (C.93) and (C.95)	[48, 49] (book level; classical moment theory)	Moments μ_0–μ_4 of the self response from Eq. (11.37) (equal-time averages, exact at every n_A); band of Appendix C.10 on the real line, valid at $T = 0$ and sharp on $\omega \geq 0$ where all canonical nodes are ≥ 0. $\mathfrak{T}$: exact moments, so exact in n_A and n_T when the moments are; no assumption on $\mathcal{K}$. Leading correction on a qualifier, the target model and method (Sect. 9.4.2): the skewness of the moment problem, not a step of n_A.	Absolute, not relative. Silicon, $q = 100\,\mathrm{keV}$: width 0.670 at E_R (band 0.191–0.861); weight above threshold fixed to 10 % only for $\xi \leq -1.84$ ($\omega \leq 28\,\mathrm{meV}$); band/tail weight 6 at $\xi = 3$, 63 at $\xi = 5$. Half-line band narrower by at least 6.8 % (100 keV) and 24.5 % (50 keV) in the windows where a node is negative (Eq. (C.96)). Plot (Fig. 11.19), fallback: $x = \xi$; y (log) = ϱ_{MK} and $1 - F$ for silicon at 100 keV and a Gaussian; dashed tail form; shaded $\xi \leq -1.84$.	**tail form exceeded by** 34 % **at $\xi = 3$,** 20 % **at $\xi = 6$**. Target model and method: tail form exceeded by 34 % at $\xi = 3$, 20 % at $\xi = 6$ (skewness of the moment problem, relative $1/\xi$); sharp half-line band (Stieltjes canonical representations, not derived in the book).	Sect. 11.5
Incoherent spin absorption (Limit Check 11.15), Eq. (11.43)	[50, Eqs. (25)–(31)] (their $g_N = g_{aNN}/2m_N$, $g_p\lambda_j = g_{ai}/2m_N$, ξ_j the isotope-averaged $\lambda_j^2 J(J+1)/M$, $D_j = g_{\mathrm{ph},\kappa}/3$)	Spin-current row $\mathcal{J}^{\alpha\alpha}$ of Eq. (8.10) reduced to nuclei (Eq. (8.8)); $(\nabla\phi)\cdot\boldsymbol{s}$ term has no final state at $\omega = m_\phi$ (spin splittings $\lesssim 0.1\,\mu\mathrm{eV}$; phonon through $p_\phi^2\langle u_x^2\rangle \approx 10^{-18}$); unpolarized, degenerate spins factorized from the lattice, $\langle J^a J^b\rangle = \delta^{ab} J(J+1)/3$ per nucleus, distinct terms vanish exactly; $\langle f|\boldsymbol{P}_I|0\rangle = \mathrm{i}M_I\omega\langle f|\boldsymbol{u}_I|0\rangle$, one phonon exact in the harmonic crystal; Eq. (11.41) with Eq. (8.19); host density of states for the spin isotope. $\mathfrak{T}$: $\mathcal{K}$ = one-phonon (incoherent, exact), $n_A = 0$, $n_B = 0$, $n_T = 0$ at 10 mK.	Harmonic crystal, zero field, $T \ll \omega$. Silicon (^{29}Si, $\delta^M = -0.0317$): within 10 % of the single-defect rate for $m_\phi < 55.0\,\mathrm{meV} = 0.865\,\omega_{\max}$ (PBE); band integral unchanged. Plot (Fig. 11.21): $x = m_\phi$ (meV); $y = g_{29}/g_{\mathrm{ph},\kappa}$, exact single defect Eq. (11.44) (solid), first order (dotted), formula one (dashed); shaded within 10 %.	**−3.5 to +10 % below** 55 meV **(Si); 1.78 and 0.72 near the top of the optical band**. Mass of the spin carrier: $g_{29}/g_{\mathrm{ph},\kappa} = 1 + \delta^M h_{\mathrm{iso}}(\omega)$, Eq. (11.45); quadrupole alignment of $J > \frac{1}{2}$ nuclei at mK; anharmonic multiphonon terms of $\mathcal{J}^{\alpha\alpha}$.	Sect. 11.6

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Lifshitz mass-defect Green function (Limit Check 11.16), Eq. (11.44)	[51, 52] (classical result; equations not verified against the original texts)	Harmonic host with unchanged force constants; one site with mass $\bar{M}(1-\delta_i^M)$; Dyson equation for $(\Phi-\omega^2\mathsf{M})^{-1}$ closes on the site; cubic site, G_h a multiple of the identity; $\mathrm{Re}\,\mathsf{G}_\mathrm{h}$ by Kramers–Kronig from $g_{\mathrm{ph},\kappa}$. $\mathfrak{T}$: $n_A=0$; exact for one defect.	One defect in a perfect host; pair terms below 10 % except within a meV of the sharp edges of the density of states (natural Si around ^{30}Si: below 2.2 % up to 55 meV); local mode for $\delta^M>0.22$ (Si). Plot (Fig. 11.22): $x=\omega$ (meV); $y=g_i/3$ for $\delta^M=-0.067$, -0.5, $+0.3$ (local mode at 65.1 meV, weight 0.51) and the host; shaded where the leading pair term exceeds 10 % (natural mixture; 1 % of the strong defects).	**pair term $\leq 2.2\,\%$ (natural Si, to 55 meV); $g_\mathrm{md}=2.0\times10^{-4}$.** Pair term, first order in the second defect: $\mathsf{G}_\mathrm{h}\to\mathsf{G}_\mathrm{h}+\sum_j x_j t_j \times[\partial\mathsf{G}_\mathrm{h}/\partial(\bar{M}\omega^2)-\mathsf{G}_\mathrm{h}^2]$, $t_j=-\delta_j^M\bar{M}\omega^2/(1+\delta_j^M\bar{M}\omega^2\mathsf{G}_\mathrm{h})$; isotope-scattering self-energy, Eq. (E.81); changes of force constants at the defect.	Sect. 11.6
Daily modulation from the anisotropic response (Limit Check 11.17), Eqs. (11.49) and (11.50)	[53, Eqs. (6), (10), (17), (25), (37)–(41), 54, Eqs. (18)–(24), (28)–(31)]	Eq. (8.16) (Eq. (A.43)) for a density row with $\tilde{V}(\lvert\boldsymbol{q}\rvert)$, without the angular average; elastic kinematics, $v_\mathrm{min}=\omega/q+q/2m_\chi$; laboratory halo $f_\mathrm{gal}(\boldsymbol{v}+\boldsymbol{v}_\mathrm{E})$, Radon transform $\hat{f}_\mathrm{gal}$ of the truncated Maxwellian; Legendre series in $\hat{\boldsymbol{q}}\cdot\hat{\boldsymbol{v}}_\mathrm{E}$ and the addition theorem Eq. (C.21); $\mathcal{H}_0=\eta$. Evenness from inversion or time reversal (Appendix A.4); point-group invariants (Appendix C.3); for the nuclear response the incoherent approximation Eq. (11.26) with site tensors, isotropic at cubic sites at every order and T. Map: their $\hat{f}_{\ell m}$ and $T_{\ell m}$ (or $W_{\ell m}$) $\to 2\pi\mathcal{H}_\ell$ (up to the normalization of $Y_{\ell 0}$) and $S_{\ell m}$ (or the loss-function components); their $D_d^{ij}\to D_\kappa$. $\mathfrak{T}$: panel 1 $\mathcal{X}$ = random-phase head, model bands, $n_B=0$; panel 2 $\mathcal{K}$ = multiphonon (incoherent), $n_T=0$, $n_A=0$, $n_B=0$.	Velocity-independent couplings; the first anisotropic term alone holds away from the kinematic edge. Silicon loss function, 5 eV, 10 MeV: $\ell=4$ within 10 % from 2.7 keV ($v_\mathrm{min}\leq 590\,\mathrm{km\,s^{-1}}$) to the end of the model, except where $\mathcal{H}_4$ or L_4 passes close to zero (near 3.4, 8 and 10 keV); sapphire nuclear above 80 meV, 10 MeV: $\ell\leq 2$ and $\ell\leq 4$ within 10 % to 40 keV (edge 53 keV). Plot (Fig. 11.24): $x=q$ (keV, log); $y=$ anisotropic part of the rate at fixed q for a wind along [100], [111] (panel 1) and along and across c (panel 2); solid full (all fitted $\ell\leq 8$; direct angular integral), dashed first term, dotted next order; shaded where $\ell\leq 4$ is within 10 % (floor 10^{-4}).	**$\boldsymbol{\mathcal{H}_6/\mathcal{H}_0\lesssim 10^{-2}$ for $v_\mathrm{min}\lesssim 600\,\mathrm{km\,s^{-1}}$.** Next multipoles; local fields and first-principles bands for the electronic components (Appendix E.16); coherent terms for the nuclear response below a few q_BZ (Sect. 11.3); velocity-dependent couplings; odd parts when inversion and time reversal are both broken ($O(q^3)$ for densities).	Sect. 11.7

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Single-phonon absorption (Limit Check 11.18), Eqs. (11.53), (11.55) and (11.57)	[55, Eqs. (2), (7), (14), (16), (40)–(44)]; [38, Eqs. (2), (3)]	Absorption form of Eq. (8.19) at $(\boldsymbol{p}_\phi, m_\phi)$; acoustic branches closed ($c_\nu v \ll 1$); zone-center optical phonons; rank-one screening Eq. (8.36) with phonon and clamped-nuclei parts, Sherman–Morrison → Eq. (11.54): vertices shifted by $(v_C \chi_{aZ}/\varepsilon_\infty)\boldsymbol{p}$; electron row → $Z_\kappa - Z^*_\kappa/\varepsilon_\infty$; coupling-to-mass invariance; two-atom eigenvector → d (Eq. (11.56)); line weight of Eq. (11.17) → Eq. (11.57). Dark photon: Sect. 8.4.5. $\mathfrak{T}$: $\mathcal{K}$ = one-phonon, $n_B = n_g = n_A = n_T = 0$; $\mathcal{X}$: static DFPT or measured Z^*, ε_∞; n_D not applicable.	$\omega \ll E_g$, no local fields, harmonic, $T \ll \omega_L$; constant-width line for $\lvert m_\phi - \omega_L \rvert < 5\gamma_L$. Away from near-cancellations. GaAs electron-coupled scalar: order of magnitude only, $\partial \ln \mathcal{R}/\partial \ln(Z^*/\varepsilon_\infty) = 8.8$; rigid-ion couplings ($Q = \pm 3$) high by 8.2 (6.7 to 9.0 for $Z^* = 2.16$ to 2.09); PBE inputs low by 250. Plot (Fig. 11.26): $x = m_\phi$ across the GaAs optical band; y = events per kg-yr per $g^2_{\phi ee}$, residue, rigid-ion and PBE couplings; widths $10^{-2}\omega_\nu$ (modelling choice).	**core kinetic row $\sim 10^{-4}$ per u in d (estimate); isotopic mass 5 % in d; static vertex** 0.3 %. Kinetic-energy row of the core electrons, $-\langle \mathcal{T} \rangle_\kappa / m_e$ per site (not computed); isotopic mean masses (0.01 u on Ga) and isotope disorder (Sect. 11.6); frequency-dependent phonon self-energy beyond $5\gamma_L$; static electronic vertex $(\omega_L/E_g)^2$, 0.3 % of d in GaAs.	Sect. 11.8
Bulk acoustic resonator (Limit Check 11.19), Eq. (11.60)	[56, Eqs. (2), (3), (5)–(8)]; the source writes m_V^2 for $m_\phi \omega_n$ (equal on resonance)	Classical field uniform over the body ($L_s \ll 2\pi/m_\phi$); quasi-static internal field of an ellipsoid, $\zeta \boldsymbol{E}'/[1 + N(\varepsilon_0 - 1)]$ (Eq. (11.59)); piezoelectric polarization $e^{pz}_{zzz}\partial_z u_z$; plano-convex Gaussian modes normalized to $\int \rho_T \boldsymbol{U}_n^2 = 1/2\omega_n$; odd n only; golden rule with a Lorentzian line ($32 = 2\pi \times 8 \times 2/\pi$); halo average → $\mathfrak{Q}_{\text{eff}}$ (Eq. (11.61)) and $\mathfrak{Q}_{\text{DM}} = (\pi/2)\lvert \mathrm{d}\eta/\mathrm{d}v \rvert$ (Eq. (11.62)). $\mathfrak{T}$: $\mathcal{K}$ = one-phonon (one mode of the body), $n_B = n_g = n_A = 0$, $\mathcal{X}$: static DFPT tensors; coherent field.	$\mathfrak{Q}_p \gg \mathfrak{Q}_{\text{DM}}$ or $\ll$ (within 10 % for $\mathfrak{Q}_p/\mathfrak{Q}_{\text{DM}} > 7.9$ or < 0.2; 1.8 too high at equality); records $\gg \min(t_{\text{coh}}, 1/\gamma_n)$; shield factor $\min\{1, (m_\phi R_s)^2\}$. Plot (Fig. 11.28): $x = \mathfrak{Q}_p/\mathfrak{Q}_{\text{DM}}$; $y = \mathfrak{Q}_{\text{eff}}/\mathfrak{Q}_{\text{DM}}$ for the halo of Appendix A.8 ($\mathfrak{Q}_{\text{DM}} = 1.34 \times 10^6$ at $v = 256\,\text{km/s}$), against the min prescription.	**$\times 1.8$ at $\mathfrak{Q}_p = \mathfrak{Q}_{\text{DM}}$; ≤ 10 % for $\mathfrak{Q}_p/\mathfrak{Q}_{\text{DM}} > 7.9$ or < 0.2.** Convolution of field and mode lines at $\mathfrak{Q}_p \sim \mathfrak{Q}_{\text{DM}}$; annual modulation of $\mathfrak{Q}_{\text{DM}}$ through η; short records (coherent regime, Sect. 14.6); off-diagonal piezoelectric elements and finite-element mode shapes (Appendix E.26).	Sect. 11.8

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Resonant-mass drive (Limit Check 11.20), Eqs. (11.63) and (11.64)	[57, Eqs. (1), (3), (4), (6)–(8)]	Coulomb Hamiltonian with $M_\kappa \to \infty$ depends on α, m_e only through a_0 and $\alpha^2 m_e \to a \propto a_0$; laboratory displacement $\boldsymbol{u} + h_{\mathrm{mod}}\boldsymbol{\xi}$, stress $C : (\nabla \boldsymbol{u}_{\mathrm{lab}} - h_{\mathrm{mod}}\not{\mathrm{K}})$; projection on a mode; free bar in closed form, Eq. (11.65). $\mathfrak{T}$: classical elastic response at $\boldsymbol{p}_\phi \to 0$; $n_B = 0$ (adiabatic electrons), $\mathcal{K}$ not applicable.	Field uniform over the body, $m_\phi v L \ll 1$ (to $\omega \approx 2 \times 10^7 \omega_1$ for a bar with $c_{\mathrm{b}} = 5\,\mathrm{km/s}$); $\omega \ll E_{\mathrm{g}}$; linear elasticity; cubic or isotropic dilation. Plot (Fig. 11.29): $x = \omega/\omega_1$; $y = \lvert u(L_{\mathrm{b}}/2)\rvert / h_{\mathrm{mod}} L_{\mathrm{b}}$, exact free bar against the lowest mode with the ω^2 drive ($\mathfrak{Q} = 100$); within 10 % for $\omega < 2.1\omega_1$.	**relativistic $\mathbf{d\ln(a/a_0)/d\ln\alpha}$: $\mathbf{-1.97 \times 10^{-2}}$ (Cu), $\mathbf{-2.2 \times 10^{-4}}$ (Si)**. Relativistic shift of a/a_0; proton-to-electron mass ratio -1.12×10^{-3} (Cu) [58, Table II]; tensor h_{ij} in non-cubic crystals; modulation of C (parametric, order $h_{\mathrm{mod}} u$); higher modes above ω_1.	Sect. 11.8
Atomic Migdal effect at dipole order (Limit Check 12.1), Eqs. (12.11) to (12.13) and (12.15)	[59, Eqs. (55), (57), (73), (86)–(92)]; tables of [60]	Migdal entry of the joint response for one atom (Appendix D.33.5); exact center-of-mass separation, $\boldsymbol{X} = \boldsymbol{R}_{\mathrm{cm}} - (m_e/m_A)\sum_e \boldsymbol{\rho}_e$ (Eqs. (12.9) and (12.10)), internal Hamiltonian with reduced mass and mass polarization; first order in the dark-matter coupling, all orders in q; free recoil of the atom; expansion to $O(q_e^2)$; one Slater determinant, Pauli-blocked transitions removed. Checks: closure and the exact first moment $(N_e m_e/M)E_R$ (Eq. (12.14); Ge tables hold 96.8 %). $\mathfrak{T}$: $n_B = 2$, $n_D = 2$, $\mathfrak{M}$ = sudden, $\mathcal{K}$ = impulse (planned as $n_D = 1$, amplitude order).	$\xi_D = q_e a_0 \ll 1$. Within 10 % of the all-multipole single-ionization probability for $\xi_D \leq 0.127$ (Ge; $m_\chi \leq 14\,\mathrm{GeV}$ at the endpoint) and ≤ 0.188 (Si; 9 GeV); hydrogen, all final states, to 0.42. In a crystal: core shells only (radius $\ll$ bond, binding $\gg$ valence width), above the impulse onset ($x_R \geq 4$). Plot (Fig. 12.3): $x = \xi_D$ (log); y = all multipoles over dipole; hydrogen exact with $1 - \frac{5}{8}\xi_D^2$; Si, Ge from the Dirac–Hartree–Fock tables with their velocity-independent part (1.6, 1.3×10^{-4}) subtracted; shaded $\xi_D \leq 0.127$. The control parameter is ξ_D, not the suddenness ratio, since the free-atom result needs no suddenness.	$\boldsymbol{O(\xi_D^2)}$**: ≤ 10 % for $\boldsymbol{\xi_D \leq 0.127}$ (Ge), 0.188 (Si)**. Outer-shell bound–bound terms in the exclusive probability, $O(\xi_D^2)$ with the outer radius; in a crystal the conduction-band final states near the core edge (order of magnitude near threshold, Sect. 10.6); deceleration $[1 - (\omega_{\mathrm{ph}}/\Delta E)^2]^{-2}$ (3×10^{-6}, Ge $3d$); impulse width $x_R^{-1/2}$ in ω_l; $M_\kappa + \delta M_\kappa$ (10^{-4}).	Sect. 12.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Migdal excitation of bound final states (Limit Check 12.2), Eqs. (12.16) and (12.17)	[61, Eqs. (1), (6)]; oscillator strengths [62]	Eq. (12.11) restricted to bound f; dipole order with Eq. (4.36); every bound level below the ionization energy I, every continuum state above it; rate Eq. (12.12) with lines. $\mathfrak{T}$: $n_B = 2$, $n_D = 2$, $\mathfrak{M}$ = sudden, $\mathcal{K}$ = impulse (planned as $n_D = 1$).	$\xi_D \ll 1$, isolated atom. Hydrogen: dipole sum over all levels within 10 % of the exact bound probability for $\xi_D \leq 0.24$. Helium lines $n \leq 3$: 84 % of all lines (0.220 against $0.263\,(v_A/\alpha)^2$). Plot (Fig. 12.5): $x = \xi_D$ (log); $y = P_{\mathrm{exc}}/\xi_D^2$ for hydrogen, exact ($n \leq 40$, $\ell \leq 12$, n^{-3} tail), dipole all levels (0.717), dipole $n \leq 3$ (0.644), oscillator-strength share (0.565); shaded $\xi_D \leq 0.24$. Marked as a lower bound.	$\leq 10\,\%$ **for $\xi_D \leq 0.24$ (H); lines $n \leq 3$ hold** 84 % **(He)**. Omitted levels (positive, so a finite list is a lower bound); $O(\xi_D^2)$ multipoles; in a condensed target the atom's environment (excimer formation, wavefunctions in the liquid); in a crystal the bound states are excitons (share $\geq$ their share of the edge response, Sect. 10.5; Appendix E.15).	Sect. 12.2
Crystal Migdal effect through the loss function, soft limit (Limit Check 12.3), Eqs. (12.21) and (12.22)	[63, Eq. (6)]	Migdal entry Eq. (12.6); diagonal $\boldsymbol{G} = \boldsymbol{G}'$ with $S^{ee}_{\boldsymbol{GG}} = (2p^2/e^2)L(\boldsymbol{p}, \omega_e)$ at the unreduced $\boldsymbol{p}$ (Eq. (12.18), Eq. (E.40)); incoherent impulse lattice factor $2\pi n_\kappa \delta(\omega_l - |\boldsymbol{q} - \boldsymbol{p}|^2/2M_\kappa)$; soft limit $p \ll q$ (Eq. (12.19)); frozen-core ion $v_\kappa = -Z_{\mathrm{ion}}(p)e^2/p^2$ (Eq. (12.20)); $e^2/\pi = 4\alpha$; coefficient checked by the f-sum (Worked example 12.3; Appendix D.33.4). $\mathfrak{T}$: $n_B = 2$, $\mathfrak{M}$ = sudden, $\mathcal{K}$ = impulse (soft limit), $\mathcal{X}$ = RPA with local fields on the diagonal.	p in the first zone for the scalar form; the diagonal of the inverse matrix beyond it. Silicon, scalar RPA ε against local fields, per-p Migdal weight within 10 % for $p \leq 1.3\,q_{\mathrm{BZ}} = 2.9\,\mathrm{keV}$, which holds 8 % of P_{Mig}; median $p = 2.5\,q_{\mathrm{BZ}}$; total ratio 1.35. Plot (Fig. 12.6): $x = p/q_{\mathrm{BZ}}$; $y =$ scalar-RPA over local-field weight $p^2 Z_{\mathrm{ion}}^2 \int_{E_{\mathrm{g}}} \mathrm{d}\omega L/\omega^4$ (Knapen–Kozaczuk–Lin tables, smoothed over 0.7 keV); dashed cumulative $P_{\mathrm{Mig}}(< p)$; shaded $p \leq 1.3\,q_{\mathrm{BZ}}$. Elements $\boldsymbol{G} \neq \boldsymbol{G}'$ in neither curve (order one per the source).	**$\times$1.35 (local fields, Si); free-recoil bracket 0.04 (100 eV) to 0.42 (1 keV)**. Local fields on the diagonal; then $\boldsymbol{G} \neq \boldsymbol{G}'$ (Eq. (12.23); Appendix E.18); exact free-recoil bracket $b/[(\omega_e - b + a)(\omega_e + a)]$, $b = \boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa$, $a = p^2/2M_\kappa$ (Eq. (12.24)), relative $-2p^2/M_\kappa\omega_e + \frac{9}{5}v_N^2 p^2/\omega_e^2$ (leading term 0.04 at $E_R = 100\,\mathrm{eV}$, 0.42 at 1 keV; full bracket: soft limit low by 0.93 at 100 eV, 0.62 at 500 eV); no term in p/q.	Sect. 12.3

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Migdal effect in semiconductors with local fields, free recoil (Limit Check 12.4), Eqs. (12.23) and (12.25)	[64, Eqs. (1), (7), (11), (12)]	As Limit Check 12.3, keeping $\boldsymbol{G} \neq \boldsymbol{G}'$: isotropic average $(\boldsymbol{k}+\boldsymbol{G})\cdot(\boldsymbol{k}+\boldsymbol{G}')/3$, symmetrized RPA matrix $\tilde{\varepsilon} = 1 - v_{\mathrm{C}}^{1/2} R^0 v_{\mathrm{C}}^{1/2}$, constant $Z_{\mathrm{ion}} \to L_{\mathrm{Mig}} = Z_{\mathrm{ion}}^2 \mathcal{F}_{\mathrm{Mig}}/\Omega$; free recoil into Eq. (11.35) with $v_{\mathrm{min}} = \omega_e/q + q/2\mu_{\chi N}$. $\mathfrak{T}$: $n_B = 2$, $\mathfrak{M}$ = sudden, $\mathcal{K}$ = impulse (soft limit), $\mathcal{X}$ = RPA + local fields.	Si: soft limit within 1.1×10^{-3} of the exact free-recoil rate for $m_\chi = 3$–$150\,\mathrm{MeV}$, within 10 % up to 1 GeV (0.91); probability at fixed E_R within 10 % for $E_R \lesssim 160\,\mathrm{eV}$ ($m_\chi \lesssim 0.5\,\mathrm{GeV}$ at the end of the recoil spectrum). Plot (Fig. 12.7): $x = m_\chi$, 3 MeV–1 GeV (top axis E_R^{max}); y = soft-limit free-recoil rate ($\omega_e \geq 1.11\,\mathrm{eV}$, heavy mediator) over (thick solid) the rate with the exact bracket Eq. (12.24), angle-averaged, and (thin solid) the same for the probability at E_R^{max}; dotted, the leading correction alone; dashed, over the soft-limit rate with the incoherent all-order response of a bound nucleus (below 10^{-5} at 3–110 MeV); dash-dotted (right axis), the share at momenta with $v_N p \geq \omega_e + p^2/2M_\kappa$, where the exact amplitude has a pole, left out of both rates; all with the diagonal loss function with local fields and $Z_{\mathrm{ion}}(p)$ in place of $\mathcal{F}_{\mathrm{Mig}}$; shaded $m_\chi \lesssim 0.5\,\mathrm{GeV}$.	**1.1×10^{-3} for $m_\chi = 3$–150 MeV; 0.91 at 1 GeV; probability 0.62 at $E_R = 500\,\mathrm{eV}$.** Motion of the nucleus during the transition: $\frac{9}{5} v_N^2 p^2/\omega_e^2$ and higher orders of Eq. (12.24) (probability low by 0.62 at $E_R = 500\,\mathrm{eV}$); near 1 keV the pole $\omega_e - b + a = 0$ (electronic stopping of the moving ion) holds 3.7 % of the probability and the expansion fails; below the onset the lattice factor at $\boldsymbol{q} - \boldsymbol{p}$ (Sect. 12.4); $\boldsymbol{G} \neq \boldsymbol{G}'$ not evaluated (Appendix E.18).	Sect. 12.3
Per-mass Migdal rate with the valence ionization factor (Limit Check 12.5), Eq. (12.26)	[63, Eqs. (1), (6), (A36)]	Eq. (12.25) with ω_e dropped from v_{min}; $\int \mathrm{d}q\, q^3 \eta(q/2\mu_{\chi N}) = 4\mu_{\chi N}^4 \langle v^3 \rangle$; $\sigma_N = A^2 \sigma_n \mu_{\chi N}^2/\mu_{\chi n}^2$, $A_\kappa/M_\kappa = 1/m_u$, equal $\mu_{\chi N}$ for all species ($m_\chi \ll M_\kappa$). $\mathfrak{T}$: as Limit Check 12.3.	$\omega_e \ll \mu_{\chi N} v^2/2$, isotropic L, soft limit, impulse. Si, $q \geq 92.5\,\mathrm{keV}$ in both: total within 10 % for $m_\chi \geq 66\,\mathrm{MeV}$; 1.25 at 30 MeV, 1.85 at 20 MeV. Plot (Fig. 12.9): $x = m_\chi$; y = per-mass form over Eq. (12.25) with exact v_{min}, total ($\omega_e \geq 1.11\,\mathrm{eV}$) and at $\omega_e = 10\,\mathrm{eV}$; shaded $m_\chi \geq 66\,\mathrm{MeV}$.	**≤ 10 % for $m_\chi \geq 66$ MeV; $\times 1.25$ at 30, $\times 1.85$ at** 20 MeV. Exact $v_{\mathrm{min}} = \omega_e/q + q/2\mu_{\chi N}$; then the corrections of Limit Checks 12.3 and 12.4. Convention: Z_{ion} counts the electrons not in L.	Sect. 12.3

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Wannier picture of the boost (Limit Check 12.6), Eqs. (12.28) and (12.29)	[65, Eqs. (2.7), (2.8)]	Boost amplitude Eq. (8.53) with $\boldsymbol{d} = -\boldsymbol{F}_e/\Delta E$; rigid Wannier following $\nabla_{\boldsymbol{X}_I} V_{eN} \to -\boldsymbol{F}_e \hat{P}_I^{\mathrm{W}}$ (Eq. (12.27)), projector on the initial-state side fixed by the acoustic sum rule (site sum = full boost; Hermitian ordering gives one half); one-body local h, $\boldsymbol{F}_e = m_e[h,[h,\boldsymbol{r}]] \to$ Eq. (12.28); flat manifold $\to$ the source. $\mathfrak{T}$: $n_B = 2$, $\mathfrak{M}$ = sudden, first order in q_e.	Exact for flat manifolds (cores, isolated atom). Bound within 10 % for $W_{\mathrm{v}} \leq 0.025\, E_{\mathrm{g}}$; Si valence ($W_{\mathrm{v}} \approx 12\,\mathrm{eV}$) bracket 1.6, 0.54, 0.25 at $\omega_{cv} = 20$, 50, 100 eV. Plot (Fig. 12.10): tight-binding chain (81 sites, one valence and one conduction orbital, equal widths, intrasite dipole $0.3a$); $x = W_{\mathrm{v}}/E_{\mathrm{g}}$; y = Wannier-picture over rigid-following probability; band from Eq. (12.29) at $\omega_{cv} = E_{\mathrm{g}}$; shaded $x \leq 0.025$; model ratio within 10 % to $x = 0.38$.	**unsized**. Non-rigid vertex $\nabla_{\boldsymbol{X}_I} V^{\mathrm{scf}} + \boldsymbol{F}_e \hat{P}_I^{\mathrm{W}}$ (open; Appendix E.18).	Sect. 12.3

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Migdal effect with phonon final states (Limit Check 12.7), Eqs. (12.34) and (12.35)	[66, Eqs. (3.1), (3.6), (3.8), (4.1), (4.5)]; [67, Eq. (21)]; [68, Eqs. (3.11), (4.1), (4.6)–(4.8)]	Two time orderings Eq. (12.30) (Eq. (2.54)); expansion in $1/\omega_e$ Eq. (12.31): $[A,B]=0$, $[[A,H_N],B]=(\boldsymbol{q}\cdot\boldsymbol{p}/M_\kappa)e^{i(\boldsymbol{q}-\boldsymbol{p})\cdot\boldsymbol{X}_I}$ for any potential (Eq. (12.32) = Eq. (12.5)); diagonal $\boldsymbol{G}=\boldsymbol{G}'$, frozen-core vertex, incoherent lattice factor (Eq. (11.26)); unit weight of $s_\kappa^{\rm inc}\to \mathrm{d}P/\mathrm{d}\omega_e =$ Eq. (12.21) at every q; for the coherent zero-phonon process valid for $q\gtrsim 2q_{\rm BZ}$ (zone average), while at $q\ll q_{\rm BZ}$ Eq. (8.32) holds (1/14.9 in Si; Table 12.3). The lattice argument printed as $\boldsymbol{q}-\boldsymbol{k}_e+\boldsymbol{K}$ in [68, Eqs. (3.8), (3.11), (4.1), (4.4)] is $\boldsymbol{q}-\boldsymbol{k}_e-\boldsymbol{K}$, as in [66, Eq. (3.8)] and Eq. (12.6). [68, Eq. (4.8)], $\mathrm{d}\sigma/\mathrm{d}q=(q/\mu_{\chi n}^2v^2)\sum_i\sigma_n^{(i)}F_{i,i}$ with $(c_i)^2=\pi\sigma_n^{(i)}/\mu_{\chi n}^2$ and $F_{1,1}=F_M\to A^2$, is twice the standard $q\sigma_nA^2/(2\mu_{\chi n}^2v^2)$ for $\mathcal{O}_1$. $\mathfrak{T}$: $n_B=2$, $\mathfrak{M}=$ sudden, $\mathcal{K}=$ multiphonon all orders (harmonic, incoherent), $\mathcal{X}=$ RPA with local fields on the diagonal.	$v_Np/\omega_e\ll 1$ and $p^2/M_\kappa\omega_e\ll 1$; no condition on p/q; whole small-q region ($q\le 92.5\,$keV in Si) to 10^{-3}. Plot (Fig. 12.11): Si nucleus in an isotropic Einstein well $\omega_0=\langle\omega^{-1}\rangle^{-1}=30.4\,$meV, $\omega_e=1.2\,$eV, direction of $\boldsymbol{p}$ averaged; $x=v_Np/\omega_e$ (log); $y=$ leading over exact second order summed over final states, $p=5.7\,$keV ($q=$ 2–300 keV) and $p=22.5\,$keV ($q\le 700\,$keV); free-nucleus exact (dashed); $1/(1+\frac{9}{5}x^2)$ (dotted); shaded $x\le 0.27$ (within 10 %); Si at $p=5.7\,$keV: $q\approx 1.5\,$MeV.	$\lesssim \mathbf{3\times10^{-3}}$ **(Si, $\boldsymbol{\omega_e}=1.2\,$eV).** Eq. (12.33): $1-2p^2/M_\kappa\omega_e+2\langle\omega_{\rm ph}^2\rangle/\omega_e^2+\frac{9}{5}v_N^2p^2/\omega_e^2$ (-2.1×10^{-3}, $+2.8\times10^{-3}$, $\le 5.1\times10^{-4}$ in Si at $\omega_e=1.2\,$eV); then local fields and $\boldsymbol{G}\neq\boldsymbol{G}'$ (Limit Check 12.3; Appendix E.18); frame of the transition density (lattice at $\boldsymbol{q}-\boldsymbol{p}$ or $\boldsymbol{q}$; Sect. 12.1.2).	Sect. 12.4

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Phonon-mediated Migdal effect (Limit Check 12.8): Limit Check 12.7 with the virtual-phonon propagator, Eq. (12.37)	[69, Eqs. (3.1)–(3.4)]	Eq. (12.30) at first order in $\boldsymbol{p}$: $A = 1 - \mathrm{i}\boldsymbol{p}\cdot\boldsymbol{u}_I$, normal modes, displacement rule $a_\nu B = B(a_\nu + \beta_\nu) \to$ Eq. (12.36), exact in ω_ν/ω_e for every final state; cubic site $N^{-1}\sum_\nu e^\alpha e^{\beta *}\delta(\omega-\omega_\nu) = \delta_{\alpha\beta} g_{\mathrm{ph},\kappa}/3$; reduction $\omega_e \gg \omega_{\mathrm{ph}}$ (Eq. (D.197)). $\mathfrak{T}$: $n_B = 2$, $\mathfrak{M}$ = sudden with bound denominators $\omega_e^2 - \omega_\nu^2$, $\mathcal{K}$ = multiphonon all orders at $\boldsymbol{q}$, $\mathcal{X}$ = RPA with local fields. Leading correction one step of n_B, $2 \to 4$: the phonon energies in the electronic denominators, $1 + O\big((\omega_{\mathrm{ph}}/\omega_e)^2\big)$ as Sect. 8.5 counts it.	Reduced form within 10 % for $\omega_e \geq 4.2\,\omega_0$ (Einstein, exact sum) and $\omega_e \geq 0.20\,\mathrm{eV}$ (Si PBE spectrum, against the propagator); Si gap $1 - 2.9\times10^{-3}$; spectrum average 3×10^{-4}. Plot (Fig. 12.13): $q = 10\,\mathrm{keV}$, $p = 5.7\,\mathrm{keV}$, Einstein $\omega_0 = 30.4\,\mathrm{meV}$; $x = \omega_e$ (log, 0.03–5 eV); y = reduced over exact (circles), propagator over exact (solid; two-phonon feature at $2\omega_0$), $1 - 2\omega_0^2/\omega_e^2$ (dotted), reduced over propagator with the Si spectrum (dashed); shaded $\omega_e \geq 4.2\,\omega_0$.	**$2\langle\omega_{\mathrm{ph}}^2\rangle/\omega_e^2 = 2.9\times10^{-3}$ at the Si gap; at order p^2 phonon-emitting fraction $\times$2.4 (5 keV).** The reduction, one step of n_B ($2 \to 4$, Sect. 8.5); order p^2: lattice factor at $\boldsymbol{q}-\boldsymbol{p}$ and two-phonon intermediate states; $\boldsymbol{G} \neq \boldsymbol{G}'$ kept in the source, dropped here.	Sect. 12.4
Migdal-type absorption of a scalar coupled to nucleons (Limit Check 12.9), Eq. (12.56)	[70, Eqs. (13)–(17)]	Absorption rate Eq. (8.19) with the row $-\sum_\kappa c_\kappa n_\kappa(\boldsymbol{p}_\phi)$; acceleration form $\omega_{fi}^2\langle f|\boldsymbol{X}_I|i\rangle = \langle f|\nabla_{\boldsymbol{X}_I} V|i\rangle/M_\kappa$ (Eq. (12.54)); translation invariance $\sum_I \nabla_{\boldsymbol{X}_I} V = -\sum_j \nabla_{\boldsymbol{r}_j} V \to$ $-\tilde{c} m_e \langle f|n_e(\boldsymbol{p}_\phi)|i\rangle$ (Eq. (12.55)); $S_{ee} = (2p_\phi^2/e^2)L$ (Eq. (4.45)); dilute limit $L \to \varepsilon_2 = \boldsymbol{n}_{\mathrm{at}}\sigma_\gamma/\omega \to$ the source with $\tilde{c} = g_{\phi nn}/m_p$. Atom: center-of-mass boost Eq. (12.10) at momentum p_ϕ. $\mathfrak{T}$: $n_D = 2$ (rate order; planned as $n_D = 1$), no Born–Oppenheimer truncation (exact identities), $\mathcal{X}$ = measured loss function, $\mathcal{K}$ not applicable, $\omega_\phi \gg \omega_{\max}$.	Si: within 10 % for $m_\phi \geq 57\,\mathrm{eV}$; $1 + 1.2\times10^{-3}$ at 1 keV; atomic formula $\approx 400\times$ high at 4.2 eV, 7.7× low at 17 eV. Plot (Fig. 12.15): $x = m_\phi$ (1.2 eV–10 keV); top, $\mathcal{R}/g^2$ for $c_\kappa = gA_\kappa$ from L and from ε_2/n_r (atomic formula with the measured attenuation $\boldsymbol{n}_{\mathrm{at}}\sigma_\gamma = \omega\varepsilon_2/n_r$); bottom, ratio $n_r/|\varepsilon|^2$ with $1 + \frac{3}{2}\omega_p^2/\omega^2$; Franta 300 K below 40 eV, Henke above; shaded $m_\phi \geq 57\,\mathrm{eV}$.	**$\frac{3}{2}\omega_p^2/\omega^2$: 1.2×10^{-3} at 1 keV; ≤ 10 % above** 57 eV. Medium factor $n_r/|\varepsilon|^2 = 1 + \frac{3}{2}\omega_p^2/\omega^2$ (ω_p of the electrons bound below ω, Eq. (12.57)); multipoles $O(kr)$; for $c_\kappa \not\propto M_\kappa$ the Umklapp remainder Eq. (E.100) (Appendix E.18).	Sect. 12.6

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Phonon-assisted electron-recoil rate and loss function (no box), Eqs. (13.15) and (13.16); onset exponent, Eq. (13.19)	—	Published form: none. Golden rule Eq. (2.67) with the four-path amplitude Eq. (13.9) into the final states Eq. (13.6); sums over $\boldsymbol{k}$ and $\boldsymbol{Q}$ (the N^{-1} of Eq. (A.25) against the second zone average), spin 2, thermal phonon factors $n_{\boldsymbol{Q}\nu}+1$ and $n_{\boldsymbol{Q}\nu}$, $T \ll E_\mathrm{g} \to$ Eq. (13.15); Eq. (4.45) $\to$ Eq. (13.16); rate Eq. (10.4) with $\bar{L} \to \bar{L}^{\mathrm{PA}}$. Zero-phonon limit ($n_g = 0$): Eq. (10.1). Onset: two free pockets (Eq. (6.43)) $\to$ x^2 (Eq. (13.17)); direction average about an isolated minimum (Eq. (13.18)) $\to x^3$; C_{4v} selection rule $\to x^4$ for the Si longitudinal acoustic kink. $\mathfrak{T}$: $n_g = 2$ (rate order), $\mathcal{X}$ = IP (unscreened), $\mathcal{K}$ = one-phonon.	Inside the off-shell window $\omega \le E_{\min}(\boldsymbol{q}) - \delta_\mathrm{w}$ ($\delta_\mathrm{w} = 0.3\,\mathrm{eV}$); $T \ll E_\mathrm{g}$; harmonic phonons. Exponent: isolated quadratic minimum of the onset with $\lvert\boldsymbol{Q}^*\rvert \ge 0.31\,\text{Å}^{-1}$ (Si: $\lvert\boldsymbol{q}\rvert \le 1.33\,\mathrm{keV}$ for the nearest valley with parabolic valleys, 1.27 keV with the computed bands). No Limit Check; Fig. 13.4 shows the exponents in a silicon model at $\lvert\boldsymbol{q}\rvert = 1.0\,\mathrm{keV}$.	**$\boldsymbol{\theta_{\mathrm{ep}} = 0.010$–$0.016$ (Si, absorption edge)**. Next order in the electron–phonon vertex, $n_g = 4$, relative θ_{ep} (Eq. (8.40)); screening of the density line and of the pair (Sect. 13.4); resonant denominators at the edge of the window (Sect. 13.6). Power laws of the onset: within 10 % for $x < 1.3\,\mathrm{meV}$ (transverse acoustic) and $x < 2.4\,\mathrm{meV}$ (longitudinal acoustic) at $\lvert\boldsymbol{q}\rvert = 1.0\,\mathrm{keV}$.	Sect. 13.3
Phonon-assisted absorption as the optical limit (Limit Check 13.1), Eq. (13.28); edge Eq. (6.44)	[71, Eq. (1)]; [72]	Continuity Eq. (4.29) between the first-order-dressed states $\to$ Eq. (13.26); $\boldsymbol{j}_e(\boldsymbol{q}) \to \hat{\boldsymbol{V}}$ at $\boldsymbol{q} \to 0 \to$ $M^{\pm}(\boldsymbol{k},\boldsymbol{q},\boldsymbol{Q}) = (q/\omega)M^{\pm}(\boldsymbol{k},\boldsymbol{Q};\hat{\boldsymbol{q}})$ (Eq. (13.27)), containing Eq. (13.10); into Eq. (13.16), $w_\pm \to n_{\boldsymbol{Q}\nu} + \frac{1}{2} \pm \frac{1}{2}$ for $T \ll E_\mathrm{g} \to$ Eq. (6.42) with $\hat{\boldsymbol{e}}_\gamma = \hat{\boldsymbol{q}}$; cubic invariance of the summed tensor $\to$ longitudinal = transverse (Eq. (4.52)). Edge: parabolic pockets, constant M_ν (Eq. (6.43)). $\mathfrak{T}$: $n_g = 2$ (rate order), $\mathcal{X}$ = IP, $\boldsymbol{q} \to 0$.	Below the direct gap (off-shell window at $\boldsymbol{q} \to 0$); local potential, or the nonlocal contact term of the vertex kept; corrections $O(q^2)$ with inversion symmetry. Edge form within 15 % of the measured 300 K absorption from 55 to 490 meV above E_g with one scale. Plot (Fig. 13.6): $x = \omega$ (1.05–3.45 eV), α_{abs} log scale; Franta 300 K data; two-branch edge (ω_{TA}, ω_{TO}, weights 5.5:4.0, $E_\mathrm{g}(300\,\mathrm{K}) = 1.123\,\mathrm{eV}$, scale fitted on 1.18–1.30 eV); shaded within 15 %.	**data/edge = 1.4 at** 36 meV**, 2 at** 17 meV **above $\boldsymbol{E_\mathrm{g}}$ (Si,** 300 K**).** Electron–hole attraction (excitonic thresholds E_x^b lower, Sect. 13.8) and thermal broadening near the gap; $\boldsymbol{k}$-dependence of $M^{\pm}$ and of the bands above 0.5 eV; second indirect threshold (2.45 eV at 300 K) and E_{dir}. At finite $\boldsymbol{q}$: band dispersion $q^2/2m_e$ in the path-A denominator (0.98 eV at 1 keV against 2.2 eV), Umklapp classes, phonons at $\boldsymbol{k}_\mathrm{c} - \boldsymbol{q}$.	Sect. 13.5

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Inverse temperature transfer of absorption data (no box), Eqs. (13.29) and (13.30)	—	Published form: none. Branch decomposition of Eq. (6.42) with temperature-independent spectra and a rigid gap shift $\Delta E_g(T) \to$ Eq. (13.29); solved for the cold spectrum with the detailed-balance ratio $n_\nu/(n_\nu+1) = \mathrm{e}^{-\omega_\nu/T}$ (Eq. (6.45)) $\to$ Neumann series Eq. (13.30). $\mathfrak{T}$: $n_g = 2$; occupations exact.	Exact for spectra resolved by branch (sixty terms reproduce the two-branch edge to 10^{-15}); ill-posed for one unresolved spectrum (error growth $\mathrm{e}^{\omega_{\mathrm{TO}}/T} = 6 \times 10^3$ per step at 77 K). Branch weight ±25 %: < 1 % from 77 K, −29/+18 percent from 300 K at 15 meV. No Limit Check; Fig. 13.7 shows the transfer for silicon.	**≈ 12 % from 77 K, ≈ 40 % from** 295 K. Temperature dependence of the branch strength beyond the occupations (fitted $\mathrm{e}^{-c_1 T}$, $c_1 = 1.7 \times 10^{-3}\,\mathrm{K}^{-1}$ [73]); band renormalization beyond a rigid shift (Sects. 6.3 and 13.7); non-phonon-assisted parts of warm data (excitonic companions, thermal broadening), $\approx 1.4\,\mathrm{cm}^{-1}$ below the cold threshold from 300 K data. Untransferred 77 K data: ×2.3–2.6, 1.7–1.9, 1.3–1.5, 1.1–1.3 at 15, 25, 50, 100 meV (lower values with the measured fall of the branch strength).	Sect. 13.5
Zero-point renormalization by special displacements (Limit Check 13.2), Eqs. (13.37) and (E.57); adiabatic AHC Eq. (6.22)	[74, Eqs. (3)–(6), (10), (20), (24)]; [75]	Williams–Lax average of the clamped supercell over the Gaussian of the normal coordinates, $\sigma_j^2 = (2n_j+1)/2\omega_j$ (Eq. (13.37)); one configuration $x_j = \varsigma_j \sigma_j$, signs alternating along the frequency-ordered modes: diagonal second-order terms equal the average, linear and cross terms cancel between effectively degenerate neighbors as the grid becomes dense; odd orders removed by the mirror configuration; higher even orders by concentration of $\sum_j x_j^2/\sigma_j^2$ at N_m. Second order $\to$ adiabatic AHC energies (Eq. (6.22)) and the adiabatic sideband $S^{\mathrm{PA}}_{\mathrm{ad}}$ (Eq. (13.38)). $\mathfrak{T}$: $n_g = 2$ for the comparison, all orders in the displacements for the quadrature; adiabatic; harmonic.	Supercells large enough that neighbors in the frequency list are effectively degenerate: in the model of Fig. 13.9, within 2 meV of the converged AHC shift (−45.2 meV) from $5 \times 5 \times 5$ cells; silicon within 10 % from $4 \times 4 \times 4$ (Worked example E.7). Away from phonon-assisted edges (Eq. (13.39)).	**fourth order** 1 % **(−0.4 meV, model); nonadiabatic** 8 meV **on the Si gap; sideband $(1+\omega_\nu/x)^3$ near the edge.** Terms of fourth order in the displacements (exact minus second-order average, Fig. 13.9); the nonadiabatic correction Eq. (13.40) (−64.3 → −56.2 meV, 6.1 at the valence maximum, Eq. (13.41)); the phonon energy missing from the sideband delta function (Eq. (13.39)): onset low by ω_ν, within 10 % only for $x > 31\,\omega_\nu$.	Sect. 13.7

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Nonadiabatic correction to the zero-point renormalization (no box), Eqs. (13.40) and (13.41)	—	Published form: none for the closed form; computed values [76, Table VI]. Fan–Migdal term Eq. (6.19) at $T = 0$, $\omega = \varepsilon_{n\boldsymbol{k}}$, minus its adiabatic form $\to$ Eq. (13.40); intraband (degenerate) term of a parabolic band with a constant optical vertex $g_{\mathrm{O}} \to -\Omega\lvert g_{\mathrm{O}}\rvert^2(2m^*)^{3/2}\sqrt{\omega_{\mathrm{O}}}/4\pi$ (Eq. (13.41)). $\mathfrak{T}$: $n_g = 2$.	Parabolic band, $\lvert g_{\mathrm{O}}\rvert$ constant for $Q \lesssim (2m^*\omega_{\mathrm{O}})^{1/2}$; interband terms relative $\omega_{\boldsymbol{Q}\nu}/\lvert\varepsilon_n - \varepsilon_m\rvert$. Si estimate 3.2 meV at the valence maximum ($\lvert g_{\mathrm{O}}\rvert = 0.25$ eV) against 6.1 meV computed. Window: none with the measured gap; 0.0023 MeV (onset) if a clamped gap plus a computed renormalization is used.	**factor ≈ 2 (estimate against calculation)**. Vertex varying across the three valence bands and with Q; intervalley terms at the conduction minimum (1.9 meV); variation of the renormalization across the pair states reached by the zero-phonon edge, ≤ 10 meV, ≤ 0.002 MeV on the edge.	Sect. 13.7
One-magnon rate at linear spin-wave order (Limit Check 14.1), Eqs. (14.1) and (14.4)	[77, Eqs. (5)–(12), 18, Eqs. (34), (38)]	Spin row of Eq. (8.3) in Eq. (8.16); projection on the ground multiplet of each ion, $\boldsymbol{s}_e \to F^{\mathrm{mag}}\sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_j}\boldsymbol{S}_j$ (Eq. (7.14)), orbital moment quenched; Holstein–Primakoff Eq. (7.7) with $S^{\pm}$ to order $\lvert\boldsymbol{S}\rvert^{1/2}$ and the quadratic Hamiltonian Eq. (7.9); Bogoliubov Eq. (7.10); transverse matrix elements at $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}$ with the sublattice phase $\to$ $w^{\mathrm{M}}_{\boldsymbol{q}}$ (Eq. (D.214)) and Eq. (14.1); contraction Eq. (14.2). $F^{\mathrm{mag}} = 1 \to$ the published rate; $\mathcal{O}_4$, heavy mediator, $\bar{\sigma}_e$ of Eq. (14.3) and the orientation-averaged kernel Eq. (A.42) $\to$ Eq. (14.4). $\mathfrak{T}$: $\mathfrak{T}_0$ with the magnetic reference state of linear spin-wave theory (relative order $\lvert\boldsymbol{S}\rvert^0$ in $1/\lvert\boldsymbol{S}\rvert$), clamped nuclei.	Energies far below the charge gap of the magnet; $T \ll \omega^{\mathrm{M}}$ (exponent -5×10^3 at 10 mK for the 4.54 meV gap of NiO); collinear two-sublattice order with B a translate of A; $\Delta S/\lvert\boldsymbol{S}\rvert < 0.1$, i.e. $\lvert\boldsymbol{S}\rvert \geq 0.78$ for $\Delta S = 0.078$. Isotropic exchange: with a 5 meV threshold the NiO rate changes by less than 3 % above 100 keV when a 4.54 meV anisotropy gap is added. Plot (Fig. 14.2): $x = 1/(2\lvert\boldsymbol{S}\rvert)$ from 0 to 1; weights in units of $\lvert\boldsymbol{S}\rvert(1+2\Delta S)$: 1 (formula), $1-\Delta S/\lvert\boldsymbol{S}\rvert$, $W_{\parallel}/W_{\perp}$, $1-\Delta S^2/[\lvert\boldsymbol{S}\rvert(1+2\Delta S)]$; lattice and exchange ratios of NiO; shaded $\Delta S/\lvert\boldsymbol{S}\rvert < 0.1$.	**about** 7.8 % **for NiO;** 3.1 % **at** $\lvert\boldsymbol{S}\rvert = \frac{5}{2}$, 16 % **at** $\lvert\boldsymbol{S}\rvert = \frac{1}{2}$. Sum-rule defect $\Delta S/\lvert\boldsymbol{S}\rvert$ of the one-magnon weight: the full square roots of Eq. (7.7) give $(\lvert\boldsymbol{S}\rvert - \Delta S)(1+2\Delta S)$ (Eq. (D.217)) with $\langle S^z\rangle$ and $W_{\parallel}$ held at their linear spin-wave values, an estimate of the leading correction (the three-magnon part lowers the one-magnon weight further; the ground-state change at the same order is not included); the weight moves into the two-magnon continuum, $W_{\parallel}/W_{\perp} = 0.073$ for NiO (Sect. 14.2). Also: magnetic form factor (×0.89 at 1 MeV in NiO), anisotropy gaps below 50 keV (×29 at 10 keV, 5 meV threshold, one-constant model, an upper value since only one branch of NiO reaches 4.54 meV).	Sect. 14.1

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Longitudinal and transverse spin weights (no box), Eq. (14.6)	—	Published form: none for the ratio; zero-point reduction of the simple-cubic antiferromagnet [78]. Ground state of the quadratic spin-wave Hamiltonian (vacuum of Eq. (D.112)): $\langle d^\dagger_{\mathrm{A}i} d_{\mathrm{A}i}\rangle = \Delta S$ (Eq. (D.215)); Wick's theorem with vanishing anomalous average on one site $\rightarrow W_\parallel = \Delta S(1+\Delta S)$ (Eq. (D.216)); $W_\perp = \lvert\boldsymbol{S}\rvert(1+2\Delta S)$. $\mathfrak{T}$: linear spin-wave theory, relative order $\lvert\boldsymbol{S}\rvert^0$; $T = 0$.	Collinear two-sublattice antiferromagnet, $T \ll T_\mathrm{N}$ and $T \ll \omega^\mathrm{M}$; ΔS converged to 5×10^{-5} on a 100^3 grid (NiO). No Limit Check; Worked example 14.1 evaluates NiO ($W_\parallel/W_\perp = 0.0731$).	**unsized**. Magnon interactions of the next order in $1/\lvert\boldsymbol{S}\rvert$ and the change of the ground state at that order; the sum rule bounds neither weight separately (Eq. (D.217)).	Sect. 14.1
Absorption into the uniform mode: the Larmor line (Limit Check 14.2, panel a), Eqs. (14.20) and (14.21); $\lambda_S = g_J - 1$, Eq. (14.19)	[79, Eqs. (2.4), (2.5), 80, Eqs. (27), (30), (34), (35), 81, Eqs. (25)–(27)]	Spin row of Eq. (8.10) in Eq. (8.19) $\rightarrow$ Eq. (14.17); $\lvert\boldsymbol{p}_\phi\rvert a \rightarrow 0$ and $F^\mathrm{mag}(0) = 1$, so $\boldsymbol{s}_e(0) = \sum_j \lambda_{S,j}\boldsymbol{S}_j$ (Eq. (14.18)); exchange invariant under spin rotations, uniform λ_S and g_ion: $[H, S^-_\mathrm{tot}] = \omega_\mathrm{Lar} S^-_\mathrm{tot}$, weight $\langle S^+_\mathrm{tot} S^-_\mathrm{tot}\rangle = 2\langle S^z_\mathrm{tot}\rangle/(1-\mathrm{e}^{-\beta\omega_\mathrm{Lar}}) \rightarrow$ Eq. (14.20). One sublattice at $T = 0$ gives the single-branch rate of Mitridate et al.; $g_{aee}/m_e = 1/f$, $\lambda_S = 1$ gives the golden-rule $(\pi/2)\lvert V\rvert^2\delta$ of Chigusa et al. Projection theorem within a J multiplet $\rightarrow$ Eq. (14.19). $\mathfrak{T}$: clamped nuclei, no electron–phonon vertex; exact in the magnetic many-body state, at every order in $1/\lvert\boldsymbol{S}\rvert$ and at every temperature. Leading correction on a qualifier, the target model (Sect. 9.4.2): the shape of the sample, applied over demagnetizing field.	Exact for any spin and temperature, ordered or paramagnetic, for a saturated sphere or an unbounded medium. Isotropic exchange, no anisotropy, uniform λ_S and g_ion. $\lambda_S = g_J - 1$ only within one J multiplet; quenched $3d$ ions have $\lambda_S = 1 - O(\xi^2_\mathrm{ion}/\Delta^2_\mathrm{CF})$ while $g_\mathrm{ion} - 2 = O(\xi_\mathrm{ion}/\Delta_\mathrm{CF})$.	$\leq 10\,\%$ **in line position and weight for $\boldsymbol{B_0 \geq 4.8\,\mu_0 M_\mathrm{s}}$ (thin film, field in plane); none for a saturated sphere**. Target model, the shape of the sample: demagnetization, Eqs. (14.24) and (14.25): the line moves to ω_Kit and the transverse weights become elliptical, averaged factor $(\omega_{\mathrm{Kit},x} + \omega_{\mathrm{Kit},y})/2\omega_\mathrm{Kit}$ (1.74 for a YIG film at 10 µeV, Worked example 14.3); anisotropy fields move the line; nonuniform $\lambda_{S,j}$ gives gapped branches weight $\propto (\lambda_{S,j} - \lambda_{S,j'})^2$; non-ellipsoidal samples spread the weight over magnetostatic modes.	Sect. 14.3

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Antiferromagnetic-resonance weight (Limit Check 14.2, panel b), Eqs. (14.26) and (14.27)	[79, Eqs. (2.4), (2.5)] (Larmor statement, which this weight replaces)	Published form: none for the weight; general multi-sublattice rate [80, Eq. (27)]. Two sublattices, easy axis, $\boldsymbol{k} = 0$ in Eq. (7.9) with $\lvert\boldsymbol{S}\rvert A_0 = \omega_E + \omega_A$, $\lvert\boldsymbol{S}\rvert B_0 = \omega_E$; Eq. (7.11) $\to$ Eq. (14.26); S^x_{tot} through the inverse of Eq. (7.10), $(u_0^{\mathrm{M}} - v_0^{\mathrm{M}})^2 = \omega_A/\omega_{\mathrm{AFMR}} \to$ Eq. (14.27); axial field splits the line into $\omega_{\mathrm{AFMR}} \pm g\mu_B B_0$ with unchanged weights below the spin flop. $\mathfrak{T}$: $\mathfrak{T}_0$ with the magnetic reference state of linear spin-wave theory (relative order $\lvert\boldsymbol{S}\rvert^0$ in $1/\lvert\boldsymbol{S}\rvert$), clamped nuclei, $T = 0$. Leading correction on a qualifier, the target model (Sect. 9.4.2): the spin-wave order and anisotropy over exchange.	Linear spin-wave theory, $T = 0$, collinear two-sublattice order, uniaxial anisotropy along the field; B_0 below the spin flop $[B_A(2B_E - B_A)]^{1/2}$. Square-root form for $\omega_A \le 0.42\,\omega_E$.	**zero-point correction 5–23 % for $\lvert\boldsymbol{S}\rvert = \frac{3}{2}$ at $\omega_A/\omega_E = 2$ to 0.3 (eight-site cluster); shape $\sim \omega_M/\omega_E = 1.6 \times 10^{-2}$ (MnF_2).** Target model, the spin-wave order: zero-point fluctuations beyond linear spin-wave order (exact diagonalization of the eight-site easy-axis cluster in Fig. 14.6b); demagnetizing field of the small precessing net moment; for $\lvert\boldsymbol{S}\rvert = \frac{5}{2}$ at small ω_A/ω_E the zero-point correction is of order $1/2\lvert\boldsymbol{S}\rvert = 0.2$ and not sized in the book.	Sect. 14.3

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Data-driven spin response (Limit Check 14.3), Eqs. (14.28) and (14.29)	[82, Eqs. (8), (11), (26), dictionary Eq. (B4)]	Magnetic neutron cross-section Eq. (C.165) per formula unit → Eq. (14.28), with $\chi''_{ab} = 2(1 - \mathrm{e}^{-\beta\omega})S_{ab}$ (a factor 4 from $\boldsymbol{\sigma} = 2\boldsymbol{s}$ and $\frac{1}{2}$ from their $\pi/\mathcal{V}$ normalization) and $e^4/[8m_e^2(2\pi)^3] = r_{\mathrm{cl}}^2/4\pi$; quenched ion, $\boldsymbol{L} = (g_{\mathrm{ion}} - 2)\boldsymbol{S}$ at first order in spin–orbit coupling, $\lambda_S = 1$ to second order (Sect. 14.3); dipole approximation (Eq. (C.167)), moment form factor $\langle j_0\rangle + (1 - 2/g_{\mathrm{ion}})\langle j_2\rangle \to C_{\mathrm{sn}}(q)$ of Eq. (14.29); $C_{\mathrm{sn}} = 1 \to$ the published form. Thermal population removed by $1 - \mathrm{e}^{-\omega/T_{\mathrm{d}}}$ (Eq. (4.18), free magnons). $\mathfrak{T}$: none added; the response is measured, with its truncation that of the neutron coupling (Born approximation, spin–spin term).	Transverse couplings only at one $\boldsymbol{q}$ without a model ($\mathfrak{D} \propto P_T$: $\mathcal{O}_9$, magnetic-dipole and anapole spin–spin terms); there $\mathcal{O}_4$ is bounded from below and the $\hat{q}_a\hat{q}_b$ couplings are not covered; within the localized-moment model, transverse data at several $\boldsymbol{q} + \boldsymbol{G}$ fix the full tensor (Eq. (14.31), Sect. 14.4). Spin-only reading within 10 % for $g_{\mathrm{ion}} - 2 \le 0.094$ at $q \le 5.3\,\mathrm{keV}$. Below $m_\chi = 134\,\mathrm{keV}$ every momentum transfer lies inside the first zone of YIG (inscribed radius 708 eV), where the data are extrapolated (Berlin et al. to $q_{1\mathrm{BZ}} = 2\pi/a = 987\,\mathrm{eV}$ for their a); phonon admixture grows as $(q/30\,\mathrm{keV})^2$. Plot (Fig. 14.7): $x = g_{\mathrm{ion}} - 2$ from 0 to 0.45; C_{sn} at $q \to 0$ ($4/g_{\mathrm{ion}}^2$), 2.6, 5.3 and 10 keV with Ni^{2+} radial integrals; formula 1; shaded $g_{\mathrm{ion}} - 2 \le 0.094$.	**−17 % for Ni^{2+} at $g_{\mathrm{ion}} = 2.2$ ($C_{\mathrm{sn}}(0) = 0.83$); −27 % at $g_{\mathrm{ion}} = 2.34$.** Orbital moment of the ion: $C_{\mathrm{sn}}(0) = [\lambda_S/(g_{\mathrm{ion}}/2)]^2$; for J multiplets $[(g_J - 1)/(g_J/2)]^2$, from $\frac{1}{9}$ (Er^{3+}) to 25 (Sm^{3+}). The $\langle j_2\rangle$ term adds −2.7 % at 5.3 keV and −9.6 % at 10 keV (Ni^{2+}, $g_{\mathrm{ion}} = 2.2$).	Sect. 14.4
Chiral-phonon detection probability as the band average of $\lvert\boldsymbol{L}^{\mathrm{ph}}\rvert$ (Limit Check 14.4, panel a), Eqs. (14.40) and (14.41)	[83, Eqs. (6), (7), 84, Eqs. (3), (4), (11), (12)]	Published form: detection probability $\langle\lvert\boldsymbol{L}^{\mathrm{ph}}\rvert\rangle$ over the lowest band times the one-phonon rate. Nucleon-number row in Eq. (8.16); one-phonon term Eq. (11.1) resolved by final state, each term times $\delta^3(\boldsymbol{L} - \boldsymbol{L}^{\mathrm{ph}}_{\boldsymbol{Q}\nu})$ with Eq. (14.40); created phonons assumed to sample the lowest band uniformly and to keep their chirality in decay → the published form. Moment Eq. (14.41): one-phonon expectation value of the two-phonon operator $\sum(eZ^*/2M)\boldsymbol{u} \times \boldsymbol{P}$. $\mathfrak{T}$: $\mathcal{K}$ = one-phonon, $n_A = 0$, $T = 0$.	GaAs (PBE force constants of Sect. 5.5): rate-weighted lowest-band average within 10 % of the band average for $m_\chi \ge 2.2\,\mathrm{MeV}$; inversion or time reversal broken. Decay does not conserve $\boldsymbol{L}^{\mathrm{ph}}$: $[H, L_z] = \frac{\mathrm{i}}{2}\boldsymbol{u} \cdot [J_z, D]\boldsymbol{u}$. Plot (Fig. 14.10a): $x = m_\chi$ from 0.1 to 5 MeV; formula: band average 0.124; fuller: one-phonon-rate-weighted average for created lowest-band phonons and for all created phonons (heavy mediator, nucleon number, orientation-averaged kernel); shaded: lowest band within 10 %.	**−10 % for $m_\chi \ge 2.2\,\mathrm{MeV}$ (GaAs, lowest band); below 1 % of the rate in that band for $m_\chi < 0.8\,\mathrm{MeV}$; force constants 0.124 against 0.07, grid, masses and weighting ≤ 14 %, long-range dipole part −14 % to +40 %.** Phonons created in the other branches (70 % of the one-phonon rate at large mass) and their three-phonon decay into the band, which changes $\boldsymbol{L}^{\mathrm{ph}}$ (Appendix E.19, Eq. (E.103)); the force constants.	Sect. 14.5

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Metastable line of a single-molecule magnet (Limit Check 14.4, panel b), Eqs. (14.43) and (14.44)	[85, Eqs. (3), (4), rate Eqs. (9), (10), (17), (18)]	Spin row with λ_S (Eq. (14.18)); independent molecules of density n_{mol} with Eq. (14.42); $S^{\pm}$ matrix elements in Eq. (4.13) $\to$ Eq. (14.43); all molecules at $m = -\lvert\boldsymbol{S}\rvert \to$ one line at ω_{meta} with weight $2\lvert\boldsymbol{S}\rvert$ (Eq. (14.44)). Dark photon through the absorption coefficient: $\zeta^2\sigma_1/\lvert\varepsilon\rvert^2$ (Eq. (8.39)) = published rate $\times\, n_r/\lvert\varepsilon\rvert^2$; axion-like particle in a field quadratic in $g_{a\gamma\gamma}B_0$ (Eq. (10.33)). $\mathfrak{T}$: exact single-molecule spin Hamiltonian; intermolecular couplings dropped.	$\omega_{\mathrm{meta}}/T \geq 3.0$ (line within 10 % of $2\lvert\boldsymbol{S}\rvert$): $T \leq 4.2\,\mathrm{K}$ for Mn_{12}-acetate at 1 T ($\omega_{\mathrm{meta}} = 1.098\,\mathrm{meV}$); observation time short against the relaxation over the barrier (4.6 meV at 1 T). Plot (Fig. 14.10b): $x = \omega_{\mathrm{meta}}/T$ from 0.3 to 30; formula 1; fuller: the line and the total of the well, $2\lvert\langle S_z\rangle\rvert$, over $2\lvert\boldsymbol{S}\rvert$; shaded $x \geq 3.0$.	**−8 % at 4 K, −38 % at 8 K (Mn_{12}-acetate, 1 T).** Thermal population of the upper well: stimulated emission and weight moved to the line 0.18 meV lower; relaxation over the barrier; intermolecular dipolar fields and hyperfine broadening of the line.	Sect. 14.5
Static magnetoelectric response of a multiferroic (Limit Check 14.5), Eqs. (14.45) and (14.46)	[86, Eqs. (5), (10)]	Axion coupling of the source = the $\mathcal{J}_e^{\alpha\alpha}$ row of Eq. (8.10) at $\boldsymbol{p}_\phi \to 0$; Kubo formula $\to$ Eq. (14.45); Eq. (14.34): static part needs inversion and time reversal broken; Lehmann sum over the uniform modes; one mode with damping added by hand $\to$ Eq. (14.46); Ginzburg–Landau stiffness $\propto T_{\mathrm{CM}} - T$ $\to$ static response $\propto (T_{\mathrm{CM}} - T)^{-1}$. $\mathfrak{T}$: linear spin-wave theory about an ordered reference that breaks both symmetries ($T \ll T_{\mathrm{CM}}$); Landau expansion near T_{CM}.	Landau form within 10 % of Weiss mean field only for $T \geq 0.94\,T_{\mathrm{CM}}$ (model ferromagnet, $\lvert\boldsymbol{S}\rvert = \frac{5}{2}$), inside the critical region; spin waves controlled for $T \leq 0.37\,T_{\mathrm{CM}}$. Needs spin–orbit coupling that breaks conservation of S^z_{tot}. Plot (Fig. 14.11): $x = T/T_{\mathrm{CM}}$; Landau, mean field, spin waves (simple cubic, gap $0.02\,T_{\mathrm{CM}}$); shaded $T \leq 0.37\,T_{\mathrm{CM}}$; hatched $T > 0.9\,T_{\mathrm{CM}}$.	**Landau over spin waves 1.4 at $0.3\,T_{\mathrm{CM}}$, 4.5 at $0.1\,T_{\mathrm{CM}}$ (model); no size near T_{CM}.** Critical fluctuations near T_{CM}, $R = \chi_\times\Phi(\omega\tau)$ with $\chi_\times$ and Φ not known; the static limit needs magnon relaxation faster than $1/m_\phi$.	Sect. 14.5

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Transverse magnetization of a driven nuclear-spin ensemble (Limit Check 14.6), Eq. (14.52)	[87, Eq. (1), 88, Eq. (1) and Eqs. (S22)–(S25), (S41), (S53)]	Spin row of a nucleus, $-\boldsymbol{b}_\phi \cdot \boldsymbol{J}^{\mathrm{nuc}}$: gradient coupling $(g_{ai}/m_N)\nabla\phi \cdot \boldsymbol{J}^{\mathrm{nuc}}$ (Sect. 11.6.2) or Schiff-moment shift $\pm\Delta\varepsilon$ along $\boldsymbol{E}^*$; Bloch equations Eq. (14.51); rotating frame at m_ϕ, counter-rotating component dropped, $\Omega_{\mathrm{Rabi}} = b_\perp/2$; $M_z = M_0 \to$ linear equation for $\tilde{M}_+ \to$ Eq. (14.52); $t \ll T_2$, $\delta\omega_{\mathrm{Lar}} \to 0 \to \Omega_{\mathrm{Rabi}} M_0 t$ (Budker et al., with $d_n \varepsilon_S E^*$ the shift of each level; their envelope $\sin(\delta\omega_{\mathrm{Lar}} t)/\delta\omega_{\mathrm{Lar}}$ is half as wide as $2\sin(\delta\omega_{\mathrm{Lar}} t/2)/\delta\omega_{\mathrm{Lar}}$); $t \gg T_2$ $\to \Omega_{\mathrm{Rabi}} M_0 T_2$, times u_{sp} for an inhomogeneous line (Aybas et al.). $\mathfrak{T}$: first order in the coupling (rate order 2 for the absorbed power), nuclear-spin sector with phenomenological T_1, T_2; field of fixed amplitude.	Drive of fixed amplitude, $t \ll t_{\mathrm{coh}}$, or $T_2 \ll t_{\mathrm{coh}}$ with a Rayleigh amplitude; $\Omega_{\mathrm{Rabi}} t \ll 1$, $\Omega_{\mathrm{Rabi}}^2 T_1 T_2 \ll 1$. Plot (Fig. 14.12): $x = t/t_{\mathrm{coh}}$ from 10^{-3} to 10^3; $y = \lvert M_\perp\rvert_{\mathrm{rms}}/(\Omega_{\mathrm{Rabi}} M_0 T_2)$; formula $1 - \mathrm{e}^{-t/T_2}$ with its limits t/T_2 and 1; fuller result the Gaussian-field average Eq. (14.53) for the standard halo, Larmor frequency at the peak of the line, $T_2 = 10$ and $0.1\,t_{\mathrm{coh}}$, with a Monte Carlo check; shaded $t \le 1.6\,t_{\mathrm{coh}}$.	$\le 10\,\%$ **for $t \le 1.6\,t_{\mathrm{coh}}$ at $T_2 = 10\,t_{\mathrm{coh}}$; 0.6 % at all t for $T_2 = 0.1\,t_{\mathrm{coh}}$.** Wandering phase of the Gaussian field: for $T_2 \gg t_{\mathrm{coh}}$ the magnetization random-walks, $\langle\lvert M_\perp\rvert^2\rangle = 2\pi F_\phi(\omega_{\mathrm{Lar}})\Omega_{\mathrm{rms}}^2 M_0^2 t$, and saturates at $\pi F_\phi \Omega_{\mathrm{rms}}^2 M_0^2 T_2$ (0.39 of the fixed-drive value at $T_2 = 10\,t_{\mathrm{coh}}$). Saturation $M_z = M_0/(1 + \Omega_{\mathrm{Rabi}}^2 T_1 T_2)$; transient nonlinearity $(\Omega_{\mathrm{Rabi}} t)^2/6$.	Sect. 14.6
Gaussian-field signal and its power spectrum (Limit Check 14.7), Eqs. (14.47), (14.48) and (14.54)	[89, Eqs. (8), (10), (24)]	Halo field as a sum of plane waves with random phases → Gaussian field with the two-point function of Sect. 14.6.1 and the spectrum Eq. (14.47), weight at $\boldsymbol{k} = -m_\phi \boldsymbol{v}$ (sign of Eq. (A.6)); $\boldsymbol{k}$-integral $\to (\pi\rho_\chi/m_\phi^2)F_\phi$; complex amplitude in each bin Gaussian → exponential bins Eq. (14.54) with mean $\propto F_\phi$ for $t_{\mathrm{int}} \to \infty$; Gaussian average of the absorbed energy with the Fejér kernel Eq. (14.50) → Eq. (8.19) averaged over the line. $\mathfrak{T}$: target in linear response, any many-body approximation; first order in the coupling.	$t_{\mathrm{int}} \gg \min(t_{\mathrm{coh}}, 1/\Gamma_{\mathrm{tg}})$ for the golden-rule mean; stationary coupling; nonrelativistic, noninteracting field with the standard halo. Plot (Fig. 14.14): $x = t_{\mathrm{int}}/t_{\mathrm{coh}}$ from 10^{-2} to 10^3; y = mean over the field divided by the long-record value, for the periodogram at the line peak ($= T_2 \to \infty$) and for spins with $T_2 = t_{\mathrm{coh}}$, $0.1\,t_{\mathrm{coh}}$; Monte Carlo points (1500 realizations); formula 1; shaded $t_{\mathrm{int}} \ge 13\,t_{\mathrm{coh}}$.	$\le 10\,\%$ **for $t_{\mathrm{int}} \ge 13\,t_{\mathrm{coh}}$ (sharp target); $1 - 1.25\,t_{\mathrm{coh}}/t_{\mathrm{int}}$ above.** Fejér leakage of a finite record; for $t_{\mathrm{int}} \ll t_{\mathrm{coh}}, 1/\Gamma_{\mathrm{tg}}$ the mean grows as t_{int}^2, ratio $t_{\mathrm{int}}/[2\pi F_\phi(\omega_{\mathrm{pk}})]$. Substructure and streams of the halo change F_ϕ (not sized). Bunching of counts, $\mathrm{Var}\,N = \bar{N} + \bar{N}^2/N_{\mathrm{coh}}$.	Sect. 14.6

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Sensitivity scalings of the exponential-bin likelihood (Limit Check 14.8), Eqs. (14.55) and (14.56)	[87, Eqs. (A7), (A8), 89, Eqs. (29), (54), (63), (69)]	Likelihood Eq. (14.55) of exponential bins; expected ratio for data without signal, $\mathsf{t} = 2\sum_k [\ln(1+x_k) - x_k/(1+x_k)] \simeq \sum_k x_k^2$; one bin for $t_{\text{int}} \ll t_{\text{coh}} \to \sigma_{\text{sig}} \propto \mathcal{S}_{\text{n}}^{1/2} t_{\text{int}}^{-1/2}$; $t_{\text{int}}\Delta\omega_{\text{eq}}/2\pi$ bins for $t_{\text{int}} \gg t_{\text{coh}} \to \mathsf{t} = 2\pi t_{\text{int}} \sigma_{\text{sig}}^4/(\Delta\omega_{\text{eq}}\mathcal{S}_{\text{n}}^2)$ and $\sigma_{\text{sig}} \propto (t_{\text{int}} t_{\text{coh}})^{-1/4}$. Random amplitude contained in the likelihood; the fixed-amplitude correction of Centers et al. [90, Eq. (7)] applies to fixed-amplitude analyses only. $\mathfrak{T}$: none (statistics of a Gaussian signal).	White noise; away from the crossover: within 10 % of $t_{\text{int}}^{-1/2}$ for $t_{\text{int}} \le t_{\text{coh}}$ and of $(t_{\text{int}} t_{\text{coh}})^{-1/4}$ for $t_{\text{int}} \ge 100\, t_{\text{coh}}$ (Rayleigh amplitude). Plot (Fig. 14.15): $x = t_{\text{int}}/t_{\text{coh}}$ from 10^{-2} to 10^3; $y = (\sigma_{\text{sig}}^2 t_{\text{coh}}/\mathcal{S}_{\text{n}})^{1/2}$ at the median 95 % limit, exact Neyman construction with N_{eff} equal bins, Rayleigh and fixed amplitude; the two power laws; crossover (local slope $-3/8$) marked.	**fixed amplitude too strong by 1.99 in the coupling below t_{coh}, 1.76 at t_{coh}, 1.015 at $100\, t_{\text{coh}}$**. Crossover at $23\, t_{\text{coh}}$ (Rayleigh), $5.4\, t_{\text{coh}}$ (fixed); local slope down to -0.67 between; asymptotes meet at 136 and $8.7\, t_{\text{coh}}$. For data at the detection threshold the same construction gives 2.73 (Centers et al.: 2.7 frequentist, up to 10 with other priors, 8.4 for gradient couplings with random direction). Box line with N_{eff} bins in place of the Fejér-convolved halo line.	Sect. 14.6
Loss function of a superconductor: conserving response → bare bubble → $\mathfrak{g}_{\text{I}} L^{\text{nor}}$ → Lindhard (Limit Check 15.1), Eqs. (15.1), (15.6) and (15.7)	[4, Eqs. (1), (S.13), 91, Eq. (3.4), 92, Eqs. (8)–(10), 93, Eq. (2) and text]	Density row of Eq. (8.16), $S = (2q^2/e^2)L$ at $T = 0 \to$ Eq. (10.4) with $\bar{L} \to L^{\text{cons}}$; proper response = time-dependent mean field in $(\boldsymbol{\eta}, \mathsf{s}_1, \mathsf{s}_2)$, Eq. (7.34), Coulomb as RPA in $\boldsymbol{\eta} \to$ Eq. (15.1); constant $\mathcal{N}_{\text{F}}$, linear band, s_1 decoupled by particle–hole symmetry → bubbles from the pair function F_μ, Eqs. (15.4) and (15.5) (Appendix D.39), gap equation → Eq. (15.6); drop the ladder → bare bubble, $\operatorname{Im}\varepsilon^{\text{bare}} = \mathfrak{g}_{\text{I}} \operatorname{Im}\varepsilon_{\text{Lin}}$ exactly for $v_{\text{F}} q \ge (\omega^2 - 4\Delta_{\text{BCS}}^2)^{1/2}$, Eq. (15.7); normal real part → $\mathfrak{g}_{\text{I}} L^{\text{nor}}$; $\Delta_{\text{BCS}} \to 0 \to L^{\text{nor}}$, whose unscreened form below $q v_{\text{F}}$ is $m_e^2 \omega/\pi q$. $\mathfrak{T}$: BCS reference, $\mathcal{X}$ = conserving generalized RPA (pairing ladder + Coulomb RPA), clean, quasiclassical, $T = 0$, nuclei clamped.	$q \ll k_{\text{F}}, k_{\text{TF}}$; $\omega \gg c_{\text{LA}} q$ (ionic response $(c_{\text{LA}}/v_{\max})^2 = 6.6 \times 10^{-5}$ at the edge); electron mean free path $\gg \xi_{\text{coh}}$ (1.6–2.4 µm) for the structure within Δ_{BCS} of the edge. "Within 10 %" = smaller over larger ≥ 0.9: Lindhard for $y \ge 7.9$ at $q = \omega/v_{\max}$ ($y \ge 6.3$ at four times that q); $\mathfrak{g}_{\text{I}} L^{\text{nor}}$ for $y \ge 2.25$; bare bubble for $y \ge 9.1$. Plot (Fig. 15.3): $x = y$ at $q = \omega/v_{\max}$ from 2 to 50 (top axis $\Delta_{\text{BCS}}/q v_{\text{F}} = 0.392/y$); L/L^{cons} for bare bubble, its first-order form, $\mathfrak{g}_{\text{I}} L^{\text{nor}}$ and L^{nor}, solid at the edge and dashed at four times the edge momentum; shaded $y \ge 7.9$.	$\boldsymbol{\mathfrak{g}_{\text{I}}(y) - 1}$**: 0.15 at $y = 5$, 0.05 at $y = 10$; ladder: $L^{\text{cons}}/(\mathfrak{g}_{\text{I}} L^{\text{nor}}) \in [0.980, 1.085]$ for $y \ge 3$ (clean)**. Lindhard → superconductor: the case-I factor, $\mathfrak{g}_{\text{I}} - 1 \simeq (2\ln 2y - 1)/y^2$. Bare → conserving: the pair-field ladder, vertex fraction +0.44, +0.21, +0.052, +0.004 at the edge of $y = 2.2$, 3, 5, 10, with a shift of $\operatorname{Re}\varepsilon$ opposite to Eq. (15.8). Beyond the clean limit: impurity scattering (Sect. 15.3.3).	Sect. 15.1

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
First BCS correction to the screening (no box), Eq. (15.8)	—	Published form: none. $\delta \operatorname{Im}\varepsilon = \operatorname{Im}\varepsilon_{\mathrm{Lin}}[\mathfrak{g}_{\mathrm{I}}\Theta(y-2)-1]$ from Eq. (15.7); $\operatorname{Im}\varepsilon_{\mathrm{Lin}} = (\pi/2)(k_{\mathrm{TF}}^2/q^2)(\omega/qv_{\mathrm{F}})$ continued to $\omega\to\infty$; half-range Kramers–Kronig Eq. (4.25) $\to$ $\delta\operatorname{Re}\varepsilon = (k_{\mathrm{TF}}^2/q^2)(\Delta_{\mathrm{BCS}}/qv_{\mathrm{F}})\mathfrak{s}_{\mathrm{BCS}}(y)$; $\operatorname{Re}\varepsilon \gg \operatorname{Im}\varepsilon \to$ $L^{\mathrm{bare}} \simeq \mathfrak{g}_{\mathrm{I}}L^{\mathrm{nor}}[1-2(\Delta_{\mathrm{BCS}}/qv_{\mathrm{F}})\mathfrak{s}_{\mathrm{BCS}}]$. $\mathfrak{T}$: bare quasiparticle bubble with Coulomb RPA, clean, quasiclassical, $T=0$; first order in $\Delta_{\mathrm{BCS}}/qv_{\mathrm{F}}$.	$2(\Delta_{\mathrm{BCS}}/qv_{\mathrm{F}})\mathfrak{s}_{\mathrm{BCS}} \ll 1$ and $\omega \ll qv_{\mathrm{F}}$. In halo kinematics the term is at most $2(v_{\max}/v_{\mathrm{F}})\mathfrak{s}_{\mathrm{BCS}}(y)/y$: 3.9 % at $y=10$, 17 % at $y=5$, 53 % at $y=3$, >1 at $y=2.2$ (aluminum, $v_{\max}/v_{\mathrm{F}}=0.392$). No Limit Check; it is the dotted curve of Fig. 15.3.	**bare bubble at the edge 0.935 vs 0.961 ($y=10$), 0.796 vs 0.835 ($y=5$)**. Resummation of the screening near the edge (Sect. 15.2); the pair-field ladder, which at the edge shifts $\operatorname{Re}\varepsilon$ by a comparable amount of the opposite sign and recovers the first-order coefficient only for $v_{\mathrm{F}}q \gtrsim 10^3\Delta_{\mathrm{BCS}}$ (0.954, 1.048, 1.053 against $\mathfrak{s}_{\mathrm{BCS}}(5)=1.053$ at $v_{\mathrm{F}}q/\Delta_{\mathrm{BCS}} = 10^2, 10^3, 10^4$).	Sect. 15.1
Resummed loss function on the kinematic edge (Limit Check 15.2), Eqs. (15.10) and (15.11)	[92, Eqs. (8), (9)]	Published form: $L = \operatorname{Im}\varepsilon^{\mathrm{bare}}/\lvert\varepsilon^{\mathrm{bare}}\rvert^2$ of the bare BCS dielectric function, evaluated numerically. Absorptive part $\mathfrak{g}_{\mathrm{I}}\operatorname{Im}\varepsilon_{\mathrm{Lin}}$, exact for $\omega < qv_{\mathrm{F}}$ (Eq. (15.7)); real part shifted by the Kramers–Kronig term of Eq. (15.8); Lindhard function at $q \ll k_{\mathrm{F}}$, Eq. (15.9), with $\mathfrak{f}_{\mathrm{L}}(\mathfrak{w})$ and $\mathfrak{r}_\varepsilon = \pi\mathfrak{w}/2\operatorname{Re}\mathfrak{f}_{\mathrm{L}}$ kept; $L = \operatorname{Im}\varepsilon/\lvert\varepsilon\rvert^2$ without expansion $\to$ Eq. (D.219) with $\mathfrak{g}=\mathfrak{g}_{\mathrm{I}}$, $\delta=\delta_{\mathfrak{s}} \to$ Eq. (15.10); first order in $\delta_{\mathfrak{s}}$ and $\mathfrak{r}_\varepsilon^2$ at $\omega \ll qv_{\mathrm{F}} \to$ Eq. (15.8); conserving response Eq. (15.6) $\to$ $\mathfrak{g} = \mathfrak{g}_{\mathrm{I}}(1+\mathfrak{v})$ and its own shift, Eq. (15.11). $\mathfrak{T}$: $\mathcal{X}$ = bare quasiparticle bubble with Coulomb RPA (the box formula) and RPA with the pair-field ladder (the comparison); BCS mean-field reference, clean, quasiclassical, $T=0$.	$\omega < qv_{\mathrm{F}}$, $q \ll k_{\mathrm{F}}$, constant density of states; clean, $\mathfrak{l}_e \geq 10\,\mu\mathrm{m}$ for the clean values to hold to 4 % at $y \geq 2.2$. The bare form is at least 0.9 of the conserving one for $y \geq 9.1$ on the kinematic edge, and the Kramers–Kronig integral to infinity reproduces the directly evaluated bare bubble to 1.5 %. Plot (Fig. 15.4): $x=y$ from 2 to 30 on the edge $q=\omega/v_{\max}$ ($v_{\max}/v_{\mathrm{F}}=0.392$); $y = L/L^{\mathrm{nor}}$; conserving (clean), bare (direct), resummed bare (Kramers–Kronig to infinity and cut at qv_{F}), first order; conserving with Born impurities at $\mathfrak{l}_e = 1\,\mu\mathrm{m}$ and 30 nm; shaded $y \geq 9.1$.	**bare over conserving 0.38, 0.53, 0.75, 0.92 at $y=2.2$, 3, 5, 10 on the edge**. The pair-field ladder, vertex fraction 0.44, 0.21, 0.052, 0.004 at the same y, whose shift of $\operatorname{Re}\varepsilon$ has the opposite sign; the spread of the Kramers–Kronig prescriptions, 0.33 to 0.40 of $\mathfrak{g}_{\mathrm{I}}L^{\mathrm{nor}}$ at $y=2.2$. Edge remainder about $\mathfrak{g}_{\mathrm{I}}L^{\mathrm{nor}}$ two-sided, -2 % to $+9$ % for $y \geq 3$; energies $y<3$ excluded (conserving value varies by more than 10 % with $\mathfrak{l}_e$ from 30 nm to 100 µm). Beyond the clean limit: impurities, $L^{\mathrm{cons}}/\mathfrak{g}_{\mathrm{I}}L^{\mathrm{nor}}$ at $y=2.2$ from 0.87 (clean) to 0.19 (30 nm), within 1.008–1.091 at $y \geq 3$ (Sect. 15.3.3); local dirty limit Eq. (15.18).	Sect. 15.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Drude and Mermin loss functions of a metal with collisions (no box), Eqs. (15.15) and (15.17)	[8, Eq. (19)]; [2, Eqs. (168), (172)]; [4, Eqs. (S.17), (S.18)]; [94]	Mermin function Eq. (4.62); $q \ll k_F$ in Eq. (4.56) $\to R^0 = -2\mathcal{N}_F \mathfrak{f}_L(\mathfrak{w})$ (Eqs. (15.9) and (15.14)) $\to$ Eq. (15.15). Diffusive, $q\mathfrak{l}_e \ll 1$ (then $\omega \le q v_{max} < q v_F \ll \gamma_{rel}$): $\mathfrak{f}_L \simeq -1/3\mathfrak{w}_\gamma^2$ $\to$ diffusion pole Eq. (15.16) $\to$ $L = \omega\gamma_{rel}/\omega_p^2$ for any $\omega/D_{diff}q^2$, the Drude tail (the Fröhlich form with core term 1 and no gap at $\omega \ll \omega_p$). Ballistic, $q\mathfrak{l}_e \gg 1$: $\mathfrak{f}_L \simeq 1 + i\pi\mathfrak{w}_\gamma/2 \to$ the Lindhard corner $\pi\omega q/(2k_{TF}^2 v_F)$, γ_{rel}-independent at first order. $\mathfrak{T}$: $\mathcal{X}$ = RPA for a homogeneous medium with number-conserving relaxation (not a many-body approximation), $n_B = 0$.	$q \ll k_F$, $q \ll k_{TF}$; weak disorder, $k_F\mathfrak{l}_e \gg 1$; $qd_{film} \gg 1$ for bulk screening. Diffusive form within 10 % for $q\mathfrak{l}_e \le 0.64$, ballistic form within 10 % for $q\mathfrak{l}_e \ge 9.8$ ($\omega/\gamma_{rel} = 0.034$). No Limit Check; Fig. 15.5b shows the crossover.	**1.43 against the ballistic 0.79 at $q\mathfrak{l}_e = 1.5$;** 0.2 % **for the Drude form at $q\mathfrak{l}_e = 0.086$.** The full crossover Eq. (15.15) between the limits; corrections of order q/k_F; the Lindhard function at $\omega + i\gamma_{rel}$ gives twice the Drude tail at every $q\mathfrak{l}_e \ll 1$. Outside: strong disorder, $k_F\mathfrak{l}_e \approx 1$ (amorphous WSi, Sect. 15.3.5); sheet screening for $qd_{film} \ll 1$, Eq. (15.19); superconducting dirty film, Eq. (15.18).	Sect. 15.3
Phonon-assisted (Holstein) absorption of a superconductor (Limit Check 15.3), Eqs. (15.23) and (15.25)	[95, Eqs. (13), (16), (17)]; Allen [96], as quoted there	Current row at $\boldsymbol{p}_\phi \to 0$, through σ_1 in Eq. (10.31); one band, $\hat{\boldsymbol{e}} \cdot \hat{\boldsymbol{V}} = \sum(\hat{\boldsymbol{e}} \cdot \boldsymbol{v}_{\boldsymbol{k}})\Psi_{\boldsymbol{k}}^\dagger \Psi_{\boldsymbol{k}}$ (Eq. (15.20)) commutes with $H_{MF} \to$ the second-order amplitude collapses exactly to Eq. (15.23) (Eq. (D.223)), case-I factor from the phonon vertex, $g_\nu(-\boldsymbol{k} - \boldsymbol{Q}, \boldsymbol{Q}) = g_\nu(\boldsymbol{k}, \boldsymbol{Q})$; $\sigma_1 = (\pi e^2/\omega\mathcal{V}) \sum_f |T_{fi}|^2 \delta$; Fermi-surface approximation, independent ξ, $\xi' \to 2x\mathfrak{g}_I(x/\Delta_{BCS})$; $\alpha^2 F_{tr}$ of Eq. (15.24) $\to$ Eq. (15.25); $\Delta_{BCS} \to 0 \to$ Allen's normal-metal form; Debye $\alpha^2 F_{tr} = 2\lambda_{tr}\omega'^4/\omega_D^4 \to$ the published Eqs. (16), (17). $\mathfrak{T}$: $n_g = 2$; BCS mean-field reference, $T = 0$; final pair without the ladder of Eq. (7.34).	$T = 0$; ω below the interband threshold (about 0.5 eV in aluminum); pure crystal; weak coupling; ladder $(\Delta_{BCS}/v_F Q)^2$, averaged over the phonon-assisted integrand, 4×10^{-4} at $\omega = 2.2\,\Delta_{BCS}$. Normal-metal formula within 10 % for $\omega \ge 11.4\,\Delta_{BCS}$ (ω'^4 law), $7.4\,\Delta_{BCS}$ (ω'^2 law). Plot (Fig. 15.6): $x = \omega/\Delta_{BCS}$ from 2 to 400; $y = \sigma_1^S/\sigma_1^N$; case I with the ω'^4 and ω'^2 laws; the case-II factor and $\mathfrak{g}_{II}$ for comparison; formula 1; shaded $\omega \ge 11.4\,\Delta_{BCS}$.	$\boldsymbol{\sigma_1^S/\sigma_1^N = 0.42\ (\omega'^4)}$ **or 0.73 (ω'^2) at** 1 meV**; $\times$0.52 from $1/(1+\lambda_{tr})^2$, known for the normal metal at $\omega \ll \omega_D$ [97, Eq. (3)], open for the superconducting form**. Shape of $\alpha^2 F_{tr}$ below a few meV (the plane-wave vertex of Fig. 6.7 fails the acoustic sum rule and puts 1.25×10^4 times the ω'^4 weight below 1 meV); strong coupling, pair weight $N^{El}N^{El\prime} + P^{El}P^{El\prime}$ (Appendix E.20); electron–electron absorption, unsized; impurity-assisted absorption in films ($\times 1.4 \times 10^6$ at 1 meV for the 50 nm film of Worked example 15.3).	Sect. 15.4

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Impurity-assisted (Mattis–Bardeen) absorption (no box), Eqs. (15.22) and (D.224)	[98]	Exact impurity eigenstates α with time-reversed partners $\bar{\alpha}$, same gap (Sect. 15.3); current odd under time reversal → unit Nambu matrix → pair amplitude $u^{\mathrm{B}}_{\alpha}v^{\mathrm{B}}_{\beta} - v^{\mathrm{B}}_{\alpha}u^{\mathrm{B}}_{\beta}$ (case II); dirty limit $\gamma_{\mathrm{rel}} \gg \Delta_{\mathrm{BCS}}$: $\lvert V^{e}_{\alpha\beta}\rvert^2$ constant over a few Δ_{BCS}, ξ_{α}, ξ_{β} independent → case-II integral of Eq. (7.30) = $\mathfrak{g}_{\mathrm{II}}$, closed form Eq. (D.224); no pair-field ladder (odd velocity against even pair vertices). $\mathfrak{T}$: first order in the current with exact impurity states; BCS mean field; $T = 0$.	$\mathfrak{l}_e \ll \xi_{\mathrm{coh}}$; $T = 0$; below the interband threshold. $\mathfrak{g}_{\mathrm{II}}$ within 10 % of 1 for $\omega \geq 8.1\,\Delta_{\mathrm{BCS}}$. No Limit Check; dotted curve of Fig. 15.6, Fig. 15.7a.	**unsized**. Correlation of the momenta of α and β when $\mathfrak{l}_e$ approaches ξ_{coh}, where the clean limit (no absorption at $\boldsymbol{q} \to 0$) and the dirty limit meet; strong coupling as in the row above.	Sect. 15.4
Spin-current absorption in a clean superconductor (no box), Eqs. (15.26) and (15.27)	—	Published form: none; projections use Eq. (10.32) with the phonon-assisted σ_1 [95, Eq. (22)]. $\mathcal{J}^{zz}_{e} = \frac{1}{2}\sum v^{z}_{\boldsymbol{k}}\Psi^{\dagger}_{\boldsymbol{k}}\boldsymbol{\eta}\Psi_{\boldsymbol{k}}$ (Eq. (D.220)); its part $(\Delta_{\mathrm{BCS}}/E_{\boldsymbol{k}})\gamma^{\dagger}_{\boldsymbol{k}1}\gamma^{\dagger}_{\boldsymbol{k}2}$ creates pairs at $\boldsymbol{q} = 0$ (Eq. (D.221)); spin-rotation invariance, $S_{\mathcal{J}\mathcal{J}} = 3S_{\mathcal{J}^{zz}\mathcal{J}^{zz}}$; golden rule → Eq. (15.27); integral $(\pi/2)(\omega_p^{\mathrm{intra}})^2$. Dirty limit: $\sigma_n\mathfrak{g}_{\mathrm{I}}$. $\mathfrak{T}$: first order in the spin current; BCS mean field without spin–orbit coupling; $T = 0$.	$\mathfrak{l}_e \gg \xi_{\mathrm{coh}}$ (1.6–2.4 µm: bulk high-purity crystals); no spin–orbit coupling; $T = 0$. Above the phonon-assisted σ_1^{N} up to 10.9 meV in aluminum; $g_{aee} = 6.1 \times 10^{-13}$ with the factor $1/(1+\lambda_{\mathrm{ep}})$ (5.1×10^{-13} bare) against 5.6×10^{-9} (our recomputation with the phonon-assisted σ_1^{S}) at 0.72 meV for three events per kg-yr. Total weight exact (spin f-sum rule, zero spin stiffness of the singlet); coherent weight near the edge times $(1 + F_1^a/3)/(1+\lambda_{\mathrm{ep}})$. Normal metal: spin current relaxed by electron–electron collisions, $\sigma_1^{\mathcal{J}} \approx$ 5–45 eV at 0.72 meV (estimate). No Limit Check; Fig. 15.7b.	$\boldsymbol{g_{aee} = 5.4}$**–**$\boldsymbol{7.1 \times 10^{-13}}$ **for** $\boldsymbol{\lvert F_1^a/3\rvert \leq 0.1}$ **(Leggett form** $\boldsymbol{(1+F)(1-f)/(1+Ff)}$**)**. For $F_1^a < 0$ a bound spin-one $\ell = 1$ mode below $2\Delta_{\mathrm{BCS}}$; triplet p-wave vertex; strong coupling beyond the residue; spin–orbit coupling, which mixes current and spin current; the phonon-assisted spin-current part near the edge. For $\mathfrak{l}_e \lesssim \xi_{\mathrm{coh}}$ the dirty form $\sigma_n\mathfrak{g}_{\mathrm{I}}$ applies: for the 50 nm film $g_{aee} = 1.5 \times 10^{-12}$ against 2.0×10^{-12} from its Mattis–Bardeen σ_1.	Sect. 15.4

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Dirac-material rates with the background in the total ε (Limit Check 16.1), Eqs. (16.1) and (16.2)	[5, Eqs. (2.6), (3.7), (3.11), (3.13), (3.14), (4.4), (4.5)]; [99, Eqs. (2.5)–(2.7), (2.25)–(2.27), (3.3), (3.13), (3.14)]	Published form: vacuum $\bar{\sigma}_e$, pair transitions with a unit-charge vertex ($\mathrm{Im}\,\varepsilon_\mathrm{D}$), in-medium factor $1/\varepsilon_\mathrm{pub}$ with $\varepsilon_\mathrm{pub} = 1 + \varepsilon_\mathrm{D}/\varepsilon_\mathrm{bg}$; absorption with $\Pi = \omega^2(1 - \varepsilon_\mathrm{pub})$. Book: total-charge coupling, Eq. (8.38) with the total ε; proper response split into Dirac bands and background, $1 - v_\mathrm{C}\tilde{R}_\mathrm{bg} \to \varepsilon_\mathrm{bg}$ (local, static) $\to$ Eq. (D.225); first-order $\boldsymbol{k}\cdot\boldsymbol{p}$ cone Eq. (7.38), bubble Eq. (D.134) with a hard cutoff $\to$ Eq. (7.44); $L = \mathrm{Im}\,\varepsilon_\mathrm{D}/|\varepsilon_\mathrm{bg} + \varepsilon_\mathrm{D}|^2 \to L_\mathrm{pub} = \varepsilon_\mathrm{bg}^2 L$ (Eq. (D.228)), $\mathit{\Gamma}_\mathrm{pub} = \varepsilon_\mathrm{bg}\mathit{\Gamma}$ (Eq. (D.232)); anisotropic: $\hat{\boldsymbol{q}}\cdot\varepsilon_\mathrm{bg}\cdot\hat{\boldsymbol{q}}$ against the per-component factor (Eq. (D.229)), $\varepsilon_{\mathrm{bg},ii}^2$ on the axes. Reach: $\bar{\sigma}_e^\mathrm{lim} \times \varepsilon_\mathrm{bg}^2$, $\zeta^\mathrm{lim} \times \varepsilon_\mathrm{bg}^{1/2}$ for isotropic models (Eq. (D.233)), not also $\times|1 + \varepsilon_\mathrm{D}/\varepsilon_\mathrm{bg}|^{-2}$, which the published $|\varepsilon_\mathrm{pub}|^2$ contains. The source's Eq. (3.14), and the 00 entry of its Eq. (3.13), print $q_x^2 v_z^2/\varepsilon_{\mathrm{bg},xx}$ for $q_x^2 v_x^2/\varepsilon_{\mathrm{bg},xx}$; its table column of gaps holds the half-gap Δ_D (text below its Table 1). $\mathfrak{T}$: $\mathcal{X}$ = RPA for the Dirac electrons; static local background; $T = 0$, chemical potential in the gap.	Published form within 10 % only for $\varepsilon_\mathrm{bg} \le 1.049$ (scattering), 1.10 (absorption). Total-ε form: $g_\mathrm{deg}\tilde{\alpha}\ln[v_\mathrm{D}\Lambda_\mathrm{D}/\max(\Delta_\mathrm{D}, v_\mathrm{D}q)] \lesssim 1$ ($\tilde{\alpha} < 1$ for $\varepsilon_\mathrm{bg} > \alpha/v_\mathrm{D}$, 18.2 at $v_\mathrm{D} = 4\times10^{-4}$); ω below the remote-band gap and away from polar phonons; q small against the reciprocal lattice; $\lvert E_\mathrm{F}\rvert < \Delta_\mathrm{D}$. Plot (Fig. 16.2): $x = \varepsilon_\mathrm{bg}$ from 1 to 1000 (log); y = rates over their value at $\varepsilon_\mathrm{bg} = 1$; scattering at $m_\chi = 100\,\mathrm{keV}$ (light mediator) and absorption at $m_\phi = 50\,\mathrm{meV}$, gapped benchmark $v_\mathrm{D} = 4\times10^{-4}$, $\Delta_\mathrm{D} = 2.5\,\mathrm{meV}$, $g_\mathrm{deg} = 2$, $\Lambda_\mathrm{D} = 0.2\,\mathrm{keV}$; shaded $\varepsilon_\mathrm{bg} \le 1.049$; hatched $\tilde{\alpha} < 1$.	**published form $\times\boldsymbol{\varepsilon_\mathrm{bg}^2}$ (scattering), $\times\boldsymbol{\varepsilon_\mathrm{bg}}$ (absorption): 1600 and 40 at $\boldsymbol{\varepsilon_\mathrm{bg} = 40}$; 5200–6400 for $\mathbf{ZrTe_5}$ scattering (measured-structure row, $\boldsymbol{\Delta_\mathrm{D}} = 11.8\,\mathrm{meV}$, book halo); 4800–6600 over the rows at 11.8 and 15.6 meV and the halos (238, 250.5, 544) and (220, 232, 600) $\mathrm{km\,s^{-1}}$; 4300 with $\boldsymbol{v_0} = 220$, $\boldsymbol{v_\mathrm{esc}} = 544\,\mathrm{km\,s^{-1}}$.** Of the total-$\varepsilon$ form: the electron–hole ladder, Sommerfeld factor $x/(1 - \mathrm{e}^{-x})$, $x = \pi\tilde{\alpha}/b$, above 1.5 for pair energies below 1.37 times the edge in $ZrTe_5$ (60 %, 12 %, 3 % of the rate at 20, 100, 1000 keV, measured-structure row); self-energy and vertex of relative order $\tilde{\alpha}$ with an unfixed coefficient; leading weak-screening form $\mathrm{Im}\,\varepsilon_\mathrm{D}/\varepsilon_\mathrm{bg}^2$ overstates the $ZrTe_5$ rate by 6–16 %; $\tilde{\alpha} \gtrsim 1$ open.	Sect. 16.1

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
One-dimensional plasmon of linear channels (Limit Check 16.2), Eqs. (16.7) and (16.9)	[100, Eqs. (3)–(6)]	Published form: $S_{1D} = g\lvert q\rvert\delta(\omega - v\lvert q\rvert)$ with $g = [(2\pi v_F + g_4 - g_2)/(2\pi v_F + g_4 + g_2)]^{1/2}$, one right- and one left-moving mode. Density row; embedding Eq. (16.3) with $F_\perp = J_0(q_\perp r_t)$; N_{ch} linear channels, forward density–density coupling $g_2 = g_4 = v_C^{1D}$ (Eq. (16.7)); random-phase response, exact for this model → Eq. (16.9). The charge mode carries all channels; the $N_{ch} - 1$ neutral modes do not couple. The source has $N_{ch} = 1$ with the measured four-channel g_c: its S_{1D} and rate (Eq. (6)) are low by 4 at fixed g_c (by 2.1 at a fixed kernel, $q_\parallel = 10\,\text{eV}$). $\mathfrak{T}$: $\mathcal{X}$ = RPA, $d = 1$, $n_B = 0$.	Linear bands: $q_\parallel$ small against the scale k_F of band curvature; $\omega < 2v_D/r_t = 1.7\,\text{eV}$ (next subbands) and $q_\parallel \ll 1/r_t$ for a nanotube; independent tubes ($q_\perp$ above the inverse spacing); $T = 0$, no backscattering. Plot (Fig. 16.5): $x = q_\parallel/k_F$ (0.01–1.9, log); y = plasmon frequency and pole share of the f-sum for $N_{ch} = 4$ parabolic channels over the linear-band values; kernel of a tube with $r_t = 0.7\,\text{nm}$, $\varepsilon_{env} = 1$; $v = v_D = 3.02 \times 10^{-3}$; $k_F = 1/r_t = 282\,\text{eV}$ (model); shaded $q_\parallel < 1.36\,k_F$.	$\lesssim 10\,\%$ **for $q_\parallel < 1.36\,k_F$;** $+4.0\,\%$ **in ω_{pl} at $q_\parallel = k_F$.** Band curvature, $\omega_{pl}^{par}/\omega_{pl} = 1 + (q_\parallel^2/2k_F^2)g_c^2[\frac{1}{4} + g_c^2/3(1 - g_c^2)]$, with weight lost to the pair band (pole share 0.97 at k_F); then the subbands at $2v_D/r_t$, the Coulomb interaction between tubes of a forest, backscattering, and damping of the plasmon by contacts and disorder.	Sect. 16.2
Acoustic (demon) plasmon of two components and its Landau width (Limit Check 16.3), Eqs. (16.10) to (16.12)	[101]; [102, Eqs. (1), (2)]	Published form: the first-principles loss function of Sr_2RuO_4 (demon at 80–800 meV) in the loss-function rate; the two-dimensional width is derived here. Density row; two species; light species static, Lindhard at $q \ll k_{F,l}$ (f_3, f_2); heavy species dynamic, $n_h q^2/m_h\omega^2$; zero of $\varepsilon \to v_a = \omega_{p,h}/k_{TF,l}$; residue $\pi/\partial_\omega\varepsilon \to$ share $(q^2/(q^2 + k_{TF,l}^2))^2$ (3D), $(q_\parallel/(q_\parallel + q_s))^2$ (2D); $\Gamma_a = 2\,\text{Im}\,\varepsilon/\partial_\omega\varepsilon \to$ Eq. (16.12). $\mathfrak{T}$: $\mathcal{X}$ = RPA with two components, $n_B = 0$.	$v_{F,h} \ll v_a \ll v_{F,l}$; $q \ll k_{TF,l}$ (3D), $q_\parallel \ll q_s$ (2D); $q \ll k_F$. Plot (Fig. 16.6): $x = v_a/v_{F,l}$ (0.02–0.6, log); $y = \Gamma_a/\omega_a$ (full width at half maximum); formulas against the width with the full ν dependence of the light Lindhard function, heavy carriers cold, at $q = 0.02\,k_{TF,l}$ (3D) and $0.02\,q_s$ (2D); shaded $v_a/v_{F,l} < 0.157$ (3D) and < 0.53 (2D).	$\lesssim 10\,\%$ **for $v_a/v_{F,l} < 0.157$ (3D) and < 0.53 (2D);** $+3.5\,\%$ **in ω_a from the heavy band at $v_a/v_{F,h} = 3.16$.** Dispersion of the heavy band, $\frac{3}{4}(v_{F,h}/v_a)^2$ in ω_a^2 (2D); finite $q_\parallel/q_s$ (3 % in the width at 0.02); interband transitions and local fields of a real crystal, which a first-principles loss function contains.	Sect. 16.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Effective medium of a layered stack (Limit Check 16.4), Eqs. (16.14) to (16.16)	[9, Eq. (16), thin film Eq. (19)]	Density row; random-phase response with the local fields of the stacking axis kept (q_z unreduced); layers with local $\varepsilon_A(\omega)$, $\varepsilon_B(\omega)$; $ka_s \to 0$: tangential field and normal displacement continuous $\to$ $\varepsilon_\perp$ arithmetic, $\varepsilon_\parallel$ harmonic mean $\to$ Eq. (16.14), identical to the source's Eq. (16) at any filling (stated there for $d_A/a_s \ll 1$). In the plane, exact at any ka_s: Eq. (16.15); its expansion Eq. (16.16); one layer ($d_B \to \infty$) = the source's Eq. (19) up to the real constant d_A; thin sheets: Eq. (16.13). $\mathfrak{T}$: $\mathcal{X}$ = RPA with layer local fields, $n_B = 0$.	$ka_s \ll 1$; local layer response, $k \lesssim 1/d_A$. Plot (Fig. 16.7): $x = ka_s$ (0.05–12, log); y = frequency of the loss peak over the effective-medium value; toy stack of the source ($d_A = 1\,\mathrm{nm}$, $\varepsilon_A = 10[1 - \omega_p^2/\omega(\omega + \mathrm{i}\gamma)]$, $\omega_p = 0.5\,\mathrm{eV}$, $\gamma = 0.1\,\omega_p$; $d_B = 4\,\mathrm{nm}$ of vacuum); $\boldsymbol{k}$ in the plane (Eq. (16.15), with Eq. (16.16)) and at 30° from the axis (exact periodic matching); shaded $ka_s < 1.94$.	$\leq 10\,\%$ **for $ka_s < 1.94$ at** 30° **and < 7.2 in the plane; 0.7 and** $2.8\,\%$ **at $ka_s = 1$**. In the plane $(ka_s)^2 f^2(1-f)^2(\varepsilon_A - \varepsilon_B)^2/(12\varepsilon_\parallel\varepsilon_\perp^2)$ in $1/\varepsilon$ (Eq. (16.16)); nonlocal $\varepsilon_A(\boldsymbol{k},\omega)$ at $k \gtrsim 1/d_A$; interface states.	Sect. 16.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Vacuum ejection from graphene and nanotubes with plane-wave final states (Limit Check 16.5), Eqs. (16.17) to (16.19) and (16.21)	[103, Eqs. (1)–(5), (7)]; [104, Eqs. (16)–(18), (23)]	Density row; occupied Bloch states of the sheet, final state hole + free electron above Φ_{wf}; plane wave → $M_{\mathrm{PW}} = \tilde{\psi}_{i\boldsymbol{k}_\parallel}(\boldsymbol{p}-\boldsymbol{q})$ (Eq. (2.17)) → Eq. (16.18); Eq. (8.16) with Eq. (1.34), $\rho_T \mathcal{V} = N_{\mathrm{cell}} M_{\mathrm{cell}}$ → Eq. (16.19) = the source's Eq. (5); the source writes the state sum as $2N_{\mathrm{C}}A_{\mathrm{cell}}\int \mathrm{d}^2k_\parallel/(2\pi)^2$ per band with a nearest-neighbor cluster function for $\tilde{\psi}$; tube: $\boldsymbol{k}_\parallel$ around the circumference → n/r_{t} (the second source; its text prints $E_r = k'^2/2m_\chi$ for $k'^2/2m_e$). Orientation average and Eq. (A.42) → Eq. (16.21), which for a full atomic shell is Eq. (2.23). $\mathfrak{T}$: $\mathcal{X}$ = IP, frozen orbitals, plane-wave final state; $n_B = 0$, $n_g = 0$.	$\omega > \Phi_{\mathrm{wf}} = 4.3\,\mathrm{eV}$ ($m_\chi \geq 1.22\,\mathrm{MeV}$, $q \geq 1.62\,\mathrm{keV}$); one-step ejection without rescattering in the layer. Plot (Fig. 16.9): $x = E_{\mathrm{ej}}$ from 0.3 to 300 eV (log); y = OPW spectrum (Eq. (16.20)) over the plane-wave spectrum; π electrons of graphene, hydrogenic $2p_z$ with $Z_{\mathrm{eff}} = 4.03$ orthogonalized to itself, nearest-neighbor binding energies ($t = 2.8\,\mathrm{eV}$), orientation-averaged, standard halo, heavy mediator, $m_\chi = 10$ and 100 MeV; Fermi factor ($Z_{\mathrm{eff}} = 1$) dotted; shaded $E_{\mathrm{ej}} < 0.60\,\mathrm{eV}$ and $> 94\,\mathrm{eV}$.	**OPW factor 1.1 to 3.5 for $0.6 < E_{\mathrm{ej}} < 94\,\mathrm{eV}$ (4.3 at the end of the 10 MeV spectrum); total rate ×1.44 (10 MeV) and ×1.84 (100 MeV), heavy mediator; final-state factor between 1 and 7.3, 3.4, 2.6 at 10, 50,** 100 eV. Orthogonality to all occupied states, including the σ bands and the 1s shell (Appendix E.21); final-state interaction with the screened hole and the sheet potential, not to be multiplied onto the OPW factor; squared-amplitude ratios 0.46–25 at 50–100 eV and q = 5–40 keV (0.4–4.3 for the more diffuse Clementi–Raimondi exponent $1.568/a_0$) are not rate corrections. The first source's state sum counts each band twice unless each cluster function carries weight $\frac{1}{2}$.	Sect. 16.3
Interband response of a semiconducting nanotube, one-dimensional massive Dirac bands in the random-phase approximation (no box), Eqs. (16.22) and (16.23)	—	Published form: none. Subband $h = v_{\mathrm{D}}k\sigma_x + \Delta_n\sigma_z$, $\Delta_n = nv_{\mathrm{D}}/3r_{\mathrm{t}}$ from zone folding, $g_{\mathrm{deg}} = 4$; Lorentz invariance and current conservation → $q_\parallel^2 \mathfrak{p}(s)$; $q_\parallel \to 0$ overlap $q_\parallel^2 v_{\mathrm{D}}^2 \Delta_n^2/4\varepsilon_k^4$ → Eq. (16.22); weight $g_{\mathrm{deg}}v_{\mathrm{D}}$, metallic limit $g_{\mathrm{deg}}v_{\mathrm{D}}q_\parallel^2\delta(s)$; RPA with Eq. (16.7), embedding Eq. (16.3); $q_\parallel \to 0$, dilute aligned array → Eq. (16.23). $\mathfrak{T}$: $\mathcal{X}$ = RPA, $d = 1$, $n_B = 0$, $T = 0$, undoped.	ω below the next subband pair and within the linear range of the cone; $q_\parallel \ll 1/a_{\mathrm{cc}}$; independent tubes; field along the axis for Eq. (16.23). No Limit Check; checked against direct integration (machine precision) and the metallic limit.	**unsized**. Excitons below each edge (Bethe–Salpeter, Sect. 4.9), which move the weight of the van Hove singularity; transitions between subbands through higher azimuthal harmonics of $F_\perp$; curvature of the tube and band nonlinearity at $\omega \sim t$.	Sect. 16.3

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Quantum-dot scattering rate and its bulk limit, with the prefactor of the published rate corrected (Limit Check 16.6), Eqs. (16.24) and (16.25)	[105, Eqs. (2)–(6), (10)–(12)]; [7, Eqs. (3.13), (3.17)]	Dilute dots, each screened by its host; envelope-function states $u_\kappa\psi_{nlm}$ in a hard-walled sphere, gap Eq. (16.24); matrix element = band-edge cell factor × envelope overlap (q small against the inverse lattice constant); closure $\sum_e \lvert\langle\psi^e\lvert e^{i\boldsymbol{q}\cdot\boldsymbol{r}}\rvert\psi^h\rangle\rvert^2 = 1$; dense levels → $S^{\mathrm{QD}} \to \varphi_{\mathrm{QD}} S^{\mathrm{bulk}}(E_g \to \Delta E_{\mathrm{QD}})$. Eq. (A.44) with $\bar{S} = (2\pi/\Omega)\langle\lvert f_{\mathrm{QD}}\rvert^2\rangle$ (the source's Eq. (10)) → Eq. (16.25); bulk limit → Eq. (D.171) with the scissored gap. The source's Eq. (6) exceeds Eq. (16.25) by $\mu_{\chi e}/4\omega$ (4.05×10^4 at 10 MeV, 3 eV) with its Eq. (10), and has mass dimension 2 instead of 1 with its Eq. (12) (= the crystal form factor of the second source, Eq. (3.17)); the plotted bulk rates come from the code of the second source. $\mathfrak{T}$: $\mathcal{X}$ = IP (envelopes), $n_B = 0$; strong confinement, $R \ll a_x$.	Level spacing small against the resolution; $R \ll a_x$ (about 20 nm in PbS); ω well above the dot gap. Plot (Fig. 16.10): $x = (\omega - \Delta E_{\mathrm{QD}})/E_{\mathrm{conf}}$ from 0.5 to 150 (log); y = dot response integrated from the gap over the same integral of the scissored bulk; hard-walled sphere, equal masses, $qR = 3$ and 8; unscissored bulk dashed; $1 - 9\pi/8kR$ dotted; the PbS range ($d = 3.3$ nm, 2.5–10 eV, $E_{\mathrm{conf}} = 0.79$ eV) marked; shaded $x \geq 109$ ($kR \geq 33$).	**0.76 to 1.03 of the scissored bulk at 3–11 confinement energies ($qR = 8$; 0.70–0.97 at $qR = 3$), 2.4–66 nearer the gap (hard-wall model); within 10 % only above $109\,E_{\mathrm{conf}}$.** Surface term of the level count, $-9\pi/8kR$ at $q \to 0$, which the scissor does not remove; finite walls, surface and ligand states (smaller depletion); exciton correlation for $R \gtrsim a_x$; size dispersion; nonparabolic bands far from the edge.	Sect. 16.3
Dark-photon absorption by quantum dots in a host: the depolarization factor (no box), Eq. (16.26)	[105, Eq. (16)]	Published form: $\sigma_{A'}v = \kappa^2\sigma_A/\lvert\varepsilon\rvert^2$ per atom, with ε a model of the dot material [106]. Current row at $\boldsymbol{p}_\phi \to 0$, Eq. (10.31); dot radius $\ll 1/m_\phi v \to$ quasi-static sphere in a uniform field, $E_{\mathrm{in}} = 3\varepsilon_{\mathrm{out}}E_0/(\varepsilon_{\mathrm{in}} + 2\varepsilon_{\mathrm{out}})$; dilute Maxwell Garnett medium, first order in φ_{QD}, real $\varepsilon_{\mathrm{out}} \to$ $L = 9\varphi_{\mathrm{QD}}\,\mathrm{Im}\,\varepsilon_{\mathrm{in}}/\lvert\varepsilon_{\mathrm{in}} + 2\varepsilon_{\mathrm{out}}\rvert^2 \to$ the factor Eq. (16.26) against $\mathrm{Im}\,\varepsilon_{\mathrm{in}}/\lvert\varepsilon_{\mathrm{in}}\rvert^2$. $\mathfrak{T}$: local $\varepsilon_{\mathrm{in}}$, $\varepsilon_{\mathrm{out}}$; first order in ζ and in φ_{QD}.	$\varphi_{\mathrm{QD}} \ll 1$ (Maxwell Garnett to $\varphi_{\mathrm{QD}} \lesssim 0.1$); local dielectric functions; $m_\phi v R \ll 1$. No Limit Check; Worked example 16.3.	**×6.00 for PbS ($\varepsilon_{\mathrm{in}} = 17.63$) in silica; 4.62–6.71 for $\mathrm{Re}\,\varepsilon_{\mathrm{in}} = 10$–25; 0.566 for an illustrative $\varepsilon_{\mathrm{in}} = 1 + \mathrm{i}$ at** 5 eV. Size and sign over 2.5–10 eV need $\varepsilon_{\mathrm{in}}(\omega)$ of real dots of the same size (open); higher multipoles at finite $m_\phi v R$; Bruggeman mixing at higher filling.	Sect. 16.3

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Frenkel-exciton rate binned in ω: incoherent sum of molecular form factors (Limit Check 16.7), Eqs. (16.27), (16.30) and (16.32)	[107, Eqs. (11), (13), (16)–(19), 108, Eqs. (8), (9)]	Density row; Eq. (4.104) in orbitals localized on the molecules; zero intermolecular overlap → charge-transfer pairs decouple, K^{d} intramolecular, $g_{\mathrm{deg}}K^{\mathrm{x}}$ between molecules $= \mathfrak{J}$ → molecular singlets $E_{\mathfrak{r}}$, form factors Eq. (16.27), Frenkel Hamiltonian Eq. (16.28), structure factor Eq. (16.29); window holding complete manifolds, $\varepsilon_{\mathrm{mac}}$ constant, completeness of A^{λ} → Eq. (16.30) (the Davydov oscillator-strength sum rule at finite q); bare vertex, $\varepsilon_{\mathrm{mac}} = 1$, line at $E_{\mathfrak{r}}$; Eq. (8.16) with Eq. (1.34), Radon transform of Sect. 11.7 → Eq. (16.32); direction average with Eq. (11.49) → the first source's Eq. (19) per molecule (detection efficiency one); the second source's Eq. (8) is the direction-resolved form. $\mathfrak{T}$: $\mathcal{X}$ = BSE in a molecular basis, zero overlap; $n_B = 0$, $n_g = 0$ (Condon).	Exciton bands narrow against the scale qv_0 on which $\eta(v_{\mathrm{min}})$ changes; no coupling of the window to other manifolds; $qa_{\mathrm{mol}} \gg 1$ (no macroscopic or local-field screening of the vertex); no charge-transfer admixture. Plot (Fig. 16.11): $x = m_\chi$ from 0.75 to 1000 MeV (log); y = rate of the full Frenkel response of anthracene (49 π singlets per molecule, $\mathfrak{J}$ by Ewald with transition charges, macroscopic term $(e^2/\Omega q^2)FF^{\dagger}$ in the Hamiltonian, levels shifted by 0.27 eV to keep the 3.1 eV optical line) over the binned rate, heavy and light mediator, direction-averaged standard halo; dash-dotted: without the macroscopic term; shaded: heavy-mediator ratio within 10 %.	**+6.3 % (heavy, $m_\chi \geq 30$ MeV), up to +27 % between 1.2 and** 7.5 MeV**, +38 % at 0.88**–1.0 MeV**, +29 to +35 % (light, $m_\chi \geq 2$ MeV), in a π-electron model**. Intensity borrowing from the strong B_b transition, first order in $\mathfrak{J}/\Delta E$ (Eq. (16.31)); exciton-band dispersion against the variation of η near threshold; the macroscopic term, which moves weight up in energy (without it the local fields alone give ×8.8 at 1.1 MeV). Model omits σ electrons (unscreened $\mathfrak{J}$). Charge-transfer admixture θ^2_{CT} and the periodic Bethe–Salpeter difference (Appendix E.21), unsized.	Sect. 16.4

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Dark-photon absorption by dilute donors, host screening counted once (Limit Check 16.8), Eqs. (16.34), (16.37) and (16.39)	[109, Eqs. (9), (10), (12) and the absorption rate after Eq. (12)]	Published form: $\mathcal{R} = \kappa^2(\rho_\chi/\rho_T)\, L^{\mathrm{D}}(m_\phi)$ with L^{D} the hydrogenic ionization continuum, their Eq. (12), times $(E_{\mathrm{eff}}/E_0)^2 = 4.84$, $E_{\mathrm{b}}^{\mathrm{D}} = 45\,\mathrm{meV}$, $m^* = 0.3\,m_e$, $a^* = 23$ a.u. Total-charge row; donors dilute and uncorrelated, each screened by the host below its gap (real $\varepsilon_{\mathrm{host}}$), first order in n_{D} in Eq. (8.35) $\to$ Eq. (16.34); multivalley effective-mass states (Appendix D.23), envelope slowly varying $\to$ Eq. (16.35); intravalley transitions $\to$ $\tilde{n}_e = \mathfrak{f}_{\mathrm{D}}^{\mathrm{loc}} n_e/\varepsilon_{\mathrm{host}}$ (Eq. (16.36)), $\mathfrak{f}_{\mathrm{D}}^{\mathrm{loc}} = 1$; $q \to 0$, $L = (e^2/2q^2)S \to$ Eq. (16.39) = the source's absorption rate with $L^{\mathrm{D}} = \varepsilon_2^{\mathrm{D}}/\varepsilon_{\mathrm{host}}^2$ (one screening factor). Single isotropic valley, hydrogenic envelope $\to$ Eq. (16.37), checked against the Coulomb partial-wave sum (10^{-5}) and at $q \to 0$ against the hydrogen dipole continuum (Worked example 4.5). The source's Eq. (12) prints $3\tilde{q}$ for $3\tilde{q}^2$: absorption ($q \to 0$) unaffected; with ε held fixed and the halo of Table 1.1, a scattering rate with the printed bracket is 1.47, 0.91, 0.59, 0.56 of the correct one at $m_\chi = 0.02$, 0.1, 1, 10 MeV (heavy mediator) and 1.46, 1.24, 1.07, 1.05 (light mediator). Their m^*, a^* imply $1/2m^*a^{*2} = 85.7\,\mathrm{meV}$; their continuum carries 0.30 of the f-sum of mass m^*, 4.85 per m_e after the factor 4.84, against 3.85 for the whole donor. $\mathfrak{T}$: $\mathcal{X}$ = IP (envelopes), $n_B = 0$, $n_g = 0$; first order in ζ and in n_{D}.	$n_{\mathrm{D}} \ll n_{\mathrm{c}} \approx 3 \times 10^{18}\,\mathrm{cm}^{-3}$; $\lvert\varepsilon^{\mathrm{D}}\rvert \ll \varepsilon_{\mathrm{host}}$; m_ϕ below the gap; $q \ll \lvert\boldsymbol{k}_j - \boldsymbol{k}_{j'}\rvert$ for the single-valley form. Plot (Fig. 16.14): $x = n_{\mathrm{D}}$ from 10^{13} to $3 \times 10^{18}\,\mathrm{cm}^{-3}$ (log); $y =$ loss function of the Clausius–Mossotti medium, $\varepsilon_{\mathrm{eff}} = \varepsilon_{\mathrm{host}}(1+2y)/(1-y)$, $y = \varepsilon^{\mathrm{D}}/3\varepsilon_{\mathrm{host}}$, over $\varepsilon_2^{\mathrm{D}}/\varepsilon_{\mathrm{host}}^2$; anisotropic effective-mass donor with a model central cell (3.2 Å, tuned to 45.59 meV) and with the effective-mass envelope; $m_\phi = 60\,\mathrm{meV}$ and the $2p_0$ line center (34.11 meV) at full widths 4.2 µeV and 0.1 meV; $\varepsilon_{\mathrm{host}} = 11.4$; expansion dotted; shaded $n_{\mathrm{D}} < 2.6 \times 10^{17}\,\mathrm{cm}^{-3}$; hatched $n_{\mathrm{D}} > 0.3\,n_{\mathrm{c}}$.	$\leq 10\,\%$ **for $n_{\mathrm{D}} < 2.6 \times 10^{17}\,\mathrm{cm}^{-3}$ (continuum, both ground states;** $8.8 \times 10^{17}\,\mathrm{cm}^{-3}$ **with the central cell); at a** 4.2 µeV **line center only below** $4.8 \times 10^{15}\,\mathrm{cm}^{-3}$**; impurity band at $n_{\mathrm{D}}/n_{\mathrm{c}} \approx 0.3$–$0.6$ unsized**. Donor self-screening, $1 - 4\,\mathrm{Re}\,y$ in the continuum and $1/[1 + (4/9)(\varepsilon_2^{\mathrm{D}}/\varepsilon_{\mathrm{host}})^2]$ at a line center; central cell ($2p$ line strengths $\times 1/3.2$ to $1/3.4$ against the effective-mass envelope, $3p$ $\times 1/1.9$ to $1/2.7$); one envelope common to A_1 and E in the valley-orbit lines; $\mathfrak{f}_{\mathrm{D}}^{\mathrm{loc}} - 1$ (host local fields, not computed); pairs and clusters near n_{c} (Appendix E.21).	Sect. 16.5

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Color-center creation by a nuclear recoil, free recoil with a step at the displacement threshold (Limit Check 16.9, panel a), Eqs. (16.40) and (16.41)	[110, Eqs. (1), (2), (5)–(7)]	Nucleon row; final-state class = recoil of species κ with deposit ω along $\hat{\boldsymbol{q}}$ (Sect. 8.7); hand-off Eq. (8.69) with $P_{\text{casc}} \to Y_{\text{CC}} = p_{\text{FP}} p_{\text{surv}} p_q f_{\text{opt}}$ (Eq. (16.40)); $q \gg q_{\text{BZ}} \to$ self response only (Sect. 11.3); $x_R \gg 1 \to$ impulse Gaussian Eq. (11.34); heavy mediator, orientation average $\to$ Eq. (16.41); $\sigma_\omega \to 0 \to$ free recoil Eq. (11.35); $p_{\text{FP}} \to \Theta(E_R - E_{\text{d}}(\hat{\boldsymbol{r}}_R))$, $p_{\text{surv}} = p_q = f_{\text{opt}} = 1 \to$ the source's Eq. (1) with $E_r^{\text{min}} = q_{\text{min}}^2/2m_1$ (its Eq. (5)); $[f_P Z + f_N(A-Z)]^2 \sigma_p/\mu_{\chi p}^2 = \sigma_{N,\kappa}/\mu_{\chi\kappa}^2$ with the per-nucleon σ_p of Eq. (2.27). $\mathfrak{T}$: seed $\mathcal{K}$ = impulse, $n_B = 0$, $T = 0$; the yield lies outside $\mathfrak{T}$.	$E_R^{\text{max}} \geq 1.22$–$1.36\, E_{\text{d}}$ for the Doppler width ($m_\chi \geq 0.218\,\text{GeV}$ at 1.36 for fluorine in LiF at $E_{\text{d}} = 27\,\text{eV}$); yield factors equal to one. Plot (Fig. 16.16a): $x = E_R^{\text{max}}/E_{\text{d}}$ from 1 to 30 (log); $y =$ impulse rate (Eq. (16.41), step at $\omega = E_{\text{d}}$) over the step rate; $\langle\omega\rangle_{\text{F}} = 77.2$ and $33.5\,\text{meV}$; dotted: directional threshold of the source's Eq. (2), $a = 0.4$, over the step; shaded $x \geq 1.36$.	**rate ratio within** 10 % **for** $\boldsymbol{x \geq 1.36}$ **(**77.2 meV**) and 1.22 (**33.5 meV**), within** 2 % **for** $\boldsymbol{x \geq 1.96}$**; nonzero below** $\boldsymbol{x = 1}$**; directional threshold (**$\boldsymbol{a = 0.4}$**) away from the step by more than** 10 % **up to** $\boldsymbol{x = 4.8}$. Zero-point (Doppler) width $\sigma_\omega = (E_R\langle\omega\rangle_{\text{F}})^{1/2}$, 3.5–5.4 % of E_{d}; the threshold surface $E_{\text{d}}(\hat{\boldsymbol{r}}_R)$ and the smooth p_{FP}; the factors p_{surv}, p_q, f_{opt} (Sect. 16.6.4), unsized.	Sect. 16.6
Bond breaking in a molecular solid, classical free-recoil rate with the dissociation threshold (Limit Check 16.9, panel b), Eqs. (16.43) to (16.45)	[111, Eqs. (3.3), (3.11), (3.12), (3.17), (4.3), (4.5)]	Nucleon row on atom 1; $\boldsymbol{R}_1 = \boldsymbol{R}_{\text{CM}} + (\mu_{12}/M_1)\boldsymbol{r}$; intermolecular forces on $\boldsymbol{R}_{\text{CM}}$ only (free rotor) $\to$ $H = H_{\text{CM}} + H_{\text{int}} \to$ $S_{11} = S_{\text{CM}} * S_{\text{int}}(\boldsymbol{q}_{12})$ (Eq. (16.43)); dissociation = S_{int} above E_{bond}; impulse approximation of both factors $\to$ Eq. (16.45) over the Gaussian radial ground state (the source's Eq. (3.12), width $(\mu_{12}\omega_{\text{vib}})^{-1/2}$); $\boldsymbol{p}_{12} \to 0$, free center of mass $\to q \geq q_{\text{min}}$ (Eq. (16.44)), $\omega = q^2/2M_1 \to$ the source's Eq. (4.3); plane-wave final states with $\lvert\boldsymbol{q}_{12} + \boldsymbol{p}_{12}\rvert^2/2\mu_{12} > E_{\text{bond}} \to$ the improved Born form of its Eqs. (3.17), (4.5). $\mathfrak{T}$: $\mathcal{K}$ = impulse (internal and center of mass), $n_B = 0$, $T = 0$.	$(q_{\text{max}}/q_{\text{min}})^2 \geq 2.65$ ($m_\chi \geq 42.9\,\text{MeV}$ for H_2, $E_{\text{bond}} = 4.75\,\text{eV}$, $\omega_{\text{vib}} = 0.546\,\text{eV}$); free rotor; $p_{\text{esc}} = 1$. Plot (Fig. 16.16b): $x = (q_{\text{max}}/q_{\text{min}})^2$ from 1 to 30 (log), $q_{\text{max}} = 2\mu_{\chi 1} v_{\text{max}}$; y = impulse rate over the classical rate; dashed: improved Born over classical; shaded $x \geq 2.65$.	**within** 10 % **for** $\boldsymbol{x \geq 2.65}$ **(impulse) and 3.61 (improved Born,** $\boldsymbol{m_\chi \geq 50.5\,\text{MeV}}$**); mass scale lowered by a factor 0.84 (impulse) and 0.75 (Born) at the classical rate of** $\boldsymbol{x = 1.1}$. Zero-point momentum of the bond, rms $(\mu_{12}\omega_{\text{vib}}/6)^{1/2} = 6.53\,\text{keV}$ along $\hat{\boldsymbol{q}}$; initial kinetic energy $\omega_{\text{vib}}/4$ (impulse vs Born); center-of-mass Doppler shift of v_{min} 2×10^{-6} (negligible); cage escape p_{esc} and electronic (σ^*) dissociation, unsized.	Sect. 16.6

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Two-level response to a classical strain or field, $\tan\delta = \tan\delta_0 \tanh(\omega/2T)$, $Q^{-1} = \pi C_j \tanh(\omega/2T)$ (Limit Check 16.10), Eqs. (16.51) and (16.52)	[112, Eqs. (4), (12), (13), (15), 113, Eq. (20), electric form]	Double well → Eq. (16.46) in $\|\mathrm{L}\rangle$, $\|\mathrm{R}\rangle$ (Eq. (C.169) without bath); flat $P(\Delta_{\mathrm{as}}, \lambda_{\mathrm{tun}}) = \bar{P}$, $\Delta_{\mathrm{as}} \geq 0$ → Eq. (16.48); $\delta\Delta_{\mathrm{as}} = 2\gamma_{\mathrm{TLS}} u_{\mathrm{d}} + 2\boldsymbol{p}_{\mathrm{TLS}} \cdot \boldsymbol{E}$ → Eq. (16.50); transverse part $(\Delta_0/\Delta_{\mathrm{tl}})\tau_x$ (Eq. (16.47)), golden rule with $p_0 - p_1 = \tanh(\Delta_{\mathrm{tl}}/2T)$, width $\ll \omega, T \to \chi''$ per system; ensemble weight Eq. (16.49) → Eq. (16.51); orientation average $p^2/3$, Heaviside–Lorentz → Eq. (16.52); modulus $\rho_T c_j^2 \to \pi C_j$. Relaxation part dropped. The second source prints the acoustic form with a factor $\frac{1}{2}$ against its own Eq. (16). $\mathfrak{T}$: $n_B = 0$ (rigid wells), independent systems, linear response.	Low power ($I \ll I_{\mathrm{c}}$); independent systems; width $\ll \omega, T$; T below the relaxation crossover. Plot (Fig. 16.18): $x = T$ from 3 mK to 30 K (log); $y =$ loss over its $T = 0$ value: $\tanh(\omega/2T)$ (dashed) and resonant + one-phonon relaxation, Eq. (16.54) with Eq. (16.53), vitreous silica ($\rho_T = 2.2$ g cm^{-3}, $c_{\mathrm{LA}} = 5.8$, $c_{\mathrm{TA}} = 3.8$ km s^{-1}, assumed $\gamma_{\mathrm{l}} = 1$ eV, $\gamma_{\mathrm{l}}/\gamma_{\mathrm{t}} = 1.6$, $K_{\mathrm{ph}} = 4.63 \times 10^4$ eV^{-2}), at 0.5 and 5 GHz; plateau $\frac{1}{2}$ dotted; shaded $T < 1.36$ K (5 GHz).	**$\leq 10\,\%$ for $T < 1.36$ K at 5 GHz and < 0.43 K at 0.5 GHz; 4×10^{-9} at 10 mK, 5 GHz.** Relaxation absorption of the population, $(\pi^3/12)K_{\mathrm{ph}}T^3/\omega$ at low T, plateau $\frac{1}{2}$; crossover $\propto K_{\mathrm{ph}}^{-1/4}$; saturation $[1 + I/I_{\mathrm{c}}]^{-1/2}$; multiphonon relaxation above 2 K; interactions between systems, unsized.	Sect. 16.7
Dark-photon absorption on two-level systems (Limit Check 16.11), Eq. (16.55)	[113, Eq. (20), electric form]	Published form: the electric loss of the tunneling model, $\pi\bar{P}p^2/3\varepsilon_r$; the rate form is Eq. (10.31). Chain: Eq. (10.31) at $\boldsymbol{p}_\phi \to 0$ ($p_\phi \approx 3 \times 10^{-8}$ eV at 25 μeV), $\varepsilon = \varepsilon_r(1 + \mathrm{i}\tan\delta) \to L = \tan\delta/[\varepsilon_r(1 + \tan^2\delta)]$; Eq. (16.52) → Eq. (16.55); one thermal factor (net absorption); field far below saturation (Rabi frequency $< 6 \times 10^{-7}\,\mathrm{s}^{-1}$ at the reach of Worked example 16.7). $\mathfrak{T}$: first order in ζ, $n_B = 0$, local ε.	m_ϕ inside the band where $\bar{P}$ is flat (tested near 6 GHz); $\tan\delta_0 \ll 1$. Plot (Fig. 16.19): $x = m_\phi$ from 10^{-10} to 10^{-3} eV (log); $y = \zeta$ for 3 events per kg yr, thermal SiO_2 ($\tan\delta_0 = 3 \times 10^{-4}$, $\varepsilon_r = 3.9$, $\rho_T = 2.2$ g cm^{-3}), formula (dashed) and $\varepsilon_2/\|\varepsilon\|^2$ with Eq. (16.54) (solid), at 10 and 100 mK; shaded $m_\phi > 1.15$ neV (10 mK).	**$\leq 10\,\%$ for $m_\phi > 1.15$ neV at 10 mK and > 115 neV at 100 mK; $\tan^2\delta_0 \approx 10^{-7}$.** Relaxation absorption of systems with $\Delta_{\mathrm{tl}} \approx T$, ratio $(\pi^3/6)K_{\mathrm{ph}}T^4/m_\phi^2$; the upper edge of the flat band and the vibrational absorption above 0.46 meV (Limit Check 16.12).	Sect. 16.7

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Scattering off a random medium in the first Born approximation and its validity (Limit Check 16.13), Eqs. (16.64), (16.65) and (16.67)	[114, Eqs. (2), (9)–(12), (14)]	Nucleon row; $q \ll$ inverse nuclear spacing $\rightarrow$ smooth $n(\boldsymbol{r})$; first Born approximation for the body $\rightarrow$ Eq. (16.64); $n = n_0 + \delta n$, configurational average $\rightarrow$ homogeneous + fluctuation terms; exponential correlation, $R \gg \xi_c \rightarrow$ Eq. (16.65). Validity: optical potential Eq. (16.66), $\lvert V_{\text{opt}}\rvert \ll E_{\text{kin}} \rightarrow$ Eq. (16.67). Test: Born reflectance $(V_{\text{opt}}/4E_n)^2$ of a planar face against the exact step reflectance. $\mathfrak{T}$: first order in the dark-matter coupling of the whole body; $\mathcal{K}$ = elastic.	$\sigma_{\chi N} \ll \sigma_{\text{crit}} = m_\chi^4 v^4/4\pi n_0^2$ over the dominant velocities ($1.16 \times 10^{-38}\,\text{cm}^2$ for Ti–6Al–4V, 20 eV, $v = 10^{-3}$). Plot (Fig. 16.21): $x = \lvert V_{\text{opt}}\rvert/E_n$ from 10^{-3} to 10 (log); y = exact over Born reflectance, repulsive and attractive; shaded $x < 0.092$; marker at 1.86 ($\sigma_{\chi N} = 4 \times 10^{-38}\,\text{cm}^2$, 20 eV, $v = 10^{-3}$).	$\boldsymbol{\leq 10\,\%}$ **for** $\boldsymbol{\lvert V_{\text{opt}}\rvert/E_n < 0.092}$ **(repulsive),** $\boldsymbol{< 0.110}$ **(attractive); relative correction** $\boldsymbol{\pm V_{\text{opt}}/E_n}$. Distortion of the wave by the mean medium (phase $k \rightarrow k'$), refraction and total reflection above $V_{\text{opt}} = E_n$; projection on the face normal; force above σ_{crit} (Appendix E.19), unsized.	Sect. 16.7
Glass absorption below the Ioffe–Regel frequency, $C^{\text{IR}} = A^{\text{IR}} + B^{\text{IR}}\omega^2$ (Limit Check 16.12), Eqs. (16.61) to (16.63)	[115, Eq. (1), 116, Eq. (2)]	Charge row of the glass vibrations; harmonic modes of one configuration, rank-one macroscopic $\varepsilon \rightarrow$ Eq. (16.60); narrow modes $\rightarrow$ Eq. (16.61); $\omega < \omega_{\text{IoR}}$, plane-wave modes $\boldsymbol{e}_{I\nu} = \hat{\boldsymbol{e}}_\nu e^{i\boldsymbol{k}\cdot\boldsymbol{x}_I}(M_I/N\bar{M})^{1/2} \rightarrow$ masses cancel, Eq. (16.62) with branch weights c_ν^{-3}; small-k expansion of S_ν^Z (global neutrality: constant; local neutrality: k^2) $\rightarrow$ Eq. (16.63); rate $(\rho_\chi/\rho_T)\zeta^2\,\text{Im}(-1/\varepsilon)$ (Eq. (10.31)). $\mathfrak{T}$: $n_A = 0$ (harmonic), $\mathcal{X}$ = Born charges, local ε.	$\omega < \omega_{\text{IoR}}$ ($\approx 35\,\text{cm}^{-1}$, 4.3 meV, in silica); harmonic; $\varepsilon_2 \ll \varepsilon_r$. Plot (Fig. 16.20): $x = m_\phi$ from 0.06 to 15 meV (log); y = events per kg yr per ζ^2 with $A^{\text{IR}} = 4000\,\text{cm}^{-2}$, $B^{\text{IR}} = 0.3$ (room-temperature fit in spectroscopic units, $\tilde{\omega}$ in cm^{-1}, B^{IR} dimensionless), Debye g ($\Theta_{\text{D}} = 330\,\text{K}$), $\varepsilon_r = 3.9$: $A^{\text{IR}} + B^{\text{IR}}\tilde{\omega}^2$ (solid), A^{IR} alone (dashed), two-level rate of thermal SiO_2 (dotted); shaded $\omega < \omega_{\text{IoR}}$.	$\boldsymbol{B^{\text{IR}}\omega^2/A^{\text{IR}} \leq 9\,\%}$ **for** $\boldsymbol{\omega < \omega_{\text{IoR}}}$ **in silica**. The curvature term (local neutrality); Rayleigh scattering of the plane waves, relative order ω^3; boson-peak excess of $g(\omega)$ over Debye; anharmonic and relaxational absorption at millikelvin temperatures, unsized.	Sect. 16.7

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Two-level relaxation by one phonon (Jäckle) and by conduction electrons (Black), with the superconducting factor (Limit Check 16.14, panel a), Eqs. (16.53), (16.68) to (16.70) and (16.72)	[112, Eqs. (7)–(9), 117, Eqs. (4), (5)]	Two configurations, coupling $\frac{1}{2}\sigma_z X$ linear in the bath coordinate (Eq. (C.169)); golden rule in the eigenbasis, first order in J_s, emission plus absorption → Eq. (16.68). Phonons: coupling $\gamma u_\mathrm{d}\sigma_z$ ($\delta\Delta_\mathrm{as} = 2\gamma u_\mathrm{d}$), isotropic continuum, long wavelength, orientation-averaged γ_l^2, γ_t^2 → Eq. (16.69) → Eq. (16.53) = the first source's Eq. (7) and the second's Eqs. (4), (5) ($8\pi^3/\rho h^4 = 1/2\pi\rho$). The first source's text writes $\delta\varepsilon = \gamma u_\mathrm{d}$, while its Eq. (7) holds for $\delta\varepsilon = 2\gamma u_\mathrm{d}$; with $\delta\varepsilon = \gamma u_\mathrm{d}$ the rate would be four times smaller (Appendix D.42). Electrons: ohmic $J_\mathrm{s} = 2\pi\alpha_\mathrm{ohm}\omega$, $\omega \ll \omega_\mathrm{cut}$ → Eq. (16.70) = its Eq. (8) with $W_\mathrm{el}^2 = 4\alpha_\mathrm{ohm}$. Superconductor: case-I coherence factor, $\Delta_\mathrm{tl} < 2\Delta_\mathrm{BCS}$, quasiparticle scattering only → Eq. (D.235); $T, \Delta_\mathrm{tl} \ll \Delta_\mathrm{BCS}$ → Eq. (16.72); $\Delta_\mathrm{tl} \ll T$ → $2/(1+\mathrm{e}^{\Delta_\mathrm{BCS}/T})$, its Eq. (9). $\mathfrak{T}$: defect configurations at $n_B = 0$; harmonic phonons, $n_A = 0$; rate of second order in the defect–bath coupling; T kept exactly.	$\Gamma_1 \ll \Delta_\mathrm{tl}$ (normal aluminum with $\alpha_\mathrm{ohm} = 0.28$, $\Delta_0 = 0.3\,\Delta_\mathrm{tl}$: below 1.5 K); Δ_tl/c_ν below the inverse defect size; superconducting factor for $\Delta_\mathrm{tl} \le 0.7\,T$. Plot (Fig. 16.23a): $x = \Delta_\mathrm{tl}/T$ from 0.01 to 10 (log), curves ending at $2\Delta_\mathrm{BCS}(T)$; y = Eq. (D.235) over $2/(1+\mathrm{e}^{\Delta_\mathrm{BCS}/T})$ at $T/T_c = 0.2$, 0.5, 0.8, aluminum, weak-coupling gap; dotted: $\sinh u\,K_1(u)$; shaded where within 10 %.	**superconducting factor within** 10 % **for $\boldsymbol{\Delta_\mathrm{tl} \le 0.69}$–$\boldsymbol{0.73\,T}$; 0.86 at $\boldsymbol{\Delta_\mathrm{tl} = T}$, 0.71 at $\boldsymbol{2T}$**. The Bessel factor $\sinh u\,K_1(u) = 1 + (u^2/2)[\ln(u/2) + \gamma_\mathrm{E} - \frac{1}{6}] + \ldots$, $u = \Delta_\mathrm{tl}/2T$ (Appendix D.42); higher orders in the bath coupling when $\Gamma_1 \to \Delta_\mathrm{tl}$ (Limit Check 16.15); multiphonon relaxation above a few kelvin, unsized.	Sect. 16.8

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Ohmic coupling of a tunneling defect in the Born limit, $\alpha_{\mathrm{ohm}} = 4K_{\mathrm{b}}$ (Limit Check 16.14, panel b), Eq. (16.71)	[118, Eq. (4) and the single-spin density of states of its supplement]	Contact potentials $\pm V_{\mathrm{d}}$ over Ω in the two configurations; electron–hole pairs as a Gaussian bath, second order in V_{d}; constant density of states $\mathcal{N}_{\mathrm{F}}$ per spin $\rightarrow J_{\mathrm{s}}^{\mathrm{el}} = 8\pi K_{\mathrm{b}}\omega \rightarrow$ Eq. (16.71) (Appendix D.42); with Eq. (16.68) the source's rate $4\pi K\Delta_0^2 \coth(\Delta_{\mathrm{tl}}/2T)/\Delta_{\mathrm{tl}}$, $K = K_{\mathrm{b}}$. Beyond Born: eigenphase form Eq. (E.73); one s channel per spin, unitarized contact t-matrix, $\tan\delta = \mp\pi\mathcal{N}_{\mathrm{F}}V_{\mathrm{d}}\Omega \rightarrow$ Eq. (D.234). With a both-spin density $n_0 = 2\mathcal{N}_{\mathrm{F}}$ and $K = (n_0 V_{\mathrm{d}}\Omega)^2$ the Born result reads $\alpha_{\mathrm{ohm}} = K$. $\mathfrak{T}$: electrons at clamped configurations, $n_B = 0$; Born in V_{d}.	$\pi K_{\mathrm{b}}^{1/2} \ll 1$; one channel per spin; $T, \Delta_{\mathrm{tl}} \ll \omega_{\mathrm{cut}}$. Plot (Fig. 16.23b): $x = K_{\mathrm{b}}$ from 10^{-4} to 1 (log); $y = \alpha_{\mathrm{ohm}}/4K_{\mathrm{b}}$ from Eq. (D.234); dashed $1 - \frac{2}{3}\pi^2 K_{\mathrm{b}}$; dots at the fits $K_{\mathrm{b}} = 0.07$, 0.11; shaded $K_{\mathrm{b}} \leq 0.017$.	$\leq 10\,\%$ **for $K_{\mathrm{b}} \leq 0.017$; ratio 0.70 at $K_{\mathrm{b}} = 0.07$ and 0.60 at 0.11**. Saturation of the phase shift, $1 - \frac{2}{3}\pi^2 K_{\mathrm{b}}$; several channels for a displaced (hopping) scatterer, unsized; the real part of the local Green's function, absorbed into V_{d}.	Sect. 16.8
Ohmic scaling limit of a symmetric tunneling defect, small-coupling relaxation rate $\tilde{\Delta}_0^2/\gamma_0$ (Limit Check 16.15), Eqs. (16.73) to (16.75)	[119, Eqs. (8)–(10)]	Spin–boson model with ohmic bath (Eqs. (C.169) and (C.171)); the source's K is α_{ohm}, not K_{b}. Incoherent rate second order in Δ_0, exact in α_{ohm} (noninteracting blips) $\rightarrow$ Eq. (C.174) = the source's Eqs. (8), (9) without electron-assisted tunneling; $\Delta_0, T \ll \omega_{\mathrm{cut}} \rightarrow$ Eq. (C.175); $\Delta_{\mathrm{as}} = 0$, duplication and reflection formulas, Δ_{r} of Eq. (C.176) $\rightarrow$ Eq. (16.73); $\alpha_{\mathrm{ohm}} \ll 1$ $\rightarrow$ the slow pole $\tilde{\Delta}_0^2/\gamma_0$ of its Eq. (10). Ratio $\Gamma(1+\alpha_{\mathrm{ohm}})/\Gamma(1-\alpha_{\mathrm{ohm}})$, checked by numerical time integration at $\omega_{\mathrm{cut}} = 10^4\,T$. $\mathfrak{T}$: defect at $n_B = 0$; bath linear and Gaussian.	$\Delta_{\mathrm{r}} \ll T \ll \omega_{\mathrm{cut}}$, $T > T_{\mathrm{coh}}$ (Eq. (16.75)); $\alpha_{\mathrm{ohm}} < \frac{1}{2}$ for the anti-Arrhenius law. Plot (Fig. 16.24): $x = \alpha_{\mathrm{ohm}}$ from 0 to 0.5; $y =$ incoherent rate over $\tilde{\Delta}_0^2/\gamma_0$; open circles: numerical time integral; dashed $1 - 2\gamma_{\mathrm{E}}\alpha_{\mathrm{ohm}}$; shaded $\alpha_{\mathrm{ohm}} \leq 0.091$.	$\leq 10\,\%$ **for $\alpha_{\mathrm{ohm}} \leq 0.091$; 0.71 at 0.28, 0.56 at 0.44**. $\Gamma(1+\alpha_{\mathrm{ohm}})/\Gamma(1-\alpha_{\mathrm{ohm}}) = 1 - 2\gamma_{\mathrm{E}}\alpha_{\mathrm{ohm}} + O(\alpha_{\mathrm{ohm}}^2)$; below T_{coh} coherent tunneling (the blip approximation fails at $T \lesssim \Delta_{\mathrm{r}}$); electron-assisted tunneling (the source's Δ_1), unsized.	Sect. 16.8

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Thermally activated hopping and tunneling through one barrier: transition-state theory with Wigner and parabolic-barrier transmission, instanton below the crossover (Limit Check 16.16), Eqs. (16.76) to (16.78)	[120, Eqs. (1)–(4), (69)–(71), 117, Eq. (8)]	Published form: $kZ_r = (\kappa/2\pi\beta)(Z^{\ddagger})\mathrm{e}^{-\beta V_\mathrm{b}}$, κ_W, κ_PB, and k_inst of the source's Eq. (4). One reaction coordinate, harmonic reactant; classical motion over the barrier → $\Gamma = \Gamma_0 \mathrm{e}^{-V_\mathrm{b}/T}$ (Appendix E.11); inverted-parabola barrier → Eq. (16.76), divergent at Eq. (16.77); below $T_\times$ stationary-action periodic orbit, $S_\mathrm{inst} \to 2\lambda_\mathrm{tun}$ as $T \to 0$. Freezing: activated rate equal to $V_\mathrm{b}\lvert\dot{T}\rvert/T^2$ → Eq. (16.78) = the second source's Eq. (8) with $\tau_0 = 1/\Gamma_0$. Test case: symmetric Eckart barrier, exact transmission. $\mathfrak{T}$: Born–Oppenheimer surface, $n_B = 0$; no dissipation in the barrier crossing.	Wigner: $T \geq 2.1\,T_\times$; parabolic: $T \geq 1.5\,T_\times$; instanton: $T \lesssim 0.5\,T_\times$. Plot (Fig. 16.25): $x = T/T_\times$ from 0.25 to 4 (log); y = approximate over exact rate (log) for the Eckart barrier with $V_\mathrm{b} = 6\omega^{\ddagger}/\pi$; shaded where Wigner is within 10 %.	**Wigner within** 10 % **for $T \geq 2.1\,T_\times$; exact/classical 6.2 at $T_\times$, 2.2 at $1.5\,T_\times$; instanton** 18 % **low at 0.25–0.5 $T_\times$.** Next term of κ_PB, $(7/5760)(\omega^{\ddagger}/T)^4$; anharmonicity of the barrier top; higher terms of the instanton expansion in $\hbar$; dissipation by the bath during the crossing, unsized.	Sect. 16.8
Heat release of a glass cooled from T_hi to T_lo (Limit Check 16.17), Eq. (16.79)	[117, Eqs. (1), (6)–(8), 121, Table 1]	Standard tunneling model, density $\bar{P}$ uniform in Δ_as and λ_tun; instantaneous quench, populations of T_hi; relaxation by one phonon at T_lo (Eq. (16.53)); $\Delta_0 \ll \Delta_\mathrm{as}$ → excess $\int \mathrm{d}\Delta_\mathrm{tl}\,(\Delta_\mathrm{tl}/2)[\tanh(\Delta_\mathrm{tl}/2T_\mathrm{lo}) - \tanh(\Delta_\mathrm{tl}/2T_\mathrm{hi})] = \pi^2(T_\mathrm{hi}^2 - T_\mathrm{lo}^2)/12$; uniform in $\ln\Gamma_1$ with density $\frac{1}{2} \to 1/2t$ → Eq. (16.79) = the source's Eq. (6); start above $T_\mathrm{frz} \to T_\mathrm{hi} \to T_\mathrm{frz}$, its Eq. (7). $\mathfrak{T}$: independent defects; harmonic phonons; golden-rule relaxation.	$\Gamma_\mathrm{max}^{-1} \ll t \ll \Gamma_\mathrm{min}^{-1}$ at $\Delta_\mathrm{tl} \sim T_\mathrm{hi}$. Plot (Fig. 16.26): $x = t/\tau_\mathrm{min}$ from 10^{-2} to 10^{18} (log); $y = t\dot{Q}_\mathrm{TLS}$ over the prefactor, model with λ_tun cut over a range 14, $\Delta_0 \leq 0.1\,T_\mathrm{lo}$, exact Δ_tl, $T_\mathrm{lo}/T_\mathrm{hi} = 1/480$; shaded where within 10 %.	$\leq$ 10 % **for $1.8\,\tau_\mathrm{min} \leq t \leq 7.1 \times 10^{10}\,\tau_\mathrm{min}$ (10.6 of 12.2 decades); within 2×10^{-4} from 10^2 to $10^8\,\tau_\mathrm{min}$.** The ends of the distribution in λ_tun; finite cooling rate (freezing, Eq. (16.78)); measured $\bar{P}$ from heat release four times below the specific-heat value in vitreous silica (stress-lowered barriers); interactions and avalanches (Sect. 16.8.6), unsized.	Sect. 16.8

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Partial (valence) f-sum of the loss function (Limit Check 17.1), Eq. (17.16)	[122, Table I] (measured ω_p^{val})	Published form: $\int_{\mathrm{val}} \omega L \,\mathrm{d}\omega = (\pi/2)(\omega_p^{\mathrm{val}})^2$. Chain: f-sum of the loss function Eq. (4.46); $\varepsilon = \varepsilon_{\mathrm{val}} + \varepsilon_{\mathrm{core}} - 1$ with additive groups (no Pauli transfer between shells), valence absorption below the first core edge, $\varepsilon_{\mathrm{core}}(\omega) \to \varepsilon_{\mathrm{core}}(0)$ (Kramers–Kronig, Eq. (C.41)) over the valence spectrum; the large-frequency tail of $\varepsilon_{\mathrm{m}}^{-1} - 1/\varepsilon_{\mathrm{core}}(0)$, with the model function $\varepsilon_{\mathrm{m}} = \varepsilon_{\mathrm{core}}(0) + \varepsilon_{\mathrm{val}} - 1$ (equal to ε below the core edge to $O(\omega^2/\omega_{\mathrm{core}}^2)$; for the true ε an intermediate asymptote), gives $(\pi/2)(\omega_p^{\mathrm{val}})^2/\varepsilon_{\mathrm{core}}(0)^2$. $\mathfrak{T}$: clamped nuclei, $q \to 0$, exact response.	Upper limit between the valence spectrum and the first core edge. Silicon: within 10 % for ω_c from 32.7 eV ($2\omega_p^{\mathrm{val}}$) to the L edge at 99.8 eV; 1.000 just below the edge. Plot (Fig. 17.6): $x = \omega_c$, 3 eV–30 keV (log); y = running $\int_0^{\omega_c} \omega L$ and $\int_0^{\omega_c} \omega\varepsilon_2$ over $(\pi/2)(\omega_p^{\mathrm{val}})^2$ from measured optical constants (Franta et al., below 35 eV) and the Henke tables above; dashed the formula, dotted with $\varepsilon_{\mathrm{core}}(0) = 1.0154$; shaded the valid range.	**−3 % core screening, +3 % shell transfer (Si)**. Core screening $\varepsilon_{\mathrm{core}}(0)^{-2} = 0.970$; oscillator strength moved from blocked core transitions into the valence region (ε_2-sum 1.032 below the edge); both known to the 1.4 % normalization of the data; at finite q valid only while $qv_{\max} < \omega_{\mathrm{core}}$.	Sect. 17.5
Pair energy and Fano factor as cascade moments (Limit Check 18.1), Eqs. (18.10) and (18.11)	[122, Eqs. (2)–(3)]	Published form: $\varepsilon(\omega) = \omega/\bar{n}(\omega)$ and $F(\omega) = \mathrm{Var}\, N_{\mathrm{eh}}/\bar{n}$ at each deposit, constant at high energy. Chain: energy identity Eq. (18.5) (no capture, excitons or escaping photons before counting) → mean and variance at fixed ω Eq. (18.10) (Eq. (D.242)); heat split $E_{\mathrm{heat}} = E_< + E_>$ → Eq. (18.11) (Eq. (D.243)); additive deposits solved by Eqs. (D.240) and (D.241) under the energy-only random-k reduction (Sect. 18.1.4); large-ω slopes, existence of the leading term Eq. (18.15) (Appendix D.44), constant terms b_N, c_N fitted. $\mathfrak{T}$: detector.	Silicon model ($E_{\mathrm{g}} = 1.160$ eV, $\omega_{\mathrm{AB}} = 63$ meV, $A_{\mathrm{AB}} = 5.703\,\mathrm{eV}^3$, $\theta_{\mathrm{ii}} = 1$, $B_{\mathrm{h}} = 12$ eV, one-sided seed): both ratios within 10 % of the limits for $\omega \geq 14.9\,E_{\mathrm{g}} = 17.3$ eV. Plot (Fig. 18.3): $x = \omega/E_{\mathrm{g}}$ from 1 to 60 (log); $y = \varepsilon(\omega)/\varepsilon_{\mathrm{eh}}$ and $F(\omega)/F_{\mathrm{eh}}$; dashed with the constant terms $b_N = 0.35$, $c_N = -0.02$; shaded $\omega \geq 14.9\,E_{\mathrm{g}}$.	**$b_N\varepsilon_{\mathrm{eh}}/\omega$: 10 % at $\omega = 10E_{\mathrm{g}}$; residual after it ≤ 0.9 % there**. Constant terms of the mean and variance (not derived; Sect. 18.2.6); periodic remainder from the lattice of losses in a one-phonon model; step structure below $15E_{\mathrm{g}}$.	Sect. 18.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Linear mean yield (no box), Eqs. (18.14) and (18.15)	[123, Chap. XI] (key renewal theorem)	Tagged-unit identity Eq. (D.244) (energy conservation and Markov property only) $\rightarrow$ walk of $\ln x$ along the line of one unit; scale-free share law approached as $(E_{\mathrm{g}}/x)^{\varkappa}$ (Dirichlet law without a hole cutoff; Beta law of the two electrons with the cutoff, where the phonon ratio falls as $x^{-3/2}$), non-lattice, finite mean, regular near threshold $\rightarrow$ coupling to the scale-free walk and key renewal theorem for its undershoot (Appendix D.44). $\mathfrak{T}$: detector.	Any energy-conserving Markov cascade satisfying the four conditions of Appendix D.44; random-k cascade with $\varkappa = 1$. Fails for fixed-proportion splits (log-lattice).	**unsized**. Constant term b_N (0.35 in silicon, one-sided seed; 13 % of $\bar{n}$ at 10 eV) and the variance law need second-order renewal estimates (Sect. 18.2.6).	Sect. 18.2
Random-parking (Rényi) limit (Limit Check 18.2), ρ_∞ and F_{park}	[124, Eqs. (2), (19), (23), (44)]	Published form: $\rho_\infty = \int_0^\infty \mathrm{e}^{-2\,\mathrm{Ein}(t)}\mathrm{d}t$, $F = 0.0510387$, $F_{\mathrm{unc}} = \mathrm{Var}\,g/(1+\langle g\rangle)^2$. The source prints $\rho_\infty = 0.7475987$ (the integral, at 30 digits: 0.747597920) and $F_{\mathrm{unc}} = 0.0439766$, a dropped digit of 0.04439766, as its own $F_{\mathrm{nnp}} = F_{\mathrm{unc}} + 0.007685 = 0.05208$ requires (the printed value would give 0.05166). Chain: random-k cascade (Eq. (18.8)) with constant electron density of states, delta-function hole density of states, $\theta_{\mathrm{ii}} = 1$, $\lambda_{\mathrm{AB}} \rightarrow 0 \rightarrow$ Rényi parking of unit cars; mean recursion by Laplace transform (Worked example 18.2); second moment by Eq. (C.104). $\mathfrak{T}$: detector.	Within 10 % (both $E_{\mathrm{g}}/\varepsilon_{\mathrm{eh}}$ and F_{eh}) for $\lambda_{\mathrm{AB}} < 0.40$ at $w_{\mathrm{AB}} = 0.0543$. Plot (Fig. 18.5): $x = \lambda_{\mathrm{AB}}$ from 10^{-3} to 10 (log); $y = F_{\mathrm{eh}}/F_{\mathrm{park}}$, $(E_{\mathrm{g}}/\varepsilon_{\mathrm{eh}})/\rho_\infty$, $F_{\mathrm{ren}}/F_{\mathrm{park}}$; grid step $E_{\mathrm{g}}/74$; shaded $\lambda_{\mathrm{AB}} < 0.40$.	$\boldsymbol{\pm 0.8\,\lambda_{\mathrm{AB}}}$ **for** $\boldsymbol{\lambda_{\mathrm{AB}} \leq 0.003}$**;** $-3.2\,\%$ **(**$\boldsymbol{\rho}$**),** $+3.7\,\%$ **(**$\boldsymbol{F}$**) at 0.1**. Phonon emission above threshold; the parabolic density of states moves the $\lambda_{\mathrm{AB}} \rightarrow 0$ values to $1.585\,E_{\mathrm{g}}$ and 0.081.	Sect. 18.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Renewal (uncorrelated final-state) Fano estimate (Limit Check 18.3), Eqs. (18.12) and (18.13)	[124, Eq. (7)]	Published form: $F_{\mathrm{unc}} = \langle \delta\epsilon^2 \rangle / \epsilon^2$. Chain: independent, identically distributed costs per pair, single renewal chain Eq. (C.109) with the energy constraint ($\varepsilon_{\mathrm{eh}}$ in the denominator) $\rightarrow$ Eq. (18.12); costs built from one terminal electron and one terminal hole with the computed residual variances and Poisson phonon heat $\rightarrow$ Eq. (18.13). Compared with the full Eq. (18.11). $\mathfrak{T}$: detector.	Lower value of F_{eh} in every computed case (ratio 0.870 parking; 0.696, 0.837 parabolic without phonons at $\theta_{\mathrm{ii}} = 1$, 3/2; 0.193 Si; 0.418 Si at $\theta_{\mathrm{ii}} = 3/2$; 0.126 Ge, hole band cut at 12.6 eV); nowhere within 10 %; the general bound is open (Sect. 18.2.6). Plot (Fig. 18.6): $x = \lambda_{\mathrm{AB}}$ from 10^{-3} to 30 (log); $y = F_{\mathrm{ren}}/F_{\mathrm{eh}}$; band 0.9–1 empty.	$\boldsymbol{F_{\mathrm{eh}}/F_{\mathrm{ren}} = 1.15}$ **(parking) to 5.2 (Si), 7.9 (Ge)**. All correlations between carriers (siblings, electron and hole of a pair, distant carriers), non-identical cost laws, and the non-Poisson phonon heat with its covariance $2\mathcal{V}_{<>}$.	Sect. 18.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Step model and Ramanathan–Kurinsky model as limits of the seeded cascade distribution (Limit Check 18.4), Eqs. (18.18), (18.20) and (18.21)	[122, Eqs. (5)–(10), (14), 7, Eq. (5.1)]	Published form: $n(\omega) = 1 + \int_0^{\omega - E_g} \mathrm{d}E\, P(E,\omega)\langle N(E)\rangle$ with $P(E,\omega)$ the Beta split of the electron share, and $N_{eh} = 1 + \lfloor(\omega - E_g)/\varepsilon_{eh}\rfloor$. Chain: hand-off Eq. (8.70) → seed distribution Eq. (18.16) with the energy-only reduction applied after the first collisions (first-generation momenta dropped) → Eqs. (18.17) and (18.18) from the product form Eq. (18.9); seed replaced by a process-independent, electron–hole symmetric split Eq. (18.20) fitted to photon yields; one yield function for electrons and holes; random-k kernels with free-particle densities of states and one phonon (Eq. (18.8)), $\theta_{ii} = 1$, hole energies below the valence bandwidth → the source's Eqs. (5), (8), (10); large-ω renewal limit with the constant term of the mean set to zero → its Eq. (14). Step model: every pair after the first costs exactly ε_{eh}, deterministic renewal chain (Appendix C.11) → Eq. (18.21), Fano factor zero. $\mathfrak{T}$: detector.	Prompt pair number independent of the seed. Silicon (cascade with $A_{AB} = 5.703\,\mathrm{eV}^3$, $E_g = 1.160\,\mathrm{eV}$): step model within 0.1 in d_{TV} of the dark-matter seed for $\omega < 4.37\,\mathrm{eV}$; splits with $\alpha_{RK} = 1$ and $\to \infty$ also for $\omega > 12.95\,\mathrm{eV}$. Plot (Fig. 18.8): $x = \omega$ from 1.4 to 21.2 eV; $y = d_{TV}$ to the cascade seeded by the dark-matter kick at q_0 (empirical-pseudopotential pair weights), for the step model, the splits $\alpha_{RK} \to \infty$, 1, $\to 0$ and the photon seed; shaded the two regions.	**$d_{TV} \leq 0.062$ (energy-only model, one k grid; unresolved in the full band, where the photon seed moves 0.11–0.14 between grids) for a photon seed in place of the dark-matter seed (largest at 5.67 eV; mean 4.6 %); splits with $\alpha_{RK} = 1, \to \infty$ up to 0.21, 0.39, one-sided split 0.67; step model up to 0.86**. Leading correction: the momentum-transfer dependence of the seed (photon against dark-matter shares); then the constant term of the mean, 0.008 (uniform split) to 0.35 (one-sided seed), which decays in d_{TV} only as $\omega^{-1/2}$; first-generation valley and momentum memory, dropped by the energy-only reduction (negligible within the deformation-potential model of a full-band cascade). Against that full-band cascade at matched pair energy: energy-only cascade 0.12–0.20 at 6–8 eV; splits $\alpha_{RK} = 1, \to \infty$ 0.11–0.12, 0.45–0.50; step model 0.79–0.83.	Sect. 18.3

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Klein relation (Limit Check 18.6), Eqs. (18.25) to (18.27)	[125, Eq. (3)]; [126, Eq. (1), Sect. II]	Published form: $\varepsilon_{\mathrm{eh}} = \frac{14}{5} E_{\mathrm{g}} + r_K \omega_K$, with $r_K \omega_K$ fitted (about 0.5–0.6 eV). Chain: energy identity Eq. (18.5) with two terminal carriers per pair (one-sided pair seed) → Shockley balance Eq. (18.25), exact in the energy-only reduction; (K1) threshold $\theta_{\mathrm{ii}} = \frac{3}{2}$, Eq. (18.3); (K2) terminal population uniform in $\boldsymbol{k}$ on $[0, \theta_{\mathrm{ii}} E_{\mathrm{g}}]$ with $\rho \propto x^{1/2}$, $\bar{E}_{\mathrm{res}} = \frac{3}{5} \theta_{\mathrm{ii}} E_{\mathrm{g}}$ Eq. (18.26). $\mathfrak{T}$: detector (cascade, incoherent golden-rule events).	Within 10 % of the cascade ($\theta_{\mathrm{ii}} = \frac{3}{2}$, $w_{\mathrm{AB}} = 0.0543$) for $0.65 \le \lambda_{\mathrm{AB}} \le 84$; exact at $\lambda_{\mathrm{AB}} = 3.5$ (the $\theta_{\mathrm{ii}} = \frac{3}{2}$ cascade recalibrated to silicon, $\lambda_{\mathrm{AB}} = 3.61$: 0.1 %). Plot (Fig. 18.10): $x = \lambda_{\mathrm{AB}}$, 10^{-3}–100 (log); $y = \varepsilon_{\mathrm{eh}}/E_{\mathrm{g}}$ of the cascade and of $\frac{14}{5} + r_K \omega_K / E_{\mathrm{g}}$ with the cascade's phonon term; $2\bar{E}_{\mathrm{res}}/E_{\mathrm{g}}$; shaded the 10 % range.	**+36 % at $\boldsymbol{\lambda_{\mathrm{AB}} \to 0}$ ($\boldsymbol{\theta_{\mathrm{ii}} = \frac{3}{2}}$), +39 % at $\boldsymbol{\theta_{\mathrm{ii}} = 1}$.** (K2) overstates the phonon-free value of $\varepsilon_{\mathrm{eh}}/E_{\mathrm{g}}$ (exact 2.056 and 1.585) because the jamming population of residuals sits near the band edge ($\bar{E}_{\mathrm{res}} = 0.528 E_{\mathrm{g}}$, $0.293 E_{\mathrm{g}}$); with strong phonon loss the residual rises above $0.9 E_{\mathrm{g}}$ and the relation undershoots (−10 % at $\lambda_{\mathrm{AB}} = 84$).	Sect. 18.4
Monte Carlo planes of the random-k cascade (Limit Check 18.5), Eq. (18.24)	[122, Eqs. (8), (12)–(13)]	Published form: Eq. (18.24), least-squares planes through event sampling at deposits of about 100 eV. Chain: energy-only reduction (Sect. 18.1.4) → random-k kernels Eq. (18.8) with $\rho \propto x^{1/2}$, $\theta_{\mathrm{ii}} = 1$, one phonon of 63 meV, hole band cut at 12 eV → moment recursions of Sect. 18.2.1, slopes for seed energies 20–40E_{g}. The source adds a plasmon channel and hole re-splitting. $\mathfrak{T}$: detector.	Both ratios within 2 % (the source's Monte Carlo scatter) for $3.65 \le A_{\mathrm{AB}} \le 7.27$ at $E_{\mathrm{g}} = 1.124$ eV; at five points ($A_{\mathrm{AB}} = 3$, 5.2, 7 at $E_{\mathrm{g}} = 1.124$ eV; $E_{\mathrm{g}} = 1.0$, 1.2 eV at 5.2) ≤1.2 % (ε), ≤2.4 % (F), largest at $A_{\mathrm{AB}} = 3$; slopes $\partial\varepsilon_{\mathrm{eh}}/\partial A_{\mathrm{AB}} = 0.090$, $\partial F_{\mathrm{eh}}/\partial A_{\mathrm{AB}} = 0.0016$, $\partial\varepsilon_{\mathrm{eh}}/\partial E_{\mathrm{g}} = 1.70$, $\partial F_{\mathrm{eh}}/\partial E_{\mathrm{g}} = -0.030$ against 0.084, 0.0015, 1.7, −0.028 (within 8 %); $F_{\mathrm{eh}} = 0.1144$ at $A_{\mathrm{AB}} = 5.2$, $E_{\mathrm{g}} = 1.124$ eV against the source's asymptotic 0.115. Plot (Fig. 18.9): $x = A_{\mathrm{AB}}$, 1–12; $y =$ cascade over plane for $\varepsilon_{\mathrm{eh}}$ and F_{eh}; points at $E_{\mathrm{g}} = 1.0$, 1.2 eV; shaded the 2 % range.	**−1.3 to −2.4 % in $\boldsymbol{F}$ at the five points; −6 to −7 % at $\boldsymbol{A_{\mathrm{AB}} = 1}$, −4 % at 12.** Curvature of the cascade in A_{AB} that a plane cannot follow; the plasmon channel and hole re-splitting of the source, not separable at this precision. Agreement of two implementations of one model, not evidence for the model.	Sect. 18.4

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Linear law of the pair energy (Limit Check 18.7), Eqs. (18.28) and (18.29)	[126, Sect. II] (empirical $1.54E_g + 1.89\,\mathrm{eV}$)	Published form: $\varepsilon_{eh} = 1.54E_g + 1.89\,\mathrm{eV}$ (fit to measurements). Chain: cascade as in Limit Check 18.5 without hole cutoff, one constant $A_{AB} = 5.703\,\mathrm{eV}^3$ fitted to silicon Eq. (18.23); 60 solutions on $0.003 \le \lambda_{AB} \le 100$, $0.01 \le w_{AB} \le 0.1$ (slopes for seed energies 12.5–25E_g, 30–60E_g for $\lambda_{AB} > 5$) fitted by Eq. (18.28) (observed, not derived) → Eq. (18.29) with $k_2 = 1.2$–$2.5\,\mathrm{eV}$. $\mathfrak{T}$: detector.	Fit within 0.61 % (rms 0.21 %) inside the fit region; the model within −20 to +7 % of measurement (model over measured, minus one) in the validated range $0.007 \le \lambda_{AB} \le 14.4$, $0.016 \le w_{AB} \le 0.054$ (Table 18.2). Plot (Fig. 18.11): $x = E_g$, 0.12–20 eV (log); $y = \varepsilon_{eh}$; the law with $k_2(\mathrm{Si}) = 1.88\,\mathrm{eV}$ and the in-region k_2 band, the empirical line, the cascade for 24 crystals (inside/outside the fit region) and the measurements; shaded E_g of the measured set.	**residuals −19.8 to +7.0 % (rms 10.5 %) against measurement**. The universal constant A_{AB} (light-atom and polar crystals under-predicted); outside the validated range (halides, fluorides: $w_{AB} < 0.006$; lead salts, InSb: $\lambda_{AB} > 200$) no measured anchor; the energy-only reduction and the plasmon channel unquantified.	Sect. 18.4
Binary electron–hole threshold of a slow recoil (no box), Eqs. (19.6) and (19.7)	—	Published form: none in this notation; the kinematic bound is that of slow-ion stopping in a free electron gas. Chain: inner limit $\omega \le k v_N$ of the stopping integral Eq. (4.54) → support of the free-gas loss function above $\omega_-(k) = k^2/2m_e - k v_F$ (Sect. 4.2) with a gap added by hand → largest single transfer $2m_e v_N(v_N + v_F)$, Eq. (19.6) (equivalently head-on reflection in the ion frame) → set equal to E_g and drop v_N^2 against $v_N v_F$ → Eq. (19.7). Valence density $32/a^3$ (four electrons per atom). $\mathfrak{T}$: a support statement, independent of the order in the ion's charge (Born parameter $Z_\kappa\alpha/v_F = 15$ in silicon).	Valence electrons no faster than v_F; binary collision with one electron; $v_N \ll v_F$ (silicon: $v_N = v_F/43$ at threshold). Silicon $E_{R,\mathrm{th}} = 352\,\mathrm{eV}$ ($E_g = 1.17\,\mathrm{eV}$, $a = 5.431\,\text{Å}$), germanium 399 eV ($E_g = 0.744\,\mathrm{eV}$, $a = 5.658\,\text{Å}$). No Limit Check; Fig. 19.3 marks the threshold against the silicon data.	**−4.4 % (Si), −3.1 % (Ge) from the v_N^2 term; momentum tail unsized**. Keeping v_N^2 gives 336 eV and 387 eV. The valence momentum tail beyond k_F makes the threshold soft: in the pseudopotential model of silicon 5.6×10^{-3} of the valence electrons can still be promoted at 100 eV (Eq. (19.8)), an understatement of the true tail. Close collisions (level crossings, Eq. (19.9)) excite below it.	Sect. 19.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Ionization yield of silicon recoils below the binary threshold (no box), Eqs. (19.11) to (19.13)	[127, Table II, Eq. (1)]	Published form: measured Y_q and F_{NR} at 0.10–3.91 keV with 3.8 eV per pair and the phonon energy of the recoil set equal to E_R. Chain (specified, not evaluated): partition trajectories (Sect. 19.1) $\rightarrow$ excitation events from the close-collision kernel Eq. (19.10) at every close approach, with binary, level-crossing and momentum-tail channels kept apart $\rightarrow$ each excitation converted by $P_{\mathrm{casc}}(N_{\mathrm{eh}}\vert\omega_e)$ (Chap. 18), Eq. (19.11) $\rightarrow$ Migdal part added once, Eq. (19.14) $\rightarrow$ reduced to the measurement's convention. $\mathfrak{T}$: off the perturbative axes (classical nuclei, electrons propagated in time).	Every step of Appendix E.23; test points 0.10, 0.22, 0.46 keV (silicon) and 254 eV (germanium, 77 K). The chain reduces to Lindhard's yield Eq. (16.42) only with $\bar{n}(\omega_e) = \omega_e/\varepsilon_{\mathrm{eh}}$ and continuous friction, which overstate the silicon data by 1.5 at 100 eV.	**unsized**. Exchange–correlation spread of the kernel (adiabatic functionals place charge-transfer crossings too early); stored energy in the measurement's convention, a few percent (Sect. 19.4); Migdal part $2.3\text{–}4.4 \times 10^{-4}$ in Y_q, at most 0.4 % at 100 eV.	Sect. 19.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Lindhard quenching (Limit Check 19.1), Eqs. (16.42) and (19.19)	[128, Eqs. (3)–(7)]; [129, Eqs. (5)–(7)]; [130, Eq. (3)]; [43, 131]	Published form: $y_L = k_L g_L/(1+k_L g_L)$, $g_L = 3\epsilon_L^{0.15} + 0.7\epsilon_L^{0.6} + \epsilon_L$, $k_L = 0.133 Z^{2/3} A^{-1/2}$, $\epsilon_L = 11.5 Z^{-7/3} E_R/\mathrm{keV}$. Chain: partition Eq. (19.4) with $E_{\mathrm{st}} = 0 \to$ binary collisions with uncorrelated targets at rest and memoryless ion transport $\to$ first-collision balance Eq. (19.18) for the damage energy $\to$ (A) no recoils from electrons, (B) $U = 0$, (C) small electronic transfers (first order), (D) nuclear and electronic collisions separate, (E) derivative at $E \to$ screened Thomas–Fermi cross-section Eq. (19.16) and friction Eq. (19.17) $\to$ Eq. (19.19) $\to$ numerical solution fitted by g_L. Small-ϵ_L law Eq. (19.20). Conversion $Y_q = y_L$ assumes $N_{\mathrm{eh}} = y_L E_R/\varepsilon_{\mathrm{eh}}$, full collection, no trapping. $\mathfrak{T}$: detector (classical binary transport; electronic loss first order in the electronic transfer, coefficient fitted).	$\epsilon_L \gtrsim 0.1$ ($E_R \gtrsim 4\,\mathrm{keV}$ in silicon), where the formula describes the silicon data. Fit against our solution of Eq. (19.19) (Thomas–Fermi impulse cross-section, $x_L^{1/3}$ below $x_L = 2\times10^{-3}$): within 3 % below 100 eV, 4 % to 5 % low from 1 to 20 keV, within 5 % for $\epsilon_L \geq 0.1$. Plot (Fig. 19.4): $x = E_R$, 0.01–100 keV (log; top axis ϵ_L); y = formula over solution, band for the two continuations of the cross-section; data over formula; constant-binding model over formula; shaded $\epsilon_L \geq 0.1$; line at $E_{R,\mathrm{th}}$.	**−8 to −38 % (data over formula, 0.1–3.9 keV); fit within 3–5 % of its own equation**. Below a few keV: stored energy (Sect. 19.4), the binary threshold and near-gap conversion (Sect. 19.2), trapping (Sect. 19.5); the solution is sensitive to the continuation of the cross-section below $x_L = 2\times10^{-3}$: +13 % at 1 keV for the local slope 0.363, −11 % and +40 % for exponents 0.30 and 0.40; the $\epsilon_L^{1/6}$ law follows from the $x_L^{1/3}$ continuation (the Thomas–Fermi tail $x_L^{1/2}$ keeps y_L finite). Constant binding $U = 0.15\,\mathrm{keV}$ cuts the yield off at 298 eV, against data at 100 eV. Modified models book the binding of secondaries as ionization, and their $\frac{1}{2} k_L \epsilon_L^{3/2} \epsilon_{\mathrm{d}}''$ term contradicts (D).	Sect. 19.3

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Bethe stopping from the loss function (Limit Check 19.2), Eqs. (19.21) and (19.22)	[132, Eq. (34.5), Sect. 34.2.1]	Published form: $-\mathrm{d}E/\mathrm{d}x = Kz^2(Z/A)\beta^{-2}[\frac{1}{2}\ln(2m_e\beta^2\gamma^2W_{\max}/I^2) - \beta^2 - \delta/2]$. Chain: dielectric stopping Eq. (4.54) (first order in Z_1, heavy projectile, $T = 0$) → split at q_0, with $q_0 v$ above the loss spectrum and q_0 below the inverse atomic size → optical limit $L(0,\omega)$ below q_0, f-sum Eq. (4.46) saturated from q_0 to the Bethe ridge $2m_e v$ above → q_0 cancels → Eq. (19.21) with $\ln \bar{I}$ Eq. (19.22); the source's bracket reduces to $\ln(2m_e v^2/I) + O(v^4)$ for $W_{\max} = 2m_e\beta^2\gamma^2$, $\delta = 0$. $\mathfrak{T}$: first Born, $n = 1$ in Z_1.	$2m_e v^2 \gg \bar{I}$ and $Z_1\alpha/v \ll 1$. Electron gas of the silicon valence density (Mermin, relaxation $0.1\omega_p$, $\bar{I} = 17.9\,\mathrm{eV}$): within 10 % for $v \geq 1.84 v_\mathrm{F}$, 5 % for $v \geq 2.27 v_\mathrm{F}$, 1 % above $4.3 v_\mathrm{F}$. Silicon $\bar{I}$ from the measured loss function 161 eV, adopted 173.0 eV. Plot (Fig. 19.6): $x = v/v_\mathrm{F}$, 0.02–30 (log; top axis $Z_1\alpha/v$ for $Z_1 = 14$); $y = -\mathrm{d}E/\mathrm{d}x$ for $Z_1 = 1$; dielectric, Bethe, friction; shaded the 10 % region; band of silicon recoils.	$\leq$10 % **for $v \geq 1.84 v_\mathrm{F}$**. Shell corrections (the f-sum not exhausted at q_0), Barkas (Z_1^3) and Bloch terms, density effect (relativistic). Silicon recoils ($v = 0.013$–$0.13\, v_\mathrm{F}$, $Z_1\alpha/v = 117$–1170) lie in the friction regime, where $\bar{I}$ plays no role.	Sect. 19.3
Displacement count and defect energy loss (Limit Check 19.3), Eqs. (19.24) to (19.26)	[133, Eq. (1)]	Published form: $N_\mathrm{FP} = 0$ for $T_\mathrm{d} < E_\mathrm{d}$, 1 for $E_\mathrm{d} \leq T_\mathrm{d} < 2E_\mathrm{d}/0.8$, $0.8\,T_\mathrm{d}/2E_\mathrm{d}$ above (Norgett–Robinson–Torrens, as quoted). Chain: partition Eq. (19.4) → mean stored energy over end basins Eq. (19.23) → binary collisions with uncorrelated targets at rest (Sect. 19.3) and the Kinchin–Pease rules (displaced above E_d; replacement when the projectile keeps less than E_d), no binding energy, electronic losses removed so that the argument is T_d → first-collision balance Eq. (19.24) → hard spheres: Eq. (19.25) exactly; Thomas–Fermi cross-section of Eq. (19.16) ($x_L^{1/3}$ below $x_L = 2\times10^{-3}$, i.e. $\mathrm{d}\sigma \propto T^{-4/3}\mathrm{d}T$ near threshold): $\xi_\mathrm{d} = 0.70$ → times $p_\mathrm{surv}E_\mathrm{FP}$: Eq. (19.26). $\mathfrak{T}$: adiabatic ($n_B = 0$), classical binary transport.	Linear range $T_\mathrm{d} \geq 2.5E_\mathrm{d}$; binary, uncorrelated collisions, random medium. Screened count: $2E_\mathrm{d}N_\mathrm{FP}/T_\mathrm{d} = 0.72$ at $10E_\mathrm{d}$, 0.70 ($\pm3\times10^{-3}$ for $E_\mathrm{d} = 12.5$–$36\,\mathrm{eV}$) above $100E_\mathrm{d}$; a direct Monte Carlo of the same rules (power law) agrees within 1.6 standard errors. Plot (Fig. 19.7): $x = T_\mathrm{d}/E_\mathrm{d}$, 1–1000 (log); $y = 2E_\mathrm{d}N_\mathrm{FP}/T_\mathrm{d}$; screened ($E_\mathrm{d} = 36$, 12.5 eV), power law, hard spheres, formula; shaded $T_\mathrm{d} \geq 2.5E_\mathrm{d}$.	**+11 to +14 % (formula over screened count, $T_\mathrm{d} \geq 10E_\mathrm{d}$); −20 % against hard spheres**. The factor depends on cross-section and counting rule. A binding energy per displacement and recombination in dense cascades both lower the count (metals: a third to a quarter of the formula). Below $2.5E_\mathrm{d}$ the direction dependence of E_d and the zero-point width of the step (0.72 eV at 12.5 eV, Sect. 19.1) set p_FP. Read through Eq. (19.26), the stored-energy bounds of Limit Check 19.5 give $p_\mathrm{surv}E_\mathrm{FP}/E_\mathrm{d} \leq 0.19$ (Si, 10 keV), 0.27 (Si, 100 keV), ≥ 0.23 (Ge, not conservative) with the factor 0.8.	Sect. 19.4

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Unit heat quenching (Limit Check 19.4), $Y_{\mathrm{ph}}(0) = 1$; Eqs. (19.29) and (19.30)	[134, p. 4]	Published form: the heat quenching factor, the ratio of the heat signals of nuclear and electron recoils of equal energy, set to one. Chain: balance law Eq. (19.27) (complete reservoirs, carrier energies from μ_F, configurational E_{st}, ΔE_{zp} in the phonons) $\rightarrow$ signal = efficiency $\times$ (deposit $-$ missing energy at t_{win}), absolute efficiency cancels $\rightarrow$ Eq. (19.29) $\rightarrow$ gap energy of the pairs written with the returned fraction $f_{\mathrm{ret}}(t_{\mathrm{win}})$, Eq. (19.30) $\rightarrow \delta_{\mathrm{coll}} = 0$, $f_{\mathrm{ret}} = 1$ for both recoils, $E_{\mathrm{st}} = E_{\mathrm{esc}} = \Delta E_{\mathrm{zp}} = 0$. $\mathfrak{T}$: detector (energy bookkeeping, no expansion).	All pairs recombined or collected with their gap energy returned as phonons within the window, no stored, trapped or escaped energy, equal collection. Silicon, 100 eV ($Y_q = 0.102$): within 1 % of one for a held fraction $1 - f_{\mathrm{ret}} \leq 0.035$ (0.036 at 0.89 keV, 0.041 at 3.91 keV). Plot (Fig. 19.9): $x = f_{\mathrm{ret}}$ from 0 to 1; $y = Y_{\mathrm{ph}}(0)$ for the measured Y_q at 0.10, 0.89, 3.91 keV, and at 0.10 keV with $E_{\mathrm{st}} = 0.05\,\omega$; shaded $f_{\mathrm{ret}} \geq 0.965$; marker at the asymptotic $f_{\mathrm{ret}} = 0.941$ of the illustrative kinetics of Fig. 18.12.	**+41 % at $f_{\mathrm{ret}} = 0$ (100 eV); +1.7 % for 5.9 % of the gap energy trapped or radiated.** Gap energy held at the end of the window, $(1 - Y_q)(E_{\mathrm{g}}/\varepsilon_{\mathrm{eh}})(1 - f_{\mathrm{ret}})$ at first order, invalid beyond small $1 - f_{\mathrm{ret}}$ (first order 1.280 against exact 1.407 at $f_{\mathrm{ret}} = 0$); stored energy $-E_{\mathrm{st}}(t_{\mathrm{win}})/\omega$, of opposite sign; collection offset $-\delta_{\mathrm{coll}}$.	Sect. 19.5

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Stored energy from the phonon-yield ratio (Limit Check 19.5), Eqs. (19.33) and (19.35) to (19.37)	[135, Eq. (4), Table II]; [130, Eqs. (1), (4), Sects. 2, 3.2, 4]	Published form: $f_{\mathrm{DF}} = 1 - Y_{\mathrm{ph}}$ read as stored energy, with $E_R = E_\phi - E_Q e V_{\mathrm{bias}}/\varepsilon_{\mathrm{cal}}$ on a phonon scale from bulk electron recoils of unit yield, $\varepsilon_{\mathrm{cal}} = 3.82\,\mathrm{eV}$; Si $Y_{\mathrm{ph}} = 0.952^{+0.009}_{-0.007}$ (95 % C.L.), Ge $f_{\mathrm{DF}} = (6.08 \pm 0.18)\,\%$. Chain: balance law Eq. (19.27) with bias work Eq. (19.28) → primary and Luke phonons collected with ϵ_{prim}, ϵ_{Luke}; Luke phonons follow the true $\varepsilon_{\mathrm{eh}}$ (g_{Luke}), scale and subtraction the adopted $\varepsilon_{\mathrm{cal}}$ (g_{cal}), signals Eq. (19.32) → exact identity Eq. (19.33), calibration identity Eq. (19.34) → first order Eq. (19.35) with C_{cal} → exact bound identity Eq. (19.36) (Appendix D.45) → (a) $\delta_{\mathrm{coll}} \geq 0$, (b) $\mathfrak{r}_{\mathrm{coll}} \geq 1$, (c) other missing energy of the nuclear recoil at least $(1 + f_{\mathrm{ion}} g_{\mathrm{cal}})/(1 + g_{\mathrm{cal}})$ times that of the electron recoil, (d) $E_{\mathrm{st}}/\omega \leq 1 - f_{\mathrm{ion}}$, (e) $\varepsilon_{\mathrm{eh}} \leq \varepsilon_{\mathrm{cal}} \Rightarrow E_{\mathrm{st}}/\omega \leq f_{\mathrm{DF}} \rightarrow 95\,\%$ C.L.: lower edge of the Si interval; Ge $\hat{f} + 1.645\sigma$ (Appendix C.14). $\mathfrak{T}$: detector (bookkeeping with multiplicative collection).	Electron recoils fully collected; one efficiency per phonon population; calibration energies only (Si 10–100 keV, 4 V/cm; Ge ^{206}Pb at 103 keV). Bounds 0.055 (Si; 0.049 under (a)–(d) with $\varepsilon_{\mathrm{eh}} = 3.75\,\mathrm{eV}$, attained at $\mathfrak{r}_{\mathrm{coll}} = 1$), 0.064 (Ge); the sign of $\mathfrak{r}_{\mathrm{coll}} - 1$ is not fixed by the balance law; the mean-field transport estimate gives 0.81–1.02 with energy conserved (0.82–1.01 at the mean Luke energy of silicon electrons at 1 V/cm), up to 1.5 only for a selective sink not identified in a detector, so collection alone (1.128) is not reached by it but not excluded until computed kernels test real detectors (Sect. 20.5.3, Appendix E.25); the two-bias measurement decides. Plot (Fig. 19.10): $x = \mathfrak{r}_{\mathrm{coll}}$ from 0.8 to 1.3; (a) Y_{ph} at 4 and 40 V for $E_{\mathrm{st}} = 0$ and $0.042\,\omega$ ($f_{\mathrm{ion}} = y_C(20\,\mathrm{keV}) = 0.324$, band 0.269–0.497), measured band, dot at $\mathfrak{r}_{\mathrm{coll}} = 1.128$; (b) 95 % bound on E_{st}/ω from Eq. (19.37), the same with $\varepsilon_{\mathrm{cal}} = \varepsilon_{\mathrm{eh}}$, Ge point; shaded $\mathfrak{r}_{\mathrm{coll}} \geq 1$.	**-0.0064 in Y_{ph} from $\varepsilon_{\mathrm{cal}} \neq \varepsilon_{\mathrm{eh}}$ (Si, 4 V); bound 0.081 at $\mathfrak{r}_{\mathrm{coll}} = 0.9$; first order against exact $\leq 3.0 \times 10^{-3}$ for deficits in $[0, 0.05]$.** Second-order terms $\delta_{\mathrm{coll}} d_{\mathrm{NR}}$ and those in d_{ER}. On $\mathfrak{r}_{\mathrm{coll}} < 1$ the inferred stored energy grows linearly, Eq. (19.37). The datum is degenerate: collection alone ($\mathfrak{r}_{\mathrm{coll}} = 1.128$), stored energy alone (0.042), or a gap offset with $\rho_g - 1 = 0.21$ each reproduce it. A negative zero-point change widens the bounds by about 10^{-3} (0.056, 0.065). Polarity-dependent trapping in Ge (detectors differ by 2.06 standard deviations).	Sects. 19.4 and 19.5
Gap-energy release law (no box), Eqs. (19.30) and (19.31)	—	Published form: none. Chain: Eq. (19.29) with the pair energy at t_{win} written through the returned fraction $f_{\mathrm{ret}}(t)$; linear pair kinetics of Eq. (18.37) give $1 - f_{\mathrm{ret}}$ as a sum of exponentials, equal for nuclear and electron recoils; in dense tracks two-body loss, $\mathrm{d}n/\mathrm{d}t \propto -n^2$ (exciton annihilation) or $-n^3$ (Auger), gives Eq. (19.31). $\mathfrak{T}$: rate equations, first order in each process.	Linear law: dilute pairs, every transition first order in the pair's own state, zero field. Power laws: no diffusion during the window ($4 D_X t_{\mathrm{win}}$ small against the squared track radius), no linear decay competing.	**unsized**. Diffusion of the track (saturating two-body loss); electron–hole droplets; the rates at millikelvin, none measured (Sect. 18.5.4); excitons hold $E_{\mathrm{g}} - E_x^b$, an offset of at most 1.7 % of the held energy in silicon.	Sect. 19.5

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Two-bias inversion (no box), Eqs. (19.39) and (D.246)	—	Published form: none. Chain: Eq. (19.33) multiplied by its denominator, homogeneous and linear in $(1-\delta_{\mathrm{coll}})(1-d_{\mathrm{NR}})$, $1-d_{\mathrm{ER}}$, $\mathfrak{r}_{\mathrm{coll}}$, with $g_{\mathrm{Luke}}/g_{\mathrm{cal}}$ independent of bias, Eq. (D.245) $\rightarrow$ divided by $1-d_{\mathrm{ER}}$, Eq. (19.39) $\rightarrow$ two biases, Cramer's rule, Eq. (D.246). $\mathfrak{T}$: detector (exact within the signal model).	$\mathfrak{r}_{\mathrm{coll}}$, δ_{coll} and the missing energies independent of bias over the scan; f_{ion} measured at both biases; both biases collect all pairs within the window (a zero-bias point does not qualify). Same $\varepsilon_{\mathrm{cal}}$ at both biases and $\varepsilon_{\mathrm{eh}}$ known. Silicon at 4 and 40 V: collection alone predicts 0.919 at 40 V, stored energy alone 0.947; an error of 0.005 on the 40 V ratio moves $\mathfrak{r}'$ by 0.020–0.026.	**unsized**. Field dependence of $\mathfrak{r}_{\mathrm{coll}}$ through the Luke birth spectrum, residual of order $(\mathrm{d}\mathfrak{r}_{\mathrm{coll}}/\mathrm{d}V_{\mathrm{bias}})\Delta V_{\mathrm{bias}}$ (Appendix E.25); field dependence of trapping. The heat quenching it returns still mixes δ_{coll} and E_{st}.	Sect. 19.5
Defecton diffusion in a phonon gas (Limit Check 19.6), Eqs. (19.47) and (19.48)	[136, Eqs. (12), (49)–(61), (64)]	Published form: $D_T=(a\bar{c}/3\alpha_{\mathrm{P}}\sigma_{\mathrm{P}})[\varepsilon(p)/\theta_p]^2(\theta_p/T)^9$, $\theta_p=\bar{c}/2a$, $\alpha_{\mathrm{P}}=1.27$. Chain: narrow tight-binding band Eq. (19.46) ($A_{\mathrm{dfn}}a\ll\bar{c}$) $\rightarrow$ one-phonon absorption and emission kinematically forbidden $\rightarrow$ two-phonon (Rayleigh) scattering, amplitude $\propto\omega$ for both the quadratic coupling and the linear coupling at second order (the second difference of the band survives) $\rightarrow$ collision rate $\propto T^7$, momentum change per collision $\sim T/\bar{c}\ll 1/a$, $\langle q^2\rangle\propto T^9$ Eq. (19.47) $\rightarrow$ crystal momentum diffusing at a constant rate $D_1=\langle q^2\rangle/6$ with the detailed-balance drift, separable band, Green–Kubo integral solved in closed form $\rightarrow$ Eq. (19.48), $D_9=\Delta_{\mathrm{dfn}}^2/12\langle q^2\rangle$; equals the published form for $\varepsilon(p)=\Delta_{\mathrm{dfn}}/4$. $\mathfrak{T}$: defect on the Born–Oppenheimer surface, phonons at second order in the coupling ($n_B=2$ in the rate), dilute defectons.	$\Delta_{\mathrm{dfn}}\ll T\ll\theta_p$ as published; in the band model within 10 % for $T\geq 0.28\,\Delta_{\mathrm{dfn}}$, 1 % for $T\geq 0.93\,\Delta_{\mathrm{dfn}}$; no strain gradient with $\lvert\nabla\phi\rvert a>\Delta_{\mathrm{dfn}}$ (localization radius $R_0=a(V_0/\Delta_{\mathrm{dfn}})^{1/3}$); defecton–defecton scattering negligible. Plot (Fig. 19.12): $x=T/\Delta_{\mathrm{dfn}}$, 10^{-2}–$10^{1.5}$ (log); $y=D/D_9$; band model, effective-mass law, leading correction; shaded $T\geq 0.28\,\Delta_{\mathrm{dfn}}$.	$\leq$10 % **for $T\geq 0.28\,\Delta_{\mathrm{dfn}}$**. The bracket of Eq. (19.48) expands as $1-\frac{5}{576}(\Delta_{\mathrm{dfn}}/T)^2$; at $T\ll\Delta_{\mathrm{dfn}}/6$ the law becomes $6T^2/\langle q^2\rangle\propto T^{-7}$ (twice the coefficient of the published estimate, Eq. (64)); local exponent -7.9 at $T=0.1\,\Delta_{\mathrm{dfn}}$. Above θ_p the Rayleigh law fails. The band model assumes a $\boldsymbol{k}$-independent momentum diffusion.	Sect. 19.6

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Crowdion profile and kink mass (Limit Check 19.7), Eq. (19.49); Eqs. (C.180), (C.181) and (C.184)	[137, Eqs. (2), (3)]; [138]	Published form: $u(z) = (2a/\pi)\arctan \mathrm{e}^{-\mu(z-z_0)}$, $E_0 = (2a/\pi)(2V_0\beta)^{1/2}$ with $\mu = 1/d_{\mathrm{kink}}$, $\beta = K_{\mathrm{ch}}$, $V_0 = V_{\mathrm{s}}$. Chain: Frenkel–Kontorova chain Eq. (C.177) → continuum limit $u_{n+1} - 2u_n + u_{n-1} \to a_{\mathrm{ch}}^2 u'' \to$ sine-Gordon equation Eq. (C.179) → static kink Eq. (C.180), energy Eq. (C.181) → Lorentz invariance with $c_{\mathrm{ch}} \to M_{\mathrm{cr}} = E_{\mathrm{kink}}/c_{\mathrm{ch}}^2$ Eq. (19.49); continuum profile summed on the atoms with Poisson summation → E_{PN} Eq. (C.184) (the published migration energy, Eq. (6) of the same source, is a factor 2 lower). Compared with the discrete chain relaxed numerically at the minimum and the saddle. $\mathfrak{T}$: classical static chain on a model substrate potential; Born–Oppenheimer.	$d_{\mathrm{kink}} \gg a_{\mathrm{ch}}$; sinusoidal substrate potential. All three quantities within 10 % of the discrete chain for $d_{\mathrm{kink}} \geq 1.74\,a_{\mathrm{ch}}$ (set by E_{PN}); energy within 10 % above $0.57\,a_{\mathrm{ch}}$, mass above $0.75\,a_{\mathrm{ch}}$. Tungsten ($V_{\mathrm{s}} \approx 1$ eV, $K_{\mathrm{ch}}a_{\mathrm{ch}}^2 = 50$–$100$ eV): $d_{\mathrm{kink}} = 1.59$–$2.25\,a_{\mathrm{ch}}$, $M_{\mathrm{cr}} = 0.127$–$0.090\,M_\kappa$. Plot (Fig. 19.13): $x = d_{\mathrm{kink}}/a_{\mathrm{ch}}$, 0.35–3; $y =$ discrete over continuum for E_{kink}, M_{cr}, E_{PN}; leading correction to the energy; shaded $d_{\mathrm{kink}} \geq 1.74\,a_{\mathrm{ch}}$.	**-1.4 % in E_{kink} at $d_{\mathrm{kink}} = a_{\mathrm{ch}}$; E_{PN} -22 % there, -7.5 % at $2a_{\mathrm{ch}}$.** Energy: the k^4 term of the discrete spring energy, $-E_{\mathrm{kink}}a_{\mathrm{ch}}^2/72d_{\mathrm{kink}}^2$, followed within 2×10^{-4} above $1.5\,a_{\mathrm{ch}}$; mass +3 % at one spacing. A non-sinusoidal substrate (double-sine for tungsten) raises the barrier to the meV of atomistic calculations.	Sect. 19.6
Hecht collection (Limit Check 20.1), Eq. (20.4); Eqs. (20.5) and (20.6)	[139], [140, Eqs. (1)–(4)]	Published form: $Q/Q_0 = (\mu\tau V/d^2)(1 - \mathrm{e}^{-d^2/\mu\tau V})$. Chain: Shockley–Ramo theorem Eq. (20.1) with the planar weighting potential $z/d_{\mathrm{det}} \to$ induced charge proportional to the drifted path, Eq. (20.2) → independent capture events: survival $\mathrm{e}^{-s/\lambda_{\mathrm{d}}}$ along the path → charge created at one face, one carrier type crosses → mean cut-off path, Eq. (20.4) → $\lambda_{\mathrm{d}} = v_{\mathrm{dr}}\tau_{\mathrm{tr}}$ with $v_{\mathrm{dr}} = \mu_{\mathrm{mob}}\mathcal{E}$, $\mathcal{E} = V_{\mathrm{bias}}/d_{\mathrm{det}} \to$ published form. Bulk events: both carriers, Eq. (20.5), depth average Eq. (20.6). $\mathfrak{T}$: incoherent classical drift after the prompt cascade; independent capture events.	Exponential survival (trap occupation fixed during the drift); uniform field; for the voltage form, $v_{\mathrm{dr}} \propto \mathcal{E}$ and a field-independent τ_{tr}. With the saturating interpolation $v_{\mathrm{dr}} = \mu_{\mathrm{mob}}\mathcal{E}/(1 + \mu_{\mathrm{mob}}\mathcal{E}/v_{\mathrm{sat}})$, $v_{\mathrm{sat}}\tau_{\mathrm{tr}} = 2d_{\mathrm{det}}$: within 1 % for $\mu_{\mathrm{mob}}\mathcal{E} \leq 0.0101\,v_{\mathrm{sat}}$, 10 % for $\leq 0.117\,v_{\mathrm{sat}}$. Plot (Fig. 20.3): $x = \mu_{\mathrm{mob}}\tau_{\mathrm{tr}}V_{\mathrm{bias}}/d_{\mathrm{det}}^2$, 10^{-2}–10^2 (log); $y =$ collected fraction over the published form; saturating drift and leading correction; shaded $\mu_{\mathrm{mob}}\mathcal{E} \leq 0.1\,v_{\mathrm{sat}}$.	**$-\mu_{\mathrm{mob}}\mathcal{E}/v_{\mathrm{sat}}$ (relative) at low bias; -8.9 % at $\mu_{\mathrm{mob}}\mathcal{E} = 0.1\,v_{\mathrm{sat}}$.** From $\lambda_{\mathrm{d}} = \mu_{\mathrm{mob}}\tau_{\mathrm{tr}}\mathcal{E}(1 - \mu_{\mathrm{mob}}\mathcal{E}/v_{\mathrm{sat}})$; at high bias the collection saturates at $\kappa(1 - \mathrm{e}^{-1/\kappa})$, $\kappa = v_{\mathrm{sat}}\tau_{\mathrm{tr}}/d_{\mathrm{det}}$ (0.787 at $\kappa = 2$). With a capture cross-section per unit path independent of speed, λ_{d} does not depend on the bias.	Sect. 20.1

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Bias work and the Luke phonon energy (Limit Check 20.2), Eq. (20.3); Eq. (19.28)	[130, Eq. (1)]	Published form: $E_\phi = E_R + n_Q e V_b$, Luke part $n_Q e V_b$. Chain: Green's reciprocity → Shockley–Ramo theorem Eq. (20.1) → work of the sources $\sum_k V_k \Delta Q_k$ → Eq. (20.3) with the vacuum potential φ_{bias} of the electrodes (exact with space charge) → two planar electrodes: $eV_{\text{bias}}\mathfrak{q}$ per pair, Eq. (19.28) → trapping, depth average Eq. (20.6) → full collection, $\mathfrak{q} = 1$ → published form. Luke heat $= W_{\text{bias}} - \Delta E_{\text{es}}$ (Appendix D.45). $\mathfrak{T}$: classical electrostatics in a linear dielectric; quasistatic drift.	Two planar electrodes (with more, the read charge follows φ_s and the work φ_{bias}); ΔE_{es} negligible (1.3×10^{-5} eV for a pair 10 µm apart in silicon). Within 1 % for $d_{\text{det}} \le 0.030\,\lambda_{\text{d}}$ (equal drift lengths), $d_{\text{det}} \le 0.015\,\lambda_{\text{d}}^e$ ($\lambda_{\text{d}}^h = \lambda_{\text{d}}^e/3$). Plot (Fig. 20.4): $x = d_{\text{det}}/\lambda_{\text{d}}^e$, 10^{-3}–10 (log); $y = W_{\text{bias}}/N_{\text{eh}} e V_{\text{bias}}$; two drift-length ratios with leading corrections; shaded $d_{\text{det}} \le 0.030\,\lambda_{\text{d}}$.	$\boldsymbol{-d_{\text{det}}/6\lambda_{\text{d}}^e - d_{\text{det}}/6\lambda_{\text{d}}^h}$**; −3.3 % at $\boldsymbol{d_{\text{det}} = 0.1\,\lambda_{\text{d}}}$ (equal drift lengths).** Next order $+d_{\text{det}}^2/24\lambda_{\text{d}}^2$ per carrier, so the leading term is accurate to a fraction $d_{\text{det}}/4\lambda_{\text{d}}$ of itself. The planar reconstruction subtracts the Luke heat of partly collected pairs exactly through the read charge.	Sect. 20.1
Fano factor of the measured charge (Limit Check 20.3), Eq. (20.7); Eq. (C.100)	[122, Eqs. (A6)–(A7)]	Published form: $F_{\text{meas}} = F/\eta + (1-\eta)$, from the variance $Fn + \eta(1-\eta)n$ over the mean ηn, with η the collection efficiency (= f_{det}). Chain: induced charge per pair $\mathfrak{q}_i \in [0,1]$, independent and identically distributed given the event position and independent of N_{eh} → compound sum at fixed position → law of total variance over the position → Eq. (20.7) → all-or-nothing, position-independent collection: binomial thinning $fF_{\text{eh}} + 1 - f$, Eq. (C.100) → published form, which leaves the pair variance unscaled by η^2. A Monte Carlo of 4×10^5 events (binomial pair number, mean 26.6, Fano factor 0.114) agrees with Eq. (20.7) to 0.0004. $\mathfrak{T}$: carriers drift independently after the prompt cascade (no recombination or screening in the track).	Published form within 10 % of exact thinning for $f_{\text{det}} \ge 0.923$; exact thinning needs all-or-nothing loss with one probability everywhere. At fixed position $\sigma_{\mathfrak{q}}^2 \le \bar{\mathfrak{q}}(1-\bar{\mathfrak{q}})$, so partial charges give less than thinning. Plot (Fig. 20.5): $x = f_{\text{det}}$, 0.5–1; $y = F_{\text{meas}}$ at $F_{\text{eh}} = 0.114$; published form, thinning, midplane partial charges, uniform depth at $\bar{N}_{\text{eh}} = 26.7$; shaded $f_{\text{det}} \ge 0.923$.	$\boldsymbol{+F_{\text{eh}}(1/f_{\text{det}} - f_{\text{det}})}$ **against thinning, 0.012 at $\boldsymbol{f_{\text{det}} = 0.95}$.** The source's quoted values, 0.130 at 98 % and 0.160 at 95 %, lie close to exact thinning (0.133 and 0.159 at $F = 0.115$; the printed form gives 0.137 and 0.171). Beyond thinning: partial charges lower F_{meas} at fixed position (0.125 against 0.158 at 95 %), and depth-dependent collection adds the position term, $1.22 \times 10^{-4}\bar{N}_{\text{eh}}$ at 95 % for uniform depth (Worked example 20.1).	Sect. 20.1

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Down-conversion and the scaling spectrum (Limit Check 20.5), Eqs. (20.14), (20.18) and (20.19); Eq. (20.17)	[141, Eqs. (6), (8), Table I]; [142, Sect. 2.2.2]; [143, Tables 3, 4]	Published form: $\Gamma_{\mathrm{A}} = A\nu^5$ (longitudinal), $\Gamma_{\mathrm{I}} = B\nu^4$, with the statement that the details of a high-frequency birth distribution are rapidly lost. Chain: three-phonon linewidth Eq. (5.70) at $T = 0$ → acoustic sum rule removes the zeroth order of the vertex in each wavevector, Eq. (20.13) → phase space $\propto \omega^2$ → Eq. (20.14); transverse decay closed in the isotropic continuum → isotope mixing faster than longitudinal decay ($B_{\mathrm{iso}}/A_{\mathrm{anh}} = 33\,\mathrm{THz}$ in Si) → one mean rate $f_{\mathrm{LA}}\Gamma_{\mathrm{anh}}$ → energy-conserving fragmentation Eq. (20.17) with a self-similar daughter distribution → one scale $E_*(t)$, Eq. (20.18) → scaling form Eq. (20.19). Compared with the solution of Eq. (20.17) for three hard births (line at 62 meV, E^3 spectrum, optical-decay pairs), $P_{\mathrm{d}}(y) = 60y^3(1-y)^2$, exact matrix exponential on a logarithmic grid of step 0.01. Mean energy $0.667\,E_*$; median $0.66\,E_*$, 80 % of the energy within 0.36–0.97 E_*. $\mathfrak{T}$: three-phonon decay at lowest order ($n_A = 2$ in the rate), isotope scattering at second order in the mass defects, incoherent transport, $n_T = 0$.	$t \gg 1/(f_{\mathrm{LA}} A_{\mathrm{anh}} f_{\mathrm{top}}^5)$ (0.19 ns Si, 1.3 ns Ge); isotope mixing faster than decay (fails for sapphire, $B_{\mathrm{iso}}/A_{\mathrm{anh}} = 0.02$–$0.21\,\mathrm{THz}$); no surfaces or absorption; continuum rates (the transverse branches of Si are linear only below 1.4 THz, Limit Check 20.4). Mean energy within 10 % of the scaling law for $t \geq 0.46\,\mathrm{ns}$ for all three births. Memory: births above E_{mem} agree to 10^{-3} once $E_* \leq 0.51$–$0.53\,E_{\mathrm{mem}}$. Plot (Fig. 20.7): $x = t$, 3×10^{-3}–10^4 ns (log); $y = \langle E_{\mathrm{ath}} \rangle / 0.667 E_*$; three births and the undecayed line; shaded $t \geq 0.46\,\mathrm{ns}$.	≤10 % **for $t \geq 0.46$ ns (Si).** Before the first decays the mean is the birth energy, $\langle E_{\mathrm{ath}} \rangle / 0.667 E_* = (f_{\mathrm{LA}} A_{\mathrm{anh}} f_0^5 t)^{1/5}/0.667$ for a line at f_0. The numbers rest on A_{anh} (Si: 7.41–11.5 $\times 10^{-56}\,\mathrm{s}^4$, E_* changes by 0.92; Ge: ×10 spread, factor 0.63) and on f_{LA} (0.07–0.10, E_* within 6 %); the shape constants depend on P_{d}, here a symmetric stand-in for the isotropic-continuum distribution of Appendix E.12.	Sect. 20.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Rayleigh law of isotope scattering (Limit Check 20.4), Eq. (20.16); Eqs. (20.15) and (E.81)	[143, Eq. (4), Table 4]	Published form: $\Gamma_{\mathrm{iso}} = B_{\mathrm{iso}}\nu^4$, $B_{\mathrm{iso}} = 4\pi^3 g_{\mathrm{md}}\Omega_{\mathrm{at}}\langle c^{-3}\rangle$ (Tamura's result; written with Γ_{md} and $1/\langle c^3\rangle$ in the source). Chain: mass defects about the mean mass, mass variance $g_{\mathrm{md}} \rightarrow$ self-energy of the mass disorder at lowest order (Sect. 11.6); cubic site symmetry makes the width independent of the initial mode $\rightarrow$ Eq. (20.15) $\rightarrow$ Debye density of states $3\Omega_{\mathrm{at}}\omega^2\langle c^{-3}\rangle/2\pi^2 \rightarrow$ published form. Compared with Eq. (20.15) for the PBE density of states of Sect. 5.2, integrated on rays from Γ (32 polar, 64 azimuthal points, 800 radial steps, 0.1 meV bins), Debye reference with the PBE sound speeds. Si: $g_{\mathrm{md}} = 2.007 \times 10^{-4}$, $\langle c^{-3}\rangle = 4.90 \times 10^{-12}\,\mathrm{s^3\,m^{-3}}$ (measured tensor), $B_{\mathrm{iso}} = 2.44 \times 10^{-42}\,\mathrm{s^3}$ (measured 2.42–2.56, polarization spread). $\mathfrak{T}$: second order in the mass defects, harmonic lattice with mass-independent force constants.	Wavelengths long against the lattice spacing; independent isotopes; single scattering (pair terms of Sect. 11.6 below 2.2 % to 55 meV in natural Si). Within 10 % for $E_{\mathrm{ath}} \leq 5.6$ meV (1.36 THz, 0.10 ω_{D}). Plot (Fig. 20.6): $x = E_{\mathrm{ath}}$, 0–30 meV; $y = \Gamma_{\mathrm{iso}}$ over the Rayleigh law; leading correction; shaded $E_{\mathrm{ath}} \leq 5.6$ meV.	$\mathbf{+9.6\,(E_{ath}/\omega_D)^2}$**;** $+5$ % **at** 1 THz**,** $+10$ % **at** 5.6 meV. The transverse branches bend over: the density of states exceeds the Debye form, up to a factor 8.1 at the van Hove peak near 17.6 meV. The PBE $\langle c^{-3}\rangle$ is 22 % above the measured one (soft transverse branches); the absolute rate uses the measured tensor.	Sect. 20.2

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Transfer-kernel bound on observable differences (no box), Eq. (20.21)	—	Published form: none; the statement that only the soft fraction of the birth spectra survives transport. Chain: linear transport, observable linear in the delivered energies with weights in $[0, 1]$ $\rightarrow$ response $R_O(E)$ to unit energy born at $E \rightarrow$ no absorption before t_{mem}: responses to births above E_{mem} differ by at most $\Delta_{\mathrm{h}}(t_{\mathrm{mem}}) \rightarrow \int \Delta\hat{\rho}_0 = 0$, subtract the midpoint of the response above $E_{\mathrm{mem}} \rightarrow$ Eq. (20.21); $= \Delta_{\mathrm{soft}} + \Delta_{\mathrm{h}}$ when $\Delta\hat{\rho}_0$ has one sign below E_{mem}. Ratio observables linearized. $\mathfrak{T}$: as the scaling spectrum; transport linear in the deposit (occupations $\sim 4 \times 10^{-9}$).	Linear transport; no sensor absorption before t_{mem} (bulk events farther than $r_{\mathrm{rms}}(t_{\mathrm{mem}})$ from an instrumented face: 68 µm at 100 ns, 0.54 mm at 1 µs in Si); Δ_{h} from the scaling solution, $E_{\mathrm{m}} \geq E_*(t_{\mathrm{mem}})/0.52$.	**bound 0.15 against the largest model difference 0.012 at** 1 µs **(Si, the births of Fig. 20.8); it reduces to $f_{\mathrm{s}} = 0.017$ only if E_{mem} lies below every hard birth**. With $E_{\mathrm{m}} = 21.6$ meV the bound counts the part of the cubic spectrum below E_{mem} as well as the soft fraction; a ratio observable adds its linearization error, quadratic in the difference.	Sect. 20.2
Impedance law as the averaged interface transmission (Limit Check 20.6), Eq. (C.161); Eqs. (20.27) and (20.28)	[144]	Published form: $t_{12}(0) = 4Z_{\mathrm{ac},1}Z_{\mathrm{ac},2}/(Z_{\mathrm{ac},1} + Z_{\mathrm{ac},2})^2$. Chain: welded plane interface between isotropic media (Voigt averages) $\rightarrow$ conserved $Q_\parallel$, Eq. (20.22) $\rightarrow$ SH in closed form Eq. (20.23), L–SV by the four boundary conditions Eq. (20.24) and the flux fractions Eq. (20.25) $\rightarrow$ flux-weighted angle and branch average Eq. (20.27) $\rightarrow$ at equal speeds on both sides no refraction, no critical angle and no conversion, so $\bar{t}_{12} \rightarrow t_{12}(0)$. Unitarity of the scattering matrix at fixed $(\omega, Q_\parallel)$ gives the reciprocity Eq. (20.28); diffuse limit Eq. (20.29). $\mathfrak{T}$: harmonic continuum elasticity, wavelength between the lattice spacing and the film thickness.	Within 10 % in both directions for $c_{\mathrm{LA}}(\mathrm{film})/c_{\mathrm{LA}}(\mathrm{Si}) = 1.146$–$1.242$ in the aluminum-like family (density $2.70\,\mathrm{g\,cm^{-3}}$, $c_{\mathrm{T}}/c_{\mathrm{LA}} = 0.498$ of aluminum at zero temperature), where $\mathfrak{n}_{\mathrm{film}} \approx \mathfrak{n}_{\mathrm{Si}}$. Aluminum on silicon (ratio 0.737): $t_{12}(0) = 0.994$ against $\bar{t}_{\mathrm{Si}\rightarrow\mathrm{Al}} = 0.938$ and $\bar{t}_{\mathrm{Al}\rightarrow\mathrm{Si}} = 0.368$; reciprocity residual 2×10^{-15}. Plot (Fig. 20.10): $x = c_{\mathrm{LA}}(\mathrm{film})/c_{\mathrm{LA}}(\mathrm{Si})$, 0.35–2; y = transmission; impedance law, both averages, leading correction, diffuse mismatch; shaded 1.146–1.242.	**factor $\mathfrak{n}_{\mathrm{Si}}/\mathfrak{n}_{\mathrm{Al}} = 0.392$ on the escape from aluminum; the corrected form within** 6.9 % **of the average over the plotted range**. Leading correction: total reflection of the slower medium's phonons beyond the critical angles, written as reciprocity applied to the impedance law on the slower side, $t_{12}(0)\min(1, \mathfrak{n}_{\mathrm{out}}/\mathfrak{n}_{\mathrm{in}})$ (0.390 against 0.368 for the escape from aluminum). Not included: silicon's anisotropy ($\bar{t}_{\mathrm{esc}} = 0.327$–$0.405$ between its [100] and [111] speeds), interlayers (normal-incidence minimum 0.844 for amorphous SiO_2), partial diffuseness; room-temperature aluminum constants ($\bar{t}_{\mathrm{esc}} = 0.327$).	Sect. 20.3

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Lumped film absorption (no box), Eqs. (20.31) and (20.32)	[142, Eq. (17)]	Published form: escape unabsorbed with probability $\exp(-4d_\mathrm{f}/\mathfrak{l}_\mathrm{f})$ after one round trip. Chain: round trips at fixed angle Eq. (20.30) $\to$ direction randomized at every pass, survival $2E_3(2d_\mathrm{f}/\mathfrak{l}_\mathrm{f})$, escape $\bar{t}_\mathrm{esc}$, Eq. (20.31) $\to \bar{t}_\mathrm{esc} = 1$ and the angle average moved into the exponent ($\langle 1/\cos\theta\rangle = 2$) gives the published form. Escape time Eq. (20.32) from the same counting in time. $\mathfrak{T}$: kinetic (rate) description of the film phonons, frequency-independent interface.	At $\bar{t}_\mathrm{esc} = 1$: within 10 % of Eq. (20.31) for $d_\mathrm{f}/\mathfrak{l}_\mathrm{f} < 0.074$ and above 0.55, above it everywhere (at most 14 %, at $d_\mathrm{f}/\mathfrak{l}_\mathrm{f} = 0.24$). Film randomized in direction and branch faster than it escapes; wavelengths below d_f; no loss at the top face.	**factor $1/\bar{t}_\mathrm{esc} = 2.72$ in the thin-film limit for aluminum on silicon, 1.82 for** 60 nm **at $\mathfrak{l}_\mathrm{f} = v_s t_0$.** The escape probability: the published form returns every unabsorbed phonon after one round trip, while $\bar{t}_\mathrm{esc} = 0.368$ (acoustic) or 0.282 (diffuse) makes it return $1/\bar{t}_\mathrm{esc}$ times; $t_l = 0.187$–0.244 ns for 60 nm (0.223–0.281 ns with room-temperature aluminum constants). Without in-film mixing, the 56 % of the flux beyond the transverse critical angle never escapes.	Sect. 20.3
Pair-breaking efficiency (Limit Check 20.7), Eq. (20.38); Eqs. (20.35) to (20.37), (20.39) and (20.40)	[145, pp. 2, 5, Fig. 4]; [146]	Published form: $\eta_s = 2\Delta/h\nu_s$ for $2\Delta \le h\nu_s \le 4\Delta$, independent of the phonon loss time; $\eta_s = 0.57$–0.6 in infinite volume. Chain: golden rule with the case-I factors Eq. (7.29) at $v_\mathrm{F} q \gg E_\mathrm{ath} \to$ split Eq. (20.35), pair-breaking time Eq. (20.36) (normal width times $\mathfrak{g}_\mathrm{I}$, width $\propto E_\mathrm{ath}$ for $\alpha^2F = b_\alpha E_\mathrm{ath}^2$), emission density Eq. (20.37) $\to$ a pair-breaking emission needs $E \ge 3\Delta_\mathrm{BCS} \to N_\mathrm{qp} = 2$ below $4\Delta_\mathrm{BCS}$, Eq. (20.38) $\to$ above: lumped film with exponential escape after t_l, Eq. (20.39) $\to$ mean-number recursion Eq. (20.40) (dilute quasiparticles, $T = 0$, no recombination during the cascade). $\mathfrak{T}$: detector stage (kinetics, first order in each rate).	Exact below $4\Delta_\mathrm{BCS}$ for $T \ll T_c$, any film and any α^2F. Above: lumped film, $T = 0$, $\alpha^2F \propto E_\mathrm{ath}^2$ (aluminum up to about 10 meV); phonon and photon seeds differ by at most 0.007 (at $5\Delta_\mathrm{BCS}$). Plot (Fig. 20.12): $x = E_\mathrm{ath}/\Delta_\mathrm{BCS}$, 2–10; $y = Y_\mathrm{qp}$; $2\Delta_\mathrm{BCS}/E_\mathrm{ath}$ dashed; cascade (photon seed) at $t_l/t_0 = 0.5$, 1, 2, 4, 8, ∞; kinetic-equation points read off Fig. 4 of the source (digitized to ± 0.005); shaded $E_\mathrm{ath} < 4\Delta_\mathrm{BCS}$.	**≤ 0.011 for $t_l/t_0 \le 4$; 0.036–0.041 at $t_l/t_0 = 8$ (kinetic equations above the cascade, 5–10Δ_BCS).** The terms of the Chang–Scalapino equations that the cascade drops: finite temperature ($T/T_c = 0.1$), measured α^2F, recombination and reabsorption of $2\Delta_\mathrm{BCS}$ phonons; the readout probe adds about 2 %, and counting the excess energy of the quasiparticles about 4 %. Above $4\Delta_\mathrm{BCS}$ escape dominates: over the arrival band 13.6–43.8 Δ_BCS the yield falls by 21.5 % at $t_l/t_0 = 0.5$ and by 1.9 % at 8, and lies 47–58 % below the no-escape value at $t_l/t_0 = 0.5$.	Sect. 20.4

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Collection ratio of Luke phonons, one film type (no box), Eq. (20.45); mean-field kernels Eqs. (20.43) and (20.44)	—	Published form: none. Linear transport → collection kernels Eq. (20.41) and the functionals Eq. (20.42) → mean-field limit (diffusion across the crystal faster than absorption, diffuse grey faces): position-independent absorption rate Eq. (20.43), triangular solve of Eq. (20.44) for the time-integrated spectrum → one film type, primaries absorbed with their mean yield $\bar{Y}_{\mathrm{prim}}$, absorbed fraction common to all phonons above $2\Delta_{\mathrm{BCS}}$ → $\mathfrak{r}_{\mathrm{coll}} - 1 = s_{\mathrm{ng}} - (1 - \kappa_{\mathrm{sub}})s_{\mathrm{sub}}$. $\mathfrak{T}$: incoherent propagation, $n_A = 2$ for three-phonon decay; lumped films with the Kaplan-cascade yield of Sect. 20.4 and film escape at the t_l/t_0 of Sect. 20.3; every phonon a film emits after its first pair break is returned to the crystal (energy conserved).	Diffusion time across the crystal short against the absorption time (15 µs for a 4 mm silicon slab with 10 % film coverage); faces grey and diffuse; one film type; absorption per bounce independent of energy above threshold (exact for thick films, within 0.011 with escape). No Limit Check; Fig. 20.13 shows the estimate: energy conserved, one or two films, 60 nm to thick, 0.81–1.02 over 0.75–4.7 meV and 0.82–1.01 at 0.84 meV; escape moves one film by < 0.002 at a bare-face loss of 10^{-3} (numerical, not exact); a uniform sink of 0.3 per hit keeps it ≤ 1.03; the lumped film's selective sink gives an upper envelope 1.05–1.10 (300 nm) and 1.30–1.53 (60 nm); 1.128 is not reached by the transport estimate, and not excluded until computed kernels test real detectors (Appendix E.25); the two-bias measurement of Sect. 19.5 decides.	**0.008 from the extra bare-face loss of a two-film detector; 10^{-3} for a thick film**. A second film type that absorbs only part of the Luke spectrum lengthens its dwell time ($\mathfrak{r}_{\mathrm{coll}} = 0.992$ with sub-gap phonons absorbed) and adds its own near-gap yield (up to +0.09 when film phonons are deleted; with them returned the ratio stays ≤ 1.02); the fate of phonons that leave a film (returned here; a selective sink would raise the ratio to the envelope above); position dependence of the kernels (position spread) beyond the mean field; near-gap transparency of thin films.	Sect. 20.5

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
High-voltage readout ceiling (no box), Eq. (20.50)	—	Published form: none; the phonon energy of a biased detector is $E_{\mathrm{ph}} = E_R + N_{\mathrm{eh}} e V_{\mathrm{bias}}$ with $\langle N_{\mathrm{eh}} \rangle = Y_q E_R / 3.8\,\mathrm{eV}$ [127, Eq. (1)]. Chain: equal total phonon energy fixes the electron-recoil energy $E_R(1 + g Y_q')/(1 + g) \rightarrow$ difference of mean detected pairs $f_{\mathrm{det}}(E_R/\varepsilon_{\mathrm{eh}})(1 - Y_q')/(1 + g) \rightarrow$ the Luke share at fixed E_{ph} is linear in $N_{\mathrm{det}} \rightarrow$ variances $F_{\mathrm{eff}} \langle N_{\mathrm{det}} \rangle$ with the cascade and partial-charge terms of Eq. (20.7) bracketed by the binomial and mean-charge limits $\rightarrow$ separation Eq. (20.46). $\mathfrak{T}$: Gaussian separation measure; primary phonon energy equal to the recoil energy (Limit Check 19.4).	Ideal readout of the Luke share (contrast one, no baseline noise, no position spread); event position spread of the collection neglected (the position term of Eq. (20.7), $1.2 \times 10^{-4} \bar{N}_{\mathrm{eh}}$ at 95 % collection, is below 1 % of the Fano factor at ten pairs); measured Y_q, F_{NR} at 0.1–0.89 keV. No Limit Check; Fig. 20.14 shows it against E_R.	**factor 1.23 between the binomial and mean-charge limits at** 220 eV **(1.22 against 1.50)**. The spread of induced charge per pair, between all-or-nothing and equal partial charges; baseline noise and position spread lower the ceiling further; the yield and Fano factor below the binary threshold are measured, not derived (Sect. 19.2).	Sect. 20.5
Mean inverse speed of the standard halo (Limit Check 21.1), Eqs. (D.3) and (D.4)	[7, Eqs. (B.4), (B.5)]	Published form: the same two branches at a fixed Earth speed, with $v_0 = 230$, $v_{\mathrm{esc}} = 600$ and $v_{\mathrm{E}} = 240\,\mathrm{km\,s^{-1}}$. Chain: halo functional Eq. (21.1) with an angle-averaged response $\rightarrow$ angular average of the energy delta function, $\Theta(v - v_{\mathrm{min}})/2qv$, and integration by parts, Eq. (21.5), with η of Eq. (1.28) $\rightarrow$ laboratory distribution Eq. (1.8) of the truncated Maxwellian Eq. (1.3) $\rightarrow$ the integrals of Appendix D.1 $\rightarrow$ the exposure average of η over $\lvert \boldsymbol{v}_{\mathrm{E}}(t) \rvert$ (Appendix A.8) replaced by its value at $v_{\mathrm{E}}^{\mathrm{ref}} = 250.5\,\mathrm{km\,s^{-1}}$. $\mathfrak{T}$: not applicable (halo side only).	Exact for the standard halo at one Earth speed. As a stand-in for the yearly average: the average exceeds it by less than 1 % for $v_{\mathrm{min}} \leq 430.3\,\mathrm{km\,s^{-1}}$ and by less than 10 % for $v_{\mathrm{min}} \leq 735.6\,\mathrm{km\,s^{-1}}$. Plot (Fig. 21.2): $x = v_{\mathrm{min}}$, 0–812 $\mathrm{km\,s^{-1}}$; top $y = \eta$ in units of $1/c$, closed form at $v_{\mathrm{E}}^{\mathrm{ref}}$ (solid), yearly average with the exact $\lvert \boldsymbol{v}_{\mathrm{E}}(t) \rvert$ on 7305 days (dashed), no escape cut (dotted); bottom, ratio of the yearly average to the closed form and the second-order correction; shaded $v_{\mathrm{min}} \leq 735.6\,\mathrm{km\,s^{-1}}$.	+1 % **at** 430 $\mathrm{km\,s^{-1}}$**,** +10 % **at** 736 $\mathrm{km\,s^{-1}}$**,** +20 % **at** 760 $\mathrm{km\,s^{-1}}$. The second-order term in $\delta v_{\mathrm{E}} = \lvert \boldsymbol{v}_{\mathrm{E}}(t) \rvert - v_{\mathrm{E}}^{\mathrm{ref}}$, Eq. (21.7), with $\langle \delta v_{\mathrm{E}} \rangle = 1.61\,\mathrm{km\,s^{-1}}$ and $\langle \delta v_{\mathrm{E}}^2 \rangle = 108.3\,\mathrm{km^2\,s^{-2}}$; within 1 % of the yearly average up to 786.3 $\mathrm{km\,s^{-1}}$. Above $v_{\mathrm{max}} = 794.5\,\mathrm{km\,s^{-1}}$ the closed form vanishes while the yearly average stays positive up to 810.4 $\mathrm{km\,s^{-1}}$.	Sect. 21.1

Formula	Source (eq.)	Approximation chain	Validity	Leading correction	Section
Halo-independent bound for a speed-only response (Limit Check 21.2), Eq. (21.12); Eqs. (21.11) and (21.13)	[147, Eqs. (A1)–(A2)]; [148, Sect. 2]; [149, Eqs. (2.17), (2.18)]	Published form: a limit on $\rho_\chi \sigma g_1 / m_\chi$ for the step halo $g = g_1 \Theta(v_1 - v)$, one bin; for electrons, counts linear in steps of $\tilde{\eta}$. Chain: halo functional Eq. (21.3) $\rightarrow$ counts $\int \mathsf{k}_i \, \mathrm{d}\Xi$ with the stream kernel Eq. (21.9) $\rightarrow$ semi-infinite program Eq. (C.111) for $\tilde{\eta}(u_0)$, Eq. (21.10) $\rightarrow$ angle-averaged response, fixed $\boldsymbol{v}_\mathrm{E}$: $\mathsf{k}_i = \mathcal{A}_i(v)/v$ with the nondecreasing step column Eq. (21.11) $\rightarrow$ one stream at u_0 and the one-bin certificate $\lambda_j = 1/\mathcal{A}_j(u_0)$ give Eq. (21.12) exactly (Appendix D.46). Directional: circle average Eq. (D.248) $\rightarrow$ Eq. (21.13), sidereal mean over declination, galactic support. $\mathfrak{T}$: the label of the response tables that give k_i; the bound itself adds none.	Exact for a response independent of $\hat{\boldsymbol{q}}$ with nonnegative bin weights and constant $\boldsymbol{v}_\mathrm{E}$. Silicon at 2 MeV, one- and two-pair bins of the cascade of Sect. 18.3, crystal at 45° N with [001] up, integrated over sidereal time, model anisotropy of Sect. 11.7: within 10 % of the linear program for every u_0 on a $5\,\mathrm{km\,s^{-1}}$ grid from 340 to $790\,\mathrm{km\,s^{-1}}$ except 340, 345, 365 and $470\,\mathrm{km\,s^{-1}}$; within 2.0 % for $u_0 \geq 560\,\mathrm{km\,s^{-1}}$. Plot (Fig. 21.5): $x = u_0$, 340–800 $\mathrm{km\,s^{-1}}$; y = program over formula; formula (1), program (dots), first-order correction (dashed); shaded where within 10 %.	**+2 to +15 % below** $560\,\mathrm{km\,s^{-1}}$ **(sampled from** $340\,\mathrm{km\,s^{-1}}$**), at most +2.0 % above; +0.4 % at** $790\,\mathrm{km\,s^{-1}}$. First order in the anisotropy: the smallest sidereal-mean kernel of the tightest bin among the declinations allowed at u_0, which reproduces the program to 1.8 %. Largest near the onset speed of a bin, where every reachable pair has $v_\mathrm{min} \approx u_0$ and carries the full anisotropy; a search binned in sidereal time keeps the crystal-frame anisotropy (kernel down to 0.33 of its average in the model).	Sect. 21.2

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